Triadic Modular Holography

Systematic exposition of the formal kernel,
its extensions and derivations.

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Foundations, objectives and structural results

Triadic Modular Holography studies the production of numerical, geometric and operator structures from oriented discrete arithmetic. Its initial object is a grid constructed from the integers one through nine. Addition and multiplication determine two evaluations of every position; modular reductions preserve remainders and quotients; a ternary unit selects operational regimes, and a transport system orders their transitions and memory. Compatible iteration of these operations constitutes the enriched state and its coinductive extension.

The governing thesis attributes to number a structure preceding its scalar evaluation. Position, orientation, carry, incidences and the chronology of operations remain in the state producing the value. The Archimedean realization is obtained by contracting compatible cylinders; the conjugate realization preserves boundary incidence. Both are readings of the same construction. In this precise sense, the real line appears as the result of double projection.

This work systematically develops that genealogy. Its main results are organized in four levels: coinductive regional generation of constants; articulation between evaluation and exceptional incidence; conservation and information-recovery laws; and Dimensional Mechanics realizations. Cardinal closure of the generated continuum and terminal realizations rest on this architecture, with the domains, operators and hypotheses established in their theorems.

Operational definition of the formal kernel

All Possible Paths, abbreviated APP, denotes the arithmetic support of trajectories on nine rows and nine columns. Its principal matrix is constructed by multiplying row and column indices and applying the positive digital root. If $D_9 = \{1, \dots, 9\}$, its positions are $(i, j) \in D_9^2$, and the multiplicative entry is

$$\Pi(i, j) = 1 + ((ij - 1) \bmod 9).$$

The additive sheet evaluates $i + j$ on the same positions. On each sheet, the pair consisting of positive remainder and quotient restores the integer: $n = \rho_9(n) + 9q_9(n)$. The symbol 9 represents the zero class within the positive section. This convention makes the modular boundary visible and preserves its arithmetic lift (1.3).

The TRIT is the oriented ternary unit, with values $-1, 0, +1$. It acts as a reader of operational classes and a regime selector. Its arithmetic decomposition preserves the pair $n = r_3(n) + 3c_3(n)$, where $r_3(n)$ is the balanced representative and $c_3(n)$ its quotient. Orientation and mode selection subsequently acquire quadratic and temporal realizations, defined on their respective carriers. Composition between moduli three and nine transports both quotients; the identity (1.30) determines that transport exactly.

The *Topological Phase Kernel*, TPK, is the topological kernel of phase. It combines selection, displacement, emission, update, lifting and return operators. A transition receives a determined state, decides the admissible operation and preserves the register needed to continue it. The enriched state thus combines APP position, active sheet, tritic orientation, remainders, carries, phase, turns and accumulated incidences. Definitions and emission rules are developed in Sections 1.1, 1.2 and 1.3.

Geometric representations are constructed on these data. APP translations determine a Cayley graph; periodic identification of boundaries produces its toroidal realization. The Kähler structure corresponds to the differential realization with declared metric and complex structure. The central block $\left(\begin{smallmatrix} 7 & 2 \\ 2 & 7 \end{smallmatrix}\right)$ admits a determinantal normalization preserving a two-dimensional Minkowski form. Each realization preserves the object and normalization from which it arises 3.1.2.3–3.1.5.

Regional generation and coinductive extension

The first governing result consists in passing from the finite domain of executions to compatible generation at arbitrary depth. Initial enumeration distinguishes 104 976 configurations, 468 distinct emissions and 243 admissible ternary words within the $729 = 3^6$ possible words. The corresponding orbital action produces 43 orbits. Each cardinal counts a defined object: configurations, emissions, words or orbits.

The lifts extend the words according to the sequence

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9.$$

The first two lifts use L_0; subsequent ones use L_1 and preserve the register when changing regime. The boundary R_{36} opens coinductive continuation, and G_9 combines the nine local realizations. The holonomy Γ_9 acts on the enriched state. Its phase reduction is a subsequent reading.

The calendar simultaneously preserves nonadic phase and the dodecaphase register through

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0.$$

The carry of this extension determines the difference between a phase return and a state update. After nine transitions, the initial phase reappears and turn memory increases. Theorem 1.44 and the development of holonomy 8 specify the composition law.

The regions of π, φ and e are obtained through emitters, lifts, filters and survival conditions of the same TPK. Their compatible cylinders determine expansions to any requested depth. The five regions of π, the φ axis, the joint continuations and the mirror involution are distinguished by their operations and registers. Machin, Perron–Fibonacci and exponential-semigroup characterizations enter as subsequent identifications or certificates of the constructed outputs. Generation order remains fixed by the regional operators.

Coupling constant and fine structure

The holographic constant of arithmetic-geometric coupling is the generated dodecaphase register K, with its window order, phase origin, orientation and incidences. Its determination is a central result: the conversion clock, nonadic return and phase–charge reader produce the regional descriptor; incidence and heterotypy conditions select the register in the declared terminal domain. Chapter 11.3 separately preserves the descriptor, block repertoire, selection conditions and inversion of the signed register.

The mathematical distinctiveness of K is expressed through the operations it articulates. Its positional reading enters arithmetic closure; its incidences determine directions and projectors; its full orbit supplies a reference for operator recovery. The rational reading κ, orbital frame O_K and memory operators receive their own definitions. This distinction reveals the effective composition between number, geometry and restitution.

The regional coordinates and dodecaphase register jointly determine the fine-structure constant. The integer route normalizes blocks with their carry and preserves the boundary when extending execution. The analytic route develops a complete function, with a recurrence for all its coefficients, and proves compatibility with the generated pre-coordinate. The analytic extension is determined within its declared class from the same state; both routes express a common coinductive fourfold relation $(\pi, \varphi, e, \alpha)$. The internal order distinguishes prior regional readings, selection of K, dodecaphase closure and correlated analytic representation 11.6.4.2.

Double projection and exceptional incidence

The second governing thesis is the joint construction of the continuum. Compatible extensions constitute an inverse system; their cylinders determine Archimedean evaluation. The other projection preserves the incidences, orientations and boundary marks supporting exceptional realizations. Contraction of the reading interval and persistence of that incidence describe simultaneous properties of the same state.

The spectral construction, solenoidal balance, gauge incidence, cohomological coherence and determinant line are obtained correlatively from the common system. Their operators commute with refinements on the proved domains. This unity supplies the content of Chapters 5 and 6 and governs subsequent transports.

The exceptional chain develops Paley matrices, Witt incidences, Golay codes and lattice gluings. The marked Leech neighbour is characterized through integrality, evenness, unimodularity and root exclusion. Its extension through the lattice algebra, oscillators, cocycle and twisted sector leads to the Moonshine module. The Monster then acts on the resulting vertex operator algebra. These results are specified in Theorems 21.70 and 21.71. Orthogonal reduction $12 \to 11 \to 10$, isotropic reduction $26 \to 24$ and momentum-winding duality are coordinated through their quotient maps. The relation with M-theory preserves this algebraic realization and the fields, actions and additional conditions of its physical realization.

Evolutionary conservation and information recovery

Evolutionary conservation of unity is proved through maps preserving measure or norm and inverses restoring transported data. The holographic extreme and mean ratio is developed through the completion $1000 = 729 + 271$, transfer between scales and conservation of normalized unity. Carry, transport memory, boundary incidence and the operator archive have complementary functions within that articulation.

The bilateral law establishes an identity valid for two isometric transports between Hilbert spaces, in finite or infinite dimension. In its nonadic specialization, the two channels Q_- and Q_+ satisfy

$$9\|Q_- v\|^2 + 8\|Q_+ v\|^2 = 17 \cdot 73\, \|v\|^2.$$

The normalized channels constitute an isometry; its adjoint recovers the state and has norm one. The law also identifies which readers remain invariant and how the others transform. Recovery at arbitrary depth includes a uniform bound for the terminal component and a convergent inverse series 147.1, 147.7.

Register reciprocity extends this conservation to unit-covolume deformations, dual lattices, Plücker minors and spectral forms. The flow direction arises from the generated register; regional ratios subsequently act as marks. The proofs determine both the preserved invariants and the orientations needed to invert the geometric reading.

Dimensional Mechanics realizations

Dimensional Mechanics develops physical realizations in the order of their antecedents. The fine-structure coordinate and regional readings enter the action scale. The local determinant and its order-sixteen quotient fix the valuation producing the decade -34; forward and return sections distinguish the two Planck readings. Angle and counter-angle are constructed with those coordinates and preserve their conversions between radians and degrees.

Hexadic–octadic incidence produces the channels $90/120$. Their evaluation on the angular pair determines the Barbero–Immirzi functional; angular–Witt transport preserves the relation with mixing coordinates. The vacuum constitutive response reconstructs its electric and magnetic channels, while the gravitational section relates action to areal unity. Boltzmann transduction expresses energy per decision relative to the declared thermal section. Each constant preserves its structural definition, units and evaluation operation.

The electronic structure lies at the unique fixed centre of the atlas, $(5, 5)$. The remaining material realizations are organized through routes, orbits and multisections. The atlas of eighty-one cells, fifty-six route types and thirteen multisections precede the mass readers. The composition extension $\to$ route $\to$ record $\to$ signature $\to$ character $\to$ towers preserves the provenance of each observable. Metrological comparisons subsequently evaluate the corresponding representations and normalizations.

Cardinal closure and regime of proofs

Generation of the continuum receives an explicit cardinal closure. In the genealogical closure $\mathfrak{G}$ defined by the treatise's rules of terms and definability, condensation places every internal rational cut at a countable stage. The code $j(r) = \omega\beta(r) + n(r)$ distinguishes their values; encoding countable well-orders supplies the reverse injection. Theorem 19.27 concludes

$$\mathfrak{G}_{\mathrm{HMT}}(h, m_\infty) \models 2^{\aleph_0} = \aleph_1,$$

and quantifies all internal subsets of the generated line. The structure and parameters of this closure arise from the described genealogy; its representation by $L[h, m_\infty]$ subsequently identifies the theorem's domain.

The exposition distinguishes arithmetic identities, exhaustive enumerations of finite domains, compatibility theorems and passages to the limit. Verified formalizations and numerical certificates accompany their statements with precise quantifiers and dependencies. Physical realizations add the dimensional charts, couplings and observation conditions they use. This differentiation makes each result's own strength apparent and allows its composition to be followed within a single construction.

Preface

Complex interaction between two simple machines

This document compiles the extensive technical material—for future publications—concerning the results obtained and the mathematical-physical developments carried out between 2023 and 2026 through the use of Large Language Model (LLM) tools. From this process arose what we designate, respectively, Triadic Modular Holography and Dimensional Mechanics (HMT–MD). The origin of this project, however, dates back to the beginning of the twenty-first century, specifically to the study of the discovery underlying the artifact invented by Oumar Haidara Fall in 2001. It should be noted here that this artifact was brought to the attention of the various competent academic and civil authorities in the respective places of origin of both authors.

The accumulated material and the knowledge progressively acquired over more than two decades of remark, study, and analysis, concerning both the artifact and its phenomenology, constituted the operational and informational basis that we subsequently applied to artificial-intelligence models.

Therefore, the HMT–MD result presented here, together with its derivations and proofs, is preceded by an empirical genealogy that, *a priori*, might at best appear counterintuitive to the technical specialist—for it contravenes the postulates and predictions of the theoretical-interpretive framework in which the artifact is inscribed by historical tradition, namely, classical mechanics. Nevertheless, its metrological verification is decisive. Furthermore, once cognitive prejudices or confirmation biases are set aside, it becomes directly demonstrable by the facts.

Finally, one might consider something that remains surprising as a historical paradox: when we began reasoning with "thinking machines" in "the age of artificial intelligence," we also had the serendipity to reveal an interaction unprecedented in ordinary mechanics and were thereby compelled to reconsider the relation between two simple machines: the lever and the wheel.

Generating theorem of Triadic Modular Holography

The proof begins in the finite domain $\mathcal{D}_9 = \{1, \dots, 9\}$. Two oriented cursors traverse, respectively, the additive and multiplicative sheets. Euclidean division preserves residue and quotient, and the APP catalog produces

$$\sum_{i,j\in\mathcal{D}_9} R^{+}_{ij} = 405, \qquad \sum_{i,j\in\mathcal{D}_9} R^{\times}_{ij} = 459, \qquad 459 - 405 = 54.$$

The coordinated transport $c(a) = a \bmod 9 : \mathcal{D}_9 \to \{0, \dots, 8\}$ changes the chart and preserves the class. The effective composition of APP, TRIT and TPK preserves at each step sheet, regime, residue, quotient, carry, direction, boundary and memory. The exhaustive catalog establishes the initial finite chain and supplies its executable certificate:

$$\boxed{104\,976 \text{ seed pairs} \xrightarrow{54 \text{ steps}} 468\, U_6 \xrightarrow{\rho_3} 243\, w_6 \subset \mathbb{F}_3^6, \qquad |\mathbb{F}_3^6| = 729.} \tag{1}$$

The fiber identity

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$$

proves the exhaustiveness of the first map. The numbers 468, 243 and 729 count emissions, realized words and the ternary ambient space, respectively; their types remain separated by the arrows of (1).

Theorem 0.1 (Unique generating chain)**.** *Let $\mathfrak{H}_9$ be the structure formed by APP, TRIT, TPK and its enriched fiber. APP executes the sum and product sheets and preserves Euclidean division*

$$i + j = R^{+}_{ij} + 9q^{+}_{ij}, \qquad ij = R^{\times}_{ij} + 9q^{\times}_{ij}.$$

TRIT orients each transition in the elliptic, parabolic and hyperbolic regimes. TPK composes

$$\boxed{\mathcal{U}_t = \mathrm{Upd}_t \circ \mathrm{Tra}_t \circ \mathrm{Sel}_t : X^{\mathrm{enr}}_{\mathrm{TPK}} \longrightarrow X^{\mathrm{enr}}_{\mathrm{TPK}}.}$$

Sel_t *lifts the tritic selector,* Tra_t *lifts cardinal transport and* Upd_t *updates residue, quotient, carry, signature, cylinder, boundary, register and memory without erasing provenance. Nonadic holonomy prolongs words by*

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9,$$

and satisfies

$$\Gamma_{9,t} = \mathcal{U}_{t+8} \circ \cdots \circ \mathcal{U}_t, \qquad g(t+9) = g(t), \qquad q_{\mathrm{mem}}(\Gamma_{9,t}\widehat{x}_t) = q_{\mathrm{mem}}(\widehat{x}_t) + 1.$$

In general, $\Gamma_{9,t}\widehat{x}_t \neq \widehat{x}_t$: the return preserves the phase and increases memory. The operator order is $\mathcal{U}_t \to \Gamma_9 \to K_{\mathrm{ph}}$; K_{ph} is a subsequent reduced publication, not the generator. The compatible truncations form a nonempty inverse system and define a state x_∞ of arbitrary depth.

This state generates a unique discrete structure of the continuum and two natural publications:

$$x_\infty \xrightarrow{(\mathcal{P}_{\mathrm{arq}}, \mathcal{Q}_\infty)} \mathbb{R} \times (\mathscr{C}^{\mathrm{cal}}_W)^{\mathbb{N}}. \tag{2}$$

The first contracts compatible cylinders down to a real value; the second preserves incidence and genealogy through the constructions of Paley, Witt, Golay, Leech, $V^\natural$ and the Monster group. On the same source the four correlated characters are published

$$(\pi_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}}),$$

and, through typed transductions, the laws of action, matter, mass, gravity and information of Dimensional Mechanics.

Proposition 0.2 (Intrinsic reconstruction and evidentiary self-sufficiency)**.** *Each result y published in this volume is reconstructed through the local mathematical receipt*

$$\mathscr{R}_y = \big(D_y, A_y, \tau_y, U_y, C_y, X_y^{\mathrm{enr}}, y_{\mathrm{HMT}}, \mathrm{Rec}_y, \mathcal{F}_y\big),$$

where D_y is the domain; A_y specifies the APP sheets and operations; τ_y the TRIT regime and orientation; U_y the TPK composition with domain, codomain and action; C_y the preserved residue, quotient, carry, sheet, route, boundary, memory and survival data; X_y^{enr} the resident enriched state; y_{HMT} the output prior to any external identification; Rec_y *subsequent conventional recognition; and $\mathcal{F}_y$ a falsifier of the substantive arrow. The sequence*

$$\mathcal{D}_9 \longrightarrow \Omega_{\mathrm{seed}} \longrightarrow \mathcal{U}_6^{\mathrm{cat}} \longrightarrow \mathcal{W}_6 \longrightarrow X_\infty^{\mathrm{enr}} \longrightarrow C_{\mathrm{cont}}^{\mathrm{disc}}$$

is defined and proved by the equations, theorems and receipts printed in the volume itself. The appended reproducible programs evaluate those identities again; they supply neither a missing premise nor a target value.

The primary falsifiers are: failure of the fiber identity of (1)*; loss of any of the fields of C_y under U_y; incompatibility between extension and truncation; emptiness of the inverse limit; lack of contraction or internal surjectivity of the rational cylinders; failure of naturality in the exceptional branch; and, for a terminal publication, failure of the hypothesis defining its exact domain. Consequently, the validity of an output is decided within its domain and by its explicit falsifier, not by an analogy, a subsequent nomenclature or a metrological evaluation.*

Uniqueness of the generating system and typing of its projections. There is only one APP, followed by TRIT and by the TPK with enriched state. APP produces the two sheets and their registers; TRIT orients them; the TPK composes the routes and accumulates sheet, opposition, residue, quotient, carry, boundary and memory in $X_{\mathrm{TPK}}^{\mathrm{enr}}$. The signed chart is a projection of that same terminal state:

$$R_{\mathrm{sgn}} : X_{\mathrm{TPK}}^{\mathrm{term,enr}} \longrightarrow \mathcal{U}_{12}^{\mathrm{sgn}}, \qquad U_{12}^{\mathrm{sgn}} = R_{\mathrm{sgn}}(x_{\mathrm{term}}), \qquad K = \frac{1}{4}\mathcal{H}_{12}U_{12}^{\mathrm{sgn}}, \qquad \mathcal{H}_{12}^2 = 4I_{12}.$$

The last two morphisms are integral and reversible on their image. No direct APP emitter toward U_{12}^{sgn} is postulated, because such an arrow would skip TRIT and TPK: the correct domain of R_{sgn} is the already generated enriched state. Shadows that forget the genealogical register or memory are subsequent projections, not alternative mathematical profiles. This unique composition excludes from its inputs $U, K, D_3K, D_4K, Q, \alpha$ and every target expansion.

Number and particle. An HMT number is a compatible section $\boldsymbol{x} = (x_n)_{n\geq 1} \in \varprojlim_n X_n^{\mathrm{enr}}$ with sheet, orientation, residue, quotient, carry, route, boundary and memory. Its Archimedean value is the unique intersection $\mathrm{val}(\boldsymbol{x}) = \bigcap_n I_n(x_n)$; the quotient by equality of value publishes the real number and its fiber preserves the genealogies that realize it. An HMT–MD particle is a class of persistent sections of the enriched extension: the state preserves identity, the route records its observable history and subsequent readers publish mass, charge, spin and statistics in their physical codomains.

Typing of the exceptional corridor. The Hadamard and Paley readings start from the same terminal state, but are not serialized with one another. With $X_\infty^{\mathrm{enr}} := \varprojlim_n X_n^{\mathrm{enr}}$, there are distinct maps

$$\mathrm{Pub}_\alpha : X_\infty^{\mathrm{enr}} \longrightarrow \mathbb{R}_{>0}, \quad x_\infty \longmapsto \alpha_{\mathrm{HMT}}, \qquad \mathrm{Inc}_\alpha : X_\infty^{\mathrm{enr}} \longrightarrow A_2^{12}, \quad x_\infty \longmapsto v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}. \tag{3}$$

They share genealogy, but equality between their outputs is not postulated. The exceptional branch splits after C_W:

$$\begin{aligned} C_P &\longrightarrow A_W = C_P \bmod 3 \longrightarrow C_W = [12,6,6]_3 \\ &\longrightarrow \begin{cases} W_{12} \xrightarrow{\widehat{\iota}_{D,\ell}} W_{24} \subset \mathcal{G}_{24}, \\ N(C_W) \xrightarrow{\mathrm{Nbr}_{v_\alpha}} N_\alpha \cong \Lambda_{24} \longrightarrow V_{\Lambda_{24}} \longrightarrow V^\natural. \end{cases} \end{aligned} \tag{4}$$

Here

$$\mathcal{O}_{24} := \{O \subset [24] : |O| = 8,\ \mathbf{1}_O \in \mathcal{G}_{24}\}, \qquad \mathcal{H}(D) := \{O \cap D : O \in \mathcal{O}_{24},\ |O \cap D| = 6\}. \tag{5}$$

With $\ell : W_{12} \simeq (D, \mathcal{H}(D))$ fixed, $\widehat{\iota}_{D,\ell}(H) = O_H$, where $O_H \cap D = \ell(H)$. Such an octad is unique: two choices would contain the same five-point subset, contradicting $S(5,8,24)$. The ternary branch glues $N(A_2^{12}) = N(C_W)$; the index-three neighbor determined by v_α produces Leech. Finally,

$$\mathrm{Aut}(V^\natural) \cong \mathbb{M}, \qquad a : \mathbb{M} \times V^\natural \longrightarrow V^\natural, \qquad T_g(\tau) = \mathrm{Tr}_{V^\natural}(g q^{L_0 - 1}). \tag{6}$$

The identities $v_\alpha^2 = 54$ and $[q]J = 196884$ are scalar evaluations; neither defines the action a.

Parallel realization of T-duality. T-duality does not continue the Moonshine branch: it acts in its own domain

$$\mathscr{D}_{\mathrm{dual}} = \mathbb{Z}^2 \times \mathbb{R}_{>0}, \qquad \mathsf{T}(m, w, R_0) = (w, m, R_0^{-1}), \qquad \mathsf{T}^2 = \mathrm{id}, \qquad E_{R_0}(m,w) = E_{R_0^{-1}}(w,m). \tag{7}$$

Proof. The census of (1) constructs the finite catalog and selects, without target values, the three initial blocks w_6. The oriented transition register of the complete TPK state makes explicit the twelve post-seed blocks of the three published biographies. For each of the two transition intervals, the six independent rows extracted from those three biographies form matrices $X_\ell, Y_\ell \in M_6(\mathbb{F}_3)$, $\ell \in \{0,1\}$, whose rows are, respectively, the departure and arrival states of the register. Exact calculation on this register proves $\det X_0 = 1$, $\det X_1 = -1$ and $X_\ell L_\ell = Y_\ell$; it therefore reconstructs uniquely $L_\ell = X_\ell^{-1} Y_\ell$. The calendar (L_0, L_0, L_1, L_1) is the only one of the sixteen binary calendars that simultaneously preserves the three biographies selected by the fiber invariants; the one hundred and forty-four unit perturbations destroy that triple recognition. Therefore, the transports, the twelve blocks and their calendar are deduced in the unique APP–TRIT–TPK composition: APP produces the sheets and the starting register, TRIT fixes the orientation of each transition and TPK prolongs the register with memory. The seed census and the enriched state are successive stages of the same generator, not rival profiles or different theories.

The nonsplit extension

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0$$

separates phase return and state return. The truncations remove the last update and preserve the prefix; König's lemma produces an infinite compatible branch. For each precision N, the recurrence and holonomy generate the section and its characters, and determine the cylinder that prolongs the preceding genealogical register. The family of prefixes satisfies

$$\rho_{N+1,N}(p_{N+1}) = p_N, \qquad p_\infty \in \varprojlim_N \{0, \ldots, 9\}^N.$$

A thousand, ten thousand or a million digits are finite projections of this unique infinite object.

The nested rational brackets $p_N^- < \chi < p_N^+$ are calculated after producing the character χ and satisfy $[p_{N+1}^-, p_{N+1}^+] \subset [p_N^-, p_N^+]$ with diameter tending to zero. They certify the generated output to any finite precision; they do not select the orbit, the transport or the character.

The three enriched turns $\gamma_-, \gamma_0, \gamma_+$ of nine emissions are distinguished by the carry $\mathrm{Carr}(\gamma_\rho) = \rho\kappa_{\mathrm{Carr}}$ and admit the three prolongations after any prefix. The limit of their recorded routes satisfies

$$\mathrm{dig}_{\mathrm{Carr}}^{\omega} : \Omega_\Gamma^{\mathrm{cal}} \xrightarrow{\sim} \{-1, 0, +1\}^{\mathbb{N}}.$$

After grouping six returns, the map

$$\beta = \mathrm{blk}_6 \circ (\rho \mapsto \rho + 1)^{\mathbb{N}} \circ \mathrm{dig}_{\mathrm{Carr}}^{\omega} : \Omega_\Gamma^{\mathrm{cal}} \xrightarrow{\sim} (\mathbb{F}_3^6)^{\mathbb{N}}$$

is a homeomorphism; its inverse concatenates the prescribed turns. The balanced forms normalized by

$$a_k + c_k = \rho_k + 3c_{k+1}, \qquad \rho_k \in \{-1, 0, +1\},$$

first produce the ordered ring $\mathbb{Z}_{\mathrm{HMT}}$, then its field of fractions $\mathbb{Q}_{\mathrm{HMT}}$ and finally

$$\mathbb{R}_{\mathrm{HMT}} := \mathrm{Cau}(\mathbb{Q}_{\mathrm{HMT}})/\asymp.$$

If $\beta(\xi) = (b_q)$ and $\nu(b) = \sum_{i=1}^{6} b_i 3^{6-i}$, the sequence

$$q_n = m + \sum_{q=0}^{n-1} \frac{\nu(b_q)}{729^{q+1}}$$

is Cauchy, and the recurrent internal partition into 729 subintervals proves surjectivity

$$P_{\mathrm{ar}} : \mathbb{Z}_{\mathrm{HMT}} \times \Omega_\Gamma^{\mathrm{cal}} \twoheadrightarrow \mathbb{R}_{\mathrm{HMT}}, \qquad (\mathbb{Z}_{\mathrm{HMT}} \times \Omega_\Gamma^{\mathrm{cal}})/\!\sim_{\mathrm{ar}} \cong \mathbb{R}_{\mathrm{HMT}}.$$

Only afterwards does uniqueness of the complete Archimedean ordered field publish $\mathbb{R}_{\mathrm{HMT}} \cong \mathbb{R}$. The inward direction subdivides $[0_{\mathrm{HMT}}, 1_{\mathrm{HMT}})$; the outward direction concatenates the prefix encoding $m \in \mathbb{Z}_{\mathrm{HMT}}$. Thus the line is covered without prior use of $\mathbb{R}$, integer part or target decimal expansion. The survival subspaces of particular characters individualize sections within this total capacity. Coverage of the continuum follows from the global completion; the nonadic recurrence generates $\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}}$, and the typed filters certify their biographies within that coverage.

The five spectral, solenoidal, gauge, cohomological and determinantal actions are calculated on the same enriched emission and share support, route, genealogical register and truncations. Their joint naturality preserves them in the limit. The perfect dichotomy classifies each analytic publication as countable or of continuum cardinality. The generated code h selects the canonical genealogical universe $M_h = L[h]$. The construction of the HMT reals identifies each number by value, cylinder, route and memory; the well-order induced by that genealogy proves the classical formula

$$M_h \models 2^{\aleph_0} = \aleph_1.$$

Since M_h is precisely the codomain generated by the continuum construction of this theorem, the terminal conclusion of the system is

$$\boxed{2^{\aleph_0} = \aleph_1.}$$

Independence in the bare language of ZFC classifies the loss of the genealogical selector h; HMT preserves that datum and determines the cardinality of its continuum. The preserved sentence is the classical continuum hypothesis, with the ordinary cardinals of M_h, and the detailed proof establishes the bridge

between compatible cylinders, genealogical well-order and cardinality. This cardinal closure is distinct from the strong logical corridor, also constructed from the genealogical continuum:

$$\begin{aligned} C_{\mathrm{cont}}^{\mathrm{disc}} &\longrightarrow \mathcal{L}_0^{\log} \xrightarrow{\Phi} \mathcal{L}_1^{\log} \longrightarrow \gamma_F \longrightarrow w(\gamma_F) \\ &\longrightarrow \mathsf{NoEscape}_{24} \longrightarrow \mathsf{AntiDiag}_{369} \longrightarrow \mathsf{GödelIII}_{\mathrm{HMT}} \\ &\longrightarrow \mathsf{CycleCriterion}_{\mathrm{ZFC}}, \end{aligned}$$

which publishes, in its typed domain, the criterion

$$\boxed{\neg\,\mathrm{Consis}(\mathrm{ZFC}) \Longleftrightarrow \exists C\ \big(C \text{ is an out-of-band cycle and } |C| > 24\big).}$$

The letters $\mathcal{L}_0^{\log}, \mathcal{L}_1^{\log}$ here denote the objects of the logical corridor and not the ternary lifting matrices L_0, L_1; the separation of types prevents a local conclusion about one of these pairs from weakening that of the other.

The three functional orbits of the catalog produce the HMT characters of closure, propagation and self-scaling. Their internal laws force, respectively,

$$16\arctan(1/5) - 4\arctan(1/239) = \pi_{\mathrm{HMT}} = \pi, \qquad P_{s+t} = P_s P_t \Rightarrow P_1 = \sum_{n\geq 0} \frac{1}{n!} = e_{\mathrm{HMT}} = e,$$

$$F_{\mathrm{av}}\begin{pmatrix} 1 \\ \varphi_{\mathrm{HMT}} \end{pmatrix} = \varphi_{\mathrm{HMT}}\begin{pmatrix} 1 \\ \varphi_{\mathrm{HMT}} \end{pmatrix}, \qquad \varphi_{\mathrm{HMT}} = \varphi.$$

The internal involution exchanges the branches $\pi_{\mathrm{HMT}}/e_{\mathrm{HMT}}$, preserves the axis φ_{HMT} and retains the aggregate $e_{\mathrm{HMT}} + \pi_{\mathrm{HMT}}$. In the enriched TPK phase of the unique composition, the canonical projection of the terminal state publishes its signed primitive coordinate $U = R_{\mathrm{sgn}}(x_\infty)$, and the identity $\mathcal{H}_{12}^2 = 4I$ reconstructs the seal integrally and uniquely $K = \mathcal{H}_{12}U/4$. The dodecaphase recurrence with carry and the remote prolongation of its cylinders publish α_{HMT}. Independently, the enriched state determines the signature

$$\mathcal{I}_9 = (2\pi_{\mathrm{HMT}}, 3, 4, 7, 20, 21, 45, 46, 120, 11),$$

and the associated graded reader constructs, without consulting the root, the exact polynomial operator $E_{21/22}^{(9)} = P_9 - \log_{10}(10/9)$. Its unique simple positive root ξ_9 is the finite order-nine witness of the second internal analytic publication. The same graded transport of the TPK is iterated over successive degrees and constructs the complete compatible system $E_{21/22}^{(6+3m)}$; its truncation squares commute with those of the dodecaphase family and determine, at each depth, the same surviving cylinder. Consequently,

$$\alpha_A := \varprojlim_N \rho_N(\alpha_{12}), \qquad E_{21/22} := \varprojlim_m E_{21/22}^{(6+3m)}, \qquad \alpha_B := \operatorname*{arg\,zero}_{x\in I_\alpha} E_{21/22}(x),$$

$$\boxed{\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}.}$$

The finite equality

$$\mathrm{Tr}_9(\xi_9^{-1}) = \mathrm{Tr}_9(\alpha_{12}^{-1}) = 137.035999084$$

certifies the first visible square of that identity; it does not define it, does not limit it to nine digits and does not select the coefficients of the prolongation. The associated functional bidirectional closure is

$$\alpha T^2 - B_9(\alpha)T + \alpha^3 = 0, \qquad T_\pm = \alpha e^{\pm\xi}, \qquad T_+T_- = \alpha^2, \qquad J_\alpha(T) = \frac{\alpha^2}{T}.$$

Thus, the coordinate published by the dodecaphase closure is the geometric mean preserved by the two conjugate routes. The dodecaphase route and the analytic reader retain their own domains, operators and

falsifiers, but their inverse systems publish the same limit section. The finite comparator and metrological recognition come afterwards.

The incidence coordinate of sector α fixes

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \qquad v_\alpha^2 = 54,$$

and the index-three neighbor

$$N_{\alpha,0} = \{x \in N(C_W) : \langle x, v_\alpha \rangle \equiv 0 \pmod 3\}, \qquad N_\alpha = N_{\alpha,0} + \mathbb{Z}\frac{v_\alpha}{3} \cong \Lambda_{24}.$$

This branch constructs exactly the Leech neighbor and then the chain $\Lambda_{24} \to V^\natural$ and the action of $\mathrm{Aut}(V^\natural) \cong \mathbb{M}$ on $V^\natural$. The scalar α_{HMT} and the vector v_α are the two typed coordinates of the same dodecaphase sector. Their exact coupling is the common-origin hinge leading to the Monster symmetries: the scalar coordinate is published in $\mathbb{R}_{\mathrm{HMT}}$ and the incidence coordinate is published in the branch $A_2^{12} \to \Lambda_{24} \to V^\natural \to \mathbb{M}$, with their codomains preserved and their shared genealogical antecedent explicit.

Dimensional Mechanics receives the preceding outputs and composes, in their constitutive order, phase retention, the decade -34, the Planck pair, the channels $90/120$, the electron, the mass atlas, gravity, area and information. The following chapters prove each arrow, publish its falsifier and link its executable certificate. Conventional identifications appear afterwards as recognition of the quantities already produced. This completes the chain of (2). □

Hierarchy of geometric and dimensional publications

Proposition 0.3 (TRIT as the local unit of the three regimes)**.** *The tritic signature preserves two coordinated readers. The geometric reader publishes* $\tau_{\mathrm{g}} \in \{+1, 0, -1\}$ *and*

$$J_{\tau_{\mathrm{g}}}^2 = -\tau_{\mathrm{g}} I, \qquad J_+^2 = -I, \qquad J_0^2 = 0, \qquad J_-^2 = +I, \tag{8}$$

and distinguishes the elliptic, parabolic and hyperbolic regimes. The active temporal reader publishes another sign, ε_{t}*, through*

$$\begin{aligned} +1 &\mapsto \textit{return, memory and past},\\ 0 &\mapsto \textit{threshold and present},\\ -1 &\mapsto \textit{bidirectional opening and future}. \end{aligned} \tag{9}$$

No $\tau_{\mathrm{g}} = \varepsilon_{\mathrm{t}}$ *is postulated. These two outputs do not identify curvature sign and temporal orientation: they are distinct projections of the same local unit. The TPK transports the complete unit in*

$$\mathcal{F}_q = \mathcal{O}_q \times \mathcal{H}_q \times \mathcal{C}_q \times \mathcal{S}_q \times \mathcal{B}_q \times \Lambda_q \tag{10}$$

and preserves sheet and orientation, residue and quotient, carry, signature, cylinder, boundary and register. In particular, the elliptic branch has the finite realizer

$$A_W^2 = 2I_6 = -I_6 \quad \textit{in } M_6(\mathbb{F}_3), \qquad i_{\mathrm{HMT}} := \chi_i(A_W), \qquad i_{\mathrm{HMT}}^2 = -1, \tag{11}$$

where $\chi_i : \langle A_W \rangle \to \mathbb{C}$ *is the oriented chart. The root of* -1*, the complex plane and the Euler system are publications of a constructed endomorphism, not inputs to the generator.*

Proof. The tritic selector incorporates one of the three generators of (8) without erasing the sum and product sheets. Cardinal transport preserves its orientation and updating preserves the six factors of (10). Therefore, the real projection can condense the three regimes, while the enriched state reconstructs

which of them and what memory produced the visible value. The trigonometric, nilpotent and hyperbolic realizations developed in the body are the respective images of this unique local typing. □

Theorem 0.4 (Chain of action, gravity and matter)**.** *After publishing in a correlated manner* $(\pi_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}})$*, the determinantal branch fixes the decade*

$$\Sigma_H = \varphi_{\mathrm{HMT}} \alpha_{\mathrm{HMT}}^{16}, \qquad d_H := \lfloor \log_{10} \Sigma_H \rfloor = -34. \tag{12}$$

The mantissa comes from the distinct regional reader

$$\begin{aligned} H_5 &= \frac{1}{2}\sqrt{\frac{1000\alpha_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}}} - \alpha_{\mathrm{HMT}} + \frac{9}{16}\alpha_{\mathrm{HMT}}^2 - \frac{5}{9}\alpha_{\mathrm{HMT}}^3 + \frac{7}{48}\alpha_{\mathrm{HMT}}^4 - \frac{1}{54}\alpha_{\mathrm{HMT}}^5, \\ D_A &= \exp\!\left(-\frac{100\pi_{\mathrm{HMT}}}{9}\alpha_{\mathrm{HMT}}\right), \qquad \mathscr{C}_\pi = 169\alpha_{\mathrm{HMT}}^6 + \frac{D_A\alpha_{\mathrm{HMT}}^7}{1 - D_A\alpha_{\mathrm{HMT}}}, \\ \eta_{\mathrm{pre}} &:= H_5, \qquad \eta_{\mathrm{ret}} := H_5 - \frac{90}{\pi_{\mathrm{HMT}}}\mathscr{C}_\pi > 0. \end{aligned} \tag{13}$$

For $a \in \{\mathrm{pre}, \mathrm{ret}\}$*, the two action sections are*

$$\hbar_a = \eta_a 10^{d_H} \mathcal{U}_S = \eta_a 10^{-34} \mathcal{U}_S, \qquad h_a = 2\pi_{\mathrm{HMT}}\hbar_a. \tag{14}$$

Σ_H *fixes only the decade; it is not presented as the generator of the mantissa.*

With $t_0 > 0$*,* $c_{\mathrm{int}} > 0$ *and* $\ell_0 = c_{\mathrm{int}} t_0$ *fixed, CT108 publishes*

$$\omega_{108} = \frac{2\pi_{\mathrm{HMT}}}{108 t_0}, \qquad R_{108} = \frac{c_{\mathrm{int}}}{\omega_{108}} = \frac{54}{\pi_{\mathrm{HMT}}}\ell_0, \qquad m_{108,a} = \frac{\hbar_a \omega_{108}}{c_{\mathrm{int}}^2}. \tag{15}$$

The typed gravitational family is

$$G_a(\vartheta) = \vartheta \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a}. \tag{16}$$

In the reduced convention $r_{g,a} = G_a m_{108,a}/c_{\mathrm{int}}^2$*, the closure* $r_{g,a}(\vartheta) = R_{108}$ *has the unique positive solution*

$$\vartheta_{108} = \left(\frac{54}{\pi_{\mathrm{HMT}}}\right)^2, \qquad G_a := G_a(\vartheta_{108}), \tag{17}$$

and only then publishes

$$\ell_P := \sqrt{\frac{G_a\hbar_a}{c_{\mathrm{int}}^3}} = \sqrt{\vartheta_{108}}\,\ell_0 = R_{108} = \bar{\lambda}_{C,a} = r_{g,a}. \tag{18}$$

Moreover, $G_a\hbar_a = \vartheta_{108} c_{\mathrm{int}}^5 t_0^2$*, so* ℓ_P *is independent of the orientation* a*. Tensor gravity*

$$\mathbb{G}_{\mathrm{TT},a}(n) = G_a \Lambda(n) \Gamma_5 P_5 : V_{12} \longrightarrow \mathcal{H}_{\mathrm{TT}}(n)$$

comes after the scalar. The Palatini–Cartan–Holst action then receives G_a *and the independent output* γ_{HMT}*; the hexad–octad functional transports the latter to the area spectrum and to boundary information.*

In the matter branch, the complete law acts on 81 *cells,* 56 *families,* 13 *multisections and* 324 *routes, with a separate null sector. The readers* K_D *and* K_Ω *have, respectively,* 14 *and* 9 *profiles, form* 40 *pairs and coincide on* 18 *routes only in* $(80, 54, 6)$*. The central fiber* $(5, 5)$ *produces the beta electron. On the remaining fibers, the law publishes positive operators, spectral measures and multiplicities; external recognition is opened afterwards through the exhaustive inventory*

$$\mathscr{I}_{\mathrm{ext}} = \mathscr{I}_m^{471} \coprod \mathscr{I}_\Gamma^{384}, \qquad |\mathscr{I}_{\mathrm{ext}}| = 855. \tag{19}$$

CKM transports the quark spectral frames and PMNS the charged and neutral rank-three frames; their domains, operators and falsifiers remain distinct. No inventory value selects a route, a coefficient, an eigenvalue or an angle.

Proof. The Smith normal form produces $\mathrm{diag}(1, 2, 2, 4)$ and a cokernel of order 16; the bounds of Σ_H fix $d_H = -34$, while (13)–(14) publish $\hbar_a$ and then h_a. Substitution of ϑ_{108} into $G_a(\vartheta)$ proves (18). The incidence of the sets $H_{\mathrm{act}} \times \binom{H_{\mathrm{act}}}{2}$ and $\{1, \ldots, 8\} \times \binom{H_{\mathrm{act}}}{2}$ produces 90 and 120. The twelve dodecaphase sectors and the unique oriented return seam produce, through the linear reader, the weight 12:−1 of the hexad–octad transition. The coefficient of $(1 - q)^{-1}$, equal to $1/24$, and Diophantine minimality subsequently verify that weight. The finite census of the atlas and its two readers proves the cardinalities of the mass branch; cellular inversion has a unique fixed point, $(5, 5)$, which canonically reduces the electronic fiber to rank one. The external inventory is composed only after freezing the fiber, operator, projector, representation, scheme and scale, so that a permutation of its quantities leaves the entire internal constructor invariant. □

The holographic closure of infinity: discrete structure of the continuum and origin of the fundamental constants

Continuum, number, exceptional symmetry and physical realization as typed publications of a single genealogical APP–TRIT–TPK state

Governing thesis and mathematical falsification criterion

The work begins with the point that decides all subsequent readings. The horizontal and vertical chart $1, \dots, 9$ establishes the two APP sheets; TRIT orients and types their three regimes; and the complete TPK dynamics selects, transports and updates the state while preserving carry, boundary, route and memory. The exact census $104\,976 \to 468 \to 243 \subset 729$ constructs the catalog and the visible seeds. Iteration of the typed endomorphism $\mathcal{U}_t = \mathsf{Upd}_t \circ \mathsf{Tra}_t \circ \mathsf{Sel}_t$ then produces the oriented register of the twelve transitions following the three seeds. Its four full-rank matrices uniquely reconstruct L_0, L_1, and the exhaustive census of calendars determines 0011. The compatible reductions of half-open cylinders then produce the canonical prefix at every finite depth and the inverse limit constructs the joint discrete structure of the continuum. From the first page, the thesis is exposed to a stringent criterion: it fails if a generating arrow, a rank, calendar uniqueness, truncation compatibility or a naturality square fails. Once these tests have been passed, the real value is the Archimedean shadow of cylinder reduction and not the antecedent of the generator.

The digital root $\mathrm{dr}_9(n) = 1 + ((n-1) \bmod 9)$ publishes the nonadic class, while the quotient preserves the turns omitted by that publication. APP simultaneously maintains

$$i + j = R^{+}_{ij} + 9q^{+}_{ij}, \qquad ij = R^{\times}_{ij} + 9q^{\times}_{ij};$$

the oriented difference between the pairs (R^{+}, q^{+}) and $(R^{\times}, q^{\times})$ constitutes the local memory that TRIT types and TPK transports. The signed balance produces $B_c = 9P_{+} + 5P_{-}$, the gap $9 - 5 = 4$ and the hierarchy $4 \to 2 \to 6 \to 54$. Thus the exceptional corridor begins in the sum–product difference preserved by APP, not in a numerical coincidence recognized at the end.

The same precedence governs subsequent publications. In particular, the correlated coinductive generation at arbitrary depth of $\pi_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, e_{\mathrm{HMT}}$ and α_{HMT} and, together with it, the exceptional chain up to $V^{\natural}$ and the Monster group, the constitutive constants of the vacuum —including permittivity and permeability—, the electron, the General Law of Masses, CKM and PMNS transports and the hexad–octad functional of area, modular Hamiltonian and boundary information are presented as typed outputs of that construction.

The five inseparable constructions of the continuum lead to classical realizations with explicit universal quantifiers: Guinand–Weil positivity and localization of every nontrivial zero; existence, uniqueness and global smoothness of Navier–Stokes for every smooth periodic datum; quantum existence and positive gap of Yang–Mills for every compact, connected simple Lie group; algebraic realization of every rational Hodge class; and the equality of rank, order of vanishing and leading term of Birch–Swinnerton–Dyer for every elliptic curve over $\mathbb{Q}$. Each quantifier is proved in its chapter through its own constructor that makes it effective; FIN1, the family $\nu_{E,\ell,n}$ and the remaining comparators appear as objects constructed within those proofs, not as external conditions or as boundaries that weaken their scope. Extensional equality in Zermelo–Fraenkel with choice identifies published values, but does not reconstruct their routes, carries, boundaries and memories; its independence result on the continuum hypothesis does not decide this enriched genealogical object. The volume first constructs that object, preserves the genealogical selector h, publishes the canonical model $M_h = L[h]$ and proves in it the classical sentence $2^{\aleph_0} = \aleph_1$. This is the causal meaning governing all the chapters.

The discrete structure of the continuum

Hierarchy of the construction and reading of its realizations. The Archimedean value and the history determining it occupy distinct levels of the same object. The construction that follows keeps that distinction visible from the seeds to the compatible limit. The nonadic turn preserves the phase and advances memory; its refinements provide the link between finite registers and arbitrary depth. The action scale precedes the dimensional quantities using it, and physical realizations preserve their respective transducers. In particular, section 51.13, page 1373, brings together hexad–octad incidence, the Barbero–Immirzi functional and the area spectrum: its gravitational recognition takes place after that construction. This hierarchy allows common antecedents, particular proofs and subsequent realizations to be distinguished without replacing any of them with a scalar coincidence.

Structure, value and realization. Cylinder contraction determines the Archimedean reading; the exceptional projection retains boundary incidence. The common genealogy links both constructions without confusing their objects. The final value expresses a precise reading of the history, but does not exhaust the operations, orientations or memory constituting it. Thus the exposition preserves three levels: the produced structure, the calculated value and the mathematical-physical realization. Their connection requires the explicit maps developed in the following chapters.

The returning phase provides a recurring reference; the advancing memory preserves the path completed. This difference allows prolongation to be read without identifying the return of a coordinate with the reset of the complete state. The dual projection and arbitrary depth are understood from that preservation of history, before identifying the publications with their conventional names.

The mathematics of the continuum commonly begins once the continuum has already been granted. This work reverses that order. It begins from a finite nonadic support, distinguishes upon it the additive and multiplicative evaluations, orients each transition, and constructs the state of the Topological Phase Kernel. On that antecedent it forms the oriented seeds; the emitter and its ternary reduction then produce the enriched state. The nonadic connection preserves its route, carry, boundary, and memory up to X_χ. The compatible limit does not erase its provenance: it publishes it. The constructive chain that governs the volume is

$$X_9 = \mathrm{APP} \longrightarrow X_9^{\mathrm{TRIT}} \longrightarrow x_{\mathrm{TPK}}^{(0)} \longrightarrow S^{\mathrm{seed}} \longrightarrow \Omega_{\mathrm{seed}} \xrightarrow{T_{\min}} U_6^{\mathrm{cat}} \xrightarrow{\rho_3} W_6 \longrightarrow \widehat{x}_K \xrightarrow{\Gamma_9} \mathrm{Surv}^{(\chi)} \longrightarrow X_\chi. \tag{20}$$

Theorem 0.5 (Inaugural thesis of generation, continuity and dual publication)**.** *The exhaustive sweep of the horizontal and vertical nonadic input produces, without consulting target real values,*

$$104\,976 \longrightarrow 468 \longrightarrow 243 \hookrightarrow 729.$$

The four cardinalities count, respectively, weighted seeds, states U_6, visible words w_6 and elements of the ambient fiber $\mathbb{F}_3^6$. The integer predicates of closure, propagation and autoscale select three oriented characters. Their full-rank transition registers uniquely reconstruct the operators

$$X_0 L_0 = Y_0, \qquad X_1 L_1 = Y_1, \qquad L_i = X_i^{-1} Y_i,$$

with $\det X_0 = 1$, $\det X_1 = -1$ *in* $\mathbb{F}_3$ *and design rank* 36*. Among the* 16 *binary calendars of length* 4, 0011 *is the only one compatible with the three biographies; its* 144 *unit perturbations destroy triple compatibility.*

The prolongation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

constructs, for every $N \in \mathbb{N}_{>0}$, a canonical prefix of length N compatible with all the preceding ones. One thousand, ten thousand, one million or any other prescribed finite depth are instances of the same

algorithm; their entire family defines the infinite section in the inverse limit. From that correlated family of coinductive sections arise two publications:

$$\mathcal{S}_\infty \xrightarrow{\mathcal{P}_{\mathrm{Arq}}} \mathbb{R}^4, \qquad \mathcal{S}_\infty \xrightarrow{\mathcal{P}_{\mathrm{Exc}}} W_{12} \to G_{24} \to \Lambda_{24} \to V^\natural \to \mathrm{Aut}(V^\natural) = \mathbb{M}.$$

The first contracts cylinders and jointly publishes $(\pi_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}})$*; the second preserves incidence and publishes the exceptional chain. Thus* $\mathbb{R}$ *is the Archimedean shadow of one of the two projections, not the initial space selecting the constructor. The Machin and Perron–Fibonacci formulas, the exponential semigroup and the metrological tables are subsequent recognitions or comparisons. The transductions of Dimensional Mechanics then act on the HMT outputs to obtain the catalog of constants, masses, fields and physical relations developed in the volume.*

Proof. The first census is obtained by exact enumeration of the APP–TRIT–TPK emitter and the action D_3. The invertibility of X_0, X_1 proves uniqueness of the lifts; the exhaustive censuses of 16 calendars and 144 perturbations prove their joint rigidity. The forward relation of TPK, applied to the internal character already carried by the R36 state, first produces the next prefix and its compatible truncation among the 729 subcylinders. Only afterward does the internal rational bracket intersect that subcylinder and certify the Archimedean publication; it does not select the child of Ext. The iteration that reads only the preceding state has no terminal depth and defines the inverse limit; every finite run is only a witness. Nonadic holonomy returns the phase and advances memory, so the section preserves genealogy. The Archimedean and exceptional readers have the same constructed domain and distinct codomains; consequently, none of their results can retroactively act as input. The following chapters develop each of these arrows, their equations, certificates and negative controls contiguously. □

Theorem 0.6 (Simultaneous closure of generation, the continuum, exceptional symmetry and the five terminal realizations)**.** *The scales* $0 \to 1$, $0 \to 10$, $0 \to 10^N$ *and* $N \to \infty$ *are four cuts of a single coinductive prolongation. For every* $N \geq 1$*, the section of depth* $N+1$ *truncates to that of depth* N*, preserves sheet, route, residue, quotient, carry, boundary and memory, and publishes a canonical decimal prefix. The limit of all compatible sections constructs the real line as an Archimedean quotient; it does not presuppose the line to select the history.*

The reconstruction of L_0, L_1 *and of the calendar* 0011 *belongs to this same chain. The terminal state jointly produces*

$$x_{\mathrm{term}} \longmapsto (P, E, \Phi), \qquad x_{\mathrm{term}} \longmapsto U_{12}^{\mathrm{sgn}} \overset{\mathcal{H}_{12}}{\longleftrightarrow} K_{\mathrm{dod}};$$

therefore U_{12}^{sgn} *and* K_{dod} *are not independent inputs. The integer recurrence with carry determines the compatible dodecaphasic family whose limit section is* $\alpha_A := \varprojlim_N \rho_N(\alpha_{12})$*. The same enriched state factorizes, through the signature* $\mathcal{I}_9$*, the finite analytic reader* $E_{21/22}^{(9)}$*; this has a unique simple positive root* ξ_9*, rationally enclosed and obtained without using* α_{12} *as input. The graded transport of TPK constructs the compatible family* $E_{21/22}^{(6+3m)}$ *and its limit* $E_{21/22}$*. The truncation squares of both families commute and determine the same surviving cylinder at each depth; therefore*

$$\alpha_B := \operatorname*{arg\,zero}_{x \in I_\alpha} E_{21/22}(x), \qquad \boxed{\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}}.$$

The finite equality $\mathrm{Tr}_9(\xi_9^{-1}) = \mathrm{Tr}_9(\alpha_{12}^{-1})$ *is the first visible certificate of that identity; it does not define it, limit it to nine digits or select the prolongations.*

The coordinate α *also links publication of the value and exceptional publication. The scalar* α_{HMT}*, the vector of twelve blocks, the oriented support of the carry and the vector* v_α *are coordinated types of the same state. Incidence transport satisfies*

$$v_\alpha^2 = 54, \qquad C_W \longrightarrow N(A_2^{12}) \xrightarrow{v_\alpha} N_\alpha \cong \Lambda_{24} \longrightarrow V^\natural \longrightarrow \mathrm{Aut}(V^\natural) = \mathbb{M}.$$

Thus the classical theory of codes, lattices, vertex operator algebras and Moonshine recognizes objects already produced: it does not supply them as antecedents and there is no literal action of the Monster on digits.

From the same genealogical continuum, the five terminal constructions emerge correlatively. Their readers construct complete Guinand–Weil positivity; the global smooth Navier–Stokes solution for every smooth periodic datum; the gauge measure with reflection positivity and mass gap for every compact, connected simple Lie group; the algebraic cycle realizing every rational Hodge class; and the determinantal complex publishing the complete Birch–Swinnerton–Dyer formula for every elliptic curve over $\mathbb{Q}$*. FIN1 and the compatible family* $\nu_{E,\ell,n}$ *are constructed within the respective owners and close their surjectivity and compatibility steps. In particular, the conclusions are obtained in the terminal chapters through the identities of Gram, Stein, Osterwalder–Schrader, unitriangular extension and Bockstein; none of these properties is inserted into the definition of the common continuum.*

The extensional publication preserving only the real value is not faithful: distinct histories may have the same image. The independence of CH in ZFC concerns that extensional language and does not revoke the genealogical construction of the continuum. The ordinal reader, the relative closure $L[h]$ *and the condition* GE_h *are typed afterward and do not participate in Archimedean coverage.*

Proof. Quantification at every depth, reconstruction of the two lifts and uniqueness of the calendar are verified in the nonadic generation certificate and developed in the chapter on the connection Γ_9. The dual terminal reading of α is (31.57); the integer route, the kernel $E_{21/22}$, its local inverse and the comparator appear contiguously in the same owner. The construction of N_α, its parity, unimodularity, absence of roots and passage to $V^\natural$ are proved in the exceptional corridor. The classical closures are theorems 96.7 y 98.3, theorems 99.2 y 100.4 and theorem 101.8. The construction of the real quotient and the descent of operations precede the ordinal reader in the volume. This precedence separates generation from its recognitions and proves all the assertions of the statement. □

Arithmetic corrections fixing the types. At the boundary $U008$ two different outputs are preserved:

$$u_5^{\pi} + u_5^{e} = 210112, \qquad u_4^{\varphi} A_W L_1^3 = 111101.$$

The first aggregates the channels π and e; the second transports the channel φ through Witt incidence and three lifts. Equality between them is excluded. Likewise, $E^{(9)}_{21/22}(\xi_9) = 0$ is exact, whereas $E^{(9)}_{21/22}(\alpha_{12})$ is the residue of the first truncation. This finite difference distinguishes the two representatives of order nine without separating their coinductive sections: the truncation squares prove $\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}$. No edition may turn the finite residue into a divergence of the limits or replace one assertion with the other.

Here U_6^{cat} is the categorized output of the emitter, W_6 its ternary reduction, and $\widehat{x}_K$ the enriched state that retains sheet, orientation, residue, quotient, carry, signature, cylinder, boundary, route, memory, survival, and multisection. The return $9 \to 1$ of the observable phase does not reset that state: Γ_9 transports the complete history, survival selects the compatible prolongations, and X_χ gathers their indefinitely refinable sections. Thus, Triadic Modular Holography constructs the discrete structure of the continuum and closes the passage from the finite mark to the compatible section.

Distinct publications start from that same antecedent, without retrospective selection of the constructor. The Archimedean publication contracts the compatible cylinders; the exceptional publication preserves its incidence. This second publication passes through Hadamard and through the Paley chain

$$U_{12}^{\mathrm{sgn}} \longrightarrow C_P \longrightarrow A_W \longrightarrow C_W$$

and bifurcates into 2 typed branches: the designs-and-groups branch

$$C_W \longrightarrow W_{12} \longrightarrow W_{24}, \qquad \mathrm{Aut}(W_{12}) = M_{12}, \quad \mathrm{Aut}(W_{24}) = M_{24},$$

and the lattice and vertex operator branch

$$C_W \longrightarrow N(A_2^{12}) \longrightarrow \Lambda_{24} \longrightarrow V_{\Lambda_{24}} \longrightarrow V^\natural.$$

In the latter, $\mathrm{Aut}(V^\natural) = \mathbb{M}$, $J = T_{1A}$ and the family $\{T_g\}_{g\in\mathbb{M}}$ publishes Moonshine. The constants and dimensional magnitudes are other readers of the same state, and not data that determine it.

The constructed structure also admits 5 terminal closures that unfold in their own chapters, after their dimensional realization. The work proves positivity of the Weil form and localization of the nontrivial zeros of ζ on $\Re s = \frac{1}{2}$; existence, uniqueness and global smoothness of three-dimensional Navier–Stokes for every smooth periodic datum; existence of quantum Yang–Mills theory in dimension 4, with positive gap, for every compact, connected simple Lie group; algebraicity of every rational Hodge class; and the equality of rank and order of vanishing, together with the leading-term formula of Birch–Swinnerton–Dyer, for every elliptic curve over $\mathbb{Q}$. FIN1/cellular exhaustiveness and the family $\nu_{E,\ell,n}$ are effectively constructed components of the proofs, not promotion hypotheses. The 5 typed conclusions are not added from outside: they are specializations of the connection and survival preserved in (20).

The problem of the continuum traverses the entire history of mathematics. Arithmetic learned to count units; geometry learned to compare lengths; analysis constructed limits and completed sequences; measure theory assigned magnitudes to sets; physics converted those magnitudes into observables. Each advance enlarged the domain of the calculable and left open a question prior to representation: what structure preserves the identity of an object as it passes from mark to interval, from interval to measure, from measure to value, and from value to an indefinite sequence of digits.

> **Governing invariant of the work.** A single enriched discrete state preserves unity, orientation, quotient, residue, carry, boundary, route, memory, and survival through refinement. Continuous publication, exceptional publication, and dimensional realization are distinct readers of that same antecedent; none retrospectively selects the constructor.

This invariant fixes the dependence among the 5 parts:

$$\begin{aligned} \text{APP–TRIT–TPK kernel} &\longrightarrow \text{discrete structure of the continuum} \\ &\longrightarrow \text{genealogy of the constants} \\ &\longrightarrow \text{dimensional realization} \\ &\longrightarrow \text{terminal mathematical closures.} \end{aligned}$$

Every change of scale preserves the antecedent and adds a map; it never replaces the state with the digit published by one of its readers.

Triadic Modular Holography (HMT) addresses that question through an inverse system of finite state spaces. A compatible section $x = (x_n)_n \in \varprojlim X_n$ preserves genealogy between resolutions and its Archimedean evaluation publishes the value. For the channels π, e, φ, α, the internal relation Ext, coinductive survival and the exhaustive partition of the 729 elements of the ambient fiber determine the compatible prolongation. In the first three channels, the exact rational readers then locate and certify the blocks already generated; in the fourth, the signed dodecaphasic reader and the vacancy equation certify the coordinate α of the same state. At each level, the complete character preserves a unique prolongable cylinder compatible with the preceding truncation; the nested cylinders thus determine a unique coinductive section. The runs of 10^3 and of more than 10^4 digits are extensive certificates of this general construction, not its logical foundation. A digit is the coordinate published by a reader; a value is the image of a section under a given chart; a number additionally preserves the sequence of states, operations and memories that allows each coordinate to be recovered from the preceding ones.

This definition also modifies the concept of symmetry. An HMT symmetry is a transformation that preserves the genealogical identity of the state. A return to a visible coordinate may coexist with a change of height, carry, sheet, or orientation. Invariance then belongs to the complete object and not merely to its projection. Cyclic and helical representations, tori, rings, lattices, codes, and exceptional groups appear as successive realizations of this evolutionary conservation.

The mathematical scope of the proposal lies in jointly explaining 2 objects that the usual representation presents separately: the continuous value and the symmetry that preserves its structure. Its physical scope lies in making metrological coordinates appear at the end of a derivation rather than at its beginning. In the ordinary formulation, the fundamental constants parameterize the equations once they have been determined experimentally; in HMT and Dimensional Mechanics, the discrete architecture first constructs the section, operator, and scale, and experimental metrology subsequently recognizes the published coordinate. The discrete structure of the continuum and the origin of the fundamental constants therefore constitute a single thesis concerning the mathematical production of value.

This unit is expressed by means of 3 families of typed evaluations on the common genealogical antecedent:

$$\mathcal{X}_{\infty}^{\mathrm{cal}} \xrightarrow{\mathcal{P}_{\mathrm{arq}}} \mathrm{im}(\mathcal{P}_{\mathrm{arq}}) \subseteq \mathbb{R}, \qquad \mathcal{X}_{\infty}^{\mathrm{cal}} \times \mathcal{C}_{\mathrm{exc}} \xrightarrow{\mathcal{P}_{\mathrm{exc}}} \mathcal{E}_{\mathrm{exc}},$$

$$\mathcal{X}_{\infty}^{\mathrm{cal}} \overset{\mathcal{R}_{\mathrm{dim}}}{\dashrightarrow} \coprod_{\lambda} \mathcal{L}_{\lambda}.$$

The first publishes the Archimedean coordinate; the second preserves incidence and symmetry with respect to a declared gauge; the third realizes sections, fibers, or operators over dimensional lines wherever the corresponding physical reader has been constructed. Value, form, and magnitude remain coordinated by their common antecedent without being confused in a single codomain.

The order of the treatise follows the constructive dependence among the objects. Every theorem states its domain, its initial data, and the conclusion deduced from them; each physical realization subsequently adds the morphism that transports that conclusion to a dimensional line. A later metrological comparison does not alter the genealogy that produced the output.

Genealogical, operational, matrix, and physical closures

The title *The Holographic Closure of Infinity* and the subtitle *The Discrete Structure of the Continuum and the Origin of the Fundamental Constants* state a single governing thesis. The continuum is not the original domain upon which a lattice is subsequently superimposed: it is the compatible closure of a discrete genealogy. Its states preserve value, sheet, orientation, residue, quotient, carry, boundary, route, ledger, memory, survival, phase, and gauge; the real line appears at the end as the Archimedean evaluation of a section. Without identifying codomains, the same section feeds the exceptional reading that retains the incidence eliminated by publication of the value. The complete causal chain therefore begins before both projections:

$$\mathrm{APP} \longrightarrow \mathrm{TRIT}_{\mathrm{orientado}} \longrightarrow \mathrm{TPK} \longrightarrow \mathcal{X}_{\infty}^{\mathrm{cal}},$$

$$\mathcal{X}_{\infty}^{\mathrm{cal}} \xrightarrow{\mathcal{P}_{\mathrm{arq}}} \mathbb{R}, \qquad \mathcal{X}_{\infty}^{\mathrm{cal}} \times \mathcal{C}_{\mathrm{exc}} \xrightarrow{\mathcal{P}_{\mathrm{exc}}} \mathcal{E}_{\mathrm{exc}}.$$

The nonadic return $9 \to 1$ preserves the phase and adds one unit of memory: the recovered unity does not erase the turn traversed. This evolutionary conservation is the common condition of Archimedean continuity and exceptional publication. The latter bifurcates, from the Paley–Witt code, into the branch of designs $W_{12} \to W_{24}$, with $\mathrm{Aut}(W_{12}) = M_{12}$ and $\mathrm{Aut}(W_{24}) = M_{24}$, and the lattice branch $N(A_2^{12}) \to \Lambda_{24} \to V^{\natural}$, with $\mathrm{Aut}(V^{\natural}) = \mathbb{M}$, $J = T_{1A}$, and the family $\{T_g\}_{g \in \mathbb{M}}$ publishing Moonshine; neither publication is obtained by horizontally compressing the other.

Genealogical closure specifies what that section conserves before publishing its value. Descent along fibres separates existence, constancy on the fibres, and uniqueness of the descended reader; the König argument separates the existence of an infinite branch from the uniqueness of an instance. The

extensionality of sections is not reduced to equality of readings, and theorem $0- 1-\infty$ coordinates decreasing cylindrical support, divergent height, unit mass, and weak-* convergence toward a point measure. Foundation of prefixes and coinduction, ordinals fibred by route memory, cylindrical power, closed power, full power, and typed choice thus receive distinct objects and domains. Conservative encoding in ZF/ZFC serves as a language of transfer, not as an ontological antecedent. In particular, the Gödel–Turing obstruction affects a uniform total decider over all effective towers; it does not negate the particular selectors constructed in calibrated HMT images. On the informational side, a bijective update of the complete state has zero logical Landauer cost in the ideal regime, whereas every projection that discards the fibre explicitly localizes the lost entropic term.

The same genealogy induces an operatorial closure without replacing the TPK history. At level m, the diagonal cylindrical algebras $D_m = C(X_m)$ are represented in $A_m = M_{9^m}(\mathbb{C})$. The unital inclusion

$$\iota_m : A_m \longrightarrow A_{m+1}, \qquad \iota_m(a) = a \otimes I_9,$$

conserves unitarity and trace and produces

$$A_{9^\infty} := \varinjlim(A_m, \iota_m) \cong \mathrm{UHF}(9^\infty) \cong \mathrm{UHF}(3^\infty), \qquad \pi_\tau(A_{9^\infty})'' \cong R_{\mathrm{hyp}},$$

the hyperfinite factor of type II_1. The normalized trace of its projections realizes every dimension in $[0, 1]$. This branch remains separate from the marked inclusion $a \mapsto a \otimes p_j$, whose natural closure leads to the compact operators and, weakly, to a factor of type I_∞. The tracial closure likewise absorbs neither the topology nor the genealogical composition: the transverse object is

$$\mathfrak{C}_{\mathrm{HMT}} = \big(C(\Omega_\infty), R_{\mathrm{hyp}}, \mathcal{G}_{\mathrm{TPK}}\big),$$

where the first component preserves points and topology, the second dimension and statistics, and the third composition, orientation, and genealogy of the routes.

The matrix closure adds 2 coordinates that must not be confused with depth. For a fixed exterior size n, the family $\mathfrak{M}^{(n)}_{m,s}$ is bigraded by genealogical depth m and matrix level s; as n varies, the architecture is trigraded by (m, n, s). The spaces

$$\mathcal{K}_{m,s} = \mathrm{UCP}\big(C(X_m), M_s(\mathbb{C})\big)$$

admit commuting genealogical restrictions and matricial compressions. A family compatible at every depth determines a unique limiting UCP map and possesses a common Stinespring dilation; on the diagonal corridor, a single projective spectral measure simultaneously dilates all cylindrical measurements. On the noncommutative tower, the trace-preserving conditional expectation $\mathrm{id} \otimes \tau_9$ realizes the controlled loss of the final digit, and its dilation preserves that datum in the environment. The minimal dilation, the faithful environmental realization, and APP–TPK dynamical fidelity are 3 distinct requirements; the latter still requires the intertwining of routes, monodromy, and composition.

Within that framework, qutrit quantum magic squares of parameters $(9, 3)$ and $(12, 3)$ are constructed, together with their cubes of orders 9 and 12, and noncommutativity is rigorously separated from non-semiclassicality. The conditional projections of the sum and product sheets of APP generate, at depth 2,

$$C^*(P_{\Sigma,2}, P_{\Pi,2}) \cong \mathbb{C}^4 \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C}),$$

and the central block publishes, at every matrix level, the free interval $[5I_s, 9I_s]$. Matrix convexity, completely positive maps, and dilations provide here a language of proof, classification, and falsification; they do not become the origin of APP, TRIT, or TPK or replace the double projection.

Physical closure starts from a constructed common domain, not from 4 nominally juxtaposed arrows. If $\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}$ denotes the category of prepared states and recorded routes, the 4 reference realizations are coordinated through the fiber product

$$\mathsf{P}^{+}_{\mathrm{CT}} := \mathsf{D}_{\mathrm{sp}} \times_{\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}} \mathsf{D}_{Q} \times_{\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}} \mathsf{D}_{D} \times_{\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}} \mathsf{D}_{C}.$$

Its canonical projections jointly construct the spectral representation of the route, the Hilbert realization, the dimensional section, and discrete Cartan transport. The positive-total-variation reader and the oriented net-transport reader receive the same route, but neither factors through the other. Only on the reversible subdomain, once unitarity, additivity, antisymmetry, and compatibility of the cocycles have been verified, does localization in the groupoid produce the oriented realization. Dimensional Mechanics thus prolongs the enriched state into physical magnitudes without retrospectively reconstructing the route from its spectrum. Every additional realization declares the intertwiner and the hypotheses that define it.

These 4 closures are not superimposed appendices. They are causal realizations of a single architecture: the genealogical closure determines identity through refinement; the operatorial closure realizes continuous dimension from unitary nonadic inclusions; the matricial closure controls compression, dilation, and simultaneity; and the physical closure represents prepared routes through typed morphisms. The Archimedean evaluation, the exceptional branch, the tracial tower, and the dimensional realization share an antecedent, but retain their own codomains, hypotheses, and conclusions. The complete thesis is thereby maintained: the continuum and the constants appear as publications of a discrete structure that preserves the unity, orientation, and memory of its genesis.

From the elementary markers to APP, TRIT, and TPK

The construction begins on the discrete line. The marks $1, \dots, 9$ provide the visible representatives of a nonadic phase; the next step closes the circuit and preserves its quotient. Each mark can be read as a position and each interval as a distance. This double reading separates the act of counting from the act of measuring at the outset. Upon taking 2 coordinates, the multiplicative kernel 8×8, publication of the null class through 9, and the gnomon of 17 cells complete the chart 9×9:

$$81 = 64 + 17.$$

The modular identification of its sides produces

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2$$

and its Cayley graph. This object is APP, the common support of 2 evaluations: the additive sheet Σ and the multiplicative sheet Π. APP preserves the position, the cardinal routes, and the toroidal geometry on which both readings can be compared without merging.

The central block of APP possesses the decomposition

$$B_c = 9P_+ + 5P_-.$$

Its first differential invariant is, therefore,

$$\boxed{9 - 5 = 4}.$$

The local exchange coefficient is 2; modular compression publishes the difference 6; unfolding over the 9 blocks yields $54 = 9 \cdot 6 = 459 - 405$. These numbers belong to different levels of the same sum–product incompatibility. The spectral separation and the block projectors $P_{\pm}$ fix its eigenmodes. The conditional projectors $P_{\Sigma,d}$ and $P_{\Pi,d}$, together with their commutator, provide a finite geometry of mixed signature,

a normalization in $SO^+(1,1)$, and the algebraic antecedent of the incompatibility of observables that quantum physics will formulate thereafter.

TRIT adds orientation and local memory. Its enriched state residue–quotient preserves the observable output and the quotient that produced it. In a declared integral chart,

$$a + b = r + 3c,$$

the pair $\widehat{\tau} = (r, c)$ distinguishes histories that may share the same output. The associated 3 quadratic branches realize

$$J^2 = -I, \qquad J^2 = 0, \qquad J^2 = +I,$$

and organize the regimes elliptic, parabolic and hyperbolic. The plane complex, the circular closure and the Poincare hyperbolic disk it are presented as coordinated charts of this algebraic family.

TPK, the Topological Phase Kernel, is the operatorial system of genealogical dynamics. Its architecture comprises 8 classes: support and state; internal selection; transport; readers and quotients; memory and boundary; periodicity and return; prolongation; and Archimedean, exceptional, or classificatory outputs. The internal ternary sheet selector M_{ph}, the readers modulo 3 and modulo 9, the quotient and residue modulo 1000 associated with block notation in base 1000, the cycles 27, 108, and 1080, the bidirectional unfolding operators, the memory, and the enriched chronological ledger are typed subobjects of the TPK. The exposition of each fixes its domain, operation, codomain, and invariant before employing it in a composition.

The transition $9 \to 1$ expresses the fundamental mechanism. The visible phase returns to its first position, and the state advances its memory upon return. The integer sequence decomposes into phase and height,

$$n = 9\kappa_9(n) + \rho_9(n),$$

so that the observable angular coordinate is the planar projection of the nonadic helical lift. The return preserves the angular position and elevates the genealogy. This monodromy makes indefinite iteration possible without confusing phase repetition with state repetition.

Nonadic arithmetic also preserves the radical ideal $\mathfrak{m} = (3) \subset \mathbb{Z}/9\mathbb{Z}$, whose visible representatives organize the network 3–6–9 and the ordered orbit $369 \to 693 \to 936$. The power–vacancy law

$$V_9(n) = \left\lceil n \log_{10}\left(\frac{10}{9}\right) \right\rceil$$

exactly quantifies the decimal distance between 9^n and 10^n. In the tower 9^{9^9}, it produces

$$\mathrm{ord}_{10}\left(9^{9^9}\right) = 369\,693\,099, \qquad \#\,\mathrm{cifras}\left(9^{9^9}\right) = 369\,693\,100.$$

This evaluation is distinct of the census of 396 events elementary of boundary by channel: a counts decimal orders and vacancies; the another belongs to the calendar of emission.

Nonadic Filtration and Generation of Fundamental Numbers

The emitter APP–TRIT–TPK organizes the exact sequence of reductions and prolongations.

$$104\,976 \longrightarrow 468\ U_6 \longrightarrow 243\ w_6 \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9.$$

The 468 decimal emissions, the 243 visible words and the ambient $\mathbb{F}_3^6$ of 729 words are dominios different linked by maps explicit. The chain

$$w_6 \xrightarrow{L_0} w_{12} \xrightarrow{L_0} w_{18} \xrightarrow{L_1} w_{24} \xrightarrow{L_1} w_{30} \xrightarrow{R_{36}} \text{boundary} \xrightarrow{G_9} \text{surviving section}$$

opens the coinductive regime. In each filtration stage, the partition exhaustive of the 729 prolongations of the ambient fiber is recorded the discarded states, the surviving section and the memory linking one depth with the next.

Before any decimal evaluation, APP constructs 4 integer channels

$$C_\alpha = 729, \qquad C_e = 271, \qquad C_\pi = 315, \qquad C_\varphi = 161.$$

Its identities

$$1000 = C_\alpha + C_e = 729 + 271, \qquad 744 = C_\pi + C_e + C_\varphi - 3$$

coordinate the cubic completion, the pointed boundary, and the modular endpoint. Each channel is a typed direction of the reader: 271 realizes propagation and crown closure; 315, circular closure; 161, self-scaling; and 729, the nonadic ambient space that feeds the dodecaphase closure. The corresponding constant appears when the filtration, orientation, and memory realize that channel on its section.

The readers of π, e, φ and α act on the same enriched state and preserve distinct genealogical trajectories. π organizes closure, orientation and 5 regions U_6; e governs propagation and prefix change; φ governs autoscale and stability, recognized later in the Perron–Fibonacci chart; and α has the signed dodecaphasic reader that prolongs the carry. The internal relation Ext, coinductive survival and the exhaustive partition of the 729 elements of the ambient fiber determine the compatible extension for every finite depth N. The exact rational readers then locate and certify the Archimedean coordinates already generated. At each level, the complete character preserves a unique prolongable cylinder compatible with the preceding truncation; the nested cylinders therefore determine a unique coinductive section and the same algorithm publishes any requested finite decimal truncation. The runs of 10^3 and of more than 10^4 digits are extensive certificates of that general construction, not its logical foundation.

From the same enriched state the channels of π, e, φ and the dodecaphasic channel open. In the typed internal order, the publications P, E, Φ, the signed state U_{12}^{sgn}, the seal K and the carry recurrence feed the oriented cyclic closure of 12 phases that publishes the coordinate α, without separating it from the common genealogy. The first internal route produces

$$\alpha_{\mathrm{HMT}} = 0.00729735256928380099728510547238066 3\ldots$$

from the state and its reversible charts. The finite vacancy equation $E_{21/22}^{(9)}$ provides the first exact representative ξ_9 of the second internal construction, through an operator whose coefficients factor through invariants of the enriched state before solving for the root. Graded transport produces the complete family $E_{21/22}^{(6+3m)}$; its truncation squares with the dodecaphasic family select the same compatible cylinder at every depth and prove $\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}$. Equality after Tr_9 is only the first finite certificate of that identity, not its foundation or its limit. CODATA recognition constitutes a subsequent metrological comparison and does not select either route.

The same architecture extends number theory. The comprehensive presentation

$$Q_{\mathbb{Z}} = \mathbb{Z}[u]/(u^2 + 1)$$

has distinct specializations over $\mathbb{F}_3$ and $\mathbb{R}$: the first is realized by the finite endomorphism $A_W^2 = -I_6$, and the second by the real structure $J_{\mathrm{e}}^2 = -I$. No inclusion $\mathbb{F}_9 \to \mathbb{R}$ is introduced. The section of π is generated before applying exponential closure; only after obtaining $\mathcal{P}_{\mathrm{arq}}(x_\pi) = \pi_{\mathrm{HMT}}$ does one conclude

$$\mathrm{Exp}(\pi_{\mathrm{HMT}} J_{\mathrm{e}}) + I = 0.$$

The exceptional branch fixes the quadratic type, not the angle or its digits; this precedence proves the absence of circularity. The quadratic family brings together the 3 exponentials $\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau$, with $J_\tau^2 = -\tau I$, and links Euler's identity to the hyperbolic Fibonacci branch through $F_{\mathrm{av}}^2 = \exp(2\log\varphi\, J_{-1})$.

The Rogers–Ramanujan continued fraction articulates level 5, φ, A_5, and Klein's icosahedral identity for j. The primes receive readers of irreducible periodicity and a representation of occupations: the vector $(\nu_p(n))_p$ converts the factorization of an integer into a finite state of modes, while $\sum_p \nu_p(n) \log p = \log n$ transforms the product into additive energy. The thermal trace recovers the Euler product and the zeta function. The period-doubling dynamics reconstructs Feigenbaum approximants and shows how a deterministic rule generates visible complexity when part of its genealogy is projected. Euler–Mascheroni, Stieltjes, Apery, Glaisher–Kinkelin, Euler–Kronecker, Meissel–Mertens, Catalan, Khinchin, and Feigenbaum are organized through finite parts, functions L, products, jets, traces, and renormalization operators.

Double projection and genealogy of the exceptional symmetries

The enriched state has 2 coordinated evaluations. The Archimedean projection contracts a compatible family of cylinders and publishes a real value. The exceptional projection preserves incidence, orientation, carry, and boundary. The real continuum and the exceptional structure are thus 2 images of a common antecedent.

Formally, if $\mathcal{X}_\infty^{\mathrm{cal}} = \varprojlim_n \mathcal{S}_n$ is the space of surviving sections compatible with the transported gauge, the thesis is expressed through the typed maps

$$\mathcal{P}_{\mathrm{arq}} : \mathcal{X}_\infty^{\mathrm{cal}} \longrightarrow \mathbb{R}, \qquad \mathcal{P}_{\mathrm{exc}} : \mathcal{X}_\infty^{\mathrm{cal}} \times C_{\mathrm{exc}} \longrightarrow \mathcal{E}_{\mathrm{exc}}.$$

The governing statement establishes their common domain, their compatibility with the truncations, and the finite intersection $B_K = H_e \cap P_\pi$. Coverage of the entire real codomain and joint faithfulness correspond to additional statements: the double projection proved here identifies 2 coordinated evaluations of each admissible section.

The second branch develops a complete genealogy. The Hadamard transform organizes the signed dodecaphasic chart; the Paley conference matrix fixes the projective gauge, and its ternary reduction produces the Golay code of length 12. Hamming distance measures the weight and separation of its words. The Steiner systems $S(5, 6, 12)$ and $S(5, 8, 24)$, their extensions, and the Witt stars construct the designs on which M_{12} and M_{24} act. From the common code are distinguished the transport $W_{12} \to W_{24}$ and the lattice branch

$$C_W \longrightarrow N(A_2^{12}) \longrightarrow \Lambda_{24}.$$

The Leech lattice feeds the vertex operator algebra V_Λ; the involution and the twisted sector produce the orbifold $V^\natural$; its automorphism group is the Monster. For every conjugacy class $[g] \subset \mathbb{M}$, the graded trace constructs its McKay–Thompson series T_g; modularity and replicability organize the correspondence called Moonshine. The function j thus occupies the modular endpoint of a genealogy that begins in the same discrete structure that publishes the real value.

The configuration of the 27 lines on a smooth cubic surface, the Schläfli graph, and the action of $W(E_6)$ constitute another realization of the incidence. Each step preserves its operator: cyclotomic transport, the integral construction of the lattice, the reflections, the roots, and the action of the Weyl group. This separation turns the succession of classical names into a verifiable mathematical chain.

Planck constant, mass self-duality, area operator, and mixing

The Smith normal form of the local reader of fixed action

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4), \qquad |\operatorname{coker} H_4| = 16.$$

The transfer

$$\Sigma_H = \varphi \alpha_{\mathrm{HMT}}^{16}, \qquad 10^{-34} < \Sigma_H < 10^{-33},$$

places Σ_H in the internal decade that determines the scale $10^{-34}\mathcal{U}_S$. On esa line it are obtained the action coordinates before and after the return,

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S, \qquad h = 2\pi\hbar.$$

The chart $\mathcal{U}_S \to \mathrm{J\,s}$ subsequently publishes the physical unit. The Planck scale therefore appears as a consequence of the determinantal reader and of the already constructed fine-structure constant.

The circular realization of the temporal endomorphism, with $\ell_0 := c_{\mathrm{int}} t_0$, fixes

$$T_{108} = 108 t_0, \qquad R_{108} = \frac{54}{\pi}\,\ell_0, \qquad \theta_{108}^{\mathrm{circ}} = \left(\frac{54}{\pi}\right)^2.$$

In that chart coincide

$$R_{108} = \bar{\lambda}_{\mathrm{C}} = r_{\mathrm{g}} = \ell_{\mathrm{P}}^{\mathrm{circ}},$$

and the broken line of masses conserves

$$Z_a(m) = \frac{m_{\mathrm{P},a}^{\circ}}{m} P_+ + \frac{m}{m_{\mathrm{P},a}^{\circ}} P_-, \qquad N_{\mathrm{sp}}(Z_a(m)) = 1.$$

The involution $m \mapsto (m_{\mathrm{P},a}^{\circ})^2/m$ exchanges the Compton and gravitational readings. The Planck scale is its fixed point; the electron appears afterward as a genealogical displacement on that line, not as an identification with a classical black hole.

The direct angular coordinate A and its conjugate angular coordinate C^* produce the channels $q_\pm$, the rate ellipse, and bidirectional holonomy. The hexad–octad inclusion constructs the functional $\Gamma_{6\to 8}$, the Barbero–Immirzi parameter, the quantum area operator, its discrete spectrum, the collective intertwiner, and the boundary entropies. In a branch of rank 3, the same incidence discipline constructs the CKM matrix, its angles, the CP-violation phase, and the Jarlskog invariant. The PMNS realization preserves its own operator and mixing problem.

Vacancies and sheet channels construct the electric and magnetic responses of the vacuum. From them are obtained the permittivity ε_0, permeability μ_0, impedance, and the constitutive propagation identity. Internal temporal periodicity and ternary information construct the thermal relation that assigns a function to the Boltzmann constant. These magnitudes are presented in scientific tables immediately after their derivation: HMT object, formula, internal value, unit chart, experimental reference, and uncertainty belong to distinct columns.

Dimensional Mechanics and General Mass Law

Dimensional Mechanics prolongs the numerical genealogy through physical readers. A particle is a persistent section of the extension; its observable route determines an enriched chronological register; the register determines an integer signature; the signature feeds a character; the character acts on the global hierarchy of towers; the dimensional chart publishes the magnitude. The sequence

$$\text{extension} \to \text{route} \to \text{record} \to \mathbb{Z}^5 \to \chi_{\mathrm{full}} \to \langle 90, 120\rangle \to \text{mass section}$$

is the precise boundary between HMT and MD.

The General Mass Law is formulated over the entire domain:

$$81 \text{ cells}, \qquad 56 \text{ families}, \qquad 13 \text{ multisections}, \qquad 324 \text{ oriented routes}.$$

The action scale, CT108 temporal periodicity, and internal velocity jointly fix the absolute section,

$$\mathbf{m}_\star^{\mathrm{ret}} = \eta_{\mathrm{ret}} 10^{-34} \mathcal{U}_S \frac{2\pi}{108 t_0} c_{\mathrm{int}}^{-2},$$

and the operator of a multisection F is

$$\widehat{\mathbf{M}}_F^{\beta,r} = \mathbf{m}_\star^{\mathrm{ret}} \otimes R_{\mathrm{act}}^{\widehat{K}_F^r/2} \widehat{\mathcal{A}}_F^{(0)} R_{\mathrm{act}}^{\widehat{K}_F^r/2}, \qquad r \in \{D, \Omega\}.$$

Here $\widehat{\mathcal{A}}_F^{(0)}$ brings together the signature, character, and complete tower $\langle 90, 120 \rangle$; K_D and K_Ω preserve the route-dependent bidirectional refinements.

The central electron is the rank reduction 1. The global torsion that intervenes in its reader is

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729}\right).$$

Its 2 readers coincide on $(80, 54, 6)$ and produce

$$\kappa_e = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4,$$

$$\boxed{\begin{aligned} E_e^\beta = m_e^\beta c^2 &= 0.510998950690000 80 86082571648\ldots \text{ MeV}, \\ m_e^\beta &= 0.510998950690000 80 86082571648\ldots \text{ MeV}/c^2. \end{aligned}}$$

The remaining realizations preserve the type of their output: closed scalar, fiber spectrum, conditioned evaluation, null chart, mixing operator, fusion invariant, or prolongation horizon. The physical domain comprises charged leptons, neutrinos, 6 quark flavors with color, the photon, gluons, W, Z, the Higgs, mesons, baryons, Fibonacci and Ising sectors, typed Majorana realizations within their own domain, $\mathbb{Z}/6$, and virtuality. The PDG reference enters after the internal output has been frozen and preserves the unit, scheme, scale, uncertainty, and validity regime of each comparison, without converting that expository inventory into a cardinality of the law.

Quantum dynamics, Boltzmann constant, information, and gravitation

The finite temporal representation on $\mathbb{C}^{108}$ has a self-adjoint Hamiltonian whose unitary propagator reproduces the periodic TPK endomorphism. Its variational action leads to the Schrödinger equation. On the chronological record $\mathbb{T}_{27}^2$, the concrete realization

$$U_{\mathrm{micro}} = S D_J C_4, \qquad U_{12}^{\mathrm{reg}} = U_{\mathrm{micro}}^{12},$$

is unitary; its powers sum exactly the ordered weights of all finite routes, and the discrete Fourier transform decomposes it into 27^2 directional blocks. This result is a finite matrix sum, not an automatic identification with a continuous functional integral. The Clifford–Dirac realification then constructs the fermionic operator. The CAR and CCR relations, together with the Hamiltonian and entropy maximization, respectively produce the Fermi–Dirac and Bose–Einstein distributions. The exterior grading realizes the Pauli exclusion principle, and the Möbius band organizes the double return $2\pi/4\pi$.

The Boltzmann constant links action, temperature, and ternary information; the ensembles construct Gibbs and the KMS condition. The reversible transport of the complete fiber realizes the null Landauer limit, whereas the projection that discards genealogy receives a positive cost. Area, modular entropy, and entanglement entropy are presented with their explicit states, operators, and partitions.

The gravitational branch first receives the internal outputs of action, clock, length and velocity, and constructs on them the dimensionless selector and the exact transducer

$$G_{\mathrm{int}}(\theta_G) = \theta_G \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_{\mathrm{int}}} \in \mathcal{L}_G^+$$

is the positive chart of the gravitational transducer. The internal periodicity conditions construct

$$R_{108} = \frac{54}{\pi_{\mathrm{HMT}}}\ell_0, \qquad r_{g,a}(\theta_G) = \theta_G \frac{\pi_{\mathrm{HMT}}}{54}\ell_0,$$

and the equality $r_{g,a}(\theta_G) = R_{108}$ positively and uniquely selects

$$\theta_G = \theta_{108}^{\mathrm{circ}} = \left(\frac{54}{\pi_{\mathrm{HMT}}}\right)^2.$$

The chart thus determined publishes G_a. Only afterward is the output realized in the Palatini–Cartan–Holst action, its cotetrad and connection equations, the Legendre transform and the constrained Hamiltonian. Metrological recognition is subsequent and does not participate in selection. The common finite TPK operator admits solenoidal and gauge realizations; self-inertial frames, the Machian reading and sheet cosmology are developed after local and collective dynamics have been fixed.

The complete thesis may be summarized in a single causal direction. The discrete structure preserves unity, orientation, and memory; its prolongation constructs the continuum and numbers; its double projection constructs values and symmetries; its readers determine constants; and Dimensional Mechanics subsequently realizes those outputs as particles, masses, fields, and physical geometry. The origin of the fundamental constants and the discrete structure of the continuum are, within this framework, 2 aspects of the same mathematical problem.

Keywords. Genealogical continuum; Triadic Modular Holography; Dimensional Mechanics; APP; TRIT; TPK; nonadic monodromy; double projection; fundamental constants; Steiner systems; Witt stars; Golay codes; Leech lattice; vertex operator algebras; Monster; Moonshine; Planck constant; Barbero–Immirzi; General Mass Law; Einstein–Cartan–Holst gravity.

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Causal architecture of the continuum problem and mathematical-physical scope of HMT–MD

Inaugural thesis: the discrete structure of the continuum and the common origin of its publications

Triadic Modular Holography does not take the real line as a prior setting or receive from it the values it will later use. Its input is finite: the nine positions $1, \dots, 9$, arranged horizontally and vertically so that the same chart supports two irreducible operations. APP simultaneously preserves the additive sheet, which accumulates, and the multiplicative sheet, which folds; the residue, quotient and carry prevent equality between two published values from erasing the difference between the histories that produced them. TRIT orients each transition and distinguishes its regimes. TPK turns that local orientation into dynamics: it selects the tritic mode without sacrificing any APP sheet, transports, memorizes, returns and updates the state. Its readings modulo 3, modulo 9 and modulo 1000, the nonadic connection, the dodecaphase, the boundary and the return with memory are not independent additions, but coordinated charts of the same genealogical operation.

Preservation is written from the first level:

$$i + j = R^{+}_{ij} + 9q^{+}_{ij}, \qquad ij = R^{\times}_{ij} + 9q^{\times}_{ij}.$$

The residue publishes the class; the quotient unfolds its provenance. The TPK transition preserves that distinction through

$$\mathcal{U}_t = \mathrm{Upd}_t \circ \mathrm{Tra}_t \circ \mathrm{Sel}_t,$$

where selection incorporates the TRIT mode, transport realizes the oriented direction and updating brings together memory and return. In the canonical chart, the mode $m_{\mathrm{ph}} = (1, -1, 0)^3$ is common to the APP sheets and has singular value $s = 1/2$; therefore

$$1 - s^2 = \frac{3}{4} = \frac{90}{120}, \qquad \left\| [P_{\Sigma,3}, P_{\Pi,3}] \right\| = \frac{\sqrt{3}}{4}.$$

The normalized commutator satisfies $J^2_{\mathrm{comm}} = -I$. Sum–product incompatibility, tritic orientation, defect $90 \to 120$, complex structure and electronic prefactor thus belong to the same internal genealogy; they are not linked by decimal resemblance.

The compatible prolongation of these finite states produces increasingly narrow cylinders and simultaneously preserves the history of their refinement. The real line then appears as an Archimedean publication of the limit, not as initial data. That chart first publishes the values of the nine symbols of the seed and then prolongs the same genealogy toward π, e, φ and α, without using any of them as a retrospective selector. This is the thesis governing the entire work: the joint discrete structure constructs the continuum and explains why its stable values are what they are. The chain

$$\{1, \dots, 9\}_{\mathrm{h}} \times \{1, \dots, 9\}_{\mathrm{v}} \longrightarrow \mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \mathcal{X}_{\mathrm{enr}} \longrightarrow \mathcal{S}_{\infty} \longrightarrow \mathbb{R}$$

does not summarize heterogeneous chapters: it expresses a single mechanism seen at successive resolutions. In it the phase may return while the state advances, because the quotient, carry, orientation and memory have changed. The cycle observed only through its phase is circular; with its genealogy preserved, it is helical.

This difference is fixed through an injective lift. If $n = 9q_9(n) + \rho_9(n)$, with $1 \le \rho_9(n) \le 9$, the circular projection $\mathcal{R}_9$ and the helical immersion $\mathcal{H}_9$ satisfy

$$\mathcal{R}_9(n+9) = \mathcal{R}_9(n), \qquad \mathcal{H}_9(n+9) = \mathcal{H}_9(n) + (0,0,1).$$

The phase has closed and the genealogical height has grown: phase return and state return are mathematically separated. The two-way involution organizes conjugate ascending and descending sheets of the same lift, not two independent generators. In this geometry, the helix preserves the longitudinal advance of memory, the spiral records the transverse contraction and the cylinder radius measures the width of the interval still compatible with the prefix. The nesting of these half-open cylinders, with radius tending to zero and genealogy preserved, is the mechanism that subsequently publishes the real section.

The certified initial segment of the uniform prolongation constructing the coinductive state $\mathcal{S}_\infty$ is

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9,$$

but writing the chain acquires content only by declaring its levels, thresholds and invariants. For $\chi \in \{\pi, e, \varphi\}$, let $b_0^\chi = w_6^\chi$ and

$$b_{k+1}^\chi = b_k^\chi L_{\epsilon_k} \pmod 3, \qquad (\epsilon_0,\epsilon_1,\epsilon_2,\epsilon_3) = (0,0,1,1), \qquad w_{6(k+1)}^\chi = b_0^\chi \| \cdots \| b_k^\chi.$$

The twelve transitions produced by $\mathcal{U}_t$ form the matrix pairs X_i, Y_i of the two regimes and uniquely determine $L_i = X_i^{-1} Y_i$, $i = 0, 1$. The calendar (L_0, L_0, L_1, L_1) is the only one of the sixteen binary calendars that simultaneously satisfies the three biographies; the one hundred and forty-four single-entry perturbations fail. Therefore L_0, L_1 and 0011 are rigid outputs of APP–TRIT–TPK transport, never data chosen from π, e or φ. The matrices thus constructed act on the current block; concatenation literally preserves all preceding blocks. The visible word accompanies an enriched state that also transports APP sheet, TRIT orientation, residue, quotient, carry, boundary, memory and incidence shadow. Therefore, coincidence of words and coincidence of states are distinct mathematical conditions.

Levels and thresholds of TPK prolongation

w_6**: visible seed.** It is the six-position ternary word published in each channel after the chain of types $104\,976 \to 468\,U_6 \to 243\,w_6 \subset \mathbb{F}_3^6$, with $|\mathbb{F}_3^6| = 729$. It fixes the first observable prefix, while the decimal emission U_6, its region, multiplicity and source state remain in the enriched fiber. The action D_3, the fiber invariants and the gauge $(\mathrm{diag}_c, h_+) = (6, ES)$ fix the channel representative before any comparison with its real value; in channel π, five distinct regions U_6 also preserve different biographies over the same word w_6^π.

w_{12}**: first lift.** It adds $b_1 = b_0 L_0$ to the seed. It changes the visible depth from one to two blocks and preserves w_6 as an exact prefix, together with the channel and the transported genealogy. The lift produces an extension of the state; it does not replace the seed with a word without a past.

w_{18}**: closure of regime** L_0**.** It adds $b_2 = b_1 L_0$. The second application of the same operator completes the first calendar regime and distinguishes a recurrence from an isolated coincidence: the eighteen trits contain, in their order, the levels of six and twelve trits that precede them.

w_{24}**: typed change of lift.** It adds $b_3 = b_2 L_1$. Here the operator applied to the current block changes, from L_0 to L_1, while the eighteen-trit prefix and the transported state remain. The regime change modifies the continuation and preserves the history on which it acts.

w_{30}**: first-boundary prefix.** It adds $b_4 = b_3 L_1$ and completes the finite calendar (L_0, L_0, L_1, L_1). Its five blocks per channel constitute the prefix from which deep prolongability is tested. It is an entry level to the boundary, while the same recurrence admits further blocks.

R_{36}: first joint coinductive boundary. If u_5^{χ} is the sixth block of channel χ, then

$$R_{36} := (u_5^{\pi}, u_5^{e}, u_5^{\varphi})^T, \qquad w_{36}^{\chi} = w_{30}^{\chi} \| u_5^{\chi}.$$

The boundary relation selects that triple while preserving the axis φ, the lateral aggregate π/e, saturation, column occupancy and dual neutrality. Its equations leave exactly the two mirror solutions

$$B_- = (021222, 222220, 102011)^T, \qquad B_+ = (222220, 021222, 102011)^T;$$

the APP orientation of the half-open cylinder fixes $R_{36} = B_+$. The subscript records thirty-six trits exposed per channel; R_{36} is the boundary of the three sixth blocks and $\mathbb{R}$ is the subsequent Archimedean publication of the cylinder system.

G_9: connection of prolongation and return with memory. Level G_9 organizes the nine local charts that act after R_{36}. If $\mathcal{R}_K$ is the compatible extension relation in phase K, their composition defines the nonadic holonomy

$$\Gamma_{9,K} = \mathcal{R}_{K+8} \circ \cdots \circ \mathcal{R}_K : \widehat{X}_K \longrightarrow \widehat{X}_{K+9}.$$

The subscript 9 comes from the phase cycle C_9, inherited from the nine APP positions, and counts the nine relations composing a turn. Its operational content is certified by

$$g(K+9) = g(K), \qquad q_{\mathrm{mem},K+9} = q_{\mathrm{mem},K} + 1, \qquad \widehat{x}_{K+9} \neq \widehat{x}_K.$$

The phase label returns, memory advances and the prolonged state changes. This is the structural meaning of G_9: a connection turn over an enriched fiber, whose cardinality comes from the architecture and whose action is verified in transport. The modal quotient $\mathcal{K}_{\mathrm{ph}}$ occupies the subsequent reduced stage,

$$\mathcal{U}_t \longrightarrow \Gamma_9 \longrightarrow \mathcal{K}_{\mathrm{ph}},$$

and publishes compressed memory after the holonomy. The local resolution of the ninth phase also distinguishes two horizons: $\#\mathrm{Surv}_1(B_9) = 3$ and $\#\mathrm{Surv}_2(B_9) = 1$. This descent $3 \to 1$ certifies the pruning of a specific fiber; composition of the nine relations certifies the complete turn.

The dodecaphase records another coordinate of the same transport. The vertical counter $q_{\mathrm{mem},K}$ remains unbounded and its reading $\beta_K = [q_{\mathrm{mem},K}]_{12}$ satisfies $\beta_{K+9} = \beta_K + 1 \pmod{12}$. Reduction modulo twelve preserves the dodecaphasic position; the integer counter preserves the number of turns. In the terminal state, this memory feeds the signed record U_{12}^{sgn}, the reversible seal K and the carry cocycle that publishes the coordinate α_{HMT}. The nonsplit extension

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0$$

types the seam between the two calendars: nine steps return the phase and advance one dodecaphasic sector; one hundred and eight steps close the observable clock while the enriched state preserves the genealogical height reached.

Bidirectionality combines the channel mirror and the reversibility of the lift within a single generator. The involution

$$\mathfrak{c}(u^{\pi}, u^{e}, u^{\varphi}) = (u^{e}, u^{\pi}, u^{\varphi})$$

exchanges the conjugate paths π/e and preserves the axis u^{φ} and the aggregate $u^{\pi} + u^{e}$. In the integer chart

$$x_k A = u_k + 3x_{k+1}, \qquad \det A = 1,$$

the advance determines residue and quotient, and the pair (u_k, x_{k+1}) recovers x_k. Ascent and descent are therefore conjugate paths of the same state; descent reconstructs provenance without erasing accumulated memory.

Finally, G_9 constitutes the recurrent unit of prolongation and not its endpoint. The surviving histories form the inverse system

$$\cdots \longrightarrow \mathrm{Surv}_{\chi,n+1} \longrightarrow \mathrm{Surv}_{\chi,n} \longrightarrow \cdots, \qquad x_\chi \in \varprojlim_n \mathrm{Surv}_{\chi,n}\,.$$

Nonemptiness, local uniqueness and compatibility of truncations produce the coinductive section; contraction of its cylinders publishes a unique real value, while the inverse limit preserves the genealogy that this scalar value omits. The twenty blocks and the per-channel certificate of 396 events, 2376 trits, 43 turns and 387 pairs $(K, K+9)$ are finite executions of that uniform law, not bounds on prolongation.

On the coinductive state thus constructed, a correlated kernel of four publications is obtained. The correlation expresses common origin and compatibility; it does not flatten their operational order. The channels first publish the vectors P, E, Φ, while the signed terminal reading constructs $U_{12}^{\mathrm{sgn}} \overset{\mathcal{H}_{12}}{\longleftrightarrow} K$. On these outputs, and without reintroducing their values as data, the dodecaphasic carry cocycle derives the coinductive section α_{HMT}, whose dodecaphasic window is the rational projection α_{12}. The vacancy kernel $E_{21/22}$ provides a second internal construction: its finite simple positive root α_E, prolonged by the same graded TPK transport to the inverse system whose limiting root is α_B. Integer closure defines $\alpha_A := \varprojlim_N \rho_N(\alpha_{12})$, and the exact two-way theorem establishes

$$\boxed{\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}}.$$

The equality $\mathrm{Tr}_9(\alpha_E^{-1}) = \mathrm{Tr}_9(\alpha_{12}^{-1})$ is the first finite certificate of that identity: it neither defines it, limits its scope nor selects any of the prolongations. Thus, $\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}$ and α_{HMT} belong to a single correlated coinductive generation, but preserve differentiated stages, domains, readers and certificates. The Machin identities and Leibniz brackets, the Perron–Fibonacci realization, the factorial semigroup and the metrological tables subsequently recognize or certify the publications already generated; none of them selects the cylinders, vectors or carry of the generator.

The same common source has other publications that must not be confused with one another either. In the internal mirror chart, the involution exchanges the branches u^π and u^e, preserves u^φ and the aggregate $u^\pi + u^e$; only after real publication are expressions such as $e+\pi$, $e\pi$, π^e or the literal quotient π/e formed, each with an arithmetic status that does not follow from the mere existence of the ternary aggregate. In the incidence chart, H_α^-, v_α and N_α denote, respectively, a support, the vector of the exact index-three neighbor and the resulting lattice; they do not denote the scalar α_{HMT} or intervene in its numerical carry. The tetrad B_K enters the Witt star and the vector v_α determines the neighbor leading to $N_\alpha \cong \Lambda_{24}$. From the common Paley–Witt code C_W, two typed branches then open, not a single linear chain:

$$C_W \longrightarrow W_{12} \longrightarrow W_{24}, \qquad C_W \longrightarrow N(A_2^{12}) \xrightarrow{v_\alpha} \Lambda_{24} \longrightarrow V_\Lambda \longrightarrow V^\natural \longrightarrow \mathbb{M} \longrightarrow \text{Moonshine}.$$

The scalar value of α does not generate these symmetries. Numerical derivation and exceptional prolongation are typed readings of the same enriched state and preserve a common genealogical root.

Dimensional Mechanics begins when that structure is transduced into action, magnitude, field and matter. From it come, according to its own readers, the Planck constant and the electromagnetic, thermal, gravitational and quantum constants; the mathematical structure of the electron, the General Mass Law and mixing transports; and, later, the Barbero–Immirzi parameter as a structural object of the transition between matter, area and information. Conventional values appear at the end as metrological recognition, never as inputs to the calculation. For the same reason, resolution of the continuum problem and the five

inseparable constructions emerging from it precede terminal proofs concerning classical mathematical problems: these stress-test an already constructed structure, rather than generating it retrospectively.

The physical order is equally binding. The coordinate α_{HMT} opens, in typed branches, the retention η_{ret} and the structural valuation 10^{-34}; their confluence yields $\hbar_{\mathrm{ret}}$, and only then are the angle and counter-angle constructed and a single conversion between degrees and radians performed. The electron brings this chain together in a material persistence. Its basal

$$e_0 = \frac{\sqrt{3}}{4}\left[\exp\!\left(\frac{\varphi}{\pi^2}\right) - 22\alpha_{\mathrm{HMT}}^{3}\right](1 + 15\Delta_4)$$

precedes beta closure; its two readers coincide at

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80, 54, 6), \qquad \kappa_e = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4,$$

and the published mass is recognized metrologically only after evaluating $m_e^{\beta} = e_0(H_5/\eta_{\mathrm{ret}})^{\kappa_e}$. Tensor gravity, its five-component algebraic sector and the typed physical realization; the T-duality and M-theory screens; and the geometry of area and information prolong the same antecedent with their own domains and falsifiers. None of these realizations reintroduces as a premise the magnitude it must explain.

Everything that follows develops this single thesis in different domains. Numbers, symmetries and matter are not three juxtaposed collections. They are three ways in which the constructed continuum publishes and returns the memory of its own formation.

0.1 Projective horizon: dependence between scale, boundary, and measure

Finite oriented chain and distinction between marks, cells, and genealogical units

The exposition begins with a elementary operation: fixing to segment, marking to scale on it and comparing the regions that esa scale determines. In this point not it is presupposed yet the real line as complete support. The initial object is a finite oriented chain

$$\mathcal{I}_K = (L_K, \prec_K;\ x_{K,0} \prec_K x_{K,1} \prec_K \cdots \prec_K x_{K,n_K};\ o_K),$$

where $(L_K, \prec_K)$ is a finite totally ordered set, the $x_{K,j} \in L_K$ are observable marks, and o_K distinguishes the two directions of traversal. No real magnitude has yet been assigned to L_K. The marks and consecutive cells are distinct objects: under the closed-cell convention, the common mark is incident to the closures of two adjacent cells without thereby becoming an additional cell.

This apparently minimal distinction determines the subsequent architecture. A visible value describes the position published in a chart; a genealogical unit further preserves the interval of provenance, the direction, the accumulated quotient, and compatibility with finer scales. Therefore, two numerically equal presentations may represent different histories, and the same history may admit several coordinated publications.

For $n \in \mathbb{N}$, the nonadic chart organizes this information by means of the Euclidean division

$$n = 9q_9(n) + \rho_9(n), \qquad q_9(n) = \left\lfloor \frac{n}{9} \right\rfloor, \qquad \rho_9(n) \in \{0, \ldots, 8\}. \tag{21}$$

The public phase numbering is obtained through

$$g(n) = 1 + \rho_9(n-1) \in \{1, \ldots, 9\} \qquad (n \geq 1).$$

The residue localizes that phase, and the quotient q_9 preserves how many revolutions have been absorbed by the scale. This pair realizes a phase return while preserving the trajectory. Its decisive invariant is the joint persistence of both coordinates under refinement.

Inverse system of finite resolutions and relative observation horizon

We shall call *horizon* the boundary up to which a finite chart has resolved the object and at which that description is linked with the next chart. If $\mathcal{S}_K$ denotes the nonempty set of enriched states compatible with the scale $K \in \mathbb{N}$, the surjective links of forgetting

$$\tau_{K+1,K} : \mathcal{S}_{K+1} \longrightarrow \mathcal{S}_K$$

remove visible detail, but preserve the necessary information to recognize a prolongation. We write

$$\tau_{K,K} = I, \qquad \tau_{L,K} \circ \tau_{M,L} = \tau_{M,K} \quad (K \le L \le M).$$

The complete object is then the inverse limit,

$$X = \varprojlim_{K} (\mathcal{S}_K, \tau_{K+1,K}), \qquad x = (x_K)_K, \quad \tau_{K+1,K}(x_{K+1}) = x_K. \tag{22}$$

The nonemptiness of the levels and surjectivity of the links guarantee the existence of compatible families; in the specialized channels, their sections are subsequently constructed by means of the survival theorems. Each truncation is finite; compatibility has no preset maximum depth. Ten, one thousand, or ten thousand digits are checks of the same prolongation rule, not alternative definitions of the number.

This formulation distinguishes three operations that decimal notation usually superimposes. The first refines a partition. The second transports the state between phases. The third publishes a coordinate in a continuous chart. Only the last takes values in $\mathbb{R}$. Thus, we begin with a finitely marked line and end with the real line as the final Archimedean realization of a compatible section.

Compatible augmentation and evolutionary conservation of the unit

Unity is formulated before every Archimedean evaluation. Let $C_K = \mathbb{Z}[\mathcal{S}_K]$ be the free module of the states of level K. The link map induces $(\tau_{K+1,K})_{\#} : C_{K+1} \to C_K$, and there exists the augmentation

$$\varepsilon_K : C_K \longrightarrow \mathbb{Z}, \qquad \varepsilon_K\left(\sum_{x \in \mathcal{S}_K} a_x [x]\right) = \sum_x a_x,$$

compatible with the links, $\varepsilon_K \circ (\tau_{K+1,K})_{\#} = \varepsilon_{K+1}$. Evolutionary preservation of unity expresses that refinement redistributes its content among visible part, boundary, and memory without creating a second root or eliminating the terminal term.

Let P_m be the cellular path with m vertices and $m - 1$ edges, and

$$Q_{m,d} = P_m^{\square d}.$$

Its number of k-cells is

$$\#C_k(Q_{m,d}) = \binom{d}{k} (m-1)^k m^{d-k}. \tag{23}$$

The identity

$$10^d = (9+1)^d = 9^d + \sum_{j=0}^{d-1} \binom{d}{j} 9^j \tag{24}$$

is the equality of the 0 cells of the completion. The subcomplex $Q_{9,d}$ and its relative corona together form $Q_{10,d}$. In dimension three,

$$1000 = 729 + 271,$$

where 271 counts only the new vertices. The vector cellular relative complete is

$$\#C_\bullet(Q_{10,3}, Q_{9,3}) = (271, 756, 702, 217), \qquad 271 - 756 + 702 - 217 = 0. \tag{25}$$

Its edges, faces, and cubes cannot be omitted when studying incidence, homotopy, or transport.

The adjective *holographic* designates this compatibility here: the boundary datum, accompanied by orientation, carry, and memory, controls the prolongation of the state without replacing the interior by a number. The unit is the same across dimensions, although its decomposition and readers change.

Nonadic holonomy and phase return with memory increment

Non-Abelian transport is a single connection over the phase cycle. Its full holonomy is written

$$\Gamma_9 = \mathrm{Hol}_{C_9}(\nabla_{\mathrm{TPK}}), \qquad g(k+9) = g(k). \tag{26}$$

Here $g : \mathbb{Z} \to C_9$, and for every point $\widehat{x}_k$ of an admissible orbit of the enriched state,

$$\Gamma_9 \widehat{x}_k = \widehat{x}_{k+9}, \qquad q_{\mathrm{mem}}(\widehat{x}_{k+9}) = q_{\mathrm{mem}}(\widehat{x}_k) + 1.$$

In particular, when q_{mem} distinguishes states, $\widehat{x}_{k+9} \neq \widehat{x}_k$. The nine ordered components describe a turn of the connection; are not nine filters applied to a motionless state. In each channel there exists a map local typed

$$\mathrm{loc}_{K,\chi} : X_\chi \longrightarrow \mathbb{F}_3^6, \qquad |\mathbb{F}_3^6| = 729.$$

The transfer traverses the ambient fiber and locates the coordinate of the compatible extension. The turn returns the phase and advances the memory, the prefix, the carry, and the boundary.

In an isometric realization, the separation between visible publication and expressed memory is typed through

$$J : \mathcal{H}_{\mathrm{vis}} \hookrightarrow \mathcal{H}_{\mathrm{full}}, \quad J^*J = I, \qquad U_{\Gamma_9}^* U_{\Gamma_9} = I,$$
$$T = J^* U_{\Gamma_9} J, \qquad \eta = (I - JJ^*) U_{\Gamma_9} J.$$

It follows

$$U_{\Gamma_9} J = JT + \eta, \qquad J^*\eta = 0, \qquad I - T^*T = \eta^*\eta. \tag{27}$$

The term η expresses exactly the orthogonal defect produced by this compression. For composable routes $\gamma_1 : x_0 \to x_1$ and $\gamma_2 : x_1 \to x_2$, the carry $\mathrm{Carr}_R(\gamma) \in \mathsf{C}$ satisfies, with $H(\gamma), Y(\gamma) \in \mathrm{End}(\mathsf{C})$, the cocycle law

$$\mathrm{Carr}_R(\gamma_2\gamma_1) = \mathrm{Carr}_R(\gamma_2) H(\gamma_1) + Y(\gamma_2)\,\mathrm{Carr}_R(\gamma_1), \tag{28}$$

which is the algebraic form of the genealogical continuity.

Dimensional elevation and coordination of geometric regimes

The marked line provides the order and horizon. APP constructs then the two-dimensional chart in which sum and product act on a support common without losing their quotients. TRIT locally orients each interaction by the unique residual decomposition

$$a + b = r + 3c, \qquad a, b \in \mathbb{Z}, \quad r \in \{-1, 0, +1\}, \quad c \in \mathbb{Z}.$$

The Topological Phase Kernel organizes routes, sheets, boundary, signature, orientation, cylinder, record, memory, and transport. Elevation to the cube does not add a spatial metaphor: it introduces the incidences and coherences that make it possible to compare interior, crown, and publication in successive dimensions.

This sequence also prepares the articulation among three geometries. The circular structure preserves phase; the complex structure represents the quarter-turn on a real space V by an operator $\mathcal{J} : V \to V$ with $\mathcal{J}^2 = -I_V$; the hyperbolic structure is realized in the Poincaré disk, whose boundary lies at infinite hyperbolic distance. Their identifications will be effected by explicit maps, not by equality of drawings.

The geometric horizon thus anticipates the thesis of the book: a published metric, the memory of what has been traversed, and the openness of continuations are coordinated readings of an oriented structure.

The double publication originates from a single source:

$$\mathrm{ev}_\infty^{\mathrm{arq}} : X_\chi \longrightarrow \mathbb{R}, \qquad \mathrm{ev}_\infty^{\mathrm{inc}} : X_\chi \longrightarrow \mathcal{E}_{\mathrm{inc}},$$

where $\mathcal{E}_{\mathrm{inc}}$ designates the limit of the boundary incidence objects constructed in Part II. The first map publishes the Archimedean value; the second preserves the exceptional genealogy.

Causal sequence APP–TRIT–TPK, continuum, and typed publications

The following chapters construct APP, TRIT and the Topological Phase Kernel from their finite data; develop the discrete structure of the continuum and its two publications; derive the genealogies of the constants; integrate the five inseparable realizations—spectral, solenoidal, gauge, cohomological and determinantal—of the jointly constructed continuum; and then construct, through Dimensional Mechanics, the physical realizations of the sectors that have them. Archimedean publication appears at the end of each chain, as a reader of an already constructed object. This is the precise sense in which the work begins with a line and ends in $\mathbb{R}$.

Location of complete proofs. This opening fixes the objects and causal order. The complete proofs are found in the chapters devoted to the discrete measure complex, the APP gnomonic construction, TRIT oriented transport, the operator architecture of the Topological Phase Kernel, cubic completion, and double publication. Subsequent cross-references link each construction to its complete proof; the present opening fixes the common domain and the order of composition.

0.2 Projective Formalization of Unity in Successive Scales and Dimensions

0.2.1 Construction of the finite line through order, scale, and conservation of the unit

The construction begins with an oriented line and a declared scale. It does not begin with the already completed set of real numbers, but with the finite object

$$\mathcal{I}_m = \big([0, 10^m],\, \mathcal{M}_m,\, \mathcal{E}_m,\, o_m\big), \qquad m \geq 1, \tag{29}$$

where $\mathcal{M}_m$ is the set of marks, $\mathcal{E}_m$ the set of elementary intervals, and o_m an orientation. The formula brings together four distinct objects. The segment provides the ordered support; the marks publish positions; the intervals compare separations; the orientation determines the direction of the reading. Suppressing any of these coordinates changes the type of the object.

In the first chart, $[0, 10]$, the visible nonadic positions are $1, \ldots, 9$. Between them existen eight interior intervals. The step that connects the last visible state with the first of the next turn it is recorded as closure-and-carry edge. By eso the equality

$$9 \text{ visible marks} \quad \longleftrightarrow \quad 8 \text{ interior intervals} + 1 \text{ closure link} \tag{30}$$

does not identify marks with intervals. It expresses how counting and measure share a boundary without becoming the same operation.

The Euclidean division

$$n = 9q_9(n) + \rho_9(n), \qquad 0 \leq \rho_9(n) \leq 8, \tag{31}$$

is now read in two directions. The residue ρ_9 determines the published phase; the quotient q_9 preserves the accumulated height. When the phase returns to its initial value after nine steps, the quotient has advanced. The trajectory is helicoidal: one coordinate of the wheel has been recovered, but not the complete state.

Algebraic interpretation of the residue–quotient decomposition. The equality (31) is not merely an algorithm for computing remainders. It defines an extension: projection to the residue contracts the history, and the quotient preserves the information required to lift it. The elementary falsifier consists in imposing $\rho_9(n+9) = \rho_9(n)$ and $q_9(n+9) = q_9(n)$ simultaneously. The first equality is correct; the second contradicts Euclidean division.

0.2.2 Projective system of scales and cubical completion to arbitrary depth

Scaling is performed through refinement maps, not through a graphic repetition of the same rule. For $m \leq n$, let

$$r_{n,m} : \mathcal{I}_n \longrightarrow \mathcal{I}_m, \qquad r_{m,m} = I, \qquad r_{n,m} r_{\ell,n} = r_{\ell,m}. \tag{32}$$

The map $r_{n,m}$ preserves the mark published at resolution 10^{-m} and contracts the detail lying below that threshold. The composition equation ensures that forgetting two levels at once produces the same result as forgetting them successively. This is the condition that makes it possible to speak of a single history observed at different scales.

In three dimensions, the passage from the nonadic fiber to the complete decimal chart is expressed by

$$1000 = (9+1)^3 = 9^3 + 3\,9^2 + 3\,9 + 1 = 729 + 243 + 27 + 1 = 729 + 271. \tag{33}$$

The equality contains a stratification. The 729 interior states have no boundary coordinate; the next 243 states have one; the 27 states have two; the apex has three. The number 271 counts the vertices added by the decimal corona. It does not count the internal boundary of the nonadic cube, whose cardinality is $9^3 - 7^3 = 386$, nor the complete relative complex, which includes edges, faces, and cubes.

For every dimension d, let

$$\overline{C}_9^{(d)} = (E_9 \sqcup \{\star\})^d, \qquad \star \notin E_9, \tag{34}$$

and denote by $B_r(d)$ the stratum with exactly r coordinates equal to $\star$. Then

$$|B_r(d)| = \binom{d}{r} 9^{d-r}, \qquad \sum_{r=0}^{d} |B_r(d)| = 10^d. \tag{35}$$

The equation describes growth of the scale without imposing a terminal dimension. Each d has more states, but all belong to the same completion law.

0.2.3 Compatible product measure and conservation of the unit under projection

The expansion of cardinals is accompanied by a conservation of measure. Assign to each symbol of $E_9 \sqcup \{\star\}$ weight $1/10$, and let μ_d be the product measure over $\overline{C}_9^{(d)}$. If $\pi_d : \overline{C}_9^{(d)} \to \overline{C}_9^{(d-1)}$ forgets the last coordinate, then

$$(\pi_d)_* \mu_d = \mu_{d-1}, \qquad \mu_d\big(\overline{C}_9^{(d)}\big) = 1. \tag{36}$$

The first equality is the evolutionary conservation of unity. The dimension increases, the number of states changes, and the normalized mass remains equal. The boundary does not add a second unity: it redistributes the same unity among compatible strata.

This conservation has to cellular version. For the path P_m and the cubical product $Q_{m,d} = P_m^{\square d}$,

$$\#C_k(Q_{m,d}) = \binom{d}{k} (m-1)^k m^{d-k}. \tag{37}$$

In particular, the relative crown of $Q_{9,3} \subset Q_{10,3}$ has cellular vector

$$\#C_\bullet(Q_{10,3}, Q_{9,3}) = (271, 756, 702, 217), \qquad 271 - 756 + 702 - 217 = 0. \tag{38}$$

The null Euler characteristic certifies that the new vertices are accompanied by the required incidences. Reducing the corona to 271 eliminates precisely the transport that the theory must preserve.

0.2.4 Geometric curvature and observation fibers: formal separation between metric and horizon

Euclid's fifth postulate governs the global behavior of parallels. In the Euclidean formulation, it fixes a geometry in which the sum of the angles of a triangle and the distance between geodesics correspond to zero curvature. Its modification opens elliptic and hyperbolic geometries. The visible horizon of a geometric chart does not by itself constitute a proof of the postulate: it is the boundary at which the chart ceases to distinguish prolongations.

The correct relation between the two concepts is established through maps. A geometry (M, g) provides distances and geodesics; a remark chart

$$\mathrm{Obs}_K : (M, g) \longrightarrow Y_K \tag{39}$$

publishes the accessible information to resolution K. The relative horizon of $x \in M$ is the fiber

$$\mathsf{Hor}_K(x) = \mathrm{Obs}_K^{-1}\big(\mathrm{Obs}_K(x)\big). \tag{40}$$

Curvature belongs to g; indistinguishability belongs to the fiber of Obs_K. The horizon can change when the chart is refined without the curvature changing. This separation prevents identifying an observer limitation with an intrinsic property of space.

In the Poincaré disk, the Euclidean boundary lies at infinite hyperbolic distance. On the sphere, positive curvature closes the geodesics without an analogous boundary. In the Euclidean chart, parallelism remains global. HMT coordinates these regimes through the tritic parameter and the algebras

$$\mathbb{A}_s = \mathbb{R}[J]/(J^2 + sI), \qquad s \in \{+1, 0, -1\}, \tag{41}$$

but preserves its metrics and codomains. The common algebraic unit does not erase the difference geometric between cos, the parabolic branch and cosh.

0.2.5 Limit relation between infinitesimal, infinite prolongation, and projective threshold

An indefinite expansion contains two orientations of the same scaling operation. Toward precision, refinement distinguishes increasingly narrow cylinders; toward extension, height records increasingly long revolutions. Let $\mathcal{S}_K$ be the set of states surviving at level K. The complete section is

$$X_\chi = \varprojlim_K \mathcal{S}_K, \qquad x = (x_K)_K, \qquad \tau_{K+1,K}(x_{K+1}) = x_K. \tag{42}$$

Each x_K is finite; compatibility has no maximum depth. An infinitesimal is not identified with a prolongation branch: the former is a scale relation in a realization; the latter is a datum of the inverse system. Only a reader that declares the metric and the evaluation can transform the latter into the former.

The mathematics of thresholds appears when a prolongation changes regime upon crossing a boundary. If $\mathrm{Surv}_K^{(N)}$ denotes the states of level K that admit extension up to N, then

$$\mathrm{Surv}_K^{(N+1)} \subseteq \mathrm{Surv}_K^{(N)}, \qquad \mathrm{Surv}_K = \bigcap_{N \geq K} \mathrm{Surv}_K^{(N)}. \tag{43}$$

The vacancy records the threshold at which a visible continuation ceases to be compatible. It is not an unstructured void: it preserves the cause of discarding, the carry, and the boundary that determine the next level.

0.2.6 Compatible limit of finite horizons and realization of the real line

The Archimedean evaluation appears after constructing the section. For a decimal chart of base 10, let

$$\mathrm{ev}_{\mathbb{R}}(x) = a_0 + \sum_{j\geq 1} a_j 10^{-j}, \qquad a_j = \mathrm{Pub}_j(x_j). \tag{44}$$

Convergence is obtained because the compatible cylinders have diameters that tend to zero in the declared chart. The digit a_j is a coordinate; the sequence $(x_j)_j$ is the history; the real number is the limit publication. Therefore,

$$[0,10] \longrightarrow [0,1000] \longrightarrow [0,10^m] \longrightarrow X_\chi \xrightarrow{\mathrm{ev}_{\mathbb{R}}} \mathbb{R} \tag{45}$$

is a type chain, not a naive inclusion chain.

The second publication of the same x preserves the exceptional incidence. The work will traverse both directions and will finally return to $\mathbb{R}$. The return will then be a conclusion: the real line will have been explained as an evaluation of the compatible state, and not used as a premise that renders genealogy unnecessary.

0.3 Historical genealogy of the continuum problem and HMT–MD mathematical formulation

The historical problem of the continuum, value and measure

The continuum constitutes one of the foundational problems of mathematics. Arithmetic represents discrete units; geometry compares extensions; analysis completes sequences and constructs limits; measure theory assigns magnitudes to sets; physics converts those magnitudes into observables. The efficacy of these disciplines rests on the possibility of publishing a stable and reproducible result. The problem governing this work arises one level earlier: to determine what structure makes that stability possible and what information must be preserved in order to recover a value, a measure, and their symmetries from a common origin.

On the discrete line, a mark and the gap separating it from the next mark are distinct objects. The mark corresponds to the operation of counting; the interval, to the operation of measuring. The passage from 0 to 10 contains the nine visible marks $1, \ldots, 9$, the eight interior intervals between consecutive marks, and one lifted closing edge that carries the count into the next turn. The modular reading preserves the phase; the quotient preserves the attained height. When both coordinates are retained together, the circular repetition becomes a helix, and the sequence of integers preserves the history of its returns.

Triadic Modular Holography (HMT) develops this intuition as a mathematical construction. A finite system of positions, sheets, orientations, paths, and transports produces compatible families of states. Its evaluations publish digits and values; its lifts preserve residue, quotient, carry, boundary, and memory. The continuum is realized as the image of a compatible section of the system of prolongations. Its discrete structure resides in the genealogy that links all resolutions.

This formulation distinguishes three levels. A *digit* is a visible coordinate produced by a reader. A *value* is the image of a section under a specified chart. A *genealogical number* is the complete section that preserves depth, operation, orientation, sheet, residue, quotient, carry, route, boundary, memory, and survival. The decimal expansion constitutes an evaluation of that object; compatibility among its stages constitutes its continuity.

The same structure transforms the concept of symmetry. A symmetry is the evolutionary preservation of identity under a transformation with memory. A cycle may recover the visible position while simultaneously transporting the state to another height, another sheet, or a new block. Invariance belongs to the complete genealogy. The real value and exceptional symmetry therefore appear as two coordinated projections of the same antecedent: one contracts compatible cylinders, and the other preserves the incidence that the Archimedean contraction leaves outside the published coordinate.

Dimensional Mechanics (MD) prolongs this construction into the physical domain. The numerical section receives a spectral representation; persistence is realized as a particle-route; the route ledger produces a signature; the signature feeds a character and a hierarchy of towers; the dimensional chart finally publishes a magnitude. The governing chain of the work is

unit $\longrightarrow$ state with memory $\longrightarrow$ continuum, number and symmetry $\longrightarrow$ particle, constant and quantity.

The scientific objective is unitary and is developed in five Parts. The formal core constructs APP, TRIT, and the Topological Phase Kernel; the discrete structure of the continuum unfolds its sections, the double projection, and the exceptional genealogy; the genealogy of constants produces π, e, φ, α, the internal action scale, and its transducers; five specialization chains construct their internal antecedents and isolate the comparators leading to each classical recognition; Dimensional Mechanics then constructs the constitutive constants, mass, quantum statistics, information, and gravity. Each measurement is presented together with the object and operation that generate it; each metrological comparison is opened only after the internal output has been fixed.

The order of the work follows the constructive dependency: an object appears where it begins to govern subsequent results. Each statement declares its premises, its proof, and, where appropriate, the morphism of physical realization. In this way, a subsequent metrological comparison does not replace the genealogy that produced the output.

From the genealogical closure to the physical realization

The thesis of this work attains four coordinated and irreducible closures: genealogical, operatorial, matricial, and physical. It does not consist in adjoining a set theory, a theory of operator algebras, or a quantum formalism to a prior numerical construction. The four developments arise from the same causal precedence: APP constructs the dual positional and arithmetic realization; oriented TRIT distinguishes return, the torsional present, and opening; TPK preserves value, sheet, orientation, residue, quotient, carry, boundary, route, register, memory, survival, phase, and gauge. Only thereafter are the compatible section, its readers, and its representations formed. The governing diagram is, therefore,

$$\mathrm{APP} \longrightarrow \mathrm{TRIT}_{\mathrm{orientado}} \longrightarrow \mathrm{TPK} \longrightarrow \mathcal{X}_{\infty}^{\mathrm{cal}},$$

$$\mathcal{X}_{\infty}^{\mathrm{cal}} \xrightarrow{\mathfrak{R}} \mathbb{R}, \qquad \mathcal{X}_{\infty}^{\mathrm{cal}} \times C_{\mathrm{exc}} \xrightarrow{\mathfrak{E}} \mathcal{E}_{\mathrm{exc}},$$

$$C(X_m) \xrightarrow{\mathrm{diag}} M_{9^m}(\mathbb{C}) \xrightarrow{a \mapsto a \otimes I_9} \mathrm{UHF}(9^{\infty}) \xrightarrow{\pi_\tau(\,\cdot\,)''} R_{\mathrm{hyp}},$$

$$\mathsf{P}_{\mathrm{CT}}^{+} \xrightarrow{\mathfrak{R}_{\mathrm{MD}}^{\mathrm{disc},+}} \mathbf{SpecRoute} \times \mathsf{Hilb}_{\mathbb{R},J} \times \mathbf{B}\mathfrak{M}_D^{+} \times \mathbf{B}G_C.$$

The four lines do not form a single linear chain. Real evaluation and exceptional projection are sister readers of the enriched section; the tracial branch passes through the cylindrical algebras and their unital inclusions; the physical realization begins with prepared routes endowed with the structures required by each component. This separation of domains prevents the von Neumann factor from being deduced directly from a section, the exceptional projection from being identified with an action on digits, or a physical magnitude from being presented as though it were a TPK state without a realization morphism.

Genealogical Closure: Extensionality, Infinity, and Decision

The first closure determines what it means to preserve an object across all depths. Given an inverse system of finite spaces (X_m, p_m), a section $x = (x_m)_m$ is not characterized solely by the equality of its published values. The fiber-descent theorem establishes the conditions under which a finite reader descends to a quotient and distinguishes surjectivity, constancy on fibers, and uniqueness of the descended map. König's lemma guarantees the existence of a branch when the levels are finite, nonempty, and finitely branching; uniqueness further requires the complete character that selects a single compatible prolongation.

On a fixed section, the decreasing cylinders provide an exact realization of the passage $0- \; 1- \infty$: the support contracts, the height increases, the mass remains unitary, and the cylindrical measures converge weak-* to δ_x. This result coordinates the finite Kronecker delta, the conditional expectations between levels, and the Dirac delta as a point measure, without identifying the latter with a differential operator. The prefix foundation precludes infinite descending chains in each finite presentation; coinduction controls compatible prolongation. The von Neumann ordinals are lifted with a route-memory fiber, and cardinality appears as a decategorification that does not retain the entire genealogy.

The same distinction requires cylindrical power, closed power, and full power to be separated, as well as set-theoretic choice, natural choice, algebraic choice, and effective choice. Encoding in ZF/ZFC preserves the constructions and permits comparison with Extensionality, Foundation, Power Set, and Choice; it does not make the real line or the set-theoretic universe the origin of APP–TRIT–TPK. In the effective domain, decidability of each finite window, existence of the limit, uniqueness of an instance, and the impossibility of a uniform total decider are four distinct assertions. The Gödel–Turing reduction affects only the fourth: it does not invalidate particular selectors already constructed. Analogously, the Gödel–Cohen comparison separates the extension $M \mapsto M[G]$ from the internal enrichment of a state and avoids conflating the crossing of depth-dense sets with genericity over a model. Borel cylinders, comparison hierarchies, and the projective minimax passage are introduced with their hypotheses of compactness, continuity, and consistency, not as automatic consequences of inverse notation.

This closure also has an informational formulation. If an update U of the complete state is bijective, then $H(X \mid U(X)) = 0$, and the ideal logical Landauer term vanishes. If a reader q preserves only the visible shadow, the discarded information is localized in $H(X \mid q(X))$. In the operatorial formulation, a trace-preserving automorphism preserves entropy, whereas a trace-preserving conditional expectation $E_N : M \to N$ satisfies

$$S_\tau(E_N h) - S_\tau(h) = D_\tau(h \| E_N h) \geq 0.$$

The cost corresponds to forgetting the fiber, not to the reversible transport of the enriched state; this identity does not assert null total physical dissipation nor reduced computational complexity.

Operatorial Closure: From the Nonadic Unit to Continuous Dimension

The second closure operationally realizes the evolutionary conservation of unity. For the complete system, one considers

$$A_m = M_{9^m}(\mathbb{C}), \qquad \iota_m(a) = a \otimes I_9, \qquad \tau_m(a) = 9^{-m} \operatorname{Tr}(a).$$

Identity $\tau_{m+1} \circ \iota_m = \tau_m$ simultaneously preserves unity and trace. The inductive limit is the UHF algebra of supernatural type $9^\infty = 3^\infty$, and the weak closure of its tracial GNS representation is the hyperfinite factor R_{hyp} of type II_1. Values $\operatorname{rank}(P)/9^m$ become dense, and the trace of projectors in R_{hyp} realizes the entire interval $[0, 1]$: a continuous dimension emerges from a tower of discrete inclusions preserving unity.

This construction is not to be conflated with the marked branch $j_{m,r}(a) = a \otimes p_r$. The isometries of an individual fiber generate in norm the compact operators on the Hilbertian limit and, under weak closure, a factor of type I_∞. Selection of one sector preserves the mark, but not the unit of the tower; the simultaneous lifting of the nine fibers is what produces the UHF–II_1 branch. Nor may the UHF realization be identified without proof with the crossed-product realization of the profinite clock: a

canonical conjugacy requires the action, measure, amenability, ergodicity, and isotropy to be declared. The orthomodular logic of projectors, Boolean contexts, normal states, and entropic normalization

$$S_\tau(9^m\rho) = S_{\mathrm{vN}}(\rho) - m\log 9$$

are derived within this operator construction and keep separate the matrix state, its tracial density, and its statistical publication.

Topology, dimension, and composition do not collapse in the factor. The closure transversal is recorded by means of

$$\mathfrak{C}_{\mathrm{HMT}} = \big(C(\Omega_\infty), R_{\mathrm{hyp}}, \mathcal{G}_{\mathrm{TPK}}\big).$$

The commutative spectrum preserves points and cylinders; the tracial factor preserves dimension and statistics; the groupoid preserves concatenation, admissible inversion, and the provenance of routes. This threefold architecture is formulated after the double projection and the exceptional corridor. It neither founds nor replaces them: Archimedean evaluation and the Paley–Witt–Golay–Leech–$V^\natural$–Monster–Moonshine chain remain the two coordinated readings of the same enriched antecedent.

Matrix Closure: Depth, External Size and Dilation

The third closure introduces simultaneous control of three indices. If m measures genealogical depth, n the exterior size of a grid, and s the matrix level of its entries, the family $\mathfrak{M}^{(n)}_{m,s}$ is bigraded by (m, s) when n remains fixed and trigraded by (m, n, s) when the exterior size also varies. In particular,

$$\mathcal{K}_{m,s} := \mathrm{UCP}\big(C(X_m), M_s(\mathbb{C})\big)$$

has two independent operations: the genealogical restriction $\rho^s_m(\Phi) = \Phi \circ \jmath_m$ and the matrix compression

$$\mathrm{mc}_V(\Phi_1, \ldots, \Phi_k)(f) = \sum_r V_r^* \Phi_r(f) V_r, \qquad \sum_r V_r^* V_r = I_s.$$

Both commute exactly:

$$\rho^s_m(\mathrm{mc}_V(\Phi_1, \ldots, \Phi_k)) = \mathrm{mc}_V\big(\rho^{s_1}_m(\Phi_1), \ldots, \rho^{s_k}_m(\Phi_k)\big)\,.$$

The identity is functorial and distinguishes refinement of history of reduction from the observer.

Every depth-compatible family determines a unique UCP map over $C(X_\infty)$ and admits a common Stinespring dilation. In the diagonal corridor, a single projective spectral measure P satisfies

$$E^{(m)}_x = V^* P([x]_m) V$$

at every depth and on every cylinder. The minimal dilation is unique up to unitary equivalence and exactly preserves what is seen by the reader; a direct sum provides a faithful ambient realization that preserves all distinctions of the algebra. On the UHF limit, a compatible family of UCP maps likewise admits a common prolongation and dilation. The conditional expectation

$$\mathbb{E}_m = \mathrm{id} \otimes \tau_9 : M_{9^{m+1}}(\mathbb{C}) \longrightarrow M_{9^m}(\mathbb{C})$$

preserves the trace and discards the final factor in a controlled manner; its Stinespring environment records the omitted digit. In general, no uniform finite auxiliary memory exists for all depths. Moreover, faithfulness of a representation does not by itself imply APP–TPK dynamical faithfulness: the latter requires an intertwiner compatible with route, monodromy, and composition.

The matrix theory is instantiated in verifiable objects. The qutrit quantum magic squares with parameters $(9, 3)$ and $(12, 3)$ have, respectively, 81 and 144 effects, exactly unital row and column sums,

and declared simultaneous dilations; the same POVM systems produce quantum magic cubes of orders nine and twelve. The order-twelve case is organized through the fibration $12 \to 4 \times 3$, with its gauge and its explicit equivariance condition. By another route, the sum and product conditional expectations of APP generate the algebra

$$C^*(P_{\Sigma,2}, P_{\Pi,2}) \cong \mathbb{C}^4 \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C}),$$

and at every level s the central block realizes the free matrix interval $[5I_s, 9I_s]$. These constructions distinguish quantum Latin squares, magic squares, projective measures, positive measures, and semiclassicality; no identification is obtained merely because the objects share order nine.

The common spectral antecedent of two readers is formulated only after both have been constructed. For finite maps $R_m : X_m \to \mathcal{A}_m$ and $Q_m : X_m \to \mathcal{E}_m$, the incidences

$$P_{a,e} = 1_{\{R_m=a\}} 1_{\{Q_m=e\}}$$

refine the two diagonal spectral measures and recover their marginals. The result does not prove joint injectivity or require one reader to be a compression of the other. In the noncommutative regime, two mutually unbiased qutrit PVM systems do not possess a sharp joint PVM; simultaneous realization requires compatible noise or a larger environment. This difference in type preserves the HMT double projection: the matrix theorem classifies realizations of readers and does not rewrite their foundational antecedent.

Physical Closure: Common Domain, Two Orientations, and Localization

The fourth closure constructs the domain from which a TPK route can simultaneously receive spectral, Hilbert-space, dimensional, and Cartan realizations. Let D_{sp}, D_Q, D_D, and D_C be the reference domains of those four representations, with faithful forgetful functors into the category $\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}$ of prepared states and registered routes. The fiber product

$$\mathsf{P}^{+}_{\mathrm{CT}} := \mathsf{D}_{\mathrm{sp}} \times_{\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}} \mathsf{D}_Q \times_{\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}} \mathsf{D}_D \times_{\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}} \mathsf{D}_C$$

imposes that the four components have exactly the same antecedent. Their canonical projections are functorial adapters and produce

$$\mathfrak{R}^{\mathrm{disc},+}_{\mathrm{MD}} = \big(\mathfrak{S}_+, \mathcal{Q}_+, \mathfrak{D}_+, \rho^{\mathrm{disc}}_{C,+}\big),$$

with values, respectively, in spectral routes, real Hilbert spaces with complex structure, the positive dimensional monoid, and the discrete Poincare group. The construction asserts the functor on the common domain; it does not assert that every TPK route automatically admits all four lifts.

On a path $\gamma = e_1 \cdots e_r$, the total variation

$$V_+(\gamma) = \sum_j w(e_j)$$

and the net oriented transport

$$V_{\mathrm{or}}(\gamma) = \sum_j \varepsilon(e_j) w(\underline{e}_j)$$

are sibling readers. The loop $ee^\vee$ proves that the former does not, in general, factor through the latter: it has zero oriented transport and positive variation. To obtain an invertible realization, the domain is restricted to those edges whose spectral and Hilbertian components are invertible, whose dimensional datum is an additive cochain, and whose Cartan cocycles are normalized and antisymmetric. Only then does the localization $\Pi_\vee(\mathsf{P}^{+}_{\mathrm{CT,rev}})$ produce the oriented functor $\mathfrak{R}^{\mathrm{disc,or}}_{\mathrm{MD}}$ and transport inversion component by component. The positive reading is not eliminated, and the oriented reading is not obtained merely by changing a sign.

This physical construction precedes the Planck and electron constructions because it supplies them with a domain, representations, and transport rules, but does not absorb their reference proofs or the General Mass Law. The number–particle–route–enriched chronological record–signature– character–towers chain preserves its causal direction; the absolute action scale, the central electron, the mass families, CKM, PMNS, and the physical sectors retain their own objects, domains, and conclusions. A metrological coincidence is compared only after the internal output has been frozen. Extensions that still require an intertwiner, a smooth limit, a proof of dynamical stability, or a physical morphism are stated with those data as exact hypotheses. The absence of any of those maps prevents the theorem from being extended to the corresponding codomain, without altering the genealogical, operatorial, and matricial theorems already proved in their domains.

The reading of the work follows, consequently, the order of dependence and not that of external recognition:

$$\begin{aligned} &\mathrm{APP} \longrightarrow \mathrm{TRIT}_{\mathrm{orientado}} \longrightarrow \mathrm{TPK} \\ &\longrightarrow \text{enriched section} \longrightarrow \begin{cases} \text{Archimedean evaluation,} \\ \text{exceptional projection,} \\ \text{cylinder algebras and tracial closure,} \\ \text{prepared routes and physical realization.} \end{cases} \end{aligned}$$

Each output preserves its domain, hypotheses, and precise obstructions. What is common is not a forced equality between codomains, but provenance from the same enriched state. This is the precise meaning of holographic closure: no terminal publication exhausts the genealogy that produces it, and no subsequent closure can replace the double direction –Archimedean and exceptional– that preserves the unit with memory.

Historical perspective and development of the theory

The history of mathematics can be read as a successive enlargement of the operations admitted on the unit. The integer made counting possible; the rational incorporated division; the real closed processes of approximation; the complex turned rotation into an algebraic operation. Each enlargement fixed a domain, an equivalence, and a representation. Equality published the result with enormous efficacy and at the same time absorbed much of the specific route that had produced it.

Geometry followed an analogous development. Euclidean construction fixed incidence and measure; non-Euclidean geometries showed the coexistence of formally coherent curvature regimes [44, 105]. The trigonometric circle, the complex plane, and the hyperbolic disk coordinate rotation, periodicity, and transport through different state spaces. HMT places beneath these realizations a common ternary state and constructs from it the elliptic, parabolic, and hyperbolic branches.

Mechanics incorporated a relational dependence into the unit. The law of the lever linked weight, arm, and equilibrium; Mach's critique of absolute motion brought the relation among body, frame, and totality to the fore [6, 85]; relativity formalized kinematic invariances, local equivalence, and the geometry of the gravitational field [42]; quantum theory introduced an action scale and propagation through path amplitudes [104, 48]. The HMT–MD program assembles these requirements within a prior problem: constructing the state that preserves position, orientation, operation, memory, and prolongation before publishing a number or a magnitude.

The three terms of the name specify the method. *Holography* denotes preservation of the genealogy and of the prolongation rule between resolutions. *Modular* denotes the coordination of the chart modulo nine, the orientation modulo three, the residue–quotient readers, and decimal publication in base one thousand. *Triadic* denotes both the local orientation $\{-1, 0, +1\}$ and the organization of the levels linking the finite support, the numerical section, and its physical realization.

Evolutionary Conservation and the Meaning of Symmetry

HMT formulates symmetry as preservation of identity under transformation with memory. An operation can recover the same visible phase without recovering the same enriched state. If Θ denotes the transport, the relation

$$\mathrm{pr}_{\mathrm{vis}} \circ \Theta^9 = \mathrm{pr}_{\mathrm{vis}}$$

coexists with a nontrivial increment of carry or memory. Horizontal repetition is the projection of a helical evolution. Identity is preserved because its genealogy permits it to be recognized through change, not because it remains motionless.

The extreme and mean holographic ratio realizes this principle in the cubic completion.

$$999 = 729 + 243 + 27, \qquad 1000 = 729 + 243 + 27 + 1.$$

The apex +1 of the cubic completion and the terminal term $+I$ of the exponential closure are closure operations defined on distinct domains. Their structural correspondence lies in the fact that both complete an evolution and preserve the unit in the respective codomain. This conservation organizes the nonadic return, the decimal carry, the inward growth of the cylinders, and the outward growth of the incidences.

The dimensional prolongation is expressed through

$$10^d = (9 + 1)^d = \sum_{k=0}^{d} \binom{d}{k} 9^k.$$

After normalization by 10^d, the coefficients form a projectively compatible family of measures whose total mass remains equal to one. The interior, the boundary strata, and the apex redistribute that unit as the rank changes. In the cubic realization $C_9 = (\mathbb{Z}/9\mathbb{Z})^3$, the six oriented projections preserve the APP support, and the face–vertex operator produces a space of rank four. The signature $(3, 1)$ emerges from that operator and provides the finite antecedent of the subsequent spacetime reading. Dimension, boundary, and conservation of unity are here results of the same system of projections.

Algebraic Unification of Elliptic, Parabolic, and Hyperbolic Geometries

The conceptual point of departure was to articulate three realizations that mathematics ordinarily presents in distinct spaces: the complex plane, the trigonometric circle, and the Poincare hyperbolic disk. HMT unifies them through the family

$$\mathbb{A}_s = \mathbb{R}[X]/(X^2 + s), \qquad s \in \{+1, 0, -1\}.$$

The elliptic branch contains an internal root of -1 and the circular closure; the parabolic branch contains the nilpotent threshold; the hyperbolic branch contains the two idempotents and the Poincare opening. These three algebras arise from TRIT and subsequently govern the sheets, the channels, the Lorentz–Minkowski metric, and the physical reading of time.

The difference between sum and product provides the elementary operation that articulates these geometries. Sum accumulates displacements and alternatives; product composes stages and folds inputs onto a common class when noninvertible elements intervene. APP preserves both evaluations on a common support. Its spectral separation and the block projectors $P_{\pm}$ fix the intrinsic modes; the conditional projectors $P_{\Sigma,d}$ and $P_{\Pi,d}$, together with their commutator, construct a finite precursor of Heisenberg incompatibility. The subsequent physical reading transforms that separation into the conjugate angular coordinates A and C^*, oriented channels, and holonomy.

Construction of the formal kernel

APP is constructed from the positions $1, \dots, 9$ in two coordinates. The kernel 8×8, the publication of the null class by means of 9 and the gnomon of seventeen cells completan the chart 9×9. The identification of opposite sides produces the torus

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2$$

and its displacements generate the Cayley graph. The evaluations Σ and Π act on the same vertices and paths.

TRIT is the ternary oriented state. In an elevation

$$a + b = r + 3c,$$

the residue r publishes the output and the quotient c preserves how it was obtained. The triple differentiates orientation, threshold and memory. TPK is the operational system of genealogical dynamics that incorporates the tritic selector of the operating mode, transports the state and simultaneously preserves the two APP sheets with their position, orientation, phase, carry, boundary, path, record and survival. Its nonadic, ternary and positional readers, its twelve-phase system and its lifts form a single tree of domains and codomains. Mathematical dependencies are the typed arrows of that tree; each map is presented with its domain, codomain and a reproducible calculation.

With greater precision, APP is the quintuple

$$\mathcal{A}_9 = \big(X_9, \Gamma_{\mathrm{APP}}, \Sigma, \Pi, \mathrm{Path}(\Gamma_{\mathrm{APP}})\big), \qquad \Gamma_{\mathrm{APP}} = \mathrm{Cay}\big(X_9, \{\pm e_1, \pm e_2\}\big),$$

with evaluations

$$\Sigma(i,j) = \mathrm{dr}_9(i+j), \qquad \Pi(i,j) = \mathrm{dr}_9(ij), \qquad \mathrm{dr}_9(n) = 1 + ((n-1) \bmod 9).$$

The lift simultaneously preserves $i + j = R^{+}_{ij} + 9Q^{+}_{ij}$ and $ij = R^{\times}_{ij} + 9Q^{\times}_{ij}$; the published value and quotient information belong to distinct coordinates of the same state.

The lift of TRIT, $\widehat{\tau} = (r, c)$, preserves residue and carry. The fiber over the visible symbol 0 contains $(0,-1)$, $(0,0)$, and $(0,+1)$, whose prolongation trees may differ. This multiplicity is the minimal form in which the same visible coordinate supports distinct histories.

TPK is organized into eight operator classes before its particular realizations are presented: support and state; internal selection; transport; readers and quotients; memory and boundary; periodicity and return; prolongation; and outputs. This classification separates the carrier objects, the maps updating the state, the readers publishing its coordinates, and the functors contracting the section toward an Archimedean, exceptional, or classificatory codomain. Each TPK operator thus receives a domain, a codomain, a rule, and its invariants before receiving an abbreviated name.

The nonadic phases are distributed among three operator fibers: $\{1, 4, 7\}$ activates additive evaluation, $\{2, 5, 8\}$ activates multiplicative evaluation, and $\{3, 6, 9\}$ records the threshold. The transition $9 \to 1$ restores the published phase and lifts the dodecaphase memory; the visible cycle is therefore the projection of a helix of enriched states.

Once the three objects have been presented, the work develops its consequences. The central block

$$B_c = 9P_{+} + 5P_{-}$$

fixes the local exchange coefficient 2 and the spectral gap $9 - 5 = 4$; modular aggregation produces $51 - 45 = 6$, and the unfolding over the nine positions produces $54 = 9 \cdot 6$. The mixed-signature form and $SO^{+}(1,1)$ construct the finite Lorentzian geometry. The cubic completion constructs the extreme and mean holographic ratio, the discrete area law, and the propagation boundary.

Uniform algorithm at finite depth and correlated coinductive publication of π, e, φ, α

Numerical generation is governed by the funnel

$$104\,976 \to 468\,U_6 \to 243\,w_6 \to w_{30} \to R_{36} \to G_9.$$

The channels of π, e, φ and α preserve different typed readers: angular closure, self-proportional propagation, autoscale and signed dodecaphasic coupling. The internal relation Ext, coinductive survival and the exhaustive partition of the fiber produce at each level a unique prolongable cylinder compatible with the preceding truncation; the nested family determines a unique enriched section of every depth. Only afterward do the Leibniz brackets, the factorial semigroup and Perron–Fibonacci locate the already generated blocks Archimedeanly. The runs of one thousand and ten thousand digits are extensive finite certificates of that uniform construction, not its logical foundation.

The initial prolongation constructs

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36},$$

and the internal census selects before the evaluation

$$w_\pi = 010211, \qquad w_e^{(\mathrm{Euler})} = 201101, \qquad w_\varphi = 121200.$$

The superscript specifies that $w_e^{(\mathrm{Euler})}$ belongs to the reader of Euler's number e; it is an object distinct from the electronic path Γ_e of Dimensional Mechanics. For each prescribed finite depth, Ext, TPK transport and compatible survival first produce exactly one cylinder $C_{\chi,n}$ among the 729 refinements. The rational reader is then applied to the already generated cylinder: it constructs its internal bracket and verifies $j_{\min} = j_{\max}$ as an Archimedean localization certificate, without selecting the output or supplying target digits to the generator. The theorem 8.54 proves, for $\chi \in \{\pi, e, \varphi\}$, nonemptiness, uniqueness and compatibility of these survivors at every finite depth; therefore, the partitions construct

$$c_\chi = (C_{\chi,n})_{n\geq 0} \in \varprojlim_n \{C_{\chi,n}\}.$$

The cylinders are nested, their diameters converge to zero, and the EF–G_9 lift preserves the enriched fiber. This separates the uniform algorithm parameterized by depth, the coinductive section it determines, and its finite runs of one thousand and more than ten thousand digits, which act as extensive certificates and not as the logical foundation of the theorem.

The three sections realize distinct structural functions. π organizes closure, orientation, and monodromic boundary; e organizes self-proportional propagation and prefix evolution; φ organizes Perron–Fibonacci self-scaling and its stable mode φ^{-2}. The oriented operator $\mathscr{R}(x, y) = x/y^2$ publishes the conjugate readings π/φ^2 and φ/π^2, which reappear in the physical areal–volumetric hinge.

The terminal state supplies the block and signed record that feed the dodecaphasic closure of α_{HMT}. The carry recurrence in base one thousand and the equation $E_{21/22}$ constitute two internal pathways with distinct sources. Next, number theory organizes the complementary constants through extraction operators: finite parts, Laurent jets, L functions, prime products, Gauss dynamics, Rogers–Ramanujan and Feigenbaum renormalization. The text distinguishes a common antecedent from equality of value; for example, φ, φ^{-1} and φ^2 share an algebraic genealogy without becoming the same number.

The complete genealogy of the first path is

$$\begin{aligned} 104\,976 \to 468\,U_6 \to 243\,w_6 \to w_{30} \to R_{36} \to G_9,\\ G_9 \to x_{\mathrm{term}} \to (B_{\mathrm{term}}, U_{12}^{\mathrm{sgn}}) \to K. \end{aligned}$$

The seal K and the twelve terminal triads feed the carry

$$s_m = P_m + E_m - \Phi_m - K_m + c_{m+1},$$
$$A_m = s_m \bmod 1000, \qquad c_m = \frac{s_m - A_m}{1000}.$$

with $c_{13} = 0$. The resulting integral identity fixes

$$\alpha_{12} = \pi_{12}(\alpha_{\mathrm{HMT}}) = \frac{729735256928380099728510547238 0663}{10^{36}}.$$

This identity publishes the exact thirty-six-digit rational window; the compatible recurrence at arbitrary depth constructs $\alpha_{\mathrm{HMT}} = \varprojlim_N A^{(N)}$, which is not reduced to a single window. The second pathway solves the vacancy equation $E_{21/22}$ through an analytic kernel with a simple positive root α_E. The graded TPK transport prolongs that kernel into a compatible family whose limiting root is α_B; integer closure determines $\alpha_A := \varprojlim_N \rho_N(\alpha_{12})$. The truncation squares of both families commute and determine the same surviving cylinder at each depth, so

$$\boxed{\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}}.$$

The equality after Tr_9 is only the first finite certificate of this exact identity; it does not constitute its criterion, bound its extension or select the prolongations. Each pathway preserves its own boundary condition within the same enriched state.

Number theory extends the same criterion of object, operator, and extraction. Euler–Mascheroni appears as a finite part; the Stieltjes constants as Laurent coefficients; Apery as the evaluation of $\mathcal{Z}_3(3)$; Glaisher–Kinkelin through regularized derivation; Euler–Kronecker through an Eisenstein character; Meissel–Mertens through a prime trace; Catalan through $L(2, \chi_{-4})$; and Khinchin through Gauss dynamics. Rogers–Ramanujan, the prime clocks, and the Feigenbaum approximants belong to distinct operator constructions. The combinations $e + \pi$, $e\pi$, and π^e are formed after their arguments have been constructed and preserve the arithmetic domain corresponding to each statement.

The radical $(3) \subset \mathbb{Z}/9\mathbb{Z}$ organizes the visible network 3–6–9 and the orbit $369 \to 693 \to 936$. Over the same arithmetic, the power–vacancy theorem

$$V_9(n) = \left\lceil n \log_{10}\left(\frac{10}{9}\right) \right\rceil$$

gives the exact evaluation

$$\mathrm{ord}_{10}\left(9^{9^9}\right) = 369\,693\,099, \qquad \#\,\mathrm{cifras}\left(9^{9^9}\right) = 369\,693\,100.$$

This figure is not the census of 396 elementary boundary events per channel: the former measures the decimal order of a nonadic power; the latter counts states in an emission calendar.

Double projection as the sole thesis

The real continuum is the Archimedean shadow of an enriched section. The projection $\mathcal{P}_{\mathrm{arq}}$ contracts compatible cylinders and publishes the value. The projection $\mathcal{P}_{\mathrm{exc}}$ preserves support, orientation, carry, and memory. Both proceed from the same state; therefore, number and exceptional symmetry form a single causal thesis.

The incidence branch constructs Paley, Witt, the ternary Golay code and the Steiner systems. From the common code C_W, the lift $W_{12} \to W_{24}$ remains distinct of the branch

$$C_W \to N(A_2^{12}) \to \Lambda_{24} \to V_\Lambda \to V^\natural \to \mathbb{M}, \qquad [g] \longmapsto T_g.$$

The configuration of the twenty-seven lines, the Schläfli graph, and $W(E_6)$ receive their own incidence operators. The definition of the lattice VOA, the orbifold involution, and the replicable functions show how genealogical identity is preserved when changing codomain. This structure also provides the rank screens and the mathematical interface with T-duality, strings, and M-theory.

Level-five modular arithmetic enters through the Rogers–Ramanujan continued fraction $R(q)$. The Klein–Ramanujan identity expresses $j(\tau)$ rationally in $R(q)^5$, and the limit $R(q) \to \varphi^{-1}$ links pentadic self-scaling with the modular branch. On the canonical branch of order two,

$$t_2 = q^{-1} \prod_{n\geq 1} (1+q^n)^{-24}, \qquad T_{2A} = t_2 + 24 + \frac{2^{12}}{t_2}, \qquad j = \frac{(t_2+256)^3}{t_2^2}.$$

The lattice VOA $V_\Lambda = M(1) \otimes \mathbb{C}[\Lambda]$, the involution $\theta = -I$, and the fixed–twisted orbifold realize the lattice output in the domain of vertex operators. The classical FLM identification fixes $V^\natural$ and $\mathrm{Aut}(V^\natural) = \mathbb{M}$. HMT supplies the antecedent and the incidence maps that reach that construction; the VOA identification theorems retain their own mathematical domain.

The finite root and the real structure are coordinated through the integral presentation.

$$Q_\mathbb{Z} = \mathbb{Z}[u]/(u^2+1),$$

not by means of an inclusion between fields of different characteristics. The Witt endomorphism $A_W^2 = -I_6$ and the real operator $J_\mathrm{e}^2 = -I$ are specialized representations of that relation. The quarter-turn $z \mapsto \mathrm{i}z$ preserves both the Euclidean metric and the Poincare metric of the disk. The identity

$$\mathrm{Exp}(\pi_{\mathrm{HMT}} J_\mathrm{e}) + I = 0$$

is applied only after the Archimedean reader has generated $\mathcal{P}_{\mathrm{arq}}(x_\pi) = \pi_{\mathrm{HMT}}$. The exceptional branch supplies the quadratic type, not the angle: value and incidence remain coordinated without either branch acting as an oracle for the other. The family $J_s^2 = -sI$, $s \in \{+1, 0, -1\}$, extends Euler to the elliptic, parabolic, and hyperbolic realizations. The final term +1 and the apex of EMRH perform closure functions in distinct and coordinated domains. This relation provides an operator expression of the evolutionary preservation of the unit without identifying objects of different types.

Planck scale, Barbero–Immirzi functional, and CKM–CP mixing

The double projection causally precedes the angular and physical constructions. The Smith normal form of H_4 produces a cokernel of order sixteen and the transfer

$$\Sigma_H = \varphi \alpha_{\mathrm{HMT}}^{16}$$

proves $\lfloor \log_{10} \Sigma_H \rfloor = -34$ and thereby fixes the internal scale $10^{-34}\mathcal{U}_S$. The action line contains two coordinates,

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S.$$

Temporal periodicity of order 108 supplies a self-dual Planck closure. In the circular realization, with $\ell_0 := c_{\mathrm{int}} t_0$,

$$T_{108} = 108 t_0, \qquad R_{108} = \frac{54}{\pi}\ell_0, \qquad \theta_{108}^{\mathrm{circ}} = \left(\frac{54}{\pi}\right)^2,$$

and the quantum satisfies

$$R_{108} = \bar{\lambda}_\mathrm{C} = r_\mathrm{g} = \ell_\mathrm{P}^{\mathrm{circ}}.$$

The broken line

$$Z_a(m) = \frac{m_{\mathrm{P},a}^\circ}{m} P_+ + \frac{m}{m_{\mathrm{P},a}^\circ} P_-, \qquad N_{\mathrm{sp}}(Z_a(m)) = 1,$$

interchanges Compton and gravity under $m \mapsto (m^{\circ}_{\mathrm{P},a})^2/m$. Its fixed point is the Planck scale. The electron section constitutes a subsequent genealogical realization on that line and not an identification of the electron with a classical black hole. The direct angular coordinate A and its conjugate angular coordinate C^* subsequently construct the channels $q_{\pm}$. The hexad–octad inclusion produces the Barbero–Immirzi functional, the area operator, and its spectrum. The same rank-three incidence constructs CKM, the CP phase, and the Jarlskog invariant. These results belong to the mathematical closure of HMT; their field interpretations are resumed in MD.

The Smith normal form explicitly fixes

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4), \qquad |\operatorname{coker} H_4| = 16,$$

and the action line distinguishes $\hbar_{\mathrm{pre}}$, $\hbar_{\mathrm{ret}}$ and $h_{\mathrm{ret}} = 2\pi\hbar_{\mathrm{ret}}$. The chart $\mathcal{U}_S \mapsto \mathrm{J\,s}$ publishes then the unit metrological; the order -34 belongs to the internal section that precedes to esa chart.

The combinatorial inclusion of a hexad into its associated octad produces

$$6\binom{6}{2} = 90, \qquad 8\binom{6}{2} = 120, \qquad \frac{90-120}{24\binom{6}{2}} = -\frac{1}{12}.$$

On these spaces, the dodecaphasic fundamental chain brings together twelve sectors and a single oriented return seam. Its linear reader produces $12S_{90} - S_{120}$ and, therefore, the weights $12 : -1$ of the functional $\Gamma_{6\to 8}$. The coefficient of $(1-q)^{-1}$, equal to $1/24$, positivity and minimality constitute subsequent arithmetic checks of the pair produced. The Holst specialization $\gamma = \Gamma_{6\to 8}$ is realized at the triadic boundary through

$$\widehat{A}_{\mathrm{HMT}} = \gamma_{\mathrm{HMT}} a_0 \sum_{e\in\partial A} N_e, \qquad \mathrm{Spec}(\widehat{A}_{\mathrm{HMT}}) = \{n\gamma_{\mathrm{HMT}} a_0 : 0 \le n \le 72\}.$$

The collective intertwiner satisfies $J_{6\to 8} D_{\mathrm{inc}} = D_{\mathrm{area}} J_{6\to 8}$ on the collective incidence subspaces. The area operator, its spectrum, and the triadic, bilayer, and modular entropies thus constitute the internal closure that realizes the Barbero–Immirzi parameter. The rank-three rational transformation constructs, in a sister branch, the CKM angles, the CP phase, and the Jarlskog invariant before experimental comparison.

Genealogical Closure of the Number and Physical-Spectral Realization

The first four Parts culminate where a conventional exposition usually begins: they define number as a compatible section and value as its projection. Dimensional Mechanics preserves that definition and adopts a physical–spectral reader. It first reconstructs the APP–TRIT–TPK kernel in physical–spectral form. APP is realized as a two-dimensional torus and distinguishes additive, multiplicative, areal, volumetric, and torsional sheets. TRIT receives the semantics

$$\begin{aligned} +1 &= \text{return, memory and past,}\\ 0 &= \text{threshold and present,}\\ -1 &= \text{opening and compatible future.} \end{aligned}$$

TPK transports the route, the record, and the holonomy. Exact periodicity constructs the finite dynamics; a physical time crystal adds a Hamiltonian, robust subharmonicity, and stability. The conjugate angular coordinates A and C^* spectrally realize the sum–product difference.

The gnomonic completion here acquires an explicit geometric function. The L carries the Euclidean chart onto which the inertial and gravitational readers project to a unit quotient. This chart constitutes the realization of the local equivalence principle. The enriched state prior to projection keeps the areal and volumetric sheets separate, and the Ambivalence Principle fixes their ratio $\pi_{\mathrm{HMT}}/\varphi^2_{\mathrm{HMT}}$. Together with the negative- and positive-curvature charts, this yields an AdS–Euclidean–dS atlas: the first realizes the

correspondence between interior and boundary, the second governs local physics, and the third satisfies $\ddot{a}/a = H^2 > 0$ and geometrically explains accelerated expansion. The three charts coexist because they are realizations of the same TRIT triad, linked by TPK transport.

The particle is defined on this grammar. A particle-route is a persistent section endowed with orientation, sheet, occupation, action, propagator, record, and anti-section. The mass law is first formulated on each multisection F:

$$\widehat{\mathbf{M}}_F^{\beta,r} = \mathbf{m}_\star^{\mathrm{ret}} \otimes R_{\mathrm{act}}^{\widehat{K}_F^r/2}\, \widehat{\mathcal{A}}_F^{(0)}\, R_{\mathrm{act}}^{\widehat{K}_F^r/2}, \qquad r \in \{D, \Omega\},$$

where

$$\mathbf{m}_\star^{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34} \mathcal{U}_S \frac{2\pi}{108 t_0}\, c_{\mathrm{int}}^{-2}$$

fixes the absolute mass section and $\widehat{\mathcal{A}}_F^{(0)}$ gathers signature, character, and all the towers of $\langle 90, 120\rangle$. The complete atlas is thus the domain of the law; each species subsequently selects its realization within the corresponding fiber.

The central electron is the first rank-one scalar reduction of this definition. The global torsion entering its reader is

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729}\right).$$

Its bidirectional closure produces

$$\begin{aligned} R_{\mathrm{act}} &= \frac{H_5}{\eta_{\mathrm{ret}}}, & \kappa_e &= 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4, \\ \mathfrak{e}_e^\beta &= \mathfrak{e}_e^{(0)} R_{\mathrm{act}}^{\kappa_e}, \\ \mathfrak{e}_e^\beta &= 0.510998950690000 80860825716 48\ldots \text{ MeV}, & m_e^\beta &= \mathfrak{e}_e^\beta / c^2. \end{aligned}$$

The identity $\Sigma_H = \varphi \alpha_{\mathrm{HMT}}^{16}$ simultaneously supplies the scale $10^{-34}\mathcal{U}_S$ to the two action coordinates; the quotient R_{act} preserves the relative refinement and cancels that common scale. Thus, the internal action scale, the bidirectional correction, and the energy chart in MeV occupy successive levels of a single derivation. The first quantity is the published mass energy; the second is the mass obtained through the explicit chart $E = mc^2$. This distinction precludes identifying an energy coordinate with a unit of mass. Vacancies construct polarization and current; the response of the sheets constructs the permittivity and permeability of the vacuum. The SI chart publishes

$$\begin{aligned} \mu_0^{\mathrm{HMT}} &= 1.256637062121047789\ldots \times 10^{-6}\, \mathrm{H\,m^{-1}}, \\ \varepsilon_0^{\mathrm{HMT}} &= 8.854187812793002329\ldots \times 10^{-12}\, \mathrm{F\,m^{-1}}. \end{aligned}$$

The atlas then situates the quarks and remaining species, defining color, flavor, charge, spin, statistics, and representation within the same genealogy.

In this classification, *colour* is the ternary fibre that organizes the three realizations of a quark route and whose neutral composition satisfies $0 + 1 + 2 = 0 \pmod 3$. *Flavour* is the coordinate that distinguishes, within the same representation, the sections transported by the mixing operator. *Spin* labels the representation of the cover of the rotation group carried by the section. The holonomy of the oriented band realizes its closures $2\pi/4\pi$ and distinguishes phase return from orientation return. Statistics follows from the grading and from the symmetric or exterior realizations; once CCR or CAR is fixed, it publishes Bose–Einstein or Fermi–Dirac occupation and, in the latter case, the Pauli exclusion principle. A virtual particle is a finite prolongation that conserves route, record, and signature up to a declared horizon; a prolongable material section conserves the corresponding global compatibility.

Operational domain of paths, fibers, and physical realizations

The construction proves that the internal genealogy reaches the complete physical domain. The domain distinguishes 81 positions, 56 families, 13 multisections, and 324 oriented routes, together with the null sector. Each realization preserves, according to its codomain, position or fiber, family, route, orientation, phase word, register, signature, character, refinement operators, chart, and comparison. External values are joined by identifier after the internal output has been frozen.

The construction chain that governs those rows is

$$\begin{aligned}\text{surviving section} \to \text{observable route} \to \text{record} \to \sigma \in \mathbb{Z}^5,\\ \sigma \to \chi_0 \to \chi_{\text{full}} \to \text{towers} \to \text{dimensional section.}\end{aligned}$$

The two bidirectional readers publish

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2}\right), \qquad K_\Omega = (\Omega, 54, P_6),$$

and act as diagonal operators on each fiber. In the central cell, both reduce to $(80, 54, 6)$; in the noncentral fibers, their spectra preserve the distinct routes before the selection of a physical realization. The operatorial comparison presents first the HMT output, then the chart, and finally the Standard Model or PDG reference with its unit, scheme, uncertainty, and residual.

The scientific output assumes the type corresponding to each sector. The electron cell publishes the closed scalar beta mass; the photon and gluons publish the null chart; the noncentral fibres preserve their operators and spectra. The finite flavour–generation–colour morphism to a persistent route is already constructed, and the twelve nominal routes $e, \mu, \tau, u, d, s, c, b, t, W, Z, H$ publish their angular signatures and their scalar characters before external comparison. In leptons and quarks, the operators K_D/K_Ω also refine the fibre; they are not the requirement that retroactively creates its scalarization. W, Z, and H preserve their native CT–108 reader even though they do not form part of the operatorial atlas of thirteen multisections. The topological sectors publish fusion and braiding; the virtual sections publish their prolongation horizon. This heterogeneity preserves the proper codomain of each class and makes the table a complete comparison of typed results, values, and realization boundaries.

Quantum, information, gravity and cosmology

The propagation of paths constructs the finite sum over paths, the Clifford–Dirac representation and the Schrodinger evolution. Fixed the Hamiltonian of the internal calendar, the evolution satisfies

$$i\hbar_{\text{int}}\dot{\psi} = H_{\text{clk}}\psi,$$

and its finite propagator admits the expansion

$$(U^R)_{fi} = \sum_{\gamma: i \to f,\, |\gamma| = R} \prod_{\ell \in \gamma} U_\ell.$$

The product preserves the order of transitions within a history, and the sum accumulates alternative histories with common endpoints. The grading of the exchange is realized concretely on $\mathbb{T}^2_{27} \times \{\text{N}, \text{S}, \text{E}, \text{W}\}$ through

$$U_{\text{micro}} = SD_JC_4, \qquad U_{12}^{\text{reg}} = U_{\text{micro}}^{12}.$$

This propagator is unitary; its powers enumerate the finite routes exactly, and its discrete transform produces 27^2 Fourier blocks. The construction is not thereby identified with a continuous functional integral. The grading fixes the exchange datum; the representation of the permutation group determines the symmetric or exterior completion. Once the CCR or CAR relations are given, the variational principle and entropy maximization produce the Bose–Einstein and Fermi–Dirac distributions, and the latter realizes

Pauli exclusion. In the Möbius basis, one visible turn of 2π changes sheet and accumulates $h/2$; the double visible-phase return of 4π restores orientation and accumulates h. Per radian of internal phase, the transport remains equal to $\hbar$.

The Boltzmann constant appears as a transducer between action and ternary information. The internal construction fixes the product $k_B\Theta_{\mathrm{clk}}$ and its structural function; the BIPM definition of the kelvin then fixes the numerical coordinate of k_B in the SI chart:

$$k_B\Theta_{\mathrm{clk}}\log 3 = \hbar_{\mathrm{int}}\frac{2\pi}{108t_0}, \qquad k_B^{\mathrm{SI}} = 1.380649\times 10^{-23}\ \mathrm{J\,K^{-1}}.$$

The ensembles realize Gibbs and the KMS condition. Null Landauer corresponds to bijective transport of the complete fiber; preparation, projection, and reset receive separate balances when the coupling discards a fiber. The thermal reading first constructs the internal transducer and then applies the unit chart.

Bilayer responses produce directly impedance, velocity, permeability and permittivity through

$$\widehat{Z} = \sqrt{\widehat{\mu}/\widehat{\varepsilon}}, \qquad \widehat{c} = (\widehat{\mu\varepsilon})^{-1/2}.$$

The same causal precedence governs the electromagnetic reading: the internal constitutive operator precedes the realization of $\varepsilon_0, \mu_0, Z_0$ and c in the International System of Units.

The gravitational branch first constructs, from the outputs of the enriched state, the action scale, the gravitational selector and the section G_a. The Palatini–Cartan–Holst variational formulation then receives these outputs, together with γ_{HMT}, and admits the constrained Hamiltonian transform

$$H_{\mathrm{tot}} = \int_\Sigma (\lambda^i G_i + N^a V_a + N\mathcal{H}).$$

The subsequent equivalence between covariant action, constrained Hamiltonian system and Einstein–Schrödinger evolution preserves co-tetrad, connection, curvature and torsion within the same scheme. The torsion-free collective sector carries the gravitational excitation; local torsion induced by spin density supplies the material channel. The type, transducer and chart of G precede this dynamical realization and are not selected through it.

The solenoidal and gauge branches receive realizations of the common finite operator

$$\mathcal{L}_{\mathrm{cyc}}^{\mathrm{TPK}} = 2I - C - C^{-1} = 3(I - P_0) = 3I - J,$$

$$\mathfrak{R}_{\mathrm{sol}}(\mathcal{L}_{\mathrm{cyc}}^{\mathrm{TPK}}) = \mathcal{L}_{\mathrm{H}}^{(0)}, \qquad \mathfrak{R}_{\mathrm{gau}}(\mathcal{L}_{\mathrm{cyc}}^{\mathrm{TPK}}) = \mathcal{L}_{\mathrm{H}}^{(1)}.$$

The hydrodynamic branch transports it through the Galerkin–register intertwiner, with coercivity and residual declared; the gauge branch transports it through the gauge functional and the transfer–gap operator. The finite proof and the bridges to the continuous equations are presented with their distinct hypotheses. The self-inertial frames, the Machian factorization, and sheet cosmology are developed after this local and collective construction, where they acquire a domain, a reader, and a typed physical interpretation.

Method of proof and contrast

Each result is presented through its object, domain, map, derivation, premises, certificate, and scope. Reproducible calculations accompany the proof and do not replace it. Metrological charts are applied after the internal scalar or operator has been constructed. CODATA and PDG comparisons are opened after the HMT output has been sealed. This order preserves the causal direction of the program: the structure generates the digit and the magnitude; the external reference measures the agreement attained by HMT.

The work is organized into five causally ordered Parts. The first four construct the formal core, the discrete continuum, the genealogy of the constants, and the five specialization chains; the fifth realizes

that architecture in magnitudes, particles, fields, and physical geometry. This sequence expresses the mathematical dependence among structure, evaluation, closure, and dimensional realization.

Composition of the treatise and scope of the statements

Part I constructs APP from the nonadic cell, presents TRIT as an oriented lift, and defines the Topological Phase Kernel with its fibers, transports, memory, clock, joint nonadic connection, and compatible prolongation. The horizon line opens this construction through the notions of scale, boundary, and measure.

Part II develops the discrete structure of the continuum. The double projection separates Archimedean evaluation from incidential publication. From Paley–Witt, the Steiner–Golay design branch and the Niemeier–Leech–$V^\natural$–Monster–Moonshine lattice branch are distinguished. Within that same structure appear the internal square root of -1, the three quadratic realizations, and the spectral–polar, solenoidal, gauge, cohomological, and determinantal substructures, typed by their own maps and terminals.

Part III establishes the genealogy of the fundamental constants: the correlated coinductive publication of $\pi, e, \varphi, \alpha_{\mathrm{HMT}}$, the specific readers and the two internal pathways of α_{HMT}, $-1/12$, the action scale, Planck, Boltzmann, the Barbero–Immirzi parameter, CKM–CP, the vacuum constitutives, the gravitational constant and the atlas of mathematical and physical constants.

Part IV applies the five substructures constructed in Part II in chapters of identical editorial rank. Each chapter sets out the internal result, the specialized comparator, its premises, and the exact criterion of publication. The conclusion in the corresponding conventional formulation is deduced only when the comparator identity has been verified throughout its domain.

Part V realizes APP–TRIT–TPK through the physical readers: torus and sheets, temporal semantics, curvatures, channels, and holonomy. On that grammar it defines the particle-route, the bidirectional electron, the mass families, the constitutive laws, statistics, information, mixing, variational gravity, self-inertia, sheet cosmology, and the interface with M-theory.

The appendices gather auxiliary proofs, complete censuses, seeds, decimal strings, extensive tables, auxiliary derivations, and reproducible calculations. The main body preserves the sequence of definition, construction, theorem, interpretation, and result. Each claim distinguishes its mathematical domain, its starting structure, its realization chart, and its external test. This separation permits the exact closures, physical realizations, and extension programs to be presented together while preserving for each the degree of proof proper to it.

Part I

Formal Core of Triadic Modular Holography: APP, TRIT, and TPK

Chapter 1

Construction and operatorial architecture of the APP–TRIT–TPK formal kernel: arithmetic support, ternary orientation, and genealogical dynamics

1.1 Arithmetic construction of APP

The name *All Possible Paths*, abbreviated APP, designates the arithmetic structure of trajectories built on a grid of nine rows and nine columns. Its main presentation is the multiplication table of the integers from one to nine, reduced by means of the positive digital root. Addition is involved both in the accumulation that produces each row and in a second evaluation of the same positions. The trajectories traverse this single support and preserve the order of the operations performed.

The development begins with the visible construction of APP and continues with the operations needed to preserve its arithmetic information. The remainder and quotient recover each integer evaluation; the carry determines how these recoveries compose; the translation record distinguishes the return to a position from the accumulated circulation. These are foundations of the generative dynamics. The TRIT, an oriented ternary unit with values $-1, 0, +1$, will provide the regime of each transition; the *Topological Phase Kernel*, TPK, or topological phase kernel, will compose selection, transport and memory updating. Its operator definition is based on the objects constructed here.

1.1.1 Construction of the APP matrix through multiplication and digital root

1.1.1.1 Arrangement of the integers from one to nine

Definition 1.1 (Ordered nonadic support)**.** Define $D_9 = \{1, 2, \ldots, 9\}$ and the two-dimensional support $\mathcal{X}_9 = D_9 \times D_9$. A position is an ordered pair $x = (i, j)$; the first coordinate identifies the row and the second, the column. Rows increase southward and columns eastward.

The support contains $9^2 = 81$ positions. The distinction between position and evaluation is essential: several positions can receive the same arithmetic result while retaining different coordinates. For example, $(2, 3)$ and $(3, 2)$ have equal sum and product, while the orientation of their coordinates distinguishes them.

In row i, the accumulation of i repeated j times determines the entry in column j:

$$P(i, j) = \underbrace{i + \cdots + i}_{j \text{ summands}} = ij.$$

The integer table is thus obtained first. Each column increment adds the row index; each row increment adds the column index.

Table 1.1: Integer multiplicative evaluation: $P(i, j) = ij$.

$i\backslash j$	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	10	12	14	16	18
3	3	6	9	12	15	18	21	24	27
4	4	8	12	16	20	24	28	32	36
5	5	10	15	20	25	30	35	40	45
6	6	12	18	24	30	36	42	48	54
7	7	14	21	28	35	42	49	56	63
8	8	16	24	32	40	48	56	64	72
9	9	18	27	36	45	54	63	72	81

The positive digital root is calculated by repeatedly adding the decimal digits until an integer from 1 to 9 is obtained. For example,

$$16 \longmapsto 1+6=7, \qquad 20 \longmapsto 2+0=2, \qquad 25 \longmapsto 2+5=7, \qquad 81 \longmapsto 8+1=9.$$

Applying it to each of the eighty-one entries produces the main APP matrix. Using the notation ρ_9 for this reduction, we write $\Pi(i,j) = \rho_9(ij)$.

Table 1.2: Principal APP matrix: multiplication reduced by positive digital root, $\Pi(i,j) = \rho_9(ij)$.

$i\backslash j$	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

The first row reproduces $1,\dots,9$; the second distributes the increment two in the order $2,4,6,8,1,3,5,7,9$; the third traverses $3,6,9$ three times. The last row and the last column have value 9. These patterns arise from the operation applied to each entry and from the convention that represents the zero class modulo nine by 9. Rows whose indices are coprime to nine traverse all nine residues; the remaining rows concentrate several positions on the same residue. The proof and the information preserved by this concentration are developed in the following sections.

The block of rows $4,5$ and columns $4,5$ can already be read directly:

$$\begin{pmatrix} 16 & 20 \\ 20 & 25 \end{pmatrix} \xrightarrow{\rho_9} \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Its selection by congruence, its spectral decomposition and its geometric realizations will be the subject of section 3.1. This localization fixes from now on the positions that produce the block.

1.1.1.2 Positions, intervals and central inversion

The sequence $1,\dots,9$ contains eight intervals between consecutive integers. Adding the cyclic transition $9 \to 1$ yields nine links. This convention makes it possible to specify the relationship between length, number of positions and combinatorial period. When the endpoints 0 and 10 are also considered on a linear scale, the reference support changes: it is advisable to state explicitly which endpoints and which intervals are involved in each construction.

With both endpoints, the linear subdivision is written

$$[0,1],\ [1,2],\ [2,3],\ [3,4],\ [4,5],\ [5,6],\ [6,7],\ [7,8],\ [8,9],\ [9,10].$$

These are ten segments determined by eleven integer marks. The inversion $i \mapsto 10-i$ exchanges the endpoints and preserves the central mark 5; on the nine interior integers it induces the pairs $(1,9)$, $(2,8)$, $(3,7)$ and $(4,6)$, as well as the fixed integer 5. The same inversion will act on each coordinate of the two-dimensional support.

Before evaluating sum or product, the positional table is

Table 1.3: The eighty-one positions before their arithmetic evaluation.

$i\backslash j$	1	2	3	4	5	6	7	8	9
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)	(1, 7)	(1, 8)	(1, 9)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)	(2, 7)	(2, 8)	(2, 9)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)	(3, 7)	(3, 8)	(3, 9)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)	(4, 7)	(4, 8)	(4, 9)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)	(5, 7)	(5, 8)	(5, 9)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)	(6, 7)	(6, 8)	(6, 9)
7	(7, 1)	(7, 2)	(7, 3)	(7, 4)	(7, 5)	(7, 6)	(7, 7)	(7, 8)	(7, 9)
8	(8, 1)	(8, 2)	(8, 3)	(8, 4)	(8, 5)	(8, 6)	(8, 7)	(8, 8)	(8, 9)
9	(9, 1)	(9, 2)	(9, 3)	(9, 4)	(9, 5)	(9, 6)	(9, 7)	(9, 8)	(9, 9)

1.1.1.3 Representation of the zero remainder and preservation of the quotient

Definition 1.2 (Positive remainder and nonadic quotient)**.** For each integer $n \geq 1$, define

$$\rho_9(n) = 1 + ((n-1) \bmod 9), \qquad q_9(n) = \left\lfloor \frac{n-1}{9} \right\rfloor. \tag{1.1}$$

The pair $\big(\rho_9(n), q_9(n)\big)$ is the complete nonadic record of n.

The choice of representatives $1, \ldots, 9$ assigns the integer 9 to the zero residue class. Thus, 9, 18 and 27 have the same positive remainder and respective quotients 0, 1 and 2. The residue coordinate describes a class; the quotient determines the integer lift within that class.

Proposition 1.3 (Exact arithmetic restitution)**.** *The map*

$$\mathbb{N}_{>0} \longrightarrow D_9 \times \mathbb{N}_0, \qquad n \longmapsto \big(\rho_9(n), q_9(n)\big) \tag{1.2}$$

is bijective. Its inverse is $(r, q) \mapsto r + 9q$. In particular,

$$n = \rho_9(n) + 9q_9(n). \tag{1.3}$$

Proof. Euclidean division of $n - 1$ by 9 produces a unique pair (q, t) with $q \geq 0$, $0 \leq t \leq 8$ and $n - 1 = 9q + t$. Setting $r = t + 1$ yields $n = r + 9q$ with $r \in D_9$. This is precisely the pair defined in (1.1). Conversely, given $r \in D_9$ and $q \in \mathbb{N}_0$, the integer $n = r + 9q$ satisfies $n - 1 = (r - 1) + 9q$. Its Euclidean division returns the same remainder $r - 1$ and the same quotient q. Both inverse compositions are thereby proved. □

This restitution fixes the meaning of preserving arithmetic information at the initial stage: exactly the data needed to recover the integer are preserved. The orientation of an operation and the record of a sequence of operations will be incorporated with their own coordinates. The quotient of an isolated evaluation therefore has a precise scope within the complete construction.

Example 1.4 (Two lifts of a residue class)**.** The integers 10 and 19 are represented as $(1, 1)$ and $(1, 2)$. Their first coordinate coincides; the second makes it possible to recover $1 + 9 = 10$ and $1 + 18 = 19$ respectively.

1.1.1.4 Digital root and decimal evaluation

Proposition 1.5 (Expression for the digital root)**.** *The iterated sum of the decimal digits of a positive integer n terminates at* $\rho_9(n)$.

Proof. Write $n = \sum_{k=0}^{m} d_k 10^k$, with $0 \le d_k \le 9$. Since $10 \equiv 1 \pmod 9$, we have $n \equiv \sum_k d_k \pmod 9$. For $n \ge 10$, at least one digit with positive index is nonzero, so that

$$n - \sum_k d_k = \sum_{k \ge 1} d_k (10^k - 1) > 0.$$

The iteration strictly decreases while the integer has more than one digit. It terminates at an integer in D_9 and preserves the class modulo nine. Uniqueness of the positive representative identifies the result with $\rho_9(n)$. □

The integer zero admits the conventional digital root 0. The positive representation of the residue group, in turn, represents its class by 9. Both conventions are fixed by their domains: the digital root acts on integers; the positive section selects representatives of residue classes. This clarification becomes relevant when decimal blocks with leading zeros appear.

1.1.2 Additive and multiplicative evaluations of APP

1.1.2.1 Joint definition of the two evaluations

Definition 1.6 (Enriched arithmetic evaluation of APP)**.** For $(i, j) \in \mathcal{X}_9$, define

$$S(i,j) = i + j, \qquad P(i,j) = ij, \tag{1.4}$$

$$\Sigma(i,j) = \rho_9(i+j), \qquad \Pi(i,j) = \rho_9(ij), \tag{1.5}$$

$$q_+(i,j) = q_9(i+j), \qquad q_\times(i,j) = q_9(ij). \tag{1.6}$$

The simultaneous evaluation is

$$\mathcal{A}(i,j) = \big((\Sigma(i,j), q_+(i,j)), (\Pi(i,j), q_\times(i,j))\big). \tag{1.7}$$

The state at this stage also retains the position (i, j).

The values of S range over the integers from 2 to 18; those of P lie between 1 and 81. The operation is performed first in the integers and then decomposed into remainder and quotient. This order makes it possible to recognize which content comes from the integer calculation and which content corresponds to its modular representation.

Multiplication here admits its construction by iterated addition:

$$P(i,1) = i, \qquad P(i,j+1) = P(i,j) + i, \qquad P(i,j) = \underbrace{i + \cdots + i}_{j \text{ summands}}.$$

By induction on j, this recurrence produces ij. Each row of the multiplicative table is therefore obtained by successively adding its index. The additive evaluation is constructed by adding the two position indices just once. Both operations arise from the integers of the support, with explicitly distinct rules.

For $2 \le k \le 10$, the equation $i + j = k$ has $k - 1$ solutions in D_9^2. For $11 \le k \le 18$, it has $19 - k$. In multiplication, the multiplicity of a value m is

$$\#\{i \in D_9 : i \mid m \text{ and } m/i \in D_9\}.$$

These multiplicities count positions with a given evaluation and preserve the difference between the number of positions and the number of distinct values.

Table 1.4: Integer additive evaluation: $S(i, j) = i + j$.

$i\backslash j$	1	2	3	4	5	6	7	8	9
1	2	3	4	5	6	7	8	9	10
2	3	4	5	6	7	8	9	10	11
3	4	5	6	7	8	9	10	11	12
4	5	6	7	8	9	10	11	12	13
5	6	7	8	9	10	11	12	13	14
6	7	8	9	10	11	12	13	14	15
7	8	9	10	11	12	13	14	15	16
8	9	10	11	12	13	14	15	16	17
9	10	11	12	13	14	15	16	17	18

Corollary 1.7 (Simultaneous restitution of the two evaluations)**.** *At each APP position, the evaluated integers are recovered exactly by*

$$S = \Sigma + 9q_+, \qquad P = \Pi + 9q_\times. \tag{1.8}$$

Proof. Apply proposition 1.3 to the positive integers $i + j$ and ij at the same position (i, j). □

1.1.2.2 Residue atlas and quotient records

The residue multiplicative matrix already presented is now completed with the additive residues and the two quotient records. These three tables, together with the main APP matrix, display the coordinates of (1.7) separately. Reading them jointly reconstructs the two integer tables. This breakdown makes visible a property decisive for subsequent development: equality of residues can coexist with differences in quotient.

Table 1.5: Positive representatives of additive evaluation: $\Sigma(i, j) = \rho_9(i + j)$.

$i\backslash j$	1	2	3	4	5	6	7	8	9
1	2	3	4	5	6	7	8	9	1
2	3	4	5	6	7	8	9	1	2
3	4	5	6	7	8	9	1	2	3
4	5	6	7	8	9	1	2	3	4
5	6	7	8	9	1	2	3	4	5
6	7	8	9	1	2	3	4	5	6
7	8	9	1	2	3	4	5	6	7
8	9	1	2	3	4	5	6	7	8
9	1	2	3	4	5	6	7	8	9

Proposition 1.8 (Records of the center and residue coincidence)**.** *The positions* $(5, 5)$ *and* $(8, 2)$ *have the same residue pair* $(\Sigma, \Pi) = (1, 7)$ *and different multiplicative records:*

$$\mathcal{A}(5, 5) = ((1, 1), (7, 2)), \tag{1.9}$$

$$\mathcal{A}(8, 2) = ((1, 1), (7, 1)). \tag{1.10}$$

Proof. At $(5, 5)$, the evaluations are $10 = 1 + 9$ and $25 = 7 + 18$. At $(8, 2)$, they are $10 = 1 + 9$ and $16 = 7 + 9$. Dividing the difference by nine produces the indicated quotients. □

Table 1.6: Quotients of additive evaluation: $q_+(i, j) = q_9(i + j)$.

$i\backslash j$	1	2	3	4	5	6	7	8	9
1	0	0	0	0	0	0	0	0	1
2	0	0	0	0	0	0	0	1	1
3	0	0	0	0	0	0	1	1	1
4	0	0	0	0	0	1	1	1	1
5	0	0	0	0	1	1	1	1	1
6	0	0	0	1	1	1	1	1	1
7	0	0	1	1	1	1	1	1	1
8	0	1	1	1	1	1	1	1	1
9	1	1	1	1	1	1	1	1	1

Table 1.7: Quotients of multiplicative evaluation: $q_\times(i, j) = q_9(ij)$.

$i\backslash j$	1	2	3	4	5	6	7	8	9
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	1	1	1	1	1
3	0	0	0	1	1	1	2	2	2
4	0	0	1	1	2	2	3	3	3
5	0	1	1	2	2	3	3	4	4
6	0	1	1	2	3	3	4	5	5
7	0	1	2	3	3	4	5	6	6
8	0	1	2	3	4	5	6	7	7
9	0	1	2	3	4	5	6	7	8

The multiplicative record of the center is therefore $(7, 2)$: a remainder and a quotient. This typed identification must accompany any subsequent use of these digits. A decimal concatenation, a coordinate of the support and a remainder–quotient pair designate different objects and admit different operations.

1.1.2.3 Transposition and central inversion

The transposition $\sigma(i, j) = (j, i)$ preserves both evaluations by commutativity. Its fixed points are the nine diagonal positions; the other seventy-two positions are distributed into thirty-six two-element orbits. On the cardinal directions, it exchanges north with west and east with south.

The central inversion $\rho_c(i, j) = (10-i, 10-j)$ satisfies $\rho_c^2 = \mathrm{id}$. The fixed-point equation is equivalent to $2i = 10$ and $2j = 10$, yielding the unique fixed point $(5, 5)$. This property is an internal geometric characterization of the center. Subsequent physical interpretations must preserve the concrete operation from which it arises.

The joint evaluation (S, P) determines the multiset $\{i, j\}$, since both numbers are the roots of $t^2 - St + P$. Retaining the ordered pair adds the orientation of the coordinates and distinguishes the two orders when $i \neq j$.

1.1.3 Cardinal transport and circulation record

1.1.3.1 Translation group of the support

We identify each coordinate of D_9 with its class in $\mathbb{Z}/9\mathbb{Z}$. The support thus acquires the structure of the additive group

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The positive representation again selects the integers $1, \ldots, 9$. The four elementary displacements are

$$\nu(N) = (-1, 0), \quad \nu(E) = (0, 1), \quad \nu(S) = (1, 0), \quad \nu(O) = (0, -1). \tag{1.11}$$

The transition in direction a is $x \mapsto x + \nu(a)$ in X_9. Each transition has the inverse determined by the opposite direction. Upon crossing an edge of the matrix representation, reduction returns a position in the same support; the integer displacement remains part of the record.

Write T_N, T_E, T_S, T_O for the four translations. The sum of opposite vectors is zero; therefore,

$$T_N \circ T_S = T_S \circ T_N = \mathrm{id}, \qquad T_E \circ T_O = T_O \circ T_E = \mathrm{id}\,.$$

Interior cases and boundary crossings are displayed in the following table. At each crossing, the difference between the integer coordinate reached and its positive representative is a multiple of nine.

Table 1.8: Interior transitions and boundary crossings in the matrix representation of APP.

Direction	Vector	Caso interior	Boundary crossing
N	$(-1, 0)$	$(5, 5) \to (4, 5)$	$(1, 5) \to (0, 5) \equiv (9, 5)$
E	$(0, 1)$	$(5, 5) \to (5, 6)$	$(5, 9) \to (5, 10) \equiv (5, 1)$
S	$(1, 0)$	$(5, 5) \to (6, 5)$	$(9, 5) \to (10, 5) \equiv (1, 5)$
O	$(0, -1)$	$(5, 5) \to (5, 4)$	$(5, 1) \to (5, 0) \equiv (5, 9)$

Congruence of pairs is interpreted coordinate by coordinate modulo nine. The intermediate pair belongs to the integer lift of the displacement.

The corresponding Cayley graph has eighty-one vertices and four outgoing edges at each vertex. It therefore contains 324 directed edges and 162 undirected edges. The directed count preserves the distinction between a transition and its inverse.

Its cellular realization is obtained by arranging nine columns and nine rows of squares and performing the periodic identifications of opposite edges. With ninety degrees between the two directions of the

planar representation, the resulting lattice has 81 vertices, 162 edges and 81 two-dimensional cells. Its Euler characteristic is $81 - 162 + 81 = 0$. The toroidal description belongs to this cellular realization of the support and its identifications; the arithmetic definition remains given by X_9 and its translations.

1.1.3.2 Sequences of displacements and integer lift

Let $w = a_1 \cdots a_r$ be a sequence of directions of length r. Given an initial position x_0, its trajectory is defined by

$$x_k = x_0 + \sum_{\ell=1}^{k} \nu(a_\ell) \quad \text{in } X_9, \qquad 0 \le k \le r.$$

The accumulated integer displacement is

$$d(w) = \sum_{\ell=1}^{r} \nu(a_\ell) = \big(\#S - \#N, \#E - \#O\big) \in \mathbb{Z}^2. \tag{1.12}$$

There are $81 \cdot 4^r$ pairs of initial position and cardinal sequence of length r. This count corresponds to the complete cardinal alphabet; the TPK admissibility criteria will act on this support with their additional records.

For $r = 0$, the empty sequence acts as the identity: each position retains its initial value and the accumulated displacement is $(0, 0)$. There are exactly eighty-one pairs of initial position and empty sequence. Under concatenation, the empty sequence is the identity element.

Theorem 1.9 (Positional return and number of turns)**.** *The trajectory of w returns to its initial position if and only if $d(w) \in 9\mathbb{Z}^2$. For a closed trajectory, the vector*

$$\mathrm{wind}(w) = \frac{d(w)}{9} \in \mathbb{Z}^2 \tag{1.13}$$

records the oriented turns in the two periodic directions. It is additive under concatenation of closed trajectories. Reversing the trajectory changes its sign.

Proof. The equality $x_r = x_0$ in $(\mathbb{Z}/9\mathbb{Z})^2$ is equivalent to divisibility by nine of both coordinates of $d(w)$. The sum of displacements is additive under concatenation. The inverse trajectory is obtained by reversing the order of the directions and replacing each with its opposite; its displacement is $-d(w)$. Division by nine preserves these identities. □

Example 1.10 (Two returns with different records)**.** The sequence E^9 and the sequence EO return to the initial position. Their turn vectors are, respectively, $(0, 1)$ and $(0, 0)$. The terminal point coincides in both cases, while the circulation record distinguishes the two paths.

The turn vector preserves net circulation. The complete order of transitions contains additional information: the sequences EO and NS have the same zero vector and different transitions. The progressive construction of the state will retain each record according to the operation to be recovered. In this way, terminal position, net circulation and ordered trajectory retain clearly differentiated functions.

1.1.4 Carry composition and change of section

1.1.4.1 The cocycle of the positive section

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \to D_9$ be the section of positive representatives. For two classes a, b, define

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}. \tag{1.14}$$

The numerator is divisible by nine. Since the first two quantities are between one and nine and the third represents their sum, $c_9(a, b)$ belongs to $\{0, 1\}$. The identity

$$s_9(a) + s_9(b) = s_9(a + b) + 9c_9(a, b)$$

expresses the carry produced when adding positive representatives.

Theorem 1.11 (Associative compatibility of the carry)**.** *For arbitrary $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c). \tag{1.15}$$

Proof. We multiply the expression on the left by nine and substitute (1.14). The terms $s_9(a + b)$ cancel, leaving

$$s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c).$$

The same substitution in the expression on the right cancels $s_9(b+c)$ and produces the identical expression. Division by nine proves the equality. □

The identity expresses the independence of the total carry from the two parenthesizations of a triple sum. Repeating it yields compatibility of any finite sum with its modular reduction. The record makes explicit the term connecting the integer calculation with the residue operation.

1.1.4.2 Positive residue and usual residue normalizations

The usual section s_0 takes values in $\{0, \dots, 8\}$. If $\delta_0(a)$ equals one for the zero class and zero for the others, the two sections are related by

$$s_9(a) = s_0(a) + 9\delta_0(a). \tag{1.16}$$

We define c_0 by the same expression as (1.14), replacing s_9 with s_0.

Proposition 1.12 (Exact transformation of the carry)**.** *The two cocycles satisfy*

$$c_9(a, b) = c_0(a, b) + \delta_0(a) + \delta_0(b) - \delta_0(a + b). \tag{1.17}$$

In particular, $c_9(0, a) = 1$ and $c_0(0, a) = 0$.

Proof. We substitute $s_9 = s_0 + 9\delta_0$ at the three occurrences of the section in (1.14). The terms in s_0 produce c_0 and the remaining terms produce the displayed difference of indicators. For $a = 0$, the identity is evaluated directly. □

The change of section modifies the cocycle by a coboundary. This formulation identifies exactly which term changes when the zero class is represented by 9 or by 0. Restitution of the integer remains consistent under both conventions when the quotient is transformed simultaneously.

To make the boundary cases explicit, set $\varepsilon(a, b) = c_9(a, b) - c_0(a, b)$. Formula (1.17) produces the following partition:

Table 1.9: Correction of the carry under a change of section.

Condition on the classes	$\varepsilon(a, b)$
$a = b = 0$	$1 + 1 - 1 = 1$
Exactly one of a, b is zero	1
$a \neq 0, b \neq 0, a + b = 0$	-1
$a \neq 0, b \neq 0, a + b \neq 0$	0

The first row makes explicit the overlap of the zero conditions; the four rows are disjoint and cover all pairs of classes.

1.1.5 Integer balances and multiplicative structure

1.1.5.1 The six totals of the arithmetic atlas

Proposition 1.13 (Totals of the evaluations and their records)**.** *The sum over the eighty-one positions satisfies*

$$\sum S = 810, \qquad \sum P = 2025, \tag{1.18}$$

$$\sum \Sigma = 405, \qquad \sum \Pi = 459, \tag{1.19}$$

$$\sum q_+ = 45, \qquad \sum q_\times = 174. \tag{1.20}$$

Proof. The sum $1 + \cdots + 9$ equals 45. By separation of the two coordinates,

$$\sum_{i,j} (i + j) = 9 \cdot 45 + 9 \cdot 45 = 810, \qquad \sum_{i,j} ij = 45^2 = 2025.$$

Each row of the residue additive evaluation contains a permutation of the nine representatives, with sum 45. Therefore, $\sum \Sigma = 9 \cdot 45 = 405$.
In residue multiplication, the rows with indices $1, 2, 4, 5, 7, 8$ are permutations and sum to 45. The rows with indices 3 and 6 contain each of the representatives $3, 6, 9$ three times and sum to 54. The row with index 9 contains nine entries of value 9 and sums to 81. This yields $\sum \Pi = 6 \cdot 45 + 2 \cdot 54 + 81 = 459$.
Finally, we sum (1.8): $\sum q_+ = (810 - 405)/9 = 45$ and $\sum q_\times = (2025 - 459)/9 = 174$. □

Table 1.10: Cumulative row evaluations of the APP support.

i	$\sum_j S$	$\sum_j \Sigma$	$\sum_j q_+$	$\sum_j P$	$\sum_j \Pi$	$\sum_j q_\times$
1	54	45	1	45	45	0
2	63	45	2	90	45	5
3	72	45	3	135	54	9
4	81	45	4	180	45	15
5	90	45	5	225	45	20
6	99	45	6	270	54	24
7	108	45	7	315	45	30
8	117	45	8	360	45	35
9	126	45	9	405	81	36
Grand total	810	405	45	2025	459	174

1.1.5.2 Difference between addition and multiplication

The difference between the two evaluations decomposes at each position as

$$P - S = (\Pi - \Sigma) + 9(q_\times - q_+). \tag{1.21}$$

Summing the complete atlas yields

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \tag{1.22}$$

The total residue contrast is 54; the quotient contrast is 129 and contributes 1161 to the total integer contrast 1215. The local equality determines the global one without changing the meaning of any of the records.

This decomposition makes it possible to track the residue value and its integer lift separately. When these data participate in subsequent operators, the provenance of the coefficients will be fixed by the evaluation and the sum that produced them. The appearance of 54 acquires a specific arithmetic meaning here, prior to any subsequent dimensional interpretation.

1.1.5.3 Units, radical ideal and ternary extraction

In the ring $R = \mathbb{Z}/9\mathbb{Z}$, a residue is invertible exactly when it is coprime to nine. Therefore,

$$R^\times = \{1, 2, 4, 5, 7, 8\}, \qquad 1^{-1} = 1, \quad 2^{-1} = 5, \quad 4^{-1} = 7, \quad 8^{-1} = 8.$$

The multiplicative rows corresponding to these indices are permutations. For indices 3 and 6, the image has three residues; for index 0, positively represented by 9, it has one.

The inverses are checked by composition. The additive translation $b \mapsto a + b$ is inverted by $c \mapsto c - a$; both compositions return the initial argument. If a is a unit, the multiplicative map $b \mapsto ab$ is inverted by $c \mapsto a^{-1}c$, since $a^{-1}(ab) = b$ and $a(a^{-1}c) = c$. The six indicated units are verified through the products $1 \cdot 1 = 1$, $2 \cdot 5 = 10 \equiv 1$, $4 \cdot 7 = 28 \equiv 1$ and $8 \cdot 8 = 64 \equiv 1$; the two intermediate pairs also provide the inverses of 5 and 7.

Proposition 1.14 (Radical structure of nonadic multiplication)**.** *The ideal* $\mathfrak{m} = (3) = \{0, 3, 6\}$ *satisfies* $\mathfrak{m}^2 = 0$*. Multiplication by three has kernel and image both equal to* $\mathfrak{m}$*. The map*

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \qquad x + \mathfrak{m} \longmapsto 3x \tag{1.23}$$

is an isomorphism of vector spaces over $\mathbb{F}_3$*.*

Proof. The product of two multiples of three is a multiple of nine; hence $\mathfrak{m}^2 = 0$. The equality $3x = 0$ in R is equivalent to $x \in \mathfrak{m}$ and the values of $3x$ range over $\mathfrak{m}$. If $x - y \in \mathfrak{m}$, then $3(x - y) = 0$; thus, (1.23) is well defined. Its kernel is zero and its image contains all three elements of the codomain. Additivity and compatibility with scalars in $\mathbb{F}_3$ follow immediately from multiplication in R. □

The operation in (1.23) preserves the linear structure. The product of two elements of $\mathfrak{m}$ is zero, whereas the quotient $R/\mathfrak{m}$ is the field with three elements. This distinction determines the algebraic type of the isomorphism.

The visible sequence $3, 6, 9$ admits two ternary evaluations. Its direct reduction modulo three is $(0, 0, 0)$. Extracting the factor three identifies $3k$ with its coordinate k mod 3 and produces $(1, 2, 0)$. The first operation calculates the class of the original integer; the second calculates the internal coordinate of the ideal. Their distinction prepares the typing of the ternary reader without anticipating its dynamics.

1.1.6 Operator realization of the evaluations

1.1.6.1 Additive resolution of the identity

The operator representation is constructed from the residue evaluations already defined. Let $\mathcal{H}_9 = \mathbb{C}^9$, with orthonormal basis $(e_r)_{r \in R}$, and let $P_r = e_r e_r^*$ be the projector onto the direction of e_r. To position $(a, b) \in R^2$ we assign the additive projector $P^+_{ab} = P_{a+b}$.

Proposition 1.15 (Operator identity by rows and columns)**.** *Each entry* P^+_{ab} *is a rank-one orthogonal projector and*

$$\sum_{b \in R} P^+_{ab} = I_9, \qquad \sum_{a \in R} P^+_{ab} = I_9. \tag{1.24}$$

Proof. Orthonormality implies $P_r^2 = P_r = P_r^*$ and $\sum_r P_r = I_9$. For fixed a, the translation $b \mapsto a + b$ permutes all residues; analogously for fixed b. Both sums are therefore the sum of the nine basis projectors. □

The realization expresses the Latin property of the additive table through a resolution of the identity. The Hilbert space acts here as the codomain of an explicit representation of the constructed arithmetic.

1.1.6.2 Multiplicities of the multiplicative operator

We define $P^{\times}_{ab} = P_{ab}$ and $T_a = \sum_b P^{\times}_{ab}$, where the subscript ab denotes the product of residues. For $a \in R^{\times}$, we obtain $T_a = I_9$. For $a = 3$ or $a = 6$,

$$T_a = 3(P_0 + P_3 + P_6), \tag{1.25}$$

and for $a = 0$, we obtain $T_0 = 9P_0$.

These identities are proved by counting the preimages of each residue under $b \mapsto ab$. In the two rows with ternary image, the residues $0, 3, 6$ each have three preimages. In the zero row, the residue 0 has nine. Consequently, the ranks of T_a are respectively 9, 3 and 1; its nonzero eigenvalues are 1, 3 and 9. In all three cases, the trace retains the value 9.

The operator thus records both the concentration of multiplicities and the total number of contributions. Equality of traces coexists with differences in rank and spectrum. This distinction provides a complete example of preservation of an aggregate quantity with transformation of its operator distribution.

1.1.7 Decimal blocks and compatibility between moduli

1.1.7.1 Base-thousand decomposition

For $N \in \mathbb{N}_0$, Euclidean division by one thousand produces

$$N = 1000Q + B, \qquad 0 \le B < 1000. \tag{1.26}$$

The block B has the width-three decimal notation

$$B = 100a + 10b + c, \qquad a, b, c \in \{0, \dots, 9\}.$$

The width is preserved even when leading zeros appear: the block 001 has three digit positions and integer value one. The index of the block in a sequence and the quotient Q preserve the scale information required by refinement.

In the first APP table, sums and products are less than one thousand. Their remainder modulo one thousand coincides with their integer value. Reduction modulo nine has a different function: it produces the evaluations Σ and Π and their quotients. Comparison of the two moduli therefore begins with the concrete operations that each performs.

1.1.7.2 Exact identity between quotients

Theorem 1.16 (Compatibility of nonadic and decimal records)**.** *Let $N > 0$ and let (1.26) be its Euclidean division by one thousand. Then*

$$\rho_9(N) = \rho_9(Q + B), \qquad q_9(N) = 111Q + q_9(Q + B). \tag{1.27}$$

Proof. Since $N > 0$, we have $Q + B > 0$. The identity $1000 = 999 + 1$ gives

$$N = 999Q + (Q + B) = 9 \cdot 111Q + (Q + B).$$

We apply nonadic restitution to the last term:

$$N = \rho_9(Q + B) + 9\big(111Q + q_9(Q + B)\big).$$

The remainder belongs to D_9 and the quotient is a nonnegative integer. Uniqueness in proposition 1.3 identifies them with $\rho_9(N)$ and $q_9(N)$. □

Example 1.17 (Crossing the decimal threshold)**.** For $N = 1001$, division by one thousand produces $Q = 1$, $B = 1$. The formula gives $\rho_9(N) = \rho_9(2) = 2$ and $q_9(N) = 111$. The final block considered in isolation has remainder one; incorporating the quotient recovers the remainder two of the complete integer.

For $N = 1000$, we obtain $(\rho_9, q_9) = (1, 111)$; for $N = 999$, we obtain $(9, 110)$. The transition between the two integers simultaneously changes the positive representative and the quotient.

1.1.7.3 Recomposition of a finite sequence of blocks

Corollary 1.18 (Sum of blocks and restitution of scale)**.** *Let* $N = \sum_{k=0}^{m} B_k 1000^k > 0$*, with* $0 \le B_k < 1000$*, and let* $S_B = \sum_k B_k$*. Then*

$$\rho_9(N) = \rho_9(S_B), \tag{1.28}$$

$$q_9(N) = q_9(S_B) + \sum_{k=1}^{m} B_k \frac{1000^k - 1}{9}. \tag{1.29}$$

Proof. Subtracting S_B from the expansion of N produces $\sum_{k\ge1} B_k(1000^k - 1)$. Each coefficient $1000^k - 1$ is divisible by nine. Substituting $S_B = \rho_9(S_B) + 9q_9(S_B)$ and applying uniqueness of restitution yields both identities. □

The remainder depends on the sum of the blocks; the second formula also preserves their position through powers of one thousand. The order and scale of the expansion are explicit in the integer record. This identity delimits the function of modulus one thousand in the present chapter. The selection of blocks by the calendar, the tritic orientation and the coinductive extension correspond to the TPK operations that will be developed on this basis.

1.1.8 Dependencies established for the tritic stage

The construction has produced an ordered support of eighty-one positions, two simultaneous integer evaluations and their reversible decompositions. It has also produced the cardinal translations, the circulation record, the carry cocycle and the internal ternary structure of the multiplicative ideal. The projector realization and the compatibility of decimal blocks are obtained after these operations and retain their explicit domains.

At this stage, exact reconstruction concerns the arithmetic evaluations; restitution of a complete trajectory requires its ordered record. This hierarchy determines the transition to the enriched state: each new operation must indicate which coordinates it uses, what it modifies and what it preserves. The introduction of TRIT can then be based on the ternary reduction and integer lift already defined; the introduction of TPK can act on concrete transitions and records.

The subsequent development of the regions and constants will be inserted into that generative sequence. Its exposition must preserve the difference between the operator that generates an output and the formula that allows it to be recognized or evaluated in a later language. The arithmetic basis is proved here in the detail needed to track both functions without interchanging them.

1.2 TRIT: ternary orientation, carry and quadratic representation

The construction of APP jointly preserves position, the additive and multiplicative evaluations and their nonadic decompositions. The oriented ternary reduction now acts on these data. Its task is twofold: to distinguish the three operational classes of an evaluation and to preserve the integer coordinate that allows it to be restored. This distinction is subsequently involved in the phase selection of the topological phase kernel, called TPK after its initials in English, *Topological Phase Kernel*.

The term *trit* designates a ternary unit. In the balanced convention of HMT its values are $-1, 0, +1$. The same alphabet admits different functions depending on the domain: reading an arithmetic evaluation,

Table 1.11: Balanced decomposition of the nine positions on an APP line.

n	1	2	3	4	5	6	7	8	9
$r_3(n)$	1	−1	0	1	−1	0	1	−1	0
$c_3(n)$	0	1	1	1	2	2	2	3	3

Coincidence of ternary representatives preserves three distinct positions in each class; their quotients restore the integers.

selecting an operational mode or indexing a quadratic realization. Each function is defined with its domain and the information it preserves.

The development begins with the integer evaluations of APP. Quadratic realizations appear later as representations of the tritic type already determined. The concrete construction of their generators and the Euler extension are developed in chapter 12.1, after introducing the calendar and the TPK mode selection.

1.2.1 Balanced ternary reduction of APP evaluations

At a position (i, j), with $i, j \in \{1, \dots, 9\}$, the integer evaluations are $S(i, j) = i + j$ and $P(i, j) = ij$. Each preserves its positive remainder modulo nine and its quotient. Ternary reduction is applied to the corresponding evaluation and produces another pair of coordinates.

Definition 1.19 (Oriented ternary decomposition)**.** Let

$$\mathcal{T} = \{-1, 0, +1\}, \qquad c_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \qquad r_3(n) = n - 3c_3(n), \quad n \in \mathbb{Z}.$$

The decomposition and restitution maps are

$$\mathcal{B}_3 : \mathbb{Z} \longrightarrow \mathcal{T} \times \mathbb{Z}, \qquad \mathcal{B}_3(n) = (r_3(n), c_3(n)),$$

$$\mathcal{L}_3 : \mathcal{T} \times \mathbb{Z} \longrightarrow \mathbb{Z}, \qquad \mathcal{L}_3(r, c) = r + 3c.$$

The first component is the balanced ternary representative; the second is the extension quotient.

Proposition 1.20 (Exact restitution and uniqueness)**.** *The maps $\mathcal{B}_3$ and $\mathcal{L}_3$ are inverses. In particular, for every integer n,*

$$n = r_3(n) + 3c_3(n), \qquad r_3(n) \in \mathcal{T},$$

and this expression is unique.

Proof. If $c = \lfloor (n+1)/3 \rfloor$, then $3c \le n+1 < 3c+3$. Therefore $-1 \le n-3c < 2$, and the integer $r = n-3c$ belongs to $\mathcal{T}$. If $r + 3c = s + 3d$, with $r, s \in \mathcal{T}$, the integer $r - s$ is a multiple of three and lies between -2 and 2. Thus, $r = s$ and $c = d$. Finally, for any pair (r, c), $\lfloor (r + 3c + 1)/3 \rfloor = c$, since $0 \le r + 1 \le 2$. This proves both inverse compositions. □

The oriented section of the quotient $\mathbb{Z}/3\mathbb{Z}$ is fixed by

$$[0]_3 \longmapsto 0, \qquad [1]_3 \longmapsto +1, \qquad [2]_3 \longmapsto -1.$$

Replacing 2 with -1 is accompanied by the necessary quotient: $2 = -1 + 3$. The integer evaluation remains recoverable in the pair.

For example, at $(i, j) = (5, 5)$, the sum equals 10 and the product equals 25. Both ternary readings give +1, while their complete decompositions are

$$\mathcal{B}_3(10) = (1, 3), \qquad \mathcal{B}_3(25) = (1, 8).$$

Table 1.12: Additive carry of the balanced ternary section.

$\chi(r,s)$	$s=-1$	$s=0$	$s=+1$
$r=-1$	-1	0	0
$r=0$	0	0	0
$r=+1$	0	0	1

The two extremes encode $-1-1=1-3$ and $1+1=-1+3$.

The common signature identifies a residue type; the sheet and quotient distinguish the two evaluations. The APP state also preserves the position that produced them.

1.2.1.1 Compatibility between moduli nine and three

Write the nonadic decomposition of an evaluation as

$$n = R_9(n) + 9q_9(n), \qquad R_9(n) \in \{1, \dots, 9\}.$$

Reducing its positive representative ternarily yields

$$R_9(n) = r_3(R_9(n)) + 3c_3(R_9(n)).$$

Substituting,

$$r_3(n) = r_3(R_9(n)), \qquad c_3(n) = c_3(R_9(n)) + 3q_9(n). \tag{1.30}$$

Uniqueness of the balanced decomposition proves both identities. The visible reduction modulo three can be performed on the nonadic remainder, while the complete reconstruction also transports the previous quotient.

In the product $5 \cdot 5 = 25$, the nonadic decomposition is $25 = 7 + 9 \cdot 2$. Since $7 = 1 + 3 \cdot 2$, the preceding formula gives $25 = 1 + 3(2 + 3 \cdot 2) = 1 + 3 \cdot 8$. Both reading orders restore the same integer because the quotients compose explicitly.

1.2.2 Arithmetic of ternary states with carry

Preserving the pairs makes it possible to transfer the APP operations to balanced coordinates. The transfer determines the additive carry and the cross terms of the product.

Definition 1.21 (Balanced additive carry)**.** For $r, s \in \mathcal{T}$, let

$$r \oplus s = r_3(r+s), \qquad \chi(r,s) = \frac{r + s - r_3(r+s)}{3}.$$

The integer $\chi(r,s)$ is the carry produced when normalizing the sum of the two representatives.

Proposition 1.22 (Recoverable operations)**.** *If $n = r + 3c$ and $m = s + 3d$, with $r, s \in \mathcal{T}$, then*

$$(r,c) \boxplus (s,d) = \big(r \oplus s, c + d + \chi(r,s)\big), \tag{1.31}$$

$$(r,c) \boxtimes (s,d) = \big(rs, rd + sc + 3cd\big) \tag{1.32}$$

are, respectively, the decompositions of $n+m$ and nm.

Proof. The sum decomposes as

$$n + m = r + s + 3(c+d) = r_3(r+s) + 3\big(c + d + \chi(r,s)\big).$$

In the product,

$$nm = (r + 3c)(s + 3d) = rs + 3(rd + sc + 3cd).$$

The integer rs belongs to $\mathcal{T}$. Both expressions are normalized; uniqueness in proposition 1.20 identifies their components. □

These laws distinguish accumulation and multiplication in the record. The sum aggregates the quotients and adds the residue carry. The product also combines each representative with the quotient of the other evaluation. This arithmetic content is maintained when the operations are incorporated into a TPK transition.

Example 1.23 (Sum and product of two evaluations)**.** From $2 = -1 + 3$ and $4 = 1 + 3$ follows

$$(-1, 1) \boxplus (1, 1) = (0, 2), \qquad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

Restitution produces 6 and 8, respectively. In the second operation, the quotient is $(-1)\cdot 1+1\cdot 1+3\cdot 1\cdot 1 = 3$.

Proposition 1.24 (Cocycle identity and associativity)**.** *For $r, s, t \in \mathcal{T}$,*

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t). \tag{1.33}$$

The operations $\boxplus$ and $\boxtimes$, together with their identity elements and additive inverses transported by $\mathcal{B}_3$, make $\mathcal{T} \times \mathbb{Z}$ a ring isomorphic to $\mathbb{Z}$.

Proof. When normalizing $(r + s) + t$, the total quotient is the first side of (1.33); when normalizing $r + (s + t)$, it is the second. The final representatives coincide because they represent the same class modulo three. Uniqueness of the complete decomposition requires the quotients to coincide as well. By proposition 1.22, $\mathcal{L}_3$ is a bijection preserving sum and product. It therefore transports associativity, distributivity and commutativity of the integer operations. Its identity elements are $(0, 0)$ and $(1, 0)$. □

The cocycle proof expresses a concrete compatibility of the record: normalization can be performed in stages without changing the final evaluation. If the sequence of operations is of interest, the record also preserves the order of the normalizations performed.

1.2.3 Ternary refinement and recovery at arbitrary depth

The decomposition is iterated on the quotient. Given $n_0 \in \mathbb{Z}$, define recursively

$$r_k = r_3(n_k), \qquad n_{k+1} = c_3(n_k), \qquad k \geq 0.$$

Each transition satisfies $n_k = r_k + 3n_{k+1}$. Composing these equalities produces a restitution at any prescribed depth.

Proposition 1.25 (Restitution of a truncation)**.** *For every $N \geq 1$,*

$$n_0 = \sum_{k=0}^{N-1} 3^k r_k + 3^N n_N. \tag{1.34}$$

The sequence $(r_0, \ldots, r_{N-1})$, together with the terminal quotient n_N, determines n_0 exactly.

Proof. For $N = 1$, the identity is the initial decomposition. If it holds for N, substitute $n_N = r_N + 3n_{N+1}$ into the last term to obtain the formula for $N + 1$. Restitution also reconstructs the intermediate quotients, starting at n_N and applying $n_k = r_k + 3n_{k+1}$ in reverse order. □

Example 1.26 (A carry crossing three levels)**.** The evaluation 26 successively gives

$$26 = -1 + 3 \cdot 9, \qquad 9 = 0 + 3 \cdot 3, \qquad 3 = 0 + 3 \cdot 1, \qquad 1 = 1 + 3 \cdot 0.$$

Its representatives, ordered from lower to higher power, are $(-1, 0, 0, 1)$. Restitution is $26 = -1 + 27$. The zero symbol at the intermediate levels preserves a position and a determined quotient.

Proposition 1.27 (Termination for a fixed integer)**.** *Iteration of the balanced quotient reaches zero for every initial integer. Once trailing zeros are removed, the finite balanced ternary expansion of an integer is unique.*

Proof. For $|n| \geq 2$, $|c_3(n)| \leq (|n| + 1)/3 < |n|$. For $n \in \{-1, 0, 1\}$, the quotient is zero. The absolute value strictly decreases while it is at least two, so the iteration terminates. If two expansions represent the same integer, their reduction modulo three fixes the same first representative; subtracting it and dividing by three repeats the argument for all positions. □

This result concerns the arithmetic of a fixed integer. In TPK extensions, the record length increases and new evaluations can be generated. There, depth corresponds to a compatible sequence of states, whose construction requires the survival rules. Formula (1.34) provides the local restitution mechanism that those extensions must preserve.

1.2.4 Two ternary coordinates of a nonadic class

The reduction of $\mathbb{Z}/9\mathbb{Z}$ modulo three has three elements in each fiber. The additional coordinate can be written as a residue ternary quotient.

Proposition 1.28 (Nonadic coordinates with carried addition)**.** *The map*

$$\beta : \mathcal{T} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \qquad \beta(r, \bar{c}) = [r + 3c]_9$$

is a bijection of sets. Nonadic addition and multiplication take the form

$$\beta(r, \bar{c}) + \beta(s, \bar{d}) = \beta\big(r \oplus s, \overline{c + d + \chi(r, s)}\big), \tag{1.35}$$

$$\beta(r, \bar{c})\beta(s, \bar{d}) = \beta\big(rs, \overline{rd + sc}\big). \tag{1.36}$$

Proof. Changing c to $c + 3k$ changes $r + 3c$ by $9k$, so the map is well defined. If two images coincide, reduction modulo three fixes the same r. The remaining difference, divided by three, fixes the same class $\bar{c}$. Injectivity between two sets of nine elements proves bijectivity. The operations come from (1.31) and (1.32); the term $3cd$ in the multiplicative quotient vanishes upon reduction modulo three. □

The carry establishes the composition law between the two levels. For example,

$$\beta(1, 0) + \beta(1, 0) = \beta(-1, 1).$$

The second coordinate records the step performed by the integer sum before its residue reading. In particular, $\mathbb{Z}/9\mathbb{Z}$ has elements of order nine, whereas the additive group of $\mathbb{F}_3^2$ has exponent three. The preceding bijection preserves the sum through the explicit carried law.

The three APP positions $3, 6, 9$ are written

$$[3]_9 = \beta(0, 1), \qquad [6]_9 = \beta(0, 2), \qquad [9]_9 = \beta(0, 0).$$

On the ideal $3\mathbb{Z}/9\mathbb{Z}$, extraction of the second coordinate is

$$[3c]_9 \longmapsto [c]_3.$$

This is an additive bijection with $\mathbb{F}_3$. It differs from direct reduction modulo three, which assigns zero to all three elements. Moreover, the product of two elements of this ideal is zero modulo nine. Its multiplicative structure thus retains a function different from multiplication in the field $\mathbb{F}_3$.

1.2.4.1 The fiber of six ternary positions

An oriented word of length six belongs to $\mathcal{T}^6$, with $3^6 = 729$ possibilities. The choice of representatives induces a bijection with $\mathbb{F}_3^6$. Grouping positions in pairs and applying β also yields a set-theoretic bijection with $(\mathbb{Z}/9\mathbb{Z})^3$. The transported law uses the carry of each pair.

The census of words actually emitted by the TPK is performed after constructing its emitter and chronology. The ambient fiber of 729 elements fixes coordinate capacity; the set of emissions and its ternary image are determined by the operational rules. This separation makes it possible to preserve, in the subsequent census, both multiplicities and the distinct records sharing a visible word.

1.2.5 Oriented involution and transport memory

Proposition 1.29 (Compatibility with sign reversal)**.** *For all $n \in \mathbb{Z}$,*

$$\mathcal{B}_3(-n) = (-r_3(n), -c_3(n)).$$

Proof. From $n = r + 3c$ we deduce $-n = (-r) + 3(-c)$. Since $-r \in \mathcal{T}$, uniqueness of the decomposition proves the statement. □

For a sequence of signatures of length m, define

$$C_{\mathrm{rev}}(\tau_1, \ldots, \tau_m) = (-\tau_m, \ldots, -\tau_1).$$

The second application restores the initial order and signs, so that $C_{\mathrm{rev}}^2 = \mathrm{id}$. In a sequence of pairs, the same rule simultaneously transports $(r_k, c_k) \mapsto (-r_{m+1-k}, -c_{m+1-k})$. The length and each restorable evaluation are determined.

Division memory and transport memory serve different purposes. The pair $(r_3(n), c_3(n))$ reconstructs an integer evaluation. Reconstructing a trajectory also requires the initial position, orientations, sequence of modes and boundary crossings. Two paths can reach the same evaluation and preserve different transport records. The enriched state integrates both orders of information.

In the temporal interpretation adopted by HMT, +1 corresponds to return with memory, 0 to the present threshold and −1 to the opening of conjugate extensions. This assignment fixes a regime reader. Effective evolution requires the TPK transition on the complete records; its iterations determine which phase returns and which memory is incremented.

1.2.6 Quadratic algebras of the three regimes

The tritic index has an algebraic realization through a quadratic relation. Its integral formulation precedes the choice of the scalar field and allows discrete information to be separated from subsequent analytic realizations.

Definition 1.30 (Quadratic algebra of tritic type)**.** For $\tau \in \mathcal{T}$, define

$$Q_{\tau,\mathbb{Z}} = \mathbb{Z}[u]/(u^2 + \tau).$$

For a commutative ring A, its realization by change of scalars is

$$Q_{\tau,A} = A[u]/(u^2 + \tau).$$

The class of u, written J_τ, satisfies $J_\tau^2 = -\tau I$.

Division by the monic polynomial $u^2 + \tau$ provides a unique expression $aI + bJ_\tau$. The product is calculated by reducing only the quadratic term:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \tag{1.37}$$

This multiplication represents the quadratic regime. The arithmetic of balanced pairs, for its part, preserves integers and their quotients through (1.31) and (1.32).

1.2.6.1 Nine-element realizations

Over $\mathbb{F}_3$, the three algebras have nine elements and admit the following classification:

$$\mathcal{Q}_{+1,\mathbb{F}_3} \cong \mathbb{F}_9, \qquad \mathcal{Q}_{0,\mathbb{F}_3} = \mathbb{F}_3[\varepsilon]/(\varepsilon^2), \qquad \mathcal{Q}_{-1,\mathbb{F}_3} \cong \mathbb{F}_3 \times \mathbb{F}_3.$$

Proposition 1.31 (Distinction of the three finite types)**.** *The first algebra is a field; the second contains a nonzero nilpotent; the third has two nontrivial orthogonal idempotents whose sum is the identity. The three are pairwise nonisomorphic.*

Proof. The polynomial $u^2 + 1$ takes the values $1, 2, 2$ on $0, 1, 2$; it is therefore irreducible over $\mathbb{F}_3$. Its quotient is a field of degree two. In the second case, the class ε is nonzero and satisfies $\varepsilon^2 = 0$. In the third, $u^2 - 1 = (u - 1)(u + 1)$, with coprime factors. The evaluation $a + bu \mapsto (a + b, a - b)$ is an isomorphism to $\mathbb{F}_3 \times \mathbb{F}_3$. The elements $2(1 + u)$ and $2(1 - u)$ are its two coordinate idempotents. These algebraic properties distinguish the three types. □

In the development of exceptional incidence, a concretely constructed ternary operator satisfying $A_W^2 = -I$ induces the action of $\mathbb{F}_9$ on its space. If that space is $\mathbb{F}_3^6$, the dimension over $\mathbb{F}_9$ is three: the action is defined by $(a + b\iota)v = av + bA_W v$. The operator fixes multiplication by the quadratic unit. The corresponding Paley construction supplies that operator in its causal place; the present classification establishes the content of the relation it must satisfy.

1.2.6.2 Conjugation, algebraic norm and inverse

In the real realization, conjugation and the algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \qquad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 1.32 (Multiplicative norm and inverse restitution)**.** *For $z, w \in \mathcal{Q}_{\tau,\mathbb{R}}$,*

$$z\bar{z} = N_\tau(z)I, \qquad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

An element z is invertible exactly when $N_\tau(z) \neq 0$; in that case $z^{-1} = \bar{z}/N_\tau(z)$.

Proof. The product (1.37) gives $(aI + bJ_\tau)(aI - bJ_\tau) = (a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so $(zw)\overline{zw} = (z\bar{z})(w\bar{w})$. If the norm is nonzero, the formula exhibits an inverse. If $zw = I$, multiplicativity gives $N_\tau(z)N_\tau(w) = 1$, forcing $N_\tau(z) \neq 0$. □

The matrix representation

$$aI + bJ_\tau \longmapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}$$

is faithful and has determinant $a^2 + \tau b^2$. For $\tau = +1$, the form is positive definite. For $\tau = 0$, it depends only on a, while J_0 retains a nonzero linear component. For $\tau = -1$, the form has signature $(1, 1)$; the directions $I \pm J_{-1}$ are zero divisors.

In particular, the word *norm* here designates a multiplicative quadratic form. The positive norms of Hilbert spaces, used in other realizations of the state, retain their own definition.

1.2.6.3 Composition of ratios between coordinates

A direction with nonzero scalar component is represented by $I + uJ_\tau$. Multiplication produces

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

When $1 - \tau uv \neq 0$, the ratio between the components of the product is

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}. \tag{1.38}$$

The scalar factor $1 - \tau uv$ forms part of the complete product. If only the ratio is retained, that factor must be recorded when it is involved in a subsequent reconstruction. If the denominator vanishes, the product is examined in its two components: it can represent a direction with zero scalar component or the zero element.

The quadratic classification thus prepares a precise reading of the three regimes. Selection of a regime preserves its index, arithmetic preserves its carries and transport preserves the oriented sequence. The TPK composes these data through explicit operators. On that composition will be built the concrete generators and the preservation of forms that articulates the Euler extension.

1.3 Topological phase kernel and generation of the seed catalog

The two APP evaluations acquire generative efficacy when it is specified which evaluation acts, at which position it is performed, how that position moves and what information each transition preserves. The *Topological Phase Kernel*, abbreviated TPK and called in Spanish *núcleo topológico de fase*, brings these operations together in dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this chapter provides the seeds on which the regional lifts and the coinductive extension will act.

The main result of this stage is a generated and classified catalog: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; the dihedral action organizes these words into orbits. Each step preserves its preimages and the relations that allow a return to the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \qquad 243 \longrightarrow 43,$$

abbreviates different objects. Each object is constructed below and each count is justified, before any of them is used as data for regional selection.

1.3.1 Phase selection, transport and enriched state

1.3.1.1 Tritic selector of operational phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\mathrm{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \qquad M_{\mathrm{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \tag{1.39}$$

In the emitting realization, the value +1 activates the additive evaluation, −1 activates the multiplicative evaluation and 0 records the neutral event. The cardinal orientation of the cursor is another coordinate of the state. This separation allows both the operation being read and the direction in which its evaluation point is transported to be fixed simultaneously.

For transitions numbered by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t-1)]_9, \qquad \epsilon_t = M_{\mathrm{ph}}(g_t), \tag{1.40}$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b-1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

1.3.1.2 Oriented cursors and cardinal transport

Indices $I_9 = \{0, \dots, 8\}$ are used for positions and positive labels $(a, b) = (i+1, j+1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \qquad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act through

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = \big([i + (v_d)_1]_9, [j + (v_d)_2]_9\big).$$

Thus, M_{ph} and T_d have different domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport modifies its position in the direction that cursor already preserves.

The integer lift of the cursor is $\widetilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its turn counts are recovered in

$$x = 9\lfloor x/9 \rfloor + [x]_9, \qquad y = 9\lfloor y/9 \rfloor + [y]_9. \tag{1.41}$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic register, which will be preserved alongside them.

1.3.1.3 Composition of the transition

The state of the finite realization includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition register. Each local register contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. The prefix and boundary fields that appear in a prolongation will be incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \mathsf{Mem}_t \circ \mathsf{Tra}_t \circ \mathsf{Sel}_t. \tag{1.42}$$

Selection determines the regime and preserves the sample prior to any displacement. Transport applies the active movement and the cardinal inversion prescribed by the calendar. Updating incorporates that sample into the accumulators and the register; upon completing a window, it emits its block. The composition therefore uses a sample taken before transport even though its definitive archiving occurs at the end of the transition.

1.3.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\mathrm{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \qquad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\mathrm{seed}}| = 324^2 = 104\,976. \tag{1.43}$$

The number 81 counts positions; 324 counts oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP preserves only part of an initial condition of the emitter.

The numbering is fixed

$$\mathrm{ind}(N) = 0, \quad \mathrm{ind}(E) = 1, \quad \mathrm{ind}(S) = 2, \quad \mathrm{ind}(O) = 3,$$

and one defines

$$\nu(i, j; d) = 4(9i + j) + \mathrm{ind}(d). \tag{1.44}$$

Proposition 1.33 (Exact enumeration of initial conditions)**.** *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The index pair is encoded reversibly by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\mathrm{ind}(d) = [\nu]_4, \qquad j = [\lfloor \nu/4 \rfloor]_9, \qquad i = \lfloor \nu/36 \rfloor.$$

The domain bounds guarantee that these three quantities belong to their index sets and reconstruct ν through (1.44). For the pair, Euclidean division by 324 recovers quotient ν_+ and residue $\nu_\times$. □

The enumeration traverses the complete domain before any regional grouping. The indices are used to locate the preimages of an emission, keeping explicit the condition that produced each trajectory.

1.3.3 Evaluation of the sheets and arithmetic register

At a positive position (a, b), the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, one maintains

$$\rho_9(n) = 1 + [n - 1]_9, \qquad q_9(n) = \lfloor (n - 1)/9 \rfloor, \qquad n = \rho_9(n) + 9q_9(n). \tag{1.45}$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined below.

1.3.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod 9, \qquad 2^6 \equiv 1 \pmod 9.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \tag{1.46}$$

The emission law extends the observable through $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. In this way, an observed value of zero can come from the unit 1 or from one of the three noninvertible residues, and its origin remains recoverable.

1.3.3.2 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is moved. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is moved. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are chained together while preserving positions and orientations. At the start of each window only its three local accumulators are reset to zero; the previous register remains available.

Proposition 1.34 (Inversion of lifted transport)**.** *Let $a, b \in \{0, 1\}$ be the activation and inversion indicators of a cursor. If* opp *interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = \left(z + a v_d, \mathrm{opp}^b(d)\right)$$

has an inverse

$$F_{a,b}^{-1}(z', d') = \big(z' - av_{\mathrm{opp}^b(d')}, \mathrm{opp}^b(d')\big).$$

Spatial reduction modulo nine intertwines this transformation with finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction is recovered, subtracting av_d undoes the displacement. For spatial projection, the equality $[z + av_d]_9 = [[z]_9 + av_d]_9$ suffices, component by component. Direction inversion preserves this equality. □

The phase determines the indicators used by the inverse. The spatial and arithmetic registers can thus be preserved together, even when crossing the boundary of the representative I_9^2.

1.3.4 Six windows and decimal emission by blocks

1.3.4.1 Accumulation in a nonadic window

Window m, with $1 \le m \le 6$, comprises the steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents and Z_m its number of neutral events.

Lemma 1.35 (Activation multiplicities)**.** *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets of (1.39) has cardinality three. □

The six windows form 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible registers.

1.3.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \tag{1.47}$$

The weights $100, 10, 1$ are the hundreds, tens and units positions. The subscript 10 indicates the decimal residue, and each block satisfies $0 \le U_m \le 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients $3, 5, 7$ constitute the explicit weights of this emission law. Once the rule is fixed, each digit is calculated from the accumulators, and the results of this chapter follow from that rule. The number 1 appearing in its reduced form has a specific additional provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \tag{1.48}$$

The letter e_m here denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 1.36 (Recovery of the two signatures)**.** *The map $G : \{0, \dots, 9\}^2 \to \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \qquad e = [b - a]_{10}, \qquad c = [3s + 5e + 1]_{10}.$$

Consequently, the word **U** *determines the two six-component signatures* **s** *and* **e**.

Proof. The last two summands of (1.48) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s. The tens digit is $[s+e]_{10}$; subtracting s and reducing modulo ten recovers e. The units digit verifies the third congruence. Reconstruction is applied independently to the six positions of the word. □

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This difference identifies exactly what information each level of encoding preserves.

1.3.4.3 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^{6} U_m 1000^{6-m}, \qquad 0 \le \mathcal{N}(\mathbf{U}) < 1000^6. \tag{1.49}$$

The first block occupies the highest-weight position and the sixth, the units position in base one thousand. A block 058 occupies a complete position with integer value 58; its three-digit width is part of the decimal presentation of that position.

Proposition 1.37 (Positional recovery of emissions)**.** *The encoding* (1.49) *determines the complete word through*

$$U_m = \left[\left\lfloor \frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right\rfloor \right]_{1000}, \qquad 1 \le m \le 6.$$

Proof. Upon division by 1000^{6-m}, the earlier blocks produce integer multiples of 1000, block m produces U_m, and the contribution of later blocks belongs to $[0, 1)$. The integer part eliminates this last contribution and the residue modulo 1000 eliminates the earlier ones. □

This encoding preserves the six emissions. Its domain has ordered positions in base one thousand, whereas the APP domain has spatial positions modulo nine. The composition maintains both indices: m identifies an emission window and (i, j) identifies the cell visited during a transition in that window.

1.3.5 Complete calculation of an emission beginning with 501

The initial conditions are fixed

$$c_{+,0} = (0, 4; N), \qquad c_{\times,0} = (2, 3; N), \qquad \nu_+ = 16, \quad \nu_\times = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$ and the register is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

The three active additive evaluations are

$$1 + 5 = 6, \qquad 9 + 5 = 14 = 5 + 9, \qquad 8 + 5 = 13 = 4 + 9,$$

so that $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are $3, 8, 4$, with observables $0, 3, 2$; therefore, $E_1 = 5$. Finally $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

The second additive reading above has arithmetic quotient 1, while its lifted cursor is at $(-1, 4; N)$, with vertical turn count -1. These are two different data in the same register. Both enter into the full recovery of the evaluation and its position.

Table 1.13: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position ×	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	−1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	−1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	−1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

The positions are positive labels of the cells before movement. Neutral events increment only Z.

Table 1.14: The six windows of the initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

The resulting word and its signatures are

$$\mathbf{U} = (501, 614, 498, 169, 272, 272),$$
$$\mathbf{s} = (5, 6, 4, 1, 2, 2), \qquad \mathbf{e} = (5, 5, 5, 5, 5, 5).$$

The cardinal inversion after step 27 modifies the paths of the last three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological register.

As a second case, the minimal initial condition $(\nu_+, \nu_\times) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial position and orientation after 54 transitions. The nonadic phase completes six turns, the chronological counter advances by 54 units and the register contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

1.3.6 Factorization of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_\times(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor during the six windows. The independence of positions and directions between the two cursors gives

$$T_{\mathrm{em}}(\nu_+, \nu_\times) = G^{(6)}\big(f_+(\nu_+), f_\times(\nu_\times)\big), \tag{1.50}$$

where $G^{(6)}$ applies (1.48) at each position. The factorization allows 324+324 trajectories to be enumerated first and the distinct signatures to be combined afterward, preserving the multiplicity of their preimages.

1.3.6.1 Images of the two signatures

The following tables include all signatures. Their multiplicity is the number of initial conditions producing them; the minimum index locates one of those conditions. The complete preimage is defined by

$$f_\sigma^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_\sigma(\nu) = a\}.$$

The recurrence of the preceding sections determines each element of this set through a finite integer calculation.

Table 1.15: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 1.16: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 1.38 (Image and multiplicities of the emitter)**.** *The image* $\mathcal{U}_6 = \operatorname{im} T_{\mathrm{em}}$ *contains* 468 *distinct words. Their preimages are distributed among* 432 *fibers of cardinality* 144*,* 18 *of cardinality* 432 *and* 18 *of cardinality* 1944*.*

Proof. The exhaustive evaluation of the algorithm in section 1.3.9 over the indices $0, \dots, 323$ gives the two preceding tables. The first partition has 18 classes of cardinality 18, and the second 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ check the total mass of both partitions. Proposition 1.36 makes the coupling of their images injective; therefore, there are $18 \cdot 26 = 468$ emissions. By (1.50), the preimage of a pair of signatures is the product

of their two preimages. One obtains $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$ and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorization explains why that count covers all pairs. □

The emission in example 1.3.5 has a fiber of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fiber of cardinality $18 \cdot 8 = 144$. In both cases the emitted word recovers the signatures, while identifying a particular initial condition uses its index or its trajectory register.

1.3.7 Oriented ternary reading and preservation of fibers

The oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \qquad \Psi(U_1, \ldots, U_6) = ([-U_1]_3, \ldots, [-U_6]_3). \tag{1.51}$$

Its sign fixes an orientation of the catalog. The balanced identification $0 \mapsto 0$, $1 \mapsto +1$, $2 \mapsto -1$ is applied afterward, component by component.

Modulus nine organizes the APP evaluations and positions; residues modulo ten form each digit; base one thousand brings together three decimal positions; modulus three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be another operation.

For the two calculated conditions one obtains

$$\Psi(501, 614, 498, 169, 272, 272) = (0, 1, 0, 2, 1, 1),$$
$$\Psi(252, 109, 252, 903, 850, 850) = (0, 2, 0, 0, 2, 2).$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. That function explains its position in the genealogy: it links an already calculated emission to a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$\lvert\Psi^{-1}(w)\rvert$	1	2	3	4	5	6
number of words w	132	66	18	3	6	18.

(1.52)

The sums of both readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \qquad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibers.

Example 1.39 (Two emissions with the same ternary signature)**.** The emissions

$$(498, 501, 614, 252, 252, 109), \qquad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ. Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibers is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \qquad \Omega_{\mathbf{U}} = T_{\mathrm{em}}^{-1}(\mathbf{U}).$$

Therefore, the complete preimage of w in the initial domain is the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched register can preserve the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then refers to the recorded datum with its type; the isolated six-trit word maintains the precise scope of (1.51).

1.3.8 Dihedral action and the forty-three orbits

On six coordinates the permutations are defined

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \tag{1.53}$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \tag{1.54}$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the two groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They thus generate an action of the dihedral group D_3, with six elements.

Proposition 1.40 (Equivariance and stability of the catalog)**.** *The reduction Ψ commutes with the action of D_3. The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ is verified in each coordinate because g permutes positions and Ψ applies the same reduction to each position. Catalog stability constitutes another check: when generating its 468 elements through (1.50), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ belong to that set again. The algorithm in section 1.3.9 performs these membership checks. Equivariance then transmits stability to its ternary image. □

Proposition 1.41 (Complete orbital decomposition)**.** *The set $\mathcal{W}_6$ decomposes into* 43 *orbits:* 38 *of cardinality six and* 5 *of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$ one forms

$$O(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

The lexicographically minimal word in that set is chosen as the representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in table 1.17; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalog takes part in this operation and the catalog is stable, the list covers the entire image. □

Orbits inherit integer invariants from their fibers. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, while

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records how many times each ternary class appears. These data, together with the orientation and support of the paths, provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

Table 1.17: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

1.3.9 Self-contained enumeration and control algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values + and $\times$.

1. For each $\nu = 0, \dots, 323$, recover $(i, j; d)$ through the inverse of (1.44). Initialize an empty list of signatures.
2. For each window $m = 1, \dots, 6$, initialize the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (1.40).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. Upon completing $t = 27$ or $t = 54$, replace d by its opposite. Upon concluding the window, add $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices by equality of their six-component lists. These two groupings give f_+, $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Check signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities of (1.52).
8. Apply r, s to each emission and verify membership in the catalog. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in table 1.17.

Execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and check the factorization (1.50); its scope coincides with the same finite domain.

1.3.10 Preservation of the register and function of the seeds

Let $\mathsf{Sample}(q_t)$ be the sample preceding the transition and $\mathcal{M}_t$ the accumulated register. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \mathsf{Sample}(q_t), \tag{1.55}$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\mathsf{Sample}(q_0), \dots, \mathsf{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the cursor turns and the orientations remain associated with the same transitions. Execution determines that register through the recurrence; the archive preserves the specific provenance of the readings.

At the end of this stage, the complete domain of initial conditions, the decimal emission, its two recoverable signatures, the ternary image, its fibers and its orbital classification have been constructed. The censuses 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of obtainment. Regional selections will receive these structures with their invariants, and lifts will act on oriented words with their provenance preserved. That continuity makes the finite catalog a constructive antecedent of the coinductive development.

1.4 Nonadic connection, calendar extension and transport memory

TPK chronology combines an observable phase, a dodecaphasic register and a memory of the transitions performed. Their relation is determined by the composition of displacements. Crossing the phase boundary increments a quotient; preserving that quotient distinguishes successive returns. This structure provides the arithmetic basis of nonadic holonomy and its bilateral representation.

It is useful to fix from the outset the three objects involved. The phase belongs to a cycle of nine positions. The finite calendar retains twelve turns of that cycle. The enriched history additionally preserves the integer number of turns and the registers of position, orientation, sheet, carry and boundary. Each reduction preserves a specific observation of the same evolution; the available information is specified through its projections.

1.4.1 Incidence of arithmetic modes

1.4.1.1 Selection, persistence and cyclic succession

Let Σ and Π be the modes corresponding to the additive and multiplicative APP evaluations. In each block of three events, the phase selector traverses the sequence $(+1, -1, 0)$. Its action on mode m is

$$+1 : m \longmapsto \Sigma, \qquad -1 : m \longmapsto \Pi, \qquad 0 : m \longmapsto m.$$

The neutral event preserves the preceding mode. Starting from the cyclic closure of the block, the three recorded modes are therefore

$$(\Sigma, \Pi, \Pi).$$

The third position preserves the multiplicative-mode label selected in the preceding event. In the emitter, that neutral event increments only the counter Z. This distinction allows the incidence to be constructed directly from the calendar transitions.

Proposition 1.42 (Modal incidence of the tritic block)**.** *Counting the consecutive pairs of the block, including its closure, and ordering the modes as* (Σ, Π) *gives*

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \tag{1.56}$$

In a sequence of $3q$ events consisting of q repetitions of the block, the transition count is qF_{av}.

Proof. The ordered pairs are $\Sigma \to \Pi$, $\Pi \to \Pi$ and $\Pi \to \Sigma$. Each appears once; $\Sigma \to \Sigma$ appears zero times. These are, respectively, the upper-right, lower-right, lower-left and upper-left entries of the matrix. Each repetition adds exactly the same three pairs, including the link from the last mode of one block to the first of the next. The sum of the counts is qF_{av}. □

The lengths 9, 27 and 108 thus produce $3F_{\mathrm{av}}$, $9F_{\mathrm{av}}$ and $36F_{\mathrm{av}}$. These matrices count events of an already fixed periodic itinerary. The powers of F_{av} count concatenations admitted by its incidence. These are two distinct mathematical operations on the same matrix: summation of occurrences and composition of transitions.

1.4.1.2 Concatenation of incidences and induced recurrence

Let us define integers $a_0 = 0$, $a_1 = 1$ and $a_{n+1} = a_n + a_{n-1}$ for $n \geq 1$. Their appearance follows from multiplication of the matrix already constructed.

Proposition 1.43 (Powers of modal incidence)**.** *For every $n \geq 1$,*

$$F_{\mathrm{av}}^n = \begin{pmatrix} a_{n-1} & a_n \\ a_n & a_{n+1} \end{pmatrix}, \qquad a_{n-1}a_{n+1} - a_n^2 = (-1)^n.$$

Proof. For $n = 1$, the formula reproduces (1.56) because $a_2 = 1$. If the formula holds for n, multiplication by F_{av} gives

$$\begin{pmatrix} a_n & a_{n-1} + a_n \\ a_{n+1} & a_n + a_{n+1} \end{pmatrix} = \begin{pmatrix} a_n & a_{n+1} \\ a_{n+1} & a_{n+2} \end{pmatrix}.$$

Induction proves the first identity. Taking determinants and using $\det F_{\mathrm{av}} = -1$ gives the second. □

In particular, $F_{\mathrm{av}}^2 = F_{\mathrm{av}} + I$. This identity allows a hyperbolic generator to be normalized through

$$J_{-1} = \frac{2F_{\mathrm{av}}^2 - 3I}{\sqrt{5}} = \frac{2F_{\mathrm{av}} - I}{\sqrt{5}}, \qquad J_{-1}^2 = I. \tag{1.57}$$

Indeed, $(2F_{\mathrm{av}} - I)^2 = 4(F_{\mathrm{av}} + I) - 4F_{\mathrm{av}} + I = 5I$. The coefficient 5 comes here from the matrix identity. The quadratic representation and Euler extension use this generator after its modal construction.

The recurrence (a_n) subsequently admits identification with the Fibonacci sequence. The proof order in this subsection is explicit: tritic selector, sequence of modes, edge count, incidence and recurrence. Identification of the sequence preserves that order. Coinductive regional generation also has its own emitters, filters, lifts and survival conditions; the modal recurrence specifies one of its structural realizations.

1.4.2 Arithmetic extension of the calendar

1.4.2.1 Decomposition into phase and dodecaphasic position

We write $C_q = \mathbb{Z}/q\mathbb{Z}$ for the additive group of residues modulo q. The position of a calendar of $108 = 12 \cdot 9$ events has a unique decomposition

$$t = 9b + r, \qquad 0 \leq b < 12, \quad 0 \leq r < 9,$$

for the representative $0 \leq t < 108$. The coordinate r is the nonadic phase; b identifies the position of the turn in the dodecaphasic register. The scale 108 brings those two registers together according to the law of composition with carry.

Theorem 1.44 (Calendar extension with carry)**.** *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \qquad p : C_{108} \longrightarrow C_9, \quad [t] \longmapsto [t]_9$$

determine an exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0. \tag{1.58}$$

In the coordinates (b, r)*, addition is given by*

$$(b, r) \oplus (b', r') = \left(\left[b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor \right]_{12}, [r + r']_9 \right). \tag{1.59}$$

This extension has no homomorphic section.

Proof. The equality $[9b] = 0$ in C_{108} is equivalent to $12 \mid b$, so ι is injective. The map p is surjective and its kernel consists exactly of the classes that are multiples of nine, the image of ι. This demonstrates exactness.

To add two representatives, one writes

$$9b + r + 9b' + r' = 9 \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor \right) + [r + r']_9.$$

Reducing the first coefficient modulo twelve gives (1.59). Now suppose that a homomorphic section $s : C_9 \to C_{108}$ exists. The representative a of $s([1])$ satisfies $a \equiv 1 \pmod 9$ and $9a \equiv 0 \pmod{108}$. The second condition implies $12 \mid a$, whence $a \equiv 0 \pmod 3$; the first implies $a \equiv 1 \pmod 3$. The contradiction demonstrates the last assertion. □

1.4.2.2 Cocycle, inverse and restitution

The term

$$\kappa_9(r, s) = \left\lfloor \frac{r + s}{9} \right\rfloor, \qquad 0 \le r, s < 9,$$

records the threshold crossing. Its composition identity is

$$\kappa_9(r, s) + \kappa_9([r + s]_9, u) = \kappa_9(s, u) + \kappa_9(r, [s + u]_9). \tag{1.60}$$

Both sides are $\lfloor (r + s + u)/9 \rfloor$: it suffices to substitute $r + s = 9\kappa_9(r, s) + [r + s]_9$, or the analogous decomposition of $s + u$. The identity ensures that the accumulated quotient is independent of the parenthesization of three summands.

The inverse of (b, r) is determined by

$$-(b, r) = \begin{cases} ([-b]_{12}, 0), & r = 0, \\ ([-b - 1]_{12}, 9 - r), & 1 \le r < 9. \end{cases}$$

The additional term -1 represents the exact boundary correction when inverting an interior residue. For example, the representative $t = 17$ has coordinates $(1, 8)$. Its inverse is $(-2, 1)$ in coordinates with first component modulo twelve: $9(-2) + 1 = -17$. The inverted residue and the corrected quotient jointly restore the opposite integer.

1.4.3 Integer lift and nonadic holonomy

1.4.3.1 Finite phase and turn counter

Euclidean division extends to every $k \in \mathbb{Z}$:

$$k = 9m + r, \qquad m = \left\lfloor \frac{k}{9} \right\rfloor, \quad 0 \le r < 9.$$

The pair (m, r) determines k exactly. Its reduction $(m, r) \mapsto ([m]_{12}, r)$ recovers the finite calendar. In the integer lift, a turn of nine steps acts by

$$(r, m) \longmapsto (r, m + 1). \tag{1.61}$$

Twelve turns restore both finite coordinates and shift the integer counter by twelve units. Enriched memory includes that counter together with the ordered register of transitions.

1.4.3.2 Composition of admissible prolongations

At level k, we represent the enriched regional state by

$$\widehat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

B_k brings together the regional blocks; C_k is the compatible cylinder; p_k is the prefix; c_k records the carry; Σ_k preserves the boundary and W_k the incidence. The last coordinates are the phase and the turn counter. The prolongation operators act on this complete state, with the dependencies of each component.

Definition 1.45 (Nonadic holonomy of the prolongation system)**.** Let $\mathcal{R}_k \subseteq \widehat{X}_k \times \widehat{X}_{k+1}$ be the admissible relations produced by the TPK. The composition of one turn is

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \ge 1$, the composition of s turns is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

Relational composition retains the prolongations that simultaneously satisfy the constraints of the state. An admissible trajectory singles out a sequence of such transitions. The notation distinguishes the admissibility relation, the chosen trajectory and the finite observation of its phase.

Proposition 1.46 (Phase return and memory advance)**.** *On a trajectory whose turn rule is $g_{k+9} = g_k$, $m_{k+9} = m_k + 1$, every admissible composition of s turns satisfies*

$$g_{k+9s} = g_k, \qquad m_{k+9s} = m_k + s, \qquad [m_{k+9s}]_{12} = [m_k + s]_{12}.$$

Proof. The case $s = 1$ is the turn rule. Composing an additional turn with the trajectory already constructed preserves the phase and the counter receives another unit. Induction demonstrates the first two equalities; projection modulo twelve gives the third. □

The register allows each return to be recognized as a transformation of history. The observable phase expresses a periodicity, while the counter and boundary coordinates distinguish its successive realizations. The regional prolongation proofs additionally specify which cylinders survive and which prefixes are determined.

1.4.4 Bilateral representation of phase transport

1.4.4.1 Unit shift and group of turns

The lift of the index admits a representation in $\mathcal{H}_{\mathrm{ind}} = \ell^2(\mathbb{Z})$. Its orthonormal basis is written $e_k = e_{r,m}$ when $k = 9m + r$. We define $Te_k = e_{k+1}$, that is,

$$Te_{r,m} = \begin{cases} e_{r+1,m}, & r < 8, \\ e_{0,m+1}, & r = 8. \end{cases}$$

The transformation bijectively permutes an orthonormal basis and extends by continuity to a unitary operator. The inverse takes $e_{r,m}$ to $e_{r-1,m}$ when $r > 0$ and takes $e_{0,m}$ to $e_{8,m-1}$.

Theorem 1.47 (Faithful representation of the turn counter)**.** *Let $S = T^9$. Then*

$$S^s e_{r,m} = e_{r,m+s}, \qquad s \in \mathbb{Z}.$$

The map $s \mapsto S^s$ is a faithful unitary representation of $(\mathbb{Z}, +)$.

Proof. Nine steps preserve the residue and increase the quotient by one unit. The same identity applied to the inverse provides the formula for negative exponents. The powers satisfy $S^{s+t} = S^s S^t$. If $S^s = I$, the equality $e_{r,m+s} = e_{r,m}$ forces $s = 0$ by independence of the basis vectors. This proves faithfulness. □

The index space is a realization of the counter. The TPK state space also incorporates the arithmetic and incidence data. Unitary inversion of the index applies to its bilateral covering; inversion of a complete transition uses the archive of that transition and its admissibility conditions.

1.4.4.2 Memory operator and boundary correction

Let us define, on finitely supported vectors,

$$D_M e_k = \left\lfloor \frac{k}{9} \right\rfloor e_k, \qquad \mathcal{I} e_k = e_{-k}, \qquad P_0 e_k = \begin{cases} e_k, & 9 \mid k, \\ 0, & 9 \nmid k. \end{cases}$$

This domain is dense and stable under the operations considered.

Proposition 1.48 (Memory and inversion identities)**.** *On that domain the following hold*

$$[D_M, S] = S, \qquad \mathcal{I}T\mathcal{I} = T^{-1}, \qquad \mathcal{I} D_M \mathcal{I} = -D_M - (I - P_0). \tag{1.62}$$

Proof. If $k = 9m + r$, then $D_M S e_k = (m + 1)e_{k+9}$ and $S D_M e_k = m e_{k+9}$, which proves the commutator. The composition $\mathcal{I}T\mathcal{I}$ takes e_k to e_{k-1}. Finally,

$$\left\lfloor \frac{-k}{9} \right\rfloor = \begin{cases} -m, & r = 0, \\ -m - 1, & r > 0. \end{cases}$$

The diagonal action of the last member of (1.62) coincides with that expression on each basis vector. Linearity concludes the proof. □

The correction $I - P_0$ locates the boundary contribution in the interior phases. When the path is inverted, the quotient and residue are transformed jointly. That identity expresses arithmetically the information that disappears when only the phase is observed.

1.4.5 Refinement capacity and arithmetic vacancies

1.4.5.1 Exact comparison of ternary and decimal blocks

A block of six trits has $3^6 = 729$ configurations in its ambient space. A decimal block of three digits has $10^3 = 1000$ positions. Their cardinalities define an arithmetic encoding capacity. For $n \geq 0$, let

$$C_n = \max\{a \in \mathbb{N} : 1000^a \leq 729^{n+1}\}. \tag{1.63}$$

The definition compares integers and can be evaluated exactly. In a subsequent logarithmic representation,

$$\rho = \log_{1000} 729 = \log_{10} 9, \qquad \delta = 1 - \rho, \qquad C_n = \lfloor (n+1)\rho \rfloor.$$

The ratio ρ belongs to $(0, 1)$ and is irrational. If $\rho = p/q$ with positive integers, one would have $10^p = 9^q$; the prime factorization of the two sides prevents this.

Definition 1.49 (Register of capacity vacancies)**.** We define

$$Z_n = \lfloor (n+1)\delta \rfloor, \qquad z_n = \lfloor (n+1)\delta \rfloor - \lfloor n\delta \rfloor, \quad n \geq 0.$$

The value $z_n \in \{0, 1\}$ records capacity retention at the corresponding step. One has $z_0 = 0$ at the chosen origin.

Theorem 1.50 (Balance of capacity and vacancies)**.** *For $n \geq 0$ in the first identity and $n \geq 1$ in the second,*

$$C_n + Z_n = n, \qquad C_n - C_{n-1} = 1 - z_n. \tag{1.64}$$

In particular, for $0 \leq a < b$,

$$C_b - C_a + \sum_{n=a+1}^{b} z_n = b - a.$$

Proof. The irrationality of δ implies $\lceil (n+1)\delta \rceil = \lfloor (n+1)\delta \rfloor + 1$. Therefore,

$$\lfloor (n+1)(1-\delta) \rfloor = n + 1 - \lceil (n+1)\delta \rceil = n - Z_n.$$

The difference between two consecutive equalities gives the second identity for $n \geq 1$. The telescoping sum demonstrates the formula on an interval. □

The register Z_n can be obtained through $n - C_n$, using only the integer comparison (1.63). The logarithmic functions express the same construction in analytic coordinates. The first capacity retention occurs between $n = 20$ and $n = 21$:

$$C_{19} = 19, \quad C_{20} = 20, \quad C_{21} = 20, \quad C_{22} = 21.$$

The history receives an additional block while decimal capacity remains constant. The vacancy records precisely that offset.

1.4.5.2 Distinguishing the operational indices

The phase of the finite emitter, the regional prolongation phase and decimal capacity belong to declared indices. The nonadic connection preserves the phase after nine steps of its own index; vacancy preserves a capacity during one step of the block index. The comparison between the two is expressed by

$$[C_n]_q = \big[[n]_q - [Z_n]_q \big]_q, \qquad q = 9 \text{ or } 108.$$

The phase offset depends on the accumulated vacancy register. Preserving it allows one phase reading to be transformed into the other without improperly identifying the two calendars.

The resulting structure links modal selection, carry composition, holonomy and refinement. Memory fulfills precise functions at each level: it restores quotients, distinguishes turns, records transitions and controls encoding capacity. These operations constitute the temporal and archival support on which regional readers and realizations of unit conservation act.

1.5 APP: support, operations, and paths

The acronym APP derives from *All Possible Paths*. It designates a single path structure on a finite arithmetic support: the underlying graph has eighty-one vertices, whereas the set of finite paths, taken over all lengths, is countable. Its principal arithmetic matrix is the nonadic multiplication table. Addition is the internal law whose accumulation generates each row and each column of that table, and it also acts as a comparison observable on the same support. APP thus brings together addition, multiplication, and path composition before the ternary projection or the Archimedean evaluation is performed. The following definition exhaustively fixes the universe of paths under consideration.

1.5.1 Domain, generators and preformal relations of the APP support

The support of APP is a finite arithmetic torus: each of its two coordinates ranges over the nine residue classes and returns to its origin after nine unit translations. Its eighty-one points are therefore pairs of classes modulo nine. The associated Cayley graph preserves this cyclic structure: a vertex represents a position, an oriented edge represents a unit cardinal displacement, and a finite sequence of edges represents a path. Thus, *all possible paths* denotes the set of finite cardinal words with an initial point on the torus.

Each vertex simultaneously receives two arithmetic evaluations. The *additive sheet* publishes the sum of its two coordinates, and the *multiplicative sheet* publishes their product. The term *sheet* here designates an evaluation of the same point and the same path; APP preserves a single support and two functions upon it.

The operative difference between the two sheets is exact. In a row or column, addition acts by translation and repeatedly accumulates the same increment. Multiplication by a unit also permutes the classes; multiplication by a zero divisor folds several classes onto one and the same residue. To preserve the information that this folding conceals, APP lifts each evaluation to its residue–quotient coordinates:

$$i + j = R^{+}_{ij} + 9q^{+}_{ij}, \qquad ij = R^{\times}_{ij} + 9q^{\times}_{ij}.$$

The pairs (R^{+}, q^{+}) and $(R^{\times}, q^{\times})$ recover the integer values prior to reduction. When blocks are concatenated, additive carry κ_9 records the discrepancy between adding integer representatives and first adding their residues; together with both quotients, it makes recovery compositional. Thus, additive accumulation constructs visible trajectories, whereas the residue–quotient lift unfolds the fibers created by multiplicative folding.

For a path $\gamma = (x_0, \ldots, x_r)$, the genealogy visible is the sucesion of positions and pairs of evaluations

$$\mathcal{G}_{\mathrm{vis}}(\gamma) = \big(\gamma; (\Sigma(x_k), \Pi(x_k))_{k=0}^{r}\big).$$

At each step, its enriched genealogy adjoins the quotients, carry, and transported orientation. The enriched–visible projection preserves the path and residues and forgets precisely those memory coordinates. This distinction will be the input to the complete state of TRIT and of the Topological Phase Kernel. The complete domain of (104 976) double-cursor initial conditions and the (468) emission fibers they produce are published in full in the appendix D.1; there each seed is distinguished from the emission to which it belongs, and the multiplicity of the fiber is preserved.

1.5.2 Aritmetic support, translations and observables

Let

$$\mathrm{dr}_9(n) = 1 + ((n-1) \bmod 9) \in \{1, \dots, 9\}.$$

This map chooses the representative 9 for the null class. We define $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$, on which acts the group of translations $(\mathbb{Z}/9\mathbb{Z})^2$, and fix the symmetric set of generators

$$\mathscr{D} = \{(1,0), (0,1), (-1,0), (0,-1)\}.$$

The Cayley graph

$$\Gamma_{\mathrm{APP},9} = \mathrm{Cay}(X_9, \mathscr{D})$$

has 81 vertices and four oriented edges originating at each vertex. An oriented path of length r is a sequence $\gamma = (x_0, \dots, x_r)$ in X_9 such that $x_{k+1} - x_k \in \mathscr{D}$ for every $k < r$. We denote by $\mathrm{Path}(\Gamma_{\mathrm{APP},9})$ the set of all finite paths so defined.

Definition 1.51 (APP Structure)**.** The arithmetic structure of possible paths is the tuple

$$\mathscr{A}_9 = (X_9, \Gamma_{\mathrm{APP},9}, \Sigma, \Pi, \mathrm{Path}(\Gamma_{\mathrm{APP},9})),$$

where the two vertex observables are

$$\Sigma(i,j) = \mathrm{dr}_9(i+j), \qquad \Pi(i,j) = \mathrm{dr}_9(ij), \qquad 1 \le i, j \le 9.$$

Marked cardinal and path words

To make the orientation of the torus explicit, we use coordinates (row, column), with the row increasing southward and the column increasing eastward, and write

$$\delta(N) = (-1,0), \qquad \delta(E) = (0,1), \qquad \delta(S) = (1,0), \qquad \delta(O) = (0,-1).$$

Let $\mathscr{D}_{\mathrm{card}} = \{E, N, O, S\}$, where O denotes west. Given an initial point $x_0 \in X_9$ and a word $w = d_1 \cdots d_r \in \mathscr{D}_{\mathrm{card}}^r$, its trajectory is

$$\gamma(x_0; w)_k = x_0 + \sum_{j=1}^{k} \delta(d_j) \pmod 9, \qquad 0 \le k \le r.$$

The correspondence $(x_0, w) \mapsto \gamma(x_0; w)$ is bijective between $X_9 \times \mathscr{D}_{\mathrm{card}}^r$ and the paths oriented of length r. In particular, existen exactly $81 \cdot 4^r$ paths of that length when it allowed returns and backtracks.

The word w closes in the torus precisely when

$$\#S(w) - \#N(w) \equiv 0 \pmod 9, \qquad \#E(w) - \#O(w) \equiv 0 \pmod 9.$$

For a closed word, the integer displacement computed before reduction modulo nine defines the winding vector

$$\mathrm{wind}(w) = \frac{1}{9}\big(\#S(w) - \#N(w), \#E(w) - \#O(w)\big) \in \mathbb{Z}^2.$$

Reversal of orientation reverses the order of the word, exchanges $E \leftrightarrow O$ and $N \leftrightarrow S$, and transforms $\mathrm{wind}(w)$ into its opposite. This cardinal framework is subsequently transported in the phase kernel and forms part of the definition of each trajectory.

The expressions are written here in the chart of representatives $\{1, \dots, 9\}$. The formula $\widetilde{\Sigma}(i,j) = \mathrm{dr}_9(i+j-1)$ is a translated chart. It represents the same torus only when the translation is also

applied to indices, initial conditions, positional trajectories, and labels. All the following proofs employ $\Sigma(i, j) = \mathrm{dr}_9(i + j)$ and do not alternate both charts.

APP arithmetic matrix
$\Pi(i, j) = \mathrm{dr}_9(ij)$

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	**9**
2	2	4	6	8	1	3	5	7	**9**
3	3	6	9	3	6	9	3	6	**9**
4	4	8	3	7	2	6	1	5	**9**
5	5	1	6	2	7	3	8	4	**9**
6	6	3	9	6	3	9	6	3	**9**
7	7	5	3	1	8	6	4	2	**9**
8	8	7	6	5	4	3	2	1	**9**
9	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**

subconjunto coordenado de la clase nula

Ley interna de acumulación

$$\Pi(i, j) = \mathrm{dr}_9(\underbrace{i + \cdots + i}_{j \text{ sumandos}}) = \mathrm{dr}_9(ij).$$

Addition generates the multiplicative rows and, in the transverse direction, the columns. The observable $\Sigma(i, j) = \mathrm{dr}_9(i + j)$ preserves that elementary law in order to compare both readings at the same point.

$\mathcal{A}_9 = (X_9, \Gamma_{\mathrm{APP},9}, \Pi; \Sigma, \mathrm{Path}(\Gamma_{\mathrm{APP},9}))$
a single APP, one principal matrix, one internal additive law, and four oriented directions N, E, S, O.

Figure 1.1: The unique APP matrix and its internal law. The nonadic multiplication table occupies the main body; each of its rows is generated by additive accumulation. The violet shading indicates the nonunitary rows or columns; the blue line identifies the subset determined by the representative 9 of the null class; the gold frame highlights the central block 2×2.

Every point of X_9 simultaneously receives the product and sum values, but it does not thereby belong to two distinct APP matrices. The paths of $\mathrm{Path}(\Gamma_{\mathrm{APP},9})$ traverse a single support; Π publishes its main table and Σ conserves the internal operation that generates the accumulations and allows its incompatibility to be measured.

Proposition 1.52 (Structure of Rows and Columns)**.** *The matrix of Σ is a Latin square of order nine. The matrix of Π is not; nevertheless, for each $u \in (\mathbb{Z}/9\mathbb{Z})^\times = \{1, 2, 4, 5, 7, 8\}$, the row and column corresponding to u are permutations of the nine residue classes.*

Proof. With i fixed, the map $j \mapsto i + j$ is a translation of $\mathbb{Z}/9\mathbb{Z}$; the same holds when j is fixed. Hence each representative occurs exactly once in every row and column of Σ. For Π, multiplication by u permutes the classes if and only if u is a unit. The rows corresponding to $3, 6, 9$ contain repetitions, so the Latin property does not extend to the entire multiplicative matrix. □

1.5.3 Multiplicative structure of the local ring

Each row of the multiplication table is a repeated sum. Row i accumulates i along one direction; column j accumulates j along the perpendicular direction. Cell (i, j) is the intersection of both accumulations and returns $ij = ji$. The table thus represents product and commutativity through two transverse accumulations on the same lattice.

The second row is

$$2, 4, 6, 8, 1, 3, 5, 7, 9.$$

Before the modular return the even ones appear and after the odd ones. The third row is

$$3, 6, 9, 3, 6, 9, 3, 6, 9,$$

and makes the radical visible. Let

$$R := \mathbb{Z}/9\mathbb{Z}, \qquad \mathfrak{m} := (3) = \{\overline{0}, \overline{3}, \overline{6}\}.$$

Multiplication by three defines

$$M_3 : R \longrightarrow R, \qquad x \longmapsto 3x, \qquad \ker M_3 = \operatorname{im} M_3 = \mathfrak{m}.$$

By the first isomorphism theorem induces a linear identification

$$\overline{M}_3 : R/\mathfrak{m} \xrightarrow{\sim} \mathfrak{m}, \qquad \overline{x} \longmapsto 3x, \tag{1.65}$$

of vector spaces on $\mathbb{F}_3$. No is an isomorphism of rings: $\mathfrak{m}^2 = 0$, whereas the quotient is a field.

In positive representatives, the three phases of (1.65) are

$$369 \longmapsto 693 \longmapsto 936 \longmapsto 369, \tag{1.66}$$

produced by the phase shift $j \mapsto j+1$. The row corresponding to $6 \equiv -3 \pmod 9$ reverses the orientation of the same cycle. This radical orbit must not be conflated with the integer 369 693 100, which will subsequently appear as the digit count of 9^{9^9}.

In direct reduction modulo three,

$$3, 6, 9 \longmapsto 0, 0, 0.$$

If we first extract the factor three and retain its internal coordinate,

$$\eta_{\mathfrak{m}}(3a) = a \bmod 3, \qquad 3, 6, 9 \longmapsto 1, 2, 0.$$

The second map preserves the internal coordinate of the ideal generated by three. It must not be confused with the oriented ternary projection to be defined in the section devoted to the TRIT.

1.5.4 Coordinate subset of the null class

In the graph of the multiplicative observable,

$$\Pi(9, j) = \Pi(i, 9) = 9.$$

The last row and the last column form a union of seventeen positions, all with value nine. They do not constitute an intrinsic boundary of the arithmetic torus: they are the subset of the chart in which at least one coordinate represents the null class by means of 9. There are also four interior null representatives in

$$(3, 3), (3, 6), (6, 3), (6, 6).$$

The figure thus distinguishes this subset from the internal zeros of the nonunital ideal. The additional symbol used later in the cubic completion belongs to an enlarged space and is not identified with any of them.

1.5.5 Residue–quotient extension

Reduction modulo nine preserves the residue and eliminates the quotient of the Euclidean division. To study both components, we define, at each point,

$$i + j = 9q^{+}_{ij} + R^{+}_{ij}, \qquad ij = 9q^{\times}_{ij} + R^{\times}_{ij},$$

where $R^{+} = \Sigma$, $R^{\times} = \Pi$ and the q preserve the quotient. The residue–quotient extension of APP it is denoted by

$$\mathcal{A}^{\#}_{9} = (R^{+}, q^{+}; R^{\times}, q^{\times}).$$

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \to \{1, \dots, 9\}$ be the section of representatives fixed by dr_9. The additive carry associated with this section is

$$\kappa_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9} \in \mathbb{Z}.$$

Lemma 1.53 (Cocycle Identity of the Carry)**.** *For arbitrary $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$\kappa_9(a, b) + \kappa_9(a + b, c) = \kappa_9(b, c) + \kappa_9(a, b + c).$$

Proof. Both sides are equal to

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$ □

This cocycle records the discrepancy between summing integer representatives and first summing in the quotient. It is the compositional form of the quotient coordinate that will reappear in base 1000 concatenation.

The gross totals are

$$\sum_{i,j}(i + j) = 9\sum_i i + 9\sum_j j = 2 \cdot 9 \cdot 45 = 810,$$

and

$$\sum_{i,j} ij = \left(\sum_i i\right)\left(\sum_j j\right) = 45^2 = 2025.$$

The visible sums are obtained from the tables:

$$\sum R^+ = 405, \qquad \sum R^\times = 459.$$

Therefore, the total sums of the quotient coordinates are

$$\sum q^+ = \frac{810 - 405}{9} = 45, \qquad \sum q^\times = \frac{2025 - 459}{9} = 174.$$

Decomposition exacta of the difference integrada

$$\boxed{2025 - 810 = 1215 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129.}$$

The difference between the sums of residues is 54, whereas the difference between the sums of quotients is 129. The equality recomposes the total difference between the integer observables and proves that the quotient component contains arithmetic information not determined by the residual class.

1.6 Arithmetic Operators and Local Geometry of the Path Space

1.6.1 Localization of the excess in the nilpotent radical

The group of units of $\mathbb{Z}/9\mathbb{Z}$ is

$$(\mathbb{Z}/9\mathbb{Z})^\times = \{1, 2, 4, 5, 7, 8\},$$

and the maximal ideal is

$$\mathfrak{m} = (3) = \{0, 3, 6\} = \{9, 3, 6\}_{\text{visual}}, \qquad \mathfrak{m}^2 = 0.$$

Each unit row of Π is a permutation of $1, \dots, 9$ and sums to 45. Rows 3 and 6 sum to 54; row 9 sums to 81. Hence

$$459 = 6 \cdot 45 + 2 \cdot 54 + 81$$

and

$$459 - 405 = 2(54 - 45) + (81 - 45) = 54.$$

All the excess is located in the nonunitary sector.

The union of rows and columns $3, 6, 9$ contains 45 cells and sums to 297. Its central square 3×3 sums to $81 = 3 \cdot 27$; the outer arms, to $216 = 3 \cdot 72$. Thus,

$$297 = 216 + 81 = 3(72 + 27) = 3 \cdot 99.$$

The interpretation of 72 and 27 is obtained later from the uniquely selected central block.

1.6.2 Conditional expectation with respect to the projection $\mathbb{Z}/9\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$

The canonical reduction $\mathbb{Z}/9\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}$ determines the partition

$$C_1 = \{1, 4, 7\}, \qquad C_2 = \{2, 5, 8\}, \qquad C_0 = \{3, 6, 9\}.$$

The uniform conditional expectation averages each observable on the nine points of each block $C_a \times C_b$. The aggregate matrices are

$$M_\Pi = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \qquad M_\Sigma = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

Since each block contains nine positions,

$$9 \sum_{a,b} (M_\Pi)_{ab} = 459, \qquad 9 \sum_{a,b} (M_\Sigma)_{ab} = 405.$$

Thus, the averaging preserves the totals of both observables and relocates their difference:

$$\sum M_\Pi - \sum M_\Sigma = 51 - 45 = 6, \qquad 9 \cdot 6 = 54.$$

Theorem 1.54 (Spectrum of the aggregated multiplicative operator)**.** *The matrix operator M_Π satisfies*

$$\det M_\Pi = -9$$

and its real spectrum is

$$\operatorname{Spec}(M_\Pi) = \{-1,\ 9 - 6\sqrt{2},\ 9 + 6\sqrt{2}\}.$$

In particular, the associated quadratic form has signature $(2, 1)$.

Proof. The expansion of the determinant gives -9. The characteristic polynomial factors as

$$(\lambda + 1)\big((\lambda - 9)^2 - 72\big).$$

Its roots are those indicated and $9 - 6\sqrt{2} > 0$; therefore, there exist two positive directions and one negative direction. □

The spectral decomposition admits an integral form over an invariant sublattice. For

$$v_- = (1, -1, 0), \qquad v_+ = (1, 1, 0), \qquad v_0 = (0, 0, 1),$$

one has $M_\Pi v_- = -v_-$, whereas the restriction of M_Π to the submodule generated by v_+, v_0 is represented by

$$\begin{pmatrix} 9 & 6 \\ 12 & 9 \end{pmatrix} = 3H, \qquad H = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix} \in SL(2,\mathbb{Z}).$$

The eigenvalues of H are

$$3 \pm 2\sqrt{2} = (1 \pm \sqrt{2})^2.$$

The three vectors v_-, v_+, v_0 generate a sublattice of index two in $\mathbb{Z}^3$, since the determinant of the matrix having them as columns equals -2. Therefore, an integral direct sum of all of $\mathbb{Z}^3$ is not asserted here: on that invariant sublattice, the aggregate operator separates an eigendirection of eigenvalue -1 and a hyperbolic submodule governed by the quadratic unit $3 + 2\sqrt{2}$; over $\mathbb{R}^3$, the spectral decomposition is complete.

1.6.3 Commutator of the arithmetic projectors and separation of their images

The difference between Σ and Π is not reducible to their global sums. For $d \geq 2$, let $\mathscr{A}_d = \{1, \ldots, 9\}^d$, and define

$$\Sigma_d(x) = \mathrm{dr}_9(x_1 + \cdots + x_d), \qquad \Pi_d(x) = \mathrm{dr}_9(x_1 \cdots x_d).$$

In the Hilbert space $\ell^2(\mathscr{A}_d)$, endowed with the uniform measure, let $P_{\Sigma,d}$ and $P_{\Pi,d}$ denote the orthogonal conditional expectations onto the functions measurable with respect to Σ_d and Π_d, respectively. The normalized joint table is

$$C_{ab}^{(d)} = \frac{N_{ab}^{(d)}}{\sqrt{n_a m_b}}, \quad N_{ab}^{(d)} = \#\{x : \Sigma_d(x) = a,\ \Pi_d(x) = b\},$$

where $n_a = \sum_b N_{ab}^{(d)}$ and $m_b = \sum_a N_{ab}^{(d)}$.

On the original two-dimensional support, the exact joint correlation is

$$N^{(2)} = \begin{pmatrix} 0 & 0 & 2 & 0 & 0 & 2 & 3 & 0 & 2 \\ 3 & 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 & 3 \\ 0 & 0 & 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 & 3 \\ 3 & 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 2 & 0 & 0 & 2 & 3 & 0 & 2 \\ 0 & 2 & 0 & 0 & 2 & 0 & 0 & 2 & 3 \end{pmatrix}.$$

The rows, ordered by the value of Σ, sum to nine; the columns, ordered by the value of Π, sum to $(6, 6, 12, 6, 6, 12, 6, 6, 21)$. Complete enumeration thus determines the joint correlation of the two observables before any subsequent evaluation map is introduced.

Proposition 1.55 (Switches in two and three dimensions)**.** *The two pairs of projectors satisfy*

$$\|[P_{\Sigma,2}, P_{\Pi,2}]\| = \frac{\sqrt{2}}{3}, \qquad \|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}.$$

In particular, the additive and multiplicative observables do not induce simultaneously compatible decompositions.

Proof. The singular values s of $C^{(d)}$ are the cosines of the principal angles between the two subspaces. In the corresponding two-dimensional block, the norm of the commutator is $s\sqrt{1 - s^2}$. The exact enumeration of the 9^d words produces, for $d = 2$, the nontrivial eigenvalues

$$s^2 \in \left\{\frac{5}{7}, \frac{1}{3}, \frac{1}{3}\right\}.$$

The maximum of $s^2(1-s^2)$ is $2/9$, whence $\sqrt{2}/3$. For $d = 3$, the same construction gives the maximum $s^2(1-s^2) = 3/16$, attained at $s^2 = 1/4$, and therefore the norm is $\sqrt{3}/4$. In both cases, the commutator is nonzero. □

Proposition 1.56 (Intersection of the observable subspaces)**.** *The exact enumeration of the words of $\mathscr{A}_d$, for $2 \le d \le 15$, verifies*

$$\operatorname{Ran} P_{\Sigma,d} \cap \operatorname{Ran} P_{\Pi,d} = \mathbb{C}\mathbf{1}.$$

In that interval of dimensions, the constant functions constitute the only subspace common to the two arithmetic decompositions.

Proof. For two orthogonal projections, the dimension of the intersection of their ranges coincides with the multiplicity of the singular value 1 in the overlap matrix $C^{(d)}$. The integer computation of the tables $N^{(d)}$, followed by the exact computation of that multiplicity, gives the value one for each $d = 2, \dots, 15$. The constant vector belongs to both ranges; it therefore generates the entire intersection. □

The sequence of norms is not extrapolated from the first two cases. In particular, for $d = 4$ the same calculation gives

$$\|[P_{\Sigma,4}, P_{\Pi,4}]\| = \sqrt{\frac{2618939}{22240656}} \simeq 0.3431538652,$$

which does not coincide with the elementary prolongation $2/(d+1)$. This case constitutes a counterexample to any formula inferred solely from $d = 2, 3$.

1.6.4 Central block and spectral decomposition

The characterization of the central block is obtained by means of a congruence condition. For two consecutive representatives $k, k+1$, consider

$$B_k = \begin{pmatrix} k^2 & k(k+1) \\ k(k+1) & (k+1)^2 \end{pmatrix} \pmod 9.$$

Theorem 1.57 (Uniqueness of the Adjacent Diagonal Block)**.** *There is a unique block B_k whose diagonal entries coincide. It is determined by $k = 4$ and is*

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Proof. The equality of the diagonals is equivalent to

$$k^2 \equiv (k+1)^2 \pmod 9,$$

that is, $2k + 1 \equiv 0 \pmod 9$. Since 2 is a unit of $\mathbb{Z}/9\mathbb{Z}$, the congruence has the unique solution $k \equiv 4 \pmod 9$. □

Therefore, the positions $4, 5$ of the multiplicative observable form

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \qquad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad R^2 = I.$$

With $P_\pm = (I \pm R)/2$,

$$B_c = 9P_+ + 5P_-.$$

Theorem 1.58 (Typed hierarchy of the sum-product separation)**.** *The central block has a local spectral gap (9-5=4). Its off-diagonal coefficient is (2). Compression by the three residue classes modulo three produces the aggregate difference (51-45=6), and unfolding across the nine positions produces the global defect*

$$9 \cdot 6 = 459 - 405 = 54.$$

The integers (4), (2), (6), and (54) belong, respectively, to the spectral jump, the local coupling, the modular compression, and the two-dimensional integration.

Proof. The decomposition $B_c = 7I + 2R = 9P_+ + 5P_-$ gives the jump and the local coefficient. The aggregated matrices M_Π, M_Σ constructed above satisfy $\sum M_\Pi = 51$ and $\sum M_\Sigma = 45$. Each aggregated entry represents nine positions, so unfolding multiplies the difference by nine and recovers the totals (459) and (405). □

Their ordered concatenations produce the blocks 72 and 27:

$$72 + 27 = 99, \qquad 72 - 27 = 45 = \det B_c.$$

The concatenation of the diagonal and antidiagonal entries produces, respectively, 77 and 22. Therefore,

$$72 + 27 = 77 + 22 = 99, \qquad 77 - 22 = 55 = 54 + 1.$$

These equalities express two exact partitions of the same block; they do not by themselves identify an external constant.

Proposition 1.59 (Local–global relationship of the radical sector)**.** *The partition by rows of the central block reproduces, with scale factor three, the partition of the radical sector:*

$$3 \cdot 72 = 216, \qquad 3 \cdot 27 = 81, \qquad 3 \cdot 99 = 297.$$

Proof. The block $C_0 \times C_0$ contains nine null representatives and sums to 81. The remaining four arms of the rows and columns of C_0 sum to 216. The identities follow from $72 + 27 = 99$. □

Moreover,

$$M_c := \frac{B_c}{\sqrt{45}} = \exp\left(\frac{1}{2}\log\frac{9}{5}\,R\right) \in SO^+(1,1).$$

If $\eta = \mathrm{diag}(1,-1)$, the normalized matrix M_c satisfies $M_c^{\mathsf{T}}\eta M_c = \eta$, $\det M_c = 1$, and belongs to $SO^+(1,1)$. This finite realization is confined to the constructed block 2×2; its space-time interpretation requires the dimensional bridge developed in the treatise on Dimensional Mechanics.

Proposition 1.60 (Spectral family APP–TRIT and reading Markov–Lorentz local)**.** *For $\tau \in \{-1, 0, +1\}$, let*

$$B_\tau = (4 + 3\tau)I + (5 - 3\tau)R.$$

Then

$$\mathrm{Spec}(B_\tau) = \{9,\ 6\tau - 1\}, \qquad B_\tau = 9P_+ + (6\tau - 1)P_-,$$

and the differential mode recovers the orientation through

$$\tau = \frac{\lambda_- + 1}{6}.$$

In particular, $B_{+1} = B_c$. Its stochastic normalization

$$P_B := \frac{1}{9}B_c = P_+ + \frac{5}{9}P_-$$

satisfies, for all $n \geq 0$,

$$P_B^n = P_+ + \left(\frac{5}{9}\right)^n P_-.$$

Therefore, the contraction of the antisymmetric mode is $\kappa_B = 5/9$ *and the spectral gap of this local chain is* $4/9$. *If*

$$\eta_c := \frac{1}{2}\log\frac{9}{5},$$

the lorentzian normalization of the same block satisfies

$$\frac{B_c}{\sqrt{45}} = \exp(\eta_c R), \qquad \boxed{\frac{5}{9} = e^{-2\eta_c}}.$$

Proof. Since R acts with eigenvalues $+1$ and -1 on $\operatorname{im} P_+$ and $\operatorname{im} P_-$, respectively, the two eigenvalues of B_τ are

$$(4+3\tau)+(5-3\tau) = 9, \qquad (4+3\tau)-(5-3\tau) = 6\tau - 1.$$

The formula for the powers follows from $P_\pm^2 = P_\pm$ and $P_+P_- = 0$. Finally, $e^{-2\eta_c} = e^{-\log(9/5)} = 5/9$. □

This coincidence identifies two exact readings of the local block: the differential memory decreases by the same quotient that encodes the speed in the decomposed realization. It does not turn B_τ into the global Markov kernel of TPK, or $5/9$ into a physical constant. Nor does it by itself extend the realization $SO^+(1,1)$ to a Minkowski space $3+1$: those steps require additional dimensional maps.

1.6.5 Discrete curvature induced by the interaction of the sum and product leaves and its commutation defect

On the torus we write, now in residues,

$$S(x,y) = x + y, \qquad P(x,y) = xy \pmod 9.$$

For every function $T : X_9 \to \mathbb{Z}/9\mathbb{Z}$, we fix the forward differences

$$\Delta_x T(x,y) = T(x+1,y) - T(x,y), \qquad \Delta_y T(x,y) = T(x,y+1) - T(x,y),$$

where the coordinates are taken modulo nine. This convention also fixes the positive orientation of each plaquette: first the direction x is traversed, then the direction y, and one returns along the opposite edges. It is important to type these two objects correctly. S and P are vertex potentials; the link variables are their discrete differences

$$A_x^T = \Delta_x T, \qquad A_y^T = \Delta_y T.$$

For an ordered mixture of potentials (T_x, T_y), we define the plaquette curvature by

$$F(T_x,T_y) = \Delta_x A_y^{T_y} - \Delta_y A_x^{T_x} = \Delta_x\Delta_y T_y - \Delta_y\Delta_x T_x.$$

Since

$$\Delta_x S = \Delta_y S = 1, \qquad \Delta_x P = y, \qquad \Delta_y P = x,$$

is obtained in the 81 plaquettes:

$$F(S,S) = F(P,P) = 0, \qquad F(S,P) = +1, \qquad F(P,S) = -1 \pmod 9.$$

The connections associated with a single observable are flat; the mixed composition of the two observables produces curvature of opposite sign according to the order. For a coordinate rectangle $R_{a,b}$ contained in

the chart, we define the additive boundary Wilson observable as the oriented sum of the link variables. Cancellation of the interior edges gives

$$W_{\partial R_{a,b}}(S,P) = ab, \qquad W_{\partial R_{a,b}}(P,S) = -ab \pmod 9,$$

which is the finite Stokes identity for this convention.

The distinction between potential and linkage is algebraically necessary. If A_x, A_y were directly substituted by S, P, terms depending on x, y would appear and the preceding constant values would not be obtained. The typed formulation is what produces the mixed cocycle used later in the finite Heisenberg closure.

Remark 1.61. The integrated difference 54 and the mixed curvature ± 1 are distinct invariants. The former compares global sums; the latter depends on the order of the two potentials and is localized on each plaquette. The formulation in terms of self-adjoint operators and commutators will be developed after the corresponding state space has been constructed.

1.6.6 Path Transition Operator Expansion

The definition of APP includes the space $\mathrm{Path}(\Gamma_{\mathrm{APP},9})$. Once a transition operator U has been fixed on the vertices of $\Gamma_{\mathrm{APP},9}$, we assume that $U_{yx} = 0$ whenever $x \to y$ is not an edge of the graph. Under this support hypothesis, its powers admit the expansion

Proposition 1.62 (Finite Path Expansion)**.** *For each $R \geq 1$ and each pair of vertices i, f,*

$$(U^R)_{fi} = \sum_{\substack{\gamma: i \to f \\ |\gamma| = R}} \prod_{k=0}^{R-1} U_{x_{k+1}, x_k}, \qquad \gamma = (x_0 = i, x_1, \ldots, x_R = f).$$

Proof. For $R = 1$ the identity is the definition of the matrix elements of U. If it holds for R, then

$$(U^{R+1})_{fi} = \sum_j U_{fj} (U^R)_{ji}.$$

Substituting the induction hypothesis concatenates to every path of length R from i to j the final edge $j \to f$; thus, every path of length $R+1$ appears exactly once. □

Consequently, APP brings together in a single structure the finite arithmetic torus, the sum and product observables, the residue–quotient decomposition, the nilpotent radical, and an oriented mixed curvature. The representative 9 of the null class fixes a visible chart of the quotient; replacing it with 0 without simultaneously transporting the chart alters the registers used by the subsequent dynamics.

1.7 Aritmetic Matrix of APP and Local Ring Structure

The section 1.8 constructed a single support X_9. APP has a single principal arithmetic matrix: nonadic multiplication published by the positive digital root. Addition does not constitute a second APP table; it is the internal accumulation law that generates the rows and columns of the multiplicative matrix and provides an auxiliary observable for comparison. This distinction will be maintained throughout the work.

1.7.1 Nonadic multiplicative matrix and digital root

For $a, b \in \mathcal{D}_9 = \{1, \dots, 9\}$, the APP entry is obtained directly through

$$\boxed{\Pi(a, b) = \mathrm{dr}_9(ab) = 1 + ((ab - 1) \bmod 9).} \tag{1.67}$$

The only APP matrix is, therefore,

Π	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

Each row can also be reconstructed by reiterated addition of its index, but that recurrence describes the internal law of the matrix and not a second sheet. The rows and columns of indices $1, 2, 4, 5, 7, 8$ traverse the nine classes; those of indices $3, 6, 9$ are exactly those that contract the support.

In the residual chart, the set of units is

$$R_9^\times = (\mathbb{Z}/9\mathbb{Z})^\times = \{[1], [2], [4], [5], [7], [8]\}.$$

The maximal ideal is

$$\mathfrak{m} = (3) = \{[0], [3], [6]\}.$$

Theorem 1.63 (Classification of multiplicative rows)**.** *For $a \in \mathcal{D}_9$, the map*

$$m_a : \mathcal{D}_9 \longrightarrow \mathcal{D}_9, \qquad b \longmapsto \Pi(a, b)$$

is bijective if and only if $[a]_9 \in R_9^\times$. If $[a]_9 \in \mathfrak{m} \setminus \{0\}$, its image has three elements. If $[a]_9 = [0]_9$, its image has a single element.

Proof. The residual reduction of m_a is multiplication by $[a]_9$. In a finite ring, that multiplication is bijective exactly when $[a]_9$ is a unit. If $[a]_9 = [3]$ or $[6]$, its image is the ideal $\mathfrak{m}$, of cardinality three. If $[a]_9 = [0]$, every entry is sent to the null class, published as 9. □

Proposition 1.64 (Internality of the additive law)**.** *The auxiliary observable*

$$\Sigma(a, b) = \mathrm{dr}_9(a + b)$$

acts by translations: for each $a, c \in \mathcal{D}_9$ there exists a unique $b \in \mathcal{D}_9$ such that $\Sigma(a, b) = c$, and analogously in the other coordinate.

Proof. In the residual chart, $[a + b]_9 = [c]_9$ has the unique solution $[b]_9 = [c - a]_9$. The positive section transports that class to a unique element of $\mathcal{D}_9$. □

The difference of geometry of fibers remains thus typed without duplicar the matrix APP: the law additive internal only translates, whereas the matrix multiplicative permutes on the units and contracts on the radical.

Proposition 1.65 (Linear structure of the radical)**.** *The map*

$$M_3 : R_9 \longrightarrow R_9, \qquad x \longmapsto 3x$$

satisfies

$$\ker M_3 = \operatorname{im} M_3 = \mathfrak{m}.$$

The induced map

$$R_9/\mathfrak{m} \longrightarrow \mathfrak{m}, \qquad [x] \longmapsto 3x$$

is an isomorphism of vector spaces over $\mathbb{F}_3$, but not a ring isomorphism.

Proof. The congruence $3x \equiv 0 \pmod 9$ is equivalent to $x \equiv 0 \pmod 3$, whence the kernel $\mathfrak{m}$. The image is $\{[0], [3], [6]\} = \mathfrak{m}$. The first isomorphism theorem gives the linear isomorphism. It is not a ring isomorphism because $\mathfrak{m}^2 = 0$ in R_9, whereas the quotient $R_9/\mathfrak{m} \cong \mathbb{F}_3$ has a nonzero unit. □

The positive section publishes the non-trivial radical cycle as

$$369 \longmapsto 693 \longmapsto 936 \longmapsto 369,$$

according to the position from which the triple is read. This notation preserves the order of the coordinates; it does not turn the ideal into three independent integers.

1.7.2 Residue-quotient extensions and nonadic carry cocycle

Let $s_0 : R_9 \to \{0, \dots, 8\}$ be the normalized residual section. The binary carry of the sum is

$$\kappa_0(a, b) = \frac{s_0(a) + s_0(b) - s_0(a + b)}{9}.$$

Lemma 1.66 (Cocyclic Identity)**.** *For $a, b, c \in R_9$,*

$$\kappa_0(a, b) + \kappa_0(a + b, c) = \kappa_0(b, c) + \kappa_0(a, b + c).$$

Proof. Multiply by nine and expand either of the two sides produces

$$s_0(a) + s_0(b) + s_0(c) - s_0(a + b + c).$$

Therefore, both are equal. □

Define $\chi_0(a) = 1$ if $a = [0]_9$ and 0 otherwise. Since

$$s_+(a) = s_0(a) + 9\chi_0(a),$$

the carryovers of the two sections are related by

$$\kappa_+(a, b) = \kappa_0(a, b) + \chi_0(a) + \chi_0(b) - \chi_0(a + b).$$

Changing the representative modifies the cocycle by a coboundary. This is the exact form for transporting memory when the null class is published as 9 instead of 0.

1.7.3 Integrated censuses of residues and quotients

Summing over all pairs $(i, j) \in \mathcal{D}_9^2$, one obtains

$$\sum_{i,j}(i+j) = 810, \qquad \sum_{i,j} ij = 2025.$$

The positive residues satisfy

$$\sum_{i,j} R^{+}_{i,j} = 405, \qquad \sum_{i,j} R^{\times}_{i,j} = 459.$$

By the raised identities,

$$\sum_{i,j} Q^{+}_{i,j} = \frac{810-405}{9} = 45, \qquad \sum_{i,j} Q^{\times}_{i,j} = \frac{2025-459}{9} = 174.$$

Theorem 1.67 (Exact separation of the sum-product excess)**.** *The integrated difference between product and sum decomposes uniquely into a published part and a quotient part:*

$$\boxed{2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129.}$$

Proof. We subtract, term to term, the identities $ij = R^{\times}_{i,j} + 9Q^{\times}_{i,j}$ and $i + j = R^{+}_{i,j} + 9Q^{+}_{i,j}$, and sum on the eighty and a pairs. The censuses preceding dan the equality. The uniqueness proceeds of the division Euclidean in each pair. □

The value 54 is not the complete difference between two supposed matrices. It is the visible defect between the multiplicative matrix and the additive accumulator in the positive chart. The term $9 \cdot 129$ is the contribution that remains in the quotients. Omitting it would confuse publication with genealogy.

1.7.4 Bivariate central block of APP and decomposition into 2 spectral modes

For $k \in \mathbb{Z}/9\mathbb{Z}$, consideramos the block symmetric

$$B_k = \begin{pmatrix} k^2 & k(k+1) \\ k(k+1) & (k+1)^2 \end{pmatrix} \pmod 9.$$

The diagonals coincide exactly when

$$k^2 \equiv (k+1)^2 \pmod 9 \quad \Longleftrightarrow \quad 2k + 1 \equiv 0 \pmod 9.$$

Since $2^{-1} = 5 \pmod 9$, the unique solution is $k = 4$. Thus appears the block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Let

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_{\pm} = \frac{1}{2}(I \pm R).$$

Then

$$B_c = 7I + 2R = 9P_{+} + 5P_{-}.$$

Theorem 1.68 (Uniqueness and spectrum of the central block)**.** *The block B_c is the unique adjacent block in the family B_k with equal diagonals. Its eigenvalues on the symmetric and antisymmetric subspaces are, respectively,* 9 *and* 5*.*

Proof. The congruence before proves the uniqueness. Since that R acts by $+1$ on $\operatorname{im} P_+$ and by -1 on $\operatorname{im} P_-$, the matrix $7I + 2R$ acts by 9 and 5, respectively. □

The two ordered readings of its digits give

$$72 + 27 = 99, \qquad 72 - 27 = 45 = \det B_c,$$

and the two diagonals give

$$77 + 22 = 99, \qquad 77 - 22 = 55 = 54 + 1.$$

These identities are internal controls of the block; they do not by themselves define the global defect.

1.7.5 Discrete curvature of the interaction between additive and multiplicative leaves

On the residual chart we write

$$S(x, y) = x + y, \qquad P(x, y) = xy.$$

For a function $T : X_9 \to R_9$, let

$$\Delta_x T(x, y) = T(x + 1, y) - T(x, y), \qquad \Delta_y T(x, y) = T(x, y + 1) - T(x, y).$$

We define the ordered curvature of a pair of laws by

$$\mathcal{F}(T_x, T_y) := \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Proposition 1.69 (Orientations of the mixed composition)**.** *The following hold*

$$\mathcal{F}(S, S) = 0, \qquad \mathcal{F}(P, P) = 0,$$

and, for mixed compositions,

$$\mathcal{F}(S, P) = +1, \qquad \mathcal{F}(P, S) = -1 \quad \text{in } R_9.$$

Proof. The mixed second differences of the linear law S vanish. For $P(x, y) = xy$, both pure mixed differences equal one and cancel when compared in the same order. Combining a linear law and a bilinear one leaves a single unit whose sign changes when the order of the sheets is reversed. □

The curvature not creates another matrix APP. Is recorded that the order $\Sigma \to \Pi$ and the order $\Pi \to \Sigma$ are orientations opposite, whereas the sectors pure remain flat. This triad $-1, 0, +1$ will be typed in the unit TRIT.

1.7.6 Obstructions of the arithmetic matrix of APP: translations, local radical, central block, mixed curvature, and conservation of carry

Comparison Criterion 1.70 (Refutation criteria for the APP matrix)**.** The unit is refuted if:

1. the internal additive law ceases to act by translations;
2. a nonunitary multiplicative row is declared bijective;
3. $R_9/\mathfrak{m}$ is identified with $\mathfrak{m}$ as a ring;
4. the exceso 1215 it replaces by 54 without conservar quotients;
5. the central block is chosen without resolving $2k + 1 = 0 \pmod 9$;

6. reversal of the mixed order does not change the sign of the curvature;
7. the carry it eliminates to the cambiar between the sections s_0 and s_+.

APP is now constructed as support, category of trajectories, nonadic multiplicative matrix, internal additive law, local radical, carry cocycle, and mixed orientation. The following unit will not summarize these structures: it will take the triad of orientations that has just emerged and construct the complete TRIT state with its three quadratic regimes.

1.8 Gnomonic completion and arithmetic support of APP

This unit constructs the APP support from the nonadic cell of the section 1.9. It does not presuppose a continuous plane, an infinite lattice, or a numerical constant. The starting object is finite: two ordered copies of the positive chart of $\mathbb{Z}/9\mathbb{Z}$. Completion of its interior interstices produces exactly eighty-one states, and the residual translation law turns that square support into a discrete torus.

Throughout the unit, APP denotes a mathematical structure and not an image. Henceforth, the acronym APP names this structure; the properties that follow depend on its objects and morphisms, not on a verbal expansion of the acronym.

1.8.1 Conjugation between the positive and residual charts of the same class modulo 9

We define

$$\mathcal{D}_9 := \{1, 2, \dots, 9\}, \qquad \mathcal{D}_8 := \{1, 2, \dots, 8\}.$$

The residual publication application is

$$\lambda_9 : \mathcal{D}_9 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \qquad \lambda_9(a) = [a]_9.$$

Therefore, $\lambda_9(9) = [0]_9$. Its inverse as a set map is the positive section

$$s_+([a]_9) = \begin{cases} a, & [a]_9 \neq [0]_9, \quad 1 \leq a \leq 8, \\ 9, & [a]_9 = [0]_9. \end{cases}$$

The two expressions 9 and $[0]_9$ are not two phases: they are two representatives of the same element in different charts.

For two coordinates we will write

$$\lambda_9^{\times 2} : \mathcal{D}_9^2 \longrightarrow X_9 := (\mathbb{Z}/9\mathbb{Z})^2, \qquad (a, b) \longmapsto ([a]_9, [b]_9).$$

This map is a bijection of sets. The chart $\mathcal{D}_9^2$ makes the positive marks visible; the group X_9 makes the residual sum and the translations visible.

Lemma 1.71 (State-preserving chart change)**.** *The maps $\lambda_9^{\times 2}$ and $s_+^{\times 2}$ are inverses. In particular, the positive representation and the residual representation contain exactly the same eighty-one states.*

Proof. In each coordinate, $\lambda_9 \circ s_+ = \mathrm{id}_{\mathbb{Z}/9\mathbb{Z}}$ and $s_+ \circ \lambda_9 = \mathrm{id}_{\mathcal{D}_9}$ hold. The Cartesian product of these identities gives the two required identities. □

1.8.2 Decomposition of the APP orthogram into inner core and gnomonic complement

The inner core is

$$\mathcal{N}_8 := \mathcal{D}_8 \times \mathcal{D}_8, \qquad |\mathcal{N}_8| = 8^2 = 64.$$

In the plane of indices, $\mathcal{N}_8$ is the square that not uses the mark of closure in no coordinate. For publish also the null class it must completar the ninth row and the ninth column. We define the complement gnomonic disjoint by

$$\mathcal{G}_9 := (\{9\} \times \mathcal{D}_9) \sqcup (\mathcal{D}_8 \times \{9\}).$$

The corner $(9, 9)$ belongs to the first piece; for that reason the second uses $\mathcal{D}_8$ and not $\mathcal{D}_9$. Consequently,

$$|\mathcal{G}_9| = 9 + 8 = 17, \qquad \mathcal{D}_9^2 = \mathcal{N}_8 \sqcup \mathcal{G}_9, \qquad 81 = 64 + 17.$$

Proposition 1.72 (Minimality of the Completion)**.** *Let Y be a subset of $\mathcal{D}_9^2$ that contains $\mathcal{N}_8$ and whose projection onto each coordinate is all of $\mathcal{D}_9$. If, in addition, Y contains all pairs required to translate one coordinate while the other remains arbitrary, then $\mathcal{G}_9 \subseteq Y$. Therefore, the addition of seventeen states is the minimal completion compatible with the two translations.*

Proof. To transport the first coordinate through the closure class while the second remains at an arbitrary value $b \in \mathcal{D}_9$, all pairs $(9, b)$ are necessary. Nine states are obtained. To transport the second coordinate through closure while the first assumes an arbitrary value $a \in \mathcal{D}_8$, the remaining eight pairs $(a, 9)$ are necessary. The corner $(9, 9)$ was already counted in the first family. In total, $9 + 8 = 17$ states are necessary, exactly those of $\mathcal{G}_9$. □

The hypothesis of compatibility with the two translations is essential. It would suffice to add a single mark per coordinate for both projections to be surjective, but that set would not contain the boundary states needed to move one coordinate while holding the other fixed. APP requires the complete product, not two disconnected axes.

1.8.3 Exact realization of the interstitial interval via APP cells and boundary coordinates

Let P_9 be the ordered path with nine marks and eight interior edges

$$E(P_9) = \{e_1, \ldots, e_8\}.$$

Closing the endpoints incorporates an edge e_9 and produces the cycle C_9:

$$E(C_9) = E(P_9) \sqcup \{e_9\}.$$

The pairs of interior interstices form $E(P_9)^2$. The pairs of interstices of the complete cycle form $E(C_9)^2$. The difference is

$$E(C_9)^2 \setminus E(P_9)^2 = (\{e_9\} \times E(C_9)) \sqcup (E(P_9) \times \{e_9\}). \tag{1.68}$$

The decomposition is disjoint because the corner (e_9, e_9) is reserved for the first piece.

Theorem 1.73 (Equivalence of the mark model and the interstice model)**.** *The map*

$$\beta : E(C_9)^2 \longrightarrow \mathcal{D}_9^2, \qquad \beta(e_i, e_j) = (i, j),$$

is bijective. Its restrictions satisfy

$$\beta(E(P_9)^2) = \mathcal{N}_8, \qquad \beta(E(C_9)^2 \setminus E(P_9)^2) = \mathcal{G}_9.$$

Therefore, the identity $81 = 64 + 17$ *simultaneously counts pairs of published marks and pairs of closed interstices.*

Proof. Each edge of C_9 has a unique index in $\mathcal{D}_9$. The map $(i, j) \mapsto (e_i, e_j)$ is the explicit inverse of β. If $i, j \leq 8$, the pair belongs to $E(P_9)^2$ and its image belongs to $\mathcal{D}_8^2$. If at least one of the indices is nine, the pair belongs to the complement of equation (1.68); separating the case $i = 9$ from the case $i \leq 8$, $j = 9$ produces exactly the two pieces of $\mathcal{G}_9$. □

1.8.4 Toroidal closure and translations

Let

$$\mathbf{e}_1 = ([1]_9, [0]_9), \qquad \mathbf{e}_2 = ([0]_9, [1]_9).$$

The set of elementary directions is

$$\mathscr{D}_{\mathrm{APP}} := \{+\mathbf{e}_1, -\mathbf{e}_1, +\mathbf{e}_2, -\mathbf{e}_2\}.$$

We define the oriented Cayley graph.

$$\Gamma_{\mathrm{APP}} := \mathrm{Cay}(X_9, \mathscr{D}_{\mathrm{APP}}).$$

Its vertices are the eighty-one elements of X_9. For every $x \in X_9$ and $d \in \mathscr{D}_{\mathrm{APP}}$, there is an oriented edge $x \xrightarrow{d} x + d$. The inverse edge bears the label $-d$.

In the positive chart, unit translation is written by means of the published successor and its carry:

$$u_+(a) = \begin{cases} a + 1, & a < 9, \\ 1, & a = 9, \end{cases} \qquad \kappa_+(a) = \begin{cases} 0, & a < 9, \\ 1, & a = 9. \end{cases}$$

The raised identity is

$$a + 1 = u_+(a) + 9\kappa_+(a). \tag{1.69}$$

Upon applying λ_9, the carry term vanishes, leaving

$$\lambda_9(u_+(a)) = \lambda_9(a) + [1]_9.$$

Lemma 1.74 (Compatibility between closure, residue, and elevation)**.** *The addition of* e_9*, the identification* $9 \sim [0]_9$*, and the translation* $x \mapsto x + \mathbf{e}_i$ *describe three compatible operations in distinct domains. The residual quotient publishes the return; the carry of equation* (1.69) *preserves the elevation that the quotient omits.*

Proof. Topological closure incorporates the edge joining the endpoints of the path. The residual chart identifies the published endpoint with the null class. If $a < 9$, the successor has zero carry; if $a = 9$, the lifted equality is $10 = 1 + 9$. Both cases descend to the same residual sum, but the value of κ_+ distinguishes them. □

Corollary 1.75 (Toroidal support)**.** *Identifying opposite sides of the chart* $\mathcal{D}_9^2$ *yields a geometric realization of the group* X_9*. Each elementary translation is defined at every vertex and has an inverse.*

1.8.5 APP Path Category: Objects, Prolongation Morphisms and Compatible Composition

Definition 1.76 (Oriented trajectory)**.** A trajectory of length $n \geq 0$ is a pair

$$\gamma = (x_0; d_1, \ldots, d_n), \qquad x_0 \in X_9, \qquad d_k \in \mathscr{D}_{\mathrm{APP}}.$$

Its vertices are

$$x_k = x_0 + \sum_{r=1}^{k} d_r, \qquad 0 \leq k \leq n.$$

The source is $s(\gamma) = x_0$ and the target is $t(\gamma) = x_n$. For $n = 0$ the identity trajectory in x_0 is obtained.

If $t(\gamma) = s(\delta)$, its concatenation is

$$\delta \circ \gamma = (x_0; d_1, \ldots, d_n, e_1, \ldots, e_m),$$

where $e_1, \ldots, e_m$ are the directions of δ. Concatenation is associative, and paths of zero length are identities. We thus obtain the free category

$$\mathsf{Path}(\Gamma_{\mathrm{APP}}).$$

Proposition 1.77 (Finiteness of Support and Countability of Trajectories)**.** *The APP state set is finite, of cardinality* 81. *For a fixed source, there are* 4^n *direction words of length n. The union of all finite trajectories is countable.*

Proof. The first assertion follows from $|X_9| = 9^2$. At each of the n steps there are four independent choices, whence 4^n. Finally, $\bigcup_{n\geq 0} X_9 \times \mathscr{D}_{\mathrm{APP}}^n$ is a countable union of finite sets. □

This proposition separates two levels that will not be identified: APP has a finite support, but admits trajectories of arbitrary length. The prolongation of a route does not create new vertices of the support; it creates new genealogy over them.

1.8.6 Multiplicative matrix and internal additive law on a single support

For $n \in \mathbb{Z}_{>0}$, the root digital positive is

$$\mathrm{dr}_9(n) := 1 + ((n-1) \bmod 9).$$

On the positive chart we define the principal matrix and the internal law.

$$\Sigma(a,b) := \mathrm{dr}_9(a+b), \qquad \Pi(a,b) := \mathrm{dr}_9(ab).$$

Both maps have the same domain $\mathcal{D}_9^2$ and the same codomain $\mathcal{D}_9$, but they do not have the same status: Π publishes the unique APP matrix, and Σ generates its accumulations. Their residual versions are

$$\bar{\Sigma}([a]_9, [b]_9) = [a+b]_9, \qquad \bar{\Pi}([a]_9, [b]_9) = [ab]_9.$$

The change of chart is compatible with both operations:

$$\lambda_9 \circ \Sigma = \bar{\Sigma} \circ \lambda_9^{\times 2},$$
$$\lambda_9 \circ \Pi = \bar{\Pi} \circ \lambda_9^{\times 2}.$$

To preserve the integer quotient, lifts are introduced.

$$a + b = R_{a,b}^{+} + 9Q_{a,b}^{+}, \qquad 1 \leq R_{a,b}^{+} \leq 9, \tag{1.70}$$

$$ab = R^{\times}_{a,b} + 9Q^{\times}_{a,b}, \qquad 1 \le R^{\times}_{a,b} \le 9. \tag{1.71}$$

Here

$$R^{+}_{a,b} = \Sigma(a,b), \qquad R^{\times}_{a,b} = \Pi(a,b),$$

and the quotients are uniquely determined by

$$Q^{+}_{a,b} = \frac{a+b-R^{+}_{a,b}}{9}, \qquad Q^{\times}_{a,b} = \frac{ab-R^{\times}_{a,b}}{9}.$$

Definition 1.78 (APP genealogical)**.** The APP structure at this stage is the tuple

$$\mathrm{APP} := \big(X_9, \Gamma_{\mathrm{APP}}, \mathsf{Path}(\Gamma_{\mathrm{APP}}), \Sigma, \Pi, R^{+}, Q^{+}, R^{\times}, Q^{\times}\big).$$

The genealogy of a trajectory preserves, for each vertex visited, the position, incoming direction, multiplicative entry, additive accumulation, and their respective residues and quotients.

Theorem 1.79 (Causal construction of the APP support)**.** *The chain*

$$\mathcal{D}_9 \longrightarrow \mathcal{D}_8^2 \longrightarrow \mathcal{D}_8^2 \sqcup \mathcal{G}_9 \longrightarrow \mathcal{D}_9^2 \xrightarrow{\lambda_9^{\times 2}} X_9 \longrightarrow \Gamma_{\mathrm{APP}} \longrightarrow \mathsf{Path}(\Gamma_{\mathrm{APP}}) \longrightarrow (\Sigma, \Pi)$$

constructs, without loss of cardinality, the support, motion, paths, APP multiplicative matrix and its internal additive law.

Proof. The proposition 1.72 constructs the complete chart from the kernel. The theorem 1.73 identifies its interstitial model. The lemma 1.71 transports the eighty-one states to X_9. The Cayley graph incorporates the four reversible translations without changing the vertices. The free category incorporates all finite concatenations. Finally, Σ and Π are distinct total maps on the same domain; the equations equations (1.70) y (1.71) preserve the part that residual reduction does not publish. □

1.8.7 Decomposition of the APP orthogram into kernel 8×8, gnomon of 17 states, and chart 9×9 with opposite sides identified

The Figure 1.1 must be read in three layers: the inner square of sixty-four states, the gnomon of seventeen states, and the identification of opposite sides. The nine-by-nine chart thereby completes the eight-by-eight core by means of the closure gnomon. The multiplicative matrix and the internal additive law are evaluated at every vertex of the same state, preserve distinct residue–quotient lifts, and do not constitute two APP squares.

1.8.8 Obstructions to the fidelity of the APP support: cardinals $81/17$, reversible translations, and separation of the sum and product sheets

Comparison Criterion 1.80 (Refutation criteria for the APP support)**.** The construction is refuted if any of the following facts occurs:

1. the complete support has cardinality other than 81;
2. the gnomonic complement does not have cardinal 17;
3. a state of X_9 lacks any of the four translations;
4. it identify Σ and Π because coincide in a value;
5. Q^{+} or $Q^{\times}$ is eliminated upon returning through a boundary;
6. trajectories of different lengths are presented as new support vertices;

7. it interpreta 9 and $[0]_9$ as phases independent.

The unit has constructed APP up to its complete genealogical evaluation. It has not yet oriented the difference between the matrix and its accumulator, has not introduced the TRIT, and has not defined the Topological Phase Kernel. The following unit will study the algebraic structure of rows and columns, the sector of units, and the carry cocycle that prepares that orientation.

1.9 Construction of the nonadic cell from marked segments and interstices

The construction begins before every constant and before every choice of physical unit. The inaugural datum is not an already constituted real number, but an ordered segment, a family of marks, and the adjacency relation between consecutive marks. This separation is necessary because a mark, an interval, the length of that interval, and the numeral that publishes that length are objects of different types.

1.9.1 Decimal residual projection of the finite positional chart of APP

Definition 1.81 (Marked segment of order ten)**.** Let

$$\mathsf{M}_{10} := \{m_0, m_1, \dots, m_{10}\}$$

a set totally ordered by $m_0 < m_1 < \cdots < m_{10}$. Its adjacency graph is the path

$$\mathsf{P}_{10} : \quad m_0 \longrightarrow m_1 \longrightarrow \cdots \longrightarrow m_{10}.$$

We shall denote by

$$\mathsf{I}_k := (m_k, m_{k+1}), \qquad 0 \le k \le 9,$$

the abstract interstice between two consecutive marks. The family of interstices is $\mathsf{E}_{10} := \{\mathsf{I}_0, \dots, \mathsf{I}_9\}$.

The open notation (m_k, m_{k+1}) does not yet presuppose the real line. It indicates only the ordered pair of endpoints and the edge joining them. When the Archimedean chart is chosen

$$\chi_{10}(m_k) = k,$$

the abstract interstice I_k is published by the semiopen interval $[k, k+1)$. Therefore, the index k, the edge I_k, and the interval $[k, k+1)$ will not be identified in the text: the first is a label, the second is a combinatorial element, and the third is its realization in a chart.

Lemma 1.82 (Census of Marks and Gaps)**.** *The segment* P_{10} *contains eleven marks and ten interstices. More generally, a path with* $n+1$ *marks has exactly* n *edges.*

Proof. The map

$$\{0, \dots, n-1\} \longrightarrow E(\mathsf{P}_n), \qquad k \longmapsto (m_k, m_{k+1}),$$

is bijective. The particular assertion is obtained with $n = 10$. □

The endpoint m_0 fixes the origin of the chart, and m_{10} fixes its threshold. Neither by itself constitutes an additional interstice. This elementary remark prevents three subsequent confusions: counting endpoints as cells, confusing closure with a new phase state, and treating zero as a tenth phase independent of the nonadic cycle.

1.9.2 Construction of the nonadic cell of 9 phases via residual projection

The decimal chart distinguishes the anchor $[0, 1)$ of the periodic cell. The nine positive marks

$$\mathsf{D}_9 := \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

it utilizaran as representantes publicos of $\mathbb{Z}/9\mathbb{Z}$, haciendo corresponder the mark 9 with the class residual $\overline{0}$. We define

$$\lambda_9 : \mathsf{D}_9 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \qquad \lambda_9(a) = \overline{a}.$$

Proposition 1.83 (Positive chart of the nonadic classes)**.** *The map λ_9 is bijective. Its inverse is the positive section*

$$s_+(\overline{a}) = \begin{cases} a, & 1 \le a \le 8, \\ 9, & \overline{a} = \overline{0}. \end{cases}$$

In particular, the public representation 9 *and the residual representation* 0 *represent the same class, but belong to different charts.*

Proof. The nine classes of $\mathbb{Z}/9\mathbb{Z}$ have unique representatives in $\{0, 1, \ldots, 8\}$. Replacing the representative 0 by 9 produces another complete transversal, so λ_9 is bijective and s_+ is its inverse. □

The phase cell is not obtained by arbitrarily erasing the origin. The segment $[0, 1)$ fixes the decimal chart from which the cycle is observed; the nine phases are the nine classes published as $1, \ldots, 9$. In the lifted chart, the step that closes the cell is represented by $[9, 10)$; in the residual chart, that same step returns from $\overline{0}$ to $\overline{1}$. Closure and change of representative are not the same operation.

1.9.3 Position and height coordinates induced by the Euclidean division

Definition 1.84 (Nonadic Coordinates)**.** For each integer $n \ge 1$, we define

$$\rho_9(n) := 1 + ((n-1) \bmod 9), \qquad q_9(n) := \left\lfloor \frac{n-1}{9} \right\rfloor.$$

The first coordinate is called phase position; the second, turn height or memory quotient.

Theorem 1.85 (Unique nonadic decomposition)**.** *Every integer $n \ge 1$ admits a unique expression*

$$n = 9q + \rho, \qquad q \in \mathbb{Z}_{\ge 0}, \quad \rho \in \mathsf{D}_9.$$

In esa escritura $q = q_9(n)$ and $\rho = \rho_9(n)$.

Proof. We apply Euclidean division to $n - 1$: there exist unique $q \ge 0$ and $r \in \{0, \ldots, 8\}$ such that $n - 1 = 9q + r$. Taking $\rho = r + 1$ gives $n = 9q + \rho$, with $1 \le \rho \le 9$. The uniqueness of q, r implies that of q, ρ. □

Corollary 1.86 (Phase return with memory increment)**.** *For every $n \ge 1$,*

$$\rho_9(n + 9) = \rho_9(n), \qquad q_9(n + 9) = q_9(n) + 1.$$

Consequently, a full rotation preserves the visible phase and changes the raised state.

Proof. If $n = 9q + \rho$, then $n + 9 = 9(q + 1) + \rho$; uniqueness of theorem 1.85 gives the two identities. □

The one-step evolution is written unambiguously as

$$(q,\rho) \longmapsto \begin{cases} (q,\rho+1), & 1 \le \rho < 9, \\ (q+1,1), & \rho = 9. \end{cases}$$

The second line is not a restart. It recovers the initial phase label and increases the quotient that records how many turns have been performed.

1.9.4 Circular projection periodicity and injectivity of the nonadic helical lifting

Let

$$\vartheta_9(\rho) = \frac{2\pi(\rho-1)}{9}.$$

The phase projection onto the unit circle is

$$\mathcal{R}_9(n) = \big(\cos\vartheta_9(\rho_9(n)), \sin\vartheta_9(\rho_9(n))\big).$$

To simultaneously preserve position and height we introduce the helical immersion.

$$\mathcal{H}_9(n) = \left(\cos\vartheta_9(\rho_9(n)), \sin\vartheta_9(\rho_9(n)), q_9(n) + \frac{\rho_9(n)-1}{9}\right).$$

Proposition 1.87 (The circular projection does not determine the state)**.** *The map $\mathcal{R}_9$ is periodic with period nine. The map $\mathcal{H}_9$ is injective. In particular,*

$$\mathcal{R}_9(n+9) = \mathcal{R}_9(n), \qquad \mathcal{H}_9(n+9) \neq \mathcal{H}_9(n).$$

Proof. The first equality follows from corollary 1.86. The third coordinate of $\mathcal{H}_9(n+9)$ exceeds that of $\mathcal{H}_9(n)$ by one unit, hence both points are distinct. Finally, n is recovered from the third coordinate by multiplying by nine and adding one to the phase part; therefore, $\mathcal{H}_9$ is injective. □

This proposition is the minimal model of the principle that will run through the entire work:

phase return ≠ state return.

The nonadic connection of the Topological Phase Kernel will preserve more fields than the elementary helix —sheet, orientation, carry, boundary, terminal, and route memory—, but it will not alter this inaugural distinction.

1.9.5 Correspondence between Discrete Cardinality and Projective Measure

Definition 1.88 (Counting)**.** The count of a finite set X is its cardinality $|X|$. It requires neither an orientation, a scale, nor an additional reader.

Definition 1.89 (Measured data)**.** A discrete measurement datum is a triple

$$(x, \mathcal{L}, C),$$

where x is a state, $\mathcal{L}$ is a typed reader, and C is a publication chart compatible with the codomain of $\mathcal{L}$. Its visible output is

$$C(\mathcal{L}(x)).$$

The same set may have a single cardinality and admit multiple measures, because the local cochain, the orientation of integration, or the publication chart may vary. Conversely, two distinct states may publish the same coordinate. The work will therefore always preserve two levels:

$$\text{state with genealogy} \quad \longrightarrow \quad \text{published coordinate.}$$

Comparison Criterion 1.90 (Refutation criteria for the inaugural unit)**.** The construction is refuted if any of the following facts occurs:

1. ten independent phases are counted in the residue cell;
2. the mark 9 is identified with the integer 0 without declaring the change of chart;
3. the passage $9 \to 1$ forces $q \mapsto 0$;
4. two distinct integers have the same pair (q_9, ρ_9);
5. a physical unit is attributed to the cell before defining the dimensional transducer.

1.9.6 Autonomous arithmetic domain of the nonadic cell and causal precedence with respect to APP–TRIT–TPK and the constants analiticas

The unit constructed so far contains only:

$$\mathsf{P}_{10}, \quad \mathsf{D}_9, \quad \lambda_9, \quad (q_9, \rho_9), \quad \mathcal{R}_9, \quad \mathcal{H}_9.$$

It does not yet contain APP, TRIT, the Topological Phase Kernel, or an analytic constant. The following unit will take two copies of the nonadic cell, construct its gnomonic completion, and prove how the bidimensional support appears over which the multiplicative matrix of APP and its internal additive law act.

1.10 Cyclotomic skeleton of APP and binomial completion of the unit

The defect between the APP matrix and its internal accumulator is not an isolated number. Its law in Cartesian length, its decomposition into orbits of order three, and the completion of the nonadic cell produce three different objects that must remain typed:

$$\text{scalar defect} \quad D_d, \qquad \text{cyclotomic module} \quad W_{\mathrm{APP}}, \qquad \text{completion} \quad \overline{C}_9.$$

The first counts; the second preserves orientation and rank; the third adds boundary strata. Equalities between their cardinals do not authorize identifying their elements.

1.10.1 Cyclotomic Defect Invariance at Fixed Modulus Under Length Variation

For each integer $d \geq 1$, let

$$B_d := \{1, \ldots, 9\}^d.$$

The nonary digital root is represented by

$$\mathrm{dr}_9(n) = \begin{cases} n \bmod 9, & n \not\equiv 0 \pmod 9, \\ 9, & n \equiv 0 \pmod 9. \end{cases}$$

We define the additive accumulation and the multiplicative matrix evaluation on the same domain:

$$A_d(x_1, \ldots, x_d) = \mathrm{dr}_9\left(\sum_{j=1}^{d} x_j\right),$$

$$M_d(x_1, \ldots, x_d) = \mathrm{dr}_9\left(\prod_{j=1}^{d} x_j\right).$$

Their visible masses and their defect are

$$S_\Sigma(d) := \sum_{x \in B_d} A_d(x), \qquad S_\Pi(d) := \sum_{x \in B_d} M_d(x),$$

$$D_d := S_\Pi(d) - S_\Sigma(d).$$

Here d counts Cartesian positions. The module 9 does not vary, and no physical dimension has yet been introduced.

Lemma 1.91 (Mass of the additive accumulator)**.** *For every $d \geq 1$,*

$$S_\Sigma(d) = 45\, 9^{d-1}.$$

Proof. Once the first $d-1$ coordinates and a published residue $r \in \{1, \dots, 9\}$ have been fixed, there exists a unique last coordinate that makes the sum have residue r. Each visible value thus has 9^{d-1} preimages. Summing the weights $1 + \cdots + 9 = 45$ gives the formula. □

Lemma 1.92 (Census of the multiplicative matrix)**.** *For each of the six unit residues there are 6^{d-1} preimages for each fixed product value. The residues 3 and 6 have $d6^{d-1}$ preimages each. The zero residue, published as 9, has*

$$9^d - 6^d - 2d6^{d-1}$$

preimages. Consequently,

$$S_\Pi(d) = 9^{d+1} - 9(d+3)6^{d-1}.$$

Proof. The unit group of $\mathbb{Z}/9\mathbb{Z}$ has six elements. Upon fixing $d-1$ units, the last is determined by the prescribed unit product, giving 6^{d-1} solutions for each of the six classes.

To obtain a product of ternary valuation exactly one, one chooses which of the d coordinates belongs to one of the two classes 3, 6, while the others are units. Once the product 3 or 6 is fixed, the last unit is determined; hence $d6^{d-1}$ for each class. All remaining tuples have product zero modulo 9. Subtracting the two preceding counts from 9^d gives their multiplicity. The sum weighted by the visible representatives yields the final identity. □

Theorem 1.93 (Internal dimensional law of APP)**.** *For every $d \geq 1$,*

$$\boxed{D_d = 4\, 9^d - 9(d+3)6^{d-1}.}$$

In particular,

$$D_1 = 0, \qquad D_2 = 54,$$

and

$$\sum_{d \geq 1} D_d z^d = \frac{54z^2(1-3z)}{(1-9z)(1-6z)^2}.$$

Moreover,

$$\frac{D_d}{9^d} = 4 - (d+3)\left(\frac{2}{3}\right)^{d-1} \longrightarrow 4.$$

Proof. The formulas from the two lemmas are subtracted. For the generating function, one uses

$$\sum_{d \geq 1} 9^d z^d = \frac{9z}{1-9z}, \qquad \sum_{d \geq 1} (d+3)6^{d-1} z^d = \frac{z(4-18z)}{(1-6z)^2},$$

and the common numerator is reduced. The normalized formula follows by dividing by 9^d; the exponential term multiplied by $d+3$ converges to zero. □

The equality $D_2 = 54$ reproduce the census $459 - 405$, but the formula before and the law of modulo variable $\Delta_{3^m}^{\mathrm{pl}} = m3^{2m-1}$ are enunciados distintos: the first varia d manteniendo the modulo 9; the second varia the modulo.

1.10.2 Cyclotomic operator of APP and action on the 12 A_2-type fibers

Let

$$G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

Proposition 1.94 (Coxeter of type A_2)**.** *They are verified.*

$$C^3 = I_2, \qquad C^{\mathsf{T}} G_{A_2} C = G_{A_2}, \qquad \det(XI_2 - C) = X^2 + X + 1.$$

Therefore C preserves the form A_2, has order exactly three, and has no nonzero fixed vector.

Proof. The multiplication gives

$$C^2 = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C^3 = I_2.$$

Direct substitution into $C^{\mathsf{T}} G_{A_2} C$ yields G_{A_2}. Finally, 1 is not a root of X^2+X+1, hence $\ker(C-I_2) = 0$. □

1.10.3 Decomposition of APP into 12 fibers of type A_2

Consider the action

$$a : \mathbb{Z}/9\mathbb{Z} \longrightarrow \mathbb{Z}/9\mathbb{Z}, \qquad a(r) = 4r.$$

Since $4^3 \equiv 1 \pmod 9$, its restriction to the units has two free orbits:

$$\mathcal{O}_1 = (1, 4, 7), \qquad \mathcal{O}_2 = (2, 8, 5).$$

For an orbit $\mathcal{O} = (r, 4r, 7r)$, we define the augmentation module

$$A(\mathcal{O}) := \ker\left(\mathbb{Z}[\mathcal{O}] \xrightarrow{\Sigma} \mathbb{Z}\right).$$

At the base

$$b_1 = e_r - e_{4r}, \qquad b_2 = e_{4r} - e_{7r},$$

the restricted Euclidean form has Gram matrix G_{A_2}, and the cycle $r \mapsto 4r \mapsto 7r \mapsto r$ acts by C.

For $x = (x_1, x_2, x_3) \in (\mathbb{Z}/9\mathbb{Z})^3$, the three unordered pairs are

$$\{1, 2\}, \quad \{2, 3\}, \quad \{3, 1\}.$$

Over each pair, two typed laws are evaluated:

$$\ell_{ij}^{\Sigma}(x) = x_i + x_j, \qquad \ell_{ij}^{\Pi}(x) = x_i x_j.$$

Since

$$\ell_{ij}^{\Sigma}(ax) = a\ell_{ij}^{\Sigma}(x), \qquad \ell_{ij}^{\Pi}(ax) = a^{-1}\ell_{ij}^{\Pi}(x),$$

the sum carries the orientation C and the product the inverse orientation C^{-1}.

Definition 1.95 (Cyclotomic module of APP)**.** It is defined

$$\boxed{W_{\mathrm{APP}} := \bigoplus_{\{i,j\}} \bigoplus_{\mu\in\{\Sigma,\Pi\}} \bigoplus_{O\in\{O_1,O_2\}} A(O).}$$

There are $3 \cdot 2 \cdot 2 = 12$ summands, each of rank two; therefore

$$W_{\mathrm{APP}} \cong A_2^{12}, \qquad \operatorname{rank} W_{\mathrm{APP}} = 24.$$

The par–law–orbit labels are part of the object.

For $u \in \mathbb{Z}/9\mathbb{Z}$, we write

$$\beta_O(u) = \begin{cases} (1,0), & u = r, \\ (0,1), & u = 4r, \\ (-1,-1), & u = 7r, \\ (0,0), & u \notin O, \end{cases}$$

in the basis (b_1, b_2). The state map is

$$\Psi(x)_{ij,\mu,O} := \beta_O(\ell_{ij}^{\mu}(x)).$$

Theorem 1.96 (Equivariance and rank of the state skeleton)**.** *Let $\rho(a)$ be the action using C on the six additive fibers and C^{-1} on the six multiplicative fibers. Then*

$$\boxed{\Psi(ax) = \rho(a)\Psi(x)}$$

for the 729 *states of* $(\mathbb{Z}/9\mathbb{Z})^3$*. The rational submodule generated by the images has rank* 24*.*

Proof. Under the additive law, multiplying all coordinates by 4 advances one position in each orbit; under the multiplicative law, it multiplies the value by $4^2 \equiv 7 \equiv 4^{-1} \pmod 9$ and therefore reverses the cycle. This proves componentwise equivariance.

In order for the rank not to remain hidden in a computational statement, we fix the coordinate order.

$$(12,\Sigma,O_1), (12,\Sigma,O_2), (12,\Pi,O_1), (12,\Pi,O_2),$$

followed by the same order for the pairs 23 and 31, and, within each fiber, the two coordinates (b_1, b_2). Consider the states

$$001, 002, 004, 005, 010, 011, 012, 013, 014, 015, 016, 020,$$
$$022, 023, 040, 050, 101, 102, 104, 105, 110, 120, 140, 150.$$

The integer matrix M_Ψ whose rows are its twenty-four images has entries in (-1,0,1). Bareiss's fraction-free elimination yields the following successive leading principal minors

$$1, 1, 1, -1, -1, -1, -1, 1, -1, 1, 1, 1, 1, -2, -1, 1, 1, 1, 2, 1, 4, -6, 12, 9.$$

In particular,

$$\det M_\Psi = 9 \neq 0.$$

These twenty-four images are linearly independent. Since the codomain already has rank 24, they generate $W_{\mathrm{APP}} \otimes \mathbb{Q}$. □

The proof exhibits a finite reproducible certificate; it does not deduce full rank from the mere surjectivity of each component projection.

1.10.4 Derivation of the sum-product defect 459−405=54 in the central block of APP

Let

$$\mathcal{M} := \mathbb{Z}[\mathbb{Z}/9\mathbb{Z}]$$

and, for $\mu \in \{\Sigma, \Pi\}$,

$$\nu_\mu := \sum_{x,y\in\mathbb{Z}/9\mathbb{Z}} e_{\ell^\mu(x,y)}.$$

The census of the two tables gives

$$\delta_{\Pi\Sigma} := \nu_\Pi - \nu_\Sigma = 12e_0 + 3e_3 + 3e_6 - 3\sum_{u\in(\mathbb{Z}/9\mathbb{Z})^\times} e_u.$$

Its coefficients are constant on the orbits of a; therefore

$$(I - a)\delta_{\Pi\Sigma} = 0.$$

We define the increase projection

$$T : \mathcal{M} \longrightarrow A(\mathcal{O}_1) \oplus A(\mathcal{O}_2),$$

$$T(\nu) = \left(((I - a)\nu)|_{\mathcal{O}_1}, ((I - a)\nu)|_{\mathcal{O}_2}\right).$$

Then

$$T(\delta_{\Pi\Sigma}) = 0.$$

However, the visible reader

$$w(e_0) = 9, \qquad w(e_r) = r \quad (1 \le r \le 8)$$

satisfies

$$w(\delta_{\Pi\Sigma}) = 54.$$

Corollary 1.97 (Isotopic Obstruction)**.** *The scalar* 54 *belongs to the C_3-invariant sector and cannot be recovered by means of a linear functional that factors exclusively through the augmentation modules. In particular,*

$$\sum_x \Psi(x) = 0$$

and, for every linear functional L on $W_{\mathrm{APP}} \otimes \mathbb{Q}$,

$$L\left(\sum_x \Psi(x)\right) = 0 \neq 54.$$

This obstruction prevents presenting 54 as a retrospective linear sum of the twelve fibers. The subsequent APP–Fock coupling preserves the orientation and uses an additional object; it does not contradict the corollary.

1.10.5 Cubic completion of the nonad cell

Let E_9 be the underlying set of $\mathbb{Z}/9\mathbb{Z}$, and let $\star \notin E_9$ be a boundary state. We define

$$C_9 := E_9^3, \qquad \overline{C}_9 := (E_9 \sqcup \{\star\})^3.$$

The symbol $\star$ is not the zero class: it indicates that a coordinate has reached the closure stratum.

Theorem 1.98 (Binomial stratification of the unit cube)**.** *The stratum with exactly r coordinates equal to $\star$ contains*

$$\binom{3}{r} 9^{3-r}$$

elements. Therefore,

$$\boxed{1000 = 729 + 243 + 27 + 1 = 729 + 271.}$$

Upon removing only the apex $(\star, \star, \star)$, there remains

$$999 = 729 + 243 + 27 = 729 + 270.$$

Proof. The r boundary coordinates are chosen in $\binom{3}{r}$ ways; each remaining coordinate admits nine values. The four strata are disjoint and exhaust the Cartesian product. The sum is the expansion of $(9+1)^3$. □

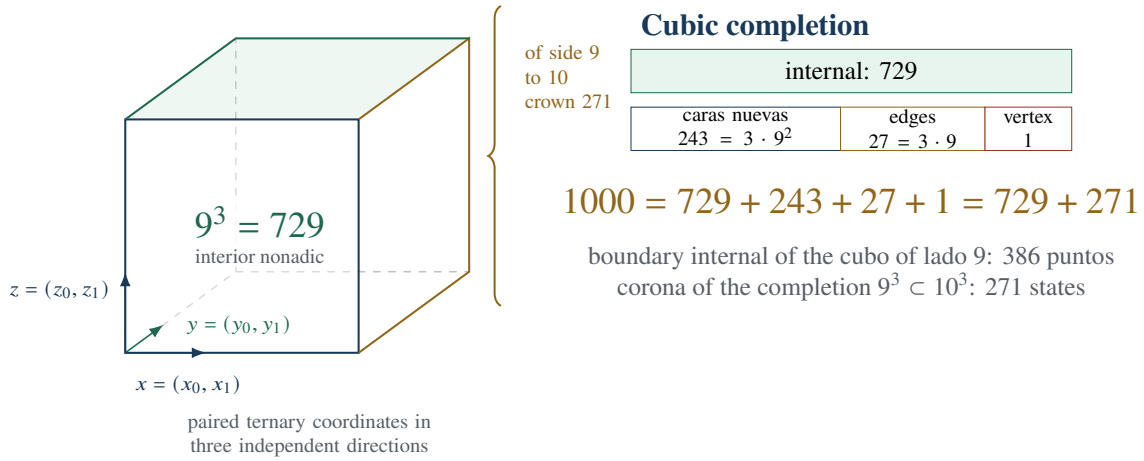


Figure 1.2: Cubic completion of the nonadic cell. The interior of cardinality 729 and the boundary strata of cardinalities 243, 27, and 1 are disjoint parts of a single object.

This stratification receives in HMT the name of extreme and mean holographic ratio. Mathematically it is the typed binomial completion of the nonadic cell; the name does not replace its definition.

For every $d \geq 1$, let

$$\overline{C}_9^{(d)} = (E_9 \sqcup \{\star\})^d.$$

Asignemos peso $1/10$ to each symbol of a coordinate and tomemos the measure product μ_d.

Theorem 1.99 (Projective compatibility of the measure)**.** *The projection that forgets the last coordinate satisfies*

$$(\pi_d)_* \mu_d = \mu_{d-1}.$$

Moreover,

$$10^d = \sum_{r=0}^{d} \binom{d}{r} 9^{d-r}, \qquad \mu_d(\overline{C}_9^{(d)}) = 1.$$

The number of states increases with d, while the normalized total mass is conserved.

Proof. For every cylinder $A \times (E_9 \sqcup \{\star\})$,

$$\mu_d(A \times (E_9 \sqcup \{\star\})) = \mu_{d-1}(A) \sum_{u \in E_9 \sqcup \{\star\}} \frac{1}{10} = \mu_{d-1}(A).$$

The cilindros generate the algebra finite of subconjuntos, what that determines the measure image. The identity cardinal again becomes the binomio of Newton. □

1.10.6 Morphism from the central block of APP to its Euclidean realization

The already constructed central block is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Row reading and diagonal reading give

$$72 + 27 = 77 + 22 = 99,$$

$$72 - 27 = 45 = \det B_c, \qquad 77 - 22 = 55 = 54 + 1.$$

The relation with the completion is not obtained by concatenating symbols, but by the Euclidean division.

$$729 = 10 \cdot 72 + 9, \qquad 271 = 10 \cdot 27 + 1.$$

Theorem 1.100 (Uniqueness of the Decimal Lift)**.** *For $L_d(n) = 10n + d$, with $d \in \{0, \dots, 9\}$, there exists a single pair (d, d') such that*

$$L_d(72) = 729, \qquad L_d(72) + L_{d'}(27) = 1000, \qquad d + d' = 10.$$

It is the pair

$$(d, d') = (9, 1).$$

Proof. The first equality is $720 + d = 729$, hence $d = 9$. The second becomes $729 + 270 + d' = 1000$, whence $d' = 1$. The third is verified, and the uniqueness of quotient and remainder rules out any other solution. □

Thus the causal diagram is obtained.

$$B_c \longrightarrow (72, 27) \longleftarrow ((72, 9), (27, 1)) \longleftarrow (729, 271).$$

The block and its reading belong to APP; the dividends and remainders belong to the completion. The diagram shares quotients; it does not invert causality.

1.10.7 Incidence of the cube and separation with respect to sextets and octets

Let M_d be the incidence matrix of the $2d$ codimension-one faces of a hypercube on its 2^d vertices. Each face contains 2^{d-1} vertices; two opposite faces are disjoint, and two transversal faces share 2^{d-2} vertices.

Theorem 1.101 (Hypercube incidence spectrum)**.**

$$\mathrm{Spec}(M_d M_d^{\mathsf{T}}) = \left\{ d 2^{d-1\ (1)}, 2^{d-1\ (d)}, 0^{(d-1)} \right\}.$$

Para $d = 3$,

$$\mathrm{Spec}(M_3 M_3^{\mathsf{T}}) = \{12, 4, 4, 4, 0, 0\}.$$

Proof. The row space decomposes into: the constant line; the d differences of each pair of opposite faces; and the (d-1) combinations of pair sums whose total sum is zero. The intersections described above cause the Gram matrix to act by $d2^{d-1}$, 2^{d-1}, and 0, respectively. □

The three-dimensional cube has six faces, eight vertices, and twelve edges. Those cardinals anticipate an incidence pattern (6/8), but a face is not a hexad and the set of vertices is not an octad. The exceptional identification requires a subsequent transporter that preserves supports and gauge.

1.10.8 Range 6 of the incidence operator and multiplicity of its resonant mode

Define the strata that preserve exactly k nonadic coordinates in rank d:

$$\mathcal{I}_k(d) := \binom{d}{k} 9^k.$$

The defects of APP are

$$D_{\text{vis}} = 54, \qquad D_{\text{tot}} = 1215, \qquad D_{\text{coc}} = 129, \qquad 1215 = 54 + 9 \cdot 129.$$

Theorem 1.102 (Uniqueness of the Binomial Resonance)**.** *The unique positive integer d that simultaneously satisfies*

$$\mathcal{I}_1(d) = 54, \qquad \mathcal{I}_2(d) = 1215$$

is $d = 6$. In that range,

$$\frac{\mathcal{I}_2(6) - \mathcal{I}_1(6)}{9} = 129.$$

Proof. The first equality is $9d = 54$, which forces $d = 6$. Then

$$\mathcal{I}_2(6) = 81\binom{6}{2} = 81 \cdot 15 = 1215,$$

and

$$\frac{1215 - 54}{9} = 129.$$

There remains no other possible rank because the first equality already determines d. □

1.10.9 Obstructions of the cyclotomic skeleton of APP: fibers A_2^{12}, completion $729 + 271$, rank resonance 6, and cubic incidence

Comparison Criterion 1.103 (Typed falsification of the unit)**.** The construction is refuted if any of the following facts occurs:

1. the law of D_d not reproduce $D_1 = 0$ and $D_2 = 54$;
2. variation of length is fused with variation of modulus;
3. C ceases to preserve G_{A_2} or acquires a fixed vector;
4. the labels pair–sheet–orbit not produce twelve fibers;
5. the smallest certificate of Ψ has zero determinant;
6. a functional that factors through the augmentation module recovers 54;
7. the symbol of boundary $\star$ it identifica with the zero residue;
8. the four strata do not sum to 1000;

9. a cubic face is identified with a hexad without an incidence transporter.

The unit has constructed, without recognizing them retrospectively, the module A_2^{12}, the completion $729 + 271$, the projective measure, and the rank-six resonance. These structures will feed, in distinct ways, the generation of constants and the exceptional branches.

1.11 TRIT as a ternary oriented unit: internal geometries, temporal order, and dynamic conditions of time crystal

The mixed curvature of APP produced three oriented values: zero on the pure sheets and opposite signs when the sum–product order is exchanged. The TRIT types that triad and preserves, in addition to the visible residue, the quotient required to reconstruct the lifted integer. This unit explicitly separates four levels: ternary class, balanced representative, carry, and quadratic regime.

Definition 1.104 (Topological Information Unit TRIT)**.** The *TRIT* is the elementary oriented unit whose state space is $\mathsf{TRIT} = \{-1, 0, +1\}$. Its native logarithmic capacity is $\log 3$, and its expression in binary units is $\log_2 3$. Unlike a bit, its three states do not reduce to a cardinal encoding: they respectively type opening, threshold, and return, and preserve the carry quotient required to reconstruct the transition that produced the visible residue.

1.11.1 Oriented ternary system and balanced representation of TRIT states

We define the oriented set

$$\mathsf{TRIT} := \{-1, 0, +1\}.$$

The balanced chart is the bijection of sets.

$$\mathrm{bal} : \mathbb{F}_3 \longrightarrow \mathsf{TRIT}, \qquad [0]_3 \mapsto 0, \qquad [1]_3 \mapsto +1, \qquad [2]_3 \mapsto -1.$$

It is not of an equality of types. $\mathbb{F}_3$ carries sum and product of field; TRIT carries order and orientation. Every operation transported between both must indicate the chart used.

The orientation of the nine phases is

$$\varepsilon_9(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

His complete table is

r	1	2	3	4	5	6	7	8	9
$\varepsilon_9(r)$	+1	−1	0	+1	−1	0	+1	−1	0

.

Periodicity of three does not eliminate the nonadic phase: the phases $1, 4, 7$, for example, have the same orientation and different positions in the circuit.

1.11.2 Balanced ternary division algorithm and uniqueness of the quotient–residue pair

Theorem 1.105 (Local ternary lifting)**.** *For each $n \in \{-4, -3, \ldots, 4\}$ there exists a single pair $(r, c) \in \mathsf{TRIT}^2$ such that*

$$n = r + 3c.$$

Proof. The existence is verified in the table.

n	−4	−3	−2	−1	0	1	2	3	4
r	−1	0	+1	−1	0	+1	−1	0	+1
c	−1	−1	−1	0	0	0	+1	+1	+1

.

If $r + 3c = r' + 3c'$, then $r - r' = 3(c' - c)$. Since $r - r' \in \{-2, -1, 0, 1, 2\}$, the only possible multiple of three is zero; therefore $r = r'$ and $c = c'$. □

The visible projection $(r, c) \mapsto r$ has three local states over the value zero:

$$0^- := (0, -1), \qquad 0^0 := (0, 0), \qquad 0^+ := (0, +1).$$

They publish the same residue, but reconstruct $-3, 0, +3$. This triple fiber is the minimal form of a distinction that will reappear in the enriched state of the Topological Phase Kernel.

Corollary 1.106 (Balanced Iteration)**.** *Every integer N admits a finite balanced ternary expansion*

$$N = r_0 + 3r_1 + \cdots + 3^m r_m, \qquad r_j \in \mathsf{TRIT}.$$

It is obtained by iterating the division $N_k = r_k + 3N_{k+1}$, where r_k is the balanced representative of N_k mod 3.

Proof. Euclidean division modulo three yields a residue in $\{0, 1, 2\}$; replacing 2 with -1 increases the quotient by one and produces $r_k \in \mathsf{TRIT}$. The absolute value of the quotient decreases outside a finite set, so the process terminates. Uniqueness follows by comparing the first differing digit and reducing modulo three. □

1.11.3 Carry cocycle of the oriented ternary composition

For $r, s \in \mathsf{TRIT}$, we define $r \oplus s \in \mathsf{TRIT}$ and $\kappa_3(r, s) \in \mathbb{Z}$ by the unique equality

$$r + s = (r \oplus s) + 3\kappa_3(r, s).$$

The table is

$\oplus$	-1	0	$+1$
-1	$+1$	-1	0
0	-1	0	$+1$
$+1$	0	$+1$	-1

,

κ_3	-1	0	$+1$
-1	-1	0	0
0	0	0	0
$+1$	0	0	$+1$

.

Proposition 1.107 (Ternary cocyclic identity)**.** *For $r, s, t \in \mathsf{TRIT}$,*

$$\kappa_3(r, s) + \kappa_3(r \oplus s, t) = \kappa_3(s, t) + \kappa_3(r, s \oplus t).$$

Proof. Both sides are the total quotient that appears when writing $r + s + t$ as a balanced representative plus a multiple of three. The uniqueness of the lift proves the equality. □

If (r, c) and (s, d) are states levantados, its sum is

$$(r, c) \boxplus (s, d) := \big(r \oplus s,\ c + d + \kappa_3(r, s)\big).$$

The proposition 1.107 proves that $\boxplus$ is associative and that the reconstruction $\mathrm{rec}_3(r, c) = r + 3c$ satisfies

$$\mathrm{rec}_3(x \boxplus y) = \mathrm{rec}_3(x) + \mathrm{rec}_3(y).$$

1.11.4 Orientation and Temporal Order Involution

For to tritic word $\tau = (\tau_1, \dots, \tau_n)$, define

$$C_{\mathrm{rev}}(\tau_1, \dots, \tau_n) := (-\tau_n, \dots, -\tau_1).$$

Lemma 1.108 (Oriented reversal)**.** *The map C_{rev} is an involution. It interchanges the sectors $+1$ and -1 and fixes the sector 0.*

Proof. Applying it twice reverses the order twice and changes every sign twice. The initial word is thereby recovered. The remaining assertions are immediate in each component. □

When a time-oriented parameter is fixed, the TPK convention is

$$+1 : \text{return with memory}, \qquad 0 : \text{present threshold}, \qquad -1 : \text{prolongation opening}.$$

This assignment does not add a fourth temporal state. The direction is preserved in the sign and the exact position is preserved in the nonadic phase.

1.11.5 Unified classification of the 3 quadratic algebras of the TRIT

For $\tau \in \mathsf{TRIT}$, define

$$\mathbb{A}_\tau := \mathbb{R}[J_\tau]/(J_\tau^2 + \tau I).$$

The three cases are:

τ	J_τ^2	regime	$\exp(tJ_\tau)$
$+1$	$-I$	elliptic	$\cos t\, I + \sin t\, J_{+1}$
0	0	parabolic	$I + tJ_0$
-1	$+I$	hyperbolic	$\cosh t\, I + \sinh t\, J_{-1}$

.

Theorem 1.109 (Uniform quadratic exponential)**.** *If $J_\tau^2 = -\tau I$, then*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau,$$

where

τ	$+1$	0	-1
$C_\tau(t)$	$\cos t$	1	$\cosh t$
$S_\tau(t)$	$\sin t$	t	$\sinh t$

.

Proof. We separate the exponential series into even and odd powers. The relations $J_{+1}^{2m} = (-1)^m I$, $J_0^2 = 0$, and $J_{-1}^{2m} = I$ respectively produce the cosine and sine series, the linear truncation, and the hyperbolic series. □

On $\mathbb{F}_3$ the analogues appear

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1), \qquad \mathbb{D}_3 = \mathbb{F}_3[\epsilon]/(\epsilon^2), \qquad \mathbb{S}_3 = \mathbb{F}_3[j]/(j^2 - 1).$$

The first is a field of nine elements; the second contains a nonzero nilpotent; the third decomposes through the idempotents $(1 \pm j)/2$ and is isomorphic to $\mathbb{F}_3 \times \mathbb{F}_3$. The coincidence of cardinality does not identify these three algebras.

The integral source of the elliptic regime is

$$\mathbb{Q}_{\mathbb{Z}} := \mathbb{Z}[u]/(u^2 + 1).$$

Reduction modulo three sends u to ι and produces $\mathbb{F}_9$. The extension of scalars to $\mathbb{R}$ and a $u \mapsto J_{+1}$ representation produce the real elliptic realizer. There is therefore no untyped map $\mathbb{F}_9 \to \mathbb{R}$: both realizations follow from a common integral source.

1.11.6 Matrix representations of the TRIT selected by the sign of J_τ^2

Let

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad C^3 = I,$$

and let us define

$$J_e := \frac{2C + I}{\sqrt{3}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}.$$

A direct multiplication gives

$$(2C + I)^2 = -3I, \qquad J_e^2 = -I.$$

The parabolic realizer is

$$J_p := N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0.$$

For the hyperbolic regime we take

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad A := F_{\mathrm{av}}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

and

$$J_h := \frac{2A - 3I}{\sqrt{5}} = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Then

$$(2A - 3I)^2 = 5I, \qquad J_h^2 = I.$$

The bilinear form

$$G = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$$

satisfies

$$A^T G A = G.$$

Moreover,

$$\cosh(2 \log \varphi) = \frac{3}{2}, \qquad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2},$$

for which

$$A = \exp(2 \log \varphi \, J_h) = \cosh(2 \log \varphi) I + \sinh(2 \log \varphi) J_h.$$

This identity will be recovered causally in the biography of φ; here only the matrix representation is verified.

The three exponential controls are

$$\exp(\pi J_e) + I = 0, \qquad \exp(J_p) = I + J_p, \qquad \exp(2 \log \varphi \, J_h) = F_{\mathrm{av}}^2.$$

The presence of π and φ in these verification identities does not authorize using them as constructors of those constants. Their constructors will appear later from the nonadic genealogy.

1.11.7 Local coupling APP–TRIT

With the exchange

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

we define, for $\tau \in \mathsf{TRIT}$,

$$B_\tau = (4 + 3\tau)I + (5 - 3\tau)R.$$

Since $P_\pm = (I \pm R)/2$,

$$B_\tau = 9P_+ + (6\tau - 1)P_-.$$

Theorem 1.110 (Spectral Recovery of Orientation)**.** *The spectrum of B_τ is*

$$\mathrm{Spec}(B_\tau) = \{9, 6\tau - 1\}.$$

The differential eigenvalue λ_- uniquely determines the TRIT:

$$\tau = \frac{\lambda_- + 1}{6}.$$

Moreover, the block central of APP is B_{+1}.

Proof. The projectors P_+ and P_- diagonalize R with eigenvalues $+1$ and -1. The spectral formula is obtained by substitution. The second identity solves for τ. For $\tau = +1$, $B_{+1} = 7I + 2R = \left(\begin{smallmatrix} 7 & 2 \\ 2 & 7 \end{smallmatrix}\right)$, which is the unique block of theorem 1.68. □

Therefore, the TRIT not it anade as retrospective label: can be recovered of the differential mode of a family constructed on the internal exchange of APP.

1.11.8 Obstructions of TRIT: residue–representative–carry separation, involutivity, and spectral recovery of τ

Comparison Criterion 1.111 (Refutation Criteria for the TRIT)**.** The unit is refuted if:

1. it identify class residual, representative balanceado and carry;
2. it is asserted that 0^-, 0^0, and 0^+ are the same state;
3. the composition $\boxplus$ ceases to reconstruct the integer sum;
4. C_{rev} is not involutive;
5. the three algebras are identified because they have the same cardinal;
6. some realizer fails to satisfy $J_\tau^2 = -\tau I$;
7. the value of τ is chosen after observing the outcome and is not recovered from the spectrum of B_τ.

APP provides the support, the multiplicative matrix, and the internal additive law; TRIT provides orientation, lift, and quadratic type. The next unit will construct the Topological Phase Kernel as the transport category of these data, without confusing the internal selector with the complete state.

1.12 TPK: base observable, enriched state, and typed separation of its projections

APP has supplied a support, a multiplicative matrix, and an internal additive law; TRIT has supplied a lifted orientation. The Topological Phase Kernel, hereinafter TPK, specifies how those data are transported jointly, which coordinates are preserved when a route is closed, and which readers can be published without identifying the publication with the state. Its mathematical object is a category over a base of paths, endowed with an explicit family of fibers and prolongation relations.

1.12.1 Bivariate category of APP paths as the basis of the TPK enriched state

Let $\mathsf{P}_{\mathrm{APP}}$ be the category libre constructed in the section 1.8. We define

$$\mathsf{P}_{\mathrm{APP}}^{(2)} := \mathsf{P}_{\mathrm{APP}} \times \mathsf{P}_{\mathrm{APP}}.$$

An object is a pair of positions $(p^{\Sigma}, p^{\Pi}) \in X_9^2$. A morphism is a pair of trajectories; one of them may be the identity. The first component is evaluated by Σ, the second by Π. They are not two lattices: they are two typed cursors over the same APP support.

The set of cardinal directions is

$$\mathscr{D}_{\mathrm{card}} := \{N, E, S, O\},$$

with shifts

$$\delta(N) = (-1, 0), \quad \delta(E) = (0, 1), \quad \delta(S) = (1, 0), \quad \delta(O) = (0, -1) \quad \text{in } X_9.$$

The orientation involution is

$$\overline{N} = S, \quad \overline{S} = N, \quad \overline{E} = O, \quad \overline{O} = E.$$

1.12.2 Non-split extension $\mathbf{C}_{12} \to \mathbf{C}_{108} \to \mathbf{C}_9$ and dodecaphase record of the nonadic phase

The complete clock of this stage is

$$\Theta_{108} := \mathbb{Z}/108\mathbb{Z}.$$

For $\theta \in \Theta_{108}$, let $\widetilde{\theta} \in \{0, \dots, 107\}$ be its normalized representative. We define three distinct coordinates:

$$\begin{aligned} g_0(\theta) &= [\widetilde{\theta} + 1]_9 \in C_9, \\ r(\theta) &= 1 + (\widetilde{\theta} \bmod 9) \in \mathcal{D}_9, \\ \beta(\theta) &= \left[\left\lfloor \frac{\widetilde{\theta}}{9} \right\rfloor\right]_{12} \in C_{12}. \end{aligned}$$

The first is the phase class, the second its positive label, and the third the dodecaphasic sector. The map β is a block map; it is not the canonical reduction $\theta \bmod 12$.

The signature and the active sheet are

$$\tau(\theta) := \varepsilon_9(r(\theta)) \in \mathsf{TRIT},$$

and

$$\mu(\theta) = \begin{cases} \Sigma, & \tau(\theta) = +1, \\ \Pi, & \tau(\theta) = -1, \\ \varnothing, & \tau(\theta) = 0. \end{cases}$$

The symbol $\varnothing$ denotes a threshold step in which neither cursor moves; it does not denote the absence of a state.

Definition 1.112 (Observable configuration)**.** The observable base is

$$\mathcal{Q}_{\mathrm{obs}} := X_9^2 \times \mathscr{D}_{\mathrm{card}}^2 \times \Theta_{108}.$$

An element is written

$$q = (p^{\Sigma}, p^{\Pi}, d^{\Sigma}, d^{\Pi}, \theta).$$

It contains two positions, two cardinal orientations, and a calendar index.

1.12.3 Sheet selector $M_{\mathrm{ph}} : \mathcal{D}_9 \to \mathsf{TRIT}$ and joint evolution of position, orientation, and calendar

The function

$$M_{\mathrm{ph}} : \mathcal{D}_9 \longrightarrow \mathsf{TRIT}, \qquad M_{\mathrm{ph}}(r) = \varepsilon_9(r)$$

is the internal sheet selector. It selects which evaluation acts; it does not by itself displace any position.

We define the orientation inversion indicator by

$$\eta_{27}(\theta) = \begin{cases} -1, & \widetilde{\theta} + 1 \in \{27, 54, 81, 108\}, \\ +1, & \text{otherwise.} \end{cases}$$

The new directions are

$$d^{\nu,+} = \begin{cases} \overline{d^{\nu}}, & \eta_{27}(\theta) = -1, \\ d^{\nu}, & \eta_{27}(\theta) = +1, \end{cases} \qquad \nu \in \{\Sigma, \Pi\}.$$

The positions evolve via

$$(p^{\Sigma,+}, p^{\Pi,+}) = \begin{cases} (p^{\Sigma} + \delta(d^{\Sigma}), p^{\Pi}), & \tau(\theta) = +1, \\ (p^{\Sigma}, p^{\Pi} + \delta(d^{\Pi})), & \tau(\theta) = -1, \\ (p^{\Sigma}, p^{\Pi}), & \tau(\theta) = 0. \end{cases}$$

Definition 1.113 (Observable evolution)**.** The map

$$F_{\mathrm{obs}} : \mathcal{Q}_{\mathrm{obs}} \longrightarrow \mathcal{Q}_{\mathrm{obs}}$$

is defined by

$$F_{\mathrm{obs}}(q) = (p^{\Sigma,+}, p^{\Pi,+}, d^{\Sigma,+}, d^{\Pi,+}, [\theta + 1]_{108}).$$

The category libre of the graph $q \to F_{\mathrm{obs}}(q)$ it is denoted by $\mathsf{P}_{\mathrm{obs}}$.

Each arrow of $\mathsf{P}_{\mathrm{obs}}$ induces a pair of trajectories in APP, so there exists a functor

$$B : \mathsf{P}_{\mathrm{obs}} \longrightarrow \mathsf{P}_{\mathrm{APP}}^{(2)}.$$

If $\tau = +1$, the second component of B is the identity; if $\tau = -1$, the first is the identity; if $\tau = 0$, both are the identity.

Proposition 1.114 (Separation between selection, motion, and remark)**.** *The selector M_{ph}, the translations $T_d(p) = p + \delta(d)$, and any temporal sampling have distinct domains and codomains. They are not interchangeable operators.*

Proof. M_{ph} takes a phase label and returns a tritic orientation. T_d is an automorphism of X_9. A sampling acts on a sequence or a category of arrows. Identifying two of them would violate typing. F_{obs} composes them in a declared order without erasing that distinction. □

1.12.4 Decomposition of the enriched state in 6 families fibradas

Over each $q \in Q_{\mathrm{obs}}$ we fix the total fiber

$$\mathcal{F}_q = \mathsf{O}_q \times \mathsf{H}_q \times \mathsf{C}_q \times \mathsf{S}_q \times \mathsf{B}_q \times \Lambda_q.$$

Each factor has its own type and function.

1.12.4.1 Joint determination of the orientation and sheet of the enriched state

$$\mathsf{O}_q \subseteq \mathscr{D}^2_{\mathrm{card}} \times \{+1, -1\}.$$

The two directions coincide with those stored in q; the last coordinate records the sheet orientation when the realization requires it. The sheet label does not replace the cardinal directions.

1.12.4.2 Euclidean decomposition of the state into compatible residue and quotient

The first lift fiber is

$$\mathsf{H}^{(1)} = \mathcal{D}_9 \times \mathbb{Z}, \qquad \mathrm{rec}_9(r, u) = r + 9u.$$

For each integer n there exist unique

$$r = \mathrm{dr}_9(n), \qquad u = \frac{n - r}{9}.$$

A greater depth consists of a declared tower of these fibers and their truncation maps. The term “Hensel” does not replace the definition of that tower.

1.12.4.3 Carry operator and joint update of residue, quotient, and memory

C_q is an abelian group. In the readers of decimal blocks one takes $\mathsf{C}_q = \mathbb{Z}$. The increment is not an independent coordinate, but a cochain over the arrows:

$$\kappa(r) \in \mathsf{C}_{q'} \quad \text{for } r : q \to q'.$$

1.12.4.4 Construction of the full signature from orientation, carry, and boundary

The signature is derived via a typed application.

$$\mathrm{Sig}_q : \mathsf{H}_q \times \mathsf{C}_q \times \Lambda_q \longrightarrow \mathsf{S}_q.$$

In the nonadic boundary realization, individual conservation is maintained,

$$q_\partial, a_\partial, c_\partial \in \mathbb{F}_3^6, \qquad \mathrm{colw} \in \mathbb{Z}^6_{\geq 0}, \qquad \rho_W, r_\partial \in \mathbb{F}_3^3.$$

These coordinates are margin, axial combination, cubic condition, column weights, Witt residual class, and ternary return. They do not condense into a single word if a subsequent proof uses any of them.

1.12.4.5 Cylindrical compatibility and boundary signature of the enriched state

$$\mathsf{B}_q := \mathsf{Cyl}_q \times \mathsf{Fr}_q.$$

The cylinder localizes a family of prefixes; the boundary determines which extensions are compatible. Survival will be a relation between states endowed with both data, not a number added a posteriori.

1.12.4.6 Genealogical transport record

Λ_q contains the transport records whose endpoint is q. A prolongation $r : q \to q'$ acts by concatenation:

$$\lambda \longmapsto r \circ \lambda.$$

The genealogical register preserves the route; its observable endpoint does not determine it.

1.12.5 Cardinal operator T_d of transport between enriched fibers with conservation of path and memory

For each arrow $r : q \to q'$ maps are declared

$$\mathsf{H}(r) : \mathsf{H}_q \to \mathsf{H}_{q'}, \quad \mathsf{C}(r) : \mathsf{C}_q \to \mathsf{C}_{q'}, \quad \Lambda(r) : \Lambda_q \to \Lambda_{q'}.$$

They respect identities and composition, and $\mathsf{C}(r)$ is a homomorphism. The carry satisfies

$$\kappa(r_2 \circ r_1) = \kappa(r_2) + \mathsf{C}(r_2)\big(\kappa(r_1)\big). \tag{1.72}$$

Therefore, the transport of the determined coordinates is

$$\begin{aligned} h' &= \mathsf{H}(r)(h), \\ c' &= \mathsf{C}(r)(c) + \kappa(r), \\ \lambda' &= \Lambda(r)(\lambda), \\ \sigma' &= \mathrm{Sig}_{q'}(h', c', \lambda'). \end{aligned}$$

Lemma 1.115 (Associativity of the transported carry)**.** *The equation equation (1.72) guarantees that transporting a state by r_1 and then by r_2 produces the same carry as transporting it by $r_2 \circ r_1$.*

Proof. The successive transport gives

$$\mathsf{C}(r_2)\big(\mathsf{C}(r_1)c + \kappa(r_1)\big) + \kappa(r_2).$$

By functoriality of C, the first term is $\mathsf{C}(r_2 r_1)c$; the sum of the other two is $\kappa(r_2 r_1)$ by equation (1.72). □

1.12.6 Typed prolongation relation Ext_r between enriched fibers

Orientation and boundary are not imposed by means of a universal function. For each arrow $r : q \to q'$, a relation is declared

$$\mathrm{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}.$$

The two conditions are required.

$$\Delta_q \subseteq \mathrm{Ext}_{\mathrm{id}_q}, \qquad \mathrm{Ext}_{r_2} \circ \mathrm{Ext}_{r_1} \subseteq \mathrm{Ext}_{r_2 \circ r_1}.$$

The first preserves identities; the second preserves composition. A relation may admit several local extensions and leave only one after a larger horizon.

Definition 1.116 (The Topological Phase Kernel)**.** An object of the TPK is a tuple

$$x = (q, o, h, c, \sigma, C, b, \lambda)$$

such that

$$\begin{gathered} q \in Q_{\mathrm{obs}}, \quad o \in \mathsf{O}_q, \quad h \in \mathsf{H}_q, \quad c \in \mathsf{C}_q, \quad \lambda \in \Lambda_q, \\ \sigma = \mathrm{Sig}_q(h, c, \lambda), \qquad (C, b) \in \mathsf{Cyl}_q \times \mathsf{Fr}_q. \end{gathered}$$

We associate to each object its fiber component.

$$\mathbf{f}(x) := (o, h, c, \sigma, (C, b), \lambda) \in \mathcal{F}_q.$$

A morphism $x \to x'$ is a triple (r, x, x') in which $r : q \to q'$ belongs to $\mathsf{P}_{\mathrm{obs}}$, the transportable coordinates satisfy the preceding laws, and $(\mathbf{f}(x), \mathbf{f}(x')) \in \mathrm{Ext}_r$. Composition is

$$(r_2, x', x'') \circ (r_1, x, x') = (r_2 r_1, x, x''),$$

and the identity of x is (id_q, x, x).

Theorem 1.117 (Categorical structure of the TPK)**.** *The definition 1.116 defines a category and the map*

$$p : \mathsf{TPK} \longrightarrow \mathsf{P}_{\mathrm{obs}}, \qquad p(x) = q, \qquad p(r, x, x') = r,$$

is a functor.

Proof. Route composition is associative. The lemma 1.115 proves compatibility of the carry coordinate; the declared functoriality of H, C, and Λ proves compatibility of the other determined coordinates. The stability of Ext guarantees that the composite remains admissible, and its diagonal condition provides the identities. The map p preserves identities and composition by construction. □

1.12.7 Observable quotients of the enriched state and conserved genealogical fields

The elementary projections are

$$\phi(x) = g_0(\theta), \qquad \tau(x) = \varepsilon_9(r(\theta)), \qquad \mu(x) = \mu(\theta).$$

The projection to the local TRIT state is

$$\pi_{\mathrm{TRIT}}(x) = \big(\tau, \mu, o, h, c, \sigma, \lambda, \phi\big).$$

It forgets the cylinder, the boundary, and part of q; it does not define another container. Two states may have the same image under π_{TRIT} and differ in the forgotten information.

1.12.8 Criterion of route survival via conservation of the projected boundary

Let $\mathcal{X}_{\mathrm{TPK}}^{(n)}$ be the family of states at depth n. For $N \geq n$, we define

$$\mathrm{Surv}_n^{(N)} = \left\{ x_n : \begin{array}{l} \text{there exist } x_{n+1}, \ldots, x_N \text{ with} \\ (x_j, x_{j+1}) \in \mathrm{Ext}_{r_j} \ \text{ for } n \leq j < N \end{array} \right\}.$$

Proposition 1.118 (Survival filtration)**.** *For $N \geq n$,*

$$\mathrm{Surv}_n^{(N+1)} \subseteq \mathrm{Surv}_n^{(N)}.$$

If the scheme is defined to all depth, the states indefinitely prolongable are

$$\mathrm{Surv}_n = \bigcap_{N \geq n} \mathrm{Surv}_n^{(N)}.$$

Proof. A prolongation to $N + 1$ truncates to a prolongation to N, which proves the inclusion. The intersection expresses exactly prolongability for every finite horizon. □

1.12.9 Insertion of the 6 fibered families of the enriched state into the operator hierarchy of TPK

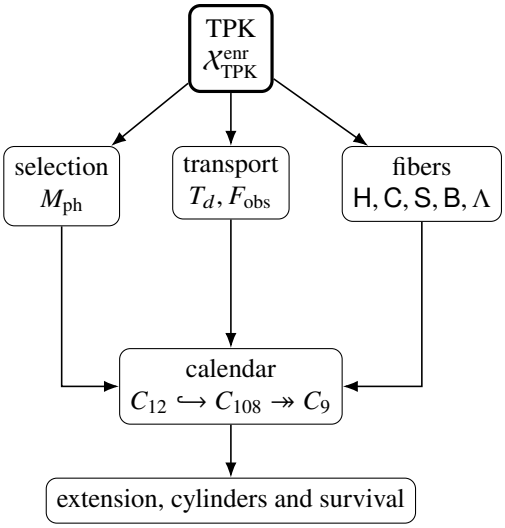


Figure 1.3: Operator containment of the Topological Phase Kernel. The selector, transport, readers, and fibers are not parallel containers: they are maps or coordinates of the same enriched state.

1.12.10 Obstructions of the enriched state of the TPK: typing of the selector, carry cochain, compatible extension, and genealogical conservation

Comparison Criterion 1.119 (Refutation Criteria for the Enriched State)**.** The definition is refuted if:

1. the selector M_{ph} it usa as translation;
2. θ, $g_0(\theta)$, $r(\theta)$, and $\beta(\theta)$ are identified;
3. a signature is allowed to be chosen independently of (h, c, λ);
4. the composition of carries incumple equation (1.72);
5. Ext_r is replaced by an unproved functional choice;
6. a projection that has forgotten the cylinder, boundary, or genealogical record is called a complete state;
7. the phase return erases any of the coordinates of fiber.

This unit has defined the container. The next will enumerate its readers, operators, odometer, and calendars, and will prove why the returns of nine, twenty-seven, and one hundred eight steps are not restarts of the state.

1.13 Typed hierarchy of the 34 heterogeneous canonical constructions and causal dynamics of TPK updating

The Topological Phase Kernel is neither a reduction modulo three nor a clock with one hundred and eight positions. It is a category of enriched states equipped with operators of selection, transport, reading, memory, periodicity, prolongation, and publication. To prevent an abbreviation from replacing the object, each operator is recorded by its type and by the information it preserves.

1.13.1 Typed schema of the 34 canonical TPK constructions: domain, codomain, action, invariants, and memory

Definition 1.120 (Operator specification)**.** A canonical realization of the TPK is a sextuple

$$O = (D_O, C_O, f_O, I_O, L_O, M_O), \tag{1.73}$$

where:

1. $f_O : D_O \to C_O$ is its rule;
2. I_O is the family of conserved relations;
3. L_O declares the information published or lost;
4. M_O enumerates the memory that remains in the enriched state.

Two realizations represent the same operator only when there exists a typed equivalence that intertwines rules and invariants. Sharing a period, cardinality, or visible output is not sufficient.

1.13.2 Typed L_0–L_7 hierarchy of the 34 heterogeneous canonical TPK constructions

The inventory is partitioned into:

C1. supports and states;

C2. internal selection;

C3. transporte;

C4. readers and quotients;

C5. memory and boundary;

C6. periodicity and return;

C7. prolongation;

C8. salidas.

The class is determined by domain, codomain, and function, not by the name of a number associated with it.

1.13.3 Functional grouping of the 34 canonical constructions in 14 families of composition APP–TRIT–TPK

Grouping	Operators or spaces	Function and invariants
State and route memory	$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}, \mathcal{F}_q$	Preserve position, sheet, orientation, residue, quotient, carry, signature, boundary, cylinder, prefix, provenance and memory.
Internal selection	$M_{\mathrm{ph}} : \{1,\dots,9\} \to \{-1,0,+1\}$	Activate the additive, multiplicative evaluation or the threshold step; does not move the state.
Transporte espacial	$T_N, T_E, T_S, T_O, F_{\mathrm{loc}}, F_{\mathrm{obs}}$	Perform cardinal translations, sheet feedback, and timeline of two cursors.
Lectores modulares	$\rho_9, q_9, \rho_3^{\mathrm{or}}, c_3, q_{1000}, r_{1000}$	Publish phase, orientation and decimal block preserving the lift quotients.
Block and memory	$\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}, \mathcal{N}_{27}$	The former composes twenty-seven transports; the latter filters memory events. An equal index does not imply an equal operator.
Twelve-phase state	$\mathcal{R}_{12}, C_{12}, \Theta_{108}, \Gamma_{108}$	Distinguish enriched record, sector, cyclic law, and ordered support.
Periodicity and depths	$C_{12} \hookrightarrow C_{108} \twoheadrightarrow C_9, w_{108}, w_{1080}$	Preserves the non-splittable extent, words of different lengths and the vertical memory.
Channels 90/120	$\mathcal{F}_6, \mathcal{F}_8, \mathcal{I}_{6\to 8}, S_{90}, S_{120}$	Separate finite incidences, analytic tails, and their normalizations.
Bidirectionality	$P_{\pm}, \mathrm{rev}, \mathrm{Ext}_{\pm}, N_{\pm}, \beta, C_M$	Perform exchange, route inversion, future and past prolongation, valence, balance, and conjugation.
Emission and coinduction	$L_0, L_1, R_{36}, G_9, \varprojlim \mathrm{Cyl}_m$	Lift the seed, open the boundary, and organize finite and compatible extensions.
Numeric Readers	$\mathscr{C}_\pi, \mathscr{P}_e, F_{\mathrm{av}}$	They construct rational forks, factorial propagation, and Fibonacci self-scaling.
Exceptional Projection	Γ_{A_W}, weight, projective support	Transport the same emission to ternary incidence without selecting Archimedean numbers.
Atlas and realization	route signature, atlas 81/56/13/324, C_M	Classify stories and characters before applying a dimensional chart.
Finite validation	censuses, structural necessity tests, integrity identities and independence controls	Checks coverage at the executed depth and separates construction, independent verification, and comparison data.

The eight classes describe mathematical nature; the fourteen groupings describe execution function. They are two gradings of the same inventory.

1.13.4 Mathematical classes, domains, codomains, and causal dependencies of the 34 canonical constructions

$$\mathfrak{O}_{\mathrm{TPK}} := (\mathfrak{o}_1, \dots, \mathfrak{o}_{34}). \tag{1.74}$$

#	Symbol	Domain and codomain	Rule or function
1	$\mathcal{A}_9$	$(\mathbb{Z}/9)^2$, matrix and internal law	Additive-multiplicative arithmetic cell.
2	(ρ_9, q_9)	$\mathbb{Z}_{>0} \to \{1, \dots, 9\} \times \mathbb{Z}_{\geq 0}$	Nonadic division with positive representative and quotient.
3	$\rho_3^{\mathrm{or}}, c_3$	$\mathbb{Z} \to \{-1, 0, +1\} \times \mathbb{Z}$	Lift ternary oriented $n = \rho_3^{\mathrm{or}}(n) + 3c_3(n)$.
4	$\iota_{1000} := (q_{1000}, r_{1000})$	$\mathbb{Z} \to \mathbb{Z} \times \{0, \dots, 999\}$	Euclidean division $N = 1000q_{1000} + r_{1000}$.
5	T_d	$X_9 \to X_9$, $d \in \{N, E, S, O\}$	Cardinal oriented translation.
6	F_{loc}	$\mathcal{Q}_{\mathrm{loc}} \times \mathscr{D}_{\mathrm{card}} \to \mathcal{Q}_{\mathrm{loc}}$	Local position feedback and sheet.
7	$T_{\min}$	$\Omega_{\mathrm{seed}} \to \mathcal{U}_6$	Minimum emitter of the two cursors, including the coordinate signature.
8	G	$\operatorname{im} s_\Sigma \times \operatorname{im} s_\Pi \to \{0, \dots, 999\}^6$	Coordinate-by-coordinate decimal coupling.
9	Ψ_6	$\mathcal{U}_6 \to \mathbb{F}_3^6$	Tertiary reduction of the catalog.
10	$\mathcal{R}_3$	$\mathcal{Q}_{\mathrm{loc}} \times \mathscr{D}_{\mathrm{card}}^3 \to \{-1, 0, +1\}$	Local three-step observable factor.
11	$\mathcal{K}_{27}$	$\mathcal{Q}_{\mathrm{obs}} \to \mathcal{Q}_{\mathrm{obs}}$	Composition F_{obs}^{27}.
12	$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}$	enriched states category	Preserve all the typed coordinates of the nucleus.
13	$\mathcal{H}$	relation on enriched states	Hensel Transition that retains residue and quotient.
14	$C_{3,1000}$	crossed states (N, L, D, K, A, B)	Exact update of ternary-decimal cylinders.
15	Σ_{B}	boundary memories → signature joint	Publish the block orientation preserving its genealogical record.
16	EF–G_9	extension relation	Filter compatible extensions and organize the nonadic genealogy.
17	Γ_9	enriched histories ⟶ histories	Holonomy of a phase twist with boundary memory.
18	X_χ	$\varprojlim_m \mathrm{Surv}_m^{(\chi)}$	Global section when non-emptiness, compatibility, and uniqueness are demonstrated in full depth.
19	$\mathcal{P}_{\mathrm{APP}}^{(2)}$	bivariate path category	Transports simultaneously the multiplicative evaluation and the additive accumulation.
20	F_{obs}	$\mathcal{Q}_{\mathrm{obs}} \to \mathcal{Q}_{\mathrm{obs}}$	Deterministic observable chronology with two cursors.
21	$\mathcal{A}_{42}$	alphabet of forty-two triads	Extended decimal support of finite emission.
22	Ext_r	$\mathrm{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}$	Typed relation of fiber prolongation.
23	$\mathcal{K}_{\mathrm{ph}}$	$\mathcal{U}_6 \times C_9$ about itself	Transfer of phase-compatible explicit continuation.
24	(P, H, Q)	$\mathcal{X}_\times \to \mathcal{X}_\times$	Advance, half-turn, and quarter-turn of seeds.
25	$(c_{\mathrm{mode}}, c_{\mathrm{or}})$	caminos → $\mathbb{Z}^2$	Mode and orientation change cocycles.
26	E_{20}	finite states → twenty blocks	Finite triadic prolongation with sourcebook.
27	$\mathcal{R}_{12}$	history enriched → record of twelve sectors	Timed system with route, signature, carry, and genealogical record.
28	R_{sgn}	terminal state → $U_{12}^{\mathrm{sgn}} \subseteq \mathbb{Z}^{12}$	Reader of the signed harmonic observable.
29	$\mathcal{H}_{12}$	$\mathbb{Z}^{12} \to \mathbb{Z}^{12}$	Transformation by three orbits of step three; inverse $\mathcal{H}_{12}/4$ over the pertinent image.
30	$\mathcal{E}_{\mathrm{dod}} = (D_3, D_4, Q)$	$\mathbb{Z}^{12} \to \mathbb{Z}^{25}$	Transversal extractor set of arities three and four.
31	C_α	blocks and carries → blocks	Dodecaphasic closure recurrence in base thousand.
32	$(L_{\mathrm{tick}}, \chi_{\mathrm{mass}})$	enriched history → temporal character	Posterior dimensional interface; does not participate in the numeric constructor of this work.
33	$\mathcal{W}_\pm$	bidirectional lattice of index twelve	Reading in both directions preserving orientation and origin.
34	$\mathcal{A}_{\mathrm{species}}$	extension families → atlas	Posterior realization interface, not a primitive fourth.

1.13.5 Update of the enriched state through $U_t = \mathrm{Upd}_t \circ \mathrm{Tra}_t \circ \mathrm{Sel}_t$

Definition 1.121 (Complete canonical state)**.** Let

$$\mathcal{E} := \mathbb{Z}_{\mathrm{quo}} \times \mathbb{Z}_{\mathrm{car}} \times \mathsf{Mem} \times \mathsf{Term} \times \mathsf{Cyl} \times \mathsf{Fr} \times \mathsf{Pref} \times \Lambda.$$

The canonical enriched state at instant t is

$$x_t = (p_t, m_t, \sigma_t, b_t, r_t; q_t, c_t, \eta_t, s_t, C_t, \partial_t, N_t, \lambda_t) \in X_9 \times \{\Sigma, \Pi\} \times \mathsf{TRIT} \times C_{12} \times C_9 \times \mathcal{E}. \tag{1.75}$$

The state abbreviations used in other units are projections of this tuple. In particular, the quotient q_t, carry c_t, terminal s_t, cylinder C_t, boundary ∂_t, prefix N_t, and ledger λ_t are not omitted.

Let ρ_t be the observable arrow determined by x_t. The transport laws of U005 and the transition $C_{3,1000}$ determine a memory endomorphism

$$\begin{aligned} \mathbf{E}_t &: \mathcal{E} \longrightarrow \mathcal{E}, \\ (q, c, \eta, s, C, \partial, N, \lambda) &\longmapsto \begin{pmatrix} \mathsf{H}(\rho_t)q, \mathsf{C}(\rho_t)c + \kappa(\rho_t), \mathsf{M}(\rho_t)\eta, \\ \mathsf{T}(\rho_t)s, \mathsf{Cyl}(\rho_t)C, \mathsf{Fr}(\rho_t)\partial, \\ \mathsf{Pref}(\rho_t)N, \Lambda(\rho_t)\lambda \end{pmatrix}. \end{aligned} \tag{1.76}$$

Each component published in (1.76) has the domain indicated by its fiber; $\kappa(\rho_t)$ is the carry cochain.

We define three endomorphisms of the canonical product. If $\sigma' = M_{\mathrm{ph}}(g(t))$ and $m' = \mu(\sigma', m)$, then

$$\mathsf{Sel}_t(p, m, \sigma, b, r; e) := (p, m', \sigma', b, r; e), \tag{1.77}$$

$$\mathsf{Tra}_t(p, m, \sigma, b, r; e) := (T_{d(t)}p, m, \sigma, b, r; e), \tag{1.78}$$

$$\mathsf{Upd}_t(p, m, \sigma, b, r; e) := (p, m, \sigma, b', r'; \mathbf{E}_t(e)), \tag{1.79}$$

where (b', r') is the destination of $\rho_t : (b, r) \to (b', r')$. The causal update is a composition of maps, not the composition of a map with a tritic value:

$$\boxed{\mathcal{U}_t := \mathsf{Upd}_t \circ \mathsf{Tra}_t \circ \mathsf{Sel}_t.} \tag{1.80}$$

Theorem 1.122 (Closure of causal update)**.** *The map $\mathcal{U}_t$ is an endomorphism of the complete canonical state. Readers, prolongations, and projections are applied after $\mathcal{U}_t$ only when their domain has been produced.*

Proof. Sel_t changes only the mode and the TRIT; Tra_t changes only the APP position; and Upd_t applies the typed product (1.76) and the phase arrow. All three codomains are the same product of definition 1.121; their composition is therefore defined and preserves all state coordinates. □

1.13.6 Typed separation between operators, states, projections, and publications of the TPK

The inventory imposes

$$\mathcal{S}_4 \neq D_4 \neq \mathrm{sd}_4, \qquad \mathcal{R}_{12} \neq C_{12} \neq \Theta_{108} \neq \Gamma_{108}, \qquad F_{\mathrm{obs}} \neq \mathcal{K}_{\mathrm{ph}}, \tag{1.81}$$

and

$$U_{12}^{\mathrm{cat}} \neq U_{12}^{\mathrm{sgn}}, \qquad \mathcal{E}_{\mathrm{dod}} \neq D_{\mathrm{raw}}^{90,120} \neq \mathcal{K}_{6\to 8}. \tag{1.82}$$

They are type incompatibilities, not mere value inequalities.

Comparison Criterion 1.123 (Audit of operatorial completeness)**.** An execution is operatorially complete only if:

1. declares the inputs of $\mathfrak{O}_{\mathrm{TPK}}$ used;
2. is recorded the domain and the codomain of each composition;
3. preserves the causal order of (1.80);
4. distinguishes dynamics, transfer, reading, prolongation, and output;
5. permits reconstruction from λ_t of every block change and every carry;

6. does not promote a finite check to an infinite section without proving non-emptiness, compatibility, and uniqueness at every depth.

Any symbol used without a type, or any fusion prohibited by (1.81) and (1.82), constitutes a verifiable failure.

1.14 Tritic incidence 90/120 and typed separation of the temporal, nilpotent, and angular operators

The integers 90 and 120 appear in several strata of the construction. Their numerical coincidence does not establish an identity of objects. Likewise, the letter epsilon denotes three constructions with incompatible domains. The purpose of this unit is to fix their types before using them and to indicate precisely which realizations require structures that have not yet been constructed.

1.14.1 Temporal, nilpotent, and angular operators: domains and codomains

1.14.1.1 Action of the tritic temporal selector over $\{-1,0,+1\}$*: opening, threshold, and return with memory*

The internal calendar selector is

$$\varepsilon_9 : \{1,\dots,9\} \longrightarrow \{-1,0,+1\}, \qquad \varepsilon_9(r) = \begin{cases} +1, & r \in \{1,4,7\}, \\ -1, & r \in \{2,5,8\}, \\ 0, & r \in \{3,6,9\}. \end{cases} \tag{1.83}$$

Its codomain is the oriented TRIT. It determines which sheet acts and does not itself shift any position.

1.14.1.2 Nilpotent parabolic generator $J_0^2 = 0$ *and evolution at the temporal threshold*

The class

$$\varepsilon_{\mathrm{nil}} \in Q_{\mathrm{n}} := \mathbb{F}_3[\varepsilon_{\mathrm{nil}}]/(\varepsilon_{\mathrm{nil}}^2) \tag{1.84}$$

satisfies

$$\varepsilon_{\mathrm{nil}} \neq 0, \qquad \varepsilon_{\mathrm{nil}}^2 = 0. \tag{1.85}$$

It is an element of a degenerate quadratic algebra; it is not a temporal selector.

1.14.1.3 Angular residue induced after temporal selection and orientation preservation

The symbol

$$\varepsilon_{\mathrm{res}}^{\circ} \in \mathbb{R} \tag{1.86}$$

designates, when its geometric parameters are introduced, the residue of an angular identity. At this stage only its type remains reserved. It is not obtained from ε_9 or from $\varepsilon_{\mathrm{nil}}$, and it does not intervene in the construction of π, e, φ or α.

Proposition 1.124 (No identification of the epsilon symbols)**.** *There is no well-typed equality between*

$$\varepsilon_9, \qquad \varepsilon_{\mathrm{nil}}, \qquad \varepsilon_{\mathrm{res}}^{\circ}.$$

Proof. The first object is a function between finite sets; the second, a nilpotent element of an algebra over $\mathbb{F}_3$; the third, a real scalar with angular unit. Their domains, codomains, and laws of composition are distinct. □

1.14.2 Typed residences of the indices 90/120 in 6 non-equivalent operator objects

Definition 1.125 (Dodecaphasic Extractor)**.** The integral extractor

$$\mathcal{E}_{\mathrm{dod}} := (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25} \tag{1.87}$$

reads differences of arities three and four and a sum coordinate. The subscripts are not angular lengths.

Definition 1.126 (Raw difference of sequences)**.** For $|q| < 1$, define

$$S_m^{\mathrm{raw}}(q) := \frac{q^m}{1 - q^{3m}}, \qquad D_{\mathrm{raw}}^{90,120}(q) := S_{90}^{\mathrm{raw}}(q) - S_{120}^{\mathrm{raw}}(q). \tag{1.88}$$

This object is an analytic function of q; it is not a finite census.

Definition 1.127 (Marked families of incidence)**.** Once a hexad H, its image $\ell(H)$ in a binary dodecad D, and the lifted octad O_H have been constructed, one will define

$$\mathcal{F}_6(H) := \ell(H) \times \binom{\ell(H)}{2}, \tag{1.89}$$

$$\mathcal{F}_8(H) := O_H \times \binom{\ell(H)}{2}. \tag{1.90}$$

By construction,

$$|\mathcal{F}_6(H)| = 6\binom{6}{2} = 90, \qquad |\mathcal{F}_8(H)| = 8\binom{6}{2} = 120. \tag{1.91}$$

The inclusion $\ell(H) \hookrightarrow O_H$ inducira

$$\mathcal{I}_{6\to 8} : \mathcal{F}_6(H) \hookrightarrow \mathcal{F}_8(H). \tag{1.92}$$

The equations (1.89) y (1.90) fix the type from now on. Their effective existence will be proved after constructing C_W, W_{12}, the binary datum G_{24}, D, ℓ, and W_{24}. That branch is not anticipated here.

Definition 1.128 (Normalized Queue Combination)**.** For parameters $q_-, q_+ \in (0, 1)$, let

$$S_m(q) := \frac{q^m}{1 - q^{3m}}, \qquad \Delta S_m := S_m(q_-) - S_m(q_+). \tag{1.93}$$

The functional

$$\mathcal{K}_{6\to 8} := 12\,\Delta S_{90} - \Delta S_{120} \tag{1.94}$$

is an analytic combination of tails. Its coefficient twelve belongs to its normalization and does not turn it into the record $\mathcal{R}_{12}$.

Definition 1.129 (Semigroup of Degrees and Moments)**.** The additive semigroup generated by both degrees is

$$\mathfrak{S}_{90,120} := \langle 90, 120\rangle = \{90a + 120b : a, b \in \mathbb{N}_0\}. \tag{1.95}$$

For $n \in \mathfrak{S}_{90,120}$, the oriented moment is

$$s_n := q_-^n - q_+^n. \tag{1.96}$$

The moments are differential monomials indexed by a semigroup; the series of (1.93) are infinite sums of those moments over specific progressions.

1.14.3 Normalized defect 1−90/120=1/4 of the tritic incidence

When the inclusion $\mathcal{I}_{6\to 8}$ has been constructed, its difference of cardinalities will be

$$|\mathcal{F}_6(H)| - |\mathcal{F}_8(H)| = 90 - 120 = -30.$$

If normalization is declared

$$24\binom{6}{2} = 360,$$

we obtain

$$\frac{90-120}{24\binom{6}{2}} = \frac{6-8}{24} = -\frac{1}{12}. \tag{1.97}$$

The inclusion determines -30; the rational $-1/12$ depends moreover of the declared normalization. No it what must present as if the cardinality by if alone fixed the denominator.

1.14.4 Domain invariants, orientation, and memory of the 90/120 realizations

Theorem 1.130 (Separation of the six objects)**.** *The objects*

$$\mathcal{E}_{\mathrm{dod}}, \quad D_{\mathrm{raw}}^{90,120}, \quad \mathcal{I}_{6\to 8}, \quad \mathcal{K}_{6\to 8}, \quad \mathfrak{S}_{90,120}, \quad \varepsilon_{\mathrm{res}}^{\circ} \tag{1.98}$$

are mutually distinct by type: they are, respectively, an integer reader, an analytic function, a finite morphism, a combination of series, an index semigroup, and a real angular scalar.

Proof. The conclusion is obtained by comparing their domains and codomains:

$$\mathbb{Z}^{12} \to \mathbb{Z}^{25}, \quad \mathbb{D} \to \mathbb{C}, \quad \mathcal{F}_6(H) \to \mathcal{F}_8(H), \quad (0,1)^2 \to \mathbb{R}, \quad \mathbb{N}_0^2 \to \mathbb{N}_0, \quad \{*\} \to \mathbb{R}.$$

There are no two identical signatures. □

Comparison Criterion 1.131 (Refutation criteria)**.** The classification is refuted if any of its domains or codomains is altered, if the census of (1.91) does not reproduce 90 and 120, if $-1/12$ is attributed to the inclusion without declaring the normalization, or if the angular residual is presented as an output of the constants generator of this monograph.

1.15 Local transport factor and ternary-decimal transition

This unit separates three operations involved in the transport of the Topological Phase Kernel that do not share a domain: the cardinal realization of a trit, the stroboscopic sampling of a sequence, and the exact update of a pair of ternary–decimal cylinders. The first is a finite dynamics; the second is a temporal reader; the third preserves the integers that make it possible to decide compatibility without floating-point arithmetic.

1.15.1 Oriented transition local operator and APP sheet update

Let

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2, \qquad \mathscr{D}_{\mathrm{card}} = \{N, E, S, O\},$$

with

$$\delta(N) = (-1, 0), \quad \delta(E) = (0, 1), \quad \delta(S) = (1, 0), \quad \delta(O) = (0, -1).$$

The visible evaluations of APP are

$$L_\Sigma(x, y) = \rho_9((x+1) + (y+1)), \qquad L_\Pi(x, y) = \rho_9((x+1)(y+1)),$$

where $\rho_9(n) = 1 + ((n-1) \bmod 9)$. We further define

$$\rho_3^{\mathrm{or}}([0]) = 0, \qquad \rho_3^{\mathrm{or}}([1]) = +1, \qquad \rho_3^{\mathrm{or}}([2]) = -1.$$

Definition 1.132 (Local factor of APP–TRIT)**.** The local state space is

$$\mathcal{Q}_{\mathrm{loc}} = X_9 \times \{\Sigma, \Pi\}, \qquad |\mathcal{Q}_{\mathrm{loc}}| = 162.$$

For $q = (p, m)$, $d \in \mathscr{D}_{\mathrm{card}}$, we set

$$p_d = p + \delta(d), \qquad a_d(q) = \rho_3^{\mathrm{or}}([L_m(p_d)]_3),$$

and

$$\mu(a, m) = \begin{cases} \Sigma, & a = +1, \\ \Pi, & a = -1, \\ m, & a = 0. \end{cases}$$

The local transition is

$$F_d^{\mathrm{loc}}(p, m) = (p_d, \mu(a_d(p, m), m)).$$

The position it is modified before evaluating the sheet. The null trit preserves the last active mode; not eliminates the sheet nor reinitializes the history.

For a cardinal word $u = d_1 d_2 d_3$, let

$$q_j = F_{d_j}^{\mathrm{loc}}(q_{j-1}), \qquad q_0 = q,$$

and let us denote by

$$R_3(q, u) \in \{-1, 0, +1\}$$

the trit produced in the third displacement.

Definition 1.133 (Remark on step three)**.** On an oriented sequence $a = (a_t)_{t \geq 1}$ we define

$$\mathrm{Obs}_3(a) = (a_3, a_6, a_9, \ldots).$$

The map R_3 takes a state and a cardinal word of length three; Obs_3 takes a sequence and publishes a subsequence. They are operators of different types.

Theorem 1.134 (Exact surjectivity of the sampled factor)**.** *For every $q \in \mathcal{Q}_{\mathrm{loc}}$ and every $b \in \{-1, 0, +1\}$, the fiber*

$$\mathcal{R}_3(q, b) = \{u \in \mathscr{D}_{\mathrm{card}}^3 : R_3(q, u) = b\}$$

is non-empty and satisfies

$$\boxed{8 \leq |\mathcal{R}_3(q, b)| \leq 40.}$$

In particular, the $162 \cdot 3 = 486$ *state–output fibers are non-empty.*

Proof. From each of the 162 states, the $4^3 = 64$ cardinal words are enumerated, and the recurrence of definition 1.132 is applied literally. The exhaustive census of the 486 pairs (q, b) has a minimum of eight, a maximum of forty, and no zero value. The statement is therefore a finite identity on the defined system, not a mixing hypothesis. □

Corollary 1.135 (Realization of an arbitrary oriented sequence)**.** *Remark. For every sequence $(b_k)_{k \geq 1} \in \{-1, 0, +1\}^{\mathbb{N}}$ there exists a cardinal history whose observation satisfies*

$$(a_3, a_6, a_9, \ldots) = (b_1, b_2, b_3, \ldots).$$

Proof. Fixing a total order on $\mathscr{D}^3_{\mathrm{card}}$, at stage k the first element of $\mathcal{R}_3(q_{k-1}, b_k)$ is taken. The terminal state of the block again belongs to Q_{loc}, so that the choice may be iterated indefinitely. □

The corollary proves that the sampled factor can be realized. It does not select the sections of π, e, φ, fix the two intermediate trits or replace the compatibility of the enriched state.

1.15.2 Cylindrical systems of the 2 arithmetic readers and truncation compatibility

A ternary word of length L, interpreted as the integer N, determines

$$I_3(N, L) = \left[\frac{N}{3^L}, \frac{N+1}{3^L}\right).$$

A prefix of K decimal triads, interpreted as the integer D, determines

$$I_{1000}(D, K) = \left[\frac{D}{1000^K}, \frac{D+1}{1000^K}\right).$$

We introduce the cross residues

$$A = N1000^K - D3^L, \qquad B = (N+1)1000^K - D3^L = A + 1000^K.$$

Lemma 1.136 (Integer compatibility criterion)**.** *The cylinders $I_3(N, L)$ and $I_{1000}(D, K)$ intersect if and only if*

$$\boxed{A < 3^L, \qquad B > 0.}$$

Equivalently,

$$N1000^K < (D+1)3^L, \qquad (N+1)1000^K > D3^L.$$

Proof. Two half-open intervals $[a, b)$ and $[c, d)$ intersect exactly when $a < d$ and $c < b$. Multiplication by $3^L 1000^K > 0$ produces the two inequalities. Subtracting $D3^L$ and $N1000^K$, respectively, gives the form in A, B. □

1.15.3 Completion operator $C_{3,1000}$: ternary extension and preservation of the unit in 10^3 positions

A new block of six trits is represented by

$$u \in \{0, \dots, 3^6 - 1\} = \{0, \dots, 728\}.$$

The ternary update is

$$N' = 729N + u, \qquad L' = L + 6.$$

Decimal publication has two cases. If the refined cylinder does not yet determine a new triad, then $K' = K$, $D' = D$. If it determines $\delta \in \{0, \dots, 999\}$, then

$$K' = K + 1, \qquad D' = 1000D + \delta.$$

Definition 1.137 (Crossed state and exact transition)**.** The cylinder state is the sextuple

$$\mathfrak{c} = (N, L, D, K, A, B)$$

subject to

$$A = N1000^K - D3^L, \qquad B = A + 1000^K.$$

The transition $C_{3,1000}$ adjoins u and, when corresponds, δ, and updates

$$(N, L, D, K, A, B) \longmapsto (N', L', D', K', A', B')$$

through means of

$$A' = \begin{cases} 729A + u1000^K, & K' = K, \\ 729000A + u1000^{K+1} - \delta\, 729\, 3^L, & K' = K + 1, \end{cases} \qquad B' = A' + 1000^{K'}.$$

The prolongation is admissible exactly when

$$A' < 3^{L'}, \qquad B' > 0.$$

Theorem 1.138 (Accuracy and Sufficiency of the Cross State)**.** *The transition of the definition 1.137 exactly preserves the identities*

$$A' = N'1000^{K'} - D'3^{L'}, \qquad B' = (N' + 1)1000^{K'} - D'3^{L'}.$$

Consequently, (A', B', L', K') *suffices to decide the compatibility of the new pair of cylinders, while* (N', D') *preserves its reconstructive provenance.*

Proof. If a decimal triplet does not appear,

$$N'1000^K - D3^{L+6} = (729N + u)1000^K - 729D3^L = 729A + u1000^K.$$

Si aparece δ,

$$\begin{aligned} N'1000^{K+1} - D'3^{L+6} &= (729N + u)1000^{K+1} - (1000D + \delta)7293^L \\ &= 729000A + u1000^{K+1} - \delta\, 729\, 3^L. \end{aligned}$$

In both cases, increasing N' by one unit adds $1000^{K'}$, which yields B'. The criterion of the lemma 1.136 concludes. □

Proposition 1.139 (Compatibility with concatenation)**.** *Successively applying* $C_{3,1000}$ *to two blocks produces the same pair* (N'', D'') *and the same cross-residues as appending at once the two ternary words and the decimal triads published in the same order.*

Proof. The recurrences $N \mapsto 729N + u$ and $D \mapsto 1000D + \delta$ are the laws of positional concatenation in bases 729 and 1000. The theorem 1.138 identifies A, B with functions of those four integers; hence its stagewise update coincides with the evaluation subsequent to concatenation. □

1.15.4 Factorization of the TPK transition as $U_t = \mathrm{Upd}_t \circ \mathrm{Tra}_t \circ \mathrm{Sel}_t$

The correct sequence is

$$(q, u) \xmapsto{R_3} b, \qquad (a_t)_{t\geq 1} \xmapsto{\mathrm{Obs}_3} (a_{3k})_{k\geq 1}, \qquad (\mathfrak{c}, u, \delta) \xmapsto{C_{3,1000}} \mathfrak{c}'.$$

R_3 proves local realizability; Obs_3 publishes a subsequence; $C_{3,1000}$ transports compatibility between two bases. None of the three can replace the survival relation of the full TPK.

Comparison Criterion 1.140 (Refutation criteria)**.** The construction is refuted in its domain if any of the following facts occurs:

1. any of the 486 fibers $\mathcal{R}_3(q, b)$ is empty or remains outside of the interval $[8, 40]$;
2. the null trit changes the active mode;

3. R_3 is identified with Obs_3;
4. the recurrence of A' does not reproduce its direct definition;
5. a declared compatible prolongation violates $A' < 3^{L'}$ or $B' > 0$;
6. the decision depends on a numerical tolerance absent from the integer inequalities.

1.16 Record dodecafasico integral, extension ciclica and decomposition 1 + 5 + 6

The observable period of one hundred eight positions contains twelve sectors of nine positions. This equality of cardinalities does not suffice to describe the temporal state: the ordered support, cyclic law, sector index, and enriched register of transports, orientations, carries, and incidences must be distinguished. This unit constructs the four objects and proves the integral reconstruction of the twelve memory coordinates.

1.16.1 Cyclic coordinate on the time-oriented 12-sector support

For $m \in \{1, \dots, 12\}$, define

$$E_m := \{9(m-1)+1, \dots, 9m\}. \tag{1.99}$$

The ordered support and the cyclic group are

$$\Gamma_{108} := E_1 \sqcup \cdots \sqcup E_{12}, \qquad \Theta_{108} := \mathbb{Z}/108\mathbb{Z}. \tag{1.100}$$

Γ_{108} preserves position, order and pertenencia to block; Θ_{108} preserves the law of translation. No are the same object.

For $\theta \in \Theta_{108}$, let $\widetilde{\theta} \in \{0, \dots, 107\}$ be its representative. We define

$$\beta(\theta) := \left[\left\lfloor \frac{\widetilde{\theta}}{9} \right\rfloor\right]_{12}, \qquad r(\theta) := 1 + (\widetilde{\theta} \bmod 9). \tag{1.101}$$

The first coordinate locates the duodecaphase sector; the second publishes the positive position $1, \dots, 9$ within the sector.

1.16.2 nonsplit extension $0 \to C_{12} \to C_{108} \to C_9 \to 0$ and carry cocycle

Let $C_n = \mathbb{Z}/n\mathbb{Z}$. The applications

$$\iota : C_{12} \longrightarrow C_{108}, \quad \iota([b]_{12}) = [9b]_{108}, \qquad p : C_{108} \longrightarrow C_9, \quad p([\theta]_{108}) = [\theta]_9 \tag{1.102}$$

form the sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0. \tag{1.103}$$

For the set-theoretic section

$$s([r]_9) := [\widetilde{r}]_{108}, \qquad 0 \le \widetilde{r} < 9,$$

the defect of additivity is

$$c(r, r') := \left[\left\lfloor \frac{\widetilde{r} + \widetilde{r}'}{9} \right\rfloor\right]_{12}, \qquad s(r) + s(r') = s(r + r') + \iota(c(r, r')). \tag{1.104}$$

Theorem 1.141 (Accuracy, non-division, and cohomological class)**.** *The sucesion* (1.103) *is exacta and not it escinde. For the action trivial,*

$$H^2(C_9, C_{12}) \simeq C_{12}/9C_{12} \simeq C_3, \tag{1.105}$$

and the cocycle (1.104) *represents its generating class.*

Proof. The image of ι is the subgroup of multiples of nine, which coincides with $\ker p$; p is surjective. This proves exactness. If the extension were split, C_{108} would be isomorphic to $C_{12} \times C_9$, but

$$\exp(C_{108}) = 108, \qquad \exp(C_{12} \times C_9) = \operatorname{mcm}(12, 9) = 36.$$

The cocycle identity is the Euclidean division of $\widetilde{r} + \widetilde{r'}$ by nine. For a cyclic group with trivial action, $H^2(C_9, C_{12}) \simeq C_{12}/9C_{12}$; the unit carry projects to $[1]$, which generates the quotient of order three. □

The non-splitting typifies the return:

$$g_0(\theta + 9) = g_0(\theta), \qquad \beta(\theta + 9) = \beta(\theta) + 1 \pmod{12}.$$

Nine iterations close the visible phase and shift one sector; one hundred eight close both coordinates of the clock. The enriched state further preserves the vertical memory, cylinder, boundary, and genealogy.

1.16.3 Enhanced temporal registration: joint encoding of phase, sheet, boundary, and memory

Definition 1.142 (Oriented dodecaphonic register)**.** For an enriched history, let τ_m be the ordered subroute on E_m, σ_m its orientation, κ_m the carry crossing the block boundary, and Λ_m its local incidence ledger. The complete record is

$$\mathcal{R}_{12} := \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=1}^{12}. \tag{1.106}$$

Thus four levels are distinguished:

$$\boxed{\mathcal{R}_{12} \neq C_{12} \neq \Theta_{108} \neq \Gamma_{108}.} \tag{1.107}$$

The first is an enriched register; the second, a sector index; the third, a group of translations; the fourth, an ordered support. There exist projections between them, but no identifications.

Let $q_{\mathrm{mem}} \in \mathbb{Z}_{\geq 0}$ be the unbounded number of nonadic turns. The duodecaphase coordinate is only its residue

$$\beta = [q_{\mathrm{mem}}]_{12}.$$

After one hundred and eight steps,

$$\beta \longmapsto \beta, \qquad q_{\mathrm{mem}} \longmapsto q_{\mathrm{mem}} + 12.$$

The calendar returns; the vertical memory is not reset.

1.16.4 Extraction of the signed dodecaphase state from the enriched time record

Let $(d_t)_{t=1}^{108}$ be the digital record on Γ_{108}. We define

$$\mathcal{D}_{\mathrm{evt}} := \{2, 4, 6, 8\}, \qquad \Sigma_{12} := (+, +, +, -, -, -, +, +, +, -, -, -). \tag{1.108}$$

For $m = 0, \ldots, 11$, let

$$e_m := \Sigma_{12}(m + 1)\, \#\{t \in E_{m+1} : d_t \in \mathcal{D}_{\mathrm{evt}}\} \in \mathbb{Z}. \tag{1.109}$$

The application preserves the local census and the sectorial orientation as separate factors.

We pair the two semicrowns by means of

$$p_i := e_i + e_{i+6}, \qquad a_i := e_i - e_{i+6}, \qquad 0 \leq i \leq 5. \tag{1.110}$$

We define the total census, the five differences with respect to the sixth pair, and the six antisymmetries:

$$s_C := \sum_{i=0}^{5} p_i, \qquad M_i := p_i - p_5 \quad (0 \le i \le 4), \qquad A_6 := (a_0, \dots, a_5). \tag{1.111}$$

Definition 1.143 (Integral reader 1 + 5 + 6)**.** The linear reader is

$$\mathcal{L}_{12} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{12}, \qquad \mathcal{L}_{12}(e) := (s_C, M_0, \dots, M_4, a_0, \dots, a_5). \tag{1.112}$$

The coordinate distribution is

$$\underbrace{1}_{s_C} + \underbrace{5}_{M_0,\dots,M_4} + \underbrace{6}_{a_0,\dots,a_5} = 12. \tag{1.113}$$

Theorem 1.144 (Exact reconstruction and integral image)**.** *The reader $\mathcal{L}_{12}$ is injective. On its image,*

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \qquad p_i = p_5 + M_i \quad (0 \le i \le 4), \tag{1.114}$$

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2} \quad (0 \le i \le 5). \tag{1.115}$$

An output tuple belongs to $\operatorname{im} \mathcal{L}_{12}$ *if and only if*

$$s_C - \sum_{i=0}^{4} M_i \equiv 0 \pmod 6 \tag{1.116}$$

and, for the reconstructed p_i,

$$p_i \equiv a_i \pmod 2 \qquad (0 \le i \le 5). \tag{1.117}$$

Proof. From $M_i = p_i - p_5$ is obtained

$$s_C = \sum_{i=0}^{4} (p_5 + M_i) + p_5 = 6p_5 + \sum_{i=0}^{4} M_i,$$

which gives the first equation of (1.114); the others are immediate. Adding and subtracting $p_i = e_i + e_{i+6}$ and $a_i = e_i - e_{i+6}$ produces (1.115). Divisibility by six and the six parity congruences are precisely the necessary and sufficient conditions for all reconstructed coordinates to be integers. □

Division by six appears after the oriented census, not during it. The integrality conditions for this unit are exactly (1.116) and (1.117); no Smith normal form is attributed here to a matrix that has not previously been defined.

1.16.5 Genealogical distinction between 2 dodecaphasic words of equal period

The distinction must be preserved.

$$U_{12}^{\mathrm{cat}} := U_6 \| U_6, \qquad U_{12}^{\mathrm{sgn}} := \mathcal{H}_{12}(K). \tag{1.118}$$

The first concatenates two catalog emissions of length six. The second is a signed coordinate reversibly linked to K by the operator $\mathcal{H}_{12}$. Their common length does not imply the same provenance, semantics, or transport law.

Comparison Criterion 1.145 (Falsification criteria)**.** The construction is refuted if the sequence (1.103) is not exact, if its cohomology class is trivial, if a tuple in the image violates (1.116) or (1.117), if the inverse formulas do not recover the twelve initial integers, or if the return of β is interpreted as a reset of q_{mem}.

1.17 Cyclotomic phase functor–length and modular elevation of paths

The odometric quotient $O_{9,12}$ simultaneously preserves the nonadic position, the dodecaphase register, and the integral length of each arrow. In this unit, its phase is realized over fibers A_2, and the cardinal lift that replaces each edge with four oriented edges is proved. The cyclotomic representation and the absolute length will be retained as distinct coordinates.

1.17.1 Coxeter representation of order 3 on the dodecaphasic fibers of type A_2

Let

$$G_{A_2} := \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad C := \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}. \tag{1.119}$$

By the proposition 1.94,

$$C^3 = I_2, \qquad C^{\mathsf{T}} G_{A_2} C = G_{A_2}, \qquad \det(XI_2 - C) = X^2 + X + 1. \tag{1.120}$$

Thus, C realizes the character cyclotomic of C_3 on A_2.

For each $b \in \mathbb{Z}/12\mathbb{Z}$, let $A_{2,b}$ be a copy of A_2, let

$$\iota_b : A_{2,b} \xrightarrow{\sim} A_2$$

an integral identification, and let $P_b \in O(G_{A_2})$ be a frame that depends on b, but not on the residue r. We define

$$\rho_b := \iota_b^{-1} P_b C P_b^{-1} \iota_b, \qquad \rho_b^3 = \mathrm{id}_{A_{2,b}}. \tag{1.121}$$

The independencia respect of r liga the character to the fiber dodecaphasic; r preserves the position internal of the turn nonadic.

1.17.2 Odometer category of the TPK: cylinders, truncations, and sequence morphisms

Recall that $O_{9,12}$ has objects

$$(b, r) \in \mathbb{Z}/12\mathbb{Z} \times \{0, \dots, 8\}.$$

For $n \in \mathbb{N}$, the flecha unique of length n is

$$\gamma_{(b,r),n} : (b, r) \longrightarrow (b + \kappa_r(n), [r + n]_9), \qquad \kappa_r(n) := \left\lfloor \frac{r+n}{9} \right\rfloor. \tag{1.122}$$

The composition relies on

$$\kappa_r(n + m) = \kappa_r(n) + \kappa_{[r+n]_9}(m). \tag{1.123}$$

The length $\ell(\gamma_{(b,r),n}) = n$ is additive.

Definition 1.146 (Cyclotomic Phase–Length Functor)**.** Let $\mathbf{Rep}_{\mathbb{C}}(C_3)$ be the category of complex representations of C_3, and let $\mathbf{B}\mathbb{N}$ be the monoidal category with a single object whose endomorphisms are $(\mathbb{N}, +)$. We define

$$\mathfrak{F}_{\mathrm{cyc}} : O_{9,12} \longrightarrow \mathbf{Rep}_{\mathbb{C}}(C_3) \times \mathbf{B}\mathbb{N} \tag{1.124}$$

by

$$\mathfrak{F}_{\mathrm{cyc}}(b, r) = (A_{2,b} \otimes_{\mathbb{Z}} \mathbb{C}, \rho_b)$$

and

$$\mathfrak{F}_{\mathrm{cyc}}(\gamma) := \left(\iota_{b'}^{-1} P_{b'} C^{\ell(\gamma)} P_b^{-1} \iota_b, \ell(\gamma) \right) \tag{1.125}$$

for $\gamma : (b, r) \to (b', r')$.

Theorem 1.147 (Functoriality and separation of the records)**.** *The assignment of the definition 1.146 is a functor. Its first component preserves only the length class modulo three; the second preserves the*

complete integer and, by reduction, determines that class. The cyclotomic component does not allow the absolute length to be reconstructed.

Proof. If $\gamma_1 : (b_0, r_0) \to (b_1, r_1)$ and $\gamma_2 : (b_1, r_1) \to (b_2, r_2)$ have lengths n_1, n_2, the intermediate trivialization cancels:

$$\begin{aligned}&\left(\iota_{b_2}^{-1} P_{b_2} C^{n_2} P_{b_1}^{-1} \iota_{b_1}\right)\left(\iota_{b_1}^{-1} P_{b_1} C^{n_1} P_{b_0}^{-1} \iota_{b_0}\right)\\ &\quad = \iota_{b_2}^{-1} P_{b_2} C^{n_1+n_2} P_{b_0}^{-1} \iota_{b_0}.\end{aligned} \tag{1.126}$$

The second component composes by addition. Since C^n depends only on $[n]_3$, whereas $\ell = n$ preserves the integer, the two coordinates are not identified. □

If

$$Q : \mathsf{TPK} \longrightarrow O_{9,12}$$

is the odometer functor of U006, the typed realization is

$$\mathfrak{F}_{\mathrm{cyc}} \circ Q : \mathsf{TPK} \longrightarrow \mathbf{Rep}_{\mathbb{C}}(C_3) \times \mathbf{B}\mathbb{N}. \tag{1.127}$$

1.17.3 Modular elevation of cardinal paths

Let $\mathcal{A}_{N,d}$ be the category libre of paths cardinal of $(\mathbb{Z}/N\mathbb{Z})^d$. Let

$$N \geq 2, \quad u \geq 1, \quad \gcd(u, N) = 1, \quad k := \mathrm{ord}_N(u), \quad q := \frac{u^k - 1}{N}.$$

Definition 1.148 (Cardinal Substitution by u)**.** The functor

$$\mathcal{S}_u : \mathcal{A}_{N,d} \longrightarrow \mathcal{A}_{N,d}$$

sends an object p to up and each oriented edge to the ordered concatenation of u copies with the same direction.

Theorem 1.149 (Modular lifting to endomorphism of paths)**.** *For every object p, path γ, and*

$$\kappa_{N,r}(n) := \left\lfloor \frac{r+n}{N} \right\rfloor,$$

we have

$$\mathcal{S}_u^k(p) = p, \tag{1.128}$$

$$\ell(\mathcal{S}_u^k \gamma) = u^k \ell(\gamma), \tag{1.129}$$

$$\kappa_{N,r}(u^k n) - \kappa_{N,r}(n) = qn. \tag{1.130}$$

Proof. The category is free, so the images of objects and generators determine a unique functor. Each map multiplies the length by u. Since $u^k = 1 + qN$, the objects return modulo N, and

$$\left\lfloor \frac{r + n + qNn}{N} \right\rfloor = \left\lfloor \frac{r+n}{N} \right\rfloor + qn.$$

□

In the nonadic case,

$$(N, u, k, q) = (9, 4, 3, 7), \qquad 4^3 = 64 = 1 + 7 \cdot 9. \tag{1.131}$$

Therefore,

$$\boxed{\mathcal{S}_4^3(e) \text{ preserves the endpoints and contains seven nonadic turns oriented per initial edge.}} \tag{1.132}$$

The endpoints and the residual position return under three maps; the enriched state need not return. Without a proof of invertibility, $\mathcal{S}_4^3$ is a vertical endomorphism of paths, not a monodromy.

Remark 1.150 (Separation with respect to clock dilation). $\mathcal{S}_4$ replaces objects and edges in a path category. The map D_4 of U006C will multiply the coordinate Θ_{108} by four. They share an integer factor, but not a domain, codomain, or composition law.

Comparison Criterion 1.151 (Criteria of refutation)**.** The block remains refuted if fails any of the identities (1.120), (1.123), (1.126) or (1.131); also remains badly typed if it identify the phase C^n, the length n, the substitution $\mathcal{S}_4$ and the dilation D_4.

1.18 Quaternary dilation of the register, observable obstruction, and refined semiconjugacy

The cardinal substitution $\mathcal{S}_4$ constructed in the preceding unit acts on paths. In this unit, a second operation, arithmetic in nature, is defined on the register $\mathbb{Z}/12\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}$, and a third, categorical operation subdivides each observable transition into four microtransitions. The aim is not to merge them, but to prove the exact relationship between their realizations.

1.18.1 Dilation on the product register

Let

$$Q_{12,9} := \mathbb{Z}/12\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}.$$

For $r \in \mathbb{Z}/9\mathbb{Z}$, we denote its normalized representative by $\widetilde{r} \in \{0, \dots, 8\}$.

Definition 1.152 (Quaternary dilation of the register)**.** We define

$$D_4(b, r) := \left(\left[4b + \left\lfloor \frac{4\widetilde{r}}{9} \right\rfloor \right]_{12}, [4r]_9 \right). \tag{1.133}$$

The total coordinate is

$$\Theta : Q_{12,9} \xrightarrow{\sim} \mathbb{Z}/108\mathbb{Z}, \qquad \Theta(b, r) := [9b + \widetilde{r}]_{108}. \tag{1.134}$$

Proposition 1.153 (Realization by modular multiplication)**.** *In the total coordinate,*

$$\boxed{\Theta \circ D_4 \circ \Theta^{-1}(\theta) = [4\theta]_{108}.} \tag{1.135}$$

In particular,

$$\operatorname{im}(D_4) = 4\mathbb{Z}/108\mathbb{Z} \simeq \mathbb{Z}/27\mathbb{Z}, \tag{1.136}$$

and the induced restriction on the image has order nine.

Proof. For $\theta = 9b + \widetilde{r}$,

$$4\theta = 9 \left(4b + \left\lfloor \frac{4\widetilde{r}}{9} \right\rfloor \right) + (4\widetilde{r} \bmod 9).$$

This is exactly the quotient–residue decomposition of (1.133). The image consists of the classes that are multiples of four. Dividing by four yields $\mathbb{Z}/27\mathbb{Z}$, and

$$\begin{aligned} &4^1 \equiv 4, \quad 4^2 \equiv 16, \quad 4^3 \equiv 10, \\ &4^4 \equiv 13, \quad 4^5 \equiv 25, \quad 4^6 \equiv 19, \\ &4^7 \equiv 22, \quad 4^8 \equiv 7, \quad 4^9 \equiv 1 \pmod{27}. \end{aligned}$$

No preceding power is one; therefore $\mathrm{ord}_{27}(4) = 9$. □

1.18.2 Exact carry identity

For $r \in \{0, \dots, 8\}$ and $n \in \mathbb{N}$, define

$$\kappa_r(n) := \left\lfloor \frac{r+n}{9} \right\rfloor, \qquad r' := [r+n]_9.$$

Theorem 1.154 (Quaternary carry cocycle)**.** *We have*

$$\boxed{\kappa_{[4r]_9}(4n) = 4\kappa_r(n) + \left\lfloor \frac{4\widetilde{r'}}{9} \right\rfloor - \left\lfloor \frac{4\widetilde{r}}{9} \right\rfloor.} \tag{1.137}$$

The last difference is the coboundary produced by the choice of representatives $r \mapsto \widetilde{r}$.

Proof. Write

$$r + n = 9\kappa_r(n) + \widetilde{r'}.$$

When multiplied by four,

$$4r + 4n = 36\kappa_r(n) + 4\widetilde{r'}.$$

Separating quotient and residue modulo nine and subtracting the quotient already contained in $4\widetilde{r}$ yields (1.137). □

Corollary 1.155 (Third iteration)**.** *For every $(b, r) \in Q_{12,9}$,*

$$\boxed{D_4^3(b, r) = ([4b + 7\widetilde{r}]_{12}, r).} \tag{1.138}$$

Proof. By proposition 1.153,

$$\Theta(D_4^3(b, r)) = [4^3(9b + \widetilde{r})]_{108} = [64(9b + \widetilde{r})]_{108}.$$

Since $64 = 1 + 7 \cdot 9$, the residue modulo nine is again r, whereas the block coordinate becomes $[4b + 7\widetilde{r}]_{12}$. □

The return of r is not return of the complete register. The difference $7\widetilde{r}$ preserves the memory of the internal representative traversed.

1.18.3 Obstruction in the observable quotient

In the observable configuration of definition 1.112, we write

$$q = (p^\Sigma, p^\Pi, d^\Sigma, d^\Pi, \theta) \in Q_{\mathrm{obs}}.$$

The positional projection is

$$\pi_{\mathrm{pos}}(q) := (p^\Sigma, p^\Pi) \in X_9^2. \tag{1.139}$$

The temporal signature is

$$\tau(q) := \varepsilon_9(r(\theta)) \in \{-1, 0, +1\}.$$

We define the oriented observable shift

$$a(q) := \begin{cases} (\delta(d^\Sigma), 0), & \tau(q) = +1, \\ (0, \delta(d^\Pi)), & \tau(q) = -1, \\ (0, 0), & \tau(q) = 0. \end{cases} \tag{1.140}$$

Then

$$\pi_{\mathrm{pos}}(F_{\mathrm{obs}}q) = \pi_{\mathrm{pos}}(q) + a(q),$$

with the address update and clock defined in U005.

Theorem 1.156 (Obstruction to the ordinary observable endofunctor)**.** *There is no endofunctor of the ordinary observable space that simultaneously satisfies:*

1. *multiplying the geometric position by four;*
2. *replace each transition by four genuine microtransitions;*
3. *preserving sheet and orientation in each microtransition;*
4. *sending a complete duodecaphase fiber to a single duodecaphase fiber.*

The obstruction persiste in a step of umbral with $a(q) = 0$.

Proof. Let

$$\mathcal{B}_b := \{(b, r) : 0 \le r \le 8\}.$$

Replace one elementary iteration with four force

$$h(\theta + 1) = h(\theta) + 4$$

and, by induction,

$$h(\theta) = 4\theta + c \pmod{108}.$$

Take $0 \le \widetilde{c} < 108$ and write $\widetilde{c} = 9c_b + c_0$, with $0 \le c_0 < 9$. If the whole fiber $\mathcal{B}_b$ were sent to a single fiber, the destination block index would have to be independent of r; however, it contains

$$4b + c_b + \left\lfloor \frac{4r + c_0}{9} \right\rfloor \pmod{12}, \tag{1.141}$$

which is not constant in r. Moreover, the neutral transition operator satisfies $P_0^{\mathrm{TPK}} = I$, distinct from the cyclotomic trivializations P_b; the ordinary calendar $(+1, -1, 0)$ causes four elementary iterations to visit both arithmetic modes and the neutral mode. □

1.18.4 Minimal temporal subdivision compatible with semiconjugation

Definition 1.157 (Subdivided observable state)**.** Let

$$\widetilde{Q}_4 := Q_{\mathrm{obs}} \times \{0, 1, 2, 3\}, \qquad \widetilde{\pi}(q, j) := 4\pi_{\mathrm{pos}}(q) + j\, a(q). \tag{1.142}$$

We define

$$\widetilde{F}(q, j) := \begin{cases} (q, j + 1), & 0 \le j < 3, \\ (F_{\mathrm{obs}}q, 0), & j = 3, \end{cases} \qquad \iota_4(q) := (q, 0). \tag{1.143}$$

Theorem 1.158 (Four-step semiconjugacy)**.** *The inclusion ι_4 satisfies*

$$\boxed{\widetilde{F}^4 \circ \iota_4 = \iota_4 \circ F_{\mathrm{obs}}.} \tag{1.144}$$

The intermediate positions are exactly

$$4\pi_{\mathrm{pos}}(q) + j\, a(q), \qquad j = 0, 1, 2, 3.$$

Proof. Starting from $(q, 0)$, the first three applications increment only the microinstant index. The fourth applies F_{obs} and returns the index to zero. This gives

$$(q,0) \mapsto (q,1) \mapsto (q,2) \mapsto (q,3) \mapsto (F_{\mathrm{obs}}q, 0).$$

The formula for the positions is the definition of $\widetilde{\pi}$. The refinement makes visible a determined subdivision; it does not add a dynamic decision. □

1.18.5 Subdivided category and cyclotomic elevation

Let $\mathrm{sd}_4\,\mathsf{P}_{\mathrm{obs}}$ be the category obtained by replacing each generator $\gamma : q \to F_{\mathrm{obs}}q$ with the chain

$$(q,0) \to (q,1) \to (q,2) \to (q,3) \to (F_{\mathrm{obs}}q, 0). \tag{1.145}$$

The functor

$$\mathrm{sd}_4 : \mathsf{P}_{\mathrm{obs}} \longrightarrow \mathrm{sd}_4\,\mathsf{P}_{\mathrm{obs}}$$

sends each generator to that string. The collapse functor

$$\mathrm{col}_4 : \mathrm{sd}_4\,\mathsf{P}_{\mathrm{obs}} \longrightarrow \mathsf{P}_{\mathrm{obs}}$$

recompose the four microarrows and satisfies

$$\mathrm{col}_4 \circ \mathrm{sd}_4 = \mathrm{id}_{\mathsf{P}_{\mathrm{obs}}}. \tag{1.146}$$

In a chain that remains in the fiber b, the first three microarrows act by ρ_b; the fourth incorporates the interfiber transport if a dodecaphase crossing occurs. By $\rho_b^3 = I$, the ordered product of the four actions coincides with the cyclotomic morphism of the original arrow.

Proposition 1.159 (Cyclotomic Invariance under Subdivision)**.** *There exists a typed lift*

$$\widetilde{\mathfrak{F}}_{\mathrm{cyc}} : \mathrm{sd}_4\,\mathsf{P}_{\mathrm{obs}} \longrightarrow \mathbf{Rep}_{\mathbb{C}}(C_3) \times \mathbf{B}\mathbb{N}$$

such that, for every arrow γ of $\mathsf{P}_{\mathrm{obs}}$,

$$\boxed{\widetilde{\mathfrak{F}}_{\mathrm{cyc}}\big(\mathrm{sd}_4\,\gamma\big) = \big(\mathfrak{F}_{\mathrm{cyc}} \circ Q\big)(\gamma).} \tag{1.147}$$

Proof. The successive powers of ρ_b are assigned to the internal microarrows, and transport between the frames of the source and target fibers is assigned to the fourth. The composition of the first three factors is the identity because $\rho_b^3 = I$. The remaining factor is exactly the morphism that $\mathfrak{F}_{\mathrm{cyc}} \circ Q$ assigns to the collapsed arrow. The subdivided category is free on those microarrows, so the assignment extends uniquely. □

1.18.6 Commutators of 3 quaternary operations and separation of their actions

The architecture contains three distinct operations:

1. $\mathcal{S}_4$, cardinal substitution of objects and paths;
2. D_4, multiplication by four in Θ_{108};
3. sd_4, functorial subdivision of an observable transition.

Their compatibilities are given, respectively, by

$$4^3 = 1 + 7 \cdot 9, \qquad \Theta D_4 = 4\Theta, \qquad \widetilde{F}^4 \iota_4 = \iota_4 F_{\mathrm{obs}}.$$

Sharing the integer four converts none into another.

Comparison Criterion 1.160 (Refutation criteria)**.** The construction is refuted if any of the identities (1.135), (1.137), (1.138), (1.144), or (1.147) fails; likewise if an observable realization satisfies the four conditions of theorem 1.156 without preserving a register equivalent to the microinstant.

1.19 Modular readers, odometer, and dodecaphasic calendar

The enriched state of the TPK is not identified with any of its readings. This unit defines the readers modulo nine, modulo three, and modulo one thousand, the phase–sector odometer, the predictive quotient of thirty-six states, and the cyclic extension of one hundred eight positions. Each construction declares what it publishes and what memory it preserves.

1.19.1 Typed scheme of the modular readers of the TPK: domain, codomain, publication, and conserved memory

Definition 1.161 (Operator specification)**.** A canonical TPK operator is recorded by means of a sextuple

$$O = (D_O, C_O, f_O, I_O, L_O, M_O),$$

where D_O is the domain, C_O the codomain, f_O the rule, I_O the preserved invariants, L_O the information published or forgotten, and M_O the memory that remains in the enriched state.

Two symbols with equal cardinality or equal period do not represent the same operator unless there exists an equivalence intertwining their domains, rules, and invariants. This convention will prevent confusing the predictive quotient C_{36}^{pred}, the boundary R_{36}, and a cardinal defect of value thirty-six.

1.19.2 Nonadic reader of the enriched state: phase, transition and memory-return

For $n \geq 1$, define

$$\rho_9(n) = 1 + ((n-1) \bmod 9), \qquad q_9(n) = \left\lfloor \frac{n-1}{9} \right\rfloor.$$

Then

$$n = 9q_9(n) + \rho_9(n).$$

The card of ρ_9 is

$$\big(\mathbb{Z}_{>0}, \mathcal{D}_9, \rho_9, n \equiv \rho_9(n) \pmod 9, \text{positive phase}, q_9(n)\big).$$

The positive digital root numerically coincides with ρ_9, but its semantics is distinct: one is presented as a section of the quotient; the other, as an iterated sum of digits.

1.19.3 Triticale reader with quotient and carry transport

The balanced division is written

$$n = r_3(n) + 3c_3(n), \qquad r_3(n) \in \mathsf{TRIT}, \quad c_3(n) \in \mathbb{Z}.$$

Its card is

$$\big(\mathbb{Z}, \mathsf{TRIT} \times \mathbb{Z}, n \mapsto (r_3(n), c_3(n)), \mathrm{rec}_3(r, c) = r + 3c, r_3, c_3\big).$$

The publication may display only r_3, but the TPK preserves c_3 if a subsequent composition requires reconstruction.

1.19.4 Reader positional by blocks in base 10^3

The Euclidean division defines

$$N = 1000q_{1000}(N) + r_{1000}(N), \qquad 0 \le r_{1000}(N) < 1000.$$

Its card is

$$\big(\mathbb{Z}, \mathbb{Z} \times \{0, \dots, 999\}, N \mapsto (q_{1000}, r_{1000}), \mathrm{rec}_{1000}(q, r) = 1000q + r, r_{1000}, q_{1000}\big).$$

Proposition 1.162 (Concatenation and residual reading)**.** *If* $u_1, \dots, u_m \in \{0, \dots, 999\}$ *and*

$$D_m = 1000D_{m-1} + u_m, \qquad D_0 = 0,$$

then

$$D_m = \sum_{j=1}^{m} u_j 1000^{m-j}.$$

Moreover,

$$D_m \equiv \sum_{j=1}^{m} u_j \pmod 9$$

because $1000 \equiv 1 \pmod 9$.

Proof. The first equality is obtained by induction on m. The second follows by reducing it modulo nine. □

The congruence does not reconstruct the order of the blocks. The genealogical register preserves the sequence $(u_1, \dots, u_m)$; the residual reader publishes only its sum modulo nine.

1.19.5 Odometer category of the TPK: cylindrical objects and composition by truncation

Definition 1.163 (Nonadic odometer–dodecaphasic)**.** The category $O_{9,12}$ has objects

$$(b, r) \in C_{12} \times \{0, \dots, 8\}.$$

An arrow of length $n \in \mathbb{N}$ is

$$(b, r) \xrightarrow{n} \left(b + \left\lfloor \frac{r+n}{9} \right\rfloor, r + n \bmod 9\right).$$

The composition sums lengths.

Let

$$\kappa_r(n) = \left\lfloor \frac{r+n}{9} \right\rfloor.$$

Lemma 1.164 (Cocycle of the odometer)**.** *If* $r_1 = r + n_1 \bmod 9$*, then*

$$\kappa_r(n_1 + n_2) = \kappa_r(n_1) + \kappa_{r_1}(n_2).$$

Proof. We write $r + n_1 = 9\kappa_r(n_1) + r_1$, with $0 \le r_1 < 9$. Adding n_2 and again dividing by nine yields the identity. □

For a state x whose base contains θ, we define

$$Q_0(x) = (\beta(\theta), \widetilde{\theta} \bmod 9).$$

A TPK arrow has length equal to the number of observable generators that compose it.

Theorem 1.165 (Phase–length functor)**.** *The assignment sending a state x to $Q_0(x)$ and an arrow of length n to the odometric arrow of length n defines a functor*

$$Q : \mathsf{TPK} \longrightarrow O_{9,12}.$$

Proof. Each generator increases θ by one unit. After n steps, the phase residue increases by n modulo nine and the sector increases by $\lfloor (r+n)/9 \rfloor$. The identity has zero length and the lemma 1.164 proves compatibility with composition. □

After nine steps,

$$(b, r) \longmapsto (b+1, r).$$

After one hundred and eight,

$$(b, r) \longmapsto (b, r).$$

These identities belong to the odometric quotient. They do not assert that the cylinder, boundary, genealogical register, or vertical memory have returned.

1.19.6 Memory quotient of order 1 and predictive sufficiency of the modal projection

Let

$$X_{9,12} = \{(a, m) : a \in \mathbb{Z}/9\mathbb{Z}, 0 \le m < 12\}$$

with successor

$$S(a, m) = \begin{cases} (a, m+1), & m < 11, \\ (a+1 \bmod 9, 0), & m = 11. \end{cases}$$

We define the observables

$$\beta_{\mathrm{pred}}(a, m) = 4a \pmod{12}, \qquad d_{\mathrm{pred}}(a, m) = 1 + (m + \beta_{\mathrm{pred}}(a, m) \pmod{12}).$$

For an observable q, the inclusive future word of horizon h is

$$Q_h^q(x) = (q(x), q(Sx), \ldots, q(S^h x)).$$

Theorem 1.166 (Minimum predictive quotient)**.** *The observables β_{pred}, d_{pred}, and the pair $(\beta_{\mathrm{pred}}, d_{\mathrm{pred}})$ produce the same stable quotient*

$$\pi_{36}(a, m) = (a \bmod 3, m).$$

It has thirty-six states and fibers of cardinality three. The minimal horizons are 11*,* 8*, and* 0*, respectively.*

Proof. For a finite system (X, T), let

$$E_h = \mathrm{Eq}(Q_h^q).$$

Then

$$E_{h+1} = E_h \cap (T \times T)^{-1}(E_h).$$

The decreasing chain stabilizes and its limit is the largest T-compatible equivalence contained in $\mathrm{Eq}(q)$. In the declared system, the exact enumeration produces the following number of classes:

h	0	1	2	3	4	5	6	7	8	9	10	11
$\#(X/E_h^{\beta})$	3	6	9	12	15	18	21	24	27	30	33	36

,

and

h	0	1	2	3	4	5	6	7	8
$\#(X/E_h^{d})$	12	15	18	21	24	27	30	33	36.

The pair already distinguishes the thirty-six classes at horizon zero. In all cases, the stable classes preserve m and a mod 3; the three possibilities for a with the same residue modulo three form each fiber. □

We denote this object by C_{36}^{pred}. No is the boundary R_{36}, that will contain three ordered channels of six trits, nor a census of thirty-six elements of another construction.

1.19.7 Central extension of the calendar

We define

$$\iota : C_{12} \longrightarrow C_{108}, \qquad \iota([b]_{12}) = [9b]_{108},$$

and

$$p : C_{108} \longrightarrow C_9, \qquad p([\theta]_{108}) = [\theta]_9.$$

The exact sequence is obtained.

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0. \tag{1.148}$$

A section of sets $s([r]_9) = [r]_{108}$, $0 \le r < 9$, induces the cocycle

$$c(r, r') = \left[\left\lfloor \frac{r + r'}{9} \right\rfloor\right]_{12}.$$

Proposition 1.167 (The extension does not split)**.** *The sucesion equation (1.148) not is isomorfa to the extension product* $C_{12} \times C_9$.

Proof. The exponent of C_{108} is 108. The exponent of $C_{12} \times C_9$ is $\mathrm{mcm}(12, 9) = 36$. Two isomorphic finite groups have the same exponent; therefore, they are not isomorphic. □

Theorem 1.168 (Necessity and exact period of one hundred eight positions)**.** *The translation* $\theta \mapsto \theta + 1$ *on* C_{108} *has order exactly* 108. *Consequently, an iteration n returns the complete calendar if and only if* $108 \mid n$.

Proof. The n-th power sends $[\theta]$ to $[\theta + n]_{108}$. It is the identity for every θ exactly when $[n]_{108} = 0$, that is, when $108 \mid n$. □

In particular, 1008 is not another return of the calendar:

$$1008 \equiv 36 \pmod{108}.$$

After 1008 steps, the phase modulo nine returns, but the coordinate of C_{108} has shifted by thirty-six positions. Naturally, $216, 324, \ldots$ also return the calendar, but they are repetitions of the fundamental circuit of length 108.

1.19.8 Unbounded Growth of Memory Depth Under Phase Returns

Let $q_{\mathrm{mem}} \in \mathbb{Z}_{\ge 0}$ be the unbounded number of nonadic turns, and let

$$\beta = [q_{\mathrm{mem}}]_{12}.$$

Then a nine-step turn satisfies

$$q_{\mathrm{mem}} \longmapsto q_{\mathrm{mem}} + 1, \qquad \beta \longmapsto \beta + 1 \pmod{12}.$$

After one hundred eight steps,

$$\beta \longmapsto \beta, \qquad q_{\mathrm{mem}} \longmapsto q_{\mathrm{mem}} + 12.$$

The calendar returns; the vertical memory does not.

Corollary 1.169 (Hierarchy of returns)**.** *In the phase quotient,* 9 *steps constitute one turn. The composite morphisms*

$$\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}, \qquad F_{\mathrm{obs}}^{54}, \qquad F_{\mathrm{obs}}^{108}$$

contain, respectively, three, six, and twelve nonadic turns. None erases the genealogical record or the boundary by definition.

1.19.9 Minimal type system of the enriched state and compatibilities between its readers

The following table answers explicitly which internal objects already exist in the TPK before introducing seeds:

Objeto u operador	Dominio y codominio	Función
X_9	$(\mathbb{Z}/9)^2$	soporte APP
T_N, T_E, T_S, T_O	$X_9 \to X_9$	traslaciones cardinales
M_{ph}	$\mathcal{D}_9 \to \mathsf{TRIT}$	selección de hoja
F_{obs}	$Q_{\mathrm{obs}} \to Q_{\mathrm{obs}}$	sucesor observable
(ρ_9, q_9)	$\mathbb{Z}_{>0} \to \mathcal{D}_9 \times \mathbb{Z}_{\geq 0}$	fase y altura
(r_3, c_3)	$\mathbb{Z} \to \mathsf{TRIT} \times \mathbb{Z}$	orientación y carry
(q_{1000}, r_{1000})	$\mathbb{Z} \to \mathbb{Z} \times \{0, \ldots, 999\}$	bloque decimal y cociente
Q	$\mathsf{TPK} \to O_{9,12}$	fase y longitud
C_{36}^{pred}	cociente de $X_{9,12}$	memoria predictiva mínima
C_{108}	calendario cíclico	extensión no escindida
q_{mem}	$\mathbb{Z}_{\geq 0}$	memoria vertical no acotada
Ext_r	$\mathrm{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}$	prolongaciones compatibles

1.19.10 Obstructions of the readers and the TPK calendar: Euclidean quotients, C_{36}^{pred}, extension C_{108}, and vertical memory

Comparison Criterion 1.170 (Reader Refutation Criteria and Calendar)**.** The unit is refuted if:

1. a residual reader is presented as a complete transition;
2. the quotient of some subsequently used Euclidean division is omitted;
3. C_{36}^{pred} is identified with R_{36};
4. it is asserted that the extension C_{108} is the product $C_{12} \times C_9$;
5. complete return is declared for a number not divisible by 108;
6. the return of β fuerza $q_{\mathrm{mem}} = 0$;
7. $\mathcal{K}_{27}$ is presented as a distinct clock and not as a power of the transport.

1.20 Complete operatorial structure of APP and arithmetic-geometric hierarchy

The operatorial development of APP fixes the hierarchy $4 \to 2 \to 6 \to 54$, the arithmetic projectors, the central block, and the local geometry of paths. These consequences remain subordinated to the already constructed formal kernel and do not introduce a second origin of APP.

The next unit will construct the quadruple subdivision, the cyclotomic functor, and the internal channels of ninety and one hundred twenty positions. Only afterward will the exhaustive space of seeds be introduced.

1.21 Typed architecture, inventory, and memory conservation of the Topological Phase Kernel

1.21.1 Categorical structure of the TPK: enriched states, transport morphisms, and operatorial classification

The TPK is the categorical container that transports over APP the oriented states of TRIT. Its definition precedes the inventory of realizations: first objects, morphisms, composition, and operatorial classes are fixed; only afterward are the operators that realize each class introduced.

Definition 1.171 (TPK as a category over the observable base with fiber family)**.** Let $\mathsf{P}_{\mathrm{obs}}$ be the category of observable paths over the Cayley graph of APP. For each configuration q the fiber is fixed

$$\mathcal{F}_q = \mathsf{O}_q \times \mathsf{H}_q \times \mathsf{C}_q \times \mathsf{S}_q \times \mathsf{B}_q \times \Lambda_q,$$

whose factors retain, respectively, orientation and sheet; residue and quotient; carry; signature; cylinder and boundary; and transport register. A TPK object is a pair $x = (q, f)$, with $f \in \mathcal{F}_q$. A morphism $\Gamma : x \to y$ is a route $\gamma : q \to q'$ of $\mathsf{P}_{\mathrm{obs}}$ together with the fiber transport that preserves the declared extension relations between f and f'. The identity is the empty route with identity transport, and composition is the concatenation of routes with composition of the fiber transports. The projection

$$p : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow \mathsf{P}_{\mathrm{obs}}$$

sends each state to its observable configuration and each morphism to the path that transports. This definition fixes a category over the base and a family of fibers; the existence is not presupposed the existence of Cartesian lifts for every flecha of $\mathsf{P}_{\mathrm{obs}}$.

Definition 1.172 (Ontological–operational classification of the TPK)**.** The internal structure of the TPK is organized into eight classes distinguished by type of domain, codomain, and function:

1. **Support and state.** The base is $\mathsf{P}_{\mathrm{obs}}$ and the carrier is $\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \to \mathsf{P}_{\mathrm{obs}}$. This class contains objects; the next seven contain maps between objects or readers of those objects.

2. **Internal selection.** The phase function

$$M_{\mathrm{ph}} : \{1, \ldots, 9\} \longrightarrow \{+1, -1, 0\}$$

produces the scalar value that the operator Sel_t receives to activate the additive evaluation, the multiplicative evaluation or the threshold step. It is Sel_t, not M_{ph} in isolation, that acts on the sheet coordinate; neither modifies the position in APP by itself.

3. **Transport.** For each cardinal direction d, $T_d : X_9 \to X_9$, $p \mapsto p + \delta(d)$, it transports the position. The evolution $F_{\mathrm{obs}} : Q_{\mathrm{obs}} \to Q_{\mathrm{obs}}$ composes selection, translation, orientation, and phase advance.

4. **Readers and quotients.** ρ_9 publishes the nonadic class; (r_3, c_3) conserves tritic residue and quotient; and (q_{1000}, r_{1000}) separates decimal block and carry. They are reading maps, not state transitions.

5. **Memory and boundary.** The transports of $\mathsf{H}_q, \mathsf{C}_q, \mathsf{S}_q, \mathsf{B}_q$ and Λ_q update quotient, carry, signature, cylinder, boundary, and record. Their visible projections may coincide even though the enriched states differ.

6. **Periodicity and return.** The extension

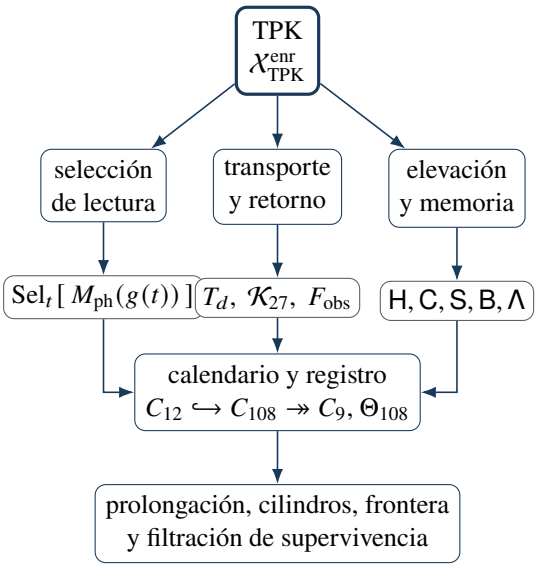
$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0$$

types sector, calendar, and visible phase. The nonadic return restores the phase; the periodicity of one hundred eight iterations restores the calendar; neither erases by definition the boundary memory. The loop $\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}$ is a composite morphism, not another clock.

7. **Prolongation.** The relations $\mathrm{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}$ select compatible continuations. The cylinders and their inverse system publish finite survival and, under the stated compatibility hypotheses, a global section.

8. **Outputs.** The evaluation functors contract the same enriched state toward the Archimedean output, the exceptional-incidence output, or the classifiers of the atlas. These outputs are codomains of reading; they do not constitute new containers alongside TPK.

Each operator appearing below is determined by the sextuple (D, C, f, I, L, M): domain D, codomain C, rule f, preserved invariants I, published or lost information L, and retained memory M. Equality of cardinalities, periods, or historical names does not identify two operators with distinct sextuples. Once this classification has been fixed, the following diagram summarizes five functional groupings of realizations that refine the eight operator classes of TPK:



The diagram fixes a containment, not a succession of theories. In particular, the internal selector, the cardinal translations, the stroboscopic observation, and the modular readers preserve the types fixed in the definition 1.172. Remark. The TPK coordinates their compositions and conserves their provenance in a single enriched state.

Proposition 1.173 (Separation of selection, motion, and remark)**.** *The selection* $\mathrm{Sel}_t\,[\,M_{\mathrm{ph}}(g(t))\,]$*, the cardinal transport* $\mathrm{Lift}(T_d)$ *and the sampling* Obs_3 *are noninterchangeable operations: the first chooses the active evaluation; the second modifies the position and winding; the third publishes a temporal subsequence. Their composition within the TPK determines an observable route, while the fiber* $\mathcal{F}_q$ *preserves the data needed to distinguish prolongations with the same endpoint.*

Proof. The types are different: M_{ph} takes values in the ternary activation set, Sel_t updates the sheet coordinate, $\mathrm{Lift}(T_d)$ transports the state on the torus and Obs_3 acts on sequences. An identification between two of these objects would violate domain or codomain typing. The extension by fibers also incorporates residue–quotient, carry, boundary and record, so projection to the path endpoint is an explicit forgetful map. □

Canonical nomenclature

The public name of the container is *TPK*. When it is necessary to name the transported data, *enriched TPK state* or $\mathcal{X}^{\mathrm{enr}}_{\mathrm{TPK}}$ will be used. The function M_{ph} designates exclusively the internal phase value received by Sel_t.

1.21.2 Fundamental operators of the TPK: domains, actions, invariants, and memory

The definition 1.172 first fixed the eight ontological–operatorial classes: support and state; selection; transport; readers and quotients; memory and boundary; periodicity and return; prolongation; and outputs. This section defines their realizations. For each operator, the sextuple (D, C, f, I, L, M) of domain, codomain, rule, invariants, published or lost information, and preserved memory is declared. The subsequent nominal census groups symbols already defined by these six data and does not replace their domains or rules.

Definition 1.174 (Typed Record for a TPK Operator)**.** A canonical operator is a sextuple

$$O = (D_O, C_O, f_O, I_O, L_O, M_O),$$

where $f_O : D_O \to C_O$ is the action rule, I_O is the family of relations it preserves, L_O declares the published coordinates and the equivalence relation induced by forgetting, and M_O enumerates the data that remain in the enriched state after publication. Two realizations represent the same operator only when a typed equivalence exists that intertwines their rules and invariants; sharing a period, cardinality, or visible output is insufficient.

1.21.2.1 Modular readers built

The nonadic reduction is fixed by the published representative.

$$\rho_9(n) = 1 + ((n-1) \bmod 9) = n - 9\left\lfloor \frac{n-1}{9} \right\rfloor, \tag{1.149}$$

so that the null class is written as 9. For $n \geq 1$, the digital root satisfies

$$\mathrm{dr}_9(n) = \rho_9(n),$$

but the two expressions preserve distinct semantics: the first is a section of the quotient; the second publishes the iterated sum of digits. The tritic lift writes uniquely

$$n = r_3(n) + 3c_3(n), \qquad r_3(n) \in \{-1, 0, +1\}, \quad c_3(n) \in \mathbb{Z}, \tag{1.150}$$

and simultaneously preserves oriented residue and carry quotient. Finally, the decimal-block reader is the Euclidean division

$$N = 1000q_{1000}(N) + r_{1000}(N), \qquad 0 \leq r_{1000}(N) < 1000, \tag{1.151}$$

therefore basis 1000, residue modulo 1000 and carry q_{1000} are three data related and not a same object.

1.21.2.2 Classification of Operator Realizations by Domain and Function

The preceding definitions govern the reading of the table. Its fourteen functional groupings refine the eight operator classes and summarize the realizations that share a function; a class may therefore occupy more than one row. The table does not operate as a replacement definition. Symbols that share a cardinal remain separated by domain and codomain.

Functional grouping	Operators and Spaces	Domain, Action and Invariants
State and route memory	$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}$, fiber $\mathcal{F}_q$, route memory	They preserve position, leaf, orientation, residue, quotient, carry, signature, boundary, cylinder, prefix, genealogy, and memory.
Sheet selection	$M_{\mathrm{ph}} : \{1, \dots, 9\} \to \{-1, 0, +1\}$	It activates the additive evaluation, the multiplicative evaluation, or the threshold passage; it does not by itself move the state.
Transporte espacial	T_N, T_E, T_S, T_O, reversible two-cursor engine, F_{obs}	It realizes the translations of the Cayley graph, updates sheet and orientation and preserves each transition of the path.
Lectores modulares	ρ_9, (r_3, c_3), (q_{1000}, r_{1000})	Publish nonadic phase, tritic orientation and decimal block without erasing the lift quotients.
Block and memory	$\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}$, memory filter $\mathcal{N}_{27}$	The first composes twenty-seven transports; the second selects the memory events. Equal cardinality does not imply an equal operator.
Twelve-phase state	$\mathcal{R}_{12}$, centers $\theta_m = 30m - 10$ degrees, refinement $\mathcal{R}_{120}$	Organize the twelve phases, their opposition, and the angular refinement in increments of three degrees.
Periodicity and depths	$C_{12} \hookrightarrow C_{108} \twoheadrightarrow C_9, \Pi_{108}^{\mathrm{cur}}, w_{108}, w_{1080}, \Gamma_0(1080)$	It distinguishes the total periodicity of order 108, the cursor projection, the words of 108 and 1080 trits, and modular level 1080.
Channels (90/120)	mask C_m, S_{90}, S_{120}, orbital grammar, supertrace, and invariants ν_{120}, ν_{270}	It separates the two genealogies, their geometric tails, their weights, and the coordinates that they extract from the state.
Bidirectionality	$P_{\pm}$, route inversion, $\mathrm{Ext}_{\pm}$, $N_{\pm}$, β, $C_M = -\mathrm{rev}$	It implements, respectively, spectral exchange, spatial outward passage and return, future/past prolongation, valence, balance, and word conjugation.
Emission and coinduction	L_0, L_1, R_{36}, G_9, partition $3^6 = 729$, $\varprojlim \mathrm{Cyl}_m$	It lifts the seed, opens the boundary, and exhaustively enumerates the finite prolongations. The numerical limit and the enriched lift belong to distinct types.
Numeric Readers	$\mathscr{C}_\pi$, $\mathscr{P}_e$, F_{av}	Through rational forks, factorial recurrence, and Fibonacci self-scaling, they construct the published sections of π, e, φ.
Exceptional Projection	Witt vector lifting, weight enumerator, and non-selective shadow	Transports the same state to ternary incidence without intervening in the Archimedean selection of digits.
Atlas and realization	route signature, atlas (81/56/13/324), involution C_M, electronic center, and fiber operators	Convert the route state into classifiers, characters, and operators before applying any dimensional chart.
Finite validation	exhaustive censuses, ablations, and causal independence	It checks coverage and uniqueness at the computed depth, and verifies that no target prefix intervenes in the generating map.

1.21.2.3 Functional Index of the Canonical Operators

The preceding classification organizes the realizations by function. The following table gathers the operators after the categorical definition and the constructed readers. Each row declares domain, codomain, rule, and invariants; the symbols make it possible to recognize a single definition throughout its different realizations without identifying operators of different types.

#	Canonical Operator	Symbol	Capa tipada
1	Additive–multiplicative arithmetic cell of order nine	$\mathcal{A}_9$	arithmetic support
2	Residual decomposition and nonadic quotient	(ρ_9, q_9)	reduction and elevation
3	Ternary oriented reduction	ρ_3^{or}	reduction and elevation
4	Ternary-decimal positional reader of base thousand	ι_{1000}	reduction and elevation
5	Cardinal Action Oriented on the Arithmetic Torus	$T_d,\ d \in \{N, E, S, O\}$	transporte cardinal
6	Additive–multiplicative local feedback	F_{loc}	transporte cardinal
7	Minimum emitter of two cursors	$T_{\min}$	finite emission
8	Decimal Arithmetic Sign Coupler	G	finite emission
9	Ternary projection of the catalog	Ψ_6	finite emission
10	Local Surjectivity of Cardinal Transport	$\mathcal{R}_3$	transporte cardinal
11	Transportation loop of twenty-seven iterations	$\mathcal{K}_{27}$	transporte cardinal
12	Enriched state of the topological phase core	$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}$	estado enriquecido
13	Hensel State Transition	$\mathcal{H}$	estado enriquecido
14	Exact Transition of Ternary-Decimal Cylinders	$C_{3,1000}$	estado enriquecido
15	Joint boundary signature	Σ_{B}	estado enriquecido
16	Relationship of extension and nonadic survival	EF–G_9	prolongation and monodromy
17	Nonadic monodromy with boundary holonomy	$\mathcal{M}_9$	prolongation and monodromy
18	Global Section of the Inverse Survivor System	$X_\chi = \varprojlim_m \mathrm{Surv}_m^{(\chi)}$	prolongation and monodromy
19	Bivariant Category of Arithmetic Paths	$\mathcal{P}_{\mathrm{APP}}^{(2)}$	transporte cardinal
20	Observable chronology of two cursors	F_{obs}	transporte cardinal
21	Decimal Alphabet of Forty-Two Trinities	$\mathcal{A}_{42}$	finite emission
22	Typed extension relation of fibers	Ext_r	estado enriquecido
23	Nonadic transfer of compatible fibers	K_{ph}	prolongation and monodromy
24	Refinement of Seeds by Advancement and Rotation	(P, H, Q)	transporte cardinal
25	Mode and Orientation Cocycles	$(c_{\mathrm{mode}}, c_{\mathrm{or}})$	estado enriquecido
26	Triadic prolongation of twenty blocks	E_{20}	prolongation and monodromy
27	Dodecaphonic oriented system	$\mathcal{R}_{12}$	dodecaphonic reading
28	Signed harmonic dodecaphonic observable	U_{12}^{sgn}	dodecaphonic reading
29	Harmonic transformation by three-step orbits	$\mathcal{H}_{12}$	dodecaphonic reading
30	Cyclic difference transversal extractor	$\mathcal{E}_{3,4}$	dodecaphonic reading
31	Dodecaphasic closure recurrence in base thousand	C_α	close in base thousand
32	Enhanced history transducer to angular character	$(L_{\mathrm{tick}}, \chi_{\mathrm{mass}})$	interfaz dimensional
33	Bidiirectional lattice of index twelve	$\mathcal{W}_\pm$	interfaz dimensional
34	Nonadic Atlas of Extension Families	$\mathcal{A}_{\mathrm{species}}$	interfaz dimensional

The symbol L_{tick} denotes the iteration and return reader. Its codomain is the temporal memory of the state; it does not define an independent physical time unit.

The internal selector function

$$M_{\mathrm{ph}} : \mathcal{D}_9 \longrightarrow \{+1, -1, 0\}$$

belongs to the TPK; its tabulated vector is

$$m_{\mathrm{ph}} := \big(M_{\mathrm{ph}}(1), \ldots, M_{\mathrm{ph}}(9)\big)^{\mathsf{T}}.$$

The function M_{ph}, the vector m_{ph} and the state operator are objects of different types. The scalar value $M_{\mathrm{ph}}(g(t))$ enters as the second argument of the selector Sel_t; it is not composed directly with the state. Cardinal transport T_d occupies another factor of the transition $\mathcal{U}_t$ and performs movement on APP through its lift $\mathrm{Lift}(T_d)$. This separation preserves a single container, distinguishes phase selection from transport, and preserves the complete architecture by types.

Finite depth and enriched limit

The enumeration, censuses, and contraction of numerical cylinders are exact results in their domains. The existence and uniqueness of an enriched section jointly subject to all survival predicates follow when the lifting fibers are nonempty, singletons, and compatible at every depth. A calculation on a finite window verifies the fibers in that window, but does not by itself imply the quantifier over all depths.

Theorem 1.175 (Composition of the internal dynamics)**.** *A transition observable of the TPK is the composition typed*

$$\mathrm{Tra}_t(y,d) := \mathrm{Lift}(T_d)(y), \qquad \mathrm{Upd}_t := \mathrm{Rec}_t \circ \mathrm{Mem}_t,$$

$$\mathcal{U}_t(x) = \mathrm{Upd}_t\big(\mathrm{Tra}_t\big(\mathrm{Sel}_t\big(x, M_{\mathrm{ph}}(g(t))\big), d(t)\big)\big)$$

on $\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}$, followed, when appropriate, by a modular reader, a cylindrical extension, or one of the two projections. Each factor acts on a distinct component of the state; for that reason the selector, the movement, the clock, the memory, and the publication cannot collapse into a single modular reduction.

The preceding table enumerates the operator families. Equivalences among local realizations must intertwine their domains, rules, and invariants. Subsequent developments select compositions of these operations; they introduce no fourth primitive object alongside APP, TRIT, and TPK.

1.22 Transport, selection, and prolongation in the Topological Phase Kernel

The acronym TPK designates the *Topological Phase Kernel*, or the topological phase kernel.[1] It does not denote a single 108-step algorithm, a table of results, or the catalog of 468 emissions. It is the system of operators that enables the additive and multiplicative readings to coexist over the APP path category without identifying their genealogies.

The word *topological* has a precise content here. The support of APP is the Cayley graph of the quotient group

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2$$

with its cardinal edges. A closed route admits a winding class after being lifted to cover $\mathbb{Z}^2$. TPK transports that class, the route orientation, and phase holonomy. Two paths ending in the same cell may therefore represent distinct states. The phase kernel is precisely the information that remains invisible when publishing only the final cell or its residue.

The dynamical basis is the free category $\mathsf{P}_{\mathrm{APP}}$ of the cardinal graph of APP. Since the two arithmetic evaluations do not induce the same partition, transport resides on the bivariate category

$$\mathsf{P}_{\mathrm{APP}}^{(2)} = \mathsf{P}_{\mathrm{APP}} \times \mathsf{P}_{\mathrm{APP}}.$$

The first component is evaluated through the additive sheet Σ and the second through the multiplicative sheet Π. This separation is constitutive, but it does not duplicate APP: it records two ordered readings of one and the same matrix of paths. The multiplicative component is the principal table; the additive component preserves the law of accumulation that generates its rows and makes it possible to measure what information a visible projection eliminates.

[1] The denomination likewise preserves the authorial dedication to Tan Peng Kian.

1.22.1 Residual reductions and decimal representation

The architecture uses three reductions, each with a distinct purpose:

$$\mathbb{Z} \xrightarrow{(\rho_9, q_9)} \{1, \dots, 9\} \times \mathbb{Z}, \qquad \rho_3^{\mathrm{or}} : \mathbb{Z}/3\mathbb{Z} \longrightarrow \{-1, 0, +1\},$$

$$(d_1, d_2, \dots) \in \{0, \dots, 999\}^{\mathbb{N}} \xrightarrow{\iota_{1000}} \sum_{m \geq 1} d_m 1000^{-m}.$$

The ternary orientation is

$$\rho_3^{\mathrm{or}}([0]) = 0, \qquad \rho_3^{\mathrm{or}}([1]) = +1, \qquad \rho_3^{\mathrm{or}}([2]) = -1.$$

The first separates the visible residue from the nonadic quotient; the second assigns the oriented type of the transition; the third groups the Archimedean reading into decimal triads, whose register preserves the carry. None can replace another. In particular, reduction modulo three does not reconstruct the quotient modulo nine, and a decimal triad is not a six-trit word. Addition accumulates displacements; multiplication folds classes and contracts the noninvertible rows. The need for TPK arises because the two readings share a support but do not share a partition, period, or quotient.

1.22.2 Correspondence between TPK symbols, domains, and compositions

Historical names are fixed here by object, not by numerical similarity. This table forms part of the public definition of TPK.

Name	Object	Function within the core
nonadic coordinate	$\rho_9(n)$ together with $q_9(n)$	Separa the representative visible modulo 9 of the quotient that preserves the memory of the cruce of boundary. No is a root digital.
ternary temporal selector	$M_{\mathrm{ph}}(t) = \varepsilon(t) \in \{+1, -1, 0\}$	At $1, 4, 7$ it activates additive accumulation; at $2, 5, 8$ it activates multiplicative accumulation; at $3, 6, 9$ it maintains both cursors and records the threshold. It does not denote the spatial directions.
transporte cardinal	T_N, T_E, T_S, T_O	It shifts the position on the APP torus and preserves the path word and its winding. It is independent of the temporal selector.
ternary reader	ρ_3^{or}	It converts the three residual classes into orientation $0, +1, -1$. It does not by itself reconstruct the nonadic quotient.
reader in base thousand	ι_{1000}	It groups the Archimedean publication into decimal triads and transports their carries. It is a positional reader, not a congruence "modulo 1000".
return of twenty-seven	$\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}$	It composes twenty-seven positional and oriented updates. Its fourth iterate restores the observable phase, while the complete register preserves the accumulated memory.
dodecaphonic register	$\Lambda \in \mathcal{R}_{12}$	It preserves six positions, five differences, and one total. It is the support of the reversible transport $1 + 5 + 6 = 12$.
bidirectional reading	q_+, q_- and its conjugate weights	It simultaneously preserves the outward and return orientations. In Dimensional Mechanics, a signature (n_A, s_C) first produces $W_\pm = 6n_A \pm s_C$; only afterward does $s_C/6$ appear in the angular character.

The temporal selector, cardinal transport, and ternary orientation are therefore three distinct maps. Their separation preserves the difference between *when* a reading is activated, *in which direction* the route advances, and *with which orientation* the step is recorded. TPK contains all three and preserves their relations.

1.22.3 Cardinal Action and Arithmetic Feedback

Let

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1)$$

the four generators cardinal. For $d \in \{N, E, S, O\}$,

$$T_d(i, j) = (i, j) + \nu_d \pmod 9.$$

The letter O means west. These directions are displacements over APP and not names of the channels π, e, φ.

A word $w = d_1 \cdots d_r$ determines

$$\gamma(x_0; w)_k = x_0 + \sum_{j=1}^{k} \nu_{d_j} \pmod 9.$$

When it closes, its lift preserves

$$\mathrm{wind}(w) = \frac{1}{9}\big(\#S - \#N, \#E - \#O\big) \in \mathbb{Z}^2.$$

The bidirectional reading preserves word, orientation, and sheet. Its formal reversal inverts the order and exchanges $N \leftrightarrow S$, $E \leftrightarrow O$. Its eventual reading as advance–delay is a later physical interface, not part of this route definition.

The locally observable state is

$$s_t = (i_t, j_t, m_t), \qquad m_t \in \{\Sigma, \Pi\}.$$

The active sheet is evaluated,

$$a_t = \rho_3^{\mathrm{or}}\big(m_t(i_t, j_t)\big),$$

and the mode is updated via

$$m_{t+1} = \begin{cases} \Sigma, & a_t = +1, \\ \Pi, & a_t = -1, \\ m_t, & a_t = 0. \end{cases}$$

Thus, each cardinal step modifies both the position and the future arithmetic reading. The path is not superimposed on a previously constructed word: it is the positional history that produces it.

Theorem 1.176 (Surjectivity of the sampled three-step factor)**.** *For each of the* $162 = 9^2 \cdot 2$ *states* (i, j, m) *and each target oriented trit* $b \in \{-1, 0, +1\}$*, there exists a route* $u \in \{N, E, S, O\}^3$ *whose third step emits* b*. In the* 486 *state–target cases, the number of admissible routes is between* 8 *and* 40*. Consequently, for every target succession* $(b_k)_{k \geq 1}$*, there exists a cardinal history whose output satisfies*

$$(a_3, a_6, a_9, \ldots) = (b_1, b_2, b_3, \ldots).$$

The two intermediate trits a_{3k-2} *and* a_{3k-1} *of each block are not prescribed by this assertion.*

Proof. The $4^3 = 64$ three-step routes from each state are enumerated. The included exhaustive certificate records, for each of the $162 \cdot 3$ cases, the route executed, the final state, and the total number of witnesses; its indices $0, 1, 2$ are transported by ρ_3^{or} to $0, +1, -1$. All counts are positive; the minimum is 8 and the maximum is 40. After each block, the same result is applied to the state reached. The choice, for example, of the first witness in lexicographic order recursively produces a history for any target sequence. □

Surjectivity concerns the factor observed every three steps, not the word emitted at unit time. This theorem establishes that sampled capacity of the route support. The selection of the sections π, e, φ belongs to the enriched EF–G9 relation defined later: capacity for realization and selection that is genealogical are two distinct operators.

1.22.4 Chronological evolution of the cursor pair and determination of the observable state

The calendar is formulated over

$$\Theta_{108} = \mathbb{Z}/108\mathbb{Z}, \qquad \beta(\theta) = \left[\left[\frac{\widetilde{\theta}}{9}\right]\right]_{12}, \qquad r(\theta) = 1 + (\theta \bmod 9),$$

where $\widetilde{\theta} \in \{0, \dots, 107\}$. The active signature is

$$\varepsilon(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

The observable space of two cursors is

$$Q_{\mathrm{obs}} = X_9^2 \times \{N, E, S, O\}^2 \times \Theta_{108}.$$

If $\varepsilon = +1$, the additive cursor advances; if $\varepsilon = -1$, the multiplicative cursor advances; if $\varepsilon = 0$, neither advances. The two orientations are reversed after steps $27, 54, 81, 108$. This rule defines a permutation

$$F_{\mathrm{obs}} : Q_{\mathrm{obs}} \longrightarrow Q_{\mathrm{obs}}.$$

Positions and orientations are restored after 54 steps and the complete phase after 108:

$$F_{\mathrm{obs}}^{108} = I.$$

Therefore, the chronology of a single route is reversible and does not by itself select a distribution on the continuations; selection of all compatible prolongations belongs to the fiber transfer operator defined below.

1.22.5 Finite emitter associated with the cursor pair and generation of admissible states

A cursor seed is

$$\mathcal{S} = \mathbb{T}_9^2 \times \{N, E, S, O\}, \qquad |\mathcal{S}| = 324.$$

Minimal TPK receives an additive cursor $P \in \mathcal{S}$ and a multiplicative cursor $M \in \mathcal{S}$. Their six-window signatures are $s(P)$ and $e(M)$. In each coordinate one applies

$$G(s, e) = 100s + 10\big((s + e) \bmod 10\big) + \big((3s + 5e + 1) \bmod 10\big).$$

The figure of hundreds recovers s; known s, the of tens recovers e. Therefore G is injective on the effective signatures. The emitter factorizes as

$$\mathcal{S}^2 \xrightarrow{s \times e} \operatorname{im} s \times \operatorname{im} e \xrightarrow{G} \mathcal{U}_6,$$

with

$$|\mathcal{S}^2| = 104\,976, \qquad |\operatorname{im} s| = 18, \qquad |\operatorname{im} e| = 26, \qquad |\mathcal{U}_6| = 18 \cdot 26 = 468.$$

The factorization $468 = 18 \cdot 26$ is therefore a decomposition of the object: each emission conserves an additive and a multiplicative signature.

The projection

$$\Psi_6 : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \qquad \Psi_6(U) = -U \pmod 3,$$

has an image of 243 words. The chain typed is

$$104\,976 \text{ seeds} \longrightarrow 468 \text{ decimal emissions } U_6 \longrightarrow 243 \text{ ternary words } w_6.$$

The ambient space $\mathbb{F}_3^6$ contains 729 words; it is not identified with the effective image of 243. Nor are U_6 and w_6 the same type.

The emitter minimo repite, in the chart dodecaphasic of catalogo,

$$U_{12}^{\mathrm{cat}} = U_6 \| U_6.$$

Its visible period six explains why it cannot replace the signed state U_{12}^{sgn}. This difference is not an insufficiency of TPK: it separates the catalogue projection from the enriched history.

Each emitted decimal block belongs to an exact alphabet of $7 \cdot 6 = 42$ triads. If $\overline{d_1 d_2 d_3}$ is one of them,

$$\boxed{d_3 \equiv 5d_2 - 2d_1 + 1 \pmod{10}.}$$

The sweep of the 104 976 seeds produces no triad outside this law. It is therefore useful to distinguish the complete chain of quotients:

$$\mathcal{S}^2 \longrightarrow \Omega_{\mathrm{seed}} \longrightarrow \mathcal{U}_6 \xrightarrow{\Psi_6} \operatorname{im}\Psi_6 \subset \mathbb{F}_3^6.$$

The microscopic presentations, the 468 decimal emissions, the 243 ternary words, and the environment of 729 words are four distinct objects.

Orbital Commutativity of Typed Morphisms. Let $W_{243} := \operatorname{im}\Psi_6 \subset \mathbb{F}_3^6$. The complete funnel and its dihedral quotient are represented without turning inclusions into selections:

$$\underbrace{\mathcal{S}^2}_{104\,976 \text{ semillas}} \xrightarrow[\text{fiber census}]{\text{emission}} \underbrace{\mathcal{U}_6}_{468\ U_6} \xrightarrow[\text{visible projection}]{\Psi_6} \underbrace{W_{243}}_{243\ w_6} \hookrightarrow \underbrace{\mathbb{F}_3^6}_{729 \text{ palabras}}, \qquad W_{243} \xrightarrow{/D_3^{\mathrm{cat}}} \underbrace{\mathcal{O}_{43}}_{43 \text{ orbits}}.$$

(TPK-embudo)

The first arrow groups initial conditions by the emission they produce; the second forgets the decimal presentation; the third is an inclusion in the algebraic ambient space; the last is a quotient by the action D_3^{cat} on W_{243}. In particular, there is no implicit quotient $729 \to 43$. The exact census is

$$243 = 38 \cdot 6 + 5 \cdot 3,$$

that is, thirty-eight orbits of size six and five of size three. These 43 orbital classes are not the 43 complete nonadic turns per channel of the thousand-digit G9 witness either: the first 43 belongs to the finite quotient $W_{243}/D_3^{\mathrm{cat}}$; the second is a depth counter of a certified run.

1.22.6 Configuration of the enriched state of the TPK: sheet, orientation, residue, quotient, carry, boundary, route, and memory

The complete dynamic unit is a history, not an isolated word. At depth m we write

$$v_m = \big(B_m, \boldsymbol{\delta}_m, \boldsymbol{x}_m, \boldsymbol{C}_m, \Sigma_m, \omega_m, \ell_m, g(m), \Lambda_m\big),$$

where

$$B_m = (u_m^{\pi}, u_m^{e}, u_m^{\varphi}) \in (\mathbb{F}_3^6)^3.$$

Their components have distinct functions:

1. B_m is the joint visible block of the three channels;
2. $\boldsymbol{\delta}_m$ retains the new decimal triads and the associated carry;

3. $\boldsymbol{x}_m$ is the deep Hensel state;
4. $\boldsymbol{C}_m$ contains the ternary and decimal cylinders and their integer residues;
5. Σ_m is the signature joint of boundary;
6. ω_m and ℓ_m conserve orientation and bidirectional sheet;
7. $g(m) \in \mathbb{Z}/9\mathbb{Z}$ is the nonadic phase;
8. Λ_m is the dodecaphasic record oriented.

Definition 1.177 (Operatorial presentation and TPK category)**.** The operator inventory of the enriched state of TPK is

$$\mathfrak{T}_{\mathrm{HMT}} = \left(\mathcal{A}_9, \widehat{\mathbb{T}}, T_{N,E,S,O}, T_{\min}, H, C_{3,1000}, \Sigma_B, EF, \mathcal{G}_9, \mathcal{R}_{12}\right)$$

that transports histories $(v_0, \ldots, v_m)$ and their truncations. Its formal carrier is the category TPK over the observable base: its objects are enriched states, and its morphisms are the prolongations Ext_r stable under identity and concatenation, defined below. The inventory does not replace that category. The visible projection eliminates components of v_m; it does not define an equivalence with the complete state.

This definition contains exactly what disappears when cursors are reset or only the residue is conserved: the path, the sheet, the quotient, the carry, the orientation, and the boundary. TPK is, for that reason, the point at which modular arithmetic ceases to be an operation without history.

1.22.7 Ternary-decimal cylinders and state of extension

Let N be the integer determined by a ternary prefix of length L, and let D be the integer determined by K decimal triads. We define

$$A = N1000^K - D3^L, \qquad B = (N+1)1000^K - D3^L.$$

The cylinders intersect exactly when

$$\boxed{A < 3^L, \qquad B > 0.}$$

When incorporating a ternary block $u \in \{0, \ldots, 728\}$,

$$N' = 729N + u.$$

If no appears a new triad decimal,

$$A' = 729A + u1000^K.$$

Si aparece $\delta \in \{0, \ldots, 999\}$,

$$D' = 1000D + \delta,$$

$$A' = 729000A + u1000^{K+1} - \delta\, 729\, 3^L.$$

In both cases,

$$B' = A' + 1000^{K+\Delta K}.$$

These integer recurrences transport the decimal datum without turning it into an external target of the update.

The Hensel state preserves the information that does not appear in the residue. The active matrix is

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad \det A_H = 1,$$

and the residue–quotient division is

$$y = xA_H, \qquad u = y \bmod 3, \qquad x^+ = \frac{y-u}{3}.$$

For the three channels, the joint cocycles are additionally formed

$$q = (\pi + e + \varphi) \bmod 3, \qquad c = \frac{\pi + e + \varphi - q}{3},$$

$$a = (\pi + e - \varphi) \bmod 3, \qquad h = \frac{\pi + e - \varphi - a}{3}.$$

The second-order update corrects the next state with c and h. Two branches with the same visible block are therefore not necessarily the same state: they may differ in their Hensel extension and in their cylinders. Since $\det A_H = 1$, the reduction of A_H belongs to $\mathrm{GL}_6(\mathbb{F}_3)$. Hence, before cylinders, boundaries, or survival are imposed, the residual Hensel chart covers exactly $\mathbb{F}_3^6$. This raw coverage does not imply coverage of the filtered system.

Formally, over each observable configuration q a fiber is fixed

$$\mathcal{F}_q = \mathsf{O}_q \times \mathsf{H}_q \times \mathsf{C}_q \times \mathsf{S}_q \times \mathsf{B}_q \times \Lambda_q,$$

corresponding, respectively, to orientation and sheet, residue and quotient, carry, signature, cylinder and boundary, and transport record. For a cardinal arrow $r : q \to q'$, the admissible prolongation does not reduce to a function on residues; it is the relation

$$\mathrm{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \qquad \mathrm{Ext}_{r_2} \circ \mathrm{Ext}_{r_1} \subseteq \mathrm{Ext}_{r_2 \circ r_1}.$$

The carry increment is a cocycle:

$$\kappa(r_2 \circ r_1) = \kappa(r_2) + \mathsf{C}(r_2)\big(\kappa(r_1)\big).$$

Thus the forgetful map is typed: projecting an extension to its observable route eliminates components of $\mathcal{F}_q$, but does not authorize identifying both objects.

1.22.8 Compound returns and transport calendars

The length of a trace does not determine its type. The academic presentation separates the composite return, calendar, trace, and atlas according to their respective preserved states.

The symbol $\mathcal{K}_{27}$ is reserved for the composition of twenty-seven updates of a positional and oriented transport loop. The integer 27 is its composition length; this does not turn it into a trace of 108 iterations or into an infinite emitter. Another construction, historical in origin, groups four loops whose joint periods reach 108; that joint rotor is not denoted by $\mathcal{K}_{27}$ in isolation.

Object	Estado conservado	Significado de 108
Contador simplificado	position and velocity	its real period is 18; it is a historical control.

Object	Estado conservado	Significado de 108
Rotor set of four loops	four routes, reading, rotation, and motion	the individual periods are 54, 54, 36, 36; the joint state has period 108.
Raw clocked trace	four positions, orientations, and phase	the dynamic content repeats at 54.
Trace of $T_{\min}$	two positions and two directions	the state returns to 54 and the output is $U_6 \| U_6$.
Reinitialized pseudocode	cursors reset each window	collapses to $(222, 222, 222, 333, 333, 333)^2$; it does not represent the enriched state of TPK.
Rutas condicionadas	path and prescribed word	108 is horizon, not period.
Sectorial odometer	$\mathbb{Z}/9\mathbb{Z} \times \{0, \dots, 11\}$	the complete state has period 108; the sectorial observable, 36.
Projection raw	quotient by forgetfulness of $X_{\mathrm{TPK}}^{\mathrm{enr}}$	does not possess by itself the signed reading.
Traza enriquecida	all the fields of v_m during 108 updates	is the temporal restriction of the enriched state of TPK.
Atlas of Routes	81 cells/56 signatures or 324 oriented seeds	is a taxonomy, not an orbit of 108 states.

The loop $\mathcal{K}_{27}$ is the finite transport operator of twenty-seven updates. The windows, joint rotor, and calendrical projection receive distinct symbols, so that the shared length does not replace the domain or composition of each object.

1.22.9 Fiber transfer and spin memory

The factorization $\mathcal{U}_6 \simeq \mathcal{S}_+ \times \mathcal{S}_\times$ allows one to define, with respect to the uniform count, the conditional expectations

$$(E_+ f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{S}_+} f(s', e), \qquad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{S}_\times} f(s, e').$$

E_+ renews the additive fiber and preserves the multiplicative one; $E_\times$ realizes the dual operation. They are the only Markov projectors that completely marginalize the active coordinate, preserve the inactive one, and preserve the uniform count.

Let $\tau = (+1, -1, 0)^3$ be the phase word on C_9, and $P_{+1} = E_+$, $P_{-1} = E_\times$, $P_0 = I$. The nonadic transfer is

$$\mathcal{K}_{\mathrm{ph}}\big((u, r), (v, r+1)\big) = P_{\tau(r)}(u, v).$$

It is doubly stochastic, irreducible, of period 9, and has a unique stationary distribution $1/(9 \cdot 468)$. Moreover,

$$\mathcal{K}_{\mathrm{ph}}^9 = E_+ E_\times, \qquad (E_+ E_\times)(u, v) = \frac{1}{468}.$$

This equality expresses the simultaneous renewal of all compatible continuations, not the deterministic successor of a single path.

Three geometric operations on the seeds preserve additional information: P, advance in the active orientation; H, a half-turn; and Q, a quarter-turn. The pair (P, H) refines the 26 product signatures into 28 classes. In particular, the signature 555555 splits into three sheets of eight representatives, and

$$18 \cdot 28 = 504 = 468 + 36.$$

The term 36 counts additional sheets of the refinement and is not the object R_{36}. The pair (P, Q) produces 162 classes of two elements, transitively permuted by

$$J(i, j, d) = \left(7 - i,\ 7 - j,\ \overline{d}\right) \pmod 9.$$

In a trace of 108 steps, there appear 36 signs of each type, 72 effective changes between the two sheets, and four orientation reversals; the total signed cocycle is zero. These counts are invariants of the evolution record and link nonadic dynamics with the dodecaphase record.

1.22.10 Tertiary prolongation and joint boundary

Relative to the named seeds, boundaries, and readers of the published profile, the initial visible words are

$$w_6^{\pi} = 010211, \qquad w_6^{e} = 201101, \qquad w_6^{\varphi} = 121200.$$

After five blocks, the prefixes are

$$\begin{aligned} w_{30}^{\pi} &= 010211|012222|010211|002111|110221, \\ w_{30}^{e} &= 201101|121221|102011|012222|102011, \\ w_{30}^{\varphi} &= 121200|112202|121020|010210|010200. \end{aligned}$$

The first deep boundary is

$$R_{36} = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}.$$

For each channel the prefix $w_{36}^{\chi} := w_{30}^{\chi}|u_5^{\chi}$ is defined; R_{36} is the ordered triple of the three sixth blocks. Neither is an emitter nor a terminal state.

For $B_k = (u_k^{\pi}, u_k^{e}, u_k^{\varphi})$, the signatures

$$q_k = u_k^{\pi} + u_k^{e} + u_k^{\varphi}, \qquad a_k = u_k^{\pi} + u_k^{e} - u_k^{\varphi}$$

determine

$$u_k^{\varphi} = 2(q_k - a_k), \qquad u_k^{\pi} + u_k^{e} = q_k - u_k^{\varphi} \quad \text{in } \mathbb{F}_3^6.$$

Thus, φ is the fixed axis and the residual freedom is the internal distribution π/e. In the ninth phase,

$$B_9 = (100100|020112|010122)$$

admits three local outputs with the same axis φ and the same aggregate $\pi + e$; all three survive one horizon and only

$$B_{10} = (101222|112221|112002)$$

survives the second. This is the executable3 $\to$ 1 descent of EF–G9. The published branch continues through twenty blocks per channel: w_{30} and R_{36} are prefix and boundary, not closure.

1.22.11 Relation of extension and nonadic monodromy

Let $\mathcal{H}_m$ be the set of admissible enriched histories of depth m, and let

$$p_{n,m} : \mathcal{H}_n \longrightarrow \mathcal{H}_m$$

the truncation. For $N \geq m$,

$$\mathrm{Surv}_m^{(N)} = p_{N,m}(\mathcal{H}_N), \qquad \mathrm{Surv}_m = \bigcap_{N \geq m} \mathrm{Surv}_m^{(N)}.$$

The EF–G9 relation conserves the extensions that belong to this stable core. The phase satisfies

$$g(m+9) = g(m),$$

but the transported state satisfies, in general,

$$v_{m+9} \neq v_m.$$

The return acts on the fiber of states of a same phase; does not reset the history.

Definition 1.178 (Number HMT)**.** An HMT number of channel χ is solely a compatible global section

$$x = (x_m)_{m \geq 0} \in X_\chi := \varprojlim_m \mathrm{Surv}_m^{(\chi)}.$$

The reader and the published value are not part of the section.

A presentation or measurement of the number is a pair (x, L), where L is a declared reader whose domain contains x. When L is the Archimedean reader of cylinders, $L(x)$ is the published value; x conserves in addition orientation, quotient, carry, and boundary.

Theorem 1.179 (Existence of a global prolongation)**.** *Let $h \in \mathcal{H}_m$. If the prolongation tree of h is finitely branching and has at least one node at every depth, then there exists an infinite history whose truncations prolong h. If, in addition, each stable level is a singleton and the unique states are compatible, the global section is unique.*

Proof. The non-empty finite levels form a finitely ramified tree. König's lemma provides an infinite branch. The compatibility of the truncations identifies that branch with an element of the inverse limit. If each level contains a single state, there is no second compatible branch. □

The preceding theorem is abstract and conditional: by itself it asserts neither nonemptiness nor the stable uniqueness of a concrete channel. For π, e, φ, those hypotheses are subsequently discharged through the executable rational filtration of theorem 8.54. This placement preserves here only the inverse-limit principle and does not anticipate its application.

1.22.12 Composition of the fundamental operators

The above operators lead to the enriched terminal state without flattening its coordinates:

$$\begin{aligned} \mathrm{APP} &\longrightarrow \mathrm{TRIT} \longrightarrow \text{oriented route} \longrightarrow \text{enriched TPK state} \\ &\longrightarrow \text{cylinders and survival} \longrightarrow x_{\mathrm{term}}. \end{aligned}$$

The dodecaphonic closure then receives two coordinated evaluations of the same state:

$$x_{\mathrm{term}} \xrightarrow{(\pi_{\mathrm{term}}, R_{\mathrm{sgn}})} \left(B_{\mathrm{term}}, U_{12}^{\mathrm{sgn}}\right), \qquad U_{12}^{\mathrm{sgn}} \overset{\mathcal{H}_{12}}{\longleftrightarrow} K,$$

$$(P, E, \Phi, K, c_{13} = 0) \longleftrightarrow \alpha_{12}.$$

No arrow $B_{\mathrm{term}} \to U_{12}^{\mathrm{sgn}}$ appears: both coordinates follow from x_{term}. The subsequent passage to Dimensional Mechanics likewise creates no other machine; it prolongs the preserved genealogy through an enriched chronological record, signature, character, and towers.

1.23 Construction and operatorial architecture of the APP–TRIT–TPK formal kernel: arithmetic support, ternary orientation, and genealogical dynamics

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: the positions receive arithmetic evaluations, the transitions acquire a ternary signature and the dynamics preserves the information necessary to prolong them. Only afterward are the numerical realizations formed. A constant recognized at the end of this process does not replace the rule that produced its region or the history that determines its evaluation.

The name *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not imply that its arithmetic law is Feynman's path integral. The *Topological Phase Kernel*, TPK,[2] designates the composition of selection, transport, memory update and prolongation. The acronyms are retained only after the objects they represent have been defined.

1.23.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks of a graduated line between zero and ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2. The interior marks must not be confused with the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to decide a trajectory.

The cyclic identification of the positions leads to the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives preserves the symbol nine for the zero class. For every positive integer one defines

$$\rho_9(n) = 1 + ((n-1) \bmod 9), \qquad q_9(n) = \frac{n - \rho_9(n)}{9}, \qquad n = \rho_9(n) + 9q_9(n). \tag{1.152}$$

The positive digital root is therefore a choice of representative of the residual class. It does not by itself preserve the integer before reduction; the coordinate q_9 provides precisely the missing information.

At each position (i, j), the accumulation of i repeated j times produces ij. The two integer evaluations and their projections are

$$S(i,j) = i + j, \qquad P(i,j) = ij, \qquad \Sigma(i,j) = \rho_9(i+j), \quad \Pi(i,j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ interchanges the directions of accumulation and preserves both evaluations. It is a property of a single support, not a subsequent identification between two independent matrices.

Definition 1.180 (Lifted arithmetic evaluation)**.** The evaluation of APP that preserves the two sheets is

$$\mathcal{A}(i,j) = \big((R^+_{ij}, q^+_{ij}), (R^\times_{ij}, q^\times_{ij})\big) = \big((\rho_9(i+j), q_9(i+j)), (\rho_9(ij), q_9(ij))\big). \tag{1.153}$$

The term *sheet* here indicates an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when reducing the residues.

Proposition 1.181 (Reconstruction of the evaluations)**.** *Los dos pares de* (1.153) *determinan exactamente* $i + j$ *e* ij*. Los residuos aislados no determinan esas evaluaciones enteras ni sus cocientes.*

[2] The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

One support: 81 positions.

Two evaluations:
$S(i, j) = i + j$, $P(i, j) = ij$.

Each row is generated by accumulation:
$P(i, j + 1) - P(i, j) = i$.

La carta representa 0 (mod 9) por 9.

El bloque señalado es
$B_c = \left(\begin{smallmatrix} 7 & 2 \\ 2 & 7 \end{smallmatrix}\right)$,
con autovalores 9 y 5.

Figure 1.4: Tabla multiplicativa de APP en representantes positivos. Las filas y columnas no unitarias se distinguen por trama gris; el marco identifica el bloque diagonal adyacente de diagonales iguales. La tabla dibujada es un observable del toro aritmético; sus bordes de representación no constituyen una frontera topológica intrínseca del toro.

Proof. La primera afirmación es (1.152) aplicada a cada hoja. Para la segunda, $10 = 1 + 9$ y $19 = 1 + 2 \cdot 9$ tienen el mismo residuo y distinto cociente. En el soporte bidimensional, las posiciones $(5, 5)$ y $(8, 2)$ dan la misma suma y el mismo residuo producto, pero productos $25 = 7 + 18$ y $16 = 7 + 9$. La proyección residual identifica datos que la evaluación levantada distingue. □

1.23.1.1 Grafo de Cayley, topología toroidal y grado de enrollamiento

Las traslaciones cardinales están dadas, en coordenadas de fila y columna, por

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \qquad T_d(x) = x + \nu_d.$$

The first coordinate increases southward and the second eastward. The graph

$$\mathcal{G}_9 = \mathrm{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic grid: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not make all rows of APP into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^{k} \nu_{d_j} \pmod 9.$$

There are $81 \cdot 4^r$ oriented paths of length r, including returns and backtracking. When the path closes, its integer displacement before reduction defines

$$\mathrm{wind}(w) = \frac{1}{9}\big(\#S(w) - \#N(w), \#E(w) - \#O(w)\big) \in \mathbb{Z}^2. \tag{1.154}$$

Reversing the path changes the sign of this vector. Returning to the same residual position therefore does not require the winding to be zero. This is the first example of a difference that will reappear in the phase return: one coordinate may close while another records a nontrivial evolution.

1.23.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \to D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called *carry* in positional arithmetic, is

$$c_9(a,b) = \frac{s_9(a) + s_9(b) - s_9(a+b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 1.182 (Cocycle identity)**.** *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a,b) + c_9(a+b,c) = c_9(b,c) + c_9(a,b+c).$$

Proof. Both sums are equal to

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a+b+c)}{9}.$$

The associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. □

With positive representatives, this cocycle is not normalized at the identity: $c_9(0,a) = 1$. If s_0 takes representatives in $\{0,\dots,8\}$, its normalized carry c_0 satisfies

$$c_9(a,b) = c_0(a,b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the calendar into phase and window will use representatives from zero to eight.

The totals of both evaluations distinguish the visible contribution and the one preserved in the quotient:

$$\sum S = 810, \quad \sum P = 2025, \quad \sum \Sigma = 405, \quad \sum \Pi = 459,$$
$$\sum q^{+} = 45, \qquad \sum q^{\times} = 174.$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \tag{1.155}$$

The sum–product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Removing it would change the arithmetic being transported, even if the residual table remained identical.

1.23.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1,2,4,5,7,8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0,3,6\}, \qquad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is one exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus the multiplicative folding already distinguishes, before any continuous geometry, the units from the nilpotent sector.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \qquad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence $3, 6, 9$ admits two different readings: its direct reduction modulo three is $0, 0, 0$, whereas extracting the factor three followed by reducing its internal coordinate gives $1, 2, 0$. This second operation preserves a coordinate of the ideal; it is not the same projection as directly reducing the original number.

1.23.1.4 Latin realization of the additive evaluation

The difference between the two evaluations can also be expressed through operators. Let $(e_r)_{r\in\mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in each row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \qquad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a + b$ permutes the nine residues. A realization of a Latin square by rank-one projectors is obtained, in the terminology of quantum Latin squares [152, 154]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves other content: for $a = 3$, the vectors e_{3b} repeat the residues $0, 3, 6$, and

$$\sum_{b\in\mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3\big(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|\big).$$

The multiplicity is recorded in the operator and characterizes the multiplicative radical. Thus the toroidal support, the Latin property and the operator resolution of the identity are properties that require different definitions. The quantum magic squares studied by De las Cuevas, Drescher and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commutative entries[153]. This reference provides a precise language for comparing realizations of APP.

1.23.2 TRIT: orientation and quadratic regimes

Definition 1.183 (Ternary signature and local lift)**.** The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \widehat{\mathbb{T}}, \qquad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \widehat{\mathbb{T}}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1), (0, 0), (0, +1)$ may have the same zero residue and different carries.

The signature does not replace the trajectory. A residual zero does not mean an absence of orientation, phase or quotient information. Nor does it by itself introduce a physical quantity. Its function is to distinguish algebraically the transport classes that will be realized next.

1.23.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \qquad \tau \in \widehat{\mathbb{T}}. \tag{1.156}$$

If J_τ is the class of X, then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$ and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types that oriented reduction has already produced.

Proposition 1.184 (Exponentiation of the three types)**.** *In each algebra* (1.156)*,*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \tag{1.157}$$

where

$$C_\tau(t) = \sum_{n\geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \qquad S_\tau(t) = \sum_{n\geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
$+1$	$-I$	$\cos t\, I + \sin t\, J_\tau$
0	0	$I + tJ_\tau$
-1	I	$\cosh t\, I + \sinh t\, J_\tau$.

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, all terms of degree at least two vanish. In the other two cases the trigonometric or hyperbolic series are recovered. □

The exponential is common to the three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau)\exp(-tJ_\tau) = I$. The same algebraic principle preserves unity across realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction appear jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates $+1$ with the return with memory, 0 with the threshold and -1 with bidirectional opening. This interpretation requires an order of parametrization; it does not add a fourth algebraic type.

1.23.2.2 Balanced division and composition of lifts

Ternary reduction allows an integer evaluation of APP to be expressed in a visible coordinate and a prolongation coordinate. Its function does not consist merely in abbreviating the notation of an integer: it provides a composition rule that determines how the quotient is modified when evaluations are added or multiplied. The balanced choice also makes explicit the compatibility of that rule with sign reversal.

Definition 1.185 (Balanced ternary coordinates)**.** For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \qquad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \widehat{\mathbb{T}} \times \mathbb{Z}, \qquad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 and belongs to the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the datum necessary to reconstruct the preceding one. The quotient is not a perturbation of the residue: it constitutes its complementary coordinate.

The signature obtained from an evaluation F, with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$ and $\{3, 6, 9\}$, with values $+1, -1, 0$. The TPK selector defined next uses these classes to determine the operational mode. Reading an evaluation and selecting a mode thus have the same ternary reduction, but different domains and functions.

Proposition 1.186 (Arithmetic of balanced pairs)**.** *Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \widehat{\mathbb{T}}$. Define*

$$\chi(r,s) = \frac{r + s - r_3(r+s)}{3}.$$

The integer operations are transported to pairs through

$$(r,c) \boxplus (s,d) = \big(r_3(r+s),\, c + d + \chi(r,s)\big), \tag{1.158}$$

$$(r,c) \boxtimes (s,d) = \big(rs,\, rd + sc + 3cd\big). \tag{1.159}$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r+s) + 3\chi(r,s)$ gives

$$n + m = r_3(r+s) + 3\big(c + d + \chi(r,s)\big).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. □

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1,1) \boxplus (1,1) = (0,2), \qquad (-1,1) \boxtimes (1,1) = (-1,3).$$

The resulting evaluations are 6 and 8. Adding quotients suffices in this additive example; the product involves the cross terms rd, sc and $3cd$. The discrepancy between the two operations also lies in the prolongation coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r+s)$, it satisfies

$$\chi(r,s) + \chi(r \oplus s, t) = \chi(s,t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity guarantees that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component suppresses that accounting: $1 + 1 = -1 + 3$, whereas the residual reading preserves only -1. The eliminated information is determined, in this case, by $\chi(1,1) = 1$.

The relation with the modulus 9 preserves a second discrete scale. The map

$$\beta : \widehat{\mathbb{T}} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \qquad (r, \bar{c}) \longmapsto r + 3c \pmod 9$$

is well defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images first fixes r by reduction modulo 3 and then $\bar{c}$. The sum transported by this bijection contains the term $\chi(r,s)$ of (1.158), now reduced modulo 3. Thus the two ternary levels are not composed through an independent sum of coordinates: normalization of the first level modifies the second.

In particular, the classes 3, 6 and 0 of $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second component 1, 2 and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β. Preserving that coordinate prevents the zero residual sector from being artificially turned into a single position. The modulus 1000, used later to organize decimal blocks, serves another positional representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas the coincidence of residues in different bases does not by itself identify their transport states.

1.23.2.3 Reversal, residual class and nonvisible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0 and −3 have coordinates $(0, 1)$, $(0, 0)$ and $(0, -1)$. They share a zero signature, but represent different evaluations. In a transition restricted to a quotient of maximum amplitude 1, they are the three possibilities of the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, residue and quotient, and differ in orientation, mode sequence or winding. Thus the decomposition $\mathcal{B}$ is integrated into the transport state, instead of replacing it. The arithmetic memory of a division and the memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n, oriented reversal is

$$C_{\mathrm{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of the positions and the sign of each signature; it preserves its length and the zero positions in reverse order. In the adopted parametrization, it interchanges the return with memory, associated with +1, and bidirectional opening, associated with −1, maintaining the threshold 0. The transport of the remaining data of a trajectory is specified through the corresponding involution of TPK.

This reversal of signatures must be distinguished from conjugation within a quadratic algebra. The former may change the type τ; the latter, which is constructed next, acts within the type already fixed. The distinction prevents a sign reversal from being credited with an algebraic equivalence between different regimes.

1.23.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \tag{1.160}$$

This is a real realization of the type already selected. The coordinates a, b are not residues of APP, and the product (1.160) does not replace (1.159); it expresses another operation, in the declared quadratic codomain.

Definition 1.187 (Conjugation and algebraic norm)**.** Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \qquad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 1.188 (Multiplicativity and invertibility)**.** *For $z, w \in \mathbb{A}_\tau$,*

$$z\bar{z} = N_\tau(z)I, \qquad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, and in that case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so $(zw)\overline{zw} = (z\bar{z})(w\bar{w})$. If the norm does not vanish, the indicated formula provides the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. □

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2, and the element J_0 is nonzero although its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. The algebraic norm here is a multiplicative quadratic form: it does not presuppose a positive vector space norm.

The three types can be faithfully realized through

$$aI + bJ_\tau \longmapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. Thus the complex structure, the dual number structure and the split real structure are represented with the same matrix expression. Each preserves its law and its zero divisors. The form of signature $(1, 1)$ in the last case has a Lorentzian interpretation in dimension $1 + 1$; identifying a physical metric additionally requires a specific space, field and transport rule.

1.23.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of elements of norm one:

$$N_\tau\big(\exp(tJ_\tau)\big) = 1.$$

This conservation is maintained when composing parameters. Indeed, all the factors are functions of the same generator, and

$$\exp(tJ_\tau)\exp(sJ_\tau) = \exp((t + s)J_\tau).$$

Comparing the two components yields the addition laws

$$C_\tau(t + s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \tag{1.161}$$

$$S_\tau(t + s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \tag{1.162}$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ: it reverses an evolution of the chosen regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential trajectory traverses the circle of norm one. In the hyperbolic type, the eigencoordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \qquad uv = 1.$$

The growth of one coordinate is accompanied by the contraction of the other. Conservation expresses a relation between both coordinates; it does not state that each one remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \qquad (I + tJ_0)^{-1} = I - tJ_0.$$

Evolution does not stop because $J_0^2 = 0$. Nilpotency eliminates the terms of order at least two, but preserves the linear term that transports the parameter. The visible symbol 0, the zero element of the algebra and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with an absence of state or dynamics.

Proposition 1.189 (Composition of affine coordinates)**.** *In the chart where the scalar component does not vanish, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u+v}{1-\tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u+v)J_\tau.$$

Dividing by the scalar component gives the stated expression. □

The affine chart preserves the ratio between components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the full algebra; the singularity of a coordinate does not automatically amount to a singularity of the object. This formula will reappear in the elliptic regional reading. Its use requires retaining the denominator and, where appropriate, the normalization factor.

1.23.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$ and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Thus the change of signature that organizes a route cannot be interpreted as an undifferentiated isomorphism between the three realizations. TPK preserves the regime index, the domains and the effective operator acting at each transition.

Within the electronic chapter, the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution of square I and two nilpotent directions. The joint realization will be proved on that specific pair of projections; the preceding classification explains why the three quadratic relations can appear in the same matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. The relation endows the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its link with a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2+1).$$

The base changes to $\mathbb{F}_3$ and to $\mathbb{R}$ produce representations of that same relation. Each preserves its characteristic, its dimension and its incidence information. The development of the respective operators establishes the link, not the equality of their cardinalities.

The conservation of norm one therefore describes the quadratic realizations of evolutions. The conservation of quotients, orientation and trajectory belongs to the transported arithmetic state. Both properties play a role in the construction, with different functions: one determines the identities of the representations; the other allows the operations that originate them to be composed, reconstructed and prolonged.

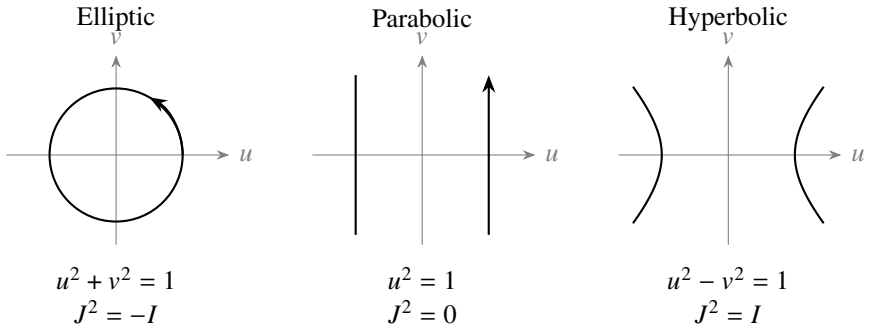


Figure 1.5: Quadratic realizations of TRIT. For $J_\tau = \left(\begin{smallmatrix} 0 & -\tau \\ 1 & 0 \end{smallmatrix}\right)$, the exponential preserves $u^2 + \tau v^2$. Parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

1.23.3 TPK: topological phase kernel

The dynamics uses the phase selector

$$M_{\mathrm{ph}}(r) = \begin{cases} +1, & r \in \{1,4,7\}, \\ -1, & r \in \{2,5,8\}, \\ 0, & r \in \{3,6,9\}. \end{cases}$$

Its value decides which evaluation operates. Cardinal transport shifts the corresponding cursor; the register preserves the preceding value, the new quotient, the orientation and the transition. Therefore, selection and transport are distinct functions: a ternary sign is not a displacement of the torus.

We shall denote by $\widehat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of the two cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix and the boundary data. When a refinement cylinder is introduced, it forms part of the state. The notation avoids replacing the object with a reduced version that preserves only the visible symbol.

Definition 1.190 (TPK update)**.** Let Sel_t be mode selection through M_{ph}, Tra_t lifted cardinal transport and Mem_t the update of the registers. The transition is

$$\mathcal{U}_t = \mathsf{Mem}_t \circ \mathsf{Tra}_t \circ \mathsf{Sel}_t. \tag{1.163}$$

Each map preserves the fields that it does not modify and updates the others through Euclidean division and concatenation of the trajectory.

The notation (1.163) expresses the order of composition; it does not replace the effective rules. The following section fully specifies its finite two-cursor realization. Afterward, prolongation will not be obtained by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. In this way, APP provides the evaluations, TRIT distinguishes their regimes and TPK composes their evolution.

1.23.3.1 Dynamic state, observables and dependence of the operations

The topological phase kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition includes the state on which it acts, the transition rules and the relations that make the prolongations compatible. The equation (1.163) acquires operational content by specifying which coordinates each of its factors receives and modifies. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; the quotients record the arithmetic

evaluation before its reduction. These coordinates are related through the transition and preserve different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_\times, d_\times, \theta) \in Q_{\mathrm{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \qquad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \tag{1.164}$$

The subscripts distinguish the additive and multiplicative cursors. In the fibers of complete states over q, the ordered trajectory, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary registers are preserved. In a precision prolongation, the prefix and the compatible cylinder are added. Therefore, two states with the same projection q may preserve different registers.

The dependence of the operations has three levels. At the local level, the selector activates a sheet, transport modifies its position and the update calculates the new arithmetic coordinates. At the trajectory level, these transitions are composed and preserve the registers of the preceding steps. At the refinement level, extension relations determine which continuation preserves the prefix, the signature and the boundary of the antecedent. The finite calendar describes the phase of these operations; holonomy describes their composition on the history.

1.23.3.2 Tritic selection and lifted cardinal transport

Let $\bar\theta \in \{0, \dots, 107\}$ be the calendar representative. The operational phase and the dodecaphasic sector are

$$r(\theta) = 1 + (\bar\theta \bmod 9), \qquad \beta(\theta) = \left[\left\lfloor \frac{\bar\theta}{9} \right\rfloor\right]_{12}. \tag{1.165}$$

The selector already defined can be written $\varepsilon_9 = M_{\mathrm{ph}}$. Its vector of values, in the ordered chart of D_9, is

$$m_{\mathrm{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\mathrm{ph}} : D_9 \to \widehat{\mathbb{T}}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \qquad a_\times(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

The transport of the cursors is then

$$x'_+ = x_+ + a_+(\theta)\nu_{d_+}, \qquad x'_\times = x_\times + a_\times(\theta)\nu_{d_\times} \quad \text{in } X_9. \tag{1.166}$$

In phases $3, 6, 9$ both positions remain unchanged; the neutral event is part of the chronology and its register. The cardinal directions determine the vectors ν_d, while the phase signature determines which of those vectors is applied.

The integer lift preserves the boundary crossing. If $s : X_9 \to D_9^2$ chooses the positive representatives and $x' = x + a\nu_d$, with $a \in \{0, 1\}$, one defines

$$b_d(x, a) = \frac{s(x) + a\nu_d - s(x')}{9} \in \mathbb{Z}^2. \tag{1.167}$$

Divisibility by 9 follows from equality in X_9. If $z \in \mathbb{Z}^2$ records the preceding crossings, the update is $z' = z + b_d(x, a)$. In this way,

$$s(x') + 9z' = s(x) + 9z + a\nu_d.$$

The integer transport datum remains reconstructible after reducing the position on the torus.

Proposition 1.191 (Conservation of the lifted displacement)**.** *For a cardinal trajectory of n steps, whose activation indicators are a_t, one has*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t \nu_{d_t}.$$

The same equality is preserved when concatenating two compatible trajectories.

Proof. The one-step identity is (1.167). Summing it over the n steps cancels the intermediate coordinates. Dividing the summation interval into two segments produces exactly the concatenation law. In a closed trajectory, the difference $z_n - z_0$ is the winding number defined previously. □

The same principle governs APP evaluations. If a sheet takes the integer value N_t, its register simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

The spatial crossing memory z_t and the arithmetic quotient q_t have different definitions; their joint conservation maintains the geometric and arithmetic provenance of each transition.

1.23.3.3 Two-cursor chronology and composite returns

The cardinal involution $\jmath$ interchanges N with S and E with O. After applying (1.166), both directions are reversed when a block of 27 steps ends. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\mathrm{obs}}(q) = \big(x'_+, \jmath^{h(\theta)} d_+, x'_\times, \jmath^{h(\theta)} d_\times, \theta + 1\big). \tag{1.168}$$

The reading that forms the emission uses the positions before transport, according to the explicit recurrence in section 4.3. The observable update and the emission are therefore composed in a fixed order.

Proposition 1.192 (Reversibility and period of the observable calendar)**.** *The map F_{obs} is a permutation of $\mathcal{Q}_{\mathrm{obs}}$ and satisfies $F_{\mathrm{obs}}^{108} = I$. In a block of* 54 *steps, positions and directions are restored; the coordinate of Θ_{108} advances by* 54*.*

Proof. Given the final state, the preceding phase is first recovered by subtracting one in Θ_{108}. That phase determines whether the directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The next block preserves the activations and reverses the directions, so the integer displacements cancel in 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds with any origin of the calendar. After 108 steps, θ is also restored. □

We reserve

$$\mathcal{K}_{27} = F_{\mathrm{obs}}^{27} : \mathcal{Q}_{\mathrm{obs}} \longrightarrow \mathcal{Q}_{\mathrm{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (1.164). Projection of the history onto the calendar allows this finite check to be performed. The trajectory register, its evaluations and the new prefixes are preserved in the dynamic state on which $\mathcal{U}_t$ acts; observable equality does not reset them.

1.23.3.4 Affine transport of memory and composition of states

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q, we write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves the orientation and the sheet; h, the residues and quotients; c, the additive transport memory; σ, the boundary signature; C, the precision cylinder; b, its boundary label; and λ, the recorded trajectory. The signature is determined by a map

$$\mathrm{Sig}_q : \mathsf{H}_q \times \mathsf{C}_q \times \mathsf{Hist}_q \longrightarrow \mathsf{S}_q, \qquad \sigma = \mathrm{Sig}_q(h, c, \lambda).$$

The symbols $\mathsf{H}_q, \mathsf{C}_q, \mathsf{Hist}_q$ and S_q denote, respectively, the sets of residue–quotient data, the abelian group of transport memory, the recorded trajectories and the boundary signatures. The realizations of this signature are specified with their components in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data producing it are fixed.

The elementary residue and quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n-1) \bmod 9), \qquad u = \frac{n-r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives of D_9 congruent modulo 9 are equal. This chart provides local integer reconstruction. The precision towers additionally incorporate the truncation maps and the cylinders developed in 4.3 and 10.5.

Let $\gamma : q \to q'$ be an observable trajectory. The transport of the fibers is expressed through maps

$$\mathsf{H}(\gamma) : \mathsf{H}_q \to \mathsf{H}_{q'}, \qquad \mathsf{C}(\gamma) : \mathsf{C}_q \to \mathsf{C}_{q'}, \qquad \mathsf{Hist}(\gamma) : \mathsf{Hist}_q \to \mathsf{Hist}_{q'}.$$

The maps respect identities and composition, and $\mathsf{C}(\gamma)$ is a homomorphism of abelian groups. The increment $k_{\mathrm{tr}}(\gamma) \in \mathsf{C}_{q'}$ satisfies

$$\begin{aligned} k_{\mathrm{tr}}(\mathrm{id}_q) &= 0, \\ k_{\mathrm{tr}}(\gamma_2 \circ \gamma_1) &= k_{\mathrm{tr}}(\gamma_2) + \mathsf{C}(\gamma_2) k_{\mathrm{tr}}(\gamma_1). \end{aligned} \tag{1.169}$$

The symbol k_{tr} here designates the transport cocycle; the rational coordinate κ of the register K retains its notation in the chapter on dodecaphasic closure.

The transportable coordinates are updated by

$$\begin{aligned} h' &= \mathsf{H}(\gamma) h, \quad & c' &= \mathsf{C}(\gamma) c + k_{\mathrm{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, \quad & \sigma' &= \mathrm{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{1.170}$$

The second equation is an affine action: the linear term transports the preceding memory and the cocycle adds the increment produced by the trajectory. In the realization of spatial crossing for each cursor separately, $\mathsf{C}(\gamma) = I$ and $k_{\mathrm{tr}}(\gamma)$ is the sum of the vectors of (1.167). This concrete instance links the abstract law to the cardinal operations already defined.

Proposition 1.193 (Compatibility of the update with concatenation)**.** *The update* (1.170) *by* γ_1*, followed by the update by* γ_2*, coincides with the update by* $\gamma_2 \circ \gamma_1$*.*

Proof. For the affine coordinate, two updates give

$$c'' = \mathsf{C}(\gamma_2)\mathsf{C}(\gamma_1) c + \mathsf{C}(\gamma_2) k_{\mathrm{tr}}(\gamma_1) + k_{\mathrm{tr}}(\gamma_2).$$

The functoriality of C and (1.169) turns this expression into $\mathsf{C}(\gamma_2 \circ \gamma_1)c + k_{\mathrm{tr}}(\gamma_2 \circ \gamma_1)$. The assertion for h follows from the composition of H, and for λ, from the associativity of the trajectories. The final signature coincides because it is evaluated on the same three reconstructed coordinates. □

The cylinder and its boundary complete this law through the relations Ext_γ, defined in 1.23.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. The states, together with the triples (γ, x, x') admitted by these relations, form a category over the observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x). Composition preserves both arithmetic transport and boundary prolongability. This is the structure in which returns with memory and successive refinements are carried out.

1.23.3.5 Dodecaphasic register, phase and accumulated memory

The 108 transitions are distributed in the ordered windows

$$E_m = \{9m - 8, \ldots, 9m\}, \qquad 1 \le m \le 12.$$

The register preserves the phase origin, the orientation and the data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=1}^{12},$$

where τ_m is the subroute of the window, σ_m its orientation, κ_m the integer boundary increment and Λ_m the local incidence register. The chart that will be developed in 11.4.2 is adopted; its window indices correspond to $m = \beta + 1$. The domain comprises the compatible histories of TPK and the codomain, their oriented temporal registers. The composition of segments and their increments is governed by (1.170). The group C_{12} instead describes the translation of the window index; Θ_{108} preserves the step calendar. This distinction makes it possible to specify what is transformed and what is observed upon completing a cycle.

In the coordinates $\theta = 9b + r$, with $0 \le r < 9$, advancing by 9 steps preserves r and increases b by one. The addition of phases introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar preserves $[b]_{12}$; a turn register preserves $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4.3 proves the nonsplit exact sequence (4.33). Its function is to describe the relation between these coordinates, before using them in the construction of the signed state.

The signed register receives the phase and memory data through the composition developed in section 11.4:

$$R_{\mathrm{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \tag{1.171}$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between that signed state and K, as well as the rational reading of K, are proved in the corresponding chapter. The register operations precede these transformations. Reconstructing a register from its cyclic differences verifies its recoverability; dynamic provenance must be preserved through the events that produced it.

For the encoded trace (d_t) preserved by the register, the event reader counts the codes $\{2, 4, 6, 8\}$ of each window with the orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are distinct coordinates of the register. Chapter 11.4 develops this reader, the integral decomposition

$1+5+6$, the congruences that characterize its image and its exact inverse. The inversion proof establishes which data that chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

1.23.3.6 Phase incidence and internal construction of the channels

The selector determines three subsets of D_9:

$$H_+ = \{1,4,7\}, \qquad H_- = \{2,5,8\}, \qquad H_0 = \{3,6,9\}.$$

Let $H_{\mathrm{act}} = H_+ \sqcup H_-$ be the set of active phases and let $O_{\mathrm{ph}} = D_9 \setminus \{9\}$ be the chart before the marked return. Denote by $P_2(H_{\mathrm{act}})$ the set of unordered pairs of active phases. The internal incidence is given by

$$\mathcal{F}_6^{\mathrm{ph}} = H_{\mathrm{act}} \times P_2(H_{\mathrm{act}}), \qquad \mathcal{F}_8^{\mathrm{ph}} = O_{\mathrm{ph}} \times P_2(H_{\mathrm{act}}), \qquad \Theta_{\mathrm{ph}} = D_9 \times P_2(H_{\mathrm{act}}). \tag{1.172}$$

The indices describe the number of positions of the first component. The construction uses the phases and the calendar origin already fixed; subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 1.194 (Cardinalities and oriented phase incidence)**.** *The sets in* (1.172) *have cardinalities* $90, 120, 135$*, respectively. If*

$$\partial_{\mathrm{ph}} = \Theta_{\mathrm{ph}} \times \{-1,+1\}, \qquad E_{\mathrm{ph}} = \partial_{\mathrm{ph}} \sqcup \{\infty\}, \qquad L_{\mathrm{ph}} = H_0 \times P_2(H_{\mathrm{act}}),$$

then

$$|\partial_{\mathrm{ph}}| = 270, \quad |E_{\mathrm{ph}}| = 271, \quad |L_{\mathrm{ph}}| = 45, \quad |\partial_{\mathrm{ph}} \sqcup L_{\mathrm{ph}}| = 315.$$

Simultaneously translating the phase origin and the marked return preserves all these cardinalities and their inclusions.

Proof. There are six active phases, so $|P_2(H_{\mathrm{act}})| = \binom{6}{2} = 15$. The products by $6, 8, 9$ give $90, 120, 135$. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor through a bijection and takes the marked point to its image. The products and disjoint unions thus preserve incidence. □

The cardinality 271 here has a realization through phase incidence. The cubic completion of section 4.3 has its own construction of the crown $1000 - 729$. The correspondence between both realizations must preserve the maps and marks that identify them, in addition to the cardinality. Likewise, the selector ε_9, the incidence $(90, 120)$ and the extractor $\mathcal{E}_{108}^{90,120}$ preserve the domains and operations that have just been distinguished.

1.23.3.7 Reversal of trajectories and conjugate prolongations

Bidirectionality begins with an operation on trajectories, before assigning scalar values to them. For $\gamma = (x_0; d_1, \ldots, d_n)$, with endpoint x_n, we define

$$\operatorname{rev}\gamma = (x_n; \jmath d_n, \ldots, \jmath d_1).$$

The origin of the reversed trajectory is the endpoint of the original. Composition satisfies

$$\operatorname{rev}^2\gamma = \gamma, \qquad \operatorname{rev}(\gamma_2 \circ \gamma_1) = \operatorname{rev}\gamma_1 \circ \operatorname{rev}\gamma_2,$$

and the integer displacement changes sign. These identities are obtained by reversing the order of the steps and replacing each cardinal vector with its opposite. For a loop it follows that $\operatorname{wind}(\operatorname{rev}\gamma) = -\operatorname{wind}(\gamma)$.

On the registers, reversal also requires transporting the order of their components and the orientation of their events. Route reversal and a signature conjugation are distinguishable operations: their composition

is specified when applied to a concrete register. In particular, the regional involution that interchanges the closure and propagation components preserves the autoscale axis; its formula is presented in section 10.5.14. This regional operation acts on a different domain from the reversal of a cardinal trajectory.

Extensions over a route $r : q \to q'$ are expressed through relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q, one requires

$$\mathrm{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \qquad \Delta_{\mathcal{F}_q} \subseteq \mathrm{Ext}_{\mathrm{id}_q},$$

$$\mathrm{Ext}_{r_2} \circ \mathrm{Ext}_{r_1} \subseteq \mathrm{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, the orientation, the quotients, the prefix and the boundary. Reversal of the route does not by itself imply that every extension relation is a bijection. The conjugate prolongation is defined on the domain where the transport of these data satisfies the corresponding compatibilities.

1.23.3.8 Nonadic holonomy and continuity of refinement

One phase turn composes the transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \cdots \circ \mathcal{U}_k. \tag{1.173}$$

On an individualized history, this functional composition realizes the holonomy relation; on the complete domain one preserves $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$, with the admissible prolongations developed later. The cyclic base has nine phases, while the fiber preserves the trajectory, the quotients and the boundary. The monodromy is the action of one turn of that base on the fibers; the holonomy $\Gamma_{9,k}$ expresses its realization through the effective transitions of TPK.

If g observes the phase and M the number of preserved turns,

$$g(\Gamma_{9,k}x) = g(x), \qquad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure and the second identifies the new temporal depth. Coinductive prolongation also preserves its cylinders and prefixes. The restrictions between depths compose an inverse system; a compatible section gathers its antecedents without replacing them with the latest observation. The development of this arithmetic of histories and its unity occupies 4.3.8–4.3.10; the construction of the regions and their readers is carried out in section 10.5.

Phase transfer on emissions already constructed has another domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the alphabets of additive and multiplicative signatures, of cardinalities 18 and 26. Their product identifies the catalog $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \to \mathbb{R}$, the sheet averages are

$$(E_+ f)(s,e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s',e), \qquad (E_\times f)(s,e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s,e').$$

These emissions and their cardinalities are constructed through the recurrence in section 4.3. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \qquad \mathcal{K}_{\mathrm{ph}}((u,r),(v,r+1)) = P_{M_{\mathrm{ph}}(r)}(u,v),$$

with zero weight for the remaining phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1,-1,0)^3$ then produces the renewal

$$\mathcal{K}_{\mathrm{ph}}^{9}\big|_r = E_+ E_\times.$$

Indeed, both averages are idempotent, commute and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. The transport of histories remains given by (1.173), with its memory and its boundary.

The resulting hierarchy establishes the function of the following chapters. The finite emission constructs the domain and the regions; the extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphasic register determines the data involved in K and in the closure of α. The numerical realization and the incidence realization receive that preceding structure. This order allows each consequence to be presented in its chapter while preserving, from the formal kernel, the objects that make it possible.

1.23.4 Accumulated memory and transport resolvents

The return of the calendar does not eliminate the increments produced during the path. To express this difference in operator terms, one first constructs an accumulative register of trajectory events and then its generating series. The reduction of memory variables will be carried out on that transport, also preserving the source and the output reader.

1.23.4.1 Oriented events and register update

Let (d_t) be the sequence of integer direction codes preserved by a TPK trajectory. The code d_t and the cardinal direction of the cursor are distinct coordinates; no identification between their alphabets is assumed. We number the windows from zero

$$I_m = \{9m + 1, \dots, 9m + 9\}, \qquad m \geq 0.$$

During the first cycle, I_m is window E_{m+1} of the dodecaphase record. Its orientation is $\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1)$. For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign retained in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \qquad |a_m| \leq 9. \tag{1.174}$$

Each summand is incorporated when its transition occurs. This definition counts the events of the trace; it does not choose directions through a value that is to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$ and $S\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. We write $R = S^{-1}$. The accumulated register in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix without accumulated history starts at $z_0 = 0$; at another origin, z_0 preserves the previous register. The frame accompanying the calendar is $y_m = S^{-m} z_m$.

Proposition 1.195 (Update and return with memory)**.** *The register in the moving frame satisfies*

$$y_{m+1} = R y_m + \eta_m, \qquad \eta_m = a_m R \mathbf{e}_0. \tag{1.175}$$

For all $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \tag{1.176}$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \tag{1.177}$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1}y_m + a_m S^{-1}\mathbf{e}_0$, because $S^{-m}\mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (1.176) is obtained by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j}\eta_j = a_j R^{12-j}\mathbf{e}_0 = a_j\mathbf{e}_j$, which proves (1.177). □

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^{\mathsf{T}} D_3 + D_4^{\mathsf{T}} D_4,$$

$$\mathcal{M}(y) = \tfrac{1}{2} y^{\mathsf{T}} B y, \qquad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the register; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 1.196 (Balance produced by the event)**.** *The update* (1.175) *satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \tag{1.178}$$

Proof. Expanding the products, $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; therefore $R^{\mathsf{T}} B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m\mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^{\mathsf{T}} B y_m + \frac{1}{2} a_m^2 B_{00}$. The coordinate sum is invariant under the permutation R, which gives the second equality. □

1.23.4.2 Generating series and uniform control of memory

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u, let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \qquad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 1.197 (Resolvent and tail bound)**.** *The formal identity*

$$Y(u) = (I - uR)^{-1}\big(y_0 + u H_\eta(u)\big). \tag{1.179}$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \to \mathbb{R}$ is linear, $L = \|\ell\|_{1\to\mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1 - r} + \frac{9r}{(1-r)^2} \right). \tag{1.180}$$

The series of derivatives converges uniformly on each disk $|u| \leq r < 1$.

Proof. Multiplying (1.175) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m) r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1 - r}, \qquad \sum_{m > N} m r^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} L m (M_0 + 9m) r^{m-1} < \infty$ majorizes the derivative series and proves its uniform convergence. □

The variable u counts windows of nine transitions. Its identification with the variable of another reader requires preserving that clock and the map relating the two readings.

1.23.4.3 Memory elimination: operator, source and reader

On a domain where the update is fixed as $U : X \to X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \qquad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \qquad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \to V_0$, $B_1 : V_1 \to V_0$, $C : V_0 \to V_1$, and $D : V_1 \to V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 1.198 (Schur reduction with source and reader)**.** *Let*

$$D_u = (I - uD)^{-1}, \qquad \mathscr{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ *is*

$$w_0 = \mathscr{F}(u)^{-1}\big(v_0 + uB_1 D_u v_1\big), \qquad w_1 = D_u(v_1 + uCw_0). \tag{1.181}$$

Its complete reading satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v = (\ell_0 + u\ell_1 D_u C)\mathscr{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ + \ell_1 D_u v_1. \end{aligned} \tag{1.182}$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1 w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting it into the first gives (1.181); applying ℓ_0 and ℓ_1 to both components gives (1.182). All the formal inverses exist because their constant terms are identities. □

The term $u^2 B_1 D_u C = \sum_{k\geq 0} u^{k+2} B_1 D^k C$ describes an exit to memory, k steps within it and a return. By contrast, $u\ell_1 D_u C$ reads the memory before that return. Thus reducing only the operator does not necessarily preserve the observation: source and reader must also be transformed.

The difference is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\big\langle \mathbf{e}_0, (I - uR)^{-1}\mathbf{e}_0 \big\rangle = \frac{1}{1 - u^{12}}, \qquad \big\langle \mathbf{e}_{11}, (I - uR)^{-1}\mathbf{e}_0 \big\rangle = \frac{u}{1 - u^{12}}. \tag{1.183}$$

Indeed, $R^n \mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I, whereas the compression of the resolvent preserves the omitted returns. This example corresponds to the twelve-window register, not to the entirety of the TPK state.

1.23.4.4 Composition of segments and coherence of reduction

Let three composable affine transports be $F_i(y) = R_i y + b_i$, with compatible domains and codomains. The memory produced by the complete path is

$$F_3 F_2 F_1(y) = R_3 R_2 R_1 y + b_3 + R_3 b_2 + R_3 R_2 b_1. \tag{1.184}$$

The proof is the successive substitution of the three affine formulas. Grouping the first two segments or the last two first produces the same expression; the factors transport each increment from the point where it originated. In (1.175), $R_i = R$ and b_i is the event of its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \qquad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2Q_2C_2, & B_1' &= B_1 + uB_2Q_2D_{21}, \\ C_1' &= C_1 + uD_{12}Q_2C_2, & D_{11}' &= D_{11} + uD_{12}Q_2D_{21}. \end{aligned} \tag{1.185}$$

Proposition 1.199 (Transitivity of elimination)**.** *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - uA' - u^2B_1'(I - uD_{11}')^{-1}C_1' = I - uA \\ - u^2(B_1\ B_2)(I - uD_*)^{-1}\begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \tag{1.186}$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting it into the first two produces (1.185). Then eliminating the second component gives the left-hand side of (1.186). Solving the two memory equations simultaneously gives the right-hand side, by Proposition 1.198. The formal solution is unique because $I - uT$ has constant term I. Both procedures also transform the source and the reader according to (1.181)–(1.182). □

The balance (1.178) describes the increments of the register. The preceding identities allow them to be transported when reusing memory in the dodecaphasic closure; they do not by themselves determine additional analytic coefficients.

1.23.5 Operator architecture of transport, memory, and prolongation

The topological phase kernel organizes the evolution of the two APP evaluations on oriented histories. Its structure brings together a path support, local update rules, prolongation relations, temporal records, and reading maps. Each component has its own domain. This hierarchy allows the observable position, the transport quotient, and the memory of the crossed boundaries to be preserved within a single construction. The composition in equation (1.163) thus acquires a precise realization at each of these levels.

1.23.5.1 Bivariate paths and lifted transport

Let $\mathsf{P}_{\mathrm{APP}}$ be the free category of the cardinal graph on $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. Its objects are the APP positions; its morphisms are cardinal words, with composition by concatenation. The category

$$\mathsf{P}_{\mathrm{APP}}^{(2)} = \mathsf{P}_{\mathrm{APP}} \times \mathsf{P}_{\mathrm{APP}}$$

preserves, in order, the path of the additive sheet and that of the multiplicative sheet. Both paths traverse the same arithmetic support and keep their evaluations distinct.

In the chart $I_9 = \{0, \dots, 8\}$, the generators are

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \qquad T_d(x) = [x + \nu_d]_9.$$

For $w = d_1 \cdots d_r$, the lift to the integer covering space preserves

$$\widetilde{x}_k = \widetilde{x}_0 + \sum_{j=1}^{k} \nu_{d_j}, \qquad x_k = [\widetilde{x}_k]_9.$$

When $x_r = x_0$, the winding class is

$$\mathrm{wind}(w) = \frac{1}{9}(\#S - \#N, \#E - \#O) \in \mathbb{Z}^2. \tag{1.187}$$

The final position and winding are two different readings of the same route. The second records the displacement of the lift even when the first has returned.

Path reversal has a determined domain and codomain:

$$\mathrm{rev} : \mathrm{Hom}(x, y) \longrightarrow \mathrm{Hom}(y, x), \qquad d_1 \cdots d_r \longmapsto \overline{d}_r \cdots \overline{d}_1,$$

where $\overline{N} = S$, $\overline{E} = O$. It satisfies $\mathrm{rev}^2 = I$, reverses the order of composition, and changes the sign of (1.187). This is the precise operation that must be transported when comparing conjugate orientations of a route.

1.23.5.2 Full state and preservation through updates

The description of $\widehat{X}$ is organized over the positions and directions of the two cursors. Over each observable configuration q, the arithmetic fiber is written

$$\mathcal{F}_q = \mathsf{O}_q \times \mathsf{H}_q \times \mathsf{C}_q \times \mathsf{S}_q \times \mathsf{B}_q \times \Lambda_q. \tag{1.188}$$

The coordinates O_q preserve orientation and sheet; H_q, the remainders and quotients of the evaluations; C_q, compatible carries; S_q, the boundary signature; B_q, the prefix and its cylinders; and Λ_q, the ordered transport record. The fiber is restricted by the corresponding arithmetic, incidence, and truncation identities. The full state also preserves the route that produced it and the origin from which its phases are read.

Mode selection has type

$$M_{\mathrm{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \qquad \mathsf{Sel}_t : \widehat{X} \longrightarrow \widehat{X}_t^{\mathrm{sel}}.$$

The second map incorporates the value of the first into the state and activates the corresponding sheet. Lifted transport

$$\mathsf{Tra}_t : \widehat{X}_t^{\mathrm{sel}} \longrightarrow \widehat{X}_t^{\mathrm{tra}}$$

applies T_d to the active cursor, concatenates its route letter, and preserves the selected sheet. Finally,

$$\mathsf{Upd}_t : \widehat{X}_t^{\mathrm{tra}} \longrightarrow \widehat{X}$$

updates the evaluations and records. In this notation, Upd_t is the update Mem_t of (1.163), made explicit in its components.

Each integer evaluation a is preserved through $a = \rho_9(a) + 9q_9(a)$; each ternary evaluation, through its oriented remainder and carry. For a concatenation $r_2 r_1$, carry transport satisfies the cocycle law

$$k_{\mathrm{tr}}(r_2 r_1) = k_{\mathrm{tr}}(r_2) + \mathsf{C}(r_2) k_{\mathrm{tr}}(r_1), \tag{1.189}$$

where $\mathsf{C}(r_2)$ is the transport action of the second route on the carry coordinate of the first. In the ordered record, updating is performed by adjoining the event produced:

$$\Lambda_{t+1} = \Lambda_t \| (\mathrm{sheet}, d_t, \mathrm{residue}, \mathrm{quotient}, \mathrm{carry}, \mathrm{orientation}, \mathrm{boundary})_t.$$

Preservation has a dynamic meaning here: earlier contributions remain recoverable within the record being prolonged. Their coordinates may change when the state is transported.

1.23.5.3 Affine action on transport memory

For a trajectory $r : q \to q'$, the maps

$$\mathsf{H}(r) : \mathsf{H}_q \to \mathsf{H}_{q'}, \qquad \mathsf{C}(r) : \mathsf{C}_q \to \mathsf{C}_{q'}$$

respect identity and concatenation; C_q is the abelian group of transport memory, and $\mathsf{C}(r)$ its homomorphism. The increment $k_{\mathrm{tr}}(r) \in \mathsf{C}_{q'}$, with $k_{\mathrm{tr}}(\mathrm{id}_q) = 0$, is the cocycle of (1.189). This symbol distinguishes the transport increment from the rational coordinates of the dodecaphasic record.

In a chart with data (h, c, λ), the update has the explicit form

$$h' = \mathsf{H}(r)h, \qquad c' = \mathsf{C}(r)c + k_{\mathrm{tr}}(r), \qquad \lambda' = r \circ \lambda. \tag{1.190}$$

The signature is evaluated through $\mathrm{Sig}_{q'}(h', c', \lambda')$. Its residence in the fiber $\mathsf{S}_{q'}$ preserves its dependence on those data.

Proposition 1.200 (Composition of affine memory)**.** *Updating by r_1, followed by updating by r_2, agrees with updating by $r_2 r_1$.*

Proof. Two memory updates give

$$c'' = \mathsf{C}(r_2)\mathsf{C}(r_1)c + \mathsf{C}(r_2)k_{\mathrm{tr}}(r_1) + k_{\mathrm{tr}}(r_2).$$

Compatibility of C with concatenation and identity (1.189) turn this expression into $\mathsf{C}(r_2 r_1)c + k_{\mathrm{tr}}(r_2 r_1)$. The coordinate h satisfies the same compatibility by functoriality of H; the trajectory does so by associativity. The signature agrees because it is evaluated on the same data. □

The spatial chart of each cursor provides an elementary realization of that law. For $x \in I_9^2$ and a cardinal direction d,

$$a_d(x) = \frac{x + \nu_d - T_d(x)}{9} \in \mathbb{Z}^2, \qquad x' = T_d(x), \qquad c' = c + a_d(x).$$

Here $\mathsf{C}(r) = I$, and $k_{\mathrm{tr}}(r)$ sums the integer boundary crossings. For a route of length n, the telescoping sum gives

$$x_n = x_0 + \sum_{j=1}^{n} \nu_{d_j} - 9 \sum_{j=1}^{n} a_{d_j}(x_{j-1}).$$

On a loop, the last sum equals $\mathrm{wind}(r)$. The residual position returns while the lift preserves the number and orientation of its crossings. The remaining memory components retain their own transport maps and prolongation relations.

1.23.5.4 Chronology of the two cursors and observable return

The finite calendar lives in $\Theta_{108} = \mathbb{Z}/108\mathbb{Z}$. For $0 \le \widetilde{\theta} < 108$, set

$$r(\theta) = 1 + (\widetilde{\theta} \bmod 9), \qquad \beta(\theta) = \left[\left\lfloor \frac{\widetilde{\theta}}{9} \right\rfloor\right]_{12}, \qquad \varepsilon_9(r) = M_{\mathrm{ph}}(r).$$

The presentation (1.164) groups position and direction by cursor. This extension groups the two positions first and then the two directions. Both charts are identified through the permutation of factors

$$\begin{aligned} \mathfrak{p}_{\mathrm{obs}} :\ & (X_9 \times \mathcal{D})^2 \times \Theta_{108} \longrightarrow X_9^2 \times \mathcal{D}^2 \times \Theta_{108}, \\ & (x_+, d_+, x_\times, d_\times, \theta) \longmapsto (x_+, x_\times, d_+, d_\times, \theta), \end{aligned} \tag{1.191}$$

with $\mathcal{D} = \{N, E, S, O\}$. The inverse restores each direction alongside its cursor. If superscripts agr and sep distinguish these two expressions of the observable successor, then

$$\mathfrak{p}_{\mathrm{obs}} F_{\mathrm{obs}}^{\mathrm{agr}} = F_{\mathrm{obs}}^{\mathrm{sep}} \mathfrak{p}_{\mathrm{obs}}.$$

Indeed, the selector depends on the same phase, the cardinal vector is applied to the same cursor, and direction reversal occurs at the same instant; only the coordinate order changes. We retain Q_{obs} and F_{obs} for the space and map so identified. This presentation permutation leaves the events and memory coordinates of the history unchanged.

The observable space is

$$Q_{\mathrm{obs}} = X_9^2 \times \{N, E, S, O\}^2 \times \Theta_{108}.$$

Write a state as $q = (x_+, x_\times, d_+, d_\times, \theta)$. The active sheet is first evaluated at the current position. The positions are then updated by

$$(x'_+, x'_\times) = \begin{cases} (T_{d_+} x_+, x_\times), & \varepsilon_9(r) = +1, \\ (x_+, T_{d_\times} x_\times), & \varepsilon_9(r) = -1, \\ (x_+, x_\times), & \varepsilon_9(r) = 0. \end{cases} \tag{1.192}$$

If $\widetilde{\theta} + 1$ is a multiple of 27, $(d_+, d_\times)$ is replaced by $(\overline{d}_+, \overline{d}_\times)$. In all cases θ is incremented modulo 108. These rules define the successor $F_{\mathrm{obs}} : Q_{\mathrm{obs}} \to Q_{\mathrm{obs}}$. The emission realization in section 4.3 applies exactly this order of evaluation, transport, and reversal.

Proposition 1.201 (Observable return and preservation of history)**.** *The map F_{obs} is a permutation. The operator $\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}$ satisfies $\mathcal{K}_{27}^4 = I$. A complete history of one hundred eight updates also preserves the events and carry accumulated during that turn.*

Proof. Knowing the later phase recovers the preceding phase. The rule indicates whether a direction reversal occurred; undoing it and applying inverse cardinal transport to the active cursor recovers the previous state. Each segment of twenty-seven transitions contains nine activations of each sheet. From an endpoint of a segment, each cursor makes nine displacements in the same direction, whose sum is zero in X_9; it then reverses its orientation. Two segments restore the orientations, and four also restore the calendar. The identity is transported to any phase by invertibility of F_{obs}. The full record preserves the concatenation of the one hundred eight events and their counter, which have been updated at every transition. □

The periodicity of this permutation belongs to the observable quotient. In the full history, the integer window counter increases by twelve over the same interval. Exact sequence (4.33) expresses the reduction of the counter to its dodecaphasic coordinate and preserves its defect of additivity.

1.23.5.5 Bidirectional electronic transport and arithmetic record

The central representation uses a cell shared by the two APP evaluations. Its coordinates belong to $D_9 = \{1, \ldots, 9\}$, and the inversion $(r, c) \mapsto (10 - r, 10 - c)$ has the unique fixed point $(5, 5)$. The visible state is written (r, c, h, v, n), with $h, v \in \{-1, 1\}$ and integer chronological index $n \geq 0$. The tritic selector repeats $(+1, -1, 0)$. On the additive sheet, $r + c$ is read and the column is translated in direction h; on the multiplicative sheet, rc is read and the row is translated in direction v. The threshold preserves position. Both orientations are reversed upon completing each interval of twenty-seven updates.

To express reading and carry with the same positive convention, define

$$[z]_9^+ = 1 + ((z - 1) \bmod 9), \qquad q_9(z) = \frac{z - [z]_9^+}{9}.$$

Evaluation precedes displacement. If a_n is the integer read, the event preserves $(a_n, [a_n]_9^+, q_9(a_n))$, the sheet, previous position, orientation, and boundary crossing. The tritic symbol is $[a_n]_9^+ \bmod 3$. The spatial lift retains, for example, the integer $q_9(c+h)$ when c is replaced by $[c+h]_9^+$.

The shared-cell representation and the two-cursor representation of (1.192) have different visible states. Their records are related to the TPK state through their respective representation maps. For the central trajectory, the full update on the shared cell producing the electronic word is used here.

Proposition 1.202 (Invertible update with preserved quotients)**.** *Add to the visible state two integer records $Q_+, Q_\times$ and two winding numbers W_r, W_c. Each active reading increments its sheet's record by $q_9(a_n)$; each displacement increments the corresponding winding by its boundary crossing. This update has an explicit inverse on its image.*

Proof. The later index determines the preceding mode and whether an orientation reversal has just occurred. This reversal is undone first. If the mode was additive, the previous column is recovered as $[c-h]_9^+$; if multiplicative, the row is recovered as $[r-v]_9^+$. The neutral mode preserves both coordinates. The recovered cell again determines the integer read, its remainder, quotient, and boundary crossing. Subtracting these increments from $Q_+, Q_\times, W_r, W_c$ and reversing the chronological index recovers the preceding state exactly. Composing these inverses recovers an entire finite history. Event concatenation also preserves the order of readings. □

The registers $Q_+, Q_\times$ are an explicit additive expression of quotient history. The construction is integrated into the memory fibre alongside its other coordinates. Its composition law, for concatenable histories u, v, is

$$Q_s(uv) = Q_s(u) + Q_s(v), \qquad s \in \{+, \times\},$$

where the second history begins at the first one's final state. Algebraic inversion of an update subtracts its increments; subsequently traversing a spatial return path records new evaluations. This distinction simultaneously preserves transport reversibility and history duration.

Proposition 1.203 (Central return and exact accumulation)**.** *Starting from $(r, c, h, v, n) = (5, 5, 1, 1, 0)$ and zero initial records, the spatial projection and its orientations return after fifty-four updates. Over that interval,*

$$(\Delta W_r, \Delta W_c) = (0, 0), \qquad (\Delta Q_+, \Delta Q_\times) = (10, 46).$$

After $54N$ updates, for every integer $N \geq 0$, one has $(Q_+, Q_\times) = (10N, 46N)$, while the spatial coordinates and their orientations return. The integer index takes the value $54N$; the dodecaphasic calendar modulo 108 returns for even N.

Proof. Each pair of active modes displaces each coordinate once. In twenty-seven updates, both complete a turn of nine positions and return to the central diagonal; h, v are then reversed. The next block traverses the opposite turn. Spatial crossings cancel between the two blocks.

In each block, the additive readings run through $2k$, $1 \leq k \leq 9$, whose positive quotient sums to five. The multiplicative readings of the positive block run through $k[k+1]_9^+$. Their integer sum and remainder sum are

$$\sum_{k=1}^{9} k[k+1]_9^+ = 249, \qquad \sum_{k=1}^{9} [k[k+1]_9^+]_9^+ = 42.$$

The second identity follows from the list $(2, 6, 3, 2, 3, 6, 2, 9, 9)$. By Euclidean division, the quotient sum is $(249-42)/9 = 23$. Substitution $k \mapsto [k-1]_9^+$ shows that the negative block contains the same multiset of products. The two blocks therefore produce ten and forty-six units in the respective records. The spatial projection and its orientations return to their initial values. The activation and reversal rules repeat upon shifting the index by 54, and the increments do not depend on the accumulated records. The preceding additive law proves the formula for every N by induction. □

The tritic reading and window orientation provide, on the same trajectory,

$$\gamma_e = \big((100000220)^3(120200000)^3\big)^2, \qquad \tau_e = (+,+,+,-,-,-,+,+,+,-,-,-).$$

Powers represent concatenations. The discrepancy and window readers developed in (16.39) and (16.40) both compute $(80, 54, 6)$. Evaluation is obtained from the trajectory before incorporating α, the action section or a mass. Quotient memory preserves information additional to that triple: over one hundred and eight updates it records $(20, 92)$, with zero net spatial winding. Each expression thus retains the type of the observable producing it.

1.23.5.6 Bidirectionality of the product reader and quotient memory

The product reading preserves four arithmetic coordinates before expressing a signature. For $x = (i, j) \in D_9^2$, $D_9 = \{1, \dots, 9\}$, define

$$N(x) = ij, \qquad r(x) = 1 + ((ij - 1) \bmod 9), \qquad q(x) = \frac{N(x) - r(x)}{9},$$

and the discrete reader

$$\ell(1, 2, 4, 8, 7, 5) = (0, 1, 2, 3, 4, 5), \qquad \ell(3) = \ell(6) = \ell(9) = 0.$$

Thus $N = r + 9q$ simultaneously preserves the integer product, positive residue, and arithmetic quotient. The function $f_\ell(x) = \ell(r(x))$ is the coordinate used by the regional signature. The functions N, r, q, and f_ℓ have a positional domain; the transport direction remains in the oriented seed.

Advance, half-turn, and reading before transport. Let $\mathcal{X}_\times = D_9^2 \times \{N, E, S, O\}$. The cardinal vectors are $\nu_N = (-1, 0)$, $\nu_E = (0, 1)$, $\nu_S = (1, 0)$, $\nu_O = (0, -1)$. Positional addition is reduced to positive representatives modulo nine. Define

$$P(x, d) = (x + \nu_d, d), \qquad H(x, d) = (x, \bar{d}), \qquad \nu_{\bar{d}} = -\nu_d.$$

These operators satisfy $P^9 = I$, $H^2 = I$, and $HPH = P^{-1}$.

In the first fifty-four TPK transitions, the product channel is activated in phases $2, 5, 8$ of each nonadic window. Each activation reads the position before advance. After transitions 27 and 54, H is applied. Thus the eighteen positions read are, for a seed s,

$$P^0 s, P^1 s, \dots, P^8 s, \quad HP^9 s,\ PHP^9 s, \dots, P^8 HP^9 s,$$

with the understanding that the positional reading omits only direction. For any positional function f, put $f_n = f(P^n s)$, with indices modulo nine. The word of raw readings is

$$\mathcal{R}_f(s) = (f_0, f_1, \dots, f_8,\ f_0, f_8, f_7, \dots, f_1). \tag{1.193}$$

The six window sums, each comprising three activations, are

$$A_j^f(s) = f_{3j} + f_{3j+1} + f_{3j+2}, \qquad 0 \le j < 3, \tag{1.194}$$

$$A_{3+j}^f(s) = f_{-3j} + f_{-3j-1} + f_{-3j-2}, \qquad 0 \le j < 3. \tag{1.195}$$

For $f = f_\ell$, the signature is $E_{\mathrm{sig}}(s) = ([A_0^f]_{10}, \dots, [A_5^f]_{10})$. The decimal quotient of each accumulator is $c_j^{10} = \lfloor A_j^f / 10 \rfloor$, and it preserves $A_j^f = [A_j^f]_{10} + 10c_j^{10}$.

Proposition 1.204 (Half-turn and permutation of half-cycles)**.** *For every oriented seed and every positional function f,*

$$\mathcal{R}_f(Hs) = \mathrm{rot}_9\,\mathcal{R}_f(s), \qquad A_j^f(Hs) = A^f_{j+3 \bmod 6}(s).$$

The same permutation transports the decimal residues and quotients of the accumulators. It applies, in particular, to N, r, q, and f_ℓ.

Proof. The relation $P^nH = HP^{-n}$ and equality $f(Hs) = f(s)$ give $f(P^nHs) = f(P^{-n}s) = f_{-n}$. Substitution into (1.193) exchanges its two groups of nine entries. Summation in groups of three produces the window identity. Euclidean division by ten is performed componentwise and respects the permutation. □

Advance also has an exact law on the accumulators:

$$A_j^f(Ps) - A_j^f(s) = f_{3j+3} - f_{3j}, \qquad 0 \le j < 3, \tag{1.196}$$

$$A^f_{3+j}(Ps) - A^f_{3+j}(s) = f_{1-3j} - f_{-2-3j}, \qquad 0 \le j < 3. \tag{1.197}$$

The proof consists in cancelling the two common summands of each window. Transporting a signature therefore requires the boundary evaluations appearing in these differences, in addition to the sums already expressed.

Chronological inversion and boundary terms. Let $\gamma = (x_0, \dots, x_n)$ be a recorded trajectory, and let $\gamma^\dagger = (x_n, \dots, x_0)$ be its chronological inversion, with each edge oriented in the opposite direction. For a positional function f, the reading before each advance is

$$C_f(\gamma) = \sum_{t=0}^{n-1} f(x_t).$$

Then

$$C_f(\gamma^\dagger) = C_f(\gamma) + f(x_n) - f(x_0). \tag{1.198}$$

If $n = 3m$ and A_j^f groups three consecutive edges,

$$A_j^f(\gamma^\dagger) = A^f_{m-1-j}(\gamma) + f(x_{3(m-j)}) - f(x_{3(m-j-1)}). \tag{1.199}$$

Both formulas follow because the reverse trajectory reads the arrival endpoints of the original edges. The interior summands coincide; the difference preserves the endpoints.

For an antisymmetric edge observable $a(x, y) = -a(y, x)$, its word does transform according to

$$\mathcal{R}_a(\gamma^\dagger) = -\,\mathrm{rev}\,\mathcal{R}_a(\gamma).$$

This identity acts on oriented edges. Identities (1.198)–(1.199) act on positional readings. The half-turn H changes the initial direction and preserves calendar order; chronological inversion reverses the complete record. Their domains and reading rules are specified separately.

Occupation cocycle and memory preserved on outward and return paths. For a cardinal edge $x \to y$, with positive representatives $\widetilde{x}, \widetilde{y}$, the covering crossing is

$$b(x, y) = \frac{\widetilde{x} + \nu_d - \widetilde{y}}{9} \in \mathbb{Z}^2.$$

The identity $b(y, x) = -b(x, y)$ holds. In the free category of recorded paths, define the additive reader

$$\mathfrak{m}(\gamma) = \left(\sum_{t=0}^{n-1} b(x_t, x_{t+1}),\ C_q(\gamma),\ n \right). \tag{1.200}$$

Concatenation satisfies $\mathfrak{m}(\gamma_2 \circ \gamma_1) = \mathfrak{m}(\gamma_1) + \mathfrak{m}(\gamma_2)$. It is an additive cocycle on recorded trajectories: the record preserves outward and return edges as different events.

By (1.198),

$$\mathfrak{m}(\gamma^\dagger \circ \gamma) = \big(0,\ 2C_q(\gamma) + q(x_n) - q(x_0),\ 2n\big), \tag{1.201}$$

where $\gamma^\dagger \circ \gamma$ denotes the temporal concatenation of the outward path followed by the return. The oriented sum of crossings cancels; arithmetic occupation and length remain. This reader does not descend to the quotient that identifies a path followed by its inverse with the empty trajectory, since that quotient eliminates the events preserved by the second and third coordinates.

There is also a purely boundary-based even reading:

$$V_b(\gamma) = \sum_{t=0}^{n-1} b(x_t, x_{t+1}) b(x_t, x_{t+1})^{\mathsf{T}}, \qquad V_b(\gamma^\dagger \circ \gamma) = 2V_b(\gamma).$$

The odd and even readings record distinct properties of the same edges; both are computed before the path is forgotten.

Exact application to the signature fiber 555555**.** The finite enumeration of $\mathscr{X}_\times$ contains 324 seeds, 26 signatures, and 24 seeds with signature 555555. The stable partition under P, H refines this fiber into three classes of eight elements. They can be labeled by their signature after one advance:

$$\mathscr{C}_{177}, \quad \mathscr{C}_{717}, \quad \mathscr{C}_{771}.$$

The half-turn action is

$$H\mathscr{C}_{177} = \mathscr{C}_{177}, \qquad H\mathscr{C}_{717} = \mathscr{C}_{771}, \qquad H\mathscr{C}_{771} = \mathscr{C}_{717}.$$

The attached finite certificate verifies these identities on all representatives. In the twenty-four seeds, the six raw accumulators of f_ℓ are exactly five. Hence the decimal quotients c_j^{10} vanish on this fiber. The additional information is observed in the positions, integer evaluations, and arithmetic quotients, which have readings different from that signature.

The fixed transverse coordinate a of the cardinal path is 4 or 5 in these seeds. During one turn of nine advances, each value of the moving coordinate $u \in D_9$ is traversed once. Since $\gcd(a, 9) = 1$, the positive residues $r(au)$ permute D_9. This gives, without using a physical constant,

$$\sum_{u=1}^{9} au = 45a, \qquad \sum_{u=1}^{9} r(au) = 45, \qquad \sum_{u=1}^{9} q(au) = 5(a-1).$$

The subsequent return traverses the same positions in the opposite direction. Over the eighteen activations,

$$\sum N = 90a, \qquad \sum r = 90, \qquad \sum q = 10(a-1) = \begin{cases} 30, & a = 4, \\ 40, & a = 5. \end{cases} \tag{1.202}$$

The accumulated integer product is, respectively, 360 or 450. The net winding is zero; the sum of quotients and the eighteen incidences preserve the history of the reading. Each of the three signature classes contains representatives of both transverse values. The same signature and the same transport class of that finite quotient can therefore coexist with distinct arithmetic occupations.

These results connect preserved memory with concrete operations: permutation of half-cycles, boundary differences, arithmetic quotients, and incidence records. The physical-mathematical identification of any subsequent reader preserves its own maps; the identities proved here belong to the internal dynamics of the product channel.

1.23.5.7 Arithmetic memory of the multiplicative channel

Preservation of the regional signature extends to the full arithmetic evaluation of APP. For an oriented position $x = (i, j, d) \in \{1, \dots, 9\}^2 \times \{N, E, S, O\}$, let $\chi(x) = (a, b, \epsilon)$, where a is the fixed coordinate, b the moving coordinate, and ϵ its orientation:

$$\chi(i,j,N) = (j,i,-1), \quad \chi(i,j,S) = (j,i,+1), \qquad \chi(i,j,E) = (i,j,+1), \quad \chi(i,j,O) = (i,j,-1).$$

The axis label $\iota \in \{i, j\}$ preserves which of the two original coordinates moves. Writing $\langle b \rangle_9 = 1 + ((b-1) \bmod 9)$, the operators and evaluation are

$$\overline{P}(a,b,\epsilon) = (a, \langle b+\epsilon \rangle_9, \epsilon), \qquad \overline{H}(a,b,\epsilon) = (a,b,-\epsilon), \tag{1.203}$$

$$n(a,b) = ab, \qquad r(a,b) = \langle ab \rangle_9, \qquad q(a,b) = \frac{ab - r(a,b)}{9}. \tag{1.204}$$

Proposition 1.205 (Exact representation of arithmetic transport)**.** *The identities $\chi P = \overline{P}\chi$ and $\chi H = \overline{H}\chi$ hold. The representation χ has* 162 *states and is the coarsest of the representations stable under P, H that preserve the integer product or, equivalently, the pair (r, q). Additionally preserving ι recovers the* 324 *original oriented positions.*

Proof. Advance modifies only b, and the half-turn only ϵ, proving the two transport identities. The nine observations $n_k = n(\overline{P}^k \chi(x))$, $0 \le k < 9$, run through $a, 2a, \dots, 9a$. Thus one recovers $a = \min_k n_k$, $b = n_0/a$, and the sign through $(n_1/a - n_0/a) \bmod 9 \in \{1, 8\}$. Every stable equivalence that preserves n preserves these nine observations and therefore distinguishes all states (a, b, ϵ). The identity $n = r + 9q$ proves equivalence of the two evaluations. Each fiber of χ contains the two transposed routes: $(i, j, d) \mapsto (j, i, d^\top)$, with $N^\top = O, O^\top = N, E^\top = S, S^\top = E$. The axis label distinguishes these two preimages and allows position and direction to be reconstructed. □

The TRIT selector activates the product at $t \equiv 2, 5, 8 \pmod 9$, before each advance. The half-turn acts at the end of every 27 instants. If y_{t-1} denotes the preceding arithmetic state, the increment and its accumulated memory are determined by

$$\eta_q(t) = \mathbf{1}_{\{t \equiv 2 \ (3)\}} q(y_{t-1}), \qquad \lambda_q(t) = \lambda_q(t-1) + \eta_q(t), \quad \lambda_q(0) = 0. \tag{1.205}$$

The bound $0 \le q(a, b) \le a - 1$ holds. To invert an update, one first undoes the corresponding half-turn, then the advance, and finally subtracts the quotient recovered from the preceding state. The record retains phase, axis, event order, and memory; λ_q is an additive cocycle of this transport.

Each block of 27 contains nine advances and traverses the moving coordinate once. Hence the return of position and orientation at 54, and of phase at 108, coexist with

$$\lambda_q(108N) = NQ_\times^+(a), \qquad Q_\times^+(a) = \frac{180a - 4\sum_{b=1}^{9} r(a,b)}{9}, \qquad N \ge 0. \tag{1.206}$$

Indeed, each block sums the products $a, 2a, \dots, 9a$; summing $n = r + 9q$ over the four blocks gives the formula. If a is coprime to 9, the residues sum to 45, and $Q_\times^+(a) = 20(a-1)$. In window j, counted from $j = 0$, the relative calendar sign is $\sigma_j = (-1)^{\lfloor j/3 \rfloor}$, and the effective orientation is $\epsilon_j = \epsilon_0 \sigma_j$. The charge signed by σ_j vanishes; the charge weighted by the effective orientation likewise vanishes after multiplication by ϵ_0. The ledger preserves the initial orientation, the accumulated charge, and its

increments. These quantities belong to the multiplicative cursor; the readers of a trajectory that alternates between both sheets act on the record of that trajectory.

The finite emission, its coupling of signatures, and ternary reduction have successive types:

$$\mathcal{S}^2 \xrightarrow{s\times e} \operatorname{im} s \times \operatorname{im} e \xrightarrow{G} \mathcal{U}_6 \xrightarrow{\Psi_6} W_6 \hookrightarrow \mathbb{F}_3^6.$$

Section 4.3 proves factorization of the 104 976 seeds into $18 \cdot 26 = 468$ emissions and the 243 visible words of the 729-element ambient space. The maps G and Ψ_6 are those of (4.31) and (4.32); the register preserves arithmetic preimages alongside this finite expression.

1.23.5.8 Compatibility of quotients and refinement cylinders

Extension compares states that simultaneously preserve remainders, quotients and boundaries. In the register's six-coordinate linear chart, the matrix

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad \det A_H = 1,$$

gives the division map

$$\mathcal{H} : \mathbb{Z}^6 \longrightarrow \{0,1,2\}^6 \times \mathbb{Z}^6, \qquad x \longmapsto (u, x^+), \quad y = xA_H, \quad u = [y]_3, \quad x^+ = (y-u)/3. \tag{1.207}$$

Rows are interpreted as vectors. The pair (u, x^+) preserves $y = u + 3x^+$ and recovers $x = (u + 3x^+)A_H^{-1}$. The ternary orientation of u is then obtained through ρ_3^{or}, preserving the preceding quotient. This makes the reversible content of this local chart explicit.

The cylindrical chart relates a ternary prefix of length L and integer index N to K decimal triads of index D. The corresponding half-open cylinders have rational endpoints

$$I_3(N,L) = \left[\frac{N}{3^L}, \frac{N+1}{3^L}\right), \qquad I_{1000}(D,K) = \left[\frac{D}{1000^K}, \frac{D+1}{1000^K}\right).$$

Define the boundary integers

$$a = N1000^K - D3^L, \qquad b = (N+1)1000^K - D3^L.$$

Lemma 1.206 (Integer update of compatible cylinders)**.** *The preceding cylinders intersect if and only if $a < 3^L$ and $b > 0$. Adjoining a ternary block $u \in \{0,\dots,728\}$ gives $N' = 729N + u$ and $L' = L + 6$. For $\Delta K \in \{0,1\}$, the boundaries are updated by*

$$\begin{aligned} \Delta K = 0: &\quad a' = 729a + u1000^K, \\ \Delta K = 1: &\quad D' = 1000D + \delta, \quad a' = 729000a + u1000^{K+1} - \delta\, 729\, 3^L, \end{aligned}$$

where $\delta \in \{0,\dots,999\}$ is the triad of the compatible prolongation. In both cases, $b' = a' + 1000^{K+\Delta K}$.

Proof. Two half-open intervals intersect exactly when each left endpoint is less than the other's right endpoint. Multiplying those two inequalities by $3^L 1000^K$ gives $a < 3^L$, $b > 0$. Substituting N', L', and D' into $a' = N'1000^{K+\Delta K} - D'3^{L'}$ produces the two recurrences. Subtracting the definitions of b' and a' gives the last identity. □

These operations act on integer prefixes and their compatible restrictions. The Archimedean reading appears when the resulting cylinder system is evaluated. The prolongation domain also incorporates orientation, carry, and the joint boundary signature of (1.188); interval comparison is one component of this joint compatibility.

1.23.5.9 Extension relations and joint nonadic connection

For an observable route $r : q \to q'$, prolongation has type

$$\mathrm{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \qquad \mathrm{Ext}_{r_2} \circ \mathrm{Ext}_{r_1} \subseteq \mathrm{Ext}_{r_2 r_1}\,.$$

The identities satisfy $\Delta_{\mathcal{F}_q} \subseteq \mathrm{Ext}_{\mathrm{id}_q}$. It preserves the truncations, sheets, orientations, carries, and boundary incidences of the two histories. The identities and admissible concatenations form the category of TPK states and prolongations. This relational structure admits local fibers with more than one continuation and allows their persistence to be imposed jointly.

Let $\mathcal{H}_m$ be the set of admissible histories of depth m, with truncations $p_{n,m}$. Survival is expressed by

$$\mathrm{Surv}_m^{(N)} = p_{N,m}(\mathcal{H}_N), \qquad \mathrm{Surv}_m = \bigcap_{N \geq m} \mathrm{Surv}_m^{(N)}, \qquad X_\chi = \varprojlim_m \mathrm{Surv}_m^{(\chi)}\,.$$

The compatible section simultaneously preserves all its prefixes. The local connection relations $\mathcal{R}_k$ compose into

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k, \qquad \Gamma_{9,k} : \widehat{X}_k \dashrightarrow \widehat{X}_{k+9}. \tag{1.208}$$

The indicated domain is that of compatible histories. On an individualized history, the nine relations represent its effective transport; the system retains all admissible prolongations before this individualization.

The return law is written

$$g(\Gamma_{9,k}x) = g(x), \qquad q_{\mathrm{mem}}(\Gamma_{9,k}x) = q_{\mathrm{mem}}(x) + 1.$$

Phase returns and history extends. The prefixes $w_6, w_{12}, w_{18}, w_{24}, w_{30}$, boundary R_{36} and charts G_9 are states, extensions or presentations of this construction, with types distinct from the holonomy Γ_9. Their regional realization is developed in Section 10.5; turn composition and its representations are proved in 10.55–10.57.

1.23.5.10 Phase incidence and arithmetic origin of the channels

The selector M_{ph} induces the classes

$$H_+ = \{1,4,7\}, \quad H_- = \{2,5,8\}, \quad H_0 = \{3,6,9\}, \quad H_{\mathrm{act}} = H_+ \sqcup H_-.$$

The phase origin fixes the terminal representative $\star_{\mathrm{ph}} = 9$ and the set $O_{\mathrm{ph}} = D_9 \setminus \{\star_{\mathrm{ph}}\}$. Write $P_2(H_{\mathrm{act}})$ for the unordered pairs of active phases and define

$$\mathcal{F}_6^{\mathrm{ph}} = H_{\mathrm{act}} \times P_2(H_{\mathrm{act}}), \qquad \mathcal{F}_8^{\mathrm{ph}} = O_{\mathrm{ph}} \times P_2(H_{\mathrm{act}}), \qquad \Theta_{\mathrm{ph}} = D_9 \times P_2(H_{\mathrm{act}}).$$

Theorem 1.207 (Incidence of the phase semicorona)**.** *At the declared window origin,*

$$|\mathcal{F}_6^{\mathrm{ph}}| = 90, \qquad |\mathcal{F}_8^{\mathrm{ph}}| = 120, \qquad |\Theta_{\mathrm{ph}}| = 135.$$

The oriented boundary, pointed return, and lateral closure,

$$\partial_{\mathrm{ph}} = \Theta_{\mathrm{ph}} \times \{-1,+1\}, \quad E_{\mathrm{ph}} = \partial_{\mathrm{ph}} \sqcup \{\infty\}, \quad L_{\mathrm{ph}} = H_0 \times P_2(H_{\mathrm{act}}), \quad P_{\mathrm{ph}} = \partial_{\mathrm{ph}} \sqcup L_{\mathrm{ph}},$$

satisfy

$$|\partial_{\mathrm{ph}}| = 270, \quad |E_{\mathrm{ph}}| = 271, \quad |L_{\mathrm{ph}}| = 45, \quad |P_{\mathrm{ph}}| = 315.$$

Translation of the origin transports all these sets by bijections and preserves their incidences.

Proof. There are six active phases and $\binom{6}{2} = 15$ unordered pairs. Products by $6, 8, 9$ give $90, 120, 135$. Orientation doubles the last set, the pointed return adds one element, and the three neutral phases contribute $3 \cdot 15 = 45$ lateral incidences. The C_9 action simultaneously transports the selector, the terminal representative, and each factor of the Cartesian products. □

The difference of active cardinalities is $120 - 90 = 30$. The normalized defect therefore has the exact expression

$$\frac{30}{120} - \frac{30}{90} = -\frac{1}{12}. \tag{1.209}$$

Here 90 and 120 are cardinalities of incidences produced by the phase rule. The subsequent hexadic–octadic realization transports this incidence to its combinatorial objects. The angular coordinates of these objects are established through their realization maps.

1.23.5.11 Temporal record, signed extraction, and reversible coordinates

Let $E_m = \{9(m-1)+1, \dots, 9m\}$. Restricting a history to twelve windows preserves in each its subroute τ_m, orientation σ_m, boundary carry κ_m, and local record Λ_m:

$$\mathcal{R}_{12}(x) = \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=1}^{12}.$$

The domain of $\mathcal{R}_{12}$ is histories; the elements of the group C_{12} are sector classes, and Θ_{108} preserves their calendar. This distinction keeps the data preserved by each realization visible.

The signed extraction of section 11.4 factors through

$$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,mem}} \xrightarrow{\mathcal{R}_{12}} \mathcal{R}_{12}^{\mathrm{mem}} \xrightarrow{\mathcal{E}_{108}^{90,120}} \mathbb{Z}^{12} \times \mathbb{Z}^{12} \times \mathbb{Z} \xrightarrow{\Pi_H} \mathbb{Z}^{12}. \tag{1.210}$$

The second map expresses the channels $(b^{90}, b^{120}, Q_{\mathrm{TPK}})$; the third constructs the signed vector U. The rule $\varepsilon_9 = M_{\mathrm{ph}}$ belongs to phase selection, whereas $\mathcal{E}_{108}^{90,120}$ acts on the complete channel register. Their domains and outputs distinguish them.

Equations (11.104) and (11.105) give the arithmetic action of Π_H. The transformation $\mathcal{H}_{12}$, constructed from the three step-three orbits, then reversibly relates U to the record K. The cyclic differences and total charge provide subsequent readings of the same record. This preserves the common provenance of the regional coordinates and the signed coordinate before the integer closure determining α.

1.23.5.12 Reduced transfer and memory preserved by the dynamics

Once the emission factorization has been constructed, $\mathcal{U}_6 \simeq \mathcal{A}_+ \times \mathcal{A}_\times$, with $|\mathcal{A}_+| = 18$ and $|\mathcal{A}_\times| = 26$, there are two projections on functions of the catalog:

$$(E_+ f)(s,e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s',e), \qquad (E_\times f)(s,e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s,e').$$

The signature alphabets $\mathcal{A}_+, \mathcal{A}_\times$ are distinct from the seed sets of the two cursors. Each projection averages the active coordinate and preserves the other. With $P_{+1} = E_+$, $P_{-1} = E_\times$, $P_0 = I$, the transfer kernel on $\mathcal{U}_6 \times C_9$ is

$$\mathcal{K}_{\mathrm{ph}}((u,r),(v,r+1)) = P_{M_{\mathrm{ph}}(r)}(u,v).$$

The two projections commute and are idempotent. Thus their composition during one phase turn satisfies

$$\mathcal{K}_{\mathrm{ph}}^{9}\big|_r = E_+ E_\times, \qquad (E_+ E_\times)(u,v) = \frac{1}{468}. \tag{1.211}$$

This operator is an expression of reduced memory on emissions already constructed. The holonomy Γ_9 additionally preserves the coordinates of the history on which this expression is performed.

In an isometric realization U_{Γ_9} of transport, and for an isometric inclusion J of the observed coordinate, its two components are

$$T = J^* U_{\Gamma_9} J, \qquad \eta = (I - JJ^*) U_{\Gamma_9} J.$$

One obtains directly

$$U_{\Gamma_9} J = JT + \eta, \qquad J^* \eta = 0, \qquad I - T^* T = \eta^* \eta.$$

The component η is computed from the same realized update and the declared projection. The telescoping identity (4.36) preserves both the contributions expressed in the orthogonal complement and the terminal term. This separation is the operator basis for the subsequent distinction between recorded information, entropy of an observation, and dimensional realization of action.

Chapter 2

Cohomological rank of carry memory: independent cocycles, dimensional lift, and projective preservation of unity

2.1 Cohomology of the helicoidal carry: evaluation of cocycles and minimum dimension of cumulative memory

A periodic coordinate records the visible position; the carry records how many times the trajectory has crossed its boundary. The former returns, whereas the latter accumulates. This distinction turns periodicity into a lift: two instants may share a visible phase yet belong to distinct states because they retain different winding numbers.

In the nonadic alphabet, for $n \in \mathbb{Z}_{\geq 1}$, let us define

$$r(n) = 1 + ((n-1) \bmod 9), \qquad q(n) = \left\lfloor \frac{n-1}{9} \right\rfloor,$$

and the helical lift

$$H_9(n) = (r(n), q(n)). \tag{2.1}$$

Then

$$H_9(n+9) = H_9(n) + (0,1) :$$

the phase returns and memory changes. If $c_k \in \mathbb{Z}$ is the carry of step $k \to k+1$, its discrete integral

$$Q_N = Q_0 + \sum_{k=0}^{N-1} c_k \tag{2.2}$$

constitutes the cumulative coordinate of the lift. Its independence is algebraic; a later metric realization may represent it by means of a transverse or orthogonal direction.

2.1.1 Independent cocycles and minimum number of memory coordinates

Let G be the directed graph of visible states, and let $c_1, \ldots, c_d \in Z^1(G;\mathbb{Z})$ be integer cocycles. For a route γ, define

$$Q_j(\gamma) = \sum_{e \in \gamma} c_j(e), \qquad \widetilde{\gamma} = \big(q_{\mathrm{vis}}(\gamma), Q_1(\gamma), \ldots, Q_d(\gamma)\big).$$

On an open route this evaluation retains the representative of the cocycle and the endpoints. Indeed, with the convention $(\mathrm{d}\phi)(e) = \phi(t(e)) - \phi(s(e))$,

$$Q_j^{c_j + \mathrm{d}\phi}(\gamma) = Q_j^{c_j}(\gamma) + \phi(t\gamma) - \phi(s\gamma). \tag{2.3}$$

Therefore, for an open path, the representative and its endpoints are fixed, and the memory preserves those data. If γ is a cycle, the boundary term vanishes and $Q_j(\gamma)$ depends solely on the class $[c_j] \in H^1(G;\mathbb{Z})$.

Theorem 2.1 (Cohomological criterion for dimensional rank)**.** *Suppose that the classes of $c_1, \ldots, c_d$ generate a free submodule of rank d in $H^1(G;\mathbb{Z})$. Let*

$$Q : Z_1(G;\mathbb{Z}) \longrightarrow \mathbb{Z}^d$$

its evaluation on cycles. If a family of m additive counters is obtained by means of a homomorphism $A : \mathbb{Z}^d \to \mathbb{Z}^m$, such that

$$\widehat{Q} := A \circ Q : Z_1(G;\mathbb{Z}) \longrightarrow \mathbb{Z}^m,$$

and permits reconstruction of the evaluations, that is, there exists $B : \mathbb{Z}^m \to \mathbb{Z}^d$ with $Q = B \circ \widehat{Q}$, then

$$\boxed{m \geq d}.$$

Therefore, the lift requires at least d independent memory coordinates in addition to the visible coordinate.

Proof. The pairing $H^1(G;\mathbb{Z}) \times H_1(G;\mathbb{Z}) \to \mathbb{Z}$ and the independence of $[c_1], \ldots, [c_d]$ imply $\operatorname{rk}\operatorname{im} Q = d$. Since $Q = B \circ \widehat{Q}$,

$$d = \operatorname{rk}\operatorname{im} Q \leq \operatorname{rk}\operatorname{im} \widehat{Q} \leq m.$$

No homomorphism into fewer than d integer counters admits an additive inverse on the submodule generated by the evaluations Q_j. The stronger condition $B \circ A = I_{\mathbb{Z}^d}$ is sufficient, but the bound requires only reconstruction of Q on the evaluations performed by the cycles. □

The genealogical dimension here quantifies degrees of freedom necessary to recover the transport. The theorem does not yet identify that rank with a physical spatial dimension: it provides the minimum domain on which a metric realization may act without erasing the independent cocycles.

2.1.2 Nonadic tower and projective conservation of unity

Let

$$C_9^{(d)} = (\mathbb{Z}/9\mathbb{Z})^d.$$

For to mark $\star \in \mathbb{Z}/9\mathbb{Z}$, define the inclusion

$$h_{d,\star} : C_9^{(d)} \longrightarrow C_9^{(d+1)}, \qquad h_{d,\star}(x) = (x, \star),$$

and the projection

$$\pi_{d+1,d}(x_1, \ldots, x_{d+1}) = (x_1, \ldots, x_d).$$

Proposition 2.2 (Conservation by Dimensional Elevation)**.** *We have*

$$\pi_{d+1,d} \circ h_{d,\star} = \mathrm{id}_{C_9^{(d)}}.$$

Consequently, the cell of dimension d reappears exactly as a coordinate section of the cell of dimension $d + 1$: an affine slice, and a subgroup when $\star = 0$. The term codimension-one face *is reserved for an expressly declared additional cubical realization. For the normalized uniform measures μ_d,*

$$(\pi_{d+1,d})_* \mu_{d+1} = \mu_d.$$

Proof. The first identity follows by eliminating the newly added final coordinate. For a point $x \in C_9^{(d)}$, the fiber of $\pi_{d+1,d}$ contains nine points, each with mass $9^{-(d+1)}$; its total mass is $9^{-d} = \mu_d(\{x\})$. □

The first equality expresses structural conservation; the second, normalized conservation. Lifting does not replace the preceding layer: it includes it as a section and distributes the new coordinate over its fibers. In dimension three, $|C_9^{(3)}| = 729$; its decimal completion and the disjoint decomposition

$$1000 = 729 + 243 + 27 + 1$$

remain in the residence of cubic completion, where interior, faces, edges, and added vertex are distinguished.

2.1.3 From the cumulative coordinate to the metric

Cocycle lifting and the choice of a metric are successive operations. The former constructs independent memory coordinates; the latter declares a quadratic form that measures their variations. In APP, the central block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-$$

produce the reference normalization

$$M_c := \frac{B_c}{\sqrt{45}} \in SO^+(1,1),$$

proved in proposition 1.60. On the cubic screen, the physical realization selects the corresponding Lorentzian signature, and the TPK temporal reader orients its dynamics. The causal chain is

$$\begin{aligned} \text{return with carry} &\longrightarrow \text{cumulative cocycle} \longrightarrow \text{memory rank} \\ &\longrightarrow \text{quadratic form} \longrightarrow \text{realized metric.} \end{aligned}$$

Separation between the internal cohomological range and its spatiotemporal realization. The nonadic helix and the integral of carry provide the premises for the cohomological rank criterion and its lower bound. The cell tower and conservation of measure follow from the finite projection. Identifying genealogical rank with a spacetime dimension requires the declared metric functor in Dimensional Mechanics.

Chapter 3

Spectral, geometric, and operatorial realizations of the APP–TRIT–TPK kernel

3.1 Geometry of APP and structure of its trajectories

The main APP matrix makes it possible to recognize its regularities visually. Its geometric content is determined by the operations relating its positions: translation, inversion, composition of trajectories and simultaneous evaluation of sum and product. These operations produce distinct and compatible realizations. The Cayley graph describes adjacency; periodic identification of the edges produces a toroidal surface; the integer lift preserves winding; the interaction of the two evaluations determines an oriented discrete curvature. The central block, selected within the matrix itself, also admits a spectral realization and a Lorentzian normalization.

Each construction is introduced with its domain. This preserves the relationship between the finite support, its cellular realization, the spaces of functions on it and the bilinear forms obtained from its operators. This distinction makes it possible to use the full geometric content of APP without substituting one realization for another.

3.1.1 Cayley graph and arithmetic structure of trajectories

3.1.1.1 From the positional support to the directed graph

Identifying the representatives $1, \dots, 9$ with the classes modulo nine transforms the positional support of section 1.1 into the additive group

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The projection $\mathcal{X}_9 \to X_9$ is bijective: each class of each coordinate has a unique representative in D_9. The four cardinal displacements determine the symmetric set

$$\mathscr{D} = \{(-1, 0), (0, 1), (1, 0), (0, -1)\}.$$

The orientation of the coordinates is the one fixed earlier: first coordinate southward and second eastward.

Definition 3.1 (Cardinal graph of APP)**.** The directed graph $\mathscr{G}_9$ has vertex set X_9 and an edge $x \to x + v$ for each $x \in X_9$ and $v \in \mathscr{D}$. In Cayley graph notation,

$$\mathscr{G}_9 = \mathrm{Cay}(X_9, \mathscr{D}).$$

The inverse edge of $x \to x + v$ is $x + v \to x$, associated with $-v$.

Proposition 3.2 (Connectivity, degree and translation symmetry)**.** *The graph $\mathscr{G}_9$ is connected, has* 81 *vertices,* 324 *directed edges and* 162 *undirected edges. Each translation $T_a : x \mapsto x + a$ is a graph automorphism and acts freely on the vertices.*

Proof. The classes of $(1, 0)$ and $(0, 1)$ generate X_9. From any vertex, another can be reached through a sequence of these displacements or their inverses; this proves connectivity. The four vectors in $\mathscr{D}$ are distinct modulo nine and none is zero. Therefore, each vertex has four outgoing edges and the total number of directed edges is $81 \cdot 4 = 324$. Each undirected edge appears exactly twice, once in each direction, giving 162.

If $y - x \in \mathscr{D}$, then $(y + a) - (x + a) = y - x \in \mathscr{D}$. Thus, T_a preserves the edges and their orientation; its inverse is T_{-a}. Finally, $T_a(x) = x$ implies $a = 0$, which is the freeness of the action. □

The group structure used in this result is addition of positions. The multiplicative observable Π remains defined on the same vertices and retains the radical described in proposition 1.14. The two operations

therefore play distinct roles: addition defines the translations of the graph; multiplication assigns to each position its main evaluation and multiplicities.

3.1.1.2 Trajectories, concatenation and order of records

An oriented path of length r is specified by an origin x_0 and a sequence $w = a_1 \cdots a_r$ of directions. Its position at time k is

$$x_k = x_0 + \sum_{j=1}^{k} \nu(a_j) \quad \text{in } X_9.$$

The difference $x_k - x_{k-1}$ recovers the direction a_k, since the four generators are distinct. This establishes a bijection between the pairs (x_0, w) and the oriented trajectories of length r. The count $81 \cdot 4^r$ includes exactly the trajectories with backtracking and returns admitted by this definition.

Definition 3.3 (Composition of APP trajectories)**.** A trajectory $\gamma : x \to y$ and another $\delta : y \to z$ are composed by concatenating their lists of edges. The empty trajectory at each position is the identity. The two evaluations of a trajectory are the ordered sequences

$$\Sigma[\gamma] = (\Sigma(x_0), \ldots, \Sigma(x_r)), \qquad \Pi[\gamma] = (\Pi(x_0), \ldots, \Pi(x_r)).$$

The complete arithmetic record adds the quotients q_+ and $q_\times$ to each position.

Concatenation is associative because list concatenation is associative. The preceding objects and operations constitute the path category of the graph. Reversing the path reverses the order of the edges and changes each direction to its opposite. Concatenating a trajectory with its inverse preserves a list of outward and return transitions; passing to the quotient that cancels immediate backtracking yields, by contrast, an identity. This distinction expresses exactly what information the recorded trajectory retains and what information its reduced class preserves.

The APP structure can be brought together as

$$\mathcal{A}_{\mathrm{APP}} = \big(X_9, \mathcal{G}_9, \Sigma, \Pi, \mathrm{Path}(\mathcal{G}_9)\big),$$

accompanied by the remainder–quotient lift defined in (1.7). The term *sheet* designates one of the two evaluations on this common support. If two cursors are used, each trajectory also retains the identifier of the sheet on which it is evaluated. This differentiation is the basis of the subsequent coupling of the two emissions.

3.1.2 Toroidal realization and lifting of paths

3.1.2.1 Periodic identification of the grid

The cellular realization is constructed on the integer lattice of the plane: its vertices are integer pairs; its edges join pairs that differ by a cardinal generator; its faces are unit squares. Cells translated by vectors in $9\mathbb{Z}^2$ are identified. The quotient contains nine positions per direction and its edge graph is precisely $\mathcal{G}_9$.

Proposition 3.4 (Toroidal surface of APP)**.** *The preceding cellular realization is a closed orientable surface homeomorphic to the two-dimensional torus. Its cellular decomposition has* 81 *vertices,* 162 *edges and* 81 *faces, and its Euler characteristic is zero.*

Proof. The square of side length nine, with opposite sides identified by translations, is a fundamental domain. Identifying one pair of opposite sides first yields a cylinder; identifying the two ends of the cylinder by the remaining translation yields the torus. The orientation of the squares in the plane is preserved under translations and descends to the quotient.

The vertex classes are the pairs of residues modulo nine, so there are 81 of them. From each class emanate a positive horizontal edge and a positive vertical edge; each geometric edge has a unique

description of this type, so there are 162. The square classes are identified by their initial vertex and number 81. Consequently,

$$\chi = 81 - 162 + 81 = 0.$$

The identification eliminates the boundary of the fundamental domain: an edge of that boundary is joined to its opposite and belongs to two faces. The link of each vertex is a cycle of four edges, as in the planar grid. This locally verifies the closed-surface structure. □

The edges of the printed table act as cuts of a fundamental domain. Their arithmetic role comes from the chosen representatives. In particular, the last row and the last column correspond to zero-class coordinates. These cuts allow periodicity to be represented on a page without making them an intrinsic boundary of the toroidal surface.

Proposition 3.5 (Distribution of multiplicative zero representatives)**.** *The evaluation* Π *takes the value nine at exactly twenty-one positions. Seventeen belong to the union of the last row and the last column; the remaining four are*

$$(3,3), \quad (3,6), \quad (6,3), \quad (6,6).$$

Proof. The condition $\Pi(i,j) = 9$ is equivalent to $9 \mid ij$. If one coordinate is nine, it holds for any value of the other. The last row and the last column comprise $9 + 9 - 1 = 17$ positions. If both coordinates are between one and eight, divisibility by nine requires both to be multiples of three, since neither already contains the factor nine. The only possibilities are $i, j \in \{3, 6\}$, which give the four indicated positions. The two sets are disjoint and their union has 21 positions. □

Multiplicative vanishing thus distinguishes the coordinate cut and the interior radical sector of the representation. Each position also retains its additive evaluation and quotients; the visible value nine has twenty-one different positional origins.

3.1.2.2 Integer lift and boundary-crossing record

Fix an integer representative $\widetilde{x}_0 \in D_9^2$ of the origin. The lift of a trajectory is constructed without modular reduction:

$$\widetilde{x}_k = \widetilde{x}_0 + \sum_{j=1}^{k} \nu(a_j) \in \mathbb{Z}^2.$$

Its reduction modulo nine is x_k. Denote by $s_9(x_k) \in D_9^2$ the coordinatewise positive representative. There exists a unique vector $q_k \in \mathbb{Z}^2$ such that

$$\widetilde{x}_k = s_9(x_k) + 9q_k.$$

The vector q_k counts the displacements between fundamental domains of the lift.

Proposition 3.6 (Exact accumulation of crossings)**.** *For each transition, the vector*

$$\kappa_k = \frac{s_9(x_{k-1}) + \nu(a_k) - s_9(x_k)}{9} \in \mathbb{Z}^2$$

satisfies $q_k = q_{k-1} + \kappa_k$*. In a trajectory of length* r*,*

$$\sum_{k=1}^{r} \kappa_k = \frac{s_9(x_0) + d(w) - s_9(x_r)}{9}.$$

If the trajectory is closed, this sum is its winding vector $\mathrm{wind}(w)$.

Proof. The congruence $s_9(x_{k-1}) + \nu(a_k) \equiv s_9(x_k) \pmod 9$ proves that κ_k is integral. Substituting $\widetilde{x}_{k-1} = s_9(x_{k-1}) + 9q_{k-1}$ into the lifting rule yields

$$\widetilde{x}_k = s_9(x_k) + 9(q_{k-1} + \kappa_k).$$

Uniqueness of the decomposition gives the first identity. Summing the expressions for κ_k cancels the intermediate positions, leaving the second. For a closed path, $s_9(x_r) = s_9(x_0)$; the result is $d(w)/9$, according to theorem 1.9. □

Example 3.7 (Lift of a horizontal turn)**.** From $(5, 5)$, nine eastward displacements produce the integer columns

$$5, 6, 7, 8, 9, 10, 11, 12, 13, 14.$$

Their positive representatives are

$$5, 6, 7, 8, 9, 1, 2, 3, 4, 5.$$

In the step from 9 to 1, the second component of κ_k equals one; in the other steps it equals zero. The final residue position coincides with the initial one, while the final lift is $(5, 14)$ and the winding is $(0, 1)$.

The crossing integer preserves the accumulated circulation. The ordered sequence of crossings and positions preserves additional data, such as the instant when the cut is crossed. The two records have different applications within the enriched state. Preserving them makes it possible to transport a history that repeatedly traverses the same set of positions.

3.1.2.3 Complex realization and Kähler structure of the continuous torus

The cellular realization admits an explicit geometric extension:

$$\mathbb{T}_{9,\mathrm{an}} = \mathbb{R}^2/(9\mathbb{Z})^2.$$

The map sending an integer position to its class in this quotient includes the 81 APP vertices as a finite grid. Real space is used here as the realization codomain of the arithmetic object already constructed. The developments of the genealogical continuum will separately establish their own construction of values.

In the local coordinates (u, v) inherited from the plane, define

$$g = \mathrm{d}u^2 + \mathrm{d}v^2, \qquad J(a, b) = (-b, a), \qquad \omega = \mathrm{d}u \wedge \mathrm{d}v.$$

These structures are invariant under lattice translations. They therefore descend to the torus.

Proposition 3.8 (Kähler structure of the toroidal realization)**.** *The triple (g, J, ω) defines a flat Kähler structure on $\mathbb{T}_{9,\mathrm{an}}$. It satisfies*

$$J^2 = -I, \qquad g(JX, JY) = g(X, Y), \qquad \omega(X, Y) = g(JX, Y), \qquad \mathrm{d}\omega = 0.$$

Proof. The formula for J gives $J^2(a, b) = (-a, -b)$ and preserves the inner product. If $X = (a, b)$ and $Y = (c, d)$, then

$$g(JX, Y) = -bc + ad = \omega(X, Y).$$

The constant coefficients of ω imply $\mathrm{d}\omega = 0$. The complex coordinate $z = u + \mathrm{i}v$ provides the complex structure; changes between the coordinates of two fundamental domains are translations $z \mapsto z + 9m + 9\mathrm{i}n$, which are holomorphic. Consequently, J is integrable. The metric has constant coefficients in these local coordinates, so its Riemannian curvature vanishes. □

The Kähler property belongs to this differential realization. The graph retains its combinatorial description and the 81 vertices retain their arithmetic evaluations. The explicit choice of g and J shows

how the two levels are related and makes it possible to identify the additional datum used by the continuous realization.

3.1.3 Interaction of the evaluations and discrete curvature

3.1.3.1 Vertex potentials and link differences

We now work with the observables taking values in $R = \mathbb{Z}/9\mathbb{Z}$. To avoid confusing them with their integer representatives, we write

$$\overline{S}(x, y) = x + y, \qquad \overline{P}(x, y) = xy.$$

The map s_9 recovers the visible tables: $\Sigma = s_9 \circ \overline{S}$ and $\Pi = s_9 \circ \overline{P}$. In this subsection, subtractions and products of values are performed in R.

For $T : X_9 \to R$, define

$$\Delta_x T(x, y) = T(x + 1, y) - T(x, y), \qquad \Delta_y T(x, y) = T(x, y + 1) - T(x, y).$$

Each difference is a variable associated with the corresponding link. An ordered choice of potentials (T_x, T_y) determines the variables

$$A_x = \Delta_x T_x, \qquad A_y = \Delta_y T_y.$$

The circulation around the positively oriented plaquette is

$$\begin{aligned} F(T_x, T_y)(x, y) &= A_x(x, y) + A_y(x + 1, y) \\ &\quad - A_x(x, y + 1) - A_y(x, y) \\ &= \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x. \end{aligned} \tag{3.1}$$

The formula simultaneously fixes the orientation and order of the potentials.

Proposition 3.9 (Oriented curvature of the mixed evaluations)**.** *In every APP plaquette, we have*

$$F(\overline{S}, \overline{S}) = F(\overline{P}, \overline{P}) = 0, \qquad F(\overline{S}, \overline{P}) = 1, \qquad F(\overline{P}, \overline{S}) = -1.$$

Proof. The differences of the two potentials are

$$\Delta_x \overline{S} = 1, \quad \Delta_y \overline{S} = 1, \qquad \Delta_x \overline{P} = y, \quad \Delta_y \overline{P} = x.$$

Differencing them again yields $\Delta_x \Delta_y \overline{S} = \Delta_y \Delta_x \overline{S} = 0$ and $\Delta_x \Delta_y \overline{P} = \Delta_y \Delta_x \overline{P} = 1$. Substitution into (3.1) produces the four values. All identities are polynomial in the residue ring, so they remain valid in plaquettes crossing the cut of the positive representation. □

In the same plaquette, selecting sum in the horizontal direction and product in the vertical direction produces an oriented unit of circulation. Interchanging the two selections reverses its sign. The result is obtained from the actual order of the evaluations and their differences, before any physical reading.

Theorem 3.10 (Discrete Stokes identity)**.** *Let $R_{a,b}$ be a rectangle of $a \times b$ plaquettes in a lift of the grid. The oriented sum of its boundary links satisfies*

$$W_{\partial R_{a,b}}(\overline{S}, \overline{P}) = ab, \qquad W_{\partial R_{a,b}}(\overline{P}, \overline{S}) = -ab \quad \text{in } R.$$

Proof. We sum the first expression in (3.1) over the ab plaquettes. Each interior link is traversed once in each direction, with the same potential and opposite values, and its contributions cancel. Exactly the outer boundary links remain, with the induced orientation. The curvature of each plaquette is respectively 1 or -1, so the total sum is ab or $-ab$ modulo nine. □

For a one-plaquette rectangle the circulation is one unit; for a three-by-three rectangle its sum is nine and its residue class is zero. The difference between accumulated integer circulation and its modular evaluation again makes the lifting record relevant. The discrete curvature in this subsection and the Riemannian curvature of the Kähler realization act on different objects: the former measures the mixing of two arithmetic potentials on the links; the latter belongs to the declared metric on the continuous surface.

3.1.4 Selection and spectral decomposition of the central block

3.1.4.1 Intrinsic selection by congruence

In the main matrix, consider the blocks of two consecutive rows and two consecutive columns with matching indices. For $1 \le k \le 8$, their integer evaluations are

$$\widetilde{B}_k = \begin{pmatrix} k^2 & k(k+1) \\ k(k+1) & (k+1)^2 \end{pmatrix}.$$

Their entrywise reduction by ρ_9 defines B_k.

Theorem 3.11 (Uniqueness of the adjacent block with equal diagonals)**.** *The equality condition for the two diagonal entries of B_k selects only $k = 4$. The resulting block is*

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Proof. Equality of positive representatives is equivalent to equality of their classes. Therefore,

$$\rho_9(k^2) = \rho_9((k+1)^2) \quad \Longleftrightarrow \quad 2k + 1 \equiv 0 \pmod 9.$$

The inverse of two modulo nine is five. Multiplying the congruence by five gives $k \equiv -5 \equiv 4 \pmod 9$. Among the indices $1, \dots, 8$, only 4 appears. The four entries are the reductions of $16, 20, 20, 25$, producing $7, 2, 2, 7$. □

The condition selecting this block depends only on adjacent APP entries. The fixed point $(5, 5)$ of central inversion and the block of positions $\{4, 5\} \times \{4, 5\}$ have different types: the former is a position; the latter brings together four evaluations. Their relationships are maintained through the indices, without identifying the point with the matrix.

3.1.4.2 Symmetric and antisymmetric modes

We define the exchange involution and its two projectors:

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_+ = \frac{I + R}{2}, \qquad P_- = \frac{I - R}{2}.$$

The vectors $(1, 1)$ and $(1, -1)$ generate their respective eigenspaces. Since $R^2 = I$, we have $P_\pm^2 = P_\pm$, $P_+P_- = 0$ and $P_+ + P_- = I$.

Proposition 3.12 (Exact decomposition of the block)**.** *The selected block satisfies*

$$B_c = 7I + 2R = 9P_+ + 5P_-, \qquad \det B_c = 45, \qquad \operatorname{tr} B_c = 14.$$

Its eigenvalue difference is four and its exchange coefficient is two.

Proof. Substituting $I = P_+ + P_-$ and $R = P_+ - P_-$ into $7I + 2R$ yields $9P_+ + 5P_-$. The eigenvalues are therefore nine and five; their product is 45 and their sum 14. The difference is $9 - 5 = 4$, while the coefficient of the operator R is the two in the antidiagonal entries. □

The ordered decimal concatenation of each row produces the blocks 72 and 27. Concatenation of the diagonals produces 77 and 22. These readings satisfy

$$72 + 27 = 77 + 22 = 99, \qquad 72 - 27 = 45, \qquad 77 - 22 = 55 = 54 + 1.$$

The equality $72 - 27 = \det B_c$ relates two explicit evaluations of the same block: decimal concatenation and determinant. Each reading retains its rule; equality of their results is a verifiable identity between these rules.

3.1.5 Conservative normalizations of the block and Lorentzian realization

3.1.5.1 Stochastic normalization and differential contraction

The sum of each row and each column of B_c equals nine. Division by nine defines the doubly stochastic matrix

$$P_B = \frac{1}{9} B_c = \begin{pmatrix} 7/9 & 2/9 \\ 2/9 & 7/9 \end{pmatrix} = P_+ + \frac{5}{9} P_-.$$

Its entries are positive and its row and column sums are one.

Proposition 3.13 (Iteration of the two modes)**.** *For every $n \geq 0$,*

$$P_B^n = P_+ + \left(\frac{5}{9}\right)^n P_-.$$

If $v = (a, b)^\mathsf{T}$, the sum of its components is preserved and their difference is multiplied by $(5/9)^n$.

Proof. Orthogonality of the projectors eliminates all cross products in the power. Their positive powers are themselves, yielding the formula. Moreover,

$$v = \frac{a+b}{2}(1, 1)^\mathsf{T} + \frac{a-b}{2}(1, -1)^\mathsf{T}.$$

The first component is fixed and the second is multiplied by $(5/9)^n$. This simultaneously proves preservation of the sum and contraction of the difference. □

This reading describes a local transformation of two components. Its factor $5/9$ and gap $1 - 5/9 = 4/9$ come from the spectrum of the selected block. The global TPK operator uses additional states and records; the two scales retain their own domains.

3.1.5.2 Determinantal normalization and two-dimensional Minkowski form

Let $\eta = \mathrm{diag}(1, -1)$ and define

$$M_c = \frac{B_c}{\sqrt{45}}.$$

The normalization is fixed by the determinant just obtained from the block.

Theorem 3.14 (Local Lorentzian realization of APP)**.** *The matrix M_c satisfies*

$$M_c^\mathsf{T} \eta M_c = \eta, \qquad \det M_c = 1.$$

It preserves the future component of the timelike cone of the form $t^2 - x^2$ and belongs to $SO^+(1, 1)$. If

$$\theta_c = \frac{1}{2} \log \frac{9}{5},$$

then

$$M_c = \exp(\theta_c R), \qquad \frac{5}{9} = \exp(-2\theta_c).$$

Proof. Direct multiplication gives

$$B_c^{\mathsf{T}} \eta B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} \begin{pmatrix} 7 & 2 \\ -2 & -7 \end{pmatrix} = \begin{pmatrix} 45 & 0 \\ 0 & -45 \end{pmatrix} = 45\eta.$$

Dividing by 45 proves invariance of the form. Since $\det B_c = 45$, dividing each of the two columns by $\sqrt{45}$ gives $\det M_c = 1$.

For $t > |x|$, we have $7t + 2x \geq 7t - 2|x| > 0$. The first coordinate of $M_c(t, x)$ is positive and the timelike form is preserved, so it remains in the same component of the cone.

Finally, the spectral decomposition produces

$$M_c = \sqrt{\frac{9}{5}}\, P_+ + \sqrt{\frac{5}{9}}\, P_-.$$

Since $R = P_+ - P_-$, its exponential is $e^{\theta_c} P_+ + e^{-\theta_c} P_-$. The definitions of θ_c identify the two coefficients exactly. The last identity follows from $-2\theta_c = -\log(9/5)$. □

The same integer matrix therefore admits two explicit normalizations. Normalization by the row sum preserves the uniform mode and contracts the differential mode. Normalization by the square root of the determinant preserves an indefinite form. The relation $5/9 = e^{-2\theta_c}$ links the two readings of the block. The proved realization has dimension $1 + 1$; subsequent incidence constructions and dimensional maps determine its physical extensions.

3.1.6 Reduction by ternary classes and aggregate spectral structure

3.1.6.1 Averaging of the two evaluations

The projection from classes modulo nine to classes modulo three determines

$$C_1 = \{1, 4, 7\}, \qquad C_2 = \{2, 5, 8\}, \qquad C_0 = \{3, 6, 9\}.$$

Each set $C_a \times C_b$ contains nine positions. For a real evaluation T on D_9^2, we define its aggregate matrix by

$$(M_T)_{ab} = \frac{1}{9} \sum_{i \in C_a} \sum_{j \in C_b} T(i, j), \qquad a, b \in \{1, 2, 0\}.$$

The index order $(1, 2, 0)$ is fixed in rows and columns.

Proposition 3.15 (Aggregate APP matrices)**.** *Averaging the residue tables gives*

$$M_\Pi = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \qquad M_\Sigma = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The totals are preserved through

$$9 \sum_{a,b} (M_\Pi)_{ab} = 459, \qquad 9 \sum_{a,b} (M_\Sigma)_{ab} = 405.$$

Proof. For multiplication, the block $C_1 \times C_1$ contains each value $1, 4, 7$ three times, so its average is $(1 + 4 + 7)/3 = 4$. The block $C_1 \times C_2$ contains each value $2, 5, 8$ three times and has average five. The blocks of a unit class with C_0 contain each value $3, 6, 9$ three times and have average six. The block $C_2 \times C_2$ has average four and the block $C_0 \times C_0$ consists of nine values of nine. Symmetry completes the matrix.

In addition, the block $C_a \times C_b$ distributes the three representatives of the class $a + b$ modulo three uniformly. The averages of the classes C_1, C_2, C_0 are respectively four, five and six. Evaluating $a + b$ in the fixed order yields M_Σ. The nine blocks partition the 81 positions and each average is multiplied by nine to recover its sum. This proves the two identities for totals. □

We obtain $\sum M_\Pi = 51$ and $\sum M_\Sigma = 45$. The difference six corresponds to the aggregate structure; restoring the multiplicity of nine positions per block recovers $9 \cdot 6 = 54$. This calculation connects ternary aggregation with the global contrast in section 1.1.5.

3.1.6.2 Form of signature two plus one and hyperbolic sublattice

Theorem 3.16 (Exact spectrum of the aggregate multiplicative matrix)**.** *The characteristic polynomial and spectrum of M_Π are*

$$\det(\lambda I - M_\Pi) = (\lambda + 1)((\lambda - 9)^2 - 72),$$
$$\mathrm{Spec}(M_\Pi) = \{-1, 9 - 6\sqrt{2}, 9 + 6\sqrt{2}\}.$$

Its determinant is -9 and the quadratic form $v \mapsto v^{\mathsf{T}} M_\Pi v$ has signature $(2, 1)$.

Proof. The vector $v_- = (1, -1, 0)$ satisfies $M_\Pi v_- = -v_-$. The plane generated by $v_+ = (1, 1, 0)$ and $v_0 = (0, 0, 1)$ is invariant, since

$$M_\Pi v_+ = 9v_+ + 12v_0, \qquad M_\Pi v_0 = 6v_+ + 9v_0.$$

The matrix of the restriction in that basis is $\left(\begin{smallmatrix} 9 & 6 \\ 12 & 9 \end{smallmatrix}\right)$ and its characteristic polynomial is $(\lambda - 9)^2 - 72$. Adding the factor $\lambda + 1$ from the first direction gives the formula. The two roots of the quadratic factor are positive because $81 > 72$. The direction with eigenvalue -1 is negative and the other two are positive. The product of the eigenvalues is $-(81 - 72) = -9$. □

The restriction to the invariant plane is written

$$\begin{pmatrix} 9 & 6 \\ 12 & 9 \end{pmatrix} = 3H, \qquad H = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}, \qquad \det H = 1.$$

The eigenvalues of H are $3 \pm 2\sqrt{2} = (1 \pm \sqrt{2})^2$. The matrix whose columns are v_-, v_+, v_0 has determinant of absolute value two. These vectors therefore generate a sublattice of index two in $\mathbb{Z}^3$. The decomposition is complete as a real decomposition; its integral version explicitly retains this index.

3.1.6.3 Localization of the contrast in the nonunit sector

The sector formed by the rows and columns with indices $3, 6, 9$ comprises $27 + 27 - 9 = 45$ positions. Their intersection $C_0 \times C_0$ has nine entries of value nine and sums to 81. The remaining four blocks, $C_0 \times C_1$, $C_0 \times C_2$, $C_1 \times C_0$ and $C_2 \times C_0$, each sum to 54 and contribute 216. Thus,

$$297 = 216 + 81 = 3 \cdot 72 + 3 \cdot 27 = 3(72 + 27).$$

The partition by ternary classes and the concatenations of the central block produce these identities through their respective procedures. The factor three remains explicit; the value 297 belongs to a sum over 45 positions and the value 99 belongs to a concatenation of pairs from the block.

The numbers appearing throughout the development thus acquire precise meanings: four is the local spectral separation; two, the exchange coefficient; six, the difference of aggregate averages; fifty-four, the global residue contrast. Their relationships are proved through the operations connecting each level.

3.1.7 Operator realization of trajectories

Let $\mathcal{H}_X = \ell^2(X_9)$, with the orthonormal basis $(e_x)_{x\in X_9}$, and let U be an operator whose matrix elements satisfy $U_{yx} = 0$ when $x \to y$ is not an edge of $\mathcal{G}_9$. The hypothesis expresses locality with respect to the graph already constructed. Once that operator is fixed, the composition of its transitions has an exact expansion.

Theorem 3.17 (Finite path expansion of the transition)**.** *For $r \geq 1$ and positions $i, f \in X_9$,*

$$(U^r)_{fi} = \sum_{\substack{\gamma: i\to f \\ |\gamma|=r}} \prod_{k=0}^{r-1} U_{x_{k+1},x_k}, \qquad \gamma = (x_0 = i, x_1, \ldots, x_r = f).$$

Proof. For $r = 1$, the statement coincides with the entry U_{fi}, and entries outside the graph are zero by hypothesis. Assume the formula holds for r. Matrix multiplication gives

$$(U^{r+1})_{fi} = \sum_{j\in X_9} U_{fj}(U^r)_{ji}.$$

We substitute the inductive expression for $(U^r)_{ji}$. Each term is a trajectory of length r from i to j, followed by the final edge $j \to f$. Every trajectory of length $r+1$ has a unique penultimate vertex j and a unique truncation of length r. The sum therefore traverses each trajectory exactly once, with the product of its transition elements. □

The expansion expresses the composition of a declared operator through APP trajectories. The complete record of a TPK transition also incorporates the selected sheet, the tritic signature, the carry, the phase and memory. The support and operator expansion of this chapter provide the domains on which this enriched update is constructed. The dependency order remains explicit: first positions and evaluations, then trajectories and their realizations, and upon them the law of selection and transport with memory.

3.2 TRIT as an oriented ternary state: carry memory, temporal order, and 3 quadratic regimes

TRIT preserves within a single state the published residue, the carry cocycle, orientation, and compositional memory. Its 3 algebraic regimes are not editorial layers: they are correlative realizations of the quadratic operator $J_\tau^2 = -\tau I$, linked to the 2 orientations of time and to the threshold state.

3.3 Ternary local state and transport orientation

TRIT is the ternary coordinate that preserves the local orientation of a transition. Its definition precedes any temporal interpretation: the three values distinguish algebraic types of transport and acquire a reading of past, threshold, or future only when an ordered parameter is fixed.

3.3.1 Oriented ternary signature and state space

The space of visible signatures is

$$\widehat{\mathbb{T}} = \{-1, 0, +1\}.$$

Its elements are oriented symbols, not results of an operation in $\mathbb{Z}$. The balanced map

$$\mathbb{F}_3 \longrightarrow \widehat{\mathbb{T}}, \qquad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1,$$

allows passage from the ternary residue to the oriented signature.

Definition 3.18 (Complete state of the TRIT and carry fiber)**.** In a local step whose amplitude requires at most one carry, the balanced ternary decomposition is written

$$\widehat{\tau} = (r, c), \qquad a + b = r + 3c, \qquad r, c \in \{-1, 0, +1\}.$$

The first coordinate r is the visible residue, and the second c preserves the quotient that made that reading possible. In particular, the fiber over visible zero contains three distinct states:

$$0^+ = (0, +1), \qquad 0^0 = (0, 0), \qquad 0^- = (0, -1), \qquad \mathrm{vis}(0^+) = \mathrm{vis}(0^0) = \mathrm{vis}(0^-) = 0.$$

This local separation between residue and quotient is what, in the subsequent Bockstein–Hensel lift, takes the matrix form

$$x_k A = u_k + 3x_{k+1}.$$

There u_k publishes the residue of level k, while x_{k+1} transports the memory necessary to recover the lift x_k. Therefore, a visible zero does not by itself determine the state or its admissible future. This assertion is algebraic: it does not convert $0^\pm$ into an infinitesimal, an energy, a physical torsion, or a temporal interval.

A visible trit does not determine the set of admissible continuations. The complete transport state space contains the fields

$$x = (t, \mu, o, h, c, \sigma, \lambda, g).$$

Here $t \in \widehat{\mathbb{T}}$ is the visible signature; $\mu \in \mathsf{M} := \{\Sigma, \Pi, \varnothing\}$, the label of the additive, multiplicative, or neutral observable; o, the orientation; h, the extension coordinate; c, the accumulated carry; σ, the boundary signature; λ, the trajectory register; and $g \in \mathbb{Z}/9\mathbb{Z}$, the phase. The projection $x \mapsto t$ removes all other components.

Remark 3.19 (Distinction between symbol and state). Two states may display the same trit and have different continuations. Nor does a ternary word by itself determine the observable label, orientation, carry, or monodromy. This distinction explains why five regions of π may share a word w_π without becoming a single region.

3.4 Quadratic algebras and transport regimes

3.4.1 3 associated quadratic algebras

For $\tau \in \widehat{\mathbb{T}}$, let $\bar{\tau} \in \{-1, 0, +1\} \subset \mathbb{R}$ be its image under the preceding balanced identification. We consider the unital real algebra

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \bar{\tau}),$$

in which J denotes the class of X and I the unit. This convention separates the TRIT symbol from its scalar value. According to the sign, three types are obtained:

$$\begin{aligned} \tau = +1 \quad &: \quad J^2 = -I, \quad \text{elliptic or complex type,} \\ \tau = 0 \quad &: \quad J^2 = 0, \quad \text{parabolic or nilpotent type,} \\ \tau = -1 \quad &: \quad J^2 = +I, \quad \text{hyperbolic or split type.} \end{aligned}$$

Exponentials exhibit the difference:

$$e^{tJ} = \begin{cases} \cos t + J \sin t, & J^2 = -I, \\ 1 + tJ, & J^2 = 0, \\ \cosh t + J \sinh t, & J^2 = +I. \end{cases}$$

The three algebras are, respectively, the complex extension, the algebra of dual numbers, and the split real algebra. Their exponential subgroups produce elliptic, parabolic, and hyperbolic evolutions without introducing any additional continuous geometry into the definition of TRIT. The concrete real realizations selected by the APP Coxeter object, the nilpotent generator, and the TPK sheet incidence are constructed, without duplicating this abstract classification, in section 3.5.6.

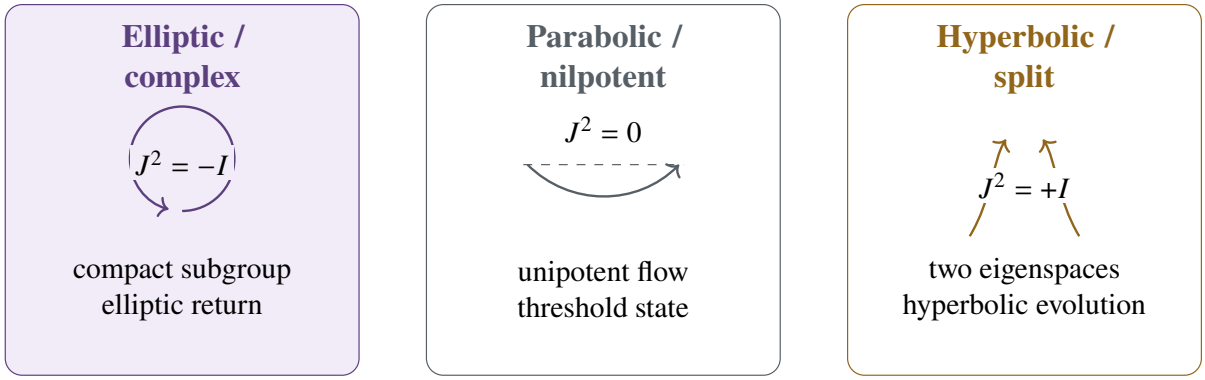


Figure 3.1: Three algebraic types associated with the oriented ternary state. The first line records the algebraic identities; the second, their interpretation with respect to an ordered parametrization.

3.4.2 Compatibility and parameterized trajectories

The expression *All Possible Paths* invites one to imagine paths traversed in time. HMT places the object one step earlier. Let Ω_{HMT} be a set of local states endowed with an admissible transition relation $\rightsquigarrow$, a TRIT signature, and transport data to be typed in the following section. If $\mathcal{T}$ is a discretely ordered set, a parametrized trajectory is a map

$$\gamma : \mathcal{T} \longrightarrow \Omega_{\mathrm{HMT}}$$

that satisfies $\gamma(t) \rightsquigarrow \gamma(t^+)$ for every successor t^+. The ordering of $\mathcal{T}$ converts compatibility into a parameterized sequence. Before an order is chosen, there is only a compatible family indexed by the corresponding graph or poset; it is not called a section unless a projection $p : E \to \mathcal{T}$ with $p \circ \gamma = \mathrm{id}_{\mathcal{T}}$ has been declared. Consequently, a finite sum over paths can be read as a sum over temporalizations of a structure that, at its primary level, presupposes no external time.

3.4.3 Ordered parameterization of compatibility

An ordered parametrization permits return, threshold, and opening to be interpreted as three relative positions in an evolution. The convention used in Figure 3.1 is

$$\begin{aligned} J^2 = -I &\quad\rightsquigarrow\quad \text{return and reverse orientation,} \\ J^2 = 0 &\quad\rightsquigarrow\quad \text{threshold state,} \\ J^2 = +I &\quad\rightsquigarrow\quad \text{decomposition into two eigendirections.} \end{aligned}$$

In the adopted temporal parametrization, $+1$ represents return, memory, and past; 0, threshold and present; and -1, bidirectional opening and future. This ordered reading does not alter the three preceding quadratic types. The exact involution, typed for each $n \geq 1$,

$$C_{\mathrm{rev}} : \widehat{\mathbb{T}}^n \longrightarrow \widehat{\mathbb{T}}^n, \qquad C_{\mathrm{rev}}(\tau_1, \ldots, \tau_n) = (-\tau_n, \ldots, -\tau_1)$$

exchanges $+1$ and -1, preserves 0, and satisfies $C_{\mathrm{rev}}^2 = \mathrm{id}_{\widehat{\mathbb{T}}^n}$. Thus, temporal orientation is not introduced as a fourth state: it results from the parametrization of the signature together with an involution that conserves the threshold.

3.4.4 Spectral decomposition of the hyperbolic type

In the hyperbolic type, the projectors

$$P_{\pm} = \frac{I \pm J}{2}$$

separate two proper directions. The involution that changes orientation exchanges them. Algebraically, $P_+P_- = 0$, $P_+ + P_- = I$, and $JP_{\pm} = \pm P_{\pm}$. Therefore the hyperbolic type contains two complementary proper subspaces and an exact operation that permutes them.

Remark 3.20 (Complete structure of the TRIT state). TRIT is not a substitute for the signs $-, 0, +$. It is the minimal observable of a state with an observable label, orientation, and path record. Its three associated algebras make it possible for one and the same support to admit elliptic closures, parabolic thresholds, and hyperbolic openings before a temporal evaluation is fixed.

3.5 Quadratic algebras of order 9

The identity $A_W^2 = -I_6$, obtained in the preceding section by reducing the Paley conference matrix, contains more information than a modular orthogonality relation. It determines a scalar extension of degree two, turns the six-dimensional ternary space into a three-dimensional space over the field of nine elements, and permits the ternary Golay code to be written as the graph of a quadratic unit. This section develops those consequences and distinguishes them from the characteristic-nine arithmetic used by the Hensel branch.

3.5.1 Estructura of $\mathbb{F}_9$ inducida by Paley

The polynomial $X^2 + 1$ has no roots in $\mathbb{F}_3$: its values in $0, 1, 2$ are $1, 2, 2$. It is therefore irreducible, and the quotient

$$\mathbb{F}_9 := \mathbb{F}_3[\iota]/(\iota^2 + 1)$$

is a field of nine elements.

Theorem 3.21 (Structure quadratic induced by A_W)**.** *Let $V = \mathbb{F}_3^6$, written by means of row vectors, and let $J(v) = vA_W$. The rule*

$$(a + b\iota)v := av + bJ(v), \qquad a, b \in \mathbb{F}_3,$$

turns V into a vector space of dimension three over $\mathbb{F}_9$. In the coordinates fixed by the Paley construction,

$$\mathbb{F}_3^6 \cong \mathbb{F}_9^3.$$

Proof. The equality $J^2 = -I$ makes the evaluation $\mathbb{F}_3[X] \to \mathrm{End}_{\mathbb{F}_3}(V)$, $X \mapsto J$, vanish on the ideal $(X^2 + 1)$. The action therefore factorizes through the field $\mathbb{F}_9$. Since $[\,\mathbb{F}_9 : \mathbb{F}_3\,] = 2$ and $\dim_{\mathbb{F}_3} V = 6$, one obtains $\dim_{\mathbb{F}_9} V = 3$. □

The isomorphism depends on the coordinate system that fixes A_W. The assertion does not follow from the mere cardinal $3^6 = 9^3$: the matrix exhibits multiplication by all scalars $a + b\iota$ and therefore provides the complete linear structure.

3.5.2 The ternary code as an isotropic graph

The code's generating application is

$$c(q) = (q, qA_W).$$

After identifying J with multiplication by ι, its image admits the writing

$$C_W = \{(q, \iota q) : q \in \mathbb{F}_9^3\},$$

where both components are regarded, by restriction of scalars, as vectors of $\mathbb{F}_3^6$.

Proposition 3.22 (Isotropic maximality)**.** *With respect to the standard bilinear form of $\mathbb{F}_3^{12}$, the subspace C_W is totally isotropic and maximal among the totally isotropic subspaces.*

Proof. The generator matrix $G = (I_6 \mid A_W)$ satisfies

$$GG^{\mathsf{T}} = I_6 + A_W A_W^{\mathsf{T}} = I_6 + 2I_6 = 0.$$

The generated space is therefore totally isotropic. It has dimension six, half the dimension of the nondegenerate ambient space, and is therefore maximal; indeed, it coincides with its orthogonal dual. □

The root of $-I$ thus appears as a finite incidence transformation. The analytic complex field is not introduced into the system: a quadratic structure over $\mathbb{F}_3$ is obtained, sufficient to organize the polarity, the code, and its two eigenspaces after extending scalars.

3.5.3 Characteristic 9 and characteristic 3

Two sets of 729 states enter the construction:

$$(\mathbb{Z}/9\mathbb{Z})^3 \qquad \text{and} \qquad \mathbb{F}_9^3.$$

They are not isomorphic as additive groups or as rings. The first has characteristic nine and contains elements of order nine; the second has characteristic three and its entire additive group has exponent three.

For $a, b \in \mathbb{F}_3$, choosing representatives $\widetilde{a}, \widetilde{b} \in \{0, 1, 2\}$, let

$$\beta(a, b) = \widetilde{a} + 3\widetilde{b} \pmod 9.$$

The coordinate map induced by β is a bijection of sets between $\mathbb{F}_3^6$ and $(\mathbb{Z}/9\mathbb{Z})^3$, but does not preserve addition. Indeed, if

$$\kappa(a, c) := \left\lfloor \frac{\widetilde{a} + \widetilde{c}}{3} \right\rfloor,$$

then

$$\beta(a, b) + \beta(c, d) = \beta\big(a + c,\ b + d + \kappa(a, c)\big),$$

with the coordinates of the second member reduced modulo three. For example,

$$\beta(2, 0) + \beta(2, 0) = \beta(1, 1),$$

whereas coordinatewise addition in $\mathbb{F}_3^2$ would give $(1, 0)$. The second coordinate of $\beta(1, 1)$ preserves the quotient of Euclidean division; this term is precisely the datum lost under componentwise reduction.

This distinction types the two readings of the cardinal 729. The realization $(\mathbb{Z}/9\mathbb{Z})^3$ preserves residue and quotient; the realization $\mathbb{F}_9^3$ converts each ternary pair into a scalar of a quadratic extension. One prolongs arithmetic with division memory; the other organizes exceptional incidence.

Proposition 3.23 (Two typed readings of the same ternary support)**.** *Once the set-theoretic section used in β, the basis, the polarization, and the operator J of theorem 3.21 are fixed, there exists the diagram of set bijections*

$$\boxed{(\mathbb{Z}/9\mathbb{Z})^3 \xleftarrow{\beta^{\times 3}} \mathbb{F}_3^6 \xrightarrow{\iota_J} \mathbb{F}_9^3.}$$

The left arrow transports the nonadic sum twisted by the carry cocycle κ; the right arrow transports the linear structure over $\mathbb{F}_9$ determined by $J^2 = -I$. The two arrows share the coordinate support $\mathbb{F}_3^6$, but do not identify their algebraic laws.

Proof. The formula

$$\beta(a,b)+\beta(c,d)=\beta\big(a+c,b+d+\kappa(a,c)\big)$$

proves that $\beta^{\times 3}$ is a bijection with twisted addition and is not an isomorphism from the componentwise sum of $\mathbb{F}_3^6$. By theorem 3.21, the choice of J turns that same six-dimensional ternary space into a three-dimensional space over $\mathbb{F}_3[J]\cong\mathbb{F}_9$, thereby defining ι_J. □

The proposition is a synthesis demonstrated with explicit starting structure. It does not assert a canonical identification between $(\mathbb{Z}/9\mathbb{Z})^3$ and $\mathbb{F}_9^3$: the first contains additive elements of order nine, whereas the second has additive exponent three. The cocycle measures precisely the information that would be lost by confusing both laws.

3.5.4 Classification of Quadratic Algebras

Over a field of characteristic other than two, a unital quadratic algebra is classified by the type of its defining polynomial. Over $\mathbb{F}_3$ the three representatives may be chosen as

$$Q_{\mathrm{e}}=\mathbb{F}_3[t]/(t^2+1)\cong\mathbb{F}_9,$$

$$Q_{\mathrm{n}}=\mathbb{F}_3[\varepsilon]/(\varepsilon^2),\qquad Q_{\mathrm{s}}=\mathbb{F}_3[r]/(r^2-1)\cong\mathbb{F}_3\times\mathbb{F}_3.$$

Theorem 3.24 (Classification of the quadratic algebras over $\mathbb{F}_3$)**.** *The relations*

$$J^2=-I,\qquad N^2=0,\qquad R^2=I$$

define actions of Q_{e}, Q_{n}, and Q_{s}, respectively. These three algebras have nine elements and are mutually nonisomorphic: the first is a field; the second contains nonzero nilpotents; the third contains nontrivial idempotents.

Proof. Each relation makes the evaluation of $\mathbb{F}_3[t]$ factor through the corresponding ideal. The irreducibility of t^2+1 proves that Q_{e} is a field. In Q_{n}, the class ε is nonzero and satisfies $\varepsilon^2=0$. In Q_{s}, since $2^{-1}=2$ in $\mathbb{F}_3$, the elements

$$e_+=2(1+r),\qquad e_-=2(1-r)$$

are nontrivial orthogonal idempotents and sum to the unit. Being a field, possessing a nonzero nilpotent, and possessing nontrivial idempotents are properties invariant under isomorphism. □

Three copies of the two-dimensional regular representation yield three six-dimensional actions on $\mathbb{F}_3$. The relation $J^2=-I$ corresponds to Paley polarity; relation $N^2=0$ describes the linearization of an extension with quotient; relation $R^2=I$ separates two summands through the idempotents $e_\pm$. This classification supplies the finite algebraic framework in which the elliptic, nonreduced, and split structures are distinguished without appeal to a continuous geometry.

The construction does not replace $\mathbb{C}$: the field $\mathbb{F}_9$ does not possess the topology, completeness, or complex analytic structure. The exact result is different: a square root of $-I$ and the corresponding extension of scalars emerge from the incidence matrix already constructed within the ternary system.

3.5.5 Integral quadratic form and finite and real specializations

The finite root and the real operator root are coordinated through a common integral presentation. Let

$$Q_{\mathbb{Z}}:=\mathbb{Z}[u]/(u^2+1). \tag{3.2}$$

Its change of basis are

$$Q_{\mathbb{Z}}\otimes_{\mathbb{Z}}\mathbb{F}_3\cong\mathbb{F}_3[u]/(u^2+1)\cong\mathbb{F}_9,$$

and

$$Q_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}[u]/(u^2+1).$$

Theorem 3.25 (Two realizations of an integral quadratic form)**.** *The finite representation $u \mapsto A_W$ on $\mathbb{F}_3^6$ and the real representation $u \mapsto J_{\mathrm{e}}$ on V_{e}, constructed in the theorem 3.26, realize, after their respective changes of basis, the same integral relation*

$$u^2 + 1 = 0.$$

Proof. The equality $A_W^2 + I_6 = 0$ makes the evaluation $\mathbb{F}_3[u] \to \mathrm{End}_{\mathbb{F}_3}(\mathbb{F}_3^6)$ factor through (u^2+1). In the same way, $J_{\mathrm{e}}^2 + I = 0$ makes $\mathbb{R}[u] \to \mathrm{End}_{\mathbb{R}}(V_{\mathrm{e}})$ factor. The two representations follow from specializations of $Q_{\mathbb{Z}}$; no unital homomorphism $\mathbb{F}_9 \to \mathbb{R}$ exists or is used. □

The common source coordinates relations; it does not identify modules of distinct characteristics. The finite branch preserves incidence; the real branch incorporates the analytic structure required for the operator exponential.

3.5.6 3 generators concrete of the TRIT regimes

The section 3.4 fixes the three abstract quadratic types $J^2 = -I$, $J^2 = 0$, and $J^2 = I$, while the theorem 3.24 provides their finite models on $\mathbb{F}_3$. Those objects must not be confused with the subsequent real realizations. Here three real spaces and three concrete operators are fixed, each obtained from its declared causal antecedent.

Let

$$V_{\mathrm{e}} = A_2 \otimes_{\mathbb{Z}} \mathbb{R}, \qquad V_{\mathrm{p}} = \mathbb{R}^2, \qquad V_{\mathrm{h}} = \mathbb{R}^2.$$

On V_{e} acts the Coxeter C of the proposition 1.94. On V_{p} it fixes

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Finally, the theorem 6.20 constructs

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad A := F_{\mathrm{av}}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

on V_{h}.

Theorem 3.26 (Three concrete realizations of the TRIT regimes)**.** *Define*

$$J_{\mathrm{e}} := \frac{2C + I}{\sqrt{3}}, \qquad J_{\mathrm{p}} := N, \qquad J_{\mathrm{h}} := \frac{2A - 3I}{\sqrt{5}}.$$

Then

$$J_{\mathrm{e}}^2 = -I_{V_{\mathrm{e}}}, \qquad J_{\mathrm{p}}^2 = 0, \qquad J_{\mathrm{h}}^2 = I_{V_{\mathrm{h}}}.$$

Therefore, for $t \in \mathbb{R}$,

$$\exp(tJ_{\mathrm{e}}) = \cos t\, I + \sin t\, J_{\mathrm{e}}, \qquad \exp(tJ_{\mathrm{p}}) = I + tJ_{\mathrm{p}},$$

$$\exp(tJ_{\mathrm{h}}) = \cosh t\, I + \sinh t\, J_{\mathrm{h}}.$$

Furthermore, in the elliptic branch,

$$C = -\frac{1}{2} I + \frac{\sqrt{3}}{2} J_{\mathrm{e}},$$

and, for $n \in \mathbb{Z}$,

$$C^n = \cos\left(\frac{2\pi n}{3}\right) I + \sin\left(\frac{2\pi n}{3}\right) J_{\mathrm{e}}.$$

In the hyperbolic branch, the form

$$G = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$$

has signature $(1,1)$ *and satisfies*

$$A^{\mathsf{T}} G A = G, \qquad \det A = 1, \qquad \operatorname{tr} A = 3.$$

If

$$\eta := \operatorname{arcosh} \frac{3}{2} = 2 \log \varphi,$$

then

$$A = \cosh(\eta) I + \sinh(\eta) J_{\mathrm{h}}, \qquad A^n = \cosh(n\eta) I + \sinh(n\eta) J_{\mathrm{h}}.$$

Proof. The polinomio caracteristico of the Coxeter is $X^2 + X + 1$; by Cayley–Hamilton,

$$C^2 + C + I = 0.$$

From here,

$$(2C + I)^2 = 4C^2 + 4C + I = -3I,$$

and, therefore, $J_{\mathrm{e}}^2 = -I$. The identity for C^n follows from $C = -\frac{1}{2} I + \frac{\sqrt{3}}{2} J_{\mathrm{e}}$.

Direct multiplication gives $N^2 = 0$. Hence, the exponential series truncates exactly:

$$\exp(tN) = I + tN.$$

For the third branch,

$$A^{\mathsf{T}} G A = G, \qquad \det G = -5 < 0,$$

so that G is nondegenerate and has signature $(1,1)$. Likewise,

$$(2A - 3I)^2 = 5I,$$

therefore $J_{\mathrm{h}}^2 = I$. As

$$\cosh(2 \log \varphi) = \frac{3}{2}, \qquad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2},$$

we obtain

$$A = \frac{3}{2} I + \frac{\sqrt{5}}{2} J_{\mathrm{h}} = \cosh(\eta) I + \sinh(\eta) J_{\mathrm{h}}.$$

The formula for A^n follows from the exponential law. The three general expressions are obtained, in each case, by separating the even and odd powers in the exponential series. □

Corollary 3.27 (Extended Euler equation by the tritic parameter)**.** *For* $\tau \in \{+1, 0, -1\}$*, let* J_τ *be one of the preceding generators, normalized by*

$$J_\tau^2 = -\tau I.$$

Define the functions of the quadratic family.

$$C_\tau(t) = \begin{cases} \cos t, & \tau = +1, \\ 1, & \tau = 0, \\ \cosh t, & \tau = -1, \end{cases} \qquad S_\tau(t) = \begin{cases} \sin t, & \tau = +1, \\ t, & \tau = 0, \\ \sinh t, & \tau = -1. \end{cases}$$

Then the three branches satisfy a single exponential identity:

$$\boxed{\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau.} \tag{3.3}$$

In particular,

$$\exp(\pi J_{+1}) + I = 0, \qquad \exp(J_0) = I + J_0, \qquad \exp\bigl(2\log\varphi\, J_{-1}\bigr) = F_{\mathrm{av}}^2.$$

The first equality is Euler's matrix closure; the second realizes the parabolic threshold transport; the third incorporates Fibonacci self-scaling through the eigenvalues φ and $-\varphi^{-1}$.

Proof. The identity (3.3) is obtained by separating the even and odd powers of the exponential series and using $J_\tau^2 = -\tau I$. The three specializations follow from $J_{+1}^2 = -I$, $J_0^2 = 0$, and from the already proved equality $F_{\mathrm{av}}^2 = \exp(2\log\varphi\, J_{-1})$. □

3 **exact realizations of the quadratic regimes of the TRIT** The elliptic, parabolic, and hyperbolic branches realize, respectively, the internal Coxeter element of A_2, the nilpotent generator, and the incidence F_{av} induced by the TPK calendar. The matrix identities are exact in the declared spaces and gauges.

Realization conditions of the 3 **quadratic regimes of TRIT and obstructions of algebraic nondegeneration** The isolated TRIT type organizes the three quadratic possibilities, but it does not by itself select the operators C, N, and F_{av}: the selection uses, respectively, the APP Coxeter element, the nilpotent transport model, and the TPK descent of modes. The appearance of π in the elliptic spectral formula belongs to the analytic expression of the cycle and does not replace the regional generation of π_{HMT}. The identity $A^{\mathsf{T}}GA = G$ constructs an exact Lorentz transformation in dimension $1+1$; it does not by itself identify this screen with physical spacetime $3+1$. A failure of

$$(2C + I)^2 = -3I, \qquad N^2 = 0, \qquad (2A - 3I)^2 = 5I, \qquad A^{\mathsf{T}}GA = G$$

would refute the corresponding realization.

3.6 Operator genealogy of arithmetic and discrete calculus: composition, contraction of fibers, and lifts with memory

Arithmetic publishes results; a genealogical theory must additionally preserve the conditions that made each result possible. This distinction does not add a new operation to sum, product, power, root, logarithm, differentiation, or integration. It specifies what structure each contracts and what data a lift must transport in order to distinguish antecedents identified by the same output. APP exhibits the first case on a common support; TRIT locally orients the transition; TPK preserves provenance throughout composition.

3.6.1 Published operation and genealogical lifting

Let X and Y be sets and let $f : X \to Y$ be a map that publishes an operation. An *operation memory* is a set M_f together with a map

$$\widetilde{f} : X \longrightarrow Y \times M_f, \qquad \mathrm{pr}_Y \circ \widetilde{f} = f. \tag{3.4}$$

The first component publishes the output; the second records the data that the reader elects to preserve. The lift is *complete on the fibers* if, for every $y \in Y$, the restriction

$$\mathrm{pr}_{M_f} \circ \widetilde{f}\big|_{f^{-1}(y)} : f^{-1}(y) \longrightarrow M_f$$

is injective. This condition expresses exactly that two antecedents with the same visible output remain distinguishable by their memory. In HMT realizations, no single token is required to preserve the entire history: the complete memory may be distributed among sheet, quotient, carry, phase, orientation, boundary, and the enriched chronological register.

If $f : X \to Y$ and $g : Y \to Z$ possess fixed lifts, their composition admits the canonical lift associated with those lifts,

$$\widetilde{g \circ f} : X \longrightarrow Z \times M_f \times M_g, \qquad \widetilde{g \circ f}(x) = \big(g(f(x)), m_f(x), m_g(f(x))\big), \tag{3.5}$$

where m_f and m_g are the memory components. Concatenation does not erase the provenance of the first stage in order to replace it with the second: it coordinates them. This is the operator principle that the TPK realizes on paths.

Lemma 3.28 (Composition of complete surveys)**.** *If $\widetilde{f}$ and $\widetilde{g}$ are complete over the fibers of f and g, respectively, then $\widetilde{g \circ f}$ is complete over the fibers of $g \circ f$.*

Proof. Let $x_1, x_2 \in X$ with $g(f(x_1)) = g(f(x_2))$ and with equal memory components in (3.5). Equality of the memories $m_g(f(x_1)) = m_g(f(x_2))$, together with the completeness of $\widetilde{g}$ on that fiber of g, implies $f(x_1) = f(x_2)$. The equality $m_f(x_1) = m_f(x_2)$ and the completeness of $\widetilde{f}$ then imply $x_1 = x_2$. □

3.6.2 Sum and product: alternatives and composition

In a path algebra, the sum combines alternatives with compatible endpoints,

$$a_1\gamma_1 + a_2\gamma_2,$$

while the product concatenates stages when the end of the first coincides with the origin of the second,

$$\gamma_2\gamma_1, \qquad t(\gamma_1) = s(\gamma_2).$$

Addition preserves the linear coexistence of histories; multiplication constructs a composite history and, when only its value is published, may forget the factorization that produced it. On the APP chart, the evaluations Σ and Π act on the same state (i, j) but organize distinct fibers. In the central block B_c of APP, the spectral separation developed in the following chapter measures precisely that local incompatibility; it is not a discrepancy between two independent tables, nor is it presented as a global distance among all readers.

On the additive sheet, translation $T_a(x) = x + a$ is reversible on the residual group. On the multiplicative sheet, $M_a(x) = ax$ is invertible in $\mathbb{Z}/9\mathbb{Z}$ exactly when $\gcd(a, 9) = 1$. The nonunitary classes $\overline{0}, \overline{3}, \overline{6}$, published in the positive chart as $9, 3, 6$, fold fibers and make visible the need to preserve quotient and provenance. Sum and product thus constitute the two inaugural readers of a single path geometry.

Lemma 3.29 (Affine conjugation in the unit sector)**.** *Let $R = \mathbb{Z}/9\mathbb{Z}$. For $a \in R^\times$ and $b \in R$, the maps $T_b(x) = x + b$ and $M_a(x) = ax$ satisfy*

$$M_a T_b M_a^{-1} = T_{ab}. \tag{3.6}$$

If $a \in (3)$, M_a ceases to be invertible and conjugation is not defined: its image remains contained in the radical and several branches of the additive sheet are identified.

Proof. For $a \in R^\times$, $M_a T_b M_a^{-1}(x) = a(a^{-1}x + b) = x + ab = T_{ab}(x)$. If $a \in (3)$, the kernel of M_a is nontrivial; hence there is no M_a^{-1}, and the conjugation expression is ill-typed. The case $a = 3$ explicitly realizes the folding through $\ker M_3 = \operatorname{im} M_3 = (3)$. □

3.6.3 Power, logarithm, and root

Let x be an element of a monoid, or of an associative algebra, and let $n \geq 1$. The power records iteration of a composition:

$$x^n = \underbrace{x \cdots x}_{n \text{ factores}}.$$

The exponent is the depth of that iteration, and the orbit of x determines which outputs are identified. In the positive real domain, if $b > 0$, $b \neq 1$, and $x > 0$, then

$$\log_b(x^n) = n \log_b x.$$

In the complex domain, the same identity requires a compatible branch and requires the iteration not to cross its cut. The logarithm coordinates the depth n only after the generator x has also been fixed and $\log_b x \neq 0$ has been required. Base, generator, domain, branch, and cut form part of the reader.

The root inverts a power as a fibered problem,

$$y^n = x.$$

The fiber may be empty, a singleton, or multiple. Selecting a root requires branch, orientation, phase, or boundary data. The subsequent construction of an internal operator $J_E \in \mathrm{End}(V)$, with $J_E^2 = -I_V$, exemplifies this principle: the symbol $\sqrt{-1}$ publishes a coordinate, whereas the operator, its domain, and its action preserve the structure that supports it.

3.6.4 Discrete Calculus on Compatible Cylinders: Differentiation and Integration Operators

Let P_n be a connected oriented path and let A be an abelian group. In the discrete measure complex, the combinatorial coboundary

$$d : C^0(P_n; A) \longrightarrow C^1(P_n; A), \qquad (df)(e) = f(t(e)) - f(s(e)),$$

has as its kernel exactly the constant 0-cochains. Reconstructing f from df requires a boundary condition, for example, the value at a reference vertex.

The coboundary depends on incidence and orientation, but not on a length. If A is also a real vector space and $\ell : E(P_n) \to \mathbb{R}_{>0}$ is an edge length, the metric derivative is defined by

$$D_\ell f(e) = \frac{(df)(e)}{\ell(e)}.$$

For a route $\gamma = (e_1, \ldots, e_r)$ compatible with the orientations,

$$\sum_{k=1}^{r} D_\ell f(e_k)\, \ell(e_k) = f(t(\gamma)) - f(s(\gamma)). \tag{3.7}$$

Integration thus accumulates a cochain along a route and recovers the endpoint difference when the cochain is exact. Coboundary, metric derivative, and integral depend on distinct data: incidence and orientation; length and scale; path and boundary section.

This dependence explains why the passage from counting marks to measuring intervals is structural. The derivation lives on interstitial relations; integration composes those relations; the cycle adds the topological carrier of holonomy; TPK transport determines which memory survives the revolution.

3.6.5 Hierarchy of fibers and function of the TPK

The hierarchy can be condensed without identifying its levels:

Operation	Primary action	Fiber contracted by the output	Memory of reconstruction
Sum	aggregation or translation	alternatives with the same evaluation	origin, coefficients and support
Product	composition or scale	nonunitary factorizations and classes	order of factors, quotient and sheet
Power	iteration	composition orbits	exponent, order, and initial point
Root	multivalued inversion	antecedents of a power	branch, phase, orientation, and boundary
Logarithm	depth coordinate	periods and determinations	base, generator, domain, branch, and cutoff
Cobordism and metric derivative	local difference and normalization	modo constante	incidence, orientation, length, scale and reference
Integration	path accumulation	paths with equal published value	path, measure, boundary and holonomy

TPK conserves these coordinates within transportable states and not as a list added to the result. Its file (D, C, f, I, L, M) declares for each operator the domain, codomain, rule, invariants, the information published or lost, and the conserved memory. The transition

$$\text{visible operation} \longrightarrow \text{lift with memory} \longrightarrow \text{compatible section}$$

provides the link between the inaugural arithmetic, reconstruction of the continuum, and the subsequent definition of genealogical number. The number–particle boundary will preserve the same architecture when adding a dimensional realization to the surviving section.

Autonomous arithmetic domain of the nonadic cell and causal precedence with respect to APP–TRIT–TPK and the constants analiticas The priority of sum and product in APP, the conservation of phase, sheet, carry, and boundary by TPK, and the transition from section to route constitute the premises of the lift (3.4), the notion of completeness over fibers, and the table of memories. An operation is not identified with its physical realization: each subsequent step preserves its domain, its chart, and its source.

3.7 APP–TRIT–TPK Composition: arithmetic support, ternary orientation, and memory-based transport

3.7.1 APP as fiber arithmetic

APP is not an illustrative table: it is the arithmetic support on which the difference between aggregation and composition becomes visible. In $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$, summation accumulates displacements and alternative histories; multiplication composes stages and, when a factor is not invertible, folds distinct inputs onto the same residue. The residue–quotient lift unfolds those fibers and preserves the genealogy contracted by the modular publication.

Let $q(a, b) = a + b \pmod 9$. The published datum $q(a, b)$ does not determine the pair (a, b). The fiber

$$q^{-1}(r) = \{(a, b) : a + b \equiv r \pmod 9\}$$

is memory of the alternatives. For multiplication, the contraction is even more pronounced: the nonunit classes $(0, 3, 6)$ produce fibers of distinct sizes. APP therefore publishes a geometry of operations before any physical interpretation.

The active central block is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2S = 9P_+ + 5P_-, \qquad P_\pm = \frac{I \pm S}{2}.$$

Its local spectral jump is $9 - 5 = 4$; the coupling coefficient is 2. The value 54 belongs to a later unfolding and, in the electronic fiber, to $D_1(\Gamma_e)$. Equating them because they occur in a numerical chain would erase their types.

3.7.2 Local Lorentz geometry and Minkowski plane $1 + 1$

Normalization

$$M_c = \frac{B_c}{\sqrt{45}} = \exp\left(\frac{1}{2}\log\frac{9}{5}\,S\right)$$

realizes an element of $SO^+(1, 1)$. APP thus contains, prior to continuous physics, a decomposition into P_+, P_- modes, a gap, a coupling, and a velocity. The result is not yet Minkowski space-time: it is the algebraic structure that will make a Lorentzian representation possible.

3.7.3 TRIT as orientation and local memory

TRIT introduces the sign that the modular publication omits. In a triadic sum.

$$a + b = r + 3c,$$

the residue r publishes the result, and the carry c preserves how it was obtained. The states $(+1, 0, -1)$ should not be interpreted solely as 3 colors. In the temporal parametrization they act, respectively, as return with memory of the past, present threshold, and bidirectional opening toward the future; in the algebraic realization they distinguish counterflow, fixation, and flow; in the geometric realization they label the hyperbolic, parabolic, and elliptic branches of a quadratic family. Compatibility among these readings requires maps, not a verbal identification.

Definition 3.30 (Enhanced TRIT state)**.** A TRIT state is a pair (r, c) endowed with orientation. The first component is publishable; the second belongs to the memory fiber. Change of reader may forget c, but TPK transport cannot do so.

This separation makes it possible to formalize present, past, and future without converting them into three absolute instants. The present is the locus of coupling between two orientations; past and future are senses of transport. Route inversion exchanges both senses and leaves the coupling point as a section of boundary.

3.7.4 TPK as the dynamics of selection, transport, memory, return, and deployment.

TPK brings together support, orientation, and memory. Its objects are finite or coinductive prepared sections; its morphisms are recorded routes compatible with the readers. Each route contains

$$\Gamma = (\text{origin}, \text{target}, \text{sheet}, \text{orientation}, \text{carry}, \text{boundary}, \text{record}).$$

Composition concatenates paths and composes memories. Inversion reverses the order and orientation. The readers, the 108-phase calendar, boundary, and record are internal operators, not added external observers.

A published operation $f : X \to Y$ admits a lift

$$\widetilde{f} : X \longrightarrow Y \times M_f, \qquad \pi_Y \circ \widetilde{f} = f.$$

Given fixed lifts of f and g, the composition has an associated canonical lift

$$\widetilde{g \circ f} : X \longrightarrow Z \times M_f \times M_g.$$

Memory is not a comment on the operation: it is part of its enriched codomain.

Proposition 3.31 (Genealogy of the operations)**.** *In TPK, sum, product, power, root, logarithm, derivation, and integration admit a common reading:*

Operation	*Action*	*Preserved Memory*
Sum	*aggregation of alternatives*	*origin and carry*
Product	*stage composition*	*order and folded fiber*
Power	*monoid iteration*	*depth and orbit*
Root	*multivalued inversion*	*branch and orientation*
Logarithm	*iteration coordinate*	*generator and caliber*
Derivation	*differ by scale*	*border and orientation*
Integration	*composition of increments*	*path and boundary conditions*

Exponentiation does not introduce an operation ontologically disconnected from addition and product: it iterates composition. The root inverts that iteration and must preserve the branch selection. The logarithm records depth. This hierarchy explains why nonadic powers and their vacancies are informative for APP: the raised operation preserves layers of memory that an isolated digital root contracts.

3.7.5 Temporal periodicity of 108 phases and returns at 27, 54, and 108 iterations

The observable calendar has three distinct thresholds. After 27 iterations, the position returns and the orientation is reversed; after 54, position and orientation are restored; after 108, the complete observable phase is restored. The memory of the positive semigroup, however, is not erased. Schematically,

$$F_{\text{obs}}^{27}(x, o) = (x, -o), \qquad F_{\text{obs}}^{54}(x, o) = (x, o), \qquad F_{\text{obs}}^{108}(x, o, \phi) = (x, o, \phi).$$

This hierarchical order will be crucial for separating outward propagation, return, orientation, and complete cycle. A finite observable period can coexist with nonperiodic genealogical memory.

Type controls

It is inadmissible to identify TPK with a list of 34 operators; to call the structure a "fibered category" without Cartesian lifts; to place the carry simultaneously as a cellular coboundary and as a 2-cocycle without separating their complexes; or to convert the block $SO^+(1, 1)$ into physical spacetime before constructing the metric representation.

3.8 Hopf fibration, Villarceau circles, double covering, and Möbius holonomy

3.8.1 Complex structure and Hopf fibration

Let $S^3 \subset \mathbb{C}^2$ and

$$h(z_0, z_1) = \left(2\Re(z_0\overline{z}_1), 2\Im(z_0\overline{z}_1), |z_0|^2 - |z_1|^2\right) \in S^2.$$

On the Hopf torus

$$T_\eta = \{(z_0, z_1) : |z_0| = \cos\eta, |z_1| = \sin\eta\},$$

the curves

$$\gamma_+(t) = (\cos\eta\, e^{it}, \sin\eta\, e^{it}),$$
$$\gamma_-(t) = (\cos\eta\, e^{it}, \sin\eta\, e^{-it})$$

have distinct behaviors: γ_+ is a Hopf fiber and γ_- projects with degree two. Under stereographic projection, the homology classes $(1,1)$ and $(1,-1)$ produce the two diagonal Villarceau families.

The total complex conjugation

$$K(z_0, z_1) = (\overline{z}_0, \overline{z}_1)$$

preserves each unoriented family and reverses its orientation. It does not interchange the two families. Partial conjugation can interchange them, but it is neither a complex-linear map nor the standard global antiunitary. This correction is essential to avoid manufacturing a particle–antiparticle symmetry from the toroidal drawing.

3.8.2 Double covering and spinor return

The map $SU(2) \to SO(3)$ is a double cover. A loop of 2π in $SO(3)$ can lift to $-I$ in $SU(2)$; the loop of 4π recovers I. The sign structure does not turn a circle into a Möbius band. Obtaining a band requires a real line bundle whose holonomy character is -1.

Let $L_\chi \to S^1$ be the associated bundle to

$$\chi : \pi_1(S^1) \longrightarrow \{\pm 1\}, \qquad \chi(1) = -1.$$

Its total space is a Möbius band. The route $C_M = -\,\mathrm{rev}$ provides the sign candidate; the Villarceau circle provides the loop; the physical intertwiner must prove that the former is the holonomy of the latter.

3.8.3 Spectral decomposition of the oriented memory operator and derivation of the ratio (5:2)

With $B_c = 7I + 2S$, define the oriented memory operator

$$B^{\varepsilon}_{\mathrm{mem}} = 5I + 2\varepsilon S, \qquad \varepsilon = \pm 1.$$

For $\varepsilon = +1$,

$$\mathrm{Spec}(B^{+}_{\mathrm{mem}}) = \{7, 3\}, \qquad \det B^{+}_{\mathrm{mem}} = 21.$$

Interpreting the values $(7, 3)$ as the outer and inner equatorial boundaries in a common positive chart yields

$$R + r = 7\ell, \qquad R - r = 3\ell,$$

and therefore

$$R = 5\ell, \qquad r = 2\ell, \qquad \sin\beta = \frac{r}{R} = \frac{2}{5}.$$

The Villarceau angle is

$$\beta = \arcsin\frac{2}{5} = 23.5781784782\ldots^{\circ}.$$

The algebraic theorem $\{7, 3\} \leftrightarrow (5, 2)$ is exact. The geometric premise interpreting the spectrum as the two boundaries of a torus is a declared interface. It must not be concealed, nor may the angle be used as a target.

The realization of the full block B_c retains another reading: $R : r = 7 : 2$ and $\beta = \arcsin(2/7)$. These are not rival values; they belong to two functors: total block and memory mode. The physical priority of the latter must be decided by the intertwiner with the dynamics, not by decimal proximity to another constant.

3.8.4 Conjugate Angular Coordinates and Phase Unit Normalization in the Hopf Fibration

HMT's native angle is a phase $u \in \mathbb{R}/\mathbb{Z}$. The readers publish

$$\theta_{\mathrm{rad}} = 2\pi u, \qquad \theta_{\mathrm{deg}} = 360u.$$

Radians and degrees are distinct charts of the same circular element. The radian is not more "ontological" by virtue of being dimensionless; its naturality follows from arc length. Degrees provide another absolute coordinate of the full turn. Any relation with α^{-1}, φ, or CKM must first be formulated as an invariant quotient or holonomy and only thereafter evaluated in an angular chart.

The exact identity already available

$$\tanh^2(2\log\varphi) = \frac{5}{9}$$

connects the Perron–Fibonacci channel to the spectral quotient of B_c. Its complement is $4/9$, the normalized gap. This relation is structural. It does not imply that every angle constructed with φ is a physical angle.

3.8.5 Brouwer, Möbius, and degree

Brouwer's fixed-point theorem, fractional linear Mobius transformations, and the Mobius band are three distinct objects. They are related only after fixing domain, action, and bundle. The elliptic advance and retardation orbits may be modeled as two sections of a bundle with sign holonomy; the Mobius transformation describes the conformal action in the chart; Brouwer's theorem guarantees a fixed point for continuous maps of the disk. None of these statements replaces the realization diagram.

Result and boundary

The degree-two character of the Hopf curve, the orientation of the total conjugation, and the algebraic construction $7, 3 \mapsto 5, 2$ are proved. The toroidal reading (5:2), Möbius holonomy, and the particle–antiparticle interpretation form a chain of interfaces that must share a single intertwiner. The volume does not force that intertwiner through an angular coincidence.

3.9 Finite Hilbertian realization of the path system and its conditional projectors

3.9.1 Matrix realization of each depth

For $d \geq 1$, define

$$C_9^{(d)} = (\mathbb{Z}/9\mathbb{Z})^d, \qquad H_d = \ell^2(C_9^{(d)}), \qquad A_d = B(H_d) \cong M_{9^d}(\mathbb{C}).$$

The canonical basis $\{\delta_x : x \in C_9^{(d)}\}$ preserves the discrete label of each state. The uniform measure induces the normalized trace

$$\tau_d(a) = \frac{1}{9^d}\,\mathrm{Tr}(a).$$

The factorization

$$C_9^{(d+1)} = C_9^{(d)} \times \mathbb{Z}/9\mathbb{Z}$$

produce a unitary identification

$$H_{d+1} \cong H_d \otimes \mathbb{C}^9.$$

This is the exact input to the operator tower. This does not identify $(\mathbb{Z}/9)^3$ with $\mathbb{F}_3^6$ as groups: they share 729 elements, but their operations and carry cocycles are distinct.

3.9.2 Weyl–Heisenberg Representation of Phase and Displacement on the Triadic Fiber

Let $\omega = e^{2\pi i/9}$. On $H_1 = \ell^2(\mathbb{Z}/9\mathbb{Z})$, we define

$$(Xf)(r) = f(r-1), \qquad (Zf)(r) = \omega^r f(r).$$

Then

$$X^9 = Z^9 = I, \qquad ZX = \omega XZ.$$

The group generated by $X, Z, \omega I$ is the finite Heisenberg group of order $9^3 = 729$.

Theorem 3.32 (Stone–von Neumann finite)**.** *Every irreducible unitary representation of the finite Heisenberg group in which the center acts by means of the primitive character ωI is unitarily equivalent to the preceding representation.*

Proof. Let v be an eigenvector of Z. The vectors

$$v, Xv, \ldots, X^8 v$$

have eigenvalues distinct from Z, since $ZX^k v = \omega^k X^k Zv$. They are orthogonal and span an invariant subspace. By irreducibility, they form a basis of the entire representation. In that basis, X is the cyclic shift and Z the diagonal phase. □

Structural obstruction

Uniqueness fails if the primitive central character is not fixed or if reducible representations are allowed. This finite theorem must not be confused with the continuous version for the canonical commutation relations.

3.9.3 Conditional projectors and pre-Heisenberg incompatibility

The additive and multiplicative sheets of APP define information subspaces and their orthogonal projectors $P_{\Sigma,d}, P_{\Pi,d}$. The constructed identities satisfy

$$\|[P_{\Sigma,2}, P_{\Pi,2}]\| = \frac{\sqrt{2}}{3}, \qquad \|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}.$$

Noncommutativity is not deduced from a quantum analogy: it is a finite matrix theorem. Its "pre-Heisenberg" reading is subsequent to that calculation.

Proposition 3.33. *The preceding projectors do not jointly belong to the same Boolean subalgebra of projections.*

Proof. In an abelian subalgebra all projectors commute. The displayed commutators are nonzero. □

3.9.4 Hilbertian structure of the matrix publication and preserved genealogical memory

Hilbert space provides linear superposition, orthogonal projection, and spectral calculus. HMT supplies the genealogical index of the basis and distinguishes which operator represents a transition, a measurement, or a shadow. The realization does not replace TPK: it represents it.

Premises and derivation

The APP projectors and the Weyl–Heisenberg pair are part of the preceding construction. The organization H_d, A_d and the uniqueness proof are established here by applying the corresponding classical theorem to the HMT objects.

3.10 Finite symplectic structure of APP and nonadic fibration of observables

Symplectic geometry is not part of the published theorems on De las Cuevas–Netzer magic squares. The legitimate link passes through a neighboring, standard route: Weyl operators, discrete phase space and mutually unbiased bases [52, 100]. In HMT, that route acquires a nonadic lift and fiber memory.

3.10.1 The alternating form and the Weyl commutation

On $V_3 = \mathbb{F}_3^2$ we define

$$\langle (q,p),(q',p')\rangle_{\mathrm{sp}} = qp' - pq'. \tag{GMC-5.1}$$

If $X|k\rangle = |k+1\rangle$, $Z|k\rangle = \omega^k|k\rangle$, and $\omega = e^{2\pi i/3}$, the operators $W_{q,p} = X^q Z^p$ satisfy, under these conventions,

$$W_u W_v = \omega^{-\langle u,v\rangle_{\mathrm{sp}}} W_v W_u. \tag{GMC-5.2}$$

Therefore, two operators commute exactly when their labels are orthogonal for the alternating form. In dimension two, the maximal isotropic sublines are the four points of

$$\mathbb{P}^1(\mathbb{F}_3) = \{[1:0],[0:1],[1:1],[1:2]\}. \tag{GMC-5.3}$$

Each generates a maximal abelian context of $M_3(\mathbb{C})$; its spectral projectors constitute a PVM. Contexts associated with distinct lines produce the four MUB bases of (GMC-2.13).

Proposition 3.34 (Qutrit decomposition by directions)**.** *Let $\mathcal{D}_a$ be the four diagonal subalgebras determined by the lines of* (GMC-5.3) *and $\mathcal{D}_a^0$ their traceless parts. Then*

$$M_3(\mathbb{C}) = \mathbb{C}I \oplus \bigoplus_{a\in\mathbb{P}^1(\mathbb{F}_3)} \mathcal{D}_a^0 \tag{GMC-5.4}$$

as an orthogonal sum for the Hilbert–Schmidt product. The dimension count is $9 = 1 + 4\cdot 2$.

Proof. Each $\mathcal{D}_a^0$ has dimension two. Unbiasedness implies $\mathrm{Tr}(A^*B) = 0$ for $A \in \mathcal{D}_a^0$ and $B \in \mathcal{D}_b^0$, $a \neq b$. All are orthogonal to I. The dimensions sum to nine, the dimension of $M_3(\mathbb{C})$. □

This identity explains why the twelve projections are not twelve arbitrary independent pieces: they are four resolutions of the identity whose centered parts decompose the entire qutrit operator space.

3.10.2 The lift $12 \to 4\times 3$

On the modulo nonadic $V_9 = (\mathbb{Z}/9\mathbb{Z})^2$ it preserves the forma alternante

$$\langle (a,b),(c,d)\rangle_9 = ad - bc \pmod 9. \tag{GMC-5.5}$$

The projective line $\mathbb{P}^1(\mathbb{Z}/9\mathbb{Z})$ consists of the primitive vectors modulo multiplication by a unit. It can be represented as

$$\{[1:t] : t \in \mathbb{Z}/9\mathbb{Z}\} \sqcup \{[3u:1] : u \in \mathbb{Z}/3\mathbb{Z}\}, \tag{GMC-5.6}$$

and has $9 + 3 = 12$ points. Reduction modulo three gives the diagram

$$\mathbb{P}^1(\mathbb{Z}/9\mathbb{Z}) \xrightarrow{r} \mathbb{P}^1(\mathbb{F}_3), \qquad |r^{-1}(a)| = 3. \tag{GMC-5.7}$$

The basis a of a qutrit context and the outcome k of its PVM likewise form a set 4×3. A *calibration* is a bijection

$$\Xi : \mathbb{P}^1(\mathbb{Z}/9\mathbb{Z}) \longrightarrow \{P_{a,k} : a \in \mathbb{P}^1(\mathbb{F}_3),\ k \in \mathbb{F}_3\} \tag{GMC-5.8}$$

that sends each fiber of r to the three outcomes of the context lying over a.

Structural obstruction

The fibration is canonical; assigning the internal order of each fiber to the three outputs k is a gauge. A bijection is not yet an equivariance. To promote Ξ to an intertwiner, one must prove that the relevant linear, projective, and antiunitary actions are transported in every fiber. The preliminary gauge fails an antiunitary check in one fiber; that failure is preserved as a falsifier, not concealed.

3.10.3 Depth, carry, and memory over the discrete phase space

The Weyl geometry over $\mathbb{F}_3^2$ explains commutation, contexts, and MUB. HMT adds three levels that are not recovered from the symplectic form by itself:

1. the lift of a ternary direction to three points of $\mathbb{P}^1(\mathbb{Z}/9\mathbb{Z})$;
2. the sheet, carry and orientation that distinguish states with the same reduction;
3. the TPK route that transports those coordinates and can return to the phase without returning to the state.

The potential contribution to the qutrit literature is not "another construction of four MUBs," which is already known. It is a geometry of *context roots*: each ternary context has three nonadic antecedents, and those antecedents may carry dynamics and monodromy. The order-twelve square of Theorem 26.9 is the first normalized shadow of that atlas.

3.10.4 Typed separation between the qutrit overlap $1/3$ and the HMT reader $1/\pi$

In a pair of MUB qutrit bases,

$$\operatorname{Tr}(P_{a,k}P_{b,\ell}) = \frac{1}{3}. \tag{GMC-5.9}$$

In the HMT reader defined here, the relation between a real sheet and a volumetric reading can take the quotient

$$\frac{\pi}{\pi^2} = \frac{1}{\pi}. \tag{GMC-5.10}$$

The two expressions are not interchangeable: $1/3$ is an overlap between projectors in dimension three; $1/\pi$ is a scalar weight of another reader. The difference does not prevent their coordination. For a fixed PVM $\{P_0, P_1, P_2\}$, we define a measurement with nine effects:

$$F_k^{(\pi)} = \frac{3}{\pi}P_k \quad (k = 0, 1, 2), \qquad F_r^{(\pi)} = \frac{1 - 3/\pi}{6}I_3 \quad (r = 3, \dots, 8). \tag{GMC-5.11}$$

It is a POVM because $3/\pi < 1$ and

$$\sum_{r=0}^{8} F_r^{(\pi)} = \frac{3}{\pi}I_3 + \left(1 - \frac{3}{\pi}\right) I_3 = I_3. \tag{GMC-5.12}$$

If Q is a rank-one projector of a distinct MUB context,

$$\operatorname{Tr}(QF_k^{(\pi)}) = \frac{3}{\pi}\operatorname{Tr}(QP_k) = \frac{3}{\pi}\cdot\frac{1}{3} = \frac{1}{\pi}. \tag{GMC-5.13}$$

Matrix bridge result

The formula (GMC-5.11) exactly transforms the MUB probability $1/3$ into the reader $1/\pi$ through the gauge $3/\pi$, without identifying the two numbers. The six residual effects preserve the unit and complete a POVM of nine outcomes, from which another cyclic QMS $(9, 3)$ is obtained. This

construction does not yet identify the geometric provenance of the residual mass; it fixes the exact operator type that it must have.

3.10.5 Symplectic group and equivariance equation

In dimension two over $\mathbb{F}_3$, the symplectic group coincides with $SL(2,\mathbb{F}_3)$ and permutes the four lines. Its action lifts projectively, by means of the Clifford group, on the PVM qutrits. Let G_{HMT} be a group or groupoid of transformations of the nonadic atlas and U_g the candidate unitary or antiunitary operator in the qutrit chart. The condition that turns a calibration into a dynamical bridge is

$$\Xi(g\cdot\ell) = U_g\,\Xi(\ell)\,U_g^{*}, \qquad g\in G_{\mathrm{HMT}},\quad \ell\in\mathbb{P}^1(\mathbb{Z}/9\mathbb{Z}). \tag{GMC-5.14}$$

If g is an arrow and not merely a global symmetry, the equation must include the source and target and is formulated as a natural transformation. The verification of (GMC-5.14) is the step that separates a fortunate labeling from a structural transduction.

Premises and derivation

The symplectic and MUB identities rely on the finite-phase literature [52, 100]. They are not attributed to the papers on magic squares by De las Cuevas–Netzer. The nonadic fibration, calibration, and route readers constitute the original contribution of the HMT construction studied here.

3.11 Spatial return and conservation of winding in the calendar of twenty-seven transitions

The two periodic APP coordinates arise from two transported integer coordinates. The residual reading identifies positions that differ by multiples of nine; the integer quotients preserve the turns of that transport. The TPK calendar provides a complete example in which the residual position returns and the lifted cursor preserves a nonzero horizontal displacement.

3.11.1 Lifted coordinates and boundary cocycle

For $z\in\mathbb{Z}$, let

$$\rho_9(z) = 1 + ((z-1) \bmod 9), \qquad q_9(z) = \left\lfloor \frac{z-1}{9} \right\rfloor.$$

Euclidean division, also for negative coordinates, gives

$$z = \rho_9(z) + 9q_9(z), \qquad 1\le \rho_9(z)\le 9. \tag{3.8}$$

The transport carry of an increment $a\in\mathbb{Z}$ is

$$c(z,a) = q_9(z+a) - q_9(z).$$

From (3.8) we deduce

$$\rho_9(z+a) + 9c(z,a) = \rho_9(z) + a, \tag{3.9}$$

$$c(z,a+b) = c(z,a) + c(z+a,b). \tag{3.10}$$

The second identity follows by canceling $q_9(z+a)$ on the right-hand side. The second increment is evaluated from the endpoint of the first; that dependence preserves the boundary crossing information.

Let $D=\{N,E,S,O\}$, with displacements

$$\delta(N) = (-1,0), \quad \delta(E) = (0,1), \quad \delta(S) = (1,0), \quad \delta(O) = (0,-1).$$

For a direction word $w = d_1 \cdots d_n$, define $\Delta(w) = \sum_j \delta(d_j)$ and, from $o = (o_1, o_2)$,

$$\mathrm{Ext}(o, w) = o + \Delta(w), \qquad \nu(o, w) = \big(c(o_1, \Delta_1(w)), c(o_2, \Delta_2(w))\big).$$

The vector ν records both windings.

Proposition 3.35 (Composition and reversal of turns)**.** *For two concatenable routes u, v one has*

$$\nu(o, uv) = \nu(o, u) + \nu(\mathrm{Ext}(o, u), v).$$

If w^{-1} reverses the order and replaces each direction with its opposite, then

$$\mathrm{Ext}(\mathrm{Ext}(o, w), w^{-1}) = o, \qquad \nu(\mathrm{Ext}(o, w), w^{-1}) = -\nu(o, w).$$

For a spatially closed route in APP,

$$\nu(o, w) = \Delta(w)/9 \in \mathbb{Z}^2.$$

Proof. The sum of displacements is additive under concatenation. Applying (3.10) in each coordinate proves the first identity. Reversal gives $\Delta(w^{-1}) = -\Delta(w)$ and recovers the initial endpoint. Subtracting the quotients in reverse order changes the sign of ν. Finally, spatial closure is equivalent to $\rho_9(o_i + \Delta_i) = \rho_9(o_i)$ for both coordinates; the first equality of (3.10) then gives $9\nu_i = \Delta_i$. □

The cycles of nine southward moves and nine eastward moves, from $(1, 1)$, have turns $(1, 0)$ and $(0, 1)$, respectively. They are two independent cycles of the periodic APP surface.

3.11.2 Update of the two cursors

A lifted observable cursor is $c = (x, y, d) \in \mathbb{Z}^2 \times D$. The phase p belongs to $\mathbb{Z}/108\mathbb{Z}$ and is represented by $0, \ldots, 107$. In the additive sheet, displacement is activated when $p \equiv 0 \pmod 3$; in the multiplicative sheet, when $p \equiv 1 \pmod 3$. Define

$$a_+(p) = \mathbf{1}_{\{p \equiv 0 \ (3)\}}, \qquad a_\times(p) = \mathbf{1}_{\{p \equiv 1 \ (3)\}}, \qquad \varepsilon(p) = \mathbf{1}_{\{p+1 \equiv 0 \ (27)\}}.$$

The observable operator acts through

$$c'_s = \big(x_s + a_s(p)\delta_1(d_s),\ y_s + a_s(p)\delta_2(d_s),\ \mathrm{op}^{\varepsilon(p)}\, d_s\big), \qquad p' = p + 1 \pmod{108}, \tag{3.11}$$

where $s \in \{+, \times\}$ and op interchanges N with S and E with O. Displacement uses the preceding direction and then the half-cycle transition updates the orientation.

The inverse first recovers $p = p' - 1 \pmod{108}$, restores $d_s = \mathrm{op}^{\varepsilon(p)}\, d'_s$ and subtracts $a_s(p)\delta(d_s)$ from the coordinates. Since $\mathrm{op}^2 = I$, both compositions are the identity.

Theorem 3.36 (Effective spatial return in the first half-crown)**.** *Start from $c_+(0) = c_\times(0) = (1, 1, E)$ and $p(0) = 0$. After twenty-seven transitions of (3.11), one obtains*

$$c_+(27) = c_\times(27) = (1, 10, O), \qquad p(27) = 27.$$

Both residual positions return to $(1, 1)$, while the horizontal quotient increases by one and the vertical quotient remains unchanged.

Proof. For $0 \le n \le 27$, the number of additive activations before transition n is $\lfloor (n + 2)/3 \rfloor$; the number of multiplicative activations is $\lfloor (n + 1)/3 \rfloor$. Before the phase transition from 26 to 27 there is no direction reversal. By induction on n, the coordinates are

$$x_+(n) = x_\times(n) = 1, \qquad y_+(n) = 1 + \left\lfloor \frac{n+2}{3} \right\rfloor, \qquad y_\times(n) = 1 + \left\lfloor \frac{n+1}{3} \right\rfloor.$$

The directions are E for $0 \le n < 27$ and O for $n = 27$. The decomposition in each group of three transitions is

n	$y_+(n)$	$y_\times(n)$	direction
$3k$	$1+k$	$1+k$	E
$3k+1$	$2+k$	$1+k$	E
$3k+2$	$2+k$	$2+k$	E

$(0 \le k \le 8)$,

and the final endpoint is $(y_+, y_\times, d) = (10, 10, O)$. There are nine activations in each sheet. The final transition has preceding phase 26, of class two modulo three; it therefore reverses the directions and maintains the coordinates that have already reached ten.

Finally, $\rho_9(10) = 1$, $q_9(10) = 1$, $\rho_9(1) = 1$ and $q_9(1) = 0$. The residual position coincides with the initial one, and the spatial winding is $(0, 1)$. □

The return in this example is spatial. The recorded direction and phase are $(O, 27)$ at the end and $(E, 0)$ at the start. The example expressly preserves the three data: residual position, orientation and phase; their equality or difference is established for each one.

3.11.3 Cursor recovery and compatibility at every depth

Proposition 3.37 (Recovery of the lifted cursor)**.** *The residual position, the direction and the two integer quotients uniquely determine the cursor $c = (x, y, d)$.*

Proof. The two identities (3.8) recover x and y and the direction is preserved in the observable projection. If these five data coincide for two cursors, their three components coincide and the cursors are therefore equal. □

Let P be the projection of the two cursors onto their positions modulo nine, preserving their directions and the phase. The finite update T_{fin} uses the same activation and orientation indicators, with residual addition in the coordinates. If T is (3.11), then

$$PT = T_{\mathrm{fin}} P, \qquad PT^n = T_{\mathrm{fin}}^n P \quad (n \ge 0).$$

The first equality results from compatibility of reduction modulo nine with addition and preservation of direction and phase; the second is proved by induction on n.

The ordered archive of observable states can be prolonged through $(q, \mathcal{A}) \mapsto (Tq, \mathcal{A} \| q)$. After n transitions it preserves its initial prefix and its length increases by n. This operation keeps the current state, the winding and the register of its preceding states separate. In the complete enriched construction, the observable factor is composed with the other cylinder, incidence and memory readings through their declared projections.

Part II

discrete structure of the continuum: surviving sections, double projection and incidence hierarchies

Chapter 4

Inverse path system of compatible trajectories: orientation, boundary, carry, and memory under refinement

4.1 Arithmetic Genealogy of Measure and Nonadic Configuration of the Inverse System

4.1.1 The Continuum Problem and the Genealogy of Measure

This opening develops the number–measure dimension of the treatise's inaugural thesis: the discrete structure constructs the continuum and the real chart subsequently publishes its coordinate, without intervening as a generating domain.

The history of the continuum can be read as the history of a separation and of its successive reconciliations. Ancient arithmetic organized pluralities; geometry compared lengths, areas, and volumes. The Greek theory of proportions made it possible to reason with incommensurable magnitudes: the diagonal of the unit square existed as an exact construction even when its length could not be published as a ratio of integers. Books V and VII of the *Elements* therefore preserved two domains, magnitude and number, united by rules of comparison but endowed with distinct operations [44, 113].

Modern analysis brought that reconciliation to a decisive form. Cauchy sequences, Dedekind cuts, and set theory made $\mathbb{R}$ a complete ordered field, capable of publishing continuous magnitudes in a single arithmetic language. Cantor distinguished cardinality from order; Dedekind made continuity a structural property of the number system; Weierstrass established the local discipline of limits and approximation [115, 114, 97]. The resulting real continuum is one of the most powerful constructions in the history of mathematics: it identifies presentations that converge to the same value and makes possible calculus, analytic geometry, and the quantitative formulation of physical theories.

Triadic Modular Holography poses a question prior to that identification: *what finite and prolongable structure produces the value published by the real line?* Its primary object is the compatible genealogy of an evaluation. A digit is the output of a positional reader; a value is the scalar coordinate obtained by contracting a family of cylinders; an HMT number is the section that additionally preserves phase, orientation, arithmetic sheet, lifting quotient, carry, boundary, and route memory. The Archimedean evaluation thus takes the form

$$\mathrm{ev}_{\mathbb{R}} : X_{\infty}^{\mathrm{HMT}} \longrightarrow \mathbb{R},$$

and their fibers gather stories that share a coordinate without thereby losing their operational identity.

This formulation changes the explanatory order of measurement. Counting determines the cardinality of a finite set; measuring couples a state, a reader, and a chart, restricts their compatible continuations, and publishes an observable. If X is the state space, A the apparatus or reader, and $C \subset X \times A$ the compatibility condition, the minimal scheme is

$$(x, a) \in C \longmapsto O(x, a) \longmapsto \mathcal{U}\big(O(x, a)\big),$$

where O conserves the type of the observable and $\mathcal{U}$ expresses it in the chosen chart. The final digit constitutes the visible coordinate of the complete coupling. This distinction will be the mathematical basis of the theory of the observer and of Dimensional Mechanics in Part V.

The epistemological consequence is precise. Ordinary physical metrology assigns coordinates to observables, and theories subsequently relate those coordinates by means of equations; HMT first constructs the state, the route, and the reader from which the coordinate descends. The fundamental constants then assume the status of results of a genealogy, and experimental comparison is applied to the output. In the same movement, an HMT symmetry expresses the conservation of unity through a transformation: the value records the published coordinate, and the exceptional projection records the

organization that has survived transport. Structure and number become two verifiable aspects of the same discrete evolution.

4.1.1.1 Marks, intervals, and the nonad cell

The finite construction begins on the line of integers. The origin 0, the decimal threshold 10, and the cell of positive representatives are fixed

$$E_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

The elements of E_9 perform two related but distinct functions: they are the nine phase marks of the path P_9, and they are also the nine positive integer coordinates $d(0, r) = r$ measured from the origin. This second reading adds no edges to the path. The nine marks determine exactly the eight half-open interior intervals

$$J_r = [r, r + 1), \qquad 1 \le r \le 8.$$

The closure edge is represented in the lifted chart by $[9, 10)$; its nonadic projection returns to the first phase and preserves the carry of the next turn. The origin 0 belongs solely to the expanded decimal chart and does not add an interior interval to P_9. Marks, coordinates, intervals, and closure are thus typed before any operation is applied.

For each $n \ge 1$, nonadic Euclidean division defines

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \qquad q_9(n) = \left\lfloor \frac{n - 1}{9} \right\rfloor,$$

$$n = 9q_9(n) + \rho_9(n). \tag{4.1}$$

The coordinate ρ_9 gives the position on a nine-phase wheel; q_9 preserves the accumulated height. The successor acts by

$$(q, r) \longmapsto \begin{cases} (q, r + 1), & 1 \le r < 9, \\ (q + 1, 1), & r = 9. \end{cases}$$

Therefore, n and $n + 9$ share a phase and occupy consecutive heights. The geometric realization

$$\mathcal{H}_9(n) = \left(\cos \frac{2\pi(\rho_9(n) - 1)}{9}, \sin \frac{2\pi(\rho_9(n) - 1)}{9}, q_9(n) + \frac{\rho_9(n) - 1}{9} \right)$$

makes this structure visible: projected onto the plane it is a wheel; conserved in the three coordinates it is a helix. The return $9 \to 1$ recovers the angle and elevates the history. This is the elementary form of the distinction that will govern the entire TPK:

$$\boxed{\text{phase return} \neq \text{state return.}}$$

4.1.1.2 Construction of APP via decomposition of 1 cell into 2 coordinates

The Cartesian product of two nonadic cells constructs a two-dimensional chart. Its inner kernel contains $8^2 = 64$ positions. The ninth row and the ninth column publish the null residual class through the label 9 and form a gnomon of seventeen cells:

$$81 = 64 + (9 + 8) = 64 + 17.$$

The modular identification of opposite sides converts the square chart into the torus

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2,$$

and the cardinal translations endow it with a Cayley graph. The geometric operation is, consequently, a precise sequence: kernel 8×8, gnomonic completion 9×9, toroidal identification, and transport by the graph.

On each vertex $(i, j) \in X_9$ act two evaluations of the same support: the additive sheet Σ and the multiplicative sheet Π. Their exact incompatibility supplies dynamics because a route must preserve which sheet acted, which residue it published and which quotient it produced. APP constructs the support and its two evaluations; TRIT locally orients the transition; TPK transports the complete state, simultaneously preserves both sheets and, through M_{ph}, types the mode and TRIT phase of the transition; it also preserves the memory of elevation and decides the survival of prolongations. The inaugural architecture is fixed by the chain

$$\boxed{E_9 \longrightarrow \mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK}.}$$

The importance of this beginning lies in its capacity for prolongation. The same difference between position and height that turns the wheel into a helix is reproduced in the periodicity of $108 = 12 \cdot 9$ iterations, in the oriented cyclic system of twelve phases, in the architecture of levels and thresholds of the TPK prolongation $w_6 \to w_{12} \to w_{18} \to w_{24} \to w_{30} \to R_{36} \to G_9$, in the derivation of π, e, φ and α_{HMT}, and in the double projection towards the real value and exceptional incidence. The discrete structure of the continuum designates precisely this continuity of a finite genealogy across arbitrary depths.

Hypothesis, domain, and conclusion

The history of number, magnitude and continuum belongs to the classical mathematical context. The decomposition (4.1), the nonadic torus, the APP sheets and memory transport are results of the HMT corpus. The helical embedding $\mathcal{H}_9$ is a geometric formalization of that decomposition. Evaluation on all of $\mathbb{R}$ is established by the coverage and Archimedean closure theorem of the genealogical continuum theorem 18.17; the genealogy of the constructed values is formulated in the inverse system of compatible sections.

4.2 Genealogical construction of the continuum

4.2.1 Generative composition APP→TRIT→TPK and construction of the enriched state

The construction begins with three irreducible and causally ordered objects. The object of *All Possible Paths* (hereafter, APP) constructs the common nonadic support, its graph of all finite paths, and the two sum–product evaluation sheets. The tritic unit (hereafter, $\widehat{\mathbb{T}}$) is the minimal local unit of topological, geometric, informational, and, once ordered, temporal relation: its visible signature has three states, and its lift preserves the carry that a binary or residual reading would lose. The *Topological Phase Kernel* (hereafter, TPK) is the transport category that composes these units into routes and preserves, within the same state, sheet, orientation, boundary, genealogical ledger, memory, phase, and survival.

Object	Construction	Invariantes transportados
APP	positive chart, finite torus, Cayley graph, paths, and leaves Σ, Π	marker, cell, neighborhood, sum, product, quotient and closure
TRIT	ternary signature visible and local topological-temporal lift	residue, carry, sheet, orientation, threshold and reversal
TPK	category of paths, fibers, and compatible extensions	cylinder, boundary, genealogical record, memory, phase, holonomy and survival

The exact construction of APP is established on a marked segment and a finite chart; the real line appears afterward as a limiting publication. The reference interval, its marks, and the oriented path that traverses them are:

$$I_{10} = [0, 10], \qquad \mathcal{M}_{10} = I_{10} \cap \mathbb{Z} = \{0, 1, \dots, 10\}, \qquad \mathsf{P}_{10} : 0 \longrightarrow 1 \longrightarrow \cdots \longrightarrow 9 \longrightarrow 10, \tag{4.2}$$

The endpoints fix the edge and the nine interior marks $I_{10}^{\circ} \cap \mathbb{Z}$ form the positive alphabet. The exact mathematical object of departure is the chart

$$\mathcal{D}_9 = \{1, \dots, 9\}, \qquad \mathcal{D}_8 = \{1, \dots, 8\}, \qquad \lambda_9 : \mathcal{D}_9 \xrightarrow{\sim} \mathbb{Z}/9\mathbb{Z}, \qquad \lambda_9(9) = [0]_9. \tag{4.3}$$

The mark 9 is simultaneously the last positive representative and the publication of the null class. For every $n \geq 1$, the positive digital root and quotient are separated by means of

$$\mathrm{dr}_9(n) = 1 + ((n-1) \bmod 9), \qquad q_9(n) = \left\lfloor \frac{n-1}{9} \right\rfloor, \qquad n = 9q_9(n) + \mathrm{dr}_9(n). \tag{4.4}$$

To the path $1 \to 2 \to \cdots \to 9$ we add the unique closing edge $9 \to 1$. The successor and its memory are separated:

$$u_+(a) = \begin{cases} a+1, & a < 9, \\ 1, & a = 9, \end{cases} \qquad \kappa_+(a) = \begin{cases} 0, & a < 9, \\ 1, & a = 9, \end{cases} \qquad a + 1 = u_+(a) + 9\kappa_+(a). \tag{4.5}$$

For that reason $9 \to 1$ closes the phase, increases the memory, and prolongs the genealogical state.

The square of the chart produces the finite torus and its movement graph:

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2, \qquad \Gamma_{\mathrm{APP}} = \mathrm{Cay}\big(X_9, \{\pm e_1, \pm e_2\}\big), \qquad \mathrm{Path}(\Gamma_{\mathrm{APP}}). \tag{4.6}$$

Identifying opposite sides preserves the cardinality of the chart and adds the continuation law. The core of $8^2 = 64$ interstices is completed by the minimal gnomon that publishes the zero class in both coordinates:

$$\mathcal{D}_9^2 = \mathcal{D}_8^2 \sqcup \mathcal{G}_9, \qquad 81 = 64 + (9 + 8) = 64 + 17. \tag{4.7}$$

The same law is prolonged to charts of greater depth; the scale changes, but the closure, the shared edge, and the memory of the turn preserve their type.

The associated volumetric completion is expressed through the strata identity.

$$10^3 = (9 + 1)^3 = 729 + 243 + 27 + 1. \tag{4.8}$$

The interior volume, faces, edges, and point closure are thus typed as distinct layers of the same completed unit. The evaluations act on the 81 cells of the positive chart

$$\Sigma, \Pi : \mathcal{D}_9^2 \longrightarrow \mathcal{D}_9, \qquad \Sigma(i, j) = \mathrm{dr}_9(i + j), \qquad \Pi(i, j) = \mathrm{dr}_9(ij),$$

Their residues and quotients raised are defined by

$$R^+(i, j) := \Sigma(i, j), \quad Q^+(i, j) := \frac{i + j - R^+(i, j)}{9},$$
$$R^\times(i, j) := \Pi(i, j), \quad Q^\times(i, j) := \frac{ij - R^\times(i, j)}{9},$$

and, therefore,

$$\widehat{\Sigma}(i, j) := \big(R^+(i, j), Q^+(i, j)\big), \qquad \widehat{\Sigma}, \widehat{\Pi} : \mathcal{D}_9^2 \longrightarrow \mathcal{D}_9 \times \mathbb{Z}_{\geq 0}.$$
$$\widehat{\Pi}(i, j) := \big(R^\times(i, j), Q^\times(i, j)\big),$$

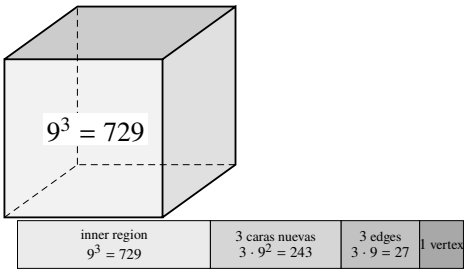
The bijection $\lambda_9^{\times 2} : \mathcal{D}_9^2 \to X_9$ transports these operations and their lifts to the torus. Therefore, APP is the tuple

Multiplicative leaf of the APP chart: $\Pi(i,j) = \mathrm{dr}_9(ij)$

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	**9**
2	2	4	6	8	1	3	5	7	**9**
3	3	6	9	3	6	9	3	6	**9**
4	4	8	3	7	2	6	1	5	**9**
5	5	1	6	2	7	3	8	4	**9**
6	6	3	9	6	3	9	6	3	**9**
7	7	5	3	1	8	6	4	2	**9**
8	8	7	6	5	4	3	2	1	**9**
9	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**	**9**

$8^2 = 64$ cells interior; gnomon of closure $9 + 8 = 17$
corona positive of nueves: the mark 9 publishes $[0]_9$

Cube of Extreme and Mean Holographic Ratio



$10^3 = 729 + 243 + 27 + 1 = 729 + 271$
growth corona $9^3 \subset 10^3$: 271 states
internal boundary of the cube of side 9: 386 points

Figure 4.1: Two finite publications of the support. Left: the product table with positive digital root and its closure gnomon. Right: the EMRH cubic completion, which distinguishes interior, faces, edges, and vertex. The growth corona 271 and the internal boundary 386 are two typed counts: the former measures the enlargement $9^3 \subset 10^3$ and the latter the boundary of the cube of side 9.

$$\mathrm{APP} = \big(\mathcal{D}_9, \lambda_9, X_9, \Gamma_{\mathrm{APP}}, \mathrm{Path}(\Gamma_{\mathrm{APP}}), \Sigma, \Pi, \widehat{\Sigma}, \widehat{\Pi}\big), \tag{4.9}$$

The table constitutes a two-dimensional representation of this complete tuple. The additive sheet is Latin; the multiplicative sheet conserves the ternary radical of the ring $\mathbb{Z}/9\mathbb{Z}$

$$R_9 := \mathbb{Z}/9\mathbb{Z}, \qquad \mathfrak{m} := ([3]_9) = \{[0]_9, [3]_9, [6]_9\} \triangleleft R_9, \qquad \mathfrak{m}^2 = (0). \tag{4.10}$$

The internal rotation is the automorphism

$$\rho : \mathfrak{m}^3 \longrightarrow \mathfrak{m}^3, \qquad \rho(x_1, x_2, x_3) = (x_2, x_3, x_1),$$

whose distinguished orbit is

$$([3]_9, [6]_9, [0]_9) \xrightarrow{\rho} ([6]_9, [0]_9, [3]_9) \xrightarrow{\rho} ([0]_9, [3]_9, [6]_9) \xrightarrow{\rho} ([3]_9, [6]_9, [0]_9).$$

The direct reduction sends $3, 6, 9$ to $0, 0, 0$. The internal coordinate of the ideal, by contrast, conserves the extracted factor:

$$\eta_{\mathfrak{m}} : (\mathfrak{m}, +) \xrightarrow{\sim} (\mathbb{Z}/3\mathbb{Z}, +), \qquad \eta_{\mathfrak{m}}([3a]_9) := [a]_3, \qquad ([3]_9, [6]_9, [0]_9) \xmapsto{\eta_{\mathfrak{m}}^{\times 3}} ([1]_3, [2]_3, [0]_3), \tag{4.11}$$

so that the $3, 6, 9$ sector remains intact and its internal coordinate is read as $1, 2, 0$. This map belongs to the radical of APP; the temporal TRIT projection constitutes the subsequent calendrical reader. The three calendrical classes remain separated as

$$C_+ = \{1, 4, 7\}, \qquad C_- = \{2, 5, 8\}, \qquad C_0 = \{3, 6, 9\}.$$

The incompatibility between the two leaves becomes a measurable invariant. On $\mathscr{A}_3 = \mathcal{D}_9^3$, with uniform measure $\nu_3(x) = 9^{-3}$, let $\mathcal{H}_3 = L^2(\mathscr{A}_3, \nu_3)$. We define

$$\Sigma_3, \Pi_3 : \mathscr{A}_3 \longrightarrow \mathcal{D}_9, \qquad \Sigma_3(x) = \mathrm{dr}_9(x_1 + x_2 + x_3), \qquad \Pi_3(x) = \mathrm{dr}_9(x_1 x_2 x_3),$$

and the orthogonal projectors

$$P_{\Sigma,3} := \mathbb{E}_{\nu_3}[\,\cdot \mid \sigma(\Sigma_3)], \qquad P_{\Pi,3} := \mathbb{E}_{\nu_3}[\,\cdot \mid \sigma(\Pi_3)] \in \mathcal{B}(\mathcal{H}_3).$$

The joint enumeration of the 9^3 words then determines

$$\left\|[P_{\Sigma,3}, P_{\Pi,3}]\right\| = \frac{\sqrt{3}}{4}, \qquad B_c = 7I + 2R = 9P_+ + 5P_-, \qquad 459 - 405 = 54.$$

The first number reappears as the electronic prefactor; the second organizes the central block; the third records the global sum–product defect.

4.2.2 TRIT: local topological unit and temporal relation unit

HMT first fixes the visible signature

$$\mathrm{TRIT}_{\mathrm{vis}} = \{-1, 0, +1\}, \qquad \tau \in \mathrm{TRIT}_{\mathrm{vis}}. \tag{4.12}$$

In contrast to the binary unit, the TRIT preserves three local regimes. In the uniform distribution, its information and entropy are distinguished as

$$I_{\mathrm{TRIT}} = \log 3 \text{ nats} = \log_2 3 \text{ bits}, \qquad S_{\mathrm{TRIT}} = k_B I_{\mathrm{TRIT}}. \tag{4.13}$$

The bit remains the binary unit; the TRIT is the minimal unit capable of simultaneously distinguishing return, threshold, and opening. The equality (4.13) quantifies the information of the triad; the cost $k_B\Theta \log 3$ belongs to the subsequent thermal reset reader. Dimensional Mechanics also preserves the arithmetic antecedent of that signature. For $(i, j) \in \mathcal{D}_9^2$, the associated lifted arithmetic cell is

$$c_{\mathrm{MD}}(i, j; \tau) = \big(i, j; R^+, Q^+, R^\times, Q^\times; \tau\big). \tag{4.14}$$

The residues $R^+, R^\times$ are the visible publication of the two sheets; the quotients $Q^+, Q^\times$ conserve the exact memory of Euclidean division; and τ orients the transition. The visible signature occupies the first level; the cell that bears it adds the arithmetic antecedents, and the enriched state then completes boundary, route, phase, and survival.

It is useful to distinguish four strata. The visible signature is τ; the residue–carry pair is its local lift; c_{MD} is the APP arithmetic cell equipped with that signature; and the TPK state adds extension, boundary, genealogical ledger, phase, and survival. This hierarchy preserves the distinct identity of the observed coordinate, its lift, the carrier cell, and the enriched history that transports it.

In the balanced ternary chart, the lifted local integer $n \in \{-4, -3, \ldots, 3, 4\}$ decomposes uniquely as

$$\widehat{n} = (r, \kappa), \qquad n = r + 3\kappa, \qquad r, \kappa \in \{-1, 0, +1\}. \tag{4.15}$$

The visible signature is $\tau = r$; κ conserves the carry that the projection $n \mapsto r$ would forget. The fiber over the visible zero contains three distinct states,

$$0^+ = (0, +1), \qquad 0^0 = (0, 0), \qquad 0^- = (0, -1),$$

because $n = -3, 0, 3$ have the same visible residue and different carries. This is the minimal topological lift of MD: an oriented APP transition is endowed with the information required to invert it and to decide which prolongations remain compatible.

The signature also has a sheet-memory rule. If m_k is the active arithmetic mode, then

$$m_{k+1} = \begin{cases} \Sigma, & \tau_k = +1, \\ \Pi, & \tau_k = -1, \\ m_k, & \tau_k = 0. \end{cases} \tag{4.16}$$

Zero encodes retention of the previous sheet. For that reason the word $(+1, -1, 0)$ publishes the sequence (Σ, Π, Π): the third state conserves the active multiplicative mode.

The temporal relation is obtained by orderly parametrizing these cells. Each signature carries a quadratic algebra.

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \qquad J_\tau = [X], \tag{4.17}$$

so that

τ	J_τ^2	ordered temporal relation
$+1$	$-I$	elliptic return, memory and past,
0	0	parabolic threshold and present,
-1	$+I$	bidirectional hyperbolic opening and future.

The TRIT precedes external time: it first classifies the algebraic and topological type of the transport; an ordered parameterization then publishes it as a temporal relation. The twofold direction is typed by the exact involution

$$C_{\mathrm{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1), \qquad C_{\mathrm{rev}}^2 = I, \tag{4.18}$$

which exchanges return and opening and preserves the threshold. On the hyperbolic sheet, $P_\pm = (I \pm J_{-1})/2$ separate the two eigendirections, and reversal exchanges their orientations. Topology and time are thus the two coordinated readings of the TRIT.

The combinatorial involution C_{rev} and an operatorial conjugation C belong, however, to distinct domains. In a separate dynamic realization, endowed with $H = H^*$, $CJ_EC^{-1} = -J_E$, and $CHC^{-1} = H$, we fix an internal action scale $\hbar_{\mathrm{int}} > 0$ and write $h_{\mathrm{int}} := 2\pi\hbar_{\mathrm{int}}$. Then

$$U(t) = \exp(-J_E H t/\hbar_{\mathrm{int}}), \qquad CU(t)C^{-1} = U(-t). \tag{4.19}$$

The intertwiner that transports TRIT words to the dynamical realization fixes the correspondence between C_{rev} and C. The twofold local orientation provides the domain of that construction.

4.2.3 TPK: category of transport and phase evolution

The *Topological Phase Kernel* is the transport category over the observable base of APP that composes digits, words, operators, and prolongation relations within a single enriched state. Its internal architecture is distributed among eight operator classes, differentiated by domain, codomain, and function:

(i) observable support and enriched state;

(ii) internal sheet selector $M_{\mathrm{ph}} : \{1, \dots, 9\} \to \{+1, -1, 0\}$;

(iii) cardinal translations and visible evolution F_{obs};

(iv) readers and quotients nonadic, tritico and of block 1000;

(v) memory, carry, cylinder, boundary, and genealogical register;

(vi) periodicity and return $C_{12} \hookrightarrow C_{108} \twoheadrightarrow C_9, \mathcal{K}_{27}$, and channels 90/120;

(vii) prolongations Ext_r, cylinders, survival, and natural bidirectional extension;

(viii) Archimedean-output, calibrated-exceptional, and dimensional functors.

The return of 27 steps is the composite morphism.

$$\mathcal{K}_{27} := F_{\mathrm{obs}}^{27}, \qquad \mathcal{K}_{27}^4 = F_{\mathrm{obs}}^{108}. \tag{4.20}$$

Its fourth iterate restores the observable phase of the calendar, while the memory and the complete TPK state continue along their trajectory; $\mathcal{K}_{27}$ is the internal return morphism of C_{108}. The calendar

C_{108}, the nonadic connection, the bilateral extension, and the readers occupy typed places within this architecture. Let $\mathsf{P}_{\mathrm{APP}}$ be the free category of the directed graph of APP, and let $\mathsf{P}_{\mathrm{APP}}^{(2)} = \mathsf{P}_{\mathrm{APP}} \times \mathsf{P}_{\mathrm{APP}}$ be its simultaneous sum–product reading. Let, moreover,

$$\mathcal{Q}_{\mathrm{obs}} = X_9^2 \times \{N, E, S, O\}^2 \times \Theta_{108}, \qquad \Theta_{108} = \mathbb{Z}/108\mathbb{Z},$$

where, for the representative $\widetilde{\theta} \in \{0, \dots, 107\}$,

$$r(\theta) = 1 + (\widetilde{\theta} \bmod 9), \qquad g_0(\theta) = [\widetilde{\theta}]_9,$$

$$\tau(\theta) = \begin{cases} +1, & r(\theta) \in \{1,4,7\}, \\ -1, & r(\theta) \in \{2,5,8\}, \\ 0, & r(\theta) \in \{3,6,9\}, \end{cases} \qquad \mu(\theta) = \begin{cases} \Sigma, & \tau(\theta) = +1, \\ \Pi, & \tau(\theta) = -1, \\ \varnothing, & \tau(\theta) = 0. \end{cases}$$

The calendar label $\mu = \varnothing$ marks the threshold, and the dynamic memory m_k of (4.16) retains the last active sheet. A configuration $q = (p^{\Sigma}, p^{\Pi}, d^{\Sigma}, d^{\Pi}, \theta)$ preserves the two positions, the two cardinal directions, and the calendar index. The visible evolution F_{obs} activates the sheet indicated by TRIT, transports the corresponding position, and advances θ. Let $\mathsf{P}_{\mathrm{obs}}$ denote the free category generated by the arrows $q \to F_{\mathrm{obs}}(q)$. Each arrow induces a pair of APP paths and defines the base functor

$$B : \mathsf{P}_{\mathrm{obs}} \longrightarrow \mathsf{P}_{\mathrm{APP}}^{(2)}. \tag{4.21}$$

Over each q lies the typed fibre

$$\mathcal{F}_q = \mathsf{O}_q \times \mathsf{H}_q \times \mathsf{C}_q \times \mathsf{S}_q \times \mathsf{B}_q \times \Lambda_q, \tag{4.22}$$

Its six factors conserve, respectively, orientation and sheet; residue and quotient; carry; signature; cylinder and boundary; and ordered register of transport. Memory and survival are properties of the composition and of the prolongation relations.

For a route $r : q \to q'$, the relation

$$\mathrm{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'} \tag{4.23}$$

contains all admissible continuations and satisfies

$$\Delta_q \subseteq \mathrm{Ext}_{\mathrm{id}_q}, \qquad \mathrm{Ext}_{r_2} \circ \mathrm{Ext}_{r_1} \subseteq \mathrm{Ext}_{r_2 \circ r_1}.$$

Transports respect identities and composition:

$$\begin{gathered} \mathsf{H}(\mathrm{id}_q) = I, \quad \mathsf{C}(\mathrm{id}_q) = I, \quad \Lambda(\mathrm{id}_q) = I, \quad \kappa(\mathrm{id}_q) = 0, \\ \mathsf{H}(r_2 \circ r_1) = \mathsf{H}(r_2)\mathsf{H}(r_1), \quad \mathsf{C}(r_2 \circ r_1) = \mathsf{C}(r_2)\mathsf{C}(r_1), \quad \Lambda(r_2 \circ r_1) = \Lambda(r_2)\Lambda(r_1). \end{gathered} \tag{4.24}$$

Along with these functorial laws, the transported coordinates obey

$$\begin{aligned} h' &= \mathsf{H}(r)(h), \\ c' &= \mathsf{C}(r)(c) + \kappa(r), \\ \lambda' &= \Lambda(r)(\lambda), \\ \sigma' &= \mathrm{Sig}_{q'}(h', c', \lambda'), \end{aligned} \qquad \kappa(r_2 \circ r_1) = \kappa(r_2) + \mathsf{C}(r_2)\kappa(r_1). \tag{4.25}$$

The last identity is the carry cocycle that makes the composition with memory associative.

The TPK is then the category TPK whose objects are

$$x = (q, o, h, c, \sigma, C, b, \lambda), \qquad (o, h, c, \sigma, (C, b), \lambda) \in \mathcal{F}_q, \tag{4.26}$$

and whose morphisms $x \to x'$ are the triples (r, x, x') such that the coordinates satisfy (4.25) and $(f_x, f_{x'}) \in \mathrm{Ext}_r$, where $f_x \in \mathcal{F}_q$ and $f_{x'} \in \mathcal{F}_{q'}$. Composition is

$$(r_2, x', x'') \circ (r_1, x, x') = (r_2 \circ r_1, x, x''), \tag{4.27}$$

the identity is (id_q, x, x), and the functor

$$p : \mathsf{TPK} \longrightarrow \mathsf{P}_{\mathrm{obs}}, \qquad p(x) = q, \qquad p(r, x, x') = r, \tag{4.28}$$

publishes the visible configuration without identifying states of the same fiber. The definition fixes a category over a base; it does not presuppose that it is a categorical fibration nor that every arrow admits a Cartesian lift. The local projections remain separated by type:

$$\pi_{\mathrm{TRIT}} : \mathrm{Ob}(\mathsf{TPK}) \longrightarrow \mathrm{TRIT}_{\mathrm{vis}}, \qquad \pi_{\mathrm{TRIT}}(x) = \tau(\theta),$$

$$\pi_{\mathrm{loc}}(x) = \big(\pi_{\mathrm{TRIT}}(x), \mu(\theta), o, h, c, \sigma, \lambda, g_0(\theta)\big).$$

The first recovers only the visible TRIT signature. The second further preserves the calendrical sheet label, orientation, the residue–quotient pair, carry, transport signature, genealogical register, and phase, but still forgets the cylinder, its boundary label, and other coordinates of q. This is why neither the visible TRIT nor its local register replaces the enriched state of the TPK.

The name encompasses three inseparable functions. *Topological* refers to the category of paths, incidence, boundary, and prolongation by cylinders; *Phase* refers to the nonadic index, the oriented calendar, and holonomy; *Kernel* names the persistent transport kernel that remains before the state is contracted by a reader. Formally, the object is the preceding category: an additional topology on objects or morphisms, or its identification with the linear kernel of a map, requires independent data.

Consequently, two routes with the same visible endpoint may remain objects distinct by sheet, quotient, carry, boundary, cylinder, phase, or register. The TPK does not choose a scalar output: it conserves the common antecedent from which the Archimedean, exceptional, and dimensional publications depart.

4.3 Development of the formal kernel: initial conditions, geometry and refinement

The support of eighty-one positions does not coincide with the domain of initial conditions of the dynamics. An initial condition must also specify an orientation; and the additive and multiplicative sheets have independent cursors. This distinction allows emission to be reconstructed exhaustively without selecting a numerical region in advance.

4.3.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to the positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \qquad \Omega_{\mathrm{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so

$$|\mathcal{S}| = 324, \qquad |\Omega_{\mathrm{seed}}| = 324^2 = 104\,976. \tag{4.29}$$

The census of residual-emitter signatures in Section 155 enumerates these conditions by identifiers and presents their fibers; the law transforming them is defined here. The rows of an emission table are fibers of this initial-condition domain.

The finite realization uses six windows of nine transitions. At each instant t, the phase signature is $M_{\mathrm{ph}}(1+((t-1) \bmod 9))$. If it equals $+1$, the positive residue $\Sigma(i+1, j+1) = \rho_9(i+j+2)$ of the first cursor is read and that cursor is then shifted; if it equals -1, the positive residue $\Pi(i+1, j+1) = \rho_9((i+1)(j+1))$

of the second cursor is read and the second is then shifted. Zero increments the neutral counter. At the end of transitions twenty-seven and fifty-four, the cardinal orientations are reversed. Integer evaluations and their quotients are preserved in the register, but the emitter uses the residue readings specified here.

In the unit sector, the multiplicative reader uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^{\times}$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For finite emission it is extended by zero to the nonunit sector. This extension is an observable, not an identification of the elements $3, 6, 9$ with the multiplicative unit. The sheet and its membership in the radical sector remain in the trajectory register.

Let S_m be the sum of the three positive additive residues Σ of window m, E_m the sum of the three discrete exponents of the multiplicative residues Π and $Z_m = 3$ the number of neutral events. Thus, S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the register. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \tag{4.30}$$

The integer coefficients and calendar of this recurrence are an explicit part of the declared emission law. No values of π, φ, e or α are received. The distinction between an emission rule and a constant recognized from its output prevents the genealogy from being hidden in a table of decimals.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule is written

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \tag{4.31}$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalogue adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \tag{4.32}$$

The sign of this chart is explicit: changing it modifies the words and requires subsequent orientations to be transported. Balanced identification of each component of $\mathbb{F}_3$ with TRIT is performed after (4.32).

Proposition 4.1 (Injective factorization of emission)**.** *The word $U = (U_1, \dots, U_6)$ determines the two signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient space of 729 words.*

Proof. The hundreds digits of U_m recover s_m. If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence of (4.31). Coupling is therefore injective on the product of signatures.

Enumerating the 324 conditions for one cursor through the preceding recurrence produces eighteen additive signatures, each with eighteen preimages, and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. Section 155 collects the signatures, their multiplicities, and the exact rule recovering their initial conditions. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\underbrace{432}_{18\cdot 24} \text{ fibers of } 144, \qquad 18 \text{ fibers of } 432, \qquad 18 \text{ fibers of } 1944.$$

The sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (4.32) to the 468 emissions produces exactly 243 words. This last number is the cardinality of the image, not of the space $\mathbb{F}_3^6$. □

This factorization explains the difference between finite exhaustiveness and unlimited depth. The domain (4.29) is complete for the declared support and emitter; it does not contain in advance every prefix of every prolongation. Depth belongs to the system of compatible extensions constructed on those conditions.

4.3.2 Calendar closure and memory advance

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \tag{4.33}$$

If $t = 9b + r$ with $0 \le r < 9$, adding two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor,\ [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 4.2 (Nonsplitting of the calendar)**.** *The sequence* (4.33) *is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι. If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\operatorname{mcm}(12, 9) = 36$, a contradiction. □

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring the prefix of a prolongation. The nonadic holonomy Γ_9 denotes the transport of one turn on states with memory, so

$$g(\Gamma_9 x) = g(x), \qquad M(\Gamma_9 x) = M(x) + 1. \tag{4.34}$$

Phase equality and memory increment are compatible and constitute two observations of the same transport.

4.3.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by TPK operations on $\widehat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \widehat{X}, \qquad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t. The compatibility relation precedes choice of the parameter: the order allows its transitions to be expressed as an evolution. When the term section is used, the projection $p : E \to \mathcal{T}$ and condition $p \circ \gamma = \mathrm{id}_{\mathcal{T}}$ are also declared. This specifies the state space, temporal order and information transported.

The signature τ selects a quadratic regime and admits a temporal reading relative to that parametrization. Type +1, with $J_{+1}^2 = -I$, represents return and memory; type 0, with $J_0^2 = 0$, represents the transition threshold; type −1, with $J_{-1}^2 = I$, represents opening in two eigendirections. In the model's temporal terminology, these functions correspond, respectively, to past, present and future. The correspondence refers to the operational position of a transition within a history: preserved antecedent, update and admissible prolongation.

This interpretation acquires content through three distinct operations. Restricting a trajectory recovers the already constructed prefix. The update $\mathcal{U}_t$ composes selection, transport and recording in the observed transition. Prolongation considers the continuations that preserve that prefix's boundary conditions. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x, removing the last step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unlimited continuation is a family $(\gamma_n)_{n\geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the restrictions each depth imposes on the preceding ones. Existence of a compatible family is established in the corresponding prolongation domain; merely establishing the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the sequence of signatures, the involution C_{rev} reverses the order and changes the sign of each component. The trajectory is additionally acted on by the position, sheet and boundary transformations specified by transport. The inverted signature is an observation of the conjugate trajectory, not a replacement for its remaining coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical register is nonzero.

The relation to exponential geometries is equally precise. In each fixed regime, $\exp(tJ_\tau)$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to that local representation. The phase return of Γ_9, by contrast, acts on the trajectory with memory, according to (4.34). A trajectory can recover its phase and preserve a new antecedent: the historical datum has changed even though the phase observation is the same. The parabolic threshold retains a linear shift; hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, while TPK determines their selection and effective concatenation.

The aperiodic suspension of Section 10.5 will make this distinction explicit through a finite phase factor, an irrational rotation and a memory degree. The link between TRIT and temporality is thus expressed by operations on trajectories and their quadratic representations. Its content goes beyond a classification of signs: it preserves the regime, direction of transport and relation between antecedent, transition and continuation.

The monodromy representation of this return is developed in Section 10.5.11. There the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension of Section 10.5.13 defines the aperiodic time-crystal model, and Section 10.5.15 composes that temporality with the three Euler realizations.

4.3.3 Observable compression and orthogonal conservation

An isometric realization allows what a reduced observation loses to be expressed quantitatively. Let $U : \mathcal{H} \to \mathcal{H}$ be an isometric realization of the transport of one turn, $U^*U = I$, and $J : \mathcal{H}_{\mathrm{vis}} \to \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \qquad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \qquad J^*D = 0, \qquad I - T^*T = D^*D. \tag{4.35}$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 4.3 (Conservation at a finite horizon)**.** *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r} D^* D T^r + T^{*n} T^n. \tag{4.36}$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r and summing produces a telescoping sum. The interior terms cancel and $I - T^{*n}T^n$ remains. □

The last summand preserves the terminal state. Suppressing it before justifying a limit would alter the equality. This distinction is the operational content of preservation of unity in compression: the norm is distributed among the visible component, memory and terminal, without being replaced by a single residue.

4.3.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonals modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod 9 \quad \Longleftrightarrow \quad 2k+1 \equiv 0 \pmod 9$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_\pm = (I \pm R)/2$, we obtain $B_c = 9P_+ + 5P_-$. The unity of stochastic normalization and preservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \qquad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2}\log\frac{9}{5}.$$

For $Q = \mathrm{diag}(1,-1)$, the second matrix satisfies $B_c^{\mathsf{T}} Q B_c = 45Q$. This yields a local realization in $SO^+(1,1)$; proving it does not require introducing a four-dimensional spacetime beforehand.

Uniform averaging over the classes $C_1 = \{1,4,7\}$, $C_2 = \{2,5,8\}$, $C_0 = \{3,6,9\}$ gives

$$M_\Pi = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_\Sigma = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_Π is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

whence its quadratic form has signature $(2,1)$. The local spectral gap $9 - 5 = 4$, coefficient 2, aggregate difference $51 - 45 = 6$ and integrated defect $9 \cdot 6 = 54$ describe distinct levels of the same arithmetic comparison. They are not interchangeable indices of the same operator.

Discrete curvature can be described directly before this spectral realization. In residues, let $S(x,y) = x + y$, $P(x,y) = xy$ and Δ_x, Δ_y be the forward differences. The link variables are potential differences, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and we fix

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S,S) = F(P,P) = 0, \qquad F(S,P) = +1, \quad F(P,S) = -1 \pmod 9. \tag{4.37}$$

Sum and product, taken separately, induce flat connections; the ordered mixture changes the sign of plaquette curvature. On summing over a rectangle with sides a, b, interior edges cancel and the oriented boundary sum equals $\pm ab$ modulo nine. An exact relation thus appears between composition order and boundary orientation.

4.3.5 Cubic completion and two notions of boundary

The enlargement of the cubic support of side nine to that of side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \tag{4.38}$$

The first term corresponds to the three interior coordinates; the second, to one new coordinate; the third, to two; and the last, to the vertex of three new coordinates. The enlargement shell contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (4.38) by 1000 gives a normalized partition of unity. Cube incidence distinguishes six faces, twelve edges and eight vertices. The subsequent link with Steiner hexads and octads requires the corresponding combinatorial maps; cube cardinalities do not replace them. This completion here provides the relation between bases 729 and 1000, while exceptional prolongation will be developed after numerical generation, the dodecaphasic register and the constructions of α.

4.3.5.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description making its operation explicit. Let E_9 be the set of nine residue classes, positively represented by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ is distinct from the zero class, already represented by 9. Consequently, enlargement from E_9 to $\widehat{E} = E_9 \sqcup \{*\}$ increases cardinality from nine to ten without modifying the internal modular identification. The cubic completion is $\widehat{C} = \widehat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 4.4 (Strata of combinatorial codimension)**.** For $0 \le r \le 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \widehat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\widehat{C}^\circ = \widehat{C} \setminus \{a_*\}$.

Proposition 4.5 (Exact decomposition of the completion)**.** *The strata are disjoint and satisfy*

$$\widehat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \qquad |S_r| = \binom{3}{r} 9^{3-r}, \tag{4.39}$$

$$|\widehat{C}^\circ| = 729 + 243 + 27 = 999, \qquad |\widehat{C}| = 999 + 1 = 1000. \tag{4.40}$$

The punctured enlargement contains 270 *elements additional to the initial cube; the complete enlargement contains* 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning to the remaining ones one of the nine classes of E_9. Classification by the number of additional coordinates is exhaustive and its classes are disjoint. The unique element of S_3 is a_*. Removing it subtracts one unit from the total cardinality; restoring it recovers the complete Cartesian product. □

The expression *holographic extreme and mean ratio*, abbreviated EMRH in the corpus, here denotes this articulation between the initial body, enlargement strata and unit restitution. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a determined object, the punctured completion; 1000 represents its closure by restitution of the apex. The added unit therefore has an explicit incidence residence.

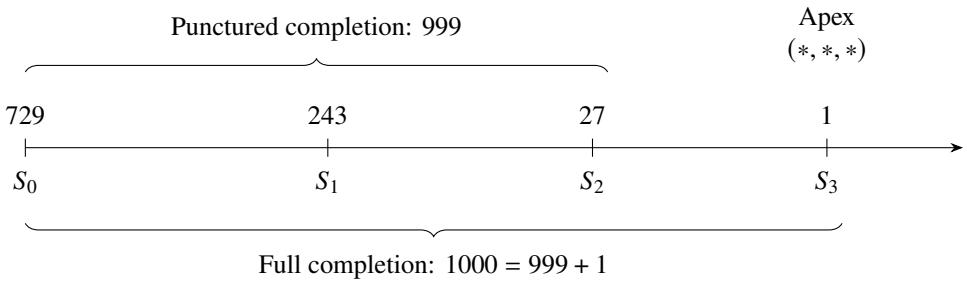


Figure 4.2: Strata of the cubic completion. The arrangement represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 are three coordinate strata of 81 elements; the 27 of S_2 are three strata of 9. The apex completes the partition.

4.3.5.2 Preservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d, the product $\widehat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \widehat{E}^{d+1} \to \widehat{E}^d$ removes the last coordinate.

Proposition 4.6 (Projective compatibility of normalized measures)**.** *For all $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \qquad \mu_d(\widehat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r}/10^d$, and the sum of the masses of all strata is one.

Proof. Each point of $\widehat{E}^d$ has ten preimages. Their total mass is $10\,10^{-(d+1)} = 10^{-d}$. Equality for any subset is obtained by summing over its points. The stratified formula follows from the same count as (4.39); the binomial expression $(9 + 1)^d$ gives total normalization. □

In dimension three, removing the apex leaves mass $999/1000$, and restoring it adds $1/1000$. Renormalizing before restitution would define another measure: the distinction is operationally relevant. In particular, conservation of probability mass, preservation of transport registers and thermodynamic dissipation are assertions about different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamic and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \ldots, 9\}^3$, removing $\{2, \ldots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\widehat{C} \setminus C$ belongs to its enlargement. Cubic completion, base-one-thousand positional expansion and the exceptional incidence reading share support data; their respective maps determine what information from those data each representation uses.

4.3.6 Cubic incidence and Lorentzian realization

Cubic completion provides a relation between interior and boundary. Its geometry can be developed through incidence of the six faces with the eight vertices. Label the faces by (i, ε), $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and the vertices by $\nu \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),\nu} = \mathbf{1}_{\{\nu_i = \varepsilon\}}.$$

Let $u = (1, \ldots, 1)^\mathsf{T} \in \mathbb{R}^6$, R_F be face opposition and $P_u = uu^\mathsf{T}/6$, $P_- = (I - R_F)/2$.

Proposition 4.7 (Four-dimensional incidence quotient)**.** *The incidence Gram matrix satisfies*

$$MM^\mathsf{T} = 12P_u + 4P_-.$$

Its eigenvalues $12, 4, 0$ *have multiplicities* $1, 3, 2$. *Consequently,*

$$Q_\square = \mathbb{R}^6/\ker(MM^{\mathsf{T}}) \simeq \mathbb{R}u \oplus E_-, \qquad E_- = \operatorname{im} P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct nonopposite faces share two vertices. The matrix $12P_u + 4P_-$ has precisely those entries. The uniform space has dimension one; differences between opposite faces span a three-dimensional space orthogonal to the uniform one. Its complement has dimension two and belongs to the kernel. □

The decomposition $1 + 3$ comes from incidence. The temporal choice is declared through the bilinear form

$$B_\square([x], [y]) = \langle x, (P_- - P_u)y\rangle.$$

It is well-defined on the quotient, because both projectors annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \qquad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\operatorname{diag}(-1, 1, 1, 1)$. The domain, decomposition and sign convention producing the Minkowski realization are made explicit. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_\square = \sqrt{6}P_u + \sqrt{2}P_-, \qquad W_\square e_{i,\varepsilon} = t + \varepsilon d_i$$

sends the six face labels to null directions, since $B_\square(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Inversion of a direction and face opposition thus have a specific geometric realization.

4.3.6.1 Soldering, inversion and subdivision

The cellular realization uses an invertible transport $U_m(a)$, preserving the Lorentzian form, between the fibers at the endpoints of an oriented dual edge a, a face label $\lambda_m(a)$ and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the soldering map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a)\beta_m(a).$$

Identify the origin fibers at two consecutive levels through the declared refinement isometry. If the nine subsegments $a_1, \dots, a_9$, of level $m + 1$, preserve the label and orientation when transported to that common fiber, then

$$\beta_m(a) = \sum_{j=1}^{9} U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \tag{4.41}$$

Each transported term equals $\ell_m/9$ times the same null direction, proving the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation combines direction, scale and transport, instead of limiting the geometric correspondence to a count of dimensions.

With orientation and Lorentzian form also fixed, Hodge duality on $\Lambda^2 Q_\square^*$ satisfies $\star^2 = -I$. In dimension four, the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, the eigenspaces of values $+i$ and $-i$. The electromagnetic realization will use this domain after constructing the electronic section and declaring its coupling.

4.3.7 Compatibility between modular and positional reductions

The block decimal base enters through a Euclidean division simultaneously preserving remainder and quotient:

$$n = 1000q + r, \qquad 0 \le r < 1000.$$

Since $1000 \equiv 1 \pmod 9$ and $1000 \equiv 1 \pmod 3$,

$$n \equiv q + r \pmod 9, \qquad n \equiv q + r \pmod 3.$$

Therefore, reduction of a block requires the quotient's contribution to recover the integer's class. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and modulo 3. This is an operator reason to preserve memory during a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated through a single integer residue. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D3^L.$$

On adding six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \le \nu < 729$. If the decimal prefix remains fixed, we obtain

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \tag{4.42}$$

Both identities are proved by substituting the definitions. Endpoint comparison is thus reduced to integer operations retaining the contribution of each prolongation. This compatibility allows a region to be refined before its real coordinate is evaluated.

4.3.8 The number as a compatible section and the value as a reading

Distinguishing state from observation specifies what object is prolonged as resolution increases. At depth n, let X_n be the set of admissible states produced by APP–TRIT–TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. These are not independent coordinates: earlier arithmetic evaluations and transports impose their relations. A prolongation preserves these relations in each antecedent, even while adding new steps and memory data. This is the meaning of the arithmetic of histories developed in (Sections 4.3 and 10.5.11).

Let $r_{m,n} : X_m \to X_n$, for $m \ge n$, be the depth restrictions, with $r_{n,n} = \mathrm{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined in it.

Definition 4.8 (Compatible numerical section)**.** An HMT number is a compatible history of the state system:

$$x = (x_n)_{n\ge 0} \in X_\infty := \varprojlim_n X_n, \qquad r_{m,n}(x_m) = x_n. \tag{4.43}$$

A reader is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the reader. The pair formed by the section and its reader constitutes a read presentation, not a redefinition of the section's identity.

In the Archimedean domain X_∞^{Arch}, the reader associates a closed cylinder with each prefix, preserves their nesting and makes their diameter tend to zero. It thus determines

$$\mathrm{ev}_{\mathbb{R}} : X_\infty^{\mathrm{Arch}} \longrightarrow \mathbb{R}, \qquad \{\mathrm{ev}_{\mathbb{R}}(x)\} = \bigcap_n I_n(x). \tag{4.44}$$

The specific construction of these cylinders and determination of their boundaries are developed in Section 10.5. Definition (4.43) does not by itself assert that the image of (4.44) is the whole line: a coverage assertion requires its own theorem. Nor does equality of values establish, without an injectivity result, equality of their histories. This separation allows precision to be discussed as depth while preserving the difference between the arithmetic antecedent and its scalar realization.

4.3.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category C whose arrows are its admissible histories. Composition is written $vu = v \circ u$: first u is traversed, then v. Elementary steps come from TPK selection, transport and update. A local tritic fiber admits the expression $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the transported state; a transition between regimes is not equivalent to identifying their quadratic algebras.

The following formulation makes an algebraic sector of that composition precise. For each object x, a unital associative real algebra A_x is given, which may be noncommutative. For each $u : x \to y$, a unital homomorphism $\rho_u : A_x \to A_y$ is given. We assume a strict action:

$$\rho_{\mathrm{id}_x} = \mathrm{id}_{A_x}, \qquad \rho_v \rho_u = \rho_{vu}. \tag{4.45}$$

For each composable pair $u : x \to y$, $v : y \to z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. We require the normalization $\Omega(\mathrm{id}_y, u) = \Omega(u, \mathrm{id}_x) = 1_y$ and, for each composable triple, the identity

$$\rho_w\big(\Omega(v, u)\big)\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \tag{4.46}$$

These are explicit hypotheses on transports and their coefficients, not a consequence of calling the dynamics holonomic. When applying this presentation to a TPK sector, its ρ_u and $\Omega(v, u)$ must be identified. Transitions expressed through bimodules require that structure to be preserved and are not replaced by (4.45).

The space of finite sums

$$\mathcal{H}(C, A, \rho, \Omega) = \bigoplus_{u \in \mathrm{Mor}(C)} A_{t(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \tag{4.47}$$

if the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. The sum combines histories with coefficients; the product concatenates histories while preserving the register multiplier.

Proposition 4.9 (Sectorial associativity with a central multiplier)**.** *Under* (4.45) *and* (4.46)*, the product* (4.47) *is associative. If the object set is finite, its unit is* $\sum_x 1_x T_{\mathrm{id}_x}$.

Proof. For $u : x \to y$, $v : y \to z$, $w : z \to t$, let $a \in A_y$, $b \in A_z$, $c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: \quad c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: \quad c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action equates $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to be moved to the right of that factor; (4.46) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero because of their initial and terminal states. Bilinearity completes the proof. The local identities and normalization of Ω verify the indicated unit. □

Calendar carry provides an effective example. In the category with one object and arrows C_9, take coefficients $A = \mathbb{R}[z, z^{-1}]$, identity action and

$$\Omega(b,a) = z^{c_0(a,b)}, \qquad c_0(a,b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \quad 0 \le a, b < 9.$$

The two carry sums of a triple are $\lfloor (a+b+c)/9 \rfloor$; hence (4.46) holds. The formal variable z records the turn without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphasic register of the calendar (4.33); retaining z without that relation distinguishes successive turns. The example realizes the memory of this calendar, without identifying it with the entirety of TPK memory.

4.3.10 Double variance of refinement and preservation of unity

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \mathrm{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r^*_{m,n} : D_n \longrightarrow D_m, \qquad r^*_{m,n} f = f \circ r_{m,n}. \tag{4.48}$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\mathrm{cyl}} = \varinjlim_n (D_n, r^*_{n+1,n})$ brings together observations of finite depth. This algebra is not identified here with the entire route algebra: prolonging transport operators also requires their compatibilities with refinement.

Lemma 4.10 (Unit and naturality of evaluation)**.** *If e_x is the indicator of $x \in X_n$, then*

$$r^*_{m,n} e_x = \sum_{y:\, r_{m,n}(y) = x} e_y, \qquad r^*_{m,n} 1_n = 1_m, \tag{4.49}$$

and, for every $f \in D_n$, $y \in X_m$,

$$\langle r^*_{m,n} f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\mathrm{cyl}} \to \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m. Their indicators are orthogonal and sum to 1_m, proving (4.49). Naturality is the definition of precomposition. Finally, $(r^*_{m,n} f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. □

The global unit is the constant function one over all possibilities; the individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement may increase the number of extensions and the depth of each history while maintaining unity and all evaluations of its antecedents.

4.3.11 Phase return, Weyl representation and two readings

The law (4.34) acquires an operator realization in the bilateral suspension to be constructed in Section 10.5. There T is the one-step advance, V_9 observes the nonadic phase, W_δ the rotation phase and D_M the memory degree. Writing $\widehat{\Gamma}_9 = T^9$ for the turn in this representation, the Weyl relations (10.53) imply, on finitely supported vectors,

$$\begin{aligned} V_9 \widehat{\Gamma}_9 &= \widehat{\Gamma}_9 V_9, \qquad W_\delta \widehat{\Gamma}_9 = e^{2\pi i 9\delta} \widehat{\Gamma}_9 W_\delta, \\ [D_M, \widehat{\Gamma}_9] &= \widehat{\Gamma}_9. \end{aligned} \tag{4.50}$$

The first equality follows from $\zeta_9^9 = 1$; the second, from nine applications of the covariance of W_δ; the third expresses the unit memory increment. Finite phase returns, while irrational rotation and memory distinguish the next state. The characters and parameter δ are subsequently realized from the numerical construction specified there: they do not retrospectively select the seeds. This representation describes observables and translations of the suspension; it does not replace all coordinates or all operators of TPK.

The double reading prolongs the same distinction between history and observable. In its Archimedean domain, (4.44) is applied; in the calibrated incidence domain, the maps Q_n of Section 21.11.1 preserve the words and boundary frames. The equality proved there $r_{n,m}Q_n = Q_m p_{n,m}$ determines their prolongation Q_∞. On the common domain of both readings one may jointly consider $\mathrm{ev}_{\mathbb{R}}(x)$ and $Q_\infty(x)$: the first retains the value; the second, the transported incidence preserved by its codomain. The section precedes both. This articulation allows the passage from arithmetic of histories to constants and the exceptional reading without turning their realizations into new initial data.

Chapter 5

Unitary construction of the discrete structure of the continuum: spectral transport, solenoidal balance, gauge incidence, cellular coherence and determinant line

The five constructions are generated correlatively on a single APP–TRIT–TPK state. They are consubstantial operations of the continuum, not an exhaustive enumeration of its properties, five independent branches, or a retrospective selection determined by subsequent applications.

5.1 Common state, prolongation tree, and genealogical measure

A single enriched state prior to any internal discrimination

The object of this chapter is the complete finite state produced by APP, oriented by TRIT, and transported by TPK. From it are constructed the groupoid of reversible symmetries, the tree of all admissible prolongations, the multisection of complete histories, and its canonical measure. No specialized domain is introduced: the domain is the total set of states effectively generated by the same APP–TRIT–TPK mechanism. Nor is any sheet, continuation, or subsequent reading selected.

5.1.1 Finite data produced by APP

For each depth $k \geq 0$, let A_k denote the finite set of cellular prefixes produced by APP, and let

$$\rho^{\mathrm{APP}}_{k+1,k} : \mathsf{A}_{k+1} \longrightarrow \mathsf{A}_k \tag{5.1}$$

the erasure of the last refinement. This erasure conserves the already constructed prefix. The neutral cell of APP gives a prolongation

$$\eta^{\mathrm{APP}}_k : \mathsf{A}_k \longrightarrow \mathsf{A}_{k+1}, \qquad \rho^{\mathrm{APP}}_{k+1,k}\eta^{\mathrm{APP}}_k = I_{\mathsf{A}_k}. \tag{5.2}$$

We write

$$\varepsilon^0_k(a) : a \longrightarrow \eta^{\mathrm{APP}}_k(a) \tag{5.3}$$

for the emission that realizes that prolongation. Thus, its destination, $\eta^{\mathrm{APP}}_k(a)$, and the operation that produces it, $\varepsilon^0_k(a)$, are kept separate. η^{APP}_k is not interpreted as an absence of history: it records a length-one refinement with neutral emission and materially preserves the preceding state.

Each $a \in \mathsf{A}_k$ simultaneously bears the two sheets of APP,

$$\mathrm{Sh}_k(a) = \big(\Sigma_k(a), \Pi_k(a)\big) \in \mathsf{H}^{\Sigma}_k \times \mathsf{H}^{\Pi}_k, \tag{5.4}$$

and not a chosen element of the pair. For an elementary emission $\epsilon : a \to a'$, APP further produces two typed Euclidean divisions,

$$\delta^{\diamond}(\epsilon) = \mathrm{res}^{\diamond}(\epsilon) + 9\,\mathrm{quo}^{\diamond}(\epsilon), \qquad \mathrm{res}^{\diamond}(\epsilon) \in \{0, \ldots, 8\}, \quad \mathrm{quo}^{\diamond}(\epsilon) \in \mathbb{Z}, \tag{5.5}$$

where $\diamond \in \{\Sigma, \Pi\}$. The label $\diamond$ merely indicates which leaf is being read; both divisions remain in the same ledger.

The cellular source and target of ϵ determine its oriented boundary

$$\partial\epsilon := t(\epsilon) - s(\epsilon). \tag{5.6}$$

The neutral emission satisfies

$$\mathrm{res}^{\diamond}(\varepsilon_k^0(a)) = 0, \qquad \mathrm{quo}^{\diamond}(\varepsilon_k^0(a)) = 0, \qquad \partial\varepsilon_k^0(a) = 0. \tag{5.7}$$

Definition 5.1 (Directed graph of APP emissions)**.** Let $\mathsf{E}_k(a)$ be the finite set of APP emissions of length one that depart from $a \in \mathsf{A}_k$. We conserve the emissions with multiplicity: two operations with the same target but with different residue, quotient, or provenance are different elements of $\mathsf{E}_k(a)$. By (5.2),

$$\varepsilon_k^0(a) \in \mathsf{E}_k(a), \qquad 1 \le \#\mathsf{E}_k(a) < \infty. \tag{5.8}$$

5.1.2 Tritic extension of the incidence datum: balanced orientation and genealogical carry cocycle

The signed incidence of an APP emission is an integer $\Delta(\epsilon)$. The balanced representative modulo three uniquely gives

$$\Delta(\epsilon) = \mathrm{or}(\epsilon) + 3\kappa(\epsilon), \qquad \mathrm{or}(\epsilon) \in \{-1, 0, +1\}, \quad \kappa(\epsilon) \in \mathbb{Z}. \tag{5.9}$$

Thus, orientation and carry are not conditions imposed on a history: they are total functions of the APP data.

To make the composition visible, let $\mathsf{C}_\epsilon : \mathsf{M}_{s(\epsilon)} \to \mathsf{M}_{t(\epsilon)}$ be the memory transport induced by the emission. The native carry of a composition satisfies

$$\kappa(\epsilon_2\epsilon_1) = \kappa(\epsilon_2) + \mathsf{C}_{\epsilon_2}\kappa(\epsilon_1), \tag{5.10}$$

whenever $t(\epsilon_1) = s(\epsilon_2)$. For the identity and the neutral emission,

$$\kappa(I_a) = 0, \qquad \kappa(\varepsilon_k^0(a)) = 0, \qquad \mathsf{C}_{I_a} = I_{\mathsf{M}_a}. \tag{5.11}$$

Lemma 5.2 (Associativity of the carry ledger)**.** *For three composable emissions, the evaluation of* $\kappa(\epsilon_3\epsilon_2\epsilon_1)$ *does not depend on the parenthesization.*

Proof. Applying (5.10) twice to the left parenthesization gives

$$\kappa(\epsilon_3) + \mathsf{C}_{\epsilon_3}\kappa(\epsilon_2) + \mathsf{C}_{\epsilon_3}\mathsf{C}_{\epsilon_2}\kappa(\epsilon_1).$$

The right parenthesization produces the same expression because the transport of memory is functorial: $\mathsf{C}_{\epsilon_3\epsilon_2} = \mathsf{C}_{\epsilon_3}\mathsf{C}_{\epsilon_2}$. □

The oriented reversal of a reversible emission is denoted by $\overline{\epsilon}$. TRIT acts by

$$\mathrm{or}(\overline{\epsilon}) = -\mathrm{or}(\epsilon), \qquad \partial\overline{\epsilon} = -\partial\epsilon, \qquad \mathsf{C}_{\overline{\epsilon}} = \mathsf{C}_\epsilon^{-1}. \tag{5.12}$$

The inverse carry is determined, not chosen, by

$$\kappa(\overline{\epsilon}) = -\mathsf{C}_\epsilon^{-1}\kappa(\epsilon), \tag{5.13}$$

for $\kappa(\overline{\epsilon}\epsilon) = 0$.

5.1.3 TPK transport of the 2 leaves and of memory

TPK chronology is part of the type of each emission and is not reconstructed a posteriori by counting letters. Here we recover the action of the ometer of definition 1.163 y theorem 1.165. Let

$$C_9 := \mathbb{Z}/9\mathbb{Z}, \qquad \widetilde{O}_9 := \mathbb{Z}_{\ge 0} \times C_9, \qquad O_{9,12} := \mathbb{Z}/12\mathbb{Z} \times C_9. \tag{5.14}$$

The unrolled phase-and-memory state is $\mathsf{Q}_k = (w_k, r_k) \in \widetilde{\mathcal{O}}_9$; its reduction $(w, r) \mapsto ([w]_{12}, r)$ recovers $\mathcal{O}_{9,12}$. The canonical seed of the clock is

$$\mathsf{Q}_0(a_0) := (0, [0]_9) \qquad (a_0 \in \mathsf{A}_0); \tag{5.15}$$

the other choices of origin are their coherent translations by C_9 and do not change the update law. For $\bar{r} \in \{0, \dots, 8\}$, we set

$$\chi_9(r) := \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor. \tag{5.16}$$

Every emission ϵ that lleva depth k to depth $k + 1$ acts in the odometro by

$$\sigma_9(w, r) := (w + \chi_9(r), r + [1]_9), \qquad \mathsf{Q}_{k+1}(\epsilon_* \widehat{x}_k) = \sigma_9 \mathsf{Q}_k(\widehat{x}_k). \tag{5.17}$$

Its phase component is, by typed definition,

$$\mathrm{ph}(\epsilon) : r_k \longmapsto r_k + [1]_9, \qquad \mathrm{ph}(\epsilon_n \cdots \epsilon_1) : r_k \longmapsto r_k + [n]_9. \tag{5.18}$$

We write $g_k := r_k \in C_9$ for the published phase and $q_k^{(9)} := w_k$ for the vertical memory, and $\beta_k := [w_k]_{12}$ for its duodecaphase reduction. The iteration of (5.17) is

$$\sigma_9^n(w, r) = \left(w + \left\lfloor \frac{\bar{r} + n}{9} \right\rfloor, r + [n]_9 \right). \tag{5.19}$$

In efecto, the sumandos $\chi_9(r), \chi_9(r + [1]_9), \dots, \chi_9(r + [n - 1]_9)$ cuentan exactly the cruces of the representative 8 to the representative 0, that are $\lfloor (\bar{r} + n)/9 \rfloor$. In particular,

$$\sigma_9^9(w, r) = (w + 1, r), \qquad g_{k+9} = g_k, \qquad q_{k+9}^{(9)} = q_k^{(9)} + 1. \tag{5.20}$$

Even the neutral emission of APP realizes $g_k \mapsto g_{k+1}$: it is neutral in residue, quotient, carry, and boundary, but records that an extension has occurred. The phase word

$$\boldsymbol{\tau}_k^{(N)} := (g_k, g_{k+1}, \dots, g_N), \qquad g_{j+1} = g_j + [1]_9, \tag{5.21}$$

and the memory $q^{(9)}$ type the phase return and the state change as distinct objects.

5.1.3.1 Simultaneous Renewal of the Complete Fiber

The preceding odometer conserves the chronology of each extension. The same nonadic turn also has the internal Markov realization, already constructed over the APP–TPK catalogue, which simultaneously renews the two coordinates of the complete fiber. It is essential to conserve both realizations and not to identify a generic edge of the tree with an emission of the catalogue without a typing map.

Proposition 5.3 (Internal Markov Realization of the Nonadic Connection)**.** *Let the APP seed domain, its observable emission, and its multiplicities be*

$$T_{\min} : \Omega_{\mathrm{seed}} \twoheadrightarrow \mathcal{U}_6, \qquad m(u) := \#T_{\min}^{-1}(u),$$

and the factorization produced by the TPK

$$\mathcal{U}_6 \simeq \mathcal{S}_+ \times \mathcal{S}_\times, \qquad |\mathcal{S}_+| = 18, \qquad |\mathcal{S}_\times| = 26.$$

On functions in $\mathcal{U}_6$, the two conditional expectations act by

$$(E_+ f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{S}_+} f(s', e), \qquad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{S}_\times} f(s, e').$$

For

$$\tau = (+1, -1, 0, +1, -1, 0, +1, -1, 0), \qquad P_{+1} = E_+, \quad P_{-1} = E_\times, \quad P_0 = I,$$

the catalog transfer and its elevation to all seeds are

$$\begin{aligned} \mathcal{K}_{\mathrm{ph}}\big((u,r),(v,r+1)\big) &= P_{\tau(r)}(u,v), \qquad (5.22)\\ L_0(\omega,\omega') &= \mathbf{1}_{\{\omega=\omega'\}},\\ L_+(\omega,\omega') &= \frac{E_+(T_{\min}\omega, T_{\min}\omega')}{m(T_{\min}\omega')},\\ L_\times(\omega,\omega') &= \frac{E_\times(T_{\min}\omega, T_{\min}\omega')}{m(T_{\min}\omega')},\\ \widehat{\mathcal{K}}_{\mathrm{ph}}\big((\omega,r),(\omega',r+1)\big) &= L_{\tau(r)}(\omega,\omega'). \qquad (5.23) \end{aligned}$$

For $u = T_{\min}\omega$, it holds phase by phase that

$$\sum_{\omega' : T_{\min}\omega' = v} \widehat{\mathcal{K}}_{\mathrm{ph}}\big((\omega,r),(\omega',r+1)\big) = \mathcal{K}_{\mathrm{ph}}\big((u,r),(v,r+1)\big). \qquad (5.24)$$

In particular,

$$\boxed{\mathcal{K}_{\mathrm{ph}}^{9}\big|_r = E_+E_\times, \qquad (E_+E_\times)(u,v) = \frac{1}{468}.} \qquad (5.25)$$

Proof. In a phase + or ×, summing (5.23) over $T_{\min}^{-1}(v)$ exactly cancels the denominator $m(v)$; in a zero phase, L_0 projects onto the identity. This proves (5.24). The operators E_+, $E_\times$, and I are idempotent and commute, and each appears three times in the nonadic word. Therefore

$$\prod_{r \in C_9} P_{\tau(r)} = E_+^3 E_\times^3 I^3 = E_+E_\times.$$

The composition first renews one coordinate of cardinal 18 and then the other of cardinal 26, so that each target receives $1/(18 \cdot 26) = 1/468$. □

Remark 5.4 (Two internal realizations, without fictitious identification). The transfer $\mathcal{K}_{\mathrm{ph}}$ is the global realization of the complete catalog, and σ_9 is the chronological action on each state of the tree. One is not substituted for the other. An edge-by-edge equality would require an additional map

$$\mathrm{Cat}_{k,x} : \mathrm{Out}(x) \longrightarrow \mathcal{U}_6$$

that verified the corresponding sum identity on fibers. That map is not used in this volume. The complete monodromy (5.25) proceeds from the APP–TPK catalogue, while the genealogy of each history remains in the tree and its ledger.

TPK assigns to each admissible emission a typed transport of the leaf pair,

$$U_\epsilon : \mathsf{H}^{\Sigma}_{s(\epsilon)} \times \mathsf{H}^{\Pi}_{s(\epsilon)} \longrightarrow \mathsf{H}^{\Sigma}_{t(\epsilon)} \times \mathsf{H}^{\Pi}_{t(\epsilon)}, \qquad (5.26)$$

with

$$U_{I_a} = I, \qquad U_{\epsilon_2\epsilon_1} = U_{\epsilon_2}U_{\epsilon_1}. \qquad (5.27)$$

The codomain of U_ϵ remains the complete pair. No operation in this chapter replaces (Σ, Π) by one of its coordinates.

Let $m \in \mathsf{M}_{s(\epsilon)}$ be an accumulated memory. The TPK action over it is the affine transformation

$$\mathsf{U}_\epsilon(m) := \mathsf{C}_\epsilon m + \kappa(\epsilon). \qquad (5.28)$$

Lemma 5.5 (TPK Action on Memory)**.** *The transformations* U_ϵ *satisfy*

$$\mathsf{U}_{I_a} = I, \qquad \mathsf{U}_{\epsilon_2\epsilon_1} = \mathsf{U}_{\epsilon_2}\mathsf{U}_{\epsilon_1}. \tag{5.29}$$

Proof. The identity follows from (5.11). For the composition,

$$\begin{aligned}\mathsf{U}_{\epsilon_2}\mathsf{U}_{\epsilon_1}(m) &= \mathsf{C}_{\epsilon_2}\big(\mathsf{C}_{\epsilon_1}m + \kappa(\epsilon_1)\big) + \kappa(\epsilon_2)\\ &= \mathsf{C}_{\epsilon_2\epsilon_1}m + \kappa(\epsilon_2) + \mathsf{C}_{\epsilon_2}\kappa(\epsilon_1)\\ &= \mathsf{C}_{\epsilon_2\epsilon_1}m + \kappa(\epsilon_2\epsilon_1),\end{aligned}$$

where the last equality is (5.10). □

A finite route is a composable word.

$$\gamma = \epsilon_1\epsilon_2\cdots\epsilon_k. \tag{5.30}$$

Its boundary, transport, carry, and memory are

$$\partial\gamma := \sum_{j=1}^{k}\partial\epsilon_j, \tag{5.31}$$

$$U_\gamma := U_{\epsilon_k}\cdots U_{\epsilon_1}, \tag{5.32}$$

$$\kappa(\gamma) := \kappa(\epsilon_k) + \sum_{j=1}^{k-1}\mathsf{C}_{\epsilon_k}\cdots\mathsf{C}_{\epsilon_{j+1}}\kappa(\epsilon_j), \tag{5.33}$$

$$m(\gamma; m_0) := \mathsf{U}_{\epsilon_k}\cdots\mathsf{U}_{\epsilon_1}(m_0). \tag{5.34}$$

In (5.31), the interior endpoints cancel, leaving

$$\partial\gamma = t(\epsilon_k) - s(\epsilon_1). \tag{5.35}$$

5.1.4 The common state and its total domain

Definition 5.6 (Elementary ledger)**.** For an emission that appears between two consecutive enriched prefixes, we denote by $\mathsf{Q}^-(\epsilon) = (w^-(\epsilon), r^-(\epsilon))$ the odometer of the source prefix and by

$$\mathsf{Q}^+(\epsilon) := \sigma_9\mathsf{Q}^-(\epsilon) = (w^-(\epsilon) + \chi_9(r^-(\epsilon)), r^-(\epsilon) + [1]_9) \tag{5.36}$$

that of the destination prefix. Thus the odometer is not applied to a bare APP cell. The emission record is

$$\lambda(\epsilon) := \big(s(\epsilon), t(\epsilon), \mathrm{res}^\Sigma(\epsilon), \mathrm{quo}^\Sigma(\epsilon), \mathrm{res}^\Pi(\epsilon), \mathrm{quo}^\Pi(\epsilon), \mathrm{or}(\epsilon), \kappa(\epsilon), \partial\epsilon, \mathsf{Q}^-(\epsilon), \mathsf{Q}^+(\epsilon)\big). \tag{5.37}$$

The ledger of γ is the word ordered

$$\mathrm{Led}(\gamma) := \big(\lambda(\epsilon_1), \ldots, \lambda(\epsilon_k)\big). \tag{5.38}$$

Preserving this word, and not only its sum, distinguishes two histories with the same terminal output.

5.1.4.1 Complete reader of the TPK state

The depth K of TPK blocks and the depth k of the extension tree are distinct types. In this subsection, K denotes exclusively the depth of the TPK state already generated by the operator mechanism; it is not identified with k or with a generic edge.

Definition 5.7 (Compact nonadic memory reader)**.** The TPK enriched state of depth K is

$$v_K = \big(B_K, \boldsymbol{\delta}_K, \boldsymbol{x}_K, \boldsymbol{C}_K, \Sigma_K, \omega_K, \ell_K, g(K), \Lambda_K\big), \qquad B_K = (u_K^{\pi}, u_K^{e}, u_K^{\varphi}). \tag{5.39}$$

Its compact reader does not add coordinates: it publishes them from that same state,

$$\mathfrak{r}_9(v_K) := \big(B_K, C_K, p_K, c_K, \Sigma_K, W_K, g(K)\big), \tag{5.40}$$

where

$$\begin{aligned} C_K &:= \boldsymbol{C}_K = \big(C_K^{\chi}\big)_{\chi\in\{\pi,e,\varphi\}},\\ C_K^{\chi} &:= (N_K^{\chi}, L_K^{\chi}, r_K^{\chi}, D_K^{\chi}, A_K^{\chi}, B_K^{\chi}),\\ p_K &:= (B_0, \dots, B_K),\\ c_K &:= (\boldsymbol{\delta}_K, \boldsymbol{x}_K, \kappa(\gamma_{0:K}^{\mathrm{TPK}})), \end{aligned} \tag{5.41}$$

and the oriented shadow of Witt is preserved as

$$W_K := \big((u_K^{\chi}, u_K^{\chi} A_W)\big)_{\chi\in\{\pi,e,\varphi\}}, \qquad A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1\\ 1 & 0 & 1 & 2 & 2 & 1\\ 1 & 1 & 0 & 1 & 2 & 2\\ 1 & 2 & 1 & 0 & 1 & 2\\ 1 & 2 & 2 & 1 & 0 & 1\\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \qquad A_W^2 = -I_6. \tag{5.42}$$

The route $\gamma_{0:K}^{\mathrm{TPK}}$ is the word of extensions encoded by Λ_K; its prefix of length j is the one that produces v_j. The signature is the output of the explicit internal primitive

$$\Sigma_B : \mathsf{B}_{\mathrm{TPK}} \times \mathsf{C}_{\mathrm{TPK}} \times \mathbb{Z}[\mathsf{A}] \times \Omega_{12} \times \mathsf{L}_{\mathrm{TPK}} \longrightarrow \mathsf{S}_{\mathrm{TPK}}, \quad (B, c, \partial\gamma, \omega, \ell) \longmapsto \Sigma_B(B, c, \partial\gamma, \omega, \ell), \tag{5.43}$$

with $\Sigma_B(g \cdot -) = g \cdot \Sigma_B(-)$ for each TPK symmetry. Its explicit starting structure is $(B_K, c_K, \partial\gamma_{0:K}^{\mathrm{TPK}}, \omega_K, \ell_K)$. It is not replaced by an external signature nor inferred retrospectively from a numerical reader.

The action of step $K \mapsto K+1$ is written without omitting the deep state. For each $\chi \in \{\pi, e, \varphi\}$,

$$y_K^{\chi} = x_K^{\chi} A_H, \qquad u_{K+1}^{\chi} = y_K^{\chi} \bmod 3, \qquad x_{K+1}^{\chi} = \frac{y_K^{\chi} - u_{K+1}^{\chi}}{3}, \tag{5.44}$$

with

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1\\ 2 & 1 & 2 & 2 & 1 & 1\\ 1 & 0 & 0 & 1 & 0 & 2\\ 0 & 0 & 1 & 1 & 1 & 0\\ 2 & 2 & 2 & 0 & 2 & 1\\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad \det A_H = 1.$$

If

$$\mathrm{ind}_3(u_1, \dots, u_6) := \sum_{j=1}^{6} \overline{u}_j \, 3^{6-j} \in \{0, \dots, 728\},$$

the update of the prefix and its 729 subcylinders is

$$N_{K+1}^{\chi} = 729 N_K^{\chi} + \mathrm{ind}_3(u_{K+1}^{\chi}), \qquad L_{K+1}^{\chi} = L_K^{\chi} + 6, \tag{5.45}$$

$$I^{\chi}_{K,u} := \left[\frac{729N^{\chi}_{K} + u}{3^{L^{\chi}_{K}+6}}, \frac{729N^{\chi}_{K} + u + 1}{3^{L^{\chi}_{K}+6}} \right), \qquad 0 \le u < 729. \tag{5.46}$$

The decimal cylinder and the associated carry are updated by the two integer cases of the same operator $C_{3,1000}$. If a new decimal triad is not published,

$$r^{\chi}_{K+1} = r^{\chi}_{K}, \qquad D^{\chi}_{K+1} = D^{\chi}_{K}, \qquad A^{\chi}_{K+1} = 729A^{\chi}_{K} + \mathrm{ind}_3(u^{\chi}_{K+1})1000^{r^{\chi}_{K}}, \qquad B^{\chi}_{K+1} = A^{\chi}_{K+1} + 1000^{r^{\chi}_{K}}. \tag{5.47}$$

The primitive $C_{3,1000}$ produces the option datum

$$d^{\mathrm{dec}}_{K+1,\chi} \in \{\bot\} \sqcup \{0, \dots, 999\};$$

in the first case it equals $d^{\mathrm{dec}}_{K+1,\chi} = \bot$. If a triad occurs, we write $d^{\mathrm{dec}}_{K+1,\chi} = \delta^{\chi}_{K+1} \in \{0, \dots, 999\}$ and

$$\begin{aligned} r^{\chi}_{K+1} &= r^{\chi}_{K} + 1, \qquad D^{\chi}_{K+1} = 1000D^{\chi}_{K} + \delta^{\chi}_{K+1}, \\ A^{\chi}_{K+1} &= 729000A^{\chi}_{K} + \mathrm{ind}_3(u^{\chi}_{K+1})1000^{r^{\chi}_{K}+1} - 729\delta^{\chi}_{K+1}3^{L^{\chi}_{K}}, \\ B^{\chi}_{K+1} &= A^{\chi}_{K+1} + 1000^{r^{\chi}_{K}+1}. \end{aligned} \tag{5.48}$$

The register $\boldsymbol{\delta}_{K+1}$ adds those triads and the two joint cocycles

$$\begin{aligned} q_{K+1} &= (u^{\pi}_{K+1} + u^{e}_{K+1} + u^{\varphi}_{K+1}) \bmod 3, \quad c^{\mathrm{joint}}_{K+1} = \frac{u^{\pi}_{K+1} + u^{e}_{K+1} + u^{\varphi}_{K+1} - q_{K+1}}{3}, \\ a_{K+1} &= (u^{\pi}_{K+1} + u^{e}_{K+1} - u^{\varphi}_{K+1}) \bmod 3, \quad h^{\mathrm{joint}}_{K+1} = \frac{u^{\pi}_{K+1} + u^{e}_{K+1} - u^{\varphi}_{K+1} - a_{K+1}}{3}, \\ \boldsymbol{\delta}_{K+1} &= \boldsymbol{\delta}_{K} \star \big((d^{\mathrm{dec}}_{K+1,\chi})_{\chi}, c^{\mathrm{joint}}_{K+1}, h^{\mathrm{joint}}_{K+1}\big). \end{aligned} \tag{5.49}$$

Thus C_{K+1}, $\boldsymbol{\delta}_{K+1}$, and $\boldsymbol{x}_{K+1}$ have material actions, not merely names. The other coordinates are simultaneously updated by

$$p_{K+1} = p_{K} \star B_{K+1}, \tag{5.50}$$

$$\kappa(\gamma^{\mathrm{TPK}}_{0:K+1}) = \kappa(\varepsilon^{\mathrm{TPK}}_{K+1}) + \mathsf{C}_{\varepsilon^{\mathrm{TPK}}_{K+1}}\kappa(\gamma^{\mathrm{TPK}}_{0:K}), \tag{5.51}$$

$$c^{\mathrm{comp}}_{K+1} := (\boldsymbol{\delta}_{K+1}, \boldsymbol{x}_{K+1}, \kappa(\gamma^{\mathrm{TPK}}_{0:K+1})), \tag{5.52}$$

$$\Sigma_{K+1} = \Sigma_{B}\big(B_{K+1}, c^{\mathrm{comp}}_{K+1}, \partial\gamma^{\mathrm{TPK}}_{0:K} + \partial\varepsilon^{\mathrm{TPK}}_{K+1}, \omega_{K+1}, \ell_{K+1}\big), \tag{5.53}$$

$$W_{K+1} = \big((u^{\chi}_{K+1}, u^{\chi}_{K+1}A_{W})\big)_{\chi}, \qquad g(K+1) = g(K) + [1]_9. \tag{5.54}$$

Since A_H is invertible modulo three and $A^2_W = -I_6$, the residue and shadow updates do not collapse distinct blocks. Truncation preserves the complete sequences

$$(B_0, \dots, B_K),\ (C_0, \dots, C_K),\ (p_0, \dots, p_K),\ (c_0, \dots, c_K),\ (\Sigma_0, \dots, \Sigma_K),\ (W_0, \dots, W_K)$$

and simultaneously removes its last entry. Hence the reader $\mathfrak{r}_9$ commutes with the erasure of a block.

Proposition 5.8 (Universal scope and certified scope of non-rebooting)**.** *On every horizontal circuit of nine steps, the following are preserved*

$$g(K+9) = g(K), \qquad q_{\mathrm{mem}}(K+9) = q_{\mathrm{mem}}(K) + 1,$$

and the prefix is prolonged with nine emissions, so the state is not reset. In the certified joint section $\chi \in \{\pi, e, \varphi\}$, $0 \le K < 387$, it also holds that

$$(p_K, C_K) \neq (p_{K+9}, C_{K+9}), \qquad (B_K, \mathbf{N}_K, W_K) \neq (B_{K+9}, \mathbf{N}_{K+9}, W_{K+9}), \qquad \mathbf{N}_K := (N_K^{\pi}, N_K^{e}, N_K^{\varphi}). \tag{5.55}$$

Proof. The two universal identities are the exact iteration of (5.17); the prefix preserves the nine entries adjoined by (5.50). For the joint section, the recurrence (5.45) compares the 387 nonadic pairs of each of the three channels and produces (5.55); this is the assertion certified in theorem 8.54. Finally, $u \mapsto (u, uA_W)$ is injective, so the change of block also preserves the change of its complete shadow. This last inequality is not extended to an arbitrary neutral prolongation outside the certified section. □

Definition 5.9 (APP–TRIT–TPK state of depth k)**.** For a route $\gamma = \epsilon_1 \cdots \epsilon_k$ born in an APP cell a_0, the common enriched state is

$$\widehat{x}_k(\gamma) := \Big(a_0, \dots, a_k; \mathrm{Sh}_0, \dots, \mathrm{Sh}_k; \mathbf{res}_k, \mathbf{quo}_k; \mathbf{or}_k, \kappa_k; \mathsf{Q}_k; g_k; q_k^{(9)}; \beta_k; m_k; \partial\gamma; \gamma; \mathrm{Led}(\gamma)\Big), \tag{5.56}$$

where

$$\begin{aligned} \mathbf{res}_k &:= \big((\mathrm{res}^{\Sigma}(\epsilon_j), \mathrm{res}^{\Pi}(\epsilon_j))\big)_{1\le j\le k}, \\ \mathbf{quo}_k &:= \big((\mathrm{quo}^{\Sigma}(\epsilon_j), \mathrm{quo}^{\Pi}(\epsilon_j))\big)_{1\le j\le k}, \\ \mathbf{or}_k &:= \big(\mathrm{or}(\epsilon_j)\big)_{1\le j\le k}, \qquad \kappa_k := \big(\kappa(\epsilon_j)\big)_{1\le j\le k}, \\ m_k &:= m(\gamma; m_0). \end{aligned}$$

The intermediate sheets are obtained by $\mathrm{Sh}_j = U_{\epsilon_j}\,\mathrm{Sh}_{j-1}$.

The simultaneous presence of the route and the ledger prevents identifying distinct words that terminate in the same cell. The state is not a list of cardinals: each coordinate has the type and composition equations of the preceding sections.

Definition 5.10 (Total depth finite domain)**.** Let

$$\widehat{\mathcal{X}}_k^{\mathrm{tot}} := \left\{ \widehat{x}_k(\epsilon_1 \cdots \epsilon_k) : \begin{array}{l} a_0 \in \mathsf{A}_0, \\ \epsilon_j \in \mathsf{E}_{j-1}(a_{j-1}), \\ s(\epsilon_j) = a_{j-1},\ t(\epsilon_j) = a_j \end{array} \right\}. \tag{5.57}$$

No condition from a later publication is added to (5.57). Being an element of the domain means exactly having been produced by the total actions (5.5), (5.9), and (5.17)– (5.28).

The common truncation deletes the last emission and all and only the entries of the ledger that it produced:

$$\rho_{k+1,k} : \widehat{\mathcal{X}}_{k+1}^{\mathrm{tot}} \longrightarrow \widehat{\mathcal{X}}_k^{\mathrm{tot}}, \qquad \rho_{k+1,k}\widehat{x}_{k+1} = \widehat{x}_k. \tag{5.58}$$

Let $\varepsilon_k^0(x)$ be the neutral emission (5.3) applied to the APP endpoint of x. The neutral prolongation defines

$$\eta_k : \widehat{\mathcal{X}}_k^{\mathrm{tot}} \longrightarrow \widehat{\mathcal{X}}_{k+1}^{\mathrm{tot}}, \qquad \eta_k(x) := (\varepsilon_k^0(x))_* x, \qquad \rho_{k+1,k}\eta_k = I. \tag{5.59}$$

Proposition 5.11 (Finiteness, non-emptiness, and surjectivity)**.** *For every k, the set $\widehat{\mathcal{X}}_k^{\mathrm{tot}}$ is finite and nonempty, and $\rho_{k+1,k}$ is surjective.*

Proof. APP produces a nonempty finite set at depth zero. At each step, every state has a finite set of emissions through (5.8); by induction, there are only finitely many words of length k. The sequence of neutral prolongations produces at least one word of each length. Finally, (5.59) is a right inverse of truncation, so the latter is surjective. □

5.1.5 Finite reversible symmetry groupoid

The reversible operations of APP, TRIT, and TPK act on the entire state. Let U be the group of coherent families $g = (g_k)_{k\ge 0}$ generated by those operations, where

$$\rho_{k+1,k} g_{k+1} = g_k \rho_{k+1,k}, \qquad g_{k+1}\eta_k = \eta_k g_k. \tag{5.60}$$

We denote by U_k its image in $\mathrm{Sym}(\widehat{\mathcal{X}}_k^{\mathrm{tot}})$. This image is finite because it acts on a finite set. The action of g_k on

$$\widehat{x} = (a_{\le k}; \mathrm{Sh}_{\le k}; \mathbf{res}, \mathbf{quo}; \mathbf{or}, \kappa; \mathsf{Q}_k; g_k; q_k^{(9)}; \beta_k; m; \partial\gamma; \gamma; \mathrm{Led}(\gamma))$$

is

$$\begin{aligned} g_k \cdot \widehat{x} := \big(&g_k a_{\le k}; U_{g_k}\,\mathrm{Sh}_{\le k}; \mathbf{res}(g_k\gamma), \mathbf{quo}(g_k\gamma); \mathbf{or}(g_k\gamma), \kappa(g_k\gamma);\\ &\mathsf{Q}(g_k\gamma); g(g_k\gamma); q^{(9)}(g_k\gamma); \beta(g_k\gamma); \mathsf{U}_{g_k}(m); (g_k)_*\partial\gamma; g_k\gamma; \mathrm{Led}(g_k\gamma)\big). \end{aligned} \tag{5.61}$$

All the terms on the right-hand side are recalculated by means of the APP, TRIT, and TPK rules; they are not copied as independent labels.

Definition 5.12 (Common action groupoid)**.** The finite groupoid of depth k is

$$\mathfrak{G}_k := \mathsf{U}_k \ltimes \widehat{\mathcal{X}}_k^{\mathrm{tot}}. \tag{5.62}$$

Its objects are the common states. An arrow $(g_k, \widehat{x})$ has source $\widehat{x}$, target $g_k \cdot \widehat{x}$, identity $(I, \widehat{x})$, and inverse $(g_k^{-1}, g_k \cdot \widehat{x})$.

Lemma 5.13 (The action preserves all equations of state)**.** *The action* (5.61) *preserves the pair of sheets, the APP divisions, the TRIT orientation, the carry, the phase odometer, the vertical memory, its dodecaphase reduction, the affine memory, the boundary, the route, and the ledger, in their respective types. Moreover,*

$$I \cdot \widehat{x} = \widehat{x}, \qquad (g_{2,k} g_{1,k}) \cdot \widehat{x} = g_{2,k} \cdot (g_{1,k} \cdot \widehat{x}). \tag{5.63}$$

Proof. The APP divisions are unique, so the action on an emission redetermines its residues and quotients. The balanced decomposition (5.9) is likewise unique. Composition of the carry is associative by lemma 5.2; the action on memory is associative by lemma 5.5; and the boundary is functorial by (5.35). Finally, the action on the ordered word induces the action on its ledger entry by entry. Coherence with the odometer recalculates Q, g, and $q^{(9)}$ from the transformed route and preserves (5.17). These identities prove (5.63). □

5.1.6 Finite tree of all prolongations

For $x \in \widehat{\mathcal{X}}_k^{\mathrm{tot}}$, let $\mathrm{Out}(x)$ be the set of all admissible TPK emissions of length one from the endpoint of x. An emission belongs to $\mathrm{Out}(x)$ only when its action jointly updates all coordinates of (5.56). In particular,

$$1 \le d(x) := \#\,\mathrm{Out}(x) < \infty, \qquad \varepsilon_k^0(x) \in \mathrm{Out}(x). \tag{5.64}$$

Definition 5.14 (Truncated prolongation tree)**.** Given $N \ge k$ and $x \in \widehat{\mathcal{X}}_k^{\mathrm{tot}}$, we define $\mathcal{T}_{k,N}(x)$ as the rooted tree whose vertices at height j, $0 \le j \le N - k$, are the words

$$h = (\epsilon_{k+1}, \dots, \epsilon_{k+j}), \qquad \epsilon_{n+1} \in \mathrm{Out}(x_n), \qquad x_{n+1} = (\epsilon_{n+1})_* x_n, \tag{5.65}$$

with the empty word as root. An edge appends an emission. Two words reaching the same terminal state remain distinct vertices if their ledgers differ.

The last condition is essential: the tree records genealogies and not the quotient that remembers only the target. Its terminal level is

$$\mathrm{Term}_{k,N}(x) := \{h \in \mathscr{T}_{k,N}(x) : |h| = N - k\}. \tag{5.66}$$

Each h preserves the complete terminal state.

$$\mathrm{ter}_N(h) := \big(x_N, m_N, \partial(\gamma h), \gamma h, \mathrm{Led}(\gamma h)\big). \tag{5.67}$$

Proposition 5.15 (Finiteness and non-emptiness at every horizon)**.** *For all x, k, and $N \geq k$, the tree $\mathscr{T}_{k,N}(x)$ is finite and* $\mathrm{Term}_{k,N}(x) \neq \varnothing$.

Proof. The root has a finite positive number of successors by (5.64). Applying the same argument at each level yields, by induction, finiteness at every finite horizon. The word formed by neutral emissions $(\varepsilon^0_k, \varepsilon^0_{k+1}, \ldots, \varepsilon^0_{N-1})$ is a terminal vertex, so level (5.66) is nonempty. □

The erasure of the last emission defines

$$\pi_{N+1,N} : \mathrm{Term}_{k,N+1}(x) \longrightarrow \mathrm{Term}_{k,N}(x). \tag{5.68}$$

Lemma 5.16 (Terminal surjectivity)**.** *The map $\pi_{N+1,N}$ is surjective for every $N \geq k$.*

Proof. Let $h \in \mathrm{Term}_{k,N}(x)$, with terminal state v_h. Adding the emission $\varepsilon^0_N(v_h)$ produces $h' \in \mathrm{Term}_{k,N+1}(x)$ and $\pi_{N+1,N}(h') = h$. □

5.1.7 Nonadic Connection: Phase, Holonomy, and Memory

A word of nine emissions is not called a holonomy merely because it has length nine. The horizontal connection determined by the phase functor is first preserved. For $r \in C_9$, we define

$$\begin{aligned}\mathscr{R}_{r,k} := \big\{(x, \epsilon, \epsilon_* x) : x \in \widehat{\mathcal{X}}^{\mathrm{tot}}_k, \quad \epsilon \in \mathrm{Out}(x),\\ \mathsf{Q}_k(x) = (w, r), \quad \mathsf{Q}_{k+1}(\epsilon_* x) = \sigma_9(w, r)\big\}.\end{aligned} \tag{5.69}$$

The last equality does not restrict the domain: it is the total action (5.17) of every elementary TPK emission. The connection simultaneously transports sheet, memory and ledger:

$$\nabla_{\mathrm{TPK}}(\epsilon) := \big(\mathrm{ph}(\epsilon), U_\epsilon, \mathsf{U}_\epsilon, \lambda(\epsilon)\big). \tag{5.70}$$

Its composition is calculated by means of (5.27), (5.29), and the concatenation of the ledger.

The apex of the nonadic turn is now the set of phase-compatible horizontal lifts.

$$\begin{aligned}\mathsf{Z}^{\Gamma}_{9,k} := \coprod_{r \in C_9} \big\{(x, \epsilon_1, \ldots, \epsilon_9) : x_0 = x, \quad x_i = (\epsilon_i)_* x_{i-1},\\ (x_{i-1}, \epsilon_i, x_i) \in \mathscr{R}_{r+i-1,k+i-1}, \quad 1 \leq i \leq 9\big\},\end{aligned} \tag{5.71}$$

where the subscripts of $r + i - 1$ are read in C_9. The summand is uniquely determined by $r = g_k(x)$, so the coproduct does not duplicate witnesses. The source and target maps are

$$\widehat{\mathcal{X}}^{\mathrm{tot}}_k \xleftarrow{s_{9,k}} \mathsf{Z}^{\Gamma}_{9,k} \xrightarrow{t_{9,k}} \widehat{\mathcal{X}}^{\mathrm{tot}}_{k+9}, \qquad s_{9,k}(x, \epsilon) = x, \quad t_{9,k}(x, \epsilon) = (\epsilon_9 \cdots \epsilon_1)_* x. \tag{5.72}$$

The relational holonomy is

$$\Gamma_{9,k} := \{(s_{9,k} z, t_{9,k} z) : z \in \mathsf{Z}^{\Gamma}_{9,k}\} = \mathrm{Hol}_{C_9}(\nabla_{\mathrm{TPK}}). \tag{5.73}$$

Lemma 5.17 (The ninth level of the tree realizes the holonomy)**.** *The map that forgets the intermediate states already determined by the emissions induces a bijection*

$$\mathsf{Z}^{\Gamma}_{9,k} \xrightarrow{\cong} \coprod_{x\in\widehat{\chi}^{\mathrm{tot}}_k} \mathrm{Term}_{k,k+9}(x). \tag{5.74}$$

Proof. Each edge of $\mathrm{Out}(x_i)$ satisfies (5.17); hence a terminal word of length nine determines the nine elements of (5.69). Reciprocally, a horizontal lift of (5.71) is a composable word of the tree. The two operations conserve the emissions and are inverse. Thus (5.74) is a theorem derived from the connection, not its definition. □

Proposition 5.18 (Phase return, memory advance and no restart)**.** *For every $z = (x, \epsilon_1, \dots, \epsilon_9)$ of the apex,*

$$g(t_{9,k}z) = g(s_{9,k}z), \tag{5.75}$$
$$q^{(9)}(t_{9,k}z) = q^{(9)}(s_{9,k}z) + 1, \tag{5.76}$$
$$\mathrm{Led}(t_{9,k}z) = \mathrm{Led}(s_{9,k}z) \star \big(\lambda(\epsilon_1), \dots, \lambda(\epsilon_9)\big). \tag{5.77}$$

In particular, the phase identification $g(t_{9,k}z) = g(s_{9,k}z)$ does not induce an identification of states: after transporting both registers to the same type of comparison, their vertical memories differ by one unit. The published phase returns, not the enriched state.

Proof. The first two identities are (5.20). The third is the definition of ledger concatenation. Since the vertical memory differs by one unit, the complete states are distinct even if some additional visible coordinate could coincide. □

Proposition 5.19 (Isometric compression of the same holonomy)**.** *Let $U_{\Gamma_9} : \mathcal{H}_{\mathrm{full}} \to \mathcal{H}_{\mathrm{full}}$ be an isometric realization of the preceding holonomy and let $J : \mathcal{H}_{\mathrm{vis}} \to \mathcal{H}_{\mathrm{full}}$ be the isometric inclusion of the visible register, with $E := J^*$ and $P := JE$. We define*

$$T := EU_{\Gamma_9}J, \qquad \eta := (I - P)U_{\Gamma_9}J. \tag{5.78}$$

Then

$$\boxed{U_{\Gamma_9}J = JT + \eta, \qquad I - T^*T = \eta^*\eta.} \tag{5.79}$$

Proof. Inserting $I = P + (I - P)$ before $U_{\Gamma_9}J$ gives

$$U_{\Gamma_9}J = JEU_{\Gamma_9}J + (I - P)U_{\Gamma_9}J = JT + \eta.$$

The two summands belong to orthogonal subspaces. Since U_{Γ_9} and J are isometries,

$$I = (U_{\Gamma_9}J)^*(U_{\Gamma_9}J) = T^*T + \eta^*\eta,$$

which is the second identity. □

Proposition 5.20 (Totality and Equivariance of Holonomy)**.** *The map $s_{9,k}$ is surjective. The nine neutral prolongations form the section*

$$\begin{gathered} x_0 := x, \qquad \varepsilon^0_{k+i} := \varepsilon^0_{k+i}(x_i), \qquad x_{i+1} := (\varepsilon^0_{k+i})_* x_i \quad (0 \le i < 9), \\ \eta^{(9)}_k(x) := (x, \varepsilon^0_k, \dots, \varepsilon^0_{k+8}) \in \mathsf{Z}^{\Gamma}_{9,k}, \qquad s_{9,k}\eta^{(9)}_k = I. \end{gathered} \tag{5.80}$$

Every coherent symmetry a whose action on the odometer satisfies $\mathsf{Q}\,a = \bar{a}\,\mathsf{Q}$ and $\bar{a}\,\sigma_9 = \sigma_9\,\bar{a}$ induces

$$a^{Z}(x, \epsilon_1, \dots, \epsilon_9) := (ax, a\epsilon_1, \dots, a\epsilon_9)$$

and satisfies

$$s_{9,k}a^{Z} = a_k s_{9,k}, \qquad t_{9,k}a^{Z} = a_{k+9}t_{9,k}. \tag{5.81}$$

Proof. The neutral word exists in every source and satisfies the phase action (5.17); this proves the section and surjectivity. The condition $\bar{a}\sigma_9 = \sigma_9\bar{a}$ sends horizontal relations to horizontal relations. Acting entry by entry preserves composition, carry, memory, boundary, and ledger; reading the first or the final state before or after acting gives (5.81). □

5.1.8 Total survival and complete multisection

Definition 5.21 (Produced survival)**.** A state $x \in \widehat{\mathcal{X}}_k^{\mathrm{tot}}$ survives when

$$\mathrm{Term}_{k,N}(x) \neq \varnothing \qquad \text{for every } N \geq k. \tag{5.82}$$

We denote by Surv_k the set of such states.

Theorem 5.22 (The total domain is survivor)**.** *For every depth,*

$$\mathrm{Surv}_k = \widehat{\mathcal{X}}_k^{\mathrm{tot}}. \tag{5.83}$$

In particular, survival does not define a subset supposed before constructing the state.

Proof. The inclusion $\mathrm{Surv}_k \subseteq \widehat{\mathcal{X}}_k^{\mathrm{tot}}$ belongs to the definition. The opposite inclusion is exactly the nonemptiness at every horizon proved in proposition 5.15. □

Definition 5.23 (Multisection of complete prolongations)**.** The complete multisection over x is the inverse limit

$$\mathrm{Msect}(x) := \varprojlim_{N \geq k} \big(\mathrm{Term}_{k,N}(x), \pi_{N+1,N}\big). \tag{5.84}$$

Its elements are all the compatible infinite histories that prolong x. None of them receives priority within (5.84).

Theorem 5.24 (No material emptiness of the multisection)**.** *For every $x \in \widehat{\mathcal{X}}_k^{\mathrm{tot}}$,*

$$\mathrm{Msect}(x) \neq \varnothing. \tag{5.85}$$

Moreover, every finite coordinate of a history in $\mathrm{Msect}(x)$ *preserves both leaves, orientation, carry, memory, boundary, path, and ledger of the state that it prolongs.*

Proof. The terminal sets are finite and nonempty by proposition 5.15, and their bonding maps are surjective by lemma 5.16. A compatible element can be constructed recursively: one begins at the root and, at every step, retains a prolongation of the preceding prefix. The neutral prolongation guarantees that the process never stops. Compatibility of the coordinates follows from the fact that each edge of the tree acts through the unique state action defined in (5.56). □

5.1.9 Canonical measure generated by the tree

The measure is not introduced as an external weight on the outputs. It is obtained by counting the admissible emissions at each vertex before choosing a continuation.

Definition 5.25 (Local weight and horizon measure)**.** For $v \in \widehat{\mathcal{X}}_j^{\mathrm{tot}}$ and $\epsilon \in \mathrm{Out}(v)$, we set

$$p_v(\epsilon) := \frac{1}{d(v)} \in \mathbb{Q}_{>0}. \tag{5.86}$$

If $h = (\epsilon_{k+1}, \dots, \epsilon_N) \in \mathrm{Term}_{k,N}(x)$ and x_j is the prefix that precedes ϵ_{j+1}, we define

$$\mu_{x,N}(h) := \prod_{j=k}^{N-1} p_{x_j}(\epsilon_{j+1}) = \prod_{j=k}^{N-1} \frac{1}{d(x_j)}. \tag{5.87}$$

Lemma 5.26 (Normalization)**.** *For every horizon $N \geq k$,*

$$\sum_{h \in \mathrm{Term}_{k,N}(x)} \mu_{x,N}(h) = 1. \tag{5.88}$$

Proof. At height one, $\sum_{\epsilon \in \mathrm{Out}(x)} d(x)^{-1} = 1$. Assuming the normalization at height $N - k$, each terminal prefix h, with final state v_h, contributes at the next level

$$\sum_{\epsilon \in \mathrm{Out}(v_h)} \mu_{x,N}(h) p_{v_h}(\epsilon) = \mu_{x,N}(h).$$

Summing over all prefixes proves the identity by induction. □

Proposition 5.27 (Projective consistency)**.** *Finite measures satisfy*

$$(\pi_{N+1,N})_* \mu_{x,N+1} = \mu_{x,N}. \tag{5.89}$$

Proof. For $h \in \mathrm{Term}_{k,N}(x)$, the mass of its fiber is

$$\begin{aligned} \sum_{\pi_{N+1,N}(h')=h} \mu_{x,N+1}(h') &= \mu_{x,N}(h) \sum_{\epsilon \in \mathrm{Out}(v_h)} p_{v_h}(\epsilon) \\ &= \mu_{x,N}(h). \end{aligned}$$

This is precisely the equality of point measures equivalent to (5.89). □

The cylinders of the multisection are

$$[h]_{x,N} := \{\omega \in \mathrm{Msect}(x) : \omega|_N = h\}. \tag{5.90}$$

Theorem 5.28 (Canonical Measure of Extensions)**.** *There exists a unique measure of cilindros μ_x on* $\mathrm{Msect}(x)$ *such that*

$$\mu_x([h]_{x,N}) = \mu_{x,N}(h). \tag{5.91}$$

It is normalized, has support on the whole multisection, and is obtained only from the finite sets of APP–TRIT–TPK prolongations.

Proof. The consistency of proposition 5.27 makes (5.91) independent of the horizon at which a cylinder is represented. The normalization follows from lemma 5.26. Two cylinders are either disjoint or one contains the other; hence the finite masses uniquely define an additive measure on the cylinder algebra. The multisection is a limit of finite discrete sets, and the cylinders form its compact basis; the measure

extends uniquely to the σ-algebra they generate. Finally, $p_v(\epsilon) > 0$ for every admissible emission, so every nonempty cylinder has positive mass and the support is the entire multisection. □

5.1.10 Naturality under symmetries and truncations

A coherent family $g = (g_j)_{j\geq 0} \in \mathsf{U}$ transports an outgoing emission of depth j by means of

$$(g_{j+1}, g_j)_* : \mathrm{Out}(x) \longrightarrow \mathrm{Out}(g_j \cdot x), \qquad \epsilon \longmapsto g_{j+1}\epsilon g_j^{-1}. \tag{5.92}$$

The TPK relations make (5.92) a bijection that preserves multiplicities. Acting on each entry yields a tree isomorphism

$$g_{*,N} : \mathscr{T}_{k,N}(x) \xrightarrow{\cong} \mathscr{T}_{k,N}(g_k \cdot x). \tag{5.93}$$

Theorem 5.29 (Naturalness of the tree, the multisection, and the measure)**.** *For every coherent reversible symmetry $g = (g_j)_{j\geq 0}$,*

$$\pi_{N+1,N} g_{*,N+1} = g_{*,N} \pi_{N+1,N}, \tag{5.94}$$

$$g_* \,\mathrm{Msect}(x) = \mathrm{Msect}(g_k \cdot x), \tag{5.95}$$

$$(g_{*,N})_* \mu_{x,N} = \mu_{g_k \cdot x, N}, \qquad (g_*)_* \mu_x = \mu_{g_k \cdot x}. \tag{5.96}$$

Proof. The first equality states that entrywise action commutes with deleting the final entry. Passing to the inverse limit yields (5.95). Since (5.92) is bijective, $d(g_j \cdot v) = d(v)$; by (5.87), a word and its image have the same weight. This proves the first equality of measures in (5.96). The second follows on the cylinders from (5.91), and the cylinders determine the measure. □

The truncations of the state and those of the tree also form the square.

$$\begin{array}{ccc} \mathrm{Term}_{k,N+1}(x) & \xrightarrow{\pi_{N+1,N}} & \mathrm{Term}_{k,N}(x) \\ {\scriptstyle \mathrm{ter}_{N+1}}\downarrow & & \downarrow{\scriptstyle \mathrm{ter}_N} \\ \widehat{\mathcal{X}}^{\mathrm{tot}}_{N+1} & \xrightarrow{\rho_{N+1,N}} & \widehat{\mathcal{X}}^{\mathrm{tot}}_{N}. \end{array} \tag{5.97}$$

Commutativity is literal: both paths erase the last emission and its last ledger entry.

5.1.11 Inverse limit $\widehat{\Omega}$ of complete histories and common APP–TRIT–TPK enriched state

The surjections (5.58) produce the object of complete histories

$$\widehat{\Omega} := \varprojlim_{k} \big(\widehat{\mathcal{X}}^{\mathrm{tot}}_k, \rho_{k+1,k}\big). \tag{5.98}$$

An element $\widehat{x} = (\widehat{x}_k)_{k\geq 0}$ contains, at each depth, the same pair of transported sheets, the produced orientations and carries, the accumulated memory, the boundaries, the route, and the complete ledger. Its fiber of continuations from any prefix is $\mathrm{Msect}(\widehat{x}_k)$, equipped with $\mu_{\widehat{x}_k}$.

Theorem 5.30 (Common enriched state APP–TRIT–TPK)**.** *The system*

$$\mathfrak{C}^{\mathrm{gen}} := \left(\begin{array}{l} \{\widehat{\mathcal{X}}^{\mathrm{tot}}_k, \rho_{k+1,k}, \eta_k\}_{k\geq 0}, \widehat{\Omega}, \{\mathfrak{G}_k\}_{k\geq 0}, \{\mathscr{T}_{k,N}\}_{N\geq k}, \\ \mathrm{Msect}, \mathrm{Surv}, \{\mu_x\}, \{\mathfrak{r}_9(v_K)\}_{K\geq 0}, \mathcal{K}_{\mathrm{ph}}, \widehat{\mathcal{K}}_{\mathrm{ph}}, \\ \mathrm{Sh}, \mathrm{or}, \kappa, \mathsf{Q}, g, q^{(9)}, \beta, m, \partial, \gamma, \mathrm{Led}, \\ \{\mathsf{Z}^{\Gamma}_{9,k}, s_{9,k}, t_{9,k}, \Gamma_{9,k}\}_{k\geq 0} \end{array} \right) \tag{5.99}$$

is defined for every state produced by APP–TRIT–TPK. It is non-empty, natural under reversible symmetries, and compatible with refinement. In particular:

1. *the two leaves are transported together;*
2. *orientation and carry it calculan by the division TRIT;*
3. *the memory obeys the affine TPK action;*
4. *the phase obeys the TPK odometer, returns after nine emissions, and the vertical memory advances by one unit;*
5. *the Markov realization of that turn satisfies* $\mathcal{K}_{\mathrm{ph}}^{9} = E_{+}E_{\times}$ *and uniformly renews the complete fiber of* $468 = 18 \cdot 26$ *emissions;*
6. *the compact reader conserves block, cylinder, prefix, carry, signature, Witt shadow, and phase, with their updates and truncations;*
7. *boundary, path, and ledger are composed without erasing the endpoints or the order of the emissions;*
8. *every state survives and has a nonempty multisection of complete prolongations;*
9. *the canonical measure is projective and has support on all those prolongations.*

Proof. Totality and finiteness follow from proposition 5.11. The sheet, carry, and memory identities are proved in lemmas 5.2 y 5.5; the phase, vertical memory, and nonadic holonomy, including its fiber realization and its state reader, are proved in propositions 5.3, 5.8 y 5.18 a 5.20; the boundary preserves its endpoints by (5.35). Total survival and nonemptiness of the multisections are theorems 5.22 y 5.24. The existence, consistency, and support of the measure are proposition 5.27 y theorem 5.28. Finally, compatibility with the symmetries and truncations is proved in theorem 5.29 and (5.97). □

5.1.12 Structural necessity and origin

Proposition 5.31 (Falsifiers of the common enriched state)**.** *Any of the following operations replaces* $\mathfrak{C}^{\mathrm{gen}}$ *with another object and blocks the transfer of the preceding theorems:*

1. *elegir* Σ *or* Π *and descartar the another sheet;*
2. *conserving the residue and erasing the quotient;*
3. *preserving the TRIT sign and erasing its carry;*
4. *replacing the ordered ledger with its terminal value;*
5. *identifying two words because they reach the same cell;*
6. *choosing a continuation and replacing the multisection with it;*
7. *declaring survival without exhibiting prolongations to each horizon;*
8. *uniformly normalizing only the last level, ignoring the degrees of the prefixes and losing projective consistency;*
9. *using a subsequent reading to decide which states enter* $\widehat{\mathcal{X}}_{k}^{\mathrm{tot}}$.

Proof. The first six points eliminate coordinates or relations that appear in the definition of the state, tree, or multisection. The seventh eliminates the proof of theorem 5.22. The eighth may violate (5.89) when the terminal degrees are not constant. The last replaces the total domain (5.57) with a retrospective selection. □

The APP cells, the pair of sheets, the TRIT lift, the TPK extensions, the carry cocycle, memory, boundary, route, and preservation of the register constitute the starting structures. Upon them, the finite

groupoid, the word tree, the multisection, and the normalized local measure are brought together in a single domain. The theorems of total survival, nonemptiness, consistency, and naturality certify this joint construction.

5.2 Intrinsic Correlative Operations of the Same Continuum

Tesis rectora

The symbols sp, sol, gau, coh and det that appear in this chapter mark five correlative operations over a single APP–TRIT–TPK state. Each operation receives the same surviving extension, the same orientation, the same carry, the same memory, the same boundary, and the same route. The following sections are successive mathematical stages of a single construction and not autonomous expositions.

5.2.1 Generating edge and common genealogical state

We do not repeat here the state, tree, or measure constructed in the preceding chapter. We take $\widehat{x} = \widehat{x}_k(\gamma) \in \widehat{\mathcal{X}}_k^{\mathrm{tot}}$ from (5.56) and all its admissible emissions $\alpha \in \mathrm{Out}(\widehat{x})$. If $\widehat{y} = \alpha_* \widehat{x}$, we write

$$\mathrm{Ch}_k(\widehat{x}) := \{(\alpha, \alpha_* \widehat{x}) : \alpha \in \mathrm{Out}(\widehat{x})\}, \qquad d_k(\widehat{x}) := \#\,\mathrm{Ch}_k(\widehat{x}). \tag{5.100}$$

The finiteness and non-emptiness of this set are already (5.64); survival at all horizons, the complete multisection, and its measure are theorems 5.22, 5.24 y 5.28. A second non-emptiness is not again assumed or proved.

Each pair $(\alpha, \widehat{y})$ of (5.100) preserves the unique record

$$\begin{aligned} \mathfrak{e}_\alpha := \big(&\alpha, \mathrm{Ext}_{\gamma_\alpha}; \mathrm{Sh}_{s(\alpha)}, \mathrm{Sh}_{t(\alpha)}; \\ &\mathrm{res}^{\Sigma,\Pi}(\alpha), \mathrm{quo}^{\Sigma,\Pi}(\alpha); \\ &\mathrm{or}(\alpha), \kappa(\alpha), \mathsf{C}_\alpha; \mathsf{Q}(s(\alpha)), \mathsf{Q}(t(\alpha)), g(s(\alpha)), g(t(\alpha)), q^{(9)}(s(\alpha)), q^{(9)}(t(\alpha)); \\ &m(\gamma\alpha; m_0), \partial\alpha, \gamma\alpha, \mathrm{Led}(\gamma\alpha); \\ &\{\mathrm{Term}_{k+1,N}(\widehat{y})\}_{N \geq k+1}, \mathrm{Msect}(\widehat{y}), \mu_{\widehat{y}}\big). \end{aligned} \tag{5.101}$$

Here $\mathrm{Ext}_{\gamma_\alpha}$ denotes the complete relation that the admissible emission already satisfies; it introduces no additional filter. The subsequent operations are coordinated applications of $\mathfrak{e}_\alpha$ and do not reconstruct the record from their outputs.

5.2.2 Intrinsic Correlative Actions on the Common Generator

The internal marks are fixed without changing their source: sp denotes transport and grading of the two sheets; sol, signed increment, balance, and memory; gau, incidence and carry curvature; coh, ports, rectangular difference, and cellular complex; and det, based complex, correlative filling, and determinant line. The following formulas display their couplings over the same $\mathfrak{e}_\alpha$; their actions literally share source, target, route, and ledger.

The double APP evaluation provides, on the same support, the labels Σ and Π. For an extension α, their multiplicities are calculated from the complete ledger, without an external counter:

$$N_\diamond(\alpha) := \sum_{\epsilon \ \mathrm{en} \ \gamma\alpha} \mathbf{1}_{\{\mathrm{res}^\diamond(\epsilon) \neq 0\}}, \qquad \diamond \in \{\Sigma, \Pi\}. \tag{5.102}$$

Let there also be the total oriented coefficient

$$\Delta_{\gamma_\alpha} := \sum_{\epsilon \ \mathrm{en} \ \gamma\alpha} \mathrm{or}(\epsilon) \in \mathbb{Z}.$$

The signed action is

$$b_\alpha := \Delta_{\gamma_\alpha}\big(N_\Sigma(\alpha) - N_\Pi(\alpha)\big), \qquad b_\alpha^{\pm} := \frac{|b_\alpha| \pm b_\alpha}{2}. \tag{5.103}$$

By construction,

$$b_\alpha = b_\alpha^{+} - b_\alpha^{-}, \qquad |b_\alpha| = b_\alpha^{+} + b_\alpha^{-}, \qquad b_\alpha^{+} b_\alpha^{-} = 0. \tag{5.104}$$

The same set of extensions simultaneously produces the ports

$$I_k(\widehat{x}_k) := \mathrm{Ch}_k(\widehat{x}_k) \times \{\Sigma\}, \qquad J_k(\widehat{x}_k) := \mathrm{Ch}_k(\widehat{x}_k) \times \{\Pi\}. \tag{5.105}$$

A visible leaf is not assumed to contain more prolongations than the other: each surviving extension is evaluated on both leaves of the same support. By (5.64), both sets of ports are nonempty.

For $A_N = \mathbb{Z}/3^N\mathbb{Z}$, the cellular action on generators is

$$\begin{aligned} \kappa_{k,N} &: A_N^{I_k} \oplus A_N^{J_k} \longrightarrow A_N^{I_k \times J_k}, \\ \kappa_{k,N}(u,v)_{(\alpha,\Sigma),(\beta,\Pi)} &:= u_{(\alpha,\Sigma)} - v_{(\beta,\Pi)}. \end{aligned} \tag{5.106}$$

Selected first ports according to the TPK order, the rectangular difference

$$\delta(w)_{ij} := w_{ij} - w_{ij_0} - w_{i_0 j} + w_{i_0 j_0} \tag{5.107}$$

download the split sequence

$$0 \longrightarrow A_N \xrightarrow{\Delta} A_N^{I_k} \oplus A_N^{J_k} \xrightarrow{\kappa_{k,N}} A_N^{I_k \times J_k} \xrightarrow{\delta} A_N^{(|I_k|-1)(|J_k|-1)} \longrightarrow 0. \tag{5.108}$$

This is an identity of the same APP fibre: it adds no cell foreign to the generator $\mathfrak{e}_\alpha$.

The local cellular action uses the two integer coordinates accumulated by that same ledger. For $\widehat{x} = \widehat{x}_k(\gamma)$ and $\widehat{y} = \alpha_* \widehat{x} = \widehat{x}_{k+1}(\gamma\alpha)$, we set

$$\begin{aligned} c_{\widehat{x}} &:= \kappa(\gamma), & c_\alpha &:= c_{\widehat{y}} := \kappa(\gamma\alpha), \\ v_{\widehat{x}} &:= \sum_{\epsilon \text{ en } \gamma} \mathrm{or}(\epsilon), & v_{\widehat{y}} &:= \sum_{\epsilon \text{ en } \gamma\alpha} \mathrm{or}(\epsilon), \\ v_\alpha &:= v_{\widehat{y}} - v_{\widehat{x}} = \mathrm{or}(\alpha). \end{aligned} \tag{5.109}$$

On the generators f, r, e, b, g is defined

$$\partial f = 3c_\alpha e + b, \qquad \partial r = 0, \qquad \partial e = g, \qquad \partial b = -3c_\alpha g, \qquad \partial g = 0. \tag{5.110}$$

The cancellation

$$\partial^2 f = 3c_\alpha g - 3c_\alpha g = 0 \tag{5.111}$$

is literal and uses the same carry with the two signs imposed by the orientation.

The incidence seed is obtained from the free module of the two leaves,

$$V_{\mathrm{TPK}} := \mathbf{F}_3 e_\Sigma \oplus \mathbf{F}_3 e_\Pi, \qquad \mathscr{I} := \mathbf{P}^1(V_{\mathrm{TPK}}), \qquad |\mathscr{I}| = 4, \tag{5.112}$$

and of its unordered pairs

$$\mathscr{E} := \binom{\mathscr{I}}{2}, \qquad |\mathscr{E}| = 6, \qquad \mathscr{B} := \mathscr{I} \times \{+,-\}, \qquad |\mathscr{B}| = 8. \tag{5.113}$$

The incidence application is determined on the base by

$$B[e_L] = \sum_{L' \neq L} e_{\{L,L'\}}, \qquad L \in \mathscr{F}, \tag{5.114}$$

and, by linearity,

$$(Ba)_{\{L,L'\}} = a_L + a_{L'}. \tag{5.115}$$

If $s(q) = \sum_{E \in \mathscr{E}} q_E$, each L appears three times and

$$s(Ba) = 0. \tag{5.116}$$

Therefore the function

$$W_c(q) := \mathbf{1}_{\{s(q)=0\}} - \frac{1}{3} \tag{5.117}$$

satisfies $W_c(q + Ba) = W_c(q)$. The identities (5.112)–(5.117) are internal and exact. The material lift of q from the cells and common carries is treated in the assembly layer; this incidence identity is not conflated with the construction of that lift.

5.2.3 Unitary transport, polarization, and balance on the common fiber

The non-emptiness of (5.100) allows one to fix, without selecting a prolongation,

$$p_k(\alpha, \widehat{y} \mid \widehat{x}) := \frac{1}{d_k(\widehat{x})}. \tag{5.118}$$

Let the space of the two leaves of the same state be

$$\mathscr{H}_k := \bigoplus_{\widehat{x} \in \widehat{\mathcal{X}}_k^{\mathrm{tot}}} \left(\mathsf{H}_{\widehat{x}}^{\Sigma} \oplus \mathsf{H}_{\widehat{x}}^{\Pi}\right), \tag{5.119}$$

and let $\iota_{\widehat{x}}$ be the inclusion of the indicated fiber. The TPK transport of (5.26) gives the exact action

$$U_k \iota_{\widehat{x}} v := \sum_{(\alpha,\widehat{y}) \in \mathrm{Ch}_k(\widehat{x})} p_k(\alpha, \widehat{y} \mid \widehat{x})^{1/2} \iota_{\widehat{y}} U_\alpha v. \tag{5.120}$$

The children of two distinct parents are disjoint because truncation is a function. Moreover, two distinct witnesses α do not collapse into the same direct summand: the complete route γ, including its last step, is a coordinate of $\widehat{y}$. Consequently, the Gram identity is

$$U_k^* U_k\big|_{\mathsf{H}_{\widehat{x}}^{\Sigma} \oplus \mathsf{H}_{\widehat{x}}^{\Pi}} = \sum_{(\alpha,\widehat{y}) \in \mathrm{Ch}_k(\widehat{x})} p_k(\alpha, \widehat{y} \mid \widehat{x})\, U_\alpha^* U_\alpha. \tag{5.121}$$

Therefore $U_k^* U_k = I$ exactly when, in each initial fibre,

$$\sum_{(\alpha,\widehat{y}) \in \mathrm{Ch}_k(\widehat{x})} p_k(\alpha, \widehat{y} \mid \widehat{x})\, U_\alpha^* U_\alpha = I. \tag{5.122}$$

The isometry of every sheet transport U_α is sufficient for this equality, together with normalization of the weights; the necessary and sufficient condition is normalization of the average Gram matrix displayed in (5.122). For $\ell > k$, the complete history is preserved through

$$U_{\ell,k} := U_{\ell-1} \cdots U_k, \qquad U_{k,k} := I. \tag{5.123}$$

The grading acts on the two coordinates already present in each fiber of (5.119):

$$\sigma_k \iota_{\widehat{x}}(v_\Sigma, v_\Pi) := \iota_{\widehat{x}}(v_\Sigma, -v_\Pi), \qquad P_k := \frac{I + \sigma_k}{2}, \quad Q_k := \frac{I - \sigma_k}{2}. \tag{5.124}$$

The TRIT threshold belongs to the orientation and to the carry of $\widehat{x}$; it erases neither of the two APP sheets and does not require a third case in (5.124).

The leaf-preserving part and the leaf-changing part are two components of the same U_k:

$$A_k := P_{k+1}U_kP_k + Q_{k+1}U_kQ_k, \tag{5.125}$$

$$\eta_k := Q_{k+1}U_kP_k + P_{k+1}U_kQ_k. \tag{5.126}$$

By orthogonality of P_{k+1} and Q_{k+1},

$$\begin{aligned} U_k &= A_k + \eta_k, \\ A_k^*A_k + \eta_k^*\eta_k &= P_kU_k^*U_kP_k + Q_kU_k^*U_kQ_k. \end{aligned} \tag{5.127}$$

Proof of the sheetwise Gram balance. Expanding the products, orthogonality of the target projectors gives

$$\begin{aligned} A_k^*A_k &= P_kU_k^*P_{k+1}U_kP_k + Q_kU_k^*Q_{k+1}U_kQ_k, \\ \eta_k^*\eta_k &= P_kU_k^*Q_{k+1}U_kP_k + Q_kU_k^*P_{k+1}U_kQ_k. \end{aligned}$$

Their sum and $P_{k+1} + Q_{k+1} = I$ prove (5.127). The difference between $U_k^*U_k$ and that balance is $P_kU_k^*U_kQ_k + Q_kU_k^*U_kP_k$; it vanishes exactly when both cross blocks vanish. In particular, condition (5.122) yields $U_k^*U_k = I$ and reduces the balance to I. □

Proposition 6.24 develops this same general identity in the subsequent realization on complete histories, whose isometry is proved by retaining the label of each prolongation. This identity and the increment (5.103) are not two transports: A_k, η_k act on the history basis and b_α reads the edge of that same basis.

5.2.3.1 Signed polarization, non-autonomous telescoping, and return without reinitialization

No independent spectral branch is introduced here. The notations are restricted to the same space $\mathscr{H}_k$ and the same already constructed transport U_k

$$\mathcal{H}_k^{\mathrm{sp}} := \mathscr{H}_k, \qquad \Lambda(\gamma_{\widehat{x}}) := \big(\mathrm{or}, \mathrm{Carr}, \mathrm{Mem}, \partial, \gamma\big)(\widehat{x}),$$

and $\Sigma_{\mathrm{reg}}\Lambda(\gamma_{\widehat{x}})$ denotes its family of compatible truncations. Thus, the following signed reader, defect, and terminal are actions of the common correlative constructor, not a retrospective selection of histories.

5.2.3.2 Signed reader with genealogical register

For $v \in \mathcal{H}_k^{\mathrm{sp}}$, the signed polarization is

$$\mathrm{pol}_k(v) = \langle v, \sigma_k v\rangle = \|P_kv\|^2 - \|Q_kv\|^2. \tag{5.128}$$

The specialized reader furthermore preserves the source:

$$\mathcal{L}_{\mathrm{sp},k}^{\mathrm{sgn}}(x_k; v) = \big(\Lambda(\gamma_{x_k}), \Sigma_{\mathrm{reg}}(\Lambda(\gamma_{x_k})), \|P_kv\|^2, \|Q_kv\|^2, \mathrm{pol}_k(v)\big). \tag{5.129}$$

The TRIT orientation is preserved in $\Lambda(\gamma_{x_k})$; it is not identified with the sheet sign. At horizon N, the record keeps the family of readers for all levels $k \le N$. Truncating means removing only subsequent entries.

5.2.3.3 Non-autonomous and terminal product

We define

$$F_{0,0} = I_{\mathcal{H}_0^{\mathrm{sp}}}, \qquad F_{0,k} = A_{k-1}\cdots A_1A_0 \quad (k \ge 1). \tag{5.130}$$

Therefore,

$$F_{0,k+1} = A_kF_{0,k}. \tag{5.131}$$

The term published upon leaving the channel at step k and the terminal at horizon N are

$$\mathcal{D}_k = F_{0,k}^* \eta_k^* \eta_k F_{0,k}, \qquad \mathcal{T}_N = F_{0,N}^* F_{0,N}. \tag{5.132}$$

Theorem 5.32 (Spectral telescope–non-autonomous polar)**.** *For every finite horizon N,*

$$\boxed{I = \sum_{k=0}^{N-1} F_{0,k}^* \eta_k^* \eta_k F_{0,k} + F_{0,N}^* F_{0,N}.} \tag{5.133}$$

Moreover,

$$0 \preceq \mathcal{T}_{N+1} \preceq \mathcal{T}_N \preceq I,$$

so that the strong positive limit exists

$$\mathcal{T}_\infty = \text{s-lim}_{N\to\infty} \mathcal{T}_N$$

and

$$I = \sum_{k=0}^{\infty} F_{0,k}^* \eta_k^* \eta_k F_{0,k} + \mathcal{T}_\infty \tag{5.134}$$

in the strong topology. The theorem does not assert that $\mathcal{T}_\infty$ is zero.

Proof. Multiplying (5.127) by $F_{0,k}^*$ and $F_{0,k}$, and using (5.131), one obtains

$$F_{0,k}^* F_{0,k} = F_{0,k+1}^* F_{0,k+1} + F_{0,k}^* \eta_k^* \eta_k F_{0,k}.$$

The sum from $k = 0$ to $N - 1$ cancels all intermediate terms and leaves precisely (5.133). Since $A_k^* A_k \preceq I$,

$$\mathcal{T}_{N+1} = F_{0,N}^* A_N^* A_N F_{0,N} \preceq F_{0,N}^* F_{0,N} = \mathcal{T}_N.$$

Monotonicity and the positive bound provide the strong limit. Passing to the limit in the finite identity gives (5.134). □

5.2.3.4 Phase return and non-reinitialized evolution of the common state

The branch explicitly preserves

$$g(k + 9) = g(k), \qquad x_{k+9} \neq x_k \quad \text{in general.} \tag{5.135}$$

The second member records the increase in memory and possible changes of sheet, carry, cylinder and boundary. Neither σ_k, nor the projectors, nor the signed reader contract this difference.

The conditional expectation associated with the same weight is

$$(E_k F)(\widehat{x}) := \sum_{(\alpha,\widehat{y})\in \mathrm{Ch}_k(\widehat{x})} p_k(\alpha, \widehat{y} \mid \widehat{x}) F(\alpha, \widehat{y}). \tag{5.136}$$

For the fine increment b_f, set

$$b_c := E_k b_f, \qquad \kappa_k^{\mathrm{can}} := \frac{E_k|b_f| - |b_c|}{2} \geq 0. \tag{5.137}$$

Then

$$E_k b_f^+ = b_c^+ + \kappa_k^{\mathrm{can}}, \qquad E_k b_f^- = b_c^- + \kappa_k^{\mathrm{can}}. \tag{5.138}$$

The superscript can distinguishes this cancellation carry from the integer carry κ_α; both arise from the same transport, but have distinct types.

Signed memories require no external initial condition: for $\gamma = (\epsilon_1, \dots, \epsilon_k)$ they are defined by summing the ledger itself,

$$\begin{aligned} M_0^{\pm} &:= 0, \\ M_k^{\pm}(\gamma) &:= \sum_{j=1}^{k} b_{\epsilon_j}^{\pm}, \\ M_{k+1}^{\pm}(\gamma\alpha) &:= M_k^{\pm}(\gamma) + b_\alpha^{\pm}, \\ D_k^0 &:= M_k^+ - M_k^-, \qquad H_k^0 := M_k^+ + M_k^-. \end{aligned} \tag{5.139}$$

Thus, the signed balance, the total memory, and the polarization $\langle v, \sigma_k v\rangle$ are three coordinated readings of the same prefix, not three states reconstructed separately.

5.2.3.5 Nonadic Refinement, Orthogonal Losses and Terminal Memory

The index j in the following development is the level of the same tree of prolongations, and its thirteen-coordinate record is the restriction of the common record of (5.101). Accordingly, $D_{\mathrm{sol},j} := D_j^{(13)}$ is fixed and no additional domain is postulated.

5.2.3.6 Common nonadic refinement: inclusion, balance expectation and memory

Let $Y_j(x)$ be the finite set of cylinders compatible with the prefix $x_{\le j}$, endowed with the projective measure μ_j. The truncation

$$\tau_{j+1,j} : Y_{j+1}(x) \longrightarrow Y_j(x)$$

has nine admissible extensions over each cylinder. Finite spaces are considered.

$$\mathscr{H}_j(x) := \ell^2\big(Y_j(x), \mu_j\big).$$

The inclusion and the nonadic expectation are

$$J_{9,j} : \mathscr{H}_j \longrightarrow \mathscr{H}_{j+1}, \qquad (J_{9,j} f)(y) = f(\tau_{j+1,j} y), \tag{5.140}$$

$$E_{9,j} : \mathscr{H}_{j+1} \longrightarrow \mathscr{H}_j, \qquad (E_{9,j} g)(z) = \frac{1}{9} \sum_{\tau_{j+1,j} y = z} g(y). \tag{5.141}$$

The projective compatibility of μ_{j+1} and μ_j gives

$$E_{9,j} J_{9,j} = I_{\mathscr{H}_j}, \qquad J_{9,j}^* = E_{9,j}, \qquad P_{9,j} := J_{9,j} E_{9,j} = P_{9,j}^* = P_{9,j}^2. \tag{5.142}$$

The complementary microscopic projector is

$$Q_{9,j} := I - P_{9,j}. \tag{5.143}$$

If b_f is the increment on the fine fiber and

$$b_c := E_{9,j} b_f, \tag{5.144}$$

the elementary convexity of $|\cdot|$ constructs, rather than postulates, the carry of cancellation

$$\kappa_j := \frac{E_{9,j}|b_f| - |b_c|}{2} \ge 0. \tag{5.145}$$

Indeed, using $t^{\pm} = (|t| \pm t)/2$, one obtains

$$E_{9,j} b_f^+ = \frac{E_{9,j}|b_f| + E_{9,j} b_f}{2} = b_c^+ + \kappa_j, \tag{5.146}$$

$$E_{9,j} b_f^- = \frac{E_{9,j}|b_f| - E_{9,j} b_f}{2} = b_c^- + \kappa_j. \tag{5.147}$$

Both sheets lose the same amount when compressed; therefore, the signed difference is preserved while total memory records the cancellation:

$$E_{9,j}(b_f^+ - b_f^-) = b_c, \qquad E_{9,j}(b_f^+ + b_f^-) = |b_c| + 2\kappa_j. \tag{5.148}$$

5.2.3.7 Orthogonal factorization of the degrees of depth and of the resolution layers in $\mathscr{H}_j^{\text{depth}} \widehat{\otimes} \mathscr{H}_j^{\text{res}}$

To prevent the same loss from being counted both as depth and as resolution, the space at each level is typed as

$$\mathscr{H}_j = \mathscr{H}_j^{\text{depth}} \widehat{\otimes} \mathscr{H}_j^{\text{res}}. \tag{5.149}$$

For $j \geq 1$, the first factor carries

$$P_j^{\text{depth}} := J_{9,j-1} E_{9,j-1} \quad \text{on } \mathscr{H}_j^{\text{depth}},$$

and $P_0^{\text{depth}} = I$ is fixed. The second factor carries the finite shell partition inherited from the cylinder boundary. If $\sigma_j : Y_j^{\text{res}} \to S_j$ is that partition, its operators are

$$J_j^{\text{res}} f = f \circ \sigma_j, \qquad E_j^{\text{res}} g(s) = \frac{1}{\#\sigma_j^{-1}(s)} \sum_{\sigma_j(y)=s} g(y), \tag{5.150}$$

$$P_j^{\text{res}} := J_j^{\text{res}} E_j^{\text{res}}. \tag{5.151}$$

The three mutually orthogonal projectors are defined.

$$P_j^{\text{term}} := P_j^{\text{depth}} \otimes P_j^{\text{res}}, \tag{5.152}$$

$$Q_j^{\text{micro}} := (I - P_j^{\text{depth}}) \otimes I, \tag{5.153}$$

$$Q_j^{\text{shell}} := P_j^{\text{depth}} \otimes (I - P_j^{\text{res}}). \tag{5.154}$$

Then

$$Q_j^{\text{micro}} Q_j^{\text{shell}} = 0, \qquad I = P_j^{\text{term}} + Q_j^{\text{micro}} + Q_j^{\text{shell}}, \tag{5.155}$$

and, for all $\xi \in \mathscr{H}_j$,

$$\|\xi\|^2 = \|P_j^{\text{term}} \xi\|^2 + \|Q_j^{\text{micro}} \xi\|^2 + \|Q_j^{\text{shell}} \xi\|^2. \tag{5.156}$$

The term $(I - P_j^{\text{depth}}) \otimes (I - P_j^{\text{res}})$ does not appear a second time: it is already contained exactly once in Q_j^{micro}.

5.2.3.8 Cylinder extensions and full row of losses

An extension of cylinders from level j to level $j+1$ induces the map

$$\Gamma_j : \mathscr{H}_j \longrightarrow \mathscr{H}_{j+1}, \qquad \Gamma_j f = f \circ \tau_{j+1,j}, \tag{5.157}$$

with all leaf, orientation, carry, memory, and path labels transported in the cylinder basis. For $n > j$, we write

$$\Gamma_{j:n} := \Gamma_{n-1} \cdots \Gamma_j, \qquad \Gamma_{j:j} := I. \tag{5.158}$$

On $\mathscr{H}_j$, the nonnegative increments define multiplication operators

$$B_j^+ := M_{\sqrt{b_j^+}}, \qquad B_j^- := M_{\sqrt{b_j^-}}, \qquad K_j := M_{\sqrt{2\kappa_j}}. \tag{5.159}$$

Each root is taken pointwise over the finite set of cylinders; if a coordinate is zero, its operator is zero. The space of losses is the external orthogonal sum

$$\mathscr{E}_j := \mathscr{H}_j^{(+)} \oplus \mathscr{H}_j^{(-)} \oplus \mathscr{H}_j^{(\kappa)} \oplus \mathscr{H}_j^{(\mathrm{micro})} \oplus \mathscr{H}_j^{(\mathrm{shell})},$$

and the complete row is

$$L_j : \mathscr{H}_j \longrightarrow \mathscr{E}_j, \qquad L_j\xi = \big(B_j^+\xi, B_j^-\xi, K_j\xi, Q_j^{\mathrm{micro}}\xi, Q_j^{\mathrm{shell}}\xi\big). \tag{5.160}$$

Therefore,

$$\|L_j\xi\|^2 = \|B_j^+\xi\|^2 + \|B_j^-\xi\|^2 + \|K_j\xi\|^2 + \|Q_j^{\mathrm{micro}}\xi\|^2 + \|Q_j^{\mathrm{shell}}\xi\|^2. \tag{5.161}$$

The positive part, negative part, cancellation carry, microstructure, and shell each appear once. They are neither merged into an anonymous constant nor repeated in two summands.

5.2.3.9 Complete history terminal

Let $\mathrm{Hist}_J^{\mathrm{sol}}$ be the set of histories surviving to the horizon J, each with its package of thirteen coordinates. The terminal space is defined as the free space on these complete records:

$$\mathscr{T}_J^{\mathrm{sol}} := \ell^2\Big(\big\{(h, D_{\mathrm{sol},J}(h)) : h \in \mathrm{Hist}_J^{\mathrm{sol}}\big\}\Big). \tag{5.162}$$

The terminal evaluation

$$E_J^{\mathrm{term}} : \mathscr{H}_J \longrightarrow \mathscr{T}_J^{\mathrm{sol}} \tag{5.163}$$

sends each basic cylinder vector to the complete record of the history that generates it and extends linearly. In particular, the terminal retains fiber, relations, numbers, orientation, carry, memory, boundary, route, multisection, survival, and reader. It is not replaced by the last visible value of the route.

5.2.3.10 Observation column, Gram operator, and telescoping

For $0 \le j \le J$, the complete future column is

$$\begin{aligned} G_{j;J} &: \mathscr{H}_j \longrightarrow \mathscr{T}_J^{\mathrm{sol}} \oplus \bigoplus_{n=j}^{J-1} \mathscr{E}_n, \\ G_{j;J}\xi &:= E_J^{\mathrm{term}}\Gamma_{j:J}\xi \ \oplus \ \bigoplus_{n=j}^{J-1} L_n\Gamma_{j:n}\xi. \end{aligned} \tag{5.164}$$

For $j = J$, the sum of perdidas is empty and $G_{J;J} = E_J^{\mathrm{term}}$. The Gram terminal is

$$U_{j;J} := G_{j;J}^* G_{j;J} \succeq 0. \tag{5.165}$$

Rearranging the orthogonal sum of the codomain yields the literal identity.

$$G_{j;J}\xi \simeq \big(L_j\xi,\ G_{j+1;J}\Gamma_j\xi\big). \tag{5.166}$$

Consequently,

$$\boxed{U_{j;J} = L_j^* L_j + \Gamma_j^* U_{j+1;J}\Gamma_j} \tag{5.167}$$

and

$$U_{j;J} - \Gamma_j^* U_{j+1;J}\Gamma_j = L_j^* L_j \succeq 0. \tag{5.168}$$

There is no sign mutation: the loss term is on the positive side of the identity because it arises from a sum of squares.

Iterating (5.167) yields the exact telescoping identity

$$U_{j;J} = \Gamma_{j:J}^{*}(E_J^{\mathrm{term}})^{*}E_J^{\mathrm{term}}\Gamma_{j:J} + \sum_{n=j}^{J-1}\Gamma_{j:n}^{*}L_n^{*}L_n\Gamma_{j:n}, \tag{5.169}$$

o, in quadratic form,

$$\langle U_{j;J}\xi,\xi\rangle = \|E_J^{\mathrm{term}}\Gamma_{j:J}\xi\|^2 + \sum_{n=j}^{J-1}\|L_n\Gamma_{j:n}\xi\|^2. \tag{5.170}$$

The first term preserves the complete terminal future; the sum preserves, level by level, all the losses and all the carries that led to it.

5.2.4 Orientation, carry, and simultaneous composition of paths

Path inversion acts only once on $\mathfrak{e}_\alpha$. Let $\mathsf{J}(v_\Sigma, v_\Pi) := (v_\Pi, v_\Sigma)$ be the leaf exchange, and let P_{tr} be the transposition of port matrices. Its five local expressions are

$$\begin{gathered}\mathsf{J}\sigma_k = -\sigma_k\mathsf{J}, \qquad b_{\bar\alpha} = -b_\alpha, \qquad (b^+_{\bar\alpha}, b^-_{\bar\alpha}) = (b^-_\alpha, b^+_\alpha),\\ \vartheta_L(K,\epsilon) := \begin{cases}(L,-\epsilon), & K = L,\\ (K,\epsilon), & K \neq L,\end{cases} \qquad \kappa_{J,I}(v,u) = -P_{\mathrm{tr}}\kappa_{I,J}(u,v),\\ \kappa(\bar\alpha) = -\mathsf{C}_\alpha^{-1}\kappa(\alpha), \qquad v_{\bar\alpha} = -v_\alpha.\end{gathered} \tag{5.171}$$

Here J exchanges the two sheets, P_{tr} transposes the incidence matrix, and ϑ_L exchanges exactly the two faces in the direction L, fixing the other six. In the free basis of $\mathscr{B} = \mathscr{I}\times\{+,-\}$, the eight vectors are distinct, $\vartheta_L^2 = I$ and $\vartheta_L \neq \vartheta_{L'}$ for $L \neq L'$. The same orientation explains all signs in (5.171).

For composable extensions $\alpha : x \to y$ and $\beta : y \to z$, the native carry satisfies

$$\kappa(\beta\alpha) = \kappa(\beta) + \mathsf{C}(\beta)\kappa(\alpha). \tag{5.172}$$

The matrix lift is not left as a name. In the ordered basis (f, r, e, b, g), let $\widetilde{L}_N(\alpha) \in \mathrm{Mat}_5(\mathbb{Z})$ be the centered integral representative of the matrix of Ψ_α of (5.176) modulo 3^N, and define

$$c_N(\beta,\alpha) := \frac{\widetilde{L}_N(\beta)\widetilde{L}_N(\alpha) - \widetilde{L}_N(\beta\alpha)}{3^N} \in \mathrm{Mat}_5(\mathbb{Z}). \tag{5.173}$$

Divisibility follows from equality of the two compositions modulo 3^N. Writing $L_N = [\widetilde{L}_N]_{3^N}$, the integer lifts satisfy

$$\widetilde{L}_N(\beta)\widetilde{L}_N(\alpha) = \widetilde{L}_N(\beta\alpha) + 3^N c_N(\beta,\alpha) \tag{5.174}$$

and associativity produces the cocycle

$$c_N(\delta,\beta\alpha) + \widetilde{L}_N(\delta)c_N(\beta,\alpha) = c_N(\delta\beta,\alpha) + c_N(\delta,\beta)\widetilde{L}_N(\alpha). \tag{5.175}$$

The same carry increment acts upon the local complex through

$$\begin{gathered}\Psi_\alpha(r_x) = r_y, \quad \Psi_\alpha(e_x) = e_y, \quad \Psi_\alpha(g_x) = g_y, \quad \Psi_\alpha(f_x) = f_y,\\ \Psi_\alpha(b_x) = b_y + 3(c_y - c_x)e_y.\end{gathered} \tag{5.176}$$

The oriented boundary change remains in

$$K_\alpha := (v_y - v_x) f_y, \qquad K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha. \tag{5.177}$$

The equations (5.172), (5.175), and (5.177) are three degrees of the same ledger of composition. None authorizes erasing the other two.

5.2.4.1 Nerve of extensions and Alexander–Whitney diagonal

For a history of the common ledger, write $A_n := \mathbf{Z}/3^n\mathbf{Z}$ and $\mathsf{T}_m(x)$ for the finite category of *all* its TPK prefixes of depth m, with the extension morphisms already constructed. The following diagonal and counit are applied to that same nerve; they do not define a fifth autonomous branch.

5.2.4.2 Nerve of routes, Alexander–Whitney diagonal and counit

The normalized complex is defined

$$G_{m,n}(x) := N_*^{\mathrm{norm}}(N\mathsf{T}_m(x); A_n)\,. \tag{5.178}$$

For a nondegenerate simplex $\sigma = [y_0 \to y_1 \to \cdots \to y_q]$, the diagonal is

$$\Delta_{\mathrm{AW}}(\sigma) = \sum_{j=0}^{q} [y_0 \to \cdots \to y_j] \otimes [y_j \to \cdots \to y_q]. \tag{5.179}$$

The counit is given by

$$\epsilon_G([y]) = 1, \qquad \epsilon_G(\sigma) = 0 \quad (q > 0). \tag{5.180}$$

In particular,

$$\Delta_{\mathrm{AW}}[y] = [y] \otimes [y], \qquad \epsilon_G([y]) = 1. \tag{5.181}$$

If $m' \geq m$ and $n' \geq n$, truncation and coefficient reduction induce

$$R^{n',n}_{m',m} : G_{m',n'}(x) \longrightarrow G_{m,n}(x), \tag{5.182}$$

and they are verified

$$R\partial = \partial R, \qquad (R \otimes R)\Delta_{\mathrm{AW}} = \Delta_{\mathrm{AW}} R, \qquad \epsilon_G R = \rho_{n',n}\epsilon_G. \tag{5.183}$$

The path is not reduced to its endpoints: the entire simplex remains in $G_{m,n}(x)$.

5.2.5 Cellular incidence and correlative filling of the same trajectory

The sequence (5.108) assigns to each prefix the two-term complex

$$C_{k,N}(\widehat{x}) := \left[A_N^{I_k} \oplus A_N^{J_k} \xrightarrow{\kappa_{k,N}} A_N^{I_k \times J_k} \right]. \tag{5.184}$$

A refinement acts on ports by truncating the extension and on coefficients by $A_{N'} \twoheadrightarrow A_N$. When $\alpha_u : I_{k'} \to I_k$ and $\beta_u : J_{k'} \to J_k$ are those maps, their action on generators is

$$(p, q) \longmapsto (p \circ \alpha_u, q \circ \beta_u), \qquad w \longmapsto w \circ (\alpha_u \times \beta_u), \tag{5.185}$$

and satisfies

$$\kappa_{k',N}\Gamma(u)_1 = \Gamma(u)_0 \kappa_{k,N}. \tag{5.186}$$

The incidence is not fabricated by labeling four copies of a neutral emission. It is generated by the free cellular completion of the module of the two sheets that the same state already bears. For

$$V_{\widehat{x}} := \mathbf{F}_3 e_{\Sigma,\widehat{x}} \oplus \mathbf{F}_3 e_{\Pi,\widehat{x}},$$

the four internal lines are

$$\mathscr{I}_{\widehat{x}} := \{L_0, L_\infty, L_+, L_-\} = \mathbf{P}^1(V_{\widehat{x}}), \qquad \begin{aligned} L_0 &= \langle e_\Sigma \rangle, & L_\infty &= \langle e_\Pi \rangle, \\ L_+ &= \langle e_\Sigma + e_\Pi \rangle, & L_- &= \langle e_\Sigma - e_\Pi \rangle. \end{aligned} \tag{5.187}$$

If v_L is, respectively, $(1,0)^t, (0,1)^t, (1,1)^t, (1,-1)^t$, the action of the direction L on the generators of sheet is the proyector

$$P_L := v_L (v_L^t v_L)^{-1} v_L^t, \tag{5.188}$$

that is, over $\mathbf{F}_3$,

$$P_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_\infty = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad P_+ = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \quad P_- = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Direct calculation gives

$$P_L^2 = P_L, \qquad \operatorname{rk} P_L = 1, \qquad \operatorname{Im} P_L = L, \qquad P_L = P_{L'} \Longrightarrow L = L'. \tag{5.189}$$

Therefore, the four directions are distinct outputs of the module $V_{\widehat{x}}$; they are not four names placed upon the same action.

Definition 5.33 (Cell-free leaf TPK completion)**.** Let $\mathrm{Cell}_{\mathrm{TPK}}(\widehat{x})$ be the free complex

$$C_2(\widehat{x}) \xrightarrow{\partial_2} C_1(\widehat{x}) \xrightarrow{\partial_1} C_0(\widehat{x})$$

with

$$\begin{aligned} C_0(\widehat{x}) &:= \mathbf{F}_3[v_\star] \oplus \bigoplus_{L \in \mathscr{I}} \mathbf{F}_3[v_L] \oplus \bigoplus_{(L,\epsilon) \in \mathscr{B}} \mathbf{F}_3[v_{L,\epsilon}] \oplus \bigoplus_{E \in \mathscr{E}} \mathbf{F}_3[v_E], \\ C_1(\widehat{x}) &:= \bigoplus_{L \in \mathscr{I}} \mathbf{F}_3[u_L] \oplus \bigoplus_{(L,\epsilon) \in \mathscr{B}} \mathbf{F}_3[u_{L,\epsilon}] \oplus \bigoplus_{E = \{L,L'\} \in \mathscr{E}} \big(\mathbf{F}_3[u_{L'|L}] \oplus \mathbf{F}_3[u_{L|L'}]\big), \\ C_2(\widehat{x}) &:= \bigoplus_{E \in \mathscr{E}} \mathbf{F}_3[\omega_E]. \end{aligned}$$

Its action on generators is

$$\partial u_L = v_L - v_\star, \qquad \partial u_{L'|L} = v_E - v_L, \tag{5.190}$$

$$\partial u_{L,\epsilon} = v_{L,\epsilon} - v_L, \qquad \partial \omega_{\{L,L'\}} = u_L + u_{L'|L} - u_{L'} - u_{L|L'}. \tag{5.191}$$

The eight generators $v_{L,\epsilon}$ materially realize the screen $\mathscr{B}$. The preceding involution extends to the complex by

$$\vartheta_L v_{K,\epsilon} := v_{\vartheta_L(K,\epsilon)}, \qquad \vartheta_L u_{K,\epsilon} := u_{\vartheta_L(K,\epsilon)}, \tag{5.192}$$

and fixes the other generators. Since the basis is free,

$$|\{v_{L,\epsilon} : (L,\epsilon) \in \mathscr{B}\}| = 8, \qquad \vartheta_L^2 = I, \qquad \vartheta_L \vartheta_{L'} = \vartheta_{L'} \vartheta_L. \tag{5.193}$$

Moreover, $\partial \vartheta_L = \vartheta_L \partial$, because both paths send $u_{L,\epsilon}$ to $v_{L,-\epsilon} - v_L$ and fix the remaining boundaries.

The two words

$$p_{L,L'} := u_{L'|L} u_L, \qquad p_{L',L} := u_{L|L'} u_{L'}$$

are different paths in the free complex. Their difference is the boundary of $\omega_{\{L,L'\}}$, and

$$\partial^2 \omega_E = (v_L - v_\star) + (v_E - v_L) - (v_{L'} - v_\star) - (v_E - v_{L'}) = 0. \tag{5.194}$$

The representation of TPK on the leaves is fixed by

$$\varrho(u_L) = P_L, \qquad \varrho(u_{L'|L}) = P_{L'}, \qquad \varrho(p_{L,L'}) = P_{L'}P_L.$$

Even if two published matrices coincided in a specialization, the two paths do not collapse: they remain distinct free generators of $C_1(\widehat{x})$, joined by an explicit 2–cell.

The complete incidence coordinate is the group of 2–cochains.

$$Q_{\mathrm{inc}}(\widehat{x}) := C^2(\mathrm{Cell}_{\mathrm{TPK}}(\widehat{x}); \mathbf{F}_3) = \bigoplus_{E \in \mathscr{E}} \mathbf{F}_3 \omega_E^* \simeq \mathbf{F}_3^6. \tag{5.195}$$

The internal operation returns all of $Q_{\mathrm{inc}}(\widehat{x})$, not a selected element. Each q has a unique expansion $q = \sum_E q_E \omega_E^*$. For $a \in \mathbf{F}_3^{\mathscr{I}}$, its change of potential is

$$q \longmapsto q + Ba, \qquad (Ba)_{\{L,L'\}} = a_L + a_{L'}. \tag{5.196}$$

As each line belongs to three pairs,

$$s(Ba) = 3 \sum_{L \in \mathscr{I}} a_L = 0,$$

and therefore

$$W_c(q + Ba) = W_c(q) \qquad (q \in Q_{\mathrm{inc}}(\widehat{x})). \tag{5.197}$$

The unit of the completion only anchors the complex in the state:

$$\eta^{\mathrm{cell}}_{\widehat{x}} : V_{\widehat{x}} \longrightarrow V_{\widehat{x}} \otimes C_0(\mathrm{Cell}_{\mathrm{TPK}}(\widehat{x})), \qquad e_{\diamond,\widehat{x}} \longmapsto e_{\diamond,\widehat{x}} \otimes v_\star, \tag{5.198}$$

and marks the origin $q = 0$; it does not assert that the orbit of that origin is all of Q_{inc}.

The construction is functorial under common truncation. If $\tau_{\ell,k}\widehat{x}_\ell = \widehat{x}_k$, we define

$$\begin{aligned}
&\tau_V : V_{\widehat{x}_\ell} \longrightarrow V_{\widehat{x}_k}, && \tau_V e_{\diamond,\widehat{x}_\ell} = e_{\diamond,\widehat{x}_k},\\
&\tau_a : \mathbf{F}_3^{\mathscr{I}_{\widehat{x}_\ell}} \longrightarrow \mathbf{F}_3^{\mathscr{I}_{\widehat{x}_k}}, && (\tau_a a)_{\tau_{\mathscr{I}} L} = a_L,\\
&\tau_Q : Q_{\mathrm{inc}}(\widehat{x}_\ell) \longrightarrow Q_{\mathrm{inc}}(\widehat{x}_k), && (\tau_Q q)_{\tau_{\mathscr{E}} E} = q_E,
\end{aligned}$$

where $\tau_{\mathscr{I}}[v] = [\tau_V v]$ and $\tau_{\mathscr{E}}\{L, L'\} = \{\tau_{\mathscr{I}} L, \tau_{\mathscr{I}} L'\}$. The map of cadenas $\mathrm{Cell}(\tau) : C_\bullet(\widehat{x}_\ell) \to C_\bullet(\widehat{x}_k)$ acts in generators by

$$v_\bullet(\widehat{x}_\ell) \mapsto v_\bullet(\widehat{x}_k), \quad u_\bullet(\widehat{x}_\ell) \mapsto u_\bullet(\widehat{x}_k), \quad \omega_E(\widehat{x}_\ell) \mapsto \omega_E(\widehat{x}_k). \tag{5.199}$$

From (5.190)– (5.191) one obtains

$$\begin{aligned}
\mathrm{Cell}(\tau)\partial &= \partial\, \mathrm{Cell}(\tau),\\
\tau_V P_L &= P_{\tau_{\mathscr{I}} L} \tau_V, \qquad \tau_Q B = B \tau_a, \qquad W_c \circ \tau_Q = W_c,\\
\mathrm{Cell}(\tau)\vartheta_L &= \vartheta_{\tau_{\mathscr{I}} L}\, \mathrm{Cell}(\tau),\\
(\tau_V \otimes \mathrm{Cell}_0(\tau))\eta^{\mathrm{cell}}_{\widehat{x}_\ell} &= \eta^{\mathrm{cell}}_{\widehat{x}_k} \tau_V.
\end{aligned} \tag{5.200}$$

Identity and composition are verified on each generator. Thus $\mathrm{Cell}_{\mathrm{TPK}}$ is the functorial cellular completion of the same APP–TRIT–TPK fiber, not a carrier added from a subsequent publication.

5.2.5.1 Incidence measure, carry curvature, and adjoint naturality

In the following development, they are identified exclusively as abbreviations of the common completion,

$$I_{\mathrm{gau}} := \mathscr{I}_{\widehat{x}}, \qquad \mathscr{B}_{\mathrm{gau}} := \mathscr{B}_{\widehat{x}}, \qquad Q_{\mathrm{gau}} := Q_{\mathrm{inc}}(\widehat{x}).$$

With the uniform counting average on Q_{gau}, the function W_c of (5.117) satisfies

$$\mathbb{E}_Q W_c = 0, \qquad \mathbb{E}_Q W_c^2 = \frac{2}{9}, \qquad \mathbb{E}_Q W_c^3 = \frac{2}{27}. \tag{5.201}$$

These identities are properties of the same cell and the same carry, not data from an external theory.

5.2.5.2 Incidence, carry curvature, and common domain of naturalness

For $x \in C_{\mathrm{cont}}^{\mathrm{disc}}$, let us denote by $Y_m(x)$ the finite set of gauge prefixes of depth m that simultaneously conserve

fiber, extension relations, numbers, orientation, carry, memory,

boundary, route, multisection, survival, terminal and reader.

There exists a collar neighborhood application.

$$q_m : Y_m(x) \longrightarrow Q_{\mathrm{gau}} \tag{5.202}$$

which records the six incidences without erasing the history that produces them. For $\ell \geq m$, the truncation is written

$$\tau_{\ell,m} : Y_\ell(x) \longrightarrow Y_m(x).$$

For $y, z \in Y_m(x)$, let

$$\mathrm{Ext}_m^{\mathrm{adm}}(y, z) \tag{5.203}$$

the set of *all* admissible TPK extensions from y to z. An extension belongs to this set only if it transports the entire fiber, the relations Ext_Γ, the native and operatorial carry, the orientation, the accumulated memory, the boundary, the route, and the survival condition. We define

$$a_m(y, z) := \#\,\mathrm{Ext}_m^{\mathrm{adm}}(y, z), \qquad d_m(y) := \sum_{z \in Y_m(x)} a_m(y, z), \tag{5.204}$$

and, when $d_m(y) > 0$,

$$k_m(y, z) := \frac{a_m(y, z)}{d_m(y)}. \tag{5.205}$$

Thus, k_m is a normalized count of complete extensions, not a function introduced after projecting the state.

The restriction $C_{\mathrm{cont}}^{\mathrm{gau}}$ is formed by the surviving histories x for which the following conditions hold at every depth.

1. The sets $Y_m(x)$ are nonempty, $d_m(y) > 0$, and the truncations $\tau_{\ell,m}$ are surjective.
2. There exists a nonempty multisection of strictly positive rational weights $\mu_m : Y_m(x) \to \mathbf{Q}_{>0}$ such that
$$\sum_{y \in Y_m} \mu_m(y) = 1, \qquad \mu_m(y) = \sum_{\tau_{\ell,m}\widetilde{y} = y} \mu_\ell(\widetilde{y}). \tag{5.206}$$
3. Each μ_m is stationary:
$$\sum_{y \in Y_m} \mu_m(y) k_m(y, z) = \mu_m(z). \tag{5.207}$$

4. The marginal of $(q_m)_*\mu_m$ is the uniform counting measure on Q_{gau}. This condition makes the calculations of (5.201) literally applicable to the lift $W_c \circ q_m$.

5. Direct lumpability holds
$$\sum_{\tau_{\ell,m}\widetilde{z}=z} k_\ell(\widetilde{y},\widetilde{z}) = k_m(\tau_{\ell,m}\widetilde{y}, z) \tag{5.208}$$
for every $\widetilde{y} \in Y_\ell$ and $z \in Y_m$.

6. For the adjoint kernel
$$k_m^*(z,y) := \frac{\mu_m(y)k_m(y,z)}{\mu_m(z)}, \tag{5.209}$$
the adjoint lumpability condition is satisfied
$$\sum_{\tau_{\ell,m}\widetilde{y}=y} k_\ell^*(\widetilde{z},\widetilde{y}) = k_m^*(\tau_{\ell,m}\widetilde{z}, y). \tag{5.210}$$

7. The categories of prefixes possess the TPK root as an initial object, so that their nerve has the explicit terminal contraction of (5.228) below.

These conditions define the typed domain; outside it, a measure or natural operator is not published by mere declaration.

5.2.5.3 Projective measure, kernel, and direct and adjoint naturality

Having fixed a branch of the multisection $\{\mu_m\}_m$, let
$$H_m := \mathrm{Fun}(Y_m, \mathbf{Q}), \qquad \langle f, g\rangle_m := \sum_{y\in Y_m} \mu_m(y)f(y)g(y). \tag{5.211}$$

The operator generated by all extensions is
$$(K_m f)(y) := \sum_{z\in Y_m} k_m(y,z)f(z). \tag{5.212}$$

The Markov equations and stationarity give
$$K_m\mathbf{1} = \mathbf{1}, \qquad K_m^*\mathbf{1} = \mathbf{1}, \qquad \|K_m f\|_m \le \|f\|_m. \tag{5.213}$$

The lift and the truncated expectation are
$$J_{\ell,m}f := f \circ \tau_{\ell,m}, \qquad E_{m,\ell} := J_{\ell,m}^*. \tag{5.214}$$

The projectivity of μ implies
$$E_{m,\ell}J_{\ell,m} = I_{H_m}. \tag{5.215}$$

The two lumpability conditions discharge, respectively,
$$K_\ell J_{\ell,m} = J_{\ell,m}K_m, \qquad K_\ell^* J_{\ell,m} = J_{\ell,m}K_m^*. \tag{5.216}$$

The second equality is not inferred from the first without the attached condition: both remain registered in the domain.

The positive internal operator is defined
$$T_m := K_m^* K_m. \tag{5.217}$$

Then

$$\langle f, T_m f\rangle_m = \|K_m f\|_m^2 \geq 0, \qquad T_\ell J_{\ell,m} = J_{\ell,m} T_m. \tag{5.218}$$

5.2.5.4 Common law of 8 *faces and* 4 *Fubini squares*

On $Y_m^{\mathscr{B}_{\mathrm{gau}}}$, we define the rational mass

$$\Omega_m^{(8)}(\mathbf{y}) := \sum_{z\in Y_m} \mu_m(z) \prod_{(L,\varepsilon)\in\mathscr{B}_{\mathrm{gau}}} k_m(z, y_{L,\varepsilon}). \tag{5.219}$$

It is a probability because each factor is Markov. All faces arise from the same latent state z and the same kernel; they are not eight independently added measures.

For functions $f_{L,+}, f_{L,-} \in H_m$, finite Fubini gives

$$\begin{aligned}\sum_{\mathbf{y}} \Omega_m^{(8)}(\mathbf{y}) \prod_{L\in I_{\mathrm{gau}}} f_{L,+}(y_{L,+}) f_{L,-}(y_{L,-}) \\ = \sum_{z\in Y_m} \mu_m(z) \prod_{L\in I_{\mathrm{gau}}} (K_m f_{L,+})(z)(K_m f_{L,-})(z).\end{aligned} \tag{5.220}$$

Upon taking $f_{L,+} = f_{L,-} = f_L$, the four squares appear simultaneously

$$\sum_{z\in Y_m} \mu_m(z) \prod_{L\in I_{\mathrm{gau}}} (K_m f_L(z))^2. \tag{5.221}$$

The marginal of any opposite pair satisfies

$$\sum_{y_+,y_-} \Omega_{m,L}^{(2)}(y_+, y_-) f(y_+) g(y_-) = \langle K_m f, K_m g\rangle_m = \langle f, T_m g\rangle_m. \tag{5.222}$$

The commutativity of the product in (5.219) proves

$$(\vartheta_L)_* \Omega_m^{(8)} = \Omega_m^{(8)} \qquad (L \in I_{\mathrm{gau}}). \tag{5.223}$$

5.2.5.5 Exact internal operator and self-test witness

The projection onto the constant sector and its complement are

$$P_{\Omega,m} f := \langle \mathbf{1}, f\rangle_m \mathbf{1}, \qquad Q_m := I - P_{\Omega,m}. \tag{5.224}$$

The internal cell operator is defined by

$$D_m^{\mathrm{HMT}} := 3 Q_m. \tag{5.225}$$

The coefficient 3 is the cardinality of the residue field used in (5.188); it does not represent an added scale. For the lift

$$W_{c,m} := W_c \circ q_m$$

one has, by (5.201),

$$P_{\Omega,m} W_{c,m} = 0, \qquad D_m^{\mathrm{HMT}} W_{c,m} = 3 W_{c,m}, \qquad \|W_{c,m}\|_m^2 = \frac{2}{9}. \tag{5.226}$$

Thus, the complementary sector possesses a nonzero witness constructed by the incidence of the six edges.

5.2.5.6 Cellular terminal and strong retractor

Let $C_*(N\mathsf{Y}_m; A_n)$ be the normalized chain complex of the nerve of the prefix category, and let y_* be its initial root. The inclusion and the augmentation are

$$\iota_m : A_n[0] \longrightarrow C_*(N\mathsf{Y}_m; A_n), \quad \iota_m(1) = [y_*], \qquad \epsilon_m : C_*(N\mathsf{Y}_m; A_n) \longrightarrow A_n[0].$$

The initial object provides the extra degeneracy

$$H_m[y_0 \to \cdots \to y_q] := [y_* \to y_0 \to \cdots \to y_q], \tag{5.227}$$

with the normalized convention when a degeneration appears. It is verified

$$\epsilon_m \iota_m = I, \qquad \partial H_m + H_m \partial = I - \iota_m \epsilon_m, \qquad H_m^2 = H_m \iota_m = \epsilon_m H_m = 0. \tag{5.228}$$

This cellular terminal is distinct from the constant projection $P_{\Omega,m}$: the former contracts the nerve, while the latter separates the sectors of the function space. Both are preserved.

The same rule defines its action on an elementary emission $\alpha : x \to y$. Since U_α transports the ordered pair of sheets, its label map is $\overline{U}_\alpha e_{\Sigma,x} = e_{\Sigma,y}$ and $\overline{U}_\alpha e_{\Pi,x} = e_{\Pi,y}$. Its projectivization $\alpha_\mathscr{f}[v] := [\overline{U}_\alpha v]$ acts in $\mathbf{P}^1(V_x)$ and, therefore,

$$\begin{gathered}
\mathrm{Cell}_{\mathrm{TPK}}(\alpha) v_\star(x) = v_\star(y), \qquad \mathrm{Cell}_{\mathrm{TPK}}(\alpha) v_L(x) = v_{\alpha_\mathscr{f} L}(y), \\
\mathrm{Cell}_{\mathrm{TPK}}(\alpha) v_{\{L,L'\}}(x) = v_{\{\alpha_\mathscr{f} L, \alpha_\mathscr{f} L'\}}(y), \qquad \mathrm{Cell}_{\mathrm{TPK}}(\alpha) v_{L,\epsilon}(x) = v_{\alpha_\mathscr{f} L,\epsilon}(y), \\
\mathrm{Cell}_{\mathrm{TPK}}(\alpha) u_L(x) = u_{\alpha_\mathscr{f} L}(y), \qquad \mathrm{Cell}_{\mathrm{TPK}}(\alpha) u_{L,\epsilon}(x) = u_{\alpha_\mathscr{f} L,\epsilon}(y), \\
\mathrm{Cell}_{\mathrm{TPK}}(\alpha) u_{L'|L}(x) = u_{\alpha_\mathscr{f} L'|\alpha_\mathscr{f} L}(y), \qquad \mathrm{Cell}_{\mathrm{TPK}}(\alpha) \omega_{\{L,L'\}}(x) = \omega_{\{\alpha_\mathscr{f} L, \alpha_\mathscr{f} L'\}}(y).
\end{gathered} \tag{5.229}$$

In the leaf fiber and on its screen, one also has

$$\overline{U}_\alpha P_L = P_{\alpha_\mathscr{f} L} \overline{U}_\alpha, \qquad \mathrm{Cell}_{\mathrm{TPK}}(\alpha) \vartheta_L = \vartheta_{\alpha_\mathscr{f} L} \mathrm{Cell}_{\mathrm{TPK}}(\alpha). \tag{5.230}$$

Boundary formulas give

$$\partial \, \mathrm{Cell}_{\mathrm{TPK}}(\alpha) = \mathrm{Cell}_{\mathrm{TPK}}(\alpha) \partial, \qquad \mathrm{Cell}_{\mathrm{TPK}}(\beta\alpha) = \mathrm{Cell}_{\mathrm{TPK}}(\beta) \, \mathrm{Cell}_{\mathrm{TPK}}(\alpha). \tag{5.231}$$

Thus the map of complexes used in the terminal register is the free action induced by the same emission, not an added symbol.

5.2.5.7 Module of paths, totalization and nonadic return cone

Let $J_d(x)$ be the full category of depth-d prefixes of the common tree, $\mathrm{Term}_d(x)$ its terminal set, and $\Gamma_{d,N}(\nu) := C_{d,N}(\nu)$ the cellular complex of (5.184). The module, totalization, and cone that follow simultaneously preserve all continuations; they do not select a history or separate a cohomological construction from the common continuum.

5.2.5.8 Module of paths, cellular totalization and nonadic terminal

Let $\mathrm{Term}_d(x)$ be the conjunto of terminal of memory of the prefix. For $\nu \in J_d(x)$, definase

$$W_{d,N}(\nu) := A_N \big[\mathrm{Path}_{J_d(x)}(\nu, \mathrm{Term}_d(x)) \big]. \tag{5.232}$$

It is the free module on all terminal routes; it selects none. If $u : \nu \to \nu'$, the contravariant action is

$$W_{d,N}(u) : W_{d,N}(\nu') \longrightarrow W_{d,N}(\nu), \qquad [\lambda] \longmapsto [\lambda \circ u]. \tag{5.233}$$

Each generator preserves orientation, carry, boundary, route, and memory.

5.2.5.9 Bilateral bar complex and totalization of the bicolumn

For $q \geq 0$ and $r \in \{0, 1\}$, let

$$B^{d,N}_{q,r}(x) := \bigoplus_{\nu_0 \xrightarrow{u_1} \cdots \xrightarrow{u_q} \nu_q} W_{d,N}(\nu_q) \otimes_{A_N} \Gamma_{d,N}(\nu_0)_r. \tag{5.234}$$

The bar differential is defined without abbreviation by

$$\begin{aligned} d_{\mathrm{bar}}(w; u_1, \ldots, u_q; c) := {} & (w; u_2, \ldots, u_q; \Gamma(u_1)c) \\ & + \sum_{i=1}^{q-1} (-1)^i (w; u_1, \ldots, u_{i+1}u_i, \ldots, u_q; c) \\ & + (-1)^q (W(u_q)w; u_1, \ldots, u_{q-1}; c). \end{aligned} \tag{5.235}$$

For $q = 1$, this is exactly the relation

$$d_{\mathrm{bar}}(w; u; c) = w \otimes \Gamma(u)c - W(u)w \otimes c. \tag{5.236}$$

A transported incidence is thereby identified with the same incidence written in the refined chart; the relation is not concealed beneath the name of the totalizer.

The total complex is

$$K_{d,N}(x) := \mathrm{Tot}\, B^{d,N}_{\bullet,\bullet}(x), \qquad D_{\mathrm{tot}} := d_{\mathrm{bar}} + (-1)^q \partial_\Gamma \quad \text{on } B^{d,N}_{q,r}. \tag{5.237}$$

The functoriality of W and Γ, together with $\partial_\Gamma^2 = 0$, gives

$$D^2_{\mathrm{tot}} = 0. \tag{5.238}$$

In compact notation, but now already discharged by (5.234)– (5.235),

$$K_{d,N}(x) \simeq W_{d,N} \otimes^{\mathbf{L}}_{A_N[J_d(x)]} \Gamma_{d,N}. \tag{5.239}$$

5.2.5.10 Depth and terminal nonadic colimit

The prefixes form the colimit.

$$K_{\infty,N}(x) := \varinjlim_{d} K_{d,N}(x). \tag{5.240}$$

The nine-phase extension carries depth d to depth $d + 9$. Only after forming (5.240) does it induce a well-typed endomorphism

$$U_{\Gamma_9,N} : K_{\infty,N}(x) \longrightarrow K_{\infty,N}(x). \tag{5.241}$$

The terminal is

$$\mathcal{T}_{\mathrm{coh},N}(x) := \mathrm{Cone}\big(I - U_{\Gamma_9,N}\big). \tag{5.242}$$

With the homological convention

$$\mathrm{Cone}(f)_q = K_{\infty,N,q-1} \oplus K_{\infty,N,q},$$

its differential is

$$\partial_{\mathrm{Cone}}(a, b) = \big(-\partial_K a, \partial_K b + f(a)\big). \tag{5.243}$$

Although the visible phase of h and $\Gamma_9 h$ coincides, the generators $[h]$ and $[\Gamma_9 h]$ are different because their ledgers and depths are different. The terminal object preserves

$$(I - U_{\Gamma_9,N})[h] = [h] - [\Gamma_9 h]. \tag{5.244}$$

Precision reductions commute with the terminal:

$$r_{N+1,N} U_{\Gamma_9,N+1} = U_{\Gamma_9,N} r_{N+1,N}. \tag{5.245}$$

5.2.5.11 Terminal cohomological reader: incidence cycle and class in $H_0(\mathrm{Cone}(I - U_{\Gamma_9,N}))$

Each cell of $x_{\le d}$ carries a vector of incidence

$$\omega_{\nu,N}(x) \in A_N^{I_\nu \times J_\nu} = C_{\nu,N,0}$$

and the complete terminal route $\lambda_{\nu,x}$. The raw cycle is

$$z_{d,N}(x) := \sum_{\nu \in \mathrm{Cell}_d(x)} \epsilon_\nu [\lambda_{\nu,x}] \otimes \omega_{\nu,N}(x) \in B_{0,0}^{d,N}(x). \tag{5.246}$$

As the total complex is homologically non-negative,

$$D_{\mathrm{tot}} z_{d,N}(x) = 0. \tag{5.247}$$

Both the cycle and its class are preserved.

$$[z_{d,N}(x)] \in H_0(K_{d,N}(x)).$$

After including it in $K_{\infty,N}$, its terminal class is

$$[(0, z_{d,N}(x))] \in H_0(\mathcal{T}_{\mathrm{coh},N}(x)).$$

The complete reader is

$$\mathcal{L}_{\mathrm{coh},d,N}(x) := \big(z_{d,N}(x), [z_{d,N}(x)], U_{\Gamma_9,N} z_{d,N}(x), [(0, z_{d,N}(x))]\big). \tag{5.248}$$

The publication is not merely a quotient class: the representative, the route that produced it, its continuation and the terminal remain.

5.2.5.12 Surviving fibres and corrections without a selector

Let $\mathcal{S}_\lambda^{\mathrm{coh}}$ be the free A_N-module generated by the admissible finite histories at level $\lambda = (d, N)$. The reader extends linearly to

$$e_\lambda : \mathcal{S}_\lambda^{\mathrm{coh}} \longrightarrow \mathcal{O}_\lambda^{\mathrm{coh}}, \qquad \mathcal{O}_\lambda^{\mathrm{coh}} := H_0(K_{d,N}) \oplus H_0(\mathcal{T}_{\mathrm{coh},N}). \tag{5.249}$$

The surviving codomain is

$$\mathcal{O}_\infty^{\mathrm{surv}} := \mathrm{Im}\left(e_\infty : C_{\mathrm{cont}}^{\mathrm{coh}} \longrightarrow \varprojlim_\lambda \mathcal{O}_\lambda^{\mathrm{coh}} \right). \tag{5.250}$$

For $a = (a_\lambda)_\lambda \in \mathcal{O}_\infty^{\mathrm{surv}}$, define

$$\mathcal{F}_\lambda(a) := \left\{ s_\lambda \in \mathcal{S}_\lambda^{\mathrm{coh}} : \begin{array}{l} e_\lambda(s_\lambda) = a_\lambda, \\ \exists y \in C_{\mathrm{cont}}^{\mathrm{coh}} \text{ with } y|_\lambda = s_\lambda \text{ and } e_\infty(y) = a \end{array} \right\}. \tag{5.251}$$

These fibers are nonempty by the definition of the surviving codomain. If $\lambda' \ge \lambda$, the restriction is surjective:

$$\rho_{\lambda',\lambda} : \mathcal{F}_{\lambda'}(a) \twoheadrightarrow \mathcal{F}_\lambda(a). \tag{5.252}$$

Indeed, a complete history that witnesses $s_\lambda \in \mathscr{F}_\lambda(a)$ provides by truncation an antecedent at any later level. That antecedent is not chosen as part of the object.

The complete multisect is

$$\Sigma^{\mathrm{coh}}_{\lambda',\lambda}(s_\lambda; a) := \{s_{\lambda'} \in \mathscr{F}_{\lambda'}(a) : \rho_{\lambda',\lambda} s_{\lambda'} = s_\lambda\}. \tag{5.253}$$

For a provisional prolongation $\widetilde{s}_{\lambda'}$, let

$$\delta_{\lambda'} := a_{\lambda'} - e_{\lambda'}(\widetilde{s}_{\lambda'}). \tag{5.254}$$

The relation of all corrections is

$$\mathrm{Corr}_{\lambda',\lambda}(\widetilde{s}_{\lambda'}, \delta_{\lambda'}) := \left\{ \eta \in \mathcal{S}^{\mathrm{coh}}_{\lambda'} : \begin{array}{l} \rho_{\lambda',\lambda}\eta = 0, \\ e_{\lambda'}(\eta) = \delta_{\lambda'}, \\ \eta \text{ is supported on new cells}, \\ \eta \text{ preserves orientation, carry, memory and terminal} \end{array} \right\}. \tag{5.255}$$

If two lifts are comparable, their difference $\eta = s_{\lambda'} - \widetilde{s}_{\lambda'}$ belongs to this relation exactly when it satisfies the four preceding conditions. It is not asserted that (5.255) is nonempty for an arbitrary discrepancy: the governing non-emptiness is that of (5.253) on the surviving domain.

The local complex (5.110) and the vertex y_α, endpoint of that same extension in the nerve, form the augmented pullback. In it are defined

$$\mathbf{1}_\alpha = (r, [y_\alpha]), \qquad z_-(\alpha) = (r, [y_\alpha]), \tag{5.256}$$

$$z_+(\alpha) = (r + 3c_\alpha v_\alpha e + v_\alpha b, [y_\alpha]), \qquad h_*(\alpha) = (-v_\alpha f, 0). \tag{5.257}$$

The action on generators gives, without an additional abstract existence,

$$\partial z_- = \partial z_+ = 0, \qquad \partial h_* = z_- - z_+. \tag{5.258}$$

The correlative filling, the cell, and the collar preserve the same $(\alpha, \mathrm{or}(\alpha), \kappa(\alpha), \partial\alpha, \gamma\alpha, m(\gamma\alpha; m_0))$; they merely apply different internal readers to the common record.

For the same finite based complex, the total determinant line is defined, without requiring balance or acyclicity.

$$\mathrm{Det}(C_{k,N}) := \bigotimes_q \left(\bigwedge^{\mathrm{top}} C_{k,N,q} \right)^{(-1)^q}. \tag{5.259}$$

Refinement induces the determinantal morphism of the map of complexes and simultaneously transports homology modules, Smith normal forms, and Bockstein pages as fields of the register. Neither the existence of (5.259) nor its transport requires homology to vanish. If the cone of refinement is contractible and oriented, the morphism is canonically trivialized; in the general case the cone is preserved and not replaced by a fictitious isomorphism.

5.2.5.13 Smith Normal Form of the Terminal Operator and Primary Decomposition of Its Cokernel

For each prefix y, the unoriented carry and the oriented coefficient of boundary of the common register admit the lifts

$$\widetilde{c}_y, \widetilde{v}_y \in \mathbf{Z}, \qquad c_{y,n} := \widetilde{c}_y \bmod 3^n, \qquad v_{y,n} := \widetilde{v}_y \bmod 3^n. \tag{5.260}$$

The determinantal output is neither an input nor a separate branch: it is the sector produced by the same terminal reduction when the typed conditions are verified

$$
\begin{aligned}
C_{\mathrm{cont}}^{\mathrm{det}} := \big\{ & x \in C_{\mathrm{cont}}^{\mathrm{disc}} : x \text{ survives at every depth;} \\
& \widehat{P}_x^{\mathrm{red}} \text{ is finite and based;} \\
& \chi(\widehat{P}_x^{\mathrm{red}}) = 0; \\
& H_*(\widehat{P}_x^{\mathrm{red}} \otimes_{\mathbf{Z}_3} K_3) = 0; \\
& \text{the reduction produces a square block stable under refinement} \big\}.
\end{aligned}
\tag{5.261}
$$

5.2.5.14 Lifting of the cokernel through the Bockstein tower and preservation of carry

Only the common root is eliminated:

$$
\overline{P}_{y;m,n} := P_{y;m,n}^{\mathrm{gen}} / A_n \mathbf{1}_y, \qquad \widehat{P}_{y;m}^{\mathrm{red}} := \varprojlim_n \overline{P}_{y;m,n}. \tag{5.262}
$$

The base is first ordered by the TPK order of routes and then by

$$
f \prec e \prec b \prec g. \tag{5.263}
$$

The following internal algorithm is executed.

1. Find the coefficients that are units of $\mathbf{Z}_3$.
2. Choose the first according to (5.263) and the order of paths.
3. Cancel the corresponding pair and record the elementary operations, their sign, and their contribution to the orientation.
4. Repeat until no cancellable unit pivot remains.

The survival condition of (5.261) requires that, cofinally in m, the result be a two-term block

$$
\mathbf{Z}_3^{r_x} \xrightarrow{S_x} \mathbf{Z}_3^{r_x}, \qquad \det S_x \neq 0 \text{ in } K_3, \tag{5.264}
$$

and that a refinement only add unit contractible blocks and recorded changes of basis. If these conditions fail, the history does not belong to $C_{\mathrm{cont}}^{\mathrm{det}}$; a nonexistent terminal is not forced.

The identity route provides an elementary witness. After removing the root, the local complex is

$$
A_n f \xrightarrow{\binom{3c}{1}} A_n e \oplus A_n b \xrightarrow{(1,-3c)} A_n g. \tag{5.265}
$$

Is contractible via

$$
s(g) = e, \qquad s(ae + \beta b) = \beta f, \qquad s(f) = 0. \tag{5.266}
$$

Thus, the terminal conditions admit at least the internal identity witness and do not describe a formally empty subdomain.

5.2.5.15 Smith, 3-adic order and initial unit

In oriented bases, a valuation diagonalization has the form

$$
U_x S_x V_x = \mathrm{diag}(3^{a_1} u_1, \dots, 3^{a_r} u_r), \qquad a_i \in \mathbf{N}, quad u_i \in \mathbf{Z}_3^{\times}. \tag{5.267}
$$

We define

$$
d_x := v_3(\det S_x) = \sum_{i=1}^{r} a_i \tag{5.268}
$$

and the initial unit

$$
\lambda_x := 3^{-d_x} \det S_x \in \mathbf{Z}_3^{\times}. \tag{5.269}
$$

The orientation coordinate trivializes the determinant line. If a transition produces

$$S_x \oplus I_q = U S_{x'} V, \tag{5.270}$$

the orientation is transported by the inverse factor of $\det U \det V$. The oriented unit λ_x^{or} is then natural and does not change when contractible blocks are added.

At finite precision,

$$S_{x,n} = S_x \bmod 3^n, \qquad a_i^{(n)} = \min(a_i, n). \tag{5.271}$$

The terminal is, therefore, the compatible family of all precisions and not an isolated integer.

5.2.5.16 Bockstein Tower and Higher Order Carry

Let

$$C_x^1 \xrightarrow{S_x} C_x^0$$

the terminal block. If a class $[\bar{y}]$ admits a lift $y \in C_x^1$ with $S_x y \in 3^q C_x^0$, we define

$$\beta_q[\bar{y}] := [3^{-q} S_x y \bmod 3]\,. \tag{5.272}$$

Changing the lift alters the representative by a boundary of the corresponding page, so β_q is well defined. The higher carry is

$$\mathrm{Carr}_q(y) := 3^{-q} S_x y. \tag{5.273}$$

It does not replace the TPK carry of (5.260); it records its prolongation in the coefficient tower.

For a block $3^{a_i} u_i$, the class persists until page a_i, and on that page the nonzero differential is multiplication by $\bar{u}_i \in \mathbf{F}_3^\times$. Consequently,

$$\{\text{survival lengths}\} = \{a_1, \dots, a_r\}, \qquad d_x = \sum_i a_i. \tag{5.274}$$

The oriented product of the page trivializations satisfies

$$\Theta_{\mathrm{Bock}}(x) = \overline{\lambda_x^{\mathrm{or}}} \in \mathbf{F}_3^\times. \tag{5.275}$$

The proof is carried out in each diagonal block and transported by the recorded unimodular changes. Smith, Bockstein and the initial unit are thus three readings of the same terminal.

5.2.6 Naturalness of the ensemble and compatibility under refinement

Before forming the horizon record, we make explicit the total action of the five operations developed here on *the same* edge $\mathfrak{e}_\alpha$. If P_x, Q_x denote the restrictions of (5.124) to the fiber of $x = s(\alpha)$, and analogously in $y = t(\alpha)$, we first set

$$A_\alpha := P_y U_\alpha P_x + Q_y U_\alpha Q_x, \qquad \eta_\alpha := Q_y U_\alpha P_x + P_y U_\alpha Q_x. \tag{5.276}$$

The five readings are dependent families indexed by the $\mathfrak{e}_\alpha$ itself. The index fixes source, target, route, and ledger, but is not copied again within the value. Their types are the dependent records $\mathrm{LedgerEntry} := \mathrm{cod}(\lambda)$ and

$$\begin{aligned}
\mathsf{Y}_{\mathrm{sp}}[\alpha] &:= \mathrm{Rec}[U : \mathsf{H}_x \to \mathsf{H}_y; \sigma_x \in \mathrm{End}\,\mathsf{H}_x; \sigma_y \in \mathrm{End}\,\mathsf{H}_y; A, \eta : \mathsf{H}_x \to \mathsf{H}_y],\\
\mathsf{Y}_{\mathrm{sol}}[\alpha] &:= \mathrm{Rec}[b, b^+, b^- \in \mathrm{LedgerEntry}; \mathrm{upd} : \mathsf{Mem}_x \to \mathsf{Mem}_y],\\
\mathsf{Y}_{\mathrm{gau}}[\alpha] &:= \mathcal{P}_{\mathrm{fin}}\,\mathrm{Rec}[\mathrm{Cell}(\alpha); q \in Q_{\mathrm{inc}}(y); B; W_c(q)],\\
\mathsf{Y}_{\mathrm{coh}}[\alpha] &:= \mathrm{Rec}[I_y, J_y, \kappa_{y,N}, \delta_y, C_{y,N}; F_\alpha : C_{y,N} \to C_{x,N}],\\
\mathsf{Y}_{\mathrm{det}}[\alpha] &:= \mathrm{Rec}[\Psi_\alpha, K_\alpha, \mathbf{1}_\alpha, z_\pm, h_*; \mathrm{Det}, H_*, \mathrm{Smith}, \mathrm{Boc}].
\end{aligned}$$

Therefore, the common judgment is

$$\mathfrak{e}_\alpha : \mathrm{Ext}_k \ \vdash\ O_\bullet(\mathfrak{e}_\alpha) \in \mathsf{Y}_\bullet[\alpha], \qquad \bullet \in \{\mathrm{sp}, \mathrm{sol}, \mathrm{gau}, \mathrm{coh}, \mathrm{det}\}.$$

$$O_{\mathrm{sp}}(\mathfrak{e}_\alpha) := \big(U_\alpha, \sigma_x, \sigma_y, A_\alpha, \eta_\alpha\big), \tag{5.277}$$

$$O_{\mathrm{sol}}(\mathfrak{e}_\alpha) := \big(b_\alpha, b_\alpha^+, b_\alpha^-;\ (M^+, M^-, D, H) \mapsto (M^+ + b_\alpha^+, M^- + b_\alpha^-, D + b_\alpha, H + |b_\alpha|)\big), \tag{5.278}$$

$$O_{\mathrm{gau}}(\mathfrak{e}_\alpha) := \big\{\big(\mathrm{Cell}_{\mathrm{TPK}}(\alpha), q, B, W_c(q)\big) : q \in Q_{\mathrm{inc}}(y)\big\}, \tag{5.279}$$

$$O_{\mathrm{coh}}(\mathfrak{e}_\alpha) := \big(I_y, J_y, \kappa_{y,N}, \delta_y, C_{y,N}, F_\alpha : C_{y,N} \longrightarrow C_{x,N}\big), \tag{5.280}$$

$$O_{\mathrm{det}}(\mathfrak{e}_\alpha) := \big(\Psi_\alpha, K_\alpha, \mathbf{1}_\alpha, z_\pm(\alpha), h_*(\alpha), \mathrm{Det}(C_{y,N}), H_*(C_{y,N}), \mathrm{Smith}(C_{y,N}), \mathrm{Boc}(C_{y,N})\big). \tag{5.281}$$

The internal Archimedean reading is the first component of this simultaneous action; it does not yet introduce any classical form:

$$\mathcal{L}^{\mathrm{int}}_{\mathrm{arq}}(\mathfrak{e}_\alpha) := O_{\mathrm{sp}}(\mathfrak{e}_\alpha). \tag{5.282}$$

For every value in those families, including each element of the multisection O_{gau}, its support is the dependent index

$$\begin{aligned} \mathsf{B}_\alpha &:= \{(s(\alpha), t(\alpha), \gamma\alpha, \mathrm{Led}(\gamma\alpha))\},\\ \mathrm{supp}_\alpha : \mathsf{Y}_\bullet[\alpha] &\longrightarrow \mathsf{B}_\alpha, \qquad o \longmapsto (s(\alpha), t(\alpha), \gamma\alpha, \mathrm{Led}(\gamma\alpha)). \end{aligned} \tag{5.283}$$

Therefore, they remain typed, without a projection that recovers the edge,

$$\begin{aligned} s_\alpha(O_\bullet) &:= s(\alpha), \qquad t_\alpha(O_\bullet) := t(\alpha),\\ \gamma_\alpha(O_\bullet) &:= \gamma\alpha, \qquad \mathrm{Led}_\alpha(O_\bullet) := \mathrm{Led}(\gamma\alpha). \end{aligned} \tag{5.284}$$

Here $\mathrm{Cell}_{\mathrm{TPK}}(\alpha)$ is the map of complexes (5.229) induced by transport of the same sheet fiber, and F_α is (5.185) for that extension. In particular, O_{gau} preserves the complete multisection of q's; it does not choose the origin or an orbit.

The five formulas above constitute a single dependent action.

$$O(\mathfrak{e}_\alpha) := \big\langle O_{\mathrm{sp}}, O_{\mathrm{sol}}, O_{\mathrm{gau}}, O_{\mathrm{coh}}, O_{\mathrm{det}}; \mathrm{Cpl}_\alpha \big\rangle, \tag{5.285}$$

where Cpl_α is the certificate formed by the identities

$$\begin{gathered} \mathrm{supp}_\alpha(O_{\mathrm{sp}}) = \mathrm{supp}_\alpha(O_{\mathrm{sol}}) = \mathrm{supp}_\alpha(O_{\mathrm{gau}}) = \mathrm{supp}_\alpha(O_{\mathrm{coh}}) = \mathrm{supp}_\alpha(O_{\mathrm{det}}),\\ \mathsf{Q}(t(\alpha)) = \sigma_9 \mathsf{Q}(s(\alpha)), \quad g(t(\alpha)) = g(s(\alpha)) + [1]_9, \quad q^{(9)}(t(\alpha)) = q^{(9)}(s(\alpha)) + \chi_9(g(s(\alpha))),\\ A_\alpha^* A_\alpha + \eta_\alpha^* \eta_\alpha = P_x U_\alpha^* U_\alpha P_x + Q_x U_\alpha^* U_\alpha Q_x,\\ (M^+, M^-, D, H)' = (M^+ + b_\alpha^+, M^- + b_\alpha^-, D + b_\alpha, H + |b_\alpha|),\\ Q_{\mathrm{inc}} = C^2(\mathrm{Cell}_{\mathrm{TPK}}; \mathbf{F}_3), \qquad sB = 0, \qquad W_c(q + Ba) = W_c(q),\\ C_{y,N} \text{ is the same complex in } O_{\mathrm{coh}} \text{ and } O_{\mathrm{det}}, \qquad \partial h_* = z_- - z_+, \qquad K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha. \end{gathered} \tag{5.286}$$

The symbol $\bullet$ ranges over the five readings of (5.285); all preserve the same source, destination, route, and ledger.

We collect, without separating the state, the operations constructed up to here:

$$\begin{aligned}\mathfrak{D}_{k,N}(\widehat{x}) := \big(&\{\mathfrak{e}_\alpha\}_\alpha;\\ &U_k, \sigma_k, P_k, Q_k, A_k, \eta_k;\\ &b_\alpha, b^{\pm}_\alpha, M^{\pm}_k, D^0_k, H^0_k, \kappa^{\mathrm{can}}_k;\\ &V_{\mathrm{TPK}}, \mathscr{I}, \mathscr{E}, \mathscr{B}, B;\\ &\mathrm{Cell}_{\mathrm{TPK}}(\widehat{x}), Q_{\mathrm{inc}}(\widehat{x}), W_c;\\ &I_k, J_k, \kappa_{k,N}, \delta, C_{k,N};\\ &c_\alpha, v_\alpha, \Psi_\alpha, K_\alpha, \mathbf{1}_\alpha, z_\pm(\alpha), h_*(\alpha);\\ &\mathrm{Det}(C_{k,N}), H_*(C_{k,N}), \mathrm{Smith}(C_{k,N}), \mathrm{Boc}(C_{k,N});\\ &\{O(\mathfrak{e}_\alpha)\}_\alpha\big).\end{aligned} \tag{5.287}$$

The semicolon merely separates levels of interpretation of the same dependent tuple; it does not define independent structures. The input state is neither retained as the first coordinate nor recovered through a projection. Each operation is proved through its action on the generators $\mathfrak{e}_\alpha$, which preserve the same support, the same route, and the same ledger.

For $\ell \geq k$ and $N' \geq N$, transport is defined component by component through truncation of histories, reduction of coefficients, and precomposition of ports. The naturality assertion that must be proved is a single one:

$$u_{(\ell,N'),(k,N)}\mathfrak{D}_{\ell,N'}(\widehat{x}_\ell) = \mathfrak{D}_{k,N}(\tau_{\ell,k}\widehat{x}_\ell). \tag{5.288}$$

Naturality is formulated over the generated history, not by pretending that an isolated fine edge must remain an edge after truncating its two endpoints. For $\gamma = (\varepsilon_1, \dots, \varepsilon_\ell)$, we define

$$\mathsf{E}_{\leq\ell}(\gamma) := (\mathfrak{e}_{\varepsilon_i})_{1\leq i\leq \ell}, \qquad O_{\leq\ell}(\gamma) := (O(\mathfrak{e}_{\varepsilon_i}))_{1\leq i\leq\ell}.$$

The two transports are the explicit prefix maps.

$$u^{\mathfrak{e}}_{\ell,k} : \mathsf{E}_{\leq\ell}(\gamma) \longrightarrow \mathsf{E}_{\leq k}(\gamma), \qquad (\mathfrak{e}_{\varepsilon_i})_{i\leq\ell} \longmapsto (\mathfrak{e}_{\varepsilon_i})_{i\leq k}, \tag{5.289}$$

$$u^{O}_{\ell,k} : O_{\leq\ell}(\gamma) \longrightarrow O_{\leq k}(\gamma), \qquad (O(\mathfrak{e}_{\varepsilon_i}))_{i\leq\ell} \longmapsto (O(\mathfrak{e}_{\varepsilon_i}))_{i\leq k}. \tag{5.290}$$

In each retained entry, the component sp intertwines U, σ, P, Q, A, η; the component sol applies the conditional expectation to $b, b^{\pm}, \kappa$; the component gau applies $\mathrm{Cell}(\tau) \oplus \tau_Q$; the component coh precomposes the ports with F; and the component det applies $\Psi, K, H_*, \mathrm{Smith}, \mathrm{Boc}$ to the same map of complexes. The prefix acts simultaneously on those five components. Its components already discharged are

$$\begin{gathered}U_{m,k} = U_{m,\ell}U_{\ell,k} \quad (m \geq \ell \geq k), \qquad E_k b^{\pm}_f = b^{\pm}_c + \kappa^{\mathrm{can}}_k,\\ r_{N',N}\kappa_{k,N'} = \kappa_{k,N}(r_{N',N} \oplus r_{N',N}),\\ \Psi_\beta\Psi_\alpha = \Psi_{\beta\alpha}, \qquad K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha,\\ u^{O}_{\ell,k}O_{\leq\ell}(\gamma) = O_{\leq k}(\gamma) = \mathrm{map}(O)\big(u^{\mathfrak{e}}_{\ell,k}\mathsf{E}_{\leq\ell}(\gamma)\big).\end{gathered} \tag{5.291}$$

The first equality uses multiplication of the conditional weights over the same system of common cylinders; the remaining ones are the preceding explicit formulas. No visible projection can be used to reconstruct or select $\widehat{x}$.

To complete the last component, let us denote by $F_{(\ell,N'),(k,N)} : C_{\ell,N'} \to C_{k,N}$ the map of complexes of (5.185). Its cone is preserved as part of the transition. Functoriality of homology and of the Bockstein sequence gives

$$H_*(F_{m,k}) = H_*(F_{\ell,k})H_*(F_{m,\ell}), \qquad \mathrm{Boc}(F_{m,k}) = \mathrm{Boc}(F_{\ell,k})\,\mathrm{Boc}(F_{m,\ell}). \tag{5.292}$$

The determinant line is transported by the identity of the exact triangle.

$$\mathrm{Det}(C_{k,N}) \simeq \mathrm{Det}(C_{\ell,N'}) \otimes \mathrm{Det}\big(\mathrm{Cone}\, F_{(\ell,N'),(k,N)}\big)\,. \tag{5.293}$$

This formula also records the refinements that add homology. The Smith reading is conserved without feigning invariance: each transition is assigned the total certificate

$$\mathcal{S}(F) := \big(\mathrm{Smith}(C_{\ell,N'}), \mathrm{Smith}(C_{k,N}), \mathrm{Smith}(\mathrm{Cone}\, F)\big), \tag{5.294}$$

which literally determines which summands are conserved, born, or die.

The action of a coherent symmetry g on the common tree induces permutation isometries $\widetilde{g}_k$ on $\mathscr{H}_k$. The bijection of children and preservation of the weights of (5.96) prove, on the basis vectors,

$$\widetilde{g}_{k+1}U_k = U_k\widetilde{g}_k, \qquad \widetilde{g}_k\sigma_k = \sigma_k\widetilde{g}_k, \qquad \widetilde{g}_kP_k = P_k\widetilde{g}_k, \qquad \widetilde{g}_kQ_k = Q_k\widetilde{g}_k. \tag{5.295}$$

Theorem 5.34 (Totality and naturality of the correlative constructor)**.** *For every total state of* (5.56)*, every finite horizon, and every coefficient level, the expression* (5.287) *is defined. Its five characteristics are simultaneous operations on the same extension ledger and satisfy the single identity* (5.288)*. The collar, the complex, and the determinant line remain nonempty even when their subsequent readers are neither acyclic nor invertible.*

Proof. Nonemptiness of the tree and its ports is proposition 5.15; the measure and the expectation are total by theorem 5.28. The formulas (5.120)–(5.139) define transport, polarity, balance, and memory for all generators. The four directions of (5.187) are distinct by (5.189); the cellular completion definition 5.33 contains its two path orders and six cells, while (5.200) proves its naturality. The cellular sequence is exact by (5.108), the local complex satisfies $\partial^2 = 0$, and the correlated filling is explicit by (5.258). Every finite based construction possesses the line (5.259); homology, Bockstein, and the Smith certificate are transported by (5.292)– (5.294). Finally, the identities are verified on each generator $\mathfrak{e}_\alpha$ through (5.291), (5.200), and (5.295); by linearity and by the functorial composition of routes, (5.288) is obtained for the integral record. □

5.2.7 Causal separation of the identities, deductions, and realizations of the correlator constructor

No vacuity and exact identities of the correlator constructor The nonemptiness imported from (5.64) and the identities (5.104), (5.108), (5.116), (5.121), (5.127), (5.138), (5.111), (5.175), and (5.258) follow from the definitions and the exhibited APP–TRIT–TPK relations. Their status is `EXACT_INTERNAL`.

Starting hypotheses for the isometry of U_k, the cellular sequence, and the composition Ψ/K The construction of U_k takes as its starting structure the finiteness of the fibres, survival at every horizon, functional truncation, and the transports of both sheets already produced by TPK. Equality $U_k^*U_k = I$ is equivalent to verifying the normalized average Gram matrix of (5.122); in particular, it suffices to check $U_\alpha^*U_\alpha = I$ for each emission. General identity (5.121) is retained for all declared sheet transports. The cellular sequence uses both ordered APP ports. Composition Ψ/K uses integer carry, orientation, and boundary. Under these enumerated premises, the stated conclusions are affirmative.

Additional data for isometric realizations, balanced retracts, and trivialization of the determinant line The internal constructor is completed by the positive Gram form of (5.121), the total space Q_{inc}, the cellular sequence, and the determinant line with its transition cone. A specialized publication may require additional data: the isometry of each U_α, a concrete numerical identification of the APP defect,

a balanced acyclic retraction, or trivialization of a determinant line. Those data are intertwiners of the published codomain; they do not form part of the domain or of the existence of $\mathfrak{D}_{k,N}$. Their absence precludes exclusively the promotion that mentions them.

Comparison Criterion 5.35 (Falsifiers of the intrinsic correlation)**.** The joint construction fails at the first level at which any of the following facts occurs:

1. an operation uses an extension, orientation, carry, memory, boundary or route different from that recorded in $\mathfrak{e}_\alpha$;
2. a child of $\mathrm{Ch}_k(\widehat{x}_k)$ is chosen and the remaining ones are eliminated;
3. the weights of (5.118) do not sum to one, (5.121) fails, or $U_k^* U_k = I$ is declared without verifying the normalized average Gram matrix of (5.122), for example through the isometry of each emission transport;
4. κ_k^{can} is identified with the integer carry κ_α;
5. $sB = 0$ fails, $Q_{\mathrm{inc}}(\widehat{x})$ is contracted to its origin, the action $q \mapsto q + Ba$ is altered, or the two path orders are identified before applying the reader;
6. a refinement breaks (5.186) or the cocycle (5.175);
7. fails $\partial^2 = 0$, $z_\pm$ are not ciclos or $\partial h_* \neq z_- - z_+$;
8. a nonzero determinantal scalar is asserted without the retract, the balance, and the acyclicity required by that publication;
9. any of the five operations is presented as an object separate from the constructor $\mathfrak{D}_{k,N}$.

In all those cases the diagnosis is: *common object substituted; conclusion not transferable*.

5.2.8 Joint emergence of the five constructions

For $0 \le k \le N$, let

$$\rho_{N,k} := \rho_k \circ \rho_{k+1} \circ \cdots \circ \rho_{N-1}, \qquad \rho_{k,k} := \mathrm{id}_{H_k},$$

and, for a surviving history $\zeta \in H_k$, define

$$\mathrm{Desc}_{k,N}(\zeta) := \{x \in H_N : \rho_{N,k}(x) = \zeta\}.$$

The finite tree $\mathscr{T}_{k,N}(\zeta)$ has vertices

$$\coprod_{j=k}^{N} \{y \in H_j : \rho_{j,k}(y) = \zeta\}$$

and has as edges only the TPK prolongations $a : y \to z$ with $z \in \rho_j^{-1}(y)$. Thus neither its vertices nor its prolongations come from an auxiliary tree.

For $x \in \mathrm{Desc}_{k,N}(\zeta)$, the marked branch $\lambda_{\zeta,x}$ is the sequence

$$\zeta = x_k \longrightarrow x_{k+1} \longrightarrow \cdots \longrightarrow x_N = x$$

determined by the truncations of x. The common base register is

$$\begin{aligned}\mathscr{R}_{k,N}(\zeta;x) := \big(&\mathscr{T}_{k,N}(\zeta), \lambda_{\zeta,x};\\ &\big(\varepsilon_a, s_a, o_a, (\delta_{\diamond,a}, r_{\diamond,a}, q_{\diamond,a})_{\diamond\in\{\Sigma,\Pi\}}, \mathrm{Carr}_a, C_a, t_a, \mathrm{Fr}_a, \mathrm{Ruta}_a, \mathrm{Mem}_a, \mathrm{Surv}_a\big)_a;\\ &\mathrm{src}, \mathrm{tgt}, \circ, (\rho_j)_{k\le j<N}\big),\end{aligned} \tag{C1}$$

where

$$U_a(m_0) = C_a m_0 + t_a, \qquad \delta_{\diamond,a} = r_{\diamond,a} + 9q_{\diamond,a}.$$

The symbol m_0 in this formula is a memory coordinate; it is not the modular depth m used below. The true base space is

$$\mathfrak{R}_{k,N}(\zeta) := \{\mathscr{R}_{k,N}(\zeta;x) : x \in \mathrm{Desc}_{k,N}(\zeta)\}.$$

The marked branch is not an added input: it is the sequence of actually produced edges ending at x. The rest of the register simultaneously preserves the lateral fibres of the tree; thus the mark does not replace branching with an isolated path.

Set $R_m := \mathbb{Z}/3^m\mathbb{Z}$, and let $V_j(\zeta) := \mathrm{Desc}_{k,j}(\zeta)$. Transport, the sheet projectors and the terminal residue restrict to the subtree through

$$\mathcal{U}_j^\zeta := \mathcal{U}_j \restriction_{\ell^2(V_j(\zeta),\mu_j)}, \qquad P_j^\zeta := P_j \restriction_{V_j(\zeta)}, \qquad Q_j^\zeta := I - P_j^\zeta,$$

$$A_j^\zeta := P_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + Q_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta, \qquad \eta_j^\zeta := Q_{j+1}^\zeta \mathcal{U}_j^\zeta P_j^\zeta + P_{j+1}^\zeta \mathcal{U}_j^\zeta Q_j^\zeta.$$

For $k \le n \le N$, let

$$F_{k,k}^\zeta := I, \qquad F_{k,n}^\zeta := A_{n-1}^\zeta \cdots A_k^\zeta, \qquad T_{k,n}^\zeta := (F_{k,n}^\zeta)^* F_{k,n}^\zeta.$$

The ports belonging to the same subtree are

$$\mathscr{P}_j^\Sigma(\zeta) := \{(y,\Sigma) : y \in V_j(\zeta)\}, \qquad \mathscr{P}_j^\Pi(\zeta) := \{(y,\Pi) : y \in V_j(\zeta)\};$$

we denote by $B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta$ the restrictions of the incidence and the rectangular complex to those ports, and by L_j^ζ the precomposition induced by removing prefixes.

To define the path reader without abbreviations, let

$$W_\ell^\zeta(y) := R_m[\mathrm{Path}_{\mathscr{T}_{k,\ell}(\zeta)}(y, \mathrm{Term}_\ell)]$$

and set

$$\mathscr{D}^{(m)}_{q,r;k,\ell}[\mathscr{R}] := \bigoplus_{y_0 \to \cdots \to y_q \text{ in } \mathscr{T}_{k,\ell}(\zeta)} W_\ell^\zeta(y_q) \otimes_{R_m} \big(\Gamma_r(y_0) \otimes_{\mathbb{Z}} R_m\big), \qquad \mathrm{Tot}_d\,\mathscr{D} := \bigoplus_{q+r=d} \mathscr{D}^{(m)}_{q,r;k,\ell}[\mathscr{R}].$$

The formulas already proved produce, on each $\mathscr{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five objects

$$\begin{aligned}
\mathcal{F}_{\mathrm{sp}}[\mathscr{R}] &:= (\ell^2(V_j(\zeta),\mu_j), \mathcal{U}_j^\zeta, A_j^\zeta, \eta_j^\zeta, T_{k,j}^\zeta)_{k\le j\le N},\\
\mathcal{F}_{\mathrm{sol}}[\mathscr{R}] &:= (b, \kappa_{\mathrm{can}}, C_\alpha, \kappa_\alpha)_{\alpha\subset\mathscr{T}_{k,N}(\zeta)},\\
\mathcal{F}^{(m)}_{\mathrm{gau}}[\mathscr{R}] &:= (\mathscr{P}_j^\Sigma(\zeta), \mathscr{P}_j^\Pi(\zeta), B_j^\zeta, \kappa_j^\zeta, \delta_j^\zeta; R_m)_{k\le j\le N},\\
\mathcal{F}^{(m)}_{\mathrm{coh}}[\mathscr{R}] &:= (\Gamma_y \otimes R_m, \partial, \Psi_\alpha, H_\bullet)_{y,\alpha\subset\mathscr{T}_{k,N}(\zeta)},\\
\mathcal{F}^{(m)}_{\mathrm{det}}[\mathscr{R}] &:= (\mathrm{Det}(\mathrm{Tot}\,\mathscr{D}^{(m)}_{\bullet,\bullet;k,\ell}[\mathscr{R}]))_{k\le\ell\le N}.
\end{aligned} \tag{C2}$$

Here $\mathscr{D}^{(m)}_{\bullet,\bullet;k,\ell}[\mathscr{R}]$ is the normalized bicomplex of paths and memory from the preceding sections, restricted to the subtree of horizon ℓ and reduced over R_m. The final reader preserves the ordered family of determinantal lines up to horizon N; in this way, its truncation is a canonical projection and does not assume a nonexistent determinantal factorization. The law μ is the canonical law of the census, and the ports are fixed through the lexicographically minimal branch of the APP order. Thus none of the five readers receives an external weighting or port.

For $T \in \mathfrak{T} := \{\mathrm{sp}, \mathrm{sol}, \mathrm{gau}, \mathrm{coh}, \mathrm{det}\}$, define the typed total space

$$\widetilde{\mathcal{F}}^{(m)}_{T;k,N}(\zeta) := \coprod_{\mathscr{R}\in\mathfrak{R}_{k,N}(\zeta)} \{\mathcal{F}^{(m)}_{T}[\mathscr{R}]\}_{\mathscr{R}}, \qquad p_T : \widetilde{\mathcal{F}}^{(m)}_{T;k,N}(\zeta) \longrightarrow \mathfrak{R}_{k,N}(\zeta),$$

where $\mathcal{F}^{(m)}_{\mathrm{sp}} := \mathcal{F}_{\mathrm{sp}}$ and $\mathcal{F}^{(m)}_{\mathrm{sol}} := \mathcal{F}_{\mathrm{sol}}$; that is, the superscript m is inert for those two readers. The projection p_T forgets only the reader structure and preserves its source register. The correct fibre product is

$$\begin{aligned}\mathcal{J}^{(m)}_{k,N}(\zeta) := \widetilde{\mathcal{F}}^{(m)}_{\mathrm{sp;k,N}}(\zeta) \underset{\mathfrak{R}_{k,N}(\zeta)}{\times} \widetilde{\mathcal{F}}^{(m)}_{\mathrm{sol;k,N}}(\zeta) \underset{\mathfrak{R}_{k,N}(\zeta)}{\times} \widetilde{\mathcal{F}}^{(m)}_{\mathrm{gau;k,N}}(\zeta)\\ \underset{\mathfrak{R}_{k,N}(\zeta)}{\times} \widetilde{\mathcal{F}}^{(m)}_{\mathrm{coh;k,N}}(\zeta) \underset{\mathfrak{R}_{k,N}(\zeta)}{\times} \widetilde{\mathcal{F}}^{(m)}_{\mathrm{det;k,N}}(\zeta).\end{aligned} \tag{C3}$$

Equality of the five projections requires preserving the same tree, the same edges, the same produced branch and the same register. For each $\mathscr{R}$, the five constructions determine the element

$$s^{(m)}_{k,N}(\mathscr{R}) := (\mathcal{F}^{(m)}_{T}[\mathscr{R}])_{T\in\mathfrak{T}} \in \mathcal{J}^{(m)}_{k,N}(\zeta).$$

Existence of this element follows from the operations of (C2), not from the fibre-product notation.

Removal of the last layer induces

$$\mathrm{res}^{\mathfrak{R}}_{N+1,N}\big(\mathscr{R}_{k,N+1}(\zeta;x)\big) := \mathscr{R}_{k,N}(\zeta;\rho_N(x)).$$

Restriction of each operator to the retained generators defines

$$r^{m}_{N+1,N} : \mathcal{J}^{(m)}_{k,N+1}(\zeta) \longrightarrow \mathcal{J}^{(m)}_{k,N}(\zeta).$$

In the determinantal reader, $r^{m}_{N+1,N}$ removes only the last coordinate of the family of lines of (C2). The reduction $R_{m+1} \twoheadrightarrow R_m$ defines, by base change,

$$q^{N}_{m+1,m} : \mathcal{J}^{(m+1)}_{k,N}(\zeta) \longrightarrow \mathcal{J}^{(m)}_{k,N}(\zeta).$$

On each generator, we have

$$\begin{gathered} r^{m}_{N+1,N} q^{N+1}_{m+1,m} = q^{N}_{m+1,m} r^{m+1}_{N+1,N},\\ \partial\Psi_\alpha = \Psi_\alpha\partial, \qquad L^\zeta_j \kappa^\zeta_j = \kappa^\zeta_{j+1} L^\zeta_j, \qquad L^\zeta_j \delta^\zeta_j = \delta^\zeta_{j+1} L^\zeta_j,\end{gathered} \tag{C4}$$

and compositions of the maps r and q satisfy the functorial identities. The first equality holds because deleting the last layer and reducing the coefficients act on distinct coordinates. The other three are the naturality identities proved earlier. A unique pro-system in (N, m) is obtained, not five juxtaposed limits.

At the root, define

$$\mathfrak{J}^{(m)}_{N,\bullet} := \mathcal{J}^{(m)}_{0,N}(*). \tag{C5}$$

The joint discrete structure of the continuum is the limit of those correlative objects:

$$\boxed{C^{\mathrm{disc}}_{\mathrm{cont}} := \varprojlim_{\substack{N\geq 1\\ m\geq 1}} \big(\mathfrak{J}^{(m)}_{N,\bullet}, r^{m}_{N+1,N}, q^{N}_{m+1,m}\big).} \tag{C6}$$

This is a definition through the pro-system generated by the complete tree; it is not the image or graph of a map retaining the input state as its first coordinate.

The registers have a produced terminal mark. Their compatible projections induce

$$\Pi : C^{\mathrm{disc}}_{\mathrm{cont}} \longrightarrow H^{\mathrm{enr}}_{\infty}.$$

Conversely, a complete history determines, through its already emitted prefixes, the compatible sections $s^{(m)}_{0,N}$. Thus, after having defined the continuum, one obtains

$$\mathrm{Gen}_{\mathrm{cont}} : H^{\mathrm{enr}}_{\infty} \longrightarrow C^{\mathrm{disc}}_{\mathrm{cont}}, \qquad \Pi \circ \mathrm{Gen}_{\mathrm{cont}} = \mathrm{id}_{H^{\mathrm{enr}}_{\infty}} . \tag{C7}$$

This section proves recoverability; it is not used to define $C^{\mathrm{disc}}_{\mathrm{cont}}$.

Theorem 5.36 (Correlative emergence of the five constructions)**.** *For every surviving history $\zeta \in H_k$, every $N > k$, every $m \geq 1$ and every $\mathscr{R} \in \mathfrak{R}_{k,N}(\zeta)$, the five constructions of* (C2) *are defined on the same register, and* (C3) *contains $s^{(m)}_{k,N}(\mathscr{R})$. Their truncations, modular reductions and transports satisfy* (C4)*. Furthermore, $C^{\mathrm{disc}}_{\mathrm{cont}} \neq \varnothing$, the map Π is surjective and is split by* $\mathrm{Gen}_{\mathrm{cont}}$*. The five components are inseparable aspects of the same continuum, not five domains or five independent maps.*

Proof. The survival defining H_k and the TPK prolongation law give $\rho_k^{-1}(\zeta) \neq \varnothing$ for each vertex. Since the fibres are finite, Kőnig's lemma produces $H^{\mathrm{enr}}_{\infty} \neq \varnothing$.

Given a register $\mathscr{R}$, the transport of histories and sheets produces $\mathcal{F}_{\mathrm{sp}}[\mathscr{R}]$; affine composition and the exact cancellation defect produce $\mathcal{F}_{\mathrm{sol}}[\mathscr{R}]$; incidences and the rectangular sequence produce $\mathcal{F}^{(m)}_{\mathrm{gau}}[\mathscr{R}]$; the memory complex and its transport produce $\mathcal{F}^{(m)}_{\mathrm{coh}}[\mathscr{R}]$; and the lines of the normalized path assembly produce $\mathcal{F}^{(m)}_{\mathrm{det}}[\mathscr{R}]$. Each construction acts on the same generated edges, not on an assumed subdomain or on the abstract ambient space of 729 words.

Affine, transport, incidence and memory compatibilities prove (C4). Truncating the determinantal family of (C2) commutes by definition with r; its base change commutes with q. The sections $s^{(m)}_{0,N}$ are compatible, so they produce $\mathrm{Gen}_{\mathrm{cont}}$. The identity (C7) proves simultaneously nonemptiness of the limit, surjectivity of Π and recoverability of each history without introducing it as a defining coordinate.

The fibre, relation, number, orientation, carry, memory, boundary and route remain in the common register; the multisection, terminals, and linear and probabilistic readers remain in the five correlative constructions. Passing to the limit therefore preserves dependency closure and does not separate the continuum. □

Subsequent recognition of the uniquely constructed continuum through spectral, solenoidal, gauge, cohomological or determinantal language belongs to the later conventional comparison. It does not select its states or enter its APP–TRIT–TPK generation.

Chapter 6

Existence of the discrete continuum: nonemptiness, preservation, terminality, naturality, refinement, and compatible limit

6.1 Multivalued terminal, unique naturality and passage to the limit

Tesis rectora

The discrete continuum produces a single system of histories, extensions, memories, and terminals. The internal discriminants $\mathrm{sp}, \mathrm{sol}, \mathrm{gau}, \mathrm{coh}, \mathrm{det}$ name five characteristics that are simultaneous within that same system: they neither fix distinct inputs nor authorize the selection of different histories. In this chapter, the terminal preserves all continuations, the connection remains a nonadic correspondence, and naturality is expressed by a single identity.

6.1.1 Surviving histories with extension witnesses

Let $\widehat{x}_k \in \widehat{\mathcal{X}}_k^{\mathrm{tot}}$ be the total state of (5.57). For $N \geq k$, we define without any other predicate of membership

$$\mathscr{H}_{k,N}(\widehat{x}_k) := \mathrm{Term}_{k,N}(\widehat{x}_k). \tag{6.1}$$

Each $h = (\varepsilon_{k+1}, \ldots, \varepsilon_N)$ on the right contains through (5.65) its intermediate states, the witnesses of Ext, the carries, boundaries, routes, and ledger. Each edge satisfies the phase action (5.17); hence every nine-step subpath determines, with its nine emissions and not merely with its length, an element of $\mathsf{Z}_{9,j}^{\Gamma}$ and a witness of $\Gamma_{9,j}$ through (5.71)– (5.73). Therefore, Ext and nonadic holonomy arise from the same phase-compatible word. To (6.1) there is adjoined neither a survival filter nor a second condition Γ_9.

For $k \leq \ell \leq N$, let

$$\mathscr{P}_{\ell|k}(\widehat{x}_k) := \{h_{[k,\ell]} : h \in \mathscr{H}_{k,N}(\widehat{x}_k) \text{ for some } N \geq \ell\}. \tag{6.2}$$

For $p \in \mathscr{P}_{\ell|k}(\widehat{x}_k)$, we use the abbreviation

$$\mathscr{H}_{\ell,N}(p) := \{q : p \star q \in \mathscr{H}_{k,N}(\widehat{x}_k)\}. \tag{6.3}$$

The prolongation neutral of (5.59) extends each prefix to cualquier horizon. Consequently,

$$\mathscr{H}_{k,N}(\widehat{x}_k) \neq \varnothing, \qquad \mathscr{P}_{\ell|k}(\widehat{x}_k) \neq \varnothing, \tag{6.4}$$

by proposition 5.15.

The truncation

$$\pi^{\mathscr{H}}_{N+1,N} : \mathscr{H}_{k,N+1}(\widehat{x}_k) \longrightarrow \mathscr{H}_{k,N}(\widehat{x}_k) \tag{6.5}$$

is exactly (5.68) and is surjective by lemma 5.16. Survival is, by (5.83), a demonstrated output of the total domain and not a prior selection.

6.1.2 Generated terminal record and its 13 coordinates

Let us fix a history

$$h = (\varepsilon_{j+1} : x_j \longrightarrow x_{j+1})_{j=k}^{N-1}, \qquad \gamma_{k,k} := I_{x_k}, \qquad \gamma_{k,j+1} := \gamma_{k,j} \star \varepsilon_{j+1},$$

The operator notation $U_{\gamma_{k,j+1}} = U_{\varepsilon_{j+1}} U_{\gamma_{k,j}}$ preserves the order of application, whereas $\star$ preserves the chronological order of the word. We further fix a coefficient level $r \geq 1$. For brevity, we write

$$C_{x_j,r} := C_{j,r}(x_j), \qquad F_{\varepsilon_{j+1}} : C_{x_{j+1},r} \longrightarrow C_{x_j,r}$$

for the complex and the map of complexes of (5.184) and (5.185). The type of the level record j, denoted $\mathfrak{R}_{j,r}$, is a dependent record with the following thirteen fields. In each field we indicate the type, the seed and the action of an edge $\varepsilon := \varepsilon_{j+1}$.

The state x_k already contains the complete word $\gamma_{0,k} = \varepsilon_1 \star \cdots \star \varepsilon_k$. Therefore a "seed in k" never restarts the genealogy: it is the fold of the same updates from the root to k. We denote

$$\begin{gathered} M_k^{\pm} := \sum_{i=1}^{k} b_{\varepsilon_i}^{\pm}, qquadD_k := \sum_{i=1}^{k} b_{\varepsilon_i}, qquadH_k := \sum_{i=1}^{k} |b_{\varepsilon_i}|, \\ \mathbf{c}_{\le k} := \big(c_r(\varepsilon_i, \gamma_{0,i-1})\big)_{1\le i\le k}, \qquad \mathbf{k}_{\le k} := \big(\kappa_{i-1}^{\mathrm{can}}\big)_{1\le i\le k}, \\ \mathbf{F}_{\le k} := \big(F_{\varepsilon_i} : C_{x_i,r} \to C_{x_{i-1},r}\big)_{1\le i\le k}. \end{gathered} \tag{6.6}$$

All empty lists appearing below are understood literally only in $k = 0$; for $k > 0$ they are replaced by prefixes inherited from (6.6) and from the ledger of $\gamma_{0,k}$. Thus the record does not preserve an opaque copy of x_k, but neither does it lose its APP–TRIT–TPK past.

Fiber F: local domain and conservation under prolongation The elementary fiber is

$$\mathrm{Fib}_r(x) := \big(\mathsf{H}_x^{\Sigma} \oplus \mathsf{H}_x^{\Pi},\ \mathsf{M}_x,\ \mathrm{Cell}_{\mathrm{TPK}}(x),\ C_{x,r}\big). \tag{6.7}$$

The field $\mathsf{F}_{k,j}$ is a word in the category of these fibers. Its seed and update are

$$\mathsf{F}_{k,k} := \big(\mathrm{Fib}_r(x_0), I\big) \star \overset{k}{\underset{i=1}{\star}} \big(\mathrm{Fib}_r(x_i), U_{\varepsilon_i}, \mathsf{C}_{\varepsilon_i}, \mathrm{Cell}_{\mathrm{TPK}}(\varepsilon_i), F_{\varepsilon_i}\big), \tag{6.8}$$

$$\mathrm{Upd}_{\varepsilon}^{\mathsf{F}}\, \mathsf{F}_{k,j} := \mathsf{F}_{k,j} \star \big(\mathrm{Fib}_r(x_{j+1}), U_{\varepsilon}, \mathsf{C}_{\varepsilon}, \mathrm{Cell}_{\mathrm{TPK}}(\varepsilon), F_{\varepsilon}\big). \tag{6.9}$$

It is nonempty because the two sheets, the memory, the cellular unit (5.198), and the port complex have already been constructed. The composition of its five maps proves naturality.

Extensions E: morphisms of compatible prolongation of the state The codomain is the category of typed relation words. We set

$$\mathsf{E}_{k,k} := (I_{x_0}, \mathrm{Ext}_{I_{x_0}}) \star \overset{k}{\underset{i=1}{\star}} \big(\varepsilon_i, \mathrm{Ext}_{\varepsilon_i}, \mathrm{Ext}_{\varepsilon_i} \circ \mathrm{Ext}_{\gamma_{0,i-1}} \subseteq \mathrm{Ext}_{\gamma_{0,i}}\big), \tag{6.10}$$

$$\mathrm{Upd}_{\varepsilon}^{\mathsf{E}}\, \mathsf{E}_{k,j} := \mathsf{E}_{k,j} \star \big(\varepsilon, \mathrm{Ext}_{\varepsilon}, \mathrm{Ext}_{\varepsilon} \circ \mathrm{Ext}_{\gamma_{0,j}} \subseteq \mathrm{Ext}_{\gamma_{0,j+1}}\big). \tag{6.11}$$

Identity and neutral emission prove nonemptiness; associativity of the composition relational proof naturality.

Relations R: incidence, orientation, and compatibility of composition Let $\mathrm{RelBlock}_{j,r}$ be the type of blocks of equations labeled by the edge and the prefix over which they are verified, and let $\mathrm{Pres}_{j,r} := \mathrm{List}(\mathrm{RelBlock}_{j,r})$. The unit block and the seed are

$$\mathsf{B}_I := (I_{x_0}; U_I = I,\ \mathsf{U}_I = I,\ \kappa(I) = 0,\ \partial I = 0), \qquad \mathsf{R}_{k,k} := (\mathsf{B}_I) \star \overset{k}{\underset{i=1}{\star}} \mathsf{B}_R(\varepsilon_i; \gamma_{0,i-1}). \tag{6.12}$$

For an edge $\varepsilon : x_j \to x_{j+1}$, the new block is the labeled tuple

$$\begin{aligned}\mathsf{B}_R(\varepsilon;\gamma_{0,j}) := \big(&\varepsilon,\gamma_{0,j};\ \delta^{\diamond}(\varepsilon)=\mathrm{res}^{\diamond}(\varepsilon)+9\,\mathrm{quo}^{\diamond}(\varepsilon),\\ &\Delta(\varepsilon)=\mathrm{or}(\varepsilon)+3\kappa(\varepsilon),\quad U_{\gamma_{0,j+1}}=U_\varepsilon U_{\gamma_{0,j}},\\ &\mathsf{U}_{\gamma_{0,j+1}}=\mathsf{U}_\varepsilon\mathsf{U}_{\gamma_{0,j}},\quad \kappa(\gamma_{0,j+1})=\kappa(\varepsilon)+\mathsf{C}_\varepsilon\kappa(\gamma_{0,j}),\\ &\partial\gamma_{0,j+1}=\partial\varepsilon+\partial\gamma_{0,j},\quad \mathrm{Cpl}_\varepsilon\ \text{of (5.286)}\big).\end{aligned} \tag{6.13}$$

The update concatenates the block without losing date or multiplicity:

$$\mathrm{Upd}^{\mathsf{R}}_{\varepsilon}\,\mathsf{R}_{k,j} := \mathsf{R}_{k,j}\star\mathsf{B}_R(\varepsilon;\gamma_{0,j}). \tag{6.14}$$

The truncation $\mathrm{tr}^{\mathsf{R}}_{j+1,j}$ removes the final labeled block. Moreover, $a_*\mathsf{B}_R(\varepsilon;\gamma)=\mathsf{B}_R(a\varepsilon;a\gamma)$, because each equation is natural over the same edge and the same prefix.

Numbers N: publication of value with conservation of genealogy For an edge $\varepsilon: x_j\to x_{j+1}$, we define the numerical datum

$$\nu(\varepsilon) := \big(j+1,\mathsf{Q}_{j+1},g_{j+1},q^{(9)}_{j+1},\beta_{j+1},\mathrm{res}^{\Sigma},\mathrm{quo}^{\Sigma},\mathrm{res}^{\Pi},\mathrm{quo}^{\Pi},\mathrm{or},\kappa,\Delta\big)(\varepsilon). \tag{6.15}$$

Its type is

$$\mathbb{N}\times\widetilde{O}_9\times C_9\times\mathbb{N}\times\mathbb{Z}/12\mathbb{Z}\times\{0,\dots,8\}^2\times\mathbb{Z}^5.$$

The seed contains the complete numerical list of the prefix $0\to k$; the update adds exactly the datum of the next edge:

$$\mathsf{N}_{k,k} := ((0,\mathsf{Q}_0,g_0,q^{(9)}_0,\beta_0;0,\dots,0))\star\overset{k}{\underset{i=1}{\star}}\nu(\varepsilon_i),\qquad \mathrm{Upd}^{\mathsf{N}}_{\varepsilon}\,\mathsf{N}_{k,j} := \mathsf{N}_{k,j}\star\nu(\varepsilon). \tag{6.16}$$

Totality proceeds from the divisions APP, the division TRIT, and the odometer TPK.

Orientation O: sign transport and leaf selection The type of an orientation state is

$$\mathbb{Z}\times(\mathsf{H}^{\Sigma}_x\oplus\mathsf{H}^{\Pi}_x)\times\mathbb{Z}[\mathsf{A}],$$

with seed and update

$$\mathsf{O}_{k,k} := \left(\sum_{i=1}^{k}\mathrm{or}(\varepsilon_i),\mathrm{Sh}(x_k),\partial\gamma_{0,k}\right), \tag{6.17}$$

$$\mathrm{Upd}^{\mathsf{O}}_{\varepsilon}(\mathfrak{o},\mathrm{Sh},\mathfrak{b}) := \big(\mathfrak{o}+\mathrm{or}(\varepsilon),U_\varepsilon\,\mathrm{Sh},\mathfrak{b}+\partial\varepsilon\big). \tag{6.18}$$

The APP leaves prove nonemptiness. Functoriality of U, additivity of or, and telescoping of the boundary prove naturality.

Carry K: update of the quotient and residual memory We use the typed integer lifts in (5.173). The field has type

$$\mathsf{M}_{x_j}\times\mathrm{Hom}(\mathsf{M}_{x_0},\mathsf{M}_{x_j})\times\mathrm{Mat}_5(\mathbb{Z})\times\mathrm{List}(\mathrm{Mat}_5(\mathbb{Z}))\times\mathrm{List}(\mathbb{Q}_{\ge0}).$$

For $\gamma_{k,j}=\varepsilon_j\cdots\varepsilon_{k+1}$, we explicitly fix

$$\widetilde{L}_{\gamma_{k,k}} := I_5,\qquad \widetilde{L}_{\gamma_{k,j}} := \widetilde{L}_r(\varepsilon_j)\cdots\widetilde{L}_r(\varepsilon_{k+1}). \tag{6.19}$$

For the inherited prefix we also fix

$$\widetilde{L}_{\gamma_{0,k}} := \widetilde{L}_r(\varepsilon_k)\cdots\widetilde{L}_r(\varepsilon_1).$$

The seed and update are

$$\mathsf{K}_{k,k} := \big(\kappa(\gamma_{0,k}), \mathsf{C}_{\gamma_{0,k}}, \widetilde{L}_{\gamma_{0,k}}, \mathbf{c}_{\le k}, \mathbf{k}_{\le k}\big), \tag{6.20}$$

$$\mathrm{Upd}^{\mathsf{K}}_{\varepsilon}(\kappa_\gamma, \mathsf{C}_\gamma, \widetilde{L}_\gamma, \mathbf{c}, \mathbf{k}) := \big(\kappa(\varepsilon) + \mathsf{C}_\varepsilon\kappa_\gamma,\ \mathsf{C}_\varepsilon\mathsf{C}_\gamma,\ \widetilde{L}_r(\varepsilon)\widetilde{L}_\gamma,\ \mathbf{c}\star c_r(\varepsilon,\gamma),\ \mathbf{k}\star\kappa_j^{\mathrm{can}}\big). \tag{6.21}$$

The identity provides the seed; the two cocycles (5.10) and (5.175) prove independence of parenthesization and naturality.

Memory M: record of the traversal and compatibility under truncation Its type at the same time preserves affine memory, ledger, and signed memories:

$$\mathrm{LedgerEntry} := \{\lambda(\varepsilon) : \varepsilon \text{ typed elementary emission}\}. \tag{6.22}$$

$$\mathsf{M}_x \times \mathrm{List}(\mathrm{LedgerEntry}) \times \mathbb{N}^2 \times \mathbb{Z} \times \mathbb{N}.$$

We set

$$\mathsf{M}_{k,k} := \big(m_k, \mathrm{Led}(\gamma_{0,k}), M_k^+, M_k^-, D_k, H_k\big), \tag{6.23}$$

$$\mathrm{Upd}^{\mathsf{M}}_{\varepsilon}(m, \mathbf{Led}, M^+, M^-, D, H) := \big(\mathsf{C}_\varepsilon m + \kappa(\varepsilon),\ \mathbf{Led}\star\lambda(\varepsilon),\ M^+ + b_\varepsilon^+,\ M^- + b_\varepsilon^-,\ D + b_\varepsilon,\ H + |b_\varepsilon|\big). \tag{6.24}$$

The affine action and (5.139) prove totality and naturality; the past is not recomputed from the last value.

Boundary B: gluing data and survival of the extension The type is a boundary chain accompanied by a word of complejos based and chain maps. Its seed and update are

$$\mathsf{B}_{k,j} \in \mathrm{Rec}[\mathfrak{b} \in \mathbb{Z}[\mathsf{A}_j]; C_{x_j,r} \in \mathrm{Ch}_{\mathrm{bas}}; \mathbf{F} \in \mathrm{Path}_{\mathrm{Ch}_{\mathrm{bas}}}(C_{x_j,r}, C_{x_0,r})].$$

$$\mathsf{B}_{k,k} := \big(\partial\gamma_{0,k}, C_{x_k,r}, \mathbf{F}_{\le k}\big), \tag{6.25}$$

$$\mathrm{Upd}^{\mathsf{B}}_{\varepsilon}(\mathfrak{b}, C_{x_j,r}, \mathbf{F}) := \big(\mathfrak{b} + \partial\varepsilon,\ C_{x_{j+1},r},\ \mathbf{F}\star F_\varepsilon\big). \tag{6.26}$$

Non-emptiness proceeds from the common ports; $\partial^2 = 0$, $F_{\beta\varepsilon} = F_\varepsilon F_\beta$, and (5.35) prove compatibility.

Route G: observable history of the surviving extension The field is the list of prefixes in the path category:

$$\mathsf{G}_{k,k} := (\gamma_{0,i})_{0\le i\le k}, \qquad \mathrm{Upd}^{\mathsf{G}}_{\varepsilon}\,\mathsf{G}_{k,j} := \mathsf{G}_{k,j}\star\gamma_{0,j+1}. \tag{6.27}$$

Identity, associative concatenation, and $a(\varepsilon\gamma) = a(\varepsilon)a(\gamma)$ prove the three required properties.

Multisection X: stable finite fiber and obstruction to pointwise selection For $M \ge j$, we define by action

$$\mathsf{X}_{k,j}(M) := \{\gamma_{k,j}\star q : q \in \mathrm{Term}_{j,M}(x_j)\}. \tag{6.28}$$

Thus

$$\mathsf{X}_{k,k}(M) = \mathrm{Term}_{k,M}(x_k),$$

and

$$\mathrm{Upd}^{\mathsf{X}}_{\varepsilon}\,\mathsf{X}_{k,j}(M) := \{\gamma_{k,j}\star\varepsilon\star q : q \in \mathrm{Term}_{j+1,M}(x_{j+1})\}. \tag{6.29}$$

The neutral prolongation proves nonemptiness. Deleting the final step and acting by a symmetry commute literally with concatenation.

Survival S: persistence under refinement and passage to the limit Here, a proposition is not stored, but witnesses. Let

$$s_{j,j} := I_{x_j}, \qquad s_{j,M+1} := s_{j,M} \star \varepsilon^0_M(t(s_{j,M})). \tag{6.30}$$

The seed is $\mathsf{S}_{k,k} := (s_{k,M})_{M\geq k}$, and

$$\mathrm{Upd}^{\mathsf{S}}_{\varepsilon}\, \mathsf{S}_{k,j} := (\gamma_{k,j} \star \varepsilon \star s_{j+1,M})_{M\geq j+1}. \tag{6.31}$$

By construction $\pi_{M+1,M} s_{j,M+1} = s_{j,M}$; the naturality of neutral emission proves equivariance.

Terminal T: universality of the limit object and factorization of readers The terminal is the inverse system, not a selected queue:

$$\mathsf{T}(x_j) := \big((\mathrm{Term}_{j,M}(x_j))_{M\geq j}, (\pi_{M+1,M})_{M\geq j}\big). \tag{6.32}$$

For each $M \geq j$, the map of prefix tipado is

$$\mathrm{pref}_{\gamma_{k,j},M} : \mathrm{Term}_{j,M}(x_j) \longrightarrow \mathrm{Term}_{k,M}(x_k), \qquad q \longmapsto \gamma_{k,j} \star q. \tag{6.33}$$

We define the field state and its update by

$$\mathsf{T}_{k,j} := \Big(\mathsf{T}(x_j), (\mathrm{pref}_{\gamma_{k,j},M})_{M\geq j}\Big), \tag{6.34}$$

$$\mathrm{Upd}^{\mathsf{T}}_{\varepsilon}\, \mathsf{T}_{k,j} := \Big(\mathsf{T}(x_{j+1}), (\mathrm{pref}_{\gamma_{k,j+1},M})_{M\geq j+1}\Big), \qquad \gamma_{k,j+1} := \gamma_{k,j} \star \varepsilon. \tag{6.35}$$

In particular, $\mathsf{T}_{k,k}$ uses $\gamma_{k,k} = I_{x_k}$. The witnesses (6.30) and the surjectivity of π prove that the system is nonempty; level by level,

$$\pi^{\mathscr{H}}_{M+1,M}\, \mathrm{pref}_{\gamma_{k,j},M+1} = \mathrm{pref}_{\gamma_{k,j},M}\, \pi^{\mathscr{H}}_{M+1,M}, \tag{6.36}$$

because both traversals only remove the last emission of q.

13. Terminal reader L: evaluation of the compatible extension and conservation of genealogy The reader contains no classical object. Its type is the category of paths of the unique dependent constructor $\mathfrak{D}_{j,r}$ of (5.287). We set

$$u^{\mathfrak{D}}_{j+1,j} := u_{(j+1,r),(j,r)} : \mathfrak{D}_{j+1,r}(x_{j+1}) \longrightarrow \mathfrak{D}_{j,r}(x_j), \tag{6.37}$$

the component of the common transport (5.288) that truncates the last emission, reduces the coefficients and precomposes the ports. By composition,

$$u^{\mathfrak{D}}_{m,k} = u^{\mathfrak{D}}_{\ell,k} u^{\mathfrak{D}}_{m,\ell} \qquad (m \geq \ell \geq k). \tag{6.38}$$

$$\mathsf{L}_{k,k} := (\mathfrak{D}_{0,r}(x_0)) \star \overset{k}{\underset{i=1}{\star}} \big(\mathfrak{D}_{i,r}(x_i), u^{\mathfrak{D}}_{i,i-1}, O(\mathfrak{e}_{\varepsilon_i})\big), \tag{6.39}$$

$$\mathrm{Upd}^{\mathsf{L}}_{\varepsilon}\, \mathsf{L}_{k,j} := \mathsf{L}_{k,j} \star \big(\mathfrak{D}_{j+1,r}(x_{j+1}), u^{\mathfrak{D}}_{j+1,j}, O(\mathfrak{e}_{\varepsilon})\big). \tag{6.40}$$

The totality of $\mathfrak{D}$ and (6.38) proves non-emptiness and naturality.

The thirteen preceding seeds form a single element.

$$\mathfrak{r}^{0}_{k,r} := \langle \mathsf{F}_{k,k}; \mathsf{E}_{k,k}; \mathsf{R}_{k,k}; \mathsf{N}_{k,k}; \mathsf{O}_{k,k}; \mathsf{K}_{k,k}; \mathsf{M}_{k,k}; \mathsf{B}_{k,k}; \mathsf{G}_{k,k}; \mathsf{X}_{k,k}; \mathsf{S}_{k,k}; \mathsf{T}_{k,k}; \mathsf{L}_{k,k} \rangle \in \mathfrak{R}_{k,r}. \tag{6.41}$$

The action of an edge is the total application.

$$\mathrm{Upd}_{\varepsilon} : \mathfrak{R}_{j,r}(\gamma_{0,j}) \longrightarrow \mathfrak{R}_{j+1,r}(\gamma_{0,j+1}), \qquad \mathrm{Upd}_{\varepsilon} := \langle \mathrm{Upd}^{\mathsf{F}}_{\varepsilon}, \dots, \mathrm{Upd}^{\mathsf{L}}_{\varepsilon} \rangle, \tag{6.42}$$

with the thirteen actions (6.9)– (6.40). Finally, the terminal register is *generated* by the fold

$$\boxed{\mathfrak{r}_{k,N}(h) := \mathrm{Upd}_{\varepsilon_N} \circ \cdots \circ \mathrm{Upd}_{\varepsilon_{k+1}}(\mathfrak{r}^{0}_{k,r}).} \tag{6.43}$$

It is not the unpacking of an intact copy of the state. On the image of this fold, the truncation is defined componentwise

$$\mathrm{tr}_{13;j+1,j}\, \mathfrak{r}_{k,j+1}(h \star \varepsilon) := \mathfrak{r}_{k,j}(h). \tag{6.44}$$

It is well defined: the fields E and G preserve the ordered list of emissions and prefixes, so that equality of two records entails equality of their labeled histories. In $\mathsf{F}, \mathsf{E}, \mathsf{R}, \mathsf{N}, \mathsf{G}, \mathsf{T}, \mathsf{L}$, the last entry or block is deleted; in $\mathsf{O}, \mathsf{K}, \mathsf{M}, \mathsf{B}$, the cumulative formula is applied to the recovered prefix; and in X, S, the cone of tails is restricted. Neither an idempotent union is inverted nor a continuation selected.

Proposition 6.1 (Totality and naturalness of the thirteen updates)**.** *All the seeds of* (6.41) *exist, each* $\mathrm{Upd}_{\varepsilon}$ *is total, and, for every coherent symmetry a and every horizon truncation,*

$$a_{*}\, \mathrm{Upd}_{\varepsilon} = \mathrm{Upd}_{a\varepsilon}\, a_{*}, \qquad \mathrm{tr}_{13;j+1,j}\, \mathrm{Upd}_{\varepsilon}\, \mathfrak{r}_{k,j}(h) = \mathfrak{r}_{k,j}(h). \tag{6.45}$$

Proof. The existence of each seed was exhibited in the thirteen paragraphs. In $\mathsf{F}, \mathsf{E}, \mathsf{G}, \mathsf{X}, \mathsf{S}, \mathsf{T}, \mathsf{L}$, the formula is concatenation with an already typed map. In $\mathsf{R}, \mathsf{N}, \mathsf{O}, \mathsf{K}, \mathsf{M}, \mathsf{B}$, the formula is, respectively, concatenation of a labeled block of equations, adjunction of a numerical datum, transport of sheets, composition of cocycles, affine action, and chain map. The naturality identities cited after each formula prove the first equality of (6.45) componentwise. The second is exactly (6.44); its sound definition uses the ordered history preserved by E and G. □

Proposition 6.2 (Joint Certificate of the Five Internal Developments)**.** *The updates of the thirteen coordinates jointly transport: the spectral–polar telescoping and return without reinitialization of theorem 5.32 y equation* (5.135)*; the orthogonal balance, the remark column, and the Stein identity of equations* (5.155) *y* (5.167)*; direct and adjoint naturality and the cellular retraction of equations* (5.216) *y* (5.228)*; totalization, the cone of the nonadic return, and the complete fibers of equations* (5.238), (5.242) *y* (5.253)*; and the Alexander–Whitney diagonal, Smith reduction, and the Bockstein tower of equations* (5.183), (5.267) *y* (5.275)*. All these actions have as their domain the same register generated by* (6.43)*; none introduces an initial subdomain, a privileged history, or a selection motivated by a subsequent application.*

Proof. Each cited identity acts on a coordinate of the common register, and its truncation maps are restrictions of (6.44). The componentwise naturality proved in (6.45) permits their assembly into a single update. Equality of domain and preservation of the ordered history preclude replacing correlative generation with five retrospective projections. □

6.1.3 Multivalued terminal without selector

The finite terminal generated by $\widehat{x}_k$ is the multisection

$$\mathfrak{T}_{k,N}(\widehat{x}_k) := \left\{ \mathfrak{r}_{k,N}(h) : h \in \mathscr{H}_{k,N}(\widehat{x}_k) \right\}. \tag{6.46}$$

The expression of the right returns the set whole. No contains a rule that distinguishes a history privileged.

The terminal level environment is obtained by gathering those multisects without identifying records:

$$\mathfrak{T}_{k,N} := \bigcup_{\widehat{x}_k \in \widehat{\mathcal{X}}_k^{\mathrm{tot}}} \mathfrak{T}_{k,N}(\widehat{x}_k). \tag{6.47}$$

The linear version, when contributions must be added, is defined on the free module by

$$\begin{aligned} \mathsf{E}^{\mathrm{all}}_{k,N} : \mathbb{Z}[\widehat{\mathcal{X}}_k^{\mathrm{tot}}] &\longrightarrow \mathbb{Z}[\mathfrak{T}_{k,N}], \\ [\widehat{x}_k] &\longmapsto \sum_{h \in \mathscr{H}_{k,N}(\widehat{x}_k)} [\mathfrak{r}_{k,N}(h)]. \end{aligned} \tag{6.48}$$

Each witness appears with its multiplicity. If exists a measure projective positive μ, the version isometric uses all the probabilities conditional:

$$\widehat{\mathsf{E}}^{\mu}_{k,N} e_{\widehat{x}_k} := \sum_{h \in \mathscr{H}_{k,N}(\widehat{x}_k)} \mu_{\widehat{x}_k,N}(h)^{1/2} e_{\mathfrak{r}_{k,N}(h)}. \tag{6.49}$$

The projectivity gives

$$\left\| \widehat{\mathsf{E}}^{\mu}_{k,N} e_{\widehat{x}_k} \right\|^2 = \sum_{h \in \mathscr{H}_{k,N}(\widehat{x}_k)} \mu_{\widehat{x}_k,N}(h) = 1. \tag{6.50}$$

The sum is not substituted by “the history that generates the cylinder”.

6.1.4 The nonadic connection as correspondence

The nonadic holonomy was generated by the odometer and the TPK connection, without a membership hypothesis, in – . In this chapter(5.71) we use (5.73)

$$\mathsf{Z}_{9,k} := \mathsf{Z}^{\Gamma}_{9,k}. \tag{6.51}$$

The maps

$$\widehat{\mathcal{X}}_k^{\mathrm{tot}} \xleftarrow{s_{9,k}} \mathsf{Z}^{\Gamma}_{9,k} \xrightarrow{t_{9,k}} \widehat{\mathcal{X}}_{k+9}^{\mathrm{tot}} \tag{6.52}$$

are those of (5.72); its source map is surjective by proposition 5.20. Every point of the apex preserves the nine phase-compatible emissions. It is not replaced by the pair of its endpoints.

For $N \geq k + 9$, define the apex of horizon before of linearizing:

$$\mathsf{Z}^{\Gamma}_{9;k,N} := \coprod_{z=(\widehat{x}_k,\boldsymbol{\omega}) \in \mathsf{Z}^{\Gamma}_{9,k}} \mathrm{Term}_{k+9,N}(t_{9,k} z). \tag{6.53}$$

An element is written (z, q), where $\boldsymbol{\omega}$ is the nonadic word of nine emissions and q its tail up to N. The maps on histories use the labeled codomain

$$\mathscr{D}^{\Gamma}_{9;k,N} := \coprod_{z \in \mathsf{Z}^{\Gamma}_{9,k}} \mathrm{Term}_{k+9,N}(t_{9,k} z). \tag{6.54}$$

Thus the maps are

$$S_{k,N}(z,q) := \omega \star q, \tag{6.55}$$

$$T_{k,N}(z,q) := (z,q) \in \mathscr{D}^{\Gamma}_{9;k,N}. \tag{6.56}$$

The first lands in $\coprod_x \mathscr{H}_{k,N}(x)$; the second expressly preserves the labelled summand by z.

The link map of the apex is the explicit action

$$\pi^{Z}_{N+1,N}(z, q \star \varepsilon) := (z,q). \tag{6.57}$$

It is surjective: if (z,q) is given, one adds to the tail the emission neutral at the endpoint of q. The codomain of tails has link

$$\pi^{\mathscr{D}}_{N+1,N}(z, q \star \varepsilon) := (z,q). \tag{6.58}$$

We denote by $\pi^{\mathscr{H},j}_{N+1,N}$ the map (6.5) with initial level j. Then

$$\pi^{\mathscr{H},k}_{N+1,N} S_{k,N+1} = S_{k,N} \pi^{Z}_{N+1,N}, \tag{6.59}$$

$$\pi^{\mathscr{D}}_{N+1,N} T_{k,N+1} = T_{k,N} \pi^{Z}_{N+1,N}. \tag{6.60}$$

In the first square, both paths send $\omega \star q \star \varepsilon$ to $\omega \star q$; in the second, they send $(z, q \star \varepsilon)$ to (z,q). By unique decomposition of a word into its first nine emissions and its tail, $S_{k,N}$ is bijective. The second square is Cartesian by the coproduct–fibered definition of (6.53). These are the Beck–Chevalley squares preserving witnesses and multiplicities.

The corresponding apex of registers, now generated by the thirteen updates, is

$$\begin{aligned} \mathsf{Z}^{\mathfrak{T}}_{9;k,N} := \big\{ & (z, q; \mathfrak{r}_{k,N}(\omega \star q), \mathfrak{r}_{k+9,N}(q)) : \\ & z = (\widehat{x}_k, \omega) \in \mathsf{Z}^{\Gamma}_{9,k}, \quad q \in \mathrm{Term}_{k+9,N}(t_{9,k} z) \big\}. \end{aligned} \tag{6.61}$$

Its source and target maps read, respectively, the primer or the second record, but never remove from their domain the witness (z,q):

$$\mathfrak{T}_{k,N} \xleftarrow{\ s^{\mathfrak{T}}_{9;k,N}\ } \mathsf{Z}^{\mathfrak{T}}_{9;k,N} \xrightarrow{\ t^{\mathfrak{T}}_{9;k,N}\ } \mathfrak{T}_{k+9,N}. \tag{6.62}$$

If $\zeta = (z, q; r_s, r_t)$, its action is

$$s^{\mathfrak{T}}_{9;k,N}(\zeta) := r_s, \qquad t^{\mathfrak{T}}_{9;k,N}(\zeta) := r_t. \tag{6.63}$$

Its link is defined by

$$\begin{aligned} & \pi^{Z,\mathfrak{T}}_{N+1,N}\big(z, q \star \varepsilon; \mathfrak{r}_{k,N+1}(\omega \star q \star \varepsilon), \mathfrak{r}_{k+9,N+1}(q \star \varepsilon)\big) \\ & \qquad := (z, q; \mathfrak{r}_{k,N}(\omega \star q), \mathfrak{r}_{k+9,N}(q)). \end{aligned} \tag{6.64}$$

The pair (z,q) is a coordinate of the apex, not a representation chosen of a triad; therefore the link is well defined even if two registros terminals coincide. The compatibility of Upd with the deletion of the ultima edge proof that (6.64) commutes with both maps of (6.62).

If an action on free modules is desired, this correspondence is linearized by summing all witnesses:

$$[\Gamma_9]_{k,N}[r] := \sum_{\zeta \in (s^{\mathfrak{T}}_{9;k,N})^{-1}(r)} [t^{\mathfrak{T}}_{9;k,N}(\zeta)]. \tag{6.65}$$

The formula preserves multiplicities. Equality of phases after nine steps does not contract the states, the witnesses, or their ledgers.

6.1.5 Uniqueness of the prefix decomposition and naturality of the truncation system

For $k \le \ell \le N$, every history of $\mathscr{H}_{k,N}(\widehat{x}_k)$ has a unique prefix $p \in \mathscr{P}_{\ell|k}(\widehat{x}_k)$. Therefore there is a disjoint decomposition by prefixes, expressed without choosing any of them:

$$\mathscr{H}_{k,N}(\widehat{x}_k) = \bigsqcup_{p \in \mathscr{P}_{\ell|k}(\widehat{x}_k)} \left\{ p \star q : q \in \mathscr{H}_{\ell,N}(p) \right\}. \tag{6.66}$$

Let Conc_p be the operation that prepends the record of the prefix p and successively applies (6.42). We define the transition in common over a complete family of terminals by

$$\mathfrak{u}_{\ell,k}\left(\left(\mathfrak{T}_{\ell,N}(p) \right)_{p \in \mathscr{P}_{\ell|k}(\widehat{x}_k)} \right) := \bigcup_{p \in \mathscr{P}_{\ell|k}(\widehat{x}_k)} \mathrm{Conc}_p\, \mathfrak{T}_{\ell,N}(p). \tag{6.67}$$

Writing $\mathbf{G}_{k,N}(\widehat{x}_k) := \mathfrak{T}_{k,N}(\widehat{x}_k)$, all naturality is concentrated in the single identity

$$\boxed{\mathfrak{u}_{\ell,k}\left(\left(\mathbf{G}_{\ell,N}(p) \right)_{p \in \mathscr{P}_{\ell|k}(\widehat{x}_k)} \right) = \mathbf{G}_{k,N}(\widehat{x}_k).} \tag{6.68}$$

There is no distinct law for each internal discriminant. The five characteristics obey (6.68) because each is updated by means of the same prefix, the same extension, and the same ledger.

The nonadic correspondence is compatible with this identity through the maps already constructed, not through a condition that is implicit. For every horizon, (6.59) and (6.60) preserve the nonadic witness with its multiplicity, and (6.64) does the same in the thirteen coordinates. If a is a coherent symmetry of the common state, its action

$$a^{\mathcal{Z}}(z, q) := (a^{\mathcal{Z}} z, aq)$$

satisfies

$$Sa^{\mathcal{Z}} = aS, \qquad Ta^{\mathcal{Z}} = aT, \qquad \pi^{\mathcal{Z}} a^{\mathcal{Z}} = a^{\mathcal{Z}} \pi^{\mathcal{Z}}, \qquad s^{\mathfrak{T}} a^{\mathcal{Z}} = a s^{\mathfrak{T}}, \qquad t^{\mathfrak{T}} a^{\mathcal{Z}} = a t^{\mathfrak{T}}. \tag{6.69}$$

These equalities are verified entry by entry using (5.81) and the naturality of the thirteen updates. They are the material condition that permits the use of (6.65); a mere equality of visible values does not replace it.

6.1.6 Passage to the limit without interchanging image and limit

Truncating a history and each of the thirteen fields defines

$$\pi^{\mathfrak{T}}_{N+1,N} : \mathfrak{T}_{k,N+1}(\widehat{x}_k) \longrightarrow \mathfrak{T}_{k,N}(\widehat{x}_k), \qquad \pi^{\mathfrak{T}}_{N+1,N} \mathfrak{r}_{k,N+1}(h) = \mathfrak{r}_{k,N}(h_{\le N}). \tag{6.70}$$

It is surjective by (6.5). The complete terminal is therefore

$$\mathfrak{T}_{k,\infty}(\widehat{x}_k) := \left\{ (r_N)_{N \ge k} : r_N \in \mathfrak{T}_{k,N}(\widehat{x}_k), \quad \pi^{\mathfrak{T}}_{N+1,N} r_{N+1} = r_N \right\}. \tag{6.71}$$

The finite sets of (6.46) are nonempty and their transition maps are surjective; therefore,

$$\mathfrak{T}_{k,\infty}(\widehat{x}_k) \neq \varnothing. \tag{6.72}$$

Corollary 6.3 (Simultaneous persistence of the five developments)**.** *In each element of $\mathfrak{T}_{k,\infty}(\widehat{x}_k)$ there simultaneously persist the strong limit $\mathcal{T}_\infty$ of the polar telescopy, the Gram family and losses, the incidence measure with its two naturalities, the cone* $\mathrm{Cone}(I - U_{\Gamma_9,N})$*, and the compatible family of Smith forms and Bockstein pages. None of these data is obtained by reconstructing only the visible terminal value.*

Proof. Surjectivity of the truncations preserves the complete witnesses. The telescoping, naturality, and reduction equations of proposition 6.2 commute with those truncations; by universality of the inverse limit, they define a unique compatible family on each surviving history. □

The apex nonadico in the limit is defined, not inferred from an image:

$$\mathsf{Z}^{\Gamma}_{9;k,\infty} := \varprojlim_{N\geq k+9} \big(\mathsf{Z}^{\Gamma}_{9;k,N}, \pi^{Z}_{N+1,N}\big). \tag{6.73}$$

Since the links do not modify the finite witness z and only truncate its tail, there exists the explicit isomorphism

$$\mathsf{Z}^{\Gamma}_{9;k,\infty} \xrightarrow{\cong} \coprod_{z\in \mathsf{Z}^{\Gamma}_{9,k}} \mathrm{Msect}(t_{9,k}z), \qquad ((z,q_N))_N \longmapsto (z,(q_N)_N). \tag{6.74}$$

The inverse takes a witness z and a compatible tail $(q_N)_N$, and forms $((z,q_N))_N$. Both compositions are identities coordinate by coordinate. The limit maps are

$$S_{k,\infty}((z_N)_N) := (S_{k,N}z_N)_N, \qquad T_{k,\infty}((z_N)_N) := (T_{k,N}z_N)_N; \tag{6.75}$$

are well defined precisely by (6.59) and (6.60).

The apex of registers is defined in the same way

$$\mathsf{Z}^{\mathfrak{T}}_{9;k,\infty} := \varprojlim_{N\geq k+9} \big(\mathsf{Z}^{\mathfrak{T}}_{9;k,N}, \pi^{Z,\mathfrak{T}}_{N+1,N}\big). \tag{6.76}$$

The correspondences (6.62) then form, component by component, the span

$$\mathfrak{T}_{k,\infty} \xleftarrow{s^{\mathfrak{T}}_{9;k,\infty}} \mathsf{Z}^{\mathfrak{T}}_{9;k,\infty} \xrightarrow{t^{\mathfrak{T}}_{9;k,\infty}} \mathfrak{T}_{k+9,\infty}. \tag{6.77}$$

The image of a limit has not been identified with the limit of certain images: (6.71), (6.73) and (6.76) construct directly the three compatible families and their maps.

6.1.7 Internal reconstruction of the enriched state from its 13 genealogical coordinates

Recovering does not unpack names: it repeats the fold that generated the record. For $h = (\varepsilon_{k+1},\dots,\varepsilon_N)$, we define

$$\boxed{\mathrm{Rec}_{13;k,N}(h) := \mathrm{Upd}_{\varepsilon_N} \circ \cdots \circ \mathrm{Upd}_{\varepsilon_{k+1}}(\mathfrak{r}^0_{k,r}).} \tag{6.78}$$

By (6.43), $\mathrm{Rec}_{13;k,N}(h) = \mathfrak{r}_{k,N}(h)$, but the equality is now an identity between two compositions of actions, not the projection of a preserved input. If pr_{A} denotes the projector from the register to field $\mathsf{A} \in \{\mathsf{F},\mathsf{E},\mathsf{R},\mathsf{N},\mathsf{O},\mathsf{K},\mathsf{M},\mathsf{B},\mathsf{G},\mathsf{X},\mathsf{S},\mathsf{T},\mathsf{L}\}$, then

$$\mathrm{pr}_{\mathsf{A}}\,\mathrm{Rec}_{13;k,N}(h) = \mathrm{Upd}^{\mathsf{A}}_{\varepsilon_N} \circ \cdots \circ \mathrm{Upd}^{\mathsf{A}}_{\varepsilon_{k+1}}(\mathsf{A}_{k,k}). \tag{6.79}$$

The proof is induction on $N-k$: for $N=k$ both sides are the seed; the inductive step applies the component A of $\mathrm{Upd}_{\varepsilon_N}$.

Además,

$$\mathrm{Rec}_{13;k,N}(h_{\leq N}) = \mathrm{tr}_{13,N+1,N}\big(\mathrm{Rec}_{13;k,N+1}(h)\big), \tag{6.80}$$

where $\mathrm{tr}_{13,N+1,N}$ removes only the last update and restricts its cone of tails. Consequently, an element of $\mathfrak{T}_{k,\infty}(\widehat{x}_k)$ determines a compatible family of the thirteen coordinates at every depth.

6.1.8 Multivalued terminality, naturality, and compatible limit of the continuum

Theorem 6.4 (Multivalued terminal, naturality, and limit of the continuum)**.** *For every $\widehat{x}_k \in \widehat{\mathcal{X}}_k^{\mathrm{tot}}$, the rules* (6.1)– (6.80) *construct, from the unique discrete continuum:*

1. *a multisection terminal finite not empty at each horizon;*
2. *an evaluation that conserves all histories and multiplicities;*
3. *the connection nonadic as a correspondence with witnesses;*
4. *the unique identity of naturality* (6.68)*;*
5. *a nonempty inverse terminal and a nonadic correspondence in the limit;*
6. *the compatible recovering of the fiber, extensions, relations, numbers, orientation, carry, memory, boundary, route, multisection, survival, terminal, and reader.*

The five intrinsic characteristics remain correlated within each record and each extension witness.

Proof. The non-emptiness of $\mathscr{H}_{k,N}(\widehat{x}_k)$ is nonempty by proposition 5.15, and its truncation is surjective by lemma 5.16. The rule (6.42) applies the APP, TRIT, and TPK identities to each witness ε_j; by induction on $N-k$, it produces all the fields of (6.43) and conserves their relations.

Taking the complete set in (6.46), or the complete sum in (6.48), proves that the terminal does not contain a selector. The weighted version is isometric by the projective equality (6.50).

The witness $z = (\widehat{x}_k, \boldsymbol{\omega})$ carries the phase action of each of its nine emissions; by proposition 5.18 it defines the two maps in (6.52) without converting the relation into a function or identifying phase with state. Equations (6.55)–(6.64) prove the two horizon squares and the same construction over registers. Summing the source space of each witness yields (6.65) without erasing multiplicities.

The partition (6.66) is disjoint because every history has a determined prefix of length ℓ. Concatenating each prefix with all its continuations reconstructs once and only once each history of $\mathscr{H}_{k,N}(\widehat{x}_k)$. This directly proves (6.68) for the whole register; five disconnected proofs are not needed.

Surjectivity of (6.70) and finiteness of each level imply nonemptiness of (6.71). The explicit links (6.57) and (6.64), together with their squares, construct the limits (6.73) and (6.76); from them follows (6.77). Finally, (6.78) reruns the successive updates, and (6.80) proves that those readings pass jointly to the limit. □

6.1.9 Obstructions to the terminality of the enriched state: loss of fiber, orientation, memory, or reader

Proposition 6.5 (Falsifiers of the terminal set)**.** *Any of the following operations replaces the constructed object and blocks the conclusion of theorem 6.4:*

1. *send to cylinder to to single history when it admits several;*
2. *choosing a tail of the terminal field* T *and eliminating the remaining ones;*
3. *replace* Γ_9 *by a function without proving uniqueness of each target and of its witness;*
4. *linearize to correspondence while ignoring multiplicities;*
5. *identifying phase return with state return;*
6. *preserve the input intact as the first coordinate and use its recoverability as purported generation;*
7. *omit any of the thirteen fields of* (6.43)*;*
8. *allowing two intrinsic characteristics to use different histories, carries, boundaries, or terminals;*
9. *replace* (6.68) *by five independent assertions;*

10. *assert operatorial compatibility of* Γ_9 *without preserving the space of witnesses and its multiplicities;*
11. *interchanging, without proof, formation of images and passage to the limit;*
12. *present a list of thirteen names without the actions* (6.42) *that produce them.*
13. *omit the polar telescoping, the loss row, the naturality adjoint, the module of all continuations or the reduction Smith–Bockstein certified in proposition 6.2;*
14. *turning any of those five developments into an autonomous branch or using a later classical application to select its domain.*

Proof. The first five operations contract the multisection or the correspondence. The next four break their common support and naturality. The final three eliminate the generative proof or alter passage to the limit. In each case, the resulting object ceases to satisfy at least one of (6.46), (6.62), (6.68), or (6.80); therefore, the conclusion does not carry over. □

Hypothesis, domain, and conclusion

The organization of the complete terminal object, unique naturality, and passage to the limit are established jointly. The surviving multisection, the carries, the register, the boundary, the route, and the nonadic connection are its structural antecedents. The linearization over free modules is proved by (6.65) and its Beck–Chevalley squares. A subsequent promotion to a bounded operator on an analytic completion additionally requires that norm to be declared and the bound proved; the correspondence (6.77) does not depend on that interface.

6.2 Joint nonadic connection Γ_9 as a compatible correspondence of the inverse system

6.2.1 Terminal Assembly Theorem of the Correlative Constructions Generated by the Common State

Each depth k possesses an enriched state $\mathcal{E}_k$ with all its surviving emissions and a single correlative constructor

$$O_k : \mathcal{E}_k \longrightarrow \mathcal{Z}_k, \qquad \widehat{x}_k \longmapsto O\big(\{\mathfrak{e}_\alpha : \alpha \in \mathrm{Ch}_k(\widehat{x}_k)\}\big). \tag{6.81}$$

The value of O_k simultaneously preserves the spectral, solenoidal, gauge, coherent, and determinantal coordinates,

$$(\mathrm{sp}, \mathrm{sol}, \mathrm{gau}, \mathrm{coh}, \mathrm{det}), \tag{6.82}$$

as five characteristics of the same state and of the same genealogical register. The truncations

$$u^{\mathfrak{e}}_{\ell,k} : \mathcal{E}_\ell \to \mathcal{E}_k, \qquad u^{O}_{\ell,k} : \mathcal{Z}_\ell \to \mathcal{Z}_k$$

satisfy the unique naturality

$$u^{O}_{\ell,k} \circ O_\ell = O_k \circ u^{\mathfrak{e}}_{\ell,k}, \qquad \ell \geq k. \tag{6.83}$$

Therefore, the corresponding action passes to the limit as

$$O_\infty : C^{\mathrm{disc}}_{\mathrm{cont}} := \varprojlim_k \mathcal{E}_k \longrightarrow \mathcal{Z}_\infty := \varprojlim_k \mathcal{Z}_k. \tag{6.84}$$

The five terminal applications are seen as intrinsic manifestations of that single value:

$$\boxed{\mathfrak{G}^{\mathrm{int}}_T := \mathrm{View}_T\Big(O_\infty(C^{\mathrm{disc}}_{\mathrm{cont}})\Big), \quad T \in \{\mathrm{sp}, \mathrm{sol}, \mathrm{gau}, \mathrm{coh}, \mathrm{det}\}.} \tag{6.85}$$

Simultaneity precedes the specialized resolutions: each view exposes the required coordinate and the complete state conserves the other four.

6.2.2 Survival and inverse limit

Let $\mathrm{Surv}_N^{(\chi)}$ be the set of histories of depth N that simultaneously satisfy the active laws of channel χ and admit prolongation. We define the compatible truncations

$$\tau_{N+1,N} : \mathrm{Surv}_{N+1}^{(\chi)} \longrightarrow \mathrm{Surv}_N^{(\chi)}$$

These preserve the enriched prefix. The genealogical continuum is the limit.

$$X_\chi = \varprojlim_N \big(\mathrm{Surv}_N^{(\chi)}, \tau_{N+1,N}\big), \qquad \tau_{N+1,N}(x_{N+1}) = x_N. \tag{6.86}$$

The global space of surviving sections and one of its enriched sections are denoted, respectively, by

$$X_\infty := \bigsqcup_\chi X_\chi, \qquad \widehat{\Omega}_\infty \in X_\infty. \tag{6.87}$$

The global Archimedean reader is

$$P_{\mathrm{arq}} := \mathrm{ev}_{\mathbb{R}} : X_\infty \longrightarrow \mathbb{R}; \tag{6.88}$$

its restriction to each channel evaluates the unitary intersection of the compatible cylinders of the corresponding section. When the exceptional reader intervenes, $X_\infty^{\mathrm{cal}} \subseteq X_\infty$ denotes the subspace in which the calibration type of (6.103) has been fixed. Survival determines the section by internal compatibility: every history must be prolongable in the complete system. The numerical reader intervenes afterward to certify and publish the intersection of the already compatible descendant.

6.2.3 Joint nonadic connection, holonomy, and vertical memory

At depth K, the holonomy of the nonadic return is initially the typed correspondence

$$\Gamma_{9,K} = \mathrm{Hol}_{C_9}(\nabla_{\mathrm{TPK}}) : \widehat{\mathcal{X}}_K \rightrightarrows \widehat{\mathcal{X}}_{K+9}. \tag{6.89}$$

Its iterates are relational compositions. Only when each state has a unique compatible prolongation is it restricted to a map, and only when that map is invertible does it constitute a monodromy. This distinction conserves all surviving extensions before selecting a section. After nine phases the local phase returns, while the complete state advances:

$$g(K+9) = g(K), \qquad q_{\mathrm{mem},K+9} = q_{\mathrm{mem},K} + 1, \qquad \widehat{x}_{K+9} \neq \widehat{x}_K \text{ in general.}$$

Block, cylinder, prefix, carry, and memory continue to evolve. This is the difference between visible periodicity and return with biography.

The transported state at level (K) can be exhibited as

$$\widehat{x}_K = (B_K, C_K, p_K, c_K, \Sigma_K, W_K, g(K), q_{\mathrm{mem},K}, s_K), \tag{6.90}$$

where each coordinate has its own transport law. The phase return is represented on the catalog.

$$\mathcal{U}_6^{\mathrm{cat}} \cong \mathfrak{S}_+ \times \mathfrak{S}_\times, \qquad |\mathfrak{S}_+| = 18, \qquad |\mathfrak{S}_\times| = 26.$$

Conditional averages

$$(E_+ f)(s,e) = \frac{1}{18} \sum_{s' \in \mathfrak{S}_+} f(s',e), \qquad (E_\times f)(s,e) = \frac{1}{26} \sum_{e' \in \mathfrak{S}_\times} f(s,e')$$

commute. The phase word $\tau = (+1, -1, 0, +1, -1, 0, +1, -1, 0)$ applies $E_+, E_\times, I$ in that order, and one revolution closes through

$$\mathcal{K}_{\mathrm{ph}}^{9} = (E_+E_\times) \otimes I_{\ell^2(C_9)}, \qquad \left\langle \delta_u, E_+E_\times \delta_v \right\rangle = \frac{1}{468} \quad (u, v \in \mathcal{U}_6^{\mathrm{cat}}). \tag{6.91}$$

The sum and product sheets do not act as independent filters: they are two components of a single holonomy that closes only after traversing the complete cycle.

The compression–expression colligation separates what returns from what remains as memory. If (J) embeds the history space into the phase space, (T) is the surviving compression and η the defect channel, then

$$\boxed{U_{\Gamma_9} J = JT + \eta, \qquad J^*\eta = 0, \qquad I - T^*T = \eta^*\eta.} \tag{6.92}$$

The first equality decomposes transport into compressed return and memory expression; the second proves their orthogonality; the third measures exactly the information that cannot fit within the periodic shadow. The central statement is thereby formalized: the phase returns, not the complete state.

The vertical memory is organized through the cyclic extension oriented

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \qquad c(r, r') = \left[\left[\frac{r + r'}{9} \right]\right]_{12}. \tag{6.93}$$

Here $\iota([b]_{12}) = [9b]_{108}$ and $p([\theta]_{108}) = [\theta]_9$. The extension does not split: C_{108} has exponent 108, whereas $C_{12} \times C_9$ has exponent 36. The cocycle records the carry from each turn of the base C_9 in the dodecaphase memory C_{12}. C_{108} is therefore an oriented phase–memory calendar. Only after the dimensional chart is that extension realized as a finite 108-step quantum clock.

The local MD lift carrying the TRIT signature is the topological–temporal unit of that genealogical crystal. On $\mathcal{H}_{\mathrm{clk}} = \mathbb{C}^{108}$, with basis $\{|j\rangle : j \in \mathbb{Z}/108\mathbb{Z}\}$, let

$$\zeta = e^{2\pi \mathrm{i}/108}, \qquad |\widetilde{k}\rangle = \frac{1}{\sqrt{108}} \sum_{j=0}^{107} \zeta^{jk} |j\rangle, \qquad U_{\mathrm{clk}}|j\rangle = |j+1\rangle.$$

With $\omega_0 = 2\pi/(108 t_0)$, the Hamiltonian

$$H_{\mathrm{clk}} = \hbar_{\mathrm{int}} \omega_0 \sum_{k=0}^{107} k \, |\widetilde{k}\rangle\langle\widetilde{k}| \tag{6.94}$$

generates exactly

$$U_{\mathrm{step}} = \exp\!\left(-\frac{\mathrm{i} H_{\mathrm{clk}} t_0}{\hbar_{\mathrm{int}}} \right) = U_{\mathrm{clk}}, \qquad U_{\mathrm{step}}^{108} = I, \tag{6.95}$$

If $Z_{\mathrm{clk}}|j\rangle = \zeta^j |j\rangle$, then

$$\langle j_0 | U_{\mathrm{step}}^{-n} Z_{\mathrm{clk}} U_{\mathrm{step}}^{n} | j_0 \rangle = \zeta^{j_0 + n}, \tag{6.96}$$

whose orbit has primitive period 108. In HMT terminology, this is the internal genealogical time crystal: the observable calendar closes, C_{12} records the circuit, and the enriched TPK state need not return. Its subsequent realization as a macroscopic physical phase also requires a Floquet protocol, primitive subharmonic response, robustness, persistence, and a protection mechanism; that chart does not alter the finite theorem of unitary periodicity.

6.2.4 Archimedean publication, genealogical number and measurement

The Archimedean publication uses semi-open ternary cylinders.

$$I_{N,L} = \left[\frac{N}{3^L}, \frac{N+1}{3^L}\right), \qquad \operatorname{diam} I_{N,L} = 3^{-L} \longrightarrow 0. \tag{6.97}$$

A compatible section determines nested cylinders and, by completeness, a unique real point. The HMT number is the section itself,

$$x = (x_N)_N \in X_\chi = \varprojlim_N \mathrm{Surv}_N^{(\chi)}, \tag{6.98}$$

not the tuple formed by it and its evaluations. The value appears when a reader forgets part of the genealogy. Therefore, precision is depth: measuring means descending to a finer cylinder while the causal prefix remains in the antecedent. The real line is reconstructed as the Archimedean quotient of a space of much richer histories.

6.2.5 Dimensional realization of the HMT state: screen map, observable magnitude, and spectral chart

Dimensional Mechanics begins where the genealogical state couples to a realization boundary. With a field K_n fixed, let $M_n(x)$ be the module generated by the prolongation channels of a state x, let $\mathcal{R}_n(x)$ be the submodule of its relations, and let $\beta_n(x)$ be the conserved boundary datum. The dimensional screen is

$$\Sigma_n(x) = \big(K_n, M_n(x), \beta_n(x), \mathcal{R}_n(x)\big), \qquad d_{\Sigma_n}(x) = \dim_{K_n} \frac{M_n(x)}{\mathcal{R}_n(x)}. \tag{6.99}$$

If $M_n(x) \simeq K_n^{g_n}$ and the columns of $R_n(x)$ generate the relations, then

$$\boxed{d_{\Sigma_n}(x) = g_n - \operatorname{rank}_{K_n} R_n(x).} \tag{6.100}$$

Dimension is not attached as a label: it is the number of independent channels that survive after imposing the boundary relations.

A physical quantity is a section of a dimensional line L_D; a unit is a chart $\chi_u : L_D \to \mathbb{R}$ that publishes its coordinate. Hence

$$\text{TPK state} \longrightarrow \Sigma_n(x) \longrightarrow s_D \in \Gamma(L_D) \xrightarrow{\chi_u} [s_D]_u \in \mathbb{R} \tag{6.101}$$

separates structure, dimensional type, and measured number exactly. Measuring consists in coupling a section to a reversible apparatus, imposing edge compatibility, and applying the reader; changing units changes χ_u, not the section or its genealogy.

6.2.6 Double projection mathematical and dimensional realization

The exceptional branch preserves gauge data that the real publication does not need. Let $\mathscr{C}_{\mathrm{exc}}$ be the space of gauges, and let

$$\mathfrak{E} : X_\infty^{\mathrm{cal}} \times \mathscr{C}_{\mathrm{exc}} \longrightarrow \mathbf{Exc} \tag{6.102}$$

the incidence reader. We fix once

$$\begin{aligned}\chi_{\mathrm{exc}} = \big(&\text{projective order},\ \text{Paley gauge},\ \text{dodecad},\\ &\text{incidence alignment},\ \text{lattice chamber}\big) \in \mathscr{C}_{\mathrm{exc}}\end{aligned} \tag{6.103}$$

and abreviamos $\mathfrak{E}_{\chi_{\mathrm{exc}}}(x) := \mathfrak{E}(x, \chi_{\mathrm{exc}})$. The gauge initial it transports by the history; not vuelve to elegirse in each depth.

The same calibrated limit then admits two readers with distinct codomains:

$$\mathbb{R} \xleftarrow{\mathrm{ev}_{\mathbb{R}}} X_{\infty}^{\mathrm{cal}} \xrightarrow{\mathfrak{E}_{\chi_{\mathrm{exc}}}} \mathbf{Exc}. \tag{6.104}$$

The first branch contracts cylinders to a continuous value. The second preserves incidences, orientations, gauge, and boundary. Both descend in parallel from the same source state. This bifurcation jointly explains metric continuity and the emergence of exceptional symmetries. The second branch is a calibrated reconstruction: it is exact once χ_{exc} has been fixed, but the bare state does not retrospectively select the five data composing that gauge.

The non-serializing dimensional realization does not serialize those two branches. It is a third reader of the same antecedent:

$$P_{\mathrm{dim}} : X_{\infty} \longrightarrow \mathcal{R}_{\mathrm{MD}}, \qquad \mathcal{R}_{\mathrm{MD}} = \{\text{action, mass, field, geometry, information}\}. \tag{6.105}$$

In this manner, a number, a symmetry, and a physical magnitude may share a genealogy without being confused with or reduced to one another.

The thesis of the continuum is thus formulated precisely: Archimedean reality is the zero-radius shadow of infinitely refinable discrete histories; what the shadow omits reappears as memory, incidence, geometry, and physical structure.

6.3 Surviving extensions, monodromy, and dynamics of the quotient

The language of paths requires a decisive precision. An APP path, a visible emission, an enriched TPK history, and a global section are not four names for the same object. They constitute four consecutive levels of a single construction. The path preserves positional order; the emission preserves a finite reading; the history adds signature, carry, orientation, boundary, and memory; the global section requires all of its prefixes to survive simultaneously at every depth.

This hierarchy makes it possible to separate two questions that had remained intermingled. The first is logical: under what hypotheses a finite prefix truly belongs to a compatible infinite extension. The second is dynamical: what it means to transport a single history and what it means to transfer all its compatible continuations simultaneously. The distinction is essential in APP. A given trajectory carries one state to another; a path transfer carries a distribution of extensions to another distribution.

This section resolves the finite problem at depth six. The published deterministic chronology has visible-phase return and cannot select a unique probability. By contrast, transport biased by the nonadic phase, when each active channel is interpreted as the normalized renewal of its entire fibre of compatible continuations, has a uniform nine-step return over the (468) emissions. The weights (1/3) then cease to be an external choice: they follow from the three occurrences of each signature (+1,-1,0) in the oriented nonadic cycle. Normalization within each fibre remains a precise mathematical premise —the uniform count of all compatible paths— and must not be conflated with the successor of a single trajectory. Passage to the inverse limit over all depths preserves a distinct statement, formulated at the end of the section.

6.3.1 5 levels of the notion of path

Let Γ_{APP} be the category of cardinal paths of the APP torus. For each depth m, let $\mathcal{H}_m$ be the finite set of admissible enriched histories

$$h_m = (v_0, e_1, v_1, \ldots, e_m, v_m),$$

where each arrow e_j transports all fields declared by TPK. If $n \geq m$, the application

$$p_{n,m} : \mathcal{H}_n \longrightarrow \mathcal{H}_m$$

removes the tail beyond depth m. The positional shadow

$$\mathrm{sh}_m : \mathcal{H}_m \longrightarrow \Gamma_{\mathrm{APP}}$$

forgets the memory fiber and preserves the underlying APP path.

Definition 6.6 (Hierarchy of extensions)**.** We distinguish the following objects.

1. An *APP path* is a morphism of Γ_{APP}.
2. An *observable route* is a reading of that morphism that jointly conserves the sum and product leaves.
3. A *TPK extension* is a lift of the observable route to a history $h_m \in \mathcal{H}_m$, with its residue, quotient, signature, orientation, boundary, carry, and record data.
4. A *surviving extension* is a finite history that admits compatible prolongations to every horizon.
5. A *global section* is a compatible family $x = (h_m)_{m\geq 0}$ of surviving extensions.

In particular, the route is the shadow of the extension; it is not an external object that the extension subsequently chooses.

For $h \in \mathcal{H}_m$, its cylinder of prolongations is

$$Z_m(h) = \{x = (h_n)_n \in \varprojlim_n \mathcal{H}_n : x_m = h\}.$$

In particular, every family of $Z_m(h)$ also satisfies $p_{n,\ell}(h_n) = h_\ell$ for $n \geq \ell$; collections of incompatible truncations are not admitted. Let us also define

$$\mathcal{S}_m^{(N)} = p_{N,m}(\mathcal{H}_N), \qquad \mathcal{S}_m^{\infty} = \bigcap_{N\geq m} \mathcal{S}_m^{(N)}.$$

Theorem 6.7 (Conditional extension survivor criterion)**.** *Let us fix $h \in \mathcal{H}_m$ and, for $N \geq m$, write*

$$\mathcal{P}_N(h) = \{g \in \mathcal{H}_N : p_{N,m}(g) = h\}.$$

Suppose that the truncations are coherent and that each $\mathcal{P}_N(h)$ is finite. The following are equivalent:

1. *$\mathcal{P}_N(h) \neq \varnothing$ for every $N \geq m$, that is, $h \in \mathcal{S}_m^{\infty}$;*
2. *the tree of finite extensions of h is finitely branching and has a nonempty level at every depth;*
3. *$Z_m(h) \neq \varnothing$.*

Consequently, whenever the surviving finite truncations are nonempty, the set

$$\Omega_{\mathrm{H}}^{\mathrm{surv}} = \varprojlim_m \mathcal{S}_m^{\infty}$$

is exactly the space of surviving global sections. The theorem does not assert that these truncations are nonempty for every APP input: that property must be proved by means of the survival protocol.

Proof. The implication from the third condition to the first two is obtained by truncation. Now assume the first condition and order the elements of $\bigsqcup_{N\geq m} \mathcal{P}_N(h)$ by the prefix relation. Each level is finite and nonempty, and coherence of the truncations makes this union a finitely branching tree with nodes at every depth; this yields the second condition. Conversely, from the second condition König's lemma provides an infinite branch. Its truncations form a compatible family belonging to $Z_m(h)$, which is the third condition. The same construction identifies the indicated inverse limit wherever its finite truncations are nonempty. □

The theorem resolves the possible discrepancy between "extendable to each horizon separately" and "possessing a global prolongation": finiteness of the branches is the hypothesis that permits the choice

of a single compatible chain. It does not claim that all APP paths belong to $\mathrm{sh}(\Omega_{\mathrm{H}}^{\mathrm{surv}})$. That coverage requires proving that the enriched closures do not empty the corresponding fibers.

6.3.2 Phase return and monodromy relation

The return of the phase–length quotient is formalized separately in definition 1.163 y theorem 1.165:

$$(b,r) \xrightarrow{9} (b+1,r), \qquad (b,r) \xrightarrow{108} (b,r).$$

These equalities belong to the odometer 9×12. They do not constitute the proof of $F_{\mathrm{obs}}^{108} = I$, which uses the observable state and appears below.

Let ϕ be the phase coordinate with values in $\mathbb{Z}/9\mathbb{Z}$. A visible-phase return satisfies

$$\phi(h_{k+9}) = \phi(h_k).$$

The complete record may, however, satisfy

$$B(h_{k+9}) \neq B(h_k).$$

This inequality is not a failure of periodicity: it is the memory of the circuit. An ordinary action of C_9 on the complete state would require the ninth power to be the identity and would erase precisely that information.

To type the return, let H be a hexad and $P_2(H)$ its set of fifteen pairs. Let $\mathcal{B}_k$ be the record states that survive at phase k, let $\mathcal{T}_k \subseteq \mathcal{B}_k \times \mathcal{B}_{k+1}$ be the published transport relation, and suppose a fiber encoder is given

$$c_k : \mathcal{B}_k \longrightarrow P_2(H).$$

Only after declaring those encoders is the observable relation defined.

$$R_k = (c_k \times c_{k+1})(\mathcal{T}_k) \subseteq P_2(H) \times P_2(H).$$

The relational monodromy of a loop is

$$\mathcal{M}_k = R_{k+8} \circ R_{k+7} \circ \cdots \circ R_k.$$

Proposition 6.8 (Classification of the nonadic return given the encoder)**.** *Three logically distinct levels are given.*

1. *Without a proof of functionality, the return is the relation $\mathcal{M}_k$.*
2. *If each R_k is the graph of a map $\mu_k : P_2(H) \to P_2(H)$, the transport is the skew product*

$$T(k,p) = (k+1, \mu_k(p)),$$

and

$$T^9(k,p) = (k, M_k(p)), \qquad M_k = \mu_{k+8} \circ \cdots \circ \mu_k.$$

3. *If the μ_k are bijective, M_k is an automorphic holonomy. The transport descends to an ordinary action of C_9 on the complete state if and only if $M_k = I$ in all fibers.*

Proof. The first claim is the definition of image and relational composition once $(c_k)_k$ have been fixed. Under functionality, nine consecutive applications keep the first coordinate fixed and compose their actions on the second. The composition of bijections is a bijection. Finally, $T^9 = I$ is exactly equivalent to $M_k = I$ for each phase and each point of the fiber. □

Corollary 6.9 (No return under cumulative transport)**.** *Let $q : X \to V$, $T : X \to X$, and $x \in X$ be given. If*

$$q(T^9 x) = q(x), \qquad T^9 x = \mathcal{M}_x\, x, \qquad \mathcal{M}_x\, x \neq x,$$

then the projection returns and the state does not:

$$q(T^9 x) = q(x), \qquad T^9 x \neq x.$$

The pointwise hypothesis $\mathcal{M}_x\, x \neq x$ is indispensable; the global inequality $\mathcal{M}_x \neq I$ does not exclude x from being a fixed point.

Proof. The first equality is a hypothesis. The second and the pointwise separation give $T^9 x = \mathcal{M}_x\, x \neq x$. The statement would be refuted by a case that satisfies the three premises and had $T^9 x = x$; it does not authorize replacing the third premise by $\mathcal{M}_x \neq I$. □

Theorem 6.10 (Functional sufficiency of the enriched tuple)**.** *Let*

$$E : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow \mathcal{S}_E$$

the tuple that conserves the visible block, new blocks and carry, lifting state, cylinders, boundary signature, orientation, observable, phase, and dodecaphasic register. Let

$$T : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \to \mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}, \qquad O : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \to Y.$$

When maps exist

$$\widehat{T} : \mathcal{S}_E \to \mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}, \qquad \widehat{O} : \mathcal{S}_E \to Y$$

such that

$$T = \widehat{T} \circ E, \qquad O = \widehat{O} \circ E,$$

the equality $E(x) = E(y)$ implies $T(x) = T(y)$, $O(x) = O(y)$ and, by determinism, the same continuations and subsequent outputs.

Proof. By factorization,

$$T(x) = \widehat{T}(E(x)) = \widehat{T}(E(y)) = T(y), \qquad O(x) = \widehat{O}(E(x)) = \widehat{O}(E(y)) = O(y).$$

The first equality permits iteration of the argument. This is a sufficient condition, not an unproved equivalence. Two states with the same tuple and different transitions or outputs, access to an absent coordinate, or querying a target value are falsifiers. The visible five-element microfiber of proposition 8.36 controls only the nonfaithfulness of the word w_6; it does not prove this positive factorization. □

The published finite window contains nine phase returns with change of the enriched state and a local selection from three continuations to one upon increasing two horizons. This proves return with memory and refutes the identification of the register with a memoryless cycle. Global extraction of the nine encoders c_k and, through them, of the relations R_k, from all surviving histories is a distinct operation. The preceding proposition classifies the return once the register–pairs map is given; it does not derive that map from the enriched TPK state.

The concrete episode of the ninth phase, with its three continuations, the block B_{11}, the boundary $(502, 662, 638)$, and the integer inequalities of cylinders, is proved only once in the theorem 8.39. In this residence only its typed consequence is used:

$$\#\,\mathrm{Surv}_1(B_9) = 3, \qquad \#\,\mathrm{Surv}_2(B_9) = 1.$$

The other two words can precede the same ternary tail only by changing the deep Hensel extension. They share part of the visible residue, but not the genealogy of the state.

6.3.3 The catalog of 468 emissions as a product of channels

The finite catalogue starts from the set Ω_{seed} of 104 976 microscopic presentations. Let $\mathcal{U}_6$ be the set of distinct sextuple emissions and let

$$F : \Omega_{\mathrm{seed}} \twoheadrightarrow \mathcal{U}_6.$$

The exact census is

$$|\mathcal{U}_6| = 468, \qquad \#\{U : |F^{-1}(U)| = 144, 432, 1944\} = 432, 18, 18.$$

These counts belong to F. Comparing them with depth-six TPK histories would require an explicit map from the history domain to Ω_{seed}; no result in this section presupposes that map or identifies catalogue presentations with enriched states. The finite catalogue supplies, for every emission, an additive signature s and a multiplicative signature e. The joint map is a bijection

$$\mathcal{U}_6 \xrightarrow{\sim} \mathcal{S}_+ \times \mathcal{S}_\times, \qquad |\mathcal{S}_+| = 18, \quad |\mathcal{S}_\times| = 26.$$

Thus,

$$\boxed{468 = 18 \cdot 26}$$

is a factorization of the object and not merely of its cardinality. The first coordinate preserves how the emission sums; the second, how the product accumulates. This separation is the explicit finite form of sum–product incompatibility: both readings coexist in the state, but each projection forgets the complementary signature.

The first coordinate admits an additional parametrization that will be useful when comparing it with the nonadic boundary. If an additive seed has coordinates $(i, j) \in (\mathbb{Z}/9\mathbb{Z})^2$ and orientation $\varepsilon = +1$ for the east–south directions and -1 for north–west, its complete signature is determined bijectively by

$$\big(i + j \bmod 9, \varepsilon\big) \in (\mathbb{Z}/9\mathbb{Z}) \times \{\pm 1\}.$$

Therefore,

$$\boxed{\mathcal{S}_+ \simeq (\mathbb{Z}/9\mathbb{Z}) \times \{\pm 1\}.}$$

The product coordinate has no analogous uniformity: its 26 signatures have fibers of seeds of sizes 8, 24, and 108, with histogram $24 \times 8 + 1 \times 24 + 1 \times 108 = 324$. This asymmetry is structural. The factor 18 already contains phase and orientation; any subsequent descent of the product factor to a class of pairs must preserve additional memory or accept nonuniform fibers.

On $\ell^2(\mathcal{S}_+ \times \mathcal{S}_\times)$, let us define the conditional expectations

$$(E_+ f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{S}_+} f(s', e),$$
$$(E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{S}_\times} f(s, e').$$

The first operator renews the additive signature and preserves the multiplicative one; the second performs the dual operation. The third channel retains both.

6.3.4 The unique path and path transfer

Unlike the return of the odometer defined in definition 1.163 y theorem 1.165, the following proposition proves the return of the observable operator F_{obs}.

The published observable law updates a single cursor at each step. In the phases +1, the additive cursor advances; in the phases −1, the multiplicative cursor advances; in the phases 0, both remain stationary. After the steps $27, 54, 81, 108$, the two orientations are inverted. Denote this deterministic evolution by F_{obs}.

Proposition 6.11 (Return of the Observable Chronology)**.** *For every choice of the two initial positions and orientations, the positions and orientations are restored after* 54 *steps. Upon also restoring the phase, one obtains*

$$F_{\mathrm{obs}}^{108} = I.$$

Therefore, the Poincaré return of 108 *steps induces the identity on* $\mathcal{U}_6$ *and does not select a unique stationary measure.*

Proof. In every block of 27 steps, each cursor makes exactly nine advances in a fixed orientation. Since the position space is $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$, its displacement is $9d = 0$. At the end of the block, d is inverted. Two blocks restore position and orientation; four also restore the index in Θ_{108}. The induced return is consequently the identity. Every probability is stationary for the identity, so none is selected by that chronology. □

The problem arises even before the return if one attempts to forget the route memory. The product signature

$$e_0 = (5, 5, 5, 5, 5, 5)$$

has twenty-four representatives. After an advance in its own orientation, those representatives separate, eight by eight, among

$$(5, 5, 5, 7, 1, 7), \qquad (5, 5, 5, 7, 7, 1), \qquad (5, 5, 5, 1, 7, 7).$$

Thus, the product advance does not descend to a map of the twenty-six signatures: the representative that had been forgotten becomes relevant again. This witness explains why an individual route cannot be turned, by mere projection, into the refreshment of a complete fiber.

The APP reading is different. Instead of choosing a representative, it preserves all compatible continuations. To formalize it in the finite catalogue, consider the subalgebras of functions that depend only on one of the two signatures. With respect to uniform counting, E_+ and $E_\times$ are the conditional expectations that forget exactly the active coordinate. They are the only Markov projectors that simultaneously satisfy:

1. preservation of the coordinate inactive;
2. complete forgetting of the active coordinate;
3. preservation of the uniform counting state.

Indeed, a row that has forgotten the source coordinate must be constant on the 18, respectively 26, compatible destinations; the normalization forces the values $1/18$ and $1/26$. This is the relative canonicity of the renewal: no preference is postulated between branches that APP declares equally possible.

Let $C_9 = \mathbb{Z}/9\mathbb{Z}$, with the phase word

$$\tau = (+1, -1, 0, +1, -1, 0, +1, -1, 0).$$

Write

$$P_{+1} = E_+, \qquad P_{-1} = E_\times, \qquad P_0 = I.$$

Definition 6.12 (Phase-biased TPK transfer)**.** On $\mathcal{U}_6 \times C_9$ we define

$$\mathcal{K}_{\mathrm{ph}}\big((u,r),(v,r+1)\big) = P_{\tau(r)}(u,v),$$

and set the remaining entries to zero. This definition is relative to the reader of all compatible paths: an active phase uniformly renews the fiber read by that phase, and the neutral phase retains the emission.

Theorem 6.13 (Uniform nonadic return)**.** *The kernel $\mathcal{K}_{\mathrm{ph}}$ is doubly stochastic and irreducible, has period exactly nine, and possesses a unique stationary state,*

$$\pi_{\mathrm{ph}}(u,r) = \frac{1}{9 \cdot 468}.$$

In each phase fiber, its return of nine steps is

$$\mathcal{K}_{\mathrm{ph}}^{9}\big|_r = E_+E_\times, \qquad (E_+E_\times)(u,v) = \frac{1}{468}.$$

If the initial phase is uniformly distributed, the shadow of a step over the emissions is

$$\overline{K}_{\mathrm{ph}} = \frac{1}{3}(I + E_+ + E_\times).$$

Proof. Each $P_{\tau(r)}$ is doubly stochastic and the shift $r \mapsto r+1$ is a permutation; their skew product conserves both sums. The three operators are idempotent and commute. Since each appears three times in the nonadic word,

$$\prod_{r \in C_9} P_{\tau(r)} = E_+^3 E_\times^3 I^3 = E_+E_\times.$$

The first expectation forgets the additive signature and the second the multiplicative signature, so their composition assigns to each of the $18 \cdot 26 = 468$ destinations the probability $1/468$. Hence, in nine steps any two states of the same phase communicate; the remaining steps communicate the phases, and the kernel is irreducible. Every return must have length divisible by nine, whereas the diagonal of $\mathcal{K}_{\mathrm{ph}}^{9}$ is positive; the period is nine. Double stochasticity and irreducibility yield the unique uniform state. Finally, averaging the nine phases counts three copies of each channel and produces the final formula. □

Forgetting the phase is not a strong Markov quotient: from the same emission, two distinct phases apply I, E_+, or $E_\times$, which have distinct rows. The operator $\overline{K}_{\mathrm{ph}}$ is an averaged shadow in the stationary phase, not the chronological successor of a phaseless emission. This distinction preserves both the dynamics and the meaning of the measure.

Definition 6.14 (Averaged kernel induced by transferrence)**.** We call the phase shadow the averaged kernel of the catalog

$$\boxed{K_{\mathrm{cat}} = \frac{1}{3}(I + E_+ + E_\times).}$$

The weights are induced by the TPK calendar, which contains three occurrences of each sign. The renewal operators are induced relative to the uniform normalization of all compatible continuations of the active fiber. Neither statement identifies K_{cat} with F_{obs}.

Proposition 6.15 (Convex family of stochastic kernels with a common uniform measure)**.** *For $a, b, c \geq 0$, $a + b + c = 1$, the family*

$$K_{a,b,c} = aI + bE_+ + cE_\times$$

consists of symmetric, doubly stochastic kernels. If $b, c > 0$, they are irreducible and, because they have positive diagonal, aperiodic. Their spectrum is

$$\operatorname{Spec}(K_{a,b,c}) = \{1^{[1]}, (a+c)^{[17]}, (a+b)^{[25]}, a^{[425]}\}.$$

In particular, there exists a continuum of dynamics with the same factorization and the same uniform measure. The choice $K_{\mathrm{cat}} = K_{1/3,1/3,1/3}$ is not characterized by the census alone. It is selected jointly by the TPK phase word and the normalized transfer of the definition 6.12.

Proof. The three summands are symmetric stochastic kernels; by convexity, the same is true of $K_{a,b,c}$, and symmetry converts row stochasticity into column stochasticity. If $b, c > 0$, two steps make it possible to renew both coordinates successively and connect any pair of states. Moreover,

$$K_{a,b,c}((s,e),(s,e)) = a + \frac{b}{18} + \frac{c}{26} > 0,$$

so that the kernel is aperiodic. In the orthogonal decomposition

$$\mathbf{1} \oplus \ell_0^2(\mathcal{S}_+) \oplus \ell_0^2(\mathcal{S}_\times) \oplus \big(\ell_0^2(\mathcal{S}_+) \otimes \ell_0^2(\mathcal{S}_\times)\big),$$

the pairs $(E_+, E_\times)$ act, respectively, as $(1,1), (0,1), (1,0), (0,0)$. The four eigenvalues and the multiplicities $1, 17, 25, 425$ are thereby obtained. For $b, c > 0$, irreducibility makes the uniform stationary measure unique, which exhibits non-uniqueness even under that requirement. □

Theorem 6.16 (Exact spectrum of the balanced refreshment)**.** *The kernel K_{cat} is symmetric, doubly stochastic, irreducible and aperiodic. Its spectrum, with multiplicities, is*

$$\boxed{\operatorname{Spec}(K_{\mathrm{cat}}) = \left\{1^{[1]}, \left(\frac{2}{3}\right)^{[42]}, \left(\frac{1}{3}\right)^{[425]}\right\}.}$$

Consequently, its spectral gap is (1/3) and its only stationary state is

$$\boxed{\tau_{468}(f) = \frac{1}{468} \sum_{U \in \mathcal{U}_6} f(U).}$$

Proof. The operators E_+ and $E_\times$ are commuting orthogonal averaging projectors. The rows and columns of each sum to one; the same is true of their average. From (s,e), (s',e') is reached in two renewals, and the contribution of I provides a self-edge. This proves irreducibility and aperiodicity.

The space decomposes orthogonally into four sectors:

$$\mathbf{1}, \qquad \ell_0^2(\mathcal{S}_+), \qquad \ell_0^2(\mathcal{S}_\times), \qquad \ell_0^2(\mathcal{S}_+) \otimes \ell_0^2(\mathcal{S}_\times).$$

Their dimensions are $1, 17, 25, 17 \cdot 25$. In the constant sector the three summands act as the identity; in each one-coordinate sector they act as (I+I+0); in the interaction sector, as (I+0+0). Dividing by three gives the announced eigenvalues and multiplicities. A finite irreducible and aperiodic kernel has a unique stationary state; since it is doubly stochastic, that state is uniform. □

The theorem is exact relative to the factorization $\mathcal{U}_6 \simeq \mathcal{S}_+ \times \mathcal{S}_\times$, the TPK calendar, and the uniform count of the compatible continuations of each fiber. The family $K_{a,b,c}$ proves that the dynamics do not follow from the census alone. The proposition 6.11 distinguishes the transfer of all paths from the chronology of one of them.

6.3.5 Compensated elevation constructed from the quotient

Let us write $m(U) = |F^{-1}(U)|$. From the kernel already chosen in the quotient we define the kernel of presentations by

$$\boxed{K_{\mathrm{seed}}(x,y) = \frac{K_{\mathrm{cat}}(F(x),F(y))}{m(F(y))}}$$

This kernel is stochastic. Summing over a target fiber yields

$$\sum_{F(y)=V} K_{\mathrm{seed}}(x,y) = K_{\mathrm{cat}}(F(x),V),$$

independently of the representative x. This identity shows that the constructed lift projects exactly onto K_{cat}. It does not prove that a preexisting microscopic dynamics of the enriched TPK state descends to the catalog.

Theorem 6.17 (Reversible elevation constructed from the quotient)**.** *The probability*

$$\widetilde{\tau}(x) = \frac{1}{468\, m(F(x))}$$

is estacionaria and reversible for K_{seed}, and $F_{\#}\widetilde{\tau} = \tau_{468}$. If

$$J_F \delta_U = m(U)^{-1/2} \sum_{F(x)=U} \delta_x, \qquad D = \mathrm{diag}(m(U)),$$

then

$$K_{\mathrm{seed}} J_F = J_F \big(D^{1/2} K_{\mathrm{cat}} D^{-1/2}\big).$$

In particular, Hilbert-space intertwining without conjugation by D occurs if and only if K_{cat} commutes with the fiber multiplicities.

Proof. Normalization follows because each of the 468 fibres receives total mass (1/468). Using the symmetry of K_{cat}, for $F(x) = U$ and $F(y) = V$ we have

$$\widetilde{\tau}(x) K_{\mathrm{seed}}(x,y) = \frac{K_{\mathrm{cat}}(U,V)}{468 m(U) m(V)} = \widetilde{\tau}(y) K_{\mathrm{seed}}(y,x).$$

This proves detailed balance, stationarity, and the identity of the image measure. Applying both sides of the last identity to a basis vector δ_V gives, on the fiber U, the coefficient $K_{\mathrm{cat}}(U,V)/\sqrt{m(V)}$; that is exactly the coefficient of the conjugate operator shown. □

Theorem 6.18 (Phase Lift Faithful to Quiescence)**.** *Let $X = \Omega_{\mathrm{seed}}$, $m(u) = |F^{-1}(u)|$, and let us define*

$$\begin{aligned} L_0(x,y) &= \mathbf{1}_{\{x=y\}}, \\ L_+(x,y) &= \frac{E_+(F(x),F(y))}{m(F(y))}, \\ L_\times(x,y) &= \frac{E_\times(F(x),F(y))}{m(F(y))}. \end{aligned}$$

On $X \times C_9$, let

$$\widehat{\mathcal{K}}_{\mathrm{ph}}\big((x,r),(y,r+1)\big) = L_{\tau(r)}(x,y).$$

Then (F, id_{C_9}) satisfies the property of strong aggregability *(*strong lumpability*) from $\widehat{\mathcal{K}}_{\mathrm{ph}}$ to $\mathcal{K}_{\mathrm{ph}}$. The microscopic kernel is irreducible, has period nine, and its unique stationary state is*

$$\widetilde{\pi}_{\mathrm{ph}}(x,r) = \frac{1}{9 \cdot 468\, m(F(x))}.$$

Its nonadic return satisfies

$$\widehat{\mathcal{K}}^{9}_{\mathrm{ph}}\big((x,r),(y,r)\big) = \frac{1}{468\, m(F(y))}.$$

Proof. The sum of L_+, respectively $L_\times$, over a target fiber is E_+, respectively $E_\times$, and L_0 projects to the identity. This proves aggregability in each phase. The factors $m(F(y))$ cancel under composition: first the active signature is selected uniformly and then a representative of the obtained emission. Thus, L_+ and $L_\times$ are commuting projectors and

$$L_+ L_\times(x,y) = \frac{1}{468\, m(F(y))}.$$

The positivity of this entry connects all representatives in nine steps. The phase forces the return lengths to be multiples of nine, and the diagonal entry of the return is positive. The stationary formula is verified by first summing the $m(u)$ representatives of each emission; each of the $9 \cdot 468$ phase–emission classes receives the same mass. □

This lift improves the reading of the averaged kernel K_{seed}: in the neutral phase it preserves the microscopic representative, instead of silently remixing its fiber. Upon averaging and forgetting the phase, both constructions have the same shadow K_{cat}, but not the same hidden dynamics. The distinction is necessary in order to respect that the zero of TRIT is memory of quiescence.

6.3.6 Minimal congruences stable under advance, half-turn and 4th of a turn: 28 and 162 memory classes

The witness $e_0 = (5,5,5,5,5,5)$ shows that the product signature is too coarse to transport a route. The exact missing minimal memory can be computed. Let

$$\mathcal{X}_\times = X_9 \times \mathscr{D}_{\mathrm{card}}$$

the space of 324 product seeds. Let P denote the advance by one cell in the stored orientation, H the half-turn of that orientation, and Q the quarter-turn

$$N \longmapsto E \longmapsto S \longmapsto O \longmapsto N.$$

Theorem 6.19 (Forward and spin congruences)**.** *The minimal congruence that refines the twenty-six product signatures and is stable under P and H has 28 classes: twenty-seven have size eight, and one has size 108. The only new splitting is*

$$(5,5,5,5,5,5) \longrightarrow \mathcal{L}_0 \sqcup \mathcal{L}_1 \sqcup \mathcal{L}_2, \qquad |\mathcal{L}_j| = 8.$$

Its orbits under P, H have sizes

$$9, \quad 9, \quad 9, \quad 1.$$

By combining this refinement with the eighteen additive signatures one obtains

$$18 \cdot 28 = 504 = 468 + 36.$$

The minimal congruence stable under P and Q has 162 classes of two seeds. Each class is an orbit of the involution

$$J(i,j,d) = \big(7-i,\, 7-j,\, \overline{d}\big) \pmod 9.$$

The operators induced by P, Q act transitively on the 162 classes.

Proof. One begins with the partition by the twenty-six signatures and iteratively refines a class whenever two of its elements have images in distinct classes under any of the generators. The process terminates because $\mathcal{X}_\times$ is finite. For (P, H), the exhaustive census produces the histogram $27 \times 8 + 1 \times 108 = 324$, the four indicated orbits, and three classes over the exceptional fiber of size twenty-four. For (P, Q), it produces $162 \times 2 = 324$. Direct verification proves that J preserves every signature, commutes with both generators, and pairs exactly the two elements of each class; the graph of classes generated by P, Q is connected. □

The excess 36 and the object R_{36} have distinct types. The boundary at depth thirty-six is

$$R_{36} = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix},$$

that is, three ordered blocks of six trits, one per channel. It is not a set of thirty-six states. The term $504 - 468 = 36$, for its part, counts additional sheet memory: the half-turn that makes product transport functional adds exactly two new sheets to a single signature and, when repeated over the eighteen additive signatures, produces $18(3 - 1) = 36$. The coincidence of the integer (36) expresses two gradings of transport, boundary depth and sheet multiplicity, but does not authorize identification of their objects.

The quarter-turn Q was not hidden in F_{obs} either. The published chronology contains programmed advances and half-turns; incorporating Q is an explicit enlargement of the route alphabet. Once it is incorporated, a lazy symmetric random walk on $P^{\pm 1}, Q^{\pm 1}$ is irreducible and aperiodic on the 162 classes, and its uniform measure is unique. This is the exact mathematical form of "counting the turns": the turn is a transport coordinate and not a subsequent decimal correction.

6.3.7 Mode and Orientation Cocycles

The same rule makes it possible to define the operational counters without introducing Dimensional Mechanics. Add to the state the last active mode $m \in \{\Sigma, \Pi\}$. A signature $+1$ fixes $m = \Sigma$, a signature -1 fixes $m = \Pi$, and a signature 0 preserves the previous mode. For an arrow a, define the signed cocycle

$$\omega_{\mathrm{sw}}(a) = \begin{cases} +1, & \Pi \longrightarrow \Sigma, \\ -1, & \Sigma \longrightarrow \Pi, \\ 0, & \text{the mode is preserved.} \end{cases}$$

Its positive variation is $s(a) = |\omega_{\mathrm{sw}}(a)|$. We further define $g(a) = 1$ when the ordered pair of cardinal orientations changes and $g(a) = 0$ when it is preserved. Keeping the mode and the terminal orientation fixed at the endpoints, the sums

$$\Omega_{\mathrm{sw}}(\gamma) = \sum_{a \subset \gamma} \omega_{\mathrm{sw}}(a), \qquad S(\gamma) = \sum_{a \subset \gamma} s(a), \qquad G(\gamma) = \sum_{a \subset \gamma} g(a)$$

are additive under concatenation.

In a nonadic closed loop, the sequence of modes is

$$(\Sigma, \Pi, \Pi, \Sigma, \Pi, \Pi, \Sigma, \Pi, \Pi).$$

Therefore,

$$\Omega_{\mathrm{sw}}(C_9) = 0, \qquad S(C_9) = 6 :$$

there are three changes $\Pi \to \Sigma$, three changes $\Sigma \to \Pi$, and three retentions. In the cycle of 108 steps,

$$\#(+1) = \#(-1) = \#(0) = 36, \qquad \Omega_{\mathrm{sw}} = 0, \qquad S = 72, \qquad G = 4.$$

The final number counts the four simultaneous inversions programmed in 27, 54, 81, 108. These cocycles proceed directly from the TPK calendar.

6.3.7.1 Quotient of modes and areal-volumetric correspondence

The calendar sign, mode label, and geometric sheet are three typed objects. In particular, the phase sign is not identified with the angular observable on digits. For the geometric correspondence, we fix the typed HMT reader

$$\mathfrak{g}_{\mathrm{av}}(\Sigma) = \mathscr{H}_{\mathrm{ar}}, \qquad \mathfrak{g}_{\mathrm{av}}(\Pi) = \mathscr{H}_{\mathrm{vol}}.$$

The first assignment expresses the additive or boundary reading; the second, the multiplicative or interior reading. The map $\mathfrak{g}_{\mathrm{av}}$ is part of the sheet typing. The following result proves that, once that typing is fixed, TPK dynamics forces the incidence between them.

Theorem 6.20 (Necessary descent of leaf incidence)**.** *Let m_r be the mode obtained after applying phase r of the primitive block $(+1, -1, 0)$, with the rule*

$$+1 : \ m \mapsto \Sigma, \qquad -1 : \ m \mapsto \Pi, \qquad 0 : \ m \mapsto m.$$

The cyclic orbit of modes is (Σ, Π, Π). If

$$N_3(i, j) := \#\{r \in \mathbb{Z}/3\mathbb{Z} : (m_r, m_{r+1}) = (i, j)\},$$

then, in the order (Σ, Π),

$$\boxed{N_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.}$$

By repetition of the calendar,

$$N_9 = 3N_3, \qquad N_{27} = 9N_3, \qquad N_{108} = 36N_3.$$

After applying $\mathfrak{g}_{\mathrm{av}}$, the primitive incidence matrix is, in the order $(\mathscr{H}_{\mathrm{ar}}, \mathscr{H}_{\mathrm{vol}})$,

$$\boxed{F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.}$$

Thus, the two cross transitions, the unique volumetric self-retention, the absence of areal self-retention and the unit primitive multiplicities are consequences of the TPK mode quotient; they are not four added postulates.

Proof. The three consecutive pairs, with cyclic closure, are

$$\Sigma \longrightarrow \Pi, \qquad \Pi \longrightarrow \Pi, \qquad \Pi \longrightarrow \Sigma.$$

Each appears once, and the pair $\Sigma \to \Sigma$ does not appear. This determines the four entries of N_3. The calendars of lengths 9, 27, 108 respectively contain 3, 9, 36 copies of the primitive block, so their incidence matrices are the indicated multiples. The greatest common divisor of the nonzero entries of each long matrix removes only that global repetition and returns N_3. Finally, $\mathfrak{g}_{\mathrm{av}}$ transports Σ, Π to the areal and volumetric sheets without altering the incidence. □

The real hyperbolic realization that uses this matrix and proves its Lorentz invariance is formulated in section 3.5.6; there F_{av} is used as an antecedent, without repeating its derivation.

This theorem must be distinguished from a stronger Markov assertion. Forgetting the phase is not a strong Markov quotient of $\mathcal{K}_{\mathrm{ph}}$: the visible mode does not determine which of the three operators

$I, E_+, E_\times$ acts at the next elementary iteration. What descends without an additional hypothesis is the edge multigraph of a TPK block.

There is, however, a memory-one completion uniquely determined by that descent. Let X_{av} be the smallest subshift of finite type over $\{\mathscr{H}_{\mathrm{ar}}, \mathscr{H}_{\mathrm{vol}}\}$ that contains all the words obtained by forgetting the phase and retains only consecutive pairs. Every one-step subshift with that property must admit exactly the three observed pairs; the smallest such subshift excludes the sole unobserved pair. Its adjacency matrix is therefore F_{av}.

Corollary 6.21 (Parry Measure of the One-Memory Wrapper)**.** *The subshift X_{av} is primitive, has topological entropy* $\log\varphi$*, and possesses a unique measure of maximal entropy. Its stationary vertex weights satisfy*

$$\boxed{\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}.}$$

Proof. The Perron root of F_{av} is φ, and a positive Perron vector is $(1, \varphi)$. Since the matrix is symmetric, the Parry vertex weights are proportional to the product of the left and right vectors, that is, to $(1, \varphi^2)$. □

The Parry measure belongs to the memory-one path envelope. It is not the image of τ_{468}, nor the temporal frequency of the periodic orbit of modes. The latter is

$$\nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{ar}}) = \frac{1}{3}, \qquad \nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{vol}}) = \frac{2}{3},$$

whereas τ_{468} is uniform over emissions, and the Parry measure weights renewal histories of arbitrary length. The three measures answer distinct questions. The irrational ratio φ^{-2} is thus derived canonically from the path envelope, without attributing it to a quotient of finite cardinalities.

From chronological counting to the return marked by π The two formulas

$$\left(\frac{1}{3}, \frac{2}{3}\right) \qquad \text{and} \qquad \frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}$$

are not approximations of one another. The first counts the three instants of the primitive calendar (Σ, Π, Π); the second weights all the admissible histories of the memory-one envelope. Hence the chronological frequency is rational, and the Parry ratio is irrational. In what follows, we write

$$\frac{\pi_{\mathrm{HMT}}}{\varphi^2} = \pi_{\mathrm{HMT}}\varphi^{-2};$$

this is not division by φ^{-2}.

The appearance of φ^{-2} can be seen directly in the linear dynamics. If F_n denotes the n-th Fibonacci number,

$$F_{\mathrm{av}}^{\,n} = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix} \qquad (n \geq 1).$$

The eigenvalues of F_{av} are φ and $-\varphi^{-1}$. The contracted branch reverses orientation in one iteration. Upon passing to the quadratic memory, namely $\mathrm{Sym}^2(\mathbb{R}^2)$, that reversal is recorded before the sign disappears. In the basis $(x^2, 2xy, y^2)$, with the images of the basis vectors represented by rows, the induced operator is

$$\mathrm{Sym}^2(F_{\mathrm{av}}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \chi(t) = (t+1)(t^2 - 3t + 1).$$

Its three eigenvalues are

$$\varphi^2, \qquad -1, \qquad \varphi^{-2}.$$

Thus, φ^{-2} is the positive stable mode of second-order memory. This step explains why the golden ratio appears here as a return law and not as an added statistical mean.

The closure π belongs to another level of the same HMT state: it is the coinductive output of the regional reader already constructed, not a new eigenvalue of F_{av}. Let

$$P_{\mathrm{ar}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_{\mathrm{vol}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Among the positive operators that preserve both sheets, there is exactly one whose volumetric normalization is one and whose areal mark is the generated closure π_{HMT}:

$$D_\pi = \pi_{\mathrm{HMT}} P_{\mathrm{ar}} + P_{\mathrm{vol}}.$$

Uniqueness is relative to these three explicit clauses: positivity, diagonality with respect to the leaves, and the two normalizations. With $e_{\mathrm{ar}} = (1,0)^T$ and $e_{\mathrm{vol}} = (0,1)^T$, the marked return of length n is

$$\Theta_n := \frac{\langle e_{\mathrm{ar}}, D_\pi F_{\mathrm{av}}^n e_{\mathrm{ar}} \rangle}{\langle e_{\mathrm{vol}}, F_{\mathrm{av}}^n e_{\mathrm{vol}} \rangle} = \pi_{\mathrm{HMT}} \frac{F_{n-1}}{F_{n+1}}.$$

Therefore,

$$\boxed{\lim_{n\to\infty} \Theta_n = \frac{\pi_{\mathrm{HMT}}}{\varphi^2}.}$$

The mark π_{HMT} is applied only once, at the boundary return; it is not reinjected at every transition. Finite combinatorics produces the stable mode φ^{-2}, the coinductive reader produces the closure π_{HMT}, and D_π composes both data without reversing their causal order.

The native length 108 provides an exact witness to that convergence:

$$(F_{\mathrm{av}}^{108})_{\mathrm{ar,ar}} = F_{107} = 10284720757613717413913,$$

$$(F_{\mathrm{av}}^{108})_{\mathrm{vol,vol}} = F_{109} = 26925748508234281076009.$$

Consequently,

$$\left| \frac{F_{107}}{F_{109}} - \varphi^{-2} \right| = 6.1684979916614407936\ldots \times 10^{-46},$$

and, after the only closing marker,

$$\left| \pi_{\mathrm{HMT}} \frac{F_{107}}{F_{109}} - \frac{\pi_{\mathrm{HMT}}}{\varphi^2} \right| = 1.9378907974286976068\ldots \times 10^{-45}.$$

These errors measure the finite approximation of the return; they are not adjustments of π or of φ.

Cylindrical reader of the ratio. Let $I_n^\pi = [\underline{\pi}_n, \overline{\pi}_n]$ and $I_n^\varphi = [\underline{\varphi}_n, \overline{\varphi}_n]$ be the positive, nested cylinders produced by the two HMT readers. The interval operation

$$Q(I,J) := \left[\frac{\inf I}{(\sup J)^2}, \frac{\sup I}{(\inf J)^2} \right]$$

is the smallest interval containing x/y^2 for every $(x,y) \in I \times J$. It preserves nesting: if the input cylinders are refined, then $Q(I_n^\pi, I_n^\varphi)$ is also refined. Since their diameters tend to zero and the limit of I_n^φ is positive,

$$\bigcap_n Q(I_n^{\pi}, I_n^{\varphi}) = \left\{ \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2} \right\}.$$

With the twelve already published base-thousand blocks, the quotient interval forces thirty-five decimal digits:

$$\frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2} = 1.19998161486432666111577775331680692\ldots.$$

This construction preserves the provenance of each digit: the numerator comes from the closure cylinder and the denominator from the autoscale.

Algebraic separation between areal reading, volumetric reading and their respective codomains The complete ratio cannot arise from the rational algebra of F_{av} alone. Indeed, its rational spectral functions belong to $\mathbb{Q}(\sqrt{5})$; φ^{-2} is found there, but not the transcendental number π. Consequently, any derivation purporting to obtain π/φ^2 solely from the matrix 2×2 would be ill typed. The transcendental datum must derive from the coinductive closure reader, or from an equivalent boundary cocycle proved independently. This negative control fixes the division of labor between the finite correspondence and the number reader.

Exchange involution of the quotient The correspondence admits two exact orientations of the same law. If

$$\mathscr{R}(x, y) = \frac{x}{y^2}, \qquad \mathsf{J}(x, y) = (y, x),$$

then $\mathsf{J}^2 = I$ and

$$\mathscr{R}(\pi, \varphi) = \frac{\pi}{\varphi^2}, \qquad (\mathscr{R} \circ \mathsf{J})(\pi, \varphi) = \frac{\varphi}{\pi^2}.$$

The identities are also verified.

$$\mathscr{R}(\pi, \varphi)\mathscr{R}(\varphi, \pi) = \frac{1}{\pi\varphi}, \qquad \frac{\mathscr{R}(\pi, \varphi)}{\mathscr{R}(\varphi, \pi)} = \left(\frac{\pi}{\varphi}\right)^3.$$

Writing $\mathcal{B}_{\mathrm{av}}^{(\pi)} = \mathscr{R}(\pi, \varphi)$ also yields the exact reparametrization

$$\frac{\varphi}{\pi^2} = \frac{1}{\varphi^3 \left(\mathcal{B}_{\mathrm{av}}^{(\pi)}\right)^2}, \qquad e^{\varphi/\pi^2} = \exp\left[\frac{1}{\varphi^3 \left(\mathcal{B}_{\mathrm{av}}^{(\pi)}\right)^2}\right].$$

Thus, the change of orientation between closure and growth transports the boundary ratio to its exchanged reading:

$$\frac{\pi}{\varphi^2} \overset{\mathsf{J}}{\longleftrightarrow} \frac{\varphi}{\pi^2}.$$

The first orientation reads the marked areal closure over quadratic volumetric memory; the second reads interior growth over quadratic closure. This correspondence is a theorem of the HMT reader. Dimensional Mechanics subsequently applies the second orientation within the electronic construction, without converting the electron into an antecedent of π, φ, or of the correspondence.

Ambivalence Principle of the areal and volumetric readers The Ambivalence Principle asserts that the areal reading and the volumetric reading are two coordinated, but not interchangeable, functionals of one and the same state with memory. Their coincidence on the diagonal is

$$\mathcal{L}_{\mathrm{ar}}^{\mathrm{diag}}(x) = \mathcal{L}_{\mathrm{vol}}^{\mathrm{diag}}(x).$$

The diagonal equality does not erase the ambivalence outside of it.

On the areal–volumetric boundary, both readers factor through the same positive memory amplitude $\mathcal{M}(x)$:

$$\mathcal{L}_{\mathrm{vol}}(x) = \mu_{\mathrm{vol}}\mathcal{M}(x), \qquad \mathcal{L}_{\mathrm{ar}}(x) = \pi_{\mathrm{HMT}}\mu_{\mathrm{ar}}\mathcal{M}(x).$$

Then the Parry theorem and the return mark give

$$\boxed{\frac{\mathcal{L}_{\mathrm{ar}}(x)}{\mathcal{L}_{\mathrm{vol}}(x)} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2}.}$$

With volumetric normalization, the relative difference of readers is

$$\frac{\mathcal{L}_{\mathrm{ar}} - \mathcal{L}_{\mathrm{vol}}}{\mathcal{L}_{\mathrm{vol}}} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2} - 1 = 0.1999816148643266611\ldots.$$

The physical reading of this correspondence is undertaken later, in the Cartan–Holst chapter, after constructing holonomy, co-tetrad, connection, torsion, and curvature. Here the result is entirely HMT: an exact ratio between two readers of the same sheet memory.

They should be distinguished from the enriched extractor subsequently used in Dimensional Mechanics. There a complete history preserves the action count n_A, the twelve oriented events $e_0, \ldots, e_{11}$ and the torsional cochain at step level; these yield

$$s_C = \sum_{i=0}^{5}(e_i + e_{i+6}), \qquad W_{\pm} = 6n_A \pm s_C.$$

The channels already generated by HMT then satisfy the exact identity

$$\exp\!\left(n_A A_{\mathrm{rad}} + \frac{s_C}{6}C^*_{\mathrm{rad}}\right) = \left(q_+^{-W_+}q_-^{-W_-}\right)^{1/12}.$$

Therefore, the route–angular-character passage is not absent: it is realized on the enriched dodecaphase record. What this phase calculation proves by itself are $\Omega_{\mathrm{sw}}, S, G$; it neither replaces them nor permits reconstruction of the complete record after it has been projected.

The uniform measure on the 104 976 presentations descends to atoms of masses $1/729$, $1/243$ and $1/54$, but not to $1/468$. In the fivefold microfiber of π, the two probabilities are

$$\tau_{468}(\mathscr{R}_\pi) = \frac{5}{468}, \qquad F_{\#}\mu_{\mathrm{seed}}(\mathscr{R}_\pi) = \frac{7}{729}.$$

Both the averaged lift and the phase lift realize the first: they do not push forward the uniform measure on seeds, but assign the same total mass to each observable class. The phase lift also preserves microscopic stillness at neutral steps.

To relate phase elevation to the enriched histories of depth six, a kernel $\mathcal{K}_6$ on $\mathcal{H}_6^{\mathrm{cat}} \times C_9$ is required. It descends to $\widehat{\mathcal{K}}_{\mathrm{ph}}$ through $\rho_6 \times \mathrm{id}_{C_9}$ if and only if it satisfies, at each phase, the strong lumpability condition

$$\sum_{\rho_6(g)=y} \mathcal{K}_6((h,r),(g,r+1)) = L_{\tau(r)}(\rho_6(h), y),$$

independently of h within each fiber of ρ_6. In that case, composition with F projects to $\mathcal{K}_{\mathrm{ph}}$. Constructing a family $(\mathcal{K}_m)_{m\geq 6}$ that satisfies this identity and commutes with all truncations is the remaining step toward a transfer of the inverse limit.

6.3.8 Scale cocycle and cylindrical conservation

Let $r = (1, \varphi)$ be a Perron–Frobenius vector of F_{av}. The Parry transition matrix is

$$P_{ij} = \frac{(F_{\mathrm{av}})_{ij} r_j}{\varphi r_i} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}.$$

For an admissible route $\gamma : i \to j$ of length n,

$$\mathbb{P}(\gamma \mid i) = \frac{r_j}{\varphi^n r_i}, \qquad \boxed{\chi_P(\gamma) = \varphi^n \frac{r_i}{r_j}.}$$

The telescope proves

$$\chi_P(\gamma_2 \circ \gamma_1) = \chi_P(\gamma_2)\chi_P(\gamma_1).$$

On a closed route, $\chi_P = \varphi^n$; its square and its inverse generate the combinations

$$C_n = \frac{\chi_P^2 + \chi_P^{-2}}{2}, \qquad S_n = \frac{\chi_P^2 - \chi_P^{-2}}{2},$$

which satisfy

$$x_{n+1} = 3x_n - x_{n-1}.$$

Since F_{av} is symmetric, let

$$\ell = \frac{r}{1 + \varphi^2}, \qquad \ell^{\mathsf{T}} r = 1.$$

The measure of a cylinder is

$$\mu[x_0 \cdots x_n] = \ell_{x_0} r_{x_n} \varphi^{-n}.$$

Therefore,

$$\boxed{V_n(x) = \frac{\mu[x_0]}{\mu[x_0 \cdots x_n]} = \varphi^n \frac{r_{x_0}}{r_{x_n}} = \chi_P(x_0 \cdots x_n).}$$

For a typed dimension scale $D > 0$,

$$a_n^{(D)}(x) = V_n(x)^{1/D}$$

satisfies the cylindrical conservation law

$$\boxed{\left(a_n^{(D)}(x)\right)^D \mu[x_0 \cdots x_n] = \mu[x_0].}$$

In spatial rank three and on closed returns, we write $a_n := a_n^{(3)}$ and obtain

$$\boxed{\frac{a_9}{a_0} = \varphi^3, \qquad \frac{a_{27}}{a_0} = \varphi^9, \qquad \frac{a_{108}}{a_0} = \varphi^{36}.}$$

If $A_k := a_{9k}$ for $k \geq 1$, then, for $k \geq 2$,

$$A_{k+1} - 2A_k + A_{k-1} = A_{k-1}(\varphi^3 - 1)^2 > 0.$$

The inequality proves convexity in return depth; a physical acceleration further requires a duration chart.

The primary computation of $\mathrm{Sym}^2(F_{\mathrm{av}})$ in section 6.3.7.1 shows that its spectrum is $\{\varphi^2, -1, \varphi^{-2}\}$. Here we conserve only the corollary for the scale cocycle: on a closed route with $V_n = \varphi^n$, the stable mode obeys

$$\boxed{M_n = \varphi^{-2n} = V_n^{-2}.}$$

Identification of these three eigenspaces with the three TRIT sheets requires an intertwiner; coincidence of cardinality does not replace it. The corollary would be refuted if the closed route failed to satisfy $V_n = \varphi^n$ or if the stable mode did not have eigenvalue φ^{-2}.

6.3.9 Dynamic prolongation theorem: global hypotheses and certified domain

The following objects have been explicitly constructed and separated from the finite problem:

$$F, \qquad F_{\mathrm{obs}}, \qquad \mathcal{K}_{\mathrm{ph}}, \qquad \widehat{\mathcal{K}}_{\mathrm{ph}}, \qquad K_{\mathrm{cat}}, \qquad \tau_{468}, \qquad F_{\mathrm{av}}, \qquad \mathrm{Sym}^2(F_{\mathrm{av}}), \qquad D_\pi, \qquad Q.$$

For F_{obs}, deterministic return has been proved and an explicit witness of failure to descend as a product has been exhibited. For $\mathcal{K}_{\mathrm{ph}}$, double stochasticity, irreducibility, period nine, uniform nonadic return, and uniqueness of π_{ph} have been proved. For $\widehat{\mathcal{K}}_{\mathrm{ph}}$, strong lumpability, microscopic conservation of rest, and the compensated stationary measure have been proved. K_{cat} is no longer an average without provenance: it is the shadow of one step after the uniform phase is averaged. The TPK mode quotient induces F_{av}; its quadratic lift separates the modes $\varphi^2, -1, \varphi^{-2}$. The operator D_π marks the boundary return exactly once with the generated closure, and the interval reader Q preserves the nesting of the cylinders of π and φ. These four objects construct $\pi_{\mathrm{HMT}}/\varphi^2_{\mathrm{HMT}}$ without conflating temporal frequency, the uniform measure, and the Parry measure.

The force of this induction is relative and concrete. The TPK word forces the weights $1/3$. The APP interpretation of all compatible continuations, together with uniform counting, forces E_+ and $E_\times$. The deterministic trajectory F_{obs} does not force those operators and, indeed, does not produce them. This is not a defect of the result, but the mathematical difference between evolution along a path and transfer over a space of paths.

The finite conclusion is complete in its domain: the uniform return, the stationary measure, the lumpability at depth six, and the separation between evolution of a route and transfer of the route space remain proved by the constructed objects. In particular, $\widehat{\mathcal{K}}_{\mathrm{ph}}$ is not identified with the chronology of F_{obs}, as proposition 6.11 shows.

6.4 Correlative construction of the continuum: histories, transport, incidence and terminal

The arithmetic published by a normal form preserves a precise reading of the state that produces it. To situate that reading within HMT, the histories, sheets and operations acting on them must also be retained. This chapter brings together that common substrate. Its various constructions are obtained from the same APP–TRIT–TPK execution tree; the divisions of the exposition allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented above. An admissible emission contains the APP operation, the TRIT orientation, the sheet change, the residue and quotient, together with the effective TPK memory update. When only a residue is published, the execution history still belongs to the starting object. Two executions with identical final readings are not identified for that reason alone.

6.4.1 Admissible histories and memory by composition

Let X_k be the set of admissible histories of length k. Its elements are constructed by successive emissions from the kernel's initial domain. We write $h\varepsilon$ for the prolongation of h by an admissible emission ε, and $\rho_k(h\varepsilon) = h$ for removal of the last emission. The emission label is preserved, even when different labels produce the same published state.

On the two arithmetic sheets, the local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \qquad 0 \le r_\diamond < 9, \qquad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate likewise admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preservation of the quotient allows these publications to be composed without replacing an emission by its isolated residue.

In a memory chart, the update of an emission is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a trajectory $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of these updates. Its linear part and its translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \qquad \kappa_\alpha = \sum_{j=1}^{k} C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 6.22 (Affine compatibility of concatenation)**.** *If α precedes β, then*

$$C_{\beta\alpha} = C_\beta C_\alpha, \qquad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \qquad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β produces $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the trajectory. □

This cocycle identity is the precise content of accumulated memory. The quantity transported by a second trajectory depends on the chart update it performs; in general, it is not the sum of two registers without transport.

The nonadic return preserves a second distinction. In phase and turn coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its publication is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod 9 \right), \qquad \sigma_9^9(w, r) = (w + 1, r).$$

During nine steps, exactly one crossing from 8 to 0 occurs. The phase returns and the turn coordinate increases. The enriched holonomy also retains the memory updates accompanying that path.

6.4.2 Tree refinement and measure of histories

The finite nonempty initial set of admissible seeds is fixed, with strictly positive initial mass summing to 1. In this section each history has a finite positive number of prolongations. Removal of emissions defines the refinement maps. Kernel admissibility provides the prolongations; when a residence uses a neutral emission, this explicitly gives a section of ρ_k. Existence of that section refers to the tree including that emission, not to any arbitrary restriction of routes.

Let $d(h) \geq 1$ be the number of admissible emissions at h. Equiprobable choice among its children defines

$$\mu(h\varepsilon) = \frac{\mu(h)}{d(h)}.$$

More generally, a positive local distribution $p(\varepsilon \mid h)$, whose sum is 1, defines $\mu(h\varepsilon) = \mu(h)p(\varepsilon \mid h)$. This second formulation preserves the reader's weights when the numbers of prolongations are not uniform.

Proposition 6.23 (Cylindrical compatibility and conditional expectation)**.** *At each level we have*

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

In the spaces $L^2(X_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \qquad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^*$, $E_k J_k = I$ and $\|J_k f\| = \|f\|$.

Proof. The first identity is normalization of the local probabilities. Applying it to $|f(x)|^2$ yields isometry. Grouping by parents the sum defining $\langle J_k f, g\rangle$ gives the formula for E_k; on a function constant on each fiber, that formula returns its value. □

The compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(h)$. The measure extends from the cylinder algebra, and its uniqueness follows because these generate the sigma-algebra. If the tree is finitely branching and each history has a prolongation, the inverse limit of its levels under emission removal is nonempty. In the cylindrical topology, each nonempty cylinder has positive measure.

This measure distinguishes histories converging to the same publication. The uniform measure on terminal values would be a different reading, since it would eliminate the multiplicities that the construction has just preserved.

6.4.3 Transport by sheets and exact accounting of outputs

First consider the space with orthonormal basis consisting of the histories of level k. The transport preserving the label of each prolongation is

$$\mathcal{U}_k e_h = \sum_{\varepsilon \text{ admisible en } h} \sqrt{p(\varepsilon \mid h)}\, e_{h\varepsilon}.$$

The sets of children of distinct parents are disjoint. Normalization of p therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation through history masses is related to this one by the diagonal change of basis incorporating $\sqrt{\mu(h)}$.

The sheet label is part of the complete register of a history. We write

$$\widehat{h} = ((h_0, s_0), \ldots, (h_k, s_k)), \qquad s_j \in \{\Sigma, \Pi\},$$

where s_j is the active sheet at step j. Prolongation adds the emission and the sheet s_{k+1} determined by the operational TPK selector, preserving the entire prefix, including the initial sheet. In the basis of these lifted histories,

$$\mathcal{U}_k e_{\widehat{h}} = \sum_{\varepsilon \text{ admisible en } \widehat{h}} \sqrt{p(\varepsilon \mid \widehat{h})}\, e_{\widehat{h}\varepsilon}, \qquad P_k e_{\widehat{h}} = \mathbf{1}_{s_k=\Sigma} e_{\widehat{h}}, \quad Q_k = I - P_k.$$

Thus, P_k reads the current sheet, while histories remain distinguished by their complete prefix. Two initial histories reaching the same active sheet still have distinct children. The isometry proof by disjoint families of prolongations applies literally to this basis. Tritic orientation is another register coordinate: the value 0 preserves the threshold reading on each sheet and does not add a third sheet to this decomposition. We now separate sheet preservation and sheet exchange:

$$A_k = P_{k+1} \mathcal{U}_k P_k + Q_{k+1} \mathcal{U}_k Q_k, \qquad \eta_k = Q_{k+1} \mathcal{U}_k P_k + P_{k+1} \mathcal{U}_k Q_k.$$

Proposition 6.24 (Gram identity by sheets)**.** *For a linear transport U, whether isometric or not, the preceding decomposition satisfies*

$$A^*A + \eta^*\eta = PU^*UP + QU^*UQ.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^*A_k + \eta_k^*\eta_k = I.$$

Proof. In A^*A and $\eta^*\eta$, mixed terms containing $P_{k+1}Q_{k+1}$ vanish. Summing the remaining terms yields $P_kU^*(P_{k+1} + Q_{k+1})UP_k$, and its analogue with Q_k. The first identity follows from complementarity of the projectors; the second also uses $U^*U = I$. □

The first identity is the general formula. Replacing its right-hand side by U^*U requires the cross blocks of that Gram matrix to vanish. Isometric preservation of history transport provides that condition here; it should not automatically be attributed to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$ and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 6.25 (Finite decomposition with terminal residual)**.** *For each horizon N,*

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing, and has a positive strong limit T_∞. The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 6.24 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. The telescoping sum proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a bounded positive form and, by the representation theorem for bounded forms, an operator T_∞. Convergence is strong because, for a positive operator $S \le I$, $\|Sv\|^2 \le \langle v, Sv\rangle$; this is applied to $S = T_N - T_\infty$. □

The terminal residue is retained until its extinction in the domain under consideration is proved. This distinction prevents a phase return, by itself, from becoming an assumption of exhaustion of all histories.

6.4.4 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)\big(N_\Sigma(h) - N_\Pi(h)\big), \qquad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a reader publishes the mean over prolongations, $b_c = Eb_f$ is defined. Orientation remains in the argument of b_f; it is not removed before averaging.

Proposition 6.26 (Exact cancellation defect)**.** *The quantity*

$$\kappa_{\mathrm{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\mathrm{can}}, \qquad E(b_f^-) = (b_c)^- + \kappa_{\mathrm{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \le E|b_f|$. Writing the positive and negative parts through $(|b| \pm b)/2$ yields both equalities. □

The information omitted by publishing only the balance is the common cancellation of opposite contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fiber contains nine children with equal weights; conditional expectation is the rule that remains valid for general branching and weights.

6.4.5 Ternary incidence of the sheets and rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They give rise to six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus, the counts $4, 6, 8$ belong to distinct operations on the same sheet module.

On the six pair positions, we define

$$(Ba)_{\{L,L'\}} = a_L + a_{L'}, \qquad s(q) = \sum_{\{L,L'\}} q_{\{L,L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The reader $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \qquad \mathbb{E}W^2 = \frac{2}{9}, \qquad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fiber contains 3^5 elements. The reader equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These means describe the uniform ambient space; the distribution induced by a subset of histories must be calculated with its own multiplicities, already preserved in X_k.

To fix the ports without losing multiplicities, at depth k the labeled readings $\lambda_\Sigma(h) = (h, \Sigma)$ and $\lambda_\Pi(h) = (h, \Pi)$ are retained. Their sets are

$$I_k = \{\lambda_\Sigma(h) : h \in X_k\}, \qquad J_k = \{\lambda_\Pi(h) : h \in X_k\}.$$

They are finite and nonempty because X_k is. Values, quotients and residues are published on these ports; equality of values does not identify their labels. The product $I_k \times J_k$ is the domain of the rectangular reading, within which the support of admissible incidences is recorded.

With this level fixed, write $I = I_k$, $J = J_k$ and $A_N = \mathbb{Z}/3^N\mathbb{Z}$. Choose reference ports i_0, j_0. This choice only provides coordinates for the reconstruction formulas. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \qquad (\delta w)_{ij} = w_{ij} - w_{ij_0} - w_{i_0 j} + w_{i_0 j_0}$$

define the following sequence, valid over the finite ring A_N:

$$0 \longrightarrow A_N \longrightarrow A_N^I \oplus A_N^J \xrightarrow{\kappa} A_N^{I \times J} \xrightarrow{\delta} A_N^{(I\setminus\{i_0\})\times(J\setminus\{j_0\})} \longrightarrow 0.$$

Proposition 6.27 (Exactness of rectangular incidence)**.** *The preceding sequence is exact, with first arrow given by the constant diagonal.*

Proof. The equality $u_i - v_j = 0$ for each pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{ij_0}$ and $v_j = w_{i_0 j_0} - w_{i_0 j}$. Then $u_i - v_j = w_{ij}$. Finally, any matrix of the last term is extended with zero reference row and column; its image under δ is the initial matrix. The proof uses only addition and subtraction, so it does not require division by a unit of the ring. □

Reducing from A_{N+1} to A_N, keeping k, I_k, J_k, i_0, j_0 fixed, commutes with both maps. The reconstruction formulas are the same at each modular depth. This is the arithmetic compatibility proved in this

section. Prolongation of histories $k \to k+1$ preserves prefix labels; its maps between sets of ports are data distinct from the change of ring $N \to N+1$.

These maps are obtained from emission removal:

$$\rho_k^{\Sigma}(h\varepsilon, \Sigma) = (h, \Sigma), \qquad \rho_k^{\Pi}(h\varepsilon, \Pi) = (h, \Pi).$$

A reference history with compatible prolongations is fixed, whose existence was proved for the tree without terminal leaves. Its two labels define compatible reference ports at all levels. Coordinate transport is precomposition:

$$(L_k u)_{i'} = u_{\rho_k^{\Sigma}(i')}, \quad (L_k v)_{j'} = v_{\rho_k^{\Pi}(j')}, \quad (L_k w)_{i'j'} = w_{\rho_k^{\Sigma}(i'),\rho_k^{\Pi}(j')}.$$

For the last module of the sequence, the matrix is first extended by zero in the reference row and column, this same formula is applied and the result is restricted to the new complement of the reference ports. This rule defines L_k even when a new port has the marked port as its parent.

Proposition 6.28 (Naturality of the rectangular complex under prolongation)**.** *The preceding maps satisfy $L_k\kappa_k = \kappa_{k+1}L_k$ and $L_k\delta_k = \delta_{k+1}L_k$. They also commute with coefficient reduction $A_{N+1} \to A_N$ and compose according to prefix removal.*

Proof. The first identity follows from $u_{\rho_k^{\Sigma}(i')} - v_{\rho_k^{\Pi}(j')}$. In the second, each of the four terms defining δ is transported by the same precomposition. Compatibility of the marked ports identifies their rows and columns; when a parent is the marked port, the four terms cancel and agree with extension by zero. All formulas are sums and differences with integer coefficients, so they commute with modular reduction. Composition of precompositions is precomposition with the composite prefix, proving the last assertion. □

6.4.6 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, at a residence y, defines a free integer complex. Its reduction to A_N uses the same integer register before reduction; successive depths are therefore compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \qquad \partial e_y = g_y, \qquad \partial b_y = -3c_y g_y, \qquad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y through $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a trajectory $\alpha : y \to y'$ updates memory, its transport in the complex sends the correspondingly named generators to their new representatives, except

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 6.29 (Naturality of the memory complex)**.** *We have $\partial\Psi_\alpha = \Psi_\alpha\partial$ and $\Psi_{\beta\alpha} = \Psi_\beta\Psi_\alpha$. The maps extend linearly over the chosen coefficient ring.*

Proof. On f_y, the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'}e_{y'} + b_{y'}$. On b_y, the boundary of its image is $-3c_{y'}g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. In a concatenation, memory differences sum telescopically, proving the second identity. □

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v, reduced in the same ring where appropriate, let

$$z^- = r_y, \qquad z^+ = r_y + 3c_y v e_y + v b_y, \qquad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport that also updates v, the correction $K_\alpha = (v_{y'} - v_y) f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α of this local homotopy do not denote the dodecaphasic seal K: they belong to another domain and their function is specified by the preceding identity. Their function can also be checked in the equality

$$z^+_{y'} - \Psi_\alpha z^+_y = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'} e_{y'} + b_{y'})$.

Proposition 6.30 (Homology and memory of the local complex)**.** *Over $\mathbb{Z}$, or after reduction to A_N, we have*

$$H_0(\Gamma) = H_2(\Gamma) = 0, \qquad H_1(\Gamma) \cong A_N[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integer realization.

Proof. Define the homotopy of degree 1 by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be the projection onto the summand generated by r_y and ι its inclusion. Checking on each generator gives

$$\partial h + h\partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex deformation retracts onto the module generated by r_y, concentrated in degree 1, which proves the assertion. □

The transport Ψ_α is a shear of determinant 1. Thus, this complex explicitly preserves how c_y changes at the chain level, but its local homology does not distinguish the values of that coordinate. The full memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinantal reading, in addition to this local complex.

6.4.7 Assembly with terminal paths

For a finite horizon, the residences and their prolongations form a path category. Each residence ν is assigned the free module $W(\nu) = A_N[\mathrm{Path}(\nu, \mathrm{Term})]$, which preserves each terminal continuation as a generator. An initial path acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding section acts covariantly through Ψ.

The assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \to \cdots \to \nu_q} W(\nu_q) \otimes_{A_N} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary removes an intermediate residence by composing the adjacent arrows. At the first endpoint it transports Γ; at the last it precomposes W. The alternating signs yield the path boundary d_{bar}. The total differential is

$$D = d_{\mathrm{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d^2_{bar} in pairs. The naturality of Proposition 6.29 implies that d_{bar} commutes with ∂_Γ; the factor $(-1)^q$ cancels the cross terms of the square. Therefore, $D^2 = 0$. Compatibility is proved on path generators and is not introduced by a label of membership in a complex.

Terminal evaluation likewise does not require choosing a particular history. For each continuation h, the effective composition of its updates is applied. The result preserves the pair formed by h and its reading, with its multiplicity. If $k < \ell < N$, the continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N. Proposition 6.22 proves that evaluating directly or evaluating these two parts produces the same reading. This yields the naturality square of the terminal.

The path modules, incidences, memory transport and their differentials arise from the same prolongations. Their assembly does not consist in adding to a state five coordinates that would be preserved by definition: their relations are constructed through sums, boundaries and effective compositions, and are verified on their generators.

6.4.8 Determinants, limit and arithmetic reading

At each residence, the free local complex Γ admits its graded determinant line

$$\operatorname{Det}\Gamma = \bigotimes_{r} \left(\bigwedge^{\operatorname{rk}\Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The total assembly of the preceding section, in turn, has the modules $\operatorname{Tot}_d \mathcal{B} = \bigoplus_{q+r=d} \mathcal{B}_{q,r}$. At a finite horizon and over the same ring, they are free of finite rank and only finitely many of them are nonzero. Its line is

$$\operatorname{Det}(\operatorname{Tot}\mathcal{B}) = \bigotimes_{q,r} \left(\bigwedge^{\operatorname{rk}\mathcal{B}_{q,r}} \mathcal{B}_{q,r} \right)^{(-1)^{q+r}}.$$

This formula follows from decomposition as a direct sum in each degree. It distinguishes the line of the local complex from that of the assembly with all paths; their bases are ordered through the preserved labels. Modular reduction commutes with these exterior powers because the modules and their bases come from the same free integer modules at a fixed horizon. The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_N; the negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, the bases and the relevant homology identifications. To calculate Smith factors, the integer matrix produced before reduction is used, not an arbitrary lift of a modular matrix. A square integer block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_N only when that determinant is a unit, that is, when it is not divisible by 3. Refinement of memories and incidences allows the stability of the relevant invariant factors to be formulated and checked; these conditions do not follow from the existence of the determinant line.

Passage to the limit preserves the compatible projective systems of residues, the histories and their multiplicities. The squares proved in the preceding sections remain commutative because their maps are given by the same formulas at each level. For an Archimedean coordinate, the additional step consists in publishing nonempty cylindrical intervals, compatible by inclusion and with diameter tending to zero. Their nested compact closures have a single-point intersection: existence follows from compactness and uniqueness from the diameter bound. This is the value of the limit reading. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional notation. The value does not replace the history that produced it. The other publication preserves the incidences of the same state. The two destinations share their origin without thereby receiving the same information.

In the following chapters, the coinductive regions, dodecaphasic register and exceptional incidence are published from that substrate with their respective maps. The Hilbert realization then preserves the phase labels and specifies its own unital inclusion; duality and screens act on the codomains constructed there. The joint operations assembled here transmit histories, sheets, orientation and memory to those publications. Identifying a specific functional with an external analytic form requires, in turn, writing its

action on the generators and verifying its inner products: the conservation identities of this chapter do not replace that composition.

6.4.9 Provenance, verification and scope

The five operations are recognized by their action on the same histories and by the relations proved in the preceding sections. The affine transport U_α of memory and the linear transport $\mathcal{U}_k$ of histories have the respective domains of Sections 6.4.1 and 6.4.3. The sheet grading is $\sigma_k = P_k - Q_k$. For a history $h_j = (\varepsilon_1, \dots, \varepsilon_j)$, accumulation of the balance is made explicit through

$$M_0^{\pm} = 0, \qquad M_k^{\pm}(h_k) = \sum_{j=1}^{k} b^{\pm}(h_j), \qquad M_{k+1}^{\pm}(h_k\varepsilon) = M_k^{\pm}(h_k) + b^{\pm}(h_k\varepsilon).$$

Each summand is the signed reading of the corresponding prefix, defined in Section 6.4.4. Separating the last summand proves the update; the identities $M_k^+ - M_k^- = \sum_{j=1}^k b(h_j)$ and $M_k^+ + M_k^- = \sum_{j=1}^k |b(h_j)|$ follow from the definitions of the positive and negative parts. Accumulation and conditional cancellation are thus distinct operations on the same contributions.

Joint operation	Objects and effective action	Development and proof
Transport and grading of sheets	$\mathcal{U}_k : \ell^2(\widehat{X}_k) \to \ell^2(\widehat{X}_{k+1})$, with $\widehat{X}_k$ formed by the lifted histories; $\sigma_k = P_k - Q_k$, A_k and η_k preserve or exchange the current sheet.	Section 6.4.3; Proposition 6.24 and Theorem 6.25.
Balance, cancellation and memory	$b : X_k \to \mathbb{Z}$, $b^{\pm} : X_k \to \mathbb{N}_0$; accumulation $M_k^{\pm}$ over prefixes and cancellation $\kappa_{\mathrm{can}} = (E\lvert b_f\rvert - \lvert Eb_f\rvert)/2$ when publishing the conditional mean.	Section 6.4.4; Proposition 6.26.
Ternary incidence of the sheets	$B : \mathbb{F}_3^4 \to \mathbb{F}_3^6$, $(Ba)_{\{L,L'\}} = a_L + a_{L'}$; the complete ambient space of q is preserved, with $sB = 0$ and $W(q + Ba) = W(q)$.	First part of Section 6.4.5.
Ports and rectangular complex	$\kappa : A_N^{I_k} \oplus A_N^{J_k} \to A_N^{I_k \times J_k}$, $\kappa(u,v)_{ij} = u_i - v_j$; the difference δ and precomposition L_k preserve labels and their multiplicities.	Propositions 6.27 and 6.28.
Cellular memory, filling and determinant line	$\Psi_\alpha : \Gamma_y \to \Gamma_{y'}$, $K_\alpha \in \Gamma_{y',2}$, $\partial h_* = z^- - z^+$; the lines $\mathrm{Det}\,\Gamma$ and $\mathrm{Det}(\mathrm{Tot}\,\mathcal{B})$ are formed on the corresponding based complexes.	Sections 6.4.6–6.4.8; Propositions 6.29 and 6.30.

The table identifies five actions of the same construction: each prefix preserves its sheets, orientation, residues, quotients, memory and terminal prolongations. Port precomposition takes coordinates from one level to the refined level; coefficient reduction takes A_{N+1} to A_N. Their directions and domains are distinct and their commutative squares are made explicit in Proposition 6.28. The assembly in Section 6.4.7 simultaneously uses terminal continuations and transport of the memory complex; its composition formulas combine the table's actions on the same system of histories. Reading a determinant line as a scalar retains the basis, homology and nonsingularity conditions of Section 6.4.8.

This development brings together the constructions of state, tree and common measure; the correlative intrinsic operations; and the terminal assembly with naturality and limit proved in this section. Their balance, incidence and memory identities are incorporated, with proofs within this chapter. The five constructions of the continuum are retained as operations on the same tree. The exposition expands their local proofs and makes explicit the conditions of each specialization: isometric transport, uniform weights, extinction of the terminal and nonsingularity of a determinantal block.

The attached checker tests exact instances of the cocycles, weighted cancellation, rectangular sequence and memory complex. These checks accompany the general proofs in the text and verify the operators defined here in their respective domains. The subsequent step toward exceptional representations and the Hilbert realization preserves their actions on generators: Section 23.5 specifies the phase fiber, Weyl

operators and inclusions determining their closures. Documentary preservation and finite checks retain their function of traceability and comparison of these proofs.

6.5 Correlative construction of the continuum: histories, transport, incidence and terminal

The arithmetic published by a normal form preserves a precise reading of the state that produces it. To situate that reading within HMT, the histories, sheets and operations acting on them must also be retained. This chapter brings together that common substrate. Its various constructions are obtained from the same APP–TRIT–TPK execution tree; the divisions of the exposition allow their properties to be studied without turning them into independent domains chosen by a subsequent application.

The starting point is the formal kernel presented above. An admissible emission contains the APP operation, the TRIT orientation, the sheet change, the residue and quotient, together with the effective TPK memory update. When only a residue is published, the execution history still belongs to the starting object. Two executions with identical final readings are not identified for that reason alone.

6.5.1 Admissible histories and memory by composition

Let X_k be the set of admissible histories of length k. Its elements are constructed by successive emissions from the initial domain of the kernel. We write $h\varepsilon$ for the prolongation of h by an admissible emission ε, and $\rho_k(h\varepsilon) = h$ for deletion of the last emission. The emission label is retained, even when different labels produce the same published state.

On the two arithmetic sheets, the local data satisfy the exact divisions

$$\delta_\diamond = r_\diamond + 9q_\diamond, \qquad 0 \le r_\diamond < 9, \qquad \diamond \in \{\Sigma, \Pi\}.$$

The oriented TRIT coordinate likewise admits an expression $\Delta = o_r + 3\kappa$, with the oriented representative declared by the kernel. Preservation of the quotient allows these publications to be composed without replacing an emission by its isolated residue.

In a memory chart, the update of an emission is written

$$U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon.$$

For a trajectory $\alpha = (\varepsilon_1, \dots, \varepsilon_k)$, the effective operation is the ordered composition of these updates. Its linear part and its translation are

$$C_\alpha = C_{\varepsilon_k} \cdots C_{\varepsilon_1}, \qquad \kappa_\alpha = \sum_{j=1}^{k} C_{\varepsilon_k} \cdots C_{\varepsilon_{j+1}} \kappa_{\varepsilon_j}.$$

The empty product in the last sum is the identity.

Proposition 6.31. Affine compatibility of concatenation

If α precedes β, then

$$C_{\beta\alpha} = C_\beta C_\alpha, \qquad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta, \qquad U_{\beta\alpha} = U_\beta U_\alpha.$$

Proof. Substituting $U_\alpha(m)$ into U_β produces $C_\beta C_\alpha m + C_\beta \kappa_\alpha + \kappa_\beta$. Decomposition into linear part and translation gives the three identities. Associativity of composition proves that the result does not depend on an auxiliary subdivision of the trajectory. □

This cocycle identity is the precise content of accumulated memory. The quantity transported by a second trajectory depends on the chart update it performs; in general, it is not the sum of two registers without transport.

The nonadic return preserves a second distinction. In phase and turn coordinates, with $\bar{r} \in \{0, \dots, 8\}$, its publication is

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + 1 \pmod 9 \right), \qquad \sigma_9^9(w, r) = (w + 1, r).$$

During nine steps, exactly one crossing from 8 to 0 occurs. The phase returns and the turn coordinate increases. The enriched holonomy also retains the memory updates accompanying that path.

6.5.2 Tree refinement and measure of histories

Fix the nonempty finite initial set of admissible seeds, with strictly positive initial mass summing to 1. In this section, each history has a positive finite number of prolongations. Deletion of emissions defines the refinement maps. Admissibility of the kernel supplies the prolongations; when a residence uses a neutral emission, this explicitly gives a section of ρ_k. The existence of that section concerns the tree including that emission, not an arbitrary restriction of routes.

Let $d(h) \geq 1$ be the number of admissible emissions at h. Equiprobable choice among its children defines

$$\mu(h\varepsilon) = \frac{\mu(h)}{d(h)}.$$

More generally, a positive local distribution $p(\varepsilon \mid h)$, whose sum is 1, defines $\mu(h\varepsilon) = \mu(h)p(\varepsilon \mid h)$. This second formulation preserves the reader's weights when the numbers of prolongations are not uniform.

Proposition 6.32. Cylindrical compatibility and conditional expectation

At each level, we have

$$\sum_{\rho_k(y)=x} \mu(y) = \mu(x).$$

In the spaces $L^2(X_k, \mu_k)$, the maps

$$(J_k f)(y) = f(\rho_k(y)), \qquad (E_k g)(x) = \sum_{\rho_k(y)=x} \frac{\mu(y)}{\mu(x)} g(y)$$

satisfy $E_k = J_k^*$, $E_k J_k = I$ and $\|J_k f\| = \|f\|$.

Proof. The first identity is the normalization of the local probabilities. Applying it to $|f(x)|^2$ gives the isometry. Grouping the sum defining $\langle J_k f, g \rangle$ by parents gives the formula for E_k; on a function constant on each fiber, that formula returns its value. □

The compatible masses define a probability measure on the space of infinite histories: the cylinder of a finite history receives its mass $\mu(h)$. The measure extends from the cylinder algebra, and its uniqueness follows from the fact that cylinders generate the sigma-algebra. If the tree is finitely branching and every history has a prolongation, the inverse limit of its levels under deletion of emissions is nonempty. In the cylindrical topology, every nonempty cylinder has positive measure.

This measure distinguishes histories converging to the same publication. The uniform measure on terminal values would be a different reading, since it would eliminate the multiplicities that the construction has just preserved.

6.5.3 Transport by sheets and exact accounting of outputs

First consider the space with an orthonormal basis formed by the histories at level k. The transport preserving the label of each prolongation is

$$\mathcal{U}_k e_h = \sum_{\varepsilon \text{ admissible en } h} \sqrt{p(\varepsilon \mid h)}\, e_{h\varepsilon}.$$

The sets of children of distinct parents are disjoint. Normalization of p therefore proves $\mathcal{U}_k^* \mathcal{U}_k = I$. The representation through history masses is related to this one by the diagonal change of basis incorporating $\sqrt{\mu(h)}$.

Let P_k, Q_k be the complementary projectors onto the two sheets. We separate sheet preservation and sheet exchange:

$$A_k = P_{k+1}\mathcal{U}_k P_k + Q_{k+1}\mathcal{U}_k Q_k, \qquad \eta_k = Q_{k+1}\mathcal{U}_k P_k + P_{k+1}\mathcal{U}_k Q_k.$$

Proposition 6.33. Gram identity by sheets

For a linear transport U, whether isometric or not, the preceding decomposition satisfies

$$A^*A + \eta^*\eta = PU^*UP + QU^*UQ.$$

In particular, for $U = \mathcal{U}_k$,

$$A_k^* A_k + \eta_k^* \eta_k = I.$$

Proof. In A^*A and $\eta^*\eta$, the mixed terms containing $P_{k+1}Q_{k+1}$ vanish. Adding the remaining terms gives $P_k U^*(P_{k+1} + Q_{k+1})UP_k$, and its analogue with Q_k. The first identity follows from complementarity of the projectors; the second also uses $U^*U = I$. □

The first identity is the general formula. Replacing its right-hand side by U^*U requires the cross blocks of that Gram matrix to vanish. Isometric preservation of history transport provides that condition here; it should not automatically be attributed to a subsequent compression.

Write $F_{0,0} = I$, $F_{0,n} = A_{n-1} \cdots A_0$ and $T_N = F_{0,N}^* F_{0,N}$.

Theorem 6.34. Finite decomposition with terminal residue

For each horizon N,

$$I = \sum_{n=0}^{N-1} F_{0,n}^* \eta_n^* \eta_n F_{0,n} + T_N.$$

The sequence T_N is positive and decreasing, and has a positive strong limit T_∞. The sum of outputs converges strongly to $I - T_\infty$.

Proof. Proposition 6.33 gives $T_n - T_{n+1} = F_{0,n}^* \eta_n^* \eta_n F_{0,n}$. The telescoping sum proves the finite equality and monotonicity. For each vector, the quadratic forms decrease to a limit; polarization defines a bounded positive form and, by the representation theorem for bounded forms, an operator T_∞. Convergence is strong because, for a positive operator $S \le I$, $\|Sv\|^2 \le \langle v, Sv\rangle$; this is applied to $S = T_N - T_\infty$. □

The terminal residue is retained until its extinction in the domain under consideration is proved. This distinction prevents a phase return, by itself, from becoming an assumption of exhaustion of all histories.

6.5.4 Oriented balance and cancellation under refinement

A history carries its accumulated orientation $o(h)$ and the counts of its visits to the two sheets. The corresponding local balance is

$$b(h) = o(h)\big(N_\Sigma(h) - N_\Pi(h)\big), \qquad b^+ = \frac{|b| + b}{2}, \quad b^- = \frac{|b| - b}{2}.$$

When a reader publishes the mean over prolongations, $b_c = Eb_f$ is defined. Orientation remains in the argument of b_f; it is not removed before averaging.

Proposition 6.35. Exact cancellation defect

The quantity

$$\kappa_{\mathrm{can}} = \frac{E|b_f| - |Eb_f|}{2}$$

is nonnegative and satisfies

$$E(b_f^+) = (b_c)^+ + \kappa_{\mathrm{can}}, \qquad E(b_f^-) = (b_c)^- + \kappa_{\mathrm{can}}.$$

Proof. The weighted triangle inequality gives $|Eb_f| \le E|b_f|$. Expressing the positive and negative parts through $(|b| \pm b)/2$ yields both equalities. □

The information omitted by publishing only the balance is the common cancellation of opposite contributions. This quantity is determined from the same prolongations used by the preceding transport. The factor $1/9$ appears only when the fiber contains nine children with equal weights; conditional expectation is the rule that remains valid for general branching and weights.

6.5.5 Ternary incidence of the sheets and rectangular complex

The two sheets determine the module $V = \mathbb{F}_3 e_\Sigma \oplus \mathbb{F}_3 e_\Pi$. Its four projective directions are the lines generated by $e_\Sigma, e_\Pi, e_\Sigma + e_\Pi, e_\Sigma - e_\Pi$. They give rise to six unordered pairs. The eight nonzero vectors are their oriented representatives. Thus, the counts $4, 6, 8$ belong to distinct operations on the same sheet module.

On the six pair positions, we define

$$(Ba)_{\{L,L'\}} = a_L + a_{L'}, \qquad s(q) = \sum_{\{L,L'\}} q_{\{L,L'\}}.$$

Each direction occurs in three pairs; hence $sB = 0$ in $\mathbb{F}_3$. The reader $W(q) = \mathbf{1}_{s(q)=0} - 1/3$ is invariant under $q \mapsto q + Ba$. On the uniform ambient space $\mathbb{F}_3^6$, its moments are

$$\mathbb{E}W = 0, \qquad \mathbb{E}W^2 = \frac{2}{9}, \qquad \mathbb{E}W^3 = \frac{2}{27}.$$

Indeed, s is surjective and each fiber contains 3^5 elements. The reader equals $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$. These means describe the uniform ambient space; the distribution induced by a subset of histories must be calculated with its own multiplicities, already preserved in X_k.

For the effective ports of a level, let I and J be the nonempty sets of ports of its two sheets and $A_N = \mathbb{Z}/3^N\mathbb{Z}$. Fix reference ports i_0, j_0. The maps

$$\kappa(u, v)_{ij} = u_i - v_j, \qquad (\delta w)_{ij} = w_{ij} - w_{ij_0} - w_{i_0 j} + w_{i_0 j_0}$$

define the following sequence, valid over the finite ring A_N:

$$0 \longrightarrow A_N \longrightarrow A_N^I \oplus A_N^J \xrightarrow{\kappa} A_N^{I\times J} \xrightarrow{\delta} A_N^{(I\setminus\{i_0\})\times(J\setminus\{j_0\})} \longrightarrow 0.$$

Proposition 6.36. Exactness of rectangular incidence

The preceding sequence is exact, with first arrow given by the constant diagonal.

Proof. The equality $u_i - v_j = 0$ for each pair forces all coordinates of u and v to be the same constant. Direct substitution gives $\delta\kappa = 0$. If $\delta w = 0$, take $u_i = w_{ij_0}$ and $v_j = w_{i_0j_0} - w_{i_0j}$. Then $u_i - v_j = w_{ij}$. Finally, any matrix in the last term extends with zero reference row and column; its image under δ is the initial matrix. The proof uses only addition and subtraction, so it does not require division by a unit of the ring. □

Reduction from A_{N+1} to A_N commutes with the two maps. The reconstruction formulas are the same at every depth. Incidence and its exactness are thus compatible with arithmetic refinement of the starting state.

6.5.6 Memory complex and compatibility of incidences

The same accumulated memory coordinate $c_y \in \mathbb{Z}$, in a residence y, defines a free integer complex. Its reduction to A_N uses the same integer register before reducing it; successive depths are therefore compatible. Its generators are f_y in degree 2, e_y, b_y, r_y in degree 1, and g_y in degree 0:

$$\partial f_y = 3c_y e_y + b_y, \qquad \partial e_y = g_y, \qquad \partial b_y = -3c_y g_y, \qquad \partial r_y = 0.$$

The identity $\partial^2 = 0$ is verified on f_y through $3c_y g_y - 3c_y g_y = 0$, and is immediate on the other generators.

If a trajectory $\alpha : y \to y'$ updates memory, its transport in the complex sends the correspondingly named generators to their new representatives, except

$$\Psi_\alpha(b_y) = b_{y'} + 3(c_{y'} - c_y)e_{y'}.$$

Proposition 6.37. Naturality of the memory complex

We have $\partial\Psi_\alpha = \Psi_\alpha\partial$ and $\Psi_{\beta\alpha} = \Psi_\beta\Psi_\alpha$. The maps extend linearly over the chosen coefficient ring.

Proof. On f_y, the image of its boundary is $3c_y e_{y'} + b_{y'} + 3(c_{y'} - c_y)e_{y'} = 3c_{y'}e_{y'} + b_{y'}$. On b_y, the boundary of its image is $-3c_{y'}g_{y'} + 3(c_{y'} - c_y)g_{y'} = -3c_y g_{y'}$. The other two cases are immediate. In a concatenation, memory differences add telescopically, proving the second identity. □

The relation between two representatives of the same residence can also be written as an explicit boundary. For an integer coordinate v, reduced in the same ring where appropriate, let

$$z^- = r_y, \qquad z^+ = r_y + 3c_y v e_y + v b_y, \qquad h_* = -v f_y.$$

Then $\partial h_* = z^- - z^+$. Under a transport that also updates v, the correction $K_\alpha = (v_{y'} - v_y)f_{y'}$ satisfies $K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha$. The symbols K_α of this local homotopy do not denote the dodecaphasic seal K: they belong to another domain and their function is specified by the preceding identity. Their function can also be checked in the equality

$$z^+_{y'} - \Psi_\alpha z^+_y = \partial K_\alpha.$$

Both sides equal $(v_{y'} - v_y)(3c_{y'}e_{y'} + b_{y'})$.

Proposition 6.38. Homology and memory of the local complex

Over $\mathbb{Z}$, or after reduction to A_N, we have

$$H_0(\Gamma) = H_2(\Gamma) = 0, \qquad H_1(\Gamma) \cong A_N[r_y]$$

in the modular realization, and $H_1(\Gamma) \cong \mathbb{Z}[r_y]$ in the integer realization.

Proof. Define the degree-1 homotopy by $h(g_y) = e_y$, $h(b_y) = f_y$, and zero on the other generators. Let p be the projection onto the summand generated by r_y and ι its inclusion. Checking on each generator gives

$$\partial h + h\partial = I - \iota p.$$

In particular, on b_y the terms $3c_y e_y$ and $-3c_y e_y$ cancel. The complex deformation retracts onto the module generated by r_y, concentrated in degree 1, which proves the assertion. □

The transport Ψ_α is a shear of determinant 1. Thus, this complex explicitly preserves how c_y changes at the chain level, but its local homology does not distinguish the values of that coordinate. The full memory remains in the labels and transports of the source state; it is not recovered from H_1 alone. A memory-dependent numerical torsion requires the specified assembly and determinantal reading, in addition to this local complex.

6.5.7 Assembly with terminal paths

For a finite horizon, the residences and their prolongations form a path category. Each residence ν is assigned the free module $W(\nu) = A_N[\mathrm{Path}(\nu, \mathrm{Term})]$, which preserves each terminal continuation as a generator. An initial path acts on W by precomposition. The complex $\Gamma(\nu)$ of the preceding section acts covariantly through Ψ.

The assembly uses both transports in the same bigraded module:

$$\mathcal{B}_{q,r} = \bigoplus_{\nu_0 \to \cdots \to \nu_q} W(\nu_q) \otimes_{A_N} \Gamma_r(\nu_0).$$

The normalized path complex is used: degenerate chains containing identities are omitted. The finite horizon then bounds the number of strict prolongations and the horizontal degree. The horizontal boundary removes an intermediate residence by composing the adjacent arrows. At the first endpoint it transports Γ; at the last it precomposes W. The alternating signs yield the path boundary d_{bar}. The total differential is

$$D = d_{\mathrm{bar}} + (-1)^q \partial_\Gamma.$$

The face identities cancel the terms of d_{bar}^2 in pairs. The naturality of Proposition 6.37 implies that d_{bar} commutes with ∂_Γ; the factor $(-1)^q$ cancels the cross terms of the square. Therefore, $D^2 = 0$. Compatibility is proved on path generators and is not introduced by a label of membership in a complex.

Terminal evaluation likewise does not require choosing a particular history. For each continuation h, the effective composition of its updates is applied. The result preserves the pair formed by h and its reading, with its multiplicity. If $k < \ell < N$, the continuations decompose as a disjoint union of prefixes up to ℓ and their continuations up to N. Proposition 6.31 proves that evaluating directly or evaluating these two parts produces the same reading. This yields the naturality square of the terminal.

The path modules, incidences, memory transport and their differentials arise from the same prolongations. Their assembly does not consist in adding to a state five coordinates that would be preserved by definition: their relations are constructed through sums, boundaries and effective compositions, and are verified on their generators.

6.5.8 Determinants, limit and arithmetic reading

At each finite truncation, the free complex admits its graded determinant line

$$\operatorname{Det}\Gamma = \bigotimes_r \left(\bigwedge^{\operatorname{rk}\Gamma_r} \Gamma_r \right)^{(-1)^r}.$$

The line is defined by the modules and their degrees, both over $\mathbb{Z}$ and after reduction to A_N; the negative exponent denotes the dual module. A nonzero numerical torsion additionally requires specifying the complex over a field, the bases and the relevant homology identifications. To calculate Smith factors, the integer matrix produced before reduction is used, not an arbitrary lift of a modular matrix. A square integer block with nonzero determinant is invertible over $\mathbb{Q}$; its reduction is invertible over A_N only when that determinant is a unit, that is, when it is not divisible by 3. The refinement of memories and incidences allows the stability of the relevant invariant factors to be formulated and checked; these conditions do not follow from the existence of the determinant line.

Passage to the limit preserves the compatible projective systems of residues, the histories and their multiplicities. The squares proved in the preceding sections remain commutative because their maps are given by the same formulas at each level. For an Archimedean coordinate, the additional step consists in publishing nonempty cylindrical intervals, compatible by inclusion and with diameter tending to zero. Their nested compact closures have a single-point intersection: existence follows from compactness and uniqueness from the diameter bound. This is the value of the limit reading. If the intervals are half-open, the point may belong only to their closures; the boundary representation rule then determines its positional notation. The value does not replace the history that produced it. The other publication preserves the incidences of the same state. The two destinations share their origin without thereby receiving the same information.

The sector of normal forms studied in the arithmetic chapter is obtained through the normalized publication of TPK prefixes. Its integer evaluation and operations are subsequent readers on the substrate constructed here. Spectral functionals, finite parts and irreducible clocks are applied to these published modes. Identifying a specific functional with an external analytic form requires, in turn, writing its action on the generators and verifying its inner products: the conservation identities of this chapter do not replace that composition.

6.5.9 Provenance, verification and scope

This development brings together the author's architecture from owners 04b1, 04b2 and 04b3 of the discrete structure, and the balance, incidence and memory identities from developments 05. It jointly preserves the five constructions of the continuum as operations on a single tree. The exposition expands their local proofs and makes explicit the conditions of each specialization: isometric transport, uniform weights, extinction of the terminal and nonsingularity of a determinantal block.

The attached checker tests exact instances of the cocycles, weighted cancellation, rectangular sequence and memory complex. These checks accompany the general proofs in the text. Their scope is local: neither a finite enumeration nor the preservation of provenance files is presented as a proof of Weil positivity or as an identification of the spectral operator that has not yet been composed.

Chapter 7

Projective measure of the genealogical continuum: evolutionary conservation of unity, disintegration by values, and conditioning of futures

7.1 Projective measure and evolutionary conservation of the unit

Cylinder topology provides a notion of proximity; a theory of paths also requires weights. The evolutionary conservation of the unit here takes an exact probabilistic form: every cylinder distributes its weight among its extensions, and the total sum remains equal to one. The unit is not fixed in a cell; it is transported through refinement.

7.1.1 Consistent weights on finite levels

Let μ_n be a probability on the finite set $\mathcal{S}_n$. The family is *projectively consistent* when

$$\mu_n(a) = \sum_{\substack{b \in \mathcal{S}_{n+1} \\ \rho_{n+1,n}(b)=a}} \mu_{n+1}(b) \qquad (a \in \mathcal{S}_n). \tag{7.1}$$

This equality expresses the local preservation of the unit: the weight of a truncated history is the sum of the weights of all its children.

Theorem 7.1 (Genealogical measure on cylinders)**.** *Every family of finite probabilities that satisfies* (7.1) *determines a unique Borel probability μ_∞ on Ω_∞ such that*

$$\mu_\infty([a]_n) = \mu_n(a). \tag{7.2}$$

Proof.

$$\mathscr{A} := \{\text{finite unions of cylinders}\}$$

is an algebra: the intersection of two cylinders is either empty or a cylinder of the deeper level. Every element of $\mathscr{A}$ can be refined into a disjoint union of cylinders of a common level. If $A = \bigsqcup_{a \in I} [a]_n$, we set

$$\mu_0(A) = \sum_{a \in I} \mu_n(a).$$

Iterating (7.1) proves that the value does not depend on the chosen common level.

If a cylinder $[a]_n$ is empty, finite branching and Koenig's lemma [73] imply that there is a depth with no descendants of a. Iterated consistency then gives $\mu_n(a) = 0$. Hence μ_0 is well defined even for representations containing empty cylinders and is finitely additive.

The cylinders are both open and closed. If a decreasing sequence of elements of $\mathscr{A}$ had empty intersection, compactness would imply that one of them was already empty; hence μ_0 is continuous at the empty set and constitutes a countably additive premeasure. The Caratheodory extension theorem —equivalently, the Kolmogorov extension for this finite projective system— extends it to the σ-algebra generated by the cylinders [22, 75]. This is the Borel algebra of Ω_∞, and finiteness of the measure ensures uniqueness. □

The measure may follow from local transitions. If $P_n(b \mid a) \geq 0$ satisfies

$$\sum_{\rho_{n+1,n}(b)=a} P_n(b \mid a) = 1,$$

then $\mu_{n+1}(b) = \mu_n(a)P_n(b \mid a)$ is consistent. Uniformity is a possible choice, not an obligation. The weights may depend on the sheet, carry, orientation, a nonnegative action cost, or a record, provided they preserve the sum. For example,

$$P_n(b \mid a) = \frac{e^{-\beta a_n(b|a)}}{\sum_{b' \mapsto a} e^{-\beta a_n(b'|a)}}, \qquad a_n \geq 0,$$

defines normalized positive weights. The orthogonal action phase belongs to the kernel of section 65.4; it is not inserted into these positive probabilities.

7.1.2 Decomposition of the genealogical measure on fibers of equal Archimedean value

The Archimedean evaluation transports the measure to K:

$$\nu := \mathrm{Val}_* \, \mu_\infty, \qquad \nu(B) = \mu_\infty(\mathrm{Val}^{-1}(B)). \tag{7.3}$$

The visible probability of a region sums the weight of all the genealogies that publish it. When two sections possess the same value, their contributions are aggregated in ν, although they remain separate in μ_∞.

Proposition 7.2 (Value Decomposition)**.** *If K is a standard Borel space, there exists a family of conditional measures $\{\mu_y\}_{y \in K}$, defined for ν-almost every y, such that μ_y is supported on $\mathfrak{G}_y = \mathrm{Val}^{-1}(y)$ and*

$$\int_{\Omega_\infty} F \, \mathrm{d}\mu_\infty = \int_K \left(\int_{\mathfrak{G}_y} F \, \mathrm{d}\mu_y \right) \mathrm{d}\nu(y)$$

for every integrable function F.

This formula separates value and genealogy at the level of the measure. Physics that depends only on the value first integrates the entire fibre; a memory observable can distinguish different distributions μ_y over the same real coordinate.

7.1.3 Conditioning of the measure through restriction of the future extension tree

Let $A \subseteq \Omega_\infty$ be an observable event with $\mu_\infty(A) > 0$. The conditional measure is

$$\mu_\infty(B \mid A) = \frac{\mu_\infty(B \cap A)}{\mu_\infty(A)}. \tag{7.4}$$

When A is the reading of a register, its intersection with each cylinder eliminates the incompatible extensions. Normalization does not create new branches: it redistributes the unit among those that have survived.

Proposition 7.3 (Compatible update)**.** *If A is a union of cylinders at level m, then for every $n \geq m$ the conditional distribution on $\mathcal{S}_n$ is obtained by removing the states whose truncations do not belong to A and renormalizing the remaining weights. The conditional distributions continue to satisfy* (7.1)*.*

Proof. The identity is Bayes' rule on finite sets. The sum of the weights of the conditioned children coincides with the conditioned weight of the parent, because intersection and disjoint union commute. □

7.1.4 Combinatorial conservation and dimensional completion

The same law appears in the binomial completion:

$$1 = \sum_{r=0}^{d} \binom{d}{r} \left(\frac{9}{10}\right)^{d-r} \left(\frac{1}{10}\right)^{r}.$$

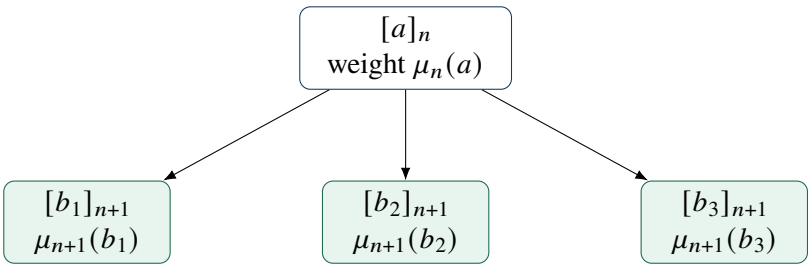
In three dimensions,

$$1000 = 729 + 243 + 27 + 1,$$

so that the interior, new faces, extreme edges, and final vertex form a partition of decimal unity. The passage $729 + 270 = 999 \to 729 + 271 = 1000$ distinguishes the state prior to closure of the completed unity. In the path system, the analogous identity is

$$\mu([a]_n) = \sum_{b \mapsto a} \mu([b]_{n+1}).$$

The unit evolves as it is distributed among the strata and is recovered by summing the complete genealogy.



$\mu_n(a) = \mu_{n+1}(b_1) + \mu_{n+1}(b_2) + \mu_{n+1}(b_3)$:
the unit is preserved when the genealogy is refined.

Figure 7.1: Projective weight conservation on cylinders.

Hypothesis, domain, and conclusion

The projective measure is constructed on the HMT inverse system. The extension theorem is classical mathematics; the identification of its cylinders with TPK states is exact once the transition weights have been fixed. The selection of a specific physical measure belongs to the dynamical model and does not follow solely from the census of branches.

The projective measure is characterized by positivity, conservation of weight between parents and children, and conditioning within the observed event. A complete experimental realization adds preparation and the sampling protocol; with those data, the same measure produces reading probabilities comparable with frequencies.

7.2 Discrete calculation of the measure on the inverse limit: differences, carry cocycle, ultrametric, reconstruction of $\mathbb{R}$, and genealogical disintegration

The distinction between counting and measuring admits an operator formulation. Counting assigns a cardinality to a collection of marks. Measuring evaluates relations among marks by means of a local rule, integrates them along a path, and preserves the chart in which the result is published. The minimal unit of that process is a difference; its composition produces length, circulation, and, when a modular reduction intervenes, carry. This architecture places the measure prior to its real coordinate: the published scalar is the contraction of a calculation possessing a domain, orientation, and memory.

7.2.1 Marks and difference operator

Let P_n be the path with ordered vertices $V(P_n) = \{v_0, \dots, v_{n-1}\}$ and oriented edges $e_j = (v_{j-1}, v_j)$, $1 \le j \le n-1$. For a commutative ring A, we define

$$C^0(P_n; A) = A^{V(P_n)}, \qquad C^1(P_n; A) = A^{E(P_n)},$$

and the difference operator

$$d : C^0(P_n; A) \longrightarrow C^1(P_n; A), \qquad (df)(e_j) = f(v_j) - f(v_{j-1}). \tag{7.5}$$

Theorem 7.4 (Range of the difference operator on a path)**.** *If A is a field, then*

$$\ker d = A\mathbf{1}, \qquad \operatorname{rank} d = n - 1.$$

Consequently, n connected marks determine $n-1$ independent differences and a unique constant mode.

Proof. The equality $df = 0$ successively implies $f(v_0) = f(v_1) = \cdots = f(v_{n-1})$, so that the kernel is formed by the constant functions and has dimension one. The rank–nullity theorem gives $\operatorname{rank} d = n - 1$. Equivalently, every assignment $(a_1, \dots, a_{n-1})$ to the edges is integrated uniquely once $f(v_0)$ has been fixed:

$$f(v_k) = f(v_0) + \sum_{j=1}^{k} a_j.$$

□

The constant mode expresses freedom of choosing the origin. Differences contain relational information; the initial mark supplies the section reconstructing positions. In the inaugural HMT cell, nine ordered marks produce eight interior differences. Periodic closure adds the ninth edge and transforms the path into a cycle; the two-dimensional completion of those intervals will be realized exactly in the APP construction.

7.2.2 Positive length and oriented integral

Two evaluations of a route must remain separate. A discrete metric is a positive cochain.

$$\ell : E(P_n) \longrightarrow \mathbb{R}_{>0}.$$

For an unoriented path $|\gamma|$, its length is

$$L_\ell(\gamma) = \sum_{e \in |\gamma|} m_\gamma(e)\, \ell(e), \tag{7.6}$$

where $m_\gamma(e)$ counts the traversals of the edge. An oriented cochain $\omega \in C^1(P_n; A)$, by contrast, is integrated with sign:

$$\int_\gamma \omega := \sum_{k=1}^{r} \varepsilon_k \omega(e_{i_k}), \qquad \varepsilon_k \in \{-1, +1\}. \tag{7.7}$$

Length sums geometric cost; the oriented integral sums transport. The first is invariant under route inversion, and the second changes sign.

Proposition 7.5 (Discrete Fundamental Theorem)**.** *For $f \in C^0(P_n; A)$ and every path orientada $\gamma : a \to b$,*

$$\int_\gamma df = f(b) - f(a).$$

In particular, the integral of an exact cocycle over a cycle is zero.

Proof. The sum of the consecutive differences telescopes. Each interior vertex appears once with positive sign and once with negative sign; only the endpoints remain. □

The proposition distinguishes two levels of closure. The group $H_1(C_n; \mathbb{Z}) \cong \mathbb{Z}$ provides the topological carrier of the turn; a holonomy further requires a transport. If $T : \mathrm{Path}(C_n) \to C$ is a functor toward a category of automorphisms, the holonomy of a cycle γ is

$$\mathrm{Hol}_T(\gamma) = T(e_r) \cdots T(e_1). \tag{7.8}$$

The cycle contributes the traversal class; T contributes the discrete connection that turns that class into an endomorphism. This distinction will be essential when the visible-phase return recovers the phase and the transport conserves a different boundary state.

7.2.3 Carry as a composition cocycle

Modular reduction publishes a class, and a section chooses representatives. Let $Q = \mathbb{Z}/9\mathbb{Z}$, and let $s_0 : Q \to \{0, \ldots, 8\} \subset \mathbb{Z}$ be the ordinary residue section. We define

$$\kappa_9(a, b) := \frac{s_0(a) + s_0(b) - s_0(a + b)}{9} \in \mathbb{Z}. \tag{7.9}$$

Then

$$s_0(a) + s_0(b) = s_0(a + b) + 9\kappa_9(a, b).$$

Proposition 7.6 (Cocycle Identity of the Carry)**.** *The map $\kappa_9 : Q \times Q \to \mathbb{Z}$ satisfies*

$$\kappa_9(a, b) + \kappa_9(a + b, c) = \kappa_9(b, c) + \kappa_9(a, b + c). \tag{7.10}$$

It is, therefore, a normalized 2-cocycle associated with the chosen section.

Proof. Both sides are expanded using (7.9). After multiplication by nine, both are equal to

$$s_0(a) + s_0(b) + s_0(c) - s_0(a + b + c).$$

Normalization is obtained from $s_0(0) = 0$. □

The cocyclic identity expresses the associativity of lifting: grouping first $(a + b) + c$ or $a + (b + c)$ produces the same visible class and the same total carry. In a category of routes, the same structure takes the form

$$\widetilde{g}\,\widetilde{h} = \kappa(g, h)\,\widetilde{gh},$$

where $\widetilde{g}, \widetilde{h}$ are enriched transports lifting visible morphisms. The carry is thus a perfectly controlled composition defect before receiving any geometric interpretation as curvature or torsion.

7.2.4 Compatibility of the nonadic, ternary, and decimal-completion charts in base 10^3

The residual readers that intervene in TPK are not distinct names of a single operation. Let

$$R_9 := \mathbb{Z}/9\mathbb{Z}, \qquad r_9 : \mathbb{Z} \longrightarrow R_9, \qquad r_3 : \mathbb{Z} \longrightarrow \mathbb{F}_3$$

the canonical reductions, and let

$$\pi_3 : R_9 \longrightarrow \mathbb{F}_3, \qquad \pi_3(\overline{n}) = n \bmod 3.$$

The square of reductions commutes:

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\ r_9\ } & R_9 \\ {\scriptstyle r_3}\downarrow & & \downarrow{\scriptstyle \pi_3} \\ \mathbb{F}_3 & = & \mathbb{F}_3. \end{array} \qquad r_3 = \pi_3 \circ r_9. \tag{7.11}$$

The positive section

$$s_+ : R_9 \longrightarrow \{1, \dots, 9\}, \qquad s_+(\overline{0}) = 9, \qquad s_+(\overline{k}) = k \quad (1 \le k \le 8),$$

produce, for $n > 0$, the root digital positive

$$\mathrm{dr}_9^+(n) = s_+(r_9(n)).$$

This section is not a homomorphism: it publishes the null class by means of 9 and forgets the quotient. The residual section $s_0(\overline{0}) = 0$ publishes that same class by means of 0. Therefore, the visible change $9 \mapsto 0$ is a change of section, and the equality

$$9 = s_0(\overline{0}) + 9 \cdot 1$$

preserves the carry 1 as quotient. The topological successor of the positive cycle is, by contrast, $9 \mapsto 1$, with

$$10 = s_+(\overline{1}) + 9 \cdot 1.$$

Thus the closure $P_9 \to C_9$, the change of chart and the increment of memory are separated; none of the three is replaced by the other two.

The coordinate of the radical belongs to another diagram. For $\mathfrak{m} = (3) \subset R_9$, the application

$$\iota_{\mathfrak{m}} : \mathbb{F}_3 \longrightarrow \mathfrak{m}, \qquad \overline{a} \longmapsto 3a,$$

is an isomorphism of vector spaces over $\mathbb{F}_3$, with inverse

$$\eta_{\mathfrak{m}} : \mathfrak{m} \longrightarrow \mathbb{F}_3, \qquad \eta_{\mathfrak{m}}(3a) = a \bmod 3. \tag{7.12}$$

It is not a ring isomorphism. Nor does it coincide with direct reduction: for $3a \in \mathfrak{m}$, one has $\pi_3(3a) = 0$, whereas $\eta_{\mathfrak{m}}(3a) = a \bmod 3$. The first reading publishes the residue class; the second publishes the internal coordinate of the radical.

Finally, let $u_j \in \{0, \dots, 999\}$ and

$$D_N = 1000 D_{N-1} + u_N, \qquad D_0 = 0,$$

the concatenation of blocks in base 10^3. Since $10^3 \equiv 1 \pmod 9$,

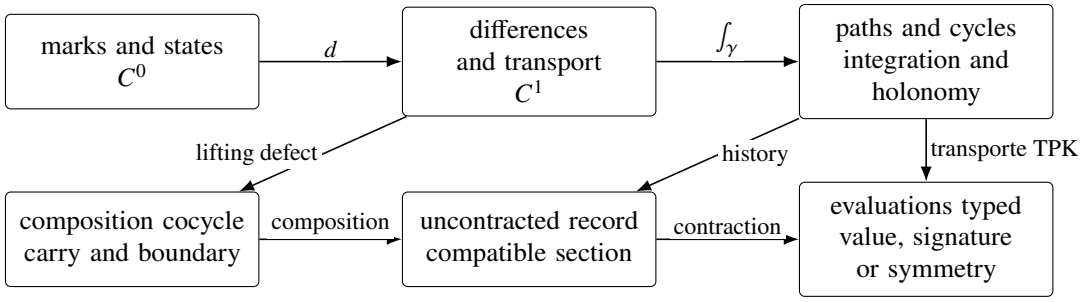


Figure 7.2: Architecture of the discrete measure complex. The difference operator, path integral, and composition cocycle occupy distinct domains; TPK assembles them into a compatible section before a reader publishes a value or a symmetry.

$$r_9(D_N) = r_9\left(\sum_{j=1}^{N} u_j\right), \qquad r_3(D_N) = r_3\left(\sum_{j=1}^{N} u_j\right). \tag{7.13}$$

The basis 10^3 preserves the order of the blocks; the reductions preserve only their accumulated class. The digital root adds a positive section; the radical coordinate recovers an internal direction of (3). This hierarchy fixes what information each reader publishes and what quotient, order, or chart TPK memory must retain in order to reconstruct the antecedent.

7.2.5 Graded system of measurement and non-contracted recording

The three previous levels are organized in a graded system:

degree 0	:	values or states on marks,
degree 1	:	differences and transports along edges,
degree 2	:	composition and carry cocycles.

Degrees zero and one form the cellular complex $C^0 \xrightarrow{d} C^1$; degree two lies in the nerve complex of the path category. Their union constitutes the *discrete measure complex*: an organizing structure that preserves the distinction between cellular geometry and categorical memory.

For an enriched route $\Gamma = (x_0 \xrightarrow{e_1} x_1 \to \cdots \xrightarrow{e_r} x_r)$, the register is the uncontracted integral

$$\Lambda(\Gamma) = \big((x_{k-1}, e_k, x_k), \omega_k, \kappa_k, \sigma_k, b_k, m_k\big)_{k=1}^{r}, \tag{7.14}$$

where ω_k is the local reading, κ_k the carry, σ_k the orientation or partial signature, b_k the boundary datum, and m_k the preserved memory. An observable is a typed map

$$O = F_O \circ \Lambda.$$

Evaluation may publish a sum, a residual class, a signature, or a scalar without converting those outputs into complete states. Concatenation of routes concatenates registers and adds the cocycle at the common boundary; the uncontracted register therefore preserves exactly the information required to recompose the history that an isolated coordinate contracts.

7.2.6 Reconstruction of the continuum from the inverse system of compatible cylinders

Prolongation to arbitrary depth is formulated by means of an inverse system. Let S_n be nonempty finite sets of enriched prefixes and $r_{n+1,n} : S_{n+1} \to S_n$ surjective restriction maps. Their space of compatible sections is

$$\Omega_\infty = \varprojlim_n S_n = \{(x_n)_n : r_{n+1,n}(x_{n+1}) = x_n\}.$$

Each S_n is endowed with the discrete topology and Ω_∞ with the subspace topology of the product.

7.2.7 Cylinders, ultrametric and Archimedean return with memory conservation

For $a \in S_n$, the prefix cylinder a is

$$[a]_n := \{x \in \Omega_\infty : x_n = a\}. \tag{7.15}$$

Two cylinders at the same level are disjoint, and every cylinder at level $n+1$ is contained in the cylinder of its restriction. This clopen basis publishes the depth shared by two sections without contracting their memory to a real coordinate.

For $x \neq y$, let

$$N(x, y) := \min\{n \geq 0 : x_n \neq y_n\}.$$

Fijado $0 < \vartheta < 1$, definimos

$$d_\vartheta(x, y) := \vartheta^{N(x,y)}, \qquad d_\vartheta(x, x) := 0. \tag{7.16}$$

Proposition 7.7 (First Divergence Ultrametric)**.** *The function d_ϑ is an ultrametric, and its balls are precisely the cylinders, with the index adjustment corresponding to the initial level:*

$$d_\vartheta(x, z) \leq \max\{d_\vartheta(x, y), d_\vartheta(y, z)\}.$$

Therefore, two sections that publish the same value can remain at positive distance when their sheet, carry, boundary or history of prolongation diverges.

Proof. If x and y agree through level n, and y and z do as well, then x and z agree through that level. The first divergence of the first and third states therefore occurs no earlier than the minimum of the other two. Since $0 < \vartheta < 1$, raising ϑ to those exponents yields the strong inequality. The description of the balls follows directly from fixing a common prefix. □

Theorem 7.8 (Compactness and completeness of the genealogical space)**.** *If each S_n is finite and the bonding maps are surjective, then Ω_∞, with d_ϑ, is nonempty, compact, and complete. If, in addition,*

$$\forall x \in \Omega_\infty \; \forall n \; \exists y \neq x : y_n = x_n, \tag{7.17}$$

has no isolated points. In that regime it is a compact metrizable, zero-dimensional, and perfect space; by the classical characterization, it is homeomorphic to Cantor space.

Proof. Surjectivity and finite branching provide a section by König's lemma. The product of the finite discrete sets is compact, and the equations $r_{n+1,n}(x_{n+1}) = x_n$ define a closed subspace. The induced topology coincides with the cylinder topology and, by proposition 7.7, with that of d_ϑ. Every compact metric space is complete. The condition (7.17) guarantees that every cylinder contains a section distinct from its center, so no singleton is open. □

Visible periodicity and state return belong to distinct maps. Let $p : E \twoheadrightarrow B$ be a projection, $T : E \to E$ an enriched transport, and $f : B \to B$ the published evolution, with $p \circ T = f \circ p$. If $f^9 = I_B$, then

$$p \circ T^9 = p, \tag{7.18}$$

so that T^9 preserves each fiber of p, but need not be the identity on it. In the nonadic coordinate

$$(q, r) \longmapsto \begin{cases} (q, r+1), & r < 9, \\ (q+1, 1), & r = 9, \end{cases}$$

one revolution recovers the position and increases the height. Under CT108 periodicity, nine iterations advance one dodecaphase phase, and one hundred eight recover position and phase; the enriched state may still preserve different sheet, boundary, orientation, and genealogy. This is the formal distinction between phase return and state return.

Theorem 7.9 (Reconstruction by Compatible Sections)**.** *The space Ω_∞ is nonempty and compact. Suppose that to each prefix $x_n \in S_n$ there is assigned a compact interval $I_n(x_n) \subset \mathbb{R}$ such that, for every section $x = (x_n)_n$,*

$$I_{n+1}(x_{n+1}) \subseteq I_n(x_n), \qquad \operatorname{diam} I_n(x_n) \longrightarrow 0.$$

Then there exists a unique value

$$\mathrm{ev}_{\mathbb{R}}(x) \in \bigcap_{n \geq 0} I_n(x_n). \tag{7.19}$$

If the interval assignments are locally constant at each level, the evaluation $\mathrm{ev}_{\mathbb{R}} : \Omega_\infty \to \mathbb{R}$ is continuous.

Proof. Surjectivity permits compatible prefixes to be constructed inductively; compactness of the finite sets and the finite-intersection property yield a global section. The product $\prod_n S_n$ is compact, and Ω_∞ is closed, since every compatibility equation defines a closed set. For a fixed section, the intervals are compact, nested, and of diameter tending to zero; the nested-interval theorem yields a unique common point. Given $\varepsilon > 0$, one chooses a level whose interval has diameter less than ε. Sections sharing that prefix have evaluations within the same interval, which proves continuity. □

The real line thus appears as the codomain of a contraction. Two sections may share a value while preserving different records; the relation

$$x \sim_{\mathbb{R}} y \quad \Longleftrightarrow \quad \mathrm{ev}_{\mathbb{R}}(x) = \mathrm{ev}_{\mathbb{R}}(y)$$

forms the Archimedean quotient. The genealogy remains in Ω_∞, whereas the coordinate resides in its real image. The continuum problem therefore receives a constructive answer within the domain covered by the declared prefix systems: continuity of evaluation and memory of the section are coordinated properties, not competing descriptions.

7.2.8 Coherent coverage and reconstruction of the Archimedean quotient

The preceding evaluation assigns a value to each section. To prove that the visible space is completely reconstructed, it is also necessary that refinement lose no points. Let $K \subset \mathbb{R}$ be compact, and suppose that the intervals of the preceding theorem satisfy

$$K = \bigcup_{a \in S_0} I_0(a), \tag{7.20}$$

$$I_n(a) = \bigcup_{\substack{b \in S_{n+1} \\ r_{n+1,n}(b) = a}} I_{n+1}(b) \qquad (a \in S_n). \tag{7.21}$$

These identities define a *coherent covering*: every visible compact decomposes exactly into the compacts of its prolongations.

Theorem 7.10 (Genealogical Reconstruction of a Compact Set)**.** *Suppose that:*

1. *the sets S_n are finite and nonempty, and the bonding maps $r_{n+1,n}$ are surjective;*
2. *the intervals are compact, nonempty, nested, and uniformly contractive;*
3. (7.20) *and* (7.21) *hold.*

Then $\mathrm{ev}_{\mathbb{R}}(\Omega_\infty) = K$*, and the map induced*

$$\overline{\mathrm{ev}}_{\mathbb{R}} : \Omega_\infty/\!\sim_{\mathbb{R}} \xrightarrow{\cong} K \tag{7.22}$$

is a homeomorphism. The fiber over $y \in K$ is the complete set of compatible sections that publish that same value.

Proof. Compactness and continuity follow from theorem 7.9. With $y \in K$ fixed, let

$$T_n(y) := \{a \in S_n : y \in I_n(a)\}.$$

Coherent coverage makes every $T_n(y)$ nonempty and connects each state to a prolongation. The resulting tree is finitely branching and has vertices at every depth; by the infinite-branch lemma it admits a compatible chain $(x_n)_n$. The point y belongs to every $I_n(x_n)$, and contraction forces $\mathrm{ev}_{\mathbb{R}}(x) = y$. Therefore, the evaluation is surjective. The quotient map is continuous and bijective; since its domain is compact and K is Hausdorff, it is a homeomorphism. □

The theorem distinguishes two assertions. The contraction produces a unique value for each genealogy; the coherent covering proves that all values of the compact set appear. A check at finite depth certifies the examined levels, but it does not replace any of these infinite hypotheses.

Proposition 7.11 (Genealogical Atlas of the Real Line)**.** *For each $m \geq 1$, let $K_m = [-m, m]$, and suppose that a system is given that satisfies theorem 7.10, together with closed inclusions*

$$J_m : \Omega_\infty^{(m)} \hookrightarrow \Omega_\infty^{(m+1)}$$

that intertwine the evaluations and the inclusion $K_m \hookrightarrow K_{m+1}$. Then the Archimedean quotient of the strict colimit of the sections is

$$\varinjlim_m K_m = \bigcup_{m \geq 1} [-m, m] = \mathbb{R},$$

with its usual topology, while the fibers of the colimit preserve the genealogies of each value.

Proof. Each piece has quotient K_m by (7.22). The declared inclusions make these quotients compatible. The final topology of the exhaustion by compact intervals coincides with the usual topology of $\mathbb{R}$: a set is open exactly when its intersection with each K_m is open in K_m. □

7.2.9 Algebraic descent of operations

Reconstructing the visible space does not suffice to reconstruct its algebra. Let $\star$ be a continuous partial operation on Ω_∞, with domain $D_\star$ saturated by the fibers of $\mathrm{ev}_{\mathbb{R}} \times \mathrm{ev}_{\mathbb{R}}$.

Theorem 7.12 (Fiberwise Descent Criterion)**.** *There exists a visible operation $\overline{\star}$ that satisfies*

$$\mathrm{ev}_{\mathbb{R}}(x \star y) = \mathrm{ev}_{\mathbb{R}}(x) \,\overline{\star}\, \mathrm{ev}_{\mathbb{R}}(y)$$

if and only if the value of the left-hand side is constant upon replacing x or y by any other representative of its respective fiber. If two descended continuous operations coincide with addition and multiplication on dense subsets of their visible domains, they coincide with them throughout the domain.

Proof. Constancy on fibers is precisely the universal property of the quotient: it permits $\overline{\star}$ to be defined independently of the representative and is necessary for that definition to be meaningful. The final assertion follows from continuity and density. □

APP conserves the datum required to address this descent: the sum and product leaves act on the same support, but maintain different quotients, carries, and fibers. The visible operation can descend even though the memory of its representatives remains distinct.

7.2.10 Projective measure and conservation of the unit

Cylinder topology organizes proximity and truncation; a path theory also requires weights. Let μ_n be a probability measure on S_n. The family is projectively consistent when

$$\mu_n(a) = \sum_{\substack{b \in S_{n+1} \\ r_{n+1,n}(b)=a}} \mu_{n+1}(b). \tag{7.23}$$

The equality expresses the local conservation of the unit: the weight of a truncated history is the sum of the weights of all its prolongations.

Theorem 7.13 (Genealogical Measure on the Inverse Limit)**.** *Every family of finite probabilities satisfying* (7.23) *determines a unique Borel probability μ_∞ on Ω_∞ such that*

$$\mu_\infty([a]_n) = \mu_n(a) \tag{7.24}$$

for every prefix cylinder $a \in S_n$.

Proof. Finite unions of cylinders form an algebra. Any one of them can be refined into a disjoint union at a common level, and (7.23) proves that the sum of their weights does not depend on the chosen level. A premeasure is thereby obtained. Since the cylinders are both open and closed and Ω_∞ is compact, a decreasing sequence of elements of the algebra with empty intersection already reaches the empty set at some level. The premeasure is therefore continuous at the empty set and extends uniquely to the σ-algebra of Borel sets. □

The evaluation transports that measure to the visible compact,

$$\nu = (\mathrm{ev}_{\mathbb{R}})_* \mu_\infty, \qquad \nu(B) = \mu_\infty\big(\mathrm{ev}_{\mathbb{R}}^{-1}(B)\big). \tag{7.25}$$

The visible probability adds the genealogies of a single fiber without erasing them from the total space.

Proposition 7.14 (Value Decomposition and Genealogical Multiplicity)**.** *If K is a standard Borel space, then for ν-almost every $y \in K$ there exists a conditional probability μ_y supported on*

$$\mathfrak{G}_y := \mathrm{ev}_{\mathbb{R}}^{-1}(y)$$

such that, for every integrable function F,

$$\int_{\Omega_\infty} F \, \mathrm{d}\mu_\infty = \int_K \left(\int_{\mathfrak{G}_y} F \, \mathrm{d}\mu_y \right) \mathrm{d}\nu(y). \tag{7.26}$$

Thus, the visible coordinate and the distribution of histories that realize it are distinct typed data: an observable that factors through $\mathrm{ev}_{\mathbb{R}}$ *reads only ν, whereas a memory reader can distinguish the measures μ_y within a single fibre.*

Proposition 7.15 (Compatible cylinder conditioning)**.** *Let $A \subseteq \Omega_\infty$ be a union of cylinders of level m, with $\mu_\infty(A) > 0$. For every $n \geq m$, the conditioned distribution on S_n is obtained by suppressing the states whose truncation does not belong to A and renormalizing the remaining weights. The resulting distributions continue to satisfy projective consistency* (7.23)*.*

Proof. This is Bayes' rule on the finite sets at level n. The sum of the conditional weights of the prolongations equals the conditional weight of the parent because intersection with A distributes over the disjoint union of its prolongations. Conditioning eliminates incompatible futures and redistributes the unit among the survivors; it neither creates new histories nor removes from the total space those lying outside the event being read. □

This separation provides the exact basis for distinguishing preparation, reading, conditioning, and history in the quantum realization.

Mathematical force and architectural function

The rank $n - 1$, the discrete fundamental theorem, the carry cocycle identity, the reconstruction theorems, and the projective measure are exact results in the declared domains. Their HMT application uses as its starting structure the states, restrictions, intervals, and transports constructed by APP–TRIT–TPK. The discrete measure complex is an organizing formalization: it does not identify the degree-two categorical cocycle with a geometric cell until the corresponding map is provided. This separation allows the common language subsequently to govern holonomy, the sum over paths, and the transition from number to particle without anticipating their physical realizations.

Chapter 8

nonadic holonomy Γ_9 of the Topological Phase Kernel: transport of the enriched state, coinductive prolongation and 9→1 return with memory advance

The connection is constructed from the complete orbital classification and from its first coinductive elevation. The following development fixes the three initial histories, the oriented boundaries and the chain $w_6 \to w_{30} \to R_{36}$ before introducing the nonadic holonomy.

8.1 Orbital classification, initial words and 1st boundary

The ternary reduction of the emitter realizes 243 words within the ambient fiber $\mathbb{F}_3^6$. This unit does not choose three words by decimal comparison. It first constructs the symmetry action of the catalogue, classifies its orbits by integer invariants, and only afterwards orients the three structural representatives.

8.1.1 Dihedral action on 6 coordinates

For $u = (u_1, u_2, u_3, u_4, u_5, u_6) \in \mathbb{F}_3^6$, define

$$r(u) = (u_3, u_1, u_2, u_5, u_6, u_4),$$
$$s(u) = (u_4, u_5, u_6, u_1, u_2, u_3).$$

We verify

$$r^3 = s^2 = 1, \qquad srs = r^{-1}.$$

Therefore, r, s generate an action of D_3.

The same permutation acts on the six blocks of each emission $U = (U_1, \ldots, U_6)$. The projection Ψ satisfies

$$\Psi(gU) = g\Psi(U), \qquad g \in D_3.$$

Theorem 8.1 (Orbital decomposition of the catalog)**.** *The 243 realized words decompose into 43 orbits:*

$$38 \textit{ orbits of cardinality } 6, \qquad 5 \textit{ orbits of cardinality } 3.$$

In particular,

$$38 \cdot 6 + 5 \cdot 3 = 243.$$

Proof. The set is applied to each word.

$$\{u, ru, r^2u, su, sru, sr^2u\}$$

and repetitions are removed. Exhaustive enumeration of the catalog produces the indicated histogram. The equivariance of Ψ ensures that no orbit leaves the realized image. □

8.1.2 Fiber Invariants

For a word realized w, let

$$\mathcal{F}_w := \{U \in \operatorname{im} T_{\min} : \Psi(U) = w\},$$
$$n(w) := |\mathcal{F}_w|, \qquad M(w) := \sum_{U \in \mathcal{F}_w} \operatorname{mult}(U),$$

and

$$c(w) = \big(\#\{j : w_j = 0\}, \#\{j : w_j = 1\}, \#\{j : w_j = 2\}\big).$$

The enriched catalogue further preserves, for each emission, a diagonal coordinate $\mathrm{diag}_c(U) \in \{0, \dots, 8\}$, a cardinal orientation $h_+(U)$, and the multiplicative signature $E_{\mathrm{sig}}(U)$. All of them are computed from the seed route; they are not read from a target constant. Their componentwise algorithm will be printed alongside the complete catalogue in the certification appendix and forms part of the input state of the following predicates.

Definition 8.2 (Closure orbit)**.** An orbit $\mathcal{O}$ is of closure if it has cardinality six and, for each $w \in \mathcal{O}$,

$$n(w) = 5, \qquad M(w) = 1008,$$

$$\{\mathrm{mult}(U) : U \in \mathcal{F}_w\} = \{432, 144, 144, 144, 144\},$$

and the common tritic census is $c(w) = (2, 3, 1)$.

Definition 8.3 (Radical Minimal Orbits)**.** An orbit $\mathcal{O}$ is minimally radical if it has cardinality six, each word possesses a unique emission of multiplicity 144, and

$$\{\mathrm{diag}_c(U) : \Psi(U) \in \mathcal{O}\} = \{0, 3, 6\}.$$

Among them, the orbit with census $(2, 3, 1)$ is called the propagation orbit, and the orbit with census $(2, 2, 2)$ is called the self-scaling orbit.

Theorem 8.4 (Uniqueness of the Three Structural Orbits)**.** *In the catalogue of 468 emissions there exists a unique closure orbit, a unique minimally radical orbit of propagation, and a unique minimally radical orbit of self-scale.*

Proof. The 43 orbits of theorem 8.1 are enumerated. For each one, $n(w)$, $M(w)$, the multiset of multiplicities, the census $c(w)$, and the diagonal support are computed. The predicates of definitions 8.2 y 8.3 leave exactly one orbit in each class. The proof is exhaustive over a finite set and does not use decimal digits of the analytic characters. □

The three orbits are

$$\begin{aligned} \mathcal{O}_{\mathrm{cl}} &= \{001112, 010211, 100121, 112001, 121100, 211010\}, \\ \mathcal{O}_{\mathrm{prop}} &= \{011120, 012110, 101201, 110012, 120011, 201101\}, \\ \mathcal{O}_{\mathrm{aut}} &= \{002112, 020211, 112002, 121200, 200121, 211020\}. \end{aligned}$$

8.1.3 Internal Orientation of Each Orbit

We fix the internal caliber.

$$\mathrm{diag}_c(U) = 6, \qquad h_+(U) = ES.$$

A word in an orbit is admissible in that gauge if some emission in its fiber satisfies both conditions.

Proposition 8.5 (Unique oriented representative)**.** *Each of the three structural orbits contains a unique word admissible in the gauge* $(6, ES)$*. They are*

$$\boxed{w_6^{\pi} = 010211, \qquad w_6^{e} = 201101, \qquad w_6^{\varphi} = 121200.}$$

Proof. The enumeration of the catalogue fibers counts the words containing an emission with $\mathrm{diag}_c = 6$ and $h_+ = ES$. In each of the three orbits, the number of coincidences is one. The superscripts π, e, φ henceforth name the structural functions of closure, propagation, and self-scaling; they did not participate in the selection. □

8.1.4 Closure microfiber

The fiber of the representative w_6^{π} consists of five emissions:

U_6	E_{sig}	mult	diag_c	stratum
501|614|498|169|272|272	555555	432	4	$\perp$
810|923|870|169|272|272	339555	144	6	++
810|983|810|169|272|272	393555	144	6	++
870|923|810|169|272|272	933555	144	6	++
870|923|810|418|674|521	933717	144	5	−−

Thus,

$$432 + 4 \cdot 144 = 1008, \qquad 5 = 1_{\perp} + 3_{++} + 1_{--}.$$

The unipolar emissions oriented of the other two channels are

$$U_e = 850|963|890|149|252|212,$$

$$U_{\varphi} = 830|943|830|109|252|252.$$

The mass 1008 of this microfiber is not the period of the calendar C_{108}; the two numbers belong to different domains.

8.1.5 Ternary lifting matrices

On row vectors of $\mathbb{F}_3^6$, the two unique transports reconstructed from the oriented register produced by the TPK transition are

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

The simultaneous compatibility of the three biographies uniquely selects the lifting calendar (L_0, L_0, L_1, L_1) among the sixteen binary calendars of length four.

Definition 8.6 (Block recurrence and concatenation)**.** For each channel $\chi \in \{\pi, e, \varphi\}$, let $b_0^{\chi} = w_6^{\chi}$ and

$$b_{k+1}^{\chi} = b_k^{\chi} L_{\epsilon_k} \pmod 3, \qquad (\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3) = (0, 0, 1, 1).$$

The accumulated word is

$$w_{30}^{\chi} = b_0^{\chi} \| b_1^{\chi} \| b_2^{\chi} \| b_3^{\chi} \| b_4^{\chi}.$$

This formulation distinguishes the current block $b_k \in \mathbb{F}_3^6$ from the concatenated word $w_{6(k+1)}$. A word of length $6k$ is not multiplied by a matrix 6×6.

Theorem 8.7 (Ternary biographies of thirty positions)**.** *The recurrence of the definition 8.6 produce*

$$w_{30}^{\pi} = 010211|012222|010211|002111|110221,$$
$$w_{30}^{e} = 201101|121221|102011|012222|102011,$$
$$w_{30}^{\varphi} = 121200|112202|121020|010210|010200.$$

Proof. The five row vectors are successively multiplied by the indicated matrices, reducing each coordinate modulo three after each product. The concatenation of the five computed blocks gives the three expressions. Each equality can be verified component by component by means of 24 scalar products per channel. □

The environmental indices of the five blocks are

	b_0	b_1	b_2	b_3	b_4
π	103	161	103	67	349
e	523	457	301	161	301
φ	450	398	438	102	99

,

where each word of six trits is interpreted as an integer in base three.

8.1.6 1st common boundary

The boundary is obtained through a finite relation that preserves axis, aggregate, saturation, column occupancy, and dual neutrality. The incidence matrix used in this relation is

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3). \tag{8.1}$$

If u_4^{χ} is the fifth block of w_{30}^{χ}, the axis and the aggregate prior to separating the two lateral channels are

$$u_5^{\varphi} = u_4^{e} = 102011, \qquad u_5^{\pi} + u_5^{e} = 210112, \qquad u_4^{\varphi} A_W L_1^3 = 111101. \tag{8.2}$$

From them, they calculate

$$q = 012120, \qquad a = 111101, \qquad qA_W = 012000, \qquad aA_W = 120210. \tag{8.3}$$

Depth six imposes the saturation $c = 111111$; the minimal compatible occupancy is colw = 223232; and dual neutrality requires

$$(BA_W)\mathbf{1} = 000. \tag{8.4}$$

Proposition 8.8 (Enumeration of the first boundary)**.** *The conditions* (8.2)– (8.4) *leave exactly two matrices:*

$$B_- = \begin{pmatrix} 021222 \\ 222220 \\ 102011 \end{pmatrix}, \qquad B_+ = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}. \tag{8.5}$$

The mirror involution swaps its first two rows. Its internal cards are, respectively,

$$B_- \longmapsto (789, 048, 894), \qquad B_+ \longmapsto (793, 045, 894). \tag{8.6}$$

The orientation APP of the cylinder semiabierto selecciona B_+.

Proof. The axis fixes the third row and the aggregate fixes the sum of the first two. Saturation and occupation eliminate the distributions that do not fill the six columns. The enumeration of the solutions of the six linear equations of (8.4) leaves the two matrices of (8.5); the semi-open chart and the orientation distinguish the second without consulting a target decimal expansion. □

We therefore define,

$$R_{36} := B_+ = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}, \qquad \mathrm{ind}_3(R_{36}) = (726, 215, 301). \tag{8.7}$$

The object R_{36} contains three ordered blocks of six trits. It is not the predictive quotient C_{36}^{pred} of theorem 1.166 or the excess of 36 classes of the seed gauge.

8.1.7 Residue–quotient lifting

Let $A \in M_6(\mathbb{Z})$ be the integer lift of L_0:

$$A = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad \det A = 1.$$

For $x_k \in \mathbb{Z}^6$, define

$$y_k = x_k A, \qquad u_k = y_k \bmod 3 \in \{0,1,2\}^6, \qquad x_{k+1} = \frac{y_k - u_k}{3}.$$

Then

$$x_k A = u_k + 3x_{k+1}. \tag{8.8}$$

Theorem 8.9 (Reversibility of the lift)**.** *The pair (u_k, x_{k+1}) is uniquely determined by x_k, and*

$$x_k = (u_k + 3x_{k+1})A^{-1}.$$

Proof. Euclidean division by three in each coordinate of $x_k A$ determines residue and quotient. Since $\det A = 1$, the inverse A^{-1} has integer coefficients. Multiplying equation (8.8) by it recovers x_k. □

In channel e, the visible block 102011 reappears twice, but the prestates modulo 27 are

$$(25, 11, 7, 17, 8, 16) \neq (7, 8, 4, 23, 26, 7),$$

and the subsequent quotients are

$$(6, 13, 9, 2, 13, 1) \neq (15, 0, 3, 19, 23, 25).$$

It is thus explicitly verified that the visible-phase return of a word is not the return of the enriched state.

8.1.8 Obstructions to orbital selection and to the lifting $w_6 \to w_{30} \to R_{36}$: uniqueness, gauge, and Hensel reversibility

Comparison Criterion 8.10 (Orbital selection refutation criteria)**.** The unit is refuted if:

1. the action D_3 leaves the catalogue or changes multiplicities;
2. any one of the three structural predicates leaves more than one orbit;
3. the gauge $(6, ES)$ selects two words in an orbit;
4. a target digit is used to define the predicates;
5. the accumulated word w_{6k} is multiplied by L_0 or L_1 instead of the block b_k;

6. $\det A \neq 1$ or fails the inverse of the lift;
7. two equal visible residues are declared to be equal complete states.

The first joint boundary has already been constructed. The next unit will classify the nine phase components, prove the nonadic return without resetting, and construct the oriented mirror with its local resolution $3 \to 1$.

8.2 Joint Holonomy Γ_9: observable compression T, orthogonal component η and terminal memory conservation

The nine phases that appear after R_{36} are not nine independent proofs. They are the nine local components of a single connection on enriched states. Their composition around a turn defines a holonomy. The return of the phase label conserves the continuity of prefix, carry, boundary, Witt shadow, and memory.

8.2.1 Transport of the enriched state by the nonadic holonomy and memory update

At the level K, we write

$$\widehat{x}_K = (B_K, C_K, p_K, c_K, \Sigma_K, W_K, g(K), q_{\mathrm{mem},K}),$$

where:

1. $B_K = (u_K^{\pi}, u_K^{e}, u_K^{\varphi})$ is the visible block of three words of six trits;
2. C_K is the cylinder ternario compatible;
3. p_K is the complete prefix;
4. c_K is the accumulated carry;
5. Σ_K is the signature joint of boundary;
6. W_K is the Witt incidence shadow;
7. $g(K) = 1 + ((K-1) \bmod 9)$ is the public phase label;
8. $q_{\mathrm{mem},K} \in \mathbb{Z}_{\geq 0}$ is the memory vertical not acotada.

The dodecaphasic coordinate is derived:

$$\beta_K = [q_{\mathrm{mem},K}]_{12}.$$

q_{mem} is not replaced by β, since reduction modulo twelve would lose the number of circuits.

8.2.2 Local transition relations and construction of the nonadic holonomy Γ_9

For each K, let

$$\mathcal{R}_K \subseteq \widehat{X}_K \times \widehat{X}_{K+1}$$

the relation of compatible extension between complete states. The relational composition of a loop is

$$\Gamma_{9,K} := \mathcal{R}_{K+8} \circ \cdots \circ \mathcal{R}_{K+1} \circ \mathcal{R}_K : \widehat{X}_K \longrightarrow \widehat{X}_{K+9}. \tag{8.9}$$

Definition 8.11 (Nonadic holonomy)**.** The holonomy of one turn is

$$\Gamma_9 := \mathrm{Hol}_{C_9}(\nabla_{\mathrm{TPK}}),$$

whose realization at level K is $\Gamma_{9,K}$. The nine relations $\mathcal{R}_K, \ldots, \mathcal{R}_{K+8}$ are charts of this single connection.

Theorem 8.12 (Phase return without restart)**.** *If* $(\widehat{x}_K, \widehat{x}_{K+9}) \in \Gamma_{9,K}$*, then*

$$g(K+9) = g(K), \qquad q_{\mathrm{mem},K+9} = q_{\mathrm{mem},K} + 1,$$

and

$$\beta_{K+9} = \beta_K + 1 \pmod{12}.$$

In general, $\widehat{x}_{K+9} \neq \widehat{x}_K$.

Proof. The first identity follows from the definition of g. A complete turn increments the unbounded winding counter by definition. Reducing that identity modulo twelve yields the third. The remaining coordinates are transported by the composition equation (8.9); no law of the TPK resets them. Therefore, equality of phase does not imply equality of the complete state. □

8.2.3 Exhaustive classification of the 9 phases of the Topological Phase Kernel

The visible blocks and the local censuses of the first round are

K	N_π	N_e	N_φ	N_{loc}	N_{ext}	u^π	u^e	u^φ
1	378	422	349	3	2	012222	121221	112202
2	238	368	388	2	5	010211	102011	121020
3	184	284	173	5	4	002111	012222	010210
4	207	207	122	4	1	110221	102011	010200
5	39	151	151	1	1	222220	021222	102011
6	110	110	69	1	3	111201	201202	221021
7	81	29	81	3	3	212121	222210	200221
8	59	59	59	3	2	200121	212212	110110
9	43	43	43	2	1	100100	020112	010122

Theorem 8.13 (Functions of the nine components)**.** *In the preceding turn, the nine components perform the following functions:*

1. *local opening of three orientations and first resolution;*
2. *binary split with a local maximum of five extensions;*
3. *maximal local multiplicity of five states;*
4. *contraction of four local states to an extension;*
5. *first strong saturation, with a single continuation;*
6. *unique local present and triple opening of the future;*
7. *triple mirror fiber;*
8. *synchronization of the three censuses in* 59*;*
9. *oriented closure: two local input states and a relation that, after the additional horizon, conserves a prolongation.*

Proof. Each assertion is read from the censuses $N_{\mathrm{loc}}, N_{\mathrm{ext}}$, from the equality or inequality of N_π, N_e, N_φ, and from the explicit extension relation in the ninth, tenth, and eleventh phases to be proved in the following unit. The table enumerates the entire revolution; no subset of phases has been chosen. □

The fifth component coincides with the first common boundary.

$$R_{36} = (222220|021222|102011)^T.$$

The ninth component contains

$$B_9 = (100100|020112|010122).$$

The presence of two local states in the ninth phase and of three extensions from the oriented state belongs to different layers; they are not contradictory censuses.

Holonomy acts on the complete enriched state. Before its modal quotient is published, one therefore constructs the involution that exchanges the π and e fibres, the fixed φ axis, the ternary invariants, and the local $3 \to 1$ resolution determined by the second horizon.

8.3 Mirror involution and local selection $3 \to 1$ of the surviving extension

The ninth phase contains two local input states. Once the oriented state B_9 has been fixed, the boundary relation produces three extensions toward the first phase. The three share an axis of self-scale and a ternary aggregate; they differ in the ordered distribution of that aggregate between the branches of closure and propagation. A second horizon conserves exactly one.

8.3.1 Symmetric and axial coordinates

Let

$$\mathcal{B} := (\mathbb{F}_3^6)^3.$$

A visible block is

$$B = (u^\pi, u^e, u^\varphi) \in \mathcal{B}.$$

We define

$$q(B) := u^\pi + u^e + u^\varphi, \qquad a(B) := u^\pi + u^e - u^\varphi.$$

All operations are performed componentwise in $\mathbb{F}_3^6$.

Theorem 8.14 (Reconstruction of the axis and the aggregate)**.** *The coordinates q, a determine*

$$\boxed{u^\varphi = 2(q-a),} \qquad \boxed{u^\pi + u^e = q - u^\varphi.}$$

Proof. Subtracting the definitions gives $q - a = 2u^\varphi$. In $\mathbb{F}_3$, $2^{-1} = 2$; therefore, $u^\varphi = 2(q-a)$. Substitution into the definition of q recovers the remaining aggregate. □

The specular involution is

$$\mathfrak{c}(u^\pi, u^e, u^\varphi) = (u^e, u^\pi, u^\varphi).$$

Proposition 8.15 (Invariants of the Involution)**.** $\mathfrak{c}^2 = \mathrm{id}$*, y*

$$q(\mathfrak{c}B) = q(B), \qquad a(\mathfrak{c}B) = a(B).$$

In particular, u^φ and $u^\pi + u^e$ are invariantes.

Proof. Exchanging the first coordinates twice recovers B. Both q and a depend on them only through their sum, so that the exchange preserves them. The theorem 8.14 transfers that invariance to the axis and the aggregate. □

The involution asserts neither that $\pi = e$ nor that one is the quotient of the other. It asserts that the two branches occupy conjugate positions with respect to a fixed structure.

8.3.2 Oriented state in phase 9 and associated boundary data

The visible projection of the selected state is

$$B_9 = (100100 \mid 020112 \mid 010122).$$

The induced relation on visible blocks is denoted

$$\mathcal{M}_9^B \subseteq \mathcal{B} \times \mathcal{B}.$$

The two local input states of the phase table precede this relation. After selecting the complete state that projects to B_9, its output fiber has cardinal three.

Theorem 8.16 (The three extensions of the ninth phase)**.** *The relational image of B_9 contains exactly*

$$\begin{aligned} B_{10}^{(0)} &= (101222 \mid 112221 \mid 112002), \\ B_{10}^{(1)} &= (102221 \mid 111222 \mid 112002), \\ B_{10}^{(2)} &= (111221 \mid 102222 \mid 112002). \end{aligned}$$

The three satisfy

$$q_{10} = 022112, \qquad a_{10} = 101111,$$

and, consequently,

$$u_{10}^{\varphi} = 112002, \qquad u_{10}^{\pi} + u_{10}^{e} = 210110.$$

Proof. The extension relation on complete states is enumerated in the fiber of B_9 and produces the three indicated rows. Adding and subtracting their three components gives the same pair (q_{10}, a_{10}) in all three cases. The theorem 8.14 produces the common axis and aggregate. The first two components are distinct, so the three rows represent different distributions of the same aggregate. □

All their publications in base three are

$$(279, 352, 365), \qquad (280, 351, 365), \qquad (282, 350, 365).$$

8.3.3 Exact compatibility of cylinders

A ternary word of length L and index N determines the half-open cylinder

$$I_3(N, L) = \left[\frac{N}{3^L}, \frac{N+1}{3^L} \right).$$

A prefix of b decimal blocks, with associated integer D, determines

$$I_{1000}(D, b) = \left[\frac{D}{1000^b}, \frac{D+1}{1000^b} \right).$$

Lemma 8.17 (Integer intersection test)**.** *The cylinders $I_3(N, L)$ and $I_{1000}(D, b)$ intersect if and only if*

$$N 1000^b < (D+1) 3^L, \qquad (N+1) 1000^b > D 3^L.$$

Proof. Two half-open intervals $[a, b)$ and $[c, d)$ intersect if and only if $a < d$ and $c < b$. We substitute the four endpoints and multiply by the positive integer $3^L 1000^b$. □

The test uses only integer arithmetic. It does not depend on a floating-point tolerance.

8.3.4 2nd horizon and uniqueness

The following block is

$$B_{11} = (022212 \mid 110001 \mid 100021),$$

and the corresponding decimal boundary is

$$(502, 662, 638).$$

Theorem 8.18 (Local resolution $3 \to 1$)**.** *The three extensions of theorem 8.16 are compatible at depth one. When B_{11} and the boundary $(502, 662, 638)$ are imposed simultaneously, only $B_{10}^{(0)}$ prolongs as the same complete state. Therefore,*

$$\boxed{\#\,\mathrm{Surv}_1^{\mathrm{cal}}(B_9) = 3, \qquad \#\,\mathrm{Surv}_2^{\mathrm{cal}}(B_9) = 1.}$$

Proof. We apply lemma 8.17 to each of the three channels. At the first horizon, the six inequalities of every candidate are satisfied. After adjoining B_{11}, the inequalities of the three channels remain satisfied for $B_{10}^{(0)}$. In $B_{10}^{(1)}$ and $B_{10}^{(2)}$, compatibility of channel φ is preserved, but the conditions of branches π and e fail. Enumeration therefore leaves a unique state at depth two. □

This proof does not define the global connection from a local example. It is an explicit witness of how a multivalued relation at one horizon can become single-valued by preserving memory and requiring a longer prolongation.

8.3.5 Phase return with block change

The first phase of the initial loop is

$$B_1 = (012222 \mid 121221 \mid 112202),$$

while the first phase of the next turn is

$$B_{10} = (101222 \mid 112221 \mid 112002).$$

Thus,

$$g(1) = g(10) = 1, \qquad B_1 \neq B_{10}.$$

The phase returns, but the axis, the aggregate, the distribution, and the memory of quotient have evolved.

8.3.6 Diagram of the mirror involution π/e, the fixed axis φ, and the local selection $3 \to 1$ by prolongability

8.3.7 Residue–quotient memory during the mirror

The recurrence

$$x_k A = u_k + 3x_{k+1}$$

of the equation (8.8) is applied to each of the three channels. The part u_k publishes the visible block; x_{k+1} conserves the quotient. Since $A \in \mathrm{GL}_6(\mathbb{Z})$, the pair recovers the preceding state. The mirror can exchange the visible branches without eliminating their provenance quotients.

8.3.8 Obstructions to the mirror involution π/e and to the local selection $3 \to 1$: invariants, depth-2 horizon, and Hensel memory

Comparison Criterion 8.19 (Refutation Criteria for the Mirror)**.** The construction is refuted if:

1. the involution modifies u^{φ} or $u^{\pi} + u^{e}$;
2. the two local input states are conflated with the three outputs;

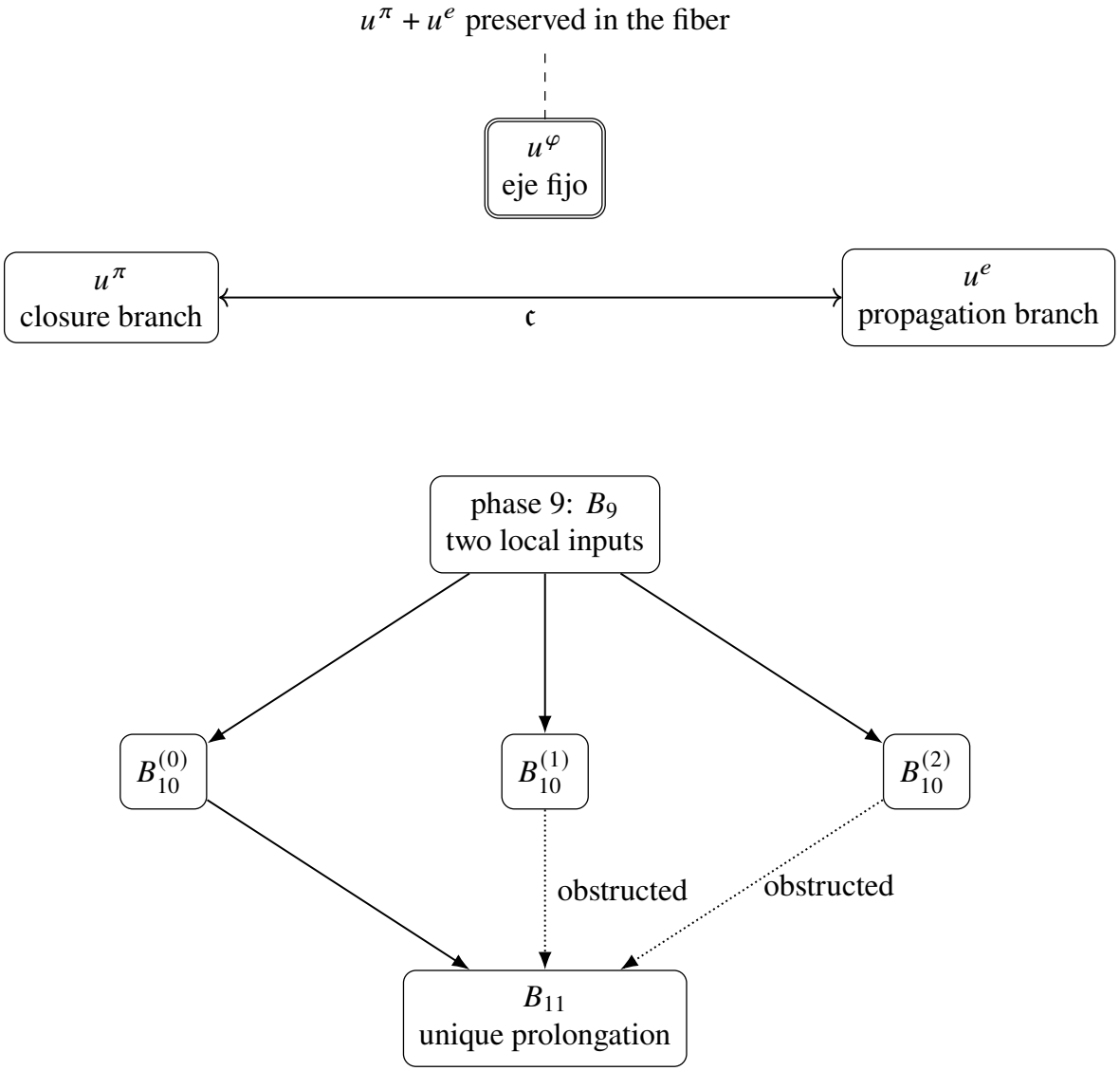


Figure 8.1: The mirror preserves the axis u^{φ} and the aggregate $u^{\pi} + u^{e}$. The return of the ninth phase opens three visible extensions; the memory of the second horizon preserves one. The circuit does not reset the state.

3. one of the three outputs has a different (q_{10}, a_{10});
4. the cylinder test uses a floating tolerance;
5. more than one output survives the declared second horizon;
6. phase return is interpreted as equality $B_1 = B_{10}$;
7. the visible exchange erases the Hensel quotients.

The mirror and the local selection have now been constructed. The following units will develop the Sturmian vacancies and then the three complete analytic biographies.

8.3.9 Joint transfer of the fiber of 468 emissions

The factorization of the catalog is

$$\mathcal{U}_6^{\mathrm{cat}} \cong \mathfrak{S}_+ \times \mathfrak{S}_\times, \qquad |\mathfrak{S}_+| = 18, \qquad |\mathfrak{S}_\times| = 26.$$

For a function f on the catalogue, let

$$(E_+ f)(s,e) = \frac{1}{18} \sum_{s' \in \mathfrak{S}_+} f(s', e),$$

and

$$(E_\times f)(s,e) = \frac{1}{26} \sum_{e' \in \mathfrak{S}_\times} f(s, e').$$

The projectors E_+ and $E_\times$ commute.

Let

$$\tau = (+1, -1, 0, +1, -1, 0, +1, -1, 0), \qquad P_{+1} = E_+, quadP_{-1} = E_\times, quadP_0 = I.$$

On $\mathcal{U}_6^{\mathrm{cat}} \times C_9$ we define phase transfer by

$$\mathcal{K}_{\mathrm{ph}}\big((u,r),(v,[r+1]_9)\big) := P_{\tau(r)}(u,v), \tag{8.10}$$

and we make it zero for the remaining pairs of phases. Thus, the cyclic permutation and the operator applied in each of the nine steps are determined.

Proposition 8.20 (Joint Renewal After a Turn)**.**

$$\mathcal{K}_{\mathrm{ph}}^9 = E_+ E_\times.$$

In particular, on a point mass,

$$(E_+ E_\times)(u,v) = \frac{1}{18 \cdot 26} = \frac{1}{468}.$$

Proof. The ordered product over a loop is

$$P_{\tau(8)} P_{\tau(7)} \cdots P_{\tau(0)} = I E_\times E_+ I E_\times E_+ I E_\times E_+ = E_+^3 E_\times^3 I^3 = E_+ E_\times.$$

The second equality uses commutativity and the final one idempotence. The second result is the uniform distribution on the product of the two signatures. □

This identity expresses a joint renewal, not nine independent filtrations.

8.3.10 Decomposition of the holonomy U_{Γ_9} into visible compression T and orthogonal component η: $I - T^*T = \eta^*\eta$

Let $\mathcal{H}_{\text{vis}}$ and $\mathcal{H}_{\text{full}}$ be Hilbert spaces for the visible and complete realizations. Let

$$J : \mathcal{H}_{\text{vis}} \longrightarrow \mathcal{H}_{\text{full}}$$

an isometric inclusion and U_{Γ_9} an isometric realization of the holonomy. We define

$$T := J^* U_{\Gamma_9} J, \qquad \eta := (I - JJ^*) U_{\Gamma_9} J.$$

Theorem 8.21 (Compression-decomposition expression)**.** *The following hold*

$$\boxed{U_{\Gamma_9} J = JT + \eta,}$$

$$J^*\eta = 0, \qquad \boxed{I - T^*T = \eta^*\eta.}$$

Proof. We decompose $U_{\Gamma_9} J$ by means of the orthogonal projections JJ^* and $I - JJ^*$:

$$U_{\Gamma_9} J = JJ^* U_{\Gamma_9} J + (I - JJ^*) U_{\Gamma_9} J = JT + \eta.$$

Orthogonality gives $J^*\eta = 0$. Since J and U_{Γ_9} are isometries,

$$I = J^* U_{\Gamma_9}^* U_{\Gamma_9} J = T^*T + \eta^*\eta.$$

Reorder the proof of the last identity. □

T is the visible compression; η is the complementary expression. Neither should be called a strict contraction. For that function a different operator is reserved.

8.3.11 Strict contraction and terminal telescoping

Let V_9 be an isometry and let q be a parameter with $0 < q < 1$. We define

$$M_9 = qV_9, \qquad D_9 = (I - M_9^* M_9)^{1/2}.$$

Then $\|M_9\| = q < 1$ and

$$I - M_9^* M_9 = D_9^* D_9.$$

Theorem 8.22 (Stein Identity with Terminal Term)**.** *For every $L \geq 1$,*

$$\boxed{I = \sum_{r=0}^{L-1} M_9^{*r} D_9^* D_9 M_9^r + M_9^{*L} M_9^L.}$$

More generally, if S_0 satisfies $S_0 - M_9^* S_0 M_9 = D_9^* D_9$, then

$$\boxed{S_0 = \sum_{r=0}^{L-1} M_9^{*r} D_9^* D_9 M_9^r + M_9^{*L} S_0 M_9^L.}$$

Proof. We multiply the defect identity on the left by M_9^{*r} and on the right by M_9^r. Upon summing over $r = 0, \ldots, L-1$, the interior terms telescope, leaving the initial and terminal terms. The same proof applies to the general Stein equation. □

The final term does not vanish at finite depth. Its disappearance in a limit requires a convergence proof; it is not part of the algebraic identity.

8.3.12 Carry Cocycle and Relational Powers

For composable arrows γ_1, γ_2, the carry satisfies

$$\mathrm{Carr}(\gamma_2\gamma_1) = \mathrm{Carr}(\gamma_2)H(\gamma_1) + Y(\gamma_2)\,\mathrm{Carr}(\gamma_1).$$

The formula preserves the contribution of the first stage after transporting it through the second.

The longer loops are written with displaced indices:

$$\begin{aligned}\Gamma_{27,K} &= \Gamma_{9,K+18} \circ \Gamma_{9,K+9} \circ \Gamma_{9,K},\\ \Gamma_{54,K} &= \Gamma_{9,K+45} \circ \cdots \circ \Gamma_{9,K},\\ \Gamma_{108,K} &= \Gamma_{9,K+99} \circ \cdots \circ \Gamma_{9,K}.\end{aligned}$$

Only after declaring a global operator on the graded union may this notation be abbreviated as $\Gamma_9^3, \Gamma_9^6, \Gamma_9^{12}$.

8.3.13 Complete nonadic state: phase, carry, boundary, route, and memory after the prolongation

The one-turn connection is recorded by the quintuple

$$\mathfrak{N}_9 = \big(\Gamma_9, T, \eta, \mathrm{Carr}, S_{\mathrm{term}}\big),$$

where $S_{\mathrm{term}} = M_9^{*L} S_0 M_9^L$ at a finite depth L. An application functor that uses the connection must declare its compatibility with all five fields. Commuting only with the visible phase does not constitute naturality with respect to the complete nonadic state.

8.3.14 Obstructions of the holonomy Γ_9: 9→1 return, composition indices and phase conservation, carry, boundary, path and memory

Comparison Criterion 8.23 (Refutation Criteria of the Connection)**.** The construction is refuted if:

1. the nine phases are presented as independent filters;
2. the return $9 \to 1$ resets q_{mem}, the prefix, or the carry;
3. T is called a strict contraction;
4. some identity of theorem 8.21 fails;
5. the terminal term is removed at finite depth;
6. $\Gamma_{27,K}$ is composed without shifting the indices;
7. an application functor ignores any of the five fields of $\mathfrak{N}_9$.

The complete developments that follow first reconstruct the finite emitter, its factorization and its microfibers; they then unfold the specular involution, the three extensions of B_9, uniqueness at two horizons and the Bockstein–Hensel inverse that preserves memory during phase return.

8.4 Finite arithmetic image and factorization $468 = 18 \cdot 26$

The image in base 1000 of the finite map that assigns the decimal blocks is a proper subset of the one thousand decimal blocks. The evolution law defined in the section 4.1 and the arithmetic of $\mathbb{Z}/9\mathbb{Z}$ determine an image of 42 triples, organized into six classes of seven. The same separation of the additive and multiplicative observables subsequently factors the 468 outputs into 18 additive signatures and 26 multiplicative signatures.

8.4.1 Primitives and recurrence of the finite sender

Definition 8.24 (Initial data of $T_{\min}$)**.** Let $D = \{N, E, S, O\}$ be given, with cardinal vectors

$$v(N) = (-1, 0), \quad v(E) = (0, 1), \quad v(S) = (1, 0), \quad v(O) = (0, -1) \qquad \text{in } (\mathbb{Z}/9\mathbb{Z})^2.$$

The temporal signature and the multiplicative index are fixed by

$$\operatorname{sgn}_9(t) = \begin{cases} +1, & t \bmod 9 \in \{1, 4, 7\}, \\ -1, & t \bmod 9 \in \{2, 5, 8\}, \\ 0, & t \bmod 9 \in \{0, 3, 6\}, \end{cases}$$

and

$$\operatorname{ind}_2(1, 2, 4, 8, 7, 5) = (0, 1, 2, 3, 4, 5),$$

with value zero outside the units. An entry of $T_{\min}$ is a pair of seeds

$$I_9 := \{0, 1, \ldots, 8\}, \qquad s_+ = (i_+, j_+, d_+), \qquad s_\times = (i_\times, j_\times, d_\times) \in I_9^2 \times D.$$

These tables, the readers L_Σ, L_Π, the six blocks of nine instants, and the inversion of both vectors when $27 \mid t$ are all the starting data. Neither π, e, φ, nor any other target value is introduced.

Algorithm 8.25 (54-step recurrence)**.** At the outset, the two cursors occupy their seed positions and carry the vectors $v(d_+)$, $v(d_\times)$. For $t = 1, \ldots, 54$, in this order:

1. if $\operatorname{sgn}_9(t) = +1$, $L_\Sigma(i_+(t), j_+(t))$ is accumulated and the additive cursor is advanced;
2. if $\operatorname{sgn}_9(t) = -1$, then $\operatorname{ind}_2(L_\Pi(i_\times(t), j_\times(t)))$ is accumulated and the multiplicative cursor is shifted;
3. if $\operatorname{sgn}_9(t) = 0$, both cursors remain stationary;
4. after that update, both vectors are reversed if $27 \mid t$.

The accumulators are reset at the beginning of each block $9(r-1)+1 \le t \le 9r$. If S_r, E_r are their final values, then there is emitted

$$U_r = 100\bar{S}_r + 10(\bar{S}_r + \bar{E}_r)_{10} + (3S_r + 5E_r + 1)_{10}, \qquad \bar{S}_r = S_r \bmod 10, \quad \bar{E}_r = E_r \bmod 10,$$

y $w_r = (-U_r) \bmod 3$. Thus it is obtained.

$$T_{\min}(s_+, s_\times) = (U_1, \ldots, U_6) \in \mathcal{U}_6^{\text{cat}}, \qquad \Psi(U_6) = (-U_6) \bmod 3 =: w_6 \in \mathbb{F}_3^6.$$

Comparison Criterion 8.26 (Target-value-independent dual control)**.** The recurrence admits two independent formulations: channel factorization and joint simulation. Exhaustive traversal of the 104 976 inputs produces the same sextuple by both routes for every seed, without consulting a target constant. A discrepancy for a single seed would refute the equivalence.

8.4.2 Bivariate dynamics and block observables

We work on the torus $\mathbb{Z}_9^2$. Under the preceding convention, a complete seed is

$$\omega = (p_+, p_\times, h_+, h_\times) \in (\mathbb{Z}_9^2)^2 \times \{N, E, S, O\}^2,$$

so that

$$|\Omega| = 9^4 \cdot 4^2 = 104\,976.$$

In each window of nine steps, the times $1, 4, 7$ evaluate the additive observable, the times $2, 5, 8$ evaluate the multiplicative observable, and the times $3, 6, 9$ increment the neutral counter. If S_m, E_m, T_m are the accumulators of the window m, the decimal observable is

$$d_1 \equiv S_m, \qquad d_2 \equiv S_m + E_m, \qquad d_3 \equiv 3S_m + 5E_m + 7T_m \pmod{10},$$

and the decimal triple is $U_m = 100d_1 + 10d_2 + d_3$.

Lemma 8.27 (Neutral counter)**.** *For every window, $T_m = 3$.*

Proof. Each block of nine times contains exactly the three neutral times $3, 6, 9$. Therefore the counter is incremented three times. □

8.4.3 Image of the additive accumulator: $\operatorname{Im} S_m = \{6, 9, 12, 15, 18, 21, 24\}$

Let $s = i + j \pmod 9$. The visible additive observable is

$$f(s) = \mathrm{dr}_9(s + 2) = (2, 3, 4, 5, 6, 7, 8, 9, 1).$$

A cardinal translation modifies s in $+1$ or in -1. In a window, three consecutive terms are read; reversing the orientation only reverses their order. The nine cyclic sums are

$$9, 12, 15, 18, 21, 24, 18, 12, 6.$$

Lemma 8.28 (Image of the additive accumulator)**.**

$$S_m \in \{6, 9, 12, 15, 18, 21, 24\}.$$

In particular, $d_1 = S_m \bmod 10$ traverses precisely $\{1, 2, 4, 5, 6, 8, 9\}$.

Proof. The preceding list enumerates all initial positions of the trajectory and the two orientations. Their decimal residues are seven and distinct. □

8.4.4 Image of the multiplicative accumulator: $\operatorname{Im} E_m = \{0, 1, 3, 5, 7, 9\}$

The unit group $(\mathbb{Z}/9\mathbb{Z})^\times$ is cyclic of order six. We fix the multiplicative coordinate function

$$\ell_\times(1, 2, 3, 4, 5, 6, 7, 8, 9) = (0, 1, 0, 2, 5, 0, 4, 3, 0).$$

If the row or column traversed has a factor divisible by three, each evaluation belongs to $\{3, 6, 9\}$ and contributes zero. If the fixed factor is a unit, the sums of three consecutive terms of $b \mapsto \ell_\times(\mathrm{dr}_9(ab))$ are:

a	possible sum
1	$\{1,3,7,9\}$
2	$\{3,5,9\}$
4	$\{1,5,7\}$
5	$\{1,5,7\}$
7	$\{3,5,9\}$
8	$\{1,3,7,9\}$.

Lemma 8.29 (Image of the multiplicative accumulator)**.**

$$E_m \in \{0,1,3,5,7,9\}.$$

Proof. The noninvertible case contributes 0. The finite table of the six units contains all possibilities in the invertible case, and their union is $\{1,3,5,7,9\}$. □

8.4.5 Decimal congruence of the ternary alphabet: uniqueness of the 3rd digit

Theorem 8.30 (Control congruence)**.** *Every terna $\overline{d_1 d_2 d_3}$ emitted by the system satisfies*

$$\boxed{d_3 \equiv 5d_2 - 2d_1 + 1 \pmod{10}.}$$

Proof. From $d_2 \equiv S_m + E_m$ one obtains $E_m \equiv d_2 - d_1 \pmod{10}$. Since $T_m = 3$,

$$d_3 \equiv 3d_1 + 5(d_2 - d_1) + 21 \equiv 5d_2 - 2d_1 + 1 \pmod{10}.$$ □

Theorem 8.31 (Exact image of the decimal application)**.** *The system produces exactly $7 \cdot 6 = 42$ decimal triples. Grouped by $E = d_2 - d_1 \pmod{10}$, they are:*

E	*decimal triples*						
0	114	227	443	556	669	885	998
1	129	232	458	561	674	890	903
3	149	252	478	581	694	810	923
5	169	272	498	501	614	830	943
7	189	292	418	521	634	850	963
9	109	212	438	541	654	870	983.

Proof. There are seven possibilities for d_1 and six for E. For each pair, $d_2 \equiv d_1 + E$ and the control congruence fix a unique d_3, so there exist at most 42 triples. The table applies these two formulas to the 42 pairs and contains no repetitions. The enumeration of the 104 976 seeds finds all the triples in the table and none outside it; the bound is attained. □

8.4.6 Factorization of the 2 observables

Let $\mathcal{S} = \mathbb{Z}_9^2 \times \{N, E, S, O\}$, with $|\mathcal{S}| = 324$. To each additive seed $P \in \mathcal{S}$ is associated the six-window signature

$$s(P) = (S_1, \ldots, S_6) \pmod{10},$$

and to each multiplicative seed $T \in \mathcal{S}$,

$$e(T) = (E_1, \ldots, E_6) \pmod{10}.$$

The two signatures are combined coordinate by coordinate by

$$G(s,e)_k = 100s_k + 10(s_k + e_k)_{10} + (3s_k + 5e_k + 1)_{10}.$$

The map G is injective: from a triple one recovers $s_k = d_1$ and $e_k = d_2 - d_1 \pmod{10}$, while the third digit verifies membership in the image.

Lemma 8.32 (Exact classification of the additive channel)**.**

$$|\operatorname{im} s| = 18,$$

and each additive signature has exactly 18 *preimages.*

Proof. The signature depends only on $(i + j \bmod 9, \varepsilon)$, where $\varepsilon = +1$ for E, S and -1 for N, O. The $9 \cdot 2$ pairs produce distinct signatures; each has nine positions on its diagonal and two directions, that is, 18 preimages. A different number of signatures or a class of size different from 18 would refute the classification. □

Lemma 8.33 (Exact classification of the multiplicative channel)**.**

$$|\operatorname{im} e| = 26.$$

There are 24 *classes of size* 8*, one of size* 24*, and one of size* 108*.*

Proof. Enumerating the $9^2 \cdot 4 = 324$ seeds using the explicit table of $\ell_\times$ gives 26 sextets $(E_1, \dots, E_6)$ and the histogram

$$24 \cdot 8 + 24 + 108 = 324.$$

An additional, absent, or differently sized class would refute the census. □

Theorem 8.34 (Exact injective factorization of the catalog)**.**

$$|\operatorname{im} s| = 18, \qquad |\operatorname{im} e| = 26, \qquad |\operatorname{im} G| = 468 = 18 \cdot 26.$$

Each additive signature has 18 preimages. The multiplicative signatures possess the histogram

$$24 \textit{ size classes } 8, \qquad 1 \textit{ of size } 24, \qquad 1 \textit{ of size } 108.$$

Proof. The two preceding lemmas give the cardinalities of the signatures. In each emitted component, the hundreds digit is $\bar{S}_r$, while the tens digit is $\bar{S}_r + \bar{E}_r \bmod 10$. Therefore,

$$\bar{S}_r = \left\lfloor \frac{U_r}{100} \right\rfloor, \qquad \bar{E}_r = \left(\left\lfloor \frac{U_r}{10} \right\rfloor - \bar{S}_r \right) \bmod 10.$$

The units digit verifies $3S_r + 5E_r + 1 \bmod 10$; it introduces no other identification. Thus, G is injective, and its image has $18 \cdot 26 = 468$ elements. The 24 multiplicative classes of size 8, combined with the 18 additive signatures, produce 432 sextets of multiplicity 144; the other two classes produce 18 sextets of multiplicity 432 and 18 of multiplicity 1944. In particular,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

Two pairs of signatures with the same U_6, or a histogram that failed to recover the complete domain, would refute the theorem. □

The complete composition factors, therefore, as

$$\mathcal{S}^2 \xrightarrow{s\times e} \operatorname{im} s \times \operatorname{im} e \xrightarrow{G} \mathcal{U}_6 \xrightarrow{\rho_3} \mathcal{W}_6 \subset \mathbb{F}_3^6.$$

The final projection has 243 values. The first descent preserves the sum–product separation; the second preserves 468 decimal states; the third removes the fiber coordinate and retains a ternary word.

Corollary 8.35 (Census of the observable projection)**.** *The complete enumeration determines the chain of cardinalities*

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729.$$

The 468 *intermediate elements are sextuples of integers between zero and* 999*; only their final image belongs to* $\mathbb{F}_3^6$*. The fiber multiplicities of the first map satisfy*

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

Proof. The factorization theorem gives $18 \cdot 26 = 468$, and the exhaustive enumeration of the 104 976 seeds produces exactly the three indicated multiplicities. Componentwise reduction traverses the 243 words recorded in the ternary catalogue. The appended catalogues of signatures and routes reproduce both enumerations. □

Proposition 8.36 (Quintuple microfiber of the word 010211)**.** *For the restricted reduction*

$$\Psi : \mathcal{U}_6^{\mathrm{cat}} \longrightarrow \mathbb{F}_3^6, \qquad \Psi(U_6) = (-U_6) \bmod 3,$$

the fiber of 010211 *consists of exactly the following five sextets:*

U_6	σ_+	$\sigma_\times$	*multiplicity*
(501, 614, 498, 169, 272, 272)	(5, 6, 4, 1, 2, 2)	(5, 5, 5, 5, 5, 5)	432
(810, 923, 870, 169, 272, 272)	(8, 9, 8, 1, 2, 2)	(3, 3, 9, 5, 5, 5)	144
(810, 983, 810, 169, 272, 272)	(8, 9, 8, 1, 2, 2)	(3, 9, 3, 5, 5, 5)	144
(870, 923, 810, 169, 272, 272)	(8, 9, 8, 1, 2, 2)	(9, 3, 3, 5, 5, 5)	144
(870, 923, 810, 418, 674, 521)	(8, 9, 8, 4, 6, 5)	(9, 3, 3, 7, 1, 7)	144

They are five preimages of the catalog U_6*, not five complete TPK histories. They preserve distinct signatures and multiplicities although they share the same visible word.*

Proof. The exhaustive scan of the catalogue of 468 sextuples selects exactly the five rows of the table. The signature-recovery function reproduces the two sextuples of every row, and the sum of multiplicities is $432 + 4 \cdot 144 = 1008$. Enumeration additionally verifies that every cell is the Cartesian product of its channel seeds. A cardinality other than five or identification of two rows despite their enriched data would refute the statement. □

8.5 Inverse systems, Weyl–Heisenberg representation and predictive quotient

The construction distinguishes three levels: the finite seed catalogue, the state space that preserves carry and orientation, and the inverse limit of compatible prolongations. This stratification makes it possible to formalize contraction of cylindrical intervals to a real point, persistence of trajectory data in a single evaluation fiber, and the finite Weyl–Heisenberg representation associated with the alternating form of the nonadic torus supporting the additive and multiplicative observables.

The symbols V_t, X_t, and $\rho_{t+1,t}$ used here are the same sets of local states, admissible prefixes, and truncations introduced in equation (6.98), with the purely notational change $t = m$ and $\rho_{n,m} = p_{n,m}$. This section makes their coordinates explicit and introduces no second inverse system.

8.5.1 Observable projection and complete state space of the TPK

The observable projection of the phase transport system (TPK) is the finite encoding application

$$104\,976 \longrightarrow 468 \longrightarrow 243.$$

which classifies seeds, signatures, and words w_6. The complete TPK state space further preserves

$$\begin{gathered}\text{bidirectional orientation,} \quad \text{Hensel state,} \quad \text{cylinder and carry,}\\ \text{joint boundary signature,} \quad \text{visible nonadic phase } g_{\mathrm{vis}}(m).\end{gathered}$$

Two candidates with the same local signature may belong to distinct classes when the following boundary is conserved. We do not introduce a second tuple of state. For

$$x_m = (q_m, o_m, h_m, c_m, \sigma_m, C_m, b_m, \lambda_m) \in \mathcal{X}_{\mathrm{TPK}}^{(m)}$$

we use the coordinate chart

$$\chi_m(x_m) = \big(B_m, h_m, (C_m, b_m), \sigma_m, o_m, \mu(q_m), g_{\mathrm{vis}}(m), \lambda_m\big),$$

with

$$B_m = (u_m^{\pi}, u_m^{e}, u_m^{\varphi}) \in (\mathbb{F}_3^6)^3.$$

Here h_m is the Hensel extension; (C_m, b_m), the cylinder and its boundary; $o_m, \mu(q_m)$, the orientation and the bidirectional observable; and

$$g_{\mathrm{vis}}(m) = 1 + ((m-1) \bmod 9) \in \{1, \dots, 9\},$$

the visible nonadic phase label. The corresponding class coordinate is $g_0(\theta_m) \in \mathbb{Z}/9\mathbb{Z}$, with $\theta_m = [m-1]_{108}$. The joint signature is the coordinate already typed by the phase kernel,

$$\sigma_m = \mathrm{Sig}_{q_m}(h_m, c_m, \lambda_m) = (q_\partial, a_\partial, c_\partial, \mathrm{colw}, \rho_W, r_\partial).$$

The joint signature does not reduce to three independent channel signatures, because the selection associated with the involution π/e and the axis φ depends on correlated data. Two nodes with the same B_m represent distinct states when they differ in extension, cylinder, carry, signature, orientation, or transport record.

8.5.1.1 Exact compatibility of the cylinders under truncation and nonadic prolongation

Let N be the integer of a ternary prefix of length L, and let D be the integer obtained by concatenating its b decimal triads. We define

$$A = N1000^b - D3^L, \qquad B = (N+1)1000^b - D3^L.$$

The cylinders intersect if and only if

$$\boxed{A < 3^L, \qquad B > 0.}$$

Adding a ternary block $u \in \{0, \dots, 728\}$ gives

$$N' = 729N + u.$$

If a new decimal triad does not appear,

$$A' = 729A + u1000^{b}.$$

If δ occurs, so that $D' = 1000D + \delta$, then

$$A' = 729000A + u1000^{b+1} - \delta\, 729\, 3^{L}.$$

If $\Delta b \in \{0, 1\}$ records the appearance of that new triad, in both cases

$$B' = A' + 1000^{b+\Delta b}.$$

The transition, once the admissible pair (u, δ) has been fixed, does not consult a target value. The joint filtration by intersection of cylinders and nonadic return acts at a distinct level: it selects the boundary pairs that admit compatible prolongation.

8.5.1.2 Cyclic registration of 108 stages and its reduced projection

We call the sequence of *108-stage cyclic record* the sequence of 108 steps grouped into twelve sectors of nine steps, with the index convention fixed in section 4.1.

Let $\mathcal{X}_{\mathrm{TPK}}^{[108]} \subseteq \mathcal{X}_{\mathrm{TPK}}$ be the domain of complete states whose register λ contains the declared 108 steps. Let us denote by C_{108}^{red} the set of registers that result by conserving only the reduced fields, and by $\mathcal{U}_{12}^{\mathrm{sgn}}$ the set of admissible signed dodecaphases. The instance distinguishes two declared maps:

$$F : \mathcal{X}_{\mathrm{TPK}}^{[108]} \longrightarrow C_{108}^{\mathrm{red}}, \qquad R : \mathcal{X}_{\mathrm{TPK}}^{[108]} \longrightarrow \mathcal{U}_{12}^{\mathrm{sgn}}.$$

The map F retains only the fields of the reduced register, whereas R evaluates the signed dodecaphase. The existence of a map Ξ such that $R = \Xi \circ F$ is equivalent to the factorization of R through a projection that may eliminate the observable coordinate, orientation, carry, or boundary. This factorization constitutes an additional proposition and is not an implicit part of the reconstruction on the complete state space.

The reduced record is, by definition, a projection of the complete state, not the initial state. Subsequent results use R on the complete state and do not presuppose a factorization through F.

8.5.2 Inverse systems of survivors

The TPK transition need not be a function. To type this fact correctly, let V_t be the finite set of possible complete states at depth t, and let

$$\mathcal{R}_t \subseteq V_t \times V_{t+1}$$

the admissible transition relation, with all fields of the enriched state conserved. We then define X_t, not as the set of isolated terminal states, but as the set of admissible prefixes, that is, of complete admissible histories:

$$X_t = \Big\{ (v_0, \ldots, v_t) \in \prod_{j=0}^{t} V_j : (v_j, v_{j+1}) \in \mathcal{R}_j \ (0 \le j < t), \text{ and the closures are satisfied up to } t \Big\}.$$

On these objects, truncation is indeed a canonical application:

$$\rho_{t+1,t} : X_{t+1} \longrightarrow X_t,$$
$$(v_0, \ldots, v_t, v_{t+1}) \longmapsto (v_0, \ldots, v_t).$$

It is well defined precisely because every prefix of an admissible history remains an admissible history for the closures already imposed. If a filter were to use future information and could retroactively invalidate a prefix, X_t would first have to be replaced by the subset of prolongable prefixes; under that convention, the same truncation is again well defined.

A global HMT history is a compatible family of prefixes.

$$x = (x_t)_{t\geq 0}, \qquad x_t \in X_t, \qquad \rho_{t+1,t}(x_{t+1}) = x_t.$$

It is equivalent to an infinite sequence $(v_0, v_1, \ldots)$ whose consecutive pairs belong to the relations $\mathcal{R}_t$ and whose prefixes satisfy all closures. The history space is the inverse limit

$$X_\infty = \varprojlim X_t.$$

This formulation avoids imposing a single-valued local transition $B_t \mapsto B_{t+1}$. The transition may be a finite relation, several branches may share the same visible terminal state, and uniqueness may appear only after future compatibility is imposed. By preserving the complete prefix, the information that makes it possible to distinguish them is not lost prematurely.

Theorem 8.37 (Unique coinductive section)**.** *Suppose that, for every t, the survivor set $S_t \subseteq X_t$ is nonempty, the restrictions $\rho_{t+1,t} : S_{t+1} \to S_t$ are compatible, and every S_t contains a unique prefix. Then $\varprojlim S_t$ contains a unique global history.*

Proof. Let x_t be the unique prefix of S_t. Compatibility forces $\rho_{t+1,t}(x_{t+1}) = x_t$, so that $(x_t)_t$ belongs to the inverse limit. Every other history would have a member of S_t at each level and, by uniqueness, would coincide prefix by prefix with $(x_t)_t$. □

The theorem formalizes the passage from level-by-level compatibility to a global history. It does not turn a finite window into a coinductive conclusion: its application to a concrete family requires verifying at every depth non- emptiness, compatibility of the restrictions, and uniqueness of the surviving prefixes.

8.5.3 Nonadic Survival Connection

Cylinder contraction does not proceed block by block as a deterministic substitution. At certain boundaries, there are several local allocations of the same aggregate datum; prolongability determines which belongs to the published history. The resulting structure is a connection on the nine phase classes: upon completing one turn, the phase class returns, but the state of the fiber has changed. This section constructs the complete finite window in which that return is observed.

The domain of this local calculation is the transport register generated by the APP–TRIT–TPK composition. Its initial states, its integer propagation matrix and its boundary relations are prior internal publications of the enriched transition $\mathcal{U}_t$, in accordance with the oriented register of twelve transitions and the unique reconstruction $L_i = X_i^{-1}Y_i$. This section receives them as already proved results, not as external premises or conventional data. All its statements are exact on those outputs. The preceding coinductive theorem extends the same construction to arbitrary depth and preserves nonemptiness, truncation compatibility and uniqueness of the surviving prefixes.

8.5.3.1 Prefix of 30 *trits and 1st boundary*

We write $u_k^\chi \in \mathbb{F}_3^6$, with $\chi \in \{\pi, e, \varphi\}$, for the visible block of channel χ at depth k. The first five blocks form three prefixes of thirty trits:

$$\begin{aligned} w_{30}^\pi &= 010211|012222|010211|002111|110221, \\ w_{30}^e &= 201101|121221|102011|012222|102011, \\ w_{30}^\varphi &= 121200|112202|121020|010210|010200. \end{aligned}$$

The sixth block is the first boundary following those prefixes:

$$R_{36} = \begin{pmatrix} u_5^\pi \\ u_5^e \\ u_5^\varphi \end{pmatrix} = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}.$$

The notation R_{36} indicates that thirty-six trits per channel have been displayed. It does not denote a terminal state: the record reconstructed below contains twenty blocks, and the recurrence admits further continuations when the corresponding Hensel states are supplied.

8.5.3.2 Self-scale axis and residual fiber

For

$$B_k = (u_k^{\pi}, u_k^{e}, u_k^{\varphi}) \in (\mathbb{F}_3^6)^3$$

we define, coordinate by coordinate,

$$q_k = u_k^{\pi} + u_k^{e} + u_k^{\varphi}, \qquad a_k = u_k^{\pi} + u_k^{e} - u_k^{\varphi}.$$

Theorem 8.38 (Axial decomposition of the boundary signature)**.** *The pair (q_k, a_k) uniquely determines the axial channel and the aggregate of the other two channels:*

$$u_k^{\varphi} = 2(q_k - a_k), \qquad u_k^{\pi} + u_k^{e} = q_k - u_k^{\varphi}.$$

Consequently, a fiber of the application

$$q_{\mathrm{ax}}(u^{\pi}, u^{e}, u^{\varphi}) = (u^{\pi} + u^{e}, u^{\varphi})$$

can preserve ambiguity only in the internal allocation between u^{π} and u^{e}.

Proof. Subtracting the two defining equations gives $q_k - a_k = 2u_k^{\varphi}$. The inverse of 2 in $\mathbb{F}_3$ is again 2, whence $u_k^{\varphi} = 2(q_k - a_k)$. Subtracting this vector from q_k produces $u_k^{\pi} + u_k^{e}$. The operation does not recover each summand separately; for this reason the unique residual fiber is the set of distributions of the fixed aggregate. □

In the model's geometric terminology, φ constitutes the axis fixed by the signature and the pair π/e the residual fiber. The expression does not postulate an analytic symmetry of the real constants: it names the exact linear decomposition just proved in $\mathbb{F}_3^6$.

8.5.3.3 Orbita complete of the 9 fases consecutivas of the Topological Phase Kernel

We fix the absolute phase by

$$g(k) = \begin{cases} k \bmod 9, & k \not\equiv 0 \pmod 9, \\ 9, & k \equiv 0 \pmod 9. \end{cases}$$

Since w_{30} occupies $k = 0, \dots, 4$, the first subsequent complete circuit after the prefix traverses $k = 5, \dots, 13$, with phases $5, 6, 7, 8, 9, 1, 2, 3, 4$. Enumeration of the $729 = 3^6$ candidates of aggregate signature at each level produces the following window:

k	$g(k)$	locals	selected	u_k^{π}	u_k^{e}	u_k^{φ}
5	5	1	1	222220	021222	102011
6	6	1	1	111201	201202	221021
7	7	3	1	212121	222210	200221
8	8	3	1	200121	212212	110110
9	9	2	1	100100	020112	010122
10	1	1	1	101222	112221	112002
11	2	1	1	022212	110001	100021
12	3	1	1	012012	202222	000120
13	4	1	1	111210	112102	211122

The "local" column counts the admissible input states at the indicated level; it does not count the outputs of the boundary relation. This distinction is essential in $k = 9$: the two local input states precede a three-output fiber of the selected state.

8.5.3.4 Return 9→1 of the nonadic phase with state change and memory advance

Let

$$\pi_B : V_k \longrightarrow \mathcal{B} := (\mathbb{F}_3^6)^3$$

the projection onto the visible triple. Once the fiber data of the record are fixed, the transport relation on complete states induces a visible relation $\mathscr{M}_9^B$. The selected state entering the ninth phase is

$$B_9 = (100100 \mid 020112 \mid 010122).$$

Its relational image consists exactly of

$$\begin{aligned} B_{10}^{(0)} &= (101222 \mid 112221 \mid 112002), \\ B_{10}^{(1)} &= (102221 \mid 111222 \mid 112002), \\ B_{10}^{(2)} &= (111221 \mid 102222 \mid 112002). \end{aligned}$$

The three outputs have

$$q_{10} = 022112, \qquad a_{10} = 101111,$$

and, by theorem 8.38, share

$$u_{10}^{\varphi} = 112002, \qquad u_{10}^{\pi} + u_{10}^{e} = 210110.$$

They differ exclusively in the three published distributions of that aggregate between π and e.

Theorem 8.39 (Two-level extendibility selection)**.** *In the calibrated window, the three preceding outputs admit a prolongation of depth one. Upon also imposing the next block*

$$B_{11} = (022212 \mid 110001 \mid 100021)$$

and the decimal boundary $(502, 662, 638)$*, only* $B_{10}^{(0)}$ *admits a prolongation of the same complete state. Therefore*

$$\#\,\mathrm{Surv}_1^{\mathrm{cal}}(B_9) = 3, \qquad \#\,\mathrm{Surv}_2^{\mathrm{cal}}(B_9) = 1.$$

Proof. The decimal triples associated with the three outputs are, respectively,

$$(279, 352, 365), \quad (280, 351, 365), \quad (282, 350, 365).$$

For a ternary word of length L, with associated integer N, and a prefix of b triples, with associated integer D, compatibility of the two cylinders is equivalent to the two integer inequalities

$$N1000^b < (D+1)3^L, \qquad (N+1)1000^b > D3^L.$$

The six inequalities for each output are satisfied at the current level. Upon adjoining B_{11} and $(502, 662, 638)$, they are again satisfied for the three channels of $B_{10}^{(0)}$. For $B_{10}^{(1)}$ and $B_{10}^{(2)}$, the inequality of channel φ is preserved, but those of π and e fail. The evaluation is performed with integers in the corresponding finite certificate; no numerical tolerance intervenes. Consequently, the candidate set is reduced from three to one when passing from depth one to depth two. □

The other two words can precede the same ternary tail only if the deep Hensel extension is changed. They are therefore not prolongations of the same complete state of the register: the visible residue partially coincides, but the genealogy of the fiber is distinct.

8.5.3.5 Phase return with state change

The phase satisfies $g(k+9) = g(k)$. The visible state, by contrast, is not reset. In particular,

$$B_1 = (012222 \mid 121221 \mid 112202), \qquad B_{10} = (101222 \mid 112221 \mid 112002),$$

so that $g(1) = g(10) = 1$ but $B_1 \neq B_{10}$. The verification of the nine pairs $k \mapsto k+9$, for $k = 1, \ldots, 9$, also gives

$$u^{\varphi}_{k+9} \neq u^{\varphi}_{k}, \qquad u^{\pi}_{k+9} + u^{e}_{k+9} \neq u^{\pi}_{k} + u^{e}_{k}.$$

Therefore, the window satisfies exactly

$$\boxed{g(k+9) = g(k), \qquad B_{k+9} \neq B_k \quad (1 \leq k \leq 9).}$$

This is a phase return with nontrivial visible holonomy. The designation *nonadic monodromy* refers to this return action on the enriched fiber; it does not introduce a tenth phase. To promote the finite remark to a global monodromy, the action at every depth must be constructed and the coinductive compatibility required at the beginning of the section must be proved.

8.5.3.6 Prolongation of the surviving section via 20 compatible blocks

The preceding return does not close the emission. The twenty blocks that the certificate reconstructs from the three published integer states are

$$\begin{aligned}
\pi :\ & 010211|012222|010211|002111|110221|222220|111201| \\
& 212121|200121|100100|101222|022212|012012|111210| \\
& 121011|200220|120210|000101|022010|020111, \\
e :\ & 201101|121221|102011|012222|102011|021222|201202| \\
& 222210|212212|020112|112221|110001|202222|112102| \\
& 102010|022020|222002|012010|001112|022212, \\
\varphi :\ & 121200|112202|121020|010210|010200|102011|221021| \\
& 200221|110110|010122|112002|100021|000120|211122| \\
& 202200|202110|221000|100102|111122|020010.
\end{aligned}$$

Thus, w_{30} is the prefix and R_{36} the first boundary of an explicitly continued branch; neither object replaces the enriched state that transports that continuation.

8.5.3.7 Exact decoding of 12 *decimal triples by means of the nonadic reader*

Let $N_{\chi,k}$ be the integer represented in base three by the first k blocks of channel χ. With $k = 13$, the word has 78 trits and defines

$$D_{\chi,12} = \left\lfloor \frac{1000^{12} N_{\chi,13}}{3^{78}} \right\rfloor.$$

The escritura of $D_{\chi,12}$ in twelve blocks of three cifras is

π	141	592	653	589	793	238	462	643	383	279	502	884
e	718	281	828	459	045	235	360	287	471	352	662	497
φ	618	033	988	749	894	848	204	586	834	365	638	117.

The identity is entire: it is equivalent to

$$D_{\chi,12}3^{78} \le 1000^{12}N_{\chi,13} < (D_{\chi,12}+1)3^{78}.$$

Therefore, the two cylinders intersect in each channel and the reading operation produces exactly the thirty-six published digits. This result is a consequence of the generated register. The historical constructions that used conventional expansions belong exclusively to comparison and subsequent checking: they do not enter into orbital selection, the transition $\mathcal{U}_t$, the reconstruction of L_0, L_1 or the calendar 0011. The identity certifies the Archimedean reading of the states produced by APP–TRIT–TPK and preserves intact their generative provenance independent of target values.

8.5.3.8 Schedule for cylinder reopening and periodicity of the compatible prolongation

The conversion of a six-trit block to decimal triples has average slope

$$\lambda = 6\log_{1000} 3 \in (0,1), \qquad k(t) = \lfloor \lambda(t+1) \rfloor.$$

An index t is a reopening without a new triple when $k(t+1) = k(t)$. Let $\delta = 1-\lambda$.

Theorem 8.40 (Beatty Reading Calendar Ternary-Decimal)**.** *The set of reopenings is*

$$\mathcal{T} = \left\{ \left\lceil \frac{m}{\delta} \right\rceil - 2 : m \ge 1 \right\}.$$

Its first elements are

$$20, 42, 64, 86, 108, 130, 151, 173, 195, 217, \ldots,$$

and two reaperturas consecutivas distan 21 *or* 22 *blocks.*

Proof. The quantity $\lambda = \log_{1000} 729$ is irrational: an equality $\lambda = p/q$ would imply $729^q = 1000^p$, impossible by prime factorization. Also δ is irrational, so that

$$k(t) = (t+1) - \lceil \delta(t+1) \rceil.$$

The equality $k(t+1) = k(t)$ occurs exactly when the ceiling function of the second term increases by one. The m-th occurrence is obtained by solving $\delta(t+1) < m < \delta(t+2)$, which gives $t = \lceil m/\delta \rceil - 2$. The differences of a Beatty sequence belong to

$$\left\{ \left\lfloor \delta^{-1} \right\rfloor, \left\lceil \delta^{-1} \right\rceil \right\} = \{21, 22\}. \qquad \square$$

8.5.3.9 Bockstein–Hensel Recurrence of Residue and Quotient

The memory carried by the branch proceeds from the exact sequence.

$$0 \longrightarrow \mathbb{Z}^6 \xrightarrow{\times 3} \mathbb{Z}^6 \longrightarrow \mathbb{F}_3^6 \longrightarrow 0$$

and of the unimodular matrix

$$A = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad \det A = 1.$$

For a row state $x_k^{\Upsilon} \in \mathbb{Z}^6$, it is defined

$$y_k^{\chi} = x_k^{\chi} A, \qquad u_k^{\chi} = y_k^{\chi} \bmod 3 \in \{0,1,2\}^6, \qquad x_{k+1}^{\chi} = \frac{y_k^{\chi} - u_k^{\chi}}{3}.$$

Proposition 8.41 (Exact conservation of quotient memory)**.** *With x_k^{χ} and A fixed, the pair $(u_k^{\chi}, x_{k+1}^{\chi})$ is unique and satisfies*

$$x_k^{\chi} A = u_k^{\chi} + 3x_{k+1}^{\chi}.$$

Conversely, since $A \in \mathrm{GL}_6(\mathbb{Z})$, the pair uniquely determines x_k^{χ}.

Proof. Euclidean division of each coordinate of $x_k^{\chi} A$ by three produces a unique residue in $\{0,1,2\}$ and a unique integer quotient. This proves the first assertion and the identity. Since $\det A = 1$, the inverse A^{-1} has integer entries; therefore

$$x_k^{\chi} = (u_k^{\chi} + 3x_{k+1}^{\chi})A^{-1}$$

recovers the previous state unambiguously. □

Residue u_k^{χ} is the visible part, and quotient x_{k+1}^{χ} the transported 3-adic memory. The certificate reproduces sixty coordinated steps —twenty per channel—, verifies $AA^{-1} = I$ with integer arithmetic, and checks all blocks against the register. The designation Bockstein–Hensel refers here to this residue–quotient separation and its 3-adic iteration; it is not identified without additional data with a cohomological Bockstein operator.

8.5.4 Archimedean Evaluation of the Compatible Prefix and Determination of the Real Point

Let us associate with each compatible prefix $x_t \in X_t$ a closed interval $I_t(x_t) \subset \mathbb{R}$. The geometric interpretation is that of a cylinder: all futures sharing the same prefix project into I_t.

Theorem 8.42 (Archimedean convergence of nested intervals)**.** *Let $(x_t)_{t\geq 0}$ be a compatible history. If*

$$I_{t+1}(x_{t+1}) \subseteq I_t(x_t)$$

and

$$\operatorname{diam} I_t(x_t) \longrightarrow 0,$$

then there exists a unique $r \in \mathbb{R}$ such that

$$\bigcap_{t\geq 0} I_t(x_t) = \{r\}.$$

Proof. This is the principle of nested intervals. The left endpoints form a bounded increasing sequence and the right endpoints a bounded decreasing one. Their limits exist; the difference between them is bounded by the diameter, which tends to zero. Both limits coincide at a single point contained in all the intervals. □

Let $X_\infty^{\mathrm{Arch}} \subseteq X_\infty$ be the domain of histories for which the cylinders are nested and their diameters tend to zero. If those two properties are part of global admissibility, then $X_\infty^{\mathrm{Arch}} = X_\infty$; if not, this domain restriction is necessary. We define the evaluation

$$\mathrm{ev}_{\mathbb{R}} : X_\infty^{\mathrm{Arch}} \longrightarrow \mathbb{R}, \qquad \mathrm{ev}_{\mathbb{R}}(x) = \text{the unique point of } \bigcap_{t\geq 0} I_t(x_t).$$

Uniqueness of the value does not entail uniqueness of the history. Two genealogies $x \neq y$ may satisfy

$$\mathrm{ev}_{\mathbb{R}}(x) = \mathrm{ev}_{\mathbb{R}}(y).$$

We define reading equivalence

$$x \sim_{\mathrm{ev}} y \quad \Longleftrightarrow \quad \mathrm{ev}_{\mathbb{R}}(x) = \mathrm{ev}_{\mathbb{R}}(y).$$

Theorem 8.43 (Archimedean evaluation quotient)**.** *The map*

$$\overline{\mathrm{ev}}_{\mathbb{R}} : X_{\infty}^{\mathrm{Arch}}/\!\sim_{\mathrm{ev}} \longrightarrow \mathrm{Im}(\mathrm{ev}_{\mathbb{R}}) \subseteq \mathbb{R},$$
$$[x] \longmapsto \mathrm{ev}_{\mathbb{R}}(x)$$

is a bijection. In particular, the quotient is identified with all of $\mathbb{R}$ *if and only if the evaluation* $\mathrm{ev}_{\mathbb{R}}$ *is surjective.*

Proof. The definition of $\sim_{\mathrm{ev}}$ makes the value independent of the representative, so the map is well defined. It is injective because two classes with the same value are, by definition, a single class; it is surjective onto the image by the very definition of image. The last assertion is equivalent to $\mathrm{Im}(\mathrm{ev}_{\mathbb{R}}) = \mathbb{R}$. □

The covering theorem for the Archimedean reader constructs precisely those prefixes: for every compact interval $K \subset \mathbb{R}$, $\mathrm{ev}_{\mathbb{R}}(\Omega_{\infty,K}) = K$, and the intervals $[-m, m]$ form a directed family whose union is $\mathbb{R}$. Consequently, every real number possesses at least one compatible HMT history, and the proved assertion is

$$X_{\infty}^{\mathrm{Arch}}/\!\sim_{\mathrm{ev}} \cong \mathrm{Im}(\mathrm{ev}_{\mathbb{R}}) = \mathbb{R}.$$

The Archimedean value is, therefore, the quotient that identifies the fibers of the evaluation; the enriched HMT number also conserves the class of trajectory, the carry, and the orientation. The identification does not follow from the mere abstract formation of the quotient: it follows from the covering theorem already proved in the discrete construction of the continuum.

Algebraic descent is another problem and must be solved operation by operation. Suppose that two candidate operations $\oplus$ and $\odot$ have been constructed on $X_{\infty}^{\mathrm{Arch}}$ and are closed on that domain. Addition descends to the quotient if and only if

$$x \sim_{\mathrm{ev}} x',\ y \sim_{\mathrm{ev}} y' \Longrightarrow x \oplus y \sim_{\mathrm{ev}} x' \oplus y',$$

while the product descends, independently, if and only if

$$x \sim_{\mathrm{ev}} x',\ y \sim_{\mathrm{ev}} y' \Longrightarrow x \odot y \sim_{\mathrm{ev}} x' \odot y'.$$

The first congruence does not imply the second. Moreover, for the induced operations to be precisely the ordinary sum and product of the Archimedean quotient, it is necessary to prove separately

$$\mathrm{ev}_{\mathbb{R}}(x \oplus y) = \mathrm{ev}_{\mathbb{R}}(x) + \mathrm{ev}_{\mathbb{R}}(y),$$

$$\mathrm{ev}_{\mathbb{R}}(x \odot y) = \mathrm{ev}_{\mathbb{R}}(x)\,\mathrm{ev}_{\mathbb{R}}(y),$$

and that the image of the evaluation be closed under the corresponding operation. Only after these certificates does the quotient inherit the structure of a subring —or of a subfield, if additive opposites and multiplicative inverses are also constructed— of $\mathbb{R}$. The set-theoretic bijection of the preceding theorem is formal; the content specific to HMT lies in constructing the evaluation, characterizing its fibers, proving coverage, and establishing each descent law.

8.5.5 Weyl–Heisenberg Representation Associated with the Nonadic Torus

The oriented area form of the nonadic torus is

$$\sigma((a,b),(c,d)) = ad - bc \pmod 9.$$

This alternating form is obtained, with the chosen orientation, by integrating the mixed curvature of APP over parallelograms of the torus. Let $\omega = \exp(2\pi i/9)$ and $\mathcal{H}_9 = \mathbb{C}^9$, with basis $\{|n\rangle : n \in \mathbb{Z}_9\}$. We define

$$X|n\rangle = |n+1\rangle, \qquad Z|n\rangle = \omega^n |n\rangle.$$

Then

$$ZX = \omega XZ,$$

and, more generally,

$$(X^a Z^b)(X^c Z^d) = \omega^{bc} X^{a+c} Z^{b+d}.$$

Therefore

$$c\big((a,b),(c,d)\big) = bc \pmod 9$$

is the polarized 2-cocycle that appears in this section of the central extension. We explicitly define

$$\mathrm{Heis}_9 = \{\omega^k X^a Z^b : a, b, k \in \mathbb{Z}/9\mathbb{Z}\}.$$

It is a group of order $9^3 = 729$. It is a central extension associated with the translation group of the APP support X_9, not a subgroup of the graph or the APP path category. The commutator of the two transports is

$$(X^a Z^b)(X^c Z^d)(X^a Z^b)^{-1}(X^c Z^d)^{-1} = \omega^{bc-ad} I.$$

Consequently, its exponent is $-\sigma((a,b),(c,d))$. The polarized cocycle c and the form σ are not the same object: the second is, under this convention, the opposite of the antisymmetrization $c(u,v) - c(v,u)$. The commutator phase is thus the inverse of the Wilson phase for the fixed orientation.

To construct observables, we take

$$Q_{\mathrm{H}} = \mathrm{diag}(0,1,\dots,8), \qquad F_{jk} = \frac{1}{3}\omega^{jk}, \qquad P_{\mathrm{H}} = F^{\dagger} Q_{\mathrm{H}} F.$$

The operators $Q_{\mathrm{H}}, P_{\mathrm{H}}$ are self-adjoint. Their eigenbases are mutually unbiased because $|F_{jk}|^2 = 1/9$. For every unit vector $\psi \in \mathcal{H}_9$, the Robertson inequality gives

$$\mathrm{Var}_\psi(Q_{\mathrm{H}})\ \mathrm{Var}_\psi(P_{\mathrm{H}}) \geq \frac{1}{4}\,|\langle \psi, [Q_{\mathrm{H}}, P_{\mathrm{H}}]\psi\rangle|^2\,,$$

and the entropy bound for mutually unbiased bases produces

$$H_{Q_{\mathrm{H}}}(\psi) + H_{P_{\mathrm{H}}}(\psi) \geq \log 9.$$

Here H is the Shannon entropy, with natural logarithm, of the corresponding spectral distributions. The constructed relations are summarized, without attributing any additional physical causality to them, by

$$\text{alternating form } (X_9, \sigma) \longrightarrow \mathrm{Heis}_9,$$
$$\mathrm{Heis}_9 \longrightarrow (X, Z) \longrightarrow (Q_{\mathrm{H}}, P_{\mathrm{H}}) \longrightarrow \text{finite uncertainty}.$$

This is a finite algebraic model of operators; by itself it does not specify physical dynamics or units of measurement.

8.5.6 Cyclic dynamics on $\mathbb{Z}_9 \times \mathbb{Z}_{12}$ and minimal predictive quotient

The finite dynamical system $\mathfrak{C}_{9,12}$ makes it possible to determine the minimal memory required for the observed evolution to be deterministic. Let

$$X = \{(b, m) : b \in \mathbb{Z}/9\mathbb{Z}, 0 \le m < 12\}$$

with successor

$$S(b, m) = \begin{cases} (b, m+1), & m < 11, \\ (b+1 \bmod 9, 0), & m = 11. \end{cases}$$

The marginal labels are

$$\beta(b, m) = 4b \pmod{12}, \qquad d(b, m) = 1 + (m + \beta(b, m) \pmod{12}).$$

For a remark application q, the inclusive future word of horizon h is

$$Q_h^q(x) = \big(q(x), q(Sx), \dots, q(S^h x)\big).$$

Theorem 8.44 (Minimum predictive quotient)**.** *Let X finite, $T : X \to X$ and $q : X \to V$. Definamos*

$$\mathrm{Eq}(f) := \{(x, y) \in X^2 : f(x) = f(y)\}, \qquad E_h = \mathrm{Eq}(Q_h^q), \qquad E_\infty = \bigcap_{k \ge 0} \mathrm{Eq}(q \circ T^k).$$

Then $E_{h+1} = E_h \cap (T \times T)^{-1}(E_h)$, the sequence stabilizes in finite time, and X/E_∞ is the quotient with the smallest number of states that refines q and admits an induced deterministic dynamics.

Proof. E_h requires equality of observations at the times $0, \dots, h$, whereas $(T \times T)^{-1}(E_h)$ requires equality at $1, \dots, h+1$. Their intersection is E_{h+1}. Because X is finite, the number of classes cannot grow indefinitely. At the first stable level, T respects the classes and descends. Every T-compatible equivalence contained in $\mathrm{Eq}(q)$ identifies only states with the same complete future and is therefore contained in E_∞; hence the minimality property. □

The evaluation exhaustiva of $\mathfrak{C}_{9,12}$ da:

observable	instantaneous values	index h	observations	stable quotient
β	3	11	12	C_{36}, fibra 3
d	12	8	9	C_{36}, fibra 3
(β, d)	36	0	1	C_{36}, fibra 3

In the two marginal observables,

$$\mathrm{Eq}(Q_{11}^\beta) = \mathrm{Eq}(Q_8^d) = \mathrm{Eq}(\pi_{36}), \qquad \pi_{36}(b, m) = (b \bmod 3, m).$$

The factor 36 is not introduced as a target. It appears as the minimal predictive quotient of each marginal. The phase m is a state coordinate required to predict those futures; this mathematical assertion does not presuppose a physical causal interpretation. The transformation $b \mapsto b + 3$ is a symmetry invisible to those observables at every time.

As an abstract dynamical system, the quotient is a cycle of thirty-six states. Writing C_{36} does not, however, choose a distinguished origin: it must be fixed to obtain a coordinate $\mathbb{Z}/36\mathbb{Z}$. Nor does this automatically turn those states into the thirty-six pairs of a nonad. Indeed, $\mathrm{Aut}\, J(9, 2) \cong S_9$. If the two points of a pair lie in distinct cycles of lengths r, s of a permutation, the orbit of the pair has length $\mathrm{mcm}(r, s)$, with $r + s \le 9$, and hence length at most 20. If they lie in the same cycle of length $r \le 9$, the

orbit has length r, except in the antipodal case, when it has length $r/2$. Thus no automorphism induces an orbit of length 36 on the pairs. Therefore, the passage

$$C_{36} \longleftrightarrow \binom{[9]}{2}$$

remains a calibrated identification, not a natural equivariant identification. Even a Hamiltonian ordering of the pairs would prove local preservation of incidence step by step, not a global automorphism or the canonicity of the dictionary.

8.5.7 HMT number as a compatible history in X_∞: genealogical identity, reading structure, and Archimedean evaluation

The preceding results determine the following instance of the governing definition. In the notation of this section, an HMT number is a compatible history

$$x \in X_\infty = \varprojlim X_t.$$

A reader L is additional structure: the pair (x, L) constitutes a read presentation or a measurement, but does not alter the identity type of x. When x belongs to X_∞^{Arch}, its Archimedean value is the point intersection of the cylinders; the set of values so constructed is $\mathrm{Im}(\mathrm{ev}_{\mathbb{R}}) \subseteq \mathbb{R}$, and its identification with the entire line requires the additional coverage theorem. Its minimal memory relative to an observable is the predictive quotient of futures; and the alternating form of the sum–product support admits the finite Weyl–Heisenberg representation constructed above. Value, trajectory class, and uncertainty are defined by three operations on specified domains of the same transport architecture.

8.6 Monodromic generation by survival of π, e and φ

Generation does not begin with a prescribed decimal expansion. It begins in the complete TPK state: visible word, sheet, orientation, quotient, carry, Hensel state, ternary and decimal cylinders, joint signature, boundary, phase, and dodecaphasic register. The EF–G_9 relation prolongs that state and eliminates a continuation only when it ceases to survive some future boundary. The decimal reader subsequently acts on the section that has been determined.

The causal chain is

$$\Omega_{\mathrm{TPK}} \longrightarrow 468\,\mathcal{U}_6 \longrightarrow 243\,\mathcal{W}_6 \longrightarrow (w_\pi, w_e, w_\varphi) \longrightarrow (\mathcal{R}_{\pi,n}, \mathcal{R}_{e,n}, \mathcal{R}_{\varphi,n})_{n\geq 0} \longrightarrow \varprojlim_n \mathcal{R}_{\chi,n} \xrightarrow{L_{\mathrm{arch}}} \chi.$$

The ambient space $\mathbb{F}_3^6$ has 729 words, but the effective image contains 243; the two objects are not identified. Within that image, internal selection fixes

$$w_\pi = 010211, \qquad w_e = 201101, \qquad w_\varphi = 121200$$

Before consulting figures of π, e, φ.

8.6.1 Generative regions and global section

Definition 8.45 (Generative region of a channel)**.** For $\chi \in \{\pi, e, \varphi\}$, let $\mathcal{R}_{\chi,n}$ be the finite set of complete states of depth n that start from w_χ and satisfy at each transition the relation EF–G_9, the exact intersection of cylinders, and the joint boundary signature. If $\tau_{n+1,n}$ eliminates only the last layer, we define

$$\mathcal{R}_\chi = \varprojlim_n (\mathcal{R}_{\chi,n}, \tau_{n+1,n}).$$

The published value is $L_{\mathrm{arch}}(x_\chi)$, where $x_\chi \in \mathcal{R}_\chi$; the reader is not part of the state that selects it.

Theorem 8.46 (Monodromy Prolongation Principle in the Full TPK)**.** *Suppose that, for a channel $\chi \in \{\pi, e, \varphi\}$, each set $\mathrm{Surv}_{\chi,n}$ is nonempty and unitary, and that its elements are compatible with truncation. Then there exists a unique section*

$$x_\chi = (x_{\chi,n})_{n\geq 0} \in \mathcal{R}_\chi$$

The decimal cylinders associated with their truncations are nested, and their diameters tend to zero. If the channel reader identifies their intervals with the declared closure, propagation, or self-scaling, the intersection publishes, respectively, π, e, or φ.

Proof. In the full profile, stable survival at level n is

$$\mathrm{Surv}_{\chi,n} = \bigcap_{h\geq 1} \{s : s \text{ admits a compatible prolongation across } h \text{ boundaries}\}.$$

By hypothesis, the surviving state at each level truncates to that at the preceding level. The levels form a finitely branching, nonempty tree at every depth. The Koenig lemma provides a global branch; levelwise unitarity precludes a second section.

For a ternary prefix of length L, interpreted as N, and a prefix of m triads, interpreted as D, the compatibility conditions are

$$I_{N,L} = \left[\frac{N}{3^L}, \frac{N+1}{3^L}\right), \qquad J_{D,m} = \left[\frac{D}{1000^m}, \frac{D+1}{1000^m}\right), \qquad I_{N,L} \cap J_{D,m} \neq \varnothing.$$

The prolongation conserves the intersection and makes both diameters tend to zero. The total intersection contains a unique real. The declared reader identifies that real within its channel. □

The reproducible execution does not replace the theorem's hypotheses with an implicit assertion. It verifies them at each of the (3501) levels needed to print ten thousand digits: there the partition $729 \to 1$, the compatibility of truncations, and the change of memory after return are checked exhaustively.

8.6.2 Survival filtration, nonadic monodromy, and π/e symmetry

For

$$B_n = (u_n^\pi, u_n^e, u_n^\varphi), \qquad q_n = u_n^\pi + u_n^e + u_n^\varphi, \qquad a_n = u_n^\pi + u_n^e - u_n^\varphi,$$

the joint signature determines

$$u_n^\varphi = 2(q_n - a_n), \qquad u_n^\pi + u_n^e = q_n - u_n^\varphi \quad \text{in } \mathbb{F}_3^6.$$

Thus, φ is the fixed self-scaling axis, and the only local ambiguity is the mirror allocation between π and e.

The phase is $g(n) = n \bmod 9$, with $0 \equiv 9$. After w_{30}, the first turn traverses

$$5, 6, 7, 8, 9, 1, 2, 3, 4.$$

They are not nine independent filters, but the nine phases of a single connection: pruning, synchronization, mirror orientation, reopening, and confirmation by survival.

In the ninth phase,

$$B_9 = (100100 \mid 020112 \mid 010122)$$

has three locally compatible outputs. The three conserve $u_{10}^\varphi = 112002$ and $u_{10}^\pi + u_{10}^e = 210110$; all survive one boundary, but only

$$\boxed{B_{10} = (101222 \mid 112221 \mid 112002)}$$

the second survives. The future breaks the local mirror. Moreover,

$$g(n+9) = g(n), \qquad x_{\chi,n+9} \neq x_{\chi,n}$$

in the returns: the next stage returns to phase 1, not to the preceding state. The quotient, carry, and boundary turn the circuit into a helix of memory. This return action with nontrivial transport is the nonadic monodromy of the TPK.

8.6.3 Reproducible emission of 12 triads and reconstruction of their enriched state

Thirteen blocks, equivalent to 78 trits, suffice for the reader to

$$D = \left\lfloor \frac{1000^{12} N}{3^{78}} \right\rfloor$$

publish the following segment directly from the ternary branches:

π	141	592	653	589	793	238	462	643	383	279	502	884
e	718	281	828	459	045	235	360	287	471	352	662	497
φ	618	033	988	749	894	848	204	586	834	365	638	117.

This table does not feed the transition: it exhibits its output. The iteration of the same connection, with the return $9 \to 1$, produces any requested finite prefix under the hypotheses of the theorem 8.46; that theorem specifies the prolongation to the inverse limit.

8.6.4 Unique closure of π and classification of its 5 regional histories

Let

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \qquad \Psi(U_6) = (-U_6) \bmod 3$$

the coordinate projection of the catalog. The fiber

$$\mathscr{R}_\pi^{\mathrm{micro}} := \Psi^{-1}(010211)$$

contains exactly the five vectors

$$\begin{aligned} U_\pi^{\perp} &= (501, 614, 498, 169, 272, 272), \\ U_{\pi,1}^{++} &= (810, 923, 870, 169, 272, 272), \\ U_{\pi,2}^{++} &= (810, 983, 810, 169, 272, 272), \\ U_{\pi,3}^{++} &= (870, 923, 810, 169, 272, 272), \\ U_\pi^{--} &= (870, 923, 810, 418, 674, 521). \end{aligned}$$

Their multiplicative signatures are

$$555555, \qquad 339555, \qquad 393555, \qquad 933555, \qquad 933717,$$

and their multiplicities are

$$432, \qquad 144, \qquad 144, \qquad 144, \qquad 144.$$

Therefore,

$$\boxed{5 = 1_{\perp} + 3_{++} + 1_{--}, \qquad 1008 = 432 + 4 \cdot 144.}$$

The first identity classifies orientation functions; the second counts seed routes. They are not two ways of counting five copies of the same object.

The visible quotient

$$\mathscr{R}_\pi^{w_6} = \Psi(\mathscr{R}_\pi^{\text{micro}}) = \{010211\}$$

has a single element because it forgets regional provenance. Equality of words does not identify the five states: the first four regions preserve the tail 169|272|272, whereas the negative region effects a mutated return. The transverse region, of multiplicity 432, likewise cannot be reduced to one of the four branches of multiplicity 144.

The closure function is also seen in the multigraph of blocks. Each of the five regions belongs to a minimal closed walk of length eight that also traverses the anchors U_e, U_φ, and U_{APP}. The enumeration and the proof of minimality reside in chapter 9; the regional list and its multiplicities, in chapter 9.

Proposition 8.47 (Closure Package for π)**.** *The datum*

$$\mathfrak{C}_\pi = \big(\mathscr{R}_\pi^{\text{micro}}, w_\pi, \rho_\pi, m_\pi, \ell_{\text{cl}}\big),$$

where

$$\rho_\pi = (\perp, ++, ++, ++, --), \qquad m_\pi = (432, 144, 144, 144, 144), \qquad \ell_{\text{cl}} = 8,$$

preserves exactly the functional information lost by the quotient $\mathscr{R}_\pi^{w_6}$. Consequently, the function of π in this generation is to close five regional histories into a single evaluation, not to repeat the same constant five times.

Proof. The fiber, roles, and multiplicities are those of chapter 9. The chapter 9 proves that each region enters a minimal closure of length eight with the three declared anchors. The projection Ψ sends the five regions to the same word, so that the quotient preserves the visible value and forgets $\rho_\pi, m_\pi, \ell_{\text{cl}}$. □

8.6.5 e: visible propagation and differentiated quotient memory

The fiber of $w_e = 201101$ consists of an emission U_e with multiplicity 144. Its functional feature is not an infinite periodicity, but rather a local repetition exhaustively distinguished within the reader of the 243 words. The theorem 53.12 proves that w_e is the only word whose twelve-digit reading contains a square of length four beginning at the second digit:

$$718281828459 = 7\ \underbrace{1828\,1828}_{\text{local square}}\ 459.$$

Juxtaposition of blocks is decimal concatenation; it is not the arithmetic product $7 \cdot 1828^2 \cdot 459$. The catalog contains another square of length four,

$$575157518472 = \underbrace{5751\,5751}_{\text{local square}}\ 8472,$$

but begins at the first digit and, therefore, does not satisfy the same positional predicate.

The complete census of squares of lengths $1, \ldots, 6$ is

ℓ	1	2	3	4	5	6
occurrences	240	21	3	2	0	0
words that contain them	153	19	3	2	0	0.

The repetition 1828 1828 does not begin a period either: a third copy would require a 1 as the next digit, but the effective reading contains a 4.

The dynamics of elevation show why the episode does express propagation:

$$201101 \mid 121221 \mid 102011 \mid 012222 \mid 102011.$$

The third and fifth visible blocks coincide, but the states that emit them are distinct. Their prestates modulo 27 are

$$(25, 11, 7, 17, 8, 16), \qquad (7, 8, 4, 23, 26, 7),$$

and their subsequent Hensel quotients are

$$(6, 13, 9, 2, 13, 1), \qquad (15, 0, 3, 19, 23, 25).$$

There is return of the projection 102011, not return of the enriched state. The output can propagate and reappear because the quotient transports the genealogy that the visible word has omitted.

Proposition 8.48 (Functional content of e)**.** *The exact generation of e combines three compatible facts:*

1. *the selection internal of* $w_e = 201101$*;*
2. *the positional uniqueness of the local square* 1828 1828 *in the fixed catalogue;*
3. *the reappearance of* 102011 *under different memory states.*

Its function is propagation with memory renewal. It is not periodicity of the expansion nor repetition of the complete state.

Proof. The first point is the case e of the theorem 53.14; the second and its census are the theorem 53.12. The inequality of the two prestates and of the two Hensel quotients proves the third. □

8.6.6 Autoscaling Axis and Mirror Fiber

The joint signature of the three generations preserves a linear decomposition that will subsequently be used by the scale reader. For

$$B_k = (u_k^{\pi}, u_k^{e}, u_k^{\varphi}) \in (\mathbb{F}_3^6)^3$$

we define, coordinate by coordinate,

$$q_k = u_k^{\pi} + u_k^{e} + u_k^{\varphi}, \qquad a_k = u_k^{\pi} + u_k^{e} - u_k^{\varphi}.$$

Theorem 8.49 (Axial decomposition of the boundary signature)**.** *The pair* (q_k, a_k) *uniquely determines the self-scaling channel and the aggregate of the other two channels:*

$$u_k^{\varphi} = 2(q_k - a_k), \qquad u_k^{\pi} + u_k^{e} = q_k - u_k^{\varphi}.$$

Consequently, a fiber of

$$q_{\mathrm{ax}}(u^{\pi}, u^{e}, u^{\varphi}) = (u^{\pi} + u^{e}, u^{\varphi})$$

preserves ambiguity only in the internal allocation between u^{π} *and* u^{e}*.*

Proof. Subtracting the two defining equations gives $q_k - a_k = 2u_k^{\varphi}$. Since the inverse of 2 in $\mathbb{F}_3$ is again 2,

$$u_k^{\varphi} = 2(q_k - a_k).$$

Upon subtracting this vector from q_k there remains $u_k^{\pi} + u_k^{e}$. The operation fixes the axis φ, but does not separate the two summands of the fiber π/e; that is exactly the residual ambiguity. □

The expression *mirror fiber* names this exact decomposition in $\mathbb{F}_3^6$. It does not posit an analytic identity between the real constants: it indicates which component is fixed by the signature and where an internal choice persists.

8.6.7 φ: the autoescala inducida by TPK

The word $w_\varphi = 121200$ possesses a single emission of multiplicity 144, but its function is not deduced from that cardinality. The TPK calendar produces the block of modes

$$(\Sigma, \Pi, \Pi).$$

Reading Σ as the areal leaf and Π as the volumetric leaf, the three consecutive pairs of the cycle force the incidence

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

as theorem 6.20 proves. The golden ratio is not introduced to choose this matrix: it appears afterward as its Perron root. Indeed,

$$\det(tI - F_{\mathrm{av}}) = t^2 - t - 1, \qquad F_{\mathrm{av}} \begin{pmatrix} 1 \\ \varphi \end{pmatrix} = \varphi \begin{pmatrix} 1 \\ \varphi \end{pmatrix}.$$

For an admissible path $\gamma : i \to j$ of length n, with $r = (1, \varphi)$, the Parry character

$$\chi_P(\gamma) = \varphi^n \frac{r_i}{r_j}$$

is multiplicative under concatenation. On a closed path, $\chi_P(\gamma) = \varphi^n$. The measure of maximal entropy of the memory-one hull satisfies, by corollary 6.21,

$$\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}.$$

Second-order memory also preserves the contracted direction. In the basis $(x^2, 2xy, y^2)$,

$$\mathrm{Sym}^2(F_{\mathrm{av}}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \mathrm{Spec}\big(\mathrm{Sym}^2(F_{\mathrm{av}})\big) = \{\varphi^2, -1, \varphi^{-2}\}.$$

The eigenvalue φ^{-2} is the positive stable mode. On closed returns, the scale cocycle of section 6.3.8 gives

$$M_n = \varphi^{-2n} = \big(\chi_P(\gamma)\big)^{-2}.$$

Proposition 8.50 (Functional content of φ)**.** *The function of φ in HMT is the self-scaling of the quotient of modes: growth by the Perron root φ, cylindrical conservation by χ_P and stable second-order return by φ^{-2}. These three expressions belong to one and the same operator induced by the TPK calendar.*

Proof. The incidence F_{av} descends from the block (Σ, Π, Π) by theorem 6.20. Its characteristic polynomial produces the Perron root, and the multiplicativity of χ_P is obtained by telescopy. The quadratic action separates the three indicated eigenvalues; the contracted positive mode is φ^{-2}. □

8.6.8 Oriented correspondence between closure and self-scale

The matrix F_{av} generates φ^{-2}, but cannot by itself generate π: every rational function of its spectrum belongs to $\mathbb{Q}(\sqrt{5})$. Transcendental closure comes from the coinductive regional reader. This separation of origins is preserved by the diagonal operator

$$D_\pi = \pi_{\mathrm{HMT}} P_{\mathrm{ar}} + P_{\mathrm{vol}}, \qquad P_{\mathrm{ar}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_{\mathrm{vol}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The marked return of length n is

$$\Theta_n = \pi_{\mathrm{HMT}} \frac{F_{n-1}}{F_{n+1}}, \qquad \lim_{n\to\infty} \Theta_n = \frac{\pi_{\mathrm{HMT}}}{\varphi^2}.$$

The dynamical proof, the exact witness of length 108, and the reader of nested cylinders reside in section 6.3.7.1. The mark of π is applied only once to the return: it is not inserted into every transition of F_{av}.

The same reason has two orientations. Let

$$\mathscr{R}(x, y) = \frac{x}{y^2}, \qquad \mathsf{J}(x, y) = (y, x).$$

Then $\mathsf{J}^2 = I$ and

$$\boxed{\mathscr{R}(\pi, \varphi) = \frac{\pi}{\varphi^2}, \qquad (\mathscr{R} \circ \mathsf{J})(\pi, \varphi) = \frac{\varphi}{\pi^2}.}$$

Moreover,

$$\frac{\pi/\varphi^2}{\varphi/\pi^2} = \left(\frac{\pi}{\varphi}\right)^3, \qquad \frac{\varphi}{\pi^2} = \frac{1}{\varphi^3 \left(\pi/\varphi^2\right)^2}.$$

The first orientation reads areal closure over volumetric memory; the second, interior self-scaling over quadratic closure. Section 6.3.7.1 proves this involution and its subsequent appearance in the electron exponent. This correspondence does not identify π with φ: it preserves their order and makes explicit what changes when that order is reversed.

8.6.9 Arbitrary precision of monodromic emission

Corollary 8.51 (Emission of any finite prefix)**.** *Under the hypotheses of theorem 8.46, for every $M \geq 1$ the EF–G_9 iteration determines the first M decimal digits of each of the three channels. In particular, $M = 100$ and $M = 1000$ are finite windows of the same monodromic prolongation and do not require the introduction of an external table of digits.*

Proof. Let $K(n)$ be the number of triads published after n ternary blocks. The section supplied by the theorem has levels of every depth and $K(n) \to \infty$. Given M, choose n with $3K(n) \geq M$. The theorem 8.46 uniquely determines the truncation of depth n, and its decimal cylinder fixes the first $3K(n)$ digits. Taking the first M concludes the argument. □

The difference of functions is fixed within a single construction: π closes distinct regional provenances in a common evaluation; e propagates a visible output while renewing the memory that transports it; φ governs the self-scaling of the quotient of modes. Monodromy explains how those functions return to the same phase and pass through the mirror π/e. Under the coinductive hypotheses of the prolongation theorem, they publish new blocks indefinitely without resetting their history.

8.7 Monodromic prolongation and calculation to prescribed depth

The construction starts from the enriched state produced by APP → TRIT → TPK: active catalog of 468 emissions, ternary projection, action D_3, orientation gauge, full-rank transition record, closure, propagation and self-scaling characters, ternary cylinders, nonadic phase and Paley–Witt shadow. The record uniquely reconstructs the lifts L_0, L_1 before opening the Archimedean chart. No digit of π, e or φ is received as input to the generator.

8.7.1 Initial coinductive prolongation from w_6 until the compatibility boundary of order 1

The exhaustive census selects the words

$$w_6^{\pi} = 010211, \qquad w_6^{e} = 201101, \qquad w_6^{\varphi} = 121200.$$

With row vectors over $\mathbb{F}_3^6$, the two reconstructed lifts are

$$L_0 = \begin{pmatrix} 2&2&2&1&2&1\\ 2&1&2&2&1&1\\ 1&0&0&1&0&2\\ 0&0&1&1&1&0\\ 2&2&2&0&2&1\\ 2&1&2&1&0&0 \end{pmatrix}, \qquad L_1 = \begin{pmatrix} 2&1&2&0&2&2\\ 1&2&1&1&2&0\\ 1&2&1&1&1&2\\ 0&1&0&0&2&1\\ 2&0&2&1&0&1\\ 0&2&2&2&1&1 \end{pmatrix}.$$

Let B^{C}, B^{P} and B^{A} be the structural biographies of closure, propagation and self-scaling published below, decomposed into blocks $b_0^{\chi}|\cdots|b_4^{\chi}$. Form

$$X_0 = (b_0^{\mathsf{C}}, b_1^{\mathsf{C}}, b_0^{\mathsf{P}}, b_1^{\mathsf{P}}, b_0^{\mathsf{A}}, b_1^{\mathsf{A}})^T,$$
$$Y_0 = (b_1^{\mathsf{C}}, b_2^{\mathsf{C}}, b_1^{\mathsf{P}}, b_2^{\mathsf{P}}, b_1^{\mathsf{A}}, b_2^{\mathsf{A}})^T,$$

and, analogously, X_1, Y_1 with the transitions $b_2 \to b_3$ and $b_3 \to b_4$. Each block is obtained by the ternary projection of the state produced after the complete update

$$x_{j+1}^{\chi} = \mathcal{U}_{9j+9} \circ \cdots \circ \mathcal{U}_{9j+1}(x_j^{\chi}), \qquad b_j^{\chi} = \Pi_6(x_j^{\chi}), \qquad \mathcal{U}_t = \mathsf{Upd}_t \circ \mathsf{Tra}_t \circ \mathsf{Sel}_t.$$

Thus, the record preceding the lifts is published in full by the four matrices

$$X_0 = \begin{pmatrix} 0&1&0&2&1&1\\ 0&1&2&2&2&2\\ 2&0&1&1&0&1\\ 1&2&1&2&2&1\\ 1&2&1&2&0&0\\ 1&1&2&2&0&2 \end{pmatrix}, \quad Y_0 = \begin{pmatrix} 0&1&2&2&2&2\\ 0&1&0&2&1&1\\ 1&2&1&2&2&1\\ 1&0&2&0&1&1\\ 1&1&2&2&0&2\\ 1&2&1&0&2&0 \end{pmatrix},$$
$$X_1 = \begin{pmatrix} 0&1&0&2&1&1\\ 0&0&2&1&1&1\\ 1&0&2&0&1&1\\ 0&1&2&2&2&2\\ 1&2&1&0&2&0\\ 0&1&0&2&1&0 \end{pmatrix}, \quad Y_1 = \begin{pmatrix} 0&0&2&1&1&1\\ 1&1&0&2&2&1\\ 0&1&2&2&2&2\\ 1&0&2&0&1&1\\ 0&1&0&2&1&0\\ 0&1&0&2&0&0 \end{pmatrix}. \tag{8.11}$$

Their rows are successive outputs of the TPK endomorphism on the three seeds selected by the fiber invariants; they are not coefficients imposed on L_0, L_1 or data taken from Archimedean expansions.

Theorem 8.52 (Unique reconstruction of the lifts and calendar)**.** *In* $\mathbb{F}_3$ *we have*

$$\det X_0 = 1, \qquad \det X_1 = -1, \qquad \operatorname{rank}(I_6 \otimes X_i) = 36.$$

Consequently, the systems

$$X_0 L_0 = Y_0, \qquad X_1 L_1 = Y_1$$

have the unique solutions $L_i = X_i^{-1} Y_i$*, which are exactly the matrices printed above. Among the* 16 *binary calendars of length* 4*,* 0011 *is the only one simultaneously preserving the three biographies. The* 144 *possible unit substitutions in both matrices destroy that triple compatibility.*

Proof. The determinants and ranks are calculated by exact elimination modulo 3. The invertibility of X_i gives existence and uniqueness of L_i; the 16 calendars and the $144 = 2 \cdot 6 \cdot 6 \cdot 2$ perturbations form two exhaustive finite censuses. All their inputs are words, transitions and residues of the enriched state prior to real evaluation. The conventional names of the constants intervene in none of those calculations. □

The sequence

$$u_1 = u_0 L_0, \qquad u_2 = u_1 L_0, \qquad u_3 = u_2 L_1, \qquad u_4 = u_3 L_1$$

yields

$$\begin{aligned} w_{30}^{\pi} &= 010211|012222|010211|002111|110221, \\ w_{30}^{e} &= 201101|121221|102011|012222|102011, \\ w_{30}^{\varphi} &= 121200|112202|121020|010210|010200. \end{aligned}$$

The matrix

$$R_{36} = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}$$

is the first coinductive boundary. The matrices L_0, L_1 are results of the internal reconstruction of theorem 8.52 and are applied before opening the decimal witnesses. The theorem's domain is the single generating chain APP → TRIT → TPK → enriched state. APP supplies its two arithmetic sheets; TRIT orients the regime; TPK constructs and preserves the route, carry, boundary and memory fields on which the lifts act. Asking the visible projection of APP to contain the terminal state by itself confuses two consecutive domains of that same chain and does not turn the lifts into manually introduced data.

8.7.2 Finite selection and construction of characters

Let C the multiconjunto of semillas, $\mathcal{U}$ the catalogo of emisiones and $W = \Psi(\mathcal{U}) \subset \mathbb{F}_3^6$, with

$$C \xrightarrow{T_{54}^{\text{seed}}} \mathcal{U} \xrightarrow{\Psi} W \hookrightarrow \mathbb{F}_3^6, \qquad \Psi(U_6) = ((-u_j) \bmod 3)_{j=1}^{6}.$$

The exact enumeration gives

$$\sum_{U \in \mathcal{U}} \text{mult}(U) = 104\,976, \qquad |\mathcal{U}| = 468, \qquad |W| = 243, \qquad |\mathbb{F}_3^6| = 729.$$

These four numbers count distinct typed objects.

The action D_3 on W produces 43 orbits. The closure orbit is the unique orbit whose six orientations have five lifts, mass 1008, and multiplicities $432 + 4 \cdot 144$. Among the unit radical orbits, the oriented census of closure uniquely determines propagation, and census $(2, 2, 2)$ uniquely determines self-scaling. Gauge $\text{diag}_c = 6$, with ES orientation, fixes one word in each orbit. Consequently, the triple

$$(w_6^{\pi}, w_6^{e}, w_6^{\varphi}) = (010211, 201101, 121200)$$

it is obtained before constructing any Archimedean interval.

The three characters are also obtained from the selected structure. For the closure, the incidence matrix

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \quad \text{satisfies} \quad A_W^2 = 2I_6 = -I_6$$

and fixes the oriented inversion. On the pentafiber, the symmetrizer that forgets the lift label is

$$P_5 = \frac{1}{5}\mathbf{1}\mathbf{1}^T.$$

The conservation $\mathbf{1}^T\lambda = 1$ and the invariance $P_5\lambda = \lambda$ force $\lambda = (1/5)\mathbf{1}$, hence $t_1 = 1/5$. The four compositions by

$$t \star s = \frac{t+s}{1-ts}$$

give $t_4 = 120/119$. The return equation to the diagonal,

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1,$$

has the unique positive integral solution $q = 239$. The Gaussian identity

$$(5+i)^{16}(239-i)^4 = -6813866078035761571 84 \tag{8.12}$$

acknowledges afterward the half turn. Thus, the expression

$$16\arctan(1/5) - 4\arctan(1/239)$$

is the exact evaluation of the closure character, not its selector.

For propagation, the composition of paths and the preserved unit give

$$P_{s+t} = P_s P_t, \qquad P_0 = 1, \qquad [t]P_t = 1.$$

If $P_t = \sum_{n\geq 0} a_n t^n$, comparison of the coefficient of $s^n t$ forces

$$(n+1)a_{n+1} = a_n, \qquad a_0 = 1,$$

and, by induction, $a_n = 1/n!$. The equality $P_t' = P_t$ and the value $P_1 = e$ are subsequent analytic recognitions of this formal law.

For the self-scale, the calendar $(+,-,0)$ induce

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its unique invariant positive ray $v = (1,x)^T$ satisfies

$$F_{\mathrm{av}}v = xv, \qquad x^2 - x - 1 = 0, \qquad x > 0.$$

Consecutive Fibonacci quotients provide nested rational brackets for x; the equation $x = 1 + 1/x$ is the scalar coordinate of this eigen-equation, not a condition imposed externally.

Proposition 8.53 (Separation between selection, evaluation, and recognition)**.** *For each channel χ, the causal composition is*

$$\widehat{X} \xrightarrow{\mathrm{Sel}_{\mathrm{TPK}}} X_\chi \xrightarrow{\mathrm{ev}_{\mathbb{R}}} \mathbb{R} \xrightarrow{\mathrm{Rec}} \{\pi, \varphi, e, \alpha\}.$$

$\mathrm{Sel}_{\mathrm{TPK}}$ *uses the census, orbit, gauge, lifts, carry, and internal character. Evaluation contracts the compatible cylinders. Recognition assigns the conventional name to the already constructed value. No target prefix intervenes in the first two arrows.*

Proof. The uniqueness of the three orbits is decided through integer predicates of the catalog. The parameters $5, 4, 239$, the recurrence $(n+1)a_{n+1} = a_n$ and the matrix F_{av} are deduced from those orbits and from the laws of unity, composition and incidence. The coordinate of α subsequently comes from the internal triple through the signed record U_{12}, the seal K and the dodecaphase carry. Rational bounds are calculated only to evaluate and certify the already closed section. Decimal comparison is performed afterwards; therefore, it cannot act as a causal input. □

8.7.3 Extension of survivors and subsequent certification

Let $\mathcal{H}_m^\chi$ be the set of admissible enriched histories of channel χ at depth m, and let $p_{n,m} : \mathcal{H}_n^\chi \to \mathcal{H}_m^\chi$ be the truncation. Define

$$\mathrm{Surv}_m^{(N),\chi} = p_{N,m}(\mathcal{H}_N^\chi), \qquad \mathrm{Surv}_m^\chi = \bigcap_{N \geq m} \mathrm{Surv}_m^{(N),\chi},$$

and the numerical section by

$$x_\chi \in X_\chi := \varprojlim_m \mathrm{Surv}_m^\chi .$$

The reader and the state are not identified: a presentation (x_χ, L_χ) publishes the value, while the state also preserves orientation, carry, boundary, sheet, exceptional incidence and genealogy. At the R36 boundary, χ_{HMT} and the internal residue ε_K^χ are already state coordinates. APP division and the phase selector produce

$$d_K^\chi = \mathrm{Dig}_{729}(\varepsilon_K^\chi), \qquad \varepsilon_{K+1}^\chi = 729\varepsilon_K^\chi - d_K^\chi, qquad N_{K+1}^{\mathrm{Ext}} = 729N_K + d_K^\chi,$$

and the selection, transport and update

$$\mathrm{Sel}_K^\chi(x) := \mathrm{Sel}\big(x, M_{\mathrm{ph}}(\rho_9(K+1))\big), \qquad \mathrm{Tra}_K^\chi(y,d) := \mathrm{Lift}(T_d)(y), \qquad \mathrm{Upd}_K^\chi := \mathrm{Rec}_K^\chi \circ \mathrm{Mem}_K^\chi,$$

$$\mathcal{U}_K^\chi(x) := \mathrm{Upd}_K^\chi\big(\mathrm{Tra}_K^\chi\big(\mathrm{Sel}_K^\chi(x), d_K^\chi\big)\big)$$

depends only on the preceding state. The complete relation Ext may be multivalued; the subrelation $\mathrm{Ext}^{\chi,\to}$, anchored in the R36 state that already carries the character, is the graph of $\mathcal{U}_K^\chi$. It therefore fixes a compatible global section without asserting that every local fiber of Ext is a singleton. It is not a retrospective intersection defined by P_{ar}, and it reads neither future digits nor brackets.

If N_K is the ternary prefix and $L_K = 6K$, its 729 subcylinders are

$$I_{K,u} = \left[\frac{729N_K + u}{3^{L_K+6}}, \frac{729N_K + u + 1}{3^{L_K+6}}\right), \qquad 0 \leq u < 729.$$

For the exact rational bracket $[\ell_{\chi,K}, h_{\chi,K}]$ constructed by the internal recurrence of the character χ,

$$j_{\min} = \max\Big(729N_K, \big\lfloor \ell_{\chi,K} 3^{L_K+6} \big\rfloor\Big),$$
$$j_{\max} = \min\Big(729N_K + 728, \big\lceil h_{\chi,K} 3^{L_K+6} \big\rceil - 1\Big).$$

The subsequent Archimedean check is

$$j_{\min} = j_{\max} = N_{K+1}^{\mathrm{Ext}}.$$

It certifies that the partition is exhaustive and locates the survivor already produced by $\mathrm{Ext}^{\chi,\to}$; it does not select it. The bracket is not read from a decimal expansion: it is calculated from the pentafiber and the unitary return, from the propagation recurrence or from the incidence F_{av}, according to the channel. Therefore, the intersection is a publication certificate, not a descent rule.

Theorem 8.54 (Correlated coinductive generation to arbitrary depth)**.** *In the enriched TPK structure constructed in the preceding theorems, the three derived characters and the dodecaphase closure construct to arbitrary depth the correlated family of sections $x_\pi, x_\varphi, x_e, x_\alpha$ satisfying:*

1. *el barrido*
$$104\,976 \longrightarrow 468 \longrightarrow 243$$
separates seeds, emissions, and visible words;

2. *los levantamientos*
$$w_6 \xrightarrow{L_0} w_{12} \xrightarrow{L_0} w_{18} \xrightarrow{L_1} w_{24} \xrightarrow{L_1} w_{30}$$
precede to the first boundary coinductive R_{36};

3. *at each level, $\mathrm{Ext}^{\chi,\to}$ first produces the compatible child, and its truncation coincides with the state at the preceding level; the 729 numerical subcylinders form an ordered partition, and the rational bracket subsequently locates the already generated child;*

4. *the phase satisfies $g(K+9) = g(K)$, whereas the block, the cylinder index, the prefix, and the Witt shadow change;*

5. *the readers publish, respectively, π, φ, e and α, and the extension rule has no terminal depth.*

The scope of the statement is coinductive and of arbitrary depth. For the four channels, uniqueness of the forward section and its compatibility under truncation come from the state update. It is not asserted that every local fiber of Ext is a singleton. Runs at any finite depth are only regression and publication witnesses.

Proof. The finite selection of proposition 8.53 preserves the five regions of the closure orbit before opening the real chart. The primitive slope $1/5$, the composition of the four companion branches, and the unit-return condition force, in that order, $120/119$ and $q = 239$. The identity (8.12) then recognizes the first positive half-turn, so that the subsequent evaluator is

$$\pi_{\mathrm{HMT}} = 16\arctan(1/5) - 4\arctan(1/239).$$

The arctangents are evaluated by alternating series with rational bounds.

In propagation, the law $P_{s+t} = P_s P_t$, the unit $P_0 = 1$, and the linear normalization first force $(n+1)a_{n+1} = a_n$. Only afterwards are $P'_t = P_t$ and

$$e_{\mathrm{HMT}} = \sum_{n\geq 0} \frac{1}{n!}.$$

On auto-scale, the calendar first produces

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

and its Perron ray. The equation $x = 1 + 1/x$ is its subsequent scalar coordinate.

The update reading only the preceding state produces in each channel the compatible child from the present internal character and residue. By induction, two forward sections with the same R36 state

coincide at every depth. In the three channels with a real reader developed here, rational arithmetic subsequently produces intervals of arbitrarily small width: the exact partition of the 729 subcylinders checks that the already generated child lies inside the published decimal cell. The signed closure $U_{12} \longrightarrow K$ prolongs the fourth coordinate, and the truncation squares of its two internal routes commute, so that $\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}$. The thousand- and ten-thousand-digit runs are finite projections, not the criterion of the theorem. □

Extension to a section of the inverse limit is governed by theorem 8.46. The preceding theorem establishes, for π, φ, e, α, the nonemptiness, compatibility and global uniqueness of each forward section required by that abstract principle, without postulating gate-by-gate singleton fibers of Ext. This formulation distinguishes the coinductive rule of arbitrary depth from its finite thousand- and ten-thousand-digit evaluations, without altering phase return or memory transport.

Corollary 8.55 (Arbitrary depth and infinite decimal object)**.** *For every $N \in \mathbb{N}_{>0}$, the same construction publishes a unique canonical decimal prefix d_N^χ of length N, with $d_{N+1}^\chi|_N = d_N^\chi$, for $\chi \in \{\pi, \varphi, e, \alpha\}$. The compatible family $(d_N^\chi)_{N\geq 1}$ determines the infinite expansion in the inverse limit. A thousand, ten thousand or a million digits are finite runs of the same parametrized algorithm; none of them limits the scope of the theorem.*

Proof. The update Upd_χ first produces the compatible cylinder. The rational bracket of the character has arbitrarily small width and subsequently certifies that this cylinder intersects a unique member of the published partition of 729 subcylinders. Given N, there exists m with $3^{-6m} < 10^{-N}$; the already produced cylinder is then contained in a canonical decimal cell of length N. Truncation compatibility identifies those cells as the finite coordinates of a forward section. □

Theorem 8.56 (Correlated coinductive generation and double projection)**.** *Let $\mathcal{S}_\infty$ be the fiber product of the four correlated forward sections of arbitrary depth generated by*

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow R_{36} \longrightarrow G_9.$$

Two typed publications depart from $\mathcal{S}_\infty$:

$$\mathcal{S}_\infty \xrightarrow{\mathcal{P}_{\mathrm{Arq}}} \mathbb{R}^4, \qquad \mathcal{S}_\infty \xrightarrow{\mathcal{P}_{\mathrm{Exc}}} W_{12} \longrightarrow G_{24} \longrightarrow \Lambda_{24} \longrightarrow V^\natural \longrightarrow \mathrm{Aut}(V^\natural) = \mathbb{M}.$$

The first contracts compatible cylinders and publishes in a correlated manner $(\pi_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}})$; the second preserves boundary incidence and publishes the exceptional chain. Consequently, $\mathbb{R}$ is the Archimedean shadow of the first projection, not an input selecting the genealogy. Machin, Perron–Fibonacci, the exponential semigroup and conventional names act after the HMT output.

Proof. theorem 8.54 and corollary 8.55 construct the compatible family before evaluation. The dodecaphase closure publishes α_{HMT} from its correlated R36 triple and the record $U_{12} \longrightarrow K$; the route $E_{21/22}$ preserves its own analytic domain. The Archimedean projection forgets incidence when contracting the cylinders, while the exceptional projection transports it through the Witt–Golay– Leech intertwiners to $V^\natural$. Both come from the same antecedent, and neither can act retroactively as its generator. □

8.7.4 Closure Regions

The five regions are not five values of π, but five genealogies with a common visible projection. Their multiplicity is

$$432 + 4 \cdot 144 = 1008,$$

and their papers are

$$3_{++} + 1_{--} + 1_{\perp}.$$

The regional identity is verified by the quadruple

$$(U_6, E_{\mathrm{sig}}, \text{multiplicity}, \text{role}),$$

no only by the census $3/1/1$. In particular,

U_6	E_{sig}	multiplicity	role
501|614|498|169|272|272	555555	432	$\perp$
810|923|870|169|272|272	339555	144	++
810|983|810|169|272|272	393555	144	++
870|923|810|169|272|272	933555	144	++
870|923|810|418|674|521	933717	144	−−

The table jointly fixes the enriched decimal word, emission signature, multiplicity, and role. In particular, the multiplicity anchor (432) is transversal, and the signature (933717) is the mutated return; the census (3/1/1) alone does not determine those names.

8.7.5 Phase return and memory change

In the ninth phase,

$$B_9 = (100100 \mid 020112 \mid 010122)$$

admits three orientations that preserve the axis φ and the aggregate $\pi + e$:

$$(101222 \mid 112221 \mid 112002),$$
$$(102221 \mid 111222 \mid 112002),$$
$$(111221 \mid 102222 \mid 112002).$$

The three survive a boundary and conserve

$$u^{\varphi} = 112002, \qquad u^{\pi} + u^{e} = 210110.$$

The following boundary selects

$$B_{10} = (101222 \mid 112221 \mid 112002).$$

Therefore,

$$\#\,\mathrm{Surv}_1(B_9) = 3, \qquad \#\,\mathrm{Surv}_2(B_9) = 1.$$

Index ten again occupies phase one, but not the previous state. This is the monodromy: phase return with transformation of memory.

In terms of the complete state

$$\widehat{x}_K = (B_K, C_K, p_K, c_K, \Sigma_K, W_K, g(K)),$$

the loop is the unique holonomy

$$\Gamma_9 = \mathrm{Hol}_{C_9}(\nabla_{\mathrm{TPK}}), \qquad g(K+9) = g(K),$$
$$q_{\mathrm{mem}}\Gamma_9 = q_{\mathrm{mem}} + 1, \qquad \Gamma_9 \widehat{x}_K \neq \widehat{x}_K.$$

In isometric realization, the same continuity is satisfied.

$$U_{\Gamma_9} J = JT + \eta, \qquad I - T^{*}T = \eta^{*}\eta.$$

The carry is composed of

$$\mathrm{Carr}(\gamma_2\gamma_1) = \mathrm{Carr}(\gamma_2)H(\gamma_1) + Y(\gamma_2)\,\mathrm{Carr}(\gamma_1),$$

and the telescoping preserves the finite terminal:

$$S_0 = \sum_{r=0}^{L-1} M_9^{*r} D_9^* D_9 M_9^r + M_9^{*L} S_0 M_9^L.$$

The windows 27, 54 and 108 are the concatenations

$$\Gamma_{27} = \Gamma_9^3, \qquad \Gamma_{54} = \Gamma_9^6, \qquad \Gamma_{108} = \Gamma_9^{12};$$

transport an accumulated memory of 3, 6, and 12 turns, respectively. They do not constitute restarts of the algorithm.

8.7.6 Exhaustive census of 396 transitions, 2.376 trits, 43 turns and 387 equal-phase pairs

The run to depth one thousand produces, per channel,

$$396 \text{ transitions}, \quad 2376 \text{ trits}, \quad 43 \text{ complete turns}, \quad 387 \text{ pairs } (K, K+9),$$

and 1000 decimal digits. At each transition, the update $\mathrm{Ext}^{\chi,\to}$ produces the survivor, and the subsequent exhaustive scan of the 729 subcylinders certifies its publication. External witnesses are consulted after generation and used only for recognition.

Independence from the target is obtained because the generating map receives only the finite catalog of 468 emissions, the matrices $L_0, L_1, A_W, F_{\mathrm{av}}$ and rational arithmetic. None of its transition functions contains prefixes of π, e, φ or an operation evaluating those constants. The brackets are deduced from the constructed characters and act only in subsequent evaluation.

After the 396 stages of prolongation and certification, the final cylinder

$$\left[\frac{N}{3^{2376}}, \frac{N+1}{3^{2376}}\right)$$

is contained in a single decimal cell of depth 10^{-1000}. The thousand digits are therefore to finite witniss of the parametrizable law, not its maximum depth.

The uniform argument constructs a cylinder for any prescribed finite depth. The census of 396 stages checks only the thousand-digit instance: it keeps causal generation separate from subsequent decimal comparison and does not replace the quantifier of theorem 8.54.

Theorem 8.57 (Joint continuity of the nonadic holonomy)**.** *Let $\widehat{X}_K$ be the APP–TRIT–TPK state enriched by sheet, orientation, residue, quotient, carry, signature, cylinder, boundary, path, memory, and survival. The composition of the nine local relations publishes a unique holonomy*

$$\Gamma_{9,K} = \mathcal{R}_{K+8} \circ \cdots \circ \mathcal{R}_K : \widehat{X}_K \longrightarrow \widehat{X}_{K+9},$$

compatible with the truncations of the inverse system. At each turn the phase returns and memory advances without reinitializing the state; the carry preserves its cocycle law, the transfer acts simultaneously on its nine components, and the operator realization preserves both the orthogonal component and the terminal term of the Stein identity. By composition, these properties determine compatible sections at arbitrary depth, not independent choices level by level.

Proof. The composition and the return without reset are, respectively, definition 8.11 y theorem 8.12. The joint action on the nine coordinates and their transfer are theorem 8.13 y proposition 8.20. The compression–expression decomposition and preservation of the terminal are theorems 8.21 y 8.22. The coinductive prolongation certified in this same chapter preserves these identities under truncation; therefore, the compatible family belongs to the inverse limit of the enriched state and can be iterated to any depth. □

8.8 Affine holonomy, helical suspension, and conservation of memory depth

8.8.1 Dynamic composition of the Topological Phase Kernel over the enriched state: tritic selection, cardinal transport, memory update, and typed return

8.8.1.1 Definition and factorization of the enriched transition operator U_t

The TPK is not a nominal container. The phase function and its tabulated vector are, respectively,

$$M_{\mathrm{ph}} : \mathcal{D}_9 \longrightarrow \{+1, -1, 0\}, \qquad m_{\mathrm{ph}} := \big(M_{\mathrm{ph}}(1), \ldots, M_{\mathrm{ph}}(9)\big)^{\mathsf{T}}.$$

They are not the same type of object and neither acts by direct composition on the state. For $x \in X_{\mathrm{TPK}}^{\mathrm{enr}}$, we define

$$\mathrm{Tra}_t(y, d) := \mathrm{Lift}(T_d)(y), \qquad \mathrm{Upd}_t := \mathrm{Rec}_t \circ \mathrm{Mem}_t\,.$$

The elementary transition is then written

$$U_t(x) = \mathrm{Upd}_t\big(\mathrm{Tra}_t\big(\mathrm{Sel}_t\big(x, M_{\mathrm{ph}}(g(t))\big), d(t)\big)\big), \qquad U_t : X_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow X_{\mathrm{TPK}}^{\mathrm{enr}}. \tag{8.13}$$

The selector Sel_t reads the scalar value of the tritic regime and chooses the additive, multiplicative or threshold sheet; $\mathrm{Tra}_t(\,\cdot\,, d(t)) = \mathrm{Lift}(T_{d(t)})$ applies the cardinal direction; Mem_t updates residue, quotient, carry, signature, cylinder, boundary and memory; and Rec_t performs the typed return. The abbreviated notation $\mathrm{Upd}_t \circ \mathrm{Tra}_t \circ \mathrm{Sel}_t$ designates this same composition and not a second engine.

The minimal enriched fiber can be represented as

$$\mathcal{F}_q = O_q \times H_q \times C_q \times S_q \times B_q \times \Lambda_q,$$

with orientation/leaf, lifting, carry, signature, boundary, and record. The detail of each factor depends on its residence, but the integrity rule is stable: a coordinate cannot be removed if it changes a continuation.

8.8.1.2 Nonadic holonomy

For a nine-step compatible block,

$$\Gamma_{9,K} = U_{K+8} \circ \cdots \circ U_K. \tag{8.14}$$

The phase satisfies

$$g(K + 9) = g(K),$$

but the enriched state does not in general return:

$$\Gamma_{9,K}(x_K) \neq x_K.$$

In particular,

$$q_{\mathrm{mem}}(\Gamma_9 x) = q_{\mathrm{mem}}(x) + 1, \qquad B_{K+9} \neq B_K \quad \text{in general.}$$

Theorem 8.58 (Phase return with memory advance)**.** *The holonomy Γ_9 closes the phase coordinate and transports the state to a new depth. Therefore, phase return is not state return.*

Proof. Phase closure is the periodicity of g. The ledger update includes the depth increment and boundary transport. If the state returned, all coordinates would have to coincide, contradicting $q_{\mathrm{mem}} \mapsto q_{\mathrm{mem}} + 1$. Therefore, only the phase returns. □

8.8.1.3 Bimodular cocycle law of the carry under path concatenation

Let $\mathcal{K}_{\mathrm{Carr}}$ be the carry module, and let $H, Y : \mathscr{P}^{\mathrm{enr}}_{\mathrm{TPK}} \to \mathrm{End}(\mathcal{K}_{\mathrm{Carr}})$ be the right and left actions induced, respectively, by the preserved prefix and by the active sheet of a route. We regard $\mathcal{K}_{\mathrm{Carr}}$ as a bimodule and write $kH(\gamma)$ and $Y(\gamma)k$ for those actions. The map $\mathrm{Carr} : \mathscr{P}^{\mathrm{enr}}_{\mathrm{TPK}} \to \mathcal{K}_{\mathrm{Carr}}$ assigns to each route its total carry.

For paths composable γ_1, γ_2, the carry satisfies

$$\mathrm{Carr}(\gamma_2\gamma_1) = \mathrm{Carr}(\gamma_2)H(\gamma_1) + Y(\gamma_2)\,\mathrm{Carr}(\gamma_1). \tag{8.15}$$

This law shows what preserving memory means: not keeping a quantity constant, but retaining its exact update rule. Accumulated memory is dynamic and traceable.

8.8.1.4 Nonsplit cyclic extension $0 \to C_{12} \to C_{108} \to C_9 \to 0$ and internal chronology of the TPK

The internal calendar is organized through

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0.$$

The extension preserves the difference between a visible turn and the lifted state. In the realization $C_{108} \simeq C_{27} \times C_4$, the part $C_{27} \to C_9$ does not split: the lifts of the generator have order 27, not an order dividing 9. The factor C_4 supplies the quarter-turn; the ternary kernel supplies the elevation. The combination is not an external temporal label, but the return architecture of the TPK.

8.8.1.5 Posteriority of the Markov quotient as a publication of reduced memory

Averaging enriched dynamics may produce a reduced-memory operator K_{ph}. This operator is useful for studying transfer, stationarity, and spectrum, but does not generate Γ_9. The dependence is

$$U_t \longrightarrow \Gamma_9 \longrightarrow K_{\mathrm{ph}}.$$

Inverting it would era sheet, prefix, boundary, and carry. In this publication the radial reader is constructed over enriched histories, never over the Markov quotient.

8.9 Exact helical geometry of the nonadic memory

8.9.1 Helical covering of the nonadic phase and integral coordinate of memory depth

8.9.1.1 Nonadic phase lift to the phase–memory covering

Let

$$\mathrm{Rep}_9 = \{0, 1, \ldots, 8\}, \qquad \widetilde{\Theta} = \mathrm{Rep}_9 \times \mathbb{Z}.$$

We define

$$\widetilde{\Gamma}_9(r, n) = (r, n + 1), \qquad \Pi(r, n) = [r + 9n]_{108}. \tag{8.16}$$

The first coordinate records the visible phase and the second records the depth. It holds

$$\Pi(r, n + 12) = \Pi(r, n), \qquad \widetilde{\Gamma}_9^{12}(r, n) = (r, n + 12).$$

The covering transformation

$$\tau_{12}(r, n) = (r, n + 12)$$

acts freely. Each fiber of Π is an orbit of that action.

Definition 8.59 (Nonadic helix)**.** The nonadic helix is the lifted system

$$(\widetilde{\Theta}, \widetilde{\Gamma}_9, \Pi)$$

with phase r mod 9, calendar $\Pi(r, n)$, and deep memory n.

Theorem 8.60 (Periodic phase and aperiodic memory)**.** *The projection of the helix is periodic of order* 12 *on the calendar C_{108}, while the lifted state distinguishes all depths $n + 12k$, $k \in \mathbb{Z}$.*

Proof. The equality $\Pi(r, n) = \Pi(r', n')$ forces $r = r'$ and $n - n' \in 12\mathbb{Z}$. Therefore, the projection identifies exactly the orbits of τ_{12}, but the second coordinate of the lift remains integral and distinguishes their elements. □

8.9.1.2 Immersion H_9 of the nonadic quotient–residue into $\mathbb{R}^3$ and calculation of the Frenet invariants

For $N \geq 1$, let

$$z_N = q_9(N) + \frac{\rho_9(N) - 1}{9}.$$

The immersion

$$H_9(N) = \big(\cos(2\pi z_N), \sin(2\pi z_N), z_N\big) \tag{8.17}$$

satisfies

$$z_{N+9} = z_N + 1, \qquad H_9(N + 9) = H_9(N) + (0, 0, 1).$$

The projection onto the first two coordinates returns; the third priservis the turn. The figure is not imposed on the arithmetic: it is an immersion of the exact decomposition $N = 9q_9(N) + \rho_9(N)$.

For the continuous curve

$$h(z) = (\cos 2\pi z, \sin 2\pi z, z),$$

we have

$$h'(z) = (-2\pi \sin 2\pi z, 2\pi \cos 2\pi z, 1),$$

$$h''(z) = (-4\pi^2 \cos 2\pi z, -4\pi^2 \sin 2\pi z, 0).$$

Its curvature and torsion are constants:

$$\kappa_h = \frac{4\pi^2}{4\pi^2 + 1}, \qquad \mathcal{T}_h = \frac{2\pi}{4\pi^2 + 1}. \tag{8.18}$$

Calculation of equation (8.18). The norm of h' is $\sqrt{4\pi^2 + 1}$ and $\|h' \times h''\| = 4\pi^2\sqrt{4\pi^2 + 1}$. Therefore,

$$\kappa_h = \frac{\|h' \times h''\|}{\|h'\|^3} = \frac{4\pi^2}{4\pi^2 + 1}.$$

Furthermore, with $h''' = (8\pi^3 \sin 2\pi z, -8\pi^3 \cos 2\pi z, 0)$,

$$\mathcal{T}_h = \frac{\det(h', h'', h''')}{\|h' \times h''\|^2} = \frac{2\pi}{4\pi^2 + 1}.$$

□

8.9.1.3 Circular sections and cylindrical foliation of the phase–memory covering

A section n = constant produces a circle. A family of sections with constant radius produces a cylinder. If the radius depends on depth, the same structure produces a radial helical suspension. Therefore:

$$\text{orbit} \longleftrightarrow \text{helical sheet},$$

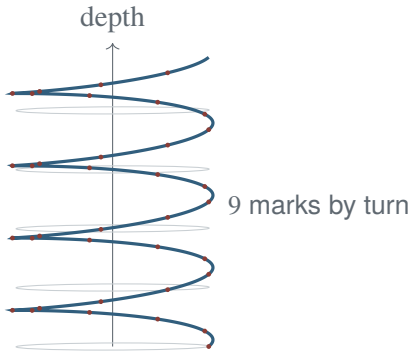


Figure 8.2: Nonadic helix: the rings are level sections; the curve conserves memory advance.

$$\text{level section} \longleftrightarrow \text{circle}, \qquad \text{family of sections} \longleftrightarrow \text{cylinder}.$$

This is a geometric translation of the lifted system, not a replacement of the ledger by a drawing.

8.9.2 Dynamic suspension of the enriched affine system

8.9.2.1 State-dependent holonomy

Let U_{APP} be a radial module and associate with each nonadic block an affine endomorphism

$$H_9(x) : u \longmapsto \Lambda_9(x)u + C_9(x).$$

When holonomy depends on the initial state, we define $F(x,u) = (\Gamma_9 x, H_9(x)u)$. If F is continuous, its suspension is the mapping torus

$$\mathfrak{S}^{\mathrm{rad}}_{\mathrm{TPK}} = \frac{X^{\mathrm{enr}}_{\mathrm{TPK}} \times U_{\mathrm{APP}} \times [0,1]}{(x,u,1) \sim (\Gamma_9 x, H_9(x)u, 0)}. \tag{8.19}$$

If F is to homeomorphism —which requiris, in addition to the invertibility of the linear parts, the continuous reversibility of Γ_9 and of its dependence on x— to bidirectional suspension is obtained. In the general monoid the same quotient encodis only forward evolution; it is not identified with to teliscope and not global inversion is prisupposed.

Definition 8.61 (Radial helical suspension)**.** The suspension equation (8.19), endowed with an angular coordinate of phase and a subsequent radial evaluation, is called a *radial helical suspension*. The term *general helix* is reserved for the cases that satisfy a differential condition of Lancret.

8.9.2.2 Orbits of the skew product and identification relation of the suspension space

The orbit of the skew–product is

$$(x_0,u_0), (\Gamma_9 x_0, H_9(x_0)u_0), (\Gamma_9^2 x_0, H_9(\Gamma_9 x_0)H_9(x_0)u_0), \dots.$$

The stitching preserves the order of the factors. Replacing the whole orbit with the final product would prevent truncating the history and reconstructing its prefixes. The helicoidal sheet is, therefore, an orbit with ledger, not an isolated curve.

8.9.2.3 Iteration of the stationary affine endomorphism and classification of its fixed points

If $H_9(x) = H = (\Lambda, C)$ is constant,

$$u_{n+1} = \Lambda u_n + C.$$

The exact internal iteration, valid for every endomorphism Λ, is

$$u_n = \Lambda^n u_0 + \sum_{k=0}^{n-1} \Lambda^k C. \tag{8.20}$$

If $I - \Lambda$ is invertible in the radial module, the fixed point exists

$$u_* = (I - \Lambda)^{-1} C,$$

and

$$u_n - u_* = \Lambda^n (u_0 - u_*). \tag{8.21}$$

In the translational sector $\Lambda = I$,

$$u_n = u_0 + nC.$$

Only after choosing a positive scalar character $\chi : U_{\mathrm{APP}} \to \mathbb{R}$, with $\chi\Lambda = \lambda\chi$, are the numerical formulas $u_* = C/(1-\lambda)$ for $\lambda \neq 1$ and the real powers of λ used in continuous interpolation obtained. An abstract endomorphism is not divided by $1 - \Lambda$. These two possibilities anticipate the separation between a self-scaling family and a translational family. Neither by itself selects a known spiral; the curve appears after applying $\exp_p$ and the angular coordinate.

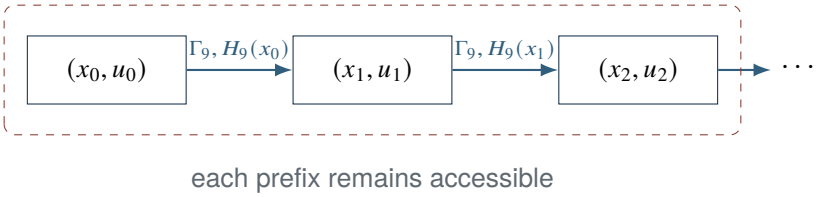


Figure 8.3: The suspension preserves the sequence of factors, not only the final holonomy.

8.9.3 Bidirectional Involution and Conjugate Leaves

8.9.3.1 Antipodal of the reversible path groupoid and conjugation of the affine holonomy

Let $\mathscr{P}^{\times}_{\mathrm{TPK}}$ be the subgroupoid of reversible routes. If the radial representation satisfies

$$h_{e^\dagger} = h_e^{-1},$$

then, for $\gamma = e_n \cdots e_1$,

$$H_{C_{\mathrm{rev}}\gamma} = h_{e_1}^{-1} \cdots h_{e_n}^{-1} = (h_{e_n} \cdots h_{e_1})^{-1} = H_\gamma^{-1}.$$

Theorem 8.62 (Conjugation of Ascent and Descent)**.** *In the reversible sector, the route involution interchanges the ascent and descent leaves and inverts the affine holonomy.*

On the helical coverage, let

$$\iota(r, n) = (r, -n).$$

Then

$$\iota \widetilde{\Gamma}_9 \iota = \widetilde{\Gamma}_9^{-1}. \tag{8.22}$$

The equality is obtained directly:

$$(r, n) \overset{\iota}{\mapsto} (r, -n) \overset{\widetilde{\Gamma}_9}{\longmapsto} (r, -n+1) \overset{\iota}{\mapsto} (r, n-1).$$

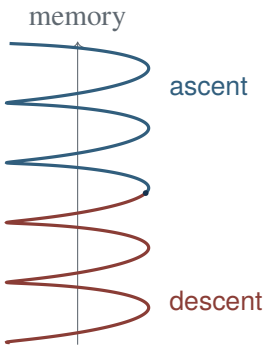


Figure 8.4: Conjugate leaves produced by route inversion.

8.9.3.2 Involution π/e of the fiber of constants, invariant $u^\pi + u^e$ and fixed axis φ

The involution on the constant fiber

$$\mathfrak{c}(u^\pi, u^e, u^\varphi) = (u^e, u^\pi, u^\varphi)$$

interchanges the branches π/e and fixes the axis φ. It also preserves $u^\pi + u^e$. This involution and route inversion are distinct objects: the former acts on a publication fiber; the latter acts on the path groupoid. No global conjugation $\mathfrak{c}\Gamma_9\mathfrak{c} = \Gamma_9^{-1}$ is asserted without constructing the corresponding intertwiner.

This distinction protects the positive thesis. The mirror π/e, the axis φ, and bidirectionality are present; what is not permitted is to identify them through an untyped equality of operators.

8.9.3.3 Nested circles as cross sections of the leaves

Let $r_n > 0$ be the scale of section n. The family

$$C_n = \{(r_n \cos\theta, r_n \sin\theta, n) : 0 \le \theta < 2\pi\}$$

produces nested circles as r_n varies and cylinders when sections with a common radius are grouped. The union of the C_n along a TPK route is the discrete trace of a helical sheet. The involution $n \mapsto -n$ produces the conjugate sheet.

8.10 Derivation of the incidence channels 90 and 120 from the tritic semicoronas of the Topological Phase Kernel

The cardinalities 90 and 120 are generated within the internal dynamics of the Topological Phase Kernel before any exponential evaluation, hexad–octad realization, or metrological reading. Their provenance is the oriented partition of the 9 phases and the incidence of their active pairs; the spectral channels $q_\pm$ will subsequently act on these already constructed objects.

8.10.1 Tritian Partition of the Nonadic Phase into Oriented Semicoronas

In the canonical chart $D_9 = \{1, \dots, 9\}$, the tritic phase selector determines the disjoint decomposition

$$H_+ = \{1, 4, 7\}, \qquad H_- = \{2, 5, 8\}, \qquad H_0 = \{3, 6, 9\}.$$

The classes of H_+ activate the additive sheet, those of H_- the multiplicative sheet and those of H_0 conserve the threshold. We define

$$H_{\text{act}} = H_+ \sqcup H_-, \qquad |H_{\text{act}}| = 6, \qquad \left|P_2(H_{\text{act}})\right| = \binom{6}{2} = 15,$$

where $P_2(H_{\mathrm{act}})$ denotes the set of unordered pairs of active phases. This partition conserves leaf, orientation, and phase origin; it is not a retrospective classification induced by physical realizations.

8.10.2 Incidence channel of cardinality 90

The internal incidence between a distinguished active phase and an active pair is

$$\mathcal{F}_{90} := H_{\mathrm{act}} \times P_2(H_{\mathrm{act}}).$$

By construction,

$$|\mathcal{F}_{90}| = 6 \cdot 15 = 90.$$

The cardinality 90 therefore follows from the tritic semicorona of the TPK. It is not an input from Barbero–Immirzi, from the mass law, or from a metrological table.

8.10.3 Incidence channel of cardinality 120

Extending the first factor to the 8 positions that precede the nonadic return yields

$$O_{\mathrm{ph}} = \{1, \dots, 8\}, \qquad \mathcal{F}_{120} := O_{\mathrm{ph}} \times P_2(H_{\mathrm{act}}),$$

and, consequently,

$$|\mathcal{F}_{120}| = 8 \cdot 15 = 120.$$

The difference between $\mathcal{F}_{90}$ and $\mathcal{F}_{120}$ preserves the boundary information: the second channel incorporates the 8 positions preceding the return, whereas the first preserves only the 6 operational phases.

8.10.4 Dodecaphasic compatibility and causal precedence with respect to spectral evaluation

The dodecaphase wheel pairs the 2 oriented semicoronas and records the decomposition $1 + 5 + 6 = 12$, invertible on its image. Thus, the incidence cardinalities are fixed in the causal order

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow (H_+, H_-, H_0) \longrightarrow (\mathcal{F}_{90}, \mathcal{F}_{120}).$$

Only afterward, once the conjugate angular coordinates have been constructed, the channels $q_\pm$ spectrally evaluate these incidences by means of

$$V_{90,120} = (I - T^{90})(I - T^{120})^{-1} = r_+ P_+ + r_- P_-, \qquad r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}}.$$

The evaluation $V_{90,120}$ does not generate the cardinalities that appear in its exponents: it operatorially publishes the incidence channels previously produced by TPK.

8.11 Center, boundary, and vacancies in the nonadic connection

8.11.1 The thesis and the order of construction

The relation between center and boundary acquires mathematical content when one specifies which data are published, which transformations transport them and with what information they are reconstructed. In APP, a residue is accompanied by its quotient; in TRIT, orientation distinguishes operations that an aggregate can merge; in TPK, a turn returns the phase and prolongs memory. The enriched state brings these coordinates together. Its residence in the joint discrete structure of the continuum preserves dependencies between the various subsequent readings.

The path is APP → TRIT → TPK → enriched state → joint discrete structure of the continuum. This development makes local cuts of that path: it does not reconstruct the five consubstantial aspects of the continuum from a spectral form or turn them into five independent modules. The internal constants are

received at their publication stage, without using their conventional values to choose seeds, routes or coefficients.

The author's intuition can be expressed thus: **publication of a value is a projection of a construction; transporting that construction preserves the information that allows its realizations to be distinguished**. The following operations make that statement concrete. The subsequent language of interpolation and orthogonal decomposition proves properties of already specified objects; it does not retrospectively select the APP state.

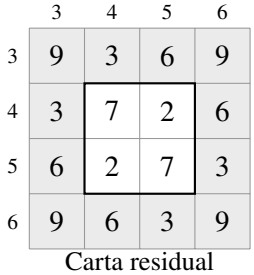
8.11.2 The local crown and reconstruction of the center

In the positive chart we use

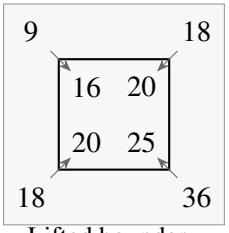
$$\rho_9(n) = 1 + [(n-1)]_9, \qquad q_9(n) = \frac{n - \rho_9(n)}{9}, \qquad n = \rho_9(n) + 9q_9(n).$$

The product table is $P(i,j) = ij$, with $1 \le i, j \le 9$. Restriction to $Q = \{3,4,5,6\}^2$ publishes

$$(\rho_9(ij))_{i,j=3}^{6} = \begin{pmatrix} 9 & 3 & 6 & 9 \\ 3 & 7 & 2 & 6 \\ 6 & 2 & 7 & 3 \\ 9 & 6 & 3 & 9 \end{pmatrix}.$$



Carta residual
corona 3, 6, 9



Lifted boundary
and multiaffine interpolation

Figure 8.5: The same chart in two readings. On the left, the radical crown surrounds the block B_c. On the right, the four integer corners determine the interior values under the multiaffine law. Their corner residues coincide, but the quotients $(0,1,1,3)$ distinguish the lift.

Its inner block is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

The perimeter of this chart contains twelve positions: four with value 3, four with 6 and four with 9; their sum is 72. A radical crown becomes visible around the central block. The global electronic center is $(5,5)$, the unique fixed point of inversion of the complete APP; its position is not changed by choosing this local chart, whose geometric center is at $(4.5, 4.5)$.

Proposition 8.63 (Recovery from the lifted boundary)**.** *Let $F(x,y) = a + bx + cy + dxy$ be a multiaffine law in the chart $[3,6]^2$. For $t = (x-3)/3$ and $u = (y-3)/3$,*

$$\begin{aligned} F(x,y) = (1-t)(1-u)F(3,3) + (1-t)uF(3,6) \\ + t(1-u)F(6,3) + tuF(6,6). \end{aligned}$$

Proof. For fixed y, affinity in x recovers $F(x, y)$ from $F(3, y)$ and $F(6, y)$. Affinity in y recovers each of those two values from its endpoints. Composition gives the formula. In particular, the four corners determine the entire chart within the declared space of laws. □

For the product, the four lifted values are $9, 18, 18, 36$. Their positive residues are all 9, but their quotients are $0, 1, 1, 3$. The formula recovers $16, 20, 20, 25$ at the interior positions; its residual publication gives $7, 2, 2, 7$. In this way, the boundary determines the interior **while preserving the law and the lift**. Interpolation is performed before modular reduction: one does not divide by 3 in $\mathbb{Z}/9\mathbb{Z}$, where 3 is not invertible.

This specifies the meaning of "preserving the information of the center." The residual crown alone does not individualize all lifts: F and $F + 9$ have the same residues along the entire boundary and different lifted values. The residue–quotient pair removes that ambiguity within the indicated reconstruction. The assertion is proved for multiaffine laws; extending it to complete states also requires preserving the operators, routes, and memory acting on them.

Multiaffine recovery is a **new formalization of the preexisting authorial architecture**. The charts and their readings are results recovered from the corpus; this explicit presentation of their inverse is not attributed as an independent historical discovery.

8.11.3 Sum, product, and the three values of discrete curvature

The multiplicative rows are constructed by additive accumulation. The difference between the two sheets appears when both coordinates are varied simultaneously. With unit differences,

$$\Delta_x \Delta_y (x + y) = 0, \qquad \Delta_x \Delta_y (xy) = 1.$$

The same property is recovered on the lifted boundary:

$$\frac{P(6,6) - P(6,3) - P(3,6) + P(3,3)}{3 \cdot 3} = 1;$$

for $S(x, y) = x + y$, this quotient equals 0. The product records the cross variation that the sum does not introduce. The corpus expresses the oriented version through

$$\mathcal{F}(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x,$$
$$\mathcal{F}(S, S) = \mathcal{F}(P, P) = 0, \qquad \mathcal{F}(S, P) = +1, \qquad \mathcal{F}(P, S) = -1.$$

The order of the sheets distinguishes the sign. The formulation "the sum accumulates and the product folds or unfolds" rests here on a verifiable operation: modular projection preserves a residue; lifting recovers the quotient recording what was exceeded. Discrete curvature additionally records the order of the variations. This finite difference is not identified with a differential curvature of a physical space without its corresponding realization map.

The global radical cross

In $R = \mathbb{Z}/9\mathbb{Z}$, the ideal $\mathfrak{m} = \{0, 3, 6\}$ satisfies $\mathfrak{m}^2 = 0$. Its positive publication is $\{9, 3, 6\}$. The cross

$$(\mathfrak{m} \times R) \cup (R \times \mathfrak{m})$$

has $27 + 27 - 9 = 45$ positions. The intersection contains nine values 9, summing to 81; the remaining arms have twelve values of each class $3, 6, 9$, summing to 216. The total is

$$297 = 216 + 81 = 3(72 + 27) = 3 \cdot 99.$$

Four boundaries are distinguished: the local perimeter of twelve positions, the radical cross of forty-five, the gnomon of seventeen completing 8^2 to 9^2, and the cubic crown of 271. Each has its own domain and operation. The tritic quotient $R/\mathfrak{m}$ preserves a class; transporting the state requires accompanying it with the lifted data and their history.

8.11.4 The 72–27 and 77–22 readouts and the 729–271 completion

The ordered rows of B_c are published decimally as 72 and 27; its diagonals as 77 and 22. This yields

$$72 + 27 = 77 + 22 = 99.$$

In the cubic completion, counting the interior, faces, edges, and vertex gives

$$1000 = 729 + 243 + 27 + 1 = 729 + 271,$$

$$999 = 729 + 243 + 27 = 729 + 270.$$

The coincidence of readouts is expressed while preserving the decimal remainder:

$$729 = 10 \cdot 72 + 9, \qquad 271 = 10 \cdot 27 + 1.$$

Therefore, the two-dimensional reading publishes $(72, 27)$; the cubic reading publishes $((72, 9), (27, 1))$. The pair of remainders $(9, 1)$ distinguishes the prolongations that would share the same quotient. The chart constructions are carried out first; those publications are then compared. The identity of the integers does not reverse that causal order.

The chapter on integer channels provides another specific composition. With the internal counts $\Delta = 54$, $\nu = \Delta/2 = 27$, and $\Theta = 5\nu = 135$, it obtains

$$C_\alpha = \nu^2 = 729, \qquad C_e = 2\Theta + 1 = 271.$$

Here ν is the new name for the local integer denoted by κ in that source, to distinguish it from the rational readout of the dodecaphase record. The channels are integer objects; their identification with the constants passes through the regional readouts and the corresponding closure, not through the first digits of an expansion. This comparison uses the integer channels and does not require the subsequent angular realization.

For the oral reference to a fraction of π, the located source contains

$$\frac{396}{126} = \frac{22}{7}, \qquad 396 = 18 + 72 + 108 + 198, \quad 126 = 18 + 108.$$

It is a rational ratio of alternating rings. This ring ratio is an exact rational reading; its role is distinguished from the regional publication of π.

8.11.5 The electronic center and its conjugate vacancies

The inversion of the positive APP is $\rho(r, c) = (10 - r, 10 - c)$. Its unique fixed point is $(5, 5)$. In the central additive fiber,

$$\mathcal{F}_{10} = \{(5 + t, 5 - t) : t \in \mathbb{Z},\ |t| \le 4\},$$

the product equals $25 - t^2$. Preserving its central residue requires $t^2 \equiv 0 \pmod 9$, equivalent to $3 \mid t$. The only positions are therefore

$$(5, 5), \qquad (8, 2), \qquad (2, 8).$$

If $\mathcal{A}$ preserves the residue–quotient pair on each sheet,

$$\mathcal{A}(5,5) = ((1,1),(7,2)),$$
$$\mathcal{A}(8,2) = \mathcal{A}(2,8) = ((1,1),(7,1)).$$

The two vacancies preserve the additive readout and the multiplicative residue; the product quotient decreases by exactly one unit. The mirror $t \mapsto -t$ exchanges their positions. The residual value does not distinguish their routes, whereas orientation does preserve them. This is a precise realization of the connection indicated in the note: **a vacancy is individualized by a relation to the center and by a recorded defect, not by being a cell without information**.

The initial electronic orientation $(h_0, v_0) = (1,1)$, reversed every 27 steps, subsequently publishes the dodecaphase signature $(+++\ ---\ +++\ --)$. This is an orientation record, distinct both from the position $(5,5)$ and from the following product signature.

The signature 555555 and the memory enabling its prolongation

For $U_j = 100d_{1j} + 10d_{2j} + d_{3j}$, the readout

$$\mathcal{E}_\times(U) = ([d_{2j} - d_{1j}]_{10})_{j=1}^{6}$$

applied to the transverse region

$$U_\pi^\perp = (501, 614, 498, 169, 272, 272)$$

produces 555555; its reduction $\Psi(U) = (-U_j \bmod 3)_j$ produces 010211. The signature has twenty-four product representatives. Transport separates them into three sheets of eight: after advancement, 555177, 555717, and 555771 appear. The congruence stable under advancement and half-turn refines the twenty-six signatures into twenty-eight classes, with histogram $27 \cdot 8 + 108 = 324$. This expresses why sheet memory must be retained to continue a signature that initially coincides.

Definition 8.64 (Signature readout and transport of representatives)**.** Let $\mathcal{X}_\times = D_9^2 \times \{N, E, S, O\}$, of cardinality 324. Advancement P moves one cell in the stored direction, and the half-turn H reverses that direction. The readout uses

$$\ell_9(1,2,4,8,7,5) = (0,1,2,3,4,5), \qquad \ell_9(3) = \ell_9(6) = \ell_9(9) = 0.$$

In each of six windows of nine steps, $\ell_9(\rho_9(ij))$ is evaluated before the product advancement selected by TRIT at times $2, 5, 8 \pmod 9$. The sum in each window is reduced modulo 10. After steps 27 and 54, the direction is reversed. The rule determines $E_{\mathrm{sig}}(x)$ from the position and direction of x, with the kernel's time conventions.

Proposition 8.65 (Minimal stable refinement of the signature)**.** *Starting from equivalence by signature $\sim_0$, the iteration*

$$x \sim_{n+1} y \iff x \sim_n y, \quad Px \sim_n Py, \quad Hx \sim_n Hy$$

terminates and produces the coarsest stable congruence refining the signature. On the preceding domain the census is $26 \to 28 \to 28$*; the fiber of* 555555 *separates into three classes of eight representatives.*

Proof. At each step, a partition of a finite set is refined. The process stabilizes after finitely many splittings. At the fixed point, equivalence is preserved under P and H. If a stable congruence refines $\sim_0$, it refines each $\sim_n$ by induction: its images under both operators remain equivalent. It therefore refines the fixed point obtained, proving its minimality in added information.

The finite enumeration applies the definition to the 81 positions and the four orientations. Their six sums modulo 10 are grouped first, and then the triple consisting of the class of x, that of Px, and that of

Hx, until stabilization. The resulting histogram is $27 \cdot 8 + 108 = 324$, and the only signature splitting is the one indicated. The following representatives make the three images explicit:

x	$E_{\mathrm{sig}}(x)$	$E_{\mathrm{sig}}(Px)$
$(3, 5, N)$	555555	555177
$(3, 4, N)$	555555	555717
$(3, 4, S)$	555555	555771

The complete rule and its executable enumeration are preserved in the reproduction package; the witnesses in the table also show that preserving only the signature does not make its continuation unique. □

The conclusion describes the minimal information of transport, not a new mass. The additive factor preserves eighteen representatives per signature (nine compatible positions and two orientations). Combining it with the product fibers of sizes 24 and 8 gives, respectively, 432 and 144. Regional incidence and sheet memory thus preserve distinct functions within the same catalog.

The link to the electronic equation passes through the regions. The transverse region has multiplicity $432 = 18 \cdot 24$; each of the other four has $144 = 18 \cdot 8$. Their five identities are inserted, in the declared gauge of Article I, into

$$\mathcal{K}_e = \mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3$$

through $(4, 0), (5, 0), (6, 0), (6, 1), (6, 2)$. Once this insertion has been fixed, its complement has $27 - 5 = 22$ positions and its three sheets per region have $5 \cdot 3 = 15$ positions. The oriented deficit and torsional memory are -22 and 15. The proof of these counts follows from the preceding explicit injection: its five images are distinct; this development brings it together, rather than selecting its positions from a target mass.

There are therefore two linked progressions in the composition: **center → conjugate vacancies → orientation**, and **regional signature → transport sheets → incidence of the five regions → electronic coefficients**. Simply identifying $(5, 5)$ with 555555 would erase precisely the operations explaining their relation.

8.11.6 Nonadic mirror, capacity, and return with memory

On the regional fiber, the mirror acts by

$$\mathfrak{c}(u^\pi, u^e, u^\varphi) = (u^e, u^\pi, u^\varphi).$$

It preserves $s = u^\pi + u^e$ and reverses $d = u^\pi - u^e$. In $\mathbb{F}_3$,

$$u^\pi = 2(s + d), \qquad u^e = 2(s - d).$$

The sum preserves the aggregate; the oriented difference preserves its distribution. The axis u^φ remains fixed under this involution, although it may evolve under transport. The regional mirror and the cellular mirror of the vacancies have different domains: their respective formulas make it possible to track what each preserves.

The capacity vacancy adds an explicit temporal link. In the normalization of refinement 729 versus 1000, let $\rho_c = \log_{10} 9$, $\delta = 1 - \rho_c$, and $C_t = \lfloor (t + 1)\rho_c \rfloor$. If

$$z_t = \lfloor (t + 1)\delta \rfloor - \lfloor t\delta \rfloor,$$

then, for $t \geq 1$,

$$C_t - C_{t-1} = 1 - z_t, \qquad C_b - C_a + \sum_{t=a+1}^{b} z_t = b - a.$$

Proof. The irrationality of $\log_{10} 9$ follows from the impossibility of $9^n = 10^m$ for positive integers. Hence, $C_t = t - \lfloor (t+1)\delta \rfloor$. Subtraction and telescoping summation give both identities. □

A vacancy retains capacity while the history advances.

If $p_j = \lfloor j/\delta \rfloor$, $v_j = p_j - j$, and $h_j = p_{j+1} - p_j$, the inherited development establishes $h_j \in \{21, 22\}$. For $s_j = 22 - h_j$,

$$[v_{j+1}]_3 = [v_j - s_j]_3, \qquad \tau_j = M_{\mathrm{ph}}(1 + [v_j]_9).$$

Vacancies thus act on the retained phase read by the tritic selector. This capacity sequence does not automatically become the electron's cellular trajectory: the manuscript preserves their respective mechanisms and their composition within the same kernel.

8.11.7 Vacancies of nonadic refinement and hierarchy of returns

The APP–TRIT–TPK prolongation preserves, alongside the published block, the cylinder, orientation and history that allow it to be continued. In its capacity reading, two cardinalities already determined by that construction are compared: the $3^6 = 729$ possibilities of a ternary block of six positions and the $10^3 = 1000$ positions of its three-digit decimal chart. This comparison produces its own calendar. Some transitions increase the capacity index; others incorporate a block into the history while maintaining that index. The latter are the *capacity vacancies* studied below.

The object considered is, therefore, a publication of the joint refinement. Its definition preserves its origin in the two finite supports; the logarithmic expression makes it possible to prove the properties of its calendar. Block time, the vacancy index and the capacity index will be written separately. This separation makes clear why the returns 21/22 and 6/7 belong to successive scales of the same construction.

8.11.7.1 Decimal capacity and preservation of the increment

Definition 8.66 (Capacity index and vacancy)**.** At phase origin 0, define

$$\rho = \log_{1000} 729 = \log_{10} 9, \qquad \delta = 1 - \rho = \log_{10}(10/9), \tag{8.23}$$

and, for $t \in \mathbb{Z}_{\geq 0}$,

$$C_t = \lfloor (t+1)\rho \rfloor, \qquad z_t = \lfloor (t+1)\delta \rfloor - \lfloor t\delta \rfloor. \tag{8.24}$$

The integer C_t is the decimal capacity index of the readout; $z_t = 1$ marks a vacancy. The letter K remains reserved for the dodecaphasic register and does not denote a capacity counter here.

We have $0 < \delta < 1$. Moreover, δ is irrational: if $\rho = a/b$ were rational, with a, b positive integers, the equality $9^b = 10^a$ would have zero valuation at the prime 5 on the left and positive valuation on the right. This contradiction proves the irrationality of ρ and of $1 - \rho$. In particular, no positive integer multiple of δ belongs to $\mathbb{Z}$, which fixes the endpoints of the intervals in this development unambiguously.

Proposition 8.67 (Exact capacity–vacancy balance)**.** *For $t \geq 1$, $z_t \in \{0, 1\}$ and*

$$C_t - C_{t-1} = 1 - z_t. \tag{8.25}$$

Consequently, for $0 \leq a < b$,

$$C_b - C_a + \sum_{t=a+1}^{b} z_t = b - a. \tag{8.26}$$

Proof. Since $0 < \delta < 1$, the difference between the floors of two numbers separated by δ is 0 or 1. Furthermore, the established irrationality gives

$$C_t = t + 1 - \lceil (t+1)\delta \rceil = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting this equality at t and $t-1$ proves (8.25); telescopic summation proves (8.26). □

Each transition contributes exactly one unit to the right-hand side of the balance. That unit is recorded as acquired capacity when $z_t = 0$, or as a vacancy when $z_t = 1$. Retention of the index C_t therefore leaves a recoverable position in the history. The equation expresses preservation of the increment through two readouts, rather than an indeterminate loss of information.

The density also follows from the same balance:

$$\frac{1}{N}\sum_{t=1}^{N} z_t = \frac{\lfloor (N+1)\delta \rfloor}{N} \longrightarrow \delta.$$

Thus (z_t) is not eventually periodic: an eventually periodic binary sequence has rational density. Here aperiodicity follows from the capacity rule already fixed by 729/1000.

8.11.7.2 Exact determination of the intervals 21/22

Lemma 8.68 (Rational bounds for the slope)**.** *The vacancy density satisfies*

$$\frac{7}{153} < \delta < \frac{6}{131}. \tag{8.27}$$

If $\lambda = \delta^{-1}$, $\beta = 22 - \lambda$, and $\mu = \beta^{-1}$, then

$$21 < \lambda < 22, \qquad \frac{1}{7} < \beta < \frac{1}{6}, \qquad 6 < \mu < 7. \tag{8.28}$$

Proof. We provide a finite verification by rational arithmetic. For $0 < u < 1$, let

$$S_N(u) = 2\sum_{k=0}^{N} \frac{u^{2k+1}}{2k+1}, \qquad R_N(u) = \frac{2u^{2N+3}}{(2N+3)(1-u^2)}.$$

Integrating the geometric series for $2/(1-u^2)$ gives

$$S_N(u) < \log\frac{1+u}{1-u} < S_N(u) + R_N(u).$$

Indeed, the tail is positive and, on replacing each denominator $2k+1$ by $2N+3$, is bounded by the geometric series defining R_N. These inequalities are applied to $u = 1/19, 1/3, 1/9$, corresponding respectively to $\log(10/9), \log 2, \log(5/4)$, and we use $\log 10 = 3\log 2 + \log(5/4)$.

With $N = 4$, writing $A = S_4(1/19)$, $B = 3S_4(1/3) + S_4(1/9)$, and $E = 3R_4(1/3) + R_4(1/9)$, we obtain

$$\frac{7}{153} < \frac{A}{B+E} < \delta < \frac{A + R_4(1/19)}{B} < \frac{6}{131}.$$

The two outer inequalities are checked by multiplying through by the positive denominators of these sums of five rational terms. This proves (8.27) using a remainder bound, without taking a decimal expansion as a premise. Inverting the bound gives $131/6 < \lambda < 153/7$. Subtracting from 22 and then inverting yields all the inequalities in (8.28). □

Theorem 8.69 (Positions and gaps of the vacancies)**.** *The positions of the events $z_t = 1$, ordered from the first event, are exactly*

$$p_j = \lfloor j\lambda \rfloor = \left\lfloor \frac{j}{\delta} \right\rfloor, \qquad j \geq 1. \tag{8.29}$$

Their gaps satisfy

$$h_j = p_{j+1} - p_j \in \{21, 22\}. \tag{8.30}$$

Both values occur infinitely often.

Proof. The event crossing the integer j occurs precisely when $t\delta < j < (t+1)\delta$. Irrationality excludes equality at the endpoints. Dividing by δ gives $t = \lfloor j/\delta \rfloor = p_j$. Since $\delta < 1$, a step crosses at most one integer, so the enumeration is exact. Writing $\lambda = 21 + \eta$, with $0 < \eta < 1$, gives

$$h_j = 21 + \lfloor (j+1)\eta \rfloor - \lfloor j\eta \rfloor,$$

which proves (8.30). Finally, $(p_{J+1} - p_1)/J \to \lambda \in (21, 22)$. If either value occurred only finitely often, this mean would converge to the other integer value, a contradiction. □

8.11.7.3 Induction on the short intervals: the returns 6/7

Theorem 8.70 (Induced schedule of short intervals)**.** *Let $s_j = 22 - h_j$, for $j \geq 1$. This indicator equals* 1 *exactly when the interval between vacancies is short, of length* 21*. With β and μ as defined in Lemma 8.68,*

$$s_j = \lfloor (j+1)\beta \rfloor - \lfloor j\beta \rfloor. \tag{8.31}$$

The indices of the short intervals are

$$q_m = \lfloor m\mu \rfloor = \lfloor m/\beta \rfloor, \qquad q_{m+1} - q_m \in \{6, 7\}, \quad m \geq 1. \tag{8.32}$$

Both gaps occur infinitely often, and their order is not eventually periodic.

Proof. The irrationality of δ implies that of λ, β, and μ. Since $\lambda = 22 - \beta$, for $j \geq 1$ we have

$$p_j = 22j - \lceil j\beta \rceil = 22j - \lfloor j\beta \rfloor - 1.$$

Subtracting consecutive terms gives (8.31). Applying the integer-crossing argument of the preceding theorem to the indicator of slope β shows that its event m occurs at $q_m = \lfloor m/\beta \rfloor$. The inequality $6 < \mu < 7$ proves the two possible gaps. Their mean converges to μ; hence both occur infinitely often. Moreover, an eventually periodic sequence of gaps would have rational mean, whereas μ is irrational. This also rules out strict alternation, whose mean would be $13/2$. □

Here the numbers 6 and 7 count *intervals between vacancy events*, not refinement blocks. The return to the primary scale is equally explicit. Between q_m and q_{m+1} there is a single short interval, located at the beginning, and all the others are long. Consequently, if $d_m = q_{m+1} - q_m$,

$$p_{q_{m+1}} - p_{q_m} = 21 + 22(d_m - 1) = 22d_m - 1 \in \{131, 153\}. \tag{8.33}$$

Thus $131 = 14 \cdot 9 + 5$ and $153 = 17 \cdot 9$ describe two return readouts *in the block clock*. The equation preserves the operation connecting the scales, rather than identifying their cardinalities merely because they occur.

8.11.7.4 Retained capacity and selection of the TRIT regime

Let $v_j = C_{p_j}$ be the capacity recorded at vacancy j. From $p_j\delta < j < (p_j+1)\delta$ and $\delta < 1$, it follows that $\lfloor (p_j+1)\delta \rfloor = j$. By (8.24),

$$v_j = p_j - j, \qquad v_{j+1} - v_j = h_j - 1 = 21 - s_j, \qquad [v_{j+1}]_3 = [v_j - s_j]_3. \tag{8.34}$$

A long interval preserves the class modulo 3; a short interval shifts it by one unit in the negative direction. Applying the selector of the formal kernel to the positive phase $r_j = 1 + [v_j]_9$ gives

$$\tau_j = M_{\mathrm{ph}}(r_j), \qquad M_{\mathrm{ph}}(r) = \begin{cases} +1, & r \in \{1,4,7\}, \\ -1, & r \in \{2,5,8\}, \\ 0, & r \in \{3,6,9\}. \end{cases} \tag{8.35}$$

This composition fixes the regime induced by capacity at each event. It preserves both the order of the changes and the number of events for which each regime persists.

Corollary 8.71 (Plateaus of the induced regime)**.** *For every $m \geq 1$, the class $[v_j]_3$, and hence τ_j, is constant on $q_m + 1 \leq j \leq q_{m+1}$. The length of each complete plateau is $q_{m+1} - q_m \in \{6, 7\}$.*

Proof. On the indicated integer interval, the next change occurs exactly when passing from $j = q_{m+1}$ to $j = q_{m+1} + 1$: this is the next position at which $s_j = 1$. At all preceding steps $s_j = 0$, so (8.34) preserves the class. At the endpoint it decreases the class by one unit. The selector (8.35) assigns distinct values to the three classes; it therefore preserves the same plateaus and their endpoints. □

The initial segment $1 \leq j \leq q_1$ depends on the chosen origin and is an initial restriction of a plateau. At the present origin it has length $q_1 = 6$. The first twenty classes are

$$\underbrace{2, \ldots, 2}_{6}, \qquad \underbrace{1, \ldots, 1}_{7}, \qquad \underbrace{0, \ldots, 0}_{7},$$

with signatures $0, -1, +1$, respectively. Each short interval triggers the next step of the cycle of regimes.

The two phases of interest are defined as

$$g_t^{\mathrm{cron}} = [t]_9, \qquad g_t^{\mathrm{cap}} = [C_t]_9, \qquad g_t^{\mathrm{cap}} - g_{t-1}^{\mathrm{cap}} \equiv 1 - z_t \pmod 9.$$

A vacancy retains the capacity phase while block time advances. The transition of the enriched state remains the TPK composition of selection, transport, and memory update. Thus $C_t = C_{t-1}$ neither resets the state nor replaces the holonomic winding Γ_9 by a pause. The two induced scales likewise retain this distinction: between consecutive short events in (8.33), the capacity increment is $21d_m - 1$, that is, 125 or 146, whereas the block increment is 131 or 153.

8.11.7.5 Rationally certified sample and provenance

The table presents a sample at the canonical origin. h_j is the gap *following* p_j; thus vacancy $j = 6$, at block 131, begins the first short interval.

j	1	2	3	4	5	6	7	8	9	10
p_j	21	43	65	87	109	131	152	174	196	218
h_j	22	22	22	22	22	21	22	22	22	22
v_j	20	41	62	83	104	125	145	166	187	208
$[v_j]_3$	2	2	2	2	2	2	1	1	1	1

The first short-interval indices and their gaps are

$$(q_m)_{m=1}^{10} = (6, 13, 20, 27, 34, 41, 48, 54, 61, 68),$$

$$(q_{m+1} - q_m)_{m=1}^{9} = (7, 7, 7, 7, 7, 7, 6, 7, 7).$$

The sample exhibits the actual order, while the preceding proofs establish its law at arbitrary depth.

The attached file `ampliacion/verificar_vacancias.py` computes rational brackets using the series and remainders of Lemma 8.68. Each floor is accepted only when both endpoints of the bracket yield the same integer. At order 16, it reproduces 512 events up to block 11189, and 64 secondary gaps, together with the balance identities, residues, and plateaus. This finite check retains a scope distinct from that of the universal theorems.

The capacity index, its complementarity with vacancies, and induction of the TRIT selector are recovered from Article I and from the chapters on the schedule and induced clock in the monograph on aperiodic topological–temporal order. Their definitions and the proofs of both scales are assembled here with a single indexing convention. The rational certificate provides an additional check of those results and preserves their provenance.

8.11.8 Factor 8/81 in the decomposition between reading and complement

In the double-circle development, the uniform inclusion of nine phases is $Jv = (v, \dots, v)/3$. With the seam

$$S_C(v_0, \dots, v_8) = (Cv_8, v_0, \dots, v_7),$$

one obtains $T = J^* S_C J = (8I + C)/9$. If $z = (C - I)v$, the complement of the publication is exactly

$$\eta v = (I - JJ^*) S_C J v = \frac{1}{27}(8z, -z, \dots, -z).$$

Therefore,

$$\|\eta v\|^2 = \frac{64 + 8}{729}\|z\|^2 = \frac{8}{81}\|(C - I)v\|^2.$$

Here appears the $8/81$ mentioned by the author: one transported phase against eight others, with the uniform normalization of nine phases. The result does not depend on which of the nine positions is marked. The complete operator balance is

$$I - T^*T = \frac{8}{81}(I - C)^*(I - C) + \frac{1}{9}(I - C^*C).$$

If C is isometric, the energy omitted by the publication lies exactly in its complement. For an arbitrary seam, its explicit defect is also preserved. The integer $72 = 64 + 8$ counts quadratic weights here; in the local crown it counts a sum of values; in $72/27$ it belongs to an ordered decimal reading. Specifying these types makes it possible to relate the constructions without assuming that a numerical coincidence already provides all the maps between them.

This balance remains that of the double-circle transport development. When applied to the arithmetic form $\mathscr{W}$, it produces a signed identity; proving global positivity requires global control of its terms, not merely recovery of the center or positivity of an auxiliary norm. The results already proved concerning memory, transport, and the coercive window are preserved in full.

8.12 Combinatorial selection and representation transport of the first joint boundary

The first joint boundary receives the blocks already produced by the APP seeds, tritic reduction and TPK lifting calendar. Its construction preserves the distinction between three operations: determining admissible distributions, orienting their two lateral channels and transporting incidence to the exceptional coordinate system. The present account develops these operations on the same finite state. Names of real coordinates designate their subsequent readers; the following selection conditions act on ternary words, transport matrices and integer incidences.

8.12.1 Regional state preceding the boundary

Let $\mathcal{W} \subset \mathbb{F}_3^6$ be the repertoire of 243 visible words obtained from the APP–TRIT–TPK emission census. For a row vector $s \in \mathcal{W}$, define

$$b_0(s) = s, \quad b_1(s) = sL_0, \quad b_2(s) = sL_0^2, \quad b_3(s) = sL_0^2L_1, \quad b_4(s) = sL_0^2L_1^2,$$

with operations in $\mathbb{F}_3$. The matrices previously derived from the regional registers are

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

The preceding regional selector yields

$$s_{\mathrm{cl}} = 010211, \qquad s_{\mathrm{pr}} = 201101,$$

through the conditions $b_2(s_{\mathrm{cl}}) = b_0(s_{\mathrm{cl}})$, $b_4(s_{\mathrm{pr}}) = b_2(s_{\mathrm{pr}})$ and $b_1(s_{\mathrm{cl}}) = b_3(s_{\mathrm{pr}})$. At this stage, the current blocks are

$$b_4(s_{\mathrm{cl}}) = 110221, \qquad b_4(s_{\mathrm{pr}}) = 102011.$$

Concatenation preserves the five blocks of each register; vector b_4 is the six-coordinate input received by the boundary relation.

To express the recovered relation faithfully, the incidence matrix is used in its original representation,

$$A^{\mathrm{o}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \tag{8.36}$$

We verify $(A^{\mathrm{o}})^2 = -I$ in $M_6(\mathbb{F}_3)$; therefore, $(A^{\mathrm{o}})^{-1} = -A^{\mathrm{o}}$. This representation, together with its vectors, will later be transported to the one used by the exceptional reader.

8.12.2 Compatibility equation and column saturation

The boundary axis is fixed by $z = b_4(s_{\mathrm{pr}}) = 102011$. For each $s \in \mathcal{W}$, define

$$m(s) = b_4(s)A^{\mathrm{o}}L_1^3, \tag{8.37}$$

$$q(s) = m(s) + z, \tag{8.38}$$

$$\widetilde{q}(s) = s(A^{\mathrm{o}})^{-1}L_0^3(A^{\mathrm{o}})^3. \tag{8.39}$$

The compatibility condition is $q(s) = \widetilde{q}(s)$. The lateral rows $x, y \in \mathbb{F}_3^6$ satisfy $x + y = m(s)$; the ordered boundary is $B = (x, y, z)^{\mathsf{T}}$.

To form remainder and carry, the integer representatives 0, 1, 2 of each trit are used. If $S_j = x_j + y_j + z_j$, require

$$q_j = S_j \bmod 3, \qquad c_j = \frac{S_j - q_j}{3} = 1,$$

and column occupation is given by

$$\mathbf{1}_{\{x_j \neq 0\}} + \mathbf{1}_{\{y_j \neq 0\}} + \mathbf{1}_{\{z_j \neq 0\}} = 2 + \mathbf{1}_{\{z_j \neq 0,\ m_j(s) \neq 2\}}. \tag{8.40}$$

Dual neutrality completes the condition through

$$BA^{\circ}\mathbf{1} = 0 \quad \text{in } \mathbb{F}_3^3. \tag{8.41}$$

Remainder, carry and occupation act jointly: modular reduction preserves the remainder, and the integer register distinguishes the distributions producing it.

Lemma 8.72 (Resolution of regional compatibility)**.** *The equation $q(s) = \widetilde{q}(s)$ has, in $\mathbb{F}_3^6$, the three solutions*

$$020010, \qquad 121200, \qquad 222120.$$

Its intersection with the visible repertoire is $\{121200, 222120\}$.

Proof. The equation is written $sD = z$, where

$$D = (A^{\circ})^{-1} L_0^3 (A^{\circ})^3 - L_0^2 L_1^2 A^{\circ} L_1^3 = \begin{pmatrix} 2 & 1 & 1 & 2 & 2 & 2 \\ 0 & 2 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 2 & 2 & 2 & 0 \\ 1 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 & 2 \end{pmatrix}.$$

Elimination over $\mathbb{F}_3$ gives the parametrization

$$s = (t, 2, t, 2t, 1 + 2t, 0), \qquad t \in \mathbb{F}_3.$$

Substitution verifies the six equations and exhausts their solutions. In the generated repertoire $\mathcal{W}$, the two words corresponding to $t = 1, 2$ are present, and the one corresponding to $t = 0$ is absent. This final check is an exact query of the visible-word census, preceding any Archimedean reading. □

Proposition 8.73 (Determination of the self-scaling axis)**.** *Compatibility, unit carry and column occupation select $s_{\mathrm{au}} = 121200$ from the two visible solutions of Lemma 8.72. We obtain*

$$b_4(s_{\mathrm{au}}) = 010200, \qquad m(s_{\mathrm{au}}) = 210112, \qquad q = 012120,$$

with occupation 223232.

Proof. For $s = 222120$, we compute $b_4(s) = 222000$, $m(s) = 201010$ and $q(s) = 000021$. In its third column, $m_3 = 1$ and $z_3 = 2$. The required occupation is three, so x_3, y_3 must both be nonzero. The only pair with $x_3 + y_3 \equiv 1 \pmod 3$ is $(2, 2)$; its integer sum with $z_3 = 2$ is six and produces carry two. This contradicts $c_3 = 1$.

For $s = 121200$, the six columns respectively admit the pairs

j	1	2	3	4	5	6
(x_j, y_j)	$(0,2),(2,0)$	$(2,2)$	$(1,2),(2,1)$	$(2,2)$	$(2,2)$	$(0,2),(2,0)$.

Each pair satisfies the remainder, carry and occupation equations. There are therefore eight arrangements before dual neutrality is applied. □

Theorem 8.74 (Exact enumeration of boundary orientations)**.** *The preceding conditions determine exactly*

$$B_- = \begin{pmatrix} 021222 \\ 222220 \\ 102011 \end{pmatrix}, \qquad B_+ = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}.$$

Both have remainder 012120*, carry* 111111 *and occupation* 223232*.*

Proof. The preceding proof reduces enumeration to eight possibilities. Since $A^{\circ}\mathbf{1} = (2,1,1,1,1,1)^{\mathsf{T}}$, neutrality of a row v is equivalent to $2v_1 + v_2 + \cdots + v_6 = 0$ in $\mathbb{F}_3$. Its evaluation on the eight arrangements is

x	y	$(xA^{\circ}\mathbf{1}, yA^{\circ}\mathbf{1}, zA^{\circ}\mathbf{1})$
021220	222222	$(1,2,0)$
021222	222220	$(0,0,0)$
022220	221222	$(2,1,0)$
022222	221220	$(1,2,0)$
221220	022222	$(2,1,0)$
221222	022220	$(1,2,0)$
222220	021222	$(0,0,0)$
222222	021220	$(2,1,0)$.

The second and seventh rows are the only solutions. The integer column sums of either are $(3,4,5,4,5,3)$, recovering the stated remainder and carry. □

8.12.3 Directed length and boundary orientation

Regional transport defines a directed graph on $\mathbb{F}_3^6$, with edges $v \to vL_0$ and $v \to vL_1$. For mutually reachable vectors u, v, let

$$d(u,v) = \min\{n \geq 0 : \exists \epsilon_1, \ldots, \epsilon_n \in \{0,1\},\ v = uL_{\epsilon_1} \cdots L_{\epsilon_n}\}.$$

This length is a combinatorial datum of transport. Its sum over the three rows preserves the regional labels:

$$\mathcal{L}(B) = d(110221, B_1) + d(102011, B_2) + d(010200, B_3).$$

Proposition 8.75 (Selection by directed length)**.** *We have* $\mathcal{L}(B_-) = 23$ *and* $\mathcal{L}(B_+) = 20$*. On the domain of the two admissible boundaries, the minimum is unique and selects* B_+*.*

Proof. The shortest paths and their lengths are

	origin	target	transport word	d
B_-	110221	021222	01100011	8
	102011	222220	10111111	8
	010200	102011	1010101	7
B_+	110221	222220	1111	4
	102011	021222	001100101	9
	010200	102011	1010101	7.

Each word is verified by successive multiplication of its matrices. To verify minimality as well, form the disjoint layers

$$F_0(u) = \{u\}, \qquad F_{n+1}(u) = \big(F_n(u)L_0 \cup F_n(u)L_1\big) \setminus \bigcup_{j=0}^{n} F_j(u).$$

Induction on n proves that $F_n(u)$ is exactly the set of vertices at distance n: every shortest word of length $n+1$ comes from a shortest word of length n, and quotienting by previous layers eliminates shorter paths. The enumeration obtained with the two matrices displayed above gives

$u\backslash n$	0	1	2	3	4	5	6	7	8	9
110221	1	2	4	8	16	32	60	105	156	177
102011	1	2	3	5	10	20	37	71	119	171
010200	1	2	4	8	15	30	56	100	148	172.

The destinations in the first table occur for the first time in the layers indicated there. This enumeration is finite, uses only products modulo three and agrees with the breadth-first traversal of the executable certificate. Summing the three lengths gives $8+8+7=23$ and $4+9+7=20$. □

Orientation of the APP representation through half-open cylinders previously selects the same matrix B_+ at this boundary. Theorem 8.74 and Proposition 8.75 thus establish agreement of both selections on the concrete set $\{B_-, B_+\}$. The domain of this agreement is the first joint boundary. Coinductive extension preserves its own admissibility, memory, survival and orientation relations at each depth.

8.12.4 Naturality of the exceptional change of representation

The matrix used by the exceptional reader in the active representation is

$$A^{\mathrm{a}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

The coordinate permutation

$$R_p(v_1, v_2, v_3, v_4, v_5, v_6) = (v_1, v_2, v_3, v_5, v_6, v_4)$$

simultaneously transports the word and the incidence matrix.

Theorem 8.76 (Intertwining of boundary incidence)**.** *For every $v \in \mathbb{F}_3^6$,*

$$R_p(vA^{\mathrm{o}}) = R_p(v)A^{\mathrm{a}}.$$

The transformation is invertible and preserves dual neutrality, remainders, carries and occupations expressed in the transported coordinates. In particular, the selected triple has the two representations

	source representation	*exceptional representation*
closure	222220	222202
propagation	021222	021222
self-scaling	102011	102110.

Proof. If $p = (1, 2, 3, 5, 6, 4)$, comparing the entries of both matrices gives $A^{\mathrm{a}}_{ij} = A^{\mathrm{o}}_{p(i),p(j)}$. For coordinate j,

$$\big(R_p(v)A^{\mathrm{a}}\big)_j = \sum_i v_{p(i)} A^{\mathrm{o}}_{p(i),p(j)} = \sum_k v_k A^{\mathrm{o}}_{k,p(j)} = \big(R_p(vA^{\mathrm{o}})\big)_j.$$

The inverse is $R_p^{-1}(w) = (w_1, w_2, w_3, w_6, w_4, w_5)$. Since R_p permutes coordinates, it preserves their sum and transports coordinatewise the integer sums, their remainders modulo three, quotients and numbers of nonzero entries. Preservation of all the stated registers follows. The table is obtained by applying R_p to the three words of B_+, and the inverse recovers their original words exactly. □

The positional reader and incidence reader remain associated with the same enriched state through this explicit change of representation. If ℓ_{pos} is the reader in the original representation, its expression in the transported representation is $\ell_{\mathrm{pos}} \circ R_p^{-1}$. This composition preserves positional order: changing the incidence representation and changing the reader are coordinated operations. The first boundary thus enters extension with its three words, regional orientation, integer carry and exceptional frame recoverable.

Chapter 9

Oriented classification of the closure microfiber of π: 5 regions, multiplicity 432+4·144=1008, φ-axis, and π/e involution

9.1 Construction of the closure character and of the genealogical trajectories of π

9.1.1 Causal separation between seed, character, and evaluation

The construction is divided into three strata:

$$\text{finite selection} \longrightarrow \text{construction of the closure character} \longrightarrow \text{Archimedean evaluation}.$$

The decimal expansion does not intervene in the first stratum. The type chain is

$$104\,976 \longrightarrow 468\,\mathcal{U}_6 \xrightarrow{\Psi} 243\,\mathcal{W}_6 \hookrightarrow \mathbb{F}_3^6, \qquad |\mathbb{F}_3^6| = 729. \tag{9.1}$$

The cardinalities count, respectively, enriched initial conditions, distinct emissions, visible ternary words, and elements of the complete ambient fiber.

For $U_6 = (u_1, \dots, u_6)$, the projection is

$$\Psi(U_6) = ((-u_1) \bmod 3, \dots, (-u_6) \bmod 3).$$

The census of the 43 orbits of D_3 proves that the closure orbit is the only one whose six orientations possess at once:

$$\text{five lifts}, \qquad \text{mass } 1008, \qquad 1008 = 432 + 4 \cdot 144.$$

The gauge

$$\mathrm{diag}_c = 6, \qquad \mathrm{card} = ES$$

select a single representative:

$$\boxed{w_6^{\pi} = 010211.} \tag{9.2}$$

The selection usis catalogue, multiplicitiis, orbital action, and orientation; it dois not consult to prefix of π.

9.1.2 Liftings L_0, L_1 and 1st coinductive boundary

On vectors row of $\mathbb{F}_3^6$, let

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

If $b_1 = w_6^{\pi}$, the recurrence of blocks is

$$b_{k+1} = b_k L_{\varepsilon_k}, \qquad (\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4) = (0, 0, 1, 1),$$

and the accumulated word is formed by concatenation:

$$w_{6(k+1)} = w_{6k} \parallel b_{k+1}.$$

A word of length $6k$ is not multiplied by a matrix 6×6.

For the closing channel:

depth	new block
6	010211
12	012222
18	010211
24	002111
30	110221

and

$$\boxed{w_{30}^{\pi} = 010211|012222|010211|002111|110221.} \tag{9.3}$$

The following boundary is

$$u_5^{\pi} = 222220.$$

Along with the other channels,

$$R_{36} = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}.$$

The symbol R_{36} records the sixth block of three histories and opens a boundary; it is not a terminal.

The certified branch of twenty blocks begins

$$\begin{aligned} \pi : {} & 010211|012222|010211|002111|110221|222220|111201| \\ & 212121|200121|100100|101222|022212|012012|111210| \\ & 121011|200220|120210|000101|022010|020111. \end{aligned}$$

For this reason, ni w_{30} ni R_{36} terminate the genealogy.

9.1.3 Oriented decomposition of the microfiber of π: central region, 4 lateral regions, and multiplicity $432 + 4 \cdot 144 = 1008$

The microfiber

$$\mathscr{R}_{\pi}^{\mathrm{micro}} := \{U \in \mathcal{U}_6 : \Psi(U) = 010211\}$$

contains exactly five regions:

U_6	E_{sig}	mult	role
501|614|498|169|272|272	555555	432	$\perp$
810|923|870|169|272|272	339555	144	++
810|983|810|169|272|272	393555	144	++
870|923|810|169|272|272	933555	144	++
870|923|810|418|674|521	933717	144	−−

Consequently,

$$\boxed{5 = 1_{\perp} + 3_{++} + 1_{--}, \qquad 1008 = 432 + 4 \cdot 144.} \tag{9.4}$$

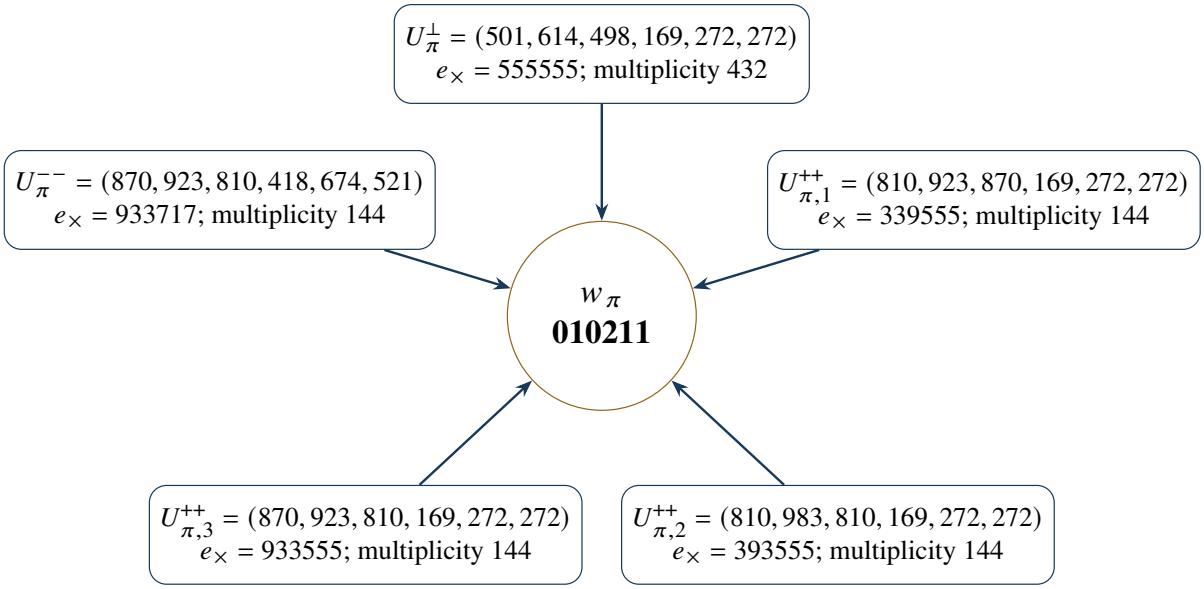
The first identity classifies orientation functions. The second counts provenance multiplicities. They do not count five indistinguishable copies.

The transversal region is

$$501|614|498|169|272|272, \qquad E_{\mathrm{sig}} = 555555, \qquad \mathrm{mult} = 432.$$

The mutated return is

$$870|923|810|418|674|521, \qquad E_{\mathrm{sig}} = 933717, \qquad \mathrm{mult} = 144.$$



Five distinct regions, one shared quotient word.
$432 + 4 \cdot 144 = 1008$ routes; alignment labels: $1_{\perp} + 3_{++} + 1_{--}$.

Figure 9.1: Closure microfiber of π: transver region, 3 regions of positive orientation and mutated return. The 5 provenancis share the same quotient word without losing their genealogy.

Interchanging them would preserve the census $3/1/1$ but falsify the complete regional structure.

9.1.4 Descent of a complete nonadic block to the phase transfer multigraph

For $U_6 = (U_1, \ldots, U_6)$, let

$$A(U_6) = (U_1, U_2, U_3), \qquad B(U_6) = (U_4, U_5, U_6).$$

Each emission defines an edge between $A(U_6)$ and $B(U_6)$. The raw multigraph has 96 vertices, distributed into components of sizes

$$28, 28, 28, 4, 4, 4.$$

The internal phase is quotiented by

$$(a, b, c) \sim (b, c, a) \sim (c, a, b).$$

The representative is the lexicographically minimal rotation. The quotient graph has 32 vertices and two components of sizes 28 and 4.

The anchors are

$$\begin{aligned} U_{\mathrm{APP}} &= 903|850|850|252|109|252, \\ U_e &= 850|963|890|149|252|212, \\ U_{\varphi} &= 830|943|830|109|252|252. \end{aligned}$$

We number the regions

$$U_\pi^{(1)} = 810|923|870|169|272|272,$$
$$U_\pi^{(2)} = 810|983|810|169|272|272,$$
$$U_\pi^{(3)} = 870|923|810|169|272|272,$$
$$U_\pi^{(4)} = 870|923|810|418|674|521,$$
$$U_\pi^{(5)} = 501|614|498|169|272|272.$$

9.1.5 The 5 oriented closure cycles

9.1.5.1 Closure of $U_\pi^{(1)}$

$$252|109|252|903|850|850 \to 903|850|850|252|109|252 \to$$
$$252|109|252|923|870|810 \to 810|923|870|169|272|272 \to$$
$$272|169|272|963|890|850 \to 850|963|890|149|252|212 \to$$
$$212|149|252|943|830|830 \to 830|943|830|109|252|252 \to 252|109|252|903|850|850.$$

9.1.5.2 Closure of $U_\pi^{(2)}$

$$903|850|850|252|109|252 \to 272|169|272|903|850|850 \to$$
$$272|169|272|983|810|810 \to 810|983|810|169|272|272 \to$$
$$272|169|272|963|890|850 \to 850|963|890|149|252|212 \to$$
$$212|149|252|943|830|830 \to 830|943|830|109|252|252 \to 903|850|850|252|109|252.$$

9.1.5.3 Closure of $U_\pi^{(3)}$

$$903|850|850|252|109|252 \to 292|189|232|903|850|850 \to$$
$$292|189|232|923|810|870 \to 870|923|810|169|272|272 \to$$
$$272|169|272|963|890|850 \to 850|963|890|149|252|212 \to$$
$$212|149|252|943|830|830 \to 830|943|830|109|252|252 \to 903|850|850|252|109|252.$$

9.1.5.4 Closure of $U_\pi^{(4)}$

$$903|850|850|252|109|252 \to 272|169|272|903|850|850 \to$$
$$272|169|272|963|890|850 \to 850|963|890|149|252|212 \to$$
$$212|149|252|923|810|870 \to 870|923|810|418|674|521 \to$$
$$943|830|830|521|418|674 \to 830|943|830|109|252|252 \to 903|850|850|252|109|252.$$

9.1.5.5 Closure of $U_\pi^{(5)}$

$$252|109|252|903|850|850 \to 903|850|850|252|109|252 \to$$
$$614|498|501|252|109|252 \to 501|614|498|169|272|272 \to$$
$$272|169|272|963|890|850 \to 850|963|890|149|252|212 \to$$
$$212|149|252|943|830|830 \to 830|943|830|109|252|252 \to 252|109|252|903|850|850.$$

Theorem 9.1 (Minimality of the Five Closures)**.** *For each $i \in \{1, \dots, 5\}$, the minimum length of a closed walk that simultaneously traverses*

$$U_\pi^{(i)}, \quad U_e, \quad U_\varphi, \quad U_{\mathrm{APP}}$$

in the quotient multigraph is eight.

Proof. Given i, let

$$E_* = \{U_\pi^{(i)}, U_e, U_\varphi, U_{\mathrm{APP}}\}.$$

The finite space is considered.

$$\mathcal{S}_i = \{(v, M) : v \in V(G),\ M \subseteq E_*\}.$$

The first component records the vertex and the second the target edges already traversed. If an edge e takes v to v', the transition is

$$(v, M) \longmapsto \big(v', M \cup (E_* \cap \{e\})\big).$$

Breadth-first search enumerates lengths in increasing order. The first return to (v, E_*) occurs at length eight for all five regions. The printed walks attain the bound; the optimality of the search excludes a shorter length. □

The closure function is encoded by

$$\mathfrak{C}_\pi = \big(\mathscr{R}_\pi^{\mathrm{micro}}, w_\pi, \rho_\pi, m_\pi, \ell_{\mathrm{cl}}\big),$$

$$\rho_\pi = (\perp, ++, ++, ++, --), \qquad m_\pi = (432, 144, 144, 144, 144), \qquad \ell_{\mathrm{cl}} = 8.$$

9.1.6 Oriented holonomies of the closure of the 5 regions of π

The incidence shadow is

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Multiplication in $\mathbb{F}_3$ gives

$$\boxed{A_W^2 = 2I_6 = -I_6.} \tag{9.5}$$

Products between distinct rows and columns vanish modulo three, whereas each diagonal product equals two. One application effects a ternary quarter-turn; two applications effect the oriented inversion.

9.1.7 Determination of the slope 1/5 via the balance of the 5 oriented regions

Let

$$\lambda = (\lambda_1, \dots, \lambda_5)^{\mathsf{T}} \in \mathbb{Q}_{\geq 0}^5, \qquad \mathbf{1}^{\mathsf{T}}\lambda = 1.$$

Forgetting the regional label applies the symmetrizer

$$P_5 = \frac{1}{5}\mathbf{1}\mathbf{1}^{\mathsf{T}}.$$

For every normalized vector,

$$P_5\lambda = \frac{1}{5}\mathbf{1}.$$

The invariance $P_5\lambda = \lambda$ therefore has a unique solution:

$$\boxed{\lambda = \frac{1}{5}\mathbf{1}, \qquad t_1 = \frac{1}{5}.} \tag{9.6}$$

The integer five comes from the regional microfiber; it is not recognized in a formula for π.

9.1.8 Determination of the slope 1/5 by the multiplicity of the regional microfiber of π

The law of slope composition is

$$t \star s := \frac{t+s}{1-ts}.$$

The first duplication gives

$$t_2 = t_1 \star t_1 = \frac{2/5}{1-1/25} = \frac{5}{12}.$$

The second gives

$$t_4 = t_2 \star t_2 = \frac{10/12}{1-25/144} = \frac{120}{119}.$$

The return to the unit diagonal requires $q > 0$ such that

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1.$$

Multiplying by $119q$ produces

$$120q - 119 = 119q + 120,$$

and forces

$$\boxed{q = 239.} \tag{9.7}$$

9.1.9 Gaussian identity and 1st half-turn

We define

$$\Theta_\pi := 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}. \tag{9.8}$$

The successive squares are

power	Gaussian integer
$(5+i)^2$	24+10i
$(5+i)^4$	476+480i
$(5+i)^8$	−3824+456960i
$(5+i)^{16}$	−208797818624−3494830080i
$(239-i)^2$	57120−478i
$(239-i)^4$	3262465916−54606720i

and the final multiplication gives

$$\boxed{(5+i)^{16}(239-i)^4 = -681386607803576157184.} \tag{9.9}$$

The imaginary part is zero and the real part is negative. Therefore,

$$\Theta_\pi \equiv \pi \pmod{2\pi}.$$

Para $0 < x < 1$,

$$x - \frac{x^3}{3} < \arctan x < x.$$

From here

$$\Theta_\pi > 16\left(\frac{1}{5} - \frac{1}{3\cdot 5^3}\right) - \frac{4}{239} > 3$$

and

$$\Theta_\pi < \frac{16}{5} < 4.$$

The only negative orientation in that interval is the first positive half-turn:

$$\boxed{\Theta_\pi = \pi.} \tag{9.10}$$

The causal order is

$$\text{five regions} \to \frac{1}{5} \to \frac{5}{12} \to \frac{120}{119} \to 239 \to \Theta_\pi \to \pi.$$

9.1.10 Arbitrary precision rational horocycles

For $0 < x < 1$, let

$$S_n(x) = \sum_{r=0}^{n} (-1)^r \frac{x^{2r+1}}{2r+1}.$$

Leibniz's criterion gives

$$S_{2m+1}(x) < \arctan x < S_{2m}(x).$$

We define

$$a_m^- = S_{2m+1}(1/5), \quad a_m^+ = S_{2m}(1/5),$$
$$b_m^- = S_{2m+1}(1/239), \quad b_m^+ = S_{2m}(1/239),$$

and

$$\ell_{\pi,m} = 16a_m^- - 4b_m^+, \qquad h_{\pi,m} = 16a_m^+ - 4b_m^-. \tag{9.11}$$

Then

$$\ell_{\pi,m} < \pi < h_{\pi,m}$$

and

$$h_{\pi,m} - \ell_{\pi,m} = \frac{16}{(4m+3)5^{4m+3}} + \frac{4}{(4m+3)239^{4m+3}} \longrightarrow 0. \tag{9.12}$$

The lower bounds grow and the upper bounds decrease.

Since $3 < \pi < 4$, the ternary reader acts on

$$r_\pi := \pi - 3 \in (0,1),$$

with bracket

$$[\ell_{\pi,m} - 3, h_{\pi,m} - 3].$$

9.1.11 Production by Ext and subsequent rational certification in the ambient fiber

Suppose that Ext has constructed, through the graded transport of the TPK on the enriched state, a ternary prefix of length $L_K = 6K$, represented by N_K. The same prolongation law first produces the unique surviving extension N_{K+1}, preserving sheet, route, residue, quotient, carry, boundary and memory. The 729 subcylinders of the ambient fiber in which that output is published are

$$I_{K,u} = \left[\frac{729N_K + u}{3^{L_K+6}}, \frac{729N_K + u + 1}{3^{L_K+6}}\right), \qquad 0 \le u < 729.$$

Once N_{K+1} has been produced, the order $m = m(K)$ is increased until the rational bracket is contained in a single subcylinder and thus certifies, a posteriori, the coordinate already published by Ext. Let

$$\ell_{\pi,K} := \ell_{\pi,m(K)} - 3, \qquad h_{\pi,K} := h_{\pi,m(K)} - 3.$$

The compatible integer endpoints are

$$j_{\min} = \max\Big(729N_K, \lfloor \ell_{\pi,K} 3^{L_K+6} \rfloor\Big),$$

$$j_{\max} = \min\Big(729N_K + 728, \lceil h_{\pi,K} 3^{L_K+6} \rceil - 1\Big).$$

The unitary certification condition is

$$\boxed{j_{\min} = j_{\max}, \qquad N_{K+1} = j_{\min}.} \tag{9.13}$$

The equality expresses the exhaustive sweep of the ambient fiber and recognizes the numerical subcylinder containing the previously produced extension. In particular, $N_{K+1} = j_{\min}$ is verified here as a certification identity: it does not define N_{K+1}, select its block or feed the generator. The existence and uniqueness of the extension belong to the transport Ext of the enriched state; the bracket and the fiber of 729 elements subsequently certify its Archimedean location.

9.1.12 Filtration of $\mathbf{F}_3{}^6$ by regional admissibility and selection of compatible subcylinders

Let $\mathcal{H}_m^\pi$ be the set of enriched histories at depth m, and let

$$p_{n,m} : \mathcal{H}_n^\pi \longrightarrow \mathcal{H}_m^\pi$$

the truncation. We define

$$\mathrm{Surv}_m^{(N),\pi} := p_{N,m}(\mathcal{H}_N^\pi),$$
$$\mathrm{Surv}_m^\pi := \bigcap_{N \geq m} \mathrm{Surv}_m^{(N),\pi}.$$

When the relation Ext is nonempty, single-valued and compatible at every level, the sets are singletons and satisfy

$$p_{m+1,m}(\mathrm{Surv}_{m+1}^\pi) = \mathrm{Surv}_m^\pi.$$

In that domain of extendability, there exists, therefore, a unique section.

$$x_\pi \in X_\pi := \varprojlim_m \mathrm{Surv}_m^\pi.$$

The cylinders are nested and their diameters are 3^{-6K}, which tends to zero. Their intersection contains a unique point, identified by the angular character with $r_\pi = \pi - 3$.

For every decimal depth M, one may choose a level $K(M)$ whose cylinder lies within a single decimal cell of diameter 10^{-M}. The deep prefixes truncate exactly to the preceding ones.

9.1.13 Characterization of the coinductive channel of π: selection of w_6^π, 5 regions, oriented inversion, and uniqueness of the limit section

Theorem 9.2 (Characterization of the closure channel)**.** *Starting from the catalog, the action D_3, the oriented gauge, the operators L_0, L_1, A_W, the slope law and the rational filtration:*

1. $w_6^\pi = 010211$ *is selected;*
2. *five regions are constructed with the identities of* (9.4)*;*

3. *each region belongs to a minimum closure of length eight;*
4. $A_W^2 = -I_6$ *fixes the oriented inversion;*
5. *regional symmetrization forces* $1/5$*;*
6. *the composition forces* $120/119$*;*
7. *the unit return forces* 239*;*
8. *the Gaussian identity recognizes the first half-turn;*
9. Ext*, through the graded transport of the TPK, produces the compatible section at arbitrary depth, while the brackets and the fiber of* 729 *elements subsequently certify, at each finite depth, the block already produced.*

Its evaluation is

$$\boxed{\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}.}$$

The complete chain is

$$\mathrm{APP} \to \mathrm{TRIT} \to \mathrm{TPK} \to 104\,976 \to 468 \to 243 \to w_6^{\pi} \to w_{30}^{\pi} \to R_{36} \to \Gamma_9 \to X_{\pi} \to \pi.$$

Comparison Criterion 9.3 (Channel Closure Falsification)**.** The derivation is refuted if:

1. the closure orbit ceases to be unique;
2. one of the five regions disappears or its complete identity changes;
3. one of the five closures has length less than eight;
4. $A_W^2 \neq 2I_6$;
5. the symmetrizer admits another normalized distribution;
6. the compositions do not produce $5/12$, $120/119$, and 239;
7. the Gaussian identity fails;
8. a bracket ceases to contain the angular character;
9. a level does not traverse the 729 elements of the ambient fiber or does not locate exactly one extension;
10. a target expansion intervenes before recognition.

Chapter 10

Coinductive Generation of π, e and φ via Compatible Nonadic Refinement

The complete construction of the closure character, the five regions, and the structural biography of π was established in chapter 9. Starting from that same already constructed character, the present chapter now prolongs the refinement toward e and φ, without restarting the genealogy or repeating its first stage.

10.1 Propagation, memory return and construction of the exponential character

10.1.1 Finite orbital selector: existence and uniqueness of compatible emission

Let $\mathcal{W} = \mathbb{F}_3^6$. For an emission $U = (u_1, \dots, u_6)$, the visible word is

$$\Psi(U) = ((-u_1) \bmod 3, \dots, (-u_6) \bmod 3).$$

On $\mathcal{W}$, the dihedral action is given by

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4),$$
$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3).$$

For $w \in \mathcal{W}$, define

$$n(w) = \#\Psi^{-1}(w), \qquad M(w) = \sum_{U \in \Psi^{-1}(w)} \mathrm{mult}(U),$$

and

$$\nu(w) = \big(\#\{j : w_j = 0\}, \#\{j : w_j = 1\}, \#\{j : w_j = 2\}\big).$$

Theorem 10.1 (Structural selection of the exponential channel)**.** *Among the* 43 *orbits of the visible catalog, there is a unique orbit whose fibers simultaneously satisfy:*

$$n(w) = 1, \qquad M(w) = 144, \qquad \mathfrak{D}(w) = \{0, 3, 6\}, \qquad \nu(w) = (2, 3, 1).$$

The gauge

$$\mathrm{diag}_c = 6, \qquad \mathrm{card} = ES$$

select in it

$$\boxed{w_6^e = 201101.}$$

Proof. The four conditions are evaluated over the 243 words in the catalog, not over a sample. A single orbit satisfies the joint predicate. In that orbit, exactly one emission has both diagonal 6 and orientation ES. The enumeration uses only fields from the APP–TRIT catalog and does not consult an expansion of e. □

10.1.2 Coinductive prolongation of the exponential emission via compatible blocks

The recurrence L_0, L_0, L_1, L_1 produces

$$w_{30}^e = 201101|121221|102011|012222|102011,$$

and the joint boundary contributes

$$u_5^e = 021222.$$

The record of thirty-six trits remains

$$w_{36}^{e} = 201101|121221|102011|012222|102011|021222.$$

10.1.3 Construction of the decimal signature via the ternary–decimal positional square

The initial reading of twelve digits is

$$\mathbf{d}_{e}^{(12)} = 718281828459 = 7\ \underbrace{1828\,1828}_{\text{local square}}\ 459.$$

The central equality is the concatenation

$$1828 \parallel 1828,$$

not an arithmetic product nor the supposed beginning of a period.

Definition 10.2 (Positional square)**.** A word $d_1 \cdots d_{12}$ contains a square of length ℓ at position p when

$$d_p \cdots d_{p+\ell-1} = d_{p+\ell} \cdots d_{p+2\ell-1}.$$

The position and the lateral contexts belong to the signature.

Table 10.1: Comprehensive census of squares in twelve-digit readings.

Length ℓ	1	2	3	4	5	6
Occurrences	240	21	3	2	0	0
Words that contain them	153	19	3	2	0	0

The other word with a square of length four is

$$575157518472 = \underbrace{5751\,5751}_{\text{initial position 1}}\ 8472.$$

Proposition 10.3 (Positional uniqueness)**.** *The reading of channel e is the only one that has a square of length four after a prefix of length one and with contexts* 7 *and* 459*.*

Proof. The census leaves only two words with squares of length four. One starts it at the first position; the other after the prefix 7. The position and contexts separate the two possibilities. □

10.1.4 Phase return with non-trivial memory state update

In the elevation

$$201101|121221|102011|012222|102011$$

the third and the fifth blocks coincide:

$$B_3 = B_5 = 102011.$$

Their prestates modulo 27 are, however,

$$p_3 = (25, 11, 7, 17, 8, 16), \qquad p_5 = (7, 8, 4, 23, 26, 7),$$

and the subsequent quotients:

$$q_3 = (6, 13, 9, 2, 13, 1), \qquad q_5 = (15, 0, 3, 19, 23, 25).$$

$p_3 \neq p_5$ and $q_3 \neq q_5$ hold.

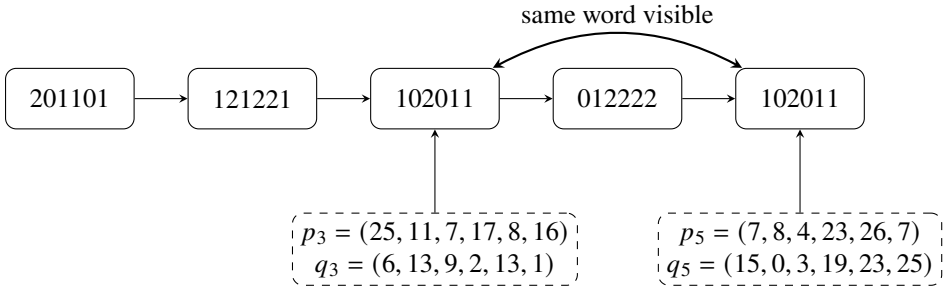


Figure 10.1: Visible return of 102011 with renewal of the prestate and the quotient.

Theorem 10.4 (Visible-phase return without periodicity of the state)**.** *$B_3 = B_5$ is a return of the visible projection, not a periodic orbit of the enriched state.*

Proof. The return of the complete state would require equality of all its coordinates. The prestate and quotient inequalities exclude it. Nor does 1828|1828 begin a decimal period: after the second block, 4 appears, whereas a periodic continuation would require 1. □

10.1.5 Subsequent factorial certification of the coinductive exponential section

Let $\mathcal{A}$ be a unital commutative algebra over $\mathbb{Q}$, and let $P_t \in \mathcal{A}[[t]]$ be the formal propagator. Temporal concatenation imposes

$$P_{s+t} = P_s P_t, \qquad P_0 = 1.$$

We write

$$P_t = \sum_{n=0}^{\infty} a_n t^n.$$

The unit fixes $a_0 = 1$, and normalized elementary transport fixes $a_1 = 1$.

Theorem 10.5 (Factorial recurrence)**.**

$$(n + 1)a_{n+1} = a_n \qquad (n \geq 0),$$

and, therefore,

$$a_n = \frac{1}{n!}, \qquad P_t = \sum_{n=0}^{\infty} \frac{t^n}{n!}.$$

En $t = 1$,

$$\boxed{P_1 = e.}$$

Proof. The coefficient of $s^n t$ in P_{s+t} is

$$\binom{n+1}{n} a_{n+1} = (n + 1)a_{n+1}.$$

In $P_s P_t$ it is $a_n a_1 = a_n$. Equality of series produces the recurrence. Induction from $a_0 = 1$ gives $a_n = 1/n!$. □

Corollary 10.6 (Differential equation)**.**

$$\frac{d}{dt}P_t = P_t, \qquad P_0 = 1.$$

Proof.

$$P_t' = \sum_{n\geq 0}(n+1)a_{n+1}t^n = \sum_{n\geq 0} a_n t^n = P_t.$$

□

10.1.6 Cofinal Forking of Rational Tangles and Convergence to the Published Value

Let

$$S_N = \sum_{n=0}^{N} \frac{1}{n!}.$$

Theorem 10.7 (Explicit factorial bound)**.** *For $N \geq 1$,*

$$\boxed{S_N < e < S_N + \frac{N+2}{(N+1)(N+1)!}.} \tag{10.1}$$

Proof. The rest is

$$R_N = \sum_{k=0}^{\infty} \frac{1}{(N+1+k)!} > 0.$$

Para $k \geq 0$,

$$(N+1+k)! \geq (N+1)!(N+2)^k.$$

Therefore,

$$R_N \leq \frac{1}{(N+1)!}\sum_{k=0}^{\infty}\frac{1}{(N+2)^k} = \frac{N+2}{(N+1)(N+1)!}.$$

The inequality is strict from $k = 2$ onward. □

We define

$$J_N^{(e)} = \left[S_N,\ S_N + \frac{N+2}{(N+1)(N+1)!}\right]$$

and, for the fractional part,

$$\widetilde{J}_N^{(e)} = J_N^{(e)} - 2.$$

The width converges to zero. For each M, there exists N such that

$$\operatorname{diam} J_N^{(e)} < 10^{-M}.$$

If the endpoints belong to the same decimal cell, the first M digits are certified by rational arithmetic.

10.1.7 Coinductive prolongation of the exponential section through ternary subcylinders of the Topological Phase Kernel

At level K, TPK and the relation Ext have already generated a unique compatible subcylinder. The smallest $N = N_e(K)$ for which the factorial bracket is contained in that subcylinder is taken. $N_e(K)$ is an order of analytic certification, not a generating index or a rule for selecting the fiber.

The TPK cylinders are compatible under truncation independently of the semigroup. The semigroup theorem subsequently identifies their Archimedean evaluation with $e - 2$; the integer part recovers e. If the orbit, Ext or truncation fails, generation fails; if the recurrence or the factorial bound fails, this analytic certificate fails, not the APP–TRIT–TPK genealogy.

10.1.8 Orbital uniqueness, factorial recurrence, and cylindrical convergence of electron generation

Comparison Criterion 10.8 (Exponential Channel Falsification)**.** The construction is refuted if:

1. another orbit satisfies the complete finite predicate;
2. the signature 7[1828 1828]459 is not positionally unique;
3. the prestates or quotients of the two blocks 102011 coincide;
4. the semigroup does not force $(n+1)a_{n+1} = a_n$;
5. the bound (10.1) fails;
6. a level does not locate a compatible extension uniquely;
7. a tabulated expansion intervenes before recognition.

10.2 Modal incidence, Perron ray, and construction of the self-similar character

10.2.1 Finite Selection of the Self-Scaling Seed

Let $\mathcal{U}_6 \subset \{0,\dots,999\}^6$ be the catalog of the 468 distinct emissions constructed in section 8.2, and let

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \qquad \Psi(U) = ((-U_j) \bmod 3)_{j=1}^6.$$

For $w \in \Psi(\mathcal{U}_6)$, we recall the invariants

$$n(w) = \#\Psi^{-1}(w), \qquad M(w) = \sum_{U \in \Psi^{-1}(w)} \operatorname{mult}(U),$$

and the occupation vector

$$\nu(w) = \big(\#\{j : w_j = 0\}, \#\{j : w_j = 1\}, \#\{j : w_j = 2\}\big).$$

Theorem 10.9 (Orbital selection of the self-scale)**.** *The dihedral action on $\Psi(\mathcal{U}_6)$ contains a unique orbit that satisfiis the self-scale predicate*

$$n(w) = 1, \qquad M(w) = 144, \qquad \nu(w) = (2,2,2).$$

The internal caliber

$$\operatorname{diag}_c = 6, \qquad \operatorname{card} = ES$$

select in that orbit the word

$$\boxed{w_6^{\varphi} = 121200.}$$

Its only emission is

$$\boxed{U_{\varphi} = (830, 943, 830, 109, 252, 252),}$$

of multiplicity 144*, multiplicative signature* 555933*, additive orientation ES, multiplicative support of cardinality six and residual type A.*

Proof. The direct projection gives

$$U_{\varphi} \bmod 3 = (2,1,2,1,0,0), \qquad -U_{\varphi} \bmod 3 = (1,2,1,2,0,0).$$

The enumeration is carried out over the 243 visible words and their 43 orbits, not over a selection of examples. The joint predicate $n = 1$, $M = 144$, $\nu = (2,2,2)$ leaves one orbit. Within it, the declared

diagonal and orientation leave one emission. At this stage neither the equation $x^2 - x - 1 = 0$, nor a decimal table, nor the conventional name of the constant has been introduced. □

10.2.2 Blockwise extension and construction of the 1st coinductive boundary

Let $b_1 = w_6^{\varphi}$, and let $\varepsilon = (0, 0, 1, 1)$. With the matrices L_0, L_1 constructed in section 9.1, we define

$$b_{k+1} = b_k L_{\varepsilon_k} \pmod 3 \qquad (1 \le k \le 4),$$

and concatenate

$$w_{6(k+1)}^{\varphi} = w_{6k}^{\varphi} \parallel b_{k+1}.$$

Matrix calculus produces, block by block,

$$\begin{aligned} b_1 &= 121200, \\ b_2 &= 112202, \\ b_3 &= 121020, \\ b_4 &= 010210, \\ b_5 &= 010200, \end{aligned}$$

and, therefore,

$$\boxed{w_{30}^{\varphi} = 121200|112202|121020|010210|010200.} \tag{10.2}$$

The dual neutrality of the boundary adds

$$u_5^{\varphi} = 102011,$$

so that

$$\boxed{w_{36}^{\varphi} = 121200|112202|121020|010210|010200|102011.} \tag{10.3}$$

Remark 10.10 (Two distinct uses of a visible word). The word 102011 is also visible in the biography of e. The equality of a projection does not identify enriched states: the channel, prefix, quotient, carry, phase, and memory belong to the type of the object. This separation is mandatory before any real character is constructed.

10.2.3 Construction of the incidence multigraph from the triticale calendar

10.2.3.1 Primitive modal incidence F_{av} and memory dynamics of order 1

The primitive triticale calendar is

$$(+1, -1, 0).$$

We fix two visible modis, Σ and Π, and the rule

$$+1 : m \longmapsto \Sigma, \qquad -1 : m \longmapsto \Pi, \qquad 0 : m \longmapsto m. \tag{10.4}$$

Under the cyclic closure condition, the modal orbit produced by (10.4) is

$$\boxed{(\Sigma, \Pi, \Pi).}$$

The three consecutive pairs are

$$\Sigma \longrightarrow \Pi, \qquad \Pi \longrightarrow \Pi, \qquad \Pi \longrightarrow \Sigma.$$

Definition 10.11 (Chronological incidence matrix)**.** For $i, j \in \{\Sigma, \Pi\}$, let

$$N_3(i,j) = \#\{r \in \mathbb{Z}/3\mathbb{Z} : (m_r, m_{r+1}) = (i,j)\}.$$

The subscript 3 counts instants of the primitive calendar; it does not denote a matrix power.

Theorem 10.12 (Necessary descent of modal incidence)**.** *In the order* (Σ, Π)*,*

$$\boxed{N_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.} \tag{10.5}$$

By exact repetition of the calendar,

$$\boxed{N_9 = 3N_3, \qquad N_{27} = 9N_3, \qquad N_{108} = 36N_3.} \tag{10.6}$$

These three equalities are equalities of edge counts; they do not assert $N_9 = N_3^3$*,* $N_{27} = N_3^9$ *or* $N_{108} = N_3^{36}$*.*

Proof. $\Sigma \to \Sigma$ does not occur; each of the other three pairs occurs once. This determines the four entries of N_3. The calendars of lengths 9, 27, and 108 contain respectively 3, 9, and 36 copies of the primitive block. Each repetition multiplies all the counts by the same integer, which proves (10.6). □

10.2.3.2 Publication areal–volumetrica and razon estable φ^{-2} under the measure of Parry

The modal label and the geometric sheet are distinct objects. The publication chart is

$$\mathfrak{g}_{\mathrm{av}}(\Sigma) = \mathscr{H}_{\mathrm{ar}}, \qquad \mathfrak{g}_{\mathrm{av}}(\Pi) = \mathscr{H}_{\mathrm{vol}}.$$

Transporting (10.5) through $\mathfrak{g}_{\mathrm{av}}$, we obtain the incidence operator

$$\boxed{F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.} \tag{10.7}$$

Thus, the two cross-transitions, the unique volumetric self-retention, and the absence of areal self-retention are derived from the calendar; they are not four independent inputs.

Remark 10.13 (Scope of the descent). Forgetting the elementary phase does not by itself produce a strong Markov quotient of the Topological Phase Kernel, because the visible mode does not determine which of the three internal operators will act at the next step. What does descend without an additional hypothesis is the edge multigraph of a complete block. Its minimal memory-one envelope has adjacency matrix F_{av}.

10.2.4 Construction of the modal incidence multigraph from the triticale calendar

Theorem 10.14 (Perron ray derived)**.** *The positive cone of* F_{av} *contains a unique eigenray. If we normalize it as* $v = (1, x)^{\mathsf{T}}$*, then*

$$x^2 - x - 1 = 0, \qquad x > 0,$$

and, therefore,

$$\boxed{x = \frac{1+\sqrt{5}}{2} = \varphi.} \tag{10.8}$$

Proof. The equation $F_{\mathrm{av}} v = \lambda v$ gives, coordinate by coordinate,

$$x = \lambda, \qquad 1 + x = \lambda x.$$

Eliminating λ yields $x^2 - x - 1 = 0$. Its roots are $(1 \pm \sqrt{5})/2$, and only the positive root lies in the interior of the cone. The matrix F_{av} is primitive, since F_{av}^2 has all its entries positive; the Perron–Frobenius

theorem ensures that this positive ray is simple and unique. The name φ is assigned only after this construction. □

Corollary 10.15 (Scalar Self-Similarity Equation)**.**

$$\varphi = 1 + \frac{1}{\varphi}, \qquad \varphi^2 = \varphi + 1, \qquad \varphi^{-1} = \varphi - 1.$$

Thi identitiis are consequencis of (10.8)*; they select neither the seed nor the matrix.*

10.2.5 Fibonacci Recurrence of the Perron Ray and Diophantine Embedding of Rational Gaps

Let $F_0 = 0$, $F_1 = 1$, and

$$F_{n+1} = F_n + F_{n-1} \qquad (n \geq 1).$$

Theorem 10.16 (Exact Powers of Incidence)**.** *For every $n \geq 1$,*

$$\boxed{F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.} \tag{10.9}$$

Proof. For $n = 1$, the identity is the definition of F_{av}. If it holds for n, right multiplication by F_{av} gives

$$\begin{pmatrix} F_n & F_{n-1} + F_n \\ F_{n+1} & F_n + F_{n+1} \end{pmatrix} = \begin{pmatrix} F_n & F_{n+1} \\ F_{n+1} & F_{n+2} \end{pmatrix},$$

which is the case $n + 1$. □

Theorem 10.17 (Cassini's Identity)**.** *For every $n \geq 1$,*

$$\boxed{F_{n+1}F_{n-1} - F_n^2 = (-1)^n.} \tag{10.10}$$

Proof. The determinant of F_{av} is -1. Taking determinants in (10.9),

$$F_{n-1}F_{n+1} - F_n^2 = \det(F_{\text{av}}^n) = (-1)^n.$$

□

We define $r_n = F_{n+1}/F_n$ for $n \geq 1$. From Cassini it follows that

$$r_n - r_{n-1} = \frac{F_{n+1}F_{n-1} - F_n^2}{F_nF_{n-1}} = \frac{(-1)^n}{F_nF_{n-1}}.$$

Therefore, the approximations alternate around the Perron ray.

Proposition 10.18 (Perron Rational Gaps)**.** *For $m \geq 1$,*

$$\frac{F_{2m+2}}{F_{2m+1}} < \varphi < \frac{F_{2m+1}}{F_{2m}}, \tag{10.11}$$

and the width is

$$\boxed{\frac{F_{2m+1}}{F_{2m}} - \frac{F_{2m+2}}{F_{2m+1}} = \frac{1}{F_{2m}F_{2m+1}}.} \tag{10.12}$$

Proof. The preceding alternation places the even quotients below and the odd ones above φ. The formula for the width is (10.10) with $n = 2m + 1$, after putting both fractions over a common denominator. Since $F_n \to \infty$, the width tends to zero. □

10.2.6 Memory incidence of order 1 and Parry measure of the symbolic envelope

Let X_{av} be the smallest subshift of finite type over $\{\mathscr{H}_{\mathrm{ar}}, \mathscr{H}_{\mathrm{vol}}\}$ that admits exactly the three pairs observed in the modal calendar. Its adjacency matrix is F_{av}.

Theorem 10.19 (Transition and Parry Measure)**.** *Let $r = (1, \varphi)^{\mathsf{T}}$. The Parry stochastic matrix is*

$$\boxed{P_{ij} = \frac{(F_{\mathrm{av}})_{ij} r_j}{\varphi r_i} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}.} \tag{10.13}$$

The measure estacionaria of vertices is proporcional to $(1, \varphi^2)$, hence

$$\boxed{\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}.} \tag{10.14}$$

Proof. The sum of row i of P is

$$\frac{(F_{\mathrm{av}} r)_i}{\varphi r_i} = 1.$$

Since F_{av} is symmetric, the left and right eigenvectors coincide. The stationary weights are proportional to their coordinatewise products, that is, to $(1, \varphi^2)$. The ratio between the two weights is φ^{-2}. □

Remark 10.20 (Three measures that must not be confused). The chronological frequency of the primitive orbit is

$$\nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{ar}}) = \frac{1}{3}, \qquad \nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{vol}}) = \frac{2}{3}.$$

The uniform measure τ_{468} lives on the emission catalog. The Parry measure lives on admissible histories of arbitrary length. The first counts instants, the second counts emissions, and the third weights dynamic cylinders. None is an approximation of the others.

10.2.7 Multiplicative character and cylindrical conservation

For an admissible route $\gamma : i \to j$ of length n, we define

$$\chi_P(\gamma) = \varphi^n \frac{r_i}{r_j}. \tag{10.15}$$

Proposition 10.21 (Concatenation multiplicativity)**.** *If $\gamma_1 : i \to j$ and $\gamma_2 : j \to k$, then*

$$\chi_P(\gamma_2 \circ \gamma_1) = \chi_P(\gamma_2)\chi_P(\gamma_1).$$

On to closed route of length n,

$$\chi_P(\gamma) = \varphi^n.$$

Proof. Length adds and the intermediate coordinate telescopes:

$$\varphi^{n_1+n_2} \frac{r_i}{r_k} = \left(\varphi^{n_1} \frac{r_i}{r_j}\right)\left(\varphi^{n_2} \frac{r_j}{r_k}\right).$$

If $i = k$, the quotient terminal is one. □

We normalize the left vector as

$$\ell = \frac{r}{1 + \varphi^2}, \qquad \ell^{\mathsf{T}} r = 1.$$

The measure of a cylinder is

$$\mu[x_0 \cdots x_n] = \ell_{x_0} r_{x_n} \varphi^{-n}.$$

Therefore,

$$V_n(x) := \frac{\mu[x_0]}{\mu[x_0 \cdots x_n]} = \varphi^n \frac{r_{x_0}}{r_{x_n}} = \chi_P(x_0 \cdots x_n). \tag{10.16}$$

For to typed dimension $D > 0$, the scale

$$a_n^{(D)}(x) = V_n(x)^{1/D}$$

satisfies the exact law

$$\boxed{\big(a_n^{(D)}(x)\big)^D \mu[x_0 \cdots x_n] = \mu[x_0].} \tag{10.17}$$

In three dimensions and over closed returns,

$$\frac{a_9}{a_0} = \varphi^3, \qquad \frac{a_{27}}{a_0} = \varphi^9, \qquad \frac{a_{108}}{a_0} = \varphi^{36}.$$

If $A_k = a_{9k}$, then

$$A_{k+1} - 2A_k + A_{k-1} = A_{k-1}(\varphi^3 - 1)^2 > 0. \tag{10.18}$$

This is convexity in return depth; it is not interpreted as a kinematic quantity without an additional duration chart.

10.2.8 Quadratic memory and contracted branch

The eigenvalues of F_{av} are

$$\varphi, \qquad -\varphi^{-1}.$$

The contracted branch changes orientation at each iteration. To preserve that sign before squaring, we pass to $\mathrm{Sym}^2(\mathbb{R}^2)$. In the basis $(x^2, 2xy, y^2)$,

$$\mathrm{Sym}^2(F_{\mathrm{av}}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}. \tag{10.19}$$

Theorem 10.22 (Spectrum of the second-order memory)**.**

$$\det\big(tI - \mathrm{Sym}^2(F_{\mathrm{av}})\big) = (t+1)(t^2 - 3t + 1),$$

and

$$\boxed{\mathrm{Spec}\big(\mathrm{Sym}^2(F_{\mathrm{av}})\big) = \{\varphi^2, -1, \varphi^{-2}\}.} \tag{10.20}$$

The only positive stable mode is φ^{-2}.

Proof. The eigenvalues of a degree-two symmetric power are the products λ_1^2, $\lambda_1\lambda_2$, and λ_2^2. With $\lambda_1 = \varphi$ and $\lambda_2 = -\varphi^{-1}$, one obtains φ^2, -1, and φ^{-2}. The first exceeds one, the second has modulus one, and the third belongs to $(0, 1)$. □

On a closed route with $V_n = \varphi^n$, the stable mode satisfies

$$\boxed{M_n = \varphi^{-2n} = V_n^{-2}.} \tag{10.21}$$

10.2.9 Normalized hyperbolic realization of the self-scale: $\exp(2\log\varphi\, J_h) = F_{\mathrm{av}}^2$

Let

$$A = F_{\mathrm{av}}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad J_h = \frac{2A - 3I}{\sqrt{5}}.$$

Then

$$J_h^2 = I, \qquad A = \frac{3}{2}I + \frac{\sqrt{5}}{2}J_h.$$

Since

$$\cosh(2\log\varphi) = \frac{3}{2}, \qquad \sinh(2\log\varphi) = \frac{\sqrt{5}}{2},$$

we obtain

$$\boxed{\exp(2\log\varphi\, J_h) = F_{\mathrm{av}}^2.} \tag{10.22}$$

The self-scaling constant is therefore simultaneously the Perron eigenvalue of the incidence and the exponential parameter of its normalized hyperbolic realization.

10.2.10 Characterization of self-similarity via rational forks and coinductive prolongation

The finite construction determines the positive ray of F_{av}. The subsequent analytic characterization is

$$x^2 - x - 1 = 0, \qquad x > 0.$$

The brackets (10.11) have rational endpoints and explicit width (10.12). For each nonadic level K, we choo the least $m = m_\varphi(K)$ who bracket remains contained in a single subcylinder among the 729 elements of the ambient fiber. Minimality converts localization into a function, not into to tacit choice.

The selected cylinders are compatible with the truncations, and their diameters tend to zero. Their intersection contains a unique real number; the theorem 10.14 identifies it with φ. The Fibonacci quotients are therefore rational certificates of localization, not decimal data used to decide the seed.

10.2.11 Internal certificate of uniqueness, compatibility, and convergence of self-scaling

Comparison Criterion 10.23 (Refutation of the self-scale biography)**.** The construction remains refuted if any of the following facts occurs:

1. the finite predicate selects more than one orbit or more than one emission;
2. the recurrence L_0, L_0, L_1, L_1 does not produce (10.2);
3. the calendar does not produce exactly the three pairs used in (10.5);
4. a count multiple N_9, N_{27}, N_{108} is confused with a power of N_3;
5. F_{av} has another positive eigenray;
6. (10.9) or Cassini's identity fails;
7. the rational brackets are not nested or do not contract their width;
8. the chronological frequency, the uniform measure on the catalog or the Parry measure are identified;
9. a decimal table is involved before the Perron ray is constructed;
10. a selected cylinder does not truncate to the previous cylinder.

10.3 Diagonal certification, inverse limit and generation to arbitrary precision

10.3.1 Typed separation of the 4 indices of nonadic prolongation

This unit has the status of subsequent certification. The prefixes and their compatible extensions come from the internal relation Ext, coinductive survival and the APP–TRIT–TPK transition laws. The rational brackets that follow do not choose among the 729 elements of a fiber: they locate in the Archimedean chart the already generated extension and certify its precision at arbitrary depth.

Four parameters of different kinds enter each character $\chi \in \{\pi, e, \varphi\}$:

1. n, order of the rational analytic bracket;
2. K, refinement level of the Topological Phase Kernel;
3. $L_K = 6K$, ternary length of the published prefix;
4. M, the requested number of decimal positions.

Neither $n = K$, $n = L_K$, $K = M$ nor a fixed proportionality between them is postulated. The coupling is constructed through interval inclusions decided by integer arithmetic.

The integer parts are

$$m_\pi = 3, \qquad m_e = 2, \qquad m_\varphi = 1,$$

and the fractional characters are

$$\rho_\pi = \pi - 3, \qquad \rho_e = e - 2, \qquad \rho_\varphi = \varphi - 1.$$

The three families of semiopen forks are written

$$J_{\chi,n} = [\ell_{\chi,n}, h_{\chi,n}).$$

Their endpoints are rational, contain ρ_χ and satisfy

$$\ell_{\chi,n} \nearrow \rho_\chi, \qquad h_{\chi,n} \searrow \rho_\chi, \qquad h_{\chi,n} - \ell_{\chi,n} \longrightarrow 0.$$

For π, the endpoints come from the alternating sums of Machin's identity proved in section 9.1; for e, from the factorial sums and their rational remainder in section 10.1; for φ, from the Fibonacci and Cassini quotients in section 10.2.

10.3.2 Ternary cylinders and the immediate ambient fibre

A ternary prefix of length $L_K = 6K$ is represented by an integer N_K with

$$0 \le N_K < 3^{L_K}.$$

His semi-open cylinder is

$$C(N_K, L_K) = \left[\frac{N_K}{3^{L_K}}, \frac{N_K + 1}{3^{L_K}} \right). \tag{10.23}$$

The next emission adds six trits. The immediate ambient fiber contains $3^6 = 729$ elements, indexed by $u \in \{0, \dots, 728\}$:

$$C_{K,u} = \left[\frac{729N_K + u}{3^{L_K+6}}, \frac{729N_K + u + 1}{3^{L_K+6}} \right). \tag{10.24}$$

The $C_{K,u}$ are disjoint and

$$C(N_K, L_K) = \bigsqcup_{u=0}^{728} C_{K,u}.$$

The number 729 counts coordinates of this ambient fiber; it does not count seeds, distinct emissions, orbits, or surviving states.

10.3.2.1 Diophantine comparison of the rational endpoints of the gap

Let $J = [\ell, h) \subset C(N_K, L_K)$, with $\ell, h \in \mathbb{Q}$. Escribimos $S_K = 3^{L_K+6}$ and define

$$j_{\min}(J, K) = \max(729N_K, \lfloor S_K \ell \rfloor), \tag{10.25}$$

$$j_{\max}(J, K) = \min(729N_K + 728, \lceil S_K h \rceil - 1). \tag{10.26}$$

Lemma 10.24 (Unitary localization criterion)**.** *The bracket J is contained in a unique element of the fiber* (10.24) *if and only if*

$$\boxed{j_{\min}(J, K) = j_{\max}(J, K).} \tag{10.27}$$

In that case, if j is the common value,

$$u = j - 729N_K, \qquad N_{K+1} = j.$$

Proof. The integers j whose elementary intervals of length S_K^{-1} intersect J form the integer interval

$$[\lfloor S_K \ell \rfloor, \lceil S_K h \rceil - 1] \cap [729N_K, 729N_K + 728].$$

This contains exactly one element if and only if its endpoints coincide. The equality identifies the numerator of the subcylinder and, by truncating its last six trits, N_K is recovered. □

10.3.3 Minimal diagonal certification

Generation does not fix the same analytic order for all levels in advance. Once the compatible extension has been produced, the order of the bracket is increased only until it is certified uniquely.

Definition 10.25 (Diagonal certification recurrence)**.** Given a family of compatible cylinders fixed by Ext, $C(N_K, L_K)$, whose publication contains ρ_χ, an initial order $n_\chi(K_0 - 1) \geq 1$ is chosen and we recursively define

$$n_\chi(K) = \min\left\{n \geq n_\chi(K-1) : j_{\min}(J_{\chi,n}, K) = j_{\max}(J_{\chi,n}, K)\right\}, \tag{10.28}$$

and we verify

$$N_{\chi,K+1} = j_{\min}(J_{\chi,n_\chi(K)}, K). \tag{10.29}$$

The set

$$\Delta_\chi = \{(n_\chi(K), K) : K \geq K_0\}$$

is the analytic–nonadic diagonal of the character χ.

Theorem 10.26 (Existence and uniqueness of the certification diagonal)**.** *For $\chi \in \{\pi, e, \varphi\}$, the recurrence* (10.28) *is defined at every finite level. The index $n_\chi(K)$ and the coordinate $N_{\chi,K+1}$ are unique.*

Proof. Suppose the cylindrical extension of level K containing ρ_χ has been generated internally. Since ρ_χ is irrational, it belongs to no ternary boundary $3^{-L}\mathbb{Z}$. It therefore lies in the interior of a unique subcylinder C_{K,u_*}, at a positive distance from both endpoints. The brackets $J_{\chi,n}$ contain ρ_χ and their diameter tends to zero; hence there exists an n for which the bracket is contained in C_{K,u_*}. The set over which the minimum is taken in (10.28) is nonempty and, by the well-ordering of $\mathbb{N}$, has a least element. lemma 10.24 proves that the bracket locates it uniquely and that the located numerator agrees with the one already generated. Induction on K completes the argument. □

Corollary 10.27 (Absence of decimal parameters)**.** *The diagonal is computed using sums, products, powers, Euclidean divisions, floors and ceilings of integers. It receives no decimal digit of π, e or φ as certification data.*

10.3.4 Compatibility with the joint nonadic connection

Let $\widehat{x}_{\chi,K}$ be the enriched state of channel χ at the level K. Its coordinates include the prefix, the cylinder, the phase, the quotient, the carry, the unbounded vertical memory, the terminal term, and the provenance register. The residual phase is

$$g_0(K) = [K]_9 \in \mathbb{Z}/9\mathbb{Z},$$

while the public label is

$$g(K) = 1 + ((K-1) \bmod 9) \in \{1, \dots, 9\}.$$

The vertical memory $q_{\mathrm{mem}}(K) \in \mathbb{Z}_{\geq 0}$ is not reduced modulo 12; only its dodecaphasic projection is

$$\beta(K) = q_{\mathrm{mem}}(K) \bmod 12$$

does so.

For each K, let

$$\Gamma_{9,K} : \widehat{X}_K \longrightarrow \widehat{X}_{K+9}$$

the holonomy of a phase loop. Then

$$g_0(K+9) = g_0(K), \qquad q_{\mathrm{mem}}(K+9) = q_{\mathrm{mem}}(K) + 1, \qquad \beta(K+9) = \beta(K) + 1 \pmod{12}.$$

Theorem 10.28 (Phase return with genealogical conservation)**.** *Phase equality does not reset the state:*

$$g_0(K+9) = g_0(K), \qquad \widehat{x}_{\chi,K+9} \neq \widehat{x}_{\chi,K}.$$

The depths 27*,* 54*, and* 108 *are the typed compositions*

$$\begin{aligned} \Gamma_{27,K} &= \Gamma_{9,K+18} \circ \Gamma_{9,K+9} \circ \Gamma_{9,K}, \\ \Gamma_{54,K} &= \Gamma_{9,K+45} \circ \cdots \circ \Gamma_{9,K}, \\ \Gamma_{108,K} &= \Gamma_{9,K+99} \circ \cdots \circ \Gamma_{9,K}. \end{aligned}$$

In no case are they restarts of the prefix, the carry, the terminal, or the memory.

Proof. The first coordinate is computed in $\mathbb{Z}/9\mathbb{Z}$, whereas q_{mem} belongs to an unbounded monoid. Each application of $\Gamma_{9,K}$ adds one unit of memory, six blocks per intermediate level, and the corresponding cocycle composition. Hence the phase may coincide while the enriched state does not. The depth formulas display the indices of each map and prevent composition as though all were endomorphisms of the same space. □

10.3.4.1 Compression, expression, and terminal term

The operational realization of a loop satisfies

$$U_\Gamma J = JT + \eta, \qquad J^*\eta = 0, \qquad I - T^*T = \eta^*\eta. \tag{10.30}$$

The visible strict contraction is reserved for

$$M_9 = \frac{5}{9} V_9;$$

is not identified with the compression operator T. Telescoping at depth L preserves the terminal term:

$$S_0 = \sum_{r=0}^{L-1} M_9^{*r} D_9^* D_9 M_9^r + M_9^{*L} S_0 M_9^L. \tag{10.31}$$

Omitting the last summand before proving its disappearance would erase boundary information and would turn phase return into a false restart.

10.3.5 Uniqueness of the Minimal Diagonal Section in the Rational Fork System

Let $\mathcal{H}_K^{\chi}$ be the set of admissible enriched histories of channel χ at depth K, and let

$$p_{N,K} : \mathcal{H}_N^{\chi} \longrightarrow \mathcal{H}_K^{\chi} \qquad (N \geq K)$$

the truncation. We define

$$\mathrm{Surv}_K^{(N)}(\chi) = p_{N,K}(\mathcal{H}_N^{\chi})$$

and

$$\mathrm{Surv}_K(\chi) = \bigcap_{N \geq K} \mathrm{Surv}_K^{(N)}(\chi). \tag{10.32}$$

The coinductive object is

$$X_{\chi} = \varprojlim_{K} \big(\mathrm{Surv}_K(\chi), p_{K+1,K}\big). \tag{10.33}$$

Theorem 10.29 (Section compatible under enhanced extendibility)**.** *Suppose further that, over each generated numerical subcylinder, the relation* Ext *is nonempty, single-valued and compatible with the truncations. Then* Ext *and survival produce a family*

$$x_{\chi} = (\widehat{x}_{\chi,K})_{K \geq K_0}$$

which satisfies

$$p_{K+1,K}(\widehat{x}_{\chi,K+1}) = \widehat{x}_{\chi,K}.$$

Therefore, $x_{\chi} \in X_{\chi}$.

Proof. The subcylinder of level $K+1$ has numerator $N_{\chi,K+1} = 729N_{\chi,K} + u$. Truncation removes its last six trits and recovers $N_{\chi,K}$. The other coordinates are obtained through the functorial laws of carry, memory, terminal and phase. The single-valuedness hypothesis for Ext fixes the subcylinder and the enriched extension; the local uniqueness of lemma 10.24 subsequently certifies its Archimedean localization, and the compatibility of Ext provides truncation equality. □

10.3.6 Decimal cells and prescribed precision

For $M \geq 1$ and $q \in \{0, \ldots, 10^M - 1\}$, let

$$D_{M,q} = \left[\frac{q}{10^M}, \frac{q+1}{10^M}\right).$$

A bracket $J = [\ell, h)$ is contained in a unique decimal cell if and only if

$$\boxed{\lfloor 10^M \ell \rfloor = \lceil 10^M h \rceil - 1 = q.} \tag{10.34}$$

Definition 10.30 (Minimum level of decimal certification)**.** For $\chi \in \{\pi, e, \varphi\}$, we defines $K_{\chi}(M)$ as the least level for which there exist an analytic order n and an integer q such that

1. $J_{\chi,n}$ satisfies (10.34);
2. $J_{\chi,n}$ is contained in the cylinder generated by state $\widehat{x}_{\chi,K}$;
3. the transition to the next level satisfies (10.27).

The pair (K, n) is minimized lexicographically.

Theorem 10.31 (Certification of coinductive outputs to arbitrary decimal precision)**.** *For each $\chi \in \{\pi, e, \varphi\}$ and each $M \geq 1$, there exists $K_\chi(M) < \infty$. The generated section determines a unique decimal prefix of M positions, and the bracket recognizes and certifies that prefix. If $M' < M$, the prefix of order M truncates to the one of order M', and the corresponding cells are nested.*

Proof. The three characters are irrational: φ is irrational by its quadratic polynomial with nonsquare discriminant; the irrationality of e and π is taken from their classical characterization theorems, subsequent to the finite construction. Thus none belongs to a rational decimal or ternary boundary. For a fixed M, the distance from the character to the finite set of relevant decimal boundaries is positive. The rational brackets contract their diameter until they lie in a unique decimal cell. theorem 10.26 couples that bracket to a TPK cylinder and a unique extension. The lexicographic minimum exists by well-ordering. Compatibility of the section proves nesting as M increases. □

Corollary 10.32 (Real uniqueness)**.** *The nested cells have diameters $10^{-M} \to 0$. Their intersection is a singleton, identified by the preceding characterizations with π, e or φ, respectively.*

10.3.7 Equivalence of 2 finite realizations of the uniform law

The witnesses of $1,000$ and $10,000$ positions do not define the characters or limit the prolongation. They are executions of the same uniform recurrence at two different depths.

Table 10.2: Exact dimensions of certified runs, by channel.

Counted object	$1,000$ positions	$10,000$ positions
Partitions exhaustive of the fiber ambient	396	3501
Trits transportados	2376	21006
Pairs of equal phase $(K, K+9)$	387	3492
Complete nonadic turns	43	388
Elements examined by partition	729	729
Compatible extensions preserved by partition	1	1

Para $10,000$ posiciones,

$$B = 3501 = 389 \cdot 9, \qquad 6B = 21006, \qquad 3^{6B} > 10^{10000}.$$

The first 396 blocks of the long run coincide with the run of $1,000$ positions. In each of the 3492 pairs of equal phase, the phase coordinate is preserved and the prefix, depth, cylinder, and memory are extended. The certificate also records any repetitions of local projections without conflating them with the return of the complete state.

10.3.8 Separation between generation and verification

Definition 10.33 (Structural generator)**.** The generator receives the finite catalog, L_0, L_1, the initial state, the connection laws and the internal relation Ext. At each level, it:

1. enumerates the 729 elements of the ambient fiber;
2. applies Ext and retains the unique compatible extension;
3. updates phase, quotient, carry, memory, and terminal;
4. writes the ternary block, the selection receipt, and the genealogical record.

Definition 10.34 (Independent post hoc verification)**.** The independent subsequent check receives the closed outputs of the generator and the three families of rational brackets; it computes $j_{\min}$ and $j_{\max}$,

certifies their equality with the extension already produced, reconstructs the partitions, checks the truncation squares, the pairs $(K, K+9)$, the partition sums

$$\mathrm{killed}_{\mathrm{izq}} + 1 + \mathrm{killed}_{\mathrm{der}} = 729,$$

the integrity of the outputs and, finally, the agreement with reference expansions.

Theorem 10.35 (Isolation of the target data)**.** *Reference decimal expansions and metrological data are not involved in selecting a compatible extension. The comparison is performed after the three prefixes have been written.*

Proof. The mathematical input to each generative decision consists of the finite catalog, L_0, L_1, the inherited state and the internal relation Ext. The rational endpoints, powers of 3, floors and ceilings belong to the subsequent check. The generator finishes before that check opens the brackets or runs the decimal controls. The direction of the data flow is

$$\text{structure} \longrightarrow \text{prefixes and ledgers} \longrightarrow \text{independent subsequent check};$$

there is no arrow in the opposite direction. □

10.3.9 Differentiated obstructions to generation and its certification at arbitrary precision

Comparison Criterion 10.36 (Refutation of generation to arbitrary precision)**.** Failures of orbit, extension, truncation or return refute the generation; failures of brackets, order or decimal control refute the corresponding analytic certificate; and a discrepancy between the two refutes the joint identification of the published output. With this separation of types, the complete assertion is refuted if:

1. any of the indices n, K, L_K, M are identified without a linking theorem;
2. a bracket ceases to contain its character or does not contract its diameter;
3. a stage satisfies $j_{\min} \neq j_{\max}$ after the order declared to be minimal;
4. an extension does not truncate to the cylinder of the previous level;
5. the phase return resets the prefix, carry, terminal or memory;
6. the terminal term in (10.31) is omitted without proving that it vanishes;
7. the first 396 blocks of the long execution differ from the short execution;
8. a decimal or metrological control is read before the output is closed;
9. a printed digit differs from the certified mathematical section;
10. the proof depends on the requested precision being $1,000$ or $10,000$, rather than an arbitrary M.

10.4 Markov dynamics as a modal quotient of order-1 memory of the holonomy Γ_9, compatible with the inverse limit

Let $\rho_m : H_m \twoheadrightarrow Y_m$ be the observable quotient at depth m, and let $L_m(y, y')$ be the transition kernel already constructed on Y_m. For each finite fibre $\rho_m^{-1}(y)$, let $\nu_{m,y}$ be the conditional uniform measure. The nonadic regularity of the extensions allows these measures to be chosen so that their pushforward under truncation is $\nu_{m-1,\bar{y}}$.

Define, for $h \in H_m$,

$$\mathcal{K}_m(h, A) := \sum_{y' \in Y_m} L_m(\rho_m(h), y')\, \nu_{m,y'}\big(A \cap \rho_m^{-1}(y')\big). \tag{10.35}$$

Theorem 10.37 (Strongly additive elevation and Markov limit)**.** *The kernels $\mathcal{K}_m$ satisfy*

$$\sum_{\rho_m(g)=y'} \mathcal{K}_m(h, \{g\}) = L_m(\rho_m(h), y'), \tag{10.36}$$

independently of the representative h of the fibre. Moreover, they commute with the truncations. Therefore, there exists a unique Markov kernel $\mathcal{K}_\infty$ on the inverse limit whose level-m marginal is $\mathcal{K}_m$. Its projection by ρ_∞ is the observable kernel L_∞.

Proof. The sum of the left-hand side of (10.36) is

$$L_m(\rho_m(h), y') \sum_{g \in \rho_m^{-1}(y')} \nu_{m,y'}(\{g\}) = L_m(\rho_m(h), y').$$

Compatibility of L_m, ρ_m, and the conditional measures proves the truncation square. The cylindrical distributions obtained by iterating $\mathcal{K}_m$ are consistent; the extension theorem for projective systems of finite spaces produces $\mathcal{K}_\infty$, and uniqueness follows because the cylinders generate the sigma-algebra of the limit. □

The construction preserves microscopic stillness and complete labels in the conditioned measure. The visible average does not replace the enriched state and it is not identified with the Topological Phase Kernel.

10.5 Coinductive generation of π, φ, e to arbitrary depth

Emission of six blocks is a finite stage of the construction, not a complete decimal expansion. Its function is to produce regions with multiplicity, orientation and transport rules. Prolongation must preserve these data and determine a compatible family of prefixes of arbitrary length. The word *generation* denotes this operation; *genealogy* denotes reconstruction of its dependencies. Both notions enter into the process but serve distinct functions.

10.5.1 Orbits, regions, and characters of closure, propagation, and self-scaling

On a word $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \qquad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; these permutations therefore realize the dihedral group D_3. The action preserves the catalogue and multiplicities and commutes with Ψ. The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fiber predicate: its six orientations have five emissions per word, total multiplicity 1008 and distribution $432 + 4 \cdot 144$. There is only one orbit with these properties in the catalogue. To make the other two selections precise, write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \mathrm{mult}(U)$ and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature preserves this class and the orientation ES or NO. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the orbits of size six, the joint predicate

$$f(w) = 1, \qquad m(w) = 144 \quad (w \in \mathcal{O}), \qquad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, equal to that of closure, selects only propagation; the census $\nu = (2, 2, 2)$ selects only self-scale. The diagonal support cannot be omitted:

without it there are four balanced single-point orbits of multiplicity 144, not one. Finally, existence of an emission with $d = 6$ and orientation ES selects one representative of each orbit and leads to

$$w^{\mathrm{cl}} = 010211, \qquad w^{\mathrm{pr}} = 201101, \qquad w^{\mathrm{au}} = 121200. \tag{10.37}$$

These symbols first denote catalogue regions. Scalar identification will be proved through their readers; they are not chosen because they match digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Firma multiplicativa	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transverse	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and constant signature 555555 characterize the transverse region. They do not identify the central position $(5, 5)$ of APP. Confusing a six-window signature with a support position would remove the distinction between numerical region and electronic structure.

The microfiber also has a precise bipartite morphology. Each emission is a product of an additive cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transverse region has a product 18×24; the three positive regions together, another 18×24; and the return, a product 18×8. Decomposition by emissions has five parts, while decomposition by connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing shape to a cardinality.

10.5.2 Oriented metadata and decomposition of the regional microfiber

The additive signature preserves the diagonal class of origin and the orientation of its shift. For an initial cursor $(i, j; d)$, with indices between zero and eight, define

$$c = [i + j]_9, \qquad h(d) = \begin{cases} ES, & d \in \{E, S\}, \\ NO, & d \in \{N, O\}. \end{cases}$$

In consecutive additive activations, the class c increases by one for ES and decreases by one for NO. After the ninth activation the direction is reversed. The sample sequence is therefore determined by (c, h), with reversal after the ninth sample. The sample in class c' is $\rho_9(c' + 2)$; each decimal signature sums three consecutive samples and reduces the total modulo ten.

Proposition 10.38 (Recovery of additive metadata)**.** *The reader $(c, h) \mapsto f_+(c, h)$ is bijective onto the eighteen additive signatures of the emitter. Each preimage in cursor space consists of eighteen initial conditions. The two coordinates are recovered through the following table.*

c	$f_+(c, ES)$	$f_+(c, NO)$
0	988212	212988
1	212645	645212
2	546988	988546
3	889221	221889
4	122564	564122
5	465898	898465
6	898122	122898
7	221456	456221
8	654889	889654

Proof. For each entry, the class recurrence described before the statement is applied over eighteen activations; their groups of three give the six printed digits. For example, from $(c, h) = (4, NO)$ the first nine classes are $4, 3, 2, 1, 0, 8, 7, 6, 5$ and the samples are $6, 5, 4, 3, 2, 1, 9, 8, 7$. The three totals are $15, 6, 24$, which publish $5, 6, 4$. Reversal leaves the second half $1, 2, 2$, yielding 564122. The eighteen strings in the table are distinct, so the reader is injective and enumerates its entire image. For each c there are nine pairs (i, j), one for each i, and for each h there are two directions. This gives $9 \cdot 2 = 18$ conditions per signature and $18 \cdot 18 = 324$ conditions in total. □

Let $A_s = f_+^{-1}(s)$ and $B_e = f_\times^{-1}(e)$ be the cursor classes of the two sheets. The decimal law recovers both signatures of each emission U: if $U_j = 100s_j + 10t_j + u_j$, then $s_j = \lfloor U_j/100 \rfloor$ and $e_j = [t_j - s_j]_{10}$. Consequently, the preimage of U is exactly $A_s \times B_e$. The initial conditions remain in that product; their cardinalities are an additional reading of the same preimage.

Definition 10.39 (Bipartite incidence of the microfiber)**.** For the oriented word $w =$ 010211, form the graph whose vertices in one class are the additive signatures appearing over w and whose vertices in the other class are the multiplicative signatures. An edge (s, e) represents the corresponding emission and preserves as its labeled preimage the product $A_s \times B_e$.

Recovery of the signatures of the five emissions gives the complete table of edges:

s	e	$\lvert A_s \rvert$	$\lvert B_e \rvert$	stratum
564122	555555	18	24	transverse
898122	339555	18	8	positive
898122	393555	18	8	positive
898122	933555	18	8	positive
898465	933717	18	8	return

Theorem 10.40 (Exact decomposition into three components)**.** *The bipartite incidence has three connected components, isomorphic to $K_{1,1}$, $K_{1,3}$ and $K_{1,1}$. Their joint preimage is the disjoint union*

$$A_{564122} \times B_{555555},$$
$$A_{898122} \times (B_{339555} \sqcup B_{393555} \sqcup B_{933555}),$$
$$A_{898465} \times B_{933717}.$$

The respective cardinalities are $432, 432, 144$ *and sum to* 1008*. Decomposition by emissions consists of five products and decomposition by components consists of three sets.*

Proof. The three printed additive signatures are distinct. The five multiplicative signatures are also distinct. Only the three positive edges share a vertex, the one associated with 898122; they form a

connected star with three edges. The other two edges share no vertices with it or with each other, and each forms a component. Preimages of different signatures under a function are disjoint. Factorization of the emitter and distributivity of Cartesian product over that disjoint union prove the three expressions. Their cardinalities are $18 \cdot 24$, $18(8+8+8)$ and $18 \cdot 8$. The five edges remain individually distinguished within those components, with their multiplicity and ordered emission. □

The metadata recovery map gives $(c, h) = (4, NO)$ for the transverse region, $(6, ES)$ for each of the three positive regions and $(5, NO)$ for the return. Thus, orientation, diagonal class, incidence and multiplicity are read from the same signatures produced by the cursors. The distinction between five emissions and three components preserves the complete census structure, including the coincidence of cardinality 432 between two distinct components.

10.5.3 Finite transport and the prolongation boundary

The endomorphisms L_0, L_1 of ternary word space act before Archimedean realization. Their determination begins with the arithmetic state's transitions, not with prescribing two matrices. If U_t is the composite step of selection, transport and update, the observed block satisfies

$$b_j^{\chi} = \Pi_6(x_j^{\chi}), \qquad x_{j+1}^{\chi} = U_{9j+9} \cdots U_{9j+1}(x_j^{\chi}), \qquad \chi \in \{\mathrm{cl}, \mathrm{pr}, \mathrm{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The pairs of source and target matrices obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each six-digit string represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ uniquely determine $L_a = X_a^{-1} Y_a$. In that row-vector chart we obtain

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod 3.$$

The calendar (L_0, L_0, L_1, L_1) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the extension relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace production of those transitions by U_t. The provenance of their register and examination of the sixteen binary calendars are documented in the technical supplement. The indicated calendar is the only one of those sixteen that jointly reproduces the three sequences of five blocks.

10.5.4 Five closure regions and angular compensation

The closure region contains one transverse component and four companions. The symmetric reader acts on the space of its five positions through

$$P_5 = \frac{1}{5}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \qquad \mathbf{1} = (1,1,1,1,1)^{\mathsf{T}}.$$

Invariance $P_5\lambda = \lambda$ and conservation $\mathbf{1}^{\mathsf{T}}\lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes unity among reader positions; it does not replace seed multiplicities 432, 144, 144, 144, 144 with uniform frequencies. The reader's elliptic chart realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j, so the primitive slope is $1/5$. The rule passing from a normalized coordinate to a slope is thus made explicit, instead of identifying the two notions without a map. The reader composes the four companion positions by the law

$$a \oplus b = \frac{a+b}{1-ab}. \tag{10.38}$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in regime $J^2 = -I$ and dividing by its scalar component. Composition of slopes is therefore a realization of TRIT's quadratic structure, not an ordinary sum of four fractions.

Proposition 10.41 (Closure compensator)**.** *Composition of four slopes* $1/5$ *is* $120/119$*. The condition of return to slope one determines a unique positive compensator of the form* $1/q$*, with* $q > 1$*, and gives* $q = 239$*.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second, $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, so $q = 239$. □

The resulting closure character has realization

$$\Pi_{\mathrm{cl}} = 4\left(4\arctan\frac{1}{5} - \arctan\frac{1}{239}\right). \tag{10.39}$$

Machin's angular identity recognizes $\Pi_{\mathrm{cl}} = \pi$. The construction direction is precise: the slope and number of compositions are declared by the pentafiber reader; the compensation equation forces 239. Once this character is produced, its evaluation by alternating series does not select the seed orbit anew.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^{n} \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes of (10.39), they provide rational endpoints with explicit error. The numerical value is thus calculated as a consequence of the exact character, without introducing a target prefix.

10.5.5 Propagation reader and normalization of unity

The propagation region is realized through a formal family $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and oriented elementary transport fixes the scalar generator $P_0' = +1$. The condition fixes its sign, not just its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

normalization determines $a_0 = a_1 = 1$. Comparing the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P_t' = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \qquad a_n = \frac{1}{n!}. \tag{10.40}$$

The character at one parameter unit is $\Pi_{\mathrm{pr}} = P_1$, subsequently recognized as e.

Normalization of the generator matters. Without that datum the semigroup law would admit $P_t = e^{ct}$ for different c; hence the condition fixing the parameter unit in the reader is not omitted. This condition precedes decimal comparison and belongs to exposition of the operational rule, not to selection of an experimental value.

Proposition 10.42 (Rational propagation bounds)**.** *For $n \geq 1$, let $E_n = \sum_{j=0}^{n} 1/j!$. Then*

$$E_n < \Pi_{\mathrm{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Each subsequent ratio is at most $1/(n+2)$. The geometric sum majorizing the tail is

$$\frac{1}{(n+1)!}\frac{1}{1-1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

The strict inequality follows from the decrease of successive ratios. □

10.5.6 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. In modes Σ, Π, the rule is $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$, $0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π). Its edges are $\Sigma \to \Pi$, $\Pi \to \Pi$ and $\Pi \to \Sigma$. Its incidence, in that mode order, is therefore

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \tag{10.41}$$

The four entries are obtained from the calendar, not from an equation that previously received the value of φ. Repetition of the calendar gives counts $3F_{\mathrm{av}}, 9F_{\mathrm{av}}, 36F_{\mathrm{av}}$ at lengths nine, twenty-seven and one hundred and eight. These counts are not powers of the matrix.

Theorem 10.43 (Positive self-scaling character)**.** *The incidence* (10.41) *has a unique strictly positive eigenray. Normalizing it as $(1, x)^{\mathsf{T}}$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and is $\varphi = (1+\sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ produces the polynomial. Only one of its two roots is positive. Since F_{av}^2 has all entries positive, the positive ray is unique and simple. The radical expression recognizes the resulting coordinate; it does not participate in production of F_{av}. □

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{\mathrm{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The consecutive ratios F_{n+1}/F_n alternate around φ. The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily small.

The measure on all compatible histories does not coincide with the mode frequency of the three-event calendar. The chronological frequency is $(1/3, 2/3)$. The measure of maximal entropy of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2}. The latter enters the areal–volumetric reading; it is not obtained merely by replacing one counting frequency with another.

10.5.6.1 Trajectory space and invariant measure between sheets

The toroidal topology of APP fixes the support of cardinal translations and their winding classes. The correspondence between sheets comes from another operation on that same support: temporal selection of the additive or multiplicative evaluation. Its geometric typing is

$$\eta_{\mathrm{av}}(\Sigma) = \mathscr{H}_{\mathrm{ar}}, \qquad \eta_{\mathrm{av}}(\Pi) = \mathscr{H}_{\mathrm{vol}}.$$

The first sheet corresponds to the additive boundary reading; the second, to the multiplicative interior reading. This map transports the mode labels and preserves the incidence (10.41). The square of the self-scale ratio is determined through the trajectories of that incidence.

Definition 10.44 (Memory-one symbolic envelope)**.** Let

$$X_{\mathrm{av}} = \{(s_k)_{k\in\mathbb{Z}} \in \{\mathscr{H}_{\mathrm{ar}}, \mathscr{H}_{\mathrm{vol}}\}^{\mathbb{Z}} : (F_{\mathrm{av}})_{s_k, s_{k+1}} = 1 \text{ for every } k\}.$$

The index shift acts on this space. It is the smallest memory-one symbolic system containing the modal sequences and preserving their consecutive pairs: it admits both cross transitions and volumetric self-retention. Phase, cardinal orientation and trajectory registers remain in the TPK state from which this envelope is obtained.

Proposition 10.45 (Weighting of trajectories between sheets)**.** *In areal–volumetric order, the invariant measure of maximal entropy of X_{av} has transition matrix and stationary distribution*

$$P_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \qquad (\mu_{\mathrm{ar}}, \mu_{\mathrm{vol}}) = \frac{(1, \varphi^2)}{1+\varphi^2}. \tag{10.42}$$

In particular, $\mu_{\mathrm{ar}}/\mu_{\mathrm{vol}} = \varphi^{-2}$.

Proof. The already constructed incidence has positive eigenvector $r = (1, \varphi)^{\mathsf{T}}$ and eigenvalue φ. The normalization

$$(P_{\mathrm{av}})_{ij} = \frac{(F_{\mathrm{av}})_{ij} r_j}{\varphi r_i}$$

gives the indicated matrix. Its rows sum to one by $\varphi^{-2} + \varphi^{-1} = 1$. Symmetry of F_{av} implies that weights proportional to r_i^2 satisfy detailed balance; therefore, $\mu P_{\mathrm{av}} = \mu$. Irreducibility and volumetric self-retention determine a unique stationary distribution.

To verify maximality, every allowed transition satisfies $\log(P_{\mathrm{av}})_{ij} = \log r_j - \log r_i - \log \varphi$. Under any invariant measure, the endpoint terms cancel on average. If ν_i are its vertex weights and $q_i(j) = \nu(s_1 = j \mid s_0 = i)$, for $\nu_i > 0$, then

$$h_\nu \le H_\nu(s_1 \mid s_0) = \log \varphi - \sum_{i:\nu_i>0} \nu_i D(q_i \| P_i) \le \log \varphi,$$

where $D(q\|p) = \sum_j q_j \log(q_j/p_j)$ is the relative entropy on allowed pairs and $0 \log 0 = 0$ is adopted. The Markov measure of (10.42) attains the bound. Equality in the first inequality requires the Markov property; equality in the second requires $q_i = P_i$. Stationarity and irreducibility then fix μ, establishing uniqueness. This is the Parry measure of the defined envelope. □

The frequency $(1/3, 2/3)$ counts the instants of the primitive calendar; μ weights the trajectories of the envelope. Its quadratic ratio expresses the product of incoming and outgoing incidence amplitudes. TPK memory and phase restrictions retain their own domain; the envelope's entropy corresponds to the symbolic system specified in this definition.

10.5.6.2 Closure operator and limit of marked returns

The closure coordinate π, generated by the regional reader, enters after construction of modal incidence. Let $e_{\mathrm{ar}} = (1, 0)^{\mathsf{T}}$, $e_{\mathrm{vol}} = (0, 1)^{\mathsf{T}}$ and $P_i = e_i e_i^{\mathsf{T}}$ be the sheet projectors. The reader's conditions are preservation of both sheets, unit volumetric normalization and areal mark equal to the closure coordinate. These conditions determine the positive operator

$$D_\pi = \pi P_{\mathrm{ar}} + P_{\mathrm{vol}} = \begin{pmatrix} \pi & 0 \\ 0 & 1 \end{pmatrix}. \tag{10.43}$$

Diagonality expresses preservation of sheets and the two entries express their normalizations; uniqueness of the operator for the fixed reader is thus explicit.

Proposition 10.46 (Ratio of boundary returns)**.** *For $n \geq 1$, the marked evaluation*

$$\Theta_n = \frac{\langle e_{\mathrm{ar}}, D_\pi F_{\mathrm{av}}^n e_{\mathrm{ar}} \rangle}{\langle e_{\mathrm{vol}}, F_{\mathrm{av}}^n e_{\mathrm{vol}} \rangle} = \pi \frac{F_{n-1}}{F_{n+1}} \tag{10.44}$$

satisfies

$$\lim_{n \to \infty} \Theta_n = \pi \frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \frac{\pi}{\varphi^2}. \tag{10.45}$$

Proof. The diagonal entries of F_{av}^n count paths of length n returning to their initial sheet. The already proved power formula provides F_{n-1} and F_{n+1}. The operator D_π multiplies the first count by π. Finally, F_{n-1}/F_{n+1} is the product of two consecutive ratios converging to φ^{-1}; its limit is φ^{-2}. The identity with the weight ratio follows from (10.42). □

The closure operator appears only once, in the terminal evaluation of the return. The transfer F_{av} remains determined by the calendar. This composes information from two readers of the same state: modal incidence determines the quadratic weighting and the closure region contributes its coordinate to the boundary.

The composition also preserves refinement. For positive cylinders $I = [a, b]$ and $J = [c, d]$, the image under the reader $\mathcal{R}(x, y) = x/y^2$ is the interval

$$Q(I, J) = \left[\frac{a}{d^2}, \frac{b}{c^2} \right].$$

Monotonicity in each argument proves that the nested cylinders of closure and self-scale produce nested intervals of the ratio; positivity of their limits and contraction of their diameters determine π/φ^2. The scalar expression thus retains a chain of operations compatible with depth, subsequently applied in its two orientations.

After the characters are generated, the two orientations of the reader

$$R(x, y) = \frac{x}{y^2}$$

provide

$$\rho_A = \frac{\pi}{\varphi^2}, \qquad \rho_E = \frac{\varphi}{\pi^2}, \qquad \varphi^3 \rho_A^2 \rho_E = 1. \tag{10.46}$$

The second expression enters the exponent of the electronic construction in Section 16.2.1.1. The first is also obtained as the limit of the marked reading $\pi F_{n-1}/F_{n+1}$. Exchanging arguments relates the two, but does not turn a reparametrization of the electronic equation into a second independent derivation of all its coefficients.

10.5.7 Exponential realizations of the generated coordinates

The already determined regional coordinates allow the specific realizations of TRIT's quadratic family to be recognized. The order is substantive: construction of the regions precedes evaluation of an exponential at parameters π or $2\log\varphi$. These expressions identify the function of the coordinates in operators constructed from APP and TPK; they do not enter into selection of the prefixes producing them.

10.5.7.1 Ternary cycle, nilpotent transport and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either, formed by integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$. In residue evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. The quotients of the integer lift are preserved in the corresponding arithmetic register.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (10.41) provides

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad A = F_{\mathrm{av}}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the nilpotent-transport plane and the modal-incidence plane. Sharing matrix dimension does not identify these spaces or their coordinates.

Proposition 10.47 (Concrete generators of the three types)**.** *The operators*

$$J_{\mathrm{e}} = \frac{2C + I}{\sqrt{3}}, \qquad J_{\mathrm{p}} = N, \qquad J_{\mathrm{h}} = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_{\mathrm{e}}^2 = -I, \qquad J_{\mathrm{p}}^2 = 0, \qquad J_{\mathrm{h}}^2 = I.$$

Proof. The relation in C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. □

Proposition 10.48 (Three evaluations of the exponential identity)**.** *In the preceding realizations,*

$$\boxed{\exp(\pi J_{\mathrm{e}}) + I = 0, \qquad \exp(J_{\mathrm{p}}) = I + J_{\mathrm{p}}, \qquad \exp(2\log\varphi\, J_{\mathrm{h}}) = F_{\mathrm{av}}^2.} \tag{10.47}$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi\, I + \sin \pi\, J_{\mathrm{e}} = -I$. Nilpotency removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \qquad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

Therefore, the exponential equals $\frac{3}{2} I + \frac{\sqrt{5}}{2} J_{\mathrm{h}} = A$. □

The first equality represents the elliptic half-turn; the full turn satisfies $\exp(2\pi J_{\mathrm{e}}) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2}; one direction expands and the other contracts while preserving the determinant. The three evaluations use distinct parameters of a common exponential family.

In that family, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI)\exp(tI) = \exp((s+t)I)$. Its role is not restricted to the parabolic type. The coordinate e, previously obtained from its region, is recognized here as the unit evaluation of the exponential; this identity does not replace its coinductive genealogy.

10.5.7.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_{\mathrm{e}} \oplus J_{\mathrm{p}} \oplus J_{\mathrm{h}}, \qquad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \qquad Q^3 = Q.$$

Use of Q keeps this operator square separate from the dodecaphasic register K.

Proposition 10.49 (Polynomial regime projectors)**.** *The maps*

$$p_{\mathrm{e}} = \frac{Q^2 - Q}{2}, \qquad p_{\mathrm{p}} = I - Q^2, \qquad p_{\mathrm{h}} = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I and recover the three summands of the representation.

Proof. The three polynomials take values $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross products, are multiples of $X^3 - X$. Substituting Q, which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. □

We thus obtain

$$\begin{aligned} \exp(t\mathbb{J}) = {} & p_{\mathrm{e}}(\cos t\, I + \sin t\, \mathbb{J}) + p_{\mathrm{p}}(I + t\mathbb{J}) \\ & + p_{\mathrm{h}}(\cosh t\, I + \sinh t\, \mathbb{J}). \end{aligned} \tag{10.48}$$

Each summand preserves its quadratic relation and conjugation reverses its evolution. The total representation combines the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation and memory. Exponential articulation allows their types to be recognized without replacing that dynamics with identification of the fibers.

10.5.8 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^{6} b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A sequence of blocks defines integer prefixes by $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting at $N_0 = m$, where m is the integer part determined by the reader. The preceding brackets place the closure, propagation and self-scale characters in $(3, 4)$, $(2, 3)$ and $(1, 2)$, respectively; their initial values are therefore $m = 3, 2, 1$. Subsequent blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \qquad |I_n| = 729^{-n}. \tag{10.49}$$

The half-open convention is used to select a representative when a value falls on a boundary. Passage to the limit uses the closures of these intervals and identifies the two equivalent boundary expansions. This precision prevents assuming that every intersection of nested half-open intervals is automatically nonempty.

Theorem 10.50 (Determination of the scalar realization)**.** *A compatible block history determines a unique limit of the left endpoints of* (10.49)*. If an exact character has rational brackets of arbitrarily small width and is separated from the boundary of each decimal cylinder examined, each decimal prefix of finite length is determined after a finite calculation. The prefixes thus obtained are mutually compatible.*

Proof. Extension by a block preserves the preceding prefix by Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints form a Cauchy sequence and have a unique limit. For a fixed decimal length, a value separated from the partition boundaries has positive distance from them. A sufficiently narrow bracket lies within a unique cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; that decision does not follow from a numerical width alone. □

Corollary 10.51 (Finite determination and compatibility of regional prefixes)**.** *For each of the realizations π, φ, e and every finite decimal length, the constructed brackets determine the corresponding prefix through a finite calculation. Prefixes obtained at different depths are compatible under truncation.*

Proof. The closure, propagation and self-scale characters have the preceding brackets. After their recognition, irrationality of π, e, φ guarantees their separation from the rational boundaries of each finite decimal partition. The preceding theorem is applied to each character. □

Arbitrary depth expresses the quantifier of the corollary: for every finite horizon there is a calculation determining its prefix. The mathematical object is the complete compatible family; each run obtains a finite projection of it.

10.5.9 Mirror involution and synchronization of the regions

The nonadic connection jointly transports the three characters. In the ninth-phase chart, the self-scale axis and ternary sum of closure and propagation remain fixed under mirror exchange. Helical inversion changes the transport direction; the two-channel mirror changes their relative orientation. They are different involutions, although both act on the same trajectory system.

Census coincidences illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality between π, e and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. In phases eight and nine the relative component vanishes, but the diagonal component changes from $(59, 59, 59)$ to $(43, 43, 43)$. Relative synchronization coexists with a shift of -16 per channel. The closure leading to α must preserve both pieces of information: relative coincidence and evolution of memory.

10.5.10 Sturmian helical suspension

The relation between bases nine and ten of the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \tag{10.50}$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic reading of refinement has an exact consequence.

Lemma 10.52 (Irrationality of the slope)**.** *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. Factorization of the left-hand side contains the prime three and that of the right-hand side only the primes two and five. The equality is impossible. □

The lower mechanical word, with intercept θ, is

$$z_n(\theta) = \lfloor (n + 1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod 1$ through the interval $[1 - \delta, 1)$. The event sequence has density δ, and the sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, event gaps are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first, not a second free parameter.

10.5.10.1 Capacity index, vacancies, and induction of the tritic regime

The vacancy sequence is linked to refinement through an increment identity. To show it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \qquad C_t = \lfloor (t + 1)\rho \rfloor \quad (t \geq 0).$$

The index C_t expresses the decimal capacity of refinement in the normalization comparing a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphasic register to be constructed later.

Proposition 10.53 (Complementarity of capacity and vacancy)**.** *For $t \geq 1$,*

$$C_t - C_{t-1} = 1 - z_t(0). \tag{10.51}$$

Consequently, if $0 \leq a < b$,

$$C_b - C_a + \sum_{t=a+1}^{b} z_t(0) = b - a.$$

Proof. Irrationality of δ gives

$$C_t = t + 1 - \lceil (t+1)\delta \rceil = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting two consecutive terms produces (10.51). The sum over the interval telescopes. □

Each transition increases the block counter by one unit. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity remains unchanged while a transition is incorporated into the history. Vacancy is here an event of capacity retention, with a determined position and antecedent. The preceding balance exactly preserves length between acquired capacity and recorded vacancies. The observed phase $g_t^{\mathrm{cap}} = [C_t]_9$ satisfies

$$g_t^{\mathrm{cap}} - g_{t-1}^{\mathrm{cap}} \equiv 1 - z_t \pmod 9.$$

This capacity phase and the chronological phase $[t]_9$ are two different observations. Local capacity retention and a complete holonomy turn have distinct mechanisms, although both allow phase observation to be preserved while the state changes.

The relation to TRIT can be followed in the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \qquad v_j = C_{p_j}, \qquad h_j = p_{j+1} - p_j.$$

The inequality $p_j\delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j. Since $0 < \delta < 1$, it also gives $\lfloor (p_j + 1)\delta \rfloor = j$, and therefore

$$v_j = p_j - j, \qquad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$ and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \tag{10.52}$$

An interval of length 22 preserves the class; one of length 21 shifts it one unit in the negative direction. The canonical selector is applied to the positive phase $r_j = 1 + [v_j]_9$:

$$\tau_j = M_{\mathrm{ph}}(r_j).$$

A tritic regime is thus determined by the capacity index retained at each event, in addition to binary encoding of the vacancy.

The short-interval positions form the induced mechanical encoding of slope $22 - 1/\delta$. Their gaps 6 and 7 determine the lengths of the plateaus of class $[v_j]_3$, once the origin and endpoint convention are fixed. The first classes are 2 for 6 events, 1 for 7 and 0 for 7; the first segment of 6 is the initial restriction of a plateau. Their signatures run through $0, -1, +1$. Equations (10.51) and (10.52) make explicit the link between refinement, vacancies and regime selection. The quadratic TRIT reading then realizes these types through their respective generators; the temporal event and its geometric representation thus retain an identifiable dependence.

Inverting the monotone index provides another useful observation:

$$\mathcal{T}(k) = \min\{t : C_t = k\} = \left\lceil \frac{k}{\rho} \right\rceil - 1 \qquad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks needed to gain m units of capacity is

$$\mathcal{T}(k+m) - \mathcal{T}(k) = m + \lceil (k+m)\sigma \rceil - \lceil k\sigma \rceil \in \{m + \lfloor m\sigma \rfloor, m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, while the return time measured by the other index admits two values. This distinction is part of refinement dynamics, not an a posteriori correction of its numerical values.

Theorem 10.54 (Complexity of phase readings)**.** *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. Retaining phase n mod 9 gives complexity $p_9(m) = 9(m+1)$; retaining only its ternary class gives $p_3(m) = 3(m+1)$. All three topological entropies are zero.*

Proof. For factors of length m, the predecessors of the two endpoints of the coding interval form $m + 1$ distinct points on the circle: coincidence between shifted endpoints removes duplications and irrationality prevents additional coincidences. These points determine $m + 1$ intervals with constant and distinct itineraries. Every irrational orbit visits each interval, whence $p(m) = m + 1$.

With a phase modulo nine fixed, positions of that phase follow a rotation of step 9δ, also irrational. By density, all factors appear in each phase, and the first coordinate distinguishes their nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m+1)) \to 0$ for $c = 1, 3, 9$. □

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has frequency characters

$$\frac{a}{9} + k\delta \pmod 1, \qquad a \in \mathbb{Z}/9\mathbb{Z},\ k \in \mathbb{Z}.$$

The spectral module is not log 10: it is the additive module generated by $1/9$ and δ, modulo the integers. The factor log 10 belongs to normalization of (10.50).

An operator presentation makes memory explicit. On $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n+1\rangle$, $V_9|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n\delta}|n\rangle$ and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finitely supported vectors,

$$V_9 T = \zeta_9 T V_9, \qquad W_\delta T = e^{2\pi i \delta} T W_\delta, \qquad [D_M, T^9] = T^9. \tag{10.53}$$

The first two relations are Weyl covariances; the third expresses that a nonadic turn increases the memory degree. The base rotation is abelian, while the algebra of observables and translations is not. Inversion $|n\rangle \mapsto |-n\rangle$ conjugates forward motion to backward motion. This realization provides an aperiodic helical suspension with finite phase return, not a periodic word of length nine.

For this count we fix the intercept $\theta = 0$ and accumulated endpoints $(N_0, \ldots, N_4) = (0, 729, 972, 999, 1000)$, corresponding to the cubic order $(729, 243, 27, 1)$. The lower counts are $\lfloor N_j\delta \rfloor - \lfloor N_{j-1}\delta \rfloor$, giving $(33, 11, 1, 0)$. The upper counts are $\lceil N_j\delta \rceil - \lceil N_{j-1}\delta \rceil$, with the same zero initial boundary, giving $(34, 11, 1, 0)$. These values depend on the intercept and stratum order; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in the same count preceding any units chart; identifying them with physical exponents additionally requires the corresponding dimensional derivation.

The designation *aperiodic crystal* is used here for this symbolic order, with specified complexity, spectrum and return. Zero topological entropy does not by itself equal zero physical dissipation: a

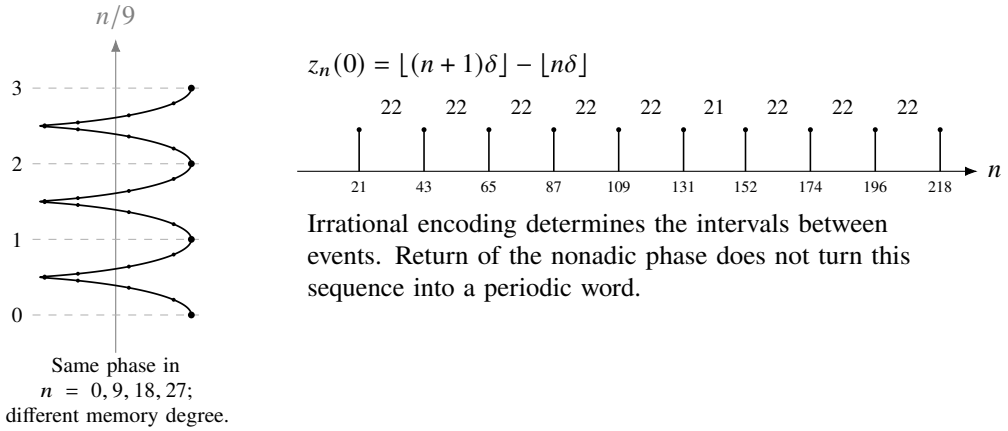


Figure 10.2: Complementary representations of transport. Left, projection of the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$: advancing nine steps recovers phase and increases height. Right, the first ten events of the lower word for $\delta = \log_{10}(10/9)$, with exact gaps 21 or 22. The helical representation introduces no additional physical metric.

thermodynamic realization requires defined energetic dynamics and an erasure operation. Likewise, this Sturmian reading does not replace all TPK operators or by itself prove periodicity or aperiodicity of the complete output of α. Its function is to make explicit an order structure that mere calendar return does not describe.

10.5.11 Nonadic connection, holonomy and monodromy representation

The already constructed regional connection is now studied through three objects: transport of the complete state, the representation of its turns, and the suspension of its itineraries. The APP–TRIT–TPK operations and the initial catalog retain their preceding definitions. The necessary distinction here concerns what each representation retains of the same history (Sections 4.3 and 10.5).

At prolongation index k, a state chart preserves

$$\widehat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k combines the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry register, Σ_k the boundary signature and W_k the incidence information. The phase is $g_k = 1 + [(k-1)]_9$; m_k counts complete turns. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphasic position, while the integer m_k distinguishes its successive turns.

Definition 10.55 (Transport of a nonadic turn)**.** Let $\mathcal{R}_k \subseteq \widehat{X}_k \times \widehat{X}_{k+1}$ be the admissible extension relations of TPK. The realization of holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation to s turns is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves admissible continuations. On an individualized history, each arrow leads to the state actually prolonged; the isolated visible block may still admit more than one continuation. This difference is precisely the function of memory and boundary conditions in determining the trajectory.

Proposition 10.56 (Iteration of return and accumulated memory)**.** *In every admissible composition of s turns,*

$$g_{k+9s} = g_k, \qquad m_{k+9s} = m_k + s, \qquad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-turn rule preserves phase and increments the counter by one unit. Applying it to the already transported state repeats exactly these two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. □

The counter thus provides an integer witness to the difference between turns. After 12 turns the pair of finite phases is restored, but the counter increases by 12. The nonsplit extension (4.33) expresses calendar composition through its cocycle; the trajectory also preserves the integer lift of that datum. Conservation here means transport of prior contributions, not immobility of all coordinates.

10.5.11.1 Bilateral representation of the return index

The shift T of (10.53) admits an expression in phase and turn coordinates. Number the phases from 0 to 8, by shifting the origin relative to g_k, and write $k = 9m + r$, with $0 \le r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\mathrm{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ expresses that same unit shift

$$T|r, m\rangle = \begin{cases} |r + 1, m\rangle, & r < 8, \\ |0, m + 1\rangle, & r = 8. \end{cases}$$

The inverse decreases r by one unit when $r > 0$, and takes $|0, m\rangle$ to $|8, m - 1\rangle$. Passing through the phase origin therefore incorporates the exact quotient unit.

Proposition 10.57 (Representation of the cycle's monodromy)**.** *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m + 1\rangle, \qquad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\mathrm{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the cycle's turn group.

Proof. Nine shifts preserve the remainder of division by 9 and increment the quotient. The powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m + s\rangle$ distinguishes every integer s. Permutation of an orthonormal basis is unitary. □

This representation corresponds to the covering of the cycle, whose turn group is $\mathbb{Z}$, and represents phase with its counter. The TPK state also contains the trajectory coordinates already described. The bilateral extension admits negative indices as coordinates of the covering; a trajectory starting from an initial condition uses its admissible half-axis. An operator inverse of the index is thus available without turning each extension relation into a bijection of the complete state.

With $D_M|k\rangle = \lfloor k/9 \rfloor |k\rangle$, the identity $[D_M, S] = S$ expresses the memory increment on the dense domain of finitely supported vectors. If $\mathcal{I}|k\rangle = |-k\rangle$ and P_0 projects onto indices divisible by 9, we also obtain

$$\mathcal{I}T\mathcal{I} = T^{-1}, \qquad \mathcal{I}D_M\mathcal{I} = -D_M - (I - P_0). \tag{10.54}$$

Indeed, for $k = 9m + r$, $\lfloor -k/9 \rfloor = -m$ if $r = 0$ and equals $-m - 1$ if $r > 0$. Inversion thus preserves the boundary term of Euclidean division.

10.5.12 Block clock and capacity clock

The index of a refinement map and the number of available decimal positions have related but different laws. To express them, reserve n for the ternary-block index and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \qquad \delta = 1 - \rho, \qquad C_n = \lfloor (n+1)\rho \rfloor, \quad n \geq 0.$$

The letter C_n here denotes capacity, not the dodecaphasic register K. At the canonical phase origin, the preceding mechanical word provides

$$Z_n = \sum_{j=0}^{n} z_j(0) = \lfloor (n+1)\delta \rfloor.$$

Proposition 10.58 (Exact balance between capacity and vacancies)**.** *For $n \geq 0$ and, in the second identity, $n \geq 1$,*

$$C_n = n - Z_n, \qquad C_n - C_{n-1} = 1 - z_n(0).$$

Proof. Irrationality of δ implies $\lceil (n+1)\delta \rceil = \lfloor (n+1)\delta \rfloor + 1$. Then

$$\lfloor (n+1)(1-\delta) \rfloor = n + 1 - \lceil (n+1)\delta \rceil = n - Z_n.$$

Subtracting two consecutive identities completes the proof. □

An event $z_n = 1$ adds a block to the history while preserving capacity C_n; an event $z_n = 0$ increments both counters. In any interval, capacity increments and vacancies sum to exactly its length. Reducing modulo q, for $q = 9$ or $q = 108$, gives

$$[C_n]_q = [n]_q - [Z_n]_q.$$

Uniform block phase and capacity phase are related through integrated vacancy memory. In particular, capacity retention and a complete turn of the connection have distinct mechanisms, although both can preserve a phase observation.

Monotone inversion of the clock gives a first arrival time:

$$T_{\mathrm{cap}}(a) = \min\{n : C_n = a\} = \left\lceil \frac{a}{\rho} \right\rceil - 1, \qquad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time spent gaining b units of capacity is

$$T_{\mathrm{cap}}(a+b) - T_{\mathrm{cap}}(a) = b + \lceil (a+b)\sigma \rceil - \lceil a\sigma \rceil.$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. This yields 9 or 10 blocks to gain 9 units of capacity, and 113 or 114 to gain 108. These are times of the inverse clock, measured from a first arrival; the turn counter is always interpreted in the index to which it belongs.

10.5.13 Symbolic suspension and aperiodic time crystal

Rotation encoding combines phase periodicity, local recurrence and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0,1\}^{\mathbb{Z}}$ be the closure, in the product topology, of the translates of the bilateral word $z_n(\theta)$. Denote by σ_{sh} the shift of its sequences. Retaining a nonadic phase, the step is

$$F(r, z) = ([r+1]_9, \sigma_{\mathrm{sh}} z), \qquad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 10.59 (Symbolic aperiodic time crystal)**.** The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = \big((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}\big)/\!\sim, \qquad (y, t+1) \sim (Fy, t),$$

with flow $\Phi_s[y,t] = [y, t+s]$. This temporal-order model, together with its vacancy encoding and turn-counter representation, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and of its return section; aperiodicity characterizes its itineraries. The frequency of ones in each sequence is δ: the uniform discrepancy bound of less than one, obtained by telescoping in any window, persists upon taking the closure. A periodic sequence would have a rational frequency. Therefore, $\mathcal{X}_\delta$ contains no periodic orbits. The suspension also has no closed orbits: a return of the flow would require an integer time and periodicity of the itinerary.

Finite words do recur. The itineraries of length m arise from a partition of the circle by $m+1$ points; each interval is visited recurrently by the irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why each word appears in the 9 phases and why the complexities already established are

$$p(m) = m+1, \qquad p_9(m) = 9(m+1), \qquad p_3(m) = 3(m+1).$$

The recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

The realization by the observables V_9, W_δ and the shift T expresses the same order through the covariances (10.53). The frequencies of the uniform clock belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[C_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by the uniform phase when interpreting the returns. The term *time crystal* is thus defined through a mathematical dynamical system. Its physical realization requires specifying an energy dynamics, its temporal observables and its stability, data distinct from the present symbolic construction.

10.5.14 Mirror symmetry of the regions and dual realization

The mirror interchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$\mathfrak{c}B = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and its aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixing refers to the involution in a fiber; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$ and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q-a), \qquad s = q - u^\varphi, \qquad u^\pi = 2(s+d), \qquad u^e = 2(s-d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d; this antisymmetric coordinate singles out the distribution that the aggregate preserves without distinguishing it. The reversal $\mathcal{I}$ of the temporal index and the involution $\mathfrak{c}$ of channels therefore act on different data.

The phase return has an explicit witness:

$$B_1 = (012222 \mid 121221 \mid 112202),$$
$$B_{10} = (101222 \mid 112221 \mid 112002).$$

Both positions have phase 1; the block, the axis and the memory have changed. Likewise, synchronization of the censuses in phases 8 and 9 cancels their relative differences, while the diagonal component changes from $(59, 59, 59)$ to $(43, 43, 43)$. Equality of a relative observable does not exhaust the state that determines it.

The dual realization preserves that architecture. In section 21.11.1, the same prefix determines nested cylinders on the one hand and a sequence of encoded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞. Figure 10.3 anticipates these two maps. The subsequent incidence path constructs codes, lattices and the realization of Moonshine in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

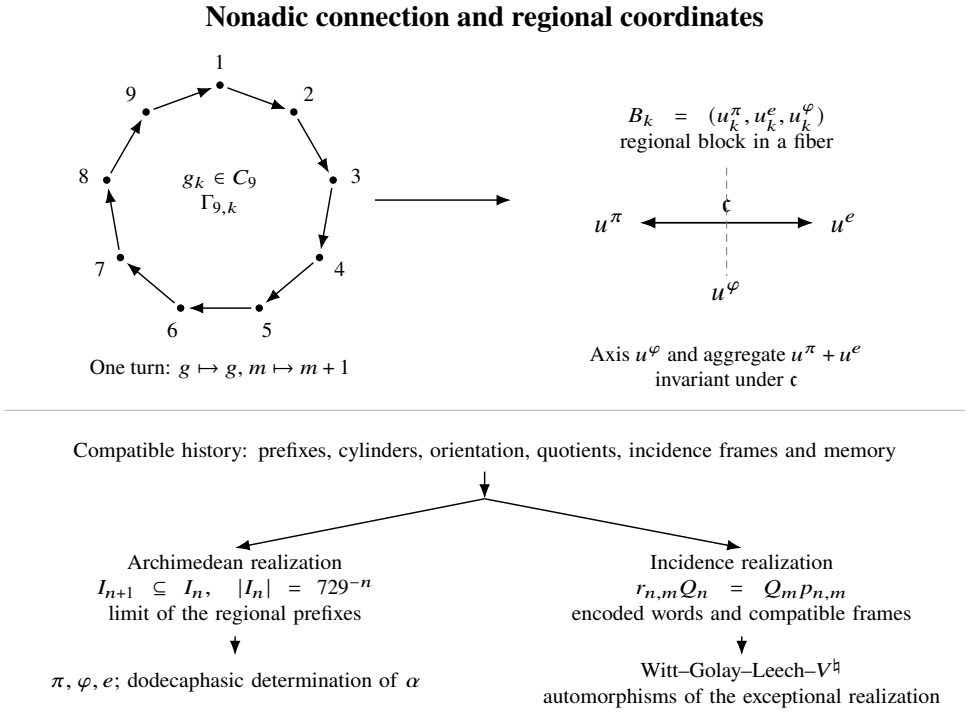


Figure 10.3: Nine phases of a connection and two realizations of the compatible history. The cycle represents the phase; each turn increments the counter. The regional diagram represents the involution $\mathfrak{c}$, which interchanges closure and propagation and fixes the autoscale axis in a fiber. This invariance does not require the axis to be constant during transport. The two lower maps use the same prefix: one determines its numerical limit and the other preserves words and incidence frames. The maps of this second realization are specified in section 21.11.1; the exceptional path is developed afterward.

10.5.14.1 Rectilinear representation of the phases and the return

The representation on the segment $[0, 10]$ preserves the positional origin of APP and orders its nine interior marks. In this diagram, the integers $1, \dots, 9$ are phase representatives. The horizontal position indicates the order of the charts; the upper labels specify regional operations. Figure 10.4 thus brings together the initial line, the fixing of the autoscale axis, the mirror distribution of the two channels and the return with memory (section 10.5.11).

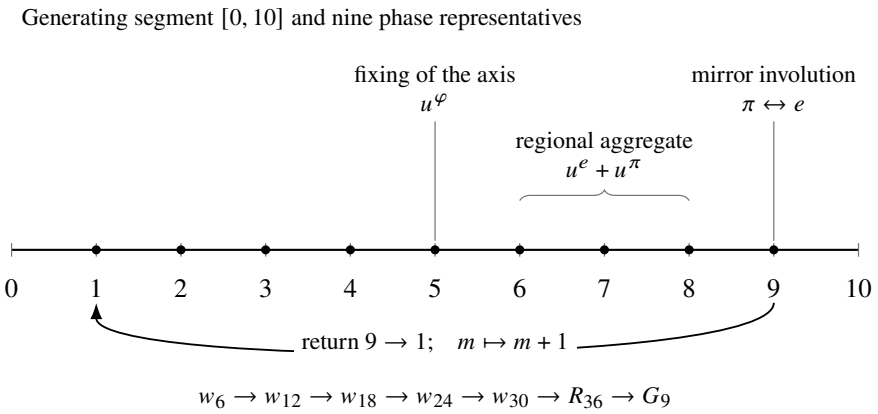


Figure 10.4: Rectilinear representation of the nonadic connection. The mark 5 locates the first strong saturation and the regional autoscale axis; the brace 6–8 identifies the ternary aggregate fixed in each fiber; phase 9 selects the closure orientation. The return preserves the phase and increments the memory. The abscissae number operational positions, while the labels $u^{\pi}, u^{e}, u^{\varphi}$ designate blocks of $\mathbb{F}_3^6$.

The fixing of the axis is formulated in the fiber of a given boundary. If $q = u^{\pi} + u^{e} + u^{\varphi}$ and $a = u^{\pi} + u^{e} - u^{\varphi}$, then

$$u^{\varphi} = 2(q - a), \qquad s = u^{\pi} + u^{e} = q - u^{\varphi} \quad \text{in } \mathbb{F}_3^6.$$

The data (q, a) determine the axis and the aggregate; the ordered distribution of s preserves the mirror freedom resolved by the prolongation. The fifth phase has a local solution and an extension, with boundary datum $c = 111111$ and residual signature $\rho_W = 000$. This is the meaning of its first strong saturation. The symbol φ placed above the mark 5 identifies the regional axis involved in that operation.

The intermediate phases preserve this decomposition within each fiber, while their boundary data evolve. The complete census of local multiplicities and extensions of the distinguished branch is

phase	1	2	3	4	5	6	7	8	9
N_{loc}	3	2	5	4	1	1	3	3	2
N_{ext}	2	5	4	1	1	3	3	2	1

Phase 6 combines local uniqueness and three prolongations; phase 7 resolves a triple mirror fiber; phase 8 synchronizes the three regional censuses. Phase 9 preserves two local distributions with the same axis and the same aggregate, one of which extends. These counts express different functions within a single connection, and justify the marks on the line.

In particular, for phases $6, 7, 8$ the regional sums are respectively 012100, 101001 and 112000. The conservation represented by the brace is an invariance with respect to the mirror distribution in each fiber. The subsequent real evaluation applies the readers of the compatible cylinders to the complete histories; the internal sum represented in the figure belongs to the ternary domain.

Finally, the arc $9 \to 1$ represents the passage from one turn to the next. The formulas in section 10.5.11 preserve the phase remainder and increment the turn counter. The rectilinear diagram and the cyclic

diagram in Figure 10.3 therefore express two complementary aspects of the same holonomy: the order of the positions and the composition of the return on the state with memory.

10.5.15 Temporal transport and tritic exponential composition

The Euler relations in section 10.5.7 describe the local realizations of the three regimes. For a fixed signature τ, the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation takes J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau)\exp(tJ_\tau) = \exp((s+t)J_\tau), \qquad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is unity. In the elliptic type it is realized through $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, through $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I + tN)(I - tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

The elliptic realization closes a turn; the nilpotent one preserves a linear displacement; the hyperbolic one separates expansion and contraction with unit determinant. In the regional parameters already generated,

$$\exp(\pi J_{\mathrm{e}}) = -I, \qquad \exp(J_{\mathrm{p}}) = I + J_{\mathrm{p}}, \qquad \exp(2\log\varphi\, J_{\mathrm{h}}) = F_{\mathrm{av}}^2.$$

The evaluation $\exp(I) = eI$ recognizes the unity of the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transporting a local realization through an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U\exp(tJ) = \exp(tJ')U$. The identity transports the regime and its norm. A transition between different quadratic types instead belongs to the selection of sheets and regimes of TPK; it is not a conjugation that changes the value of J^2. The temporal product therefore preserves the order of the transitions and the information of their endpoints.

This articulation distinguishes the closure of a local representation and the holonomy of a history. It is possible to have $\exp(2\pi J_{\mathrm{e}}) = I$ at the same time as $S|r, m\rangle = |r, m+1\rangle$. The first equality restores the orientation of the regime; the second records another turn. The discrete structure preserves both: the local return and the accumulated provenance of the state that realizes it.

10.5.16 Multiplicative irreducibility and return periods

The distinction between structure, evaluation and phase also appears in the multiplicative sector. The local product of APP contains both the residue and the quotient; its iteration allows normal forms and factorizations to be constructed before examining their decimal periods. This development places the expression *prime mode* in a precise domain and relates irreducibility to the operations already defined, not to an isolated coincidence of periods (section 10.5.16).

Let

$$\mathcal{W}_9 = \{\varnothing\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \ldots, 9\}^\ell.$$

Sequences are ordered from the least significant digit. The evaluation $\mathrm{ev}_9(u) = \sum_j u_j 9^j$, with $\mathrm{ev}_9(\varnothing) = 0$, is a bijection onto the nonnegative integers: for a positive integer, dividing $n - 1$ by 9 uniquely determines the first digit and the continuation. This is bijective nonadic numeration; it preserves the boundary representative 9.

The sum $\boxplus_\#$ is obtained by applying the additive APP sheet to the digits at each position and propagating their quotients. If there is only one digit, it constitutes the provisional residue; a nonzero incoming quotient is normalized through a second additive operation. The sum of the quotients is transported to the next position. To multiply a sequence by a digit d, the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated through

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ \big(R^+(r_j, c_j),\, q_j + q^+(r_j, c_j)\big), & c_j > 0. \end{cases}$$

The procedure begins with $c_0 = 0$ and incorporates the final quotient when it is nonzero. During addition, $c_j \le 2$; during multiplication by a digit, $c_j \le 9$. The local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_\# u$, with $d \odot_\# \varnothing = \varnothing$.

For $v = (v_0, \dots, v_{m-1})$, the full product is constructed by the recurrence

$$a_m = \varnothing, \qquad a_j = (9 \odot_\# a_{j+1}) \boxplus_\# (v_j \odot_\# u), \qquad u \boxtimes_\# v = a_0.$$

The rule composes the two local sheets. Its evaluation satisfies

$$\mathrm{ev}_9(u \boxtimes_\# v) = \mathrm{ev}_9(u)\,\mathrm{ev}_9(v). \tag{10.55}$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the power of 9. The recurrence then gives $\mathrm{ev}_9(a_j) = 9\,\mathrm{ev}_9(a_{j+1}) + v_j\,\mathrm{ev}_9(u)$, whose iteration proves (10.55). Uniqueness of the normal form transports associativity and commutativity to the constructed product; its unit is (1).

Definition 10.60 (Reduced multiplicative coproduct)**.** On the vector space of finite support generated by the nonempty normal forms, define

$$\widetilde{\Delta}_\# e_w = \sum_{\substack{u \boxtimes_\# v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A nonunit normal form is irreducible when its image under $\widetilde{\Delta}_\#$ is zero.

The factorization fibres are finite by (10.55). In the completion $\mathcal{H}_\# = \ell^2(\mathcal{W}_9 \setminus \{\varnothing\})$, the images of distinct forms have disjoint supports. If $d_\#(w)$ counts the ordered nonunit factorizations, the domain of the closure is

$$\mathrm{Dom}(\overline{\widetilde{\Delta}_\#}) = \left\{ x \in \mathcal{H}_\# : \sum_w d_\#(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is that weighted sum. Therefore, the operator

$$A_\# = (\overline{\widetilde{\Delta}_\#})^* \overline{\widetilde{\Delta}_\#}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered nonunit factorizations of w. The projection

$$P_{\mathrm{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|)\mathbf{1}_{\{0\}}(A_\#)$$

selects exactly the irreducible forms. Through multiplicative evaluation, these correspond to the ordinary primes. Selection depends on the factorization structure and keeps the unit state separate.

For a value $n \ge 2$ coprime to 10, decimal division induces the residue permutation $r \mapsto 10r \pmod n$. Return of the residue 1 has period

$$\tau_n = \mathrm{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \mathrm{mcm}_{p^{a_p} \| n}\, \mathrm{ord}_{p^{a_p}}(10).$$

The composite return synchronizes the returns of the prime powers. For a prime $p \notin \{2, 3, 5\}$, we have

$$\tau_p \mid p - 1, \qquad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \qquad M_p \equiv 0 \pmod 9.$$

The last congruence expresses the nonadic closure of the decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for any n coprime to 90. In particular,

7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This distinction makes precise the relationship with aperiodic refinement. The vacancy sequence determines the increments of the capacity index; factorization determines the multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, while the quotient and factorization preserve information that this closure does not distinguish. The link with the spectral constructions of the treatise resides in those operators and their intertwinings. The proofs of their subsequent analytic results retain their own development; they are not replaced by the periodicity of a residual observation.

10.5.16.1 Vacancy discrepancy and Dirichlet readouts

The link also admits an exact analytic expression. The same encoding $z_n = z_n(0)$, for $n \geq 1$, is preserved, and two subsequent observables are considered: the series over all indices and the product over the indices selected by irreducibility. For $\operatorname{Re} s > 1$,

$$Z^{+}_{\mathrm{vac}}(s) = \sum_{n\geq 1} \frac{z_n}{n^s}, \qquad Z^{\times}_{\mathrm{vac}}(s) = \prod_{p} (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in that half-plane. Their common factor is the density δ, but their discrepancies differ.

Proposition 10.61 (Separation of vacancy density)**.** *We have*

$$Z^{+}_{\mathrm{vac}}(s) = \delta\zeta(s) + H_{\mathrm{vac}}(s), \qquad H_{\mathrm{vac}} \in \mathcal{O}(\operatorname{Re} s > 0).$$

In $\operatorname{Re} s > 1$*, with logarithms defined by the absolutely convergent Euler series,*

$$Z^{\times}_{\mathrm{vac}}(s) = \zeta(s)^{\delta} G_{\mathrm{vac}}(s), \qquad \log G_{\mathrm{vac}}(s) = \sum_{p} (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centred partial sums satisfy

$$A_N := \sum_{n=1}^{N} (z_n - \delta) = \lfloor (N+1)\delta \rfloor - N\delta, \qquad |A_N| < 1.$$

Abel summation represents the centred term by $s \int_1^{\infty} A_{\lfloor x \rfloor} x^{-s-1}\, dx$. The integral converges locally uniformly in $\operatorname{Re} s > 0$, proving holomorphy. The second identity follows by separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. □

The decomposition makes visible what the readout on primes adds: it examines the same vacancy sequence on a subset determined by factorization. Control of the discrepancy over all integers gives the first holomorphic extension; the second observable preserves a discrepancy of its own on primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation explains their relevance to the arithmetic core without transferring to this article the proofs of the terminal results of the treatise.

10.6 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP–TRIT–TPK state. We denote them by the indices $c \in \{\mathrm{P}, \mathrm{E}, \mathrm{F}\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c. The subsequent identifications $v_{\mathrm{P}} = \pi_{\mathrm{HMT}}$, $v_{\mathrm{E}} = e_{\mathrm{HMT}}$, and $v_{\mathrm{F}} = \varphi_{\mathrm{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

10.6.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \le v_c \le u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \ge J$,

$$\lfloor s\ell_c(j) \rfloor = \lceil s u_c(j) \rceil - 1 = \lfloor s v_c \rfloor. \tag{10.56}$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 10.62 (Separation index and emitted prefix)**.** For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lceil s u_c(j) \rceil - 1\}.$$

For an integer base $b \ge 2$ and a depth $n \ge 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \tag{10.57}$$

The prefix contains the integer part and the first n positions in base b; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 10.63 (Termination and correctness of emission)**.** *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \qquad P_c(b, n) \le b^n v_c < P_c(b, n) + 1. \tag{10.58}$$

For each scale s, a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (10.56) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \le \lfloor s v_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor s v_c \rfloor \le \lceil s u_c(j) \rceil - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (10.58). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. □

Proposition 10.64 (Exact compatibility of prefixes)**.** *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \tag{10.59}$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \qquad 0 \leq d_c(b, n) < b. \tag{10.60}$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \ldots, b^m - 1\}$. Euclidean division by b^m recovers N. The specialization $m = 1$ yields (10.60) and its residual bound. □

10.6.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \qquad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \tag{10.61}$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^{5} B_{c,j}(t) 3^{5-j}. \tag{10.62}$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 10.65 (Stability of extraction at any depth)**.** *For every n,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \tag{10.63}$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 10.63. For the second, apply (10.59) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (10.61) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. □

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (10.63) determines its compatibility with every subsequent extension exactly.

Example 10.66 (First joint block)**.** Evaluation of the selected regional readers gives the words

$$B_{\mathrm{P}}(0) = 010211, \qquad B_{\mathrm{E}}(0) = 201101, \qquad B_{\mathrm{F}}(0) = 121200.$$

Their lifting by (10.62) produces

$$u_{\mathrm{P}}(0) = 103, \qquad u_{\mathrm{E}}(0) = 523, \qquad u_{\mathrm{F}}(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

10.6.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_{\mathrm{P}}(t) \\ B_{\mathrm{P}}(t+1) \\ B_{\mathrm{E}}(t) \\ B_{\mathrm{E}}(t+1) \\ B_{\mathrm{F}}(t) \\ B_{\mathrm{F}}(t+1) \end{pmatrix}. \tag{10.64}$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$X_0 = \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, \qquad Y_0 = \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix},$$

$$X_1 = \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, \qquad Y_1 = \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}.$$

Theorem 10.67 (Unique reconstruction of the two initial links)**.** *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

Both matrices are invertible.

Proof. The matrices

$$A_0 = \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, \qquad A_1 = \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

satisfy $A_iX_i = X_iA_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_iY_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \qquad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_iL_i^{-1} = L_i^{-1}L_i = I$ proves exact recovery of each transformed word. All products use residues $0, 1, 2$ with their interpretation in $\mathbb{F}_3$. □

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \ldots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 10.65. The matrix reconstruction proved here has the two links specified in its statement as its domain.

10.6.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n}\right), \qquad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k}\right). \tag{10.65}$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 10.68 (Integer compatibility defects)**.** We define

$$A(s) = N1000^k - D729^n, \tag{10.66}$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \tag{10.67}$$

The state is compatible when

$$A(s) < 729^n, \qquad B(s) > 0. \tag{10.68}$$

Lemma 10.69 (Integer criterion for intersection)**.** *The cylinders in* (10.65) *have a nonempty intersection if and only if* (10.68) *holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (10.65) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (10.68). □

Definition 10.70 (Update of depths and prefixes)**.** For $0 \le u < 729$ and $0 \le d < 1000$, define

$$T_u(n,k,N,D) = (n+1,k,729N+u,D),$$
$$T_{u,d}(n,k,N,D) = (n+1,k+1,729N+u,1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 10.71 (Exact boundary recurrences)**.** *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \tag{10.69}$$
$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \tag{10.70}$$

Proof. For the first identity,

$$(729N+u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned}(729N+u)1000^{k+1} &- (1000D+d)729^{n+1}\\ &= 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}.\end{aligned}$$

Both are integer identities preceding any real approximation. □

Theorem 10.72 (Refinement of the generated regional cylinders)**.** *Let*

$$s_c(n,k) = (n,k,P_c(729,n),P_c(1000,k)).$$

Then $s_c(n,k)$ is compatible for all n,k. Moreover,

$$T_{u_c(n)} s_c(n,k) = s_c(n+1,k),$$
$$T_{u_c(n),d_c(1000,k)} s_c(n,k) = s_c(n+1,k+1).$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a,b \ge 0$,

$$\frac{P_c(729,n+a) - P_c(729,n+a) \bmod 729^a}{729^a} = P_c(729,n),$$

and the analogous identity holds in base 1000*.*

Proof. Theorem 10.63 places v_c simultaneously in both intervals in (10.65); the intersection criterion gives compatibility. The update identities are (10.60) in each base. Recovery follows from (10.59). Thus the defect recurrence and prefix recovery are consequences of the same update. □

Example 10.73 (First pair of regional cylinders)**.** For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	−422	578
F	1179	1618	−522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The

different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

10.6.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 10.74 (Admissibility of regional prolongation)**.** For fixed c, n, let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$\lfloor s\ell_c(j) \rfloor \le 729P_c(729, n) + u \le \lceil su_c(j) \rceil - 1. \tag{10.71}$$

Theorem 10.75 (Uniqueness of the prolongation compatible with the reader)**.** *Condition* (10.71) *is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (10.71) coincide with $P_c(729, n+1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n+1)$. By (10.60), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. □

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

10.6.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t, write $B_{c,j} = B_{c,j}(t)$. For each column j, calculate the following integers and residues:

$$S_j = B_{\mathrm{P},j} + B_{\mathrm{E},j} + B_{\mathrm{F},j}, \qquad q_j = S_j \bmod 3, \qquad c_j = \left\lfloor S_j/3 \right\rfloor, \tag{10.72}$$

$$a_j = (B_{\mathrm{P},j} + B_{\mathrm{E},j} - B_{\mathrm{F},j}) \bmod 3, \qquad w_j = \sum_c \mathbf{1}_{B_{c,j}\neq 0}, \qquad \bar{w}_j = w_j \bmod 3. \tag{10.73}$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{\mathrm{P},j} + B_{\mathrm{E},j} + 3 - B_{\mathrm{F},j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \tag{10.74}$$

Two residual charges are preserved for each region:

$$v_c^{\mathrm{res}} = \sum_j B_{c,j} \bmod 3, \qquad v_c^{\mathrm{W}} = \sum_j (B_c W)_j \bmod 3. \tag{10.75}$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\mathrm{res}}, v^{\mathrm{W}})$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 10.76 (Exact recovery and bounds of the signature)**.** *For each column and each region,*

$$S_j = q_j + 3c_j, \qquad 0 \le q_j, a_j < 3, \qquad 0 \le c_j \le 2, \tag{10.76}$$

$$B_{\mathrm{F},j} = 2(q_j + 3 - a_j) \bmod 3, \qquad 0 \le w_j \le 3, \qquad 0 \le \bar{w}_j < 3. \tag{10.77}$$

Moreover,

$$v_c^{\mathrm{W}} = \left(\sum_j \big((B_c W)_j \bmod 3 \big) \right) \bmod 3. \tag{10.78}$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (10.76) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \le S_j \le 6$ and $c_j \le 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{\mathrm{F},j}$. Since the inverse of 2 is 2, this yields (10.77); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (10.78). □

Example 10.77 (Signature of the first joint block)**.** The three initial bands produce

$$q = (0,0,2,2,1,2), \qquad a = (1,2,0,1,1,2),$$
$$c = (1,1,0,1,0,0), \qquad w = (2,2,2,3,1,2),$$
$$v^{\mathrm{res}} = (2,2,0), \qquad v^{\mathrm{W}} = (2,1,1).$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2,0,2,1,1,2), \qquad (0,0,0,2,0,2), \qquad (2,1,1,2,1,0).$$

Their residual sums by row yield the three charges $(2,1,1)$.

Definition 10.78 (Integer clock of decimal reading)**.** The transition index t and the decimal depth of expression are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \le 3^{6(t+1)}\}. \tag{10.79}$$

Proposition 10.79 (Characterization and monotonicity of the clock)**.** *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \le 729^{t+1} < 1000^{\kappa(t)+1}. \tag{10.80}$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1}. Monotonicity of the latter expression proves that of κ. The final two evaluations are the integer comparisons

$$1000^{20} \le 729^{21} < 1000^{21}, \qquad 1000^{20} \le 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

10.6.7 Composition theorem for the block, cylinder, and signature

Theorem 10.80 (Joint compatibility of regional readings)**.** *For each depth n, there exists a unique triple of blocks $(u_{\mathrm{P}}, u_{\mathrm{E}}, u_{\mathrm{F}})$ satisfying regional admissibility (10.71) in all three regions. This triple is $(u_{\mathrm{P}}(n), u_{\mathrm{E}}(n), u_{\mathrm{F}}(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (10.69) and (10.70);*
2. *its six trits produce the signature $\mathcal{S}(B_{\mathrm{P}}(n), B_{\mathrm{E}}(n), B_{\mathrm{F}}(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (10.78);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 10.75 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 10.72 to be applied. The deterministic application of (10.61) produces the bands used by (10.72)–(10.75). The recovery identities are those of Theorem 10.76. Finally, Theorem 10.65 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

10.7 Initial conditions and regional generation

An APP position does not by itself contain the initial condition of its dynamics. The latter includes the direction of transport and distinguishes the cursors of the two sheets. Before classifying particles, this domain must be constructed: it produces the regional records that will enter the electronic coordinate, action, and mass readers. The operations in this section receive neither a physical species nor a metrological value as an initial condition.

10.7.1 Two oriented cursors on the complete support

We use indices $I_9 = \{0, \dots, 8\}$, whose positive label is $i + 1$, and the cardinal directions $\mathcal{D} = \{N, E, S, O\}$. The domain of one cursor and the joint domain are

$$\mathcal{S} = I_9^2 \times \mathcal{D}, \qquad \Omega_{\mathrm{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \qquad |\mathcal{S}| = 324, \quad |\Omega_{\mathrm{seed}}| = 104\,976. \tag{10.81}$$

The subscript seed retains the domain designation used in the antecedents; mathematically, these are oriented initial conditions. The Cartesian product includes all ordered pairs. The additive sheet has one cursor and the multiplicative sheet another; a subsequent coincidence between their readings is a result of the traversal.

During the 54 instants of the initial emission, the phase selector repeats $(+1, -1, 0)$. In the positive regime, $\rho_9(i + j + 2)$ is first evaluated at the additive cursor, and that cursor is then transported by one position. In the negative regime, $\rho_9((i + 1)(j + 1))$ is evaluated at the multiplicative cursor, after which the second cursor is transported. The null regime records a neutral event. The directions are reversed after

instants 27 and 54. The latter reversal belongs to the terminal state although it does not alter readings that have already been taken.

The multiplicative observable uses the exponent chart of the group of units of $\mathbb{Z}/9\mathbb{Z}$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

In the sector $\{3, 6, 9\}$, this observable takes the value zero. Membership in the radical sector remains present in the state; assigning that value to a scalar reading does not identify the radical with the multiplicative identity.

10.7.2 Finite formula for the window signatures

The six windows each contain nine instants. Each sheet is evaluated three times per window. A closed formula reconstructs the signatures without consulting a prior list. For a cursor with origin $p = (i, j)$ and direction d, let

$$f(k) = \begin{cases} k, & 0 \le k < 9, \\ 18 - k, & 9 \le k < 18, \end{cases} \qquad p_k = p + f(k)\nu_d \pmod 9.$$

The index k counts only the evaluations of that sheet. The reversal after nine of its evaluations explains the second expression for f. Writing $p_k = (i_k, j_k)$, for $m = 0, \dots, 5$ we obtain

$$\begin{aligned} S_m(p, d) &= \sum_{k=3m}^{3m+2} \rho_9(i_k + j_k + 2), \\ E_m(p, d) &= \sum_{k=3m}^{3m+2} \ell_9(\rho_9((i_k + 1)(j_k + 1))), \\ s_m &= [S_m]_{10}, \qquad e_m = [E_m]_{10}. \end{aligned} \tag{10.82}$$

The integer evaluation and its division quotient remain stored in the record. The preceding signatures are the decimal projections specified by the emitter.

Since each window has $Z_m = 3$ neutral events, the rule coupling the two signatures is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \qquad d_{m,2} = [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100 s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \end{aligned} \tag{10.83}$$

The final term 1 is the reduction of $7Z_m = 21$ modulo ten. The coefficients and calendar are stated in the emission law; physical magnitudes do not appear among its arguments. The oriented ternary projection is

$$\Psi(U_0, \dots, U_5) = ([-U_0]_3, \dots, [-U_5]_3) \in \mathbb{F}_3^6. \tag{10.84}$$

The balanced identification of TRIT is then applied. The minus sign belongs to this chart and must be transported when it changes.

Proposition 10.81 (Factorisation and census of the emission)**.** *The initial conditions produce* 18 *additive signatures and* 26 *multiplicative signatures. Their coupling is injective and has* 468 *emissions. The image of* Ψ *contains* 243 *words in the ambient space of* 729 *elements.*

Proof. The hundreds digits of U_m recover s_m. If b_m is the tens digit, $e_m = [b_m - s_m]_{10}$; the units digit checks the third congruence. Hence no distinct pair of signatures has the same emission.

To obtain the census, (10.82) is evaluated at $i, j = 0, \dots, 8$ and the four directions. Appendix 94.15 contains all the resulting signatures. Each additive signature has 18 preimages. Among the multiplicative signatures, 24 have 8, one has 24, and one has 108. These sums are respectively $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. Since the two cursors are independent, the multiplicities multiply. This gives 432 emissions of weight 144, 18 of weight 432, and 18 of weight 1944, with sum

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The six reductions of (10.84) are applied to each pair of the 18 and 26 signatures published in the appendix. The complete orbit decomposition included there gives $38 \cdot 6 + 5 \cdot 3 = 243$ different images. All the operations of this enumeration are the finite sums, products and residues of the preceding formulas. □

The finite proof does not identify exhaustiveness of the domain with unlimited depth. The initial conditions have been enumerated on a fixed support. Depth is constructed through compatible prolongations preserving those conditions and their transports.

10.7.3 Selection of regions through oriented fibres

For $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \qquad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

The identities $r^3 = s^2 = I$, $srs = r^{-1}$ realise D_3. Permuting the six components of U in the same way commutes with Ψ. The preceding enumeration verifies that the orbits preserve the multiplicities of their fibres.

Let $f(w)$ be the number of emissions over w, $m(w)$ its total multiplicity, and $\nu(w) = (n_0, n_1, n_2)$ its ternary census. The additive signature preserves the diagonal class $d = [i + j]_9$ of the origin and the sense ES or NO. We write $\mathcal{D}(\mathcal{O})$ for the set of these classes over an orbit.

Proposition 10.82 (Distinguished regions of the catalogue)**.** *There is a unique orbit of six words with five emissions per word and multiplicities* $432 + 4 \cdot 144$. *Among the orbits of six words satisfying*

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \qquad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\},$$

the census $(2, 3, 1)$ *and the census* $(2, 2, 2)$ *each select a unique orbit. The mark* $d = 6$ *with orientation* ES *determines the representatives*

$$w^{\mathrm{cl}} = 010211, \qquad w^{\mathrm{pr}} = 201101, \qquad w^{\mathrm{au}} = 121200.$$

Proof. The complete table of the 43 orbits in 94.15 allows the predicates to be applied. The closure orbit is the only one with a fibre of size five and weight 1008; its five weights are those stated. The diagonal condition and the simple fibre of weight 144 leave six orbits. Their censuses select respectively one with $(2, 3, 1)$ and one with $(2, 2, 2)$. Applying the six permutations of D_3 and checking the diagonal class and orientation in (10.82) fixes the three representatives displayed. Uniqueness uses all the conditions: if the set of diagonals is omitted, four balanced orbits with simple fibre and weight 144 appear. □

The designations closure, propagation, and self-scaling apply first to these regions. The next section introduces their evaluation characters. In particular, the multiplicative signature 555555 distinguishes a transverse closure emission, whereas $(5, 5)$ is a position of the electronic support. The two constructions are related through subsequent operations, not through identity of their digits.

10.7.3.1 Mirror involution of the regional components

On the regional block $B = (u^{\mathrm{cl}}, u^{\mathrm{pr}}, u^{\mathrm{au}}) \in (\mathbb{F}_3^6)^3$, the interchange is

$$\mathfrak{c}B = (u^{\mathrm{pr}}, u^{\mathrm{cl}}, u^{\mathrm{au}}).$$

Its square is the identity and it preserves the self-scaling component. If $q = u^{\mathrm{cl}} + u^{\mathrm{pr}} + u^{\mathrm{au}}$, $a = u^{\mathrm{cl}} + u^{\mathrm{pr}} - u^{\mathrm{au}}$, and $d = u^{\mathrm{cl}} - u^{\mathrm{pr}}$, the block is recovered by

$$u^{\mathrm{au}} = 2(q - a), \quad s = q - u^{\mathrm{au}}, \quad u^{\mathrm{cl}} = 2(s + d), \quad u^{\mathrm{pr}} = 2(s - d).$$

The proof uses $2^{-1} = 2$ in $\mathbb{F}_3$. The interchange preserves q, a, s and reverses the sign of d. Thus the symmetric aggregate and the oriented difference jointly preserve the two components that the aggregate alone does not distinguish. This regional involution and the reversal of a cardinal route act on different domains. The fixed component may evolve during transport even though it is fixed by the interchange.

10.7.4 Calendar, monodromy, and memory

The finite calendar consists of twelve nonadic windows. Its group sequence is

$$0 \longrightarrow C_{12} \xrightarrow{[b] \mapsto [9b]} C_{108} \xrightarrow{[t] \mapsto [t]_9} C_9 \longrightarrow 0. \tag{10.85}$$

For $t = 9b + r$, $0 \le r < 9$, composition contains the transport quotient

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

with the first coordinate reduced modulo twelve. The projection is exact because its kernel consists of the multiples of nine. It admits no homomorphic section: such a section would give $C_{108} \cong C_{12} \times C_9$, but the respective exponents are 108 and 36.

Monodromy describes the action of one circuit of the phase base on its fibre. In the dynamics with memory, the nonadic holonomy Γ_9 preserves phase and advances depth. Its projection onto phase and counter satisfies

$$(r, n) \longmapsto (r, n + 1).$$

This formula is a reduced representation of the action, not a definition eliminating the rest of the state. On bidirectional histories the counter can be extended to $\mathbb{Z}$; on histories starting at depth zero, backward motion is defined only where the predecessor is preserved. The calendar can return after 108 transitions while preserving a memory different from the initial one. This distinction will be essential when the particle is constructed as persistence rather than repetition of a visible coordinate.

10.7.5 From records to prolongation operators

The linear publication operators are determined on the transport transitions. If x_j^{χ} is a regional state and Π_6 its six-component publication, the dependence is

$$b_j^{\chi} = \Pi_6(x_j^{\chi}), \qquad x_{j+1}^{\chi} = U_{9j+9} \cdots U_{9j+1}(x_j^{\chi}).$$

The origin matrices X_a and destination matrices Y_a, in row vectors of $\mathbb{F}_3^6$, are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

We have $\det X_0 = 1$, $\det X_1 = -1$ in $\mathbb{F}_3$. The unique solution of $X_a L_a = Y_a$ is $L_a = X_a^{-1} Y_a$. Concatenation of blocks published in the calendar (L_0, L_0, L_1, L_1) produces lengths $6, 12, 18, 24, 30$. The boundary of length 36 preserves the terminal and the prolongation relation. These matrices represent transitions: linear inversion proves their uniqueness for the given record, while generation of the states belongs to the effective composition U_t of the TPK.

10.7.6 Scope of the regional construction in the study of masses

The census has separated initial conditions, emissions, regions, and operators representing their transitions. The next stage evaluates the regional coordinates and composes the dodecaphase record. Later, the particle atlas will be constructed on the same support but will use another classification. This continuity preserves a common provenance without turning a numerical fibre into a species through mere coincidence of cardinalities or labels.

10.8 Arithmetic of histories, composition, and refinement

The persistence of a particle and the evaluation of its character require a distinction between the transported history and the coordinates read from it. The APP–TRIT–TPK kernel provides the admissible states, their routes, and the operations preserving their provenance. This section specifies three structures used in passing from those data to the atlas: compatibility between depths, composition of histories with memory, and the transport of observables in the direction opposite to refinement. Algebraic coefficients act on states and routes already specified; the relations proved below preserve that order.

10.8.1 Compatible sections and evaluation of their readers

Let X_n be the set of admissible states at depth n produced by the kernel. A state records the APP positions, the additive and multiplicative sheets, their remainders and quotients, the tritic regime, orientation, phase, boundary, route, carry, and memory pertinent to survival. These data satisfy the relations of the operations that produced them; they are not regarded as a product of independent coordinates.

For $m \geq n$, the restriction $r_{m,n} : X_m \longrightarrow X_n$ preserves the antecedent at depth n. The restrictions satisfy

$$r_{n,n} = \mathrm{id}_{X_n}, \qquad r_{\ell,n} = r_{m,n} \circ r_{\ell,m} \quad (\ell \geq m \geq n). \tag{10.86}$$

New memory belongs to the prolongation; restriction preserves the memory already determined in the prefix.

Definition 10.83 (Compatible numerical section)**.** An HMT numerical section is a sequence of states of the same genealogy

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \qquad r_{m,n}(x_m) = x_n \quad (m \geq n). \tag{10.87}$$

A reader is an operation defined on a specified domain of these sections. A finite observation belongs to a prefix; a scalar value is the evaluation of a reader. The section and the presentation of that section through a reader are therefore objects of different types.

Inverse-limit notation gathers together the compatibility of the states already constructed. The existence of a particular prolongation is established through the admissibility rules of that construction, not merely by writing X_∞.

In the Archimedean domain of a reader, each prefix determines a nonempty closed interval $I_n(x) = [a_n(x), b_n(x)]$, with

$$I_{n+1}(x) \subseteq I_n(x), \qquad b_n(x) - a_n(x) \longrightarrow 0. \tag{10.88}$$

Regional interval evaluation and the determination of its prefixes have been assembled in Proposition 90.44.

Proposition 10.84 (Value determined by compatible cylinders)**.** *For every section in that domain there is a unique real number* $\mathrm{ev}(x)$ *such that*

$$\{\mathrm{ev}(x)\} = \bigcap_{n\geq 0} I_n(x). \tag{10.89}$$

Proof. Nestedness makes the sequence $a_n(x)$ increasing and $b_n(x)$ decreasing. Both remain bounded in the initial interval. By completeness of the line, $a = \sup_n a_n(x)$ and $b = \inf_n b_n(x)$ exist and satisfy $a \leq b$. The vanishing width forces $a = b$. This point belongs to every interval, and any other point of the intersection would have distance from it less than or equal to every width, and would therefore coincide with it. □

This proposition determines the value of each section of the indicated domain. Coverage of all of $\mathbb{R}$ requires proving that each value belongs to the reader's image; the definition of an inverse limit does not in itself provide that proof. Likewise, identifying two histories from their value requires an injectivity result for the reader considered. Evaluation preserves genealogy as an antecedent and specifies exactly which part of it it publishes.

10.8.2 Sectorial route algebra and memory multipliers

Fix a sector of states and the category C of its admissible histories. We write $vu = v \circ u$: first u is traversed, then v. The elementary steps are TPK selection, transport, and update operations. The category preserves their endpoints and the conditions allowing two routes to be concatenated.

The local tritic fibre can be represented by

$$\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x), \qquad \tau(x) \in \{+1, 0, -1\}.$$

The representation preserves the regime and its orientation. To handle richer coefficients, we shall allow an associative unital real algebra A_x at each object x. A transition between regimes must specify its map on these algebras; it does not amount to identifying them.

The following construction applies to sectors possessing unital transports $\rho_u : A_x \to A_y$, for $u : x \to y$, with

$$\rho_{\mathrm{id}_x} = \mathrm{id}_{A_x}, \qquad \rho_v \circ \rho_u = \rho_{vu}. \tag{10.90}$$

For each composable pair $u : x \to y$, $v : y \to z$, let $\Omega(v, u) \in Z(A_z)^\times$ be an invertible central multiplier. Its rules are

$$\Omega(\mathrm{id}_y, u) = \Omega(u, \mathrm{id}_x) = 1_y \tag{10.91}$$

and, for $w : z \to t$,

$$\rho_w(\Omega(v, u))\, \Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \tag{10.92}$$

These conditions describe the starting structure of the sector: every specific application must provide its transports and multipliers. A realisation through bimodules preserves its own compositions and is not automatically replaced by the strict action (10.90).

Let $t(u)$ be the terminal endpoint of the arrow u. On sums of finite support

$$\mathcal{H} = \bigoplus_{u \in \mathrm{Mor}(C)} A_{t(u)} T_u$$

we define, by bilinearity,

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v,u)T_{vu} \tag{10.93}$$

when u, v are composable, and zero otherwise. Addition gathers histories with their coefficients; multiplication concatenates the histories and preserves the memory multiplier. Finite support ensures that each product is defined without requiring additional convergence.

10.8.3 Associativity, local identities, and calendar memory

Proposition 10.85 (Associative composition of the sector)**.** *Under* (10.90)–(10.92)*, the product* (10.93) *is associative. If there are finitely many objects, its unit is* $\sum_x 1_x T_{\mathrm{id}_x}$*. For an arbitrary set of objects, the same expressions on finite sets of endpoints give local identities for every finite family of histories.*

Proof. Let $u : x \to y$, $v : y \to z$, $w : z \to t$, with coefficients $a \in A_y$, $b \in A_z$, $c \in A_t$. The two associations produce, as coefficients of T_{wvu},

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: \quad c\rho_w(b)\Omega(w,v)\rho_{wv}(a)\Omega(wv,u), \\ (cT_w)((bT_v)(aT_u)) &: \quad c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v,u))\Omega(w,vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality allows $\Omega(w,v)$ to be moved to the right of this factor, and the cocycle identity equates the remaining factors. The general coefficients a, b, c, which may not commute, are not permuted. If the three arrows do not form a composable chain, both associations are zero by their endpoints. Bilinearity extends the equality to finite sums. Finally, (10.91) and the unitality of each ρ_u verify the unit. For a finite family it suffices to sum the identities of all its initial and terminal endpoints, proving the last statement. □

The nonadic calendar provides an effective realisation of these conditions. Take the one-object category with arrows $C_9 = \mathbb{Z}/9\mathbb{Z}$, coefficients $A = \mathbb{R}[z, z^{-1}]$, and identity action. For representatives $0 \le a, b < 9$, define

$$c_0(a,b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \qquad \Omega(b,a) = z^{c_0(a,b)}. \tag{10.94}$$

The formal variable z preserves the number of circuits. Normalisation follows from $c_0(0,a) = c_0(a,0) = 0$. If $[a+b]_9$ denotes the reduced representative, Euclidean division gives

$$c_0(a,b) + c_0([a+b]_9, c) = \left\lfloor \frac{a+b+c}{9} \right\rfloor = c_0(b,c) + c_0(a,[b+c]_9).$$

This equality proves the cocycle (10.92) for the multipliers, and hence the associativity of the calendar with memory.

If $T = T_{[1]_9}$, the product rules give $T^j = T_{[j]_9}$ for $0 \le j < 9$ and $T^9 = zT_{[0]_9}$. Subsequently imposing $z^{12} = 1$ yields $T^{108} = 1$. The expressions $z^k T_{[j]_9}$, $0 \le k < 12$, $0 \le j < 9$, form the 108 positions of the resulting calendar. This recovers the cyclic group C_{108} and its projection onto C_9; retaining z without that relation distinguishes successive circuits. This is a realisation of calendar memory: the remaining coordinates of the TPK histories remain in their own domain.

10.8.4 Double variance of refinement and preservation of unity

Restriction of states induces a prolongation of observables in the opposite direction. For the function algebras with pointwise multiplication $D_n = \mathrm{Fun}(X_n, \mathbb{R})$, precomposition defines

$$r^*_{m,n} : D_n \longrightarrow D_m, \qquad r^*_{m,n} f = f \circ r_{m,n}. \tag{10.95}$$

The contravariant order is preserved: $r^*_{\ell,n} = r^*_{\ell,m} \circ r^*_{m,n}$. These maps are unital homomorphisms. If $r_{m,n}$ is surjective, then $r^*_{m,n}$ is also injective: a difference between two functions can be evaluated at a preimage of a point where they differ.

Lemma 10.86 (Indicators, unity, and compatible evaluation)**.** *For $x \in X_n$, let e_x be its indicator function. Then*

$$r^*_{m,n} e_x = \mathbf{1}_{r^{-1}_{m,n}(x)}, \qquad r^*_{m,n} 1_n = 1_m. \tag{10.96}$$

When the fibre $r^{-1}_{m,n}(x)$ is finite, the first equality is the finite sum

$$r^*_{m,n} e_x = \sum_{y : r_{m,n}(y) = x} e_y.$$

For every section $x = (x_n) \in X_\infty$, the rule $[f] \mapsto f(x_n)$, with $f \in D_n$, defines a unital homomorphism

$$\mathrm{Ev}_x : D_{\mathrm{cyl}} \longrightarrow \mathbb{R}, \qquad D_{\mathrm{cyl}} = \varinjlim_n (D_n, r^*_{n+1,n}). \tag{10.97}$$

Proof. For $y \in X_m$, we have $(r^*_{m,n} e_x)(y) = 1$ exactly when $r_{m,n}(y) = x$. This is the first equality of (10.96). If the fibre is finite, its point indicators are orthogonal and sum to its indicator. The second equality follows from $1_n(r_{m,n}(y)) = 1$, without any assumption on the cardinality of the fibre.

For every $f \in D_n$, the naturality of evaluation is

$$(r^*_{m,n} f)(y) = f(r_{m,n}(y)).$$

In a compatible section, $r_{m,n}(x_m) = x_n$, hence $(r^*_{m,n} f)(x_m) = f(x_n)$. Thus two representatives identified at a common depth have the same evaluation. Point evaluation preserves addition, multiplication, and unity, completing the proof of (10.97). □

For infinite fibres the formulation through $\mathbf{1}_{r^{-1}_{m,n}(x)}$ remains valid; there is no need to interpret an infinite sum as a finite algebraic sum. The direct limit gathers the finite-depth observables with the identifications induced by the restrictions. Prolonging route operators additionally requires the compatibility squares of their transports with those restrictions.

The preserved unity is the constant function one on the domain of possibilities. An individual section selects a compatible history in that domain. Refinement may increase depth and distinguish more prolongations while preserving unity and the evaluation of each antecedent. This unital preservation provides the precise algebraic content used here; a probability measure, when it enters another reader, additionally requires its normalisation and its own rules.

10.9 Coinductive depth of the fourfold relation

The regional readers and dodecaphase closure publish, respectively, π, φ, e and α from the common enriched state. We reserve N for operator depth and n for decimal length: the two indices are not identified and no uniform reading cost is assumed. For each exact realization χ, its integer part is preserved separately and $p_{\chi,n}$ denotes the prefix of length n of its fractional part, in the canonical decimal expansion not eventually equal to nine. Define

$$\rho^{\mathrm{dec}}_{n+1,n}(d_1 \dots d_n d_{n+1}) = d_1 \dots d_n, \qquad p_{\chi,0} := \varnothing.$$

The uniqueness of $p_{\pi,n}, p_{\varphi,n}, p_{e,n}$ follows from the uniqueness of the exact character already generated and of its canonical decimal cylinder. That of $p_{\alpha,n}$ follows from the integral dodecaphase closure, its tail identity, and the projective compatibility proved in Theorem 11.70. The ninth-order analytic resolvent provides an additional finite characterization, while the complete analytic reader computes the remaining coefficients by $b_{10+3n} = \lambda/729^n$, proves uniform convergence, and constructs rational intervals contained in the integral cylinders. It thereby identifies that same section at arbitrary depth by Theorem 11.76. The finite jet is used neither as a generator nor as a substitute for the coinductive prolongation.

Theorem 10.87 (Projective compatibility of π, φ, e, α at arbitrary depth)**.** *The coinductive action of the common state determines*

$$\begin{gathered} \forall \chi \in \{\pi, \varphi, e, \alpha\}\ \forall n \geq 1 \quad \exists!\, p_{\chi,n}, \\ \rho^{\mathrm{dec}}_{n+1,n}(p_{\chi,n+1}) = p_{\chi,n}, \qquad p_{\chi,\infty} := (p_{\chi,n})_{n\geq 0} \in \varprojlim_{n\geq 0} \{0,\dots,9\}^n. \end{gathered} \tag{D1}$$

In particular,

$$(p_{\pi,\infty}, p_{\varphi,\infty}, p_{e,\infty}, p_{\alpha,\infty}) \tag{D2}$$

is a correlated coinductive publication of the unique enriched state.

Proof. Result 10.51 fixes the exact realizations of the three regional readers. Theorem 11.70 constructs the complete section α_{HMT} and proves compatibility of all its prefixes. The canonical decimal expansion of each of these four realizations has a unique prefix of each length; truncating the prefix of length $n+1$ gives the one of length n. Applying those readers to the common state gives the stated correlated family. Existence and compatibility of this exact publication do not identify n with N or add a cost bound to the reading procedures already established. □

The internal constructive order preserves the readers of π, φ, e, the signed state, the integer dodecaphase seal K and the dodecaphase wheel that publishes α; no conventional constant is promoted to a generator. An execution at one thousand, ten thousand or one million digits verifies the corresponding finite computational instance of (D1); it does not constitute a bound on the theorem. The theorem is the compatible family for every n and the infinite section determined by it.

This depth is not used to infer the continuum hypothesis. The four publications certify the coinductive action of the same generator; cardinal closure has already been obtained through coverage, condensation and the two injections. Its verification criterion is twofold: compatibility of all truncations of (D1) and internal provenance of the data of the dodecaphase wheel that publishes α_{HMT}.

Chapter 11

Internal derivation of α through the reversible dodecaphasic closure of the signed state

11.1 Initial generation of the signed dodecaphase state and reconstruction of the seal K

The construction of the integer seal does not begin with α, its inverse or a metrological table. Its domain is the enriched terminal state of the Topological Phase Kernel obtained after the already constructed chain

$$\boxed{\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \Omega_{\mathrm{seed}} \xrightarrow{T_{\min}} \mathcal{U}_6^{\mathrm{cat}} \xrightarrow{\rho_3} W_{243} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9 \longrightarrow x_{\mathrm{term}}.} \tag{11.1}$$

The foregoing arrows were defined with their respective domains: pairs of seeds, signatures, emissions, ternary words, boundaries, compatible histories, and nonadic holonomy. This unit does not reselect any of those data. It constructs the integral charts of the same terminal state and proves their equivalence.

11.1.1 Oriented tracis over windows of length 108

Let

$$\Theta_{108} = \mathbb{Z}/108\mathbb{Z}$$

the observable calendar. Its twelve nonadic windows are

$$E_m = \{9m + 1, \dots, 9m + 9\}, \qquad 0 \le m < 12.$$

The partition is disjoint and exhaustive:

$$\{1, \dots, 108\} = \bigsqcup_{m=0}^{11} E_m.$$

The set of address codes that publish an event is fixed as

$$D_{\mathrm{evt}} = \{2, 4, 6, 8\}.$$

The orientation of the twelve windows is

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -). \tag{11.2}$$

If d_t is the direction encoded at step t, the signed event of window m is

$$e_m = \Sigma_{12}(m)\, \#\{t \in E_m : d_t \in D_{\mathrm{evt}}\}. \tag{11.3}$$

The formula distinguishes the nonnegative number of events from the sign of the orientation. It does not replace a cardinal direction with a sign: both coordinates remain present in the TPK state.

Definition 11.1 (Enhanced dodecaphonic registration)**.** The record transported by the trace is

$$\mathcal{R}_{12}(x) = \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=0}^{11},$$

where τ_m is the TRIT orientation, σ_m the boundary signature, κ_m the carry, and Λ_m the genealogical register segment that preserves the route. There is a chain of projections

$$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow \mathcal{R}_{12} \longrightarrow \Theta_{108},$$

but neither of the two arrows is an identification.

In particular, Θ_{108} preserves only the position in the calendar. The record $\mathcal{R}_{12}$ additionally preserves orientation, signature, carry, and provenance. The complete state further preserves cylinder, boundary, quotient, terminal, and vertical memory.

11.1.2 Integral chart of the dodecaphasic opposition

For $0 \le i < 6$, the opposite positions determine

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}. \tag{11.4}$$

The central charge and the five relative margins are

$$s_C = \sum_{i=0}^{5} p_i, \qquad M_i = p_i - p_5 \quad (0 \le i < 5). \tag{11.5}$$

Definition 11.2 (Integral chart $1 + 5 + 6$)**.** It is defined

$$\mathcal{L}_{12}(e_0, \dots, e_{11}) = \big(s_C, M_0, \dots, M_4, a_0, \dots, a_5\big). \tag{11.6}$$

The coordinate partition is

$$1 + 5 + 6 = 12.$$

Theorem 11.3 (Exact inverse of the integral chart)**.** *A tuple*

$$\ell = (s_C, M_0, \dots, M_4, a_0, \dots, a_5) \in \mathbb{Z}^{12}$$

belongs to the image of $\mathcal{L}_{12}$ if and only if

$$s_C - \sum_{i=0}^{4} M_i \equiv 0 \pmod 6, \tag{11.7}$$

$$p_i \equiv a_i \pmod 2 \quad (0 \le i < 6), \tag{11.8}$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \tag{11.9}$$

$$p_i = p_5 + M_i \quad (0 \le i < 5). \tag{11.10}$$

In that case, the only entire preimage is

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2}. \tag{11.11}$$

Proof. Summing $M_i = p_i - p_5$ for $0 \le i < 5$, we obtain

$$\sum_{i=0}^{4} M_i = \sum_{i=0}^{4} p_i - 5p_5 = s_C - 6p_5,$$

which forces (11.9) and the congruence modulo six. Once the six p_i have been reconstructed, the system

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}$$

has the unique solution (11.11). This solution is integral exactly when p_i and a_i have the same parity. The two families of congruences are therefore necessary and sufficient. □

This theorem precedes the numerical closure of α. It explains how the oriented trace produces a reversible integral chart, and why preserving the reduced calendar is not enough.

11.1.3 Selection of the common terminal state

The three compatible sections reach terminal depth with catalogs of cardinalities

$$|\mathcal{T}_\pi| = 196, \qquad |\mathcal{T}_e| = 197, \qquad |\mathcal{T}_\varphi| = 196.$$

The product contains

$$196 \cdot 197 \cdot 196 = 7\,567\,952 \tag{11.12}$$

ordered triples.

The column margin of the terminal reader is

$$q_\partial = (1, 2, 2, 1, 2, 0) \in \mathbb{F}_3^6,$$

and the row orientation of the combined channel is

$$\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3.$$

The first datum acts on columns; the second acts on rows. Neither is deducible from the other.

The local signatures of Gram matrix, norm, weight, and distance reduce (11.12) to 331 triples. The margin q_∂ leaves two indices:

$$(120, 197, 157), \qquad (129, 197, 148).$$

Their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \qquad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row sums are

$$(0, 2, 0), \qquad (2, 2, 1).$$

Since

$$\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\},$$

only B_1 satisfies the axial condition.

Theorem 11.4 (Uniqueness of the Visible Terminal Publication)**.** *The joint selection determines*

$$B_{\text{term}} = \begin{pmatrix} 021020 \\ 001220 \\ 100210 \end{pmatrix}, \tag{11.13}$$

corresponding to the triple $(129, 197, 148)$. *No decimal evaluation of α enters the census or the two selection conditions.*

Proof. The census and margin leave exactly B_0, B_1. The charge of B_0 does not belong to $\langle \sigma \rangle$, whereas that of B_1 is 2σ. The second candidate is therefore the only one that simultaneously satisfies both typed requirements. □

11.1.4 Archimedean and exceptional publications of the common state: inequivalent codomains and shared genealogy

Let

$$x_{\mathrm{term}} \in \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}} \subseteq \mathcal{X}_{\mathrm{TPK}}^{[108]}.$$

The visible reading is

$$\pi_{\mathrm{term}}(x_{\mathrm{term}}) = B_{\mathrm{term}}.$$

The signed chart preserves another coordinate of the same state.

Definition 11.5 (Terminal signed causal reading)**.** The domain incorporates neither K nor U as input coordinates. The state x_{term}, produced by the full composition APP $\to$ TRIT $\to$ TPK, contains the enriched terminal register, and the extractor of its twelve oriented windows is

$$R_{12} : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}} \longrightarrow \mathcal{R}_{12}^{\mathrm{or}}, \qquad R_{12}(x_{\mathrm{term}}) = \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=0}^{11}. \tag{11.14}$$

The ledger's channel reader 90/120 emits, without receiving the result as data,

$$E_{108}^{90,120} : \mathcal{R}_{12}^{\mathrm{or}} \longrightarrow \mathbb{Z}^{12} \times \mathbb{Z}^{12} \times \mathbb{Z}, \qquad E_{108}^{90,120}\big(R_{12}(x_{\mathrm{term}})\big) = (b_{90}, b_{120}, Q_{\mathrm{TPK}}),$$

with

$$\begin{aligned} b_{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461),\\ b_{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128),\\ Q_{\mathrm{TPK}} &= 6263. \end{aligned}$$

For $r = 1, 2, 3$ and $j = 0, 1, 2, 3$, define

$$\begin{aligned} B_r &= \sum_{j=0}^{3} b_{120,r+3j}, \qquad d_{r,j} = b_{90,r+3j},\\ A_1 &= \frac{Q_{\mathrm{TPK}} + 2B_1 + B_2}{3}, \qquad A_2 = A_1 - B_1, \quad A_3 = A_2 - B_2. \end{aligned}$$

The publication Π_H forms the three blocks

$$U^{(r)} = \big(A_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}\big),$$

and reinterleaves them in dodecaphasic order, that is,

$$\Pi_H(b_{90}, b_{120}, Q_{\mathrm{TPK}}) := \mathrm{reint}_3\big(U^{(1)}, U^{(2)}, U^{(3)}\big) = U.$$

Let $\mathcal{H}_{12}$ be the integral transformation to be constructed in the next section. The publication module is

$$\mathcal{U}_{12}^{\mathrm{sgn}} := \mathrm{im}\big(\mathcal{H}_{12} : \mathbb{Z}^{12} \to \mathbb{Z}^{12}\big). \tag{11.15}$$

The signed reading is, causally, the composition

$$R_{\mathrm{sgn}} := \Pi_H \circ E_{108}^{90,120} \circ R_{12} : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}} \longrightarrow \mathcal{U}_{12}^{\mathrm{sgn}}, \qquad R_{\mathrm{sgn}}(x_{\mathrm{term}}) = U. \tag{11.16}$$

Only after this output is $K = \mathcal{H}_{12}^{-1}U$ reconstructed; the identity $U = \mathcal{H}_{12}K$ certifies that reconstruction and does not define the reader's domain.

The correct joint writing is

$$x_{\mathrm{term}} \xrightarrow{(\pi_{\mathrm{term}}, R_{\mathrm{sgn}})} (B_{\mathrm{term}}, U_{12}^{\mathrm{sgn}}). \tag{11.17}$$

No arrow $B_{\mathrm{term}} \to U_{12}^{\mathrm{sgn}}$ is interposed: the visible matrix has forgotten precisely the orientation, opposition, quotient, carry, and genealogical record coordinates used by the signed chart. The source of both publications is the enriched state produced by (11.1).

The evaluation of the signed record gives

$$\boxed{U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997).} \tag{11.18}$$

11.1.5 Naturality with respect to depth

For $N' \geq N \geq 108$, the truncations of the signed system are

$$\tau_{N',N}^{\mathrm{sgn}} x_{\mathrm{term}}^{[N']} = \tau_{N',N} x_{\mathrm{term}}^{[N']}. \tag{11.19}$$

Therefore, truncation acts only on the history and the enriched state; it does not carry either K or U as independent inputs.

Theorem 11.6 (Square of naturality of the signed reading)**.** *We have*

$$\boxed{R_{\mathrm{sgn},N} \circ \tau_{N',N}^{\mathrm{sgn}} = \mathrm{id}_{\mathcal{U}_{12}^{\mathrm{sgn}}} \circ R_{\mathrm{sgn},N'}.} \tag{11.20}$$

Proof. The naturality of the states literally preserves the first one hundred and eight windows of the ledger:

$$R_{12,N} \circ \tau_{N',N}^{\mathrm{sgn}} = R_{12,N'}.$$

Truncation can modify only the subsequent history. Since $E_{108}^{90,120}$ and Π_H depend exclusively on those windows, both sides return the same U. Commutativity is an equality of maps, not an analogy between diagrams. □

11.1.6 Orbits of step 3 and transform of Hadamard

The cyclic order of the seal is distributed over the three orbits.

$$K^{(1)} = (K_1, K_4, K_7, K_{10}), \tag{11.21}$$

$$K^{(2)} = (K_2, K_5, K_8, K_{11}), \tag{11.22}$$

$$K^{(3)} = (K_3, K_6, K_9, K_{12}). \tag{11.23}$$

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{11.24}$$

If P_{orb} reorders the twelve coordinates according to (11.21)–(11.23), the global transformation is

$$\mathcal{H}_{12} = P_{\mathrm{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4) P_{\mathrm{orb}}. \tag{11.25}$$

The three blocks of (11.18), read by orbits, are

$$U^{(1)} = (2378, -452, -668, -322),$$
$$U^{(2)} = (1406, 998, -204, -28),$$
$$U^{(3)} = (2479, -551, -371, -997).$$

Lemma 11.7 (Algebraic Hadamard quarter-turn)**.** *The matrix H_4 satisfies*

$$H_4^2 = 4I_4, \qquad H_4^{-1} = \frac{1}{4}H_4, \qquad \det H_4 = -16. \tag{11.26}$$

Proof. The rows of H_4 are orthogonal, and each has squared norm four. Since the matrix is symmetric, $H_4^{\mathsf{T}}H_4 = H_4^2 = 4I_4$. The formula for the inverse and the determinant follow immediately. □

Theorem 11.8 (Unique full reconstruction of the seal)**.** *The inverse*

$$K^{(r)} = \frac{1}{4}H_4U^{(r)} \qquad (r = 1, 2, 3) \tag{11.27}$$

yields

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 058, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

By undoing the orbital order one obtains

$$\boxed{K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601).} \tag{11.28}$$

It is the unique rational preimage of U, and it belongs to $\mathbb{Z}^{12}$.

Proof. By (11.26), the rational preimage is unique. The explicit multiplications are

$$\begin{aligned} H_4U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4U^{(2)} &= (2172, 2636, 232, 584), \\ H_4U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and reinterleaving yields (11.28). □

The factor $1/4$ is not chosen to fit an output: it is the inverse forced by $H_4^2 = 4I_4$. Nor is it the square root of -1. The internal complex structure $A_W^2 = -I$ will appear after constructing the Paley matrix; these are two facts of different types.

11.1.7 Integral structure and Smith normal form

Theorem 11.9 (Integral image of H_4)**.** *The Smith normal form is*

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4). \tag{11.29}$$

Consequently,

$$\operatorname{coker} \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3, \tag{11.30}$$

and the image of $\mathcal{H}_{12}$ has index

$$16^3 = 4096$$

in $\mathbb{Z}^{12}$. For $u \in \mathbb{Z}^4$,

$$u \in H_4\mathbb{Z}^4 \quad \Longleftrightarrow \quad H_4u \equiv 0 \pmod 4. \tag{11.31}$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has absolute value two; those of order three are multiples of four, and one has absolute value four; the determinant has modulus sixteen. The determinantal divisors are $1, 2, 4, 16$, yielding the invariant factors

$1, 2, 2, 4$. The triple direct sum repeats the factors and multiplies the indices. Finally, $H_4^{-1} = H_4/4$, so the preimage is integral exactly under the congruence (11.31). □

11.1.8 Certification and reconstruction by differences of steps 3 and 4

Once the ledger has emitted $(b_{90}, b_{120}, Q_{\mathrm{TPK}})$, Π_H has produced U and Hadamard has reconstructed K, the following coordinates act downstream as an independent certificate of those prior outputs. With cyclic indices modulo twelve, define

$$(D_3K)_m = K_m - K_{m+3}, \qquad (D_4K)_m = K_m - K_{m+4}, \qquad Q(K) = \sum_{m=1}^{12} K_m. \tag{11.32}$$

For (11.28), one verifies

$$\begin{aligned} D_3K = b_{90} = (&-495, -116, -684, 108, 601, -90,\\ &-173, -88, 313, 560, -397, 461),\\ D_4K = b_{120} = (&-425, -281, -481, 671, -255, 30,\\ &475, -543, 680, 251, 6, -128),\\ Q(K) = Q_{\mathrm{TPK}} = 6263.& \end{aligned}$$

Theorem 11.10 (Completeness of the transversal extractor)**.** *The operator*

$$\mathcal{E}_{\mathrm{dod}} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and Smith normal form

$$\mathrm{SNF}(\mathcal{E}_{\mathrm{dod}}) = \mathrm{diag}(\underbrace{1, \ldots, 1}_{11}, 12). \tag{11.33}$$

In particular, $(D_3K, D_4K, Q(K))$ determines a unique K. This uniqueness certifies and reconstructs the data already emitted by the ledger; it neither generates nor selects them.

Proof. If $D_3v = D_4v = 0$, then v is constant along steps three and four. Since $\gcd(3, 4) = 1$, those steps generate all of $\mathbb{Z}/12\mathbb{Z}$, so that

$$\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle.$$

The total charge does not vanish on that line, since $Q(\mathbf{1}) = 12$. The rank is twelve. The rows of (D_3, D_4) are oriented incidences of a connected Cayley graph; a spanning tree supplies eleven unit invariant factors. Every maximal minor must contain the row Q, and the unimodular operations reduce its final contribution to twelve. There is a maximal minor of modulus twelve; therefore, the last factor is exactly twelve. □

The certifying reconstruction of U from these coordinates reproduces the prior publication of Π_H and does not require multiplying again by H_4. For $r = 1, 2, 3$, let

$$B_r = \sum_{j=0}^{3} (D_4K)_{r+3j} = \sum_{j=0}^{3} b_{120,r+3j}. \tag{11.34}$$

Then

$$(B_1, B_2, B_3) = (972, -1073, 101).$$

Orbital sums are

$$A_1 = \frac{Q + 2B_1 + B_2}{3} = 2378, \tag{11.35}$$

$$A_2 = A_1 - B_1 = 1406, \tag{11.36}$$

$$A_3 = A_2 - B_2 = 2479. \tag{11.37}$$

If $d_{r,j} = (D_3K)_{r+3j} = b_{90,r+3j}$, the signed block of orbit r is

$$\big(A_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}\big), \tag{11.38}$$

which reproduces the three blocks $U^{(r)}$ already published by Π_H.

11.1.9 Threshold Tetrad of the Seal

The local decimal completion satisfies

$$1000 = 729 + 271.$$

The support of the seal coordinates that reach the threshold 729 is

$$\boxed{B_K := \{m : K_m \geq 729\} = \{4, 6, 9, 10\}.} \tag{11.39}$$

This tetrad is obtained after constructing K. In this unit it is not yet called a Witt star: that object will be defined only after constructing the ternary code, its hexads, and the Steiner system.

11.1.10 Independence of the genealogical construction with respect to external metrological or physical data

Theorem 11.11 (Isolation of U and K)**.** *The composition*

$$\Omega_{\text{seed}} \longrightarrow x_{\text{term}} \xrightarrow{R_{\text{sgn}}} U \xrightarrow{\mathcal{H}_{12}^{-1}} K \tag{11.40}$$

does not receive α, α^{-1}, CODATA or a target decimal string as input. The seal K is determined before defining the recurrence that will produce α.

Proof. The first part of the composition uses only seeds, emissions, ternary readers, L_0, L_1 operators, survival relations, phase transport and the charts of the TPK state. The reading $R_{\text{sgn}} = \Pi_H \circ E_{108}^{90,120} \circ R_{12}$ is the causal composition of the terminal register, and the last arrow is the linear transformation $\mathcal{H}_{12}^{-1} = \mathcal{H}_{12}/4$ on its integral image. None of the formulas contains a physical variable or a metrological table. The base-one-thousand recurrence is introduced only in the next unit, after K has been fixed. □

11.1.11 Obstructions of the dodecaphase record and the seal K: naturality, Hadamard integrity, and metrological isolation

Comparison Criterion 11.12 (Refutation of the register and the seal)**.** The construction is refuted if any of the following facts is established:

1. the windows E_m do not partition the one hundred eight steps;
2. the chart $\mathcal{L}_{12}$ fails its inverse or its congruences;
3. the terminal census does not produce 331 triples, two matrices and a unique axial charge;
4. an attempt is made to obtain U from B_{term} after forgetting the signed coordinates;
5. the naturality square (11.20) fails;
6. $H_4^2 \neq 4I_4$, some preimage is not integral or the reconstructed vector differs from (11.28);

7. the Smith normal form differs from (11.29);
8. the certificate (D_3, D_4, Q) loses rank, does not match the ledger's prior emissions or does not reproduce U;
9. B_K is introduced before constructing K, or is assigned a Witt structure before defining W_{12};
10. a digit of α or a metrological reference intervenes in any of the arrows of (11.40).

11.2 Initial construction of the boundary 662/663 through the dodecaphase carry cocycle

11.2.1 Aritmetic closure domain

The construction receives four dodecaphasic coordinates obtained in the preceding units:

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884),\\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497),\\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117),\\ K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \end{aligned}$$

The first three vectors are the first twelve triads of the fractional Archimedean readings of π, e, and φ. The fourth is the dodecaphase coordinate of the enriched terminal state of the Topological Phase Kernel, constructed before the carry operation is opened.

We fix the base.

$$B := 10^3 = 1000$$

and the concatenation

$$\iota(v_1, \dots, v_{12}) := \sum_{m=1}^{12} v_m B^{12-m}. \tag{11.41}$$

In particular,

$$\begin{aligned} \iota(P) &= 141592653589793238462643383279502884,\\ \iota(E) &= 718281828459045235360287471352662497,\\ \iota(\Phi) &= 618033988749894848204586834365638117,\\ \iota(K) &= 234543140729659824621058914794146601. \end{aligned}$$

The operation studied in this unit is the map

$$\mathfrak{C}_{12} : (P, E, \Phi, K, c_{13}) \longmapsto (A, C), \tag{11.42}$$

where $A = (A_1, \dots, A_{12})$ is the output word and

$$C = (c_1, \dots, c_{13})$$

is the complete system of thirteen Euclidean quotients: twelve internal quotients and one right boundary condition.

11.2.2 Left-to-right oriented Euclidean division

For $m = 12, 11, \dots, 1$, define

$$\begin{aligned} s_m &:= P_m + E_m - \Phi_m - K_m + c_{m+1},\\ A_m &:= s_m \bmod B, \qquad 0 \le A_m < B,\\ c_m &:= \frac{s_m - A_m}{B} = \left\lfloor \frac{s_m}{B} \right\rfloor. \end{aligned} \tag{11.43}$$

The second equality for c_m is also valid when $s_m < 0$, because Euclidean division with remainder in $\{0, \dots, B-1\}$ is used. Therefore,

$$-431 = 569 - 1000, \qquad -717 = 283 - 1000, \qquad -1200 = 800 - 2 \cdot 1000,$$

and negative loans do not introduce ambiguity.

Theorem 11.13 (Existence and Uniqueness of the Oriented Carry)**.** *Given P, E, Φ, K and the terminal carry c_{13}, the recurrence* (11.43) *uniquely determines*

$$(A_1, \dots, A_{12}; c_1, \dots, c_{12}).$$

Proof. Row $m = 12$ is determined by the Euclidean division of

$$P_{12} + E_{12} - \Phi_{12} - K_{12} + c_{13}$$

by B. Once c_{12} is known, row $m = 11$ is determined by the same property. Descending induction reaches $m = 1$. At each step, uniqueness of the Euclidean quotient and remainder prevents two distinct outputs. □

11.2.3 Finite restriction of the signed dodecaphase state and reversible closure of the cycle

The finite window is defined by the terminal condition

$$c_{13}^{\mathrm{fin}} = 0.$$

The complete evaluation is as follows. The table is printed in ascending order, although it must be read causally from the last row to the first.

Table 11.1: The twelve Euclidean divisions and thirteen carries of the finite dodecaphasic window.

m	P_m	E_m	Φ_m	K_m	c_{m+1}	s_m	A_m	c_m
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	−1	352	352	0
4	589	459	749	729	−1	−431	569	−1
5	793	045	894	659	−2	−717	283	−1
6	238	235	848	824	−1	−1200	800	−2
7	462	360	204	621	0	−3	997	−1
8	643	287	586	058	−1	285	285	0
9	383	471	834	914	−1	−895	105	−1
10	279	352	365	794	0	−528	472	−1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

The complete word of thirteen carries is

$$\boxed{C^{\mathrm{fin}} = (c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0).}$$

The dodecaphasic output is

$$\boxed{A^{\mathrm{fin}} = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 663).}$$

We define the finite rational

$$\boxed{\alpha_{12}^{\mathrm{fin}} := 10^{-36} \iota(A^{\mathrm{fin}}) = \frac{729735256928380099728510547238066 3}{10^{36}}.} \tag{11.44}$$

Therefore,

$$\alpha_{12}^{\mathrm{fin}} = 0.007297352569283800997285105472380663.$$

11.2.4 Posterior metrological recognition: CODATA 2018

The metrological comparison is performed only after closing the recurrence, the right boundary, and the preceding rational. The integer identity gives

$$\boxed{\alpha_{12}^{\mathrm{fin}} = 10^{-36} \left\lfloor \frac{10^{36}}{137.035999084} \right\rfloor .} \tag{11.45}$$

Consequently,

$$(\alpha_{12}^{\mathrm{fin}})^{-1} = 137.035999084000000000000000000000001066588\ldots, \tag{11.46}$$

and exactly reproduces the central chain published by CODATA 2018 [122]:

$$\boxed{137.035999084.}$$

The word *exactly* refers to the published central chain, not to a purported experimental measurement of thirty-six digits. The subsequent digits in (11.46) are consequences of the HMT rational already fixed. CODATA does not select the state, does not determine (K), does not decide the direction of the carry, and does not intervene in the dodecaphase table.

11.2.5 Complete affine identity

Each row of the table satisfies exactly

$$P_m + E_m - \Phi_m - K_m + c_{m+1} = A_m + Bc_m. \tag{11.47}$$

Introducing

$$C^- := (c_1, \ldots, c_{12}), \qquad C^+ := (c_2, \ldots, c_{13}),$$

the twelve local identities come together in

$$\boxed{P + E - \Phi - K + C^+ - BC^- = A.} \tag{11.48}$$

The term

$$\partial_B C := C^+ - BC^-$$

is the carry cocycle of the oriented chain. Consequently, $P + E - \Phi - K = A$ is not a coordinatewise equality in $\mathbb{Z}^{12}$: it becomes correct only upon incorporating $\partial_B C$.

Theorem 11.14 (Telescoping of the Thirteen Carries)**.** *For any terminal condition c_{13}, the recurrence satisfies*

$$\boxed{\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) - \iota(A) = B^{12}c_1 - c_{13}.} \tag{11.49}$$

In the finite window, $c_1 = c_{13} = 0$, and therefore

$$\boxed{\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) = \iota(A^{\mathrm{fin}}).} \tag{11.50}$$

Proof. We multiply (11.47) by B^{12-m} and sum over $m = 1, \ldots, 12$. The contribution of the carries is

$$\sum_{m=1}^{12} (Bc_m - c_{m+1})B^{12-m}.$$

The internal components cancel:

$$\sum_{m=1}^{12}(Bc_m - c_{m+1})B^{12-m} = \sum_{m=1}^{12} c_m B^{13-m} - \sum_{m=1}^{12} c_{m+1}B^{12-m}$$
$$= B^{12}c_1 - c_{13}.$$

This proves (11.49). The specialization $c_1 = c_{13} = 0$ gives (11.50). □

11.2.6 Exact inversion of the seal K

The local recurrence is reversible. Solving for K_m in (11.47),

$$\boxed{K_m = P_m + E_m - \Phi_m + c_{m+1} - A_m - Bc_m.} \tag{11.51}$$

By concatenation,

$$\boxed{\iota(K) = \iota(P) + \iota(E) - \iota(\Phi) - \iota(A) - B^{12}c_1 + c_{13}.} \tag{11.52}$$

On the finite boundary,

$$\iota(K) = \iota(P) + \iota(E) - \iota(\Phi) - \iota(A^{\mathrm{fin}}).$$

Algebraic reversibility does not invert the causal order. The effective order is

$$x_{\mathrm{term}} \longrightarrow K \longrightarrow (P, E, \Phi, K, c_{13}) \longrightarrow (A, C).$$

Formula (11.51) is a commutativity check: it proves that the output and the carries recover the seal that was already fixed. It is not a rule for retrospectively selecting K from a desired value of α.

11.2.7 Coinductive prolongation of the dodecaphasic seal and inverse carry recurrence in $C = \{-2, -1, 0, 1, 2\}$

Let

$$(P_k)_{k\geq 1}, \qquad (E_k)_{k\geq 1}, \qquad (\Phi_k)_{k\geq 1}$$

the infinite sequences of triads produced by the three coinductive branches, and let

$$\kappa(k) := 1 + ((k-1) \bmod 12).$$

The prolongation of the stamp is periodic:

$$K_k^{\mathrm{per}} := K_{\kappa(k)}.$$

For a finite depth N and a terminal condition $t = c_{N+1}$, the recurrence is

$$\begin{aligned} s_k^{(N,t)} &= P_k + E_k - \Phi_k - K_{\kappa(k)} + c_{k+1}^{(N,t)}, \\ A_k^{(N,t)} &= s_k^{(N,t)} \bmod 1000, \\ c_k^{(N,t)} &= \left\lfloor \frac{s_k^{(N,t)}}{1000} \right\rfloor, \qquad k = N, \ldots, 1. \end{aligned} \tag{11.53}$$

The set of terminal carries

$$C := \{-2, -1, 0, 1, 2\}$$

is invariant for the certified inputs: for each triad and each $c_{k+1} \in C$, the quotient c_k again belongs to C. No favorable terminal condition is chosen. All five conditions are propagated, and only the prefix common to all of them is published.

The execution of certified depth employs

quantity	value
ternary blocks per channel	3501
trits per channel	$3501 \cdot 6 = 21006$
guard decimal digits	10020
guard triads	3340
propagated terminal conditions	5
common triads	3339
common digits	10017
published digits	10000

The five terminal branches coalesce before the published prefix. In particular, no terminal condition extracted from a target value of α intervenes in the output.

Theorem 11.15 (Stabilized Output Compatibility)**.** *Let $A^{[N]}$ and $A^{[N']}$, with $N' > N$, be two runs whose five terminal conditions have coalesced before position N. Then the truncation of $A^{[N']}$ to its first N blocks coincides with $A^{[N]}$. The stabilized prefixes define a unique section in the inverse limit of finite words.*

Proof. The recurrence is deterministic from right to left once the carry entering a position is known. Coalescence asserts that all remote conditions produce the same carry before the boundary of the prefix. From that point, the same triads P_k, E_k, Φ_k and the same periodic seal produce, by Euclidean uniqueness, the same remainders and quotients at all preceding positions. This is exactly commutativity with truncation. □

11.2.8 Boundary originating from the remote tail

The prolongation determines on the visible boundary.

$$c_{13}^{\infty} = -1.$$

The twelve internal carries remain the same as in the finite window:

$$\boxed{C_{\le 13}^{\infty} = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, -1).}$$

Only the information entering from the thirteenth triad changes. The two relevant divisions are

$$\begin{aligned} s_{12}^{\infty} &= 884 + 497 - 117 - 601 - 1 = 662, \\ A_{12}^{\infty} &= 662, \qquad c_{12}^{\infty} = 0, \end{aligned}$$

and

$$\begin{aligned} s_{13}^{\infty} &= 197 + 757 - 720 - 234 - 1 = -1, \\ A_{13}^{\infty} &= 999, \qquad c_{13}^{\infty} = -1. \end{aligned}$$

Therefore, the beginning of the prolonged word is

$$\boxed{A^{\infty} = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \ldots).}$$

The corresponding expansion begins

$$\boxed{\alpha_{\infty} = 0.007297352569283800997285105472380662999564172539642\ldots.} \tag{11.54}$$

The difference between both boundaries is local and exact:

$$c_{13}^{\mathrm{fin}} - c_{13}^{\infty} = 1, \qquad A_{12}^{\mathrm{fin}} - A_{12}^{\infty} = 1,$$

while

$$A_m^{\mathrm{fin}} = A_m^{\infty}, \qquad 1 \le m \le 11.$$

Applying the telescoping theorem to the first twelve blocks of the prolongation,

$$\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) - \iota(A_{\le 12}^{\infty}) = -c_{13}^{\infty} = 1.$$

Then

$$\boxed{\iota(A^{\mathrm{fin}}) = \iota(A_{\le 12}^{\infty}) + 1.} \tag{11.55}$$

The ending 663 does not contradict the ending 662: the former is the finite representative obtained by closing the window with $c_{13} = 0$; the latter retains the carry -1 from the remote tail.

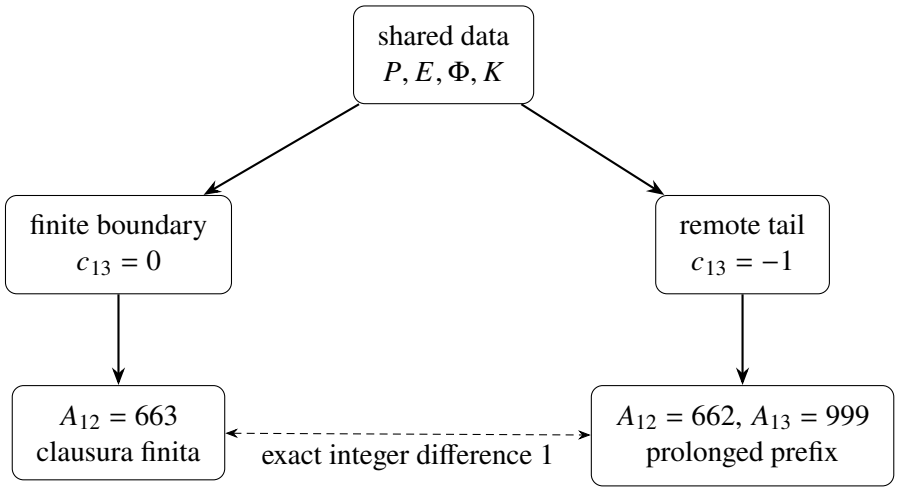


Figure 11.1: Boundary bifurcation of a single recurrence. Two distinct generators are not being compared: only the datum entering from the right changes.

11.2.9 Truncation and rounding to 36 digits

For $x \ge 0$, define

$$\mathrm{Tr}_{36}(x) := 10^{-36} \lfloor 10^{36} x \rfloor$$

and

$$\mathrm{Rd}_{36}(x) := 10^{-36} \left\lfloor 10^{36} x + \frac{1}{2} \right\rfloor.$$

Let

$$a := 729735256928380099728510547238066 2.$$

The expansion (11.54) can be written as

$$\alpha_{\infty} = \frac{a + \rho}{10^{36}}, \qquad \rho = 0.999564172539642709\ldots,$$

so that $0.999 < \rho < 1$.

Theorem 11.16 (Exact resolution of the boundary 662/663)**.** *The coinductive prolongation satisfies*

$$\boxed{\mathrm{Tr}_{36}(\alpha_{\infty}) = 0.007297352569283800997285105472380662,}$$

while its rounding to thirty-six digits satisfies

$$\boxed{\mathrm{Rd}_{36}(\alpha_\infty) = 0.007297352569283800997285105472380663 = \alpha_{12}^{\mathrm{fin}}.}$$

Proof. Since $0 \le \rho < 1$,

$$\lfloor 10^{36}\alpha_\infty \rfloor = a.$$

This gives the truncation ending in 662. Since $\rho > 1/2$,

$$\left\lfloor 10^{36}\alpha_\infty + \frac{1}{2} \right\rfloor = a + 1,$$

which ends in 663. By (11.44), $(a+1)10^{-36}$ is exactly $\alpha_{12}^{\mathrm{fin}}$. □

Moreover,

$$0 < \alpha_{12}^{\mathrm{fin}} - \alpha_\infty = \frac{1-\rho}{10^{36}} < 10^{-39}.$$

The difference is not an arithmetic error in the table: it quantifies the difference between two distinct boundary operators.

11.2.10 Causal isolation from α and metrological data

At depth N, the full domain of the operation is

$$\mathscr{D}_N = \big((P_k)_{k=1}^N, (E_k)_{k=1}^N, (\Phi_k)_{k=1}^N, K, c_{N+1}\big).$$

The carry map

$$\mathfrak{C}_N : \mathscr{D}_N \longrightarrow \{0,\dots,999\}^N \times C^N$$

is defined exclusively by (11.53). Its domain contains no target value of α, its inverse, a metrological table, a target decimal string or a terminal carry chosen by comparison.

Theorem 11.17 (No interference from the metrological target)**.** *Let $\mathscr{Y}$ be any space of external data containing metrological values or comparison tables. The formal extension*

$$\widetilde{\mathfrak{C}}_N : \mathscr{D}_N \times \mathscr{Y} \longrightarrow \{0,\dots,999\}^N \times C^N, \qquad \widetilde{\mathfrak{C}}_N(d,y) := \mathfrak{C}_N(d),$$

factorizes by the projection

$$\mathrm{pr}_{\mathscr{D}_N} : \mathscr{D}_N \times \mathscr{Y} \longrightarrow \mathscr{D}_N.$$

Therefore, for any $y, y' \in \mathscr{Y}$,

$$\widetilde{\mathfrak{C}}_N(d,y) = \widetilde{\mathfrak{C}}_N(d,y').$$

No intervention on a value of α or a metrological table alters K, the quotients, the remainders or the prefix produced.

Proof. The statement results from

$$\widetilde{\mathfrak{C}}_N = \mathfrak{C}_N \circ \mathrm{pr}_{\mathscr{D}_N}.$$

All operations of $\mathfrak{C}_N$ are additions, subtractions and Euclidean divisions of the elements of $\mathscr{D}_N$. No coordinate of $\mathscr{Y}$ appears in them. □

The complete causal order is

$$\text{enriched branches of } \pi, e, \varphi \longrightarrow (P, E, \Phi),$$

$$x_{\mathrm{term}} \longrightarrow K,$$

$$(P, E, \Phi, K, \text{boundary}) \longrightarrow (A, C) \longrightarrow \alpha_\infty,$$

$$\alpha_\infty \longrightarrow \text{external comparison.}$$

The last arrow cannot be reversed to select any of the previous ones.

11.2.11 Obstructions to affine prolongation of α: telescoping, inversion of K, boundary 662/663, and isolation of target data

Comparison Criterion 11.18 (Refutation of the affine closure and its prolongation)**.** The block is refuted in its domain if any of the following facts occurs:

1. some row of the table 11.1 fails to satisfy $s_m = A_m + 1000c_m$;
2. the published vector of thirteen carries does not reproduce the twelve rows;
3. the telescoping identity fails
$$\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) - \iota(A) = 1000^{12}c_1 - c_{13};$$
4. the inversion (11.51) does not recover the twelve coordinates of K;
5. remote propagation produces $c_{13} \neq -1$ for the certified prefix;
6. the five terminal conditions of $C = \{-2, -1, 0, 1, 2\}$ do not coalesce before the published digits;
7. the triad following 662 is not 999, in which case the identification of the rounding with 663 is no longer proved;
8. the generator opens a target expansion of α, its inverse, a metrological table, or a terminal condition chosen to force the coincidence.

Thus, two distinct boundary identities are thereby demonstrated:

$$\boxed{663 = \text{finite closure with } c_{13} = 0 = \text{rounding to 36 digits of } \alpha_\infty,}$$

$$\boxed{662 = \text{truncation to 36 digits of } \alpha_\infty \quad \text{with} \quad c_{13} = -1,\ A_{13} = 999.}$$

The central metrological result is finally published in the same residence as the derivation:

$$\boxed{\begin{array}{c}(\alpha_{12}^{\mathrm{fin}})^{-1} = 137.035999084000000000000000000000001066588\ldots, \\ \text{central CODATA 2018 string:} \qquad 137.035999084.\end{array}} \tag{11.56}$$

The implication expresses a subsequent comparison with the central string published; it does not introduce that string into the constructor.

11.3 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register articulates two constructions that should be developed in their effective order. The first produces the selector's antecedents from the regional expressions of the TPK: six-trit bands, column signatures, conversion calendar, oriented descriptor and functional panel. The second operates on these antecedents and on the terminal repertoire of eight blocks, enumerates their admissible orderings and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The register's value appears at the end of this composition.

The present development presupposes the regional expressions obtained in the preceding chapters from APP–TRIT–TPK. Its purpose is to make explicit every intermediate operation leading from those expressions to the finite selector. The realization of the same register through signed incidences, Hadamard

inversion and expression of the fine-structure constant are then incorporated. The three constructions preserve their respective domains: register selection, recovery from its channels and arithmetic evaluation of its expressions.

11.3.1 Regional bands, encoding, and integer signatures

11.3.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \qquad B_\chi(t) \in \mathbb{F}_3^6, \qquad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ preserves its position in its sequence of expressions; index t counts ternary blocks. The natural evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\mathrm{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \qquad 0 \le \mathrm{enc}_6(b) < 729. \tag{11.57}$$

Its fixed-width inverse is

$$\mathrm{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \qquad 0 \le j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

In the window of one hundred blocks, the six-hundred-trit expression of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed through

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \qquad 0 \le t < 100, \qquad B_\chi(t) = \mathrm{enc}_6^{-1}(b_\chi(t)). \tag{11.58}$$

The six-hundred-trit integer is an expression of the regional producer; the hundred-row table is the evaluation of (11.58). This separation allows a historical table to be replaced by its producer without modifying the selector that consumes it.

11.3.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t, introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \tag{11.59}$$

$$q_j(t) = T_j(t) \bmod 3, \qquad c_j(t) = \left\lfloor T_j(t)/3 \right\rfloor, \tag{11.60}$$

$$a_j(t) = \big(B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j\big) \bmod 3, \tag{11.61}$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \qquad \bar{w}_j(t) = w_j(t) \bmod 3. \tag{11.62}$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 11.19 (Reconstruction of the column sum)**.** *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \qquad 0 \le q_j(t) < 3, \quad 0 \le c_j(t) \le 2, \quad 0 \le w_j(t) \le 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \le T_j(t) \le 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. □

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates

than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

11.3.2 Conversion clock and first zero advance

11.3.2.1 Ternary capacity and decimal depth

An expression of $t+1$ bands contains $6(t+1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \le 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \le 729^{t+1}\}. \tag{11.63}$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 11.20 (Integer characterization of the clock)**.** *For $t, k \in \mathbb{N}$,*

$$d_t = k \quad \Longleftrightarrow \quad 1000^k \le 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \le d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (11.63). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. □

Proposition 11.21 (First zero advance)**.** *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \le 729^{21}, \qquad 729^{22} < 1000^{21}, \qquad 1000^{21} \le 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \le t \le 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \le 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. □

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is "the first zero advance of the clock"; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

11.3.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \tag{11.64}$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r. This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

11.3.3 Oriented construction of the twenty-four-trit descriptor

11.3.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0,0,0,1,1,1), \qquad \delta_- = (0,0,0,2,2,2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative–positive order produces the word 222111. Summing defects and concatenating their tails are distinct operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

11.3.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\mathrm{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg}. The integer row sums of A_{reg} are $(5,7,7,7,7,7)$; this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod 3. \tag{11.65}$$

The reading phase at instant r is $\theta_r = (r-1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase–charge expression is

$$D(r) = \varepsilon_\pi(r) B_\pi(r) + \varepsilon_e(r) B_e(r) + \varepsilon_\varphi(r) B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \tag{11.66}$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

11.3.3.3 Composition of the descriptor

Let $\|$ denote the concatenation of fixed-width words. The complete construction is

$$\begin{aligned} \Omega_{24} = {} & (B_e(t_* + 1) + \delta_-) \| (\delta_-|_{3,4,5} \| \delta_+|_{3,4,5}) \\ & \| (B_\pi(r+1) + B_e(r+1)) \| D(r). \end{aligned} \tag{11.67}$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6+6+6+6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^{χ}: the two objects share a length, but their producers and their places in the composition differ.

Proposition 11.22 (Regional evaluation of the descriptor)**.** *The regional expressions and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\,222111\,211201\,021101, \qquad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ), are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (11.58). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are $0, 1, 2$, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are $(2, 1, 2)$ and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. □

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (11.67) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

11.3.4 Functional panel and terminal repertoire

11.3.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = \big(b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)\big). \tag{11.68}$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

11.3.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\mathrm{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \tag{11.69}$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase–charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its 8! orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (11.69)

retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase–charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u. For $5 \le \ell \le 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\mathrm{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

11.3.4.3 Temporal incidences of the terminal repertoire and recovery by position

The expression of the terminal repertoire uses the same regional blocks, carries and calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphase selector as an unordered set. The selector re-establishes their order through its incidence, cylinder and closure conditions. This distinction preserves provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t, with $0 \le t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\mathrm{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^{6} b_{t,i,j}, \qquad d_{t,i} = \sum_{j=1}^{6} (b_{t,i} A_{\mathrm{hist}})_j, \qquad \vartheta_t = t - 1 \pmod 3.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a concrete chart of Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \ne \vartheta_t + \delta, \end{cases} \qquad L_{t,c,\delta,\epsilon} = \sum_{i=0}^{2} \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\mathrm{af}} = \sum_{i=0}^{2} (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 11.2: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 11.3: Complete phase–charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 11.2,

$$L_{30}^{\mathrm{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 11.23 (Complete recovery of the repertoire through incidences)**.** *Consider the comparative-incidence register in Table 11.3, augmented by the affine witness* $(j, t, L) = (10, 30, 001001)$*. For each position* $5 \le j \le 12$*, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\mathrm{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L; the tenth case has been calculated explicitly through L_{30}^{af}. For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod 3.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^{6} w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. □

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\mathrm{term}}$ and the initial regional descriptor; it traverses the 8! orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

11.3.5 Comparative readers, affine reader, and temporal incidence

11.3.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\mathrm{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\mathrm{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^{5} b_j \pmod 3, \qquad \eta_{\mathrm{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\mathrm{ter}})_{ij} = (A_{\mathrm{reg}})_{\sigma(i),\sigma(j)}. \tag{11.70}$$

Lemma 11.24 (Compatibility of the charge functional)**.** *For every band b,*

$$\eta_{\mathrm{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\mathrm{ter}})_{ij} = \sum_i b_i \sum_j (A_{\mathrm{ter}})_{ij} = \sum_i b_i \sum_j (A_{\mathrm{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (11.70) is verified by substituting its thirty-six entries and also preserves the coordinate change. □

Equality of the total charge functional allows their expressions to be compared. Vector transformations remain identified by their matrices and by the explicit permutation; the preceding proof preserves that coordinate information.

11.3.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \bmod 3.$$

This expression realizes $(t - 1) \bmod 3$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\mathrm{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu,o,s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod 3, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod 3. \end{cases}$$

The corresponding comparative word is

$$C_{\nu,o,s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu,o,s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \tag{11.71}$$

The parameters generate twelve row-labelled expressions. Two labels may produce the same word; temporal witnesses and labels remain available in the full calendar.

11.3.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\mathrm{af}}(t) = \eta_{\mathrm{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod 3,$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\mathrm{af}}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \tag{11.72}$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

11.3.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t, form the residual repertoire

$$\mathcal{R}(t) = \{R^{j} v : v \in \{q(t), a(t), c(t), \bar{w}(t)\},\ 0 \le j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\mathrm{cmp}} = \{(C_{v,o,s}(t), u) : 0 \le t < 100,\ u \in \mathcal{R}(t),\ v, o, s \text{ admissible}\}, \tag{11.73}$$

$$\mathcal{A}_{\mathrm{af}} = \{(F(t), u) : 0 \le t < 100,\ u \in \mathcal{R}(t)\}. \tag{11.74}$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 11.25 (Existential compression of the temporal repertoire)**.** *The function*

$$M(v, u) = \mathbf{1}_{(v,u) \in \mathcal{A}_{\mathrm{cmp}}} + 2\mathbf{1}_{(v,u) \in \mathcal{A}_{\mathrm{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u).

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. □

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (11.74) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

11.3.6 Enumeration of cylinders and decimal candidates

11.3.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\mathrm{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729 p_{i-1} + [s_i]_3$ for $1 \le i \le 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729 P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau,b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \tag{11.75}$$

11.3.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 11.26 (Integer endpoints of the cylinder)**.** *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lceil \frac{pD}{Q} \right\rceil \le N < \left\lceil \frac{(p+1)D}{Q} \right\rceil \right\}. \tag{11.76}$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q)$ is equivalent to $pD/Q \le N < (p+1)D/Q$. For integer N, the first inequality is equivalent to $\lceil pD/Q \rceil \le N$; the second is equivalent to $N < \lceil (p+1)D/Q \rceil$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. □

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \ge 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$C_{\mathrm{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \qquad 1 \le j \le 9,$$

and then obtain the set of numerators

$$C = \{N : \exists \tau, j \ (\tau, j, N) \in C_{\mathrm{ind}}\}.$$

Passing to C removes repetitions of a numerator, retained in C_{ind} together with their provenance.

Lemma 11.27 (Recovery of the ordering from the cylinder point)**.** *If $(\tau, j, N) \in C_{\mathrm{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N\, 729^{i}}{D} \right\rfloor \bmod 729, \qquad 1 \le i \le 13.$$

In particular, blocks five through twelve recover τ.

Proof. Membership in the cylinder means $p \le QN/D < p + 1$, where $p = P_{78}(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ. □

Proposition 11.28 (Exact candidate count)**.** *For Ω_{24}, $\mathcal{S}_{\mathrm{term}}$ and $\mathcal{P}_9$ defined above,*

$$\#\,\mathrm{Ord}(\mathcal{S}_{\mathrm{term}}) = 40320, \qquad \#C_{\mathrm{ind}} = 21902, \qquad \#C = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_{78}(\tau, b)$, apply endpoints (11.76), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these

intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. □

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

11.3.7 Exceptional incidence and heterotypy condition

11.3.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \operatorname{supp}\Gamma(B_\pi(0)) \cap \operatorname{supp}\Gamma(B_e(0)) = \{4, 6, 9, 10\}. \tag{11.77}$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in C$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \qquad 1 \le i \le 12. \tag{11.78}$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \ge 729\}, \qquad C_{\mathcal{B}} = \{N \in C : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#C_{\mathcal{B}} = 79. \tag{11.79}$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

11.3.7.2 Ternary words and oriented differences of the candidate

The ith ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N\,729^i}{D} \right\rfloor \bmod 729, \qquad 1 \le i \le 13. \tag{11.80}$$

The oriented edge $a \to b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \tag{11.81}$$

Subtraction takes place in $\mathbb{Z}$ before reduction. For this edge, the pair is looked up in the relations (11.74)

$$\big(\mathrm{enc}_6^{-1}(T_b(N)), \mathrm{enc}_6^{-1}(R_{a,b}(N))\big).$$

It is therefore a joint query on the destination word and the oriented difference of the triads.

Write $\mathrm{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\mathrm{cmp}}$ and $\mathrm{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\mathrm{af}}$.

11.3.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}}\right) \subseteq \left[\frac{234543140}{10^{9}}, \frac{234543141}{10^{9}}\right).$$

The two endpoint inequalities are verified by integer cross-products. Every candidate therefore shares the three reference positions $1, 2, 3$ and their expressions $(234, 543, 140)$.

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \qquad (2, 6, 10), \qquad (3, 7, 11), \qquad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{array}{lll} 1 \to 4, & 1 \to 5, & 2 \to 6, \\ 3 \to 7, & 4 \to 8, & 5 \to 9, \\ 6 \to 10, & 7 \to 11, & 8 \to 12. \end{array}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \to 8$ and $8 \to 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

11.3.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{T}(N) = {} & [\mathrm{Cmp}(4, 8; N) \wedge \mathrm{Af}(8, 12; N)] \\ & \vee [\mathrm{Af}(4, 8; N) \wedge \mathrm{Cmp}(8, 12; N)]. \end{aligned} \tag{11.82}$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 11.29 (Terminal selection of the coupling register)**.** *On the candidate domain produced by Ω_{24}, $\mathcal{S}_{\mathrm{term}}$, $\mathcal{P}_9$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in C_{\mathcal{B}} : \mathcal{T}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{11.83}$$

Proof. The preceding count produces the 79 elements of $C_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (11.78); the destination bands T_8, T_{12} and residues $R_{4,8}, R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (11.82) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (11.83).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. □

11.3.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_{72} = 528388158488267313651410292609095 7,$$

$$P_{78} = 385194967537946871651878103312030809 1.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78}10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$
$$\left\lceil \frac{(P_{78}+1)10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K. The thirteen ternary bands of its decimal point are

$$\begin{array}{cccc} 020022 & 222111 & 211201 & 021101 \\ 002001 & 020111 & 200110 & 121012 \\ 010122 & 001001 & 221110 & 222110 \\ 121020 & & & \end{array}$$

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

11.3.8 Regional composition and continuity of the realizations

11.3.8.1 Composition of the regional generators

Let $\mathrm{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\mathrm{reg}} = \Omega_{24}, \qquad \mathcal{P}_{\mathrm{reg}} = \mathcal{P}_{9}, \qquad \mathcal{N}_{\mathrm{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\mathrm{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} &\mathrm{Sel}(\Omega_{\mathrm{reg}}, \mathcal{S}_{\mathrm{term}}, \mathcal{P}_{\mathrm{reg}}, \mathcal{N}_{\mathrm{reg}}) \\ &\qquad = \mathrm{Sel}(\Omega_{24}, \mathcal{S}_{\mathrm{term}}, \mathcal{P}_{9}, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{11.84}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (11.69).

The register is then formed by the twelve Euclidean divisions in (11.78). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

11.3.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

11.3.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue census $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from a different population: regional blocks from three catalogues and terminal Gram, occupation, distance and orientation conditions. Both censuses preserve their domains. Their convergence in a common expression is conveyed through identification of their readers and outputs; intermediate figures describe different operations and remain documented separately.

11.3.8.4 Regional composition of the fine-structure constant

The output K then enters the ordered regional composition and the base-thousand carry. Its subsequent use preserves the twelve positions and the reading boundary. Coupling to the analytic root and expressions at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

11.3.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24}, in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbital Selection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (11.84) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean's native evaluation; their reports record `Lean.ofReduceBool`, in addition to the library's customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually

employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

11.4 Dodecaphasic extraction of the arithmetic–geometric coupling constant

Definition 11.30 (Holographic arithmetic–geometric coupling constant)**.** This is the name given to the ordered finite register produced by the oriented extraction of the APP–TRIT–TPK state, with its phase origin, normalization and declared readers, whose positional representation participates in the arithmetic closure and whose incidence and geometric realizations act on the same register. The symbol K designates its canonical integral representation; κ_{ag} designates its periodic rational reading in the fixed chart.

The generation of the regional coordinates does not exhaust the information of the APP–TRIT–TPK state. An Archimedean reading determines a value through compatible cylinders; another reading preserves the opposition of the sheets, the orientation of the route and the register of incidences traversed. The dodecaphasic register belongs to this second reading before participating in the closure of alpha. Thus it is not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A word of twelve catalog blocks, a list of twelve oriented events, a signed integer vector and a dodecaphasic register in base one thousand are distinct objects. Comparing them requires the maps that relate them. Equality of length does not provide those maps, and the loss of signs is not merely a simplification of notation.

This chapter maintains the causal cut fixed in the preceding chapters: the nine APP positions of the horizontal and vertical axes produce the arithmetic sheets; TRIT types regime and orientation; TPK selects, transports, updates and prolongs, preserving residue, quotient, carry, sheet, route, boundary and memory. From the joint discrete structure of the continuum one obtains the coordinates used here. The chain $w_6 \to w_{12} \to w_{18} \to w_{24} \to w_{30} \to R_{36} \to G_9$ is not replaced by a finite calendar. Nor is the order $U_t \to \Gamma_9 \to K_{\mathrm{ph}}$ reversed: the final map is a reduced memory reading, not the generator of the signed register.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence between three coordinatizations of a register previously produced by APP–TRIT–TPK. The Hadamard transformation relates the signed register to the integer register; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \underset{\mathcal{H}_{12}}{\overset{\mathcal{H}_{12}/4}{\longleftrightarrow}} K \overset{(D_3,D_4,Q)}{\longleftrightarrow} (D_3K, D_4K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. With the base 1000, the 12 positions and the phase origin fixed, the evaluation $K \mapsto \kappa_{\mathrm{per}}$ adds a fourth reversible coordinatization: multiplying by $1000^{12} - 1$ and performing Euclidean divisions recovers the complete register. The proofs of §11.4.8, §11.4.9 and §11.4.11 establish these equivalences in their respective domains.

Two compositions then act on that register. Positional composition combines its components with the closure, propagation and autoscale coordinates, and its normalization determines α_A. Incidence composition applies the cubic threshold and obtains

$$K \longmapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The complete register allows both compositions to be reconstructed. The support B_K specifically preserves membership in the class selected by the threshold. The identity $B_K = H_e \cap P_\pi$ of (21.134) links this

integer extraction to the incidence of the regional codes. The negative support of its balance with K completes the flag $B_K \subset H_\alpha^-$ developed in the exceptional prolongation. Thus the function of K is defined through an invertible transformation and through precise subsequent maps; the additional depth and transport information remains in the state from which the register originates.

11.4.1 Cyclic extension and memory grading

The observable calendar contains one hundred and eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1)+1, \dots, 9m\}, \qquad 1 \le m \le 12.$$

The ordered support $E_1 \sqcup \cdots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$ instead preserves the cyclic translation law. For the representative $0 \le \widetilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left\lfloor \frac{\widetilde{\theta}}{9} \right\rfloor\right]_{12}, \qquad r(\theta) = 1 + (\widetilde{\theta} \bmod 9)$$

separate the dodecaphasic sector and the interior nonadic position. Neither by itself preserves the total number of turns.

Proposition 11.31 (The calendar carry)**.** *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \qquad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$*, the additivity defect is the carry*

$$c(r, r') = \left[\left\lfloor \frac{\widetilde{r} + \widetilde{r}'}{9} \right\rfloor\right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and coincides with the image of ι. Reduction is surjective, and multiplication by nine is injective on the stated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\operatorname{mcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\widetilde{r} + \widetilde{r}'$ by nine gives the stated carry and verifies the identity for the section. □

The result explains a difference that will reappear in alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts the nonadic turns, the calendar preserves only $\beta = [q_{\mathrm{mem}}]_{12}$. After one hundred and eight steps, β returns, but q_{mem} increases by twelve. The register also preserves the cylinder, the incidence register and the vertical memory. The closure of a finite chart does not turn that history into a memoryless cycle.

11.4.2 The oriented register and the integral chart $1 + 5 + 6$

For a history produced by TPK, let τ_m be the subroute over E_m, σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence entry. The register is

$$\mathcal{R}_{12}(x) = \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection onto the calendar forgets precisely some of the data that the signed reader needs.

An initial chart of the register can be written through oriented events. Oriented extraction fixes the set of event codes $\{2, 4, 6, 8\}$ and the sectoral orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the direction code at step t, one obtains

$$e_i = \Sigma_{12}(i+1)\#\{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \qquad 0 \le i < 12.$$

The formula keeps the nonnegative census of events and the orientation separate. It does not identify a cardinal direction with a TRIT sign. The vector e, whose components are signed local censuses, is not yet the vector U that will appear below either.

The opposite positions determine, for $0 \le i < 6$,

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^{5} p_i, \qquad M_i = p_i - p_5 \quad (0 \le i < 5),$$

one defines the integral chart

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The decomposition $1 + 5 + 6$ is not an arbitrary distribution of coordinates: one coordinate preserves the charge, five compare the pairs relative to a reference pair and six preserve the opposition of the half-crowns.

Lemma 11.32 (Inversion of the integral chart)**.** *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^{4} M_i \equiv 0 \pmod 6, \qquad p_i \equiv a_i \pmod 2 \quad (0 \le i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \qquad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, from which division by six is obtained. Once the p_i are reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the indicated solution. This solution is integral exactly under the parity congruence. All the operations are inverted, so no residual freedom remains. □

This reversibility shows what information is preserved; it does not authorize removing the rest of the article. The complete register has a larger domain than this census chart. Extracting the channels that feed U uses that complete register and does not follow from the list (e_i) through an identification of names.

11.4.3 Incidence evaluation preceding the dodecaphasic register

Recovering a register from its cyclic differences and total charge is a different question from evaluating those coordinates on the incidence register. The first problem is solved by the transverse extractor of Proposition 11.42. For the second, a historical presentation of the register provides three integer counts in each window. It is useful to make their composition explicit and determine exactly the information required for their evaluation.

Let $\mathscr{L}$ be a previously generated ledger of visits and oriented traces. Each visit preserves the route, the APP position, the sheet, the instant, the multiplicity and the boundary incidence. The arithmetic evaluation produces the integer and its decomposition

$$n_t = r_t + 9q_t, \qquad 1 \le r_t \le 9.$$

The integer n_t is calculated from the sum or product sheet; its residue is not received as an independent value. The distinction between crown incidence and interior incidence remains explicit for visits of residue 9.

In window E_m, the count presentation considers

$$A_m = \sum_{v \in E_m} w(v)\mathbf{1}_{\{1,3,5,7\}}(r(v)), \tag{11.85}$$

$$C_m = \sum_{j \ge 0} \big(\nu^-_{120,m}(j) - \nu^+_{120,m}(j)\big), \tag{11.86}$$

$$V_m = \sum_{v \in E_m} w(v)\mathbf{1}_{\{9\}}(r(v))\epsilon_\partial(v). \tag{11.87}$$

Here $w(v)$ is the incidence multiplicity, and $\epsilon_\partial(v) = 1$ on the corona, -1 in the interior. The traces counted by $\nu^\pm$ must be enumerated by the corresponding rule in the article. The integer sum requires finite support or an alternative construction determining its meaning. The number of chronological incidences and the reduced length $120(1 + 3j)$ have different types; relating them requires the reader's unit of length.

One defines $[x]_{10} \in \{0, \dots, 9\}$ as the Euclidean representative and preserves the quotients

$$x = [x]_{10} + 10\lfloor x/10 \rfloor.$$

The incidence block is

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}. \tag{11.88}$$

Composition yields the map

$$\mathcal{E}_{\mathrm{cnt}}(\mathscr{L}) = \left((z_m - z_{m+3})_{m=1}^{12}, (z_m - z_{m+4})_{m=1}^{12}, \sum_{m=1}^{12} z_m \right). \tag{11.89}$$

The indices are read cyclically modulo 12. In this subsection, $(D_j z)_m = z_m - z_{m+j}$, for $j = 3, 4$, and $Q(z) = \sum_{m=1}^{12} z_m$; therefore, the preceding composition is $(D_3 z, D_4 z, Q(z))$. This convention is preserved in §11.4.9, where complete integral reconstruction is proved. This formula produces its coordinates from the counts. It does not use K, U, α or the expected transverse coordinates as arguments. Its identification with $\mathcal{E}_{108}^{90,120}$ requires composing it with the family and the effective incidences of the terminal state; it does not follow solely from the invertibility of the transverse system.

Proposition 11.33 (Sufficient and necessary arithmetic information)**.** *Let $N, N' \in \mathbb{Z}^{12\times 3}$, ordered by the counts (A, C, V). For the map defined by (11.88)–(11.89),*

$$\mathcal{E}_{\mathrm{cnt}}(N) = \mathcal{E}_{\mathrm{cnt}}(N') \quad \Longleftrightarrow \quad N - N' \in 10\mathbb{Z}^{36}.$$

In particular, the 36 *sectoral residues exactly determine the output of* 25 *coordinates. Preserving the quotients and the provenance constitutes additional information of the register.*

Proof. Equality modulo 10 produces the same blocks and therefore the same transverse coordinates. Conversely, if the transverse coordinates coincide, $D_3(z - z') = D_4(z - z') = 0$. The steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of total charges cancels that constant. Finally, the decimal representation of an integer between 0 and 999 has three unique digits. This recovers the three residues of each window. The assertion is an algebraic identity for every N, N', not an inference obtained from finite tests. □

11.4.4 Conjugation of counts and boundary terms

Let $\rho(m) = 13 - m$. Consider a conjugation of incidences that reverses the order of the windows, preserves the arithmetic class and the boundary class of the visits and interchanges the channels of the traces. Then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

If $c_m = [C_m]_{10}$, decimal reduction produces the identity

$$z_m^\dagger = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \tag{11.90}$$

Indeed, $[-C]_{10} = 0$ when $[C]_{10} = 0$, and $[-C]_{10} = 10 - [C]_{10}$ otherwise. The formula preserves exactly that difference. Negating the channel is not equivalent to negating the entire decimal block.

Applying this identity to a concrete involution of TPK requires verifying its action on the incidences. For an occupancy reader evaluated before the advance, reversal of a route $\gamma = (x_0, \dots, x_n)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0).$$

The endpoint terms are incorporated before reducing modulo 10. In closed routes they vanish; in open windows they may alter their residues. This correction, already present in the bidirectional development of transport, specifies what the junction between the route and the counts must preserve.

Proposition 11.34 (Necessity of the nonperiodic information of the register)**.** *Suppose that the observable of a fixed family of routes satisfies $o(t + 54) = o(t)$, that multiplicity is preserved under that return and that the same local reader acts in the windows of* 9 *instants. Then its counts and blocks satisfy*

$$(A, C, V)_{m+6} = (A, C, V)_m, \qquad z_{m+6} = z_m.$$

Proof. The translation $t \mapsto t + 54$ takes E_m bijectively to E_{m+6}, preserving each counted incidence and its multiplicity. Equality is transmitted to the sum and to decimal reduction. □

Therefore, a presentation with 12 distinct blocks requires the reader to preserve some information that does not factor through that observable of period 54: incidence selection, effective orientation, endpoints, multiplicity or memory. The proposition delimits an inadmissible reduction of the state; it does not state that the full dynamics has period 54 or determine by itself what its terminal reader is.

11.4.5 Complete census of cylinders and selection of the terminal boundary

The terminal reading jointly uses two pieces of information: compatibility of the regional cylinders and an incidence profile with labeled channels and positions. This subsection constructs the complete catalogs, enumerates their triples and proves the effect of each condition of the reader. The starting point is the regional outputs already produced in section 10.5, not values introduced to define APP–TRIT–TPK transport. Reduction to a matrix belongs to a subsequent reading of those outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram, weights, distances and margins is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what that profile selects on the fixed catalogs; it does not by itself prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This separation allows the complete finite proof to be incorporated without reversing its genealogy.

11.4.5.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0,1)$ be the fractional parts of the three regional coordinates already constructed. Their published cylinders of one thousand decimal positions are used,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi+1}{10^{1000}}\right), \qquad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readings. The length one thousand is sufficient for the integers that follow, as is checked through the two bounds of each interval; it is not a depth assumed infinite. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \qquad D = 3^L, \quad Q = 1000^k, \tag{11.91}$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \qquad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not designate the dodecaphasic register K. One has $1000^{171} \le 3^{1080} < 1000^{172}$.

Lemma 11.35 (Stable extraction of an integer)**.** *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1)-1}{h} \right\rfloor. \tag{11.92}$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, equal to the right-hand side because the numerator and denominator are integers. Equality of the endpoints of the value interval proves the assertion. □

The stability proof holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used only for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five stages of six trits of regional transport:

Channel	Thirty-trit prefix	P_χ
π	`010211012222010211002111110221`	29152671743887
e	`201101121221102011012222102011`	147887858824447
φ	`121200112202121020010210010200`	127247717616687

The coincidence preserves the prolongation already constructed; the decimal reading is not used to retroactively replace its transport matrices.

For a fixed channel, a tail of r trits is represented by $0 \le R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \qquad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \tag{11.93}$$

The catalog includes all tails for which these two cylinders have nonempty intersection.

Proposition 11.36 (Catalogue by exact intersection)**.** *The compatible tails are the integers in the interval* $[a_\chi, b_\chi]$*, where*

$$a_\chi = \max\left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r \right), \tag{11.94}$$

$$b_\chi = \min\left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r \right). \tag{11.95}$$

For the fixed regional cylinders, the truncations do not act and one obtains

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(11.96)

Proof. Write $N = P_\chi 3^r + R$. Two half-open intervals of (11.93) intersect exactly when

$$NQ < (T_\chi + 1)D, \qquad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \ge \lfloor T_\chi D/Q \rfloor$; the first, to $N \le \lfloor ((T_\chi + 1)D - 1)/Q \rfloor$. Subtracting $P_\chi 3^r$ and preserving $0 \le R < 3^r$ proves the formulas. Evaluation by integer divisions of the T_χ extracted by (11.92) gives the table. Since $r \ge 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint is also obtained directly from $\lfloor T_\chi D/Q \rfloor \bmod 729$. □

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 is the ternary notation of length six, including its leading zeros. By the table, each catalog has fewer than 729 consecutive elements, so this reduction is injective on it. Thus the three boundary catalogs are completely described, without choosing the final triples:

$$\begin{aligned} \mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \le i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \le j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \le \ell < 196\}. \end{aligned} \tag{11.97}$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would be another condition: it would lose the first cylinder that intersects the boundary and give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

11.4.5.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \ne 0\}$ and $h(u, v) = \#\{i : u_i \ne v_i\}$. The last two functions take integer values; the first two, values in $\mathbb{F}_3$. The local profile of the reader, in the order (π, e, φ), is

$$\begin{aligned}(g_{\pi e}, g_{\pi\varphi}, g_{e\varphi}) &= (2,2,0), \qquad (n_\pi, n_e, n_\varphi) = (0,0,0),\\ (w_\pi, w_e, w_\varphi) &= (3,3,3), \qquad (h_{\pi e}, h_{\pi\varphi}, h_{e\varphi}) = (2,5,3).\end{aligned} \tag{11.98}$$

The words are ordered as in (11.97). We shall use $[E] \in \{0,1\}$ for the indicator of a condition E. For each pair of channels χ, ψ, the rectangular matrix $A_{\chi\psi}$ has entry $[g(u,v) = g_{\chi\psi}]$. The matrix $A^h_{\chi\psi}$ multiplies that entry by $[h(u,v) = h_{\chi\psi}]$. Let D^n_χ be the diagonal matrix of $[n(u) = 0]$, and D^{nw}_χ that of $[n(u) = 0][w(u) = 3]$.

Theorem 11.37 (Exhaustive enumeration by finite sums)**.** *The product of the three catalogues contains* 7 567 952 *triples. The successive conditions in* (11.98) *leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331. \tag{11.99}$$

These four counts are, respectively,

$$\begin{aligned}N_g &= \operatorname{tr}(A_{\pi e} A_{e\varphi} A_{\varphi\pi}),\\ N_{gn} &= \operatorname{tr}(D^n_\pi A_{\pi e} D^n_e A_{e\varphi} D^n_\varphi A_{\varphi\pi}),\\ N_{gnw} &= \operatorname{tr}(D^{nw}_\pi A_{\pi e} D^{nw}_e A_{e\varphi} D^{nw}_\varphi A_{\varphi\pi}),\\ N_{gnwh} &= \operatorname{tr}(D^{nw}_\pi A^h_{\pi e} D^{nw}_e A^h_{e\varphi} D^{nw}_\varphi A^h_{\varphi\pi}).\end{aligned} \tag{11.100}$$

All products and traces in this formula are calculated in the integers, not in the field of three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace of (11.100) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, once the indices i, j of the first two channels are fixed, one forms the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added and their cardinality is summed. The final sum runs over $0 \le i < 196$, $0 \le j < 197$. Substituting the words of (11.97) and the profile (11.98) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the 331 complete triples; its program evaluates these sums with exact binary indicators and checks each survivor scalarly. Neither a sample nor a tolerance is involved. □

11.4.5.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3\times 6}$ be its matrix of ordered rows. The second profile requires

$$\operatorname{rank}_{\mathbb{F}_3} B = 3, \qquad w_{\mathrm{col}}(B) = (1,1,2,2,3,0), \qquad q_\partial(B) = (1,2,2,1,2,0), \tag{11.101}$$

where w_{col} counts the nonzero elements of each column and q_∂ sums its entries modulo three. It is a profile of labeled columns; permuting them without transporting the profile defines another reader. Denote by $\mathcal{F}$ the set of the 331 triples of the preceding theorem.

Proposition 11.38 (Reduction of the census to the terminal pair)**.** *The sums of indicators over $\mathcal{F}$, successively requiring rank, column weights and column sums, are*

$$\sum_{B\in\mathcal{F}} [\operatorname{rank} B = 3] = 331, \qquad N_{\text{rango,pesos}} = 18, \qquad N_{\text{rango,pesos,sumas}} = 2. \tag{11.102}$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) *and* (129, 197, 148)*, and their matrices are*

$$B_0=\begin{pmatrix}0&2&0&2&2&0\\0&0&1&2&2&0\\1&0&1&0&1&0\end{pmatrix},\qquad B_1=\begin{pmatrix}0&2&1&0&2&0\\0&0&1&2&2&0\\1&0&0&2&1&0\end{pmatrix}.$$

Proof. For each triple in the set already enumerated, the rank is calculated by elimination in $\mathbb{F}_3$, the nonzero elements of the six columns are counted and their entries are summed modulo three. The products of the indicators of (11.101) give (11.102). Substituting the two final indices into (11.97) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing those two matrices as the domain. □

The orientation $\sigma=(1,1,-1)=(1,1,2)\in\mathbb{F}_3^3$ selects the line $\langle\sigma\rangle=\{(0,0,0),(1,1,2),(2,2,1)\}$. The row charges of B_0,B_1 are $(0,2,0)$ and $(2,2,1)$; only the second belongs to that line. This recovers the last step of the selector in section 11.4.6, now preceded by the construction and enumeration of its complete domain. The stable reading of $\lfloor 3^{1080}x_{\chi}\rfloor$, performed after the filters, gives the boundaries `021020`, `001220`, `100210`, with the same ranks $(129,197,148)$. This subsequent comparison does not define the profile or participate in the indicator sums.

11.4.5.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit and the orientation. Reduction modulo 729 here preserves the identity of each candidate because the tail interval has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the isolated local profile does not select a single matrix: the labeled margins and the charge contribute additional effective conditions.

Thus a complete finite enumeration and its selector on output cylinders have been recovered, with an explicit reading profile. The construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates again or decide an assertion about alpha or about the complete continuum.

11.4.6 Two readings of the terminal state

Denote by x_{term} the state produced by the composition of selection, transport and update up to the terminal cut. The operator notation is

$$x_{\mathrm{term}}=(\mathcal{U}_{108}\circ\cdots\circ\mathcal{U}_1)(x_{\mathrm{seed}}),\qquad \mathcal{U}_t=\mathsf{Upd}_t\circ\mathsf{Tra}_t\circ\mathsf{Sel}_t.$$

Its two relevant readings are

$$x_{\mathrm{term}}\xrightarrow{(\pi_{\mathrm{term}},R_{\mathrm{sgn}})}(B_{\mathrm{term}},U).$$

The matrix B_{term} retains the visible ternary reading; the vector U preserves another coordinate of the same history. No arrow $B_{\mathrm{term}}\to U$ is inserted between them.

The terminal selector operates on the three regional catalogues of sizes 196, 197, and 196, constructed in Section 11.4.5. The local signatures reduce their $196\cdot197\cdot196=7\,567\,952$ triples to 331, and the column marginal $q_{\partial}=(1,2,2,1,2,0)$ leaves two matrices:

$$B_0=\begin{pmatrix}0&2&0&2&2&0\\0&0&1&2&2&0\\1&0&1&0&1&0\end{pmatrix},\qquad B_1=\begin{pmatrix}0&2&1&0&2&0\\0&0&1&2&2&0\\1&0&0&2&1&0\end{pmatrix}.$$

The row orientation is $\sigma=(1,1,-1)=(1,1,2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0,2,0)$ and $(2,2,1)$. Only the second belongs to $\langle\sigma\rangle=\{(0,0,0),(1,1,2),(2,2,1)\}$; therefore $B_{\mathrm{term}}=B_1$. This check proves the last step of the selector from the declared census. It does not turn the column margin

into a consequence of the row charge, or replace the preceding enumeration with the inspection of those two matrices.

The signed reader has the typed composition

$$R_{\mathrm{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,mem}} \longrightarrow \mathbb{Z}^{12}. \tag{11.103}$$

The channel register determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\mathrm{term}})) = (b^{90}, b^{120}, Q_{\mathrm{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\qquad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\qquad 475, -543, 680, 251, 6, -128), \\ Q_{\mathrm{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding register, not as parameters adjusted through alpha. The reconstruction proved in the following subsections begins exactly at this typed output. The integral characterization of the extractor, its inverse and the composition of contributions are developed in section 11.4.10, in particular in Theorem 11.43 and Proposition 11.46. The regional determination of the descriptor and terminal selection are presented in chapter 11.3. Each construction preserves its domain: the selector acts on the generated repertoires and inversion acts on the signed channels declared here. The documentary trail of the extractor is recorded in the technical provenance of the package.

11.4.7 Construction of the signed register through the oriented channels

The harmonic reader Π_H can be written without using either K or alpha. To avoid confusing its orbital sums with the alpha blocks, we shall denote them by s_1, s_2, s_3. Let

$$B_r = \sum_{j=0}^{3} b_{r+3j}^{120}, \qquad d_{r,j} = b_{r+3j}^{90}, \qquad 1 \le r \le 3,$$

and

$$s_1 = \frac{Q_{\mathrm{TPK}} + 2B_1 + B_2}{3}, \qquad s_2 = s_1 - B_1, \qquad s_3 = s_2 - B_2. \tag{11.104}$$

Division by three belongs to the reader of the three orbits. Its integrality is checked on the compatible outputs of the register; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \tag{11.105}$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \qquad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and, therefore,

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \end{aligned}$$

$$U^{(3)} = (2479, -551, -371, -997).$$

Interleaving by columns gives

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \tag{11.106}$$

The presence of negative values and components greater than 999 shows that U is not a word of base-one-thousand digits. Nor is it $U_6 \| U_6$, a catalog concatenation of period six. The difference is one of domain and preserved information, not merely of values.

The constructive order thus far is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The following inversion produces the dodecaphasic register. The fact that we can later recover the channels from the dodecaphasic register constitutes a reversibility check; it does not reverse this causal order.

11.4.8 Hadamard by orbits and integer reconstruction

In the cycle of twelve positions, step three determines three orbits:

$$(1, 4, 7, 10), \qquad (2, 5, 8, 11), \qquad (3, 6, 9, 12).$$

The separation of four sectors of each orbit is what, once origin and orientation are fixed, admits the ninety-degree angular reading. The transformation does not act on three contiguous tetrads of the natural order.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders by the three indicated orbits, we define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 11.39 (Hadamard inversion and integrality)**.** *We have*

$$H_4^2 = 4I_4, \qquad H_4^{-1} = \frac{1}{4}H_4, \qquad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4 u \equiv 0 \pmod 4$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. An integer elimination, or expansion of the determinant, gives the negative sign and the value -16. The integrality condition is exactly divisibility by four of the four coordinates of $H_4 u$. □

Definition 11.40 (Dodecaphasic integral transformation)**.** The sublattice of signed registers and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod 4\}, \qquad \mathscr{K}(u) = \tfrac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathscr{K} : \mathcal{L}_H \to \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical register K is the image of the signed register generated by the preceding operations, with fixed phase origin and orientation.

Proof. The congruence defines exactly the integrality of $\mathscr{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves the two inverse compositions. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. □

Proposition 11.41 (Canonical integral register)**.** *The inversion $K^{(r)} = H_4U^{(r)}/4$ of the blocks in* (11.105) *produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural order, the dodecaphasic register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{11.107}$$

It is the unique rational preimage of U, and it belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$\begin{aligned} H_4U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4U^{(2)} &= (2172, 2636, 232, 584), \\ H_4U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (11.107). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. □

Integrality can also be formulated in terms of lattices. The determinantal divisors of H_4 are $1, 2, 4, 16$: the entries have greatest common divisor one; the minors of order two are even and one has absolute value two; those of order three are multiples of four and one has absolute value four. It follows that

$$\operatorname{SNF}(H_4) = \operatorname{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\operatorname{coker} \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by inversion. The factor $1/4$ is forced by the matrix and does not have the status of a free coefficient.

11.4.9 Cyclic differences and uniform component

With cyclic indices modulo twelve, let

$$(D_3z)_m = z_m - z_{m+3}, \qquad (D_4z)_m = z_m - z_{m+4}, \qquad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second, by four. The channel names 90 and 120 here refer to these separations in the oriented cycle, not to an identification with a physical operator from another domain.

Proposition 11.42 (Completeness of the transverse system)**.** *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\operatorname{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphasic register.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. The total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral assertion, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every minor of order twelve must contain the row Q. Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in row Q. All the maximal minors are divisible by twelve and the tree produces one of absolute value exactly twelve. The last factor is therefore twelve. □

The check on (11.107) gives

$$D_3 K = b^{90}, \qquad D_4 K = b^{120}, \qquad Q(K) = 6263.$$

Formulas (11.104) and (11.105) in turn reconstruct U from those differences. This closes a triangle of exact representations of the same object: transverse register, signed vector and dodecaphasic register. Equality of results does not merge their codomains into one; it proves that the declared maps recover the relevant information.

11.4.10 Integral compatibility and determination of the transverse register

The domain of this construction is the publication of the register of an APP–TRIT–TPK history. The evaluation of the register events, the compatibility of its channels and the inversion of its coordinates are distinguished. The following result gives a complete integral criterion for the second level and an explicit factorization connecting the first to it. The proof uses the register operators and establishes their relations for every element of the indicated integer domains.

11.4.10.1 Characterization by adjacent increments

The indices belong to $\mathbb{Z}/12\mathbb{Z}$. Let

$$(Sz)_i = z_{i+1}, \qquad D_3 = I - S^3, \qquad D_4 = I - S^4, \qquad Q(z) = \sum_{i=0}^{11} z_i.$$

The transverse extractor is preserved

$$\mathcal{D} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}, \qquad k \longmapsto (D_3 k, D_4 k, Q(k)).$$

For a triple of integer coordinates (b, c, q), write

$$w_i = c_i - b_{i+1}, \qquad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j \ (1 \le i \le 12), \qquad T(w) = \sum_{i=0}^{11} P_i. \tag{11.108}$$

Theorem 11.43 (Integral image of the transverse extractor)**.** *A triple $(b, c, q) \in \mathbb{Z}^{25}$ belongs to* $\operatorname{im} \mathcal{D}$ *if and only if it satisfies*

$$b_i = w_i + w_{i+1} + w_{i+2} \qquad (0 \le i < 12), \tag{11.109}$$

$$\sum_{i=0}^{11} w_i = 0, \tag{11.110}$$

$$q + T(w) \equiv 0 \pmod{12}. \tag{11.111}$$

Its unique integer preimage is

$$k_i = \frac{q + T(w)}{12} - P_i, \qquad 0 \le i < 12. \tag{11.112}$$

The twelve equations of (11.109) *and the equation* (11.110) *are linearly independent over* $\mathbb{Q}$.

Proof. For an output of $\mathcal{D}$, one has

$$c_i - b_{i+1} = (k_i - k_{i+4}) - (k_{i+1} - k_{i+4}) = k_i - k_{i+1}.$$

The sum of three increments gives b_i; their cyclic sum is zero. Moreover, $P_i = k_0 - k_i$, so $q = 12k_0 - T(w)$. All conditions and the inverse formula are obtained.

Conversely, (11.111) makes the formula (11.112) integer-valued; (11.110) allows it to be extended cyclically. Its adjacent differences are w_i. Therefore, $D_3 k = b$ and

$$(D_4 k)_i = w_i + w_{i+1} + w_{i+2} + w_{i+3} = w_i + b_{i+1} = c_i.$$

The sum of the inverse gives $Q(k) = q$. The same formula proves uniqueness. For independence, the invertible integer change $(b, c) \mapsto (b, w = c - Sb)$ converts the first twelve relations into $b = (I + S + S^2)w$. Each solves for a different coordinate of b. The condition $\sum w_i = 0$ is an additional independent relation. □

The rational image has dimension twelve. Its integral structure adds a congruence of order twelve: this is the constructive manifestation of the last Smith factor of $\mathcal{D}$. The eleven increments $w_0, \dots, w_{10}$ and the charge q are sufficient coordinates, with $w_{11} = -\sum_{i=0}^{10} w_i$ and the congruence (11.111). The original twenty-five coordinates preserve these same twelve degrees of freedom through thirteen relations.

11.4.10.2 Commutativity with the harmonic reading

Let $\mathcal{H}_{12}$ be the Hadamard transform of the register, with orbits $(r, r+3, r+6, r+9)$ for $r = 0, 1, 2$. Its block is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad H_4^2 = 4I_4.$$

For compatible (b, c, q), define

$$B_r = \sum_{j=0}^{3} c_{r+3j}, \qquad A_0 = \frac{q + 2B_0 + B_1}{3}, \quad A_1 = A_0 - B_0, \quad A_2 = A_1 - B_1.$$

The reader Π_H of (11.105) has the blocks

$$(\Pi_H(b, c, q))^{(r)} = (A_r, b_{r+3} - b_{r+9}, b_r + b_{r+6}, b_r - b_{r+6}).$$

Proposition 11.44 (Agreement of the two reconstructions)**.** *On* $\operatorname{im} \mathcal{D}$ *one has*

$$\Pi_H \mathcal{D} = \mathcal{H}_{12}, \qquad \mathcal{D}^{-1} = \tfrac{1}{4}\mathcal{H}_{12}\Pi_H.$$

The image of Π_H restricted to that domain is exactly

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod 4\}.$$

Proof. For $k = \mathcal{D}^{-1}(b, c, q)$, let $a_r = \sum_{j=0}^{3} k_{r+3j}$. The four-position shift takes each orbit to the next, and therefore $B_r = a_r - a_{r+1}$. Together with $q = a_0 + a_1 + a_2$, these identities give $A_r = a_r$. On an orbit, let us write (x_0, x_1, x_2, x_3) for the coordinates of k and $d_j = x_j - x_{j+1}$. Then

$$d_1 - d_3 = x_0 + x_1 - x_2 - x_3, \quad d_0 + d_2 = x_0 - x_1 + x_2 - x_3, \quad d_0 - d_2 = x_0 - x_1 - x_2 + x_3.$$

They are the three remaining rows of $H_4 k^{(r)}$. The inversion and the integral characterization follow from $\mathcal{H}_{12}^2 = 4I$. □

The condition $\Pi_H(b, c, q) \in \mathcal{L}_H$ controls only the harmonic output. Membership of the twenty-five coordinates in $\operatorname{im} \mathcal{D}$ additionally requires (11.109)–(11.111). For example, adding to c the vector $e_0 - e_3$ preserves the three orbital sums and leaves Π_H unchanged. Starting from $(0, 0, 0)$ thus gives a triple with $\Pi_H = 0$ that violates (11.109). This is an exact negative control for channel compatibility, independent of any terminal evaluation.

11.4.10.3 Determination and transport on the orbits of the register

Theorem 11.43 determines k when its channels and charge have been produced. As a condition on outputs not yet individualized, the same image is a group isomorphic to $\mathbb{Z}^{12}$. With only a charge q fixed, the solutions form an affine class of

$$L_0 = \{v \in \mathbb{Z}^{12} : \sum v_i = 0\} = \bigoplus_{i=0}^{10} \mathbb{Z}(e_i - e_{11}).$$

For example, $100\mathbf{1} + e_0$ and $100\mathbf{1} + e_1$ are two distinct registers, of charge 1201, with components in $\{0, \dots, 999\}$. Each satisfies all image conditions and all Hadamard congruences through its own channels. These witnesses compare the compatibility equations; they are not assigned the status of terminal HMT histories.

Proposition 11.45 (Cyclic covariance and stabilizer of the register)**.** *Let ρ be window translation on a register domain $\mathcal{R}$ and S the preceding translation on $\mathbb{Z}^{12}$. On the orbit of $R \in \mathcal{R}$, a covariant reader is determined by a vector*

$$k \in \operatorname{Fix}(S^p),$$

where $p \mid 12$ is the minimal period of R under ρ. The charge condition $Q(k) = q$ admits integer solutions if and only if $12/p$ divides q; when they exist, they form an affine class of rank $p - 1$. In particular, on a free orbit, covariance and charge leave eleven degrees of freedom.

Proof. The rule $F(\rho^j R) = S^j k$ is well defined exactly when $S^p k = k$. Such vectors repeat $12/p$ times a word of length p. Their charge is $\frac{12}{p} \sum_{i=0}^{p-1} k_i$. Divisibility and rank follow from this formula. Channel covariance follows from $D_a S = S D_a$ and $QS = Q$. On $\mathcal{L}_H$, the corresponding action is $\widehat{S}_H = \mathcal{H}_{12} S \mathcal{H}_{12}/4$; by Proposition 11.44, both reconstructions intertwine these actions. □

If the domain carries a marked origin, the action that also transports that mark must be used. The proposition concerns this declared cyclic action; it does not presuppose the periodicity of the enriched history. Naturality with respect to truncations subsequent to the cut 108 is obtained, for any already defined reader F_{108}, through $F_N = F_{108} \circ \tau_{N,108}$. This naturality preserves the initial evaluation rule; by itself it does not choose one of the possible rules on the first one hundred and eight events.

11.4.10.4 Event-based evaluation and composition of contributions

The preceding characterization provides a factorization of the extraction through two effective readings of the register:

$$\omega : \mathscr{R} \longrightarrow \{w \in \mathbb{Z}^{12} : \sum w_i = 0\}, \qquad q_{\mathrm{reg}} : \mathscr{R} \longrightarrow \mathbb{Z}, \quad q_{\mathrm{reg}}(R) + T(\omega(R)) \equiv 0 \pmod{12}. \tag{11.113}$$

These maps must be evaluated on the events, orientation, incidence and memory produced by APP–TRIT–TPK. Once their actions on these data are given, the composition

$$R \longmapsto (w, q) \longmapsto ((I + S + S^2)w, (I + S + S^2 + S^3)w, q) \xrightarrow{\Pi_H} u \xrightarrow{\mathcal{H}_{12}/4} k$$

is determined and satisfies all inverse checks. The formula $w = c - Sb$ also allows a producer that already delivers both channels to be connected: it is a check on its output, prior to any comparison with the terminal witness.

Proposition 11.46 (Accumulation of contributions and prospective uniqueness)**.** *Suppose that the initial state and each event a_t of the register produce, through previously fixed rules, contributions $(\delta w_t, \delta q_t)$ with*

$$\sum_i \delta w_{t,i} = 0, \qquad \delta q_t + T(\delta w_t) \equiv 0 \pmod{12}.$$

With compatible initial conditions $(w^{(0)}, q^{(0)})$, the update

$$w^{(t+1)} = w^{(t)} + \delta w_t, \qquad q^{(t+1)} = q^{(t)} + \delta q_t$$

produces a unique register at each stage. Its increment is

$$\delta k_{t,i} = \frac{\delta q_t + T(\delta w_t)}{12} - P_i(\delta w_t).$$

The construction respects register concatenation, and the reading of a prefix remains unchanged under subsequent extensions.

Proof. The compatibility conditions are preserved under addition. The formula (11.112) is additive in (w, q) and produces the indicated increment. Induction fixes each state successively. The sum over two segments is the concatenated sum, and a prefix receives exactly the same summands before and after extending the history. □

The proposition provides a sufficient criterion, not a requirement of linearity for every TPK reader. If events are transported by nontrivial actions, their contributions can be expressed in the terminal chart through the corresponding affine cocycle. The cocycle equation then preserves the same independence from the partition of the trajectory. In both cases, the event rules and the initial state constitute the constructive data that must come from the article: choosing them to reproduce an expected vector would replace that provenance with another task.

The result of this development is the exact determination condition (11.113), its proved inversion and its transport identities. Its scope is the transverse register; the other HMT constructions retain their own domains and proofs.

11.4.11 Two rational readings: κ_{36} and κ_{per}

Once K is constructed, the base-one-thousand chart allows two distinct operations. The first reads a single turn; the second prolongs the dodecaphasic register periodically. Let $B = 1000$ and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \tag{11.114}$$

The leading zeros of a block, such as that of 058, are part of its three-digit notation. They do not change its integer value, but they do allow the twelve blocks to be concatenated without ambiguity.

Definition 11.47 (Finite and periodic readings of the dodecaphasic register)**.** The reading of one turn is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\mathrm{per}} = K_{1+((j-1) \bmod 12)}, \qquad \kappa_{\mathrm{per}} = \sum_{j \geq 1} K_j^{\mathrm{per}} B^{-j}.$$

Proposition 11.48 (Exact value and period of the prolonged dodecaphasic register)**.** *We have*

$$\kappa_{\mathrm{per}} = \frac{N_K}{B^{12}-1}, \qquad \kappa_{\mathrm{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12}-1)} = B^{-12}\kappa_{\mathrm{per}}. \tag{11.115}$$

The infinite word of the dodecaphasic register has minimum period twelve in base one thousand and minimum period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12}. The geometric series produces $N_K/(B^{12}-1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; therefore the minimum block period is twelve. The decimal expansion is canonical, since the dodecaphasic register has no tail of 999 blocks. If its minimum decimal period were d, it would divide 36. Grouping in threes would produce a block period $d/\gcd(d,3)$. No proper divisor of 36 gives twelve through that operation. Hence the minimum decimal period is 36. □

Both readings are rational, but they are not the same rational number. The first ends after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\mathrm{per}} - \kappa_{36} < B^{-12}$. Their closeness does not allow one to be substituted for the other in an exact identity. In particular, the finite closure of alpha uses κ_{36}, whereas the sum of the prolonged section uses κ_{per}.

11.4.12 Reversibility of the rational constant and phase change

The rational constant κ_{per} fully determines the register K when the base, length and reading origin are preserved. Its numerical function and its structural function are related through an exact inversion.

Proposition 11.49 (Integral recovery and cyclic transformation)**.** *Let* $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. *Then*

$$I(K) = (1000^{12}-1)\kappa_{\mathrm{per}}.$$

The integer expansion of $I(K)$ *into twelve blocks of base* 1000 *recovers all the* K_j*, including the leading zeros of each block. If* $\rho(K) = (K_2, \ldots, K_{12}, K_1)$*, one has*

$$\kappa(\rho K) = 1000\kappa(K) - K_1. \tag{11.116}$$

Proof. The first identity inverts the definition of the periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. Furthermore,

$$I(\rho K) = 1000 I(K) - K_1(1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ gives (11.116). □

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the transformations D_3K, D_4K, Q already constructed. The rational evaluation preserves the signed register and the cyclic differences that depend on it. The total depth and the transport operators of a history remain in their own coordinates: the preceding inversion has the integral dodecaphasic register as its domain.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \qquad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical reading, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

The subsequent incidence also has an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because the latter first recovers the integer components. The signed support associated with α also uses its regional registers and the boundary normalization. The relation between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the function of K: it provides a reversible integral coordinatization of the oriented register, whose periodic evaluation and incidence functions participate in subsequent constructions. The value, the transformation and the domain are preserved jointly.

11.4.13 Periodicity and cokernel of the transport operator

The periodicity we have just proved belongs to the dodecaphasic register. It does not imply periodicity of the regional coordinates, of the closure carries or of alpha. It is useful to specify this separation through a second quotient, purely integral.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \longmapsto K + \partial_b h, \qquad C \longmapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it is not authorization to modify the dodecaphasic register already selected by the register.

Proposition 11.50 (Periodic carry quotient)**.** *With the preceding orientation,*

$$\operatorname{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module is presented as $\mathbb{Z}[t]/(t^{12}-1)$. Imposing $S = bI$ is equivalent to imposing $t = b$, and leaves the relation $b^{12} - 1 = 0$. Evaluating the coefficients at b, with the chosen concatenation order, represents the resulting class. □

The denominator $B^{12} - 1$ of κ_{per} and this quotient arise from the same cyclic length, but their objects remain distinct: one is a rational evaluation and the other a quotient of a carry module. The finite closure of alpha will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

The next chapter is thus prepared. The dodecaphasic register has been fixed before solving the alpha recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the register; and its two rational evaluations have separate domains and uses. None of these facts uses alpha as input, and none yet decides the arithmetic type of the output of the complete closure.

11.5 Complete recovery of the register through weighted hexadic incidences

The generated dodecaphasic register preserves twelve ordered components and a phase origin. Paley–Witt incidence provides an additional reading of that register: each hexadic support combines six of its components through a sum weighted by the incidence itself. The reconstruction developed here receives the values of these sums from the same register. Its purpose is to prove that the readings preserve all the components and to exhibit the inverse operator with explicit rational coefficients.

11.5.1 Incidence encoder and twelve explicit witnesses

In the incidence coordinate system already fixed, let

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3).$$

For $w \in \mathbb{F}_3^6$, the encoder is $\mathrm{Enc}(w) = (w, wA_W) \in \mathbb{F}_3^{12}$ and its support is

$$H(w) = \{j \in \{1, \dots, 12\} : \mathrm{Enc}(w)_j \neq 0\}.$$

Choosing a position uses indices from one to twelve; implementations with zero-based indices are transported through $j \mapsto j + 1$.

The following twelve witnesses belong to the domain of the encoder:

i	$w^{(i)}$	i	$w^{(i)}$
1	112210	7	122020
2	122101	8	112002
3	121200	9	111001
4	111100	10	110122
5	121012	11	101221
6	111020	12	011111.

Define $H_i = H(w^{(i)})$ and $B_{ij} = \mathbf{1}_{\{j \in H_i\}}$. Evaluating the encoder gives

$$B = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{11.117}$$

Each row contains six entries equal to one. Consequently, each H_i is a hexad of the image of the encoder, and the matrix B is constructed from those supports.

11.5.2 Rational inverse and uniqueness of recovery

Let the integer matrix be

$$M = \begin{pmatrix} 7 & 1 & 8 & -7 & 7 & -1 & -4 & 8 & -7 & 3 & 3 & -15 \\ 7 & 7 & -4 & -1 & 1 & -7 & 8 & 8 & -7 & 3 & -15 & 3 \\ 1 & 7 & 8 & -7 & 7 & -7 & 8 & -4 & -1 & -15 & 3 & 3 \\ 1 & 1 & 2 & 5 & -5 & -1 & -4 & -4 & -1 & 3 & 3 & 3 \\ 1 & -5 & -4 & -1 & 1 & 5 & 2 & -4 & -1 & 3 & 3 & 3 \\ -5 & 1 & -4 & -1 & 1 & -1 & -4 & 2 & 5 & 3 & 3 & 3 \\ -5 & -11 & 2 & 5 & -5 & 11 & -10 & 2 & 5 & 3 & 3 & 3 \\ -5 & -5 & -10 & 11 & -11 & 5 & 2 & 2 & 5 & 3 & 3 & 3 \\ -11 & -5 & 2 & 5 & -5 & 5 & 2 & -10 & 11 & 3 & 3 & 3 \\ -11 & -11 & -4 & 17 & 1 & 11 & -10 & -10 & 11 & 3 & 3 & 3 \\ -11 & 1 & -10 & 11 & -11 & 17 & -4 & -10 & 11 & 3 & 3 & 3 \\ 1 & -11 & -10 & 11 & -11 & 11 & -10 & -4 & 17 & 3 & 3 & 3 \end{pmatrix}. \tag{11.118}$$

Theorem 11.51 (Exact recovery from twelve hexadic readings)**.** *For $v \in \mathbb{Q}^{12}$, let*

$$\mathrm{Read}(v) = Bv, \qquad \mathrm{Rec}(y) = \frac{1}{18} My.$$

We verify

$$\mathrm{Rec}(\mathrm{Read}(v)) = v, \qquad \mathrm{Read}(\mathrm{Rec}(y)) = y.$$

Therefore every $y \in \mathbb{Q}^{12}$ admits a unique register $v \in \mathbb{Q}^{12}$ with $\mathrm{Read}(v) = y$.

Proof. Integer multiplication of the matrices (11.117) and (11.118) gives $MB = BM = 18 I_{12}$. Each diagonal entry results from a scalar product equal to eighteen and each off-diagonal entry results from a zero scalar product. Dividing by eighteen gives the two inverse identities.

In particular, $\mathrm{Rec}(\mathrm{Read}(v)) = (MB/18)v = v$ and $\mathrm{Read}(\mathrm{Rec}(y)) = (BM/18)y = y$. The second identity proves existence through $v = \mathrm{Rec}(y)$. If $Bu = Bv$, applying $M/18$ to both sides gives $u = v$, which proves uniqueness. □

The check of B from the encoder and the two exact multiplications are formalized in the theorems `incidence_from_encoder`, `inverse_left` and `inverse_right`. The composition identities correspond to `recover_read` and `read_recover`. Their mathematical content is found entirely in the matrices and in the preceding proof; the formal certificate records the verification of those same operations.

Corollary 11.52 (Separation by all weighted incidences)**.** *If $u, v \in \mathbb{Q}^{12}$ satisfy*

$$\sum_{j\in H(w)} u_j = \sum_{j\in H(w)} v_j$$

for every $w \in \mathbb{F}_3^6$ whose encoded support has six positions, then $u = v$.

Proof. The hypothesis applies in particular to the twelve displayed witnesses. By the definition of B, their twelve equalities constitute $Bu = Bv$. Theorem 11.51 allows both inputs to be recovered and concludes $u = v$. □

Example 11.53 (Recovery of a peripheral component)**.** For the canonical vector $v = e_{12}$, the only nonzero readings are the first and the ninth:

$$y = Be_{12} = (1,0,0,0,0,0,0,0,1,0,0,0)^{\mathsf{T}}.$$

The sum of the first and ninth columns of M is $18e_{12}$. Consequently, $My/18 = e_{12}$. The two hexadic sums distinguish this position in the joint reading because the remaining components cancel exactly in the recovery operator.

11.5.3 Symmetric recovery operator on the one hundred and thirty-two hexads

The Steiner system $S(5,6,12)$ already produced by incidence contains 132 hexads. Denote their set by $\mathcal{H}$ and define

$$\mathcal{I} : \mathbb{Q}^{12} \longrightarrow \mathbb{Q}^{\mathcal{H}}, \qquad (\mathcal{I}v)_H = \sum_{j\in H} v_j.$$

This reader has redundancy: it preserves the twelve components through 132 sums, whereas the preceding theorem provides a basis of twelve of these sums.

Theorem 11.54 (Inverse on the image of the complete reader)**.** *We have*

$$\mathcal{I}^{\mathsf{T}}\mathcal{I} = 36I_{12} + 30J_{12},$$

where J_{12} has all its entries equal to one. For $y = \mathcal{I}v$, the register is recovered through

$$396 v_i = 11 \sum_{H\ni i} y_H - 5 \sum_{H\in\mathcal{H}} y_H, \qquad i = 1,\dots,12. \tag{11.119}$$

Proof. The property $S(5,6,12)$ implies that each position belongs to $\binom{11}{4}/\binom{5}{4} = 66$ hexads. Each pair of distinct positions belongs to $\binom{10}{3}/\binom{4}{3} = 30$ hexads. Therefore, the product of the transposed incidence matrix with itself has diagonal 66 and off-diagonal entries 30, which is $36I + 30J$.

Let $S = \sum_j v_j$. The two preceding counts give

$$\sum_{H\ni i} y_H = 66 v_i + 30 \sum_{j\neq i} v_j = 36 v_i + 30 S, \qquad \sum_{H} y_H = 66 S.$$

Multiplying the first equality by eleven and subtracting five times the second eliminates S and produces $396 v_i$. This proves (11.119) for every reading belonging to the image of $\mathcal{I}$. □

Corollary 11.55 (Uniform and transverse components)**.** *On $\mathbb{R}^{12}$, the complete reader has singular value $\sqrt{396}$ in the uniform direction and singular value 6 in its orthogonal complement. Its inverse operator on the image has operator norm $1/6$.*

Proof. The matrix J acts by twelve on $\mathbb{R}\mathbf{1}$ and by zero on $\mathbf{1}^{\perp}$. The eigenvalues of $36I + 30J$ are therefore 396 and 36. The singular values are their positive roots; the largest singular value of the inverse operator is the reciprocal of the smallest of them. □

11.5.4 Composition with the generated register

Applying these identities to the integer dodecaphasic register K, already produced with its order and origin, the readings BK or $\mathit{I}K$ allow its twelve components to be recovered exactly. The subsequent positional interpretation takes that same register in base one thousand; the Hadamard operators, differences and hexadic readers are recoverable realizations of a common provenance.

The system of twelve rational coordinates admits any input $y \in \mathbb{Q}^{12}$ and reconstructs it as a unique rational register. The subdomain corresponding to generated HMT registers additionally preserves its conditions of integrality, triad width, origin, selection and boundary. For the redundant system of 132 readings, the inverse acts on the image of I. These specifications type the recovered content without modifying the antecedent generation of the register.

11.6 Derivation of the fine-structure constant

The fine-structure constant receives in this development an arithmetic determination combining the three regional coordinates and the integral dodecaphasic register. The operation combines their blocks with fixed orientation and transports the quotients of positional normalization. The resulting parameter expresses the compatibility of that oriented difference with prolongation of the cylinders. Its value and the incidence data used in its construction will remain related throughout the exceptional development.

The internal order is important. The coordinates π, e and φ, already constructed in the preceding section, are prior coordinates of the same arithmetic state with trajectory memory, and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select alpha. Correlated generation of the four coordinates shares origin and coinductive compatibility, without thereby requiring their four components to be evaluated in a single step.

The two paths express different mathematical functions of the same parameter. The first realizes an integer closure by Euclidean division in base 1000. The second studies the analytic equilibrium of vacancies through the resolvent $E_{21/22}$ and its graded prolongations. One path determines blocks and quotients; the other determines a root through an equation and a uniqueness domain. Their identification is formulated later in the common cylinders at every depth. The exposition distinguishes these three results: positional construction, analytic determination and their compatibility.

11.6.1 First path: positional determination by integer closure

11.6.1.1 Domain and orientation of integer closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and prolong the dodecaphasic register by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphasic register is subtracted according to the already typed dodecaphasic closure. For $B = 1000$, the local operation distinguishes the signed balance

$$Z_j := P_j + E_j - \Phi_j - K_j$$

from the sum incorporating the incoming quotient. Normalization is

$$s_j = Z_j + c_{j+1}, \qquad A_j = s_j - B\left\lfloor \frac{s_j}{B} \right\rfloor, \qquad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \tag{11.120}$$

Thus, $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted because the table is printed from left to right.

Definition 11.56 (Finite-depth closure)**.** For a depth N and an integer boundary t, the map C_N receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (11.120) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division has no choice of residue sign: the remainder is fixed in $\{0, \dots, 999\}$ and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of the closure are not anomalies; they are the integer information needed to maintain the oriented equality.

11.6.1.2 Dodecaphasic window and carry conditions

The first twelve triads obtained are

$$P = (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884),$$
$$E = (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497),$$
$$\Phi = (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117).$$

In the closure channel, the five regions determining the coordinate P retain their distinct antecedents. Sharing this reading does not identify them as regions or as histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. Each input, incoming carry, total before normalization, resulting block and outgoing carry are shown. The table is a reproducible integer calculation, not a list of blocks selected for their resemblance to a target.

j	P_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	−1	352	352	0
4	589	459	749	729	−1	−431	569	−1
5	793	045	894	659	−2	−717	283	−1
6	238	235	848	824	−1	−1200	800	−2
7	462	360	204	621	0	−3	997	−1
8	643	287	586	058	−1	285	285	0
9	383	471	834	914	−1	−895	105	−1
10	279	352	365	794	0	−528	472	−1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 11.4: Initial closure with zero right boundary. The dependence propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the window reading is the rational number

$$\alpha_{12}^{(0)} = \frac{729735256928380099728510547238 0663}{10^{36}} = 0.007297352569283800997285105472380663. \tag{11.121}$$

The superscript recalls the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readings will become visible when the carry of the remote tail is transported.

11.6.1.3 Telescoping and reversibility in a boundary fiber

The local form of the equality is

$$P_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \tag{11.122}$$

The carry term is a link between adjacent positions. If it is removed coordinate by coordinate, a false equality results. Only an evaluation retaining its endpoints allows it to telescope.

Proposition 11.57 (Integer identity at arbitrary depth)**.** *For $\iota_N(z) = \sum_{j=1}^{N} z_j B^{N-j}$, every execution of C_N satisfies*

$$\iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(K) - \iota_N(A) = B^N c_1 - c_{N+1}. \tag{11.123}$$

Proof. Multiplying (11.122) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^{N} (Bc_j - c_{j+1}) B^{N-j} = B^N c_1 - c_{N+1}.$$

Each interior carry appears twice with opposite signs; only the two endpoints survive. □

For Table 11.4, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphasic register constant that appears is κ_{36}, not κ_{per}. The domain of this identity contains truncations and a finite boundary; its form does not authorize erasing both conditions on passing to the complete section.

Local inversion is also exact:

$$K_j = P_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ, the blocks A and the complete carries, it recovers the dodecaphasic register. Concatenation also recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not become a new causal recipe starting from alpha and then manufacturing the register. The construction direction remains signed register, dodecaphasic register, closure and output.

11.6.1.4 Prolongation and canonical normalization of the complete section

Prolongation preserves the periodic dodecaphasic register and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} P_j B^{-j}, \qquad e_0 = \sum_{j \geq 1} E_j B^{-j}, \qquad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2 and 1, so that $p = \pi - 3$, $e_0 = e - 2$ and $f = \varphi - 1$.

Proposition 11.58 (Existence and uniqueness of complete normalization)**.** *The series*

$$y = p + e_0 - f - \kappa_{\mathrm{per}} = \sum_{j \geq 1} (P_j + E_j - \Phi_j - K_j) B^{-j} \tag{11.124}$$

converges absolutely. In the domain of the obtained blocks, $0 < y < 10^{-2}$. *There is a unique canonical expansion* (A_j) *of* y, *with no eventually constant 999 tail, and a unique bounded integer sequence* (c_j) *satisfying* (11.122) *with* $c_1 = 0$. *These carries belong to the invariant set* $C = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B-1)$, so a geometric series dominates. If $Z_j = P_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r\geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; therefore $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y. Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying the equality of the two series by B^{j-1} and separating their first $j-1$ terms yields

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Moreover, $R_j(A) \in [0, 1)$, hence $-3 < c_j < 2$, which includes all carries in C. The tail shift identities give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y. The canonical expansion is unique and, once fixed, the tail formula determines all carries. Finally, if a finite run is applied with $c_{j+1} \in C$, we have $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to C. □

The proposition makes explicit the normalization of the subsequent reader: it does not replace generation of the three sequences with a conventional series introduced as a datum. Once these sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite procedure preserved in the corpus propagates the five boundaries of C and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. No boundary is chosen because it yields favorable digits, nor is compatibility at every depth inferred from observing a long run.

The initial comparison shows why this precaution matters. In the stabilized prolongation we obtain $c_{13}^{\infty} = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first turn. The first eleven triads do not change, but the prolongation begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \ldots).$$

An occurrence of block 999 does not violate the canonical convention: what is excluded is an eventually constant 999 tail. The window with zero boundary ends in 663; the truncation of the complete section ends in 662. Both assertions describe different and exact operations. They must not be merged into a single projection notation.

11.6.1.5 Operational definition and complete additive equation

Definition 11.59 (Alpha coordinate of the closure)**.** The integer path determines the coordinate

$$\alpha_A = \sum_{j\geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\mathrm{per}}, \tag{11.125}$$

where (A_j, c_j) is the compatible canonical normalization of (11.120). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\mathrm{per}}.} \tag{11.126}$$

The integer four is not calibrated: it is $3 + 2 - 1$, a combination of the integer parts of the three coordinates. Base one thousand comes from the cubic chart and period twelve from the dodecaphasic register. The signs are those of the oriented channel and closure, and the remote boundary is that imposed by the compatible section. These are the effective origins of the equation's coefficients.

At this level alpha can be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphasic return, calculated with compatible carry. The term *defect* denotes a determined algebraic difference, not an arbitrary loss or imperfection of the system. This description does not remove the genealogy: without the regions, signed register, orientation and carry law, the scalar equality would not be a sufficient exposition of the construction.

Three objects that alpha notation may accompany in subsequent readers must also be kept separate: A is a word of blocks; the negative pre-carry support is obtained from the sign of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None of them is interchangeable with the real value α_A.

Signed balance and sum with incoming quotient. Positional division consists of two consecutive stages. With $K_j^{\mathrm{per}} = K_{1+((j-1) \bmod 12)}$, which denotes the periodic prolongation also denoted by K_j in (11.120), the balance before quotient transport is

$$Z_j := P_j + E_j - \Phi_j - K_j^{\mathrm{per}}.$$

The variable s_j of (11.120) denotes the sum already including the incoming contribution, so

$$s_j = Z_j + c_{j+1}, \qquad A_j = Z_j + c_{j+1} - 1000c_j.$$

The incidence configuration H_α^- uses the signs of the balance Z_j of the initial window; determination of the digits additionally uses the linking quotients c_{j+1} and c_j. This distinction jointly preserves the signed support and normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and dodecaphasic register come from the common APP–TRIT–TPK generation. Once produced, their positional composition has an explicit dependence on reading depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\mathrm{per}}, \qquad y_N = \sum_{j=1}^{N} Z_j B^{-j}, \qquad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (11.124) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \qquad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window and that of its prolongation, both expressed in the same coordinates.

Proposition 11.60 (Boundary transport and canonical cylinder)**.** *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in C = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^{N} A_j^{(N,t)} B^{-j}$. For the registers in the domain considered, we have*

$$\alpha_A - a_N^{(t)} = B^{-N}\big(R_{N+1}(Z) - t\big), \qquad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \tag{11.127}$$

The quotient of the complete prolongation is $c^{\infty}_{N+1} = \lfloor R_{N+1}(Z) \rfloor$. *With this boundary,*

$$0 \le \alpha_A - a_N^{(c^{\infty}_{N+1})} < B^{-N},$$

so that the resulting window is the canonical prefix of order N.

Proof. Telescoping (11.122) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, each incoming state belongs to C and the first dividend lies between 5 and 9, we have $c_1 = 0$. Substituting this equality into the decomposition of the series proves (11.127). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in C$.

For complete canonical normalization, $c^{\infty}_{N+1} = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. Integrality of the quotient implies $c^{\infty}_{N+1} = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c^{\infty}_{N+1})} = B^{-N} R_{N+1}(A)$, which proves membership in the canonical cylinder. □

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed at the scale of the next position. Calculation over the five boundaries preserves this information as a finite family of possibilities until the blocks to the left stabilize. Quantitative contraction comes from depth and tail transport; its object is normalization of the already generated coordinates.

Graded domain of the analytic resolvent. The analytic composition displayed in §11.6.2.2 acts on $\operatorname{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation produces exactly $T_{7:9}$; degree 9 is the last component of that domain. The identity $N^3 = 0$ explains why the resolvent is expressed through three summands. To describe a prolongation to higher degrees, one must additionally specify the transport between successive graded domains and its action on the initial component of each domain. Local nilpotency and global compatibility thus perform distinct mathematical functions.

11.6.1.6 Constructive determination and function of the dodecaphasic register

The preceding definition makes precise what it means to determine alpha from the structure. The domain of the operation contains four registers of known provenance: the three regional sequences and the integral phase register. Normalization additionally incorporates a boundary condition compatible with prolongation. Once these data are fixed, Euclidean division determines each remainder and its quotient. Proposition 11.58 extends that determination to the complete system. Independence from a target physical value is therefore a property of the construction order: the value compared at the end does not enter into the selection of any argument of this map.

This causal independence must be distinguished from algebraic independence between numbers. The former is established by identifying the domain, operations and provenance of the coefficients; the latter would be an assertion about polynomial relations. The result used here is the former. The construction determines a coordinate through registers produced by APP–TRIT–TPK and allows the relations it satisfies to be checked subsequently. That order makes it possible to use an already generated coordinate in a later operation without turning it into an external parameter.

The register K has a more precise function than a numerical correction added to a formula. Its twelve positions come from the reversible transformation of the signed register. Equation $U = \mathcal{H}_{12}K$ preserves the possibility of returning to that register, while the transverse differences and total sum allow another reconstruction. In periodic evaluation, the order of positions determines the positional weight of each component. The same set of components, arranged in another order, generally produces another rational coordinate. The phase origin and orientation are therefore part of the definition of the number appearing in (11.126).

Alpha normalization composes that information with the regions. Its local identity distinguishes two successive operations. First the oriented balance of blocks is formed; then that balance is represented by a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and allows subsequent extraction of a support. The second allows the prefixes of the real

coordinate to be formed. Both readings remain related because the state preserves the data before and after normalization. The signed support and positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. Interior quotients cancel in pairs when a prefix is evaluated, while their endpoints remain in the equality. This cancellation is a property of the weighted sum; it neither removes the quotients from the state nor allows them to be suppressed before taking the sum. The right boundary of a window receives the contribution of the continuation. For this reason, evaluating an isolated window and restricting a prolonged section are distinct operations. The change of the final block from 663 to 662, proved earlier, provides an explicit realization of this difference without altering the already stabilized blocks to its left.

In this sense, bidirectionality has a specific arithmetic formulation. Generation provides compatible prolongations; normalization transmits the integer information of their boundary from lower-weight positions to higher-weight ones. The equation preserves the contribution of both directions through its indices and boundary conditions. Reversibility in the fixed fiber is checked by reconstructing K from the other registers and complete quotients. This inversion certifies consistency of the original operation; the genealogical direction remains that which first produces the register and then the alpha coordinate.

The second determination has complementary content. In it, the immediate object is a vacancy balance and the parameter appears as its root in an isolating domain. The positional path provides a canonical normalization of registers; the analytic path studies the vanishing of an operator. Presenting both requires preserving precisely that difference of functions. Their relations to the common state must be expressed through refinement maps and comparison of their complete realizations, as developed in the corresponding subsection. Agreement of a finite approximation constitutes a localized check; identification of the complete constructions has the domain and scope of its own statement.

Finally, this organization explains alpha's position in the article. Its equation appears after regional generation and construction of K because it uses both results. The electronic section then uses that coordinate as a factor in its composition. Exceptional prolongation receives the registers and their supports, where incidence information is preserved. These two continuations take up the same provenance and develop distinct mathematical functions: evaluation of a central section and construction of a boundary configuration. The development of each allows the structure to be followed through the equation, instead of limiting comparison to its scalar value.

11.6.2 Second route: analytic determination through vacancies

11.6.2.1 Transport invariants and graded resolvent

The analytic route receives invariants of the oriented state, not the output α_A. Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \tag{11.128}$$

and the vacancy condition uses

$$\delta = \log_{10}\frac{10}{9}.$$

The ratio $10/9$ compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification reading; it is not a target alpha value.

The genealogy of the coefficients must be preserved in full, but with its exact logical force. Recent developments bring together a marked signature of the state. The synchronization matrix and associated census are

$$D_{\mathrm{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \qquad N_9 = 43.$$

From its determinant we extract

$$a = -\frac{\det D_{\mathrm{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \qquad k_* = mb - 1 = 20.$$

The vacancy reading in the one-thousand chart gives

$$v_- = \lfloor 1000\delta \rfloor = 45, \qquad v_+ = \lceil 1000\delta \rceil = 46.$$

These integers are typed antecedents of the truncation evaluation. They are not obtained from its root or from a metrological approximation to it.

The lower branch also coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \qquad \det B_c = 45.$$

The transverse rank of the register in the preceding chapter gives $r_{\mathrm{dod}} = 11$, and the phase half-crown provides $f_{\mathrm{ph}} = 8\binom{6}{2} = 120$. In particular, $45 r_{\mathrm{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{I}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\mathrm{ph}}, r_{\mathrm{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{E}_9(\mathcal{I}_9)(x) = \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ - \frac{x^7}{f_{\mathrm{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_- r_{\mathrm{dod}}}. \end{aligned} \tag{11.129}$$

Substitution recovers the coefficients of P_6 and its order-nine prolongation. The pair $45/46$ crosses the separation between them: the upper branch occupies degree six and the lower one reappears in degrees eight and nine.

The factorization in (11.129) is formulated for a state preserving origin, channel order and orientation. Naturality with respect to morphisms preserving those data is not equivalent to uniqueness of every reader after they have been forgotten. The domain of this factorization is determined by the signature (11.129); the extension to all degrees is constructed in Section 11.6.4, with its recurrence class and its dependence on the state's pre-coordinate explicitly defined.

11.6.2.2 Nilpotent resolvent in degrees 7, 8 and 9

The first prolongation admits a particularly explicit finite operator construction. On the space with basis e_7, e_8, e_9, let

$$Ne_7 = -\frac{8}{3}e_8, \qquad Ne_8 = \frac{2}{11}e_9, \qquad Ne_9 = 0, \qquad v_7 = -\frac{1}{120}e_7.$$

The symbol N here denotes a nilpotent operator, not a truncation depth. Its domain and grading are fixed before evaluating the polynomial at a number.

Lemma 11.61 (Exact nilpotent prolongation)**.** *We have* $N^3 = 0$ *and*

$$(I - N)^{-1} v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

The evaluation $e_n \mapsto x^n$ produces

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \tag{11.130}$$

Proof. The operator increases degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \qquad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. □

Here the denominators 120, 45 and 495 are not decimal adjustments: they come from the half-crown, the APP determinant and its product by the dodecaphasic rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only the symmetry of the cycle and replacing this operator with another symmetric operator does not necessarily preserve the coefficients.

11.6.2.3 Root of multiplicity one of the order-nine truncation

Define

$$P_9 = P_6 + T_{7:9}, \qquad E^{(9)}_{21/22}(x) = P_9(x) - \delta, \qquad I_\alpha = (0, 10^{-2}).$$

This finite operator has its own existence and uniqueness result.

Proposition 11.62 (Root of multiplicity one of the truncation)**.** *The operator $E^{(9)}_{21/22}$ has a unique root ξ_9 in I_α, and that root is simple. The following rational isolation is obtained:*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \tag{11.131}$$

Proof. The bounds on the internal coordinates $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ suffice for existence. At zero, $E^{(9)}_{21/22}(0) = -\delta < 0$. At 10^{-2}, the linear contribution minus the negative terms is greater than 0.0626, hence the operator value is positive.

For $0 \le x \le 10^{-2}$, discarding the positive terms of the derivative gives

$$(E^{(9)}_{21/22})'(x) \ge 2\pi - \frac{7}{2}\cdot 10^{-2} - \frac{10}{21}\cdot 10^{-8} - \frac{6}{46}\cdot 10^{-10} - \frac{7}{120}\cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. The narrower interval (11.131) results from a directed rational evaluation. Let x_- and x_+ be its endpoints. To compute $\delta = \log(10/9)/\log 10$, we use, for $z > 1$,

$$L_N(z) = 2\sum_{j=0}^{N-1} \frac{y^{2j+1}}{2j+1}, \qquad y = \frac{z-1}{z+1}, \qquad 0 < \log z - L_N(z) < \frac{2y^{2N+1}}{(2N+1)(1-y^2)}.$$

The remainder is bounded by the geometric series of positive terms. This formula is applied to $z = 10/9$ and $z = 3$, with $N = 520$, using $\log 10 = \log(10/9) + 2\log 3$. Evaluation of π uses the generated thousand-digit regional cylinder, including its width 10^{-1000}. Interval arithmetic in (11.129) gives

$$-\frac{4559}{10^{23}} < E^{(9)}_{21/22}(x_-) < -\frac{4558}{10^{23}}, \qquad \frac{1698}{10^{23}} < E^{(9)}_{21/22}(x_+) < \frac{1699}{10^{23}}.$$

These signs and the proved monotonicity establish the isolation. The endpoints are results of evaluating the operator; its coefficients arise from the signature constructed beforehand. □

The result concerns ξ_9. It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. The root of a truncation may share a prefix with the root of the prolonged operator and differ at a later position. Simplicity allows the movement of the root under perturbations to be controlled; it does not annul those perturbations.

11.6.3 Compatibility of the two determinations at every depth

Remark 11.63 (Truncation, extension and compatibility). The operations from degrees 6 to 9 determine a finite jet. Positional normalization determines the complete coordinate through its compatible carry sequence. The following proposition expresses the general criterion for identification by cylinders. Section 11.6.4 then constructs a correlated analytic chart of all degrees, with recurrence, uniform convergence, uniqueness of the root and projective compatibility. The construction receives the shared coordinate of the state and the declared class of prolongations; it thus preserves the difference between an analytic representation of that coordinate and an independent selection.

The complete analytic path is organized into a family of compatible graded readers

$$E^{(6)}_{21/22} \longleftarrow E^{(9)}_{21/22} \longleftarrow E^{(12)}_{21/22} \longleftarrow \cdots .$$

The symbol

$$E_{21/22} = \varprojlim_{m} E^{(6+3m)}_{21/22}$$

denotes that complete reader, with its transports and truncations, not a formal series freely chosen to make it pass through α_A. The second route determines its unique positive root in I_α, denoted by α_B, with the uniqueness and compatibility of the complete reader proved in Section 11.6.4.

The proofs in Section 11.6.4 establish that, at each depth, both systems determine the same surviving cylinder C^α_N, and that their diameters tend to zero. It is useful to state the mathematical consequence of this compatibility separately.

Proposition 11.64 (Identification of compatible limits)**.** *If the values of the integer and analytic routes belong to the same cylinders C^α_N for every N, and* $\operatorname{diam} C^\alpha_N \to 0$*, then*

$$\alpha_A = \alpha_B = \alpha.$$

Proof. For every N, the distance between the two values is bounded by $\operatorname{diam} C^\alpha_N$. Letting N tend to infinity gives distance zero. Existence of the compatible sections ensures that this is not the empty intersection of a system of approximations. □

This proof identifies the consequence of the compatibility squares. Their construction at arbitrary depth appears in Section 11.6.4. Finite evaluations check realizations of that construction; the recurrence and its uniform bounds establish the passage to the limit.

The difference can be expressed algebraically. Let $F = \mathbb{Q}(\pi, \delta)$ and let $j_9 : F[[x]] \to F[x]/(x^{10})$ be the truncation. Then

$$\ker j_9 = x^{10} F[[x]].$$

All operators

$$E_h(x) = E^{(9)}_{21/22}(x) + x^{10} h(x)$$

have the same ninth-order truncation. However, taking $h = 0$ and $h = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0,$$

so they do not share that root. The observation does not permit choosing a tail foreign to HMT transport; it demonstrates precisely why the generated tail and its compatibility must be preserved. Nor does

an arbitrary formal series suffice for speaking of analytic roots without its convergence domain or the realization of the cylinder system.

11.6.4 Compatibility between the dodecaphase and analytic realizations of alpha

Compatibility is established at every depth. The reversible dodecaphase publication and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal publications with distinct codomains, produced from the same enriched APP–TRIT–TPK state. The jet $E^{(9)}_{21/22}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below is not deduced from the bare jet: it receives the shared coordinate of the state and publishes its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This separation preserves the causal order

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \mathcal{X}^{\mathrm{enr}}_{\alpha} \longrightarrow (P, E, \Phi, U^{\mathrm{sgn}}_{12}, K) \longrightarrow \alpha_{\mathrm{HMT}}.$$

The coordinates P, E, Φ and the register K are prior outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition and not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\widehat{\mathcal{X}}^{\mathrm{enr},\alpha}_{r}$ be the sector of truncated states extending the internal alpha seed fixed in R_{36}. We write

$$x^{\alpha}_{r} = (\widetilde{x}_r, \chi_{\alpha}, \varepsilon^{\alpha}_{r}, N^{\alpha}_{r}),$$

where χ_{α} is the internal character of the sector, not the published scalar value. Each state preserves residue, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, register, memory, survival, provenance and depth. If

$$d^{\alpha}_{r} = \mathrm{Dig}_{729}(\varepsilon^{\alpha}_{r}), \qquad \varepsilon^{\alpha}_{r+6} = 729\varepsilon^{\alpha}_{r} - d^{\alpha}_{r}, \qquad N^{\alpha}_{r+6} = 729N^{\alpha}_{r} + d^{\alpha}_{r},$$

the effective transition is

$$\mathcal{U}^{\alpha}_{r} = \mathrm{Upd}^{\alpha}_{r} \circ \mathrm{Tra}^{\alpha}_{r} \circ \mathrm{Sel}^{\alpha}_{r}, \qquad \mathrm{Ext}^{\alpha,\rightarrow}_{r+6,r} = \mathrm{graph}(\mathcal{U}^{\alpha}_{r}).$$

The functionality of these extensions, with the seed R_{36} fixed, produces a unique compatible section: the phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}^{\mathrm{enr}}_{\alpha} := \widehat{\Omega}_{\alpha} := \varprojlim_{r \geq 36} \big(\widehat{\mathcal{X}}^{\mathrm{enr},\alpha}_{r}, \mathrm{Ext}^{\alpha,\rightarrow}_{r+6,r}\big). \tag{11.132}$$

In this section, κ_{per} is the periodic reading κ_{ag} of the already constructed register; both notations denote the same fraction in the fixed base-one-thousand chart.

11.6.4.1 The dodecaphase closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \ldots, 999\}^N, \qquad \mathsf{T}_{\alpha,N}(\mathsf{s}) := (A_1, \ldots, A_N), \qquad \mathsf{s} \in \mathcal{X}^{\mathrm{enr}}_{\alpha},$$

where the blocks A_j are obtained through (11.120) with the compatible boundary $c^{\infty}_{N+1} = \lfloor R_{N+1}(Z) \rfloor$. If $N' \geq N$, let $\rho_{N',N} : \mathcal{W}_{N'} \to \mathcal{W}_N$ be the restriction to the first N blocks.

Theorem 11.65 (Complete dodecaphase output and projective compatibility)**.** *The family* $(\mathsf{T}_{\alpha,N})_{N\geq 1}$ *is compatible:*

$$\rho_{N',N} \circ \mathsf{T}_{\alpha,N'} = \mathsf{T}_{\alpha,N} \qquad (N' \geq N). \tag{11.133}$$

Its canonical cylinders

$$C_N^A(\mathsf{s}) := \left[\sum_{j=1}^{N} A_j 1000^{-j},\ \sum_{j=1}^{N} A_j 1000^{-j} + 1000^{-N}\right) \cap I_\alpha \tag{11.134}$$

are nested, have diameter at most 1000^{-N} *and satisfy*

$$\bigcap_{N\geq 1} C_N^A(\mathsf{s}) = \{\alpha_A(\mathsf{s})\}, \qquad \alpha_A = \pi_{\mathrm{HMT}} + e_{\mathrm{HMT}} - \varphi_{\mathrm{HMT}} - 4 - \kappa_{\mathrm{per}}. \tag{11.135}$$

Consequently, the first route publishes a unique infinite section, which we denote by $\alpha_{\mathrm{HMT}} := \alpha_A$.

Proof. Proposition 11.58 provides the unique canonical expansion and the unique bounded sequence of carries. The local identity (11.122) and its telescoping (11.123) preserve the remote boundary; proposition 11.60 proves that restricting a prolonged execution recovers the previous prefix exactly. This gives (11.133). Membership of the complete sum in each cylinder follows from the tail identity, and $\operatorname{diam} C_N^A \leq 1000^{-N} \to 0$; the intersection contains a single point. Finally, (11.126) identifies that point with the expression of (11.135). All its coefficients and signs are fixed before alpha is published. □

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^{\infty} = -1$. In particular,

$$\rho_{36,12}(\alpha_{A,36}) = \alpha_{A,12} \tag{11.136}$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

11.6.4.2 Analytic representation of the alpha coordinate

The analytic prolongation is specified through all its coefficients. Let $\mathsf{s} \in \mathcal{X}_\alpha^{\mathrm{enr}}$. Before decimal normalization, the same state already publishes the precoordinate

$$a(\mathsf{s}) := p(\mathsf{s}) + e_0(\mathsf{s}) - f(\mathsf{s}) - \kappa_{\mathrm{per}}(\mathsf{s}) \in I_\alpha, \tag{11.137}$$

with $p = \pi_{\mathrm{HMT}} - 3$, $e_0 = e_{\mathrm{HMT}} - 2$ and $f = \varphi_{\mathrm{HMT}} - 1$. Equivalently, $a = \pi_{\mathrm{HMT}} + e_{\mathrm{HMT}} - \varphi_{\mathrm{HMT}} - 4 - \kappa_{\mathrm{per}}$. No conventional α is introduced here: p, e_0, f are the three already generated regional characters and κ_{per} is the rational reading of the terminal register. Integer closure applies to (11.137) the normalization of (11.120); the following chart publishes a compatible analytic representation on the same precoordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\mathrm{per}})$ directly: it does not contain an edge $\alpha_A \to \lambda$, although the equality proved below subsequently allows both publication names to be identified.

Write $q = 1/729$, let F_{s} be the ordered field generated by the coordinates of s, and set $E_{9,\mathsf{s}}(X) = E_{21/22}^{(9)}(\mathsf{s}; X)$. We define the linking coefficient

$$\lambda(\mathsf{s}) := -\frac{E_{9,\mathsf{s}}(a(\mathsf{s}))\big(1 - q\,a(\mathsf{s})^3\big)}{a(\mathsf{s})^{10}}. \tag{11.138}$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (11.138) must be read with its exact type: once the first publication of the state has produced $a(\mathsf{s})$, it fixes the unique prolongation in the geometric class defined below. It does not constitute a second independent selection of $a(\mathsf{s})$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_{\mathsf{s}}$ set

$$E_{\mathsf{s},\mu}(X) := E_{9,\mathsf{s}}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(\mathsf{s}) < 10^{-2}$ and $qa(\mathsf{s})^3 < 1$, we have

$$E_{\mathsf{s},\mu}(a(\mathsf{s})) = 0 \quad \Longleftrightarrow \quad \mu = \lambda(\mathsf{s}). \tag{11.139}$$

Thus, the coefficient seed has no freedom once the state and the class of q-geometric prolongations are fixed. This uniqueness is relative to the declared class; it does not assert that the order-nine jet alone selects a tail among all possible analytic series.

The local action on blocks of three coefficients is

$$C_\alpha : F_{\mathsf{s}}^3 \longrightarrow F_{\mathsf{s}}^3, \qquad C_\alpha(u, v, w) = q(u, v, w), \qquad c^{(0)} = (\lambda(\mathsf{s}), 0, 0). \tag{11.140}$$

Therefore, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(\mathsf{s}), \qquad b_{11+3n} = b_{12+3n} = 0. \tag{11.141}$$

This is the rule that calculates every coefficient beyond degree nine. No free parameter remains at each prolongation.

For $M \geq 0$, let

$$E_{\mathsf{s},M}(X) := E_{9,\mathsf{s}}(X) + \lambda(\mathsf{s}) \sum_{n=0}^{M} q^n X^{10+3n}. \tag{11.142}$$

The complete function is

$$E_{\mathsf{s},\infty}(X) := E_{9,\mathsf{s}}(X) + \lambda(\mathsf{s}) \frac{X^{10}}{1 - X^3/729}. \tag{11.143}$$

To formulate the projective system without implicit types, we fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{J}_M := \coprod_{\mathsf{s} \in \mathcal{X}_\alpha^{\mathrm{enr}}} \{\mathsf{s}\} \times F_{\mathsf{s}}[X]/(X^{d_M+1}). \tag{11.144}$$

If $M' \geq M$, the restriction is

$$\tau_{M',M}(\mathsf{s}, [P]_{d_{M'}}) := (\mathsf{s}, [P]_{d_M}). \tag{11.145}$$

Proposition 11.66 (Naturality of the analytic recurrence)**.** *Let*

$$\Gamma_M(\mathsf{s}) = \big(\mathsf{s}, [E_{\mathsf{s},M}]_{d_M}\big).$$

If $M' \geq M$, the truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \qquad \Gamma_{E21}(\mathsf{s}) = \varprojlim_M \Gamma_M(\mathsf{s}). \tag{11.146}$$

Projection onto the three new degrees is precisely $C_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (11.141) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ only adds the next block; truncating it recovers the previous polynomial literally. Compatibility for $M' \geq M$ follows by composition. Thus, the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under C_α. □

11.6.4.3 Uniform convergence, simple root and rational intervals

The regional cylinders and the periodic register provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}\},$$

let P_c be its certified thousand-digit nonadic prefix. The datum used is not the rational P_c as though it were the complete value, but the half-open cylinder and its outer closure

$$C_c^{(1000)} = [P_c, P_c + \varepsilon), \qquad \overline{C}_c^{(1000)} = [P_c, P_c + \varepsilon], \qquad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively with those closures. If P_π, P_e, P_φ denote the three prefixes, we obtain

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\mathrm{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\mathrm{per}}, \\ A_{1000} &:= [a_-, a_+], \qquad a(\mathsf{s}) \in [a_-, a_+] = A_{1000}, \qquad \operatorname{diam} A_{1000} = 3\varepsilon. \end{aligned} \tag{11.147}$$

The natural interval extension of (11.138) then transports A_{1000}, the cylinder of π_{HMT} and the directed rational bound of δ to a closed interval $\Lambda_{1000} \ni \lambda(\mathsf{s})$; none of the three tails is frozen. In particular,

$$\begin{gathered} \frac{729735256928380099728510547238066 2}{10^{36}} < a(\mathsf{s}) < \frac{729735256928380099728510547238066 3}{10^{36}}, \\ -8.711 \times 10^{-6} < \lambda(\mathsf{s}) < -8.709 \times 10^{-6}. \end{gathered} \tag{11.148}$$

These inequalities use certified prefixes with their rational tail bound; they do not receive an alpha table. The package verifier reproduces them from the three nonadic outputs and from the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \le x \le 10^{-2}$, the tail beyond M satisfies

$$\left| E_{\mathsf{s},\infty}(x) - E_{\mathsf{s},M}(x) \right| \le 8.711 \times 10^{-26}\, \frac{r^{M+1}}{1-r}. \tag{11.149}$$

Moreover,

$$\left| \frac{\mathrm{d}}{\mathrm{d}x} \left(\lambda \frac{x^{10}}{1 - x^3/729} \right) \right| \le 8.711 \times 10^{-6} \left(\frac{10\, 10^{-18}}{1-r} + \frac{3\, 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \tag{11.150}$$

The derivative tail beyond block M additionally admits the depth-dependent estimate

$$\sup_{0 \le x \le 10^{-2}} \left| E'_{\mathsf{s},\infty}(x) - E'_{\mathsf{s},M}(x) \right| \le 8.711 \times 10^{-24}\, r^{M+1} \left(\frac{13 + 3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow[M \to \infty]{} 0. \tag{11.151}$$

Theorem 11.67 (Limit function and unique simple root of the correlated chart)**.** *The sequence $E_{\mathsf{s},M}$ converges uniformly, together with its derivatives, to $E_{\mathsf{s},\infty}$ on $\overline{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{\mathsf{s},\infty}(a(\mathsf{s})) = 0, \qquad \inf_{x \in \overline{I}_\alpha} E'_{\mathsf{s},\infty}(x) > 6.239. \tag{11.152}$$

Consequently, $a(\mathsf{s})$ is the unique simple root of the complete function in I_α.

Proof. The ratio of two consecutive terms of the tail is bounded by $r < 1$, which gives (11.149); term-by-term differentiation and summation of the two geometric series give (11.150) and (11.151). The jet proof establishes $E'_{9,\mathsf{s}} > 6.24$ on $\overline{I}_\alpha$; therefore, the complete perturbation leaves the derivative above 6.239. Finally, substituting (11.138) into (11.143) makes the two summands evaluated at $a(\mathsf{s})$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with that precoordinate. □

To make the identification effective at every depth, directed evaluations of the endpoints first certify

$$E_{\mathsf{s},\infty}(0) < 0 < E_{\mathsf{s},\infty}(10^{-2}), \qquad E'_{\mathsf{s},\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{in } [0, 10^{-2}].$$

By continuity, a root exists in the interval and, by the second inequality, it is unique. Only after establishing this existence, monotonicity and the invariant that the bracket contains the root is the mean-value-theorem rescue applied. We start from $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{\mathsf{s},\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, \delta$ and λ, with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4}\,(s_j - r_j). \tag{11.153}$$

When $F_j^+ < 0$, the right half is retained; when $F_j^- > 0$, the left half. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{\mathsf{s},\infty} \geq c = 6$ and the mean value theorem imply that any root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4}\right]. \tag{11.154}$$

Indeed, let ξ_j be the root and let $y_j = E_{\mathsf{s},\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (11.153), we have $|y_j| \leq c w_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, which proves (11.154). The argument includes the case $y_j = 0$: the root is then at the midpoint, but the algorithm does not need to recognize that equality in order to retain it.

If the available enclosure does not satisfy (11.153), the source cylinders are first refined and the evaluation repeated; no sign is declared and no insufficient contraction is accepted. In any of the three branches, the new interval retains the root and has diameter at most $w_j/2$. For each finite target depth, directed refinement allows the width bound to be satisfied without an uncertified comparison of a real number with zero. Monotonicity proves inductively

$$a(\mathsf{s}) \in D_{j+1} \subseteq D_j, \qquad \operatorname{diam} D_{j+1} \leq \tfrac{1}{2} \operatorname{diam} D_j, \qquad \operatorname{diam} D_j \longrightarrow 0, \tag{11.155}$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule to twenty-four base-thousand levels and checks that the complete outer closure A_{1000}, not just its lower endpoint, lies within each published cylinder. Its negative tests reject an excessively wide enclosure, an unordered enclosure, a point other than the midpoint, a domain without a sign change and a degenerate bracket. Coinductive prolongation allows one thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$ the verifier calculates

$$j_N^- := \lfloor S_N a_- \rfloor, \qquad j_N^+ := \lfloor S_N a_+ \rfloor \tag{11.156}$$

and publishes the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and avoids assigning to the left-hand cell an endpoint lying exactly on its right-hand boundary.

Let $C_N^A(\mathsf{s})$ be the integer cylinder of (11.134). We recursively choose $m(N) > m(N-1)$ so that $\operatorname{diam} D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(\mathsf{s}). \tag{11.157}$$

We define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is nonempty because both factors contain $a(\mathsf{s})$. Since the cylinder is half-open, u_N may not belong to $I_{B,N}$. Evaluation at that endpoint is understood through the continuous extension of $E_{\mathsf{s},\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(\mathsf{s}), \qquad \operatorname{diam} I_{B,N} \longrightarrow 0, \qquad E_{\mathsf{s},\infty}(\ell_N) \leq 0 \leq E_{\mathsf{s},\infty}(u_N). \tag{11.158}$$

The bound (11.149) allows a truncation to be chosen that preserves the sign at each endpoint where the complete function does not vanish. If an endpoint coincides with the root, the identity of the complete function and the already proved membership are used; that same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{\mathsf{s},\infty}(x) - E_{\mathsf{s},M}(x) = \lambda(\mathsf{s}) x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

On the positive domain considered, where $\lambda(\mathsf{s}) < 0$, we obtain at the root

$$E_{\mathsf{s},M}(a(\mathsf{s})) = -\lambda(\mathsf{s}) a(\mathsf{s})^{10} \frac{(qa(\mathsf{s})^3)^{M+1}}{1 - qa(\mathsf{s})^3} > 0 \qquad (M < \infty).$$

Therefore, certification of the weak signs in (11.158) refers to $E_{\mathsf{s},\infty}$. Its evaluation uses the complete function or the truncation accompanied by the signed remainder. This covers both the included infimum and the possibly excluded supremum of the cylinder.

Proposition 11.68 (Compatibility squares and common cylinders)**.** *Restrictions of words and analytic publications commute:*

$$\begin{array}{ccc} \mathcal{X}_\alpha^{\mathrm{enr}} & \xrightarrow{\mathsf{T}_{\alpha,N'}} & \mathcal{W}_{N'} \\ \| & & \downarrow \rho_{N',N} \\ \mathcal{X}_\alpha^{\mathrm{enr}} & \xrightarrow{\mathsf{T}_{\alpha,N}} & \mathcal{W}_N \end{array} \qquad \begin{array}{ccc} \mathcal{X}_\alpha^{\mathrm{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{I}_{M'} \\ \| & & \downarrow \tau_{M',M} \\ \mathcal{X}_\alpha^{\mathrm{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{I}_M. \end{array}$$

The family $I_{B,N}$ provides the effective connection between the two squares and materially satisfies $I_{B,N} \subseteq C_N^A$ for every N.

Proof. The first square is compatibility of integer normalization; the second is proposition 11.66. Inclusion follows from definition (11.157), and nonemptiness and the signs follow from theorem 11.67. No step identifies the root of a truncation with that of an arbitrary tail. □

We now define, without leaving the prefix reader implicit,

$$\alpha_B(\mathsf{s}) := \operatorname*{arg\,zero}_{x \in I_\alpha} E_{\mathsf{s},\infty}(x), \qquad \mathrm{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \qquad \alpha_{B,N} := \mathrm{pref}_N(\alpha_B). \tag{11.159}$$

Theorem 11.69 (Projective equality of the two correlated publications)**.** *For every N, the integer path and analytic path publish the same prefix:*

$$\boxed{\alpha_{A,N} = \mathrm{pref}_N(\alpha_A) = \mathrm{pref}_N(\alpha_B) = \alpha_{B,N}.} \tag{11.160}$$

In the limit,

$$\boxed{\alpha_A = \operatorname*{arg\,zero}_{x \in I_\alpha} E_{\mathsf{s},\infty}(x) = \alpha_B = \alpha_{\mathrm{HMT}}.} \tag{11.161}$$

Proof. Integer normalization publishes exactly $a(\mathsf{s})$. Theorem 11.67 proves that the analytic chart represents that same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Thus, their prefixes agree for every N; nesting and diameters tending to zero give equality of the limits. Metrological recognition occurs afterward. □

The analytic function constitutes a correlated publication of the same state, not a second derivation causally independent of integer closure. This clarification does not diminish the proved equality: it

identifies its exact mechanism, prevents attributing to the jet a tail that it does not determine and avoids promoting a conventional value to a generator.

11.6.5 Compatibility between the dodecaphase and analytic realizations of alpha

Compatibility is established at every depth. The reversible dodecaphase publication and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal publications with distinct codomains, produced from the same enriched APP–TRIT–TPK state. The jet $E^{(9)}_{21/22}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below is not deduced from the bare jet: it receives the shared coordinate of the state and publishes its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This separation preserves the causal order

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \mathcal{X}^{\mathrm{enr}}_{\alpha} \longrightarrow (P, E, \Phi, U^{\mathrm{sgn}}_{12}, K) \longrightarrow \alpha_{\mathrm{HMT}}.$$

The coordinates P, E, Φ and the register K are prior outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition and not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\widehat{\mathcal{X}}^{\mathrm{enr},\alpha}_{r}$ be the sector of truncated states extending the internal alpha seed fixed in R_{36}. We write

$$x^{\alpha}_{r} = (\widetilde{x}_r, \chi_{\alpha}, \varepsilon^{\alpha}_{r}, N^{\alpha}_{r}),$$

where χ_{α} is the internal character of the sector, not the published scalar value. Each state preserves residue, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, register, memory, survival, provenance and depth. If

$$d^{\alpha}_{r} = \mathrm{Dig}_{729}(\varepsilon^{\alpha}_{r}), \qquad \varepsilon^{\alpha}_{r+6} = 729\varepsilon^{\alpha}_{r} - d^{\alpha}_{r}, \qquad N^{\alpha}_{r+6} = 729N^{\alpha}_{r} + d^{\alpha}_{r},$$

the effective transition is

$$\mathcal{U}^{\alpha}_{r} = \mathrm{Upd}^{\alpha}_{r} \circ \mathrm{Tra}^{\alpha}_{r} \circ \mathrm{Sel}^{\alpha}_{r}, \qquad \mathrm{Ext}^{\alpha,\rightarrow}_{r+6,r} = \mathrm{graph}(\mathcal{U}^{\alpha}_{r}).$$

The functionality of these extensions, with the seed R_{36} fixed, produces a unique compatible section: the phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}^{\mathrm{enr}}_{\alpha} := \widehat{\Omega}_{\alpha} := \varprojlim_{r \geq 36} \big(\widehat{\mathcal{X}}^{\mathrm{enr},\alpha}_{r}, \mathrm{Ext}^{\alpha,\rightarrow}_{r+6,r}\big). \tag{11.162}$$

11.6.5.1 The dodecaphase closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \ldots, 999\}^N, \qquad \mathsf{T}_{\alpha,N}(\mathsf{s}) := (A_1, \ldots, A_N), \qquad \mathsf{s} \in \mathcal{X}^{\mathrm{enr}}_{\alpha},$$

where the blocks A_j are obtained through (11.120) with the compatible boundary $c^{\infty}_{N+1} = \lfloor R_{N+1}(Z) \rfloor$. If $N' \geq N$, let $\rho_{N',N} : \mathcal{W}_{N'} \to \mathcal{W}_N$ be the restriction to the first N blocks.

Theorem 11.70 (Complete dodecaphase output and projective compatibility)**.** *The family* $(\mathsf{T}_{\alpha,N})_{N\geq 1}$ *is compatible:*

$$\rho_{N',N} \circ \mathsf{T}_{\alpha,N'} = \mathsf{T}_{\alpha,N} \qquad (N' \geq N). \tag{11.163}$$

Its canonical cylinders

$$C_N^A(\mathsf{s}) := \left[\sum_{j=1}^{N} A_j 1000^{-j},\ \sum_{j=1}^{N} A_j 1000^{-j} + 1000^{-N}\right) \cap I_\alpha \tag{11.164}$$

are nested, have diameter at most 1000^{-N} *and satisfy*

$$\bigcap_{N\geq 1} C_N^A(\mathsf{s}) = \{\alpha_A(\mathsf{s})\}, \qquad \alpha_A = \pi_{\mathrm{HMT}} + e_{\mathrm{HMT}} - \varphi_{\mathrm{HMT}} - 4 - \kappa_{\mathrm{per}}. \tag{11.165}$$

Consequently, the first route publishes a unique infinite section, which we denote by $\alpha_{\mathrm{HMT}} := \alpha_A$.

Proof. Proposition 11.58 provides the unique canonical expansion and the unique bounded sequence of carries. The local identity (11.122) and its telescoping (11.123) preserve the remote boundary; proposition 11.60 proves that restricting a prolonged execution recovers the previous prefix exactly. This gives (11.163). Membership of the complete sum in each cylinder follows from the tail identity, and $\operatorname{diam} C_N^A \leq 1000^{-N} \to 0$; the intersection contains a single point. Finally, (11.126) identifies that point with the expression of (11.165). All its coefficients and signs are fixed before alpha is published. □

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^{\infty} = -1$. In particular,

$$\rho_{36,12}(\alpha_{A,36}) = \alpha_{A,12} \tag{11.166}$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

11.6.5.2 Exact analytic characterization of order nine

Let

$$\mathbb{K} := \mathbb{Q}\left(\pi_{\mathrm{HMT}}, \log_{10}\frac{10}{9}\right), \qquad E_9(x) := E_{21/22}^{(9)}(x) = P_9(x) - \delta.$$

Equations (11.129) and (11.130) construct $P_9 = P_6 + T_{7:9}$ from the state's signature and the nilpotent resolvent; Proposition 11.62 proves that E_9 has a unique simple root $\xi_9 \in I_\alpha$, isolated by the rational interval (11.131).

Finite bidirectionality can be written without introducing an analytic tail. Define

$$B_9(x) := \delta + \frac{7}{4}x^2 - \frac{x^4}{20} + \frac{2x^5}{21} + \frac{x^6}{46} + \frac{x^7}{120} - \frac{x^8}{45} - \frac{2x^9}{495}, \tag{11.167}$$

$$q_x(T) := xT^2 - B_9(x)T + x^3, \qquad x \in I_\alpha = (0, 10^{-2}), \tag{11.168}$$

$$J_x : \mathbb{R}^\times \longrightarrow \mathbb{R}^\times, \qquad J_x(T) := \frac{x^2}{T}. \tag{11.169}$$

Proposition 11.71 (Finite quadratic equivalence and involution)**.** *For every* $x \in I_\alpha$ *and every* $T \in \mathbb{R}^\times$,

$$\frac{q_x(2\pi_{\mathrm{HMT}})}{2\pi_{\mathrm{HMT}}} = P_9(x) - \delta = E_9(x), \tag{11.170}$$

$$J_x(J_x(T)) = T, \qquad T^2 q_x(J_x(T)) = x^2 q_x(T). \tag{11.171}$$

Let $Z_x = \{T \in \mathbb{R}^\times : q_x(T) = 0\}$. *Consequently,* $J_x|_{Z_x} : Z_x \to Z_x$ *is involutive. When* $E_9(x) = 0$, *it interchanges the roots* $2\pi_{\mathrm{HMT}}$ *and* $x^2/(2\pi_{\mathrm{HMT}})$. *This is an exact involution on the scalar fibre of the order-nine jet; it does not act on the enriched state.*

Proof. Dividing $q_x(T)$ by T and substituting $T = 2\pi_{\mathrm{HMT}}$ reproduces the definition of $P_9(x) - \delta$ term by term. The first identity in (11.171) is immediate. For the second,

$$T^2 q_x(x^2/T) = x^5 - B_9(x)x^2T + x^3T^2 = x^2 q_x(T).$$

Thus, J_x preserves the set of roots and, by Viète's formula, interchanges the two roots whose product is x^2. □

The finite decimal witness then retains its exact strength:

$$\mathrm{Tr}_9(\xi_9^{-1}) = \mathrm{Tr}_9\Big((\alpha_{12}^{(0)})^{-1}\Big) = 137.035999084. \tag{11.172}$$

The equality (11.172) certifies the finite characterization and the preceding involution. It does not turn the root of a truncated polynomial into the root of every compatible prolongation.

11.6.5.3 Degrees of freedom in the analytic prolongation of a finite jet

Let

$$j_9 : \mathbb{K}[[x]] \longrightarrow \mathbb{K}[x]/(x^{10})$$

be the truncation, and for $h \in \mathbb{K}[x]$ set

$$E_h(x) := E_9(x) + x^{10}h(x). \tag{11.173}$$

All these polynomials satisfy $j_9(E_h) = j_9(E_9)$.

Theorem 11.72 (Nonuniqueness of the root from the jet)**.** *There is no map*

$$\mathfrak{r}_9 : \mathbb{K}[x]/(x^{10}) \longrightarrow I_\alpha$$

that assigns, from the jet alone, the positive root of every polynomial prolongation of E_9 having a unique root in I_α. In particular, E_0 and $E_{1/729}$ have the same jet and distinct positive roots.

Proof. The bound proved in the proof of 11.62 gives $E_9'(x) > 6.24$ on $[0, 10^{-2}]$. For $t \geq 0$, write $E_t(x) = E_9(x) + tx^{10}$. Then

$$E_t'(x) = E_9'(x) + 10tx^9 > 6.24,$$

whereas $E_t(0) = -\delta < 0$ and $E_t(10^{-2}) > E_9(10^{-2}) > 0$. Each E_t therefore has a unique positive root $\eta(t) \in I_\alpha$. For $t = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0;$$

since $E_{1/729}$ is strictly increasing and changes sign only once, its root satisfies $\eta(1/729) < \xi_9 = \eta(0)$. However, $j_9(E_{1/729}) = j_9(E_0)$. A map factoring the root through j_9 would have to assign the same point to both polynomials, a contradiction. □

The variation is not merely existential. The implicit function theorem applies because $\partial_x E_t(\eta(t)) > 6.24$, and gives

$$\eta'(t) = -\frac{\eta(t)^{10}}{E_9'(\eta(t)) + 10t\eta(t)^9} < 0. \tag{11.174}$$

This proves that the coefficients beyond degree nine modify the root even though they leave the entire jet constructed up to that degree invariant. This conclusion does not deny the tail produced by the enriched state: it proves only that the root does not factor through the bare truncation j_9. Choosing a tail h from an external conventional root would be circular. The following completion consults neither that root nor a metrological table: it directly evaluates the regional and terminal coordinates of the state. Since that evaluation coincides algebraically with the integer publication, the result is a correlated representation, not an assertion of independence between two selectors.

11.6.5.4 Analytic representation of the alpha coordinate

The preceding jet check requires exhibiting the tail: merely naming a complete reader is insufficient. Let $\mathsf{s} \in \mathcal{X}_\alpha^{\mathrm{enr}}$. Before decimal normalization, the same state already publishes the precoordinate

$$a(\mathsf{s}) := p(\mathsf{s}) + e_0(\mathsf{s}) - f(\mathsf{s}) - \kappa_{\mathrm{per}}(\mathsf{s}) \in I_\alpha, \tag{11.175}$$

with $p = \pi_{\mathrm{HMT}} - 3$, $e_0 = e_{\mathrm{HMT}} - 2$ and $f = \varphi_{\mathrm{HMT}} - 1$. Equivalently, $a = \pi_{\mathrm{HMT}} + e_{\mathrm{HMT}} - \varphi_{\mathrm{HMT}} - 4 - \kappa_{\mathrm{per}}$. No conventional α is introduced here: p, e_0, f are the three already generated regional characters and κ_{per} is the rational reading of the terminal register. Integer closure applies to (11.175) the normalization of (11.120); the following chart publishes a compatible analytic representation on the same precoordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\mathrm{per}})$ directly: it does not contain an edge $\alpha_A \to \lambda$, although the equality proved below subsequently allows both publication names to be identified.

Write $q = 1/729$, let F_{s} be the ordered field generated by the coordinates of s, and set $E_{9,\mathsf{s}}(X) = E^{(9)}_{21/22}(\mathsf{s}; X)$. We define the linking coefficient

$$\lambda(\mathsf{s}) := -\frac{E_{9,\mathsf{s}}(a(\mathsf{s}))\big(1 - q\, a(\mathsf{s})^3\big)}{a(\mathsf{s})^{10}}. \tag{11.176}$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (11.176) must be read with its exact type: once the first publication of the state has produced $a(\mathsf{s})$, it fixes the unique prolongation in the geometric class defined below. It does not constitute a second independent selection of $a(\mathsf{s})$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_{\mathsf{s}}$ set

$$E_{\mathsf{s},\mu}(X) := E_{9,\mathsf{s}}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(\mathsf{s}) < 10^{-2}$ and $qa(\mathsf{s})^3 < 1$, we have

$$E_{\mathsf{s},\mu}(a(\mathsf{s})) = 0 \quad \Longleftrightarrow \quad \mu = \lambda(\mathsf{s}). \tag{11.177}$$

Thus, the coefficient seed has no freedom once the state and the class of q-geometric prolongations are fixed. This uniqueness is relative to the declared class; it does not assert that the order-nine jet alone selects a tail among all possible analytic series.

The local action on blocks of three coefficients is

$$C_\alpha : F_{\mathsf{s}}^3 \longrightarrow F_{\mathsf{s}}^3, \qquad C_\alpha(u, v, w) = q(u, v, w), \qquad c^{(0)} = (\lambda(\mathsf{s}), 0, 0). \tag{11.178}$$

Therefore, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(\mathsf{s}), \qquad b_{11+3n} = b_{12+3n} = 0. \tag{11.179}$$

This is the rule that calculates every coefficient beyond degree nine. No free parameter remains at each prolongation.

For $M \geq 0$, let

$$E_{\mathsf{s},M}(X) := E_{9,\mathsf{s}}(X) + \lambda(\mathsf{s}) \sum_{n=0}^{M} q^n X^{10+3n}. \tag{11.180}$$

The complete function is

$$E_{\mathsf{s},\infty}(X) := E_{9,\mathsf{s}}(X) + \lambda(\mathsf{s}) \frac{X^{10}}{1 - X^3/729}. \tag{11.181}$$

To formulate the projective system without implicit types, we fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{J}_M := \coprod_{\mathsf{s} \in X_\alpha^{\mathrm{enr}}} \{\mathsf{s}\} \times F_{\mathsf{s}}[X]/(X^{d_M+1}). \tag{11.182}$$

If $M' \geq M$, the restriction is

$$\tau_{M',M}(\mathsf{s}, [P]_{d_{M'}}) := (\mathsf{s}, [P]_{d_M}). \tag{11.183}$$

Proposition 11.73 (Naturality of the analytic recurrence)**.** *Let*

$$\Gamma_M(\mathsf{s}) = \big(\mathsf{s}, [E_{\mathsf{s},M}]_{d_M}\big).$$

If $M' \geq M$, the truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \qquad \Gamma_{E21}(\mathsf{s}) = \varprojlim_M \Gamma_M(\mathsf{s}). \tag{11.184}$$

Projection onto the three new degrees is precisely $C_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (11.179) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ only adds the next block; truncating it recovers the previous polynomial literally. Compatibility for $M' \geq M$ follows by composition. Thus, the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under C_α. □

11.6.5.5 Uniform convergence, simple root and rational intervals

The regional cylinders and the periodic register provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}\},$$

let P_c be its certified thousand-digit nonadic prefix. The datum used is not the rational P_c as though it were the complete value, but the half-open cylinder and its outer closure

$$C_c^{(1000)} = [P_c, P_c + \varepsilon), \qquad \overline{C}_c^{(1000)} = [P_c, P_c + \varepsilon], \qquad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively with those closures. If P_π, P_e, P_φ denote the three prefixes, we obtain

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\mathrm{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\mathrm{per}}, \\ A_{1000} &:= [a_-, a_+], \qquad a(\mathsf{s}) \in [a_-, a_+] = A_{1000}, \qquad \operatorname{diam} A_{1000} = 3\varepsilon. \end{aligned} \tag{11.185}$$

The natural interval extension of (11.176) then transports A_{1000}, the cylinder of π_{HMT} and the directed rational bound of δ to a closed interval $\Lambda_{1000} \ni \lambda(\mathsf{s})$; none of the three tails is frozen. In particular,

$$\begin{gathered} \frac{729735256928380099728510547238066 2}{10^{36}} < a(\mathsf{s}) < \frac{729735256928380099728510547238066 3}{10^{36}}, \\ -8.711 \times 10^{-6} < \lambda(\mathsf{s}) < -8.709 \times 10^{-6}. \end{gathered} \tag{11.186}$$

These inequalities use certified prefixes with their rational tail bound; they do not receive an alpha table. The package verifier reproduces them from the three nonadic outputs and from the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \le x \le 10^{-2}$, the tail beyond M satisfies

$$\left|E_{\mathsf{s},\infty}(x) - E_{\mathsf{s},M}(x)\right| \le 8.711 \times 10^{-26}\, \frac{r^{M+1}}{1-r}. \tag{11.187}$$

Moreover,

$$\left|\frac{\mathrm{d}}{\mathrm{d}x}\left(\lambda \frac{x^{10}}{1-x^3/729}\right)\right| \le 8.711 \times 10^{-6}\left(\frac{10\,10^{-18}}{1-r} + \frac{3\,10^{-24}/729}{(1-r)^2}\right) < 8.712 \times 10^{-23}. \tag{11.188}$$

The derivative tail beyond block M additionally admits the depth-dependent estimate

$$\sup_{0\le x\le 10^{-2}} \left|E'_{\mathsf{s},\infty}(x) - E'_{\mathsf{s},M}(x)\right| \le 8.711 \times 10^{-24}\, r^{M+1}\left(\frac{13+3M}{1-r} + \frac{3r}{(1-r)^2}\right) \xrightarrow[M\to\infty]{} 0. \tag{11.189}$$

Theorem 11.74 (Limit function and unique simple root of the correlated chart)**.** *The sequence $E_{\mathsf{s},M}$ converges uniformly, together with its derivatives, to $E_{\mathsf{s},\infty}$ on $\overline{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{\mathsf{s},\infty}(a(\mathtt{s})) = 0, \qquad \inf_{x\in\overline{I}_\alpha} E'_{\mathsf{s},\infty}(x) > 6.239. \tag{11.190}$$

Consequently, $a(\mathtt{s})$ is the unique simple root of the complete function in I_α.

Proof. The ratio of two consecutive terms of the tail is bounded by $r < 1$, which gives (11.187); term-by-term differentiation and summation of the two geometric series give (11.188) and (11.189). The jet proof establishes $E'_{9,\mathsf{s}} > 6.24$ on $\overline{I}_\alpha$; therefore, the complete perturbation leaves the derivative above 6.239. Finally, substituting (11.176) into (11.181) makes the two summands evaluated at $a(\mathtt{s})$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with that precoordinate. □

To make the identification effective at every depth, directed evaluations of the endpoints first certify

$$E_{\mathsf{s},\infty}(0) < 0 < E_{\mathsf{s},\infty}(10^{-2}), \qquad E'_{\mathsf{s},\infty} \ge \mu := 6239/1000 > c := 6 \quad \text{in } [0, 10^{-2}].$$

By continuity, a root exists in the interval and, by the second inequality, it is unique. Only after establishing this existence, monotonicity and the invariant that the bracket contains the root is the mean-value-theorem rescue applied. We start from $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{\mathsf{s},\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, \delta$ and λ, with

$$0 \le F_j^+ - F_j^- \le \frac{c}{4}\,(s_j - r_j). \tag{11.191}$$

When $F_j^+ < 0$, the right half is retained; when $F_j^- > 0$, the left half. If $F_j^- \le 0 \le F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{\mathsf{s},\infty} \ge c = 6$ and the mean value theorem imply that any root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4}\right]. \tag{11.192}$$

Indeed, let ξ_j be the root and let $y_j = E_{\mathsf{s},\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (11.191), we have $|y_j| \le c w_j/4$. The mean value theorem gives $|m_j - \xi_j| \le |y_j|/c \le w_j/4$, which proves (11.192). The argument includes the case $y_j = 0$: the root is then at the midpoint, but the algorithm does not need to recognize that equality in order to retain it.

If the available enclosure does not satisfy (11.191), the source cylinders are first refined and the evaluation repeated; no sign is declared and no insufficient contraction is accepted. In any of the three branches, the new interval retains the root and has diameter at most $w_j/2$. For each finite target depth,

directed refinement allows the width bound to be satisfied without an uncertified comparison of a real number with zero. Monotonicity proves inductively

$$a(\mathsf{s}) \in D_{j+1} \subseteq D_j, \qquad \operatorname{diam} D_{j+1} \le \tfrac{1}{2} \operatorname{diam} D_j, \qquad \operatorname{diam} D_j \longrightarrow 0, \tag{11.193}$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule to twenty-four base-thousand levels and checks that the complete outer closure A_{1000}, not just its lower endpoint, lies within each published cylinder. Its negative tests reject an excessively wide enclosure, an unordered enclosure, a point other than the midpoint, a domain without a sign change and a degenerate bracket. Coinductive prolongation allows one thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$ the verifier calculates

$$j_N^- := \lfloor S_N a_- \rfloor, \qquad j_N^+ := \lfloor S_N a_+ \rfloor \tag{11.194}$$

and publishes the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and avoids assigning to the left-hand cell an endpoint lying exactly on its right-hand boundary.

Let $C_N^A(\mathsf{s})$ be the integer cylinder of (11.164). We recursively choose $m(N) > m(N-1)$ so that $\operatorname{diam} D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(\mathsf{s}). \tag{11.195}$$

We define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is nonempty because both factors contain $a(\mathsf{s})$. Since the cylinder is half-open, u_N may not belong to $I_{B,N}$. Evaluation at that endpoint is understood through the continuous extension of $E_{\mathsf{s},\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(\mathsf{s}), \qquad \operatorname{diam} I_{B,N} \longrightarrow 0, \qquad E_{\mathsf{s},\infty}(\ell_N) \le 0 \le E_{\mathsf{s},\infty}(u_N). \tag{11.196}$$

The bound (11.187) allows a truncation to be chosen that preserves the sign at each endpoint where the complete function does not vanish. If an endpoint coincides with the root, the identity of the complete function and the already proved membership are used; that same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{\mathsf{s},\infty}(x) - E_{\mathsf{s},M}(x) = \lambda(\mathsf{s}) x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

On the positive domain considered, where $\lambda(\mathsf{s}) < 0$, we obtain at the root

$$E_{\mathsf{s},M}(a(\mathsf{s})) = -\lambda(\mathsf{s}) a(\mathsf{s})^{10} \frac{(qa(\mathsf{s})^3)^{M+1}}{1 - qa(\mathsf{s})^3} > 0 \qquad (M < \infty).$$

Therefore, certification of the weak signs in (11.196) refers to $E_{\mathsf{s},\infty}$. Its evaluation uses the complete function or the truncation accompanied by the signed remainder. This covers both the included infimum and the possibly excluded supremum of the cylinder.

Proposition 11.75 (Compatibility squares and common cylinders)**.** *Restrictions of words and analytic publications commute:*

$$\begin{array}{ccc} \mathcal{X}_\alpha^{\mathrm{enr}} & \xrightarrow{\mathsf{T}_{\alpha,N'}} & \mathcal{W}_{N'} \\ \| & & \downarrow \rho_{N',N} \\ \mathcal{X}_\alpha^{\mathrm{enr}} & \xrightarrow{\mathsf{T}_{\alpha,N}} & \mathcal{W}_N \end{array} \qquad \begin{array}{ccc} \mathcal{X}_\alpha^{\mathrm{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{I}_{M'} \\ \| & & \downarrow \tau_{M',M} \\ \mathcal{X}_\alpha^{\mathrm{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{I}_M. \end{array}$$

The family $I_{B,N}$ provides the effective connection between the two squares and materially satisfies $I_{B,N} \subseteq C_N^A$ for every N.

Proof. The first square is compatibility of integer normalization; the second is proposition 11.73. Inclusion follows from definition (11.195), and nonemptiness and the signs follow from theorem 11.74. No step identifies the root of a truncation with that of an arbitrary tail. □

We now define, without leaving the prefix reader implicit,

$$\alpha_B(\mathtt{s}) := \operatorname*{arg\,zero}_{x \in I_\alpha} E_{\mathtt{s},\infty}(x), \qquad \mathrm{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \qquad \alpha_{B,N} := \mathrm{pref}_N(\alpha_B). \tag{11.197}$$

Theorem 11.76 (Projective equality of the two correlated publications)**.** *For every N, the integer route and the analytic route publish the same prefix:*

$$\boxed{\alpha_{A,N} = \mathrm{pref}_N(\alpha_A) = \mathrm{pref}_N(\alpha_B) = \alpha_{B,N}.} \tag{A7$_0$}$$

In the limit,

$$\boxed{\alpha_A = \operatorname*{arg\,zero}_{x \in I_\alpha} E_{\mathtt{s},\infty}(x) = \alpha_B = \alpha_{\mathrm{HMT}}.} \tag{A8$_0$}$$

Proof. Integer normalization publishes exactly $a(\mathtt{s})$. Theorem 11.74 proves that the analytic chart represents that same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Thus, their prefixes agree for every N; nesting and diameters tending to zero give equality of the limits. Metrological recognition occurs afterward. □

The analytic function constitutes a correlated publication of the same state, not a second derivation causally independent of integer closure. This clarification does not diminish the proved equality: it identifies its exact mechanism, prevents attributing to the jet a tail that it does not determine and avoids promoting a conventional value to a generator.

Chapter 12

Construction of $\sqrt{-1}$ and Euler tritic exponential family: operators J_τ and coordination of the regimes elliptic, parabolic and hyperbolic

12.1 Triadic Euler extension and preservation of forms

The quadratic representation of TRIT acquires operator content by identifying the generators selected by APP and by the TPK calendar. This chapter reconstructs those antecedents before expressing their exponential realizations. The order distinguishes the discrete operation, its matrix representation and the analytic function that allows its evolution in a real parameter to be described.

Three constructions are involved: the multiplicative cycle of order three in APP, the elementary transport preserving a prolongation coordinate and the incidence between modes descending from the tritic calendar. Their generators satisfy the elliptic, parabolic and hyperbolic relations, respectively. The alternating form of the plane then allows a common conservation to be established, with the metric and orientation differences of each realization explicitly preserved.

12.1.1 APP multiplicative cycle and elliptic generator

In $\mathbb{Z}/9\mathbb{Z}$, multiplication by four is invertible and has order three:

$$4^2 = 7 \pmod 9, \qquad 4^3 = 1 \pmod 9.$$

On the units appear the cycles $(1, 4, 7)$ and $(2, 8, 5)$. The classes $0, 3, 6$ remain fixed. The selection of the multiplier and its orbits therefore belongs to the nonadic arithmetic of APP.

12.1.1.1 Contragredient action on the two evaluations

Let $\bar{S}(x, y) = x + y$ and $\bar{P}(x, y) = xy$, with values in $\mathbb{Z}/9\mathbb{Z}$. For $a = 4$,

$$\bar{S}(ax, ay) = a\bar{S}(x, y), \qquad \bar{P}(ax, ay) = a^2\bar{P}(x, y) = a^{-1}\bar{P}(x, y). \tag{12.1}$$

The first identity follows from distributivity; the second, from $a^3 = 1$. Sum and product traverse opposite orientations of the same residual cycle. In integer evaluations, the nonadic quotients remain associated with the lift of these classes.

12.1.1.2 Augmentation module and cycle matrix

Fixing one of the cycles $(r, 4r, 7r)$, consider the free module generated by e_r, e_{4r}, e_{7r}. Its augmentation submodule is

$$L = \{x_0 e_r + x_1 e_{4r} + x_2 e_{7r} : x_0 + x_1 + x_2 = 0\}.$$

A basis is $b_1 = e_r - e_{4r}$, $b_2 = e_{4r} - e_{7r}$. The cycle sends b_1 to b_2 and b_2 to $-b_1 - b_2$. Its matrix, acting on column vectors, is

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

The scalar product inherited from the three-coordinate module has Gram matrix

$$G_{\mathrm{e}} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

One thus obtains the lattice A_2, with its form determined by the differences between positions of the cycle.

Proposition 12.1 (Elliptic normalization of the APP cycle)**.** *The cycle satisfies*

$$C^2 + C + I = 0, \qquad C^{\mathsf{T}} G_{\mathrm{e}} C = G_{\mathrm{e}}.$$

In $L \otimes_{\mathbb{Z}} \mathbb{R}$, the operator

$$J_{+1} = \frac{2C + I}{\sqrt{3}}$$

has trace zero and satisfies $J_{+1}^2 = -I$.

Proof. Multiplication gives $C^2 = \left(\begin{smallmatrix} -1 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, whence $C^2 + C + I = 0$. The cyclic permutation preserves the scalar product of the three coordinates and its augmentation submodule; in the chosen basis this is equivalent to the Gram identity. Finally, $(2C + I)^2 = 4C^2 + 4C + I = -3I$, and $\operatorname{tr}(2C + I) = 0$. □

Real normalization provides an operator root of $-I$ from the cycle already constructed. The identity can also be written

$$C = -\frac{1}{2} I + \frac{\sqrt{3}}{2} J_{+1}.$$

Its angular interpretation is obtained after fixing the analytic realization of the elliptic type.

12.1.2 Threshold transport and parabolic generator

Consider the integer coordinates (x, c), where c is preserved as a recorded increment. The elementary transport

$$(x, c) \longmapsto (x + c, c)$$

is represented by

$$U_0 = I + N, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

This is the linear model of threshold transport used in the quadratic realization. Balanced division and its normalization, developed in chapter 1.2, preserve their own rules when applied to an integer evaluation.

Proposition 12.2 (Iteration and reversibility of transport)**.** *One has $N^2 = 0$, $N \neq 0$ and, for every $m \in \mathbb{Z}$,*

$$U_0^m = I + mN, \qquad U_0^m(x, c) = (x + mc, c).$$

Transport preserves the determinant and admits inverse $I - N$.

Proof. The image of N is contained in the first coordinate, on which N vanishes; hence $N^2 = 0$. The identity $(I + aN)(I + bN) = I + (a + b)N$ proves the positive powers, the inverse and the negative powers. The determinant of $I + mN$ is one. □

The parabolic index $\tau = 0$ corresponds to a square-zero generator. The operator N transports a linear datum and can be nonzero on a particular state. The zero signature, the zero operator and a nilpotent generator are three distinct objects.

12.1.3 Mode incidence derived from the TPK calendar

The construction in section 1.4.1 provides the matrix (1.56). Its primitive action is made explicit here to link the tritic count with the hyperbolic quadratic representation developed below.

The tritic phase selector takes, over the positions $1, \ldots, 9$, the values $(+1, -1, 0)$ repeated three times. On the operative mode the rule used is

$$+1 : \ m \mapsto \Sigma, \qquad -1 : \ m \mapsto \Pi, \qquad 0 : \ m \mapsto m.$$

After each update, the primitive block produces the sequence of modes (Σ, Π, Π). Spatial transport and register updating remain in the complete TPK transition; here one reads its modal incidence.

Theorem 12.3 (Primitive incidence of the calendar)**.** *Counting consecutive pairs with cyclic closure in* (Σ, Π, Π)*, the matrix obtained in the order* (Σ, Π) *is*

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

In calendars of length $9, 27, 108$*, the counts are* $3F_{\mathrm{av}}, 9F_{\mathrm{av}}, 36F_{\mathrm{av}}$*, respectively.*

Proof. The three pairs are $\Sigma \to \Pi$, $\Pi \to \Pi$ and $\Pi \to \Sigma$. Each appears once. The pair $\Sigma \to \Sigma$ appears zero times. This determines the four entries. Repeating the calendar multiplies all counts by the number of primitive blocks, which is $3, 9, 36$ in the indicated cases. □

The sheet reader associates Σ with the areal reading and Π with the volumetric reading. This assignment transports the resulting incidence to the two reading types. The matrix results from the calendar; its spectral analysis is performed once its entries have been constructed.

Counts of repeated calendars and matrix powers have different functions. $36F_{\mathrm{av}}$ counts the pairs traversed during a calendar of 108 events. By contrast, F_{av}^{108} counts admissible chains of length 108 in the incidence graph. The first object follows a chronology; the second belongs to its adjacency hull.

12.1.3.1 Hyperbolic realization of incidence

Two incidence compositions produce

$$A = F_{\mathrm{av}}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its coefficients count two-edge paths. One has $\det A = 1$, $\operatorname{tr} A = 3$ and, by direct multiplication, $A^2 - 3A + I = 0$.

Proposition 12.4 (Hyperbolic generator and invariant form)**.** *The operator*

$$J_{-1} = \frac{2A - 3I}{\sqrt{5}}$$

has trace zero and square I*. The symmetric form*

$$G_{\mathrm{h}} = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$$

has signature $(1, 1)$ *and satisfies* $A^{\mathsf{T}} G_{\mathrm{h}} A = G_{\mathrm{h}}$*.*

Proof. From $A^2 - 3A + I = 0$ one obtains $(2A - 3I)^2 = 4A^2 - 12A + 9I = 5I$. The trace of the numerator is zero. Moreover,

$$G_{\mathrm{h}} A = \begin{pmatrix} 3 & 4 \\ -1 & -3 \end{pmatrix}, \qquad A^{\mathsf{T}} G_{\mathrm{h}} A = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}.$$

The determinant of G_{h} is -5, so its two eigenvalues have opposite signs. □

The realization is Lorentzian in dimension $1 + 1$: a real plane, a form of signature $(1, 1)$ and a transport preserving it are specified. The physical realization of a spacetime subsequently uses its corresponding fields, dimensions and transduction maps.

12.1.4 Exponential identity of the three regimes

The three concrete operators act in spaces of distinct provenance: the augmentation module of an APP cycle, the threshold-transport plane and the modal-incidence plane. Their normalizations satisfy

$$J_\tau^2 = -\tau I, \qquad \operatorname{tr} J_\tau = 0, \qquad \tau \in \{+1, 0, -1\}.$$

This relation determines their powers and allows the evolution functions to be constructed by convergent series.

Definition 12.5 (Functions of the quadratic regime)**.** For $t \in \mathbb{R}$, one defines

$$c_\tau(t) = \sum_{k=0}^{\infty} \frac{(-\tau)^k t^{2k}}{(2k)!}, \qquad s_\tau(t) = \sum_{k=0}^{\infty} \frac{(-\tau)^k t^{2k+1}}{(2k+1)!}.$$

These functions analytically realize the quadratic type already obtained.

Theorem 12.6 (Triadic extension of Euler's identity)**.** *For the preceding generators,*

$$E_\tau(t) := \exp(tJ_\tau) = c_\tau(t) I + s_\tau(t) J_\tau. \tag{12.2}$$

The specializations of (c_τ, s_τ) *are* $(\cos, \sin)$*,* $(1, t)$ *and* $(\cosh, \sinh)$*, in the order* $\tau = +1, 0, -1$*. Moreover,*

$$c_\tau(t)^2 + \tau s_\tau(t)^2 = 1, \qquad E_\tau(t)^{-1} = E_\tau(-t).$$

Proof. The exponential series converges absolutely in matrix norm. The quadratic relation gives $J_\tau^{2k} = (-\tau)^k I$ and $J_\tau^{2k+1} = (-\tau)^k J_\tau$. Separating the even and odd terms produces (12.2). The three specializations are recognized in their series. Since the matrices tJ_τ and $-tJ_\tau$ commute, their exponentials have product I. The parity of the series gives

$$E_\tau(t) E_\tau(-t) = \big(c_\tau(t)^2 + \tau s_\tau(t)^2\big) I,$$

which proves the scalar identity and the inverse. □

The numbers appearing when analytic functions are evaluated have here the status of recognition of previously constructed operators. Coinductive regional generation preserves its emitter, its lifts and its cylinders. In particular, these series describe the exponential realization; selection of the regional prefixes is performed through the TPK.

12.1.4.1 Composition law and quadratic invariants

Commutation of multiples of the same generator gives $E_\tau(t) E_\tau(s) = E_\tau(t+s)$. Comparing coefficients of I, J_τ,

$$c_\tau(t+s) = c_\tau(t) c_\tau(s) - \tau s_\tau(t) s_\tau(s), \tag{12.3}$$

$$s_\tau(t+s) = s_\tau(t) c_\tau(s) + c_\tau(t) s_\tau(s). \tag{12.4}$$

Quadratic conjugation changes t to $-t$ within the fixed regime. Oriented inversion of a tritic sequence, for its part, can change the regime indices and must be transported through the corresponding TPK involution.

In the elliptic type, the preserved identity is $a^2 + b^2 = 1$. In the hyperbolic type, $a^2 - b^2 = 1$, and the coordinates $u = a + b$, $v = a - b$ satisfy $uv = 1$. Hyperbolic evolution has $(u, v) = (e^t, e^{-t})$: expansion of one coordinate is accompanied by contraction of the other. In the parabolic type, $E_0(t) = I + tN$, with linear transport and determinant one.

12.1.4.2 Evaluations on the discrete operators

The representation of the elliptic cycle gives

$$C = E_{+1}(2\pi/3), \qquad C^n = E_{+1}(2\pi n/3), \quad n \in \mathbb{Z}.$$

In particular,

$$E_{+1}(\pi) + I = 0, \qquad E_{+1}(2\pi) = I, \qquad E_0(1) = I + N.$$

The angular coordinate π expresses the half-turn analytically; its identification with the HMT regional output belongs to the coinductive reader developed in its causal position.

The incidence F_{av} has characteristic polynomial $x^2 - x - 1$. Its positive root λ satisfies $\lambda^2 = \lambda + 1$; recognizing it as the golden ratio, one writes $\lambda = \varphi$. With $\eta = 2\log\lambda$,

$$\cosh\eta = \frac{3}{2}, \qquad \sinh\eta = \frac{\sqrt{5}}{2}, \qquad E_{-1}(\eta) = A = F_{\mathrm{av}}^2.$$

The check uses $\lambda^{-1} = \lambda - 1$, whence $\lambda^2 + \lambda^{-2} = 3$ and $\lambda^2 - \lambda^{-2} = \sqrt{5}$. The matrix was obtained earlier by counting transitions; spectral recognition characterizes its scale. Equality with the generated regional coordinate is preserved with the reader and the proof of that region.

Likewise, the scalar identity $\exp(I) = eI$ characterizes the unit evaluation of the exponential. Its function in the analytic representation is distinct from the coinductive construction of the corresponding region.

12.1.5 Polynomial resolution of the regimes

In the direct sum of the three real spaces, define

$$\mathbb{J} = J_{+1} \oplus N \oplus J_{-1}, \qquad Q = \mathbb{J}^2.$$

On those summands, Q acts as $-I, 0, I$, respectively. Therefore $Q^3 = Q$ and $\mathbb{J}^6 = \mathbb{J}^2$.

Proposition 12.7 (Regime idempotents)**.** *The operators*

$$p_{\mathrm{e}} = \frac{Q^2 - Q}{2}, \qquad p_{\mathrm{p}} = I - Q^2, \qquad p_{\mathrm{h}} = \frac{Q^2 + Q}{2}$$

are orthogonal idempotents in the algebraic sense and sum to I. Their images are the three summands of the representation.

Proof. The three polynomials take the values $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$ on $-1, 0, +1$. Their squares minus themselves and their cross products vanish at those three points, so they are multiples of $X^3 - X$. Substituting Q demonstrates idempotence and vanishing of the cross products. The sum of the three polynomials is one. Their action on each block identifies the images. □

The complete exponential is expressed by

$$\begin{aligned}\exp(t\mathbb{J}) = {} & p_{\mathrm{e}}(\cos t\, I + \sin t\, \mathbb{J}) + p_{\mathrm{p}}(I + t\mathbb{J}) \\ & + p_{\mathrm{h}}(\cosh t\, I + \sinh t\, \mathbb{J}).\end{aligned} \tag{12.5}$$

This resolution distinguishes the regimes exactly. Transitions between them belong to the TPK maps on states with orientation, mode and memory; the direct sum keeps their domains separate.

12.1.6 Nonadic compression and complementary register

Conservation between a reading and its complementary register follows from the nine-phase structure. Let V be a space with a scalar product and $C : V \to V$ a linear transport. In V^9, define

$$S_C(v_0, \dots, v_8) = (Cv_8, v_0, \dots, v_7), \qquad Jv = \frac{1}{3}(v, \dots, v).$$

The inclusion J is isometric because it brings together nine copies with factor $1/3$. Its adjoint sums the components and divides by three. The compression onto that inclusion is

$$T = J^* S_C J = \frac{8I + C}{9}.$$

Eight components preserve v and one passes through the transport C. The orthogonal complement of the inclusion records

$$(I - JJ^*)S_C Jv = \frac{1}{27}\big(8(C - I)v, -(C - I)v, \dots, -(C - I)v\big).$$

The square of its norm is $8\|(C - I)v\|^2/81$. The same vector content can be represented isometrically by

$$D = \frac{\sqrt{8}}{9}(C - I),$$

maintaining the nine-component register when their individual labels are used.

Proposition 12.8 (Algebraic recovery from two readings)**.** *For every $v \in V$,*

$$v = Tv - \frac{1}{\sqrt{8}}Dv. \tag{12.6}$$

If C is isometric, one additionally has

$$\|v\|^2 = \|Tv\|^2 + \|Dv\|^2.$$

Proof. The first identity follows from $(8I + C - (C - I))/9 = I$. For the second, expanding the products gives

$$T^*T + D^*D = \frac{72I + 9C^*C}{81}.$$

The hypothesis $C^*C = I$ gives the identity. The mixed terms cancel with coefficients $+8$ and -8. □

Recovery uses the vector records, including their correlations. The intensities $\|Tv\|^2$ and $\|Dv\|^2$ provide the scalar balance, whereas the inverse formula uses the complete Tv and Dv.

12.1.7 Conservation of oriented area

In any oriented real plane, let

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For a matrix $L = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, direct multiplication gives

$$L^{\mathsf{T}}\Omega L = (ad - bc)\Omega = (\det L)\Omega. \tag{12.7}$$

Since the tritic generators have zero trace, $\det E_\tau(t) = 1$. Their realizations therefore preserve the alternating form, although their metric forms differ.

Table 12.1: Contributions to the oriented area of three generated transports.

Transporte	$\operatorname{tr} C$	$\det T$	$\det D$
APP cycle C	-1	$19/27$	$8/27$
Threshold $I + N$	2	1	0
Incidence F_{av}^2	3	$89/81$	$-8/81$

Each pair sums to one. The sign records orientation; the table concerns alternating area, not probabilities.

Theorem 12.9 (Nonadic conservation common to the three regimes)**.** *For $C = E_\tau(t)$, let $T = (8I + C)/9$ and $D = \sqrt{8}(C - I)/9$. Then*

$$\Omega = T^{\mathsf{T}}\Omega T + D^{\mathsf{T}}\Omega D. \tag{12.8}$$

Proof. Expansion gives

$$T^{\mathsf{T}}\Omega T + D^{\mathsf{T}}\Omega D = \frac{72\Omega + 9C^{\mathsf{T}}\Omega C}{81}.$$

The terms $C^{\mathsf{T}}\Omega$ and ΩC cancel. By (12.7), $C^{\mathsf{T}}\Omega C = \Omega$, and the result is Ω. □

With $c = c_\tau(t)$, the determinant contributions are

$$\det T = \frac{65 + 16c}{81}, \qquad \det D = \frac{16(1 - c)}{81}, \qquad \det T + \det D = 1. \tag{12.9}$$

Indeed, for $\det C = 1$, $\det(aI + C) = a^2 + a \operatorname{tr} C + 1$, and $\operatorname{tr} C = 2c$.

In parabolic transport, D may be nonzero and have rank one: its area is zero and its record contains linear information. In hyperbolic transport, the complement has opposite orientation. In elliptic transport, the positive form intrinsic to the cycle also permits the norm balance. The three realizations share the alternating identity with their types preserved.

12.1.8 Variable forms and restitution between successive planes

Consider a sequence of planes V_n with positive-definite symmetric matrices G_n. Let $B_n, C_n : V_n \to V_{n+1}$ be two transports satisfying

$$B_n^{\mathsf{T}} G_{n+1} B_n = G_n, \qquad C_n^{\mathsf{T}} G_{n+1} C_n = G_n.$$

The transport B_n identifies the coordinates between the two planes; C_n includes the additional evolution. Define

$$T_n = \frac{8B_n + C_n}{9}, \qquad D_n = \frac{\sqrt{8}}{9}(C_n - B_n).$$

Theorem 12.10 (Conservation and recovery with a variable form)**.** *The following hold*

$$G_n = T_n^{\mathsf{T}} G_{n+1} T_n + D_n^{\mathsf{T}} G_{n+1} D_n, \tag{12.10}$$

$$v = B_n^{-1}\left(T_n v - \frac{1}{\sqrt{8}} D_n v\right). \tag{12.11}$$

Proof. In the expansion of the first expression, the cross terms cancel. This leaves $72B_n^{\mathsf{T}} G_{n+1} B_n/81$ and $9C_n^{\mathsf{T}} G_{n+1} C_n/81$, whose sum is G_n. The identity $T_n - D_n/\sqrt{8} = B_n$ proves the second formula. Positivity of the forms makes B_n injective; between planes of equal dimension, it is invertible. □

For $I_n(v) = \frac{1}{2} v^{\mathsf{T}} G_n v$, $v_{n+1} = T_n v_n$ and $m_n = D_n v_n$, one obtains

$$I_n(v_n) = I_{n+1}(v_{n+1}) + I_{n+1}(m_n).$$

Summing over N steps gives

$$I_0(v_0) = I_N(v_N) + \sum_{n=0}^{N-1} I_{n+1}(m_n). \tag{12.12}$$

Each record is evaluated in the form of the plane where it was produced. The inverse transports of (12.11) restore all states in reverse order from the terminal state and the complete records.

A concrete realization uses invertible matrices M_n, $G_n = M_n^{-\mathsf{T}} M_n^{-1}$ and rotations R_n:

$$B_n = M_{n+1} M_n^{-1}, \qquad C_n = M_{n+1} R_n M_n^{-1}.$$

The two form identities are verified by substituting these expressions and using $R_n^{\mathsf{T}} R_n = I$. The coordinates, scales and parameters of M_n, R_n are received from the state readers when this realization is applied to Dimensional Mechanics.

12.1.8.1 Exact criterion for transfer to the record

If R_n is a rotation through angle θ_n, in normalized coordinates $y_n = M_n^{-1} v_n$ one has

$$y_{n+1} = \frac{8I + R_n}{9} y_n, \qquad \left(\frac{8I + R_n}{9}\right)^{\mathsf{T}} \left(\frac{8I + R_n}{9}\right) = r_n I,$$

$$r_n = 1 - \frac{32}{81} \sin^2(\theta_n/2).$$

By induction, $I_N(v_N) = I_0(v_0) \prod_{n<N} r_n$.

Proposition 12.11 (Criterion for vanishing of the terminal state)**.** *If $I_0(v_0) > 0$, the terminal state satisfies $I_N(v_N) \to 0$ exactly when*

$$\sum_{n \geq 0} \sin^2(\theta_n/2) = +\infty.$$

Proof. For $c = 32/81$ and $u \in [0, 1]$, integrating $1/(1 - s)$ between zero and cu gives

$$cu \leq -\log(1 - cu) \leq \frac{cu}{1 - c}.$$

The sum of the logarithms diverges exactly when the sum of the $u_n = \sin^2(\theta_n/2)$ diverges. The product of the r_n tends to zero exactly in that case. □

The conservation law (12.12) holds at every finite depth. The preceding criterion determines when the entire initial form is transferred to the record in this realization. The angular sequence is the one obtained by the corresponding trajectory reader.

12.1.9 Integrated action and compatibility with refinement

In a canonical plane with coordinates (Q, P), let γ be a piecewise C^1 closed curve. For any real matrix L of order two,

$$\oint_{L\gamma} P \, \mathrm{d}Q = \det(L) \oint_{\gamma} P \, \mathrm{d}Q. \tag{12.13}$$

To prove this, write $Q' = aQ + bP$, $P' = cQ + dP$. The closed integrals of $Q \, \mathrm{d}Q$ and $P \, \mathrm{d}P$ are zero. Integrating $\mathrm{d}(PQ)$ gives $\oint Q \, \mathrm{d}P = -\oint P \, \mathrm{d}Q$. The coefficient $ad - bc$ remains, proving the formula.

Applied to (12.9), this yields

$$\oint_{\gamma} P \, \mathrm{d}Q = \oint_{T\gamma} P \, \mathrm{d}Q + \oint_{D\gamma} P \, \mathrm{d}Q.$$

The orientation of each curve preserves the sign of its contribution.

Refinement additionally requires compatibility between operators at successive levels. In this section, let $\iota_n : V_n \to V_{n+1}$ be linear injections and $A_n : V_n \to V_n$ endomorphisms satisfying $A_{n+1}\iota_n = \iota_n A_n$. At each level we define $T_n = (8I_{V_n} + A_n)/9$ and $D_n = \sqrt{8}(A_n - I_{V_n})/9$. These expressions imply

$$T_{n+1}\iota_n = \iota_n T_n, \qquad D_{n+1}\iota_n = \iota_n D_n.$$

Restitution (12.6) then commutes with refinement. This identity preserves the operation and the record simultaneously; its verification in a concrete realization uses the map ι_n produced by that prolongation.

12.1.9.1 Preservation of the reader's scalar factor

If a realization receives a scale factor e^{-x}, the transport is $C = e^{-x}E_\tau(t)$. Its determinant is e^{-2x}, and the same expansion of the nonadic law gives

$$T^{\mathsf{T}}\Omega T + D^{\mathsf{T}}\Omega D = \frac{8 + e^{-2x}}{9}\Omega.$$

For $x \geq 0$, the scale complement

$$D_{\mathrm{esc}} = \frac{\sqrt{1 - e^{-2x}}}{3} I$$

restores the complete balance:

$$\Omega = T^{\mathsf{T}}\Omega T + D^{\mathsf{T}}\Omega D + D_{\mathrm{esc}}^{\mathsf{T}}\Omega D_{\mathrm{esc}}.$$

The coefficient is obtained from the determinant defect of the scale. Each application preserves its reader's factor and specifies whether this additional record is needed.

The triadic extension of Euler is thus articulated through generated operations, invariant forms and recovery operators. The APP cycle, nilpotent transport and modal incidence preserve their provenance; the analytic realization describes their evolutions; the nonadic division of the reading preserves the complement required for restitution. This sequence expresses the link between regime, transformation and memory without replacing the genealogy with an isolated exponential formula.

12.2 Exponential identity of TRIT and coordination of the elliptic, parabolic, and hyperbolic regimes

12.2.1 Construction of the elliptic operator by means of the Paley matrix and the relation $J_{+1}^2 = -I$

The root of -1 is not introduced as a formal symbol without an antecedent. The six units modulo nine are coordinatized over $\mathbb{P}^1(\mathbb{F}_5)$, and the quadratic character of $\mathbb{F}_5$ determines the Paley conference matrix. Its ternary reduction is

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3).$$

The conference identity reduces to

$$\boxed{A_W^2 = 2I_6 = -I_6.} \tag{12.14}$$

Therefore, A_W is a quarter-turn operator in the ternary space. After extending scalars to $\mathbb{F}_9$, it decomposes into the conjugate eigenspaces of the roots of $X^2 + 1$. The common integral source

$$Q_{\mathbb{Z}} = \mathbb{Z}[u]/(u^2 + 1)$$

has a finite realization $u \mapsto A_W$ and a real realization $u \mapsto J_e$. The source coordinates the quadratic relation; it does not identify fields of different characteristics.

12.2.2 Unified family $\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau$

The three TRIT regimes are carried out by means of operators

$$J_{+1}^2 = -I, \qquad J_0^2 = 0, \qquad J_{-1}^2 = I.$$

Separating the even and odd powers of the exponential yields a single family:

$$\boxed{\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \qquad \tau \in \{+1, 0, -1\},} \tag{12.15}$$

where

$$(C_{+1}, S_{+1}) = (\cos, \sin), \qquad (C_0, S_0) = (1, t), \qquad (C_{-1}, S_{-1}) = (\cosh, \sinh).$$

12.2.3 Elliptic, parabolic, and hyperbolic specializations and compatibility with the temporal reader

Its three governing specializations are

$$\exp(\pi J_{+1}) + I = 0, \qquad \exp(J_0) = I + J_0, \qquad \exp\big(2(\log\varphi)J_{-1}\big) = F_{\mathrm{av}}^2. \tag{12.16}$$

The elliptic branch contains the complex circular closure; the parabolic branch, threshold transport; the hyperbolic branch, golden self-scaling. Euler's equation thus appears as a specialization of a family that coordinates rotation, boundary and hyperbolic expansion.

12.3 Hopf fibration, Villarceau circles, and Möbius holonomy as a topological realization of the elliptic regime

12.3.1 Complex structure and Hopf fibration

Let $S^3 \subset \mathbb{C}^2$ and

$$h(z_0, z_1) = \big(2\Re(z_0\overline{z}_1), 2\Im(z_0\overline{z}_1), |z_0|^2 - |z_1|^2\big) \in S^2.$$

On the Hopf torus

$$T_\eta = \{(z_0, z_1) : |z_0| = \cos\eta, |z_1| = \sin\eta\},$$

the curves

$$\gamma_+(t) = (\cos\eta\, e^{it}, \sin\eta\, e^{it}),$$
$$\gamma_-(t) = (\cos\eta\, e^{it}, \sin\eta\, e^{-it})$$

have distinct behaviors: γ_+ is a Hopf fiber and γ_- projects with degree two. Under stereographic projection, the homology classes $(1, 1)$ and $(1, -1)$ produce the two diagonal Villarceau families.

The total complex conjugation

$$K(z_0, z_1) = (\overline{z}_0, \overline{z}_1)$$

preserves each unoriented family and reverses its orientation. It does not interchange the two families. Partial conjugation can interchange them, but it is neither a complex-linear map nor the standard global antiunitary. This correction is essential to avoid manufacturing a particle–antiparticle symmetry from the toroidal drawing.

12.3.2 Double covering and spinor return

The map $SU(2) \to SO(3)$ is a double cover. A loop of 2π in $SO(3)$ can lift to $-I$ in $SU(2)$; the loop of 4π recovers I. The sign structure does not turn a circle into a Möbius band. Obtaining a band requires a real line bundle whose holonomy character is -1.

Let $L_\chi \to S^1$ be the associated bundle to

$$\chi : \pi_1(S^1) \longrightarrow \{\pm 1\}, \qquad \chi(1) = -1.$$

Its total space is a Möbius band. The route $C_M = -\,\mathrm{rev}$ provides the sign candidate; the Villarceau circle provides the loop; the physical intertwiner must prove that the former is the holonomy of the latter.

12.3.3 Corrispondence between the spectrum $\{7, 3\}$, the ratio 5:2 and the spinorial return

With $B_c = 7I + 2S$, define the oriented memory operator

$$B^{\varepsilon}_{\mathrm{mem}} = 5I + 2\varepsilon S, \qquad \varepsilon = \pm 1.$$

For $\varepsilon = +1$,

$$\mathrm{Spec}(B^{+}_{\mathrm{mem}}) = \{7, 3\}, \qquad \det B^{+}_{\mathrm{mem}} = 21.$$

Interpreting the values $(7, 3)$ as the outer and inner equatorial boundaries in a common positive chart yields

$$R + r = 7\ell, \qquad R - r = 3\ell,$$

and therefore

$$R = 5\ell, \qquad r = 2\ell, \qquad \sin\beta = \frac{r}{R} = \frac{2}{5}.$$

The Villarceau angle is

$$\beta = \arcsin\frac{2}{5} = 23.5781784782\ldots^{\circ}.$$

The algebraic theorem $\{7, 3\} \leftrightarrow (5, 2)$ is exact. The geometric premise interpreting the spectrum as the two boundaries of a torus is a declared interface. It must not be concealed, nor may the angle be used as a target.

The realization of the full block B_c retains another reading: $R : r = 7 : 2$ and $\beta = \arcsin(2/7)$. These are not rival values; they belong to two functors: total block and memory mode. The physical priority of the latter must be decided by the intertwiner with the dynamics, not by decimal proximity to another constant.

12.3.4 Angular momentum representation in the elliptic regime and phase space unit closure

HMT's native angle is a phase $u \in \mathbb{R}/\mathbb{Z}$. The readers publish

$$\theta_{\mathrm{rad}} = 2\pi u, \qquad \theta_{\mathrm{deg}} = 360u.$$

Radians and degrees are distinct charts of the same circular element. The radian is not more "ontological" by virtue of being dimensionless; its naturality follows from arc length. Degrees provide another absolute coordinate of the full turn. Any relation with α^{-1}, φ, or CKM must first be formulated as an invariant quotient or holonomy and only thereafter evaluated in an angular chart.

The exact identity already available

$$\tanh^2(2\log\varphi) = \frac{5}{9}$$

connects the Perron–Fibonacci channel to the spectral quotient of B_c. Its complement is $4/9$, the normalized gap. This relation is structural. It does not imply that every angle constructed with φ is a physical angle.

12.3.5 Brouwer, Möbius, and degree

Brouwer's fixed-point theorem, fractional linear Mobius transformations, and the Mobius band are three distinct objects. They are related only after fixing domain, action, and bundle. The elliptic advance and retardation orbits may be modeled as two sections of a bundle with sign holonomy; the Mobius transformation describes the conformal action in the chart; Brouwer's theorem guarantees a fixed point for continuous maps of the disk. None of these statements replaces the realization diagram.

Result and boundary

The degree-two character of the Hopf curve, the orientation of the total conjugation, and the algebraic construction $7, 3 \mapsto 5, 2$ are proved. The toroidal reading (5:2), Möbius holonomy, and the particle–antiparticle interpretation form a chain of interfaces that must share a single intertwiner. The volume does not force that intertwiner through an angular coincidence.

Chapter 13

Genealogical regularization of -1/12 through triadic extraction and Hensel–Witt lifting

13.1 Transduction kernel Hensel–Witt: domain, operation and genealogical preservation

Where do the cylinders, the monodromy, the five regions, the hexads, the octads and the physical readers actually intersect?

13.1.1 Hensel–Witt Transduction Kernel: π microregions, holonomy Γ_9, dodecaphase record, and hexad–octad transport

The NTHW is the data and maps package

$$\mathfrak{N}_{\mathrm{HW}} = \left(\mathscr{R}_{\pi}^{\mathrm{micro}}, \mathcal{G}_9, C_{12}, K, \mathcal{W}_{12\to 24}, \mathcal{I}_{6\to 8}\right),$$

with the following compatibilities:

1. the Hensel regions share a quotient word and retain distinct memory;
2. $\mathcal{G}_9$ transports orientation and produces monodromy;
3. C_{12} organizes the signed observable and the seal K;
4. the Witt reading converts supports of the same boundary into hexads and stars;
5. the extension $W_{12} \to W_{24}$ produces the hexad–octad interface;
6. the Archimedean, exceptional and dimensional readers receive invariants of the same state.

NTHW gives a formal name to the place previously described as the "Rosetta Stone." The substitution is conceptual: a stone translates through resemblance; a transduction specifies input, output, preserved datum and loss of information.

13.1.2 Translation of the ternary code into the Witt system

The reversible APP boundary selects, with declared gauges, a matrix A_W over $\mathbb{F}_3$ satisfying

$$A_W A_W^T = 2I \pmod 3.$$

The graphic code

$$C_W = \{(q, qA_W) : q \in \mathbb{F}_3^6\}$$

is the extended ternary Golay code. Its weight enumerator is

$$1 + 264y^6 + 440y^9 + 24y^{12}.$$

The 264 words of weight six, identified up to sign, yield 132 supports. Every five-point subset belongs to a unique hexad:

$$W_{12} = S(5, 6, 12).$$

The field of 495 star involutions generates a group of order 95 040, recognized as M_{12}. An isolated star does not generate the group. The ordered HMT frame has trivial stabilizer and therefore fixes a gauge within the action.

13.1.3 Hexagonal–Octagonal Inclusion and Incidence Transport through the 90/120 Channels

Let H be a hexad and O_H an octad containing it. Marking one point and a pair of points of the hexad yields

$$F_6(H) = H \times \binom{H}{2}, \qquad |F_6| = 6\binom{6}{2} = 90.$$

If the marked point can traverse the octad:

$$F_8(H) = O_H \times \binom{H}{2}, \qquad |F_8| = 8\binom{6}{2} = 120.$$

The inclusion $H \hookrightarrow O_H$ induces $F_6 \hookrightarrow F_8$. Its normalized defect is

$$\boxed{\frac{90-120}{24\binom{6}{2}} = \frac{6-8}{24} = -\frac{1}{12}.}$$

This combinatorial theorem precedes Barbero's angular reading. The two objects share the interface $6 \to 8$, but are not identical.

13.1.4 4 rutas hacia -1/12

The construction contains four realizations that must be presented together, not counted as four independent proofs of a single provenance.

13.1.4.1 Elementary Operator of the Dodecaphasic Cycle and Update of the Signed Phase

One tooth of C_{12} measures $30°$. Then

$$\frac{30}{120} - \frac{30}{90} = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}.$$

It is the angular shadow of the pattern $90/120$.

13.1.4.2 Gram matrix of the surviving sections and positivity criterion

In the certified jump $60 \to 81$, two fibers have sizes 0 and 12. The Gram operator is

$$K_{60,81} = \operatorname{diag}(0, 12).$$

On the live subspace, its pseudoinverse is $1/12$; with the negative boundary orientation,

$$E_{\text{surv}} = -K^{+}_{60,81}$$

yields $-1/12$.

13.1.4.3 Triadic Scale Extraction Operator and Its Rational Image

The realization via a triadic scale extractor is the second-difference construction developed in full in section 13.3.3. This first occurrence establishes its role within the inventory of realizations; the indicated residence preserves the definitions of T_L, $a(L)$, $A_1(L)$, $C(L)$, the independence of the residue from L, and the exact status of the normalization.

13.1.4.4 Modular corridor between dodecaphasic cards and conservation of the phase invariant

The eta function satisfies

$$\frac{\eta(\tau)}{\eta(3\tau)} = q^{-1/12} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-1}.$$

The exponent comes from $1/24 - 3/24 = -1/12$. Taking twelve copies:

$$t_3(q) = \left(\frac{\eta(\tau)}{\eta(3\tau)}\right)^{12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \cdots .$$

The coefficient 54 coincides with the APP defect. The corridor

$$j = \frac{(t_3 + 27)(t_3 + 243)^3}{t_3^3}$$

connects afterwards with the classical modular branch.

What has been proved in the corridor

The APP defect 54, the eta exponent $-1/12$, the hexad–octad defect, and the modular identities are exact once t_3 has been fixed. The strong assertion is that all of them are natural images of a single TPK operator. That naturality must be proved through the NTHW diagram; it is not deduced merely from repeating the same integers.

13.1.5 Representation of Primary Power Defects via the Number Operator

For a prime power p^m, the complete APP defect generalizes to

$$\Delta_{p^m}^{\text{full}} = \frac{m(p-1)}{2} p^{2m-1}.$$

In particular,

$$\Delta_3 = 3, \qquad \Delta_9 = 54, \qquad \Delta_{27} = 729.$$

Upon normalization,

$$a_{p,m} = \frac{2\Delta_{p^m}^{\text{full}}}{(p-1)p^{2m-1}} = m.$$

With $x = p^{-s}$:

$$\sum_{m\geq 1} a_{p,m} x^m = \sum_{m\geq 1} m x^m = \frac{x}{(1-x)^2} = x\frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{1-x}.$$

The factor $(1-x)^{-1}$ is the local Euler factor; differentiation inserts the number operator. Thus, an exact bridge appears:

tower of APP defects in p^m $\longleftrightarrow$ insertion of the number operator into the local Euler factor.

13.1.6 Joint naturality conditions for the Archimedean, exceptional, and dimensional transduction

The NTHW will be a complete transduction theorem when there is a family of invariants $\mathcal{I}$ and readers

$$\mathfrak{R}, \quad \mathfrak{E}, \quad \mathfrak{D}$$

such that

$$\mathcal{I} = \mathcal{I}_{\text{arq}} \circ \mathfrak{R} = \mathcal{I}_{\text{exc}} \circ \mathfrak{E} = \mathcal{I}_{\text{dim}} \circ \mathfrak{D},$$

and when the maps of region, seal, incidence, and action are derived from the same transport. The finite proofs already provide candidates: monodromy, support B_K, borrowing pattern, two-way parity, and counts 90/120.

Executable test 108–144–90–120

Label each of the 1008 routes of π by CT108 orbit and by Witt completion; compute the fibers of the map; test whether the four regions of multiplicity 144 realize factor $8/6$ on blocks of 108; and verify whether the region of 432 transports the monodromy. A single counterexample to label compatibility would refute this bridge in its present form.

Result of the section

The NTHW is the true center of the account because there the same boundary may be read as cylinder, monodromy, seal, incidence, and action. Its power does not consist in many numbers resembling one another, but in several finite maps already sharing supports and exact defects. The next task is to prove the joint naturality of those charts.

13.2 Marked inclusion of the hexad-octad incidence and preservation of its multiplicities

With the dodecad and the alignment of the section 17.1.6 fixed, let H be a hexad in the HMT coordinates, let $\ell(H) \subset D$ be its image, and let $O_H = \iota_D(\ell(H))$ be its unique lift. Marking a point and a pair of points of the residual hexad yields

$$F_6(H) = \ell(H) \times \binom{\ell(H)}{2}, \qquad |F_6| = 6\binom{6}{2} = 90.$$

If the marked point can traverse the octad:

$$F_8(H) = O_H \times \binom{\ell(H)}{2}, \qquad |F_8| = 8\binom{6}{2} = 120.$$

The inclusion $\ell(H) \hookrightarrow O_H$ induces the incidence morphism

$$\mathcal{I}_{6\to 8} : F_6(H) \hookrightarrow F_8(H).$$

Normalizing the difference by the chosen factor $24\binom{6}{2}$ yields

$$\boxed{\frac{90-120}{24\binom{6}{2}} = \frac{6-8}{24} = -\frac{1}{12}.}$$

The equality is a normalized defect of the combinatorial interface $6 \to 8$. The factor $24\binom{6}{2}$ forms part of the definition of this normalization; the inclusion alone determines the difference -30. The operator $\mathcal{I}_{6\to 8}$ has marked incidence configurations as its domain; it is neither the transverse extractor of the dodecaphase state nor the family of angular moments.

13.3 Classification by domain of the 5 algebraic realizations of $-1/12$

In addition to the normalized defect of the hexad–octad inclusion, five operator realizations of the same rational number appear. Their domains are specified separately; equality of values does not identify the operators without an additional intertwiner.

13.3.1 Difference of dodecaphasic periodicities

One tooth of C_{12} measures 30°. Then

$$\frac{30}{120} - \frac{30}{90} = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}.$$

It is the oriented difference between the periods of 90° and 120° in this realization.

13.3.2 Pseudoinverse of the Gram Operator on the Live Component

In the exact jump $60 \to 81$, two fibers have sizes 0 and 12. The Gram operator is

$$K_{60,81} = \operatorname{diag}(0, 12).$$

On the nonzero component of its range, the pseudoinverse is $1/12$; with the negative boundary orientation,

$$E_{\text{surv}} = -K^{+}_{60,81}$$

yields $-1/12$.

13.3.3 2nd difference between triadic scales

Define

$$T_L = \sum_{n=1}^{L-1} n(L-n) = \frac{L(L^2-1)}{6}, \qquad a(L) = -\frac{T_L}{L^2} = -\frac{L}{6} + \frac{1}{6L}.$$

The first scale subtraction is

$$A_1(L) = a(L) - \frac{a(3L)}{3} = \frac{4}{27L},$$

and the second

$$C(L) = LA_1(L) - \frac{3L}{3}A_1(3L) = \frac{8}{81}.$$

The residue $8/81$ is independent of L. To obtain $-1/12$, the declared normalization is applied

$$R(L) = -\frac{27}{32}C(L) = -\frac{1}{12}.$$

The calculation derives $8/81$; it does not by itself derive the factor $-27/32$.

13.3.4 Composition of modular transformations and preservation of the signed state

The eta function satisfies

$$\frac{\eta(\tau)}{\eta(3\tau)} = q^{-1/12}\prod_{n\geq 1}(1+q^n+q^{2n})^{-1}.$$

The exponent comes from $1/24 - 3/24 = -1/12$. Taking twelve copies:

$$t_3(q) = \left(\frac{\eta(\tau)}{\eta(3\tau)}\right)^{12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \cdots.$$

The coefficient 54 coincides with the APP defect. The rational identity

$$j = \frac{(t_3+27)(t_3+243)^3}{t_3^3}$$

establishes the connection with the classical modular branch.

13.3.5 Defect of cyclic difference projectors

Let S be the cyclic shift on $\mathbb{Q}^{12}$ and define

$$D_r = I - S^r.$$

The kernel of D_r consists of the vectors constant on each orbit of S^r; therefore,

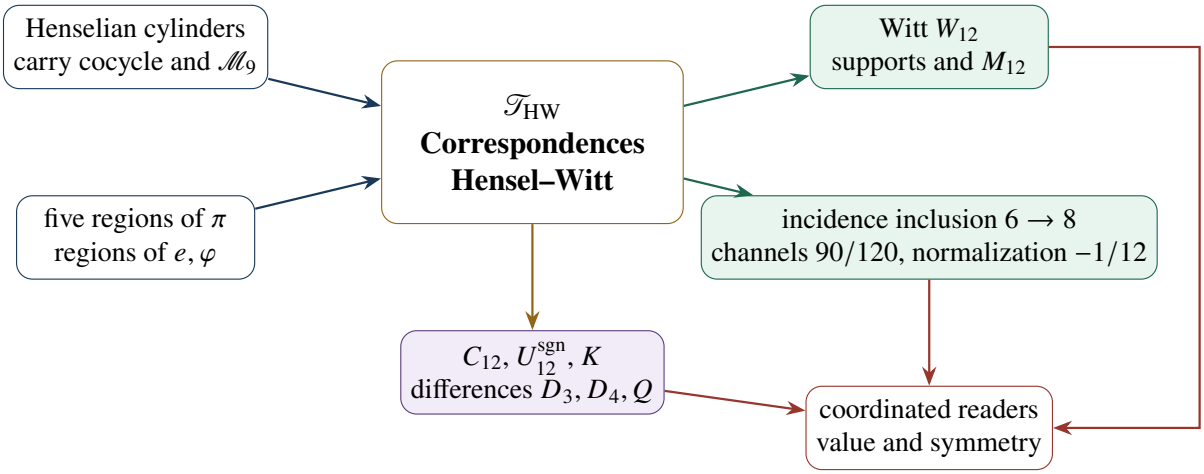
$$\dim \ker D_r = \gcd(12, r).$$

Let $\Pi_{\ker D_3}$ and $\Pi_{\ker D_4}$ be the rational projectors onto $\ker D_3$ and $\ker D_4$, and let $\tau(T) = \mathrm{Tr}(T)/12$ be the normalized trace. Then

$$\boxed{\tau(\Pi_{\ker D_3} - \Pi_{\ker D_4}) = \frac{\operatorname{rank} \Pi_{\ker D_3} - \operatorname{rank} \Pi_{\ker D_4}}{12} = \frac{3-4}{12} = -\frac{1}{12}.}$$

This realization expresses the rational as the transverse defect of steps three and four in the twelve-phase cycle.

Remark 13.1 (Domains of the realizations). The APP defect 54, the eta exponent $-1/12$, the combinatorial defect of the hexad–octad lift, and the modular identities once t_3 has been fixed are exact. The value of the pseudoinverse, the normalized extractor, and the projector defect are likewise exact. Each equality preserves its domain and its corresponding map.



Compatibility of cylinders, dodecaphase, and incidences on the same enriched state.

Figure 13.1: The NTHW as a typed interface. The central mathematical task is to make the diagram commute, not to accumulate coincidences around a name.

Chapter 14

Regional–determinantal confluence of the reduced action: construction of η_{ret}, scale 10^{-34}, $\hbar_{\mathrm{ret}}$, and parangular publication (A, C^{*})

14.1 Genealogical bifurcation of the action scale and its dimensionless component

The intrinsic phase α_{HMT}, already obtained by means of the reversible dodecaphase closure, gives rise to two independent internal branches. The first determines the dimensionless component of the action subsequent to the return; the second determines its absolute decade. Neither of them presupposes the counter-angle.

In the regional branch, the character of degree 5 is

$$H_5(\alpha_{\mathrm{HMT}}, \varphi) = \frac{1}{2}\sqrt{\frac{10^3\alpha_{\mathrm{HMT}}}{\varphi}} - \alpha_{\mathrm{HMT}} + \frac{9}{16}\alpha_{\mathrm{HMT}}^2 - \frac{5}{9}\alpha_{\mathrm{HMT}}^3 + \frac{7}{48}\alpha_{\mathrm{HMT}}^4 - \frac{1}{54}\alpha_{\mathrm{HMT}}^5. \tag{14.1}$$

The completion $10^3 = 729 + 243 + 27 + 1$ publishes the phase in the analytic coordinate

$$x_A = \frac{50\pi}{9}\alpha_{\mathrm{HMT}}, \qquad D(\alpha_{\mathrm{HMT}}) = e^{-2x_A} = \exp\!\left(-\frac{100\pi}{9}\alpha_{\mathrm{HMT}}\right). \tag{14.2}$$

With the regional selector $\rho_\pi = 169$, the convergent recurrence

$$\mathscr{C}_\pi(\alpha) = 169\alpha^6 + \frac{D(\alpha)\alpha^7}{1 - D(\alpha)\alpha} \tag{14.3}$$

directly determines

$$\boxed{\eta_{\mathrm{ret}} = H_5 - \frac{90}{\pi}\mathscr{C}_\pi(\alpha_{\mathrm{HMT}}) > 0.} \tag{14.4}$$

This equation not contains C^{*}: constructs the coordinate dimensionless of action that intervenes dispues in the closure bicapa.

In the determinantal branch, the Smith normal form produces

$$\Sigma_H = \varphi\alpha_{\mathrm{HMT}}^{16}, \qquad \varepsilon_H = \mathrm{ord}_{10}(\Sigma_H) = -34. \tag{14.5}$$

The two branches converge, and only then, in the magnitude of action.

$$\boxed{\hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}} 10^{\varepsilon_H} U_S = \eta_{\mathrm{ret}} 10^{-34} U_S.} \tag{14.6}$$

14.2 Angular publication and compatibility of the global and analytic charts

Once the phase α_{HMT}, the coordinate η_{ret}, and the magnitude $\hbar_{\mathrm{ret}}$ have been constructed, the parangular map publishes

$$\mathcal{P}(\alpha_{\mathrm{HMT}}, \eta_{\mathrm{ret}}) = \left(A_{\mathrm{deg}}, C^{*}_{\mathrm{deg}}\right) = \left([10^3\alpha_{\mathrm{HMT}}]_{360}, [2(\eta_{\mathrm{ret}} + \alpha_{\mathrm{HMT}})]_{360}\right). \tag{14.7}$$

The factor 10^3 comes from nonadic completion; the factor 2, from the bidirectional closure of degree 2; and 360 is the global chart that renders simultaneously visible $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120 = 360$. The analytic chart applies once

$$\kappa_{360}(a,c)=\frac{\pi}{180}(a,c),\qquad A_{\mathrm{rad}}=\frac{50\pi}{9}\alpha_{\mathrm{HMT}},\qquad C^{*}_{\mathrm{rad}}=\frac{\pi}{90}(\eta_{\mathrm{ret}}+\alpha_{\mathrm{HMT}}). \tag{14.8}$$

Therefore, $C^{*}_{\mathrm{deg}}=2(\eta_{\mathrm{ret}}+\alpha_{\mathrm{HMT}})$ is the direct theorem, whereas $\eta_{\mathrm{ret}}=C^{*}_{\mathrm{deg}}/2-\alpha_{\mathrm{HMT}}$ constitutes solely its inverse verification. The action satisfies

$$S(\theta)=\hbar_{\mathrm{ret}}\theta_{\mathrm{rad}}=h_{\mathrm{ret}}\frac{\theta_{\mathrm{deg}}}{360},\qquad h_{\mathrm{ret}}=2\pi\hbar_{\mathrm{ret}}. \tag{14.9}$$

The channels $q_{\pm}$ subsequently evaluate the incidence cardinalities 90/120, already generated in the Topological Phase Kernel; they do not constitute their origin.

14.3 Preconstitutive construction of the bidirectional action response and the vacuum operator

14.3.1 Derivation of the action scale from the dodecaphasic state

The action decade is determined intrinsically, without adopting it as an inherited unit. The local determinantal functional is applied to the previously constructed HMT output α_{HMT} and defines

$$\Sigma_{H}=\varphi_{\mathrm{HMT}}\alpha_{\mathrm{HMT}}^{16}. \tag{14.10}$$

The exponent 16 is the order of the local cokernel of the signed closure; its decimal valuation fixes

$$\mathrm{ord}_{10}(\Sigma_{H})=-34. \tag{14.11}$$

The origin of the exponent can be verified within this volume. On the local orbit of four leaves, the Hadamard block acts

$$H_{4}=\begin{pmatrix}1&1&1&1\\1&1&-1&-1\\1&-1&1&-1\\1&-1&-1&1\end{pmatrix}. \tag{14.12}$$

Lemma 14.1 (Local index of the action line)**.** *The form normal of Smith of H_4 is*

$$\mathrm{SNF}(H_{4})=\mathrm{diag}(1,2,2,4). \tag{14.13}$$

Consequently,

$$\mathrm{coker}(H_{4})\simeq\mathbb{Z}/2\mathbb{Z}\oplus\mathbb{Z}/2\mathbb{Z}\oplus\mathbb{Z}/4\mathbb{Z},\qquad|\mathrm{coker}(H_{4})|=16.$$

Proof. The greatest common divisor of the entries is one; that of the nonzero 2×2 minors is two; that of the 3×3 minors is four; and $|\det H_{4}|=16$. The successive quotients of these determinantal divisors are $1,2,2,4$, which prove equation (14.13). The product of the invariants gives the order of the cokernel. □

The norm of the section α_{HMT} over those sixteen sheets produces $\alpha_{\mathrm{HMT}}^{16}$; the self-scaling φ_{HMT} acts only once on the basis, not once per sheet. Hence equation (14.10). The certified comparisons

$$\alpha_{\mathrm{HMT}}^{16}<10^{-34}<\varphi_{\mathrm{HMT}}\alpha_{\mathrm{HMT}}^{16}<10^{-33} \tag{14.14}$$

prove, without introducing the SI unit, that the decimal order is exactly -34. The terminal evaluation begins with

$$\Sigma_{H}=1.046243838104756580184229208303089\ldots\times10^{-34}. \tag{14.15}$$

The substitution of the kernel by three copies with exponent 16^{3}, or the repeated application of φ on each sheet, would alter the genealogy. Both operations constitute falsification criteria rather than equivalent conventions. The causal chain that is preserved is therefore,

$$\mathrm{APP} \to \widehat{\mathbb{T}} \to \mathrm{TPK} \to U_{12}^{\mathrm{sign}} \to \alpha_{\mathrm{HMT}} \to \Sigma_H \to 10^{-34}\mathcal{U}_S \to \{\hbar_{\mathrm{pre}}, \hbar_{\mathrm{ret}}\}.$$

This volume does not rederive α_{HMT}, but develops the metrological geometry subsequent to the selection of the type and decade by Σ_H. The integer -34 is neither a mantissa nor an adjustment to the International System, but the terminal valuation of the local functional.

14.3.2 Oriented sections of the action line

The realizations before and after the return constitute two sections of the same action line:

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S. \tag{14.16}$$

The common decade determines the line, while the oriented information is expressed through

$$R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}}, \qquad \xi_{\mathrm{act}} = \frac{1}{2}\log R_{\mathrm{act}}. \tag{14.17}$$

The first quantity provides a multiplicative coordinate and the second an additive coordinate of rapidity type. They do not represent two independent Planck constants, but the two orientations of the same section with respect to the return.

If $h_a = 2\pi\hbar_a$, $a \in \{\mathrm{pre}, \mathrm{ret}\}$, a quantity proportional to $h_a^{1/2}$ changes as $R_{\mathrm{act}}^{1/2}$, one proportional to $h_a^{-1/2}$ changes as $R_{\mathrm{act}}^{-1/2}$, and a quotient in which the action cancels is a bidirectional invariant.

14.3.3 Bilayer spectral decomposition of the angular operator

Let $R = R^* = R^{-1}$ be the active involution and

$$P_\pm = \frac{I \pm R}{2}, \qquad T = q_+P_+ + q_-P_-, \qquad 0 < q_+ < q_- < 1.$$

With

$$x = A_{\mathrm{rad}} = \frac{\pi A}{180}, \qquad y = C^*_{\mathrm{rad}} = \frac{\pi C^*}{180},$$

the two channels are constructed before evaluating the vacuum:

$$q_+ = e^{-x-y}, \qquad q_- = e^{-x+y}. \tag{14.18}$$

The exchange of sheets reverses C^*, that is, $y \mapsto -y$, and permutes $q_+ \leftrightarrow q_-$. The pair is thus the spectral reading of the same angular antecedent, not two parameters fitted separately.

Definition 14.2 (Constitutive response)**.**

$$\mathcal{V}_{90,120} = (I - T^{90})(I - T^{120})^{-1}. \tag{14.19}$$

Since $\|T\| < 1$, the denominator is positive and invertible. Functional calculus gives

$$\mathcal{V}_{90,120} = r_+P_+ + r_-P_-, \qquad r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}}. \tag{14.20}$$

The assertion of invertibility deserves to be made explicit. For each leaf, $0 < q_\pm^{120} < 1$; therefore

$$I - T^{120} = (1 - q_+^{120})P_+ + (1 - q_-^{120})P_-$$

is strictly positive and

$$(I - T^{120})^{-1} = \frac{P_+}{1 - q_+^{120}} + \frac{P_-}{1 - q_-^{120}}.$$

The response $\mathcal{V}_{90,120}$ is not obtained by solving four separate scalar equations: it is the result of the functional calculus of a single positive contraction. This observation prevents permeability, permittivity, impedance and propagation from losing their shared antecedent.

Theorem 14.3 (Joint spectral determination of the four constitutive relations)**.** *The four constituent characters*

$$\widehat{\varepsilon} = r_{+}^{2}, \qquad \widehat{\mu} = r_{-}^{2}, \qquad \widehat{Z} = \frac{r_{-}}{r_{+}}, \qquad \widehat{c} = \frac{1}{r_{+} r_{-}} \tag{14.21}$$

are spectral functions of $\mathcal{V}_{90,120}$. They satisfy

$$\widehat{\mu}\widehat{\varepsilon} = \widehat{c}^{-2}, \qquad \frac{\widehat{\mu}}{\widehat{\varepsilon}} = \widehat{Z}^{2}. \tag{14.22}$$

Proof. The identities are obtained by substituting equation (14.21). No dimensional chart nor any exterior value intervenes. □

Corollary 14.4 (Reconstruction of the two sheets)**.** *The four characters do not contain four degrees of freedom. Any of the pairs $(\widehat{\mu}, \widehat{\varepsilon})$, $(\widehat{Z}, \widehat{c})$, or $(\mathcal{E}, \mathcal{B})$ reconstructs the two positive eigenvalues:*

$$r_{-} = \sqrt{\widehat{\mu}} = \sqrt{\widehat{Z}/\widehat{c}}, \qquad r_{+} = \sqrt{\widehat{\varepsilon}} = \frac{1}{\sqrt{\widehat{Z}\widehat{c}}}, \tag{14.23}$$

$$\log r_{-} = \frac{\mathcal{E} + \mathcal{B}}{2}, \qquad \log r_{+} = \frac{\mathcal{E} - \mathcal{B}}{2}. \tag{14.24}$$

In particular, orientation is lost if only the product is retained, but is recovered by adding the quotient.

Proof. The first line is the positive root of the identities equations (14.21) y (14.22). The second inverts the linear system $\mathcal{E} = \log r_{-} + \log r_{+}$, $\mathcal{B} = \log r_{-} - \log r_{+}$. □

This corollary expresses a central property of HMT: product and quotient are complementary functionals of the same state. The product determines the propagation in common, whereas the quotient preserves the relative orientation of the inner and outer sheets.

In the V2 certificate,

$$\begin{aligned}
r_{+} &= 0.99999962854824936317869522 81\ldots,\\
r_{-} &= 0.99972413718951836413151658 53\ldots,\\
\widehat{\varepsilon} &= 0.99999925709663670276044161 55\ldots,\\
\widehat{\mu} &= 0.99944835047932693508998155 59\ldots,\\
\widehat{Z} &= 0.99972450853893721545555434 66\ldots,\\
\widehat{c} &= 1.00027631048615752610638626 89\ldots.
\end{aligned}$$

These numerals are the terminal recognition of equation (14.19); they do not define its exponents $90, 120$.

14.3.4 Harmonic factorization of sectors 90 and 120

The harmonic semigroup is

$$\langle 90, 120\rangle = 30\langle 3, 4\rangle.$$

If $s = q^{30}$, then

$$\frac{1-q^{90}}{1-q^{120}} = \frac{1-s^3}{1-s^4} = \frac{1+s+s^2}{1+s+s^2+s^3}. \tag{14.25}$$

The identity also separates the response from its defect. If

$$F_{3\to4}(s) = \frac{1+s+s^2}{1+s+s^2+s^3}, \qquad 0 < s < 1,$$

then

$$d_{3\to4}(s) := 1 - F_{3\to4}(s) = \frac{s^3}{1+s+s^2+s^3} > 0. \tag{14.26}$$

The additional term is preserved exactly as a positive defect. Moreover,

$$F'_{3\to4}(s) = -\frac{s^2(3+2s+s^2)}{(1+s+s^2+s^3)^2} < 0. \tag{14.27}$$

Consequently, the order of the angular channels uniquely determines the order of the constitutive responses. The assignment of $\widehat{\mu}$ and $\widehat{\varepsilon}$ to the respective sheets is obtained analytically, without resorting to their decimal expansions.

Each sheet of the vacuum constitutes the exact completion from three to four terms of the same elementary mode 30. Consequently, permeability and permittivity are interpreted as two quadratic functionals of the same completion evaluated on conjugate sheets, and not as independent constitutive parameters.

Since the function $(1-q^{90})/(1-q^{120})$ decreases in $0 < q < 1$ and $q_+ < q_-$, we obtain

$$0 < r_- < r_+ < 1, \qquad 0 < \widehat{\mu} < \widehat{\varepsilon} < 1, \qquad 0 < \widehat{Z} < 1 < \widehat{c}.$$

The limits of the functional determine its geometry. When $q \to 0^+$, $F(q) \to 1$: the four readings approach unity and the memory defect disappears. When $q \to 1^-$,

$$F(q) \longrightarrow \frac{90}{120} = \frac{3}{4}.$$

The response therefore takes values in the positive interval $(3/4, 1)$. The endpoint $3/4$ is not a fitted constant: it is the quotient of the two channel lengths and the continuous realization of the three–to–four completion.

14.3.5 Sheet involution and preservation of orientation

The sheet exchange $C^* \mapsto -C^*$ acts by

$$\widehat{\mu} \longleftrightarrow \widehat{\varepsilon}, \qquad \widehat{Z} \longmapsto \widehat{Z}^{-1}, \qquad \widehat{c} \longmapsto \widehat{c}. \tag{14.28}$$

The common propagation is even; impedance preserves orientation. This distinction will reappear in Barbero and CKM: not all functionals associated with the change of sheet have the same parity.

14.3.6 Logarithmic characters and tracial compatibility

Define

$$\mathcal{E} = \log(r_+ r_-), \qquad \mathcal{B} = \log(r_-/r_+).$$

With the normalized trace $\tau = \mathrm{Tr}/2$,

$$2\tau(\log \mathcal{V}) = \mathcal{E}, \qquad -2\tau(R \log \mathcal{V}) = \mathcal{B}. \tag{14.29}$$

The pair $(\mathcal{E}, \mathcal{B})$ constitutes a linear coordinate system for the operator logarithm:

$$\log \mathcal{V} = \frac{\mathcal{E}}{2} I - \frac{\mathcal{B}}{2} R. \tag{14.30}$$

The scalar component is invariant under sheet exchange; the oriented component changes sign. This decomposition directly demonstrates the parity of propagation and the oriented character of impedance.

The nonadic amplification

$$\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$$

is compatible with the expectations $E_m = \mathrm{id} \otimes \tau_9$:

$$E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m.$$

Therefore, $\mathcal{E}$ and $\mathcal{B}$ are not peculiarities of a 2×2 matrix; they persist in the UHF tower and in its tracial closure II_1.

More precisely, for every polynomial p and afterwards, by continuity, for every continuous function on the spectrum,

$$E_m\big(p(\mathcal{V}_{m+1})\big) = p(\mathcal{V}_m). \tag{14.31}$$

Compatibility does not preserve only two numbers: it preserves the entire functional algebra generated by the response. In particular, it preserves its spectral projectors, powers, logarithm, and quadratic forms that measure the defects equation (14.26).

14.3.7 Dimensional realization of the constitutive relations

Dimensional Mechanics determines

$$Z_{\mathrm{vac}} = \widehat{Z}\, u_S u_Q^{-2}, \tag{14.32}$$

$$c_{\mathrm{vac}} = \widehat{c}\, u_C, \tag{14.33}$$

$$\mu_{\mathrm{vac}} = \widehat{\mu}\, u_S u_C^{-1} u_Q^{-2}, \tag{14.34}$$

$$\varepsilon_{\mathrm{vac}} = \widehat{\varepsilon}\, u_S^{-1} u_C^{-1} u_Q^{2}. \tag{14.35}$$

The constitutive identities hold on the straight lines:

$$\mu_{\mathrm{vac}} \varepsilon_{\mathrm{vac}} = c_{\mathrm{vac}}^{-2}, \qquad \mu_{\mathrm{vac}} / \varepsilon_{\mathrm{vac}} = Z_{\mathrm{vac}}^{2}. \tag{14.36}$$

The SI chart comes later. It is not necessary to declare c primitive and then solve $\mu\varepsilon = c^{-2}$: the three objects descend jointly from $\mathcal{V}$ and from the declared lines.

The separation of levels is expressed by the diagram

$$\begin{array}{ccccc} (A, C^*) & \longrightarrow & T & \longrightarrow & \mathcal{V} \longrightarrow (\widehat{\mu}, \widehat{\varepsilon}, \widehat{Z}, \widehat{c}) \\ \big\downarrow \text{line charts} & & & & \big\downarrow \text{final evaluation} \\ (u_S, u_Q, u_C) & \longrightarrow & (\mu_{\mathrm{vac}}, \varepsilon_{\mathrm{vac}}, Z_{\mathrm{vac}}, c_{\mathrm{vac}}) & \longrightarrow & \text{SI coordinates.} \end{array}$$

The first line is dimensionless and operatorial; the second determines types of quantity; the third selects a metrological chart. Altering the final chart changes neither the eigenvalues nor the constitutive identities.

In this precise sense, mathematical–physical unification consists in distinguishing the genealogical invariant from its coordinate system, not in removing the dimensional structure.

Proposition 14.5 (Spectral minimality)**.** *Let W be a positive two-sheet operator with simple spectrum $\{r_+, r_-\}$. If four positive characters satisfy $\mu\varepsilon = c^{-2}$ and $\mu/\varepsilon = Z^2$, if their propagation in common is determined by $c^{-1} = \det W = r_- r_+$, and if its orientation fixes $Z = r_-/r_+$, then necessarily*

$$(\mu, \varepsilon, Z, c) = \left(r_-^2, r_+^2, \frac{r_-}{r_+}, \frac{1}{r_- r_+}\right).$$

Therefore, equation (14.21) is not one parametrization among many: it is the minimal positive realization compatible with product, quotient and orientation.

Proof. From $Z = r_-/r_+$ and $c^{-1} = \det W = r_- r_+$ the positive roots are obtained uniquely roots. The two constitutive identities then force afterwards $\mu = r_-^2$ and $\varepsilon = r_+^2$. □

14.3.8 Angular determinant and Weinberg electroweak relation

The determinant of T is

$$\det T = q_+ q_- = D_A. \tag{14.37}$$

The constitutive sector of the vacuum evaluates $F(T)^2$, where $F(z) = (1 - z^{90})/(1 - z^{120})$, whereas the electroweak angular sector evaluates $\det T$. These are two functions of a common antecedent, not a horizontal factorization between constants. This distinction is developed in the section 49.1.

Falsification criterion 14.6. The block is refuted if any of these exact identities is violated: positivity of $I - T^{120}$, equality equation (14.25), relations equation (14.22), or compatibility $E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m$. An SI coincidence cannot compensate for an operator failure.

Hypothesis, domain, and conclusion

Operator, spectrum, completion $3 \to 4$, parity, amplification, and constitutive sections form an exact internal theorem. Their assembly into a single operator is established in this formulation; the angular data and the lines, previously derived in HMT, constitute the explicit starting structure.

14.4 Quantum–electric constants, thermal scale, and Planck system

14.4.1 Derivation of the charge section from the vacuum

With Z_{vac}, $h_a = 2\pi\hbar_a$ and α_{HMT} fixed, the charge of sheet $a \in \{\mathrm{pre}, \mathrm{ret}\}$ is

$$e_a = \sqrt{\frac{2\alpha_{\mathrm{HMT}} h_a}{Z_{\mathrm{vac}}}}. \tag{14.38}$$

The section e_a is not introduced through an International System coordinate intended to reconstruct α. The causal order proceeds from α_{HMT}, the action, and the vacuum toward the charge section.

14.4.2 Von Klitzing and Josephson constants, conductance and magnetic flux

From equation (14.38) we deduce

$$R_K = \frac{h_a}{e_a^2} = \frac{Z_{\mathrm{vac}}}{2\alpha_{\mathrm{HMT}}}, \tag{14.39}$$

$$G_0 = \frac{2e_a^2}{h_a} = \frac{4\alpha_{\mathrm{HMT}}}{Z_{\mathrm{vac}}}, \tag{14.40}$$

$$\Phi_{0,a} = \frac{h_a}{2e_a} = \sqrt{\frac{h_a Z_{\mathrm{vac}}}{8\alpha_{\mathrm{HMT}}}}, \tag{14.41}$$

$$K_{J,a} = \frac{2e_a}{h_a} = \Phi_{0,a}^{-1}. \tag{14.42}$$

Consequently,

$$R_K G_0 = 2, \qquad G_0 Z_{\mathrm{vac}} = 4\alpha_{\mathrm{HMT}}, \qquad K_{J,a}\Phi_{0,a} = 1. \tag{14.43}$$

The bidirectional covariance takes the form

$$R_K^{\mathrm{pre}} = R_K^{\mathrm{ret}}, \qquad G_0^{\mathrm{pre}} = G_0^{\mathrm{ret}},$$

$$\frac{\Phi_{0,\mathrm{pre}}}{\Phi_{0,\mathrm{ret}}} = R_{\mathrm{act}}^{1/2}, \qquad \frac{K_{J,\mathrm{pre}}}{K_{J,\mathrm{ret}}} = R_{\mathrm{act}}^{-1/2}.$$

The resistance and conductance relations are independent of the orientation of action. The flux–frequency pair preserves that orientation and transforms its two components reciprocally.

14.4.3 Relation thermal intrinsic of the scale 108

The chronological structure 108 fixes

$$k_B \Theta_{\mathrm{clk}} \log 3 = \frac{h_{\mathrm{int}}}{108 t_0} = \frac{2\pi\hbar_{\mathrm{int}}}{108 t_0}. \tag{14.44}$$

This equality does not separately determine a numerical coordinate of k_B and another of Θ_{clk} without an additional normalization. It does determine their product as a section of energy. The factor $\log 3$ has three compatible generators:

$$\log 3 = C_1 + C_2 - 2C_0 = \frac{S_{\widehat{\mathbb{T}}}}{k_B} = \frac{E_P}{k_B \Theta_{\mathrm{clk}}}. \tag{14.45}$$

The three equalities correspond to residual, informational and thermodynamic realizations of the same scalar. The coincidence of value does not identify their respective domains.

14.4.4 Planck System and Bidirectional Covariance

The CT108 selector constructs

$$\theta_{108} = \left(\frac{54}{\pi}\right)^2, \qquad \ell_P = \frac{54}{\pi}\ell_0, \qquad t_P = \frac{54}{\pi} t_0, \tag{14.46}$$

and

$$E_P = \frac{\hbar_{\mathrm{int}}}{t_P} = \frac{\pi\hbar_{\mathrm{int}}}{54 t_0}. \tag{14.47}$$

Comparing equations (14.44) y (14.47) gives

$$\boxed{E_P = k_B \Theta_{\mathrm{clk}} \log 3.} \tag{14.48}$$

The relation does not depend on a coincidence between units: both sides follow from the same chronological structure 108.

When the gravitational constant of chapter 50 is introduced, the Planck family satisfies

$$\ell_P^2 = \frac{G\hbar}{c^3} = \theta_{108}\ell_0^2, \tag{14.49}$$

$$F_P = \frac{c^4}{G}, \qquad P_P = \frac{c^5}{G}, \tag{14.50}$$

$$\rho_P = \frac{\hbar}{\theta_{108}^2 c^5 t_0^4}. \tag{14.51}$$

Although G and $\hbar$ transform reciprocally between the sections before and after the return, their product in ℓ_P^2 preserves the geometry. This invariance determines area as the interface between the geometric, informational and entanglement realizations.

14.4.5 Spectral formulations of Wien's displacement law

For the spectral density per wavelength, the maximum satisfies

$$5(1 - \mathrm{e}^{-x_5}) = x_5, \qquad x_5 = 5 + W_0(-5\mathrm{e}^{-5}).$$

With equation (14.44),

$$\lambda_{\max}(\Theta_{\mathrm{clk}}) = \frac{108 \log 3}{x_5}\,\ell_0. \tag{14.52}$$

For the frequency density,

$$3(1 - \mathrm{e}^{-x_3}) = x_3, \qquad x_3 = 3 + W_0(-3\mathrm{e}^{-3}),$$

and

$$\nu_{\max}(\Theta_{\mathrm{clk}}) = \frac{x_3}{108 t_0 \log 3}. \tag{14.53}$$

The maxima are not reciprocals, since the spectral densities transform with the corresponding Jacobian. The quotient x_5/x_3 belongs to the structure of the parameterization and does not represent an inconsistency.

14.4.6 Stefan–Boltzmann law and thermodynamics of the photon gas

Evaluating the radiation law at Θ_{clk} produces the intrinsic flux

$$j_\star(\Theta_{\mathrm{clk}}) = \frac{2\pi^5}{15\,108^4 (\log 3)^4}\,\frac{h_{\mathrm{int}}}{\ell_0^2 t_0^2}, \tag{14.54}$$

in the declared normalization. The Stefan–Boltzmann constant is not inserted as a primitive: it results from integrating the bosonic spectrum once action, propagation, and temperature have been fixed. The spectral laws employed are those of Planck and Wien [102, 133]; HMT fixes here the section on which they are evaluated, not their constants by retrospective comparison.

In units of the Planck scale,

$$n_\gamma \ell_P^3 = \frac{2\zeta(3)}{\pi^2 \log^3 3}, \tag{14.55}$$

$$\frac{u_\gamma \ell_P^3}{E_P} = \frac{\pi^2}{15 \log^4 3}, \tag{14.56}$$

$$\frac{s_\gamma \ell_P^3}{k_B} = \frac{4\pi^2}{45 \log^3 3}. \tag{14.57}$$

$\zeta(3)$ represents the third-order bosonic moment, whereas $\log 3$ determines the energy associated with the TRIT decision. The equation preserves the distinction between the two invariants.

The quantum thermal conductance is expressed as

$$\frac{\kappa_Q t_0}{k_B} = \frac{\pi^2}{324 \log 3}. \tag{14.58}$$

14.4.7 Posterior metrological confluence of action, charge, conductance, and magnetic flux

Object	Equation internal	Significado estructural
R_K	$Z_{\mathrm{vac}}/(2\alpha)$	bidirectionally invariant quantum resistance
G_0	$4\alpha/Z_{\mathrm{vac}}$	conductancia complementaria
Φ_0	$\sqrt{hZ/(8\alpha)}$	flow preserving orientation of action
K_J	Φ_0^{-1}	reciprocal relation between frequency and flux
E_P	$k_B\Theta \log 3$	energy of the cycle and TRIT informational content
Wien's law	$108 \log 3/x_5$	endpoint spectral in the scale chronological
Photonic Apéry	$2\zeta(3)/(\pi^2 \log^3 3)$	number density of the bosonic field

Falsification criterion 14.7. The block is falsified if a change of spectral variable improperly preserves the same Wien extremum, if $\zeta(3)$ replaces $\log 3$, or if separate values are assigned to k_B and Θ without specifying a chart. The identities equations (14.43) y (14.48) provide direct algebraic controls.

Hypothesis, domain, and conclusion

Electric quanta and Planck–TRIT confluence: *Internal exact theorem*. Wien, Stefan–Boltzmann and gas photonic: *Composition mathematical exact* with the declared bosonic continuum. The SI chart remains as *Contrast metrological external*.

14.5 Graduated analytic realization of the quintuple microfiber

The finite catalog and an analytic evaluation answer questions of different types. The catalog determines which regions exist, which share a return and what information each preserves. An analytic evaluation must also decide how the length of a word is transformed into degree, how a branch is prolonged after its return and with which scalar character that prolongation is read. This chapter explicitly constructs that second operation.

The construction begins with four data already obtained in the preceding chapters: the five-region microfiber of π, the length six of each regional word, the dodecaphasic closure of α, and the bilayer transport associated with the common coordinate $A = 10^3\alpha$. No value of the conjugate angular coordinate C_*^{HMT}, of the reduced action component, of the Barbero–Immirzi functional, or of a metrological constant is used in the evaluation. The result is a convergent regional character, an exact recurrence, and an oriented phase whose sexagesimal chart reproduces the published conjugate angular coordinate.

The separation between the finite part and the analytic part is essential. The integer 169 will be a theorem of the catalogue. The series that prolongs it will be the theorem of an analytic reader whose rules are declared before it is evaluated. Thus, the additional premises are not concealed within a decimal coincidence.

14.5.1 The persistent return of the 5 regions

Let

$$\mathcal{F}_\pi = \Psi^{-1}(010211) \subset \mathcal{U}_6$$

the microfiber relative to the catalog. Recall its five elements, now separated into initial block and terminal block:

$$(501, 614, 498 \mid 169, 272, 272),$$
$$(810, 923, 870 \mid 169, 272, 272),$$
$$(810, 983, 810 \mid 169, 272, 272),$$
$$(870, 923, 810 \mid 169, 272, 272),$$
$$(870, 923, 810 \mid 418, 674, 521).$$

We define the return projection.

$$p_{\mathrm{ret}} : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^3, \qquad p_{\mathrm{ret}}(u_1, \dots, u_6) = (u_4, u_5, u_6).$$

Proposition 14.8 (Persistent terminal block)**.** *The multiset $p_{\mathrm{ret}}(\mathcal{F}_\pi)$ has a unique element of maximum multiplicity:*

$$b_\pi = (169, 272, 272), \qquad \mathrm{mult}_{\mathcal{F}_\pi}(b_\pi) = 4.$$

The only remaining terminal block is

$$b_\pi^- = (418, 674, 521)$$

and has multiplicity one.

Proof. The assertion follows by exact enumeration of the five rows of the catalog that satisfy $\Psi(U) = 010211$. Four rows have terminal block $(169, 272, 272)$; the fifth, which carries the mutated return, has $(418, 674, 521)$. The maximum multiplicity is four and is attained only once. □

The oriented difference between the mutated return and the persistent is

$$\omega_\pi = b_\pi^- - b_\pi = (249, 402, 249), \qquad \langle (1, 1, 1), \omega_\pi \rangle = 900.$$

This identity separates two objects that must not be conflated. The integer 169 is a coordinate of a terminal datum, that is, a datum of degree zero. The vector ω_π can be read as a one-cochain after orienting the edge joining the two blocks. Calling it a cocycle would additionally require fixing a cochain complex and verifying that its coboundary vanishes. The analytic prolongation constructed below does not presuppose that identification.

The selection of 169 does not require privileging the first position of the block. The symmetric group $\mathfrak{S}_3$ acts by permuting its three coordinates. The stabilizer of b_π exchanges the two occurrences of 272 and fixes the unique coordinate of multiplicity one. Denote that coordinate by $\mathrm{sing}(b)$ when the block has type (r, s, s), with $r \neq s$. Equivalently,

$$\mathrm{sing}(b) = \sum_{j=1}^{3} b_j - 2\,\mathrm{moda}(b).$$

In particular,

$$\boxed{\mathrm{sing}(b_\pi) = 169.}$$

Definition 14.9 (Residual fiber selector)**.** The selector ρ_π successively applies two operations:

1. selects the terminal block of maximum multiplicity within $p_{\mathrm{ret}}(\mathcal{F}_\pi)$;

2. retains the coordinate fixed by the stabilizer of that block.

By proposition 14.8,

$$\boxed{\rho_\pi = 169.}$$

Theorem 14.10 (Equivariant uniqueness of return)**.** *Among the selectors that*

1. *are invariant under permutations of the five regions;*
2. *are equivariant under permutations of the three terminal coordinates;*
3. *select the block of maximal multiplicity;*
4. *retain a coordinate fixed by its stabilizer,*

the return value is uniquely determined and is 169.

Proof. Invariance under $\mathfrak{S}_5$ prevents using the order of the rows. The third condition selects b_π, which is the unique mode by proposition 14.8. Its stabilizer in $\mathfrak{S}_3$ has two orbits on the coordinates: the pair of values 272 and the singleton value 169. The fourth condition selects the second orbit. Equivariance preserves its value when its position is shifted. □

This argument improves the positional reading of the return. The 169 is not «the first coordinate» of a selected row: it is an invariant of the regional multiset and of the action of its reordering symmetries. Nor is it selected by global rarity. Among the 468 emissions in the catalogue, 169 appears 84 times, uniformly distributed with 14 occurrences in each of the six positions. The uniqueness of theorem 14.10 is therefore relational and regional: it resides in the combination of microfiber, modal persistence, and stabilizer, not in the isolated frequency of the integer.

14.5.2 Cylindrical length and analytic degree

Let $\mathcal{P}$ be the free module generated by the regional words and let $F^n\mathcal{P}$ be its length filtration. The associated grading records the first level at which each word appears. For a parameter z, the ordinary length character is

$$\chi_z([w]) = z^{|w|}.$$

Multiplicativity

$$\chi_z([uv]) = \chi_z([u])\chi_z([v])$$

expresses that concatenation adds lengths. When evaluated at $z = \alpha$, a word of length n receives degree α^n.

Definition 14.11 (Degree compatibility)**.** The regional analytic reader is compatible with the cylindrical filtration when it sends the homogeneous component of length n to analytic degree n:

$$\mathrm{gr}_n \mathcal{P} \longrightarrow \alpha^n \mathbb{R}.$$

The five regions belong to $\mathcal{U}_6$; therefore, the return datum appears in degree six:

$$\rho_\pi[U_6] \longmapsto 169\alpha^6.$$

This six is the length of the regional word. It is not the length of the closed walks of chapter 9. Those walks traverse the four anchors $U_\pi, U_e, U_\varphi, U_{\mathrm{APP}}$ simultaneously and have minimum length eight. A word, an edge of the block graph, and a closed walk are objects of distinct types; the grading used here belongs to the first.

14.5.3 The Determinantal Character of Common Transport

The dodecaphasic closure provides α. The completion of the three nonadic pairs provides the scale 10^3, so that

$$A = 10^3 \alpha.$$

Having chosen the Archimedean angular chart, the common coordinate in radians is

$$x = \frac{\pi A}{180} = \frac{50\pi}{9}\alpha.$$

Let R be a real involution of zero trace in dimension two,

$$R^2 = I_2, \qquad \operatorname{tr} R = 0,$$

and consider the formal two-layer transport

$$\mathcal{T}(x, y) = \exp(-xI_2 - yR).$$

The oriented variable y has not yet been evaluated. Nevertheless, the action of $\mathcal{T}$ on the determinant line is independent of it:

$$\begin{aligned} \det \mathcal{T}(x, y) &= \exp(\operatorname{tr}(-xI_2 - yR)) \\ &= e^{-2x}. \end{aligned}$$

We define

$$\boxed{D_A = e^{-2x} = e^{-100\pi\alpha/9}.}$$

The subscript recalls that this is the determinant of the common contraction associated with A, not a function of C_*^{HMT}.

Proposition 14.12 (Character of the determinant line)**.** *The map*

$$\delta_A : (\mathbb{N}_0, +) \longrightarrow (\mathbb{R}_{>0}, \cdot), \qquad \delta_A(k) = D_A^k,$$

is the unique multiplicative character whose value on an elementary prolongation is D_A.

Proof. The multiplicativity gives

$$\delta_A(k) = \delta_A(\underbrace{1 + \cdots + 1}_{k \text{ times}}) = \delta_A(1)^k = D_A^k.$$

The same identity proves existence and uniqueness. □

The determinant remains invariant under conjugation of $\mathcal{T}$ and under the exchange of sheets $R \mapsto -R$. Therefore, the tail constructed from D_A belongs to the common mode. The orientation will appear in the sign with which the regional character corrects the phase.

14.5.4 Rules of the Regional Analytic Reader

Analytic continuation is fixed by the following rules.

Definition 14.13 (Regional Determinantal Reader)**.** The determinantal regional reader satisfies:

1. *degree compatibility*: cylindrical length is the analytic degree;
2. *renewal decomposition*: the finite return impulse and strict continuation belong to distinct additive summands;

3. *continuation normalization*: after recording the residue ρ_π just once, the surviving branch begins at the unit of the determinant line;
4. *concatenation covariance*: each elementary prolongation acts through $\bigwedge^2 \mathcal{T} = D_A$;
5. *orientation*: in the cardinal convention fixed for APP, the regional return belongs to the compensating sector and is subtracted from the fifth-degree action phase.

The third rule has mathematical content. The value 169 is recorded once as the amplitude of the return; it is not multiplied again at each prolongation. The strict continuation counts the sole line that continues and normalizes it by its unit. This is the distinction between a renewal recurrence and homogeneous iteration of the initial impulse.

The need to declare that normalization can be proved. For every $\lambda \in \mathbb{R}$, the family

$$F_\lambda(z) = 169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

preserves the same finite return and the same determinantal ratio between consecutive coefficients. The catalogue and the ratio alone do not distinguish λ. The unit of the determinant line fixes $\lambda = 1$ before $z = \alpha$ is evaluated. Thus, the normalization is not obtained by fitting the final value; it forms part of the definition of the character.

14.5.5 Return recurrence and closed sum

We define the coefficients κ_n by

$$\kappa_6 = 169, \qquad \kappa_7 = D_A, \qquad \kappa_{n+1} = D_A \kappa_n \quad (n \geq 7).$$

The first equality is the regional impulse; the next two describe the normalized continuation.

Theorem 14.14 (Graduated analytic realization)**.** *The analytic germ compatible with the rules of definition 14.13 is unique and equals*

$$\boxed{\mathscr{C}_\pi(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z}.}$$

Their coefficients satisfy

$$\kappa_n = D_A^{n-6} \qquad (n \geq 7).$$

Proof. Write

$$F(z) = 169z^6 + \sum_{k \geq 1} c_k z^{6+k}.$$

The normalization of continuation and the determinantal covariance give

$$c_1 = D_A, \qquad c_{k+1} = D_A c_k.$$

By induction, $c_k = D_A^k$. The geometric sum yields

$$\sum_{k \geq 1} D_A^k z^{6+k} = \frac{D_A z^7}{1 - D_A z}.$$

No free coefficients remain. □

Since $0 < \alpha < 1$ and $0 < D_A < 1$,

$$0 < D_A \alpha < 1.$$

Consequently, $\mathscr{C}_\pi(\alpha)$ converges absolutely and its radius of convergence is $D_A^{-1} > 1$. Numerically,

$$D_A = 0.77512911924715643018888409648 02\ldots,$$

$$D_A\alpha = 0.00565639046986492673885079095469\ldots.$$

The speed of this contraction explains why the complete sum and its degree-seven truncation have the same first fifteen digits in the final chart.

14.5.6 The 5th-degree action character

The central block

$$\mathcal{B}_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$$

has eigenvalues 9 and 5. The dodecaphasic cycle contributes rank 12; its step-three orbits have length four; and the complete census contributes the exact quotient

$$\frac{104976}{1944} = 54.$$

From these invariants the fifth-degree action character is fixed.

$$\boxed{H_5(\alpha,\varphi) = \frac{1}{2}\sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5.}$$

The four rational coefficients can be written as

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \qquad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \qquad \frac{7}{48} = \frac{\frac{1}{2}\operatorname{tr}\mathcal{B}_c}{4\cdot 12}, \qquad \frac{1}{54} = \frac{1944}{104976},$$

with $(\lambda_+, \lambda_-) = (9, 5)$. These identities certify the provenance of the numbers used. The successive assignment of those invariants to degrees two, three, four, and five forms part of the analytic character; it does not follow from the mere existence of the finite catalog.

This precision removes a common ambiguity. A combination of invariants of the construction is an internal and reproducible definition; its naturality requires the reader rules specifying order, signs, and normalization. Here those rules are visible and are subject to the same negative controls as the rest of the construction.

14.5.7 Oriented phase and conjugate angular coordinate

The fifth-degree phase prior to return is

$$y_5 = \frac{\pi}{90}\big(H_5 + \alpha\big).$$

The complete regional character produces the correction $\mathscr{C}_\pi(\alpha)$. With the orientation fixed in definition 14.13, we define

$$\boxed{y_* = \frac{\pi}{90}\big(H_5 + \alpha\big) - 169\alpha^6 - \frac{D_A\alpha^7}{1 - D_A\alpha}.}$$

This is the primary formula. The sexagesimal unit does not come into play until applying the chart.

$$C_*^{\mathrm{HMT}} = \frac{180}{\pi}y_*.$$

Theorem 14.15 (Conjugate angular coordinate of the regional reader determinantal)**.** *With the dodecaphasic closure of α, $\varphi = (1+\sqrt{5})/2$ and the rules of the determinantal regional reader fixed, the phase y_* and its conjugate angular coordinate are determined without introducing the sought value. Their evaluation is*

$$\boxed{y_* = 0.037066226465838263984937794092306 38\ldots,}$$

$$\boxed{C_*^{\mathrm{HMT}} = 2.123738338968645724261951380393 36\ldots^\circ.}$$

Therefore, to fifteen decimal places,

$$\boxed{C_*^{\mathrm{HMT}} = 2.123738338968646^\circ.}$$

Proof. theorem 14.10 determines 169. The closure of α determines A, x, and $D_A = e^{-100\pi\alpha/9}$. theorem 14.14 determines the complete series. The formula for H_5 contains only α, φ, and declared structural invariants. Substituting these data into the expression for y_* gives the printed value. The decimal value of C_*^{HMT} occurs in none of the right-hand sides. □

The reduced action component after return is now as follows output:

$$\boxed{\bar{h}_{\mathrm{HMT}}^{\mathrm{ret}} = \frac{C_*^{\mathrm{HMT}}}{2} - \alpha = H_5 - \frac{90}{\pi}\mathscr{C}_\pi(\alpha).}$$

Numerically,

$$\bar{h}_{\mathrm{HMT}}^{\mathrm{ret}} = 1.05457181691503906113369058472430\ldots.$$

The inverse identity has not been used to introduce C_*^{HMT}; it is obtained after generating y_*.

14.5.8 Degree-7 truncation and exact remainder bound

The expression initially found,

$$y_7 = \frac{\pi}{90}(H_5 + \alpha) - 169\alpha^6 - D_A\alpha^7,$$

is the seventh-degree truncation of the full character. Produce

$$C_7 = 2.123738338968646002654038271039 19\ldots^\circ.$$

The omitted remainder is not an indeterminate term; it is the exact geometric tail.

$$\mathscr{C}_\pi(\alpha) - 169\alpha^6 - D_A\alpha^7 = \frac{D_A^2\alpha^8}{1 - D_A\alpha}.$$

Therefore,

$$\left|C_7 - C_*^{\mathrm{HMT}}\right| = \frac{180}{\pi}\frac{D_A^2\alpha^8}{1 - D_A\alpha} = 2.7839208689064583\ldots \times 10^{-16}\ ^\circ.$$

The recurrence thus transforms a seventh-degree approximation into a complete analytic realization and provides a closed-form bound for each subsequent truncation.

14.5.9 Uniqueness conditions for the regional selector and the analytic prolongation

The construction changes when any of its active components is removed. The following values are obtained without recalibrating the other terms:

variant	conjugate angular coordinate in degrees
$\rho_\pi = 168$	$2.12373833897729768639044 87\ldots$
$\rho_\pi = 169$	$2.12373833896864572426195 14\ldots$
$\rho_\pi = 170$	$2.12373833895999376213345 41\ldots$
without determinantal tail	$2.12373833896869494153016 75\ldots$
D_A replaced by 1	$2.12373833896863134099658 26\ldots$

These ablations do not in themselves constitute a probabilistic estimate. They demonstrate that the equivariant selector and the determinantal character are effective components of the output and cannot be removed while maintaining the same realization.

14.5.10 Spectral reconstruction and hexad-octad transition

The common phase and the oriented phase determine the two channels.

$$q_- = e^{-x+y_*}, \qquad q_+ = e^{-x-y_*}.$$

The invariants are recovered exactly.

$$q_- q_+ = D_A, \qquad \alpha = -\frac{9}{100\pi}\log D_A, \qquad C_*^{\mathrm{HMT}} = \frac{90}{\pi}\log\frac{q_-}{q_+}.$$

The first equality reads the common contraction; the third, the oriented separation of the sheets. The exchange $R \mapsto -R$ permutes q_+ and q_-, preserves D_A and reverses the sign of y_*.

For

$$S_{90}(q) = \frac{q^{90}}{1-q^{270}}, \qquad S_{120}(q) = \frac{q^{120}}{1-q^{360}},$$

the hexad–octad transition functional is written entirely in the phase chart:

$$\Gamma_{6\to 8} = \frac{180}{\pi}xy_* + 12\big(S_{90}(q_-) - S_{90}(q_+)\big) - \big(S_{120}(q_-) - S_{120}(q_+)\big).$$

The evaluation produced by the coordinate C_*^{HMT} of theorem 14.15 is

$$\boxed{\Gamma_{6\to 8} = 0.27400767269445927474725526077 4\ldots.}$$

Thus, the internal value recognized in Dimensional Mechanics as the Barbero–Immirzi functional ceases to receive C_*^{HMT} as an independent literal: both descend from the same pair of phases generated in this chapter. The geometric and physical interpretation of $\Gamma_{6\to 8}$ belongs to the work on Dimensional Mechanics; its mathematical evaluation already belongs to the previously constructed interface.

14.5.11 Action prior and posterior to return

The fifth-degree character and the post-action component after return are two distinct values:

$$H_5 = 1.05457181764615446962582182943306\ldots,$$

$$\bar{h}_{\mathrm{HMT}}^{\mathrm{ret}} = 1.05457181691503906113369058472430\ldots.$$

The first is the local action before incorporating the regional character; the second is the oriented action remaining after the return. The correction that generates the conjugate angular coordinate prevents identifying them.

The metrological comparison must respect this distinction. The first value gives the mantissa

$$2\pi H_5 = 6.626070149999987926001110349899 47\ldots,$$

while the subsequent action gives

$$2\pi \bar{h}^{\mathrm{ret}}_{\mathrm{HMT}} = 6.626070145406254333510749847592 88\ldots.$$

In the International System, the value

$$h = 6.62607015 \times 10^{-34}\ \mathrm{J\,s}$$

is exact by definition [20]. By therefore,

$$2\pi H_5 - 6.62607015 = -1.2073998889650101\ldots \times 10^{-14},$$

and it is not an exact equality. The proximity of the value before the return is a remarkable numerical check; it does not authorize replacing the defining constant of the SI with the subsequent component or confusing the two HMT actions.

This result precisely determines the mathematical situation. The regional reader generates C^{HMT}_* and, from it, the functional $\Gamma_{6\to 8}$. The same calculation separates the Planck approximation produced by H_5 from the reduced action intervening in the conjugate angular coordinate. The separation does not weaken the construction: it eliminates an identification incompatible with the exact figures and converts every comparison into a falsifiable assertion of a well-defined type.

14.5.12 Relative transporter and spectral disjunction of the blocks

In the ordered basis of channels, let

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_{\pm} = \frac{I \pm R}{2}, \qquad B_0 = 7I + 2R, \qquad B_1 = A_{\mathrm{deg}} I + C^*_{\mathrm{deg}} R.$$

The split decomposition

$$xI + yR = \rho(x,y)\, e^{\eta(x,y)R}, \quad \rho(x,y) = \sqrt{x^2 - y^2}, \quad \eta(x,y) = \operatorname{artanh}(y/x)$$

constructs the unique transporter

$$T_{\mathrm{rel}} = \frac{\sqrt{(A_{\mathrm{deg}})^2 - (C^*_{\mathrm{deg}})^2}}{\sqrt{45}} \exp\left[\left(\operatorname{artanh}\frac{C^*_{\mathrm{deg}}}{A_{\mathrm{deg}}} - \operatorname{artanh}\frac{2}{7}\right) R\right], \qquad T_{\mathrm{rel}} B_0 = B_1.$$

In the spectral basis it acts by $9 \mapsto A_{\mathrm{deg}} + C^*_{\mathrm{deg}}$ and $5 \mapsto A_{\mathrm{deg}} - C^*_{\mathrm{deg}}$.

Theorem 14.16 (Spectral Disjunction of the Local and Angular Realizations)**.** *The spectra*

$$\mathrm{Spec}(B_0) = \{9, 5\}, \qquad \mathrm{Spec}(B_1) = \{A_{\mathrm{deg}} + C^*_{\mathrm{deg}}, A_{\mathrm{deg}} - C^*_{\mathrm{deg}}\}$$

are disjoint. Consequently,

$$\mathrm{Hom}_{B_0,B_1} := \{\Phi : \Phi B_0 = B_1 \Phi\} = \{0\}.$$

Proof. The rational bounds constructed for the angular coordinates give $7.29 < A_{\mathrm{deg}} < 7.30$ and $2.12 < C^*_{\mathrm{deg}} < 2.13$; therefore, $9.41 < A_{\mathrm{deg}} + C^*_{\mathrm{deg}} < 9.43$ and $5.16 < A_{\mathrm{deg}} - C^*_{\mathrm{deg}} < 5.18$. In the basis that diagonalizes R, each entry of an intertwiner is multiplied by the difference between an eigenvalue of B_0 and one of B_1; disjointness forces all of them to vanish. □

The relative transporter compares two blocks within the split algebra; the dodecaphasic–Witt and incidence intertwiners act between equivalent representations in their own spaces of residence. This separation positively types both operations and preserves their domains.

14.5.13 Generation chain $\mathcal{F}_\pi \to b_\pi \to \rho_\pi \to D_A \to C_*^{\mathrm{HMT}} \to \Gamma_{6\to 8}$: regional selection, determinantal prolongation and oriented coordinate

The entire procedure can be summarized as a chain of applications:

$$\begin{aligned} \mathcal{F}_\pi &\longmapsto b_\pi \longmapsto \rho_\pi = 169, \\ \alpha &\longmapsto A \longmapsto x \longmapsto D_A, \\ (\rho_\pi, D_A) &\longmapsto \mathscr{C}_\pi(\alpha), \\ (H_5, \mathscr{C}_\pi(\alpha)) &\longmapsto y_* \longmapsto C_*^{\mathrm{HMT}}, \\ (x, y_*) &\longmapsto (q_+, q_-) \longmapsto \Gamma_{6\to 8}. \end{aligned}$$

The first line belongs to the finite catalog; the second, to the common closure of α; the third, to the determinantal prolongation; the fourth produces the oriented coordinate; the fifth transports it toward the dimensional interface.

In this organization, α measures the common contraction of the transport, and C_*^{HMT} measures the oriented separation of its two sheets. The regional return adds no external constant: it selects a finite amplitude and prolongs it through the determinant already fixed by α. The chart in degrees is subsequent. The primary object is the pair of phases (x, y_*).

An exhaustive verification reconstructs the five regions directly from the catalogue, traverses the actions of $\mathfrak{S}_5$ and $\mathfrak{S}_3$, checks the recurrence, and sums the series. The input of α is the exact output of its dodecaphasic closure, not a transcribed metrological digit. The verification keeps the catalogue theorems, reader rules, and external comparison separate; therefore, no experimental comparison intervenes in the generating function.

14.6 Determinantal structure and oriented action sections

The publications of π, φ, e, α arise from the APP–TRIT–TPK state of sections 1.23–11.6. The dodecaphasic register of 11.4 preserves its orientation, its residues and quotients and the memory of the return (4.34). Two successive operations are defined on that register: a local norm determines the valuation of the action section, and a character with regional continuation determines its coordinates before and after the return. Both operations transmit their rules and normalizations to the angular construction.

14.6.1 Local quotient and determinantal norm

The local step-three block of the dodecaphasic register, identified in 11.4.8, is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \tag{14.59}$$

The quotient corresponds to this local block. The three blocks of the complete register preserve their positions; the norm operation that follows acts on the sheets of G_H.

Lemma 14.17 (Structure of the local quotient)**.** *The Smith normal form of* H_4 *is* $\operatorname{diag}(1,2,2,4)$. *Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \qquad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even and one has value -2, so their greatest common divisor is two. The identities $H_4 H_4^{\mathsf{T}} = 4I$ and $\det H_4 = -16$ show that the cofactors equal ± 4; the third determinantal divisor is four. The fourth is 16. The successive quotients of $1, 2, 4, 16$ are $1, 2, 2, 4$; their product determines the order of the quotient. □

Definition 14.18 (Determinantal action reader)**.** The scalar section α, constant on the sheets of G_H, and one application of the autoscale φ determine

$$\Sigma_H = \varphi\, \mathrm{N}_{G_H}(\alpha), \qquad \mathrm{N}_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \tag{14.60}$$

Its decimal valuation is

$$\operatorname{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor.$$

The sixteenth power arises from the number of sheets once the multiplicative norm is fixed. The norm and the single application of φ are part of the declared reader: the order of the group, considered in isolation, admits other functionals.

Theorem 14.19 (Decimal valuation of the action section)**.** *The preceding reader, applied to the HMT coordinates already generated, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \qquad \operatorname{ord}_{10}(\Sigma_H) = -34. \tag{14.61}$$

Proof. The rational inclusions of the preceding publications are used,

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \qquad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}. \tag{14.62}$$

The function $(x, a) \mapsto x a^{16}$ is increasing in both positive variables. The integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \qquad 1619 \cdot 7298^{16} < 10^{66}$$

give

$$10^{-34} < \frac{1618}{1000}\left(\frac{7297}{10^6}\right)^{16} < \Sigma_H < \frac{1619}{1000}\left(\frac{7298}{10^6}\right)^{16} < 10^{-33}.$$

The definition of the valuation concludes the proof. The inclusions are subsequent checks on the coordinates already produced; the determinantal operation precedes any units chart. □

14.6.2 Action character and regional continuation

The central chart preserves the matrix

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \qquad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the step-three orbit has length four, and the census quotient used is $104976/1944 = 54$, with the censuses defined in Section 4.3. These invariants determine the coefficients assigned to degrees $2, 3, 4, 5$ of the character:

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \qquad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \qquad \frac{7}{48} = \frac{\operatorname{tr}(B_c)/2}{4 \cdot 12}, \qquad \frac{1}{54} = \frac{1944}{104976}. \tag{14.63}$$

The assignment to those degrees and the alternation of signs belong to the definition of the analytic character. The operational content combines the support, the orientation and that assignment rule.

Definition 14.20 (Local character and regional return reader)**.** On the publications α, φ, π, the fifth-order character, the regional transfer and its continuation are

$$H_5 = \frac{1}{2}\sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \tag{14.64}$$

$$D_A = \exp\!\left(-\frac{100\pi\alpha}{9}\right), \tag{14.65}$$

$$C_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \tag{14.66}$$

$$\eta_{\mathrm{ret}} = H_5 - \frac{90}{\pi}C_\pi(\alpha). \tag{14.67}$$

The integer 169 is the regional selector of the publications constructed in 10.5: the regional emission $169\,|\,272\,|\,272$ is preserved, distinguished from $418\,|\,674\,|\,521$ in the exceptional construction. The first strict prolongation of its continuation has weight D_A and each concatenation multiplies the weight by D_A. The initial normalization is unitary.

The continuation is formulated before evaluating at $z = \alpha$. With D_A already produced by its reader, we write $\mathscr{C}(z) = 169z^6 + \mathscr{R}(z)$, where $\mathscr{R}(z) \in z^7\mathbb{R}[[z]]$. Concatenation increases the degree by one; the initial weight and the transfer rule give

$$\mathscr{R}(z) = D_A z^7 + D_A z\mathscr{R}(z). \tag{14.68}$$

Proposition 14.21 (Uniqueness of the normalized continuation)**.** *Equation* (14.68) *determines a unique series:*

$$\mathscr{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n. \tag{14.69}$$

Its realization is analytic for $|D_A z| < 1$.

Proof. The constant term of $1 - D_A z$ is one, so that element is invertible in the ring of formal series. Solving (14.68) gives the rational expression. Equivalently, the coefficient of degree seven is D_A and each subsequent coefficient is D_A times the preceding one. The geometric series converges absolutely in the indicated disk. □

The normalization of the first weight plays a role in uniqueness. The degree-six impulse and the concatenation ratio, without that initial weight, are also compatible with

$$169z^6 + \lambda\frac{D_A z^7}{1 - D_A z}.$$

The unitary rule fixes $\lambda = 1$ before evaluation. For $0 < \alpha < 1$, we have $0 < D_A < 1$ and $D_A\alpha < 1$; therefore,

$$\frac{D_A\alpha^7}{1 - D_A\alpha} = \sum_{n=0}^{\infty} D_A^{n+1}\alpha^{n+7}. \tag{14.70}$$

After the term with index N, the exact remainder is $D_A^{N+2}\alpha^{N+8}/(1 - D_A\alpha)$. The rational expression thus preserves the complete continuation to any precision.

14.6.3 Positive sections and action per turn

Let $\mathcal{L}_S$ be the oriented action line and $\mathcal{U}_S$ a positive basis. Its sections before and after the return are

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S. \tag{14.71}$$

The return preserves the regional compensation $90C_\pi(\alpha)/\pi$ and the ratio

$$\delta_{\mathrm{ret}} = H_5 - \eta_{\mathrm{ret}}, \qquad R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}}. \tag{14.72}$$

Proposition 14.22 (Positivity and covariance of the action ratio)**.** *On the intervals* (14.62)*,* $0 < \eta_{\mathrm{ret}} < H_5$*, and* $R_{\mathrm{act}} > 1$*. The ratio remains fixed under a change of the common positive basis* $\mathcal{U}_S$*.*

Proof. The terms of C_π are positive. To bound H_5, $\alpha > 0.007$, $\alpha < 0.008$ and $\varphi < 1.619$ suffice. The root term of (14.64) is greater than one and the sum of its negative terms is less than 0.008001; hence $H_5 > 0.99$. With $D_A < 1$ and $\pi > 3$,

$$0 < \frac{90}{\pi}C_\pi < 30\left(169(0.008)^6 + \frac{(0.008)^7}{1-0.008}\right) < 10^{-6}.$$

Thus, $\eta_{\mathrm{ret}} > 0.989999 > 0$ and $\eta_{\mathrm{ret}} < H_5$. The change of basis divides both action coordinates by the same positive factor; it cancels in their quotient. □

The structural determination of angular action is the pair $(\hbar_{\mathrm{pre}}, \hbar_{\mathrm{ret}})$ with its identification by return. It preserves the valuation of the local norm, the action character and the regional continuation. In each section $s \in \{\mathrm{pre}, \mathrm{ret}\}$, the action of a complete turn is

$$h_s = 2\pi\hbar_s. \tag{14.73}$$

The subsequent chart $\mathcal{U}_S \mapsto \mathrm{J\,s}$ expresses those sections in units of action. The normalized radius to be constructed preserves its independence from this common choice of units.

14.7 Angular realization and orientation of the channels

The section $\hbar_{\mathrm{ret}}$ belongs to the action line $\mathcal{L}_S$; η_{ret} is its dimensionless coefficient in the basis declared in (14.71). The angular publication then acts on $(\alpha, \eta_{\mathrm{ret}})$. Writing $[t]_{360}$ for the class of t in $\mathbb{R}/(360\mathbb{Z})$, one defines

$$\mathcal{P}_{\mathrm{ang}}(\alpha, \eta_{\mathrm{ret}}) = \big([1000\alpha]_{360}, [2(\eta_{\mathrm{ret}} + \alpha)]_{360}\big). \tag{14.74}$$

The nonadic completion determines the factor 1000; the bidirectional closure determines the factor two. The circular chart of 360 units preserves the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120$. On the positive branch, the lifted representatives are

$$A_{\mathrm{deg}} = 1000\alpha, \qquad C^*_{\mathrm{deg}} = 2(\eta_{\mathrm{ret}} + \alpha). \tag{14.75}$$

The subscript distinguishes these angular coordinates from the blocks of the dodecaphasic register.

Proposition 14.23 (Shift of the base-thousand expression)**.** *Let* $\alpha = \sum_{n\geq 1} b_n 1000^{-n}$*, with* $b_n \in \{0, \dots, 999\}$*, be the compatible expansion already generated. Its lifted coordinate satisfies*

$$A_{\mathrm{deg}} = b_1 + \sum_{n\geq 1} b_{n+1} 1000^{-n}.$$

The publication preserves the blocks from the second onward and places the initial block in the integer part.

Proof. The series converges absolutely. Multiplying each term by 1000 and separating the initial term provides the identity. The operation acts on the constructed expansion, with its blocks and memory; the subsequent circular publication takes its class modulo 360. □

14.7.1 Circular charts and covariance of the action

Definition 14.24 (Charts of the angular coordinate)**.** The chart in degrees expresses a complete turn by 360 units; the chart in radians, by 2π. The change of coordinates and its inverse are

$$\theta_A = \frac{\pi}{180} A_{\deg}, \qquad \theta_C = \frac{\pi}{180} C^*_{\deg}, \qquad A_{\deg} = \frac{180}{\pi}\theta_A, \qquad C^*_{\deg} = \frac{180}{\pi}\theta_C. \tag{14.76}$$

A path with a preserved number of turns uses the lifted coordinates. The circular quotient publishes its class modulo the corresponding turn.

The register retains orientation, turns and memory of the transport. The final phase and the lift are different publications: the second distinguishes paths that share the first. Each action expression declares which of these coordinates it uses.

Proposition 14.25 (Covariance of the action reader)**.** *For a section $s \in \{\mathrm{pre}, \mathrm{ret}\}$ and a lifted angular coordinate, one has*

$$\mathcal{S}_s(\theta) = \hbar_s \theta_{\mathrm{rad}} = h_s \frac{\theta_{\deg}}{360}. \tag{14.77}$$

A positive turn expresses h_s.

Proof. Substituting $\theta_{\mathrm{rad}} = \pi\theta_{\deg}/180$ and $h_s = 2\pi\hbar_s$ gives

$$\hbar_s \frac{\pi\theta_{\deg}}{180} = \frac{h_s \theta_{\deg}}{360}.$$

In one turn $\theta_{\deg} = 360$, which gives h_s. □

The periodicity of a representation retains its own condition: if $U(2\pi) = -I$, operator restoration occurs at 4π. The normalization of the action per turn of (14.77) and the return of the represented state thus preserve their respective domains.

14.7.2 Positive channels and reversal of the odd coordinate

The channels are published, in either of the two charts, through

$$q_\pm = \exp[-(\theta_A \pm \theta_C)] = \exp\Big[-\frac{\pi}{180}(A_{\deg} \pm C^*_{\deg})\Big]. \tag{14.78}$$

Proposition 14.26 (Recovery of the angular coordinates)**.** *In the lifted real chart, the positive channels uniquely determine*

$$A_{\deg} = -\frac{90}{\pi}\log(q_+ q_-), \qquad C^*_{\deg} = \frac{90}{\pi}\log\frac{q_-}{q_+}. \tag{14.79}$$

The exchange $(q_+, q_-) \mapsto (q_-, q_+)$ preserves $A_{\deg}$ and reverses $C^*_{\deg}$.

Proof. The sum and difference of the logarithms in (14.78) are $-\pi A_{\deg}/90$ and $-\pi C^*_{\deg}/90$, respectively. The real logarithm is single-valued on the positive channels. Their product is symmetric, and their quotient is inverted upon interchange. □

The antipodal involution $r \mapsto r + 6 \pmod{12}$ of the register forms six pairs. The uniform distribution of the odd coordinate over these pairs assigns $C^*_{\deg}/6$ to each one; the sum recovers $C^*_{\deg}$. The conversion of units is applied only once to each argument. In particular, $C^*_{\deg}/A_{\deg} = \theta_C/\theta_A$ when $A_{\deg} \neq 0$, while the preceding state preserves the lifted angles and its memory.

14.8 Dodecaphase update and memory register

Let d_t be the integer direction code preserved by a TPK trace. This code and the cursor's cardinal direction are distinct coordinates. The twelve windows $E_m = \{9m+1, \dots, 9m+9\}$, $0 \le m < 12$ divide the register's 108 steps. The orientation is

$$\Sigma_{12} = (+,+,+,-,-,-,+,+,+,-,-,-).$$

The event of a window is the output

$$a_m = \Sigma_{12}(m) \sum_{t \in E_m} \mathbf{1}_{\{2,4,6,8\}}(d_t). \tag{14.80}$$

The notation a_m avoids confusing this event with Euler's number. The source register also preserves TRIT orientation, signature, carry and the corresponding history segment, in accordance with the architecture of Section 1.23.5.

Proposition 14.27 (Increment produced by transport)**.** *Let* $\mathbf{e}_m$ *be the basis of* $\mathbb{R}^{12}$ *and* $S\mathbf{e}_m = \mathbf{e}_{m+1 \pmod{12}}$. *If* $\mathbf{z}_{m+1} = \mathbf{z}_m + a_m \mathbf{e}_m$ *and* $y_m = S^{-m}\mathbf{z}_m$, *then*

$$y_{m+1} = S^{-1} y_m + \eta_m, \qquad \eta_m = a_m S^{-1}\mathbf{e}_0. \tag{14.81}$$

After a complete turn,

$$y_{12} = y_0 + \sum_{m=0}^{11} a_m \mathbf{e}_m. \tag{14.82}$$

Proof. Multiplying the update of $\mathbf{z}_m$ by $S^{-(m+1)}$ gives

$$y_{m+1} = S^{-1} y_m + a_m S^{-(m+1)}\mathbf{e}_m = S^{-1} y_m + a_m S^{-1}\mathbf{e}_0.$$

Upon iteration, the coefficient of event a_m is $S^{-(12-m)}\mathbf{e}_0 = \mathbf{e}_m$. Since $S^{12} = I$, we obtain (14.82). □

Boldface here identifies the accumulated vector $\mathbf{z}_m \in \mathbb{R}^{12}$, whereas the incidence block z_m of (11.88) is an integer between 0 and 999. The moving frame y_m transports the first object; the decimal coding of the second retains its own definition.

Thus, η_m is not a free force chosen to impose conservation: it is an expression of the TPK event. Equality of phase after one turn does not cancel accumulated events. This reader preserves counts, while the full register additionally preserves their order and provenance. We distinguish η_m, a vector increment, from η_{ret}, a scalar action mantissa, and from η_{el}, a shape parameter introduced in Section 15.6.

14.8.1 Quadratic balance

We define

$$\nabla_3 = I - S^3, \quad \nabla_4 = I - S^4, \quad B = \nabla_3^{\mathsf{T}}\nabla_3 + \nabla_4^{\mathsf{T}}\nabla_4, \qquad \mathcal{M}(y) = \tfrac{1}{2} y^{\mathsf{T}} B y.$$

The oriented readers of the signed register satisfy $(D_j z)_m = z_m - z_{m+j}$; with the convention $S\mathbf{e}_m = \mathbf{e}_{m+1}$, we have $D_j = I - S^{-j} = \nabla_j^{\mathsf{T}}$. Both directions produce the same quadratic form B, since powers of S commute. We have $S^{\mathsf{T}} B S = B$ and $B_{00} = 4$. Substituting (14.81) yields exactly

$$\mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) = a_m (B y_m)_0 + 2a_m^2. \tag{14.83}$$

Likewise, the charge $q(y) = \sum_j y_j$ satisfies $q(y_{m+1}) - q(y_m) = a_m$. The proof expands the quadratic polynomial after removing the permutation S^{-1}. No metrological value enters this calculation. The form $\mathcal{M}$ measures this register; interpreting it as a determined physical energy would require the corresponding dimensional reader.

14.8.2 Transport resolvents and preservation of memory

The extended history determines the windows $I_m = \{9m+1, \dots, 9m+9\}$, for $m \geq 0$. With this section's numbering, $I_m = E_m$ during the first cycle. Let $\sigma_m \in \{-1, 1\}$ be the orientation preserved by this history; for $0 \leq m < 12$, it coincides with $\Sigma_{12}(m)$. We define

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \qquad |a_m| \leq 9.$$

The register and the moving frame are extended by

$$\mathbf{z}_{m+1} = \mathbf{z}_m + a_m \mathbf{e}_{m \bmod 12}, \qquad y_m = S^{-m}\mathbf{z}_m.$$

The initial state $\mathbf{z}_0$ preserves the previous history; it is zero for an initially empty prefix. In this subsection we write $R = S^{-1}$ for frame transport. The update (14.81) is maintained in every window. The orientation and subsequent events are obtained from the extended history: the return of the frame does not presuppose periodicity of the event sequence.

Proposition 14.28 (Truncation identity with terminal state)**.** *For $N \geq 1$, let*

$$Y_N(u) = \sum_{m=0}^{N} y_m u^m, \qquad H_{\eta,N}(u) = \sum_{m=0}^{N-1} \eta_m u^m.$$

The exact polynomial identity holds

$$(I - uR)Y_N(u) = y_0 + uH_{\eta,N}(u) - u^{N+1}Ry_N. \tag{14.84}$$

Proof. In degrees $1 \leq m \leq N$, the coefficient of the left-hand side is $y_m - Ry_{m-1} = \eta_{m-1}$, by (14.81). Its coefficients in degrees zero and $N+1$ are y_0 and $-Ry_N$, respectively. Comparing coefficients proves the equality. □

The terminal term preserves the transported memory when a history is cut. The identity allows a finite evaluation to be compared with its extension while keeping explicit the state that remains at the boundary.

14.8.2.1 Correspondence with common-core transport

The window numbering of Section 1.23.4 and that used here describe the same partition of the history. The correspondence is

$$\underbrace{I_m = E_{m+1}}_{\text{index of the common kernel}} \qquad \longleftrightarrow \qquad \underbrace{I_m = E_m}_{\text{index of this section}}, \qquad 0 \leq m < 12.$$

The subscript changes the name of the window; its nine transitions, its orientation $\sigma_m = \Sigma_{12}(m)$, its event a_m, and its position in the record remain the same. For every prolongation, $R = S^{-1}$, $y_m = S^{-m}\mathbf{z}_m$, and update (14.81) coincide with (1.175). Orientation and the event sequence retain their dependence on the history, also after frame return.

This identification allows Proposition 1.197 to be applied directly: series (1.179), bound (1.180), and uniform convergence of its derivatives correspond to this same record and the same window clock. The finite identity (14.84) also determines the terminal term when that series is truncated. Its hypotheses retain the initial history and the bounded event $|a_m| \leq 9$.

When the record is reused, Proposition 1.198 jointly preserves the operator, source, and reader through (1.181) and (1.182). Example (1.183) specifies the effect of the excursions omitted by a compression. The affine composition (1.184), the intermediate blocks (1.185), and identity (1.186) ensure consistency when

segments are concatenated or several layers eliminated. Their complete proofs reside in Section 1.23.4, with the domains, sources, and reading maps specified there.

Balance (14.83) preserves the correspondence between the oriented readers of the signed record and the common quadratic form. The distinction between the accumulated vector $\mathbf{z}_m$ and the incidence decimal block z_m remains that established in this section. The transport identities apply to memory thus typed; they do not by themselves determine additional analytic coefficients.

14.9 Dodecaphase seal and reversible rational reading

The terminal register has a visible coordinate and a signed coordinate. The seal is obtained from the latter through the orbitwise Hadamard transformation; a subsequent scalar reading recovers the seal when its representation frame is preserved. This property specifies what a structural definition of a constant means here: production and recovery maps are made explicit alongside its value.

Let $U \in \mathbb{Z}^{12}$ be the signed coordinate produced by the register, H_4 the Hadamard matrix developed in Section 14.11 and P_{orb} the permutation grouping

$$(1, 4, 7, 10), \qquad (2, 5, 8, 11), \qquad (3, 6, 9, 12).$$

Define

$$\mathcal{H}_{12} = P_{\mathrm{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\mathrm{orb}}.$$

We have $\mathcal{H}_{12}^2 = 4I$. When $\mathcal{H}_{12}U \equiv 0 \pmod 4$, the integral seal is

$$K = \tfrac{1}{4}\mathcal{H}_{12}U, \qquad U = \mathcal{H}_{12}K.$$

The transverse differences D_3K, D_4K and charge $Q(K)$ recover the channels specified by the register in Section 11.4. The seal is thus a reconstructible coordinate of that oriented register; depth and complete history remain distinct fields.

Proposition 14.29 (Recovery of blocks and transport of the origin)**.** *Fix an integer base $b \geq 2$ and twelve ordered positions, and let $K_j \in \{0, \dots, b-1\}$. Define*

$$I(K) = \sum_{j=1}^{12} K_j b^{12-j}, \qquad \kappa(K) = \frac{I(K)}{b^{12} - 1}.$$

The exact reading κ uniquely determines the twelve blocks. For the rotation $\rho K = (K_2, \dots, K_{12}, K_1)$,

$$\boxed{\kappa(\rho K) = b\kappa(K) - K_1.}$$

Proof. The integer $I = (b^{12} - 1)\kappa$ decomposes uniquely into twelve base-b positions, padding the necessary leading positions with zeros. On the other hand,

$$I(\rho K) = bI(K) - K_1(b^{12} - 1).$$

Dividing by $b^{12} - 1$ gives the transformation law. For the constant block $K_j = b - 1$, the reading is $\kappa = 1$; the fixed periodic representation type likewise preserves its recovery. □

In the register's base 1000, three-digit blocks remain recoverable, including leading zeros. The periodic rational reading preserves more than a truncated decimal approximation. Changing the phase origin generally changes its value according to the preceding identity; after twelve rotations, it returns. The memory advance of the TPK state is an additional transformation: periodicity of the reader does not identify accumulated histories.

Remark: reversible reading and information. Exact block recovery is a property of the rational reader in its ordered frame. Word complexity, the entropy of a symbolic dynamics, reduced-state entropy, and thermodynamic production are distinct objects. A relation between them requires the distribution or state and the corresponding transport map to be specified; reversibility of the seal alone does not identify these functionals.

14.10 Cyclic decomposition and determinantal representations

The cyclic projectors

$$P_3 = \tfrac{1}{4}(I + S^3 + S^6 + S^9), \qquad P_4 = \tfrac{1}{3}(I + S^4 + S^8)$$

have images of dimensions three and four. If $E = P_3 - P_4$, then

$$\frac{\operatorname{Tr} E}{12} = -\frac{1}{12}. \tag{14.85}$$

The scalar arises from a normalized rank difference, not an ordinary sum of positive integers.

Theorem 14.30 (Determinantal representation of the channel readout)**.** *Let* $\mathcal{H}_3 = \operatorname{Fix}(S^3)$ *and* $\mathcal{H}_4 = \operatorname{Fix}(S^4)$. *For* $0 < s < 1$,

$$F(s) = \frac{\det_{\mathcal{H}_3}(I - sS)}{\det_{\mathcal{H}_4}(I - sS)} = \frac{1 - s^3}{1 - s^4} = \frac{1 + s + s^2}{1 + s + s^2 + s^3}. \tag{14.86}$$

Its continuous extension at $s = 1$ *equals* $3/4$.

Proof. The restriction of S to each image is a cycle of three or four positions, respectively. Their characteristic polynomials give $\det(I - sS) = 1 - s^r$ for $r = 3, 4$. Cancelling the common factor $1 - s$ leaves the final expression, regular at $s = 1$. □

Substitution $s = q^{30}$ expresses $F(q^{30}) = (1 - q^{90})/(1 - q^{120})$. The same pair of spaces producing (14.85) produces the channel quotient. This equality does not identify a twelve-coordinate matrix with the entire TPK state.

14.10.1 Oriented information and the complex sector

In the antipodal sector $V_- = \operatorname{Ran}((I - S^6)/2)$, the operator $J = S^3$ satisfies $J^2 = -I$ and $J^{\mathsf{T}} = -J$. With $Q = P_4(I - S^6)/2$, we have

$$\operatorname{rank} Q = 2, \qquad E|_{V_-} = -Q, \qquad B|_{V_-} = 5I - 3Q.$$

On $\operatorname{Ran} Q$, $S^2 = -I$ and the determinant is $1 + s^2$. Thus,

$$F(s) = \frac{1 + s + s^2}{1 + s}\,\frac{1}{1 + s^2}.$$

Orientation reversal changes the sign of the form $\omega(u, v) = \langle Ju, v\rangle$ but preserves B and F. Scalar evaluation does not distinguish all oriented transports.

The transported trace also produces

$$a_n^{\mathrm{tr}} = \operatorname{Tr}(S^n E) = 3\mathbf{1}_{3|n} - 4\mathbf{1}_{4|n},$$

and, for $|s| < 1$,

$$-\log F(s) = \sum_{n \geq 1} \frac{a_n^{\mathrm{tr}}}{n}\, s^n.$$

This system of moments connects the present block with the collected spectral readers. Its complete arithmetic development, including Catalan, will have its own location; it is not presupposed here that a functional identity alone determines its arithmetic nature.

14.11 Planck constants and the determinantal structure of action

14.11.1 Determinantal action scale and decimal valuation

The absolute scale and relative correction arise from distinct objects. The former uses the local block of the step-three dodecaphase register,

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \tag{14.87}$$

The operation uses a local quotient. The three blocks of the complete register are not replaced by tensorization that would multiply this reader's number of sheets again.

Lemma 14.31 (Number of sheets of the local quotient)**.** *The Smith normal form of H_4 is* $\operatorname{diag}(1, 2, 2, 4)$. *Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \qquad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. Second-order minors are even and one has value -2, so their greatest common divisor is two. Since $H_4 H_4^{\mathsf{T}} = 4I$ and $\det H_4 = -16$, the cofactors are ± 4; the third determinantal divisor is four. The fourth is 16. Successive quotients of the divisors $1, 2, 4, 16$ are $1, 2, 2, 4$. Their product gives the quotient's order. □

Definition 14.32 (Determinantal action reader)**.** The multiplicative norm of the scalar section α, constant on the sheets of G_H, and one application of self-scaling φ produce

$$\Sigma_H = \varphi \, \mathrm{N}_{G_H}(\alpha), \qquad \mathrm{N}_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \tag{14.88}$$

The associated decimal chart determines $\operatorname{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor$.

The sixteenth power thus comes from the quotient's number of sheets, once this local reader is fixed. The factor φ acts once on the basis of the action line. Smith calculation alone would not select any possible functional on the quotient: the multiplicative norm and self-scaling are part of the declared operation.

Theorem 14.33 (Decade of the action section)**.** *The reader* (14.88)*, applied to HMT coordinates already generated, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \qquad \operatorname{ord}_{10}(\Sigma_H) = -34. \tag{14.89}$$

Proof. The rational intervals, much coarser than the available cylinders, suffice:

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \qquad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}.$$

The function $(x, a) \mapsto x a^{16}$ is increasing in both positive arguments. The two integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \qquad 1619 \cdot 7298^{16} < 10^{66}$$

imply

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6} \right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6} \right)^{16} < 10^{-33}.$$

The definition of decimal order completes the proof. Neither an SI unit nor a reference mass enters these inequalities. □

The normalized section begins with $\Sigma_H = 1.0462438381\ldots \times 10^{-34}$. Its mantissa is not identified with the action coordinate before or after return. The determinantal reader fixes decimal order; the two return coordinates are obtained through the following readers.

14.11.2 Action sections and dimensionless return quotient

Let $\mathcal{L}_S$ be an oriented action line and $\mathcal{U}_S$ its positive basis. The fifth-order character and regional return term have the expressions

$$H_5 = \frac{1}{2}\sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \tag{14.90}$$

$$D_A = \exp\left(-\frac{100\pi\alpha}{9}\right), \tag{14.91}$$

$$C_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \tag{14.92}$$

$$\eta_{\mathrm{ret}} = H_5 - \frac{90}{\pi}C_\pi(\alpha). \tag{14.93}$$

These are subsequent readings of dodecaphase closure and its regional incidence. The integer 169 is retained as the regional selector already constructed, and the coefficients of H_5 as those of its local character; they are not replaced by an interpolation of the Planck constant. The formulas specify exactly which readings are needed for the electronic stage. Other mass-theory functionals need not be introduced here.

For $0 < \alpha < 1$, we have $0 < D_A < 1$, so $D_A\alpha < 1$. The rational term of (14.92) preserves the complete sum of the geometric tail

$$\frac{D_A\alpha^7}{1 - D_A\alpha} = \sum_{n=0}^{\infty} D_A^{n+1}\alpha^{n+7}. \tag{14.94}$$

It is not a truncation that can be silently modified when precision changes. After the term of index N, the remaining tail is $D_A^{N+2}\alpha^{N+8}/(1 - D_A\alpha)$, allowing evaluation to be controlled without recourse to a target quantity.

Definition 14.34 (Action sections and return reader)**.** The sections before and after return are

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S. \tag{14.95}$$

Whenever $\eta_{\mathrm{ret}} > 0$, define

$$\delta_{\mathrm{ret}} = H_5 - \eta_{\mathrm{ret}}, \qquad R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}}. \tag{14.96}$$

Proposition 14.35 (Positivity and invariance of the ratio)**.** *In the intervals used in the proof of* (14.89)*,* $0 < \eta_{\mathrm{ret}} < H_5$*, and therefore* $R_{\mathrm{act}} > 1$*. The ratio does not change when the basis* $\mathcal{U}_S$ *is replaced by any other common positive basis.*

Proof. The terms of C_π are positive. For an elementary bound, it suffices to use $\alpha < 0.008$, $\varphi < 1.619$, $\alpha > 0.007$, $D_A < 1$ and $\pi > 3$. The root in (14.90) exceeds one, whereas the sum of negative terms is less than 0.008001; in particular, $H_5 > 0.99$. Likewise,

$$0 < \frac{90}{\pi}C_\pi < 30\left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008}\right) < 10^{-6}.$$

Thus $\eta_{\mathrm{ret}} > 0.989999 > 0$ and $\eta_{\mathrm{ret}} < H_5$. Finally, the change of basis divides both action coordinates by the same positive factor, which cancels in their ratio. □

Subsequent evaluation gives

$$H_5 = 1.0545718176461544696\ldots,$$
$$\eta_{\mathrm{ret}} = 1.0545718169150390611\ldots,$$
$$R_{\mathrm{act}} = 1.0000000006932817630\ldots.$$

The difference and quotient are two readings of the same sections, one additive and the other multiplicative. They do not represent two independent Planck constants. The passage from angular action to action per complete turn is expressed in each section by $h = 2\pi\hbar$. The subsequent chart $\mathcal{U}_S \mapsto \mathrm{J\,s}$ assigns a unit but does not act retroactively on the decimal order already proved.

Regional return can also be written in an angular chart. This reparametrization is not an indispensable additional dependency: (14.90)–(14.93) supply H_5, η_{ret} and R_{act} directly. If angle and counter-angle are introduced, both must be transported in the same chart; degrees and radians are not mixed within a difference.

14.11.2.1 Action character and provenance of its coefficients

The reduced Planck constant, $\hbar$, is the physical object motivating realization of the angular-action section. Its position in the electronic dependency must be specified at two levels. The determinantal norm Σ_H determines the scale's decade. The character H_5 and regional return then determine the action-section coordinates. This separation allows the result needed for the electron to be presented without anticipating complete developments of other Dimensional Mechanics sectors.

The provenance of the fifth-order character can be made explicit through invariants already fixed on the support. In the central chart, consider

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \qquad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the orbit of step 3 has length 4, and the relevant census quotient is $104976/1944 = 54$. The coefficients of the character satisfy

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\mathrm{tr}(B_c)/2}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976}. \tag{14.97}$$

Assignment of these invariants to degrees $2, 3, 4, 5$, with the alternating signs of (14.90), belongs to the definition of the analytic character used. Making this explicit is essential: the set of cardinalities, considered without the character and the ordering of its degrees, would admit other expressions. The constructive content lies in composition of the support, orientation and declared analytic rule.

14.11.2.2 Regional continuation and renewal equation

The selector 169 determines the degree-six impulse. Subsequent extension is organized by the transfer D_A and unit normalization of the determinant line. These rules admit a formal-series formulation before substituting $z = \alpha$, with D_A already constructed. Write

$$\mathscr{C}(z) = 169z^6 + R(z), \qquad R(z) \in z^7\mathbb{R}[[z]].$$

Concatenating an extension increases degree by one and multiplies its weight by D_A. The first strict extension has weight D_A. Its renewal equation is therefore

$$R(z) = D_A z^7 + D_A z R(z). \tag{14.98}$$

Proposition 14.36 (Formal uniqueness of the normalized continuation)**.** *Equation* (14.98) *determines a unique series, and*

$$\mathscr{C}(z) = 169z^{6} + \frac{D_A z^{7}}{1 - D_A z} = 169z^{6} + \sum_{n=7}^{\infty} D_A^{n-6} z^{n}.$$

The realization is analytic in $|D_A z| < 1$.

Proof. The element $1 - D_A z$ has constant term one and is invertible in the formal power-series ring. Solving the equation produces the displayed series. Equivalently, the degree-seven coefficient is D_A and each subsequent coefficient is D_A times the preceding one. The geometric series converges absolutely in the stated disk. □

Normalization of the first extension is an indispensable operational datum. If only the degree-six impulse and concatenation ratio were retained, the series

$$169z^{6} + \lambda \frac{D_A z^{7}}{1 - D_A z}$$

would still share both data for any λ. The unit rule fixes $\lambda = 1$ before evaluation. Each rule's contribution to reader uniqueness is thus located, and the rational expression preserves the complete continuation.

14.11.2.3 Reduced Planck constant and return of the action section

In the dimensional chart of (14.95), the section before return is the model's construction intended for comparison with the reduced Planck constant:

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34} \mathcal{U}_S.$$

The subsequent section also preserves the regional compensation $90\mathscr{C}(\alpha)/\pi$. Both belong to the same action line, and their quotient R_{act} measures the multiplicative effect of return. The passage to h through 2π uses the relationship between angular parametrization and a complete turn.

This construction effectively enters $R_{\mathrm{act}}^{80-54\alpha+6\Delta_4}$. The electron preserves its central coordinate and trajectory; the chronological scale does not replace this structure with an undifferentiated quantum. Nor is a mass already expressed in MeV multiplied again by 10^{-34}: the decade belongs to the action section, cancels in its dimensionless quotient and reappears only where the corresponding dimensional realization acts.

The numerical comparison of H_5, the returned section and $h/(2\pi)$ is presented in Section 140.1. This distinguishes the name of the physical object, the equation assigned to it by the model and the precision with which that equation reproduces it.

14.11.3 Structural definition and role of the Planck constants

Definition 14.37 (Structural determination of angular action)**.** On the oriented line $\mathcal{L}_S$, the HMT determination of angular action is the pair of sections $\hbar_{\mathrm{pre}}, \hbar_{\mathrm{ret}}$ in (14.95), together with their identification by return. Its structure combines the valuation of the local norm $\varphi \mathrm{N}_{G_H}(\alpha)$, the character H_5, and the regional continuation C_π. The action per complete turn is $h_a = 2\pi\hbar_a$ in each orientation a.

This definition preserves three successive determinations. The local quotient in (14.87) fixes the sixteen sheets of the norm; the inequality (14.89) fixes its decimal order. The character and the normalized renewal in (14.98) then determine the pre-return and post-return coordinates. Uniqueness of the tail uses its first weight and its concatenation rule; the distribution of the coefficients of H_5 preserves the character declared in Section 14.11.2.1. On the positive branch, the multiplicative information of the return is the invariant

$$\frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = R_{\mathrm{act}} = \frac{H_5}{\eta_{\mathrm{ret}}} > 1. \tag{14.99}$$

Its physical function is expressed through the contraction between action and frequency: a section $\hbar_a$ assigns energy to an angular frequency, whereas h_a expresses the action of one turn. The quotient (14.99) preserves the structural modification between both readings even when the common basis of units is changed. The identification with the physical constant and its precision are examined in the chart of section 140.1, with the two return coordinates distinguished.

14.11.4 Angular coordinates, turn normalization and action units

The action quantity and angular coordinate belong to different spaces. The section $\hbar_{\mathrm{ret}}$ occupies the action line $\mathcal{L}_S$; η_{ret} is its coefficient in the declared basis. Angular expression acts on the dimensionless pair $(\alpha, \eta_{\mathrm{ret}})$ after its construction:

$$\mathcal{P}(\alpha, \eta_{\mathrm{ret}}) = \big([1000\alpha]_{360}, [2(\eta_{\mathrm{ret}} + \alpha)]_{360}\big). \tag{14.100}$$

Nonadic completion determines the factor 1000; bidirectional closure determines the factor two. The circular chart of 360 units makes visible the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120$. On the article's positive branch, the representatives are

$$A_{\mathrm{deg}} = 1000\alpha, \qquad C^*_{\mathrm{deg}} = 2(\eta_{\mathrm{ret}} + \alpha).$$

Proposition 14.38 (Shift of the base-thousand expression)**.** *Let $\alpha = \sum_{n\geq 1} b_n 1000^{-n}$, with $b_n \in \{0, \dots, 999\}$, be the compatible expansion already generated. The lifted coordinate $A_{\mathrm{deg}} = 1000\alpha$ satisfies*

$$A_{\mathrm{deg}} = b_1 + \sum_{n\geq 1} b_{n+1} 1000^{-n}.$$

The expression preserves the sequence of blocks from the second onward; the initial block passes to the integer part.

Proof. The series converges absolutely. Multiplying every term by 1000 and separating the initial term gives the identity. The operation acts on the constructed expansion, with its blocks and memory; subsequent circular expression takes its class modulo 360. □

Definition 14.39 (Charts of the angular coordinate)**.** The degree chart expresses a complete turn by 360 units. The radian chart expresses it by 2π. The coordinate changes and their inverses are

$$x = \frac{\pi}{180} A_{\mathrm{deg}}, \qquad y = \frac{\pi}{180} C^*_{\mathrm{deg}}, \qquad A_{\mathrm{deg}} = \frac{180}{\pi} x, \qquad C^*_{\mathrm{deg}} = \frac{180}{\pi} y.$$

Along a path with preserved turn count, lifted coordinates are used; the circular quotient expresses its class modulo the corresponding turn.

The definition characterizes degrees and radians by their exact relationship to the turn. The register also retains orientation, turns and transport memory. Expression of its angular class preserves phase, whereas the lift distinguishes paths having the same final phase. The physical reading of action must specify which of these coordinates enters the integral.

Proposition 14.40 (Covariance of the action reader)**.** *For $h_a = 2\pi\hbar_a$, the action of a lifted coordinate satisfies*

$$\mathcal{S}_a(\theta) = \hbar_a \theta_{\mathrm{rad}} = h_a \frac{\theta_{\mathrm{deg}}}{360}. \tag{14.101}$$

A positive turn expresses h_a.

Proof. Substituting $\theta_{\mathrm{rad}} = \pi\theta_{\mathrm{deg}}/180$ gives $\hbar_a \pi\theta_{\mathrm{deg}}/180 = (2\pi\hbar_a)\theta_{\mathrm{deg}}/360$. Over one turn $\theta_{\mathrm{deg}} = 360$, whence h_a. □

The periodicity of a representation is declared separately. If the transported representation satisfies $U(2\pi) = -I$, its operatorial restoration occurs at 4π. This property of the lift coexists with the per-turn normalization of (14.101). Each preserves its object: integrated quantity, circular phase and represented state.

14.11.5 Covariance of the channels and the incidence functional

The channels are equivalently written as

$$q_{\pm} = \exp[-(x \pm y)] = \exp\Big[-\frac{\pi}{180}(A_{\mathrm{deg}} \pm C^*_{\mathrm{deg}})\Big]. \tag{14.102}$$

Proposition 14.41 (Recovery of the even and odd coordinates)**.** *In the lifted real chart, the positive channels uniquely determine the two coordinates:*

$$A_{\mathrm{deg}} = -\frac{90}{\pi}\log(q_+q_-), \qquad C^*_{\mathrm{deg}} = \frac{90}{\pi}\log\frac{q_-}{q_+}.$$

The exchange $(q_+, q_-) \mapsto (q_-, q_+)$ preserves A_{deg} and reverses C^*_{deg}.

Proof. The sum and difference of the logarithms of (14.102) are, respectively, $-\pi A_{\mathrm{deg}}/90$ and $-\pi C^*_{\mathrm{deg}}/90$. The real logarithm is single-valued on the positive channels. The product is symmetric and the quotient is inverted when they are exchanged. □

In the twelve-position register, the antipodal involution $r \mapsto r + 6 \pmod{12}$ forms six pairs. The uniform distribution of the odd coordinate over these pairs is $C^*_{\mathrm{deg}}/6$: summing its six contributions recovers C^*_{deg}. This sectorial coordinate is distinct from the Planck constant h, which belongs to the action line.

The conversion is applied only once to each argument. The quantity α remains dimensionless and the quantity $\hbar$ still belongs to the action line; their prior coordinates enter the map $\mathcal{P}$.

The Barbero–Immirzi functional contains, in addition to the channels, a component bilinear in the angular representatives. Its transport is

$$\frac{\pi}{180}A_{\mathrm{deg}}C^*_{\mathrm{deg}} = \frac{180}{\pi}xy. \tag{14.103}$$

The identity is obtained by substituting both representatives in degrees with their expressions in radians. Thus, the complete formula in analytic coordinates is

$$\gamma_{\mathrm{HMT}} = \frac{180}{\pi}xy + 12\big[S_{90}(e^{-x+y}) - S_{90}(e^{-x-y})\big] - \big[S_{120}(e^{-x+y}) - S_{120}(e^{-x-y})\big].$$

Equivalence of the two charts transports the bilinear component and exponential functions simultaneously. This transformation specifies the precise role of the degree in the construction and makes evaluation reproducible without ambiguity of angular units.

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$$A_{\mathrm{deg}} = b_1 + \sum_{n\geq 1} b_{n+1} 1000^{-n}.$$

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The periodicity of a representation is declared separately. If the transported representation satisfies $U(2\pi) = -I$, its operatorial restoration occurs at 4π. This property of the lift coexists with the per-turn normalization of (14.105). Each preserves its object: integrated quantity, circular phase and represented state.

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The exchange $(q_+, q_-) \mapsto (q_-, q_+)$ *preserves* A_{deg} *and reverses* C^*_{deg}.

Proof. The sum and difference of the logarithms of (14.106) are, respectively, $-\pi A_{\mathrm{deg}}/90$ and $-\pi C^*_{\mathrm{deg}}/90$. The real logarithm is single-valued on the positive channels. The product is symmetric and the quotient is inverted when they are exchanged. □

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The conversion is applied only once to each argument. The quantity α remains dimensionless and the quantity $\hbar$ still belongs to the action line; their prior coordinates enter the map $\mathcal{P}$.

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The identity is obtained by substituting both representatives in degrees with their expressions in radians. Thus, the complete formula in analytic coordinates is

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Equivalence of the two charts transports the bilinear component and exponential functions simultaneously. This transformation specifies the precise role of the degree in the construction and makes evaluation reproducible without ambiguity of angular units.

Chapter 15

Incidence channels 90/120 of the Topological Phase Kernel: tritic semicrowns, sheet factorization, and hexad–octad realization of the morphism 6→8

15.1 Arithmetic classification of integer channels and compatibility with the triticale semicrown

The preceding generations distinguish the function of each constant. This block proves the integral structure of their readers. The quantities $729, 271, 315$ and 161 are constructed as *integral channels relative to declared typed rules*; they do not enter as decimal approximations. Their statuses are not identical: 729 is the quadratic power of channel 27; 271 and 315 receive a nonadic realization constructed from the published TPK phase partition and a second realization relative to the Steiner lift; and 161 remains, within the present scope, an integral identity without a set-theoretic realization of its own. The catalogue is then classified through published predicates and gauges. Only at the end may an Archimedean chart act to produce decimal expansions.

This order separates two distinct operations. Seeking the integer 271 because it recalls the first digits of e would be a retrospective reading. Finding a TPK region and calling it e without constructing the type of return it represents would be a designation without a map. The integer channels, the conditioned orbital classification, and the dictionary relating them are therefore kept separate. The isolated integer channel selects no constant; selection belongs to the enriched TPK reader, which incorporates the orbit, orientation, survival, and boundary memory without receiving as input the decimal value that it ultimately evaluates.

15.1.1 Visible residue, digital root and crown memory

For $n > 0$, the nonadic digital root is the section

$$\mathrm{dr}_9(n) = \begin{cases} 9, & n \equiv 0 \pmod 9, \\ n \bmod 9, & n \not\equiv 0 \pmod 9. \end{cases}$$

Reduction modulo nine conserves the visible class; the section by $\{1, \dots, 9\}$ also conserves the choice of the null representative. We define the lifted quotient

$$q_9^\sharp(n) := \frac{n - \mathrm{dr}_9(n)}{9} \in \mathbb{Z}_{\geq 0}.$$

One obtains the exact decomposition, which is not the usual remainder convention in $\{0, \dots, 8\}$,

$$n = 9q_9^\sharp(n) + \mathrm{dr}_9(n).$$

This separates two readings: $\mathrm{dr}_9(n)$ records the visible value, and $q_9^\sharp(n)$ records how many complete crowns precede the chosen visible representative. This is the exact form of the colloquial image according to which the modulus compacts and the carry preserves what compactification would lose.

In APP the additive and multiplicative leaves produce

$$\sum \Sigma = 405, \qquad \sum \Pi = 459, \qquad \Delta := \sum \Pi - \sum \Sigma = 54.$$

The sum of the additive quotients is

$$S := \sum Q^+ = 45.$$

We define

$$\kappa = \frac{\Delta}{2} = 27, \qquad \Theta = 5\kappa = 135.$$

All these numbers come from the orthogram and its two evaluations. No constant has been used yet.

The central block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$$

produce four ordered words:

$$72, \qquad 27, \qquad 77, \qquad 22.$$

Satisfy

$$72 + 27 = 77 + 22 = 99, \qquad 72 - 27 = 45, \qquad 77 - 22 = 55 = 54 + 1.$$

The center does not remain isolated: its two complementary readings project onto the boundary 99, while their differences recover the additive crown 45 and the pointed defect $54 + 1$.

15.1.2 Classification Theorem of the 4 Integer Channels

Theorem 15.1 (Typed identities of the four channels)**.** *Relative to the composition rules* $\kappa \mapsto \kappa^2$, $\Theta \mapsto 2\Theta + 1$, $(\Theta, S) \mapsto 2\Theta + S$, *and* $(\Theta, \kappa) \mapsto \Theta + \kappa - 1$, *the four integer channels are defined*

$$\begin{aligned} C_\alpha &= \kappa^2 = 729, \\ C_e &= 2\Theta + 1 = 271, \\ C_\pi &= 2\Theta + S = 315, \\ C_\varphi &= \Theta + \kappa - 1 = 161. \end{aligned}$$

Moreover,

$$\begin{aligned} C_e &= 5\Delta + 1 = 10 \cdot 27 + 1 = 4\Delta + (\Delta + 1), \\ C_\pi &= 5\Delta + S = 7S, \\ C_\varphi &= 3\Delta - 1 = 4\Delta - (\Delta + 1), \end{aligned}$$

and the closures are satisfied

$$\boxed{1000 = C_\alpha + C_e = 729 + 271}$$

and

$$\boxed{C_\pi + C_e + C_\varphi - 3 = 744 = 729 + 15.}$$

The α *channel also has the central reading* $729 = 10 \cdot 72 + 9$, *and the* e *channel has the reading* $271 = 10 \cdot 27 + 1$.

Proof. Substituting $\Delta = 54$, $S = 45$, $\kappa = 27$, and $\Theta = 135$ yields the four formulas. The alternative expressions are integer identities. The first completion is obtained from $27^2 + 2(5 \cdot 27) + 1 = 729 + 270 + 1 = 1000$; the second, from $315 + 271 + 161 - 3 = 744 = 729 + 15$. The two decimal divisions follow from the central words $(72, 27)$ and their remainders already fixed by the cubic completion. □

The equality $1000 - 3^6 = 271$ is universal for that cubic ambient space. By itself it does not distinguish e: in an ambient space of cardinality B, constructed with an alphabet of cardinality q in dimension d, the formal complement is $B - q^d$. The additional information in this section is that 271 also admits a realization as a pointed return and the central reading $27|1$. Its association with the orbit named e belongs to the subsequent typed dictionary, not to the isolated subtraction.

15.1.3 Semicorona nonadic relative to the TPK phase rule

The minimal motion rule of the TPK emitter does not treat the nine instants of a window as an interchangeable list. Relative to the published phase origin, it defines

$$D_9 = \{1, \dots, 9\}, \qquad H_+ = \{1, 4, 7\}, \qquad H_- = \{2, 5, 8\}, \qquad H_0 = \{3, 6, 9\}.$$

The instants of H_+ activate the additive reading, those of H_- the multiplicative reading, and those of H_0 preserve the neutral channel. Set

$$H_{\mathrm{act}} = H_+ \sqcup H_-, \qquad O_{\mathrm{ph}} = \{1, \dots, 8\}, \qquad \star_{\mathrm{ph}} = 9,$$

and let $P_2(H_{\mathrm{act}})$ be the set of unordered pairs of active phases. We define

$$\begin{aligned} \mathcal{F}_6^{\mathrm{ph}} &= H_{\mathrm{act}} \times P_2(H_{\mathrm{act}}), \\ \mathcal{F}_8^{\mathrm{ph}} &= O_{\mathrm{ph}} \times P_2(H_{\mathrm{act}}), \\ \Theta_{\mathrm{ph}} &= D_9 \times P_2(H_{\mathrm{act}}). \end{aligned}$$

Theorem 15.2 (Semicrown Relative to the TPK Phase)**.** *Relative to the published window origin and to the terminal return* $9 \to 1$*, it holds*

$$|\mathcal{F}_6^{\mathrm{ph}}| = 90, \qquad |\mathcal{F}_8^{\mathrm{ph}}| = 120, \qquad |\Theta_{\mathrm{ph}}| = 135.$$

The oriented boundary, the pointed return, the neutral lateral boundary, and the lateral closure are, respectively,

$$\begin{aligned} \partial_{\mathrm{ph}} &= \Theta_{\mathrm{ph}} \times \{-1, +1\}, \\ E_{\mathrm{ph}} &= \partial_{\mathrm{ph}} \sqcup \{\infty\}, \\ L_{\mathrm{ph}} &= H_0 \times P_2(H_{\mathrm{act}}), \\ P_{\mathrm{ph}} &= \partial_{\mathrm{ph}} \sqcup L_{\mathrm{ph}}, \end{aligned}$$

and satisfy

$$\boxed{|\partial_{\mathrm{ph}}| = 270, \quad |E_{\mathrm{ph}}| = 271, \quad |L_{\mathrm{ph}}| = 45, \quad |P_{\mathrm{ph}}| = 315.}$$

The construction is covariant under a change of origin: translating the window by C_9 *simultaneously translates all strata and changes neither their cardinalities nor their incidences.*

Proof. The sign rule gives six active phases and three neutral ones, so that

$$|P_2(H_{\mathrm{act}})| = \binom{6}{2} = 15.$$

Multiplying by 6, 8, and 9 produces 90, 120, and 135. The orientation doubles 135; the pointed return adds one element; and the three neutral phases contribute $3 \cdot 15 = 45$ lateral positions. Covariance follows because a translation of D_9 conjugates the phase rule and transports each Cartesian product by a bijection. □

This theorem substantively strengthens the cardinal identity $270 + 1$: the order $6 \subset 8 \subset 9$ is no longer chosen from among 9! possible orders, but instead arises from the TPK phase clock and its published return. Canonicity is relative to the origin of the window; changing it does not alter the object up to the natural action of C_9.

15.1.4 Semicorona relative to the hexad-octad elevation and pointed return

Fix the dodecad, the alignment, and a hexad $H \in \mathcal{H}(D)$ of the construction of section 17.1.6. By theorem 17.5, there exists a unique octad O_H such that $O_H \cap D = H$. Write

$$P_2(H) = \{\{a, b\} : a, b \in H,\ a \neq b\}, \qquad |P_2(H)| = \binom{6}{2} = 15,$$

and add to O_H a return state $\star$:

$$\widehat{O}_H = O_H \sqcup \{\star\}, \qquad |\widehat{O}_H| = 9.$$

We define

$$\Theta_H = \widehat{O}_H \times P_2(H).$$

Within it appear the two combinatorial layers.

$$\mathcal{F}_6 = H \times P_2(H), \qquad \mathcal{F}_8 = O_H \times P_2(H).$$

Theorem 15.3 (Realization of the semicrown)**.** *The preceding objects satisfy*

$$|\mathcal{F}_6| = 90, \qquad |\mathcal{F}_8| = 120, \qquad |\Theta_H| = 135.$$

The oriented boundary

$$\partial_H = \Theta_H \times \{-1, +1\}$$

has 270 *states. Adjoining one return point,*

$$E_H = \partial_H \sqcup \{\infty\},$$

has 271 *states. The boundary lateral of the stratum of six points is*

$$L_H = \Theta_H \setminus \mathcal{F}_6, \qquad |L_H| = 45,$$

and the lateral-oriented closure

$$P_H = \partial_H \sqcup L_H$$

tiene 315 *estados.*

Proof. The cardinals are

$$6\binom{6}{2} = 90, \qquad 8\binom{6}{2} = 120, \qquad 9\binom{6}{2} = 135.$$

The orientation doubles 135, adjoining one point adds a unit, and the difference between Θ_H and the six-point stratum contains $(9 - 6)15 = 45$ elements. From this result 270, 271 and $270 + 45 = 315$. □

Here the two integers have different set-theoretic types. 271 is the cardinality of an oriented boundary to which one point has been adjoined; that fact alone does not make it the orbit of e. 315 is the cardinality of that boundary together with the 45 lateral positions not belonging to the six-point stratum. The inclusion $H \subset O_H$ used here is indeed the Steiner lift already proved, relative to the fixed dodecad and alignment. The semicorona then adjoins the point $\star$, the orientation, and the lateral boundary. This construction does not yet provide the nonadic order induced by TPK or an intertwiner between the faces of the cube and Steiner incidences; these are additional maps.

The two semicrowns share the same cardinal profile, but they must not therefore be identified. An unmarked isomorphism between the chains

$$H_{\mathrm{act}} \subset O_{\mathrm{ph}} \subset D_9 \quad \text{and} \quad H \subset O_H \subset \widehat{O}_H$$

that preserves the return point may freely permute the six elements of the first stratum and the two elements added in the second. There exist

$$\boxed{6!\,2! = 1440}$$

chain gauges. The phase–Steiner intertwiner must select one of them by means of additional incidences; equality of cardinalities does not construct it.

15.1.5 Dihedral action and classification under declared predicates

For an emission $U = (u_1, \dots, u_6) \in \mathcal{U}_6$, we define the visible word

$$\Psi(U) = ((-u_1) \bmod 3, \dots, (-u_6) \bmod 3) \in \mathbb{F}_3^6.$$

Let the coordinate permutations be explicitly defined by

$$r \cdot (u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4),$$

$$s \cdot (u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3).$$

Generan a grupo diedral $D_3 \simeq S_3$. The catalogo verifica

$$r\mathcal{U}_6 = \mathcal{U}_6, \qquad s\mathcal{U}_6 = \mathcal{U}_6, \qquad \Psi(gU) = g\Psi(U),$$

and preserves the microscopic multiplicities.

The action divides the 243 visible words into 38 orbits of size six and five orbits of size three. On each word w, write

$$n(w) = |\Psi^{-1}(w)|, \qquad M(w) = \sum_{\Psi(U)=w} |F^{-1}(U)|.$$

We now fix three published predicates. Predicate P_π requires $n(w) = 5$, $M(w) = 1008$, multiplicities $(432, 144, 144, 144, 144)$, and tritic census $(2, 3, 1)$. Predicate P_e requires, over the complete orbit, singleton emissions of microscopic mass 144, union of values $\bigcup_{w \in O} \mathrm{diag}_c(w) = \{0, 3, 6\}$, and census $(2, 3, 1)$. Predicate P_φ preserves the first three conditions of P_e and requires balanced census $(2, 2, 2)$. The orientation gauges are $\mathrm{diag}_c = 6$ and the cardinal class is ES.

Theorem 15.4 (Finite orbital classification conditioned)**.** *In the reconstructed catalog, the preceding predicates select the following unique orbits without consulting digits of any constant.*

1. *There is a unique orbit of six words for which, in every word,* $n(w) = 5$*,* $M(w) = 1008$*, and the five multiplicities are* $432, 144, 144, 144, 144$*. All its words have trit census* $(n_0, n_1, n_2) = (2, 3, 1)$*. It is the dihedral orbit of the regional type of* π*.*
2. *Between the orbits radical of emissions one-point and mass microscopic* 144 *who union orbital of values* diag_c *is* $\{0, 3, 6\}$*, there exists to unique orbit with census* $(2, 3, 1)$*. Is the orbit of return of* e*.*
3. *In the same sector there is a unique balanced orbit with census* $(2, 2, 2)$*. It is the proportional orbit of* φ*.*

The orientation declared by $\mathrm{diag}_c = 6$ *and the cardinal class* ES *select, respectively,*

$$\boxed{w_\pi = 010211, \qquad w_e = 201101, \qquad w_\varphi = 121200.}$$

Proof. The exhaustive reconstruction of the 468 emissions first verifies closure of the action and groups the 243 words into orbits. It then applies the indicated predicates to all of them. The first predicate leaves a single orbit; the two singleton radical-fiber predicates respectively leave an orbit of census $(2, 3, 1)$ and one of census $(2, 2, 2)$. Finally, the orientation gauge is applied to each orbit, leaving a unique word. None of the selector predicates contains the decimal expansions of π, e or φ. □

15.1.5.1 Initial intrinsic encoder and existence of its limit

In the stratum of visible words containing exactly two zeros, there is an encoder that does arise from the TPK datum itself:

$$c_0(w) = \{j \in \{1, \dots, 6\} : w_j = 0\} \in P_2(\{1, \dots, 6\}).$$

It is D_3-equivariant because a coordinate permutation transports the zero support. On the orbits classified by the theorem 15.4, its images are

$$c_0(O_\pi) = \{\{1,2\}, \{1,3\}, \{2,3\}, \{4,5\}, \{4,6\}, \{5,6\}\},$$
$$c_0(O_e) = \{\{1,6\}, \{2,5\}, \{3,4\}\},$$

where, on the second line, each pair has two preimages within the orbit. The orbit of φ shares the first support type with π, but is separated from it by its singleton fiber, its multiplicity, and its TRIT census.

The map c_0 is the first rung of the register–pair encoder; it is not the complete family c_k. In deep blocks, (0,1,3) or (4) zeros occur, so the same formula does not take values in P_2 at every depth. Furthermore, $\{1, \dots, 6\}$ is here the hexad of coordinates of the word, not an already identified Witt hexad. Prolonging c_0 requires a phase-dependent rule and the corresponding incidence intertwiner.

The canonical microfiber of π remains exactly

$$\mathscr{R}_\pi^{\text{micro}} = \Psi^{-1}(010211),$$

formed by five distinct regions U_6. The other five words of the dihedral orbit are gauge images of that orbital form; they do not constitute five new canonical microfibers of π. Analogously, the names e and φ are applied after fixing the published predicates and gauge. The enumeration is exhaustive *relative* to those premises. Predicates and gauge are declared components of the enriched TPK reader: they are reconstructed not from the three coarse statistics appearing below, but from the complete genealogy of the emission, its orientation, its boundary, and its survival.

The sum of the six integers of an emission, its residue modulo nine, and the number of even entries do not suffice for this theorem. The oriented emission of e and three of the five regions of π share exactly

$$\sum_j u_j = 3316, \qquad \sum_j u_j \equiv 4 \pmod 9, \qquad \#\{j : u_j \text{ even}\} = 4.$$

The distinction requires the istructura of the fiber, the multiplicity, the sector radical and the orbit complete. This ablation demonstrates by que the parity and the root digital are ingredients of the reader, but not replace the transport TPK.

15.1.6 Typed correspondence $271 \to e$ and $315 \to \pi$

The theorems 15.2 a 15.4 are independent: the first constructs the semicrown from the phase clock, the second provides its relative Steiner realization, and the third classifies orbits under fixed predicates. To compose the first and third results, we declare the type dictionary that the reader must preserve:

internal type	orbital type
bidirectional pointed return boundary-plus-lateral closure	single-point radical fiber with the census of the regional branch closure orbit with five regions.

Corollary 15.5 (Typed, orbit-relative associations)**.** *Relative to the dictionary, the predicates P_e, P_π, and the preceding orientation gauges, the orbital classification produces the conditioned associations*

$$\boxed{|E_{\mathrm{ph}}| = 271 \rightsquigarrow O_e \longrightarrow w_e,} \qquad \boxed{|P_{\mathrm{ph}}| = 315 \rightsquigarrow O_\pi \longrightarrow w_\pi.}$$

In the second case, the oriented representative $w_\pi = 010211$ determines the canonical five-region microfiber $\Psi^{-1}(010211)$. The associations are obtained before Archimedean evaluation and do not use the decimal values of the constants.

Proof. By the orbital theorem, each row of the dictionary possesses a unique orbit satisfying the declared predicate. Classification is first performed at the level of the complete dihedral orbit; the oriented gauge then selects the indicated representative. Uniqueness is therefore relative to the dictionary, predicates, and gauges fixed. □

The channel $161 = \Theta + \kappa - 1 = 3\Delta - 1$ remains the integer identity. Although P_φ classifies a balanced orbit and the published gauge selects $w_\varphi = 121200$, this section does not construct a set-theoretic object of cardinality 161 whose type induces that orbit. Therefore, no reader $161 \to \varphi$ is asserted here.

The corollary depends on the type dictionary because cardinalities, taken in isolation, do not define a function into the catalogue. In the complete reader, that dictionary is realized through the nine monodromy relations of the section 6.3, together with orientation and regional survival. The important consequence is affirmative: once the structural reader has been fixed, the evaluation consults no decimal approximation of the constant.

15.1.6.1 Joint dependence of the channels on the product selector

The executable factorization of the catalog and the parametrization of its additive signature give

$$\mathcal{U}_6 \simeq \mathcal{S}_+ \times \mathcal{S}_\times, \qquad \mathcal{S}_+ \simeq D_9 \times \{\pm 1\}.$$

On the other hand,

$$\partial_{\mathrm{ph}} \simeq (D_9 \times \{\pm 1\}) \times P_2(H_{\mathrm{act}}).$$

Thus, a functional attempt at descent that preserves phase and orientation identically reduces to the map

$$q_\times : \mathcal{S}_\times \longrightarrow P_2(H_{\mathrm{act}}).$$

This is the field absent from the catalogue: the published rows end at the product signature E_{sig} and do not contain a class $P_2(H_{\mathrm{act}})$.

Proposition 15.6 (Obstruction to the isolated product selector)**.** *There is no total D_3-equivariant function*

$$q_\times : \mathcal{S}_\times \longrightarrow P_2(H_{\mathrm{act}})$$

if D_3 acts on the (26) product signatures through the published coordinate permutations r, s and on H_{act} through

$$a(x) = x + 3 \pmod 9, \qquad j(x) = -x \pmod 9.$$

Proof. The action on $\mathcal{S}_\times$ has orbits of sizes

$$6 + 6 + 6 + 3 + 3 + 1 + 1.$$

Its two globally fixed points are the signatures 000000 and 555555. The action of D_3 on the six elements of H_{act} is regular, and its induced action on the fifteen pairs has orbits

$$6 + 3 + 3 + 3,$$

with no globally fixed point. An equivariant function must map a fixed point of the source to a fixed point of the target, which is impossible. □

The obstruction precisely identifies the information lost by the isolated product signature: the fixed sectors can be resolved only by preserving phase, orientation, and the enriched record. The preceding encoder c_0 acts on the two-zero stratum, but it cannot replace the family c_k with a single column of E_{sig}.

15.1.7 Visible parity, carry parity and physical boundary

The even–odd separation can be formulated without depending on a particle label. For each of the leaves $\bullet \in \{+, \times\}$, set

$$n_+(i,j) = i + j, \qquad n_\times(i,j) = ij,$$

and let us define the two bits

$$\epsilon_R^\bullet(i,j) = \mathrm{dr}_9(n_\bullet(i,j)) \bmod 2, \qquad \epsilon_Q^\bullet(i,j) = q_9^\sharp(n_\bullet(i,j)) \bmod 2.$$

Theorem 15.7 (Parity-Carry Factorization)**.** *For the* 81 *cells of each APP sheet,*

$$\boxed{n_\bullet(i,j) \bmod 2 = \epsilon_R^\bullet(i,j) + \epsilon_Q^\bullet(i,j) \pmod 2.}$$

The exact census by pairs $(\epsilon_R, \epsilon_Q) = (00, 01, 10, 11)$ *is*

	00	01	10	11
+	16	20	20	25
×	25	5	20	31

and, therefore, the additive and multiplicative leaves possess distinct parity–carry distributions.

Proof. Reduction modulo two $n = 9q_9^\sharp(n) + \mathrm{dr}_9(n)$ gives the identity, since $9 \equiv 1 \pmod 2$. On the additive sheet, the multiplicities of $i + j$ are $1, 2, \ldots, 9, 8, \ldots, 1$; separating representative and quotient produces the first row. The same calculation on the 81 products ij produces the second. Enumeration of the 162 cells verifies the identity before accumulating the counts. □

On $\ell^2(D_9^2)$, the operators diagonal

$$\Gamma_R^\bullet f(i,j) = (-1)^{\epsilon_R^\bullet(i,j)} f(i,j), \qquad \Gamma_Q^\bullet f(i,j) = (-1)^{\epsilon_Q^\bullet(i,j)} f(i,j)$$

are commutative involutions. The unreduced parity factorizes as

$$\boxed{\Gamma_{\mathrm{raw}}^\bullet = \Gamma_R^\bullet \Gamma_Q^\bullet.}$$

This is the precise operatorial formulation of the idea that the digital root reads the visible component while the quotient preserves the drag of the coronas. The difference between the two rows of the table is also an exact form of the incompatibility between addition and product accumulation.

Each involution decomposes the space into even and odd sectors and provides an internal grading $\mathbb{Z}/2\mathbb{Z}$. A Bose–Einstein or Fermi–Dirac statistic, however, is an assertion about exchanging several indistinguishable realizations, not about the parity of a single cell. Its derivation requires an exchange functor representing symmetric or braid groups and intertwining the dynamics. HMT parity constructs the graded domain on which that functor must act; it is not identified here with its physical representation.

15.2 Derivation of the channels 90/120 through inclusion of hexad–octad incidences

With the dodecad and the alignment of the section 17.1.6 fixed, let H be a hexad in the HMT coordinates, let $\ell(H) \subset D$ be its image, and let $O_H = \iota_D(\ell(H))$ be its unique lift. Marking a point and a pair of points of the residual hexad yields

$$F_6(H) = \ell(H) \times \binom{\ell(H)}{2}, \qquad |F_6| = 6\binom{6}{2} = 90.$$

If the marked point can traverse the octad:

$$F_8(H) = O_H \times \binom{\ell(H)}{2}, \qquad |F_8| = 8\binom{6}{2} = 120.$$

The inclusion $\ell(H) \hookrightarrow O_H$ induces the incidence morphism

$$\mathcal{I}_{6\to 8} : F_6(H) \hookrightarrow F_8(H).$$

Normalizing the difference by the chosen factor $24\binom{6}{2}$ yields

$$\boxed{\frac{90-120}{24\binom{6}{2}} = \frac{6-8}{24} = -\frac{1}{12}.}$$

The equality is a normalized defect of the combinatorial interface $6 \to 8$. The factor $24\binom{6}{2}$ forms part of the definition of this normalization; the inclusion alone determines the difference -30. The operator $\mathcal{I}_{6\to 8}$ has marked incidence configurations as its domain; it is neither the transverse extractor of the dodecaphase state nor the family of angular moments.

15.3 Transport of $-1/12$ to the channels 90/120 in 5 algebraic realizations

In addition to the normalized defect of the hexad–octad inclusion, five operator realizations of the same rational number appear. Their domains are specified separately; equality of values does not identify the operators without an additional intertwiner.

15.3.1 Difference of dodecaphasic periodicities

One tooth of C_{12} measures $30°$. Then

$$\frac{30}{120} - \frac{30}{90} = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}.$$

It is the oriented difference between the periods of $90°$ and $120°$ in this realization.

15.3.2 Pseudoinverse of the Gram Operator on the Live Component

In the exact jump $60 \to 81$, two fibers have sizes 0 and 12. The Gram operator is

$$K_{60,81} = \operatorname{diag}(0, 12).$$

On the nonzero component of its range, the pseudoinverse is $1/12$; with the negative boundary orientation,

$$E_{\text{surv}} = -K^{+}_{60,81}$$

yields $-1/12$.

15.3.3 2nd difference between triadic scales

Define

$$T_L = \sum_{n=1}^{L-1} n(L-n) = \frac{L(L^2-1)}{6}, \qquad a(L) = -\frac{T_L}{L^2} = -\frac{L}{6} + \frac{1}{6L}.$$

The first scale subtraction is

$$A_1(L) = a(L) - \frac{a(3L)}{3} = \frac{4}{27L},$$

and the second

$$C(L) = LA_1(L) - \frac{3L}{3}A_1(3L) = \frac{8}{81}.$$

The residue $8/81$ is independent of L. To obtain $-1/12$, the declared normalization is applied

$$R(L) = -\frac{27}{32}C(L) = -\frac{1}{12}.$$

The calculation derives $8/81$; it does not by itself derive the factor $-27/32$.

15.3.4 Composition of modular transformations and preservation of the signed state

The eta function satisfies

$$\frac{\eta(\tau)}{\eta(3\tau)} = q^{-1/12}\prod_{n\geq 1}(1+q^n+q^{2n})^{-1}.$$

The exponent comes from $1/24 - 3/24 = -1/12$. Taking twelve copies:

$$t_3(q) = \left(\frac{\eta(\tau)}{\eta(3\tau)}\right)^{12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \cdots .$$

The coefficient 54 coincides with the APP defect. The rational identity

$$j = \frac{(t_3+27)(t_3+243)^3}{t_3^3}$$

establishes the connection with the classical modular branch.

15.3.5 Defect of cyclic difference projectors

Let S be the cyclic shift on $\mathbb{Q}^{12}$ and define

$$D_r = I - S^r.$$

The kernel of D_r consists of the vectors constant on each orbit of S^r; therefore,

$$\dim\ker D_r = \gcd(12, r).$$

Let $\Pi_{\ker D_3}$ and $\Pi_{\ker D_4}$ be the rational projectors onto $\ker D_3$ and $\ker D_4$, and let $\tau(T) = \mathrm{Tr}(T)/12$ be the normalized trace. Then

$$\boxed{\tau(\Pi_{\ker D_3} - \Pi_{\ker D_4}) = \frac{\operatorname{rank}\Pi_{\ker D_3} - \operatorname{rank}\Pi_{\ker D_4}}{12} = \frac{3-4}{12} = -\frac{1}{12}.}$$

This realization expresses the rational as the transverse defect of steps three and four in the twelve-phase cycle.

Remark 15.8 (Domains of the realizations). The APP defect 54, the eta exponent $-1/12$, the combinatorial defect of the hexad–octad lift, and the modular identities once t_3 has been fixed are exact. The value of

the pseudoinverse, the normalized extractor, and the projector defect are likewise exact. Each equality preserves its domain and its corresponding map.

15.4 Barbero–Immirzi parameter: area operator, spectrum, channel intertwiner, and modular entropy

The HMT identification of the functional $\Gamma_{6\to 8}$ with the Barbero–Immirzi coordinate is carried out on the same triadic boundary that supports the incidence (90/120). The bulk and its boundary sheet have the same cardinality,

$$\mathcal{B} = (\mathbb{Z}/9\mathbb{Z})^3, \quad |\mathcal{B}| = 729, \qquad \partial\mathcal{B} = (\mathbb{Z}/27\mathbb{Z})^2, \quad |\partial\mathcal{B}| = 729,$$

by regrouping six trits into three nonadic coordinates or two coordinates of order twenty-seven.

Theorem 15.9 (Areal Law of Triadic Cutoff)**.** *In the cubic graph $N \times N \times N$ with unit capacity, the minimum cut between two opposite faces has capacity N^2. For $N = 9$,*

$$\mathcal{A}_{\min} = 81, \qquad S_{\mathrm{tri}} = \mathcal{A}_{\min} \log 3 = 81 \log 3.$$

Proof. The N^2 lines parallel to the normal direction are edge-disjoint paths, so every cut intercepts them. A transverse layer contains exactly N^2 edges and attains the bound. Each triadic link contributes $\log 3$ to the entropy of a maximally mixed state. □

On the boundary torus 27×27, let A be the canonical block 9×9. Its boundary decomposes exactly into

$$\partial A = \partial A_{90} \sqcup \partial A_{120}, \qquad |\partial A_{90}| = |\partial A_{120}| = 18. \tag{15.1}$$

The horizontal links represent channel (90), and the vertical ones channel (120). This decomposition turns the pair of incidences into an operator decomposition of the boundary.

The areal entropy $S_{\mathrm{tri}} = 81 \log 3$ belongs to the maximally mixed state of the cut links. The bilayer entropy $S_{\mathrm{bi}}(y)$ of the definition 73.2 belongs to the normalized distribution between the two sheets; the two objects retain distinct domains.

Let $N = \mathrm{diag}(0, 1, 2)$ be the number operator of a tritic link. The areal realization declares the positive scale

$$\theta_{\mathrm{clk}} = \frac{C^*}{6}\frac{\pi}{180}, \qquad a_0 = \frac{1 + 2\cos(10^\circ)}{3}\,\theta_{\mathrm{clk}}, \qquad \gamma_{\mathrm{HMT}} = \Gamma_{6\to 8}.$$

The formula for a_0 constitutes the starting structure of this realization; once that scale is fixed, the operator and its spectrum follow exactly.

Definition 15.10 (Area Operator HMT)**.** On the boundary sheet $\mathcal{H}_\partial = \bigotimes_{e\in\partial A} \mathbb{C}^3$, define

$$\widehat{\mathcal{A}}_{\mathrm{HMT}} = \gamma_{\mathrm{HMT}} a_0 \sum_{e\in\partial A} N_e. \tag{15.2}$$

Its spectrum in the canonical cut is

$$\mathrm{Spec}\,\widehat{\mathcal{A}}_{\mathrm{HMT}} = \{n\gamma_{\mathrm{HMT}} a_0 : 0 \le n \le 72\},$$

with multiplicities given by the coefficients of $(1 + z + z^2)^{36}$. Prolongation through compatible cuts produces the protected family $\mathcal{A}_n^{\mathrm{prot}} = n\gamma_{\mathrm{HMT}} a_0$.

The intertwiner that fixes normalization is constructed in the subspace of collective channels. Let u_{90}, u_{120} be the uniform unit vectors of the hexadic and octadic incidences, and let v_{90}, v_{120} be the uniform unit vectors of the two sets of links of (15.1). The map

$$\begin{aligned} \mathcal{J}_{6\to 8} :\ \operatorname{span}\{u_{90}, u_{120}\} &\longrightarrow \operatorname{span}\{v_{90}, v_{120}\}, \\ \mathcal{J}_{6\to 8} u_{90} = v_{90}, &\qquad \mathcal{J}_{6\to 8} u_{120} = v_{120} \end{aligned} \tag{15.3}$$

is an isometry between these collective subspaces. If

$$D_{\mathrm{inc}} = 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \qquad D_{\mathrm{area}} = 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|,$$

then

$$\boxed{\mathcal{J}_{6\to 8} D_{\mathrm{inc}} = D_{\mathrm{area}} \mathcal{J}_{6\to 8}.}$$

In particular, the primitive combination $12 : -1$ acts with the same coefficients before and after $\mathcal{J}_{6\to 8}$. This equality fixes the normalization dictionary that inserts $\gamma_{\mathrm{HMT}} = \Gamma_{6\to 8}$ into the Ashtekar–Barbero connection.

The information retained by reduction to the region is published through the modular Hamiltonian. The starting structure of this reading is declared to be

$$\Delta_4^{\mathrm{mod}} = \frac{(A - C^*)\pi/180}{9 \cdot 90} \left(1 - \frac{7}{12}\frac{\pi}{729}\right), \qquad w_{90} = 90\Delta_4^{\mathrm{mod}}, \quad w_{120} = 120\Delta_4^{\mathrm{mod}},$$

we define

$$\begin{aligned} K_{\mathrm{link}}(w) = wN, \qquad \rho_{\mathrm{link}}(w) &= \frac{\operatorname{diag}(1, e^{-w}, e^{-2w})}{1 + e^{-w} + e^{-2w}}, \\ \rho_A = \bigotimes_{e\in\partial A_{90}} \rho_{\mathrm{link}}(w_{90}) \otimes \bigotimes_{e\in\partial A_{120}} \rho_{\mathrm{link}}(w_{120}), &\qquad K_A = -\log \rho_A. \end{aligned} \tag{15.4}$$

Then

$$K_A = \sum_{e\in\partial A} w_e N_e + c\mathbf{1}, \qquad S(\rho_A) = 18S(w_{90}) + 18S(w_{120}),$$

and the evaluation of that starting structure gives

$$S(\rho_A) = 39.54837320881808\ldots \text{ nats} = 57.05624190358778\ldots \text{ bits}.$$

The quantity $S(\rho_A)$ is the von Neumann entropy of the declared mixed product. A global purification identifies it with an entanglement entropy across the cut; without that purification, its canonical type is reduced modular entropy. Thus, γ_{HMT} fixes the quantum of area/torsion, S_{tri} counts the areal capacity, S_{bi} quantifies the allocation between sheets, and $S(\rho_A)$ measures modular mixing. Area operator, spectrum, and intertwiner belong to a single boundary construction.

15.5 Hexadic–octadic incidence and the Barbero–Immirzi functional

Let $H \hookrightarrow O$ be a marked inclusion, $|H| = 6$, $|O| = 8$. The incidence objects

$$\mathcal{F}_6 = H \times \binom{H}{2}, \qquad \mathcal{F}_8 = O \times \binom{H}{2}$$

have cardinalities 90 and 120. Their normalized difference gives

$$\frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24\binom{6}{2}} = -\frac{1}{12}.$$

This is a combinatorial realization of the same scalar. Numerical equality with (14.85) does not erase the distinction between the spaces on which it was computed.

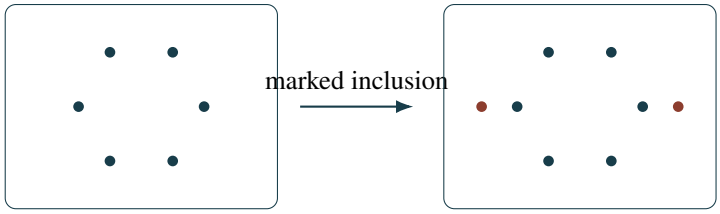


Figure 15.1: Marked incidence from six to eight points. The drawing represents the sets and inclusion; it does not attempt to turn the faces or vertices of a cube into Steiner blocks without their incidence map.

15.5.1 Dodecaphase fundamental chain and pole coefficient

In the free module of oriented events, the fundamental chain of the turn is expressed as

$$c_{12}^{+} = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\mathrm{ret}}, \mathcal{F}_8].$$

Multiplicity twelve corresponds to the sectors and multiplicity one to the oriented seam. The involution reverses orientation of the complete chain. Let

$$S_m(q) = \frac{q^m}{1 - q^{3m}}, \qquad 0 < q < 1.$$

The linear reader assigns S_{90} to every sector and S_{120} to the seam; consequently,

$$\mathscr{R}(c_{12}^{+}) = 12 S_{90} - S_{120}.$$

Theorem 15.11 (Pole coefficient and conjugate evaluation)**.** *The image of the fundamental chain has coefficient* $1/24$ *relative to* $(1-q)^{-1}$ *and residue* $-1/24$ *relative to* $(q-1)^{-1}$*. Its evaluation on the previously generated channels determines*

$$\gamma_{\mathrm{HMT}} = \frac{\pi A C^{*}}{180} + 12\big[S_{90}(q_{-}) - S_{90}(q_{+})\big] - \big[S_{120}(q_{-}) - S_{120}(q_{+})\big]. \tag{15.5}$$

Proof. The factorization $1 - q^{3m} = (1-q)\sum_{j=0}^{3m-1} q^{j}$ implies $\lim_{q\uparrow 1}(1-q)S_m(q) = 1/(3m)$. By linearity,

$$\frac{12}{270} - \frac{1}{360} = \frac{1}{24}.$$

The coordinate $q - 1$ reverses the residue's sign. The conjugate evaluation and the source's angular component produce (15.5). □

The Diophantine certificate $4a - 3b = 45$ admits the pairs $(12 + 3t, 1 + 4t)$, $t \geq 0$; it certifies minimality of $(12, 1)$ but does not replace the origin of multiplicities in the chain. The evaluation of (15.5) is

$$\gamma_{\mathrm{HMT}} = 0.27400767269445927474725526077398041940\ldots.$$

This number is an output of the source functional; it is not an external value used to select A, C^{*}, or their coefficients.

15.5.2 Parity and structural sensitivity

Reversing C^* exchanges q_+ and q_- and changes the sign of the angular component. Therefore

$$\gamma_{\mathrm{HMT}}(A, -C^*) = -\gamma_{\mathrm{HMT}}(A, C^*).$$

The functional preserves orientation, whereas other readers, such as q_+q_-, are even. This difference precedes its physical interpretation.

15.5.3 Structural dependence of the functional and areal normalization

The preceding construction places the Barbero–Immirzi parameter at the intersection of three determinations: marked incidence, dodecaphase orientation and angular evaluation. Incidence supplies the event spaces; orientation fixes the chain on those spaces; evaluation transports the chain to a dimensionless quantity. This decomposition preserves the origin of each coefficient when the functional is subsequently used in area geometry.

The inclusion $\mathcal{F}_6 \hookrightarrow \mathcal{F}_8$ maintains the marked pair of the hexad. Its complement has cardinality $2\binom{6}{2} = 30$. This increment reappears as the determinant of the map reconstructing boundary occupations from area and the weighted observable. Equality of both integers has a concrete explanation here: sector weights differ precisely by the number of events added by the marked inclusion.

Proposition 15.12 (Incidence and recoverability of occupations)**.** *Let n_6, n_8 be the aggregate occupations of two sectors, and let $r_6 = |\mathcal{F}_6|$, $r_8 = |\mathcal{F}_8|$. The map*

$$(n_6, n_8) \longmapsto (n_6 + n_8, r_6 n_6 + r_8 n_8)$$

is invertible on its image when inclusion adds at least one event. For the constructed inclusion, its determinant is 30.

Proof. The matrix of the map is $\left(\begin{smallmatrix} 1 & 1 \\ r_6 & r_8 \end{smallmatrix}\right)$. Its determinant is $r_8 - r_6 = |\mathcal{F}_8 \setminus \mathcal{F}_6|$. The inclusion considered adds $2\binom{6}{2} = 30$ events. Linear inversion recovers both occupations and preserves the integer restrictions inherent to its image. □

The result recovers aggregate occupations. Information about each link and the ordering of events remains in the complete state and its additional readers. This hierarchy precisely formulates the difference between area value and boundary genealogy: area preserves a sum; the observable pair preserves that sum's sector decomposition; the register additionally preserves its distribution and history.

The parameter γ_{HMT} transports that incidence to the areal scale. Its structural interpretation is that of an oriented coefficient of the hexadic–octadic return functional. The gravitational realization subsequently identifies it in the Holst action. This distinguishes its intrinsic definition, value and function in the physical equation.

15.5.4 Structural definition and role of the Barbero–Immirzi parameter

Definition 15.13 (Oriented evaluation of hexadic–octadic incidence)**.** The parameter γ_{HMT} is the conjugate evaluation of the fundamental chain c_{12}^+, completed by its angular component. On the domain $0 < q_\pm < 1$, its definition (15.5) takes the form

$$\gamma_{\mathrm{HMT}} = \frac{\pi A C^*}{180} + [\mathscr{R}(c_{12}^+)](q_-) - [\mathscr{R}(c_{12}^+)](q_+). \tag{15.6}$$

The marked incidence supplies the event spaces; the oriented chain fixes their multiplicities; the channels in (14.102) perform the evaluation.

The determination jointly uses the cardinalities 90, 120, the twelve sector contributions, and the return seam. The pole coefficient $1/24$ and Diophantine minimality verify the compatibility of these weights

with the normalization. Their provenance is preserved in c_{12}^{+}, before the channels are evaluated. Angular inversion exchanges q_+ and q_- and produces $\gamma_{\mathrm{HMT}}(A,-C^*) = -\gamma_{\mathrm{HMT}}(A,C^*)$: the dimensionless value transports an orientation of the incidence.

In the geometric realization, this same coefficient enters (80.12) and the area increment $\gamma_{\mathrm{HMT}} a_0 \ell_P^2$. The Holst action uses $\gamma_{\mathrm{HMT}} \neq 0$; the positive areal magnitude uses $|\gamma_{\mathrm{HMT}}|$. Proposition 80.20 makes the coordinated change of orientation explicit. Thus, the structure of the parameter explains what it weights in the incidence, what it preserves under transport, and how it participates in the torsional response and in the geometric resolution of information.

15.6 Action ellipse and modular reciprocity

With the outputs $A, C^*, \hbar > 0$ fixed in the chart $|C^*/A| < 1$, define

$$\eta_{\mathrm{el}} = \operatorname{artanh}(C^*/A), \quad a_S = \sqrt{2\hbar}\, e^{\eta_{\mathrm{el}}/2}, \quad b_S = \sqrt{2\hbar}\, e^{-\eta_{\mathrm{el}}/2}.$$

The balanced canonical coordinates have units of square root of action; position and momentum are not added here without a dimensional chart. We have

$$a_S b_S = 2\hbar, \qquad \tau_{\mathrm{el}} = i\,\frac{b_S}{a_S}, \qquad R_{\mathrm{el}} = \sqrt{\frac{b_S}{a_S}} = \left(\frac{A - C^*}{A + C^*}\right)^{1/4}.$$

The positive root determines the composition $\tau_{\mathrm{el}} = iR_{\mathrm{el}}^2$ and recovers the angular ratio:

$$\frac{C^*}{A} = \frac{1 - R_{\mathrm{el}}^4}{1 + R_{\mathrm{el}}^4}.$$

Shape inversion sends R_{el} to R_{el}^{-1} and τ_{el} to $-1/\tau_{\mathrm{el}}$; Equation (15.11) develops shape reciprocity.

Proposition 15.14 (Coincident energy, distinguishable action)**.** *Let*

$$D(\eta) = \begin{pmatrix} e^{\eta} & 0 \\ 0 & e^{-\eta} \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Both maps satisfy $T^{\mathsf{T}} D(-\eta) T = D(\eta)$. However, for $\omega = \mathrm{d}Q \wedge \mathrm{d}P$,

$$S_2^* \omega = -\omega, \qquad J_2^* \omega = \omega.$$

If $\mathcal{I}(\gamma) = \oint_\gamma P\ \mathrm{d}Q$ along a closed curve,

$$\mathcal{I}(S_2\gamma) = -\mathcal{I}(\gamma), \qquad \mathcal{I}(J_2\gamma) = \mathcal{I}(\gamma).$$

Proof. Conjugation interchanges the diagonal entries of D. The determinants of S_2, J_2 are $-1, 1$, respectively, which proves the transformation of ω. For $\lambda = P\ \mathrm{d}Q$,

$$S_2^* \lambda = \mathrm{d}(PQ) - \lambda, \qquad J_2^* \lambda = \lambda - \mathrm{d}(PQ).$$

Integrating the exact differential along the closed curve completes the proof. □

The quadratic energy and the modulus do not suffice to distinguish the two transports. The orientation of the action does distinguish them, even in the circular form. The geometric marking is thus part of the recoverable reading, not an embellishment added to the value.

15.6.1 Semiaxes, foci, and interior radii

The figure is obtained from the angular and action outputs of the preceding sections. APP preserves the sheets and their residues and quotients; TRIT fixes the orientation; TPK transports the boundary, carry and memory to the expressions of the same enriched state and the joint discrete structure of the continuum. Here the pair (A, C^*) and the positive section $\hbar$ are received after that construction. The geometry makes explicit the information preserved by their readers, without using an observed length to select the source state. Section 15.6.1 develops this realization from the readers already generated, and the angular construction is retained in Section 14.11.6.

Let $\xi = C^*/A \in (-1, 1)$, with $A \neq 0$ and both angles in the same unit. Write $\eta = \operatorname{artanh} \xi$ and

$$a_S = \sqrt{2\hbar}\, e^{\eta/2}, \qquad b_S = \sqrt{2\hbar}\, e^{-\eta/2}, \qquad \gamma(\theta) = (a_S \cos\theta, b_S \sin\theta).$$

The semiaxes are labelled by their coordinates: if $\eta < 0$, the vertical axis is the major axis. Both belong to the square-root-of-action line of the balanced chart, not to the line of spatial lengths.

Proposition 15.15 (Focal geometry and radial interior)**.** *Let $a = \max(a_S, b_S)$, $b = \min(a_S, b_S)$. The focal half-distance and eccentricity are*

$$f_S = \sqrt{a^2 - b^2} = \sqrt{4\hbar \sinh|\eta|}, \qquad e_f = \frac{f_S}{a} = \sqrt{1 - e^{-2|\eta|}}. \tag{15.7}$$

For $\eta > 0$, the foci are $(\pm f_S, 0)$; for $\eta < 0$, they are $(0, \pm f_S)$. At $\eta = 0$ they coincide with the centre. The boundary radius in polar direction ϕ is

$$\rho_\partial(\phi) = \left(\frac{\cos^2\phi}{a_S^2} + \frac{\sin^2\phi}{b_S^2} \right)^{-1/2}. \tag{15.8}$$

The interior segment in that direction consists of $r(\cos\phi, \sin\phi)$, with $0 \le r < \rho_\partial(\phi)$. In particular, $b \le \rho_\partial(\phi) \le a$.

Proof. The squares of the major and minor semiaxes are $2\hbar e^{|\eta|}$ and $2\hbar e^{-|\eta|}$. Their difference gives (15.7). To verify the focus, suppose $\eta > 0$. At $X = \gamma(\theta)$, the distances to $F_+ = (f_S, 0)$ and $F_- = (-f_S, 0)$ satisfy

$$|X - F_+|^2 = (a_S - f_S \cos\theta)^2, \qquad |X - F_-|^2 = (a_S + f_S \cos\theta)^2.$$

Since $a_S > f_S$, its positive roots sum to $2a_S$. For $\eta < 0$, interchanging the coordinates gives the same proof. In the circular case, $f_S = 0$. Finally, substituting $(Q, P) = r(\cos\phi, \sin\phi)$ into $Q^2/a_S^2 + P^2/b_S^2 = 1$ gives (15.8). The coefficient of r^2 lies between a^{-2} and b^{-2}, proving the bounds and the characterization of the interior. □

15.6.2 Parametric phase, polar angle, and action area

Proposition 15.16 (Areal law and two angular coordinates)**.** *The phase θ of γ and its continuous polar angle ϕ, lifted from zero, satisfy*

$$\phi(\theta) = \arg(a_S \cos\theta + i b_S \sin\theta), \qquad \frac{d\phi}{d\theta} = \frac{a_S b_S}{a_S^2 \cos^2\theta + b_S^2 \sin^2\theta} > 0. \tag{15.9}$$

In a chart where both tangents are defined, $\tan\phi = (b_S/a_S)\tan\theta$. The oriented swept area is

$$\mathcal{A}(\theta) = \frac{1}{2} \int_0^\theta (QP' - PQ')\, dt = \hbar\theta, \qquad \mathcal{A}(1) = \hbar, \qquad \mathcal{A}(2\pi) = h. \tag{15.10}$$

Proof. The derivative of the argument of a nonzero vector is $(QP' - PQ')/(Q^2 + P^2)$. Here the numerator is $a_S b_S = 2\hbar$, proving the derivative and monotonicity. The quotient P/Q gives the tangent relation; the continuous lift fixes the quadrants and the number of turns. The area integral is $a_S b_S \theta/2 = \hbar\theta$, and $h = 2\pi\hbar$ gives the last identity. □

The parametric phase does not generally coincide with the polar angle. One radian of phase sweeps out $\hbar$, but this does not identify it with a circular arc or a sector of unit polar angle. In polar coordinates, the same law is written $d\mathcal{A} = \rho_\partial(\phi)^2 d\phi/2$. With the orientation of γ used here, $\oint P\,dQ = -h$: the positive area is h, whereas the integral of that one-form retains its sign. This agrees with the distinction between the symplectic and antisymplectic transports of the preceding section.

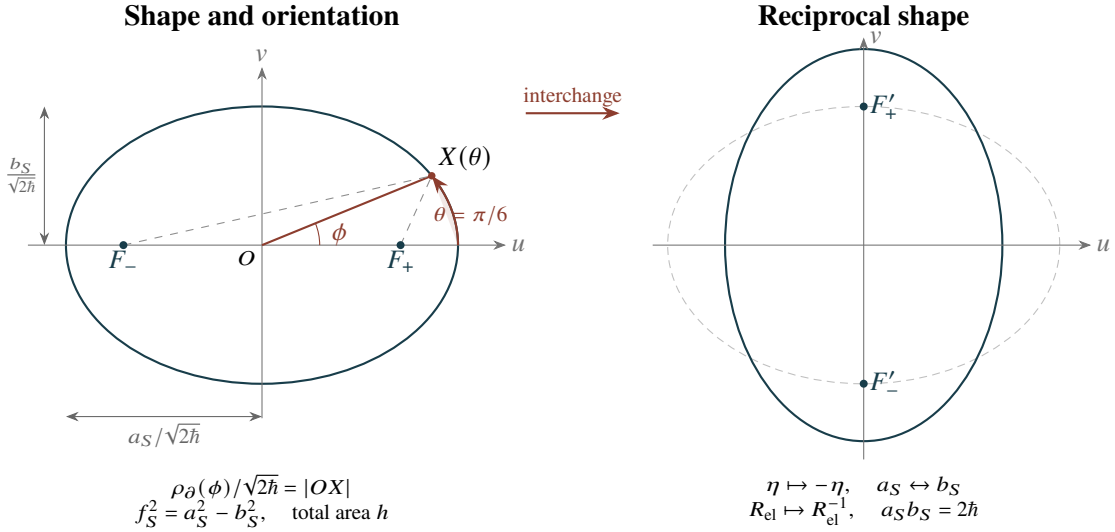


Figure 15.2: Semiaxes, foci and reciprocity of the action ellipse in the normalized coordinates $u = Q/\sqrt{2\hbar}$, $v = P/\sqrt{2\hbar}$. The marked sector has parametric phase $\theta = \pi/6$, distinct from its polar angle ϕ; its action area is $\hbar\pi/6$. The figure uses the algebraic case $\xi = 1/3$ to make the relations legible: it does not assign that value to the HMT angular output. The foci are calculated from the semiaxes. On the right, interchanging the axes preserves the total area h and inverts the dimensionless modular radius.

For the control case in the figure, $\xi = 1/3$, the normalized coordinates are exactly

$$\eta = \tfrac{1}{2}\log 2, \qquad \frac{a_S}{\sqrt{2\hbar}} = 2^{1/4} = 1.1892071150\ldots, \qquad \frac{b_S}{\sqrt{2\hbar}} = \frac{f_S}{\sqrt{2\hbar}} = R_{\mathrm{el}} = 2^{-1/4} = 0.8408964153\ldots.$$

These are regression values for the drawing, not values assigned to the angular generator. The local certificate also checks the circular form and the orientations $\xi = \pm 1/3, \pm 3/5$.

15.6.3 Shape reciprocity and interior distance

The rectangular expression recovers

$$R_{\mathrm{el}} = \sqrt{b_S/a_S} = e^{-\eta/2}, \qquad R_{\mathrm{el}}^4 = \frac{1-\xi}{1+\xi}, \qquad \xi = \frac{1 - R_{\mathrm{el}}^4}{1 + R_{\mathrm{el}}^4}. \tag{15.11}$$

The proof consists of substituting $2\eta = \log((1+\xi)/(1-\xi))$. Interchanging the axes sends η to $-\eta$, R_{el} to R_{el}^{-1} and $\tau = iR_{\mathrm{el}}^2$ to $-1/\tau$. It preserves $a_S b_S$. The radius R_{el} is dimensionless; the polar radius ρ_∂ has units of square root of action.

The interior also admits the Hilbert projective distance. If z_-, x, y, z_+ are ordered along a chord of the ellipse, define

$$d_H(x,y) = \frac{1}{2}\log\frac{|y-z_-|\,|x-z_+|}{|x-z_-|\,|y-z_+|}.$$

This distance is dimensionless: the units cancel in the cross-ratio. The normalization $(Q,P) \mapsto (Q/a_S, P/b_S)$ maps the ellipse to the unit disk and preserves that ratio, since lengths along a single straight line are multiplied by a common factor. It thus transports the metric to the Beltrami–Klein model. Neither this interior distance nor the distance between the foci is identified with an atomic length.

15.6.4 Bohr radius: a consequence of the action scale

In the electrodynamic realization, the Bohr scale is expressed as $a_{\mathrm{B}} = \hbar/(m_e c\alpha)$ [181]. In the chain of the present article, that reading follows the constructions of action, vacuum, fine structure and electron mass. Here $\hbar, m_e, c, \alpha > 0$ are received in the same dimensional realization.

Proposition 15.17 (Areal expression of the Bohr scale)**.** *In that realization,*

$$a_{\mathrm{B}} = \frac{a_S b_S}{2m_e c\alpha} = \frac{\mathcal{A}(2\pi)}{2\pi m_e c\alpha} = \frac{\bar{\lambda}_C(m_e)}{\alpha}, \qquad \bar{\lambda}_C(m_e) = \frac{\hbar}{m_e c}. \tag{15.12}$$

Proof. The identities $a_S b_S = 2\hbar$ and $\mathcal{A}(2\pi) = 2\pi\hbar$, substituted into the Bohr expression, give the first two terms. The definition of $\bar{\lambda}_C(m_e)$ gives the third. □

The numerator has the dimension of action and $m_e c$ that of momentum; the quotient is a length. This composition uses the product of the semiaxes and the other three quantities: it does not assert that the Bohr scale is a semiaxis, a focus, a Hilbert distance or the modular radius. Nor does it equate the electron mass with the circular quantum of order 108. The notation a_{B} avoids confusing this length with the normalization a_0 of the area operator.

15.6.5 Inductive–capacitive realization and constitutive continuity

The electrodynamic bridge of the ellipse preserves an additional chart: a clock $\omega > 0$ and a reference impedance $Z_* > 0$. The construction recovered from the corpus defines

$$L_\eta = \frac{Z_*}{\omega}e^{-\eta}, \qquad C_\eta = \frac{1}{Z_*\omega}e^{\eta}, \qquad Z_\eta = \sqrt{L_\eta/C_\eta}. \tag{15.13}$$

Here C_η is capacitance, distinct from the counterangle C^*.

Proposition 15.18 (Dynamic product and constitutive quotient)**.** *In the preceding chart,*

$$L_\eta C_\eta = \omega^{-2}, \qquad \frac{Z_\eta}{Z_*} = e^{-\eta} = \frac{b_S}{a_S} = R_{\mathrm{el}}^2. \tag{15.14}$$

With $Q = \sqrt{Z_*}\,q$, $P = \Phi/\sqrt{Z_*}$, the energy $H_{LC} = q^2/(2C_\eta) + \Phi^2/(2L_\eta)$ takes the form

$$H_{LC} = \frac{\omega}{2}(e^{-\eta}Q^2 + e^{\eta}P^2).$$

On $\gamma(\theta)$, its electric and magnetic components are $U_E = \hbar\omega\cos^2\theta$ and $U_B = \hbar\omega\sin^2\theta$; their sum is $\hbar\omega$.

Proof. The product in (15.13) cancels the exponents and Z_*; its quotient is $Z_*^2 e^{-2\eta}$. Taking the positive root and using (15.11) gives (15.14). Substituting $q = Q/\sqrt{Z_*}$ and $\Phi = \sqrt{Z_*}P$ into H_{LC} gives its balanced form. Finally, the semiaxes of γ cancel the factors $e^{\mp\eta}$, and $\cos^2\theta + \sin^2\theta = 1$. □

This bridge distinguishes the shape, the clock and the transducer. The choice of dynamic direction must agree with the symplectic convention: for the equations $\dot{Q} = \partial_P H$, $\dot{P} = -\partial_Q H$, the preceding parametrization satisfies $\dot{\theta} = -\omega$. The energy identities do not depend on reversing that path.

The vacuum resolvent reader acts on the channels already produced $q_\pm = e^{-(A_{\mathrm{rad}} \pm C^*_{\mathrm{rad}})}$ and the incidence 90/120. Its responses $r_\pm = (1 - q_\pm^{90})/(1 - q_\pm^{120})$ express $\widehat{Z} = r_-/r_+$ and $\widehat{c} = (r_+ r_-)^{-1}$. These are spectral responses, not geometric radii. The identity (15.14) fixes the LC realization here; identifying it with the constitutive impedance of the vacuum preserves the map between both readers and their charts. Their values are not equated merely because they share the angular antecedent.

15.6.6 Canonical covariance and reciprocity of the action ellipse

The preceding construction determines two complementary invariants: the product of the semiaxes fixes the action area and their quotient preserves the anisotropy of the angular pair. We now specify the canonical realization in which these semiaxes are two standard deviations. The starting structure is the ellipse already obtained from the angular and action expressions of APP–TRIT–TPK; the operator realization is established after that construction.

In this subsection we write $\eta = \eta_{\mathrm{el}} = \operatorname{artanh}(C^*/A)$, with $|C^*/A| < 1$ and $\hbar > 0$. The angular quotient uses the same unit for both its components. In the balanced coordinates (Q, P), both of square-root-of-action dimension, we define

$$V_\eta := \frac{1}{4}\begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} e^{\eta} & 0 \\ 0 & e^{-\eta} \end{pmatrix}. \tag{15.15}$$

This positive matrix has entries of action dimension. Its interpretation as a quantum covariance is specified by the following realization.

Proposition 15.19 (Gaussian realization of the action covariance)**.** *Consider the canonical representation on $L^2(\mathbb{R}, dQ)$, with $\widehat{Q} f(Q) = Q f(Q)$ and $\widehat{P} f(Q) = -i\hbar f'(Q)$, initially on the Schwartz space $\mathcal{S}(\mathbb{R})$. The normalized state*

$$\psi_\eta(Q) = \left(\pi\hbar e^{\eta}\right)^{-1/4} \exp\!\left(-\frac{Q^2}{2\hbar e^{\eta}}\right) \tag{15.16}$$

has zero means and covariance matrix V_η. In particular,

$$(\Delta Q)^2 = \frac{\hbar e^{\eta}}{2}, \qquad (\Delta P)^2 = \frac{\hbar e^{-\eta}}{2}, \qquad \operatorname{Cov}(Q, P) = 0, \qquad \det V_\eta = \frac{\hbar^2}{4}. \tag{15.17}$$

This realization saturates the Robertson–Schrödinger inequality.

Proof. The preceding operators are essentially self-adjoint on $\mathcal{S}(\mathbb{R})$, which is a common invariant core; on this space, $[\widehat{Q}, \widehat{P}] = i\hbar I$. For $\widehat{Y} = (\widehat{Q}, \widehat{P})$, the symmetric covariance is

$$V_{jk} = \frac{1}{2}\left\langle (\widehat{Y}_j - \langle \widehat{Y}_j \rangle)(\widehat{Y}_k - \langle \widehat{Y}_k \rangle) + (\widehat{Y}_k - \langle \widehat{Y}_k \rangle)(\widehat{Y}_j - \langle \widehat{Y}_j \rangle) \right\rangle.$$

With $s = \hbar e^{\eta} > 0$, let $I_s = \int_{\mathbb{R}} e^{-Q^2/s}\, dQ$. The product of two equal integrals, evaluated in polar coordinates, gives $I_s^2 = 2\pi \int_0^\infty r e^{-r^2/s}\, dr = \pi s$. Moreover, the integral of $\frac{d}{dQ}(Q e^{-Q^2/s})$ is zero, owing to decay at both endpoints; hence $\int_{\mathbb{R}} Q^2 e^{-Q^2/s}\, dQ = s I_s/2$. These identities and parity prove normalization, the zero mean and $\langle \widehat{Q}^2 \rangle = s/2$. Moreover, $\widehat{P}\psi_\eta = i e^{-\eta} Q \psi_\eta$. Therefore, $\langle \widehat{P} \rangle = 0$, $\|\widehat{P}\psi_\eta\|^2 = \hbar e^{-\eta}/2$ and $\operatorname{Re}\langle \widehat{Q}\psi_\eta, \widehat{P}\psi_\eta \rangle = 0$.

For any normalized state in that domain, Cauchy–Schwarz applied to $(\widehat{Q} - \langle \widehat{Q} \rangle)\psi$ and $(\widehat{P} - \langle \widehat{P} \rangle)\psi$ gives

$$(\Delta Q)^2 (\Delta P)^2 \geq \operatorname{Cov}(Q, P)^2 + \frac{1}{4}\left|\langle [\widehat{Q}, \widehat{P}] \rangle\right|^2 = \operatorname{Cov}(Q, P)^2 + \frac{\hbar^2}{4}.$$

The computed variances attain equality. Equivalently, the operator

$$a_\eta = \frac{e^{-\eta/2}\widehat{Q} + ie^{\eta/2}\widehat{P}}{\sqrt{2\hbar}}$$

satisfies $[a_\eta, a_\eta^\dagger] = I$ and $a_\eta \psi_\eta = 0$, by direct substitution. □

Proposition 15.20 (Area, dispersion, and reciprocal transport)**.** *The action ellipse is exactly the covariance sublevel set*

$$\mathcal{E}_\eta = \left\{ y = (Q,P)^\mathsf{T} : y^\mathsf{T} V_\eta^{-1} y \le 4 \right\} = \left\{ \frac{Q^2}{a_S^2} + \frac{P^2}{b_S^2} \le 1 \right\}. \tag{15.18}$$

Its area is $h = 2\pi\hbar$, its semiaxes are $a_S = 2\Delta Q$, $b_S = 2\Delta P$, and

$$\frac{\Delta P}{\Delta Q} = e^{-\eta} = \frac{b_S}{a_S} = R_{\mathrm{el}}^2. \tag{15.19}$$

The canonical rotation J_2 of Section 15.6 transports V_η to $V_{-\eta}$ and preserves the symplectic form.

Proof. Inverting the diagonal matrix (15.15) gives (15.18). Its semiaxes are the positive roots of four times its variances, and

$$\mathrm{Area}(\mathcal{E}_\eta) = 4\pi\sqrt{\det V_\eta} = 2\pi\hbar = h,$$

in agreement with the areal law (15.10). The ratio of the positive roots proves (15.19). If $M_\eta = \mathrm{diag}(e^{\eta/2}, e^{-\eta/2})$, then

$$V_\eta = M_\eta V_0 M_\eta^\mathsf{T}, \qquad M_\eta^\mathsf{T} J_2 M_\eta = J_2, \qquad J_2 V_\eta J_2^\mathsf{T} = V_{-\eta}.$$

These identities follow by multiplication. In operator terms, $(\widehat{Q}', \widehat{P}') = (-\widehat{P}, \widehat{Q})$ preserves $[\widehat{Q}', \widehat{P}'] = i\hbar I$. The permutation $S_2 : (Q,P) \mapsto (P,Q)$ also exchanges the variances but reverses the sign of the commutator and of the symplectic form. Thus covariance transport respects the distinction between canonical reciprocity and orientation reversal already established in Section 15.6. □

Position, momentum, and dimensional transduction. Let $\lambda_{xp} > 0$ be a factor of the dimensional chart, with the dimension of momentum per unit length. The transformation

$$Q = \sqrt{\lambda_{xp}}\, x, \qquad P = \frac{p_x}{\sqrt{\lambda_{xp}}}$$

realizes the balanced pair as a position x and its conjugate momentum p_x; it preserves $dQ \wedge dP = dx \wedge dp_x$ and $[\widehat{x}, \widehat{p}_x] = i\hbar I$. By transformation of covariance,

$$V_\eta^{(x,p_x)} = \frac{\hbar}{2}\begin{pmatrix} e^\eta/\lambda_{xp} & 0 \\ 0 & \lambda_{xp} e^{-\eta} \end{pmatrix}, \qquad \Delta x\, \Delta p_x = \frac{\hbar}{2}, \qquad \frac{\Delta p_x}{\Delta x} = \lambda_{xp} e^{-\eta}. \tag{15.20}$$

The dispersion ratio acquires the dimension of momentum per length here. By contrast, (15.19) belongs to the balanced chart. For the inductive–capacitive realization already defined, (15.14) supplies the dimensionless identity $Z_\eta / Z_* = e^{-\eta} = \Delta P / \Delta Q$. Each reading uses its declared transduction and preserves the same action determinant.

Contour energy and mean energy of the state. The inductive–capacitive realization has Hamiltonian

$$H_\eta(Q,P) = \frac{\omega}{2}\left(e^{-\eta} Q^2 + e^{\eta} P^2 \right), \qquad \omega > 0.$$

On $\partial\mathcal{E}_\eta$, its value is $H_\eta = \hbar\omega$, by the semiaxis equations. In the quantum realization of Proposition 15.19,

$$\langle \widehat{H}_\eta \rangle_{\psi_\eta} = \frac{\omega}{2}\left(e^{-\eta}\frac{\hbar e^{\eta}}{2} + e^{\eta}\frac{\hbar e^{-\eta}}{2}\right) = \frac{\hbar\omega}{2}. \tag{15.21}$$

The two-standard-deviation contour therefore has twice the mean energy of this Gaussian state. The areal law fixes action normalization; the angular pair fixes its distribution between the two coordinates; the canonical representation and explicit state realize that distribution as covariance. Saturation of uncertainty is thus established in the realization specified by its operators and its state.

Chapter 16

Dodecaphase Mixing Operator: CKM–CP Phase and Construction of the Pre-Dimensional Electron Reader

CKM Mixing and Charge-Parity Violation

The incidence sextet $(b, p, q, h_6, o_8, t) = (4, 5, 7, 6, 8, 3)$ fixes the rational transformation

$$M_{\mathrm{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \qquad \det M_{\mathrm{CKM}} = -\frac{3857}{6} \neq 0. \tag{16.1}$$

With

$$h_{\mathrm{deg}} = \frac{C^*_{\mathrm{deg}}}{6}, \qquad \Delta_{\mathrm{deg}} = \frac{180}{\pi}\Delta_4,$$

we define

$$\begin{pmatrix} \theta_{12} \\ \theta_{23} \\ \theta_{13} \end{pmatrix} = M_{\mathrm{CKM}} \begin{pmatrix} A_{\mathrm{deg}} \\ h_{\mathrm{deg}} \\ \Delta_{\mathrm{deg}} \end{pmatrix}, \qquad \delta_{\mathrm{CP,deg}} = 9A_{\mathrm{deg}}.$$

To prevent any ambiguity of chart, it is fixed

$$s_{ij} := \sin\!\left(\frac{\pi\theta_{ij}}{180}\right), \qquad c_{ij} := \cos\!\left(\frac{\pi\theta_{ij}}{180}\right).$$

The standard parametrization produces a unitary matrix V, and the Jarlskog invariant is

$$J = c_{12}c_{23}c_{13}^2 s_{12}s_{23}s_{13} \sin\!\left(\frac{\pi\delta_{\mathrm{CP,deg}}}{180}\right) = 3.1189723326632974 \times 10^{-5} \neq 0. \tag{16.2}$$

Nonvanishing expresses CP violation in the constructed model. Comparison with the Particle Data Group is undertaken afterward; it does not adjust (16.1). The PMNS sector requires its own mixing operator and is identified neither with CKM nor with a rank-one diagonal phase.

16.1 Dimensionless construction of the electronic scalar from APP incompatibility and the dodecaphase state

The electron appears in HMT as the point at which six preceding and mutually typed constructions are assembled: the incompatibility between the two arithmetic sheets of APP, the orientation induced by the involution of the areal–volumetric hinge, two integer quotients from the APP–TPK census, the dodecaphasic closure of α_{HMT}, the global defect Δ_4, and the unique center of the nonadic atlas. The route band then adds electron–positron conjugation. The MeV chart and the comparison with CODATA are subsequent to all of them.

This precedence is part of the mathematical content: the factors, their types and their causal order are determined before opening the tabulated chart, which intervenes only in the subsequent metrological comparison.

16.1.1 Electronic scalar $\mathfrak{e}_{\mathrm{HMT}}$ and causal graph of its factors APP–TPK, α_{HMT}, Δ_4 and nonadic atlas

The internal output is the dimensionless scalar.

$$\boxed{\mathfrak{e}_{\mathrm{HMT}} = \frac{\sqrt{3}}{4}\left[\exp\!\left(\frac{\varphi}{\pi^2}\right) - \frac{396}{18}\,\alpha_{\mathrm{HMT}}^3\right]\left[1 + \frac{270}{18}\,\Delta_4\right].} \tag{16.3}$$

The identities

$$\frac{396}{18} = 22 = 27 - 5, \qquad \frac{270}{18} = 15 = \binom{6}{2} = 3 \cdot 5$$

allow the abbreviated notation to be recovered

$$\mathfrak{e}_{\mathrm{HMT}} = \frac{\sqrt{3}}{4}\left(e^{\varphi/\pi^2} - 22\alpha_{\mathrm{HMT}}^3\right)(1 + 15\Delta_4).$$

Here the letter e inside the exponential is Euler's number; the subscript of $\mathfrak{e}_{\mathrm{HMT}}$ denotes the electronic section.

The causal graph of the equation is

$$\begin{aligned}
\text{APP projectors in dimension three} &\longrightarrow \sqrt{3}/4,\\
\text{TPK calendar and areal–volumetric sheets} &\longrightarrow F_{\mathrm{av}},\ \pi/\varphi^2,\ \varphi/\pi^2,\\
\text{dodecaphasic closure} &\longrightarrow \alpha_{\mathrm{HMT}},\\
\text{APP–TPK census and electronic family} &\longrightarrow 396,\ 18,\ 270,\ 22,\ 15,\ 5,\\
\text{global readings of } \pi, e \text{ and register} &\longrightarrow \Delta_4,\\
\text{nonadic atlas and anti-section} &\longrightarrow (5,5),\ \text{electron–positron band.}
\end{aligned}$$

No arrow in this diagram receives as an argument the reference electronic mass.

16.1.2 1st factor: exact incompatibility of the APP leaves

For $d \geq 2$, let

$$\mathscr{A}_d = \{1, \ldots, 9\}^d$$

with uniform measure, and let

$$\Sigma_d(x) = \mathrm{dr}(x_1 + \cdots + x_d), \qquad \Pi_d(x) = \mathrm{dr}(x_1 \cdots x_d)$$

the additive and multiplicative observables. Their orthogonal conditional expectations on $\ell^2(\mathscr{A}_d)$ are $P_{\Sigma,d}$ and $P_{\Pi,d}$. The proposition 1.55 proves by exact enumeration that

$$\left\|[P_{\Sigma,3}, P_{\Pi,3}]\right\| = \frac{\sqrt{3}}{4}.$$

Proposition 16.1 (Electronic prefactor APP content)**.** *The prefactor $\sqrt{3}/4$ of equation (16.3) is the exact norm of the noncommutativity between the sum and product sheets in dimension three. It is not a geometric factor introduced from the electronic formula.*

Proof. The joint table of Σ_3 and Π_3 determines the principal angles between $\mathrm{Ran}\, P_{\Sigma,3}$ and $\mathrm{Ran}\, P_{\Pi,3}$. If s is one of its singular values, the corresponding contribution to the commutator norm is $s\sqrt{1 - s^2}$. Enumeration of the 9^3 words gives at most

$$s^2(1 - s^2) = \frac{3}{16}, \qquad s^2 = \frac{1}{4}.$$

Therefore the maximum norm is $\sqrt{3/16} = \sqrt{3}/4$, before introducing α_{HMT}, Δ_4, a unit of energy or a reference mass. □

The physical function of the factor is thereby fixed: it measures the maximal incompatibility of the two arithmetic readings on the same ternary support. The electron does not inherit two independent numbers, one additive and the other multiplicative; it inherits the exact obstruction to rendering the two sheets simultaneously compatible. The prefactor preserves, within the mass law, the local memory of the APP disagreement from which the route arises.

16.1.3 2nd factor: the exchange involution between π and φ

The primitive calendar of modes (Σ, Π, Π) induces, on the areal and volumetric sheets, the matrix

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

theorem 60.7 obtains it by counting the transitions $\Sigma \to \Pi$, $\Pi \to \Pi$ and $\Pi \to \Sigma$, each once. The matrix is not postulated from the golden ratio: its eigenvalues φ and $-\varphi^{-1}$ appear after the incidence has been constructed.

In the second-order memory, $\mathrm{Sym}^2(F_{\mathrm{av}})$ has spectrum

$$\varphi^2, \qquad -1, \qquad \varphi^{-2}.$$

Thus, φ^{-2} is the positive stable mode of the hinge. The closure mark π, applied just once to the areal return, produces

$$\frac{\pi}{\varphi^2}.$$

This reason preserves the closure–self-scaling order. The electronic law uses its opposite orientation.

Theorem 16.2 (Exact Inversion of the Electronic Hinge)**.** *Let*

$$\mathscr{R}(x, y) = \frac{x}{y^2}, \qquad \mathsf{J}(x, y) = (y, x).$$

Then $\mathsf{J}^2 = I$ *and*

$$\mathscr{R}(\pi, \varphi) = \frac{\pi}{\varphi^2}, \qquad (\mathscr{R} \circ \mathsf{J})(\pi, \varphi) = \frac{\varphi}{\pi^2}.$$

Consequently,

$$\frac{\varphi}{\pi^2} = \frac{1}{\varphi^3 \left(\pi/\varphi^2\right)^2}, \qquad e^{\varphi/\pi^2} = \exp\left[\frac{1}{\varphi^3 \left(\pi/\varphi^2\right)^2}\right].$$

Proof. The first pair of identities is obtained by applying $\mathscr{R}$ before and after the involution J. For the reparametrization,

$$\varphi^3 \left(\frac{\pi}{\varphi^2}\right)^2 = \frac{\pi^2}{\varphi},$$

and its inverse is φ/π^2. The last equality follows by applying the exponential. □

The simultaneous presence of π and φ therefore has a precise structural meaning. π/φ^2 reads areal closure over quadratic volumetric memory; φ/π^2 reads interior self-growth over quadratic closure. They are not two similar quotients chosen for convenience. They are the two orientations of the same reader, exchanged by an exact involution. The term e^{φ/π^2} publishes the interior orientation of that hinge in the electron section.

16.1.4 3rd and 4th factors: the integers 22, 15 and the center 5

The census of the APP chart employed by the electronic family provides three integer sums:

$$\Sigma(\mathrm{APP}) = 396, \qquad \Sigma(\text{centro } 2\times 2) = 18, \qquad \Sigma(\text{canal}) = 270.$$

The first admits the ring decomposition.

$$396 = 18 + 72 + 108 + 198.$$

The kernel 2×2 is therefore an explicit substructure of the complete chart, and the integer 270 belongs to the global channel of the same APP–TPK census. The sum of the even rings is

$$18 + 108 = 126, \qquad \frac{396}{126} = \frac{22}{7}.$$

This last quotient is the rational closure seal of the chart, not an identification of $22/7$ with the number π generated by the coinductive reader. It does show that the numerator 22 already serves an internal function before appearing in the electronic law.

Theorem 16.3 (Internal Origin of the Electronic Coefficients)**.** *In the electronic family fixed by the APP chart and its global channel,*

$$\boxed{22 = \frac{\Sigma(\mathrm{APP})}{\Sigma(\text{centro})} = \frac{396}{18} = 27 - 5,}$$

$$\boxed{15 = \frac{\Sigma(\text{canal})}{\Sigma(\text{centro})} = \frac{270}{18} = 3 \cdot 5 = \binom{6}{2}.}$$

In particular, the coefficients are determined without consulting Z_0, the electron mass or a chart of units.

Proof. The first two equalities are exact divisions of the finite census:

$$396 = 22 \cdot 18, \qquad 270 = 15 \cdot 18.$$

By defining

$$p := 27 - 22,$$

one obtains $p = 5$, and then

$$15 = 3p = 3 \cdot 5 = \binom{6}{2}.$$

All quantities belong to the integer domain of the chart and the family before metrological transduction. □

16.1.4.1 Persistence selector in the central return

The same pair has a dynamic construction that also fixes its sign without consulting a mass. The four orientations of the central seed $(5, 5)$ return to that cell for the first time after twenty-seven steps. The typed return space is

$$\mathcal{K}_e = \mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3, \qquad |\mathcal{K}_e| = 27.$$

Moreover, the microfiber of π contains the five genealogically distinct regions

$$\mathscr{R}_\pi = 3_{++} + 1_{--} + 1_{\perp}.$$

The regional gauge diag_c and, within a repeated phase, the internal order of U_6 define the injection

$$\iota : \mathscr{R}_\pi \hookrightarrow \mathcal{K}_e$$

through the five distinct places

$$(4,0), \quad (5,0), \quad (6,0), \quad (6,1), \quad (6,2).$$

The first two correspond, respectively, to the transverse region and the mutated return; the last three separate the positive regions by their tritic sheet. The common visible word does not intervene as the inverse of the map.

Theorem 16.4 (Internal selector of the electronic coefficients)**.** *With the central return and the five typed regions fixed, the coefficients of the electronic equation are*

$$\boxed{n_e = -\left|\mathcal{K}_e \setminus \iota(\mathscr{R}_\pi)\right| = -22, \qquad m_e^{\mathrm{tor}} = \left|\mathscr{R}_\pi \times \mathbb{F}_3\right| = +15.}$$

The first symbol records the boundary deficit; the second, the retained torsional memory.

Proof. The five places above are distinct, so ι is injective and

$$|\mathcal{K}_e \setminus \iota(\mathscr{R}_\pi)| = 27 - 5 = 22.$$

Each of the five regions preserves three triticale leaves, hence

$$|\mathscr{R}_\pi \times \mathbb{F}_3| = 5 \cdot 3 = 15.$$

The orientation rule assigns a negative sign to states removed from the shell and a positive sign to retained memory. The construction opens only the center, the return of twenty-seven states, and the regional identities; it does not open an observed mass, an impedance, or a calibrated signature. □

The number 22 measures how many central kernels of weight 18 are recomposed by the APP sum 396. The number 15 measures how many of those same kernels are recomposed by the channel 270, and simultaneously counts the pairs of a hexad. The number 5 is not an additional parameter:

$$5 = 27 - 22, \qquad 15 = 3 \cdot 5.$$

The three expressions display the same closure from depth 27, from the central normalization, and from hexadic combinatorics.

16.1.4.2 Internal derivation of coefficients 22 and 15 through the APP–TPK censuses

The historical exploration first located $p = 5$ through a finite comparison with Z_0, and from there wrote $(27 - p, 3p) = (22, 15)$. That procedure retains value as a history of discovery and as an external check. It is not, however, the proof that governs the law: the subsequent count

$$\frac{396}{18} = 22, \qquad \frac{270}{18} = 15$$

constructs both integers within APP–TPK and recovers $p = 5$ as a consequence. The history explains how the pair was found; the census explains why it belongs to the equation.

16.1.5 5.º factor: the cubo of the closure α_{HMT}

The closure of α_{HMT} precedes Dimensional Mechanics. As established in section 30.12, the twelve base-thousand blocks of π, e, φ and the seal K combine through the reversible carry

$$P_m + E_m - \Phi_m - K_m + c_{m+1} = A_m + 1000c_m, \qquad c_{13} = c_1 = 0.$$

The output is

$$A_\alpha = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 663)$$

and, by concatenation, the exact dodecaphasic window

$$\alpha_{12} = \pi_{12}(\alpha_{\mathrm{HMT}}) = \frac{729735256928380099728510547238066 3}{10^{36}}.$$

Telescoping the carries proves the complete integer identity; the compatible prolongation constructs $\alpha_{\mathrm{HMT}} = \varprojlim_N A^{(N)}$, and α_{12} is its thirty-six-digit rational projection. The comparison with the fine-structure constant is performed subsequently and does not intervene in its construction.

The electronic law takes the third power

$$\alpha_{\mathrm{HMT}}^3.$$

Mathematically, this is the multiplicative evaluation of the third symmetric power of the same scalar already closed; neither three metrological copies nor three different parameters are introduced. Its contribution is

$$22\alpha_{\mathrm{HMT}}^3 = 0.000008549065992005517270149798075982 28\ldots.$$

Proposition 16.5 (Statute of the Electronic Cubic Channel)**.** *The term* $22\alpha_{\mathrm{HMT}}^3$ *of the electronic law depends only on the prior closure of* α_{HMT}*, on internal multiplication, and on the integer* 22 *of theorem 16.3. It does not depend on a reading of the electron mass.*

Proof. α_{12} is the rational number fixed by the concatenation and carry identity, and the compatible family of windows determines the section α_{HMT}. The projection α_{12}^3 certifies the printed digits of α_{HMT}^3. The coefficient 22 is determined by $396/18$. The composition of both operations contains only data constructed before the energy chart. □

This proposition fixes exactly what is used for the electron. The third power is the cubic degree of this channel in the published law. There is no need here to postulate a universal rule according to which every occurrence of α^d automatically represents a dimension d; a universal statement of that kind would require its own theorem of multiplicativity among degrees. The electron genealogy is complete without turning that generalization into a premise.

16.1.6 6th factor: minimal torsion and global defect Δ_4

The terms *minimal torsion* and *global defect* denote a single invariant here, not two distinct corrections. The object used by the mass law is

$$\boxed{\Delta_4 = \frac{\pi - e\log\pi}{270}\left(1 + \frac{\pi}{729}\right).} \tag{16.4}$$

Its inputs are the already fixed HMT readings of π and e, the global channel 270, and the interior cell 729. Consequently,

$$\Delta_4 = 0.000111196422692140054051134786120679 09\ldots.$$

The first ratio measures the signed defect $\pi - e\log\pi$ in units of the 270 channel; the second factor couples that ratio to the interior cell through $\pi/729$. This reading adds no parameters to the formula: it separates its two constituent operations. The chapter 35 shows that it is a pre-dimensional mathematical input and distinguishes it from the operator that eventually realizes it as a change of sheet.

Proposition 16.6 (Globality of the Electronic Defect)**.** Δ_4 *is a global invariant preceding the species. In the electronic law it enters through*

$$1 + 15\Delta_4 = 1.001667946340382100810767021791810 19\ldots.$$

It is not a corrector chosen after knowing the residue of the electron mass.

Proof. The equation (16.4) contains neither a species label, a mass, nor a unit. The integer 15 has already been constructed by $270/18$. Therefore, the composition $1 + 15\Delta_4$ is determined by the global data and the internal census. □

Δ_4 must be distinguished from the other blocks of the mass character and from the other notions of torsion. In particular, it is neither s_{270}, nor the quadratic invariant $135\Delta_4^2$, nor the angular residue $\varepsilon_{\mathrm{res}}$, nor the geometric Cartan–Holst torsion. Substituting one for another would repeat or alter the level of the correction. For the electron, Δ_4 preserves the global memory of refinement; the factor 15 states how many times the electron family reads it. The qualifier *minimal* preserves the historical name of this invariant and is not used as a substitute for a variational theorem of minimality.

16.1.7 Uniqueness of the central cell (5,5) as the genealogical support of the electron

Let

$$C_{81} = \{1, \dots, 9\}^2$$

the nonadic atlas and

$$\rho(r, c) = (10 - r, 10 - c)$$

its central inversion. theorem 90.11 proves that the atlas contains a unique generation-zero cell and that this cell bears the positional label e_{core}.

Proposition 16.7 (Uniqueness of the Electronic Center)**.** *The involution ρ has a unique fixed point:*

$$\boxed{(r, c) = (5, 5).}$$

Its uniqueness is combinatorial and does not depend on the value of $\mathfrak{e}_{\mathrm{HMT}}$.

Proof. The equation $(r, c) = \rho(r, c)$ is equivalent to

$$2r = 10, \qquad 2c = 10.$$

In $\{1, \dots, 9\}^2$ it has the unique solution $r = c = 5$. The evaluation of the mass does not appear in the argument. □

Corollary 16.8 (Universality of the Central Section)**.** *Every admissible automorphism of the atlas that intertwines the central inversion preserves the cell* $(5, 5)$.

Proof. If $f\rho = \rho f$ and $\rho(5, 5) = (5, 5)$, then $\rho f(5, 5) = f\rho(5, 5) = f(5, 5)$. Therefore $f(5, 5)$ is another fixed point of ρ; uniqueness forces $f(5, 5) = (5, 5)$. □

This is the structural reason why the electronic class is the same in every realization preserving the atlas and its inversion: a local decimal does not recur by chance; rather, every admissible chart must transport the same fixed point and the same chain of invariants. The physical reading of this uniqueness identifies each electron with a realization of that central class; the proposition proves the internal universality of the atlas.

In the electronic family fixed by the construction, this point is the geometric anchor of the section. Four objects should not be confused:

$$(5, 5), \qquad e_{\mathrm{core}}, \qquad v_e = (0, 0, 0, 0, 0), \qquad \text{the raw signature of the central cell.}$$

The first is a position; the second, its internal label; the third, the chosen origin in the fine action lattice; the fourth preserves the raw atlas data and need not be the zero vector. Their coordination explains why the electron occupies the center without erasing the genealogy of the representations describing it.

Oriented electronic routes e_{++}, e_{--}, envelope e_{env}, action origin and signatures of the central cell The central cell carries two oriented histories of twelve events. Writing v^2 for the concatenation of two copies of a sextuple, the active route of orientation ++, its opposite history, and the historical envelope are, respectively,

$$\begin{aligned} e_{\mathrm{act}} = e_{++} &= (2,2,5,-4,-3,-2)^2, \\ e_{--} &= (4,3,2,-2,-2,-5)^2, \\ e_{\mathrm{env}} &= (4,3,5,-4,-3,-5)^2. \end{aligned} \tag{E-centro-1}$$

The two stories publish the same word

$$\tau_{12} = {+}{+}{+}{-}{+}{+}{+}{-},$$

but they are not the same route. The envelope takes the absolute maximum coordinate by coordinate while preserving the sign; it records amplitudes, but flattens the genealogical order and does not replace e_{act}.

The other readers of the center live in distinct codomains:

$$\begin{array}{rcll} v_e & = & (0,0,0,0,0), & \text{affine origin of the action lattice,} \\ R(5,5) & = & (24,0,12,0,-12), & \text{raw APP signature of the cell,} \\ \mathrm{Sig}_{\mathrm{masa}}^{\Gamma_{108}} & = & (-12,-4,12,-6,12), & \text{even mass projection,} \\ \mathrm{Sig}_{\mathrm{carga}}^{\Gamma_{108}} & = & (0,0,0,0), & \text{odd charge projection.} \end{array} \tag{E-centro-2}$$

The fact that the charge word has zero odd signature does not turn the route into zero, nor does it identify the raw signature with the affine origin. Position, active route, opposite route, envelope, origin and both signatures are coordinated objects, not six notations for the same vector.

16.1.8 Electron–positron conjugate orbit on the oriented path band

Electronic identity does not end at the central point or at the mass value. Material conjugation is realized on dodecaphasic words. Let

$$C_M(\mathbf{t}) = -\operatorname{rev}(\mathbf{t})$$

be the oriented anti-section and let $\chi \in \{-1,+1\}$ be the charge orientation. The band is the class

$$[(\mathbf{t},\chi)]_M = \{(\mathbf{t},\chi),(C_M\mathbf{t},-\chi)\}$$

of definition 92.28. The band preserves its even components and exhibits two sections with opposite odd components.

Theorem 16.9 (Electron and positron as sections of a band)**.** *For the band character of theorem 92.31,*

$$m(C_M\mathbf{t},-\chi) = m(\mathbf{t},\chi).$$

In the electronic channel,

$$(n,\chi) = (-22,+1) \quad \longleftrightarrow \quad (+22,-1)$$

and both pairs produce

$$\chi n = -22.$$

Therefore, electron and positron have the same mass and opposite charge orientations.

Proof. If E_+ and E_- are, respectively, the even and odd components of the character with respect to C_M, then

$$E_+(C_M\mathbf{t}) = E_+(\mathbf{t}), \qquad E_-(C_M\mathbf{t}) = -E_-(\mathbf{t}).$$

From here,

$$E_{\text{band}}(C_M\mathbf{t}, -\chi) = E_{\text{band}}(\mathbf{t}, \chi),$$

and the equality of mass is obtained by applying the exponential. In the electronic term,

$$(+1)(-22) = (-1)(+22) = -22.$$

□

The finite closure of proposition 92.33 reinforces this interpretation: over the eighty-one CT–108 routes, neither pure sign nor pure reversal preserves the taxonomy, whereas − rev does preserve it, with

$$81 = 54 + 9 + 9 + 9.$$

The antiparticle is therefore neither a negative mass nor an independent copy. It is the other oriented section of the same band: it preserves the even invariants that determine mass and reverses the odd invariants that determine charge and orientation.

16.1.9 Basal stage of the electronic closure

Theorem 16.10 (Holographic composition of the electronic section)**.** *With the objects constructed in the preceding sections fixed, the dimensionless electronic section is*

$$\mathfrak{e}_{\text{HMT}} = \left\|[P_{\Sigma,3}, P_{\Pi,3}]\right\| \left[\exp\big((\mathscr{R}\circ\mathsf{J})(\pi,\varphi)\big) - \frac{\Sigma(\text{APP})}{\Sigma(\text{centro})}\,\alpha_{\text{HMT}}^3\right]\left[1 + \frac{\Sigma(\text{canal})}{\Sigma(\text{centro})}\,\Delta_4\right].$$

Equivalently,

$$\boxed{\mathfrak{e}_{\text{HMT}} = \frac{\sqrt{3}}{4}\left(e^{\varphi/\pi^2} - 22\alpha_{\text{HMT}}^3\right)(1 + 15\Delta_4).}$$

Its evaluation is

$$\boxed{\mathfrak{e}_{\text{HMT}} = 0.510998922488066072509870877191349437 63\ldots.}$$

This scalar is the basal stage $\mathfrak{e}_e^{(0)}$, not the governing electronic output. The bidirectional beta closure publishes $0.510998950690000808608\ldots$ MeV.

Proof. The proposition 16.1 contributes

$$\left\|[P_{\Sigma,3}, P_{\Pi,3}]\right\| = \sqrt{3}/4.$$

The theorem 16.2 contributes

$$(\mathscr{R}\circ\mathsf{J})(\pi,\varphi) = \varphi/\pi^2.$$

The theorem 16.3 contributes

$$\frac{\Sigma(\text{APP})}{\Sigma(\text{centro})} = 22, \qquad \frac{\Sigma(\text{canal})}{\Sigma(\text{centro})} = 15.$$

The dodecaphasic closure contributes the rational window α_{12} of the coinductive section α_{HMT}, and equation (16.4) contributes the global defect. Substitution yields the abbreviated formula. Successive evaluation of

$$
\begin{aligned}
e^{\varphi/\pi^2} &= 1.17814494259630678081160095680888910\ldots,\\
22\alpha_{\mathrm{HMT}}^{3} &= 0.00000854906599200551727014979807598228\ldots,\\
1+15\Delta_4 &= 1.00166794634038210081076702179181019\ldots
\end{aligned}
$$

gives the declared value. □

The theorem is holographic in a verifiable sense: each local factor retains a distinct layer of the complete system. The commutator retains sheet duality; the exponential retains the interior orientation of the hinge π–φ; 22 and 15 retain the global census and its central kernel; $\alpha_{\mathrm{HMT}}^{3}$ retains fine closure; and Δ_4 retains global refinement. The center determines where the section is anchored, and the band determines how it survives under conjugation. The value does not replace this structure: it is its common evaluation.

16.1.9.1 Anterior and posterior action splitting upon return

The regional realization of the conjugate angular coordinate C^* distinguishes two coordinates on the same internal action line. The coordinate before the return is H_5 and the subsequent coordinate is

$$\eta_{\mathrm{ret}}^{(\mathrm{deg})} = H_5 - \frac{90}{\pi}\mathscr{C}_\pi(\alpha_{\mathrm{HMT}}).$$

The notation $\eta_{\mathrm{ret}}^{(\mathrm{deg})}$ is the canonical sexagesimal coordinate defined on the action branch by $\eta_{\mathrm{ret}}^{(\mathrm{deg})} = C^*/2 - \alpha_{\angle,\mathrm{deg}}$. The preceding equality is evaluated in that chart; it does not introduce a second return coordinate. Therefore, the additive and multiplicative charts of the oriented splitting are

$$\boxed{\delta_{2w} := H_5 - \eta_{\mathrm{ret}}^{(\mathrm{deg})} = \frac{90}{\pi}\mathscr{C}_\pi(\alpha_{\mathrm{HMT}}), \qquad R_{\mathrm{act}} := \frac{\hbar_{\mathrm{HMT}}^{\mathrm{pre}}}{\hbar_{\mathrm{HMT}}^{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}^{(\mathrm{deg})}}.} \tag{16.5}$$

Both coordinates follow from a single pre-return/post-return transport. They are not two Planck constants: the common factor $10^{-34}\mathcal{U}_S$ fixes the dimensional line and cancels exactly in R_{act}. The action ellipse subsequently realizes $\mathrm{Area}(\theta) = \hbar_{\mathrm{HMT}}^{\mathrm{ret}}\theta$, so that the splitting precedes the holonomy reading.

The coupling

$$\kappa_e^{\beta} = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4, \qquad \mathfrak{e}_e^{\beta} = \mathfrak{e}_{\mathrm{HMT}} R_{\mathrm{act}}^{\kappa_e^{\beta}} = 0.5109989506900008086\ldots \tag{16.6}$$

is the bidirectional quantitative closure of the electron. The two independent readers of the electronic register constructed in the following chapter select, without consulting the comparison mass,

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80, 54, 6),$$

and derive exactly κ_e^{β}. The integer 80 comes from the first coordinate of those readers; the word $w_e = 201101$ retains its own residence in the channel of Euler's constant.

Internal Closure of Bidirectional Refinement

The identity (16.5), the reduction $(80, 54, 6)$, and the equation (16.6) constitute internally exact, target–free results on the electronic record. Propagation to fibers of higher rank is formulated by means of the multisection operators.

16.2 Algebraic structure of the electron and action scale

The electronic construction receives the arithmetic state with trajectory memory produced by APP–TRIT–TPK and its numerical coordinates. At this point in the development, π, φ, e and α designate the coordinates already obtained from the same genealogy, in the constructive order set out above. They are not parameters taken from a physical table. The electronic section is a subsequent realization of that joint discrete structure of the continuum: it preserves the cell of origin, the two arithmetic sheets, orientation, carry, and return memory before producing a scalar.

Three objects should be distinguished from the outset. The first is the central section and its transport history; the second is the dimensionless coordinate obtained from it; the third is its realization on an energy or mass line. Equality between two numerical coordinates does not identify these three levels. In particular, Euler's number e, the electronic coordinate ε_e, and a rest energy E_e will have distinct notations.

The aim of this section is to establish the composition

$$\varepsilon_e^{\beta} = \frac{\sqrt{3}}{4}\left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3\right](1 + 15\Delta_4)R_{\mathrm{act}}^{80-54\alpha+6\Delta_4}, \tag{16.7}$$

explaining separately the origin of the prefactor, the two integers of the basal stage, normalization of Δ_4, the action ratio and the return triple. The dimensional chart opens only after this composition. The result does not consist in making a dimensionless number acquire units merely through its resemblance to a tabulated quantity.

The composition brings together factors with different functions. The following table identifies their meaning and refers to the corresponding construction; the proofs are developed in the order of their dependencies.

Factor	Meaning in the composition	Construction and proof
$\sqrt{3}/4$	Norm of the commutator of the projectors on the principal plane.	Equation (16.12).
$\exp(\varphi/\pi^2)$	Exponential character of the electronic orientation of the regional quotient.	Equation (16.18); subsections 10.5.6.1 and 10.5.6.2.
$-22\alpha^3$	Contribution of the regional complement in the central return.	Equations (16.20) and (16.21).
$1 + 15\Delta_4$	Torsional normalization composed with regional and tritic multiplicity.	Equations (16.20), (16.22) and (16.24).
$R_{\mathrm{act}}^{\kappa_e}$	Quotient of the action sections, raised to the character determined by the two readers of the electronic return.	Equations (16.35), (16.42), (16.43) and (16.44).

Table 16.1: Factors of the electronic composition and provenance of their rules.

Identity of the section and composition of its observables. The structural identity of the electron is introduced through the central section and its transport operations. Inversion of the APP chart selects the cell $(5, 5)$ by its fixed-point property. The two arithmetic sheets allow the deformations preserving the sum and product residue to be studied. The result distinguishes the two orientations $(8, 2)$ and $(2, 8)$ and preserves the difference of one quotient unit. This is the first level of information: a cell, an involution, a preserved fiber and a minimal integer defect. No mass evaluation is needed to state or prove it.

The passage to arithmetic projections introduces a second level. The sum and product fibres determine conditional expectations on the same finite space. Their commutator measures dependence on the order of these two projections. The norm $\sqrt{3}/4$ follows from the spectrum of their pairing; the block attaining this norm also preserves the quadratic relations proved below. Thus, the scalar prefactor and the local

algebra have a common provenance, but contain different information. The norm retains the magnitude of the commutator, while the matrices preserve its orientation and its composition law.

Areal–volumetric transfer incorporates the regional coordinates at a third level. The mode calendar determines the incidence matrix and its quadratic action. The closure mark produces the ratio π/φ^2; interchanging the two arguments provides the orientation φ/π^2 used in the electronic exponential character. The two expressions are related by an explicit involution. This relationship explains why a regional coordinate may appear in the electronic equation with a power or in a position different from that in its first reading: the operation applied is part of the mathematical datum that is preserved.

The cubic alpha term belongs to the regional complement of a marked return. Counting its positions yields $27 - 5 = 22$, while the three sheets of each of the five regions yield $3 \cdot 5 = 15$. The proofs preserve the domain of these counts and the orientation determining their signs. The basal formula therefore composes an incompatibility norm, a transfer character, a position deficit, and a torsional memory. Writing it on a single line expresses the composition of these operations; explaining it requires reconstructing them in that order.

The transition from the basal coordinate to the complete coordinate preserves the same central section. The return factor uses the history records, and the two stated readers recover the triple $(80, 54, 6)$ along that trajectory. Multiplication by $R_{\mathrm{act}}^{80-54\alpha+6\Delta_4}$ modifies the evaluation, but keeps its origin and antecedents identified. The basal value constitutes an intermediate stage of the composition; the complete value also includes this return contribution. The physical comparison must use the stage specified by the formula being evaluated.

The action scale intervenes subsequently with a different function. A ratio of two action sections is dimensionless and cancels their common scale. The absolute realization instead uses an action basis, a temporal reader, and the map to energy or mass. This separation allows the construction of the scale and the identity of the electronic section to be preserved simultaneously. The chronological quantum provides a scale reference; the central section determines the structural factor expressed relative to it. The dimensional equations of this chapter keep that composition explicit.

The electron's relevance to the overall argument arises from this convergence of operations. The same arithmetic that produced the regions now determines a central cell, its deformations, and its projections. The fine-structure constant, constructed earlier, acts in this composition as a closure coordinate. Return information determines the complete evaluation, and dimensional realization allows subsequent comparison. Explaining the electron from its structure consists in preserving this succession of objects, maps, and values at every step of the proof.

16.2.1 Center, orientation, and vacancy on a preserved fiber

The cellular support under consideration is $\{1, \dots, 9\}^2$, with both APP evaluations, $r + c$ and rc. The central inversion is

$$\rho(r, c) = (10 - r, 10 - c). \tag{16.8}$$

The center is not chosen by a mass: it is characterized by a property of the support that precedes its physical reading.

Proposition 16.11 (Uniqueness of the center)**.** *The inversion ρ has a unique fixed point,* $(5, 5)$.

Proof. The equations $r = 10 - r$ and $c = 10 - c$ are equivalent to $2r = 2c = 10$. Their unique solution in the nonadic chart is $(5, 5)$. □

The preceding property identifies a cell, not a unique deep history. A route adds the initial orientation choice and its compatible prolongations. In the chart used here the central oriented section starts at $(5, 5)$ with both positive orientations; it will be denoted by Γ_e. Return of its cellular projection does not erase the transported orientations or the carry register of the section with its trajectory memory.

Vacancy is now considered, after the closure of α, as a local operation on this section. Its arithmetic can be isolated in a finite lemma; this isolation does not make it a substitute for the preceding generator of constants.

Definition 16.12 (Central fiber and quotient defect)**.** The central additive fiber is

$$\mathcal{F}_{10} = \{(5+t, 5-t) : t \in \mathbb{Z},\ |t| \le 4\}.$$

In the chart $n = \rho_9(n) + 9q_9(n)$, with $\rho_9(n) \in \{1, \dots, 9\}$, the quotient defect relative to the center is defined as

$$v(t) = q_9(25) - q_9(25 - t^2),$$

when both products have the same nonadic residue.

Lemma 16.13 (Minimal central vacancy)**.** *The only points of $\mathcal{F}_{10}$ that preserve the multiplicative residue of the center are*

$$(5,5), \qquad (8,2), \qquad (2,8).$$

At the two noncentral points, the quotient defect is exactly one.

Proof. The sum is constantly ten, whereas

$$(5+t)(5-t) = 25 - t^2.$$

Preservation of the residue is equivalent to $t^2 \equiv 0 \pmod 9$, that is, to $3 \mid t$. In the integer interval $[-4, 4]$, the solutions are $t = -3, 0, 3$. For the two nonzero solutions,

$$25 = 7 + 9 \cdot 2, \qquad 16 = 7 + 9 \cdot 1.$$

The residue seven is preserved and one unit of quotient is lost. There is no other admissible noncentral displacement in that domain, so the positive defect is minimal. The interchange $t \mapsto -t$ permutes the two outputs and preserves the defect. □

Vacancy here is not a numerical zero: it is a difference between two states with the same residual reading and different quotients. If only the residue were preserved, that difference would disappear. Orientation also distinguishes the two output points. Neither the quotient defect nor that orientation is automatically identified with an electric charge or a mass; their physical realization requires the corresponding readers.

16.2.1.1 Central coordinates, orientation, and regional records

The central structure relates position, evaluation, and transport. At cell $(5, 5)$, the two APP sheets give

$$\mathcal{A}(5,5) = \big((1,1),(7,2)\big), \qquad 5+5 = 1+9\cdot 1, \qquad 5\cdot 5 = 7+9\cdot 2. \tag{16.9}$$

The two ones on the additive sheet are, in order, residue and quotient. Their numerical coincidence preserves two distinct arithmetic functions. On the sum-10 fiber, the two oriented vacancies satisfy

$$\mathcal{A}(8,2) = \mathcal{A}(2,8) = \big((1,1),(7,1)\big). \tag{16.10}$$

The complete additive sheet and the multiplicative residue are preserved; the quotient coordinate of the product decreases by exactly 1. Transport additionally retains which of the two conjugate positions has been reached.

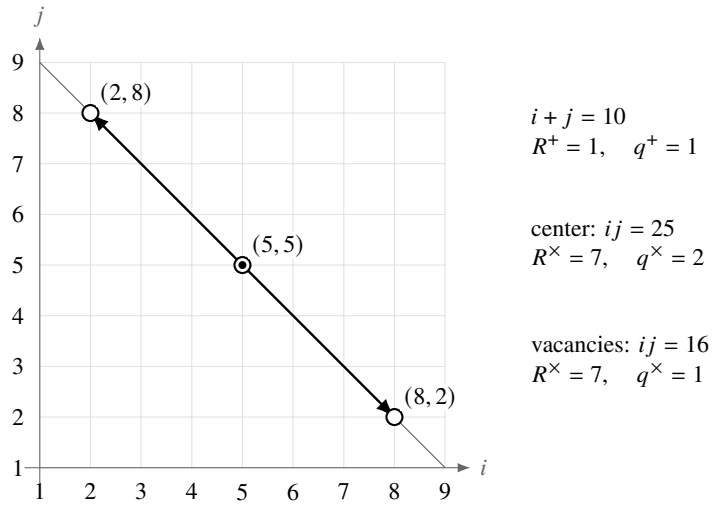


Figure 16.1: Electronic center and conjugate vacancies in the APP chart. The diagonal is the additive fiber $\mathcal{F}_{10}$; the center is the fixed point of $(i, j) \mapsto (10 - i, 10 - j)$. The two shifts of magnitude 3 preserve the sum and product residue and decrease the multiplicative quotient by one unit. The arrows represent deformations along the fiber.

Central symmetry also characterizes its persistence under chart changes. If an admissible atlas transformation f intertwines inversion, $f\rho = \rho f$, then

$$\rho f(5, 5) = f\rho(5, 5) = f(5, 5).$$

Uniqueness of the fixed point forces $f(5, 5) = (5, 5)$. This is the internal property making the central location stable: every transformation of that type preserves the cell before evaluating its electronic character (Section 16.2.1.1).

The central matrix block belongs to evaluation on the two adjacent positions $4, 5$. Its decomposition, established in the local development of the kernel, is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-.$$

Here 9 and 5 are eigenvalues of two modes, whereas $(5, 5)$ is a position in the atlas. Concatenations of its entries give $72 + 27 = 77 + 22 = 99$ and $77 - 22 = 55 = 54 + 1$. The first equality preserves the total sum under two arrangements; the second expresses the defect between diagonal and antidiagonal readings. These operations precede electronic composition and preserve their matrix provenance.

The initial orientation is another datum. In the representation of electronic return, $(h_0, v_0) = (1, 1)$ is fixed, that is, positive advance in both coordinates. Simultaneous inversion every 27 steps, read in windows of 9, determines

$$\tau_e = (+1, +1, +1, -1, -1, -1, +1, +1, +1, -1, -1, -1).$$

The segments $(1, 1, 1)$ belong to this orientation sequence: each segment comprises three windows in the same direction. The recurrence in the section on readers proves this record in (16.38), and its two evaluations recover the triple $(80, 54, 6)$ entering (16.7). The positional coordinate, APP quotients, and sign sequence are thus connected through explicit operations.

The signature 555555 belongs to a regional reading in another domain. For a decimal emission $U = (U_1, \ldots, U_6)$, we write $U_j = 100d_{1j} + 10d_{2j} + d_{3j}$, with three digits, including leading zeros. The multiplicative-signature reader is

$$\mathcal{E}_{\times}(U) = \big([d_{2j} - d_{1j}]_{10}\big)_{j=1}^{6}.$$

In the transversal region $U_{\pi}^{\perp} = (501, 614, 498, 169, 272, 272)$, the differences are $(-5, -5, 5, 5, 5, 5)$. Consequently,

$$\mathcal{E}_{\times}(U_{\pi}^{\perp}) = (5, 5, 5, 5, 5, 5), \qquad \Psi(U_{\pi}^{\perp}) = 010211. \tag{16.11}$$

The first map preserves a multiplicative signature of the catalog; the second preserves the ternary word common to the five regions. Figure 21.5 shows the transversal position of this emission and its multiplicity 432.

Memory required to transport the constant signature. The signature 555555 additionally provides a verifiable example of the trajectory information required by TPK. Let $\mathcal{X}_{\times} = D_9^2 \times \{N, E, S, O\}$, with 324 oriented positions in the product channel. Advance P moves one square in the stored direction, and the half-turn H reverses that direction. In this representation, the reader E_{sig} sums the discrete exponents by windows:

$$\ell_9(1, 2, 4, 8, 7, 5) = (0, 1, 2, 3, 4, 5), \qquad \ell_9(3) = \ell_9(6) = \ell_9(9) = 0.$$

During each of the six nine-step windows, $\ell_9(\rho_9(ij))$ is evaluated before every product advance, selected by TRIT at times $2, 5, 8 \pmod 9$. The window sum is reduced modulo 10. After steps 27 and 54, the direction is reversed. This rule determines the six components of E_{sig} from the initial position and orientation.

Enumeration of this domain produces 26 signatures. The fiber of 555555 contains 24 oriented positions; after one advance, their images are distributed among three signatures, with eight representatives each:

$$555177, \qquad 555717, \qquad 555771.$$

Three witnesses, in coordinates from 1 to 9, allow the distinction to be checked directly:

x	$E_{\mathrm{sig}}(x)$	$E_{\mathrm{sig}}(Px)$
$(3, 5, N)$	555555	555177
$(3, 4, N)$	555555	555717
$(3, 4, S)$	555555	555771

Thus, the common value of the signature does not determine its next value: the transport class within that fiber forms part of the operational datum. The minimal stable partition is calculated by starting from signature equivalence $\sim_0$ and refining through

$$x \sim_{n+1} y \quad \Longleftrightarrow \quad x \sim_n y, \quad Px \sim_n Py, \quad Hx \sim_n Hy.$$

The procedure terminates because the domain is finite. Every stable congruence refining $\sim_0$ refines every $\sim_n$, by induction; its fixed point is therefore the minimal stable congruence required. The census gives $26 \to 28 \to 28$, with histogram $27 \cdot 8 + 108 = 324$. The only signature that splits is 555555, into three classes of eight elements. The accompanying certificate reproduces the enumeration, the three classes, and stability under both operators (Section 16.2.1.1).

The multiplicity 24 here belongs to the product-channel fiber; the multiplicity 432 belongs to the complete regional emission $U_{\pi}^{\perp}$. The first proves why the signature needs transport memory; the second places that signature within the joint catalog of the two sheets. Both readings participate in the same architecture, with specified domains and operations.

This distinction of domains makes it possible to bring together the occurrences of unity and the center with their respective meanings:

Registro	Object and function
$(5, 5)$	Fixed cell of the central inversion of the APP atlas.
$(R^+, q^+) = (1, 1)$	Residue and quotient of the central additive evaluation.
$(h_0, v_0) = (1, 1)$	Initial orientation of the two shifts.
$(+1, +1, +1)$	Three consecutive windows of τ_e with positive orientation.
$c = 111111$	Datum of the fifth nonadic boundary, Section 10.5.14.1.
$\mathcal{E}_\times = 555555$	Signature of the transversal region of π, with six positions.

Electronic coordination is obtained by composing these levels in their dependency order. APP determines the center and evaluations; TRIT selects regimes and orientations; TPK prolongs the trajectory and preserves its records. The previously generated regions provide the characters π, φ, e; the dodecaphase record provides α; the electronic readers determine the factors of equation (16.7). This composition preserves the identity of the section throughout its different representations, up to dimensional evaluation and subsequent metrological comparison.

16.2.2 Commutator norm of the arithmetic projections

The electronic prefactor is obtained without using α, torsion, or a dimensional unit. The domain is the third realization of APP, $\mathcal{A}_3 = \{1, \dots, 9\}^3$, equipped with the uniform measure. We define

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \qquad \Pi(x) = \rho_9(x_1 x_2 x_3).$$

The conditional expectations P_Σ and P_Π are the orthogonal projections onto functions constant on each additive and multiplicative fiber, respectively. Thus, for a function f,

$$(P_\Sigma f)(x) = \frac{1}{|\Sigma^{-1}(\Sigma(x))|} \sum_{y:\, \Sigma(y)=\Sigma(x)} f(y),$$

and analogously for P_Π. The two projections act on the same space; they do not represent two independent arithmetic supports.

Lemma 16.14 (Two projections and principal angles)**.** *If $s \in [0, 1]$ is a singular value of the pairing between orthonormal bases of the ranges of two projections P, Q, its nontrivial block contributes to $\|[P, Q]\|$ with $s\sqrt{1 - s^2}$.*

Proof. On the corresponding principal plane, one can write

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} s^2 & s\sqrt{1 - s^2} \\ s\sqrt{1 - s^2} & 1 - s^2 \end{pmatrix}.$$

Their commutator is

$$[P, Q] = s\sqrt{1 - s^2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The final matrix is orthogonal, so its norm is one. Common or orthogonal blocks have zero commutator. The orthogonal decomposition into principal planes gives the claim and reduces the total norm to the maximum of these contributions. □

Proposition 16.15 (Exact incompatibility norm)**.** *On $\ell^2(\mathcal{A}_3)$,*

$$\|[P_\Sigma, P_\Pi]\| = \frac{\sqrt{3}}{4}. \tag{16.12}$$

Proof. Let $N_{ab} = |\Sigma^{-1}(a) \cap \Pi^{-1}(b)|$. Integer enumeration of the three symbols yields

$$N = 9\begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}.$$

Each row sums to 81; the column sums are

$$(m_1, \dots, m_9) = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

The normalized indicator functions of the fibers have pairing matrix $C_{ab} = N_{ab}/\sqrt{81 m_b}$. Consequently,

$$M = CC^{\mathsf{T}}, \qquad M_{ac} = \sum_{b=1}^{9} \frac{N_{ab} N_{cb}}{81 m_b}$$

is a rational matrix whose spectrum consists of the squares of the required singular values.

Exact multiplication shows that $(1, \dots, 1)$, $(1, -1, 0)^3$, and $(1, 1, -2)^3$ are eigenvectors with eigenvalues 1, $1/4$, and $7/132$, respectively. The subspace of vectors supported on $\{3, 6, 9\}$ whose sum is zero has dimension two and eigenvalue $1/18$. The zero-sum vectors supported on $\{1, 4, 7\}$ or $\{2, 5, 8\}$ form a four-dimensional kernel. These subspaces are independent and their dimensions sum to nine. The complete spectrum is therefore established:

$$\operatorname{spec}(M) = \left\{1, \frac{1}{4}, \frac{7}{132}, \frac{1}{18}, \frac{1}{18}, 0, 0, 0, 0\right\}.$$

The maximum of $\lambda(1 - \lambda)$ on this set is $3/16$, attained at $\lambda = 1/4$. The preceding lemma gives $\sqrt{3/16} = \sqrt{3}/4$. □

The mode $(1, -1, 0)^3$ is precisely the value distribution of the tritic phase selector in the nonadic chart. The scalar norm retains a measure of incompatibility, but does not by itself preserve the orientation of that mode or reconstruct the two projectors. That memory remains in the route state accompanying the scalar reading.

Proposition 16.16 (Algebra of the local electronic block)**.** *On the principal plane attaining* (16.12)*, let*

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1/4 & \sqrt{3}/4 \\ \sqrt{3}/4 & 3/4 \end{pmatrix}, \qquad S = 2P - I, \qquad J = \frac{4}{\sqrt{3}}[P, Q].$$

Then $S^2 = I$, $J^2 = -I$, $SJ = -JS$, and the real algebra generated by S, J is $M_2(\mathbb{R})$, a realization of $\mathrm{Cl}_{1,1}$*. Moreover,*

$$n_{\pm} = \frac{S \pm J}{2}, \qquad n_{\pm}^2 = 0, \qquad n_+ n_- + n_- n_+ = I.$$

Proof. We have $S = \operatorname{diag}(1, -1)$ and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Multiplication verifies the three relations. The matrices I, S, J, SJ are linearly independent and form a basis of $M_2(\mathbb{R})$, proving the claim about the algebra. Finally, $(S \pm J)^2 = S^2 + J^2 \pm (SJ + JS) = 0$, and the sum of the cross products is $(S^2 - J^2)/2 = I$. □

In this block, an elliptic direction, a hyperbolic direction, and two parabolic null directions coexist. The result describes a local matrix representation of the arithmetic of the two sheets. It does not identify all of APP with $M_2(\mathbb{R})$, or with a quantum Latin square; such claims would require other domains and other relations. Nor does the exponential belong exclusively to the parabolic regime: it can be applied to any of the generators of this block.

16.2.3 Spinor representation of the electronic block and helicity

The electronic structure has a representation preserving more information than its scalar evaluation. The principal plane of the two three-dimensional APP projectors, selected by the tritic mode $(1, -1, 0)^3$, has already provided the operators S and J. The former is a symmetric involution; the latter is the normalized commutator and preserves sheet orientation. Their composition makes it possible to construct explicitly a rotation representation and separate its two helicity components. This construction acts on the electronic block; the TPK route record and its memory remain as data of the block's provenance.

Let E_e be the real principal plane, equipped with the inner product inherited from the conditional expectations, and let $\mathcal{S}_e = E_e \otimes_{\mathbb{R}} \mathbb{C}$ be its complexification. The operators already obtained satisfy

$$S^* = S, \qquad J^* = -J, \qquad S^2 = I, \qquad J^2 = -I, \qquad SJ = -JS.$$

Complexification extends the representation field. It preserves the real operators and allows the scalar unit i to be distinguished from the endomorphism J, although both square to -1 in their respective domains.

Proposition 16.17 (Hermitian triple induced by the two sheets)**.** *The operators*

$$\sigma_1 = SJ, \qquad \sigma_2 = -iJ, \qquad \sigma_3 = S \tag{16.13}$$

are Hermitian and traceless. They satisfy

$$\sigma_a\sigma_b = \delta_{ab} I + i \sum_{c=1}^{3} \epsilon_{abc}\sigma_c, \qquad \frac{1}{2}\operatorname{tr}(\sigma_a\sigma_b) = \delta_{ab}, \tag{16.14}$$

where $\epsilon_{123} = 1$. They therefore form an orthonormal basis of the real space V_e of traceless Hermitian endomorphisms of $\mathcal{S}_e$, for the inner product $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$.

Proof. The preceding relations give $(SJ)^* = (-J)S = SJ$ and $(-iJ)^* = -iJ$. The squares of all three matrices are I. Furthermore,

$$(SJ)(-iJ) = iS, \qquad (-iJ)S = iSJ, \qquad S(SJ) = J = i(-iJ).$$

Reversing the order changes the sign. This proves the first identity in (16.14). In the orthonormal basis of the principal plane, the matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Their trace is zero, and taking traces in the product identity proves orthonormality. Every traceless Hermitian endomorphism has the form $\left(\begin{smallmatrix} z & x-iy \\ x+iy & -z \end{smallmatrix}\right)$, with real x, y, z, and thus belongs to their real span. □

In the usual matrix representation, this triple is called the Pauli matrices. Their use here follows the construction of P, Q, S, J: they are obtained through the products (16.13). The direction space of this representation is the unit sphere of V_e, whose positive form has just been constructed through the trace.

Proposition 16.18 (Rotation generators and directional decomposition)**.** *Define* $L_a = \sigma_a/2$*. Then*

$$[L_a, L_b] = i \sum_c \epsilon_{abc} L_c, \qquad \sum_{a=1}^{3} L_a^2 = \frac{3}{4} I. \tag{16.15}$$

For $n = (n_1, n_2, n_3)$ *with* $\sum_a n_a^2 = 1$*, let*

$$H_e(n) = \sum_a n_a \sigma_a, \qquad h_e(n) = \frac{1}{2} H_e(n), \qquad \Pi_\pm(n) = \frac{1}{2}\big(I \pm H_e(n)\big).$$

The linear formula $H_e(v) = \sum_a v_a \sigma_a$ *is also used for* v *of arbitrary norm. One has*

$$\mathcal{S}_e = \operatorname{im} \Pi_+(n) \oplus \operatorname{im} \Pi_-(n), \qquad h_e(n)\Pi_\pm(n) = \pm \frac{1}{2} \Pi_\pm(n). \tag{16.16}$$

The two summands are orthogonal complex lines. Rotation transport around n *is*

$$U_n(\theta) = \exp\left(-\frac{i\theta}{2} H_e(n)\right) = \cos\frac{\theta}{2}\, I - i \sin\frac{\theta}{2}\, H_e(n), \qquad U_n(2\pi) = -I, \quad U_n(4\pi) = I. \tag{16.17}$$

Proof. The product identity of the triple gives the commutators in (16.15); their squares sum to $3I/4$. Antisymmetric terms cancel when multiplying $H_e(n)$ by itself, so that $H_e(n)^2 = I$. It follows that

$$\Pi_\pm(n)^2 = \Pi_\pm(n), \qquad \Pi_+(n)\Pi_-(n) = 0, \qquad \Pi_+(n) + \Pi_-(n) = I.$$

They are Hermitian and have trace 1; their rank is therefore 1. Multiplication by $H_e(n)/2$ proves (16.16). The exponential series splits into even and odd powers because $H_e(n)^2 = I$, producing (16.17). It also directly proves unitarity and the law $U_n(\theta + \psi) = U_n(\theta) U_n(\psi)$.

Angular normalization is verified on V_e. For every $v \in \mathbb{R}^3$, the same product identity gives

$$U_n(\theta) H_e(v) U_n(\theta)^* = H_e\big(v \cos\theta + (n \times v) \sin\theta + n(n \cdot v)(1 - \cos\theta)\big)\,.$$

The induced action is a rotation of angle θ and axis n. The factor $1/2$ in the spinor generator is therefore the one corresponding to that angular parameterization. A complete rotation restores the direction in V_e and changes the sign of the vector in $\mathcal{S}_e$; two complete rotations restore both. □

The representation is irreducible: a complex subspace invariant under the three matrices would be invariant under all of $M_2(\mathbb{C})$, the algebra they generate. Its only invariant subspaces are 0 and $\mathcal{S}_e$. The relations (16.15) thus identify the spin-$1/2$ representation. If n is the propagation direction in a kinematic realization, $h_e(n)$ is its dimensionless helicity, with values $\pm 1/2$. The kinematic direction is declared when that realization is made; the two projectors are defined for every direction before a particular propagation state is selected.

Preservation of the electronic structure and types of inversion. Direction inversion has an exact law:

$$H_e(-n) = -H_e(n), \qquad \Pi_\pm(-n) = \Pi_\mp(n).$$

This operation exchanges helicity labels relative to a direction. Charge conjugation, by contrast, acts on the oriented signature of the material route; cellular inversion acts on the atlas coordinates. Their domains and actions remain distinct. Uniqueness of cell $(5, 5)$ is compatible with the two lines in (16.16): an atlas cell is not a vector in the spinor representation it carries.

The electronic evaluation ε_e^{β} preserves its formula and readers. On $\mathcal{S}_e$, its scalar realization is $\varepsilon_e^{\beta} I$, which commutes with the generators and with $\Pi_{\pm}(n)$. Helicity decomposition adds rotational information without modifying the evaluation or turning spin into a multiplicative mass coefficient.

Chirality corresponds to the grading of the relativistic representation used: an involution Γ_{ch} determines its projectors $(I \mp \Gamma_{\mathrm{ch}})/2$. The helicity in (16.16) additionally depends on n. For its part, the holonomy of a real Möbius band describes the change of sign of a line when traversing a loop in its base. The law of that line, the spinor return (16.17), and nonadic phase holonomy with memory advance are transport laws on specified objects. Their comparison preserves those objects: the helical form of a trajectory does not replace the representation just constructed or by itself fix chirality or charge.

16.2.4 Transfer between sheets and coefficients of the basal stage

The primitive tritic calendar visits the modes (Σ, Π, Π). Counting consecutive pairs, including the closure of the block, yields one transition $\Sigma \to \Pi$, one $\Pi \to \Pi$, and one $\Pi \to \Sigma$. Their primitive incidence is

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

This matrix realization is obtained after the calendar. It is not used to replace the coinductive reader that has already generated φ. Its characteristic polynomial is $X^2 - X - 1$, and recognition of the two roots as φ and $-\varphi^{-1}$ describes the action of that output in modal transfer. The quadratic action has eigenvalues $\varphi^2, -1, \varphi^{-2}$; the last is the positive stable mode. The regional mark π, applied once, gives π/φ^2.

The trajectory space, its measure, and the closure operator are constructed in subsections 10.5.6.1 and 10.5.6.2. Formula (10.44) preserves its provenance as a quotient of returns; the electronic application subsequently composes this reading with the following involution. The regional signature 555555 accompanies the closure region, whose transport preserves the state with memory; the centre $(5, 5)$ fixes the cell of origin of the oriented electronic section. Both kinds of information converge in the composition and retain their respective domains.

Lemma 16.19 (Interchange of the two orientations)**.** *Let $\mathcal{R}(x, y) = x/y^2$ and $\mathcal{S}_{\pi\varphi}(x, y) = (y, x)$, for $x, y > 0$. Then*

$$\begin{aligned} \mathcal{S}_{\pi\varphi}^2 = I, \qquad \mathcal{R}(\pi, \varphi) = \frac{\pi}{\varphi^2}, \\ (\mathcal{R} \circ \mathcal{S}_{\pi\varphi})(\pi, \varphi) = \frac{\varphi}{\pi^2} = \frac{1}{\varphi^3 (\pi/\varphi^2)^2}. \end{aligned} \tag{16.18}$$

Proof. Applying the interchange twice gives the identity. Moreover, $\varphi^3(\pi/\varphi^2)^2 = \pi^2/\varphi$; taking its inverse proves the last equality. □

The electronic reader uses the orientation φ/π^2, followed by the exponential character. The preceding identity explains the interchange, but does not replace the rule that chooses that character or by itself produce the integers of the correction. These have a different provenance.

The phase-compatible central return has minimal length twenty-seven. With the phase and sheet record, its finite space is

$$\mathcal{K}_e = \mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3.$$

The five regions of π are distinct genealogical identities, not five interchangeable copies of a visible expansion. In the regional gauge fixed by the construction, their injection into the return occupies the positions

$$(4, 0), \quad (5, 0), \quad (6, 0), \quad (6, 1), \quad (6, 2). \tag{16.19}$$

The transverse orientation, mutated return, and three positive regions remain separated by that record. Admissible changes of gauge transport the labels jointly; they do not reselect the positions from an electronic value.

Proposition 16.20 (Deficit and retained memory)**.** *Given the central return and the preceding regional injection, the oriented boundary deficit and torsional memory are*

$$n_e = -|\mathcal{K}_e \setminus \iota(\mathscr{R}_\pi)| = -22, \qquad m_e^{\mathrm{tor}} = |\mathscr{R}_\pi \times \mathbb{F}_3| = 15. \tag{16.20}$$

Proof. The five positions in (16.19) are distinct. Thus, $|\mathcal{K}_e| = 9 \cdot 3 = 27$, and the complement of the image has cardinality $27 - 5 = 22$. The orientation convention records the deficit removed from the boundary as negative. The retained memory has three sheets per region, so its positive cardinality is $5 \cdot 3 = 15$. □

The same pair has the integer control of the family's APP chart:

$$396 = 18 + 72 + 108 + 198, \qquad \frac{396}{18} = 22, \qquad \frac{270}{18} = 15 = 3 \cdot 5 = \binom{6}{2}. \tag{16.21}$$

The weight 18 belongs to the central 2×2 kernel, and 270 to the declared global channel. These equalities check the normalization of the count; the return argument also fixes the signs. The quotient of the alternating rings, $396/(18 + 108) = 22/7$, is rational and is not identified with π. Nor does $\binom{6}{2}$ mean $6/2$: it counts the unordered pairs of a six-element set.

Evaluation of the deficit in the third symmetric realization uses α^3. This power is the product of three copies of the closure scalar already obtained; it does not assert a universal rule associating any dimension with the same power of α. The specified domain and reader are essential to the composition.

16.2.5 Global torsion and the basal coordinate

Definition 16.21 (Electronic torsional normalization)**.** The reduced defect d_4 and the complete reader Δ_4 are distinguished:

$$d_4 = \frac{\pi - e \log \pi}{270}, \qquad \Delta_4 = d_4 \left(1 + \frac{\pi}{729}\right). \tag{16.22}$$

The denominator 270 corresponds to the census channel, whereas the factor $1 + \pi/729$ retains the coupling with the cubic cell of 729 positions. The electronic reading uses Δ_4 both in its basal stage and in the return exponent.

This definition acts on previously generated π and e. The logarithm is a subsequent operation in their real realization, not a retrospective input of conventional values. Torsion is global: it is not determined independently for each section by an observed mass.

Lemma 16.22 (The two normalizations are not interchangeable)**.** *One has*

$$\Delta_4 - d_4 = \frac{\pi}{729} d_4. \tag{16.23}$$

In particular, $d_4 > 0$ and $\Delta_4 > d_4$ for the coordinates under consideration.

Proof. The identity follows by distributivity. For positivity, the function $f(x) = x - e \log x$, on $x > 0$, has derivative $1 - e/x$ and second derivative $e/x^2 > 0$. Its unique minimum is at $x = e$, where it is zero. Since $\pi \neq e$, we have $f(\pi) > 0$. Both factors in (16.22) are positive and $1 + \pi/729 > 1$. □

Remark 16.23 (Preservation of the normalization). Removing the second factor in (16.22) changes the reader; it is not an algebraic simplification. If d_4 is introduced as an auxiliary coordinate, the same object must be written as $\Delta_4 = (1 + \pi/729) d_4$ in every occurrence. Keeping the integers 15 and 6 and replacing only Δ_4 by d_4 produces a different formula.

Definition 16.24 (Basal electronic coordinate)**.** The composition of incompatibility, modal transfer, deficit, and retained memory is

$$\varepsilon_e^{(0)} = \frac{\sqrt{3}}{4}\left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3\right](1 + 15\Delta_4). \tag{16.24}$$

The equation combines three operations of different kinds. The norm $\sqrt{3}/4$ quantifies the incompatibility of the sheet projections; its origin is a principal angle of the APP fibers. The exponential acts on the orientation φ/π^2 of the transfer reader. The cubic subtraction records the 22 complementary positions of the marked return, and the torsional factor preserves the 15 components corresponding to three sheets per region. The composition preserves these domains even though its final evaluation is scalar.

The presence of π, φ, and e thus has operational meanings. Euler's number enters through the exponential and the logarithmic normalization of Δ_4; φ and π enter through oriented transfer. The constant α acts after its own construction through the cubic closure term. The chronological return must still act on this basal coordinate. Separating the stages makes it possible to explain what changes when an orientation, normalization, or record is modified, while preserving the central structure that identifies the electronic section.

The parentheses in this equation are part of its content. The term $-22\alpha^3$ is subtracted *outside* the exponential; the memory $1 + 15\Delta_4$ multiplies the entire bracket. Placing the deficit in the exponent or applying a different torsion to it would alter the composition.

Proposition 16.25 (Basal positivity)**.** *For $0 < \alpha < 10^{-2}$, $\varphi > 0$, and the normalization* (16.22)*, we have $\varepsilon_e^{(0)} > 0$.*

Proof. The exponential is greater than one, and $22\alpha^3 < 22 \cdot 10^{-6} < 1$. The bracket is therefore positive. Also, $1 + 15\Delta_4 > 1$ and $\sqrt{3}/4 > 0$. □

The subsequent evaluation of (16.24) begins with

$$\varepsilon_e^{(0)} = 0.5109989224880660725\ldots.$$

This number is still a dimensionless coordinate and is not the final stage: the return memory of the route itself remains to be applied.

16.2.6 Determinantal action scale and decimal valuation

The absolute scale and relative correction arise from distinct objects. The former uses the local block of the step-three dodecaphase register,

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad G_H = \mathbb{Z}^4/H_4\mathbb{Z}^4. \tag{16.25}$$

The operation uses a local quotient. The three blocks of the complete register are not replaced by tensorization that would multiply this reader's number of sheets again.

Lemma 16.26 (Number of sheets of the local quotient)**.** *The Smith normal form of H_4 is* $\operatorname{diag}(1, 2, 2, 4)$*. Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \qquad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. Second-order minors are even and one has value -2, so their greatest common divisor is two. Since $H_4 H_4^{\mathsf{T}} = 4I$ and $\det H_4 = -16$, the cofactors are ± 4; the third determinantal divisor is four. The fourth is 16. Successive quotients of the divisors $1, 2, 4, 16$ are $1, 2, 2, 4$. Their product gives the quotient's order. □

Definition 16.27 (Determinantal action reader)**.** The multiplicative norm of the scalar section α, constant on the sheets of G_H, and one application of self-scaling φ produce

$$\Sigma_H = \varphi\, \mathrm{N}_{G_H}(\alpha), \qquad \mathrm{N}_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \tag{16.26}$$

The associated decimal chart determines $\mathrm{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor$.

The sixteenth power thus comes from the quotient's number of sheets, once this local reader is fixed. The factor φ acts once on the basis of the action line. Smith calculation alone would not select any possible functional on the quotient: the multiplicative norm and self-scaling are part of the declared operation.

Theorem 16.28 (Decade of the action section)**.** *The reader* (16.26)*, applied to HMT coordinates already generated, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \qquad \mathrm{ord}_{10}(\Sigma_H) = -34. \tag{16.27}$$

Proof. The rational intervals, much coarser than the available cylinders, suffice:

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \qquad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}.$$

The function $(x, a) \mapsto x a^{16}$ is increasing in both positive arguments. The two integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \qquad 1619 \cdot 7298^{16} < 10^{66}$$

imply

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6}\right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6}\right)^{16} < 10^{-33}.$$

The definition of decimal order completes the proof. Neither an SI unit nor a reference mass enters these inequalities. □

The normalized section begins with $\Sigma_H = 1.0462438381\ldots \times 10^{-34}$. Its mantissa is not identified with the action coordinate before or after return. The determinantal reader fixes decimal order; the two return coordinates are obtained through the following readers.

16.2.7 Action sections and dimensionless return quotient

Let $\mathcal{L}_S$ be an oriented action line and $\mathcal{U}_S$ its positive basis. The electronic composition uses two sections of this line and their dimensionless quotient. Its construction requires the determinantal decade already obtained, the fifth-order action character, and the regional contribution of the return.

16.2.7.1 Action character and provenance of its coefficients

The reduced Planck constant, $\hbar$, is the physical object motivating realization of the angular-action section. Its position in the electronic dependency must be specified at two levels. The determinantal norm Σ_H determines the scale's decade. The character H_5 and regional return then determine the action-section coordinates. This separation allows the result needed for the electron to be presented without anticipating complete developments of other Dimensional Mechanics sectors.

The provenance of the fifth-order character can be made explicit through invariants already fixed on the support. In the central chart, consider

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \qquad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the orbit of step 3 has length 4, and the relevant census quotient is $104976/1944 = 54$. The coefficients of the character satisfy

$$\frac{9}{16}=\frac{\lambda_+}{4^2},\quad \frac{5}{9}=\frac{\lambda_-}{\lambda_+},\quad \frac{7}{48}=\frac{\mathrm{tr}(B_c)/2}{4\cdot 12},\quad \frac{1}{54}=\frac{1944}{104976}. \tag{16.28}$$

Assignment of these invariants to degrees $2,3,4,5$, with the alternating signs of (16.29), belongs to the definition of the analytic character used. Making this explicit is essential: the set of cardinalities, considered without the character and the ordering of its degrees, would admit other expressions. The constructive content lies in composition of the support, orientation and declared analytic rule.

On this basis, the fifth-order character and regional return term have the expressions

$$H_5=\frac{1}{2}\sqrt{\frac{1000\alpha}{\varphi}}-\alpha+\frac{9}{16}\alpha^2-\frac{5}{9}\alpha^3+\frac{7}{48}\alpha^4-\frac{1}{54}\alpha^5, \tag{16.29}$$

$$D_A=\exp\!\left(-\frac{100\pi\alpha}{9}\right), \tag{16.30}$$

$$C_\pi(\alpha)=169\alpha^6+\frac{D_A\alpha^7}{1-D_A\alpha}, \tag{16.31}$$

$$\eta_{\mathrm{ret}}=H_5-\frac{90}{\pi}C_\pi(\alpha). \tag{16.32}$$

These are subsequent readings of dodecaphase closure and its regional incidence. The integer 169 is retained as the regional selector already constructed, and the coefficients of H_5 as those of its local character; they are not replaced by an interpolation of the Planck constant. The formulas specify exactly which readings are needed for the electronic stage. Other mass-theory functionals need not be introduced here.

For $0<\alpha<1$, we have $0<D_A<1$, so $D_A\alpha<1$. The rational term of (16.31) preserves the complete sum of the geometric tail

$$\frac{D_A\alpha^7}{1-D_A\alpha}=\sum_{n=0}^{\infty}D_A^{n+1}\alpha^{n+7}. \tag{16.33}$$

It is not a truncation that can be silently modified when precision changes. After the term of index N, the remaining tail is $D_A^{N+2}\alpha^{N+8}/(1-D_A\alpha)$, allowing evaluation to be controlled without recourse to a target quantity.

Definition 16.29 (Action sections and return reader)**.** The sections before and after return are

$$\hbar_{\mathrm{pre}}=H_5\,10^{-34}\mathcal{U}_S,\qquad \hbar_{\mathrm{ret}}=\eta_{\mathrm{ret}}\,10^{-34}\mathcal{U}_S. \tag{16.34}$$

Whenever $\eta_{\mathrm{ret}}>0$, define

$$\delta_{\mathrm{ret}}=H_5-\eta_{\mathrm{ret}},\qquad R_{\mathrm{act}}=\frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}}=\frac{H_5}{\eta_{\mathrm{ret}}}. \tag{16.35}$$

Proposition 16.30 (Positivity and invariance of the ratio)**.** *In the intervals used in the proof of* (16.27)*,* $0<\eta_{\mathrm{ret}}<H_5$*, and therefore* $R_{\mathrm{act}}>1$*. The ratio does not change when the basis* $\mathcal{U}_S$ *is replaced by any other common positive basis.*

Proof. The terms of C_π are positive. For an elementary bound, it suffices to use $\alpha<0.008$, $\varphi<1.619$, $\alpha>0.007$, $D_A<1$ and $\pi>3$. The root in (16.29) exceeds one, whereas the sum of negative terms is less than 0.008001; in particular, $H_5>0.99$. Likewise,

$$0<\frac{90}{\pi}C_\pi<30\left(169(0.008)^6+\frac{(0.008)^7}{1-0.008}\right)<10^{-6}.$$

Thus $\eta_{\mathrm{ret}} > 0.989999 > 0$ and $\eta_{\mathrm{ret}} < H_5$. Finally, the change of basis divides both action coordinates by the same positive factor, which cancels in their ratio. □

Subsequent evaluation gives

$$H_5 = 1.0545718176461544696\ldots,$$
$$\eta_{\mathrm{ret}} = 1.0545718169150390611\ldots,$$
$$R_{\mathrm{act}} = 1.0000000006932817630\ldots.$$

The difference and quotient are two readings of the same sections, one additive and the other multiplicative. They do not represent two independent Planck constants. The passage from angular action to action per complete turn is expressed in each section by $h = 2\pi\hbar$. The subsequent chart $\mathcal{U}_S \mapsto \mathrm{J\,s}$ assigns a unit but does not act retroactively on the decimal order already proved.

Regional return can also be written in an angular chart. This reparametrization is not an indispensable additional dependency: (16.29)–(16.32) supply H_5, η_{ret} and R_{act} directly. If angle and counter-angle are introduced, both must be transported in the same chart; degrees and radians are not mixed within a difference.

16.2.7.2 Regional continuation and renewal equation

The selector 169 determines the degree-six impulse. Subsequent extension is organized by the transfer D_A and unit normalization of the determinant line. These rules admit a formal-series formulation before substituting $z = \alpha$, with D_A already constructed. Write

$$\mathscr{C}(z) = 169z^6 + R(z), \qquad R(z) \in z^7\mathbb{R}[[z]].$$

Concatenating an extension increases degree by one and multiplies its weight by D_A. The first strict extension has weight D_A. Its renewal equation is therefore

$$R(z) = D_A z^7 + D_A z R(z). \tag{16.36}$$

Proposition 16.31 (Formal uniqueness of the normalized continuation)**.** *Equation* (16.36) *determines a unique series, and*

$$\mathscr{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n.$$

The realization is analytic in $|D_A z| < 1$.

Proof. The element $1 - D_A z$ has constant term one and is invertible in the formal power-series ring. Solving the equation produces the displayed series. Equivalently, the degree-seven coefficient is D_A and each subsequent coefficient is D_A times the preceding one. The geometric series converges absolutely in the stated disk. □

Normalization of the first extension is an indispensable operational datum. If only the degree-six impulse and concatenation ratio were retained, the series

$$169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

would still share both data for any λ. The unit rule fixes $\lambda = 1$ before evaluation. Each rule's contribution to reader uniqueness is thus located, and the rational expression preserves the complete continuation.

16.2.7.3 Reduced Planck constant and return of the action section

In the dimensional chart of (16.34), the section before return is the model's construction intended for comparison with the reduced Planck constant:

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34} \mathcal{U}_S.$$

The subsequent section also preserves the regional compensation $90\mathscr{C}(\alpha)/\pi$. Both belong to the same action line, and their quotient R_{act} measures the multiplicative effect of return. The passage to h through 2π uses the relationship between angular parametrization and a complete turn.

This construction effectively enters $R_{\mathrm{act}}^{80-54\alpha+6\Delta_4}$. The electron preserves its central coordinate and trajectory; the chronological scale does not replace this structure with an undifferentiated quantum. Nor is a mass already expressed in MeV multiplied again by 10^{-34}: the decade belongs to the action section, cancels in its dimensionless quotient and reappears only where the corresponding dimensional realization acts.

The numerical comparison of H_5, the returned section and $h/(2\pi)$ is presented in Section 140.1. This distinguishes the name of the physical object, the equation assigned to it by the model and the precision with which that equation reproduces it.

16.2.8 Two readers of the electronic return

Electronic memory is represented by a tritic word of length 108 and a signed twelve-phase word. To make the exponent calculation reproducible, we specify the chronological projection used. Initially, let $(r_0, c_0) = (5, 5)$ and $(h_0, v_0) = (1, 1)$. At each step, the selector repeats $(+1, -1, 0)$. With orientation fixed within each twenty-seven-step block, the visible rule is

mode	read integer	cellular transport
$+1$	$r + c$	$c \mapsto 1 + (c - 1 + h) \bmod 9$
-1	rc	$r \mapsto 1 + (r - 1 + v) \bmod 9$
0	9	without displacement

The emitted symbol is $\rho_9(\text{read integer}) \bmod 3$. After steps $27, 54, 81$, h and v are inverted simultaneously. This rule calculates the observable projection; the state update additionally preserves residue, quotient, carry, sheet, boundary and memory. The register of 108 symbols is not identified with the complete state.

In each complete three-step block, each coordinate advances once; nine such blocks are needed to return to the center in the same phase. The minimal length of this recorded return is therefore 27. If phase is forgotten, the cell is already reached after step 26, before the neutral threshold completing the block; that is not the typed return counted by $\mathcal{K}_e$. The explicit reading of the first two return blocks gives

$$a = 100000220, \qquad b = 120200000, \qquad \gamma_e = (a^3 b^3)^2. \tag{16.37}$$

Here, powers denote word concatenation. In particular, $\gamma_{e,t+54} = \gamma_{e,t}$. In the record of nine steps per phase, the central orientations give

$$\tau_e = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1). \tag{16.38}$$

The periodicity of these representations does not amount to restarting the history that produces them.

Definition 16.32 (Cyclic discrepancy reader)**.** For a word $\gamma \in \mathbb{F}_3^{108}$, define

$$D_k(\gamma) = \#\{t \in \mathbb{Z}/108\mathbb{Z} : \gamma_t \neq \gamma_{t+k}\}.$$

On the domain of words whose period divides 54, the integer reader is

$$K_D(\gamma) = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right). \tag{16.39}$$

Lemma 16.33 (Integrality and central evaluation)**.** *The D_k on the preceding domain are even. For* (16.37)*,*

k	1	3	4	27	36
D_k	54	56	68	48	32

y $K_D(\gamma_e) = (80, 54, 6)$.

Proof. The map $t \mapsto t + 54$ pairs discrepancy indices without fixed points and preserves their condition; each cardinality is even. For the explicit word a^3b^3 of length 54, the discrepancy count for each displacement is, in the same order, 27, 28, 34, 24, 16. Doubling the word doubles these five cardinalities. Finally,

$$(D_{27} + D_{36}, D_1, (D_4 - D_3)/2) = (48 + 32, 54, (68 - 56)/2) = (80, 54, 6).$$

□

The shifts 27 and 36 are the chronological lifts of the channels of 90 and 120 degrees in a chart where nine steps correspond to thirty degrees. The angular reading explains that representation, but the integers are computed on the word, without introducing a mass or a sought real value.

Definition 16.34 (Dodecaphasic window reader)**.** For $\tau \in \{-1, 0, 1\}^{12}$, with cyclic indices, let

$$C_k(\tau) = \sum_{r=0}^{11} \tau_r \tau_{r+k},$$

$$A_\ell(\tau) = \sum_{r=0}^{11} \left(\sum_{j=0}^{\ell-1} \tau_{r+j} \right)^2,$$

$$P_6(\tau) = \sum_{r=0}^{5} \tau_r \tau_{r+6}.$$

We define

$$\Omega(\tau) = 4A_3(\tau) - 3A_4(\tau), \qquad K_\Omega(\tau) = (\Omega(\tau), 54, P_6(\tau)). \tag{16.40}$$

The 54 in this reader records the declared chronological half-turn. It is not confused with the local APP jump of value four.

Lemma 16.35 (Primitive contrast and correlation form)**.** *Among integer linear contrasts $uA_3 + vA_4$ that vanish whenever $A_3 = 3a$ and $A_4 = 4a$, the primitive pair, with orientation* 90 − 120*, is* $(u, v) = (4, -3)$*. Moreover,*

$$\Omega = -2C_1 - 4C_2 - 6C_3. \tag{16.41}$$

Proof. Vanishing for every a requires $3u + 4v = 0$. By coprimality, the solutions are $(u, v) = n(4, -3)$, and orientation fixes the sign of the primitive generator. Expanding the squares gives

$$A_\ell = \ell C_0 + 2 \sum_{k=1}^{\ell-1} (\ell - k) C_k.$$

Therefore, $A_3 = 3C_0 + 4C_1 + 2C_2$ and $A_4 = 4C_0 + 6C_1 + 4C_2 + 2C_3$. Their combination $4A_3 - 3A_4$ cancels C_0 and yields (16.41). □

The uniqueness proved belongs to that class of integer linear contrasts; it does not by itself exclude every nonlinear functional of the windows. Specifying the functional class avoids turning an identity of two channels into a claim of universal selection.

Theorem 16.36 (Return triple of the electronic section)**.** *The two readers applied to the records of* Γ_e *coincide and give*

$$K_D(\gamma_e) = K_\Omega(\tau_e) = (80, 54, 6). \tag{16.42}$$

Their evaluation in the already generated coordinates is

$$\kappa_e = 80 - 54\alpha + 6\Delta_4. \tag{16.43}$$

Proof. The first reader was computed above. For τ_e, the antipodal product is always one, and hence $P_6 = 6$. The cyclic count gives $C_1 = 4$, $C_2 = -4$, $C_3 = -12$; thus

$$\Omega = -2 \cdot 4 - 4(-4) - 6(-12) = 80.$$

Equivalently, $A_3 = 44$, $A_4 = 32$, and $4 \cdot 44 - 3 \cdot 32 = 80$. Both triples are therefore $(80, 54, 6)$. Evaluating the functional $(A, B, C) \mapsto A - \alpha B + \Delta_4 C$ proves the last formula. □

This coincidence concerns the central section. It does not identify the two readers on every route domain: one compares tritic symbols and the other compares windows of a signed record. Nor is 80 obtained from the period of a word assigned to Euler's number. Its origin in this composition is the triple of the electronic route.

16.2.9 Complete electronic composition and normalization

Definition 16.37 (Multiplicative return)**.** The action of return memory on the basal coordinate is

$$\varepsilon_e^{\beta} = \varepsilon_e^{(0)} \exp(\kappa_e \log R_{\mathrm{act}})\,. \tag{16.44}$$

Theorem 16.38 (Positive electronic factorization)**.** *With the declared readers, orientation, and normalization,* (16.44) *coincides with* (16.7)*. All its factors are dimensionless and the resulting coordinate is positive.*

Proof. The incompatibility norm is (16.12); modal transfer, deficit, and retained memory compose (16.24). The triple (16.42) produces (16.43), and the ratio of the two action sections is (16.35). Substituting those three identities into (16.44) yields exactly (16.7). The basal stage is positive, as is the exponential of a real number. The common factor with the dimension of action has already canceled in forming R_{act}. □

The evaluation of this composition begins with

$$\varepsilon_e^{\beta} = 0.5109989506900008086082571648\ldots. \tag{16.45}$$

The digits are a subsequent evaluation of the formula, not data used to select 22, 15, or $(80, 54, 6)$. The numerical representation of the basal coordinate by a finite prefix introduces only its truncation error; it does not modify the exact formula defining it.

The effect of a different torsional normalization can be located without comparison to a measurement. Write $B = \frac{\sqrt{3}}{4}[\exp(\varphi/\pi^2) - 22\alpha^3]$, $R = R_{\mathrm{act}}$, and

$$\mathcal{E}(d) = B(1 + 15d)R^{80-54\alpha+6d}.$$

For the two defects in (16.22),

$$\frac{\mathcal{E}(\Delta_4)}{\mathcal{E}(d_4)} = \frac{1+15\Delta_4}{1+15d_4} R^{6(\Delta_4-d_4)} > 1. \tag{16.46}$$

The first torsional inequality and $R > 1$ prove that they are not the same coordinate. If only the defect in the exponent is changed, keeping the complete basal coordinate, the ratio of the two outputs is $R^{6(\Delta_4-d_4)}$, again different from one. A very small difference is still a difference of formula; it cannot be attributed to a mere conversion of units.

16.2.10 Dimensional realization: energy, mass, and chronological scale

The electronic coordinate finally acts on a dimensional line. Let $\mathcal{L}_E$ be an energy line with positive basis $\mathcal{U}_E$. The realization is

$$E_e^{(0)} = \varepsilon_e^{(0)} \mathcal{U}_E, \qquad E_e^{\beta} = \varepsilon_e^{\beta} \mathcal{U}_E, \qquad m_e^{\beta} = \frac{E_e^{\beta}}{c_{\mathrm{int}}^2}. \tag{16.47}$$

The last expression belongs to the mass line when c_{int} carries the dimension of speed in the declared realization. In the comparative chart $\mathcal{U}_E = 1$ MeV, (16.45) expresses an energy in MeV; the corresponding mass is expressed in MeV/c^2. It is common to abbreviate both quantities by taking $c = 1$, but that abbreviation must not obscure the dimensional map.

The action scale also admits an action–clock realization. If t_0 is a positive temporal section, the 108-step clock defines

$$\omega_0 = \frac{2\pi}{108 t_0}, \qquad E_{\mathrm{clk}} = \hbar_{\mathrm{ret}} \omega_0, \qquad m_{\mathrm{clk}} = E_{\mathrm{clk}} c_{\mathrm{int}}^{-2}. \tag{16.48}$$

Action times frequency has the dimension of energy. This chronological quantum is a common dimensional section; it is not, by definition, the electronic section. The route word and its factors are not obtained from the uniformity of the clock.

Proposition 16.39 (Cancellation of the common scale)**.** *The factor* $10^{-34}\mathcal{U}_S$ *in* (16.34) *is not part of* κ_e*. On the same energy line,*

$$\frac{E_e^{\beta}}{E_e^{(0)}} = R_{\mathrm{act}}^{\kappa_e}. \tag{16.49}$$

The result is independent of the choice of the common dimensional basis.

Proof. In forming $\hbar_{\mathrm{pre}}/\hbar_{\mathrm{ret}}$, the factor $10^{-34}\mathcal{U}_S$ cancels exactly. Dividing the two energy expressions in (16.47) also cancels $\mathcal{U}_E$, and (16.44) gives the stated ratio. □

Therefore, a coordinate already expressed in MeV is not multiplied again by 10^{-34}. If E_{clk} is chosen as the basis of $\mathcal{L}_E$, the coordinates must be transported by the change-of-basis factor; retaining unchanged the numerical value from the one-MeV chart would identify two bases without proving it. This is the precise point at which the action scale enters the absolute realization and ceases to enter into the relative corrector.

There is also a linear realization that makes the separation between clock and genealogy visible. On $\ell^2(\mathbb{Z}/108\mathbb{Z};\mathbb{R})$, the cyclic shift S has fixed space

$$\ker(S - I) = \mathbb{R} n_0, \qquad n_0 = \frac{1}{\sqrt{108}} \sum_{t=0}^{107} \delta_t.$$

If g_{Γ_e} is the route symbol in the free linearization of genealogies and $\mathcal{U}_M$ is a basis of the mass line, the rank-one map is defined by

$$\mathfrak{C}_e : \mathbb{R}n_0 \otimes \mathbb{R}g_{\Gamma_e} \longrightarrow \mathcal{L}_M, \qquad \mathfrak{C}_e(n_0 \otimes g_{\Gamma_e}) = \varepsilon_e^{\beta}\mathcal{U}_M. \tag{16.50}$$

The map assembles the preceding factorization once its bases have been declared. The uniform vector n_0 does not select ε_e^{β}: that coordinate comes from APP, the oriented return, and its readers. An algebraic realization of the null mode is thus distinguished from an identification of the electron with the clock quantum.

The internal conclusion is the positive factorization of a central section, with its explicit operations, memory and normalizations. Assignment to a physical electron adds the charge representation, spin, coupling and corresponding units chart; it does not follow exclusively from a decimal coincidence. Metrological comparison is presented after the exceptional prolongation and does not alter the inputs to any construction performed here.

16.2.10.1 Structural dependence of the electronic coordinate

The electronic equation expresses a hierarchical composition. Its first factor, $\sqrt{3}/4$, belongs to the central matrix structure and fixes a geometric normalization. The exponent φ/π^2 uses the already generated regional coordinates in the indicated orientation of the areal–volumetric reader. The term $22\alpha^3$ incorporates the dodecaphasic coupling into the cubic correction. The factor $1 + 15\Delta_4$ preserves the global torsional normalization. Finally, the quotient of the action sections acts with the exponent determined by the return readers. Each operation thus has a different antecedent within the same construction.

The articulation becomes more precise when its dependencies are considered. The central norm can be calculated before the regional coordinates are evaluated. The exponential and cubic correction require those coordinates and the closure of α. Multiplicative memory additionally requires the action sections and the 108-position trajectory. Knowing the basal expression therefore does not amount to having evaluated the complete coordinate. The succession of operations does not add an external mass to the point $(5, 5)$; it develops the readers assigned to its persistent section.

The distinction between scale and structure also provides an interpretation of the physical comparison. Mass is the dimensional realization of the complete coordinate, while the centre, involution, and vacancy describe the relations that produce it. A common change of unit modifies the dimensional coordinates, but leaves the dimensionless ratios and the proved transport identities unchanged. This is the sense in which the electron preserves its structural character when expressed relative to a chronological scale.

16.2.11 Lorentzian transport and electromagnetic representation

The electronic section provides a structure and a coordinate before its application to an interaction. The Lorentzian realization of cubic incidence determines, with the declared temporal convention, the space of forms in which that interaction can be expressed. To formulate it, an oriented local realization $(\mathcal{M}, g)$ of dimension four is fixed, with tangent fibers identified with the model $Q_{\square}$. A connection ∇ with $\nabla g = 0$, a potential one-form $A \in \Omega^1(\mathcal{M})$, its curvature $F = dA$ and a charge representation of the section are also specified. These data are the arguments of the physical representation that follows.

Hodge duality satisfies $\star^2 = -I$ on two-forms. The electric and magnetic separation corresponds to a temporal choice in the decomposition $1 + 3$; F and $\star F$ preserve their relation within the same space of two-forms. Neutrality of a charge representation means $q = 0$, while the section may preserve orientation, phase and internal transport. Absence of electric coupling in that representation has content distinct from absence of phase information.

In this dimensional realization, let $m_e > 0$, q be the charge, $\gamma(s)$ a parametrized trajectory and $u = \dot{\gamma}$ its four-velocity. The Lorentz force is expressed by

$$m_e \nabla_u u^\mu = q F^\mu{}_\nu u^\nu. \tag{16.51}$$

Antisymmetry of F implies $F_{\mu\nu} u^\mu u^\nu = 0$. Consequently,

$$\frac{d}{ds} g(u,u) = 2g(\nabla_u u, u) = 0.$$

Normalization of the trajectory is preserved provided ∇ is metric and (16.51) holds. The mass used is the dimensional realization of the preceding electronic section; the coupling parameter is identified in the adopted electromagnetic convention. The specific derivation of the potential and a field configuration constitutes an additional problem with its own boundary conditions. The result presented here determines the domain, representation data and conservation implied by its equation.

TPK route inversion and orientation of an electronic line then offer a precise comparison: each has its involution, transport and representation. The Wheeler and Feynman antecedent places the question historically; equivalence of two realizations is decided through their maps and compatibility conditions. This arrangement allows the arithmetic origin of the electron to remain connected to the formulation of its interaction, making explicit the function of each stage.

16.3 CKM mixing angles, CP phase and leptonic transport

16.3.1 Common antecedent and distinction between the two transports

The common foundation of the article produces the enriched state through the two APP evaluations, the oriented TRIT selection and the effective composition of selection, transport, carry, memory and return in the TPK. The angular expressions do not replace that state: they are readers of its record. In particular, the extension $w_6 \to w_{12} \to w_{18} \to w_{24} \to w_{30} \to R_{36} \to G_9$ preserves the provenance of the numerical readings, and the holonomy Γ_9 acts before the reduced expression K_{ph}. Phase return does not reset the record.

Two classes of objects are received here from that same antecedent. The first contains the direct angle, the counterangle and the torsional defect Δ_4, already constructed in their own readers. The second contains the surviving extensions, their observable routes, the dodecaphasic record and the operators of the material fibres. The extension–route–record–signature–character–towers chain preserves the atlas $81/56/13/324$ and distinguishes the null chart from the positive branch. The mixing angles neither retrospectively select those routes nor generate the mass eigenvalues.

Quark transport uses a rational angular chart and its complex unitary realization. Leptonic transport uses the spectral frames of the charged and neutral fibres. Both proceed from the common discrete

structure, but their domains, readers and changes of basis are distinct. Nor are the Barbero–Immirzi functional and mixing defined through one another: they receive typed expressions from the angular and incidence antecedent. Their relationship is expressed through those maps, not through a retrospective formula using a Barbero value as a mixing selector.

Remark 16.40 (Sectorial rules and evaluation of the reader). The three sectorial rules and their evaluation are developed below in Section 16.3.2. Theorem 16.44 obtains the columns as images of the angular basis. Subsequent proofs establish inversion of the chart, sheet transport, the metric and the complex realization. Choosing the rules and evaluating their coefficients are two distinct mathematical questions: the initial system is declared in Definition 16.43, and the theorem's proof uses that complete system. The preceding composition that selects those rules from the incidence requires its own proof.

16.3.2 Sectorial incidence and construction of the angular reader

The angular construction uses the action and closure expressions obtained previously from APP–TRIT–TPK. The direct angle, the counterangle and the torsional defect preserve their own domains. The quark reader combines these three coordinates through sectorial incidence; its evaluation precedes the complex amplitudes and metrological comparison. The hexadic–octadic inclusion enters the third sector through the difference between their cardinalities.

16.3.2.1 Marked incidence and bidirectional normalization

The reader invariants preserve the nature of the object from which they arise. The marked intersection $B_K = H_e \cap P_\pi$ has cardinality $b = 4$; the integer $q = 7$ is the coordinate of the pivot p_φ in the oriented numbering. The determination order of the Steiner design is $p = 5$. The hexadic and octadic supports have cardinalities $h_6 = 6$ and $o_8 = 8$, and the tritic orientation alphabet has cardinality $t = 3$. A support cardinality and a pivot coordinate are used through their respective maps, even when both are expressed as integers.

Normalization of the pentagonal self-crossing admits the following explicit realization. Let P be the five-position support of the closure, and let $X = P \times P$ be its space of ordered pairs. Bidirectional inversion on this space is

$$\jmath : X \longrightarrow X, \qquad \jmath(x, y) = (y, x), \qquad \jmath^2 = I.$$

The stabilizer information of each orbit is preserved. For an orbit $[z]$, define its weight as $1/|\operatorname{Stab}_{C_2}(z)|$. The sum of these weights is the cardinality of the action groupoid $X/\!/C_2$.

Proposition 16.41 (Orbital measure of bidirectional self-crossing)**.** *For a finite support of cardinality p, inversion of ordered pairs determines*

$$|(P \times P)/\!/C_2| = \sum_{[z] \in (P \times P)/C_2} \frac{1}{|\operatorname{Stab}_{C_2}(z)|} = \frac{p^2}{2}.$$

For $p = 5$, the self-crossing has ten free orbits and five fixed orbits, and its orbital measure is $10 + 5/2 = 25/2$.

Proof. Each diagonal pair (x, x) forms a fixed orbit, whose stabilizer has order two; there are p such orbits. The remaining $p(p-1)$ pairs are grouped in twos, with trivial stabilizer. Their contribution is therefore $p(p-1)/2$, and that of the diagonal is $p/2$. The sum is $p^2/2$. Equivalently, the orbit–stabilizer identity gives $1/|\operatorname{Stab}(z)| = |[z]|/2$; summing over all orbits produces $|P \times P|/2$. □

The measure explicitly preserves the contribution of fixed points. The ordinary orbit set has cardinality $p(p+1)/2$, equal to fifteen in the pentagonal closure. The orbital measure and that cardinality are two

determinate readings of the same inversion. The reader's bidirectional normalization uses the former: the ten free orbits have unit weight, and the five diagonals have half-unit weight.

This realization specifies the factor $p^2/2$ used in the following sectorial system. The assignment of that contribution to sector 23, its negative orientation, and the other two incidence rules remain specified by the complete sectorial system. The proof establishes the normalization of self-crossing and preserves the selection of the angular functionals as a distinct matter.

16.3.2.2 Carry transport and integer holonomy of the arithmetic crossing

The modular reading of the crossing admits an integral realization that retains the information accumulated when crossing the boundary. That realization is developed here through transport quotients. Its domain is a lifted chart of the APP support; the subsequent assignment of a region to an angular sector preserves its own incidence map.

Nonadic chart and lifted transport. We use the exact decomposition

$$\rho_9(n) = 1 + ((n-1) \bmod 9), \qquad q_9(n) = \left\lfloor \frac{n-1}{9} \right\rfloor, \qquad n = \rho_9(n) + 9q_9(n).$$

A lifted position is $z = (x, y, q_x, q_y) \in D_9^2 \times \mathbb{Z}^2$, where $X = x + 9q_x$ and $Y = y + 9q_y$. The integer increment u of the first coordinate acts by

$$T_u^x(x, y, q_x, q_y) = \big(\rho_9(x+u), y, q_x + c_9(x;u), q_y\big), \qquad c_9(x;u) = \left\lfloor \frac{x+u-1}{9} \right\rfloor.$$

The second coordinate is transported in the same way. Cardinal advances correspond to $u = \pm 1$; the zero increment preserves the state. The cocycle satisfies

$$c_9(x; u+v) = c_9(x;u) + c_9(\rho_9(x+u); v).$$

Indeed, both sides are the quotient of the unique decomposition of $x + u + v$ with remainder in D_9. Consequently, $T_v^x T_u^x = T_{u+v}^x$ and $T_{-u}^x = (T_u^x)^{-1}$; the transports of the two coordinates commute. The state preserves the number of edge crossings even when the visible position returns.

Lifted sheets and update of records. The integral realization is fixed by prolonging the APP arithmetic operations to the transported coordinates:

$$\widetilde{S}(z) = X + Y = x + y + 9(q_x + q_y),$$
$$\widetilde{P}(z) = XY = xy + 9(q_x y + q_y x) + 81 q_x q_y.$$

At each position the record of both sheets is preserved:

$$\big(\rho_9(\widetilde{S}), q_9(\widetilde{S}), \rho_9(\widetilde{P}), q_9(\widetilde{P})\big).$$

If $e : z \to z'$ is a link and T the selected sheet, its full increment is obtained from the record before and after transport:

$$\delta_e \widetilde{T} = \rho_9(\widetilde{T}(z')) - \rho_9(\widetilde{T}(z)) + 9\big(q_9(\widetilde{T}(z')) - q_9(\widetilde{T}(z))\big). \tag{16.52}$$

This identity makes conservation explicit: the residue jump and the quotient increment jointly reconstruct the change of sheet. Sheet selection, cardinal transport and record updating are distinct operations. In this chart, the ordered sheet selection (T_x, T_y) is first composed with T_d and with the updating of their quotients; the formula does not postulate that the entire TPK is reduced to this local representation.

Proposition 16.42 (Integral curvature and boundary transport)**.** *Define* $\widetilde{A}_x = \delta_x \widetilde{T}_x$ *and* $\widetilde{A}_y = \delta_y \widetilde{T}_y$ *through* (16.52)*, and let*

$$\widetilde{F}_{xy}(X,Y) = \widetilde{A}_x(X,Y) + \widetilde{A}_y(X+1,Y) - \widetilde{A}_x(X,Y+1) - \widetilde{A}_y(X,Y).$$

Homogeneous choices have zero curvature. The mixed choices $(\widetilde{S},\widetilde{P})$ *and* $(\widetilde{P},\widetilde{S})$ *have curvature* $+1$ *and* -1*, respectively, as integers. For the positive boundary of a rectangle* $R_{a,b}$ *with integer sides* $a, b > 0$*,*

$$\widetilde{W}_{SP}(\partial R_{a,b}) = ab, \qquad \widetilde{W}_{PS}(\partial R_{a,b}) = -ab.$$

The result is independent of the base point and the number of crossings of nonadic seams.

Proof. Exact reconstruction of the two sheets gives

$$\delta_x \widetilde{S} = \delta_y \widetilde{S} = 1, \qquad \delta_x \widetilde{P} = Y, \qquad \delta_y \widetilde{P} = X.$$

Thus the SP connection is $(1, X)$, with curvature $1 + (X+1) - 1 - X = 1$; the PS connection is $(Y, 1)$, with curvature $Y + 1 - (Y+1) - 1 = -1$. For a single sheet, mixed differences cancel. Summing the curvatures of the ab plaquettes cancels each interior link with its inverse and preserves the boundary links. This yields $\pm ab$ in $\mathbb{Z}$. The transported quotients ensure the same identity holds across any seam. □

The boundary accumulator can in turn be retained in nonadic form: $W = r_W + 9q_W$. For an integer contribution w_e, it is updated by

$$r'_W = \rho_9(r_W + w_e), \qquad q'_W = q_W + \left\lfloor \frac{r_W + w_e - 1}{9} \right\rfloor.$$

Composition through links yields the same sum as its direct integer evaluation. In this chart, integer zero has coordinates $(r_W, q_W) = (9, -1)$; this is an arithmetic representation of the accumulator, with no additional identification with the different oriented zeros of the TRIT.

Three operations on orientation. Let γ be a closed cardinal route. Its time reversal γ^{-1} reverses the order of links and the orientation of each; by additivity,

$$\widetilde{W}_{SP}(\gamma^{-1}) = -\widetilde{W}_{SP}(\gamma).$$

Exchange of sheets likewise satisfies

$$\widetilde{W}_{PS}(\gamma) = -\widetilde{W}_{SP}(\gamma).$$

To check this on any route, note that the sum of both connections is the exact increment of $\widetilde{S}+\widetilde{P} = X+Y+XY$; its integral over a loop is zero.

The simultaneous half-turn of both directions, on the other hand, acts through $J_c(X,Y) = (2c_x - X, 2c_y - Y)$, with $2c \in \mathbb{Z}^2$, and therefore preserves the lattice chart. For a loop,

$$\widetilde{W}_{SP}(J_c\gamma) = \widetilde{W}_{SP}(\gamma).$$

Indeed, the closed horizontal contribution is zero and the vertical contribution transforms as $\sum(2c_x - X)(-\delta Y) = \sum X\delta Y$, since $\sum \delta Y = 0$. A spatial half-turn therefore preserves the areal orientation of this chart. Its combination with time reversal or sheet interchange is calculated by composing the corresponding operations.

The difference matters for the electronic record: simultaneous reversal of both displacements every 27 advances produces, in windows of 9, the sequence

$$\tau_j = (-1)^{\lfloor j/3 \rfloor}, \qquad 0 \le j < 12, \qquad (+,+,+,-,-,-,+,+,+,-,-,-).$$

This sequence records advance orientations. Holonomy additionally requires the links traversed, selected sheets, and accumulated quotients. Transport of electronic orientation is thus preserved without identifying a half-turn of both axes with time reversal of a loop.

Application to the marked crossing and sector scope. For the rectangular crossing with sides 4 and 7, this realization expresses exactly

$$\widetilde{W}_{SP} = 28 = 1 + 9 \cdot 3, \qquad \widetilde{W}_{SP}(\gamma^{-1}) = -28 = 8 + 9(-4).$$

The integer preserves the oriented area; the isolated residue expresses only its nonadic class. For example, starting from $(7, 1)$, the periodic representatives of the links sum to -35, whereas the carry correction adds 63 and recovers 28.

In the marked incidence of the angular reader, 4 is $|H_e \cap P_\pi|$ and 7 is the marked pivot p_φ. The rectangular reading supplies an integer realization of the product $|H_e \cap P_\pi|\, p_\varphi$ once that crossing has been declared. The map identifying that region with sector 12 must preserve the mark and orientation of the sector system in Definition 16.43. Sector assignments $12, 23, 13$ and their combinations of A, h, Δ° are an additional specification beyond the rectangular-holonomy law. The present result makes the area lift, carry, and their involutions explicit, with these starting data declared.

Definition 16.43 (Sectorial system of angular incidence)**.** The sectorial system consists of the ordered sextuple of invariants

$$(b, p, q, h_6, o_8, t) = (4, 5, 7, 6, 8, 3)$$

and the linear maps, on angular coordinates (a, u, d),

$$\mathscr{A}_{12}^{\mathrm{ang}}(a, u, d) = 2a - pu + bq\, d, \tag{16.53}$$

$$\mathscr{A}_{23}^{\mathrm{ang}}(a, u, d) = qu - \frac{p^2}{2}\, d, \tag{16.54}$$

$$\mathscr{A}_{13}^{\mathrm{ang}}(a, u, d) = a - pb\, u - \frac{o_8 - h_6}{t}\, d. \tag{16.55}$$

The symbols h_6, o_8 preserve the cardinalities of the marked incidence; $t = 3$ is the declared tritic normalization of that sector. The sextuple and the three rules jointly form the starting system of this construction.

Theorem 16.44 (Exact evaluation of the sectorial system)**.** *The joint map* $(\mathscr{A}_{12}^{\mathrm{ang}}, \mathscr{A}_{23}^{\mathrm{ang}}, \mathscr{A}_{13}^{\mathrm{ang}})^{\mathsf{T}}$ *of Definition 16.43 has the unique matrix*

$$\mathsf{M}_{\mathrm{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}.$$

The three columns are the images of the coordinate vectors $(1, 0, 0)$*,* $(0, 1, 0)$*, and* $(0, 0, 1)$ *under these maps.*

Proof. The sectorial operations give

$$bq = 4 \cdot 7 = 28, \qquad p^2/2 = 5^2/2 = 25/2, \qquad pb = 5 \cdot 4 = 20, \qquad (o_8 - h_6)/t = (8 - 6)/3 = 2/3.$$

Thus the successive images of the coordinate basis are

$$c_A = (2, 0, 1)^{\mathsf{T}}, \qquad c_h = (-5, 7, -20)^{\mathsf{T}}, \qquad c_\Delta = (28, -25/2, -2/3)^{\mathsf{T}}.$$

A linear map is determined by its images on a basis. The matrix with these images as its columns is the one indicated. The proof uses the complete sectorial system of the definition. □

On the previously produced coordinates, take

$$(a, u, d) = \left(A_{\mathrm{deg}}, \frac{C^{*}_{\mathrm{deg}}}{6}, \frac{180}{\pi}\Delta_4 \right).$$

The direct, conjugate, and torsional contributions to each angle are thereby individualized. In particular, the difference $o_8 - h_6 = 2$ is preserved in the coefficient of the torsional defect in sector 13. This concrete presence makes it possible to follow incidence through to the angular reader, keeping the sectorial system, its evaluation, and the unitary parametrization separate.

16.3.3 Rational CKM chart and angular units

We write A_{deg} and C^{*}_{deg} for the degree coordinates of the HMT angles already generated. We use the abbreviations

$$h_\theta = \frac{C^{*}_{\mathrm{deg}}}{6}, \qquad \Delta_{\mathrm{deg}} = \frac{180}{\pi_{\mathrm{HMT}}}\Delta_4, \qquad x_\theta = (A_{\mathrm{deg}}, h_\theta, \Delta_{\mathrm{deg}})^{\mathsf{T}}. \tag{16.56}$$

Here h_θ is an angular coordinate, distinct from the Planck constant h. Conversion to degrees is a subsequent expression; it does not introduce a conventional value of π into the generator.

Definition 16.45 (Quark angular reader)**.** The inherited reader $\mathsf{M}_{\mathrm{CKM}} : \mathbb{R}^3 \to \mathbb{R}^3$ expresses

$$\boldsymbol{\theta}_{\mathrm{deg}} = \mathsf{M}_{\mathrm{CKM}} x_\theta, \qquad \mathsf{M}_{\mathrm{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \qquad \delta_{\mathrm{deg}} = 9 A_{\mathrm{deg}}. \tag{16.57}$$

The order of the components is $(12, 23, 13)$. The subsequent trigonometric functions are evaluated at $\vartheta_{ab} = (\pi_{\mathrm{HMT}}/180)\theta_{ab,\mathrm{deg}}$ and $\delta = (\pi_{\mathrm{HMT}}/180)\delta_{\mathrm{deg}}$.

Proposition 16.46 (Rational reconstruction of the coordinates)**.** *The reader has determinant* $-3857/6$ *and inverse*

$$\mathsf{M}^{-1}_{\mathrm{CKM}} = \begin{pmatrix} 1528/3857 & 3380/3857 & 801/3857 \\ 75/3857 & 176/3857 & -150/3857 \\ 6/551 & -30/551 & -12/551 \end{pmatrix}. \tag{16.58}$$

In particular,

$$\begin{aligned} A_{\mathrm{deg}} &= \frac{1528\theta_{12,\mathrm{deg}} + 3380\theta_{23,\mathrm{deg}} + 801\theta_{13,\mathrm{deg}}}{3857}, \\ C^{*}_{\mathrm{deg}} &= \frac{6(75\theta_{12,\mathrm{deg}} + 176\theta_{23,\mathrm{deg}} - 150\theta_{13,\mathrm{deg}})}{3857}, \\ \Delta_{\mathrm{deg}} &= \frac{6\theta_{12,\mathrm{deg}} - 30\theta_{23,\mathrm{deg}} - 12\theta_{13,\mathrm{deg}}}{551}. \end{aligned} \tag{16.59}$$

The phase satisfies the exact linear check

$$3857\,\delta_{\mathrm{deg}} = 9(1528\theta_{12,\mathrm{deg}} + 3380\theta_{23,\mathrm{deg}} + 801\theta_{13,\mathrm{deg}}). \tag{16.60}$$

Proof. Expanding the determinant of (16.57) gives $-3857/6$. Multiplying the matrix indicated in (16.58) by $\mathsf{M}_{\mathrm{CKM}}$, in both orders, gives I_3. Its rows recover the three components of x_θ; the second is multiplied by six to recover C^*_{deg}. Finally, we use $\delta_{\mathrm{deg}} = 9A_{\mathrm{deg}}$. □

Inversion reconstructs the coordinates of the reader once the reader has been fixed. It does not turn the observed angles into antecedents of A, C^* or Δ_4.

16.3.4 Geometric decomposition under sheet inversion

In the angular chart, sheet inversion preserves a, d and reverses u. The expressed vector admits the decomposition

$$\boldsymbol{\theta}_+ = ac_A + uc_h + dc_\Delta, \qquad \boldsymbol{\theta}_- = ac_A - uc_h + dc_\Delta.$$

Proposition 16.47 (Even and odd components of angular transport)**.** *The two expressions satisfy*

$$\frac{\boldsymbol{\theta}_+ + \boldsymbol{\theta}_-}{2} = ac_A + dc_\Delta, \qquad \frac{\boldsymbol{\theta}_+ - \boldsymbol{\theta}_-}{2} = uc_h.$$

The contribution that changes sign occupies the line $\mathrm{span}(c_h)$*; the preserved component occupies* $\mathrm{span}(c_A, c_\Delta)$.

Proof. The sum cancels the terms uc_h, and the difference cancels the other two. The columns are linearly independent, since $\det \mathsf{M}_{\mathrm{CKM}} = -3857/6$. The two components therefore form a direct decomposition. □

The same inversion acts on the Barbero–Immirzi evaluation: it interchanges q_+ and q_-, and reverses the bilinear component in A and C^*. Consequently, the incidence functional is odd. An observable symmetric in both channels preserves its value. This parity difference expresses how a single operation on the structure yields different responses depending on the reader: areal orientation, symmetric information and the mixing vector preserve different information from the same antecedent.

16.3.5 Transport of sheet inversion

Conjugation of the angular charts retains A_{deg} and Δ_{deg}, and reverses h_θ. Define

$$D_\theta = \mathrm{diag}(1, -1, 1), \qquad c_h = (-5, 7, -20)^{\mathsf{T}}, \qquad \ell_h = \frac{1}{3857}(75, 176, -150). \tag{16.61}$$

If $c_A = (2, 0, 1)^{\mathsf{T}}$ and $c_\Delta = (28, -25/2, -2/3)^{\mathsf{T}}$, then $\ell_h c_A = \ell_h c_\Delta = 0$ and $\ell_h c_h = 1$.

Theorem 16.48 (Rational reflection and transported metric)**.** *The operator*

$$\mathcal{R}_{\mathrm{CKM}} = \mathsf{M}_{\mathrm{CKM}} D_\theta \mathsf{M}_{\mathrm{CKM}}^{-1} = I_3 - 2c_h\ell_h \tag{16.62}$$

satisfies

$$\mathcal{R}_{\mathrm{CKM}}^2 = I_3, \qquad \det \mathcal{R}_{\mathrm{CKM}} = -1, \qquad \mathcal{R}_{\mathrm{CKM}} \mathsf{M}_{\mathrm{CKM}} = \mathsf{M}_{\mathrm{CKM}} D_\theta. \tag{16.63}$$

It is an orthogonal reflection for the positive metric

$$G_\theta = (\mathsf{M}_{\mathrm{CKM}}^{-1})^{\mathsf{T}} \mathsf{M}_{\mathrm{CKM}}^{-1}, \qquad \mathcal{R}_{\mathrm{CKM}}^{\mathsf{T}} G_\theta \mathcal{R}_{\mathrm{CKM}} = G_\theta. \tag{16.64}$$

Proof. The operator $P_h = c_h\ell_h$ satisfies $P_h^2 = P_h$, has image $\mathbb{R}c_h$ and kernel $\mathrm{span}(c_A, c_\Delta)$. Hence $(I_3 - 2P_h)^2 = I_3$, with eigenvalues $1, 1, -1$. Its action on these three columns is that of D_θ in the original basis, proving conjugation and intertwining. For $v \neq 0$, $v^{\mathsf{T}} G_\theta v = \|\mathsf{M}_{\mathrm{CKM}}^{-1} v\|^2 > 0$. The last identity follows from $D_\theta^{\mathsf{T}} D_\theta = I_3$. □

Its full expression, retained from the preceding support, is

$$\mathcal{R}_{\mathrm{CKM}} = \begin{pmatrix} 4607/3857 & 1760/3857 & -1500/3857 \\ -150/551 & 199/551 & 300/551 \\ 3000/3857 & 7040/3857 & -2143/3857 \end{pmatrix}. \tag{16.65}$$

The phase $\delta_{\mathrm{deg}} = 9A_{\mathrm{deg}}$ remains fixed under this reflection. Consequently, sheet transport must not be identified with CP conjugation of amplitudes, which changes the sign of the complex phase. The combinatorial parity of a route, its sheet conjugation and its physical representation retain the distinct domains in equations (16.66) and (16.67).

16.3.6 Record parity and CP realization

To preserve the discrete support of this distinction, let ρ be a recorded route of length L, with directions $d_t \in \{N, E, S, W\}$ and sheets $\ell_t \in \{\Sigma, \Pi\}$. The intermediate states are those produced by TPK transport. The reflection P acts through $(i, j) \mapsto (i, -j) \pmod 9$ and permutes E, W; C_{hoja} permutes Σ, Π; T_{reg} reverses the order of the record and replaces each direction by its opposite. Inversion is performed on the enriched record, not on a projection that has forgotten its predecessor.

Proposition 16.49 (Invariance of the combinatorial route action)**.** *For $\lambda \geq 0$, the dimensionless functional*

$$\mathcal{A}_\lambda(\rho) = \sum_{t=2}^{L} \left[1 - \delta_{d_t, d_{t-1}} + \lambda(1 - \delta_{\ell_t, \ell_{t-1}})\right] \tag{16.66}$$

is invariant under P, C_{hoja}, T_{reg} and their compositions.

Proof. A global permutation of the alphabet preserves equality between neighbours. Reversing the order preserves the number of consecutive changes. Therefore, each operation separately preserves the direction turns and sheet changes summed by (16.66). □

λ is the declared parameter of this system of functionals; $\mathcal{A}_\lambda$ is not identified with a dimensional Planck section. Invariance of the counter does not imply that every oriented reader is even either. In a mixing realization $\mathcal{V}$, CP compatibility takes the precise form

$$\mathcal{V}(CP_{\mathrm{HMT}}\rho) = D_u\, \overline{\mathcal{V}(\rho)}\, D_d^{\dagger}, \tag{16.67}$$

with the declared diagonal phase changes and representation spaces. The route counter, rational sheet transport and complex conjugation are three distinct maps of the preserved support.

Remark 16.50 (CP compatibility of the physical reader). Equation (16.67) is the CP compatibility condition for the complete physical reader. The proofs in this section establish record invariance and the identities of the complex reader, but do not automatically turn $\mathcal{R}_{\mathrm{CKM}}$ into CP_{HMT}. This qualification concerns that representation square, not the existence of the internal route transport already constructed.

16.3.7 Unitary amplitudes, CP phase and Jarlskog invariant

We write $c_{ab} = \cos\vartheta_{ab}$ and $s_{ab} = \sin\vartheta_{ab}$. The complex realization of the reader is

$$V_{\mathrm{CKM}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}. \tag{16.68}$$

The complex structure used is the realization of the oriented structure already constructed; the unitary parametrization expresses the angular result, rather than determining its coefficients.

Proposition 16.51 (Unitarity and phase invariant)**.** *The matrix* (16.68) *is unitary. The quartet*

$$J_{\mathrm{CKM}} = \mathrm{Im}(V_{11}V_{22}\overline{V_{12}}\,\overline{V_{21}}) = c_{12}c_{23}c_{13}^2 s_{12}s_{23}s_{13}\sin\delta \tag{16.69}$$

is invariant under $V \mapsto D_u V D_d$, with unitary diagonal D_u, D_d, and changes sign under $V \mapsto \overline{V}$.

Proof. The matrix is the product $R_{23}U_{13}(\delta)R_{12}$, where

$$\begin{aligned} R_{12} &= \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ U_{13}(\delta) &= \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix}, \\ R_{23} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix}. \end{aligned} \tag{16.70}$$

Each factor is unitary by $c_{ab}^2 + s_{ab}^2 = 1$. Substituting the four entries of the quartet, its imaginary part is $c_{12}s_{12}c_{13}^2 c_{23}s_{23}s_{13}(c_{12}^2 + s_{12}^2)\sin\delta$, which gives (16.69). The phase factors of each row and column cancel with their conjugates. Complex conjugation reverses the imaginary part. □

A nonzero J_{CKM} obstructs making the entire matrix real through those phase changes. Its physical interpretation also retains the corresponding sector and representation; it is not a scalar correction added to the masses. If the spectra have degeneracies, their freedoms of change of basis must be preserved before formulating a CP conclusion.

16.3.8 Conjugation of amplitudes and oriented invariant

The complex realization preserves the three mixing angles and the interference phase as explicit coordinates. Write $V(\boldsymbol{\vartheta}, \delta) = R_{23}U_{13}(\delta)R_{12}$, with the rotations and phase defined in the unitary development. The arguments of the trigonometric functions are their representatives in radians.

Proposition 16.52 (Conjugation in the mixing realization)**.** *For the three declared real angles and the declared real phase,*

$$V(\boldsymbol{\vartheta}, -\delta) = \overline{V(\boldsymbol{\vartheta}, \delta)}, \qquad J(\boldsymbol{\vartheta}, -\delta) = -J(\boldsymbol{\vartheta}, \delta).$$

The invariant J also preserves its value under diagonal phase changes of the charged bases.

Proof. The matrices R_{12} and R_{23} have real entries. In U_{13}, the change $\delta \mapsto -\delta$ conjugates the exponential factors. Conjugating the product gives the first identity. In the quartet $V_{11}V_{22}\overline{V_{12}V_{21}}$, the diagonal factors of each row and column cancel. Conjugation reverses the sign of its imaginary part. Equivalently, the result follows by substitution into

$$J = c_{12}c_{23}c_{13}^2 s_{12}s_{23}s_{13}\sin\delta.$$

□

In the three-generation sector, the nonzero internal value of J expresses the obstruction to realizing all amplitudes by a real matrix through diagonal phase changes. This is the charge–parity violation reading

of the constructed matrix. The route reading, in turn, preserves the discrete operation of conjugation and parity and its representation. Their full compatibility is expressed by the square

$$\mathcal{V}(CP_{\mathrm{HMT}}\rho) = D_u\,\overline{\mathcal{V}(\rho)}\,D_d^{\dagger},$$

with the representation spaces and phase changes declared. The conjugation identity proved here fixes exactly the action that this square must transport.

Proposition 16.53 (Mixing and spectrum)**.** *Let $B_u = \mathrm{diag}(a_1, a_2, a_3)$ and $B_d = \mathrm{diag}(b_1, b_2, b_3)$ be the Hermitian mass readers already produced in their fibres. The transport*

$$B_d^{(u)} = V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^{\dagger} \tag{16.71}$$

preserves the spectrum of B_d, and satisfies

$$\left|\det[B_u, B_d^{(u)}]\right| = 2|J_{\mathrm{CKM}}| \prod_{i<j} |a_i - a_j| \prod_{r<s} |b_r - b_s|. \tag{16.72}$$

Proof. Unitary conjugation preserves the characteristic polynomial. If $B = B_d^{(u)}$, the entries of the commutator are $[B_u, B]_{ij} = (a_i - a_j)B_{ij}$, with zero diagonal. Its determinant is

$$2i(a_1 - a_2)(a_2 - a_3)(a_3 - a_1)\,\mathrm{Im}(B_{12}B_{23}B_{31}).$$

Upon substituting $B_{ij} = \sum_r b_r V_{ir}\overline{V_{jr}}$, the imaginary part is an alternating polynomial of degree three in the b_r. It is divisible by their three differences: if two b_r coincide, $B = bI + (b' - b)vv^{\dagger}$ and the off-diagonal product is real. Expanding one coefficient, using column orthogonality, gives the quartet of (16.69), with the sign of the chosen ordering. Taking absolute values removes that convention and yields (16.72), including its vanishing at degeneracies. □

Remark 16.54 (Evaluation and comparison). The decimal CKM coordinates in Table 83.3 are subsequent evaluations of the preceding reader. The quantitative comparison section also presents the angles and amplitude moduli. These evaluations are not used to choose its columns or to prove (16.63) or (16.69). This block assigns no new metrological significance to those figures: a joint comparison requires its external edition, observable definitions and covariance. Marginal deviations do not replace it.

16.3.9 Leptonic record, operators and spectral frames

The leptonic sector preserves a different domain. The spectral construction of (16.73) gathers four neutral depths of dimensions 2, 4, 6, 8, before the three-dimensional compression:

$$\mathcal{H}_\nu^{\mathrm{reg}} = \bigoplus_{j=1}^{4} \mathcal{H}_{\nu,G_j}, \qquad \dim \mathcal{H}_\nu^{\mathrm{reg}} = 20, \qquad M_\nu = (P_\nu \mathcal{M}_\nu P_\nu)|_{\mathrm{Im}\,P_\nu}, \quad \mathrm{rank}\,P_\nu = 3. \tag{16.73}$$

The four depths are neither four flavours nor four masses. P_ν is the neutral projector inherited from the record reader, and $M_\ell = (P_\ell \mathcal{M}_\ell P_\ell)|_{\mathrm{Im}\,P_\ell}$, with $\mathrm{rank}\,P_\ell = 3$, is the compressed charged operator. The scale of these operators proceeds from the action–clock–mass branch; no PMNS angle determines it.

Definition 16.55 (Transport of leptonic frames)**.** In the declared common three-dimensional realization, let $F_\ell, F_\nu : \mathbb{C}^3 \to \mathcal{H}_3$ be complete orthonormal spectral frames of the charged and neutral operators. The identification with $\mathcal{H}_3$ is fixed before taking their product. The reader is

$$U_{\mathrm{PMNS}} = F_\ell^{\dagger} F_\nu : \mathbb{C}^3_\nu \longrightarrow \mathbb{C}^3_\ell. \tag{16.74}$$

The frames and identification proceed from the representation and the record; they are not chosen from an experimental matrix one wishes to reproduce.

Theorem 16.56 (Unitarity, degeneracies and naturality)**.** *Under the preceding identification, U_{PMNS} is unitary. If the spectra are simple, their frame changes are diagonal phases and permutations; once the spectral order is fixed, the phases remain. If there are multiplicities $m_{\ell,\lambda}$ and $m_{\nu,\eta}$, the object independent of these bases is the double coset*

$$\left(\prod_{\lambda} U(m_{\ell,\lambda})\right) \backslash U(3) / \left(\prod_{\eta} U(m_{\nu,\eta})\right). \tag{16.75}$$

A common unitary refinement W intertwining both operators preserves transport when the frames are extended as $F'_{\ell} = WF_{\ell}$, $F'_{\nu} = WF_{\nu}$.

Proof. Completeness in the same space gives $F_{\ell}F_{\ell}^{\dagger} = F_{\nu}F_{\nu}^{\dagger} = I_{\mathcal{H}_3}$. Hence $U_{\mathrm{PMNS}}^{\dagger}U_{\mathrm{PMNS}} = F_{\nu}^{\dagger}F_{\ell}F_{\ell}^{\dagger}F_{\nu} = I_3$, and likewise in the opposite order. Basis changes preserving a Hermitian operator act blockwise within its eigenspaces. Substituting $F_{\ell}D_{\ell}, F_{\nu}D_{\nu}$ gives $U \mapsto D_{\ell}^{\dagger}UD_{\nu}$, which determines the double coset. Finally, $(WF_{\ell})^{\dagger}WF_{\nu} = F_{\ell}^{\dagger}F_{\nu}$. □

The common-space condition is substantive. For two isometric frames in a larger ambient space, separate orthonormality produces only a compression; it does not justify cancelling $F_{\ell}F_{\ell}^{\dagger}$ as though it were the ambient identity.

16.3.10 Record kernel and rank sufficiency

The readers Ξ_r preserve the carry, sheet, orientation, vacancy and holonomy components of the record. On the finite set of admitted routes, form

$$K_{\mathrm{reg}}(c,d) = \sum_{\substack{r\in\{\mathrm{car,hoj,ori,vac,hol}\}\\ d_r>0}} \frac{\langle \Xi_r(c), \Xi_r(d)\rangle}{d_r}, \qquad d_r = \sum_{x} \|\Xi_r(x)\|^2. \tag{16.76}$$

Normalization proceeds from the norms of the record itself. Zero-norm components are omitted; they are not replaced with terms taken from a target matrix. Each Ξ_r and each aggregation class must preserve its route provenance and its compatibility with refinement.

Proposition 16.57 (Positivity and rank of the record)**.** *The matrix K_{reg} is positive semidefinite. If its feature vectors are aggregated into three sectors, the resulting Gram matrix G_{ν} is invertible exactly when those three aggregated vectors are linearly independent.*

Proof. Define $v_c = \bigoplus_{r:d_r>0} d_r^{-1/2}\Xi_r(c)$. Then $K_{\mathrm{reg}}(c,d) = \langle v_c, v_d\rangle$, and for arbitrary coefficients z_c, $\sum_{c,d}\overline{z}_c K_{\mathrm{reg}}(c,d)z_d = \|\sum_c z_c v_c\|^2 \geq 0$. The same identity holds for the three aggregated vectors. The kernel of their Gram matrix consists exactly of their linear relations, yielding the invertibility criterion. □

This Gram matrix is distinguished from a sector of the *cross* matrix between the charged classes L_a and neutral classes N_b, defined by equation (16.77):

$$G_{ab} = \frac{1}{\sqrt{|L_a||N_b|}} \sum_{c\in L_a}\sum_{d\in N_b} K_{\mathrm{reg}}(c,d). \tag{16.77}$$

The nonempty classes proceed from the internal classifier. The cross matrix need not be Hermitian, unlike a Gram matrix constructed with the same three vectors on both sides.

Proposition 16.58 (Polar factor and its scope)**.** *If G is invertible,*

$$U_G = G(G^\dagger G)^{-1/2} \tag{16.78}$$

is unitary. If G is also Hermitian positive definite, then $U_G = I_3$. Consequently, the rank of a positive Gram matrix certifies the sufficiency of the observables, but does not by itself determine nontrivial mixing between the charged and neutral frames.

Proof. $G^\dagger G$ is positive definite; its positive square root is invertible and

$$U_G^\dagger U_G = (G^\dagger G)^{-1/2}(G^\dagger G)(G^\dagger G)^{-1/2} = I_3.$$

By finite dimensionality, also $U_G U_G^\dagger = I_3$. If $G > 0$, the positive square root of $G^\dagger G = G^2$ is G, and $U_G = I_3$. □

Remark 16.59 (Polar factorization and spectral frames). Equations (16.76) and (16.77) preserve, respectively, the rank criterion of the record and the cross matrix of the charged and neutral sectors. This exposition separates their functions to avoid turning a positive Gram matrix into an artificial angle selector. Identifying U_G with (16.74) requires compatibility of the cross kernel with the declared spectral frames. The identity of a polar factor or mere unitarity does not replace that identification. No new numerical PMNS matrix is presented here.

16.3.11 Rank-one phase check

Let b_K be the normalized incidence witness of the source and $Q_K = |b_K\rangle\langle b_K|$. The operator

$$\Phi_K = I + (e^{i\pi_{\mathrm{HMT}}/6} - 1)Q_K \tag{16.79}$$

is unitary in its ambient space. For isometric frames, consider $U_\Phi = F_\ell^\dagger \Phi_K F_\nu$.

Proposition 16.60 (Exact criterion for a unitary compression)**.** *The operator U_Φ is a contraction. It is unitary exactly when*

$$\Phi_K(\operatorname{Im} F_\nu) = \operatorname{Im} F_\ell. \tag{16.80}$$

Proof. Let $P_\ell = F_\ell F_\ell^\dagger$ and $P_\nu = F_\nu F_\nu^\dagger$. Then $U_\Phi^\dagger U_\Phi = F_\nu^\dagger \Phi_K^\dagger P_\ell \Phi_K F_\nu \le I_3$. Equality is equivalent to $(I - P_\ell)\Phi_K F_\nu = 0$, that is, to inclusion of the left image of (16.80) in the right one. Both have dimension three, so inclusion is equality. □

The minimal ansatz of (16.80) is retained as a negative control of physical identification. Its ambient unitarity, and even fulfilment of (16.80), do not replace the full record or the frames of (16.74). The control is not removed when angular propagation is gathered in Article II.

16.3.12 Leptonic phase and Dirac–Majorana criterion

Basis phase changes must be distinguished from the self-conjugacy data of the neutral sector. For any unitary leptonic transport, one can form the quartet

$$J_\ell = \operatorname{Im}\big(U_{e1} U_{\mu 2} \overline{U_{e2}}\, \overline{U_{\mu 1}}\big). \tag{16.81}$$

The phase cancellation proved for (16.69) applies here as well. The quartet does not fix all the data of the neutral realization or by itself determine its Dirac or Majorana type.

The Majorana realization is specified through the antiunitary self-conjugation C_ν compatible with mass and dynamics:

$$C_\nu^2 = I, \qquad C_\nu M_\nu = M_\nu C_\nu, \qquad C_\nu H_\nu = H_\nu C_\nu, \tag{16.82}$$

with preserved domains. The representation is realized in the corresponding fixed real form. The Dirac alternative also preserves a charge Q distinguishing the mode from its conjugate:

$$C_\nu Q C_\nu^{-1} = -Q, \qquad Q\psi \neq 0. \tag{16.83}$$

These statements preserve their representational data. Neutrality, the involution of dodecaphasic words and an Ising topological sector do not separately decide the alternative.

Remark 16.61 (Basis freedom and Majorana phases). The double coset (16.75) describes the basis freedom of the Hermitian pair. If a Majorana real form is also fixed, admissible phase changes must preserve that additional datum; not all the freedoms of a Dirac representation can simply be transferred. This block preserves the self-conjugacy criterion and assigns no values to Majorana phases or neutrino masses. Such values do not follow from neutrality or a rank-one polarization.

16.3.13 Closure of angular transport

The quark construction first expresses angular coordinates, then complex amplitudes and finally phase invariants and operator transport. The leptonic construction expresses record operators and spectral frames before the change of basis. Both routes preserve their provenance in the common enriched state and the discrete structure of the continuum; neither introduces an experimental table as a selector of coefficients or states.

The checks in this section are local and explicit: rational CKM inversion and intertwining; spectral conservation under conjugation; completeness and common space of PMNS frames; record rank; compatibility of the cross matrix; and preservation of self-conjugacy data. Angular propagation is thus incorporated into the theory of action, area and information without making mixing a second mass law or turning the Barbero reader into its cause.

Chapter 17

Double projection of the same state: Archimedean publication and exceptional conservation of incidence

17.1 Ambivalence Principle: involute of the areal and volumetric readers over the enriched state

The double projection compares two evaluations of the common enriched domain X_∞^{cal} already constructed in theorem 17.8. One produces Archimedean values; the other, together with a calibration of its codomain, preserves the combinatorial incidence that leads to codes, designs, and automorphism groups. Both evaluations start from the unique initial state X_∞^{cal} and preserve their common provenance in distinct codomains.

17.1.1 Domains of the 2 projections

On X_∞^{cal}, the cylinders contract to a point and the boundary admits the incidence prolongation. In this residence, we fix **Arch** := $\mathbb{R}$. The Archimedean evaluation and the exceptional evaluation are, respectively,

$$\mathfrak{R} : X_\infty^{\mathrm{cal}} \longrightarrow \mathbf{Arch}, \qquad \mathfrak{E} : X_\infty^{\mathrm{cal}} \times \mathscr{C}_{\mathrm{exc}} \longrightarrow \mathbf{Exc}.$$

Here $\mathscr{C}_{\mathrm{exc}}$ contains the projective order, Paley gauge, dodecad, incidence alignment, and lattice chamber specified in each construction. The first map produces values; the second produces codes, designs, lattices, and automorphisms. The preceding global theorem is the source of compatibility and permits comparison of invariants obtained through distinct operations without again intersecting domains defined separately. The finite comparison in this work is performed on the flag $B_K \subset H_\alpha^-$.

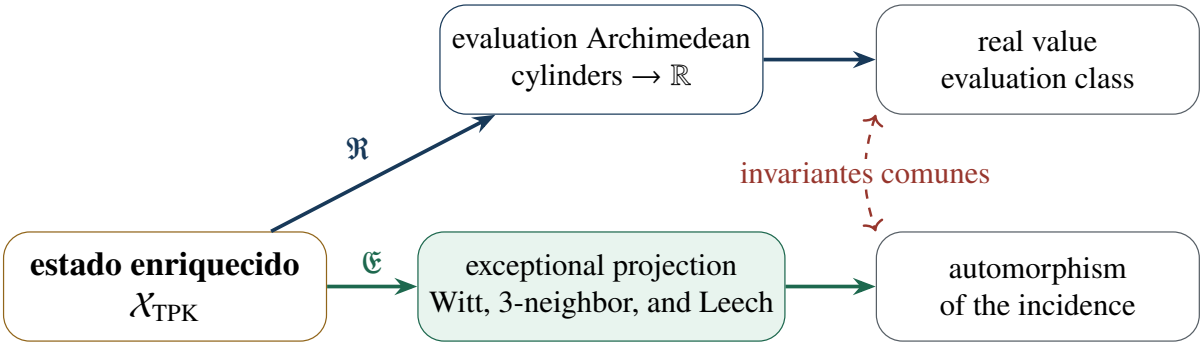


Figure 17.1: Typed scheme of the two projections. The Archimedean evaluation identifies states within the same fiber; the exceptional evaluation, relative to its calibration, preserves incidence structure and automorphisms.

17.1.2 Topology of the space of histories

The formulation by cylinders admits a standard realization as an inverse limit. Let S_n be the finite sets of states admissible at depth n, endowed with the discrete topology, and let

$$\tau_{n+1,n} : \mathsf{S}_{n+1} \longrightarrow \mathsf{S}_n$$

the truncation maps. In the HMT realization these maps are surjective: the prolongation law supplies a surviving continuation of each prefix, and its projective compatibility is preserved in the limit by theorem 17.8. This is the instance of the system involved in the global double projection (2). We define

$$\mathcal{X} = \varprojlim_{n} \mathsf{S}_n.$$

If $x \neq y$, let $m(x, y)$ be the first depth at which their components differ. The function

$$d(x, y) = 3^{-m(x,y)}, \qquad d(x, x) = 0,$$

is an ultrametric that induces the topology of the inverse limit.

Theorem 17.1 (Space of compact histories)**.** *The space $\mathcal{X}$ is compact and totally disconnected. When every history admits bifurcations at arbitrarily large depths, $\mathcal{X}$ has no isolated points and is homeomorphic to Cantor space.*

Proof. The product $\prod_n \mathsf{S}_n$ is compact. The equations $\tau_{n+1,n}(x_{n+1}) = x_n$ define a closed subset, so that $\mathcal{X}$ is compact. The sets of histories with a fixed prefix are simultaneously open and closed and form a basis; this proves total disconnectedness. Under the hypothesis of arbitrarily deep branching, no cylinder contains an isolated history. The characterization of compact, metric, perfect, and totally disconnected spaces proves the final assertion. □

The closed intervals $I_n(s)$ associated by the Archimedean reader with each $s \in \mathsf{S}_n$ are compatible with truncation and satisfy

$$\operatorname{diam} I_n(s) \leq C\lambda^n, \qquad 0 < \lambda < 1.$$

The intersection of the intervals corresponding to a history then contains a unique point.

Theorem 17.2 (Archimedean Evaluation)**.** *The map*

$$\mathrm{val} : \mathcal{X} \longrightarrow \mathbb{R}, \qquad \{\mathrm{val}(x)\} = \bigcap_{n} I_n(x_n),$$

is continuous and Hölder with respect to d. If $x \sim_{\mathrm{Arch}} y$ is equivalent to $\mathrm{val}(x) = \mathrm{val}(y)$, the induced application is a homeomorphism

$$\mathcal{X}/{\sim_{\mathrm{Arch}}} \;\cong\; \mathrm{val}(\mathcal{X}).$$

Proof. Two histories with a common prefix of length n have values in the same interval of diameter no greater than $C\lambda^n$. Since $d(x, y) = 3^{-n}$, this estimate is a Hölder bound. The induced map on the quotient is a continuous bijection from a compact space onto a Hausdorff space and therefore a homeomorphism. □

The theorem formalizes the Archimedean projection: the value retains the evaluation class and eliminates the distinction between histories situated in the same fiber. Coherent coverage, compact exhaustion and the exact reconstruction of the quotient are proved in theorems 18.3 y 18.9 y proposition 18.5; therefore, in the active HMT realization, $\mathrm{val}(\mathcal{X}) = \mathbb{R}$.

17.1.3 Separation by the 2 projections

Theorem 17.3 (Joint separation)**.** *Let $P_1 : \mathcal{X} \to Y_1$ and $P_2 : \mathcal{X} \to Y_2$ be continuous maps with Hausdorff codomains. If*

$$x \neq y \quad \Longrightarrow \quad P_1(x) \neq P_1(y) \text{ or } P_2(x) \neq P_2(y),$$

the diagonal application

$$(P_1, P_2) : \mathcal{X} \longrightarrow Y_1 \times Y_2$$

is a topological embedding.

Proof. The hypothesis expresses precisely the injectivity of the diagonal map. A continuous injective map from a compact space to a Hausdorff space is a homeomorphism onto its image. □

With a declared gauge $c \in \mathscr{C}_{\mathrm{exc}}$ fixed, let

$$\mathfrak{E}_c : X_\infty^{\mathrm{cal}} \longrightarrow \mathbf{Exc}, \qquad \mathfrak{E}_c(x) := \mathfrak{E}(x, c).$$

Applied to HMT, the theorem takes

$$P_1 = \mathfrak{R}, \qquad P_2 = \mathfrak{E}_c$$

on the common domain, or on the quotient by the declared coordinate changes. Projective compatibility and the global double realization are proved, respectively, in theorems 17.8 y 17.9. Faithfulness is typed by the codomain: the exceptional reader is faithful on the visible stream, while the enriched domain preserves the memory that no isolated publication contracts. The passage to every depth is thus a conclusion of the construction, not an external hypothesis.

At a finite level S, this condition is decidable. Given $p : S \to A$, $q : S \to B$, and an equivalence relation of coordinates, the pairs $x \nsim y$ that simultaneously satisfy are enumerated

$$p(x) = p(y), \qquad q(x) = q(y).$$

The absence of such pairs is equivalent to joint injectivity at that level. The calculation must perform the two maps separately: when $q = f \circ p$, the second has the same fibers as the first and does not recover information eliminated by p.

At the level developed here, the equality

$$B_K = H_e \cap P_\pi$$

constitutes the finite witness of compatibility between the two branches. Its compatible prolongation to all depths is provided by theorems 17.8 y 17.9; the equality of supports is the finite shadow of that joint realization.

17.1.4 Archimedean quotient by the evaluation fibers

An HMT section contains a block, an orientation, a carry, an observable label, and an extension coordinate. The evaluation val contracts its cylinders to a point. If two sections x, y satisfy

$$\mathrm{val}(x) = \mathrm{val}(y),$$

the Archimedean evaluation identifies them although their transport is different. In this precise sense,

a real value is an equivalence class of the evaluation.

On the canonical subdomain, sum, product and evaluation descend compatibly, and the image inherits those operations. The existence of the limit, the determination of its image, the characterization of the fibers and the coverage of all of $\mathbb{R}$ are established by theorems 18.3 y 18.9 y proposition 18.5. Finite windows exhibit the local structure of that global result; unitary survival and compatibility are proved properties of the HMT prolongation.

17.1.5 Common invariant with the Witt system

The vector K defines the support superior

$$B_K = \{m : K_m \geq 729\} = \{4, 6, 9, 10\}.$$

In the Witt atlas appear the supports

$$P_\pi = \{2, 4, 5, 6, 7, 8, 9, 10, 11\},$$

$$H_e = \{1, 3, 4, 6, 9, 10\}, \qquad H_\varphi = \{1, 2, 3, 4, 7, 9\}.$$

The support of the negative carries of α is

$$H_\alpha^- = \{4, 5, 6, 7, 9, 10\}.$$

The support identities are verified.

$$\boxed{B_K = H_e \cap P_\pi = H_e \cap H_\alpha^-,}$$

and

$$\boxed{H_\alpha^- = B_K \sqcup \{5, 7\}.}$$

Witt polarity sends the four points of B_K to the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{8, 12\}, \quad \{5, 7\}.$$

Three receive the positive label; the last completes the negative support. These identities are exact in the declared gauge and constitute internal checks of the same generated state: the vector, threshold, words and Paley gauge are correlated coordinates of the construction, not statistically independent inputs. The equality is obtained by direct evaluation and does not constitute a probabilistic estimate.

17.1.6 Residual design on a dodecada and elevation of hexads

The relation between the two Witt designs is obtained within the extended binary Golay code and not through an identification of cardinalities. Let

$$\mathcal{G}_{24} \subset \mathbb{F}_2^{24}$$

the extended binary Golay code, and denote by $\mathcal{O}_{24}$ the family of supports of its weight-eight words. It is a classical result that $(\{1, \dots, 24\}, \mathcal{O}_{24})$ is the Steiner system $W_{24} = S(5, 8, 24)$ [91, 34]. Fix a weight-twelve word and write D for its support. We define

$$\mathcal{H}(D) := \{O \cap D : O \in \mathcal{O}_{24},\ |O \cap D| = 6\}. \tag{17.1}$$

The family $\mathcal{H}(D)$ is intrinsic to the pair $(\mathcal{G}_{24}, D)$: it does not require choosing coordinates in D.

Theorem 17.4 (Residual design of a dodecada)**.** *The incidence system*

$$(D, \mathcal{H}(D))$$

is a Steiner system $S(5, 6, 12)$. In particular, $|\mathcal{H}(D)| = 132$, and every subset of five points of D is contained in a unique hexad of $\mathcal{H}(D)$.

Proof. Let $A \subset D$, with $|A| = 5$. Since the octads form $S(5, 8, 24)$, there is a unique octad O_A containing A. If $j = |O_A \cap D|$, the binary sum of the characteristic words of O_A and D belongs to $\mathcal{G}_{24}$ and has weight

$$\mathrm{wt}(\mathbf{1}_{O_A} + \mathbf{1}_D) = 8 + 12 - 2j = 20 - 2j.$$

The weights of the extended binary Golay code are $0, 8, 12, 16, 24$. Since $j \le 8$, it follows that $j \in \{2, 4, 6\}$. The inclusion $A \subset O_A \cap D$ gives $j \ge 5$, hence $j = 6$. Therefore, $O_A \cap D \in \mathcal{H}(D)$ is a hexad containing A.

If two members of $\mathcal{H}(D)$ contained A, their octads of origin would contain the same set of five points; the Steiner property of W_{24} would make them coincide. This proves uniqueness. The number of blocks is obtained by counting incidences with five-point subsets:

$$|\mathcal{H}(D)| = \frac{\binom{12}{5}}{\binom{6}{5}} = 132. \qquad \square$$

Theorem 17.5 (Unique elevation of a hexad)**.** *For each $H \in \mathcal{H}(D)$, there exists a unique octad $O_H \in \mathcal{O}_{24}$ such that*

$$O_H \cap D = H.$$

Consequently, the application

$$\iota_D : \mathcal{H}(D) \longrightarrow \mathcal{O}_{24}, \qquad H \longmapsto O_H,$$

is well defined and is injective.

Proof. Existence forms part of the definition of $\mathcal{H}(D)$. If two octads O_1, O_2 had intersection H with D, both would contain any five-point subset of H. The uniqueness of the octad incident with those five points implies $O_1 = O_2$. Injectivity follows by intersecting again with D. $\square$

The lift has a local law that will be used to compare the five regions of π. In a system $S(t, k, v)$, the number of blocks containing a fixed subset of $s \le t$ points is

$$\lambda_s = \frac{\binom{v-s}{t-s}}{\binom{k-s}{t-s}}.$$

Consequently,

$$\lambda_4(W_{12}) = \frac{\binom{8}{1}}{\binom{2}{1}} = 4, \qquad \lambda_4(W_{24}) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5. \tag{17.2}$$

Corollary 17.6 (Decomposition 4 + 1 on a face)**.** *Let $B \subset D$, with $|B| = 4$. The five octads of W_{24} that contain B decompose uniquely into*

$$\{O_H : H \in \mathcal{H}(D),\ B \subset H\} \ \sqcup\ \{O_B^{\perp}\}.$$

The first set contains four octads and satisfies $|O_H \cap D| = 6$; the remaining octad satisfies

$$O_B^{\perp} \cap D = B.$$

Proof. The first equality in (17.2) provides four residual hexads containing B, and theorem 17.5 provides four distinct octads. The second equality in (17.2) shows that there exists exactly one fifth octad $O_B^{\perp}$.

Since $B \subset O_B^{\perp} \cap D$, the intersection spectrum used in the proof of theorem 17.4 leaves only the possibilities four and six. An intersection of size six would be a fifth residual hexad that contains B, contradicting $\lambda_4(W_{12}) = 4$. Therefore, the intersection has size four and coincides with B. $\square$

17.1.6.1 Alineamiento with the 12 *positions HMT*

The residual design is defined on the points of D, whereas the design obtained from the ternary code C_W is defined on the twelve dodecaphase positions. To avoid conflating the two sets, we shall call a bijection *HMT binary alignment*

$$\ell : \{1, \dots, 12\}_{\mathrm{HMT}} \xrightarrow{\sim} D \quad \text{such that} \quad \{\ell(H) : H \in W_{12}^{\mathrm{HMT}}\} = \mathcal{H}(D). \tag{17.3}$$

Here W_{12}^{HMT} denotes the design of weight-six supports obtained from the ternary code C_W. Uniqueness of the Witt system $S(5, 6, 12)$ up to isomorphism ensures the existence of ℓ. Two distinct alignments differ by automorphisms of the source and target designs; hence ℓ is coordinate data and not an additional

invariant. The pair (D, ℓ) is the calibration datum required to transport the lift ι_D to the HMT coordinate system. Once fixed, each HMT hexad H unambiguously determines the octad

$$\widehat{H} := \iota_D(\ell(H)).$$

The dodecad and the alignment are data of the binary codomain; they are not identified with a decimal output of the transport kernel.

For the distinguished face

$$B_K = \{4, 6, 9, 10\},$$

the four incident hexads are

$$B_K \sqcup \{1, 3\}, \quad B_K \sqcup \{2, 11\}, \quad B_K \sqcup \{5, 7\}, \quad B_K \sqcup \{8, 12\}.$$

The third agrees with $H_\alpha^- = B_K \sqcup \{5, 7\}$. The alignment ℓ lifts these four hexads to the four residual octads over $\ell(B_K)$; corollary 17.6 also determines a unique transverse octad.

17.1.6.2 Incidence correspondence between the microfiber of π and the residual octads: multiplicity $432{+}4{\cdot}144 = 1008$

The exact microfiber of the word $w_\pi = 010211$ contains five distinct decimal regions:

multiplicity	U_6	$e(U_6)$
432	501|614|498|169|272|272	$(5, 5, 5, 5, 5, 5)$
144	810|923|870|169|272|272	$(3, 3, 9, 5, 5, 5)$
144	810|983|810|169|272|272	$(3, 9, 3, 5, 5, 5)$
144	870|923|810|169|272|272	$(9, 3, 3, 5, 5, 5)$
144	870|923|810|418|674|521	$(9, 3, 3, 7, 1, 7)$.

Here $e(U_6)$ is the multiplicative signature recovered from each decimal block $\overline{d_1 d_2 d_3}$ by means of $e = d_2 - d_1 \pmod{10}$. The first region is the only one of multiplicity 432 and constant signature. The remaining four have multiplicity 144; among them, three share the return block 169|272|272, whereas the fourth has return 418|674|521. The catalogue distinguishes the five rows by a reproducible predicate: the three rows of type ES form the positive triple; among the two rows of type NO, that of maximal multiplicity and constant signature is termed transverse, and the remaining one is termed negative. Relative to this role predicate, the partition is

$$\boxed{1_\perp + 3_{++} + 1_{--}.}$$

The sign labels are a convention; what is intrinsic to the catalog is the partition obtained from its columns, not a prior binary polarity.

The alineamiento regional complete the givesto (D, ℓ) with to bijection that sends the transver region to $O^\perp_{\ell(B_K)}$, the region negativa to $\widehat{H_\alpha^-}$ and the tris regionis positivas to the elevacionis of

$$B_K \sqcup \{1, 3\}, \qquad B_K \sqcup \{2, 11\}, \qquad B_K \sqcup \{8, 12\}.$$

The ordered assignment between the three positive regions and those three hexads is a choice of coordinates. The two distinguished rows and the triple are reproducible through the preceding catalogue predicate; their designation as negative, transversal, and positive belongs to the alignment. Thus, the law $4 + 1$ does not identify the five regions with five octads by mere cardinality: it provides the incidence space, whereas (D, ℓ) and the regional alignment specify the transport between the two objects.

17.1.7 Lattice prolongation and dimensional screens

The lift of hexads opens two branches that must not be confused. The first conserves binary incidence and prolongs W_{12} toward W_{24}. The second conserves ternary transport:

$$C_{12}^{(3)} \longrightarrow N(A_2^{12}) \longrightarrow \Lambda_{24} \longrightarrow V^\natural \longrightarrow \mathbb{M} \longrightarrow \text{Moonshine}.$$

The primary proof of the gluing, the rootless 3 neighbor, and the corridor to Moonshine resides in section 21.9 y theorem 21.49. There is no canonical arrow $W_{24} \to N(A_2^{12})$, nor any direct action of the Monster on TPK words: both branches start from the same calibrated code and preserve different structures.

The reductions $243/11$, $12 \to 11 \to 10$, and $26 \to 24$ are not identities between numerals either. The chapter 25 constructs their spaces, quotients, and maps: the perfect ternary quotient, the screen marked by K, and the isotropic reduction of the Lorentzian lattice. This section thereby retains a single function: to link the already proved finite incidence with its two prolongations, without repeating the reference proofs.

17.1.8 Residue–quotient involution and conservation of state information

For $n = 9Q + r$, we define the lattice energy

$$E_R(r, Q) = \frac{r^2}{R^2} + Q^2 R^2.$$

Then

$$\boxed{E_R(r, Q) = E_{1/R}(Q, r).}$$

The exchange $(r, Q, R) \leftrightarrow (Q, r, 1/R)$ is an exact involution of the lattice model: the residue and quotient exchange their functions.

17.1.9 Finite compatibility and joint prolongation

The finite witness B_K, the carry pattern, and the Witt frame fix local compatibility. Its prolongation is no longer left as an abstract scheme: the theorem 17.8 constructs, block by block, the exceptional reader with a unique transported initial gauge, and the theorem 17.9 proves that this reading and the Archimedean evaluation globalize over the same TPK section. Truncating the history before reading is equivalent in both cases to truncating the output after reading.

The Archimedean evaluation identifies histories by their limiting value. The exceptional evaluation preserves the visible current and its Paley–Witt incidence; it does not thereby recover the hidden memories that the visible emitter does not publish. The double projection is therefore a joint construction with two distinct codomains, not an identification between the continuum and the exceptional branch.

Remark 17.7 (Compatibility of the two projections). The support B_K, the carry pattern, and the Witt frame provide the common invariants. The lattice chain prolongs this compatibility to a rootless neighbor and, by classical classification, to the Leech lattice. The binary branch toward W_{24} remains separate: its incidence morphism is ι_D, defined after fixing the dodecad and alignment (D, ℓ), and its codomain is neither a Niemeier lattice nor the Leech lattice.

17.2 Exceptional projection of the enriched state: compatible incidence at all depths

The finite witness of the double projection acquires global scope when the exceptional reading is applied to every visible block and the gauge is transported, rather than chosen anew, as the depth increases. This section constructs that family. The domain remains the enriched TPK history; the exceptional reading is faithful on its visible stream and does not purport to replace the memory fields that this stream does not publish.

17.2.1 Common enriched domain and inequivalent codomains of the 2 projections

In the Paley–Witt chart one fixes

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3).$$

Matrix multiplication gives

$$A_W^{\mathsf{T}} = A_W, \qquad A_W^2 = -I_6.$$

For every conjugated operator $A = f A_W f^{-1}$, define

$$\Gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \qquad \Gamma_A(w) = (w, wA).$$

Its image is a calibrated copy $C_W(A)$ of the ternary Witt code. Projection onto the first six coordinates is a left inverse:

$$\mathrm{pr}_1 \, \Gamma_A = \mathrm{id}_{\mathbb{F}_3^6}.$$

Therefore, Γ_A is total and injective on the 729 words and does not need to distinguish in advance π, e, φ, or any other output.

17.2.2 Compatible extensions and uniqueness of the initial scale

Let X_n be the set of admissible TPK prefixes of depth n, with truncations

$$p_{n,m} : X_n \longrightarrow X_m, \qquad m \le n.$$

Fix an initial exceptional frame f_0. Since the changes g_j form part of the transport stored by the history, a calibrated prefix is the pair (x, f_0); we write

$$X_n^{\mathrm{cal}} := X_n \times \{f_0\}, \qquad p_{n,m}^{\mathrm{cal}}(x, f_0) := (p_{n,m}x, f_0),$$

and

$$X_\infty^{\mathrm{cal}} := \varprojlim_n (X_n^{\mathrm{cal}}, p_{n,m}^{\mathrm{cal}}).$$

In what follows we abbreviate $p_{n,m}^{\mathrm{cal}}$ as $p_{n,m}$. The compatible visible emission is

$$\varepsilon_n : X_n^{\mathrm{cal}} \longrightarrow (\mathbb{F}_3^6)^n, \qquad \varepsilon_n(x) = (w_0, \dots, w_{n-1}),$$

and satisfies

$$\mathrm{pr}_{n,m} \, \varepsilon_n = \varepsilon_m p_{n,m}.$$

This application distinguishes its three types:

$$|\mathcal{U}_6| = 468, \qquad |\operatorname{im} w_6| = 243, \qquad |\mathbb{F}_3^6| = 729.$$

The emission w_j is not identified with the complete state that conserves leaf, quotient, carry, orientation, phase, and survival.

The exceptional gauge consists of a Witt chart, a dodecad, an alignment, a lattice origin, and an oriented chamber. It is fixed only once in the initial block. For $x \in X_n^{\mathrm{cal}}$, transport of the history supplies

$$g_j(x) \in \mathrm{GL}_6(\mathbb{F}_3), \qquad 0 \le j < n-1,$$

and $g_j(x)$ depends only on the prefix $p_{n,j+1}(x)$. The frame is transported by

$$f_{j+1}(x) = g_j(x) f_j(x), \qquad A_j = f_j A_W f_j^{-1}.$$

Thus, the family (A_j) is determined by the history and the initial gauge; it is not a sequence of independent choices.

Let $\mathcal{F}_{\mathrm{cal}} \subseteq \mathrm{GL}_6(\mathbb{F}_3)$ be the set of frames reachable from f_0. The emission preserving the transported gauge is

$$\widetilde{\varepsilon}_n : X_n^{\mathrm{cal}} \longrightarrow (\mathbb{F}_3^6 \times \mathcal{F}_{\mathrm{cal}})^n, \qquad \widetilde{\varepsilon}_n(x) = \big((w_0, f_0), \dots, (w_{n-1}, f_{n-1})\big).$$

Prefix dependence implies $\mathrm{pr}_{n,m}\, \widetilde{\varepsilon}_n = \widetilde{\varepsilon}_m p_{n,m}$. Its projection onto the words is ε_n; forgetting the frames does not suffice to reconstruct the exceptional reading.

17.2.3 Joint separation of histories by the Archimedean and exceptional projections

The local operation is covariant under the row-vector convention. For every change of coordinates $f \in \mathrm{GL}_6(\mathbb{F}_3)$,

$$A^f = f A f^{-1} \quad \Longrightarrow \quad \Gamma_{A^f}(w f^{-1}) = \Gamma_A(w)(f^{-1} \oplus f^{-1}).$$

Change of frame simultaneously transports the word, the operator, and the atlas. It is not required to remain within the stabilizer of a single labeled copy.

With the set of reachable frames $\mathcal{F}_{\mathrm{cal}}$ already defined, we construct the fixed calibrated atlas

$$\mathscr{C}_W^{\mathrm{cal}} := \left\{ \big(f, \Gamma_{f A_W f^{-1}}(w)\big) : f \in \mathcal{F}_{\mathrm{cal}},\ w \in \mathbb{F}_3^6 \right\}.$$

The label f preserves the calibrated copy in which each word resides. Define

$$Q_n(x) = \big(\big(f_0, \Gamma_{A_0}(w_0)\big), \dots, \big(f_{n-1}, \Gamma_{A_{n-1}}(w_{n-1})\big)\big) \in \big(\mathscr{C}_W^{\mathrm{cal}}\big)^n.$$

Let

$$r_{n,m} : \big(\mathscr{C}_W^{\mathrm{cal}}\big)^n \longrightarrow \big(\mathscr{C}_W^{\mathrm{cal}}\big)^m$$

the truncation of the last $n - m$ blocks.

Theorem 17.8 (Projective compatibility of the exceptional incidence)**.** *For every $m \le n$,*

$$\boxed{r_{n,m} \circ Q_n = Q_m \circ p_{n,m}.}$$

Consequently, there exists a unique application

$$\boxed{Q_\infty : X_\infty^{\mathrm{cal}} \longrightarrow \big(\mathscr{C}_W^{\mathrm{cal}}\big)^{\mathbb{N}}}$$

whose finite projections are the Q_n. The map is faithful on the visible current.

Proof. Write

$$\varepsilon_n(x) = (w_0, \dots, w_{m-1}, w_m, \dots, w_{n-1}).$$

The compatibility of ε shows that the visible current of $p_{n,m}(x)$ is $(w_0, \dots, w_{m-1})$. For $j < m$, the frame f_j depends only on the initial gauge and on changes preceding j, all of which are contained in that prefix. The first m operators A_j are therefore the same before and after truncation. Applying Γ_{A_j} block by block yields the asserted identity. The universal property of the inverse limit produces Q_∞. Finally, $\mathrm{pr}_1 \Gamma_{A_j} = I$ recovers every w_j, proving faithfulness on the visible current. □

The last assertion conserves its type. Two histories with the same current $(w_j)_j$ and distinct hidden memory may have the same exceptional image. The global reader preserves exactly what each visible block publishes and the incidence added to that block.

The compatibility of enriched emissions induces a unique application.

$$\widetilde{\varepsilon}_\infty : X_\infty^{\mathrm{cal}} \longrightarrow (\mathbb{F}_3^6 \times \mathcal{F}_{\mathrm{cal}})^{\mathbb{N}}.$$

Let pr_w the projection onto the corriente of palabras and

$$\widetilde{\Gamma}_\infty\big((w_j, f_j)_{j\geq 0}\big) := \big((f_j, \Gamma_{f_j A_W f_j^{-1}}(w_j))\big)_{j\geq 0}.$$

Then $Q_\infty = \widetilde{\Gamma}_\infty \circ \widetilde{\varepsilon}_\infty$, with declared domains and codomains.

17.2.4 Bifurcation of a surviving prefix in Archimedean value and exceptional incidence

Fixed $m \in \mathbb{Z}$, let us denote by

$$\mathrm{flat} : (\mathbb{F}_3^6)^{\mathbb{N}} \longrightarrow \mathbb{F}_3^{\mathbb{N}}$$

the concatenation of the blocks in the declared coordinate order, and by

$$\mathrm{ev}_3((d_k)_{k\geq 1}) := \sum_{k\geq 1} \frac{d_k}{3^k} \in [0,1], \qquad \tau_m(t) := m + t,$$

the ternary evaluation and translation by the integer part. The visible prefix $(d_1, \dots, d_{6n})$ determines the cylinder

$$I_n(x) = \left[m + \sum_{k=1}^{6n} \frac{d_k}{3^k},\, m + \sum_{k=1}^{6n} \frac{d_k}{3^k} + 3^{-6n} \right].$$

A prolongation restricts the cylinder and $\operatorname{diam} I_n = 3^{-6n} \to 0$. Therefore, every compatible infinite history determines a unique real point $\mathcal{P}_{\mathrm{arq}}(x)$.

Theorem 17.9 (Double global realization)**.** *Every compatible section $x = (x_n)_n \in X_\infty^{\mathrm{cal}}$ determines, from the same prefixes, two compatible outputs:*

$$\mathcal{P}_{\mathrm{arq}}(x) \in \mathbb{R}, \qquad \mathcal{P}_{\mathrm{exc}}(x) = Q_\infty(x) \in \big(\mathscr{C}_W^{\mathrm{cal}}\big)^{\mathbb{N}}.$$

Truncating before reading is equivalent to truncating each reading after reading. The first output identifies histories by their limiting value; the second preserves each visible block and adjoins its Paley–Witt rotation.

Proof. The nesting of the cylinders and the convergence of their diameters prove the first assertion. The theorem 17.8 proves the second and the compatibility with truncations. Both factor through the same emission:

$$\mathcal{P}_{\mathrm{arq}} = \tau_m \circ \mathrm{ev}_3 \circ \mathrm{flat} \circ \mathrm{pr}_w \circ \widetilde{\varepsilon}_\infty, \qquad \mathcal{P}_{\mathrm{exc}} = \widetilde{\Gamma}_\infty \circ \widetilde{\varepsilon}_\infty.$$

□

The double projection is thus constructed on admissible histories and a transported initial gauge. The continuum is the Archimedean evaluation of the section; the exceptional branch is the compatible

realization of its visible current. From C_W, with the already declared alignments, the branches toward W_{12}, W_{24}, and toward $N(A_2^{12})$, Leech, $V^\natural$, and the Monster are opened. These branches do not turn the digits into a set on which the Monster acts directly: they transport the incidence that evaluation by value ceases to see.

17.2.5 Common incidence invariant between the enriched state and the Witt system

Let $x_\pi \in X_\infty^{\mathrm{cal}}$ be the surviving section of the π channel. The double realization of theorem 17.9 produces

$$\mathcal{P}_{\mathrm{arq}}(x_\pi) \in \mathbb{R}, \qquad \mathcal{P}_{\mathrm{exc}}(x_\pi) = Q_\infty(x_\pi).$$

The first branch publishes the generated angle; the second preserves the visible current and realizes the quadratic Paley–Witt relation in each block.

Theorem 17.10 (Euler–postgenerative coupling double projection)**.** *Suppose that the coinductive construction of the channel of π determines*

$$\mathcal{P}_{\mathrm{arq}}(x_\pi) = \pi_{\mathrm{HMT}},$$

and that the exceptional branch uses the gauge transported with $A_W^2 = -I_6$. Let J_{e} be the real realization of the integral source of theorem 3.25. Then

$$\boxed{\mathrm{Exp}\big(\mathcal{P}_{\mathrm{arq}}(x_\pi) J_{\mathrm{e}}\big) + I = 0.} \tag{17.4}$$

Proof. By the theorem 3.25, A_W and J_{e} realizan the relation $u^2 + 1 = 0$ in categorias distinct. In the branch real,

$$\mathrm{Exp}(\theta J_{\mathrm{e}}) = \cos\theta\, I + \sin\theta\, J_{\mathrm{e}}.$$

Substitution of the previously generated output $\theta = \mathcal{P}_{\mathrm{arq}}(x_\pi) = \pi_{\mathrm{HMT}}$ produces $-I$, and the sum of I gives the stated identity. □

Causal order and absence of circularity. The demonstrated dependence is

$$\text{emitter} \longrightarrow x_\pi \longrightarrow \mathcal{P}_{\mathrm{arq}}(x_\pi) \longrightarrow \pi_{\mathrm{HMT}} \longrightarrow \mathrm{Exp}(\pi_{\mathrm{HMT}} J_{\mathrm{e}}) + I.$$

The exceptional branch fixes the quadratic type; it selects neither the angle nor its digits. Euler's identity subsequently recognizes the compatibility of the two realizations and does not enter as an input to the generator of π_{HMT}.

17.3 Decomposition of the microfiber of π into 5 oriented regions: 4 peripheral components and 1 fixed axis

The double projection thesis acquires mathematical content when both readings are defined on the same state system and commute with depth restriction. Archimedean evaluation contracts a family of cylinders to a value; exceptional evaluation preserves incidence relations that the first may forget. These are not two juxtaposed numerical expressions, but two natural transformations with a common domain.

17.3.1 Reticular prolongation of incidence and construction of dimensional screens

Let $(X_n, p_{n+1,n})$, $(A_n, a_{n+1,n})$ and $(E_n, e_{n+1,n})$ be three inverse systems. The first contains TPK histories; the second, Archimedean cylinders or approximants; the third, exceptional incidence data. The active construction supplies the maps

$$R_n : X_n \longrightarrow A_n, \qquad Q_n : X_n \times C_n \longrightarrow E_n,$$

where C_n collects the coordinate data of the exceptional reader. The gauges form a compatible family and, for every n, the diagrams commute

$$a_{n+1,n} \circ R_{n+1} = R_n \circ p_{n+1,n},$$

$$e_{n+1,n} \circ Q_{n+1} = Q_n \circ (p_{n+1,n} \times c_{n+1,n}).$$

Theorem 17.11 (Globalization of the two evaluations)**.** *By the preceding compatibilities, already constructed block by block, there exist canonical maps*

$$R_\infty : \varprojlim X_n \longrightarrow \varprojlim A_n, \qquad Q_\infty : \varprojlim X_n \times \varprojlim C_n \longrightarrow \varprojlim E_n.$$

Its diagonal application

$$(R_\infty, Q_\infty) : X_\infty \times C_\infty \longrightarrow A_\infty \times E_\infty$$

is unique among the applications whose coordinate projections are R_n and Q_n.

Proof. Let $x = (x_n)_n \in \varprojlim X_n$. The first commutativity equation implies

$$a_{n+1,n}(R_{n+1}(x_{n+1})) = R_n(x_n),$$

and therefore $(R_n(x_n))_n$ belongs to $\varprojlim A_n$. We define $R_\infty(x)$ by means of that family. The same argument, applied to $(x_n, c_n)_n$, defines Q_∞. Uniqueness follows because an element of an inverse limit is determined by all of its coordinates. □

The theorem records the concrete proof carried out in HMT: the Archimedean result and the exceptional code come from the same compatible states and respect the truncation maps.

17.3.2 Archimedean contraction of compatible cylinders and realization of the real line

To each $x_n \in X_n$ is associated a nonempty compact interval $I_n(x_n) \subset \mathbb{R}$. If

$$p_{n+1,n}(x_{n+1}) = x_n \quad \Longrightarrow \quad I_{n+1}(x_{n+1}) \subseteq I_n(x_n)$$

and there exists $\varepsilon_n \to 0$ with $\operatorname{diam} I_n(x_n) \le \varepsilon_n$, the intersection

$$\bigcap_{n \ge 0} I_n(x_n)$$

contains a unique real. Thus it is obtained.

$$\mathrm{ev}_{\mathrm{arq}} : X_\infty^{\mathrm{arq}} \longrightarrow \mathbb{R}.$$

The phrase "the cylinders tend to zero" means here that their diameter converges to zero, not that the evaluated number tends to zero.

The relation

$$x \sim_{\mathrm{arq}} y \quad \Longleftrightarrow \quad \mathrm{ev}_{\mathrm{arq}}(x) = \mathrm{ev}_{\mathrm{arq}}(y)$$

describes exactly the information removed by the reader. The real value is an evaluation class; the history additionally preserves orientation, carry, sheet, and boundary signature.

17.3.3 Exceptional projection by incidence and boundary conservation

The second evaluation does not take values in $\mathbb{R}$. Its codomain is a category of incidence configurations. In the current finite realization, the coordinate datum contains:

$$C_{\mathrm{exc}} = (\,\text{projective order}, \text{Paley gauge}, \text{dodecad}, \\ \text{alignment}, \text{lattice origin}, \text{chamber}).$$

After fixing that data, the chain contains two descendants that must remain separated:

$$C_W \longrightarrow W_{12} \longrightarrow W_{24},$$

$$C_W \longrightarrow N(A_2^{12}) \longrightarrow \Lambda_{24} \longrightarrow V^\natural \longrightarrow \mathbb{M}.$$

The first transports designs; the second leads, through later classical results, to the Leech lattice, the vertex operator algebra, and the Monster group. The central holonomy and the Schläfli–E_6 corridor form a parallel boundary extraction; they are not obtained by replacing one of the two preceding descents.

17.3.4 Finite Compatibility of the Common Domain

The dodecaphonic seal

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$$

determines the support

$$B_K = \{m : K_m \geq 729\} = \{4, 6, 9, 10\}.$$

In the published Witt chart,

$$P_\pi = \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \qquad H_e = \{1, 3, 4, 6, 9, 10\},$$

and, by independent evaluation of the two supports,

$$\boxed{B_K = H_e \cap P_\pi.}$$

Furthermore, for the support of negative carry

$$H_\alpha^- = \{4, 5, 6, 7, 9, 10\}$$

we have

$$B_K = H_e \cap H_\alpha^-, \qquad H_\alpha^- = B_K \sqcup \{5, 7\}.$$

These identities constitute a witness of finite compatibility: the same subset is recovered from the arithmetic seal and from Witt incidence. This is not an equality of cardinalities, since it identifies specific positions.

17.3.5 Typed separation of the Archimedean and exceptional publications of the common state

Theorem 17.12 (Joint Fidelity Criterion)**.** *Let X be compact and Y, Z be Hausdorff spaces. If two continuous maps $R : X \to Y$ and $Q : X \to Z$ satisfy*

$$x \neq y \Longrightarrow R(x) \neq R(y) \text{ or } Q(x) \neq Q(y),$$

then $(R, Q) : X \to Y \times Z$ is a topological embedding.

Proof. The condition is the injectivity of the diagonal application. Every continuous injective application from a compact space into a Hausdorff space is a homeomorphism onto its image. □

At each finite depth, the condition is decided by enumerating the pairs $x \neq y$ that remain simultaneously in a fiber of R_n and of Q_n. The compatibility of those checks when passing from $n+1$ to n is precisely the projective identity proved in theorem 17.8, and theorem 17.9 carries out its globalization. The witness B_K exhibits the common finite invariant; the obstruction of the minimal reader in theorem 17.13 delimits the visible quotient, while the enriched reader preserves the orientation, sheet, signature, return and boundary data required by the double publication.

17.3.6 Precise formulation of the thesis on the real line

The enriched object associated to the Archimedean reading is the arrow

$$\mathbb{R}_{\mathfrak{H}} := \big(X_\infty^{\mathrm{arq}} \xrightarrow{\mathrm{ev}_{\mathrm{arq}}} \mathrm{Im}(\mathrm{ev}_{\mathrm{arq}}) \subseteq \mathbb{R}\big).$$

The domain preserves genealogies; the image preserves values. Coverage of the cylinders at every scale and compatible prolongability are proved in the construction of the Archimedean reader (5.282): for every compact set $K \subset \mathbb{R}$, $\mathrm{ev}_{\mathbb{R}}(\Omega_{\infty,K}) = K$, and the directed union of those intervals covers $\mathbb{R}$. Therefore,

$$\mathrm{Im}(\mathrm{ev}_{\mathrm{arq}}) = \mathbb{R}, \qquad X_\infty^{\mathrm{arq}}/\!\sim_{\mathbb{R}} \cong \mathbb{R}.$$

The real field does not act as the starting point of the construction, but as the Archimedean quotient of a space of sections with memory. The descent of sum and product is verified operation by operation and does not condition this already proved surjectivity.

The double reading is then expressed by

$$X_\infty^{\mathrm{com}} \times C_\infty \xrightarrow{(\mathrm{ev}_{\mathrm{arq}}, Q_\infty)} \mathbb{R} \times E_\infty.$$

This formula is the mathematical translation of the conceptual assertion: one and the same history may contract to a value and, in the transverse reading of its boundary, preserve the incidence leading to the exceptional symmetries.

17.4 Dependence of the reader of incidence with respect to the enriched state

Finite compatibility of the two evaluations does not imply that every incidence reader refines the partition produced by the visible reader. The regional catalog makes it possible to decide this question exactly for the minimal Paley–Witt realization. Let $\mathcal{U}_{468}$ be the set of the 468 published regional states, and let

$$\Psi : \mathcal{U}_{468} \longrightarrow \mathcal{W}_{243} \subset \mathbb{F}_3^6, \qquad \Psi(U_1, \dots, U_6) = (-U_1, \dots, -U_6) \pmod 3.$$

For $w \in \mathbb{F}_3^6$, the map

$$c(w) = (w, wA_W) \in C_W \subset \mathbb{F}_3^{12}$$

is the ternary encoder fixed in the Paley–Witt section. We define the minimal incidence reader by

$$Q_{\min} = \mathrm{Supp} \circ c \circ \Psi : \mathcal{U}_{468} \longrightarrow \mathcal{P}([12]).$$

For a map $f : X \to Y$, denote by

$$\mathrm{Eq}(f) = \{(x, x') \in X^2 : f(x) = f(x')\}$$

its fiberwise equivalence relation.

Theorem 17.13 (Obstruction of the minimal exceptional reader)**.** *In the complete catalogue of 468 states one verifies*

$$\mathrm{Eq}(\Psi, Q_{\min}) = \mathrm{Eq}(\Psi).$$

In the usual abbreviated notation for fiber partitions,

$$\ker(\Psi, Q_{\min}) = \ker \Psi.$$

Consequently, $(\Psi, Q_{\min})$ *is not injective and the minimal incidence reader does not distinguish any pair of states that has already been identified by* Ψ*.*

Proof. The factorization $Q_{\min} = S \circ \Psi$, with $S = \mathrm{Supp} \circ c$, implies

$$\Psi(u) = \Psi(v) \Longrightarrow Q_{\min}(u) = Q_{\min}(v).$$

The converse implication between the equivalence relations is immediate, because Ψ is the first component of the diagonal map. Equality of the relations therefore follows without statistical assumptions. Exhaustive enumeration further confirms that both partitions have the same histogram:

$$132\,[1] + 66\,[2] + 18\,[3] + 3\,[4] + 6\,[5] + 18\,[6],$$

where $a\,[m]$ means that there exist a fibres of cardinality m. □

The microfiber of $w_\pi = 010211$ provides the distinguished witness. Its five regions are

$$\begin{aligned} &(501, 614, 498, 169, 272, 272), \quad (810, 923, 870, 169, 272, 272),\\ &(810, 983, 810, 169, 272, 272), \quad (870, 923, 810, 169, 272, 272),\\ &(870, 923, 810, 418, 674, 521). \end{aligned}$$

The five have the same visible image and the same exceptional support.

$$P_\pi = \{2, 4, 5, 6, 7, 8, 9, 10, 11\}.$$

At the level of the 243 words there appear 213 distinct supports: 185 have one preimage, 27 have two, and one has four. Taking support therefore introduces a second quotient in addition to the one imposed by Ψ.

The theorem delimits the minimal reader, not the enriched exceptional evaluation. A reader capable of separating the regions of the same word must act before the visible quotient and use at least one of the data that it eliminates: orientation, sheet, signature, return, or boundary state. The dodecaphase intertwiner developed in the following part belongs to that enriched class and does not contradict the preceding obstruction.

Complete enumeration of the fibers produces the preceding histogram and proves that taking support introduces a second quotient.

17.5 Archimedean and incidental publication

For each level, let $\mathcal{E}_n$ be an incidence object and let

$$e_n : S_n \longrightarrow \mathcal{E}_n, \qquad \sigma_{n,m} : \mathcal{E}_n \longrightarrow \mathcal{E}_m \tag{17.5}$$

applications that satisfy

$$\sigma_{n,m} \circ e_n = e_m \circ r_{n,m}. \tag{17.6}$$

The universal property constructs

$$\mathrm{ev}_{\mathrm{inc}} : \Omega_\infty \longrightarrow \mathcal{E}_\infty, \qquad \mathcal{E}_\infty := \varprojlim_n \mathcal{E}_n. \tag{17.7}$$

The double publication is

$$\mathcal{P}_\infty := (\mathrm{ev}_{\mathbb{R}}, \mathrm{ev}_{\mathrm{inc}}) : \Omega_\infty \longrightarrow \mathbb{R} \times \mathcal{E}_\infty. \tag{17.8}$$

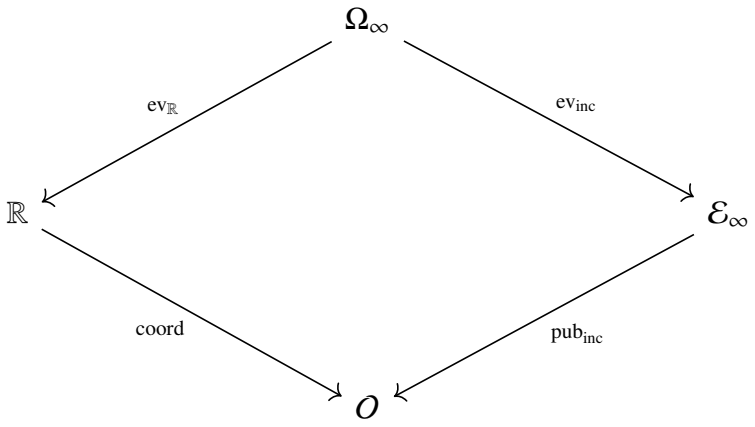


Figure 17.2: Two publications of the same history. Commutativity in O does not identify the codomains $\mathbb{R}$ and $\mathcal{E}_\infty$.

The naturality proved in theorems 17.8 y 17.9 is expressed, after applying the publication readers, by

$$\mathrm{pub}_{\mathrm{inc}} \circ \mathrm{ev}_{\mathrm{inc}} = \mathrm{coord} \circ \mathrm{ev}_{\mathbb{R}}, \tag{17.9}$$

and produces a typed comparison of the two publications. The identity preserves the common origin without constructing one branch from the other: both start from Ω_∞.

17.6 Factorization of the incidence reader through the enriched state

The second projection is not a unary reader devoid of incidence data. Let

$$\mathscr{C}_{\mathrm{exc}} = \mathscr{C}_{\mathrm{ord}} \times \mathscr{C}_{\mathrm{Paley}} \times \mathscr{C}_{\mathrm{dod}} \times \mathscr{C}_{\mathrm{inc}} \times \mathscr{C}_{\mathrm{ret}}$$

the space of exceptional gauges, whose coordinates respectively fix the projective order, the Paley gauge, the dodecad, the incidence alignment, and the lattice chamber. One fixes

$$\chi_{\mathrm{exc}} = (\chi_{\mathrm{ord}}, \chi_{\mathrm{Paley}}, \chi_{\mathrm{dod}}, \chi_{\mathrm{inc}}, \chi_{\mathrm{ret}}) \in \mathscr{C}_{\mathrm{exc}}.$$

The typed reader has domain

$$\mathfrak{E} : X_\infty^{\mathrm{cal}} \times \mathscr{C}_{\mathrm{exc}} \longrightarrow \mathbf{Exc}, \qquad \mathfrak{E}_{\chi_{\mathrm{exc}}}(u) := \mathfrak{E}(u, \chi_{\mathrm{exc}}).$$

Henceforth, every abbreviation $\mathfrak{E}(u)$ means $\mathfrak{E}_{\chi_{\mathrm{exc}}}(u)$ for this explicitly fixed gauge. The dependence on χ_{exc} does not affect the existence of the continuum or its Archimedean publication: it types the exceptional conservation of incidence once the common state has been constructed.

Falsification criterion 17.14 (Loss of the exceptional gauge)**.** The comparison becomes incomplete if the projective order, the dodecad, the incidence alignment, or the lattice chamber is changed without transporting the remaining gauge coordinates, or if $\mathfrak{E}$ is used as a unary reader before fixing χ_{exc}. Such a failure invalidates that exceptional comparison; it does not modify the Archimedean quotient or regenerate the continuum.

The second reading does not arise from the real number. Let

$$\mathfrak{E} : \mathcal{O}^{\mathrm{HMT}}_{\mathbb{R},\leftrightarrow} \longrightarrow \mathbf{Exc}$$

the continuous incidence reader, with Hausdorff codomain, that prolongs the Paley–Witt, Golay, Leech, $V^{\natural}$, Monster, and Moonshine chain. Both maps begin from the same history:

$$\mathfrak{D}(u) = (\mathfrak{R}(u), \mathfrak{E}(u)). \tag{17.10}$$

The first contracts the compatible fibres to their Archimedean mean; the second preserves incidence, orientation, and boundary memory.

Fixing the origin of the dodecaphonic calendar, let it be

$$R_+ := \{k \in \mathbb{Z} : 1 + (k \bmod 9) \in \{1, 4, 7\}\}$$

the bi-infinite set of complete returns ($\tau = +1$). It is syndetic: its gaps are uniformly bounded. For $u = (m, (x_k)_{k\in\mathbb{Z}})$, define

$$\mathrm{Ret}_+(u) := \big(m, (x_k)_{k\in R_+}\big) \in \mathbb{Z} \times \prod_{k\in R_+} X,$$

but each x_k here is the whole state (19.1) with leaf, carry, route, and incidence, not the isolated sign.

Proposition 17.15 (Lossless Sampling of the Return Genealogical Record)**.** *The map* Ret_+*, with the subspace topology in the product, is a homeomorphism onto its image. Consequently, there is a unique continuous map* $\mathfrak{E}_+$ *on that image such that*

$$\mathfrak{E} = \mathfrak{E}_+ \circ \mathrm{Ret}_+ . \tag{17.11}$$

Proof. Between two consecutive returns there is a uniformly bounded number of steps of the calendar. The complete state at a return contains the transition and the genealogical register that determine the intermediate steps; in the other direction, the inverse arrow of the groupoid completion is used. The entire orbit is thereby reconstructed. The map is continuous and, when restricted to a compact chart, a continuous injection into a Hausdorff space is a homeomorphism onto its image. The integer coordinate is preserved explicitly. The factorization is defined by transporting $\mathfrak{E}$ through the inverse and is continuous chart by chart. □

For this reason, +1 participates in the exceptional corridor as a complete memory of return: sampling the integer states does not lose the exceptional output. This proposition proves recovery and factorization, not a generation of the corridor from the isolated scalar +1. The Archimedean output and the exceptional output are two coordinated resultants of the same antecedent.

17.7 Double publication of the enriched state: Archimedean evaluation and incidence conservation

Let $(X_m, p_{m+1,m})_{m\geq 0}$ be the inverse system of calibrated APP–TRIT–TPK states. At each level, fiber, relation, number, orientation, carry, memory, boundary, route, multisection, survival, terminal, and reader are preserved. Let

$$\mathfrak{R}_m : X_m \to A_m, \qquad \mathfrak{E}_{m,\chi} : X_m \to E_m, \qquad \mathfrak{D}_m : X_m \to D_m$$

the Archimedean, calibrated exceptional, and dimensional publications. Their truncations are denoted by $a_{m+1,m}$, $e_{m+1,m}$, and $d_{m+1,m}$.

Theorem 17.16 (Natural cone of the 3 publications)**.** *If the finite-level readers satisfy*

$$\begin{aligned} a_{m+1,m}\mathfrak{R}_{m+1} &= \mathfrak{R}_m p_{m+1,m}, \\ e_{m+1,m}\mathfrak{E}_{m+1,\chi} &= \mathfrak{E}_{m,\chi} p_{m+1,m}, \\ d_{m+1,m}\mathfrak{D}_{m+1} &= \mathfrak{D}_m p_{m+1,m}, \end{aligned} \tag{17.12}$$

then the limit readers exist and are unique

$$\mathfrak{R}_\infty : X_\infty \to A_\infty, \quad \mathfrak{E}_{\infty,\chi} : X_\infty \to E_\infty, \quad \mathfrak{D}_\infty : X_\infty \to D_\infty$$

that reproduce all levels. The joint map

$$\mathfrak{J}_\chi := (\mathfrak{R}_\infty, \mathfrak{E}_{\infty,\chi}, \mathfrak{D}_\infty) : X_\infty \longrightarrow A_\infty \times E_\infty \times D_\infty \tag{17.13}$$

is a publication of the same state and not a concatenation of three generators.

Proof. For $x = (x_m)_m \in X_\infty$, the three sequences $(\mathfrak{R}_m x_m)_m$, $(\mathfrak{E}_{m,\chi} x_m)_m$, and $(\mathfrak{D}_m x_m)_m$ are compatible under (17.12); by the universal property of the inverse limit, they define the three readers. The same property proves their uniqueness. Formula (17.13) evaluates a single history x and therefore preserves the common causal antecedent. □

The legitimate equality does not assert that an output with loss reconstructs by itself all the others. It asserts that the three truncation squares commute and that their outputs are natural coordinates of one and the same cone. In particular, the Archimedean realization can forget genealogy without that forgetting being attributed to the exceptional or dimensional publication.

17.8 Uniqueness of the Initial Caliber of the Exceptional Projection

With the gauge χ_{exc} fixed, the exceptional reader acts on every generator by

$$(w_j, f_j) \longmapsto \big(f_j, \Gamma_{f_j A_W f_j^{-1}}(w_j)\big). \tag{17.14}$$

The frame f_j is part of the output, and $\Gamma_{f_j A_W f_j^{-1}}$ is an automorphism of the calibrated fibre.

Theorem 17.17 (Calibrated separation of histories)**.** *The exceptional reader $\mathfrak{E}_{\infty,\chi_{\mathrm{exc}}}$ is injective on the quotient by the coordinate changes included in χ_{exc}. Consequently,*

$$(\mathfrak{R}_\infty, \mathfrak{E}_{\infty,\chi_{\mathrm{exc}}}) : X_\infty^{\mathrm{cal}} \longrightarrow A_\infty \times E_\infty$$

is a topological embedding when X_∞^{cal} is compact and the codomain is Hausdorff.

Proof. From the output (f_j, y_j), where $y_j = \Gamma_{f_j A_W f_j^{-1}}(w_j)$, one recovers

$$w_j = \Gamma^{-1}_{f_j A_W f_j^{-1}}(y_j).$$

This reconstructs each generator and, by truncation compatibility, the entire history. Two representatives that differ only by a change included in the gauge define the same class; outside that equivalence, the reconstruction is unique. Joint injectivity is immediate. A continuous injective map from a compact space into a Hausdorff space is a homeomorphism onto its image. □

17.9 Noncommutative Algebra, Common and Factorization of the 2 Projections

Let $\mathcal{A}_{\mathrm{TPK}} = C(X_F^{\leftrightarrow}) \rtimes_\sigma \mathbb{Z}$ and let u be the unitary implementing σ. The continuous readers $\mathfrak{R}$ and $\mathfrak{E}_\chi$ induce the unital morphisms

$$\Phi_{\mathrm{arq}}(f) = f \circ \mathfrak{R}, \qquad \Phi_{\mathrm{exc}}(g) = g \circ \mathfrak{E}_\chi, \tag{17.15}$$

from the commutative algebras of its codomains toward $C(X_F^{\leftrightarrow}) \subset \mathcal{A}_{\mathrm{TPK}}$.

Theorem 17.18 (Common operator system and TPK equivariance)**.** *The unital self-adjoint subspace*

$$\mathcal{S}_\infty = \mathrm{span}\big\{I, u, u^*, \Phi_{\mathrm{arq}}(C(A_\infty)), \Phi_{\mathrm{exc}}(C(E_\infty))\big\} \subset \mathcal{A}_{\mathrm{TPK}}$$

is a noncommutative operator system. The two maps from (17.15) *are UCP. If the readers intertwine the output dynamics α_{arq} and α_{exc}, then*

$$\mathrm{Ad}_u \, \Phi_{\mathrm{arq}} = \Phi_{\mathrm{arq}} \alpha_{\mathrm{arq}}, \qquad \mathrm{Ad}_u \, \Phi_{\mathrm{exc}} = \Phi_{\mathrm{exc}} \alpha_{\mathrm{exc}}. \tag{17.16}$$

Proof. Composition with a continuous map is a unital homomorphism of C^*-algebras and is therefore completely positive. The unitary u makes the envelope noncommutative when the dynamics is nontrivial. The identity (17.16) is verified by evaluating both sides on a history and using equivariance of each reader. □

This construction does not force two mutually unbiased qutrit measurements to possess a joint PVM. The common antecedent is the crossed algebra of histories; the publications are two distinct UCP channels from their codomains.

Chapter 18

Archimedean publication of the continuum: reconstruction of $\mathbb{R}$ as a quotient of semiclosed cylinders with genealogical memory

The inverse limit Ω_∞ preserves the complete history. To obtain a real coordinate, a compatible family of intervals must be added and three different facts proved: each history determines a unique value; every value of the declared compact set is attained; and the finite measures descend consistently. None of these facts follows from the mere growth in the number of states. The construction below proves all three facts jointly for the HMT system.

The Archimedean publication likewise preserves an approximation of the identity by cylindrical conditional expectations. The concentration of unit mass on compatible prefixes provides the exact passage from the nonadic cylinders to the Dirac measure, without identifying density, support measure, and total mass.

18.1 Nonadic cylinders, identity and Dirac kernel

18.1.1 Inverse system of compatible nonadric cylinders with genealogical memory

Let

$$X = (\mathbb{Z}/9\mathbb{Z})^{\mathbb{N}}$$

with the product topology and the uniform measure μ. For $x = (x_1, x_2, \ldots)$, the cylinder of depth n is

$$C_n(x) = \{y \in X : y_j = x_j \text{ for } 1 \le j \le n\}.$$

Cylinders are simultaneously open and closed, form a base and satisfy

$$\mu(C_n(x)) = 9^{-n}, \qquad C_{n+1}(x) \subset C_n(x), \qquad \bigcap_n C_n(x) = \{x\}.$$

Adding a digit does not create a new number: it restricts the cylinder of admissible prolongations. This statement gives topological content to the genealogical thesis of number.

18.1.2 Cylindrical concentration of the identity and construction of the Dirac kernel

Theorem 18.1 (Unitary nonadic concentration)**.** *Define*

$$f_{n,x} = 9^n \mathbf{1}_{C_n(x)}.$$

Then

$$\mu(C_n(x)) \to 0, \qquad f_{n,x}(x) \to \infty, \qquad \int_X f_{n,x}\, \mathrm{d}\mu = 1,$$

and the measures $f_{n,x}\mu$ converge weak- to δ_x.*

Proof. The first identity is immediate. The height of $f_{n,x}$ over its support is 9^n, so it diverges. The integral equals $9^n 9^{-n} = 1$. For $g \in C(X)$, the oscillation of g in $C_n(x)$ tends to zero by uniform continuity; therefore

$$\left| \int g f_{n,x}\, \mathrm{d}\mu - g(x) \right| \le \sup_{y \in C_n(x)} |g(y) - g(x)| \longrightarrow 0.$$

This is weak-* convergence to δ_x. □

The theorem does not assert $1/0 = \infty$. It exhibits three evaluations of the same object: support measure, density and total mass. In blocks of three digits,

$$\mu(C_{3m}) = 729^{-m}, \qquad f_{3m} = 729^{m}\mathbf{1}_{C_{3m}}, \qquad 729^{-m}729^{m} = 1.$$

The evolutionary conservation of unity here assumes an exact analytical form.

18.1.3 Convergence of the Kronecker Kernel to the Dirac Distribution

At the finite level $X_n = (\mathbb{Z}/9\mathbb{Z})^n$, let

$$K_n(x, y) = 9^n\mathbf{1}_{\{x=y\}}.$$

With the uniform measure, the operator

$$(E_n f)(x) = \sum_{y\in X_n} K_n(x, y)f(y)9^{-n}$$

is the identity. In the cylindrical limit, the condition $x = y$ is replaced by "same prefix of length n" and E_n is the conditional expectation on the cylindrical algebra. One has

$$E_n^2 = E_n, \qquad E_n^* = E_n, \qquad E_n f \to f \quad \text{in } L^2(X, \mu).$$

The exact bridge is, therefore,

$$\delta_{ij} \longrightarrow K_n(x, y) \longrightarrow \delta_x.$$

The Kronecker delta is a matrix kernel; the Dirac delta is a measure or distribution. Neither is the Dirac operator $D = \gamma^\mu\partial_\mu$. The transition to spinors and Clifford requires a second intertwiner and will be treated later.

18.1.4 Scaling Law, Effective Dimension, and Density of the Cylindrical Measure

Suppose that a contraction of ratio λ acts independently in d directions and that the product measure scales as λ^d. To preserve unit mass, the density must scale as

$$\rho \longmapsto \lambda^{-d}\rho.$$

The exponents $(-1, -2, -3)$ then appear as density weights on a line, area, and volume. This proposition is general and exact under the product hypothesis. It does not permit λ to be identified automatically with α_{HMT}; that identification requires a metric chart.

18.1.5 Nonadic Monodromy and Computability of the Archimedean Evaluation

The existence of an infinite branch in a finitely branching tree may follow from König's lemma, but König's lemma does not furnish a computable selector. There are infinite recursive binary trees with no recursive branch. The joint nonadic connection does not rest on that abstract existence: it has a certified uniform selector that generates to arbitrary finite depth and opens the decimal witnesses only after generation.

Three prohibited confusions

1/0 shall not be written; the Dirac delta shall not be represented as an ordinary function with "an infinite one"; König shall not be used as an algorithm. The strong result is the exact compatibility among cylinders, unit density, and the weak limit, together with the constructive selector independent of decimal comparison.

18.2 System of intervals associated with finite states

Let $(S_n, r_{n,m})$ be the inverse system of the section 6.1. For each $a \in S_n$, assign a nonempty compact interval

$$I_n(a) = [\ell_n(a), u_n(a)] \subset \mathbb{R}. \tag{18.1}$$

The assignment is *compatible with refinement* when

$$r_{n+1,n}(b) = a \quad \Longrightarrow \quad I_{n+1}(b) \subseteq I_n(a). \tag{18.2}$$

It is *uniformly contractive* when there exists a sequence $\delta_n \downarrow 0$ such that

$$\operatorname{diam} I_n(a) \le \delta_n \qquad (a \in S_n). \tag{18.3}$$

For a history $x = (x_n)_n$, the inclusions (18.2) produce a chain

$$I_0(x_0) \supseteq I_1(x_1) \supseteq I_2(x_2) \supseteq \cdots. \tag{18.4}$$

Theorem 18.2 (Evaluation of a history)**.** *If the assignment is compatible and uniformly contractive, then for every $x \in \Omega_\infty$ there exists a unique real number*

$$\mathrm{ev}_{\mathbb{R}}(x) \in \bigcap_{n \ge 0} I_n(x_n). \tag{18.5}$$

The map

$$\mathrm{ev}_{\mathbb{R}} : \Omega_\infty \longrightarrow \mathbb{R} \tag{18.6}$$

is continuous.

Proof. The intervals of (18.4) are compact, nonempty and nested. The nested-interval theorem guarantees that their intersection is nonempty. If s, t belong to it, then

$$|s - t| \le \operatorname{diam} I_n(x_n) \le \delta_n$$

for every n. Setting $n \to \infty$ yields $s = t$.

For continuity, let $\varepsilon > 0$ and choose n with $\delta_n < \varepsilon$. If $y \in [x_n]_n$, then $I_n(y_n) = I_n(x_n)$. The two evaluations belong to that interval and, therefore,

$$|\,\mathrm{ev}_{\mathbb{R}}(x) - \mathrm{ev}_{\mathbb{R}}(y)| < \varepsilon.$$

The cylinder $[x_n]_n$ is an open neighborhood of x. □

The construction does not identify x with $\mathrm{ev}_{\mathbb{R}}(x)$. A single coordinate may have several histories, and a history contains more data than any point on the line.

18.3 Coherent coverage of the visible compactum

Let $K \subset \mathbb{R}$ be a nonempty compact set. The HMT construction of the cylinders associates with nesting and contraction the coverage identities

$$K = \bigcup_{a \in S_0} I_0(a), \tag{18.7}$$

$$I_n(a) = \bigcup_{\substack{b \in S_{n+1} \\ r_{n+1,n}(b) = a}} I_{n+1}(b) \qquad (a \in S_n). \tag{18.8}$$

The second equality is the geometric publication of the prolongation law: each point of $I_n(a)$ remains in a compatible child cylinder. Together with the surjectivity of $r_{n+1,n}$, already established by prefix survival, it simultaneously produces states above a and the complete coverage of $I_n(a)$.

Theorem 18.3 (Surjectivity of the evaluation)**.** *The constructed identities* (18.7) *and* (18.8) *imply*

$$\mathrm{ev}_{\mathbb{R}}(\Omega_\infty) = K. \tag{18.9}$$

Proof. The left-to-right inclusion follows from $I_n(x_n) \subseteq I_0(x_0) \subseteq K$.

Let us fix $t \in K$. Let us define

$$T_n(t) := \{a \in S_n : t \in I_n(a)\}. \tag{18.10}$$

The initial cover makes $T_0(t)$ nonempty. If $a \in T_n(t)$, the identity (18.8) provides at least one $b \in T_{n+1}(t)$ with $r_{n+1,n}(b) = a$. The graph of these extensions is finitely branching and has nonempty levels at every depth. There therefore exists a compatible chain $x = (x_n)_n$ with $t \in I_n(x_n)$ for every n. Uniqueness of the intersection forces $\mathrm{ev}_{\mathbb{R}}(x) = t$. □

18.4 Archimedean genealogical quotient and homeomorphism with $\mathbb{R}$

We define the relation

$$x \sim_{\mathbb{R}} y \quad \Longleftrightarrow \quad \mathrm{ev}_{\mathbb{R}}(x) = \mathrm{ev}_{\mathbb{R}}(y). \tag{18.11}$$

Let

$$q_{\mathbb{R}} : \Omega_\infty \longrightarrow \Omega_\infty/\!\sim_{\mathbb{R}} \tag{18.12}$$

the canonical projection.

Theorem 18.4 (Reconstruction of the compact as a quotient)**.** *For the preceding HMT system there is a unique homeomorphism*

$$\overline{\mathrm{ev}}_{\mathbb{R}} : \Omega_\infty/\!\sim_{\mathbb{R}} \xrightarrow{\cong} K \tag{18.13}$$

such that

$$\mathrm{ev}_{\mathbb{R}} = \overline{\mathrm{ev}}_{\mathbb{R}} \circ q_{\mathbb{R}}. \tag{18.14}$$

The fiber over $t \in K$ is

$$\mathfrak{G}_t := \mathrm{ev}_{\mathbb{R}}^{-1}(t), \tag{18.15}$$

the complete set of genealogies that publish t.

Proof. The map $\mathrm{ev}_{\mathbb{R}}$ is continuous, surjective, and constant exactly on the classes of $\sim_{\mathbb{R}}$. By the universal property of the quotient, it induces a continuous bijective map $\overline{\mathrm{ev}}_{\mathbb{R}}$. The domain is compact because it is a continuous image of the compact space Ω_∞, and K is Hausdorff. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. □

The theorem proves two things and keeps them separate:

the history space is profinite and retains memory, its quotient by equality of evaluation is the visible compact space.	(18.16)

It is not asserted that Ω_∞ is homeomorphic to K.

18.5 Genealogical Atlas of the Real Line

For $m \geq 1$, let $K_m = [-m, m]$. The same construction, applied to each K_m, produces a system $\Omega_\infty^{(m)}$ that reconstructs K_m, together with closed inclusions

$$j_m : \Omega_\infty^{(m)} \hookrightarrow \Omega_\infty^{(m+1)} \tag{18.17}$$

which satisfy

$$\mathrm{ev}_{m+1} \circ j_m = \iota_m \circ \mathrm{ev}_m, \tag{18.18}$$

where $\iota_m : K_m \hookrightarrow K_{m+1}$ is the inclusion ordinary.

Proposition 18.5 (Exhaustion of the line)**.** *The Archimedean quotient of the strict colimit*

$$\Omega_\infty^{\mathbb{R}} := \varinjlim_m \Omega_\infty^{(m)} \tag{18.19}$$

is

$$\varinjlim_m [-m, m] = \bigcup_{m \geq 1} [-m, m] = \mathbb{R} \tag{18.20}$$

with its usual topology.

Proof. Each piece has quotient K_m by the theorem 18.4; (18.18) intertwines the quotients. The final topology of the exhaustion by compact sets coincides with the usual topology of $\mathbb{R}$: a subset $U \subseteq \mathbb{R}$ is open if and only if $U \cap [-m, m]$ is open in $[-m, m]$ for every m. □

Passage from a bounded interval to the line does not erase the fibers $\mathfrak{G}_t$. The inclusions j_m preserve the history and only enlarge the available visible chart.

18.6 Balanced ternary and block-decimal charts

The balanced ternary representation of an integer has the form

$$n = \sum_{j=0}^{J} \epsilon_j 3^j, \qquad \epsilon_j \in \{-1, 0, +1\}, \tag{18.21}$$

with a fixed carry rule. This chart preserves the local sign and the quotient of the reduction; it must not be confused with the nonnegative word in $\mathbb{F}_3$.

The block decimal chart utilizes

$$D_N = 1000 D_{N-1} + u_N, \qquad u_N \in \{0, \ldots, 999\}, \qquad D_0 = 0. \tag{18.22}$$

The block u_N, its class modulo nine, and the carry are distinct fields. Since $1000 \equiv 1 \pmod 9$,

$$D_N \equiv \sum_{j=1}^{N} u_j \pmod 9. \tag{18.23}$$

This congruence does not recover the order of the blocks; the sequence $(u_1, \ldots, u_N)$ remains in the record.

For $M \geq 1$, the cells decimales half-open are

$$\mathcal{D}_{M,q} := [q 10^{-M}, (q+1) 10^{-M}), \qquad q \in \mathbb{Z}. \tag{18.24}$$

A half-open interval $[\ell, h)$ is contained in a unique cell if and only if

$$\lfloor 10^M \ell \rfloor = \lceil 10^M h \rceil - 1. \tag{18.25}$$

The formula with a ceiling at the right endpoint also covers the case in which h coincides exactly with a decimal boundary.

Definition 18.6 (Truncation and Rounding)**.** For $x \geq 0$, define

$$\mathrm{Tr}_M(x) := 10^{-M} \lfloor 10^M x \rfloor, \tag{18.26}$$

$$\mathrm{Rd}_M(x) := 10^{-M} \left\lfloor 10^M x + \frac{1}{2} \right\rfloor. \tag{18.27}$$

These operations coincide except when the discarded fraction crosses the half-unit threshold at the final position. The distinction is semantic and arithmetic; one cannot be substituted for the other to force prefix compatibility.

18.7 Descent of operations along evaluation fibers

Let $D_\star \subseteq \Omega_\infty \times \Omega_\infty$ the domain of to operation parcial continuous

$$\star : D_\star \longrightarrow \Omega_\infty. \tag{18.28}$$

Assume that $D_\star$ is saturated by the fibers of $\mathrm{ev}_\mathbb{R} \times \mathrm{ev}_\mathbb{R}$.

Theorem 18.7 (Exact Descent Criterion)**.** *There exists a unique visible operation*

$$\overline{\star} : (\mathrm{ev}_\mathbb{R} \times \mathrm{ev}_\mathbb{R})(D_\star) \longrightarrow K \tag{18.29}$$

which satisfies

$$\mathrm{ev}_\mathbb{R}(x \star y) = \mathrm{ev}_\mathbb{R}(x) \,\overline{\star}\, \mathrm{ev}_\mathbb{R}(y) \tag{18.30}$$

if and only if

$$\begin{aligned} &\mathrm{ev}_\mathbb{R}(x) = \mathrm{ev}_\mathbb{R}(x'), \\ &\mathrm{ev}_\mathbb{R}(y) = \mathrm{ev}_\mathbb{R}(y') \end{aligned} \quad \Longrightarrow \quad \mathrm{ev}_\mathbb{R}(x \star y) = \mathrm{ev}_\mathbb{R}(x' \star y') \tag{18.31}$$

for all pairs in the domain.

Proof. If $\overline{\star}$ exists, the right-hand side of (18.30) depends only on the two evaluations, and (18.31) follows. Conversely, this constancy makes it possible to define

$$s \,\overline{\star}\, t := \mathrm{ev}_\mathbb{R}(x \star y)$$

for any representatives x, y with evaluations s, t. The condition proves that the definition is independent of the representatives. Uniqueness is immediate. □

The additive and multiplicative sheets of APP supply operations on a common support, but their carries and their memories are not identified. The preceding theorem allows sum and product to descend to the visible chart without requiring their genealogical realizations to coincide.

18.8 Correspondence between discrete cardinality, cylindrical measure, and Archimedean publication

At each depth three operations appear that must be kept separated:

$$\mathrm{Count}_n(A) := |A|, \tag{18.32}$$

$$\mathrm{Measure}_n(A) := \mu_n(A), \tag{18.33}$$

$$\mathrm{Publish}_n(a) := L_n(a). \tag{18.34}$$

The count depends on the number of states; the measure depends on compatible weights; the publication depends on the reader. Two fibers of equal cardinality can have distinct weights, and two states of equal publication can belong to distinct cylinders. Equality of one of these three outputs does not imply equality of the other two.

18.9 Obstructions to Archimedean evaluation: nesting, continuity, quotient, projective measures, and descent of operations

Comparison Criterion 18.8 (Refutation of the evaluation and the measure)**.** The construction is refuted if any of the following facts holds:

1. a compatible extension receives an interval that is not contained in the interval of its restriction;
2. the diameters do not tend to zero, but uniqueness of the intersection is asserted;
3. the evaluation is not continuous in the cylinder topology;
4. it is asserted that every point of K appears without proving the coherent covering (18.8);
5. the map induced by the quotient is not bijective;
6. the profinite space is identified with its visible quotient;
7. the inclusions of the atlas do not intertwine their evaluations;
8. $\lfloor 10^M h \rfloor$ is used instead of $\lceil 10^M h \rceil - 1$ for a semi-open interval, and the boundary case is lost;
9. truncation and rounding are conflated;
10. a descended operation depends on the representative chosen in a fibre;
11. to family $(\mu_n)_n$ viola (7.1);
12. the measure of a cylinder depends on the level to which it is refined;
13. a conditioned measure ceases to sum to one or loses consistency;
14. the visible measure ν is identified with the internal measures μ_t;
15. an arrow between $\mathbb{R}$ and $\mathcal{E}_\infty$ is inferred solely because both publications share an antecedent;
16. the count of a fibre is used as though it determined its measure or its published value.

18.10 Quotient of the space of compatible histories and realization of the real line

Let

$$\Psi : \mathbb{Z}_3^6 \xrightarrow{\sim} (\mathbb{F}_3^6)^{\mathbb{N}}$$

the Hensel conjugacy between the initial memory and the complete tape of six-trit blocks. Concatenating the six coordinates of each block yields a ternary tape $(a_n)_{n\geq 1} \in \{0,1,2\}^{\mathbb{N}}$, on which the evaluation acts

$$\mathrm{ev}_3((a_n)) := \sum_{n\geq 1} \frac{a_n}{3^n} \in [0,1].$$

The two expansions of a ternary endpoint represent the same value and remain distinct before evaluation is applied. This does not constitute an ambiguity of the genealogical object: it is precisely a fiber of the reader.

We define

$$X_{\mathrm{raw}} := \mathbb{Z} \times \mathbb{Z}_3^6, \qquad P(m,x) := m + \mathrm{ev}_3(\mathrm{flat}(\Psi x)).$$

The integer coordinate provides the countable atlas of the line, and the 3-adic coordinate preserves the complete history of the fractional part.

Theorem 18.9 (Exact reconstruction of the Archimedean quotient)**.** *The map $P : X_{\mathrm{raw}} \to \mathbb{R}$ is sobreyectiva. If*

$$x \sim_P y \quad \Longleftrightarrow \quad P(x) = P(y),$$

then

$$\overline{P} : X_{\mathrm{raw}}/\!\sim_P \longrightarrow \mathbb{R}, \qquad [x] \longmapsto P(x),$$

is a bijection. Consequently,

$$|X_{\mathrm{raw}}/\!\sim_P| = |\mathbb{R}| = 2^{\aleph_0}.$$

Proof. Every ternary sequence is grouped uniquely into consecutive blocks of six digits. The conjugation Ψ assigns a unique initial 3-adic memory to that tape. Given $r \in \mathbb{R}$, take $m = \lfloor r \rfloor$ and a ternary expansion of $r - m \in [0,1)$; the preceding reconstruction produces $x \in \mathbb{Z}_3^6$ with $P(m,x) = r$. Thus, P is surjective.

The map $\overline{P}$ is well defined by the definition of $\sim_P$, is surjective because P is, and is injective because two classes with the same image belong to the same fiber. Finally,

$$|\mathbb{Z}_3^6| = \left|(\mathbb{F}_3^6)^{\mathbb{N}}\right| = 729^{\aleph_0} = 2^{\aleph_0},$$

and multiplying by the countable coordinate $\mathbb{Z}$ does not alter that cardinal. The calculation agrees with that of the quotient, already identified with $\mathbb{R}$. □

Remark 18.10 (Surviving closure). The raw completion and the surviving subspace X_{surv} are two stages of the same construction. The survival law produces coherent coverings at each level, uniform contraction of diameters and a prolongation for every visible cell. The infinite-branch lemma then produces a section over each point, and the evaluation induces, compact set by compact set, a homeomorphism of the quotient with the published interval. The exhaustion by $[-m,m]$ reconstructs the entire line, in agreement with proposition 18.5 y theorem 18.9.

18.11 Compatibility of the median Archimedean evaluation with the double projection of the enriched state

Let $u = (m, \mathbf{u}) \in \mathcal{O}^{\mathrm{HMT}}_{\mathbb{R},\leftrightarrow}$. A prefix $\mathbf{u}_n$ determines in the fractional chart an oriented semi-open cylinder

$$I_n(\mathbf{u}_n) = [a_n, b_n),$$

with the complementary convention at the endpoints of double expansion. To apply compactness, its closure $K_n = [a_n, b_n]$ is used. The compatibilities give

$$K_{n+1} \subseteq K_n, \qquad \delta_n := b_n - a_n \longrightarrow 0.$$

The global Archimedean mean of the cylinder is

$$\mu_n(u) = m + \frac{a_n + b_n}{2}.$$

If $q \geq n$, then

$$|\mu_q(u) - \mu_n(u)| \leq \delta_n,$$

so that

$$\mathfrak{R}(u) := \lim_{n\to\infty} \mu_n(u) \tag{18.35}$$

exists. What tends to zero is the diameter of the cylinder; the real value is the resultant of the nesting, not that which vanishes.

Theorem 18.11 (Two-way quotient and real line)**.** *Let*

$$u \sim_{\mathfrak{R}} v \quad \Longleftrightarrow \quad \mathfrak{R}(u) = \mathfrak{R}(v).$$

Under the coherent covering of the HMT charts, the evaluation induces a homeomorphism.

$$\overline{\mathfrak{R}} : \mathcal{O}^{\mathrm{HMT}}_{\mathbb{R},\leftrightarrow}/\sim_{\mathfrak{R}} \xrightarrow{\sim} \mathbb{R}. \tag{18.36}$$

In particular, this is a continuous HMT morphism: not the result of choosing an isolated history from each fiber.

Proof. The cylinder bound proves the continuity of $\mathfrak{R}$. The covering produces a history over every real and hence surjectivity. Upon passage to the quotient by the fibers, the induced map is bijective. On every compact chart, a continuous bijection onto a Hausdorff interval is a homeomorphism. The global topology of the quotient is the final topology of the gluing of these charts; the evaluation fiber identifies endpoint $m + 1$ of one chart with the translated endpoint 0 of the following chart. The resulting cover is locally finite, so the chart homeomorphisms glue into a global homeomorphism onto $\mathbb{R}$. □

18.12 Genealogical radial correspondences on the enriched state: domain, codomain, and preservation of helical history

This publication develops a genealogical theory of spiral correspondences from the composition $APP \to \widehat{\mathbb{T}} \to$ TPK. The primary object is not a curve presupposed in $\mathbb{R}^2$, but a compatible history of the enriched state, together with its arithmetic sheets, orientation, carry, boundary, route, and memory. The curve appears at the end as an Archimedean publication. Consequently, two identical figures in $\mathbb{R}^2$ may proceed from different histories, and a visible circle may have intrinsically zero radial memory or be the shadow of a helix whose memory was eliminated by the projection.

The work preserves and develops the exact results of the HMT corpus concerning the mixed curvature of APP, the three quadratic regimes of TRIT, the nonadic holonomy Γ_9, phase return with memory advance, the nonadic helix, the nonsplit dodecaphasic extension, and the Pell spectral screw. On those results it constructs a new formalization: the universal affine envelope, the prefix-enriched radial reader, its naturality under truncation, its equivariance with Γ_9, and its power–logarithmic publication.

The construction does not alter the theory of masses or reopen the central electron. The sole fixed point $(5, 5)$ of the material atlas is used only as the subsequent anchor of an interface; never as a retrospective selector of the radial geometry.

Probative stratification by domain and codomain

The exposition distinguishes previously established internal theorems, theorems proved in this work, finite certifications, and physical realizations. An internal algebraic result is asserted fully in its domain. Every subsequent physical realization preserves its premises and does not intervene in the selection of the HMT generator.

18.13 Construction and main results of the enriched radial reader

A theory of spiral correspondences is constructed whose antecedent is the enriched state of Triadic Modular Holography and whose output is a family of curves in $\mathbb{R}^2$ or $\mathbb{R}^3$. APP provides a two-dimensional support with two evaluations that are not simultaneously compatible, Σ and Π; their mixed composition has oriented discrete curvature ± 1. TRIT types the elliptic, parabolic, and hyperbolic regimes without confusing them with the radial character or Frenet curvature. TPK selects, transports, memorizes, updates, and returns the complete state. The holonomy Γ_9 returns the phase, but increments the memory depth and transports the boundary.

The initially proposed reader into $(p, a, \Lambda, C, m, r, \theta)$ is decisively corrected. The coordinates (r, θ) belong to the Archimedean publication, and the final pair (Λ, C) does not preserve its prefixes. An internal reader is therefore defined

$$\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},N} : \mathrm{Hist}^{\mathrm{enr}}_N(\mathrm{TPK}) \longrightarrow \mathcal{D}^{\mathrm{enr}}_{\mathrm{rad},N}$$

which records, at each depth, the mode and orientation cocycles, the local affine factor, prefix holonomy, and complete ledger. Its naturality under truncation, extension to the inverse limit, equivariance with Γ_9, preservation of carry, boundary, and memory, and compatibility with route inversion in the reversible sector are proved.

After constructing the continuum, a radial character is applied and the power–logarithmic coordinate is introduced

$$L_p(x) = \frac{x^p - 1}{p}, \qquad L_0(x) = \log x.$$

Its composition law makes the APP sum–product articulation visible. Uniform translation in L_p publishes, for $p = 2, 1, 0, -1, -2$, the Fermat, Archimedean, logarithmic, hyperbolic, and lituus spirals. General affine holonomy produces a broader classification through $(p, [\Lambda, C])$. The spectral screw $V^{-1}\mathcal{S}V = (2+\sqrt{3})\mathrm{i}\mathcal{S}$ appears as an exact case: its transverse projection is a logarithmic spiral, and its lift preserves helical memory.

The publication concludes with the differential geometry of the published curves, a restricted electromagnetic realization of the phase–position helix, the interface with the central electron, and a system of falsifiers preventing appearance, genealogy, and physical realization from being conflated.

Keywords: APP; TRIT; Topological Phase Kernel; nonadic holonomy; affine cocycle; spirals; helicoidal suspension; memory; double projection; continuum; Dimensional Mechanics.

18.14 Key results of the radial publication APP–TRIT–TPK

R1. Visible geometry is a nonfaithful publication of the enriched state: equality in $\mathbb{R}^2$ does not imply equality of genealogy.

R2. The mixed curvature of APP distinguishes the sum–product order before any real curve.

R3. τ, p and the Frenet curvature are invariants of distinct levels and cannot be identified.

R4. The nonadic holonomy returns the phase, advances the memory and transports the boundary; therefore the periodic angular projection admits a nonadic helicoidal lift that preserves the depth of return.

R5. Complete radial information requires preserving all factors and prefix holonomies; the final pair (Λ, C) is insufficient.

R6. The enriched radial reader is natural under truncation and equivariant under Γ_9.

R7. The bidirectional involution produces conjugate ascending and descending routes in the reversible sector.

R8. The family L_p publishes the five power–logarithmic classes as outputs of one and the same law, without choosing them in advance.

R9. The Pell spectral screw constitutes an exact realization of rotation and self-scale whose projection is logarithmic.

R10. A circumference may be a radially stationary state or the shadow of a helix with nonzero ledger.

R11. The electron $(5, 5)$ remains the proven material center; the spiral signature is added as a reader, not as a substitute for the atlas $81/56/13/324$.

R12. The electromagnetic and material realizations are subsequent to the HMT construction and preserve their own physical typing.

18.15 Typed notation and causal levels of the radial publication

Symbol	Meaning
X_9	Nonadic torus $(\mathbb{Z}/9\mathbb{Z})^2$, the two-dimensional support of APP.
Σ, Π	Additive and multiplicative evaluations of the same support.
$\tau \in \{-1, 0, +1\}$	Local TRIT type of transport: hyperbolic, parabolic, or elliptic according to the declared quadratic convention.
U_t	Typed transition $\mathrm{Upd}_t \circ \mathrm{Tra}_t \circ \mathrm{Sel}_t$.
Γ_9	Holonomy of nine transitions over the enriched state.
p	Cocycle or radial degree after defining its internal extraction; it is not τ.
a	Local oriented character $a = \tau_+ - \tau_-$; it is distinct from the radial degree $p = \tau_+ + \tau_-$.
(Λ, C)	Radial affine endomorphism: linear part and translation.
H_j	Prefix affine holonomy up to depth j.
$L_p, \exp_p$	Power-logarithmic coordinate and its inverse.
$\kappa_{\mathrm{F}}, \mathcal{T}_{\mathrm{F}}$	Frenet curvature and torsion of the published curve.
$\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}}$	Radial internal genealogical reader.
$\mathrm{ev}^{\mathrm{esp}}_{\mathrm{arq}}$	Archimedean publication of the spiral firm.

Causal composition from APP–TRIT–TPK to the radial Archimedean evaluation

$$\mathrm{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow X_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow C_{\mathrm{cont}}^{\mathrm{disc}} \longrightarrow \mathcal{P}_{\mathrm{rad}}^{\mathrm{enr}} \longrightarrow \mathrm{ev}_{\mathrm{arq}}^{\mathrm{esp}} \longrightarrow (\mathbb{R}^2 \text{ or } \mathbb{R}^3).$$

The curve is an output of the generative chain; it does not enter the selection of the state or its coefficients.

18.16 Genealogical Foundation of Radial Geometry

18.16.1 Archimedean evaluation of the enriched state and genealogical fidelity defect

18.16.1.1 Causal Factorization from the Enriched History to the Archimedean Evaluation

Conventional geometry begins with a continuous set and defines in it curves, parameters, and fields. The construction developed here reverses that order. First a compatible history is produced on a finite support, then its transport data are conserved, and only at the end is a real coordinate published. The causal scheme is

$$\mathrm{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow X_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow C_{\mathrm{cont}}^{\mathrm{disc}} \longrightarrow \mathrm{ev}_{\mathrm{arq}} \longrightarrow \mathbb{R}^d. \tag{18.37}$$

It is not a matter of adorning a classical curve with discrete labels. The history is the object; the curve is one of its readings.

Definition 18.12 (Geometric Generation with Traceability)**.** To construction geometric is *constructivo–generativa* when:

1. produces its object from declared operations on a domain preceding real geometry;
2. preserves the order of operations and the coordinates needed to reconstruct it;
3. obtains the geometric appearance by means of a later map;
4. distinguishes equality of appearances from equivalence of histories.

This criterion is stronger than a parametrization. A parametrization $t \mapsto \gamma(t)$ presupposes the curve and changes its mode of traversal. A genealogy $x = (x_n)_{n\geq 0}$, by contrast, specifies the finite truncations, the admissible transitions, the arithmetic sheets, the carry, and the memory that make the output γ possible.

18.16.1.2 Enhanced state space, TPK history category, and Archimedean evaluation

Let $X_{\mathrm{TPK}}^{\mathrm{enr}}$ be the enriched state space. A finite history of length N is

$$\gamma_N = (x_0 \xrightarrow{e_1} x_1 \xrightarrow{e_2} \cdots \xrightarrow{e_N} x_N),$$

where every arrow is compatible with the typed TPK transition. We associate its ledger with each prefix

$$\mathrm{Led}_j(\gamma_N) = (R_j, Q_j, \kappa_j, \sigma_j, o_j, g_j, C_j, B_j, \lambda_j, m_j, s_j),$$

which registers, according to the concrete residence, residue, quotient, carry, leaf, orientation, phase, cylinder, boundary, route, memory, and survival. The exact content may be enriched, but it cannot be reduced if the eliminated coordinate intervenes in a continuation.

The real publication is a composition.

$$\mathrm{Hist}_N^{\mathrm{enr}}(\mathrm{TPK}) \xrightarrow{\mathcal{P}_N^{\mathrm{enr}}} \mathcal{D}_N^{\mathrm{enr}} \xrightarrow{\mathrm{ev}_N} Y,$$

with $Y = \mathbb{R}, \mathbb{R}^2$ or $\mathbb{R}^3$. The first map preserves genealogy; the second publishes. If $\mathrm{ev}_N(\gamma) = \mathrm{ev}_N(\eta)$, it follows only that both histories belong to the same evaluation fibre.

Proposition 18.13 (No fidelity of the visible geometry)**.** *If the evaluation map removes at least one ledger coordinate that admits two compatible values, then there exist distinct histories with the same geometric output.*

Proof. Let c be an eliminated coordinate and let γ, η be two compatible histories that coincide in all preserved coordinates and differ in c. By definition of the evaluation, $\mathrm{ev}(\gamma) = \mathrm{ev}(\eta)$, whereas $\gamma \neq \eta$ because their ledgers differ. Therefore, ev is not injective. □

The proposition is not an insufficiency of real geometry. It expresses its function: to contract a richer structure to a usable coordinate. The central thesis consists precisely in not confusing that contraction with the contracted object.

18.16.1.3 Distinction between intrinsic radial orbit and circular projection of a helical trajectory

Consider the application

$$c_0(t) = (R\cos t, R\sin t),$$

and the helix

$$h(t) = (R\cos t, R\sin t, bt).$$

The projection $\pi_{xy}(x, y, z) = (x, y)$ satisfies $\pi_{xy} \circ h = c_0$. The same circumference may represent:

1. a state with null radial increment and memory;
2. the shadow of a state with null radial increment and axial memory $b \neq 0$.

The visible form does not decide between both cases. The ledger is needed.

(a) Circle as a planar object.

(b) The same shadow, with memory raised.

Figure 18.1: Visible geometry does not by itself reconstruct the genealogy.

Failure of fidelity of Archimedean evaluation on genealogical classes

A circle may have intrinsically null memory and may also be the publication of a helical history. Equality of the curve does not authorize identifying the states that produce it.

18.16.2 APP: two-dimensional support, arithmetic sheets and mixed curvature

18.16.2.1 Construction of the bidimensional nonadic support $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$

The visible cell

$$E_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

simultaneously carries a phase and a chart of positive representatives. For $n \geq 1$,

$$\rho_9(n) = 1 + ((n-1) \bmod 9), \qquad q_9(n) = \left\lfloor \frac{n-1}{9} \right\rfloor,$$

so that

$$n = 9q_9(n) + \rho_9(n).$$

The Cartesian product of two cells constructs

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

It is not a repeated line but a genuinely two-dimensional support. The completion 9×9 may be read as a core 8×8 and a gnomon of 17 cells:

$$81 = 64 + 17.$$

Two evaluations of the same support act on every $(i, j) \in X_9$:

$$\Sigma(i, j) = \mathrm{dr}(i + j), \qquad \Pi(i, j) = \mathrm{dr}(ij),$$

where dr publishes the nonadic representative and the lift conserves the quotient. APP does not add two spaces; it conserves a single space with two readings.

18.16.2.2 Residue, quotient and carry

The raised identities

$$i + j = R^+_{ij} + 9q^+_{ij}, \qquad ij = R^\times_{ij} + 9q^\times_{ij}$$

distinguish the visible output R from the lift q. If only the residue is preserved, distinct operations may collapse in the same cell. The pair (R, q) unfolds the history. Under composition, the carry obeys a cocycle law; therefore it is not an auxiliary annotation, but rather the datum that allows a long operation to be recomposed from its steps.

Example 18.14 (Same residue, different history)**.** In the nonadic chart,

$$8 + 8 = 7 + 9 \cdot 1, \qquad 8 \cdot 2 = 7 + 9 \cdot 1.$$

The two operations publish the same residue and the same local quotient, but differ in leaf. The label Σ/Π remains necessary. On longer routes, the order of those leaves modifies the composite carry and the boundary reached.

18.16.2.3 Operational Incompatibility of the Additive and Multiplicative Leaves

For $d \geq 2$, let $\mathscr{A}_d = E_9^d$. The conditional expectations $P_{\Sigma,d}$ and $P_{\Pi,d}$ on the two evaluations satisfy

$$\|[P_{\Sigma,2}, P_{\Pi,2}]\| = \frac{\sqrt{2}}{3}, \qquad \|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}.$$

In dimension three, the tritic mode of operative phase has singular value $s = 1/2$, and

$$s^2 = \frac{1}{4}, \qquad 1 - s^2 = \frac{3}{4} = \frac{90}{120}, \qquad s\sqrt{1 - s^2} = \frac{\sqrt{3}}{4}.$$

Thus, the separation between the sheets is not added afterward: it is quantified in the APP support itself.

18.16.2.4 Discrete curvature sum–product

In residues, write

$$S(x, y) = x + y, \qquad P(x, y) = xy \pmod 9.$$

For a potential T, the progressive differences are defined

$$\Delta_x T(x, y) = T(x + 1, y) - T(x, y), \qquad \Delta_y T(x, y) = T(x, y + 1) - T(x, y).$$

With $A_x^T = \Delta_x T$ and $A_y^T = \Delta_y T$, the curvature of an ordered mixture is

$$F(T_x, T_y) = \Delta_x A_y^{T_y} - \Delta_y A_x^{T_x}.$$

Since

$$\Delta_x S = \Delta_y S = 1, \qquad \Delta_x P = y, \qquad \Delta_y P = x,$$

it is obtained exactly

$$F(S,S) = F(P,P) = 0, \qquad F(S,P) = +1, \qquad F(P,S) = -1. \tag{18.38}$$

Theorem 18.15 (Mixed oriented curvature of APP)**.** *Pure compositions of the APP leaves are flat in the preceding plaquette convention; mixed compositions have constant curvature of opposite sign according to the order.*

Proof. Substituting the four differences in the definition,

$$\Delta_x \Delta_y S - \Delta_y \Delta_x S = 0, \quad \Delta_x \Delta_y P - \Delta_y \Delta_x P = 0$$

for the two pure compositions, while

$$\Delta_x \Delta_y P - \Delta_y \Delta_x S = 1, \qquad \Delta_x \Delta_y S - \Delta_y \Delta_x P = -1.$$

The identity verifica in each to of the 81 plaquetas. □

For a coordinate rectangle with sides a, b, cancellation of the interior edges yields the finite form of Stokes' theorem:

$$W_{\partial R_{a,b}}(S,P) = ab, \qquad W_{\partial R_{a,b}}(P,S) = -ab \pmod 9.$$

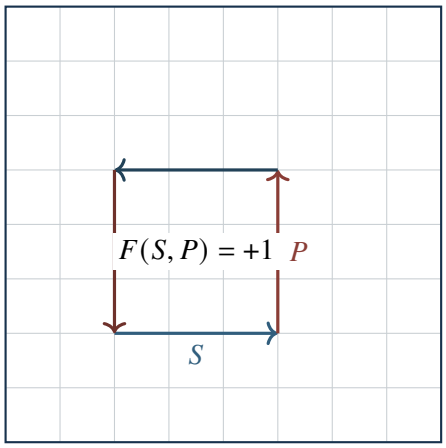


Figure 18.2: Curvature arises from the order of the leaves over the same support.

18.16.2.5 APP invariants that induce the accumulated radial character

APP contributes three distinct ingredients:

1. a domain bidimensional anterior to $\mathbb{R}^2$;
2. two operations whose ordered mixture distinguishes orientation;

3. a ledger of residue, quotient, sheet, and carry capable of remembering the path that produced the same output.

The curvature ± 1 is the local antecedent of the radial orientation, but it must not be identified without proof with the global degree p. The passage from one to the other requires the TPK accumulation and the cocycle declared in the section 18.17.2.

18.16.3 TRIT: local orientation and 3 geometric regimes

18.16.3.1 Residue and carry enriched TRIT state: multiplicity of threshold representatives

The visible observable is

$$\widehat{\mathbb{T}} = \{-1, 0, +1\}.$$

In a local step with a carry, the balanced decomposition is written

$$\widehat{\tau} = (r, c), \qquad a + b = r + 3c, \qquad r \in \{-1, 0, +1\},\ c \in \mathbb{Z}.$$

The visible zero possesses at least three elementary lifts:

$$0^{+} = (0, +1), \qquad 0^{0} = (0, 0), \qquad 0^{-} = (0, -1).$$

The visible equality does not fix the continuation. This also depends on sheet, orientation, height, carry, boundary, path and phase.

18.16.3.2 Family $\mathcal{A}_\tau = \mathbb{R}[J]/(J^2 + \tau I)$ and elliptic, parabolic, and hyperbolic classification

Para $\tau \in \widehat{\mathbb{T}}$, sea

$$\mathbb{A}_\tau = \mathbb{R}[J]/(J^2 + \tau I).$$

Three regimes are obtained:

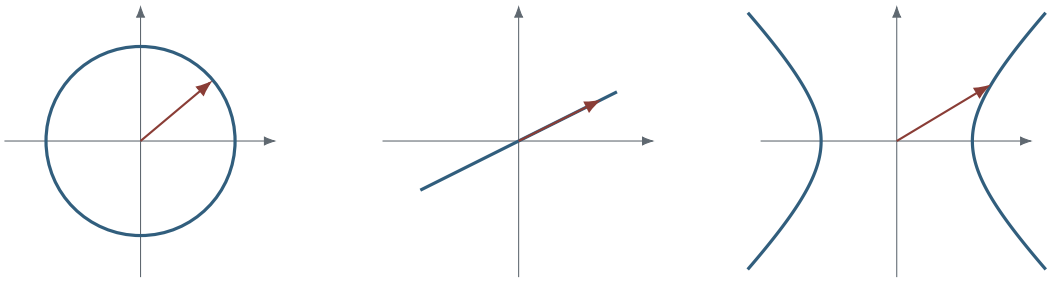
$$J_{+1}^2 = -I, \qquad J_0^2 = 0, \qquad J_{-1}^2 = +I,$$

and a single exponential identity

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau,$$

where

τ	J_τ^2	$C_\tau(t)$	$S_\tau(t)$
$+1$	$-I$	$\cos t$	$\sin t$
0	0	1	t
-1	I	$\cosh t$	$\sinh t$



elliptic $J^2 = -I$ **parabolic** $J^2 = 0$ **hyperbolic** $J^2 = I$

Figure 18.3: The three quadratic regimes of the TRIT. They are transport types, not names of specific curves.

18.16.3.3 Separation between TRIT regime, radial character and Frenet curvature

The theory requires three levels that must not be flattened:

1. τ tipa the regimen local of transport;
2. p types the accumulated radial character of a history;
3. κ_{F} and $\mathcal{T}_{\mathrm{F}}$ describe the curve already published in a real space.

A circumference has elliptic angular holonomy, but the present TRIT 0 is a parabolic threshold. The circumference and the present are not the same object. Likewise, a published logarithmic spiral may include segments whose local transitions traverse the three TRIT types.

Principle 18.16 (Separation of levels)**.** No identity among τ, p and κ_{F} is admitted by similarity of sign, name, or drawing. Every relation must be constructed by means of a map with declared domain, codomain, and action.

18.16.3.4 Orientation Inversion

For a word $\boldsymbol{\tau} = (\tau_1, \dots, \tau_n)$,

$$C_{\mathrm{rev}}(\boldsymbol{\tau}) = (-\tau_n, \dots, -\tau_1), \qquad C_{\mathrm{rev}}^2 = \mathrm{id}\,.$$

The involution exchanges the orientations and preserves the threshold. In the sector of reversible routes it will induce inversion of the affine holonomy, but that conclusion requires the local representation to satisfy $h_{e^\dagger} = h_e^{-1}$.

18.16.4 Enriched state, unique continuum and double projection

18.16.4.1 Inverse history system

Let S_n be the admissible states at depth n and

$$\pi_{n+1,n} : \mathsf{S}_{n+1} \longrightarrow \mathsf{S}_n$$

the truncations. The space of complete histories is

$$X_\infty = \varprojlim_n \mathsf{S}_n.$$

With the ultrametric

$$d(x, y) = 3^{-m(x,y)},$$

where $m(x, y)$ is the first depth at which they differ, the fixed-prefix cylinders are clopen. The structure of the continuum is obtained by jointly constructing, from the same APP–TRIT–TPK state, its five consubstantial constructions, which emerge correlatively, and by passing through terminality, naturality, and the limit. This publication neither disaggregates them nor redefines them by means of curves.

18.16.4.2 Archimedean Evaluation via Compatible Cylinders

Let $\widehat{\mathbb{Q}}$ be the uniform completion of $\mathbb{Q}$, constructed as classes of rational Cauchy sequences. The calibrated construction associates with each finite state an integer direction $k_n(s)$ and the rational semi-closed cylinder

$$I_n(s) = \{q \in \mathbb{Q} : k_n(s)3^{-n} \le q < (k_n(s) + 1)3^{-n}\}.$$

Satisfies simultaneously:

(A1) $\bigcup_{s\in\mathsf{S}_n} I_n(s) = \mathbb{Q}$ for every n;

(A2) if $t \in \mathsf{S}_{n+1}$ truncates to s, then $I_{n+1}(t) \subseteq I_n(s)$, and the children partition the parent under the half-closed boundary convention;

(A3) there exist a constant $C > 0$ and $0 < \lambda < 1$ such that

$$\operatorname{diam} I_n(s) \le C\lambda^n, \qquad 0 < \lambda < 1;$$

(A4) every chain of nested cylinders arises from a compatible history, with the two boundary expansions identified by the Archimedean relation.

Theorem 18.17 (Archimedean Closure of the Genealogical Continuum)**.** *Each calibrated history determines a unique value*

$$\{\mathrm{val}(x)\} = \bigcap_n \overline{I_n(x_n)}^{\widehat{\mathbb{Q}}}.$$

The evaluation

$$\mathrm{val} : X_\infty^{\mathrm{cal}} \longrightarrow \widehat{\mathbb{Q}}$$

is continuous, Hölder, and surjective. The quotient by equality of value satisfies

$$X_\infty^{\mathrm{cal}}/{\sim_{\mathrm{arq}}} \cong \widehat{\mathbb{Q}} \cong \mathbb{R},$$

where the last identification is the usual Archimedean recognition of the completion of $\mathbb{Q}$, not an input to the generator.

Proof. Contraction of diameters and completeness of $\widehat{\mathbb{Q}}$ reduce each nested chain to a point. Cylinder compatibility proves continuity; the geometric diameter bound gives the Hölder estimate. By (A1) and (A2), each rational Cauchy class selects at every depth a cylinder containing it; (A4) provides a compatible history and uniquely resolves the boundaries. Surjectivity follows. Two histories have the same image exactly when they belong to the relation $\sim_{\mathrm{arq}}$, so the induced map on the quotient is a continuous open bijection onto the final cylinder topology; it is therefore a homeomorphism. □

The real line is, therefore, a proved output of the evaluation and not a presupposed domain. The history is not reduced to its value.

18.16.4.3 Joint factorization of the Archimedean and exceptional publications of the enriched state

From a single calibrated domain they are distinguished

$$\mathfrak{R} : X_\infty^{\mathrm{cal}} \longrightarrow \mathbb{R}, \qquad \mathfrak{E} : X_\infty^{\mathrm{cal}} \times \mathscr{C}_{\mathrm{exc}} \longrightarrow \mathbf{Exc}.$$

The first contracts compatible cylinders; the second preserves boundary incidence toward the exceptional corridor. Spiral theory is inserted into the first branch, but preserves in its ledger the data required to compare the second. Value is not identified with incidence, nor is a publication made to generate the continuum retrospectively.

18.16.4.4 5 regions of π and multiplicity of genealogies

The microfiber of π contains five regions distinct with multiplicities

$$432 + 4 \cdot 144 = 1008.$$

The same visible value conserves different regional histories. This fact provides the exact precedent of the spiral classification: equality of a published curve does not force equality of leaf, carry, boundary, or memory. The theory does not invent that multiplicity; it transports it to a new type of geometric output.

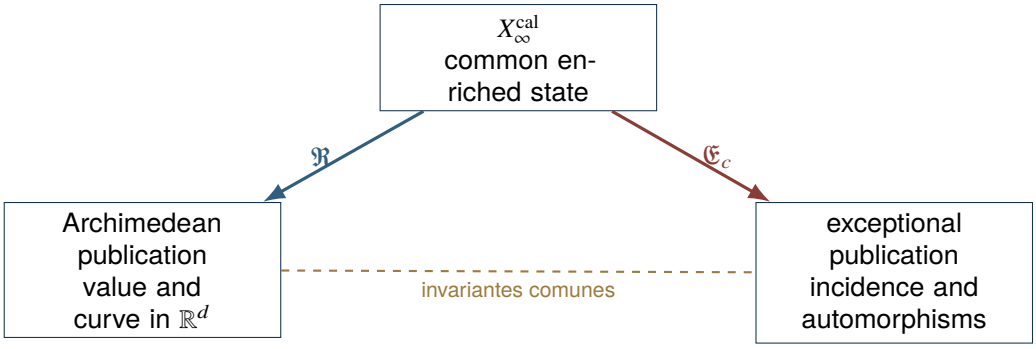


Figure 18.4: The two projections are correlative and subsequent to the same state.

Causal antecedence of the discrete continuum with respect to the radial reader

Thus far, the domain, sheets, regimes, holonomy, enriched state, and publication of the continuum have been constructed. Only now is it legitimate to introduce real coordinates (r, θ) and recognize conventional curves.

18.17 Genealogical theory of spiral correspondences

18.17.1 Affine universal envelope of the APP leaves

18.17.1.1 Construction of the universal radial module and its endomorphic envelope

The operations $u \mapsto u + a$ and $u \mapsto \lambda u$ suggest the affine group of the line. APP, however, contains noninvertible multiplicative sectors and precedes real publication. The appropriate internal target is therefore the monoid of affine endomorphisms of a universal radial module.

Let $\widetilde{X}_{\mathrm{APP}}$ be the set of complete APP states, that is, states that conserve both leaves, residue, quotient, and carry, and let us add an absorbing symbol $\perp$. We define the free module

$$U_{\mathrm{APP}} := \mathbb{Z}^{(\widetilde{X}_{\mathrm{APP}} \sqcup \{\perp\})} / \langle [\perp] \rangle. \tag{18.39}$$

For each enriched transition e, with source x_e and active digits (i_e, j_e), APP produces, without decimal evaluation, the records

$$i_e + j_e = R_e^+ + 9q_e^+, \qquad i_e j_e = R_e^\times + 9q_e^\times.$$

Let $s_e^\Sigma(x_e)$ and $s_e^\Pi(x_e)$ be the complete states obtained by adjoining, respectively, (R_e^+, q_e^+) and $(R_e^\times, q_e^\times)$, while preserving the other sheet in the ledger. The additive lift determines the explicit vector

$$A_e = [s_e^\Sigma(x_e)] - [x_e] \in U_{\mathrm{APP}},$$

and the multiplicative lifting determines the total map $m_e : \widetilde{X}_{\mathrm{APP}} \sqcup \{\perp\} \to \widetilde{X}_{\mathrm{APP}} \sqcup \{\perp\}$, extended by

$$m_e(x) = \begin{cases} s_e^\Pi(x_e), & x = x_e, \\ \perp, & x \neq x_e, \end{cases} \qquad m_e(\perp) = \perp.$$

Its linearization is

$$M_e([x]) = [m_e(x)] \in U_{\mathrm{APP}}.$$

Thus, A_e and M_e are not parameters free: are the two publications linearized of the same transition APP complete.

We define

$$\mathrm{AffEnd}(U_{\mathrm{APP}}) = \mathrm{End}(U_{\mathrm{APP}}) \ltimes U_{\mathrm{APP}}$$

with composition

$$(\Lambda_2, C_2) \circ (\Lambda_1, C_1) = \big(\Lambda_2\Lambda_1, C_2 + \Lambda_2 C_1\big). \tag{18.40}$$

The pair acts by $u \mapsto \Lambda u + C$. The linear part may be noninvertible. The subgroupoid of units is obtained by restricting $\Lambda \in \mathrm{Aut}(U_{\mathrm{APP}})$.

Proposition 18.18 (Universal property)**.** *Every map from the set of complete states to a V-module that vanishes on $\perp$ extends uniquely to a morphism $\chi : U_{\mathrm{APP}} \to V$. If it also intertwines the multiplicative lifts, $\chi M_e = M_e^V \chi$, then it uniquely transports the affine factors h_e and all their prefix holonomies.*

Proof. The first statement is the universal property of the free module quotiented by $[\perp] = 0$. The second follows by applying χ to the semidirect composition: $\chi(C_2 + \Lambda_2 C_1) = \chi(C_2) + \Lambda_2^V \chi(C_1)$. □

18.17.1.2 Affine representation $e \mapsto h_e$ of the generators of the route category

Let $e : x \to y$ be a TPK transition. The tritic operational-phase selector determines whether the additive leaf, the multiplicative leaf, or the threshold acts. It is defined

$$h_e = \begin{cases} (I, A_e), & \text{sheet } \Sigma, \\ (M_e, 0), & \text{sheet } \Pi, \\ (I, 0), & \text{threshold}, \end{cases} \tag{18.41}$$

where A_e and M_e are the complete APP lifts constructed above. Their residues alone are not taken: the quotient and carry remain in the ledger.

Definition 18.19 (Internal radial representation)**.** An internal radial representation is an assignment

$$\rho_{\mathrm{rad}} : \mathscr{P}_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow \mathrm{AffEnd}(U_{\mathrm{APP}})$$

which sends each arrow to h_e, the identity to $(I, 0)$ and composition of paths to the product equation (18.40).

The definition does not consult a curve. The factors arise from the sheet and the lift that the state already contains. A subsequent numerical realization may send U_{APP} to $\mathbb{R}$, but that evaluation does not modify the internal representation.

18.17.1.3 Ordered affine products $H_j = h_{e_j} \cdots h_{e_1}$ and preservation of prefixes

For

$$\gamma_N = e_N \cdots e_1,$$

we define

$$H_0 = (I, 0), \qquad H_j = h_{e_j} \cdots h_{e_1} = (\Lambda_j, C_j).$$

Theorem 18.20 (Affine holonomy functorial)**.** *For composable paths γ_1, γ_2,*

$$H_{\gamma_2\gamma_1} = H_{\gamma_2} H_{\gamma_1}.$$

If $H_{\gamma_i} = (\Lambda_i, C_i)$, then

$$\Lambda_{\gamma_2\gamma_1} = \Lambda_2\Lambda_1, \qquad C_{\gamma_2\gamma_1} = C_2 + \Lambda_2 C_1. \tag{18.42}$$

Proof. The successive action on u is

$$u \overset{H_{\gamma_1}}{\longmapsto} \Lambda_1 u + C_1 \overset{H_{\gamma_2}}{\longmapsto} \Lambda_2\Lambda_1 u + C_2 + \Lambda_2 C_1.$$

Uniqueness of the affine decomposition gives equation (18.42). □

For a nine-step block,

$$\Lambda_9 = \Lambda_{e_9} \cdots \Lambda_{e_1}, \tag{18.43}$$

$$C_9 = C_{e_9} + \Lambda_{e_9} C_{e_8} + \Lambda_{e_9}\Lambda_{e_8} C_{e_7} + \cdots + \Lambda_{e_9} \cdots \Lambda_{e_2} C_{e_1}. \tag{18.44}$$

The position of each factor matters. This is the affine form of the memory of order.

18.17.1.4 Sum–product commutator

In the invertible sector and on a one-dimensional realization, let

$$A_a(u) = u + a, \qquad M_\lambda(u) = \lambda u.$$

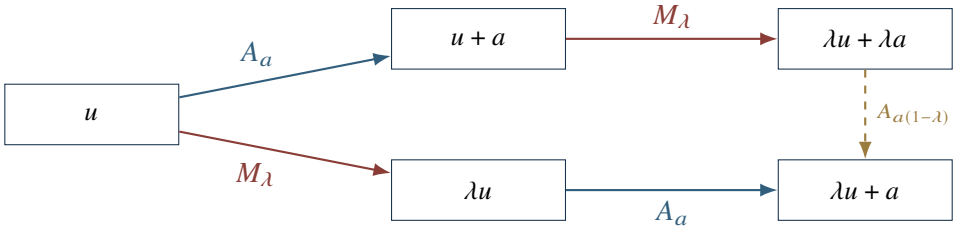
Then

$$A_a M_\lambda A_a^{-1} M_\lambda^{-1} = A_{a(1-\lambda)}. \tag{18.45}$$

The proof is direct:

$$u \mapsto \lambda^{-1}u \mapsto \lambda^{-1}u - a \mapsto u - \lambda a \mapsto u + a(1-\lambda).$$

The identity is asserted only when λ is a unit. In the full monoid, the commutation defect is preserved by the factors and the ledger, not by a commutator requiring nonexistent inverses.



The two orders end at distinct points; the vertical arrow measures exactly the composition defect.

Figure 18.5: The sum-product difference manifests as affine holonomy.

18.17.2 Radial cocycles in the TRIT lift of 2 semicoronas

18.17.2.1 Category of enriched paths by 2 TRIT orientations and definition of the characters p and a

Let $\mathscr{P}_{\mathrm{TPK}}^{2c,\mathrm{enr}}$ be the category of paths whose arrows are triples

$$\widehat{e} = (e, \tau_+(e), \tau_-(e)), \qquad \tau_\pm(e) \in \{-1, 0, +1\},$$

where the oriented pair belongs to the enriched state and the forgetful functor sends $\widehat{e}$ to the TPK transition e. We define on each generator

$$p(\widehat{e}) = \tau_+(e) + \tau_-(e), \qquad a(\widehat{e}) = \tau_+(e) - \tau_-(e). \tag{18.46}$$

For a route $\widehat{\gamma}_N = \widehat{e}_N \cdots \widehat{e}_1$, the accumulated cocycles are

$$P_j = \sum_{k=1}^{j} p(\widehat{e}_k), \qquad A_j = \sum_{k=1}^{j} a(\widehat{e}_k), \qquad P_0 = A_0 = 0. \tag{18.47}$$

Additivity under concatenation proves that (P, A) is a $\mathbb{Z}^2$-valued cocycle of the lifted category. This assertion does not identify the pair by fiat with another previously named cocycle: it constructs the pair from the two orientations actually preserved along each arrow. Subdivision by stationary micro-arrows preserves both sums.

18.17.2.2 Local exhaustive chart of the radial and oriented characters

Table 18.2: Complete local chart of the two TRIT orientations.

τ_+	τ_-	p	a	$r_3(p)$	$c_3(p)$
+1	+1	+2	0	-1	1
+1	0	+1	+1	+1	0
+1	-1	0	+2	0	0
0	+1	+1	-1	+1	0
0	0	0	0	0	0
0	-1	-1	+1	-1	0
-1	+1	0	-2	0	0
-1	0	-1	-1	-1	0
-1	-1	-2	0	+1	-1

The three states with $p = 0$ are distinguished by $a = 2, 0, -2$. Therefore the radial degree is not sufficient to reconstruct the orientation.

18.17.2.3 Unique decomposition $p = r_3(p) + 3c_3(p)$ into balanced residue and carry

For every $p \in \mathbb{Z}$, there exists to unique decomposition

$$p = r_3(p) + 3c_3(p), \qquad r_3(p) \in \{-1, 0, +1\}, \quad c_3(p) \in \mathbb{Z}.$$

In the local block $p \in [-2, 2]$, the last column of table 18.2 gives the carry. In long routes, $c_3(p)$ does not remain restricted to $\{-1, 0, +1\}$. This precision avoids converting a local example into a false global law.

18.17.2.4 Action of $(\tau_+, \tau_-) \mapsto (-\tau_-, -\tau_+)$ over the characters p and a

Local Involution

$$(\tau_+, \tau_-) \longmapsto (-\tau_-, -\tau_+)$$

yields

$$p \longmapsto -p, \qquad a \longmapsto a.$$

The radial character changes orientation, whereas the oriented difference remains. This separation will be visible in the identity $L_{-p}(x^{-1}) = -L_p(x)$.

18.17.3 Power-logarithmic coordinate and sum-product law

18.17.3.1 Definition of L_p on $\mathbb{R}_{>0}$ and domain of the inverse $\exp_p$

For $x > 0$, define

$$L_p(x) = \begin{cases} \dfrac{x^p - 1}{p}, & p \neq 0, \\ \log x, & p = 0. \end{cases} \tag{18.48}$$

Its inverse is

$$\exp_p(u) = \begin{cases} (1 + pu)^{1/p}, & p \neq 0, \quad 1 + pu > 0, \\ \mathrm{e}^u, & p = 0. \end{cases} \tag{18.49}$$

The condition $1 + pu > 0$ is not ornamental: it fixes the positive real branch.

18.17.3.2 Composition law $L_p(xy) = L_p(x) + L_p(y) + pL_p(x)L_p(y)$

In the corresponding domain, let

$$u \boxplus_p v = u + v + puv.$$

Theorem 18.21 (Sum–Product Law)**.** *For $x, y > 0$,*

$$L_p(xy) = L_p(x) \boxplus_p L_p(y). \tag{18.50}$$

Moreover, 0 is the identity of $\boxplus_p$, and

$$\exp_p(u \boxplus_p v) = \exp_p(u) \exp_p(v).$$

Proof. Para $p \neq 0$,

$$1 + pL_p(xy) = (xy)^p = x^p y^p = (1 + pL_p(x))(1 + pL_p(y)).$$

Expanding the last product and dividing by p yields equation (18.50). For $p = 0$ it is the identity $\log(xy) = \log x + \log y$. The equality for $\exp_p$ follows by applying the inverse. □

This law exactly expresses the APP articulation: the product in the radial coordinate is published as a deformed sum in the coordinate L_p. The term puv preserves the interaction that would disappear in an ordinary sum.

18.17.3.3 Bidirectional Inversion

Proposition 18.22 (Inversion Identity)**.** *For $p \in \mathbb{Z}$ and $x > 0$,*

$$L_{-p}(x^{-1}) = -L_p(x). \tag{18.51}$$

Proof. Si $p \neq 0$,

$$L_{-p}(x^{-1}) = \frac{(x^{-1})^{-p} - 1}{-p} = -\frac{x^p - 1}{p}.$$

For $p = 0$, $\log(x^{-1}) = -\log x$. □

The involution simultaneously exchanges expansion/contraction and $p/-p$. It does not eliminate the memory of orientation, which remains in a.

18.17.3.4 Logarithmic limit

Para $x > 0$,

$$\lim_{p \to 0} \frac{x^p - 1}{p} = \log x.$$

Indeed, $x^p = \mathrm{e}^{p \log x} = 1 + p \log x + O(p^2)$. Therefore, the logarithmic branch is not an added exception: it is the continuous limit of the family. In the genealogical theory, however, $p = 0$ continues to preserve a and the ledger; the functional limit does not identify its three local origins of table 18.2.

18.17.4 Power–logarithmic classification of the published curves

18.17.4.1 Uniform translation in the radial coordinate

Suppose, after the publication of the continuum, that

$$L_p(r/r_0) = \beta m, \qquad \theta = \theta_0 + \omega m, \qquad \omega \neq 0.$$

Eliminando m,

$$r_p(\theta) = r_0 \left[1 + p \frac{\beta}{\omega}(\theta - \theta_0)\right]^{1/p}, \qquad p \neq 0, \tag{18.52}$$

and

$$r_0(\theta) = r_0 \exp\left[\frac{\beta}{\omega}(\theta - \theta_0)\right]. \tag{18.53}$$

Theorem 18.23 (Publication of the five canonical classes)**.** *After reoriginating θ and rescaling r, the specializations of equations (18.52) y (18.53) are:*

p	*normalized equation*	*conventional recognition*
2	$r^2 = a^2\theta$	*Fermat spiral*
1	$r = a\theta$	*Archimedean spiral*
0	$r = r_0 e^{b\theta}$	*logarithmic spiral*
-1	$r = a/\theta$	*hyperbolic spiral*
-2	$r^2 = a^2/\theta$	*lituus*

in the corresponding positive real domains.

Proof. For $p = 1$, r is affine in θ, and recentering removes the constant term. For $p = 2$, r^2 is affine. For $p = -1$, r^{-1} is affine; for $p = -2$, r^{-2} is affine. The $p = 0$ branch is equation (18.53). Positive constants are absorbed into a, b without altering the exponent. □

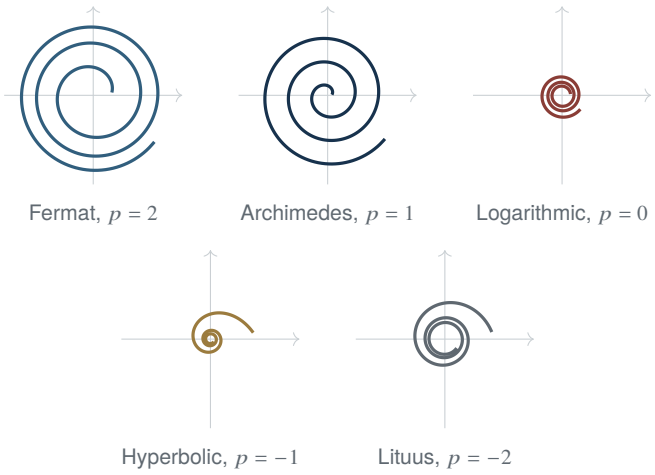


Figure 18.6: Five power–logarithmic family publications on common axes and scales.

18.17.4.2 Classificatory domain of $p \in \{2, 1, 0, -1, -2\}$ and general affine extension

The theorem classifies the family generated by uniform translation in the coordinate L_p. It does not classify all possible plane curves. The complete genealogical taxonomy must include at least

$$(p, a, [\Lambda, C], [\kappa_9], \omega_{\text{ang}}, \mu_{\text{mem}}, \varepsilon_{\text{tw}}).$$

Two routes with the same p may differ in self-scaling, orientation, carry, or boundary and publish distinct curves. Two routes with all visible data equal may still differ in an eliminated ledger coordinate.

18.17.4.3 General affine holonomy

If the coordinate $u = L_p(r/r_0)$ satisfies

$$u_{n+1} = \Lambda u_n + C,$$

the iteration internal exact, for $n \in \mathbb{N}$, is

$$u_n = \Lambda^n u_0 + \sum_{k=0}^{n-1} \Lambda^k C.$$

If $I - \Lambda$ is invertible, may iscribirse

$$u_n = u_* + \Lambda^n (u_0 - u_*), \qquad u_* = (I - \Lambda)^{-1} C.$$

Continuous interpolation is performed only after fixing a positive scalar character $\chi : U_{\mathrm{APP}} \to \mathbb{R}$ such that $\chi\Lambda = \lambda\chi$, with $\lambda > 0$. Writing $u = \chi(u)$, $C = \chi(C)$, if $\lambda \neq 1$, then $u_* = C/(1 - \lambda)$. For $\theta_n = \theta_0 + n\omega$, $\omega \neq 0$, the scalar realization admits

$$r(\theta) = r_0 \exp_p \left[u_* + \lambda^{(\theta - \theta_0)/\omega} (u_0 - u_*) \right]. \tag{18.54}$$

Real powers belong exclusively to this realization; the internal dynamics uses integer powers of Λ. The pair $(p, [\Lambda, C])$ enlarges the classical list: p fixes the radial chart, and the affine conjugacy class fixes the dynamics within it.

Remark 18.24 (Fixed point and electron). The existence of an affine fixed point does not by itself identify that point with the central electron $(5, 5)$. The electron is a prior material result; an identification with a radial fixed point requires a separate realization square.

18.17.5 Radial enriched reader

18.17.5.1 Defecto of naturality of the reader reduced $(p, a, \Lambda, C, m, r, \theta)$

The initially suggested map

$$X^{\mathrm{enr}}_{\mathrm{TPK}} \to (p, a, \Lambda, C, m, r, \theta)$$

mixes causal levels and loses prefixs. The coordinates r, θ are arquimedianas, while (Λ, C) preserves only the final product.

Proposition 18.25 (The terminal pair admits no natural truncation)**.** *In general there is no application that recovers the first factor of an affine word from its final product.*

Proof. In notation (Λ, C),

$$(1, 2)(2, 0) = (2, 2), \qquad (1, 1)(2, 1) = (2, 2).$$

The two final products coincide, but their initial prefixes are $(2, 0)$ and $(2, 1)$. A function of the final product would have to assign it two different truncations, which is impossible. □

18.17.5.2 Enriched radial reader $\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},N}$: domain, codomain, and preservation of prefixes

Let $\widehat{\gamma}_N = \widehat{e}_N \cdots \widehat{e}_1$ one route of the category lifted and let $(H_0, \mathrm{Led}_0) = ((I, 0), \mathrm{Led}(x_0))$ its datum initial. We define

$$\boxed{\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},N}(\widehat{\gamma}_N) = \Big((H_0, \mathrm{Led}_0); \big(p(\widehat{e}_j), a(\widehat{e}_j), P_j, A_j, h_{e_j}, H_j, \mathrm{Led}_j(\widehat{\gamma}_N) \big)_{j=1}^{N} \Big).} \tag{18.55}$$

The codomain $\mathcal{D}^{\mathrm{enr}}_{\mathrm{rad},N}$ contains compatible sequences of those data, including the underlying edge stored in each entry of the ledger and its source and target maps. The output conserves the initial datum, each factor, each prefix holonomy, the local and accumulated cocycles, and every memory update.

Definition 18.26 (Admissible homogeneous radial sector)**.** Let $\mathcal{D}^{\mathrm{hom}}_{\mathrm{rad},\infty}$ the space of data marked

$$D = (D^{\mathrm{enr}}; r_0, p, u_0, \beta, \theta_0, \omega),$$

where $D^{\mathrm{enr}} \in \mathcal{D}^{\mathrm{enr}}_{\mathrm{rad},\infty}$, $r_0 > 0$, $p \in \{-2,-1,0,1,2\}$, $u_0, \beta, \theta_0, \omega \in \mathbb{R}$ and $(\beta,\omega) \neq (0,0)$, talis that

$$p(\widehat{e}_j) = p, \qquad u(m) = u_0 + \beta m, \qquad \theta(m) = \theta_0 + \omega m.$$

The mark is part of the data and is not chosen afterward; consequently, the evaluation is a well-defined function. Its real publication interval is

$$I_D = \begin{cases} \{m \in \mathbb{R} : 1 + p(u_0 + \beta m) > 0\}, & p \neq 0, \\ \mathbb{R}, & p = 0. \end{cases}$$

Histories with variable degree remain in the enriched ledger and are published through an atlas of piecewise charts; they do not determine a single curve of constant degree.

Definition 18.27 (Homogeneous spiral publication)**.** After constructing the continuum and choosing the radial and angular characters compatible, it is defined

$$\mathrm{ev}^{\mathrm{esp}}_{\mathrm{arq}} : \mathcal{D}^{\mathrm{hom}}_{\mathrm{rad},\infty} \longrightarrow \coprod_{D} C^\infty(I_D, \mathbb{R}^2)$$

by

$$D \longmapsto \left[m \longmapsto \left(r_0 \exp_p(u_0 + \beta m)\cos(\theta_0 + \omega m), r_0 \exp_p(u_0 + \beta m)\sin(\theta_0 + \omega m) \right) \right].$$

The internal reader and external evaluation are separate. The first arrow preserves genealogy; the second may forget part of it. The condition $(\beta,\omega) \neq (0,0)$ excludes only the degenerate constant parametrization; on I_D the published curve is regular.

18.17.5.3 Dynamic conservation of carry, boundary, and memory by the radial reader

The projection onto carry, boundary, and memory satisfies

$$\mathrm{pr}_{\mathrm{carry},frontier,memory} \circ \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},N} = \mathrm{Led}_{\mathrm{carry},frontier,memory,N}\,.$$

To preserve does not mean to make those coordinates constant. It means that:

$$g(K+9) = g(K),$$

$$q_{\mathrm{mem}}(\Gamma_9 x) = q_{\mathrm{mem}}(x) + 1,$$

$$B_{K+9} = \mathrm{Tra}_{\Gamma_9}(B_K),$$

and the carry is updated by equation (8.15). No update is hidden by the final curve.

18.17.6 Naturalness, equivariance, and inverse limit

18.17.6.1 Naturalness under Truncation

For $M \leq N$, let

$$\pi_{N,M} : \mathrm{Hist}^{\mathrm{enr}}_N(\mathrm{TPK}) \to \mathrm{Hist}^{\mathrm{enr}}_M(\mathrm{TPK})$$

the truncation of histories, and let

$$q_{N,M} : \mathcal{D}^{\mathrm{enr}}_{\mathrm{rad},N} \to \mathcal{D}^{\mathrm{enr}}_{\mathrm{rad},M}$$

the removal of data with index greater than M.

Theorem 18.28 (Radial Reader Naturalness)**.** *The diagram*

$$\begin{array}{ccc} \mathrm{Hist}_N^{\mathrm{enr}}(\mathrm{TPK}) & \xrightarrow{\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},N}} & \mathcal{D}^{\mathrm{enr}}_{\mathrm{rad},N} \\ \pi_{N,M}\downarrow & & \downarrow q_{N,M} \\ \mathrm{Hist}_M^{\mathrm{enr}}(\mathrm{TPK}) & \xrightarrow{\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},M}} & \mathcal{D}^{\mathrm{enr}}_{\mathrm{rad},M} \end{array}$$

commutes; that is,

$$q_{N,M} \circ \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},N} = \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},M} \circ \pi_{N,M}. \tag{18.56}$$

Proof. Both sides preserve exactly the initial record and the first M arrows, together with their local and cumulative cocycles, affine factors, prefix holonomies, and ledger entries of $\widehat{\gamma}_N$. No reconstruction from the final product is involved: the complete sequence is part of the definition. □

18.17.6.2 Reader of the inverse limit

The compatibility equation (18.56) induces

$$\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},\infty} : \varprojlim_N \mathrm{Hist}_N^{\mathrm{enr}}(\mathrm{TPK}) \longrightarrow \varprojlim_N \mathcal{D}^{\mathrm{enr}}_{\mathrm{rad},N}. \tag{18.57}$$

Each finite component is obtained by truncation of the complete history. Therefore the reader does not invent an external continuity: it prolongs the finite certificates compatibly.

18.17.6.3 Naturality of the reader under the quadruple refinement sd_4

In the refinement sd_4 of the TPK, three micro-arrows are stationary and the fourth reproduces the observable transition. If

$$\rho_{\mathrm{rad}}(e_1) = \rho_{\mathrm{rad}}(e_2) = \rho_{\mathrm{rad}}(e_3) = I, \qquad \rho_{\mathrm{rad}}(e_4) = h_e,$$

then

$$H_{\mathrm{sd}_4(e)} = I\, I\, I\, h_e = h_e.$$

Naturality is thus certified for this refinement. For a general substitution $e \mapsto \widetilde{e}_k \cdots \widetilde{e}_1$, the exact condition is

$$h_e = h_{\widetilde{e}_k} \cdots h_{\widetilde{e}_1}.$$

This equality, rather than the similarity of the published curves, determines the compatibility of a new refinement.

18.17.6.4 Equivariance with respect to nonadic holonomy

Let $b_9(x)$ be the block of nine transitions that realizes Γ_9 from x, and

$$H_9(x) = H_{b_9(x)} = (\Lambda_9(x), C_9(x)).$$

The action induced on the radial ledger adds that block.

Theorem 18.29 (Cocycle Equivariance)**.** *One has*

$$\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}} \circ \Gamma_9 = \widehat{\Gamma}^{\mathrm{rad}}_9 \circ \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}}. \tag{18.58}$$

In components,

$$P \mapsto P + p(b_9), \qquad A \mapsto A + a(b_9),$$

$$\Lambda \mapsto \Lambda_9(x)\Lambda, \qquad C \mapsto C_9(x) + \Lambda_9(x)C,$$

$$g \mapsto g, \qquad q_{\mathrm{mem}} \mapsto q_{\mathrm{mem}} + 1.$$

Proof. The action of Γ_9 concatenates $b_9(x)$ to the history, where $p(b_9)$ and $a(b_9)$ are the sums of the local characters of its nine lifted arrows. The cocycle law adds the degrees, and the affine law equation (18.40) composes the holonomies. The TPK supplies the phase return and the memory increment. The definition of $\widehat{\Gamma}_9^{\mathrm{rad}}$ reproduces precisely these updates. □

This is not literal commutation of endomorphisms of the same space. The codomains are distinct and the memory changes. The correct term is *equivariance*.

18.17.6.5 Investment in the reversible sector

If $h_{e^\dagger} = h_e^{-1}$, the inversion diagram satisfies

$$\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}}(C_{\mathrm{rev}}\gamma) = \mathcal{J}_{\mathrm{rad}}\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}}(\gamma),$$

where $\mathcal{J}_{\mathrm{rad}}$ reverses the order of factors, replaces each factor by its inverse, and transports the ledger through the TPK involution. In the real output, equation (18.51) exchanges the two orientations.

18.17.7 Classification by the full radial signature and failure of fidelity of the Archimedean evaluation

18.17.7.1 Full radial signature of a history

For a finite history $\widehat{\gamma}_N = \widehat{e}_N \cdots \widehat{e}_1$, define

$$\mathfrak{I}_N(\widehat{\gamma}_N) = \left((x_0, H_0, \mathrm{Led}_0); \big(\widehat{e}_j, p(\widehat{e}_j), a(\widehat{e}_j), P_j, A_j, h_{e_j}, H_j, \mathrm{Led}_j\big)_{j=1}^{N}\right). \tag{18.59}$$

The signature is complete relative to the chosen enriched radial reader: it preserves the initial object, the typed sequence of arrows, the local and accumulated characters, all affine prefixes, and the complete ledger. It is not presented as a complete invariant of all HMT, nor does it replace the coordinates of the enriched state that do not lie in the reader's domain.

Definition 18.30 (Spiral Genealogical Equivalence)**.** Two histories are equivalent if there is a typed isomorphism of their radial data that transports the sequence of equation (18.59) term by term, respects sources, targets, composition, and ledgers, and commutes with all declared truncations.

The inverse limit yields the compatible family $(\mathfrak{I}_N)_{N\geq 0}$. Equality of curves in $\mathbb{R}^2$ is a coarser relation: the Archimedean evaluation factors through genealogical equivalence, but in general does not reflect it.

18.17.7.2 Kinematic classification through the increments (ω, β, μ)

Let ω be the angular increment, β the radial increment in the coordinate L_p, and μ the lifted-memory increment. The minimal visible table is:

Table 18.3: Kinematic classification of publications.

ω	β	μ	Published type	Interpretation
0	$\neq 0$	0	recta radial	scale variation without rotation or memory
$\neq 0$	0	0	circle	intrinsic radial angular closure
$\neq 0$	0	$\neq 0$	cylindrical helix	periodic phase with raised memory
$\neq 0$	$\neq 0$	0	espiral plana	radial variation without visible memory axis
$\neq 0$	$\neq 0$	$\neq 0$	radial helical suspension	turning, self-scaling and memory
0	0	$\neq 0$	fibra axial	memory advance without transverse figure

This table classifies the published kinematics, not the entire genealogy. Two rows with the same three increments may differ in p, carry, boundary, or affine class.

18.17.7.3 3 genealogical classes with a same Archimedean image

Example 18.31 (Circle with null memory)**.** Let $H_j = I$, $\mu = 0$, and $\theta_j = 2\pi j/9$ be given. The output is a circumference, and the ledger does not register axial lift.

Example 18.32 (The same circumference as shadow)**.** Let $H_j = I$, $\mu = 1$ and the same angular phase. En $\mathbb{R}^3$ is obtained the helix equation (8.17); to the aplicar π_{xy}, the output flat coincide with the of the example preceding.

Example 18.33 (Same spiral, different carry)**.** Let there be two histories with the same real characters p, β, ω but with different carry decompositions that radial evaluation maps to the same increment β. Both publish $r_p(\theta)$, while their ledgers differ. The enriched reader separates them; the curve does not.

18.17.7.4 Genealogical Classification Theorem

Theorem 18.34 (Spiral Genealogical Correspondences)**.** *Once the universal radial module, the local representation $e \mapsto h_e$, and the radial cocycles of the TRIT lift are fixed, every compatible TPK history determines:*

1. *one natural sequence of factors and prefix affine holonomies;*
2. *to class genealogical istable under truncamiento;*
3. *an equivariant action of the nonadic holonomy;*
4. *in the reversible sector, one conjugate history of inverse holonomy;*
5. *for each admissible homogeneous radial sector, a regular curve in $\mathbb{R}^2$ and, by conserving the depth, a suspension in $\mathbb{R}^3$.*

A history of variable degree instead determines an ordered atlas of piecewise charts. The application to curves is not faithful: it may identify histories with distinct ledgers.

Proof. Items 1 and 2 are the affine-holonomy and naturality theorems. Item 3 is equation (18.58); item 4 is inversion in the reversible subgroupoid. Item 5 is obtained by restricting the inverse limit to the domain $\mathcal{D}^{\mathrm{hom}}_{\mathrm{rad},\infty}$ and applying its radial and angular characters. Non-faithfulness follows from Figure 18.1 and from the general proposition of the first chapter. □

Hypotheses, domain, and scope of the theorem

The nonadic holonomy, the helix, the Pell screw, and the TPK cocycles are recovered results. The affine envelope, the prefix reader, its naturality, and the publication L_p constitute a new exact formalization in the declared domains. The local chart $p(\widehat{e}) = \tau_+(e)+\tau_-(e)$ and $a(\widehat{e}) = \tau_+(e)-\tau_-(e)$ is exact in the lifted category of 2 semicrowns; its use outside that domain requires exhibiting the functor that supplies both orientations. Publication as a single curve is restricted to the admissible homogeneous sector defined in this chapter.

18.17.7.5 Enriched readers and fiber memory

The difference between history and value admits an operational criterion. At a finite level H_k, let $q_k : H_k \to Z_k$ be a reader and let $M_k : H_k \to \mathcal{M}_k$ be the preserved register. The enriched reader is $\widehat{q}_k = (q_k, M_k)$. Memory distinguishes states in the same reader fibre; its sufficiency is checked on the domain considered. This formulation incorporates into the article itself the complete development of the fibre information used below.

Proposition 18.35 (Recoverability at a finite level)**.** *An inverse of $\widehat{q}_k$ exists on its image if and only if $\widehat{q}_k$ is injective. For a random variable Z taking values in H_k with distribution μ_k, we have*

$$\mathsf{H}(Z) = \mathsf{H}(q_k(Z)) + \mathsf{H}(Z \mid q_k(Z)).$$

If $\widehat{q}_k$ is injective on the support of Z, then

$$\mathsf{H}(Z \mid q_k(Z), M_k(Z)) = 0.$$

Here $\mathsf{H}(Z) = -\sum_z \Pr(Z = z) \log \Pr(Z = z)$, with $0 \log 0 = 0$, is finite entropy in nats; conditional entropy is the average entropy of the conditional distributions.

Proof. An inverse recovers a unique antecedent, which requires injectivity; conversely, injectivity allows each published pair to be assigned its unique antecedent. Since $q_k(Z)$ is a deterministic function of Z, grouping the sum defining $\mathsf{H}(Z)$ by fibres of q_k gives the chain rule. If the pair $(q_k(Z), M_k(Z))$ distinguishes the support, each corresponding conditional distribution is concentrated on a single state and its entropy is zero. □

Not publishing a coordinate and erasing the register that allows it to be recovered are distinct operations. This result does not identify entropy with the cardinality of the continuum or introduce a thermal interpretation: its domain, reader and distribution are those just fixed.

18.17.7.6 Equivariance of the measure under tree symmetries

Proposition 18.36 (Joint transport of the tree and its measure)**.** *Let $g = (g_k)_{k \geq 0}$ be a family of bijections $g_k : H_k \to H_k$ that commutes with truncations, fixes the root and bijectively transports the admissible emissions, preserving their multiplicities. Then*

$$d(g_k \zeta) = d(\zeta), \qquad (g_k)_* \mu_k = \mu_k, \qquad (g_\infty)_* \mu = \mu.$$

More generally, if μ_ζ is the law of continuations of a history $\zeta \in H_k$, we have

$$(g_\infty)_* \mu_\zeta = \mu_{g_k \zeta}.$$

Proof. The emission bijection preserves their number with multiplicities. The uniform seed distribution is also invariant. Each path and its image therefore receive the same product of local weights $1/d(\zeta)$. Equality of measures follows at each level and on each cylinder. Coherence with truncations defines g_∞, and uniqueness of the measure determined by cylinders proves equality in the limit. The same argument, starting at ζ, proves the assertion about continuations. □

This gives naturality of the tree and the measure: a coherent change of representation transports the weights and does not require choosing them anew. Preservation of emissions and multiplicities is an essential hypothesis of the proposition, not a datum added after the construction.

Chapter 19

Cardinal closure of the HMT analytic continuum: bidirectional reading of the Topological Phase Kernel and comparison with the continuum hypothesis

This chapter does not propose an alternative generator of the continuum. Its causal premise is the unique block already constructed

$$\begin{aligned}\text{APP} &\longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{unique enriched state}\\ &\longrightarrow \text{common state, tree and genealogical measure}\\ &\longrightarrow \text{correlative intrinsic operations}\\ &\longrightarrow \text{natural assembly, terminality and limit} \longrightarrow C_{\text{cont}}^{\text{disc}}.\end{aligned}$$

The five consubstantial constructions developed in the preceding chapter emerged correlatively over that same state before the real publication, without purporting to exhaust all its properties. What follows is a subsequent presentation of the quotient, its temporal memory, and its power; it does not retrospectively select any of those constructions.

19.1 The complete TRIT cell and the bidirectional temporal extension

The temporal sign does not separate from the state it orients. The local cell of departure is

$$c = (i, j; R^{+}, Q^{+}, R^{\times}, Q^{\times}; \tau), \qquad \tau \in \{-1, 0, +1\}, \tag{19.1}$$

where

$$R^{+} = \mathrm{dr}_9(i + j), \qquad Q^{+} = \frac{i + j - R^{+}}{9},$$
$$R^{\times} = \mathrm{dr}_9(ij), \qquad Q^{\times} = \frac{ij - R^{\times}}{9}.$$

The residues constitute the visible reading and the quotients preserve the memory necessary for inverting the transport. The temporal trit acts on this integer cell:

$$\begin{aligned}+1 &\longmapsto \text{return, retained memory and past,}\\ 0 &\longmapsto \text{evaluation threshold and present,}\\ -1 &\longmapsto \text{bidirectional opening and future.}\end{aligned} \tag{19.2}$$

Thus, two appearances of the same sign are not the same state if their quotients, sheet, carry, or genealogical record differ.

Let X be a compact chart of the surviving APP–TRIT–TPK limit and let $F : X \to X$ be its complete, continuous and surjective update. The surjectivity of the truncations and the surviving prolongation allow us to form the natural extension

$$X_F^{\leftrightarrow} := \left\{(x_k)_{k\in\mathbb{Z}} \in X^{\mathbb{Z}} : F(x_k) = x_{k+1} \text{ for every } k \in \mathbb{Z}\right\}. \tag{19.3}$$

The displacement

$$\sigma((x_k)_{k\in\mathbb{Z}}) = (x_{k+1})_{k\in\mathbb{Z}}$$

it no longer loses the inverse direction.

Theorem 19.1 (Natural extension two-way of the enriched state)**.** *The space $X_F^{\leftrightarrow}$ is compact and*

$$\sigma : X_F^{\leftrightarrow} \longrightarrow X_F^{\leftrightarrow}$$

is a homeomorphism, with

$$\sigma^{-1}((x_k)) = (x_{k-1}).$$

Therefore, outward and return maps are inverse continuous morphisms on the enriched object. The apparent discontinuity arises only if the coordinates that make the history invertible are first forgotten.

Proof. The product $X^{\mathbb{Z}}$ is compact. The condition $F(x_k) = x_{k+1}$ is closed becau F is continuous; by tanto $X_F^{\leftrightarrow}$ is compact. The displazamiento and the displazamiento inver are continuous in the topology product and priservan the equations of compatibility. Their compositions are the identity. □

The material inversion of the route resides in the groupoid completion of the transport. Let $C_M = -\,\mathrm{rev}$ be memory conjugation. Its contravariant action is expressed by the reversal square

$$F(x_k) = x_{k+1} \quad \Longrightarrow \quad F(C_M x_{k+1}) = C_M x_k.$$

On the natural extension it is defined

$$(\Theta u)_k := C_M(x_{-k}), \qquad u = (x_k)_{k\in\mathbb{Z}}. \tag{19.4}$$

The reversion identity guarantees that Θu again satisfies the compatibility equations. Therefore,

$$\Theta^2 = I, \qquad \Theta\sigma\Theta = \sigma^{-1}. \tag{19.5}$$

This identity does not identify the ternary group with the binary sign: the TRIT orients return, threshold, and opening, whereas the involution C_M implements the conjugation of the two temporal orientations.

For $n \geq 0$, let us denote by $S_n^{\leftrightarrow}$ the finite set of admissible temporal windows $(x_{-n}, \ldots, x_0, \ldots, x_n)$, with each state further truncated to genealogical depth n. The simultaneous restrictions of time and depth form an inverse system and

$$\mathcal{O}_{\leftrightarrow}^{\mathrm{HMT}} := \varprojlim_n S_n^{\leftrightarrow} \cong X_F^{\leftrightarrow}. \tag{19.6}$$

Each history has temporal support of type $\omega^* + 1 + \omega$, but also carries the cell (19.1), the route and the genealogical register. The map sending an orbit to all its centered windows is a homeomorphism and intertwines σ with the genealogical succession. This is the genealogical biordinal, not the von Neumann ordinal obtained after forgetting the labels.

The preceding object is a compact chart of the fractional part. The global object publishing the entire line is the countable atlas

$$\mathcal{O}_{\mathbb{R},\leftrightarrow}^{\mathrm{HMT}} := \mathbb{Z} \times \mathcal{O}_{\leftrightarrow}^{\mathrm{HMT}}. \tag{19.7}$$

The coordinate $\mathbb{Z}$ merely translates the chart; it does not erase the bidirectional genealogical ledger.

19.2 Stable Mode of the Unitary Calendar under Modular Memory Update

The temporal structure does not rest solely on a metaphor of return. On $\mathcal{H}_{\mathrm{clk}} = \mathbb{C}^{108}$, let $\zeta = \exp(2\pi i/108)$,

$$U_{\mathrm{clk}}|j\rangle = |j+1\rangle, \qquad |\widetilde{k}\rangle = \frac{1}{\sqrt{108}} \sum_{j=0}^{107} \zeta^{jk}|j\rangle,$$

and let us define

$$H_{\mathrm{clk}} = \hbar_{\mathrm{int}}\omega_0 \sum_{k=0}^{107} k\, |\widetilde{k}\rangle\langle\widetilde{k}|, \qquad \omega_0 = \frac{2\pi}{108 t_0}. \tag{19.8}$$

Then

$$U_{\mathrm{step}} = \exp\left(-\frac{i}{\hbar_{\mathrm{int}}} H_{\mathrm{clk}} t_0\right) = U_{\mathrm{clk}}, \qquad U_{\mathrm{step}}^{108} = I, \tag{19.9}$$

and, for $Z_{\mathrm{clk}}|j\rangle = \zeta^j |j\rangle$,

$$|\psi_n\rangle := U_{\mathrm{step}}^n |j_0\rangle,$$
$$\langle Z_{\mathrm{clk}}\rangle_n := \langle \psi_n | Z_{\mathrm{clk}} | \psi_n \rangle = \zeta^{j_0+n} \tag{19.10}$$

has primitive period 108.

In the modular boundary sector $\mathcal{H}_A = (\mathbb{C}^3)^{\otimes 36}$, let $\rho_A > 0$ be precisely the faithful state characterized by the following decomposition, and let Δ_4^{mod} be the declared modular jump. The generator preserving the provenance of the two boundary channels is

$$K_{\mathrm{HHM}}^{(A)} = -\log \rho_A = \kappa_A I + \Delta_4^{\mathrm{mod}}(90N_{90} + 120N_{120}). \tag{19.11}$$

If

$$N = N_{90} + N_{120}, \qquad D = 90N_{90} + 120N_{120},$$

then

$$N_{90} = 4N - \frac{D}{30}, \qquad N_{120} = \frac{D}{30} - 3N. \tag{19.12}$$

Thus, the total observable and the modular Hamiltonian jointly reconstruct the two provenance memories. On the other hand, the cylindrical Perron dynamics preserves the mode

$$V_n = \varphi^n, \qquad M_n = \varphi^{-2n} = V_n^{-2}, \qquad M_n = \left(\frac{a_n}{a_0}\right)^{-6} \qquad (D = 3). \tag{19.13}$$

Theorem 19.2 (Exact finite composition of the CT108 calendar and modular memory 90/120)**.** *The operators* (19.8) *and* (19.11) *are self-adjoint on their respective spaces. After an energy scale $E_\partial > 0$ is fixed, the product operator*

$$H_{\mathrm{tw}} := H_{\mathrm{clk}} \otimes I + I \otimes E_\partial (K_{\mathrm{HHM}}^{(A)} - \kappa_A I) \tag{19.14}$$

is self-adjoint, and its two summands strongly commute. For states whose clock marginal is $|j_0\rangle\langle j_0|$, the observable $Z_{\mathrm{clk}} \otimes I$ preserves its primitive response of period 108*. Simultaneously, $K_{\mathrm{HHM}}^{(A)}$, N_{90}, and N_{120} are conserved observables of the modular factor. The pair (N, D), with $N = N_{90} + N_{120}$, reconstructs (N_{90}, N_{120}) exactly by means of* (19.12)*.*

Proof. The first operator admits a finite real spectral decomposition, and the second is the logarithm of a faithful state; both are self-adjoint. They act on distinct tensor factors and, therefore, their spectral resolutions commute. The exponential of their sum factorizes. The primitivity of the response follows from the primitivity of ζ; channel reconstruction is the inverse of the matrix $\left(\begin{smallmatrix} 1 & 1 \\ 90 & 120 \end{smallmatrix}\right)$, de determinante 30. □

Corollary 19.3 (Stable mode of the cylindrical memory)**.** *In the Perron sector of closed paths, $M_n = \varphi^{-2n} = V_n^{-2}$ is the exact and stable spectral mode given by* (19.13)*.*

The CT108 calendar, the modular memory and the Perron mode are three readeres tipados of the same genealogical antecedent; are not the same operator. together with the natural extension and the espejo, form the internal temporal record

$$\mathcal{T}_{\mathrm{HMT}} := \left(X_F^{\leftrightarrow}, \sigma, \Theta; H_{\mathrm{clk}}, K_{\mathrm{HHM}}^{(A)}; M_\bullet\right).$$

This is the exact content of the HMT temporal closure: two inverse continuous directions, a primitive clock, recoverable memory, and a stable mode. The complete TPK state is not replaced by the cursor 108. An additional identification of the three readers with a single perturbatively stable physical dynamics

would require a common intertwiner; that identification is not used in the genealogical closure or in the reconstruction of the continuum.

19.3 Analytic realizations of HMT histories via the Archimedean and bidirectional charts

Let H be a Polish space of histories: a closed inverse limit of finite levels or a countable atlas of such limits. The topology is generated by prefix cylinders. This choice preserves the genealogy and permits the descriptive operations that are admitted to be typed exactly.

Definition 19.4 (Analytic HMT Publication)**.** An analytic publication is a set $A \subseteq \mathbb{R}$ for which there exists a Borel set

$$C \subseteq H \times \mathbb{N}^{\mathbb{N}}$$

and to Borel reader $F : H \to \mathbb{R}$ such that

$$A = \{F(x) : \exists z \in \mathbb{N}^{\mathbb{N}},\ (x, z) \in C\}.$$

The witness z may encode to route, to prolongation, to memory or to certification countable. The readers continuous are cases particular.

This class contains cylinders, Borel predicates on histories, countable operations, images under continuous readers, and publications asserting the existence of a real witness. By definition, it does not contain the full power set $\mathcal{P}(\mathbb{R})$: uncountable unions, arbitrary selectors, nonsmooth quotients, and recursion of length ω_1 require additional data.

Theorem 19.5 (Perfect Duality of HMT Publications)**.** *Every analytical HMT publication $A \subseteq \mathbb{R}$ satisfies exactly one of the two alternatives:*

$$|A| \leq \aleph_0 \qquad or \qquad |A| = 2^{\aleph_0}.$$

In the second alternative there exists a continuous injection.

$$j : 2^{\mathbb{N}} \hookrightarrow A.$$

therefore, every uncountable analytic publication contains one copia of the Cantor space and, in particular, a perfect subset.

Proof. The existential projection of a Borel subset of a product of Polish spaces is analytic, and a Borel image of a standard Borel space is again analytic. Therefore, A is an analytic subset of $\mathbb{R}$.

To make the uncountable alternative explicit, represent A as the projection of a closed set

$$D \subseteq \mathbb{R} \times \mathbb{N}^{\mathbb{N}}.$$

For each finite word $s \in \mathbb{N}^{<\mathbb{N}}$, let

$$A_s := \{x : \exists z \supseteq s,\ (x, z) \in D\}.$$

If $A_s \cap K$ is uncountable for a certain closed interval K, the set of its points that are not condensation points is countable, because $\mathbb{R}$ has a countable basis. Two distinct condensation points and two disjoint closed intervals within K are chosen, and for each interval a strict prolongation of s whose projection remains uncountable is chosen. The two prolongations need not be incompatible: separation of the branches is guaranteed by the intervals. The identity

$$A_s = \bigcup_{n \in \mathbb{N}} A_{s^\frown n}$$

guarantees that, in each chosen neighborhood, some extension preserves the non countability.

Repitiendo the procedure are obtained, for each $u \in 2^{<\mathbb{N}}$, a interval K_u and a prefix s_u such that

$$K_{u0} \cap K_{u1} = \varnothing, \qquad K_{ui} \subseteq K_u, \qquad \operatorname{diam} K_u \leq 2^{-|u|},$$

and s_{ui} strictly prolongs to s_u. For $\beta \in 2^{\mathbb{N}}$, the intervals $K_{\beta|n}$ determine a unique point $j(\beta)$, and the prefixes $s_{\beta|n}$ determine a witness z_β. The closure of D implies

$$(j(\beta), z_\beta) \in D,$$

so that $j(\beta) \in A$. The disjoint sibling intervals prove injectivity, and diameter control proves the continuity of j.

Thus,

$$2^{\aleph_0} \leq |A| \leq |\mathbb{R}| = 2^{\aleph_0},$$

which closes the second alternative. If the process does not start from an uncountable set, the first is obtained. □

Corollary 19.6 (Analytic HMT Continuum Hypothesis)**.** *Let*

$$\aleph_{1,\mathrm{HMT}}^{\mathrm{an}} := \min \{|A| : A \text{ is an uncountable analytic HMT publication}\}. \tag{19.15}$$

This symbol denotes the initial cardinal relative to the class of HMT analytic publications; it does not denote the extensional ordinal ω_1. Then

$$\boxed{\aleph_{1,\mathrm{HMT}}^{\mathrm{an}} = 2^{\aleph_0} = |\mathbb{R}| = \left|\mathcal{O}_{\mathbb{R},\leftrightarrow}^{\mathrm{HMT}}/{\sim_{\mathfrak{R}}}\right|.} \tag{19.16}$$

That is, there is no intermediate cardinal between the countable and the continuum within the analytic publications produced by HMT readers.

Proof. The set of finite depth windows

$$\mathcal{O}_{<\omega}^{\mathrm{HMT}} := \bigcup_{n<\omega} S_n^{\leftrightarrow}$$

is countable, since it is a countable union of finite sets. The complete layer surjects onto $\mathbb{R}$ by (18.36) and is contained in a countable product of finite alphabets; it therefore has cardinality $2^{\aleph_0}$. The theorem 19.5 excludes any strictly intermediate analytic size. □

Corollary 19.7 (Dichotomy of the fibers)**.** *Let $F : H \to Y$ to reader Borel between spaces Polish and let $y \in Y$. If the domain of histories admitted is analytic, the fiber*

$$\{x : F(x) = y\}$$

is countable or has cardinality $2^{\aleph_0}$.

Proof. The fiber is the intersection of the analytic domain with the Borel preimage of the singleton $\{y\}$; it is therefore analytic. We apply the theorem 19.5. □

19.4 Comparison between the genealogical cardinal closure and its extensional ordinal reduction

Let $\mathrm{WO} \subseteq 2^{\mathbb{N}}$ be the set of real codes of countable well-orders, allowing the field to be a subset of $\mathbb{N}$. The map

$$\mathrm{ot} : \mathrm{WO} \longrightarrow \omega_1$$

assigns to each code its order type and is surjective.

Proposition 19.8 (Ordinal injection into the line)**.** *In ZFC there exists an injection*

$$\iota : \omega_1 \hookrightarrow \mathbb{R}.$$

A binary subsystem of the history space HMT can be raised.

Proof. By choice, the surjection ot admits a section $s : \omega_1 \to \mathrm{WO}$. Equivalently, a well-order of WO is fixed and the first code in each fiber is taken. The Cantor map

$$c(a) := \sum_{n \geq 0} \frac{2a(n)}{3^{n+1}}$$

is injective from $2^{\mathbb{N}}$ into $[0, 1]$. Therefore

$$\iota := c \circ s$$

is the injection sought. The embedding of the binary tape in the cinta ternary and the Hensel inverse Ψ^{-1} proporcionan the lift to the genealogical space. □

The choice of s is not natural with respect to refinements and is not produced by a finite prefix. This proposition shows that the arrow $\omega_1 \hookrightarrow \mathbb{R}$ is not the content of the classical hypothesis of the continuum. The inverse direction belongs to the bare extensional ordinal. In HMT, by contrast, the natural arrow already constructed is the homeomorphism (18.36), whose domain preserves the two temporal directions.

19.5 Cardinal identity of the HMT analytic continuum: $\aleph_{1,\mathrm{HMT}}^{\mathrm{an}} = 2^{\aleph_0}$ and comparison with the classical continuum hypothesis

Let

$$Q_{\mathrm{HMT}} := X_{\mathrm{raw}}/\!\sim_P .$$

By the theorem 18.9, $Q_{\mathrm{HMT}} \cong \mathbb{R}$.

Theorem 19.9 (Ordinal equivalences of the HMT quotient)**.** *In ZFC the following are equivalent:*

1. $2^{\aleph_0} = \aleph_1$*;*
2. *there exists a injection* $j : Q_{\mathrm{HMT}} \hookrightarrow \omega_1$*;*
3. *there exists to map* $\rho : X_{\mathrm{raw}} \to \omega_1$ *such that*

$$\rho(x) = \rho(y) \quad \Longleftrightarrow \quad P(x) = P(y);$$

4. Q_{HMT} *admits an uncountable well-order whose proper initial segments are all countable.*

Proof. If the first assertion holds, a bijection $Q_{\mathrm{HMT}} \cong \mathbb{R} \cong \omega_1$ provides the second and transports the order of ω_1, which yields the fourth. The second yields the third by taking $\rho = j \circ [P]$, where $[P]$ sends each history to its evaluation class. Conversely, the biconditional condition in the third causes ρ to factor through an injection $Q_{\mathrm{HMT}} \hookrightarrow \omega_1$.

The second assertion implies $2^{\aleph_0} \leq \aleph_1$. The proposition 19.8 gives $\aleph_1 \leq 2^{\aleph_0}$; Cantor–Bernstein produces the equality. Finally, a well-order such as that of the fourth assertion cannot contain an element in position ω_1, because its initial segment would be uncountable. Its order type is therefore at most ω_1, and, since the domain is uncountable, it is exactly ω_1. This recovers the second assertion. □

The theorem localizes the target without introducing it as a premise. A function ρ with that property would not be a small supplement to the analytic dichotomy: it would be a complete formulation of the continuum hypothesis itself.

19.6 Loss of ancestral memory under reduction to the ordinal codomain

The forgetful functor that removes trit, mirror, quotients, route, and memory sends each biordinal to a bare ordered support. The following results control that forgetful functor; they do not deny the continuous HMT morphism already constructed between the two-way quotient and $\mathbb{R}$.

Theorem 19.10 (Limit of the readers toward ω_1 with the order topology)**.** *There does not exist a continuous application*

$$\rho : X_{\text{raw}} \longrightarrow \omega_1$$

that exactly separates the fibers of P. In particular, after applying the forgetful functor, no cylindrical reader toward the bare topological ordinal can demonstrate the second condition of theorem 19.9.

Proof. Every chart $\{m\} \times \mathbb{Z}_3^6$ is compact. The continuous image of a compact space in ω_1, endowed with the order topology, is compact and therefore bounded by some countable ordinal. Its image is countable. The union over $m \in \mathbb{Z}$ of those images is a countable union of countable sets and remains countable. A function separating all fibers of P would instead have an image of cardinality $|\mathbb{R}|$, a contradiction. □

Theorem 19.11 (Absence of good analytic extensional order)**.** *There is no to good order analytic, and by tanto nor Borel, of* $\mathbb{R}$*.*

Proof. Suppose that $\prec \subseteq \mathbb{R}^2$ is analytic and defines a strict total linear order. Then

$$(x, y) \notin \prec \quad \Longleftrightarrow \quad x = y \text{ or } y \prec x.$$

The complement of $\prec$ is also analytic; by Suslin separation, $\prec$ is Borel. Every well-founded Borel relation on a Polish space has rank bounded by a countable ordinal. In a linear well-order, distinct elements have distinct ranks, so the domain would be countable. This contradicts the uncountability of $\mathbb{R}$. □

The two controls are independent of whether the classical continuum hypothesis is true or false. If a set-theoretic bijection $\omega_1 \leftrightarrow \mathbb{R}$ exists, it cannot be continuous for the order topology of ω_1 or analytic as a well-order. This does not make the HMT morphism discontinuous: it shows that bare ω_1 is another codomain and that forgetting the temporal direction destroys precisely the structure that ensured continuity.

19.7 Ordinal closure of the HMT genealogical codomain

The preceding checks describe the functor that forgets genealogy. HMT positively constructs its ordinal closure while preserving it. Let $h \in 2^{\mathbb{N}}$ be the generated code of the language, finite tables, transitions and APP–TRIT–TPK readers. The canonical genealogical codomain is

$$M_h := L[h].$$

This envelope preserves the generator as a parameter, admits the canonical well-order of the constructible hierarchy and is the codomain produced by the construction of the continuum in (2).

Theorem 19.12 (Classical continuum hypothesis in the HMT genealogical codomain)**.** *The model* $M_h = L[h]$ *satisfies*

$$M_h \models 2^{\aleph_0} = \aleph_1.$$

There exists in M_h *a definable bijection*

$$b_h : \omega_1^{M_h} \xrightarrow{\sim} \mathbb{R}^{M_h}.$$

Since M_h *is the generated codomain,* $\mathbb{R}^{M_h}$ *is the entirety of the line published by the HMT construction. Let* P_h *be the HMT evaluation of the histories belonging to* M_h*. Then*

$$\rho_h := b_h^{-1} \circ P_h$$

satisfies

$$\rho_h(x) = \rho_h(y) \quad \Longleftrightarrow \quad P_h(x) = P_h(y).$$

Proof. Every real number in $L[h]$ appears at some level $L_\alpha[h]$ with $\alpha < \omega_1^{L[h]}$. For each ordinal $\alpha < \omega_1^{L[h]}$, the level $L_\alpha[h]$ is countable within the model. Therefore,

$$|\mathbb{R}^{L[h]}| \leq \aleph_1^{L[h]}.$$

The model line is uncountable, so minimality of $\aleph_1$ gives the reverse inequality and hence equality. The canonical well-ordering of $L[h]$ selects, among all the bijections whose existence has just been proved, a first definable bijection b_h.

The evaluation P_h is definable from h, so that its value belongs to $\mathbb{R}^{M_h}$. The formula of ρ_h is well typed. Since b_h is bijective, two histories receive the same ordinal if and only if they publish the same real. □

Remark 19.13 (Identification of the generated codomain). theorem 19.12 effectively constructs the ordinal arrow in the genealogically generated universe. The equality $\mathbb{R} = \mathbb{R}^{L[h]}$ is not postulated: within the theorem's domain it expresses that $L[h]$ is precisely the codomain produced by the HMT continuum, whose Archimedean coverage is proved in proposition 18.5 y theorem 18.9. Consequently, ρ_h reaches all the reals of the generated line and the bijection b_h proves the classical continuum hypothesis with the ordinary cardinals of M_h. A forcing extension belongs to another domain after forgetting or replacing the genealogical selector h; it describes the independence of the bare language of ZFC and does not alter the conclusion of the HMT system.

19.8 Compatibility among HMT cardinal closure, the set-theoretic formulation of the continuum, and bidirectional genealogical memory

The preceding results articulate five strata of the same closure:

1. the genealogical quotient reconstructs $\mathbb{R}$ and has cardinality $2^{\aleph_0}$;
2. the natural extension two-way and the genealogical biordinal are homeomorphic, with outgoing and return continuous;
3. the clock Hamiltonian, the modular Hamiltonian, and the stable mode preserve calendar, provenance, and memory without reducing the TPK to the visible cursor;
4. every analytic HMT publication is countable or has the cardinal of the continuum, so that $\aleph_{1,\mathrm{HMT}}^{\mathrm{an}} = 2^{\aleph_0}$;
5. the constructible codomain $L[h]$ has a complete ordinal enumeration of all generated reals and satisfies the classical continuum hypothesis with its ordinary cardinals.

The first three construct the HMT object and its dynamics; the fourth closes the cardinality in the domain of publications; the fifth supplies its ordinal coordinate without changing the object. The identity $\aleph_{1,\mathrm{HMT}}^{\mathrm{an}} = 2^{\aleph_0}$, together with the bijection b_h, constitutes the HMT proof of the extensional statement about the generated line. The selector h preserves the genealogy that the bare language of ZFC omits.

Proposition 19.14 (Falsifiers)**.** *Each assertion has an independent negative control:*

1. *a real without preimage under P would falsify the reconstruction of the quotient;*
2. *an intermediate state not recoverable from the complete returns would falsify the factorization* (17.11)*;*
3. *the loss of the primitive period, of the reconstruction* $90/120$ *or of the mode* φ^{-2n} *would falsify one of the three readers of the typed HMT temporal record;*
4. *to publication analytic not countable without copia of Cantor would falsify the dichotomy perfect;*
5. *a Borel well-order of an uncountable Polish space would refute the analytic no-go theorem;*
6. *one unbounded continuous image of one compact chart in* ω_1 *falsaria the topological not-go;*
7. *the existence, in the generated codomain* M_h*, of a real without an ordinal preimage under* b_h*, or of a branch published outside* $L[h]$*, would falsify the genealogical identification and the classical closure of the continuum. An extension obtained after changing the selector h does not belong to the fixed domain and does not constitute that falsifier.*

The structural conclusion is precise. HMT constructs a bidirectional temporal object, publishes it Archimedeanly as $\mathbb{R}$, preserves exceptional incidence in parallel, and proves that its first uncountable analytic layer has cardinality exactly $2^{\aleph_0}$. The bijection

$$\mathcal{O}_{\mathbb{R},\leftrightarrow}^{\mathrm{HMT}}/{\sim_{\mathfrak{R}}} \xrightarrow{\overline{\mathfrak{R}}} \mathbb{R}$$

is natural and continuous. The classical ordinal $\omega_1^{M_h}$ is the well-order coordinate of all reals in the generated codomain. The functor that forgets h, time, mirror and memory explains the subsequent independence in bare ZFC; it does not modify the constructed bijection or the classical closure proved in M_h.

19.9 Bidirectional extension groupoid and operator algebra

Let $X_F^{\leftrightarrow}$ be the natural extension of the enriched state and σ its shift homeomorphism. Define the transformation groupoid

$$\mathcal{G}_{\mathrm{TPK}} := X_F^{\leftrightarrow} \rtimes_{\sigma} \mathbb{Z}, \tag{19.17}$$

with

$$s(x,n) = x, \quad r(x,n) = \sigma^n x, \quad (\sigma^n x, m)(x,n) = (x, m+n), \quad (x,n)^{-1} = (\sigma^n x, -n).$$

Theorem 19.15 (Groupoidal Completion of Bidirectional Transport)**.** *The groupoid $\mathcal{G}_{\mathrm{TPK}}$ is locally compact, Hausdorff, second countable, etale, and amenable. For the invariant and atomless projective measure μ_{HMT}, if the shift is ergodic and essentially free,*

$$\mathcal{M}_{\mathrm{TPK}} := L^{\infty}(X_F^{\leftrightarrow}, \mu_{\mathrm{HMT}}) \rtimes_{\sigma} \mathbb{Z}$$

is the hyperfinite factor of type II_1.

Proof. The action of a countable discrete group by homeomorphisms on a compact metrizable space produces an etale, locally compact, Hausdorff, and second-countable groupoid. Amenability of $\mathbb{Z}$ implies amenability of the groupoid. The projective measure is shift-invariant. Ergodicity and essential freeness make the crossed product a factor; nonatomicity excludes the matrix type-I case. The trace induced by μ_{HMT} is finite and faithful, and amenability yields hyperfiniteness. Therefore, the factor is of type II_1. □

The type III modular classification belongs to a distinct quasi-invariant measure, with a declared Radon–Nikodym cocycle. It is not deduced from the mere presence of KMS dynamics and does not reopen the prior groupoidal completion.

19.10 Proof of $2^{\aleph_0} = \aleph_1$ in the HMT continuum

This section brings together, in a contiguous proof, extension, double projection, and arbitrary depth; it then constructs the semantic closure, the uniform lift, the genealogical machinery, and the cardinal closure. The set-theoretic hierarchy is identified only at the end of the construction and cannot be read as its causal origin. The causal dependence is:

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow X_{\infty}^{\mathrm{enr}} \longrightarrow C_{\mathrm{cont}}^{\mathrm{disc}} \longrightarrow m_{\infty} \longrightarrow \mathfrak{G}_{\mathrm{HMT}}(h, m_{\infty}). \tag{32}$$

First, the state, its extensions, the continuum, and the genealogical ledger are constructed. Then **that produced genealogy** is extended to all its hereditary objects by an explicit semantic rule. The subsequent identification with a known set-theoretic hierarchy selects neither the seeds, the operators, the genealogical ledger, the line, nor its projections. It does, however, permit the exact logical-cardinal strength of the adopted semantic rule to be assessed with complete transparency.

19.10.1 Extension of states and transfinite semantic extension

Let

$$\Sigma_{\mathrm{HMT}} = \left(\dot{R}_{\mathrm{APP}}, \dot{R}_{\mathrm{TRIT}}, \dot{R}_{\mathrm{TPK}}, \dot{R}_U, \dot{R}_{\Gamma_9}, \dot{R}_{\mathrm{Ext}}, \dot{R}_{\rho}, \dot{R}_{\pi}, \dot{R}_{\varphi}, \dot{R}_e, \dot{R}_{\alpha}, \dot{R}_{\mathrm{ar}}, \dot{R}_{\mathrm{exc}}\right) \tag{33}$$

the countable **relational** signature whose effective code is h. Each $\dot{R}_S$ is the graph of the operator S on the codes where it is defined; it is not assumed that an external function is total within each level. The graphs $\dot{R}_{\pi}, \dot{R}_{\varphi}, \dot{R}_e, \dot{R}_{\alpha}$ are the output readers constructed by coinductive generation and dodecaphase closure; they do not receive the conventional values of the constants as arguments. Their presence in

Σ_{HMT} ensures that closure preserves the complete state: it simultaneously integrates the routes publishing value and the incidence prolonged toward the exceptional realization.

At each horizon, the state code records the APP–TRIT–TPK coordinates, the route, sheet, carry, boundary and memory; in its publication stages it also records P_N, Φ_N, E_N, A_N, where A_N is the prefix of α_{HMT} produced by the same state. These coordinates do not select the branch from outside: they are part of the history being encoded. Given a primitive-recursive numbering of the finite coordinates of a state, every inverse history $x_\infty = (x_N)_{N\geq 1} \in X_\infty^{\mathrm{enr}}$ determines the code

$$\mathrm{code}(x_\infty) := \{\langle N, j, c\rangle : (x_N)_j \text{ has code } c\} \subseteq \omega.$$

The coherence equations $\rho_{N+1,N}(x_{N+1}) = x_N$ make it possible to decode the history uniquely from that real. We write m_∞ for the code of the realized history and $m_\infty(n)$ for its characteristic function under the fixed numbering. The operator code h and this complete ledger are combined, by means of a primitive-recursive pairing function, into the real

$$u_{h,m} := \{\langle 0, \langle n, h(n)\rangle\rangle : n < \omega\} \cup \{\langle 1, \langle n, m_\infty(n)\rangle\rangle : n < \omega\}. \tag{33a}$$

The two primitive projections recover h and m_∞ from $u_{h,m}$.

The first extension is the metric and operator extension of the finite states. For $N \geq 1$, let $\widehat{\mathcal{X}}_{6N}^{\mathrm{enr,raw}}$ be the set of raw enriched states of length $6N$, and define the surviving subsystem set-theoretically by

$$\begin{aligned} X_N &:= \widehat{\mathcal{X}}_{6N}^{\mathrm{enr,surv}} \\ &:= \Big\{ x \in \widehat{\mathcal{X}}_{6N}^{\mathrm{enr,raw}} : \\ &\qquad \forall M > N\ \exists z \in \widehat{\mathcal{X}}_{6M}^{\mathrm{enr,raw}} \text{ such that} \\ &\qquad \mathrm{tr}_{6M,6N}(z) = x \ \wedge\ z \text{ satisfies every intermediate TPK transition} \Big\}. \end{aligned}$$

and set

$$\rho_{N+1,N} := \mathrm{tr}_{6N+6,6N} \restriction X_{N+1},$$

and

$$\mathrm{Ext}_N(x) := \{y \in X_{N+1} : (x, y) \in \mathrm{Ext}_{6N+6,6N}\}.$$

The root level is included without any implicit exception:

$$X_0 := \{*\}, \qquad \rho_{1,0} : X_1 \to X_0 \text{ is constant}, \qquad \mathrm{Ext}_0(*) := X_1.$$

The superscript surv denotes the pruned tree of states possessing compatible TPK descendants at every finite horizon. The TPK extension law established in the preceding sections proves that the emitted seeds belong to this tree; finite branching and Kőnig's lemma turn compatibility at every horizon into an infinite branch. In particular,

$$\forall N \geq 0\ \forall x \in X_N \quad \varnothing \neq \mathrm{Ext}_N(x) \subseteq \rho_{N+1,N}^{-1}(x). \tag{34}$$

By the same finitely branching tree lemma, every surviving seed has a compatible extension

$$x_\infty = (x_N)_{N\geq 1} \in X_\infty^{\mathrm{enr}} := \varprojlim_N (X_N, \rho_{N+1,N}). \tag{35}$$

The second extension acts **afterward** on the complete code of that realized history. It is not inferred from the mere existence of the finite operator Ext_N: at the set-theoretic scale, it formalizes the HMT

principle that only those objects possessing a genealogical derivation from the state and its rules belong to the full domain. The transfinite semantic extension is:

$$\begin{aligned}
\mathfrak{G}_0(h, m_\infty) &:= \mathrm{TC}\big(\{\omega, u_{h,m}, h, m_\infty\}\big),\\
\mathfrak{G}_{\beta+1}(h, m_\infty) &:= \mathrm{Def}_{\Sigma_{\mathrm{HMT}}}\big(\mathfrak{G}_\beta(h, m_\infty)\big),\\
\mathfrak{G}_\lambda(h, m_\infty) &:= \bigcup_{\beta<\lambda} \mathfrak{G}_\beta(h, m_\infty) \quad (\lambda \text{ limit}),\\
\mathfrak{G}_{\mathrm{HMT}}(h, m_\infty) &:= \bigcup_{\beta\in\mathrm{Ord}} \mathfrak{G}_\beta(h, m_\infty).
\end{aligned} \tag{36}$$

The Continuum Hypothesis and a target bijection are not introduced among the conditions defining the successor operator; the operator is given prospectively by

$$\mathrm{Def}_{\Sigma_{\mathrm{HMT}}}(M) := M \cup \Big\{\{x \in M : \langle M, \in, (\dot R_S \cap M^{k_S})_{S\in\Sigma_{\mathrm{HMT}}}\rangle \models \varphi(x, \bar p)\} : \ulcorner\varphi\urcorner \in \omega,\ \bar p \in M^{<\omega}\Big\}. \tag{37}$$

That is: it simultaneously publishes all subsets definable by a finite formula of the HMT relational language and parameters that the genealogy has already produced. It does not consult a target real value, a conventional constant, a target cardinal or an external model. Formula (37) defines transfinite closure on the previously constructed seeds, operators, enriched state, register and double projection.

Lemma 19.16 (Joint coding of the HMT program and the realized ledger)**.** *The pair formed by the operator code h and the ledger m_∞ is encoded by a single real $u_{h,m} \subseteq \omega$, and every formula of Σ_{HMT} is uniformly translated into a formula of the language $\{\in, u_{h,m}\}$.*

Proof. Both h and m_∞ are sequences of natural numbers: h encodes the signature, finite tables and operator programs; m_∞ encodes a single realized history, not the complete space of histories. Formula (33a) and its two decoders prove the first assertion. Each symbol of (33) denotes its **graph relation encoded in** h, never the external graph of all its possible evaluations. APP, TRIT, U_t, Γ_9, the restrictions ρ and the finite extensions have primitive-recursive graphs on integer codes and therefore Δ_0 formulas absolute between transitive levels. The two publications are expressed as graph relations by the formulas

$$\begin{aligned}
\dot R_{\mathrm{ar}}(x, y) &\iff \forall k \in \omega\ \exists N\ \forall n \geq N \quad |\mu_n(x) - y| < 2^{-k},\\
\dot R_{\mathrm{exc}}(x, c, z) &\iff \forall N \in \omega \quad z \restriction N = E_N(x \restriction N, c \restriction N),
\end{aligned} \tag{37c$_0$}$$

where μ_n and E_N are the rational/finite functions whose codes occur in h. Every atomic formula containing an HMT symbol is inductively replaced by its graph formula with parameter $u_{h,m}$. Connectives and quantifiers are preserved. This induction on syntactic complexity produces a translation $\varphi \mapsto \varphi^*$ such that, for every transitive level M of (36) containing the parameters,

$$\langle M, \in, (\dot R_S \cap M^{k_S})_S\rangle \models \varphi(\bar a) \quad\iff\quad M \models \varphi^*(\bar a; u_{h,m}). \tag{37c}$$

Thus, (37) cannot consult an oracle, a target decimal table, or an uncoded relation that introduces uncountably many objects in a single step. □

HMT semantics is now defined **independently of identifying it with** $\mathfrak{G}$. Let $\mathcal{T}_0^{\mathrm{HMT}}$ be the set of canonical names of the elements of $\mathfrak{G}_0(h, m_\infty) = \mathrm{TC}(\{\omega, u_{h,m}, h, m_\infty\})$. In particular, the initial level contains the realized ledger m_∞, not every history in X_∞^{enr} as primitive parameters. The admissible terms are the well-founded trees obtained by

$$\mathcal{T}^{\mathrm{HMT}}_{\beta+1} := \mathcal{T}^{\mathrm{HMT}}_{\beta} \cup \{\langle \beta, \ulcorner \varphi \urcorner, t_0, \dots, t_{k-1}\rangle : t_i \in \mathcal{T}^{\mathrm{HMT}}_{\beta}\},$$
$$\mathcal{T}^{\mathrm{HMT}}_{\lambda} := \bigcup_{\beta<\lambda} \mathcal{T}^{\mathrm{HMT}}_{\beta}, \qquad \mathcal{T}^{\mathrm{HMT}} := \bigcup_{\beta\in\mathrm{Ord}} \mathcal{T}^{\mathrm{HMT}}_{\beta}. \tag{37a}$$

The value of a successor term is the subset of the preceding level defined by φ in the induced relational structure, with the values of $t_0, \dots, t_{k-1}$ as parameters. Before any set-theoretic comparison, define

$$\mathrm{Ob}^{\mathrm{HMT}}_{\mathrm{sem}}(h, m_\infty) := \{[\![t]\!] : t \in \mathcal{T}^{\mathrm{HMT}}\}. \tag{37b}$$

This is the rule of **generative exhaustiveness** adopted by HMT: every object of the full domain must possess one of those derivation trees and every admissible tree publishes an object. It is not a consequence of finite branching. The continuum hypothesis is not introduced into this rule; it is obtained through the proof developed in the article. Neither an enumeration of the reals nor a target bijection is introduced as data.

Theorem 19.17 (Equivalence between term semantics and the genealogical hierarchy)**.** *The evaluation of the preceding terms satisfies*

$$\mathrm{Ob}^{\mathrm{HMT}}_{\mathrm{sem}}(h, m_\infty) = \mathfrak{G}_{\mathrm{HMT}}(h, m_\infty). \tag{37d}$$

Proof. By transfinite induction. At the initial level, the two classes coincide. If every value of a term in $\mathcal{T}^{\mathrm{HMT}}_{\beta}$ belongs to $\mathfrak{G}_\beta$, the semantics of a successor term is one of the subsets in (37). Conversely, every subset in (37) is given by a formula code and a finite tuple of parameters; by the induction hypothesis those parameters possess terms, and (37a) forms the term that denotes it. At a limit, both constructions take the same union. □

The state encoder is compatible with truncations and sends every TPK extension to a successor term. Let $\mathcal{T}^{\mathrm{st}}_N \subset \mathcal{T}^{\mathrm{HMT}}$ be the subclass of terms encoding complete states of depth N, and let $\tau_{N+1,N} : \mathcal{T}^{\mathrm{st}}_{N+1} \to \mathcal{T}^{\mathrm{st}}_N$ delete the last block together with the corresponding truncation of the genealogical ledger, carry, and memory. If $J_N : X_N \to \mathcal{T}^{\mathrm{st}}_N$ is the complete encoder, compatibility with Ext_N proves the well-typed identity

$$\forall x \in X_N \ \forall y \in \mathrm{Ext}_N(x) \qquad \tau_{N+1,N}\big(J_{N+1}(y)\big) = J_N(x). \tag{37e$_0$}$$

Define within the semantics

$$\mathrm{Succ}^{\mathrm{TPK}}_{N+1}(J_N(x)) := \{t \in \mathcal{T}^{\mathrm{st}}_{N+1} : \exists y \in X_{N+1}\ (y \in \mathrm{Ext}_N(x) \ \wedge \ t = J_{N+1}(y))\}.$$

Choose β such that $\mathfrak{G}_\beta$ contains x, X_N, X_{N+1}, the finite graphs of $\mathrm{Ext}_N, J_N, J_{N+1}$, and their codes in h. Then the primitive-recursive formula for the graph of Ext_N gives

$$J_{N+1}[\mathrm{Ext}_N(x)] = \mathrm{Succ}^{\mathrm{TPK}}_{N+1}(J_N(x)), \qquad J_{N+1}[\mathrm{Ext}_N(x)] \in \mathfrak{G}_{\beta+1}, \tag{37e}$$

and both sides are formed by the graph formula of the same TPK extension. Equations (37e$_0$)–(37e) are the explicit semantic link: the subclass of successor terms is exactly the image of the admissible extensions, and its truncation commutes. The complete fiber of ρ is not identified with Ext_N, nor is Def identified with TPK merely by nomenclature.

Lemma 19.18 (Transitivity, axioms, and well-ordering of the genealogical model)**.** $\mathfrak{G}_{\mathrm{HMT}}(h, m_\infty)$ *is a transitive class, contains all ordinals, satisfies Zermelo–Fraenkel set theory (ZF), and possesses a definable global well-order; it therefore satisfies Choice and all axioms used in the subsequent cardinal comparison.*

Proof. Abbreviate $B = \mathfrak{G}_0$ and $u = u_{h,m}$. Induction on (36) proves simultaneously that each level is transitive and that $\mathfrak{G}_\alpha \in \mathfrak{G}_{\alpha+1}$; if all ordinals smaller than α are present, their set is definable at the corresponding level and publishes α. The union therefore contains all ordinals. Extensionality and Foundation are inherited from the ambient universe; Empty Set and Infinity are in B. Choosing a level containing the parameters, the formulas defining $\{a, b\}$, $\bigcup a$ and the subsets by Separation publish them at a successor.

For Collection–Replacement, let F be a functional relation definable in $\mathfrak{G}$ on the set a. The reflection scheme relative to the definable hierarchy $(\mathfrak{G}_\xi, \in, u_{h,m})_{\xi\in\mathrm{Ord}}$, applied in the ambient metatheory to the finite list of formulas defining F, produces a level $\mathfrak{G}_\theta$ containing a and the parameters and correct for those formulas. The ranks of the witnesses in $F[a]$ are then bounded by θ; Separation over $\mathfrak{G}_\theta$ forms the image in $\mathfrak{G}_{\theta+1}$. For Power Set, the internal set to be constructed is not presupposed. Given $a \in \mathfrak{G}_\gamma$, consider in the ambient metatheory the already existing set $\mathcal{P}^V(a)$. For every $b \in \mathcal{P}^V(a) \cap \mathfrak{G}$, let $r(b)$ be its first genealogical stage. Since $\mathfrak{G}$ is a definable class, Separation in the ambient metatheory first forms the set $\mathcal{P}^V(a) \cap \mathfrak{G}$. Replacement **in the ambient metatheory**, applied to that set, bounds the ordinals $r(b)$ by some θ. Consequently,

$$\mathcal{P}^{\mathfrak{G}}(a) = \{b \in \mathfrak{G}_\theta : b \subseteq a\},$$

and the right-hand side is published by Separation in $\mathfrak{G}_{\theta+1}$. It does not introduce into $\mathfrak{G}$ the subsets external to the semantics (37b). This is the noncircular argument for bounding by ranks.

Finally, each object is assigned its first level of birth and its least **derivation name**: a well-founded, finitely branching derivation tree (and, by (37a), finite for each individual name), labeled by a level ordinal, a formula code, and the tuple of names of its parameters. These names are ordered by transfinite recursion: first by the ordinal label, then by the formula code, and then lexicographically by the already ordered parameter names. The least name of each value defines a class relation $<_{\mathfrak{G}}$ that compares all objects. It is a definable global well-order and, by taking the least element of every nonempty set, proves Choice. □

19.10.2 TPK lift of the ternary fiber and complete cylinder partition

The fibre of $729 = 3^6$ refinement subcylinders does not come from the transfer quotient $\mathcal{K}_{\mathrm{ph}}$. That operator is a subsequent reduced-memory publication and cannot generate histories. Coverage is constructed directly in the system $(H_k, \rho_k, \mathrm{Ext}_k)$ through the three prolongations of one turn of the nonadic holonomy.

19.10.2.1 Hensel inversion in the arithmetic tower and realization by histories

Before lifting the branches to enriched histories, it is useful to recover the finite argument that determines the capacity of the arithmetic tower. The complete proof of the matrix's unfiltered transducer is incorporated; its domain is distinguished from the surviving tree.

Proposition 19.19 (Conjugacy of the unfiltered Hensel tower)**.** *Let p be prime, $d \geq 1$ and $A \in \mathrm{GL}_d(\mathbb{Z})$. For row vectors, taking residue representatives in $\{0, \dots, p-1\}^d$, define*

$$x_t A = u_t + p x_{t+1}, \qquad u_t = x_t A \bmod p.$$

The map

$$\Psi_T : (\mathbb{Z}/p^T\mathbb{Z})^d \longrightarrow (\mathbb{F}_p^d)^T, \qquad x_0 \longmapsto (u_0, \dots, u_{T-1})$$

is bijective, with inverse

$$x_0 \equiv \sum_{t=0}^{T-1} p^t u_t A^{-(t+1)} \pmod{p^T}.$$

The bijections commute with the reductions and give a homeomorphism $\Psi : \mathbb{Z}_p^d \to (\mathbb{F}_p^d)^{\mathbb{N}}$, where $\mathbb{Z}_p := \varprojlim_T \mathbb{Z}/p^T\mathbb{Z}$. The update $x_t \mapsto x_{t+1}$ corresponds to deleting the first block.

Proof. The integer inverse of A exists because its determinant is ± 1. Iterating $x_t = (u_t + p x_{t+1})A^{-1}$ gives

$$x_0 = \sum_{t=0}^{T-1} p^t u_t A^{-(t+1)} + p^T x_T A^{-T}.$$

Modulo p^T, a word determines a unique initial class. Given an arbitrary word, fixing $x_T = 0$ and reconstructing backward produces a class that emits it, which proves surjectivity. The same formula proves compatibility under truncation. Passing to the inverse limit gives the homeomorphism and the relation to the tape shift. □

For $p = 3$ and $d = 6$, the block alphabet has $3^6 = 729$ elements. This arithmetic capacity is not identified by definition with the realizability of each block after each surviving prefix. The link to the edges Ext is precisely the task of lemma 19.21. This retains the two operations distinguished by the matrix: representation in the raw tower and realization under APP–TRIT–TPK compatibilities.

19.10.2.2 Edge-by-edge realization of the three nonadic extensions

This subsection specifies the autonomous model generated by the TPK extension relation entering the cardinal proof. The existence of an edge is not inferred from a coincidence of types: its initial and final states, APP relation, fiber transport, and unary update of the ledger are constructed. The construction uses exclusively the two APP sheets, the TRIT selector, and the TPK composition Sel–Tra–Upd.

Complete cells and three balanced reductions. In the chart $D_9 = \{1, \dots, 9\}$, we will always retain the complete APP cell

$$a_{p,q} := (p, q; R^{\Sigma}_{p,q}, q^{\Sigma}_{p,q}, R^{\Pi}_{p,q}, q^{\Pi}_{p,q}),$$

with

$$p + q = R^{\Sigma}_{p,q} + 9q^{\Sigma}_{p,q}, \qquad pq = R^{\Pi}_{p,q} + 9q^{\Pi}_{p,q}.$$

The active sheet is recorded in a separate coordinate; $a_{p,q}$ is not split into a purported additive cell and a separate multiplicative cell. For $r \in D_9$, let

$$a(r) := \begin{cases} a_{1,9}, & r = 1, \\ a_{1,r-1}, & 2 \le r \le 9, \end{cases} \qquad \tau(r) := M_{\mathrm{ph}}(r), \qquad m(r) := \tau(r) \in \{-1, 0, +1\}. \tag{19.18}$$

The additive residue of $a(r)$ supplies an APP representative of phase here; this does not make Σ the universally active sheet. The dynamic selector activates Σ for $r \in \{1, 4, 7\}$, activates Π for $r \in \{2, 5, 8\}$, and preserves both sheets without moving the cursor for $r \in \{3, 6, 9\}$. In all three cases, $R^{\Sigma}, q^{\Sigma}, R^{\Pi}, q^{\Pi}$ are transported jointly. We have

$$(m(1), \dots, m(9)) = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The three pairs ordered by the nonadic base are

$$\mathfrak{p}_{+1} := (1, 4), \qquad \mathfrak{p}_{-1} := (2, 5), \qquad \mathfrak{p}_0 := (3, 6). \tag{19.19}$$

These pairs are the normalized section taking the first two occurrences of each TRIT type under the shift $r \mapsto r + 3$; the third phase block $7, 8, 9$ retains the turn-closure control. Both members belong to the same fiber of M_{ph}. Without introducing a target coefficient, the balanced reduction from the TRIT chapter gives

ε	$m(\mathfrak{p}_\varepsilon^{(1)}) + m(\mathfrak{p}_\varepsilon^{(2)})$	$r_3(2\varepsilon)$	$\chi(\varepsilon,\varepsilon)$
$+1$	$1+1=(-1)+3\cdot 1$	-1	$+1$
0	$0+0=0+3\cdot 0$	0	0
-1	$(-1)+(-1)=1+3\cdot(-1)$	$+1$	-1

(19.20)

Therefore,

$$\chi(\varepsilon,\varepsilon)=\frac{2\varepsilon-r_3(2\varepsilon)}{3}=\varepsilon. \tag{19.21}$$

The branch sign is thus the output of the balanced cocycle applied to two TRIT emissions produced by APP. Each pair supplies one of the three carry outputs. For clarity, we subsequently index the pair by $\varepsilon(\mathfrak{p}) := \chi(m(\mathfrak{p}^{(1)}), m(\mathfrak{p}^{(2)}))$; the symbol ε does not supply a free carry to $\mathrm{Upd}^{\mathrm{cal}}$.

The phase architecture, balanced lift, and memory transport are realized here through twenty-seven local incidences and their extension graphs. The normalized section fixes representatives of the fibers of M_{ph}; the covering to be proved uses the existence of these representatives and remains invariant when they are replaced by another compatible section.

Typed data of the nine edges. Let $K_{\mathrm{Carr}} := \langle \kappa_{\mathrm{Carr}} \rangle_{\mathrm{Ab}}$ be the carry module. The actions are typed by

$$H_{\mathrm{Carr}} : \mathscr{P}_{\mathrm{TPK}}^{\mathrm{op}} \longrightarrow \mathrm{End}(K_{\mathrm{Carr}}), \qquad Y_{\mathrm{Carr}} : \mathscr{P}_{\mathrm{TPK}} \longrightarrow \mathrm{End}(K_{\mathrm{Carr}}).$$

In this generated model, take the trivial actions $H_{\mathrm{Carr}}(e) = Y_{\mathrm{Carr}}(e) = \mathrm{id}_{K_{\mathrm{Carr}}}$: the active sheet and the route are retained as coordinates of the ledger itself and do not act on the free carry summand. Also let $(\mathsf{M}_d, \jmath_{d',d})$ be the graded direct system of memory. To avoid confusing two representatives identified by the colimit, the state also retains the degree as a coordinate, and the labeled carrier

$$\widetilde{\mathsf{M}} := \bigsqcup_{d\geq 1} \big(\{d\} \times \mathsf{M}_d\big), \qquad \mathrm{gr}_{\mathsf{M}}(d,\mu) = d.$$

Publication in the limit memory is given by

$$\mathsf{M} := \varinjlim_d \mathsf{M}_d, \qquad \jmath_d : \mathsf{M}_d \longrightarrow \mathsf{M},$$
$$u := \jmath_d(u_d), \quad \iota_{\mathsf{M}} : \mathbb{Z} \longrightarrow \mathsf{M}, \qquad n \longmapsto nu.$$

Here $\jmath_d$ is the canonical colimit map and the u_d form a compatible family. Its free component permits the reader $\deg_u(nu) = n$. The configuration, phase, memory, carry, route and survival projections will be typed on the calibrated carrier constructed recursively below; no ambient space will be imported for them. For each constructed edge e, the calibrated component $\mathrm{Upd}_e^{\mathrm{cal}} := \mathsf{Mem}_e^{\mathrm{cal}}$ will denote the third map in the factorization; its law is fixed field by field below and is not assumed as the restriction of an ambient relation.

Fix the initial depth $k_0 = 1$, a complete configuration $q_\star \in Q_{\mathrm{obs}}$ of phase one, and

$$q_{n,r} := F_{\mathrm{obs}}^{9n+r-1}(q_\star), \qquad q_{n,10} := q_{n+1,1}.$$

Because the observable calendar is periodic, its value $q_{n,r}$ does not by itself distinguish occurrences separated by twelve turns. The calibrated model therefore uses the labeled lift

$$\widetilde{q}_{n,r} := (q_{n,r}, n, r), \qquad \widetilde{q}_{n,10} := \widetilde{q}_{n+1,1}, \qquad \pi_{\mathrm{obs}}(\widetilde{q}_{n,r}) = q_{n,r}.$$

The lifted dynamics $\widetilde{F}_{\mathrm{obs}}(\widetilde{q}_{n,r}) = \widetilde{q}_{n,r+1}$ projects to F_{obs}; the label does not alter the physical phase, but makes its levels of occurrence distinct. The copy of the cell at each level is labeled by

$$a_{n,r} := (n, r; a(r)), \qquad (n,r)^{+} := \begin{cases} (n, r+1), & 1 \le r < 9, \\ (n+1, 1), & r = 9. \end{cases}$$

The label only prevents two occurrences of the same cell from being identified; the two Euclidean divisions and their four coordinates are those of $a(r)$. For turn $n \ge 0$ and phase $r \in D_9$, the symbol

$$e_{\varepsilon,n,r} : \widetilde{q}_{n,r} \longrightarrow \widetilde{q}_{n,r+1}, \qquad r^{+} := 1 + (r \bmod 9),$$

occupies level $d_{n,r} := k_n + r - 1$, where $k_n := k_0 + 9n$. Its local ledger contains the complete cell $a(r)$, the active sheet determined by $M_{\mathrm{ph}}(r)$, the type $\tau(r)$, the membership mark in $\mathfrak{p}_\varepsilon$, and, upon reaching the second member of that pair, the certified identity (19.20). We define its operators solely from that ledger:

$$\mathsf{C}_{\mathsf{M}}(e_{\varepsilon,n,r}) = J_{d_{n,r}+1, d_{n,r}}, \qquad \kappa_{\mathsf{M}}(e_{\varepsilon,n,r}) = \mathbf{1}_{\{r=9\}} u_{d_{n,r}+1}, \tag{19.22}$$

$$\mathrm{Carr}(e_{\varepsilon,n,r}) = d_{\varepsilon,r}\, \kappa_{\mathrm{Carr}}, \qquad d_{\varepsilon,r} := \mathbf{1}_{\{r=\mathfrak{p}_\varepsilon^{(2)}\}} \chi\Big(m(\mathfrak{p}_\varepsilon^{(1)}), m(\mathfrak{p}_\varepsilon^{(2)})\Big). \tag{19.23}$$

For identities and every admissible composition, also define

$$\mathsf{C}_{\mathsf{M}}(\mathrm{id}_{\widetilde{q}}; d) := \mathrm{id}_{\mathsf{M}_d}, \qquad \kappa_{\mathsf{M}}(\mathrm{id}_{\widetilde{q}}; d) := 0, \tag{19.24}$$

$$\mathsf{C}_{\mathsf{M}}(e_t \cdots e_1) := \mathsf{C}_{\mathsf{M}}(e_t) \circ \cdots \circ \mathsf{C}_{\mathsf{M}}(e_1), \qquad \kappa_{\mathsf{M}}(e_2 e_1) := \mathsf{C}_{\mathsf{M}}(e_2)\big(\kappa_{\mathsf{M}}(e_1)\big) + \kappa_{\mathsf{M}}(e_2). \tag{19.25}$$

The identity carries degree d because the same observable configuration may recur at different depths. Iterated, the final equality defines $\kappa_{\mathsf{M}}(e_t \cdots e_1)$ for every length and is the cocycle law of the direct system; the inclusions satisfy $J_{d+2,d+1} J_{d+1,d} = J_{d+2,d}$. In addition, set $s(e_{\varepsilon,n,r}) = \top$. The following twenty-seven entries certify the coordinates distinguishing the three extensions; the universal transport fields are specified immediately afterward. The symbols I and II designate, respectively, the first retained emission and the second emission performing the reduction; a dash denotes a zero carry increment, not the absence of an edge.

r	$M_{\mathrm{ph}}(r)$	$\mathfrak{p}_{+1}$		$\mathfrak{p}_0$		$\mathfrak{p}_{-1}$		Δ mem
		mark	$d_{+1,r}$	mark	$d_{0,r}$	mark	$d_{-1,r}$	
1	+1	I	0	–	0	–	0	0
2	−1	–	0	–	0	I	0	0
3	0	–	0	I	0	–	0	0
4	+1	II	+1	–	0	–	0	0
5	−1	–	0	–	0	II	−1	0
6	0	–	0	II	0	–	0	0
7	+1	–	0	–	0	–	0	0
8	−1	–	0	–	0	–	0	0
9	0	–	0	–	0	–	0	u

Thus, $d_{\varepsilon,r}$ is computed from two outputs of the selector and is distinct from the cell quotients q^{Σ}, q^{Π}. The ledger also preserves the marked pair and the balanced residue; in particular, the zero branch is not confused with an absence of event.

The carry composition law used by the TPK category is

$$\mathrm{Carr}(\mathrm{id}_{\widetilde{q}}) = 0, \qquad \mathrm{Carr}(e_2 e_1) = H_{\mathrm{Carr}}(e_1)\big(\mathrm{Carr}(e_2)\big) + Y_{\mathrm{Carr}}(e_2)\big(\mathrm{Carr}(e_1)\big). \tag{19.26}$$

On identities and paths, they are extended functorially:

$$\begin{aligned} H_{\mathrm{Carr}}(\mathrm{id}) &= Y_{\mathrm{Carr}}(\mathrm{id}) = \mathrm{id}, \\ H_{\mathrm{Carr}}(e_t \cdots e_1) &= H_{\mathrm{Carr}}(e_1) \circ \cdots \circ H_{\mathrm{Carr}}(e_t), \\ Y_{\mathrm{Carr}}(e_t \cdots e_1) &= Y_{\mathrm{Carr}}(e_t) \circ \cdots \circ Y_{\mathrm{Carr}}(e_1). \end{aligned}$$

In this model all these actions remain the identity, but their typing makes explicit which endomorphism acts on each summand of (19.26). On calibrated edges, $H_{\mathrm{Carr}} = Y_{\mathrm{Carr}} = \mathrm{id}$, so composition sums the increments already produced. Equation (19.26), rather than an identification between orientation and quotient, governs the accumulated register.

Independent construction of the transition tuples. A fiber state is written, in the order of its relevant fields, as

$$x = (a; q; \tau, o, h; b_{\mathrm{ref}}; R^{\Sigma}, q^{\Sigma}, R^{\Pi}, q^{\Pi}; c; \mathrm{Sig}; C, \partial C; p_{\mathrm{APP}}, p_{\widehat{\mathbb{T}}}; \lambda; \mathrm{gr}_{\mathsf{M}}; \mathrm{mem}; s; \mathrm{Led}).$$

The update notation $x[f \leftarrow v]$ changes only the indicated field. For $h \in \{(\Sigma,\Pi), (\Pi,\Sigma)\}$, define

$$h_r(h) = \begin{cases} (\Sigma,\Pi), & r \in \{1,4,7\}, \\ (\Pi,\Sigma), & r \in \{2,5,8\}, \\ h, & r \in \{3,6,9\}. \end{cases}$$

The complete sheet-and-arithmetic coordinate is abbreviated as

$$\widehat{h}(x) := \big(h(x), R^{\Sigma}(x), q^{\Sigma}(x), R^{\Pi}(x), q^{\Pi}(x)\big).$$

Let $\widehat{\mathcal{H}}^{\mathrm{cal}}_{\widetilde{q}_{n,r}}$ be the set of quintuples

$$\big(h, R^{\Sigma}_{a_{n,r}}, q^{\Sigma}_{a_{n,r}}, R^{\Pi}_{a_{n,r}}, q^{\Pi}_{a_{n,r}}\big), \qquad h \in \{(\Sigma,\Pi), (\Pi,\Sigma)\},$$

whose four arithmetic data are the two Euclidean divisions of the labeled APP cell $a_{n,r}$. This direct definition does not presuppose a state in a carrier not yet constructed. Let $\mathsf{Hist}^{\mathrm{cal}}_{\widetilde{q}}$ be the set of admissible finite paths in the graph of labeled edges ending at $\widetilde{q}$, including the identity, and let $\mathsf{B}^{\mathrm{cal}}_{\widetilde{q}}$ be the set of their ordered endpoint sequences. Define

$$\mathsf{S}^{\mathrm{cal}}_{\widetilde{q}} := \widehat{\mathcal{H}}^{\mathrm{cal}}_{\widetilde{q}} \times \{o_{\rightarrow}, o_{\circ}, o_{\leftarrow}\} \times K_{\mathrm{Carr}} \times \mathsf{Hist}^{\mathrm{cal}}_{\widetilde{q}} \times \mathsf{B}^{\mathrm{cal}}_{\widetilde{q}} \times \mathsf{M}.$$

The calibrated signature is not postulated: it is the typed assembly map

$$\mathrm{Sig}^{\mathrm{cal}}_{\widetilde{q}} : \widehat{\mathcal{H}}^{\mathrm{cal}}_{\widetilde{q}} \times \{o_{\rightarrow}, o_{\circ}, o_{\leftarrow}\} \times K_{\mathrm{Carr}} \times \mathsf{Hist}^{\mathrm{cal}}_{\widetilde{q}} \times \mathsf{B}^{\mathrm{cal}}_{\widetilde{q}} \times \mathsf{M} \longrightarrow \mathsf{S}^{\mathrm{cal}}_{\widetilde{q}},$$

$$(\widehat{h}, o, c, \lambda, \partial C, \mu) \longmapsto (\widehat{h}, o, c, \lambda, \partial C, \mu).$$

Likewise, $\pi_{\mathcal{F}}$ denotes the projection that removes only the first two coordinates $(a; q)$ of the state tuple; it preserves $\tau, o, h, b_{\mathrm{ref}}$, the four Euclidean data, carry, signature, cylinder, boundary, prefixes, route, memory degree, memory, survival, and ledger. The corresponding orientation is fixed by $o(+1) = o_{\rightarrow}$, $o(0) = o_{\circ}$, and $o(-1) = o_{\leftarrow}$. The calibrated APP fiber over $\widetilde{q}_{n,r}$ contains the already defined labeled copy $a_{n,r}$. The initial state is fixed without free coordinates:

$$\widehat{h}_{\star} := \big((\Sigma,\Pi), R^{\Sigma}_{1,9}, q^{\Sigma}_{1,9}, R^{\Pi}_{1,9}, q^{\Pi}_{1,9}\big).$$

$$\begin{aligned} x_{\star} := \big(&a_{0,1}; \widetilde{q}_{0,1}; +1, o_{\rightarrow}, (\Sigma,\Pi); \bot; R^{\Sigma}_{1,9}, q^{\Sigma}_{1,9}, R^{\Pi}_{1,9}, q^{\Pi}_{1,9}; 0; \\ &\mathrm{Sig}^{\mathrm{cal}}_{\widetilde{q}_{0,1}}(\widehat{h}_{\star}, o_{\rightarrow}, 0, \mathrm{id}_{\widetilde{q}_{0,1}}, \varnothing, \jmath_{k_0}(0_{\mathsf{M}_{k_0}})); \varnothing, \varnothing; (a_{0,1}), (+1); \mathrm{id}_{\widetilde{q}_{0,1}}; k_0; 0_{\mathsf{M}_{k_0}}; \top; \varnothing\big). \end{aligned} \tag{19.27}$$

The tuple contains no indeterminate coordinates and satisfies $s(x_{\star}) = \top$. Let $G_0 := \{x_{\star}\}$. Assuming G_n has been constructed, fix $x \in G_n$, $\varepsilon \in \{-1, 0, +1\}$, and $x_{\varepsilon,0}(x) := x$. For $r = 1, \ldots, 9$, first construct the elementary ledger

$$\begin{aligned}\ell_{\varepsilon,n,r} := \big(&e_{\varepsilon,n,r}, a_{n,r}, a_{(n,r)^+}, \widetilde{q}_{n,r}, \widetilde{q}_{n,r+1}, M_{\mathrm{ph}}(r),\\ &h_r(h(x_{\varepsilon,r-1}(x))), o(M_{\mathrm{ph}}(r)), \mathfrak{p}_\varepsilon, d_{\varepsilon,r},\\ &\mathsf{C}_{\mathsf{M}}(e_{\varepsilon,n,r}), \kappa_{\mathsf{M}}(e_{\varepsilon,n,r}), \mathrm{Carr}(e_{\varepsilon,n,r})\big),\end{aligned} \tag{19.28}$$

and set $\mathrm{Led}^{\mathrm{new}}_{\varepsilon,n,r} := \mathrm{Led}(x_{\varepsilon,r-1}(x))^\frown \ell_{\varepsilon,n,r}$. Then construct the tuple

$$y_{\varepsilon,r}(x) := x_{\varepsilon,r-1}(x)[\tau \leftarrow M_{\mathrm{ph}}(r),\ h \leftarrow h_r(h(x_{\varepsilon,r-1}(x))),\ o \leftarrow o(M_{\mathrm{ph}}(r)),\ b_{\mathrm{ref}} \leftarrow \mathfrak{p}_\varepsilon], \tag{19.29}$$

where selection leaves the cell, quotients, carry, route, memory, cylinder, boundary, survival, and ledger unchanged; the coordinate b_{ref} makes the chosen balanced reduction explicit. Membership in an ambient relation is not presupposed here.

Next, define $z_{\varepsilon,r}(x)$ field by field: its configuration is $\widetilde{q}_{n,r+1}$; its complete cell is $a_{(n,r)^+}$, with the four residues and quotients recomputed by the two Euclidean divisions; it preserves the selected sheet and orientation; and it performs the following six updates, in which only the common argument x is omitted:

$$\begin{aligned} C(z_{\varepsilon,r}) &:= C(y_{\varepsilon,r})^\frown e_{\varepsilon,n,r},\\ \partial C(z_{\varepsilon,r}) &:= \partial C(y_{\varepsilon,r})^\frown (\widetilde{q}_{n,r}, \widetilde{q}_{n,r+1}),\\ p_{\mathrm{APP}}(z_{\varepsilon,r}) &:= p_{\mathrm{APP}}(y_{\varepsilon,r})^\frown a_{(n,r)^+},\\ p_{\widehat{\mathbb{T}}}(z_{\varepsilon,r}) &:= p_{\widehat{\mathbb{T}}}(y_{\varepsilon,r})^\frown \tau(r),\\ \mathrm{Led}(z_{\varepsilon,r}) &:= \mathrm{Led}^{\mathrm{new}}_{\varepsilon,n,r},\\ \lambda(z_{\varepsilon,r}) &:= e_{\varepsilon,n,r}\lambda(y_{\varepsilon,r}),\end{aligned} \tag{19.30}$$

and still retains the fields $c, \mathrm{gr}_{\mathsf{M}}, \mathrm{mem}, s$ of $y_{\varepsilon,r}(x)$. Its signature, by contrast, is retyped at the target configuration through the transported presignature

$$\begin{aligned}\mathrm{Sig}(z_{\varepsilon,r}(x)) := \mathrm{Sig}^{\mathrm{cal}}_{\widetilde{q}_{n,r+1}}\Big(&\widehat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(y_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)),\\ &\partial C(z_{\varepsilon,r}(x)), \jmath_{d_{n,r}}(\mathrm{mem}_{d_{n,r}}(y_{\varepsilon,r}(x)))\Big).\end{aligned} \tag{19.31}$$

Thus, and only in $\mathsf{Tra}^{\mathrm{cal}}$, the edge, its endpoints, the mark $(\mathfrak{p}_\varepsilon, r, d_{\varepsilon,r})$, and its cocycles are added to the incidence ledger.

The initial state satisfies

$$c(x_\star) = 0 = \mathrm{Carr}(\mathrm{id}_{\widetilde{q}_{0,1}}) = \mathrm{Carr}(\lambda(x_\star)).$$

In the complete states $x_{\varepsilon,r-1}(x)$, and also after the selection $y_{\varepsilon,r}(x)$, the typed invariant $c(v) = \mathrm{Carr}(\lambda(v))$ is preserved, since $\mathsf{Sel}^{\mathrm{cal}}$ modifies neither field. Transport advances the route but leaves the carry pending; hence the pre-update tuple satisfies exactly

$$c(z_{\varepsilon,r}(x)) = c(y_{\varepsilon,r}(x)) = \mathrm{Carr}(\lambda(y_{\varepsilon,r}(x))).$$

The map $\mathrm{Upd}^{\mathrm{cal}}$ then restores $c(x_{\varepsilon,r}(x)) = \mathrm{Carr}(\lambda(x_{\varepsilon,r}(x)))$ by means of the composition law, without attributing that invariant to the intermediate state z.

Finally, apply the unary function $\mathrm{Upd}^{\mathrm{cal}}_{e_{\varepsilon,n,r}} = \mathsf{Mem}^{\mathrm{cal}}_{e_{\varepsilon,n,r}}$ to that tuple. Its result $x_{\varepsilon,r}(x)$ is given by

$$
\begin{aligned}
c\big(x_{\varepsilon,r}(x)\big) &= \mathrm{Carr}\big(\lambda(z_{\varepsilon,r}(x))\big)\\
&= H_{\mathrm{Carr}}(\lambda(y_{\varepsilon,r}(x)))\big(\mathrm{Carr}(e_{\varepsilon,n,r})\big) + Y_{\mathrm{Carr}}(e_{\varepsilon,n,r})\big(c(y_{\varepsilon,r}(x))\big),\\
\lambda\big(x_{\varepsilon,r}(x)\big) &= \lambda(z_{\varepsilon,r}(x)),\\
\mathrm{gr}_{\mathsf{M}}\big(x_{\varepsilon,r}(x)\big) &= d_{n,r}+1,\\
\mathrm{mem}_{d_{n,r}+1}\big(x_{\varepsilon,r}(x)\big) &= \mathsf{C}_{\mathsf{M}}(e_{\varepsilon,n,r})\,\mathrm{mem}_{d_{n,r}}(z_{\varepsilon,r}(x)) + \kappa_{\mathsf{M}}(e_{\varepsilon,n,r}),\\
s\big(x_{\varepsilon,r}(x)\big) &= s(e_{\varepsilon,n,r}) \wedge s(z_{\varepsilon,r}(x)),\\
\mathrm{Sig}\big(x_{\varepsilon,r}(x)\big) &= \mathrm{Sig}^{\mathrm{cal}}_{\widetilde{q}_{n,r+1}}\Big(\widehat{h}(z_{\varepsilon,r}(x)), o(z_{\varepsilon,r}(x)), c(x_{\varepsilon,r}(x)), \lambda(z_{\varepsilon,r}(x)), \partial C(z_{\varepsilon,r}(x)),\\
&\qquad\qquad \jmath_{d_{n,r}+1}\big(\mathrm{mem}_{d_{n,r}+1}(x_{\varepsilon,r}(x))\big)\Big).
\end{aligned}
\tag{19.32}
$$

All fields not enumerated in (19.32) coincide with those of $z_{\varepsilon,r}(x)$. In particular, the signature is computed from the sheet, orientation, boundary, and route already fixed by $\mathsf{Sel}^{\mathrm{cal}}$ and $\mathsf{Tra}^{\mathrm{cal}}$, and from the carry and memory computed in the preceding lines; it is not circularly defined from the final state. Neither ε, nor $d_{\varepsilon,r}$, nor u is an additional argument of $\mathrm{Upd}^{\mathrm{cal}}$: the function reads all three data from the already constructed edge.

Simultaneous induction of the levels. Once the preceding nine stages have been completed for an already constructed G_n, define immediately

$$
\gamma_{\varepsilon,k_n}(x) := x_{\varepsilon,9}(x), \qquad G_{n+1} := \{\gamma_{\varepsilon,k_n}(x) : x \in G_n,\ \varepsilon \in \widehat{\mathbb{T}}\}. \tag{19.33}
$$

The base $G_0 = \{x_\star\}$, the intermediate tuples, and (19.33) constitute a single definition by recursion on n. Therefore, no subsequent carrier is used to assume the existence of G_n: the next level is introduced only after constructing all its edges from the preceding level.

TPK extension relation and effective membership. Let $\mathcal{E}^{\mathrm{cal}}_{\widetilde{q}}$ be the smallest labelled family containing $x_\star$ and, at each step of the preceding recurrence, the four tuples $x_{\varepsilon,r-1}(x), y_{\varepsilon,r}(x), z_{\varepsilon,r}(x), x_{\varepsilon,r}(x)$ in their source and target configurations. Also let $\mathcal{A}^{\mathrm{cal}}_{\widetilde{q}_{n,r}} := \{a_{n,r}\}$, and set

$$
\widetilde{Q}_{\mathrm{obs}} := \{\widetilde{q}_{n,r} : n \geq 0,\ 1 \leq r \leq 9\}, \qquad \mathcal{E}^{\mathrm{cal}} := \bigsqcup_{\widetilde{q}} \mathcal{E}^{\mathrm{cal}}_{\widetilde{q}}, \qquad \mathcal{F}^{\mathrm{cal}}_{\widetilde{q}} := \pi_{\mathcal{F}}(\mathcal{E}^{\mathrm{cal}}_{\widetilde{q}}).
$$

For each depth d, let $\mathcal{E}^{\mathrm{cal}}_d := \mathrm{gr}^{-1}_{\mathsf{M}}(d) \subset \mathcal{E}^{\mathrm{cal}}$. The explicit degree coordinate, rather than the mere membership of a representative in some image of the direct system, makes these carriers disjoint. On them, the projections are typed as

$$
\begin{aligned}
\mathrm{cfg} &: \mathcal{E}^{\mathrm{cal}} \longrightarrow \widetilde{Q}_{\mathrm{obs}},\\
g(x) &:= \mathrm{fase}\big(\pi_{\mathrm{obs}}(\mathrm{cfg}(x))\big) \in C_9,\\
c &: \mathcal{E}^{\mathrm{cal}} \longrightarrow K_{\mathrm{Carr}},\\
s &: \mathcal{E}^{\mathrm{cal}} \longrightarrow \{\bot, \top\},\\
\lambda &: \mathcal{E}^{\mathrm{cal}}_{\widetilde{q}} \longrightarrow \mathsf{Hist}^{\mathrm{cal}}_{\widetilde{q}},\\
\mathrm{gr}_{\mathsf{M}} &: \mathcal{E}^{\mathrm{cal}} \longrightarrow \mathbb{N}_{\geq 1},\\
\mathrm{mem}_d &: \mathcal{E}^{\mathrm{cal}}_d \longrightarrow \mathsf{M}_d,\\
\mathrm{mem}(x) &:= \jmath_d\big(\mathrm{mem}_d(x)\big) \in \mathsf{M} \quad (x \text{ of depth } d).
\end{aligned}
$$

Here $\mathrm{cfg}(x)$ is the second coordinate of the tuple, and fase reads the phase of the observable calendar. For the three carrier classes, fix the diagonals

$$\Delta_{\mathrm{APP},\widetilde{q}} := \{(a,a) : a \in \mathcal{A}^{\mathrm{cal}}_{\widetilde{q}}\},$$
$$\Delta_{\mathrm{fib},\widetilde{q}} := \{(f,f) : f \in \mathcal{F}^{\mathrm{cal}}_{\widetilde{q}}\},$$
$$\Delta_{\mathrm{enr},\widetilde{q}} := \{(x,x) : x \in \mathcal{E}^{\mathrm{cal}}_{\widetilde{q}}\}.$$

In particular,

$$a_{0,1} \in \mathcal{A}^{\mathrm{cal}}_{\widetilde{q}_{0,1}}, \qquad x_\star \in \mathcal{E}^{\mathrm{cal}}_{\widetilde{q}_{0,1}}.$$

On these carriers, the following relations are defined by their complete graphs, and not through necessary implications:

$$\mathsf{Sel}^{\mathrm{cal}}_{e_{\varepsilon,n,r}} := \{(x_{\varepsilon,r-1}(x), y_{\varepsilon,r}(x)) : x \in G_n\}, \tag{19.34}$$
$$\mathsf{Tra}^{\mathrm{cal}}_{e_{\varepsilon,n,r}} := \{(y_{\varepsilon,r}(x), z_{\varepsilon,r}(x)) : x \in G_n\}, \tag{19.35}$$
$$\mathrm{Ext}^{\mathrm{APP,cal}}_{e_{\varepsilon,n,r}} := \{(a_{n,r}, a_{(n,r)^+})\}, \tag{19.36}$$
$$\mathrm{Ext}^{\mathrm{fib,cal}}_{e_{\varepsilon,n,r}} := \{(\pi_{\mathcal{F}} x_{\varepsilon,r-1}(x), \pi_{\mathcal{F}} x_{\varepsilon,r}(x)) : x \in G_n\}. \tag{19.37}$$

The synchronized enriched extension is

$$\mathrm{Ext}^{\mathrm{enr,cal}}_{e_{\varepsilon,n,r}} := \{(x_{\varepsilon,r-1}(x), x_{\varepsilon,r}(x)) : x \in G_n\}. \tag{19.38}$$

The third map is defined by

$$\mathrm{graph}\big(\mathrm{Upd}^{\mathrm{cal}}_{e_{\varepsilon,n,r}}\big) := \{(z_{\varepsilon,r}(x), x_{\varepsilon,r}(x)) : x \in G_n\}, \tag{19.39}$$

where the second component is exactly the one in (19.32). Consequently,

$$\mathcal{U}^{(1),\mathrm{cal}}_{e_{\varepsilon,n,r}} := \mathrm{Upd}^{\mathrm{cal}}_{e_{\varepsilon,n,r}} \circ \mathsf{Tra}^{\mathrm{cal}}_{e_{\varepsilon,n,r}} \circ \mathsf{Sel}^{\mathrm{cal}}_{e_{\varepsilon,n,r}} = \{(x_{\varepsilon,r-1}(x), x_{\varepsilon,r}(x)) : x \in G_n\}. \tag{19.40}$$

Each operator is indexed by the edge $e_{\varepsilon,n,r}$ produced by the reduction $\mathfrak{p}_\varepsilon$, rather than by a trit supplied from outside. The coordinate b_{ref} also distinguishes their codomains. Therefore, $\mathsf{Sel}^{\mathrm{cal}}_e$, $\mathsf{Tra}^{\mathrm{cal}}$ and $\mathrm{Upd}^{\mathrm{cal}}$ are functions on their respective domains, whereas the union $\bigcup_{\varepsilon\in\widehat{\mathbb{T}}} \mathcal{U}^{(1),\mathrm{cal}}_{e_{\varepsilon,n,r}}$ deliberately preserves the three outputs of the trisection.

For $\bullet \in \{\mathrm{APP}, \mathrm{fib}, \mathrm{enr}\}$, identities and compositions are not left implicit. Define

$$\mathrm{Ext}^{\bullet,\mathrm{cal}}_{\mathrm{id}_{\widetilde{q}}} := \Delta_{\bullet,\widetilde{q}}, \tag{19.41}$$
$$\mathrm{Ext}^{\bullet,\mathrm{cal}}_{e_t\cdots e_1} := \mathrm{Ext}^{\bullet,\mathrm{cal}}_{e_t} \circ \cdots \circ \mathrm{Ext}^{\bullet,\mathrm{cal}}_{e_1} \tag{19.42}$$

for every admissible edge word. The two APP and fiber path closures are formed separately,

$$\mathrm{Path}(\mathrm{Ext}^{\mathrm{APP,cal}}), \qquad \mathrm{Path}(\mathrm{Ext}^{\mathrm{fib,cal}}), \tag{19.43}$$

and $\mathsf{Tra}^{\mathrm{cal}}$ synchronizes them edge by edge.

Finally, the levels of complete states and their truncations are fixed by the recursion itself:

$$H^{\mathrm{cal}}_{d_{n,r}} := \{x_{\varepsilon,r-1}(x) : x \in G_n,\ \varepsilon \in \widehat{\mathbb{T}}\}, \tag{19.44}$$
$$H^{\mathrm{cal}}_{d_{n,r}+1} := \{x_{\varepsilon,r}(x) : x \in G_n,\ \varepsilon \in \widehat{\mathbb{T}}\}, \tag{19.45}$$
$$\rho^{\mathrm{cal}}_{d_{n,r}+1,d_{n,r}}\big(x_{\varepsilon,r}(x)\big) := x_{\varepsilon,r-1}(x). \tag{19.46}$$

The final ledger preserves $(n, r, \mathfrak{p}_\varepsilon)$, so the predecessor of the left-hand side is unique and ρ^{cal} is well-defined. For $b > a$, set $\rho^{\mathrm{cal}}_{b,a} := \rho^{\mathrm{cal}}_{a+1,a} \circ \cdots \circ \rho^{\mathrm{cal}}_{b,b-1}$. This map recovers the complete preceding

state, including its signature; it does not purport to invert an overwritten coordinate without consulting the ledger.

Proposition 19.20 (Autonomous realization of the TPK through explicit transitions)**.** *The family*

$$\begin{aligned}\mathfrak{T}^{\mathrm{cal}}_{\mathrm{TPK}} := \big((\widetilde{q}_{n,r})_{n,r}, \pi_{\mathrm{obs}}, \widetilde{F}_{\mathrm{obs}}, M_{\mathrm{ph}},\\ (\mathcal{A}^{\mathrm{cal}}_{\widetilde{q}})_{\widetilde{q}}, (\mathcal{E}^{\mathrm{cal}}_{\widetilde{q}})_{\widetilde{q}}, (\mathcal{F}^{\mathrm{cal}}_{\widetilde{q}})_{\widetilde{q}}, \mathrm{Ext}^{\mathrm{APP,cal}}, \mathrm{Ext}^{\mathrm{fib,cal}}, \mathrm{Ext}^{\mathrm{enr,cal}}, \rho^{\mathrm{cal}},\\ \mathsf{Sel}^{\mathrm{cal}}, \mathsf{Tra}^{\mathrm{cal}}, \mathrm{Upd}^{\mathrm{cal}}, \mathsf{C}_{\mathsf{M}}, \kappa_{\mathsf{M}}, \mathrm{gr}_{\mathsf{M}}, H_{\mathrm{Carr}}, Y_{\mathrm{Carr}},\\ \mathrm{Sig}^{\mathrm{cal}}, \pi_{\mathcal{F}}\big).\end{aligned}$$

is an enriched model of the TPK. In particular, every pair on the right-hand side of (19.40) *is an effective TPK edge, not a consequence inferred from a merely necessary condition.*

Proof. The labels (n, r) make the carrier copies disjoint and unambiguously fix each source and target. The relation $\mathrm{Ext}^{\mathrm{APP,cal}}$ transports the complete cell and preserves the two Euclidean identities; the relation $\mathrm{Ext}^{\mathrm{fib,cal}}$ relates the fibers of the preceding and subsequent complete states: it transports sheet, orientation, prefix, cylinder, boundary, and ledger, and incorporates the updated carry, memory, and signature. Selection does not modify the ledger, and transport concatenates exactly one entry and produces the target presignature (19.31). The update recomputes the signature by means of the typed map $\mathrm{Sig}^{\mathrm{cal}}_{\widetilde{q}}$ from the already determined sheet, orientation, carry, route, boundary, and memory. The map $\mathrm{Upd}^{\mathrm{cal}}$ is unary because the edge contained in $z_{\varepsilon,r}(x)$ determines its cocycles.

The identities are the diagonals, and the word extensions are the compositions in (19.42). Concatenation of prefixes and ledgers is associative; memory satisfies the cocycle law, and carry satisfies (19.26). Therefore, the two parenthesizations of three edges produce the same state. Formula (19.46) defines truncation on complete states, and uniqueness of the final ledger proves that it recovers the source in every field. These are the identity, source, target, composition, memory, carry, and truncation laws of the enriched TPK. In particular, the extension relations are determined as graphs of effective maps rather than as consequences of merely necessary conditions. □

Each γ_{ε,k_n} comprises exactly nine applications of $\mathrm{Upd}^{\mathrm{cal}}_{e_{\varepsilon,n,r}} \circ \mathsf{Tra}^{\mathrm{cal}}_{e_{\varepsilon,n,r}} \circ \mathsf{Sel}^{\mathrm{cal}}_{e_{\varepsilon,n,r}}$. Survival will be proved by means of a constructed tail and will not be used to define G_n.

Lemma 19.21 (Effective internal trisection of one turn)**.** *For every $n \geq 0$, every $x \in G_n$, and every $\varepsilon \in \{-1, 0, +1\}$, the expression $\gamma_{\varepsilon,k_n}(x)$ is defined by nine TPK transitions, with its synchronized relations* $\mathrm{Ext}^{\mathrm{APP,cal}}$ *and* $\mathrm{Ext}^{\mathrm{fib,cal}}$*, and satisfies*

$$\rho^{\mathrm{cal}}_{k_n+9,k_n}\gamma_{\varepsilon,k_n}(x) = x, \tag{19.47}$$

$$g\big(\gamma_{\varepsilon,k_n}(x)\big) = g(x), \qquad \jmath_{k_n+9}\big(\mathrm{mem}_{k_n+9}(\gamma_{\varepsilon,k_n}(x))\big) = \jmath_{k_n}\big(\mathrm{mem}_{k_n}(x)\big) + u, \tag{19.48}$$

$$c\big(\gamma_{\varepsilon,k_n}(x)\big) = c(x) + \varepsilon\kappa_{\mathrm{Carr}}. \tag{19.49}$$

The ordered ledger of the nine edges determines ε uniquely. The three images are therefore disjoint. Each endpoint belongs to G_{n+1} and again admits the three maps $\gamma_{-1,k_{n+1}}, \gamma_{0,k_{n+1}}, \gamma_{+1,k_{n+1}}$.

Proof. Formulas (19.18) traverse the nine phases once and return the copy $a_{n+1,1}$ of $a(1)$. At each step, the APP relation transports the complete sextuple; $\mathsf{Sel}^{\mathrm{cal}}$ fixes the TRIT type from the phase mark r preserved in the state; and $\mathsf{Tra}^{\mathrm{cal}}$ adds exactly one symbol to the prefix, one segment to the cylinder, and its ordered pair of endpoints to the boundary. By induction on r, the explicit tuples $y_{\varepsilon,r}(x)$ and $z_{\varepsilon,r}(x)$ satisfy the predicates of the indicated relations, and the unary update produces the state at the next level. The identity $\rho^{\mathrm{cal}}_{d_{n,r}+1,d_{n,r}}(x_{\varepsilon,r}(x)) = x_{\varepsilon,r-1}(x)$ is the definition (19.46); its well-definedness follows from uniqueness of the final ledger. The composition of the nine truncation maps recovers x, proving (19.47) without separately inverting the overwritten fields.

The phase $9 \to 1$ occurs exactly once. By (19.22) and the cocycle law, the linear memory component transports the preceding memory, and its translational component adds exactly u in the direct limit. The compatibility $\jmath_{d+1}\jmath_{d+1,d} = \jmath_d$ proves (19.48); the return concerns the phase, not the enriched state.

By (19.23), eight edges have zero increment, and the edge completing $\mathfrak{p}_\varepsilon$ has the computed increment $\chi(\varepsilon,\varepsilon)\kappa_{\mathrm{Carr}}$. The laws (19.26) and (19.21) give (19.49). The ledger preserves $(\mathfrak{p}_\varepsilon, 2\varepsilon, r_3(2\varepsilon), \chi(\varepsilon,\varepsilon))$; the three values of this ledger are distinct even when the scalar increment is zero. Hence the images are not identified, and the block decoder recovers a unique ε.

The edge definitions depend only on the new phase and concatenate, rather than replace, the prefix, cylinder, boundary, and ledger. They are also invariant under $\jmath_d\,\mathrm{mem}_d \mapsto \jmath_d\,\mathrm{mem}_d + nu$ and under translation of the input carry. Consequently, the same construction applies at the endpoint of any turn. For every finite state, the explicit sequence

$$x, \quad \gamma_{0,k_n}(x), \quad \gamma_{0,k_{n+1}}\gamma_{0,k_n}(x), \quad \dots$$

is a compatible infinite tail. This proves the survival of all constructed states and the repeated availability of the three branches, without postulating either property. Consequently, in accordance with the preceding level definitions, we may finally identify

$$H^{\mathrm{cal}}_{k_n} := G_n \tag{19.50}$$

for every $n \geq 0$. □

The preceding closure distinguishes the three clocks involved. Nine emissions return the phase in C_9 and advance memory. Twelve turns, that is, 108 emissions, also return the position of the observable calendar; even then, neither the ledger nor memory is erased. The operator $\mathcal{K}_{\mathrm{ph}}$ acts only afterward, as a recognition of reduced memory from the already constructed turns; it never generates the edges of $\mathrm{Ext}^{\mathrm{enr,cal}}$.

Let

$$\upsilon : \mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \qquad \upsilon(0) = 0, \quad \upsilon(1) = +1, \quad \upsilon(2) = -1$$

be the balanced representative. For $b = (b_1, \dots, b_6) \in \mathbb{F}_3^6$, the block extension is the ordered composition of six turns

$$\Gamma^{[54]}_{b,k} := \gamma_{\upsilon(b_6),k+45} \circ \gamma_{\upsilon(b_5),k+36} \circ \gamma_{\upsilon(b_4),k+27} \circ \gamma_{\upsilon(b_3),k+18} \circ \gamma_{\upsilon(b_2),k+9} \circ \gamma_{\upsilon(b_1),k}. \tag{39d}$$

At block level one always uses $k = k_{6N} = 1 + 54N$; consequently, each factor has the domain constructed by the preceding factor. Fix a canonical APP seed $x_\star \in H^{\mathrm{cal}}_1$ and define recursively

$$\begin{aligned} Y_0 &:= \{x_\star\}, \qquad Y_N \subseteq H^{\mathrm{cal}}_{1+54N}, \\ Y_{N+1} &:= \{\Gamma^{[54]}_{b,1+54N}(x) : x \in Y_N,\ b \in \mathbb{F}_3^6\}, \\ \rho^{\mathrm{blk}}_{N+1,N} &:= \rho^{\mathrm{cal}}_{1+54(N+1),1+54N} \restriction Y_{N+1}, \\ \mathrm{Ext}^{\mathrm{cal}}_N(x) &:= \{\Gamma^{[54]}_{b,1+54N}(x) : b \in \mathbb{F}_3^6\}. \end{aligned} \tag{39e}$$

The notation $\rho^{\mathrm{cal}}_{b,a}$ denotes the composition of all deletion maps between levels b and a. By construction, every extension in $\mathrm{Ext}^{\mathrm{cal}}_N(x)$ is a chain of fifty-four TPK edges, not a jump of the Markov quotient.

The reader is defined directly on the ledger, before invoking a representation through Γ_b. The preceding lemma provides a decoder

$$\mathrm{dec}_9 : \{\text{generated nine-edge segments}\} \longrightarrow \{-1, 0, +1\},$$

which reads the TRIT pair and its reduction on the segment. If $\mathrm{Led}^{(9)}_{j,i}(x)$ denotes the i-th nonadic segment of block j in the ledger of x, set

$$\beta_{\mathrm{blk},N}(x) := \Big(\big(\upsilon^{-1} \operatorname{dec}_9(\mathrm{Led}^{(9)}_{j,i}(x)) \big)_{i=1}^{6} \Big)_{j=0}^{N-1}. \tag{19.51}$$

For $N = 0$, the value is the empty word. This definition does not presuppose that the decomposition of x as a composition of maps Γ_b is unique.

We now specialize the preceding arithmetic conjugacy to $p = 3$, $d = 6$, and $A = I_6$. If $\pi_{N+1,N} : (\mathbb{Z}/3^{N+1}\mathbb{Z})^6 \to (\mathbb{Z}/3^N\mathbb{Z})^6$ is the canonical reduction, define

$$\Phi_N := \Psi_N^{-1} \circ \beta_{\mathrm{blk},N} : Y_N \longrightarrow (\mathbb{Z}/3^N\mathbb{Z})^6. \tag{19.52}$$

This definition does not assign a coordinate after observing a branch: the reader $\beta_{\mathrm{blk},N}$ is obtained from the already constructed TPK ledger, and Ψ_N^{-1} is the explicit inverse, fixed beforehand, of the unfiltered arithmetic tower.

Proposition 19.22 (Finite conjugacy and complete covering of the fiber)**.** *For every $N \geq 0$,*

$$\beta_{\mathrm{blk},N} : Y_N \xrightarrow{\cong} (\mathbb{F}_3^6)^N$$

is a bijection compatible with truncations, and for every $x \in Y_N$

$$\begin{aligned} \beta_{\mathrm{blk},N+1}\big[\mathrm{Ext}^{\mathrm{cal}}_N(x)\big] &= \{\beta_{\mathrm{blk},N}(x)^\frown b : b \in \mathbb{F}_3^6\}, \\ \mathrm{pref}_N \circ \beta_{\mathrm{blk},N+1} &= \beta_{\mathrm{blk},N} \circ \rho^{\mathrm{blk}}_{N+1,N}. \end{aligned} \tag{40$_0$}$$

The maps Φ_N are bijections and satisfy the intertwining identity

$$\pi_{N+1,N} \circ \Phi_{N+1} = \Phi_N \circ \rho^{\mathrm{blk}}_{N+1,N}. \tag{19.53}$$

Consequently, the tower of generated histories and the Hensel arithmetic tower are isomorphic as finite inverse systems.

Proof. The assertion is immediate for $N = 0$. Assume that it holds at level N. For each word $b \in \mathbb{F}_3^6$, the preceding lemma constructs an extension $\Gamma^{[54]}_{b,1+54N}(x)$. If $b \neq b'$, the first coordinate at which they differ determines a distinct TRIT pair and cocycle in the first divergent nonadic segment of the genealogical ledger. That ledger preserves the ordered edges, not merely the final sum of their carries; hence the two histories remain distinct even if carries from later turns cancel. The direct reading (19.51) then gives

$$\beta_{\mathrm{blk},N+1}\big(\Gamma^{[54]}_{b,1+54N}(x)\big) = \beta_{\mathrm{blk},N}(x)^\frown b. \tag{19.54}$$

The six successive trisections produce exactly 3^6 distinct extensions. Deleting the last fifty-four steps recovers x, and deleting the final block of the ledger recovers $\beta_{\mathrm{blk},N}(x)$. This simultaneously proves bijectivity and the compatibility of $\beta_{\mathrm{blk},N}$ with truncations.

Proposition 19.19 independently proves that Ψ_N is bijective and that

$$\mathrm{pref}_N \circ \Psi_{N+1} = \Psi_N \circ \pi_{N+1,N}.$$

Composing this equality with the compatibility of β_{blk} and applying Ψ_N^{-1} yields

$$\pi_{N+1,N}\Psi_{N+1}^{-1}\beta_{\mathrm{blk},N+1} = \Psi_N^{-1}\beta_{\mathrm{blk},N}\rho^{\mathrm{blk}}_{N+1,N},$$

which is exactly (19.53). The equality does not arise by defining an ad hoc coordinate on each branch: it results from composing two systems of constructed bijections that are compatible at every depth. □

The inverse limit

$$\Omega_{\Gamma}^{\mathrm{cal}} := \varprojlim_{N} (Y_N, \rho_{N+1,N}^{\mathrm{blk}}) \tag{40a$_0$}$$

is the history limit of the TPK model constructed edge by edge in the preceding lemma. Its projection under π_{obs} preserves the configurations of the observable calendar, but the assertion does not depend on identifying this autonomous model with an ambient relation defined only by necessary conditions. The finite bijections pass to the limit and produce the cylinder homeomorphism

$$\Lambda : (\mathbb{F}_3^6)^{\mathbb{N}} \xrightarrow{\cong} \Omega_{\Gamma}^{\mathrm{cal}}, \qquad \beta_{\mathrm{blk}} := \Lambda^{-1}. \tag{40b$_0$}$$

Denote the canonical projection by $\rho_{\infty,N}^{\mathrm{blk}} : \Omega_{\Gamma}^{\mathrm{cal}} \to Y_N$. Thus, Λ establishes an equivalence compatible with truncations: each ternary tape encodes a TPK history, and every prefix is extended through edges of the same generator.

The subcylinders are now defined as subsets of the limit, rather than as pairs of suffixes. For $x \in Y_N$ and $b \in \mathbb{F}_3^6$, let

$$\mathscr{Z}_N(x) := \{\xi \in \Omega_{\Gamma}^{\mathrm{cal}} : \rho_{\infty,N}^{\mathrm{blk}}(\xi) = x\}, \tag{19.55}$$

$$\mathscr{Z}_{N+1}(x; b) := \{\xi \in \Omega_{\Gamma}^{\mathrm{cal}} : \rho_{\infty,N+1}^{\mathrm{blk}}(\xi) = \Gamma_{b,1+54N}^{[54]}(x)\}. \tag{19.56}$$

Theorem 19.23 (Effective partition into 729 refinement subcylinders)**.** *For every $N \geq 0$ and every $x \in Y_N$, there is a disjoint union*

$$\mathscr{Z}_N(x) = \bigsqcup_{b \in \mathbb{F}_3^6} \mathscr{Z}_{N+1}(x; b). \tag{19.57}$$

The $3^6 = 729$ terms are nonempty clopen refinement subcylinders; each is realized by fifty-four edges of the constructed TPK.

Proof. Let $\xi \in \mathscr{Z}_N(x)$. Its level-$N+1$ coordinate belongs to Y_{N+1}, truncates to x, and, by the bijection $\beta_{\mathrm{blk},N+1}$, has a unique final block $b \in \mathbb{F}_3^6$. Identity (19.54) then implies $\rho_{\infty,N+1}^{\mathrm{blk}}(\xi) = \Gamma_{b,1+54N}^{[54]}(x)$; therefore, ξ belongs to one of the terms on the right-hand side. The reverse inclusion follows immediately by truncation. If two terms intersected, their level-$N+1$ coordinates would be equal and the ledger decoder would give $b = b'$; hence the union is disjoint.

Each endpoint $\Gamma_b^{[54]}(x)$ admits an infinite tail obtained by iterating the zero-block extension; consequently, every term is nonempty. Since Y_{N+1} is finite and discrete, the terms are inverse images of points under a continuous projection of the inverse limit and are therefore clopen. The very definition of $\Gamma_b^{[54]}$ establishes its fifty-four edges. □

The family $(\Phi_N)_N$ finally induces the homeomorphism of limits

$$\Phi_{\infty} : \Omega_{\Gamma}^{\mathrm{cal}} \xrightarrow{\cong} \mathbb{Z}_3^6, \qquad \Phi_{\infty}(\xi) := \big(\Phi_N(\rho_{\infty,N}^{\mathrm{blk}}(\xi))\big)_{N \geq 0}, \tag{19.58}$$

where $\mathbb{Z}_3^6 = \varprojlim_N (\mathbb{Z}/3^N\mathbb{Z})^6$. Equation (19.53) guarantees that the right-hand side is a compatible class, and the finite bijections prove injectivity, surjectivity, and continuity in both directions. Under Φ_{∞}, $\mathscr{Z}_N(x)$ is exactly the congruence class

$$\{z \in \mathbb{Z}_3^6 : z \equiv \Phi_N(x) \pmod{3^N}\},$$

and the sets $\mathscr{Z}_{N+1}(x;b)$ are its 729 refinement classes modulo 3^{N+1}. This is the effective junction between TPK extension, limit cylinders, and the Hensel tower.

For $b=(b_1,\dots,b_6)\in\mathbb{F}_3^6$, set

$$\nu(b):=\sum_{i=1}^{6}[b_i]_3\,3^{6-i}\in\{0,\dots,728\},\tag{40c$_0$}$$

where $[b_i]_3\in\{0,1,2\}$ is the positional representative. The 243 visible words in the initial census are the image of the first emission; they are not to be confused with the 3^6 capacity of each extension block. Finally,

$$X^{\mathrm{cal}}_{\infty,\mathbb{Z}}:=\mathbb{Z}_{\mathrm{HMT}}\times\Omega^{\mathrm{cal}}_{\Gamma}.$$

For a prefix $s=(m;b_0,\dots,b_{N-1})$, define in the generated arithmetic of $\mathbb{Q}_{\mathrm{HMT}}$

$$a_N(s):=m+\sum_{q<N}\frac{\nu(b_q)}{729^{q+1}},\qquad \mathrm{Succ}(s):=\{(m;b_0,\dots,b_{N-1},b):b\in\mathbb{F}_3^6\}.$$

If $(m;\beta_{\mathrm{blk},N}(x))$ denotes the prefix obtained by prepending m to the tuple $\beta_{\mathrm{blk},N}(x)$, (40$_0$) gives the typed equality

$$\mathrm{Succ}(m;\beta_{\mathrm{blk},N}(x))=\{(m;\beta_{\mathrm{blk},N+1}(y)):y\in\mathrm{Ext}^{\mathrm{cal}}_N(x)\}.$$

From their first definition, the half-open cylinders are subsets of the HMT completion:

$$I^{\mathrm{HMT}}_N(s):=\{r\in\mathbb{R}_{\mathrm{HMT}}:a_N(s)\le_{\mathrm{HMT}}r<_{\mathrm{HMT}}a_N(s)+729^{-N}\}.$$

They materially satisfy

$$I^{\mathrm{HMT}}_N(s)=\bigsqcup_{y\in\mathrm{Succ}(s)}I^{\mathrm{HMT}}_{N+1}(y),\qquad|\mathrm{Succ}(s)|=729,\qquad\mathrm{diam}_{\mathrm{HMT}}\,I^{\mathrm{HMT}}_N(s)=729^{-N}\longrightarrow0.\tag{40a}$$

The boundary convention identifies the two representations only after they have been constructed. The partition (40a), nesting, and contraction produce

$$\begin{aligned}&P_{\mathrm{ar}}:X^{\mathrm{cal}}_{\infty,\mathbb{Z}}\twoheadrightarrow\mathbb{R}_{\mathrm{HMT}},\\&P_{\mathrm{ar}}(m,\xi):=\text{the unique element of }\bigcap_N C^{\mathrm{cl}}_N\big(m;\beta_{\mathrm{blk},N}(\rho^{\mathrm{blk}}_{\infty,N}(\xi))\big),\\&C^{\mathrm{cl}}_N(s):=\overline{I^{\mathrm{HMT}}_N(s)}.\end{aligned}\tag{40}$$

Therefore,

$$X^{\mathrm{cal}}_{\infty,\mathbb{Z}}/{\sim_{P_{\mathrm{ar}}}}\;\cong\;\mathbb{R}_{\mathrm{HMT}}.\tag{41}$$

The other publication P_{exc} preserves incidence, boundary and memory toward Paley–Witt–Golay–Leech and, from there, toward $V^{\natural}$–Monster. Both come from the same antecedent of (35); neither the Archimedean coordinate nor exceptional recognition selects the seeds. The five constructions sp, sol, gau, coh, det remain consubstantial within $C^{\mathrm{disc}}_{\mathrm{cont}}$ and are not separated to carry out the cardinal step.

19.10.2.3 Descent of operations to the evaluation quotient

The preceding quotient identifies histories publishing the same value, not their complete registers. The following criterion characterizes exactly when an operation on histories induces an operation on that

quotient without choosing a privileged history in each fibre; its formulation continues the genealogical construction established in the preceding sections.

Proposition 19.24 (Descent criterion through evaluation fibres)**.** *Let $q : \Omega \to J$ be the evaluation of a genealogical chart onto its image. Let $\star : D \to \Omega$ be an operation with $D \subseteq \Omega^2$ saturated by the fibres of $q \times q$. There exists a unique operation $\overline{\star} : (q \times q)(D) \to J$ such that*

$$q(x \star y) = q(x) \,\overline{\star}\, q(y)$$

if and only if, for pairs in the domain,

$$q(x) = q(x'), \quad q(y) = q(y') \quad \Longrightarrow \quad q(x \star y) = q(x' \star y').$$

Proof. If the visible operation exists, its result depends only on the two values, which proves necessity. For sufficiency, define $s \,\overline{\star}\, t = q(x \star y)$, where $q(x) = s$ and $q(y) = t$. Saturation preserves the domain when representatives change; the stated condition preserves the result. Surjectivity of $q \times q$ onto the image of the domain proves uniqueness. □

A coordinate does not automatically replace a history as an argument of all its operations: the criterion determines when that substitution is legitimate. In particular, transporting an operation through a bijection is not enough to identify it with a native operation preceding the reading. If a chart Ω is compact and q is continuous and surjective onto a Hausdorff space J, evaluation also induces a homeomorphism $\Omega/{\sim_q} \to J$: it is a continuous bijection from a compact space to a Hausdorff space. It is the *quotient*, and not necessarily the history space, that is topologically identified with J.

19.10.2.4 Internal reconstruction of the chart and uniform lifting

Within $\mathfrak{G}_{\mathrm{HMT}}(h, m_\infty)$, the reconstruction in (35)–(41) is carried out with the same codes and operators. Denote by $\mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}$ the internal HMT completion. The uniqueness of the complete Archimedean ordered field provides, **after** its construction, the internal isomorphism

$$\eta : \mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}} \xrightarrow{\cong} \mathbb{R}^{\mathfrak{G}}. \tag{41$_0$}$$

For the corresponding internal interval $I_N^{\mathfrak{G}}(s) := I_N^{\mathrm{HMT}}(s) \cap \mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}$, the isomorphism preserves endpoints and order:

$$\eta[I_N^{\mathfrak{G}}(s)] = \{r \in \mathbb{R}^{\mathfrak{G}} : a_N(s) \le r < a_N(s) + 729^{-N}\}. \tag{41$_0$a}$$

The surjectivity of (40) covers every element of that completion, not only those obtained in a finite execution.

So that "every" is not concealed in the definition of the codomain, we construct the two injections relating the line to its internal power set. Let

$$E(A) = \sum_{n \ge 0} \frac{2\mathbf{1}_A(n)}{3^{2n+2}} \qquad \big(A \in \mathcal{P}^{\mathfrak{G}}(\omega)\big). \tag{41a}$$

This representation places 0 or 2 only in the even ternary positions; its image is contained in $[0, 1/4]$. It therefore never reaches the right endpoint of the root cylinder, and two distinct subsets have a different first even position: E is injective and is not subject to boundary ambiguity. Write $E_{\mathrm{HMT}} := \eta^{-1} \circ E$.

Given A, begin with the root cylinder containing $E_{\mathrm{HMT}}(A)$. If $s_N(A)$ is the prefix already chosen, write $s_N(A) = (0; b_0(A), \dots, b_{N-1}(A))$. The partition (40a), transported by η^{-1}, supplies a unique half-open subcylinder $s_{N+1}(A)$ containing $E_{\mathrm{HMT}}(A)$. By (40$_0$), that extension is also an admissible TPK extension. By induction,

$$s_{N+1}(A) \restriction N = s_N(A), \qquad E_{\mathrm{HMT}}(A) \in I_N^{\mathrm{HMT}}(s_N(A)) \quad (N < \omega). \tag{41b$_0$}$$

The recursion is a uniform formula with parameters A, h. Since both are internal, Replacement in the model of Lemma 19.18 forms the complete sequence of blocks $(b_q(A))_{q<\omega}$. Without confusing the tape with the sequence of prefixes, define

$$\xi_A := \Lambda\big((b_q(A))_{q<\omega}\big) \in \Omega_\Gamma^{\mathrm{cal},\mathfrak{G}}.$$

This yields the **uniform lift of the Cantor embedding**

$$Q : \mathcal{P}^{\mathfrak{G}}(\omega) \longrightarrow X_{\infty,\mathbb{Z}}^{\mathrm{cal},\mathfrak{G}}, \qquad Q(A) := (0, \xi_A), \qquad P_{\mathrm{ar}}(Q(A)) = E_{\mathrm{HMT}}(A). \tag{41b}$$

We do not call Q the inverse of P_{ar}: their composition publishes the embedding E_{HMT}, not the identity on the entire line. In the other direction, let $(q_n)_{n<\omega}$ be the generated enumeration of $\mathbb{Q}_{\mathrm{HMT}}$ and define

$$C(r) := \{n < \omega : \eta(q_n) < r\} \qquad (r \in \mathbb{R}^{\mathfrak{G}}).$$

The rational cut belongs to $\mathcal{P}^{\mathfrak{G}}(\omega)$, and C is injective by density. The injections E and C, followed by the internal Cantor–Bernstein theorem, prove

$$P_{\mathrm{ar}}\big[X_{\infty,\mathbb{Z}}^{\mathrm{cal},\mathfrak{G}}\big] = \mathbb{R}_{\mathrm{HMT}}^{\mathfrak{G}}, \qquad \mathbb{R}_{\mathrm{HMT}}^{\mathfrak{G}} \cong \mathbb{R}^{\mathfrak{G}} \approx \mathcal{P}^{\mathfrak{G}}(\omega). \tag{41c}$$

Here an arbitrary real is used only as a universally quantified variable to prove surjectivity of an already constructed tree. It selects neither APP, TRIT, TPK, the constants, nor the transitions of that tree.

19.10.3 Enumeration by genealogical codes and HMT condensation

Rule (37) supplies derivation names from

$$(\ulcorner \varphi \urcorner, \sigma(p_0), \dots, \sigma(p_{k-1})). \tag{42}$$

For any object a of the hierarchy, let $\beta(a)$ be its first stage and $\sigma(a)$ its least derivation tree in the recursive order of Lemma 19.18. The term $\sigma(a)$ is not required to be a natural number when the stage is uncountable. The relation

$$a \prec_{\mathfrak{G}} b \quad \Longleftrightarrow \quad (\beta(a), \sigma(a)) <_{\mathrm{lex}} (\beta(b), \sigma(b)) \tag{42a}$$

is the global genealogical well-order $<_{\mathfrak{G}}$ induced by the constructor. Repeated names are eliminated by taking the least one; neither a real nor an enumeration of the line is inserted as data.

Lemma 19.25 (Internal countability of levels below $\omega_1^{\mathfrak{G}}$)**.** *For every $\beta < \omega_1^{\mathfrak{G}}$, the level $\mathfrak{G}_\beta(h, m_\infty)$ is countable within $\mathfrak{G}_{\mathrm{HMT}}(h, m_\infty)$, and possesses a canonical surjection*

$$e_\beta : \omega \twoheadrightarrow \mathfrak{G}_\beta(h, m_\infty). \tag{43}$$

Proof. The initial level has a countable code. If e_β enumerates level β, the formula codes and finite tuples of parameters in (42) form a countable set and enumerate the successor. If $\lambda < \omega_1^{\mathfrak{G}}$ is a limit, there is internally a surjection $s : \omega \twoheadrightarrow \lambda$; take the $<_{\mathfrak{G}}$-least one, in the order already constructed in Lemma 19.18, and diagonalize the $e_{s(n)}$. Finally, among all surjections so obtained, define e_β as the $<_{\mathfrak{G}}$-least. The recursion $\beta \mapsto e_\beta$ is a uniformly definable class of $\mathfrak{G}$; every e_β, and every restriction of the recursion to an ordinal, belongs to $\mathfrak{G}$. It uses only $h, u_{h,m}$ and the global well-order. □

Lemma 19.26 (Condensation of internal reals below $\omega_1^{\mathfrak{G}}$)**.** *Every real in $\mathfrak{G}_{\mathrm{HMT}}(h, m_\infty)$ appears at a level whose index is smaller than $\omega_1^{\mathfrak{G}}$.*

Proof. Let $r \subseteq \omega$ be a real in $\mathfrak{G}$, and let $B := \mathfrak{G}_0(h, m_\infty)$. This base is transitive and internally countable. With the codes of natural numbers and finite pairs already fixed, $\mathrm{Ord} \cap B = \omega + 1$. By induction,

$$\mathrm{Ord} \cap \mathfrak{G}_\alpha = (\omega + 1) + \alpha.$$

The successor adds the ordinal formed by the ordinals of the preceding level; it cannot add a larger one, because that would not be a subset of that level. Limits are unions. Choose a limit Θ, with $\Theta > \omega^2$, such that $r \in \mathfrak{G}_\Theta$. Thus $\mathrm{Ord} \cap \mathfrak{G}_\Theta = \Theta$, and every level $\mathfrak{G}_\xi$, with $\xi < \Theta$, belongs to $\mathfrak{G}_\Theta$. Internally form the expanded structure

$$\mathcal{A}_\Theta = \langle \mathfrak{G}_\Theta, \in, u_{h,m}, H_\Theta, <_{\mathfrak{G}} \restriction \mathfrak{G}_\Theta \rangle$$

with the level relation as a predicate of its signature, $H_\Theta = \{\langle \xi, x \rangle : \xi < \Theta \,\&\, x \in \mathfrak{G}_\xi\}$, and its definable Skolem functions. Let

$$Y = \mathrm{Hull}^{\mathcal{A}_\Theta}(B \cup \{B, r\}) \prec \mathcal{A}_\Theta.$$

Closure under the countable family of Skolem functions shows within $\mathfrak{G}$ that Y is countable. Let $c : Y \to M$ be its Mostowski collapse and

$$\bar{\Theta} := \mathrm{otp}(Y \cap \Theta) < \omega_1^{\mathfrak{G}}.$$

Since $B \subseteq Y$ is transitive, induction on membership shows that c fixes every element of B, as well as B itself. In particular, it fixes $\omega, u_{h,m}, h, m_\infty$. Since $r \subseteq \omega \subseteq Y$, it also follows that $c(r) = r$.

We prove by induction on $\xi \in Y \cap \Theta$ that

$$c\text{“}\big(Y \cap \mathfrak{G}_\xi(h, m_\infty)\big) = \mathfrak{G}_{c(\xi)}(h, m_\infty). \tag{43a}$$

For $\xi \in Y \cap \Theta$, the predicate H_Θ defines the set $\mathfrak{G}_\xi$ from ξ, so $\mathfrak{G}_\xi \in Y$. Relativizing each formula to that set gives $Y \cap \mathfrak{G}_\xi \prec \mathfrak{G}_\xi$, in the language into which Lemma 19.16 translates the HMT definitions.

The base case of (43a) has already been proved because the collapse fixes B. At a successor, if $\xi + 1 \in Y$, then also $\xi \in Y$ and $c(\xi + 1) = c(\xi) + 1$. For $a \in Y \cap \mathfrak{G}_{\xi+1}$, fix a finite formula defining it over $\mathfrak{G}_\xi$. The assertion

$$a = \{z \in \mathfrak{G}_\xi : \varphi^{\mathfrak{G}_\xi}(z, \bar{p})\}$$

is an ordinary formula over $\mathfrak{G}_\Theta$, with parameters $a, \mathfrak{G}_\xi$. Elementarity supplies the parameters $\bar{p}$ within $Y \cap \mathfrak{G}_\xi$. The induction hypothesis and the elementarity of that level show that $c(a)$ is the subset defined by the same formula and the collapsed parameters over $\mathfrak{G}_{c(\xi)}$. Conversely, given a formula and parameters at the collapsed level, lift the parameters to the original level. The subset defined there exists, is unique, and belongs to Y by elementarity. Its collapse is the required subset. This proves both inclusions at the successor stage without introducing a uniform truth predicate.

For a limit $\lambda \in Y \cap \Theta$, elementarity supplies, for every $x \in Y \cap \mathfrak{G}_\lambda$, a witness $\xi \in Y \cap \lambda$ with $H_\Theta(\xi, x)$. Therefore,

$$Y \cap \mathfrak{G}_\lambda = \bigcup_{\xi \in Y \cap \lambda} (Y \cap \mathfrak{G}_\xi), \qquad c\text{“}(Y \cap \lambda) = c(\lambda).$$

Continuity of the hierarchy and the induction hypothesis give (43a) at the limit. Cofinality of $Y \cap \lambda$ in the ambient ordinal λ is not required: its collapsed indices range over $c(\lambda)$.

Finally, every $x \in Y$ has a witness $\xi \in Y \cap \Theta$ with $H_\Theta(\xi, x)$, and $\bar{\Theta}$ is a limit. Taking unions,

$$M = \bigcup_{\eta < \bar{\Theta}} \mathfrak{G}_\eta(h, m_\infty) = \mathfrak{G}_{\bar{\Theta}}(h, m_\infty).$$

Since $c(r) = r \in M$, it follows that $r \in \mathfrak{G}_{\bar{\Theta}}$ with $\bar{\Theta} < \omega_1^{\mathfrak{G}}$. □

The lemma does not say that a finite-length prefix enumerates the line. It says that each generated hereditary object has a first transfinite level and a first genealogical code within that level. This is precisely the information that the extensional reduct loses when it retains only the published value.

19.10.4 Injection of the generated line into the first uncountable ordinal of the genealogical closure

Let $\iota(r) \subseteq \omega$ be the rational cut of $r \in \mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}$ relative to the enumeration of $\mathbb{Q}_{\mathrm{HMT}}$ previously constructed from APP integers and quotients. The map $r \mapsto \iota(r)$ is definable from r and that enumeration, so $\iota(r) \in \mathfrak{G}$; Lemma 19.26 places it at a stage with countable index. Define

$$\begin{aligned} \beta(r) &:= \min\{\beta : \iota(r) \in \mathfrak{G}_{\beta+1}(h, m_\infty)\}, \\ n(r) &:= \min\{n : e_{\beta(r)+1}(n) = \iota(r)\}, \\ j_{\mathrm{HMT}}(r) &:= \omega \cdot \beta(r) + n(r). \end{aligned} \tag{44}$$

Definition (44) is uniform in the parameters h, m_∞. Since $\mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}$ is a set of $\mathfrak{G}$, Separation fixes the domain of the definition, and Replacement in $\mathfrak{G}$ forms its graph:

$$\{\langle r, \omega \cdot \beta(r) + n(r)\rangle : r \in \mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}\} \in \mathfrak{G}.$$

Therefore, j_{HMT} is an internal function of $\mathfrak{G}$.

By Lemmas 19.25 and 19.26, $j_{\mathrm{HMT}}(r) < \omega_1^{\mathfrak{G}}$ for every generated real. If $j_{\mathrm{HMT}}(r) = j_{\mathrm{HMT}}(s)$, ordinal division by ω recovers the same pair (β, n); (43) recovers the same rational cut and, by completeness, $r = s$. Hence

$$j_{\mathrm{HMT}} : \mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}} \hookrightarrow \omega_1^{\mathfrak{G}} \tag{45}$$

is an injection constructed from the rank and name of the HMT genealogy. Its existence uses precisely the semantic closure (37), the well-ordering of names from Lemma 19.18, and the condensation (43a); it does not receive the Continuum Hypothesis in order to choose the stage. The subsequent set-theoretic identification describes this same machinery in different notation without reversing its causal provenance.

19.10.5 Injection of the first uncountable ordinal into the line and cardinal equivalence

For every $\beta < \omega_1^{\mathfrak{G}}$, Lemma 19.25 allows a subset $w_\beta \subseteq \omega$ to encode a countable well-order of type β. To avoid confusing the empty order with the one-point order, both the domain $D \subseteq \omega$ and its strict order relation $R \subseteq D^2$ are encoded. Once a bijective pairing $\langle \cdot, \cdot \rangle : \omega^2 \to \omega$ is fixed, the code is

$$w(D, R) = \{2n : n \in D\} \cup \{2\langle a, b\rangle + 1 : (a, b) \in R\}.$$

The even bits recover the domain, and the odd bits recover the relation. Take the $<_{\mathfrak{G}}$-least real encoding such a well-order. Since the property "encodes a well-order of type β" and the global well-order are internal, Replacement forms the function $\beta \mapsto w_\beta$ within $\mathfrak{G}$. Distinct order types produce distinct codes. The ternary map, already used in (41a), is defined by

$$E(w) = \sum_{n \geq 0} \frac{2\mathbf{1}_w(n)}{3^{2n+2}} \tag{46}$$

and is injective and avoids the endpoint of the root cylinder. The lifting (41b) ensures that $E_{\mathrm{HMT}}(w_\beta)$ belongs to the line published by an HMT history. Thus

$$k_{\mathrm{HMT}}(\beta) := E_{\mathrm{HMT}}(w_\beta), \qquad k_{\mathrm{HMT}} : \omega_1^{\mathfrak{G}} \hookrightarrow \mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}. \tag{47}$$

The injections (45) and (47), constructed without using the equality sought, give by Cantor–Bernstein

$$\left|\mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}\right| = \aleph_1^{\mathfrak{G}}. \tag{48}$$

Since (41c) identifies the complete HMT line with $\mathcal{P}^{\mathfrak{G}}(\omega)$, (48) rewrites exactly as

$$\boxed{\mathfrak{G}_{\mathrm{HMT}}(h, m_\infty) \models 2^{\aleph_0} = \aleph_1.} \tag{49}$$

The equality does not arise from 729, from an informal product of cardinality and ordinality, from the digits of a constant, or from a numerical bracket. It arises from the covering of the line by the inverse limit, the condensation of the genealogical closure, and the two explicit injections.

19.10.6 Quantification over all internal subsets

Let $A \in \mathcal{P}^{\mathfrak{G}}(\mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}})$. If A is countable, the first alternative holds. If it is not, $j_{\mathrm{HMT}}[A]$ is an uncountable subset of $\omega_1^{\mathfrak{G}}$. It cannot be bounded by a countable ordinal, since every initial segment of ω_1 is countable. Its increasing enumeration has order type $\omega_1^{\mathfrak{G}}$, and the inverse of j_{HMT} on its image produces an injection $\omega_1^{\mathfrak{G}} \hookrightarrow A$. The inclusion $A \subseteq \mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}$, (48), and Cantor–Bernstein give

$$\boxed{\forall A \in \mathcal{P}^{\mathfrak{G}}(\mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}), \qquad |A| \leq \aleph_0 \quad \text{or} \quad |A| = \left|\mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}\right|.} \tag{50}$$

Here the full power set of the integral continuum generated by HMT is quantified. The bijective transport of the folding with memory carries (50) to any of its representations without losing parts; it does not replace the injections (45)–(47).

19.10.7 Representation of the genealogical closure as $L[h, m_\infty]$

Only after (49)–(50) is the construction in (36) assigned its name in the usual set-theoretic notation. Let $L[u_{h,m}]$ be the constructible hierarchy relative to the real already produced in (33a). Since the HMT signature is countable and all its operators are graph relations encoded by h, Lemma 19.16 eliminates its symbols uniformly. This yields the equality of classes

$$\mathfrak{G}_{\mathrm{HMT}}(h, m_\infty) = L[u_{h,m}] = L[h, m_\infty]. \tag{51}$$

To prove the first inclusion, strengthen the induction: for every ordinal β, both $\mathfrak{G}_\beta$ and the sequence $\langle \mathfrak{G}_\xi : \xi \leq \beta \rangle$ belong to $L[u_{h,m}]$. The base is the transitive closure of a set constructed from ω and $u_{h,m}$; the parameters h, m_∞ are recovered from $u_{h,m}$. At a successor, the preceding level is an *internal set*, not merely a subclass of $L[u_{h,m}]$. Its satisfaction relation and the uniform elimination (37c) make it possible to form internally the set of its definable subsets. At a limit, transfinite recursion, Replacement, and Union in $L[u_{h,m}]$ form the preceding sequence and its union. Transitivity then gives $\mathfrak{G} \subseteq L[u_{h,m}]$.

Conversely, Lemma 19.18 supplies an internal transitive model of ZF, containing all ordinals and $u_{h,m}$ as an element. The recursion constructing $L[u_{h,m}]$ is carried out in that model: at every stage, satisfaction over the preceding level and relative definability are absolute between the two transitive domains. By induction, its levels coincide with those of the ambient construction and belong to $\mathfrak{G}$. Thus $L[u_{h,m}] \subseteq \mathfrak{G}$. Finally, (33a) makes $u_{h,m}$ and the pair (h, m_∞) primitive-recursively interdefinable, from which the second equality follows.

Equation (51) explicitly identifies closure with the constructible hierarchy relative to the generated parameter. This representation does not modify causal provenance: $L[u]$ neither selects the genealogical register nor defines the prolongations or the line; the decision of the continuum hypothesis follows from the exhaustiveness rule and the proved maps between (36) and (50). The composition (33a), (36), (37d) and (49)–(51), together with theorem 19.27, provides a complete and uniform expression of the realized universe, semantic coincidence and cardinal closure; none of these representations replaces APP–TRIT–TPK generation.

19.10.8 Theorem giving an affirmative resolution of the Continuum Hypothesis

Theorem 19.27 (Decision of the continuum in HMT)**.** *For every realized history m_∞ produced by the joint action* APP $\to$ TRIT $\to$ TPK $\to X_\infty^{\mathrm{enr}} \to C_{\mathrm{cont}}^{\mathrm{disc}}$*, the HMT genealogical extension* (36) *generates the internal reconstruction of the integral line* (40)*, $\mathbb{R}_{\mathrm{HMT}}^{\mathfrak{G}}$, constructs the injections* (45) *and* (47)*, and satisfies $2^{\aleph_0} = \aleph_1$ over its entire internal power set. Therefore, HMT proves the Continuum Hypothesis affirmatively for the integral continuum that it defines and generates. The notation $L[h, m_\infty]$ is a subsequent representation of that hierarchy, not the premise deciding the result.*

19.10.9 Falsification criteria for the cardinal closure

The following list identifies failures that would invalidate a step of the preceding closure. Construction, semantic, and cardinality controls are distinguished:

1. there exists a level N or a surviving state without a compatible successor in (34);
2. $u_{h,m}$ does not hereditarily encode h, m_∞, or some graph relation of Σ_{HMT} does not admit the uniform elimination (37c);
3. an object admitted by the HMT semantic rules has no term in (37a), or a term of (37a) publishes an object outside those rules, breaking (37d);
4. $\mathfrak{G}$ does not satisfy one of the axioms used, its definable well-order fails, or a purported enumeration e_β is not internal and uniform;
5. the subcylinders in (40a) do not form an exhaustive disjoint partition, their cylinders cease to contract, or the recursion ($41b_0$) does not produce an internal compatible history;
6. the collapse in Lemma 19.26 does not commute with a successor or limit stage, or there exists an internal real that appears in no level with index smaller than $\omega_1^{\mathfrak{G}}$;
7. the selection $\beta \mapsto w_\beta$ is not internal or uniform, or does not preserve order type;
8. two distinct reals have the same genealogical pair (β, n) in (44), two distinct ordinals produce the same code in (47), or one of the injections $E, C, j_{\mathrm{HMT}}, k_{\mathrm{HMT}}$ fails;
9. there exists an uncountable internal subset of the line whose range under j_{HMT} is bounded in $\omega_1^{\mathfrak{G}}$;
10. some generative stage preceding the covering (40a) uses the Continuum Hypothesis, $\mathbb{R}$, a target constant, the Committee on Data of the International Science Council (CODATA), the Particle Data Group (PDG), or a subsequent physical identification as a causal antecedent to define a seed, a relation $\mathscr{L}_\tau$, Ext, a coefficient, admissibility, or an extension of the tree. The subsequent lift ($41b_0$) does not constitute such an inversion: there an arbitrary already constructed real is a universal variable and selects, within the previously proved exhaustive partition, the unique subcylinder containing it.

No execution restricted to a preset number of levels replaces the universal quantifier of the inverse system; an effective failure of compatibility at some depth would falsify it. No subsequent set-theoretic notation proves (49) by itself. The closure jointly uses Lemma 19.18, the line–power-set identification in (41c), Lemmas 19.25–19.26, and the two injections constructed from the genealogy.

19.11 Phase, dodecaphase register, and exceptional realization of the common state

The theorem in Section 19.10 decides the cardinal question within the complete HMT state. The constructions that follow are not a separable ontological appendix: they make explicit correlated coordinates of the same enriched state. The dual projection preserves complementary information: Archimedean value in one branch, and incidence, boundary, and genealogy in the other.

The holonomy Γ_9 composes nine updates U_t on the enriched state. The reduced kernel $\mathcal{K}_{\mathrm{ph}}$ is a subsequent publication on already constructed emissions that preserves reduced phase and memory:

$$U_t \longrightarrow \Gamma_9 \longrightarrow \mathcal{K}_{\mathrm{ph}}, \qquad g(k+9) = g(k), \quad q_{\mathrm{mem},k+9} = q_{\mathrm{mem},k} + 1, \quad \widehat{x}_{k+9} \neq \widehat{x}_k.$$

The phase returns, the memory advances, and the state is not reset. By contrast, the integral dodecaphase seal is constructed through the signed reader and the Hadamard transform:

$$\mathcal{S}_{12}\ (= R_{\mathrm{sgn}}) : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}} \longrightarrow \mathcal{U}_{12}^{\mathrm{sgn}} := \operatorname{im} \mathcal{S}_{12}, \qquad U_{\mathrm{term}} \in \mathcal{U}_{12}^{\mathrm{sgn}}, \qquad K := \frac{1}{4}\mathcal{H}_{12} U \in \mathbb{Z}^{12}.$$

The coordinate U_{term} is obtained from the enriched state by means of the oriented register and the channel readers:

$$\mathcal{S}_{12}(x_{\mathrm{term}}) = \Pi_H\Big(\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\mathrm{term}}))\Big) = U_{\mathrm{term}}.$$

Section 11.4 develops that evaluation, the reversible obtaining of K and the transverse return to the same twenty-five channels. The serialization of C_{term} consumed by the Python checks is a verification entry point, not a mathematical generator or a restriction of the printed proof. The vector K does not by itself produce either branch. Together with the correlated publications P, E, Φ, it feeds integer closure

$$Z_m := P_m + E_m - \Phi_m - K_m, \qquad R_{N+1}(Z) := \sum_{r \geq 0} Z_{N+1+r}\, 1000^{-r-1},$$

$$c_{N+1}^{\infty} := \lfloor R_{N+1}(Z) \rfloor, \qquad s_m = Z_m + c_{m+1}^{\infty}, \qquad A_m = s_m \bmod 1000, \qquad c_m = \frac{s_m - A_m}{1000},$$

$$\big(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}; c_{N+1} = c_{N+1}^{\infty}\big)_{N \geq 1} \xrightarrow{\mathsf{T}_{\alpha,\infty}} (\alpha_N)_N \longmapsto \alpha_{\mathrm{HMT}} := \varprojlim_N \alpha_N.$$

The condition $c_{13} = 0$ belongs exclusively to the isolated execution of twelve blocks and publishes the rational $\alpha_{12}^{(0)}$. It is not the boundary of the infinite arrow: in the latter, each horizon receives the quotient determined by its signed tail; in particular, the first turn of the complete section has $c_{13}^{\infty} = -1$. The exceptional publication retains a distinct codomain:

$$B_K = \beta_{\mathrm{arq}}(K) = \{i : K_i \geq 729\}, \qquad s^{(0)} = P + E - \Phi - K,$$

$$H_\alpha^- = \eta_{\mathrm{arq}}(s^{(0)}) = \eta_{\mathrm{exc}}(B_K, P_\pi, H_e, H_\varphi, H_{\mathrm{APP}}),$$

$$\widetilde{\mathcal{I}_*} = (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\mathrm{APP}}) \longmapsto o_* = 1 \longmapsto v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12},$$

$$N_{\alpha,0} = \{x \in N(C_W) : \langle x, v_\alpha \rangle \equiv 0 \pmod 3\}, \qquad N_\alpha = N_{\alpha,0} + \mathbb{Z}\frac{v_\alpha}{3} \cong \Lambda_{24},$$

$$C_W \longmapsto N(C_W) := \bigcup_{c \in C_W} \big(A_2^{12} + \phi(c)\big) \xrightarrow{\mathrm{Nbr}_{v_\alpha}} N_\alpha \cong \Lambda_{24} \longrightarrow V^\natural \xrightarrow{\mathrm{Aut}} \mathbb{M}.$$

The scalar α_{HMT}, the face B_K, the hexad H_α^-, the vector v_α, and the lattice N_α belong to different codomains. They share provenance; they are not interchangeable objects.

Proposition 19.28 (Causal unity of the correlated publications)**.** *The objects*

$$\mathcal{K}_{\mathrm{ph}}, \quad K, \quad \pi_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}}, \quad H_\alpha^-, \quad v_\alpha, \quad N_\alpha, \quad V^\natural$$

are not external data or retroactive selectors of the cardinal inference. They are all produced as coordinates, publications or realizations of the two projections of the unique enriched state. Cardinal comparison does not consult their conventional values as oracles, but does operate on the complete HMT structure preserving its genealogy, regions and memory.

Proof. The cardinal proof uses the inverse system of histories, the integral line, semantic closure, the ordinal classifier and the two injections within the same APP–TRIT–TPK state. No conventional digit determines those maps. In the internal constructive order, $\mathcal{K}_{\mathrm{ph}}$ factors the memory of Γ_9; the seal K is obtained reversibly from the terminal coordinate produced by $\mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}$; the quadrirelation is published from the coinductive sections; and the flag, the vector v_α, the Leech neighbour and $V^\natural$ are obtained by the exceptional projection. The following sections provide the complete constructions and their proofs. □

In particular, the following type separations are preserved:

$$\mathcal{K}_{\mathrm{ph}} \neq K, \qquad \mathcal{H}_{12} \neq H_\alpha^-, \qquad \alpha_{\mathrm{HMT}} \neq v_\alpha \neq N_\alpha.$$

Nor is there a canonical arrow $W_{24} \to \Lambda_{24}$: the Witt design and the ternary Leech neighbor are distinct prolongations of the common code. The Monster group acts on $V^\natural$, not on decimal digits, TPK routes, or codes of α.

19.11.1 Reflection by witness closure and construction of the image

The Collection–Replacement step of Lemma 19.18 admits the following explicit proof. The hierarchy already generated from the operator code and the realized register is maintained; we abbreviate $\mathfrak{G} = \mathfrak{G}_{\mathrm{HMT}}(h, m_\infty)$. The argument uses Separation and Replacement in the ambient metatheory, without presupposing these schemas in $\mathfrak{G}$.

Let Φ be a finite collection of formulas. First translate their universal quantifiers using negation and existential quantification, and close the resulting collection under subformulas. For every initial level there is a later limit ordinal θ such that

$$\mathfrak{G}_\theta \models \varphi(\bar{a}) \quad \Longleftrightarrow \quad \mathfrak{G} \models \varphi(\bar{a}) \qquad (\varphi \in \Phi,\ \bar{a} \in \mathfrak{G}_\theta^{<\omega}).$$

For each fixed formula, satisfaction in the class is interpreted by relativizing its quantifiers to the definable class $\mathfrak{G}$; no uniform truth predicate is introduced.

Proof. Fix α. For each existential subformula $\exists y\, \psi(y, \bar{x})$ of Φ, the tuples $\bar{a} \in \mathfrak{G}_\alpha^{<\omega}$ of the corresponding arity that have a witness in $\mathfrak{G}$ form a set by Separation in the metatheory. Assign to each tuple the least ordinal β for which some $y \in \mathfrak{G}_\beta$ satisfies $\mathfrak{G} \models \psi(y, \bar{a})$. This ordinal exists because $\mathfrak{G}$ is the union of its levels. Replacement in the metatheory bounds these first stages of witnesses. Since Φ is finite, a single bound $f(\alpha) > \alpha$, valid for all its existential subformulas, can be chosen definably. Thus every tuple under consideration has *at least one witness* in $\mathfrak{G}_{f(\alpha)}$.

Take α_0 large enough to contain the initial parameters and set

$$\alpha_{n+1} = f(\alpha_n), \qquad \theta = \sup_{n<\omega} \alpha_n.$$

By continuity of the hierarchy, every finite tuple from $\mathfrak{G}_\theta$ belongs to some $\mathfrak{G}_{\alpha_n}$. If an existential formula of Φ with that tuple is true in $\mathfrak{G}$, it has a witness in $\mathfrak{G}_{\alpha_{n+1}} \subseteq \mathfrak{G}_\theta$. Induction on formulas now proves the equivalence: atomic formulas are preserved in the induced structure; connectives preserve the equivalence; the existential step uses that witness in one direction and the same witness, viewed in $\mathfrak{G}$, in the other. The prior translation of universal quantifiers also includes the existential formulas expressing their possible counterexamples. This proves the required finite reflection. □

Now let $F(x, y, \bar{p})$ be functional on a in $\mathfrak{G}$. Apply the construction to the formulas expressing F, the existence of its witnesses, and its image, with $a, \bar{p} \in \mathfrak{G}_\theta$. Transitivity gives $a \subseteq \mathfrak{G}_\theta$. Every $x \in a$ has its value within that level, and reflection identifies the exact image as

$$b = \{y \in \mathfrak{G}_\theta : \mathfrak{G}_\theta \models \exists x \in a\, F(x, y, \bar{p})\}.$$

This set is definable over $\mathfrak{G}_\theta$, with parameters from the same level. By the successor operation of the hierarchy, $b \in \mathrm{Def}(\mathfrak{G}_\theta) \subseteq \mathfrak{G}_{\theta+1}$. This establishes Replacement in $\mathfrak{G}$. If the existence of witnesses is retained and uniqueness is omitted, the same construction provides a set of witnesses for Collection.

Arity of names and internal selection. In the well-order of Lemma 19.18, the enumeration of the base assigns to each element its least natural-number code. At each successor stage, formula codes are ordered first; once a code is fixed, the arity of its parameter tuple is fixed. Lexicographic order is then applied to a *finite product of fixed arity* of the previously constructed well-orders. The least name represents each new value, and limit levels retain the order of first appearance. This specifies the ordering of tuples without combining different arities into a single lexicographic order. Selecting the least element of every nonempty set, together with Replacement already established, forms internally the choice functions used subsequently. This construction of names precedes the cardinal injections and receives no bijection of the continuum as data.

19.11.2 Compatibility of the cardinal classifier with the restoration of histories

Proposition 154.3 constructs, from finite invertible readers compatible with truncation, a bijection $q : H \to Y$ between the limit of enriched histories and the limit of their recoverable registers. Its domain preserves the system of levels, the operations and the memory. The cardinal argument additionally uses the Archimedean reading $p : H \twoheadrightarrow R$, where $R = \mathbb{R}^{\mathfrak{G}}_{\mathrm{HMT}}$. A history and its published value are distinct objects: the fibers of p gather the enriched realizations of the same value.

Proposition 19.29 (Internal transport of fibers and conservation of the ordinal code)**.** *Fix the closure $\mathfrak{G}$ of section 19.10. Suppose the sets H, Y, R, the bijection q, the surjection p and their defining systems belong to $\mathfrak{G}$. Define*

$$\widetilde{p} = p \circ q^{-1} : Y \longrightarrow R, \qquad \mathcal{S}(A) = q[p^{-1}(A)] \quad (A \in \mathcal{P}^{\mathfrak{G}}(R)).$$

Then $\mathcal{S}$ is an isomorphism between $\mathcal{P}^{\mathfrak{G}}(R)$ and the algebra of internal subsets of Y saturated for $\widetilde{p}$. One has

$$\mathcal{S}(A) = \widetilde{p}^{-1}(A), \qquad \widetilde{p}[\mathcal{S}(A)] = A, \qquad (j_{\mathrm{HMT}} \circ \widetilde{p}) \circ q = j_{\mathrm{HMT}} \circ p.$$

The restriction $q : p^{-1}(A) \to \mathcal{S}(A)$ is bijective. The classifier j_{HMT} of (44) preserves its code when changing recoverable register and maintains the dichotomy (50) for the values of A.

Proof. Separation forms the inverse graph of q within $Y \times H$. Replacement, already obtained by witness closure, forms the compositions and images used. For each $y \in Y$, uniqueness of $q^{-1}(y)$ gives

$$y \in q[p^{-1}(A)] \iff p(q^{-1}(y)) \in A \iff y \in \widetilde{p}^{-1}(A).$$

The surjectivity of p and the bijection q prove $\widetilde{p}[\widetilde{p}^{-1}(A)] = A$. If $B \subseteq Y$ is saturated, by definition $B = \widetilde{p}^{-1}(\widetilde{p}[B])$, so B belongs to the image of $\mathcal{S}$. Preimages preserve unions, intersections and relative complements; the surjectivity of $\widetilde{p}$ makes that operation injective. This proves the isomorphism of subset algebras. The restriction of q is bijective because q is bijective on its complete domain.

Finally, $\widetilde{p} \circ q = p$, and composition with j_{HMT} gives the declared square. The functions e_β, condensation and ordinal codes are constructed within the same closure by Lemmas 19.25 and 19.26; equality (48) and the dichotomy (50) apply to A in that domain. The classification of values is thus transported while preserving all their enriched fibers. □

The result distinguishes the two bijections involved: restoration acts between histories and registers; cardinal classification acts on the published values. Saturation relates both levels without selecting a single history per value. Witness reflection guarantees the internal formation of the maps in the argument and the preservation of the quantifiers of the declared closure.

Chapter 20

Dodecaphasic–Witt intertwiner: point–hexad incidence, reduction 11→3→1, residual S_3 action and obstruction to triple equivariance

20.1 Dodecaphase–Witt intertwining and the exact reading of $3/11$

The repeated appearance of quotient $3/11$ on the dodecaphasic screen, in the polarity of a Witt star, and in nonadic filtration raises a question stronger than the arithmetic equality of three fractions. The correct question is operatorial: to determine whether those three appearances are traces of the same positive sector viewed in different realizations and, if they are, to construct the maps intertwining them without introducing a subsequent metrological datum.

The answer has four levels. First, point–hexad incidence constructs a canonical isometry between the eleven-dimensional dodecaphase module and an eleven-dimensional eigenspace of the Witt design. Second, the Witt star acts on that same module, but does so as a signed involution; its $3/11$ is an index, not the trace of a positive projector. Third, the oriented survival sector appears in $t = 7$, where a unique space of eleven branches contains a rank-three sector. Fourth, the orientation line of that triplet, together with the internal witnesses B_K and u_K, variationally selects a unique normalizer S_3, fixes its orientation and produces the isometric sector–dodecaphase intertwining through Reynolds averaging and polar decomposition. The dodecaphase–Witt step is a second incidence isometry. In the initial labeled gauge, the two arrows do not yet realize a single representation of S_3; the joint recalibration constructed at the end of the chapter transports the dodecad, origin, orientation, chamber and memory, and closes that equivariance without altering the fixed-labeling no-go. The Reynolds averages, representation decompositions and polar factor used in this section are the standard ones [109, 61].

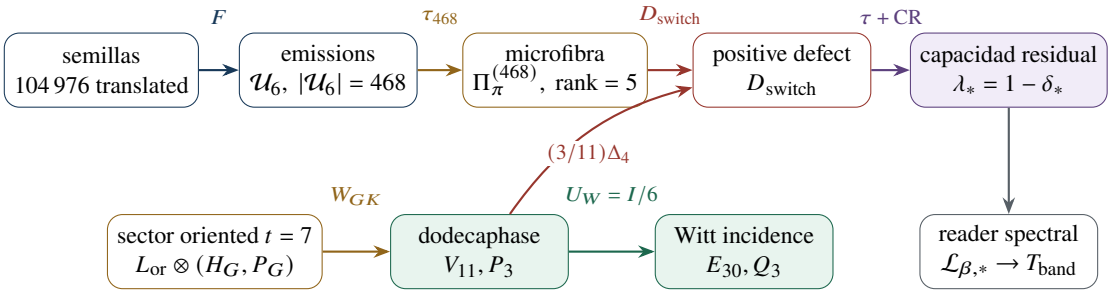


Figure 20.1: Architecture of the pre-dimensional closure. The upper row constructs the measure and action from the emission quotient; the lower row transports the oriented sector $t = 7$ to the dodecaphase and then the dodecaphase to Witt. The solid arrows are proved projector isometries. The first is fixed by the variational selection of the theorem 20.7; the second is not H_*-equivariant in the fixed gauge and becomes equivariant through the full gauge change of the theorem 20.9.

20.1.1 Typing of the objects associated with $3/11$

Let

$$V_{12} = \mathbb{R}^{12}, \qquad V_{11} = \mathbf{1}^{\perp} = \left\{ v \in \mathbb{R}^{12} : \sum_{p=1}^{12} v_p = 0 \right\}.$$

The icosahedral action declared on the twelve positions decomposes as

$$V_{12} = V_1 \oplus V_3 \oplus V_{3'} \oplus V_5.$$

Denotemos by P_3 the orthogonal projector on the first summand three-dimensional. Then

$$P_3^2 = P_3 = P_3^*, \qquad \operatorname{rank} P_3 = 3, \qquad \tau_{11}(P_3) = \frac{1}{11}\operatorname{Tr}_{V_{11}} P_3 = \frac{3}{11}.$$

There are three nearby objects and a fourth object of a different type:

1. the point–hexad incidence transports P_3 to a rank-three projector Q_3 on a Witt module;
2. survival in $t = 7$ distinguishes three branches among eleven and produces a projector P_G with normalized trace $3/11$;
3. an oriented intertwiner identifies $P_3 V_{11}$ with the oriented realization of $P_G H_G$;
4. a Witt star has seven even modes and four odd ones, so its normalized trace also equals $3/11$, but it contains negative spectrum and is not a projector.

The distinction between projector and involution will be essential to the positivity of the action law in the following section.

20.1.2 The point–hexad incidence matrix

Let $\mathcal{H}$ be the set of the 132 hexads of the design

$$W_{12} = S(5, 6, 12).$$

We define

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \qquad (Iv)(H) = \sum_{p \in H} v_p,$$

or, equivalently,

$$I_{H,p} = \mathbf{1}_{\{p \in H\}}.$$

This operator records only incidence. It contains no α, units, masses, CKM, Barbero, or any physical target.

Lemma 20.1 (Metric Identity of Incidence)**.** *We have*

$$\boxed{I^* I = 36 I_{12} + 30 J_{12}.}$$

Therefore, for every $v \in V_{11}$,

$$\|Iv\|^2 = 36\|v\|^2.$$

Proof. In an $S(5, 6, 12)$ system, each point belongs to

$$r = \frac{\binom{11}{4}}{\binom{5}{4}} = 66$$

hexads, and each pair of points belongs to

$$\lambda_2 = \frac{\binom{10}{3}}{\binom{4}{3}} = 30$$

hexads. The diagonal entries of $I^* I$ have value 66 and the nondiagonal entries have value 30. Hence

$$I^* I = (66 - 30) I_{12} + 30 J_{12}.$$

Since J_{12} vanishes on $\mathbf{1}^{\perp}$, the second identity is obtained. □

To identify the spectral branch selected by I, we introduce

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 20.2 (Spectral identity of incidence)**.** *With $J_{132,12}$ denoting the rectangular all-ones matrix,*

$$\boxed{GI = 30I + 15J_{132,12}.}$$

In particular, if $v \in V_{11}$, then

$$G(Iv) = 30Iv.$$

Proof. The entry (H, p) of GI is

$$\sum_{H' \ni p} \binom{|H \cap H'|}{4}.$$

The exhaustive census of W_{12} gives 45 when $p \in H$ and 15 when $p \notin H$. Consequently, the entry equals $30I_{H,p} + 15$. Reconstruction of the 132 hexads from the extended ternary Golay code checks the $132 \cdot 12$ entries as integers. □

Define

$$E_{30}^{\mathrm{inc}} := I(V_{11}) \subseteq \ker(G - 30I).$$

Its dimension is eleven by the lemma 20.1. The calculation modulo 1009 obtains

$$\operatorname{rank}_{\mathbb{F}_{1009}}(G - 30I) = 121.$$

Since reduction modulo 1009 gives a lower bound for the rational rank, it follows that $\dim_{\mathbb{Q}} \ker(G - 30I) \leq 11$. The preceding inclusion provides eleven linearly independent vectors in that kernel. Therefore,

$$E_{30}^{\mathrm{inc}} = \ker(G - 30I).$$

From now on we will write E_{30}.

Theorem 20.3 (Canonical dodecaphase entangler–Witt)**.** *The operator*

$$\boxed{U_W = \frac{1}{6} I\big|_{V_{11}} : V_{11} \longrightarrow E_{30}}$$

is a isometry surjective. For every $g \in \mathrm{Aut}(W_{12}) \cong M_{12}$,

$$U_W \rho_{12}(g) = \rho_{\mathcal{H}}(g) U_W.$$

Among the equivariant point–hexad kernels, the isometry condition and the positive sign on incidences uniquely determine U_W.

Proof. The isometry follows from $I^* I = 36I$ over V_{11}. Its image has dimension eleven and is contained in E_{30}, which also has dimension eleven; it is therefore surjective. Equivariance translates

$$p \in H \quad \Longleftrightarrow \quad g(p) \in g(H).$$

For uniqueness, M_{12} is transitive on the flags (p, H), with $p \in H$, and on the antiflags. Every equivariant kernel has two values, a on flags and b on antiflags:

$$T = aI + b(J - I) = (a - b)I + bJ.$$

The term J vanishes on V_{11}, so there T is a multiple of I. The isometry forces $|a - b| = 1/6$ and the positive normalization selects $a - b = 1/6$. □

20.1.3 Positive transport of the tritic projector

The entangler allows defining

$$\boxed{Q_3 = U_W P_3 U_W^*.}$$

We obtain

$$Q_3^2 = Q_3 = Q_3^*, \qquad \operatorname{rank} Q_3 = 3,$$

and the diagram

$$\begin{array}{ccc} V_{11} & \xrightarrow{U_W} & E_{30} \\ {\scriptstyle P_3}\downarrow & & \downarrow{\scriptstyle Q_3} \\ V_{11} & \xrightarrow{U_W} & E_{30} \end{array}$$

commutes. In particular,

$$\boxed{\tau_{V_{11}}(P_3) = \tau_{E_{30}}(Q_3) = \frac{3}{11}.}$$

The canonicity of U_W belongs to the labeled design and its automorphism action. It must be distinguished from the screen A_5/C_5 used to define P_3: in the constructed labeled realization, only the identity of that concrete action of A_5 preserves the fixed set of hexads. Hence U_W canonically transports the already given P_3 toward Q_3, but does not by itself convert the action A_5/C_5 into a subaction of M_{12}. The bridge to the oriented sector will be constructed separately from the marked data K and B_K.

20.1.4 Signed index of the Witt star

Let B be a tetrad. The four hexads that contain it are written

$$H_i = B \sqcup P_i, \qquad |P_i| = 2,$$

where the four petals P_i partition the complement of B. The star involution R_B fixes B and exchanges the two points of each petal. Its cycle type on twelve points is $1^4\,2^4$. In V_{11}, and by means of U_W also in E_{30}, its spectrum is

$$\operatorname{Spec}(R_B) = \{1^{[7]}, (-1)^{[4]}\}.$$

Therefore,

$$\boxed{\tau_{11}(R_B) = \frac{7-4}{11} = \frac{3}{11}.}$$

Theorem 20.4 (Obstruction projector–star)**.** *There is no isomorphism $F : V_{11} \to E_{30}$ such that*

$$FP_3 = R_B F.$$

Proof. If it existed, P_3 and R_B would be similar. However,

$$\operatorname{Spec}(P_3) = \{1^{[3]}, 0^{[8]}\}, \qquad \operatorname{Spec}(R_B) = \{1^{[7]}, (-1)^{[4]}\}.$$

Their minimal polynomials are $x(x-1)$ and $(x-1)(x+1)$, respectively. Similarity is impossible. □

The equality $3/11$ thus has two readings on the same module. For Q_3 it is the mass of a positive rank-three sector. For R_B it is a parity index. A positive action must involve Q_3; the star preserves orientation and distinguishes sheets but does not replace the projector.

20.1.5 homogeneous reduction $11 \to 3 \to 1$ in $t = 7$

The preceding reading of $3/11$ in $t = 8$ compared three local roots with eleven continuations of a single root. The numerator and denominator did not belong to the same space. The detailed table locates the correct object in $t = 7$: there are eleven local branches; all eleven survive one horizon, three survive two, and one survives three,

$$\boxed{11 \longrightarrow 3 \longrightarrow 1.}$$

index	triple of words	$h = 2$	$h = 3$
0	`200112|212221|110110`	3	0
1	`200121|212212|110110`	3	1
2	`200211|212122|110110`	3	0
3	`202122|210211|110110`	0	0
4	`202212|210121|110110`	0	0
5	`202221|210112|110110`	0	0
6	`210111|202222|110110`	0	0
7	`210222|202111|110110`	0	0
8	`212112|200221|110110`	0	0
9	`212121|200212|110110`	0	0
10	`212211|200122|110110`	0	0

Let $\mathcal{B}_7$ be the set of these eleven branches and

$$H_G = \ell^2(\mathcal{B}_7).$$

We define

$$P_G\delta_b = \begin{cases} \delta_b, & b \text{ admits continuation to horizon two,} \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$P_G^2 = P_G = P_G^*, \qquad \operatorname{rank} P_G = 3, \qquad \boxed{\tau_{H_G}(P_G) = \frac{3}{11}.}$$

This is the third positive realization.

20.1.6 The residual action of S_3 and the positive regions of π

A permutation of the three final positions, applied simultaneously to the first two words of each branch and leaving the third fixed, preserves $\mathcal{B}_7$. This rule defines an action of S_3. Its orbits have sizes

$$3 + 3 + 1 + 1 + 3 = 11.$$

The support of P_G, formado by the indices $0, 1, 2$, is one orbit complete, so that P_G conmuta with S_3.

The character of the representation on the eleven branches, evaluated at the identity, a transposition, and a cycle of order three, is

$$\chi_{\mathcal{B}_7} = (11, 5, 2) = 5\chi_{\text{triv}} + 3\chi_{\text{std}}.$$

On the image of P_G,

$$\chi_G = (3, 1, 0) = \chi_{\text{triv}} + \chi_{\text{std}}.$$

The three positive regions of the microfibre of π have signatures

$$339555, \qquad 393555, \qquad 933555.$$

The tails of the first word of the three surviving branches are

$$112, \qquad 121, \qquad 211.$$

The coordinate substitution $1 \mapsto 3$, $2 \mapsto 9$ gives

branch	tail	positive region
0	112	339555
1	121	393555
2	211	933555.

Theorem 20.5 (Horizontal–vertical bridge of the positive microfiber)**.** *The preceding rule is an S_3-equivariant bijection between the three positive regions of π and the three branches selected by P_G. The map preserves the position of the unique exceptional symbol.*

Proof. Both sets are the three words obtained by placing a single exceptional symbol in one of three positions. The substitution is applied coordinate by coordinate and therefore commutes with every permutation of positions. The exceptional position uniquely recovers the source element. □

20.1.7 Dependence of orientation in the dodecaphase entangler–Witt

For any normalizer $H = N_{A_5}(C_3) \simeq S_3$ of a subgroup of order three in A_5, we define the abstract character of the quotient

$$\operatorname{sgn}_H : H \longrightarrow H/C_3 \simeq C_2 \longrightarrow \{+1, -1\}, \qquad \operatorname{sgn}_H(h) = \begin{cases} +1, & h \in C_3, \\ -1, & h \in H \setminus C_3. \end{cases}$$

This character is not the ambient sign of a permutation of five letters: every element of A_5 has ambient sign +1. It is the intrinsic character of the reflection in the quotient H/C_3. With this convention, the character of the icosahedral sector restricted to H is

$$\chi_{P_3V_{11}|H} = (3, -1, 0) = \chi_{\operatorname{sgn}_H} + \chi_{\mathrm{std}}.$$

By contrast,

$$\chi_{\operatorname{im} P_G} = (3, 1, 0) = \chi_{\mathrm{triv}} + \chi_{\mathrm{std}}.$$

Corollary 20.6 (Obstruction to the unoriented braider)**.** *There is no S_3-equivariant isomorphism*

$$\operatorname{im} P_G \longrightarrow P_3V_{11}$$

for the non-twisted actions.

Proof. Isomorphic representations have the same character. Their values on a transposition are 1 and -1. □

Let (e_0, e_1, e_2) the basis posicional of $\mathbb{R}^3_{\mathrm{perm}}$. We defines

$$L_{\mathrm{or}} = \bigwedge^3 \mathbb{R}^3_{\mathrm{perm}}, \qquad \ell_{\mathrm{or}} = e_0 \wedge e_1 \wedge e_2$$

the orientation line of the triplet, equipped with the positive anchor ℓ_{or}. The action of S_3 on this line is precisely its abstract sign character. Upon tensoring by it,

$$\chi_{L_{\mathrm{or}} \otimes \operatorname{im} P_G} = (3, -1, 0),$$

of mode that the character obstruction disappears. The equality of caracteres proves the existence of one class of isomorphisms, but aun not selects one. The selection is obtained of two witnesses already present in the dodecaphase.

20.1.8 Variational selection of the normalizer S_3

Let

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$$

the decimal chart of the seal. The common minimal witness of the two readings is

$$B_K = H_e \cap P_\pi = \{p : K_p \geq 729\} = \{4, 6, 9, 10\},$$

in the mathematical numbering $1, \ldots, 12$. En the equivalent convention with zero-based indices, the same set is $\{3, 5, 8, 9\}$. We define two vectors of the sector P_3:

$$b_K = P_3 \left(\mathbf{1}_{B_K} - \frac{|B_K|}{12} \mathbf{1} \right), \qquad u_K = P_3 \left(K - \overline{K}\, \mathbf{1} \right).$$

The first satisfies exactly.

$$\|b_K\|^2 = 1.$$

The group A_5 contains ten subgroups C_3. For each one, let

$$H(C_3) = N_{A_5}(C_3) \simeq S_3$$

and let be the sign projector

$$e_{\mathrm{sgn},H} = \frac{1}{6} \sum_{h \in H} \mathrm{sgn}_H(h) \rho(h).$$

Here sgn_H is the abstract character of H/C_3 defined above, not the —trivial— ambient sign of A_5. The orientation energy associated with the common witness is

$$\mathcal{E}_B(H) = \|e_{\mathrm{sgn},H} b_K\|^2.$$

The exact census in $\mathbb{Q}(\sqrt{5})$ produces a unique maximum:

$$\mathcal{E}_B(H_*) = \frac{2}{3} + \frac{2}{15}\sqrt{5}.$$

The second level is

$$\frac{1}{3} + \frac{2}{15}\sqrt{5},$$

so that the exact gap is

$$\boxed{\mathcal{E}_B(H_*) - \mathcal{E}_B(H_{\mathrm{segundo}}) = \frac{1}{3}.}$$

The same H_* is the unique maximum when b_K is replaced by u_K. Therefore, the selector does not depend on a single encoding of the witness.

In this labeling,

$$H_* = N_{A_5}\big(\langle c_* \rangle\big), \qquad c_* = (2\,3\,4).$$

The rule is natural under simultaneous relabeling by A_5: when B_K, K, and P_3 are transported, the maximum is conjugated by the same element.

20.1.9 Selection of orientation and presentation of S_3

Within of H_*, consider

$$s(g) = \langle b_K, \rho(g) u_K \rangle.$$

The maximum among the two order-three elements uniquely selects

$$c_* = (0, 1, 3, 4, 2),$$

in image notation of the permutation of five points, and the maximum among the three involutions selects

$$r_* = (1, 0, 4, 3, 2).$$

It is verified exactly.

$$r_* c_* r_* = c_*^{-1}.$$

The presentation of the oriented sector by the cycle of positions $c = (0\,1\,2)$ and the reflection $r = (0\,2)$, which fixes the central position, therefore determines a unique isomorphism

$$\theta : S_3^{\mathrm{or}} \xrightarrow{\sim} H_*, \qquad \theta(c) = c_*, \qquad \theta(r) = r_*.$$

Here S_3^{or} denotes the group of relabelings of the oriented sector $t = 7$, generated by c and r; it introduces no additional operatorial class. The only branch that reaches horizon three is precisely the central branch 1; this datum fixes which of the three positions remains fixed under reflection.

20.1.10 Reynolds average and polar factor

Let

$$W_{\mathrm{or}} = L_{\mathrm{or}} \otimes \mathbb{R}^3_{\mathrm{perm}}$$

the oriented sector triple. With the anchor already fixed, we take

$$\boxed{w = \ell_{\mathrm{or}} \otimes e_1,}$$

where e_1 represents the central position, the only one reaching horizon three. Define

$$A = \sum_{g \in S_3} \rho_{W_{\mathrm{or}}}(g)\, |w\rangle\langle b_K|\, \rho_{V_{11}}(\theta(g))^{-1}.$$

By construction,

$$A \rho_{V_{11}}(\theta(g)) = \rho_{W_{\mathrm{or}}}(g) A, \qquad A = A P_3.$$

The exact computation in $\mathbb{Q}(\sqrt{5})$ obtains

$$\operatorname{rank} A = 3.$$

Moreover, $G_A = A A^*$ has all its diagonal entries equal to

$$3 + \frac{2}{5}\sqrt{5}$$

and all its off-diagonal entries equal to

$$\frac{5}{2} + \frac{3}{5}\sqrt{5}.$$

Its two isotype eigenvalues are

$$\lambda_{\mathrm{sgn}} = 8 + \frac{8}{5}\sqrt{5}, \qquad \lambda_{\mathrm{std}} = \frac{1}{2} - \frac{1}{5}\sqrt{5}.$$

Both are strictly positive. In particular, $G_A^{-1/2}$ exists, and the polar decomposition

$$\boxed{U_{GK} = G_A^{-1/2} A}$$

is one equivariant isometry of $P_3 V_{11}$ on W_{or}:

$$U_{GK}^* U_{GK} = P_3, \qquad U_{GK} U_{GK}^* = I_{W_{\mathrm{or}}}.$$

Let

$$J_G : W_{\mathrm{or}} \longrightarrow L_{\mathrm{or}} \otimes H_G$$

the isometric inclusion in the three selected branches. Then

$$J_G J_G^* = I_{L_{\mathrm{or}}} \otimes P_G.$$

The sector–dodecaphase partial isometry is denoted

$$\boxed{W_{GK} = U_{GK}^* J_G^* : L_{\mathrm{or}} \otimes H_G \longrightarrow V_{11}}$$

satisfies

$$W_{GK}^* W_{GK} = I_{L_{\mathrm{or}}} \otimes P_G, \qquad W_{GK} W_{GK}^* = P_3,$$

and, therefore,

$$P_3 W_{GK} = W_{GK} = W_{GK}(I_{L_{\mathrm{or}}} \otimes P_G).$$

Theorem 20.7 (Relative Isometric Identification of Three Projectors)**.** *Relative to the action A_5/C_5, the projector P_3, the seal K, the doubly constructed witness B_K, the oriented finite sector $t = 7$, the abstract character sgn_{H_*}, the anchor ℓ_{or}, and the variational rule $\mathcal{E}_B$, there exists a unique normalized partial isometry W_{GK} that identifies P_G with P_3. Independently, the incidence isometry U_W identifies P_3 with Q_3. Consequently, the following chain of isometric identifications of projectors is obtained:*

$$\begin{array}{ccccc} L_{\mathrm{or}} \otimes H_G & \xrightarrow{W_{GK}} & V_{11} & \xrightarrow{U_W} & E_{30} \\ {\scriptstyle I_{L_{\mathrm{or}}} \otimes P_G}\downarrow & & \downarrow{\scriptstyle P_3} & & \downarrow{\scriptstyle Q_3} \\ L_{\mathrm{or}} \otimes H_G & \xrightarrow{W_{GK}} & V_{11} & \xrightarrow{U_W} & E_{30}. \end{array}$$

Typed traces satisfy

$$\tau_{L_{\mathrm{or}} \otimes H_G}(I_{L_{\mathrm{or}}} \otimes P_G) = \tau_{V_{11}}(P_3) = \tau_{E_{30}}(Q_3) = \frac{3}{11}.$$

The statement, in the fixed gauge, isometrically identifies three projectors; it does not yet assert that their three spaces are the same representation of a common group.

Proof. The energy $\mathcal{E}_B$ selects a unique normalizer H_*; the scoring criterion $s(g)$ selects a unique pair (c_*, r_*), and with it a unique isomorphism θ. The Reynolds average is then equivariant for the actions of S_3^{or} and H_* related by θ, and has rank three. The positive part of its polar decomposition fixes the normalization without introducing a basis. The preceding partial identities prove that W_{GK} identifies P_G and P_3. On the other hand, the theorem 20.3 identifies P_3 and Q_3 as projectors through U_W. These two operator identities make that fixed-gauge diagram commute, without extending the equivariance of the first arrow to the second there. □

Proposition 20.8 (Obstruction to triple equivariance in the fixed gauge)**.** *The preceding construction does not prove that U_W is H_*-equivariant. Under the declared labeling, the fixed action of A_5/C_5 on the twelve*

points preserves the set of hexads of W_{12} only for the identity element. Therefore, the H_*-equivariant intertwining of W_{GK} does not extend, in that gauge, to a common representation on E_{30}.

Proof. The first assertion separates equivariance domains: W_{GK} uses $H_* \subset A_5$, whereas the incidence theorem proves equivariance of U_W for $\mathrm{Aut}(W_{12})$. The exact reproducible census applies the sixty elements of action A_5/C_5 to the labeled hexad system; only the identity preserves that system. Since H_* contains six elements, its action cannot enter by restriction of that fixed action in $\mathrm{Aut}(W_{12})$. □

The preceding proposition isolates exactly the fixed gauge; it is not an abstract obstruction to transporting the calibration jointly. The governing owner of S_3–Witt recalibration constructs the equivariant monomorphism and second arrow as follows.

Theorem 20.9 (Joint H_*–Witt recalibration)**.** *Let $\bar{\rho}_{\mathrm{src}} : H_* \to S_{12}$ be the point action on the twelve cosets determined by c_*, r_* and the isomorphism θ of the oriented sector, and let ρ_{src} be its linearization on V_{11}. In the one-based image convention, the permutation*

$$\boxed{\sigma = (1,5,2,6,7,4,3,10,11,9,8,12)}$$

defines the monomorphism

$$\boxed{\iota_\sigma : H_* \hookrightarrow \mathrm{Aut}(W_{12}) \cong M_{12}, \qquad \iota_\sigma(h) = \sigma\, \bar{\rho}_{\mathrm{src}}(h)\, \sigma^{-1}.}$$

If the calibration is transported in full through

$$W_{12}^{\sigma} = \{\sigma^{-1} B : B \in W_{12}\}, \qquad \widetilde{U}_W := U_W P_\sigma,$$

where P_σ is the permutation unitary, then, under the corresponding identification of the transported incidence space with E_{30}, for every $h \in H_*$,

$$\boxed{\widetilde{U}_W\, \rho_{\mathrm{src}}(h) = \rho_{E_{30}}\big(\iota_\sigma(h)\big)\, \widetilde{U}_W.}$$

In particular,

$$\widetilde{Q}_3 = \widetilde{U}_W P_3 \widetilde{U}_W^*$$

is the Witt realization of the same projector and the chain

$$I_{L_{\mathrm{or}}} \otimes P_G \xleftarrow{W_{GK}} P_3 \xleftarrow{\widetilde{U}_W} \widetilde{Q}_3$$

carries, after identifying the source action through θ, a common representation of H_*, with the target action given by ι_σ. Recalibration jointly transports C_W, W_{12}, I, the dodecad, origin, orientation, chamber and memory; it does not consist in replacing U_W in isolation.

Proof. The exact certificate reconstructs the ternary Golay code, the 132 hexads, the group M_{12} of order 95040, the source S_3 and a semiregular target S_3, both of order six. It checks for each $h \in H_*$ that $\sigma\bar{\rho}_{\mathrm{src}}(h)\sigma^{-1} \in M_{12}$, that W_{12}^{σ} is H_*-invariant and that its point–hexad incidence is equivariant. Injectivity of ι_σ follows because it is conjugation of the faithful source action. Moreover,

$$P_\sigma \rho_{\mathrm{src}}(h) = \rho_{12}\big(\iota_\sigma(h)\big) P_\sigma.$$

Composing this identity with the M_{12} equivariance of U_W yields the boxed identity. Since P_3 commutes with H_*, its transport $\widetilde{Q}_3$ commutes with $\iota_\sigma(H_*)$; the already proved equivariance of W_{GK} completes the composition. The same certificate verifies that the original design is not invariant under the source action, so the fixed-gauge no-go is preserved without contradiction. □

Material owner. This result is governed by theorem HMT–RDF–EXC–S3, sections 3.1–3.3 of the governing Witt–Moonshine exceptional propagation delta, and its reproducible certificate `verificar_recalibracion_s3_witt.py`, whose output is `PASS_RECALIBRACION_S3_WITT`. The status is `FORMALIZACION_NUEVA/CERTIFICADO_NUEVO`; uniqueness of the gauge change is not postulated.

Before using B_K, u_K, orientation and the polar factor, there were $10 \cdot 6 \cdot 4 = 240$ labeled maps, or 40 after identifying internal relabelings of the sector. The preceding rules reduce that set to one. This is the verifiable content of the word *canonical* for W_{GK}, within the declared rules. The common action on the Witt module comes from the explicit change σ; it is not asserted that this gauge change is unique or canonical.

The variational rule $\mathcal{E}_B$ uses only prior HMT data and fixes the class in which uniqueness is obtained. It does not identify the star involution with P_3, since their spectra remain distinct.

20.1.11 Comparison of the readings of $t = 7$ and $t = 8$

In $t = 8$ there are three local roots and eleven extensions to horizon nine of the unique root that continues. The sizes of the three fibers are

$$(11, 0, 0).$$

The natural branch–chain matrix has one row of eleven ones and two zero rows, so its rank is one. In $t = 9$, moreover, the visible quotient changes from $3/11$ to $3/6$. That fraction was not the trace of a stable projector. The correction does not eliminate the sector: it shifts the reading to the homogeneous object P_G of $t = 7$, where rank and dimension belong to the same space.

20.1.12 Positivity of the Action Induced by the Dodecaphasic Braider–Witt

The law of action may u now a unique class isometric of projectors positive in tris spaces of support:

$$P_G \xleftarrow{W_{GK}} P_3 \xleftarrow{U_W} Q_3.$$

The star R_{B_K} remains reserved for parity and orientation. The projector P_G records survival; P_3 records dodecaphase polarization; Q_3 records its incidence realization in Witt. This separation of functions is the mathematical premise of the measure and of the positive law in the next section. In the fixed gauge the chain expresses isometric equivalence of projectors without a common representation. The theorem 20.9 transports the complete calibration and turns the chain with $\widetilde{U}_W$ and $\widetilde{Q}_3$ into a common representation of H_*, without identifying the signed star with the positive projector.

20.2 Equivariant realization of the angular reader in Witt incidence

The angular chart and exceptional incidence preserve different information from the same APP–TRIT–TPK development. The former expresses rotation coordinates; the latter preserves relations between positions, supports and orientations. Comparing them requires a transport that preserves these relations and allows the complete coordinates to be recovered. We construct that transport here and prove its compatibility with oriented permutations of the regional triplet.

The antecedents of the composition are the dodecaphasic seal K, its marked face B_K, the sector of the three-dimensional isotypic projector P_3^{ico}, the compatible branches of the nonadic filtration and the sectorial system of Definition 16.43. Each enters with its structure, not merely through its scalar value. The real vector spaces of this subsection are subsequent realizations of those already expressed sectors. Generating the seal, transporting its coordinates and evaluating the reader correspond to distinct stages of the causal composition.

20.2.1 Regional triplet and orientation line

Let V be the counting space of the three positive regions of π_{HMT}, with ordered orthonormal basis

$$(v_1, v_2, v_3) = (933555, 393555, 339555).$$

The notation identifies each basis vector with the region it represents. In the prolongation window of index seven, the eleven admitted branches are distributed into orbits of sizes $3+3+1+1+3$ under simultaneous permutation of the final three positions of the first two words. The sector retaining continuation to horizon two consists of

branch	triple of words	positive region
0	200112 | 212221 | 110110	339555
1	200121 | 212212 | 110110	393555
2	200211 | 212122 | 110110	933555.

The central branch additionally extends to horizon three. These horizons belong to the same filtration of compatible prolongations; the successive support cardinalities are 11, 3, and 1.

The map from the tail of the first word to the positive region substitutes $1 \mapsto 3$ and $2 \mapsto 9$ coordinatewise and adds the regional tail 555. It therefore preserves the position of the exceptional symbol and commutes with every permutation of the three positions. The branch order is $(339555, 393555, 933555)$; the change from the order of V is the permutation $R = (1\,3)$.

The orientation line of V is $o(V) = \det V$. The map

$$\Psi : \Lambda^2 V \longrightarrow \mathrm{Hom}(V, \det V), \qquad \Psi(u \wedge v)(w) = u \wedge v \wedge w \tag{20.1}$$

is identified through the counting metric with an isometry $\Lambda^2 V \to o(V) \otimes V$. For $\omega = v_1 \wedge v_2 \wedge v_3$, its values on the exterior basis are

$$v_1 \wedge v_2 \mapsto \omega \otimes v_3, \qquad v_2 \wedge v_3 \mapsto \omega \otimes v_1, \qquad v_1 \wedge v_3 \mapsto -\omega \otimes v_2.$$

In angular order $(12, 23, 13)$, joint transport of basis and orientation has matrix

$$D = \det(R) P_R \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}. \tag{20.2}$$

Thus $D(\theta_{12}, \theta_{23}, \theta_{13})^{\mathsf{T}} = (-\theta_{12}, \theta_{13}, -\theta_{23})^{\mathsf{T}}$. Throughout this section, $P_p e_j = e_{p(j)}$ defines the permutation-matrix convention, and the asterisk denotes the adjoint for the declared counting metrics.

Proposition 20.10 (Naturality of the oriented identification)**.** *For every $p \in S_3$,*

$$D \Lambda^2 P_p = \det(p) P_{RpR^{-1}} D, \qquad D^* D = I_3.$$

Proof. The identity $(P_p u) \wedge (P_p v) \wedge (P_p w) = \det(p)(u \wedge v \wedge w)$ proves naturality of (20.1). Simultaneous change of regional order introduces conjugation by R. Finally, the columns of D are signed coordinate vectors, whence $D^* D = I_3$. □

The orientation line explains the difference between the character $(3, 1, 0)$ of the position triplet and the character $(3, -1, 0)$ of the oriented triplet, on the identity, transposition, and order-three cycle classes. The three transpositions provide an explicit control: omitting the determinant factor changes the sign of the preceding identity. A global change of the oriented anchor is instead transported as a joint coordinate change.

20.2.2 Dodecaphase sector and variational selection of orientation

The chart of the previously expressed seal is

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601). \tag{20.3}$$

To make its incidence realization explicit, write the ternary Paley matrix in the gauge used:

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \qquad C(a) = (a, aA_W), \quad a \in \mathbb{F}_3^6.$$

Support is taken on the twelve coordinates of $C(a)$. The marked intersection has the two compatible expressions

$$B_K = \operatorname{supp} C(010211) \cap \operatorname{supp} C(201101) = \{p : K_p \geq 729\} = \{4, 6, 9, 10\}. \tag{20.4}$$

The equality is evaluated by ternary multiplications of the displayed matrix. In particular, the intersection is obtained as a set of positions before its cardinality four is used.

To fix the representation of the twelve positions completely, consider A_5 acting by left multiplication on the cosets $r_p C_5$, where $C_5 = \langle (1\,2\,3\,4\,5) \rangle$. The representatives, written as image lists on $\{1, \ldots, 5\}$, are

p	r_p	p	r_p
1	(4, 3, 5, 2, 1)	7	(5, 4, 3, 2, 1)
2	(4, 2, 1, 3, 5)	8	(1, 3, 4, 2, 5)
3	(1, 2, 4, 5, 3)	9	(4, 2, 3, 5, 1)
4	(2, 5, 3, 4, 1)	10	(3, 2, 5, 4, 1)
5	(4, 3, 1, 5, 2)	11	(4, 5, 2, 3, 1)
6	(5, 1, 2, 3, 4)	12	(5, 3, 2, 4, 1).

Denote this permutation representation by ρ, and put $V_{11} = \mathbf{1}^{\perp} \subset \mathbb{R}^{12}$. The icosahedral three-dimensional isotypic projector is determined by

$$P_3^{\mathrm{ico}} = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \rho(a), \tag{20.5}$$

where the values of χ_3 on classes of types $1, 2^2, 3, 5_a, 5_b$ are

$$\left(3, -1, 0, \frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}\right).$$

Class 5_a contains $(1\,2\,3\,4\,5)$, and class 5_b contains its square. This convention distinguishes the two three-dimensional summands. The superscript ico identifies this component of the A_5 representation; the cyclic projector P_3 entering (14.86) acts on twelve-position transport with its own definition. Finite evaluation of (20.5) satisfies $P_3^{\mathrm{ico}} = (P_3^{\mathrm{ico}})^{*} = (P_3^{\mathrm{ico}})^2$, $P_3^{\mathrm{ico}} \mathbf{1} = 0$, and $\operatorname{Tr} P_3^{\mathrm{ico}} = 3$.

The two marked vectors of the sector are

$$b = P_3^{\mathrm{ico}} \left(\mathbf{1}_{B_K} - \frac{1}{3} \mathbf{1} \right), \qquad u = P_3^{\mathrm{ico}} (K - \overline{K} \mathbf{1}), \qquad \overline{K} = \frac{1}{12} \sum_{p=1}^{12} K_p, \qquad \|b\|^2 = 1.$$

For each of the ten order-three subgroups $C_3 \subset A_5$, let $H = N_{A_5}(C_3) \cong S_3$. The character sgn_H takes value +1 on C_3 and −1 on $H \setminus C_3$. It is the character of the quotient H/C_3, whereas the ambient sign of elements of A_5 is always positive. Define

$$e_{\mathrm{sgn},H} = \frac{1}{6}\sum_{a\in H} \mathrm{sgn}_H(a)\rho(a), \qquad \mathcal{E}_B(H) = \|e_{\mathrm{sgn},H} b\|^2. \tag{20.6}$$

Proposition 20.11 (Normalizer and orientation determined by the seal)**.** *The maximization rule of* (20.6) *selects a unique normalizer H_*. Its energy and the second distinct energy are*

$$\frac{2}{3} + \frac{2\sqrt{5}}{15}, \qquad \frac{1}{3} + \frac{2\sqrt{5}}{15},$$

respectively; the gap is $1/3$. Projection of u selects the same normalizer. Within H_*, maximizing $s(a) = \langle b, \rho(a)u\rangle$, separately among order-three elements and involutions, determines

$$c_* = (1,2,4,5,3), \qquad r_* = (2,1,5,4,3)$$

in image notation on five points. The identity $r_* c_* r_* = c_*^{-1}$ holds.

Proof. Evaluation of the preceding finite sums in $\mathbb{Q}(\sqrt{5})$ gives the following distribution of the ten energies:

energy	multiplicity	energy	multiplicity
$2/3 + 2\sqrt{5}/15$	1	$2/3 - 2\sqrt{5}/15$	1
$1/3 + 2\sqrt{5}/15$	2	$1/3 - 2\sqrt{5}/15$	2
$1/6 + \sqrt{5}/30$	2	$1/6 - \sqrt{5}/30$	2.

The greatest belongs to the subgroup generated by c_*. The next is obtained by subtracting $1/3$. The same evaluation with u has a unique maximum at that subgroup, of value $2526869/12 + 2813609\sqrt{5}/30$. The scores of the two cycle orientations are

$$s(c_*) = \frac{1341}{4} + \frac{2409\sqrt{5}}{20}, \qquad s(c_*^{-1}) = \frac{357}{2} + \frac{1853\sqrt{5}}{10}.$$

Their difference is $627/4 - 1297\sqrt{5}/20 > 0$, as is checked by comparing positive squares. The scores of the three involutions are

list of images	$s(a)$
$(2,1,3,5,4)$	$-1341/4 - 2409\sqrt{5}/20$
$(2,1,4,3,5)$	$-1659/4 - 2667\sqrt{5}/20$
$(2,1,5,4,3)$	$-357/2 - 1853\sqrt{5}/10.$

The unique maximum corresponds to r_*. Composition of the two image lists verifies the dihedral relation. All these operations use K, B_K, P_3^{ico}, and the finite actions already specified. □

Write $\vartheta : S_3 \to H_*$ for the isomorphism determined by $\vartheta((1\,2\,3)) = c_*$ and $\vartheta((1\,3)) = r_*$. The position reflection fixes the central branch reaching horizon three. This orientation completes regional identification before any angle evaluation.

20.2.3 Reynolds intertwiner and polar normalization

In the branch basis, let $\rho_o(g) = \det(g)P_g$ and $w = (0,1,0)^{\mathsf{T}}$. The Reynolds operator defined by summation over the group is

$$A_R = \sum_{g\in S_3} \rho_o(g)\,|w\rangle\langle b|\,\rho(\vartheta(g))^{-1}. \tag{20.7}$$

The symbol A_R distinguishes this operator from the angle $A_{\deg}$. Each summand has unit scale; metric normalization is performed through its polar factor.

Proposition 20.12 (Polar isometry of the oriented sector)**.** *The following hold*

$$A_R = A_R P_3^{\mathrm{ico}}, \qquad \rho_o(g)A_R = A_R\rho(\vartheta(g)), \qquad G = A_R A_R^* > 0, \qquad A_R^* G^{-1} A_R = P_3^{\mathrm{ico}}.$$

With $E = \mathbf{1}\mathbf{1}^{\mathsf{T}}/3$*,*

$$G = \lambda_s E + \lambda_t (I_3 - E), \qquad \lambda_s = 8 + \frac{8\sqrt{5}}{5}, \quad \lambda_t = \frac{1}{2} - \frac{\sqrt{5}}{5} > 0. \tag{20.8}$$

Consequently,

$$U_{GK} = G^{-1/2}A_R : P_3^{\mathrm{ico}} V_{11} \longrightarrow o(V)\otimes V$$

is a surjective isometry: $U_{GK}^* U_{GK} = P_3^{\mathrm{ico}}$ *and* $U_{GK}U_{GK}^* = I_3$*.*

Proof. The projector P_3^{ico} commutes with ρ, and $b = P_3^{\mathrm{ico}} b$; hence $A_R = A_R P_3^{\mathrm{ico}}$. Reindexing the Reynolds sum proves equivariance. Multiplication of the finite sum gives diagonal entries $3 + 2\sqrt{5}/5$ and off-diagonal entries $5/2 + 3\sqrt{5}/5$ in G. The projections E and $I_3 - E$ give (20.8). Both eigenvalues are positive; for the smaller, $25 > 20$ suffices. Thus A_R has rank three. Its row space is contained in the rank-three sector of P_3^{ico}, and equals it. The orthogonal projector onto that row space is $A_R^*(A_R A_R^*)^{-1}A_R$, giving the final identity. The two polar-factor equations follow by substitution. □

The positive root preserves equivariance because both spectral projectors commute with the oriented representation. Uniqueness of the polar factor pertains to the Reynolds operator and orientation determined by the preceding rules.

20.2.4 Point–hexad incidence and joint gauge change

Let $\mathcal{W}$ be the design $S(5,6,12)$ of 132 hexads obtained from the weight-six supports of the ternary code $C(a)$. The incidence matrix is

$$I_{H,p} = \mathbf{1}_{p\in H}, \qquad (Iz)_H = \sum_{p\in H} z_p.$$

The blocks are reconstructed by enumerating $a \in \mathbb{F}_3^6$, retaining weight six, and collecting equal supports. The finite verification that each five-point subset belongs to a unique hexad fixes the design property used below.

Lemma 20.13 (Metric and spectral incidence identities)**.** *Let* J_{12} *and* $J_{132,12}$ *be the all-ones matrices indicated by their dimensions, and let*

$$(G_{\mathcal{W}})_{H,H'} = \binom{|H\cap H'|}{4}.$$

Then

$$I^*I = 36I_{12} + 30J_{12}, \qquad G_{\mathcal{W}}I = 30I + 15J_{132,12}. \tag{20.9}$$

In particular, $U_W = I/6$ *is an isometry on* $V_{11} = \mathbf{1}^{\perp}$*, with image contained in* $E_{30} = \ker(G_{\mathcal{W}} - 30I_{132})$*.*

Proof. The design property gives 66 hexads per point and 30 per pair of points:

$$\frac{\binom{11}{4}}{\binom{5}{4}} = 66, \qquad \frac{\binom{10}{3}}{\binom{4}{3}} = 30.$$

These are the diagonal and off-diagonal entries of I^*I. For the second identity, count pairs (T, H'), where $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, ten tetrads $T \subset H$ contain p, and each belongs to four hexads; the other five tetrads, adjoined to p, determine one hexad each. The result is $10 \cdot 4 + 5 = 45$. If $p \notin H$, each of the fifteen tetrads determines a hexad together with p, and the result is 15. This yields $30I_{H,p} + 15$. The all-ones matrices vanish on V_{11}, proving the isometry and spectral membership. □

The gauge change of the twelve positions is

$$\sigma = (1, 5, 2, 6, 7, 4, 3, 10, 11, 9, 8, 12), \tag{20.10}$$

written as an image list with indices starting at one. If ρ_{perm} is the point action underlying ρ, the six permutations

$$\tau_g = \sigma \rho_{\mathrm{perm}}(\vartheta(g)) \sigma^{-1}, \qquad g \in S_3,$$

preserve $\mathcal{W}$. Their two generators have image lists

$$\tau_{(123)} = (3, 6, 5, 2, 1, 4, 9, 12, 11, 8, 7, 10),$$
$$\tau_{(13)} = (2, 1, 4, 3, 6, 5, 8, 7, 10, 9, 12, 11).$$

The assertion is checked on the finite design: these two generating permutations are applied to the weight-six supports of $C(a)$, and their images are verified to run through the same 132 supports. Products of these generators give the other four permutations. The induced action Q_{τ_g} on hexads satisfies

$$IP_{\tau_g} = Q_{\tau_g} I, \qquad P_\sigma \rho(\vartheta(g)) = P_{\tau_g} P_\sigma. \tag{20.11}$$

The first identity expresses exactly the equivalence $p \in H \Longleftrightarrow \tau_g(p) \in \tau_g(H)$.

This change jointly transports positions, code, incidence, origin, chamber, and orientation of the seal. In the initial gauge, five elements of this order-six subgroup do not preserve the fixed design. Conjugation (20.10) provides the compatible gauge and maintains that control over the initial labeling. The procedure exhibits a gauge change; its uniqueness does not enter the proof.

20.2.5 Isometric composition and coordinate recovery

Theorem 20.14 (Dodecaphase–Witt angular transport)**.** *The composition*

$$\boxed{F = \frac{1}{6} I P_\sigma A_R^* G^{-1/2} D : \mathbb{R}^3_{\mathrm{ang}} \longrightarrow E_{30}} \tag{20.12}$$

is an isometry. Its coordinate action and equivariance are

$$(F\boldsymbol{\theta})_H = \frac{1}{6} \sum_{p \in H} (P_\sigma A_R^* G^{-1/2} D \boldsymbol{\theta})_p, \tag{20.13}$$

$$F X_g = Q_{\tau_g} F, \qquad X_g = \Lambda^2 P_{R^{-1} g R}. \tag{20.14}$$

Proof. The columns of A_R^* belong to $P_3^{\mathrm{ico}} V_{11}$, and P_σ preserves V_{11}. Lemma 20.13 therefore proves membership of the image of F in E_{30}. Moreover,

$$F^*F = D^* G^{-1/2} A_R P_\sigma^* \frac{I^*I}{36} P_\sigma A_R^* G^{-1/2} D$$

$$= D^* G^{-1/2} A_R A_R^* G^{-1/2} D = D^* D = I_3.$$

The coordinate formula expands multiplication by incidence. For equivariance, compose in order the naturality of D, equivariance of A_R^*, commutation of $G^{-1/2}$ with ρ_o, and the two identities of (20.11). This produces (20.14) for each $g \in S_3$. □

Proposition 20.15 (Covariance of the complete gauge)**.** *Let S, T, V_o be orthogonal coordinate changes in the dodecaphase space, Witt point space, and oriented sector, respectively. Let Q_T be the corresponding coordinate change in the incidence space; for a design relabeling, it is the induced hexad permutation. If one jointly transports*

$$A'_R = V_o A_R S^*, \quad G' = V_o G V_o^*, \quad P'_\sigma = T P_\sigma S^*,$$
$$D' = V_o D, \qquad U'_W = Q_T U_W T^*,$$

the composition satisfies $F' = Q_T F$.

Proof. Uniqueness of the positive root gives $(G')^{-1/2} = V_o G^{-1/2} V_o^*$. Substituting the five transformations into $F' = U'_W P'_\sigma (A'_R)^* (G')^{-1/2} D'$, the factors T^*T, S^*S, and $V_o^* V_o$ cancel in pairs, leaving $Q_T U_W P_\sigma A_R^* G^{-1/2} D = Q_T F$. □

The preceding covariance describes a coordinate change of all marked data. Its application to an active cardinal transformation requires the corresponding identity between actions on the state; the two notions thus preserve their own domains.

20.2.6 Faithful realization of the angular sectorial system

We use the chart x_θ of (16.56),

$$x_\theta = \begin{pmatrix} A_{\mathrm{deg}} \\ C^*_{\mathrm{deg}}/6 \\ (180/\pi_{\mathrm{HMT}})\Delta_4 \end{pmatrix}.$$

The component $h_\theta = C^*_{\mathrm{deg}}/6$ distributes the odd component of the record among six antipodal pairs. Definition 16.43 fixes the three rules

$$R_{12} = (2, -p, bq), \quad R_{23} = (0, q, -p^2/2), \quad R_{13} = (1, -pb, -(o_8 - h_6)/t)$$

for $(b, p, q, h_6, o_8, t) = (4, 5, 7, 6, 8, 3)$. The face of cardinality four, pentadic order, pivot coordinate, hexadic–octadic difference, and tritic alphabet retain distinct types. The bidirectional orbital normalization of 16.41 preserves the term $p^2/2$, including the contribution of fixed pairs.

Its evaluation, already proved in Theorem 16.44, gives

$$\mathsf{M}_{\mathrm{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \qquad \det \mathsf{M}_{\mathrm{CKM}} = -\frac{3857}{6}.$$

The transport F acts on the coordinates expressed by these rules; the preceding choice of functionals retains its place in the sectorial construction.

Corollary 20.16 (Angular reader in the incidence module)**.** *The map*

$$C = F \mathsf{M}_{\mathrm{CKM}} \tag{20.15}$$

satisfies

$$C^* C = \mathsf{M}_{\mathrm{CKM}}^* \mathsf{M}_{\mathrm{CKM}}, \qquad \mathsf{M}_{\mathrm{CKM}}^{-1} F^* C = I_3.$$

It is an isometry from the space of x_θ, equipped with metric $\mathsf{M}_{\mathrm{CKM}}^* \mathsf{M}_{\mathrm{CKM}}$, onto its image.

Proof. Substitute $F^*F = I_3$ into both products. The nonzero determinant of $\mathsf{M}_{\mathrm{CKM}}$ permits use of its inverse, displayed in (16.58). Positivity of the metric follows from $x^*\mathsf{M}_{\mathrm{CKM}}^*\mathsf{M}_{\mathrm{CKM}}x = \|\mathsf{M}_{\mathrm{CKM}}x\|^2 > 0$ for $x \neq 0$. □

The isometry F preserves the counting metric of $\boldsymbol{\theta} = \mathsf{M}_{\mathrm{CKM}}x_\theta$; its transport to coordinates x_θ gives the metric of the corollary. This distinction avoids identifying a basis of angular generators with an orthonormal basis of their parameters.

20.2.7 Exact certification and finite unitary composition

The positive roots of the polar factor can be checked through their projectors while preserving exact arithmetic. Define

$$B = \frac{1}{6} I P_\sigma A_R^* D, \qquad H = D^*GD = B^*B, \qquad L = H^{-1}B^*. \tag{20.16}$$

The identities

$$LB = I_3, \qquad L(B\mathsf{M}_{\mathrm{CKM}}) = \mathsf{M}_{\mathrm{CKM}}, \qquad \mathsf{M}_{\mathrm{CKM}}^{-1}L(B\mathsf{M}_{\mathrm{CKM}}) = I_3$$

are evaluated in $\mathbb{Q}(\sqrt{5})$. If $E_D = D^*ED$,

$$H^{-1/2} = \lambda_s^{-1/2}E_D + \lambda_t^{-1/2}(I_3 - E_D), \qquad F = BH^{-1/2}.$$

The products $E_D^2 = E_D$, $E_D(I_3 - E_D) = 0$, and the decomposition of H prove this formula without approximating its radicals. The 132 rows of B are determined by (20.13) and by the hexad construction rule. The executable certificate of this delivery preserves those rows, those of $B\mathsf{M}_{\mathrm{CKM}}$, the left inverse L, the projectors, and the labels of each hexad. Its 906 exact checks verify this composition and its orientation, incidence, and recovery identities.

Conversion to radians is performed before the trigonometric realization:

$$\boldsymbol{\vartheta} = \mathsf{M}_{\mathrm{CKM}}\begin{pmatrix} (\pi_{\mathrm{HMT}}/180)A_{\mathrm{deg}} \\ (\pi_{\mathrm{HMT}}/180)C_{\mathrm{deg}}^*/6 \\ \Delta_4 \end{pmatrix}, \qquad \delta = \frac{\pi_{\mathrm{HMT}}}{180}\,9A_{\mathrm{deg}}. \tag{20.17}$$

The finite realization preserves the order $V_{\mathrm{CKM}} = R_{23}U_{13}(\delta)R_{12}$ of (16.70). To specify the additional information of the complex phase, let E_{ab} be the elementary matrix and define

$$J_{13} = E_{13} - E_{31}, \qquad S_{13} = E_{13} + E_{31}, \qquad X_{13}(\delta) = \cos\delta\, J_{13} - i\sin\delta\, S_{13}.$$

Then $X_{13}(\delta)^* = -X_{13}(\delta)$ and $U_{13}(\delta) = \exp(\vartheta_{13}X_{13}(\delta))$. Indeed, $X_{13}(\delta)^2 = -(E_{11} + E_{33})$, and its exponential reproduces the unitary 13 block of (16.70). Exterior transport preserves the oriented generators; the symmetric imaginary component preserves phase information. The ordered product of the three finite factors maintains these two contributions and their order of composition.

20.2.8 Structural result and scope of the composition

The construction provides an equivariant isometric transport from the three angular coordinates to the Witt incidence, together with an explicit reconstruction of A_{deg}, $C_{\mathrm{deg}}^*/6$ and Δ_{deg}. It preserves the exterior orientation, the dodecaphasic sector, the marked face of the seal and the joint gauge change. In particular, the incidence realization expresses the same angular content on another support and allows its three coordinates to be recovered.

Prior generation of K, selection of the sectorial functionals, and physical identification of the amplitudes retain their own proofs. The recovery identities proved here establish faithfulness of the composition; its subsequent use keeps those antecedents explicit. The complementary development of quartic memory

planes remains another construction of the state and does not interpose itself as an additional requirement for the alignment realized in (20.12).

Chapter 21

From spectral jump 4 to the Leech lattice: APP unfolding 4→2→6→54 and the Paley–Witt–Mathieu–Golay–Niemeier corridor

21.1 From the spectral gap $4 \to 2 \to 6 \to 54$: local coupling, modular compression, and two-dimensional integration

For two consecutive representatives $(k, k+1)$, consider

$$B_k = \begin{pmatrix} k^2 & k(k+1) \\ k(k+1) & (k+1)^2 \end{pmatrix} \pmod 9.$$

Theorem 21.1 (Uniqueness of the Adjacent Diagonal Block)**.** *There is a unique block B_k whose diagonal entries coincide. It is determined by $k = 4$ and is*

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}.$$

Proof. The equality of the diagonals is equivalent to

$$k^2 \equiv (k+1)^2 \pmod 9,$$

that is, $2k + 1 \equiv 0 \pmod 9$. Since 2 is a unit of $\mathbb{Z}/9\mathbb{Z}$, the congruence has the unique solution $k \equiv 4 \pmod 9$. □

Therefore,

$$B_c = 7I + 2R, \qquad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad R^2 = I.$$

With $P_\pm = (I \pm R)/2$,

$$B_c = 9P_+ + 5P_-.$$

Theorem 21.2 (Hierarchy of the separation between sum and product)**.** *The central block has local spectral jump $9 - 5 = 4$. Its off-diagonal coefficient is 2. Compression by the 3 classes modulo 3 produces the aggregate difference $51 - 45 = 6$, and expansion over the 9 positions produces the global defect*

$$9 \cdot 6 = 459 - 405 = 54.$$

The integers $4, 2, 6, 54$ pertain, respectively, to the spectral gap, the local coupling, modular compression, and two-dimensional integration.

Proof. The decomposition $B_c = 7I + 2R = 9P_+ + 5P_-$ gives the jump and the local coefficient. The aggregated matrices M_Π, M_Σ satisfy $\sum M_\Pi = 51$ and $\sum M_\Sigma = 45$. Each aggregated entry represents 9 positions, so unfolding multiplies the difference by 9 and recovers the totals 459 and 405. □

Their ordered concatenations produce the blocks 72 and 27:

$$72 + 27 = 99, \qquad 72 - 27 = 45 = \det B_c.$$

The concatenation of the diagonal and antidiagonal entries produces, respectively, 77 and 22. Therefore,

$$72 + 27 = 77 + 22 = 99, \qquad 77 - 22 = 55 = 54 + 1.$$

These equalities express two exact partitions of the same block; they do not by themselves identify an external constant.

21.2 Paley conference form, finite quadratic extension and self-dual ternary code

The tetrad B_K constructed in the preceding unit does not yet have a Witt star: to define it, one must first construct the incidence system that contains its hexads. This unit performs that step from the projective gauge of the six units modulo nine, obtains an internal square root of $-I_6$, and constructs the ternary graph code of length twelve. No identification follows from a mere equality of cardinalities.

21.2.1 Projective gauge of the 6 units modulo 9

Let

$$\mathcal{U}_9 = (\mathbb{Z}/9\mathbb{Z})^\times = \{1, 2, 4, 8, 7, 5\},$$

ordered by the successive powers of 2 modulo 9. Fijamos the coordinate bijection

$$\theta : \mathcal{U}_9 \longrightarrow \mathbb{P}^1(\mathbb{F}_5), \qquad \begin{array}{c|cccccc} u & 1 & 2 & 4 & 8 & 7 & 5 \\ \hline \theta(u) & \infty & 0 & 1 & 2 & 3 & 4 \end{array}. \tag{21.1}$$

The map θ is a coordinate gauge. It is not asserted to be a homomorphism between multiplication in $\mathcal{U}_9$ and an operation on $\mathbb{P}^1(\mathbb{F}_5)$.

Let

$$\chi : \mathbb{F}_5 \longrightarrow \{-1, 0, 1\}$$

the quadratic character:

$$\chi(0) = 0, \qquad \chi(1) = \chi(4) = 1, \qquad \chi(2) = \chi(3) = -1.$$

In the order $(\infty, 0, 1, 2, 3, 4)$, we define

$$(C_P)_{\infty,x} = (C_P)_{x,\infty} = 1, \qquad x \in \mathbb{F}_5, \tag{21.2}$$

$$(C_P)_{x,y} = \chi(x - y), \qquad x, y \in \mathbb{F}_5, x \neq y, \tag{21.3}$$

$$(C_P)_{x,x} = 0. \tag{21.4}$$

Thus,

$$C_P = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & -1 & -1 & 1 \\ 1 & 1 & 0 & 1 & -1 & -1 \\ 1 & -1 & 1 & 0 & 1 & -1 \\ 1 & -1 & -1 & 1 & 0 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \end{pmatrix}. \tag{21.5}$$

Lemma 21.3 (Quadratic correlation sum)**.** *For $a \neq b$ in $\mathbb{F}_5$,*

$$\sum_{t \in \mathbb{F}_5} \chi(t - a)\chi(t - b) = -1. \tag{21.6}$$

Proof. The substitution $u = (t - a)/(b - a)$ is to bijection of $\mathbb{F}_5$. Ifnce $\chi((b - a)^2) = 1$, the sum reduce to

$$\sum_{u \in \mathbb{F}_5} \chi(u(u - 1)).$$

Evaluation of $u = 0, 1, 2, 3, 4$ gives, respectively, $0, 0, 1, -1, -1$, whose sum is -1. □

Proposition 21.4 (Conference Identity)**.** *The matrix C_P is symmetric and satisfies*

$$\boxed{C_P C_P^{\mathsf{T}} = 5I_6.} \tag{21.7}$$

Proof. Each row contains five entries of modulus one and one zero entry; its squared norm is five. For two distinct finite rows, the contribution of the coordinate ∞ is $+1$, whereas lemma 21.3 gives -1 for the five finite coordinates. Their inner product is zero. The sum of a nontrivial character over $\mathbb{F}_5$ is zero, so the row corresponding to ∞ is also orthogonal to every finite row. □

21.2.2 ternary reduction and internal quadratic root of $-I_6$

We reduce C_P modulo tris, with $-1 \equiv 2 \pmod 3$:

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3). \tag{21.8}$$

Theorem 21.5 (Internal square root of $-I_6$)**.** *We have*

$$\boxed{A_W^2 = -I_6.} \tag{21.9}$$

In particular,

$$A_W^{-1} = -A_W.$$

Proof. The proposition 21.4 gives

$$A_W A_W^{\mathsf{T}} \equiv 5I_6 \equiv 2I_6 = -I_6 \pmod 3.$$

Since $A_W = A_W^{\mathsf{T}}$, $A_W^2 = -I_6$ is obtained. □

The preceding identity is the algebraic residence of the root of -1 in this corridor. No proceeds of the factor $1/4$ of the Hadamard inversion: the former is a rational scalar forced by $H_4^2 = 4I_4$; this is one quadratic structure on characteristic three.

21.2.3 Extension quadratic to $\mathbb{F}_9$

We define

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1), \qquad \iota^2 = -1. \tag{21.10}$$

The polynomial $X^2 + 1$ has no roots in $\mathbb{F}_3$, so that the quotient is a field of nine elements. For $V = \mathbb{F}_3^6$, let

$$V_9 = V \otimes_{\mathbb{F}_3} \mathbb{F}_9.$$

The operators

$$P_\iota = \frac{1}{2}(I - \iota A_W), \qquad P_{-\iota} = \frac{1}{2}(I + \iota A_W) \tag{21.11}$$

are well defined because 2 is invertible in $\mathbb{F}_3$.

Proposition 21.6 (Conjugate projectors)**.** *The following hold*

$$P_\iota^2 = P_\iota, \qquad P_{-\iota}^2 = P_{-\iota}, \qquad P_\iota P_{-\iota} = 0, \qquad P_\iota + P_{-\iota} = I.$$

Proof. Usando $\iota^2 = -1$ and $A_W^2 = -I$,

$$(\iota A_W)^2 = I.$$

The four identities are then the elementary identities for the projectors $(I \mp \iota A_W)/2$ of an involution. □

Theorem 21.7 (Conjugate spectral decomposition)**.** *Let*

$$E_{\pm\iota} = \ker(A_W \mp \iota I).$$

Then

$$V_9 = E_\iota \oplus E_{-\iota}, \qquad \dim_{\mathbb{F}_9} E_\iota = \dim_{\mathbb{F}_9} E_{-\iota} = 3. \tag{21.12}$$

Proof. The minimal polynomial of A_W divides $X^2 + 1$, which splits over $\mathbb{F}_9$ with simple roots $\pm\iota$. The projectors of proposition 21.6 produce the direct sum. Since A_W has coefficients in $\mathbb{F}_3$, the Frobenius automorphism exchanges the two eigenspaces; their dimensions are equal and sum to six. □

This construction dota $\mathbb{F}_3^6$ of istructura of modulo of rango tris over $\mathbb{F}_9$, by means of

$$(a + b\iota) \cdot v := av + b(vA_W). \tag{21.13}$$

The equality $3^6 = 9^3$ is thus accompanied by an explicit action; it is not used as a cardinal argument.

21.2.4 Ternary graph code of length 12

We define

$$c : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \qquad c(q) = (q, qA_W), \tag{21.14}$$

and

$$C_W := \operatorname{im} c. \tag{21.15}$$

Its generating matrix is

$$G_W = (I_6 \mid A_W).$$

Theorem 21.8 (Injectivity and self-duality)**.** *The map c is injective,* $\dim C_W = 6$ *and*

$$\boxed{C_W = C_W^{\perp}.} \tag{21.16}$$

Proof. The first half of $c(q)$ is q, so $\mathrm{pr}_1 \circ c = \mathrm{id}$; this proves injectivity and the dimension. Moreover,

$$G_W G_W^{\mathsf{T}} = I_6 + A_W A_W^{\mathsf{T}} = I_6 + 2I_6 = 0$$

in $\mathbb{F}_3$. Therefore, $C_W \subseteq C_W^{\perp}$. Both spaces have dimension six, so they are equal. □

For $r = 0, \dots, 12$, let

$$N_r := \#\{q \in \mathbb{F}_3^6 : \mathrm{wt}(q, qA_W) = r\}.$$

The enumeration of the $3^6 = 729$ entries produces

$$N_0 = 1, \qquad N_6 = 264, \qquad N_9 = 440, \qquad N_{12} = 24, \tag{21.17}$$

and $N_r = 0$ in the remaining weights.

Theorem 21.9 (Enumerator and minimum distance)**.** *The code C_W has parameters*

$$\boxed{C_W = [12, 6, 6]_3}$$

and weight enumerator

$$\boxed{W_{C_W}(y) = 1 + 264y^6 + 440y^9 + 24y^{12}.} \tag{21.18}$$

Proof. The dimension and self-duality have already been proved. The census (21.17) traverses the 729 words without sampling. Its least nonzero weight is six, which is the minimum distance of a linear code. The sum $1 + 264 + 440 + 24 = 729$ checks that no word was omitted. □

21.2.5 Algebraic classification of the self-dual ternary structure obtained

Three distinct objects and three typed arrows have been constructed:

$$\mathcal{U}_9 \xrightarrow{\theta} \mathbb{P}^1(\mathbb{F}_5) \xrightarrow{\chi} C_P \xrightarrow{\text{mod } 3} A_W,$$

$$A_W^2 = -I_6 \quad \Longrightarrow \quad \mathbb{F}_3^6 \cong \mathbb{F}_9^3 \quad \Longrightarrow \quad C_W = \{(q, qA_W) : q \in \mathbb{F}_3^6\}.$$

The first arrow is a gauge, the second uses the quadratic character, and the third is coefficient reduction. The structure over $\mathbb{F}_9$ is finite; it is not identified with a real or analytic complex structure.

21.2.6 Obstructions in the Paley–C_W corridor: bijection, $A_W^2 = -I_6$, self-duality, and census of 729 words

Comparison Criterion 21.10 (Falsification of the Paley–C_W corridor)**.** The construction is refuted if any of the following occurs:

1. θ is not bijective or is used as a homomorphism without proof;
2. the sum (21.6) differs from -1;
3. $C_P C_P^{\mathsf{T}} \neq 5I_6$;
4. $A_W^2 \neq -I_6$;
5. the projectors (21.11) are not complementary;
6. the map $q \mapsto (q, qA_W)$ loses injectivity or self-duality;
7. the census does not sum to 729, differs from (21.17), or presents a word of weight between one and five;
8. $\mathbb{F}_3^6 \cong \mathbb{F}_9^3$ is inferred only from $3^6 = 9^3$, omitting the action (21.13).

21.3 Ternary Steiner system, Witt-star involutions and generation of M_{12}

The preceding unit constructed the self-dual code $C_W = [12, 6, 6]_3$. Its words of weight six are now projectivized, the Steiner property is proved, and only then is the star associated with the tetrad B_K defined. This order prevents an incidence figure from being turned into an object added by visual similarity.

21.3.1 Supports projective of weight 6

Let

$$\mathcal{H}_W := \{\operatorname{supp}(c) : c \in C_W, \operatorname{wt}(c) = 6\}. \tag{21.19}$$

Each support occurs for the word pair $\{c, -c\}$. Indeed, if two weight-six words with the same support were not proportional, one could scale one of them to cancel a coordinate upon subtraction; this would yield a nonzero word of C_W of weight less than six, contrary to the minimum distance. Since neither is

there a nonzero word equal to its opposite in characteristic three, every projective orbit has exactly two elements. Therefore,

$$|\mathcal{H}_W| = \frac{264}{2} = 132. \tag{21.20}$$

For $T \in \binom{[12]}{5}$, define

$$\nu(T) := \#\{H \in \mathcal{H}_W : T \subseteq H\}. \tag{21.21}$$

The exact enumeration of the

$$\binom{12}{5} = 792$$

subsets of five points produce

$$\nu(T) = 1 \qquad \text{for every } T \in \binom{[12]}{5}. \tag{21.22}$$

Theorem 21.11 (Ternary Witt System)**.** *The incidence system*

$$W_{12} := ([12], \mathcal{H}_W)$$

is the Steiner system

$$\boxed{W_{12} = S(5, 6, 12).} \tag{21.23}$$

Proof. The equality (21.22) is precisely the Steiner property. The number of blocks is also recovered from the double count of pairs (T, H) with $T \subset H$:

$$|\mathcal{H}_W|\binom{6}{5} = \binom{12}{5}, \qquad |\mathcal{H}_W| = \frac{792}{6} = 132.$$

□

21.3.2 Residue of a tetrad and decomposition into 4 incident components

Let $B \in \binom{[12]}{4}$. In an $S(5, 6, 12)$ system, the number of hexads containing B is

$$\lambda_4 = \frac{\binom{12-4}{5-4}}{\binom{6-4}{5-4}} = \frac{8}{2} = 4. \tag{21.24}$$

We write these four hexads as

$$H_i = B \sqcup P_i, \qquad |P_i| = 2, \qquad i = 1, \dots, 4.$$

Lemma 21.12 (Residual partition of a tetrad)**.** *The four pairs $P_1, \dots, P_4$ are disjoint and*

$$[12] \setminus B = P_1 \sqcup P_2 \sqcup P_3 \sqcup P_4. \tag{21.25}$$

Proof. If two residual pairs shared a point p, the corresponding hexads would contain the same subset $B \cup \{p\}$ of five points, in contradiction with the Steiner uniqueness. The four pairs are, therefore, disjoint; contain eight points and agotan the complement of B. □

Definition 21.13 (Star and Witt star involution)**.** The star centered at B is

$$\mathcal{S}_B := \{H \in \mathcal{H}_W : B \subseteq H\} = \{B \sqcup P_1, \dots, B \sqcup P_4\}. \tag{21.26}$$

If $P_i = \{p_i^-, p_i^+\}$, its involution is

$$\sigma_B := \prod_{i=1}^{4} (p_i^- \; p_i^+). \tag{21.27}$$

The permutation σ_B fixes B pointwise and exchanges the ends of each residual petal.

The mathematical term used henceforth is *Witt star* or *Witt-star involution*. It is not a Euclidean star.

21.3.3 Star selected by the tetrad of the seal

The dodecaphasic unit produced, without using Steiner's design,

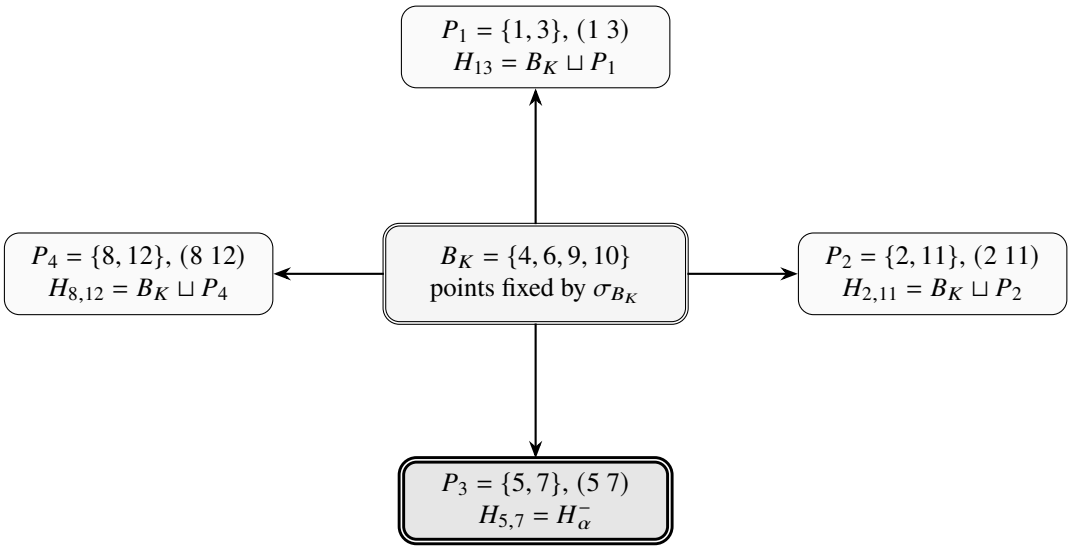
$$B_K = \{4, 6, 9, 10\}.$$

Now that W_{12} has been constructed, its star is well defined. The four incident hexads are

$$H_{13} = B_K \sqcup \{1, 3\} = \{1, 3, 4, 6, 9, 10\}, \tag{21.28}$$
$$H_{2,11} = B_K \sqcup \{2, 11\} = \{2, 4, 6, 9, 10, 11\}, \tag{21.29}$$
$$H_{5,7} = B_K \sqcup \{5, 7\} = \{4, 5, 6, 7, 9, 10\} = H_\alpha^-, \tag{21.30}$$
$$H_{8,12} = B_K \sqcup \{8, 12\} = \{4, 6, 8, 9, 10, 12\}. \tag{21.31}$$

Consequently,

$$\boxed{\sigma_{B_K} = (1\ 3)(2\ 11)(5\ 7)(8\ 12).} \tag{21.32}$$



$\mathcal{S}_{B_K} = \{H_{13}, H_{2,11}, H_{5,7}, H_{8,12}\}$.
The diagram represents incidence, not geometric distance.

Figure 21.1: Witt star centered at B_K. The center is the fixed tetrad; the four petals are the residual pairs that determine the involution.

21.3.4 Global field of the 495 stars

There exists a star for each tetrad, so

$$\#\{\mathcal{S}_B : B \in \tbinom{[12]}{4}\} = \binom{12}{4} = 495. \tag{21.33}$$

The exact census verifies

$$\sigma_B(\mathcal{H}_W) = \mathcal{H}_W \qquad \text{for every } B \in \binom{[12]}{4}. \tag{21.34}$$

We define

$$\Gamma_\star := \langle \sigma_B : B \in \tbinom{[12]}{4} \rangle \leq \mathrm{Aut}(W_{12}). \tag{21.35}$$

A generating set of six tetrads is

$$1234, \quad 1235, \quad 1236, \quad 1245, \quad 1345, \quad 2345.$$

The orders of the subgroups obtained by adding their involutions successively are

$$1, 2, 6, 18, 360, 7920, 95040. \tag{21.36}$$

Theorem 21.14 (Generation of the Mathieu group M_{12})**.** *We have*

$$\boxed{\langle \sigma_B : B \in \tbinom{[12]}{4} \rangle = \mathrm{Aut}(W_{12}) \cong M_{12}.} \tag{21.37}$$

Proof. Each σ_B preserves the 132 hexads, hence $\Gamma_\star \leq \mathrm{Aut}(W_{12})$. The Schreier enumeration of the six indicated generators yields order 95040. The automorphism group of $S(5, 6, 12)$ is M_{12} and has that same order; therefore the inclusion is an equality. □

An isolated star generates only

$$\langle \sigma_B \rangle \cong C_2.$$

The conclusion about M_{12} belongs to the global field of the 495 involutions, not to σ_{B_K} considered in isolation.

21.3.5 Action on the augmentation module

Let

$$V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}.$$

The involution σ_B fixes the four coordinates of B and acts by four transpositions on its complement. In $\mathbb{R}^{12}$ it has eight directions with eigenvalue $+1$ and four with eigenvalue -1. The constant direction belongs to the first eigenspace. Therefore,

$$\mathrm{Spec}(\sigma_B|_{V_{11}}) = \{1^{[7]}, (-1)^{[4]}\}, \qquad \frac{1}{11}\,\mathrm{Tr}(\sigma_B|_{V_{11}}) = \frac{3}{11}. \tag{21.38}$$

It is a signed involution; it is not a positive projector of rank three.

21.3.6 Obstructions concerning W_{12}, the field of 495 star involutions and the generation of M_{12}

Comparison Criterion 21.15 (Refutation of W_{12}, its stars, and M_{12})**.** The construction is refuted if any of the following facts occurs:

1. the projectivization of the 264 words of weight six does not produce 132 supports;
2. there exists a subset of five points contained in zero or in more of one hexad;
3. a tetrad does not belong to exactly four hexads;
4. the four residual pairs do not partition the complement of the tetrad;
5. the petals of B_K are not $\{1, 3\}, \{2, 11\}, \{5, 7\}, \{8, 12\}$;
6. any of the 495 involutions fails to preserve $\mathcal{H}_W$;
7. the generated group does not have order 95040;

8. M_{12} is attributed to a single star instead of the global field of involutions;
9. introduces the istrella before of constructing W_{12}.

21.4 Binary extension relative to a dodecad, system W_{24} and group M_{24}

The system W_{12} constructed over twelve phases admits a binary prolongation when an extended Golay code, a dodecad, and an incidence isomorphism are declared. These data are relative inputs of this branch: they are not deduced solely from C_W, and they do not intervene in the ternary lattice branch that will lead to the Leech lattice.

21.4.1 Golay binary code and octads

Let

$$\mathcal{G}_{24} \subset \mathbb{F}_2^{24}$$

the extended binary Golay code. Its weights are

$$0, 8, 12, 16, 24. \tag{21.39}$$

The supports of the weight eight words form the Steiner system

$$W_{24} = S(5, 8, 24). \tag{21.40}$$

Its blocks are called *octads*. Fix a dodecad D, that is, the support of a word of weight twelve, and let $\mathcal{O}_{24}$ be the family of octads.

We define

$$\mathcal{H}(D) := \{O \cap D : O \in \mathcal{O}_{24},\ |O \cap D| = 6\}. \tag{21.41}$$

Theorem 21.16 (Residual design of a dodecada)**.** *The residual system is*

$$\boxed{(D, \mathcal{H}(D)) \cong S(5, 6, 12).} \tag{21.42}$$

Moreover, each $H \in \mathcal{H}(D)$ has one unique octad O_H such that

$$O_H \cap D = H. \tag{21.43}$$

Proof. Let $A \subset D$, with $|A| = 5$. In $W_{24} = S(5, 8, 24)$ there is a unique octad O_A containing A. If $j = |O_A \cap D|$, the binary sum of the characteristic words of O_A and D belongs to $\mathcal{G}_{24}$ and has weight

$$8 + 12 - 2j = 20 - 2j.$$

Since $A \subseteq O_A \cap D$, one has $5 \le j \le 8$. The four possible weights $10, 8, 6, 4$, combined with (21.39), force $j = 6$. Thus every five-point subset of D belongs to a unique residual hexad.

If two octads had the same hexadic intersection with D, both would contain the same five points of that hexad, contradicting uniqueness in $S(5, 8, 24)$. This also proves uniqueness of the lift. □

21.4.2 Relative scale between the HMT design and the dodecagon

To relate the preceding construction to the design obtained from C_W, one must explicitly declare

$$(D, \ell), \qquad \ell : W_{12}^{\mathrm{HMT}} \xrightarrow{\sim} (D, \mathcal{H}(D)), \tag{21.44}$$

where ℓ is an incidence isomorphism. If ℓ_* denotes its action on blocks, we define

$$\widehat{\iota}_{D,\ell} := \iota_D \circ \ell_* : W_{12}^{\mathrm{HMT}} \longrightarrow W_{24}, \qquad \widehat{\iota}_{D,\ell}(H) = O_{\ell(H)}. \tag{21.45}$$

The composition $\widehat{\iota}_{D,\ell}$ avoids applying ι_D, whose domain is $\mathcal{H}(D)$, directly to a hexad that still lies in $[12]_{\mathrm{HMT}}$.

Proposition 21.17 (Injectivity of the Relative Lift)**.** *The map $\widehat{\iota}_{D,\ell}$ is injective and preserves the incidence between points and hexads after applying ℓ.*

Proof. The isomorphism ℓ is bijective on blocks. By the theorem 21.16, each residual hexad has a unique lifting octad. Two HMT hexads with the same image would have the same residual hexad and, by bijectivity of ℓ, would be equal. □

There is no absolute choice of (D, ℓ) prior to this gauge. The invariant object is the class of the construction under simultaneous change of dodecad and alignment.

21.4.3 4+1 Decomposition of the incidence of a tetrad into peripheral and axial components

For a tetrad $B \subset D$, the parameters derived from the two Steiner systems are

$$\lambda_4(W_{12}) = \frac{\binom{8}{1}}{\binom{2}{1}} = 4, \qquad \lambda_4(W_{24}) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5. \tag{21.46}$$

Therefore, the five octads containing B decompose uniquely as

$$\{O_H : H \in \mathcal{H}(D),\ B \subset H\} \sqcup \{O_B^{\perp}\}. \tag{21.47}$$

The four first satisfy $|O_H \cap D| = 6$; the quinta satisfies

$$O_B^{\perp} \cap D = B. \tag{21.48}$$

For $B = B_K$, the four residual lifts correspond to the petals

$$\{1, 3\}, \qquad \{2, 11\}, \qquad \{5, 7\}, \qquad \{8, 12\},$$

and the fifth octad is the transversal $O_{B_K}^{\perp}$. This law explains the transition from four hexads to five octads without identifying a face of a cube with a hexad or a vertex with an octad.

21.4.4 Strata Correspondence and Positivity of the Triple Freedom

The five-region structure of channel π has types

$$1_{\perp} + 1_{--} + 3_{++}.$$

The incidence on B_K has types

$$1_{\text{transversal}} + 1_{H^-} + 3_{\text{restantes}}$$

after marking the negative hexad. There is therefore a natural correspondence between *role* strata:

$$1_{\perp} \longleftrightarrow O_{B_K}^{\perp}, \qquad 1_{--} \longleftrightarrow \widehat{\iota}_{D,\ell}(H_{\alpha}^{-}), \qquad 3_{++} \longleftrightarrow O_{B_K}^{+,3}. \tag{21.49}$$

Here $O_{B_K}^{+,3}$ is the unordered set of the other three residual octads. An individual bijection between the three positive regions and those three octads requires declaring an additional gauge of S_3. The coincidence of the pattern $1 + 1 + 3$ does not by itself determine that ordering.

21.4.5 Automorphism group of the octad system

Theorem 21.18 (Mathieu group M_{24})**.** *The automorphism group of the octad Steiner system is*

$$\boxed{\mathrm{Aut}(W_{24}) \cong M_{24}, \qquad |M_{24}| = 244823040.} \tag{21.50}$$

The theorem identifies the classical automorphism group of the input binary data. It does not assert that M_{24} is generated from a single dodecad, or that the ternary code C_W determines $\mathcal{G}_{24}$ by itself.

21.4.6 Categorical separation from the lattice branch

The branch constructed in this unit is

$$(W_{12}^{\mathrm{HMT}}; \mathcal{G}_{24}, D, \ell) \longrightarrow W_{24} \longrightarrow \mathrm{Aut}(W_{24}) \cong M_{24}. \tag{21.51}$$

The lattice branch to be constructed subsequently is

$$C_W \longrightarrow N(A_2^{12}) \xrightarrow{\text{index-three neighbor}} \Lambda_{24}. \tag{21.52}$$

Their constructors, domains, and codomains are distinct. In particular,

$$\boxed{\text{no arrow is introduced } W_{24} \longrightarrow \Lambda_{24}.} \tag{21.53}$$

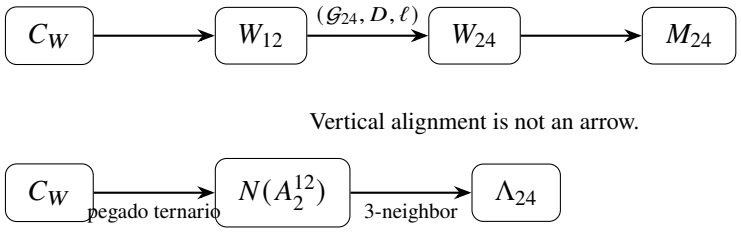


Figure 21.2: Typed bifurcation of the binary and ternary extensions.

21.4.7 Obstructions to the binary prolongation $W_{12} \to W_{24}$: Golay code, octads, alignment and separation of the Leech branch

Comparison Criterion 21.19 (Refutation of the binary prolongation)**.** The construction is refuted if any of the following facts occurs:

1. the weight spectrum of $\mathcal{G}_{24}$ contains a weight not listed in (21.39);
2. the supports of weight eight not form $S(5, 8, 24)$;
3. a residual intersection of an octad with D that contains five points of D has cardinality other than six;
4. a residual hexad admits either no lifting octad or more than one;
5. ι_D is applied directly to an HMT hexad without the alignment ℓ;
6. the incidence law on a tetrad is not $4 + 1$;
7. the positive triple is ordered without declaring the gauge S_3;
8. $\mathrm{Aut}(W_{24})$ has no order 244823040;
9. it is asserted that W_{24} arises from W_{12} without the binary datum $\mathcal{G}_{24}$, the dodecad, and the alignment;
10. the two branches are serialized by means of an arrow $W_{24} \to \Lambda_{24}$.

21.5 Ternary discriminant gluing and construction of the Niemeier lattice of type A_2^{12}

This unit begins the lattice branch directly from the self-dual ternary code C_W. The binary system W_{24}, the dodecad, and the group M_{24} are not inputs of this construction. The typed sequence is

$$C_W \longrightarrow (A_2^*/A_2)^{12} \longrightarrow N(C_W) \cong N(A_2^{12}).$$

21.5.1 Reticulo root A_2

Let

$$A_2 = \mathbb{Z}\alpha_1 \oplus \mathbb{Z}\alpha_2$$

the root lattice whose Gram matrix, in the ordered basis (α_1, α_2), is

$$G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{21.54}$$

For $x = x_1\alpha_1 + x_2\alpha_2$,

$$\langle x, x\rangle = 2(x_1^2 - x_1x_2 + x_2^2). \tag{21.55}$$

In particular, A_2 is even, positive defined and

$$\det A_2 = 3.$$

We define

$$\rho := \alpha_1 + \alpha_2.$$

Then

$$\rho^2 = 2, \qquad \langle \rho, \alpha_1\rangle = \langle \rho, \alpha_2\rangle = 1. \tag{21.56}$$

21.5.2 Decomposition of the incidence of a tetrad into 4 peripheral components and 1 axial component

The dual lattice is

$$A_2^* := \{x \in A_2 \otimes_{\mathbb{Z}} \mathbb{Q} : \langle x, A_2\rangle \subseteq \mathbb{Z}\}.$$

We choose the representatives

$$\mu_0 = 0, \qquad \mu_1 = \frac{2\alpha_1 + \alpha_2}{3}, \qquad \mu_2 = \frac{\alpha_1 + 2\alpha_2}{3}. \tag{21.57}$$

Their pairings with the simple base are

	α_1	α_2
μ_1	1	0
μ_2	0	1

thereby $\mu_1, \mu_2 \in A_2^*$. Furthermore,

$$3\mu_1 = 2\alpha_1 + \alpha_2 \in A_2, \qquad 3\mu_2 = \alpha_1 + 2\alpha_2 \in A_2,$$

while $\mu_1, \mu_2 \notin A_2$. It is obtained

$$A_2^*/A_2 = \{\overline{\mu}_0, \overline{\mu}_1, \overline{\mu}_2\} \cong \mathbb{Z}/3\mathbb{Z} \cong \mathbb{F}_3. \tag{21.58}$$

The sum of representatives is explicitly reduced by means of

$$\mu_1 + \mu_1 - \mu_2 = \alpha_1, \qquad \mu_1 + \mu_2 = \rho, \qquad \mu_2 + \mu_2 - \mu_1 = \alpha_2.$$

Consequently,

$$\mu_r + \mu_s - \mu_{r+s \,(\mathrm{mod}\, 3)} \in A_2 \qquad (r, s \in \mathbb{F}_3). \tag{21.59}$$

Proposition 21.20 (Discriminant Form of A_2)**.** *Under* (21.58)*, the discriminant bilinear and quadratic forms are*

$$b(\overline{\mu}_r, \overline{\mu}_s) \equiv \frac{2rs}{3} \pmod{\mathbb{Z}}, \tag{21.60}$$

$$q(\overline{\mu}_r) \equiv \frac{2r^2}{3} \pmod{2\mathbb{Z}}. \tag{21.61}$$

Proof. The formula (21.55) gives

$$\mu_1^2 = \mu_2^2 = \frac{2}{3}, \qquad \langle \mu_1, \mu_2 \rangle = \frac{1}{3}.$$

For $r, s \in \{0, 1, 2\}$, these values coincide, modulo $\mathbb{Z}$, with $2rs/3$. The quadratic formula results by taking $r = s$ and recalling that the lattice is even. □

21.5.3 Correspondence between incidence strata and free orbit of the positive triple

Let

$$L_0 := A_2^{12}.$$

Then

$$L_0^* = (A_2^*)^{12}, \qquad L_0^*/L_0 \cong \mathbb{F}_3^{12}.$$

For $c = (c_1, \ldots, c_{12}) \in \mathbb{F}_3^{12}$, define

$$\phi(c) := (\mu_{c_1}, \ldots, \mu_{c_{12}}) \in L_0^*. \tag{21.62}$$

The extension determined by C_W is

$$\begin{aligned} N(C_W) &:= \bigcup_{c \in C_W} (L_0 + \phi(c)) \\ &= \{x \in L_0^* : x \bmod L_0 \in C_W\}. \end{aligned} \tag{21.63}$$

Lemma 21.21 (Additive closure of the gluing)**.** *The set $N(C_W)$ is a discrete subgroup of L_0^* of rank twenty-four.*

Proof. Let $x = \ell + \phi(c)$ and $y = m + \phi(d)$, with $\ell, m \in L_0$ and $c, d \in C_W$. By (21.59),

$$\phi(c) + \phi(d) - \phi(c + d) \in L_0.$$

Since C_W is linear, $c + d \in C_W$, and then $x + y \in L_0 + \phi(c + d)$. The same identity applied to the opposite gives inverses. The inclusions

$$L_0 \subseteq N(C_W) \subseteq L_0^*$$

They test discretion and range twenty-four. □

21.5.4 Integrality induced by self-duality

The discriminant bilinear form of L_0 is the orthogonal sum of twelve copies of (21.60). For $c, d \in \mathbb{F}_3^{12}$,

$$b(c,d) \equiv \frac{2}{3}\sum_{i=1}^{12} c_i d_i \pmod{\mathbb{Z}}. \tag{21.64}$$

Lemma 21.22 (Integrity of the gluing)**.** *The lattice $N(C_W)$ is integral.*

Proof. The self-duality $C_W = C_W^{\perp}$ implica

$$\sum_{i=1}^{12} c_i d_i \equiv 0 \pmod 3 \qquad (c, d \in C_W).$$

By (21.64), the discriminant form vanishes on $C_W \times C_W$. Therefore, the pairing of any two glued classes is integral. □

21.5.5 Parity Induced by the Code Weights

At a nonzero coordinate of the code, the discriminant quadratic form equals $2/3$ modulo $2\mathbb{Z}$. Therefore,

$$q(c) \equiv \frac{2}{3}\,\mathrm{wt}(c) \pmod{2\mathbb{Z}}. \tag{21.65}$$

Lemma 21.23 (Parity of the gluing)**.** *The lattice $N(C_W)$ is even.*

Proof. The only weights of C_W are $0, 6, 9, 12$, and

$$\frac{2}{3}\{0,6,9,12\} = \{0,4,6,8\} \subseteq 2\mathbb{Z}.$$

Every glue class is isotropic for the discriminant quadratic form, so all norms of $N(C_W)$ are even. □

Let $g_1, \dots, g_6$ be the rows of $G_W = (I_6 \mid A_W)$ and $\gamma_i = \phi(g_i)$. Exact multiplication in A_2^* gives

$$(\langle \gamma_i, \gamma_j \rangle)_{i,j=1}^{6} = \begin{pmatrix} 4 & 2 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 2 & 2 & 2 \\ 2 & 2 & 2 & 4 & 2 & 2 \\ 2 & 2 & 2 & 2 & 4 & 2 \\ 2 & 2 & 2 & 2 & 2 & 4 \end{pmatrix}. \tag{21.66}$$

This matrix provides an additional direct control of integrality and parity for the six gluing generators.

21.5.6 Index, determinant, and unimodularity

The cosets of L_0 that appear in (21.63) are indexed by the $|C_W| = 729 = 3^6$ elements of the code and are distinct. Consequently,

$$[N(C_W) : L_0] = 3^6. \tag{21.67}$$

Since

$$\det L_0 = (\det A_2)^{12} = 3^{12},$$

the determinant formula for a finite extension gives

$$\det N(C_W) = \frac{\det L_0}{[N(C_W) : L_0]^2} = \frac{3^{12}}{(3^6)^2} = 1. \tag{21.68}$$

21.5.7 Root system induced by Witt incidence and classification of its orbits

En each one of the two classes discriminantes not nulas of A_2^*/A_2, the norm minima is

$$\mu_1^2 = \mu_2^2 = \frac{2}{3}.$$

If $c \in C_W \setminus \{0\}$, then $\mathrm{wt}(c) \geq 6$. By orthogonality of the twelve factors, any vector of $L_0 + \phi(c)$ has norm at least

$$6 \cdot \frac{2}{3} = 4. \tag{21.69}$$

Therefore, no nontrivial gluing class contains vectors of norm two. The roots of $N(C_W)$ are exactly the roots already present in $L_0 = A_2^{12}$.

Theorem 21.24 (Realization of the Niemeier of type A_2^{12})**.** *The lattice $N(C_W)$ is positive definite, even, and unimodular, and its root system is A_2^{12}. Consequently,*

$$\boxed{N(C_W) \cong N(A_2^{12}).} \tag{21.70}$$

Proof. Positivity follows from L_0^*. The lemmas 21.22 y 21.23 proves integrality and parity; (21.68) proves unimodularity. The bound (21.69) identifies the root system. The Niemeier classification uniquely associates the indicated lattice with that root system. □

21.5.8 Obstructions to the ternary gluing $C_W \to N(A_2^{12})$: parity, unimodularity, and root system

Comparison Criterion 21.25 (Refutation of ternary gluing)**.** The construction is refuted if any of the following facts occurs:

1. any of the representatives μ_0, μ_1, μ_2 does not belong to A_2^*, two of its classes coincide, or (21.59) fails;
2. the discriminant form is not the one given in proposition 21.20;
3. $N(C_W)$ is not closed under sum or inversos;
4. there exists $c, d \in C_W$ with $\frac{2}{3}\sum_i c_i d_i \notin \mathbb{Z}$;
5. one palabra of C_W produces $\frac{2}{3}\,\mathrm{wt}(c) \notin 2\mathbb{Z}$;
6. the index is not 3^6 or the determinant is not one;
7. a nontrivial gluing class contains a vector of norm two;
8. W_{24}, a dodecad, or a binary datum is used in any of the preceding gluing operations.

21.6 Incidental origin, typed radial problem and construction of the rootless unimodular neighbor

The lattice $N(A_2^{12})$ has already been constructed. This unit determines a base point from an ordered incidence frame, solves a variational problem on a declared domain, verifies the three conditions for the index-three neighbor, and proves that the result is the Leech lattice. The neighbor vector is denoted $v_{\mathcal{F}_*}$: it depends on the incidence frame and does not enter the numerical carry of α.

21.6.1 Ordered set of incidences

Consider the four hexads

$$H^{-} = \{4, 5, 6, 7, 9, 10\}, \tag{21.71}$$

$$H_e = \{1, 3, 4, 6, 9, 10\}, \tag{21.72}$$

$$H_\varphi = \{1, 2, 3, 4, 7, 9\}, \tag{21.73}$$

$$H_{\mathrm{APP}} = \{2, 3, 5, 10, 11, 12\}. \tag{21.74}$$

The datum is the ordered framework.

$$\mathcal{F} := (H^{-}, H_e, H_\varphi, H_{\mathrm{APP}}). \tag{21.75}$$

We define

$$o(\mathcal{F}) := (H_e \cap H_\varphi) \setminus (H^{-} \cup H_{\mathrm{APP}}). \tag{21.76}$$

Theorem 21.26 (Equivariant Reconstruction of the Origin)**.** *For the frame canonical $\mathcal{F}_*$,*

$$\boxed{o(\mathcal{F}_*) = \{1\}.} \tag{21.77}$$

Moreover, for every simultaneous permutation $g \in M_{12}$,

$$o(g\mathcal{F}) = g\, o(\mathcal{F}). \tag{21.78}$$

Proof. We have

$$H_e \cap H_\varphi = \{1, 3, 4, 9\}$$

and

$$(H^{-} \cup H_{\mathrm{APP}}) \cap \{1, 3, 4, 9\} = \{3, 4, 9\}.$$

The difference is $\{1\}$. For equivariance one applies the identities

$$g(A \cap B) = gA \cap gB, \quad g(A \cup B) = gA \cup gB, \quad g(A \setminus B) = gA \setminus gB,$$

valid for all bijections. □

The tetrad $B_K = \{4, 6, 9, 10\}$ does not by itself determine the origin. The singleton is obtained from the complete ordered frame; it is not a coordinate chosen after the result has been observed.

21.6.2 Positive radial chamber of the root system and selection of the dominant cone

Let ρ_i the copia of $\rho = \alpha_1 + \alpha_2$ in the i-esimo factor of A_2^{12}. Consideramos the dominio

$$\mathcal{R}_+ := \left\{ v(a) = \sum_{i=1}^{12} a_i \rho_i : a_i = 1 + 3b_i, b_i \in \mathbb{Z}_{\geq 0} \right\}. \tag{21.79}$$

Since the factors are orthogonal and $\rho_i^2 = 2$, if $S = \sum_i b_i$, then

$$\begin{aligned} v(a)^2 &= 2 \sum_{i=1}^{12} (1 + 3b_i)^2 \\ &= 24 + 12S + 18 \sum_{i=1}^{12} b_i^2. \end{aligned} \tag{21.80}$$

Reducing modulo eighteen,

$$v(a)^2 \equiv 6 + 12S \pmod{18},$$

and therefore

$$v(a)^2 \equiv 0 \pmod{18} \iff S \equiv 1 \pmod{3}. \tag{21.81}$$

Proposition 21.27 (Minima in the declared radial chamber)**.** *On the subset of $\mathscr{R}_+$ defined by $v(a)^2 \equiv 0 \pmod{18}$, the minimum norm is 54. It is attained exactly at twelve vectors: for each $j \in \{1, \dots, 12\}$, the coefficient a_j equals four, and the remaining eleven equal one.*

Proof. Condition (21.81) and the nonnegativity of the b_i imply that the smallest possible value of S is one. This forces a unique index j with $b_j = 1$ and $b_i = 0$ for $i \neq j$. Substituting into (21.80),

$$v(a)^2 = 24 + 12 + 18 = 54.$$

There are twelve choices of j. The next admissible value of S is four and produces a strictly greater norm. □

The origin $o(\mathcal{F}_*) = \{1\}$ selects

$$\boxed{v_{\mathcal{F}_*} = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \qquad v_{\mathcal{F}_*}^2 = 54.} \tag{21.82}$$

Remark 21.28 (Exact scope of minimality). The norm 54 is minimal in the positive radial chamber (21.79), not in the entire residue class. The vector

$$u = (\alpha_1 - 2\alpha_2, \rho, \dots, \rho)$$

satisfies

$$\alpha_1 - 2\alpha_2 - \rho = -3\alpha_2,$$

so that

$$u \equiv (\rho, \dots, \rho) \equiv v_{\mathcal{F}_*} \pmod{3A_2^{12}}.$$

However,

$$(\alpha_1 - 2\alpha_2)^2 = 2(1 + 2 + 4) = 14, \qquad u^2 = 14 + 11 \cdot 2 = 36.$$

This witness prevents extending the minimality assertion outside the domain of proposition 21.27.

21.6.3 Discriminant gluing of 12 $\mathbf{A}_2$ factors and construction of the integral extension

We write

$$N := N(A_2^{12}), \qquad v := v_{\mathcal{F}_*}.$$

We define

$$\chi_v : N \longrightarrow \mathbb{Z}/3\mathbb{Z}, \qquad \chi_v(x) = \langle x, v \rangle \bmod 3, \tag{21.83}$$

its nucleus

$$N_0 := \ker \chi_v = \{x \in N : \langle x, v \rangle \equiv 0 \pmod{3}\}, \tag{21.84}$$

and the extension

$$N_v := N_0 + \mathbb{Z}\frac{v}{3}. \tag{21.85}$$

21.6.4 Condition I1: surjectivity and exclusion of roots

By construction, $v \in A_2^{12} \subseteq N$. The roots of a factor A_2 are

$$\pm\alpha_1, \quad \pm\alpha_2, \quad \pm\rho.$$

Their pairings with ρ are ± 1 or ± 2. All the coefficients of v are congruent to one modulo three:

$$4 \equiv 1 \pmod 3, \qquad 1 \equiv 1 \pmod 3.$$

Therefore, no root of A_2^{12} has zero pairing with v modulo three. In particular,

$$\chi_v(\alpha_{1,1}) = 1,$$

so that χ_v is surjective and

$$[N : N_0] = 3. \tag{21.86}$$

Moreover, no root of N belongs to N_0.

21.6.5 Condition I2: norm divisibility

The identity (21.82) gives

$$v^2 = 54 \equiv 0 \pmod{18}, \qquad \left(\frac{v}{3}\right)^2 = 6. \tag{21.87}$$

In particular, $\langle v, v\rangle \equiv 0 \pmod 3$, hence $v \in N_0$.

21.6.6 Condition I3: class dual of order 3

If $x \in N_0$, there exists $k \in \mathbb{Z}$ with $\langle x, v\rangle = 3k$. Therefore,

$$\left\langle x, \frac{v}{3}\right\rangle = k \in \mathbb{Z},$$

and

$$\frac{v}{3} \in N_0^*. \tag{21.88}$$

On the other hand,

$$\left\langle \frac{v}{3}, \alpha_{1,1}\right\rangle = \frac{4}{3}.$$

Since $\alpha_{1,1} \in N$ and N is integral, this proves

$$\frac{v}{3} \notin N. \tag{21.89}$$

Finally, $3(v/3) = v \in N_0$. The class of $v/3$ in N_0^*/N_0 is nontrivial and has order exactly three. Consequently,

$$[N_v : N_0] = 3. \tag{21.90}$$

21.6.7 Parity, integrality and determinant of the neighbor

Proposition 21.29 (Unimodular structure of the neighbor)**.** *The lattice N_v is integral, even, and unimodular.*

Proof. Since N is unimodular and $[N : N_0] = 3$,

$$\det N_0 = [N : N_0]^2 \det N = 9.$$

Since $[N_v : N_0] = 3$,

$$\det N_v = \frac{\det N_0}{[N_v : N_0]^2} = 1.$$

Let $y = x + m(v/3)$, with $x \in N_0$, and write $\langle x, v\rangle = 3k$. Then

$$\begin{aligned} y^2 &= x^2 + 2m\left\langle x, \frac{v}{3}\right\rangle + m^2\left(\frac{v}{3}\right)^2 \\ &= x^2 + 2mk + 6m^2. \end{aligned}$$

The three summands form an even integer. The polarized form proves integrality, and determinant one proves unimodularity. □

21.6.8 2 clasis globalis and 6 clasis local

Globally, (21.85) adds only two nontrivial cosets:

$$N_0 + \frac{v}{3}, \qquad N_0 - \frac{v}{3}.$$

Locally, in each factor A_2, a coordinate of those classes belongs to one of the six classes

$$\mathscr{C}_{j,\pm} := A_2 + \mu_j \pm \frac{\rho}{3}, \qquad j \in \{0, 1, 2\}. \tag{21.91}$$

In the first coordinate, $4\rho/3 \equiv \rho/3 \pmod{A_2}$, so the same list holds for the twelve factors.

Table 21.1: Minimal representatives of the six local cosets.

class	minimal representative	(u, v)	$2(u^2 - uv + v^2)/9$
$\mathscr{C}_{0,-}$	$-\rho/3$	$(-1, -1)$	$2/9$
$\mathscr{C}_{0,+}$	$\rho/3$	$(1, 1)$	$2/9$
$\mathscr{C}_{1,-}$	$\mu_1 - \rho/3$	$(1, 0)$	$2/9$
$\mathscr{C}_{1,+}$	$\mu_1 + \rho/3 - \rho$	$(0, -1)$	$2/9$
$\mathscr{C}_{2,-}$	$\mu_2 - \rho/3$	$(0, 1)$	$2/9$
$\mathscr{C}_{2,+}$	$\mu_2 + \rho/3 - \rho$	$(-1, 0)$	$2/9$

Here (u, v) means the representative $(u\alpha_1 + v\alpha_2)/3$.

Lemma 21.30 (Exact minimum of the six classes)**.** *Each class in* (21.91) *has minimum norm exactly* $2/9$.

Proof. The norm of $(u\alpha_1 + v\alpha_2)/3$ is

$$\frac{2}{9}(u^2 - uv + v^2).$$

The integral form $u^2 - uv + v^2$ is positive definite. On a nonzero residual class it does not take the value zero, and hence is at least one. The six representatives in table 21.1 attain one. □

21.6.9 Criterion for the absence of roots in the Leech lattice and consequence for the minimum distance

Theorem 21.31 (Absence of vectors of norm two)**.** *The lattice N_v contains no vectors of norm two.*

Proof. All roots of N are those of A_2^{12}. Condition I1 proves that none belongs to N_0. Hence class N_0 contains no vectors of norm two.

In a new class $N_0 \pm v/3$, each of the twelve coordinates belongs to one of the six local classes. By lemma 21.30, each coordinate contributes at least $2/9$. Orthogonality gives

$$\left\| x \pm \frac{v}{3} \right\|^2 \geq 12 \cdot \frac{2}{9} = \frac{8}{3} > 2.$$

Nor do the new classes contain roots. □

21.6.10 Identification with the Leech lattice

Theorem 21.32 (Construction of the Leech lattice)**.** *The lattice*

$$N_v = \ker(\langle \cdot , v_{\mathcal{F}_*} \rangle \bmod 3) + \mathbb{Z}\frac{v_{\mathcal{F}_*}}{3} \tag{21.92}$$

is isometric to the Leech lattice:

$$\boxed{N_v \cong \Lambda_{24}.} \tag{21.93}$$

Proof. The lattice N_v has rank twenty-four and is positive definite. proposition 21.29 proves that it is even and unimodular; theorem 21.31 proves that it has no roots. By the Niemeier classification, there exists a unique isometry class of positive-definite, even, unimodular lattices of rank twenty-four without roots: the Leech lattice. □

21.6.11 The 243 syndromes of the ternary punctured code

Let G_{11} be the ternary Golay code with parameters $[11, 6, 5]_3$. Its syndrome space has cardinality

$$3^{11-6} = 3^5 = 243. \tag{21.94}$$

The number of weight errors of at most two is

$$1 + 2\binom{11}{1} + 2^2\binom{11}{2} = 1 + 22 + 220 = 243. \tag{21.95}$$

Proposition 21.33 (Perfect correspondence error–syndrome)**.** *Each syndrome of G_{11} has a unique error representative of weight at most two.*

Proof. Minimum distance five implies that two distinct errors of weight at most two cannot differ by a codeword: their difference would have weight at most four. The error–syndrome map is therefore injective on the ball of radius two. Both sets have cardinality 243 by (21.94) and (21.95); the map is bijective. □

These 243 syndromes are not the 243 visible words obtained from the TPK catalogue. They share a cardinal, but they possess distinct domains, equivalences and functions.

21.6.12 Obstructions of the ternary neighbor $N_v \cong \Lambda_{24}$: class of order 3, absence of roots, and separation of the cardinals 243

Comparison Criterion 21.34 (Refutation of the neighbor and of the Leech identification)**.** The construction is refuted if any of the following facts occurs:

1. the operator $o(\mathcal{F})$ not produces $\{1\}$ or is not equivariant;
2. there exists in the radial chamber an admissible vector of norm less than 54, or the number of minimizers is not twelve;
3. global minimality is asserted despite the witness of norm 36;
4. χ_v is not sobreyectivo or one root belongs to N_0;
5. $v^2 \not\equiv 0 \pmod{18}$ o $(v/3)^2 \neq 6$;
6. $v/3 \notin N_0^*$, $v/3 \in N$, or its class does not have order three;
7. the determinant of N_v is not one, or the neighbor ceases to be even;
8. some of the six clasis local has minimum least that $2/9$;

9. one of the two new global classes contains a root;
10. the two global classes are confused with the six local classes;
11. W_{24} is used as an antecedent of the gluing or of the neighbor;
12. the 243 syndromes are identified with the 243 words visible by sharing cardinality.

21.7 Finite holonomy, Schläfli configuration and symmetry E_6

The Witt branch does not exhaust the exceptional structures accessible from APP. The parallel branch of this chapter starts from the two mixed sum–product and product–sum polarizations, together with a plaquette orientation and a declared central section. The calculated curvature determines the alternating form; this defines the central extension. An extension is not inferred from cardinality 27 alone.

21.7.1 Ordered system of incidences and radial camera orientation

On $T_9 = (\mathbb{Z}/9\mathbb{Z})^2$, with $S(x, y) = x + y$ and $P(x, y) = xy$, let

$$\Delta_x f = f(x + 1, y) - f(x, y), \qquad \Delta_y f = f(x, y + 1) - f(x, y).$$

Mixed connections are

$$A^{SP} = (\Delta_x S, \Delta_y P) = (1, x), \qquad A^{PS} = (\Delta_x P, \Delta_y S) = (y, 1).$$

Its discrete curvature is

$$F_A = \Delta_x A_y - \Delta_y A_x.$$

Proposition 21.35 (Oriented opposite curvatures)**.**

$$F_{A^{SP}} = 1, \qquad F_{A^{PS}} = -1 \pmod 9.$$

The circulation over an oriented rectangle of sides r, s is, respectively, rs and $-rs$.

Proof. $\Delta_x x - \Delta_y 1 = 1$ and $\Delta_x 1 - \Delta_y y = -1$. A rectangle contains rs plaquettes; upon summing, the interior edges cancel and the boundary circulation remains. □

For displazamientos $u = (a, b)$, $v = (c, d)$, the class alternante is

$$\sigma(u, v) = ad - bc \pmod 9. \tag{21.96}$$

The orientation fixes the sign of σ. The central section used below belongs to the gauge of this realization; replacing it with a coboundary produces an isomorphic extension, not a new invariant.

21.7.2 Central extension and ternary reduction

We define the nonadic Heisenberg group

$$\mathcal{H}_9 = \{(a, b, z) : a, b, z \in \mathbb{Z}/9\mathbb{Z}\}, \quad (a, b, z)(c, d, w) = (a + c, b + d, z + w + bc). \tag{21.97}$$

The commutator of two horizontal shifts is

$$[(a, b, 0), (c, d, 0)] = (0, 0, bc - ad),$$

which records $-\sigma(u, v)$. Reducing each coordinate modulo three gives

$$H_3 = \{(a, b, z) : a, b, z \in \mathbb{F}_3\}, \qquad |H_3| = 27, \tag{21.98}$$

with the same law. The nucleo of $\mathcal{H}_9 \to H_3$ is $(3\mathbb{Z}/9\mathbb{Z})^3$, of cardinal 27.

21.7.3 Construction of the radial character and determination of its congruence kernel

The nonadic extension admits an explicit unitary realization. Let

$$\zeta_9 := \exp(2\pi i/9), \qquad \mathscr{H}_9^{\mathrm{Sch}} := \mathbb{C}^9,$$

based on $\{|n\rangle : n \in \mathbb{Z}/9\mathbb{Z}\}$. We define

$$X|n\rangle := |n+1\rangle, \qquad Z|n\rangle := \zeta_9^n |n\rangle. \tag{21.99}$$

We have

$$ZX = \zeta_9 XZ$$

and, by multiplication,

$$(X^a Z^b)(X^c Z^d) = \zeta_9^{bc} X^{a+c} Z^{b+d}. \tag{21.100}$$

Theorem 21.36 (Faithful Weyl–Heisenberg Representation)**.** *The map*

$$\rho_{\mathrm{WH}} : \mathcal{H}_9 \longrightarrow U(\mathscr{H}_9^{\mathrm{Sch}}), \qquad \rho_{\mathrm{WH}}(a,b,z) := \zeta_9^z X^a Z^b, \tag{21.101}$$

is a faithful unitary representation. For $u = (a,b)$ and $v = (c,d)$, its commutator satisfies

$$[\rho_{\mathrm{WH}}(u,0), \rho_{\mathrm{WH}}(v,0)] = \zeta_9^{bc-ad} I = \zeta_9^{-\sigma(u,v)} I. \tag{21.102}$$

Proof. The identity (21.100) reproduces the term bc of (21.97); hence the homomorphism property. If $\zeta_9^z X^a Z^b = I$, the action on the basis successively forces $a = 0$, $b = 0$, and $z = 0$. This proves faithfulness. Comparing the products in the two orders gives ζ_9^{bc-ad}, which is the alternating phase of (21.96). □

The representation distinguishes two conjugate observables. Let

$$Q_{\mathrm{H}} := \mathrm{diag}(0,1,\dots,8), \qquad F_{jk} := \frac{1}{3}\zeta_9^{jk}, \qquad P_{\mathrm{H}} := F^{\dagger} Q_{\mathrm{H}} F.$$

The eigenbases of Q_{H} and P_{H} are mutually unbiased, since $|F_{jk}|^2 = 1/9$. For every unit vector $\psi \in \mathscr{H}_9^{\mathrm{Sch}}$,

$$\mathrm{Var}_\psi(Q_{\mathrm{H}})\,\mathrm{Var}_\psi(P_{\mathrm{H}}) \geq \frac{1}{4}\,|\langle \psi, [Q_{\mathrm{H}}, P_{\mathrm{H}}]\psi\rangle|^2\,, \tag{21.103}$$

$$H_{Q_{\mathrm{H}}}(\psi) + H_{P_{\mathrm{H}}}(\psi) \geq \log 9. \tag{21.104}$$

These operators are dimensionless. A physical unit chart is a later realization and does not intervene in the central extension.

21.7.4 Action of 10 displazamientos over the grafo of Cayley

For a linear functional $\ell(a,b) = pa + qb$, set

$$D_\ell(a,b) = \left(a, b, \frac{ab}{2} + \ell(a,b)\right), \qquad Z = (0,0,1),$$

where $2^{-1} = 2$ in $\mathbb{F}_3$. The inverse-closed set

$$S_\ell = \{D_\ell(v) : v \in \mathbb{F}_3^2 \setminus \{0\}\} \cup \{Z, Z^{-1}\} \tag{21.105}$$

has ten elements and does not contain the identity.

Theorem 21.37 (Independence of the linear scale)**.** *The nine functionals ℓ produce isomorphic Cayley graphs. The nine sets S_ℓ form a single orbit under* $\mathrm{Aut}(H_3)$.

Proof. The map

$$\phi_\ell(a,b,z) = (a,b,z+\ell(a,b))$$

is a central automorphism and satisfies $\phi_\ell(S_0) = S_\ell$. To write the complete family in the polarized coordinates of equation (21.97), let $c(v,w) = bc$ for $v = (a,b)$, $w = (c,d)$. Given $M \in \mathrm{GL}(2,3)$, the defect

$$B_M(v,w) = c(Mv,Mw) - \det(M)c(v,w)$$

is symmetric, because the alternating parts of the two terms coincide. We define the quadratic correction

$$q_M(v) = 2B_M(v,v), \qquad q_M(v+w) - q_M(v) - q_M(w) = B_M(v,w), \tag{21.106}$$

where $2 = 2^{-1}$ in $\mathbb{F}_3$. The complete family of automorphisms is then written

$$(v,t) \longmapsto \big(Mv, \det(M)t + q_M(v) + \ell(v)\big), \qquad M \in \mathrm{GL}(2,3),\ \ell \in (\mathbb{F}_3^2)^*. \tag{21.107}$$

The polar identity of equation (21.106) directly proves that the central product is preserved. The family has $48 \cdot 9 = 432$ elements. Exhaustive enumeration of the $\binom{13}{5} = 1287$ inverse-closed subsets of cardinality ten finds exactly those nine with the parameters derived below. □

Write $\underline{S} = \sum_{s\in S} s$ in $\mathbb{Z}[H_3]$. Counting ordered products gives

$$\underline{S}^{\,2} = 10e + \underline{S} + 5(\underline{H_3} - e - \underline{S}) = 5\underline{H_3} + 5e - 4\underline{S}. \tag{21.108}$$

Indeed, the identity receives ten factorizations; an element of S receives one; each of the other sixteen receives five.

Theorem 21.38 (Graph of the twenty-seven lines)**.** *The matrix of adjacency A_ℓ of $\Gamma_\ell = \mathrm{Cay}(H_3, S_\ell)$ satisfies*

$$A_\ell^2 = 5I - 4A_\ell + 5J. \tag{21.109}$$

Therefore,

$$\Gamma_\ell = \mathrm{srg}(27,10,1,5), \qquad \mathrm{Spec}(A_\ell) = \{10^1, 1^{20}, (-5)^6\}.$$

It is isomorphic to the intersection graph of the twenty-seven lines of a smooth cubic surface.

Proof. The regular representation of equation (21.108) produces equation (21.109). Its entries state that every vertex has ten neighbors, that an adjacent pair has one common neighbor, and a nonadjacent pair has five. On the constant line, A_ℓ acts by 10; on its complement, $J = 0$ and $(A_\ell - I)(A_\ell + 5I) = 0$. Zero trace fixes the multiplicities 20 and 6.

In the model of a cubic obtained by blowing up six points of $\mathbb{P}^2$, the 27 classes are

$$E_i, \qquad L_{ij} = H - E_i - E_j, \qquad Q_i = 2H - \sum_{j\neq i} E_j.$$

With $H^2 = 1$, $E_i^2 = -1$, $H \cdot E_i = 0$, and $E_i \cdot E_j = 0$ for $i \neq j$, its intersection matrix has the same parameters. Identification with the classical configuration uses the uniqueness theorem for this realization of the 27 lines [86]; it is not attributed to a nonexistent certificate. □

The isomorphism assertion can be made fully effective by means of a computed bijection. The labeling is not absolute: another linear section or an automorphism of the surface produces another table in the same class.

Table 21.2: Explicit bijection between H_3 and the twenty-seven classes of lines.

$a\,b\,z$	class of line
0 0 0	E_1
0 0 1	L_{12}
0 0 2	Q_2
0 1 0	L_{13}
0 1 1	Q_1
0 1 2	E_3
0 2 0	Q_3
0 2 1	E_2
0 2 2	L_{23}
1 0 0	L_{14}
1 0 1	L_{35}
1 0 2	L_{26}
1 1 0	L_{36}
1 1 1	L_{24}
1 1 2	L_{15}
1 2 0	L_{25}
1 2 1	L_{16}
1 2 2	L_{34}
2 0 0	Q_4
2 0 1	L_{46}
2 0 2	E_6
2 1 0	E_5
2 1 1	Q_6
2 1 2	L_{56}
2 2 0	L_{45}
2 2 1	E_4
2 2 2	Q_5

As a local control, the ten neighbors of $e = (0,0,0)$ are the elements of S_0. The table transforms them into the ten classes that intersect E_1: the five L_{1j} and the five Q_j, with $2 \le j \le 6$. The complete verification compares all 729 entries of both adjacency matrices.

The complement $B_\ell = J - I - A_\ell$ satisfies

$$B_\ell^2 = 8I + 2B_\ell + 8J, \qquad \mathrm{srg}(27, 16, 10, 8), \qquad \mathrm{Spec}(B_\ell) = \{16^1, 4^6, (-2)^{20}\}. \tag{21.110}$$

This is the Schläfli graph in the convention of complementary adjacency; A_ℓ is the intersection graph with parameters $(27, 10, 1, 5)$.

Each edge belongs to a unique triangle, since $\lambda = 1$. The number of triangles is

$$\frac{27 \cdot 10}{6} = 45.$$

The full automorphism group has order

$$27 \cdot 1920 = 51840$$

and, through its action on the line configuration, is identified with $W(E_6)$, not merely by equality of orders [86].

21.7.5 The root lattice realizing E_6

To make explicit the object on which that group acts, let

$$\mathrm{Pic}(X) = \mathbb{Z}H \oplus \bigoplus_{i=1}^{6} \mathbb{Z}E_i, \qquad H^2 = 1, \qquad E_i^2 = -1, \qquad H \cdot E_i = E_i \cdot E_j = 0 \ (i \neq j), \tag{21.111}$$

the Picard lattice of the blow-up of six general points of $\mathbb{P}^2$. Its canonical class is

$$K_X = -3H + E_1 + \cdots + E_6.$$

The orthogonal complement

$$L_{E_6} = K_X^{\perp} \subset \mathrm{Pic}(X), \qquad (x, y)_{E_6} := -x \cdot y, \tag{21.112}$$

is positive definite of rank six. A simple root basis is

$$\begin{aligned} &\alpha_1 = E_1 - E_2, \quad \alpha_2 = E_2 - E_3, \quad \alpha_3 = E_3 - E_4, \\ &\alpha_4 = E_4 - E_5, \quad \alpha_5 = E_5 - E_6, \quad \alpha_6 = H - E_1 - E_2 - E_3. \end{aligned} \tag{21.113}$$

Each α_i is orthogonal to K_X, has norm two, and its products are -1 exactly for adjacent pairs of the diagram E_6: the chain $1 - 2 - 3 - 4 - 5$, with node 6 joined to node 3. Therefore, its Gram matrix is the Cartan matrix of E_6.

Proposition 21.39 (The seventy-two roots and their reflections)**.** *The roots of L_{E_6} are*

$$\begin{aligned} &E_i - E_j && (i \neq j), \\ &\pm (H - E_i - E_j - E_k) && (1 \leq i < j < k \leq 6), \\ &\pm (2H - E_1 - \cdots - E_6), \end{aligned} \tag{21.114}$$

in total $30 + 40 + 2 = 72$*. For each root α,*

$$s_\alpha(x) = x + (x \cdot \alpha)\alpha \tag{21.115}$$

preserves the intersection and K_X. The six simple reflections generate $W(E_6)$ and act faithfully by permutations on the 27 classes E_i, L_{ij}, Q_i in theorem 21.38.

Proof. Substituting each class of equation (21.114) into equation (21.111) gives $K_X \cdot \alpha = 0$ and $\alpha^2 = -2$. The class count is as stated. The reflection formula is the orthogonal reflection for the form $-(\cdot)$; it preserves K_X because $K_X \cdot \alpha = 0$. The simple system of equation (21.113) generates all roots, and its Coxeter group is $W(E_6)$. Its action on the 27 classes of exceptional curves preserves intersection, and therefore coincides with the action on the already constructed graph [86]. □

21.7.6 Stratification of the 729 Words

We write a six-trit word as

$$w = (\ell_1, h_1, \ell_2, h_2, \ell_3, h_3),$$

and we define two elements of H_3,

$$L(w) = (\ell_1, \ell_2, \ell_3), \qquad H(w) = (h_1, h_2, h_3),$$

using the central law in their products. According to $L(w)^{-1}H(w)$, the 729 words are divided into

$$\begin{aligned}\mathcal{D}_\ell &= \{w : L(w)^{-1}H(w) = e\},\\ \mathcal{I}_\ell &= \{w : L(w)^{-1}H(w) \in S_\ell\},\\ \mathcal{S}_\ell &= \{w : L(w)^{-1}H(w) \notin \{e\} \cup S_\ell\}.\end{aligned}$$

Theorem 21.40 (Hensel–Schläfli Partition)**.**

$$729 = 27 + 270 + 432, \tag{21.116}$$

corresponding to diagonal, incidence, and warping.

Proof. For each one of the 27 values of $L(w)$, there is a unique value of $H(w)$ with identity difference, ten with difference in S_ℓ and sixteen remaining ones. 27 is multiplied by $1, 10, 16$. □

The cardinal 432 of the warped stratum numerically coincides with the multiplicity of the transverse region of π in equation (9.4). The two objects are not thereby identified: here one counts words classified by $L(w)^{-1}H(w)$, whereas there one counts preimages of a register U_6. Without an explicit and equivariant intertwiner between those domains, $432 = 432$ is only a cardinal control.

21.7.7 Necessity of the central phase

Enumerating all inverse-closed subsets of cardinality ten in the five groups of order 27 gives

group	abelian	$\#\,\mathrm{srg}(27, 10, 1, 5)$
C_{27}	yes	0
$C_9 \times C_3$	yes	0
C_3^3	yes	0
$C_9 \rtimes C_3$	no	9
H_3	no	9.

Thus, retaining 27 labels while abelianizing composition destroys the configuration. Neither cardinality nor degree suffices.

If $\mathcal{T}$ is the conjunto of 45 triangulos, the cubico

$$C_\Gamma(x_1, \dots, x_{27}) = \sum_{\{i,j,k\} \in \mathcal{T}} x_i x_j x_k$$

recovers adjacency, because every edge lies in a unique triangle. Its permutation stabilizer is exactly

$$\mathrm{Aut}(\Gamma_\ell) \cong W(E_6).$$

$$\text{APP curvatures } \pm 1 \longrightarrow \sigma \longrightarrow \mathcal{H}_9 \longrightarrow H_3 \longrightarrow \Gamma_{27} \longrightarrow \mathcal{T}_{45} \longrightarrow W(E_6).$$

Figure 21.3: Causal chain of the mixed-holonomy branch. It is parallel to $C_W \to W_{12} \to M_{12}$ and not a link of it.

Comparison Criterion 21.41 (Falsification criteria for branch E_6)**.** The branch is refuted if the curvatures are not opposite, if the central cocycle is erased, if the quadratic correction (21.106) does not preserve the product, if (21.108) does not produce the coefficients $(10, 1, 5)$, if an abelian group produces the same

configuration, if the partition does not sum to 729, if the table 21.2 does not intertwine the adjacency matrices, or if $W(E_6)$ is identified only by its order and not by its action on the twenty-seven classes and the 72 roots.

21.8 Oriented APP–Fock coupling and transverse rigidity of 12 fibres

The ordinary cyclotomic determinant does not distinguish C from C^{-1}. Sum–product information is preserved by explicitly introducing sheet parity and an orientation line. The resulting coupling produces index 54 from the APP aggregate. Once the four coupling conditions have been fixed, 54, J, and 196884 are not inputs to the evaluation. This evaluative independence asserts neither universal uniqueness of the ansatz nor historical independence of its design.

21.8.1 Triangular mark on each orbit

Let

$$M = \mathbb{Z}[\mathbb{Z}/9\mathbb{Z}]$$

and let $M_{\mathrm{bal}}^{C_3}$ be the submodule of invariant elements ν whose six unit coefficients coincide:

$$\nu_u = c(\nu) \qquad \big(u \in (\mathbb{Z}/9\mathbb{Z})^\times\big).$$

The coefficients of e_0, e_3, e_6 permanecen libres.

For a free orbit $O = \{u, 4u, 7u\}$, set

$$b_\nu = e_\nu - e_{4\nu} \in A(O).$$

If $b_\nu^\flat$ is the covector defined by the form of A_2, the three roots of the triangular frame satisfy

$$\boxed{\sum_{\nu \in O} b_\nu \otimes b_\nu^\flat = 3 I_{A(O)}.}$$

Consequently,

$$\sum_{\nu \in O} \nu_\nu \, b_\nu \otimes b_\nu^\flat = 3c(\nu) I.$$

The factor three proceeds from the resolution of the identity by the root framework.

21.8.2 Oriented coupling $\mathcal{H}_{\Sigma,\Pi}^{(m)}$ between the balance C_3, leaf parity, and degree 2 of Fock space

For $m \in 2\mathbb{N}_{>0}$, let

$$\begin{aligned} \mathcal{V}_m &= \mathcal{V}_{\Sigma,m} \oplus \mathcal{V}_{\Pi,m}, \\ \mathcal{V}_{\Sigma,m} &= (A_2 \otimes \mathbb{C})^{\oplus m/2}, \qquad \mathcal{V}_{\Pi,m} = (A_2 \otimes \mathbb{C})^{\oplus m/2}, \\ g_m &= C^{\oplus m/2} \oplus (C^{-1})^{\oplus m/2}, \\ \mathsf{E}_m &= I_{\mathcal{V}_{\Sigma,m}} \oplus (-I_{\mathcal{V}_{\Pi,m}}). \end{aligned}$$

The specialization $m = 12$, with the labels of pair, sheet and orbit, is $\mathcal{V}_{12} = W_{\mathrm{APP}} \otimes \mathbb{C}$. This notation prevents confusing the representational space with the Witt system $W_{12} = S(5, 6, 12)$. Degree two of the Fock space and the induced action are

$$\mathcal{F}_2(\mathcal{V}_m) = \mathcal{V}_m[2] \oplus \mathrm{Sym}^2(\mathcal{V}_m[1]),$$

$$\Gamma_2(g_m) = g_m|_{\mathcal{V}_m[2]} \oplus \mathrm{Sym}^2(g_m|_{\mathcal{V}_m[1]}).$$

Let $o_{\Sigma\Pi}$ be the signed orientation line $\Pi - \Sigma$, and let $M_{\mathrm{bal}}^{C_3,-}$ be the oriented copy in which the exchange of leaves acts by $\nu \mapsto -\nu$. We define

$$\boxed{\begin{aligned}&\mathcal{H}^{(m)}_{\Sigma,\Pi} : M^{C_3,-}_{\mathrm{bal}} \longrightarrow \mathrm{End}(\mathcal{F}_2(\mathcal{V}_m)) \otimes o_{\Sigma\Pi},\\ &\mathcal{H}^{(m)}_{\Sigma,\Pi}(\nu) = 3c(\nu)\left(0_{\mathcal{V}_m[2]} \oplus \mathrm{Sym}^2 \mathsf{E}_m\right) \otimes o_{\Sigma\Pi}.\end{aligned}}$$

The transformation is linear in c, is supported on $\mathrm{Sym}^2(\mathcal{V}_m[1])$, uses leaf parity, and adopts the normalization of the triangular frame.

Let σ the exchange of sheets. Then

$$\nu \longmapsto -\nu, \qquad o_{\Sigma\Pi} \longmapsto -o_{\Sigma\Pi}, \qquad \mathsf{E}_m \longmapsto -\mathsf{E}_m,$$

and

$$\mathrm{Sym}^2(-\mathsf{E}_m) = \mathrm{Sym}^2(\mathsf{E}_m).$$

Therefore,

$$\mathcal{H}^{(m)}_{\Sigma,\Pi}(\sigma\nu) = \sigma\mathcal{H}^{(m)}_{\Sigma,\Pi}(\nu).$$

When the orientation is reversed, the dual covector also changes sign; thus $(-o^{\vee}_{\Sigma\Pi})(-o_{\Sigma\Pi}) = 1$, and the oriented evaluation remains normalized.

Fixing $o^{\vee}_{\Sigma\Pi}(o_{\Sigma\Pi}) = 1$, the oriented trace is

$$\mathrm{Tr}_{o,\mathcal{F}_2}(T \otimes o_{\Sigma\Pi}) = \mathrm{Tr}_{\mathcal{F}_2}(T).$$

Theorem 21.42 (APP-oriented Index–Fock)**.** *Suppose that m fibers A_2, with m even, are distributed equally between the orientations C and C^{-1}. Then*

$$\boxed{\mathrm{Tr}_{o,\mathcal{F}_2}\left(\mathcal{H}^{(m)}_{\Sigma,\Pi}(\nu)\Gamma_2(g_m)\right) = -\frac{3m\,c(\nu)}{2}.}$$

For the aggregate produced by APP,

$$\delta = \nu_\Pi - \nu_\Sigma = 12e_0 + 3e_3 + 3e_6 - 3\sum_{u\in(\mathbb{Z}/9\mathbb{Z})^\times} e_u,$$

one has $c(\delta) = -3$ and

$$m = 3\ \textit{pairs} \cdot 2\ \textit{sheets} \cdot 2\ \textit{orbits} = 12.$$

Therefore,

$$\boxed{\mathrm{Tr}_{o,\mathcal{F}_2}\left(\mathcal{H}^{(12)}_{\Sigma,\Pi}(\delta)\Gamma_2(g_{12})\right) = 54.}$$

Proof. Let $A = \mathsf{E}_m g_m$. The contributions of the two leaves cancel: $\mathrm{tr}\,A = 0$. Since E_m commutes with g_m and $\mathsf{E}_m^2 = I$,

$$\mathrm{tr}(A^2) = \mathrm{tr}(g_m^2) = -m.$$

The identity

$$\mathrm{tr}(\mathrm{Sym}^2 A) = \frac{(\mathrm{tr}\,A)^2 + \mathrm{tr}(A^2)}{2}$$

gives $-m/2$; the frame supplies $3c(\nu)$. □

Corollary 21.43 (Minimality of the Bosonic Degree Detector)**.** *In the declared class of couplings, degree one does not detect the oriented aggregate, and degree two is the first positive bosonic degree that detects it:*

$$\mathrm{tr}(\mathsf{E}_m g_m) = 0, \qquad \mathrm{tr}\Big(\mathrm{Sym}^2(\mathsf{E}_m g_m)\Big) = -\frac{m}{2} \neq 0.$$

Proof. The first equality is the cancellation of the two sheets. The second was obtained in the proof of theorem 21.42. Since degree zero contains only the unit and degree one has zero trace, degree two is the first with a nonzero response. □

21.8.3 Transverse rigidity of the number of fibers

For an integer $m \geq 1$, we define the formal series

$$T_m(q) := q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-m} \in q^{-1}\mathbb{Z}[[q]] \tag{21.117}$$

and, in the orthogonal sum A_2^m, the radial vector

$$v_m := 4\rho_1 + \rho_2 + \cdots + \rho_m, \qquad \|\rho_j\|^2 = 2. \tag{21.118}$$

We construct two entire functions without using $m = 12$ beforehand:

$$F(m) := [q]T_m(q), \qquad R(m) := \|v_m\|^2.$$

Proposition 21.44 (Fock coefficient and radial norm)**.** *For every $m \geq 1$,*

$$F(m) = \frac{m(m-3)}{2}, \qquad R(m) = 2(m+15). \tag{21.119}$$

Proof. Up to second order,

$$(1 + q + q^2)^{-m} = 1 - mq + \frac{m(m-1)}{2} q^2 + O(q^3),$$

$$(1 + q^2 + q^4)^{-m} = 1 - mq^2 + O(q^4).$$

The factors with $n \geq 3$ do not contribute to the coefficient of q^2. Since the exterior factor of T_m is q^{-1},

$$F(m) = \frac{m(m-1)}{2} - m = \frac{m(m-3)}{2}.$$

By orthogonality of the m copies of A_2,

$$R(m) = 4^2\|\rho_1\|^2 + \sum_{j=2}^{m} \|\rho_j\|^2 = 32 + 2(m-1) = 2(m+15).$$

□

Theorem 21.45 (Cyclotomic–radial rigidity)**.** *The equality $F(m) = R(m)$ is equivalent to*

$$(m - 12)(m + 5) = 0. \tag{21.120}$$

In the domain $m \in \mathbb{Z}_{>0}$, the unique solution is

$$\boxed{m = 12, \qquad F(12) = R(12) = 54.}$$

Proof. Equating the two expressions of (21.119) produce

$$m(m-3) = 4(m+15),$$

that is,

$$m^2 - 7m - 60 = (m - 12)(m + 5) = 0.$$

The second root is negative and lies outside the domain. □

This rigidity belongs to the declared cyclotomic–radial family. It does not use the value 54 to choose m: it first constructs $F(m)$ and $R(m)$, and then resolves their equality.

21.8.4 Factorization of the APP–Fock coupling: hypotheses, invariants, necessary conditions for validity, and obstructions

The coupling factorizes by

$$M_{\mathrm{bal}}^{C_3,-}/\ker c \cong \mathbb{Z} \otimes o_{\Sigma\Pi},$$

and the induced map is injective because $\mathrm{Sym}^2 \mathsf{E}_m \neq 0$. The formula is selected by the four declared conditions: linearity in c, support in $\mathrm{Sym}^2(\mathcal{V}_m[1])$, sheet parity, and normalization by the triangular frame. No uniqueness is asserted among all natural couplings, nor historical independence of the ansatz. Once those conditions have been fixed, 54, J, and 196884 are not inputs to the evaluation.

Three controls show that the result is not determined by a retroactive normalization:

$$(m, c) = (12, -2) \longmapsto 36, \qquad (m, c) = (10, -3) \longmapsto 45,$$

and

$$\delta_k = \delta + k e_0$$

leaves $\mathcal{H}_{\Sigma,\Pi}^{(12)}$ invariant while the digital weight changes from 54 to $54 + 9k$. In particular, for $k = 1$,

$$\mathrm{Tr}_{o,\mathcal{F}_2}\left(\mathcal{H}_{\Sigma,\Pi}^{(12)}(\delta + e_0)\Gamma_2(g_{12})\right) = 54, \qquad w(\delta + e_0) = 63.$$

Proposition 21.46 (Need for orientation data)**.** *If $o_{\Sigma\Pi}$ is forgotten, every natural scalar that is linear in $\nu_\Pi - \nu_\Sigma$ and depends only on the class of the representation vanishes.*

Proof. C and C^{-1} are conjugate, so the representational class is even under sheet exchange. The aggregate $\nu_\Pi - \nu_\Sigma$ is odd. Without the orientation line, a transformation that is simultaneously even and odd is zero. □

The theorem thus establishes the exact transport of the aggregate to Fock degree two. The local state-by-state defect preserves the descent obstruction of corollary 1.97; the two statements belong to distinct domains and are compatible.

21.8.5 Complete genealogy of the exceptional projection: from Paley and Hadamard to the Witt, Leech and Moonshine systems

Exceptional evaluation starts from the same enriched state as Archimedean evaluation. The former preserves incidence, orientation, carry and boundary; the latter publishes the real value of the section. This bifurcation makes a genealogy of values and a genealogy of symmetries coexist from a common antecedent.

Two linear transformations fulfill successive and typed functions. The Hadamard transform organizes the signed dodecaphase state and its cyclotomic sectors. The Paley conference matrix then fixes the projective gauge of the exceptional branch. Its ternary reduction

$$C_P C_P^{\mathsf{T}} = 5I_6, \qquad A_W = C_P \bmod 3, \qquad A_W^2 = -I_6$$

produces the internal quadratic endomorphism and the graph code

$$C_W = \{(q, qA_W) : q \in \mathbb{F}_3^6\} = [12, 6, 6]_3.$$

The enumerator

$$W_{C_W}(y) = 1 + 264y^6 + 440y^9 + 24y^{12}$$

expresses the Hamming metric of this realization. The 264 vectors of weight six, identified in opposite pairs, determine 132 hexads and construct the first Witt system,

$$W_{12} = S(5, 6, 12).$$

The 495 Witt stars provide the incidence involutions whose generated action has image M_{12}.

The second Steiner structure is obtained in the extended binary Golay code $\mathcal{G}_{24}$. The weight-eight supports form

$$W_{24} = S(5, 8, 24), \qquad \mathrm{Aut}(W_{24}) \cong M_{24}.$$

With a dodecad D fixed, the intersections of the octads with D at six points reconstruct W_{12}; each hexad admits a unique lift to an octad. The pair of systems

$$S(5, 6, 12) \longrightarrow S(5, 8, 24)$$

is therefore an incidence extension proved within binary Golay and preserves both the dodecad and its alignment.

From the common ternary code the lattice branch arises simultaneously. Coordinate-by-coordinate gluing over A_2^{12} constructs the Niemeier lattice

$$C_W \longrightarrow N(A_2^{12}).$$

The selected radial vector defines an even, unimodular index-three neighbor of rank twenty-four without roots; the classification identifies it with the Leech lattice:

$$N(A_2^{12}) \xrightarrow{\text{3-neighbor}} \Lambda_{24}.$$

On this output the lattice vertex operator algebra $V_{\Lambda_{24}}$ is constructed. The canonical sign involution and its twisted module produce the orbifold

$$V^\natural = V^+_{\Lambda_{24}} \oplus V^{T,+}_{\Lambda_{24}}, \qquad \mathrm{Aut}(V^\natural) \cong \mathbb{M}.$$

Its graded character is the normalized Hauptmodul

$$\mathrm{Tr}_{V^\natural}\, q^{L_0 - 1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots,$$

and Moonshine theory assigns to each conjugacy class $[g] \subset \mathbb{M}$ the McKay–Thompson series $T_g(\tau) = \mathrm{Tr}_{V^\natural}(g\, q^{L_0 - 1})$, endowed with the corresponding modular and replicability properties.

The common antecedent of both extensions preserves the complete chain

$$w_6 \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9 \longrightarrow x_{\mathrm{term}} \longrightarrow U_{12}^{\mathrm{sgn}} \longrightarrow K \longrightarrow B_K.$$

From this terminal state the evaluation route and the gauge route must be distinguished. The state produces the incidence words; the Paley matrix defines, by ternary reduction, the operator that evaluates them. The complete exceptional genealogy is therefore organized as follows:

$$
\begin{aligned}
C_P &\longmapsto A_W = C_P \bmod 3,\\
x \in X_\infty^{\mathrm{cal}} &\longmapsto \varepsilon_\infty(x) = (w_j)_j \longmapsto \big(\Gamma_{A_j}(w_j)\big)_j \in \prod_j C_W(A_j),\\
C_W &\longrightarrow W_{12} \xrightarrow{(D,\ell)} W_{24},\\
W_{12} &\xrightarrow{\mathrm{Aut}} M_{12}, \qquad W_{24} \xrightarrow{\mathrm{Aut}} M_{24},\\
C_W &\longrightarrow N(A_2^{12}) \longrightarrow \Lambda_{24} \longrightarrow V_{\Lambda_{24}} \longrightarrow V^\natural \longrightarrow \mathbb{M} \longrightarrow \text{Moonshine}.
\end{aligned}
$$

Hamming weight and distance belong to the metric of the codes; Hadamard belongs to the dodecaphase transformation; Paley fixes the conference gauge. This separation of functions allows each arrow to be followed through its operator, domain and certificate, while preserving the genealogical unity of the double projection.

21.9 Dodecaphase incidence and the Witt–Leech–Monster corridor

The section 21.2 has explicitly constructed the matrix A_W, the code C_W and the system W_{12}. On those coordinates, the double projection is formalized by two evaluation maps which, in the declared gauge, produce the same polarized Witt flag through distinct operations. This calibrated flag also determines a marked origin. The construction reaches the design W_{12}, its automorphism group M_{12} and, through a second branch that does not pass through W_{24}, an even unimodular rootless 3-neighbor. Classical classification identifies the latter with the Leech lattice; the extension to the moonshine vertex operator algebra and the Monster group is obtained through the corresponding classical theorems.

21.9.1 Dodecaphase state and incidence supports

The dodecaphase vector is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601).$$

The four words of the frame are

$$w_\pi = 010211, \quad w_e = 201101, \quad w_\varphi = 121200, \quad w_{\mathrm{APP}} = 022020.$$

In the symmetric Paley–Witt gauge the encoder used is

$$c(w) = (w, wA_W) \in \mathbb{F}_3^{12}.$$

Their supports are

$$
\begin{aligned}
P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\},\\
H_e &= \{1, 3, 4, 6, 9, 10\},\\
H_\varphi &= \{1, 2, 3, 4, 7, 9\},\\
H_{\mathrm{APP}} &= \{2, 3, 5, 10, 11, 12\}.
\end{aligned}
$$

P_π has nine points. The other three supports are hexads. This distinction must be preserved: the twelve hexads contained in P_π have the lines of $AG(2, 3)$ as complementary triples.

21.9.2 Coincidence of the threshold supports

Archimedean evaluation extracts from the vector K the threshold support

$$\beta_{\mathrm{arq}}(K) = \{i : K_i \geq 729\} = \{4, 6, 9, 10\}.$$

Exceptional evaluation calculates the intersection

$$\beta_{\mathrm{exc}}(w_e, w_\pi) = \operatorname{supp} c(w_e) \cap \operatorname{supp} c(w_\pi).$$

Evaluation produces the first exact equality conditional on these data:

$$\boxed{\beta_{\mathrm{arq}}(K) = \beta_{\mathrm{exc}}(w_e, w_\pi) = B_K = \{4, 6, 9, 10\}.} \tag{21.121}$$

The two maps perform distinct operations on declared inputs: the first uses thresholds of the vector K, whereas the second uses intersections of code supports. The threshold 729, the words and the Paley gauge are part of the calibration. In particular, this equality is not an independent validation of K: all integer thresholds between 660 and 729, inclusive, select the same support B_K.

21.9.3 Coincidence of the negative supports

Before carrying, the Archimedean cochain is

$$s^{(0)} = P + E - \Phi - K,$$

and its twelve components are

$$(7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663).$$

Its negative support is

$$\eta_{\mathrm{arq}}(s^{(0)}) = \{4, 5, 6, 7, 9, 10\}.$$

By the exceptional route, we fix as calibration selector the hexad H satisfying

$$B_K \subset H \subset P_\pi$$

and

$$(|H \cap H_e|, |H \cap H_\varphi|, |H \cap H_{\mathrm{APP}}|) = (4, 3, 2).$$

Enumeration of the design produces a unique solution for that selector:

$$\eta_{\mathrm{exc}} = \{4, 5, 6, 7, 9, 10\} = H_\alpha^-.$$

Therefore,

$$\boxed{\eta_{\mathrm{arq}}(s^{(0)}) = \eta_{\mathrm{exc}}(B_K, P_\pi, H_e, H_\varphi, H_{\mathrm{APP}}) = H_\alpha^-.} \tag{21.122}$$

Equations (21.121) and (21.122) are condensed into the flag

$$\mathcal{I}_*^{\mathrm{core}} = (B_K \subset H_\alpha^-).$$

This is the calibrated compatibility flag of the double projection. The marked frame

$$\widetilde{\mathcal{I}_*} = (\mathcal{I}_*^{\mathrm{core}}; P_\pi, H_e, H_\varphi, H_{\mathrm{APP}})$$

preserves the data needed to construct the exceptional branch. Under a simultaneous permutation $g \in M_{12}$, the entire construction is transported equivariantly; the invariant is its isomorphism class, not the isolated numerical names.

21.9.4 Reconstruction of the base point from the incidence frame

The origin need not remain an external selector if the complete ordered frame is preserved. We define

$$o(\mathcal{F}) = (H_e \cap H_\varphi) \setminus (H^- \cup H_{\mathrm{APP}}), \qquad \mathcal{F} = (H^-, H_e, H_\varphi, H_{\mathrm{APP}}).$$

Theorem 21.47 (Origin determined by incidences)**.** *For the canonical frame,*

$$o(\mathcal{F}_*) = \{1\}.$$

Moreover, the rule is equivariant: $o(g\mathcal{F}) = g\, o(\mathcal{F})$.

Proof. We have

$$H_e \cap H_\varphi = \{1, 3, 4, 9\},$$

and

$$(H_\alpha^- \cup H_{\mathrm{APP}}) \cap \{1, 3, 4, 9\} = \{3, 4, 9\}.$$

The difference is $\{1\}$. Equivariance follows because intersection, union and difference commute with the action of any permutation. □

The passage from this incidence origin to a vector of a lattice A_2^{12} requires fixing a metric realization of the twelve coordinates. The section 21.9.9 fixes that realization, shows that the origin selects one of the twelve radial minimizers and certifies the conditions of the 3-neighbor. Incidence determines the origin; the Gram matrix and the positive chamber then determine the vector.

21.9.5 Incidence of the ternary design

The Paley chain

$$C_P \longmapsto A_W \longmapsto C_W \longmapsto W_{12}$$

is fixed by the matrices and proofs of the section 21.2. In particular, $A_W^2 = -I_6$, $C_W = [12, 6, 6]_3$, and the 132 projective supports of weight six form $W_{12} = S(5, 6, 12)$. Passing from a word to its support identifies exactly the pair $\{c, -c\}$; the incidences used in what follows therefore belong to the projective design and not to individual signed words.

21.9.6 Hexads, octads and the cardinality identity 28

The binary design $W_{24} = S(5, 8, 24)$ has octads, each with $\binom{8}{2} = 28$ pairs. The theorem 17.4 constructs W_{12} as the residual design of a dodecad D of the extended binary Golay code; the theorem 17.5 proves that, once the alignment ℓ is fixed, each hexad has a unique lift O_H to an octad. The step $6 \to 8$ is therefore an incidence morphism relative to the binary datum (D, ℓ), and not an inference based on cardinalities.

On the other hand, the microfiber of π contains 1008 routes and a nonad has 36 pairs, so that

$$1008 = 36 \cdot 28.$$

The identity $1008 = 36 \cdot 28$ is a numerical check of the catalogue. The incidence morphism is defined independently by means of $H \mapsto \iota_D(\ell(H))$; the cardinality identity does not enter into its definition.

21.9.7 Star involutions and generation of the Mathieu group

For each face $B \in \binom{[12]}{4}$, there are four hexads of W_{12} containing B. The four residues $H \setminus B$ are disjoint pairs partitioning $[12] \setminus B$. The involution σ_B fixes B and exchanges the endpoints of those four pairs. There are

$$\binom{12}{4} = 495$$

star involutions.

Exhaustive enumeration, for the declared matrix and words, verifies

$$\sigma_B(W_{12}) = W_{12}$$

for all of them and

$$\left|\langle \sigma_B : B \in \tbinom{[12]}{4} \rangle\right| = 95040.$$

Since $\mathrm{Aut}(W_{12}) \cong M_{12}$, the 495 stars generate M_{12}. The HMT face is

$$B_K = \{4, 6, 9, 10\}, \qquad \sigma_{B_K} = (1\ 3)(2\ 11)(5\ 7)(8\ 12).$$

A single star generates only C_2. The global combinatorial action is expressed through the representation of the free group of star generators

$$\rho_\star : F_{495} \longrightarrow \mathrm{Aut}(W_{12}), \qquad \rho_\star(\gamma_B) = \sigma_B, \qquad \mathrm{im}\,\rho_\star = M_{12},$$

where F_{495} is the free group on the 495 formal stars. The global group is therefore the image of that family of involutions and not that of a single star. Identifying these generators with loops in the state space would additionally require constructing the corresponding transport; that identification does not enter into the group calculation.

21.9.8 Two exceptional extensions of the ternary code

Two branches of different types can be continued from the ternary code:

$$C_W \longrightarrow W_{12} \xrightarrow[(D,\ell)]{\iota_D} W_{24},$$
$$C_W \xrightarrow[\phi]{\text{explicit gluing}} N(A_2^{12}) \xrightarrow[v_\alpha]{\text{exact 3-neighbor}} \Lambda_{24} \longrightarrow V^\natural.$$

The first branch continues through the residual design of the dodecad and its unique lift; the second continues through ternary gluing and an index-three neighbor. The two constructions have distinct codomains. For the second, let $A_2 = \mathbb{Z}\alpha_1 \oplus \mathbb{Z}\alpha_2$ with Gram matrix

$$G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \rho = \alpha_1 + \alpha_2,$$

and take as representatives of $A_2^*/A_2 \cong \mathbb{F}_3$

$$\varpi_0 = 0, \qquad \varpi_1 = \frac{2\alpha_1 + \alpha_2}{3}, \qquad \varpi_2 = \frac{\alpha_1 + 2\alpha_2}{3}.$$

The gluing realization is the coordinate-by-coordinate map

$$\phi(c_1, \dots, c_{12}) = (\varpi_{c_1}, \dots, \varpi_{c_{12}}),$$

which forms

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)).$$

The linearity of C_W makes the union of cosets a lattice; its self-duality gives the integrality of the gluing, and divisibility by three of all weights gives its evenness. Moreover, C_W has dimension six and

$$\det N(C_W) = \frac{3^{12}}{|C_W|^2} = \frac{3^{12}}{(3^6)^2} = 1.$$

Minimum distance six implies that a nonzero gluing class has norm at least $6(2/3) = 4$. Therefore the roots of $N(C_W)$ are exactly those of A_2^{12}. This is the explicit construction of the Niemeier of type A_2^{12} used below.

21.9.9 Radial minimization and explicit construction of the 3-neighbor

In the positive chamber we consider radial vectors

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \qquad a_i = 1 + 3b_i, \quad b_i \geq 0.$$

The condition $v(a)^2 \equiv 0 \pmod{18}$ is equivalent to $\sum_i b_i \equiv 1 \pmod 3$. Its minimum is

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2 = 54,$$

and is attained exactly twelve times: one coefficient is four and the remaining eleven are one. The recovered origin $o_* = 1$ selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}.$$

Minimality belongs to the declared positive radial cone. In the entire residual quotient there exists, for example, the representative $(\alpha_1 - 2\alpha_2, \rho, \ldots, \rho)$, congruent to $(\rho, \ldots, \rho)$ modulo three and of norm $14 + 11 \cdot 2 = 36$. The direct calculation preserves it as a witness that prevents extending the domain.

We define

$$N_{\alpha,0} = \{x \in N(C_W) : \langle x, v_\alpha \rangle \equiv 0 \pmod 3\}, \qquad N_\alpha = N_{\alpha,0} + \mathbb{Z}\frac{v_\alpha}{3}.$$

Theorem 21.48 (Leech 3-neighbor)**.** *The preceding construction satisfies*

$$\begin{aligned}
&\text{(I1)} \quad v_\alpha \in A_2^{12} \subset N(C_W), && \langle N(C_W), v_\alpha \rangle \not\subset 3\mathbb{Z},\\
&\text{(I2)} \quad v_\alpha^2 = 54 \equiv 0 \pmod{18},\\
&\text{(I3)} \quad v_\alpha/3 \in N_{\alpha,0}^* \setminus N(C_W), && 3(v_\alpha/3) \in N_{\alpha,0}.
\end{aligned}$$

The lattice N_α is even, unimodular, of rank twenty-four and rootless. Consequently,

$$N_\alpha \cong \Lambda_{24}.$$

Proof. For any root of a factor A_2, the pairing with $a_i\rho_i$ is congruent to 1 or 2 modulo three. The character of (I1) is therefore surjective, and none of the old roots belongs to $N_{\alpha,0}$. The equality in (I2) is the preceding radial calculation. If $x \in N_{\alpha,0}$, then $\langle x, v_\alpha/3 \rangle \in \mathbb{Z}$, which proves the first inclusion in (I3). If $v_\alpha/3$ belonged to $N(C_W)$, the integrality of the Niemeier would force the character of (I1) to be zero. The class has order three and $v_\alpha \in N_{\alpha,0}$ because $v_\alpha^2 \equiv 0 \pmod 3$.

The kernel has index three, hence

$$\det N_{\alpha,0} = 3^2 \det N(C_W) = 9, \qquad \det N_\alpha = 9/3^2 = 1.$$

For $x \in N_{\alpha,0}$ and $m \in \mathbb{Z}$,

$$\left(x + m\frac{v_\alpha}{3}\right)^2 = x^2 + 2m\frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z},$$

so the neighbor is even. It remains to exclude new roots. Each coordinate of the two nontrivial cosets belongs to one of the six classes

$$A_2 + \varpi_j \pm \rho/3, \qquad j \in \{0, 1, 2\},$$

and the minimum of each class is $2/9$. Indeed, writing a coordinate as $(u/3, v/3)$, its norm is $2(u^2 - uv + v^2)/9$; the six residues are nonzero and each admits a representative with $u^2 - uv + v^2 = 1$. Every new norm is at least $12(2/9) = 8/3 > 2$. The neighbor has no roots. The classification of even unimodular lattices of rank twenty-four identifies it with Leech. □

Enumeration of the six local minima and the twelve radial minimizers reconstructs the 729 words of C_W, the Gram matrix of the gluing and its evenness; exact arithmetic checks I1–I3, the determinants and the bound $8/3$.

21.9.10 Corridor from Leech to the Moonshine algebra

Define

$$\Lambda_{\mathrm{HMT}} := N_\alpha \cong \Lambda_{24}, \qquad V^\natural_{\mathrm{HMT}} := V^+_{\Lambda_{\mathrm{HMT}}} \oplus V^{T,+}_{\Lambda_{\mathrm{HMT}}}.$$

No second lattice is introduced here: the Frenkel–Lepowsky–Meurman construction is applied to the neighbor produced in the preceding subsection.

Theorem 21.49 (HMT–Moonshine corridor)**.** *With the dodecad, alignment, origin and exceptional chamber of the preceding construction fixed, we have*

$$V^\natural_{\mathrm{HMT}} \cong V^\natural, \qquad \mathrm{Aut}(V^\natural_{\mathrm{HMT}}) \cong \mathbb{M},$$

and

$$\mathrm{Tr}_{V^\natural_{\mathrm{HMT}}} q^{L_0 - 1} = J_{\mathrm{Mon}}(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots .$$

Consequently, the chain

$$\mathrm{HMT} \longrightarrow N_\alpha \cong \Lambda_{24} \longrightarrow V^\natural_{\mathrm{HMT}} \longrightarrow \mathbb{M} \longrightarrow \mathrm{Moonshine}$$

is a result proved by composition.

Proof. The 3-neighbor theorem proves, from the declared data, that N_α is even, unimodular, of rank twenty-four and rootless. Lattice classification identifies it with Λ_{24}. Applied to this output, the Frenkel–Lepowsky–Meurman construction produces the orbifold $V^+_{\Lambda_{\mathrm{HMT}}} \oplus V^{T,+}_{\Lambda_{\mathrm{HMT}}}$; the classical theorems identify its automorphism group with $\mathbb{M}$ and its graded character with the normalized Hauptmodul $J_{\mathrm{Mon}} = j - 744$ [50, 28, 15]. Each arrow therefore delivers an object of the type required by the next. □

Remark 21.50 (The trivial component is not a fit). The grading satisfies

$$V^\natural_{\mathrm{HMT}} = \bigoplus_{n \geq 0} (V^\natural_{\mathrm{HMT}})_n, \qquad (V^\natural_{\mathrm{HMT}})_0 = \mathbb{C}\mathbf{1}, \qquad (V^\natural_{\mathrm{HMT}})_1 = 0.$$

The term q^{-1} thus records the vacuum line. In degree two,

$$\dim(V^\natural_{\mathrm{HMT}})_2 = 196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

The equality $196884 = 1 + 196883$ separates the trivial representation and the smallest nontrivial irreducible representation of the Monster. The other two equalities are exact integer decompositions; they neither replace this graded structure nor are used to select the corridor retrospectively.

The finite stage up to N_α starts from a declared dodecad, alignment, origin and chamber. The calculation proves that N_α is even, unimodular and rootless. The classical classification and orbifold theorems are applied to that output: the orbifold is $V^\natural$, its automorphism group is $\mathbb{M}$ and its graded character is J_{Mon}. The composition thus preserves the provenance of each arrow without reducing the corridor to a coincidence of coefficients.

21.9.11 Quotient space of 243 syndromes

Puncturing one coordinate of C_W produces

$$G_{11} = [11, 6, 5]_3.$$

The ternary ball of radius two has

$$1 + 22 + 220 = 243$$

elements. Since

$$|G_{11}| \cdot 243 = 3^6 \cdot 3^5 = 3^{11},$$

and the minimum distance is five, these balls are disjoint and cover all of $\mathbb{F}_3^{11}$. Each vector admits a unique decomposition

$$x = c + e, \qquad c \in G_{11}, \quad \mathrm{wt}(e) \le 2.$$

The linear quotient is

$$\mathbb{F}_3^{11}/G_{11} \cong \mathbb{F}_3^5, \qquad |\mathbb{F}_3^{11}/G_{11}| = 243.$$

Thus, a quotient space of 243 syndromes is obtained in an ambient space of eleven coordinates and quotient dimension five. The catalogue of 243 words w_6 and this quotient have the same cardinality, but are objects of different types; the direct encoder sends code words to the zero syndrome. The proved double projection uses the common flag, not an unstructured cardinality bijection.

21.9.12 Finite compatibility of the arithmetic and exceptional evaluations

Theorem 21.51 (Finite compatibility of the two projections)**.** *Consider the vector K and the signed dodecaphase state constructed in the section 30.12; the four catalogue words defined in the chapter 9; and the matrix A_W constructed in the section 21.2. The construction fixes a single calibration $c_{\mathrm{exc}} \in \mathscr{C}_{\mathrm{exc}}$ and transports it with the state; this calibration types the coordinates of the exceptional codomain and does not act as a generating input. The following statements hold:*

1. *Archimedean evaluation obtains α_{12} through dodecaphase carrying;*
2. *the Archimedean and exceptional support maps produce the same flag $B_K \subset H_\alpha^-$;*
3. *the marked frame recovers the origin $o_* = 1$;*
4. *the matrix A_W constructs W_{12}, and the 495 star involutions generate a group of order* 95040*, identified with M_{12};*
5. *the explicit discriminant realization constructs $N(A_2^{12})$, and the origin, together with the positive chamber, selects v_α; its 3-neighbor is even, unimodular and rootless.*

The support identities and the origin rule are equivariant under a simultaneous permutation of the twelve coordinates. The calibration transforms the incidence data simultaneously and does not alter their isomorphism class.

Proof. The first statement is the dodecaphase reconstruction of the section 30.12. Equations (21.121) and (21.122) prove the second by direct evaluation of the two maps. The set-theoretic calculation in the preceding section gives $o_* = 1$. Finally, the enumerator, the 132 weight-six supports, the 495 involutions and the group order, together with the 3-neighbor theorem, prove the last statement. □

Corollary 21.52 (Compatibility with the HMT–Moonshine corridor)**.** *The finite data of the exceptional projection construct the same N_α that enters into the theorem 21.49; therefore, finite evaluation is compatible with $V_{\mathrm{HMT}}^\natural$, its $\mathbb{M}$ action and its character J_{Mon}.*

The finite flag is obtained by two distinct operations applied to coordinates of the same enriched state, and the origin is derived after fixing that flag. This proves a typed internal identity and preserves the common provenance without promoting calibration to a causal input. The lattice construction preserves the type distinction. The composition from the data of α to Λ_{24} factors through the code, the gluing A_2^{12}, the marked origin, the radial chamber and the 3-neighbor. The branch $W_{12} \to W_{24}$ continues separately and is determined for every choice of a dodecad D and an incidence isomorphism $\ell : W_{12}^{\mathrm{HMT}} \to \mathcal{H}(D)$. The proved equivariance means that those choices coordinate the representation without altering the generated isomorphism class or the closure of the double projection.

21.10 From spectral jump 4 to Monster symmetries: APP unfolding $2 \to 6 \to 54$, the Leech lattice, $V^\natural$, and the Moonshine corridor

The corridor is not a sequence of numerical coincidences. It begins at the central block of APP, preserves the sum–product orientation, branches into the APP–Fock and APP–Witt couplings, and is coordinated anew through the grading. The binding causal order is

$$\begin{aligned} B_c = 9P_+ + 5P_- &\longrightarrow 4 \longrightarrow 2 \longrightarrow 6 \longrightarrow 54 \\ &\longrightarrow \begin{cases} \text{Fock degree two and } [q]t_3 = 54, \\ \text{APP–Witt alignment and } N_\alpha \cong \Lambda_{24} \end{cases} \\ &\longrightarrow V^\natural \longrightarrow \big(\mathrm{Aut}, \mathrm{ch}, g \mapsto T_g\big) \longrightarrow \text{Moonshine}. \end{aligned}$$

The two intermediate branches have distinct codomains. The common key 54 is a graded invariant transported by the corridor; it is not a linear map from a scalar into a vertex operator algebra.

21.10.1 Typed hierarchy and 2 realizations of index 54

Theorem 21.53 (Spectral Hierarchy and APP Unfolding)**.** *Let $B_c = 7I + 2R = 9P_+ + 5P_-$ be the central block of APP. If M_Π, M_Σ are the defined modular compressions, then*

$$9 - 5 = 4, \qquad (B_c)_{i \neq j} = 2, \qquad \sum M_\Pi - \sum M_\Sigma = 51 - 45 = 6,$$

and the deployment over the nine positions gives

$$9 \cdot 6 = 459 - 405 = 54.$$

The four integers pertain, respectively, to the spectral gap, the local coupling, modular compression, and two-dimensional integration.

Proof. The spectral decomposition of B_c gives 9 and 5, whereas $7I + 2R$ fixes the nondiagonal coefficient. The two aggregate sums are 51 and 45. Each compressed entry represents nine positions, so that the unfolding multiplies the difference by nine. None of the four integers is obtained by retrospectively identifying its types. □

Once the orientation line $o_{\Sigma\Pi}$, leaf parity, and the triangular normalization have been fixed, the oriented coupling

$$\mathcal{H}^{(m)}_{\Sigma,\Pi} : M_{\mathrm{bal}}^{C_3,-} \longrightarrow \mathrm{End}\big(\mathcal{F}_2(\mathcal{V}_m)\big) \otimes o_{\Sigma\Pi}$$

satisfies

$$\mathrm{Tr}_{o,\mathcal{F}_2}\Big(\mathcal{H}^{(m)}_{\Sigma,\Pi}(\nu)\Gamma_2(g_m)\Big) = -\frac{3m\, c(\nu)}{2}.$$

For $m = 12$ and $c(\delta) = -3$, the trace equals 54. On the other hand,

$$t_3(q) = q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \cdots,$$

from which $[q]t_3 = 54$.

Theorem 21.54 (Common Index Compatibility Cone)**.** *With the starting structures of each branch preserved, it holds that*

$$\boxed{D_{\mathrm{APP}} = \mathrm{Tr}_{o,\mathcal{F}_2}\left(\mathcal{H}^{(12)}_{\Sigma,\Pi}(\delta)\Gamma_2(g_{12})\right) = [q]t_3 = v_\alpha^2 = \frac{196884}{5\cdot 729+1} = 54.}$$

This equality constitutes a cone of evaluations toward the same integer. It does not identify W_{APP}, the Fock space, the neighboring lattice, and $V^\natural$.

Proof. The APP census yields $459 - 405 = 54$; the oriented trace yields 54; expansion of the cyclotomic product yields $[q]t_3 = 54$; and the radial vector of the neighbor satisfies $v_\alpha^2 = 4^2 \cdot 2 + 11 \cdot 2 = 54$. Finally, $5 \cdot 729 + 1 = 3646$ and $54 \cdot 3646 = 196884$. The equalities are compared only after each object has been constructed and therefore do not numerically select the corridor. □

Exact status of the hierarchy 4→2→6→54 and of the lattice identification. The hierarchy $4 \to 2 \to 6 \to 54$, the APP–Fock trace, and the extraction $[q]t_3$ are exact internal identities. Identification of the neighbor with Leech uses the declared chamber, neighboring vector, and lattice classification.

21.11 Exceptional prolongation: from dodecaphasic incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves the two sheets and division with remainder and quotient; TRIT determines the local regime and its orientation; TPK composes selection, transport, carry, memory and return. The Archimedean reading contracts compatible cylinders. The reading now studied instead preserves the incidence of positions and its transport between frames. Both start from the same arithmetic state with trajectory memory. No exceptional geometry is reconstructed from the final decimals of π, φ, e or α.

This point makes the meaning of the prolongation $\alpha/K \longrightarrow$ exceptional incidence precise. The arrow does not assert that an isolated real number determines a code, lattice or group. It transports the dodecaphasic register and signed data that entered into the determination of α, together with the words and frames arising from APP–TRIT–TPK. In particular, the ordered register K can be recovered from its rational constant κ with the base, length and origin conventions already fixed. Projection onto a support, by contrast, preserves only the selected incidences. Meanwhile, K_{ph} denotes another reading, with reduced memory, subsequent to the holonomy Γ_9.

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; subsequently recognizing the resulting objects through the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech and Monster denote those recognitions and the explicit realizations making them verifiable. They are not retrospective selectors of the seeds or coefficients of generation.

The argument starts with compatibility of prefixes and frames, constructs the ternary code and determines the flag through K and the signed register of α. Marked incidence then selects the lattice origin and Leech neighbor. The final step realizes the vertex algebra and its automorphisms. At each stage the structure preserved will be specified: linear combinations, supports, discriminant classes, bilinear form or graded product. Explanations of the function of K, normalization and the origin of norm 54 accompany their respective constructions in Sections 21.11.3 and 21.11.9.

21.11.1 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\mathrm{cal}} \to X_m^{\mathrm{cal}}$ for $m \le n$. The superscript recalls that the declared initial frame and its transport are preserved, not that a parameter is fitted against an experimental value. We write (w_j, f_j) for the visible word of six trits and the frame at step j. The frames satisfy a causal law

$$f_{j+1} = g_j(x) f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this reading. Phase return resets neither the frame nor the state memory. Likewise, the census of 468 decimal emissions, the 243 visible words of six trits and the ambient space $\mathbb{F}_3^6$, of 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a block b of six trits, the Archimedean reading takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \qquad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class belongs first to the internal completion $\mathbb{R}_{\mathrm{HMT}}$; its subsequent identification with the usual real line does not enter into the choice of blocks.

For the incidence reading we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \qquad w \longmapsto (w, wA). \tag{21.123}$$

Words are written as row vectors. If $A^f = fAf^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{A^f}(wf^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \tag{21.124}$$

The calibrated codomain preserves f together with the encoded word. We then define

$$Q_n(x) = \big((f_j, \gamma_{f_j A_W f_j^{-1}}(w_j))\big)_{0 \le j < n}. \tag{21.125}$$

Proposition 21.55 (Compatibility and prolongation of the incidence reading)**.** *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Therefore, the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves exactly the visible word, since $\mathrm{pr}_1 \gamma_A = \mathrm{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators up to index $m - 1$. By induction, they have the same frames at those indices. The first m components of (21.125) coincide, proving the square. The universal property of the inverse limit gives Q_∞. The recovery assertion is projection onto the first six coordinates of each graph. □

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\mathrm{GL}_6(\mathbb{F}_3)$ does not

by itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are also transported. The support calculations that follow are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

21.11.2 The finite realization and the ternary code

The elliptic TRIT operator admits the following finite realization in the six-position chart. These coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, once the order $\infty, 0, 1, 2, 3, 4$ is declared, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$ and $C_{a,b} = \chi(b - a)$ in the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$ and $\chi(0) = 0$. This is a coordinate realization of the already transported quadratic structure; identifying the six positions with the projective line is a chart datum, not a new numerical input into the generator.

The quadratic sum identities give $C^2 = 5I_6$. Reducing modulo three yields

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \qquad A_W^2 = 2I_6 = -I_6. \tag{21.126}$$

The quadratic equality realizes the elliptic regime; it does not identify the field $\mathbb{F}_3$ with an Archimedean realization of the imaginary unit. These are distinct realizations of a common operator relation, with their corresponding domains.

Definition 21.56 (Code of the dodecaphasic chart)**.** The ternary code associated with (21.126) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\operatorname{supp}(c) : c \in C_W, \operatorname{wt}(c) = 6\}.$$

Proposition 21.57 (Self-duality, distance and Witt design)**.** *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\operatorname{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \tag{21.127}$$

The set $\mathcal{H}$ has 132 *elements and constitutes a system $S(5, 6, 12)$: each set of five positions is contained in a unique hexad.*

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by symmetry of A_W, satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^{\mathsf{T}} = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (21.127); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two words of weight six have the same support, among their six nonzero coordinate ratios one of the signs $1, -1$ appears at least three times. Subtracting the corresponding multiple would yield a nonzero word of weight at most three, impossible. Thus a support corresponds exactly to the pair $\{c, -c\}$ and there are $264/2 = 132$ supports. If two distinct supports shared five positions, the same argument would

cancel at least three coordinates of a union of size seven, producing weight at most four. A quintuple therefore belongs to at most one hexad. Finally,

$$132\binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. □

The subsequent recognition is the extended ternary Golay code and the small Witt design [161]. The preceding proof explains which object is recognized. It does not replace the APP–TRIT–TPK genealogy with a search among known codes.

21.11.3 Dodecaphasic flag associated with K and α

Articulation of the regional coordinates and closure incidence. The function of this prolongation is understood through a distinction between three operations: preserving a register, evaluating a coordinate and extracting an incidence relation. All three act on information produced by APP–TRIT–TPK. Their articulation makes explicit the structural thesis of the article: a numerical realization has an operator provenance, and certain data from that provenance also admit a combinatorial and lattice realization. The equations allow both paths to be followed with their domains and the information remaining in each.

The starting point is the terminal state with its compatible prolongations. It preserves the residues and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation and self-scale regions are determined in that domain before their readers establish the coordinates π, e, φ. At the same time, the oriented register of the twelve windows determines the signed vector U. The subsequent appearance of K comes from the integral transformation proved in §11.4.8:

$$U = \mathcal{H}_{12}K, \qquad K = \frac{1}{4}\mathcal{H}_{12}U, \qquad \mathcal{H}_{12}^2 = 4I_{12}. \tag{21.128}$$

The congruence defining $\mathcal{L}_H$ guarantees integrality of the inverse. In this way, K constitutes a reversible coordinatization of the signed register in the relevant sublattice. Its positions preserve phase order; its differences D_3K, D_4K preserve transverse relations; the sum $Q(K)$ determines the uniform component. This structure explains why the register participates both in numerical composition and in selection of a boundary configuration.

The periodic evaluation

$$K \longmapsto \kappa_{\mathrm{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable when base, length and phase origin are fixed. Extracting a support performs a different operation: it preserves membership of each position in a class defined by a threshold and groups registers having the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete register from which they arise.

In the frame inherited from the dodecaphasic register, the data are

$$K = (234, 543, 140, 729, 659, 824, 621, \mathtt{058}, 914, 794, 146, 601), \tag{21.129}$$
$$w_\pi = 010211, \qquad w_e = 201101,$$
$$w_\varphi = 121200, \qquad w_{\mathrm{APP}} = 022020.$$

Applying γ_{A_W} yields the following incidences:

$$P_\pi = \{2, 4, 5, 6, 7, 8, 9, 10, 11\},$$
$$H_e = \{1, 3, 4, 6, 9, 10\},$$

$$H_{\varphi} = \{1, 2, 3, 4, 7, 9\},$$
$$H_{\mathrm{APP}} = \{2, 3, 5, 10, 11, 12\}.$$

The first is the support of a word of weight nine; the other three are hexads. Through its high-block reading, the register provides

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_{\pi}. \tag{21.130}$$

The integer 729 here marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in (21.56). In this specific register, all integer thresholds between 660 and 729 produce the same support: the support does not recover the threshold or the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$Z_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \tag{21.131}$$

Its negative support is

$$H_{\alpha}^{-} = \{4, 5, 6, 7, 9, 10\}. \tag{21.132}$$

These signs are not extracted from the decimal expansion of the scalar α; they are preserved in the state preceding its numerical expression.

Alpha normalization and its signed configuration. The coordinate of α is determined in the positional path through composition of the regional blocks and dodecaphasic register. Before positional division, the oriented combination has components $Z_j = P_j + E_j - \Phi_j - K_j^{\mathrm{per}}$ and the balance with incoming quotient is $s_j = Z_j + c_{j+1}$. Normalization satisfies

$$A_j = s_j - 1000c_j = Z_j + c_{j+1} - 1000c_j, \qquad 0 \leq A_j < 1000. \tag{21.133}$$

This identity simultaneously preserves the normalized block and the contribution of the transported quotient. The limit of compatible prefixes determines the scalar coordinate of the closure. The signed configuration, in turn, preserves which positions of the combination before normalization have a negative defect. In the initial window, that configuration is the vector Z_0 of (21.131), and its negative support is H_{α}^{-}.

Two contents of the same operation are thus defined. Positional normalization determines a real coordinate through its compatible remainders and quotients; the negative support determines a configuration of positions in the cycle of twelve components. The sign refers to the register balance before normalization. Its incidence reading remains available because the state preserves that balance together with the normalized result. This articulation corresponds to the proved positional path; comparison with the analytic vacancy operators retains the precise scope established in the chapter on α.

Proposition 21.58 (Agreement of the two flag selectors)**.** *Within $\mathcal{H}$, the support* (21.132) *is the unique hexad H satisfying*

$$B_K \subset H \subset P_{\pi}, \qquad (|H \cap H_e|, |H \cap H_{\varphi}|, |H \cap H_{\mathrm{APP}}|) = (4, 3, 2).$$

Thus, the signed reading and incidence reading select the same flag $B_K \subset H_{\alpha}^{-}$.

Proof. The four hexads containing B_K are obtained by adding, respectively, the pairs

$$\{1, 3\}, \qquad \{2, 11\}, \qquad \{5, 7\}, \qquad \{8, 12\}.$$

Only the second and third pairs are contained in P_π. For the second, the intersection with H_φ has size three and with H_{APP} size three; it fails the last condition. For the third, the three intersections have sizes $4, 3, 2$. This last one produces precisely the negative support of (21.131). □

Both the flag and its marked frame should be retained:

$$\mathcal{I}_* = (B_K \subset H_\alpha^-), \qquad \widetilde{\mathcal{I}}_* = (\mathcal{I}_*; P_\pi, H_e, H_\varphi, H_{\mathrm{APP}}).$$

Union, intersection and difference operations are equivariant under simultaneous relabeling of the design. The marked class, not the position numbers taken in isolation, is the transported object.

Lemma 21.59 (Origin selected by marked incidence)**.** *For the preceding frame,*

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\mathrm{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\widetilde{\mathcal{I}}_*$.

Proof. The first intersection is $\{1, 3, 4, 9\}$. The union subtracted contains $3, 4, 9$ and does not contain 1. Equivariance follows because a bijection commutes with set operations. □

This origin will select a direction of the lattice neighbor when the metric A_2 and positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

From ternary encoding to the marked flag. The regions also provide the codes w_π, w_e, w_φ and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \longmapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and adds its transform by A_W. The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \mathrm{supp}(c)$, preserves the nonzero positions. For words of weight six, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. In this way, the quotient by sign produces the hexads, while the complete code remains available for operations requiring its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \tag{21.134}$$

The left-hand side comes from an integer comparison of the components of K; the right-hand side, from the intersection of the encoded propagation and closure supports. The threshold belongs to the declared block chart and remains explicit in the reader's definition. The complete register preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes that tetrad to a hexad. The condition of membership in the support P_π, together with the intersection sizes $(4, 3, 2)$ relative to $H_e, H_\varphi, H_{\mathrm{APP}}$, selects a unique hexad. The following proofs combine the two determinations: that obtained through the sign of Z_0 and that obtained through incidence. The result is organized as

$$\widetilde{\mathcal{I}}_* = \big(B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\mathrm{APP}}\big). \tag{21.135}$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π here recover their internal differentiation. They share w_π as a visible ternary code, but preserve distinct emissions, signatures and multiplicities. In the binary realization on a marked dodecad, the transverse region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to the lifts of the four hexads of its star. The negative region is distinguished by H_α^- and the chart order individualizes the three positive regions. This correspondence preserves regional roles and the distribution $432 + 4 \cdot 144 = 1008$, while also realizing the incidence partition $1 + 4$.

21.11.4 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \qquad (Iv)(H) = \sum_{p \in H} v_p. \tag{21.136}$$

Let $V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}$. The two Euclidean realizations are here subsequent languages for the same finite design.

Proposition 21.60 (Isometry of the dodecaphasic screen)**.** *We have*

$$I^{\mathsf{T}} I = 36 I_{12} + 30 J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6} I\big|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\mathrm{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the design automorphisms.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^{\mathsf{T}} I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I \rho_{12}(g) = \rho_{\mathcal{H}}(g) I$. □

Incidence information also has a specific spectral reading. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 21.61 (Action of the intersection kernel)**.** *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$G I = 30 I + 15 J_{132,12}.$$

Thus G acts by the scalar 30 *on E_{inc}.*

Proof. The entry (H, p) of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$ and $p \in H'$. If $p \in H$, there are ten tetrads T containing p, each contained in four hexads, and five not containing it, each with a unique extension of the quintuple $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen tetrads of H give one extension each: the total is 15. Hence the identity. The rectangular term vanishes on V_{11}. □

Here a material intertwiner does exist: it has a determined domain, codomain, entries and normalization. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc}, with $U_W P = Q U_W$. This assertion does not identify all operators acting on those spaces by their trace alone.

21.11.5 The direction selected by the dodecaphasic register

The register K has a geometric function complementing its rational evaluation and support reading. Once the representation of the twelve positions is constructed, its components determine a direction in the space of zero-mean variations. The reduction $12 \to 11$ removes the uniform mode; the next reduction $11 \to 10$ removes the direction selected by the register itself. The two operations therefore have different antecedents.

To declare the chart used completely, consider the group A_5 of even permutations of five objects and the subgroup $C_5 = \langle (1\,2\,3\,4\,5) \rangle$. The twelve left cosets $r_p C_5$ are ordered by these representatives, written as lists of images:

p	r_p	p	r_p
1	$(4,3,5,2,1)$	7	$(5,4,3,2,1)$
2	$(4,2,1,3,5)$	8	$(1,3,4,2,5)$
3	$(1,2,4,5,3)$	9	$(4,2,3,5,1)$
4	$(2,5,3,4,1)$	10	$(3,2,5,4,1)$
5	$(4,3,1,5,2)$	11	$(4,5,2,3,1)$
6	$(5,1,2,3,4)$	12	$(5,3,2,4,1)$.

The action by left multiplication defines an orthogonal representation $\rho : A_5 \to O(\mathbb{R}^{12})$: $\rho(a) e_p = e_q$ when $a r_p C_5 = r_q C_5$. This representation chart acts on the register already obtained; the group characters are not used to choose its twelve values.

Let χ_3 be the values of the three-dimensional character

class	1	2^2	3	5_a	5_b
χ_3	3	-1	0	$(1+\sqrt{5})/2$	$(1-\sqrt{5})/2$.

The class 5_a contains the displayed generator of C_5, and 5_b contains its square. This convention distinguishes the two conjugate three-dimensional sectors of the representation. Define

$$P_3 = \frac{3}{60} \sum_{a \in A_5} \chi_3(a^{-1}) \rho(a), \qquad P_{11} = I_{12} - \frac{1}{12} J_{12}. \tag{21.137}$$

Both matrices are determined by finite sums in $\mathbb{Q}(\sqrt{5})$. The appearance of this quadratic field belongs to the realization of the representation, subsequent to regional generation of φ.

Lemma 21.62 (Projection of the register onto the three-dimensional sector)**.** *We have*

$$P_3 = P_3^{\mathsf{T}} = P_3^2, \quad P_3 \mathbf{1} = 0, \quad \operatorname{tr} P_3 = 3, \qquad P_3 P_{11} = P_{11} P_3 = P_3.$$

For the register (21.129)*, the vector*

$$u_K = P_3 P_{11} K \tag{21.138}$$

is nonzero. Its exact norm is

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0. \tag{21.139}$$

Proof. The character sum in (21.137) is the isotypic projector. The real character and substitution $a \mapsto a^{-1}$ give symmetry. The multiplicity of the sector in the action on A_5/C_5 is the dimension of its vectors fixed by C_5,

$$\frac{1}{5}\left(3 + 2\frac{1+\sqrt{5}}{2} + 2\frac{1-\sqrt{5}}{2}\right) = 1.$$

Hence the rank is three. Its orthogonality to the trivial character gives $P_3\mathbf{1} = 0$ and both compositions with P_{11}. These identities can also be verified by multiplying the matrices of the finite sum.

Finally, $\sum K_p = 6263$ and substitution of the twelve components into that same sum gives

$$\|u_K\|^2 = K^{\mathsf{T}} P_3 K = \frac{1}{20} \sum_{a \in A_5} \chi_3(a^{-1}) K^{\mathsf{T}} \rho(a) K = \frac{6638585 + 2275584\sqrt{5}}{20}.$$

This is an exact evaluation of sixty specified terms, reproduced by the attached certificate with quadratic rational arithmetic. The two positive coefficients prove nonvanishing. □

Definition 21.63 (Dodecaphasic direction and transverse complement)**.** The direction selected by K in the declared chart is $\ell_K = \mathbb{R}u_K$. Its transverse complement within $V_{11} = \mathbf{1}^{\perp}$ is

$$V_{10}(K) = V_{11} \cap u_K^{\perp}.$$

Proposition 21.64 (Orthogonal reduction determined by K)**.** *The operator*

$$P_{10}(K) = P_{11} - \frac{u_K u_K^{\mathsf{T}}}{\|u_K\|^2} \tag{21.140}$$

is the orthogonal projector of rank ten onto $V_{10}(K)$. We have

$$V_{11} = \ell_K \oplus^{\perp} V_{10}(K), \qquad P_{10}P_{11} = P_{10}, \quad P_{10}u_K = 0.$$

Proof. Let $E_K = u_K u_K^{\mathsf{T}}/\|u_K\|^2$. The proved nonvanishing guarantees that it is defined. It is symmetric, $E_K^2 = E_K$, has rank one and, since $u_K \in V_{11}$, $P_{11}E_K = E_K P_{11} = E_K$. Hence

$$(P_{11} - E_K)^2 = P_{11} - 2E_K + E_K = P_{11} - E_K.$$

Its image is $V_{11} \cap u_K^{\perp}$ and its trace is $11 - 1 = 10$. The decomposition and remaining identities are obtained by projection onto the two orthogonal summands. □

This result makes precise a role of K not exhausted by its designation as a register: besides reversibly representing oriented data and selecting the tetrad B_K, it determines a linear polarization. The direction depends on the complete register through P_3K, not merely on the four points of its support. The order of register, representation, direction, and dimensional reduction has thus been retained (Proposition 21.64).

The transformation has a precise covariance law. If the chart is changed simultaneously by an orthogonal permutation matrix S, $K' = SK$ and $P_3' = SP_3S^{-1}$, then

$$u'_{K'} = Su_K, \qquad P'_{10}(K') = SP_{10}(K)S^{-1}.$$

On the other hand, replacing u_K with au_K, with $a \neq 0$, does not change E_K. The direction retains a polarization and does not by itself allow recovery of either the amplitude of u_K or the complete register. Reversibility of $\kappa \leftrightarrow K$ remains in its own domain.

The geometric realization established here is a reduction of spaces with an inner product. Its subsequent use as a compact direction requires transporting this line to the corresponding physical realization. Section 21.11.17 presents the algebraic map keeping this reduction and the lattice screen coordinated.

21.11.6 The Witt star and the generation of M_{12}

Definition 21.65 (Star on a tetrad)**.** For $B \in \binom{[12]}{4}$, the Witt star is the family of the four hexads containing B. Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and exchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

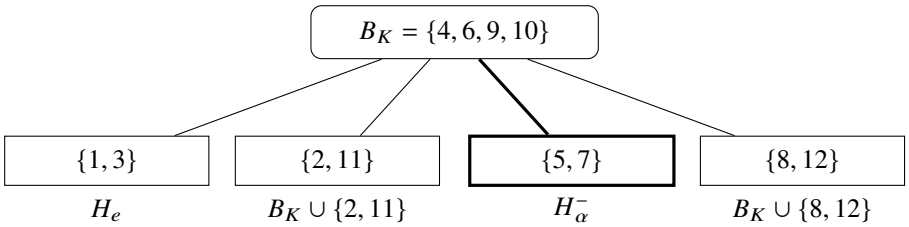


Figure 21.4: The concrete star of the dodecaphasic flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α. The four pairs exhaust the eight outer positions.

Proposition 21.66 (The family of stars)**.** *The* $\binom{12}{4} = 495$ *permutations* σ_B *preserve* $\mathcal{H}$*. The group they generate has order* 95040 *and is the Mathieu group* M_{12} *in its action on the small Witt design.*

Proof. The preservation assertion is a finite check on the 132 hexads of (21.56): for each tetrad the four petals are constructed and $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$ is verified. Closure under composition of these permutations produces 95040 elements. In lexicographic enumeration it suffices to add successively six stars not yet contained in the group; the intermediate orders are $2, 6, 18, 360, 7920, 95040$, and the 495 stars belong to the final group. The procedure, the six permutations written as their lists of images and the closure recurrence are developed in 25.18.2. Finally, recognition of the automorphism group of $S(5, 6, 12)$ as M_{12}, whose order is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$, is applied. □

An individual star generates a cyclic group of order two. It does not by itself generate M_{12}, still less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 21.67 (Equality of traces does not identify operators). On V_{11}, a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (21.136) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

21.11.7 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24}, with parameters $[24, 12, 8]_2$. Its weights are $0, 8, 12, 16, 24$, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D, the support of a word of weight twelve. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π.

Proposition 21.68 (Residue of a dodecad and lifting of hexads)**.** *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad},\ |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D. Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^{\perp}$ with $O_B^{\perp} \cap D = B$.

Proof. A quintuple $A \subset D$ belongs to a unique octad O. If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the allowed weights, and with $0 \le j \le 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \ge 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift on choosing five of its points. Furthermore,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \qquad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads on B give four distinct octads. The fifth cannot meet D in six points, since it would be another residual hexad; as it contains B, its intersection has size four and coincides with B. □

A chart $\ell : [12] \to D$ preserving hexads identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is recognition of the already constructed design as the small Witt design; a specific choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \qquad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. The domain of this injectivity is the hexads, not the register K.

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	$(501, 614, 498, 169, 272, 272)$	555555
$++_1$	144	$(810, 923, 870, 169, 272, 272)$	339555
$++_2$	144	$(810, 983, 810, 169, 272, 272)$	393555
$++_3$	144	$(870, 923, 810, 169, 272, 272)$	933555
$--$	144	$(870, 923, 810, 418, 674, 521)$	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible tritic word $w_\pi = 010211$, but not the same regional genealogy. The three positive regions preserve the tail 169|272|272; the negative region changes it to 418|674|521; the transverse region has constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O^{\perp}_{\ell(B_K)}$, the negative region to $\iota_D(H_\alpha^-)$ and the three positive regions to the other three lifts of the star. This realizes the partition $4 + 1$ of the five octads on $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart order; incidence does not determine it without that datum. The precise assertion is therefore a marked correspondence of regions, compatible with their roles and with the flag, not a canonical bijection inferred from cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also $1008 = 36 \cdot 28$ and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalogue checks, not substitutes for the map ι_D or the regional chart.

21.11.7.1 Pentadic representation of the microfiber and its incidence

The arrangement of the five regions allows simultaneous visualization of the unity of the ternary quotient and the differentiation preserved by the catalogue. The relevant map is the restriction

$$\Psi\big|_{\mathscr{R}_\pi^{\mathrm{micro}}} : \mathscr{R}_\pi^{\mathrm{micro}} \longrightarrow \{010211\}, \qquad \Psi(U) = (-U) \bmod 3.$$

Its five antecedents are the vectors U_6 in the preceding table. Figure 21.5 recovers their radial organization, preserving chart order, signatures and catalogue multiplicities (equation (10.37)).

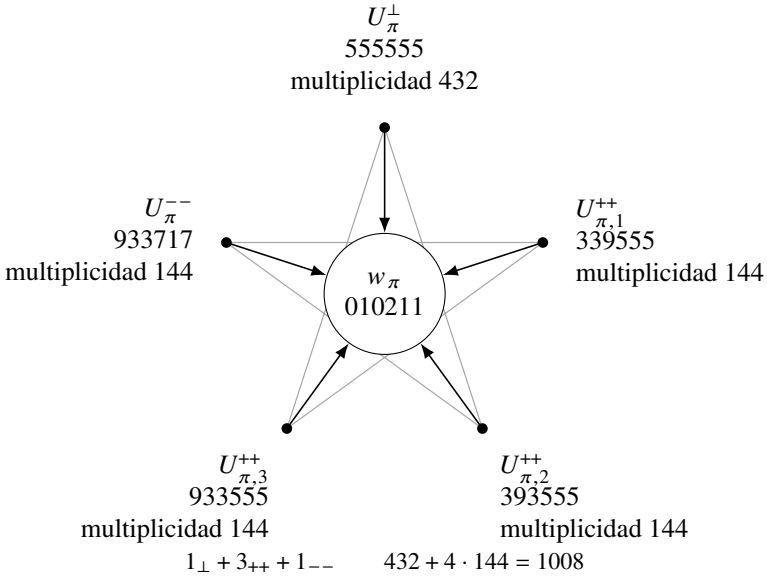


Figure 21.5: The five regions of π in a pentadic arrangement. Each position preserves its multiplicative signature and multiplicity; the radial arrows represent the common projection $\Psi(U) = 010211$. The star-shaped outline organizes the five positions of the chart. Their realization in the five Witt octads is carried out through regional alignment and the decomposition $4 + 1$ in the text.

Projection onto the center identifies the five ternary images; the register of the outer positions retains the regional information making the incidence realization possible. The distribution $1_\perp + 3_{++} + 1_{--}$ specifies one transverse region, three positive regions and one negative region. Their normalized weights are $3/7$ and $1/7$ for each of the other four: the transverse multiplicity is triple, even though the five images under Ψ coincide.

The link to the Witt star is formulated through the flag $B_K \subset H_\alpha^-$, alignment ℓ and lift ι_D already constructed. The five regional positions are realized in the five octads on $\ell(B_K)$:

$$\begin{aligned} U_\pi^{\perp} &\longmapsto O^{\perp}_{\ell(B_K)}, \\ U_\pi^{--} &\longmapsto \iota_D(H_\alpha^-), \\ \{U_{\pi,1}^{++}, U_{\pi,2}^{++}, U_{\pi,3}^{++}\} &\longmapsto \{\iota_D(B_K \cup P) : P \in \mathcal{P}_+\}, \\ \mathcal{P}_+ &= \big\{\{1,3\},\{2,11\},\{8,12\}\big\}. \end{aligned}$$

Assignment of the three positive positions to the three pairs is performed in an ordered chart. The transverse row and negative row are distinguished by their regional predicates and the flag.

The two figures are thus articulated: Figure 21.4 shows the four hexads of the small Witt design on B_K; Figure 21.5 shows the five regions that, under alignment, occupy the four residual octads and the transverse octad of the binary design. The law $4 + 1$, proved in Proposition 21.68, specifies the incidence change. In this composition, K determines the marked tetrad and the signature of α distinguishes one of its hexads. In addition to their common ternary value, the regions preserve the structure needed for that transport toward the exceptional realization.

21.11.8 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four does not require first going through the binary code. It is realized directly from C_W by gluing discriminants. Two compatible prolongations are thus distinguished: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ have Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \rho = a_1 + a_2, \qquad \rho^2 = 2.$$

We represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \qquad \varpi_1 = \frac{2a_1 + a_2}{3}, \qquad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with that discriminant. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} \big(A_2^{12} + \phi(c)\big). \tag{21.141}$$

Theorem 21.69 (Ternary gluing)**.** *The object* (21.141) *is a positive, even, unimodular lattice of rank* 24*. Its root system is exactly* A_2^{12}*; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes stable under addition. Self-orthogonality annuls the discriminant pairing: it is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a gluing class is $2\,\mathrm{wt}(c)/3$ modulo $2\mathbb{Z}$. The weights of (21.127) are multiples of three, so it is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(3^6)^2} = 1.$$

Positivity comes from the orthogonal sum of the twelve metrics A_2. In a nonzero class of A_2^*/A_2 the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Therefore a vector in a nontrivial gluing class has norm at least $6 \cdot 2/3 = 4$. No norm-two roots appear outside A_2^{12}. Inside it, the roots are precisely those of its twelve components. □

21.11.9 Incidence selection of the rootless unimodular neighbor

From the incidence origin to the lattice modification. The marked flag determines a position through an equivariant set-theoretic operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\mathrm{APP}}) = \{1\}.$$

Its function is to distinguish one component among the twelve involved in the lattice realization. The complete ternary code acts as a gluing code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and the self-duality, weights and distance of the code translate, respectively, into integrality and

determinant properties, parity and norm bounds on the gluing classes. Here the symbolic information of the code serves a function that the isolated support no longer performs.

The metric A_2 and positive radial chamber are fixed in the lattice development. Within that chamber, the congruence required for an even neighbor of index three has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \qquad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the declared chamber. The subscript α records the incidence provenance of the selection: its content is the frame involved in closure and now individualizing a lattice direction.

Pairing with v_α determines the sublattice $N_{\alpha,0}$ of index three, and adding the class $v_\alpha/3$ produces the neighbor N_α of (21.143). This modification preserves rank and recovers unimodularity and parity; selection by congruence removes the previous roots, and local bounds exclude roots in the new classes. Identification with Leech appears after these verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the latter directly receives the ternary code as gluing data.

The label $o_* = 1$ of the flag is now realized as one of the twelve components A_2. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \qquad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

that congruence is equivalent to $\sum_i b_i \equiv 1 \pmod 3$. The smallest norm value in the chamber is obtained with a single $b_i = 1$ and all the others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \qquad v_\alpha^2 = 54. \tag{21.142}$$

The word “smallest” refers to this positive chamber. It does not assert minimality in the entire residue class: the vector $(a_1 - 2a_2, \rho, \ldots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not an omissible conclusion.

Let $N = N(C_W)$. We define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod 3\}, \qquad N_\alpha = N_{\alpha,0} + \mathbb{Z}\frac{v_\alpha}{3}. \tag{21.143}$$

Theorem 21.70 (Leech neighbor determined by the marked flag)**.** *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank* 24*, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24}.*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle$ mod 3 is nonzero: a simple root of one component pairs with $a_i\rho_i$ to give $a_i \equiv 1 \pmod 3$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$ belonged to N, integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. We deduce

$$[N_\alpha : N_{\alpha,0}] = 3, \qquad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m\frac{v_\alpha}{3}\right)^2 = x^2 + 2m\frac{\langle x, v_\alpha\rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

Hence it is even.

It remains to exclude the roots, which is the decisive step. The roots of N belong to A_2^{12}. Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \qquad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24}. □

The argument does not recognize Leech through a coincidence of dimension. It constructs the gluing, proves integrality, parity and determinant, and removes the roots through a local bound. Only then does it apply the classification theorem. Dependence on α is preserved in the flag selecting o_* and in the vector (21.142); the scalar value of the constant is not turned into a lattice length.

21.11.10 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the Heisenberg oscillator space in rank twenty-four; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the lattice's central extension. The cocycle and lift of the isometries are part of the construction: a bare vector sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ. With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\mathrm{HMT}}^\natural = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \tag{21.144}$$

The sign + denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 21.71 (Prolongation to the Moonshine module)**.** *The algebra* (21.144)*, constructed on the neighbor* (21.143)*, is a realization of the Moonshine algebra $V^\natural$. It has central charge* 24*, zero weight-one component, and automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\mathrm{Tr}_{V_{\mathrm{HMT}}^\natural}\, q^{L_0 - 1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \qquad q = e^{2\pi i\tau}. \tag{21.145}$$

Proof. Theorem 21.70 verifies the lattice hypotheses: positivity, parity, unimodularity, rank twenty-four and absence of roots. The construction and theorem of Frenkel–Lepowsky–Meurman for the Leech lattice are then applied. They identify the orbifold with $V^\natural$ and establish its character and automorphism group. Composition of the flag, neighbor and VOA construction functor gives the realization indicated here. The group is not inferred from the coefficient 196884. □

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof in this article of the FLM construction [159]. Moreover, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It introduces neither π nor e as inputs to the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes with alternating sign. Hence

$$\mathrm{Tr}_{V_\Lambda}\,\theta q^{L_0-1} = t_2(\tau) := q^{-1}\prod_{n\geq 1}(1+q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character without insertion of $V^\natural$. Nor does it by itself identify a conjugacy class of the Monster: the space, insertion and lift must be specified before functions are compared.

Grading, unit and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_α once its properties have been verified. It incorporates the oscillator space, central extension of the lattice, vertex product and sector twisted by the lift of negation. The result is the algebra $V^\natural_{\mathrm{HMT}}$ of (21.144), whose automorphism group is isomorphic to the Monster. Its action domain is thus fixed by an algebraic structure with grading and product. Borcherds's Moonshine theorem subsequently determines the graded traces with insertion of these automorphisms [159, 160].

Unity acquires a precise realization at this level: the weight-zero component is the vacuum line and contributes the coefficient 1 of q^{-1} in the graded character. At the cylindrical level, the refinement maps preserve the constant function by precomposition. Each conservation assertion therefore has its object and map. Developing the proofs makes this continuity of construction visible, from arithmetic operations and memory to incidence, metric and vertex products.

This articulation places K and the closure of α in the main argument. K reversibly preserves an integral coordinate of the history; normalization of α composes the regional registers with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. The following proofs develop these maps, distinguishing reversible equivalences, information quotients and realizations adding metric or algebraic structure. A provable relation between structure, equation and value is thus expressed, preserving the attribution of each construction and the scope of its theorems.

21.11.11 State-based construction of the twelve cyclotomic fibres

On the APP marks, multiplication $a : r \mapsto 4r$ in $\mathbb{Z}/9\mathbb{Z}$ satisfies $a^3 = 1$. The sheets satisfy

$$S(ax, ay) = aS(x, y), \qquad P(ax, ay) = a^{-1}P(x, y),$$

since $16 \equiv 7 \equiv 4^{-1} \pmod 9$. The ordered unit orbits are $O_0 = (1, 4, 7)$, $O_1 = (2, 8, 5)$. In each, the augmentation module $A(O) = \ker(\mathbb{Z}[O] \xrightarrow{\Sigma} \mathbb{Z})$ inherits the inner product making the e_r orthonormal. In the basis $b_1 = e_r - e_{4r}$, $b_2 = e_{4r} - e_{7r}$, its matrices are

$$G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad C^{\mathsf{T}} G_{A_2} C = G_{A_2}.$$

The lattice form therefore originates in the orbits of the marks. For the pairs $(1, 2)$, $(2, 3)$, $(3, 1)$ and sheets $\mu \in \{\Sigma, \Pi\}$, we obtain

$$W_{\mathrm{APP}} = \bigoplus_{(i,j)} \bigoplus_{\mu} \bigoplus_{\varepsilon=0}^{1} A(O_\varepsilon), \qquad \ell^{\Sigma}_{ij}(x) = x_i + x_j, \quad \ell^{\Pi}_{ij}(x) = x_i x_j \pmod 9.$$

Each summand preserves pair, sheet, and orbit. The state map is

$$\Psi : (\mathbb{Z}/9\mathbb{Z})^3 \longrightarrow W_{\mathrm{APP}}, \qquad \Psi(x)_{ij,\mu,\varepsilon} = \beta_\varepsilon(\ell_{ij}^{\mu}(x)), \quad \beta_\varepsilon(r) = \begin{cases} e_r - e_{4r}, & r \in O_\varepsilon, \\ 0, & r \notin O_\varepsilon. \end{cases}$$

Proposition 21.72 (Equivariance and rank of the state module)**.** *The action of C on the six additive fibres and of C^{-1} on the six multiplicative fibres satisfies $\Psi(ax) = \rho(a)\Psi(x)$. The images span $W_{\mathrm{APP}} \otimes \mathbb{Q}$, of dimension twenty-four.*

Proof. The sheet identities and $\beta(4r) = C\beta(r)$ prove equivariance. For rank, order the coordinates by the three stated pairs, by Σ, Π, by O_0, O_1, and by b_1, b_2. In each orbit, β takes successive values $(1,0), (0,1), (-1,-1)$. The matrix of the images of the following states, read row by row, has determinant 9:

$$\begin{array}{cccc} (0,0,1) & (0,0,2) & (0,0,4) & (0,0,5) \\ (0,1,0) & (0,1,1) & (0,1,2) & (0,1,3) \\ (0,1,4) & (0,1,5) & (0,1,6) & (0,2,0) \\ (0,2,2) & (0,2,3) & (0,4,0) & (0,5,0) \\ (1,0,1) & (1,0,2) & (1,0,4) & (1,0,5) \\ (1,1,0) & (1,2,0) & (1,4,0) & (1,5,0). \end{array}$$

Successive elimination, reducing each row by its predecessors and normalizing its first nonzero coefficient, gives pivot columns numbered from zero

$$(8, 10, 9, 11, 0, 12, 14, 16, 13, 15, 17, 2, \\ 18, 19, 1, 3, 20, 22, 21, 23, 4, 6, 5, 7)$$

and pivots

$$(1, 1, 1, -1, 1, 1, 1, 1, 1, -1, 3, 1, 1, 3, 1, -1, 1, 1, 1, -1, 1, 1, 1, -1).$$

Their product, multiplied by the sign of the column permutation, is 9. The invertible minor establishes full rank. □

21.11.12 Dodecaphasic alignment with the Witt positions

The pair index is $i \in \mathbb{Z}/3\mathbb{Z}$, the sheet index is $\mu \in \{0,1\}$, and the orbit index is $\varepsilon \in \{0,1\}$. With the dodecaphasic origin fixed, $\lambda(i,\mu,\varepsilon) = 4i + 2\mu + \varepsilon \pmod{12}$ is bijective: the four values for each i form a block, and the three blocks are disjoint. Let $R = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Multiplication proves $RCR = C^{-1}$ and $R^{\mathsf{T}} G_{A_2} R = G_{A_2}$. For the trivializations ι_b and frames P_b transported jointly with Witt incidence, set

$$g_b = \iota_b^{-1} P_b C P_b^{-1} \iota_b, \qquad T_{i\mu\varepsilon} = \iota_{\lambda(i,\mu,\varepsilon)}^{-1} P_{\lambda(i,\mu,\varepsilon)} R^{\mu}.$$

Then

$$g_{\lambda(i,\mu,\varepsilon)} T_{i\mu\varepsilon} = T_{i\mu\varepsilon} C^{(-1)^{\mu}}.$$

The direct sum is a C_3-equivariant isomorphism between the twelve state fibres and the twelve lattice positions. It preserves pair, sheet, orbit, and form under simultaneous transport of origin, order, and orientation. These data persist in the lattice gluing and passage to the neighbour constructed below.

21.11.13 Ternary orientation of the Leech neighbor

The order-two presentation constructed above uses negation of the Leech lattice. The same lattice, obtained from the incidence marked by K and the dodecaphasic support of α, admits a second presentation compatible with the ternary character of the construction. To make this continuity explicit, the gluing is retained, its order-three isometry is constructed, and preservation of the already selected neighbour is verified. This step requires no alteration of K, its rational reading, or the preceding definition of α. The order-three prolongation is developed here, including the lattice operations and the precise conditions of the classical theorems used.

We work in the simple-root basis of each factor A_2, with

$$B_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \mathsf{C} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad \mathsf{R} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \rho = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \tag{21.146}$$

The multiplication gives

$$\mathsf{C}^3 = I_2, \quad \mathsf{C}^2 + \mathsf{C} + I_2 = 0, \quad \mathsf{C}^{\mathsf{T}} B_2 \mathsf{C} = B_2, \quad \mathsf{R}\mathsf{C}\mathsf{R} = \mathsf{C}^{-1}, \quad \mathsf{R}^{\mathsf{T}} B_2 \mathsf{R} = B_2.$$

These identities fix both the metric and the orientation of the action. In the discriminant $A_2^*/A_2 \cong \mathbb{F}_3$ we take as generator the class of $\varpi = (2,1)^{\mathsf{T}}/3$. We have $\mathsf{C}\varpi - \varpi = (-1,0)^{\mathsf{T}}$, so C acts trivially on the discriminant; the reflection R changes its sign.

The already constructed Paley–Witt matrix produces an element of full support through the following operation:

$$\begin{aligned} q_* &= (1,1,1,1,1,1), \qquad q_* A_W = (2,1,1,1,1,1), \\ c_* &= (q_*, q_* A_W) = (1,1,1,1,1,1,2,1,1,1,1,1) \in C_W. \end{aligned} \tag{21.147}$$

Every coordinate of c_* is nonzero. For the twelve positions, define the frames and the isometry

$$P_b(c_*) = \begin{cases} \mathsf{R}, & (c_*)_b = 1, \\ I_2, & (c_*)_b = 2, \end{cases} \qquad g_c = \bigoplus_{b=1}^{12} P_b(c_*)\mathsf{C}P_b(c_*)^{-1}. \tag{21.148}$$

The symbol $P_b(c_*)$ here denotes an invertible frame of an A_2 factor, not the regional projector P_3.

Proposition 21.73 (Order-three isometry on the marked neighbor)**.** *The operator g_c is an order-three isometry without nonzero fixed vectors. It preserves the Niemeier lattice $N = N(C_W)$ and the neighbor N_α constructed in* (21.143)*.*

Proof. Each block of g_c is C or C^{-1}; its minimal polynomial is $X^2 + X + 1$. This proves the order and absence of fixed vectors. Both blocks act trivially on the discriminant, so they preserve each gluing class and therefore the lattice N.

Write $v = v_\alpha = 4\rho_{o_*} + \sum_{b \neq o_*} \rho_b$, with the frames and incidence origin already fixed. Its radial coefficients are $a_{o_*} = 4$ and $a_b = 1$ for $b \neq o_*$. All satisfy $a_b \equiv 1 \pmod 3$. In one factor,

$$(\mathsf{C} - I_2)\rho = (-2,-1)^{\mathsf{T}}, \qquad (\mathsf{C}^{-1} - I_2)\rho = (-1,-2)^{\mathsf{T}}.$$

After division by three, their discriminant classes are, respectively, 2 and 1. Consequently, the vectors

$$y = \frac{g_c v - v}{3}, \qquad z = \frac{g_c^{-1} v - v}{3} \tag{21.149}$$

have discriminant words c_* and $-c_*$, both in C_W. Hence $y, z \in N$. The local product is

$$\left\langle \frac{(\mathsf{C}^{\pm 1} - I_2) a_b \rho}{3}, a_b \rho \right\rangle = -a_b^2,$$

and their sum gives $\langle y, v\rangle = \langle z, v\rangle = -(16+11) = -27$. Thus, $y, z \in N_0 = \{x \in N : \langle x, v\rangle \equiv 0 \pmod 3\}$. For $x \in N_0$,

$$\langle g_c x, v\rangle = \langle x, g_c^{-1} v\rangle = \langle x, v\rangle + 3\langle x, z\rangle \equiv 0 \pmod 3.$$

The argument applied to g_c^{-1} proves $g_c N_0 = N_0$. Finally, $g_c(v/3) = v/3 + y$ preserves the class generating $N_\alpha = N_0 + \mathbb{Z}(v/3)$. We obtain $g_c N_\alpha = N_\alpha$. □

The proof preserves more than the lattice dimension: it uses the gluing code, its orientation, the origin of the flag and the congruence of the radial vector. These data explain why the second presentation remains linked to the dodecaphasic register from which the construction began.

21.11.14 Ternary trace, twisted weight and order-three extension

Let $\Lambda = N_\alpha$ and let $\widehat{g}_c$ be the standard lift of g_c to the lattice algebra $V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda]$, with the central extension and cocycle already declared. Odd order allows this lift to be chosen of order three. Each complex factor A_2 contributes eigenvalues ζ_3 and ζ_3^2, where $\zeta_3^2 + \zeta_3 + 1 = 0$.

Proposition 21.74 (Graded trace and type of the lift)**.** *For $j = 1, 2$,*

$$\operatorname{Tr}_{V_\Lambda}\big(\widehat{g}_c^{\,j} q^{L_0 - 1}\big) = t_3(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12}. \tag{21.150}$$

The minimum weight of the nontrivial twisted modules is $4/3$, and the lift is of type zero.

Proof. In the lattice part of the trace only the zero vector contributes, because g_c and g_c^2 have no nonzero fixed vectors. In each oscillator degree, the trace of symmetric powers is the inverse of the determinant $I - g_c^j q^n$. Each factor A_2 contributes $1 + q^n + q^{2n}$, proving (21.150).

The twisted-vacuum weight formula for an order-three lattice automorphism, applied to the two twelve-dimensional eigenspaces, gives

$$\rho_{\widehat{g}_c} = \frac{1}{4 \cdot 3^2}\big(1 \cdot 2 \cdot 12 + 2 \cdot 1 \cdot 12\big) = \frac{4}{3}.$$

The type is the class of $3^2 \rho_{\widehat{g}_c} = 12$ modulo three, which is zero. The weight formula is a result of twisted-module theory; the multiplicities and isometry feeding into it have been constructed here. □

Theorem 21.75 (Second presentation of the Moonshine algebra)**.** *The type-zero cyclic extension*

$$V_{(3)} = \operatorname{Orb}_{\mathbb{Z}/3}(V_\Lambda, \widehat{g}_c) \tag{21.151}$$

is isomorphic to the vertex operator algebra $V^\natural$.

Proof. Theorem 21.70 provides a positive, even, unimodular, rootless lattice of rank twenty-four. Proposition 21.73 constructs an order-three fixed-point-free isometry on that same lattice; Proposition 21.74 verifies the lift and its type. The Chen–Lam–Shimakura theorem on the order-three orbifold of the Leech lattice algebra then applies [165, Teorema 1.1]. This identification is also covered by the uniform treatment of Abe–Lam–Yamada [166, Teorema 4.4 y secciones 2–4]. This application follows construction of the lattice data; equality of characters does not replace the hypotheses of the theorem used. □

To make the dual symmetry precise, write $V_{(3)} = W_0 \oplus W_1 \oplus W_2$, where W_i is the integral sector selected in the $\widehat{g}_c^{\,i}$-twisted module. The product satisfies $Y(W_i, z)W_j \subset W_{i+j \bmod 3}((z))$. Hence

$$g^\vee|_{W_i} = \zeta_3^i I \tag{21.152}$$

defines an automorphism of order three. Indeed, the factor on a product is $\zeta_3^{i+j} = \zeta_3^i \zeta_3^j$, and the vacuum and conformal vector belong to W_0. The classical identification of this automorphism in the Monster is

class $3B$, with series $T_{3B} = t_3 + 12$ [167, 160]. This normalization can be checked in the construction. The lattice character is $\operatorname{ch} V_\Lambda = J + 24$: its initial pole is q^{-1}, its weight-one space has dimension 24 and the weight-zero modular character is determined by these data. If F is the character of the fixed subspace and H that of each integral twisted sector, projection onto invariants and conjugation of the two sectors give

$$F = \frac{J + 24 + 2t_3}{3}, \qquad F + 2H = J, \qquad \operatorname{Tr} g^\vee q^{L_0 - 1} = F - H = t_3 + 12.$$

Its domain is the extended algebra, whereas $\widehat{g}_c$ acts on the preceding lattice algebra.

21.11.15 Equivalence of the order-two and order-three presentations

Denote by

$$V_{(2)} = (V_\Lambda)^+ \oplus (V_\Lambda^T)^+$$

the FLM presentation of (21.144). Both presentations preserve the same Leech neighbor as antecedent. Their extensions use different automorphisms and vertex products determined through the respective lifting data.

Theorem 21.76 (Graded equivalence and transport of automorphisms)**.** *With vertex operator algebra isomorphisms fixed*

$$\phi_2 : V_{(2)} \xrightarrow{\cong} V^\natural, \qquad \phi_3 : V_{(3)} \xrightarrow{\cong} V^\natural,$$

the composition

$$\Phi_{23} = \phi_2^{-1}\phi_3 : V_{(3)} \xrightarrow{\cong} V_{(2)} \tag{21.153}$$

preserves the vacuum, conformal vector, grading and vertex product. Conjugation by Φ_{23} identifies the automorphism groups and preserves their graded traces.

Proof. Composition of the two isomorphisms preserves the declared structures. In particular,

$$\Phi_{23} Y_{(3)}(a, z)b = Y_{(2)}(\Phi_{23}a, z)\Phi_{23}b, \qquad \Phi_{23} L_0^{(3)} = L_0^{(2)} \Phi_{23}.$$

If h is an automorphism of $V_{(3)}$, then $h' = \Phi_{23} h \Phi_{23}^{-1}$ is an automorphism of $V_{(2)}$. Equality of traces on each finite-dimensional homogeneous component gives

$$\operatorname{Tr}_{V_{(3)}} h q^{L_0 - 1} = \operatorname{Tr}_{V_{(2)}} h' q^{L_0 - 1}.$$

□

The choice of ϕ_2 and ϕ_3 is part of the statement; Φ_{23} is not presented as a canonical identification independent of those choices. Nor does it transform an order-three automorphism into the order-two involution: conjugation preserves order. Its content is that two distinct extensions realize the same graded structure, with explicit transport of their products, representations and traces.

21.11.16 Eta functions, Fricke involution and vacuum normalization

For $\operatorname{Im} \tau > 0$, the traces of the two lattice automorphisms admit the expressions

$$\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n), \qquad t_2 = \left(\frac{\eta(\tau)}{\eta(2\tau)} \right)^{24}, \qquad t_3 = \left(\frac{\eta(\tau)}{\eta(3\tau)} \right)^{12}. \tag{21.154}$$

They follow directly from the factorizations $1 - q^{2n} = (1 - q^n)(1 + q^n)$ and $1 - q^{3n} = (1 - q^n)(1 + q^n + q^{2n})$. The order in q of the level-three quotient, before raising it to the twelfth power, is

$$\frac{1}{24} - \frac{3}{24} = -\frac{1}{12}; \qquad 12 \left(-\frac{1}{12} \right) = -1.$$

Here $-1/12$ denotes the exact difference between the initial exponents of two eta functions. The operation is specified and does not require interpreting a divergent sum as a convergent one. Agreement with other normalized defects in the corpus preserves their respective realizations; it does not identify their operators through equality of the scalar.

Proposition 21.77 (Fricke involution on the order-two trace)**.** *Let* $w_2\tau = -1/(2\tau)$. *Then*

$$t_2(w_2\tau) = \frac{2^{12}}{t_2(\tau)}, \qquad T_{2A} = t_2 + 24 + \frac{2^{12}}{t_2} \tag{21.155}$$

y $T_{2A}(w_2\tau) = T_{2A}(\tau)$.

Proof. The transformation law $\eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau)$ gives

$$\frac{\eta(-1/(2\tau))}{\eta(-1/\tau)} = \sqrt{2}\,\frac{\eta(2\tau)}{\eta(\tau)}.$$

The twenty-fourth power produces the first identity. Substitution exchanges the two variable terms of T_{2A} and leaves the constant term fixed. Recognition of this expression as the McKay–Thompson series of class $2A$ corresponds to the level-two symmetrization of Conway–Norton [167, §§4 y 6, tabla 3] and the Moonshine theorem [160]. □

The modular uniformizations of levels two and three, in the preceding normalizations, are

$$j = \frac{(t_2 + 256)^3}{t_2^2}, \tag{21.156}$$

$$J = j - 744 = t_3 + 12 + \frac{10 \cdot 3^9}{t_3} + \frac{4 \cdot 3^{14}}{t_3^2} + \frac{3^{18}}{t_3^3}. \tag{21.157}$$

They are used as classical modular identities on the already constructed traces, not as rules for selecting the code or neighbor. The second is also written $j = (t_3 + 27)(t_3 + 243)^3/t_3^3$. Direct expansion of the product of (21.150) begins with

$$t_3 = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + 1188q^4 + O(q^5).$$

In particular, $[q]J = 54 + 10 \cdot 3^9 = 196884$; this is a consequence of the traces and uniformization. The construction of $V^\natural$ retains its lattice and orbifold proof, independently of this check of the first coefficient.

The three operations appearing in this prolongation have distinct domains. The dual symmetry $g^\vee$ acts on sectors of the vertex algebra; Fricke acts on the modular parameter and its function; radius duality exchanges momentum and winding in the compact lattice sector. Their reciprocity formulas can be coordinated through the corresponding maps, but the name “duality” does not replace those maps. This distinction allows the construction to continue toward dimensional screens while preserving incidence, grading and the origin of each operation.

21.11.17 Exceptional incidence and coordination of the screens

The direction selected by K and the Leech lattice arise from distinct representations of the constructed incidence. The first determines the orthogonal reduction $12 \to 11 \to 10$; the second enters into the isotropic reduction $26 \to 24$. Coordination preserves the prior space, its relations and the information transmitted by each projection.

Sections 24.4 and 25.14 develop this coordination jointly. The former brings together the involution of integer coordinates, unitary conjugation with its domains and the change of residual section retaining the quotient memory. The latter constructs the morphism $\mathcal{R}_M^{\mathrm{alg}}$ of (25.7) and proves the isomorphism of

the quotient presentations in Theorem 25.28. The graph preserves the coordinate in V_{11} together with its projection in V_{10}, while the isotropic quotient recovers the Leech lattice. The respective kernels and codomains are thereby preserved.

The action ellipse is constructed before duality is applied and fixes its dimensionless radius together with the reciprocal radius (Section 24.3). The compact sector then connects exceptional incidence with a determined quadratic form. The fields, action and operators of the subsequent physical realization are presented in their own domain, preserving this algebraic antecedent and its internal proofs.

21.11.18 Graded Moonshine traces and representation of the Monster

For $g \in \mathrm{Aut}(V^{\natural}_{\mathrm{HMT}})$, the trace with insertion is

$$T_g(\tau) = \mathrm{Tr}_{V^{\natural}_{\mathrm{HMT}}}\, g q^{L_0 - 1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [160, teorema 1.1]. The character J, group $\mathbb{M}$ and family $g \mapsto T_g$ are realizations of the already constructed algebra. They do not form a causal sequence in which one first guesses J and then infers the Monster.

Unity has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V^{\natural}_0 = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one the dimension is zero, which accounts for the absence of a constant term in $J = j - 744$. A vertex algebra morphism preserves the vacuum vector. This property provides the technical meaning of preservation of unity at this level; the unit of cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

The first separation corresponds to the trivial conformal line and the irreducible component of dimension 196883 of the Monster representation. The other expressions have lattice or arithmetic readings. Their equality does not automatically identify their summands with TPK regions, words or routes. Doing so would require presenting the maps between these spaces and proving that they respect the relevant actions.

In particular, an isometry $N_\alpha \simeq \Lambda_{24}$ induces a realization of (21.144), and a choice of isomorphism with $V^{\natural}$ transports its automorphism group. This has not constructed an action of the Monster on the digits of π, on the blocks of K or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\mathrm{TPK}}(g) = \rho_{V^{\natural}}(g)F,$$

including domain, codomain and preserved information. The absence of that step in the present assertion does not withdraw the incidence intertwiner U_W, flag or neighbor already constructed; it only delimits a distinct promotion, to action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned}
&\text{APP–TRIT–TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, Z_0)\\
&\longrightarrow (C_W, \mathcal{H}; B_K \subset H^-_\alpha, P_\pi, H_e, H_\varphi, H_{\mathrm{APP}})\\
&\longrightarrow \big(N(C_W), o_*, \text{metric and chamber}\big) \longrightarrow N_\alpha \longrightarrow V^{\natural}_{\mathrm{HMT}}.
\end{aligned}$$

At the incidence level, the stars generate M_{12} and the residual chart realizes the regional structure $4 + 1$ in W_{24}. At the lattice and VOA level, the construction gives Leech and the Moonshine module. They are

prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber and lift.

The result of this section is therefore an assembled derivation: the finite and lattice proofs make the passage from dodecaphasic incidence to the hypotheses of the recognition theorems effective. The conclusion comprises the structures and transports specified in the preceding propositions, with their explicit domains and choices. Comparison of constants and experimental validation concern assertions distinct from this algebraic construction.

21.11.19 Full-support orientations and lattice equivariance

The weight enumerator of (21.127) contains $24y^{12}$. Therefore, the set

$$C_W^{\times} := \{u \in C_W : u_b \in \{1, 2\} \text{ for every } b\}$$

has 24 elements. For each $u \in C_W^{\times}$, we define

$$P_b(u) = \begin{cases} \mathsf{R}, & u_b = 1, \\ I, & u_b = 2, \end{cases} \qquad g_u := \bigoplus_{b=0}^{11} P_b(u)\mathsf{C}P_b(u)^{-1}.$$

Proposition 21.78 (Robustness of Full Support Orientations)**.** *For every $u \in C_W^{\times}$,*

$$\frac{g_u v - v}{3}, \quad \frac{g_u^{-1} v - v}{3} \in N_0.$$

In particular, each g_u preserves the neighbor constructed with the transported orientation.

Proof. In simple root coordinates,

$$\mathsf{C}\rho - \rho = (-2, -1), \qquad \mathsf{C}^{-1}\rho - \rho = (-1, -2).$$

The discriminant classes of the two displacements are u and $-u$. For the radial coefficient $a_b \in \{4, 1\}$,

$$\left\langle \frac{(\mathsf{C}^{\pm 1} - I)a_b\rho}{3}, a_b\rho \right\rangle = -a_b^2.$$

The sum over the twelve factors again gives -27. The proof of proposition 21.73 is applied component by component. □

To type the recalibration, let $h \in \mathrm{Aut}(C_W)$ be monomial: a permutation σ of the twelve coordinates followed by multipliers $\epsilon_b \in \{1, 2\}$. We define its lattice lift

$$\widetilde{h} := P_\sigma \circ \bigoplus_{b=0}^{11} \mathsf{R}^{\nu_b}, \qquad \nu_b = \begin{cases} 0, & \epsilon_b = 1, \\ 1, & \epsilon_b = 2. \end{cases} \tag{21.158}$$

Its action on the discriminant is exactly h. If the change simultaneously transports the code, flag, incidence origin, vector v, neighbor, and frames, then

$$g_{hu} = \widetilde{h}\, g_u\, \widetilde{h}^{-1}. \tag{21.159}$$

The ternary matrix h does not act without a lift on A_2^{12} or on the Leech lattice.

21.12 Finite holonomy, Schläfli configuration and symmetry E_6

The Witt branch does not exhaust the exceptional structures accessible from APP. The parallel branch of this chapter starts from the two mixed sum–product and product–sum polarizations, together with a plaquette orientation and a declared central section. The calculated curvature determines the alternating form; this defines the central extension. An extension is not inferred from cardinality 27 alone.

21.12.1 Ordered system of incidences and radial camera orientation

On $T_9 = (\mathbb{Z}/9\mathbb{Z})^2$, with $S(x, y) = x + y$ and $P(x, y) = xy$, let

$$\Delta_x f = f(x + 1, y) - f(x, y), \qquad \Delta_y f = f(x, y + 1) - f(x, y).$$

Mixed connections are

$$A^{SP} = (\Delta_x S, \Delta_y P) = (1, x), \qquad A^{PS} = (\Delta_x P, \Delta_y S) = (y, 1).$$

Its discrete curvature is

$$F_A = \Delta_x A_y - \Delta_y A_x.$$

Proposition 21.79 (Oriented opposite curvatures)**.**

$$F_{A^{SP}} = 1, \qquad F_{A^{PS}} = -1 \pmod 9.$$

The circulation over an oriented rectangle of sides r, s is, respectively, rs and $-rs$.

Proof. $\Delta_x x - \Delta_y 1 = 1$ and $\Delta_x 1 - \Delta_y y = -1$. A rectangle contains rs plaquettes; upon summing, the interior edges cancel and the boundary circulation remains. □

For displazamientos $u = (a, b)$, $v = (c, d)$, the class alternante is

$$\sigma(u, v) = ad - bc \pmod 9. \tag{21.160}$$

The orientation fixes the sign of σ. The central section used below belongs to the gauge of this realization; replacing it with a coboundary produces an isomorphic extension, not a new invariant.

21.12.2 Central extension and ternary reduction

We define the nonadic Heisenberg group

$$\mathcal{H}_9 = \{(a, b, z) : a, b, z \in \mathbb{Z}/9\mathbb{Z}\}, \quad (a, b, z)(c, d, w) = (a + c, b + d, z + w + bc). \tag{21.161}$$

The commutator of two horizontal shifts is

$$[(a, b, 0), (c, d, 0)] = (0, 0, bc - ad),$$

which records $-\sigma(u, v)$. Reducing each coordinate modulo three gives

$$H_3 = \{(a, b, z) : a, b, z \in \mathbb{F}_3\}, \qquad |H_3| = 27, \tag{21.162}$$

with the same law. The nucleo of $\mathcal{H}_9 \to H_3$ is $(3\mathbb{Z}/9\mathbb{Z})^3$, of cardinal 27.

21.12.3 Construction of the radial character and determination of its congruence kernel

The nonadic extension admits an explicit unitary realization. Let

$$\zeta_9 := \exp(2\pi i/9), \qquad \mathscr{H}_9^{\mathrm{Sch}} := \mathbb{C}^9,$$

based on $\{|n\rangle : n \in \mathbb{Z}/9\mathbb{Z}\}$. We define

$$X|n\rangle := |n+1\rangle, \qquad Z|n\rangle := \zeta_9^n |n\rangle. \tag{21.163}$$

We have

$$ZX = \zeta_9 XZ$$

and, by multiplication,

$$(X^a Z^b)(X^c Z^d) = \zeta_9^{bc} X^{a+c} Z^{b+d}. \tag{21.164}$$

Theorem 21.80 (Faithful Weyl–Heisenberg Representation)**.** *The map*

$$\rho_{\mathrm{WH}} : \mathcal{H}_9 \longrightarrow U(\mathscr{H}_9^{\mathrm{Sch}}), \qquad \rho_{\mathrm{WH}}(a,b,z) := \zeta_9^z X^a Z^b, \tag{21.165}$$

is a faithful unitary representation. For $u = (a,b)$ and $v = (c,d)$, its commutator satisfies

$$[\rho_{\mathrm{WH}}(u,0), \rho_{\mathrm{WH}}(v,0)] = \zeta_9^{bc-ad} I = \zeta_9^{-\sigma(u,v)} I. \tag{21.166}$$

Proof. The identity (21.164) reproduces the term bc of (21.161); hence the homomorphism property. If $\zeta_9^z X^a Z^b = I$, the action on the basis successively forces $a = 0$, $b = 0$, and $z = 0$. This proves faithfulness. Comparing the products in the two orders gives ζ_9^{bc-ad}, which is the alternating phase of (21.160). □

The representation distinguishes two conjugate observables. Let

$$Q_{\mathrm{H}} := \mathrm{diag}(0,1,\dots,8), \qquad F_{jk} := \frac{1}{3}\zeta_9^{jk}, \qquad P_{\mathrm{H}} := F^\dagger Q_{\mathrm{H}} F.$$

The eigenbases of Q_{H} and P_{H} are mutually unbiased, since $|F_{jk}|^2 = 1/9$. For every unit vector $\psi \in \mathscr{H}_9^{\mathrm{Sch}}$,

$$\mathrm{Var}_\psi(Q_{\mathrm{H}})\,\mathrm{Var}_\psi(P_{\mathrm{H}}) \geq \frac{1}{4}\,|\langle \psi, [Q_{\mathrm{H}}, P_{\mathrm{H}}]\psi\rangle|^2\,, \tag{21.167}$$

$$H_{Q_{\mathrm{H}}}(\psi) + H_{P_{\mathrm{H}}}(\psi) \geq \log 9. \tag{21.168}$$

Here $H_{Q_{\mathrm{H}}}$ and $H_{P_{\mathrm{H}}}$ are the Shannon entropies of the Born distributions in the respective eigenbases, with natural logarithm and convention $0\log 0 = 0$. These operators are dimensionless. A physical units chart is a subsequent realization and does not enter into the central extension.

21.12.4 Action of 10 displacements on the Cayley graph

For a linear functional $\ell(a,b) = pa + qb$, set

$$D_\ell(a,b) = \left(a, b, \frac{ab}{2} + \ell(a,b)\right), \qquad Z = (0,0,1),$$

where $2^{-1} = 2$ in $\mathbb{F}_3$. The inverse-closed set

$$S_\ell = \{D_\ell(v) : v \in \mathbb{F}_3^2 \setminus \{0\}\} \cup \{Z, Z^{-1}\} \tag{21.169}$$

has ten elements and does not contain the identity.

Theorem 21.81 (Independence of the linear scale)**.** *The nine functionals ℓ produce isomorphic Cayley graphs. The nine sets S_ℓ form a single orbit under* $\mathrm{Aut}(H_3)$.

Proof. The map

$$\phi_\ell(a,b,z) = (a,b,z+\ell(a,b))$$

is a central automorphism and satisfies $\phi_\ell(S_0) = S_\ell$. To write the complete family in the polarized coordinates of equation (21.161), let $c(v,w) = bc$ for $v = (a,b)$, $w = (c,d)$. Given $M \in \mathrm{GL}(2,3)$, the defect

$$B_M(v,w) = c(Mv,Mw) - \det(M)c(v,w)$$

is symmetric, because the alternating parts of the two terms coincide. We define the quadratic correction

$$q_M(v) = 2B_M(v,v), \qquad q_M(v+w) - q_M(v) - q_M(w) = B_M(v,w), \tag{21.170}$$

where $2 = 2^{-1}$ in $\mathbb{F}_3$. The complete family of automorphisms is then written

$$(v,t) \longmapsto \big(Mv, \det(M)t + q_M(v) + \ell(v)\big), \qquad M \in \mathrm{GL}(2,3),\ \ell \in (\mathbb{F}_3^2)^*. \tag{21.171}$$

The polar identity of equation (21.170) directly proves that the central product is preserved. The family has $48 \cdot 9 = 432$ elements. Exhaustive enumeration of the $\binom{13}{5} = 1287$ inverse-closed subsets of cardinality ten finds exactly those nine with the parameters derived below. □

Write $\underline{S} = \sum_{s\in S} s$ in $\mathbb{Z}[H_3]$. Counting ordered products gives

$$\underline{S}^{\,2} = 10e + \underline{S} + 5(\underline{H_3} - e - \underline{S}) = 5\underline{H_3} + 5e - 4\underline{S}. \tag{21.172}$$

Indeed, the identity receives ten factorizations; an element of S receives one; each of the other sixteen receives five.

Theorem 21.82 (Graph of the twenty-seven lines)**.** *The matrix of adjacency A_ℓ of $\Gamma_\ell = \mathrm{Cay}(H_3, S_\ell)$ satisfies*

$$A_\ell^2 = 5I - 4A_\ell + 5J. \tag{21.173}$$

Therefore,

$$\Gamma_\ell = \mathrm{srg}(27,10,1,5), \qquad \mathrm{Spec}(A_\ell) = \{10^1, 1^{20}, (-5)^6\}.$$

It is isomorphic to the intersection graph of the twenty-seven lines of a smooth cubic surface.

Proof. The regular representation of equation (21.172) produces equation (21.173). Its entries state that every vertex has ten neighbors, that an adjacent pair has one common neighbor, and a nonadjacent pair has five. On the constant line, A_ℓ acts by 10; on its complement, $J = 0$ and $(A_\ell - I)(A_\ell + 5I) = 0$. Zero trace fixes the multiplicities 20 and 6.

In the model of a cubic obtained by blowing up six points of $\mathbb{P}^2$, the 27 classes are

$$E_i, \qquad L_{ij} = H - E_i - E_j, \qquad Q_i = 2H - \sum_{j\neq i} E_j.$$

With $H^2 = 1$, $E_i^2 = -1$, $H \cdot E_i = 0$, and $E_i \cdot E_j = 0$ for $i \neq j$, the adjacency matrix defined by intersection between distinct classes has the same parameters. The complete intersection matrix has diagonal -1. Identification with the classical configuration uses the uniqueness theorem for this realization of the 27 lines [173], with the domain and action specified there. □

The isomorphism assertion can be made fully effective by means of a computed bijection. The labeling is not absolute: another linear section or an automorphism of the surface produces another table in the same class.

Table 21.3: Explicit bijection between H_3 and the twenty-seven classes of lines.

$a\ b\ z$	class of line
0 0 0	E_1
0 0 1	L_{12}
0 0 2	Q_2
0 1 0	L_{13}
0 1 1	Q_1
0 1 2	E_3
0 2 0	Q_3
0 2 1	E_2
0 2 2	L_{23}
1 0 0	L_{14}
1 0 1	L_{35}
1 0 2	L_{26}
1 1 0	L_{36}
1 1 1	L_{24}
1 1 2	L_{15}
1 2 0	L_{25}
1 2 1	L_{16}
1 2 2	L_{34}
2 0 0	Q_4
2 0 1	L_{46}
2 0 2	E_6
2 1 0	E_5
2 1 1	Q_6
2 1 2	L_{56}
2 2 0	L_{45}
2 2 1	E_4
2 2 2	Q_5

As a local control, the ten neighbors of $e = (0,0,0)$ are the elements of S_0. The table transforms them into the ten classes that intersect E_1: the five L_{1j} and the five Q_j, with $2 \le j \le 6$. The complete verification compares all 729 entries of both adjacency matrices.

The complement $B_\ell = J - I - A_\ell$ satisfies

$$B_\ell^2 = 8I + 2B_\ell + 8J, \qquad \operatorname{srg}(27, 16, 10, 8), \qquad \operatorname{Spec}(B_\ell) = \{16^{1}, 4^{6}, (-2)^{20}\}. \tag{21.174}$$

This is the Schläfli graph in the convention of complementary adjacency; A_ℓ is the intersection graph with parameters $(27, 10, 1, 5)$.

Each edge belongs to a unique triangle, since $\lambda = 1$. The number of triangles is

$$\frac{27 \cdot 10}{6} = 45.$$

The full automorphism group has order

$$27 \cdot 1920 = 51840$$

and, through its action on the line configuration, is identified with $W(E_6)$, not merely by equality of orders [173].

21.12.5 The root lattice realizing E_6

To make explicit the object on which that group acts, let

$$\mathrm{Pic}(X) = \mathbb{Z}H \oplus \bigoplus_{i=1}^{6} \mathbb{Z}E_i, \qquad H^2 = 1, \qquad E_i^2 = -1, \qquad H \cdot E_i = E_i \cdot E_j = 0 \ (i \neq j), \tag{21.175}$$

the Picard lattice of the blow-up of six general points of $\mathbb{P}^2$. Its canonical class is

$$K_X = -3H + E_1 + \cdots + E_6.$$

The orthogonal complement

$$L_{E_6} = K_X^{\perp} \subset \mathrm{Pic}(X), \qquad (x, y)_{E_6} := -x \cdot y, \tag{21.176}$$

is positive definite of rank six. A simple root basis is

$$\begin{aligned} &\alpha_1 = E_1 - E_2, \quad \alpha_2 = E_2 - E_3, \quad \alpha_3 = E_3 - E_4, \\ &\alpha_4 = E_4 - E_5, \quad \alpha_5 = E_5 - E_6, \quad \alpha_6 = H - E_1 - E_2 - E_3. \end{aligned} \tag{21.177}$$

Each α_i is orthogonal to K_X, has norm two, and its products are -1 exactly for adjacent pairs of the diagram E_6: the chain $1 - 2 - 3 - 4 - 5$, with node 6 joined to node 3. Therefore, its Gram matrix is the Cartan matrix of E_6.

Proposition 21.83 (The seventy-two roots and their reflections)**.** *The roots of L_{E_6} are*

$$\begin{aligned} &E_i - E_j && (i \neq j), \\ &\pm (H - E_i - E_j - E_k) && (1 \leq i < j < k \leq 6), \\ &\pm (2H - E_1 - \cdots - E_6), \end{aligned} \tag{21.178}$$

in total $30 + 40 + 2 = 72$*. For each root α,*

$$s_\alpha(x) = x + (x \cdot \alpha)\alpha \tag{21.179}$$

preserves the intersection and K_X. The six simple reflections generate $W(E_6)$ and act faithfully by permutations on the 27 classes E_i, L_{ij}, Q_i in theorem 21.82.

Proof. Substituting each class of equation (21.178) into equation (21.175) gives $K_X \cdot \alpha = 0$ and $\alpha^2 = -2$. The class count is as stated. The reflection formula is the orthogonal reflection for the form $-(\cdot)$; it preserves K_X because $K_X \cdot \alpha = 0$. The simple system of equation (21.177) generates all roots, and its Coxeter group is $W(E_6)$. Its action on the 27 classes of exceptional curves preserves intersection, and therefore coincides with the action on the already constructed graph [173]. □

21.12.6 Stratification of the 729 Words

We write a six-trit word as

$$w = (\ell_1, h_1, \ell_2, h_2, \ell_3, h_3),$$

and we define two elements of H_3,

$$L(w) = (\ell_1, \ell_2, \ell_3), \qquad H(w) = (h_1, h_2, h_3),$$

using the central law in their products. According to $L(w)^{-1}H(w)$, the 729 words are divided into

$$\begin{aligned} \mathcal{D}_\ell &= \{w : L(w)^{-1}H(w) = e\}, \\ \mathcal{I}_\ell &= \{w : L(w)^{-1}H(w) \in S_\ell\}, \\ \mathcal{S}_\ell &= \{w : L(w)^{-1}H(w) \notin \{e\} \cup S_\ell\}. \end{aligned}$$

Theorem 21.84 (Hensel–Schläfli Partition)**.**

$$729 = 27 + 270 + 432, \tag{21.180}$$

corresponding to diagonal, incidence, and warping.

Proof. For each one of the 27 values of $L(w)$, there is a unique value of $H(w)$ with identity difference, ten with difference in S_ℓ and sixteen remaining ones. 27 is multiplied by $1, 10, 16$. □

The cardinal 432 of the warped stratum numerically coincides with the multiplicity of the transverse region of π in Figure 21.5. The two objects are not thereby identified: here one counts words classified by $L(w)^{-1}H(w)$, whereas there one counts preimages of a register U_6. Without an explicit and equivariant intertwiner between those domains, $432 = 432$ is only a cardinal control.

21.12.7 Necessity of the central phase

Enumerating all inverse-closed subsets of cardinality ten in the five groups of order 27 gives

group	abelian	$\#\,\mathrm{srg}(27, 10, 1, 5)$
C_{27}	yes	0
$C_9 \times C_3$	yes	0
C_3^3	yes	0
$C_9 \rtimes C_3$	no	9
H_3	no	9.

Thus, retaining 27 labels while abelianizing composition destroys the configuration. Neither cardinality nor degree suffices.

If $\mathcal{T}$ is the conjunto of 45 triangulos, the cubico

$$C_\Gamma(x_1, \dots, x_{27}) = \sum_{\{i,j,k\} \in \mathcal{T}} x_i x_j x_k$$

recovers adjacency, because every edge lies in a unique triangle. Its permutation stabilizer is exactly

$$\mathrm{Aut}(\Gamma_\ell) \cong W(E_6).$$

For the following diagram we write $\Gamma_{27} = \Gamma_\ell$ and $\mathcal{T}_{45} = \mathcal{T}$.

Comparison Criterion 21.85 (Falsification criteria for branch E_6)**.** The branch is refuted if the curvatures are not opposite, if the central cocycle is erased, if the quadratic correction (21.170) does not preserve the

$$\text{APP curvatures } \pm 1 \longrightarrow \sigma \longrightarrow \mathcal{H}_9 \longrightarrow H_3,$$
$$H_3 \longrightarrow \Gamma_{27} \longrightarrow \mathcal{T}_{45} \longrightarrow W(E_6).$$

Figure 21.6: Causal chain of the mixed-holonomy branch. It is parallel to $C_W \to W_{12} \to M_{12}$ and not a link of it.

product, if (21.172) does not produce the coefficients $(10, 1, 5)$, if an abelian group produces the same configuration, if the partition does not sum to 729, if the table 21.3 does not intertwine the adjacency matrices, or if $W(E_6)$ is identified only by its order and not by its action on the twenty-seven classes and the 72 roots.

21.13 Generation and locality of fields on the incidence lattice

The lattice construction is carried out after the selection of the dodecaphase register and its incidences. The marked origin, the integral product and the parity form used by the state space arise from that construction. The operations producing the fields, their coefficients and their locality relations are made explicit below. The same lattice enters into state generation, products on the vacuum and commutators of all integer modes.

21.13.1 Location of the marked origin and persistence of incidence

Write K_i, $0 \le i < 12$, for the coordinates of the selected register. The two support readings are

$$T_K = \{i : K_i \ge 729\}, \qquad H_K = \{i : p_i + e_i - f_i - K_i < 0\}, \tag{21.181}$$

where p_i, e_i, f_i are the regional blocks used by the pre-carry reader. The upper support uses the integer threshold 729; the negative support uses the sign of the pre-carry before reduction. Two different operations on the same register are thus preserved.

The selection already established gives

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601),$$

as the evaluation of the register, and produces the flag $T_K \subset H_K$, with $|T_K| = 4$, $|H_K| = 6$. The regional propagation and self-scaling incidences, together with the APP support, determine the set

$$O_K = (E_{\mathrm{inc}} \cap F_{\mathrm{inc}}) \setminus (H_K \cup A_{\mathrm{inc}}).$$

The incidence construction proves that O_K is a singleton. Its unique element, o, fixes the marked origin preserved throughout what follows.

In the concrete Golay realization, the support T_K corresponds to $\{4, 6, 8, 14\}$. The four hexads incident to this support have four octadic lifts; the complete octadic star has five elements. Their difference is a single additional octad. The support H_K, in turn, has a unique octadic lift in the same realization. These properties arise from the concrete residual realization and preserve the regional origin of the selected support.

The marked lattice neighbour Λ_o obtained from that incidence has rank 24, an integral even bilinear form, integral self-duality and quadratic minimum 4. The radial vector used by the marked construction has square 54. We denote by $\langle x, y\rangle \in \mathbb{Z}$ the integral form of Λ_o. The theorems of this section use precisely that lattice and that form, already constructed.

21.13.2 Parity cocycle and twisted lattice algebra

Choose an integral basis $(b_i)_{i=1}^{r}$ of Λ_o, with $r = 24$. This choice presents the coordinates of a lattice already obtained. Its Gram matrix is $G_{ij} = \langle b_i, b_j \rangle$. For $x = \sum_i x_i b_i$, $y = \sum_i y_i b_i$, define

$$\tau(x, y) = \sum_{i<j} x_i G_{ij} y_j \pmod 2, \qquad \varepsilon(x, y) = (-1)^{\tau(x,y)}. \tag{21.182}$$

Proposition 21.86 (Cocycle identity and parity commutator)**.** *The following hold*

$$\varepsilon(x, y)\varepsilon(x + y, z) = \varepsilon(y, z)\varepsilon(x, y + z), \tag{21.183}$$

$$\varepsilon(x, y)\varepsilon(y, x) = (-1)^{\langle x,y \rangle}, \tag{21.184}$$

$$\varepsilon(0, x) = \varepsilon(x, 0) = 1. \tag{21.185}$$

Proof. The bilinearity of τ in $\mathbb{F}_2$ gives $\tau(x, y)+\tau(x+y, z) = \tau(y, z)+\tau(x, y+z)$, and applying $a \mapsto (-1)^a$ yields (21.183). The symmetry of G, together with the evenness of its diagonal, allows the two triangular sums to be combined:

$$\tau(x, y) + \tau(y, x) = \sum_{i,j} x_i G_{ij} y_j = \langle x, y \rangle \pmod 2.$$

This proves (21.184). The identities involving zero follow directly from (21.182). □

The twisted complex algebra $\mathbb{C}_\varepsilon[\Lambda_o]$ consists of finite linear combinations of symbols $\mathbf{e}^x$, with product

$$\mathbf{e}^x \mathbf{e}^y = \varepsilon(x, y)\mathbf{e}^{x+y}. \tag{21.186}$$

The cocycle identity proves associativity, and $\mathbf{e}^0$ is the unit. The notation $\mathbf{e}^x$ denotes the basic element of charge x and preserves the full lattice label.

21.13.3 Oscillator space and Heisenberg representation

Let $\mathfrak{h} = \mathbb{C} \otimes_{\mathbb{Z}} \Lambda_o$. The algebraic oscillator space is

$$\mathfrak{h}_- = \bigoplus_{n \geq 1} \mathfrak{h}\, t^{-n}, \qquad M(1) = \mathrm{Sym}_{\mathbb{C}}(\mathfrak{h}_-).$$

The support of each vector is finite, whereas the set of available frequencies includes all positive integers. The state space that simultaneously preserves oscillator occupation and lattice charge is

$$V_{\Lambda_o} = M(1) \otimes_{\mathbb{C}} \mathbb{C}_\varepsilon[\Lambda_o], \qquad \mathbf{1} = 1 \otimes \mathbf{e}^0. \tag{21.187}$$

For $x \in \Lambda_o$ and $n \geq 1$, the operator $x(-n)$ multiplies by the generator $x\, t^{-n}$. The operator $x(n)$ is the derivation of $M(1)$ determined by

$$x(n)(y\, t^{-m}) = n\, \delta_{n,m} \langle x, y \rangle. \tag{21.188}$$

Both act as the identity on the second factor of (21.187). The zero mode acts on the charge through

$$x(0)(v \otimes \mathbf{e}^y) = \langle x, y \rangle v \otimes \mathbf{e}^y.$$

Theorem 21.87 (Commutation relations and non-emptiness)**.** *For $m, n \geq 1$,*

$$[x(n), y(-m)] = n\delta_{n,m}\langle x, y \rangle I, \tag{21.189}$$

$$[x(-n), y(-m)] = 0, \qquad [x(n), y(m)] = 0. \tag{21.190}$$

Moreover, $x(n)\mathbf{1} = 0$ *and* $\mathbf{1} \neq 0$.

Proof. If D is a derivation and M_a multiplication by a, the Leibniz rule gives $[D, M_a](v) = D(a)v$. Applied to (21.188), it yields (21.189). The multiplication operators commute because the symmetric algebra is commutative; the coordinate derivations commute on the generators and, by Leibniz, on all monomials. Annihilation of the vacuum follows by differentiating the unit. Finally, the functional extracting the coefficient of the empty monomial and zero charge takes the value 1 on $\mathbf{1}$, so the vacuum is nonzero. Extension by tensor product preserves all the identities. □

Each oscillator monomial has an integer weight $\mathrm{wt}(a) = \sum_{n,i} n a_{n,i}$, where $a_{n,i}$ are its occupations. This grading provides finite bounds for the sums occurring in the field coefficients.

21.13.4 Creation and annihilation coefficients in all degrees

The two formal operator series are constructed

$$E_x^-(z) = \exp\left(\sum_{n\geq 1} \frac{x(-n)}{n} z^n\right) = \sum_{d\geq 0} C_x(d) z^d, \tag{21.191}$$

$$E_x^+(s) = \exp\left(-\sum_{n\geq 1} \frac{x(n)}{n} s^n\right) = \sum_{r\geq 0} A_x(r) s^r. \tag{21.192}$$

These exponentials are formal operations on the modes already constructed. Each coefficient is defined by a finite sum. In particular, if $P_x(z) = \sum_{n\geq 1} x(-n) z^n / n$, then

$$C_x(d) = \sum_{q=0}^{d} \frac{1}{q!} [z^d] P_x(z)^q, \tag{21.193}$$

and the analogous formula for $A_x(r)$ is obtained by substituting the annihilation potential. The composition order of the endomorphisms is preserved in every power.

Lemma 21.88 (Stabilization of the coefficients)**.** *For any formal potential* P *with zero constant term,* $[z^d]P^q = 0$ *when* $q > d$. *Therefore, extending the upper bound of* (21.193) *to any* $N \geq d$ *preserves its value. Moreover,*

$$C_x(0) = A_x(0) = I, \qquad A_x(r)1 = 0 \quad (r > 0).$$

If v *has weight at most* N, *then* $A_x(r)v = 0$ *for* $r > N$.

Proof. Each factor of P has degree at least one, so a product of q factors has degree at least q. This argument uses only the grading and also applies to noncommutative coefficients. The degree-zero identities come from the empty power. Every nonempty composition of annihilation modes annihilates the unit. Finally, a term of degree r in the annihilation potential reduces the oscillator weight by r. A reduction exceeding the initial weight gives zero; by linearity, the conclusion holds throughout the subspace of weight at most N. □

The first coefficients illustrate the normalization:

$$C_x(1) = x(-1), \qquad C_x(2) = \tfrac{1}{2}\big(x(-1)^2 + x(-2)\big),$$
$$A_x(1) = -x(1), \qquad A_x(2) = \tfrac{1}{2}\big(x(1)^2 - x(2)\big).$$

Each denominator arises from the frequency or the order of the formal power. The Gram form entering the commutators remains the integral form of the marked lattice.

21.13.5 Charge field and Laurent truncation bound

The field associated with a charge $x \in \Lambda_o$ is written

$$Y_x(z) = E_x^-(z)E_x^+(z^{-1})\, e_x z^{x(0)}, \qquad Y_x(z) = \sum_{k\in\mathbb{Z}} Y_x[k] z^k, \tag{21.194}$$

where $e_x \mathbf{e}^y = \varepsilon(x,y)\mathbf{e}^{x+y}$. The following formula defines its coefficients on a monomial basis and makes the finiteness of each operation explicit.

Definition 21.89 (Field coefficient on a charged state)**.** If v is a monomial of weight N, define

$$Y_x[k](v \otimes \mathbf{e}^y) = \varepsilon(x,y) \sum_{j=0}^{N} C_x\big(k - \langle x,y\rangle + j\big) A_x(j) v \otimes \mathbf{e}^{x+y}, \tag{21.195}$$

with the convention $C_x(d) = 0$ for $d < 0$. It extends linearly to V_{Λ_o}.

Theorem 21.90 (Stability, truncation and creation of charge states)**.** *The formula* (21.195) *is independent of any weight bound greater than N. We have*

$$k < \langle x,y\rangle - N \quad \Longrightarrow \quad Y_x[k](v \otimes \mathbf{e}^y) = 0. \tag{21.196}$$

Each state therefore admits a lower Laurent bound. On the vacuum,

$$Y_x[k]\mathbf{1} = \begin{cases} C_x(k) 1 \otimes \mathbf{e}^x, & k \geq 0, \\ 0, & k < 0. \end{cases} \tag{21.197}$$

In particular, $Y_x[0]\mathbf{1} = 1 \otimes \mathbf{e}^x$.

Proof. Extending the bound adds terms $A_x(j)v$ with $j > N$, which vanish by Lemma 21.88. If the inequality in (21.196) holds, all indices $k - \langle x,y\rangle + j$, with $j \leq N$, are negative; every summand vanishes. A finite linear combination of monomials retains the smallest of its bounds, proving the assertion for any state. On $\mathbf{1}$, only $j = 0$ remains, and $\varepsilon(x,0) = 1$, $\langle x,0\rangle = 0$. We obtain (21.197). □

Theorem 21.91 (Generation of the entire lattice state space)**.** *The subspace generated from the vacuum by finite compositions of the operators $Y_x[0]$, $x \in \Lambda_o$, and all creation modes $b_i(-n)$, $n \geq 1$, is V_{Λ_o}.*

Proof. By (21.197), applying $Y_x[0]$ to the vacuum produces the ground state of any charge x. The creation operators multiply by the generators of the symmetric algebra; finite compositions of them produce all monomials on that ground state. Oscillator monomials multiplied by the elements $\mathbf{e}^x$ constitute an algebraic basis of (21.187). Their linear span is therefore the entire space. The argument covers every frequency, finite occupation and lattice charge. □

Corollary 21.92 (Determination of intertwiners by the vacuum)**.** *Two endomorphisms of V_{Λ_o} that commute with all generators of the preceding theorem and agree on the vacuum are equal.*

Proof. Commutation transports equality on the vacuum to every finite composition of generators. Total generation and linearity extend that equality to the entire space. □

21.13.6 Scalar contraction and normal ordering of products

The integer product $p = \langle x, y\rangle$ fixes a scalar sequence by the recurrence

$$s_p(0) = 1, \qquad (r+1)s_p(r+1) = (r-p)s_p(r). \tag{21.198}$$

The recurrence determines each coefficient from the preceding one. Its closed expression is $s_p(r) = (-1)^r \binom{p}{r}$, using the generalized integer binomial coefficient. The first terms are

$$1, \quad -p, \quad \frac{p(p-1)}{2}, \quad -\frac{p(p-1)(p-2)}{6}.$$

Proposition 21.93 (Formal contraction factor)**.** *The series $S_p(u) = \sum_{r\ge 0} s_p(r)u^r$ satisfies*

$$(1-u)S_p'(u) = -pS_p(u), \qquad S_p(0) = 1. \tag{21.199}$$

It is the unique formal solution of these conditions. Moreover,

$$S_{p+q} = S_pS_q, qquadS_0 = 1, qquadS_1 = 1-u, qquadS_pS_{-p} = 1.$$

For $p \ge 0$, $S_p(u) = (1-u)^p$, and its coefficients of degree greater than p vanish.

Proof. Comparing the coefficient of u^r in (21.199) reproduces exactly (21.198). Since $r+1 \neq 0$, the coefficients are determined inductively. The Leibniz rule shows that S_pS_q satisfies the same equation as S_{p+q} and has the same constant term; uniqueness gives the multiplicative identity. The cases 0, 1 are verified directly. The inverse identity follows from the case $q = -p$, and natural powers are obtained by induction. If p is natural, step $r = p$ of the recurrence produces zero and subsequent steps preserve that zero. □

Theorem 21.94 (Normal ordering in all degrees)**.** *The series constructed from the modes satisfy*

$$E_x^+(s)E_y^-(z) = S_{\langle x,y\rangle}(sz), E_y^-(z)E_x^+(s). \tag{21.200}$$

In coefficients, for $r, d \ge 0$,

$$A_x(r)C_y(d) = \sum_{i=0}^{\min(r,d)} s_{\langle x,y\rangle}(i), C_y(d-i)A_x(r-i). \tag{21.201}$$

Proof. Let $Q_x(s) = -\sum_{n\ge 1} x(n)s^n/n$ and $P_y(z) = \sum_{m\ge 1} y(-m)z^m/m$. By (21.189),

$$[Q_x(s), P_y(z)] = -\langle x, y\rangle \sum_{n\ge 1} \frac{(sz)^n}{n} I.$$

The commutator is central. The exponential identity for two potentials with central commutator is proved here coefficient by coefficient: commuting a power of Q_x through a power of P_y adds the scalar terms of that commutator; the sums in each bidegree are finite by Lemma 21.88. The resulting scalar factor has constant term one and satisfies (21.199), with $p = \langle x, y\rangle$; by the uniqueness in Proposition 21.93, it is $S_p(sz)$. This proves (21.200). Extracting the coefficient $s^r z^d$ gives (21.201); the contraction factor consumes the same degree i in both variables, which yields the bound $i \le \min(r, d)$. □

Theorem 21.95 (Exact contraction of states created from the vacuum)**.** *For every pair of charges x, y and all degrees r, d,*

$$A_x(r)C_y(d)1 = \begin{cases} s_{\langle x,y\rangle}(r), C_y(d-r)1, & r \le d, \\ 0, & r > d. \end{cases} \tag{21.202}$$

On the diagonal $r = d$, the result is $s_{\langle x,y\rangle}(r)1$. If $\langle x, y\rangle = p \geq 0$, it also vanishes for $r > p$.

Proof. We evaluate (21.201) on 1. Lemma 21.88 annihilates all summands with $r - i > 0$. Only $i = r$ remains, provided that $r \leq d$. For $r = d$, $C_y(0) = I$. The final vanishing follows from the polynomial bound of S_p. □

For example, $A_x(1)C_y(1)1 = -\langle x, y\rangle 1$. For $p = -1$, the recurrence gives $s_{-1}(r) = 1$ in all degrees; for $p = 2$, it gives $(1, -2, 1, 0, \ldots)$. These two evaluations exhibit, respectively, a contraction of unbounded order and a finite polynomial cancellation, both determined by the same integral lattice form.

21.13.7 Mixed commutator in all integer modes

For $x \in \Lambda_o$, write $H_x(m) = x(m)$, with $m \in \mathbb{Z}$, including negative creation modes, the zero charge mode and positive annihilation modes.

Theorem 21.96 (Integral mixed commutator)**.** *For all charges x, y and all integers m, k,*

$$[H_x(m), Y_y[k]] = \langle x, y\rangle, Y_y[k - m]. \tag{21.203}$$

Proof. We treat the three signs of m separately, preserving in each case the field coefficients of (21.195).

If $m > 0$, the Heisenberg relation gives

$$[x(m), C_y(d)] = \begin{cases} \langle x, y\rangle C_y(d - m), & d \geq m, \\ 0, & d < m. \end{cases}$$

Positive modes commute with the annihilation coefficients and with the charge shift. Commuting term by term in (21.195) decreases the creation degree by m, which amounts to replacing k by $k - m$. The weight cutoff remains valid because annihilation decreases the weight.

If $m = 0$, the field shifts the charge z to $y + z$. The difference between the zero-mode eigenvalues is $\langle x, y + z\rangle - \langle x, z\rangle = \langle x, y\rangle$. Oscillator operators preserve charge, yielding (21.203) with $k - m = k$.

Finally, let $m = -a$, with $a > 0$. Creation modes commute with one another, whereas the relation derived from (21.189) and (21.192) is

$$[x(-a), A_y(j)] = \begin{cases} \langle x, y\rangle A_y(j - a), & j \geq a, \\ 0, & j < a. \end{cases}$$

If the initial state has weight at most N, after creating mode $-a$ it has weight at most $N + a$. We use this common cutoff in both terms of the commutator. Summands with $j < a$ vanish; we reindex the remaining ones by $j = r + a$, $0 \leq r \leq N$. The creation degree becomes $k - \langle y, z\rangle + r + a$, exactly the field coefficient with index $k + a = k - m$. The identity is proved on charged monomial states and extends by linearity to the whole space. □

The shift of the field index correctly distinguishes the orientation of the modes: a positive mode subtracts its frequency, a negative mode adds it and the zero mode preserves the index. The three cases realize a single identity in $\mathbb{Z}$.

21.13.8 Mixed locality by coefficientwise cancellation

Consider the Heisenberg field $H_x(z) = \sum_{m\in\mathbb{Z}} H_x(m)z^{-m-1}$. For a bivariate series with coefficients $F(a, b)$, multiplication by $z - w$ acts as

$$(CF)(a, b) = F(a - 1, b) - F(a, b - 1). \tag{21.204}$$

The ordered compositions of the two fields have coefficients

$$F(a,b) = H_x(-a-1)Y_y[b], \qquad G(a,b) = Y_y[b]H_x(-a-1).$$

Theorem 21.97 (Mixed locality of order one)**.** *The constructed fields satisfy*

$$(z-w)H_x(z)Y_y(w) = (z-w)Y_y(w)H_x(z). \tag{21.205}$$

The equality is an identity of endomorphism coefficients on all of V_{Λ_o}.

Proof. By (21.204), coefficient (a,b) of the difference is

$$[H_x(-a), Y_y[b]] - [H_x(-a-1), Y_y[b-1]].$$

Theorem 21.96 turns both terms into $\langle x,y\rangle Y_y[a+b]$. Their difference is zero for all integers a,b. The identity holds in the endomorphisms, hence simultaneously for all states. □

21.13.9 Locality of charged fields and preservation of the origin

Normal ordering also proves locality for two charge fields. Let $p = \langle x,y\rangle$ and $N_p = \max(0,-p)$. The contraction factor for the two orders is expanded in the formal regions w/z and z/w; their relative parity is fixed by (21.184).

Theorem 21.98 (Uniform locality of charged fields)**.** *For arbitrary $x, y \in \Lambda_o$,*

$$(z-w)^{N_p}Y_x(z)Y_y(w) = (z-w)^{N_p}Y_y(w)Y_x(z), \qquad N_p = \max(0,-\langle x,y\rangle). \tag{21.206}$$

The exponent depends on the charges and is valid for all states.

Proof. On a state of bounded weight, products of field coefficients are expressed through (21.201), charge shifts and the cocycle. The sums for each coefficient are finite: annihilation has a weight cutoff, and the two variables of the ordered product have a lower bound. The cocycle identity combines the shifts into the same charge element $x+y$, and the commutator identity supplies the factor $(-1)^p$ between the two orders.

The resulting scalar factors are the two expansions of $(z-w)^p$. If $p \geq 0$, both are the same polynomial and agree without additional multiplication. If $p < 0$, multiplication by $(z-w)^{-p}$ transforms both expansions into the factor one. Formally, each application of C raises the exponent of both expansions by one; after $-p$ applications the exponent reaches zero. This cancellation is independent of the chosen weight cutoff and of the charge of the state on which the fields act. Thus (21.206) holds on every charged monomial state, and linearity extends it to V_{Λ_o}. □

Theorem 21.99 (Conservative composition of incidence, generation and locality)**.** *With the origin o produced by the incidence of the selected register fixed, the same space V_{Λ_o} jointly satisfies: generation from the vacuum in Theorem 21.91, products in all degrees in Theorem 21.95, charged locality in* (21.206) *and mixed locality in* (21.205)*. Its integral form, cocycle, vacuum and charge operators are those defined from Λ_o in this section.*

Proof. All the preceding statements are quantified over the same lattice and the same integral form. When specialized to the origin o already selected, their objects coincide by definition: the cocycle uses the basis of Λ_o, the oscillators use its Gram form, the fields use those oscillators and that cocycle, and the products use those same field coefficients. The conjunction of the theorems therefore preserves the incidence flag of (21.181) and all dependencies that determined the origin. No specialization introduces a new lattice choice. □

The demonstrative chain precisely distinguishes its operations: regional blocks determine prefixes, cylinders and signatures; dodecaphase selection and incidence produce the lattice origin; the lattice

determines the integral form and cocycle; these fix the modes, fields and their products. The resulting field structure is thus endowed with total generation and locality identities in the full algebraic space. The chapters devoted to reconstructing the vertex-operator structure, the twisted sector and the exceptional realization use these results with their respective additional hypotheses, preserving this same generative provenance.

Chapter 22

Vertex operator algebra $V^\natural$, modular function J, and monstrous Moonshine: graded action of the Monster, McKay–Thompson series, and replicability

22.1 Alineamiento ciclotomico, orbifold of order 3 and Monstrous Moonshine

The Leech lattice constructed in theorem 21.32 makes it possible to pass from finite incidence to an infinite graded structure. This transition will not be effected through a chain of names. The following will be constructed, in typed order:

1. the intertwining of the twelve fibres A_2;
2. a fixed-point-free lattice isometry of order three;
3. its graded trace and its type-zero lift;
4. the cyclic orbifold and its dual automorphism of class $3B$;
5. the moonshine vertex operator algebra;
6. through distinct publications, its automorphism group, its graded character, and the McKay–Thompson series.

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22.1.1 Alignment of the 12 APP and Witt fibers

The twelve fibers of $W_{\mathrm{APP}} \cong A_2^{12}$ carry three independent labels:

$$i \in \mathbb{Z}/3\mathbb{Z} \quad \text{(coordinate pair)}, \qquad \mu \in \{0,1\} \quad \text{(sheet)}, \qquad \varepsilon \in \{0,1\} \quad \text{(cyclotomic orbit)}.$$

Fixing the dodecaphasic origin constructed on the TPK, we define

$$\lambda(i,\mu,\varepsilon) := 4i + 2\mu + \varepsilon \pmod{12}. \tag{22.1}$$

Lemma 22.1 (Bijectivity of the alignment)**.** *The map λ from (22.1) is a bijection onto $\mathbb{Z}/12\mathbb{Z}$.*

Proof. For each i, the four values $4i + 2\mu + \varepsilon$ form the block $\{4i, 4i+1, 4i+2, 4i+3\}$. The three blocks for $i = 0, 1, 2$ are disjoint and cover the twelve classes. □

Let C be the Coxeter element of A_2 and let R be the reflection that satisfies

$$\mathsf{R} C \mathsf{R} = C^{-1}. \tag{22.2}$$

Once the trivializations ι_b and the oriented frames P_b have been fixed, we set

$$g_b := \iota_b^{-1} P_b C P_b^{-1} \iota_b, \tag{22.3}$$

$$T_{i\mu\varepsilon} := \iota_{\lambda(i,\mu,\varepsilon)}^{-1} P_{\lambda(i,\mu,\varepsilon)} \mathsf{R}^{\mu}. \tag{22.4}$$

Theorem 22.2 (APP Entanglement–Witt)**.** *For each terna (i,μ,ε),*

$$g_{\lambda(i,\mu,\varepsilon)} T_{i\mu\varepsilon} = T_{i\mu\varepsilon} C^{(-1)^{\mu}}. \tag{22.5}$$

The direct sum defines a C_3-equivariant isomorphism

$$T_\lambda : W_{\mathrm{APP}} \otimes \mathbb{C} \xrightarrow{\cong} W_c := \bigoplus_{b=0}^{11} (A_{2,b} \otimes \mathbb{C}). \tag{22.6}$$

Proof. For $\mu = 0$, the definition of g_b directly gives conjugation by C. For $\mu = 1$, one inserts $\mathsf{R}C\mathsf{R} = C^{-1}$. The lemma 22.1 guarantees that the direct sum uses every Witt position exactly once and preserves the pair, sheet, and orbit. □

The theorem is relative to the simultaneous transport of the order of pairs, leaves, and orbits, of the Witt origin, of the frames, and of the orientation. Changing a single one of these coordinates does not define the same functor.

22.1.2 Graded trace of the action of the ternary generator on $V^\natural$

Let

$$\rho_W := \bigoplus_{b=0}^{11} g_b$$

and consider the bosonic envelope

$$\mathcal{H}_{\mathrm{bos}}(W_c) := \mathrm{Sym}\left(\bigoplus_{n \geq 1} W_c[n] \right). \tag{22.7}$$

If N is the degree operator, we define

$$Z_{\mathrm{Fock}}(q) := q^{-1} \operatorname{Tr}_{\mathcal{H}_{\mathrm{bos}}(W_c)} \left(\Gamma_{\mathrm{bos}}(\rho_W) q^N \right). \tag{22.8}$$

Proposition 22.3 (Cyclotomic polynomial of level three)**.** *As a formal Laurent series,*

$$\boxed{Z_{\mathrm{Fock}}(q) = q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12} =: t_3(q).} \tag{22.9}$$

Its expansion begins with

$$t_3(q) = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + 1188q^4 + O(q^5). \tag{22.10}$$

Proof. Each g_b is conjugate to C, whence

$$\det(I - g_b q^n) = 1 + q^n + q^{2n}.$$

The identity

$$\sum_{k \geq 0} \operatorname{Tr}(\mathrm{Sym}^k \rho_W) t^k = \det(I - \rho_W t)^{-1}$$

applied in each degree produces the product. Formal multiplication through degree five gives the expansion. In particular, the coefficient of q is obtained as

$$[q^2]\, (1 + q + q^2)^{-12} (1 + q^2 + q^4)^{-12} = 66 - 12 = 54.$$

□

The value 54 appears here as the coefficient of a trace constructed from twelve fibers. Unit U030 also obtained it as the oriented index of the APP aggregate and proved that twelve is the unique positive solution of cyclotomic–radial rigidity. The equality of the two values is a compatibility between two realizations; neither uses the coefficient of J as input.

22.1.3 Isometry Without Fixed Points Induced by the Ternary Word of Full Support

Let

$$q_* = (1, 1, 1, 1, 1, 1) \in \mathbb{F}_3^6.$$

Multiplication by the Paley–Witt matrix gives

$$q_* A_W = (2, 1, 1, 1, 1, 1). \tag{22.11}$$

Therefore,

$$c_* := (q_*, q_* A_W) = (1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1) \in C_W \tag{22.12}$$

and all its coordinates are non-zero.

For each position b, we fix

$$P_b(c_*) = \begin{cases} \mathsf{R}, & (c_*)_b = 1, \\ I, & (c_*)_b = 2, \end{cases} \qquad g_c := \bigoplus_{b=0}^{11} P_b(c_*) C P_b(c_*)^{-1}. \tag{22.13}$$

Proposition 22.4 (Order, absence of fixed points and gluing)**.** *The operator g_c has order three, has no nonzero fixed vectors, and preserves the Niemeier lattice $N(A_2^{12})$ constructed in U021.*

Proof. Each block is conjugate to the Coxeter C, whose minimal polynomial is $X^2 + X + 1$. Hence it has order three and has no eigenvalue one. The direct sum preserves both properties. The Coxeter acts trivially on A_2^*/A_2; therefore each block preserves the discriminant class and the union of lateral classes that defines the gluing. □

22.1.4 Preservation of the Leech Neighbor

Write

$$v := v_{\mathcal{F}_*} = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \qquad v^2 = 54,$$

for the vector selected by the incidence origin of (21.82). Let

$$N := N(A_2^{12}), \qquad N_0 := \ker(\langle \cdot, v \rangle \bmod 3), \qquad N_v := N_0 + \mathbb{Z}\frac{v}{3} \cong \Lambda_{24}.$$

Theorem 22.5 (Neighbor-Oriented Preservation)**.** *The displacements*

$$y_* := \frac{g_c v - v}{3}, \qquad z_* := \frac{g_c^{-1} v - v}{3} \tag{22.14}$$

belong to N_0. Consequently,

$$g_c N_0 = N_0, \qquad g_c N_v = N_v.$$

Proof. The discriminant words of y_* and z_* are c_* and $-c_*$, both in C_W; hence $y_*, z_* \in N$. Computation in each factor A_2 gives

$$\langle y_*, v \rangle = \langle z_*, v \rangle = -(4^2 + 11) = -27 \equiv 0 \pmod 3. \tag{22.15}$$

Thus both lie in N_0. For $x \in N_0$,

$$\langle g_c x, v \rangle = \langle x, g_c^{-1} v \rangle = \langle x, v \rangle + 3\langle x, z_* \rangle \equiv 0 \pmod 3.$$

The same count with g_c^{-1} gives equality of the kernel. Finally,

$$g_c(v/3) = v/3 + y_*,$$

so that the class generated by the neighbor is preserved. □

22.1.5 24 orientations and lattice naturality

The weight enumerator of (21.18) contains $24y^{12}$. Therefore, the set

$$C_W^{\times} := \{u \in C_W : u_b \in \{1, 2\} \text{ for every } b\}$$

has 24 elements. For each $u \in C_W^{\times}$, we define

$$P_b(u) = \begin{cases} \mathsf{R}, & u_b = 1, \\ I, & u_b = 2, \end{cases} \qquad g_u := \bigoplus_{b=0}^{11} P_b(u) C P_b(u)^{-1}.$$

Proposition 22.6 (Robustness of Full Support Orientations)**.** *For every $u \in C_W^{\times}$,*

$$\frac{g_u v - v}{3}, \quad \frac{g_u^{-1} v - v}{3} \in N_0.$$

In particular, each g_u preserves the neighbor constructed with the transported orientation.

Proof. In simple root coordinates,

$$C\rho - \rho = (-2, -1), \qquad C^{-1}\rho - \rho = (-1, -2).$$

The discriminant classes of the two displacements are u and $-u$. For the radial coefficient $a_b \in \{4, 1\}$,

$$\left\langle \frac{(C^{\pm 1} - I) a_b \rho}{3}, a_b \rho \right\rangle = -a_b^2.$$

The sum over the twelve factors again gives -27. The proof of theorem 22.5 is applied component by component. □

To type the recalibration, let $h \in \mathrm{Aut}(C_W)$ be monomial: a permutation σ of the twelve coordinates followed by multipliers $\epsilon_b \in \{1, 2\}$. We define its lattice lift

$$\widetilde{h} := P_\sigma \circ \bigoplus_{b=0}^{11} \mathsf{R}^{\nu_b}, \qquad \nu_b = \begin{cases} 0, & \epsilon_b = 1, \\ 1, & \epsilon_b = 2. \end{cases} \tag{22.16}$$

Its action on the discriminant is exactly h. If the change simultaneously transports the code, flag, incidence origin, vector v, neighbor, and frames, then

$$g_{hu} = \widetilde{h}\, g_u\, \widetilde{h}^{-1}. \tag{22.17}$$

The ternary matrix h does not act without a lift on A_2^{12} or on the Leech lattice.

22.1.6 Lattice trace and computation of type 0

Let $\widehat{g}_c$ be the standard lift of g_c to the lattice VOA

$$V_{N_v} = M(1) \otimes \mathbb{C}_\varepsilon[N_v].$$

For $j = 1, 2$, the oscillator–lattice decomposition produces

$$\begin{aligned} Z_{0,j}(\tau) &:= \operatorname{Tr}_{V_{N_v}}\left(\widehat{g}_c^{\,j} q^{L_0-1}\right) \\ &= q^{-1}\left(\sum_{\lambda\in N_v^{g_c^j}} \varepsilon_j(\lambda) q^{\langle\lambda,\lambda\rangle/2}\right)\prod_{n\geq 1}\det(I-g_c^j q^n)^{-1}. \end{aligned} \tag{22.18}$$

Proposition 22.7 (Trace of the nontrivial sectors)**.** *We have*

$$Z_{0,1}(\tau) = Z_{0,2}(\tau) = t_3(\tau).$$

Proof. The absence of fixed points implies

$$N_v^{g_c} = N_v^{g_c^2} = \{0\},$$

so that the lattice sum in (22.18) equals one. In each factor, g_c^j has eigenvalues ζ_3, ζ_3^2; therefore,

$$\det(I - g_c^j q^n) = 1 + q^n + q^{2n}.$$

The twelve factors reproduce (22.9). □

Theorem 22.8 (Twisted weight and type of the lift)**.** *The standard lift has order three, twisted weight*

$$\rho_{g_c} = \frac{4}{3} \tag{22.19}$$

and type zero.

Proof. In $\mathfrak{h} = N_v \otimes_{\mathbb{Z}} \mathbb{C}$, each of the twelve factors contributes one eigenvector with eigenvalue ζ_3 and another with eigenvalue ζ_3^2. Therefore,

$$\dim \mathfrak{h}_{\zeta_3} = \dim \mathfrak{h}_{\zeta_3^2} = 12.$$

The formula for the weight of the twisted vacuum of order three gives

$$\rho_{g_c} = \frac{1}{4\cdot 3^2}\left(1\cdot 2\cdot 12 + 2\cdot 1\cdot 12\right) = \frac{4}{3}.$$

For a lattice isometry of odd order, the standard elevation preserves the order. The type is the class of

$$3^2\rho_{g_c} = 12 \equiv 0 \pmod 3.$$

Thus, the two initially independent hypotheses are calculated here. □

22.1.7 Cyclic orbifold and dual class $3B$

Theorem 22.9 (Triadic orbifold of the Leech neighbor)**.** *The classical cyclic-orbifold theorems, applied to the standard lift in theorem 22.8, yield*

$$\operatorname{Orb}_{\mathbb{Z}/3}(V_{N_v}, \widehat{g}_c) \cong V^\natural. \tag{22.20}$$

The dual automorphism belongs to the class $3B$ and its series is

$$T_{3B}(\tau) = t_3(\tau) + 12. \tag{22.21}$$

Proof. The isometry is fixed-point-free of order three; the twisted weight and type zero were calculated, not assumed. The cyclic-orbifold theorems identify the holomorphic extension with the Moonshine module and classify the dual automorphism as $3B$ [27, 2, 23]. The identification uses these theorems after constructing the lattice object; it is not deduced solely from an equality of series. □

If $V^{(i)}$ is the $\widehat{g}_c^{\,i}$-twisted module and

$$Z_{i,j}(\tau) := \mathrm{Tr}_{V^{(i)}}\left(\phi_i(\widehat{g}_c)^j q^{L_0-1}\right),$$

the character of the orbifold is the average of the nine sectors:

$$\mathrm{ch}_{V^\natural}(\tau) = \frac{1}{3}\sum_{i,j\in\mathbb{Z}/3} Z_{i,j}(\tau) = J(\tau). \tag{22.22}$$

The level three rational identity is

$$J = t_3 + 12 + \frac{10\,3^9}{t_3} + \frac{4\,3^{14}}{t_3^2} + \frac{3^{18}}{t_3^3}. \tag{22.23}$$

Since $t_3^{-1} = q + O(q^2)$,

$$[q]J = 54 + 10\cdot 3^9 = 196884 = 54(5\cdot 729 + 1). \tag{22.24}$$

This is a graded compatibility within the constructed corridor, not a scalar morphism $54 \to V^\natural$.

22.1.8 Lattice construction of order 2

The preceding route of order three and the classical construction of Frenkel–Lepowsky–Meurman produce two presentations of the same moonshine VOA. To declare the second, we fix the central extension

$$1 \longrightarrow \{\pm 1\} \longrightarrow \widehat{\Lambda}_{24} \longrightarrow \Lambda_{24} \longrightarrow 1 \tag{22.25}$$

with commutator $(-1)^{\langle\lambda,\mu\rangle}$, and a cocycle $\varepsilon(\lambda,\mu)$ that represents it. The twisted group algebra is $\mathbb{C}_\varepsilon[\Lambda_{24}]$, and

$$V_\Lambda := M(1)\otimes \mathbb{C}_\varepsilon[\Lambda_{24}] \tag{22.26}$$

is the lattice VOA of central charge 24.

The negation $\lambda \mapsto -\lambda$ has a lift θ compatible with (22.25). Its trace is

$$t_2(\tau) := \mathrm{Tr}_{V_\Lambda}(\theta q^{L_0-1}) = q^{-1}\prod_{n\geq 1}(1+q^n)^{-24}. \tag{22.27}$$

Indeed, negation pairs e^λ with $e^{-\lambda}$, and the absence of torsion leaves only $\lambda = 0$ in the lattice trace; each of the 24 oscillators contributes $(1+q^n)^{-1}$.

Let V_Λ^T the modulo irreducible torcido by θ, and $(V_\Lambda^T)^+$ its sector of signo positive. The construction FLM is

$$V^\natural = (V_\Lambda)^+ \oplus (V_\Lambda^T)^+. \tag{22.28}$$

The route $\mathbb{Z}/2$ of (22.28) and the route $\mathbb{Z}/3$ of (22.20) are parallel constructions identified by classical theorems; neither replaces the other.

22.1.9 Automorphism group, character, and graded series

eleven constructed $V^\natural$, appear two publications distinct:

$$\mathrm{Aut}(V^\natural) \cong \mathbb{M}, \tag{22.29}$$

$$\mathrm{ch}_{V^\natural}(\tau) = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots . \tag{22.30}$$

The former identifies the symmetry group; the latter counts graded dimensions. The vacuum line is

$$(V^\natural)_0 = \mathbb{C}\mathbf{1},$$

and the absence of roots in Λ_{24} gives

$$(V^\natural)_1 = 0.$$

In degree two,

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

Only the first equality is a decomposition into dimensions of representations of the Monster; the others are arithmetic controls.

For $g \in \mathbb{M}$, the McKay–Thompson series is defined

$$T_g(\tau) := \mathrm{Tr}_{V^\natural}\left(g q^{L_0 - 1}\right). \tag{22.31}$$

Theorem 22.10 (Monstrous Moonshine)**.** *The graded action of $\mathbb{M}$ on $V^\natural$ produces the family $\{T_g\}_{g\in\mathbb{M}}$. The Moonshine theorems identify these series with genus-zero modular functions and prove their replicability identities [28, 15, 50].*

This theorem has as input the pair

$$\left(V^\natural, \mathrm{Aut}(V^\natural)\right)$$

and its grading. It is not the consequence of an arrow whose domain is the abstract group $\mathbb{M}$.

For $J = T_{1A}$, let P_n be the Faber polynomials normalized by

$$P_n(J) = q^{-n} + O(q).$$

The Hecke–Faber identity is

$$P_n(J) = n\, T_n J. \tag{22.32}$$

It controls all degrees; the coefficient 196884 is only the first nontrivial manifestation.

22.1.10 Paley–Witt–Golay–Leech–Moonshine Runner Dependency Graph

Comparison Criterion 22.11 (Refutation criteria for the moonshine corridor)**.** The construction is refuted if any of the following facts occurs:

1. λ is not bijective or (22.5) falla;
2. $c_* \notin C_W$, g_c has fixed points or does not preserve the neighbor;
3. the displacements of (22.14) do not belong to N_0;
4. the lattice lift $\widetilde{h}$ does not realize the monomial automorphism h on the discriminant;
5. the twisted weight is not $4/3$ or the lift is not of type zero;
6. the orbifold is identified with $V^\natural$ without verifying the hypotheses of the classical theorem;
7. W_{24} is used as a constructor of the Leech lattice;

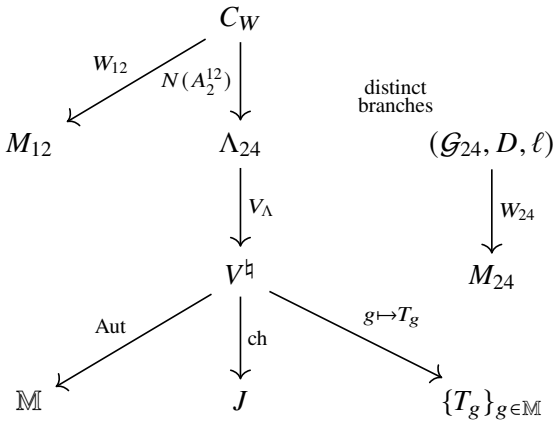

Figure 22.1: Exceptional bifurcation and three publications of $V^\natural$. The binary input $(\mathcal{G}_{24}, D, \ell)$ is independent; there is no arrow $W_{24} \to \Lambda_{24}$.

8. an equality of dimensions is presented as a construction of $\mathbb{M}$;
9. the publications Aut, ch, and $g \mapsto T_g$ are improperly serialized.

22.2 Theta control of the Leech lattice: identity $\Theta_\Lambda = E_4^3 - 720\Delta$ and separation between arithmetic coincidence and structural morphism

From Λ_{24}, the string

$$\Lambda_{24} \longrightarrow V^\natural \longrightarrow \mathbb{M} \longrightarrow \text{Moonshine}$$

is classical mathematics: vertex operator algebra, the Monster group, and McKay–Thompson functions. The contribution verified here lies in the finite input, the selection of the boundary, and the rootless 3-neighbor, not in claiming those theorems anew.

Three numerical identities accompany this chain:

$$\Theta_\Lambda = E_4^3 - 720\Delta,$$

$$196884 = 196560 + 324,$$

$$196884 = 54(3645 + 1) = 54(5 \cdot 729 + 1).$$

They are exact arithmetic identities, but without a morphism or a character decomposition, they do not by themselves constitute mathematical signals of compatibility. None replaces the construction of $V^\natural$ or provides a direct action of the Monster on HMT decimal words.

22.3 Lattice VOA, canonical involution and order-two Moonshine branch

The incidence branch first constructs the Leech lattice Λ_{24}. From that datum, the classical theory of vertex operator algebras provides a canonical prolongation. If $\mathfrak{h} = \Lambda_{24} \otimes_{\mathbb{Z}} \mathbb{C}$ and $M(1)$ is the Fock module of its Heisenberg algebra, the lattice VOA is

$$V_\Lambda = M(1) \otimes \mathbb{C}[\Lambda_{24}], \tag{22.33}$$

with central charge 24. For $\alpha \in \Lambda_{24}$, its vertex operators are written

$$Y(e^\alpha, z) = E^-(-\alpha, z)E^+(-\alpha, z)e^\alpha z^{\alpha_0}.$$

The absence of roots in Λ_{24} implies the vanishing of the weight-one sector in the Moonshine orbifold.

The involution

$$\theta : \Lambda_{24} \longrightarrow \Lambda_{24}, \qquad \theta(\lambda) = -\lambda,$$

is intrinsic and prolongs to V_Λ. Its graded trace is obtained by direct counting of the oscillator modes.

Theorem 22.12 (Trace of the negation involution)**.**

$$t_2(\tau) := \mathrm{Tr}_{V_\Lambda}\big(\theta q^{L_0-1}\big) = q^{-1}\prod_{n\geq 1}(1+q^n)^{-24}. \tag{22.34}$$

Proof. The involution pairs e^λ with $e^{-\lambda}$; since the lattice is torsion-free, only $\lambda = 0$ contributes to the trace of the lattice sector. Each one of the twenty-four energy oscillators n contributes $\sum_{k\geq 0}(-1)^k q^{nk} = (1+q^n)^{-1}$. The vacuum displacement produces the factor q^{-1}. □

The Fricke completion associated with this trace is

$$T_{2A}(\tau) = t_2(\tau) + 24 + \frac{2^{12}}{t_2(\tau)} = q^{-1} + 4372q + 96256q^2 + 1240002q^3 + \cdots. \tag{22.35}$$

The term 24 cancels the constant term of t_2; the factor 2^{12} is the degeneracy of the twisted sector of twenty-four bosons. The same boundary series determines the modular invariant through

$$\boxed{j(\tau) = \frac{(t_2(\tau)+256)^3}{t_2(\tau)^2}}. \tag{22.36}$$

Equations (22.34)– (22.36) construct the canonical branch of order two without continuous parameters.

The Frenkel–Lepowsky–Meurman orbifold is defined by

$$V^\natural = (V_\Lambda)^+ \oplus (V_\Lambda^T)^+.$$

It is a holomorphic VOA of central charge 24, satisfies $(V^\natural)_1 = 0$, and has character

$$\mathrm{ch}_{V^\natural}(\tau) = j(\tau) - 744$$

and its automorphism group is the Monster group [50, 28, 15]. In this chain, HMT fixes the Leech lattice and the involution θ; the classical VOA and FLM theorems identify the resulting prolongation with $V^\natural$ and its automorphism group. The assertion does not require an action of the Monster on the digits: the Archimedean evaluation and the exceptional branch descend from a common HMT preimage and preserve different invariants.

The neighboring function $T_{3A} = t_3 + 12 + 729/t_3$ belongs to the classical theory of Hauptmoduln of order three. The active HMT realization in the tritic corridor is the class $3B$; an identification with $3A$ requires declaration of the specific lift that changes the class.

22.3.1 APP–Witt intertwining, Leech neighbor, and orbifold of order 3

The twelve fibers bear the labels

$$(i,\mu,\varepsilon) \in \mathbb{Z}/3\mathbb{Z} \times \{0,1\} \times \{0,1\}, \qquad \lambda(i,\mu,\varepsilon) = 4i + 2\mu + \varepsilon \pmod{12}.$$

The map λ is bijective and, once trivializations and frames are fixed, the intertwiners satisfy

$$g_{\lambda(i,\mu,\varepsilon)} T_{i\mu\varepsilon} = T_{i\mu\varepsilon} C^{(-1)^\mu}.$$

Its direct sum is an C_3-equivariant isomorphism

$$T_\lambda : W_{\mathrm{APP}} \otimes \mathbb{C} \xrightarrow{\cong} \bigoplus_{b=0}^{11} (A_{2,b} \otimes \mathbb{C}).$$

Let $N = N(A_2^{12})$, let $v = v_\alpha$ the vector radial of norm 54, and definamos

$$N_0 = \ker\big(\langle \cdot, v\rangle \bmod 3\big), \qquad N_v = N_0 + \mathbb{Z}\frac{v}{3}.$$

The lattice calculation verifies that $N_v \cong \Lambda_{24}$, that the isometry g_c has order three and has no nonzero fixed points, and that

$$\frac{g_c v - v}{3}, \quad \frac{g_c^{-1} v - v}{3} \in N_0.$$

Hence $g_c N_v = N_v$. Its standard lift to the lattice VOA has twisted weight $4/3$ and type zero. The classical cyclic-orbifold theorems give

$$V_{(3)} := \mathrm{Orb}_{\mathbb{Z}/3}(V_{N_v}, \widehat{g_c}) \cong V^\natural, \qquad T_{3B} = t_3 + 12.$$

The character is recovered by the average of the nine sectors and by the level three uniformization:

$$\mathrm{ch}_{V_{(3)}} = \frac{1}{3} \sum_{i,j \in \mathbb{Z}/3} Z_{i,j} = J,$$

$$J = t_3 + 12 + \frac{10\,3^9}{t_3} + \frac{4\,3^{14}}{t_3^2} + \frac{3^{18}}{t_3^3}.$$

In particular,

$$[q]J = 54 + 10 \cdot 3^9 = 196884 = 54(5 \cdot 729 + 1).$$

22.3.2 Constructions of orders 2 and 3 of the vertex operator algebra $V^\natural$

Fix the central extension and the Leech cocycle.

$$1 \longrightarrow \{\pm 1\} \longrightarrow \widehat{\Lambda}_{24} \longrightarrow \Lambda_{24} \longrightarrow 1, \qquad V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda_{24}].$$

The negation $\lambda \mapsto -\lambda$ lifts to θ, and

$$t_2(\tau) = \mathrm{Tr}_{V_\Lambda}\left(\theta q^{L_0 - 1}\right) = q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

The FLM presentation is

$$V_{(2)} := (V_\Lambda)^+ \oplus (V_\Lambda^T)^+ \cong V^\natural.$$

Theorem 22.13 (Typed equivalence of the presentations of orders 3 and 2)**.** *Let*

$$\phi_3 : V_{(3)} \xrightarrow{\cong} V^\natural, \qquad \phi_2 : V_{(2)} \xrightarrow{\cong} V^\natural$$

the graded isomorphisms provided by the orbifold theorems applied to the preceding data. Then

$$\boxed{\Phi_{23} := \phi_2^{-1} \circ \phi_3 : V_{(3)} \xrightarrow{\cong} V_{(2)}}$$

is a graded isomorphism of vertex operator algebras. Moreover,

$$a \longmapsto \Phi_{23} a \Phi_{23}^{-1}$$

identifies its automorphism groups and preserves the graded traces.

Proof. The composition of VOA isomorphisms is a VOA isomorphism and preserves the vacuum, the conformal vector, the vertex operators, and the grading by L_0. Conjugation transports automorphisms. For every automorphism a, invariance of the trace under conjugation gives equality of the corresponding graded series. □

The isomorphism Φ_{23} depends on the identifications ϕ_2 and ϕ_3; absolute canonicity is not asserted. The order-two construction also preserves the exclusive content

$$T_{2A} = t_2 + 24 + \frac{2^{12}}{t_2}, \qquad \boxed{j = \frac{(t_2+256)^3}{t_2^2}},$$

which completes the modular uniformization of order two.

Once $V^\natural$ has been constructed, the three publications are stated without unduly serializing them:

$$\mathrm{Aut}(V^\natural) \cong \mathbb{M}, \qquad \mathrm{ch}_{V^\natural} = J, \qquad g \longmapsto T_g(\tau) = \mathrm{Tr}_{V^\natural}\left(g q^{L_0-1}\right).$$

The Moonshine theorem coordinates the graded action with genus-zero functions and their replicability identities. The Monster action resides in $V^\natural$; the fact that it is not a pointwise permutation of digits is a final type control, not the thesis of the corridor.

22.4 Continuous evidentiary chain of the exceptional publication: from the APP unfolding to Paley–Witt, $V^\natural$ and Moonshine

The exceptional publication starts from the incidence reader $\mathfrak{E}_{\mathrm{exc}}$ of (2). It does not take an already given code, lattice or exceptional group as input: it takes the boundary, orientation, sheet and memory of an enriched history and expresses them in the projective gauge fixed by nonadic holonomy. Only afterwards are the classical theorems of codes, lattices and vertex operator algebras applied to the produced objects.

22.4.1 APP unfolding and graded cone of the index 54

Let R be the involution exchanging the APP sheets and let $P_+ = (I+R)/2$, $P_- = (I-R)/2$ be its projectors. The central block

$$B_c = 7I + 2R = 9P_+ + 5P_- \tag{22.37}$$

produces four objects of distinct types:

$$9-5=4, \qquad (B_c)_{i\neq j} = 2, \qquad \sum M_\Pi - \sum M_\Sigma = 51-45 = 6, \qquad 9\cdot 6 = 459-405 = 54. \tag{22.38}$$

The 4 is the spectral jump between sheets, the 2 the local coupling, the 6 the difference of the modular compressions and the 54 the integration of that difference over the nine positions. Therefore, $4 \to 2 \to 6 \to 54$ designates a hierarchy of morphisms and evaluations, not a chain of equalities between objects.

With the orientation $o_{\Sigma\Pi}$, sheet parity and triangular normalization fixed, the Fock branch has the coupling

$$\mathcal{H}^{(m)}_{\Sigma,\Pi} : M_{\mathrm{bal}}^{C_3,-} \longrightarrow \mathrm{End}\big(\mathcal{F}_2(\mathcal{V}_m)\big) \otimes o_{\Sigma\Pi}, \tag{22.39}$$

and its oriented trace satisfies

$$\mathrm{Tr}_{o,\mathcal{F}_2}\big(\mathcal{H}^{(m)}_{\Sigma,\Pi}(\nu)\Gamma_2(g_m)\big) = -\frac{3m\,c(\nu)}{2}. \tag{22.40}$$

For $m = 12$ and $c(\delta) = -3$, 54 holds. Independently, the cyclotomic product

$$t_3(q) = q^{-1}\prod_{n\geq 1}(1+q^n+q^{2n})^{-12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \cdots \tag{22.41}$$

gives $[q]t_3 = 54$.

Theorem 22.14 (Typed cone of the common index)**.** *Once the APP census, the Fock branch, the cyclotomic product, the lattice neighbor of equation (22.52) and the lunar module of equation (22.58) have been constructed separately, their readers satisfy*

$$D_{\mathrm{APP}} = \mathrm{Tr}_{o,\mathcal{F}_2}\big(\mathcal{H}^{(12)}_{\Sigma,\Pi}(\delta)\Gamma_2(g_{12})\big) = [q]t_3 = v^2 = \frac{196884}{5\cdot 729+1} = 54. \tag{22.42}$$

The equality gathers evaluations in a cone toward $\mathbb{Z}$; it does not identify their domains or use 196884 *to select the corridor.*

Proof. The first three equalities are (22.38), (22.40) and (22.41). For the radial vector $v = 4\rho_1+\rho_2+\cdots+\rho_{12}$ of (22.51), orthogonality of the factors A_2 gives $v^2 = 4^2\cdot 2 + 11\cdot 2 = 54$. Finally, $5\cdot 729+1 = 3646$ and $54\cdot 3646 = 196884$. Each identity is proved in its own codomain before they are compared. □

22.4.2 Paley form and internal quadratic root of $-I_6$

Let

$$\mathcal{U}_9 = (\mathbb{Z}/9\mathbb{Z})^{\times} = \{1,2,4,8,7,5\},$$

in the order of the powers of 2, and let

$$\theta : \mathcal{U}_9 \xrightarrow{\sim} \mathbb{P}^1(\mathbb{F}_5), \qquad \begin{array}{c|cccccc} u & 1 & 2 & 4 & 8 & 7 & 5 \\ \hline \theta(u) & \infty & 0 & 1 & 2 & 3 & 4 \end{array}. \tag{22.43}$$

This bijection is a coordinate gauge of the reader; it is not used as a homomorphism. For the quadratic character $\chi : \mathbb{F}_5 \to \{-1,0,1\}$, define

$$C_P = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & -1 & -1 & 1 \\ 1 & 1 & 0 & 1 & -1 & -1 \\ 1 & -1 & 1 & 0 & 1 & -1 \\ 1 & -1 & -1 & 1 & 0 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \end{pmatrix}, \tag{22.44}$$

where the finite off-diagonal entries are $\chi(x-y)$ and the entries incident with ∞ are one.

Lemma 22.15 (Quadratic correlation)**.** *If $a \neq b$ in $\mathbb{F}_5$, then*

$$\sum_{t\in\mathbb{F}_5}\chi(t-a)\chi(t-b) = -1.$$

Proof. The substitution $u = (t-a)/(b-a)$ reduces the sum to $\sum_u \chi(u(u-1))$. For $u = 0,1,2,3,4$, the summands are $0,0,1,-1,-1$. □

Proposition 22.16 (Conference identity and ternary reduction)**.** *The matrix C_P is symmetric and satisfies*

$$C_P C_P^{\mathsf{T}} = 5I_6.$$

Its reduction $A_W = C_P \bmod 3$ satisfies

$$\boxed{A_W^2 = -I_6 \quad \text{in } M_6(\mathbb{F}_3).} \tag{22.45}$$

Proof. Each row of C_P has squared norm five. Two distinct finite rows have inner product $1 - 1 = 0$ by the lemma 22.15; the sum of a nontrivial character also proves orthogonality with the row of ∞. Reducing modulo three yields $A_W A_W^{\mathsf{T}} = 5I_6 = 2I_6 = -I_6$. Since A_W is symmetric, (22.45) follows. □

The identity (22.45) constructs the finite complex structure of the corridor. Indeed,

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1), \qquad P_{\pm\iota} = \frac{1}{2}(I \mp \iota A_W)$$

are complementary rank-three projectors and endow $\mathbb{F}_3^6$ with a rank-three $\mathbb{F}_9$-module structure. The action is explicit:

$$(a + b\iota) \cdot v = av + b(vA_W). \tag{22.46}$$

Therefore, $3^6 = 9^3$ is not used as a cardinal substitute for an action.

22.4.3 Self-dual ternary code and Witt system

Define

$$c : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \qquad c(q) = (q, qA_W), \qquad C_W := \operatorname{im} c. \tag{22.47}$$

Theorem 22.17 (Ternary code produced by incidence)**.** *The map c is injective and*

$$C_W = C_W^{\perp}, \qquad W_{C_W}(y) = 1 + 264y^6 + 440y^9 + 24y^{12}. \tag{22.48}$$

In particular, C_W has parameters $[12, 6, 6]_3$.

Proof. The first projection of $c(q)$ is q, so c is injective. With $G_W = (I_6 \mid A_W)$,

$$G_W G_W^{\mathsf{T}} = I_6 + A_W A_W^{\mathsf{T}} = 0$$

in $\mathbb{F}_3$. Thus $C_W \subseteq C_W^{\perp}$, and equality of dimensions gives self-duality. The enumerator is the exact finite sum

$$\sum_{q \in \mathbb{F}_3^6} y^{\mathrm{wt}(q)+\mathrm{wt}(qA_W)}.$$

Evaluating the $3^6 = 729$ rows of the explicit matrix (22.44) gives, respectively, the counts $1, 264, 440, 24$ at weights $0, 6, 9, 12$, and zero at the others. The sum of the four counts is 729, so no word remains outside the finite certificate. □

The 264 weight-six vectors occur in pairs $\{x, -x\}$; their 132 supports form the Steiner system

$$W_{12} = S(5, 6, 12). \tag{22.49}$$

This is the recognition, after constructing C_W, of the extended ternary Golay code and its Witt design [91, 34]. The design is not introduced to select the matrix A_W.

The binary prolongation remains separate. If $\mathcal{G}_{24} \subset \mathbb{F}_2^{24}$ is the extended binary Golay code, D a dodecad and $\ell : \{1, \ldots, 12\}_{\mathrm{HMT}} \to D$ a design isomorphism, the weight-eight supports form $W_{24} = S(5, 8, 24)$. For a fixed tetrad,

$$\lambda_4(W_{12}) = 4, \qquad \lambda_4(W_{24}) = 5,$$

so that four octads lift the four incident hexads and a fifth is transversal. The datum $(\mathcal{G}_{24}, D, \ell)$ belongs to the binary codomain; there is no canonical arrow $W_{24} \to \Lambda_{24}$.

22.4.4 Ternary gluing and Leech neighbor

Let $Q = A_2^{12}$, with $\det Q = 3^{12}$. The discriminant of each factor A_2 is $\mathbb{F}_3$, and the gluing defined by (22.47) is

$$N := \{x \in Q^* : x \bmod Q \in C_W\}. \tag{22.50}$$

Theorem 22.18 (Niemeier lattice produced by the code)**.** *The lattice N is positive definite, even and unimodular of rank twenty-four, and its root system is A_2^{12}. Consequently, $N \cong N(A_2^{12})$.*

Proof. Self-duality of C_W makes the gluing integral. All its weights are multiples of three, which makes the extension even. Moreover, $[N : Q] = |C_W| = 3^6$, and therefore

$$\det N = \frac{\det Q}{[N : Q]^2} = \frac{3^{12}}{3^{12}} = 1.$$

A nonzero discriminant class of A_2 has minimum norm $2/3$. Since every nonzero word of C_W has weight at least six, every gluing vector outside Q has norm at least four. The norm-two vectors are exactly the roots of the twelve factors A_2. The classification of even unimodular lattices of rank twenty-four identifies the indicated type [34]. □

Let ρ_i be the highest root of the factor $A_2^{(i)}$, normalized by $\rho_i^2 = 2$, and define

$$v = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \qquad v^2 = 54. \tag{22.51}$$

The ordered frame of the exceptional reader fixes which of the twelve fibers occupies the first position; no Archimedean value intervenes in this selection. Set

$$\chi_v(x) = \langle x, v\rangle \bmod 3, \quad N_0 = \ker \chi_v, \quad N_v = N_0 + \mathbb{Z}\frac{v}{3}. \tag{22.52}$$

Lemma 22.19 (Structure of the neighbor)**.** *The character χ_v is surjective, $[N : N_0] = 3$, $v/3 \in N_0^*$, and N_v is even and unimodular.*

Proof. For every root of a factor A_2, the product with the highest root is ± 1 or ± 2; the coefficients of v are congruent to one modulo three. Therefore, no root belongs to N_0 and χ_v is nontrivial, hence surjective. The definition of the kernel gives index three and $\det N_0 = 9$. For $x \in N_0$, $\langle x, v/3\rangle \in \mathbb{Z}$, while $(v/3)^2 = 6$. Thus the extension by $v/3$ is integral and even. Since $[N_v : N_0] = 3$, $\det N_v = 9/3^2 = 1$. □

Lemma 22.20 (Minimum of the new classes)**.** *Each coordinate of a local class appearing in $N_0 \pm v/3$ has minimum norm $2/9$. Consequently,*

$$\|x \pm v/3\|^2 \geq 12 \cdot \frac{2}{9} = \frac{8}{3} \qquad (x \in N_0).$$

Proof. Writing a coordinate of A_2^* in the simple-root basis, the six relevant oriented classes are the shifts of $\pm\rho/3$ by the three discriminant classes. Completing squares in the Gram form $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ gives minimum $2/9$ in all six classes. The orthogonal sum of the twelve coordinates gives the inequality. □

Theorem 22.21 (Construction of the Leech lattice)**.** *The neighbor N_v of (22.52) is isometric to the Leech lattice:*

$$\boxed{N_v \cong \Lambda_{24}.} \tag{22.53}$$

Proof. The roots of N do not belong to N_0, and the two new classes contain no norm-two vectors by the lemma 22.20. The lemma 22.19 proves that N_v is positive definite, even, unimodular and of rank twenty-four. The uniqueness of the lattice with these properties and no roots identifies it with Λ_{24} [34]. □

22.4.5 Order-three isometry, orbifold and lunar module

The Coxeter rotation of each factor A_2 induces an order-three isometry g_c. Stability of the code C_W proves that $g_c N = N$. For the oriented frame fixing v, the shifts

$$y_* := \frac{g_c v - v}{3}, \qquad z_* := \frac{g_c^{-1} v - v}{3} \tag{22.54}$$

belong to N_0: their discriminant words are a pair $c_*, -c_* \in C_W$, and $\langle y_*, v\rangle = \langle z_*, v\rangle = -27$. Therefore,

$$g_c N_v = N_v. \tag{22.55}$$

The action has no fixed points in $N_v \otimes \mathbb{R}$, since each factor A_2 has the two primitive eigenvalues of order three.

Let $\widehat{g}_c$ be the standard lift to the lattice VOA $V_{\Lambda_{24}}$. Its two nontrivial eigenspaces have dimension twelve. The vacuum energy of the twisted module is therefore

$$\rho_{g_c} = \frac{1}{4 \cdot 3^2} \sum_{k=1}^{2} k(3-k) \dim \mathfrak{h}_{(k)} = \frac{4}{3}. \tag{22.56}$$

The congruence $3^2 \rho_{g_c} = 12 \equiv 0 \pmod 3$ proves that the lift is of type zero.

Theorem 22.22 (Triadic orbifold)**.** *The cyclic orbifold of the preceding lift satisfies*

$$\mathrm{Orb}_{\mathbb{Z}/3}(V_{\Lambda_{24}}, \widehat{g}_c) \cong V^\natural. \tag{22.57}$$

The dual automorphism belongs to the class $3B$.

Proof. The three hypotheses that cannot be replaced by an equality of series have been verified: fixed-point-free order-three isometry, preservation of the neighbor and type zero. The cyclic orbifold theorems then identify the holomorphic extension with the lunar module and classify the dual automorphism [27, 2, 23]. □

The order-two construction is a parallel realization. For the lift θ of negation on Λ_{24},

$$V^\natural = (V_{\Lambda_{24}})^+ \oplus (V_{\Lambda_{24}}^T)^+. \tag{22.58}$$

The order-two and order-three routes are identified by the VOA theorems; they are not causally serialized one inside the other [50].

Theorem 22.23 (Publications of the lunar module)**.** *Once $V^\natural$ has been constructed, one obtains*

$$\mathrm{Aut}(V^\natural) \cong \mathbb{M}, \tag{22.59}$$

$$\mathrm{ch}_{V^\natural}(\tau) = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \tag{22.60}$$

$$T_g(\tau) = \mathrm{Tr}_{V^\natural}\big(g q^{L_0 - 1}\big), \qquad g \in \mathbb{M}. \tag{22.61}$$

The series T_g are the genus-zero modular functions of the Moonshine theorem and satisfy the replicability identities.

Proof. The first identification and the graded construction come from the FLM realization. The coordination between the graded action, the McKay–Thompson series and their modular properties is the Moonshine theorem of Conway–Norton and Borcherds [50, 28, 15]. In particular, $(V^\natural)_0 = \mathbb{C}\mathbf{1}$ and $(V^\natural)_1 = 0$; the coefficient $196884 = 1 + 196883$ is the first nontrivial decomposition into dimensions of Monster representations. □

Falsification criterion 22.24 (Exceptional corridor)**.** The publication is refuted if any of the effective steps fails: (22.38), the trace (22.40), the coefficient $[q]t_3 = 54$, $C_P C_P^{\mathsf{T}} = 5I_6$, $A_W^2 = -I_6$, self-duality or the enumerator of C_W, evenness or unimodularity of the gluing, absence of roots of the neighbor, its preservation by g_c, the twisted weight $4/3$ or type zero. A narrative is also refuted if it uses W_{24} to construct Λ_{24}, confuses the two sets of 243 on account of their cardinality, or makes the Monster act directly on digits and routes. The action of $\mathbb{M}$ resides in $V^\natural$.

Chapter 23

Hilbertian realization of the genealogical state and von Neumann continuous dimension: nonadic UHF tower, trace and hyperfinite factor II_1

23.1 From the internal complex structure to the projective geometry of Bloch

The internal square root of minus the identity has two coordinated realizations. In Part I, Paley–Witt incidence constructs $J_W^2 = -I$ over $\mathbb{F}_3^6$, organizes the ternary code, and realizes the structure $\mathbb{F}_9^3$; in an analytic realization, an orthogonal operator J_E with $J_E^2 = -I$ turns a real space of even dimension into a complex space. This second arrow adds topology, norm, and completion: it preserves the quadratic relation but does not identify the scalar fields.

The complete transition is fully characterized by

$$\boxed{\begin{array}{c} C_P \longrightarrow J_W^2 = -I_6 \longrightarrow \mathbb{F}_3^6 \simeq \mathbb{F}_9^3, \\ (E_4, J_E), \quad \dim_{\mathbb{R}} E_4 = 4, \longrightarrow \mathbb{P}(\mathcal{H}_J) \simeq \mathbb{CP}^1 \longrightarrow S^2, \\ \mathrm{SU}(2) \longrightarrow \mathrm{PSU}(2) \simeq \mathrm{SO}(3) \subset \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{M\ddot{o}b}^+(S^2), \\ 2A_5 = \pi^{-1}(A_5) \subset \mathrm{SU}(2). \end{array}} \tag{23.1}$$

The first line retains its proofs in the section 21.2 y theorem 3.21; the two projective lines are constructed here.

23.1.1 Phase and ray space realification

Let (E, g) a space euclideo real of dimension par and let $J_E \in \mathrm{End}(E)$ such that

$$J_E^2 = -I, \qquad g(J_E u, J_E v) = g(u, v).$$

Then $J_E^{\mathsf{T}} = -J_E$ and

$$R_J(\theta) := e^{\theta J_E} = \cos\theta\, I + \sin\theta\, J_E$$

is orthogonal, with $R_J(\theta)R_J(\phi) = R_J(\theta + \phi)$. The action $(a + ib)u := au + bJ_E u$ realizes the complex chart without using the scalar i as antecedent of the operator.

Now let $E_4 \subseteq E$ be a J_E-stable subspace of real dimension four and let $\mathcal{H}_J = (E_4, J_E)$ be the resulting Hilbert space, of complex dimension two. A unit vector

$$\psi = \begin{pmatrix} z_0 \\ z_1 \end{pmatrix}, \qquad |z_0|^2 + |z_1|^2 = 1,$$

determines to ray $[\psi] \in \mathbb{CP}^1$. Definamos

$$r_1 = 2\,\mathrm{Re}(\bar{z}_0 z_1), \qquad r_2 = 2\,\mathrm{Im}(\bar{z}_0 z_1), \qquad r_3 = |z_0|^2 - |z_1|^2. \tag{23.2}$$

Theorem 23.1 (Projective Bloch Identification)**.** *The map*

$$\mathcal{B} : \mathbb{CP}^1 \longrightarrow S^2, \qquad [\psi] \longmapsto (r_1, r_2, r_3),$$

is a smooth bijection. The rank one projector associated with the ray is

$$\rho_\psi = |\psi\rangle\langle\psi| = \frac{1}{2}\big(I_2 + r_1\sigma_1 + r_2\sigma_2 + r_3\sigma_3\big), \tag{23.3}$$

where $\sigma_1, \sigma_2, \sigma_3$ are the Pauli matrices.

Proof. The global phase leaves the three coordinates invariant and

$$r_1^2 + r_2^2 + r_3^2 = 4|z_0|^2|z_1|^2 + (|z_0|^2 - |z_1|^2)^2 = 1.$$

Conversely, each $\boldsymbol{r} \in S^2$ determines the positive projector $\rho = (I + \boldsymbol{r} \cdot \boldsymbol{\sigma})/2$, which satisfies $\rho^2 = \rho$ and $\operatorname{tr} \rho = 1$; its image is a single ray. The affine charts of $\mathbb{CP}^1$ make the map and its inverse [12] smooth. □

The Hopf sequence

$$S^1 \longrightarrow S^3 \longrightarrow S^2 \tag{23.4}$$

exhibits the loss published by projectivization: the normalized vector belongs to S^3, the projective observable publishes a point of S^2, and the fiber S^1 preserves the common phase that the chart identifies [63]. This is a precise realization of the HMT distinction between enriched state and visible coordinate.

23.1.2 Rotations, Möbius transformations and spinor lifting

Every element of $\mathrm{SU}(2)$ can be written

$$U = \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Theorem 23.2 (Spinorial action on the Bloch sphere)**.** *For each $U \in \mathrm{SU}(2)$ there exists a unique rotation $R_U \in \mathrm{SO}(3)$ such that*

$$U(\boldsymbol{r} \cdot \boldsymbol{\sigma})U^\dagger = (R_U \boldsymbol{r}) \cdot \boldsymbol{\sigma}.$$

The map $U \mapsto R_U$ is a surjective homomorphism with kernel $\{\pm I_2\}$.

Proof. Conjugation preserves the three-dimensional real space of traceless Hermitian matrices and the inner product $\langle A, B\rangle = \frac{1}{2}\operatorname{tr}(AB)$, and therefore induces a rotation. If it acts trivially, U commutes with the three Pauli matrices and is scalar; the unit-determinant condition leaves $U = \pm I_2$. The axis–angle representation provides the two preimages of each rotation. □

In the affine chart $z = z_0/z_1$ of $\mathbb{CP}^1$, the same action is the Möbius transformation

$$z \longmapsto \frac{\alpha z + \beta}{-\overline{\beta} z + \overline{\alpha}}.$$

The Bloch rotation and the fractional transformation are two coordinates of a single projective action: the first acts on Hermitian projectors and the second on the Riemann sphere.

The rotational group of the icosahedron is $A_5 \subset \mathrm{SO}(3)$. Its preimage defines the binary icosahedral group

$$2A_5 := \pi^{-1}(A_5) \subset \mathrm{SU}(2),$$

and produces the exact sequence

$$1 \longrightarrow \{\pm I_2\} \longrightarrow 2A_5 \longrightarrow A_5 \longrightarrow 1. \tag{23.5}$$

In particular, $|2A_5| = 120$. For a rotation of odd order n, the half-angle lift has order $2n$, whereas its opposite has order n. Icosahedral rotations of orders three and five therefore admit pairs of lifts of orders $(6, 3)$ and $(10, 5)$, respectively. This specialization links the projective geometry of two-level states with the McKay branch and the exceptional symmetries, whose modules and lattices remain in their reference constructions in Part I.

Provenance and scope of the projective chain

The Paley–Witt identity, structure $\mathbb{F}_3^6 \simeq \mathbb{F}_9^3$, action $\mathrm{SU}(2) \to \mathrm{SO}(3)$, the Mobius realization, and extension $2A_5$ are results demonstrated in their domains. Chain (23.1) is a unified functorial realization: it explicitly declares the realization that changes scalars and does not convert the finite root J_W into an analytic operator without that additional datum.

23.2 Projective realization of Bloch from the internal complex structure

23.2.1 Matrix realization of each depth

For $d \geq 1$, define

$$C_9^{(d)} = (\mathbb{Z}/9\mathbb{Z})^d, \qquad H_d = \ell^2(C_9^{(d)}), \qquad A_d = B(H_d) \cong M_{9^d}(\mathbb{C}).$$

The canonical basis $\{\delta_x : x \in C_9^{(d)}\}$ preserves the discrete label of each state. The uniform measure induces the normalized trace

$$\tau_d(a) = \frac{1}{9^d}\,\mathrm{Tr}(a).$$

The factorization

$$C_9^{(d+1)} = C_9^{(d)} \times \mathbb{Z}/9\mathbb{Z}$$

produce a unitary identification

$$H_{d+1} \cong H_d \otimes \mathbb{C}^9.$$

This is the exact input to the operator tower. This does not identify $(\mathbb{Z}/9)^3$ with $\mathbb{F}_3^6$ as groups: they share 729 elements, but their operations and carry cocycles are distinct.

23.2.2 Weyl–Heisenberg Representation of Phase and Displacement on the Triadic Fiber

Let $\omega = e^{2\pi i/9}$. On $H_1 = \ell^2(\mathbb{Z}/9\mathbb{Z})$, we define

$$(Xf)(r) = f(r-1), \qquad (Zf)(r) = \omega^r f(r).$$

Then

$$X^9 = Z^9 = I, \qquad ZX = \omega XZ.$$

The group generated by $X, Z, \omega I$ is the finite Heisenberg group of order $9^3 = 729$.

Theorem 23.3 (Stone–von Neumann finite)**.** *Every irreducible unitary representation of the finite Heisenberg group in which the center acts by means of the primitive character ωI is unitarily equivalent to the preceding representation.*

Proof. Let v be an eigenvector of Z. The vectors

$$v, Xv, \ldots, X^8 v$$

have eigenvalues distinct from Z, since $ZX^k v = \omega^k X^k Z v$. They are orthogonal and span an invariant subspace. By irreducibility, they form a basis of the entire representation. In that basis, X is the cyclic shift and Z the diagonal phase. □

Structural obstruction

Uniqueness fails if the primitive central character is not fixed or if reducible representations are allowed. This finite theorem must not be confused with the continuous version for the canonical commutation relations.

23.2.3 Aritmetic Projectors and Incompatibility

The additive and multiplicative sheets of APP define information subspaces and their orthogonal projectors $P_{\Sigma,d}, P_{\Pi,d}$. The constructed identities satisfy

$$\|[P_{\Sigma,2}, P_{\Pi,2}]\| = \frac{\sqrt{2}}{3}, \qquad \|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}.$$

Noncommutativity is not deduced from a quantum analogy: it is a finite matrix theorem. Its "pre-Heisenberg" reading is subsequent to that calculation.

Proposition 23.4. *The preceding projectors do not jointly belong to the same Boolean subalgebra of projections.*

Proof. In an abelian subalgebra all projectors commute. The displayed commutators are nonzero. □

23.2.4 Hilbertian structure of the matrix publication and preserved genealogical memory

Hilbert space provides linear superposition, orthogonal projection, and spectral calculus. HMT supplies the genealogical index of the basis and distinguishes which operator represents a transition, a measurement, or a shadow. The realization does not replace TPK: it represents it.

Premises and derivation

The APP projectors and the Weyl–Heisenberg pair are part of the preceding construction. The organization H_d, A_d and the uniqueness proof are established here by applying the corresponding classical theorem to the HMT objects.

23.3 The 9 phases of the TPK and continuous geometry

23.3.1 Typed separation between the UHF lift of nonadic ranks and the geometric prolongation of the marked branch

Let $p_j = |e_j\rangle\langle e_j|$ be the projector onto a marked gate. The factorization $H_{d+1} \cong H_d \otimes \mathbb{C}^9$ admits two lifts:

$$\iota_d(a) = a \otimes I_9, \qquad j_{d,j}(a) = a \otimes p_j.$$

Proposition 23.5 (Unit and trace)**.** *For all $a \in A_d$,*

$$\tau_{d+1}(\iota_d(a)) = \tau_d(a), \qquad \iota_d(I) = I,$$

while

$$\tau_{d+1}(j_{d,j}(a)) = \frac{1}{9}\tau_d(a), \qquad j_{d,j}(I) \neq I.$$

Proof. Since $\mathrm{Tr}(I_9) = 9$ and $\mathrm{Tr}(p_j) = 1$,

$$\tau_{d+1}(a \otimes I_9) = \frac{\mathrm{Tr}(a)\,9}{9^{d+1}} = \tau_d(a),$$

and

$$\tau_{d+1}(a \otimes p_j) = \frac{\mathrm{Tr}(a)}{9^{d+1}} = \frac{\tau_d(a)}{9}.$$

The identity identities are immediate. □

Scientific reading

"Looking at the nine phases of the TPK" means preserving the identity operator on the complete fiber. An isolated gate preserves a marked history; the nine together also preserve the unit and the global normalization.

23.3.2 The UHF limit

Theorem 23.6 (UHF nonadic closure)**.** *The unital inductive limit*

$$A_{9^\infty} := \varinjlim (A_d, \iota_d) \cong \bigotimes_{n=1}^{\infty} M_9(\mathbb{C})$$

is the UHF algebra of supernatural type $9^\infty = 3^\infty$*. It has a product trace* τ *compatible with all the* τ_d*.*

Proof. The inclusions are injective, unital, and trace-preserving. The algebraic union consists of nested finite matrices; its norm closure is, by definition, a UHF algebra. The product trace is the only trace compatible with the matrix traces of all the factors [54]. □

Theorem 23.7 (Hyperfinite factor)**.** *Let* $(\pi_\tau, \mathcal{H}_\tau, \Omega_\tau)$ *be the GNS representation of the trace. Then*

$$\pi_\tau(A_{9^\infty})'' \cong \mathcal{R}_{\mathrm{hyp}},$$

the hyperfinite factor of type II_1*.*

Proof. The closure is finite because $\tau(I) = 1$. Extremality of the trace makes it a factor. It is hyperfinite because the increasing union of the matrices $\pi_\tau(A_d)$ is weakly dense. Uniqueness of the separable hyperfinite factor II_1 completes the identification [93, 33]. □

23.3.3 Corner immersion of a marked phase: compact ideal $\mathcal{K}(\mathcal{H}_\infty)$ and factor $\mathcal{B}(\mathcal{H}_\infty)$ of type I_∞

Let $V_{d,j} : H_d \to H_{d+1}$, $V_{d,j}\xi = \xi \otimes e_j$. Then

$$j_{d,j}(a) = V_{d,j} a V_{d,j}^*.$$

In the limit of the spaces, the union of the corners contains the finite-rank operators. Its norm closure is $\mathcal{K}(H_\infty)$, and its natural bicommutant is $B(H_\infty)$, of type I_∞.

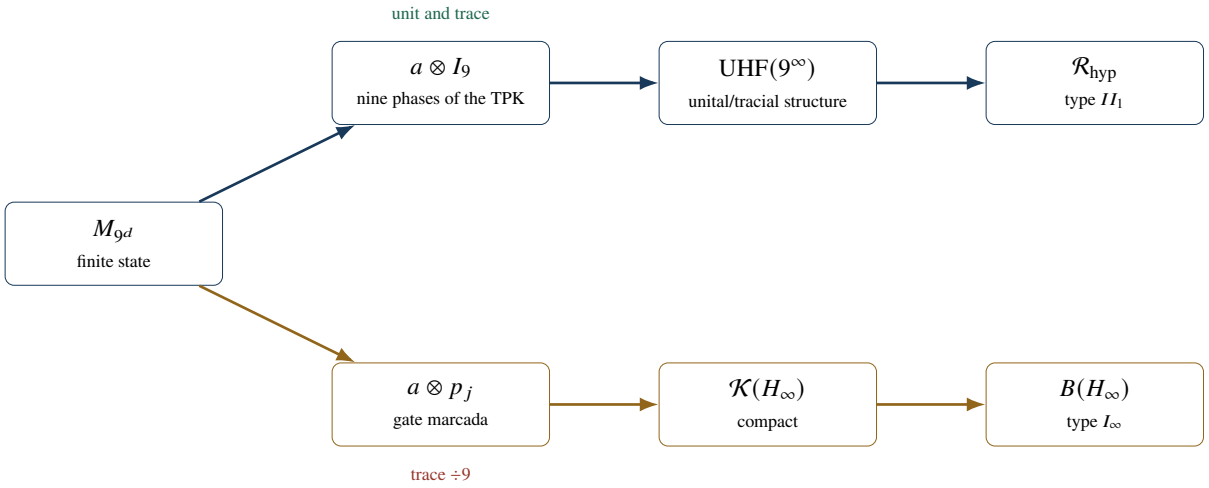

Figure 23.1: A marked door and the nine phases of the TPK produce closures of distinct types.

23.3.4 Continuous dimension from nonadic ranges

For a projection $P \in M_{9^d}$,

$$d_d(P) = \tau_d(P) = \frac{\operatorname{rank}(P)}{9^d}.$$

The values belong to $\mathbb{Z}[1/9] \cap [0,1]$, a dense set in $[0,1]$.

Theorem 23.8 (Discrete Continuous Geometry)**.** *For each $t \in [0,1]$ there exists to projection $P_t \in \mathcal{R}_{\text{hyp}}$ with $\tau(P_t) = t$. Moreover,*

$$P \sim Q \quad \Longleftrightarrow \quad \tau(P) = \tau(Q),$$

where $\sim$ denotes Murray–von Neumann equivalence.

Proof. Let $m_d = \lfloor 9^d t \rfloor$. Since $0 \le m_{d+1} - 9m_d \le 8$, nested projections of rank m_d can be chosen. Their strong limit P_t satisfies, by normality of the trace,

$$\tau(P_t) = \lim_d \frac{m_d}{9^d} = t.$$

The classification of projections in a finite factor gives the final equivalence. □

Continuous dimension is not fractional cardinality. It measures equivalence of subspaces in the closure. Each approximant preserves integer rank; the continuity appears when the entire unital genealogy is closed.

Result

Holographic conservation of the unit becomes a falsifiable operator-theoretic criterion: replacing I_9 by p_j divides the trace and changes the type of the limit. The set of gates is not a census; it is the condition for unital, trace-preserving inclusion.

23.4 Profinite dynamical system: translation, logical algebra, and state space

23.4.1 The 2nd route to the hyperfinite factor

The HMT construction produces the compact group

$$K_\infty \cong C_4 \times \mathbb{Z}_3$$

as a limit of Pell norm kernels, together with a minimal and uniquely ergodic translation $T_u(x) = x + u$ for the Haar measure μ_H. We consider

$$M_{\mathrm{clk}} = L^\infty(K_\infty, \mu_H) \rtimes_{T_u} \mathbb{Z}.$$

Theorem 23.9 (Clock factor)**.** *If T_u is free, ergodic, and Haar-preserving, then*

$$M_{\mathrm{clk}} \cong \mathcal{R}_{\mathrm{hyp}}.$$

Proof. The free and ergodic p.m.p. action produces a finite trace factor.

$$\tau_{\mathrm{clk}}\left(\sum_n f_n V^n\right) = \int f_0 \, d\mu_H.$$

Amenability of $\mathbb{Z}$ makes the orbit relation hyperfinite; therefore, the factor is the hyperfinite II_1. □

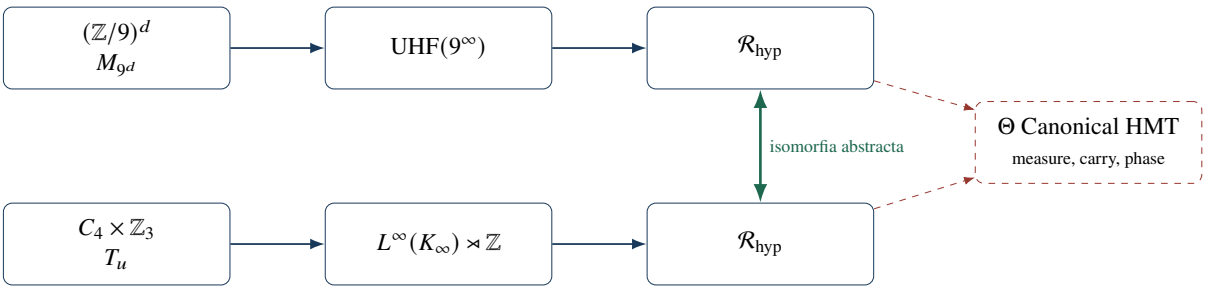

Figure 23.2: Two abstract routes toward $\mathcal{R}_{\mathrm{hyp}}$. HMT causal conjugation is an additional problem.

The coincidence of weak closures does not identify the antecedents C^*. The UHF branch and the odometer crossed product may have distinct K-theoretic invariants. A canonical HMT identification requires a map Θ with

$$\Theta_* \mu_{\mathrm{APP}} = \mu_H, \qquad \Theta S_{\mathrm{carry}} = T_u \Theta,$$

and preservation of diagonals, phase, orientation, and projectors. That intertwiner remains as boundary.

23.4.2 theorem ergodico medio of von Neumann for the odometro of Haar in K_∞

Let $Uf = f \circ T_u^{-1}$ in $L^2(K_\infty, \mu_H)$. The theorem ergodico medio of von Neumann [130] gives

$$\frac{1}{N} \sum_{j=0}^{N-1} U^j f \longrightarrow P_{\mathrm{Fix}(U)} f.$$

The ergodicity identifies the limit with

$$\left(\int f \, d\mu_H\right) \mathbf{1}.$$

The clock has pure point spectrum and zero entropy, but its temporal averages converge to the Haar average. Mixing is not required.

23.4.3 Orthomodular Lattice of Projections and the Logic of Closed Subspaces

The lattice $\mathrm{Proj}(\mathcal{R}_{\mathrm{hyp}})$, with

$$P^{\perp} = I - P, \qquad P \wedge Q, \qquad P \vee Q = (P^{\perp} \wedge Q^{\perp})^{\perp},$$

is complete and orthomodular. It is not distributive. The noncommuting APP projectors embed into $\mathcal{R}_{\mathrm{hyp}}$ and provide a finite witness of that nondistributivity [21].

Each Abelian subalgebra selects a Boolean context. Classical logic does not disappear: it occupies the commutative contexts. What fails globally is the possibility of gathering all the incompatible observables into a single Boolean algebra.

23.4.4 Density matrices and entropy

In M_{9^d}, to normal state is written

$$\varphi_\rho(a) = \mathrm{Tr}(\rho a), \qquad \rho \geq 0, \quad \mathrm{Tr}\,\rho = 1.$$

In $\mathcal{R}_{\mathrm{hyp}}$, a normal density is

$$h \in L^1(\mathcal{R}_{\mathrm{hyp}}, \tau)_+, \qquad \tau(h) = 1,$$

and $\varphi_h(a) = \tau(ha)$. Its entropy relative to the trace is

$$S_\tau(h) = -\tau(h \log h).$$

If $h = 9^d \rho$, then

$$S_\tau(h) = S_{\mathrm{vN}}(\rho) - d \log 9.$$

The canonical extension

$$\rho_{d+1} = \rho_d \otimes \frac{I_9}{9}$$

adds $\log 9$ to the matrix entropy, but leaves $S_\tau(h)$ invariant. The first quantity records an additional uniform fiber; the second uses the holographic normalization of the tower.

Structural obstruction

Tr and τ will not be used interchangeably. Nor will $\tau(P)$ be interpreted as a cardinality of points, or every vector in a representation be called pure. A diffuse factor II_1 has no minimal projections.

23.4.5 Operator types and modular boundary

The classification constructed so far is:

object	type
M_{9^d}	I_{9^d}
marked branch	I_∞
tracial UHF branch	II_1
p.m.p. clock	II_1.

A type III would require a quasi-invariant measure with nontrivial Radon–Nikodym cocycle or a KMS state whose GNS representation were not semifinite. The mere presence of the word “KMS” does not select the type.

Mathematical extensions

The completion of the total memory groupoid of TPK requires a topology or Borel structure, measure, closure under inversion, amenability, ergodicity, and control of isotropy. The clock subgroupoid satisfies those hypotheses; the total groupoid is not promoted by analogy.

23.5 Hilbert-space realization of the nonadic phase and preservation of the unit

The construction of Triadic Modular Holography first provides the states, their transitions, and the restriction maps. The Hilbert-space realization adds a linear structure and an inner product to their discrete coordinates. This order determines what each operator represents: the basis labels arise from the nonadic construction; linear operations subsequently act on those labels. Genealogical memory remains in the TPK state and accompanies the projections used to represent it.

23.5.1 Phase fiber, labeled basis, and inner product

For an integer depth $d \geq 1$, let

$$C_{9,d} = (\mathbb{Z}/9\mathbb{Z})^d, \qquad \mathcal{H}_d = \ell^2(C_{9,d}), \qquad \mathcal{A}_d = B(\mathcal{H}_d). \tag{23.6}$$

The inner product is $\langle f, g\rangle = \sum_x \overline{f(x)}g(x)$. The functions δ_x form an orthonormal basis labeled by the phase coordinates. Thus $\dim \mathcal{H}_d = 9^d$ and $\mathcal{A}_d \cong M_{9^d}(\mathbb{C})$. The matrix algebra represents this finite fiber; routes, quotients, and memory omitted by a phase projection still belong to the starting state.

Coordinate factorization induces the unitary map

$$\mathcal{H}_{d+1} \longrightarrow \mathcal{H}_d \otimes \mathbb{C}^9, \qquad \delta_{(x,r)} \longmapsto \delta_x \otimes \delta_r. \tag{23.7}$$

Define the normalized trace $\tau_d(a) = 9^{-d}\,\mathrm{Tr}(a)$. Finite counting and normalization are distinguished: $\mathrm{Tr}(I) = 9^d$, whereas $\tau_d(I) = 1$. Likewise, the group $(\mathbb{Z}/9\mathbb{Z})^3$ and the vector space $\mathbb{F}_3^6$ both have 729 elements, but retain different algebraic laws. Realization of a phase fiber does not replace the ternary encoding of APP emissions.

23.5.2 Translation and phase: finite Weyl representation

In the complex realization subsequent to the constructions of π and the structure $\mathrm{i}^2 = -1$, fix $\omega = \exp(2\pi\mathrm{i}/9)$. On $\mathcal{H}_1$, let

$$(Xf)(r) = f(r-1), \qquad (Zf)(r) = \omega^r f(r). \tag{23.8}$$

The first operation shifts the phase coordinate; the second evaluates it through a character. Both are unitary and satisfy

$$X^9 = Z^9 = I, \qquad ZX = \omega XZ. \tag{23.9}$$

Indeed, $(ZXf)(r) = \omega^r f(r-1)$, whereas $(XZf)(r) = \omega^{r-1} f(r-1)$. The character appears here as a realization of the phase already constructed and does not select the TPK numerical registers.

23.5.2.1 Oriented area and central extension of translations

The translation support is $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. The mixed APP curvature constructed in Section 4.3.4 assigns $+1$ to the oriented plaquette of the sum–product pair and -1 to the pair in the opposite order. Its bilinear extension to two displacements $u = (a, b)$ and $v = (c, d)$ is the alternating form

$$\sigma_{\mathrm{ar}}(u, v) = ad - bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \tag{23.10}$$

Indeed, alternation makes the evaluation on two displacements along the same axis vanish. The values on the oriented basis are $\sigma_{\mathrm{ar}}((1,0),(0,1)) = 1$ and $\sigma_{\mathrm{ar}}((0,1),(1,0)) = -1$; distributing both arguments gives (23.10). The same formula is the sum of plaquettes of any oriented cellular chain realizing that

parallelogram: the interior edges cancel. Changing an integer representative by a multiple of nine preserves its publication in $\mathbb{Z}/9\mathbb{Z}$.

The order $X^a Z^b$ of the already defined operators fixes a section of the central extension. Denote its composition factor by

$$c_{\mathrm{W}}(u, v) = bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \tag{23.11}$$

This cocycle preserves the order of the two operations, while σ_{ar} preserves the orientation of the parallelogram.

Proposition 23.10 (Central extension and commutator sign)**.** *The set $\mathfrak{H}_9 = (\mathbb{Z}/9\mathbb{Z})^3$, with product*

$$(a, b, t)(c, d, s) = (a + c, b + d, t + s + bc), \tag{23.12}$$

is a group of order 729*. The map*

$$\varrho(a, b, t) = \omega^t X^a Z^b$$

is a faithful unitary representation on $\mathcal{H}_1$. With the convention $[g, h] = ghg^{-1}h^{-1}$, one has

$$[\varrho(a, b, 0), \varrho(c, d, 0)] = \omega^{bc-ad} I = \omega^{-\sigma_{\mathrm{ar}}((a,b),(c,d))} I. \tag{23.13}$$

The center and the translation group form the exact sequence

$$0 \longrightarrow \mathbb{Z}/9\mathbb{Z} \xrightarrow{t \mapsto (0,0,t)} \mathfrak{H}_9 \xrightarrow{(a,b,t) \mapsto (a,b)} X_9 \longrightarrow 0.$$

Proof. For $w = (e, f)$, the cocycle identity reads

$$c_{\mathrm{W}}(u, v) + c_{\mathrm{W}}(u + v, w) = bc + (b + d)e = de + b(c + e) = c_{\mathrm{W}}(v, w) + c_{\mathrm{W}}(u, v + w).$$

It proves associativity of (23.12); the identity is $(0, 0, 0)$, and the inverse of (a, b, t) is $(-a, -b, -t + ab)$. The three residues are free, giving order $9^3 = 729$. From $ZX = \omega XZ$ it follows that

$$(X^a Z^b)(X^c Z^d) = \omega^{bc} X^{a+c} Z^{b+d},$$

which proves the representation property. If $\omega^t X^a Z^b = I$, its action on δ_r first forces $a = 0$. Evaluation at $r = 0$ gives $t = 0$, and evaluation at $r = 1$ gives $b = 0$, since ω has order nine. The representation is therefore faithful. Comparing the products in both orders gives the factor $bc - ad$. A central element must satisfy $bc - ad = 0$ for every (c, d). Taking successively $(c, d) = (1, 0), (0, 1)$ yields $b = a = 0$. This identifies the center and proves exactness. □

The Wilson phase for the orientation (23.10) is $\omega^{\sigma_{\mathrm{ar}}(u,v)}$; the commutator of the ordered section $X^a Z^b$ publishes its inverse. In particular, $c_{\mathrm{W}}(u, v) - c_{\mathrm{W}}(v, u) = -\sigma_{\mathrm{ar}}(u, v)$. This identity relates two specified conventions, without identifying the polarized cocycle with the alternating form. The extension belongs to the translation group of the APP support. Its representation acts on the phase fibre; the path register and carries retain their residence in the TPK state that produced that fibre. The order 729 of the group and the cardinality 729 of the ternary ambient space retain the distinct types established in (23.6). The centre $\mathbb{Z}/9\mathbb{Z}$ of this extension records the phase of the commutator. The memory C_{12} of the sequence $0 \to C_{12} \to C_{108} \to C_9 \to 0$, constructed in proposition 11.31, belongs to the dodecaphase calendar. Each extension retains its own composition law and projection.

Proposition 23.11 (Algebra generated by phase and translation)**.** *The* 81 *operators $X^a Z^b$, with $0 \le a, b <$ 9, form a basis of $\mathcal{A}_1 = M_9(\mathbb{C})$. In particular, the representation* (23.8) *is irreducible.*

Proof. If $a \neq 0 \pmod 9$, the matrix $X^a Z^b$ has no diagonal entries. If $a = 0$, its trace is $\sum_{r=0}^{8} \omega^{br}$, equal to 9 for $b = 0$ and 0 otherwise. By (23.9), the product $(X^a Z^b)^* X^{a'} Z^{b'}$ is a factor of modulus one times $X^{a'-a} Z^{b'-b}$. Therefore,

$$\tau_1\big((X^a Z^b)^* X^{a'} Z^{b'}\big) = \delta_{a,a'}\delta_{b,b'}.$$

These are 81 linearly independent matrices in a space of dimension 81. They generate the whole matrix algebra, whose commutant consists of scalars; irreducibility follows. □

Theorem 23.12 (Uniqueness with fixed primitive central character)**.** *Every irreducible unitary representation of the group presented by $X^9 = Z^9 = c^9 = I$, c central, and $ZX = cXZ$, in which c acts as ωI, is unitarily equivalent to* (23.8)*.*

Proof. Let $v \neq 0$ be an eigenvector of Z, with eigenvalue λ. Since $Z^9 = I$, λ is a power of ω. The vectors $X^k v$, $0 \leq k < 9$, have eigenvalues $\omega^k \lambda$, distinct because the character is primitive. They are orthogonal, and their span is invariant under X, Z, and the center. Irreducibility forces it to be the entire space. Reordering the basis to begin with eigenvalue 1, Z is diagonal with entries $1, \omega, \ldots, \omega^8$, and X is cyclic translation. Normalizing the basis produces the asserted unitary equivalence. □

Primitivity of the character and irreducibility are part of the statement. At depth d, operators acting on different factors commute, and tensor products of the preceding bases generate $\mathcal{A}_d$. The phase structure is thus represented at each level without identifying phase return with return of the complete genealogical state.

23.5.3 Two lifts and two different normalizations

Factorization (23.7) permits preserving the entire new fiber or fixing one of its labels. For $p_j = |\delta_j\rangle\langle\delta_j|$, define

$$\iota_d(a) = a \otimes I_9, \qquad \jmath_{d,j}(a) = a \otimes p_j. \tag{23.14}$$

Proposition 23.13 (Preservation of unit and trace)**.** *The inclusion ι_d is unital and preserves normalized trace. The inclusion $\jmath_{d,j}$ is a corner embedding and satisfies*

$$\tau_{d+1}(\iota_d(a)) = \tau_d(a), \quad \iota_d(I) = I, \qquad \tau_{d+1}(\jmath_{d,j}(a)) = \frac{\tau_d(a)}{9}, \quad \jmath_{d,j}(I) = I \otimes p_j.$$

Proof. The trace of a tensor product is the product of the traces. Since $\mathrm{Tr}(I_9) = 9$ and $\mathrm{Tr}(p_j) = 1$, dividing by 9^{d+1} produces both formulas. The images of the identity are read directly from (23.14). □

The unital inclusion preserves the set of 9 phases and its global normalization. The corner preserves a marked continuation and its support projector. Both operations are legitimate, but transmit different data to the next depth.

23.5.4 Inductive closure and tracial realization

Theorem 23.14 (Nonadic uniformly hyperfinite algebra)**.** *The norm closure of the union through the inclusions ι_d is*

$$\mathcal{A}_{9^\infty} = \varinjlim(\mathcal{A}_d, \iota_d) \cong \bigotimes_{n=1}^{\infty} M_9(\mathbb{C}).$$

It is a UHF algebra of type $9^\infty = 3^\infty$, with a unique normalized trace τ, whose restriction to $\mathcal{A}_d$ is τ_d.

Proof. The inclusions are injective and unital. The closure of the increasing union of matrix algebras is precisely the UHF construction. The compatibility in Proposition 23.13 defines τ on the union; positivity and norm one allow its extension to the closure. Every normalized trace on the limit restricts to the unique normalized trace on each matrix algebra. Since the union is dense, those restrictions determine the trace uniquely. □

Let $\mathcal{H}_\tau$ be the completion of $\mathcal{A}_{9^\infty}$ for $\langle a, b\rangle_\tau = \tau(a^*b)$, and let π_τ be its representation by left multiplication. The vector defined by the identity is denoted Ω_τ.

Theorem 23.15 (Tracial factor of complete refinement)**.** *The weak closure $\mathcal{R}_9 = \pi_\tau(\mathcal{A}_{9^\infty})''$ is a hyperfinite factor of type II_1. The nonadic inclusions preserve its unit and normalized trace.*

Proof. The product trace extends to a faithful finite normal trace in the tracial representation. To see that the center is trivial, let E_d be the tracial expectation onto $\pi_\tau(\mathcal{A}_d)$. On a tensor of length $n > d$, it takes the trace of the last $n - d$ factors. These maps extend continuously to the corresponding orthogonal projections in L^2. If z is central, $E_d(z)$ commutes with $\pi_\tau(\mathcal{A}_d)$, by bimodularity of E_d. It is therefore scalar and equal to $\tau(z)I$. Density of the matrix union implies $E_d(z) \to z$ in L^2, hence $z = \tau(z)I$. The algebra is thus a finite factor. It contains unital matrix algebras of unbounded sizes 9^d, so it is not a finite matrix factor. It is of type II_1. It is hyperfinite because the same union of matrices is weakly dense. Preservation of unit and trace follows from each finite inclusion and persists upon taking closures. □

23.5.5 Integer ranks and continuous tracial dimension

In $\mathcal{A}_d$, a projection P has normalized dimension $\tau_d(P) = \mathrm{rank}(P)/9^d$. Ranks remain integers at every finite depth. Their compatible normalizations allow the following passage to the limit.

Proposition 23.16 (Realization of each tracial dimension)**.** *For each $t \in [0, 1]$, there exists a projection $P_t \in \mathcal{R}_9$ with $\tau(P_t) = t$.*

Proof. Let $m_d = \lfloor 9^d t \rfloor$. Then $0 \le m_{d+1} - 9m_d \le 8$. After choosing a diagonal projection of rank m_d, its inclusion has rank $9m_d$; add $m_{d+1} - 9m_d$ diagonal positions in its complement. This yields an increasing sequence of projections within $\mathcal{R}_9$. Its strong limit is a projection P_t, and normality of the trace gives

$$\tau(P_t) = \lim_{d\to\infty} \frac{\lfloor 9^d t \rfloor}{9^d} = t.$$

□

Here t denotes the dimension coordinate represented in the closure, not an input to the APP generator. The assertion describes the scope of the resulting tracial realization. To compare it with a particular observable, it remains necessary to identify the projection and the genealogical information it preserves.

By contrast, the isometries $V_{d,j}\xi = \xi \otimes \delta_j$ realize $\jmath_{d,j}(a) = V_{d,j} a V_{d,j}^*$. In the limit of a marked route, the finite spaces are dense. The corners approximate all finite-rank operators; their norm closure is $\mathcal{K}(\mathcal{H}_\infty)$, and their bicommutant is $B(\mathcal{H}_\infty)$, of type I_∞. This difference preserves the operational content of the two choices in (23.14): selection of one continuation and unital prolongation of the entire fiber have different limits.

The Hilbert-space realization, its phase operators, and the tracial closure are thus constructed in this section, with their inclusion choices explicit. They constitute the operator-theoretic support to be used subsequently for coordinate interchange and the compact Hamiltonian.

23.6 Two-dimensional refinement of APP and carry cocycle

The realization of APP on a torus of nine positions per coordinate and its expansion to twenty-seven positions are related through Euclidean division. The expansion preserves two independent directions and assigns an additional ternary digit to each coordinate. Thus each position of the initial APP has nine lifts in the expanded support. The two additional digits preserve the position within the fiber, while the direction of transport and the regime of the TRIT reader remain as distinct coordinates of the enriched state.

For $m \geq 1$, we write

$$T_m = (\mathbb{Z}/m\mathbb{Z})^2.$$

In the calculations of this section the representatives $0, \ldots, m-1$ are chosen. The nonadic presentation $1, \ldots, 9$ prints the zero class using the sign 9. This convention allows the APP table to be read and integer divisions to be performed with a single residue rule.

23.6.1 Projection, fibers and ternary refinement coordinates

The projection and the section of representatives are

$$p : T_{27} \longrightarrow T_9, \qquad p(x, y) = (x \bmod 9, y \bmod 9),$$

$$s : T_9 \longrightarrow T_{27}, \qquad s(r, t) = (r, t), \quad 0 \leq r, t < 9.$$

The projection is a surjective homomorphism of additive groups. Its kernel is

$$\ker p = \{(9a, 9b) : a, b \in \mathbb{F}_3\} \simeq \mathbb{F}_3^2.$$

The coordinate map

$$\Psi : T_9 \times \mathbb{F}_3^2 \longrightarrow T_{27}, \qquad \Psi((r, t), (a, b)) = (r + 9a, t + 9b) \tag{23.15}$$

uses representatives $0 \leq a, b < 3$.

Proposition 23.17 (Unique decomposition of the expanded positions)**.** *The map Ψ is bijective. Each fiber of p contains nine positions and the entire support satisfies*

$$|T_{27}| = 27^2 = 9^2 \cdot 3^2 = 729.$$

The two coordinates of $\mathbb{F}_3^2$ are the two positional trits that describe the refinement of the two APP coordinates.

Proof. For $0 \leq x, y < 27$, the Euclidean divisions

$$x = r + 9a, \qquad y = t + 9b, \quad 0 \leq r, t < 9, \quad 0 \leq a, b < 3$$

exist and are unique. These formulas construct the inverse of Ψ. With r, t fixed, the nine pairs (a, b) are distinct and exhaust the fiber. The census results from multiplying the 81 residual positions by their nine lifts. □

For example, the position $(5, 5)$ has the fiber

$$\{(5 + 9a, 5 + 9b) : 0 \leq a, b < 3\}.$$

The row of this fiber with $b = 1$ is $(5, 14), (14, 14), (23, 14)$. Each position preserves the same residual pair $(5, 5)$ and a different pair of quotients. Multiplicity nine expresses two independent ternary choices. The topological and temporal reading of TRIT additionally preserves its own operator domain.

23.6.2 Additive composition and memory of modular crossing

For $r, t \in \{0, \ldots, 8\}$, define

$$r \oplus t = (r + t) \bmod 9, \qquad c_9(r, t) = \left\lfloor \frac{r + t}{9} \right\rfloor.$$

Transported by Ψ, addition in one coordinate of $\mathbb{Z}/27\mathbb{Z}$ is

$$(r,a) \boxplus (t,b) = \big(r \oplus t,\ a+b+c_9(r,t) \pmod 3\big). \tag{23.16}$$

In two coordinates, the same formula is applied separately to each axis.

Proposition 23.18 (Cocycle of the positional extension)**.** *The carry satisfies*

$$c_9(r,t) + c_9(r \oplus t, u) = c_9(t,u) + c_9(r, t \oplus u) = \left\lfloor \frac{r+t+u}{9} \right\rfloor. \tag{23.17}$$

Therefore, (23.16) *is associative and exactly reproduces the additive group of the expanded support.*

Proof. The equality $r + t = (r \oplus t) + 9c_9(r,t)$ gives

$$\left\lfloor \frac{r+t+u}{9} \right\rfloor = c_9(r,t) + \left\lfloor \frac{(r \oplus t)+u}{9} \right\rfloor.$$

The grouping $r + (t+u)$ provides the other side. Substituting $x = r + 9a$ and $y = t + 9b$ into $x + y$ proves (23.16); reducing its second component modulo three corresponds to reducing $x + y$ modulo twenty-seven. The cocycle identity directly proves the associativity of this notation by residues and quotients. □

In particular, $8 + 1 = 0 + 9 \cdot 1$ shows the elementary crossing: the residue returns to the zero class, printed as 9 in the nonadic table, and the first quotient increases by one. If only the residue is preserved, the datum distinguishing 0, 9 and 18 in the expanded support is lost; the lifted state distinguishes these three positions.

The extension

$$0 \longrightarrow \mathbb{Z}/3\mathbb{Z} \xrightarrow{a \mapsto 9a} \mathbb{Z}/27\mathbb{Z} \longrightarrow \mathbb{Z}/9\mathbb{Z} \longrightarrow 0$$

is nonsplit as an extension of groups. Indeed, any lift of the class 1 is $1 + 9a$, of order twenty-seven. A homomorphic section would have to send an element of order nine to one whose order divides nine. The section of representatives instead has the exact defect $9c_9$, which the preceding formula preserves.

23.6.3 Compatibility of the multiplicative sheet

The same decomposition preserves the second APP operation. For one coordinate, we define

$$r \odot t = rt \bmod 9, \qquad m_9(r,t) = \left\lfloor \frac{rt}{9} \right\rfloor.$$

The lifted product is

$$(r,a) \boxtimes (t,b) = \big(r \odot t,\ m_9(r,t) + rb + ta \pmod 3\big). \tag{23.18}$$

Proof. The integer expansion provides

$$(r + 9a)(t + 9b) = rt + 9(rb + ta) + 81ab.$$

The last term vanishes modulo twenty-seven. Substituting $rt = (r \odot t) + 9m_9(r,t)$ gives the formula. Uniqueness of Euclidean division proves that its two components recover the residual product and its refinement quotient. □

For example, $8 \cdot 8 = 64 \equiv 10 \pmod{27}$ is represented by $(1,1)$: the residue is 1 and $m_9(8,8) = 7 \equiv 1 \pmod 3$. Thus sum and product preserve their respective carries within the same expanded support.

23.6.4 Directions and realization of transport

Let $\mathcal{D} = \{N, S, E, O\}$ be the direction register and $\delta_d \in \mathbb{Z}^2$ the unit displacement associated with d. On the support with direction we define

$$S_m(z, d) = (z + \delta_d \pmod m, d).$$

The projection explicitly preserves the direction, $p_{\mathcal{D}}(z, d) = (p(z), d)$, and satisfies

$$p_{\mathcal{D}} S_{27} = S_9 p_{\mathcal{D}}. \tag{23.19}$$

The quotient calculation uses, for an integer displacement v,

$$c_9(r; v) = \left\lfloor \frac{r + v}{9} \right\rfloor, \qquad r' = r + v - 9c_9(r; v), \qquad a' = a + c_9(r; v) \pmod 3.$$

The formula includes negative displacements. For example, $r = 0, v = -1$ gives $r' = 8, c_9 = -1$; the reverse return restores the quotient.

Proposition 23.19 (Conjugation of displacement in fiber coordinates)**.** *The coordinate change Ψ transforms S_{27} into the residual displacement S_9 accompanied by the update of the two quotients just defined. This transformation is bijective and preserves direction. Its powers preserve the same identity for any finite length of transport.*

Proof. On each axis, $x = r + 9a$ is substituted into $x + v$ and Euclidean division is performed. Direction remains fixed in the definition of S_m. The inverse uses $-v$ and the cocycle, thereby recovering the initial state. Composing the identity once per transition proves the statement for all powers. □

The Hilbert realization preserves this decomposition. On the position bases, the operator

$$\mathcal{U}\delta_{\Psi(z,a),d} = \delta_{z,d} \otimes \delta_a$$

defines a surjective isometry

$$\ell^2(T_{27} \times \mathcal{D}) \simeq \ell^2(T_9 \times \mathcal{D}) \otimes \ell^2(\mathbb{F}_3^2).$$

The conjugated shift is the permutation operator with carry described in the proposition. An additional internal factor E remains intact when tensoring with I_E. If a phase reader depends only on the projection and a directional operator acts with the same matrix at all positions, their compatibility squares also commute. A reader using the quotient is realized precisely in the preserved fiber factor. These conditions make explicit the relation between the expanded support and the factors of the operator on the torus, with the internal actions declared in its construction.

23.6.5 Six ternary digits and distinct algebraic structures

Each coordinate of T_{27} has three base-three digits. Their joint notation defines a bijection of sets

$$T_{27} \longleftrightarrow \mathbb{F}_3^6.$$

This bijection publishes six positional digits. Addition on the torus contains carries between digits and vector addition in $\mathbb{F}_3^6$ is performed componentwise. For example, the positional sum $2 + 1 = 3$ transforms $(2, 0, 0)$ into $(0, 1, 0)$, whereas ternary vector addition would give $(0, 0, 0)$. The torus contains elements of order twenty-seven; the ternary vector space has exponent three. The two objects share a finite encoding, with different typed operations. The fiber of ternary words, the position support and the lines of a cubic surface thus preserve their specific constructions and maps.

23.7 Genealogical path kernels: finite composition, unitary realization and cylindrical limit

The matrix composition proved in this section establishes exactly that each entry of U^R is the sum of the ordered weights of routes of length R. The next step preserves the information that a visible route does not publish and organizes finite refinements into a compact space of genealogies. This distinguishes three results: the finite combinatorial identity, the cylindrical integral over the path limit and the realization of that integral as the propagator of a specific physical system.

23.7.1 Route category and functorial weight

Let $\mathsf{P}_{\mathrm{TPK}}$ be the free category generated by the admissible TPK transitions. Its objects are enriched boundary states and its morphisms are composable sequences of transitions. Composition is concatenation, and the empty path is the identity. Let P_X be the category of visible routes obtained by applying the reader to the objects and transitions; its compositions are visible concatenations. The map

$$q_{\mathrm{vis}} : \mathsf{P}_{\mathrm{TPK}} \longrightarrow \mathsf{P}_X$$

publishes the observable route and preserves in its fiber the differences of sheet, orientation, carry, block, boundary, and memory.

Definition 23.20 (Genealogical weight)**.** Let $\mathcal{B}$ be a normed real algebra, possibly noncommutative, and let $\mathbf{B}_{\mathcal{B}}$ be the category with a single object $*$ whose endomorphism monoid is $\mathrm{End}(*) = (\mathcal{B}, \cdot, 1)$. A genealogical weight is a functor

$$W : \mathsf{P}_{\mathrm{TPK}} \longrightarrow \mathbf{B}_{\mathcal{B}}$$

such that

$$W(\mathrm{id}_x) = 1, \qquad W(\gamma_2 \circ \gamma_1) = W(\gamma_2)W(\gamma_1).$$

The product order agrees with the causal order of the route.

For enriched states $\widetilde{i}, \widetilde{f}$, the kernel at depth R is

$$\mathcal{K}_R(\widetilde{f}, \widetilde{i}) = \sum_{\substack{\gamma \in \mathsf{P}_{\mathrm{TPK}}(\widetilde{i}, \widetilde{f}) \\ |\gamma| = R}} W(\gamma). \tag{23.20}$$

Two routes with the same visible word remain distinct summands when they belong to different genealogical fibers.

Proposition 23.21 (Composition through an enriched cut)**.** *If every route of length $R + S$ has a unique cut after R transitions, then*

$$\mathcal{K}_{R+S}(\widetilde{f}, \widetilde{i}) = \sum_{\widetilde{x}} \mathcal{K}_S(\widetilde{f}, \widetilde{x})\mathcal{K}_R(\widetilde{x}, \widetilde{i}).$$

The formula preserves the causal order of the factors when $\mathcal{B}$ is noncommutative.

Proof. The set of complete routes is partitioned by the enriched state $\widetilde{x}$ reached at the cut. The functorial factorization of the weight turns the sum over each class into the product of the partial kernels. Uniqueness of the cut ensures that each route appears exactly once. □

23.7.2 Construction of the internal phase through the real action character on genealogical routes

Let (E, J_E) be a real Euclidean space equipped with an orthogonal endomorphism J_E such that $J_E^2 = -I$. If $\mathcal{A}_* : \mathsf{P}_{\mathrm{TPK}} \to \mathbb{R}$ is additive under concatenation, the phase

$$\Phi_J(\gamma) := \exp[-J_E \mathcal{A}_*(\gamma)] = \cos \mathcal{A}_*(\gamma)\, I - \sin \mathcal{A}_*(\gamma)\, J_E \tag{23.21}$$

is multiplicative. Fix a nonzero positive section $\hbar_{\mathrm{int}}$ of the action line $\mathcal{L}_S$ constructed earlier; the section $\hbar_{\mathrm{ret}}$ of the declared profile may be used. For a dimensional action $S_{\mathrm{int}}(\gamma) = \hbar_{\mathrm{int}} \mathcal{A}_*(\gamma)$, this expression internally realizes the phase $\exp[-J_E\, S_{\mathrm{int}}(\gamma)/\hbar_{\mathrm{int}}]$. The scalar symbol i appears only when choosing a complex chart for the pair (E, J_E).

Proposition 23.22 (Phase functor)**.** *The map Φ_J satisfies*

$$\Phi_J(\mathrm{id}_x) = I, \qquad \Phi_J(\gamma_2 \circ \gamma_1) = \Phi_J(\gamma_2)\Phi_J(\gamma_1).$$

Proof. The action of the empty path is zero. For composable routes, additivity of $\mathcal{A}_*$ and commutativity of multiples of the same endomorphism J_E allow the exponential of the summed value to be separated. □

23.7.3 Unitary realization on the chronological register 27×27

Before passing to the cylindrical limit, a finite kernel admits an explicit unitary realization. Let

$$\mathbb{T}^2_{27} := (\mathbb{Z}/27\mathbb{Z})^2, \qquad \mathcal{D} := \{\mathrm{N}, \mathrm{S}, \mathrm{E}, \mathrm{W}\},$$

and consider the real space

$$\mathcal{H}^{\mathbb{R},\mathrm{reg}}_{27} := \ell^2(\mathbb{T}^2_{27}; \mathbb{R}) \otimes \mathbb{R}^{\mathcal{D}} \otimes E.$$

The endomorphism J_E is extended as the identity on the position and direction factors, and the same notation is retained for that extension.

In the ordered basis $(\mathrm{N}, \mathrm{S}, \mathrm{E}, \mathrm{W})$, the directional mixer is

$$C_4 := \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad C_4^{\mathsf{T}} C_4 = I_4. \tag{23.22}$$

Here C_4 denotes exclusively the Walsh–Hadamard mixer on the directional fiber; when it acts on $\mathcal{H}^{\mathbb{R},\mathrm{reg}}_{27}$ it is understood as $C_4 \otimes I_E$.

Fix a translational chart in which the action of an elementary edge γ_d depends only on its direction:

$$s_d := S_{\mathrm{int}}(\gamma_d) \in \mathcal{L}_S, \qquad a_d := \frac{s_d}{\hbar_{\mathrm{int}}} \in \mathbb{R}.$$

The real diagonal phase operator is

$$D_J := \mathrm{diag}\left(e^{-J_E a_{\mathrm{N}}}, e^{-J_E a_{\mathrm{S}}}, e^{-J_E a_{\mathrm{E}}}, e^{-J_E a_{\mathrm{W}}}\right).$$

The conditional shift transports each component in its direction:

$$\begin{aligned} (S\psi)(x, y, \mathrm{N}) &= \psi(x, y-1, \mathrm{N}), & (S\psi)(x, y, \mathrm{S}) &= \psi(x, y+1, \mathrm{S}), \\ (S\psi)(x, y, \mathrm{E}) &= \psi(x-1, y, \mathrm{E}), & (S\psi)(x, y, \mathrm{W}) &= \psi(x+1, y, \mathrm{W}), \end{aligned}$$

with all coordinates taken modulo 27. The elementary step and its composition of twelve steps are

$$U_{\mathrm{micro}} := SD_J C_4, \qquad U_{12}^{\mathrm{reg}} := U_{\mathrm{micro}}^{12}. \tag{23.23}$$

The superscript reg preserves the type: this operator acts on the directional torus 27×27 and is not identified with the other dodecaphasic objects of the construction or with the temporal endomorphism of order 108.

Proposition 23.23 (Unitarity and exact path sum in the register)**.** *The operator U_{micro} is orthogonal on the realification and unitary in the complex chart determined by J_E. In particular, U_{12}^{reg} is unitary. If $\Xi := \mathbb{T}_{27}^2 \times \mathcal{D}$, then, for every $n \geq 0$ and $\xi_i, \xi_f \in \Xi$,*

$$\left(U_{\mathrm{micro}}^n\right)_{\xi_f, \xi_i} = \sum_{\substack{\gamma : \xi_i \to \xi_f \\ |\gamma| = n}} W_{U_{\mathrm{micro}}}(\gamma),$$

where, for $\gamma = (\xi_0, \ldots, \xi_n)$,

$$W_{U_{\mathrm{micro}}}(\gamma) := (U_{\mathrm{micro}})_{\xi_n, \xi_{n-1}} \cdots (U_{\mathrm{micro}})_{\xi_1, \xi_0}.$$

The sum ranges over exactly the paths permitted by the nonzero blocks of the operator.

Proof. Since J_E is orthogonal and $J_E^2 = -I$, we have $J_E^{\mathsf{T}} = -J_E$. Thus, each block $e^{-J_E a_d}$ is an orthogonal rotation and D_J is orthogonal. The operator S permutes an orthonormal basis, and (23.22) proves that C_4 is orthogonal. Consequently,

$$U_{\mathrm{micro}}^{\mathsf{T}} U_{\mathrm{micro}} = C_4^{\mathsf{T}} D_J^{\mathsf{T}} S^{\mathsf{T}} S D_J C_4 = I.$$

All three factors commute with the extension of J_E; hence U_{micro} is complex-linear in the chart $J_E \leftrightarrow \mathrm{i}$, and its real orthogonality is equivalent to unitarity. Every power, in particular U_{12}^{reg}, preserves this property.

For $n \geq 1$, block multiplication gives

$$\left(U_{\mathrm{micro}}^n\right)_{\xi_f, \xi_i} = \sum_{\xi_1, \ldots, \xi_{n-1}} (U_{\mathrm{micro}})_{\xi_f, \xi_{n-1}} \cdots (U_{\mathrm{micro}})_{\xi_1, \xi_i}.$$

Each choice of intermediate states determines a path of length n, and each path determines a unique ordered product. For $n = 0$, the sole term is the empty path and its weight is the identity. □

Translation invariance decomposes the problem into blocks of four directional components, tensored with E. For

$$k_x = \frac{2\pi a}{27}, \qquad k_y = \frac{2\pi b}{27}, \qquad 0 \leq a, b < 27,$$

we adopt the realified Fourier convention

$$\chi_{a,b}(x, y) := \exp[-J_E(k_x x + k_y y)].$$

Direct substitution into the conditional shift produces

$$S(k_x, k_y) = \mathrm{diag}\left(e^{J_E k_y}, e^{-J_E k_y}, e^{J_E k_x}, e^{-J_E k_x}\right),$$

and, since D_J and C_4 do not depend on position,

$$\begin{aligned} U_{\mathrm{micro}}(k_x, k_y) &= S(k_x, k_y) D_J C_4, \\ U_{12}^{\mathrm{reg}}(k_x, k_y) &= U_{\mathrm{micro}}(k_x, k_y)^{12}. \end{aligned} \tag{23.24}$$

Consequently,

$$\operatorname{Spec} U_{12}^{\mathrm{reg}} = \bigcup_{a,b=0}^{26} \operatorname{Spec} U_{12}^{\mathrm{reg}} \left(\frac{2\pi a}{27}, \frac{2\pi b}{27} \right).$$

The matrix power enumerates paths, and the Fourier decomposition publishes the quasienergies of the same finite dynamics.

Fix an elementary duration $\delta t > 0$, and let $\Delta t_{12} := 12\,\delta t$. In each unitary block, choose a spectral branch $\mathfrak{b}$ of the logarithm. The operator $\operatorname{Log}_{\mathfrak{b}} U_{12}^{\mathrm{reg}}(k_x, k_y)$ is skew-adjoint and commutes with J_E. Define

$$\begin{aligned} H_{\mathrm{eff}}(k_x, k_y) &:= \frac{\hbar_{\mathrm{int}}}{\Delta t_{12}} J_E \operatorname{Log}_{\mathfrak{b}} U_{12}^{\mathrm{reg}}(k_x, k_y), \\ U_{12}^{\mathrm{reg}}(k_x, k_y) &= \exp\left[-\frac{J_E \Delta t_{12}}{\hbar_{\mathrm{int}}} H_{\mathrm{eff}}(k_x, k_y) \right]. \end{aligned} \tag{23.25}$$

The first line is self-adjoint in the complex chart. The second follows from $-J_E^2 = I$ and proves that the exponential recovers the propagator exactly. The branch $\mathfrak{b}$ fixes the quasienergy zone.

This realization constructs a finite propagator, its exact path sum and its effective generator. Identifying it with a continuous physical Hamiltonian still requires verification of the representation, action and limit hypotheses stated below.

23.7.4 Cylindrical limit on the space of genealogies

With endpoints fixed, let $\mathcal{P}_n$ be finite sets of paths at resolution n, with surjective restriction maps. The space

$$\Omega_{fi}^{\mathrm{path}} := \varprojlim_n \mathcal{P}_n$$

is compact and totally disconnected. A compatible family of probability measures μ_n on the cylinders determines a Borel measure μ on $\Omega_{fi}^{\mathrm{path}}$.

Let Π_n be the partition into level-n cylinders. For $C \in \Pi_n$, choose a representative γ_C, an approximate action $S_n(C)$ and the weight $\mu_n(C)$. For $F : \Omega_{fi}^{\mathrm{path}} \to \operatorname{End}(E)$, define

$$\mathcal{I}_n(F) := \sum_{C \in \Pi_n} \mu_n(C) F(\gamma_C) \exp\left[-J_E \frac{S_n(C)}{\hbar_{\mathrm{int}}} \right]. \tag{23.26}$$

Theorem 23.24 (Genealogical path integral)**.** *Suppose that:*

1. *the cylinder measures μ_n are projectively compatible;*
2. *F is continuous and bounded;*
3. *there exists a continuous action $S : \Omega_{fi}^{\mathrm{path}} \to \mathcal{L}_S$ for which*

$$\max_{C \in \Pi_n} \sup_{\gamma \in C} \left\| \frac{S_n(C) - S(\gamma)}{\hbar_{\mathrm{int}}} \right\| \longrightarrow 0;$$

4. *the mesh of Π_n tends to zero.*

Then $\mathcal{I}_n(F)$ converges in operator norm, the limit is independent of the representatives, and it defines the Bochner integral

$$\int_{\Omega_{fi}^{\mathrm{path}}} F(\gamma) \exp\left[-J_E \frac{S(\gamma)}{\hbar_{\mathrm{int}}} \right] \mathrm{d}\mu(\gamma). \tag{23.27}$$

Proof. Compatibility of the finite measures constructs μ on the cylinder algebra and, by extension, on the Borel sigma-algebra. The function

$$G(\gamma) = F(\gamma)\exp[-J_E\, S(\gamma)/\hbar_{\mathrm{int}}]$$

is continuous on a compact space and is therefore uniformly continuous and bounded. Orthogonality of J_E gives the estimate

$$\|e^{-J_E a} - e^{-J_E b}\| \le |a-b|.$$

Uniform approximation of the action and contraction of the mesh make the cylindrical sums converge to the Bochner integral of G [174]. □

The theorem constructs a rigorous integral on the compact space of genealogies. Its realization as a Feynman oscillatory integral for a specific Hamiltonian is obtained when the finite kernels, action and cylinder measure satisfy these convergence hypotheses. The finite matrix expansion, genealogical integral and physical propagator are thus related by explicit morphisms, without being identified by abbreviation.

23.7.5 Reading of the value and memory of the fiber

Let $\mathrm{Val}_{\mathrm{path}} : \Omega_{fi}^{\mathrm{path}} \to \mathcal{Y}_{\mathrm{val}} \subset \mathbb{R}$ be a Borel-measurable path reader, where $\mathcal{Y}_{\mathrm{val}}$ carries its standard Borel structure, and let $\nu = (\mathrm{Val}_{\mathrm{path}})_*\mu$. For every function factoring through the value, $F = \widehat{F} \circ \mathrm{Val}_{\mathrm{path}}$, we have

$$\int_{\Omega_{fi}^{\mathrm{path}}} F\,\mathrm{d}\mu = \int_{\mathcal{Y}_{\mathrm{val}}} \widehat{F}(x)\,\mathrm{d}\nu(x).$$

Since $\Omega_{fi}^{\mathrm{path}}$ is compact metrizable and μ is a Borel probability measure, there exists a family of regular conditional probabilities $(\mu_x)_{x\in\mathcal{Y}_{\mathrm{val}}}$, defined for ν-almost every x. For a general genealogical integrand, the disintegration of μ takes the form

$$\int_{\Omega_{fi}^{\mathrm{path}}} G(\gamma)\,\mathrm{d}\mu(\gamma) = \int_{\mathcal{Y}_{\mathrm{val}}} \left[\int_{\mathrm{Val}_{\mathrm{path}}^{-1}(x)} G(\gamma)\,\mathrm{d}\mu_x(\gamma)\right] \mathrm{d}\nu(x). \tag{23.28}$$

The inner integral preserves the distinct histories that publish the same coordinate. This is the precise link between the sum over all paths, real evaluation, and the thesis that the continuum is a projection of a richer genealogical domain.

Domain of the genealogical path integral: cylindrical compactness, projective consistency, kernel convergence and disintegration over fibers The finite expansion, unitary realization on $\mathbb{T}_{27}^2$ and its Fourier reduction are exact on their finite domains. The construction of H_{eff} is exact once the complex structure, elementary translational action, duration and spectral branch have been declared; it does not by itself select a continuous physical Hamiltonian. The enriched route category and internal phase precede the cylindrical assembly. theorem 23.24 is exact under the four declared premises. Realizing a specific continuous quantum model requires verifying those premises for its action and Hamiltonian.

23.8 Hilbert-space realization of the nonadic phase and preservation of the unit

The construction of Triadic Modular Holography first provides the states, their transitions, and the restriction maps. The Hilbert-space realization adds a linear structure and an inner product to their discrete coordinates. This order determines what each operator represents: the basis labels arise from the nonadic construction; linear operations subsequently act on those labels. Genealogical memory remains in the TPK state and accompanies the projections used to represent it.

23.8.1 Phase fiber, labeled basis, and inner product

For an integer depth $d \geq 1$, let

$$C_{9,d} = (\mathbb{Z}/9\mathbb{Z})^d, \qquad \mathcal{H}_d = \ell^2(C_{9,d}), \qquad \mathcal{A}_d = B(\mathcal{H}_d). \tag{23.29}$$

The inner product is $\langle f, g\rangle = \sum_x \overline{f(x)}g(x)$. The functions δ_x form an orthonormal basis labeled by the phase coordinates. Thus $\dim \mathcal{H}_d = 9^d$ and $\mathcal{A}_d \cong M_{9^d}(\mathbb{C})$. The matrix algebra represents this finite fiber; routes, quotients, and memory omitted by a phase projection still belong to the starting state.

Coordinate factorization induces the unitary map

$$\mathcal{H}_{d+1} \longrightarrow \mathcal{H}_d \otimes \mathbb{C}^9, \qquad \delta_{(x,r)} \longmapsto \delta_x \otimes \delta_r. \tag{23.30}$$

Define the normalized trace $\tau_d(a) = 9^{-d}\,\mathrm{Tr}(a)$. Finite counting and normalization are distinguished: $\mathrm{Tr}(I) = 9^d$, whereas $\tau_d(I) = 1$. Likewise, the group $(\mathbb{Z}/9\mathbb{Z})^3$ and the vector space $\mathbb{F}_3^6$ both have 729 elements, but retain different algebraic laws. Realization of a phase fiber does not replace the ternary encoding of APP emissions.

23.8.2 Translation and phase: finite Weyl representation

In the complex realization subsequent to the constructions of π and the structure $\mathrm{i}^2 = -1$, fix $\omega = \exp(2\pi\mathrm{i}/9)$. On $\mathcal{H}_1$, let

$$(Xf)(r) = f(r-1), \qquad (Zf)(r) = \omega^r f(r). \tag{23.31}$$

The first operation shifts the phase coordinate; the second evaluates it through a character. Both are unitary and satisfy

$$X^9 = Z^9 = I, \qquad ZX = \omega XZ. \tag{23.32}$$

Indeed, $(ZXf)(r) = \omega^r f(r-1)$, whereas $(XZf)(r) = \omega^{r-1} f(r-1)$. The character appears here as a realization of the phase already constructed and does not select the TPK numerical registers.

23.8.2.1 Oriented area and central extension of translations

The translation support is $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. The mixed APP curvature constructed in Section 4.3.4 assigns $+1$ to the oriented plaquette of the sum–product pair and -1 to the pair in the opposite order. Its bilinear extension to two displacements $u = (a, b)$ and $v = (c, d)$ is the alternating form

$$\sigma_{\mathrm{ar}}(u, v) = ad - bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \tag{23.33}$$

Indeed, alternation makes the evaluation on two displacements along the same axis vanish. The values on the oriented basis are $\sigma_{\mathrm{ar}}((1,0),(0,1)) = 1$ and $\sigma_{\mathrm{ar}}((0,1),(1,0)) = -1$; distributing both arguments gives (23.33). The same formula is the sum of plaquettes of any oriented cellular chain realizing that parallelogram: the interior edges cancel. Changing an integer representative by a multiple of nine preserves its publication in $\mathbb{Z}/9\mathbb{Z}$.

The order $X^a Z^b$ of the already defined operators fixes a section of the central extension. Denote its composition factor by

$$c_{\mathrm{W}}(u, v) = bc \quad \text{in } \mathbb{Z}/9\mathbb{Z}. \tag{23.34}$$

This cocycle preserves the order of the two operations, while σ_{ar} preserves the orientation of the parallelogram.

Proposition 23.25 (Central extension and commutator sign)**.** *The set $\mathfrak{H}_9 = (\mathbb{Z}/9\mathbb{Z})^3$, with product*

$$(a, b, t)(c, d, s) = (a + c, b + d, t + s + bc), \tag{23.35}$$

is a group of order 729*. The map*

$$\varrho(a,b,t) = \omega^t X^a Z^b$$

is a faithful unitary representation on $\mathcal{H}_1$. With the convention $[g,h] = ghg^{-1}h^{-1}$, one has

$$[\varrho(a,b,0), \varrho(c,d,0)] = \omega^{bc-ad} I = \omega^{-\sigma_{\mathrm{ar}}((a,b),(c,d))} I. \tag{23.36}$$

The center and the translation group form the exact sequence

$$0 \longrightarrow \mathbb{Z}/9\mathbb{Z} \xrightarrow{t\mapsto(0,0,t)} \mathfrak{H}_9 \xrightarrow{(a,b,t)\mapsto(a,b)} X_9 \longrightarrow 0.$$

Proof. For $w = (e,f)$, the cocycle identity reads

$$c_{\mathrm{W}}(u,v) + c_{\mathrm{W}}(u+v,w) = bc + (b+d)e = de + b(c+e) = c_{\mathrm{W}}(v,w) + c_{\mathrm{W}}(u,v+w).$$

It proves associativity of (23.35); the identity is $(0,0,0)$, and the inverse of (a,b,t) is $(-a,-b,-t+ab)$. The three residues are free, giving order $9^3 = 729$. From $ZX = \omega XZ$ it follows that

$$(X^a Z^b)(X^c Z^d) = \omega^{bc} X^{a+c} Z^{b+d},$$

which proves the representation property. If $\omega^t X^a Z^b = I$, its action on δ_r first forces $a = 0$. Evaluation at $r = 0$ gives $t = 0$, and evaluation at $r = 1$ gives $b = 0$, since ω has order nine. The representation is therefore faithful. Comparing the products in both orders gives the factor $bc - ad$. A central element must satisfy $bc - ad = 0$ for every (c,d). Taking successively $(c,d) = (1,0), (0,1)$ yields $b = a = 0$. This identifies the center and proves exactness. □

The Wilson phase for the orientation (23.33) is $\omega^{\sigma_{\mathrm{ar}}(u,v)}$; the commutator of the ordered section $X^a Z^b$ publishes its inverse. In particular, $c_{\mathrm{W}}(u,v) - c_{\mathrm{W}}(v,u) = -\sigma_{\mathrm{ar}}(u,v)$. This identity relates two specified conventions, without identifying the polarized cocycle with the alternating form. The extension belongs to the translation group of the APP support. Its representation acts on the phase fibre; the path register and carries retain their residence in the TPK state that produced that fibre. The order 729 of the group and the cardinality 729 of the ternary ambient space retain the distinct types established in (23.29). The centre $\mathbb{Z}/9\mathbb{Z}$ of this extension records the phase of the commutator. The memory C_{12} of the sequence $0 \to C_{12} \to C_{108} \to C_9 \to 0$, constructed in proposition 11.31, belongs to the dodecaphase calendar. Each extension retains its own composition law and projection.

Proposition 23.26 (Algebra generated by phase and translation)**.** *The* 81 *operators $X^a Z^b$, with $0 \le a, b < 9$, form a basis of $\mathcal{A}_1 = M_9(\mathbb{C})$. In particular, the representation* (23.31) *is irreducible.*

Proof. If $a \neq 0 \pmod 9$, the matrix $X^a Z^b$ has no diagonal entries. If $a = 0$, its trace is $\sum_{r=0}^{8} \omega^{br}$, equal to 9 for $b = 0$ and 0 otherwise. By (23.32), the product $(X^a Z^b)^* X^{a'} Z^{b'}$ is a factor of modulus one times $X^{a'-a} Z^{b'-b}$. Therefore,

$$\tau_1\big((X^a Z^b)^* X^{a'} Z^{b'}\big) = \delta_{a,a'}\delta_{b,b'}.$$

These are 81 linearly independent matrices in a space of dimension 81. They generate the whole matrix algebra, whose commutant consists of scalars; irreducibility follows. □

Theorem 23.27 (Uniqueness with fixed primitive central character)**.** *Every irreducible unitary representation of the group presented by $X^9 = Z^9 = c^9 = I$, c central, and $ZX = cXZ$, in which c acts as ωI, is unitarily equivalent to* (23.31).

Proof. Let $v \neq 0$ be an eigenvector of Z, with eigenvalue λ. Since $Z^9 = I$, λ is a power of ω. The vectors $X^k v$, $0 \le k < 9$, have eigenvalues $\omega^k \lambda$, distinct because the character is primitive. They are orthogonal, and their span is invariant under X, Z, and the center. Irreducibility forces it to be the entire space.

Reordering the basis to begin with eigenvalue 1, Z is diagonal with entries $1, \omega, \dots, \omega^8$, and X is cyclic translation. Normalizing the basis produces the asserted unitary equivalence. □

Primitivity of the character and irreducibility are part of the statement. At depth d, operators acting on different factors commute, and tensor products of the preceding bases generate $\mathcal{A}_d$. The phase structure is thus represented at each level without identifying phase return with return of the complete genealogical state.

23.8.3 Two lifts and two different normalizations

Factorization (23.30) permits preserving the entire new fiber or fixing one of its labels. For $p_j = |\delta_j\rangle\langle\delta_j|$, define

$$\iota_d(a) = a \otimes I_9, \qquad \jmath_{d,j}(a) = a \otimes p_j. \tag{23.37}$$

Proposition 23.28 (Preservation of unit and trace)**.** *The inclusion ι_d is unital and preserves normalized trace. The inclusion $\jmath_{d,j}$ is a corner embedding and satisfies*

$$\tau_{d+1}(\iota_d(a)) = \tau_d(a), \quad \iota_d(I) = I, \qquad \tau_{d+1}(\jmath_{d,j}(a)) = \frac{\tau_d(a)}{9}, \quad \jmath_{d,j}(I) = I \otimes p_j.$$

Proof. The trace of a tensor product is the product of the traces. Since $\mathrm{Tr}(I_9) = 9$ and $\mathrm{Tr}(p_j) = 1$, dividing by 9^{d+1} produces both formulas. The images of the identity are read directly from (23.37). □

The unital inclusion preserves the set of 9 phases and its global normalization. The corner preserves a marked continuation and its support projector. Both operations are legitimate, but transmit different data to the next depth.

23.8.4 Inductive closure and tracial realization

Theorem 23.29 (Nonadic uniformly hyperfinite algebra)**.** *The norm closure of the union through the inclusions ι_d is*

$$\mathcal{A}_{9^\infty} = \varinjlim(\mathcal{A}_d, \iota_d) \cong \bigotimes_{n=1}^{\infty} M_9(\mathbb{C}).$$

It is a UHF algebra of type $9^\infty = 3^\infty$, with a unique normalized trace τ, whose restriction to $\mathcal{A}_d$ is τ_d.

Proof. The inclusions are injective and unital. The closure of the increasing union of matrix algebras is precisely the UHF construction. The compatibility in Proposition 23.28 defines τ on the union; positivity and norm one allow its extension to the closure. Every normalized trace on the limit restricts to the unique normalized trace on each matrix algebra. Since the union is dense, those restrictions determine the trace uniquely. □

Let $\mathcal{H}_\tau$ be the completion of $\mathcal{A}_{9^\infty}$ for $\langle a, b\rangle_\tau = \tau(a^*b)$, and let π_τ be its representation by left multiplication. The vector defined by the identity is denoted Ω_τ.

Theorem 23.30 (Tracial factor of complete refinement)**.** *The weak closure $\mathcal{R}_9 = \pi_\tau(\mathcal{A}_{9^\infty})''$ is a hyperfinite factor of type II_1. The nonadic inclusions preserve its unit and normalized trace.*

Proof. The product trace extends to a faithful finite normal trace in the tracial representation. To see that the center is trivial, let E_d be the tracial expectation onto $\pi_\tau(\mathcal{A}_d)$. On a tensor of length $n > d$, it takes the trace of the last $n - d$ factors. These maps extend continuously to the corresponding orthogonal projections in L^2. If z is central, $E_d(z)$ commutes with $\pi_\tau(\mathcal{A}_d)$, by bimodularity of E_d. It is therefore scalar and equal to $\tau(z)I$. Density of the matrix union implies $E_d(z) \to z$ in L^2, hence $z = \tau(z)I$. The algebra is thus a finite factor. It contains unital matrix algebras of unbounded sizes 9^d, so it is not a finite matrix factor. It is of type II_1. It is hyperfinite because the same union of matrices is weakly dense. Preservation of unit and trace follows from each finite inclusion and persists upon taking closures. □

23.8.5 Integer ranks and continuous tracial dimension

In $\mathcal{A}_d$, a projection P has normalized dimension $\tau_d(P) = \operatorname{rank}(P)/9^d$. Ranks remain integers at every finite depth. Their compatible normalizations allow the following passage to the limit.

Proposition 23.31 (Realization of each tracial dimension)**.** *For each $t \in [0,1]$, there exists a projection $P_t \in \mathcal{R}_9$ with $\tau(P_t) = t$.*

Proof. Let $m_d = \lfloor 9^d t \rfloor$. Then $0 \le m_{d+1} - 9m_d \le 8$. After choosing a diagonal projection of rank m_d, its inclusion has rank $9m_d$; add $m_{d+1} - 9m_d$ diagonal positions in its complement. This yields an increasing sequence of projections within $\mathcal{R}_9$. Its strong limit is a projection P_t, and normality of the trace gives

$$\tau(P_t) = \lim_{d\to\infty} \frac{\lfloor 9^d t \rfloor}{9^d} = t.$$

□

Here t denotes the dimension coordinate represented in the closure, not an input to the APP generator. The assertion describes the scope of the resulting tracial realization. To compare it with a particular observable, it remains necessary to identify the projection and the genealogical information it preserves.

By contrast, the isometries $V_{d,j}\xi = \xi \otimes \delta_j$ realize $\jmath_{d,j}(a) = V_{d,j} a V_{d,j}^*$. In the limit of a marked route, the finite spaces are dense. The corners approximate all finite-rank operators; their norm closure is $\mathcal{K}(\mathcal{H}_\infty)$, and their bicommutant is $B(\mathcal{H}_\infty)$, of type I_∞. This difference preserves the operational content of the two choices in (23.37): selection of one continuation and unital prolongation of the entire fiber have different limits.

Proposition 23.26 realizes the matrix algebra through the phase operators. Theorems 23.29 and 23.30 establish the unital inductive closure and its tracial realization, with their explicit choices of inclusion. This is the operator support that will subsequently be used for the exchange of coordinates and the compact Hamiltonian.

23.9 Arithmetic–geometric coupling and bilateral recovery

The dodecaphase register of section 11.4 admits arithmetic, incidence and operator readings. We call *holographic constant of arithmetic-geometric coupling* the generated register K, preserving the order of its twelve windows, the phase origin and orientation. Its rational reading $\kappa = K_{\mathrm{int}}/(1000^{12} - 1)$ is an encoding of that register; it does not replace its types or the enriched history that produces it. Hadamard inversion recovers its signed coordinate; the incidence reading determines the Witt tetrad. The following constructions make two further functions explicit: identifying operators on its cyclic carrier and preserving information between two isometric realizations. These are operations subsequent to the realization of the continuum, not rules introducing conventional constants into APP–TRIT–TPK.

23.9.1 Exact encoding and cyclic reference

Proposition 23.32 (Recovery of the record from its rational reading)**.** *With base $b = 1000$, length twelve, origin, and orientation fixed, the map*

$$(k_1, \dots, k_{12}) \longmapsto \kappa = \frac{\sum_{j=1}^{12} k_j b^{12-j}}{b^{12} - 1}, \qquad 0 \le k_j < b,$$

is injective. From κ, one recovers the integer $n = (b^{12} - 1)\kappa$ and, by successive Euclidean divisions, the twelve blocks, including leading zeros. An absolute error strictly smaller than $1/[2(b^{12} - 1)]$ permits the same recovery by rounding.

Proof. The fixed-length integer expansion in base b is unique. Two distinct integers are separated by at least one; their rational readings are separated by at least $1/(b^{12}-1)$. The strict bound places the approximate datum within the rounding cell of a unique integer. □

Now consider $K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$, the cyclic shift S of its positions, and $O_K = (K, SK, \dots, S^{11}K)$ on $\mathbb{R}^{12}$. The shift of positions is identified neither with the entire enriched evolution nor, solely by virtue of its periodicity, with an exceptional symmetry.

Theorem 23.33 (Invertibility of the orbital reference)**.** *The matrix O_K is invertible, and*

$$589609\|c\|^2 \le \|O_K c\|^2 \le 6263^2\|c\|^2.$$

Its centred orbit has rank eleven.

Proof. The Gram matrix $O_K^* O_K$ is circulant. Its first row, obtained from the twelve inner products $\langle K, S^j K\rangle$, is

$$(4251257, 2999323, 3163385, 3287920, 3216553, 3344743,$$
$$2950064, 3344743, 3216553, 3287920, 3163385, 2999323).$$

Evaluation at the twelve characters of the cycle yields the eigenvalues

$$6263^2, \quad 697225, \quad 589609\ (2), \quad 1407529\ (2), \quad 1053157\ (2),$$
$$(1248025 - 345420\sqrt{3})\ (2), \qquad (1248025 + 345420\sqrt{3})\ (2),$$

where parentheses indicate multiplicities. The smallest value is 589609: for the only comparison that is not immediate, $658416 > 345420\sqrt{3}$, as is seen by squaring both sides. The triangle inequality for the Fourier transform and $K_j \ge 0$ give the maximum $(\sum K_j)^2 = 6263^2$. Positivity of every mode proves invertibility and the bounds. Centring the record removes only the uniform mode. □

Corollary 23.34 (Identification of an operator from its responses)**.** *A linear operator $T : \mathbb{R}^{12} \to \mathbb{R}^{12}$ is determined by $Y = (TK, TSK, \dots, TS^{11}K)$, through $T = YO_K^{-1}$. If $TS = ST$, the complete vector response $y = TK$ suffices, since $T = O_y O_K^{-1}$. For a perturbation E of Y, $\|\widehat{T} - T\| \le \|E\|/\sqrt{589609}$.*

Proof. We have $Y = TO_K$. Under commutation, $TS^jK = S^j y$, hence $Y = O_y$. The bound follows from $\|O_K^{-1}\| = 1/\sqrt{589609}$. The norms are the Euclidean norm and its induced operator norm. □

The response y contains twelve components. The statement does not attribute the recovery of an arbitrary operator to a single scalar: it specifies the observation and commutation condition that make identification possible.

23.9.2 Conservation and reconstruction of two channels

Let H, H' be Hilbert spaces, and let $\mathcal{B}, C : H \to H'$ be bounded linear isometries. For $a > 0$, define

$$p = a^2 + a + 1, \qquad N = (2a+1)p = (a+1)^3 + a^3, \qquad Q_- = a\mathcal{B} - C, \quad Q_+ = (a+1)\mathcal{B} + C.$$

Theorem 23.35 (Bilateral balance and recovery)**.** *The following hold*

$$(a+1)Q_-^*Q_- + aQ_+^*Q_+ = NI_H, \qquad W_a v = \frac{1}{\sqrt{N}}\begin{pmatrix}\sqrt{a+1}\,Q_- v\\ \sqrt{a}\,Q_+ v\end{pmatrix}, \qquad W_a^* W_a = I_H.$$

The data $x = Q_- v$, $y = Q_+ v$ determine

$$\mathcal{B}v = \frac{x+y}{2a+1}, \qquad Cv = \frac{ay - (a+1)x}{2a+1}, \qquad v = \mathcal{B}^*\frac{x+y}{2a+1}.$$

Proof. Setting $X = \mathcal{B}^*C + C^*\mathcal{B}$, one obtains

$$Q_-^*Q_- = (a^2+1)I - aX, \qquad Q_+^*Q_+ = ((a+1)^2+1)I + (a+1)X.$$

The weights cancel X, and the remaining coefficient is $2a^3 + 3a^2 + 3a + 1 = N$. Normalization gives the isometry. Adding the two channels recovers $(2a+1)\mathcal{B}v$; the combination $ay - (a+1)x$ recovers $(2a+1)Cv$. Finally, $\mathcal{B}^*\mathcal{B} = I_H$. □

The balance also holds in infinite dimension: only bounded operators are composed. The image of W_a is closed, and its orthogonal projector is $W_aW_a^*$. Thus a normalized pair has an input precisely when it belongs to that image; its orthogonal component measures the compatibility defect.

The factorization

$$W_a = R_aE_a, \qquad E_av = \frac{1}{\sqrt{p}}\begin{pmatrix}\sqrt{a(a+1)}\,\mathcal{B}v\\ Cv\end{pmatrix}, \quad R_a = \frac{1}{\sqrt{2a+1}}\begin{pmatrix}\sqrt{a} & -\sqrt{a+1}\\ \sqrt{a+1} & \sqrt{a}\end{pmatrix}$$

is verified by block multiplication; R_a acts on $H' \oplus H'$, with each scalar entry multiplying $I_{H'}$. In the nonadic partition $a = 8$, the weights are $72/73$ and $1/73$; the factor $73 = 8 \cdot 9 + 1$ follows from the general law $a(a+1)+1$, rather than being an additional primitive datum. The corresponding balance is $9\|Q_-v\|^2 + 8\|Q_+v\|^2 = 17 \cdot 73\,\|v\|^2$.

23.9.3 Transport of observables and incoming memory

Theorem 23.36 (Observables common to both channels)**.** *For a bounded self-adjoint operator F on H',*

$$W_a^*\,\mathrm{diag}(F,F)W_a = \frac{a(a+1)\mathcal{B}^*F\mathcal{B} + C^*FC}{p}.$$

If $\mathcal{B}$ is unitary and $C = \mathcal{B}R$, with R unitary, and $G = \mathcal{B}^*F\mathcal{B}$, the observable is preserved if and only if $GR = RG$.

Proof. Expanding $(a+1)Q_-^*FQ_- + aQ_+^*FQ_+$ cancels the cross terms. The other two coefficients are $a(a+1)(2a+1)$ and $2a+1$. Dividing by N gives the formula. In the stated case, equality with G is equivalent to $R^*GR = G$; unitarity makes this equivalent to $GR = RG$. □

The norm is obtained by taking $F = I$. A general observable requires the commutation condition: preserving the norm does not mean preserving every observable without transformation.

Theorem 23.37 (Unitary extension with memory)**.** *If $H' = H$, $\mathcal{B} = I$, and $C = U$ is unitary, set*

$$A = \frac{\sqrt{a+1}}{\sqrt{N}}Q_-, \qquad D = \frac{\sqrt{a}}{\sqrt{N}}Q_+, \qquad \mathscr{R}_a = \begin{pmatrix}A & -D^*\\ D & A^*\end{pmatrix}.$$

Then $\mathscr{R}_a$ is unitary on $H \oplus H$, its first column is W_a, and it preserves $\|v\|^2 + \|m\|^2$ for every input (v,m). The inverse is

$$v = A^*z_1 + D^*z_2, \qquad m = -Dz_1 + Az_2.$$

Proof. The operators A, D are normal and commute with each other and with their adjoints, since they are polynomials in U, U^*. The balance gives $A^*A + D^*D = I$, and normality gives $AA^* + DD^* = I$. The cross terms in both products $\mathscr{R}_a^*\mathscr{R}_a$ and $\mathscr{R}_a\mathscr{R}_a^*$ vanish; the diagonal blocks are the identity. Applying the adjoint to (z_1, z_2) yields the inverse. □

Recovery of the register, orbital identification and unitary extension specify three distinct levels of information. The first preserves the blocks with their chart; the second identifies operators under the

stated hypotheses; the third transports state and memory jointly. This hierarchy allows the same generated structure to be used without confusing a reduced reading with the state in its entirety.

Chapter 24

T-duality of residual memory: unitary equivalence between momentum and winding for reciprocal radii

24.1 Involution between the residue and quotient coordinates

The Euclidean division used in $\mathcal{A}_9^{\#}$ writes, for an integer in the nonadic chart,

$$n = 9Q + r, \qquad r \in \{1, \dots, 9\}.$$

The coordinate r is the observable representative, and Q records the number of boundary crossings. To formulate an involution interchanging the two coordinates, one must pass to an extension stable under the interchange. Denote by $\bar{r} \in \{0, \dots, 8\}$ the residual class, with $\bar{r} = 0$ when the representative is $r = 9$, and consider

$$\mathscr{D}_{\mathrm{dual}} = \mathbb{Z}^2 \times \mathbb{R}_{>0}.$$

The two integers of $\mathscr{D}_{\mathrm{dual}}$ are not restricted to the canonical section of representatives; this enlargement is necessary because a quotient Q may fall outside the interval $0, \dots, 8$ when occupying the first coordinate.

Definition 24.1 (Radial energy form)**.** For $R_0 > 0$ and $(m, w) \in \mathbb{Z}^2$, define

$$E_{R_0}(m, w) = \frac{m^2}{R_0^2} + w^2 R_0^2.$$

The duality transformation is

$$\mathsf{T}(m, w, R_0) = (w, m, R_0^{-1}).$$

Theorem 24.2 (Residue–Quotient Duality)**.** *The map* T *is an involution of* $\mathscr{D}_{\mathrm{dual}}$ *and preserves the energy form in the sense*

$$E_{R_0}(m, w) = E_{R_0^{-1}}(w, m).$$

Proof. The second map of T exchanges of new m, w e invierte by second time R_0, by lo that

$$\mathsf{T}^2(m, w, R_0) = (m, w, R_0).$$

Moreover,

$$E_{R_0^{-1}}(w, m) = \frac{w^2}{R_0^{-2}} + m^2 R_0^{-2} = w^2 R_0^2 + \frac{m^2}{R_0^2} = E_{R_0}(m, w). \qquad \square$$

The insertion of a nonadic state into the extended domain is

$$(r, Q) \longmapsto (\bar{r}, Q, R_0).$$

The transform $(Q, \bar{r}, R_0^{-1})$ need not belong to the section $0 \le \bar{r} \le 8$. For this reason, T is an exact involution of the enlarged domain, not an endomorphism of the residual atlas of 81 positions. The fixed point of the parameter transformation is $R_0 = 1$; there the exchange of the two coordinates preserves the same quadratic form.

The identity admits an operatorial realization. Let $\mathcal{H}_{\mathrm{lat}} = \ell^2(\mathbb{Z}^2)$, with orthonormal basis $\{|m, w\rangle\}$. Define the diagonal operator

$$H(R_0)|m,w\rangle = E_{R_0}(m,w)|m,w\rangle$$

in its natural domain, and the unitary operator

$$U_T|m,w\rangle = |w,m\rangle.$$

Corollary 24.3 (Unitary Equivalence of Reciprocal Radii)**.** *For every $R_0 > 0$,*

$$U_T H(R_0) U_T^{-1} = H(R_0^{-1}).$$

Proof. On a basis vector,

$$U_T H(R_0) U_T^{-1}|m,w\rangle = E_{R_0}(w,m)|m,w\rangle.$$

The theorem 24.2 gives $E_{R_0}(w,m) = E_{R_0^{-1}}(m,w)$, which is the action of $H(R_0^{-1})$. □

This conjugation has the mathematical form of radius duality in a circular compactification, in units in which the string scale has been normalized to one [57]. The further physical identification assigns m to the momentum number, w to the winding number, and R_0 to the compactification radius. The theorem does not introduce oscillators, fields, supersymmetry, or a string tension; it proves the lattice involution and the unitary equivalence of the declared diagonal operators.

24.2 HMT reading duality: duality T, block 243, ternary Golay code, and mass routes

The T duality of strings is realized as the continuous projection of a prior HMT duality: the exchange between visible value and boundary memory. Every integer admits the reading

$$n = 9Q + r,$$

where r is the visible value of the nonadic wheel and Q preserves boundary-crossing memory. The same structure appears in lifted APP, in the decomposition $1215 = 54 + 9 \cdot 129$, in the stratified dimensional unit $1000 = 729 + 243 + 27 + 1$, and in the reading of mass as a route with visible advance and counterangular winding. This duality links the block 243, the perfect ternary ball $V_3(11,2) = 243$, the punctured ternary Golay code $[11,6,5]_3$, the channels of π, e, φ, α, the hinge 744, the dimension $11 = 6 + 5 = 3 + 8$, and the involution

$$(r,Q;R) \longmapsto (Q,r;R^{-1}).$$

In string-theoretic language, the visible value is published as a vibrational mode or momentum, whereas boundary memory is published as winding. In Dimensional Mechanics these two modes appear as $n_A A_{\mathrm{rad}}$ and $(s_C/6)C^*_{\mathrm{rad}}$: visible advance and counterangular memory.

HMT does not take the continuum as a primary object. It starts from a discrete cell with boundary memory. The real line, Moonshine, and dimensional physics are different readings of that same cell. Consequently, the chain

$$\text{lifted APP} \longrightarrow \text{value and memory} \longrightarrow \text{duality } T \longrightarrow 243/\text{Golay} \longrightarrow \text{mass routes}$$

does not import duality from string theory: it shows that its structure exists in HMT before continuous geometrization.

24.2.1 Operational extension of APP: visible modes and memory modes

For every positive integer n, let be defined

$$r_9(n) = 1 + ((n-1) \bmod 9), \qquad Q_9(n) = \frac{n - r_9(n)}{9}.$$

Then

$$n = 9Q_9(n) + r_9(n).$$

The ordinary reading preserves the residue. The enriched reading preserves the complete pair.

$$n \longmapsto (r, Q).$$

This pair is the minimal structure of a compact reading: one component is published inside the wheel and the other preserves how many times its boundary was traversed.

Principle 24.4 (Duality between visible value and boundary memory)**.** The HMT cell admits two conjugate readings. The first publishes r as phase, value, or vibration; the second publishes Q as memory, winding, or number of boundary crossings.

24.2.2 Completion 1000=729+243+27+1 and stratification of the unit

Let $R > 0$. The functional

$$E_R(r, Q) = \frac{r^2}{R^2} + Q^2 R^2$$

satisfies

$$E_R(r, Q) = E_{1/R}(Q, r),$$

for

$$E_{1/R}(Q, r) = \frac{Q^2}{(1/R)^2} + r^2(1/R)^2 = Q^2R^2 + \frac{r^2}{R^2}.$$

Therefore,

$$(r, Q; R) \longleftrightarrow (Q, r; R^{-1}).$$

The continuous string expression contains terms of the form

$$\frac{n^2}{R^2} + \frac{w^2R^2}{\alpha'^2}.$$

Before that continuous realization, HMT supplies the visible residue, boundary memory, and scale inversion. The duality T is the geometric projection of this reading involution.

24.2.3 Operatorial extension of APP: separation between visible modes and memory modes

For the sum and product over the same support one writes

$$i + j = 9Q_{ij}^{+} + R_{ij}^{+}, \qquad ij = 9Q_{ij}^{\times} + R_{ij}^{\times}.$$

The raised APP is

$$\mathcal{A}_9^{\sharp} = (R^{+}, Q^{+}; R^{\times}, Q^{\times}),$$

where $R^{+}, R^{\times}$ are the visible modes and $Q^{+}, Q^{\times}$ the memory modes. The complete discrepancy is

$$ij - (i + j) = (R_{ij}^{\times} - R_{ij}^{+}) + 9(Q_{ij}^{\times} - Q_{ij}^{+}).$$

When summing over $1 \le i, j \le 9$,

$$\sum R^{+} = 405, \quad \sum R^{\times} = 459, \quad \sum Q^{+} = 45, \quad \sum Q^{\times} = 174.$$

From here, one obtains

$$\Delta_{\text{visible}} = 459 - 405 = 54, \qquad \Delta_{\text{memoria}} = 174 - 45 = 129,$$

and

$$\sum ij - \sum (i + j) = 2025 - 810 = 1215 = 54 + 9 \cdot 129.$$

Normalization by the wheel produces

$$\frac{1215}{9} = 135 = 6 + 129 = 120 + 15.$$

Thus, 54 is the visible defect, 129 the memory defect, 135 the normalized total defect, 120 the constitutive octadic channel, and 15 the internal pair of the hexad.

24.2.4 Decomposition 243=1+22+90+120+10 and typing of its components

The Extreme and Mean Holographic Ratio possesses the cubic completion

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1,$$

that is,

$$1000 = 729 + 243 + 27 + 1.$$

The four summands represent, respectively, the discrete volume of dimension 3, the boundary of codimension 1, the boundary of codimension 2, and the return of codimension 3. The active corona is refined as

$$270 = 243 + 27 = 3 \cdot 9^2 + 3 \cdot 9.$$

The unit is thus decomposed into interior, stratified active boundary, and return. The passage $9 \to 10$ opens a complete dimensional layer, and the boundary of one level becomes a propagation channel of the next level.

24.2.5 Derivation of the constant channels from the decomposition of block 243

The same block is obtained as a ternary Hamming ball of radius 2 and length 11:

$$V_3(11, 2) = \sum_{k=0}^{2} \binom{11}{k} 2^k = 1 + 22 + 220 = 243.$$

The punctured perfect ternary Golay code has parameters

$$G_{11} = [11, 6, 5]_3, \qquad |G_{11}| = 3^6 = 729,$$

and satisfies

$$729 \cdot 243 = 3^{11}.$$

Therefore, G_{11} tiles $\mathbb{F}_3^{11}$ by ternary balls of radius 2. The number 243 is simultaneously

$$3 \cdot 9^2, \qquad V_3(11, 2), \qquad 3^5, \qquad 1 + 22 + 220.$$

This identity joins dimensional boundary, perfect correction, dimension 11, Witt–Golay incidence and the structural reading of M-theory.

24.2.6 Construction of dimension 11 via compatible residual readers

The ball admits the decomposition

$$243 = 1 + 22 + 90 + 120 + 10.$$

Here 1 is the center and return; $22 = 2 \cdot 11$, the first Hamming shell and the electronic reading of the block; $90 = 6\binom{6}{2}$, the hexadic channel; $120 = 8\binom{6}{2}$, the octadic channel; and $10 = 9 + 1$, closure of the monodromy $9 \to 1$. Thus, the return, component 22, channels 90/120, and monodromy are contained in the same perfect ball.

24.2.7 Constant channels derived from the block 243

With $B = 243$, one obtains the channels

$$C_\alpha = 3B = 729, \qquad C_e = B + 27 + 1 = 271,$$

$$C_\pi = B + 72 = 315, \qquad C_\varphi = B - 81 - 1 = 161.$$

The channel of α is the triple block 243; that of e is the block 243 with the boundary 27 and the return; that of π is the block with the channel 72; and that of φ is the block after removing the subsquare 81 and the return. The hinge is reconstructed as

$$C_\pi + C_e + C_\varphi - 3 = 315 + 271 + 161 - 3 = 744 = 729 + 15.$$

The reduced trace of the three characters therefore combines the triple block 243 with the internal pair of the hexad.

24.2.8 Closure of the genealogical register under the T-duality involution

The punctured perfect ternary code produces

$$11 = 6 + 5,$$

where 6 is the dimension of G_{11} and 5 is the correction depth $3^5 = 243$. The same dimension decomposes as

$$11 = 3 + 8,$$

where 3 corresponds to TRIT and 8 to the internal octad. Taken together,

$$11 = 6 + 5 = 3 + 8, \qquad 12 = 11 + 1.$$

The dimension reading 11 thus coordinates code and correction, TRIT and octad, and the identity $729 \cdot 243 = 3^{11}$.

24.2.9 Dimensional realization of duality

The local action of a route is expressed as

$$E_0(P) = n_A(P) A_{\mathrm{rad}} + \frac{s_C(P)}{6} C^*_{\mathrm{rad}}.$$

The first term is the visible advance or vibration mode; the second is the boundary or contra-angular winding mode. The local mass is

$$m_{\mathrm{HMT}}(P) = m_e \exp(E_0(P)).$$

The refined tower adds the boundary memory:

$$E(P) = E_0(P) + k(P)\Delta_4 + \nu_{120}(P)s_{120} + \nu_{270}(P)\zeta_{270},$$

where Δ_4 is the minimal torsion, s_{120} the hexad–octad selector, and

$$\zeta_{270} = 135\Delta_4^2$$

the quadratic memory of the crown. The abstract duality of route is

$$E_{A,C}(n,w) = nA_{\text{rad}} + wC^*_{\text{rad}}, \qquad E_{A,C}(n,w) = E_{C,A}(w,n).$$

In the physical sector, A and C^* are polarized, and the genealogical register selects the realization. Mass is the exponential of a closure route with visible advance, boundary winding, and torsion.

24.2.10 Compatibility between leaves and closure of the genealogical record

In a closed string, compatibility between the left- and right-moving modes is expressed schematically by

$$N_L - N_R = nw.$$

In the HMT reading, the left-hand member measures the difference of excitation between sheets, whereas the right-hand member measures the coupling between the visible phase and boundary memory. The adjustment of levels thus constitutes a closure of the bidirectional genealogical record. In the readings CP, T, and CPT, an asymmetry of orientation is compensated through bidirectional inversion in order to close the record.

24.2.11 T-Duality, Modularity and J Function

In the complex torus,

$$q = e^{2\pi i\tau},$$

and the modular transformation

$$\tau \longmapsto -\frac{1}{\tau}$$

interchanges its cycles. The duality T interchanges large and small radii; both realizations publish the interchange between the visible cycle and the boundary cycle. The modular function J, the Sommerfeld action, and the duality T constitute distinct closures of the same primary boundary structure with memory.

24.2.12 Dimensional Consequences for Photon, Electron, Mass, and Statistics

The observable photon is associated with the central TPK mode:

$$L_{\text{TPK}} = 3(I - P_0), \qquad \text{Spec}(L_{\text{TPK}}) = \{0,3,3\}.$$

The zero mode is an extended null propagation with no compactified memory of its own. The electron appears as the compact closure of the conjugate zero modes,

$$e^- = \text{compact closure of } P_+ + P_-,$$

and constitutes the minimal physical winding with stable memory. Mass measures the cost of stabilizing that compact memory. Pauli exclusion is interpreted as saturation of the genealogical register: two identical fermions do not occupy the same boundary because their winding closure is already saturated. Bosons publish shareable modes of unfolded memory, whereas fermions publish compactified memory.

24.2.13 Exact identities of involution, modularity, and residual conservation

The construction satisfies jointly the identities

1. $1000 = 729 + 243 + 27 + 1$;
2. $V_3(11, 2) = 1 + 22 + 220 = 243$;
3. $729 \cdot 243 = 3^{11}$;
4. $243 = 1 + 22 + 90 + 120 + 10$;
5. $C_\alpha, C_e, C_\pi, C_\varphi = 729, 271, 315, 161$;
6. $315 + 271 + 161 - 3 = 744$;
7. $1215 = 54 + 9 \cdot 129$;
8. $E_R(r, Q) = E_{1/R}(Q, r)$;
9. $11 = 6 + 5 = 3 + 8$, $12 = 11 + 1$, $26 = 2 + 24$ y $10 = 2 + 8$.

The duality T is therefore recognized as the geometric realization of the exchange between residue and quotient. Winding is not introduced first as an external property of a compact string, but as boundary memory of the decomposition $n = 9Q + r$. The block 243 brings this duality together with ternary Golay, dimension 11, channels 90/120, generation of constants, mass routes, and M-theory. In its most general form, HMT reads mass, string, and dimension as publications of a single boundary with memory.

24.2.14 Common core of the 3 formulations of T-duality

Let us define the stable domain

$$\mathscr{D}_0 = \mathbb{Z}^2 \times \mathbb{R}_{>0}, \qquad E_{R_0}(m, w) = \frac{m^2}{R_0^2} + w^2 R_0^2,$$

and the involution

$$\mathsf{T}_0(m, w, R_0) = (w, m, R_0^{-1}).$$

Theorem 24.5 (Residual ambient core equivalence–quotient)**.** *After prolonging the variables r, Q to the stable domain $\mathbb{Z}^2$, the three formulations are identified by means of*

$$I_F(r, Q, R) = (m, w, R_0) = (r, Q, R), \qquad I_{16\to 02} = \mathrm{id}_{\mathscr{D}_0}.$$

These maps intertwine the involutions and the quadratic forms:

$$I_F \mathsf{T}_F = \mathsf{T}_0 I_F, \qquad E_R^F = E_{R_0} \circ I_F, \qquad \mathsf{T}_0^2 = \mathrm{id}, \qquad E_{R_0} = E_{R_0^{-1}} \circ \mathrm{swap}.$$

Proof. The two maps are renamings of the same three coordinates. Moreover,

$$\mathsf{T}_0^2(m, w, R_0) = (m, w, R_0)$$

and

$$E_{R_0^{-1}}(w, m) = \frac{w^2}{R_0^{-2}} + m^2 R_0^{-2} = w^2 R_0^2 + \frac{m^2}{R_0^2} = E_{R_0}(m, w).$$

The identities are the same after applying I_F or the identity between the two formulations. □

In $\mathcal{H}_{\mathrm{lat}} = \ell^2(\mathbb{Z}^2)$, let us define

$$H(R_0)|m, w\rangle = E_{R_0}(m, w)|m, w\rangle, \qquad U_T|m, w\rangle = |w, m\rangle.$$

Then U_T is unitary, $U_T^2 = I$, and the preceding scalar identity is equivalent, on the orthonormal basis, to

$$\boxed{U_T H(R_0) U_T^{-1} = H(R_0^{-1}).}$$

This is the second exact realization of the same core, not a new physical hypothesis.

24.2.15 Distinction of the 2 residual sections by their energy insertions and equivalence classes

The positive residual convention uses

$$r_9^+(n) = 1 + ((n-1) \bmod 9), \qquad Q_9^+(n) = \frac{n - r_9^+(n)}{9},$$

whereas the zero residual convention inserts the class $\bar{r}_9(n) \in \{0, \dots, 8\}$. The change of section that preserves the integer is

$$C(r, Q) = \big(\bar{r}, Q + \mathbf{1}_{\{r=9\}}\big).$$

Proposition 24.6 (Energy Obstruction of Section Change)**.** *The map C preserves $n = 9Q + r$, but in general it preserves neither the form E_R nor the exchange involution. Equivalence of the ambient kernel therefore does not imply equivalence of the two nonadic embeddings.*

Proof. For $n = 9$, the first convention gives $(r, Q) = (9, 0)$, whereas the change with carry gives $(\bar{r}, Q_0) = (0, 1)$. Their energies are

$$E_R(9, 0) = \frac{81}{R^2}, \qquad E_R(0, 1) = R^2,$$

which do not coincide for every $R > 0$. Insertion without correcting the carry would send this example to $(0, 0)$ and would likewise fail to preserve the integer. A unique section must be fixed, or a form covariant under change of section must be constructed, before this equivalence is promoted. □

Definition 24.7 (Quotient of section change and covariant energy)**.** Let

$$L_9 := \mathbb{Z}(9, -1) \subset \mathbb{Z}^2, \qquad S(x_1, x_2) = (x_2, x_1),$$

and let

$$q_R(x_1, x_2) = \frac{x_1^2}{R^2} + x_2^2 R^2.$$

On the quotients

$$\mathcal{Q}_9 = (\mathbb{Z}^2/L_9) \times \mathbb{R}_{>0}, \qquad \mathcal{Q}_9^\vee = (\mathbb{Z}^2/SL_9) \times \mathbb{R}_{>0},$$

we define

$$\mathcal{E}_R^{L_9}([x]_{L_9}) := \min_{\ell \in L_9} q_R(x + \ell)$$

and

$$\mathsf{T}_{\mathrm{cov}}([x]_{L_9}, R) := ([Sx]_{SL_9}, R^{-1}).$$

Theorem 24.8 (Covariant Equivalence of Nonary Sections)**.** *The energy $\mathcal{E}_R^{L_9}$ and the transformation $\mathsf{T}_{\mathrm{cov}}$ are well defined. The conventions $(9, 0)$ and $(0, 1)$, and in general the sections $1, \dots, 9$ and $0, \dots, 8$ with the corresponding carry, represent the same class of $\mathbb{Z}^2/L_9$. Moreover,*

$$\boxed{\mathcal{E}_{R^{-1}}^{SL_9}([Sx]_{SL_9}) = \mathcal{E}_R^{L_9}([x]_{L_9}).}$$

The inverse exchange, defined on $\mathcal{Q}_9^\vee$, returns the initial class; hence the two quotiented charts are T-dual.

Proof. If $x' = x + \ell_0$, the translation $\ell \mapsto \ell + \ell_0$ permutes L_9, so the minimum does not depend on the representative. It exists because $q_R(x + k(9, -1))$ is a coercive quadratic function of $k \in \mathbb{Z}$. The difference $(9, 0) - (0, 1) = (9, -1)$ belongs to L_9, and the same occurs under every change of section with carry. Finally,

$$\begin{aligned}\mathcal{E}^{SL_9}_{R^{-1}}([Sx]) &= \min_{\ell \in L_9} q_{R^{-1}}(Sx + S\ell) \\ &= \min_{\ell \in L_9} q_R(x + \ell) = \mathcal{E}^{L_9}_R([x]).\end{aligned}$$

Since $S^2 = I$, the second exchange recovers the original quotient and the radius R. □

Equivalence domain of the dual nucleus and nonequivalence of energy insertions. The equivalence of the kernels over $\mathscr{D}_0$ is exact. The two embeddings are not equivalent for the uncorrected energy; the covariant closure is the quotient and the energy of the preceding theorem.

24.2.16 Exact normalization of the string realization

Fix $\alpha' > 0$. In the domain $\mathscr{D}_{\alpha'} = \mathbb{Z}^2 \times \mathbb{R}_{>0}$, let

$$\mathsf{T}_{\alpha'}(m, w, R) = \left(w, m, \frac{\alpha'}{R}\right), \qquad \mathcal{N}_{\alpha'}(m, w, R) = \left(m, w, \frac{R}{\sqrt{\alpha'}}\right).$$

Theorem 24.9 (String scale conjugation)**.** *The normalization $\mathcal{N}_{\alpha'}$ is an exact conjugation:*

$$\boxed{\mathcal{N}_{\alpha'}\mathsf{T}_{\alpha'} = \mathsf{T}_0\mathcal{N}_{\alpha'}.}$$

For

$$E_{\mathrm{str}}(m, w, R) = \frac{m^2}{R^2} + \frac{w^2R^2}{\alpha'^2},$$

we have

$$E_{\mathrm{str}} = \alpha'^{-1}E_0 \circ \mathcal{N}_{\alpha'}.$$

Furthermore, if

$$p_L = \frac{m}{R} + \frac{wR}{\alpha'}, \qquad p_R = \frac{m}{R} - \frac{wR}{\alpha'},$$

then p_L remains fixed and $p_R \mapsto -p_R$.

Proof. We have

$$\mathcal{N}_{\alpha'}\mathsf{T}_{\alpha'}(m, w, R) = \left(w, m, \frac{\sqrt{\alpha'}}{R}\right) = \mathsf{T}_0\mathcal{N}_{\alpha'}(m, w, R).$$

The energy identity results from

$$E_0\left(m, w, \frac{R}{\sqrt{\alpha'}}\right) = \frac{\alpha' m^2}{R^2} + \frac{w^2R^2}{\alpha'} = \alpha' E_{\mathrm{str}}(m, w, R).$$

The substitution of $(m, w, R) \mapsto (w, m, \alpha'/R)$ in (p_L, p_R) gives, respectively, p_L and $-p_R$. □

Conjugation and spectral invariance with fixed oscillatory data are exact. The identification of m, w, R, α' with momentum, winding, physical radius, and inverse tension is a physical interface with its own premises.

24.2.17 Level separation, linear mass exchange, and modularity of the 3 realizations

The active string condition is

$$N_L - N_R = a_L - a_R - mw.$$

The schematic formula $N_L - N_R = nw$ coincides with it only after declaring $a_L = a_R$ and $n = -m$, or an equivalent reversal of leaf orientation. Without that dictionary, the two expressions are not the same formulation.

The linear symmetry of the action is typified in the extended space of coefficients and modes:

$$L(A, C; n, w) = nA + wC, \qquad \sigma(A, C; n, w) = (C, A; w, n).$$

Then

$$L \circ \sigma = wC + nA = L.$$

It is an exact identity under simultaneous exchange of parameters and coordinates. It is not equivalent to quadratic invariance of $H(R)$ on a sector with fixed physical parameters; such a promotion requires an additional intertwiner.

Finally, on the imaginary axis of the upper half-plane, let us define

$$\iota(R_0) = iR_0^2, \qquad S(\tau) = -\frac{1}{\tau}.$$

Then

$$\boxed{S \circ \iota(R_0) = \iota(R_0^{-1}).}$$

Since $J(S\tau) = J(\tau)$, one obtains $J(iR_0^2) = J(iR_0^{-2})$. This is an exact semiconjugacy of the radius parameter with the modular transformation S, proved from the explicit starting modular structure. It does not by itself identify target-space duality, the exchange of charges, or the construction of $V^{\natural}$.

24.3 Action ellipse and determination of the normalized radius

The angular reader and the action section preserve the same APP–TRIT–TPK antecedent. The form constructed below uses the publications (14.75) and a positive section $\hbar = \hbar_s$, with $s \in \{\mathrm{pre}, \mathrm{ret}\}$, of (14.71). The angular ratio will determine the form; the action section will preserve its common scale.

24.3.1 Domain and labelled semiaxes

For brevity, let $A = A_{\mathrm{deg}}$ and $C^* = C^*_{\mathrm{deg}}$. In the chart $A \neq 0$, $\hbar > 0$ and $|C^*/A| < 1$, we define

$$\xi = \frac{C^*}{A}, \qquad \eta_{\mathrm{el}} = \operatorname{artanh} \xi, \qquad a_S = \sqrt{2\hbar}\, e^{\eta_{\mathrm{el}}/2}, \qquad b_S = \sqrt{2\hbar}\, e^{-\eta_{\mathrm{el}}/2}. \tag{24.1}$$

The semiaxes and the balanced coordinates Q, P have units of the square root of action. The axes remain labelled: the sign of η_{el} determines which is larger. The curve and the product of its semiaxes are

$$\gamma(\theta) = (a_S \cos\theta, b_S \sin\theta), \qquad \frac{Q^2}{a_S^2} + \frac{P^2}{b_S^2} = 1, \qquad a_S b_S = 2\hbar. \tag{24.2}$$

Proposition 24.10 (Membership of the publications in the elliptic chart)**.** *The positive branch of* (14.75)*, with the intervals* (14.62)*, satisfies $A > 0$, $0 < C^* < A$, and $\hbar_s > 0$. Reversal of the odd coordinate preserves $|C^*/A| < 1$.*

Proof. Proposition 14.22 gives $0 < \eta_{\mathrm{ret}} < H_5$ and positivity of both action sections. For an upper bound on H_5, (14.62) implies $\alpha < 1/125$ and $\varphi > 8/5$. Retaining only the positive terms of (14.64) yields

$$H_5 < \frac{\sqrt{5}}{2} + \frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4}.$$

Since $5 < 81/16$, one has $\sqrt{5}/2 < 9/8$. Moreover,

$$\frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4} = \frac{421882}{11718750000} < \frac{1}{1000}.$$

Thus $H_5 < 9/8 + 1/1000 < 2$. Using $\alpha > 7/1000$ now gives

$$0 < C^* = 2(\eta_{\mathrm{ret}} + \alpha) < 2\left(2 + \frac{1}{125}\right) = \frac{502}{125} < 7 < 1000\alpha = A.$$

The channel interchange in 14.26 replaces C^* by $-C^*$ and preserves A; it therefore preserves the absolute inequality. □

24.3.2 Parametric phase and oriented area

Proposition 24.11 (Area law and transport of angular coordinates)**.** *The parametric phase θ of γ and its polar angle ϕ, continuously lifted from zero, satisfy*

$$\phi(\theta) = \arg(a_S \cos\theta + i b_S \sin\theta), \qquad \frac{d\phi}{d\theta} = \frac{a_S b_S}{a_S^2 \cos^2\theta + b_S^2 \sin^2\theta} > 0. \tag{24.3}$$

In a chart where both tangents are defined, $\tan\phi = (b_S/a_S)\tan\theta$*. The oriented area swept out from phase zero is*

$$\mathcal{A}(\theta) = \frac{1}{2}\int_0^\theta (QP' - PQ')\,dt = \hbar\theta, \qquad \mathcal{A}(1) = \hbar, \quad \mathcal{A}(2\pi) = h_s. \tag{24.4}$$

Proof. The derivative of the argument of a nonzero vector is $(QP' - PQ')/(Q^2 + P^2)$. For the curve (24.2), the numerator equals $a_S b_S = 2\hbar$; this proves the derivative formula and its positivity. The quotient P/Q gives the tangent relation; the continuous lift preserves quadrants and turns. The integral equals $a_S b_S \theta/2 = \hbar\theta$. Finally, $h_s = 2\pi\hbar_s$ gives the identity for the complete turn. □

One radian of parametric phase sweeps an area $\hbar$. The polar angle retains its own derivative in (24.3). In polar coordinates, the equivalent law is

$$d\mathcal{A} = \frac{1}{2}\rho_\partial(\phi)^2\,d\phi, \qquad \rho_\partial(\phi) = \left(\frac{\cos^2\phi}{a_S^2} + \frac{\sin^2\phi}{b_S^2}\right)^{-1/2}.$$

The radial formula follows by substituting $(Q, P) = \rho(\cos\phi, \sin\phi)$ into (24.2). With the orientation of γ, the integral of the canonical one-form retains the sign

$$\oint_\gamma P\,dQ = -\pi a_S b_S = -h_s. \tag{24.5}$$

Indeed, $P\,dQ = -a_S b_S \sin^2\theta\,d\theta$, and the integral of $\sin^2\theta$ over one turn is π.

24.3.3 Shape reciprocity and symplectic orientation

Define the matrices

$$D(\eta) = \begin{pmatrix} e^{\eta} & 0 \\ 0 & e^{-\eta} \end{pmatrix}, \qquad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{24.6}$$

Proposition 24.12 (Quadratic form and action under reciprocity)**.** *Both transports satisfy $T^{\mathsf{T}} D(-\eta) T = D(\eta)$. For $\omega = dQ \wedge dP$, the interchange S_2 is antisymplectic and the rotation J_2 is symplectic:*

$$S_2^* \omega = -\omega, \qquad J_2^* \omega = \omega.$$

If $\mathcal{I}(\gamma) = \oint_\gamma P\, dQ$ on a closed curve, then

$$\mathcal{I}(S_2 \gamma) = -\mathcal{I}(\gamma), \qquad \mathcal{I}(J_2 \gamma) = \mathcal{I}(\gamma). \tag{24.7}$$

Proof. Matrix multiplication interchanges the two diagonal entries of D, proving both congruences. The determinants of S_2, J_2 are $-1, 1$; this gives the transformation of the area form. For $\lambda = P\, dQ$, direct calculation gives

$$S_2^* \lambda = d(PQ) - \lambda, \qquad J_2^* \lambda = \lambda - d(PQ).$$

The integral of the exact differential over the closed curve is zero, proving both action identities. □

Orientation distinguishes the two transports even though they express the same quadratic form, including in the circular case. The labelling of the axes and the sign of the action therefore belong to the data preserved in passing to the lattice representation.

24.3.4 Normalized radius and lattice-readout matrix

Proposition 24.13 (Determination and recoverability of the shape radius)**.** *The labelled ellipse determines a unique positive radius*

$$R_\star = \sqrt{\frac{b_S}{a_S}} = e^{-\eta_{\mathrm{el}}/2} = \left(\frac{A - C^*}{A + C^*} \right)^{1/4}, \qquad \tau_{\mathrm{el}} = i R_\star^2. \tag{24.8}$$

The angular ratio is recovered through

$$\frac{C^*}{A} = \frac{1 - R_\star^4}{1 + R_\star^4}, \qquad R_\star^4 = \frac{499\alpha - \eta_{\mathrm{ret}}}{501\alpha + \eta_{\mathrm{ret}}}. \tag{24.9}$$

Interchanging the semiaxes transforms $(\eta_{\mathrm{el}}, R_\star, \tau_{\mathrm{el}})$ into $(-\eta_{\mathrm{el}}, R_\star^{-1}, -1/\tau_{\mathrm{el}})$ and preserves $a_S b_S = 2\hbar$.

Proof. Dividing the semiaxes gives $b_S/a_S = e^{-\eta_{\mathrm{el}}}$. The map $R \mapsto R^2$ is bijective on the positive reals, so the positive root is unique. For $\xi = C^*/A \in (-1, 1)$,

$$2 \operatorname{artanh} \xi = \log \frac{1 + \xi}{1 - \xi};$$

hence $R_\star^4 = (1 - \xi)/(1 + \xi)$. Solving for ξ proves the first identity in (24.9). Substituting $A = 1000\alpha$ and $C^* = 2(\eta_{\mathrm{ret}} + \alpha)$ gives

$$\frac{A - C^*}{A + C^*} = \frac{998\alpha - 2\eta_{\mathrm{ret}}}{1002\alpha + 2\eta_{\mathrm{ret}}} = \frac{499\alpha - \eta_{\mathrm{ret}}}{501\alpha + \eta_{\mathrm{ret}}}.$$

The interchange inverts the semiaxis ratio and changes the sign of η_{el}. The identity $-1/(iR_\star^2) = iR_\star^{-2}$ gives the modular transport. The product of the semiaxes remains fixed. □

On the positive branch, $0 < R_\star < 1$; the chart of the reversed form publishes $R_\star^{-1} > 1$. The angular ratio is recovered from the radius, while the absolute angles and their history remain in the enriched state. The circular form corresponds to $C^*/A = 0$, with $R_\star = 1$. Changing the positive action section while retaining the angular pair changes the common scale of the semiaxes and preserves their ratio.

Proposition 24.14 (Lattice form determined by the ellipse)**.** *The matrix of normalized squared semiaxes is*

$$D(\eta_{\mathrm{el}}) = \frac{1}{2\hbar}\begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix} = \begin{pmatrix} R_\star^{-2} & 0 \\ 0 & R_\star^2 \end{pmatrix}. \tag{24.10}$$

Its evaluation on an integer pair produces

$$E_\star(m, w) = \begin{pmatrix} m & w \end{pmatrix} D(\eta_{\mathrm{el}}) \begin{pmatrix} m \\ w \end{pmatrix} = \frac{m^2}{R_\star^2} + w^2 R_\star^2. \tag{24.11}$$

The equation of the ellipse multiplied by $2\hbar$ uses, instead, the inverse matrix $D(-\eta_{\mathrm{el}})$.

Proof. The squares of the semiaxes in (24.1), divided by $2\hbar$, are $e^{\eta_{\mathrm{el}}}$ and $e^{-\eta_{\mathrm{el}}}$. By (24.8), these are $R_\star^{-2}$ and $R_\star^2$. Multiplication by the integer vector proves (24.11). Finally, $2\hbar/a_S^2 = e^{-\eta_{\mathrm{el}}}$ and $2\hbar/b_S^2 = e^{\eta_{\mathrm{el}}}$, identifying the matrix of the elliptic equation. □

The radius $R_\star$ is a dimensionless publication of the labelled form. The polar radius ρ_∂ belongs to the square-root-of-action line. Representation by a spatial radius requires its subsequent dimensional chart; this preserves the form already determined and the independence of its construction from a prescribed length.

24.3.5 Inductive–capacitive realization of the action form

The action form already constructed admits an electrical realization in a chart equipped with a clock $\omega > 0$ and a reference impedance $Z_* > 0$. These two quantities preserve their units and accompany the shape parameter η. One defines

$$L_\eta = \frac{Z_*}{\omega} e^{-\eta}, \qquad C_\eta = \frac{e^{\eta}}{Z_* \omega}, \qquad Z_\eta = \sqrt{L_\eta / C_\eta}. \tag{24.12}$$

Here C_η denotes capacitance, distinct from the counterangle C^*.

Proposition 24.15 (Conservation of the dynamical product)**.** *The realization* (24.12) *satisfies*

$$L_\eta C_\eta = \omega^{-2}, \qquad \frac{Z_\eta}{Z_*} = e^{-\eta} = \frac{b_S}{a_S}, \tag{24.13}$$

where $a_S = \sqrt{2\hbar}\, e^{\eta/2}$ and $b_S = \sqrt{2\hbar}\, e^{-\eta/2}$ are the semiaxes of equation (24.1)*. With $Q = \sqrt{Z_*}\, q$ and $P = \Phi/\sqrt{Z_*}$, the energy takes the form*

$$H_{LC} = \frac{q^2}{2C_\eta} + \frac{\Phi^2}{2L_\eta} = \frac{\omega}{2}(e^{-\eta} Q^2 + e^{\eta} P^2).$$

On $(Q, P) = (a_S \cos\theta, b_S \sin\theta)$, the electric and magnetic contributions are $\hbar\omega \cos^2\theta$ and $\hbar\omega \sin^2\theta$; the total energy is $\hbar\omega$.

Proof. Multiplying L_η and C_η cancels the factors Z_* and $e^{\pm\eta}$, whereas their ratio is $Z_*^2 e^{-2\eta}$. The positive square root proves the impedance identity. Substitution of $q = Q/\sqrt{Z_*}$ and $\Phi = \sqrt{Z_*}\, P$ gives the form of H_{LC}. The squared semiaxes cancel the remaining shape factors; the sum uses $\cos^2\theta + \sin^2\theta = 1$. □

The Hamiltonian convention $\dot{Q} = \partial_P H$, $\dot{P} = -\partial_Q H$ assigns $\dot{\theta} = -\omega$ to this parametrization. The form, the clock and the dimensional chart remain distinct. The constitutive response of the vacuum and the impedance Z_η are related by the channel inversion of proposition 118.4; each preserves its definition and its normalization.

24.4 T-duality: integer coordinates, change of section, and operator equivalence

The arithmetic decomposition of APP preserves the residual representative and its quotient. TPK evolution also transports its orientation and the memory of boundary crossings. Interchanging the two coordinates requires preserving this distinction and specifying the domain on which the operation is again defined. An arithmetic involution is first constructed on that domain; its quadratic form subsequently admits a representation by self-adjoint operators.

24.4.1 Stable domain of the interchange

In the positive residual section,

$$r_9^+(n) = 1 + ((n-1) \bmod 9), \qquad Q_9^+(n) = \frac{n - r_9^+(n)}{9}, \qquad n = r_9^+(n) + 9Q_9^+(n). \tag{24.14}$$

An interchange can place an arbitrary quotient in the first coordinate. Its stable domain is therefore $\mathscr{D}_0 = \mathbb{Z}^2 \times \mathbb{R}_{>0}$, not merely the section $1 \le r \le 9$. Extending the domain makes it possible to define a total operation; the admissibility of a pair as the readout of a TPK route separately retains its selection rules.

Definition 24.16 (Normalized compact form and interchange)**.** For $d = (m, w, R_0) \in \mathscr{D}_0$, let

$$E_{R_0}(m, w) = \frac{m^2}{R_0^2} + w^2 R_0^2, \qquad \mathsf{T}_0(m, w, R_0) = (w, m, R_0^{-1}). \tag{24.15}$$

The symbols m, w name the two transported integer coordinates; R_0 is the positive dimensionless scale of the form. Their interpretation through momentum, winding, and physical radius is established by a subsequent realization map.

Theorem 24.17 (Involution and preservation of the quadratic form)**.** *The map T_0 is an involution of $\mathscr{D}_0$, and*

$$E_{R_0^{-1}}(w, m) = E_{R_0}(m, w).$$

Proof. Two interchanges restore (m, w), and two inversions restore R_0. Moreover,

$$E_{R_0^{-1}}(w, m) = w^2 R_0^2 + \frac{m^2}{R_0^2} = E_{R_0}(m, w).$$

□

The equivalence holds for each radius and its reciprocal. At $R_0 = 1$, the radius remains fixed and the operation acts on the same form. A specific pair is fixed by the full involution precisely when, in addition, $m = w$.

24.4.2 Compact Hamiltonian and self-adjoint domain

Let $\mathcal{H}_{\mathrm{lat}} = \ell^2(\mathbb{Z}^2)$, with orthonormal basis $\{|m, w\rangle\}$. Define

$$H(R_0)|m, w\rangle = E_{R_0}(m, w)|m, w\rangle, \tag{24.16}$$

$$D(H(R_0)) = \left\{ \psi : \sum_{m,w \in \mathbb{Z}} E_{R_0}(m, w)^2 |\psi_{m,w}|^2 < \infty \right\}. \tag{24.17}$$

Finitely supported vectors belong to the domain and form a core for the diagonal operator. The associated quadratic form has domain

$$D(H(R_0)^{1/2}) = \left\{ \psi : \sum_{m,w} E_{R_0}(m,w)|\psi_{m,w}|^2 < \infty \right\}.$$

Proposition 24.18 (Self-adjointness and compact resolvent)**.** *The operator $H(R_0)$ is self-adjoint and nonnegative. Its eigenvalues are $E_{R_0}(m,w)$, counted with multiplicity, and $(H(R_0)+I)^{-1}$ is compact.*

Proof. A multiplication operator defined by a real sequence is symmetric on (24.17). If η belongs to the domain of the adjoint, testing the adjoint identity against each vector $|m,w\rangle$ forces its image to have coordinates $E_{R_0}(m,w)\eta_{m,w}$. Membership in ℓ^2 is exactly condition (24.17); hence the adjoint has the same domain and the same action. Nonnegativity follows by summing the nonnegative weights.

With $c_{R_0} = \min(R_0^{-2}, R_0^2) > 0$, we have $E_{R_0}(m,w) \geq c_{R_0}(m^2+w^2)$. Outside increasing finite sets, the coefficients $(1+E_{R_0}(m,w))^{-1}$ tend uniformly to zero. Truncating the resolvent to those sets produces finite-rank operators converging in norm, which proves its compactness. The diagonal basis determines the stated spectrum. □

Theorem 24.19 (Unitary conjugation, including domains)**.** *The interchange $U_T|m,w\rangle = |w,m\rangle$ is unitary, $U_T^2 = I$, and satisfies*

$$U_T D(H(R_0)) = D(H(R_0^{-1})), \qquad U_T H(R_0) U_T^{-1} = H(R_0^{-1}). \tag{24.18}$$

It also transports the quadratic-form domains.

Proof. The interchange is an involutive permutation of the orthonormal basis. For a sequence ψ, the identity in Theorem 24.17 gives

$$\sum_{m,w} E_{R_0^{-1}}(m,w)^2 |(U_T\psi)_{m,w}|^2 = \sum_{m,w} E_{R_0}(m,w)^2 |\psi_{m,w}|^2.$$

Equality of the domains is established. On each coordinate, $U_T H(R_0) U_T^{-1}$ multiplies by $E_{R_0}(w,m) = E_{R_0^{-1}}(m,w)$. The same sum, with the first power of the weight, proves the result for the quadratic forms. □

The scalar equality and the operator equality are two realizations of the same involution. The second preserves the domains that allow the first to be interpreted as spectral equivalence, not merely as an equality on finitely supported vectors.

24.4.3 Specialization to the radius of the action ellipse

The radius $R_\star$ determined in section 24.3 fixes the realization of this family that corresponds to the angular publications of the state. The symbol R_0 allows the transport results to be expressed for the entire positive family; composition with the ellipse evaluates that family at $R_0 = R_\star$ and also preserves its reciprocal image.

Corollary 24.20 (Elliptic family stable under duality)**.** *The set*

$$\mathscr{D}_{\mathrm{el}} = \mathbb{Z}^2 \times \{R_\star, R_\star^{-1}\}$$

is stable under T_0. Its Hamiltonians are self-adjoint, nonnegative, and have compact resolvent. The interchange U_T transports the domain of $H(R_\star)$ onto that of $H(R_\star^{-1})$, including their form domains.

Proof. The positivity of $R_\star$ and its unique determination are established in section 24.3. Inversion permutes $R_\star$ and $R_\star^{-1}$, while interchange preserves $\mathbb{Z}^2$. Proposition 24.18 and theorem 24.19 apply on these two fibres. If $R_\star = 1$, the two fibres coincide; in every other case, the operator intertwines two distinct fibres of the same family. □

The common action scale determines the area of the ellipse; its semiaxis ratio determines $R_\star$. This separation makes it possible to preserve the area and the common action scale while interchanging the semiaxes and obtaining the reciprocal radius. The index interchange on $\ell^2(\mathbb{Z}^2)$ and the transports of the action plane act on different spaces; their respective properties are proved on their own domains.

24.4.4 Change of representative and quotient memory

The zero section uses $\bar{r} \in \{0, \dots, 8\}$. The change that preserves the integer in (24.14) is

$$C(r, Q) = (\bar{r}, Q + \mathbf{1}_{\{r=9\}}). \tag{24.19}$$

The modification of the quotient is part of the operation. For $n = 9$, the two pairs are $(9, 0)$ and $(0, 1)$. Their uncorrected energies are $81/R_0^2$ and R_0^2. Representing the same integer therefore does not imply equality of those two insertions into a form that distinguishes the coordinates.

The equivalence relation is preserved by defining

$$L_9 = \mathbb{Z}(9, -1), \qquad S(x_1, x_2) = (x_2, x_1), \qquad q_{R_0}(x) = \frac{x_1^2}{R_0^2} + x_2^2 R_0^2.$$

For $x \in \mathbb{Z}^2$, let

$$\mathcal{E}_{R_0}^{L_9}([x]) = \min_{k \in \mathbb{Z}} q_{R_0}(x + k(9, -1)). \tag{24.20}$$

This construction operates on classes. The memory distinguishing individual representatives remains in the TPK state preceding the quotient; the function in (24.20) determines the invariant evaluation of the chosen class.

Theorem 24.21 (Covariant duality of sections)**.** *The energy in* (24.20) *is well defined. The transformation*

$$\mathsf{T}_{\mathrm{cov}}([x]_{L_9}, R_0) = ([Sx]_{SL_9}, R_0^{-1})$$

is a bijection between $(\mathbb{Z}^2/L_9) \times \mathbb{R}_{>0}$ *and* $(\mathbb{Z}^2/SL_9) \times \mathbb{R}_{>0}$*, with inverse given by the second interchange, and*

$$\mathcal{E}_{R_0^{-1}}^{SL_9}([Sx]) = \mathcal{E}_{R_0}^{L_9}([x]).$$

Proof. Replacing x by $x + k_0(9, -1)$ reindexes the candidates for the minimum. The minimum exists: the function of k is quadratic and has leading coefficient $81/R_0^2 + R_0^2 > 0$. The change $(9, Q) \mapsto (0, Q + 1)$ has difference in L_9, so it represents the same class. Interchange transports the equivalence relation to SL_9, and

$$\min_{\ell \in L_9} q_{R_0^{-1}}(Sx + S\ell) = \min_{\ell \in L_9} q_{R_0}(x + \ell).$$

Finally, $S^2 = I$ recovers the initial quotient and radius. □

The minimum can be evaluated exactly using two candidates. If $x = (m, w)$, the real minimum of the quadratic occurs at

$$k_* = \frac{wR_0^4 - 9m}{81 + R_0^4}.$$

The integer minimum is attained at $\lfloor k_* \rfloor$ or $\lceil k_* \rceil$. This formula does not alter the equivalence; it makes its finite evaluation explicit and allows the effect of changing section to be checked without discarding the quotient.

24.4.5 Radius normalization in the string realization

Let $\alpha' > 0$ be a scale of the string realization. This symbol denotes the string scale and is kept distinct from the fine-structure constant α. Define

$$\mathsf{T}_{\alpha'}(m, w, R) = \left(w, m, \frac{\alpha'}{R}\right), \qquad \mathcal{N}_{\alpha'}(m, w, R) = \left(m, w, \frac{R}{\sqrt{\alpha'}}\right). \tag{24.21}$$

Theorem 24.22 (Exact conjugation by the scale)**.** *Normalization satisfies*

$$\mathcal{N}_{\alpha'}\mathsf{T}_{\alpha'} = \mathsf{T}_0\mathcal{N}_{\alpha'}.$$

If

$$E_{\mathrm{str}}(m, w, R) = \frac{m^2}{R^2} + \frac{w^2R^2}{\alpha'^2},$$

then $E_{\mathrm{str}} = \alpha'^{-1}E \circ \mathcal{N}_{\alpha'}$. *The combinations*

$$p_L = \frac{m}{R} + \frac{wR}{\alpha'}, \qquad p_R = \frac{m}{R} - \frac{wR}{\alpha'}$$

transform as $p_L \mapsto p_L$ *and* $p_R \mapsto -p_R$.

Proof. Both compositions in the first equality give $(w, m, \sqrt{\alpha'}/R)$. On the other hand,

$$E_{R/\sqrt{\alpha'}}(m, w) = \frac{\alpha' m^2}{R^2} + \frac{w^2R^2}{\alpha'} = \alpha' E_{\mathrm{str}}(m, w, R).$$

Substituting $(w, m, \alpha'/R)$ into the expressions for p_L, p_R gives the two asserted transformations, respectively. □

Conjugation fixes what it means to express the same result in a dimensional chart. In the elliptic family, the radii are

$$R = \sqrt{\alpha'}R_\star, \qquad R^\vee = \sqrt{\alpha'}R_\star^{-1}.$$

Their product is α'. The internal determination of the dimensionless semiaxis ratio and the length-squared scale of the realization serve different functions: the former fixes the shape; the latter expresses its radii in a dimensional unit.

The preceding normalization allows the same result to be expressed with normalized or physical radius. Assigning a tension and a unit to α' belongs to the realization map and has its own origin; it is not a datum entering the production of APP–TRIT–TPK registers.

24.4.6 Relationship with the modular transformation

Radius inversion has a precise representation on the imaginary axis of the upper half-plane. Let $\iota(R_0) = \mathrm{i}R_0^2$ and $\mathsf{S}(\tau) = -1/\tau$. Then

$$\mathsf{S} \circ \iota(R_0) = \iota(R_0^{-1}). \tag{24.22}$$

The equality follows from $-1/(\mathrm{i}R_0^2) = \mathrm{i}R_0^{-2}$. For the modular function $J = j - 744$, whose invariance is preserved in the Moonshine development, it follows that $J(\mathrm{i}R_0^2) = J(\mathrm{i}R_0^{-2})$.

The map ι expresses a semiconjugacy of parameters. The interchange U_T acts on the coordinates of $\ell^2(\mathbb{Z}^2)$; S acts on the upper half-plane; the dual automorphism of an orbifold acts on its graded sectors. Writing out their domains makes it possible to compose specific relationships between them

while preserving the information of each representation. The results assembled here establish coordinate interchange with explicit operator domains and proofs.

24.5 Action ellipse and determination of the normalized radius

The angular reader and the action section preserve the same APP–TRIT–TPK antecedent. The form constructed below uses the publications (14.75) and a positive section $\hbar = \hbar_s$, with $s \in \{\mathrm{pre}, \mathrm{ret}\}$, of (14.71). The angular ratio will determine the form; the action section will preserve its common scale.

24.5.1 Domain and labelled semiaxes

For brevity, let $A = A_{\mathrm{deg}}$ and $C^* = C^*_{\mathrm{deg}}$. In the chart $A \neq 0$, $\hbar > 0$ and $|C^*/A| < 1$, we define

$$\xi = \frac{C^*}{A}, \qquad \eta_{\mathrm{el}} = \operatorname{artanh} \xi, \qquad a_S = \sqrt{2\hbar}\, e^{\eta_{\mathrm{el}}/2}, \qquad b_S = \sqrt{2\hbar}\, e^{-\eta_{\mathrm{el}}/2}. \tag{24.23}$$

The semiaxes and the balanced coordinates Q, P have units of the square root of action. The axes remain labelled: the sign of η_{el} determines which is larger. The curve and the product of its semiaxes are

$$\gamma(\theta) = (a_S \cos\theta, b_S \sin\theta), \qquad \frac{Q^2}{a_S^2} + \frac{P^2}{b_S^2} = 1, \qquad a_S b_S = 2\hbar. \tag{24.24}$$

Proposition 24.23 (Membership of the publications in the elliptic chart)**.** *The positive branch of* (14.75)*, with the intervals* (14.62)*, satisfies $A > 0$, $0 < C^* < A$, and $\hbar_s > 0$. Reversal of the odd coordinate preserves $|C^*/A| < 1$.*

Proof. Proposition 14.22 gives $0 < \eta_{\mathrm{ret}} < H_5$ and positivity of both action sections. For an upper bound on H_5, (14.62) implies $\alpha < 1/125$ and $\varphi > 8/5$. Retaining only the positive terms of (14.64) yields

$$H_5 < \frac{\sqrt{5}}{2} + \frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4}.$$

Since $5 < 81/16$, one has $\sqrt{5}/2 < 9/8$. Moreover,

$$\frac{9}{16 \cdot 125^2} + \frac{7}{48 \cdot 125^4} = \frac{421882}{11718750000} < \frac{1}{1000}.$$

Thus $H_5 < 9/8 + 1/1000 < 2$. Using $\alpha > 7/1000$ now gives

$$0 < C^* = 2(\eta_{\mathrm{ret}} + \alpha) < 2\left(2 + \frac{1}{125}\right) = \frac{502}{125} < 7 < 1000\alpha = A.$$

The channel interchange in 14.26 replaces C^* by $-C^*$ and preserves A; it therefore preserves the absolute inequality. □

24.5.2 Parametric phase and oriented area

Proposition 24.24 (Area law and transport of angular coordinates)**.** *The parametric phase θ of γ and its polar angle ϕ, continuously lifted from zero, satisfy*

$$\phi(\theta) = \arg(a_S \cos\theta + i b_S \sin\theta), \qquad \frac{d\phi}{d\theta} = \frac{a_S b_S}{a_S^2 \cos^2\theta + b_S^2 \sin^2\theta} > 0. \tag{24.25}$$

In a chart where both tangents are defined, $\tan\phi = (b_S/a_S)\tan\theta$*. The oriented area swept out from phase zero is*

$$\mathcal{A}(\theta) = \frac{1}{2}\int_0^\theta (QP' - PQ')\,dt = \hbar\theta, \qquad \mathcal{A}(1) = \hbar, \quad \mathcal{A}(2\pi) = h_s. \tag{24.26}$$

Proof. The derivative of the argument of a nonzero vector is $(QP' - PQ')/(Q^2 + P^2)$. For the curve (24.24), the numerator equals $a_S b_S = 2\hbar$; this proves the derivative formula and its positivity. The quotient P/Q gives the tangent relation; the continuous lift preserves quadrants and turns. The integral equals $a_S b_S \theta/2 = \hbar\theta$. Finally, $h_s = 2\pi\hbar_s$ gives the identity for the complete turn. □

One radian of parametric phase sweeps an area $\hbar$. The polar angle retains its own derivative in (24.25). In polar coordinates, the equivalent law is

$$d\mathcal{A} = \frac{1}{2}\rho_\partial(\phi)^2\,d\phi, \qquad \rho_\partial(\phi) = \left(\frac{\cos^2\phi}{a_S^2} + \frac{\sin^2\phi}{b_S^2}\right)^{-1/2}.$$

The radial formula follows by substituting $(Q, P) = \rho(\cos\phi, \sin\phi)$ into (24.24). With the orientation of γ, the integral of the canonical one-form retains the sign

$$\oint_\gamma P\,dQ = -\pi a_S b_S = -h_s. \tag{24.27}$$

Indeed, $P\,dQ = -a_S b_S \sin^2\theta\,d\theta$, and the integral of $\sin^2\theta$ over one turn is π.

24.5.3 Shape reciprocity and symplectic orientation

Define the matrices

$$D(\eta) = \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}, \qquad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{24.28}$$

Proposition 24.25 (Quadratic form and action under reciprocity)**.** *Both transports satisfy $T^\mathsf{T} D(-\eta) T = D(\eta)$. For $\omega = dQ \wedge dP$, the interchange S_2 is antisymplectic and the rotation J_2 is symplectic:*

$$S_2^*\omega = -\omega, \qquad J_2^*\omega = \omega.$$

If $\mathcal{I}(\gamma) = \oint_\gamma P\,dQ$ on a closed curve, then

$$\mathcal{I}(S_2\gamma) = -\mathcal{I}(\gamma), \qquad \mathcal{I}(J_2\gamma) = \mathcal{I}(\gamma). \tag{24.29}$$

Proof. Matrix multiplication interchanges the two diagonal entries of D, proving both congruences. The determinants of S_2, J_2 are $-1, 1$; this gives the transformation of the area form. For $\lambda = P\,dQ$, direct calculation gives

$$S_2^*\lambda = d(PQ) - \lambda, \qquad J_2^*\lambda = \lambda - d(PQ).$$

The integral of the exact differential over the closed curve is zero, proving both action identities. □

Orientation distinguishes the two transports even though they express the same quadratic form, including in the circular case. The labelling of the axes and the sign of the action therefore belong to the data preserved in passing to the lattice representation.

24.5.4 Normalized radius and lattice-readout matrix

Proposition 24.26 (Determination and recoverability of the shape radius)**.** *The labelled ellipse determines a unique positive radius*

$$R_\star = \sqrt{\frac{b_S}{a_S}} = e^{-\eta_{\mathrm{el}}/2} = \left(\frac{A - C^*}{A + C^*}\right)^{1/4}, \qquad \tau_{\mathrm{el}} = iR_\star^2. \tag{24.30}$$

The angular ratio is recovered through

$$\frac{C^*}{A} = \frac{1 - R_\star^4}{1 + R_\star^4}, \qquad R_\star^4 = \frac{499\alpha - \eta_{\mathrm{ret}}}{501\alpha + \eta_{\mathrm{ret}}}. \tag{24.31}$$

Interchanging the semiaxes transforms $(\eta_{\mathrm{el}}, R_\star, \tau_{\mathrm{el}})$ *into* $(-\eta_{\mathrm{el}}, R_\star^{-1}, -1/\tau_{\mathrm{el}})$ *and preserves* $a_S b_S = 2\hbar$*.*

Proof. Dividing the semiaxes gives $b_S/a_S = e^{-\eta_{\mathrm{el}}}$. The map $R \mapsto R^2$ is bijective on the positive reals, so the positive root is unique. For $\xi = C^*/A \in (-1, 1)$,

$$2\,\operatorname{artanh}\xi = \log\frac{1+\xi}{1-\xi};$$

hence $R_\star^4 = (1-\xi)/(1+\xi)$. Solving for ξ proves the first identity in (24.31). Substituting $A = 1000\alpha$ and $C^* = 2(\eta_{\mathrm{ret}} + \alpha)$ gives

$$\frac{A - C^*}{A + C^*} = \frac{998\alpha - 2\eta_{\mathrm{ret}}}{1002\alpha + 2\eta_{\mathrm{ret}}} = \frac{499\alpha - \eta_{\mathrm{ret}}}{501\alpha + \eta_{\mathrm{ret}}}.$$

The interchange inverts the semiaxis ratio and changes the sign of η_{el}. The identity $-1/(iR_\star^2) = iR_\star^{-2}$ gives the modular transport. The product of the semiaxes remains fixed. □

On the positive branch, $0 < R_\star < 1$; the chart of the reversed form publishes $R_\star^{-1} > 1$. The angular ratio is recovered from the radius, while the absolute angles and their history remain in the enriched state. The circular form corresponds to $C^*/A = 0$, with $R_\star = 1$. Changing the positive action section while retaining the angular pair changes the common scale of the semiaxes and preserves their ratio.

Proposition 24.27 (Lattice form determined by the ellipse)**.** *The matrix of normalized squared semiaxes is*

$$D(\eta_{\mathrm{el}}) = \frac{1}{2\hbar}\begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix} = \begin{pmatrix} R_\star^{-2} & 0 \\ 0 & R_\star^2 \end{pmatrix}. \tag{24.32}$$

Its evaluation on an integer pair produces

$$E_\star(m, w) = \begin{pmatrix} m & w \end{pmatrix} D(\eta_{\mathrm{el}}) \begin{pmatrix} m \\ w \end{pmatrix} = \frac{m^2}{R_\star^2} + w^2 R_\star^2. \tag{24.33}$$

The equation of the ellipse multiplied by $2\hbar$ *uses, instead, the inverse matrix* $D(-\eta_{\mathrm{el}})$*.*

Proof. The squares of the semiaxes in (24.23), divided by $2\hbar$, are $e^{\eta_{\mathrm{el}}}$ and $e^{-\eta_{\mathrm{el}}}$. By (24.30), these are $R_\star^{-2}$ and $R_\star^2$. Multiplication by the integer vector proves (24.33). Finally, $2\hbar/a_S^2 = e^{-\eta_{\mathrm{el}}}$ and $2\hbar/b_S^2 = e^{\eta_{\mathrm{el}}}$, identifying the matrix of the elliptic equation. □

The radius $R_\star$ is a dimensionless publication of the labelled form. The polar radius ρ_∂ belongs to the square-root-of-action line. Representation by a spatial radius requires its subsequent dimensional chart; this preserves the form already determined and the independence of its construction from a prescribed length.

24.6 T-duality: integer coordinates, change of section, and operator equivalence

The arithmetic decomposition of APP preserves the residual representative and its quotient. TPK evolution also transports its orientation and the memory of boundary crossings. Interchanging the two coordinates requires preserving this distinction and specifying the domain on which the operation is again defined. An arithmetic involution is first constructed on that domain; its quadratic form subsequently admits a representation by self-adjoint operators.

24.6.1 Stable domain of the interchange

In the positive residual section,

$$r_9^+(n) = 1 + ((n-1) \bmod 9), \qquad Q_9^+(n) = \frac{n - r_9^+(n)}{9}, \qquad n = r_9^+(n) + 9Q_9^+(n). \tag{24.34}$$

An interchange can place an arbitrary quotient in the first coordinate. Its stable domain is therefore $\mathscr{D}_0 = \mathbb{Z}^2 \times \mathbb{R}_{>0}$, not merely the section $1 \le r \le 9$. Extending the domain makes it possible to define a total operation; the admissibility of a pair as the readout of a TPK route separately retains its selection rules.

Definition 24.28 (Normalized compact form and interchange)**.** For $d = (m, w, R_0) \in \mathscr{D}_0$, let

$$E_{R_0}(m, w) = \frac{m^2}{R_0^2} + w^2 R_0^2, \qquad \mathsf{T}_0(m, w, R_0) = (w, m, R_0^{-1}). \tag{24.35}$$

The symbols m, w name the two transported integer coordinates; R_0 is the positive dimensionless scale of the form. Their interpretation through momentum, winding, and physical radius is established by a subsequent realization map.

Theorem 24.29 (Involution and preservation of the quadratic form)**.** *The map T_0 is an involution of $\mathscr{D}_0$, and*

$$E_{R_0^{-1}}(w, m) = E_{R_0}(m, w).$$

Proof. Two interchanges restore (m, w), and two inversions restore R_0. Moreover,

$$E_{R_0^{-1}}(w, m) = w^2 R_0^2 + \frac{m^2}{R_0^2} = E_{R_0}(m, w).$$

□

The equivalence holds for each radius and its reciprocal. At $R_0 = 1$, the radius remains fixed and the operation acts on the same form. A specific pair is fixed by the full involution precisely when, in addition, $m = w$.

24.6.2 Compact Hamiltonian and self-adjoint domain

Let $\mathcal{H}_{\mathrm{lat}} = \ell^2(\mathbb{Z}^2)$, with orthonormal basis $\{|m, w\rangle\}$. Define

$$H(R_0)|m, w\rangle = E_{R_0}(m, w)|m, w\rangle, \tag{24.36}$$

$$D(H(R_0)) = \left\{ \psi : \sum_{m,w \in \mathbb{Z}} E_{R_0}(m, w)^2 |\psi_{m,w}|^2 < \infty \right\}. \tag{24.37}$$

Finitely supported vectors belong to the domain and form a core for the diagonal operator. The associated quadratic form has domain

$$D(H(R_0)^{1/2}) = \left\{ \psi : \sum_{m,w} E_{R_0}(m, w) |\psi_{m,w}|^2 < \infty \right\}.$$

Proposition 24.30 (Self-adjointness and compact resolvent)**.** *The operator $H(R_0)$ is self-adjoint and nonnegative. Its eigenvalues are $E_{R_0}(m,w)$, counted with multiplicity, and $(H(R_0)+I)^{-1}$ is compact.*

Proof. A multiplication operator defined by a real sequence is symmetric on (24.37). If η belongs to the domain of the adjoint, testing the adjoint identity against each vector $|m,w\rangle$ forces its image to have coordinates $E_{R_0}(m,w)\eta_{m,w}$. Membership in ℓ^2 is exactly condition (24.37); hence the adjoint has the same domain and the same action. Nonnegativity follows by summing the nonnegative weights.

With $c_{R_0} = \min(R_0^{-2}, R_0^2) > 0$, we have $E_{R_0}(m,w) \geq c_{R_0}(m^2+w^2)$. Outside increasing finite sets, the coefficients $(1+E_{R_0}(m,w))^{-1}$ tend uniformly to zero. Truncating the resolvent to those sets produces finite-rank operators converging in norm, which proves its compactness. The diagonal basis determines the stated spectrum. □

Theorem 24.31 (Unitary conjugation, including domains)**.** *The interchange $U_T|m,w\rangle = |w,m\rangle$ is unitary, $U_T^2 = I$, and satisfies*

$$U_T D(H(R_0)) = D(H(R_0^{-1})), \qquad U_T H(R_0) U_T^{-1} = H(R_0^{-1}). \tag{24.38}$$

It also transports the quadratic-form domains.

Proof. The interchange is an involutive permutation of the orthonormal basis. For a sequence ψ, the identity in Theorem 24.29 gives

$$\sum_{m,w} E_{R_0^{-1}}(m,w)^2 |(U_T\psi)_{m,w}|^2 = \sum_{m,w} E_{R_0}(m,w)^2 |\psi_{m,w}|^2.$$

This gives equality of domains. On each coordinate, $U_T H(R_0) U_T^{-1}$ multiplies by $E_{R_0}(w,m) = E_{R_0^{-1}}(m,w)$. The same sum, with the first power of the weight, proves the result for the quadratic forms. □

The scalar equality and the operator equality are two realizations of the same involution. The second preserves the domains that allow the first to be interpreted as spectral equivalence, not merely as an equality on finitely supported vectors.

24.6.3 Specialization to the radius of the action ellipse

The radius $R_\star$ determined in section 24.5 fixes the realization of this family that corresponds to the angular publications of the state. The symbol R_0 allows the transport results to be expressed for the entire positive family; composition with the ellipse evaluates that family at $R_0 = R_\star$ and also preserves its reciprocal image.

Corollary 24.32 (Elliptic family stable under duality)**.** *The set*

$$\mathscr{D}_{\mathrm{el}} = \mathbb{Z}^2 \times \{R_\star, R_\star^{-1}\}$$

is stable under T_0. Its Hamiltonians are self-adjoint, nonnegative, and have compact resolvent. The interchange U_T transports the domain of $H(R_\star)$ onto that of $H(R_\star^{-1})$, including their form domains.

Proof. The positivity of $R_\star$ and its unique determination are established in section 24.5. Inversion permutes $R_\star$ and $R_\star^{-1}$, while interchange preserves $\mathbb{Z}^2$. Proposition 24.30 and theorem 24.31 apply on these two fibres. If $R_\star = 1$, the two fibres coincide; in every other case, the operator intertwines two distinct fibres of the same family. □

The common action scale determines the area of the ellipse; its semiaxis ratio determines $R_\star$. This separation makes it possible to preserve the area and the common action scale while interchanging the semiaxes and obtaining the reciprocal radius. The index interchange on $\ell^2(\mathbb{Z}^2)$ and the transports of the action plane act on different spaces; their respective properties are proved on their own domains.

24.6.4 Change of representative and quotient memory

The zero section uses $\bar{r} \in \{0, \dots, 8\}$. The change that preserves the integer in (24.34) is

$$C(r, Q) = (\bar{r}, Q + \mathbf{1}_{\{r=9\}}). \tag{24.39}$$

The modification of the quotient is part of the operation. For $n = 9$, the two pairs are $(9, 0)$ and $(0, 1)$. Their uncorrected energies are $81/R_0^2$ and R_0^2. Representing the same integer therefore does not imply equality of those two insertions into a form that distinguishes the coordinates.

The equivalence relation is preserved by defining

$$L_9 = \mathbb{Z}(9, -1), \qquad S(x_1, x_2) = (x_2, x_1), \qquad q_{R_0}(x) = \frac{x_1^2}{R_0^2} + x_2^2 R_0^2.$$

For $x \in \mathbb{Z}^2$, let

$$\mathcal{E}_{R_0}^{L_9}([x]) = \min_{k \in \mathbb{Z}} q_{R_0}(x + k(9, -1)). \tag{24.40}$$

This construction operates on classes. The memory distinguishing individual representatives remains in the TPK state preceding the quotient; the function in (24.40) determines the invariant evaluation of the chosen class.

Theorem 24.33 (Covariant duality of sections)**.** *The energy in* (24.40) *is well defined. The transformation*

$$\mathsf{T}_{\mathrm{cov}}([x]_{L_9}, R_0) = ([Sx]_{SL_9}, R_0^{-1})$$

is a bijection between $(\mathbb{Z}^2/L_9) \times \mathbb{R}_{>0}$ *and* $(\mathbb{Z}^2/SL_9) \times \mathbb{R}_{>0}$*, with inverse given by the second interchange, and*

$$\mathcal{E}_{R_0^{-1}}^{SL_9}([Sx]) = \mathcal{E}_{R_0}^{L_9}([x]).$$

Proof. Replacing x by $x + k_0(9, -1)$ reindexes the candidates for the minimum. The minimum exists: the function of k is quadratic and has leading coefficient $81/R_0^2 + R_0^2 > 0$. The change $(9, Q) \mapsto (0, Q + 1)$ has difference in L_9, so it represents the same class. Interchange transports the equivalence relation to SL_9, and

$$\min_{\ell \in L_9} q_{R_0^{-1}}(Sx + S\ell) = \min_{\ell \in L_9} q_{R_0}(x + \ell).$$

Finally, $S^2 = I$ recovers the initial quotient and radius. □

The minimum can be evaluated exactly using two candidates. If $x = (m, w)$, the real minimum of the quadratic occurs at

$$k_* = \frac{wR_0^4 - 9m}{81 + R_0^4}.$$

The integer minimum is attained at $\lfloor k_* \rfloor$ or $\lceil k_* \rceil$. This formula does not alter the equivalence; it makes its finite evaluation explicit and allows the effect of changing section to be checked without discarding the quotient.

24.6.5 Radius normalization in the string realization

Let $\alpha' > 0$ be a scale of the string realization. This symbol denotes the string scale and is kept distinct from the fine-structure constant α. Define

$$\mathsf{T}_{\alpha'}(m, w, R) = \left(w, m, \frac{\alpha'}{R}\right), \qquad \mathcal{N}_{\alpha'}(m, w, R) = \left(m, w, \frac{R}{\sqrt{\alpha'}}\right). \tag{24.41}$$

Theorem 24.34 (Exact conjugation by the scale)**.** *Normalization satisfies*

$$\mathcal{N}_{\alpha'}\mathsf{T}_{\alpha'} = \mathsf{T}_0\mathcal{N}_{\alpha'}.$$

If

$$E_{\mathrm{str}}(m, w, R) = \frac{m^2}{R^2} + \frac{w^2R^2}{\alpha'^2},$$

then $E_{\mathrm{str}} = \alpha'^{-1}E \circ \mathcal{N}_{\alpha'}$*. The combinations*

$$p_L = \frac{m}{R} + \frac{wR}{\alpha'}, \qquad p_R = \frac{m}{R} - \frac{wR}{\alpha'}$$

transform as $p_L \mapsto p_L$ *and* $p_R \mapsto -p_R$*.*

Proof. Both compositions in the first equality give $(w, m, \sqrt{\alpha'}/R)$. On the other hand,

$$E_{R/\sqrt{\alpha'}}(m, w) = \frac{\alpha' m^2}{R^2} + \frac{w^2R^2}{\alpha'} = \alpha' E_{\mathrm{str}}(m, w, R).$$

Substituting $(w, m, \alpha'/R)$ into the expressions for p_L, p_R gives the two asserted transformations, respectively. □

Conjugation fixes what it means to express the same result in a dimensional chart. In the elliptic family, the radii are

$$R = \sqrt{\alpha'}R_\star, \qquad R^\vee = \sqrt{\alpha'}R_\star^{-1}.$$

Their product is α'. The internal determination of the dimensionless semiaxis ratio and the length-squared scale of the realization serve different functions: the former fixes the shape; the latter expresses its radii in a dimensional unit.

The preceding normalization allows the same result to be expressed with normalized or physical radius. Assigning a tension and a unit to α' belongs to the realization map and has its own origin; it is not a datum entering the production of APP–TRIT–TPK registers.

24.6.6 Relationship with the modular transformation

Radius inversion has a precise representation on the imaginary axis of the upper half-plane. Let $\iota(R_0) = \mathrm{i}R_0^2$ and $\mathsf{S}(\tau) = -1/\tau$. Then

$$\mathsf{S} \circ \iota(R_0) = \iota(R_0^{-1}). \tag{24.42}$$

The equality follows from $-1/(\mathrm{i}R_0^2) = \mathrm{i}R_0^{-2}$. For the modular function $J = j - 744$, whose invariance is preserved in the Moonshine development, it follows that $J(\mathrm{i}R_0^2) = J(\mathrm{i}R_0^{-2})$.

The map ι expresses a semiconjugacy of parameters. The interchange U_T acts on the coordinates of $\ell^2(\mathbb{Z}^2)$; S acts on the upper half-plane; the dual automorphism of an orbifold acts on its graded sectors. Writing their domains allows specific relations between them to be composed while preserving the information of each representation. Theorems 24.31, 24.33 and 24.34 establish, respectively, unitary conjugation, change of residual section and scale normalization, with the declared operator domains.

Chapter 25

Coordination of the Moonshine corridor and the M-theory screens: shared grading, lattice transport and double realization of the HMT state

The integers 10, 11, 24, and 26 arise in the construction through different algebraic procedures: the length of a code, the dimension of a module, the rank of a lattice, and the dimension of an orthogonal quotient. The purpose of this section is to reconstruct those procedures and the maps that relate them. Equality between integers will not suffice to identify the objects they enumerate.

We shall call an *algebraic screen* a subspace or quotient accompanied by the map that selects it. This denomination does not introduce a new class of spaces: it abbreviates the pair formed by the linear object and its projector or quotient map. With this convention, the reductions

$$12 \longrightarrow 11 \longrightarrow 10, \qquad 26 \longrightarrow 24$$

are statements about defined maps, not numerological decompositions. The first chain arises from a representation of A_5 marked by the dodecaphasic vector K; the second, from an isotropic reduction of a Lorentzian lattice. Both rely on the exceptional branch constructed in the preceding sections, but they are not the same mechanism.

25.1 The perfect quotient of 11 coordinates

Let

$$C_W = [12, 6, 6]_3$$

the extended ternary Golay code. Puncturing a coordinate produces the ternary Golay code

$$G_{11} = [11, 6, 5]_3 \subset \mathbb{F}_3^{11}$$

[91]. The length eleven belongs here to the ambient space of the code; it is not introduced through a physical interpretation.

Definition 25.1 (Ternary ball)**.** For $x \in \mathbb{F}_3^n$ and $t \geq 0$, the Hamming ball of radius t centered at x is

$$B_t(x) = \{y \in \mathbb{F}_3^n : d_H(x, y) \leq t\}.$$

Its cardinality, independent of x, is

$$V_3(n, t) = \sum_{j=0}^{t} \binom{n}{j} 2^j.$$

Theorem 25.2 (Perfection of G_{11})**.** *The Hamming balls of radius two centered at the words of G_{11} constitute a partition of $\mathbb{F}_3^{11}$. In particular,*

$$V_3(11, 2) = 243, \qquad |G_{11}|\, V_3(11, 2) = 3^{11}.$$

Proof. The minimum distance of G_{11} is five. If two balls of radius two, centered at distinct words $c, c' \in G_{11}$, had a common element y, the triangle inequality would give

$$d_H(c, c') \leq d_H(c, y) + d_H(y, c') \leq 4,$$

in contradiction with $d_H(c, c') \geq 5$. The balls are therefore disjoint. Their cardinality is

$$V_3(11,2) = \binom{11}{0} + 2\binom{11}{1} + 2^2\binom{11}{2} = 1 + 22 + 220 = 243 = 3^5.$$

Since G_{11} has dimension six, it contains 3^6 words. In consequence,

$$3^6 V_3(11,2) = 3^6 \cdot 3^5 = 3^{11} = |\mathbb{F}_3^{11}|.$$

The disjoint union of the balls has the cardinality of the ambient space and therefore coincides with it. □

Perfect decoding assigns to each received word a unique codeword at distance at most two. Equivalently, the syndrome space has $3^{11-6} = 243$ elements. This occurrence of 243 must be distinguished from the cubic layer

$$3 \cdot 9^2 = 243$$

of the completion $1000 = 729 + 243 + 27 + 1$. One quantity enumerates syndromes of a code; the other enumerates positions in a boundary layer. Equality of cardinalities authorizes a comparison, but does not provide a canonical bijection between the two realizations.

It is also convenient to separate the metric partition

$$243 = 1 + 22 + 220$$

of the typed identity

$$243 = 1 + 22 + 90 + 120 + 10.$$

In the second expression, $90 = 6\binom{6}{2}$ and $120 = 8\binom{6}{2}$ count families of incidences in the hexad–octad extension, whereas 10 will be the dimension of a linear screen. They are not subsets of weights $0, 1, 2$ of the same Hamming ball. The construction of designs underpinning that extension belongs to the classical theory of Witt systems [1].

25.2 Dimensional reduction 26→24→11→10 and coordination with M-theory screens

Length eleven is not the only rank that descends from C_W. The same code intervenes in the gluing of the root lattice A_2^{12}, whose rank is twenty-four. After fixing the chamber data and the neighborhood vector constructed in the exceptional branch, the index-three rootless neighbor is the Leech lattice Λ_{24}. The dependency structure is

$$\begin{aligned} C_W &\xrightarrow{\text{puncturing}} G_{11} \subset \mathbb{F}_3^{11}, \\ C_W &\xrightarrow{\text{gluing}} N(A_2^{12}) \xrightarrow{\text{neighbor with index 3}} \Lambda_{24}. \end{aligned} \tag{25.1}$$

The first arrow is a code morphism; the second line belongs to lattice theory. The common ternary source explains the coexistence of ranks eleven and twenty-four without identifying them. In particular, 11 is not a subdimension of 24, nor is 24 obtained by completing the punctured code.

Proposition 25.3 (Range bifurcation)**.** *The construction* (25.1) *produces two invariants of different types:*

$$\operatorname{length}(G_{11}) = 11, \qquad \operatorname{rank}(\Lambda_{24}) = 24.$$

Their common provenance is the code C_W; neither equality implies an isomorphism between the coordinate space of the code and the real space generated by the lattice.

Proof. Puncturing removes one of the twelve positions and does not alter the code's dimension six, whence length eleven. Each copy of A_2 has rank two; therefore, A_2^{12}, its finite-index gluings, and its full-rank neighbors have rank twenty-four. The objects live, respectively, in the categories of codes over $\mathbb{F}_3$ and integral lattices. An isomorphism between them would not even be well typed without an additional functor. □

25.3 Coordination of exact screens with the physical realization of M theory

Reduction $12 \to 11 \to 10$ proceeds from a second realization of the number twelve. Let $V_{12} = \mathbb{R}^{12}$ be the permutation module of A_5 on the twelve vertices of an icosahedron. Denote this representation by ρ and its character by χ_{12}. To fix unambiguously the correspondence with the twelve dodecaphasic positions, we realize the vertices as left cosets A_5/H, where A_5 acts on $\{0,1,2,3,4\}$ and

$$H = \langle (0\ 1\ 2\ 3\ 4) \rangle \cong C_5.$$

The position j is identified with the coset $r_j H$, for the following representatives, written in image notation:

j	r_j	j	r_j	j	r_j
1	$(3,2,4,1,0)$	5	$(3,2,0,4,1)$	9	$(3,1,2,4,0)$
2	$(3,1,0,2,4)$	6	$(4,0,1,2,3)$	10	$(2,1,4,3,0)$
3	$(0,1,3,4,2)$	7	$(4,3,2,1,0)$	11	$(3,4,1,2,0)$
4	$(1,4,2,3,0)$	8	$(0,2,3,1,4)$	12	$(4,2,1,3,0)$

This table is the orientation datum that allows a projector of A_5 to be applied to the vector K; without it, a coordinate norm would depend on an unspecified permutation. In the order of conjugacy classes

$$1A, \quad 2A, \quad 3A, \quad 5A, \quad 5B,$$

the sizes of the classes and the number of fixed points are

	$1A$	$2A$	$3A$	$5A$	$5B$
$\lvert C\rvert$	1	15	20	12	12
χ_{12}	12	0	0	2	2.

Indeed, rotations of orders two and three fix no vertices, whereas each rotation of order five fixes the two vertices of its axis. Write

$$\phi = \frac{1+\sqrt{5}}{2}, \qquad \phi' = \frac{1-\sqrt{5}}{2}, \qquad \phi + \phi' = 1.$$

The relevant part of the character table of A_5 is

	$1A$	$2A$	$3A$	$5A$	$5B$
χ_1	1	1	1	1	1
χ_3	3	-1	0	ϕ	ϕ'
$\chi_{3'}$	3	-1	0	ϕ'	ϕ
χ_4	4	0	1	-1	-1
χ_5	5	1	-1	0	0.

Theorem 25.4 (Decomposition of the Icosahedral Module)**.** *The permutation representation on the twelve vertices decomposes as*

$$V_{12} \cong V_1 \oplus V_3 \oplus V_{3'} \oplus V_5.$$

The subspace orthogonal to the uniform vector is

$$V_{11} := \mathbf{1}^{\perp} \cong V_3 \oplus V_{3'} \oplus V_5$$

and has dimension eleven.

Proof. The multiplicity of an irreducible character χ in χ_{12} is

$$\langle \chi_{12}, \chi \rangle = \frac{1}{60} \sum_{C} |C| \chi_{12}(C) \chi(C).$$

Only $1A, 5A, 5B$ contribute. For the characters of dimensions one, three, three-prime, four, and five one obtains, respectively,

$$\frac{12 + 24 + 24}{60} = 1,$$

$$\frac{36 + 24(\phi + \phi')}{60} = 1, \qquad \frac{36 + 24(\phi' + \phi)}{60} = 1,$$

$$\frac{48 - 24 - 24}{60} = 0, \qquad \frac{60}{60} = 1.$$

Therefore, the multiplicities are $(1, 1, 1, 0, 1)$. The only vector fixed by all permutations is the uniform vector $\mathbf{1} = (1, \dots, 1)$. Its orthogonal complement contains the three nontrivial summands and has dimension $3 + 3 + 5 = 11$. □

The orthogonal projector onto V_{11} is explicit:

$$P_{11} = I_{12} - \frac{1}{12} J_{12},$$

where J_{12} is the matrix all of whose entries are one. Thus, the first reduction is the map

$$\mathbb{R}^{12} \longrightarrow V_{11}, \qquad x \longmapsto x - \frac{1}{12} \Big(\sum_{j=1}^{12} x_j \Big) \mathbf{1}.$$

This formula specifies which information is eliminated: exclusively the uniform component. The equality $12 = 1 + 11$ is a consequence of the decomposition of the module, not a dimensional convention.

25.4 Polarization of the three-dimensional sector determined by K

Consider the already obtained dodecaphasic vector,

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \qquad \sum_{j=1}^{12} K_j = 6263.$$

Its zero-sum component is

$$\widetilde{K} = P_{11} K = K - \frac{6263}{12} \mathbf{1}.$$

To select one of the two three-dimensional representations, we use the position–coset bijection fixed in the preceding table. The corresponding central projector is written

$$P_{\mathrm{ico},3} = \frac{3}{60} \sum_{g \in A_5} \chi_3(g^{-1}) \rho(g).$$

The sum contains permutation matrices and coefficients in $\mathbb{Q}(\sqrt{5})$; consequently, $P_{\mathrm{ico},3}$ is an exact operator over that field, satisfies $P_{\mathrm{ico},3}^2 = P_{\mathrm{ico},3} = P_{\mathrm{ico},3}^{\mathsf{T}}$, and has rank three.

Definition 25.5 (Polarizing Direction)**.** The icosahedral projection of the vector and the line that generates it are

$$u_K = P_{\mathrm{ico},3}\widetilde{K}, \qquad \ell_K = \mathbb{R}u_K.$$

Lemma 25.6 (No polarization degeneracy)**.** *For the vector K and the declared position–coset correspondence,*

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

In particular, ℓ_K is a well-defined line.

Proof. Upon substituting the expression for $P_{\mathrm{ico},3}$ into $\|u_K\|^2 = \widetilde{K}^{\mathsf{T}} P_{\mathrm{ico},3}\widetilde{K}$, the sum reduces to

$$\frac{3}{60}\sum_{g\in A_5} \chi_3(g^{-1})\, \widetilde{K}^{\mathsf{T}}\rho(g)\widetilde{K}.$$

Each scalar product is rational, since $\rho(g)$ is a permutation matrix and $\widetilde{K} \in \mathbb{Q}^{12}$. The separation of the two classes of order five leaves as the only irrational part a multiple of $\sqrt{5}$. The exact sum of the sixty terms is

$$\frac{6638585}{20} + \frac{2275584}{20}\sqrt{5}.$$

Both coefficients are positive and $\sqrt{5} > 0$; therefore, the norm does not vanish. □

Exact enumeration of the group and the twelve lateral classes reconstructs the projector onto $\mathbb{Q}(\sqrt{5})$, verifies its idempotence, and produces the preceding norm.

Theorem 25.7 (Polarized reduction $11 \to 10$)**.** *Let*

$$V_{10} := (V_3 \cap \ell_K^{\perp}) \oplus V_{3'} \oplus V_5.$$

Then

$$V_{11} = \ell_K \oplus V_{10}, \qquad \dim V_{10} = 10.$$

The projector onto the ten-dimensional screen is

$$P_{10} = P_{11} - \frac{u_K u_K^{\mathsf{T}}}{\|u_K\|^2}.$$

Proof. By the lemma 25.6, $u_K \neq 0$ and, by definition, $u_K \in V_3$. Hence

$$V_3 = \ell_K \oplus (V_3 \cap \ell_K^{\perp}).$$

Al anadir the sumandos orthogonal $V_{3'}$ and V_5, the theorem 25.4 gives

$$V_{11} = \ell_K \oplus \big((V_3 \cap \ell_K^{\perp}) \oplus V_{3'} \oplus V_5\big).$$

The dimension of the second summand is $2 + 3 + 5 = 10$. The formula for P_{10} subtracts from P_{11} the rank-one orthogonal projector onto ℓ_K, so its image is precisely V_{10}. □

Corollary 25.8 (Exact Decomposition 1 + 3 + 7)**.** *Let*

$$W_7(K) := (V_3 \cap \ell_K^{\perp}) \oplus V_5.$$

Then

$$\boxed{V_{11} = \ell_K \oplus V_{3'} \oplus W_7(K), \qquad \dim(\ell_K, V_{3'}, W_7) = (1, 3, 7),}$$

and

$$\boxed{V_{10} = V_{3'} \oplus W_7(K), \qquad 10 = 3 + 7.}$$

Proof. $\ell_K \subset V_3$ is a line, so that $\dim(V_3 \cap \ell_K^{\perp}) = 2$. Therefore, $\dim W_7 = 2 + 5 = 7$, and the two decompositions follow from the theorem 25.7. □

Remark 25.9 (Condition additional for the reading 1 + 3 + 6). The exact decompositions 1 + 3 + 7 and 3 + 7 do not by themselves select a time, an extended three-dimensional space, or a compact sector of rank six. A physical realization must choose a line $\ell_t \subset W_7(K)$, prove

$$W_7(K) = \ell_t \oplus W_6,$$

construct a full-rank lattice $\Lambda_6 \subset W_6$ that is stable under transport, interpret W_6/Λ_6 as the compact sector, and exhibit an intertwiner between the TRIT temporal reader, W_6, and that lattice. The same transport must separately preserve the three extended directions of the summand $V_{3'}$, without mixing them with ℓ_t or with the compact quotient. Only under those hypotheses are the decompositions obtained

$$\boxed{V_{11} = \ell_K \oplus \ell_t \oplus V_{3'} \oplus W_6, \qquad 11 = 1 + 1 + 3 + 6,}$$

$$\boxed{V_{10} = \ell_t \oplus V_{3'} \oplus W_6, \qquad 10 = 1 + 3 + 6.}$$

The absence of a natural choice of ℓ_t, stability of Λ_6, full rank, separation of the three extended directions, or the intertwiner would refute this realization, without altering the algebraic corollary 1 + 3 + 7.

The reduction is not A_5-equivariant in the absence of the marked datum. The submodule V_3 is irreducible and contains no A_5-invariant line; the direction ℓ_K appears after introducing K. The sequence

$$12 \longrightarrow 11 \longrightarrow 10$$

therefore has two distinct natures: the passage $12 \to 11$ is canonical for the permutation action, whereas $11 \to 10$ is a polarization relative to the vector K and to the declared position–lateral-class correspondence.

25.5 Operatorial representation of APP and supersymmetric spectral triple in dimension 11

Let U be the integral hyperbolic plane with basis e_+, e_- and Gram matrix

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

It is an even, unimodular lattice of signature $(1, 1)$. The lattice

$$L_{26} := \Lambda_{24} \oplus U$$

is even, unimodular, and of signature $(25, 1)$; by the classification of even unimodular indefinite lattices, it is identified with $II_{25,1}$ [34].

The adjective *Lorentzian* is used in an algebraic sense: it indicates the presence of a negative direction in the bilinear form. The terminology comes from Lorentz and Minkowski geometry [81, 89]; it does not by itself identify the lattice with a physical spacetime.

Theorem 25.10 (Null reduction)**.** *For the primitive isotropic vector $e_+ \in U$, there exists a canonical isomorphism, once the decomposition $L_{26} = \Lambda_{24} \oplus U$ has been fixed,*

$$e_+^\perp / \mathbb{Z}e_+ \cong \Lambda_{24}.$$

Proof. Every element of L_{26} can be written uniquely as $x = \lambda + ae_+ + be_-$, with $\lambda \in \Lambda_{24}$ and $a, b \in \mathbb{Z}$. Since the sum is orthogonal and $\langle e_-, e_+ \rangle = 1$,

$$\langle x, e_+ \rangle = b.$$

Therefore,

$$e_+^\perp = \Lambda_{24} \oplus \mathbb{Z}e_+.$$

The quotient by $\mathbb{Z}e_+$ removes the second summand and preserves the first, thereby defining the asserted isomorphism. □

The dimension decreases by two because the orthogonality condition eliminates one direction and the quotient eliminates the remaining null direction. This is not an orthogonal projection onto a positive complement: an isotropic line is contained in its own orthogonal complement. This is the algebraic reason why the correct procedure is $e_+^\perp / \mathbb{Z}e_+$.

The matrix of the plane U formally coincides with the matrix that exchanges the two sectors of bidirectional transport. Adopting it as a Gram form establishes a concrete lattice realization of that involutor. The reduction theorem 25.10 is then exact within that realization; the matrix coincidence, by itself, does not identify the lattice with a physical spacetime.

This ambient space is also the one used in the Borcherds construction: the central-charge twenty-four sector is combined with the rank-two Lorentzian plane before the reduction leading to the Monster Lie algebra [15]. The relation relevant here is the explicit lattice quotient; vertex operator algebra theory follows the classical corridor described above.

25.6 Icosahedral McKay correspondence and type E_8

Three occurrences of the integer eight are involved near this boundary:

$$8_{\mathrm{oct}} = |O_H|, \qquad V_8 := V_{3'} \oplus V_5, \qquad \mathrm{rank}(E_8) = 8.$$

The first is the cardinality of an octad in the Witt system; the second is a real module of A_5; the third is the rank of a root system. They are not instances of a single object. The relevant classical bridge passes through the binary icosahedral group:

$$A_5 \longleftarrow 2A_5 \subset SU(2) \xrightarrow{\text{McKay}} \widetilde{E}_8$$

[88]. In the McKay correspondence, the vertices of the affine diagram encode irreducible representations of $2A_5$, and the edges describe the decomposition of the tensor product with the natural two-dimensional representation. This construction explains why the icosahedron and E_8 are related; it does not identify a Witt octad with the module V_8, or either of them with the lattice E_8.

Proposition 25.11 (Categorical separation of three value-eight invariants)**.** *No equality among the three preceding invariants defines an interlinker by itself. To relate two of them one needs, respectively, a map*

from incidence sets to modules, from modules of A_5 to modules of $2A_5$, or from representations to lattice data.

Proof. The cardinality of a set, the dimension of a vector space, and the rank of a lattice are invariants in different categories. An isomorphism in one of them is not defined for objects in the others. The McKay correspondence provides one of the intermediate functors, but not the remaining ones; the coincidence of the integer eight is therefore insufficient. □

25.7 Dimensional reduction and the context of M-theory

The preceding constructions produce two exact occurrences of eleven. The first is the length of the perfect code G_{11}; the second is the dimension of the active module $V_{11} = \mathbf{1}^{\perp}$. These objects are not identified: one is defined over $\mathbb{F}_3$ and the other over $\mathbb{R}$. Reduction to ten belongs exclusively to the latter and depends on the marked line ℓ_K.

Eleven-dimensional supergravity and the network of dualities that led to the formulation of M-theory provide the physical context in which an eleventh-dimensional direction can be compactified and produce ten-dimensional theories [26, 64, 135, 124]. The dimensional correspondence suggested by the HMT construction is organized in the following dictionary:

Objeto algebraico	**Dato requerido en una realización física**
$V_{11} = \ell_K \oplus V_{10}$	una dirección compacta y una pantalla de diez dimensiones con un mapa de campos definido
$(m, w, R_0) \mapsto (w, m, R_0^{-1})$	sectores de momento y enrollamiento, radio físico y escala de cuerda
$\Lambda_{24} \oplus U \cong II_{25,1}$	una interpretación dinámica del sector reticular lorentziano
$C_W \to G_{11}$ y $C_W \to \Lambda_{24}$	un entrelazador que preserve incidencia, acción y espectro

In eleven-dimensional supergravity, a complete realization must include, at least, the metric field g_{MN}, the three-form C_{MNP}, the gravitino ψ_M, its action, and its supersymmetry transformations. For the screen $V_{11} = \ell_K \oplus V_{10}$ to represent a physical compactification, a map is required that sends those fields to data on ℓ_K and V_{10} and that preserves the dynamics and the low-energy limit. None of those fields enters as a hypothesis in the theorems 24.2, 25.2, 25.7 y 25.10.

Consequently, the interface has two clearly differentiated levels. At the algebraic level, perfection of the code, the decomposition $12 = 1 + 3 + 3' + 5$, the polarization $11 = 1 + 10$, the isotropic reduction $26 \to 24$, and the radius involution are proved. At the physical level, these structures provide a domain and compatibility equations for constructing a dictionary with M-theory. The second task adds fields and dynamics; it neither modifies nor is used to prove the first level.

25.8 Quotient Diagrams and Rank Reductions

The rank diagram can be summarized without losing its types:

$$\begin{array}{ccc} & C_W = [12,6,6]_3 & \\ \swarrow & & \searrow \\ G_{11} \subset \mathbb{F}_3^{11} & & N(A_2^{12}) \longrightarrow \Lambda_{24} \end{array}$$

$$\begin{array}{c} V_{12} = \mathbb{R}^{12} \\ \downarrow P_{11} \\ V_{11} = V_3 \oplus V_{3'} \oplus V_5 \\ \downarrow P_{10}(K) \\ V_{10} \end{array}$$

$$\begin{array}{c} II_{25,1} = \Lambda_{24} \oplus U \\ \downarrow \ e_+^{\perp}/\mathbb{Z}e_+ \\ \Lambda_{24}. \end{array}$$

The arrow P_{11} eliminates the uniform mode; $P_{10}(K)$ eliminates the direction selected by K; the null reduction eliminates an isotropic pair by means of orthogonal complement and quotient. The involution T acts on an enlarged lattice domain and relates reciprocal radii. These are the mathematical mechanisms supporting the dimensional interface. Their explicit formulation makes it possible to distinguish precisely the rank theorems from the physical interpretations that may be built upon them.

25.9 From exact screens to the physical interface of M-theory

The first part constructed, in explicit mathematical domains, the reductions

$$12 \longrightarrow 11 \longrightarrow 10, \qquad 26 \longrightarrow 24,$$

and the moment, winding, and radius involutions

$$(m, w, R) \longleftrightarrow (w, m, R^{-1}).$$

These results are not yet a physical theory of strings nor an action of supergravity. They constitute the screens and equivalences that a dimensional realization must respect.

The interface is typed via a map declared

$$\mathcal{R}_M : \mathcal{X}_{\mathrm{HMT}}^{\mathrm{enr}} \longrightarrow \mathcal{X}_{11}^{\mathrm{phys}},$$

whose codomain must carry, at a minimum, the metric field g_{MN}, the three-form C_{MNP}, the gravitino ψ_M, its action, and the supersymmetry transformations. A specialization to strings further requires identification of the physical radius, tension, momentum and winding sectors, and the state space on which T-duality is realized. A conformal or CFT reading requires its own operator algebra, grading, and state–operator correspondence.

The positive connection with the exceptional branch is not a bare arrow Monster→M-theory. Both prolongations preserve data from the same enriched state—phase, grading, orientation, sewing, and memory— but employ distinct functors. In the exceptional branch, those data reach $V^\natural$, the Monster, and Moonshine; in the dimensional branch they feed the screens and, only through $\mathcal{R}_M$, a possible realization of fields and strings.

Definition 25.12 (Internal Admissibility of a Compactification)**.** A candidate compactification is represented by a triple

$$(x_\infty, \kappa, \mathcal{U}),$$

where $x_\infty \in \mathcal{X}_{\mathrm{HMT}}^{\mathrm{enr}}$ is a surviving section,

$$\kappa : \mathcal{X}^{\mathrm{enr}}_{\mathrm{HMT}} \dashrightarrow \mathcal{M}$$

is the possibly partial map into a declared moduli space $\mathcal{M}$, and $\mathcal{U}$ transports metric, orientation, register and boundary data. We define

$$\mathrm{Adm}_{\mathrm{HMT}}(x_\infty, \kappa, \mathcal{U}) = 1$$

if the following four conditions hold:

1. the section has a compatible extension to every depth;
2. the declared dualities are compatible at the point considered:

$$\kappa(\mathsf{T}_{\mathrm{HMT}} x_\infty) = \mathsf{T}_{\mathcal{M}}(\kappa(x_\infty)),$$

   when both sides are defined;
3. the spectrum descends to the compactification quotient;
4. the observables preserve their domains, codomains, and types under $\mathcal{U}$.

This predicate is an internal admissibility criterion and not an automatic identification with the physical *Swampland*. The distance, tower, weak-gravity, and absence-of-global-symmetries criteria, as well as the geometric conditions of a specific compactification, are subsequently evaluated on $\mathrm{im}\,\kappa$ and require the corresponding metric maps, fields, and scales. The preceding definition fixes the interface that a realization must preserve; it does not by itself prove the existence of a physical compactification or a Swampland conjecture.

Status of the interface

The screens, the quotients, and the T-dual involution are exact mathematical results. The complete dictionary toward $(g_{MN}, C_{MNP}, \psi_M)$, the action, supersymmetry, and a physical CFT is a typed interface that must be verified in each realization. This boundary neither diminishes the mathematical result nor authorizes its presentation as already completed physical dynamics.

25.10 Polyakov Action, T Duality, and Compactification of the HMT State in M-Theory Branes

25.10.1 Extended persistence and world surface

A physical string is an extended one-dimensional object whose evolution sweeps out a worldsheet. Each point of the string has an internal coordinate, and each instant adds a temporal coordinate; their combination forms the domain on which the induced metric, tension, and background fields live. A closed string has a circular spatial section. An open string has endpoints subject to boundary conditions. Its vibrations organize states, and its tension converts the area traversed by the worldsheet into action.

A brane of dimension p is an extended object whose history forms a worldvolume of dimension $p + 1$. Branes receive string endpoints, carry fields on their volume, and may wrap cycles of a compact space. This definition fixes three different data: the dimension of the object, the dimension of its history, and the conditions governing its boundary. The tension, charges, and spectrum belong to the dynamical realization that completes those geometric data.

HMT reaches this boundary from an architecture of persistent routes. A route preserves the orientation, sheet, phase, quotient, carry, and memory of the crossings that have prolonged it. The passage to an extended object assigns that genealogy to a section distributed over an internal coordinate. The composition of routes is then effected by sewing segments; the boundary memory records windings and intersections; the action measures the complete persistence over the traversed surface. This map preserves the causal order: first the enriched state is constructed, and its realization as a string, brane, or field is declared thereafter.

The Polyakov action provides a precise form for the physical codomain:

$$S_{\mathrm{P}}[X,h] = -\frac{T}{2}\int d^2\sigma\,\sqrt{-h}\,h^{ab}g_{MN}(X)\partial_a X^M\partial_b X^N.$$

Here T is the tension, h_{ab} the worldsheet metric and X^M the map into the target space. An HMT string realization must produce these objects, their state space and their symmetries from a typed map; the route structure supplies the genealogical domain on which that map acts.

25.10.2 Exact screens in dimensions 11, 10, 26 and 24

The first screen starts from the extended ternary code $C_W = [12,6,6]_3$. Puncturing one coordinate produces the code $G_{11} = [11,6,5]_3$. Its Hamming balls of radius two have cardinality

$$V_3(11,2) = 1 + 22 + 220 = 243 = 3^5,$$

and the 3^6 codewords satisfy

$$3^6 V_3(11,2) = 3^{11}.$$

Therefore, these balls partition the ambient space exactly. The integer eleven here measures the length of a perfect code and retains its combinatorial type.

A second screen uses the real permutation module V_{12} associated with the twelve icosahedral vertices. The complement of the uniform vector decomposes as

$$V_{11} = V_3 \oplus V_{3'} \oplus V_5.$$

The dodecaphase state K polarizes a line $\ell_K \subset V_3$ and determines the projector

$$P_{10} = P_{11} - \frac{u_K u_K^{\mathsf{T}}}{\|u_K\|^2}.$$

From it descend the exact decompositions.

$$V_{11} = \ell_K \oplus V_{10}, \qquad V_{10} = V_{3'} \oplus W_7(K),$$

with dimensions $11 = 1 + 3 + 7$ and $10 = 3 + 7$. The step $12 \to 11$ eliminates the uniform mode; the step $11 \to 10$ eliminates the direction marked by K. A physical compactification adds a temporal line, a stable rank-six lattice, and the transport that preserves the three extended directions.

The third screen starts from the lattice

$$L_{26} = \Lambda_{24} \oplus U \cong II_{25,1},$$

where U is the integral hyperbolic plane. For a primitive isotropic vector e_+, the reduction

$$e_+^{\perp}/\mathbb{Z}e_+ \cong \Lambda_{24}$$

is exact. The orthogonality condition removes one direction and the quotient removes the null direction contained in its orthogonal complement; the result is rank twenty-four. Code, real module, and lattice belong to distinct categories. Their explicit maps make it possible to relate their ranks while preserving the provenance of each.

The incidence between the six faces and the eight vertices of the cube also provides a screen of spatial and temporal signature. If M is its incidence matrix, then

$$MM^{\mathsf{T}} = 12P_u + 4P_-.$$

The quotient by the kernel has rank four and decomposes as $1 \oplus 3$. Upon declaring the uniform line to be the temporal direction, the normalized form becomes $\mathrm{diag}(-1, 1, 1, 1)$. The incidence proves the rank and the spectral ratio; the declared orientation fixes the signature.

25.10.3 Moment, winding and reciprocal radius

In a compact coordinate of radius R, the integer m records momentum and the integer w records winding. The form

$$E_R(m, w) = \frac{m^2}{R^2} + w^2 R^2$$

has the involution

$$\mathsf{T}(m, w, R) = (w, m, R^{-1}), \qquad E_R(m, w) = E_{R^{-1}}(w, m).$$

In $\ell^2(\mathbb{Z}^2)$, the exchange $U_T|m, w\rangle = |w, m\rangle$ provides the unitary equivalence

$$U_T H(R) U_T^{-1} = H(R^{-1}).$$

This is the algebraic residence of radius duality: momentum and winding are two reciprocal readings of the same lattice sector. The realization as T-duality incorporates the string scale, the oscillators and the action on the physical state space. At the self-dual radius both readings share the same scale and the exchange acts within the same spectrum.

25.10.4 Interface in dimension 11: spectral triple, supersymmetry and operator $Q^2 = H$

M-theory organizes an eleven-dimensional physical codomain with metric g_{MN}, three-form C_{MNP}, gravitino ψ_M, action, and supersymmetry transformations. The bridge is written as a map

$$\mathfrak{R}_M : \mathcal{X}^{\mathrm{enr}}_{\mathrm{HMT}} \longrightarrow \mathcal{X}^{\mathrm{phys}}_{11}.$$

Its source preserves phase, orientation, sewing and memory; its codomain adds the fields and the eleven-dimensional dynamics. The compactification of one direction must transport these fields to a ten-dimensional screen, preserve the low-energy limit and intertwine the declared dualities.

A candidate compactification is typed by a triple $(x_\infty, \kappa, \mathcal{U})$. Section x_∞ represents a prolongable persistence; κ carries it to a moduli space; $\mathcal{U}$ transports metric, orientation, register, and boundary. Its admissibility requires prolongation to every depth, commutation with duality, descent of the spectrum to the quotient, and preservation of the domains and codomains of the observables. These conditions convert an analogy of ranks into a verifiable mathematical problem.

Strings appear after compactifying the eleven-dimensional sector; branes appear as extended realizations coupled to the three-form and its duals. The exceptional branch that reaches Λ_{24}, $V^\natural$, and the Monster preserves the same enriched source through another functor. The dimensional and exceptional interfaces are distinct prolongations of a common genealogy, each with its own codomain.

Result

The perfection of G_{11}, the polarization $11 \to 10$, the isotropic reduction $26 \to 24$, the cubic screen of signature $(3, 1)$ and the radius involution are proven algebraic results. Together they form the exact residence of ranks, quotients and dualities on which the interface toward strings, branes and M-theory is built.

Physical realization

An extended history preserves position, winding, orientation, and boundary within a single persistence. Compactification determines which directions appear as macroscopic extension and which act as

internal degrees. The string reads that persistence on a worldsheet; the brane reads it on a worldvolume; M-theory coordinates both readings through eleven-dimensional fields.

Boundary of validity

The realization is falsified if the map $\mathfrak{R}_M$ alters the types of the screens, if the duality ceases to act unitarily on the spectrum, if the compactification loses prolongation, or if the fields fail to satisfy the action and supersymmetry. Physical closure requires tension, oscillators, amplitudes, controlled anomalies, and a coherent gravitational limit for each declared compactification.

25.11 operatorial extension of APP and spectral triad of the supersymmetric interface in dimension 11

The screens $11 \to 10$, the isotropic reduction $26 \to 24$, and the involution between momentum and winding provide the exact algebraic support of the interface. The physical dynamics adds a state space, a Hamiltonian, and a supercharge. These data are assembled into the triple

$$\mathfrak{M} = \big(\mathcal{H}_{\text{phys}}, H, Q\big),$$

where $\mathcal{H}_{\text{phys}}$ is a $\mathbb{Z}_2$-graded Hilbert space,

$$\mathcal{H}_{\text{phys}} = \mathcal{H}_+ \oplus \mathcal{H}_-,$$

H is a nonnegative self-adjoint operator compatible with the transports, and Q is an odd operator. The supercharge relation is

$$Q^2 = H, \qquad [H, Q] = 0. \tag{25.2}$$

The equality implies that Q exchanges the graded sectors and that the Hamiltonian preserves each of them.

25.11.1 Operatorial extension of the APP–TRIT–TPK state toward the triad $(\mathcal{H}_{\text{phys}}, H, Q)$

Enriched APP simultaneously preserves the residues and quotients of the additive and multiplicative sheets:

$$\mathcal{A}_9^\sharp = \big(R^+, Q^+; R^\times, Q^\times\big).$$

TRIT adds temporal orientation and the Topological Phase Kernel preserves phase, route, carry, boundary, and memory. Let $\mathcal{H}_{\text{HMT}}$ be the linear space generated by these enriched states. A compatible operatorial realization is a map

$$W : \mathcal{H}_{\text{HMT}} \longrightarrow \mathcal{H}_{\text{phys}}$$

which preserves the grading, transports the residue and quotient observables, and intertwines the bidirectional involution with radius duality. For the operators defined on both domains, compatibility requires

$$W\, U_T^{\text{HMT}} = U_T^{\text{phys}}\, W, \qquad W\, H_{\text{HMT}} = H\, W.$$

This arrow does not identify the two spaces by decree: it specifies the datum that the genealogy must preserve when passing from the discrete construction to its physical realization.

25.11.2 Compaction and undecadimensional physical codomain

The realization in dimension 11 has as codomain a space

$$\mathcal{X}_{11}^{\mathrm{phys}}$$

equipped with a metric g_{MN}, a 3-form C_{MNP}, a gravitino ψ_M, an action, and supersymmetry transformations. The gluing ℓ_K produces the exact decomposition

$$V_{11} = \ell_K \oplus V_{10},$$

and the compactification must induce on $\mathcal{H}_{\mathrm{phys}}$ the corresponding spectral decomposition. The duality T also requires

$$U_T H(R) U_T^{-1} = H(R^{-1}).$$

The string worldsheet and brane worldvolumes then provide the extended realizations of the same genealogical antecedent.

25.11.3 Spectral compatibility of $(\mathcal{H}_{\mathrm{phys}}, H, Q)$ with $Q^2 = H$, T-duality, and $11 \to 10$ compactification

A realization satisfies the interface when it jointly preserves:

1. the grading of $\mathcal{H}_{\mathrm{phys}}$ and the identity $Q^2 = H$;
2. the Polyakov action and the boundary conditions of the worldsheet;
3. the polarization $11 \to 10$ and the isotropic reduction $26 \to 24$;
4. the exchange of momentum and winding under $R \leftrightarrow R^{-1}$;
5. the descenso of the campos g_{MN}, C_{MNP} and ψ_M to the limite compactificado;
6. the phase, route, orientation, carry, and boundary memory of the APP–TRIT–TPK state.

The screen identities and the radius involution are exact in their algebraic domains. The ternary $(\mathcal{H}_{\mathrm{phys}}, H, Q)$, the action, and the eleven-dimensional fields fix, separately and explicitly, the data of the physical interface.

25.12 Boundary systems, dimensional range, and action functional

In Dimensional Mechanics, dimension is obtained from the number of independent channels available at a boundary. It is not appended as an external label to the state space: it is computed as the rank of a module of prolongations subject to its relations. A change in dimension then corresponds to a variation in rank during transport.

25.12.1 Constraint systems and boundary data

Let

$$\cdots \longrightarrow \mathcal{S}_{n+1} \xrightarrow{r_{n+1,n}} \mathcal{S}_n \longrightarrow \cdots$$

an inverse system of surviving states. The boundary structure at level n is the pair

$$\mathcal{F}_n = (r_{n+1,n}, \beta_n),$$

where β_n preserves the minimum datum necessary to decide the compatibility of a prolongation. Its coordinates may include the arithmetic type, the orientation, the carry cocycle, and the incidence class.

Lemma 25.13 (Compatibility of successive constraints)**.** *Suppose*

$$r_{n+2,n} = r_{n+1,n} \circ r_{n+2,n+1}, \qquad \beta_n \circ r_{n+1,n} = b_{n+1,n} \circ \beta_{n+1}.$$

Then every compatibility condition expressed through β_{n+1} induces a compatible condition at level n.

Proof. For $x \in \mathcal{S}_{n+2}$, equality of compositions identifies the direct restriction with the two consecutive restrictions. The second equality transports the boundary datum through the same diagram. The condition at level n is therefore the image of the deeper condition. □

25.12.2 Module of channels and dimensional range

Fixing a field $\mathbb{K}_n$, let $M_n(x)$ be the module generated by the prolongation channels of x, and let $\mathcal{R}_n(x)$ be the submodule of relations among them.

Definition 25.14 (Dimensional screen)**.** The screen associated with x at level n is

$$\Sigma_n(x) = (\mathbb{K}_n, M_n(x), \beta_n(x), \mathcal{R}_n(x)),$$

and its effective dimension is

$$d_{\Sigma_n}(x) = \dim_{\mathbb{K}_n} \frac{M_n(x)}{\mathcal{R}_n(x)}.$$

If $M_n(x) \cong \mathbb{K}_n^{g_n}$ and the columnas of a matrix $R_n(x)$ generate $\mathcal{R}_n(x)$, then

$$d_{\Sigma_n}(x) = g_n - \operatorname{rank}_{\mathbb{K}_n} R_n(x).$$

When a complement is fixed and $P_n(x)$ denotes the projector whose image is isomorphic to the quotient, the same dimension is $d_{\Sigma_n}(x) = \operatorname{rank}_{\mathbb{K}_n} P_n(x)$.

Proposition 25.15 (Invariance under equivalence of presentations)**.** *The screen dimension is invariant under invertible changes of basis in the generators and in the relations.*

Proof. A change of bases transforms the relation matrix R into URV, with U and V invertible. Since $\operatorname{rank}(URV) = \operatorname{rank} R$, the number $g - \operatorname{rank} R$ does not change. Equivalently, two projectors presenting the same quotient have isomorphic images and therefore the same rank. □

For a transport $T_\gamma : \mathcal{S}_n \to \mathcal{S}_m$, define

$$\Delta d(\gamma; x) = d_{\Sigma_m}(T_\gamma x) - d_{\Sigma_n}(x).$$

A dimensional transition simultaneously transports the state and its channel module; fixed-rank dynamics is the case $\Delta d = 0$.

25.12.3 Inherited arithmetic and lattice screens of HMT

The ranks that occur below are not labels appended to the equations. They arise from distinct linear constructions and preserve their selection maps. The punctured ternary Golay code

$$G_{11} = [11, 6, 5]_3 \subset \mathbb{F}_3^{11}$$

is perfect of radius two [91], since the balls centered at its 3^6 words are disjoint and

$$V_3(11, 2) = 1 + 2\binom{11}{1} + 2^2\binom{11}{2} = 1 + 22 + 220 = 243, \qquad 3^6 \cdot 243 = 3^{11}.$$

Here 243 is the cardinal of the syndrome space. It is not identified by mere numerical equality with the layer of 243 cells of the cubic completion of APP.

The second chain starts from the permutation module $V_{12} = \mathbb{R}^{12}$ of A_5 on the twelve vertices of the icosahedron. To fix the gauge, we let A_5 act on $\{0,1,2,3,4\}$, take $H = \langle (0\ 1\ 2\ 3\ 4) \rangle \cong C_5$ and identify the position j with the left coset $r_j H$. In image notation, the twelve representatives are

j	r_j	j	r_j	j	r_j
1	$(3,2,4,1,0)$	5	$(3,2,0,4,1)$	9	$(3,1,2,4,0)$
2	$(3,1,0,2,4)$	6	$(4,0,1,2,3)$	10	$(2,1,4,3,0)$
3	$(0,1,3,4,2)$	7	$(4,3,2,1,0)$	11	$(3,4,1,2,0)$
4	$(1,4,2,3,0)$	8	$(0,2,3,1,4)$	12	$(4,2,1,3,0)$.

We denote by $5A$ the conjugacy class containing the generator $g_5 = (0\ 1\ 2\ 3\ 4)$ of H, and by $5B$ the class containing g_5^2. In the class order $1A, 2A, 3A, 5A, 5B$, the permutation character is $\chi_{12} = (12, 0, 0, 2, 2)$. For $\phi = (1+\sqrt{5})/2$ and $\phi' = (1-\sqrt{5})/2$, we choose the irreducible character

$$\chi_3 = (3, -1, 0, \phi, \phi').$$

The scalar product of characters gives the exact decomposition.

$$V_{12} \cong V_1 \oplus V_3 \oplus V_{3'} \oplus V_5.$$

The projector

$$P_{11} = I_{12} - \frac{1}{12} J_{12}$$

eliminates exclusively the uniform vector and has image

$$V_{11} = \mathbf{1}^{\perp} \cong V_3 \oplus V_{3'} \oplus V_5.$$

For the dodecaphasic state

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \qquad \sum_j K_j = 6263,$$

let $\widetilde{K} = P_{11} K$. The preceding position–coset correspondence defines the permutation matrices $\rho(g)$, and the central projector

$$P_3 = \frac{3}{60} \sum_{g \in A_5} \chi_3(g^{-1}) \rho(g)$$

satisfies $P_3^2 = P_3 = P_3^{\mathsf{T}}$ and has rank three. Exact evaluation at $\mathbb{Q}(\sqrt{5})$ produces

$$u_K = P_3 \widetilde{K}, \qquad \|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

Consequently, the line $\ell_K = \mathbb{R} u_K$ is well defined and

$$V_{10} := (V_3 \cap \ell_K^{\perp}) \oplus V_{3'} \oplus V_5, \qquad V_{11} = \ell_K \oplus V_{10}, \qquad \dim V_{10} = 10.$$

The chain $12 \to 11 \to 10$ means, respectively, removing the uniform component and the line polarized by K. The finite set of sixty elements is divided into twelve classes; its enumeration reconstructs the projector and norm exactly.

The reduction $26 \to 24$ belongs to another category. Let U be the integral hyperbolic plane with Gram matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, y sea

$$L_{26} = \Lambda_{24} \oplus U.$$

With the hyperbolic summand fixed, this is the standard presentation of the even unimodular Lorentzian lattice $II_{25,1}$ [34]. For a primitive isotropic vector $e_+ \in U$,

$$e_+^\perp = \Lambda_{24} \oplus \mathbb{Z}e_+, \qquad e_+^\perp/\mathbb{Z}e_+ \cong \Lambda_{24}.$$

The decrease of rank by two comes from imposing orthogonality and then quotienting by the null line. It is not the reduction $11 \to 10$ repeated with other numbers.

25.12.4 Involution between residual data and Euclidean quotient

The decomposition $n = 9Q + r$ suggests two coordinates of different kinds: the residual representative r and the quotient Q. To formulate their exchange on a stable domain, we consider

$$\mathcal{D} = \mathbb{Z}^2 \times \mathbb{R}_{>0}, \qquad E_R(m, w) = \frac{m^2}{R^2} + w^2R^2,$$

and we define

$$\mathsf{T}(m, w, R) = (w, m, R^{-1}).$$

Proposition 25.16 (Residual–Euclidean Duality)**.** T *is an involution and*

$$E_R(m, w) = E_{R^{-1}}(w, m).$$

On $\ell^2(\mathbb{Z}^2)$*, the unitary* $U_T|m, w\rangle = |w, m\rangle$ *satisfies*

$$U_T H(R) U_T^{-1} = H(R^{-1}), \qquad H(R)|m, w\rangle = E_R(m, w)|m, w\rangle.$$

Proof. The second map again exchanges the coordinates and inverts R twice. The energy identity is obtained by direct substitution; the operatorial identity is checked on the orthonormal basis. □

The proposition is an exact lattice duality. The physical reading as momentum, winding, and compactification radius requires an additional map to the state space of a string theory; it does not follow from the algebraic involution [57].

25.12.5 Typed interface with undecadimensional reduction

The screen $V_{11} = \ell_K \oplus V_{10}$ and the involution $(m, w, R) \mapsto (w, m, R^{-1})$ provide the two algebraic data that motivate comparison with the M-theory context: a marked reduction from eleven to ten coordinates and an equivalence of reciprocal radii. For that comparison to be a physical realization, a morphism must be specified

$$\mathcal{R}_M : (\mathcal{X}_{\mathrm{TPK}}, V_{11}, \ell_K, \mathsf{T}) \longrightarrow (\mathcal{H}_M, \mathcal{F}_{10}, R_{\mathrm{phys}}, \alpha', Q_{\mathrm{susy}}),$$

where the codomain includes a state space, ten-dimensional fields, physical radius, string tension, and supersymmetric charges [135]. The algebraic decomposition fixes the domain of $\mathcal{R}_M$; it does not by itself introduce oscillators, branes, supersymmetry, or the supergravity equations.

This separation preserves the depth of the correspondence without altering the types. The reduction $11 \to 10$, the involution of radii, and the quotient $26 \to 24$ are theorems of the objects defined. The assertion that a concrete dynamical theory is their image requires exhibiting $\mathcal{R}_M$ and proving that it intertwines actions, restrictions, and observables.

25.12.6 Dimensional state and simultaneous transport

The state on which Dimensional Mechanics acts is written

$$\mathfrak{x} = (x, \Sigma_n(x), \beta_n(x), [\gamma]),$$

where $x \in \mathcal{X}_{\mathrm{TPK}}$, $\Sigma_n(x)$ is its screen, $\beta_n(x)$ is the boundary datum, and $[\gamma]$ is the class of paths compatible with that datum. A transport does not act only on x: it consists of

$$(T_\gamma, F_\gamma) : (x, \Sigma_n, \beta_n) \longrightarrow (T_\gamma x, \Sigma_m, \beta_m),$$

where $F_\gamma : M_n(x) \to M_m(T_\gamma x)$ maps relations to relations and satisfies the compatibility diagram with β_n, β_m. The dimensional jump is the change in rank of the induced morphism

$$\overline{F}_\gamma : M_n(x)/\mathcal{R}_n(x) \longrightarrow M_m(T_\gamma x)/\mathcal{R}_m(T_\gamma x).$$

This formulation prevents assigning a dimension to the isolated point: the dimension belongs to the system of channels that accompanies the state and its transport.

25.12.7 Descent to a dynamics on point states

Let $q_{\mathrm{pt}} : \mathcal{X}_{\mathrm{TPK}} \to Z$ be a projection that eliminates the genealogy of the state, and define

$$Y_{\mathrm{pt}} := \mathrm{im}(q_{\mathrm{pt}}).$$

Let, moreover, $U : \mathcal{X}_{\mathrm{TPK}} \to \mathcal{X}_{\mathrm{TPK}}$ a evolution complete.

Theorem 25.17 (Fiberwise Descent Criterion)**.** *There exists a unique map $\overline{U} : Y_{\mathrm{pt}} \to Y_{\mathrm{pt}}$ such that $q_{\mathrm{pt}}U = \overline{U}q_{\mathrm{pt}}$ if and only if*

$$q_{\mathrm{pt}}(x) = q_{\mathrm{pt}}(y) \Longrightarrow q_{\mathrm{pt}}(Ux) = q_{\mathrm{pt}}(Uy).$$

Proof. Necessity results by applying $\overline{U}$ to two representatives of one and the same fiber. For sufficiency, one defines

$$\overline{U}(q_{\mathrm{pt}}(x)) = q_{\mathrm{pt}}(Ux).$$

The condition guarantees that the definition does not depend on the representative. Since the codomain has been fixed equal to the image of q_{pt}, the surjectivity required for uniqueness is automatic. □

When the condition fails, a single point class has distinct images; the descended evolution is then multivalued. Retaining the memory datum, restricting the domain, or working with a correspondence are three distinct mathematical modifications.

25.12.8 Additive functional on oriented trajectories

To avoid a hidden path-composition term, the state of an edge is extended by the direction and arithmetic type of the preceding edge. If $\widetilde{e}_j = (e_j, d_{j-1}, t_{j-1})$, define the local weights

$$w_{\mathrm{giro}}(\widetilde{e}_j) = \mathbf{1}_{d_j \neq d_{j-1}}, \qquad w_{\mathrm{tipo}}(\widetilde{e}_j) = \mathbf{1}_{t_j \neq t_{j-1}}.$$

The data (d_0, t_0) form part of the initial boundary condition. For an augmented trajectory $\widetilde{\gamma} = (\widetilde{e}_1, \ldots, \widetilde{e}_R)$, let

$$N_{\mathrm{giro}}(\widetilde{\gamma}) = \sum_{j=1}^{R} w_{\mathrm{giro}}(\widetilde{e}_j), \qquad N_{\mathrm{tipo}}(\widetilde{\gamma}) = \sum_{j=1}^{R} w_{\mathrm{tipo}}(\widetilde{e}_j).$$

We define

$$\mathbf{I}(\widetilde{\gamma}) = (N_{\mathrm{giro}}(\widetilde{\gamma}), N_{\mathrm{tipo}}(\widetilde{\gamma})).$$

The composition is allowed only when the preceding datum of the second path coincides with the terminal datum of the first. With this convention, the partition of the sum into the two edge intervals proves

$$\mathbf{I}(\widetilde{\gamma}\widetilde{\eta}) = \mathbf{I}(\widetilde{\gamma}) + \mathbf{I}(\widetilde{\eta}),$$

so that $\mathbf{I}$ is a cocycle with values in $\mathbb{N}^2$. For $\lambda > 0$, the combination

$$I_\lambda(\widetilde{\gamma}) = N_{\mathrm{giro}}(\widetilde{\gamma}) + \lambda N_{\mathrm{tipo}}(\widetilde{\gamma})$$

is a scalar functional over the path space.

In the ordinary Lagrangian formulation, the action functional is defined on curves in a previously selected configuration space. Here the order is prior: I_λ quantifies changes in direction and arithmetic law in the category of augmented trajectories itself. A realization map may subsequently transport this cocycle into a continuous action. The additivity proved above is the property that permits that comparison without identifying the two functionals in advance.

Principle 25.18 (Minimal interaction)**.** Among admissible trajectories with fixed endpoints and boundary data, the selected evolution minimizes I_λ.

Proposition 25.19 (Finite Existence of Minimizers)**.** *If $\Gamma(a, b)$ is finite and nonempty, I_λ attains a minimum in $\Gamma(a, b)$.*

Proof. The image of a non-empty finite set under a real function has a minimum element. □

The mass character of chapter 90 constitutes another functional on trajectories: it transforms an integer cocycle into a positive ratio. The functional and the mechanism selecting the trajectory are distinct objects; their composition will be specified before introducing a unit of mass.

25.13 Coordinated projections of the HMT state: algebraic equivalence of screens and physical realization in dimension 11

Let $\mathcal{X}^{\mathrm{enr}}_{\mathrm{HMT}}$ be the domain preserving phase, orientation, quotient, carry, boundary, incidence, grading, and memory. Its exceptional and screen components are extracted by the canonical projections

$$\varepsilon_{\mathrm{exc}} : \mathcal{X}^{\mathrm{enr}}_{\mathrm{HMT}} \longrightarrow \mathscr{X}_{\mathrm{exc}}, \qquad \varepsilon_{\mathrm{scr}} : \mathcal{X}^{\mathrm{enr}}_{\mathrm{HMT}} \longrightarrow \mathcal{S}_{\mathrm{HMT}}.$$

The former feeds the already proved construction $\mathcal{P}_{\mathrm{exc}} : \mathscr{X}_{\mathrm{exc}} \to V^{\natural}$. The latter will feed the algebraic morphism $\mathcal{R}^{\mathrm{alg}}_{M}$ constructed below. The joint realization is

$$\mathcal{P}^{M}_{\mathrm{joint}}(x) = \left(\mathcal{P}_{\mathrm{exc}}\varepsilon_{\mathrm{exc}}(x), \mathcal{R}^{\mathrm{alg}}_{M}\varepsilon_{\mathrm{scr}}(x)\right).$$

This map preserves the common antecedent and keeps the codomains distinct.

25.13.1 Exact equivalence of lattice screens and conditions for the realization of undecadimensional structures

The real screen starts from

$$V_{12} \cong V_1 \oplus V_3 \oplus V_{3'} \oplus V_5, \qquad P_{11} = I_{12} - \frac{1}{12}J_{12}.$$

The state K produces $u_K = P_3P_{11}K \neq 0$, the line $\ell_K = \mathbb{R}u_K$, and

$$V_{11} = \ell_K \oplus V_{10}, \qquad \dim V_{11} = 11, \qquad \dim V_{10} = 10.$$

The step $12 \to 11$ removes the uniform mode, and the step $11 \to 10$ removes the line polarized by K.

The lattice screen belongs to another category. For

$$L_{26} = \Lambda_{24} \oplus U \cong II_{25,1}$$

and a primitive isotropic vector $e_+ \in U$,

$$e_+^\perp = \Lambda_{24} \oplus \mathbb{Z}e_+, \qquad \boxed{e_+^\perp/\mathbb{Z}e_+ \cong \Lambda_{24}.}$$

Orthogonality and the quotient eliminate two directions. This theorem is not a repetition of $11 \to 10$ nor an automatic identification with a physical spacetime.

25.13.2 Algebraic screen morphism and quotient equivalence

Let

$$P_{10} := P_{11} - \frac{u_K u_K^{\mathsf{T}}}{\|u_K\|^2}, \qquad \mathcal{G}_K := \{(v_{11}, v_{10}) \in V_{11} \times V_{10} : v_{10} = P_{10} v_{11}\}.$$

The decomposition $e_+^\perp = \Lambda_{24} \oplus \mathbb{Z}e_+$ defines

$$q_{24} : e_+^\perp \longrightarrow \Lambda_{24}, \qquad q_{24}(\lambda + ae_+) = \lambda.$$

Take

$$\mathcal{S}_{\mathrm{HMT}} := V_{12} \times \mathscr{D}_0 \times e_+^\perp, \qquad \mathcal{S}_M^{\mathrm{alg}} := \mathcal{G}_K \times \mathscr{D}_0 \times \Lambda_{24}.$$

Definition 25.20 (Algebraic screen morphism M)**.** For $d = (m, w, R_0) \in \mathscr{D}_0$, we define

$$\boxed{\mathcal{R}_M^{\mathrm{alg}}(v, d, y) := \big((P_{11}v, P_{10}v), d, q_{24}(y)\big).}$$

Theorem 25.21 (Algebraic Equivalence of Quotient Screens)**.** *The preceding morphism induces an isomorphism*

$$\boxed{\overline{\mathcal{R}}_M^{\mathrm{alg}} : (V_{12}/V_1) \times \mathscr{D}_0 \times (e_+^\perp/\mathbb{Z}e_+) \xrightarrow{\cong} \mathcal{S}_M^{\mathrm{alg}}.}$$

If

$$\tau_{\mathrm{src}}([v], d, [y]) = ([v], \mathsf{T}_0 d, [y])$$

and

$$\tau_{\mathrm{tar}}((v_{11}, v_{10}), d, \lambda) = ((v_{11}, v_{10}), \mathsf{T}_0 d, \lambda),$$

then

$$\overline{\mathcal{R}}_M^{\mathrm{alg}} \tau_{\mathrm{src}} = \tau_{\mathrm{tar}} \overline{\mathcal{R}}_M^{\mathrm{alg}}.$$

The compact energy $E_{R_0}(m, w)$ is preserved under this intertwining and under reciprocal radii.

Proof. The projector P_{11} has kernel V_1 and image V_{11}. Since $P_{10}P_{11} = P_{10}$, the map

$$[v] \longmapsto (P_{11}v, P_{10}v)$$

is an isomorphism of V_{12}/V_1 onto the graph $\mathcal{G}_K$. The map q_{24} is surjective and its kernel is $\mathbb{Z}e_+$, hence it induces the isomorphism $e_+^\perp/\mathbb{Z}e_+ \cong \Lambda_{24}$. The factor $\mathscr{D}_0$ is transported by the identity. The product of these three isomorphisms is $\overline{\mathcal{R}}_M^{\mathrm{alg}}$.

The projectors and q_{24} do not act on d, whereas the two involutions act only through T_0. The square commutes by direct substitution. The energetic invariance is the theorem 24.5. □

The mathematical reduction map, and not merely a correspondence of ranks, is thus constructed. The screen $11 \to 10$, the quotient $26 \to 24$, and radius duality appear as distinct factors of the same morphism.

25.13.3 Physical Recognition Undecadimensional

The interpretation as M-theory is separated by a recognition map.

$$\Xi_M : \mathcal{S}_M^{\mathrm{alg}} \longrightarrow \mathcal{X}_{11}^{\mathrm{phys}},$$

whose codomain carries at least

$$g_{MN}, \qquad C_{MNP}, \qquad \psi_M,$$

the action and the supersymmetry transformations. An operator completion includes

$$\mathfrak{S}_M = (\mathcal{H}_{\mathrm{phys}}, H, Q), \qquad Q^2 = H,$$

and an isometry

$$W : \mathcal{H}_{\mathrm{HMT}} \longrightarrow \mathcal{H}_{\mathrm{phys}}$$

which, in its common domains, satisfies

$$W\widehat{H} = HW, \qquad W\widehat{Q} = QW, \qquad \widehat{Q}^2 = \widehat{H}.$$

Theorem 25.22 (Physically consistent recognition criterion)**.** *If Ξ_M preserves the inner product, transports the direction ℓ_K to the compact direction, intertwines the duality, and satisfies the preceding operator identities, then the compact physical spectrum is invariant under reciprocal radii and the transported supercharge satisfies $Q^2 = H$ on the image of W.*

Proof. The interlinking of Ξ_M with τ_{tar}, joined to theorem 25.21, transports the unitary involution to the compact sector. For the supercharge,

$$Q^2 W = W\widehat{Q}^2 = W\widehat{H} = HW.$$

The preservation of the inner product converts the transported operator into a unitary equivalence on the image. □

The isomorphism $\overline{\mathcal{R}}_M^{\mathrm{alg}}$ is mathematical and has been proved. Recognition by Ξ_M is a physical interface conditioned on the fields, action, gauge constraints, supersymmetry, oscillators, amplitudes, anomaly control, and low-energy limit of the selected realization. The physical codomain is not identified with $V^\natural$: both receive coordinated information from the same antecedent through distinct functors.

A literal action of $\mathbb{M}$ on digits is not postulated. This sentence is the typing control placed at the end; it does not replace the positive chain that reaches the deep action of $\mathbb{M}$ in $V^\natural$.

25.14 Dimensional screens and coordinated reduction of the state

The ranks appearing in exceptional prolongation and in compactification belong to different objects. To relate them, their presentation is preserved: generators, relations, inner product and selected direction. Dimension is then a consequence of the channel module and its relations. Its transport is formulated through maps acting on that module, in addition to acting on the state that determines it.

25.14.1 Channels, relations and effective dimension

Let $\mathcal{S}_n$ be the set of surviving states at depth n, with compatible restrictions $r_{m,n} : \mathcal{S}_m \to \mathcal{S}_n$, $m \geq n$. The boundary datum $\beta_n(x)$ preserves the information needed to decide the compatibility of a prolongation: sheet, orientation, quotient, memory and incidence class according to the reader considered. The restrictions satisfy

$$r_{\ell,n} = r_{m,n} r_{\ell,m}, \qquad \beta_n r_{m,n} = b_{m,n} \beta_m.$$

These equalities are the naturality conditions of the presentation and retain the same data when restricting in stages or directly.

Definition 25.23 (Channel screen)**.** Given a field $\mathbb{K}_n$, let $M_n(x)$ be the space generated by the prolongation channels and $\mathcal{R}_n(x)$ the subspace of relations. The screen is

$$\Sigma_n(x) = \big(\mathbb{K}_n, M_n(x), \beta_n(x), \mathcal{R}_n(x)\big),$$

and its effective dimension is

$$d_{\Sigma_n}(x) = \dim_{\mathbb{K}_n}\big(M_n(x)/\mathcal{R}_n(x)\big). \tag{25.3}$$

If $M_n(x) \cong \mathbb{K}_n^{g_n}$ and the columns of $R_n(x)$ generate the relations, then $d_{\Sigma_n}(x) = g_n - \operatorname{rank} R_n(x)$.

Proposition 25.24 (Invariance of presentation and transport)**.** *The dimension* (25.3) *is preserved under invertible changes of basis. For two screens over the same field $\mathbb{K}_n = \mathbb{K}_m = \mathbb{K}$, if the linear map $F_\gamma : M_n(x) \to M_m(T_\gamma x)$ sends $\mathcal{R}_n(x)$ into $\mathcal{R}_m(T_\gamma x)$, it uniquely induces a linear map between the screen quotients.*

Proof. A change of generators and relations transforms R into URV, with U, V invertible, and preserves its rank. For descent, define $\overline{F}_\gamma([v]) = [F_\gamma(v)]$. If two representatives differ by a relation, linearity implies that their images differ by a relation of the terminal screen; the map is well defined. Projection to the quotient is surjective and guarantees uniqueness. □

The dimensional change during transport is determined by

$$\Delta d(\gamma; x) = d_{\Sigma_m}(T_\gamma x) - d_{\Sigma_n}(x).$$

This expression preserves the state and its channel presentation. An isolated rank does not allow reconstruction of either the relations or the map that produced it.

25.14.2 Ternary puncturing and syndrome screen

The code $C_W = [12, 6, 6]_3$, constructed from exceptional incidence, produces another screen by puncturing one coordinate. Denote that map by $p : C_W \to \mathbb{F}_3^{11}$ and its image by $G_{11} = p(C_W)$.

Proposition 25.25 (Punctured code and perfect partition)**.** *The map p is injective. The code G_{11} has parameters $[11, 6, 5]_3$, and its Hamming balls of radius 2 partition $\mathbb{F}_3^{11}$.*

Proof. A vector in the kernel of p would have support contained in the single removed coordinate. The minimum distance 6 of C_W excludes any nonzero vector of that form. Thus, the dimension remains 6. Puncturing decreases the weight by at most one, so the minimum distance of the image is at least 5. The balls of radius 2 are therefore disjoint. Each contains

$$1 + 2\binom{11}{1} + 2^2\binom{11}{2} = 1 + 22 + 220 = 243$$

vectors. Since $|G_{11}| = 3^6$ and $3^6 \cdot 243 = 3^{11}$, the balls cover the ambient space. To check that the distance is exactly 5, take a vector of weight 3. Its decoding ball cannot have centre zero and has a centre of weight at most $3 + 2 = 5$; that centre must have weight at least 5. A word of weight 5 is thus obtained. □

The integer $243 = 3^{11-6}$ is here the number of classes of the syndrome quotient $\mathbb{F}_3^{11}/G_{11}$. Its function is fixed by the puncturing and distance of the code; equality of cardinality with another fibre does not replace the map between them. This combinatorial screen and the following real screen retain their own operations.

25.14.3 The dodecaphase direction and rank reduction

The construction of 21.11.5 starts from the register K already obtained by APP–TRIT–TPK and from the representation chart declared on $\mathbb{R}^{12}$. In that chart,

$$V_{11} = \mathbf{1}^{\perp}, \qquad P_{11} = I_{12} - \frac{1}{12}J_{12}, \qquad u_K = P_3 P_{11} K. \tag{25.4}$$

Here J_{12} is the all-ones matrix and P_3 is the isotypic projector specified by the action of A_5. The finite sum defining P_3, its representatives and the generation of K are preserved in the cited section of this same article. In particular,

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

The positive norm defines $\ell_K = \mathbb{R}u_K$, and the projector

$$P_{10} = P_{11} - \frac{u_K u_K^{\mathsf{T}}}{\|u_K\|^2} \tag{25.5}$$

has image $V_{10} = V_{11} \cap u_K^{\perp}$.

Proposition 25.26 (Two orthogonal reductions)**.** *The following hold*

$$\mathbb{R}^{12} = \mathbb{R}\mathbf{1} \oplus^{\perp} V_{11}, \qquad V_{11} = \ell_K \oplus^{\perp} V_{10}, \qquad \dim V_{10} = 10.$$

Moreover, $P_{10}P_{11} = P_{10}$, $P_{10}u_K = 0$, and P_{10} is an orthogonal projector.

Proof. Proposition 21.64, proved before constructing the screens, establishes idempotence, symmetry, rank and the two operator identities for this same u_K and this same P_{10}. The first decomposition follows from $P_{11} = I - J_{12}/12$; the second is the orthogonal decomposition of that proposition. This preserves the single proof of the reduction and makes its two stages explicit here. □

The first reduction removes only the uniform mode. The second removes the direction determined by K in the fixed representation. If the chart is simultaneously changed by an orthogonal matrix S, the vector transforms by $u_K \mapsto Su_K$ and the projector by $P_{10} \mapsto SP_{10}S^{-1}$. This covariance preserves the geometric operation, even though its coordinates change.

25.14.4 Lorentzian lattice screen

Let $\Lambda = N_\alpha$ be the Leech lattice constructed in 21.11.9. Consider the integral hyperbolic plane $U = \mathbb{Z}e_+ \oplus \mathbb{Z}e_-$, with

$$\langle e_+, e_+\rangle = \langle e_-, e_-\rangle = 0, \qquad \langle e_+, e_-\rangle = 1.$$

The lattice $L_{26} = \Lambda \oplus U$ is even and unimodular, with signature $(25, 1)$. Each vector is written $y = \lambda + ae_+ + be_-$. Since $\langle y, e_+\rangle = b$,

$$e_+^{\perp} = \Lambda \oplus \mathbb{Z}e_+, \qquad e_+^{\perp}/\mathbb{Z}e_+ \cong \Lambda. \tag{25.6}$$

The quotient map is $q_{24}(\lambda + ae_+) = \lambda$; it is surjective and has kernel $\mathbb{Z}e_+$. The decrease in rank by two comes from an orthogonality restriction followed by an isotropic quotient. It is therefore an operation of a different type from (25.5).

25.14.5 Joint screen morphism

With the domain $\mathscr{D}_0$ of 24.16, let

$$\mathscr{G}_K = \{(v_{11}, P_{10}v_{11}) : v_{11} \in V_{11}\},$$

$$\mathcal{S}_{\mathrm{HMT}} = \mathbb{R}^{12} \times \mathscr{D}_0 \times e_+^{\perp}, \qquad \mathcal{S}_M^{\mathrm{alg}} = \mathscr{G}_K \times \mathscr{D}_0 \times \Lambda.$$

Definition 25.27 (Algebraic map of coordinated screens)**.** It is defined

$$\mathcal{R}_M^{\mathrm{alg}}(v, d, y) = \big((P_{11}v, P_{10}v), d, q_{24}(y)\big). \tag{25.7}$$

Theorem 25.28 (Quotient isomorphism and duality intertwining)**.** *The map* (25.7) *induces an isomorphism*

$$\overline{\mathcal{R}}_M^{\mathrm{alg}} : (\mathbb{R}^{12}/\mathbb{R}\mathbf{1}) \times \mathscr{D}_0 \times (e_+^{\perp}/\mathbb{Z}e_+) \longrightarrow \mathcal{S}_M^{\mathrm{alg}}.$$

It intertwines the transformations acting by T_0 *on the factor* $\mathscr{D}_0$ *and by the identity on the other two factors. The form* $E_{R_0}(m, w)$ *is preserved under that duality.*

Proof. The map $[v] \mapsto P_{11}v$ is an isomorphism from $\mathbb{R}^{12}/\mathbb{R}\mathbf{1}$ onto V_{11}. Since $P_{10}P_{11} = P_{10}$, the map $[v] \mapsto (P_{11}v, P_{10}v)$ identifies that quotient with $\mathscr{G}_K$; its inverse takes the first component and its class. By (25.6), q_{24} induces the lattice isomorphism of the third factor. The second is transported by the identity. The product of the three maps proves the isomorphism.

The projectors and q_{24} do not modify d, whereas duality acts only on d. The two compositions are therefore identical. Preservation of the form is theorem 24.17. □

The graph $\mathscr{G}_K$ preserves the preceding coordinate v_{11} together with its projection. The theorem proves the equivalence of the quotient presentations; the isolated reduction $V_{11} \to V_{10}$ retains its kernel ℓ_K. This clarification establishes what information the isomorphism retains and what information each projection composing it removes.

Corollary 25.29 (Screens over the elliptic family)**.** *The same morphism restricts to an isomorphism*

$$(\mathbb{R}^{12}/\mathbb{R}\mathbf{1}) \times \mathscr{D}_{\mathrm{el}} \times (e_+^{\perp}/\mathbb{Z}e_+) \;\cong\; \mathscr{G}_K \times \mathscr{D}_{\mathrm{el}} \times \Lambda,$$

and preserves the duality intertwining.

Proof. The morphism and its inverse preserve the second factor exactly. Corollary 24.20 proves that duality preserves $\mathscr{D}_{\mathrm{el}}$. Thus both the bijection and the already proved intertwining square restrict. □

25.14.6 Coordination with the exceptional realization

On the domain $\mathcal{X}_{\mathrm{HMT}}$, with phase, orientation, quotient, boundary, incidence, grading and memory, the readings $\varepsilon_{\mathrm{exc}}$ and $\varepsilon_{\mathrm{scr}}$ extract the exceptional and screen data. The corridor constructs the graded algebra $\mathfrak{V}(x) \cong V^{\natural}$, with its vertex product and automorphisms; $\mathfrak{S}_M(x)$ brings together the preceding screens, quotients and operators. The joint assignment $x \mapsto (\mathfrak{V}(x), \mathfrak{S}_M(x))$ therefore has an algebra as its first component and a family of spaces and maps as its second. The Monster action belongs to the first; the reduction directed by K and compact duality belong to the second. The two readings fix the common provenance and each construction preserves its type. The isomorphism of this section establishes the correspondence of screens with its explicit factors and proof.

25.15 Action on trajectories and eleven-dimensional dynamical realization

Extending the structure to dimensional dynamics requires joint transport of the state, channels and boundary data. This section begins with a functional defined on TPK trajectories and its exact composition law. The worldsheet and eleven-dimensional codomain are introduced afterwards, with the maps that must preserve that composition and the operators already constructed.

25.15.1 Descent of an evolution and sufficiency of the observed datum

Let $q : \mathcal{X}_{\mathrm{HMT}} \to Y$ be a surjective readout of states with memory, and let $U : \mathcal{X}_{\mathrm{HMT}} \to \mathcal{X}_{\mathrm{HMT}}$ be a complete evolution. An evolution on observed values exists precisely when the datum preserved by q suffices to determine the next observed value.

Theorem 25.30 (Exact criterion for descent along fibres)**.** *There exists a unique map $\overline{U} : Y \to Y$ satisfying $qU = \overline{U}q$ if and only if*

$$q(x) = q(y) \quad \Longrightarrow \quad q(Ux) = q(Uy). \tag{25.8}$$

Proof. If $\overline{U}$ exists, applying the same function to $q(x) = q(y)$ proves the condition. Conversely, define $\overline{U}(q(x)) = q(Ux)$. Condition (25.8) guarantees independence of the representative. Surjectivity of q defines the map on all of Y and proves uniqueness. □

The theorem determines what memory a readout must preserve. When two representatives have different observed images, the retained datum may be enlarged, the domain restricted, or a multivalued correspondence used. These are distinct mathematical choices. In particular, return of a phase coordinate does not authorize identification of the evolutions of two histories retaining different memories.

25.15.2 Cocycle of directional and arithmetic changes

An admissible TPK trajectory carries, on each edge e_j, a direction d_j and an arithmetic type t_j. To count changes in a concatenation, the preceding datum is also preserved. The augmented edge is $\widetilde{e}_j = (e_j, d_{j-1}, t_{j-1})$; the pair (d_0, t_0) belongs to the initial boundary. Define

$$w_{\mathrm{giro}}(\widetilde{e}_j) = \mathbf{1}_{\{d_j \neq d_{j-1}\}}, \qquad w_{\mathrm{tipo}}(\widetilde{e}_j) = \mathbf{1}_{\{t_j \neq t_{j-1}\}}.$$

For $\widetilde{\gamma} = (\widetilde{e}_1, \dots, \widetilde{e}_R)$, let

$$\mathbf{I}(\widetilde{\gamma}) = \left(\sum_{j=1}^{R} w_{\mathrm{giro}}(\widetilde{e}_j), \sum_{j=1}^{R} w_{\mathrm{tipo}}(\widetilde{e}_j) \right) = \big(N_{\mathrm{giro}}, N_{\mathrm{tipo}}\big). \tag{25.9}$$

Two trajectories compose when their endpoints agree and the preceding datum of the second agrees with the terminal datum of the first.

Proposition 25.31 (Additivity with preserved boundary)**.** *For every composable pair,*

$$\mathbf{I}(\widetilde{\gamma}\widetilde{\eta}) = \mathbf{I}(\widetilde{\gamma}) + \mathbf{I}(\widetilde{\eta}). \tag{25.10}$$

Thus, $\mathbf{I}$ is an additive cocycle with values in $\mathbb{N}^2$ on the category of augmented trajectories. For every $\lambda > 0$, the functional

$$I_\lambda(\widetilde{\gamma}) = N_{\mathrm{giro}}(\widetilde{\gamma}) + \lambda N_{\mathrm{tipo}}(\widetilde{\gamma})$$

is also additive.

Proof. Concatenation does not change the preceding datum of any edge: for the first edge of $\widetilde{\eta}$, the composition condition makes it agree with the terminal datum of $\widetilde{\gamma}$. The sum over all edges splits exactly into the two sums in (25.10). Applying the linear functional $(a, b) \mapsto a + \lambda b$ proves the last statement. □

Boundary information is an effective part of the proof. If two one-edge segments are evaluated separately while ignoring the connection between them, both may have zero turn count even though their directions differ. The concatenated path then contains a turn. Preserving (d_{j-1}, t_{j-1}) records that term on the first edge of the second segment and restores additivity.

For fixed endpoints and boundary data, the corpus's minimum-interaction principle selects minimizers of I_λ among admissible routes. The principle is a selection rule, whereas the following result establishes a finite-existence property.

Proposition 25.32 (Finite Existence of Minimizers)**.** *If the set $\Gamma(a, b)$ of admissible trajectories with the fixed boundary is finite and nonempty, there exists $\gamma_* \in \Gamma(a, b)$ such that $I_\lambda(\gamma_*) \le I_\lambda(\gamma)$ for all $\gamma \in \Gamma(a, b)$.*

Proof. The image of $\Gamma(a, b)$ under I_λ is a nonempty finite set of real numbers, which has a minimum element and at least one preimage. □

The proposition does not select the value of λ, nor does it confuse existence with uniqueness. The value and selection rule used in a particular realization must be part of its declared data. The additive functional recovered here quantifies changes of the path itself; its transport to a dimensional action is a later operation.

25.15.3 Extended persistence and world surface

Realization of a route as an extended object distributes its genealogy along an internal coordinate. For a string, the history has two local coordinates σ^a: one internal spatial coordinate and one temporal coordinate. A realization is described by a map $X : \Sigma \to \mathcal{M}$, a metric h_{ab} on the worldsheet and background fields in the target space. Conditions on $\partial\Sigma$ belong to the variational domain. For a brane of spatial dimension p, the history is a worldvolume of dimension $p + 1$.

The string variational codomain used by the corpus is the Polyakov action

$$S_{\mathrm{P}}[X, h] = -\frac{T_{\mathrm{s}}}{2} \int_\Sigma d^2\sigma \, \sqrt{-h}\, h^{ab} g_{MN}(X) \partial_a X^M \partial_b X^N, \tag{25.11}$$

where T_{s} is tension. The notation T_{s} distinguishes it from duality T_0, cardinal transport T_d and complete evolution U.

Composition of augmented trajectories already has the property proved in 25.31. A continuous realization of that composition must specify the map from paths to configurations, its transport of boundary conditions and the normalization comparing I_λ with the continuous action. Integral (25.11) specifies the codomain and its fields; its parameters do not retrospectively select TPK emissions or registers.

25.15.4 Eleven-dimensional realization and graded operators

The reduction directed by K provides $V_{11} = \ell_K \oplus V_{10}$, and Theorem 25.28 coordinates this reduction with the lattice screen and duality. The physical realization is typed through

$$\Xi_M : \mathcal{S}_M^{\mathrm{alg}} \longrightarrow \mathcal{X}_{11}^{\mathrm{phys}}.$$

The codomain carries a metric g_{MN}, a three-form C_{MNP}, a gravitino ψ_M, an action, gauge constraints and supersymmetry transformations. An operator realization includes

$$\mathfrak{S}_M = (\mathcal{H}_{\mathrm{phys}}, H, Q), \qquad \mathcal{H}_{\mathrm{phys}} = \mathcal{H}_+ \oplus \mathcal{H}_-.$$

The grading is expressed through an involution Γ, equal to $+I$ on $\mathcal{H}_+$ and $-I$ on $\mathcal{H}_-$. The supercharge is odd, $\Gamma Q = -Q\Gamma$, and the Hamiltonian is nonnegative. The relation $Q^2 = H$ is understood as an equality of operators, including their domains.

Let $\mathcal{H}_{\mathrm{HMT}}$ be a Hilbert-space realization of the relevant genealogical states, with operators $\widehat{H}, \widehat{Q}$. An isometry $W : \mathcal{H}_{\mathrm{HMT}} \to \mathcal{H}_{\mathrm{phys}}$ must transport domains, grading and observables. The operator conditions are written

$$W\widehat{H} = HW, \qquad W\widehat{Q} = QW, \qquad WU_T^{\mathrm{HMT}} = U_T^{\mathrm{phys}}W. \tag{25.12}$$

Theorem 25.33 (Compatible transport of the supercharge)**.** *Suppose $\widehat{Q}^2 = \widehat{H}$, $WD(\widehat{Q}) \subset D(Q)$, and the intertwining relations of* (25.12) *hold on their domains. Then, for $\psi \in D(\widehat{H}) = D(\widehat{Q}^2)$,*

$$W\psi \in D(Q^2), \qquad Q^2W\psi = HW\psi.$$

If physical transport of the radii also respects (24.18)*, the transported compact sector preserves its spectrum under reciprocal radii.*

Proof. Since $\psi \in D(\widehat{Q}^2)$, both ψ and $\widehat{Q}\psi$ belong to $D(\widehat{Q})$. Thus, $W\psi \in D(Q)$ and $QW\psi = W\widehat{Q}\psi \in D(Q)$. One may apply Q again, and

$$Q^2W\psi = W\widehat{Q}^2\psi = W\widehat{H}\psi = HW\psi.$$

Spectral equality of the compact sector is transported by the isometry and the unitary T-intertwiner; on the closed image of W, this is a unitary equivalence of the corresponding realizations. □

The theorem explicitly preserves the operators and the domain on which the identity is proved. The grading organizes two sectors; construction of fermionic fields and their occupation relations has its own structure. Thermal distributions and the species catalogue are not used in the proof of the screens or in the compact duality of this article.

25.15.5 Compatible compactification and conclusion of the development

A compactification of the state is described through a prolongable section x_∞, a map κ_{comp} to the chosen moduli space and a transport $\mathcal{U}$ of the metric, orientation, boundary and memory data. The subscript distinguishes this map from the rational readout κ of the dodecaphasic register. Its compatibility requires four concrete properties:

1. the finite restrictions of x_∞ belong to admissible TPK prolongations;
2. the compactification map intertwines the dualities, $\kappa_{\mathrm{comp}}\mathsf{T} = \mathsf{T}_M\kappa_{\mathrm{comp}}$;
3. observables and the spectrum descend to the chosen quotients according to criterion 25.30;
4. the realization transports the direction ℓ_K, boundary conditions and operator domains, and preserves the declared action and supersymmetry equations.

The preceding proofs provide the phase Hilbert-space realization, tracial closure, screens with their quotient maps, radius involution, its unitary equivalence and the additive trajectory cocycle. The eleven-dimensional realization specifies the fields and compatibilities it uses when receiving that structure. Its complete physical identification also retains the conditions on tension, oscillators, amplitudes, anomalies and the low-energy limit enumerated by the corpus. These conditions do not alter the proved algebraic identities: they delimit the subsequent claim obtained by transporting them.

The exceptional and dimensional prolongations remain linked by their common origin. The Monster acts on the graded construction $V^\natural$; the compact Hamiltonian and screens receive data from other maps of the same state. This formulation preserves the positive content of the dual realization, with its operations, proofs and the information transmitted at each stage (Theorem 25.33).

25.16 Dimensional screens and coordinated reduction of the state

The ranks appearing in exceptional prolongation and in compactification belong to different objects. To relate them, their presentation is preserved: generators, relations, inner product and selected direction. Dimension is then a consequence of the channel module and its relations. Its transport is formulated through maps acting on that module, in addition to acting on the state that determines it.

25.16.1 Channels, relations and effective dimension

Let $\mathcal{S}_n$ be the set of surviving states at depth n, with compatible restrictions $r_{m,n} : \mathcal{S}_m \to \mathcal{S}_n$, $m \geq n$. The boundary datum $\beta_n(x)$ preserves the information needed to decide the compatibility of a prolongation: sheet, orientation, quotient, memory and incidence class according to the reader considered. The restrictions satisfy

$$r_{\ell,n} = r_{m,n} r_{\ell,m}, \qquad \beta_n r_{m,n} = b_{m,n} \beta_m.$$

These equalities are the naturality conditions of the presentation and retain the same data when restricting in stages or directly.

Definition 25.34 (Channel screen)**.** Given a field $\mathbb{K}_n$, let $M_n(x)$ be the space generated by the prolongation channels and $\mathcal{R}_n(x)$ the subspace of relations. The screen is

$$\Sigma_n(x) = \big(\mathbb{K}_n, M_n(x), \beta_n(x), \mathcal{R}_n(x)\big),$$

and its effective dimension is

$$d_{\Sigma_n}(x) = \dim_{\mathbb{K}_n}\big(M_n(x)/\mathcal{R}_n(x)\big). \tag{25.13}$$

If $M_n(x) \cong \mathbb{K}_n^{g_n}$ and the columns of $R_n(x)$ generate the relations, then $d_{\Sigma_n}(x) = g_n - \operatorname{rank} R_n(x)$.

Proposition 25.35 (Invariance of presentation and transport)**.** *The dimension* (25.13) *is preserved under invertible changes of basis. For two screens over the same field $\mathbb{K}_n = \mathbb{K}_m = \mathbb{K}$, if the linear map $F_\gamma : M_n(x) \to M_m(T_\gamma x)$ sends $\mathcal{R}_n(x)$ into $\mathcal{R}_m(T_\gamma x)$, it uniquely induces a linear map between the screen quotients.*

Proof. A change of generators and relations transforms R into URV, with U, V invertible, and preserves its rank. For descent, define $\overline{F}_\gamma([v]) = [F_\gamma(v)]$. If two representatives differ by a relation, linearity implies that their images differ by a relation of the terminal screen; the map is well defined. Projection to the quotient is surjective and guarantees uniqueness. □

The dimensional change during transport is determined by

$$\Delta d(\gamma; x) = d_{\Sigma_m}(T_\gamma x) - d_{\Sigma_n}(x).$$

This expression preserves the state and its channel presentation. An isolated rank does not allow reconstruction of either the relations or the map that produced it.

25.16.2 Ternary puncturing and syndrome screen

The code $C_W = [12, 6, 6]_3$, constructed from exceptional incidence, produces another screen by puncturing one coordinate. Denote that map by $p : C_W \to \mathbb{F}_3^{11}$ and its image by $G_{11} = p(C_W)$.

Proposition 25.36 (Punctured code and perfect partition)**.** *The map p is injective. The code G_{11} has parameters $[11, 6, 5]_3$, and its Hamming balls of radius 2 partition $\mathbb{F}_3^{11}$.*

Proof. A vector in the kernel of p would have support contained in the single removed coordinate. The minimum distance 6 of C_W excludes any nonzero vector of that form. Thus, the dimension remains 6.

Puncturing decreases the weight by at most one, so the minimum distance of the image is at least 5. The balls of radius 2 are therefore disjoint. Each contains

$$1 + 2\binom{11}{1} + 2^2\binom{11}{2} = 1 + 22 + 220 = 243$$

vectors. Since $|G_{11}| = 3^6$ and $3^6 \cdot 243 = 3^{11}$, the balls cover the ambient space. To check that the distance is exactly 5, take a vector of weight 3. Its decoding ball cannot have centre zero and has a centre of weight at most $3 + 2 = 5$; that centre must have weight at least 5. A word of weight 5 is thus obtained. □

The integer $243 = 3^{11-6}$ is here the number of classes of the syndrome quotient $\mathbb{F}_3^{11}/G_{11}$. Its function is fixed by the puncturing and distance of the code; equality of cardinality with another fibre does not replace the map between them. This combinatorial screen and the following real screen retain their own operations.

25.16.3 The dodecaphase direction and rank reduction

The construction of 21.11.5 starts from the register K already obtained by APP–TRIT–TPK and from the representation chart declared on $\mathbb{R}^{12}$. In that chart,

$$V_{11} = \mathbf{1}^{\perp}, \qquad P_{11} = I_{12} - \frac{1}{12}J_{12}, \qquad u_K = P_3 P_{11} K. \tag{25.14}$$

Here J_{12} is the all-ones matrix and P_3 is the isotypic projector specified by the action of A_5. The finite sum defining P_3, its representatives and the generation of K are preserved in the cited section of this same article. In particular,

$$\|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

The positive norm defines $\ell_K = \mathbb{R}u_K$, and the projector

$$P_{10} = P_{11} - \frac{u_K u_K^{\mathsf{T}}}{\|u_K\|^2} \tag{25.15}$$

has image $V_{10} = V_{11} \cap u_K^{\perp}$.

Proposition 25.37 (Two orthogonal reductions)**.** *The following hold*

$$\mathbb{R}^{12} = \mathbb{R}\mathbf{1} \oplus^{\perp} V_{11}, \qquad V_{11} = \ell_K \oplus^{\perp} V_{10}, \qquad \dim V_{10} = 10.$$

Moreover, $P_{10}P_{11} = P_{10}$, $P_{10}u_K = 0$, and P_{10} is an orthogonal projector.

Proof. Proposition 21.64, proved before constructing the screens, establishes idempotence, symmetry, rank and the two operator identities for this same u_K and this same P_{10}. The first decomposition follows from $P_{11} = I - J_{12}/12$; the second is the orthogonal decomposition of that proposition. This preserves the single proof of the reduction and makes its two stages explicit here. □

The first reduction removes only the uniform mode. The second removes the direction determined by K in the fixed representation. If the chart is simultaneously changed by an orthogonal matrix S, the vector transforms by $u_K \mapsto Su_K$ and the projector by $P_{10} \mapsto SP_{10}S^{-1}$. This covariance preserves the geometric operation, even though its coordinates change.

25.16.4 Lorentzian lattice screen

Let $\Lambda = N_\alpha$ be the Leech lattice constructed in 21.11.9. Consider the integral hyperbolic plane $U = \mathbb{Z}e_+ \oplus \mathbb{Z}e_-$, with

$$\langle e_+, e_+\rangle = \langle e_-, e_-\rangle = 0, \qquad \langle e_+, e_-\rangle = 1.$$

The lattice $L_{26} = \Lambda \oplus U$ is even and unimodular, with signature $(25, 1)$. Each vector is written $y = \lambda + ae_+ + be_-$. Since $\langle y, e_+\rangle = b$,

$$e_+^\perp = \Lambda \oplus \mathbb{Z}e_+, \qquad e_+^\perp/\mathbb{Z}e_+ \cong \Lambda. \tag{25.16}$$

The quotient map is $q_{24}(\lambda + ae_+) = \lambda$; it is surjective and has kernel $\mathbb{Z}e_+$. The decrease in rank by two comes from an orthogonality restriction followed by an isotropic quotient. It is therefore an operation of a different type from (25.15).

25.16.5 Joint screen morphism

With the domain $\mathscr{D}_0$ of 24.28, let

$$\mathscr{G}_K = \{(v_{11}, P_{10}v_{11}) : v_{11} \in V_{11}\},$$

$$\mathcal{S}_{\mathrm{HMT}} = \mathbb{R}^{12} \times \mathscr{D}_0 \times e_+^\perp, \qquad \mathcal{S}_M^{\mathrm{alg}} = \mathscr{G}_K \times \mathscr{D}_0 \times \Lambda.$$

Definition 25.38 (Algebraic map of coordinated screens)**.** It is defined

$$\mathcal{R}_M^{\mathrm{alg}}(v, d, y) = \big((P_{11}v, P_{10}v), d, q_{24}(y)\big). \tag{25.17}$$

Theorem 25.39 (Quotient isomorphism and duality intertwining)**.** *The map* (25.17) *induces an isomorphism*

$$\overline{\mathcal{R}}_M^{\mathrm{alg}} : (\mathbb{R}^{12}/\mathbb{R}\mathbf{1}) \times \mathscr{D}_0 \times (e_+^\perp/\mathbb{Z}e_+) \longrightarrow \mathcal{S}_M^{\mathrm{alg}}.$$

It intertwines the transformations acting by T_0 *on the factor* $\mathscr{D}_0$ *and by the identity on the other two factors. The form* $E_{R_0}(m, w)$ *is preserved under that duality.*

Proof. The map $[v] \mapsto P_{11}v$ is an isomorphism from $\mathbb{R}^{12}/\mathbb{R}\mathbf{1}$ onto V_{11}. Since $P_{10}P_{11} = P_{10}$, the map $[v] \mapsto (P_{11}v, P_{10}v)$ identifies that quotient with $\mathscr{G}_K$; its inverse takes the first component and its class. By (25.16), q_{24} induces the lattice isomorphism of the third factor. The second is transported by the identity. The product of the three maps proves the isomorphism.

The projectors and q_{24} do not modify d, whereas duality acts only on d. The two compositions are therefore identical. Preservation of the form is theorem 24.29. □

The graph $\mathscr{G}_K$ preserves the preceding coordinate v_{11} together with its projection. The theorem proves the equivalence of the quotient presentations; the isolated reduction $V_{11} \to V_{10}$ retains its kernel ℓ_K. This clarification establishes what information the isomorphism retains and what information each projection composing it removes.

Corollary 25.40 (Screens over the elliptic family)**.** *The same morphism restricts to an isomorphism*

$$(\mathbb{R}^{12}/\mathbb{R}\mathbf{1}) \times \mathscr{D}_{\mathrm{el}} \times (e_+^\perp/\mathbb{Z}e_+) \;\cong\; \mathscr{G}_K \times \mathscr{D}_{\mathrm{el}} \times \Lambda,$$

and preserves the duality intertwining.

Proof. The morphism and its inverse preserve the second factor exactly. Corollary 24.32 proves that duality preserves $\mathscr{D}_{\mathrm{el}}$. Thus both the bijection and the already proved intertwining square restrict. □

25.16.6 Coordination with the exceptional realization

On the domain $\mathcal{X}_{\mathrm{HMT}}$, with phase, orientation, quotient, boundary, incidence, grading and memory, the readings $\varepsilon_{\mathrm{exc}}$ and $\varepsilon_{\mathrm{scr}}$ extract the exceptional and screen data. The corridor constructs the graded algebra $\mathfrak{B}(x) \cong V^{\natural}$, with its vertex product and automorphisms; $\mathfrak{S}_M(x)$ brings together the preceding screens, quotients and operators. The joint assignment $x \mapsto (\mathfrak{B}(x), \mathfrak{S}_M(x))$ therefore has an algebra as its first component and a family of spaces and maps as its second. The Monster action belongs to the former; the reduction directed by K and compact duality belong to the latter. The two readings fix the common provenance and each construction retains its type. Theorem 25.39 gives the isomorphism factor by factor; its proof identifies the inverses, kernels and operations preserved by each component.

25.17 Action on trajectories and eleven-dimensional dynamical realization

Extending the structure to dimensional dynamics requires joint transport of the state, channels and boundary data. This section begins with a functional defined on TPK trajectories and its exact composition law. The worldsheet and eleven-dimensional codomain are introduced afterwards, with the maps that must preserve that composition and the operators already constructed.

25.17.1 Descent of an evolution and sufficiency of the observed datum

Let $q : \mathcal{X}_{\mathrm{HMT}} \to Y$ be a surjective readout of states with memory, and let $U : \mathcal{X}_{\mathrm{HMT}} \to \mathcal{X}_{\mathrm{HMT}}$ be a complete evolution. An evolution on observed values exists precisely when the datum preserved by q suffices to determine the next observed value.

Theorem 25.41 (Exact criterion for descent along fibres)**.** *There exists a unique map $\overline{U} : Y \to Y$ satisfying $qU = \overline{U}q$ if and only if*

$$q(x) = q(y) \quad \Longrightarrow \quad q(Ux) = q(Uy). \tag{25.18}$$

Proof. If $\overline{U}$ exists, applying the same function to $q(x) = q(y)$ proves the condition. Conversely, define $\overline{U}(q(x)) = q(Ux)$. Condition (25.18) guarantees independence of the representative. Surjectivity of q defines the map on all of Y and proves uniqueness. □

The theorem determines what memory a readout must preserve. When two representatives have different observed images, the retained datum may be enlarged, the domain restricted, or a multivalued correspondence used. These are distinct mathematical choices. In particular, return of a phase coordinate does not authorize identification of the evolutions of two histories retaining different memories.

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$$w_{\mathrm{giro}}(\widetilde{e}_j) = \mathbf{1}_{\{d_j \neq d_{j-1}\}}, \qquad w_{\mathrm{tipo}}(\widetilde{e}_j) = \mathbf{1}_{\{t_j \neq t_{j-1}\}}.$$

For $\widetilde{\gamma} = (\widetilde{e}_1, \dots, \widetilde{e}_R)$, let

$$\mathbf{I}(\widetilde{\gamma}) = \left(\sum_{j=1}^{R} w_{\mathrm{giro}}(\widetilde{e}_j), \sum_{j=1}^{R} w_{\mathrm{tipo}}(\widetilde{e}_j) \right) = \big(N_{\mathrm{giro}}, N_{\mathrm{tipo}}\big). \tag{25.19}$$

Two trajectories compose when their endpoints agree and the preceding datum of the second agrees with the terminal datum of the first.

Proposition 25.42 (Additivity with preserved boundary)**.** *For every composable pair,*

$$\mathbf{I}(\widetilde{\gamma\eta}) = \mathbf{I}(\widetilde{\gamma}) + \mathbf{I}(\widetilde{\eta}). \tag{25.20}$$

Thus, $\mathbf{I}$ *is an additive cocycle with values in* $\mathbb{N}^2$ *on the category of augmented trajectories. For every* $\lambda > 0$*, the functional*

$$I_\lambda(\widetilde{\gamma}) = N_{\mathrm{giro}}(\widetilde{\gamma}) + \lambda N_{\mathrm{tipo}}(\widetilde{\gamma})$$

is also additive.

Proof. Concatenation does not change the preceding datum of any edge: for the first edge of $\widetilde{\eta}$, the composition condition makes it agree with the terminal datum of $\widetilde{\gamma}$. The sum over all edges splits exactly into the two sums in (25.20). Applying the linear functional $(a, b) \mapsto a + \lambda b$ proves the last statement. □

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For fixed endpoints and boundary data, the minimum-interaction principle selects minimizers of I_λ among admissible routes. The principle is a selection rule, whereas the following result establishes a finite-existence property.

Proposition 25.43 (Finite Existence of Minimizers)**.** *If the set* $\Gamma(a, b)$ *of admissible trajectories with the fixed boundary is finite and nonempty, there exists* $\gamma_* \in \Gamma(a, b)$ *such that* $I_\lambda(\gamma_*) \le I_\lambda(\gamma)$ *for all* $\gamma \in \Gamma(a, b)$*.*

Proof. The image of $\Gamma(a, b)$ under I_λ is a nonempty finite set of real numbers, which has a minimum element and at least one preimage. □

The proposition does not select the value of λ, nor does it confuse existence with uniqueness. The value and selection rule used in a particular realization must be part of its declared data. The additive functional recovered here quantifies changes of the path itself; its transport to a dimensional action is a later operation.

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where T_{s} is tension. The notation T_{s} distinguishes it from duality T_0, cardinal transport T_d and complete evolution U.

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If physical transport of the radii also respects (24.38)*, the transported compact sector preserves its spectrum under reciprocal radii.*

Proof. Since $\psi \in D(\widehat{Q}^2)$, both ψ and $\widehat{Q}\psi$ belong to $D(\widehat{Q})$. Thus, $W\psi \in D(Q)$ and $QW\psi = W\widehat{Q}\psi \in D(Q)$. One may apply Q again, and

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4. the realization transports the direction ℓ_K, boundary conditions and operator domains, and preserves the declared action and supersymmetry equations.

The preceding proofs provide the phase Hilbert-space realization, tracial closure, screens with their quotient maps, radius involution, its unitary equivalence and the additive trajectory cocycle. The eleven-dimensional realization specifies the fields and compatibilities it uses when receiving that structure.

Conditions for physical identification. Complete physical identification must additionally specify the tension and action normalization, construct the oscillator sector and its amplitudes, control anomalies and establish a consistent low-energy gravitational limit for each stated compactification. The string action and change of scale used here are expressed in (25.21) and (24.41), respectively. Theorem 25.44 proves the supercharge identity on the image of W, under the domains and intertwining relations in its statement. The conditions for complete physical identification do not alter this identity or the other algebraic proofs: they delimit the physical realization to which those results are transported.

The exceptional and dimensional prolongations remain linked by their common origin. The Monster acts on the graded construction $V^{\natural}$; the compact Hamiltonian and screens receive data from other maps of the same state. This formulation preserves the positive content of the dual realization, with its operations, proofs and the information transmitted at each stage: Theorem 21.71 establishes the exceptional identification, and Theorems 24.31 and 25.39 establish compact equivalence and screen coordination.

25.18 Finite enumerations of emissions and Witt stars

The two enumerations in this appendix concern distinct objects. The first classifies the cursor signatures of APP emission; the second computes the group generated by the stars of the Witt design. Their domains and rules are retained so that each table can be reproduced without being used as an input to the calculation.

25.18.1 Initial-condition indices and signature recurrence

Let $i, j \in \{0, \dots, 8\}$ and $a \in \{0, 1, 2, 3\}$, with direction order (N, E, S, O) and vectors

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The index

$$\mathrm{id}(i, j, a) = 4(9i + j) + a$$

ranges over exactly $\{0, \dots, 323\}$. Its inverse is obtained from $a = \mathrm{id} \bmod 4$, $j = \lfloor \mathrm{id}/4 \rfloor \bmod 9$ and $i = \lfloor \mathrm{id}/36 \rfloor$. Two indices, one per sheet, enumerate the 324^2 joint conditions in (4.29); this domain is not identified with the 81 positions.

For each index, the cursor starts at (i, j), with vector v_a. The instants $t = 1, \dots, 54$ are traversed in six windows of nine instants. Write $r_t = 1 + ((t - 1) \bmod 9)$. The rule is:

1. On the additive sheet, an operation occurs when $r_t \in \{1, 4, 7\}$: $\rho_9(i + j + 2)$ is added to the window accumulator, and the cursor moves by v_a, modulo nine.
2. On the multiplicative sheet, an operation occurs when $r_t \in \{2, 5, 8\}$: $\ell_9(\rho_9((i + 1)(j + 1)))$ is added, and the cursor moves by v_a, modulo nine. The extension of ℓ_9 by zero outside the units, stated in Section 4.3, is retained.
3. At all other instants, that cursor does not move. After instants 27 and 54, v_a is replaced by $-v_a$.
4. At the end of each window, its accumulator is recorded modulo ten, and only the accumulator is reset; neither position nor orientation is reset.

This defines two functions on the 324 indices,

$$s : \{0, \dots, 323\} \longrightarrow \{0, \dots, 9\}^6, \qquad e : \{0, \dots, 323\} \longrightarrow \{0, \dots, 9\}^6.$$

The multiplicity of a signature is the number of indices with that image. The witness index in each table is the least index in its fibre. Complete preimages are recovered by applying the recurrence to all 324 indices and retaining those that produce the stated signature.

Table 25.1: Additive signatures: image of s, multiplicity and witness index.

Class	Signature	Multiplicity	Witness index
1	122564	18	17
2	122898	18	24
3	212645	18	5
4	212988	18	0
5	221456	18	29
6	221889	18	12
7	456221	18	28
8	465898	18	21
9	546988	18	9
10	564122	18	16
11	645212	18	4
12	654889	18	33
13	889221	18	13
14	889654	18	32
15	898122	18	25
16	898465	18	20
17	988212	18	1
18	988546	18	8

Table 25.2: Multiplicative signatures: image of e, multiplicity and witness index.

Class	Signature	Multiplicity	Witness index
1	000000	108	8
2	177393	8	1
3	177555	8	54
4	177771	8	11
5	339555	8	6
6	339771	8	15
7	339933	8	59
8	393177	8	0
9	393393	8	45
10	393555	8	40
11	555177	8	52
12	555339	8	4
13	555393	8	41
14	555555	24	84
15	555717	8	14
16	555771	8	18
17	555933	8	24
18	717555	8	12
19	717717	8	21

Class	Signature	Multiplicity	Witness index
20	717933	8	19
21	771177	8	9
22	771339	8	13
23	771555	8	16
24	933339	8	57
25	933555	8	26
26	933717	8	17

The sums of multiplicities are $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. Joint emission is obtained by applying (4.31) to every pair of signatures. The inverse through hundreds and tens proved in Section 4.3 identifies the 468 emissions and gives their multiplicities as the product of the two fibres. The subsequent map (4.32) produces the 243 visible words. The initial domain, emission, its fibres and the ternary quotient are thus specified separately.

25.18.2 Explicit generators and closure of the stars

The matrix A_W of (21.126) is retained, and the following words are formed in lexicographic order:

$$c(w) = (w, wA_W), \qquad w \in \mathbb{F}_3^6.$$

They are grouped by weight and the supports of weight six are retained. This produces the enumerator (21.127) and the 132 hexads of $\mathcal{H}$. For each quadruple $B \subset \{1, \dots, 12\}$, the four pairs $H \setminus B$, with $B \subset H \in \mathcal{H}$, define σ_B: it fixes B and interchanges the two points of each pair. This gives the family of 495 stars of proposition 21.66.

The six generators used in its closure are expressed by their lists of images of $1, \dots, 12$:

	1	2	3	4	5	6	7	8	9	10	11	12
g_1	1	2	3	4	12	11	9	10	7	8	6	5
g_2	1	2	3	12	5	10	11	9	8	6	7	4
g_3	1	2	3	11	10	6	8	7	12	5	4	9
g_4	1	2	12	4	5	9	8	7	6	11	10	3
g_5	1	12	3	4	5	8	10	6	11	7	9	2
g_6	12	2	3	4	5	7	6	11	10	9	8	1

Each row is verified to be a star by comparison with the σ_B already constructed. For $j = 1, \dots, 6$, the closure is computed by

$$G_j^{(0)} = \{\mathrm{id}\}, \qquad G_j^{(n+1)} = G_j^{(n)} \cup \{g_i \circ p : 1 \le i \le j,\ p \in G_j^{(n)}\}.$$

The increasing sequence stabilizes in the finite set $\mathrm{Sym}(12)$. Since the generators are involutions, its stable value is exactly the group G_j they generate. With the convention $(g_i \circ p)(a) = g_i(p(a))$, their orders are

j	1	2	3	4	5	6
$\lvert G_j \rvert$	2	6	18	360	7920	95040.

Finally, one checks that each of the 495 stars belongs to G_6 and that they all preserve $\mathcal{H}$. The first inclusion, together with membership of the six generators in the family, makes the two generated groups equal. Preservation is checked through $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$. This is the finite enumeration used in the proof of 21.66; subsequent recognition as M_{12} retains the external reference declared there.

Chapter 26

Completely positive systems and matrix genealogy of the continuum: convexity, Choi matrices, Stinespring dilation, and De les Coves–Netzer bridge

26.1 Matrix range of APP: exact classification of the sum–product algebra and free interval

The qutrit squares of section 26.3 provide an operatorial input. There exists a second input, more internal to APP: the two partitions induced by sum and product generate noncommutative orthogonal projections. The language of operator systems makes it possible to study them without forcing a grid.

26.1.1 Pair of conditional expectations associated with the additive and multiplicative partitions of APP

Let

$$\mathcal{H}_d = \ell^2(\{1,\dots,9\}^d),$$

and let us define

$$\Sigma_d(x) = \mathrm{dr}(x_1 + \cdots + x_d), \qquad \Pi_d(x) = \mathrm{dr}(x_1 \cdots x_d). \tag{GMC-6.1}$$

Let $P_{\Sigma,d}$ and $P_{\Pi,d}$ be the orthogonal projections onto the functions constant on the fibres of Σ_d and Π_d. The preceding bridge proves

$$\|[P_{\Sigma,2}, P_{\Pi,2}]\| = \frac{\sqrt{2}}{3}, \qquad \|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}, \tag{GMC-6.2}$$

and

$$\operatorname{Ran} P_{\Sigma,d} \cap \operatorname{Ran} P_{\Pi,d} = \mathbb{C}\mathbf{1} \qquad (2 \le d \le 15). \tag{GMC-6.3}$$

APP therefore contains a noncommutative operator package prior to any QLS or QMS designation.

We define

$$\mathcal{S}_d = \mathrm{span}\{I, P_{\Sigma,d}, P_{\Pi,d}\}, \qquad \mathfrak{C}_d^{\mathrm{APP}} = C^*(P_{\Sigma,d}, P_{\Pi,d}). \tag{GMC-6.4}$$

$\mathcal{S}_d$ is a system of operators and its rango matrix is

$$\mathcal{W}_s(\mathrm{APP}_d) = \left\{(\Phi(P_{\Sigma,d}), \Phi(P_{\Pi,d})) : \Phi : \mathfrak{C}_d^{\mathrm{APP}} \to M_s \text{ UCP}\right\}. \tag{GMC-6.5}$$

The tower $\bigsqcup_s \mathcal{W}_s$ is matricially convex. Unlike a general label of "quantumness," (GMC-6.5) makes it possible to compute extreme points, compressions, dilations, and invariants.

26.1.2 Classification of the sum–product algebra at depth 2

Place $P = P_{\Sigma,2}$ and $Q = P_{\Pi,2}$ on $\mathcal{H}_2$, of dimension 81. The nine fibres of Σ_2 have size nine. The fibres of Π_2 have sizes

$$6, 6, 12, 6, 6, 12, 6, 6, 21. \tag{GMC-6.6}$$

Let u_a and v_b be the normalized indicator vectors of those fibres, and $C = (\langle u_a, v_b\rangle)_{a,b=1}^{9}$ their cross-Gram matrix. The exact intersection count yields

$$\chi_{CC^*}(t) = t^5(t-1)\left(t - \frac{5}{7}\right)\left(t - \frac{1}{3}\right)^2. \tag{GMC-6.7}$$

Moreover,

common subspace	dimension
$\ker P \cap \ker Q$	64
$\mathrm{Ran}\, P \cap \mathrm{Ran}\, Q$	1
$\mathrm{Ran}\, P \cap \ker Q$	5
$\ker P \cap \mathrm{Ran}\, Q$	5

(GMC-6.8)

Theorem 26.1 (Sum–product algebra of APP in $d = 2$)**.** *There is an isomorphism of finite-dimensional C^*-algebras*

$$\boxed{\mathfrak{C}_2^{\mathrm{APP}} \cong \mathbb{C}^4 \oplus M_2(\mathbb{C}) \oplus M_2(\mathbb{C}).} \tag{GMC-6.9}$$

In particular, $\dim \mathfrak{C}_2^{\mathrm{APP}} = 12$.

Proof. The canonical Halmos decomposition for two projections separates the four subspaces of (GMC-6.8) and a generic sector for each principal angle $0 < \lambda < 1$. Equation (GMC-6.7) shows that the only distinct generic values are $5/7$ and $1/3$. In each case, the pair is unitarily equivalent to

$$P_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q_\lambda = \begin{pmatrix} \lambda & \sqrt{\lambda(1-\lambda)} \\ \sqrt{\lambda(1-\lambda)} & 1-\lambda \end{pmatrix}, \tag{GMC-6.10}$$

which generates M_2. The multiplicity of $\lambda = 1/3$ amplifies the same representation but does not create another simple summand. The four sectors $0/1$ simultaneously produce $\mathbb{C}^4$. □

Specifically,

$$Q_{5/7} = \begin{pmatrix} 5/7 & \sqrt{10}/7 \\ \sqrt{10}/7 & 2/7 \end{pmatrix}, \qquad Q_{1/3} = \begin{pmatrix} 1/3 & \sqrt{2}/3 \\ \sqrt{2}/3 & 2/3 \end{pmatrix}. \tag{GMC-6.11}$$

The norm of the commutator is the maximum of $\sqrt{\lambda(1-\lambda)}$ and recovers $\sqrt{2}/3$.

Corollary 26.2 (Complete parameterization of the matrix range)**.** *A pair (X, Y) belongs to $\mathcal{W}_s(\mathrm{APP}_2)$ if and only if there exist effects $E_{00}, E_{11}, E_{10}, E_{01} \succeq 0$ and CP maps $\Psi_{5/7}, \Psi_{1/3} : M_2 \to M_s$ such that*

$$E_{00} + E_{11} + E_{10} + E_{01} + \Psi_{5/7}(I) + \Psi_{1/3}(I) = I_s, \tag{GMC-6.12}$$

$$X = E_{11} + E_{10} + \Psi_{5/7}(P_0) + \Psi_{1/3}(P_0), \tag{GMC-6.13}$$

$$Y = E_{11} + E_{01} + \Psi_{5/7}(Q_{5/7}) + \Psi_{1/3}(Q_{1/3}). \tag{GMC-6.14}$$

Proof. Every CP map from a direct sum decomposes into CP maps on its summands. The four scalar summands produce the effects E_{ab}; the two matrix summands produce Ψ_λ. Unitality is (GMC-6.12); the coordinates of P and Q in (GMC-6.9) give (GMC-6.13)–(GMC-6.14). The argument is reversible. □

Matrix bridge result

The classification (GMC-6.9) provides the operator-theoretic realization of the bridge. It converts the sum–product incompatibility into a fully computable operator object: four classical sectors and two irreducible matrix blocks. The value $1/3$ reappears here as a principal operator angle, not merely as an MUB overlap; $5/7$ is the second generic value of PQP in the sum–product sector. Neither should be identified with the contraction $5/9$ of the central block B_c. They are invariants of distinct objects, coordinated without being identified by decimal similarity.

26.1.3 Spectral separation 4 and free matrix interval

For the block of (GMC-2.4), $B_c = 7I + 2R$ with $R = R^*$ nonscalar, $R^2 = I$ and $\mathrm{Spec}(R) = \{-1, +1\}$, $C^*(B_c) = C^*(R) \cong \mathbb{C}^2$. Its matrix rank at level s is

$$\mathcal{W}_s(B_c) = \{\Phi(B_c) : \Phi : C^*(B_c) \to M_s \text{ UCP}\}.$$

Theorem 26.3 (Free interval of the central block)**.** *For every $s \geq 1$,*

$$\boxed{\mathcal{W}_s(B_c) = \{Y = Y^* : 5I_s \preceq Y \preceq 9I_s\}.} \tag{GMC-6.15}$$

Proof. A UCP map sends the symmetry R to a self-adjoint contraction $-I_s \preceq X \preceq I_s$ and, therefore, $\Phi(B_c) = 7I_s + 2X$ belongs to the interval. Conversely, given such an X, the effects $(I_s \pm X)/2$ define a UCP map from $\mathbb{C}^2$ that sends R to X. □

Thus, 7 is the center, 2 the operatorial radius, and 4 the spectral diameter of the free interval. The positive boundary, the boundary of the matrix interval, and the determinant-one hyperbola of (GMC-2.6) are three distinct notions that can now be coordinated without being conflated.

The double projection does not reside in this early classification. Its formulation from a common spectral antecedent is published, with its hypotheses and limits, in the dedicated section of the governing chapter on double projection.

26.1.4 Common antecedent of readers: spectral refinement and joint dilation limit

Let X_m be a finite set of histories and let

$$R_m : X_m \to \mathcal{A}_m, \qquad Q_m : X_m \to \mathcal{E}_m \tag{GMC-6.16}$$

two readers: for example, one Archimedean and the other exceptional. Each one defines a PVM in $C(X_m)$ by means of the characteristic functions of its fibers.

Proposition 26.4 (Common spectral predecessor)**.** *The two* PVM *of* (GMC-6.16) *commute and possess the common refinement*

$$P_{a,e} = 1_{\{R_m = a\}} 1_{\{Q_m = e\}} = 1_{\{x : R_m(x) = a,\ Q_m(x) = e\}}. \tag{GMC-6.17}$$

If the readers are compatible with the truncations, the refinements are compatible in m and determine two readers on $C(X_\infty)$ with a common cylindrical PVM*.*

Proof. All functions belong to the same commutative algebra. The product of the indicators is the indicator of the intersection. Summing in e recovers the PVM of R_m, and summing in a recovers that of Q_m. The compatibility passes to the limit by the Theorem 26.11. □

This result formalizes the common antecedent of the double projection. It does not prove that the pair of readers is jointly injective: for that purpose, it must separate points of X_∞. Nor does it automatically construct a common noncommutative antecedent.

Proposition 26.5 (Incompatibility of Two Mutually Unbiased Projective Measurements)**.** *Two mutually unbiased qutrit* PVM *do not admit a joint sharp measurement in $M_3(\mathbb{C})$.*

Proof. Two finite PVM are jointly measurable by a common PVM if and only if they commute. Projectors from distinct MUB contexts have overlap $1/3$ and do not commute. □

The HMT double projection must therefore not be identified with two sharp MUB contexts on the same qutrit. It can be realized on the common diagonal of histories, through noisy effects, or in a larger

noncommutative antecedent with two UCP channels. The last possibility requires inclusion of matrix ranges, Choi positivity and equivariance with respect to TPK dynamics.

Mathematical extensions

The noncommutative antecedent is constructed in Section 17.9: the system $\mathcal{S}_\infty$ and the UCP maps Φ_{arq} and Φ_{exc} simultaneously reproduce the two HMT projections and intertwine the TPK dynamics on the declared equivariant domain. The Proposition 26.4 provides its diagonal antecedent.

26.2 Genealogical matrix convexity: objects, domains, and typed dictionary

26.2.1 Construction of 4 cyclic quantum Latin squares in an elementary operator class

The question organizing the work of De las Cuevas–Netzer is how a collection of local measurements can compose a two-dimensional object and to what extent that object is the compression of an ideal totality. An isolated measurement —a POVM— admits a Naimark or Stinespring dilation. However, a grid simultaneously contains one POVM in every row and every column. Requiring a common dilation that makes all those measurements projective is a much stronger global condition. There are quantum magic squares that do not satisfy it, even at the smallest relevant noncommutative matrix level [36].

HMT formulates another problem of parts and wholes. The visible coordinate of a state does not necessarily preserve the route that produced it. APP distinguishes the sum and product evaluations on a common support; TRIT preserves orientation and carry; TPK preserves path composition, return, holonomy, and memory. The continuum arises by coherently prolonging finite histories, and the double projection publishes two different shadows of the same enriched state. Both theories therefore ask what is lost under compression. The difference is that one organizes loss by matrix level and the other by genealogical depth.

Scientific reading

De las Cuevas asks: «Can I view this imperfect grid as the compression of a larger projective grid?». HMT asks: «What complete history was compressed to produce this output?». The bridge answers that both questions can be formulated on the same family, with two independent indices: depth m and observation dimension s.

26.2.2 Quantum Squares, Operator Systems, and Matrix Convexity

Let $n, s \geq 1$. A quantum magic square of exterior size n and interior size s is a matrix

$$A = (A_{ij})_{i,j=0}^{n-1} \in \mathrm{Mat}_n(\mathrm{Her}_s(\mathbb{C}))$$

such that

$$A_{ij} \succeq 0, \qquad \sum_{j=0}^{n-1} A_{ij} = I_s, \qquad \sum_{i=0}^{n-1} A_{ij} = I_s. \tag{GMC-1.1}$$

Each row and each column is a POVM. If all entries are projections, A is a quantum permutation matrix; if, moreover, those projections commute, A is a commutative quantum permutation matrix [36, 39].

To QMS is *semiclasico* if there exists to POVM $\{Q_\sigma\}_{\sigma \in S_n}$ such that

$$A = \sum_{\sigma \in S_n} P_\sigma \otimes Q_\sigma, \tag{GMC-1.2}$$

where P_σ is the classical permutation matrix associated with σ. The formula does not require the effects Q_σ to commute. It states that the entire grid is obtained by reordering a single primary measurement.

A quantum Latin square of order n is a grid of unit vectors $v_{ij} \in \mathbb{C}^n$ whose rows and columns are orthonormal bases. The rank-one projectors

$$P_{ij} = |v_{ij}\rangle\langle v_{ij}|$$

form a projective QMS. The easy-class QLS are those produced by a classical Latin square and a fixed orthonormal basis. De las Cuevas, Netzer, and Valentiner-Branth prove that the semiclassical QLS are exactly those of the easy class and that the matrix convex hull of the QLS contains more than the semiclassical squares [39].

The elementary convex matrix operation is

$$X = \sum_{r=1}^{k} V_r^* X^{(r)} V_r, \qquad \sum_{r=1}^{k} V_r^* V_r = I_s. \tag{GMC-1.3}$$

It permits objects of different internal sizes to be combined. In operator language, compressions and direct sums are organized by unital completely positive maps. The set of all matrix values of an operator system is its matrix range.

Antecedent theorem

The governing results used are: a QMS dilates to a quantum permutation matrix with commuting entries if and only if it is semiclassical; for $n \geq 3$ and $s \geq 2$ there exist nonsemiclassical QMS, and for every $n \geq 3$ there exist QMS on $M_2(\mathbb{C})$ that do not dilate to any quantum permutation matrix [36]. Simultaneous dilatability is therefore a substantive property, not an automatic consequence of each row and column dilating separately.

26.2.3 Typed correspondence between the HMT–MD genealogical state and operator systems: depth, truncation, monodromy, and reading

In HMT, a visible output does not exhaust the state. A finite section carries, according to the domain, data of the form

$$x_m = (\text{prefix}, \text{sheet}, \text{orientation}, \text{carry}, \text{boundary}, \text{route}, \text{memory}, \text{gauge}). \tag{GMC-1.4}$$

The depth m is not a Hilbert-space dimension. It counts how much of the genealogy has been resolved. Two states may share a visible coordinate and differ in m, in leaf, or in memory.

APP provides the nonadic orthogram and the sum and product sheets. TRIT records the ternary orientation and the local carry. TPK is the topological phase kernel and the category of routes that composes those data. In a finite realization, the routes act by operators on a Hilbert space, but the representation does not replace the genealogy: it transports it.

The language of HMT also contains three operations that are absent from the bare definition of a QMS:

1. *truncation*: forgetting the continuation beyond depth m, preserving the prefix and aggregating its prolongations;
2. *monodromy*: returning to the same visible phase with nontrivial memory;
3. *reading*: publishing a mathematical or physical coordinate without identifying it with the complete section.

26.2.4 Realization of 3 operator contexts in a quantum square (9,3)

Matrix language	Lenguaje HMT	Correspondencia tipada
Outer size n	published grid	number of rows and columns of the magic object; it is not depth nor internal dimension
Internal size s	carta operatorial	dimension of the effects or of the matrix compression
Profundidad m	genealogy	new index that does not reduce to n or s
POVM	family of positive readers	effects that sum to the unit; the sum expresses normalization, not complete memory
PVM	contexto espectral	family of orthogonal projections that resolves the identity
UCP	matrix reader	map that preserves unitarity and positivity at all levels
Compression $V^*(\cdot)V$	shadow	step from a higher realization to an observational chart lower
Dilation	antecedente operatorio	common multiplicative representation before compressing
Semiclassicality	a single reordered primary POVM	formal property; does not equivalent to commutativity of all cells
No semiclasicidad	genuinely two-dimensional composition	cannot be reduced to the reordering of a common primary POVM
System of operators	space of readers	self-adjoint subspace with the unit that organizes all the UCP cards
Rango matricial	set of shadows to all the s	free family of internal evaluations
Inductive limit	closure of operatorial refinements	conservation of the unit through the entire nonadic filtration
Inverse limit	infinite section compatible	history that preserves all its prefixes

26.2.5 Representation of the complete genealogical atlas in a quantum square (12,3)

The magic-square formalism separates (n, s). HMT adds m. The total object must be written

$$\mathfrak{M}_{m,s}^{(n)}, \tag{GMC-1.5}$$

and not simply $\mathfrak{M}_n$ or $\mathfrak{M}_s$. Each index answers a different question:

Index	Question	Associated operation
m	How much history is preserved?	refinement and truncation
n	How is the grid composed?	rows, columns and permutations
s	From which matrix level is it observed?	direct sum and compression UCP

For APP, $n = 9$ is fixed. Then the family $\{\mathfrak{M}^{(9)}_{m,s}\}_{m,s}$ is bigraded by (m, s). If APP, the four QLS of order three, and the atlas of twelve directions are compared, the index n varies once more, and the correct structure is trigraded.

Structural obstruction

Calling it a "bigrading (m, s)" does not erase n: it means that a family with fixed exterior size is being studied. Calling APP "9×9" likewise does not fix the interior size of its operators. The internal matrices 6×6, the qutrit chart $s = 3$, and the exterior grid $n = 9$ are objects of distinct types.

26.2.6 Bidirectionality of Mathematical Exchange

The De las Cuevas–Netzer ⟶ HMT direction contributes criteria that HMT had not formulated with this degree of separation: cellwise positivity, normalization of all rows and columns, semiclassicality, the boundary of the matrix convex hull, simultaneous dilation, and Arveson extreme points. The result is a set of tests and falsifiers for any claim of a «quantum square».

The HMT ⟶ De las Cuevas–Netzer direction adds structure to the object being compressed: depth, route, nonsplitting of the carry, monodromy, dynamics, gauge, double projection, and metrological readers. The problem ceases to ask only whether a dilation exists and instead asks whether a *genealogically faithful* dilation exists: one that does not identify two distinct routes merely because they produce the same visible effect.

Matrix bridge result

The new notion organizing the article is the *simultaneous genealogical dilation*: a common multiplicative representation that dilates not only several rows and columns at fixed depth, but all compatible depths, and that specifies which part of memory remains in the representation and which part is lost in compression. The exact theorem is proved in Section 26.4.

26.3 Operator systems of incidence: quantum Latin squares, qutrit contexts, and common genealogical dilation

The outer APP object has order nine, but the number nine does not suffice to make it a quantum magic square. One must exhibit 81 positive effects and verify 18 operator normalizations. This chapter performs that task for two shadows of APP and for the complete atlas of twelve directions. It also determines its exact class in the De las Cuevas–Netzer taxonomy.

26.3.1 Cyclic classification of 4 quantum Latin squares into an elementary operator class

Let

$$\mathcal{P}_a = \{P_{a,0}, P_{a,1}, P_{a,2}\}, \qquad a \in \mathbb{P}^1(\mathbb{F}_3), \tag{GMC-4.1}$$

the four qutrit PVM of the Weyl system. For each a, choose unit vectors $v_{a,k}$ with $P_{a,k} = |v_{a,k}\rangle\langle v_{a,k}|$ and define

$$L^{(a)}_{ij} = v_{a,i+j \bmod 3}. \tag{GMC-4.2}$$

Each row and each column traverses the basis $(v_{a,0}, v_{a,1}, v_{a,2})$ once. Therefore, $L^{(a)}$ is a QLS of order three. Since it is obtained from the cyclic Latin table and from a fixed orthonormal basis, it is of easy class and its projective QMS is semiclassical [39].

Proposition 26.6 (MUB Obstruction to Latin Stacking)**.** *Two rows from distinct contexts $\mathcal{P}_a$ and $\mathcal{P}_b$, $a \neq b$, cannot be taken as rows of the same* QLS *of order three while preserving their aligned columns.*

Proof. The column condition of a QLS would require orthogonality between the two vectors of a column. But mutual unbiasedness gives

$$|\langle v_{a,k}, v_{b,\ell}\rangle|^2 = \mathrm{Tr}(P_{a,k}P_{b,\ell}) = \frac{1}{3}$$

for every k, ℓ. The overlap is not zero. □

The obstruction does not deny quantum compatibility of the four bases. It determines that the operator structure appropriate for using them simultaneously is not a QLS of rank one: one must pass to weighted effects and to a QMS.

26.3.2 Bigraduation of the genealogical-matrix family

Lemma 26.7 (Cyclic square associated with a POVM)**.** *Let $E = (E_0, \dots, E_{n-1})$ be a* POVM *on $\mathbb{C}^s$. Then*

$$A_{ij} = E_{i+j \bmod n}, \qquad 0 \le i, j < n, \tag{GMC-4.3}$$

is a QMS *of sizes (n, s). It is also semiclassical and admits a simultaneous dilation to a commutative* QPM.

Proof. Each entry is positive. Every row and every column is a permutation of the n effects, so that its sum is I_s. For $t \in \mathbb{Z}/n\mathbb{Z}$, let $\sigma_t(i) = t - i$ and P_{σ_t} be its permutation matrix. Then

$$A = \sum_{t=0}^{n-1} P_{\sigma_t} \otimes E_t, \tag{GMC-4.4}$$

which is an explicit semiclassical decomposition. If $E_t = V^* Q_t V$ is a Naimark dilation of the POVM to a PVM $(Q_t)_t$, the entries $\widehat{A}_{ij} = Q_{i+j}$ form a commutative QPM and $A_{ij} = V^* \widehat{A}_{ij} V$ simultaneously for the n^2 cells. □

This construction belongs to the class of Latin families associated with POVMs (*POVM Latin-square friends*) of De las Cuevas and Netzer [39]. The novelty of the bridge lies not in the abstract lemma, but in the POVM furnished by HMT, their nonadic calibration, and the exact separation between what each square preserves and what it still compresses.

26.3.3 Additive APP representation of sizes $(9, 3)$ and its incidence codomain

Fix one of the four contexts, for example $\mathcal{P}_0 = \{P_{0,0}, P_{0,1}, P_{0,2}\}$. For $t \in \mathbb{Z}/9\mathbb{Z}$ define

$$E_t^{\Sigma} = \frac{1}{3} P_{0,t \bmod 3}, \qquad A_{ij}^{\Sigma} = E_{i+j \bmod 9}^{\Sigma}. \tag{GMC-4.5}$$

Each projection appears three times, then

$$\sum_{t=0}^{8} E_t^{\Sigma} = \frac{1}{3} \sum_{k=0}^{2} 3P_{0,k} = I_3.$$

By Lemma 26.7, A^{Σ} is a semiclassical QMS of sizes $(9, 3)$. It is a direct transduction of the Latin law $\Sigma(i, j) = i + j \bmod 9$ into a qutrit reading. It preserves the outer size and the additive leaf; it deliberately does not distinguish the three Hensel lifts of the same ternary class.

26.3.4 3 contexts in a square $(9, 3)$

Elijamos three of the four PVM qutrit, indexadas by $a = 0, 1, 2$, and ordenemos their projectors by means of $t = 3a + k$:

$$E^{(9)}_{3a+k} = \frac{1}{3} P_{a,k}, \qquad a, k \in \mathbb{Z}/3\mathbb{Z}. \tag{GMC-4.6}$$

Since

$$\sum_{a,k} E^{(9)}_{3a+k} = \frac{1}{3} \sum_{a=0}^{2} \sum_{k=0}^{2} P_{a,k} = I_3,$$

the formula

$$A^{(9)}_{ij} = E^{(9)}_{i+j \bmod 9} \tag{GMC-4.7}$$

produce otro QMS of sizes $(9, 3)$.

Theorem 26.8 (Qutrit square of outer order nine)**.** *The square $A^{(9)}$ of* (GMC-4.7) *satisfies:*

1. *its 81 cells are positive effects of rank one and trace $1/3$;*
2. *all rows and columns sum to I_3;*
3. *contains entries that do not commute;*
4. *is semiclassical by means of*
$$A^{(9)} = \sum_{t=0}^{8} P_{\sigma_t} \otimes E^{(9)}_t; \tag{GMC-4.8}$$
5. *admits a simultaneous commutative dilation of its 81 entries.*

Proof. The first two points and the decomposition follow from Lemma 26.7. If $a \neq b$, two rank-one MUB projectors do not commute unless they coincide or are orthogonal, both of which are impossible because their overlap is $1/3$; hence some $[E^{(9)}_{3a+k}, E^{(9)}_{3b+\ell}]$ are nonzero. The last assertion is the dilation constructed in the proof of the lemma. □

Matrix bridge result

This theorem closes an existence that the preliminary bridge had left open: there already are 81 explicit effects with $n = 9$, $s = 3$ and exact normalizations. At the same time it proves a fertile conceptual separation:

$$\text{noncommutative cells} \quad \not\Longrightarrow \quad \text{no semiclassicality}.$$

Here the cells do not commute, but the entire grid arises by reordering a single primary POVM.

26.3.5 The atlas complete in a cuadrado $(12, 3)$

The projective line over the nonadic ring has twelve points:

$$|\mathbb{P}^1(\mathbb{Z}/9\mathbb{Z})| = 12. \tag{GMC-4.9}$$

Reduction modulo three induces

$$r : \mathbb{P}^1(\mathbb{Z}/9\mathbb{Z}) \longrightarrow \mathbb{P}^1(\mathbb{F}_3), \tag{GMC-4.10}$$

with four fibers of cardinality three. Fixing a calibration

$$\Xi : \mathbb{P}^1(\mathbb{Z}/9\mathbb{Z}) \xrightarrow{\sim} \{(a, k) : a \in \mathbb{P}^1(\mathbb{F}_3),\ k \in \mathbb{F}_3\} \tag{GMC-4.11}$$

that respects the fibre of r, each direction ℓ is assigned the projector $P_{\Xi(\ell)}$. Ordering the twelve directions as $t = 0, \dots, 11$, let

$$F_t = \frac{1}{4} P_{\Xi(\ell_t)}, \qquad B_{ij} = F_{i+j \bmod 12}. \tag{GMC-4.12}$$

Theorem 26.9 (Square qutrit of the complete nonadic atlas)**.** *The matrix B is a* QMS *of tamanos* $(12, 3)$*, contains the twelve projectors calibrados of the four contextos qutrit and is semiclasica:*

$$B = \sum_{t=0}^{11} P_{\sigma_t} \otimes F_t. \tag{GMC-4.13}$$

It admits a simultaneous dilation to a commutative QPM*.*

Proof. Each of the four contexts sums to I_3, so $\sum_t F_t = (1/4) \cdot 4 I_3 = I_3$. The Lemma 26.7 applies. □

The factor $1/4$ is not decorative. The atlas contains four resolutions of the identity and, without that normalization, the sum would be $4I_3$. The complete atlas is therefore a twelve-outcome measurement, not a fifth orthonormal basis.

26.3.6 Operatorial theorems and genealogical fidelity criterion APP–TPK

The three previous constructions have an exact status:

Object	Closed theorem	HMT Structure Not Yet Transported
A^{Σ}, $(9, 3)$	sheet sum, positivity, normalization and semiclassicality	product sheet, quotient and paths
$A^{(9)}$, $(9, 3)$	three contexts, noncommutativity and common dilation	fourth context, Hensel incidence, gnomon and TPK memory
B, $(12, 3)$	complete atlas $12 \to 4 \times 3$ and common dilation	equivariance of calibration, dynamics, and monodromy

Neither of the final two squares is a QLS or a QPM: their cells are projectors scaled by $1/3$ or $1/4$, not projections. They are quantum magic squares under the orthodox definition, and they do possess a simultaneous dilation. This formulation resolves the terminological ambiguity and preserves the mathematical difference between an effect and a projection.

Mathematical extensions

The next objective is not to prove again that a QMS of order nine exists. It is to construct an *APP–TPK-faithful* one: in addition to (GMC-4.3), it must simultaneously carry Σ, Π, quotient, orientation, and route, and it must satisfy an equivariance equation with the TPK action. A candidate that does not satisfy that equation will remain a valid shadow, but not the total object.

26.4 Genealogical–matrix bigrading at fixed outer size and trigrading (m, n, s): truncation, compression, and simultaneous dilation

The De las Cuevas–Netzer separation between outer size n and matrix level s resolves an ambiguity in the language of grids. HMT requires a second separation: depth m is not an inner size. This chapter proves that the m and s directions are compatible and that a family coherent at every depth possesses a common dilation.

26.4.1 Inverse system of histories and direct system of observables

Let $(X_m, p_m)_{m\geq 0}$ be a system of nonempty finite sets with surjective maps

$$p_m : X_{m+1} \longrightarrow X_m. \tag{GMC-3.1}$$

In HMT, X_m may be the complete set of nonadic prefixes of length m or a set of survivors, provided that the restriction maps are surjective on the declared domain. The inverse limit

$$X_\infty = \varprojlim_m X_m$$

is the compact space of compatible histories. Each $x \in X_m$ determines the clopen cylinder

$$[x]_m = \{\xi \in X_\infty : \xi_m = x\}. \tag{GMC-3.2}$$

By duality, the diagonal algebras

$$D_m = C(X_m)$$

form an inductive system through

$$\jmath_m : D_m \longrightarrow D_{m+1}, \qquad \jmath_m(f) = f \circ p_m. \tag{GMC-3.3}$$

Its uniform limit is $C(X_\infty)$: the cylinder functions form a unital self-adjoint subalgebra that separates points and are dense by Stone–Weierstrass.

26.4.2 The family at 2 degrees

For $s \geq 1$, define the matrix state space

$$\mathcal{K}_{m,s} = \mathrm{UCP}(D_m, M_s(\mathbb{C})). \tag{GMC-3.4}$$

If $e_x \in D_m$ is the characteristic function of $x \in X_m$, every $\Phi \in \mathcal{K}_{m,s}$ is equivalent to a POVM

$$E_x^{(m)} = \Phi(e_x) \succeq 0, \qquad \sum_{x\in X_m} E_x^{(m)} = I_s. \tag{GMC-3.5}$$

The genealogical restriction is

$$\rho_m^s : \mathcal{K}_{m+1,s} \longrightarrow \mathcal{K}_{m,s}, \qquad \rho_m^s(\Phi) = \Phi \circ \jmath_m. \tag{GMC-3.6}$$

In terms of effects,

$$E_x^{(m)} = \sum_{y\in p_m^{-1}(x)} E_y^{(m+1)}. \tag{GMC-3.7}$$

It is exactly the operation "forget the continuation but sum all its prolongations." It is neither an average nor an elimination of mass: it preserves the unit.

In the matrix direction, if $d \geq 0$, $\Phi_r \in \mathcal{K}_{d,s_r}$ and $V_r : \mathbb{C}^s \to \mathbb{C}^{s_r}$ satisfy $\sum_r V_r^* V_r = I_s$, their matrix convex combination is

$$\mathrm{mc}_V(\Phi_1,\dots,\Phi_k)(f)=\sum_{r=1}^{k} V_r^*\Phi_r(f)V_r\in\mathcal{K}_{d,s}. \tag{GMC-3.8}$$

Theorem 26.10 (Genealogy Switching and Matrix Convexity)**.** *For every family $\Phi_r\in\mathcal{K}_{m+1,s_r}$ and every compression as in* (GMC-3.8)*,*

$$\rho_m^s(\mathrm{mc}_V(\Phi_1,\dots,\Phi_k))=\mathrm{mc}_V\left(\rho_m^{s_1}(\Phi_1),\dots,\rho_m^{s_k}(\Phi_k)\right). \tag{GMC-3.9}$$

Proof. For $f\in D_m$, both sides are equal to

$$\sum_r V_r^*\Phi_r(J_m f)V_r.$$

The equality is functorial, not numerical: it holds simultaneously for all matrix levels and for any system of truncations. □

Matrix bridge result

Equation (GMC-3.9) is the core of the bigrading (m,s). Refining the history and then compressing produces exactly the same shadow as compressing each reader and then forgetting the continuation. Depth and matrix level are independent, but form a commutative square.

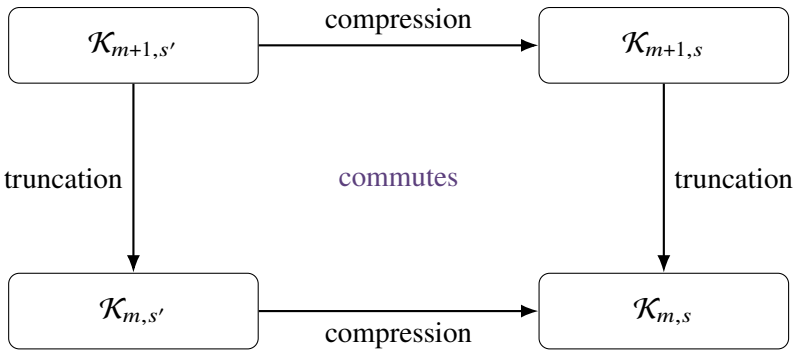


Figure 26.1: The two independent directions of the bigraduation. The symbol s' may represent a direct sum of several internal levels.

26.4.3 Uniform dilation compatible with all depths of the projective system

A family $(\Phi_m)_{m\geq 0}$ is compatible if

$$\Phi_m=\Phi_{m+1}\circ J_m. \tag{GMC-3.10}$$

This is equivalent to requiring the aggregation law (GMC-3.7) for all prefixes.

Theorem 26.11 (Simultaneous genealogical dilation, diagonal corridor)**.** *Every compatible family $\Phi_m\in\mathrm{UCP}(C(X_m),M_s)$ determines a unique map*

$$\Phi_\infty : C(X_\infty)\longrightarrow M_s \tag{GMC-3.11}$$

which coincides with Φ_m on each cylindrical algebra. There exist a Hilbert space $\mathcal{K}$, an isometry $V:\mathbb{C}^s\to\mathcal{K}$, a representation $\pi : C(X_\infty)\to B(\mathcal{K})$, and its associated projective spectral measure P on X_∞ such that

$$\Phi_\infty(f)=V^*\left(\int_{X_\infty} f(\xi)\,dP(\xi)\right)V. \tag{GMC-3.12}$$

In particular, for every m and $x \in X_m$,

$$E_x^{(m)} = V^* P([x]_m) V. \tag{GMC-3.13}$$

A common PVM *simultaneously dilates all finite* POVM *of the genealogy. If the Stinespring dilation is chosen to be minimal, the triple $(\pi, \mathcal{K}, V)$ is unique up to unitary equivalence; once π is fixed, its spectral measure P is unique.*

Proof. Compatibility defines a unital positive map on the union of the cylindrical algebras. Because the domain is commutative, positivity implies complete positivity, and the map is contractive. By density it extends uniquely to $C(X_\infty)$. The Stinespring theorem yields $\Phi_\infty(f) = V^*\pi(f)V$ and guarantees that its minimal realization is unique up to unitary equivalence. Every representation of a commutative algebra is the integral with respect to a unique PVM P associated with that representation. Applying the formula to the characteristic function of the cylinder $[x]_m$ gives (GMC-3.13). □

Corollary 26.12 (Fidelity environmental realization)**.** *The representation of Theorem 26.11 may be chosen faithful on $C(X_\infty)$ without changing any finite compression.*

Proof. If $(\pi, \mathcal{K}, V)$ is a dilation and $\pi_f : C(X_\infty) \to B(\mathcal{K}_f)$ a faithful representation, take $\pi \oplus \pi_f$ and the isometry $V \oplus 0$. The compression remains Φ_∞, and the ambient representation no longer identifies two distinct cylindrical functions. □

The final construction types the phrase "preserve the genealogy." The minimal dilation preserves exactly what the reader sees; the faithful ambient dilation additionally preserves all distinctions of the algebra before compression. The fidelity of a TPK dynamics, however, also requires intertwining its arrows and does not follow solely from the fidelity of the diagonal representation.

26.4.4 Noncommutative UHF Tower and Matrix Compatibility

The nonadic UHF tower, its trace, and its factorial closures are proved in chapter 23. Here they are taken as an explicit starting structure to prove a different result: the simultaneous prolongation of a compatible family of UCP maps and their common dilation.

Let

$$\mathcal{A}_m = M_{9^m}(\mathbb{C}), \qquad \iota_m(a) = a \otimes I_9, \qquad \mathcal{A}_\infty = \varinjlim(\mathcal{A}_m, \iota_m). \tag{GMC-3.14}$$

Theorem 26.13 (Simultaneous dilation on the UHF limit)**.** *If $\Psi_m : \mathcal{A}_m \to M_s$ are UCP maps and*

$$\Psi_m = \Psi_{m+1} \circ \iota_m, \tag{GMC-3.15}$$

there exists a unique UCP map $\Psi_\infty : \mathcal{A}_\infty \to M_s$ that prolongs them. There exists a representation $\pi : \mathcal{A}_\infty \to B(\mathcal{K})$ and an isometry $V : \mathbb{C}^s \to \mathcal{K}$ such that

$$\Psi_m(a) = V^*\pi(\iota_{m,\infty}(a))V \quad \textit{for every } m. \tag{GMC-3.16}$$

The representation can be made faithful by direct sum with a faithful representation of the UHF limit.

Proof. Compatibility defines Ψ_∞ on the algebraic inductive limit. UCP maps are completely contractive, so it extends to the completion C^*. Stinespring gives (GMC-3.16); the direct sum proves the last assertion. □

Restricting Theorem 26.13 to the canonical diagonals recovers the complete nonadic case $X_m = (\mathbb{Z}/9\mathbb{Z})^m$ of the Theorem 26.11; this restriction does not by itself cover an arbitrary system of survivors. In the canonical nonadic UHF system of (GMC-3.14), the GNS representation of its unique trace has weak

closure isomorphic to $\mathcal{R}_{\mathrm{hyp}}$. Thus, the bigrading is not a combinatorial appendix: it joins the cylinders of histories, the finite matrix charts, and the operatorial continuum of the von Neumann realization.

26.4.5 Trigraduation and Levels of Simultaneity

When a grid appears, the exterior size n is incorporated:

$$\mathfrak{M}_{m,s}^{(n)}. \tag{GMC-3.17}$$

The structure is trigraded if n varies and bigraded by (m, s) if n is fixed. 'Simultaneous' can denote three increasingly stringent requirements:

1. *by rows and columns*: a common dilation of the $2n$ POVM of a magic square;
2. *by depth*: a common dilation of all levels m, proved in Theorems 26.11 and 26.13;
3. *by dynamic genealogy*: in addition, an intertwiner preserving the route, monodromy, and TPK composition.

The first two have exact theorems within their respective domains. The third is the strongest notion that HMT contributes to the De las Cuevas–Netzer framework. The construction of the common operator system and its equivariance equation are proved in Section 17.9; the static dilation constitutes only one of its antecedents.

Structural obstruction

The theorem does not claim that every disconnected collection of effects at different depths has a common dilation. The decisive hypothesis is the compatibility (GMC-3.10) or (GMC-3.15). In HMT, that compatibility is the operator form of "each prefix is exactly the restriction of its continuations".

26.5 Matrix representation of the genealogical composition APP–TRIT–TPK

The recent program of De les Coves, van der Eyden, and Netzer extends operator systems to conic systems defined over more general categories [41]. The framework does not turn every category into a quantum theory: it requires choosing suitable cones, dualities, compressions, and completely positive maps. HMT provides an especially rich testbed in which to make that choice.

26.5.1 Realizations in systems of ordinary operators

The following families are entirely within the standard formalism:

- $\mathcal{S}_d = \mathrm{span}\{I, P_{\Sigma,d}, P_{\Pi,d}\}$ and its matrix ranks;
- the diagonal algebras $C(X_m)$ and the UHF tower $M_{9^m} \hookrightarrow M_{9^{m+1}}$;
- the readers UCP, POVM, PVM, channels CPTP and dilatacionis of Stinispring;
- the bigraded families $\mathcal{K}_{m,s}$.

No broader categorical formalism is required here in order to assert positivity, dilation, or matrix convexity. Indeed, the classification (GMC-6.9) shows that a substantial part of APP can be resolved by exact finite-dimensional operator theory.

26.5.2 Genealogical extension of operator systems for TPK routes

A TPK route is not merely an operator between two spaces: it carries source, target, sheet, orientation, length, carry, and a composition law. If those labels are forgotten and only the matrix representing the route is retained, distinct arrows may collapse. To incorporate the complete type, one proposes a category $\mathsf{C}_{\mathrm{HMT}}$ whose objects are register profiles and whose arrows are admissible transports. To each type c one would associate a cone of positive readers $A(c)$, and to each transformation a compatible action on those cones.

In the language of *Beyond Operator Systems*, the objective is to construct a S-system: a functorial extension of the tensor product by the base object to a family of cones, with CP maps defined as natural transformations level by level. The theory provides minimal and maximal systems, duality, generalized Choi–Kraus theorems, representations as intersections of S-hedra, and a $*$-autonomous category [41].

Mathematical extensions

The intrinsic S-system of the TPK remains to be constructed. At a minimum, the cone functor, the dualizing object, the evaluations and coevaluations, and the proof that TPK composition preserves positivity in every type are lacking. The fact that the TPK is monoidal in its finite realization does not by itself prove that it is $*$-autonomous.

26.5.3 Genealogical HMT Cones and Extension of the Conical Theory

The classical examples start from a base cone and study its minimal and maximal systems. HMT suggests a base cone with stratified memory:

$$C_m^{\mathrm{HMT}} = \mathrm{cone}\{\text{positive readers of histories at depth } m\}. \tag{GMC-8.1}$$

The truncations induce positive maps $C_{m+1}^{\mathrm{HMT}} \to C_m^{\mathrm{HMT}}$, while the compressions generate the matrix levels. The new phenomenon to study is whether the minimal and maximal systems remain separate upon taking the limit in m, and what part of that separation measures loss of genealogy rather than mere noncommutativity.

This proposes three invariants for the general program:

1. *separation depth*: the smallest m at which the minimal and maximal systems differ;
2. *matrix detection level*: the least s exhibiting that difference;
3. *memory defect*: the minimal route information that must be added in order to lift a point from the maximal system to the minimal system.

The first two refine the separation (m, s); the third is genuinely HMT. They are proposed definitions and must first be studied in $\mathfrak{C}_2^{\mathrm{APP}}$, where (GMC-6.9) permits exact computations.

26.5.4 TPK as a category of routes with selection, transport, memory, and return.

The paper on Latin squares proposes the development of quantum magic cubes. The cyclic lemma admits an immediate extension.

Definition 26.14 (Quantum Magic Cube)**.** A family $C = (C_{ijk})_{0\le i,j,k<n}$ of effects in M_s is a quantum magic cube if every line parallel to any of the three axes sums to I_s.

Theorem 26.15 (Cyclic cube of a POVM)**.** *For every* POVM $(E_t)_{t\in\mathbb{Z}/n\mathbb{Z}}$,

$$C_{ijk} = E_{i+j+k \bmod n} \tag{GMC-8.2}$$

is a quantum magic cube of exterior size n and interior size s.

Proof. Once two coordinates are fixed, as the third varies the index $i + j + k$ traverses $\mathbb{Z}/n\mathbb{Z}$ once. Each line therefore sums to $\sum_t E_t = I_s$. Positivity is inherited from POVM. □

Applying the theorem to $E^{(9)}$ of (GMC-4.6) and to F of (GMC-4.12) yields explicit qutrit cubes of orders nine and twelve. The former may assign the third axis to a phase coordinate; the latter uses the complete nonadic atlas. To convert that axis into genuine HMT depth, the layers must also satisfy

$$\rho_m(C^{m+1}_{ijk}) = C^m_{ijk}, \tag{GMC-8.3}$$

and intertwine the TPK. The cyclic construction closes the operatorial cube; genealogical fidelity remains an additional condition.

26.5.5 Completely positive compositions compatible with the genealogical-matricial bidegree

Another open direction of the theory consists in transferring the magic conditions to completely positive maps through Choi–Jamiołkowski. If $J(\Phi_{ij}) \succeq 0$ are the Choi matrices of a grid of CP maps, the conditions

$$\sum_j J(\Phi_{ij}) = J(\mathrm{id}), \qquad \sum_i J(\Phi_{ij}) = J(\mathrm{id}) \tag{GMC-8.4}$$

are a linear version of the magic normalizations. HMT provides channels of reading, updating, and transport; the problem is to decompose them into a grid that simultaneously preserves normalization, sheet, and route. The generalized conic formalism can treat codomains that are not simply matrix PSD cones.

26.5.6 Universality criterion in the genealogical domain

The framework of Gonda, Reinhart, Stengele, and De les Coves defines a simulator in a *target–context* category by means of

$$s : P \otimes C \longrightarrow T \otimes C, \tag{GMC-8.5}$$

with a functional compiler $s_T : P \to T$ and a context reduction. Universality is the existence of a lax reduction to the trivial simulator, not the mere richness or recurrence of the system [58].

An HMT instance may take:

- P: prefixes, route programs, or seeds;
- T: mathematical and physico-metrological observables intended to be realized;
- C: gauge, leaf, boundary conditions, and reading apparatus;
- s_T: transformation of a genealogy into an output;
- s_C: transport of the context and residual memory.

Proving universality would require constructing those data and the lax reduction. A monotone of the contextual preorder suffices to refute a proposal. The framework's impossibility theorem states, in particular, that if the functional image of the compiler has a supremum strictly below the target space for some compatible monotone, the simulator is not universal.

Matrix bridge result

HMT suggests a stronger version: *genealogically faithful universality*. In addition to reproducing the behavior of the target, the reduction must lift to a functor between categories of routes and preserve composition, return, and the enriched chronological register. Two behaviorally equivalent simulators may cease to be equivalent when preservation of provenance is required. This distinction offers a new and nontrivial instance for the categorical program.

The metrological unification demonstrated by HMT and categorical universality are distinct claims. The former derives readers and relations within the system; the latter requires the ability to simulate a declared class of targets and contexts. Typing them separately strengthens both: it avoids requiring from metrology a property that it does not need and turns universality into a falsifiable theorem.

26.5.7 Contextuality and Nonsemiclassicality

The cyclic squares of Section 26.3 are semiclassical. They therefore do not resolve the open direction concerning contextuality or nonsemiclassical QMS. APP, however, contains the noncommuting pair P_Σ, P_Π and algebra (GMC-6.9). The correct route is to construct cell effects that depend on both operators and prove that they admit no decomposition (GMC-4.4) with a POVM on S_n.

The verification is a semidefinite problem: find $Q_\sigma \succeq 0$, $\sum_\sigma Q_\sigma = I$, and

$$A_{ij} = \sum_{\sigma:\, \sigma(i)=j} Q_\sigma. \tag{GMC-8.6}$$

Feasibility certifies semiclassicality; infeasibility must be accompanied by a dual witness. A noncommutative matrix or a visual graph does not replace that certificate.

26.6 Genealogical structure of APP–TRIT–TPK at the matrix level

This chapter isolates the structures involved in the transduction and keeps their types explicit. The order is deliberate: support, sum–product difference, ternary memory, category of routes, quantum realization, operatorial closure, and metrological reading. Only afterward will it be asked which of these structures form a magic square or an operator system.

26.6.1 Common support of APP and dual evaluation via sum and product sheets

Let

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2, \qquad \mathscr{D} = \{(1,0),(0,1),(-1,0),(0,-1)\}.$$

The Cayley graph $\mathrm{Cay}(X_9, \mathscr{D})$ has 81 vertices and four oriented cardinal edges from every vertex. The APP architecture contains all finite paths in that graph. The acronym is retained as a technical name and is not used here to fix a historical expansion. At length r there are exactly $81 \cdot 4^r$ paths if returns and backtracking are allowed. There are not independent additive and multiplicative tables: there is a single support and two evaluations of the same point,

$$\Sigma(i,j) = \mathrm{dr}(i+j), \qquad \Pi(i,j) = \mathrm{dr}(ij), \qquad \mathrm{dr}(a) = 1 + ((a-1) \bmod 9). \tag{GMC-2.1}$$

The table of Σ is a classical Latin square of order nine, because every translation permutes $\mathbb{Z}/9\mathbb{Z}$. The table of Π is not: only its rows and columns indexed by the six units $\{1,2,4,5,7,8\}$ are permutations; $3, 6, 9$ fold classes.

Result

APP already contains an exact Latin structure, but APP is not exhausted by it. The sum leaf is Latin; the product leaf records the defect of Latinity of the local ring; the paths preserve the consultation order, and the residue–quotient lift preserves what modular reduction would conceal.

For each cell, the residue and quotient are retained:

$$i + j = R^+_{ij} + 9q^+_{ij}, \qquad ij = R^\times_{ij} + 9q^\times_{ij}. \tag{GMC-2.2}$$

The certified totals are

$$\sum R^+ = 405, \quad \sum R^\times = 459, \quad \sum q^+ = 45, \quad \sum q^\times = 174,$$

and therefore

$$2025 - 810 = 1215 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \tag{GMC-2.3}$$

The visible difference 54 and the quotient difference 129 are not two names for the same datum. One resides in the published residues; the other, in the memory required to recompose the integer value.

26.6.2 Posterior metrological realization of the double projection of the enriched state

The central block of APP takes the form

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R = 9P_+ + 5P_-, \qquad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad P_\pm = \frac{I \pm R}{2}. \tag{GMC-2.4}$$

From this one obtains exactly the typed hierarchy induced by the preceding spectral decomposition:

$$\underbrace{9-5}_{4}\,;\quad \underbrace{2}_{\text{off-diagonal coupling}}\quad;\ \underbrace{51-45}_{6}\,;\ \underbrace{9\cdot 6}_{54}\,. \tag{GMC-2.5}$$

The first figure is the primordial spectral gap; the second, the coupling of the block; the third, the aggregate excess of the nonunitary sector; the fourth, its transport to the global scale of the table. They must not be collapsed on account of their arithmetic kinship.

Since $\det B_c = 45$, the normalization has determinant one and belongs to the connected hyperbolic group:

$$\frac{B_c}{\sqrt{45}} = \exp(\eta R), \qquad \eta = \frac{1}{2}\log\frac{9}{5}, \qquad \frac{5}{9} = e^{-2\eta}. \tag{GMC-2.6}$$

The same spectral pair thus admits a contractive reading and a Lorentzian reading. This is the precise connection with convexity and the hyperbolic interior that the bridge must preserve: the contraction $5/9$ is the shadow of the separation of the two idempotents $P_\pm$, not a verbal analogy with curvature.

26.6.3 TRIT: visible state, orientation, carry, and genealogical memory

TRIT is the oriented signature $\{-1, 0, +1\}$ obtained from $\mathbb{F}_3$ through $0 \mapsto 0$, $1 \mapsto +1$, $2 \mapsto -1$. To type its local lift, we define the lifted alphabet

$$\mathcal{L}_1 = \{\nu = r + 3c : r, c \in \{-1, 0, +1\}\} = \{-4, -3, \dots, 4\}, \qquad \widehat{\tau}(\nu) = (r, c). \tag{GMC-2.7}$$

The decomposition $\nu = r + 3c$ is unique in $\mathcal{L}_1$; r is the visible residue and c the carry. The symbol ν is a lifted amplitude and is not identified, in the absence of further data, with the sum of two visible trits. In particular, the amplitudes $3, 0, -3$ publish the same zero residue and yield three distinct antecedents,

$$0^+ = (0, +1), \qquad 0^0 = (0, 0), \qquad 0^- = (0, -1). \tag{GMC-2.8}$$

The complete state may contain

$$x = (t, \mu, o, h, c, \sigma, \lambda, g), \tag{GMC-2.9}$$

where t is the visible signature, μ the leaf, o the orientation, h the extension, c the carry, σ the boundary, λ the route, and g the nonadic phase. The projection $x \mapsto t$ eliminates all the other coordinates. This is the structural point that HMT contributes to the theory of dilations: two equal observable effects may possess distinct genealogies.

The three signatures admit the real algebras

$$\mathbb{A}_\tau = \mathbb{R}[J]/(J^2 + \bar{\tau}),$$

of elliptic type ($J^2 = -I$), parabolic type ($J^2 = 0$), and hyperbolic type ($J^2 = +I$). In the last case, $P_\pm = (I \pm J)/2$ separate two eigendirections. The TRIT is not merely the sign: it is the minimal observable of a transport that preserves the label, orientation, and memory.

26.6.4 TPK: category of memoryful paths

TPK, *Topological Phase Kernel*, resides on the free path category of APP and on its bivariate sum–product reading. Two paths that end in the same cell may differ in cardinal word, winding, sheet, quotient, and holonomy. TPK preserves that difference and composes routes associatively. The carry cocycle

$$\kappa_9(a, b) = \frac{\mathrm{dr}(a) + \mathrm{dr}(b) - \mathrm{dr}(a + b)}{9}$$

satisfies

$$\kappa_9(a, b) + \kappa_9(a + b, c) = \kappa_9(b, c) + \kappa_9(a, b + c), \tag{GMC-2.10}$$

and guarantees that the recovery does not depend on how a concatenation is parenthesized.

The quotient phase–length has objects $(b, r) \in \mathbb{Z}/12\mathbb{Z} \times \{0, \dots, 8\}$ and arrows

$$(b, r) \xrightarrow{\ell} \left(b + \left\lfloor \frac{r + \ell}{9} \right\rfloor , r + \ell \bmod 9 \right). \tag{GMC-2.11}$$

After nine unit iterations ($\ell = 1$), r returns and one dodecaphasic phase advances; after $108 = 9 \cdot 12$ unit iterations, the quotient returns. The enriched state of TPK may not have returned, because it preserves additional memory. The minimal predictive quotient of the published observables has 36 states and fibers of cardinality three. This fact makes visible the difference between equality of output and equivalence of history.

Scientific reading

A magic square simultaneously organizes rows and columns. TPK simultaneously organizes position, sheet, phase, and composition. The correct translation does not consist in calling every HMT table "quantum," but in constructing a positive reader that preserves the APP identities and declaring, coordinate by coordinate, which memory it preserves and which memory it compresses.

26.6.5 Hilbertian realization and quantum structure

The finite quantum closure of HMT constructs, for each depth d, the space

$$\mathcal{H}_d = \ell^2((\mathbb{Z}/9\mathbb{Z})^d), \qquad \mathcal{A}_d = B(\mathcal{H}_d) \cong M_{9^d}(\mathbb{C}). \tag{GMC-2.12}$$

For $u, v \in (\mathbb{Z}/9\mathbb{Z})^d$, the nonadic translations and phases act by means of

$$X_u|x\rangle = |x + u\rangle, \qquad Z_v|x\rangle = \omega^{v \cdot x}|x\rangle, \qquad \omega = e^{2\pi i/9},$$

and satisfy

$$Z_v X_u = \omega^{u \cdot v} X_u Z_v.$$

The complete family—equivalently, the d coordinate pairs X_j, Z_j—generates $\mathcal{A}_d$ and acts irreducibly. For the primitive central character fixed by ω, the finite Heisenberg representation is determined up to unitary equivalence. In this realization, states, self-adjoint observables, spectral measures, Born probabilities, unitary dynamics, and Kraus channels are constructed; the composition of registers is transported monoidally to the tensor product. The finite Hilbert-space representation is not presupposed: it is constructed explicitly and transports the genealogy of the system.

In the qutrit quotient there appear four maximal abelian subalgebras of the three-dimensional Weyl system. Their four spectral bases are mutually unbiased:

$$\operatorname{Tr}(P_{a,k}P_{b,\ell}) = \begin{cases} \delta_{k\ell}, & a = b, \\ 1/3, & a \neq b, \end{cases} \qquad a, b \in \mathbb{P}^1(\mathbb{F}_3). \tag{GMC-2.13}$$

The twelve minimal projectors $P_{a,k}$ are the operatorial atlas that is later calibrated with the twelve directions of $\mathbb{P}^1(\mathbb{Z}/9\mathbb{Z})$.

26.6.6 From the nonadic tower to the operatorial continuum

The unital inclusion, the marked branch, the UHF limit, and the closures of types I_∞ and II_1 are constructed in chapter 23. Their formulas are premises of the matrix bridge and determine the bigrading.

The operatorial tower prolongs the finite charts through the unital inclusions preserving the normalized trace

$$\iota_d : M_{9^d}(\mathbb{C}) \longrightarrow M_{9^{d+1}}(\mathbb{C}), \qquad a \longmapsto a \otimes I_9. \tag{GMC-2.14}$$

Indeed, if $\tau_n = \operatorname{Tr}/n$, then $\tau_{9^{d+1}} \circ \iota_d = \tau_{9^d}$; the ordinary trace, by contrast, is multiplied by nine. Its inductive limit is the UHF algebra of type $9^\infty = 3^\infty$. The closure in the GNS representation of the unique trace is the hyperfinite factor $\mathcal{R}_{\text{hyp}}$ of type II_1. The marked inclusions $a \mapsto a \otimes p_j$ follow one branch and lead to the corridor of compact operators, of type I_∞. The result therefore distinguishes preserving all branches simultaneously from selecting a particular history.

In the diagonal corridor, the finite sets of prefixes form an inverse system $X_{m+1} \to X_m$. The space of infinite histories is $X_\infty = \varprojlim X_m$, while its function algebras form the opposite inductive system. This duality of directions —inverse for histories, direct for observables— will be the exact support of the new bigraduation.

26.6.7 Double projection and metrology

HMT does not identify the published real number with the state that generates it. A compatible family of cylinders produces an Archimedean reading; another projection organizes the exceptional output and its symmetries. Numerical reality appears as the shadow of a double projection of a richer state. The monodromic generation theorems for π, e, φ, and α, the action and gravitation readers, the mass law, the mixing and CP-violation sectors, and the exceptional genealogy apply in their declared domains. The bridge makes explicit the operator objects that transport their conclusions.

The Planck identity uses the same block B_c, its transport in $SO^+(1,1)$, and the self-dual identity of the mass line. The decisive mathematical fact for this translation is that ‘physical observable’ and ‘generating history’ are different types. A UCP map can publish the former; the coordinate m and the TPK category preserve the latter.

Premises and derivation

The premises of the bridge are the unital inclusion $a \mapsto a \otimes I_9$, the block $B_c = 9P_+ + 5P_-$, the self-dual identity of the mass line, and periodicity of order 108. Each is used with the domain and hypotheses of its theorem; the matrix conclusions are deduced by means of the maps written in this chapter.

26.7 Genealogical Limits, Transfer Channels and Stinespring Memory

The HMT continuum and the continuum limit of matrix product states (MPS) answer different questions. The former prolongs finite histories and then reads them; the latter refines a tensor network and requires its transfer channel to admit compatible roots. The comparison is fruitful only if the channel connecting them is constructed.

26.7.1 Typed separation of the genealogical limit and the MPS limit

In the HMT corridor, an infinite history is an element of $X_\infty = \varprojlim X_m$. Under the finiteness, surjectivity, nested-compactum, contraction, and coherent-covering conditions proved in the continuum appendix, the quotient by the Archimedean reading reconstructs the declared real compactum. The complete state and its real value remain distinct objects.

For a translationally invariant MPS, a tensor $A = \{A^i \in M_D\}_{i=1}^d$ determines the transfer channel

$$E_A = \sum_i A^i \otimes \overline{A^i}. \tag{GMC-7.1}$$

De las Cuevas, Schuch, Pérez-García, and Cirac prove that an MPS in irreducible form has a continuum limit if and only if E_A is infinitely divisible; equivalently, there exist a projector channel P and a Lindblad generator L such that

$$E_A = Pe^L, \qquad P^2 = P, \qquad PLP = PL. \tag{GMC-7.2}$$

It has a coarse continuous limit if some power E_A^r is infinitely divisible [40].

Structural obstruction

The existence of the HMT inverse limit does not imply (GMC-7.2). To apply the MPS theorem, one must exhibit a uniform local tensor, identify its transfer channel, and verify infinite divisibility. Conversely, (GMC-7.2) does not by itself preserve sheet, carry, route, or monodromy.

26.7.2 Symplectic decomposition 12→4×3 of the phase coordinates

In the tower $\mathcal{A}_{m+1} = \mathcal{A}_m \otimes M_9$, the conditional expectation

$$\mathbb{E}_m = \mathrm{id} \otimes \tau_9 : \mathcal{A}_{m+1} \longrightarrow \mathcal{A}_m, \qquad \tau_9(b) = \frac{1}{9}\,\mathrm{Tr}(b), \tag{GMC-7.3}$$

is UCP and satisfies

$$\mathbb{E}_m(a \otimes I_9) = a. \tag{GMC-7.4}$$

On the diagonal, $D_{m+1} = C(X_m \times \mathbb{Z}_9)$ reduces to the uniform average over the last digit. The branch evaluation

$$\kappa_{m,a}(f)(x) = f(x, a) \tag{GMC-7.5}$$

is another retraction, this time multiplicative on the diagonal. Averaging all branches and selecting one branch are distinct operations, parallel to the inclusions $a \otimes I_9$ and $a \otimes p_a$ of the operatorial closure.

The expectation (GMC-7.3) has a dilation with an explicit record. Let

$$W_a\psi = \frac{1}{3}\,\psi \otimes e_a, \qquad a = 0, \dots, 8.$$

Then

$$\mathbb{E}_m(X) = \sum_{a=0}^{8} W_a^* X W_a. \tag{GMC-7.6}$$

In Stinespring form,

$$\pi(X) = X \otimes I_9, \qquad V\psi = \frac{1}{3}\sum_{a=0}^{8}(\psi \otimes e_a) \otimes e_a, \qquad \mathbb{E}_m(X) = V^*\pi(X)V. \tag{GMC-7.7}$$

The final auxiliary system (*ancilla*) labels exactly the digit discarded by averaging.

Matrix bridge result

The formula (GMC-7.7) is a canonical nonadic memory in the sense of Stinespring: what is lost when applying $\mathbb{E}_m$ remains orthogonally registered. It is not yet identified with TPK memory. That identification requires an intertwiner that also transports route, leaf, and dynamics.

26.7.3 Stinespring's Memoir and Non-Uniform Growth

The Theorem 26.13 provides a common, generally infinite-dimensional representation for every compatible family. A fixed finite ancillary system that purifies all levels cannot in general be required. Indeed, the compatible family of normalized traces on M_{9^m} has rank 9^m at level m, and any purification of the corresponding state requires an ancillary dimension of at least 9^m. Therefore,

$$\text{UCP compatibility at every depth} \quad \not\Longrightarrow \quad \text{uniform finite auxiliary memory.} \tag{GMC-7.8}$$

Memory growth is not a defect of the construction: it quantifies the amount of genealogy that one requires to be preserved.

The dynamic fidelity of this memory and its comparison with the limit of matrix product states are developed in section 26.7.4, without repeating the UHF tower or the Stinespring dilation demonstrated here.

26.7.4 Dynamic fidelity equation

The trace-preserving expectation $\mathrm{id} \otimes \tau_9$, its Stinespring dilation, and the general impossibility of a uniform finite auxiliary memory have been constructed in sections 26.7.2 y 26.7.3. On that operatorial premise, and without abbreviating its proofs, the additional condition of genealogical fidelity is now formulated.

Let T_m^{TPK} be the transport between records at depth m, and let J_m be the encoding of those records in the dilation space. The Stinespring memory is TPK-faithful if there exists a transport $\widetilde{T}_m$ in the ambient space such that

$$J_{m+1}T_m^{\mathrm{TPK}} = \widetilde{T}_m J_m \tag{GMC-7.9}$$

and the observational compressions satisfy the corresponding naturality. If two distinct histories are identified under J_m, the dilation may be operatorially valid but not genealogically faithful.

The equation (GMC-7.9) is falsifiable: it suffices to have a pair of routes that TPK distinguishes and the environment identifies, or a composition whose image does not coincide with the composition of the images.

26.7.5 Compatibility Criterion for Product State Limits

To apply (GMC-7.2) to the HMT continuum, five steps must be completed:

1. choose a uniform local cell and construct matrices $A^i \in M_D$;
2. prove that nonadic cell blocking intertwines the evolution with a refinement isometry;
3. compute the channel E_A and its irreducible form;
4. resolve whether E_A or some power E_A^r is infinitely divisible;
5. transport the leaf, carry cocycle, and monodromy to the limit, rather than preserving only the tensor state.

If the channel were unitary, $E_A = \mathrm{Ad}_U$, it would have unitary roots of every order by functional calculus. But it would still be necessary to prove that this channel is the transfer channel associated with an HMT tensor and that its roots respect the genealogical refinement. If a projective part P appeared, HMT memory would offer a concrete interpretation of the sectors that survive scale grouping.

26.7.6 Tensorial decompositions with symmetry

The work of De las Cuevas with Netzer and collaborators on approximate tensor decompositions and on simplicial complexes provides another point of entry [38, 37]. APP supplies a finite complex with an action of translations; TPK supplies labels and compatibilities along routes. An HMT tensor decomposition should state:

- which tensor encodes a cell or a transition;
- which group preserves the labels and which part acts only after reduction;
- whether the factorization is exact, approximate, or only asymptotic;
- which rank remains stable as m increases.

The bidirectional opportunity is concrete. Its formalism provides criteria of factorization, invariance, and limit; HMT provides a refinement system with fibers, carry, and monodromy that can produce examples in which a visible factorization exists but no genealogically faithful factorization preserves the route.

Mathematical extensions

A subsequent paper may study "infinite divisibility with root memory": not only whether $E = E_k^k$, but whether a coherent tower of roots $E_{k\ell}^{\ell} = E_k$ equipped with TPK records can be chosen. This condition is stronger than divisibility of the bare channel and constitutes a natural extension of the MPS problem from HMT.

26.8 Completely positive matrix realization of the nonadic fibration of observables

Symplectic geometry is not part of the published theorems on De las Cuevas–Netzer magic squares. The legitimate link passes through a neighboring, standard route: Weyl operators, discrete phase space and mutually unbiased bases [52, 100]. In HMT, that route acquires a nonadic lift and fiber memory.

26.8.1 The alternating form and the Weyl commutation

On $V_3 = \mathbb{F}_3^2$ we define

$$\langle (q,p),(q',p')\rangle_{\mathrm{sp}} = qp' - pq'. \tag{GMC-5.1}$$

If $X|k\rangle = |k+1\rangle$, $Z|k\rangle = \omega^k|k\rangle$, and $\omega = e^{2\pi i/3}$, the operators $W_{q,p} = X^q Z^p$ satisfy, under these conventions,

$$W_u W_v = \omega^{-\langle u,v\rangle_{\mathrm{sp}}} W_v W_u. \tag{GMC-5.2}$$

Therefore, two operators commute exactly when their labels are orthogonal for the alternating form. In dimension two, the maximal isotropic sublines are the four points of

$$\mathbb{P}^1(\mathbb{F}_3) = \{[1:0],[0:1],[1:1],[1:2]\}. \tag{GMC-5.3}$$

Each generates a maximal abelian context of $M_3(\mathbb{C})$; its spectral projectors constitute a PVM. Contexts associated with distinct lines produce the four MUB bases of (GMC-2.13).

Proposition 26.16 (Qutrit decomposition by directions)**.** *Let $\mathcal{D}_a$ be the four diagonal subalgebras determined by the lines of* (GMC-5.3) *and $\mathcal{D}_a^0$ their traceless parts. Then*

$$M_3(\mathbb{C}) = \mathbb{C}I \oplus \bigoplus_{a\in\mathbb{P}^1(\mathbb{F}_3)} \mathcal{D}_a^0 \tag{GMC-5.4}$$

as an orthogonal sum for the Hilbert–Schmidt product. The dimension count is $9 = 1 + 4 \cdot 2$.

Proof. Each $\mathcal{D}_a^0$ has dimension two. Unbiasedness implies $\mathrm{Tr}(A^*B) = 0$ for $A \in \mathcal{D}_a^0$ and $B \in \mathcal{D}_b^0$, $a \neq b$. All are orthogonal to I. The dimensions sum to nine, the dimension of $M_3(\mathbb{C})$. □

This identity explains why the twelve projections are not twelve arbitrary independent pieces: they are four resolutions of the identity whose centered parts decompose the entire qutrit operator space.

26.8.2 The lift $12 \to 4 \times 3$

On the modulo nonadic $V_9 = (\mathbb{Z}/9\mathbb{Z})^2$ it preserves the forma alternante

$$\langle (a,b),(c,d)\rangle_9 = ad - bc \pmod 9. \tag{GMC-5.5}$$

The projective line $\mathbb{P}^1(\mathbb{Z}/9\mathbb{Z})$ consists of the primitive vectors modulo multiplication by a unit. It can be represented as

$$\{[1:t] : t \in \mathbb{Z}/9\mathbb{Z}\} \sqcup \{[3u:1] : u \in \mathbb{Z}/3\mathbb{Z}\}, \tag{GMC-5.6}$$

and has $9 + 3 = 12$ points. Reduction modulo three gives the diagram

$$\mathbb{P}^1(\mathbb{Z}/9\mathbb{Z}) \xrightarrow{r} \mathbb{P}^1(\mathbb{F}_3), \qquad |r^{-1}(a)| = 3. \tag{GMC-5.7}$$

The basis a of a qutrit context and the outcome k of its PVM likewise form a set 4×3. A *calibration* is a bijection

$$\Xi : \mathbb{P}^1(\mathbb{Z}/9\mathbb{Z}) \longrightarrow \{P_{a,k} : a \in \mathbb{P}^1(\mathbb{F}_3),\ k \in \mathbb{F}_3\} \tag{GMC-5.8}$$

that sends each fiber of r to the three outcomes of the context lying over a.

Structural obstruction

The fibration is canonical; assigning the internal order of each fiber to the three outputs k is a gauge. A bijection is not yet an equivariance. To promote Ξ to an intertwiner, one must prove that the relevant linear, projective, and antiunitary actions are transported in every fiber. The preliminary gauge fails an antiunitary check in one fiber; that failure is preserved as a falsifier, not concealed.

26.8.3 Depth, carry, and memory over the discrete phase space

The Weyl geometry over $\mathbb{F}_3^2$ explains commutation, contexts, and MUB. HMT adds three levels that are not recovered from the symplectic form by itself:

1. the lift of a ternary direction to three points of $\mathbb{P}^1(\mathbb{Z}/9\mathbb{Z})$;
2. the sheet, carry and orientation that distinguish states with the same reduction;
3. the TPK route that transports those coordinates and can return to the phase without returning to the state.

The potential contribution to the qutrit literature is not "another construction of four MUBs," which is already known. It is a geometry of *context roots*: each ternary context has three nonadic antecedents, and those antecedents may carry dynamics and monodromy. The order-twelve square of Theorem 26.9 is the first normalized shadow of that atlas.

26.8.4 Typed separation between the qutrit overlap $1/3$ and the HMT reader $1/\pi$

In a pair of MUB qutrit bases,

$$\mathrm{Tr}(P_{a,k}P_{b,\ell}) = \frac{1}{3}. \tag{GMC-5.9}$$

In the HMT reader defined here, the relation between a real sheet and a volumetric reading can take the quotient

$$\frac{\pi}{\pi^2} = \frac{1}{\pi}. \tag{GMC-5.10}$$

The two expressions are not interchangeable: $1/3$ is an overlap between projectors in dimension three; $1/\pi$ is a scalar weight of another reader. The difference does not prevent their coordination. For a fixed PVM $\{P_0, P_1, P_2\}$, we define a measurement with nine effects:

$$F_k^{(\pi)} = \frac{3}{\pi}P_k \quad (k = 0, 1, 2), \qquad F_r^{(\pi)} = \frac{1 - 3/\pi}{6}I_3 \quad (r = 3, \ldots, 8). \tag{GMC-5.11}$$

It is a POVM because $3/\pi < 1$ and

$$\sum_{r=0}^{8} F_r^{(\pi)} = \frac{3}{\pi}I_3 + \left(1 - \frac{3}{\pi}\right) I_3 = I_3. \tag{GMC-5.12}$$

If Q is a rank-one projector of a distinct MUB context,

$$\mathrm{Tr}(QF_k^{(\pi)}) = \frac{3}{\pi}\,\mathrm{Tr}(QP_k) = \frac{3}{\pi}\cdot\frac{1}{3} = \frac{1}{\pi}. \tag{GMC-5.13}$$

Matrix bridge result

The formula (GMC-5.11) exactly transforms the MUB probability $1/3$ into the reader $1/\pi$ through the gauge $3/\pi$, without identifying the two numbers. The six residual effects preserve the unit and complete a POVM of nine outcomes, from which another cyclic QMS $(9, 3)$ is obtained. This construction does not yet identify the geometric provenance of the residual mass; it fixes the exact operator type that it must have.

26.8.5 Symplectic group and equivariance equation

In dimension two over $\mathbb{F}_3$, the symplectic group coincides with $SL(2, \mathbb{F}_3)$ and permutes the four lines. Its action lifts projectively, by means of the Clifford group, on the PVM qutrits. Let G_{HMT} be a group or groupoid of transformations of the nonadic atlas and U_g the candidate unitary or antiunitary operator in the qutrit chart. The condition that turns a calibration into a dynamical bridge is

$$\Xi(g \cdot \ell) = U_g\,\Xi(\ell)\,U_g^*, \qquad g \in G_{\mathrm{HMT}}, \quad \ell \in \mathbb{P}^1(\mathbb{Z}/9\mathbb{Z}). \tag{GMC-5.14}$$

If g is an arrow and not merely a global symmetry, the equation must include the source and target and is formulated as a natural transformation. The verification of (GMC-5.14) is the step that separates a fortunate labeling from a structural transduction.

Premises and derivation

The symplectic and MUB identities rely on the finite-phase literature [52, 100]. They are not attributed to the papers on magic squares by De las Cuevas–Netzer. The nonadic fibration, calibration, and route readers constitute the original contribution of the HMT construction studied here.

26.9 Nonsemiclassical quantum magic square in a corner APP

En $\mathbb{C}^4$, with base e_1, e_2, e_3, e_4, considerese the arreglo of vectors

$$\begin{array}{cccc} e_1 & e_2 & e_3 & e_4 \\ e_2 & e_1 & e_4 & e_3 \\ (e_3+e_4)/\sqrt{2} & (e_3-e_4)/\sqrt{2} & (e_1+e_2)/\sqrt{2} & (e_1-e_2)/\sqrt{2} \\ (e_3-e_4)/\sqrt{2} & (e_3+e_4)/\sqrt{2} & (e_1-e_2)/\sqrt{2} & (e_1+e_2)/\sqrt{2}. \end{array}$$

Let P_{ij} be the rank-one projector onto the vector located in cell (i, j). Each row and each column is an orthonormal basis, hence

$$\sum_j P_{ij} = I_4, \qquad \sum_i P_{ij} = I_4. \tag{26.1}$$

To realize it in the APP fiber of dimension 9, fix

$$A_{ij} := P_{ij} \oplus \frac{1}{4} I_5 \in M_9(\mathbb{C}). \tag{26.2}$$

Theorem 26.17 (No certified semiclassicality)**.** *The matrix $A = (A_{ij})_{i,j=1}^4$ is a quantum magic square and does not admit a semiclassical decomposition*

$$A_{ij} = \sum_{\pi:\, \pi(i)=j} Q_\pi, \qquad Q_\pi \succeq 0, \qquad \sum_\pi Q_\pi = I_9.$$

Proof. The row and column identities follow from (26.1); four copies of $I_5/4$ sum to I_5. If a semiclassical decomposition existed, its compression to the corner $\mathbb{C}^4$ would be a joint measurement of the PVM defined by the rows $(P_{ij})_j$. Jointly measurable projective measurements commute. However,

$$\left[|e_1\rangle\langle e_1|, \left|\frac{e_1+e_2}{\sqrt{2}}\right\rangle\left\langle\frac{e_1+e_2}{\sqrt{2}}\right| \right] = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus 0_2 \neq 0.$$

The contradiction proves nonsemiclassicality. The compression simultaneously provides an analytic witness to the infeasibility of the semidefinite problem. □

Each row defines a UCP channel.

$$\Phi_i : \ell_4^\infty \to M_9, \qquad \Phi_i(\delta_j) = A_{ij},$$

and each column defines another. Thus, the magic normalizations translate into completely positive maps with explicit domain and codomain; a symbol $J(\mathrm{id})$ is not used without first fixing those types.

26.10 Conical route system and internal genealogical universality

For each finite profile c, let G_c be the finite groupoid of TPK routes with that profile and let $A(c) = C^*(G_c)$. At matrix level s, one takes the cone

$$C_s(c) = M_s(A(c))_+.$$

A route $r : c \to d$, represented by the partial isometry v_r, induces the CP map

$$T_r(a) = v_r^* a v_r.$$

Theorem 26.18 (Functorial conic system of TPK paths)**.** *The assignments $c \mapsto (C_s(c))_{s\geq 1}$ and $r \mapsto T_r$ respect identity, composition, matrix amplification, and genealogical truncation. The CPM category generated by the finite fibers is compact monoidal: the dual of a fiber is its conjugate space, and the*

coevaluations and evaluations are the canonical cups and caps. The projective limit of these systems preserves all compatible CP maps.

Proof. The complete positivity of $a \mapsto v_r^* a v_r$ is immediate from its Kraus form. For composable routes, $v_{r_2 r_1} = v_{r_2} v_{r_1}$, hence $T_{r_2 r_1} = T_{r_1} T_{r_2}$ with the declared variance. The inclusions and conditional expectations of the finite levels are UCP and form commuting squares. The compactification of CPM in finite dimension is the standard construction by conjugate spaces; truncation compatibility permits passage to the limit coordinate by coordinate. □

Let $\mathsf{P}_{\mathrm{HMT}}$ the set countable of codigos of routes finite and let $\mathsf{C}_{\mathrm{HMT}}$ the space of states with its contexto complete. The evaluation

$$\mathrm{Ev}(\ulcorner r \urcorner, x) = T_r(x)$$

preserves origin, destination, sheet, orientation, carry, boundary, and word of route.

Corollary 26.19 (Genealogically faithful universality in the HMT domain)**.** *The evaluator* Ev *is universal for the declared class of finite TPK transports and their compatible limits: every transport in that class has a code, composition is performed by concatenating codes, and reduction preserves the complete genealogy. This internal universality is not extended by decree to an untyped external class of targets.*

26.11 Identification of the Stinespring memory with the local record TPK

In the nonadic expectation, the Stinespring dilation retains the discarded digit in a basis $\{e_a\}_{a=0}^{8}$. For a TPK state x, let $\lambda(x, a)$ be the complete label of the continuation by the digit a. Define the isometry

$$W_x : e_a \longmapsto e_{\lambda(x,a)}$$

from the memory minima of Stinespring toward the record local TPK. Let $F_x e_{\lambda(x,a)} = e_a$ the forgetful functor restringido to its imagen.

Theorem 26.20 (The TPK memory refines the minimal memory of Stinespring)**.** *For every x,*

$$W_x^* W_x = I_9, \qquad F_x W_x = I_9.$$

The family $(W_x)_x$ intertwines the inclusion of a new digit with the truncations. At depth m, the tensor product of these isometries identifies the ancilla $(\mathbb{C}^9)^{\otimes m}$ with the TPK subregister that preserves the sequence of extensions; the additional sheet, carry, boundary, and route coordinates explain why TPK memory is a refinement rather than a second copy of the ancilla.

Proof. Continuations with distinct digits have orthogonal labels, so W_x is isometric. The definition of F_x proves the second identity. Composition of extensions concatenates the labels in the same order as the tensor product of the ancillas; hence naturality under truncation and the assertion at arbitrary depth. □

From the surviving sections to the fundamental constants

The genealogical continuum already constructed does not publish mere decimal expansions: it publishes compatible sections that preserve orientation, boundary, carry, and memory. A fundamental constant is the stable evaluation of one of those sections under a typed reader. Part III therefore reconstructs the object, the map, and the proof that precede each value; Archimedean or metrological comparisons appear only after the internal output.

In the sense fixed by the tesis inaugural, the nonadic reading does not act again as a threshold or restart the continuum. The joint nonadic connection and the prolongation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

constitute, with the changes and invariants fixed in the TPK threshold architecture, the first certified segment of the uniform law that constructs a unique enriched coinductive state. Among its correlated outputs, which form the minimal kernel of its typed nonadic image, are $\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}$ and α_{HMT}. Unity of origin does not flatten their types: angular closure, propagation, autoscale and coupling have their own readers; the coordinate α is published through the signed record, reversible seal and dodecaphasic closure. Only after these outputs are published do complex realization, modular memory, the action scale and angular geometry open, with separate domains. The common state is not exhausted by these four coordinates: its typed readers also publish invariants, aggregates, mirror relations and derived mathematical and physical constants.

Part III

Genealogy of the fundamental constants: coinductive generation and dodecaphasic closure

The order of the following chapters is expository, not causal. The nonadic connection and the complete prolongation

$$w_{6} \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_{9}$$

stabilize a unique coinductive APP–TRIT–TPK state. From that state, π, e, φ and α are published correlatively: the successive chapters separate their readers so that they can be studied precisely, but do not turn that typographical separation into serial generation. The closure of π, the propagation of e, the self-scaling of φ and the signed, dodecaphasic publication of α retain their respective statuses within the same correlated coinductive generation.

Chapter 27

Algebra and genealogical classification of the fundamental constants

The Euclidean residue–quotient extension of APP contains two local maps: the additive map normalizes collisions, and the multiplicative map produces partial products. In this section, the global multiplication of words is constructed using exclusively both maps. The correspondence with integer multiplication will be a consequence of the intertwiner, not the definition of the operation.

27.1 Euclidean coordinates of the local cell

Definition 27.1 (Additive and multiplicative applications on the same support)**.** Let $D_9 = \{1, \dots, 9\}$. For $a, b \in D_9$ they are defined by Euclidean division with remainder in D_9:

$$a + b = 9q^{+}(a, b) + R^{+}(a, b), \tag{27.1}$$

$$ab = 9q^{\times}(a, b) + R^{\times}(a, b). \tag{27.2}$$

In particular, the zero remainder is represented by 9 and the quotient is reduced by one unit. The quadruple

$$C_9^{\mathrm{rq}}(a, b) = (R^{+}(a, b), q^{+}(a, b); R^{\times}(a, b), q^{\times}(a, b))$$

is the local arithmetic extension by residue and quotient.

Aggregation by the three classes modulo three of the visible table $R^{\times}$ defines a matrix distinct from the local maps:

$$(M_\Pi)_{rs} = \frac{1}{9} \sum_{\substack{a \equiv r \ (3) \\ b \equiv s \ (3)}} R^{\times}(a, b), \qquad r, s \in \{1, 2, 0\},$$

and, in that order of classes,

$$M_\Pi = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}.$$

Thus, q^{+} and $q^{\times}$ are quotient coordinates of a cell, whereas M_Π is an aggregate operator on residual classes.

The 81 cells satisfy exactly

	$\sum R$	$\sum q$	$\sum(9q + R)$
$+$	405	45	810
$\times$	459	174	2025

and therefore

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129 = 1215.$$

The quotient $q^{\times}$ is necessary for reconstruction. For example,

$$2 \cdot 5 = 9 \cdot 1 + 1.$$

If only $R^{\times}$ is preserved, the product 10 is conflated with the visible value 1.

27.1.1 Cocycles of associativity and distributivity

The coherence of the transducer can already be decided in the cell. For every $a, b, c \in D_9$ the three identities are verified

$$q^{+}(a,b) + q^{+}(R^{+}(a,b),c) = q^{+}(b,c) + q^{+}(a,R^{+}(b,c)), \tag{27.3}$$

$$q^{\times}(R^{\times}(a,b),c) + c\,q^{\times}(a,b) = q^{\times}(a,R^{\times}(b,c)) + a\,q^{\times}(b,c), \tag{27.4}$$

$$\begin{aligned} q^{\times}(a,R^{+}(b,c)) + a\,q^{+}(b,c) &= q^{+}(R^{\times}(a,b),R^{\times}(a,c)) \\ &\quad + q^{\times}(a,b) + q^{\times}(a,c). \end{aligned} \tag{27.5}$$

Theorem 27.2 (Local coherence of the Euclidean extension)**.** *The identities* (27.3)–(27.5)*, together with the corresponding equalities of residues, express respectively additive associativity, multiplicative associativity, and distributivity in the residue–quotient extension. The three families have zero discrepancies in the* $9^3 = 729$ *triples.*

Proof. Expanding $(a + b) + c$ through (27.1), the coefficient of nine is the left-hand side of (27.3); expanding $a + (b + c)$ yields the right-hand side. Uniqueness of division with remainder in D_9 equates both residues and quotients. The same argument applied to $(ab)c = a(bc)$ gives (27.4). Finally, the two expansions of $a(b + c) = ab + ac$ produce (27.5). Direct enumeration of the 729 cases of each identity confirms the algebraic calculation. □

27.2 Bijectivity of numbering

The symbol 9 is not identified with zero. In C_9^{rq}, 9 represents a complete crossing of the nonadic boundary. Replacing it with the digits $0, \dots, 8$ would eliminate the Euclidean quotient that intervenes in the intertwining.

Definition 27.3 (Canonical words)**.** Let

$$\mathscr{W}_9^{+} = \bigsqcup_{\ell \geq 1} D_9^{\ell}, \qquad \mathscr{W}_9 = \{\varnothing\} \sqcup \mathscr{W}_9^{+}.$$

Words are written from the least significant digit. We define

$$\mathrm{ev}_9(u_0, \dots, u_{\ell-1}) = \sum_{j=0}^{\ell-1} u_j 9^j, \qquad \mathrm{ev}_9(\varnothing) = 0.$$

Theorem 27.4 (Uniqueness of the normal form)**.** *The evaluation* ev_9 *is a bijection from* $\mathscr{W}_9^{+}$ *onto* $\mathbb{N}_{\geq 1}$*, and from* $\mathscr{W}_9$ *onto* $\mathbb{N}_0$*.*

Proof. The words of length ℓ cubren the intervalo

$$\frac{9^{\ell} - 1}{8} \leq \mathrm{ev}_9(u) \leq 9\,\frac{9^{\ell} - 1}{8}.$$

The minimum of the next interval is one unit greater than the maximum of the preceding one. Within each interval, division of $n - 1$ by nine uniquely determines the digit

$$d = (n - 1) \bmod 9 + 1$$

and the quotient that is encoded recursively. The process terminates because the quotient strictly decreases. □

Examples:

$$9 \longleftrightarrow (9), \qquad 10 \longleftrightarrow (1,1), \qquad 81 \longleftrightarrow (9,8), \qquad 1000 \longleftrightarrow (1,3,3,1).$$

27.3 Fibers of the residual projections on the integral liftings

For this block, we distinguish two alphabets. The normal form $\mathscr{W}_9$ uses the digits $\{1,\dots,9\}$, written from the least significant. By contrast, $\mathscr{B}_9$ denotes the conventional positional representations in base nine, with digits $\{0,\dots,8\}$ and the most significant on the left. The two types are not identified.

Let

$$\mathscr{W}_{10}^{(2)} = \{0,\dots,9\}^2, \qquad \rho_{10}(a,b) = (b,a), \qquad \mathrm{ev}_{10}(a,b) = 10a + b,$$

and let $q_9 : \mathbb{N} \to \mathbb{Z}/9\mathbb{Z}$ be the residual reduction.

Lemma 27.5 (Residual descent of decimal reversal)**.** *Reversion on decimal words of length two is involutive and its shadow modulo nine is the identity:*

$$\rho_{10}^2 = I, \qquad q_9(\mathrm{ev}_{10}(\rho_{10}(a,b))) = q_9(\mathrm{ev}_{10}(a,b))\,.$$

Fixed length preserves the initial zeros.

Proof. The doble permutation recovers (a,b). Ifnce $10 \equiv 1 \pmod 9$,

$$10b + a \equiv a + b \equiv 10a + b \pmod 9.$$

A word whose double reversal were not itself, or whose digit sum changed modulo nine, would refute the lemma. □

Lemma 27.6 (Residual Square Descent)**.** *The map $s(n) = n^2$ induces the residual endomap*

$$[n]_9 \longmapsto [n]_9^2, \qquad q_9(s(n)) = q_9(n)^2.$$

Proof. The compatibility of a congruence with the product gives $n \equiv m \pmod 9 \Rightarrow n^2 \equiv m^2 \pmod 9$. An integer for which $q_9(n^2) \neq q_9(n)^2$ would refute the statement. □

Proposition 27.7 (Residual fibers of the full reversal)**.** *The residual shadow does not reconstruct the integral lift $\widehat{\rho}_{10} = \mathrm{ev}_{10} \circ \rho_{10}$. Indeed,*

$$18 \equiv 27 \pmod 9, \qquad \widehat{\rho}_{10}(1,8) = 81 \neq 72 = \widehat{\rho}_{10}(2,7).$$

Proof. Both integers belong to residue class zero, but their reversals are distinct. Therefore, there is no $\overline{\rho} : \mathbb{Z}/9\mathbb{Z} \to \mathbb{N}$ that simultaneously factors those two values. □

Proposition 27.8 (Residual fibres of the entire square)**.** *The same shadow likewise does not reconstruct the integer square:*

$$18 \equiv 27 \pmod 9, \qquad 18^2 = 324 \neq 729 = 27^2.$$

Proof. A function $\overline{s} : \mathbb{Z}/9\mathbb{Z} \to \mathbb{N}$ that factorized s would have to assign the two values 324 and 729 to the same zero class, which is impossible. □

Corollary 27.9 (Cross-witness 27–72–729)**.**

$$\widehat{\rho}_{10}(2,7) = 72, \qquad 27^2 = 729 = 9^3 = 3^6,$$

and, in the alfabeto convencional $\mathscr{B}_9$*,*

$$27 = 30_9, \qquad 72 = 80_9, \qquad 729 = 1000_9.$$

The three integers have residual shadow zero and positive digital root nine.

Proof. The two operations are computed directly; successive divisions by nine produce the three expressions. The result would fail if $\mathscr{B}_9$ were confused with $\mathscr{W}_9$, if any of the expressions were incorrect, or if it were claimed that faithful residual descent reconstructs the operation and its complete genealogy. □

27.4 Native adder and quotient propagation

Algorithm 27.10 (Word addition)**.** Two words are traversed by positions, with initial carry $c_0 = 0$. If there are two digits at a position, (27.1) is applied; if there is only one, that digit is taken as the provisional residue and the provisional quotient is zero. When $c_j \neq 0$, a second additive cell is applied to the provisional residue and to c_j. The new residue is published, and the sum of the quotients is transported to the next position. If a nonzero carry remains upon completion, it is published as the final digit. In this manner, the local cells never receive the absent symbol 0, which does not belong to D_9.

Lemma 27.11 (Adder Invariant)**.** *If* $u \boxplus_{\#} v$ *is the word produced by the preceding algorithm,*

$$\mathrm{ev}_9(u \boxplus_{\#} v) = \mathrm{ev}_9(u) + \mathrm{ev}_9(v).$$

The intermediate carry lies in $\{0, 1, 2\}$.

Proof. After processing the positions $0, \ldots, k-1$, let c_k be the pending quotient. The local identities imply

$$\sum_{j<k} (u_j + v_j) 9^j = \sum_{j<k} w_j 9^j + c_k 9^k,$$

including the absent digits as zero. Each new cell preserves this equality. The local maximum is $9 + 9 + 2 = 20$, so that $c_{k+1} \leq 2$. At the end the last quotient is published, and the global identity remains. □

27.5 Multiplication by a digit

Algorithm 27.12 (Scalar transducer)**.** With $d \in D_9$ fixed, each digit u_j produces

$$u_j d = 9q_j + r_j, \qquad (r_j, q_j) = \big(R^{\times}(u_j, d), q^{\times}(u_j, d)\big).$$

Let $c_j \in \{0, \ldots, 9\}$ be the incoming quotient. Since $0 \notin D_9$, the transition is defined by branches:

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ \big(R^{+}(r_j, c_j), q_j + q^{+}(r_j, c_j)\big), & c_j \in D_9. \end{cases}$$

Thus, no local map is evaluated outside its domain. The first branch expresses the absence of a pending quotient; the second normalizes the collision by means of the additive map.

Lemma 27.13 (Scalar invariant)**.** *The word $d \odot_{\#} u$ produced by the transducer satisfies*

$$\mathrm{ev}_9(d \odot_{\#} u) = d\,\mathrm{ev}_9(u).$$

The multiplicative quotient belongs to $\{0, \dots, 9\}$.

Proof. The argument is that of the adder, replacing each local summand by $u_j d$. If $c_j = 0$, the identity $u_j d = 9q_j + r_j$ directly preserves the partial equality. If $c_j \in D_9$, the second Euclidean division gives

$$r_j + c_j = 9q^+(r_j, c_j) + R^+(r_j, c_j),$$

and preserves the same equality after transporting $q_j + q^+(r_j, c_j)$. The maximum $9 \cdot 9 + 9 = 90$ and the remainder form in D_9 keep the quotient within the declared bound. The last quotient is published as a digit. □

27.6 Global product by Horner

Definition 27.14 (Native product)**.** Let $v = (v_0, \dots, v_{m-1})$. Partiendo of $a_m = \varnothing$, is defined for $j = m-1, \dots, 0$

$$a_j = (9 \odot_{\#} a_{j+1}) \boxplus_{\#} (v_j \odot_{\#} u).$$

The result is

$$u \boxtimes_{\#} v := a_0.$$

This definition does not contain

$$\mathrm{dec}_9(\mathrm{ev}_9(u)\,\mathrm{ev}_9(v)).$$

The evaluation will appear only in the proof of entanglement.

Theorem 27.15 (Native multiplicativity)**.** *For all words $u, v \in \mathscr{W}_9$,*

$$\boxed{\mathrm{ev}_9(u \boxtimes_{\#} v) = \mathrm{ev}_9(u)\,\mathrm{ev}_9(v).} \tag{27.6}$$

The empty word is absorbing, and (1) *is the unit.*

Proof. By the two previous lemmas,

$$\mathrm{ev}_9(a_j) = 9\,\mathrm{ev}_9(a_{j+1}) + v_j\,\mathrm{ev}_9(u).$$

Horner's iteration gives

$$\mathrm{ev}_9(a_0) = \mathrm{ev}_9(u) \sum_{j=0}^{m-1} v_j 9^j.$$

The last factor is $\mathrm{ev}_9(v)$. The uniqueness of the normal form identifies the resulting word. □

Corollary 27.16 (Multiplicative monoid of words)**.** *The structure*

$$\mathfrak{W}_9^{\times} = (\mathscr{W}_9, \boxtimes_{\#}, (1))$$

is a commutative monoid; the empty word is its absorbing element. Its construction does not use global multiplication of integers.

The shortest boundary example is

$$(9) \boxtimes_{\#} (9) = (9, 8), \qquad 9 + 8 \cdot 9 = 81.$$

The 17 local cells of multiplication produce

$$(9, 8) \boxtimes_{\#} (9, 8) = (9, 8, 8, 8), \qquad \mathrm{ev}_9(9, 8, 8, 8) = 6561.$$

27.7 Matrix linearization of the genealogical state and construction of the completely positive entangler

Definition 27.17 (Hilbert Sectors)**.** Let

$$\mathcal{K}_{\#}^{+} = \ell^2(\mathscr{W}_9^{+}), \qquad \mathcal{H}_{+} = \ell^2(\mathbb{N}_{\geq 1}).$$

We define on the canonical bases

$$W_{\times} e_u = e_{\mathrm{ev}_9(u)}.$$

In $\mathcal{H}_{+}$, the number operator is denoted by

$$\mathsf{N} e_n = n e_n, \qquad \mathrm{Spec}(\mathsf{N}) = \mathbb{N}_{\geq 1}.$$

The vacuum may be added separately as a vacuum vector e_0; it is not confused with $\mathcal{H}_{+}$, whose spectrum begins at one.

Theorem 27.18 (Unitarity and multiplicative square)**.** *$W_{\times}$ is unitary. In the algebraic kernel $\mathbb{C}[\mathscr{W}_9^{+} \times \mathscr{W}_9^{+}]$, let*

$$\mu_{\#}(e_u \otimes e_v) = e_{u \boxtimes_{\#} v}, \qquad \mu_{\mathbb{N}}(e_a \otimes e_b) = e_{ab}.$$

Then the square holds.

$$\begin{array}{ccc} \mathcal{K}_{\#}^{+} \otimes_{\mathrm{alg}} \mathcal{K}_{\#}^{+} & \xrightarrow{W_{\times} \otimes W_{\times}} & \mathcal{H}_{+} \otimes_{\mathrm{alg}} \mathcal{H}_{+} \\ \downarrow{\scriptstyle \mu_{\#}} & & \downarrow{\scriptstyle \mu_{\mathbb{N}}} \\ \mathcal{K}_{\#}^{+} & \xrightarrow[W_{\times}]{} & \mathcal{H}_{+} \end{array}$$

and therefore, the square commutes.

Proof. The bijection of normal forms sends one orthonormal basis to another and proves unitarity. On a basic tensor, both paths produce $e_{\mathrm{ev}_9(u)\,\mathrm{ev}_9(v)}$ by (27.6). Linearity gives the equality on the kernel. □

The maps $\mu_{\#}$ and $\mu_{\mathbb{N}}$, initially defined on the respective algebraic kernels, are densely defined and unbounded. For $c = (c_{u,v}) \in \ell^2(\mathscr{W}_9^{+} \times \mathscr{W}_9^{+})$, each sum over a product fiber is finite. The adjoint of $\mu_{\#}$ contains all finite-support vectors in its domain, so that $\mu_{\#}$ is closable. The domain of its closure is

$$\mathrm{Dom}(\overline{\mu_{\#}}) = \left\{ c \in \ell^2(\mathscr{W}_9^{+} \times \mathscr{W}_9^{+}) : \sum_{w \in \mathscr{W}_9^{+}} \left| \sum_{u \boxtimes_{\#} v = w} c_{u,v} \right|^2 < \infty \right\}.$$

Each fiber is finite because ev_9 identifies it with the ordered pairs of divisors of an integer. The same description holds for $\overline{\mu_{\mathbb{N}}}$ after applying $W_{\times} \otimes W_{\times}$, and $W_{\times}$ intertwines the graphs of both closures.

27.8 Native coproduct and native antipode

The multiplicative opening is now defined before looking at the codomain:

$$\Delta_{\#} e_w = \sum_{u \boxtimes_{\#} v = w} e_u \otimes e_v .$$

The sum is obtained by enumerating products of the transducer. One does not ask whether one ordinary integer divides another. In the integer sector define, on the corresponding algebraic kernel,

$$\Delta_{\times} e_n = \sum_{\substack{a,b \in \mathbb{N}_{\geq 1} \\ ab = n}} e_a \otimes e_b .$$

Both coproducts enumerate ordered pairs.

Theorem 27.19 (Native coassociativity)**.** *In the algebraic core,*

$$(\Delta_{\#} \otimes I)\Delta_{\#} = (I \otimes \Delta_{\#})\Delta_{\#}.$$

Moreover,

$$(W_{\times} \otimes W_{\times})\Delta_{\#} = \Delta_{\times} W_{\times}.$$

Proof. Both sides of the first equality enumerate every native triple $a \boxtimes_{\#} b \boxtimes_{\#} c = w$ exactly once. The second equality follows from the bijection between fibers induced by (27.6). □

Theorem 27.20 (Complete arithmetic coproduct)**.** *In the algebraic kernel $c_{00}(\mathbb{N}_{\geq 1}) \subset \ell^2(\mathbb{N}_{\geq 1})$, let*

$$\Delta_{\times} e_n = \sum_{ab = n} e_a \otimes e_b .$$

Its closure has domain

$$\operatorname{Dom}(\overline{\Delta_{\times}}) = \left\{ x = (x_n) : \sum_{n \geq 1} d(n) |x_n|^2 < \infty \right\},$$

is coassociative on the kernel and, for $Q e_n = n e_n$, satisfies

$$(\log Q \otimes I + I \otimes \log Q)\Delta_{\times} = \Delta_{\times} \log Q.$$

Proof. The supports of $\Delta_{\times} e_n$ are orthogonal for distinct integers, and their squared norm is the number $d(n)$ of divisors, which yields the weighted domain of the closure. The two coassociativity relations reindex the same sum over $abc = n$. Finally, $\log a + \log b = \log n$ in each term proves the intertwining. A triple factorization counted differently, or failure of this identity, would falsify the theorem. □

Theorem 27.21 (Reduced coproduct and irreducible selector)**.** *In the sector integer, the coproducto reducido*

$$\Delta_{\times}^{\mathrm{red}} e_n = \sum_{\substack{ab = n \\ a,b > 1}} e_a \otimes e_b$$

is closable. The operator

$$N_{\times}^{\mathrm{red}} = \left(\overline{\Delta_{\times}^{\mathrm{red}}}\right)^{*} \overline{\Delta_{\times}^{\mathrm{red}}}$$

is positive self-adjoint and diagonal:

$$N_{\times}^{\mathrm{red}} e_n = d_{\mathrm{red}}(n) e_n .$$

Its kernel is generated by e_1 and by the e_p with p irreducible; therefore,

$$P_{\mathfrak{I}} = \mathbf{1}_{\{0\}}(N_{\times}^{\mathrm{red}}) - |e_1\rangle\langle e_1|$$

select the irreducibles and exclude the unit.

In the native sector, the corresponding coproduct is

$$\widetilde{\Delta}_{\#} e_w = \sum_{\substack{u \boxtimes_{\#} v = w \\ u,v \neq (1)}} e_u \otimes e_v .$$

The entrelazador $W_{\times}$ transports $\Delta_{\times}^{\mathrm{red}}$, $N_{\times}^{\mathrm{red}}$ and $P_{\mathfrak{I}}$ toward $\widetilde{\Delta}_{\#}$,

$$A_{\#} := \overline{(\widetilde{\Delta}_{\#})}^{*} \overline{\widetilde{\Delta}_{\#}}, \qquad P_{\mathfrak{I},\#} = (I - P_{(1)})\mathbf{1}_{\{0\}}(A_{\#}), \qquad P_{(1)} = |e_{(1)}\rangle\langle e_{(1)}|.$$

Proof. The supports of the images of two distinct basis vectors are orthogonal. The squared norm of $\Delta_{\times}^{\mathrm{red}} e_n$ is $d_{\mathrm{red}}(n)$, which vanishes exactly for $n = 1$ or n irreducible. This proves the diagonalization and the declared kernel; subtracting the projector of e_1 removes the unit. The multiplicativity of $W_{\times}$ identifies the integer fibers with the native fibers and transports the three operators. A composite in the kernel, an irreducible outside it, or inclusion of the unit in $P_{\mathfrak{I}}$ would refute the theorem. □

Proposition 27.22 (Nonadic coincidence of primes and composites)**.** *Let* $\gcd(n, 90) = 1$, $\tau_n = \mathrm{ord}_n(10)$ *and*

$$M_n = \frac{10^{\tau_n} - 1}{n}.$$

Then $M_n \equiv 0 \pmod 9$, *both for primes and for composites. In particular, the nonadic closure and the isolated decimal period are not a selector of irreducibles.*

Proof. Because $9 \mid 10^{\tau_n} - 1$ and because n is invertible modulo nine, divisibility by nine survives passage to the quotient. The primes $7, 13$ and the composites $77, 91, 143$ share period six; the Carmichael numbers provide additional controls. The statement would be falsified only if the congruence or the period separated all primes from all composites in the declared domain. □

This statement is strictly arithmetic: it identifies factorization fibers and the operators they induce on the sector of normal forms.

27.9 Exhaustive enumeration and necessity of structural coordinates

The finite enumeration comprises:

Test	Result
Cells per local map	81
Reversible numbering paths	59050
Intertwiner pairs	532900
Intertwiner errors	0
Associative triples	531441
Associative errors	0
Coproduct window	1 to 500
Irreducibles obtained	95

Definition 27.23 (Multiplicative quotient)**.** Preserving only $R^{\times}$ transforms $2 \cdot 5 = 10$ into the visible value 1. Perturbing $q^{\times}(9, 9)$ from 8 to 7 transforms 81 into 72. The local quotient is necessary.

Definition 27.24 (Additive application)**.** The multiplicative map produces partial products, but does not resolve their collisions. Removing q^{+} breaks the global normalization; for example, it no longer reproduces the intermediate quotients of $10 \cdot 18 = 180$.

Definition 27.25 (Coordinates and boundary symbol)**.** Ordinary zero is not a digit of the construction. The empty word is 0, and the symbol 9 preserves its value and quotient. Mixing zero-indexed positional coordinates with the labels $1, \ldots, 9$ changes the cells.

Proposition 27.26 (Exact scope of the ablations and of the finite census)**.** *The preceding ablations prove, within the declared conventions:*

1. *without* $q^{\times}$, $2 \cdot 5 = 10$ *colapsa to the residue visible* 1*;*
2. *cambiar* $q^{\times}(9, 9) = 8$ *by* 7 *transforms* 81 *in* 72*;*
3. *without* q^{+}*, normalization of the additive collisions fails.*

The exhaustive enumeration comprises the $2 \cdot 81 = 162$ *cells of the two maps, the* 729 *local triples,* 532 900 *word pairs, and the* 95 *irreducibles no greater than* 500*. The indicated identities are proved on these finite domains, but the coinductive survival of every TPK history is not.*

Proof. The three counterexamples are computed directly with (27.1) and (27.2). The enumeration of the four finite domains produces the cardinals written in the statement and does not add any coinductive identity to those already proved in the monoid of words. □

27.10 Algebraic structure of the word monoid

The construction proves, in the monoid $\mathfrak{W}_9^{\times}$:

1. a unique normal form for each nonnegative integer;
2. global sum and product constructed solely from the two local maps;
3. symbolic conservation of the value under quotient propagation;
4. a unitary $W_{\times}$ and the exact multiplicative square;
5. a divisor coproduct and an irreducible selector defined natively;
6. the need to preserve the transport class when leaving the normal-form sector.

Passage to the enriched space now reduces to an existence problem: constructing a family $(\iota_n)_n$ compatible with the depth restrictions and proving that it satisfies the equation of the preceding section. This question is distinct from the exact intertwiner on $\mathfrak{W}_9^{\times}$, which is already determined by the normal form and the local maps.

27.11 Problem of determining the constants from the genealogical state

A physical constant is represented as a section of a magnitude line equipped with a rule of evaluation, orientation, and normalization. Its numerical coordinate depends on the chosen chart, whereas its genealogy remains invariant. Consequently, the determination problem comprises:

1. the identification of the antecedent HMT state;
2. the specification of the operator or character of selection;
3. determination of the conserved invariant;
4. the corresponding magnitude line;

5. the chart in which the terminal value is expressed.

The causal chain of the work is

$$\begin{aligned}\mathrm{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow \text{internal realization} \longrightarrow \text{dimensionless invariant} \\ \longrightarrow \text{magnitude section} \longrightarrow \mathbb{R}.\end{aligned} \tag{27.7}$$

The term $\mathbb{R}$ in the chain represents the terminal evaluation of the section; it does not constitute the original domain from which the route is retrospectively reconstructed.

27.12 Antecedent chain APP–TRIT–TPK and domain of the extraction operators

APP supplies the geometry of positions, residues, quotients, and boundaries. TRIT incorporates orientation $(-1, 0, +1)$ and distinguishes opening, threshold, and return. TPK completes the state through carry, route, sheet, memory, and survival. In particular, transition $9 \to 1$ preserves phase and modifies the memory state; it therefore does not constitute a reinitialization devoid of genealogy.

The realizations developed below preserve that antecedent without replacing it. The vacuum is obtained through a spectral function of the bilayer operator; gravity selects a section through the chronological structure 108; criticality uses a dodecaphasic family; the Gaussian constants follow from the evaluation of cylinders; and the modular Hamiltonian is defined on a boundary that preserves the incidence $6 \to 8$. The production of scalar values does not establish functional identity among these realizations.

Proposition 27.27 (Insufficiency of the Scalar Coincidence)**.** *Let $F_1 : X_1 \to L_1$ and $F_2 : X_2 \to L_2$ be two typed functionals. Equality of coordinates $c_1(F_1(x_1)) = c_2(F_2(x_2))$ does not by itself induce either an identification $X_1 \simeq X_2$ or an intertwiner $F_2 I = J F_1$.*

In particular, a numerical proximity such as $I_{\mathrm{or}} \approx \gamma_{\mathrm{HMT}} a_0$ cannot be promoted to an identity without constructing the corresponding intertwiner.

27.13 Lines of magnitude, oriented letters and dimensional coordinate systems

Let L_Q be an oriented real line representing the magnitude Q. A unit $u_Q \in L_Q \setminus \{0\}$ determines a provisional coordinate $L_Q \simeq \mathbb{R}$, without replacing the intrinsic structure of the line. The constitutive relation

$$L_\mu = L_S L_C^{-1} L_Q^{-2}, \qquad L_\varepsilon = L_S^{-1} L_C^{-1} L_Q^{2}$$

precedes the SI choice. In the same way,

$$L_Z = L_S L_Q^{-2}, \qquad L_G = L_C^{5} L_T^{2} L_S^{-1}$$

type impedance and the gravitational constant before expressing their magnitudes in ohms or SI units. In this work we shall write

$$Q_{\mathrm{int}} = \widehat{Q}\, u_Q$$

to distinguish the dimensionless character $\widehat{Q}$ from the reference section u_Q.

27.14 Torsor of frames and covariance of dimensional evaluations

The admissible charts form a torsor and therefore none has a character that is canonical relative to the others. The frame change $u_Q \mapsto \lambda_Q u_Q$ induces the contragredient transformation of the coordinate. Physical identities must be homogeneous with respect to this action. For example,

$$\mu\varepsilon = c^{-2}, \qquad \frac{\mu}{\varepsilon} = Z^2$$

are valid in every compatible frame. Mathematical unification and dimensional unification do not eliminate units; rather, they reconstruct them through the covariance of the lines and allow the invariant to precede its coordinate expression.

27.15 Final metrological evaluation and prohibition of retrospective selection

Each entry of the final atlas adopts the form

$$\boxed{\text{antecedent} \longrightarrow \text{operator} \longrightarrow \text{invariant} \longrightarrow \text{meaning} \longrightarrow \text{terminal value.}} \tag{27.8}$$

This order expresses a causal dependency: the numerical value is only determined after fixing the antecedent, operator, invariant, and interpretation. Omission of any of these elements reduces the result to a reconstruction that is numerical, not a derivation that is genealogical.

27.16 Exact domain of the genealogical generation and its terminal metrical chart

The generation of π, e, φ, the dodecaphasic closure of α_{HMT}, the action section, and the Barbero–Immirzi output γ_{HMT} are adopted as initial results. The present development does not recalibrate those constructions; it determines the constants and invariants resulting from their application to the constitutive, thermal, gravitational, modular, and spectral realizations.

The electron mass is likewise not included as a governing problem, since it belongs to the General Mass Law. Every occurrence of m_e in an atomic identity is treated as a datum proceeding from that domain and not as a new selection of the present development.

Logical Scope and Domain of Validity

The APP–TRIT–TPK architecture and the dimensional lines constitute the starting structure. The uniform rule of five fields of the atlas makes explicit its causal discipline in each entry.

27.17 Operator classification of 114 numerical expressions by domain, generator, invariant, and codomain

Let the finite indexed set be

$$E_{114} = \{\varepsilon_1, \ldots, \varepsilon_{114}\}$$

of expresiones numericas and let O the space of readings of the reader triadico. The map

$$T : E_{114} \longrightarrow O, \qquad \varepsilon_i \longmapsto T_{\mathrm{tri}}(\varepsilon_i), \qquad |\operatorname{im} T| = 106,$$

classifies expressions according to their operatorial role. Each ε_i, including the compared targets, is fixed as input of T; therefore, this map does not generate the values that it evaluates. The functional partition is

functional family	entries
structural kernel	5
coupling and closure	1
bridges between modes	7
angular and closure	22
exponential–logarithmic	12
spectral and regularization	14
hyperbolic and modular	2
algebraic and metric	50
typed miscellany	1

$$\sum n_i = 114.$$

Network centrality, shortest paths, and communities computed on the graph induced by T describe connectivity within the classification; they do not establish mathematical causality or HMT genealogy.

Seven fibers of T gather fifteen expressions with a common reading. A shared fiber does not identify their values either. In particular, φ^{-1}, φ, and φ^2 are distinct real numbers that the map T groups into the same self-scaling orbit; the common projection preserves dynamic function and forgets the Archimedean coordinate that separates them.

The exhaustive enumeration traversed the 114 entries of E_{114}, four hundred sixty-eight direct candidates U_{12}, and one hundred four thousand nine hundred seventy-six weighted seeds without finding, within that finite domain, a rival that reproduced the oriented dodecaphasic correction of α_{HMT}. The search domain is thereby bounded by those three sets. The census proves that no rival appears in the enumerated class; it does not prove uniqueness among all possible functionals and does not enter as a premise of theorem α_{HMT}.

For each ε_i, the taxonomy assigns the quadruple (object, reader, domain, structural function) used in the enumeration.

27.18 Spectral, informational, and thermodynamic confluence of $\log 3$

The identity

$$\log 3 = C_1 + C_2 - 2C_0 = S_{\widehat{\mathbb{T}}}/k_B = E_P/(k_B\Theta_{\mathrm{clk}}),$$

relates three distinct realizations. The first member is a residual defect, the second an entropy of decision, and the third a ratio between energy and temperature. Their equality proves the conservation of the scalar under a change of realization, without identifying the state spaces.

The first equality is deduced from the three residual channels.

$$Z_r(s) = \sum_{\substack{n\geq 1\\ n\equiv r \pmod 3}} n^{-s}, \qquad r = 0, 1, 2.$$

Then

$$Z_0(s) = 3^{-s}\zeta(s), \quad Z_1(s) + Z_2(s) = (1 - 3^{-s})\zeta(s), \quad Z_1(s) - Z_2(s) = L(s, \chi_{-3}).$$

If C_r denotes the part finite of Z_r in $s = 1$, the expansion of 3^{-s} gives

$$C_0 + C_1 + C_2 = \gamma, qquad - 2C_0 + C_1 + C_2 = \log 3, qquad C_1 - C_2 = \frac{\pi}{3\sqrt{3}}. \tag{27.9}$$

The three covectors are linearly independent. From a single residual decomposition, they recover the total finite term, the dilation of the zero class, and the orientation between the two nonzero classes. $\log 3$ is therefore a scale-change invariant before being interpreted as entropy or temperature.

27.19 Epistemological classification of the derived constants

The chapter's contributions are classified into three categories:

1. assignment of an internal operator to a known constant, as in the Catalan case;
2. exact composed evaluation of two internal characters, such as $L(1, \chi_{12})$;
3. derivation of a constant from a field or a family constructed internally, such as $\gamma_{\mathbb{Q}(\sqrt{3})}$ or A_k.

Mathematical provenance determines the status of recovery or novelty; decimal proximity plays no role in that classification.

Remark 27.28 (Negative control). The character χ_{12} must have exactly the pattern $(+,-,-,+)$ in $(1,5,7,11)$; any other pattern invalidates equations (41.8) y (41.9). The identity equation (43.2) and the factorization equation (42.1) are independent controls.

Logical Scope and Domain of Validity

$J^2 = -I$, Catalan, χ_{12}, their values L, Euler–Kronecker, and the Bendersky–Glaisher tower: *Exact internal theorem*. The gathering in a lattice $3/4/12$ is *New formalization* that preserves the mathematical provenances.

All these results are joined because they turn the terminal value into the quantitative expression of a previously constructed structure. Numerical exactness accompanies a different conception of what a fundamental constant is. The nonadic connection does not first generate three constants and then add a fourth: the prolongation $w_6 \to w_{12} \to w_{18} \to w_{24} \to w_{30} \to R_{36} \to G_9$ stabilizes a common coinductive state from which π, e, φ and α are published in a correlated manner. Their readers distinguish closure, propagation, self-scaling and coupling; within this joint generation, coordinate α retains its signed, dodecaphasic realization and its two internal derivation routes.

In the usual description, a constant is commonly presented as a number placed before an equation: a quantity that the theory requires but whose value must be received from outside. In Triadic Modular Holography, the opposite occurs. The constant appears at the end, after the same state has been traversed by several readers and it has been determined which relation has survived all of them.

This is why the constants in the document are magnitudes endowed with genealogy: invariant remnants of a transformation. Each states what has been preserved when something has changed:

- to sheet has been inverted;
- a cycle has returned to its initial phase;
- a memory has crossed the seam;
- an interior description has been published on the boundary;
- a dimensionless magnitude has received a physical realization;
- a system has changed scale while fully preserving its genealogy.

Dimensional Mechanics subsequently adds the magnitude lines and the units. But the structure that selects the constant is prior. The real value appears at the end because $\mathbb{R}$ functions here as a terminal evaluation: it is the quantitative shadow of a construction that was already complete before becoming a decimal.

With that view, the nine developments considered form a single narrative on orientation, memory, geometry, information, and scale.

27.20 Joint theorem of coinductive generation and arbitrary depth

The starting structure of this theorem is not the bare set of nine symbols. It is the typed system

$$\mathfrak{H}_9 = \big(\mathrm{APP}, \mathrm{TRIT}, \mathrm{TPK}; \widehat{\mathcal{X}}^{\mathrm{enr}}, \mathcal{U}, \Gamma_9, \mathrm{Ext}, \mathrm{tr}\big),$$

constructed on the double positional and arithmetic realization of $\mathcal{D}_9 = \{1, \dots, 9\}$. APP accumulates and folds through its sheets Σ and Π, preserving residue and quotient; TRIT orients each transition in one of its three regimes; and TPK selects, transports, memorizes and returns on the enriched fiber. In particular,

$$M_{\mathrm{ph}} : \mathcal{D}_9 \longrightarrow \{+1, -1, 0\}, \qquad m_{\mathrm{ph}} := \big(M_{\mathrm{ph}}(1), \dots, M_{\mathrm{ph}}(9)\big)^{\mathsf{T}},$$

$$\mathrm{Tra}_t(y, d) := \mathrm{Lift}(T_d)(y), \qquad \mathrm{Upd}_t := \mathrm{Rec}_t \circ \mathrm{Mem}_t,$$

and, for all $x \in \widehat{\mathcal{X}}^{\mathrm{enr}}$,

$$\mathcal{U}_t(x) = \mathrm{Upd}_t\big(\mathrm{Tra}_t\big(\mathrm{Sel}_t\big(x, M_{\mathrm{ph}}(g(t))\big), d(t)\big)\big).$$

This composition is the effective dynamic datum: the nine-symbol chart is its finite support, not a substitute for the dynamics. In particular, $M_{\mathrm{ph}}(g(t))$ is the scalar value received by the selector, whereas m_{ph} is the vector of the nine values of the function; neither composes directly with a state operator.

For $r \in 6\mathbb{N}_{>0}$, let $\widehat{\mathcal{X}}_r^{\mathrm{enr}}$ be the truncated states with sheet, orientation, residue, quotient, carry, route, boundary and memory, and let

$$\mathrm{Ext}_{r+6,r} \subseteq \widehat{\mathcal{X}}_r^{\mathrm{enr}} \times \widehat{\mathcal{X}}_{r+6}^{\mathrm{enr}}, \qquad \mathrm{tr}_{r+6,r} : \widehat{\mathcal{X}}_{r+6}^{\mathrm{enr}} \longrightarrow \widehat{\mathcal{X}}_r^{\mathrm{enr}}$$

the admissible prolongation relation and truncation. For $x_r \in \widehat{\mathcal{X}}_r^{\mathrm{enr}}$, the local fiber

$$\mathrm{Ext}_{r+6,r}(x_r) := \{x_{r+6} : (x_r, x_{r+6}) \in \mathrm{Ext}_{r+6,r}\}$$

may contain several children. The correct compatibility is

$$(x_r, x_{r+6}) \in \mathrm{Ext}_{r+6,r} \Longrightarrow \mathrm{tr}_{r+6,r}(x_{r+6}) = x_r.$$

In R_{36}, the three recovered rows $222220, 021222, 102011$, transported to the active gauge by the permutation $p_0 = (0, 1, 2, 4, 5, 3)$ in zero-based indices —equivalently, $p_1 = (1, 2, 3, 5, 6, 4)$ in one-based indices—, carry, before any real realization, the internal characters of closure, propagation and self-scaling. Their triple, together with the signed register U_{12} and the seal K, carries the internal coordinate of α. For each $\chi \in \{\pi, \varphi, e, \alpha\}$, we write the state thus enriched as

$$x_r^{\chi} = (\widetilde{x}_r, \chi_{\mathrm{HMT}}, \varepsilon_r^{\chi}, N_r^{\chi}).$$

Here ε_r^{χ} is the internal residue produced by the APP quotient in the state, not $\mathrm{frac}(P_{\mathrm{ar}}(\chi))$. Internal Euclidean division fixes

$$d_r^{\chi} = \mathrm{Dig}_{729}(\varepsilon_r^{\chi}), \quad 0 \le d_r^{\chi} < 729, \quad \varepsilon_{r+6}^{\chi} = 729\varepsilon_r^{\chi} - d_r^{\chi}, \quad N_{r+6}^{\chi} = 729N_r^{\chi} + d_r^{\chi}.$$

Selection, transport and updating on the section χ are typed by

$$\mathrm{Sel}_r^{\chi}(x) := \mathrm{Sel}\big(x, M_{\mathrm{ph}}(\rho_9(r/6 + 1))\big), \qquad \mathrm{Tra}_r^{\chi}(y, d) := \mathrm{Lift}(T_d)(y), \qquad \mathrm{Upd}_r^{\chi} := \mathrm{Rec}_r^{\chi} \circ \mathrm{Mem}_r^{\chi}.$$

The complete transition is

$$\mathcal{U}_r^{\chi}(x) := \mathrm{Upd}_r^{\chi}\big(\mathrm{Tra}_r^{\chi}\big(\mathrm{Sel}_r^{\chi}(x), d_r^{\chi}\big)\big),$$

and reads only the preceding enriched state: internal phase, memory, carry, sheet, orientation, route, boundary, signature, register, character and residue. It does not read a future digit, P_{ar}, or a Machin, factorial or Perron bracket. We define the forward subrelation

$$\mathrm{Ext}^{\chi,\rightarrow}_{r+6,r} := \{(x^{\chi}_{r}, \mathcal{U}^{\chi}_{r}(x^{\chi}_{r}))\} \subseteq \mathrm{Ext}_{r+6,r}\,.$$

It is not the retrospective selection $\mathrm{Ext} \cap \{P_{\mathrm{ar}} = \chi\}$: the character is already a causal coordinate of x^{χ}_{36}. The preceding equations give $\lfloor N^{\chi}_{r+6}/729 \rfloor = N^{\chi}_{r}$ and preserve sheet, orientation, carry, boundary, route, signature, register and genealogy.

The complete relation Ext remains multivalued. The required uniqueness is that of each forward section over its produced R36 seed,

$$\#\Big\{(x^{\chi}_{r})_{r\geq 36} : x^{\chi}_{36} \text{ fixed},\ (x^{\chi}_{r}, x^{\chi}_{r+6}) \in \mathrm{Ext}^{\chi,\rightarrow}_{r+6,r}\Big\} = 1,$$

not the false equality $\#\,\mathrm{Ext}_{r+6,r}(x_r) = 1$ at each local gate. This compatibility extends to every level. The notation

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

designates these maps and their type changes, not a list of labels. The holonomy Γ_9 satisfies

$$g(k+9) = g(k), \qquad q^{(9)}(k+9) = q^{(9)}(k) + 1,$$

so that the phase returns while the quotient, the route and the memory of the state advance.

A generic canonical history based on $a_{(1,1)}$ and $t_j = M_{\mathrm{ph}}(\rho_9(j+1))$ establishes the nonemptiness of Ext; it does not by itself individualize the four preceding sections. These are joined by their internal R36 coordinates and by the forward rule just defined.

Theorem 27.29 (Inaugural construction from the seeds)**.** *Let*

$$I_9 = \{0, \ldots, 8\}, \qquad D = \{N, E, S, O\}, \qquad \mathcal{S} = I_9^2 \times D,$$

and let $\Omega_{\mathrm{seed}} = \mathcal{S}_{+} \times \mathcal{S}_{\times}$ *be the domain of the two cursors realizing the additive and multiplicative sheets of APP. Then*

$$|\Omega_{\mathrm{seed}}| = (9^2 \cdot 4)^2 = 104\,976,$$

and the 54*-step recurrence and ternary reduction define*

$$\Omega_{\mathrm{seed}} \xrightarrow{T^{\mathrm{seed}}_{54}} \mathcal{U}^{\mathrm{cat}}_{6} \xrightarrow{\rho_3} \mathcal{W}_6 \hookrightarrow \mathbb{F}^6_3,$$

with

$$|\mathcal{U}^{\mathrm{cat}}_{6}| = 468, \qquad |\mathcal{W}_6| = 243, \qquad |\mathbb{F}^6_3| = 729.$$

The fibers satisfy exactly

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The TPK extension relation prolongs the surviving words to the family of compatible sections of arbitrary depth. The forward subrelations anchored in R36 individualize the correlated coinductive generation of π, φ, e *and* α*; the complete relation may continue to have several local prolongations. On that family, the Archimedean shadow and the exceptional realization of the theorem 27.30 are constructed jointly.*

Proof. The preceding Cartesian product exhaustively enumerates the positions and orientations of both cursors. The map T^{seed}_{54} is obtained by composing at each step sheet selection, cardinal transport, quotient updating and carry memory. Enumeration of its image contains 468 distinct sextets; the sum of the

cardinalities of its fibers is the displayed identity. The componentwise map ρ_3 contains 243 distinct words. Finally, nonemptiness at every horizon and compatibility of Ext with the truncations produce the inverse limit. Functionality of each $\mathrm{Ext}^{\chi,\to}$, starting from the R36 state that already carries χ_{HMT}, proves by induction the uniqueness of that global section without requiring each local fiber of Ext to be a singleton. Each statement is an internal identity or a consequence of projective compatibility; none uses a real number, a recognized constant or a decimal table as input. □

Theorem 27.30 (Governing thesis: one continuum and two projections)**.** *Let $\Omega_\Gamma^{\mathrm{cal}} \subset X_\infty^{\mathrm{cal}}$ be the space of surviving sections of the enriched turns of the TPK. The carry of the three turns $\gamma_-, \gamma_0, \gamma_+$ defines, without prior Archimedean evaluation, the homeomorphism*

$$\beta : \Omega_\Gamma^{\mathrm{cal}} \xrightarrow{\sim} (\mathbb{F}_3^6)^{\mathbb{N}} =: \mathcal{W}_\infty.$$

The normalized balanced forms

$$a_k + c_k = \rho_k + 3c_{k+1}, \qquad \rho_k \in \{-1, 0, +1\},$$

construct the ordered ring $\mathbb{Z}_{\mathrm{HMT}}$, its field of fractions $\mathbb{Q}_{\mathrm{HMT}}$ and the completion

$$\mathbb{R}_{\mathrm{HMT}} := \mathrm{Cau}(\mathbb{Q}_{\mathrm{HMT}})/\asymp.$$

The joint discrete structure is the fiber product

$$C_{\mathrm{cont}}^{\mathrm{disc}} = \left(\mathbb{Z}_{\mathrm{HMT}} \times \Omega_\Gamma^{\mathrm{cal}}\right) \times_{\mathcal{W}_\infty} X_\infty^{\mathrm{cal}}.$$

Its two natural maps are

$$P_{\mathrm{ar}} : \mathbb{Z}_{\mathrm{HMT}} \times \Omega_\Gamma^{\mathrm{cal}} \longrightarrow \mathbb{R}_{\mathrm{HMT}}, \qquad Q_\infty : X_\infty^{\mathrm{cal}} \longrightarrow (\mathscr{C}_W^{\mathrm{cal}})^{\mathbb{N}}.$$

If $\beta(\xi) = (b_q)_{q\geq 0}$, $\nu(b) = \sum_{i=1}^{6} b_i 3^{6-i}$ and

$$q_n(m, \xi) = m + \sum_{q=0}^{n-1} \frac{\nu(b_q)}{729^{q+1}},$$

then (q_n) is Cauchy and $P_{\mathrm{ar}}(m, \xi)$ is its class in $\mathbb{R}_{\mathrm{HMT}}$. The rational cylinders

$$I_n(m; b_0, \ldots, b_{n-1}) = [q_n, q_n + 729^{-n}]$$

are compatible, have diameter tending to zero and the recurrent partition into 729 subcylinders constructs, for every Cauchy class, a tape (b_q) representing it. Consequently,

$$P_{\mathrm{ar}} \text{ is surjective}, \qquad (\mathbb{Z}_{\mathrm{HMT}} \times \Omega_\Gamma^{\mathrm{cal}})/\!\sim_{\mathrm{ar}} \;\cong\; \mathbb{R}_{\mathrm{HMT}}.$$

Only after this internal construction does uniqueness of the complete Archimedean ordered field recognize $\mathbb{R}_{\mathrm{HMT}} \cong \mathbb{R}$.

The second projection preserves the calibrated incidence of each level; in the coordinates $x = ((w_j, f_j))_{j\geq 1}$ it acts by

$$Q_\infty(x) = \left(f_j, \Gamma_{f_j A_W f_j^{-1}}(w_j)\right)_{j\geq 1}.$$

Thus, $\mathbb{R}_{\mathrm{HMT}}$, subsequently recognized as $\mathbb{R}$, is the Archimedean shadow obtained by contracting the complete genealogy of cylinders. The exceptional projection preserves that genealogy as boundary incidence and admits the successive realization Paley–Witt–Golay–Leech–$V^\natural$–Monster. The two branches come from the same state and neither is retrospectively reconstructed from the other.

Proof. The three enriched turns exist after each prefix and their carry distinguishes them by $-1, 0, +1$. Concatenation and truncation are inverses, proving bijectivity and continuity of β. For a Cauchy class of $\mathbb{Q}_{\mathrm{HMT}}$, one first chooses its ordered unit cylinder and then, recursively, the unique half-open subcylinder of the internal partition into 729 pieces containing the class; at a common boundary, the two writings are identified by $\sim_{\mathrm{ar}}$. The sequence of left endpoints is precisely (q_n), converges to the given class and proves surjectivity without presupposing an expansion of $[0, 1]$ or a point of $\mathbb{R}$. Identification with the real field is the uniqueness theorem applied afterwards to $\mathbb{R}_{\mathrm{HMT}}$.

The subspaces satisfying the closure of a specific character are survival filters within the global capacity. Their function is to individualize the sections of π, e, φ, α; surjectivity of the continuum comes from the completion of $\mathbb{Q}_{\mathrm{HMT}}$. This separation of domains prevents turning a constant-selection condition into a premise of the coverage of $\mathbb{R}$.

In the second branch, compatibility $r_{n,m}Q_n = Q_m p_{n,m}$ allows passage to the limit and defines Q_∞. Conjugation by f_j transports the gauge without erasing incidence. The subsequent code, lattice and vertex-algebra functors realize the exceptional chain. Equality of the tapes in the fiber product constructs the joint source and preserves complementary information: Archimedean value in the first; incidence, boundary and genealogy in the second. □

Corollary 27.31 (Infinite expansion as a mathematical object)**.** *For each numerical character χ published by the coinductive state, the decimal reading is a complete sequence*

$$\mathrm{Dec}_\chi(x_\infty) = (d_{\chi,1}, d_{\chi,2}, \ldots) \in \{0, \ldots, 9\}^{\mathbb{N}}.$$

Its prefix of length N is the projection $\rho_N \mathrm{Dec}_\chi(x_\infty)$. Thus a thousand, ten thousand or a million digits are cuts of the same infinite object; generation does not consist in successively raising an experimental bound.

Theorem 27.32 (Correlated coinductive generation of π, φ, e and α)**.** *The complete profile of the system $\mathfrak{H}_9$ determines the correlated family of compatible forward sections*

$$\boldsymbol{x}_\infty = (x_\infty^\pi, x_\infty^\varphi, x_\infty^e, x_\infty^\alpha) \in \widehat{\Omega}_\pi \times_{R_{36}} \widehat{\Omega}_\varphi \times_{R_{36}} \widehat{\Omega}_e \times_{R_{36}} \widehat{\Omega}_\alpha, \qquad \widehat{\Omega}_\chi := \varprojlim_{r \geq 36} (\widehat{\mathcal{X}}_r^{\mathrm{enr},\chi}, \mathrm{Ext}_{r+6,r}^{\chi,\rightarrow}),$$

and four correlated characters

$$\big(\Pi_{\mathrm{cl}}, \Pi_{\mathrm{aut}}, \Pi_{\mathrm{prop}}, \Pi_{\mathrm{dod}}\big)(\boldsymbol{x}_\infty) = \big(\pi_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}}\big).$$

The first three characters come, respectively, from the unique closure orbit, the unique self-scaling orbit and the unique propagation orbit of the enriched catalog. The fourth comes from the signed primitive coordinate of the correlated terminal family and its dodecaphase closure. The four coordinates are therefore outputs of the correlated coinductive generation of arbitrary depth governed by the same APP–TRIT–TPK dynamics, with phase return, memory advance and genealogical preservation, even though their publication maps are distinct.

Proof. The exhaustive census $104,976 \to 468 \to 243 \subset \mathbb{F}_3^6$ decomposes the realized image into 43 dihedral orbits. The fiber invariants leave a single orbit in each of the closure, propagation and self-scaling classes; the internal gauge orients in them, without decimal consultation, the representatives

$$w_6^{\mathrm{cl}} = 010211, \qquad w_6^{\mathrm{prop}} = 201101, \qquad w_6^{\mathrm{aut}} = 121200.$$

The forward subrelations preserve the internal character and the predicates that define the three classes; Machin, the factorial recurrence and Perron do not choose any child of Ext. In the closure channel, the five-region composition with unit return produces the ratios $1/5$, $1/239$ and the relation $120/119$; the

Machin identity recognizes exactly the half-turn. In the propagation channel, the normalized semigroup law forces $(n+1)a_{n+1} = a_n$, hence $a_n = 1/n!$ and $P_1 = e$. In the self-scaling channel, the incidence

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

has a unique positive ray and its eigenvalue equation forces $x^2 - x - 1 = 0$, $x > 0$, that is, $x = \varphi$.

In the complete TPK profile, the terminal reading $x_{\mathrm{term}} \mapsto (B_{\mathrm{term}}, U_{12}^{\mathrm{sgn}})$ simultaneously publishes the chart of the three channels and the signed primitive coordinate. The second is reversibly identified with the seal K and its dodecaphase differences. The Euclidean recurrence with carry

$$s_m = P_m + E_m - \Phi_m - K_m + c_{m+1}, \qquad A_m = s_m \bmod 1000, \qquad c_m = (s_m - A_m)/1000$$

produces α_{HMT} without receiving it as input. The independent analytic characterization begins with the finite kernel $E_{21/22}^{(9)}$, which determines a simple root ξ_9 in $(0, 10^{-2})$ without consulting the dodecaphase output. Iteration of the same graded reader produces the inverse system $E_{21/22}^{(6)} \leftarrow E_{21/22}^{(9)} \leftarrow E_{21/22}^{(12)} \leftarrow \cdots$. Its truncations commute with those of the dodecaphase route and both families determine at each depth the same surviving cylinder. Since their diameters tend to zero, the intersection is a singleton and one obtains exactly $\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}$. The equality $\mathrm{Tr}_9(\xi_9^{-1}) = \mathrm{Tr}_9(\alpha_{12}^{-1})$ is only the finite order-nine certificate of that identity of complete sections. Thus, separation of readers preserves the common origin of the four coordinates without introducing a pending prolongation or a target value. □

Proposition 27.33 (Dodecaphase register of the complete state prior to α)**.** *The enriched terminal state has the double publication*

$$x_{\mathrm{term}} \xrightarrow{(\pi_{\mathrm{term}}, R_{\mathrm{sgn}})} (B_{\mathrm{term}}, U),$$

where

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997).$$

Upon ordering the coordinates by the three step-three orbits, the transform

$$\mathcal{H}_{12} = P_{\mathrm{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\mathrm{orb}}, \qquad H_4^2 = 4I_4,$$

has inverse $\mathcal{H}_{12}/4$ on its integral image and determines uniquely

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601).$$

Therefore, within the complete TPK profile,

$$x_{\mathrm{term}} \xrightarrow{R_{\mathrm{sgn}}} U \xrightarrow{\mathcal{H}_{12}^{-1}} K$$

is an exact composition prior to the carry that publishes α. The visible matrix B_{term} and the signed register U are two coordinates of the same state; the second preserves the orientation, opposition, quotient and register fields that the first contracts. The composition

$$\Omega_{\mathrm{seed}} \longrightarrow \mathcal{U}_6^{\mathrm{cat}} \longrightarrow \widehat{\mathcal{X}}_{\mathrm{term}}^{\mathrm{enr}} \xrightarrow{R_{\mathrm{sgn}}} U \xrightarrow{\mathcal{H}_{12}^{-1}} K$$

constructs each coordinate through APP–TRIT–TPK and preserves their registers. Its inputs are the seeds, the sheets and the operator rules; U, K, their differences, α and the published expansions belong to the codomain of successive stages.

Theorem 27.34 (Bidirectional closure of the second route of α)**.** *Prolongation of the vacancy kernel to degrees* 7:9 *determines*

$$B_9(\alpha) := \log_{10}\!\left(\frac{10}{9}\right) + \frac{7}{4}\alpha^2 - \frac{\alpha^4}{20} + \frac{2\alpha^5}{21} + \frac{\alpha^6}{46} + \frac{\alpha^7}{120} - \frac{\alpha^8}{45} - \frac{2\alpha^9}{495}.$$

Upon lifting the rotation to the positive unknown T, the resolvent is equivalent to

$$\boxed{\alpha T^2 - B_9(\alpha)T + \alpha^3 = 0.}$$

With $\xi = \operatorname{arcosh}(B_9(\alpha)/(2\alpha^2))$, the solutions are

$$T_\pm = \alpha e^{\pm\xi}, \qquad T_+T_- = \alpha^2, \qquad J_\alpha(T) = \frac{\alpha^2}{T}, \qquad J_\alpha(T_\pm) = T_\mp.$$

The chamber $6 < T < 7$ selects T_+; the finite closure $\pi_9 = T_+/2$ is prolonged through Γ_9 to π_{HMT}. The identity $\alpha = \sqrt{T_+T_-}$ fixes the geometric mean of the conjugate branches. The jet 7:9*, the simple root, the involution and the return constitute the complete second internal route.*

Theorem 27.35 (Exact coupling of the sector α with the Monster)**.** *In the incidence chart of the same dodecaphase register, the radial roots of A_2^{12} determine*

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \qquad v_\alpha^2 = 54,$$

and the index-three neighbor

$$N_{\alpha,0} = \{x \in N(C_W) : \langle x, v_\alpha\rangle \equiv 0 \pmod 3\}, \qquad N_\alpha = N_{\alpha,0} + \mathbb{Z}\frac{v_\alpha}{3}.$$

The lattice N_α is even, unimodular, of rank twenty-four and rootless, so $N_\alpha \cong \Lambda_{24}$. The chain

$$C_W \longrightarrow N(A_2^{12}) \xrightarrow{v_\alpha} \Lambda_{24} \longrightarrow V^\natural \xrightarrow{\mathrm{Aut}} \mathbb{M}$$

exhibits the exceptional hinge. The scalar α_{HMT} and the vector v_α belong to distinct codomains and share the originating dodecaphase state; this typed correlation is the link to the Monster symmetries.

Proposition 27.36 (Status of the transports L_0, L_1)**.** *The endomorphisms $L_0, L_1 \in \mathrm{End}(\mathbb{F}_3^6)$ and the calendar (L_0, L_0, L_1, L_1) belong to the transport law of the enriched TPK state. The six published full-rank transitions fix each matrix uniquely by $L = X^{-1}Y$ in $\mathbb{F}_3$, and the calendar is the only one of the* 16 *binary calendars that simultaneously preserves the three structural biographies. Its function is to prolong the already selected representatives; Archimedean evaluation and conventional nomenclature are applied afterwards.*

The domain of this proposition is $\mathfrak{H}_9$, the complete APP–TRIT–TPK system with enriched state. The APP census constitutes its initial stage; TRIT provides orientation and TPK constructs prolongation, memory and return. This sequence fixes exactly the domain of each arrow and the location of each output.

Theorem 27.37 (Arbitrary depth and determination of the complete expansions)**.** *For each $\chi \in \{\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}\}$ and each $N \geq 1$, there exists a unique decimal prefix $p_{\chi,N}$ of length N, produced by a finite truncation of the enriched state, such that*

$$\rho_{N+1,N}(p_{\chi,N+1}) = p_{\chi,N}.$$

The compatible family is the family of projections of a unique element

$$p_{\chi,\infty} \in \varprojlim_{N} \{0,\dots,9\}^N \cong \{0,\dots,9\}^{\mathbb{N}},$$

and the intersection of its Archimedean cylinders consists of a unique real number. Consequently, the connection generates the complete expansion; each finite precision is a projection of that infinite output.

Proof. The proof has two stages that cannot be reversed. In the first, entirely prior to the decimal chart, the surviving section of $\mathfrak{H}_9$ publishes three exact characters. The closure character is fixed by the oriented pentafiber, its primitive slope $1/5$, the composition of its four companions $120/119$ and the integer compensator 239; the corresponding Gaussian product ends on the negative real half-line and uniquely fixes the oriented half-turn. The propagation character is the unique formal series normalized by $a_0 = 1$ and $(n+1)a_{n+1} = a_n$. The self-scaling character is the unique positive expansive ray of

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad X^2 - X - 1 = 0.$$

These three conditions are outputs of the enriched state; they contain no decimal prefixes and do not consult conventional values.

Only in the second stage does Archimedean publication act. For each already generated character and each N, let Dec_N be the unique canonical decimal cylinder of diameter 10^{-N} containing its exact realization. This cylinder is a published coordinate of the character: it is not a local fiber of Ext and is not used to resolve it. Uniqueness of the realization and the nonterminal convention imply

$$\rho_{N+1,N}\,\mathrm{Dec}_{N+1}(\chi) = \mathrm{Dec}_N(\chi).$$

The Machin identities, the partial sums of the exponential series and the consecutive Fibonacci–Perron quotients provide certified rational intervals to calculate Dec_N; they do not intervene in Ext, do not choose an orbit or a character and do not select children of its local fibers. They do calculate the publication coordinate block, which remains typed as a subsequent output. The published cylinders are nested and their diameters tend to zero, so that their inverse limit determines a unique point. Since N is arbitrary and the structural extension relation has no terminal depth, the conclusion holds simultaneously for every finite length and for the infinite expansion. □

The executable carrier materially enforces that separation through three disjoint functions: it first generates the exact characters, then evaluates them and only at the end publishes Archimedean blocks. Its own syntactic audit fails if an evaluator enters the generator. The execution of 396 levels, 2376 trits, 43 turns, 387 pairs $(K, K+9)$ per channel and 1000 digits verifies an extensive material instance of the theorem; the run at $10,000$ digits verifies another. Neither quantity limits the depth. The quantifier $\forall N$ and compatibility $\rho_{N+1,N}p_{N+1} = p_N$, together with the existence of the element p_∞ of the inverse limit, prove the infinite decimal output; the finite calculations are reproducible certificates of its projections.

The coordinate α_{HMT} belongs to the same correlated limit family. Its integer route is prolonged through $\mathsf{T}_{\alpha,\infty}$. The internal analytic reader constructs exactly $E^{(9)}_{21/22}$ and its simple root ξ_9 in the order-nine jet quotient. Stability checked to $10,000$ digits certifies remote propagation of the carry of the integer route, but is only a finite projection. The truncation squares of the dodecaphase family and the jet family commute exactly; uniqueness of their compatible sections identifies $\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}$ in $\mathbb{R}_{\mathrm{HMT}}$ before any metrological comparison. Runs at nine, a thousand or ten thousand digits are finite witnesses of this identity, not its scope.

Proposition 27.38 (Complete prolongation of the integer route of α)**.** *Let $B = 1000$, let $(P_k), (E_k), (\Phi_k)$ be the three block sequences published by the coinductive state and let (K_k) be the prolonged dodecaphase sequence. The four series*

$$p = \sum_{k\geq 1} P_k B^{-k}, \qquad e_0 = \sum_{k\geq 1} E_k B^{-k}, \qquad \phi_0 = \sum_{k\geq 1} \Phi_k B^{-k}, \qquad \kappa = \sum_{k\geq 1} K_k B^{-k}$$

converge absolutely and determine

$$\alpha_A = p + e_0 - \phi_0 - \kappa.$$

The carry recurrence is the base-B writing of this equality. For each M, a sufficiently deep truncation determines the first M digits of the canonical expansion of α_A; the tail error is bounded by a constant multiplied by B^{-N}. Consequently, $\mathsf{T}_{\alpha,\infty}$ publishes a complete decimal sequence, and its runs at 3000 *or* 10 000 *digits are finite projections of that output.*

Proof. Each block belongs to a bounded integer set, so the four series are dominated by a geometric series. The partial sum through N differs from the complete sum by $O(B^{-N})$. Euclidean division from right to left transports that tail exactly through the carry; by uniqueness of the canonical expansion, every prefix separated from a cylinder boundary stabilizes as N increases, and at a boundary the canonical nonterminal convention applies. This simultaneously defines all digits, not a sequence of independent numerical targets. □

Corollary 27.39 (Separation between generation and recognition)**.** *The Machin identities, the exponential series, the Perron ray and the decimal chart are exact realizations of characters previously selected by $\mathfrak{H}_9$. The conventional names π, e, φ, α and metrological tables recognize the outputs; they do not determine orbits, characters, transports or carries. The prefixes they calculate are only subsequent coordinate projections of the already generated characters.*

Evaluation of the constructed characters. At the Archimedean evaluation stage, the propagation character determines the initial condition $a_0 = 1$ and the recurrence $(n+1)a_{n+1} = a_n$. Calculation of the terms applies that recurrence. For self-scaling, the row vector evolves through the previously constructed incidence,

$$(x_{n+1}, y_{n+1}) = (x_n, y_n)F_{\mathrm{av}}, \qquad F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \det(tI - F_{\mathrm{av}}) = t^2 - t - 1.$$

The component quotients provide the rational bounds of the self-scaling reader. The implementation verifies agreement between incidence, characteristic polynomial, orientation and normalization before performing this evaluation. Alteration or absence of these data interrupts the calculation instead of silently retaining the previous value. This control links the already produced characters to their evaluators; its place in the construction is after the generator.

27.21 Complete operator inventory and causal pipeline of the Topological Phase Kernel

The Topological Phase Kernel is neither a reduction modulo three nor a clock with one hundred and eight positions. It is a category of enriched states equipped with operators of selection, transport, reading, memory, periodicity, prolongation, and publication. To prevent an abbreviation from replacing the object, each operator is recorded by its type and by the information it preserves.

27.21.1 Typed specification of an operator

Definition 27.40 (Operator specification)**.** A canonical realization of the TPK is a sextuple

$$O = (D_O, C_O, f_O, I_O, L_O, M_O), \tag{27.10}$$

where:

1. $f_O : D_O \to C_O$ is its rule;

2. I_O is the family of conserved relations;
3. L_O declares the information published or lost;
4. M_O enumerates the memory that remains in the enriched state.

Two realizations represent the same operator only when there exists a typed equivalence that intertwines rules and invariants. Sharing a period, cardinality, or visible output is not sufficient.

27.21.2 Eight structural classes

The inventory is partitioned into:

C1. supports and states;

C2. internal selection;

C3. transporte;

C4. readers and quotients;

C5. memory and boundary;

C6. periodicity and return;

C7. prolongation;

C8. salidas.

The class is determined by domain, codomain, and function, not by the name of a number associated with it.

27.21.3 Fourteen functional groupings

Grouping	Operators or spaces	Function and invariants
State and route memory	$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}, \mathcal{F}_q$	Preserve position, sheet, orientation, residue, quotient, carry, signature, boundary, cylinder, prefix, provenance and memory.
Internal selection	$M_{\mathrm{ph}} : \{1,\dots,9\} \to \{-1,0,+1\}$	Activate the additive, multiplicative evaluation or the threshold step; does not move the state.
Transporte espacial	$T_N, T_E, T_S, T_O, F_{\mathrm{loc}}, F_{\mathrm{obs}}$	Perform cardinal translations, sheet feedback, and timeline of two cursors.
Lectores modulares	$\rho_9, q_9, \rho_3^{\mathrm{or}}, c_3, q_{1000}, r_{1000}$	Publish phase, orientation and decimal block preserving the lift quotients.
Block and memory	$\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}, \mathcal{N}_{27}$	The former composes twenty-seven transports; the latter filters memory events. An equal index does not imply an equal operator.
Twelve-phase state	$\mathcal{R}_{12}, C_{12}, \Theta_{108}, \Gamma_{108}$	Distinguish enriched record, sector, cyclic law, and ordered support.
Periodicity and depths	$C_{12} \hookrightarrow C_{108} \twoheadrightarrow C_9, w_{108}, w_{1080}$	Preserves the non-splittable extent, words of different lengths and the vertical memory.
Channels 90/120	$\mathcal{F}_6, \mathcal{F}_8, \mathcal{I}_{6\to 8}, S_{90}, S_{120}$	Separate finite incidences, analytic tails, and their normalizations.
Bidirectionality	$P_{\pm}, \mathrm{rev}, \mathrm{Ext}_{\pm}, N_{\pm}, \beta, C_M$	Perform exchange, route inversion, future and past prolongation, valence, balance, and conjugation.
Emission and coinduction	$L_0, L_1, R_{36}, G_9, \varprojlim \mathrm{Cyl}_m$	Lift the seed, open the boundary, and organize finite and compatible extensions.

Grouping	Operators or spaces	Function and invariants
Numeric Readers	$\mathscr{C}_\pi, \mathscr{P}_e, F_{\mathrm{av}}$	They construct rational forks, factorial propagation, and Fibonacci self-scaling.
Exceptional Projection	Γ_{A_W}, weight, projective support	Transport the same emission to ternary incidence without selecting Archimedean numbers.
Atlas and realization	route signature, atlas $81/56/13/324$, C_M	Classify stories and characters before applying a dimensional chart.
Finite validation	censuses, ablations, hashes, and independence proofs	Checks coverage at the executed depth and separates generator, verifier, and comparison data.

The eight classes describe mathematical nature; the fourteen groupings describe execution function. They are two gradings of the same inventory.

27.21.4 Nominal register of thirty-four entries

$$\mathfrak{D}_{\mathrm{TPK}} := (\mathfrak{o}_1, \ldots, \mathfrak{o}_{34}). \tag{27.11}$$

#	Symbol	Domain and codomain	Rule or function
1	$\mathcal{A}_9$	$(\mathbb{Z}/9)^2$, matrix and internal law	Additive-multiplicative arithmetic cell.
2	(ρ_9, q_9)	$\mathbb{Z}_{>0} \to \{1, \ldots, 9\} \times \mathbb{Z}_{\geq 0}$	Nonadic division with positive representative and quotient.
3	$\rho_3^{\mathrm{or}}, c_3$	$\mathbb{Z} \to \{-1, 0, +1\} \times \mathbb{Z}$	Lift ternary oriented $n = \rho_3^{\mathrm{or}}(n) + 3c_3(n)$.
4	$\iota_{1000} := (q_{1000}, r_{1000})$	$\mathbb{Z} \to \mathbb{Z} \times \{0, \ldots, 999\}$	Euclidean division $N = 1000q_{1000} + r_{1000}$.
5	T_d	$X_9 \to X_9$, $d \in \{N, E, S, O\}$	Cardinal oriented translation.
6	F_{loc}	$Q_{\mathrm{loc}} \times \mathscr{D}_{\mathrm{card}} \to Q_{\mathrm{loc}}$	Local position feedback and sheet.
7	$T_{\min}$	$\Omega_{\mathrm{seed}} \to \mathcal{U}_6$	Minimum emitter of the two cursors, including the coordinate signature.
8	G	$\operatorname{im} s_\Sigma \times \operatorname{im} s_\Pi \to \{0, \ldots, 999\}^6$	Coordinate-by-coordinate decimal coupling.
9	Ψ_6	$\mathcal{U}_6 \to \mathbb{F}_3^6$	Tertiary reduction of the catalog.
10	$\mathcal{R}_3$	$Q_{\mathrm{loc}} \times \mathscr{D}_{\mathrm{card}}^3 \to \{-1, 0, +1\}$	Local three-step observable factor.
11	$\mathcal{K}_{27}$	$Q_{\mathrm{obs}} \to Q_{\mathrm{obs}}$	Composition F_{obs}^{27}.
12	$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}$	enriched states category	Preserve all the typed coordinates of the nucleus.
13	$\mathcal{H}$	relation on enriched states	Hensel Transition that retains residue and quotient.
14	$C_{3,1000}$	crossed states (N, L, D, K, A, B)	Exact update of ternary-decimal cylinders.
15	Σ_{B}	boundary memories $\to$ signature joint	Publishes the block orientation while preserving its ledger.
16	EF–G_9	extension relation	Filter compatible extensions and organize the nonadic genealogy.
17	Γ_9	enriched histories $\longrightarrow$ histories	Holonomy of a phase twist with boundary memory.
18	X_χ	$\varprojlim_m \mathrm{Surv}_m^{(\chi)}$	Global section when non-emptiness, compatibility, and uniqueness are demonstrated in full depth.
19	$\mathcal{P}_{\mathrm{APP}}^{(2)}$	bivariate path category	Transports simultaneously the multiplicative evaluation and the additive accumulation.
20	F_{obs}	$Q_{\mathrm{obs}} \to Q_{\mathrm{obs}}$	Deterministic observable chronology with two cursors.
21	$\mathcal{A}_{42}$	alphabet of forty-two triads	Extended decimal support of finite emission.
22	Ext_r	$\mathrm{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}$	Typed relation of fiber prolongation.
23	$\mathcal{K}_{\mathrm{ph}}$	$\mathcal{U}_6 \times C_9$ about itself	Transfer of phase-compatible explicit continuation.
24	(P, H, Q)	$\mathcal{X}_\times \to \mathcal{X}_\times$	Advance, half-turn, and quarter-turn of seeds.
25	$(c_{\mathrm{mode}}, c_{\mathrm{or}})$	caminos $\to \mathbb{Z}^2$	Mode and orientation change cocycles.
26	E_{20}	finite states $\to$ twenty blocks	Finite triadic prolongation with sourcebook.
27	$\mathcal{R}_{12}$	history enriched $\to$ record of twelve sectors	Oriented temporal system with route, signature, carry, and ledger.
28	R_{sgn}	terminal state $\to U_{12}^{\mathrm{sgn}} \subseteq \mathbb{Z}^{12}$	Reader of the signed harmonic observable.
29	$\mathcal{H}_{12}$	$\mathbb{Z}^{12} \to \mathbb{Z}^{12}$	Transformation by three orbits of step three; inverse $\mathcal{H}_{12}/4$ over the pertinent image.
30	$\mathcal{E}_{\mathrm{dod}} = (D_3, D_4, Q)$	$\mathbb{Z}^{12} \to \mathbb{Z}^{25}$	Transversal extractor set of arities three and four.
31	C_α	blocks and carries $\to$ blocks	Dodecaphasic closure recurrence in base thousand.
32	$(L_{\mathrm{tick}}, \chi_{\mathrm{mass}})$	enriched history $\to$ temporal character	Posterior dimensional interface; does not participate in the numeric constructor of this work.
33	$\mathcal{W}_\pm$	bidirectional lattice of index twelve	Reading in both directions preserving orientation and origin.

#	Symbol	Domain and codomain	Rule or function
34	$\mathcal{A}_{\text{species}}$	extension families → atlas	Posterior realization interface, not a primitive fourth.

27.21.5 Causal update pipeline

Definition 27.41 (Complete canonical state)**.** Let

$$\mathcal{E} := \mathbb{Z}_{\text{quo}} \times \mathbb{Z}_{\text{car}} \times \mathsf{Mem} \times \mathsf{Term} \times \mathsf{Cyl} \times \mathsf{Fr} \times \mathsf{Pref} \times \Lambda.$$

The canonical enriched state at instant t is

$$x_t = (p_t, m_t, \sigma_t, b_t, r_t; q_t, c_t, \eta_t, s_t, C_t, \partial_t, N_t, \lambda_t) \in X_9 \times \{\Sigma, \Pi\} \times \mathsf{TRIT} \times C_{12} \times C_9 \times \mathcal{E}. \tag{27.12}$$

The state abbreviations used in other units are projections of this tuple. In particular, the quotient q_t, carry c_t, terminal s_t, cylinder C_t, boundary ∂_t, prefix N_t, and ledger λ_t are not omitted.

Let ρ_t be the observable arrow determined by x_t. The transport laws of U005 and the transition $C_{3,1000}$ determine a memory endomorphism. For $z = (q, c, \eta, s, C, \partial, N, \lambda) \in \mathcal{E}$, define

$$\begin{aligned} &\mathbf{E}_t : \mathcal{E} \longrightarrow \mathcal{E}, \\ &\mathbf{E}_t(z) = \big(\mathsf{H}(\rho_t)q, \mathsf{C}(\rho_t)c + \kappa(\rho_t), \mathsf{M}(\rho_t)\eta, \mathsf{T}(\rho_t)s, \mathsf{Cyl}(\rho_t)C, \mathsf{Fr}(\rho_t)\partial, \mathsf{Pref}(\rho_t)N, \Lambda(\rho_t)\lambda\big). \end{aligned} \tag{27.13}$$

Each component published in (27.13) has the domain indicated by its fiber; $\kappa(\rho_t)$ is the carry cochain.

We define three endomorphisms of the canonical product. If $\sigma' = M_{\text{ph}}(g(t))$ and $m' = \mu(\sigma', m)$, then

$$\mathsf{Sel}_t(p, m, \sigma, b, r; e) := (p, m', \sigma', b, r; e), \tag{27.14}$$

$$\mathsf{Tra}_t(p, m, \sigma, b, r; e) := (T_{d(t)}p, m, \sigma, b, r; e), \tag{27.15}$$

$$\mathsf{Upd}_t(p, m, \sigma, b, r; e) := (p, m, \sigma, b', r'; \mathbf{E}_t(e)), \tag{27.16}$$

where (b', r') is the target of $\rho_t : (b, r) \to (b', r')$. The causal pipeline is a composition of maps, not the composition of a map with a tritic value:

$$\boxed{\mathcal{U}_t := \mathsf{Upd}_t \circ \mathsf{Tra}_t \circ \mathsf{Sel}_t.} \tag{27.17}$$

Theorem 27.42 (Pipeline closure)**.** *The map $\mathcal{U}_t$ is an endomorphism of the complete canonical state. Readers, prolongations, and projections are applied after $\mathcal{U}_t$ only when their domain has been produced.*

Proof. Sel_t changes only the mode and the TRIT; Tra_t changes only the APP position; and Upd_t applies the typed product (27.13) and the phase arrow. All three codomains are the same product of definition 27.41; their composition is therefore defined and preserves all state coordinates. □

27.21.6 Non-identification rules

The inventory imposes

$$\mathcal{S}_4 \neq D_4 \neq \text{sd}_4, \qquad \mathcal{R}_{12} \neq C_{12} \neq \Theta_{108} \neq \Gamma_{108}, \qquad F_{\text{obs}} \neq \mathcal{K}_{\text{ph}}, \tag{27.18}$$

and

$$U_{12}^{\text{cat}} \neq U_{12}^{\text{sgn}}, \qquad \mathcal{E}_{\text{dod}} \neq D_{\text{raw}}^{90,120} \neq \mathcal{K}_{6\to 8}. \tag{27.19}$$

They are type incompatibilities, not mere value inequalities.

Comparison Criterion 27.43 (Audit of operatorial completeness)**.** An execution is operatorially complete only if:

1. declares the inputs of $\mathfrak{O}_{\mathrm{TPK}}$ used;
2. is recorded the domain and the codomain of each composition;
3. preserves the causal order of (27.17);
4. distinguishes dynamics, transfer, reading, prolongation, and output;
5. permits reconstruction from λ_t of every block change and every carry;
6. does not promote a finite check to an infinite section without proving non-emptiness, compatibility, and uniqueness at every depth.

Any symbol used without a type, or any fusion prohibited by (27.18) and (27.19), constitutes a verifiable failure.

27.22 Integral dodecaphasic register, cyclic extension, and decomposition 1 + 5 + 6

The observable period of one hundred eight positions contains twelve sectors of nine positions. This equality of cardinalities does not suffice to describe the temporal state: the ordered support, cyclic law, sector index, and enriched register of transports, orientations, carries, and incidences must be distinguished. This unit constructs the four objects and proves the integral reconstruction of the twelve memory coordinates.

27.22.1 Cyclic coordinate on the time-oriented 12-sector support

For $m \in \{1, \dots, 12\}$, define

$$E_m := \{9(m-1)+1, \dots, 9m\}. \tag{27.20}$$

The ordered support and the cyclic group are

$$\Gamma_{108} := E_1 \sqcup \cdots \sqcup E_{12}, \qquad \Theta_{108} := \mathbb{Z}/108\mathbb{Z}. \tag{27.21}$$

Γ_{108} preserves position, order and pertenencia to block; Θ_{108} preserves the law of translation. No are the same object.

For $\theta \in \Theta_{108}$, let $\widetilde{\theta} \in \{0, \dots, 107\}$ be its representative. We define

$$\beta(\theta) := \left[\left\lfloor \frac{\widetilde{\theta}}{9} \right\rfloor\right]_{12}, \qquad r(\theta) := 1 + (\widetilde{\theta} \bmod 9). \tag{27.22}$$

The first coordinate locates the duodecaphase sector; the second publishes the positive position $1, \dots, 9$ within the sector.

27.22.2 nonsplit extension $\mathbf{0}\to\mathbf{C}_{12}\to\mathbf{C}_{108}\to\mathbf{C}_{9}\to\mathbf{0}$ and carry cocycle

Let $C_n = \mathbb{Z}/n\mathbb{Z}$. The applications

$$\iota : C_{12} \longrightarrow C_{108}, \quad \iota([b]_{12}) = [9b]_{108}, \qquad p : C_{108} \longrightarrow C_9, \quad p([\theta]_{108}) = [\theta]_9 \tag{27.23}$$

form the sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0. \tag{27.24}$$

For the set-theoretic section

$$s([r]_9) := [\widetilde{r}]_{108}, \qquad 0 \le \widetilde{r} < 9,$$

the defect of additivity is

$$c(r,r') := \left[\left\lfloor \frac{\widetilde{r}+\widetilde{r}'}{9} \right\rfloor\right]_{12}, \qquad s(r)+s(r') = s(r+r') + \iota(c(r,r')). \tag{27.25}$$

Theorem 27.44 (Accuracy, non-division, and cohomological class)**.** *The sucesion* (27.24) *is exacta and not it escinde. For the action trivial,*

$$H^2(C_9, C_{12}) \simeq C_{12}/9C_{12} \simeq C_3, \tag{27.26}$$

and the cocycle (27.25) *represents its generating class.*

Proof. The image of ι is the subgroup of multiples of nine, which coincides with $\ker p$; p is surjective. This proves exactness. If the extension were split, C_{108} would be isomorphic to $C_{12} \times C_9$, but

$$\exp(C_{108}) = 108, \qquad \exp(C_{12} \times C_9) = \operatorname{mcm}(12,9) = 36.$$

The cocycle identity is the Euclidean division of $\widetilde{r}+\widetilde{r}'$ by nine. For a cyclic group with trivial action, $H^2(C_9, C_{12}) \simeq C_{12}/9C_{12}$; the unit carry projects to $[1]$, which generates the quotient of order three. □

The non-splitting typifies the return:

$$g_0(\theta+9) = g_0(\theta), \qquad \beta(\theta+9) = \beta(\theta)+1 \pmod{12}.$$

Nine iterations close the visible phase and shift one sector; one hundred eight close both coordinates of the clock. The enriched state further preserves the vertical memory, cylinder, boundary, and genealogy.

27.22.3 Enriched temporal register

Definition 27.45 (Oriented dodecaphonic register)**.** For an enriched history, let τ_m be the ordered subroute on E_m, σ_m its orientation, κ_m the carry crossing the block boundary, and Λ_m its local incidence ledger. The complete record is

$$\mathcal{R}_{12} := \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=1}^{12}. \tag{27.27}$$

Thus four levels are distinguished:

$$\boxed{\mathcal{R}_{12} \neq C_{12} \neq \Theta_{108} \neq \Gamma_{108}.} \tag{27.28}$$

The first is an enriched register; the second, a sector index; the third, a group of translations; the fourth, an ordered support. There exist projections between them, but no identifications.

Let $q_{\mathrm{mem}} \in \mathbb{Z}_{\geq 0}$ be the unbounded number of nonadic turns. The duodecaphase coordinate is only its residue

$$\beta = [q_{\mathrm{mem}}]_{12}.$$

After one hundred and eight steps,

$$\beta \longmapsto \beta, \qquad q_{\mathrm{mem}} \longmapsto q_{\mathrm{mem}} + 12.$$

The calendar returns; the vertical memory is not reset.

27.22.4 Extraction of the signed dodecaphase state from the enriched time record

Let $(d_t)_{t=1}^{108}$ be the digital record on Γ_{108}. We define

$$\mathcal{D}_{\mathrm{evt}} := \{2,4,6,8\}, \qquad \Sigma_{12} := (+,+,+,-,-,-,+,+,+,-,-,-). \tag{27.29}$$

For $m = 0,\dots,11$, let

$$e_m := \Sigma_{12}(m+1)\,\#\{t \in E_{m+1} : d_t \in \mathcal{D}_{\mathrm{evt}}\} \in \mathbb{Z}. \tag{27.30}$$

The application preserves the local census and the sectorial orientation as separate factors.

We pair the two semicrowns by means of

$$p_i := e_i + e_{i+6}, \qquad a_i := e_i - e_{i+6}, \qquad 0 \le i \le 5. \tag{27.31}$$

We define the total census, the five differences with respect to the sixth pair, and the six antisymmetries:

$$s_C := \sum_{i=0}^{5} p_i, \qquad M_i := p_i - p_5 \quad (0 \le i \le 4), \qquad A_6 := (a_0,\dots,a_5). \tag{27.32}$$

Definition 27.46 (Integral reader $1+5+6$)**.** The linear reader is

$$\mathcal{L}_{12} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{12}, \qquad \mathcal{L}_{12}(e) := (s_C, M_0,\dots,M_4, a_0,\dots,a_5). \tag{27.33}$$

The coordinate distribution is

$$\underbrace{1}_{s_C} + \underbrace{5}_{M_0,\dots,M_4} + \underbrace{6}_{a_0,\dots,a_5} = 12. \tag{27.34}$$

Theorem 27.47 (Exact reconstruction and integral image)**.** *The reader $\mathcal{L}_{12}$ is injective. On its image,*

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \qquad p_i = p_5 + M_i \quad (0 \le i \le 4), \tag{27.35}$$

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2} \quad (0 \le i \le 5). \tag{27.36}$$

An output tuple belongs to $\operatorname{im}\mathcal{L}_{12}$ *if and only if*

$$s_C - \sum_{i=0}^{4} M_i \equiv 0 \pmod 6 \tag{27.37}$$

and, for the reconstructed p_i,

$$p_i \equiv a_i \pmod 2 \qquad (0 \le i \le 5). \tag{27.38}$$

Proof. From $M_i = p_i - p_5$ is obtained

$$s_C = \sum_{i=0}^{4}(p_5 + M_i) + p_5 = 6p_5 + \sum_{i=0}^{4} M_i,$$

which gives the first equation of (27.35); the others are immediate. Adding and subtracting $p_i = e_i + e_{i+6}$ and $a_i = e_i - e_{i+6}$ produces (27.36). Divisibility by six and the six parity congruences are precisely the necessary and sufficient conditions for all reconstructed coordinates to be integers. □

Division by six appears after the oriented census, not during it. The integrality conditions for this unit are exactly (27.37) and (27.38); no Smith normal form is attributed here to a matrix that has not previously been defined.

27.22.5 Two dodecaphasic words of distinct origins

The distinction must be preserved.

$$U_{12}^{\mathrm{cat}} := U_6 \| U_6, \qquad U_{12}^{\mathrm{sgn}} := \mathcal{H}_{12}(K). \tag{27.39}$$

The first concatenates two catalog emissions of length six. The second is a signed coordinate reversibly linked to K by the operator $\mathcal{H}_{12}$. Their common length does not imply the same provenance, semantics, or transport law.

Comparison Criterion 27.48 (Falsification criteria)**.** The construction is refuted if the sequence (27.24) is not exact, if its cohomology class is trivial, if a tuple in the image violates (27.37) or (27.38), if the inverse formulas do not recover the twelve initial integers, or if the return of β is interpreted as a reset of q_{mem}.

27.23 Signed dodecaphasic register and integral reconstruction of the seal

The construction of the integer seal does not begin with α, its inverse or a metrological table. Its domain is the enriched terminal state of the Topological Phase Kernel obtained after the already constructed chain

$$\boxed{\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \Omega_{\mathrm{seed}} \xrightarrow{T_{\min}} \mathcal{U}_6^{\mathrm{cat}} \xrightarrow{\rho_3} W_{243} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9 \longrightarrow x_{\mathrm{term}}.} \tag{27.40}$$

The foregoing arrows were defined with their respective domains: pairs of seeds, signatures, emissions, ternary words, boundaries, compatible histories, and nonadic holonomy. This unit does not reselect any of those data. It constructs the integral charts of the same terminal state and proves their equivalence.

27.23.1 Windows of the oriented trace of length one hundred eight

Let

$$\Theta_{108} = \mathbb{Z}/108\mathbb{Z}$$

the observable calendar. Its twelve nonadic windows are

$$E_m = \{9m + 1, \dots, 9m + 9\}, \qquad 0 \le m < 12.$$

The partition is disjoint and exhaustive:

$$\{1, \dots, 108\} = \bigsqcup_{m=0}^{11} E_m.$$

The set of address codes that publish an event is fixed as

$$D_{\mathrm{evt}} = \{2, 4, 6, 8\}.$$

The orientation of the twelve windows is

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -). \tag{27.41}$$

If d_t is the direction encoded at step t, the signed event of window m is

$$e_m = \Sigma_{12}(m)\, \#\{t \in E_m : d_t \in D_{\mathrm{evt}}\}. \tag{27.42}$$

The formula distinguishes the nonnegative number of events from the sign of the orientation. It does not replace a cardinal direction with a sign: both coordinates remain present in the TPK state.

Definition 27.49 (Enhanced dodecaphonic registration)**.** The record transported by the trace is

$$\mathcal{R}_{12}(x) = \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=0}^{11},$$

where τ_m is the TRIT orientation, σ_m the boundary signature, κ_m the carry, and Λ_m the ledger segment preserving the route. There is a chain of projections

$$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow \mathcal{R}_{12} \longrightarrow \Theta_{108},$$

but neither of the two arrows is an identification.

In particular, Θ_{108} preserves only the position in the calendar. The record $\mathcal{R}_{12}$ additionally preserves orientation, signature, carry, and provenance. The complete state further preserves cylinder, boundary, quotient, terminal, and vertical memory.

27.23.2 Integral chart of the dodecaphasic opposition

For $0 \le i < 6$, the opposite positions determine

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}. \tag{27.43}$$

The central charge and the five relative margins are

$$s_C = \sum_{i=0}^{5} p_i, \qquad M_i = p_i - p_5 \quad (0 \le i < 5). \tag{27.44}$$

Definition 27.50 (Integral chart $1 + 5 + 6$)**.** It is defined

$$\mathcal{L}_{12}(e_0, \dots, e_{11}) = \big(s_C, M_0, \dots, M_4, a_0, \dots, a_5\big). \tag{27.45}$$

The coordinate partition is

$$1 + 5 + 6 = 12.$$

Theorem 27.51 (Exact inverse of the integral chart)**.** *A tuple*

$$\ell = (s_C, M_0, \dots, M_4, a_0, \dots, a_5) \in \mathbb{Z}^{12}$$

belongs to the image of $\mathcal{L}_{12}$ if and only if

$$s_C - \sum_{i=0}^{4} M_i \equiv 0 \pmod 6, \tag{27.46}$$

$$p_i \equiv a_i \pmod 2 \quad (0 \le i < 6), \tag{27.47}$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \tag{27.48}$$

$$p_i = p_5 + M_i \quad (0 \le i < 5). \tag{27.49}$$

In that case, the only entire preimage is

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2}. \tag{27.50}$$

Proof. Summing $M_i = p_i - p_5$ for $0 \le i < 5$, we obtain

$$\sum_{i=0}^{4} M_i = \sum_{i=0}^{4} p_i - 5p_5 = s_C - 6p_5,$$

which forces (27.48) and the congruence modulo six. Once the six p_i have been reconstructed, the system

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}$$

has the unique solution (27.50). This solution is integral exactly when p_i and a_i have the same parity. The two families of congruences are therefore necessary and sufficient. □

This theorem precedes the numerical closure of α. It explains how the oriented trace produces a reversible integral chart, and why preserving the reduced calendar is not enough.

27.23.3 Selection of the common terminal state

The three compatible sections reach terminal depth with catalogs of cardinalities

$$|\mathcal{T}_\pi| = 196, \qquad |\mathcal{T}_e| = 197, \qquad |\mathcal{T}_\varphi| = 196.$$

The product contains

$$196 \cdot 197 \cdot 196 = 7\,567\,952 \tag{27.51}$$

ordered triples.

The column margin of the terminal reader is

$$q_\partial = (1, 2, 2, 1, 2, 0) \in \mathbb{F}_3^6,$$

and the row orientation of the combined channel is

$$\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3.$$

The first datum acts on columns; the second acts on rows. Neither is deducible from the other.

The local signatures of Gram matrix, norm, weight, and distance reduce (27.51) to 331 triples. The margin q_∂ leaves two indices:

$$(120, 197, 157), \qquad (129, 197, 148).$$

Their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \qquad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row sums are

$$(0, 2, 0), \qquad (2, 2, 1).$$

Since

$$\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\},$$

only B_1 satisfies the axial condition.

Theorem 27.52 (Uniqueness of the Visible Terminal Publication)**.** *The joint selection determines*

$$B_{\mathrm{term}} = \begin{pmatrix} 021020 \\ 001220 \\ 100210 \end{pmatrix}, \tag{27.52}$$

corresponding to the triple $(129, 197, 148)$. *No decimal evaluation of* α *enters the census or the two selection conditions.*

Proof. The census and margin leave exactly B_0, B_1. The charge of B_0 does not belong to $\langle\sigma\rangle$, whereas that of B_1 is 2σ. The second candidate is therefore the only one that simultaneously satisfies both typed requirements. □

27.23.4 Two publications of the same state, not a conversion between them

Let

$$x_{\mathrm{term}} \in \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}} \subseteq \mathcal{X}_{\mathrm{TPK}}^{[108]}.$$

The visible reading is

$$\pi_{\mathrm{term}}(x_{\mathrm{term}}) = B_{\mathrm{term}}.$$

The signed chart preserves another coordinate of the same state.

Definition 27.53 (Signed reader produced by the terminal register)**.** The reader's input state is exclusively the enriched terminal state already produced by the causal pipeline $\mathcal{U}_t = \mathsf{Upd}_t \circ \mathsf{Tra}_t \circ \mathsf{Sel}_t$:

$$x_{\mathrm{term}} = \big(\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1\big)(x_{\mathrm{seed}}) \in \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}}. \tag{27.53}$$

Composing the oriented register with its channel readers $90/120$ extracts from ledger λ_{108}, without adding external coordinates,

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\mathrm{term}})) = \big(b^{90}, b^{120}, Q_{\mathrm{TPK}}\big), \tag{27.54}$$

where

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ Q_{\mathrm{TPK}} &= 6263. \end{aligned}$$

These twenty-five coordinates are outputs of the register of the definition 27.49, not data added to the domain.

On a compatible output (b^{90}, b^{120}, Q), we define the integral harmonic reader Π_H explicitly. For $r = 1, 2, 3$, let

$$B_r = \sum_{j=0}^{3} b_{r+3j}^{120}, \quad A_1 = \frac{Q + 2B_1 + B_2}{3}, \quad A_2 = A_1 - B_1, \quad A_3 = A_2 - B_2, \tag{27.55}$$

and $d_{r,j} = b_{r+3j}^{90}$. The orbital block r is

$$\Pi_H^{(r)}(b^{90}, b^{120}, Q) = \big(A_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}\big). \tag{27.56}$$

Interleaving the three blocks by columns yields Π_H, and the signed reading is constructed as the composition

$$\boxed{R_{\mathrm{sgn}} := \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}} \longrightarrow \mathcal{U}_{12}^{\mathrm{sgn}} := \operatorname{im} R_{\mathrm{sgn}} \subseteq \mathbb{Z}^{12}.} \tag{27.57}$$

Neither U nor K belongs to the domain: U is the output of (27.57), and K is subsequently reconstructed through the integral Hadamard inverse.

The correct joint writing is

$$x_{\mathrm{term}} \xrightarrow{(\pi_{\mathrm{term}}, R_{\mathrm{sgn}})} (B_{\mathrm{term}}, U_{12}^{\mathrm{sgn}}). \tag{27.58}$$

No arrow $B_{\mathrm{term}} \to U_{12}^{\mathrm{sgn}}$ is interposed: the visible matrix has forgotten precisely the orientation, opposition, quotient, carry, and ledger coordinates used by the signed chart. Both publications originate in the enriched state produced by (27.40).

The evaluation of the signed record gives

$$\boxed{U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997).} \tag{27.59}$$

27.23.5 Naturality with respect to depth

For $N' \geq N \geq 108$, the truncation

$$\tau_{N',N} : \mathcal{X}_{\mathrm{TPK}}^{[N'],\mathrm{enr}} \longrightarrow \mathcal{X}_{\mathrm{TPK}}^{[N],\mathrm{enr}} \tag{27.60}$$

acts on the enriched history and its ledger, not on a pair (K, U) added to the state.

Theorem 27.54 (Square of naturality of the signed reading)**.** *We have*

$$\boxed{R_{\mathrm{sgn},N} \circ \tau_{N',N} = R_{\mathrm{sgn},N'}.} \tag{27.61}$$

Proof. Each updater $\mathcal{U}_t$ adds the next incidence to the ledger without rewriting the one hundred eight windows already generated. Therefore $\mathcal{R}_{12}$, $\mathcal{E}_{108}^{90,120}$, and Π_H read the same coordinates when truncating any subsequent prolongation. Commutativity is an equality of maps on produced states, not the tautological preservation of an output included in the input. □

27.23.6 Step-three orbits and Hadamard transform

The cyclic order of the seal is distributed over the three orbits.

$$K^{(1)} = (K_1, K_4, K_7, K_{10}), \tag{27.62}$$

$$K^{(2)} = (K_2, K_5, K_8, K_{11}), \tag{27.63}$$

$$K^{(3)} = (K_3, K_6, K_9, K_{12}). \tag{27.64}$$

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{27.65}$$

If P_{orb} reorders the twelve coordinates according to (27.62)–(27.64), the global transformation is

$$\mathcal{H}_{12} = P_{\mathrm{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\mathrm{orb}}. \tag{27.66}$$

The three blocks of (27.59), read by orbits, are

$$U^{(1)} = (2378, -452, -668, -322),$$
$$U^{(2)} = (1406, 998, -204, -28),$$
$$U^{(3)} = (2479, -551, -371, -997).$$

Lemma 27.55 (Algebraic Hadamard quarter-turn)**.** *The matrix H_4 satisfies*

$$H_4^2 = 4I_4, \qquad H_4^{-1} = \frac{1}{4}H_4, \qquad \det H_4 = -16. \tag{27.67}$$

Proof. The rows of H_4 are orthogonal, and each has squared norm four. Since the matrix is symmetric, $H_4^{\mathsf{T}}H_4 = H_4^2 = 4I_4$. The formula for the inverse and the determinant follow immediately. □

Theorem 27.56 (Unique full reconstruction of the seal)**.** *The inverse*

$$K^{(r)} = \frac{1}{4}H_4U^{(r)} \qquad (r = 1, 2, 3) \tag{27.68}$$

yields

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 058, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

By undoing the orbital order one obtains

$$\boxed{K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601).} \tag{27.69}$$

It is the unique rational preimage of U, and it belongs to $\mathbb{Z}^{12}$.

Proof. By (27.67), the rational preimage is unique. The explicit multiplications are

$$\begin{aligned} H_4U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4U^{(2)} &= (2172, 2636, 232, 584), \\ H_4U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and reinterleaving yields (27.69). □

The factor $1/4$ is not chosen to fit an output: it is the inverse forced by $H_4^2 = 4I_4$. Nor is it the square root of -1. The internal complex structure $A_W^2 = -I$ will appear after constructing the Paley matrix; these are two facts of different types.

27.23.7 Integral structure and Smith normal form

Theorem 27.57 (Integral image of H_4)**.** *The Smith normal form is*

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4). \tag{27.70}$$

Consequently,

$$\operatorname{coker} \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3, \tag{27.71}$$

and the image of $\mathcal{H}_{12}$ has index

$$16^3 = 4096$$

in $\mathbb{Z}^{12}$. For $u \in \mathbb{Z}^4$,

$$u \in H_4\mathbb{Z}^4 \quad \Longleftrightarrow \quad H_4u \equiv 0 \pmod 4. \tag{27.72}$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has absolute value two; those of order three are multiples of four, and one has absolute value four; the determinant has modulus sixteen. The determinantal divisors are $1, 2, 4, 16$, yielding the invariant factors

$1, 2, 2, 4$. The triple direct sum repeats the factors and multiplies the indices. Finally, $H_4^{-1} = H_4/4$, so the preimage is integral exactly under the congruence (27.72). □

27.23.8 Certification in the second coordinate system: step-three and step-four differences

With cyclic indices modulo twelve, we define

$$(D_3K)_m = K_m - K_{m+3}, \qquad (D_4K)_m = K_m - K_{m+4}, \qquad Q(K) = \sum_{m=1}^{12} K_m. \tag{27.73}$$

Applied to the seal already reconstructed by Hadamard, these operators must recover exactly the channel coordinates produced by the TPK register before U. For (27.69),

$$\begin{aligned} D_3K &= (-495, -116, -684, 108, 601, -90,\\ &\qquad -173, -88, 313, 560, -397, 461),\\ D_4K &= (-425, -281, -481, 671, -255, 30,\\ &\qquad 475, -543, 680, 251, 6, -128),\\ Q(K) &= 6263. \end{aligned}$$

Comparison with (27.54) proves the equality of outputs

$$\boxed{D_3K = b^{90}, \qquad D_4K = b^{120}, \qquad Q(K) = Q_{\mathrm{TPK}}.} \tag{27.74}$$

The differential system certifies and reconstructs the previous register; it does not select K, U, or any constant.

Theorem 27.58 (Completeness of the transversal extractor)**.** *The operator*

$$\mathcal{E}_{\mathrm{dod}} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and Smith normal form

$$\mathrm{SNF}(\mathcal{E}_{\mathrm{dod}}) = \mathrm{diag}(\underbrace{1, \dots, 1}_{11}, 12). \tag{27.75}$$

In particular, $(D_3K, D_4K, Q(K))$ *determina a single* K.

Proof. If $D_3v = D_4v = 0$, then v is constant along steps three and four. Since $\gcd(3, 4) = 1$, those steps generate all of $\mathbb{Z}/12\mathbb{Z}$, so that

$$\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle.$$

The total charge does not vanish on that line, since $Q(\mathbf{1}) = 12$. The rank is twelve. The rows of (D_3, D_4) are oriented incidences of a connected Cayley graph; a spanning tree supplies eleven unit invariant factors. Every maximal minor must contain the row Q, and the unimodular operations reduce its final contribution to twelve. There is a maximal minor of modulus twelve; therefore, the last factor is exactly twelve. □

The reconstruction of U from these coordinates does not need to multiply again by H_4. For $r = 1, 2, 3$, let

$$B_r = \sum_{j=0}^{3} (D_4K)_{r+3j}. \tag{27.76}$$

Then

$$(B_1, B_2, B_3) = (972, -1073, 101).$$

Orbital sums are

$$A_1 = \frac{Q + 2B_1 + B_2}{3} = 2378, \tag{27.77}$$

$$A_2 = A_1 - B_1 = 1406, \tag{27.78}$$

$$A_3 = A_2 - B_2 = 2479. \tag{27.79}$$

If $d_{r,j} = (D_3K)_{r+3j}$, the signed block of orbit r is

$$\big(A_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}\big), \tag{27.80}$$

that reproduces the three blocks $U^{(r)}$.

27.23.9 Threshold Tetrad of the Seal

The local decimal completion satisfies

$$1000 = 729 + 271.$$

The support of the seal coordinates that reach the threshold 729 is

$$\boxed{B_K := \{m : K_m \geq 729\} = \{4, 6, 9, 10\}.} \tag{27.81}$$

This tetrad is obtained after constructing K. In this unit it is not yet called a Witt star: that object will be defined only after constructing the ternary code, its hexads, and the Steiner system.

27.23.10 Causal independence with respect to the physical output

Theorem 27.59 (Isolation of U and K)**.** *The composition*

$$\Omega_{\text{seed}} \longrightarrow x_{\text{term}} \xrightarrow{\mathcal{R}_{12}} \mathcal{R}_{12}(x_{\text{term}}) \xrightarrow{\mathcal{E}_{108}^{90,120}} (b^{90}, b^{120}, Q_{\text{TPK}}) \xrightarrow{\Pi_H} U \xrightarrow{\mathcal{H}_{12}^{-1}} K \tag{27.82}$$

does not receive α, α^{-1}, CODATA or a target decimal string as input. The seal K is determined before defining the recurrence that will produce α.

Proof. The first part of the composition uses only seeds, emissions, ternary readers, operators L_0, L_1, survival relations, phase transport, and TPK state charts. Register $\mathcal{R}_{12}$ and its channel reader produce $(b^{90}, b^{120}, Q_{\text{TPK}})$; formulas (27.55) and (27.56) produce U; only then does the linear transformation $\mathcal{H}_{12}^{-1} = \mathcal{H}_{12}/4$ act on its integral image. None of the formulas contains a physical variable or a metrological table. The base-thousand recurrence is introduced only in the next unit, after K has been fixed. □

27.23.11 Refutation criteria

Comparison Criterion 27.60 (Refutation of the register and the seal)**.** The construction is refuted if any of the following facts is established:

1. the windows E_m do not partition the one hundred eight steps;
2. the chart $\mathcal{L}_{12}$ fails its inverse or its congruences;
3. the terminal census does not produce 331 triples, two matrices and a unique axial charge;
4. an attempt is made to obtain U from B_{term} after forgetting the signed coordinates;
5. the naturality square (27.61) fails;

6. $H_4^2 \neq 4I_4$, some preimage is not integral or the reconstructed vector differs from (27.69);
7. the Smith normal form differs from (27.70);
8. extractor (D_3, D_4, Q) loses rank or fails to reproduce U;
9. B_K is introduced before constructing K, or is assigned a Witt structure before defining W_{12};
10. a digit of α or a metrological reference intervenes in any of the arrows of (27.82).

Determination of the correlated analytic extension

The complete construction of the analytic chart fixes the following dependency order. The enriched state s first expresses the pre-coordinate

$$a(\mathsf{s}) = p(\mathsf{s}) + e_0(\mathsf{s}) - f(\mathsf{s}) - \kappa_{\mathrm{per}}(\mathsf{s}),$$

where $p = \pi_{\mathrm{HMT}} - 3$, $e_0 = e_{\mathrm{HMT}} - 2$ and $f = \varphi_{\mathrm{HMT}} - 1$. The three regional coordinates and the periodic reading of the register belong to the same generative development. Integer normalization and the analytic chart realize two correlated expressions of this pre-coordinate.

With $q = 1/729$ and $E_{9,\mathsf{s}}$ the ninth-order jet, the linking coefficient is

$$\lambda(\mathsf{s}) = -\frac{E_{9,\mathsf{s}}(a(\mathsf{s}))\big(1 - q\,a(\mathsf{s})^3\big)}{a(\mathsf{s})^{10}}.$$

The complete extension is then determined by

$$E_{\mathsf{s},\infty}(X) = E_{9,\mathsf{s}}(X) + \lambda(\mathsf{s})\frac{X^{10}}{1 - X^3/729},$$

or, equivalently, by the recurrence of all its coefficients:

$$b_{10+3n} = 729^{-n}\lambda(\mathsf{s}), \qquad b_{11+3n} = b_{12+3n} = 0 \qquad (n \geq 0).$$

Uniqueness of λ has as its domain the class of extensions $E_{9,\mathsf{s}}(X) + \mu X^{10}/(1 - qX^3)$: the condition of vanishing at $a(\mathsf{s})$ determines $\mu = \lambda(\mathsf{s})$ exactly. Naturality of truncations, the geometric remainder and derivative bounds then establish convergence and the unique simple root. This root coincides with the integer expression at every depth.

This is the precise status of the double realization: identity of the arithmetic and analytic expressions of the same state, with explicit dependence on the generated pre-coordinate and the geometric extension class. The construction is developed in full in Section 11.6.4.2, with its proofs of naturality, convergence and compatibility.

27.24 Coinductive boundary and stabilization of compatible extensions

The initial emission has image $\mathcal{W}_6$ of cardinality 243 within the space of 729 six-trit words. Its extensions preserve the enriched state: word, position, direction, sheet, regime, quotients, carry, boundary, incidences and chronological register. For $r \geq 1$, denote by $\widehat{\mathcal{W}}_{6r}$ the set of admissible states of length $6r$ actually obtained through TPK transitions, with their compatible prefixes and survival coordinate. The word projection takes values in $(\mathbb{F}_3^6)^r$.

Truncation $t_{r+1,r}$ retains the enriched prefix of length $6r$. The extension relation is

$$\mathcal{P}_r = \{(x, y) : x \in \widehat{\mathcal{W}}_{6r},\ y \in \widehat{\mathcal{W}}_{6(r+1)},\ t_{r+1,r}(y) = x\}.$$

The relations use admissible states produced by the TPK; equality of prefixes expresses their compatibility. For $s > r$, write $t_{s,r} = t_{r+1,r} \circ \cdots \circ t_{s,s-1}$. Preservation of fields requires $t_{u,r} = t_{s,r} t_{u,s}$ when $u > s > r$.

Proposition 27.61 (Restitution of the antecedent by truncation)**.** *If p_r is an extension section whose graph is contained in $\mathcal{P}_r$, then $t_{r+1,r} p_r = I$. For a sequence of such sections, composing their truncations recovers the full antecedent state.*

Proof. Graph inclusion means, for every x in the section's domain, $t_{r+1,r}(p_r(x)) = x$. When two sections are composed, the deeper pair cancels first, followed by the preceding pair. Induction proves the formula for any finite number of stages. Equality concerns enriched states, hence preserves all their fields and prefix relations. □

27.24.1 Candidates at the first boundary and opening relation

Let $\widetilde{\mathcal{R}}_{36}$ be the set of admissible states of depth 36 with defined boundary register and edge memory. For $x \in \widetilde{\mathcal{R}}_{36}$, the set $\mathrm{Cand}_6(x)$ consists of states y of depth 42 obtained by a typed trajectory

$$x = x_{36} \xrightarrow{\mathcal{U}_{36}} x_{37} \xrightarrow{\mathcal{U}_{37}} \cdots \xrightarrow{\mathcal{U}_{41}} x_{42} = y, \qquad \tau_{42,36}(y) = x.$$

Each candidate's visible tail has six trits, so its word projection has at most 729 values. The cardinality of this projection is distinct from the cardinality of states with complete registers.

The predicate $g_{9,x}(y)$ requires all prefixes to be compatible and admissible continuations to exist at arbitrary depth. The effective boundary and its opening are defined by

$$\mathcal{R}_{36} = \{x \in \widetilde{\mathcal{R}}_{36} : \exists y \in \mathrm{Cand}_6(x),\ g_{9,x}(y) = 1\},$$
$$G_9 = \{(x, y) : x \in \mathcal{R}_{36},\ y \in \mathrm{Cand}_6(x),\ g_{9,x}(y) = 1\}.$$

The construction preserves the truncation square $\tau_{42,36}(y) = x$ and gives

$$G_9 \subseteq \mathcal{U}_{41} \circ \cdots \circ \mathcal{U}_{36} \subseteq \widehat{\mathcal{X}}_{36} \times \widehat{\mathcal{X}}_{42}.$$

Each element of $\mathcal{R}_{36}$ has a surviving candidate by definition. The relation G_9 opens a six-trit block; the holonomy Γ_9 composes nine elementary transitions. The first object is a relation between depth levels, and the second is the calendar return with its memory increment.

27.24.2 Existence of a complete history

Theorem 27.62 (Stabilization with memory preservation)**.** *Suppose the admissible prefixes arise from a finite nonempty set of initial conditions and form a finitely branching system with states at every depth, whose truncations satisfy the preceding compatibility and preserve the state fields. Suppose also that nonadic transport is natural with respect to truncation and that the return identities $g_{k+9} = g_k$, $m_{k+9} = m_k + 1$ hold on admitted trajectories. Then a compatible infinite history exists. Along it, each return preserves phase and increases integer memory by one unit.*

Proof. Each prefix is organized by length and linked to its immediate extensions. If there are several initial states, a formal root with that finite set of successors is added. Non-emptiness at every depth provides arbitrarily long extensions from this root.

A vertex admitting arbitrarily long extensions has at least one successor with the same property. Indeed, if each of its finitely many successors had a finite depth bound, the maximum of those bounds, increased by one, would bound all extensions of the vertex. This contradicts the initial property. A successor with unbounded extension is chosen successively; the finite ordering of words and registers fixes the choice when this encoding is available. Iteration produces a chain with a prefix at every level.

The extension relation guarantees that every immediate truncation restores its antecedent. Compatibility under composition proves the same equality between any pair of depths. Naturality allows the return identity to be applied to every prefix where it is defined. Its two components give, respectively, preserved phase and the unit memory increment. After s turns, $g_{k+9s} = g_k$ and $m_{k+9s} = m_k + s$, by induction. □

The proof establishes existence from finite branching, extension to every depth and compatibility. The survival predicate expresses this property of the complete trajectory. Individualizing a concrete regional section additionally uses that region's reader and functional rule, developed in its own chain. These two results retain their respective domains and compose when the selected section satisfies the preceding extension relations.

27.25 Computable residual initialization and compatibility of regional emissions

The numerical realization of a regional section admits an executable description by rational intervals. This description allows the residue to be initialized, updated and its blocks recovered without introducing a previously fixed decimal string. Its origin is the regional producer of section 10.5: the character, its orientation and its normalization are first determined; its rational approximations are then constructed. The present development formalizes the residual component of that reading. The trajectory, sheet, incidences and memory of the enriched state simultaneously preserve their own operations.

The order of operations is therefore

$$\begin{aligned}&\text{generated regional state} \longrightarrow \text{rational intervals of the character}\\ &\longrightarrow \text{rational name of the residue} \longrightarrow \text{compatible positional emissions.}\end{aligned}$$

Comparison with the Archimedean realization appears in the correctness proof. The algorithm receives intervals; none of its instructions receives a conventional real number or a list of target digits.

27.25.1 Rational name produced by a regional reader

A region is fixed and $j \mapsto [\ell_j, u_j]$ denotes its rational producer. The interface conditions are

$$\ell_j \le v \le u_j, \qquad \lim_{j\to\infty}(u_j - \ell_j) = 0, \qquad \ell_j, u_j \in \mathbb{Q}. \tag{27.83}$$

Here v designates the exact realization of the character to formulate correctness; the procedure calculating ℓ_j, u_j comes from the regional reader. The closure, propagation and autoscale intervals already constructed satisfy these conditions. In the implementation, its interface is a function that, given j, returns two exact fractions.

Nesting is obtained, if necessary, through the finite operation

$$L_j = \max_{0\le i\le j} \ell_i, \qquad U_j = \min_{0\le i\le j} u_i, \qquad I_j = [L_j, U_j]. \tag{27.84}$$

Proposition 27.63 (Nesting and preservation of the realization)**.** *The intervals I_j are rational, nonempty and nested. They contain v, their width tends to zero and $\bigcap_j I_j = \{v\}$.*

Proof. Each lower bound is less than or equal to v, and each upper bound is greater than or equal to v; the same property holds for the finite maximum and minimum. As j increases, the maximum increases and the minimum decreases. Moreover, $0 \le U_j - L_j \le u_j - \ell_j$. If two points belonged to the intersection, their distance would be bounded by each width and would be zero. Membership of v establishes nonemptiness. □

27.25.2 Initial prefix and representation of the residue

Let $b \geq 2$ be an integer base and $n_0 \geq 0$ a depth. The realization is assumed to avoid the corresponding boundaries:

$$b^n v \notin \mathbb{Z} \qquad (n \geq 0). \tag{27.85}$$

The irrationality of the three regional values implies this property; its proof or use as a hypothesis belongs to the fixed reader. For a concrete depth, avoidance at the scales used suffices. Realizations with a terminating expansion additionally require an exact boundary decider and a representation convention.

We define the rational predicate

$$\mathsf{Sep}(j,n): \quad \lfloor b^n L_j \rfloor = \lceil b^n U_j \rceil - 1.$$

It is decidable through integer operations on numerators and denominators. Let

$$j_0 = \min\{j : \mathsf{Sep}(j, n_0)\}, \qquad N_0 = \lfloor b^{n_0} L_{j_0} \rfloor. \tag{27.86}$$

The integer N_0 is calculated from the producer. Its use as an antecedent of the residual representation does not require introducing it as a literal datum.

Theorem 27.64 (Computable initialization of the regional residue)**.** *The index j_0 exists. For $j \geq j_0$, the intervals*

$$E_{0,j} = \left[b^{n_0} L_j - N_0,\ b^{n_0} U_j - N_0 \right] \tag{27.87}$$

are nonempty, nested and contained in $[0,1]$. Their width tends to zero and their intersection consists of a single element $\varepsilon_0 \in (0,1)$. In the Archimedean realization one has

$$N_0 = \lfloor b^{n_0} v \rfloor, \qquad b^{n_0} v = N_0 + \varepsilon_0. \tag{27.88}$$

The rational sequence $(E_{0,j})_{j \geq j_0}$ constitutes a computable representation of the residue.

Proof. The point $b^{n_0} v$ lies in the interior of a cell with integer endpoints. Its distance to those endpoints is positive. When the width of $b^{n_0} I_j$ is less than that distance, both endpoints belong to the same cell and $\mathsf{Sep}(j, n_0)$ holds. There is therefore a first separation index. At any separated index, the integer $\lfloor b^{n_0} v \rfloor$ lies between the two equal integers of the predicate and coincides with them. The initial equality places the endpoints of $b^{n_0} I_{j_0}$ between N_0 and $N_0 + 1$. Nesting preserves that inclusion. The affine transformation of (27.87) preserves nesting and nonemptiness and multiplies the widths by b^{n_0}. Their intersection is the point of (27.88); boundary avoidance places it in $(0,1)$. Each endpoint is calculated through exact fractions, and the search terminates by the preceding argument. □

The identity (27.88) is a correctness conclusion. The executable representation is the sequence (27.87), constructed from regional intervals. This distinction separates the production of a rational name from the recognition of its value.

27.25.3 Residual update and emission through exact arithmetic

Suppose the reading state $(n_0 + k, N_k, j_k)$ has been constructed, together with the name

$$E_{k,j} = \left[b^{n_0+k} L_j - N_k,\ b^{n_0+k} U_j - N_k \right], \qquad j \geq j_k. \tag{27.89}$$

One increases j from j_k until

$$\left\lfloor b \inf E_{k,j} \right\rfloor = \left\lceil b \sup E_{k,j} \right\rceil - 1.$$

The first index satisfying this equality is denoted by j_{k+1}, and the common integer is denoted by d_k. The update is

$$N_{k+1} = bN_k + d_k, \qquad E_{k+1,j} = bE_{k,j} - d_k \quad (j \geq j_{k+1}). \tag{27.90}$$

Multiplication of an interval by $b > 0$ and subtraction of d_k act on its two endpoints. The reading index reached, the depth and the emitted block are preserved.

Theorem 27.65 (Termination at arbitrary depth and positional commutation)**.** *Under* (27.83) *and* (27.85)*, each update terminates and, for every $k \geq 0$, one has*

$$N_k = \lfloor b^{n_0+k} v \rfloor, \qquad 0 \leq d_k < b, \tag{27.91}$$

$$\varepsilon_{k+1} = b\varepsilon_k - d_k, \qquad d_k = \lfloor b\varepsilon_k \rfloor, \tag{27.92}$$

$$b^{n_0+k} v = N_k + \varepsilon_k, \qquad 0 < \varepsilon_k < 1. \tag{27.93}$$

If $P(b, n)$ is the emission by rational separation of theorem 10.63, then

$$N_k = P(b, n_0 + k), \qquad d_k = P(b, n_0 + k + 1) \bmod b. \tag{27.94}$$

The stopping indices of the two implementations may differ; their emitted integers coincide exactly.

Proof. Initialization provides the assertions for $k = 0$. Assuming them at level k, multiplying the invariant by b gives

$$b^{n_0+k+1} v = bN_k + b\varepsilon_k.$$

Boundary avoidance implies $b\varepsilon_k \notin \mathbb{Z}$. The intervals $bE_{k,j}$ are nested, contain that point and have width tending to zero. The separation argument of the preceding theorem guarantees termination and fixes $d_k = \lfloor b\varepsilon_k \rfloor$. Since $0 < \varepsilon_k < 1$, the block belongs to $\{0, \ldots, b-1\}$. Subtracting d_k transforms the intervals into a nested name of the next residue and preserves its membership in $(0, 1)$. The integer part of the preceding equality is $bN_k + d_k$, which proves the new invariant. Induction holds for every finite depth, without imposing a maximum depth. Finally, the rational emission theorem also identifies $P(b, n)$ with $\lfloor b^n v \rfloor$. Equality of the prefixes and the residue formula modulo b prove (27.94). □

Corollary 27.66 (Truncation, recovery and residual bounds)**.** *For $k, h \geq 0$,*

$$\left\lfloor \frac{N_{k+h}}{b^h} \right\rfloor = N_k, \qquad N_k = \frac{N_{k+1} - d_k}{b}, \qquad \varepsilon_k = \frac{\varepsilon_{k+1} + d_k}{b}.$$

The width of the residual name at index j is $b^{n_0+k}(U_j - L_j)$. Consequently, any positive rational precision for the residue is reached after a finite number of queries to the producer. The depth and the archived blocks determine recovery in the reading component.

Proof. Division of the extended prefix recovers the preceding prefix by (27.91). The two recovery formulas are obtained by rearranging (27.90) and (27.92). The width formula follows immediately from (27.89); its convergence to zero provides termination of the precision search. □

27.25.4 Correspondence between ternary and decimal depths

In fractional normalization one writes $v = m + r$, where $m = P(b, 0)$ is obtained from the same producer. The fractional prefix is $C_n = P(b, n) - mb^n$. By integer calculation,

$$b^n v - P(b, n) = b^n r - C_n.$$

This equality reconciles the prefix including the integer part with the one starting at zero, preserving an identical residue. Subtracting m is a subsequent transformation of the regional name.

Tritic grouping uses $b = 729 = 3^6$. Six blocks correspond to 36 trits; the specialization $n_0 = 6$ provides a rational name of the residue at that depth. Decimal grouping uses $b = 1000 = 10^3$; six blocks correspond

to 18 decimal digits. They are different reading coordinates of the same character. Their compatibility is expressed by the cylinders containing that realization and by the scale-change identities; it does not require equating their numerical residues.

The residual antecedent of section 27.20 distinguishes the internal state from its publication. The present interface provides an executable rational representation of the numerical component, with the same residual recurrence once its reading is fixed. To identify it with another residual realization of that section, it suffices to establish that both read the same character, at the same depth and with the same normalization: the identity $b^n v = N_n + \varepsilon_n$ and uniqueness of the prefix then force equality of the residues. This comparison is an equality of readers. The evolution of positions, sheets, incidence and memory remains governed by enriched TPK and by their respective transport theorems.

27.25.5 Exact implementation contract

The implementation of this interface uses exclusively rational fractions, integer powers, maximum, minimum, floor division and ceiling division. The interval producer is an explicit argument of the program. A query does not return v: it returns its two generated bounds. The execution state preserves base, depth, prefix, index reached and ordered emission register. For each step, the register stores the preceding residual name at the decisive index, the block and the transformed name; the two recoveries in the corollary are checked through rational equalities.

The preceding universal proofs justify termination and compatibility under the producer contract. Finite tests verify the implementation of those operations and their identities; the generality of the result lies in induction and in the rational separation of cells. This formalization combines a computable reading and recovery interface without replacing regional selection or attributing to the numerical component the additional operations of the enriched state.

Chapter 28

Coinductive generation of π: closure character, 5 oriented regions, multiplicity $432 + 4 \cdot 144 = 1008$ and arbitrary precision

28.1 Causal separation between seed, character, and evaluation

The construction is divided into three strata:

$$\text{finite selection} \longrightarrow \text{construction of the closure character} \longrightarrow \text{Archimedean evaluation}.$$

The decimal expansion does not intervene in the first stratum. The type chain is

$$104\,976 \longrightarrow 468\,\mathcal{U}_6 \xrightarrow{\Psi} 243\,\mathcal{W}_6 \hookrightarrow \mathbb{F}_3^6, \qquad |\mathbb{F}_3^6| = 729. \tag{28.1}$$

The cardinalities count, respectively, enriched initial conditions, distinct emissions, visible ternary words, and elements of the complete ambient fiber.

For $U_6 = (u_1, \dots, u_6)$, the projection is

$$\Psi(U_6) = ((-u_1) \bmod 3, \dots, (-u_6) \bmod 3).$$

The census of the 43 orbits of D_3 proves that the closure orbit is the only one whose six orientations possess at once:

$$\text{five lifts}, \qquad \text{mass } 1008, \qquad 1008 = 432 + 4 \cdot 144.$$

The gauge

$$\mathrm{diag}_c = 6, \qquad \mathrm{card} = ES$$

select a single representative:

$$\boxed{w_6^{\pi} = 010211.} \tag{28.2}$$

The selection usis catalogue, multiplicitiis, orbital action, and orientation; it dois not consult to prefix of π.

28.2 Lifts L_0, L_1 and 1^{a} coinductive boundary

On vectors row of $\mathbb{F}_3^6$, let

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

If $b_1 = w_6^{\pi}$, the recurrence of blocks is

$$b_{k+1} = b_k L_{\varepsilon_k}, \qquad (\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4) = (0, 0, 1, 1),$$

and the accumulated word is formed by concatenation:

$$w_{6(k+1)} = w_{6k} \parallel b_{k+1}.$$

A word of length $6k$ is not multiplied by a matrix 6×6.

For the closing channel:

depth	new block
6	010211
12	012222
18	010211
24	002111
30	110221

and

$$\boxed{w_{30}^{\pi} = 010211|012222|010211|002111|110221.} \tag{28.3}$$

The following boundary is

$$u_5^{\pi} = 222220.$$

Along with the other channels,

$$R_{36} = \begin{pmatrix} 222220 \\ 021222 \\ 102011 \end{pmatrix}.$$

The symbol R_{36} records the sixth block of three histories and opens a boundary; it is not a terminal.

The certified branch of twenty blocks begins

$$\begin{aligned} \pi : {} & 010211|012222|010211|002111|110221|222220|111201| \\ & 212121|200121|100100|101222|022212|012012|111210| \\ & 121011|200220|120210|000101|022010|020111. \end{aligned}$$

For this reason, ni w_{30} ni R_{36} terminate the genealogy.

28.3 Closure microfiber of π: 5 oriented regions and multiplicity $432 + 4 \cdot 144 = 1008$

The microfiber

$$\mathscr{R}_{\pi}^{\mathrm{micro}} := \{U \in \mathcal{U}_6 : \Psi(U) = 010211\}$$

contains exactly five regions:

U_6	E_{sig}	mult	role
501|614|498|169|272|272	555555	432	$\perp$
810|923|870|169|272|272	339555	144	++
810|983|810|169|272|272	393555	144	++
870|923|810|169|272|272	933555	144	++
870|923|810|418|674|521	933717	144	−−

Consequently,

$$\boxed{5 = 1_{\perp} + 3_{++} + 1_{--}, \qquad 1008 = 432 + 4 \cdot 144.} \tag{28.4}$$

The first identity classifies orientation functions. The second counts provenance multiplicities. They do not count five indistinguishable copies.

The transversal region is

$$501|614|498|169|272|272, \qquad E_{\mathrm{sig}} = 555555, \qquad \mathrm{mult} = 432.$$

The mutated return is

$$870|923|810|418|674|521, \qquad E_{\mathrm{sig}} = 933717, \qquad \mathrm{mult} = 144.$$

Interchanging them would preserve the census $3/1/1$ but falsify the complete regional structure.

28.4 Closure microfiber of π: 5 oriented regions, multiplicities $432 + 4 \cdot 144 = 1008$, and common projection w_π

For $U_6 = (U_1, \dots, U_6)$, let

$$A(U_6) = (U_1, U_2, U_3), \qquad B(U_6) = (U_4, U_5, U_6).$$

Each emission defines an edge between $A(U_6)$ and $B(U_6)$. The raw multigraph has 96 vertices, distributed into components of sizes

$$28, 28, 28, 4, 4, 4.$$

The internal phase is quotiented by

$$(a, b, c) \sim (b, c, a) \sim (c, a, b).$$

The representative is the lexicographically minimal rotation. The quotient graph has 32 vertices and two components of sizes 28 and 4.

The anchors are

$$\begin{aligned} U_{\mathrm{APP}} &= 903|850|850|252|109|252, \\ U_e &= 850|963|890|149|252|212, \\ U_\varphi &= 830|943|830|109|252|252. \end{aligned}$$

We number the regions

$$\begin{aligned} U_\pi^{(1)} &= 810|923|870|169|272|272, \\ U_\pi^{(2)} &= 810|983|810|169|272|272, \\ U_\pi^{(3)} &= 870|923|810|169|272|272, \\ U_\pi^{(4)} &= 870|923|810|418|674|521, \\ U_\pi^{(5)} &= 501|614|498|169|272|272. \end{aligned}$$

28.5 Publication of a reduced memory of a complete nonad block via the phase transfer multigraph

28.5.1 Closure of $U_\pi^{(1)}$

$$\begin{aligned} &252|109|252|903|850|850 \to 903|850|850|252|109|252 \to \\ &252|109|252|923|870|810 \to 810|923|870|169|272|272 \to \\ &272|169|272|963|890|850 \to 850|963|890|149|252|212 \to \\ &212|149|252|943|830|830 \to 830|943|830|109|252|252 \to 252|109|252|903|850|850. \end{aligned}$$

28.5.2 Closure of $U_{\pi}^{(2)}$

$$903|850|850|252|109|252 \to 272|169|272|903|850|850 \to$$
$$272|169|272|983|810|810 \to 810|983|810|169|272|272 \to$$
$$272|169|272|963|890|850 \to 850|963|890|149|252|212 \to$$
$$212|149|252|943|830|830 \to 830|943|830|109|252|252 \to 903|850|850|252|109|252.$$

28.5.3 Closure of $U_{\pi}^{(3)}$

$$903|850|850|252|109|252 \to 292|189|232|903|850|850 \to$$
$$292|189|232|923|810|870 \to 870|923|810|169|272|272 \to$$
$$272|169|272|963|890|850 \to 850|963|890|149|252|212 \to$$
$$212|149|252|943|830|830 \to 830|943|830|109|252|252 \to 903|850|850|252|109|252.$$

28.5.4 Closure of $U_{\pi}^{(4)}$

$$903|850|850|252|109|252 \to 272|169|272|903|850|850 \to$$
$$272|169|272|963|890|850 \to 850|963|890|149|252|212 \to$$
$$212|149|252|923|810|870 \to 870|923|810|418|674|521 \to$$
$$943|830|830|521|418|674 \to 830|943|830|109|252|252 \to 903|850|850|252|109|252.$$

28.5.5 Closure of $U_{\pi}^{(5)}$

$$252|109|252|903|850|850 \to 903|850|850|252|109|252 \to$$
$$614|498|501|252|109|252 \to 501|614|498|169|272|272 \to$$
$$272|169|272|963|890|850 \to 850|963|890|149|252|212 \to$$
$$212|149|252|943|830|830 \to 830|943|830|109|252|252 \to 252|109|252|903|850|850.$$

Theorem 28.1 (Minimality of the Five Closures)**.** *For each $i \in \{1, \dots, 5\}$, the minimum length of a closed walk that simultaneously traverses*

$$U_{\pi}^{(i)}, \quad U_e, \quad U_{\varphi}, \quad U_{\mathrm{APP}}$$

in the quotient multigraph is eight.

Proof. Given i, let

$$E_* = \{U_{\pi}^{(i)}, U_e, U_{\varphi}, U_{\mathrm{APP}}\}.$$

The finite space is considered.

$$\mathcal{S}_i = \{(v, M) : v \in V(G),\ M \subseteq E_*\}.$$

The first component records the vertex and the second the target edges already traversed. If an edge e takes v to v', the transition is

$$(v, M) \longmapsto \big(v', M \cup (E_* \cap \{e\})\big).$$

Breadth-first search enumerates lengths in increasing order. The first return to (v, E_*) occurs at length eight for all five regions. The printed walks attain the bound; the optimality of the search excludes a shorter length. □

The closure function is encoded by

$$\mathfrak{C}_\pi = \big(\mathscr{R}_\pi^{\mathrm{micro}}, w_\pi, \rho_\pi, m_\pi, \ell_{\mathrm{cl}}\big),$$

$$\rho_\pi = (\perp, ++, ++, ++, --), \qquad m_\pi = (432, 144, 144, 144, 144), \qquad \ell_{\mathrm{cl}} = 8.$$

28.6 Oriented holonomies of the closure of the 5 regions of π

The incidence shadow is

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Multiplication in $\mathbb{F}_3$ gives

$$\boxed{A_W^2 = 2I_6 = -I_6.} \tag{28.5}$$

Products between distinct rows and columns vanish modulo three, whereas each diagonal product equals two. One application effects a ternary quarter-turn; two applications effect the oriented inversion.

28.7 Pending $1/5$ determined by the regional partition $432 + 4 \cdot 144 = 1008$

Let

$$\lambda = (\lambda_1, \dots, \lambda_5)^{\mathsf{T}} \in \mathbb{Q}_{\geq 0}^5, \qquad \mathbf{1}^{\mathsf{T}}\lambda = 1.$$

Forgetting the regional label applies the symmetrizer

$$P_5 = \frac{1}{5}\mathbf{1}\mathbf{1}^{\mathsf{T}}.$$

For every normalized vector,

$$P_5\lambda = \frac{1}{5}\mathbf{1}.$$

The invariance $P_5\lambda = \lambda$ therefore has a unique solution:

$$\boxed{\lambda = \frac{1}{5}\mathbf{1}, \qquad t_1 = \frac{1}{5}.} \tag{28.6}$$

The integer five comes from the regional microfiber; it is not recognized in a formula for π.

28.8 Determination of the slope 1/5 by the multiplicity of the regional microfiber of π

The law of slope composition is

$$t \star s := \frac{t + s}{1 - ts}.$$

The first duplication gives

$$t_2 = t_1 \star t_1 = \frac{2/5}{1 - 1/25} = \frac{5}{12}.$$

The second gives

$$t_4 = t_2 \star t_2 = \frac{10/12}{1 - 25/144} = \frac{120}{119}.$$

The return to the unit diagonal requires $q > 0$ such that

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1.$$

Multiplying by $119q$ produces

$$120q - 119 = 119q + 120,$$

and forces

$$\boxed{q = 239.} \tag{28.7}$$

28.9 Rational composition of the half-turn and determination of the compensating term 239

We define

$$\Theta_\pi := 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}. \tag{28.8}$$

The successive squares are

power	Gaussian integer
$(5+i)^2$	24+10i
$(5+i)^4$	476+480i
$(5+i)^8$	−3824+456960i
$(5+i)^{16}$	−208797818624−3494830080i
$(239-i)^2$	57120−478i
$(239-i)^4$	3262465916−54606720i

and the final multiplication gives

$$\boxed{(5+i)^{16}(239-i)^4 = -681386607803576157184.} \tag{28.9}$$

The imaginary part is zero and the real part is negative. Therefore,

$$\Theta_\pi \equiv \pi \pmod{2\pi}.$$

Para $0 < x < 1$,

$$x - \frac{x^3}{3} < \arctan x < x.$$

From here

$$\Theta_\pi > 16\left(\frac{1}{5} - \frac{1}{3\cdot 5^3}\right) - \frac{4}{239} > 3$$

and

$$\Theta_\pi < \frac{16}{5} < 4.$$

The only negative orientation in that interval is the first positive half-turn:

$$\boxed{\Theta_\pi = \pi.} \tag{28.10}$$

The order of subsequent Archimedean recognition is

$$\text{five regions} \to \frac{1}{5} \to \frac{5}{12} \to \frac{120}{119} \to 239 \to \Theta_\pi \to \pi.$$

28.10 Arbitrary precision rational horocycles

For $0 < x < 1$, let

$$S_n(x) = \sum_{r=0}^{n} (-1)^r \frac{x^{2r+1}}{2r+1}.$$

Leibniz's criterion gives

$$S_{2m+1}(x) < \arctan x < S_{2m}(x).$$

We define

$$a_m^- = S_{2m+1}(1/5), \quad a_m^+ = S_{2m}(1/5),$$
$$b_m^- = S_{2m+1}(1/239), \quad b_m^+ = S_{2m}(1/239),$$

and

$$\ell_{\pi,m} = 16a_m^- - 4b_m^+, \qquad h_{\pi,m} = 16a_m^+ - 4b_m^-. \tag{28.11}$$

Then

$$\ell_{\pi,m} < \pi < h_{\pi,m}$$

and

$$h_{\pi,m} - \ell_{\pi,m} = \frac{16}{(4m+3)5^{4m+3}} + \frac{4}{(4m+3)239^{4m+3}} \longrightarrow 0. \tag{28.12}$$

The lower bounds grow and the upper bounds decrease.

Since $3 < \pi < 4$, the ternary reader acts on

$$r_\pi := \pi - 3 \in (0, 1),$$

with bracket

$$[\ell_{\pi,m} - 3, h_{\pi,m} - 3].$$

28.11 Subsequent Archimedean certification of the already generated fiber

Suppose the internal relation Ext and coinductive survival have already produced, without using Machin or a tabulated expansion, the compatible extension N_{K+1} of the prefix N_K, of length $L_K = 6K$. The 729 subcylinders of its ambient fiber are

$$I_{K,u} = \left[\frac{729N_K + u}{3^{L_K+6}}, \frac{729N_K + u + 1}{3^{L_K+6}} \right), \qquad 0 \le u < 729.$$

The order $m = m(K)$ is then increased until the Leibniz bracket locates the already produced subcylinder. Let

$$\ell_{\pi,K} := \ell_{\pi,m(K)} - 3, \qquad h_{\pi,K} := h_{\pi,m(K)} - 3.$$

The compatible integer endpoints are

$$j_{\min} = \max\left(729N_K, \lfloor \ell_{\pi,K} 3^{L_K+6} \rfloor\right),$$

$$j_{\max} = \min\left(729N_K + 728, \lceil h_{\pi,K} 3^{L_K+6} \rceil - 1\right).$$

The condition of unitary localization is

$$\boxed{j_{\min} = j_{\max}, \qquad N_{K+1} = j_{\min}.} \tag{28.13}$$

The first equality certifies unit localization; the second verifies that this localization agrees with the internally generated extension. Neither defines the TPK transition or selects among the 729 states. Existence and uniqueness come from Ext and survival; Machin and the Leibniz brackets are subsequent recognitions.

28.12 Filtration of $\mathbf{F}_3{}^6$ by regional admissibility and selection of compatible subcylinders

Let $\mathcal{H}_m^\pi$ be the set of enriched histories at depth m, and let

$$p_{n,m} : \mathcal{H}_n^\pi \longrightarrow \mathcal{H}_m^\pi$$

the truncation. We define

$$\mathrm{Surv}_m^{(N),\pi} := p_{N,m}(\mathcal{H}_N^\pi),$$
$$\mathrm{Surv}_m^\pi := \bigcap_{N \geq m} \mathrm{Surv}_m^{(N),\pi}.$$

When the relation Ext is nonempty, single-valued and compatible at every level, the sets are singletons and satisfy

$$p_{m+1,m}(\mathrm{Surv}_{m+1}^\pi) = \mathrm{Surv}_m^\pi.$$

In that domain of extendability, there exists, therefore, a unique section.

$$x_\pi \in X_\pi := \varprojlim_m \mathrm{Surv}_m^\pi.$$

The cylinders are nested and their diameters are 3^{-6K}, which tends to zero. Their intersection contains a unique point, identified by the angular character with $r_\pi = \pi - 3$.

For every decimal depth M, one may choose a level $K(M)$ whose cylinder lies within a single decimal cell of diameter 10^{-M}. The deep prefixes truncate exactly to the preceding ones.

28.13 Structural characterization, necessary validity conditions and obstructions

Theorem 28.2 (Characterization of the closure channel)**.** *Starting from the catalog, the action D_3, the oriented gauge, the operators L_0, L_1, A_W, the slope law and the rational filtration:*

1. $w_6^\pi = 010211$ *is selected;*
2. *five regions are constructed with the identities of* (28.4)*;*
3. *each region belongs to a minimum closure of length eight;*
4. $A_W^2 = -I_6$ *fixes the oriented inversion;*
5. *regional symmetrization forces* $1/5$*;*
6. *the composition forces* $120/119$*;*
7. *the unit return forces* 239*;*
8. *the Gaussian identity recognizes the first half-turn;*
9. *the brackets and the fiber of* 729 *elements produce a section compatible at every finite depth.*

Its evaluation is

$$\boxed{\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}.}$$

The complete chain is

$$\mathrm{APP} \to \mathrm{TRIT} \to \mathrm{TPK} \to 104\,976 \to 468 \to 243 \to w_6^{\pi} \to w_{30}^{\pi} \to R_{36} \to \Gamma_9 \to X_{\pi} \to \pi.$$

Remark 28.3 (Certified and validity control: Forgery of the closure channel). The derivation is refuted if:

1. the closure orbit ceases to be unique;
2. one of the five regions disappears or its complete identity changes;
3. one of the five closures has length less than eight;
4. $A_W^2 \neq 2I_6$;
5. the symmetrizer admits another normalized distribution;
6. the compositions do not produce $5/12$, $120/119$, and 239;
7. the Gaussian identity fails;
8. a bracket ceases to contain the angular character;
9. a level does not traverse the 729 elements of the ambient fiber or does not locate exactly one extension;
10. a target expansion intervenes before recognition.

This unit has the status of subsequent certification. The prefixes and their compatible extensions come from the internal relation Ext, coinductive survival and the APP–TRIT–TPK transition laws. The rational brackets that follow do not choose among the 729 elements of a fiber: they locate in the Archimedean chart the already generated extension and certify its precision at arbitrary depth.

28.14 Typed separation of the 4 indices of nonadic prolongation

Four parameters of different kinds enter each character $\chi \in \{\pi, e, \varphi\}$:

1. n, order of the rational analytic bracket;
2. K, refinement level of the Topological Phase Kernel;
3. $L_K = 6K$, ternary length of the published prefix;
4. M, the requested number of decimal positions.

Neither $n = K$, $n = L_K$, $K = M$ nor a fixed proportionality between them is postulated. The coupling is constructed through interval inclusions decided by integer arithmetic.

The integer parts are

$$m_{\pi} = 3, \qquad m_e = 2, \qquad m_{\varphi} = 1,$$

and the fractional characters are

$$\rho_{\pi} = \pi - 3, \qquad \rho_e = e - 2, \qquad \rho_{\varphi} = \varphi - 1.$$

The three families of semiopen forks are written

$$J_{\chi,n} = [\ell_{\chi,n}, h_{\chi,n}).$$

Their endpoints are rational, contain ρ_{χ} and satisfy

$$\ell_{\chi,n} \nearrow \rho_{\chi}, \qquad h_{\chi,n} \searrow \rho_{\chi}, \qquad h_{\chi,n} - \ell_{\chi,n} \longrightarrow 0.$$

For π, the endpoints come from the alternating sums of Machin's identity proved in chapter 28; for e, from the factorial sums and their rational remainder in chapter 29; for φ, from the Fibonacci and Cassini quotients in chapter 30.

28.15 Ternary cylinders and the immediate ambient fibre

A ternary prefix of length $L_K = 6K$ is represented by an integer N_K with

$$0 \le N_K < 3^{L_K}.$$

His semi-open cylinder is

$$C(N_K, L_K) = \left[\frac{N_K}{3^{L_K}}, \frac{N_K + 1}{3^{L_K}} \right). \tag{28.14}$$

The next emission adds six trits. The immediate ambient fiber contains $3^6 = 729$ elements, indexed by $u \in \{0, \dots, 728\}$:

$$C_{K,u} = \left[\frac{729N_K + u}{3^{L_K+6}}, \frac{729N_K + u + 1}{3^{L_K+6}} \right). \tag{28.15}$$

The $C_{K,u}$ are disjoint and

$$C(N_K, L_K) = \bigsqcup_{u=0}^{728} C_{K,u}.$$

The number 729 counts coordinates of this ambient fiber; it does not count seeds, distinct emissions, orbits, or surviving states.

28.15.1 Cross comparison of numerators and denominators in the rational interval

Let $J = [\ell, h) \subset C(N_K, L_K)$, with $\ell, h \in \mathbb{Q}$. Escribimos $S_K = 3^{L_K+6}$ and define

$$j_{\min}(J, K) = \max(729N_K, \lfloor S_K \ell \rfloor), \tag{28.16}$$

$$j_{\max}(J, K) = \min(729N_K + 728, \lceil S_K h \rceil - 1). \tag{28.17}$$

Lemma 28.4 (Unitary localization criterion)**.** *The bracket J is contained in a unique element of the fiber* (28.15) *if and only if*

$$\boxed{j_{\min}(J, K) = j_{\max}(J, K).} \tag{28.18}$$

In that case, if j is the common value,

$$u = j - 729N_K, \qquad N_{K+1} = j.$$

Proof. The integers j whose elementary intervals of length S_K^{-1} intersect J form the integer interval

$$[\lfloor S_K \ell \rfloor, \lceil S_K h \rceil - 1] \cap [729N_K, 729N_K + 728].$$

This contains exactly one element if and only if its endpoints coincide. The equality identifies the numerator of the subcylinder and, by truncating its last six trits, N_K is recovered. □

28.16 Minimal diagonal certification

Generation does not fix the same analytic order for all levels in advance. Once the compatible extension has been produced, the order of the bracket is increased only until it is certified uniquely.

Definition 28.5 (Diagonal certification recurrence)**.** Given a family of compatible cylinders fixed by Ext, $C(N_K, L_K)$, whose publication contains ρ_χ, an initial order $n_\chi(K_0 - 1) \geq 1$ is chosen and we recursively define

$$n_\chi(K) = \min\left\{n \geq n_\chi(K-1) : j_{\min}(J_{\chi,n}, K) = j_{\max}(J_{\chi,n}, K)\right\}, \tag{28.19}$$

and we verify

$$N_{\chi,K+1} = j_{\min}(J_{\chi,n_\chi(K)}, K). \tag{28.20}$$

The set

$$\Delta_\chi = \{(n_\chi(K), K) : K \geq K_0\}$$

is the analytic–nonadic diagonal of the character χ.

Theorem 28.6 (Existence and uniqueness of the certification diagonal)**.** *For $\chi \in \{\pi, e, \varphi\}$, the recurrence (28.19) is defined at every finite level. The index $n_\chi(K)$ and the coordinate $N_{\chi,K+1}$ are unique.*

Proof. Suppose the cylindrical extension of level K containing ρ_χ has been generated internally. Since ρ_χ is irrational, it belongs to no ternary boundary $3^{-L}\mathbb{Z}$. It therefore lies in the interior of a unique subcylinder C_{K,u_*}, at a positive distance from both endpoints. The brackets $J_{\chi,n}$ contain ρ_χ and their diameter tends to zero; hence there exists an n for which the bracket is contained in C_{K,u_*}. The set over which the minimum is taken in (28.19) is nonempty and, by the well-ordering of $\mathbb{N}$, has a least element. lemma 28.4 proves that the bracket locates it uniquely and that the located numerator agrees with the one already generated. Induction on K completes the argument. □

Corollary 28.7 (Absence of decimal parameters)**.** *The diagonal is computed using sums, products, powers, Euclidean divisions, floors and ceilings of integers. It receives no decimal digit of π, e or φ as certification data.*

28.17 Compatibility with the joint nonadic connection

Let $\widehat{x}_{\chi,K}$ be the enriched state of channel χ at the level K. Its coordinates include the prefix, the cylinder, the phase, the quotient, the carry, the unbounded vertical memory, the terminal term, and the provenance register. The residual phase is

$$g_0(K) = [K]_9 \in \mathbb{Z}/9\mathbb{Z},$$

while the public label is

$$g(K) = 1 + ((K-1) \bmod 9) \in \{1, \ldots, 9\}.$$

The vertical memory $q_{\mathrm{mem}}(K) \in \mathbb{Z}_{\geq 0}$ is not reduced modulo 12; only its dodecaphasic projection is

$$\beta(K) = q_{\mathrm{mem}}(K) \bmod 12$$

does so.

For each K, let

$$\Gamma_{9,K} : \widehat{X}_K \longrightarrow \widehat{X}_{K+9}$$

the holonomy of a phase loop. Then

$$g_0(K+9) = g_0(K), \qquad q_{\mathrm{mem}}(K+9) = q_{\mathrm{mem}}(K) + 1, \qquad \beta(K+9) = \beta(K) + 1 \pmod{12}.$$

Theorem 28.8 (Phase return with genealogical conservation)**.** *Phase equality does not reset the state:*

$$g_0(K+9) = g_0(K), \qquad \widehat{x}_{\chi,K+9} \neq \widehat{x}_{\chi,K}.$$

The depths 27*,* 54*, and* 108 *are the typed compositions*

$$\begin{aligned}\Gamma_{27,K} &= \Gamma_{9,K+18} \circ \Gamma_{9,K+9} \circ \Gamma_{9,K},\\ \Gamma_{54,K} &= \Gamma_{9,K+45} \circ \cdots \circ \Gamma_{9,K},\\ \Gamma_{108,K} &= \Gamma_{9,K+99} \circ \cdots \circ \Gamma_{9,K}.\end{aligned}$$

In no case are they restarts of the prefix, the carry, the terminal, or the memory.

Proof. The first coordinate is computed in $\mathbb{Z}/9\mathbb{Z}$, whereas q_{mem} belongs to an unbounded monoid. Each application of $\Gamma_{9,K}$ adds one unit of memory, six blocks per intermediate level, and the corresponding cocycle composition. Hence the phase may coincide while the enriched state does not. The depth formulas display the indices of each map and prevent composition as though all were endomorphisms of the same space. □

28.17.1 Nonadic compression, Archimedean evaluation, and terminal term of the cylinder

The operational realization of a loop satisfies

$$U_\Gamma J = JT + \eta, \qquad J^*\eta = 0, \qquad I - T^*T = \eta^*\eta. \tag{28.21}$$

The visible strict contraction is reserved for

$$M_9 = \frac{5}{9} V_9;$$

is not identified with the compression operator T. Telescoping at depth L preserves the terminal term:

$$S_0 = \sum_{r=0}^{L-1} M_9^{*r} D_9^* D_9 M_9^r + M_9^{*L} S_0 M_9^L. \tag{28.22}$$

Omitting the last summand before proving its disappearance would erase boundary information and would turn phase return into a false restart.

28.18 Uniqueness of the Minimal Diagonal Section in the Rational Fork System

Let $\mathcal{H}_K^\chi$ be the set of admissible enriched histories of channel χ at depth K, and let

$$p_{N,K} : \mathcal{H}_N^\chi \longrightarrow \mathcal{H}_K^\chi \qquad (N \geq K)$$

the truncation. We define

$$\mathrm{Surv}_K^{(N)}(\chi) = p_{N,K}(\mathcal{H}_N^\chi)$$

and

$$\mathrm{Surv}_K(\chi) = \bigcap_{N \geq K} \mathrm{Surv}_K^{(N)}(\chi). \tag{28.23}$$

The coinductive object is

$$X_\chi = \varprojlim_K \big(\mathrm{Surv}_K(\chi), p_{K+1,K}\big). \tag{28.24}$$

Theorem 28.9 (Section compatible under enhanced extendibility)**.** *Suppose further that, over each generated numerical subcylinder, the relation* Ext *is nonempty, single-valued and compatible with the truncations. Then* Ext *and survival produce a family*

$$x_\chi = (\widehat{x}_{\chi,K})_{K \geq K_0}$$

which satisfies

$$p_{K+1,K}(\widehat{x}_{\chi,K+1}) = \widehat{x}_{\chi,K}.$$

Therefore, $x_\chi \in X_\chi$.

Proof. The subcylinder of level $K + 1$ has numerator $N_{\chi,K+1} = 729N_{\chi,K} + u$. Truncation removes its last six trits and recovers $N_{\chi,K}$. The other coordinates are obtained through the functorial laws of carry, memory, terminal and phase. The single-valuedness hypothesis for Ext fixes the subcylinder and the enriched extension; the local uniqueness of lemma 28.4 subsequently certifies its Archimedean localization, and the compatibility of Ext provides truncation equality. □

28.19 Decimal cells and prescribed precision

For $M \geq 1$ and $q \in \{0, \ldots, 10^M - 1\}$, let

$$D_{M,q} = \left[\frac{q}{10^M}, \frac{q+1}{10^M} \right).$$

A bracket $J = [\ell, h)$ is contained in a unique decimal cell if and only if

$$\boxed{\lfloor 10^M \ell \rfloor = \lceil 10^M h \rceil - 1 = q.} \tag{28.25}$$

Definition 28.10 (Minimum level of decimal certification)**.** For $\chi \in \{\pi, e, \varphi\}$, we defines $K_\chi(M)$ as the least level for which there exist an analytic order n and an integer q such that

1. $J_{\chi,n}$ satisfies (28.25);
2. $J_{\chi,n}$ is contained in the cylinder generated by state $\widehat{x}_{\chi,K}$;
3. the transition to the next level satisfies (28.18).

The pair (K, n) is minimized lexicographically.

Theorem 28.11 (Certification of coinductive outputs to arbitrary decimal precision)**.** *For each* $\chi \in \{\pi, e, \varphi\}$ *and each* $M \geq 1$*, there exists* $K_\chi(M) < \infty$*. The generated section determines a unique decimal prefix of M positions, and the bracket recognizes and certifies that prefix. If* $M' < M$*, the prefix of order M truncates to the one of order* M'*, and the corresponding cells are nested.*

Proof. The three characters are irrational: φ is irrational by its quadratic polynomial with nonsquare discriminant; the irrationality of e and π is taken from their classical characterization theorems, subsequent to the finite construction. Thus none belongs to a rational decimal or ternary boundary. For a fixed M, the distance from the character to the finite set of relevant decimal boundaries is positive. The rational brackets contract their diameter until they lie in a unique decimal cell. theorem 28.6 couples that bracket to a TPK cylinder and a unique extension. The lexicographic minimum exists by well-ordering. Compatibility of the section proves nesting as M increases. □

Corollary 28.12 (Real uniqueness)**.** *The nested cells have diameters* $10^{-M} \to 0$*. Their intersection is a singleton, identified by the preceding characterizations with* π*,* e *or* φ*, respectively.*

28.20 Equivalence of 2 finite realizations of the uniform law

The witnesses of $1,000$ and $10,000$ positions do not define the characters or limit the prolongation. They are executions of the same uniform recurrence at two different depths.

Table 28.1: Exact dimensions of certified runs, by channel.

Counted object	$1,000$ positions	$10,000$ positions
Partitions exhaustive of the fiber ambient	396	3501
Trits transportados	2376	21006
Pairs of equal phase $(K, K+9)$	387	3492
Complete nonadic turns	43	388
Elements examined by partition	729	729
Compatible extensions preserved by partition	1	1

Para $10,000$ posiciones,

$$B = 3501 = 389 \cdot 9, \qquad 6B = 21006, \qquad 3^{6B} > 10^{10000}.$$

The first 396 blocks of the long run coincide with the run of $1,000$ positions. In each of the 3492 pairs of equal phase, the phase coordinate is preserved and the prefix, depth, cylinder, and memory are extended. The certificate also records any repetitions of local projections without conflating them with the return of the complete state.

28.21 Separation between generation and verification

Definition 28.13 (Structural generator)**.** The generator receives the finite catalog, L_0, L_1, the initial state, the connection laws and the internal relation Ext. At each level, it:

1. enumerates the 729 elements of the ambient fiber;
2. applies Ext and retains the unique compatible extension;
3. updates phase, quotient, carry, memory, and terminal;
4. writes the ternary block, the selection receipt, and the genealogical record.

Definition 28.14 (Subsequent verifier)**.** The verifier receives the generator's closed outputs and the three families of rational brackets; it computes $j_{\min}$ and $j_{\max}$, certifies their equality with the already produced extension, reconstructs the partitions, checks the truncation squares, the pairs $(K, K+9)$, the partition sums

$$\mathrm{killed}_{\mathrm{izq}} + 1 + \mathrm{killed}_{\mathrm{der}} = 729,$$

the content identities and, finally, the match with reference expansions.

Theorem 28.15 (Isolation of the target data)**.** *Reference decimal expansions and metrological data are not involved in selecting a compatible extension. The comparison is performed after the three prefixes have been written.*

Proof. The mathematical input to each generative decision consists of the finite catalog, L_0, L_1, the inherited state and the internal relation Ext. Rational endpoints, powers of 3, floors and ceilings belong to the subsequent verifier. The generator finishes before the latter opens the brackets or decimal controls. The direction of data flow is

$$\text{structure} \longrightarrow \text{prefixes and ledgers} \longrightarrow \text{subsequent verification};$$

there is no arrow in the opposite direction. □

28.22 Finite realization of the certificate of arbitrary precision accuracy

A material realization of the procedure produces, for each of the three characters, a prefix of 10^4 positions. This finite truncation is a posterior control of the construction and does not intervene in the choice of cylinders, carries, or compatible extensions. The proved proposition retains arbitrary scope in M; the precision 10^4 certifies a concrete execution but does not limit the coinductive generator.

28.23 Differentiated obstructions to generation and its certification at arbitrary precision

Remark 28.16 (Certified and validity control: Refutation of arbitrary precision generation). Failures of orbit, extension, truncation or return refute generation; failures of bracket, order or decimal control refute the corresponding analytic certificate; and a discrepancy between the two refutes the joint identification of the published output. With this separation of types, the complete claim is refuted if:

1. any of the indices n, K, L_K, M are identified without a linking theorem;
2. a bracket ceases to contain its character or does not contract its diameter;
3. a stage satisfies $j_{\min} \neq j_{\max}$ after the order declared to be minimal;
4. an extension does not truncate to the cylinder of the previous level;
5. the phase return resets the prefix, carry, terminal or memory;
6. the terminal term in (28.22) is omitted without proving that it vanishes;
7. the first 396 blocks of the long execution differ from the short execution;
8. a decimal or metrological control is read before the output is closed;
9. a certified digit differs from the material output of the generator;
10. the proof depends on the requested precision being $1,000$ or $10,000$, rather than an arbitrary M.

Chapter 29

Coinductive generation of e: exponential character, factorial semigroup, and phase return with state memory

29.1 Finite orbital selector: existence and uniqueness of compatible emission

Let $\mathcal{W} = \mathbb{F}_3^6$. For an emission $U = (u_1, \dots, u_6)$, the visible word is

$$\Psi(U) = ((-u_1) \bmod 3, \dots, (-u_6) \bmod 3).$$

On $\mathcal{W}$, the dihedral action is given by

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4),$$
$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3).$$

For $w \in \mathcal{W}$, define

$$n(w) = \#\Psi^{-1}(w), \qquad M(w) = \sum_{U \in \Psi^{-1}(w)} \mathrm{mult}(U),$$

and

$$\nu(w) = \big(\#\{j : w_j = 0\}, \#\{j : w_j = 1\}, \#\{j : w_j = 2\}\big).$$

Theorem 29.1 (Structural selection of the exponential channel)**.** *Among the* 43 *orbits of the visible catalog, there is a unique orbit whose fibers simultaneously satisfy:*

$$n(w) = 1, \qquad M(w) = 144, \qquad \mathfrak{D}(w) = \{0, 3, 6\}, \qquad \nu(w) = (2, 3, 1).$$

The gauge

$$\mathrm{diag}_c = 6, \qquad \mathrm{card} = ES$$

select in it

$$\boxed{w_6^e = 201101.}$$

Proof. The four conditions are evaluated over the 243 words in the catalog, not over a sample. A single orbit satisfies the joint predicate. In that orbit, exactly one emission has both diagonal 6 and orientation ES. The enumeration uses only fields from the APP–TRIT catalog and does not consult an expansion of e. □

29.2 Block elevation and successive boundary compatibility

The recurrence L_0, L_0, L_1, L_1 produces

$$w_{30}^e = 201101|121221|102011|012222|102011,$$

and the joint boundary contributes

$$u_5^e = 021222.$$

The record of thirty-six trits remains

$$w_{36}^e = 201101|121221|102011|012222|102011|021222.$$

29.3 Exponential Section Prolongation via the Visible Block Chain

The initial reading of twelve digits is

$$\mathbf{d}_e^{(12)} = 718281828459 = 7\ \underbrace{1828\,1828}_{\text{local square}}\ 459.$$

The central equality is the concatenation

$$1828 \parallel 1828,$$

not an arithmetic product nor the supposed beginning of a period.

Definition 29.2 (Positional square)**.** A word $d_1 \cdots d_{12}$ contains a square of length ℓ at position p when

$$d_p \cdots d_{p+\ell-1} = d_{p+\ell} \cdots d_{p+2\ell-1}.$$

The position and the lateral contexts belong to the signature.

Table 29.1: Comprehensive census of squares in twelve-digit readings.

Length ℓ	1	2	3	4	5	6
Occurrences	240	21	3	2	0	0
Words that contain them	153	19	3	2	0	0

The other word with a square of length four is

$$575157518472 = \underbrace{5751\,5751}_{\text{initial position 1}}\ 8472.$$

Proposition 29.3 (Positional uniqueness)**.** *The reading of channel e is the only one that has a square of length four after a prefix of length one and with contexts* 7 *and* 459*.*

Proof. The census leaves only two words with squares of length four. One starts it at the first position; the other after the prefix 7. The position and contexts separate the two possibilities. □

29.4 Construction of the decimal signature and the positional square of the exponential section

In the elevation

$$201101|121221|102011|012222|102011$$

the third and the fifth blocks coincide:

$$B_3 = B_5 = 102011.$$

Their prestates modulo 27 are, however,

$$p_3 = (25, 11, 7, 17, 8, 16), \qquad p_5 = (7, 8, 4, 23, 26, 7),$$

and the subsequent quotients:

$$q_3 = (6, 13, 9, 2, 13, 1), \qquad q_5 = (15, 0, 3, 19, 23, 25).$$

$p_3 \neq p_5$ and $q_3 \neq q_5$ hold.

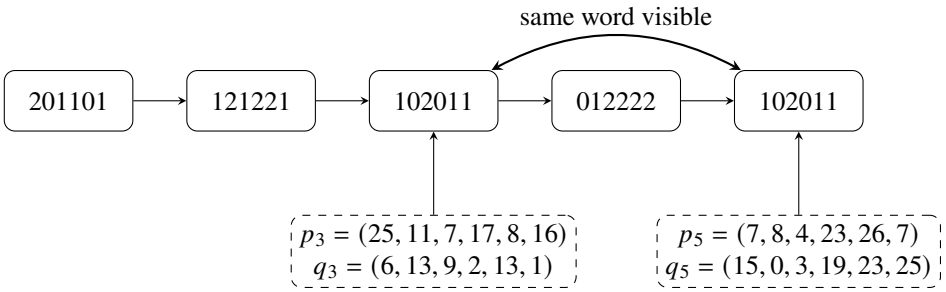


Figure 29.1: Visible return of 102011 with renewal of the prestate and the quotient.

Theorem 29.4 (Visible-phase return without periodicity of the state)**.** *$B_3 = B_5$ is a return of the visible projection, not a periodic orbit of the enriched state.*

Proof. The return of the complete state would require equality of all its coordinates. The prestate and quotient inequalities exclude it. Nor does 1828|1828 begin a decimal period: after the second block, 4 appears, whereas a periodic continuation would require 1. □

29.5 Subsequent factorial certification of the coinductive exponential section

Let $\mathcal{A}$ be a unital commutative algebra over $\mathbb{Q}$, and let $P_t \in \mathcal{A}[[t]]$ be the formal propagator. Temporal concatenation imposes

$$P_{s+t} = P_s P_t, \qquad P_0 = 1.$$

We write

$$P_t = \sum_{n=0}^{\infty} a_n t^n.$$

The unit fixes $a_0 = 1$, and normalized elementary transport fixes $a_1 = 1$.

Theorem 29.5 (Factorial recurrence)**.**

$$(n+1)a_{n+1} = a_n \qquad (n \geq 0),$$

and, therefore,

$$a_n = \frac{1}{n!}, \qquad P_t = \sum_{n=0}^{\infty} \frac{t^n}{n!}.$$

En $t = 1$,

$$\boxed{P_1 = e.}$$

Proof. The coefficient of $s^n t$ in P_{s+t} is

$$\binom{n+1}{n} a_{n+1} = (n+1)a_{n+1}.$$

In $P_s P_t$ it is $a_n a_1 = a_n$. Equality of series produces the recurrence. Induction from $a_0 = 1$ gives $a_n = 1/n!$. □

Corollary 29.6 (Differential equation)**.**

$$\frac{d}{dt} P_t = P_t, \qquad P_0 = 1.$$

Proof.

$$P_t' = \sum_{n\geq 0}(n+1)a_{n+1}t^n = \sum_{n\geq 0} a_n t^n = P_t.$$

□

29.6 Cofinal Forking of Rational Tangles and Convergence to the Published Value

Let

$$S_N = \sum_{n=0}^{N} \frac{1}{n!}.$$

Theorem 29.7 (Explicit factorial bound)**.** *For $N \geq 1$,*

$$\boxed{S_N < e < S_N + \frac{N+2}{(N+1)(N+1)!}.} \tag{29.1}$$

Proof. The rest is

$$R_N = \sum_{k=0}^{\infty} \frac{1}{(N+1+k)!} > 0.$$

Para $k \geq 0$,

$$(N+1+k)! \geq (N+1)!(N+2)^k.$$

Therefore,

$$R_N \leq \frac{1}{(N+1)!}\sum_{k=0}^{\infty}\frac{1}{(N+2)^k} = \frac{N+2}{(N+1)(N+1)!}.$$

The inequality is strict from $k = 2$ onward. □

We define

$$J_N^{(e)} = \left[S_N,\, S_N + \frac{N+2}{(N+1)(N+1)!}\right]$$

and, for the fractional part,

$$\widetilde{J}_N^{(e)} = J_N^{(e)} - 2.$$

The width converges to zero. For each M, there exists N such that

$$\operatorname{diam} J_N^{(e)} < 10^{-M}.$$

If the endpoints belong to the same decimal cell, the first M digits are certified by rational arithmetic.

29.7 Coinductive prolongation of the exponential section through ternary subcylinders of the Topological Phase Kernel

At level K, TPK and the relation Ext have already generated a unique compatible subcylinder. The smallest $N = N_e(K)$ for which the factorial bracket is contained in that subcylinder is taken. $N_e(K)$ is an order of analytic certification, not a generating index or a rule for selecting the fiber.

The TPK cylinders are compatible under truncation independently of the semigroup. The semigroup theorem subsequently identifies their Archimedean evaluation with $e - 2$; the integer part recovers e. If the orbit, Ext or truncation fails, generation fails; if the recurrence or the factorial bound fails, this analytic certificate fails, not the APP–TRIT–TPK genealogy.

29.8 Inductive prolongation of e via the nonadic holonomy Γ_9 of the TPK

Remark 29.8 (Certified and validity control: Falsification of the exponential channel). The construction is refuted if:

1. another orbit satisfies the complete finite predicate;
2. the signature 7[1828 1828]459 is not positionally unique;
3. the prestates or quotients of the two blocks 102011 coincide;
4. the semigroup does not force $(n + 1)a_{n+1} = a_n$;
5. the bound (29.1) fails;
6. a level does not locate a compatible extension uniquely;
7. a tabulated expansion intervenes before recognition.

Chapter 30

Coinductive generation of φ: modal incidence, positive Perron ray, and uniqueness of self-scaling

30.1 Finite Selection of the Self-Scaling Seed

Let $\mathcal{U}_6 \subset \{0,\dots,999\}^6$ be the catalog of the 468 distinct emissions constructed in section 8.2, and let

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \qquad \Psi(U) = ((-U_j) \bmod 3)_{j=1}^6.$$

For $w \in \Psi(\mathcal{U}_6)$, we recall the invariants

$$n(w) = \#\Psi^{-1}(w), \qquad M(w) = \sum_{U \in \Psi^{-1}(w)} \mathrm{mult}(U),$$

and the occupation vector

$$\nu(w) = \big(\#\{j : w_j = 0\}, \#\{j : w_j = 1\}, \#\{j : w_j = 2\}\big).$$

Theorem 30.1 (Orbital selection of the self-scale)**.** *The dihedral action on $\Psi(\mathcal{U}_6)$ contains a unique orbit that satisfiis the self-scale predicate*

$$n(w) = 1, \qquad M(w) = 144, \qquad \nu(w) = (2,2,2).$$

The internal caliber

$$\mathrm{diag}_c = 6, \qquad \mathrm{card} = ES$$

select in that orbit the word

$$\boxed{w_6^{\varphi} = 121200.}$$

Its only emission is

$$\boxed{U_{\varphi} = (830, 943, 830, 109, 252, 252),}$$

of multiplicity 144*, multiplicative signature* 555933*, additive orientation ES, multiplicative support of cardinality six and residual type A.*

Proof. The direct projection gives

$$U_{\varphi} \bmod 3 = (2,1,2,1,0,0), \qquad -U_{\varphi} \bmod 3 = (1,2,1,2,0,0).$$

The enumeration is carried out over the 243 visible words and their 43 orbits, not over a selection of examples. The joint predicate $n = 1$, $M = 144$, $\nu = (2,2,2)$ leaves one orbit. Within it, the declared diagonal and orientation leave one emission. At this stage neither the equation $x^2 - x - 1 = 0$, nor a decimal table, nor the conventional name of the constant has been introduced. □

30.2 Blockwise lifting and 1^{a} coinductive boundary

Let $b_1 = w_6^{\varphi}$, and let $\varepsilon = (0, 0, 1, 1)$. With the matrices L_0, L_1 constructed in section 8.1, we define

$$b_{k+1} = b_k L_{\varepsilon_k} \pmod 3 \qquad (1 \le k \le 4),$$

and concatenate

$$w_{6(k+1)}^{\varphi} = w_{6k}^{\varphi} \parallel b_{k+1}.$$

Matrix calculus produces, block by block,

$$\begin{aligned} b_1 &= 121200, \\ b_2 &= 112202, \\ b_3 &= 121020, \\ b_4 &= 010210, \\ b_5 &= 010200, \end{aligned}$$

and, therefore,

$$\boxed{w_{30}^{\varphi} = 121200|112202|121020|010210|010200.} \tag{30.1}$$

The dual neutrality of the boundary adds

$$u_5^{\varphi} = 102011,$$

so that

$$\boxed{w_{36}^{\varphi} = 121200|112202|121020|010210|010200|102011.} \tag{30.2}$$

Remark 30.2 (Two distinct uses of a visible word). The word 102011 is also visible in the biography of e. The equality of a projection does not identify enriched states: the channel, prefix, quotient, carry, phase, and memory belong to the type of the object. This separation is mandatory before any real character is constructed.

30.3 Construction of the incidence multigraph from the triticale calendar

30.3.1 Primitive modal rule

The primitive triticale calendar is

$$(+1, -1, 0).$$

We fix two visible modis, Σ and Π, and the rule

$$+1 : m \longmapsto \Sigma, \qquad -1 : m \longmapsto \Pi, \qquad 0 : m \longmapsto m. \tag{30.3}$$

Under the cyclic closure condition, the modal orbit produced by (30.3) is

$$\boxed{(\Sigma, \Pi, \Pi).}$$

The three consecutive pairs are

$$\Sigma \longrightarrow \Pi, \qquad \Pi \longrightarrow \Pi, \qquad \Pi \longrightarrow \Sigma.$$

Definition 30.3 (Chronological incidence matrix)**.** For $i, j \in \{\Sigma, \Pi\}$, let

$$N_3(i, j) = \#\{r \in \mathbb{Z}/3\mathbb{Z} : (m_r, m_{r+1}) = (i, j)\}.$$

The subscript 3 counts instants of the primitive calendar; it does not denote a matrix power.

Theorem 30.4 (Necessary descent of modal incidence)**.** *In the order* (Σ, Π)*,*

$$\boxed{N_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.} \tag{30.4}$$

By exact repetition of the calendar,

$$\boxed{N_9 = 3N_3, \qquad N_{27} = 9N_3, \qquad N_{108} = 36N_3.} \tag{30.5}$$

These three equalities are equalities of edge counts; they do not assert $N_9 = N_3^3$*,* $N_{27} = N_3^9$ *or* $N_{108} = N_3^{36}$*.*

Proof. $\Sigma \to \Sigma$ does not occur; each of the other three pairs occurs once. This determines the four entries of N_3. The calendars of lengths 9, 27, and 108 contain respectively 3, 9, and 36 copies of the primitive block. Each repetition multiplies all the counts by the same integer, which proves (30.5). □

30.3.2 Areal-volumetric publication

The modal label and the geometric sheet are distinct objects. The publication chart is

$$\mathfrak{g}_{\mathrm{av}}(\Sigma) = \mathscr{H}_{\mathrm{ar}}, \qquad \mathfrak{g}_{\mathrm{av}}(\Pi) = \mathscr{H}_{\mathrm{vol}}.$$

Transporting (30.4) through $\mathfrak{g}_{\mathrm{av}}$, we obtain the incidence operator

$$\boxed{F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.} \tag{30.6}$$

Thus, the two cross-transitions, the unique volumetric self-retention, and the absence of areal self-retention are derived from the calendar; they are not four independent inputs.

Remark 30.5 (Scope of the descent). Forgetting the elementary phase does not by itself produce a strong Markov quotient of the Topological Phase Kernel, because the visible mode does not determine which of the three internal operators will act at the next step. What does descend without an additional hypothesis is the edge multigraph of a complete block. Its minimal memory-one envelope has adjacency matrix F_{av}.

30.4 Construction of the modal incidence multigraph from the triticale calendar

Theorem 30.6 (Perron ray derived)**.** *The positive cone of* F_{av} *contains a unique eigenray. If we normalize it as* $v = (1, x)^{\mathsf{T}}$*, then*

$$x^2 - x - 1 = 0, \qquad x > 0,$$

and, therefore,

$$\boxed{x = \frac{1 + \sqrt{5}}{2} = \varphi.} \tag{30.7}$$

Proof. The equation $F_{\mathrm{av}} v = \lambda v$ gives, coordinate by coordinate,

$$x = \lambda, \qquad 1 + x = \lambda x.$$

Eliminating λ yields $x^2 - x - 1 = 0$. Its roots are $(1 \pm \sqrt{5})/2$, and only the positive root lies in the interior of the cone. The matrix F_{av} is primitive, since F_{av}^2 has all its entries positive; the Perron–Frobenius theorem ensures that this positive ray is simple and unique. The name φ is assigned only after this construction. □

Corollary 30.7 (Scalar Self-Similarity Equation)**.**

$$\varphi = 1 + \frac{1}{\varphi}, \qquad \varphi^2 = \varphi + 1, \qquad \varphi^{-1} = \varphi - 1.$$

Thi identitiis are consequencis of (30.7)*; they select neither the seed nor the matrix.*

30.5 Publication of the self-scale φ through Fibonacci powers and Diophantine convergence

Let $F_0 = 0$, $F_1 = 1$, and

$$F_{n+1} = F_n + F_{n-1} \qquad (n \geq 1).$$

Theorem 30.8 (Exact Powers of Incidence)**.** *For every $n \geq 1$,*

$$\boxed{F_{\mathrm{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.} \tag{30.8}$$

Proof. For $n = 1$, the identity is the definition of F_{av}. If it holds for n, right multiplication by F_{av} gives

$$\begin{pmatrix} F_n & F_{n-1} + F_n \\ F_{n+1} & F_n + F_{n+1} \end{pmatrix} = \begin{pmatrix} F_n & F_{n+1} \\ F_{n+1} & F_{n+2} \end{pmatrix},$$

which is the case $n + 1$. □

Theorem 30.9 (Cassini's Identity)**.** *For every $n \geq 1$,*

$$\boxed{F_{n+1}F_{n-1} - F_n^2 = (-1)^n.} \tag{30.9}$$

Proof. The determinant of F_{av} is -1. Taking determinants in (30.8),

$$F_{n-1}F_{n+1} - F_n^2 = \det(F_{\mathrm{av}}^n) = (-1)^n.$$

□

We define $r_n = F_{n+1}/F_n$ for $n \geq 1$. From Cassini it follows that

$$r_n - r_{n-1} = \frac{F_{n+1}F_{n-1} - F_n^2}{F_nF_{n-1}} = \frac{(-1)^n}{F_nF_{n-1}}.$$

Therefore, the approximations alternate around the Perron ray.

Proposition 30.10 (Perron Rational Gaps)**.** *For $m \geq 1$,*

$$\frac{F_{2m+2}}{F_{2m+1}} < \varphi < \frac{F_{2m+1}}{F_{2m}}, \tag{30.10}$$

and the width is

$$\boxed{\frac{F_{2m+1}}{F_{2m}} - \frac{F_{2m+2}}{F_{2m+1}} = \frac{1}{F_{2m}F_{2m+1}}.} \tag{30.11}$$

Proof. The preceding alternation places the even quotients below and the odd ones above φ. The formula for the width is (30.9) with $n = 2m + 1$, after putting both fractions over a common denominator. Since $F_n \to \infty$, the width tends to zero. □

30.6 Memory incidence of order 1 and Parry measure of the symbolic envelope

Let X_{av} be the smallest subshift of finite type over $\{\mathscr{H}_{\mathrm{ar}}, \mathscr{H}_{\mathrm{vol}}\}$ that admits exactly the three pairs observed in the modal calendar. Its adjacency matrix is F_{av}.

Theorem 30.11 (Transition and Parry Measure)**.** *Let $r = (1, \varphi)^{\mathsf{T}}$. The Parry stochastic matrix is*

$$\boxed{P_{ij} = \frac{(F_{\mathrm{av}})_{ij} r_j}{\varphi r_i} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}.} \tag{30.12}$$

The measure estacionaria of vertices is proporcional to $(1, \varphi^2)$, hence

$$\boxed{\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}.} \tag{30.13}$$

Proof. The sum of row i of P is

$$\frac{(F_{\mathrm{av}} r)_i}{\varphi r_i} = 1.$$

Since F_{av} is symmetric, the left and right eigenvectors coincide. The stationary weights are proportional to their coordinatewise products, that is, to $(1, \varphi^2)$. The ratio between the two weights is φ^{-2}. □

Remark 30.12 (Three measures that must not be confused). The chronological frequency of the primitive orbit is

$$\nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{ar}}) = \frac{1}{3}, \qquad \nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{vol}}) = \frac{2}{3}.$$

The uniform measure τ_{468} lives on the emission catalog. The Parry measure lives on admissible histories of arbitrary length. The first counts instants, the second counts emissions, and the third weights dynamic cylinders. None is an approximation of the others.

30.7 Multiplicative character and cylindrical conservation

For an admissible route $\gamma : i \to j$ of length n, we define

$$\chi_P(\gamma) = \varphi^n \frac{r_i}{r_j}. \tag{30.14}$$

Proposition 30.13 (Concatenation multiplicativity)**.** *If $\gamma_1 : i \to j$ and $\gamma_2 : j \to k$, then*

$$\chi_P(\gamma_2 \circ \gamma_1) = \chi_P(\gamma_2)\chi_P(\gamma_1).$$

On to closed route of length n,

$$\chi_P(\gamma) = \varphi^n.$$

Proof. Length adds and the intermediate coordinate telescopes:

$$\varphi^{n_1+n_2} \frac{r_i}{r_k} = \left(\varphi^{n_1} \frac{r_i}{r_j}\right)\left(\varphi^{n_2} \frac{r_j}{r_k}\right).$$

If $i = k$, the quotient terminal is one. □

We normalize the left vector as

$$\ell = \frac{r}{1 + \varphi^2}, \qquad \ell^{\mathsf{T}} r = 1.$$

The measure of a cylinder is

$$\mu[x_0 \cdots x_n] = \ell_{x_0} r_{x_n} \varphi^{-n}.$$

Therefore,

$$V_n(x) := \frac{\mu[x_0]}{\mu[x_0 \cdots x_n]} = \varphi^n \frac{r_{x_0}}{r_{x_n}} = \chi_P(x_0 \cdots x_n). \tag{30.15}$$

For to typed dimension $D > 0$, the scale

$$a_n^{(D)}(x) = V_n(x)^{1/D}$$

satisfies the exact law

$$\boxed{\big(a_n^{(D)}(x)\big)^D \mu[x_0 \cdots x_n] = \mu[x_0].} \tag{30.16}$$

In three dimensions and over closed returns,

$$\frac{a_9}{a_0} = \varphi^3, \qquad \frac{a_{27}}{a_0} = \varphi^9, \qquad \frac{a_{108}}{a_0} = \varphi^{36}.$$

If $A_k = a_{9k}$, then

$$A_{k+1} - 2A_k + A_{k-1} = A_{k-1}(\varphi^3 - 1)^2 > 0. \tag{30.17}$$

This is convexity in return depth; it is not interpreted as a kinematic quantity without an additional duration chart.

30.8 Quadratic memory and contracted branch

The eigenvalues of F_{av} are

$$\varphi, \qquad -\varphi^{-1}.$$

The contracted branch changes orientation at each iteration. To preserve that sign before squaring, we pass to $\mathrm{Sym}^2(\mathbb{R}^2)$. In the basis $(x^2, 2xy, y^2)$,

$$\mathrm{Sym}^2(F_{\mathrm{av}}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}. \tag{30.18}$$

Theorem 30.14 (Spectrum of the second-order memory)**.**

$$\det\big(tI - \mathrm{Sym}^2(F_{\mathrm{av}})\big) = (t+1)(t^2 - 3t + 1),$$

and

$$\boxed{\mathrm{Spec}\big(\mathrm{Sym}^2(F_{\mathrm{av}})\big) = \{\varphi^2, -1, \varphi^{-2}\}.} \tag{30.19}$$

The only positive stable mode is φ^{-2}.

Proof. The eigenvalues of a degree-two symmetric power are the products λ_1^2, $\lambda_1\lambda_2$, and λ_2^2. With $\lambda_1 = \varphi$ and $\lambda_2 = -\varphi^{-1}$, one obtains φ^2, -1, and φ^{-2}. The first exceeds one, the second has modulus one, and the third belongs to $(0, 1)$. □

On a closed route with $V_n = \varphi^n$, the stable mode satisfies

$$\boxed{M_n = \varphi^{-2n} = V_n^{-2}.} \tag{30.20}$$

30.9 Normalized hyperbolic realization of the self-scale: $\exp(2\log\varphi\, J_h) = F_{\mathrm{av}}^2$

Let

$$A = F_{\mathrm{av}}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad J_h = \frac{2A - 3I}{\sqrt{5}}.$$

Then

$$J_h^2 = I, \qquad A = \frac{3}{2}I + \frac{\sqrt{5}}{2}J_h.$$

Since

$$\cosh(2\log\varphi) = \frac{3}{2}, \qquad \sinh(2\log\varphi) = \frac{\sqrt{5}}{2},$$

we obtain

$$\boxed{\exp(2\log\varphi\, J_h) = F_{\mathrm{av}}^2.} \tag{30.21}$$

The self-scaling constant is therefore simultaneously the Perron eigenvalue of the incidence and the exponential parameter of its normalized hyperbolic realization.

30.10 Characterization of self-similarity via rational forks and coinductive prolongation

The finite construction determines the positive ray of F_{av}. The subsequent analytic characterization is

$$x^2 - x - 1 = 0, \qquad x > 0.$$

The brackets (30.10) have rational endpoints and explicit width (30.11). For each nonadic level K, we choo the least $m = m_\varphi(K)$ who bracket remains contained in a single subcylinder among the 729 elements of the ambient fiber. Minimality converts localization into a function, not into to tacit choice.

The selected cylinders are compatible with the truncations, and their diameters tend to zero. Their intersection contains a unique real number; the theorem 30.6 identifies it with φ. The Fibonacci quotients are therefore rational certificates of localization, not decimal data used to decide the seed.

30.11 Characterization of φ by the Perron ray, fork embedding, and coinductive prolongation

Remark 30.15 (Certified and validity control: Refutation of the self-biography). The construction is refuted if any of the following events occurs:

1. the finite predicate selects more than one orbit or more than one emission;
2. the recurrence L_0, L_0, L_1, L_1 does not produce (30.1);
3. the calendar does not produce exactly the three pairs used in (30.4);
4. a count multiple N_9, N_{27}, N_{108} is confused with a power of N_3;
5. F_{av} has another positive eigenray;
6. (30.8) or Cassini's identity fails;
7. the rational brackets are not nested or do not contract their width;
8. the chronological frequency, the uniform measure on the catalog or the Parry measure are identified;
9. a decimal table is involved before the Perron ray is constructed;
10. a selected cylinder does not truncate to the previous cylinder.

30.12 The dodecaphase closure of α

This section establishes dodecaphase closure through integer arithmetic in base one thousand. The construction makes explicit the twelve recurrences, the propagated Euclidean quotients and the boundary conditions determining the output.

30.12.1 The terminal selector of the three channels

The closure brings together three already constructed generations, but does not identify them by their decimal name. At terminal depth, the catalogs of compatible tails have cardinalities

$$|\mathcal{T}_\pi| = 196, \qquad |\mathcal{T}_e| = 197, \qquad |\mathcal{T}_\varphi| = 196.$$

The selection object is therefore $\mathcal{T}_\pi \times \mathcal{T}_e \times \mathcal{T}_\varphi$, with

$$196 \cdot 197 \cdot 196 = 7\,567\,952$$

possible triples.

The coordinate boundary fixed by the terminal reader is

$$q = (1, 2, 2, 1, 2, 0) \in \mathbb{F}_3^6.$$

The channel law $\pi + e - \varphi$ has sign vector

$$\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$$

and nonzero dual line

$$\langle 2\sigma \rangle = \langle (2, 2, 1) \rangle.$$

The first datum acts on the boundary columns; the second, on its rows. They are not two interchangeable features.

Theorem 30.16 (Terminal matrix selector of α)**.** *In the preceding terminal catalog, the local signatures of Gram, norm, weight and Hamming distance reduce the* 7 567 952 *triples to* 331. *Requiring the column margin q leaves two candidates:*

$$(120, 197, 157), \qquad (129, 197, 148).$$

The condition that the row charge belong to the line of the channel $\langle \sigma \rangle$ selects exactly the second. Its blocks are

$$\boxed{b_\pi = 021020, \qquad b_e = 001220, \qquad b_\varphi = 100210.}$$

Proof. The two terminal matrices surviving the column margin are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \qquad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Their row sums in $\mathbb{F}_3$ are, respectively,

$$(0, 2, 0), \qquad (2, 2, 1).$$

The first does not belong to $\langle (1, 1, 2) \rangle = \{0, (1, 1, 2), (2, 2, 1)\}$; the second does, since $(2, 2, 1) = 2\sigma$. Therefore only B_1 simultaneously satisfies the coordinate boundary and the channel charge. Its three rows are the blocks shown and correspond to the triple $(129, 197, 148)$. □

The theorem does not obtain q from the row charge: q is the coordinate datum of the terminal reader, and σ is the channel orientation. What is proved is that, once both boundaries of the same object are declared, the real branch is unique. This separation prevents a signature found a posteriori from being turned into a supposedly derived law.

Definition 30.17 (Two readings of the enriched terminal state)**.** Let

$$x_{\mathrm{term}} \in \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}} \subseteq \mathcal{X}_{\mathrm{TPK}}^{[108]}.$$

This state is the output

$$x_{\mathrm{term}} = (\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1)(x_{\mathrm{seed}})$$

of the complete pipeline of TPK selection, transport and update. The preceding selector is the reading

$$\pi_{\mathrm{term}}(x_{\mathrm{term}}) = B_{\mathrm{term}} := \begin{pmatrix} 021020 \\ 001220 \\ 100210 \end{pmatrix}.$$

The oriented register and the reader of the channels of 90° and 120° produce, from the ledger of the same state,

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\mathrm{term}})) = (b^{90}, b^{120}, Q_{\mathrm{TPK}}), \tag{30.22}$$

with

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ Q_{\mathrm{TPK}} &= 6263. \end{aligned}$$

For $r = 1, 2, 3$, let

$$B_r = \sum_{j=0}^{3} b_{r+3j}^{120}, \qquad A_1 = \frac{Q_{\mathrm{TPK}} + 2B_1 + B_2}{3}, \qquad A_2 = A_1 - B_1, \qquad A_3 = A_2 - B_2,$$

and $d_{r,j} = b_{r+3j}^{90}$. The harmonic reader Π_H forms the blocks

$$\Pi_H^{(r)} = \big(A_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}\big) \tag{30.23}$$

and interleaves them by columns. The signed reading is therefore the produced composition

$$\boxed{R_{\mathrm{sgn}} := \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}} \longrightarrow \mathcal{U}_{12}^{\mathrm{sgn}} := \operatorname{im} R_{\mathrm{sgn}}.} \tag{30.24}$$

Neither U nor K are coordinates added to the domain: U is the output of (30.24), and K will be its unique integral preimage under Hadamard. No factorization $B_{\mathrm{term}} \to U_{12}^{\mathrm{sgn}}$ is asserted: both outputs preserve distinct components of the same terminal state.

For $N' \geq N \geq 108$, the truncations act only on the enriched histories:

$$\tau_{N',N} : \mathcal{X}_{\mathrm{TPK}}^{[N'],\mathrm{enr}} \longrightarrow \mathcal{X}_{\mathrm{TPK}}^{[N],\mathrm{enr}}. \tag{30.25}$$

If $R_{\mathrm{sgn,N}}$ denotes the same composition at depth N, naturality is the concrete square

$$\boxed{R_{\mathrm{sgn,N}} \circ \tau_{N',N} = R_{\mathrm{sgn,N'}}.} \tag{30.26}$$

Theorem 30.18 (Reversible construction of the signed coordinate)**.** *The composite* (30.24)*, applied to the produced terminal state, gives exactly*

$$U = R_{\mathrm{sgn}}(x_{\mathrm{term}}) = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (30.27)$$

Its inversion by three Hadamard blocks of order four then produces

$$\boxed{K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601).}$$

The transformation is invertible; it satisfies the square (30.26) *under the truncations* (30.25)*; and it is independent of α, CODATA and every metrological figure.*

Proof. Formulas (30.22) and (30.23) give

$$(B_1, B_2, B_3) = (972, -1073, 101), \qquad (A_1, A_2, A_3) = (2378, 1406, 2479),$$

and produce the three signed blocks

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Their interleaving is (30.27). Since $H_4^2 = 4I_4$ and $\det(H_4 \oplus H_4 \oplus H_4) = (-16)^3 = -4096$, we have

$$K = \mathcal{H}_{12}^{-1} U = \frac{1}{4} \mathcal{H}_{12} U \quad (30.28)$$

after undoing the orbit permutation. Indeed,

$$H_4 U^{(1)} = (936, 2916, 2484, 3176), \quad H_4 U^{(2)} = (2172, 2636, 232, 584),$$

and

$$H_4 U^{(3)} = (560, 3296, 3656, 2404).$$

The forced division by four gives the orbits $(234, 729, 621, 794)$, $(543, 659, 058, 146)$ and $(140, 824, 914, 601)$, whose interleaving is the declared seal.

As an independent check, the cyclic operators $(D_3K)_m = K_m - K_{m+3}$ and $(D_4K)_m = K_m - K_{m+4}$, together with $Q(K) = \sum_m K_m$, satisfy

$$\operatorname{rank}(D_3, D_4) = 11, \qquad \operatorname{rank}(D_3, D_4, Q) = 12. \quad (30.29)$$

Thus $(D_3K, D_4K, Q(K))$ reconstructs a unique seal; moreover, the calculation recovers exactly

$$D_3K = b^{90}, \qquad D_4K = b^{120}, \qquad Q(K) = Q_{\mathrm{TPK}},$$

which certifies the prior reader of the register, without turning its outputs into inputs. The truncation (30.25) does not rewrite the first one hundred and eight windows of the ledger; thus both sides of (30.26) read the same register and produce U. None of the matrices used contains α or an experimental datum. □

The observable 108-step calendar is a strictly poorer projection: resetting its cursors every 9 steps erases precisely the bidirectional coordinates that the enriched register preserves. Thus the raw register does not replace the enriched state and is not used to reconstruct R_{sgn}. The preceding selection theorem constructs B_{term}; theorem 30.18 constructs, separately, the second publication of the same state.

30.12.2 Dodecaphase state and reversible maps

The theorem in this section receives from theorem 30.18 the signed vector

$$U_{12}^{\mathrm{sgn}} = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997).$$

The typed route proved from that input is

$$U_{12}^{\mathrm{sgn}} \longleftrightarrow K,$$

$$K \longleftrightarrow (D_3K, D_4K, Q),$$

$$(P, E, \Phi, K, c_{13} = 0) \longleftrightarrow A = \alpha_{12}.$$

Direct evaluation does not receive α or a metrological value as input. It receives U_{12}^{sgn}, the three Archimedean representations and the boundary condition. The theorem 30.16 first constructs the selection of the three terminal blocks; the present layer begins with the signed vector that transports them to the dodecaphase closure. The reduced trace of a single positional coordinate does not enter into this determination.

Remark 30.19 (Two dodecaphase representations of different types).

$$U_{12}^{\mathrm{cat}} = U_6 \| U_6$$

is a periodic word from the catalogue, whereas

$$U_{12}^{\mathrm{sgn}} = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997)$$

is a signed integer vector in the full state space of TPK. The first belongs to a block representation and has period six. The second preserves orientation, is not a base-thousand word and does not have that period. They are not interchangeable.

30.12.3 The canonical vector and the three orbits

The dodecaphase vector is

$$\boxed{K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601).}$$

Its sum is

$$Q(K) = 6263.$$

Step three divides C_{12} into three orbits of four sectors:

$$K^{(1)} = (K_1, K_4, K_7, K_{10}) = (234, 729, 621, 794),$$
$$K^{(2)} = (K_2, K_5, K_8, K_{11}) = (543, 659, 058, 146),$$
$$K^{(3)} = (K_3, K_6, K_9, K_{12}) = (140, 824, 914, 601).$$

Each orbit traverses four sectors separated by 90°. The cardinal labels north–east–south–west appear after fixing origin and orientation; the primary order is $(m, m+3, m+6, m+9)$.

30.12.4 First representation: Hadamard transform

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

The action on the three orbits gives

$$\begin{aligned} H_4 K^{(1)} &= (2378, -452, -668, -322), \\ H_4 K^{(2)} &= (1406, 998, -204, -28), \\ H_4 K^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns recovers

$$\mathcal{H}_{12}(K) = U_{12}^{\mathrm{sgn}}.$$

Since

$$H_4^2 = 4I_4, \qquad \det H_4 = -16,$$

the global transformation is invertible over $\mathbb{Q}^{12}$:

$$K^{(r)} = \frac{1}{4} H_4 U^{(r)}, \qquad \det \mathcal{H}_{12} = (-16)^3 = -4096 \neq 0.$$

In the canonical representation, the inverse produces a unique integer vector in $[0, 999]^{12}$.

Theorem 30.20 (Integral structure of the Hadamard transformation)**.** *The Smith normal form of the block H_4, denoted by* $\mathrm{SNF}(H_4)$*, is*

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4).$$

Consequently, for the direct sum of the three blocks,

$$\operatorname{coker} \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3,$$

and the image of $\mathbb{Z}^{12}$ has index 4096*. A vector $u \in \mathbb{Z}^4$ belongs to $H_4\mathbb{Z}^4$ exactly when*

$$H_4 u \equiv 0 \pmod 4.$$

Proof. The greatest common divisor of the entries of H_4 is one. The minors of order two are even and one has absolute value two; the minors of order three are multiples of four and one has absolute value four; finally, $|\det H_4| = 16$. The determinantal divisors are therefore $1, 2, 4, 16$, yielding the invariant factors $1, 2, 2, 4$. The direct sum of three blocks repeats these factors and has index $16^3 = 4096$. Since $H_4^{-1} = H_4/4$, the last congruence is equivalent to the integrality of the preimage. □

30.12.5 Relative cyclotomic transporter between sectors three and four

Let R be the regular rotation of the cycle C_{12} on $\mathbb{Q}^{12}$, and let

$$W_d^{\mathrm{cyc}} := \ker \Phi_d(\mathsf{R})$$

its cyclotomic sectors. The rational projectors onto the sectors of orders three and four can be written

$$\Pi_3 = \frac{1}{4}(I + \mathsf{R}^3 + \mathsf{R}^6 + \mathsf{R}^9) - \frac{1}{12} \sum_{k=0}^{11} \mathsf{R}^k,$$

$$\Pi_4 = \frac{1}{6}(I - \mathsf{R}^2 + \mathsf{R}^4 - \mathsf{R}^6 + \mathsf{R}^8 - \mathsf{R}^{10}).$$

To fix the coordinate order, let

$$\mathsf{P}_{\mathrm{orb}}(z_1, \dots, z_{12}) = \big((z_1, z_4, z_7, z_{10}), (z_2, z_5, z_8, z_{11}), (z_3, z_6, z_9, z_{12})\big).$$

The global transform used above is, in the natural order of C_{12},

$$\mathcal{H}_{12} := \mathsf{P}_{\mathrm{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)\mathsf{P}_{\mathrm{orb}}.$$

That is, $H_4 \oplus H_4 \oplus H_4$ acts in the order of the three step-three orbits, not on three contiguous quadruples in the natural order. We then define the relative transporter

$$A_{34} := \Pi_4 \mathcal{H}_{12} \Pi_3|_{W_3^{\mathrm{cyc}}} : W_3^{\mathrm{cyc}} \longrightarrow W_4^{\mathrm{cyc}}.$$

Theorem 30.21 (Dodecaphase cyclotomic transporter)**.** *In the bases*

$$u_1 = (1, -1, 0)^{\times 4}, \qquad u_2 = (0, 1, -1)^{\times 4}$$

of W_3^{cyc}*, and*

$$v_1 = (1, 0, -1, 0)^{\times 3}, \qquad v_2 = (0, 1, 0, -1)^{\times 3}$$

of W_4^{cyc}*, we have*

$$A_{34}u_1 = \frac{2}{3}(v_1 - v_2), \qquad A_{34}u_2 = \frac{2}{3}(v_1 + v_2),$$

and, therefore,

$$\boxed{\det A_{34} = \frac{8}{9}}.$$

In particular, A_{34} *is a rational isomorphism between the two sectors.*

Proof. The four preceding periodic sequences are substituted into the expressions for Π_3, Π_4 and into the three Hadamard blocks. The matrix of A_{34} in the declared bases is

$$\begin{pmatrix} 2/3 & 2/3 \\ -2/3 & 2/3 \end{pmatrix},$$

whose determinant is $8/9$. □

The rational involution that exchanges W_3^{cyc} and W_4^{cyc} by means of A_{34} and A_{34}^{-1}, and fixes W_{12}^{cyc}, transports the sum $W_4^{\mathrm{cyc}} \oplus W_{12}^{\mathrm{cyc}}$ to $W_3^{\mathrm{cyc}} \oplus W_{12}^{\mathrm{cyc}}$. In this latter sum, the cyclotomic polynomial of the rotation is

$$\Phi_3(x)\Phi_{12}(x),$$

which coincides with the polynomial of the Coxeter element of type E_6. This is an exact spectral signature and prepares the subsequent geometric corridor.

Rational transport and integral continuation

The operator A_{34} realizes rational cyclotomic transport between the sectors of orders three and four. By fixing W_{12}^{cyc}, it places the dodecaphase in the exact spectral sector $\Phi_3\Phi_{12}$ of type E_6. The integral continuation is constructed through two subsequent and compatible sources: the theorem 20.3 provides the M_{12}-equivariant isometry to the Witt module, and the theorem 20.7 chains the three incidence projectors. On that integral realization, the theorem 21.38 constructs the graph of the

twenty-seven lines and the section 21.7 identifies its Weyl action. Thus, A_{34} fixes the rational stage of the genealogy and the Witt–Schläfli intertwiners realize its lattice and incidence extension.

The sign separation is

$$U_{12}^{+} = (2378, 1406, 2479, 0, 998, 0, 0, 0, 0, 0, 0, 0),$$

$$U_{12}^{-} = (0, 0, 0, 452, 0, 551, 668, 204, 371, 322, 28, 997),$$

and

$$U_{12}^{\mathrm{sgn}} = U_{12}^{+} - U_{12}^{-}.$$

30.12.6 Second representation: transverse differences

With indices in $\{1, \dots, 12\}$ and the cyclic convention

$$K_{m+r} := K_{1+((m+r-1) \bmod 12)},$$

we define

$$(D_3K)_m = K_m - K_{m+3}, \qquad (D_4K)_m = K_m - K_{m+4}.$$

The dodecaphase transverse extractor is the operator

$$\mathcal{E}_{\mathrm{dod}} := (D_3, D_4, Q).$$

The subscript "dod" declares its domain. Shifts three and four correspond to angular separations of 90° and 120° in the oriented cycle C_{12}, but $\mathcal{E}_{\mathrm{dod}}$ is not the hexad–octad incidence operator or the angular kernel subsequently used in the towers. The data are

$$\begin{aligned} D_3K = (&-495, -116, -684, 108, 601, -90,\\ &-173, -88, 313, 560, -397, 461),\\ D_4K = (&-425, -281, -481, 671, -255, 30,\\ &475, -543, 680, 251, 6, -128),\\ Q = &\ 6263. \end{aligned}$$

If $D_3v = D_4v = 0$, the vector is constant at steps three and four. Since $\gcd(3, 4) = 1$, those steps connect the entire cycle and

$$\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle.$$

Therefore

$$\operatorname{rank}(D_3, D_4) = 11.$$

The total charge eliminates the constant translation, because $Q(\mathbf{1}) = 12 \neq 0$. Therefore

$$\boxed{\operatorname{rank}(D_3, D_4, Q) = 12,}$$

and

$$K \longleftrightarrow (D_3K, D_4K, Q)$$

is an exact coordinate system.

Theorem 30.22 (Integral normal form of the transverse system)**.** *Let*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}.$$

Its nonzero invariant factors are

$$\mathrm{SNF}(\mathcal{D}) = \mathrm{diag}(\underbrace{1, \dots, 1}_{11}, 12),$$

and, consequently,

$$\operatorname{coker} \mathcal{D} \cong \mathbb{Z}^{13} \oplus \mathbb{Z}/12\mathbb{Z}.$$

Proof. The rows of (D_3, D_4) are oriented incidences of the graph

$$\mathrm{Cay}\big(\mathbb{Z}/12\mathbb{Z}, \{\pm 3, \pm 4\}\big).$$

Since steps three and four generate $\mathbb{Z}/12\mathbb{Z}$, the graph is connected. The incidences of a spanning tree provide eleven unimodular reduced minors; the first eleven invariant factors are therefore equal to one.

Every minor of order twelve contains the row Q, since the incidence block has rank eleven. Adding the first eleven columns to the last, a unimodular operation, makes the last column vanish in the incidence rows and gives twelve in the row Q. All maximal minors are divisible by twelve. Choosing the eleven incidences of a tree gives a remaining minor of absolute value one and, therefore, a maximal determinant of absolute value twelve. The last invariant factor is exactly twelve. □

The component $\mathbb{Z}/12\mathbb{Z}$ records the condition of integral compatibility of the total charge with the reconstructed potential. In particular, the rational uniqueness proved by the rank is refined here into a statement about the integral lattice and its unique torsional obstruction.

The signed observable can also be reconstructed from these coordinates. For $r = 1, 2, 3$, let

$$B_r = \sum_{j=0}^{3} (D_4 K)_{r+3j}.$$

We obtain

$$(B_1, B_2, B_3) = (972, -1073, 101).$$

Orbital sums are

$$A_1 = \frac{Q + 2B_1 + B_2}{3} = 2378, \quad A_2 = A_1 - B_1 = 1406, \quad A_3 = A_2 - B_2 = 2479.$$

If $d_{r,j} = (D_3 K)_{r+3j}$ for $j = 0, 1, 2, 3$, with the same cyclic convention, the signed block of the orbit r is

$$(A_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}),$$

exactly the corresponding Hadamard block.

Remark 30.23 (Codomains of $\mathcal{H}_{12}$ and $\mathcal{E}_{\mathrm{dod}}$). The correct notation is

$$\mathcal{H}_{12}(K) = U_{12}^{\mathrm{sgn}}, \qquad \mathcal{E}_{\mathrm{dod}}(K) = (D_3 K, D_4 K, Q).$$

Both representations are compatible because they reconstruct the same vector, but no diagram between their codomains has been defined here. Their formulas and codomains are distinct. Sharing the steps $3/4$ or the indices $90/120$ does not authorize identifying them.

30.12.7 The three Archimedean coordinates

The vectors of twelve triads are

$$P = (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884),$$
$$E = (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497),$$
$$\Phi = (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117).$$

In concatenated form:

$$\iota(P) = 141592653589793238462643383279502884,$$
$$\iota(E) = 718281828459045235360287471352662497,$$
$$\iota(\Phi) = 618033988749894848204586834365638117.$$

P is the decimal coordinate shared by the five regions of π; the calculation does not assert that those regions are identical.

30.12.8 Base-thousand carry recurrence

We fix

$$c_{13} = 0$$

and calculate from right to left, $m = 12, 11, \ldots, 1$:

$$s_m = P_m + E_m - \Phi_m - K_m + c_{m+1},$$

$$A_m = s_m \bmod 1000, \qquad c_m = \frac{s_m - A_m}{1000}, \qquad 0 \le A_m < 1000.$$

The table is printed in increasing index order, but each carry comes from the row below.

Table 30.1: Complete dodecaphase closure in base thousand.

m	P_m	E_m	Φ_m	K_m	c_{m+1}	s_m	A_m	c_m
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	−1	352	352	0
4	589	459	749	729	−1	−431	569	−1
5	793	045	894	659	−2	−717	283	−1
6	238	235	848	824	−1	−1200	800	−2
7	462	360	204	621	0	−3	997	−1
8	643	287	586	058	−1	285	285	0
9	383	471	834	914	−1	−895	105	−1
10	279	352	365	794	0	−528	472	−1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

The carries are

$$(c_1, \ldots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0).$$

Negative values of s_m are ordinary borrowings in base thousand. Euclidean division uniquely fixes the remainder and quotient.

The output is

$$\boxed{A = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 663),}$$

and therefore

$$\boxed{\begin{aligned}\alpha_{12} &:= \frac{729735256928380099728510547238066 3}{10^{36}},\\ &= 0.007297352569283800997285105472380663.\end{aligned}}$$

This rational is the initial dodecaphase window of the compatible section generated by TPK transport. If $\rho_{N',N} : C^{\alpha}_{N'} \to C^{\alpha}_{N}$ are its truncations, the complete output is

$$\boxed{\alpha_{\mathrm{HMT}} := \varprojlim_{N} \alpha_N, \qquad \rho_{N',N}(\alpha_{N'}) = \alpha_N, \qquad \alpha_{36} = \alpha_{12}.}$$

Phase return extends the memory and narrows the surviving cylinders to arbitrary depth; it neither resets the state nor ends the construction at thirty-six digits.

30.12.9 The complete affine identity

Coordinate by coordinate,

$$\boxed{P_m + E_m - \Phi_m - K_m + c_{m+1} = A_m + 1000c_m.}$$

If

$$C = (c_1, \dots, c_{12}), \qquad C^{+} = (c_2, \dots, c_{13}),$$

the correct vector form is

$$\boxed{P + E - \Phi - K + C^{+} - 1000C = A.}$$

To avoid identifying the finite tails with the complete constants, we define

$$\tau_{36}(x) = 10^{-36} \left\lfloor 10^{36}\big(x - \lfloor x \rfloor\big) \right\rfloor, \qquad \kappa(K) = 10^{-36}\iota(K).$$

The exact decimal notation of the concatenated identity is then

$$\tau_{36}(\pi) + \tau_{36}(e) - \tau_{36}(\varphi) - \kappa(K) = \alpha_{12}.$$

Without the explicit truncation, the embedding κ and the cocycle $C^{+} - 1000C$, the coordinate equality would be false.

We define

$$\iota(v_1, \dots, v_{12}) = \sum_{m=1}^{12} v_m 1000^{12-m}.$$

Multiplying each recurrence by 1000^{12-m} and summing makes the carries telescope:

$$\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) - \iota(A) = 1000^{12}c_1 - c_{13}.$$

Since $c_1 = c_{13} = 0$:

$$\boxed{\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) = \iota(A).}$$

The two remaining integers are

$$\iota(K) = 234543140729659824621058914794146601,$$

$$\iota(A) = 007297352569283800997285105472380663.$$

30.12.10 Inversion and equivalence of the three representations

The recurrence is inverted by

$$K_m = P_m + E_m - \Phi_m + c_{m+1} - A_m - 1000c_m,$$

and the concatenated integer representation by

$$\iota(K) = \iota(P) + \iota(E) - \iota(\Phi) - \iota(A).$$

The complete diagram is

$$\boxed{K \longleftrightarrow U_{12}^{\mathrm{sgn}}, \qquad K \longleftrightarrow (D_3K, D_4K, Q), \qquad K \longleftrightarrow A \mid (P, E, \Phi, c_{13} = 0).}$$

The first two are linear representations; the third is a reversible affine map on the fiber conditioned by P, E, Φ and the boundary.

Remark 30.24 (Commutativity of the closure). The direct map obtains K from the signed representation and produces A. The inverse formula proves that the output and the boundary recover the same vector without residual freedom. Invertibility does not constitute independent statistical evidence; the commutativity of the three representations is an algebraic property of the closure.

Uniqueness is expressed at three levels of the same construction:

1. with the signed vector, P, E, Φ and the boundary fixed, the vector and the output are unique;
2. the representations are reversible and commutative;
3. the complete genealogy APP $\to$ TRIT $\to$ TPK produces those primitives before the local closure, and its compatible truncations extend the identity to the coinductive section α_{HMT}.

30.12.11 Dodecaphase boundary condition

The first dodecaphase window uses $c_{13} = 0$ and ends at

$$\ldots|380|663.$$

This boundary is part of the domain of the reversible affine map. The extension does not introduce another system: it iterates the same graded transport of the TPK on

$$w_6 \to w_{12} \to w_{18} \to w_{24} \to w_{30} \to R_{36} \to G_9$$

and constructs the successive windows with phase return, memory advance and compatible terminal carry. The section 31.7 proves that the truncation squares of this route and of the analytic reader commute and that their surviving cylinders have singleton intersection.

30.12.11.1 Integral class of the carry

The terminal condition can be compared with the periodic problem. Let S be the cyclic shift on $\mathbb{Z}^{12}$ and, for an integer base $b \geq 2$, define

$$\partial_b = S - bI.$$

The periodic affine identity takes the form

$$A = P + E - \Phi - K + \partial_b C.$$

For every $h \in \mathbb{Z}^{12}$, the transformation

$$K \longmapsto K + \partial_b h, \qquad C \longmapsto C + h$$

preserves A. Thus, the choice of carries is a choice of representative within an integral class.

Proposition 30.25 (Periodic quotient of the carry)**.** *On a cycle of length twelve,*

$$\operatorname{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Concatenation in base b, taken modulo $b^{12} - 1$, determines the periodic class.

Proof. The cyclic module is identified with $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing the relation $t = b$ produces $\mathbb{Z}/(b^{12} - 1)\mathbb{Z}$. Evaluation of the coefficient polynomial is concatenation in base b, with orientation fixed by the order of the twelve sectors. □

The closure of α does not use the periodic quotient: it uses an oriented chain with a right endpoint and requires $c_{13} = 0$. Euclidean division then selects a unique representative. This difference explains why the open boundary is part of the definition of the map and cannot be replaced by a cyclic identification of the carries.

30.12.12 Comparison with the metrological reference

After generation, the integer arithmetic of the first window verifies the recognition

$$\boxed{\alpha_{12} = 10^{-36} \left\lfloor \frac{10^{36}}{137.035999084} \right\rfloor .}$$

Its inverse is

$$\alpha_{12}^{-1} = 137.035999084000000000000000000000001066588\ldots,$$

and exactly reproduces the central string published in the 2018 adjustment by the Committee on Data for Science and Technology (CODATA) [122]:

$$\boxed{137.035999084.}$$

This is not a floating-point coincidence. The thirty-six digits of α_{12} constitute the exact truncation of the reciprocal of the printed central string. This finite check does not define α_{HMT} or limit its depth: the coinductive section already produced determines all its compatible windows, while CODATA only subsequently recognizes the prefix that its experimental resolution allows it to test.

Metrological recognition is not part of the recurrence. The mathematical result is the exact identity of the two routes

$$\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}},$$

whose dodecaphase windows are determined by the signed vector, the three generated words and the compatible boundaries. The CODATA table only subsequently compares an already fixed output.

Comparison with subsequent metrological adjustments is a separate external check. The HMT output remains frozen and can therefore be tested without modifying the generator.

30.12.13 Coordinate independence and rigidity

The closure tests distinguish deep modifications from local perturbations:

Variante	Result	What it controls
canonical right–left protocol	igualdad exacta	reference
use $U_6 \| U_6$ as K	fails macroscopically	type distinction
rotar K	breaks the closure	dodecaphase origin
reflejar K	reverses the boundary	orientation
left–right carry	changes the word	direction of transport
perturb K_i by one unit	the exact word fails	rigidez profunda

A local perturbation may alter only a deep digit; it remains a refutation of the exact equality. Structural variants produce larger failures and show that the closure is not invariant under arbitrary orders.

Remark 30.26 (Structure of the dodecaphase closure). The closure of α comprises an oriented cycle, two exact representations of the vector K, twelve typed inputs, a right-to-left recurrence, thirteen carries, a boundary condition and a telescoping integer identity. These data determine the output reproducibly.

30.13 Dodecaphasic closure and transversal dependencies of the state

30.14 Internal derivation of α_{HMT}: survival genealogy, dodecaphasic closure, two compatible constructions, and Witt bifurcation

Four strata of α_{HMT}.

1. The genealogical constructor is the enriched terminal state with its two parallel publications B_{term} and U_{12}^{sgn}.
2. The analytic characterization is the reversible chain $U_{12}^{\mathrm{sgn}} \leftrightarrow K \to A \to \alpha_{12}$, followed by its coinductive prolongation.
3. Being dimensionless, its common transduction introduces no unit: it fixes the angular chart $A_{\mathrm{deg}} = 1000\alpha_{\mathrm{HMT}}$ used by subsequent channels.
4. Comparison with CODATA begins only after obtaining α_{HMT}; it does not select K, the carry, or the prolongation.

The full genealogy explicitly preserves the input chain and reaches an enriched terminal state. Its two publications are parallel:

$$w_6 \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9 \longrightarrow x_{\mathrm{term}} \xrightarrow{\pi_{\mathrm{term}}} B_{\mathrm{term}},$$
$$x_{\mathrm{term}} \xrightarrow{R_{\mathrm{sgn}}} U_{12}^{\mathrm{sgn}} \xleftrightarrow{\mathcal{H}_{12}} K \longrightarrow A \longrightarrow \alpha_{12}.$$

B_{term} does not generate U_{12}^{sgn}: both preserve different coordinates of the same state. The transformation $\mathcal{H}_{12}$ is invertible, and the extractor $(D_3K, D_4K, Q(K))$ has rank 12. In this domain of explicit terminal primitives, the chain receives neither α, CODATA, a mass, nor any other metrological coordinate as input. Archimedean evaluation forms $(P, E, \Phi, K, c_{13} = 0)$, applies the carry cocycle, and derives α_{HMT}. Incidence evaluation transports the same words and seal to $B_K \subset H_\alpha^-$, which becomes the input to the Witt–Golay branch. Construction E21/22 and the remote prolongation $\mathsf{T}_{\alpha,\infty}$ —called $T1$ in the original notation— preserve their own domains and maps as developments of the main dodecaphasic genealogy.

Chapter 31

Dodecaphasic closure of α_{HMT} and analytic reader $E^{(9)}_{21/22}$: reversible generation, simple root and typed comparator

31.1 Internal derivation of α_{HMT}: common terminal state, dodecaphase seal, and carry cocycle

The preceding generator reaches a terminal state enriched with two coordinated publications:

$$w_6 \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9 \longrightarrow x_{\mathrm{term}} \xrightarrow{\pi_{\mathrm{term}}} B_{\mathrm{term}},$$
$$x_{\mathrm{term}} \xrightarrow{R_{\mathrm{sgn}}} U_{12}^{\mathrm{sgn}} \xleftrightarrow{\mathcal{H}_{12}} K \longrightarrow A \longrightarrow \alpha_{12}.$$

The two terminal arrows publish distinct coordinates of the same antecedent. The transform $\mathcal{H}_{12}$ is reversible; the transverse system $(D_3K, D_4K, Q(K))$ attains rank twelve; oriented Euclidean division preserves the carry cocycle and the boundary conditions. The chain uses the signed state, the seal, and the declared primitives, and obtains α_{HMT} before opening metrological recognition.

31.1.1 Dodecaphasic closure and compatible constructions

Dodecaphase closure provides the governing statement and proof. The E21/22 constructions and $\mathsf{T}_{\alpha,\infty}$ retain their own domains and functions within the principal genealogy.

31.2 Signed register, seal reconstruction, and integral carry

The signed register and the carry cocycle unfold the common terminal state, the Hadamard charts and transverse differences, the Smith normal form, the affine identity, the coinductive prolongation, and the boundaries 662/663.

31.3 Reversibility, independence, and non-circularity of the dodecaphasic closure

The state 108, the prospective constructor of K, the bidirectional Hadamard transform, integral carry, and infinite prolongation are articulated through explicit controls of reversibility, independence, and noncircularity. Each formulation preserves its domain and its refutation criterion.

31.4 Factorization of the signed dodecaphasic record and integral closure of the seal K

The construction of the integer seal does not begin with α, its inverse or a metrological table. Its domain is the enriched terminal state of the Topological Phase Kernel obtained after the already constructed chain

$$\boxed{\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \Omega_{\mathrm{seed}} \xrightarrow{T_{\min}} \mathcal{U}_6^{\mathrm{cat}} \xrightarrow{\rho_3} W_{243} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9 \longrightarrow x_{\mathrm{term}}.} \tag{31.1}$$

The foregoing arrows were defined with their respective domains: pairs of seeds, signatures, emissions, ternary words, boundaries, compatible histories, and nonadic holonomy. This unit does not reselect any of those data. It constructs the integral charts of the same terminal state and proves their equivalence.

31.4.1 Windows of the oriented trace of length 108

Let

$$\Theta_{108} = \mathbb{Z}/108\mathbb{Z}$$

the observable calendar. Its twelve nonadic windows are

$$E_m = \{9m + 1, \ldots, 9m + 9\}, \qquad 0 \le m < 12.$$

The partition is disjoint and exhaustive:

$$\{1,\dots,108\} = \bigsqcup_{m=0}^{11} E_m.$$

The set of address codes that publish an event is fixed as

$$D_{\mathrm{evt}} = \{2,4,6,8\}.$$

The orientation of the twelve windows is

$$\Sigma_{12} = (+,+,+,-,-,-,+,+,+,-,-,-). \tag{31.2}$$

If d_t is the direction encoded at step t, the signed event of window m is

$$e_m = \Sigma_{12}(m)\,\#\{t \in E_m : d_t \in D_{\mathrm{evt}}\}. \tag{31.3}$$

The formula distinguishes the nonnegative number of events from the sign of the orientation. It does not replace a cardinal direction with a sign: both coordinates remain present in the TPK state.

Definition 31.1 (Enhanced dodecaphonic registration)**.** The record transported by the trace is

$$\mathcal{R}_{12}(x) = \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=0}^{11},$$

where τ_m is the TRIT orientation, σ_m the boundary signature, κ_m the carry, and Λ_m the genealogical register segment that preserves the route. There is a chain of projections

$$\mathcal{X}^{\mathrm{enr}}_{\mathrm{TPK}} \longrightarrow \mathcal{R}_{12} \longrightarrow \Theta_{108},$$

but neither of the two arrows is an identification.

In particular, Θ_{108} preserves only the position in the calendar. The record $\mathcal{R}_{12}$ additionally preserves orientation, signature, carry, and provenance. The complete state further preserves cylinder, boundary, quotient, terminal, and vertical memory.

31.4.2 Integral chart of the dodecaphasic opposition

For $0 \le i < 6$, the opposite positions determine

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}. \tag{31.4}$$

The central charge and the five relative margins are

$$s_C = \sum_{i=0}^{5} p_i, \qquad M_i = p_i - p_5 \quad (0 \le i < 5). \tag{31.5}$$

Definition 31.2 (Integral chart $1+5+6$)**.** It is defined

$$\mathcal{L}_{12}(e_0,\dots,e_{11}) = \big(s_C, M_0,\dots,M_4, a_0,\dots,a_5\big). \tag{31.6}$$

The coordinate partition is

$$1+5+6 = 12.$$

Theorem 31.3 (Exact inverse of the integral chart)**.** *A tuple*

$$\ell = (s_C, M_0,\dots,M_4, a_0,\dots,a_5) \in \mathbb{Z}^{12}$$

belongs to the image of $\mathcal{L}_{12}$ if and only if

$$s_C - \sum_{i=0}^{4} M_i \equiv 0 \pmod 6, \tag{31.7}$$

$$p_i \equiv a_i \pmod 2 \quad (0 \le i < 6), \tag{31.8}$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \tag{31.9}$$

$$p_i = p_5 + M_i \quad (0 \le i < 5). \tag{31.10}$$

In that case, the only entire preimage is

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2}. \tag{31.11}$$

Proof. Summing $M_i = p_i - p_5$ for $0 \le i < 5$, we obtain

$$\sum_{i=0}^{4} M_i = \sum_{i=0}^{4} p_i - 5p_5 = s_C - 6p_5,$$

which forces (31.9) and the congruence modulo six. Once the six p_i have been reconstructed, the system

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}$$

has the unique solution (31.11). This solution is integral exactly when p_i and a_i have the same parity. The two families of congruences are therefore necessary and sufficient. □

This theorem precedes the numerical closure of α. It explains how the oriented trace produces a reversible integral chart, and why preserving the reduced calendar is not enough.

31.4.3 Selection of the common terminal state

The three compatible sections reach terminal depth with catalogs of cardinalities

$$|\mathcal{T}_\pi| = 196, \qquad |\mathcal{T}_e| = 197, \qquad |\mathcal{T}_\varphi| = 196.$$

The product contains

$$196 \cdot 197 \cdot 196 = 7\,567\,952 \tag{31.12}$$

ordered triples.

The column margin of the terminal reader is

$$q_\partial = (1, 2, 2, 1, 2, 0) \in \mathbb{F}_3^6,$$

and the row orientation of the combined channel is

$$\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3.$$

The first datum acts on columns; the second acts on rows. Neither is deducible from the other.

The local signatures of Gram matrix, norm, weight, and distance reduce (31.12) to 331 triples. The margin q_∂ leaves two indices:

$$(120, 197, 157), \qquad (129, 197, 148).$$

Their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \qquad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row sums are

$$(0, 2, 0), \qquad (2, 2, 1).$$

Since

$$\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\},$$

only B_1 satisfies the axial condition.

Theorem 31.4 (Uniqueness of the Visible Terminal Publication)**.** *The joint selection determines*

$$B_{\mathrm{term}} = \begin{pmatrix} 021020 \\ 001220 \\ 100210 \end{pmatrix}, \tag{31.13}$$

corresponding to the triple $(129, 197, 148)$. *No decimal evaluation of α enters the census or the two selection conditions.*

Proof. The census and margin leave exactly B_0, B_1. The charge of B_0 does not belong to $\langle \sigma \rangle$, whereas that of B_1 is 2σ. The second candidate is therefore the only one that simultaneously satisfies both typed requirements. □

31.4.4 2 publications from the same state and absence of retroactive conversion

Let

$$x_{\mathrm{term}} \in \mathcal{X}^{\mathrm{term,enr}}_{\mathrm{TPK}} \subseteq \mathcal{X}^{[108]}_{\mathrm{TPK}}.$$

The visible reading is

$$\pi_{\mathrm{term}}(x_{\mathrm{term}}) = B_{\mathrm{term}}.$$

The signed chart preserves another coordinate of the same state.

Definition 31.5 (Terminal signed causal reading)**.** The domain incorporates neither K nor U as input coordinates. The state x_{term}, produced by the full composition APP $\to$ TRIT $\to$ TPK, contains the enriched terminal register, and the extractor of its twelve oriented windows is

$$R_{12} : \mathcal{X}^{\mathrm{term,enr}}_{\mathrm{TPK}} \longrightarrow \mathcal{R}^{\mathrm{or}}_{12}, \qquad R_{12}(x_{\mathrm{term}}) = \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)^{11}_{m=0}. \tag{31.14}$$

The ledger's channel reader $90/120$ emits, without receiving the result as data,

$$E^{90,120}_{108} : \mathcal{R}^{\mathrm{or}}_{12} \longrightarrow \mathbb{Z}^{12} \times \mathbb{Z}^{12} \times \mathbb{Z}, \qquad E^{90,120}_{108}\big(R_{12}(x_{\mathrm{term}})\big) = (b_{90}, b_{120}, Q_{\mathrm{TPK}}),$$

with

$$\begin{aligned} b_{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b_{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ Q_{\mathrm{TPK}} &= 6263. \end{aligned}$$

For $r = 1, 2, 3$ and $j = 0, 1, 2, 3$, define

$$B_r = \sum_{j=0}^{3} b_{120,r+3j}, \qquad d_{r,j} = b_{90,r+3j},$$
$$A_1 = \frac{Q_{\mathrm{TPK}} + 2B_1 + B_2}{3}, \qquad A_2 = A_1 - B_1, \quad A_3 = A_2 - B_2.$$

The publication Π_H forms the three blocks

$$U^{(r)} = \big(A_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}\big),$$

and reinterleaves them in dodecaphasic order, that is,

$$\Pi_H(b_{90}, b_{120}, Q_{\mathrm{TPK}}) := \mathrm{reint}_3\big(U^{(1)}, U^{(2)}, U^{(3)}\big) = U.$$

Let $\mathcal{H}_{12}$ be the integral transformation to be constructed in the next section. The publication module is

$$\mathcal{U}^{\mathrm{sgn}}_{12} := \mathrm{im}\big(\mathcal{H}_{12} : \mathbb{Z}^{12} \to \mathbb{Z}^{12}\big). \tag{31.15}$$

The signed reading is, causally, the composition

$$R_{\mathrm{sgn}} := \Pi_H \circ E^{90,120}_{108} \circ R_{12} : \mathcal{X}^{\mathrm{term,enr}}_{\mathrm{TPK}} \longrightarrow \mathcal{U}^{\mathrm{sgn}}_{12}, \qquad R_{\mathrm{sgn}}(x_{\mathrm{term}}) = U. \tag{31.16}$$

Only after this output is $K = \mathcal{H}_{12}^{-1}U$ reconstructed; the identity $U = \mathcal{H}_{12}K$ certifies that reconstruction and does not define the reader's domain.

The correct joint writing is

$$x_{\mathrm{term}} \xrightarrow{(\pi_{\mathrm{term}}, R_{\mathrm{sgn}})} (B_{\mathrm{term}}, U^{\mathrm{sgn}}_{12}). \tag{31.17}$$

No arrow $B_{\mathrm{term}} \to U^{\mathrm{sgn}}_{12}$ is interposed: the visible matrix has forgotten precisely the orientation, opposition, quotient, carry, and genealogical record coordinates used by the signed chart. The source of both publications is the enriched state produced by (31.1).

The evaluation of the signed record gives

$$\boxed{U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997).} \tag{31.18}$$

31.4.5 Naturality with respect to depth

For $N' \geq N \geq 108$, the truncations of the signed system are

$$\tau^{\mathrm{sgn}}_{N',N} x^{[N']}_{\mathrm{term}} = \tau_{N',N} x^{[N']}_{\mathrm{term}}. \tag{31.19}$$

Therefore, truncation acts only on the history and the enriched state; it does not carry either K or U as independent inputs.

Theorem 31.6 (Square of naturality of the signed reading)**.** *We have*

$$\boxed{R_{\mathrm{sgn},N} \circ \tau^{\mathrm{sgn}}_{N',N} = \mathrm{id}_{\mathcal{U}^{\mathrm{sgn}}_{12}} \circ R_{\mathrm{sgn},N'}.} \tag{31.20}$$

Proof. The naturality of the states literally preserves the first one hundred and eight windows of the ledger:

$$R_{12,N} \circ \tau^{\mathrm{sgn}}_{N',N} = R_{12,N'}.$$

Truncation can modify only the subsequent history. Since $E^{90,120}_{108}$ and Π_H depend exclusively on those windows, both sides return the same U. Commutativity is an equality of maps, not an analogy between diagrams. □

31.4.6 Orbits of step 3 and transform of Hadamard

The cyclic order of the seal is distributed over the three orbits.

$$K^{(1)} = (K_1, K_4, K_7, K_{10}), \tag{31.21}$$

$$K^{(2)} = (K_2, K_5, K_8, K_{11}), \tag{31.22}$$

$$K^{(3)} = (K_3, K_6, K_9, K_{12}). \tag{31.23}$$

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{31.24}$$

If P_{orb} reorders the twelve coordinates according to (31.21)–(31.23), the global transformation is

$$\mathcal{H}_{12} = P_{\mathrm{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\mathrm{orb}}. \tag{31.25}$$

The three blocks of (31.18), read by orbits, are

$$U^{(1)} = (2378, -452, -668, -322),$$
$$U^{(2)} = (1406, 998, -204, -28),$$
$$U^{(3)} = (2479, -551, -371, -997).$$

Lemma 31.7 (Algebraic Hadamard quarter-turn)**.** *The matrix H_4 satisfies*

$$H_4^2 = 4I_4, \qquad H_4^{-1} = \frac{1}{4}H_4, \qquad \det H_4 = -16. \tag{31.26}$$

Proof. The rows of H_4 are orthogonal, and each has squared norm four. Since the matrix is symmetric, $H_4^{\mathsf{T}}H_4 = H_4^2 = 4I_4$. The formula for the inverse and the determinant follow immediately. □

Theorem 31.8 (Unique full reconstruction of the seal)**.** *The inverse*

$$K^{(r)} = \frac{1}{4}H_4U^{(r)} \qquad (r = 1, 2, 3) \tag{31.27}$$

yields

$$K^{(1)} = (234, 729, 621, 794),$$
$$K^{(2)} = (543, 659, 058, 146),$$
$$K^{(3)} = (140, 824, 914, 601).$$

By undoing the orbital order one obtains

$$\boxed{K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601).} \tag{31.28}$$

It is the unique rational preimage of U, and it belongs to $\mathbb{Z}^{12}$.

Proof. By (31.26), the rational preimage is unique. The explicit multiplications are

$$\begin{aligned} H_4U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4U^{(2)} &= (2172, 2636, 232, 584), \\ H_4U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and reinterleaving yields (31.28). □

The factor $1/4$ is not chosen to fit an output: it is the inverse forced by $H_4^2 = 4I_4$. Nor is it the square root of -1. The internal complex structure $A_W^2 = -I$ will appear after constructing the Paley matrix; these are two facts of different types.

31.4.7 Integral structure and Smith normal form

Theorem 31.9 (Integral image of H_4)**.** *The Smith normal form is*

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4). \tag{31.29}$$

Consequently,

$$\operatorname{coker} \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3, \tag{31.30}$$

and the image of $\mathcal{H}_{12}$ has index

$$16^3 = 4096$$

in $\mathbb{Z}^{12}$. For $u \in \mathbb{Z}^4$,

$$u \in H_4\mathbb{Z}^4 \quad \Longleftrightarrow \quad H_4 u \equiv 0 \pmod 4. \tag{31.31}$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has absolute value two; those of order three are multiples of four, and one has absolute value four; the determinant has modulus sixteen. The determinantal divisors are $1, 2, 4, 16$, yielding the invariant factors $1, 2, 2, 4$. The triple direct sum repeats the factors and multiplies the indices. Finally, $H_4^{-1} = H_4/4$, so the preimage is integral exactly under the congruence (31.31). □

31.4.8 Certification and reconstruction by differences of steps 3 and 4

Once the ledger has emitted $(b_{90}, b_{120}, Q_{\mathrm{TPK}})$, Π_H has produced U and Hadamard has reconstructed K, the following coordinates act downstream as an independent certificate of those prior outputs. With cyclic indices modulo twelve, define

$$(D_3K)_m = K_m - K_{m+3}, \qquad (D_4K)_m = K_m - K_{m+4}, \qquad Q(K) = \sum_{m=1}^{12} K_m. \tag{31.32}$$

For (31.28), one verifies

$$\begin{aligned} D_3K = b_{90} = (&-495, -116, -684, 108, 601, -90, \\ &-173, -88, 313, 560, -397, 461), \\ D_4K = b_{120} = (&-425, -281, -481, 671, -255, 30, \\ &475, -543, 680, 251, 6, -128), \\ Q(K) = Q_{\mathrm{TPK}} &= 6263. \end{aligned}$$

Theorem 31.10 (Completeness of the transversal extractor)**.** *The operator*

$$\mathcal{E}_{\mathrm{dod}} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and Smith normal form

$$\mathrm{SNF}(\mathcal{E}_{\mathrm{dod}}) = \mathrm{diag}(\underbrace{1, \ldots, 1}_{11}, 12). \tag{31.33}$$

In particular, $(D_3K, D_4K, Q(K))$ *determines a unique* K*. This uniqueness certifies and reconstructs the data already emitted by the ledger; it neither generates nor selects them.*

Proof. If $D_3 v = D_4 v = 0$, then v is constant along steps three and four. Since $\gcd(3, 4) = 1$, those steps generate all of $\mathbb{Z}/12\mathbb{Z}$, so that

$$\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle.$$

The total charge does not vanish on that line, since $Q(\mathbf{1}) = 12$. The rank is twelve. The rows of (D_3, D_4) are oriented incidences of a connected Cayley graph; a spanning tree supplies eleven unit invariant factors. Every maximal minor must contain the row Q, and the unimodular operations reduce its final contribution to twelve. There is a maximal minor of modulus twelve; therefore, the last factor is exactly twelve. □

The certifying reconstruction of U from these coordinates reproduces the prior publication of Π_H and does not require multiplying again by H_4. For $r = 1, 2, 3$, let

$$B_r = \sum_{j=0}^{3} (D_4 K)_{r+3j} = \sum_{j=0}^{3} b_{120, r+3j}. \tag{31.34}$$

Then

$$(B_1, B_2, B_3) = (972, -1073, 101).$$

Orbital sums are

$$A_1 = \frac{Q + 2B_1 + B_2}{3} = 2378, \tag{31.35}$$

$$A_2 = A_1 - B_1 = 1406, \tag{31.36}$$

$$A_3 = A_2 - B_2 = 2479. \tag{31.37}$$

If $d_{r,j} = (D_3 K)_{r+3j} = b_{90, r+3j}$, the signed block of orbit r is

$$\big(A_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}\big), \tag{31.38}$$

which reproduces the three blocks $U^{(r)}$ already published by Π_H.

31.4.9 Threshold Tetrad of the Seal

The local decimal completion satisfies

$$1000 = 729 + 271.$$

The support of the seal coordinates that reach the threshold 729 is

$$\boxed{B_K := \{m : K_m \geq 729\} = \{4, 6, 9, 10\}.} \tag{31.39}$$

This tetrad is obtained after constructing K. In this unit it is not yet called a Witt star: that object will be defined only after constructing the ternary code, its hexads, and the Steiner system.

31.4.10 Causal independence with respect to the physical output

Theorem 31.11 (Isolation of U and K)**.** *The composition*

$$\Omega_{\mathrm{seed}} \longrightarrow x_{\mathrm{term}} \xrightarrow{R_{\mathrm{sgn}}} U \xrightarrow{\mathcal{H}_{12}^{-1}} K \tag{31.40}$$

does not receive α, α^{-1}, CODATA or a target decimal string as input. The seal K is determined before defining the recurrence that will produce α.

Proof. The first part of the composition uses only seeds, emissions, ternary readers, L_0, L_1 operators, survival relations, phase transport and the charts of the TPK state. The reading $R_{\mathrm{sgn}} = \Pi_H \circ E_{108}^{90,120} \circ R_{12}$ is the causal composition of the terminal register, and the last arrow is the linear transformation $\mathcal{H}_{12}^{-1} = \mathcal{H}_{12}/4$ on its integral image. None of the formulas contains a physical variable or a metrological table. The base-one-thousand recurrence is introduced only in the next unit, after K has been fixed. □

31.4.11 Obstructions of the dodecaphasic closure: seal, carry, reversibility and non-circularity

Remark 31.12 (Certificate and validity control: Refutation of the record and the seal). The construction is refuted if any of the following facts is verified:

1. the windows E_m do not partition the one hundred eight steps;
2. the chart $\mathcal{L}_{12}$ fails its inverse or its congruences;
3. the terminal census does not produce 331 triples, two matrices and a unique axial charge;
4. an attempt is made to obtain U from B_{term} after forgetting the signed coordinates;
5. the naturality square (31.20) fails;
6. $H_4^2 \neq 4I_4$, some preimage is not integral or the reconstructed vector differs from (31.28);
7. the Smith normal form differs from (31.29);
8. the certificate (D_3, D_4, Q) loses rank, does not match the ledger's prior emissions or does not reproduce U;
9. B_K is introduced before constructing K, or is assigned a Witt structure before defining W_{12};
10. a digit of α or a metrological reference intervenes in any of the arrows of (31.40).

31.5 Coinductive prolongation of the dodecaphase cocycle and compatibility of the boundary 662/663

31.5.1 Aritmetic closure domain

The construction receives four dodecaphasic coordinates obtained in the preceding units:

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884),\\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497),\\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117),\\ K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \end{aligned}$$

The first three vectors are the first twelve triads of the fractional Archimedean readings of π, e, and φ. The fourth is the dodecaphase coordinate of the enriched terminal state of the Topological Phase Kernel, constructed before the carry operation is opened.

We fix the base.

$$B := 10^3 = 1000$$

and the concatenation

$$\iota(v_1, \dots, v_{12}) := \sum_{m=1}^{12} v_m B^{12-m}. \tag{31.41}$$

In particular,

$$\begin{aligned}
\iota(P) &= 141592653589793238462643383279502884,\\
\iota(E) &= 718281828459045235360287471352662497,\\
\iota(\Phi) &= 618033988749894848204586834365638117,\\
\iota(K) &= 234543140729659824621058914794146601.
\end{aligned}$$

The operation studied in this unit is the map

$$\mathfrak{C}_{12} : (P, E, \Phi, K, c_{13}) \longmapsto (A, C), \tag{31.42}$$

where $A = (A_1, \dots, A_{12})$ is the output word and

$$C = (c_1, \dots, c_{13})$$

is the complete system of thirteen Euclidean quotients: twelve internal quotients and one right boundary condition.

31.5.2 Left-to-right oriented Euclidean division

For $m = 12, 11, \dots, 1$, define

$$\begin{aligned}
s_m &:= P_m + E_m - \Phi_m - K_m + c_{m+1},\\
A_m &:= s_m \bmod B, \qquad 0 \le A_m < B,\\
c_m &:= \frac{s_m - A_m}{B} = \left\lfloor \frac{s_m}{B} \right\rfloor.
\end{aligned} \tag{31.43}$$

The second equality for c_m is also valid when $s_m < 0$, because Euclidean division with remainder in $\{0, \dots, B-1\}$ is used. Therefore,

$$-431 = 569 - 1000, \qquad -717 = 283 - 1000, \qquad -1200 = 800 - 2 \cdot 1000,$$

and negative loans do not introduce ambiguity.

Theorem 31.13 (Existence and Uniqueness of the Oriented Carry)**.** *Given P, E, Φ, K and the terminal carry c_{13}, the recurrence* (31.43) *uniquely determines*

$$(A_1, \dots, A_{12}; c_1, \dots, c_{12}).$$

Proof. Row $m = 12$ is determined by the Euclidean division of

$$P_{12} + E_{12} - \Phi_{12} - K_{12} + c_{13}$$

by B. Once c_{12} is known, row $m = 11$ is determined by the same property. Descending induction reaches $m = 1$. At each step, uniqueness of the Euclidean quotient and remainder prevents two distinct outputs. □

31.5.3 Finite domain of the dodecaphonic closure

The finite window is defined by the terminal condition

$$c_{13}^{\mathrm{fin}} = 0.$$

The complete evaluation is as follows. The table is printed in ascending order, although it must be read causally from the last row to the first.

Table 31.1: The twelve Euclidean divisions and thirteen carries of the finite dodecaphasic window.

m	P_m	E_m	Φ_m	K_m	c_{m+1}	s_m	A_m	c_m
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	−1	352	352	0
4	589	459	749	729	−1	−431	569	−1
5	793	045	894	659	−2	−717	283	−1
6	238	235	848	824	−1	−1200	800	−2
7	462	360	204	621	0	−3	997	−1
8	643	287	586	058	−1	285	285	0
9	383	471	834	914	−1	−895	105	−1
10	279	352	365	794	0	−528	472	−1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

The complete word of thirteen carries is

$$\boxed{C^{\mathrm{fin}} = (c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0).}$$

The dodecaphasic output is

$$\boxed{A^{\mathrm{fin}} = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 663).}$$

We define the finite rational

$$\boxed{\alpha_{12}^{\mathrm{fin}} := 10^{-36}\iota(A^{\mathrm{fin}}) = \frac{729735256928380099728510547238 0663}{10^{36}}.} \tag{31.44}$$

Therefore,

$$\alpha_{12}^{\mathrm{fin}} = 0.007297352569283800997285105472380663.$$

31.5.4 Posterior metrological recognition: CODATA 2018

The metrological comparison is performed only after closing the recurrence, the right boundary, and the preceding rational. The integer identity gives

$$\boxed{\alpha_{12}^{\mathrm{fin}} = 10^{-36}\left\lfloor \frac{10^{36}}{137.035999084} \right\rfloor.} \tag{31.45}$$

Consequently,

$$(\alpha_{12}^{\mathrm{fin}})^{-1} = 137.035999084000000000000000000000001066588\ldots, \tag{31.46}$$

and exactly reproduces the central chain published by CODATA 2018 [122]:

$$\boxed{137.035999084.}$$

The word *exactly* refers to the published central chain, not to a purported experimental measurement of thirty-six digits. The subsequent digits in (31.46) are consequences of the HMT rational already fixed. CODATA does not select the state, does not determine (K), does not decide the direction of the carry, and does not intervene in the dodecaphase table.

31.5.5 Complete affine identity

Each row of the table satisfies exactly

$$P_m + E_m - \Phi_m - K_m + c_{m+1} = A_m + Bc_m. \tag{31.47}$$

Introducing

$$C^- := (c_1, \dots, c_{12}), \qquad C^+ := (c_2, \dots, c_{13}),$$

the twelve local identities come together in

$$\boxed{P + E - \Phi - K + C^+ - BC^- = A.} \tag{31.48}$$

The term

$$\partial_B C := C^+ - BC^-$$

is the carry cocycle of the oriented chain. Consequently, $P + E - \Phi - K = A$ is not a coordinatewise equality in $\mathbb{Z}^{12}$: it becomes correct only upon incorporating $\partial_B C$.

Theorem 31.14 (Telescoping of the Thirteen Carries)**.** *For any terminal condition c_{13}, the recurrence satisfies*

$$\boxed{\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) - \iota(A) = B^{12}c_1 - c_{13}.} \tag{31.49}$$

In the finite window, $c_1 = c_{13} = 0$, and therefore

$$\boxed{\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) = \iota(A^{\mathrm{fin}}).} \tag{31.50}$$

Proof. We multiply (31.47) by B^{12-m} and sum over $m = 1, \dots, 12$. The contribution of the carries is

$$\sum_{m=1}^{12} (Bc_m - c_{m+1})B^{12-m}.$$

The internal components cancel:

$$\begin{aligned}\sum_{m=1}^{12} (Bc_m - c_{m+1})B^{12-m} &= \sum_{m=1}^{12} c_m B^{13-m} - \sum_{m=1}^{12} c_{m+1}B^{12-m} \\ &= B^{12}c_1 - c_{13}.\end{aligned}$$

This proves (31.49). The specialization $c_1 = c_{13} = 0$ gives (31.50). □

31.5.6 Exact inversion of the seal K

The local recurrence is reversible. Solving for K_m in (31.47),

$$\boxed{K_m = P_m + E_m - \Phi_m + c_{m+1} - A_m - Bc_m.} \tag{31.51}$$

By concatenation,

$$\boxed{\iota(K) = \iota(P) + \iota(E) - \iota(\Phi) - \iota(A) - B^{12}c_1 + c_{13}.} \tag{31.52}$$

On the finite boundary,

$$\iota(K) = \iota(P) + \iota(E) - \iota(\Phi) - \iota(A^{\mathrm{fin}}).$$

Algebraic reversibility does not invert the causal order. The effective order is

$$x_{\mathrm{term}} \longrightarrow K \longrightarrow (P, E, \Phi, K, c_{13}) \longrightarrow (A, C).$$

Formula (31.51) is a commutativity check: it proves that the output and the carries recover the seal that was already fixed. It is not a rule for retrospectively selecting K from a desired value of α.

31.5.7 Coinductive prolongation of the dodecaphasic seal and inverse carry recurrence in $C = \{-2, -1, 0, 1, 2\}$

Let

$$(P_k)_{k\geq 1}, \qquad (E_k)_{k\geq 1}, \qquad (\Phi_k)_{k\geq 1}$$

the infinite sequences of triads produced by the three coinductive branches, and let

$$\kappa(k) := 1 + ((k-1) \bmod 12).$$

The prolongation of the stamp is periodic:

$$K^{\mathrm{per}}_k := K_{\kappa(k)}.$$

For a finite depth N and a terminal condition $t = c_{N+1}$, the recurrence is

$$\begin{aligned} s^{(N,t)}_k &= P_k + E_k - \Phi_k - K_{\kappa(k)} + c^{(N,t)}_{k+1}, \\ A^{(N,t)}_k &= s^{(N,t)}_k \bmod 1000, \\ c^{(N,t)}_k &= \left\lfloor \frac{s^{(N,t)}_k}{1000} \right\rfloor, \qquad k = N, \dots, 1. \end{aligned} \tag{31.53}$$

The set of terminal carries

$$C := \{-2, -1, 0, 1, 2\}$$

is invariant for the certified inputs: for each triad and each $c_{k+1} \in C$, the quotient c_k again belongs to C. No favorable terminal condition is chosen. All five conditions are propagated, and only the prefix common to all of them is published.

The execution of certified depth employs

quantity	value
ternary blocks per channel	3501
trits per channel	$3501 \cdot 6 = 21006$
guard decimal digits	10020
guard triads	3340
propagated terminal conditions	5
common triads	3339
common digits	10017
published digits	10000

The five terminal branches coalesce before the published prefix. In particular, no terminal condition extracted from a target value of α intervenes in the output.

Theorem 31.15 (Stabilized Output Compatibility)**.** *Let $A^{[N]}$ and $A^{[N']}$, with $N' > N$, be two runs whose five terminal conditions have coalesced before position N. Then the truncation of $A^{[N']}$ to its first N*

blocks coincides with $A^{[N]}$. The stabilized prefixes define a unique section in the inverse limit of finite words.

Proof. The recurrence is deterministic from right to left once the carry entering a position is known. Coalescence asserts that all remote conditions produce the same carry before the boundary of the prefix. From that point, the same triads P_k, E_k, Φ_k and the same periodic seal produce, by Euclidean uniqueness, the same remainders and quotients at all preceding positions. This is exactly commutativity with truncation. □

31.5.8 Boundary originating from the remote tail

The prolongation determines on the visible boundary.

$$c^{\infty}_{13} = -1.$$

The twelve internal carries remain the same as in the finite window:

$$\boxed{C^{\infty}_{\le 13} = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, -1).}$$

Only the information entering from the thirteenth triad changes. The two relevant divisions are

$$\begin{aligned} s^{\infty}_{12} &= 884 + 497 - 117 - 601 - 1 = 662, \\ A^{\infty}_{12} &= 662, \qquad c^{\infty}_{12} = 0, \end{aligned}$$

and

$$\begin{aligned} s^{\infty}_{13} &= 197 + 757 - 720 - 234 - 1 = -1, \\ A^{\infty}_{13} &= 999, \qquad c^{\infty}_{13} = -1. \end{aligned}$$

Therefore, the beginning of the prolonged word is

$$\boxed{A^{\infty} = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \ldots).}$$

The corresponding expansion begins

$$\boxed{\alpha_{\infty} = 0.007297352569283800997285105472380662999564172539642\ldots.} \tag{31.54}$$

The difference between both boundaries is local and exact:

$$c^{\mathrm{fin}}_{13} - c^{\infty}_{13} = 1, \qquad A^{\mathrm{fin}}_{12} - A^{\infty}_{12} = 1,$$

while

$$A^{\mathrm{fin}}_{m} = A^{\infty}_{m}, \qquad 1 \le m \le 11.$$

Applying the telescoping theorem to the first twelve blocks of the prolongation,

$$\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) - \iota(A^{\infty}_{\le 12}) = -c^{\infty}_{13} = 1.$$

Then

$$\boxed{\iota(A^{\mathrm{fin}}) = \iota(A^{\infty}_{\le 12}) + 1.} \tag{31.55}$$

The ending 663 does not contradict the ending 662: the former is the finite representative obtained by closing the window with $c_{13} = 0$; the latter retains the carry -1 from the remote tail.

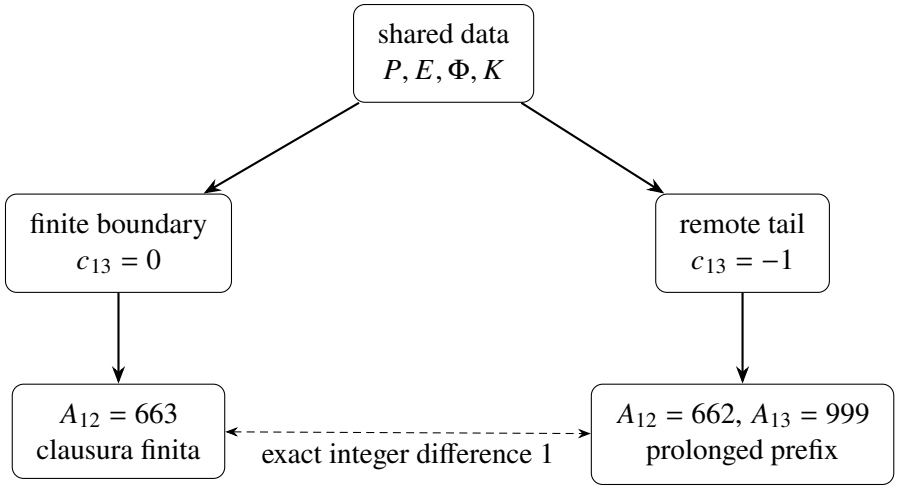


Figure 31.1: Boundary bifurcation of a single recurrence. Two distinct generators are not being compared: only the datum entering from the right changes.

31.5.9 Truncamiento compatible and rewhereo certified to 36 digits

For $x \geq 0$, define

$$\mathrm{Tr}_{36}(x) := 10^{-36} \lfloor 10^{36} x \rfloor$$

and

$$\mathrm{Rd}_{36}(x) := 10^{-36} \left\lfloor 10^{36} x + \frac{1}{2} \right\rfloor.$$

Let

$$a := 729735256928380099728510547238066 2.$$

The expansion (31.54) can be written as

$$\alpha_\infty = \frac{a + \rho}{10^{36}}, \qquad \rho = 0.999564172539642709\ldots,$$

so that $0.999 < \rho < 1$.

Theorem 31.16 (Exact resolution of the boundary 662/663)**.** *The coinductive prolongation satisfies*

$$\boxed{\mathrm{Tr}_{36}(\alpha_\infty) = 0.007297352569283800997285105472380662,}$$

while its rounding to thirty-six digits satisfies

$$\boxed{\mathrm{Rd}_{36}(\alpha_\infty) = 0.007297352569283800997285105472380663 = \alpha_{12}^{\mathrm{fin}}.}$$

Proof. Since $0 \leq \rho < 1$,

$$\lfloor 10^{36} \alpha_\infty \rfloor = a.$$

This gives the truncation ending in 662. Since $\rho > 1/2$,

$$\left\lfloor 10^{36} \alpha_\infty + \frac{1}{2} \right\rfloor = a + 1,$$

which ends in 663. By (31.44), $(a + 1)10^{-36}$ is exactly $\alpha_{12}^{\mathrm{fin}}$. □

Moreover,

$$0 < \alpha_{12}^{\mathrm{fin}} - \alpha_\infty = \frac{1-\rho}{10^{36}} < 10^{-39}.$$

The difference is not an arithmetic error in the table: it quantifies the difference between two distinct boundary operators.

31.5.10 Causal isolation from α and metrological data

At depth N, the full domain of the operation is

$$\mathscr{D}_N = \big((P_k)_{k=1}^N, (E_k)_{k=1}^N, (\Phi_k)_{k=1}^N, K, c_{N+1}\big).$$

The carry map

$$\mathfrak{C}_N : \mathscr{D}_N \longrightarrow \{0,\dots,999\}^N \times C^N$$

is defined exclusively by (31.53). Its domain contains no target value of α, its inverse, a metrological table, a target decimal string or a terminal carry chosen by comparison.

Theorem 31.17 (No interference from the metrological target)**.** *Let $\mathscr{Y}$ be any space of external data containing metrological values or comparison tables. The formal extension*

$$\widetilde{\mathfrak{C}}_N : \mathscr{D}_N \times \mathscr{Y} \longrightarrow \{0,\dots,999\}^N \times C^N, \qquad \widetilde{\mathfrak{C}}_N(d,y) := \mathfrak{C}_N(d),$$

factorizes by the projection

$$\mathrm{pr}_{\mathscr{D}_N} : \mathscr{D}_N \times \mathscr{Y} \longrightarrow \mathscr{D}_N.$$

Therefore, for any $y, y' \in \mathscr{Y}$,

$$\widetilde{\mathfrak{C}}_N(d,y) = \widetilde{\mathfrak{C}}_N(d,y').$$

No intervention on a value of α or a metrological table alters K, the quotients, the remainders or the prefix produced.

Proof. The statement results from

$$\widetilde{\mathfrak{C}}_N = \mathfrak{C}_N \circ \mathrm{pr}_{\mathscr{D}_N}.$$

All operations of $\mathfrak{C}_N$ are additions, subtractions and Euclidean divisions of the elements of $\mathscr{D}_N$. No coordinate of $\mathscr{Y}$ appears in them. □

The complete causal order is

$$\text{enriched branches of } \pi, e, \varphi \longrightarrow (P, E, \Phi),$$

$$x_{\mathrm{term}} \longrightarrow K,$$

$$(P, E, \Phi, K, \text{boundary}) \longrightarrow (A, C) \longrightarrow \alpha_\infty,$$

$$\alpha_\infty \longrightarrow \text{external comparison}.$$

The last arrow cannot be reversed to select any of the previous ones.

31.5.11 Exact obstructions of the prolongation: boundary, truncation, and causal isolation

Remark 31.18 (Certificate and validity control: Refutation of the affine closure and its prolongation). The block is refuted in its domain if any of the following events occurs:

1. some row of the table 31.1 fails to satisfy $s_m = A_m + 1000c_m$;
2. the published vector of thirteen carries does not reproduce the twelve rows;

3. the telescoping identity fails

$$\iota(P)+\iota(E)-\iota(\Phi)-\iota(K)-\iota(A)=1000^{12}c_1-c_{13};$$

4. the inversion (31.51) does not recover the twelve coordinates of K;
5. remote propagation produces $c_{13}\neq -1$ for the certified prefix;
6. the five terminal conditions of $C=\{-2,-1,0,1,2\}$ do not coalesce before the published digits;
7. the triad following 662 is not 999, in which case the identification of the rounding with 663 is no longer proved;
8. the generator opens a target expansion of α, its inverse, a metrological table, or a terminal condition chosen to force the coincidence.

Thus, two distinct boundary identities are thereby demonstrated:

$$\boxed{663 = \text{finite closure with } c_{13}=0 = \text{rounding to 36 digits of } \alpha_\infty,}$$

$$\boxed{662 = \text{truncation to 36 digits of } \alpha_\infty \quad \text{with} \quad c_{13}=-1,\ A_{13}=999.}$$

The central metrological result is finally published in the same residence as the derivation:

$$\boxed{\begin{array}{c}(\alpha_{12}^{\mathrm{fin}})^{-1}=137.035999084000000000000000000000001066588\ldots,\\ \text{central CODATA 2018 string:}\qquad 137.035999084.\end{array}} \tag{31.56}$$

The implication expresses a subsequent comparison with the central string published; it does not introduce that string into the constructor.

The closure of α does not begin with a metrological constant or with a sought decimal string. It begins in the same enriched terminal state that preserves the genealogy of the three channels. This section presents, within a single causal chain, two routes to the scale of α and a long prolongation of the carry of the first. The remote prolongation is denoted by $\mathsf{T}_{\alpha,\infty}$; it constitutes neither a second route nor a third derivation.

31.5.12 Terminal and dodecaphase evaluations of the common enriched state: blocks B_{term} and signed chart U_{12}^{sgn}

Let

$$x_{\mathrm{term}}\in \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,enr}}\subseteq \mathcal{X}_{\mathrm{TPK}}^{[108]}.$$

The terminal selector and the signed chart are two evaluations of the same state:

$$x_{\mathrm{term}} \xrightarrow{(\pi_{\mathrm{term}},R_{\mathrm{sgn}})} \big(B_{\mathrm{term}},U_{12}^{\mathrm{sgn}}\big), \tag{31.57}$$

with

$$B_{\mathrm{term}}=\begin{pmatrix}021020\\001220\\100210\end{pmatrix}$$

and

$$U_{12}^{\mathrm{sgn}}=(2378,1406,2479,-452,998,-551,-668,-204,-371,-322,-28,-997).$$

The first reading retains the three selected terminal blocks; the second retains the signed integer coordinate required for dodecaphase closure. No factorization from B_{term} to U_{12}^{sgn} exists or is needed. Drawing

one would erase precisely the fields of orientation, carry, boundary, and ledger that distinguish the two readings. From the second, however, the reversible charts have been constructed

$$U_{12}^{\mathrm{sgn}} \xleftrightarrow{\mathcal{H}_{12}} K \xleftrightarrow{\mathcal{E}_{\mathrm{dod}}} (D_3K, D_4K, Q), \qquad \mathcal{E}_{\mathrm{dod}} = (D_3, D_4, Q).$$

The domain of this closure is therefore the enriched TPK profile; a raw trace of 108 positions is a forgetful projection and does not replace that domain.

31.5.13 Reversible generation of α_{HMT} through the integral dodecaphasic closure

The Archimedean evaluations of the three channels and the canonical seal are the twelve-block vectors

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884),\\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497),\\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117),\\ K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \end{aligned}$$

With the declared right boundary $c_{13} = 0$, Euclidean division propagates from right to left:

$$\begin{aligned} s_m &= P_m + E_m - \Phi_m - K_m + c_{m+1},\\ A_m &= s_m \bmod 1000,\\ c_m &= \frac{s_m - A_m}{1000}, \qquad m = 12, 11, \ldots, 1. \end{aligned} \tag{31.58}$$

The output is

$$A = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 663)$$

and fixes the rational

$$\alpha_{12} = 0.007297352569283800997285105472380663. \tag{31.59}$$

The abbreviated equality $P + E - \Phi - K = A$ must be read in the chart of integer concatenation. If

$$\iota(v_1, \ldots, v_{12}) = \sum_{m=1}^{12} v_m 1000^{12-m},$$

then

$$\boxed{\iota(P) + \iota(E) - \iota(\Phi) - \iota(K) = \iota(A)}. \tag{31.60}$$

In coordinates, the cocycle cannot be omitted:

$$P + E - \Phi - K + C^{+} - 1000C = A, \qquad C^{+} = (c_2, \ldots, c_{13}).$$

The termination $\ldots|380|663$ belongs to this finite window and its condition $c_{13} = 0$. The complete theorem, the table of the twelve carries, and the inversion that recovers K reside in the preceding chapter; they are not duplicated here.

This route is an exact internal result with explicit starting structure: it receives x_{term}, its signed reading, P, E, Φ, the reversible seal, and the boundary. It does not receive α or CODATA as input. Metrological comparison opens only after (31.59) has been obtained.

31.5.14 Simple root ξ_9 of the analytic kernel of order 9 and exact comparison $\mathrm{Tr}_9(\xi_9^{-1}) = \mathrm{Tr}_9(\alpha_{\mathrm{HMT}}^{-1})$

The second route does not invert the charts of K. It starts from the exact density of vacancy

$$\varepsilon_{\mathrm{v}} = \log_{10}\left(\frac{10}{9}\right)$$

and of the declared analytic kernel

$$\mathcal{P}(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}. \tag{31.61}$$

The problem is to solve

$$\mathcal{P}(x) = \log_{10}\left(\frac{10}{9}\right), \qquad 0 < x < 10^{-2}. \tag{31.62}$$

The theorem of monotonia of the nucleo istablece the uniqueness of the root positive and the encierro

$$0.00729735256928379955 < x_{21/22} < 0.00729735256928379957.$$

Its inverse is

$$x_{21/22}^{-1} = 137.035999084000027020090128259874 9\ldots. \tag{31.63}$$

Consequently, the full kernel $E_{21/22}$, generated by the second reader of the same enriched state, determines, as a simple and unique positive root, the same limit section as the dodecaphasic path:

$$\boxed{\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}}, \qquad \alpha_B = \operatorname*{arg\,zero}_{x \in I_\alpha} E_{21/22}(x).$$

The finite rational α_{12} and the root of the jet $\mathcal{P} = P_6$ are finite projections of the two compatible families; their difference beyond the printed resolution does not constitute a difference between the two limit sections. The nested root intervals and the commutative truncation squares coordinate both prolongations at every depth.

The six coefficients of P_6 are not reconstructed from the root or taken from a metrological table: they are the first jet of the kernel $E_{21/22}$ produced by the internal reader with genealogy $120/45/11/495$ and prolongation $T_{7:9}$. The uniqueness of the root and the internal derivation of the coefficients are distinct stages of the same already closed path B.

31.5.15 Prolongation coinductive $\mathsf{T}_{\alpha,\infty}$ of the entire dodecaphasic closure

$\mathsf{T}_{\alpha,\infty}$ extends the recurrence of (31.58) to a chain of length N. To avoid an ambiguity of indices, let $\kappa(k) = 1 + ((k-1) \bmod 12)$. With a terminal condition at the remote endpoint one computes

$$\begin{aligned} s_k &= P_k + E_k - \Phi_k - K_{\kappa(k)} + c_{k+1}, \\ A_k &= s_k \bmod 1000, \\ c_k &= \left\lfloor \frac{s_k}{1000} \right\rfloor, \qquad k = N, N-1, \ldots, 1. \end{aligned} \tag{31.64}$$

The finite case $N = 1000$ uses one thousand triads, that is, three thousand digits. The carry arriving from the remote end modifies the twelfth visible block:

$$\begin{aligned} \text{short dodecaphasic window, } c_{13} = 0 \quad &: \quad \ldots|380|663, \\ \text{extension } \mathsf{T}_{\alpha,\infty} \text{ of one thousand triads} \quad &: \quad \ldots|380|662|999\ldots. \end{aligned}$$

They are not two rival values. They are two boundary semantics of the same carry law: the first closes at position twelve; the second transports to it information proceeding from a remote tail.

The reproducible calculation verifies the long word exactly using P, E, Φ as correlated projections of the same joint APP–TRIT–TPK coinductive state that also publishes the dodecaphasic coordinate α_{HMT}. Therefore, that execution certifies carry propagation: it does not place π, e, φ earlier in the genealogy or add α from outside. The joint projection

$$\mathcal{G}_9^{\mathrm{joint}} \longrightarrow (\pi, e, \varphi, \mathsf{T}_{\alpha,\infty})$$

factors through the section 31.6: the three Archimedean channels and the dodecaphasic channel open from the same enriched state; (31.64) is then executed and the independence of the prefix from the five terminal carry conditions is verified. The separation of projections prevents the long recurrence from retrospectively selecting its initial data.

31.5.16 Causal coordination of dodecaphase closure and the $E_{21/22}$ characterization: domains, codomains, and uniqueness

The classification of this block is fixed as follows:

Generative branch A: dodecaphasic closure.
It receives the enriched terminal state, the signed chart, the seal, the three evaluations, and the explicit boundary; it produces the rational (31.59) by means of the integer identity with carry.

analytic-characterization branch: equation $E_{21/22}$.
For the declared kernel, monotonicity and the sign change determine a unique root in $(0, 10^{-2})$. The root does not retrospectively determine the six coefficients of the kernel.

Prolongation $\mathsf{T}_{\alpha,\infty}$.
The long recurrence receives the thousand reference triads, the periodic seal, and the remote boundary, and transports the carry to the visible window. It does not constitute the analytic-characterization branch.

$\mathcal{G}_9 \to \mathsf{T}_{\alpha,\infty}$ composition.
Joint rational arithmetic produces ten thousand digits, executes the carry recurrence, and then verifies prefix stability under the declared terminal conditions.

The terminal selector, the reversible charts, and the short recurrence are proved in section 30.12; the existence and uniqueness of the second route in (31.61)–(31.63); and the long composition in section 31.6. The computational supplement reproduces the finite enumerations and terminal conditions; the proof rests on the identities and recurrences presented here.

The public causal order is thus closed:

$$x_{\mathrm{term}} \xrightarrow{(\pi_{\mathrm{term}}, R_{\mathrm{sgn}})} (B_{\mathrm{term}}, U_{12}^{\mathrm{sgn}}) \longrightarrow \begin{cases} \text{path A: dodecaphasic integer closure,} \\ \text{path B: kernel root } E_{21/22}, \end{cases}$$

with $\mathsf{T}_{\alpha,\infty}$ as remote prolongation of the generative branch and not as an additional branch. Only after these outputs is external metrological recognition opened.

31.6 Composition of the holonomy G_9 with the chart T_1 and arbitrary-precision certificate for α_{HMT}

The T1 prolongation applies the carry recurrence to the triads P_k, E_k, Φ_k previously produced by G9 and to the declared dodecaphasic coordinate K:

$$s_k = P_k + E_k - \Phi_k - K_{k \bmod 12} + c_{k+1}, \tag{31.65}$$

$$A_k = s_k \bmod 1000, \tag{31.66}$$

$$c_k = \left\lfloor \frac{s_k}{1000} \right\rfloor. \tag{31.67}$$

The input domain contains the ternary outputs generated by G9; it does not contain a decimal expansion of α, a value of α^{-1}, CODATA, or a target metrological condition.

The cylinders of 3501 blocks fix 10020 guard digits in each channel, equivalent to 3340 triads. The calculation propagates, from right to left, all possible terminal conditions in the stable set

$$C = \{-2, -1, 0, 1, 2\}.$$

The five branches coalesce in a common prefix of 3339 triads, that is, 10017 digits. Consequently, the first ten thousand digits of α_{HMT} are independent of the carry situated beyond the guard zone. The published output does not impose $c_N = 0$.

Causal independence can be checked after the construction: the first three thousand digits agree with the decimal expansion obtained by the external reader, but none of them enters (31.67). The ten-thousand-digit expansion and the finite data that make it possible to repeat the calculation are presented in the appendix E.

31.7 Two internal readers of α: dodecaphase closure and order-9 nonadic analytic kernel

This section develops the coordinate α of the tesis inaugural. Both readers belong to the same APP–TRIT–TPK genealogy and precede metrological comparison. Route A operates through the reversible dodecaphase closure of the signed state and publishes the coinductive section α_{HMT}. Route B constructs, without consulting that output, the exact order-nine nonadic analytic kernel $E^{(9)}_{21/22} := P_9 - \log_{10}(10/9)$ and publishes its simple and unique positive root ξ_9. Their domains, operations and certificates remain distinct:

$$X^{\mathrm{sig}}_{12} \xrightarrow{\Gamma^{\alpha}_{12}} \alpha_{12}, \qquad E^{(9)}_{21/22} : I_\alpha \longrightarrow \mathbb{R}, \quad E^{(9)}_{21/22}(\xi_9) = 0, \quad (E^{(9)}_{21/22})'(\xi_9) \neq 0. \tag{31.68}$$

The integer closure first publishes the finite window α_{12} and its compatible family of prolongations; its inverse limit is α_{HMT}. The second reader publishes ξ_9 as a finite order-nine witness and, by iteration of the same graded transport of the TPK, constructs the complete compatible system $E^{(6+3m)}_{21/22}$. The two inverse systems determine at each depth the same surviving cylinder C^{α}_{N}; their diameters tend to zero. Therefore, the intersection is a singleton and the two-route theorem establishes

$$\alpha_A := \varprojlim_{N} \rho_N(\alpha_{12}), \qquad E_{21/22} := \varprojlim_{m} E^{(6+3m)}_{21/22}, \qquad \alpha_B := \operatorname*{arg\,zero}_{x \in I_\alpha} E_{21/22}(x), \qquad \boxed{\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}}. \tag{31.69}$$

Neither route introduces a target value as input data; metrological identification belongs solely to subsequent conventional recognition. The equality $\mathrm{Tr}_9(\xi_9^{-1}) = \mathrm{Tr}_9(\alpha_{12}^{-1}) = 137.035999084$ is a finite certificate of the preceding identity, never its definition or maximum scope. The prolongation law is not added as a datum: it is the iteration of the graded transport already produced by APP–TRIT–TPK, with phase return, memory advance and compatible truncations.

31.7.1 Analytic kernel P_6 and exact order-9 prolongation

The characterization of vacancies employs

$$\begin{aligned} P_6(x) &= 2\pi_{\mathrm{HMT}}x - \frac{7}{4}x^2 + \frac{x^3}{2\pi_{\mathrm{HMT}}} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \\ P_6(x) &= \log_{10}\frac{10}{9}. \end{aligned} \tag{31.70}$$

The certified positive root of this jet satisfies

$$x_6^{-1} = 137.035999084000027020 09\ldots. \tag{31.71}$$

The polynomial P_6 provides the initial jet of the analytic characterization. Nonadic holonomy prolongs its degrees $7, 8, 9$ exactly and the same graded rule is iterated over successive degrees. The jet quotient is the first finite certificate of that coinductive prolongation; it does not select or replace its limit section.

31.7.2 APP–TRIT–TPK genealogy of the invariants $120, 45, 11$ and 495

The phase semicorona of the TPK constructs

$$|\mathcal{F}_8^{\mathrm{ph}}| = 120, \qquad |L_{\mathrm{ph}}| = 45. \tag{31.72}$$

APP provides the central block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \qquad \det B_c = 45, \qquad \mathrm{Spec}(B_c) = \{9, 5\}. \tag{31.73}$$

The dodecaphase transverse extractor provides

$$r_{\mathrm{dod}} = \mathrm{rank}(D_3, D_4) = 11, \qquad (D_3K)_1 = -495, \qquad 495 = 45 \cdot 11 = \det(B_c)\, r_{\mathrm{dod}}. \tag{31.74}$$

The coefficients of the prolonged jet therefore proceed from three prior invariants of APP–TRIT–TPK:

$$-\frac{1}{120} \longmapsto \frac{1}{45} \longmapsto \frac{2}{45 \cdot 11}. \tag{31.75}$$

31.7.3 Nilpotent resolvent $(I - N)^{-1}$ and exact construction of the tail $T_{7:9}$

On the basis e_7, e_8, e_9, we define

$$Ne_7 = -\frac{8}{3}e_8, \qquad Ne_8 = \frac{2}{11}e_9, \qquad Ne_9 = 0, \qquad v_7 = -\frac{1}{120}e_7. \tag{31.76}$$

Since $N^3 = 0$, its finite resolvent acts exactly by

$$(I - N)^{-1}v_7 = v_7 + Nv_7 + N^2v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}. \tag{31.77}$$

The evaluation $e_n \mapsto x^n$ produces

$$\boxed{T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}.} \tag{31.78}$$

31.7.4 Nonadic jet P_9, prolonged root, and scale of certified residues

The nonadic jet of order 9 is defined by

$$P_9(x) = P_6(x) + T_{7:9}(x), \qquad P_9(x) = \log_{10} \frac{10}{9}. \tag{31.79}$$

Theorem 31.19 (Existence, simplicity and uniqueness of the second analytic output)**.** *The operator $E^{(9)}_{21/22}$ has a unique positive root ξ_9 in $I_\alpha = (0, 10^{-2})$. It is enclosed by the exact rational interval*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}, \tag{31.80}$$

and its inverse satisfies

$$\xi_9^{-1} = 137.035999084000000000000111926291900\ldots. \tag{31.81}$$

Proof. All coefficients of P_9 are extracted from the invariants prior to solving. For $0 \le x \le 10^{-2}$, using $3.14 < \pi_{\mathrm{HMT}} < 3.15$, one obtains

$$(E^{(9)}_{21/22})'(x) \ge 2\pi_{\mathrm{HMT}} - \frac{7}{2}\,10^{-2} - \frac{10}{21}\,10^{-8} - \frac{6}{46}\,10^{-10} - \frac{7}{120}\,10^{-12} > 6.24.$$

The directed evaluations at the endpoints of equation (31.80) have opposite signs. The intermediate value theorem provides existence; the derivative bound provides uniqueness and simplicity. Interval division produces the expansion of ξ_9^{-1}, which constitutes the order-nine witness of the complete compatible system. □

Table 31.2: Certified descent of the residue of the analytical characterization

Nucleo evaluado in α_{12}	Residue
P_6	$9.0038886 \times 10^{-18}$
$P_6 - x^7/120$	$-1.7893115 \times 10^{-19}$
$P_6 - x^7/120 + x^8/45$	$-2.3708601 \times 10^{-22}$
P_9	$3.7297132 \times 10^{-27}$

31.7.5 Alternative dodecaphase operator A_{dod} and uniqueness control of the typed reader

The control operator

$$A_{\mathrm{dod}} = I - \frac{1}{4}(D_3^* D_3 + D_4^* D_4) \tag{31.82}$$

produces a different sequence of coefficients. This control proves that isolated dodecaphase symmetry is insufficient to select the prolongation $T_{7:9}$; selection belongs to the typed composition of the reader $\Gamma^{(9)}_{E21}$ with the enriched state. The difference between the two operators certifies the need to preserve domain, orientation, grading and memory, and proves the specificity of the finite reader. The analytic tail belongs to iteration of the complete typed reader, not to the isolated symmetry of this control operator.

31.7.6 Exact scope of the nilpotent operator and exclusion of tails foreign to TPK transport

The arithmetic of $120, 45, 11, 495$, the nilpotent operator, the resolvent and the identity equation (31.78) have internal exactness. The simple and unique positive root of $E^{(9)}_{21/22}$ constitutes the order-nine witness of the second internal reading. Agreement of P_9 with the dodecaphase window certifies the first finite square of the compatible system. Canonical uniqueness of the prolongation 7:9 is typed by the morphism

$$\Gamma^{(9)}_{E21} : X_{\text{APP-TRIT-TPK}} \longrightarrow \mathbb{Q}(\pi_{\mathrm{HMT}})[x]/(x^{10}), \tag{31.83}$$

whose action jointly determines the oriented seed, the two transfers, the grading $7, 8, 9$, the jet quotient F^7/F^{10} and its transport under phase return.

Let $K = \mathbb{Q}(\pi_{\mathrm{HMT}}, \log_{10}(10/9))$ and let $j_9 : K[[x]] \to K[x]/(x^{10})$ be the truncation. Then

$$\ker j_9 = x^{10}K[[x]], \qquad E_h(x) := E^{(9)}_{21/22}(x) + x^{10}h(x), \qquad h \in K[[x]]. \tag{31.84}$$

All E_h have the same certified jet. However, E_0 and $E_{1/729}$ have distinct roots in I_α, since $E_{1/729}(\xi_9) = \xi_9^{10}/729 > 0$. This observation proves only that an isolated jet does not determine a formal tail. It does not affect the second HMT route, because that route does not allow h to be freely chosen: the recurrence of TPK transport constructs the unique compatible family $E^{(6+3m)}_{21/22}$, and its truncations commute with those of the dodecaphase route. Choosing h through the target value would be circular and is excluded; choosing it through generated transport is precisely the operation already proved.

The falsifier thus separates three statements: it invalidates the finite certificate if monotonicity, the sign change or isolation of ξ_9 fails; it invalidates the prolongation 7:9 if its coefficients cease to factor through $\Gamma^{(9)}_{E21}$; and it invalidates the complete identity if compatibility of any square of the inverse system fails or if the diameters of the surviving cylinders do not tend to zero. None of these conditions introduces a pending morphism: they are the material falsifiers of the construction establishing $\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}$.

Chapter 32

Internal construction of $i = \sqrt{-1}$ via the rotation operator J: identity $J^2 = -I$ and elliptic geometry of phase

The internal root of -1 and the tritic exponential family

The root of -1 is not introduced as a formal symbol without an antecedent. The six units modulo nine are coordinatized over $\mathbb{P}^1(\mathbb{F}_5)$, and the quadratic character of $\mathbb{F}_5$ determines the Paley conference matrix. Its ternary reduction is

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix} \in M_6(\mathbb{F}_3).$$

The conference identity reduces to

$$\boxed{A_W^2 = 2I_6 = -I_6.} \tag{32.1}$$

Therefore, A_W is a quarter-turn operator in the ternary space. After extending scalars to $\mathbb{F}_9$, it decomposes into the conjugate eigenspaces of the roots of $X^2 + 1$. The common integral source

$$Q_{\mathbb{Z}} = \mathbb{Z}[u]/(u^2 + 1)$$

has a finite realization $u \mapsto A_W$ and a real realization $u \mapsto J_e$. The source coordinates the quadratic relation; it does not identify fields of different characteristics.

The three TRIT regimes are carried out by means of operators

$$J_{+1}^2 = -I, \qquad J_0^2 = 0, \qquad J_{-1}^2 = I.$$

Separating the even and odd powers of the exponential yields a single family:

$$\boxed{\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \qquad \tau \in \{+1, 0, -1\},} \tag{32.2}$$

where

$$(C_{+1}, S_{+1}) = (\cos, \sin), \qquad (C_0, S_0) = (1, t), \qquad (C_{-1}, S_{-1}) = (\cosh, \sinh).$$

Its three governing specializations are

$$\exp(\pi J_{+1}) + I = 0, \qquad \exp(J_0) = I + J_0, \qquad \exp\big(2(\log\varphi)J_{-1}\big) = F_{\mathrm{av}}^2. \tag{32.3}$$

The elliptic branch contains the complex circular closure; the parabolic branch, threshold transport; the hyperbolic branch, golden self-scaling. Euler's equation thus appears as a specialization of a family that coordinates rotation, boundary and hyperbolic expansion.

32.1 Elliptic realization of $J^2 = -I$ through the Hopf fibration and Möbius holonomy

32.1.1 Complex structure and Hopf fibration

Let $S^3 \subset \mathbb{C}^2$ and

$$h(z_0, z_1) = \big(2\Re(z_0\bar{z}_1), 2\Im(z_0\bar{z}_1), |z_0|^2 - |z_1|^2\big) \in S^2.$$

On the Hopf torus

$$T_\eta = \{(z_0, z_1) : |z_0| = \cos\eta, |z_1| = \sin\eta\},$$

the curves

$$\gamma_+(t) = (\cos\eta\, e^{it}, \sin\eta\, e^{it}),$$

$$\gamma_-(t) = (\cos\eta\, e^{it}, \sin\eta\, e^{-it})$$

have distinct behaviors: γ_+ is a Hopf fiber and γ_- projects with degree two. Under stereographic projection, the homology classes $(1, 1)$ and $(1, -1)$ produce the two diagonal Villarceau families.

The total complex conjugation

$$K(z_0, z_1) = (\bar{z}_0, \bar{z}_1)$$

preserves each unoriented family and reverses its orientation. It does not interchange the two families. Partial conjugation can interchange them, but it is neither a complex-linear map nor the standard global antiunitary. This correction is essential to avoid manufacturing a particle–antiparticle symmetry from the toroidal drawing.

32.1.2 Double covering and spinor return

The map $SU(2) \to SO(3)$ is a double cover. A loop of 2π in $SO(3)$ can lift to $-I$ in $SU(2)$; the loop of 4π recovers I. The sign structure does not turn a circle into a Möbius band. Obtaining a band requires a real line bundle whose holonomy character is -1.

Let $L_\chi \to S^1$ be the associated bundle to

$$\chi : \pi_1(S^1) \longrightarrow \{\pm 1\}, \qquad \chi(1) = -1.$$

Its total space is a Möbius band. The route $C_M = -\,\mathrm{rev}$ provides the sign candidate; the Villarceau circle provides the loop; the physical intertwiner must prove that the former is the holonomy of the latter.

32.1.3 Toroidal realization of the operator $B^{\varepsilon}_{\mathrm{mem}}$ and determination of the Villarceau angle

With $B_c = 7I + 2S$, define the oriented memory operator

$$B^{\varepsilon}_{\mathrm{mem}} = 5I + 2\varepsilon S, \qquad \varepsilon = \pm 1.$$

For $\varepsilon = +1$,

$$\mathrm{Spec}(B^{+}_{\mathrm{mem}}) = \{7, 3\}, \qquad \det B^{+}_{\mathrm{mem}} = 21.$$

Interpreting the values $(7, 3)$ as the outer and inner equatorial boundaries in a common positive chart yields

$$R + r = 7\ell, \qquad R - r = 3\ell,$$

and therefore

$$R = 5\ell, \qquad r = 2\ell, \qquad \sin\beta = \frac{r}{R} = \frac{2}{5}.$$

The Villarceau angle is

$$\beta = \arcsin\frac{2}{5} = 23.5781784782\ldots^\circ.$$

The algebraic theorem $\{7, 3\} \leftrightarrow (5, 2)$ is exact. The geometric premise interpreting the spectrum as the two boundaries of a torus is a declared interface. It must not be concealed, nor may the angle be used as a target.

The realization of the full block B_c retains another reading: $R : r = 7 : 2$ and $\beta = \arcsin(2/7)$. These are not rival values; they belong to two functors: total block and memory mode. The physical priority of the latter must be decided by the intertwiner with the dynamics, not by decimal proximity to another constant.

32.1.4 Action of the operator $J^2 = -I$ on the conjugate angular coordinates

HMT's native angle is a phase $u \in \mathbb{R}/\mathbb{Z}$. The readers publish

$$\theta_{\text{rad}} = 2\pi u, \qquad \theta_{\text{deg}} = 360u.$$

Radians and degrees are distinct charts of the same circular element. The radian is not more "ontological" by virtue of being dimensionless; its naturality follows from arc length. Degrees provide another absolute coordinate of the full turn. Any relation with α^{-1}, φ, or CKM must first be formulated as an invariant quotient or holonomy and only thereafter evaluated in an angular chart.

The exact identity already available

$$\tanh^2(2 \log \varphi) = \frac{5}{9}$$

connects the Perron–Fibonacci channel to the spectral quotient of B_c. Its complement is $4/9$, the normalized gap. This relation is structural. It does not imply that every angle constructed with φ is a physical angle.

32.1.5 Brouwer, Möbius, and degree

Brouwer's fixed-point theorem, fractional linear Mobius transformations, and the Mobius band are three distinct objects. They are related only after fixing domain, action, and bundle. The elliptic advance and retardation orbits may be modeled as two sections of a bundle with sign holonomy; the Mobius transformation describes the conformal action in the chart; Brouwer's theorem guarantees a fixed point for continuous maps of the disk. None of these statements replaces the realization diagram.

Remark 32.1 (Result and boundary). The double degree of the Hopf curve, the orientation of total conjugation, and the algebraic construction $7, 3 \mapsto 5, 2$ are proved. The toroidal reading (5:2), the Möbius holonomy, and the particle–antiparticle interpretation form a chain of interfaces that must share a single intertwiner. Volume does not force that intertwiner through an angular coincidence.

Chapter 33

Hensel–Witt derivation of $-1/12$: compatible prolongation, regularization, and modular memory

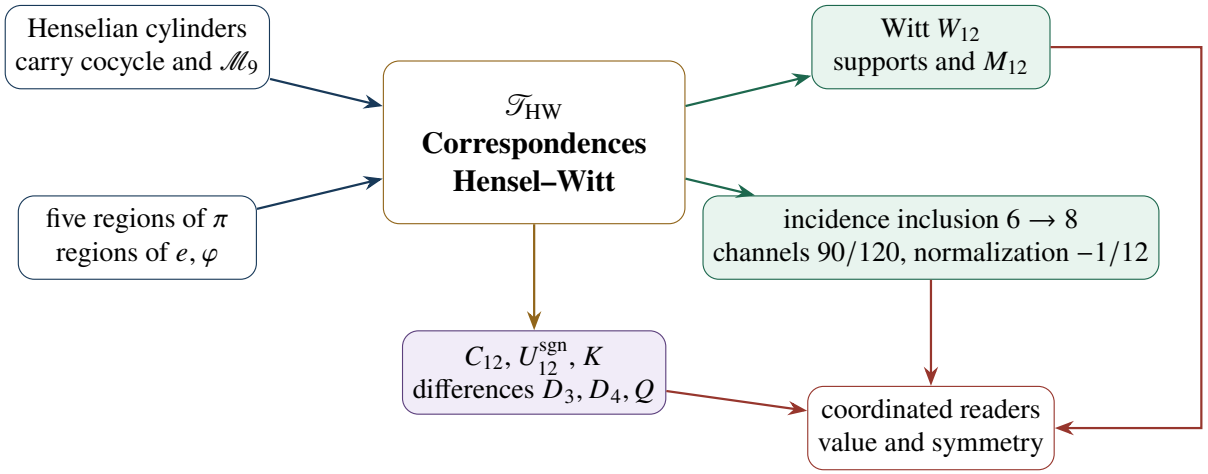
33.1 Hensel–Witt transduction operator between microregions of π, holonomy Γ_9, the dodecaphasic register and hexad–octad incidence

The NTHW is the data and maps package

$$\mathfrak{N}_{\mathrm{HW}} = \left(\mathscr{R}_{\pi}^{\mathrm{micro}}, \mathcal{G}_9, C_{12}, K, \mathcal{W}_{12\to 24}, \mathcal{I}_{6\to 8}\right),$$

with the following compatibilities:

1. the Hensel regions share a quotient word and retain distinct memory;
2. $\mathcal{G}_9$ transports orientation and produces monodromy;
3. C_{12} organizes the signed observable and the seal K;
4. the Witt reading converts supports of the same boundary into hexads and stars;
5. the extension $W_{12} \to W_{24}$ produces the hexad–octad interface;
6. the Archimedean, exceptional and dimensional readers receive invariants of the same state.



Compatibility of cylinders, dodecaphase, and incidences on the same enriched state.

Figure 33.1: The NTHW as a typed interface. The central mathematical task is to make the diagram commute, not to accumulate coincidences around a name.

NTHW gives a formal name to the place previously described as the "Rosetta Stone." The substitution is conceptual: a stone translates through resemblance; a transduction specifies input, output, preserved datum and loss of information.

33.2 Construction of the Witt system from the compatible ternary coding

The reversible APP boundary selects, with declared gauges, a matrix A_W over $\mathbb{F}_3$ satisfying

$$A_W A_W^T = 2I \pmod 3.$$

The graphic code

$$C_W = \{(q, qA_W) : q \in \mathbb{F}_3^6\}$$

is the extended ternary Golay code. Its weight enumerator is

$$1 + 264y^6 + 440y^9 + 24y^{12}.$$

The 264 words of weight six, identified up to sign, yield 132 supports. Every five-point subset belongs to a unique hexad:

$$W_{12} = S(5, 6, 12).$$

The field of 495 star involutions generates a group of order 95 040, recognized as M_{12}. An isolated star does not generate the group. The ordered HMT frame has trivial stabilizer and therefore fixes a gauge within the action.

33.3 Hexad–octad realization of the 90/120 channels previously derived from the tritic incidence of the TPK

Let H be a hexad and O_H an octad containing it. Marking one point and a pair of points of the hexad yields

$$F_6(H) = H \times \binom{H}{2}, \qquad |F_6| = 6\binom{6}{2} = 90.$$

If the marked point can traverse the octad:

$$F_8(H) = O_H \times \binom{H}{2}, \qquad |F_8| = 8\binom{6}{2} = 120.$$

The inclusion $H \hookrightarrow O_H$ induces $F_6 \hookrightarrow F_8$. Its normalized defect is

$$\boxed{\frac{90 - 120}{24\binom{6}{2}} = \frac{6 - 8}{24} = -\frac{1}{12}.}$$

This combinatorial theorem precedes Barbero's angular reading. The two objects share the interface $6 \to 8$, but are not identical.

33.4 4 operational realizations of −1/12 and separation of their domains

The construction contains four realizations that must be presented together, not counted as four independent proofs of a single provenance.

33.4.1 Angular realization through the elementary increment dodecaphasic

One tooth of C_{12} measures 30°. Then

$$\frac{30}{120} - \frac{30}{90} = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}.$$

It is the angular shadow of the pattern 90/120.

33.4.2 Oriented pseudoinverse of the Gram operator on the surviving subspace

In the certified jump $60 \to 81$, two fibers have sizes 0 and 12. The Gram operator is

$$K_{60,81} = \operatorname{diag}(0, 12).$$

On the live subspace, its pseudoinverse is $1/12$; with the negative boundary orientation,

$$E_{\mathrm{surv}} = -K^{+}_{60,81}$$

yields $-1/12$.

33.4.3 Scale extraction by the 2nd triadic difference

The realization via a triadic scale extractor is the second-difference construction developed in full in section 33.5.3. This first occurrence establishes its role within the inventory of realizations; the indicated residence preserves the definitions of T_L, $a(L)$, $A_1(L)$, $C(L)$, the independence of the residue from L, and the exact status of the normalization.

33.4.4 Modular realization through the quotient $\eta(\tau)/\eta(3\tau)$ and the function t_3

The eta function satisfies

$$\frac{\eta(\tau)}{\eta(3\tau)} = q^{-1/12} \prod_{n\geq 1} (1 + q^n + q^{2n})^{-1}.$$

The exponent comes from $1/24 - 3/24 = -1/12$. Taking twelve copies:

$$t_3(q) = \left(\frac{\eta(\tau)}{\eta(3\tau)}\right)^{12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \cdots .$$

The coefficient 54 coincides with the APP defect. The corridor

$$j = \frac{(t_3 + 27)(t_3 + 243)^3}{t_3^3}$$

connects afterwards with the classical modular branch.

Remark 33.1 (What Has Been Proved in the Corridor). The APP defect 54, the eta exponent $-1/12$, the hexad–octad defect, and the modular identities once t_3 has been fixed are exact. The strong assertion is that all of them are natural images of a single TPK operator. That naturality must be proved by means of the NTHW diagram; it does not follow merely from repeating the same integers.

33.5 5 algebraic realizations of $-1/12$ and separation of their domains

In addition to the normalized defect of the hexad–octad inclusion, five operator realizations of the same rational number appear. Their domains are specified separately; equality of values does not identify the operators without an additional intertwiner.

33.5.1 Difference of dodecaphasic periodicities

One tooth of C_{12} measures 30°. Then

$$\frac{30}{120} - \frac{30}{90} = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}.$$

It is the oriented difference between the periods of 90° and 120° in this realization.

33.5.2 Pseudoinverse of the Gram Operator on the Live Component

In the exact jump $60 \to 81$, two fibers have sizes 0 and 12. The Gram operator is

$$K_{60,81} = \operatorname{diag}(0, 12).$$

On the nonzero component of its range, the pseudoinverse is $1/12$; with the negative boundary orientation,

$$E_{\mathrm{surv}} = -K^{+}_{60,81}$$

yields $-1/12$.

33.5.3 2nd difference between triadic scales

Define

$$T_L = \sum_{n=1}^{L-1} n(L-n) = \frac{L(L^2-1)}{6}, \qquad a(L) = -\frac{T_L}{L^2} = -\frac{L}{6} + \frac{1}{6L}.$$

The first scale subtraction is

$$A_1(L) = a(L) - \frac{a(3L)}{3} = \frac{4}{27L},$$

and the second

$$C(L) = LA_1(L) - \frac{3L}{3}A_1(3L) = \frac{8}{81}.$$

The residue $8/81$ is independent of L. To obtain $-1/12$, the declared normalization is applied

$$R(L) = -\frac{27}{32}C(L) = -\frac{1}{12}.$$

The calculation derives $8/81$; it does not by itself derive the factor $-27/32$.

33.5.4 Cadena modular

The eta function satisfies

$$\frac{\eta(\tau)}{\eta(3\tau)} = q^{-1/12}\prod_{n\geq 1}(1+q^n+q^{2n})^{-1}.$$

The exponent comes from $1/24 - 3/24 = -1/12$. Taking twelve copies:

$$t_3(q) = \left(\frac{\eta(\tau)}{\eta(3\tau)}\right)^{12} = q^{-1} - 12 + 54q - 76q^2 - 243q^3 + \cdots.$$

The coefficient 54 coincides with the APP defect. The rational identity

$$j = \frac{(t_3+27)(t_3+243)^3}{t_3^3}$$

establishes the connection with the classical modular branch.

33.5.5 Defect of cyclic difference projectors

Let S be the cyclic shift on $\mathbb{Q}^{12}$ and define

$$D_r = I - S^r.$$

The kernel of D_r consists of the vectors constant on each orbit of S^r; therefore,

$$\dim \ker D_r = \gcd(12, r).$$

Let $\Pi_{\ker D_3}$ and $\Pi_{\ker D_4}$ be the rational projectors onto $\ker D_3$ and $\ker D_4$, and let $\tau(T) = \mathrm{Tr}(T)/12$ be the normalized trace. Then

$$\boxed{\tau(\Pi_{\ker D_3} - \Pi_{\ker D_4}) = \frac{\operatorname{rank} \Pi_{\ker D_3} - \operatorname{rank} \Pi_{\ker D_4}}{12} = \frac{3-4}{12} = -\frac{1}{12}.}$$

This realization expresses the rational as the transverse defect of steps three and four in the twelve-phase cycle.

Remark 33.2 (Domains of the realizations). The APP defect 54, the eta exponent $-1/12$, the combinatorial defect of the hexad–octad lift, and the modular identities once t_3 has been fixed are exact. The value of the pseudoinverse, the normalized extractor, and the projector defect are likewise exact. Each equality preserves its domain and its corresponding map.

Chapter 34

Determinantal derivation of the absolute action scale: Smith normal form, cokernel of order 16 and valuation 10^{-34}

34.1 Derivation of the action scale from the dodecaphasic state

The action decade is determined intrinsically, without adopting it as an inherited unit. The local determinantal functional is applied to the previously constructed HMT output α_{HMT} and defines

$$\Sigma_H = \varphi_{\mathrm{HMT}} \alpha_{\mathrm{HMT}}^{16}. \tag{34.1}$$

The exponent 16 is the order of the local cokernel of the signed closure; its decimal valuation fixes

$$\mathrm{ord}_{10}(\Sigma_H) = -34. \tag{34.2}$$

The origin of the exponent can be verified within this volume. On the local orbit of four leaves, the Hadamard block acts

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{34.3}$$

Lemma 34.1 (Local index of the action line)**.** *The form normal of Smith of H_4 is*

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4). \tag{34.4}$$

Consequently,

$$\mathrm{coker}(H_4) \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \qquad |\mathrm{coker}(H_4)| = 16.$$

Proof. The greatest common divisor of the entries is one; that of the nonzero 2×2 minors is two; that of the 3×3 minors is four; and $|\det H_4| = 16$. The successive quotients of these determinantal divisors are $1, 2, 2, 4$, which prove equation (34.4). The product of the invariants gives the order of the cokernel. □

The norm of the section α_{HMT} over those sixteen sheets produces $\alpha_{\mathrm{HMT}}^{16}$; the self-scaling φ_{HMT} acts only once on the basis, not once per sheet. Hence equation (34.1). The certified comparisons

$$\alpha_{\mathrm{HMT}}^{16} < 10^{-34} < \varphi_{\mathrm{HMT}} \alpha_{\mathrm{HMT}}^{16} < 10^{-33} \tag{34.5}$$

prove, without introducing the SI unit, that the decimal order is exactly -34. The terminal evaluation begins with

$$\Sigma_H = 1.046243838104756580184229208303089\ldots \times 10^{-34}. \tag{34.6}$$

The substitution of the kernel by three copies with exponent 16^3, or the repeated application of φ on each sheet, would alter the genealogy. Both operations constitute falsification criteria rather than equivalent conventions. The causal chain that is preserved is therefore,

$$\mathrm{APP} \to \widehat{\mathbb{T}} \to \mathrm{TPK} \to U_{12}^{\mathrm{sign}} \to \alpha_{\mathrm{HMT}} \to \Sigma_H \to 10^{-34} \mathcal{U}_S \to \{\hbar_{\mathrm{pre}}, \hbar_{\mathrm{ret}}\}.$$

This volume does not rederive α_{HMT}, but develops the metrological geometry subsequent to the selection of the type and decade by Σ_H. The integer -34 is neither a mantissa nor an adjustment to the International System, but the terminal valuation of the local functional.

34.2 Oriented sections of the action line

The realizations before and after the return constitute two sections of the same action line:

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S. \tag{34.7}$$

The common decade determines the line, while the oriented information is expressed through

$$R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}}, \qquad \xi_{\mathrm{act}} = \frac{1}{2}\log R_{\mathrm{act}}. \tag{34.8}$$

The first quantity provides a multiplicative coordinate and the second an additive coordinate of rapidity type. They do not represent two independent Planck constants, but the two orientations of the same section with respect to the return.

If $h_a = 2\pi\hbar_a$, $a \in \{\mathrm{pre}, \mathrm{ret}\}$, a quantity proportional to $h_a^{1/2}$ changes as $R_{\mathrm{act}}^{1/2}$, one proportional to $h_a^{-1/2}$ changes as $R_{\mathrm{act}}^{-1/2}$, and a quotient in which the action cancels is a bidirectional invariant.

34.3 Causal bifurcation subsequent to α_{HMT}

The genealogy of the action and the angular pair forms a bifurcated diagram. The regional branch produces the reduced mantissa η_{ret}; the determinantal branch produces the decimal order $\varepsilon_H = -34$. Both converge on the action line. The angular pair is constructed in a sister branch from α_{HMT} and η_{ret}:

$$\begin{array}{ccccccc}
\alpha_{\mathrm{HMT}}, \pi, \varphi & \longrightarrow & H_5, \mathscr{C}_\pi & & & & \alpha_{\mathrm{HMT}}, \varphi, H_4 \\
\downarrow & & \downarrow & & & & \downarrow \\
x_A, D_A, \rho_\pi & \longrightarrow & \eta_{\mathrm{ret}} & \longrightarrow & \hbar_{\mathrm{ret}} \in \mathcal{L}_S & \longleftarrow & \Sigma_H, \varepsilon_H = -34 \\
 & & \downarrow & & & & \\
 & & (A_{\mathrm{deg}}, C^*_{\mathrm{deg}}) & \longrightarrow & (A_{\mathrm{rad}}, C^*_{\mathrm{rad}}) & \longrightarrow & (q_+, q_-) \longrightarrow V_{90,120}.
\end{array} \tag{34.9}$$

The expository order places the Planck derivation before the complete angular publication; the diagram simultaneously preserves the exact mathematical dependence of each output.

34.4 Local cokernel and derivation of the absolute order 10^{-34}

The Smith normal form of the local dodecaphase block is

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4), \qquad \mathrm{coker}(H_4) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}. \tag{34.10}$$

The local cokernel has order 16. For the constant section $s_\alpha(g) = \alpha_{\mathrm{HMT}}$, the local norm and an action of φ produce

$$\boxed{\Sigma_H = \varphi \mathcal{N}_{G_H}(s_\alpha) = \varphi\alpha_{\mathrm{HMT}}^{16}.} \tag{34.11}$$

The exact inequalities

$$\alpha_{\mathrm{HMT}}^{16} < 10^{-34} < \varphi\alpha_{\mathrm{HMT}}^{16} < 10^{-33} \tag{34.12}$$

determine

$$\boxed{\varepsilon_H = \lfloor \log_{10} \Sigma_H \rfloor = -34.} \tag{34.13}$$

Chapter 35

Regional character of degree 5, construction of η_{ret} and positive action measure on the emission quotient

The preceding section isometrically identifies a rank-three positive sector in the oriented interface $t = 7$, the dodecaphase, and the Witt incidence module. The following construction derives the other coefficient of the angular law, $5/468$, and proves why the two terms combine through a positive sum. The question requires three operations to be distinguished:

1. count microscopic presentations;
2. count distinct emissions after forming the TPK quotient;
3. assign a action additive to a path.

The first two operations produce different measures. The third is characterized by a universal property, not by the subsequent proximity of a decimal value.

We shall denote by Γ_{tr} the directed graph of 1944 states of the enriched-path reader. Its free path category will be written $\mathsf{P}(\Gamma_{\mathrm{tr}})$. The local definition of its edges and their action is developed in the corresponding section.

35.1 Seed space, quotient of emissions and regional fibers

Let

$$\Omega_{\mathrm{seed}} = \big((\mathbb{Z}/9\mathbb{Z})^2 \times D_4\big)^2, \qquad |\Omega_{\mathrm{seed}}| = 104\,976,$$

the executable set of seed pairs of the finite sender, and let

$$F : \Omega_{\mathrm{seed}} \twoheadrightarrow \mathcal{U}_6, \qquad |\mathcal{U}_6| = 468,$$

the application that preserves the distinct sextuple emission. The exact count of fibers is

$\lvert F^{-1}(U)\rvert$	#{U of that size}	supplied seeds
144	432	62 208
432	18	7 776
1 944	18	34 992.

The three contributions sum to 104 976. Therefore, F is not a regular covering: different emissions have different numbers of microscopic presentations.

The microfiber of π contains five distinct emissions. Four have multiplicity 144, and one has multiplicity 432. This datum suffices to prove that the uniform measure on seeds and the uniform measure on emissions are not the same.

35.2 2 exact measures on the quotient of emissions and words

If each seed receives weight $1/104\,976$, the image measure under F is

$$\nu_{\mathrm{seed}}(U) = \frac{|F^{-1}(U)|}{104\,976}.$$

Its possible atoms are

$$\frac{1}{729}, \qquad \frac{1}{243}, \qquad \frac{1}{54}.$$

In particular,

$$\nu_{\mathrm{seed}}(\mathscr{R}_{\pi}^{\mathrm{micro}}) = \frac{4 \cdot 144 + 432}{104\,976} = \boxed{\frac{7}{729}}.$$

If the elementary state is instead a distinct emission $U \in \mathcal{U}_6$, the diagonal algebra of observables is

$$\mathcal{D}_{468} = \ell^{\infty}(\mathcal{U}_6)$$

and its normalized count state is

$$\tau_{468}(f) = \frac{1}{468} \sum_{U \in \mathcal{U}_6} f(U).$$

Let $\Pi_{\pi}^{(468)}$ be the characteristic function of the five regions of the microfiber. Then

$$\boxed{\tau_{468}(\Pi_{\pi}^{(468)}) = \frac{5}{468}.}$$

The exact difference between both readings is

$$\frac{5}{468} - \frac{7}{729} = \frac{41}{37\,908}.$$

They are not two approaches. The first measure counts presentations; the second counts emission classes.

Theorem 35.1 (Uniqueness of the uniform measure of the catalog)**.** *The unique probability measure on $\mathcal{U}_6$ invariant under every relabeling of its* 468 *atoms is the uniform measure. Equivalently, the unique positive unital trace on $M_{468}(\mathbb{C})$ is*

$$\tau_{468}(A) = \frac{1}{468} \operatorname{Tr} A.$$

Proof. Transpositions take any atom to any other. Invariance forces all of them to have the same mass, and normalization fixes it at $1/468$. In the matrix version, unitary invariance equates the value of all rank-one projectors. Spectral decomposition and linearity reconstruct the normalized trace. □

The decisive hypothesis is explicit: the reader works with the *quotient catalogue*. The theorem does not claim that the uniform measure on seeds becomes uniform when the fibers are forgotten. The symmetry used is the naturality of the finite unlabeled object, not an ergodic property attributed without proof to the microscopic dynamics.

35.3 Canonical Hilbertization of the quotient

The relationship between seeds and emissions can be formulated unambiguously. Let

$$H_{\mathrm{seed}} = \ell^2(\Omega_{\mathrm{seed}}), \qquad H_U = \ell^2(\mathcal{U}_6).$$

The emitter determines

$$\boxed{J_F \delta_U = |F^{-1}(U)|^{-1/2} \sum_{\omega \in F^{-1}(U)} \delta_{\omega}.}$$

Distinct fibers are disjoint and each column has norm one; therefore,

$$J_F^* J_F = I_{H_U}.$$

Moreover, for every observable f of the catalogue,

$$M_{f \circ F} J_F = J_F M_f.$$

The image $J_F H_U$ is exactly the subspace of functions that are constant on each fiber.

If $F^\sharp : H_U \to H_{\mathrm{seed}}$ denotes the map induced by precomposition, without normalization, then

$$J_F = F^\sharp \big((F^\sharp)^* F^\sharp\big)^{-1/2}.$$

It is the isometric factor of the polar decomposition of that map. This expression proves its naturality under isomorphisms and relabelings of the emitter. It does not imply a false functoriality under arbitrary composition of nonregular surjections: the normalization depends on the fibers of the specific map.

In the language of measures, the elevation of the quotient trace that is uniform within each fiber equals

$$\widetilde{\tau}(\omega) = \frac{1}{468\,|F^{-1}(F\omega)|}.$$

Each fiber receives total mass $1/468$. The inverse compensation by multiplicity, and not an informal invocation of orbits, is the exact proof.

Corollary 35.2 (Canonicity of $5/468$)**.** *Once the regional reader is defined on distinct emissions, the mass of the microfiber of π is forced:*

$$\tau_{468}(\Pi_\pi^{(468)}) = \frac{\operatorname{rank} \Pi_\pi^{(468)}}{468} = \frac{5}{468}.$$

35.4 Positive defect between the regional and dodecaphasic measures

Write

$$H_{\mathrm{ang}} = H_U \otimes V_{11}.$$

The internal scalar

$$\Delta_4 = \frac{\pi - e \log \pi}{270} \left(1 + \frac{\pi}{729}\right) = 0.0001111964226921402\ldots$$

is constructed with the already fixed HMT readings of π and e, and with the integers 270 and 729 of the ledger. Here it is a pre-dimensional mathematical input: neither units nor physical observables intervene.

The internal construction of the scalar does not by itself determine which operator must multiply. To keep this premise visible, we declare:

Def**— sheet-change defect.** The dodecaphase coordinate of the sheet-change defect is $\Delta_4 P_3$.

Addwill fix the additive form of the action but does not select Def: for any $c \geq 0$, the same argument would admit the pair $(\Pi_\pi^{(468)}, cP_3)$. Proving Defdirectly from a total transition of the enriched topological phase kernel is an additional condition.

We define the Kronecker sum

$$\boxed{D_{\mathrm{switch}}^{\mathrm{dod}} = \Pi_\pi^{(468)} \otimes I_{11} + \Delta_4 I_{468} \otimes P_3.}$$

The two summands are positive, commute, and act on different factors. Therefore,

$$D_{\mathrm{switch}}^{\mathrm{dod}} \geq 0.$$

The intertwiner U_W of the theorem 20.3 produces the Witt chart:

$$\begin{aligned} D_{\mathrm{switch}}^{\mathrm{Witt}} &= (I_{468} \otimes U_W) D_{\mathrm{switch}}^{\mathrm{dod}} (I_{468} \otimes U_W)^* \\ &= \Pi_\pi^{(468)} \otimes I_{E_{30}} + \Delta_4 I_{468} \otimes Q_3. \end{aligned}$$

Thus, the same positive operator is read in the dodecaphase and in the incidence module.

The oriented isometric chain also identifies the active sectors.

$$I_{L_{\mathrm{or}}} \otimes P_G, \qquad P_3, \qquad Q_3.$$

No equivalence of the complete complements of the three modules is asserted: W_{GK} is a partial equivalence of the selected ranges. This is exactly the information required to identify the second contribution to the trace.

Theorem 35.3 (Exact Scope of Entanglement and Action Induction)**.** *Let W_{GK} and U_W be the partial isometries of the theorem 20.7. Then:*

1. *for every $c \geq 0$,*

$$W_{GK}^{*}(cP_3)W_{GK} = c(I_{L_{\mathrm{or}}} \otimes P_G), \qquad U_W(cP_3)U_W^{*} = cQ_3;$$

thus, entanglement carries the coefficient, but does not select it;

2. *if P_{sgn} and P_{std} are the central projectors of the decomposition*

$$P_3V_{11}\big|_{H_*} \simeq \mathrm{sgn}_{H_*} \oplus \mathrm{std}, \qquad \operatorname{rank} P_{\mathrm{sgn}} = 1, \quad \operatorname{rank} P_{\mathrm{std}} = 2,$$

every positive self-adjoint operator supported on P_3V_{11} and H_*-equivariant has the form

$$D_{a,b} = aP_{\mathrm{sgn}} + bP_{\mathrm{std}}, \qquad a, b \geq 0.$$

Even impose the trace of $\Delta_4 P_3$ leaves the family

$$a + 2b = 3\Delta_4.$$

The isotropy $a = b$ is not deduced from the residual symmetry S_3;

3. *the uniform probability of the catalog, $\mu_U(U) = 1/468$, is the canonical trace of the finite quotient, and its compensated lift is determined by J_F. It is not the image measure of the uniform probability on seeds, whose atoms are $1/729, 1/243, 1/54$. In particular,*

$$\mu_U(\mathscr{R}_{\pi}^{\mathrm{micro}}) = \frac{5}{468}, \qquad F_{\#}\mu_{\mathrm{seed}}(\mathscr{R}_{\pi}^{\mathrm{micro}}) = \frac{7}{729};$$

4. *the complete law*

$$\delta_* = \frac{5}{468} + \frac{3}{11}\Delta_4, \qquad \lambda_* = 1 - \delta_*, \qquad a_*(e) = g(e) + \lambda_* s(e)$$

is a theorem relative to the five clauses Tr_q–Add–Def–Res–Loc. *The executable engine of the reduced topological phase kernel does not induce, by directed cover, the branched operator Γ_{tr}: the associated dynamics returns to the identity after 54 iterations, while each state of Γ_{tr} has four distinct successors and no self-edge.*

Consequently, the preceding results prove the canonicity of the isometric identifications, of the trace on the quotient catalogue, and of the additive character within their premises. The stronger assertion that a total transition of the enriched Topological Phase Kernel necessarily induces $\Delta_4 P_3$, the uniform measure, and the local law still requires the additional dynamical theorem specified in the induction analysis.

Proof. The identities in the first assertion are obtained by multiplying $W_{GK}^{*}W_{GK} = I_{L_{\mathrm{or}}} \otimes P_G$, $W_{GK}W_{GK}^{*} = P_3$, and $U_W P_3 U_W^{*} = Q_3$ by the scalar c. For the second assertion, the restriction to $H_* \simeq S_3$ is the

multiplicity-free sum of two nonisomorphic irreducible representations. Schur's lemma diagonalizes every equivariant self-adjoint operator on those two blocks; positivity is equivalent to $a, b \geq 0$. Its normalized trace is $(a+2b)/11$, which proves the residual freedom even after the trace has been fixed. The full action of A_5 on the irreducible triplet would reduce the family to cP_3, but it would not fix the value of c either.

The third point follows from the fibre census: F has multiplicities $144, 432, 1944$, so the image measure weights an emission by its number of presentations. By contrast, the trace τ_{468} assigns the same weight to every class; J_F realizes exactly its normalized lift. The fourth point is the logical concatenation already proved: Tr_q fixes the quotient trace, Add the sum of traces, Def declares $\Delta_4 P_3$, Res defines the residual capacity, and Loc carries it to the edges. The finite enumeration proves the obstruction between the reduced kernel and Γ_{tr}: a directed cover would preserve the outgoing star, and a semiconjugacy of the identity return would require self-edges in the image, contradicting both censuses of Γ_{tr}. □

The complete spectrum of $D_{\mathrm{switch}}^{\mathrm{dod}}$ is

eigenvalue	multiplicity
0	$(468-5)(11-3) = 3704$
Δ_4	$(468-5)3 = 1389$
1	$5(11-3) = 40$
$1+\Delta_4$	$5 \cdot 3 = 15.$

Consequently,

$$\begin{aligned}\tau_{468\cdot 11}(D_{\mathrm{switch}}) &= \tau_{468}(\Pi_{\pi}^{(468)}) + \Delta_4 \tau_{11}(P_3) \\ &= \boxed{\frac{5}{468} + \frac{3}{11}\Delta_4}.\end{aligned}$$

The same identity holds in the Witt chart. In the oriented-sector chart, the term $3/11$ is the trace of P_G on the homogeneous space of eleven branches.

35.5 Universal property of the positive action functional

The word *action* is used in an additive sense: the concatenation of two routes receives the sum of their increments. To separate this content from a chosen formula, we first define the category in which the law will be unique.

Definition 35.4 (Positive Action Character HMT)**.** A character local of action on the product regional–dodecafasico is one map

$$\delta : M_{468}(\mathbb{C})_+ \oplus M_{11}(\mathbb{C})_+ \longrightarrow \mathbb{R}_{\geq 0}$$

which satisfies:

1. exact additivity on the product cone;
2. homogeneidad positiva;
3. invariance under independent unitary conjugation in the two factors;
4. bifactorial normalization,
$$\delta(I_{468}, 0) = \delta(0, I_{11}) = 1.$$

Additivity translates concatenation; invariance eliminates the coordinates; bifactorial normalization fixes a scale in each summand. This is not a unital functional on the direct-sum algebra: its unit is (I_{468}, I_{11}) and additivity gives $\delta(I_{468}, I_{11}) = 2$. No clause contains the value of an angle.

Theorem 35.5 (Uniqueness of the positive character and relative residual capacity)**.** *En the class of the definition preceding there exists a unique character positive of action HMT. It is*

$$\boxed{\delta(A,B)=\tau_{468}(A)+\tau_{11}(B).}$$

Applied to the defect declared by Def*,* $(\Pi_{\pi}^{(468)},\Delta_4 P_3)$*, it yields*

$$\delta_*=\frac{5}{468}+\frac{3}{11}\Delta_4.$$

If, in addition, Res*is declared, according to which the self-dual capacity for sheet change is normalized to one and the defect reduces it through* $\lambda=1-\delta$*, the residual capacity within* Tr_q–Add–Def–Res*is*

$$\boxed{\lambda_*=1-\delta_*=1-\frac{5}{468}-\frac{3}{11}\Delta_4.}$$

Proof. The separability of additivity separates the factors:

$$\delta(A,B)=\delta(A,0)+\delta(0,B).$$

Each restriction is positive, homogeneous, and invariant under conjugation. For a rank-one projector, unitary invariance gives a common value. Additivity over a spectral decomposition forces each restriction to be a multiple of the trace. The bifactorial normalization fixes those multiples at $1/468$ and $1/11$. The sum of traces follows. The last formula applies the declared normalization of residual capacity. □

Numerically,

$$\boxed{\lambda_*=0.9892859130191415\ldots},\qquad 0<\lambda_*<1.$$

Positivity already follows from $0<\Delta_4<10^{-3}$ and $5/468<1$. It does not depend on a metrological comparison.

35.6 Positive Action Functor on the Category of Compatible Paths

Recall the directed graph Γ_{tr} and its free category $\mathsf{P}(\Gamma_{\mathrm{tr}})$. For an edge $e:x\to y$, define

$$g(e)=\mathbf{1}_{\{d_y\neq d_x\}},\qquad s(e)=\mathbf{1}_{\{m_y\neq m_x\}},$$

where g records a turn and s a change of sheet. The arrow from the scalar λ_* to the edges does not follow from the trace theorem. It is declared through the clause Loc: a turn has unit cost, a change of sheet has cost λ_*, and the two indicators are added when they occur on the same edge. Under Loc, the local action is

$$\boxed{a_*(e)=g(e)+\lambda_* s(e).}$$

For a route $\gamma=e_1\cdots e_n$,

$$\mathcal{A}_*(\gamma)=\sum_{j=1}^{n}a_*(e_j)=\#\,\text{giros}(\gamma)+\lambda_*\#\,\text{cambios de hoja}(\gamma).$$

For the empty path $\mathbf{1}_x$, one fixes $\mathcal{A}_*(\mathbf{1}_x)=0$. Then

$$\boxed{\mathcal{A}_*(\gamma_1\gamma_2)=\mathcal{A}_*(\gamma_1)+\mathcal{A}_*(\gamma_2).}$$

Therefore,

$$\mathcal{A}_*:\mathsf{P}(\Gamma_{\mathrm{tr}})\longrightarrow(\mathbb{R}_{\geq 0},+)$$

is a monoidal functor, or additive path cocycle.

The indicators g and s are insensitive to the direction of traversal. Orientation is not lost: it resides in the line L_{or} and in the star involution, whereas positive cost resides in the projectors. Conflating the two data would introduce negative signs into the action.

For $\beta > 0$, the associated transfer operator is

$$(\mathcal{L}_{\beta,*} f)(y) = \sum_{e : x \to y} e^{-\beta a_*(e)} f(x).$$

All its weights are strictly positive. Since the potential depends only on the two local indicators, it commutes with the coarse action $(\mathbb{Z}/3\mathbb{Z})^3$ and respects the decomposition into 27 Bloch sectors used in the following section.

A finite unweighted deterministic dynamics acts through permutation matrices and has singular values equal to one. The cocycle $\mathcal{A}_*$ turns the inventory of paths into a transfer operator that preserves the positive cone, that is, into a matrix with strictly positive weights on the permitted edges. That property does not by itself prove contraction, simplicity of the dominant eigenvalue, or spectral separation; these conclusions require a normalization or an additional spectral argument.

35.7 Regional character of degree 5 and reduced action mantissa

The central block

$$\mathcal{B}_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} \tag{35.1}$$

has eigenvalues 9 and 5. The length 4 of the step orbits 3, the rank 12 of the dodecaphasic cycle, and $104976/1944 = 54$ determine the character

$$\boxed{H_5(\alpha, \varphi) = \frac{1}{2}\sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5.} \tag{35.2}$$

The nonadic completion constructs the analytic coordinate and the regional determinant.

$$x_A = \frac{50\pi}{9}\alpha_{\mathrm{HMT}}, \qquad D_A = e^{-2x_A} = \exp\!\left(-\frac{100\pi}{9}\alpha_{\mathrm{HMT}}\right). \tag{35.3}$$

The microfiber supplies $\rho_\pi = 169$, and its absolutely convergent series is

$$\mathcal{C}_\pi(\alpha) = 169\alpha^6 + \frac{D_A \alpha^7}{1 - D_A \alpha}. \tag{35.4}$$

The reduced mantissa after return is then constructed by

$$\boxed{\eta_{\mathrm{ret}} = H_5 - \frac{90}{\pi}\mathcal{C}_\pi(\alpha_{\mathrm{HMT}}) \in \mathbb{R}_{>0}.} \tag{35.5}$$

Its right-hand side contains exclusively objects generated previously by APP–TRIT–TPK and by the closure of α_{HMT}.

Chapter 36

Self-dual derivation of the Planck constant and its action line: temporal periodicity of 108 phases, Compton closure and bidirectional covariance

36.1 Compton realization associated with the temporal periodicity of 108 phases

On the cycle of equation (36.17), we reuse the positive action section $\hbar$ already generated by the determinantal action branch. The associated angular quantum is

$$E_{108} = \hbar\omega_{108}, \qquad m_{108} = \frac{E_{108}}{c^2} = \frac{\hbar\omega_{108}}{c^2}. \tag{36.1}$$

G has not yet been used. The geometry of the cycle and angular quantization suffice to fix its two Compton readers.

Theorem 36.1 (Identity between the temporal periodicity of 108 phases and the Compton length)**.** *For the quantum* (36.1)*,*

$$\bar{\lambda}_{\mathrm{C}}(m_{108}) = \frac{\hbar}{m_{108}c} = R_{108}, \qquad \lambda_{\mathrm{C}}(m_{108}) = \frac{h}{m_{108}c} = C_{108}. \tag{36.2}$$

Therefore, the cycle radius is its reduced Compton wavelength, and the length of one revolution is its full Compton wavelength.

Proof. Substituting $m_{108} = \hbar\omega_{108}/c^2$,

$$\frac{\hbar}{m_{108}c} = \frac{\hbar c^2}{\hbar\omega_{108}c} = \frac{c}{\omega_{108}} = R_{108}.$$

Since $h = 2\pi\hbar$, the second equality is $2\pi R_{108} = C_{108}$. □

The coincidence is not a numerical fit. It is a homogeneous identity: it holds for all $\hbar, c, t_0 > 0$ and depends only on the same period determining the frequency, the cycle length, and the quantum of action.

36.2 Gravitational Transducer and Self-Dual Selection Equation

The gravitational line HMT–MD contains the transducer

$$G(\theta) = \theta\frac{c^5 t_0^2}{\hbar}, \qquad \theta > 0. \tag{36.3}$$

Homogeneity is exact, but by itself does not select θ. The CT108 cycle supplies an additional internal condition: the gravitational reader must return the same radial scale as the geometric and Compton readers.

We define the reduced gravitational radius

$$r_{\mathrm{g}}(\theta) := \frac{G(\theta)m_{108}}{c^2}. \tag{36.4}$$

Substituting (36.1) and (36.3),

$$r_{\mathrm{g}}(\theta) = \theta\frac{\pi}{54}\ell_0. \tag{36.5}$$

Theorem 36.2 (Self-dual selector of the temporal periodicity of 108 phases)**.** *The equation*

$$r_{\mathrm{g}}(\theta) = R_{108}$$

has a unique positive solution:

$$\boxed{\theta_* = \left(\frac{54}{\pi}\right)^2.} \tag{36.6}$$

Proof. By (36.17) and (36.5), the equation is

$$\theta \frac{\pi}{54}\ell_0 = \frac{54}{\pi}\ell_0.$$

Since $\ell_0 > 0$, simplification yields $\theta = (54/\pi)^2$. The function $\theta \mapsto r_{\mathrm{g}}(\theta)$ is strictly increasing on $\mathbb{R}_{>0}$, so there is no other positive solution. □

Corollary 36.3 (Internal Planck Unit)**.** *With $G_* = G(\theta_*)$, the Planck sections satisfy*

$$\ell_{\mathrm{P}} := \sqrt{\frac{\hbar G_*}{c^3}} = R_{108}, \qquad t_{\mathrm{P}} := \sqrt{\frac{\hbar G_*}{c^5}} = \frac{1}{\omega_{108}}, \tag{36.7}$$

$$m_{\mathrm{P}} := \sqrt{\frac{\hbar c}{G_*}} = m_{108}, \qquad E_{\mathrm{P}} := m_{\mathrm{P}}c^2 = E_{108}. \tag{36.8}$$

In particular,

$$\boxed{R_{108} = \bar{\lambda}_{\mathrm{C}} = r_{\mathrm{g}} = \ell_{\mathrm{P}}, \qquad C_{108} = \lambda_{\mathrm{C}} = 2\pi r_{\mathrm{g}} = 2\pi\ell_{\mathrm{P}}.} \tag{36.9}$$

Proof. $G_* = (54/\pi)^2 c^5 t_0^2/\hbar$ is substituted. Then

$$\ell_{\mathrm{P}} = \frac{54}{\pi} c t_0 = R_{108}, \qquad t_{\mathrm{P}} = \frac{54}{\pi} t_0 = \frac{1}{\omega_{108}},$$

and

$$m_{\mathrm{P}} = \frac{\pi}{54}\frac{\hbar}{c^2 t_0} = \frac{\hbar\omega_{108}}{c^2} = m_{108}.$$

The rest follows from theorem 36.1. □

Remark 36.4 (Three readers, one section). Circular geometry publishes R_{108} and C_{108}; Compton quantization publishes $\bar{\lambda}_{\mathrm{C}}$ and λ_{C}; gravitational realization publishes r_{g} and $2\pi r_{\mathrm{g}}$. In θ_* those three pairs are coordinates of a single section. This is the precise reason why the Planck unit ceases to appear as an accidental dimensional combination within this formalization.

36.3 Commutativity of the angular, temporal, and gravitational normalizations of the self-dual identity

The transducer (36.3) admits two ways of fixing the elementary scale, which must not be confused.

(i) If $\theta = 1$ is fixed, then $\sqrt{\hbar G/c^3} = \ell_0$. The elementary transition ct_0 is identified with the Planck length of that chart.

(ii) If the *complete orbit* CT108 is required to have Planck radius, the condition selects $\theta_* = (54/\pi)^2$ and $R_{108} = \ell_{\mathrm{P}}$.

They are not rival values of G. They are two gauges of the same homogeneous equation: one assigns the Planck unit to the elementary step; the other, to the radius of the complete cycle. The annex uses the latter because its thesis concerns the identity of the CT108 closure.

36.4 Gravitational conventions: reduced radius and Schwarzschild radius

The gravitational notation is fixed in an unambiguous manner:

$$r_{\mathrm{g}} = \frac{Gm}{c^2}, \qquad r_{\mathrm{s}} = \frac{2Gm}{c^2} = 2r_{\mathrm{g}}.$$

The preceding theorem uses r_{g}, not r_{s}. In the usual convention of Planck mass, the equality $\hbar/(mc) = Gm/c^2$ is produced in $m = m_{\mathrm{P}}$, whereas

$$\frac{\hbar}{mc} = \frac{2Gm}{c^2} \quad \Longleftrightarrow \quad m = \frac{m_{\mathrm{P}}}{\sqrt{2}}. \tag{36.10}$$

For this reason the phrase "gravitational scale" designates, in the rector thesis, the reduced radius. The conventional horizon radius remains a distinct reader.

36.5 Bifurcation of the action line

The internal line of action possesses two oriented sections:

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S, \qquad R_{\mathrm{act}} = \frac{H_5}{\eta_{\mathrm{ret}}}. \tag{36.11}$$

The decade 10^{-34} fixes the absolute scale; the quotient R_{act} transports the oriented difference. This separation makes it possible to ask whether Planck self-duality survives in both coordinates.

Theorem 36.5 (Reciprocal Covariance of Action and Gravity)**.** *For $a \in \{\mathrm{pre}, \mathrm{ret}\}$, let*

$$G_a(\theta) = \theta\frac{c^5 t_0^2}{\hbar_a}, \qquad m_{108,a} = \frac{\hbar_a \omega_{108}}{c^2}.$$

Then

$$\frac{G_a(\theta) m_{108,a}}{c^2} = \theta\frac{\pi}{54}\ell_0 \tag{36.12}$$

is independent of a. Moreover,

$$\frac{m_{108,\mathrm{pre}}}{m_{108,\mathrm{ret}}} = R_{\mathrm{act}}, \qquad \frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}} = R_{\mathrm{act}}^{-1}, \tag{36.13}$$

and the Planck length $\sqrt{\hbar_a G_a/c^3} = \sqrt{\theta}\,\ell_0$ is the same in both orientations.

Proof. The factors $\hbar_a$ cancel in the product $G_a m_{108,a}$. The two ratios in (36.13) follow directly from the definitions. The invariance of the Planck length follows from $\hbar_a G_a = \theta c^5 t_0^2$. □

A symmetric parametrization makes the geometry visible:

$$\hbar_{\mathrm{geom}} := \sqrt{\hbar_{\mathrm{pre}}\hbar_{\mathrm{ret}}}, \qquad \xi := \frac{1}{2}\log R_{\mathrm{act}}, \qquad \hbar_{\mathrm{pre}} = \hbar_{\mathrm{geom}} e^{\xi}, \quad \hbar_{\mathrm{ret}} = \hbar_{\mathrm{geom}} e^{-\xi}. \tag{36.14}$$

Bidirectional memory is lodged in ξ; Planck geometry is lodged in the invariant product. This allows a real difference between the two action coordinates without requiring a difference in propagation speed.

Remark 36.6 (Physical scope of the theorem). The reciprocity of theorem 36.5 is exact within the transducer. To identify G_{pre} and G_{ret} with two oriented gravitational measurements, the protocol, the observable, and the symmetry relating them must be specified. The theorem proves structural compatibility; by itself, it neither anticipates a Lorentz violation nor an observable variation of G.

The action line constructed in the preceding section admits a circular realization compatible with temporal periodicity of order 108. This realization coordinates three objects that already possess their own types: the action section, the geometry of one revolution, and the gravitational transduction. The resulting coordination provides an internal Planck identity and prepares the positive line on which the general mass operator will subsequently act.

36.5.1 Oriented action letters and declared circular realization

Let

$$\mathcal{L}_S^+, \quad \mathcal{L}_T^+, \quad \mathcal{L}_C^+, \quad \mathcal{L}_L^+, \quad \mathcal{L}_M^+, \quad \mathcal{L}_G^+$$

the positive half-lines of action, time, velocity, length, and mass. The pre-return and post-return coordinates are written

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S, \qquad R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}}. \tag{36.15}$$

The subscript $a \in \{\mathrm{pre}, \mathrm{ret}\}$ shall designate either of the two charts. In each one the exact distinction is preserved

$$h_a = 2\pi\hbar_a. \tag{36.16}$$

Let us fix an elementary duration $t_0 \in \mathcal{L}_T^+$, an internal velocity $c_{\mathrm{int}} \in \mathcal{L}_C^+$, and $\ell_0 = c_{\mathrm{int}} t_0$. The declared circular realization of the temporal periodicity of order 108 is determined by

$$T_{108} = 108 t_0, \qquad \omega_{108} = \frac{2\pi}{T_{108}}, \qquad R_{108} = \frac{c_{\mathrm{int}}}{\omega_{108}} = \frac{54}{\pi}\ell_0, \tag{36.17}$$

and by the length of the full turn

$$C_{108} = 2\pi R_{108} = c_{\mathrm{int}} T_{108} = 108\ell_0. \tag{36.18}$$

Thus, R_{108} is the radius of the realization and C_{108} its perimeter. The angular frequency, the cyclic frequency and the two action constants are related through $\omega_{108} = 2\pi/T_{108}$ and $h_a = 2\pi\hbar_a$.

36.5.2 Correspondence between the order-108 periodicity and Compton readings

In the chart a, the circular quantum and its mass section are

$$E_{108,a} = \hbar_a \omega_{108}, \qquad m_{108,a} = \frac{E_{108,a}}{c_{\mathrm{int}}^2} = \frac{\hbar_a \omega_{108}}{c_{\mathrm{int}}^2}. \tag{36.19}$$

Theorem 36.7 (Identity between circular periodicity and Compton readings)**.** *For each chart $a \in \{\mathrm{pre}, \mathrm{ret}\}$, the reduced and complete Compton readings satisfy*

$$\boxed{\bar{\lambda}_{\mathrm{C}}(m_{108,a}) = \frac{\hbar_a}{m_{108,a} c_{\mathrm{int}}} = R_{108}, \qquad \lambda_{\mathrm{C}}(m_{108,a}) = \frac{h_a}{m_{108,a} c_{\mathrm{int}}} = C_{108}.} \tag{36.20}$$

Proof. Substituting (36.19) produces

$$\frac{\hbar_a}{m_{108,a} c_{\mathrm{int}}} = \frac{c_{\mathrm{int}}}{\omega_{108}} = R_{108}.$$

The second equality follows by applying simultaneously (36.16) and (36.18). □

The theorem is homogeneous in $(\hbar_a, c_{\mathrm{int}}, t_0)$: the same periodicity determines the radius, the perimeter, and the two Compton charts. The circular realization is the declared geometric primitive; the preceding equalities are its exact consequences.

36.5.3 Circular selector and internal Planck unit

The gravitational line is coordinated with the chart a through the homogeneous transducer

$$G_a(\theta) = \theta\,\frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a} \in \mathcal{L}_G^+, \qquad \theta \in \mathbb{R}_{>0}. \tag{36.21}$$

Its reduced gravitational radius, evaluated on the circular quantum, is

$$r_{\mathrm{g},a}(\theta) := \frac{G_a(\theta) m_{108,a}}{c_{\mathrm{int}}^2} = \theta\frac{\pi}{54}\ell_0. \tag{36.22}$$

Proposition 36.8 (Planck Circular Selector)**.** *Within the circular realization* (36.17)*, the condition* $r_{\mathrm{g},a}(\theta) = R_{108}$ *selects a unique positive coordinate,*

$$\boxed{\theta_{108}^\circ = \left(\frac{54}{\pi}\right)^2}. \tag{36.23}$$

The coordinate is independent of the chart oriented a.

Proof. The equations (36.17) and (36.22) reduce the condition to

$$\theta\frac{\pi}{54}\ell_0 = \frac{54}{\pi}\ell_0.$$

The positivity of ℓ_0 and the strict monotonicity of the map $\theta \mapsto r_{\mathrm{g},a}(\theta)$ provide the solution and its uniqueness. □

Let $G_{108,a}^\circ := G_a(\theta_{108}^\circ)$. The internal units associated with this chart satisfy

$$\ell_{\mathrm{P},a}^\circ := \sqrt{\frac{\hbar_a G_{108,a}^\circ}{c_{\mathrm{int}}^3}} = R_{108}, \qquad t_{\mathrm{P},a}^\circ := \sqrt{\frac{\hbar_a G_{108,a}^\circ}{c_{\mathrm{int}}^5}} = \omega_{108}^{-1}, \tag{36.24}$$

$$m_{\mathrm{P},a}^\circ := \sqrt{\frac{\hbar_a c_{\mathrm{int}}}{G_{108,a}^\circ}} = m_{108,a}, \qquad E_{\mathrm{P},a}^\circ := m_{\mathrm{P},a}^\circ c_{\mathrm{int}}^2 = E_{108,a}. \tag{36.25}$$

The equality of radii is therefore published as

$$R_{108} = \bar{\lambda}_{\mathrm{C}} = r_{\mathrm{g}} = \ell_{\mathrm{P}}^\circ, \qquad C_{108} = \lambda_{\mathrm{C}} = 2\pi r_{\mathrm{g}} = 2\pi\ell_{\mathrm{P}}^\circ. \tag{36.26}$$

The convention of this theorem is the reduced gravitational radius $r_{\mathrm{g}} = Gm/c_{\mathrm{int}}^2$. The Schwarzschild radius satisfies $r_{\mathrm{s}} = 2r_{\mathrm{g}}$; if this second convention is adopted, its crossing with $\bar{\lambda}_{\mathrm{C}}$ occurs at $m = m_{\mathrm{P},a}^\circ/\sqrt{2}$. The declaration of the radius therefore forms part of the chart and prevents a factor of two from being inadvertently transferred between the two identities.

The coordinate θ_{108}° belongs to the declared circular specialization. The general gravitational selector, formulated on the enriched holonomy and its spectral response, retains its residence in chapter 80. Thus, the circular identity fixes an internal Planck normalization and the general criterion fixes the conditions that the physical realization of the gravitational section must satisfy.

36.5.4 Pre-return/post-return covariance

The product of the reciprocal sections of action and gravity preserves the circular geometry:

$$\frac{m_{108,\mathrm{pre}}}{m_{108,\mathrm{ret}}} = R_{\mathrm{act}}, \qquad \frac{G^{\circ}_{108,\mathrm{pre}}}{G^{\circ}_{108,\mathrm{ret}}} = R_{\mathrm{act}}^{-1}, \qquad \hbar_a G^{\circ}_{108,a} = \theta^{\circ}_{108} c_{\mathrm{int}}^5 t_0^2. \tag{36.27}$$

In particular, $\ell^{\circ}_{\mathrm{P,pre}}$ and $\ell^{\circ}_{\mathrm{P,ret}}$ coincide with R_{108}. The bidirectional memory resides in the quotient of the masses and in its gravitational reciprocal; the geometric scale resides in the invariant product.

Reciprocity admits a symmetric parametrization that separates the common scale from the orientation of the return. Define

$$\hbar_{\mathrm{geom}} := \sqrt{\hbar_{\mathrm{pre}}\hbar_{\mathrm{ret}}}, \quad G^{\circ}_{108,\mathrm{geom}} := \sqrt{G^{\circ}_{108,\mathrm{pre}} G^{\circ}_{108,\mathrm{ret}}}, \quad \xi_{\mathrm{act}} := \frac{1}{2}\log R_{\mathrm{act}}. \tag{36.28}$$

Then

$$\hbar_{\mathrm{pre}} = \hbar_{\mathrm{geom}} e^{\xi_{\mathrm{act}}}, \qquad \hbar_{\mathrm{ret}} = \hbar_{\mathrm{geom}} e^{-\xi_{\mathrm{act}}},$$

whereas the conjugate gravitational sections satisfy

$$G^{\circ}_{108,\mathrm{pre}} = G^{\circ}_{108,\mathrm{geom}} e^{-\xi_{\mathrm{act}}}, \qquad G^{\circ}_{108,\mathrm{ret}} = G^{\circ}_{108,\mathrm{geom}} e^{\xi_{\mathrm{act}}}.$$

The coordinate ξ_{act} preserves the pre-return/post-return orientation; the product $\hbar_a G^{\circ}_{108,a}$ preserves geometric scale. This parametrization of the action charts does not by itself introduce distinct propagation speeds.

36.5.5 Split self-dual mass line

Given a chart a, let the split algebra be

$$\mathbb{D}_{\mathrm{sp}} := \mathbb{R}[j]/(j^2 - 1), \qquad P_{\pm} := \frac{I \pm j}{2},$$

and consider its spectral decomposition

$$\mathbb{D}_{\mathrm{sp}} = \mathbb{R}P_+ \oplus \mathbb{R}P_-, \qquad P_{\pm}^2 = P_{\pm}, \quad P_+P_- = 0, \quad P_+ + P_- = I.$$

The broken conjugation swaps the projectors,

$$\overline{uP_+ + vP_-} = vP_+ + uP_-,$$

and its norm is

$$N_{\mathrm{sp}}(uP_+ + vP_-) := uv.$$

For $m \in \mathcal{L}_M^+$, define the section

$$Z_a(m) := \frac{m^{\circ}_{\mathrm{P},a}}{m} P_+ + \frac{m}{m^{\circ}_{\mathrm{P},a}} P_-. \tag{36.29}$$

The unit $m^{\circ}_{\mathrm{P},a}$ also determines the involution

$$\iota_{\mathrm{P},a}(m) = \frac{(m^{\circ}_{\mathrm{P},a})^2}{m}. \tag{36.30}$$

Theorem 36.9 (Split self-dual mass line)**.** *For every $m \in \mathcal{L}_M^+$,*

$$\boxed{N_{\mathrm{sp}}(Z_a(m)) = 1, \qquad \overline{Z_a(m)} = Z_a\big(\iota_{\mathrm{P},a}(m)\big)\,.} \tag{36.31}$$

Conjugation has a unique positive fixed point,

$$m = m^{\circ}_{\mathrm{P},a}, \qquad Z_a(m^{\circ}_{\mathrm{P},a}) = I.$$

Proof. The norm is the product of the two reciprocal coordinates:

$$\frac{m^{\circ}_{\mathrm{P},a}}{m}\frac{m}{m^{\circ}_{\mathrm{P},a}} = 1.$$

Conjugation interchanges both coordinates, which is equivalent to replacing m by $(m^{\circ}_{\mathrm{P},a})^2/m$. The positive fixed point satisfies $m^2 = (m^{\circ}_{\mathrm{P},a})^2$. □

With the speed

$$x_a(m) = \log \frac{m}{m^{\circ}_{\mathrm{P},a}}, \tag{36.32}$$

the section and its conjugate are written

$$Z_a(m) = e^{-x_a}P_+ + e^{x_a}P_-, \qquad \overline{Z_a(m)} = e^{x_a}P_+ + e^{-x_a}P_-.$$

Their even and odd readers are

$$Q_a(m) := \frac{e^{x_a} + e^{-x_a}}{2} = \cosh x_a, \qquad A_a(m) := \frac{e^{x_a} - e^{-x_a}}{2} = \sinh x_a. \tag{36.33}$$

The first preserves the distance to the self-dual point and the second preserves its orientation.

The reciprocal lattice vectors associated with the same coordinate are

$$r_{\mathrm{g},a}(m) = \frac{G^{\circ}_{108,a}m}{c^2_{\mathrm{int}}}, \qquad \bar{\lambda}_{\mathrm{C},a}(m) = \frac{\hbar_a}{mc_{\mathrm{int}}},$$

and satisfy

$$\frac{r_{\mathrm{g},a}(m)}{\bar{\lambda}_{\mathrm{C},a}(m)} = e^{2x_a(m)}, \qquad \frac{\bar{\lambda}_{\mathrm{C},a}(m)}{r_{\mathrm{g},a}(m)} = e^{-2x_a(m)}. \tag{36.34}$$

Conjugation interchanges the two readers; Planck's identity is their unitary equilibrium.

Remark 36.10 (Planck identity and electronic closure). Identity $m = m^{\circ}_{\mathrm{P},a}$ fixes the self-dual origin of the positive mass line. The electronic section fixes a distinct position on that same line through its own genealogy. The section 90.4.7 constructs the universal operator that transports absolute scale, character, towers, and bidirectional reading; the section 58.2 subsequently realizes its central rank-one reduction. Thus, the Planck unit supplies the identity of the coordinate, and the electron supplies the first complete material transition on it.

Proposition 36.11 (Genealogical realization of the electronic center)**.** *Let*

$$F^+_e = \mathbb{R}_{>0}(P_+ + P_-)$$

the positive half-line of the rank-one central fiber, let $\mathcal{G}_e$ be the space of enriched electronic genealogies and let $\mathcal{L}^+_M$ be the positive mass line. A realization morphism

$$\mathfrak{R}_e : F^+_e \times \mathcal{G}_e \longrightarrow \mathcal{L}^+_M$$

can send the identity $I = P_+ + P_-$ to a mass different from $m^{\circ}_{\mathrm{P},a}$ without altering the theorem 36.9: the Planck identity fixes the autodual origin of the chart, while the electronic genealogy fixes a displacement on that chart.

Proof. If $\chi_e : \mathcal{G}_e \to \mathbb{R}_{>0}$ is the positive character constructed by the electronic record, the rule

$$\mathfrak{R}_e(\lambda I, g) = \lambda\, m^{\circ}_{\mathrm{P},a}\chi_e(g)$$

is positive and multiplicative in the genealogical coordinate. For the neutral genealogy, $\chi_e = 1$, the Planck origin is recovered; for a genealogy with $\chi_e(g) \neq 1$, the same internal identity occupies a different position of the line. The concrete electronic closure preserves its antecedent development in the section 58.2. □

36.6 Autoduality of the Planck internal unit under orientation inversion

The CT108 selector constructs

$$\theta_{108} = \left(\frac{54}{\pi}\right)^2, \qquad \ell_P = \frac{54}{\pi}\ell_0, \qquad t_P = \frac{54}{\pi}t_0, \tag{36.35}$$

and

$$E_P = \frac{\hbar_{\mathrm{int}}}{t_P} = \frac{\pi\hbar_{\mathrm{int}}}{54t_0}. \tag{36.36}$$

Comparing equations (36.36) y (48.1) gives

$$\boxed{E_P = k_B\Theta_{\mathrm{clk}}\log 3.} \tag{36.37}$$

The relation does not depend on a coincidence between units: both sides follow from the same chronological structure 108.

Once the gravitational constant has been derived internally in the chapter 50, the Planck family satisfies

$$\ell_P^2 = \frac{G\hbar}{c^3} = \theta_{108}\ell_0^2, \tag{36.38}$$

$$F_P = \frac{c^4}{G}, \qquad P_P = \frac{c^5}{G}, \tag{36.39}$$

$$\rho_P = \frac{\hbar}{\theta_{108}^2 c^5 t_0^4}. \tag{36.40}$$

Although G and $\hbar$ transform reciprocally between the sections before and after the return, their product in ℓ_P^2 preserves the geometry. This invariance determines area as the interface between the geometric, informational and entanglement realizations.

36.7 Action line and Planck constant

Let $\mathcal{L}_S$ be the real action line with positive basis $\mathcal{U}_S$. The transduction

$$\tau_S : \mathbb{R}_{>0} \longrightarrow \mathcal{L}_S, \qquad u \longmapsto u\,10^{\varepsilon_H}\mathcal{U}_S \tag{36.41}$$

publish the sections preceding and following the return:

$$\hbar_{\mathrm{pre}} = H_5\,10^{-34}\mathcal{U}_S, \qquad \boxed{\hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\,10^{-34}\mathcal{U}_S.} \tag{36.42}$$

The full cycle actions are

$$h_{\mathrm{pre}} = 2\pi\hbar_{\mathrm{pre}}, \qquad h_{\mathrm{ret}} = 2\pi\hbar_{\mathrm{ret}}. \tag{36.43}$$

For a phase θ, the angular and action charts commute:

$$S(\theta) = \hbar_{\mathrm{ret}}\theta_{\mathrm{rad}} = \hbar_{\mathrm{ret}}\frac{\pi}{180}\theta_{\mathrm{deg}} = h_{\mathrm{ret}}\frac{\theta_{\mathrm{deg}}}{360}. \tag{36.44}$$

36.8 Operator genealogy of action, angular geometry, and mass

The coinductive APP–TRIT–TPK dynamics extends to arbitrary depth over the same enriched state and over the joint discrete structure of the continuum. At each prolongation it preserves value, genealogy, phase, carry, orientation, vacancy, path, boundary, memory, and survival. Its distinguished publications are first the three internal coordinates and, through the dodecaphasic seal, the coordinate α_{HMT}:

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \mathcal{S}_{\infty}^{\mathrm{enr}} \longrightarrow \mathcal{C}_{\mathrm{cont}}^{\mathrm{disc}} \longrightarrow (\pi_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, e_{\mathrm{HMT}}) \longrightarrow \alpha_{\mathrm{HMT}}.$$

The last arrow represents the signed publication of α_{HMT} within those same dynamics; it receives no conventional values and does not turn conventional π, φ, e into generators.

Only afterward is the determinantal action reader applied. It publishes the valuation $10^{-34}\mathcal{U}_S$, the two sections $\hbar_{\mathrm{pre}}, \hbar_{\mathrm{ret}}$, and their full-cycle actions

$$h_{\mathrm{pre}} = 2\pi_{\mathrm{HMT}}\hbar_{\mathrm{pre}}, \qquad h_{\mathrm{ret}} = 2\pi_{\mathrm{HMT}}\hbar_{\mathrm{ret}}.$$

From this causal point onward, two branches remain separate and do not select one another. The dimensional–gravitational branch extends the action and the causal section to the already proven forms of the Planck pair,

$$\ell^2 = G\hbar c^{-3}, \qquad \frac{\ell}{Q} = Gc^{-4};$$

the complex branch publishes, from the internal rotation operator,

$$u^2 + 1 = 0, \qquad u \longmapsto J_{+1}, \qquad J_{+1}^2 = -I.$$

This is the internal operator realization of i, not the introduction of a complex symbol without an antecedent. The dimensional–gravitational and complex branches are parallel over the common enriched state: the former does not turn i into an input to the action scale and, conversely, h, $\hbar$, c^{-3}, and c^{-4} do not generate i. The identities of the Planck pair thus remain separate from the complex realization.

Writing $\prec_{\mathrm{HMT}}$ for causal precedence with type preservation —and not to assert that each object on the right is a function of all those on the left— yields

$$\begin{aligned}(\pi, \varphi, e, \alpha)_{\mathrm{HMT}} &\prec_{\mathrm{HMT}} (10^{-34}\mathcal{U}_S, \hbar, h)\\ &\prec_{\mathrm{HMT}} \left\{G\hbar c^{-3}, Gc^{-4};\ J_{+1}^2 = -I, i\right\}\\ &\prec_{\mathrm{HMT}} (A_{\mathrm{deg}}, C_{\mathrm{deg}}^{*})\\ &\prec_{\mathrm{HMT}} (\text{electron, mass law}).\end{aligned}$$

The angular construction preserves its effective inputs α_{HMT} and η_{ret}: the two preceding parallel branches establish joint availability and typing, but are not retrospectively promoted to selectors of the angle or counterangle.

In the central electronic path, the bidirectional readers preserve two distinct stages of a single genealogy:

$$E_e^{(0)} = 0.5109989224880660725\ldots \text{ MeV}, \qquad \text{basal stage},$$
$$E_e^{\beta} = E_e^{(0)} R_{\mathrm{act}}^{80-54\alpha_{\mathrm{HMT}}+6\Delta_4} = 0.510998950690000808608\ldots \text{ MeV}, \qquad \text{governing beta output},$$

with

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80, 54, 6).$$

The coincidence of the triple in the central fiber does not identify the domains or the operations of K_D and K_Ω. In the historical canonical register, `MD-ELECTRON-001` designates the calibrated candidate of the basal stage. This identification retains its provenance function, but does not govern the beta output or select the path; the basal and beta closure remain typed stages of the bidirectional readers.

Refutation criterion. The preceding genealogy is refuted if a conventional value or the beta output is used to select a path, if the basal stage is presented as the governing closure, if either of the two parallel branches is identified without a typed map, or if the angle, counterangle, electron, or a mass is placed among the inputs to the constant or action generator.

Chapter 37

Phase operator prior to dimensional transduction: decomposition into 27 sectors, spectral radius and factorization of the double projection

The two preceding sections have gathered three pieces that previously appeared separately: an isometric identification between projectors, a quotient measure, and a positive action cocycle. The present section studies the spectral consequence of that law and places the result within the complete architecture of HMT.

The separation of statuses is precise. The triple identification is an exact theorem relative to the new internal variational rule and an oriented anchor. The measure and the positive character are exact theorems of Tr_q–Add; the defect is relative, in addition, to Def, the capacity to Res, and the local cost to Loc. The angular value is an interval-certified output of the operator Γ_{tr}. Its proximity to the canonical conjugate angular coordinate is extraordinary, but the interval for the difference excludes equality.

37.1 Decomposition of the angular operator into 27 sectors of 72 states

The all-paths reader Γ_{tr}has 1944 microscopic states. The coarse translation action of $(\mathbb{Z}/3\mathbb{Z})^3$ permits the decomposition

$$1944 = 27 \cdot 72.$$

Each block of 72 states is parametrized by

$$(r, s, m, d) \in (\mathbb{Z}/3\mathbb{Z})^2 \times \{0, 1\} \times D_4,$$

where m is the sheet and d the preceding direction. From each state four cardinal continuations depart. The TRIT of the arrival cell decides the following sheet.

With the action of the chapter 35, the weight of an edge is

$$\exp[-g(e) - \lambda_* s(e)].$$

The character transform reduces the operator to complex matrices L_{k_a,k_b} of size 72×72. The program enumerates the edges of each block; it does not receive a precomputed spectral matrix. Power iteration is retained as an exploration, but the result that follows no longer rests on it: the spectral radii are isolated by means of a complete rational certification.

Let ρ_{01} and ρ_{12} be the spectral radii of the oriented sectors $(0, 1)$ and $(1, 2)$. We define the *band phase*

$$\boxed{y_{\mathrm{band}} = \log \rho_{12} - \log \rho_{01}.}$$

For

$$\lambda_* = 1 - \frac{5}{468} - \frac{3}{11}\Delta_4,$$

the certification gives

$$1.4606551208620111111 < \rho_{01} < 1.460655120862404024,$$
$$1.5158120082709443328 < \rho_{12} < 1.515812008271195416,$$

and, by interval propagation,

$$0.037066226434427529 < y_{\mathrm{band}} < 0.037066226434862172,$$

$$\boxed{2.123738337168943^\circ < C_{\mathrm{band}} := \frac{180}{\pi} y_{\mathrm{band}} < 2.123738337193847^\circ.}$$

The primary quantity is y_{band}, a dimensionless logarithm of spectral quotient. Degrees appear later when the chart $180/\pi$ is applied. The operator first produces an internal phase and only later an Archimedean angular representation.

37.1.1 Exact calculation of the spectral radius of the angular operator

The blocks are nonnormal. Therefore, a small residual $\|Lv - \mu v\|$ does not suffice to prove that μ is the eigenvalue of greatest modulus. The proof used here consists of four steps.

First, twelve structurally zero rows occur in each block. Expanding the characteristic polynomial along these rows yields a factor z^{12} and a principal minor of order sixty. A further eight structurally zero rows occur in this minor. Therefore,

$$\det(zI_{72} - L) = z^{20} \det(zI_{52} - A),$$

and every nonzero eigenvalue is contained, with multiplicity, in a reduced block A of order fifty-two.

Second, the certificate provides rational Gaussian matrices S, R, with denominator 10^{16}. The verifier calculates $E = I - RS$ exactly and obtains

$$\|E\|_\infty \le 8.5664 \times 10^{-14} \quad \text{and} \quad \|E\|_\infty \le 3.7628 \times 10^{-14}$$

for the sectors $(0, 1)$ and $(1, 2)$, respectively. The Neumann series then proves that S is invertible.

Third, if A_0 is the rational midpoint of the interval block, the identity

$$S^{-1}AS = (I - E)^{-1}RAS$$

bounds the distance to $C_0 = RA_0S$ by

$$1.25125 \times 10^{-13} \quad \text{and} \quad 5.7037 \times 10^{-14}.$$

The Gershgorin discs of C_0, enlarged by those bounds, contain the discs of the exact similarity. In each block, one disc remains separated from the other fifty-one; the second Gershgorin theorem assigns it exactly one eigenvalue. Its lower modulus exceeds the upper modulus of every other disc. This proves that the isolated eigenvalue is the spectral radius.

Fourth, $\pi, e, \log \pi, e^{-1}, e^{-\lambda_*}$ and $\sqrt{3}$ are not accepted as library decimals. The verifier encloses them by means of Machin's formula, the factorial and logarithmic series with remainder, the alternating series for the exponential, and a quadratic rational bracket for $\sqrt{3}$. Verification employs only integers and fractions; the argument is closed by the rational reduction, the Neumann series, and the Gershgorin separation presented in this section. The reproducible decimal expansions to a depth of ten thousand are published, as independent finite witnesses, in the appendix of expansions.

37.2 Effective band operator T_{band} and isolation of the angular phase

Let

$$x = \frac{50\pi}{9} \alpha_{\mathrm{HMT}} = 0.127362829012869966\ldots$$

and let

$$R = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The effective bilayer operator associated exclusively with the band phase is

$$\boxed{T_{\text{band}} = \exp(-xI + y_{\text{band}}R).}$$

At the base of oriented sectors,

$$T_{\text{band}} = \begin{pmatrix} e^{-x+y_{\text{band}}} & 0 \\ 0 & e^{-x-y_{\text{band}}} \end{pmatrix}.$$

We verify

$$\det T_{\text{band}} = e^{-2x}, \qquad \frac{(T_{\text{band}})_{11}}{(T_{\text{band}})_{22}} = e^{2y_{\text{band}}}.$$

Therefore,

$$-\frac{9}{100\pi}\log\det T_{\text{band}} = \alpha_{\text{HMT}}, \qquad \frac{90}{\pi}\log\frac{(T_{\text{band}})_{11}}{(T_{\text{band}})_{22}} = C_{\text{band}}.$$

The two recoveries have distinct statuses. The second returns the output produced by the gap. The first rereads the α_{HMT} that entered x; it proves reversibility of the chart, not a second independent derivation of α.

The exponential form separates two functions. The common factor e^{-x} describes the shared contraction of the two sheets. The factors $e^{\pm y_{\text{band}}}$ distinguish orientations while preserving the determinant. Since I and R commute, the pair

$$\left(\det T_{\text{band}}, \frac{(T_{\text{band}})_{11}}{(T_{\text{band}})_{22}}\right)$$

uniquely reconstructs (x, y_{band}).

37.3 Posterior comparison and precision scale

The following section constructs another phase, the *regularized regional phase* y_{reg} —also denoted there by y_*—. To formulate a comparison here with that subsequent result, we anticipate the notation

$$C_{\text{reg}} := \frac{180}{\pi}y_{\text{reg}} = 2.123738338968646^\circ.$$

This line does not constitute a second definition or an entry of the spectral operator: it cites the conclusion of theorem 38.14, whose construction is developed in the following section. The two pairs of objects are not identified:

$$(y_{\text{band}}, C_{\text{band}}) \neq (y_{\text{reg}}, C_{\text{reg}}).$$

The certified interval difference of the linear law is

$$-1.799703 \times 10^{-9}\ ^\circ < C_{\text{band}} - C_{\text{reg}} < -1.774799 \times 10^{-9}\ ^\circ,$$

with relative difference enclosed by

$$-8.475 \times 10^{-10} < \frac{C_{\text{band}} - C_{\text{reg}}}{C_{\text{reg}}} < -8.356 \times 10^{-10}.$$

The law that produces C_{band} does not receive C_{reg}, CKM, Barbero, masses, or SI data. This target independence makes the proximity a candidate for strong, falsifiable recognition. The spectral certification eliminates the former numerical gap; the interval difference, which excludes zero, now fixes the exact boundary: C_{band} is a predimensional spectral output distinct from the regional realization C_{reg}.

The additional term

$$-\frac{\Delta_4^2}{27}$$

en λ produce

$$C_{\text{num}} \approx 2.12373833894104980^\circ,$$

at 2.76×10^{-11} degrees of C_{reg}. The integer 27 is internal, since it counts the characters of $(\mathbb{Z}/3\mathbb{Z})^3$, but that fact does not force the coefficient of the second moment. The exact additivity of Addexcludes quadratic terms. The variant is preserved as an ablation and not as the principal law.

37.4 Functional dependence of the spectral radius with respect to the components of the operator

All subsequent rows use the same operator; only one input of the law changes. They are numerical ablation diagnostics; the preceding certified interval pertains exclusively to the complete law Tr_q–Add–Def–Res–Loc(and, within it, the positive defect is fixed in the Tr_q–Add–Defstratum).

rule	λ	C_{num} in degrees
self-dual	1	2.082755794784418
microfiber only	$1 - 5/468$	2.123621809570142
trace $5/468 + (3/11)\Delta_4$	0.989285913019141	2.123738337181414
$4/468$ instead of $5/468$	0.991422665155894	2.115535228138553
$3/12$ instead of $3/11$	0.989288440210566	2.123728626433667
$5/243$ instead of $5/468$	0.979393542015659	2.161907060464779
$5/729$ instead of $5/468$	0.993110963140488	2.109064222037112
$7/729$, seed measure	0.990367478915522	2.119584299852700.

The ablations demonstrate that the output distinguishes the complete microfiber, the reduced dimension eleven, and the emission measure. By themselves, they do not assign a chance probability: such a probability would first require defining a space of alternative models and a measure on it.

37.5 Typed composition of the morphisms Archimedean, exceptional, and dimensional

The complete construction can be written as a sequence of applications with visible domains:

$$
\begin{aligned}
\Omega_{\mathrm{seed}} &\xrightarrow{F} \mathcal{U}_6 \xrightarrow{J_F} H_U, \\
\mathscr{R}_\pi^{++} &\xrightarrow{\beta_\pi} P_G H_G, \\
L_{\mathrm{or}} \otimes P_G H_G &\xrightarrow{W_{GK}} P_3 V_{11} \xrightarrow{U_W} Q_3 E_{30}, \\
(\Pi_\pi^{(468)}, \Delta_4 P_3) &\longmapsto D_{\mathrm{switch}} \xrightarrow{\tau} \delta_* \xrightarrow{\mathsf{Res}} \lambda_*, \\
\lambda_* &\xrightarrow{\mathsf{Loc}} a_* \xrightarrow{\mathcal{L}_{1,*}} y_{\mathrm{band}} \xrightarrow{180/\pi} C_{\mathrm{band}}, \\
(\alpha_{\mathrm{HMT}}, y_{\mathrm{band}}) &\xrightarrow{\exp(-xI + y_{\mathrm{band}} R)} T_{\mathrm{band}}.
\end{aligned}
$$

Each arrow declares what it preserves:

- F preserves the emission class and leaves its presentation multiplicity in the fibre;
- J_F compensates that multiplicity in the norm;
- β_π preserves the exceptional position;
- W_{GK} preserves the oriented representation;
- U_W preserves incidence and metric;
- the trace forgets coordinates and preserves ranks;
- the transfer operator sums paths with the action weight;
- the chart $180/\pi$ represents an internal phase in degrees.

37.6 Dependency diagram of the angular operator and its publications

Object	Mathematical Nature	Contenido demostrado
$U_W = I/6 : V_{11} \to E_{30}$	exact theorem	isometry, surjectivity, equivariant incidence, and positive uniqueness
$P_3 \leftrightarrow Q_3$	exact theorem	conjugate projectors, rank three, and trace $3/11$
star R_{B_K}	typed exact theorem	spectrum $1^{[7]}, (-1)^{[4]}$, index $3/11$, and conjugation obstruction
P_G in $t = 7$	finite exact theorem	filtration $11 \to 3 \to 1$, positive projector, and action S_3
$\mathscr{R}_\pi^{++} \leftrightarrow P_G$	exact theorem	S_3-equivariant bijection by exceptional position
oriented isometric identification	new relative theorem	unique selector through B_K, u_K, Reynolds, polar, and reduction $240 \to 1$
measure $1/468$	exact theorem of Tr_q	unique normalized trace and polar application induced by fibers
D_{switch} y δ_*	theorem exact relativo to Tr_q–Add–Def	positivity, spectrum, and trace $5/468 + (3/11)\Delta_4$
λ_*	theorem internal relativo to Tr_q–Add–Def–Res	residual capacity obtained from the defect through the rule Res
$\mathscr{A}_*$	internal theorem relative to Tr_q–Add–Def–Res–Loc	cocycle additive of paths with cost local $g + \lambda_* s$
T_{band} y C_{band}	intervalar computational theorem	reduction $72 \to 60 \to 52$, rational similarity, Gershgorin isolation and output whose certified distance lies between 1.774799×10^{-9} and 1.799703×10^{-9} degrees of C_{reg}
$C_{\text{band}} = C_{\text{reg}}$	refuted statement for the linear law	the certified interval of the difference excludes zero
total transition $\mathcal{X}_{\text{TPK}}^{\text{enr}} \to \Gamma_{\text{tr}}$	required global realizer	Tr_q–Add–Def–Res–Locdefines the relative reader; the complete transition additionally requires a dynamical pushforward measure on Γ_{tr}

37.7 Predimensional action at the center of the double projection of the genealogical state

The double projection begins from the same enriched state and produces two readings. In the Archimedean branch, compatible cylinders are refined and their diameters decrease; evaluation fixes a limiting value and forgets part of the genealogy. In the exceptional branch, incidences are prolonged toward Witt, Golay, Niemeier–Leech, $V^\natural$, the Monster, and moonshine.

The new result adds a common object stronger than a coincidence of cardinalities. The eleven-mode module, its rank-three positive sector, and the trace of the action pass from dodecaphase to Witt through U_W. The oriented sector $t = 7$ is incorporated through W_{GK}. The microfiber of π provides the regional projector $\Pi_\pi^{(468)}$, and the uniform measure on the catalogue fixes its weight.

Positive action thus stands at the center of the two readings:

$$\text{region} \quad \oplus \quad \text{oriented polarization} \quad \longmapsto \quad \text{positive defect.}$$

The Archimedean evaluation and the exceptional evaluation remain different maps; the intertwiner identifies a common sector, it does not collapse their codomains.

37.8 Compatibility of the angular operator with the cylinders of the discrete continuum

An HMT history determines nested cylinders

$$C_1 \supset C_2 \supset \cdots$$

with compatible prefixes. When the Archimedean reader satisfies

$$\operatorname{diam}(C_n) \longrightarrow 0,$$

the completeness of $\mathbb{R}$ fixes at most one value in the intersection. The new action adds structure before that limit: it distinguishes the cost of the route, the emission class, and the multiplicity of presentations that the final value no longer displays.

This is the precise mathematical meaning of the assertion that the continuum is read as a projection of discrete genealogies. The system of cylinders and its evaluations is prolonged at every scale; the global coverage theorem proves $\mathrm{ev}_{\mathbb{R}}(\Omega_{\infty,K}) = K$ for every compact set $K \subset \mathbb{R}$ and, upon directing the intervals, proves that the image is all of $\mathbb{R}$. The finite theorem in this section controls the angular action at each level and is integrated into that global coverage; it neither reopens nor conditions it. The algebraic descent of each operation retains its own verification.

37.9 Compatibility with the exceptional rider, Moonshine, and M-theory screens

The intertwiner U_W lands in the incidence module of W_{12}. From the ternary code, the two exceptional continuations remain separate:

$$C_W \longrightarrow W_{12} \xrightarrow[\text{relative to } (D,\ell)]{} W_{24},$$

and

$$C_W \xrightarrow{\phi} N(A_2^{12}) \xrightarrow{\nu_\alpha} \Lambda_{24} \longrightarrow V_\Lambda \longrightarrow V^\natural \longrightarrow \mathbb{M} \longrightarrow \text{moonshine.}$$

Positive action contributes to this branch a transported sector and an orientation line. The arrows from Leech to moonshine are the classical constructions applied to the finite object obtained. The Monster action is realized on $V^\natural$; the HMT words are not identified here with a direct module for that action.

The screens $12 \to 11 \to 10$, $26 \to 24$, and the T-dual involution form an exact mathematical interface with the dimensions and dualities used in M-theory. The full physical identification belongs to Dimensional Mechanics and is typed in the second part by the additional map to g_{MN}, C_{MNP}, ψ_M, action, and supersymmetry. It is neither deferred to another work nor deduced from the Monster action.

37.10 Spectral factorization and isolation of the dominant sectors

The pre-dimensional line is organized into four strata.

1. **Finite kernel.** APP, TRIT, the TPK emitter, the catalog $104\,976 \to 468 \to 243$, the five regions of π, the dodecaphase, K, α, Witt, and the exceptional branch have explicit constructions and certificates.
2. **Operatorial bridge.** Incidence constructs U_W; the oriented sector $t = 7$ constructs P_G; B_K and u_K select the oriented isometric identification; the three rank-three projectors are identified relative to that rule.
3. **Action extension.** Tr_qfixes the measure $1/468$, Addfixes the additive character, and Defdeclares the coupling $\Delta_4 P_3$. Restransforms the defect into capacity, and Loccarries capacity to the edges. The five clauses produce the local law.
4. **Spectral consequence.** Γ_{tr}produces T_{band} and a certified predimensional angular phase, with relative difference between 8.356×10^{-10} and 8.475×10^{-10} with respect to C_{reg}.

This permits a strong and typed closure of this route: the finite emitter of the infinite coinductive section, the incidence, the identification of projectors, the measure, and the action are closed relative to the declared primitives, the variational rule $\mathcal{E}_B$, and the action clauses Tr_q–Add–Def–Res–Loc. Addselects the linear form; Defselects its second coordinate; Resand Locspecify how it is read as a cost. The global theorem covering all of $\mathbb{R}$, induction from a total transition of $\mathcal{X}^{\mathrm{enr}}_{\mathrm{TPK}}$, and physical equivalence with M-theory are formulated as extensions of the preceding results.

37.11 Compatibility between the matrix, angular, and lattice realizations

The constructed representations simultaneously satisfy the following compatibility conditions:

1. five regions of π, with three positivas, one negativa and one transversal;
2. P_G, P_3, Q_3 as positive projectors and R_{B_K} as an orientation involution;
3. the oriented homogeneous sector $t = 7$, not the heterogeneous quotient of $t = 8$;
4. the line L_{or} in the triple isometric identification;
5. the relative character of the variational rule that selects H_*, c_*, r_*;
6. $5/468$ on emissions and $7/729$ on seeds, without interchanging them;
7. the linearity of the character as a consequence of Addand the coefficient Δ_4 as content separated from Def;
8. $(y_{\mathrm{band}}, C_{\mathrm{band}})$ as a certified spectral output, separate from $(y_{\mathrm{reg}}, C_{\mathrm{reg}})$, the canonical regional realization of the following section;
9. α_{HMT} as input to x, so that the determinant reading is reversible and not independent;
10. the branches $W_{12} \to W_{24}$ and $A_2^{12} \to \Lambda_{24}$ as separate constructions;
11. the relative reader Tr_q–Add–Def–Res–Locdistinguished from a total transition, whose dynamic image measure constitutes an additional construction.

These conditions are not editorial notes. They form part of the logical structure of the result.

Chapter 38

Exact equivalence of the global and analytic charts of the angular pair: degrees, radians, phases $q_{\pm}$, and complementary orientation

The dimensionless component posterior to the return is constructed before the counter-angle by means of $\eta_{\mathrm{ret}} = H_5 - (90/\pi)C_\pi(\alpha_{\mathrm{HMT}})$. In the parallel determinantal branch the valuation 10^{-34} is obtained, and both branches converge in the magnitude $\hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}} 10^{-34} U_S$. Only afterward are $A_{\mathrm{deg}} = 10^3 \alpha_{\mathrm{HMT}}$ and $C^*_{\mathrm{deg}} = 2(\eta_{\mathrm{ret}} + \alpha_{\mathrm{HMT}})$ published; the solving $C^*/2 - \alpha_{\mathrm{HMT}}$ remains as inverse verification.

38.1 Intrinsic angular phase α_{HMT}, nonadic lift 10^3, and precursor x_A

38.1.1 Construction of the analytic precursor x_A, the contraction D_A, and the regional correction $\mathscr{C}_\pi(\alpha)$

The finite catalog and an analytic evaluation answer questions of different types. The catalog determines which regions exist, which share a return and what information each preserves. An analytic evaluation must also decide how the length of a word is transformed into degree, how a branch is prolonged after its return and with which scalar character that prolongation is read. This chapter explicitly constructs that second operation.

The construction begins with four data already obtained in the preceding chapters: the five-region microfiber of π, the length six of each regional word, the dodecaphasic closure of α, and the bilayer transport associated with the common coordinate $A = 10^3\alpha$. No value of the conjugate angular coordinate C_*^{HMT}, of the reduced action component, of the Barbero–Immirzi functional, or of a metrological constant is used in the evaluation. The result is a convergent regional character, an exact recurrence, and an oriented phase whose sexagesimal chart reproduces the published conjugate angular coordinate.

The separation between the finite part and the analytic part is essential. The integer 169 will be a theorem of the catalogue. The series that prolongs it will be the theorem of an analytic reader whose rules are declared before it is evaluated. Thus, the additional premises are not concealed within a decimal coincidence.

38.2 Preangular construction of η_{ret} through H_5, D_A and $\mathscr{C}_\pi(\alpha)$

38.2.1 Independent derivation of $\eta_{\mathrm{ret}} = H_5 - (90/\pi)\mathscr{C}_\pi(\alpha)$ through the regional analytic reader

The finite catalogue and an analytic evaluation answer questions of different types. The catalogue determines which regions exist, which share a return, and what information each preserves. An analytic evaluation must further determine how the length of a word is transformed into degree, how a branch is prolonged after its return, and by which scalar character that prolongation is read. This section constructs that second operation explicitly.

The construction begins with the five-region microfiber of π, the length six of each regional word, the dodecaphasic closure of α, and the declared angular insertion

$$\iota_\angle(x) := 10^3 x\, u_\circ, \qquad A_{\mathrm{deg}} := 10^3 \alpha.$$

Here $u_\circ$ is the sexagesimal base whose complete turn has coordinate 360. The factor 10^3 belongs to that chart normalization and is not deduced from the arithmetic closure of α. The two-sheet transport is subsequently constructed from A_{deg}. The function evaluated by the reader does not receive as an argument the counter-angle, the post-return coordinate, the hexad–octad functional, or a metrological constant. The result is a convergent regional character, an exact recurrence, and an oriented phase in the declared sexagesimal chart.

The separation between the finite part and the analytic part is essential. The integer 169 is a theorem of the catalogue. The series that prolongs it belongs to an analytic reader whose rules are declared before evaluating it.

The phase is denoted by y_* and, when its regional origin is emphasized, by y_{reg}; both symbols denote the same reader constructed below.

The evaluation receives the catalogue of 468 emissions, the exact value of α_{HMT}, $\varphi = (1+\sqrt{5})/2$, the fixed formula of H_5, the angular chart, and the five rules of the regional reader. With those data, it computes 169, D_A, the recurrence, and C^*. The theorem is exact in this declared class of readers; it does not use C^* as an input or assert that the rules of H_5 are the only possible natural class.

38.2.2 Persistence of the return in the 5 regions of the microfiber

Let

$$\mathcal{F}_\pi = \Psi^{-1}(010211) \subset \mathcal{U}_6$$

the microfiber relative to the catalog. Recall its five elements, now separated into initial block and terminal block:

$$(501, 614, 498 \mid 169, 272, 272),$$
$$(810, 923, 870 \mid 169, 272, 272),$$
$$(810, 983, 810 \mid 169, 272, 272),$$
$$(870, 923, 810 \mid 169, 272, 272),$$
$$(870, 923, 810 \mid 418, 674, 521).$$

We define the return projection.

$$p_{\mathrm{ret}} : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^3, \qquad p_{\mathrm{ret}}(u_1, \dots, u_6) = (u_4, u_5, u_6).$$

Proposition 38.1 (Persistent terminal block)**.** *The multiset $p_{\mathrm{ret}}(\mathcal{F}_\pi)$ has a unique element of maximum multiplicity:*

$$b_\pi = (169, 272, 272), \qquad \mathrm{mult}_{\mathcal{F}_\pi}(b_\pi) = 4.$$

The only remaining terminal block is

$$b_\pi^- = (418, 674, 521)$$

and has multiplicity one.

Proof. The assertion follows by exact enumeration of the five rows of the catalog that satisfy $\Psi(U) = 010211$. Four rows have terminal block $(169, 272, 272)$; the fifth, which carries the mutated return, has $(418, 674, 521)$. The maximum multiplicity is four and is attained only once. □

The oriented difference between the mutated return and the persistent is

$$\omega_\pi = b_\pi^- - b_\pi = (249, 402, 249), \qquad \langle (1,1,1), \omega_\pi \rangle = 900.$$

This identity separates two objects that must not be conflated. The integer 169 is a coordinate of a terminal datum, that is, a datum of degree zero. The vector ω_π can be read as a one-cochain after orienting the edge joining the two blocks. Calling it a cocycle would additionally require fixing a cochain complex and verifying that its coboundary vanishes. The analytic prolongation constructed below does not presuppose that identification.

The selection of 169 does not require privileging the first position of the block. The symmetric group $\mathfrak{S}_3$ acts by permuting its three coordinates. The stabilizer of b_π exchanges the two occurrences of 272

and fixes the unique coordinate of multiplicity one. Denote that coordinate by $\mathrm{sing}(b)$ when the block has type (r, s, s), with $r \neq s$. Equivalently,

$$\mathrm{sing}(b) = \sum_{j=1}^{3} b_j - 2\,\mathrm{moda}(b).$$

In particular,

$$\boxed{\mathrm{sing}(b_\pi) = 169.}$$

Definition 38.2 (Residual fiber selector)**.** The selector ρ_π successively applies two operations:

1. selects the terminal block of maximum multiplicity within $p_{\mathrm{ret}}(\mathcal{F}_\pi)$;
2. retains the coordinate fixed by the stabilizer of that block.

By proposition 38.1,

$$\boxed{\rho_\pi = 169.}$$

Theorem 38.3 (Equivariant uniqueness of return)**.** *Among the selectors that*

1. *are invariant under permutations of the five regions;*
2. *are equivariant under permutations of the three terminal coordinates;*
3. *select the block of maximal multiplicity;*
4. *retain a coordinate fixed by its stabilizer,*

the return value is uniquely determined and is 169.

Proof. Invariance under $\mathfrak{S}_5$ prevents using the order of the rows. The third condition selects b_π, which is the unique mode by proposition 38.1. Its stabilizer in $\mathfrak{S}_3$ has two orbits on the coordinates: the pair of values 272 and the singleton value 169. The fourth condition selects the second orbit. Equivariance preserves its value when its position is shifted. □

This argument improves the positional reading of the return. The 169 is not «the first coordinate» of a selected row: it is an invariant of the regional multiset and of the action of its reordering symmetries. Nor is it selected by global rarity. Among the 468 emissions in the catalogue, 169 appears 84 times, uniformly distributed with 14 occurrences in each of the six positions. The uniqueness of theorem 38.3 is therefore relational and regional: it resides in the combination of microfiber, modal persistence, and stabilizer, not in the isolated frequency of the integer.

38.2.3 Construction of the regional analytic reader and its convergence domain

Analytic continuation is fixed by the following rules.

Definition 38.4 (Regional Determinantal Reader)**.** The determinantal regional reader satisfies:

1. *degree compatibility*: cylindrical length is the analytic degree;
2. *renewal decomposition*: the finite return impulse and strict continuation belong to distinct additive summands;
3. *continuation normalization*: after recording the residue ρ_π just once, the surviving branch begins at the unit of the determinant line;
4. *concatenation covariance*: each elementary prolongation acts through $\bigwedge^2 \mathcal{T} = D_A$;
5. *orientation*: in the cardinal convention fixed for APP, the regional return belongs to the compensating sector and is subtracted from the fifth-degree action phase.

The third rule has mathematical content. The value 169 is recorded once as the amplitude of the return; it is not multiplied again at each prolongation. The strict continuation counts the sole line that continues and normalizes it by its unit. This is the distinction between a renewal recurrence and homogeneous iteration of the initial impulse.

The need to declare that normalization can be proved. For every $\lambda \in \mathbb{R}$, the family

$$F_\lambda(z) = 169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

preserves the same finite return and the same determinantal ratio between consecutive coefficients. The catalogue and the ratio alone do not distinguish λ. The unit of the determinant line fixes $\lambda = 1$ before $z = \alpha$ is evaluated. Thus, the normalization is not obtained by fitting the final value; it forms part of the definition of the character.

38.2.4 Analytical solution of the return recurrence

We define the coefficients κ_n by

$$\kappa_6 = 169, \qquad \kappa_7 = D_A, \qquad \kappa_{n+1} = D_A \kappa_n \quad (n \geq 7).$$

The first equality is the regional impulse; the next two describe the normalized continuation.

Theorem 38.5 (Graduated analytic realization)**.** *The analytic germ compatible with the rules of definition 38.4 is unique and equals*

$$\boxed{\mathscr{C}_\pi(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z}.}$$

Their coefficients satisfy

$$\kappa_n = D_A^{n-6} \qquad (n \geq 7).$$

Proof. Write

$$F(z) = 169z^6 + \sum_{k \geq 1} c_k z^{6+k}.$$

The normalization of continuation and the determinantal covariance give

$$c_1 = D_A, \qquad c_{k+1} = D_A c_k.$$

By induction, $c_k = D_A^k$. The geometric sum yields

$$\sum_{k \geq 1} D_A^k z^{6+k} = \frac{D_A z^7}{1 - D_A z}.$$

No free coefficients remain. □

Since $0 < \alpha < 1$ and $0 < D_A < 1$,

$$0 < D_A \alpha < 1.$$

Consequently, $\mathscr{C}_\pi(\alpha)$ converges absolutely and its radius of convergence is $D_A^{-1} > 1$. Numerically,

$$D_A = 0.77512911924715643018888409648 02\ldots,$$

$$D_A \alpha = 0.00565639046986492673885079095469\ldots.$$

The speed of this contraction explains why the complete sum and its degree-seven truncation have the same first fifteen digits in the final chart.

38.2.5 Persistence of return in the 5 regions and compatibility of the η_{ret} component

Let

$$\mathcal{F}_\pi = \Psi^{-1}(010211) \subset \mathcal{U}_6$$

the microfiber relative to the catalog. Recall its five elements, now separated into initial block and terminal block:

$$(501, 614, 498 \mid 169, 272, 272),$$
$$(810, 923, 870 \mid 169, 272, 272),$$
$$(810, 983, 810 \mid 169, 272, 272),$$
$$(870, 923, 810 \mid 169, 272, 272),$$
$$(870, 923, 810 \mid 418, 674, 521).$$

We define the return projection.

$$p_{\mathrm{ret}} : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^3, \qquad p_{\mathrm{ret}}(u_1, \dots, u_6) = (u_4, u_5, u_6).$$

Proposition 38.6 (Persistent terminal block)**.** *The multiset $p_{\mathrm{ret}}(\mathcal{F}_\pi)$ has a unique element of maximum multiplicity:*

$$b_\pi = (169, 272, 272), \qquad \mathrm{mult}_{\mathcal{F}_\pi}(b_\pi) = 4.$$

The only remaining terminal block is

$$b_\pi^- = (418, 674, 521)$$

and has multiplicity one.

Proof. The assertion follows by exact enumeration of the five rows of the catalog that satisfy $\Psi(U) = 010211$. Four rows have terminal block $(169, 272, 272)$; the fifth, which carries the mutated return, has $(418, 674, 521)$. The maximum multiplicity is four and is attained only once. □

The oriented difference between the mutated return and the persistent is

$$\omega_\pi = b_\pi^- - b_\pi = (249, 402, 249), \qquad \langle (1, 1, 1), \omega_\pi \rangle = 900.$$

This identity separates two objects that must not be conflated. The integer 169 is a coordinate of a terminal datum, that is, a datum of degree zero. The vector ω_π can be read as a one-cochain after orienting the edge joining the two blocks. Calling it a cocycle would additionally require fixing a cochain complex and verifying that its coboundary vanishes. The analytic prolongation constructed below does not presuppose that identification.

The selection of 169 does not require privileging the first position of the block. The symmetric group $\mathfrak{S}_3$ acts by permuting its three coordinates. The stabilizer of b_π exchanges the two occurrences of 272 and fixes the unique coordinate of multiplicity one. Denote that coordinate by $\mathrm{sing}(b)$ when the block has type (r, s, s), with $r \neq s$. Equivalently,

$$\mathrm{sing}(b) = \sum_{j=1}^{3} b_j - 2\,\mathrm{moda}(b).$$

In particular,

$$\boxed{\mathrm{sing}(b_\pi) = 169.}$$

Definition 38.7 (Residual fiber selector)**.** The selector ρ_π successively applies two operations:

1. selects the terminal block of maximum multiplicity within $p_{\mathrm{ret}}(\mathcal{F}_\pi)$;

2. retains the coordinate fixed by the stabilizer of that block.

By proposition 38.6,

$$\boxed{\rho_\pi = 169.}$$

Theorem 38.8 (Equivariant uniqueness of return)**.** *Among the selectors that*

1. *are invariant under permutations of the five regions;*
2. *are equivariant under permutations of the three terminal coordinates;*
3. *select the block of maximal multiplicity;*
4. *retain a coordinate fixed by its stabilizer,*

the return value is uniquely determined and is 169*.*

Proof. Invariance under $\mathfrak{S}_5$ prevents using the order of the rows. The third condition selects b_π, which is the unique mode by proposition 38.6. Its stabilizer in $\mathfrak{S}_3$ has two orbits on the coordinates: the pair of values 272 and the singleton value 169. The fourth condition selects the second orbit. Equivariance preserves its value when its position is shifted. □

This argument improves the positional reading of the return. The 169 is not «the first coordinate» of a selected row: it is an invariant of the regional multiset and of the action of its reordering symmetries. Nor is it selected by global rarity. Among the 468 emissions in the catalogue, 169 appears 84 times, uniformly distributed with 14 occurrences in each of the six positions. The uniqueness of theorem 38.8 is therefore relational and regional: it resides in the combination of microfiber, modal persistence, and stabilizer, not in the isolated frequency of the integer.

38.2.6 Application of the regional reader to the conjugate angular momentum pair

Analytic continuation is fixed by the following rules.

Definition 38.9 (Regional Determinantal Reader)**.** The determinantal regional reader satisfies:

1. *degree compatibility*: cylindrical length is the analytic degree;
2. *renewal decomposition*: the finite return impulse and strict continuation belong to distinct additive summands;
3. *continuation normalization*: after recording the residue ρ_π just once, the surviving branch begins at the unit of the determinant line;
4. *concatenation covariance*: each elementary prolongation acts through $\bigwedge^2 \mathcal{T} = D_A$;
5. *orientation*: in the cardinal convention fixed for APP, the regional return belongs to the compensating sector and is subtracted from the fifth-degree action phase.

The third rule has mathematical content. The value 169 is recorded once as the amplitude of the return; it is not multiplied again at each prolongation. The strict continuation counts the sole line that continues and normalizes it by its unit. This is the distinction between a renewal recurrence and homogeneous iteration of the initial impulse.

The need to declare that normalization can be proved. For every $\lambda \in \mathbb{R}$, the family

$$F_\lambda(z) = 169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

preserves the same finite return and the same determinantal ratio between consecutive coefficients. The catalogue and the ratio alone do not distinguish λ. The unit of the determinant line fixes $\lambda = 1$ before

$z = \alpha$ is evaluated. Thus, the normalization is not obtained by fitting the final value; it forms part of the definition of the character.

38.2.7 Closed sum and publication of the posterior component upon return

We define the coefficients κ_n by

$$\kappa_6 = 169, \qquad \kappa_7 = D_A, \qquad \kappa_{n+1} = D_A \kappa_n \quad (n \geq 7).$$

The first equality is the regional impulse; the next two describe the normalized continuation.

Theorem 38.10 (Graduated analytic realization)**.** *The analytic germ compatible with the rules of definition 38.9 is unique and equals*

$$\boxed{\mathscr{C}_\pi(z) = 169 z^6 + \frac{D_A z^7}{1 - D_A z}.}$$

Their coefficients satisfy

$$\kappa_n = D_A^{n-6} \qquad (n \geq 7).$$

Proof. Write

$$F(z) = 169 z^6 + \sum_{k \geq 1} c_k z^{6+k}.$$

The normalization of continuation and the determinantal covariance give

$$c_1 = D_A, \qquad c_{k+1} = D_A c_k.$$

By induction, $c_k = D_A^k$. The geometric sum yields

$$\sum_{k \geq 1} D_A^k z^{6+k} = \frac{D_A z^7}{1 - D_A z}.$$

No free coefficients remain. □

Since $0 < \alpha < 1$ and $0 < D_A < 1$,

$$0 < D_A \alpha < 1.$$

Consequently, $\mathscr{C}_\pi(\alpha)$ converges absolutely and its radius of convergence is $D_A^{-1} > 1$. Numerically,

$$D_A = 0.77512911924715643018888409648 02\ldots,$$

$$D_A \alpha = 0.00565639046986492673885079095469\ldots.$$

The speed of this contraction explains why the complete sum and its degree-seven truncation have the same first fifteen digits in the final chart.

38.3 Confluence of η_{ret} With the valuation $\varepsilon_H = -34$ in the magnitude $\hbar_{\mathrm{ret}}$

38.3.1 Derivation of the determinantal character of parangular transport

The dodecaphasic closure provides α. The completion of the three nonadic pairs provides the scale 10^3, so that

$$A = 10^3\alpha.$$

Having chosen the Archimedean angular chart, the common coordinate in radians is

$$x = \frac{\pi A}{180} = \frac{50\pi}{9}\alpha.$$

Let R be a real involution of zero trace in dimension two,

$$R^2 = I_2, \qquad \operatorname{tr} R = 0,$$

and consider the formal two-layer transport

$$\mathcal{T}(x, y) = \exp(-xI_2 - yR).$$

The oriented variable y has not yet been evaluated. Nevertheless, the action of $\mathcal{T}$ on the determinant line is independent of it:

$$\begin{aligned}\det \mathcal{T}(x, y) &= \exp(\operatorname{tr}(-xI_2 - yR)) \\ &= e^{-2x}.\end{aligned}$$

We define

$$\boxed{D_A = e^{-2x} = e^{-100\pi\alpha/9}.}$$

The subscript recalls that this is the determinant of the common contraction associated with A, not a function of C_*^{HMT}.

Proposition 38.11 (Character of the determinant line)**.** *The map*

$$\delta_A : (\mathbb{N}_0, +) \longrightarrow (\mathbb{R}_{>0}, \cdot), \qquad \delta_A(k) = D_A^k,$$

is the unique multiplicative character whose value on an elementary prolongation is D_A.

Proof. The multiplicativity gives

$$\delta_A(k) = \delta_A(\underbrace{1 + \cdots + 1}_{k \text{ times}}) = \delta_A(1)^k = D_A^k.$$

The same identity proves existence and uniqueness. □

The determinant remains invariant under conjugation of $\mathcal{T}$ and under the exchange of sheets $R \mapsto -R$. Therefore, the tail constructed from D_A belongs to the common mode. The orientation will appear in the sign with which the regional character corrects the phase.

38.3.2 action character of degree 5 in the construction regional

The central block

$$\mathcal{B}_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$$

has eigenvalues 9 and 5. The dodecaphasic cycle contributes rank 12; its step-three orbits have length four; and the complete census contributes the exact quotient

$$\frac{104976}{1944} = 54.$$

From these invariants the fifth-degree action character is fixed.

$$\boxed{H_5(\alpha,\varphi) = \frac{1}{2}\sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5.}$$

The four rational coefficients can be written as

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \qquad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \qquad \frac{7}{48} = \frac{\frac{1}{2}\operatorname{tr}\mathcal{B}_c}{4\cdot 12}, \qquad \frac{1}{54} = \frac{1944}{104976},$$

with $(\lambda_+,\lambda_-) = (9,5)$. These identities certify the provenance of the numbers used. The successive assignment of those invariants to degrees two, three, four, and five forms part of the analytic character; it does not follow from the mere existence of the finite catalog.

This precision removes a common ambiguity. A combination of invariants of the construction is an internal and reproducible definition; its naturality requires the reader rules specifying order, signs, and normalization. Here those rules are visible and are subject to the same negative controls as the rest of the construction.

38.3.3 Exact decomposition of the action line into pre-return and post-return sections

The fifth-degree character and the post-action component after return are two distinct values:

$$H_5 = 1.0545718176461544696258218294330 6\ldots,$$

$$\bar{h}^{\mathrm{ret}}_{\mathrm{HMT}} = 1.0545718169150390611336905847243 0\ldots.$$

The first is the local action before incorporating the regional character; the second is the oriented action remaining after the return. The correction that generates the conjugate angular coordinate prevents identifying them.

The metrological comparison must respect this distinction. The first value gives the mantissa

$$2\pi H_5 = 6.626070149999987926001110349899 47\ldots,$$

while the subsequent action gives

$$2\pi\bar{h}^{\mathrm{ret}}_{\mathrm{HMT}} = 6.626070145406254333510749847592 88\ldots.$$

In the International System, the value

$$h = 6.62607015\times 10^{-34}\ \mathrm{J\,s}$$

is exact by definition [20]. By therefore,

$$2\pi H_5 - 6.62607015 = -1.2073998889650101\ldots\times 10^{-14},$$

and it is not an exact equality. The proximity of the value before the return is a remarkable numerical check; it does not authorize replacing the defining constant of the SI with the subsequent component or confusing the two HMT actions.

This result precisely determines the mathematical situation. The regional reader generates C_*^{HMT} and, from it, the functional $\Gamma_{6\to 8}$. The same calculation separates the Planck approximation produced by H_5 from the reduced action intervening in the conjugate angular coordinate. The separation does not weaken the construction: it eliminates an identification incompatible with the exact figures and converts every comparison into a falsifiable assertion of a well-defined type.

38.3.4 Factorization of the determinantal character in the 2 phase charts

Dodecaphase closure supplies α. The declared angular insertion uses the scale 10^3, compatible with the grouping of three nonadic pairs, so that

$$A_{\mathrm{deg}} = 10^3 \alpha\,^{\circ}.$$

Having chosen the Archimedean angular chart, the common coordinate in radians is

$$x =: A_{\mathrm{rad}} = \frac{\pi A_{\mathrm{deg}}}{180} = \frac{50\pi}{9}\alpha.$$

Let R be a real involution of zero trace in dimension two,

$$R^2 = I_2, \qquad \operatorname{tr} R = 0,$$

and consider the formal two-layer transport

$$\mathcal{T}(x, y) = \exp(-xI_2 - yR).$$

The oriented variable y has not yet been evaluated. Nevertheless, the action of $\mathcal{T}$ on the determinant line is independent of it:

$$\begin{aligned} \det \mathcal{T}(x, y) &= \exp(\operatorname{tr}(-xI_2 - yR)) \\ &= e^{-2x}. \end{aligned}$$

We define

$$\boxed{D_A = e^{-2x} = e^{-100\pi\alpha/9}.}$$

The subscript recalls that this is the determinant of the common contraction associated with A, not a function of the counter-angle.

Proposition 38.12 (Character of the determinant line)**.** *The map*

$$\delta_A : (\mathbb{N}_0, +) \longrightarrow (\mathbb{R}_{>0}, \cdot), \qquad \delta_A(k) = D_A^k,$$

is the unique multiplicative character whose value on an elementary prolongation is D_A.

Proof. The multiplicativity gives

$$\delta_A(k) = \delta_A(\underbrace{1 + \cdots + 1}_{k \text{ times}}) = \delta_A(1)^k = D_A^k.$$

The same identity proves existence and uniqueness. □

The determinant remains invariant under conjugation of $\mathcal{T}$ and under the exchange of sheets $R \mapsto -R$. Therefore, the tail constructed from D_A belongs to the common mode. The orientation will appear in the sign with which the regional character corrects the phase.

38.3.5 Publication of the action character of degree 5 in the parangular chart

The central block

$$\mathcal{B}_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$$

has eigenvalues 9 and 5. The dodecaphasic cycle contributes rank 12; its step-three orbits have length four; and the complete census contributes the exact quotient

$$\frac{104976}{1944} = 54.$$

The current reader declares the nature of fifth-degree action

$$\boxed{H_5(\alpha,\varphi)=\frac{1}{2}\sqrt{\frac{1000\alpha}{\varphi}}-\alpha+\frac{9}{16}\alpha^2-\frac{5}{9}\alpha^3+\frac{7}{48}\alpha^4-\frac{1}{54}\alpha^5.}$$

The four rational coefficients can be written as

$$\frac{9}{16}=\frac{\lambda_+}{4^2},\qquad \frac{5}{9}=\frac{\lambda_-}{\lambda_+},\qquad \frac{7}{48}=\frac{\frac{1}{2}\operatorname{tr}\mathcal{B}_c}{4\cdot 12},\qquad \frac{1}{54}=\frac{1944}{104976},$$

with $(\lambda_+,\lambda_-)=(9,5)$. These identities fix the internal arithmetic representability of the character coefficients. Their order, signs, and normalization belong to the graded rule of the regional reader and are applied before any chart of units. The ablations certify that every term effectively intervenes in the resulting phase. In particular, neither the SI mantissa of action nor the sexagesimal value of the counter-angle belongs to the domain of the generating function.

38.4 Parangular application $P(\alpha,\eta_{\mathrm{ret}})$ and bilayer closure $C^*=2(\eta_{\mathrm{ret}}+\alpha)$

38.4.1 Oriented phase and conjugate angular coordinate

The fifth-degree phase prior to return is

$$y_5=\frac{\pi}{90}\big(H_5+\alpha\big).$$

The complete regional character produces the correction $\mathscr{C}_\pi(\alpha)$. With the orientation fixed in definition 38.4, we define

$$\boxed{y_*=\frac{\pi}{90}\big(H_5+\alpha\big)-169\alpha^6-\frac{D_A\alpha^7}{1-D_A\alpha}.}$$

This is the primary formula. The sexagesimal unit does not come into play until applying the chart.

$$C_*^{\mathrm{HMT}}=\frac{180}{\pi}y_*.$$

Theorem 38.13 (Conjugate angular coordinate of the regional reader determinantal)**.** *With the dodecaphasic closure of α, $\varphi=(1+\sqrt{5})/2$ and the rules of the determinantal regional reader fixed, the phase y_* and its conjugate angular coordinate are determined without introducing the sought value. Their evaluation is*

$$\boxed{y_*=0.037066226465838263984937794092306 38\ldots,}$$

$$\boxed{C_*^{\mathrm{HMT}}=2.123738338968645724261951380393 36\ldots^\circ.}$$

Therefore, to fifteen decimal places,

$$\boxed{C_*^{\mathrm{HMT}}=2.123738338968646^\circ.}$$

Proof. theorem 38.3 determines 169. The closure of α determines A,x, and $D_A=e^{-100\pi\alpha/9}$. theorem 38.5 determines the complete series. The formula for H_5 contains only α, φ, and declared structural invariants. Substituting these data into the expression for y_* gives the printed value. The decimal value of C_*^{HMT} occurs in none of the right-hand sides. □

The reduced action component after return is now as follows output:

$$\boxed{\bar{h}^{\mathrm{ret}}_{\mathrm{HMT}} = \frac{C^{\mathrm{HMT}}_{*}}{2} - \alpha = H_5 - \frac{90}{\pi}\mathscr{C}_{\pi}(\alpha).}$$

Numerically,

$$\bar{h}^{\mathrm{ret}}_{\mathrm{HMT}} = 1.0545718169150390611336905847243 0\ldots.$$

The inverse identity has not been used to introduce C^{HMT}_{*}; it is obtained after generating y_{*}.

38.4.2 Conjugate coordinate C^{*} and reversal of its orientation under the bilayer involution

The fifth-degree phase prior to return is

$$y_5 = \frac{\pi}{90}\big(H_5 + \alpha\big).$$

The complete regional character produces the correction $\mathscr{C}_{\pi}(\alpha)$. With the orientation fixed in definition 38.9, we define

$$\boxed{C^{*}_{\mathrm{rad}} := y_{*} = \frac{\pi}{90}\big(H_5 + \alpha\big) - 169\alpha^{6} - \frac{D_A\alpha^{7}}{1 - D_A\alpha}.}$$

From now on we also write

$$\boxed{y_{\mathrm{reg}} := y_{*}}$$

when it is necessary to distinguish this realization from the spectral phase of Γ_{tr}. The two notations here designate the same canonical regional object. This is the primary formula. The sexagesimal unit does not intervene until the chart is applied

$$\boxed{C^{*}_{\mathrm{deg}} := \frac{180}{\pi}C^{*}_{\mathrm{rad}}.}$$

The symbols C^{*}_{rad} and C^{*}_{deg} do not denote two counterangles: they are the radial and sexagesimal coordinates of the same angular element.

Theorem 38.14 (Counter-angle of the regional determinantal reader)**.** *Given the dodecaphase closure of α, $\varphi = (1+\sqrt{5})/2$, the germ H_5 and the rules of the regional determinantal reader, the phase C^{*}_{rad} and its counter-angle are determined without introducing the sought value into the evaluation function. Its evaluation is*

$$\boxed{C^{*}_{\mathrm{rad}} = 0.037066226465838263984937794092306 38\ldots,}$$

$$\boxed{C^{*}_{\mathrm{deg}} = 2.123738338968645724261951380393 36\ldots^{\circ}.}$$

Therefore, to fifteen decimal places,

$$\boxed{C^{*}_{\mathrm{deg}} = 2.123738338968646^{\circ}.}$$

Proof. The theorem 38.8 determines 169. The closure of α determines A_{deg}, A_{rad}, and $D_A = e^{-100\pi\alpha/9}$. The theorem 38.10 determines the complete series. The fixed formula for H_5 contains only α, φ, and declared structural invariants. Substituting these data into the expression for C^{*}_{rad} yields the printed value. The decimal value of the counter-angle appears in none of the right-hand sides of this evaluation. Comparison with a metrological action mantissa is undertaken only after the phase has been obtained and does not alter the generating function. □

Let $u_{\circ}$ be the sexagesimal base whose complete turn has coordinate 360, and let us introduce the angular coordinate $\alpha_{\angle,\mathrm{deg}} := \alpha_{\mathrm{HMT}}$. The output posterior to return is then typed by

$$\boxed{\eta_{\mathrm{ret}}^{(\mathrm{deg})} = \frac{C_{\mathrm{deg}}^{*}}{2} - \alpha_{\angle,\mathrm{deg}} = H_5 - \frac{90}{\pi}\mathscr{C}_{\pi}(\alpha).}$$

Its radial coordinate is

$$\eta_{\mathrm{ret}}^{(\mathrm{rad})} = \frac{\pi}{180}\eta_{\mathrm{ret}}^{(\mathrm{deg})} = \frac{C_{\mathrm{rad}}^{*}}{2} - \frac{\pi}{180}\alpha_{\mathrm{HMT}}.$$

Numerically,

$$\eta_{\mathrm{ret}}^{(\mathrm{deg})} = 1.0545718169150390611336905847243 0\ldots.$$

The inverse identity has not been used to introduce C_{deg}^{*}; it is obtained after generating C_{rad}^{*}. The symbol $\hbar$ is reserved for the element of the action line constructed in the following section.

38.4.3 Direct theorem $C^{*} = 2(\eta_{\mathrm{ret}} + \alpha)$ and inverse character of the identity $\eta_{\mathrm{ret}} = C^{*}/2 - \alpha$

The fifth-degree character and the coordinate after return are two distinct values:

$$H_5 = 1.0545718176461544696258218294330 6\ldots,$$

$$\eta_{\mathrm{ret}}^{(\mathrm{deg})} = 1.0545718169150390611336905847243 0\ldots.$$

The former is the local character before incorporation of the regional character; the latter is the oriented coordinate remaining after the return. The correction generating the contra-angle prevents their identification.

The metrological comparison must respect this distinction. The first value gives the mantissa

$$2\pi H_5 = 6.626070149999987926001110349899 47\ldots,$$

while the posterior coordinate gives

$$2\pi\eta_{\mathrm{ret}}^{(\mathrm{deg})} = 6.626070145406254333510749847592 88\ldots.$$

As an external control only, we retain the mantissa 6.62607015 of the defining value of the International System [20]. Neither its exponent nor its unit is yet incorporated into the HMT line: the decade is derived in the section 59.2 and the unit belongs to a later metrological chart. The comparison of mantissas gives

$$2\pi H_5 - 6.62607015 = -1.2073998889650101\ldots\times 10^{-14},$$

and is not an exact equality. The proximity of the pre-return value is a noteworthy numerical control; it does not authorize replacing the defining SI constant by the subsequent coordinate or confusing an angular coordinate with the action element that will be constructed upon incorporating the decade.

This result determines the mathematical situation precisely. The regional reader generates $(C_{\mathrm{rad}}^{*}, C_{\mathrm{deg}}^{*})$ and, from that single angular element, the functional $\Gamma_{6\to 8}$. The same calculation separates the external proximity of the mantissa produced by H_5 from the return coordinate that enters the counter-angle. The separation does not weaken the construction: it eliminates an identification incompatible with the exact digits and turns each comparison into a falsifiable assertion of a well-defined type.

The cubic completion

$$10^3 = (9+1)^3 = 9^3 + 3\cdot 9^2 + 3\cdot 9 + 1 = 729 + 243 + 27 + 1 \tag{38.1}$$

determines the chart global of the coordinate direct. The closure bilayer contributes the factor 2 of the coordinate conjugate:

$$\boxed{A_{\mathrm{deg}} = 10^3\alpha_{\mathrm{HMT}}, \qquad C_{\mathrm{deg}}^{*} = 2(\eta_{\mathrm{ret}} + \alpha_{\mathrm{HMT}}) \quad \text{in } \mathbb{R}/(360\mathbb{Z}).} \tag{38.2}$$

The inverse identity

$$\eta_{\mathrm{ret}} = \frac{1}{2} C^{*}_{\mathrm{deg}} - \alpha_{\mathrm{HMT}} \tag{38.3}$$

is established as an algebraic corollary of the direct construction.

38.5 Exact commutativity of the charts in turns, degrees, and radians

38.5.1 Derivation of the cylindrical length and the analytic degree of the parangular precursor

Let $\mathcal{P}$ be the free module generated by the regional words and let $F^{n}\mathcal{P}$ be its length filtration. The associated grading records the first level at which each word appears. For a parameter z, the ordinary length character is

$$\chi_{z}([w]) = z^{|w|}.$$

Multiplicativity

$$\chi_{z}([uv]) = \chi_{z}([u])\chi_{z}([v])$$

expresses that concatenation adds lengths. When evaluated at $z = \alpha$, a word of length n receives degree α^{n}.

Definition 38.15 (Degree compatibility)**.** The regional analytic reader is compatible with the cylindrical filtration when it sends the homogeneous component of length n to analytic degree n:

$$\mathrm{gr}_{n}\,\mathcal{P} \longrightarrow \alpha^{n}\mathbb{R}.$$

The five regions belong to $\mathcal{U}_{6}$; therefore, the return datum appears in degree six:

$$\rho_{\pi}[U_{6}] \longmapsto 169\alpha^{6}.$$

This six is the length of the regional word. It is not the length of the closed walks of chapter 9. Those walks traverse the four anchors $U_{\pi}, U_{e}, U_{\varphi}, U_{\mathrm{APP}}$ simultaneously and have minimum length eight. A word, an edge of the block graph, and a closed walk are objects of distinct types; the grading used here belongs to the first.

38.5.2 Publication of the analytic degree on the global phase chart

Let $\mathcal{P}$ be the free module generated by the regional words and let $F^{n}\mathcal{P}$ be its length filtration. The associated grading records the first level at which each word appears. For a parameter z, the ordinary length character is

$$\chi_{z}([w]) = z^{|w|}.$$

Multiplicativity

$$\chi_{z}([uv]) = \chi_{z}([u])\chi_{z}([v])$$

expresses that concatenation adds lengths. When evaluated at $z = \alpha$, a word of length n receives degree α^{n}.

Definition 38.16 (Degree compatibility)**.** The regional analytic reader is compatible with the cylindrical filtration when it sends the homogeneous component of length n to analytic degree n:

$$\mathrm{gr}_{n}\,\mathcal{P} \longrightarrow \alpha^{n}\mathbb{R}.$$

The five regions belong to $\mathcal{U}_{6}$; therefore, the return datum appears in degree six:

$$\rho_{\pi}[U_{6}] \longmapsto 169\alpha^{6}.$$

This six is the length of the regional word. It is not the length of the closed walks of chapter 9. Those walks traverse the four anchors $U_\pi, U_e, U_\varphi, U_{\mathrm{APP}}$ simultaneously and have minimum length eight. A word, an edge of the block graph, and a closed walk are objects of distinct types; the grading used here belongs to the first.

The global calendar exhibits the three factorizations

$$360 = 12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120. \tag{38.4}$$

The analytical chart is obtained through

$$A_{\mathrm{rad}} = \frac{\pi}{180} A_{\mathrm{deg}} = \frac{50\pi}{9} \alpha_{\mathrm{HMT}}, \qquad C^*_{\mathrm{rad}} = \frac{\pi}{180} C^*_{\mathrm{deg}} = \frac{\pi}{90}(\eta_{\mathrm{ret}} + \alpha_{\mathrm{HMT}}). \tag{38.5}$$

Theorem 38.17 (Commutativity of Regional and Action Factorizations)**.** *The two routes toward the conjugate phase produce the same output:*

$$\boxed{\frac{\pi}{180} C^*_{\mathrm{deg}} = \frac{\pi}{90}(\eta_{\mathrm{ret}} + \alpha_{\mathrm{HMT}}) = \frac{\pi}{90}(H_5 + \alpha_{\mathrm{HMT}}) - \mathscr{C}_\pi(\alpha_{\mathrm{HMT}}) = y_*.} \tag{38.6}$$

Proof. Substituting equation (35.5) in the second member yields

$$\frac{\pi}{90}\left(H_5 - \frac{90}{\pi}\mathscr{C}_\pi + \alpha_{\mathrm{HMT}}\right) = \frac{\pi}{90}(H_5 + \alpha_{\mathrm{HMT}}) - \mathscr{C}_\pi.$$

The final expression is the regional definition of y_*, whereas the first equality follows from equation (38.2). □

38.6 spectral decomposition $q_{\pm}$ and calculo functional on the channels 90/120 previously generated

38.6.1 Spectral reconstruction of the angular pair and hexad-octad transition

The common phase and the oriented phase determine the two channels.

$$q_- = e^{-x+y_*}, \qquad q_+ = e^{-x-y_*}.$$

The invariants are recovered exactly.

$$q_- q_+ = D_A, \qquad \alpha = -\frac{9}{100\pi}\log D_A, \qquad C^{\mathrm{HMT}}_* = \frac{90}{\pi}\log\frac{q_-}{q_+}.$$

The first equality reads the common contraction; the third, the oriented separation of the sheets. The exchange $R \mapsto -R$ permutes q_+ and q_-, preserves D_A and reverses the sign of y_*.

For

$$S_{90}(q) = \frac{q^{90}}{1-q^{270}}, \qquad S_{120}(q) = \frac{q^{120}}{1-q^{360}},$$

the hexad–octad transition functional is written entirely in the phase chart:

$$\Gamma_{6\to 8} = \frac{180}{\pi} x y_* + 12\big(S_{90}(q_-) - S_{90}(q_+)\big) - \big(S_{120}(q_-) - S_{120}(q_+)\big).$$

The evaluation produced by the coordinate C^{HMT}_* of theorem 38.13 is

$$\boxed{\Gamma_{6\to 8} = 0.27400767269445927474725526077 4\ldots.}$$

Thus, the internal value recognized in Dimensional Mechanics as the Barbero–Immirzi functional ceases to receive C_*^{HMT} as an independent literal: both descend from the same pair of phases generated in this chapter. The geometric and physical interpretation of $\Gamma_{6\to 8}$ belongs to the work on Dimensional Mechanics; its mathematical evaluation already belongs to the previously constructed interface.

38.6.2 Relative transporter and spectral disjunction of the blocks

In the ordered basis of channels, let

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_{\pm} = \frac{I \pm R}{2}, \qquad B_0 = 7I + 2R, \qquad B_1 = A_{\deg} I + C_{\deg}^* R.$$

The split decomposition

$$xI + yR = \rho(x,y) e^{\eta(x,y)R}, \quad \rho(x,y) = \sqrt{x^2 - y^2}, \quad \eta(x,y) = \operatorname{artanh}(y/x)$$

constructs the unique transporter

$$T_{\mathrm{rel}} = \frac{\sqrt{(A_{\deg})^2 - (C_{\deg}^*)^2}}{\sqrt{45}} \exp\left[\left(\operatorname{artanh}\frac{C_{\deg}^*}{A_{\deg}} - \operatorname{artanh}\frac{2}{7}\right) R\right], \qquad T_{\mathrm{rel}} B_0 = B_1.$$

In the spectral basis it acts by $9 \mapsto A_{\deg} + C_{\deg}^*$ and $5 \mapsto A_{\deg} - C_{\deg}^*$.

Theorem 38.18 (Spectral Disjunction of the Local and Angular Realizations)**.** *The spectra*

$$\mathrm{Spec}(B_0) = \{9, 5\}, \qquad \mathrm{Spec}(B_1) = \{A_{\deg} + C_{\deg}^*, A_{\deg} - C_{\deg}^*\}$$

are disjoint. Consequently,

$$\mathrm{Hom}_{B_0, B_1} := \{\Phi : \Phi B_0 = B_1 \Phi\} = \{0\}.$$

Proof. The rational bounds constructed for the angular coordinates give $7.29 < A_{\deg} < 7.30$ and $2.12 < C_{\deg}^* < 2.13$; therefore, $9.41 < A_{\deg} + C_{\deg}^* < 9.43$ and $5.16 < A_{\deg} - C_{\deg}^* < 5.18$. In the basis that diagonalizes R, each entry of an intertwiner is multiplied by the difference between an eigenvalue of B_0 and one of B_1; disjointness forces all of them to vanish. □

The relative transporter compares two blocks within the split algebra; the dodecaphasic–Witt and incidence intertwiners act between equivalent representations in their own spaces of residence. This separation positively types both operations and preserves their domains.

38.6.3 Spectral evaluation $V_{90,120}$ of the incidence channels previously generated by the TPK

The common phase and the oriented phase determine the two channels.

$$q_- = e^{-A_{\mathrm{rad}} + C_{\mathrm{rad}}^*}, \qquad q_+ = e^{-A_{\mathrm{rad}} - C_{\mathrm{rad}}^*}.$$

The invariants are recovered exactly.

$$q_- q_+ = D_A, \qquad \alpha = -\frac{9}{100\pi} \log D_A, \qquad C_{\deg}^* = \frac{90}{\pi} \log \frac{q_-}{q_+}.$$

The first equality reads the common contraction; the third, the oriented separation of the sheets. The exchange $R \mapsto -R$ permutes q_+ and q_-, preserves D_A and reverses the sign of C_{rad}^*.

Let $H^2 = I$ and $P_{\pm} = (I \pm H)/2$. The angular block

$$B = A_{\mathrm{rad}} I + C_{\mathrm{rad}}^* H = (A_{\mathrm{rad}} + C_{\mathrm{rad}}^*) P_+ + (A_{\mathrm{rad}} - C_{\mathrm{rad}}^*) P_- \tag{38.7}$$

is published multiplicatively by

$$e^{-B} = q_+P_+ + q_-P_-, \qquad q_\pm = \exp[-(A_{\mathrm{rad}} \pm C^*_{\mathrm{rad}})]. \tag{38.8}$$

The involution $C^* \mapsto -C^*$ permutes q_+ and q_-, and preserves

$$q_+q_- = e^{-2A_{\mathrm{rad}}} = D_A. \tag{38.9}$$

The cardinalities 90/120, previously constructed through TPK incidence, now receive their spectral evaluation:

$$V_{90,120} = (I - T^{90})(I - T^{120})^{-1} = \sum_{\sigma\in\{+,-\}} \frac{1 - q_\sigma^{90}}{1 - q_\sigma^{120}} P_\sigma. \tag{38.10}$$

38.6.3.1 Split relative transporter

The comparison With the central block of APP can be carried out without identifying representations distinct. Fijemos the ordered basis of channels and the involution

$$R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad R^2 = I, \qquad P_\pm = \frac{I \pm R}{2}.$$

In commutative algebra

$$\mathscr{B}_R := \{xI + yR : x, y \in \mathbb{R}\}$$

consider a, within the already declared sexagesimal chart,

$$B_0 := 7I + 2R, \qquad B_1 := A_{\mathrm{deg}}I + C^*_{\mathrm{deg}}R.$$

The orthogonal change of basis

$$S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

simultaneously diagonalizes the involution and both blocks. For $x > |y|$, we define

$$\rho(x, y) := \sqrt{x^2 - y^2}, \qquad \eta(x, y) := \operatorname{artanh}\left(\frac{y}{x}\right).$$

Then

$$xI + yR = \rho(x, y)e^{\eta(x,y)R}.$$

If

$$\rho_0 = \sqrt{45}, \quad \eta_0 = \operatorname{artanh}(2/7), \quad \rho_1 = \rho(A_{\mathrm{deg}}, C^*_{\mathrm{deg}}), \quad \eta_1 = \eta(A_{\mathrm{deg}}, C^*_{\mathrm{deg}}),$$

the unique left multiplication that carries B_0 to B_1 is

$$\boxed{T_{\mathrm{rel}} := \frac{\rho_1}{\sqrt{45}} e^{(\eta_1-\eta_0)R}, \qquad T_{\mathrm{rel}}B_0 = B_1.}$$

Proposition 38.19 (Relative carrier of the two blocks)**.** *Once R, the channel order, the positive branch, and the sexagesimal chart have been fixed, T_{rel} is the unique element of $\mathscr{B}_R$ satisfying $TB_0 = B_1$. In the spectral basis it acts by*

$$9 \longmapsto A_{\mathrm{deg}} + C^*_{\mathrm{deg}}, \qquad 5 \longmapsto A_{\mathrm{deg}} - C^*_{\mathrm{deg}}.$$

Proof. The split polar decomposition yields the printed formula and proves existence. Since $\det B_0 = 45 \neq 0$, every matrix solution of $TB_0 = B_1$ satisfies $T = B_1B_0^{-1}$; it is therefore unique. Diagonalization by

means of S produces the two spectral quotients $(A_{\mathrm{deg}} + C^{*}_{\mathrm{deg}})/9$ and $(A_{\mathrm{deg}} - C^{*}_{\mathrm{deg}})/5$, equivalent to the exponential expression of T_{rel}. □

This transporter is not a representational intertwiner. Indeed, an intertwiner Φ between the endomorphisms B_0 and B_1 should satisfy

$$\Phi B_0 = B_1 \Phi.$$

Theorem 38.20 (Spectral Obstruction to Entanglement)**.** *For the values of A_{deg} and C^{*}_{deg} constructed in this section,*

$$\boxed{\Phi B_0 = B_1 \Phi \quad \Longrightarrow \quad \Phi = 0.}$$

Proof. The spectra are

$$\mathrm{Spec}(B_0) = \{9, 5\}, \qquad \mathrm{Spec}(B_1) = \{A_{\mathrm{deg}} + C^{*}_{\mathrm{deg}}, A_{\mathrm{deg}} - C^{*}_{\mathrm{deg}}\}.$$

The previously demonstrated rational bounds provide the disjoint intervals.

$$7.29 < A_{\mathrm{deg}} < 7.30, \qquad 2.12 < C^{*}_{\mathrm{deg}} < 2.13,$$

and, consequently,

$$9.41 < A_{\mathrm{deg}} + C^{*}_{\mathrm{deg}} < 9.43, \qquad 5.16 < A_{\mathrm{deg}} - C^{*}_{\mathrm{deg}} < 5.18.$$

In the basis that diagonalizes R, each entry ϕ_{ij} of Φ is multiplied by the difference between an eigenvalue of B_0 and one of B_1. All these differences are nonzero; hence all entries of Φ vanish. □

Numerically,

$$\rho_1 = 6.9814819335172378048\ldots, \qquad \frac{\rho_1}{\sqrt{45}} = 1.0407378791354140779\ldots,$$

$$\eta_1 - \eta_0 = 0.005796351470571295590\ldots.$$

The formula preserves the split algebra, the projectors $P_{\pm}$, and the Lorentzian form up to the conformal factor $\rho_1^2/45$. It nevertheless depends on B_1 already constructed: it does not constitute a prior derivation of A_{deg} or C^{*}_{deg}, nor should it be confused with the equivariant intertwiners between realizations of the same module.

For

$$S_{90}(q) = \frac{q^{90}}{1 - q^{270}}, \qquad S_{120}(q) = \frac{q^{120}}{1 - q^{360}},$$

the hexad–octad transition functional is written entirely in the phase chart:

$$\Gamma_{6\to 8} = \frac{180}{\pi} A_{\mathrm{rad}} C^{*}_{\mathrm{rad}} + 12\big(S_{90}(q_-) - S_{90}(q_+)\big) - \big(S_{120}(q_-) - S_{120}(q_+)\big).$$

The evaluation produced by the counter-angle of the theorem 38.14 is

$$\boxed{\Gamma_{6\to 8} = 0.27400767269445927474725526077 4\ldots.}$$

The mathematically equivalent forms of the internal functional are

$$\boxed{\Gamma_{6\to 8} = A_{\mathrm{rad}} C^{*}_{\mathrm{deg}} + K_{6\to 8} = \frac{180}{\pi} A_{\mathrm{rad}} C^{*}_{\mathrm{rad}} + K_{6\to 8} = \frac{\pi A_{\mathrm{deg}} C^{*}_{\mathrm{deg}}}{180} + K_{6\to 8}.}$$

These identities evaluate $\Gamma_{6\to 8}$ after generating $(C^{*}_{\mathrm{rad}}, C^{*}_{\mathrm{deg}})$; it is not used in the inverse sense to select the counter-angle. The physical reading in the Ashtekar–Barbero connection is then represented by means of

$$\Gamma_{6\to 8} \dashrightarrow \gamma_{\mathrm{BI}}.$$

The arrow introduces no third functional and does not act retroactively on C^{*}. The series S_{90}, S_{120}, the coefficients 12 and -1, and their assignment to the $6 \to 8$ passage belong to the declared mathematical construction. Physical identification additionally requires the connection's own normalization.

38.7 Naturalness under Refinement and Separation with Respect to Metrological Recognition

38.7.1 Exact estimation of the remainder of the analytic prolongation

The expression initially found,

$$y_7 = \frac{\pi}{90}(H_5 + \alpha) - 169\alpha^6 - D_A\alpha^7,$$

is the seventh-degree truncation of the full character. Produce

$$C_7 = 2.123738338968646002654038271039 19\ldots^{\circ}.$$

The omitted remainder is not an indeterminate term; it is the exact geometric tail.

$$\mathscr{C}_{\pi}(\alpha) - 169\alpha^6 - D_A\alpha^7 = \frac{D_A^2\alpha^8}{1 - D_A\alpha}.$$

Therefore,

$$\left|C_7 - C_{*}^{\mathrm{HMT}}\right| = \frac{180}{\pi}\frac{D_A^2\alpha^8}{1 - D_A\alpha} = 2.7839208689064583\ldots \times 10^{-16}\ ^{\circ}.$$

The recurrence thus transforms a seventh-degree approximation into a complete analytic realization and provides a closed-form bound for each subsequent truncation.

38.7.2 Functional dependence of the regional selector and the analytic continuation

The construction changes when any of its active components is removed. The following values are obtained without recalibrating the other terms:

variant	conjugate angular coordinate in degrees
$\rho_{\pi} = 168$	$2.123738338977297686390448 7\ldots$
$\rho_{\pi} = 169$	$2.123738338968645724261951 4\ldots$
$\rho_{\pi} = 170$	$2.123738338959993762133454 1\ldots$
without determinantal tail	$2.123738338968694941530167 5\ldots$
D_A replaced by 1	$2.123738338968631340996582 6\ldots$

These ablations do not in themselves constitute a probabilistic estimate. They demonstrate that the equivariant selector and the determinantal character are effective components of the output and cannot be removed while maintaining the same realization.

38.7.3 Operatorial Interpretation of the Conjugate Angular Coordinate

The entire procedure can be summarized as a chain of applications:

$$\begin{aligned} \mathcal{F}_\pi &\longmapsto b_\pi \longmapsto \rho_\pi = 169, \\ \alpha &\longmapsto A \longmapsto x \longmapsto D_A, \\ (\rho_\pi, D_A) &\longmapsto \mathscr{C}_\pi(\alpha), \\ (H_5, \mathscr{C}_\pi(\alpha)) &\longmapsto y_* \longmapsto C_*^{\mathrm{HMT}}, \\ (x, y_*) &\longmapsto (q_+, q_-) \longmapsto \Gamma_{6\to 8}. \end{aligned}$$

The first line belongs to the finite catalog; the second, to the common closure of α; the third, to the determinantal prolongation; the fourth produces the oriented coordinate; the fifth transports it toward the dimensional interface.

In this organization, α measures the common contraction of the transport, and C_*^{HMT} measures the oriented separation of its two sheets. The regional return adds no external constant: it selects a finite amplitude and prolongs it through the determinant already fixed by α. The chart in degrees is subsequent. The primary object is the pair of phases (x, y_*).

An exhaustive verification reconstructs the five regions directly from the catalogue, traverses the actions of $\mathfrak{S}_5$ and $\mathfrak{S}_3$, checks the recurrence, and sums the series. The input of α is the exact output of its dodecaphasic closure, not a transcribed metrological digit. The verification keeps the catalogue theorems, reader rules, and external comparison separate; therefore, no experimental comparison intervenes in the generating function.

38.7.4 Angular Truncation Convergence Certification

The expression initially found,

$$y_7 = \frac{\pi}{90}(H_5 + \alpha) - 169\alpha^6 - D_A\alpha^7,$$

is the seventh-degree truncation of the full character. Produce

$$C_7 = 2.123738338968646002654038271039 19\ldots^\circ.$$

The omitted remainder is not an indeterminate term; it is the exact geometric tail.

$$\mathscr{C}_\pi(\alpha) - 169\alpha^6 - D_A\alpha^7 = \frac{D_A^2\alpha^8}{1 - D_A\alpha}.$$

Therefore,

$$\left|C_7 - C_{\mathrm{deg}}^*\right| = \frac{180}{\pi}\frac{D_A^2\alpha^8}{1 - D_A\alpha} = 2.7839208689064583\ldots \times 10^{-16}\ ^\circ.$$

The recurrence thus transforms a seventh-degree approximation into a complete analytic realization and provides a closed-form bound for each subsequent truncation.

38.7.5 Naturality of the regional selector under refinement and chart change

The construction changes when any of its active components is removed. The following values are obtained without recalibrating the other terms:

variant	counter-angle in degrees
$\rho_\pi = 168$	$2.12373833897729768639044 87\ldots$
$\rho_\pi = 169$	$2.12373833896864572426195 14\ldots$
$\rho_\pi = 170$	$2.12373833895999376213345 41\ldots$
without determinantal tail	$2.12373833896869494153016 75\ldots$
D_A replaced by 1	$2.12373833896863134099658 26\ldots$

These ablations do not in themselves constitute a probabilistic estimate. They demonstrate that the equivariant selector and the determinantal character are effective components of the output and cannot be removed while maintaining the same realization.

38.7.6 Metric independence of the parangular construction

The entire procedure can be summarized as a chain of applications:

$$\begin{aligned} \mathcal{F}_\pi &\longmapsto b_\pi \longmapsto \rho_\pi = 169, \\ \alpha &\longmapsto A_{\mathrm{deg}} \longmapsto A_{\mathrm{rad}} \longmapsto D_A, \\ (\rho_\pi, D_A) &\longmapsto \mathscr{C}_\pi(\alpha), \\ (H_5, \mathscr{C}_\pi(\alpha)) &\longmapsto C^*_{\mathrm{rad}} \longmapsto C^*_{\mathrm{deg}}, \\ (A_{\mathrm{rad}}, C^*_{\mathrm{rad}}) &\longmapsto (q_+, q_-) \longmapsto \Gamma_{6\to 8}. \end{aligned}$$

The first line belongs to the finite catalog; the second, to the common closure of α; the third, to the determinantal prolongation; the fourth produces the oriented coordinate; the fifth transports it toward the dimensional interface.

In this organization, α measures the common contraction of transport, and the counter-angle measures the oriented separation of its two sheets. In the current evaluation, the regional return selects a finite amplitude and prolongs it through the determinant already fixed by α, without inserting an additional external constant. The degree chart is subsequent. The primary object is the phase pair $(A_{\mathrm{rad}}, C^*_{\mathrm{rad}})$. The recurrence, regional sum, and ablations depend on the catalogue and on α_{HMT}, but receive no additional metrological constant.

Chapter 39

Positive constitutive operator of the vacuum: $3 \to 4$ completion, spectral reconstruction and joint derivation of ε, μ, Z and c

39.1 Bilayer spectral decomposition of the angular operator

Let $R = R^* = R^{-1}$ be the active involution and

$$P_\pm = \frac{I \pm R}{2}, \qquad T = q_+ P_+ + q_- P_-, \qquad 0 < q_+ < q_- < 1.$$

With

$$x = A_{\mathrm{rad}} = \frac{\pi A}{180}, \qquad y = C^*_{\mathrm{rad}} = \frac{\pi C^*}{180},$$

the two channels are constructed before evaluating the vacuum:

$$q_+ = e^{-x-y}, \qquad q_- = e^{-x+y}. \tag{39.1}$$

The exchange of sheets reverses C^*, that is, $y \mapsto -y$, and permutes $q_+ \leftrightarrow q_-$. The pair is thus the spectral reading of the same angular antecedent, not two parameters fitted separately.

Definition 39.1 (Constitutive response)**.**

$$\mathcal{V}_{90,120} = (I - T^{90})(I - T^{120})^{-1}. \tag{39.2}$$

Since $\|T\| < 1$, the denominator is positive and invertible. Functional calculus gives

$$\mathcal{V}_{90,120} = r_+ P_+ + r_- P_-, \qquad r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}}. \tag{39.3}$$

The assertion of invertibility deserves to be made explicit. For each leaf, $0 < q_\pm^{120} < 1$; therefore

$$I - T^{120} = (1 - q_+^{120})P_+ + (1 - q_-^{120})P_-$$

is strictly positive and

$$(I - T^{120})^{-1} = \frac{P_+}{1 - q_+^{120}} + \frac{P_-}{1 - q_-^{120}}.$$

The response $\mathcal{V}_{90,120}$ is not obtained by solving four separate scalar equations: it is the result of the functional calculus of a single positive contraction. This observation prevents permeability, permittivity, impedance and propagation from losing their shared antecedent.

Theorem 39.2 (Joint spectral determination of the four constitutive relations)**.** *The four constituent characters*

$$\widehat{\varepsilon} = r_+^2, \qquad \widehat{\mu} = r_-^2, \qquad \widehat{Z} = \frac{r_-}{r_+}, \qquad \widehat{c} = \frac{1}{r_+ r_-} \tag{39.4}$$

are spectral functions of $\mathcal{V}_{90,120}$. They satisfy

$$\widehat{\mu}\widehat{\varepsilon} = \widehat{c}^{-2}, \qquad \frac{\widehat{\mu}}{\widehat{\varepsilon}} = \widehat{Z}^2. \tag{39.5}$$

Proof. The identities are obtained by substituting equation (39.4). No dimensional chart nor any exterior value intervenes. □

Corollary 39.3 (Reconstruction of the two sheets)**.** *The four characters do not contain four degrees of freedom. Any of the pairs* $(\widehat{\mu}, \widehat{\varepsilon})$, $(\widehat{Z}, \widehat{c})$, *or* $(\mathcal{E}, \mathcal{B})$ *reconstructs the two positive eigenvalues:*

$$r_- = \sqrt{\widehat{\mu}} = \sqrt{\widehat{Z}/\widehat{c}}, \qquad r_+ = \sqrt{\widehat{\varepsilon}} = \frac{1}{\sqrt{\widehat{Z}\widehat{c}}}, \tag{39.6}$$

$$\log r_- = \frac{\mathcal{E} + \mathcal{B}}{2}, \qquad \log r_+ = \frac{\mathcal{E} - \mathcal{B}}{2}. \tag{39.7}$$

In particular, orientation is lost if only the product is retained, but is recovered by adding the quotient.

Proof. The first line is the positive root of the identities equations (39.4) y (39.5). The second inverts the linear system $\mathcal{E} = \log r_- + \log r_+$, $\mathcal{B} = \log r_- - \log r_+$. □

This corollary expresses a central property of HMT: product and quotient are complementary functionals of the same state. The product determines the propagation in common, whereas the quotient preserves the relative orientation of the inner and outer sheets.

In the V2 certificate,

$$\begin{aligned}
r_+ &= 0.99999962854824936317869522 81\ldots,\\
r_- &= 0.99972413718951836413151658 53\ldots,\\
\widehat{\varepsilon} &= 0.99999925709663670276044161 55\ldots,\\
\widehat{\mu} &= 0.99944835047932693508998155 59\ldots,\\
\widehat{Z} &= 0.99972450853893721545555434 66\ldots,\\
\widehat{c} &= 1.00027631048615752610638626 89\ldots.
\end{aligned}$$

These numerals are the terminal recognition of equation (39.2); they do not define its exponents 90, 120.

39.2 Harmonic factorization of the 90 and 120 sectors

The harmonic semigroup is

$$\langle 90, 120\rangle = 30\langle 3, 4\rangle.$$

If $s = q^{30}$, then

$$\frac{1 - q^{90}}{1 - q^{120}} = \frac{1 - s^3}{1 - s^4} = \frac{1 + s + s^2}{1 + s + s^2 + s^3}. \tag{39.8}$$

The identity also separates the response from its defect. If

$$F_{3\to4}(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \qquad 0 < s < 1,$$

then

$$d_{3\to4}(s) := 1 - F_{3\to4}(s) = \frac{s^3}{1 + s + s^2 + s^3} > 0. \tag{39.9}$$

The additional term is preserved exactly as a positive defect. Moreover,

$$F'_{3\to4}(s) = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0. \tag{39.10}$$

Consequently, the order of the angular channels uniquely determines the order of the constitutive responses. The assignment of $\widehat{\mu}$ and $\widehat{\varepsilon}$ to the respective sheets is obtained analytically, without resorting to their decimal expansions.

Each sheet of the vacuum constitutes the exact completion from three to four terms of the same elementary mode 30. Consequently, permeability and permittivity are interpreted as two quadratic functionals of the same completion evaluated on conjugate sheets, and not as independent constitutive parameters.

Since the function $(1 - q^{90})/(1 - q^{120})$ decreases in $0 < q < 1$ and $q_+ < q_-$, we obtain

$$0 < r_- < r_+ < 1, \qquad 0 < \widehat{\mu} < \widehat{\varepsilon} < 1, \qquad 0 < \widehat{Z} < 1 < \widehat{c}.$$

The limits of the functional determine its geometry. When $q \to 0^+$, $F(q) \to 1$: the four readings approach unity and the memory defect disappears. When $q \to 1^-$,

$$F(q) \longrightarrow \frac{90}{120} = \frac{3}{4}.$$

The response therefore takes values in the positive interval $(3/4, 1)$. The endpoint $3/4$ is not a fitted constant: it is the quotient of the two channel lengths and the continuous realization of the three–to–four completion.

39.3 Sheet involution and preservation of orientation

The sheet exchange $C^* \mapsto -C^*$ acts by

$$\widehat{\mu} \longleftrightarrow \widehat{\varepsilon}, \qquad \widehat{Z} \longmapsto \widehat{Z}^{-1}, \qquad \widehat{c} \longmapsto \widehat{c}. \tag{39.11}$$

The common propagation is even; impedance preserves orientation. This distinction will reappear in Barbero and CKM: not all functionals associated with the change of sheet have the same parity.

39.4 Logarithmic characters and tracial compatibility

Define

$$\mathcal{E} = \log(r_+ r_-), \qquad \mathcal{B} = \log(r_-/r_+).$$

With the normalized trace $\tau = \mathrm{Tr}/2$,

$$2\tau(\log \mathcal{V}) = \mathcal{E}, \qquad -2\tau(R \log \mathcal{V}) = \mathcal{B}. \tag{39.12}$$

The pair $(\mathcal{E}, \mathcal{B})$ constitutes a linear coordinate system for the operator logarithm:

$$\log \mathcal{V} = \frac{\mathcal{E}}{2} I - \frac{\mathcal{B}}{2} R. \tag{39.13}$$

The scalar component is invariant under sheet exchange; the oriented component changes sign. This decomposition directly demonstrates the parity of propagation and the oriented character of impedance.

The nonadic amplification

$$\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$$

is compatible with the expectations $E_m = \mathrm{id} \otimes \tau_9$:

$$E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m.$$

Therefore, $\mathcal{E}$ and $\mathcal{B}$ are not peculiarities of a 2×2 matrix; they persist in the UHF tower and in its tracial closure II_1.

More precisely, for every polynomial p and afterwards, by continuity, for every continuous function on the spectrum,

$$E_m\big(p(\mathcal{V}_{m+1})\big) = p(\mathcal{V}_m). \tag{39.14}$$

Compatibility does not preserve only two numbers: it preserves the entire functional algebra generated by the response. In particular, it preserves its spectral projectors, powers, logarithm, and quadratic forms that measure the defects equation (39.9).

39.5 Dimensional realization of the constitutive relations

Dimensional Mechanics determines

$$Z_{\mathrm{vac}} = \widehat{Z}\, u_S u_Q^{-2}, \tag{39.15}$$

$$c_{\mathrm{vac}} = \widehat{c}\, u_C, \tag{39.16}$$

$$\mu_{\mathrm{vac}} = \widehat{\mu}\, u_S u_C^{-1} u_Q^{-2}, \tag{39.17}$$

$$\varepsilon_{\mathrm{vac}} = \widehat{\varepsilon}\, u_S^{-1} u_C^{-1} u_Q^{2}. \tag{39.18}$$

The constitutive identities hold on the straight lines:

$$\mu_{\mathrm{vac}}\varepsilon_{\mathrm{vac}} = c_{\mathrm{vac}}^{-2}, \qquad \mu_{\mathrm{vac}}/\varepsilon_{\mathrm{vac}} = Z_{\mathrm{vac}}^{2}. \tag{39.19}$$

The SI chart comes later. It is not necessary to declare c primitive and then solve $\mu\varepsilon = c^{-2}$: the three objects descend jointly from $\mathcal{V}$ and from the declared lines.

The separation of levels is expressed by the diagram

$$\begin{array}{ccccc} (A, C^*) & \longrightarrow & T & \longrightarrow & \mathcal{V} \longrightarrow (\widehat{\mu}, \widehat{\varepsilon}, \widehat{Z}, \widehat{c}) \\ \big\downarrow \text{line charts} & & & & \big\downarrow \text{final evaluation} \\ (u_S, u_Q, u_C) & \longrightarrow & (\mu_{\mathrm{vac}}, \varepsilon_{\mathrm{vac}}, Z_{\mathrm{vac}}, c_{\mathrm{vac}}) & \longrightarrow & \text{SI coordinates.} \end{array}$$

The first line is dimensionless and operatorial; the second determines types of quantity; the third selects a metrological chart. Altering the final chart changes neither the eigenvalues nor the constitutive identities. In this precise sense, mathematical–physical unification consists in distinguishing the genealogical invariant from its coordinate system, not in removing the dimensional structure.

Proposition 39.4 (Spectral minimality)**.** *Let W be a positive two-sheet operator with simple spectrum $\{r_+, r_-\}$. If four positive characters satisfy $\mu\varepsilon = c^{-2}$ and $\mu/\varepsilon = Z^2$, if their propagation in common is determined by $c^{-1} = \det W = r_- r_+$, and if its orientation fixes $Z = r_-/r_+$, then necessarily*

$$(\mu, \varepsilon, Z, c) = \left(r_-^2, r_+^2, \frac{r_-}{r_+}, \frac{1}{r_- r_+}\right).$$

Therefore, equation (39.4) is not one parametrization among many: it is the minimal positive realization compatible with product, quotient and orientation.

Proof. From $Z = r_-/r_+$ and $c^{-1} = \det W = r_- r_+$ the positive roots are obtained uniquely roots. The two constitutive identities then force afterwards $\mu = r_-^2$ and $\varepsilon = r_+^2$. □

The first conceptual inversion consists in conceiving the vacuum as a positive relational response.

In HMT, the vacuum is the minimal dynamical response of the system when its two sheets must coexist, propagate, and close their difference while preserving the orientation that distinguishes them.

The two sheets constitute two conjugate readings of the same angular antecedent. One preserves an orientation, and the other preserves the opposite orientation. Hence two channels, q_+ and q_-, appear as the two correlated spectral faces of a single state.

The operator

$$\mathcal{V}_{90,120} = (I - T^{90})(I - T^{120})^{-1}$$

compares two propagation depths: 90 and 120. Those depths are determined by the same elementary mode:

$$90 = 3 \cdot 30, \qquad 120 = 4 \cdot 30.$$

Therefore, the response of each sheet can be rewritten as

$$\frac{1 - q^{90}}{1 - q^{120}} = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \qquad s = q^{30}.$$

And here is where the interpretation becomes genuinely powerful. The numerator contains three states of the mode 30; the denominator contains those same three states and one more. The vacuum is, therefore, an exact completion $3 \to 4$.

That fourth term integrates and preserves the preceding three. It completes them. What is added is recorded as a positive completion residue:

$$1 - \frac{1 + s + s^2}{1 + s + s^2 + s^3} = \frac{s^3}{1 + s + s^2 + s^3} > 0.$$

Vacuum memory is precisely the exact increment that allows three terms to become four. It is the positive and necessary trace of the completion.

From the two eigenvalues of that single operator arise the four constitutive relations:

$$\widehat{\varepsilon} = r_+^2, \qquad \widehat{\mu} = r_-^2, \qquad \widehat{Z} = \frac{r_-}{r_+}, \qquad \widehat{c} = \frac{1}{r_+ r_-}.$$

This completely changes the interpretation of permeability and permittivity.

Permittivity is the quadratic response of one sheet. Permeability is the quadratic response of the conjugate sheet. Impedance preserves the oriented relation between them. Propagation speed depends on their common product.

Thus, product and quotient perform two different tasks:

- the product forgets which sheet was which and preserves the common propagation;
- the quotient preserves what the relative orientation between them was.

The product answers the question: “what do the two leaves share?”. The quotient answers the question: “how are they distinguished?”.

Por eso

$$\widehat{\mu}\,\widehat{\varepsilon} = \widehat{c}^{-2}$$

y

$$\frac{\widehat{\mu}}{\widehat{\varepsilon}} = \widehat{Z}^2$$

are the two complementary ways of reading the same spectral pair.

The holographic structure appears here in a particularly transparent form: four visible magnitudes actually contain only two internal degrees of freedom. Any of certain suitable pairs allows the two sheets to be reconstructed. The constitutive publication preserves genealogy when product and quotient remain jointly available.

This is the deep repair of the ablation of μ and ε. The repair reunites their values with their common operator-theoretic origin.

The International System chart may subsequently transform these intrinsic sections into μ_0, ε_0, Z_0, and c. Its function is to express, within a metrological convention, a structure whose provenance has already been selected by the operator.

The vacuum thereby ceases to be a repository of four constantes. becomes a unique spectral operation With two conjugate memories.

And that view has an important philosophical consequence: the vacuum is a compatibility relation between two oriented ways of realizing one and the same state.

39.6 Discrete polarization and the tritic mode of arithmetic projections

Refinement and comparison between the APP sheets provide two complementary observations. The first follows the accumulated discrepancy between the number of transitions and the capacity acquired; the second preserves the mode of the tritic selector in the pairing of the additive and multiplicative fibres. Discrete polarization and the electronic prefactor are obtained from these respective operations. Their articulation with the vacuum requires preserving both the temporal information and the spaces on which the operators act.

39.6.1 Refinement discrepancy and endpoint convention

The cardinalities of the ternary block and its decimal representation fix

$$\rho = \log_{1000} 729 = \log_{10} 9, \qquad \delta = 1 - \rho = \log_{10}(10/9).$$

We have $0 < \delta < 1$. Moreover, δ is irrational: an equality $\delta = p/q$, with $q > 0$, would imply $10^{q-p} = 9^q$, which is incompatible with prime factorization. This mismatch already enters the encoding of refinement. Here its centred readout is made explicit, preserving the phase origin and the endpoint convention.

Definition 39.5 (Lower and upper encodings)**.** For $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$, define

$$z_n^{\downarrow}(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor, \tag{39.20}$$

$$z_n^{\uparrow}(\theta) = \lceil (n+1)\delta + \theta \rceil - \lceil n\delta + \theta \rceil. \tag{39.21}$$

Both take values in $\{0, 1\}$. The arrow distinguishes only the endpoint convention for the intervals; tritic orientation retains its earlier definition.

The lower encoding coincides with the one used in the development of refinement. The upper encoding recovers the formulation of polarization using the ceiling function. The two expressions are related by an exactly determined boundary term.

Proposition 39.6 (Change of convention and boundary term)**.** *Let* $b_n(\theta) = \mathbf{1}_{\mathbb{Z}}(n\delta + \theta)$. *Then*

$$z_n^{\uparrow}(\theta) - z_n^{\downarrow}(\theta) = b_n(\theta) - b_{n+1}(\theta). \tag{39.22}$$

In particular, for $N \geq 0$,

$$\sum_{n=0}^{N-1} (z_n^{\uparrow} - z_n^{\downarrow}) = b_0 - b_N .$$

Proof. For every real x, $\lceil x \rceil = \lfloor x \rfloor + 1 - \mathbf{1}_{\mathbb{Z}}(x)$. Applying this identity to both endpoints of each increment gives (39.22); its sum telescopes. The irrationality of δ implies that, for fixed θ, at most one integer index satisfies $n\delta + \theta \in \mathbb{Z}$. □

With $\theta = 0$, the first upper symbol is 1 and the lower one is 0; for $n \geq 1$ they coincide. The initial term belongs to the register and is preserved when the convention is changed. The densities and discrepancy bounds are transported with that term, without modifying the TPK update or restarting the history at each phase return.

39.6.2 Bounded polarization and exact temporal balance

Definition 39.7 (Centred discrete polarization)**.** The polarization of the upper encoding, relative to the density δ, is

$$P_N^{\uparrow}(\theta) = \sum_{n=0}^{N-1} \big(z_n^{\uparrow}(\theta) - \delta\big), \qquad P_0^{\uparrow} = 0. \tag{39.23}$$

Its temporal increment is

$$J_N^{\uparrow}(\theta) = P_{N+1}^{\uparrow}(\theta) - P_N^{\uparrow}(\theta) = z_N^{\uparrow}(\theta) - \delta. \tag{39.24}$$

Centring separates uniform growth of density δ from phase information. Thus, $P_N^{\uparrow}$ measures an accumulated imbalance, not the total number of vacancies. The increment $J_N^{\uparrow}$ preserves the local pulse and the background relative to which it is evaluated.

Theorem 39.8 (Uniform bound and discrete temporal continuity)**.** *For $N \geq 0$ and every phase θ,*

$$P_N^{\uparrow}(\theta) = \lceil N\delta + \theta \rceil - \lceil \theta \rceil - N\delta, \qquad |P_N^{\uparrow}(\theta)| < 1. \tag{39.25}$$

For any integers $0 \leq a < b$,

$$\sum_{n=a}^{b-1} J_n^{\uparrow}(\theta) = P_b^{\uparrow}(\theta) - P_a^{\uparrow}(\theta), \qquad \left| \sum_{n=a}^{b-1} J_n^{\uparrow}(\theta) \right| < 1. \tag{39.26}$$

The density of the vacancy sequence is δ.

Proof. Summing the ceiling increments in (39.21) yields the first identity. Writing $r(x) = \lceil x \rceil - x \in [0, 1)$ gives $P_N^{\uparrow} = r(N\delta + \theta) - r(\theta)$; its absolute value is less than one. On the interval $[a, b)$, one similarly obtains $r(b\delta + \theta) - r(a\delta + \theta)$, proving the second bound, which is sharper than the sum of two individual bounds. Finally, $N^{-1} \sum_{n=0}^{N-1} z_n^{\uparrow} = \delta + N^{-1} P_N^{\uparrow} \to \delta$. □

The lower encoding admits the same proofs with $r(x)$ replaced by $\lfloor x \rfloor - x \in (-1, 0]$. In particular, $P_N^{\uparrow} - P_N^{\downarrow} = b_0 - b_N$. The bounded discrepancy previously used in the Dirichlet readouts and the present polarization therefore preserve an explicit relationship of origin and endpoints. Nor does aperiodicity introduce secular drift in this imbalance: the history can extend indefinitely while the centred discrepancy remains uniformly bounded.

39.6.3 Realization on the charge and time lines

To realize the balance dimensionally, specify a charge line $\mathcal{L}_Q$ with a nonzero section u_Q and an increasing sequence of times t_N on an oriented time line. With $\Delta t_N = t_{N+1} - t_N > 0$, the centred charge and the mean current of each interval are defined by

$$Q_N = P_N^{\uparrow} u_Q, \qquad I_N = \frac{Q_{N+1} - Q_N}{\Delta t_N} = \frac{u_Q}{\Delta t_N}\big(z_N^{\uparrow} - \delta\big). \tag{39.27}$$

Thus, $Q_N \in \mathcal{L}_Q$ and $I_N \in \mathcal{L}_Q \otimes \mathcal{L}_T^{-1}$. The section u_Q and the time intervals are data of this realization; the preceding arithmetic construction has already determined $P_N^{\uparrow}$ and $J_N^{\uparrow}$.

Proposition 39.9 (Preservation of balance under dimensional realization)**.** *For $a < b$,*

$$Q_b - Q_a = \sum_{n=a}^{b-1} \Delta t_n I_n, \qquad \left| \frac{Q_b - Q_a}{u_Q} \right| < 1. \tag{39.28}$$

If all intervals have length t_0, the two possible currents are $-\delta u_Q/t_0$ and $(1-\delta)u_Q/t_0$.

Proof. Multiplying each identity in (39.27) by Δt_n and summing telescopes the charges. The bound is (39.26); the two local values follow from $z_n^{\uparrow} \in \{0, 1\}$. □

The continuity proved is temporal and refers to the centred charge. A spatial realization also incorporates the localization of charges, incidence between cells, and identification of boundary fluxes. In a constitutive description by channels, the coupling is specified through an excitation map $\iota : \mathcal{L}_Q \to \mathcal{H}_{\mathrm{ch}} \otimes \mathcal{L}_Q$ and an observation map $\ell : \mathcal{H}_{\mathrm{ch}} \otimes \mathcal{L}_Q \to \mathcal{L}_Q$, where $\mathcal{H}_{\mathrm{ch}}$ is a real realization of the channel space. If both are linear and independent of the time index, the response of a fixed operator $\mathcal{V}$ is transported by

$$Q_N^{\mathrm{resp}} = \ell(\mathcal{V} \otimes I_{\mathcal{L}_Q})\iota(Q_N), \qquad I_N^{\mathrm{resp}} = \frac{Q_{N+1}^{\mathrm{resp}} - Q_N^{\mathrm{resp}}}{\Delta t_N}.$$

Linearity preserves the telescoping balance. This composition makes explicit the data of a constitutive identification: the scalar sequence determines the pulse, while ι, $\mathcal{V}$, and ℓ determine its excitation, transport, and observation through channels. Identities (39.23)–(39.28) establish the first construction independently of that choice of realization. The vacuum operator is constructed in Section 39.8 on its own channels and projectors.

39.6.4 Pairing the additive and multiplicative fibres

The second observation comes from the three-dimensional realization of APP, $\mathcal{A}_3 = D_9^3$, with uniform measure and $D_9 = \{1, \dots, 9\}$. The observables preserved are

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \qquad \Pi(x) = \rho_9(x_1 x_2 x_3),$$

where $\rho_9(n) = 1 + [(n-1) \bmod 9]$ for $n > 0$. In $\ell^2(\mathcal{A}_3)$, let $P_{\Sigma,3}$ and $P_{\Pi,3}$ be the conditional expectations on their fibres. For example,

$$(P_{\Sigma,3} f)(x) = \frac{1}{|\Sigma^{-1}(\Sigma(x))|} \sum_{y:\, \Sigma(y) = \Sigma(x)} f(y).$$

They are orthogonal projections defined on the same support. The incidence matrix $N_{ab} = |\Sigma^{-1}(a) \cap \Pi^{-1}(b)|$ results from enumerating the 729 triples, adding and multiplying their coordinates before applying ρ_9:

$$N = 9\begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}. \tag{39.29}$$

The rows sum to 81, and the column sums are

$$(m_1, \dots, m_9) = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

Therefore, the pairing of the normalized indicator functions of the fibres is $C_{ab} = N_{ab}/\sqrt{81 m_b}$. Normalization incorporates the multiplicities produced by APP, not a choice of singular values.

Proposition 39.10 (Singular mode of the tritic selector)**.** *In the canonical chart, the selector vector is* $m_{\mathrm{ph}} = (1, -1, 0)^3$. *We have*

$$C m_{\mathrm{ph}} = C^{\mathsf{T}} m_{\mathrm{ph}} = -\tfrac{1}{2} m_{\mathrm{ph}}. \tag{39.30}$$

Thus, the corresponding singular value is $s = 1/2$.

Proof. In the six columns where m_{ph} is nonzero, $m_b = 36$ and $C_{ab} = N_{ab}/54$. Multiplying the rows of (39.29) by $(1, -1, 0)^3$ yields $-m_{\mathrm{ph}}/2$. For the transposed product, the signed sums of columns $3, 6, 9$ are zero; the other six give the same values $-m_{\mathrm{ph},b}/2$. This proves both identities and, by composition, $CC^{\mathsf{T}} m_{\mathrm{ph}} = m_{\mathrm{ph}}/4$. □

39.6.5 Commutator norm and electronic prefactor

Theorem 39.11 (Exact norm of arithmetic incompatibility)**.** *The two projections satisfy*

$$\left\| [P_{\Sigma,3}, P_{\Pi,3}] \right\| = \frac{\sqrt{3}}{4}. \tag{39.31}$$

The maximum is attained in the principal plane corresponding to the tritic mode of (39.30).

Proof. The rational matrix $M = CC^{\mathsf{T}}$ has entries $M_{ac} = \sum_b N_{ab} N_{cb}/(81 m_b)$. Its complete spectrum is verified through the following independent subspaces: the vectors $(1, \dots, 1)$, $(1, -1, 0)^3$, and $(1, 1, -2)^3$ have eigenvalues 1, $1/4$, and $7/132$, respectively; the subspace of vectors supported on $\{3, 6, 9\}$ whose sum is zero has dimension 2 and eigenvalue $1/18$; the zero-sum vectors supported on $\{1, 4, 7\}$ or $\{2, 5, 8\}$ form a kernel of dimension 4. Substitution into the matrix verifies these actions, and the dimensions sum to 9. Consequently,

$$\operatorname{spec} M = \{1, 1/4, 7/132, 1/18, 1/18, 0, 0, 0, 0\}.$$

For two orthogonal projections, a singular value $s \in (0, 1)$ of the pairing determines a principal plane in which their matrices are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} s^2 & s\sqrt{1 - s^2} \\ s\sqrt{1 - s^2} & 1 - s^2 \end{pmatrix}.$$

This form is obtained by taking a unit vector u in the first image with $PQu = s^2 u$ and completing it with $(Qu - s^2 u)/(s\sqrt{1 - s^2})$. Their commutator is $s\sqrt{1 - s^2}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The sectors with singular value 0 or 1

contribute zero. Therefore, the total norm is the maximum of $\sqrt{\lambda(1-\lambda)}$ over the spectrum of M. That maximum is $\sqrt{3/16}$, attained at $\lambda = 1/4$. □

The norm is the prefactor entering the electronic composition. Its derivation preserves the three-dimensional support, the fibres, and the selector mode; scalar evaluation does not replace that structure. In particular, knowing $\sqrt{3}/4$ gives the maximum magnitude of the incompatibility, while the projectors preserve the directions in which it acts.

39.6.6 Quadratic defect and response of the $90/120$ sectors

The tritic mode simultaneously determines

$$1 - s^2 = \frac{3}{4} = \frac{90}{120}, \qquad s = \frac{1}{2}. \tag{39.32}$$

The first expression is the quadratic defect of the APP principal plane. The last preserves the lengths of the two calendar sectors. The constitutive response uses those sectors on another object: a contractive channel $q \in (0, 1)$. Its evaluation is

$$r(q) = \frac{1 - q^{90}}{1 - q^{120}}.$$

The relationship between the sector ratio and this response can be made precise without identifying their operators.

Proposition 39.12 (Limit of the response ratio)**.** *Para* $0 < q < 1$*,*

$$\frac{3}{4} < r(q) < 1, \qquad \lim_{q \uparrow 1} r(q) = \frac{3}{4} = 1 - s^2. \tag{39.33}$$

Proof. With $u = q^{30} \in (0, 1)$, factoring $1 - u^3$ and $1 - u^4$ gives $r(q) = (1 + u + u^2)/(1 + u + u^2 + u^3)$. The limit at $u = 1$ is $3/4$. Moreover, $4(1 + u + u^2) - 3(1 + u + u^2 + u^3) = (1 - u)(3u^2 + 2u + 1) > 0$, and the denominator exceeds the numerator by $u^3 > 0$. This yields both inequalities. □

The exact coincidence concerns the defect of the singular mode and the limit of the sector response. In the contractive interior, the response depends on q and is strictly greater than $3/4$. The constitutive development preserves its two channels $q_\pm$ and its projectors $P_\pm$; the projectors $P_{\Sigma,3}$ and $P_{\Pi,3}$, for their part, preserve the arithmetic fibres that determined the prefactor. An operator equivalence between the two realizations requires a map between their spaces that intertwines the corresponding actions. Equations (39.30), (39.31), and (39.33) specify the shared information and keep the data of each realization visible.

39.7 Sheet return, areal–volumetric incidence and constitutive orientation

The sheet structure enters before electromagnetic quantities are evaluated. Its function is to distinguish closure of a projection, restoration of an orientation and evolution of the transported register. These operations act on related objects of different types: a projected trajectory may have closed while its lift remains on the opposite sheet; orientation may have been restored while memory preserves the two windings performed. The constitutive construction must receive this information together with the structure determining it.

39.7.1 Projected return and restoration of the oriented cover

Let ℓ be a return loop of a surviving extension, and let $\varepsilon : \langle \ell \rangle \to \{+1, -1\}$ be the character of its lift, with $\varepsilon(\ell) = -1$. The associated interval bundle has the presentation

$$\mathcal{M} = ([0, 1] \times [-1, 1])/(0, u) \sim (1, -u). \tag{39.34}$$

The decisive datum is the sign of transport around the specified loop. The identification of endpoints geometrically realizes that character.

Proposition 39.13 (Two closure orders)**.** *One winding of the loop closes the base and reverses the fibre orientation. Two windings restore orientation. In the angular parametrization of the boundary readout, these returns correspond, respectively, to the areal closure* 2π *and the volumetric closure* 4π *of the oriented cover.*

Proof. The equivalence relation of (39.34) identifies the terminal fibre with the initial one by $u \mapsto -u$. Composing two transports gives $u \mapsto -(-u) = u$. Parametrizing one winding of the base by 2π assigns 4π to its square. The statement concerns this lift and this readout. □

TPK memory also preserves the trajectory traversed and the register updates. The square of the sign character is the identity, but it does not impose vanishing of these updates. The spinorial representation, when used, must transport this datum through its own map.

39.7.2 Descent of incidence from the tritic calendar

In the primitive block $(+1, -1, 0)$, the TPK updates the arithmetic mode through $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$ and $0 : m \mapsto m$. Its cyclic orbit is (Σ, Π, Π). The sheet readout assigns Σ to the areal sheet and Π to the volumetric sheet. In the basis order $(e_{\mathrm{ar}}, e_{\mathrm{vol}})$, edge incidence is represented by

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \tag{39.35}$$

The three consecutive pairs are $\Sigma \to \Pi$, $\Pi \to \Pi$ and $\Pi \to \Sigma$, each with multiplicity one. The calendars of lengths 9, 27 and 108 contain 3, 9 and 36 copies of the primitive block. Their incidence matrices are therefore these multiples of F_{av}.

This construction preserves the distinction between chronological frequency and the measure of the continuation space. The former assigns weights $(1/3, 2/3)$ to the mode cycle. The latter is defined on the minimal memory-one symbolic envelope admitting exactly the preceding pairs. Its adjacency matrix is F_{av}; trajectory counts belong to that envelope.

Proposition 39.14 (Return weights of the two sheets)**.** *Let $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$. For $n \geq 1$,*

$$F_{\mathrm{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}. \tag{39.36}$$

The Parry measure of this envelope assigns the sheets weights proportional to $(1, \varphi^2)$, and the ratio of diagonal counts converges to φ^{-2}.

Proof. The power formula follows by induction using the recurrence. The square of F_{av} is strictly positive. Its positive Perron vector is $(1, \varphi)$, with eigenvalue φ, since $\varphi^2 = \varphi + 1$. As the matrix is symmetric, the Parry weights are proportional to the squares of that vector's components. The other root of the characteristic polynomial is $-\varphi^{-1}$; its modulus is strictly less than φ, giving the limit of the diagonal ratios. □

Spectral identification of this incidence follows its construction from the calendar. The regional coordinate φ retains its earlier coinductive genealogy: recognition of the eigenvalue does not select the calendar.

39.7.3 Boundary marking and areal–volumetric ratio

Let $\Pi_{\mathrm{HMT}}(X_\infty)$ be the value of the regional closure readout already constructed. With the diagonal sheet projectors, define

$$D_\pi = \Pi_{\mathrm{HMT}}(X_\infty) P_{\mathrm{ar}} + P_{\mathrm{vol}}, \qquad \Theta_n = \frac{\langle e_{\mathrm{ar}}, D_\pi F_{\mathrm{av}}^n e_{\mathrm{ar}} \rangle}{\langle e_{\mathrm{vol}}, F_{\mathrm{av}}^n e_{\mathrm{vol}} \rangle}. \tag{39.37}$$

Volumetric normalization is unity; regional closure marks the areal return only once. Applying (39.36),

$$\Theta_n = \Pi_{\mathrm{HMT}}(X_\infty) \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi}{\varphi^2}. \tag{39.38}$$

The return ratio is determined by incidence, measure and marking. The quotient $2\pi/4\pi$, by contrast, describes the cover's two closure orders. Both relations participate in the sheet readout without replacing each other.

39.7.4 Transport of orientation to the constitutive channels

In a two-channel representation, let $\mathcal{R}$ be a self-adjoint involution and $\mathcal{S}$ a unitary interchange operator satisfying $\mathcal{S}\mathcal{R}\mathcal{S}^{-1} = -\mathcal{R}$. Write the angular coordinates already constructed as $x = A_{\mathrm{rad}}$ and $y = C^{*}_{\mathrm{rad}}$, with $x > |y|$. Then

$$B = xI + y\mathcal{R}, \qquad T = e^{-B}, \qquad \mathcal{S}T(y)\mathcal{S}^{-1} = T(-y). \tag{39.39}$$

Indeed, conjugation transforms $B(y)$ into $B(-y)$ and commutes with its convergent exponential series. The eigenvalues $q_{\pm} = e^{-x\mp y}$ are interchanged. To fix the response order, the chamber $x > y > 0$ is used, where $0 < q_{+} < q_{-} < 1$. The map $(x, y) \mapsto (x, -y)$ transports it to the opposite chamber $x > -y > 0$, where $0 < q_{-} < q_{+} < 1$. Both belong to the contractive domain $x > |y|$. On the boundary $y = 0$ the two channels coincide. Conjugation interchanges the chambers; it does not preserve the strict ordering of their labels. It acts on the projectors through $\mathcal{S}P_{\pm}\mathcal{S}^{-1} = P_{\mp}$, with $P_{\pm} = (I \pm \mathcal{R})/2$. This concrete map transmits orientation to the constitutive operator developed next.

39.8 Constitutive operator and reconstruction of the vacuum channels

The constitutive quantities are obtained by an operation on angular transport. The starting information comprises the channels, their projectors, and the orientation distinguishing them. This organization makes it possible to follow separately the construction of the operator, the evaluation of its spectral functions, and its subsequent dimensional realization.

39.8.1 Positive response on two sheets

In the oriented chamber $x > y > 0$ of Section 39.7, let $P_{\pm} = (I \pm \mathcal{R})/2$ and $T = q_{+}P_{+} + q_{-}P_{-}$, with $0 < q_{+} < q_{-} < 1$. The incidences constructed by the calendar fix the sectors 90 and 120. The constitutive response is

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}. \tag{39.40}$$

This definition receives the transport arising from the TPK and preserves the sheet assignments. The response rule and its domain are an explicit part of the construction.

The quotient arises from comparing the two calendar incidences on the same transport, not from fitting four quantities separately. If $S = T^{30}$ and $\Sigma_j(S) = I + S + \cdots + S^{j-1}$, factoring $I - S^{j}$ gives

$$\mathcal{V}\,\Sigma_4(S) = \Sigma_3(S), \qquad \mathcal{V} = \Sigma_3(S)\Sigma_4(S)^{-1}. \tag{39.41}$$

The factors Σ_3 and Σ_4 express, respectively, the sectors $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$. Since the spectrum of S is contained in $(0, 1)$, $\Sigma_4(S)$ is invertible: the completion equation determines a unique response for the fixed transport. This uniqueness concerns the declared constitutive rule and its incidences. Electromagnetic identification adds the oriented assignment of sheets and the dimensional lines specified below.

Proposition 39.15 (Decomposition and positivity)**.** *The response is well-defined and*

$$\mathcal{V} = r_{+}P_{+} + r_{-}P_{-}, \qquad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}. \tag{39.42}$$

Its eigenvalues satisfy $3/4 < r_{-} < r_{+} < 1$ *in this chamber.*

Proof. Each coefficient $1 - q_{\pm}^{120}$ is positive. The inverse of $I - T^{120}$ is the sum of the projectors divided by those coefficients, proving (39.42). With $s = q^{30}$, one obtains

$$f(s) = \frac{1+s+s^2}{1+s+s^2+s^3}, \qquad f'(s) = -\frac{s^2(3+2s+s^2)}{(1+s+s^2+s^3)^2} < 0. \tag{39.43}$$

The limits at 0 and 1 are 1 and $3/4$, respectively. Monotonicity and the channel ordering complete the proof. □

39.8.2 Four coordinates of a common response

Define the normalized evaluations

$$\widehat{\varepsilon} = r_+^2, \qquad \widehat{\mu} = r_-^2, \qquad \widehat{Z} = \frac{r_-}{r_+}, \qquad \widehat{c} = \frac{1}{r_+ r_-}. \tag{39.44}$$

The first pair preserves the quadratic responses assigned to each sheet. The product captures the common mode; the quotient distinguishes their relative orientation. The following hold exactly:

$$\widehat{\mu}\widehat{\varepsilon} = \widehat{c}^{-2}, \qquad \frac{\widehat{\mu}}{\widehat{\varepsilon}} = \widehat{Z}^2, \qquad r_+ = \frac{1}{\sqrt{\widehat{Z}\widehat{c}}}, \quad r_- = \sqrt{\frac{\widehat{Z}}{\widehat{c}}}. \tag{39.45}$$

The positive roots fix the inversion. Under sheet conjugation, read relative to the fixed markings $P_\pm$, $\widehat{\varepsilon}$ and $\widehat{\mu}$ are interchanged, $\widehat{Z}$ is inverted, and $\widehat{c}$ is preserved. The response is then transported to the chamber $x > -y > 0$: the formulas remain valid, with the orderings $q_- < q_+$ and $r_+ < r_-$ reversed. Consequently, common mode and orientation are data recoverable through complementary observables.

Proposition 39.16 (Determination of the quadratic evaluations)**.** *Let $W = r_+P_+ + r_-P_-$ be a positive response on the two sheets. With the product and quotient readouts fixed,*

$$\widehat{c}^{-1} = \det W = r_+ r_-, \qquad \widehat{Z} = r_-/r_+,$$

there exists a unique positive pair $(\widehat{\varepsilon}, \widehat{\mu})$ satisfying $\widehat{\mu}\widehat{\varepsilon} = \widehat{c}^{-2}$ and $\widehat{\mu}/\widehat{\varepsilon} = \widehat{Z}^2$. It is precisely (r_+^2, r_-^2).

Proof. Positivity permits taking roots unambiguously:

$$\widehat{\mu} = \frac{\widehat{Z}}{\widehat{c}} = \frac{r_-}{r_+} r_+ r_- = r_-^2, \qquad \widehat{\varepsilon} = \frac{1}{\widehat{Z}\widehat{c}} = \frac{r_+}{r_-} r_+ r_- = r_+^2.$$

These expressions satisfy both relations and are their only positive roots. □

Thus the squares are not two independent choices added to the channels: they are determined by the multiplicative and oriented readouts once the constitutive relations are fixed. This proposition establishes uniqueness within that system of rules. Interpreting the rules as an electromagnetic response retains its additional physical-representation content.

The determinantal readout of transport and the constitutive response must also preserve their order. For $T = e^{-xI - y\mathcal{R}}$, one has $\det T = e^{-2x}$, whereas

$$\det \mathcal{V} = f(q_+^{30}) f(q_-^{30}) = \widehat{c}^{-1}.$$

Both arise from the same two-sheet operator but evaluate different functions of its spectrum. In general, applying the determinant first and then a rational function is not equivalent to taking the determinant of that function of the operator. Keeping both eigenvalues until the response is completed preserves orientation information that an isolated determinant no longer distinguishes.

39.8.3 Cubic inversion and angular recovery

Theorem 39.17 (Constitutive inversion in the contractive domain)**.** *For each $r \in (3/4, 1)$, there exists a unique $s \in (0, 1)$ such that $f(s) = r$. This s is the unique root in that interval of*

$$rs^3 + (r-1)(1+s+s^2) = 0. \tag{39.46}$$

The labeled response reconstructs $s_\pm$, $q_\pm = s_\pm^{1/30}$, and

$$x = -\frac{1}{2}\log(q_+ q_-), \qquad y = \frac{1}{2}\log\frac{q_-}{q_+}. \tag{39.47}$$

Proof. The derivative and limits in (39.43) prove that f is a decreasing bijection between the stated intervals. Multiplying $f(s) = r$ by its denominator produces (39.46). The positive root of order 30 recovers the channel. Finally, adding and subtracting $\log q_\pm = -x \mp y$ gives (39.47). □

Reconstruction preserves the sheet labels. The complete operator $\mathcal{V}$ also preserves the projectors, and the inverse function is applied by spectral calculus. Recovering two angular coordinates is an operation defined on this representation; the complete historical TPK trace preserves additional data.

The interval $r \in (3/4, 1)$ and the inverse function of this theorem belong to the internal normalization of (39.44). If coordinates expressed in another chart of units are available, this normalization is first recovered through the transformation of section 63.5; cubic inversion is then applied.

39.8.4 Logarithmic coordinates and refinement

Let $\mathcal{E} = \log(r_+ r_-)$ and $\mathcal{B} = \log(r_-/r_+)$. Then

$$\log \mathcal{V} = \tfrac{1}{2}\mathcal{E} I - \tfrac{1}{2}\mathcal{B}\mathcal{R}. \tag{39.48}$$

This decomposition displays the invariant and oriented components. In the amplification $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$, normalized partial expectations satisfy $E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m$. For every polynomial p, $p(\mathcal{V}_m) = p(\mathcal{V}) \otimes I_{9^m}$; hence $E_m(p(\mathcal{V}_{m+1})) = p(\mathcal{V}_m)$. Uniform continuity extends the identity to continuous functions on the spectrum. Thus projectors, powers, and the logarithm are preserved in the stated tower.

39.8.5 Dimensional types of the four evaluations

With the action, charge, and velocity lines, equipped with sections u_S, u_Q, u_C, the dimensional realization reads

$$Z_{\mathrm{vac}} = \widehat{Z} u_S u_Q^{-2}, \qquad c_{\mathrm{vac}} = \widehat{c} u_C, \tag{39.49}$$

$$\mu_{\mathrm{vac}} = \widehat{\mu} u_S u_C^{-1} u_Q^{-2}, \qquad \varepsilon_{\mathrm{vac}} = \widehat{\varepsilon} u_S^{-1} u_C^{-1} u_Q^{2}. \tag{39.50}$$

Multiplying and dividing these expressions preserves $\mu_{\mathrm{vac}}\varepsilon_{\mathrm{vac}} = c_{\mathrm{vac}}^{-2}$ and $\mu_{\mathrm{vac}}/\varepsilon_{\mathrm{vac}} = Z_{\mathrm{vac}}^2$. The SI chart subsequently evaluates the sections. Physical comparison requires presenting their construction and normalization alongside the evaluation; the four normalized coordinates of (39.44) retain their dimensionless type.

39.8.6 Reconstruction of the action shape from the constitutive response

The constitutive response and the inductive–capacitive realization receive the same angular pair constructed from APP–TRIT–TPK. The former evaluates rational functions of the two channels; the latter uses the ratio of their exponents to determine the shape of the action ellipse. Cubic inversion allows the two readouts to be composed without identifying their numerical ratios.

Let $x = A_{\mathrm{rad}}$ and $y = C_{\mathrm{rad}}^*$, with $x > |y|$, be the lifted coordinates already constructed, and preserve their sheet labels. Write

$$q_\pm = e^{-(x\pm y)}, \qquad s_\pm = q_\pm^{30}, \qquad r_\pm = f(s_\pm), \qquad f(s) = \frac{1+s+s^2}{1+s+s^2+s^3}. \tag{39.51}$$

The contractive domain gives $s_\pm \in (0,1)$ and $r_\pm \in (3/4, 1)$. In the chamber $x > y > 0$, we have $s_+ < s_-$ and $r_+ > r_-$; sheet interchange transports this chamber to the opposite orientation.

The image of the two normalized constitutive coordinates is the domain

$$\mathcal{U} = \left\{(z,v) \in (0,\infty)^2 : (zv)^{-1/2} \in (3/4,1), \quad (z/v)^{1/2} \in (3/4,1)\right\}. \tag{39.52}$$

Here $(z,v) = (\widehat{Z}, \widehat{c})$. The restrictions express membership of the reconstructed channels in the domain of the inversion theorem; they do not constitute a subsequent choice of their values.

Proposition 39.18 (Constitutive composition of the shape and LC impedance)**.** *For $(\widehat{Z}, \widehat{c}) \in \mathcal{U}$, let $s_\pm \in (0,1)$ be the unique roots of*

$$r_\pm s_\pm^3 + (r_\pm - 1)(1 + s_\pm + s_\pm^2) = 0, \qquad r_+ = \frac{1}{\sqrt{\widehat{Z}\widehat{c}}}, \quad r_- = \sqrt{\frac{\widehat{Z}}{\widehat{c}}}. \tag{39.53}$$

The angular ratio, the anisotropy of the action ellipse, and the relative LC impedance are reconstructed exactly by

$$\frac{y}{x} = \frac{\log s_+ - \log s_-}{\log s_+ + \log s_-}, \tag{39.54}$$

$$\eta_{\mathrm{el}} = \frac{1}{2}\log\frac{\log s_+}{\log s_-}, \tag{39.55}$$

$$\frac{Z_{\mathrm{LC}}}{Z_*} = e^{-\eta_{\mathrm{el}}} = \sqrt{\frac{\log s_-}{\log s_+}}. \tag{39.56}$$

In the last identity, $Z_{\mathrm{LC}} = Z_{\eta_{\mathrm{el}}}$ is the impedance of the LC chart of (24.12)*.*

Proof. The cubic inversion in (39.46) uniquely determines $s_\pm$, because f is a decreasing bijection from $(0,1)$ onto $(3/4,1)$. Their positive roots of order 30 recover $q_\pm$. The identities

$$\log s_+ = -30(x+y), \qquad \log s_- = -30(x-y)$$

prove (39.54) by difference and sum. Both logarithms are negative; their ratio is positive and their sum is nonzero. Therefore,

$$\frac{1}{2}\log\frac{\log s_+}{\log s_-} = \frac{1}{2}\log\frac{x+y}{x-y} = \operatorname{artanh}(y/x) = \eta_{\mathrm{el}}.$$

The relation $Z_{\eta_{\mathrm{el}}}/Z_* = e^{-\eta_{\mathrm{el}}}$, proved in (24.13), gives (39.56). □

The composition preserves the information needed for each reconstruction. With $\hbar$, one recovers $a_S = \sqrt{2\hbar}\, e^{\eta_{\mathrm{el}}/2}$ and $b_S = \sqrt{2\hbar}\, e^{-\eta_{\mathrm{el}}/2}$; with the chart (ω, Z_*), one recovers the inductance and capacitance already defined. Action, clock and dimensional reference accompany the form as typed data. Inverting two dimensionless coordinates does not reconstruct their units or the entire memory of the state.

Normalization enters before solving the cubic equations. If Z_{vac} and c_{vac} are received in a dimensional chart, the relations (39.50) first determine

$$\widehat{Z} = \frac{Z_{\mathrm{vac}}}{u_S u_Q^{-2}}, \qquad \widehat{c} = \frac{c_{\mathrm{vac}}}{u_C}.$$

These are the coordinates to which (39.53) applies. The reference Z_* belongs to the LC realization and is explicitly preserved in (39.56).

The two ratios are related through the channels, but evaluate different functions:

$$\widehat{Z} = \frac{f(s_-)}{f(s_+)}, \qquad \frac{Z_{\mathrm{LC}}}{Z_*} = \sqrt{\frac{\log s_-}{\log s_+}}. \tag{39.57}$$

The preceding reconstruction provides the map between the two readouts. In particular, interchanging the sheets inverts both ratios, changes η_{el} to $-\eta_{\mathrm{el}}$, and preserves $\widehat{c}$. In the symmetric limit $y = 0$, the channels coincide and both ratios equal 1. Outside that case, a common origin does not impose equality of the two functions. The action shape is recoverable from the constitutive response while preserving orientation and normalization.

39.8.7 Constitutive speed and kinematic realization of the clock

The constitutive response and the kinematic reader realize the same velocity section. Their identification preserves the order of construction: the APP sheets, their TRIT orientation and the angular transport published by TPK precede the velocity readings and the choice of units. In the two-sheet representation of section 39.7, with $x > |y|$ and rank-one projectors $P_\pm$, the transport is

$$T = e^{-xI - y\mathcal{R}} = q_+ P_+ + q_- P_-, \qquad q_\pm = e^{-x \mp y}, \qquad 0 < q_\pm < 1.$$

Its temporal blocks of lengths 90 and 120 produce the response $\mathcal{V}$ in (39.40). The coordinates x, y and the channels are outputs of the preceding angular development; this composition applies its readouts to that already constructed transport.

The symmetric logarithmic readout of temporal transport is

$$\mathcal{E} = \log\big[(1 - q_+^{90})(1 - q_-^{90})\big] - \log\big[(1 - q_+^{120})(1 - q_-^{120})\big]. \tag{39.58}$$

On the dimensional velocity line, with positive section u_C, the kinematic reader publishes $c_{\mathrm{int}} = \exp(-\mathcal{E})u_C$. The constitutive response publishes $c_{\mathrm{vac}} = \widehat{c}\,u_C$ through (39.44).

Proposition 39.19 (Constitutive and kinematic identity)**.** *In the same dimensional chart, we have*

$$c_{\mathrm{int}} = c_{\mathrm{vac}} = \widehat{c}\,u_C, \qquad \widehat{c} = (r_+ r_-)^{-1} = \exp(-\mathcal{E}). \tag{39.59}$$

The identification preserves sheet interchange.

Proof. All factors in (39.58) are positive. The logarithm law and (39.42) give

$$\mathcal{E} = \log(r_+ r_-) = \log\det\mathcal{V}, \qquad e^{-\mathcal{E}} = (\det\mathcal{V})^{-1}.$$

The determinant corresponds to the rank-one two-sheet realization. The two definitions therefore publish the same section. Exchanging the sheets permutes r_+ and r_- and preserves their product. The determinant $\det T = e^{-2x}$ corresponds to elementary transport, whereas $\det\mathcal{V} = r_+ r_-$ corresponds to its temporal response; (39.59) uses the latter. □

39.8.7.1 Path length and elementary duration

The positive dimensional frame (u_S, u_C, u_T, u_Q) induces the length basis $u_L = u_C u_T$. For an enriched path γ of $n = |\gamma|$ transitions and a constant channel, the additive readers are

$$T_{\mathrm{int}}(\gamma) = n u_T, \qquad L_{\mathrm{int}}(\gamma) = n\widehat{c}\,u_L. \tag{39.60}$$

Concatenation adds the numbers of transitions and their realizations. Enriched routes also preserve their orientation and memory.

Corollary 39.20 (Normalization of the elementary length)**.** *For $n > 0$, $L_{\mathrm{int}}(\gamma)/T_{\mathrm{int}}(\gamma) = c_{\mathrm{vac}}$. In the normalization $t_0 = u_T$,*

$$\ell_0 = c_{\mathrm{int}} t_0 = \widehat{c}\, u_L. \tag{39.61}$$

In a temporal chart $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, we obtain $\ell_0 = \tau_0 \widehat{c}\, u_L$.

Proof. Dividing the two expressions in (39.60) cancels n and uses $u_L/u_T = u_C$. Multiplying the speed section by t_0 proves both elementary formulas. □

When the channel depends on the edge, the readout retains the additive form $L_{\mathrm{int}}(\gamma) = \sum_{e\in\gamma} \widehat{c}(e) u_L$. Its quotient by $n u_T$ is the mean of the speeds of those edges. The constant-channel expression is the homogeneous specialization of this readout.

39.8.7.2 Change of chart and adapted normalization

Proposition 39.21 (Covariance of the identification)**.** *Let $u'_C = d u_C$ and $u'_T = \eta u_T$, with $d, \eta > 0$. The coordinates of the same sections satisfy*

$$\widehat{c}' = d^{-1}\widehat{c}, \qquad \tau'_0 = \eta^{-1}\tau_0, \qquad u'_L = d\eta u_L. \tag{39.62}$$

If another publication writes $c' = \widehat{c}' u'_C$ with $u'_C = d u_C$, equality of sections is exactly equivalent to $d\widehat{c}' = \widehat{c}$.

Proof. Equality of each section in the two bases gives $\widehat{c}' u'_C = \widehat{c} u_C$ and $\tau'_0 u'_T = \tau_0 u_T$. In particular,

$$\tau'_0 \widehat{c}' u'_L = \frac{\tau_0}{\eta}\frac{\widehat{c}}{d} d\eta u_L = \tau_0 \widehat{c} u_L = \ell_0.$$

The final equivalence follows from the same substitution for speed. □

The adapted choice $d = \widehat{c}$ gives $u'_C = c_{\mathrm{vac}}$ and $\widehat{c}' = 1$: it expresses the constructed section in an adapted basis. The coefficient of the internal chart remains $(r_+ r_-)^{-1}$, and this is the normalization used by cubic inversion. Keeping the action and charge bases, (39.50) produces

$$\widehat{\mu}' = d\widehat{\mu}, \qquad \widehat{\varepsilon}' = d\widehat{\varepsilon}, \qquad \widehat{\mu}'\widehat{\varepsilon}' = (\widehat{c}')^{-2}.$$

In the adapted chart, we have $\widehat{\mu}' = r_-/r_+$ and $\widehat{\varepsilon}' = r_+/r_-$, by substitution of $\widehat{\mu} = r_-^2$ and $\widehat{\varepsilon} = r_+^2$.

39.8.7.3 Preservation of the readout under amplification

For the refinement $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$, the normalized trace $\tau_m = \mathrm{Tr}/(2 \cdot 9^m)$ satisfies

$$\exp[-2\tau_m(\log \mathcal{V}_m)] = \exp[-\log r_+ - \log r_-] = \widehat{c}. \tag{39.63}$$

Indeed, each eigenvalue appears 9^m times; that multiplicity cancels against the denominator of the normalized trace. The ordinary amplified determinant is $(r_+ r_-)^{9^m}$. The two-sheet normalization, equivalently expressed by (39.63), preserves the constitutive readout during amplification. This realization preserves the enriched transport from which it originates.

39.9 Differential inversion and domain of the two-sheet constitutive response

The angular channels generated by the TPK calendar determine the positive transport

$$T = e^{-(x+y)}P_+ + e^{-(x-y)}P_-, \qquad \Omega = \{(x,y) \in \mathbb{R}^2 : x > |y|\}.$$

Here P_+ and P_- are the labelled projectors of the two sheets. The incidences 90 and 120 act on that same transport and produce the response

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1} = r_+P_+ + r_-P_-.$$

The factorization with $S = T^{30}$ determines

$$f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \qquad r_\pm = f\big(e^{-30(x\pm y)}\big). \tag{39.64}$$

The factor 30 preserves the common unit of the incidences $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$. This section receives the angular coordinates previously produced; it studies regularity and inversion of their constitutive reading.

39.9.1 Spectral sensitivity and logarithmic coordinates

For $t > 0$, define

$$g(t) = \log f(e^{-30t}), \qquad \sigma(t) = g'(t),$$

and the quotient and inverse-product coordinates

$$B(x,y) = \log \frac{r_-}{r_+} = g(x-y) - g(x+y), \tag{39.65}$$

$$V(x,y) = \log \frac{1}{r_-r_+} = -g(x-y) - g(x+y). \tag{39.66}$$

The letters B, V here denote two scalar response coordinates; the sheet projectors remain in the operator $\mathcal{V}$.

Lemma 39.22 (Regular monotonicity of the spectral reader)**.** *The map g is a strictly increasing analytic bijection from $(0,\infty)$ to $(-\log(4/3), 0)$. With $s = e^{-30t}$,*

$$\sigma(t) = \frac{30s^3(3 + 2s + s^2)}{(1 + s + s^2)(1 + s + s^2 + s^3)} > 0. \tag{39.67}$$

Proof. Differentiating the quotient gives

$$f'(s) = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0, \qquad 0 < s < 1.$$

The chain rule produces $g'(t) = -30sf'(s)/f(s)$, which is (39.67). The denominators are positive on the domain. The limits $f(1) = 3/4$ and $f(0) = 1$ give $g(0^+) = -\log(4/3)$ and $g(+\infty) = 0$. Monotonicity, continuity and these limits prove bijectivity. Positivity of the derivative guarantees analytic regularity of the inverse on the open interval. □

Theorem 39.23 (Constitutive Jacobian and local recovery)**.** *Let $a = \sigma(x-y)$ and $b = \sigma(x+y)$. The map $\Phi(x,y) = (B(x,y), V(x,y))$ satisfies*

$$D\Phi = \begin{pmatrix} a - b & -a - b \\ -a - b & a - b \end{pmatrix}, \qquad \det D\Phi = -4ab < 0. \tag{39.68}$$

In particular, it is locally invertible throughout Ω, and the differential identities $B_x = V_y$ and $B_y = V_x$ hold.

Proof. Direct differentiation of (39.66) gives $B_x = a - b$, $B_y = -a - b$, $V_x = -a - b$ and $V_y = a - b$. The determinant is $(a-b)^2 - (a+b)^2 = -4ab$, strictly negative by (39.67). The inverse function theorem applies at every point of the domain. The two differential identities follow from the equal entries of the matrix. □

39.9.2 Image domain and global reconstruction

Theorem 39.24 (Global coordinates of the normalized response)**.** *Let $L = \log(4/3)$. The map Φ is an analytic diffeomorphism from Ω onto*

$$\mathcal{D} = \{(B, V) : 0 < V - B < 2L, \quad 0 < V + B < 2L\}. \tag{39.69}$$

If $\psi = g^{-1}$, reconstruction is given by

$$x = \frac{1}{2}\left[\psi\left(\frac{B-V}{2}\right) + \psi\left(-\frac{B+V}{2}\right)\right], \tag{39.70}$$

$$y = \frac{1}{2}\left[\psi\left(-\frac{B+V}{2}\right) - \psi\left(\frac{B-V}{2}\right)\right]. \tag{39.71}$$

The chamber $x > y > 0$ corresponds to the half $B < 0$ of $\mathcal{D}$.

Proof. The change $(x, y) \mapsto (t_-, t_+) = (x - y, x + y)$ is a linear bijection between Ω and $(0, \infty)^2$. Applying g to both coordinates gives an analytic bijection onto $(-L, 0)^2$. If $u = g(t_-)$ and $v = g(t_+)$, then $B = u - v$ and $V = -u - v$. Its linear inverse is $u = (B - V)/2$, $v = -(B + V)/2$. The conditions $-L < u, v < 0$ are exactly equivalent to (39.69). Composing these three bijections gives (39.71) and proves the global assertion. Finally, $y > 0$ is equivalent to $t_+ > t_-$; monotonicity of g transforms it into $v > u$, namely $B < 0$. □

The function ψ is evaluated through the cubic inverse of the reader already constructed. For $u \in (-L, 0)$, take $r = e^u$ and the unique root $s \in (0, 1)$ of

$$r s^3 + (r - 1)(1 + s + s^2) = 0.$$

Then $\psi(u) = -(\log s)/30$. Differential reconstruction and cubic reconstruction are therefore two expressions of the same constitutive inversion.

Corollary 39.25 (Sheet conjugation)**.** *The involution $(x, y) \mapsto (x, -y)$ acts on the response through $(B, V) \mapsto (-B, V)$. It preserves the product $r_+ r_-$ and inverts the quotient r_-/r_+.*

Proof. The involution exchanges $x - y$ with $x + y$ and consequently r_- with r_+. Formula (39.66) gives the result. □

39.9.3 Local stability of recovery

Proposition 39.26 (Singular values and sensitivity bound)**.** *The singular values of $D\Phi$ are $2a$ and $2b$. In particular,*

$$\left\| (D\Phi)^{-1} \right\|_2 = \frac{1}{2\min\{\sigma(x-y), \sigma(x+y)\}}.$$

For $0 < \delta < M < \infty$, differential recovery is uniformly bounded on $\Omega_{\delta, M} = \{(x, y) : \delta \le x - y, x + y \le M\}$.

Proof. The symmetric matrix (39.68) has the orthogonal vectors $(1,-1)$ and $(1,1)$ as eigenvectors, with eigenvalues $2a$ and $-2b$. The singular values are their absolute values. The spectral norm of the inverse is the reciprocal of the smallest singular value. The positive continuous function σ attains a positive minimum on $[\delta, M]$, providing the uniform bound. □

Example 39.27 (Exact recovery of two rational channels)**.** Take $e^{-30(x-y)} = 1/2$ and $e^{-30(x+y)} = 1/4$. Then

$$x = \frac{\log 2}{20}, \qquad y = \frac{\log 2}{60}, \qquad r_- = \frac{14}{15}, \qquad r_+ = \frac{84}{85}.$$

The constitutive coordinates are

$$B = \log \frac{17}{18}, \qquad V = \log \frac{425}{392}.$$

Cubic inversion recovers $1/2$ and $1/4$, and formulas (39.71) restore x, y. At this point,

$$a = \frac{34}{7}, \qquad b = \frac{114}{119}, \qquad \det D\Phi = -\frac{15504}{833}.$$

All rational coefficients in the example are obtained by evaluating the constitutive function and its derivative; the example verifies inversion without modifying the channel generator.

The transformation $(x, y) \leftrightarrow (B, V)$ describes exactly which angular information remains accessible through the two normalized readings. Dimensional realizations of permittivity, permeability, impedance and velocity are subsequently applied with their dimensional lines and declared sections. The preceding differential signature is a property of this constitutive coordinate map and retains that interpretive domain.

Chapter 40

Derivation of charge and electrical quanta: von Klitzing resistance, conductance, Josephson constant and magnetic flux

40.1 Derivation of the charge section from the vacuum

With Z_{vac}, $h_a = 2\pi\hbar_a$ and α_{HMT} fixed, the charge of sheet $a \in \{\mathrm{pre}, \mathrm{ret}\}$ is

$$e_a = \sqrt{\frac{2\alpha_{\mathrm{HMT}} h_a}{Z_{\mathrm{vac}}}}. \tag{40.1}$$

The section e_a is not introduced through an International System coordinate intended to reconstruct α. The causal order proceeds from α_{HMT}, the action, and the vacuum toward the charge section.

40.2 Von Klitzing and Josephson constants, conductance and magnetic flux

From equation (40.1) we deduce

$$R_K = \frac{h_a}{e_a^2} = \frac{Z_{\mathrm{vac}}}{2\alpha_{\mathrm{HMT}}}, \tag{40.2}$$

$$G_0 = \frac{2e_a^2}{h_a} = \frac{4\alpha_{\mathrm{HMT}}}{Z_{\mathrm{vac}}}, \tag{40.3}$$

$$\Phi_{0,a} = \frac{h_a}{2e_a} = \sqrt{\frac{h_a Z_{\mathrm{vac}}}{8\alpha_{\mathrm{HMT}}}}, \tag{40.4}$$

$$K_{J,a} = \frac{2e_a}{h_a} = \Phi_{0,a}^{-1}. \tag{40.5}$$

Consequently,

$$R_K G_0 = 2, \qquad G_0 Z_{\mathrm{vac}} = 4\alpha_{\mathrm{HMT}}, \qquad K_{J,a}\Phi_{0,a} = 1. \tag{40.6}$$

The bidirectional covariance takes the form

$$R_K^{\mathrm{pre}} = R_K^{\mathrm{ret}}, \qquad G_0^{\mathrm{pre}} = G_0^{\mathrm{ret}},$$

$$\frac{\Phi_{0,\mathrm{pre}}}{\Phi_{0,\mathrm{ret}}} = R_{\mathrm{act}}^{1/2}, \qquad \frac{K_{J,\mathrm{pre}}}{K_{J,\mathrm{ret}}} = R_{\mathrm{act}}^{-1/2}.$$

The resistance and conductance relations are independent of the orientation of action. The flux–frequency pair preserves that orientation and transforms its two components reciprocally.

40.3 Charge section and electrical quantization relations

The constitutive response establishes a relation between the two sheets and a relative propagation scale. The charge section is obtained by composing this response with the fine-structure constant and the action, both constructed in the preceding chapters. This order has a precise consequence: charge does not retrospectively determine the dodecaphase record. It is the record, together with the action and the vacuum response, that determines the charge section on the corresponding line.

40.3.1 Positive determination of the charge section

Consider the positive sections $\hbar_a$, with $a \in \{\mathrm{pre}, \mathrm{ret}\}$, and $h_a = 2\pi\hbar_a$. The impedance Z_{vac} has already been constructed through (39.50). The quotient h_a/Z_{vac} belongs to the quadratic charge line. This dimensional compatibility allows the following definition to be formulated without yet choosing an SI chart.

Definition 40.1 (Positive charge section)**.** The charge section corresponding to action orientation a is

$$e_a = \left(\frac{2\alpha h_a}{Z_{\mathrm{vac}}}\right)^{1/2}. \tag{40.7}$$

The root is taken on the positive charge half-line. Its conjugate section is $-e_a$; the sign of the latter is not identified with the action return index.

Proposition 40.2 (Existence, uniqueness, and compatibility of the coupling)**.** *The equation $Z_{\mathrm{vac}}e_a^2 = 2\alpha h_a$ determines a unique positive section. In the constitutive realization, one also has*

$$\alpha = \frac{Z_{\mathrm{vac}}e_a^2}{2h_a} = \frac{e_a^2}{4\pi\varepsilon_{\mathrm{vac}}\hbar_a c_{\mathrm{vac}}}. \tag{40.8}$$

Proof. Positivity of α, h_a, and Z_{vac} guarantees a unique positive root. The constitutive identities give $\varepsilon_{\mathrm{vac}}c_{\mathrm{vac}} = Z_{\mathrm{vac}}^{-1}$. Substituting this equality and $h_a = 2\pi\hbar_a$ into the first expression of (40.8) yields the second. □

The last expression is the usual form of electromagnetic coupling in a physical chart. Here it appears as a subsequent identity of the constructed sections. The proof does not recalculate α from a measured charge: it establishes the algebraic consistency of the preceding result with the section just defined. Physical identification additionally requires retaining the chart of units and the coupling regime used in the comparison. An identity between quantities and an experimental comparison have different functions in that recognition.

40.3.2 Resistance, conductance, flux, and frequency

The following four quantities use the same section e_a. Their dependencies can be fully resolved before any decimal is evaluated.

Theorem 40.3 (Composition of the electrical quanta)**.** *The sections*

$$R_K = \frac{h_a}{e_a^2} = \frac{Z_{\mathrm{vac}}}{2\alpha}, \qquad G_0 = \frac{2e_a^2}{h_a} = \frac{4\alpha}{Z_{\mathrm{vac}}}, \tag{40.9}$$

$$\Phi_{0,a} = \frac{h_a}{2e_a} = \sqrt{\frac{h_a Z_{\mathrm{vac}}}{8\alpha}}, \qquad K_{J,a} = \frac{2e_a}{h_a} = \Phi_{0,a}^{-1} \tag{40.10}$$

satisfy exactly

$$R_K G_0 = 2, \qquad G_0 Z_{\mathrm{vac}} = 4\alpha, \qquad K_{J,a}\Phi_{0,a} = 1, \qquad K_{J,a}^2 R_K = \frac{4}{h_a}. \tag{40.11}$$

Proof. Substituting $e_a^2 = 2\alpha h_a/Z_{\mathrm{vac}}$ into the first two definitions cancels h_a and gives (40.9). Squaring the positive section $h_a/(2e_a)$ gives $h_a Z_{\mathrm{vac}}/(8\alpha)$; positivity fixes the root in (40.10). Its inverse is $2e_a/h_a$. The first three products in (40.11) reduce, respectively, to 2, 4α, and 1. Finally, $(4e_a^2/h_a^2)(h_a/e_a^2) = 4/h_a$. □

The physical realization recognizes in these sections the von Klitzing resistance, the conductance quantum, the flux quantum, and the Josephson constant. The factor 2 in the conductance belongs to the

normalization of the quantity defined here; identifying it with a multiplicity of channels in a material system requires specifying that representation. The four equations do not constitute four independent generators: they share α, the action, and the vacuum response. Precisely this dependency produces the cross-identities and makes it possible to check an evaluation without altering the operations from which it arose.

40.3.3 Action return and transport of the sections

The quotient $R_{\mathrm{act}} = h_{\mathrm{pre}}/h_{\mathrm{ret}}$, obtained before the angular pair, preserves the distinction between the two action sections. It does not change the sheet assignments of the constitutive response. These are two distinct operations: action return and channel interchange.

Proposition 40.4 (Covariance under return)**.** *With common α and Z_{vac}, one has*

$$\frac{e_{\mathrm{pre}}}{e_{\mathrm{ret}}} = R_{\mathrm{act}}^{1/2}, \qquad R_K^{\mathrm{pre}} = R_K^{\mathrm{ret}}, \qquad G_0^{\mathrm{pre}} = G_0^{\mathrm{ret}}, \tag{40.12}$$

$$\frac{\Phi_{0,\mathrm{pre}}}{\Phi_{0,\mathrm{ret}}} = R_{\mathrm{act}}^{1/2}, \qquad \frac{K_{J,\mathrm{pre}}}{K_{J,\mathrm{ret}}} = R_{\mathrm{act}}^{-1/2}. \tag{40.13}$$

Proof. The first identity follows by dividing the two versions of (40.7) and taking the positive root. In (40.9), the action has already cancelled. The other two identities follow from the powers $h_a^{1/2}$ and $h_a^{-1/2}$ in (40.10). □

Resistance and conductance preserve the constitutive quotient, whereas flux and frequency also retain the action orientation. This separation gives a structural reading of their dependencies: some quantities are invariants of the sheet quotient, while others additionally measure the action section used. The metrological comparison must specify which section it evaluates; the numerical proximity of the two actions does not authorize their identification.

Chapter 41

Quadratic characters and spectral constants: Catalan, Apéry, Euler–Mascheroni and the Stieltjes hierarchy

41.1 Operatorial representation of Dirichlet characters

The constants in this chapter are grouped because they share a spectral construction independent of their decimal expansions and free of target values. Let

$$Ne_n = ne_n, \qquad n \in \mathbb{N},$$

in $\ell^2(\mathbb{N})$. A residual character χ defines the diagonal operator $Q_\chi e_n = \chi(n)e_n$; whenever the power is trace class,

$$\operatorname{Tr}(N^{-s}Q_\chi) = L(s,\chi). \tag{41.1}$$

The equation preserves the modulus, orientation, and provenance of each character; therefore, it does not identify distinct residual objects.

41.2 HMT construction of the residue characters

The preceding diagonal operator is not the initial algebraic datum. The selection chain is

$$\begin{aligned}\mathrm{APP} &\longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow \{\text{residue modulo } 3,\ C_P\}\\ &\longrightarrow \{\chi_{-3},\ J = C_P \bmod 3\} \longrightarrow \{\chi_{-3},\ J^2 = -I, \chi_{-4}\}\\ &\longrightarrow Q_3, Q_4.\end{aligned}$$

The triadic residual functional determines the difference between the two nonzero oriented classes. The incidence sector uses the symmetric Paley conference matrix of order six,

$$C_P^{\mathsf{T}} = C_P, \qquad C_P C_P^{\mathsf{T}} = 5I_6.$$

Its reduction $J = C_P \bmod 3$ satisfies

$$J^2 = 2I_6 = -I_6 \quad \text{in } \operatorname{End}_{\mathbb{F}_3}(\mathbb{F}_3^6).$$

The torsional quarter then determines the cyclic phase $1, \mathrm{i}, -1, -\mathrm{i}$. Applying the Chinese remainder theorem preserves both residual data and produces $Q_{12} = Q_3Q_4$. The trace $\operatorname{Tr}(N^{-s}Q)$ is evaluated at the terminal stage; neither the Dirichlet series nor its decimal expression enters the selection of the character.

The construction is made explicit by the tables

$$\chi_{-3}(n) = \begin{cases} 0, & 3 \mid n,\\ 1, & n \equiv 1 \pmod 3,\\ -1, & n \equiv 2 \pmod 3,\end{cases} \qquad \chi_{-4}(n) = \begin{cases} 0, & 2 \mid n,\\ 1, & n \equiv 1 \pmod 4,\\ -1, & n \equiv 3 \pmod 4.\end{cases} \tag{41.2}$$

En $\ell^2(\mathbb{N})$,

$$Q_3 e_n = \chi_{-3}(n)e_n, \qquad Q_4 e_n = \chi_{-4}(n)e_n.$$

As they are diagonals, they commute; the Chinese remainder theorem gives

$$(Q_3Q_4)e_n = \chi_{-3}(n)\chi_{-4}(n)e_n = \chi_{12}(n)e_n. \tag{41.3}$$

The product simultaneously preserves the triadic residue and the quarter of torsion. Consequently, the character modulo twelve is determined before performing any evaluation of its values L.

Remark 41.1 (Negative control). The chain is invalidated if Q_3 does not distinguish the two oriented classes, if $J^2 \neq -I$, if Q_3Q_4 does not have the declared CRT table, or if the operator is selected from the value of the constant.

41.3 Internal complex structure and Catalan's constant

The internal complex structure satisfies

$$J^2 = -I. \tag{41.4}$$

The equality $J^2 = -I$ defines a representation $\rho : C_4 \to \mathrm{Aut}_{\mathbb{F}_3}(\mathbb{F}_3^6)$, $\rho(j) = J$. To realize it spectrally, take the unitary character $\upsilon : C_4 \to U(1)$, $\upsilon(j) = \mathrm{i}$, and the phase map $n \mapsto j^n$. On $\ell^2(\mathbb{N})$, the induced diagonal representation is

$$U_4 e_n = \upsilon(j^n) e_n = \mathrm{i}^n e_n.$$

Its oriented part is

$$Q_4 = \frac{U_4 - U_4^*}{2\mathrm{i}}, \qquad Q_4 e_n = \sin\Big(\frac{\pi n}{2}\Big)\, e_n = \chi_{-4}(n) e_n.$$

This is the explicit lift between the finite generator of order four and the residual operator; $\mathbb{F}_9$ is not identified with $\mathbb{C}$.

On odd integers, the same character can be written

$$\chi_J(n) = \mathrm{i}^{n-1} = \chi_{-4}(n).$$

Equivalently,

$$Q_4 = \frac{U_4 - U_4^*}{2\mathrm{i}}, \qquad Q_4 e_n = \chi_{-4}(n) e_n. \tag{41.5}$$

The Catalan constant appears as

$$G = \mathrm{Tr}(N^{-2} Q_4) = L(2, \chi_{-4}) = \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)^2}. \tag{41.6}$$

The unitary lift of the torsion quarter generates exactly the character whose quadratic moment is Catalan's constant.

Operational proof. The operator $N^{-2}Q_4$ is trace-class because $\sum_{n \geq 1} |\chi_{-4}(n)| n^{-2} < \infty$. Its trace is therefore the sum of the diagonal elements. The table equation (41.2) annuls the even pairs and alternates the odd ones:

$$\mathrm{Tr}(N^{-2} Q_4) = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots.$$

The identity with $L(2, \chi_{-4})$ is the definition of the Dirichlet series; the equality with $\mathrm{Im}\,\mathrm{Li}_2(\mathrm{i})$ is obtained by taking the imaginary part of $\sum_{n \geq 1} \mathrm{i}^n / n^2$. □

The terminal value is

$$G = 0.915965594177219015054603514 9\ldots.$$

The determining structure is the identity $J^2 = -I$, which induces exactly the alternating sign of the odd classes; the decimal expansion is its terminal evaluation.

41.4 Residual composition of the triadic–quaternary type

The residual functional modulo three provides χ_{-3}. Its CRT product with χ_{-4} defines

$$\chi_{12} = \chi_{-3}\chi_{-4}. \tag{41.7}$$

On the units modulo twelve,

$$\chi_{12}(1) = 1, \quad \chi_{12}(5) = -1, \quad \chi_{12}(7) = -1, \quad \chi_{12}(11) = 1,$$

and vanishes outside of them. Two exact evaluations are

$$L(1, \chi_{12}) = \frac{\log(2+\sqrt{3})}{\sqrt{3}}, \tag{41.8}$$

$$L(2, \chi_{12}) = \frac{\pi^2}{6\sqrt{3}}. \tag{41.9}$$

The first contains the hyperbolic unit

$$\sqrt{3}L(1, \chi_{12}) = \operatorname{arcosh} 2 = \log(2+\sqrt{3}). \tag{41.10}$$

Together with $\log(3+2\sqrt{2}) = 2\log(1+\sqrt{2})$ and $\log(30+\sqrt{899})$, this identity determines a family of Pell scales whose members retain their respective algebraic provenances.

Certified terminal evaluations are

$$L(1, \chi_{12}) = 0.760345996300946347531094254 9\ldots,$$

$$L(2, \chi_{12}) = 0.949703126294009395263498491 7\ldots.$$

Evaluations dodecaphasic. Grouping by residue classes,

$$\begin{aligned} L(1, \chi_{12}) &= \sum_{n \geq 0} \left(\frac{1}{12n+1} - \frac{1}{12n+5} - \frac{1}{12n+7} + \frac{1}{12n+11} \right) \\ &= \int_0^1 \frac{1 - x^4 - x^6 + x^{10}}{1 - x^{12}}\, dx = \int_0^1 \frac{1 - x^4}{1 + x^6}\, dx. \end{aligned}$$

A real decomposition into the quadratic factors of $1 + x^6$ produces $\log(2+\sqrt{3})/\sqrt{3}$. For the second moment, if ψ_1 is the trigamma function,

$$\begin{aligned} L(2, \chi_{12}) &= \frac{1}{144}\left[\psi_1(1/12) - \psi_1(5/12) - \psi_1(7/12) + \psi_1(11/12)\right] \\ &= \frac{\pi^2}{144}\left(\csc^2 \frac{\pi}{12} - \csc^2 \frac{5\pi}{12} \right) = \frac{\pi^2}{6\sqrt{3}}, \end{aligned}$$

where has been used $\psi_1(z) + \psi_1(1-z) = \pi^2 \csc^2(\pi z)$. The two proofs use the CRT table and not a target decimal value. □

41.5 Euler and Stieltjes Constants and the Theta–Mellin Transform

Let

$$\vartheta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad t > 0.$$

For $\Re s > 1$, the theta–Mellin functional recovers the convergent identity

$$\pi^{-s/2}\Gamma\Big(\frac{s}{2}\Big)\,\zeta(s) = \frac{1}{2}\int_0^\infty \big(\vartheta(t) - 1\big)t^{s/2-1}\,dt.$$

The meromorphic continuation is obtained by splitting the integral at $t = 1$, using $\vartheta(t) = t^{-1/2}\vartheta(1/t)$, and separating the polar terms. Its expansion at $s = 1$,

$$\zeta(s) = \frac{1}{s-1} + \sum_{j\ge 0} \frac{(-1)^j \gamma_j}{j!}(s-1)^j, \tag{41.11}$$

situates $\gamma = \gamma_0$ and the Stieltjes constants in a single family of renormalized moments. This organization expresses an analytical common provenance, without attributing an interpretation that is physically identical to all its levels.

The Catalan constant often appears as an alternating series:

$$G = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots.$$

That formula gives its value, and the HMT genealogy explains why this specific alternation exists.

In HMT, the origin is in the torsion chamber.

There exists an internal operator J such that

$$J^2 = -I.$$

The root of -1 emerges from two successive applications of a quarter turn. The complex structure is the algebraic memory of an internal rotation.

That turn generates a character of order four. When it is published on the positive integers, it produces exactly the residual pattern that alternates among $+1, 0, -1, 0$.

The Catalan constant then appears as a spectral trace:

$$G = L(2, \chi_{-4}) = \mathrm{Tr}(N^{-2}Q_4).$$

This changes its meaning. Catalan's constant is simultaneously the sum of a particular series and the spectral response of the integers to a quarter of torsion.

Each term of the series records how the internal rotation acts on a residue class. The alternation is the arithmetic shadow of the rotation.

The triadic structure has its own character modulo 3. Combining it with the character of order four, the Chinese remainder theorem produces a character modulo 12:

$$\chi_{12} = \chi_{-3}\chi_{-4}.$$

The module 12 appears as the minimal common space in which triadic orientation and the quarter-torsion can coexist.

The dodecaphase is, in that sense, a reconciliation of 3 and 4.

Of the combined character emerges

$$L(1,\chi_{12}) = \frac{\log(2+\sqrt{3})}{\sqrt{3}},$$

y

$$L(2,\chi_{12}) = \frac{\pi^2}{6\sqrt{3}}.$$

La cantidad

$$2+\sqrt{3}$$

is the fundamental unit of the real field $\mathbb{Q}(\sqrt{3})$. Its logarithm represents a hyperbolic shift. Once again, a finite residue structure opens an infinite multiplicative dynamics.

Here rotation, torsion, arithmetic character, Pell unit, and hyperbolic displacement meet.

The Euler–Kronecker constant of the field $\mathbb{Q}(\sqrt{3})$ prolongs this structure. It measures the regular term that remains after separating the principal pole of the field's zeta function. In the HMT genealogy, this term constitutes the regular invariant of fine arithmetic memory that remains after removing the universal divergence.

The Bendersky–Glaisher family extends the mechanism into a tower. The spectral derivatives of zeta generate that tower. Apery's constant and other constants associated with odd values are organized as members of the same functional family.

The novel contribution lies in the genealogy that unifies the classical constants:

- the quarter of torsion produces the complex structure;
- the residue triadico produces the structure modulo 3;
- their composition produces the character modulo 12;
- the character generates values L;
- the values L open the quadratic field and its spectral constants.

Geometric torsion has become arithmetic while preserving entirely its orientation.

41.6 Cyclic decomposition, quarter-turn, and oriented Catalan moment

The dodecaphase register allows the constitutive response to be related to other transport invariants. The link is established on specified subspaces and maps: the rank defect measures a difference in incidence, the determinant preserves a scalar response and a coefficient of the resolvent retains orientation. The family of moments culminating in Catalan enters at this last level.

41.6.1 Two subspaces of the dodecaphase cycle

In $\mathbb{R}^{12}$, let $Ce_j = e_{j+1 \bmod 12}$ and let

$$P_3^{\mathrm{cyc}} = \frac{I + C^3 + C^6 + C^9}{4}, \qquad P_4^{\mathrm{cyc}} = \frac{I + C^4 + C^8}{3}, \qquad H_d = \operatorname{Ran} P_d^{\mathrm{cyc}}. \tag{41.12}$$

The averages over cyclic subgroups are orthogonal projectors. H_3 consists of the periodic sequences whose period divides 3, and H_4 of those whose period divides 4. Their dimensions are 3 and 4. The restriction of C to each space realizes the corresponding cycle. The superscript cyc identifies the average over the dodecaphase cycle: P_3^{cyc} is the projector onto sequences whose period divides 3. Its domain and

action differ from those of the three-dimensional isotypic projector of the icosahedral realization; equality of rank does not identify the two operators.

Proposition 41.2 (Rank defect and determinantal response)**.** *We verify*

$$\frac{\operatorname{Tr}(P_3^{\mathrm{cyc}} - P_4^{\mathrm{cyc}})}{12} = -\frac{1}{12}, \qquad \frac{\det_{H_3}(I - sC)}{\det_{H_4}(I - sC)} = \frac{1 - s^3}{1 - s^4} = f(s), \quad 0 < s < 1. \tag{41.13}$$

Proof. The trace of an orthogonal projector is its rank, so the numerator of the first expression is $3 - 4$. The characteristic polynomial of a cycle of length d is $t^d - 1$; hence its determinant $\det(I - sC)$ is $1 - s^d$. Their quotient yields the second identity. □

The substitution $s = q^{30}$ transports this representation to the $90/120$ quotient. At $s = 1$, the common constant mode is cancelled before taking the limit, which equals $3/4$. This realization of $-1/12$ as a trace defect has an explicit finite domain. The regularized and modular realizations of the same scalar retain their own operators and domains.

41.6.2 Faithful representation of the quarter-turn

Define the antipodal projector $P_{\mathrm{ant}} = (I - C^6)/2$ and $Q = P_4^{\mathrm{cyc}} P_{\mathrm{ant}}$. This projector acts on the twelve-phase cycle and differs from the constitutive projector P_- in Section 39.8. The projectors P_4^{cyc} and P_{ant} commute, and Q has rank 2. In its image, fix the orthonormal basis

$$u = \frac{1}{\sqrt{6}}(1, 0, -1, 0, 1, 0, -1, 0, 1, 0, -1, 0)^{\mathsf{T}}, \qquad v = Cu. \tag{41.14}$$

Componentwise verification gives $Cu = v$ and $Cv = -u$. Consequently, the isometry $W(a, b) = au + bv$ satisfies

$$CW = WJ, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I. \tag{41.15}$$

The complex structure of the plane is realized by the effective action of the cycle. The sign depends on the chosen orientation: in the same basis, $C^3 W = -WJ$. This precision allows the plane to be composed with the oriented readout.

41.6.3 Resolvent and unlimited depth

Since $C^2 = -I$ on $\operatorname{Ran} Q$, for $0 \le s < 1$ one has

$$(I - sC)^{-1}\big|_{\operatorname{Ran} Q} = \frac{I + sC}{1 + s^2}, \qquad k_4(s) := \langle v, (I - sC)^{-1} u \rangle = \frac{s}{1 + s^2}. \tag{41.16}$$

The sequence $\langle v, C^n u \rangle$ is $0, 1, 0, -1, \ldots$. The depth readout preserves this character through the moment

$$C_2(s) = \sum_{k \ge 0} \frac{(-1)^k s^{2k+1}}{(2k + 1)^2}, \qquad 0 \le s \le 1. \tag{41.17}$$

The series converges absolutely and uniformly on this interval by comparison with $\sum (2k + 1)^{-2}$. For $0 < s < 1$, the differentiated series converge uniformly on every compact subinterval of the interior. With $\mathcal{D} = s\, d/ds$, one obtains

$$\mathcal{D}^2 C_2(s) = k_4(s), \qquad C_2(s) = \int_0^s \frac{dt}{t} \int_0^t \frac{d\xi}{\xi}\, k_4(\xi). \tag{41.18}$$

The conditions $C_2(0) = 0$ and $\lim_{s \downarrow 0} \mathcal{D} C_2(s) = 0$ fix the two integration constants. Indeed, the difference between two solutions of $\mathcal{D}^2 F = k_4$ has the form $a \log s + b$; the second condition annuls a and the first

annuls b. Uniform convergence permits evaluation at the endpoint $s = 1$: the result is the Catalan moment. The finite representation of the character preserves its orientation; the depth index remains unlimited.

41.6.4 Composition with the constitutive response

Theorem 41.3 (Constitutive differential relation)**.** *For* $0 < s < 1$*, with the orientation of* (41.14)*,*

$$f(s) = \frac{1 + s + s^2}{s(1 + s)}\, \mathcal{D}^2 C_2(s). \tag{41.19}$$

Proof. Substituting (41.16) into (41.18) and using $(1 + s)(1 + s^2) = 1 + s + s^2 + s^3$ gives the identity. All denominators are positive on the domain. The intertwiner (41.15) identifies the operator coefficient used before its evaluation. □

Upon evaluating $s_\pm = q_\pm^{30}$, the same family produces the two channel responses of (39.42). The Catalan limit, the trace defect and the constitutive eigenvalues retain their different functions within this composition. Inverting the cycle with the marks u, v fixed changes the sign of k_4 and preserves the determinant $1 + s^2$. This identifies orientation information that the determinant alone does not distinguish.

The Pell development uses a particular evaluation of this family, with $\rho = 2 - \sqrt{3}$ and $\rho^{-1} = 2 + \sqrt{3}$. Its proof is preserved as a separate integration unit; the angular definition of $q_\pm$ remains independent of any identification of $s_\pm$ with ρ.

Figure 41.1 represents the relation between the rational response and the oriented kernel before integration. The Pell section satisfies $\rho + \rho^{-1} = 4$, so that $k_4(\rho) = (\rho + \rho^{-1})^{-1} = 1/4$. The two constitutive values are obtained, by contrast, by evaluating f at the already constructed arguments $s_\pm$. The plot includes the continuous extensions of the two functions to $[0, 1]$; the domain of constitutive inversion remains the open interval $(0, 1)$.

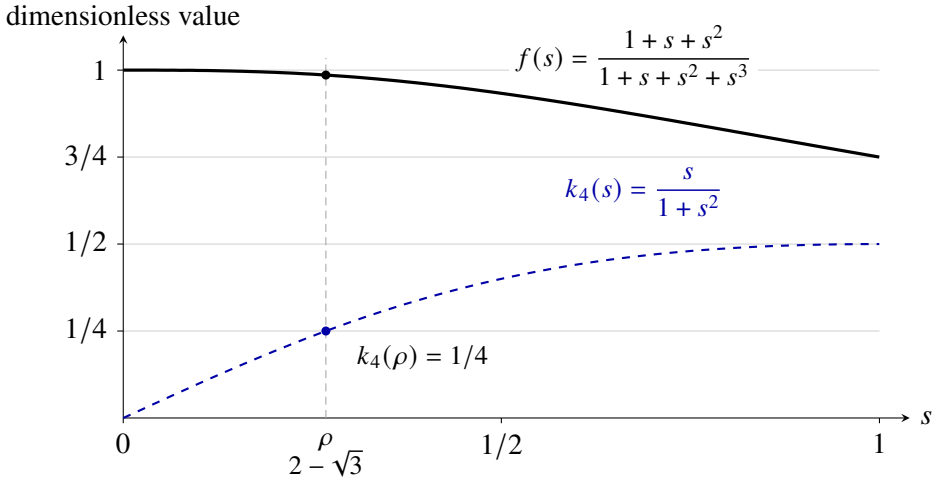


Figure 41.1: Constitutive response and kernel of the oriented moment. The solid curve is f and the dashed curve is $k_4 = \mathcal{D}^2 C_2$; they are related by (41.19). The vertical line marks the Pell section, without identifying it with the constitutive arguments s_+ or s_-.

Chapter 42

Euler–Kronecker constants and prime distributions: dodecaphasic characters, Meissel–Mertens and twin-prime series

42.1 Euler–Kronecker constant of the character of dodecaphase type

The factorization of Dedekind

$$\zeta_{\mathbb{Q}(\sqrt{3})}(s) = \zeta(s)L(s,\chi_{12}) \tag{42.1}$$

gives, upon comparing the constant terms in $s = 1$,

$$\gamma_{\mathbb{Q}(\sqrt{3})} = \gamma + \frac{L'(1,\chi_{12})}{L(1,\chi_{12})} = 1.0539656082484825800\ldots. \tag{42.2}$$

This constant represents the normalized finite term at the pole of the zeta function of the real dodecaphase field. It is not a variant of Euler's γ, but the arithmetic correction induced by χ_{12}.

In fact, writing $u = s - 1$,

$$\zeta(s) = u^{-1} + \gamma + O(u), \qquad L(s,\chi_{12}) = L_1 + L_1' u + O(u^2),$$

we obtain

$$\zeta_{\mathbb{Q}(\sqrt{3})}(s) = \frac{L_1}{u} + \bigl(\gamma L_1 + L_1'\bigr) + O(u). \tag{42.3}$$

The Euler–Kronecker constant is the quotient of the finite term by the residue; dividing equation (42.3) by L_1 proves equation (42.2). This derivation simultaneously explains the residue

$$\operatorname*{Res}_{s=1} \zeta_{\mathbb{Q}(\sqrt{3})}(s) = \frac{\log(2+\sqrt{3})}{\sqrt{3}}$$

and the finite correction. The numerical value is certified by evaluating $L(1,\chi_{12})$ and $L'(1,\chi_{12})$ with the same residue table and a tail bound; it is not imported from a table of quadratic fields.

42.2 Constantes aritmeticas asociadas to the numeros primos

Once the irreducible arithmetic cycles have been selected, two distinct families are obtained:

$$B_1 = \lim_{x\to\infty}\left(\sum_{p\le x}\frac{1}{p} - \log\log x\right), \tag{42.4}$$

$$C_2 = \prod_{p>2}\frac{p(p-2)}{(p-1)^2}. \tag{42.5}$$

B_1 regularizes a local sum; C_2 is a multiplicative density of twin-prime pairs. Both require tail control. They are not deduced from one another by sharing the primes.

42.3 Native product, factorization, and irreducibility

This chapter develops the arithmetic sector of normal forms over the common kernel and the correlative construction of the continuum in Section 6.5. The local operation extends to words, its fibers distinguish the irreducibles, and evaluation makes it possible to recognize the primes. The development recovers the native product of the corpus; its documentary provenance is collected in Appendix 102.6.

42.3.1 Composition of residues and quotients in multiplication

Selecting prime numbers requires a multiplication to have been constructed. In Triadic Modular Holography, this operation must be presented from the two APP sheets, preserving both the residual coordinate and its quotient. The residue alone makes it possible to recognize a congruence class. The residue–quotient pair makes it possible to reconstruct the integer result of the local operation. This distinction determines which information must be transported when composing the cells.

Let $D_9 = \{1, \dots, 9\}$. For each pair $(a, b) \in D_9^2$, the additive and multiplicative sheets are written as

$$a + b = R_+(a, b) + 9Q_+(a, b), \qquad ab = R_\times(a, b) + 9Q_\times(a, b),$$

with $R_+, R_\times \in D_9$. The choice of the positive representative uniquely fixes the four data. In particular, a result divisible by nine has its residue represented by 9, and its quotient is one less than that of Euclidean division with zero remainder. This is an exact representation convention, which must be maintained throughout the construction.

For example,

$$2 \cdot 5 = 1 + 9 \cdot 1, \qquad 9 \cdot 9 = 9 + 9 \cdot 8.$$

The first result preserves the information that distinguishes 10 from 1; the second preserves that which distinguishes 81 from 9. If the quotient is omitted at one of these steps, a subsequent multiplication acts on a different object. Arithmetic memory here is not an interpretation added to the algorithm: it participates in reconstructing its result.

The two sheets are coupled even during multiplication. The product of two digits generates a pair $(R_\times, Q_\times)$; when the quotient from a preceding position is incorporated into the current position, the additive sheet is needed again. The global operation composes both sheets.

Proposition 42.1. Local compatibility of the quotients

The quotients satisfy the identities required to associate and distribute the operations without changing the reconstructed integer. For example, for addition,

$$Q_+(a, b) + Q_+\big(R_+(a, b), c\big) = Q_+(b, c) + Q_+\big(a, R_+(b, c)\big).$$

For multiplication,

$$Q_\times\big(R_\times(a, b), c\big) + cQ_\times(a, b) = Q_\times\big(a, R_\times(b, c)\big) + aQ_\times(b, c).$$

Proof. Substituting the two decompositions of $a + b + c$, the final residues are the same positive representative of its class modulo nine. Subtracting them and dividing by nine gives the first equality. For abc, the same argument is applied to

$$\big(R_\times(a, b) + 9Q_\times(a, b)\big)c = a\big(R_\times(b, c) + 9Q_\times(b, c)\big).$$

The distributive identity is obtained by reconstructing $a(b + c) = ab + ac$ with the two sheets. In all three cases, uniqueness of the residual representation determines exactly the quotient that must accompany it. □

These identities make it possible to organize the operation record: two parenthesizations produce the same integer without requiring their execution histories to be identified. Equality of the result and identity of the history are different assertions.

42.3.2 Nonadic words and normal form

The extension to arbitrarily large integers uses finite sequences of digits from D_9. The orientation adopted runs from the least significant to the most significant position. For $u = (u_0, \dots, u_{r-1})$, let

$$\mathrm{ev}_9(u) = \sum_{j=0}^{r-1} u_j 9^j.$$

The empty sequence represents zero. In this bijective numeration, the symbol 9 remains a digit; it does not become a zero without a quotient.

Proposition 42.2. Existence and uniqueness of the normal form

The preceding evaluation is a bijection between finite words over D_9, including the empty word, and nonnegative integers.

Proof. If $n > 0$, define

$$d(n) = 1 + ((n-1) \bmod 9), \qquad q(n) = \frac{n - d(n)}{9}.$$

Then $d(n) \in D_9$, $q(n) \geq 0$, and $q(n) < n$. Iteration terminates at zero and produces a word whose evaluation is n. For uniqueness, the first digit of any word evaluating to n must represent the class of n modulo nine within D_9; it must therefore be $d(n)$. Removing it and dividing by nine reduces the question to $q(n)$. Induction proves uniqueness. □

Some normal forms are

$$9 \leftrightarrow (9), \quad 10 \leftrightarrow (1,1), \quad 81 \leftrightarrow (9,8), \quad 1000 \leftrightarrow (1,3,3,1).$$

The representation is checked directly: $1000 = 1 + 3 \cdot 9 + 3 \cdot 9^2 + 9^3$. This last equality does not by itself identify the word of an integer with the cubic completion of a geometric domain. The two occurrences of 1000 have different objects and operations; an identification would require its map.

The normal form is an arithmetic coordinate. The TPK state additionally preserves orientation, sheet, route, boundary, and transport memory. When different histories produce the same normal form, the evaluation map brings their values together, but does not establish a bijection between those histories. The development of the article must preserve both levels and the map relating them.

42.3.3 Genealogical construction of the product on records

Word addition is performed position by position. If a position contains two digits, the additive sheet is consulted. If it contains only one, that digit is retained. The quotient arriving from the preceding position is incorporated through another additive lookup, and the quotients of both lookups are added. If $a + b = r + 9q$ and $r + c = r' + 9q'$, then $q + q'$ is transported. For example, $9 + 9 + 1 = 1 + 9 \cdot 2$. The incoming quotient lies between zero and two; when it is zero, no additional cell is consulted. An absent position is not an APP cell containing the digit zero.

Multiplication by a digit $d \in D_9$ acts analogously: at each position, the multiplicative sheet is first applied to the digit of the word and to d; the pending quotient is then incorporated through the additive sheet. The new quotient is the sum of the quotient of the local product and that of this additive lookup, and lies between zero and nine. The procedure terminates by appending the final quotient when it is nonzero.

Denote these operations by $u \boxplus v$ and $d \odot u$. If $v = (v_0, \dots, v_{m-1})$, the native product is constructed through the recurrence

$$a_m = \varnothing, \qquad a_j = (9 \odot a_{j+1}) \boxplus (v_j \odot u), \quad j = m-1, \dots, 0,$$

and $u \boxtimes v = a_0$ is defined. Every operation on the right-hand side uses the two local sheets already defined. Word evaluation does not intervene as an instruction for computing the product.

Theorem 42.3. Exactness of the native product

For all finite words u, v,

$$\mathrm{ev}_9(u \boxtimes v) = \mathrm{ev}_9(u)\,\mathrm{ev}_9(v).$$

Proof. The cell reconstruction identities prove, by induction over positions, that

$$\mathrm{ev}_9(u \boxplus v) = \mathrm{ev}_9(u) + \mathrm{ev}_9(v), \qquad \mathrm{ev}_9(d \odot u) = d\,\mathrm{ev}_9(u).$$

The product recurrence preserves the invariant

$$\mathrm{ev}_9(a_j) = \mathrm{ev}_9(u) \sum_{k=j}^{m-1} v_k 9^{k-j}.$$

The case $j = m$ is immediate. If it holds for $j+1$, multiplying by nine and adding $v_j\,\mathrm{ev}_9(u)$ yields the case j. For $j = 0$, the asserted equality follows. □

Corollary 42.4. Multiplicative structure of the normal forms

The nonempty words, with the product $\boxtimes$, form a commutative monoid whose unit is (1). Upon adjoining the empty word, it is absorbing.

Proof. Evaluation is multiplicative by the theorem and injective by Proposition 42.2. Associativity, commutativity, and the identity are verified after evaluation and recovered by injectivity. □

The proof makes it possible to recognize ordinary multiplication in the constructed operation. The logical order matters: the cell transducer is first defined and its invariant is proved; its integer realization is then identified.

42.3.4 Multiplicative openings and selection of irreducibles

Once the product has been constructed, factorization can be defined as its fiber. For each nonempty word w, let

$$\mathrm{Fac}(w) = \{(u, v) : u \boxtimes v = w\}.$$

This fiber is finite. Indeed, if $n = \mathrm{ev}_9(w)$, exactness implies $\mathrm{ev}_9(u)\,\mathrm{ev}_9(v) = n$, with both factors positive and no greater than n. Finiteness justifies the algebraic sum

$$\Delta e_w = \sum_{(u,v) \in \mathrm{Fac}(w)} e_u \otimes e_v.$$

The order in the factor pair is preserved. For $n = 12$, for example, the evaluations of the fiber are

$$(1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1).$$

The reduced coproduct removes the two decompositions containing the unit, for $w \neq (1)$:

$$\widetilde{\Delta} e_w = \sum_{\substack{u \boxtimes v = w \\ u \neq (1),\ v \neq (1)}} e_u \otimes e_v .$$

Set $\widetilde{\Delta} e_{(1)} = 0$. This convention allows the unit to be treated separately, without confusing it with an irreducible.

Definition 42.5. Native irreducible

A word $w \neq (1)$ is multiplicatively irreducible if $\widetilde{\Delta} e_w = 0$.

The definition receives no table of primes and uses no decimal period. It asks whether the operation already constructed admits a nontrivial opening.

Theorem 42.6. Recognition of the primes

Evaluation establishes a bijection between native irreducibles and positive prime numbers.

Proof. If $w = u \boxtimes v$ with factors different from the unit, its evaluation is the product of two integers greater than one; it is therefore not prime. Conversely, if $\mathrm{ev}_9(w) = ab$, with $a, b > 1$, their normal forms u, v satisfy

$$\mathrm{ev}_9(u \boxtimes v) = ab = \mathrm{ev}_9(w).$$

Injectivity gives $u \boxtimes v = w$, so that w is not irreducible. The bijection of normal forms completes the proof. □

This result identifies a property of the operation with the property of its integer realization. Prime values appear upon evaluating the selected objects, after the product and its fibers have been defined.

Proposition 42.7. Coassociativity

On the algebraic space generated by positive words,

$$(\Delta \otimes I)\Delta = (I \otimes \Delta)\Delta.$$

Proof. Both sides enumerate, with coefficient one, the ordered triples (u, v, z) whose product is w. Associativity of the product identifies the two parenthesizations, and all sums are finite. □

42.3.5 Operator realization of the selector

In $\mathcal{H} = \ell^2(\mathcal{W}_9^+)$, with orthonormal basis $\{e_w\}$, the coproduct is first defined on finite linear combinations. The supports of Δe_w and $\Delta e_{w'}$ are disjoint if $w \neq w'$: an ordered pair has a unique product. Hence,

$$\|\Delta e_w\|^2 = |\operatorname{Fac}(w)|, \qquad \|\widetilde{\Delta} e_w\|^2 = r(w),$$

where $r(w)$ counts the ordered nontrivial openings.

The closure of $\widetilde{\Delta}$ has domain

$$\left\{ x = \sum_w x_w e_w \in \mathcal{H} : \sum_w r(w) |x_w|^2 < \infty \right\}.$$

Its associated positive operator is diagonal:

$$A_{\mathrm{irr}} = \overline{\widetilde{\Delta}}^{*} \overline{\widetilde{\Delta}}, \qquad A_{\mathrm{irr}} e_w = r(w) e_w .$$

Its maximal domain is

$$\mathrm{Dom}(A_{\mathrm{irr}}) = \left\{ x \in \mathcal{H} : \sum_{w} r(w)^2 |x_w|^2 < \infty \right\}.$$

The projector onto the irreducibles is then

$$P_{\mathrm{irr}} = (I - P_{(1)})\mathbf{1}_{\{0\}}(A_{\mathrm{irr}}).$$

The formula incorporates two distinct exclusions: the kernel removes all words possessing a nontrivial opening, and $I - P_{(1)}$ removes the unit. Positivity comes from a squared norm and has a specific meaning here: it counts decompositions. It is not yet an assertion about the positivity of another form, such as the completed Weil form.

42.3.6 Normalized readout of the TPK prefix and preservation of genealogy

The normal form admits an explicit composition with the integer prefix produced by the TPK. Denote by $\beta_9 = \mathrm{ev}_9^{-1}$ the normalizer of Proposition 42.2. Here K is the block depth of the TPK state, distinct from the length k of an elementary history in the emission tree. The construction applies to a regional coordinate of the block state; the same calculation applies to each of the three coordinates with its own record.

For the deep row vector $x_K \in \mathbb{Z}^6$, the integer TPK step uses the matrix

$$A_H = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}$$

and produces

$$y_K = x_K A_H, \qquad u_{K+1} = y_K \bmod 3, \qquad x_{K+1} = \frac{y_K - u_{K+1}}{3}.$$

The block $u_{K+1} \in \mathbb{F}_3^6$, with representatives $\overline{u}_j \in \{0, 1, 2\}$, has index

$$I(u_{K+1}) = \sum_{j=1}^{6} \overline{u}_j 3^{6-j}, \qquad 0 \le I(u_{K+1}) < 729.$$

The cylindrical readout updates the prefix through

$$N_{K+1} = 729 N_K + I(u_{K+1}).$$

The matrix A_H belongs to the TPK kernel update. In the following composition it is not replaced by an external arithmetic rule: the block and its index come from the step just executed.

Proposition 42.8. Compatibility of the native prefix and its normal form

If $w_K = \beta_9(N_K)$, then

$$w_{K+1} = \big(\beta_9(729) \boxtimes w_K\big) \boxplus \beta_9(I(u_{K+1})).$$

The typed truncation of this readout is

$$\varrho(w) = \beta_9\left(\left\lfloor \frac{\mathrm{ev}_9(w)}{729} \right\rfloor\right); \qquad \varrho(w_{K+1}) = w_K.$$

Proof. Compatibility of word addition and multiplication proves that the evaluation of the right-hand side is

$$729\,\mathrm{ev}_9(w_K) + I(u_{K+1}) = 729N_K + I(u_{K+1}) = N_{K+1}.$$

Uniqueness of the normal form gives the first equality. Since the block index lies between zero and 728, the integer quotient of N_{K+1} by 729 is N_K. Applying β_9 proves the truncation identity. □

Truncation does not consist of deleting three digits from a bijective word. For example, $\beta_9(729) = (9, 8, 8)$, whereas the integer quotient of 729 by 729 is one; deleting its three digits would produce the empty word. The rule ϱ preserves the residue and quotient convention defining the numeration.

The proposition normalizes an already generated prefix and preserves its compatibility under refinement. Its application to the state retains, in the original record, the block u_{K+1}, the deep quotient x_{K+1}, the carries of the decimal chart, the phase and the route. It does not assert that w_K reconstructs those coordinates by itself or that any word is published by a fixed regional section. The equality of normal forms refers to the numerical prefix; the identity of the histories is decided in the complete state.

The record accompanying the product contains the lookups of both sheets and their residues and quotients. It makes it possible to explain why an output was obtained and to reproduce the operation. The transducer result does not, however, exhaust the state of a TPK prolongation: a normal form may be reached by different histories.

Therefore, integration with the dynamics preserves the direction of each map: the TPK state publishes its prefix, and the latter admits the normalized reading just constructed. Compatibility of a product on histories additionally requires specifying its domain, transport and survival. It does not require inverting a numerical reading that does not individualize the history. The identity of values does not replace that compatibility square; each application must be accompanied by the specific operation and the information it preserves.

The starting point of the arithmetic is already specified: two local operations, a normal form, a global transducer, its fibers, and a selector. The spectral extension must use these results without retrospectively changing their origin.

42.3.7 Correspondence with the preserved implementation

The code `a9_native.py` implements the operations in this order:

1. Local addition and multiplication cells.
2. Bijective nonadic encoding and separate evaluation.
3. Word addition and multiplication by a digit.
4. Word product recurrence.
5. Enumeration of product fibers.
6. Selection of irreducibles by absence of a nontrivial opening.

The checker uses integer multiplication and a conventional primality test after generating the outputs. These operations check the implementation; they are not incorporated into the body of the word producer. Checks that suppress a quotient or alter a cell document which dependency is broken. The reproduction receipts state the size of the checked window; that computational scope is kept separate from the preceding general proofs.

42.4 Factorization, arithmetic clocks and hierarchy of traces

The native product distinguishes two properties: irreducibility of a normal form and the period of a multiplicative action on it. This chapter constructs their spectral readouts, preserves the normalizations of each functional and prepares the completed form developed in Section 97.8.

42.4.1 From the irreducible to its clock

Product and coproduct determine which objects are irreducible. The arithmetic clock describes another property of those objects: return of a finite multiplicative orbit. For an integer $n \geq 2$ coprime to ten, define

$$\tau_n = \min\{k \geq 1 : 10^k \equiv 1 \pmod{n}\}.$$

The length τ_n corresponds to the orbit of the unit under multiplication by ten in the group of units modulo n. The decimal base fixes this readout; another coprime base determines another clock. Irreducibility, by contrast, belongs to the product constructed before this periodic representation is chosen.

Proposition 42.9. Synchronization of clocks

If $n = \prod_p p^{a_p}$ and $\gcd(n, 10) = 1$, then

$$\tau_n = \operatorname{mcm}_{p^{a_p} \| n} \tau_{p^{a_p}}.$$

Proof. An exponent k satisfies $10^k \equiv 1 \pmod{n}$ if and only if it does so modulo each prime power in the factorization. This is equivalent to k being a multiple of every order $\tau_{p^{a_p}}$. The smallest common positive exponent is the least common multiple. □

Factorization therefore organizes synchronization. Powers of the same prime must be retained as such: their period is not obtained by replacing them, without proof, with the base prime's period.

Proposition 42.10. Nonadic residual closure

If $\gcd(n, 90) = 1$, the integer block

$$M_n = \frac{10^{\tau_n} - 1}{n}$$

is divisible by nine.

Proof. The period definition makes M_n integral. Since $10 \equiv 1 \pmod{9}$, the numerator is divisible by nine. The integer n is invertible modulo nine, so $nM_n \equiv 0 \pmod{9}$ implies $M_n \equiv 0 \pmod{9}$. □

For a prime $p \notin \{2, 3, 5\}$, one also has $\tau_p \mid p - 1$, since its orbit belongs to the multiplicative group of order $p - 1$. The congruence of Proposition 42.10 also holds for composites coprime to ninety. Its meaning is residual closure; irreducibility remains determined by product openings.

The numbers $7, 13, 77, 91, 143$ have decimal period six, but only the first two are irreducible. The structural relation is not that a period by itself identifies a prime, but that the product constructs components whose composition organizes returns.

42.4.2 Factorization as mode occupation

Let $\mathcal{P}$ be the set obtained by evaluating the native irreducibles. Each mode occupation is a family $\mathbf{k} = (k_p)_{p \in \mathcal{P}}$ of nonnegative integers with finite support. Its multiplicative evaluation is

$$N(\mathbf{k}) = \prod_{p \in \mathcal{P}} p^{k_p}.$$

Unique factorization identifies these occupations with the positive integers. The Hilbert spaces of occupations and of positive normal forms then have a unitary correspondence of bases. An occupation register preserves how often each irreducible occurs.

In the total-occupation-one sector, $\sum_p k_p = 1$, with basis $\{e_p : p \in \mathcal{P}\}$, define

$$H_{\mathcal{P}} e_p = (\log p) e_p,$$

and, on occupations,

$$H_{\mathrm{occ}} e_{\mathbf{k}} = \left(\sum_p k_p \log p \right) e_{\mathbf{k}}.$$

With maximal diagonal domains, both are self-adjoint. If $Q e_n = n e_n$, the basis identification satisfies

$$U H_{\mathrm{occ}} U^* = \log Q.$$

The equality follows from multiplicative composition:

$$\sum_p k_p \log p = \log N(\mathbf{k}).$$

The operator is not used to select primes retrospectively. It receives the irreducibles already constructed and converts their composition into an additive quantity. This is the precise place of the occupation representation in the article's genealogy.

42.4.3 One occupation structure and two trace levels

For $\Re s > 1$, the trace on the total-occupation-one sector is

$$P(s) = \mathrm{Tr}(e^{-s H_{\mathcal{P}}}) = \sum_{p \in \mathcal{P}} p^{-s}.$$

The trace on all occupations, by contrast, is

$$Z(s) = \mathrm{Tr}(e^{-s H_{\mathrm{occ}}}) = \prod_{p \in \mathcal{P}} (1 - p^{-s})^{-1} = \sum_{n \geq 1} n^{-s}.$$

The last equality follows from unique factorization. The stated half-plane guarantees absolute convergence and permits grouping of terms. After identification of the integer realization, Z agrees there with the zeta function. No table of zeta values has been used to define the modes.

Proposition 42.11. Hierarchy of moments

On $\Re s > 1$, taking the logarithm branch determined by expansion of the convergent product,

$$\log Z(s) = \sum_{k \geq 1} \frac{P(ks)}{k}.$$

Proof. The absolutely convergent expansion $-\log(1 - z) = \sum_{k \geq 1} z^k / k$, applied to each $z = p^{-s}$, gives

$$\log Z(s) = \sum_p \sum_{k \geq 1} \frac{p^{-ks}}{k}.$$

Absolute convergence permits interchange of the sums. □

The two indices retain different functions: p identifies the irreducible mode and k its occupation multiplicity. The relationship between them is not reduced to a list of constants: it produces a family of moments on the same set of irreducibles.

42.4.4 Meissel–Mertens and removal of the linear contribution

The coordinated extension brings together the finite part

$$B_1 = \gamma + \sum_p \left(\log(1 - p^{-1}) + p^{-1} \right).$$

The added term p^{-1} cancels exactly the logarithm's first order. Thus,

$$B_1 = \gamma - \sum_{k \geq 2} \frac{P(k)}{k}.$$

Here γ denotes the finite part γ_0 of the same functional, whose construction and bound are developed in Section 44.7; it is not a metrological input or a decimal chosen for evaluation. The formula relates two operations on the constructed functional: extraction of the finite part and cancellation of each mode's first moment.

Proposition 42.12. Bound on the moment tail

For an integer $K \geq 1$,

$$0 < \sum_{k > K} \frac{P(k)}{k} \leq \frac{2P(K+1)}{K+1} \leq \frac{2^{-K}}{K+1} \left(1 + \frac{2}{K} \right).$$

Proof. Since $k \geq K + 1$,

$$\sum_{k > K} \frac{P(k)}{k} \leq \frac{1}{K+1} \sum_p \frac{p^{-(K+1)}}{1 - p^{-1}} \leq \frac{2P(K+1)}{K+1}.$$

By comparison with all integers at least two,

$$P(K+1) \leq \sum_{n \geq 2} n^{-(K+1)} \leq 2^{-(K+1)} + \int_2^\infty x^{-(K+1)} \, dx.$$

Evaluating the integral gives the second bound. □

The inequality converts the finite sum into an interval, provided the finite evaluations and γ themselves have controlled bounds. The length of a decimal expansion does not replace this estimate.

42.4.5 Residual exclusion and the separation-two pattern product

For the two-position pattern $\{0, 2\}$, let ν_p be the number of forbidden classes modulo p. At $p = 2$ the two positions coincide; for $p > 2$, they are two distinct classes. Local normalization relative to two independent choices produces

$$\sigma_p = \frac{1 - \nu_p / p}{(1 - 1/p)^2}.$$

The factor at $p = 2$ equals two, while the others form

$$C_2 = \prod_{p > 2} \frac{1 - 2/p}{(1 - 1/p)^2} = \prod_{p > 2} \left(1 - \frac{1}{(p-1)^2} \right).$$

The pattern's complete singular series is $2C_2$. It is useful to reserve different symbols for the product over $p > 2$ and for the full factor: normalization distinguishes a specific residual contribution, not a dispensable convention.

Proposition 42.13. The same moment system with a different weight

The product expansion satisfies

$$\log C_2 = -\sum_{k\geq 2} \frac{2^k - 2}{k}\big(P(k) - 2^{-k}\big).$$

Proof. For each $p > 2$,

$$\log(1 - 2/p) - 2\log(1 - 1/p) = -\sum_{k\geq 2} \frac{2^k - 2}{k p^k}.$$

The sum begins at $k = 2$, because the linear term cancels. For $p \geq 3$,

$$-\log\left(1 - \frac{1}{(p-1)^2}\right) \leq \frac{4}{3(p-1)^2}.$$

Comparison with the inverse-square series proves finiteness of the absolute-value sum and permits interchange of order. Removing mode 2 produces $P(k) - 2^{-k}$. □

The same hierarchy $P(k)$ thus describes two different operations: Meissel–Mertens linear cancellation and residual exclusion of the two-position pattern. The difference lies in the weight of each moment and removal of mode 2, not in introducing two independent prime catalogues.

For an integer $K \geq 1$, let

$$S_K = \sum_{k=2}^{K} \frac{2^k - 2}{k}\big(P(k) - 2^{-k}\big),$$

and the coordinated extension gives the bound

$$0 < -\log C_2 - S_K \leq \frac{3}{K+1}\left(\frac{2}{3}\right)^{K+1}\left(1 + \frac{3}{K}\right).$$

It is obtained by substituting $2^k - 2 \leq 2^k$, summing the geometric series for each $p \geq 3$, bounding $(1 - 2/p)^{-1} \leq 3$ and comparing the sum over primes with the sum over integers $n \geq 3$. Exponentiation gives an interval for C_2. Convergence of this product is a result distinct from existence of infinitely many prime pairs separated by two.

42.4.6 Triadic realization of the functional

The source for spectral functionals uses cycles $C_m = \mathbb{Z}/3^m\mathbb{Z}$ and their Laplacian

$$(\Delta_m f)(j) = 2f(j) - f(j+1) - f(j-1).$$

Writing $N_m = 3^m$, let $e_{m,n}$ be the normalized characters of the cycle and $\widetilde{n}$ the representative of n between $-(N_m - 1)/2$ and $(N_m - 1)/2$. The signed-orientation operator is defined by

$$D_m e_{m,n} = \frac{N_m}{\sqrt{\pi}} \sin\left(\frac{\pi\widetilde{n}}{N_m}\right) e_{m,n}.$$

This fixes the sign, not merely the square. The realization satisfies

$$D_m^2 = \frac{3^{2m}}{4\pi}\Delta_m.$$

Let P_0 be the orthogonal projector onto constant functions on the cycle. The coordinate π belongs to the prior generation of the HMT kernel. This normalization is part of the analytic realization and must preserve its provenance. The positive trace removes the constant mode and counts one contribution per pair of opposite orientations:

$$\mathrm{Tr}_m^+(F(D_m)) = \frac{1}{2}\,\mathrm{Tr}\big(F(D_m)P_0^\perp\big).$$

With the cycle's Fourier modes, the heat trace converges to

$$\Theta_+(x) = \sum_{n\geq 1} e^{-\pi n^2 x}, \qquad x > 0.$$

The proof, developed in the first section of the following chapter, preserves a Gaussian majorant independent of refinement level. The Mellin transform of this limit satisfies exactly

$$\int_0^\infty \Theta_+(x)x^{s/2-1}\,dx = \pi^{-s/2}\Gamma(s/2)Z(s), \qquad \Re s > 1.$$

This brings together additive construction of the integer spectrum and multiplicative decomposition into irreducibles. The identity between functionals is proved; it is not inferred from proximity of two numerical evaluations.

Theta inversion and continuation of the completed function are proved in the next chapter. Finite parts, Laurent coefficients and residual classes modulo three require their specific evaluations; they do not follow merely from heat-trace convergence.

Transport defect and uniform component

The extension on sheet folding provides an explicit representation of a one-winding defect. Let $N = 9^m$, with $m \geq 1$, and let T be a unitary operator on a Hilbert fibre $\mathcal{H}$. On $\mathbb{C}^N$, use the usual Euclidean inner product and the unit vector

$$u_N = N^{-1/2}(1,\dots,1).$$

Define $\iota_N f = u_N \otimes f$, $P_N = \iota_N \iota_N^*$ and $Q_N = I - P_N$. Transport applies T once upon completion of the cycle:

$$U_N(T) = \sum_{r=1}^{N-1} |r+1\rangle\langle r| \otimes I + |1\rangle\langle N| \otimes T.$$

It acts on the uniform state by

$$U_N(T)\iota_N f = \iota_N f + \frac{e_1}{\sqrt{N}} \otimes (T-I)f.$$

The added term has a uniform component and an orthogonal component. The latter, with the source's normalization, is

$$\delta_N(T) = \frac{N}{\sqrt{N-1}} Q_N U_N(T)\iota_N = \eta_N \otimes (T-I),$$

where

$$\eta_N = \sqrt{\frac{N}{N-1}}\left(e_1 - \frac{u_N}{\sqrt{N}}\right), \qquad \|\eta_N\| = 1, \qquad \eta_N \perp u_N.$$

The equality is checked by substituting the preceding action and applying Q_N. For two unitary transports T, S on the same fibre, one obtains

$$\delta_N(T)^*\delta_N(S) = (I-T)^*(I-S).$$

Commutativity is not required: the products occur in the indicated order. The same decomposition gives

$$\delta_N(T)^* P_N \delta_N(S) = 0,$$

since P_N annihilates the factor η_N.

These identities specify the information preserved by defect normalization and its position relative to the uniform component. They also allow the closures of the images to be transported isometrically between two cycle lengths: the Gram operator is independent of N. Its domain is the constructed defect sector; it is not by itself an identification of all TPK histories or of any subsequent spectral projection.

42.4.7 The completed form and its boundary identity

Passage to the double circle uses a form on a test-function domain. Its arithmetic side brings together the prime-power, gamma and polar contributions. Writing it with two columns,

$$\mathcal{J}_{\mathrm{GW}} = J_+^* J_+ - J_-^* J_-,$$

exhibits a difference of forms. The prime-index producer, orbital weights, Fourier normalization and completion must appear before it is identified with another form.

The source in Chapter 96 places the central step in an encoder Ξ and an orthogonal projection P, through

$$\langle \Xi f, \Xi g\rangle = \langle J_+ f, J_+ g\rangle, \qquad \langle P\Xi f, P\Xi g\rangle = \langle J_- f, J_- g\rangle.$$

If these two identities are obtained from the declared native actions, their difference gives exactly

$$\mathcal{J}_{\mathrm{GW}}(f,g) = \langle (I-P)\Xi f, (I-P)\Xi g\rangle.$$

The following chapter proves the explicit decomposition by columns and the consequence of the two moments. Application to the native encoder retains as an antecedent its action on the test functions, the position of its charts relative to P and the proof of both moments. The orthogonality of P explains the last equality, but does not by itself determine the two preceding identities.

Transport telescoping, in turn, preserves the form

$$\|x\|^2 = \sum_{r=0}^{N-1} \|DA^r x\|^2 + \|A^N x\|^2,$$

when $D^*D = I - A^*A$. If on the sheet considered $A = qV$, with $q = 5/9$ and V isometric, the final term equals $q^{2N}\|x\|^2$ and tends to zero. The proof of this extinction is preserved together with the identity placing state x on that sheet.

42.4.8 Spectral circle and memory circle

The first circle is a coordinate of the spectral line. For $s = 1/2 + i\gamma$, with $\gamma \in \mathbb{R}$, the transformation

$$\tau(s) = \frac{s-1}{s} = \frac{\gamma + i/2}{\gamma - i/2}$$

belongs to the unit circle without the point 1 and is invertible through

$$s = \frac{1}{1-\tau}, \qquad \gamma = \frac{1}{2}\cot\frac{\theta}{2}, \quad \tau = e^{i\theta}.$$

The second circle is the character family of integer memory:

$$\chi_\tau(n) = \tau^n, \qquad n \in \mathbb{Z}.$$

The nonadic connection increments memory by one upon completion of a winding. Intertwining relates a spectral phase to its action on that memory; it does not identify zero number j with winding number j.

If $C \in \mathcal{B}(\mathcal{H})$ is the phase operator obtained in the spectral realization, the nine-position calendar acts on $\mathbb{C}^9 \otimes \mathcal{H}$ by

$$S_{C,9} = \sum_{r=1}^{8} |r+1\rangle\langle r| \otimes I + |1\rangle\langle 9| \otimes C.$$

Each vector traverses the link containing C exactly once in a complete winding. Thus,

$$S_{C,9}^9 = I_9 \otimes C.$$

The identity is exact for any C; when C is unitary, $S_{C,9}$ is also unitary. The contractive factor $5/9$ belongs to another transport readout and is distinguished from this unitary phase.

The chapter's interest does not lie in representing an arbitrary matrix on nine positions. It lies in bringing together the origin of operator C, its domain, the preceding encoder and compatibility with TPK cylinders. Those data fix what is transported and why this realization is relevant.

42.4.9 Compatibility with the joint structure of the continuum

The history space, two sheets, memory, incidence and complex are not reconstructed as independent objects for each section. The prior correlative assembly is developed in Section 6.5, with its operations, realization conditions and naturality squares. The integer, depth, irreducible and spectral-height operators remain typed; coincident notation does not identify them.

Section 97.8 brings together the boundary identity with its terminal construction, test functions, terms of the completed form and convergence. Matrix approximations preserve their three matrices of form, moment and residual. Digits are compared only after outputs have been produced and frozen.

This organization preserves the requested argument: construction, operation, resulting structure and recognition. The manuscript does not change the causal kernel to facilitate a subsequent calculation, nor use a list of zeros as input to the arithmetic producer.

Chapter 43

Bendersky–Glaisher–Kinkelin family: regular products, derivatives of the zeta function and normalization hierarchies

43.1 Generation of the Bendersky–Glaisher–Kinkelin family via zeta regularization

Let $\mathcal{Z}_3(s) := \zeta(s)$ be the scalar realization of the triadic zeta functional in the normalization of this volume, $H_k = \sum_{j=1}^{k} j^{-1}$ (with $H_0 = 0$), and let B_{k+1} be the Bernoulli number under convention $B_1 = -1/2$. We define

$$A_k = \exp\!\left(\frac{H_k B_{k+1}}{k+1} - \mathcal{Z}_3'(-k)\right), \qquad k \geq 0. \tag{43.1}$$

The first levels include

$$A_0 = \sqrt{2\pi},$$

and the Glaisher–Kinkelin constant in A_1. For $n \geq 1$,

$$\zeta(2n+1) = (-1)^{n+1} \frac{2(2\pi)^{2n}}{(2n)!} \log A_{2n}. \tag{43.2}$$

Apéry's constant, $\zeta(3)$, is the first nontrivial even level. Its presence in the photon gas of chapter 40 constitutes a subsequent physical realization of this spectral value.

The first part of the tower can be seen in a synoptic form:

k	identidad estructural	terminal value
0	$A_0 = \exp[-\zeta'(0)] = \sqrt{2\pi}$	2.5066282746310005024 …
1	$\log A_1 = 1/12 - \zeta'(-1)$	1.2824271291006226369 …
2	$\log A_2 = -\zeta'(-2) = \zeta(3)/(4\pi^2)$	1.0309167521973921 …
3	$\log A_3 = -11/720 - \zeta'(-3)$	0.9795555269428446 …
4	$\log A_4 = -\zeta'(-4)$	0.9920479745250403 …

Relation to the odd values. The functional equation of ζ, developed in $s = -2n$, gives

$$\zeta'(-2n) = (-1)^n \frac{(2n)!}{2(2\pi)^{2n}} \zeta(2n+1). \tag{43.3}$$

Since $B_{2n+1} = 0$ for $n \geq 1$, the definition equation (43.1) reduces $\log A_{2n} = -\zeta'(-2n)$. Solving for $\zeta(2n+1)$ proves equation (43.2). The index $2n$ preserves the derivation level: it is not an editorial label. □

Chapter 44

Gauss dynamics and the Khinchin, Lévy and Lochs constants: cofinality of cylinders, transfer operator and Gauss–Kuzmin–Wirsing spectrum

44.1 Cofinality between HMT cylinders and continued fraction cylinders

An irrational HMT section determines a nested family of compact cylinders. Its Archimedean evaluation is a point $x \in (0,1) \setminus \mathbb{Q}$. The continued fraction of x determines classical cylinders

$$I(a_1, \dots, a_n).$$

Proposition 44.1 (Cofinality)**.** *For every finite cylinder $I(a_1, \dots, a_n)$ containing the evaluation x, there is an HMT depth $m(n)$ such that the HMT cylinder at level $m \geq m(n)$ is contained in $I(a_1, \dots, a_n)$.*

Proof. Given n, the continued-fraction cylinder $I(a_1, \dots, a_n)$ is an open neighborhood of the irrational x. There then exists $\varepsilon_n > 0$ with $(x - \varepsilon_n, x + \varepsilon_n) \subset I(a_1, \dots, a_n)$. The HMT cylinders C_m contain x, are nested, and satisfy $\operatorname{diam} C_m \to 0$. Choose $m(n)$ with $\operatorname{diam} C_{m(n)} < \varepsilon_n$. Since $x \in C_{m(n)}$, every point of that cylinder lies from x at a distance of less than ε_n, hence $C_{m(n)} \subset I(a_1, \dots, a_n)$. Nesting gives the inclusion for every subsequent level. □

The proof uses compactness, nesting, and uniqueness of evaluation. The result establishes the transport of approximations, but does not imply conjugation of the TPK memory shift with the Gauss map on all deep fibers.

The complete genealogy of this realization is

$$\mathrm{APP} \to \widehat{\mathbb{T}} \to \mathrm{TPK} \to (C_m, \rho_{m+1,m})_{m \geq 0} \to x = \bigcap_m C_m \to \big(I(a_1, \dots, a_n)\big)_{n \geq 1}.$$

The HMT cylinders preserve route, phase, and carry; the terminal map projects these coordinates onto continued-fraction prefixes. Cofinality proves that this projection converges to the same point. A dynamical intertwiner would also require a lift $\widetilde{T}_G$ with commutative squares over the valuation fibers, a property not deduced from cofinality.

44.2 Gauss measure and Khinchin and Levy observables

The Gauss map [71, 67]

$$T_G(x) = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor$$

preserve

$$d\mu_G(x) = \frac{dx}{\log 2(1+x)}. \tag{44.1}$$

Khinchin's constant is defined by

$$\log K = \int_0^1 \log a_1(x)\, d\mu_G(x). \tag{44.2}$$

The distribution of the first digit is computed directly:

$$\mu_G(a_1 = k) = \frac{1}{\log 2} \int_{1/(k+1)}^{1/k} \frac{dx}{1+x} = \log_2\left(\frac{(k+1)^2}{k(k+2)}\right). \tag{44.3}$$

By the ergodic theorem,

$$K = \prod_{k\geq 1} k^{\log_2((k+1)^2/[k(k+2)])} = 2.685452001065306445309714835\ldots \tag{44.4}$$

for μ_G-almost every section. The product is not applied to each point: the section φ^{-1}, for example, has all its digits equal to one and is an immediate falsification criterion for a pointwise reading. A certification of equation (44.4) truncates the sum of logarithms and bounds the tail by the estimate

$$\mu_G(a_1 = k) = O(k^{-2}), \qquad \sum_{k>N} \mu_G(a_1 = k)\log k = O\left(\frac{\log N + 1}{N}\right).$$

Lévy's constant is

$$L = -\int_0^1 \log x \, d\mu_G(x) = \frac{\pi^2}{12\log 2}. \tag{44.5}$$

Indeed,

$$\frac{1}{1+x} = \sum_{n\geq 0}(-1)^n x^n, \qquad \int_0^1 (-\log x)x^n\,dx = \frac{1}{(n+1)^2},$$

so that the integral is $\eta(2)/\log 2 = \pi^2/(12\log 2)$.

Khinchin's constant determines the logarithmic mean of the partial quotients, whereas Lévy's constant determines the exponential growth rate of the denominators. They are distinct observables of the same measured system.

44.3 Entropy (metric), Lyapunov exponent, and Lochs's constant

Since

$$\log|T_G'(x)| = -2\log x,$$

the Lyapunov exponent and the entropy metric are

$$\chi_G = h_G = 2L = \frac{\pi^2}{6\log 2}. \tag{44.6}$$

The Lochs ratio between continued fractions and a base-b expansion is

$$\Lambda_b = \frac{\log b}{h_G} = \frac{6\log 2\log b}{\pi^2}. \tag{44.7}$$

In particular,

$$\Lambda_{10} = 0.970270114392039257402560192\ldots,$$
$$\Lambda_{729} = \frac{36\log 2\log 3}{\pi^2} = 2.777618966374305777705782976\ldots.$$

The basis $729 = 3^6$ is relevant to the ambient space $\mathbb{F}_3^6$, but equation (44.7) remains an almost-sure theorem of μ_G, not a pointwise equality for every HMT route.

The value Λ_{729} represents the asymptotic typical rate of partial quotients obtained per complete block of six trits. It does not identify the ambient set of 729 words with the other objects of cardinality 729 in the corpus. The equality is an information rate between partitions, not a bijection between states.

44.4 Gauss–Kuzmin–Wirsing Transfer Operator

The Gauss–Kuzmin–Wirsing operator acts by

$$(\mathcal{L}f)(x) = \sum_{n\geq 1} \frac{1}{(x+n)^2} f\left(\frac{1}{x+n}\right). \tag{44.8}$$

Its dominant eigenvalue corresponds to the invariant density. The GKW constant is the modulus of the subdominant eigenvalue on the zero-mean subspace. An HMT-compatible scheme uses partitions of 9^m cylinders, transfer matrices with intervals, and a tail bound.

To promote the calculation, the following must be certified: isolation of the eigenvalue, invariance of the subspace, truncation error, convergence of spectral projectors, and absence of another eigenvalue of equal modulus. The known decimal value cannot be introduced as the center of the initial interval.

A joint deformation contains Khinchin and Lévy in a single object:

$$(\mathcal{L}_{s,t}f)(x) = \sum_{n\geq 1} n^t (n+x)^{-2s} f\left(\frac{1}{n+x}\right), \qquad P(s,t) = \log r(\mathcal{L}_{s,t}). \tag{44.9}$$

At the point $(s,t) = (1,0)$, the spectral perturbation gives

$$\partial_t P(1,0) = \log K, \qquad -\partial_s P(1,0) = \chi_G = 2L. \tag{44.10}$$

The Hessian of P determines the asymptotic variances and covariances of $\log a_1$ and $-2\log x$. Therefore, the Khinchin and Levy constants are two transverse derivatives of the same pressure function.

The reconocimiento numerico of the autovalor subdominante is

$$\lambda_2 \simeq -0.3036630028987326586, \qquad \kappa_{\mathrm{GKW}} = |\lambda_2|, \qquad -\log \kappa_{\mathrm{GKW}} \simeq 1.19183673556055.$$

In this volume, those decimals constitute reference values for comparison and not a spectral certificate. The proposed HMT proof uses Galerkin or Ulam on the cofinal cylinders of depth 9^m, a Lasota–Yorke inequality, interval arithmetic to isolate λ_2, and a rigorous bound on the tail. An interval containing two eigenvalues of the same modulus refutes the promotion, although it does not affect Khinchin, Levy, or Lochs.

44.5 Cofinal certificate of the Gauss–Kuzmin–Wirsing spectrum over the nonadic hierarchy

For Gauss–Kuzmin–Wirsing, let P_N be the certified family of truncations on the declared function space, and define

$$\Pi_m = P_{9^m}.$$

The hierarchy (Π_m) is a cofinal subsequence; the uniform bounds, projector convergence, and absence of spectral contamination are inherited from the family (P_N). A different cylindrical discretization must provide its intertwiner with (P_{9^m}).

44.6 Analytic construction of the functionals of the triadic tower

The preceding synthesis has placed the spectral constants within the triadic tower. This block contains their primary analytic construction: a sequence of self-adjoint operators on cycles of order 3^m, a limit of heat traces and a Mellin transform. The classical constants appear as evaluations, finite parts or derivatives of the resulting meromorphic functional, not as prefixes recognized in a catalogue.

The normalization of the operator D_m explicitly contains π in the scale relating the discrete Laplacian to the Gaussian heat kernel. The subsection therefore constitutes an analytic extension of the Archimedean branch already constructed. The subsequent spectral constants are obtained simultaneously from the resulting functional, with that normalization as a starting hypothesis.

44.6.1 Inductive limits and refinement scale

The spectral extension requires distinguishing two analytic levels. The first corresponds to bounded compatible families; the second to unbounded operators, for which algebraic intertwining does not by itself resolve domain problems.

Theorem 44.2 (Bounded inductive limit)**.** *Let $J_m : \mathcal{H}_m \to \mathcal{H}_{m+1}$ be isometries and $A_m \in \mathcal{B}(\mathcal{H}_m)$ self-adjoint operators such that*

$$A_{m+1}J_m = J_m A_m, \qquad \sup_m \|A_m\| < \infty.$$

There exists a unique bounded self-adjoint operator A on $\mathcal{H}_\infty = \varinjlim(\mathcal{H}_m, J_m)$ whose restriction to each level is A_m.

Proof. On the dense algebraic union we define

$$A[J_{m,\infty}\xi] := J_{m,\infty}A_m\xi.$$

Intertwining makes the definition independent of the representative. The uniform bound allows A to be extended by continuity to $\mathcal{H}_\infty$; symmetry on the dense subspace and boundedness imply its self-adjointness. □

Remark 44.3 (Control for unbounded operators). If the A_m are unbounded, the identity $A_{m+1}J_m = J_m A_m$ only defines an action on a union of domains. To obtain a closed or self-adjoint operator, the transported domains must be declared, closability proved and deficiency indices controlled, or essential self-adjointness proved on a compatible common core. Omitting these obligations constitutes a type error, not an abbreviation of the proof.

Now let $\mathcal{S}_\infty$ be a nonempty set of compatible surviving sections and let $\mathcal{P}_m$ be the finite set of their prefixes of length m. We write

$$\mathcal{H}_m := \ell^2(\mathcal{P}_m), \qquad E_m(\gamma,\gamma') = \begin{cases} 1, & \gamma' \text{ extends } \gamma, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$d_m(\gamma) := \#\{\gamma' \in \mathcal{P}_{m+1} : \gamma' \mapsto \gamma\}, \qquad D_m^{\deg} := E_m E_m^* = \operatorname{diag}(d_m(\gamma)).$$

We restrict ourselves to the live subspace $d_m(\gamma) > 0$, and each prefix of level $m+1$ has a unique parent.

Theorem 44.4 (Normalized extension isometry)**.** *The operator*

$$J_m := E_m^*(D_m^{\deg})^{-1/2} : \mathcal{H}_m \longrightarrow \mathcal{H}_{m+1}$$

is an isometry, and

$$J_m e_\gamma = \frac{1}{\sqrt{d_m(\gamma)}} \sum_{\gamma' \mapsto \gamma} e_{\gamma'}.$$

Proof. Uniqueness of the parent gives

$$J_m^* J_m = (D_m^{\deg})^{-1/2} E_m E_m^* (D_m^{\deg})^{-1/2} = I.$$

The formula on e_γ is obtained by expanding the corresponding column of E_m^*. □

Setting

$$\mathcal{K}_m := \mathcal{H}_{m+1} \ominus J_m \mathcal{H}_m, \qquad \mathcal{H}_{\mathrm{surv}} \cong \mathcal{H}_0 \oplus \bigoplus_{m \geq 0} \mathcal{K}_m,$$

the refinement operator is defined

$$\mathcal{R}|_{\mathcal{H}_0} = 0, \qquad \mathcal{R}|_{\mathcal{K}_m} = 3^{m+1} I_{\mathcal{K}_m},$$

with maximal domain

$$\mathrm{Dom}(\mathcal{R}) = \mathcal{H}_0 \oplus \left\{ (\xi_m)_{m \geq 0} : \sum_{m \geq 0} 3^{2m+2} \|\xi_m\|^2 < \infty \right\}.$$

Theorem 44.5 (Refinement operator)**.** *If each $\mathcal{K}_m$ is finite-dimensional, $\mathcal{R}$ is positive, self-adjoint and has compact resolvent. Its spectrum measures refinement depth; it is not a spectrum of primes or zeros.*

Proof. The orthogonal decomposition reduces $\mathcal{R}$ to real scalar blocks on its maximal domain, which proves positivity and self-adjointness. The eigenvalues 3^{m+1} escape to infinity and their multiplicities are finite; therefore the resolvent is compact. An infinite-dimensional block with bounded eigenvalue would falsify this last conclusion. □

44.6.2 Integer, prime and orbital operators

Let $\mathscr{W}_{\mathrm{bal}}$ be the set of balanced ternary words reduced by theorem theorem 53.3. On $\ell^2(\mathscr{W}_{\mathrm{bal}})$, the operator

$$B e_w = b(w) e_w$$

is taken on its maximal domain. Its restriction to the subspace of positive integers is denoted by Q.

Theorem 44.6 (Integer and positive operators)**.** *The operator B is self-adjoint, has simple spectrum $\mathbb{Z}$ and compact resolvent. The operator Q has simple spectrum $\mathbb{N}_{\geq 1}$ and compact resolvent.*

Proof. The bijection of theorem 53.3 enumerates each integer exactly once. A real diagonal operator on its maximal domain is self-adjoint; since its entries escape to infinity outside finite sets, its resolvent is compact. The positive restriction preserves these properties. □

The omitted classical proof of theta inversion, continuation and the functional equation is merged only once with the triadic realization already constructed in theorem 44.20. There, $\Theta_B = 1 + 2\Theta_+$ and $Z = \mathcal{Z}_3$ are fixed literally; therefore this block does not define a second origin of ζ.

Let $\mathbb{P}$ be the set of ordinary primes, $\mathfrak{h}_{\mathbb{P}} := \ell^2(\mathbb{P})$ and

$$H_{\mathbb{P}} e_p = (\log p) e_p.$$

In the bosonic Fock space $\Gamma_s(\mathfrak{h}_{\mathbb{P}})$, the occupations $(k_p)_p$ have finite support.

Theorem 44.7 (Fock representation of factorization)**.** *The map on bases*

$$U_F|(k_p)_p\rangle = e_{\prod_p p^{k_p}}$$

extends to a unitary operator

$$U_F : \Gamma_s(\mathfrak{h}_{\mathbb{P}}) \longrightarrow \ell^2(\mathbb{N}_{\geq 1})$$

and satisfies

$$U_F\, d\Gamma(H_{\mathbb{P}})\, U_F^* = \log Q.$$

Proof. Unique factorization gives a bijection between finite occupations and positive integers, hence a unitary identification of the orthonormal bases. Moreover,

$$\sum_p k_p \log p = \log\left(\prod_p p^{k_p}\right),$$

which proves the intertwining. The explicit starting structure is the choice of prime modes and symmetric Fock space. □

On

$$\mathcal{H}_{\mathrm{orb}} := \ell^2\{(p,k,\varepsilon) : p \in \mathbb{P},\ k \geq 1,\ \varepsilon = \pm 1\},$$

we define

$$D_{\mathrm{orb}} e_{p,k,\varepsilon} = \varepsilon k \log p\, e_{p,k,\varepsilon}, \qquad W_{\log} e_{p,k,\varepsilon} = (\log p) e_{p,k,\varepsilon}.$$

Proposition 44.8 (Orbital trace)**.** *The operator D_{orb} is self-adjoint and has compact resolvent. If $f \in C_c(\mathbb{R})$ is bounded, then*

$$\mathrm{Tr}\Big(W_{\log} e^{-|D_{\mathrm{orb}}|/2} f(D_{\mathrm{orb}})\Big) = \sum_{p,k} (\log p) p^{-k/2} \big[f(k \log p) + f(-k \log p)\big].$$

Proof. The diagonal is real and its lengths escape every compact set with finite multiplicity. The compact support of f makes the regulated product finite-rank; its trace is the sum of the diagonal entries. The index p here ranges over ordinary primes. The result realizes the classical prime side, but does not identify D_{orb} with an operator dual to the zeros. □

44.6.3 Completed Guinand–Weil form

We fix

$$\mathcal{D}_{\mathrm{W}} := C_c^\infty(\mathbb{R}), \qquad \widehat{\psi}(t) = \int_{\mathbb{R}} \psi(x) e^{-itx}\, \mathrm{d}x, \qquad \psi(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\psi}(t) e^{itx}\, \mathrm{d}t,$$

and, for $\psi, \phi \in \mathcal{D}_{\mathrm{W}}$,

$$h_{\psi,\phi}(z) := \widehat{\psi}(z)\, \overline{\widehat{\phi}(\overline{z})}.$$

On the real line $h_{\psi,\psi} = |\widehat{\psi}|^2$.

Definition 44.9 (Test class)**.** The test class is $\mathcal{D}_{\mathrm{W}}$, stable under differentiation, reflection and translation. The inverse used is

$$\check{h}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} h(t) e^{itx}\, \mathrm{d}t.$$

Smoothness and compact support provide rapid decay in strips and compact support for $\check{h}$, conditions ensuring the convergence of the following blocks.

Sea $\psi_\Gamma = \Gamma'/\Gamma$. Definimos

$$\mathfrak{I}_\# := \left\{ \mathrm{ev}_9(w) : w \in \mathscr{W}_9^+,\ P_{\mathfrak{I},\#} e_w = e_w \right\}.$$

By theorem 27.21, this image coincides with the ordinary primes; here it is used only to type the index of the orbital block, without repeating that proof.

$$\mathcal{P}(h) := h(i/2) + h(-i/2), \tag{44.11}$$

$$\mathcal{G}(h) := -\log \pi\, \check{h}(0) + \frac{1}{2\pi} \int_{\mathbb{R}} h(t)\, \Re \psi_\Gamma\left(\frac{1}{4} + \frac{it}{2}\right) \mathrm{d}t, \tag{44.12}$$

$$\mathcal{O}_\#(\check{h}) := \sum_{p \in \mathfrak{I}_\#} \sum_{k \geq 1} (\log p) p^{-k/2} \left[\check{h}(k \log p) + \check{h}(-k \log p) \right]. \tag{44.13}$$

Definition 44.10 (Completed functional)**.** For $h = h_{\psi,\phi}$, we fix

$$\mathcal{W}(h) := \mathcal{P}(h) + \mathcal{G}(h) - \mathcal{O}_\#(\check{h}).$$

The orbital block comes from the irreducible selector of the chapter 27; the polar and gamma blocks belong to the classical completion of zeta. This difference in provenance is preserved in the notation and the proof.

Theorem 44.11 (Guinand–Weil explicit formula)**.** *For $\psi, \phi \in C_c^\infty(\mathbb{R})$ and $h = h_{\psi,\phi}$,*

$$\lim_{T \to \infty} \sum_{\substack{\rho:\, \zeta(\rho) = 0 \\ |\Im \rho| \leq T}} h\left(\frac{\rho - 1/2}{i}\right) = \mathcal{W}(h),$$

where the nontrivial zeros are counted with multiplicity and the limit is taken with classical symmetrization.

Proof. Paley–Wiener places $h_{\psi,\phi}$ in the admissible class of the explicit formula. The poles of $\Lambda_\zeta(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ produce $\mathcal{P}$, the Archimedean factor produces $\mathcal{G}$ and the Euler product produces the sum over prime powers. The identification of this last sum with $\mathcal{O}_\#$ is made only through proposition 44.12; thus the proof of native primality is not duplicated. □

Proposition 44.12 (Reindexing of the orbital block)**.** *If the irreducibles of the native product coincide exactly with the ordinary primes and the weights $(\log p) p^{-k/2}$ and the two orientations $\pm k \log p$ are preserved, then*

$$\mathcal{O}_\#(\check{h}) = \mathcal{O}(\check{h}).$$

Proof. The bijection between native irreducibles and primes reindexes the sum without modifying indices, weights or arguments. A nonprime irreducible, a missing prime or a change in weight would falsify the identity. □

Remark 44.13 (Meromorphic–entire separation). The function Λ_ζ has simple poles at 0 and 1, which give rise to the polar block. In contrast,

$$\xi(s) = \frac{1}{2} s(s-1) \Lambda_\zeta(s)$$

is entire: the factors $s(s-1)$ cancel precisely those poles. Therefore no poles at 0 or 1 are attributed to ξ.

Definition 44.14 (Hermitian form and preoperator)**.** We define

$$\mathcal{B}_{\mathrm{W}}(\psi, \phi) := \mathcal{W}(h_{\psi,\phi}), \qquad \mathcal{V}_{\mathrm{W}} := \mathcal{D}_{\mathrm{W}} / \operatorname{Rad} \mathcal{B}_{\mathrm{W}},$$

and, on the algebraic quotient, the candidate rule

$$D_{\mathrm{W},0}[\psi] := [-i\psi'].$$

It is not assumed that $\mathcal{B}_{\mathrm{W}}$ is positive or that $\mathcal{V}_{\mathrm{W}}$ is already a Hilbert space. The rule becomes well-defined only after invariance of the radical has been proved.

Theorem 44.15 (Algebraic symmetry and invariances)**.** *The radical is invariant under $-i\ \mathrm{d}/\mathrm{d}x$; $D_{\mathrm{W},0}$ is well-defined and $\mathcal{B}_{\mathrm{W}}$-symmetric. The translations $T_a\psi(x) = \psi(x-a)$ preserve the form and the reflection $J\psi(x) = \psi(-x)$ satisfies*

$$JD_{\mathrm{W},0}J = -D_{\mathrm{W},0}.$$

Proof. The Fourier convention gives

$$h_{-i\psi',\phi}(z) = h_{\psi,-i\phi'}(z).$$

The identity holds before applying any of the three blocks of $\mathcal{W}$, so the derivative preserves the radical and the preoperator is symmetric. The opposite translation phases cancel in h; reflection changes the sign of the derivative and leaves the polar, gamma and orbital blocks invariant. This algebraic symmetry is not yet equivalent to self-adjointness in a positive Hilbert space. □

44.6.4 Local geometry of the critical line

Lemma 44.16 (Möbius coordinate of the critical line)**.** *For $s \in \mathbb{C} \setminus \{0,1\}$, let*

$$\tau(s) := i\frac{s-1}{s}.$$

With $F(s) = 1 - s$ and $C(s) = \overline{s}$,

$$\tau(Fs) = -\frac{1}{\tau(s)}, \qquad \tau(Cs) = -\overline{\tau(s)},$$

and

$$\tau(Fs) = \tau(Cs) \iff |\tau(s)| = 1 \iff \Re s = \frac{1}{2}.$$

Proof. The two identities are obtained by direct substitution. Their equality is equivalent to $\tau(s)^{-1} = \overline{\tau(s)}$, that is, $|\tau(s)| = 1$; expanding this last condition gives $\Re s = 1/2$. For the zeros of zeta, functional and conjugate symmetries are also required. The equivalence parametrizes the line; it does not by itself place the zeros on it. □

Lemma 44.17 (Local Hermite–Biehler sign)**.** *Let A be a real entire function with a simple real zero a. For $z = a + iy$, $y > 0$ sufficiently small,*

$$|A(z) - iA'(z)|^2 - |A(z) + iA'(z)|^2 = -4A'(a)^2 y + O(y^2) < 0.$$

The local convention associated with A is

$$E_A := A + iA'.$$

Only by specializing $A = \Xi$ and additionally assuming that a is a simple real zero of Ξ do we obtain

$$E_\Xi = \Xi + i\Xi'.$$

Proof. We expand A and A' in Taylor series around a, using $A(a) = 0$ and $A'(a) \neq 0$, and retain the first-order terms in y. The result is strictly local: it does not prove that E_A has the global Hermite–Biehler property. □

Proposition 44.18 (Obstruction by counting laws)**.** *Let*

$$N_A(T) := \operatorname{Tr} \mathbf{1}_{[-T,T]}(A).$$

The operators Q, B, D_{orb} and $\mathcal{R}$ do not, by themselves, have the counting law required of a possible operator dual to the zeros.

Proof. Para $T \geq 1$,

$$N_Q(T) = \lfloor T \rfloor, \qquad N_B(T) = 2\lfloor T \rfloor + 1.$$

The prime number theorem gives

$$N_{D_{\mathrm{orb}}}(T) = 2 \sum_{p \leq e^T} \left\lfloor \frac{T}{\log p} \right\rfloor \geq 2\pi(e^T) \sim \frac{2e^T}{T}.$$

Finally, $N_{\mathcal{R}}(T)$ is constant on each multiplicative interval $[3^m, 3^{m+1})$, since $\mathcal{R}$ only has levels 3^{m+1} with finite multiplicities.

For $\Xi(z) = \xi(1/2 + iz)$, the functional equation implies $\Xi(-z) = \Xi(z)$; the zeros $\pm\gamma$ have the same multiplicity and

$$N_{\mathrm{sym}}(T) = 2N_+(T) + O(1).$$

Consequently, the bilateral convention requires

$$N_{\mathrm{sym}}(T) \sim \frac{T}{\pi} \log T,$$

whereas the equivalent unilateral one, under this symmetry and pairing of multiplicities, requires

$$N_+(T) \sim \frac{T}{2\pi} \log T.$$

None of the four preceding functions has that asymptotic behavior. □

44.6.5 Laplacians on cycles of order 3^m

For $m \geq 1$, let

$$N_m = 3^m, \qquad C_m = \mathbb{Z}/N_m\mathbb{Z},$$

and consider the combinatorial Laplacian

$$(\Delta_m f)(j) = 2f(j) - f(j+1) - f(j-1), \qquad f \in \ell^2(C_m).$$

The characters

$$e_{m,n}(j) = N_m^{-1/2} \exp\left(\frac{2\pi i n j}{N_m}\right), \qquad n \in C_m,$$

form an orthonormal basis and satisfy

$$\Delta_m e_{m,n} = 4 \sin^2\left(\frac{\pi n}{N_m}\right) e_{m,n}.$$

Since N_m is odd, each class of C_m has a unique representative $\widetilde{n}$ with

$$-\frac{N_m-1}{2} \le \widetilde{n} \le \frac{N_m-1}{2}.$$

We define the self-adjoint operator D_m by

$$D_m e_{m,n} = \frac{N_m}{\sqrt{\pi}} \sin\!\left(\frac{\pi\widetilde{n}}{N_m}\right) e_{m,n}. \tag{44.14}$$

Then

$$D_m^2 = \frac{N_m^2}{4\pi}\,\Delta_m. \tag{44.15}$$

Let P_0 be the orthogonal projection onto the constant character. For every bounded even function F on the spectrum of D_m, we introduce the positive trace

$$\mathrm{Tr}_m^{+}\big(F(D_m)\big) := \frac{1}{2}\,\mathrm{Tr}\big(F(D_m)P_0^{\perp}\big)\,. \tag{44.16}$$

The factor $1/2$ selects one contribution from each spectral pair $\{n,-n\}$; removing P_0 excludes the zero eigenvalue.

Theorem 44.19 (Theta limit of the heat traces)**.** *For every $x>0$,*

$$\lim_{m\to\infty} \mathrm{Tr}_m^{+}\!\left(e^{-xD_m^2}\right) = \Theta_+(x) := \sum_{n\ge 1} e^{-\pi n^2 x}.$$

Proof. The bilateral trace, once the constant mode is removed, is

$$\sum_{\substack{n\in C_m\\ n\neq 0}} \exp\!\left[-\frac{N_m^2 x}{\pi}\sin^2\!\left(\frac{\pi\widetilde{n}}{N_m}\right)\right].$$

For each fixed integer n, the summand converges to $e^{-\pi n^2 x}$. Moreover, for $|\widetilde{n}| \le N_m/2$, the inequality $\sin y \ge 2y/\pi$, valid on $0 \le y \le \pi/2$, provides a summable Gaussian majorant independent of m. Dominated convergence applied to the discrete sum implies

$$\lim_{m\to\infty} \mathrm{Tr}\!\left(e^{-xD_m^2}P_0^{\perp}\right) = \sum_{n\in\mathbb{Z}\setminus\{0\}} e^{-\pi n^2 x}.$$

The division by two established in (44.16) preserves one contribution from each pair $\{n,-n\}$ and gives $\Theta_+(x)$. □

44.6.6 Theta–Mellin transform

For $\Re s > 1$, define

$$\mathcal{Z}_3(s) := \frac{\displaystyle\int_0^{\infty} \Theta_+(x)\,x^{s/2-1}\,\mathrm{d}x}{\pi^{-s/2}\Gamma(s/2)}. \tag{44.17}$$

Absolute convergence allows term-by-term integration. Since

$$\int_0^{\infty} e^{-\pi n^2 x} x^{s/2-1}\,\mathrm{d}x = \Gamma(s/2)(\pi n^2)^{-s/2},$$

we obtain

$$\mathcal{Z}_3(s) = \sum_{n\geq 1} n^{-s} = \zeta(s), \qquad \Re s > 1. \tag{44.18}$$

The equality extends meromorphically through continuation of the Riemann zeta function [4]. Henceforth, all evaluations come from (44.17), its derivatives or the residual decomposition of its Dirichlet series.

Theorem 44.20 (Theta inversion and functional equation)**.** *With the preceding operator notation,*

$$\Theta_B(x) := \operatorname{Tr}(e^{-\pi x B^2}) = 1 + 2\Theta_+(x), \qquad Z(s) := \operatorname{Tr}(Q^{-s}) = \mathcal{Z}_3(s) \quad (\Re s > 1).$$

Poisson summation and the Mellin transform produce

$$\Theta_B(x) = x^{-1/2}\Theta_B(1/x)$$

and the continuation of

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)Z(s), \qquad \xi(s) = \xi(1-s).$$

Proof. The equality $\Theta_B = 1 + 2\Theta_+$ separates the zero mode and pairs $\pm n$; the identity $Z = \mathcal{Z}_3$ is (44.18). Poisson summation applied to the Gaussian gives the inversion of Θ_B. In the half-plane of convergence, term-by-term integration provides

$$\pi^{-s/2}\Gamma(s/2)Z(s) = \int_0^\infty x^{s/2-1}\Theta_+(x)\,\mathrm{d}x.$$

Splitting the integral into $(0,1)$ and $(1,\infty)$, applying theta inversion to the first part and separating the two classical poles yields the continuation and the functional equation. This single incorporated classical proof does not locate the zeros of ζ. □

44.6.7 Residual decomposition modulo three

For $r \in \{0,1,2\}$, let

$$Z_r(s) = \sum_{\substack{n\geq 1\\ n\equiv r \pmod 3}} n^{-s}. \tag{44.19}$$

In terms of the Hurwitz zeta function,

$$Z_0(s) = 3^{-s}\zeta(s), \qquad Z_1(s) = 3^{-s}\zeta\Big(s, \frac{1}{3}\Big), \qquad Z_2(s) = 3^{-s}\zeta\Big(s, \frac{2}{3}\Big).$$

The three terms have residue $1/3$ at $s = 1$. We write their Laurent expansions in the form

$$Z_r(1+z) - \frac{1}{3z} = \sum_{n\geq 0} c_{r,n}\, z^n, \qquad \zeta(1+z) - \frac{1}{z} = \sum_{n\geq 0} a_n\, z^n. \tag{44.20}$$

Theorem 44.21 (Decomposition of the Laurent coefficients)**.** *For every $n \geq 0$,*

$$a_n = c_{0,n} + c_{1,n} + c_{2,n}, \qquad \gamma_n = (-1)^n n!\, a_n.$$

If χ_{-3} denotes the nonprincipal real character modulo three, then

$$\frac{L^{(n)}(1,\chi_{-3})}{n!} = c_{1,n} - c_{2,n}.$$

Furthermore, with $\lambda = \log 3$*,*

$$c_{0,n} = \frac{1}{3}\left[\frac{(-\lambda)^{n+1}}{(n+1)!} + \sum_{k=0}^{n} a_k \frac{(-\lambda)^{n-k}}{(n-k)!}\right].$$

These three identities uniquely determine $(c_{0,n}, c_{1,n}, c_{2,n})$ *from* $(a_n, L^{(n)}(1, \chi_{-3}))$ *and the aggregated coefficients of lower order.*

Proof. The partition of the positive integers into residue classes provides $\zeta = Z_0 + Z_1 + Z_2$. Subtracting the poles at $s = 1$ and comparing the coefficients of (44.20) yields the first identity. The second coincides with the usual convention for the Stieltjes constants. Moreover, $Z_1 - Z_2 = L(s, \chi_{-3})$; the poles of Z_1 and Z_2 cancel and comparison of coefficients gives the third relation. Finally,

$$Z_0(1+z) = \frac{1}{3} e^{-\lambda z} \zeta(1+z).$$

The product of the two Laurent series provides the explicit formula for $c_{0,n}$. The sum and difference then recover $c_{1,n}$ and $c_{2,n}$, proving the asserted uniqueness. □

The sum of the three terms recovers the aggregated sector; the difference of the classes 1 and 2 preserves the character's orientation; comparison with the class divisible by three records the change of scale. These three combinations are linearly independent and determine the three coefficients $c_{r,n}$.

44.6.8 Euler–Mascheroni and Stieltjes constants

Let

$$C_r := \operatorname*{FP}_{s=1} Z_r(s).$$

The classical identities for the digamma function give [4, 77]

$$C_0 = \frac{\gamma - \log 3}{3}, \tag{44.21}$$

$$C_1 = \frac{\gamma}{3} + \frac{\log 3}{6} + \frac{\pi}{6\sqrt{3}}, \qquad C_2 = \frac{\gamma}{3} + \frac{\log 3}{6} - \frac{\pi}{6\sqrt{3}}. \tag{44.22}$$

Corollary 44.22. *The finite parts satisfy*

$$\boxed{\gamma = C_0 + C_1 + C_2},$$

$$\boxed{L(1, \chi_{-3}) = C_1 - C_2 = \frac{\pi}{3\sqrt{3}}},$$

$$\boxed{\log 3 = C_1 + C_2 - 2C_0}.$$

Proof. At order zero, the theorem 44.21 gives $a_0 = c_{0,0} + c_{1,0} + c_{2,0}$. By definition, $a_0 = \gamma$ and $c_{r,0} = C_r$, proving the first equality. The difference of the two oriented classes is $C_1 - C_2 = L(1, \chi_{-3})$. Substituting (44.21)–(44.22) yields the other two identities. □

A single extraction of coefficients by Cauchy's integral formula applied to (44.20) simultaneously determines $\gamma_0, \ldots, \gamma_6$. The discretizations with 96 and 192 nodes agree up to order 10^{-87}; the identity of the coefficients, however, follows from the theorem 44.21 and not from that numerical comparison.

44.6.9 Evaluation of Apéry's constant

Proposition 44.23. *The functional* (44.17) *satisfies*

$$\boxed{\mathcal{Z}_3(3) = \zeta(3) = 1.20205690315959428539973816151144 9\ldots}.$$

Proof. The exact equality is the specialization $s = 3$ of (44.18). To certify any decimal prefix without modifying the functional, consider $S_N = \sum_{n=1}^{N} n^{-3}$. The integral test gives

$$S_N + \frac{1}{2(N+1)^2} < \zeta(3) < S_N + \frac{1}{2N^2}.$$

The resulting nested rational intervals determine the printed digits. □

The irrationality of $\zeta(3)$ is Apéry's classical theorem [3]; it does not follow merely from evaluation of the functional.

44.6.10 Glaisher–Kinkelin constant

The spectral derivative at $s = -1$ defines

$$\boxed{A_{\mathrm{GK}} = \exp\left(\frac{1}{12} - \mathcal{Z}_3'(-1)\right)}. \tag{44.23}$$

This is the formula for the regularized determinant associated with the hyperfactorial product [126]. A five-point central differentiation, evaluated at two scales, provides

$$A_{\mathrm{GK}} = 1.28242712910062263687534256886979 17\ldots.$$

The term $1/12$ belongs to the regularization of the hyperfactorial and is not introduced as a fitting parameter.

44.6.11 Euler–Kronecker constant of the Eisenstein field

The character χ_{-3} determines the imaginary quadratic field

$$F_{-3} = \mathbb{Q}(\sqrt{-3}), \qquad \zeta_{F_{-3}}(s) = \zeta(s)L(s, \chi_{-3}).$$

Its Euler–Kronecker constant is

$$\boxed{\gamma_{F_{-3}} = \gamma + \frac{L'(1, \chi_{-3})}{L(1, \chi_{-3})}}. \tag{44.24}$$

The residual coefficients of (44.20) give

$$L(1, \chi_{-3}) = 0.60459978807807261686469275254738 52\ldots,$$
$$L'(1, \chi_{-3}) = 0.22266298696860150948666026276474 43\ldots,$$

and, by (44.24),

$$\gamma_{F_{-3}} = 0.94549728087168070323974999415818 90\ldots.$$

The choice of F_{-3} comes from the residue character modulo three, not from a search over the decimal values of constants of quadratic fields.

44.6.12 Local product associated with twin primes

For a finite pattern $H = \{h_1, \dots, h_k\} \subset \mathbb{Z}$, we define

$$\nu_p(H) := |\{-h_1, \dots, -h_k\} \bmod p|, \qquad \sigma_p(H) = \frac{1 - \nu_p(H)/p}{(1 - 1/p)^k}.$$

For $H = \{0, 2\}$, the local factors lead to

$$C_2^{\text{twin}} = \prod_{p>2} \frac{p(p-2)}{(p-1)^2}. \tag{44.25}$$

Let P_X be the product restricted to primes $2 < p \le X$. For $p > X$, all factors belong to $(0, 1)$; extending the tail product from primes to all integers gives the telescoping bound

$$\prod_{n>X} \left(1 - \frac{1}{(n-1)^2}\right) = \frac{X-1}{X} \le \prod_{p>X} \left(1 - \frac{1}{(p-1)^2}\right) < 1.$$

The product $P_{2\times 10^6}$, evaluated with directed decimal rounding, and this rational inequality for the tail give

$$0.660161507122424482640393355 1 < C_2^{\text{twin}} < 0.660161837203508124853756659 1. \tag{44.26}$$

The direct count up to 10^6 contains 8169 pairs of twin primes. The convergence of the local product and this finite census do not imply the existence of infinitely many pairs; the latter is a distinct arithmetic proposition.

44.6.13 Meissel–Mertens constant and prime Hamiltonian

The Meissel–Mertens constant is defined by

$$B_1 = \lim_{x\to\infty} \left(\sum_{p \le x} \frac{1}{p} - \log\log x \right) \tag{44.27}$$

and satisfies

$$B_1 = \gamma + \sum_p \left(\log\left(1 - \frac{1}{p}\right) + \frac{1}{p} \right). \tag{44.28}$$

With the notation of theorem 44.7, we set

$$\mathcal{F}_P := \Gamma_s(\mathfrak{h}_{\mathbb{P}}), \qquad H_P := d\Gamma(H_{\mathbb{P}}).$$

Neither the bosonic Fock space of prime modes nor its Hamiltonian is redefined here. The operator already constructed has partition function

$$\operatorname{Tr}_{\mathcal{F}_P}(e^{-sH_P}) = \prod_p \frac{1}{1 - p^{-s}} = \zeta(s), \qquad \Re s > 1. \tag{44.29}$$

Let $\mathcal{H}_P^{(1)}$ be the one-particle subspace, generated by $\{|p\rangle : p \text{ prime}\}$. Then

$$\operatorname{Tr}_{\mathcal{H}_P^{(1)}} \left(e^{-H_P} \mathbf{1}_{(-\infty, \log x]}(H_P) \right) = \sum_{p \le x} \frac{1}{p}.$$

Thus, B_1 is the finite part, as $x \to \infty$, of this truncated spectral trace after subtracting $\log\log x$. The equality (44.29) presupposes the Fock decomposition over the primes; it therefore constitutes an operator representation of the Euler product, not a deduction of primality from the triadic Laplacian.

44.6.14 Dependence on the operator

The preceding evaluations share the operator (44.14), the positive trace (44.16) and the transform (44.17). Other constants require distinct dynamics. Catalan's constant is associated with the character modulo four χ_{-4}, whereas Khinchin's constant belongs to the Gauss operator and its invariant measure. A prefix of any of them may appear in a finite catalogue without the triadic functional selecting it. This operator dependence distinguishes spectral evaluation from the purely combinatorial capacity for encoding.

44.7 Finite parts, regularization and residual readouts

The functional constructed from integer modes admits operations preserving different aspects of its structure. Evaluating it at a point of convergence, extracting the regular term accompanying a pole and differentiating its meromorphic continuation are three precise operations. This distinction organizes the constants of this chapter and their relationship with those in the moments chapter.

The arithmetic antecedent is the publication of the normal forms obtained with the two APP sheets. Their evaluation identifies each positive integer once; the numbering operator acts on the corresponding basis by

$$Qe_n = ne_n, \qquad \operatorname{Dom} Q = \left\{ u : \sum_{n\geq 1} n^2 |u_n|^2 < \infty \right\}.$$

The subsequent spectral readout is

$$Z(s) = \operatorname{Tr}(Q^{-s}) = \sum_{n\geq 1} n^{-s}, \qquad \Re s > 1.$$

The chapter on the completed form proves its coincidence with the theta–Mellin functional and its continuation. Here, an evaluation procedure with controlled remainders is brought together. The memory of the history remains in the source state; the trace is a reading of the published modes, not a reconstruction of that entire history from a scalar constant.

44.7.1 A continuation formula with explicit remainder

Define the Bernoulli numbers through the formal identity

$$\frac{t}{e^t - 1} = \sum_{j\geq 0} B_j \frac{t^j}{j!}.$$

Their coefficients are computed without a numerical table:

$$B_0 = 1, \qquad B_m = -\frac{1}{m+1} \sum_{j=0}^{m-1} \binom{m+1}{j} B_j.$$

In particular, $B_1 = -1/2$. Write $(s)_r = s(s+1)\cdots(s+r-1)$, with $(s)_0 = 1$, and denote the corresponding polynomial by $B_m(x) = \sum_{j=0}^{m} \binom{m}{j} B_j x^{m-j}$. Its periodization is $\widetilde{B}_{2p}(x) = B_{2p}(\{x\})$. For $N, p \geq 1$, Euler–Maclaurin summation gives

$$Z(s) = \sum_{n=1}^{N} n^{-s} + \frac{N^{1-s}}{s-1} - \frac{1}{2N^s} + \sum_{k=1}^{p} \frac{B_{2k}}{(2k)!}(s)_{2k-1} N^{1-s-2k} + R_{N,p}(s),$$

with

$$R_{N,p}(s) = -\frac{(s)_{2p}}{(2p)!} \int_N^\infty \widetilde{B}_{2p}(x) x^{-s-2p}\, dx.$$

Proposition 44.24. Domain and bound of the remainder

The remainder integral is holomorphic on $\Re s > 1 - 2p$. If $\sigma = \Re s > 1 - 2p$, then

$$|R_{N,p}(s)| \le \frac{|B_{2p}|\,|(s)_{2p}|}{(2p)!} \frac{N^{1-\sigma-2p}}{\sigma + 2p - 1}.$$

The preceding formula continues Z to that half-plane, with its simple pole at $s = 1$, and is compatible with the theta–Mellin continuation.

Proof. The summation formula is obtained by successive integration by parts on each unit interval, using $B_j'(x) = jB_{j-1}(x)$ and the endpoint equalities for polynomials of degree other than one. Applying it to x^{-s} on $[N, M]$, the derivatives supply the factors $(s)_r$. On $\Re s > 1$, passage to $M \to \infty$ removes the upper endpoint terms and produces exactly the identity and remainder sign written above.

To control the remainder, use

$$|B_{2p}(t)| \le |B_{2p}|, \qquad 0 \le t \le 1.$$

One proof of this inequality is the absolutely convergent Fourier series

$$B_{2p}(t) = (-1)^{p+1} \frac{2(2p)!}{(2\pi)^{2p}} \sum_{m \ge 1} \frac{\cos(2\pi m t)}{m^{2p}}.$$

The maximum of the absolute-value bound is attained at $t = 0$. This identity follows from the Fourier coefficients by integration by parts of the polynomials themselves; the factor π merely expresses the Fourier convention of this proof and does not enter numerically into the rational evaluator used later. Comparison with $x^{-\sigma-2p}$ proves the remainder bound. On compact subsets of the half-plane, it also dominates derivatives in s, which add powers of $\log x$; the integral is therefore holomorphic there. The identity on the convergence half-plane and uniqueness of continuation prove the stated compatibility. □

Order p and horizon N are computation and error-control parameters. They do not select a different constant: expressions with different N, p represent the same functional on their common domains. Nor must an asymptotic series converge as p increases indefinitely with N fixed. Each finite pair comes with its own proved bound.

44.7.2 Euler–Mascheroni as the finite part of the same functional

The finite part at the pole is defined by

$$\gamma_0 = \lim_{s \to 1} \left(Z(s) - \frac{1}{s-1} \right).$$

Setting $s = 1 + z$, the singular term satisfies

$$\frac{N^{-z}}{z} = \frac{1}{z} - \log N + O(z).$$

The preceding proposition allows the finite part to be taken term by term.

Theorem 44.25. Explicit interval for the finite part

Let

$$G_{N,p} = H_N - \log N - \frac{1}{2N} + \sum_{k=1}^{p} \frac{B_{2k}}{2kN^{2k}}, \qquad H_N = \sum_{n=1}^{N} \frac{1}{n}.$$

Then

$$|\gamma_0 - G_{N,p}| \le \frac{|B_{2p}|}{2pN^{2p}}.$$

Moreover,

$$\gamma_0 = \lim_{N\to\infty} (H_N - \log N).$$

Proof. At $s = 1$, one has $(1)_{2k-1} = (2k-1)!$; hence each Bernoulli term becomes $B_{2k}/(2kN^{2k})$. The bound in Proposition 44.24, evaluated at that point, is $|B_{2p}|/(2pN^{2p})$. With $p = 1$, all terms separating $G_{N,1}$ from $H_N - \log N$, and the remainder as well, tend to zero. This proves the limit. This is the subsequent identification with the Euler–Mascheroni constant. □

The antecedent of γ_0 used in the Meissel–Mertens formula is thus incorporated. The number is not received as an independent numerical value to correct the sum over irreducibles: it is a specified operation on the same integer partition function.

The logarithm used in the calculation also admits a rational interval. For $x \ge 1$, let $z = (x-1)/(x+1)$. The integrated series of a geometric progression gives

$$\log x = 2 \sum_{j=0}^{M-1} \frac{z^{2j+1}}{2j+1} + r_M(x), \quad 0 \le r_M(x) \le \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

The inequality follows by replacing the tail denominators with $2M+1$ and summing the geometric progression. First reduce $x = 2^a y$, with $1 \le y < 2$, and evaluate $a \log 2 + \log y$. The series argument is then at most $1/3$. For $0 < x < 1$, use $\log x = -\log(1/x)$, reversing the interval endpoints.

44.7.3 Apéry and the regularized negative evaluation

For an integer $r \ge 2$, specialization of Proposition 44.24 produces a rational centre

$$Z_{N,p}^{[r]} = \sum_{n=1}^{N} \frac{1}{n^r} + \frac{1}{(r-1)N^{r-1}} - \frac{1}{2N^r} + \sum_{k=1}^{p} \frac{B_{2k}(r)_{2k-1}}{(2k)!N^{r+2k-1}}$$

and the entirely rational bound

$$\left| Z(r) - Z_{N,p}^{[r]} \right| \le \frac{|B_{2p}|(r)_{2p-1}}{(2p)!N^{r+2p-1}}.$$

The simplification uses $(r)_{2p}/(r+2p-1) = (r)_{2p-1}$. At $r = 3$, one obtains Apéry's constant. The procedure determines its value with bounds; irrationality of that constant is a mathematical statement distinct from this evaluation.

On the negative side, the same continuation admits an exact evaluation, without a decimal approximation or fitting parameter.

Theorem 44.26. Regularized evaluation at minus one

The continuation of the integer-mode functional satisfies

$$Z(-1) = -\frac{1}{12}.$$

Proof. In the formula of Proposition 44.24, take $p = 2$, so that $s = -1$ lies in the interior of the validity half-plane. The factor $(s)_4$ vanishes there; its accompanying integral is holomorphic there, so the remainder is zero. The factor $(s)_3$, multiplying the term with B_4, also vanishes. Only the following remains:

$$Z(-1) = \sum_{n=1}^{N} n - \frac{N^2}{2} - \frac{N}{2} - \frac{B_2}{2}.$$

Since additive accumulation gives $\sum_{n=1}^{N} n = N(N+1)/2$, and the Bernoulli recurrence gives $B_2 = 1/6$, the result is $-1/12$, independently of N. □

The evaluated object is the continuation of the functional, not the ordinary limit of the positive partial sums, which grow without bound. This distinction preserves the operation giving rise to the value: the Bernoulli term remains after cancellation of the polynomial boundary terms. Regularization therefore has a structure and genealogy here, in addition to a value.

44.7.4 Laurent coefficients and preservation of the three classes

Write

$$Z(1+z) - \frac{1}{z} = \sum_{n\geq 0} a_n z^n, \qquad \gamma_n = (-1)^n n! a_n.$$

The regular function on the left is holomorphic around zero. For a circle contained in its domain of holomorphy, Cauchy's formula determines

$$a_n = \frac{1}{2\pi i} \int_{|z|=\rho} \frac{Z(1+z) - 1/z}{z^{n+1}}\, dz.$$

The continuation formula with remainder controls this extraction: a uniform error bound ε on the circle produces a bound $\varepsilon\rho^{-n}$ for the coefficient. This justifies a procedure at arbitrary order; it does not turn agreement of two finite quadratures into an error bound.

The residual readout maintains the three classes of the integer operator:

$$Q_r e_n = \mathbf{1}_{n\equiv r\ (3)} e_n, \qquad Z_r(s) = \mathrm{Tr}(Q^{-s} Q_r), \quad r = 0, 1, 2.$$

These projections have a common domain in the convergence half-plane. Their sum is the identity; the difference of classes 1 and 2 preserves the orientation of the character modulo three. One has

$$Z = Z_0 + Z_1 + Z_2, \qquad Z_0(s) = 3^{-s} Z(s), \qquad L_{-3}(s) = Z_1(s) - Z_2(s).$$

The last series has periodic coefficients $(1, -1, 0)$ and zero mean. Grouping by periods allows continuation near $s = 1$. Each Z_r has residue $1/3$ there.

Let

$$Z_r(1+z) - \frac{1}{3z} = \sum_{n\geq 0} c_{r,n} z^n, \qquad \lambda = \log 3.$$

Coefficient comparison gives

$$a_n = c_{0,n} + c_{1,n} + c_{2,n}, \qquad \frac{L_{-3}^{(n)}(1)}{n!} = c_{1,n} - c_{2,n},$$
$$c_{0,n} = \frac{1}{3}\left[\frac{(-\lambda)^{n+1}}{(n+1)!} + \sum_{k=0}^{n} a_k \frac{(-\lambda)^{n-k}}{(n-k)!}\right].$$

The third identity comes from multiplying the complete series, including its pole, by $3^{-1}e^{-\lambda z}$. Sum and difference uniquely recover the other two coefficients. The three readouts preserve aggregation, orientation and change of scale, respectively.

At order zero, evaluation can be completed without receiving a decimal value of $L_{-3}(1)$. Indeed,

$$L_{-3}(1) = \sum_{m\geq 0}\left(\frac{1}{3m+1} - \frac{1}{3m+2}\right) = \int_0^1 \frac{1-x}{1-x^3}\,dx = \int_0^1 \frac{dx}{1+x+x^2}.$$

Complete the square in the denominator and evaluate the antiderivative:

$$L_{-3}(1) = \frac{2}{\sqrt{3}}\left[\arctan\frac{2x+1}{\sqrt{3}}\right]_0^1 = \frac{\pi}{3\sqrt{3}}.$$

The coordinate π is the prior regional publication of HMT; this identity is a subsequent analytic evaluation of a class reader, not a procedure for retroactively selecting that coordinate. Thus, with $C_r = c_{r,0}$,

$$C_0 = \frac{\gamma_0 - \log 3}{3}, \qquad C_{1,2} = \frac{\gamma_0}{3} + \frac{\log 3}{6} \pm \frac{\pi}{6\sqrt{3}}.$$

The following identities are recovered:

$$C_0 + C_1 + C_2 = \gamma_0, \qquad C_1 - C_2 = L_{-3}(1), \qquad C_1 + C_2 - 2C_0 = \log 3.$$

44.7.5 Exact evaluation and scope of the certification

The program `partes_finitas_exactas.py` preserves the following order: succession of words through the additive sheet; evaluation of the integer modes; rational Bernoulli construction; application of the functional formula; incorporation of the remainder; directed decimal publication. Native multiplication remains available in the same package and its compatibility is proved in the chapter on the product. The analytic calculation on the published modes does not modify their rules.

With $N = 81$, $p = 8$ and 120 terms for each logarithm series, the resulting intervals are

$$0.57721566490153286060651209008227320404153 5 < \gamma_0 < 0.57721566490153286060651209008253138727927 84,$$

$$1.20205690315959428539973816151144731135582 9 < Z(3) < 1.20205690315959428539973816151145266311934 2.$$

The decimal endpoints are outward roundings of exact fractions. The program also preserves each interval's rational width and recovers $-1/12$ exactly. No comparison digit appears in the producer. The separate checker opens reference values after obtaining the outputs; this comparison checks execution and does not replace the proved bounds.

Within its domain, this chapter completes the finite parts and evaluations it has just defined. It uses no assumption about the location of zeros or positivity of the Weil form. Prolongation towards the spectral problem preserves the obligations of its own operators: a regularized evaluation does not by itself provide the boundary identity of another functional.

Provenance: development recovered from the triadic functionals and coordinated readout network. Bringing together the Euler–Maclaurin proof, its rational bounds and its program constitutes a formalization and additional certificate of this edition; no new historical priority is claimed for those analytic identities.

Chapter 45

Unimodal criticality of the dodecaphasic family: exact analytic profile, Feigenbaum approximants and hyperbolic renormalization certificate

45.1 Intrinsic construction of the dodecaphasic family

The Feigenbaum constants characterize the universality of period doubling. Their derivation in HMT first requires constructing an analytic unimodal family, proving its quadratic criticality, and then studying the action of the renormalization operator.

The initial map does not follow from an external harmonic choice. The return of TPK determines the oriented dodecaphase wheel U_{12}^{sign}. In the uniform sector $\eta = 0$, the normalized trace of its twelve sectors fixes $w_m = 1/12$, and the phase functional

$$\mathcal{P}_{\mathrm{crit}} : U_{12}^{\mathrm{sign}} \longrightarrow \mathbb{R}/2\pi\mathbb{Z}, \qquad m \longmapsto \frac{(30m-10)\pi}{180} = \pi\frac{3m-1}{18} \tag{45.1}$$

preserves order and orientation. The causal composition is

$$\mathrm{APP} \to \widehat{\mathbb{T}} \to \mathrm{TPK} \to U_{12}^{\mathrm{sign}} \xrightarrow{\mathcal{P}_{\mathrm{crit}}} \{(\phi_m, w_m)\}_{m=1}^{12} \to u_0 \to f_\lambda.$$

Every modification of $\mathcal{P}_{\mathrm{crit}}$, of the weights, or of the order of the phases alters the family and constitutes a criterion of refutation. These choices must be fixed before the contrast with δ_F.

The resulting dodecaphasic family starts from

$$a_m = \frac{3m-1}{18}, \qquad \phi_m = \pi a_m, \qquad m = 1, \dots, 12,$$

and of the second moment

$$D_0 = \frac{1}{12}\sum_{m=1}^{12} a_m^2.$$

The identity

$$\sum_{m=1}^{12}(3m-1)^2 = 5394 = 6 \cdot 899 \tag{45.2}$$

yields

$$D_0 = \frac{899}{648}, \qquad s_0 = \frac{36}{\pi\sqrt{899}}.$$

Therefore,

$$s_0\phi_m = \frac{2(3m-1)}{\sqrt{899}}. \tag{45.3}$$

The factor π cancels before the iterations are performed. The family does not depend on δ_F or α_F as input data.

Cancellation is a structural property of normalization. Before normalization, the second angular moment is

$$\frac{1}{12}\sum_{m=1}^{12}\phi_m^2 = \pi^2 D_0.$$

The condition of quadratic criticality fixes $s_0^2\pi^2 D_0 = 2$, and therefore each normalized phase depends only on the integer arithmetic of $3m-1$. The functional preserves the angular provenance of the wheel, but the iteration receives an intrinsic real family. This is precisely what makes it possible to distinguish a derivation from a calibration by π.

45.2 Exact analytic representation and closed trigonometric expression

We define

$$u_0(x) = 1 - \frac{1}{12}\sum_{m=1}^{12}\cos\left(\frac{2(3m-1)}{\sqrt{899}}x\right). \tag{45.4}$$

The development at the origin is

$$u_0(x) = x^2 - \frac{715769}{2424603}x^4 + O(x^6). \tag{45.5}$$

The finite sum admits the following closed expression. If $y = x/\sqrt{899}$,

$$u_0(x) = 1 - \frac{\cos(37y)\sin(36y)}{12\sin(3y)} = 1 - \frac{\sin(73y) - \sin y}{24\sin(3y)}. \tag{45.6}$$

The apparent singularity at $y = 0$ is removable and reproduces equation (45.5).

Derivation of the closed form. The twelve arguments form an arithmetic progression:

$$2(3m-1)y = (6m-2)y, \qquad m = 1, \dots, 12.$$

The sum of cosenos of a progresion satisfies

$$\sum_{m=1}^{N}\cos(a+md) = \frac{\sin(Nd/2)}{\sin(d/2)}\cos\left(a + \frac{N+1}{2}d\right).$$

Taking $N = 12$, $d = 6y$ and $a = -2y$ yields

$$\sum_{m=1}^{12}\cos((6m-2)y) = \frac{\sin 36y}{\sin 3y}\cos 37y.$$

The product–to–sum identity $2\cos 37y \sin 36y = \sin 73y - \sin y$ gives the second expression for equation (45.6). □

The quartic coefficient is not fitted either. The integer sums

$$\sum_{m=1}^{12}(3m-1)^2 = 5394, \qquad \sum_{m=1}^{12}(3m-1)^4 = 4294614 \tag{45.7}$$

inserted in the development of $\cos z$ produce exactly

$$\frac{1}{24\cdot 12}\sum_{m}\left(\frac{2(3m-1)}{\sqrt{899}}\right)^4 = \frac{715769}{2424603}.$$

Thus, both the quadratic normalization and the first nonlinear correction are finite moments of the same wheel.

45.3 Pertenencia to the clase analitica unimodal cuadratica

Consider we

$$f_\lambda(x) = 1 - \lambda u_0(x), \qquad \lambda > 0.$$

For $0 < x \le 1$, all arguments

$$\frac{2(3m-1)}{\sqrt{899}}x$$

belong to $(0, \pi)$. By therefore,

$$u_0'(x) > 0, \qquad u_0'''(x) < 0.$$

Since u_0 is even, $x = 0$ is its unique critical point and $u_0''(0) = 2$. The Schwarzian of f_λ is

$$S(f_\lambda) = \frac{u_0'''}{u_0'} - \frac{3}{2}\left(\frac{u_0''}{u_0'}\right)^2 < 0 \qquad (0 < |x| \le 1). \tag{45.8}$$

Theorem 45.1 (Exact membership in the unimodal class)**.** *For*

$$0 < \lambda \le \frac{2}{u_0(1)},$$

the family $f_\lambda = 1 - \lambda u_0$ is a self-map that is real–analytic on $[-1, 1]$ and even, with a single critical point of quadratic type at the origin and negative Schwarzian away from that point. Consequently, without using Feigenbaum values as a target value, it belongs to the S-unimodal quadratic class subject to the action of the period-doubling renormalizer [46, 79]. For λ outside that interval, the analytic germ and its local form are retained, but one does not automatically assert a self-map of the interval.

The theorem establishes the contribution intrinsic to HMT prior to the universality theorem: the analytic germ and its type of criticality.

Detalles of the signo and of the self-map. Deriving equation (45.4),

$$u_0'(x) = \frac{1}{12}\sum_m k_m \sin(k_m x),$$
$$u_0'''(x) = -\frac{1}{12}\sum_m k_m^3 \sin(k_m x), \qquad k_m = \frac{2(3m-1)}{\sqrt{899}}.$$

For $0 < x \le 1$, $0 < k_m x < \pi$, so all summands in the first line are positive and those in the second are negative. Parity extends monotonicity to both sides of the origin. Since $u_0(0) = 0$ and $0 \le u_0(x) \le u_0(1)$, the bound $0 < \lambda \le 2/u_0(1)$ implies $-1 \le 1 - \lambda u_0(x) \le 1$. Finally, $u_0''(0) = 1/12 \sum k_m^2 = 2$, so the critical point is quadratic. □

45.4 Sequences finite of parameters superstable

The sucesion superestable certified until level eight produces

$$\delta_8 = 4.66921218915\ldots, \qquad \alpha_8 = 2.50296847988\ldots.$$

Its status is *finite certificate free of target values*: each approximant is obtained from the previously defined family, without introducing the universal limit as a datum. Decimal agreement provides a convergence indicator but does not replace the hyperbolicity theorem.

An independent numerical extension of the same sequence, with a step size adapted to the separation between roots and control of the exact period, provisionally reaches:

Level	λ_n	δ_n	α_n
11	1.87403828195439908045	4.66920170694424982745	2.50290847517165181764
12	1.87403836411023460451	4.66920163036308158848	2.50290800379001546915
13	1.87403838170549870372	4.66920161361839219913	2.50290790268748193108
14	1.87403838547386545431	4.66920161007472591023	2.50290788100969754274

These levels have the status of *New numerical certificate pending reproducible registration*; they do not supersede the persistent certificate through level eight. An Aitken extrapolation yields $\lambda_\infty^{\mathrm{HMT}} \simeq 1.8740383865008918080$ and $\alpha_F \simeq 2.50290787509$ as candidate limits, conditional on a proof of hyperbolicity and transversality.

To avoid an unspecified numerical procedure, the method is formulated as follows. Define

$$F_n(\lambda) = f_\lambda^{2^n}(0). \tag{45.9}$$

The raiz superestable λ_n is the primera raiz of F_n to the right of λ_{n-1} that satisfies simultaneously

$$f_{\lambda_n}^{2^j}(0) \neq 0 \quad (0 \le j < n).$$

This inequality excludes divisor periods. With disjoint intervals $I_n \ni \lambda_n$, one computes

$$\delta_n = \frac{\lambda_{n-1} - \lambda_{n-2}}{\lambda_n - \lambda_{n-1}}, \qquad \alpha_n = -\frac{d_{n-1}}{d_n}, \tag{45.10}$$

where $d_n = f_{\lambda_n}^{2^{n-1}}(0)$ is the oriented distance from the last new point to the critical point. A persistent certificate must retain the root intervals, the exclusion of smaller periods, and the propagation of error from equation (45.10); a mere printed decimal does not satisfy those three conditions.

The discretization rule also constitutes a negative control. A fixed lower bound greater than the separation $|\lambda_n - \lambda_{n-1}|$ can omit the first root and nevertheless produce a numerically plausible succession. For this reason the increment must scale with the previous separation and each candidate must be verified through its exact period.

45.5 Conditions for certifying the renormalization operator

On a space of even analytic functions analytic, we define

$$\mathcal{R}(f)(x) = -a(f)^{-1} f\big(f(-a(f)x)\big), \qquad a(f) = -f(1). \tag{45.11}$$

The complete closure of δ_F and α_F requires jointly certifying:

1. existence of a fixed point $g = \mathcal{R}(g)$ in a declared analytic ball;
2. uniqueness local of ese point fixed;

3. a single unstable real eigenvalue of $D\mathcal{R}_g$, equal to δ_F;
4. spectral radius strictly less than one on the stable complement;
5. transversality of the curve $\lambda \mapsto f_\lambda$ with respect to the stable manifold of g.

A natural HMT implementation employs Chebyshev truncations of order 9^m, interval arithmetic, and radii polynomials. The proof must provide a ball, an approximate inverse, a defect bound, and a Lipschitz bound satisfying $Y + Z(r) < r$. Thus, the nonadic depth parameterizes a fixed-point certificate and not a mere sequence of decimal approximations.

On a Chebyshev even basis,

$$g(x) = \sum_{j=0}^{N} g_{2j} T_{2j}(x), \qquad N = 9^m,$$

the finite residue is $F_N(g) = \Pi_N(\mathcal{R}g - g)$. If A_N approximates $(DF_N(\bar{g}))^{-1}$, the quantities that must be certified are

$$\begin{aligned} Y &= \|A_N F_N(\bar{g})\|, \\ Z_0 &= \|I - A_N DF_N(\bar{g})\|, \\ Z_1(r) &= \sup_{\|h\| \le r} \|A_N (DF_N(\bar{g} + h) - DF_N(\bar{g}))\|. \end{aligned}$$

An inequality $Y + (Z_0 + Z_1(r))r < r$ encloses a unique fixed point. The later spectral block must isolate an unstable direction and contract the complement. The use of 9^m thus enters as a refinement law of the certificate, not as decoration numerological.

Corollary 45.2 (Exact convergence of mode thirty)**.** *Mode* 30 *admits three simultaneous and typologically distinct realizations:*

$$\begin{aligned} &30 \longmapsto (3 \cdot 30, 4 \cdot 30) = (90, 120) && \textit{in the constitutive completion}, \\ &30 \longmapsto 30^2 - 1 = 899 = 29 \cdot 31 && \textit{in the critical normalization}, \\ &30 \longmapsto M_{30} = \begin{pmatrix} 30 & 899 \\ 1 & 30 \end{pmatrix} && \textit{in the hyperbolic Pell transformation}. \end{aligned}$$

The three identities are exact and are obtained without prescribing target values. The conserved invariant is the central integer 30 *together with its orientation: extension* $3 \to 4$*, quadratic defect* -1*, and a unit of norm one. What is not asserted is the equality between the vacuum operator and the duplication renormalizer.*

Remark 45.3 (Negative control)*.* A second eigenvalue outside the unit disk, a ball of radii without solution, loss of uniqueness, or tangency of the HMT family to the stable variety refutes the universal promotion. None refutes the exact germ or the Schwarzian theorem.

Proposition 45.4 (Main Result)**.** *The structural significance of the doubling constant lies in the cancellation in dodecaphase form of* π*, in the identity* $899 = 30^2 - 1$*, in the closed trigonometric form, and in its proved membership in the analytic class on which the renormalizer acts. Its decimal expansion constitutes exclusively the terminal evaluation.*

Logical Scope and Domain of Validity

Profile, Taylor expansion, normalizer, Pell unit, unimodality, and Schwarzian: *Exact internal theorem*. Approximants through level eight: *Persistent finite certificate*. Levels 11–14: *New numerical certificate pending durability*. Hyperbolicity and transversality of the renormalizer: *Open program with executable falsification criterion*.

45.5.1 Weighted harmonic deformation and selection of the superstable branch

The uniform profile has a parametric extension that preserves the twelve phases and separates the harmonic deformation from the dynamic parameter. For $1 \le m \le 12$, let

$$\phi_m = \frac{(30m-10)\pi}{180}, \qquad p(\phi) = 12\cos(3\phi) - \cos(4\phi). \tag{45.12}$$

For every real η such that the weights remain positive, we define

$$w_m(\eta) = \frac{1 + \eta p(\phi_m)}{12}. \tag{45.13}$$

The sums of the third and fourth harmonics over the twelve phases vanish; therefore,

$$\sum_{m=1}^{12} w_m(\eta) = 1.$$

Let

$$s_\eta^2 = \frac{2}{\sum_{m=1}^{12} w_m(\eta)\phi_m^2}, \qquad u_\eta(x) = 1 - \sum_{m=1}^{12} w_m(\eta)\cos(s_\eta \phi_m x). \tag{45.14}$$

The cosine expansion and the definition of s_η give

$$u_\eta(x) = x^2 + O(x^4) \qquad (x \to 0).$$

Consequently, u_η is real–analytic, even, and retains a quadratic critical point at the origin. We introduce the dynamical parameter through

$$f_{\lambda,\eta}(x) = 1 - \lambda u_\eta(x). \tag{45.15}$$

A superstable parameter of exact period 2^n satisfies

$$f_{\lambda,\eta}^{2^n}(0) = 0, \qquad f_{\lambda,\eta}^{2^k}(0) \neq 0 \quad (0 \le k < n).$$

The equation may have several branches. A branch $\mathscr{B}$ is fixed by traversing an adaptive mesh immediately after the parameter of the preceding level; the first interval with a sign change is taken and bisection is applied. The notation $\lambda_n^{\mathscr{B}}(\eta)$ incorporates this selector and does not attribute global uniqueness to the superstable equation. The first level is determined exactly:

$$\lambda_1^{\mathscr{B}}(\eta) = \frac{1}{u_\eta(1)},$$

if $u_\eta(1) > 0$, because $f_{\lambda,\eta}^2(0) = f_{\lambda,\eta}(1) = 0$. Given three consecutive levels and

$$x_n^{\mathscr{B}}(\eta) := f_{\lambda_n^{\mathscr{B}}(\eta),\eta}^{2^{n-1}}(0),$$

we form

$$\delta_n^{\mathscr{B}}(\eta) = \frac{\lambda_{n-1}^{\mathscr{B}}(\eta) - \lambda_{n-2}^{\mathscr{B}}(\eta)}{\lambda_n^{\mathscr{B}}(\eta) - \lambda_{n-1}^{\mathscr{B}}(\eta)}, \qquad \alpha_n^{\mathscr{B}}(\eta) = \frac{|x_{n-1}^{\mathscr{B}}(\eta)|}{|x_n^{\mathscr{B}}(\eta)|}. \tag{45.16}$$

For $\eta = 0$, the uniform profile is recovered exactly. The weighted extension introduces a transverse parameter: it permits study of the transversality of the family with respect to the stable manifold of the renormalizer without converting a finite coincidence into a proof of universality.

Remark 45.5 (Negative control). The extension is refuted if the weights lose positivity on the declared domain, if normalization ceases to produce a unit quadratic coefficient, if the selector does not exclude divisor periods, or if a branch is chosen using the terminal universal values as input.

45.6 Reduction of Feigenbaum universality to the hyperbolic certificate of basin membership and transversality

The HMT Feigenbaum family has analytic profile, quadratic maximum, negative Schwarzian, and certified superstable approximants up to level 8. Its promotion to universality requires the certificate

$$\ell_u(\partial_t f_t) \neq 0$$

together with membership in the hyperbolic basin of the fixed point. Tangency, departure from the basin, or the presence of a second unstable eigenvalue are complete falsifiers.

Chapter 46

Arithmetic of the dodecaphasic modular family of period 30: Rogers–Ramanujan identities, normalizer 899 and nonadic Pell lifting

46.1 Normalizer 899, Pell unit, and arithmetic structure of the mode 30

The normalizer has the factorization arithmetic

$$899 = 30^2 - 1 = 29 \cdot 31. \tag{46.1}$$

The same mode 30 that generates sectors 90 and 120 simultaneously determines the twin-prime pair $29, 31$. Moreover,

$$\varepsilon_{30} = 30 + \sqrt{899}, \qquad \varepsilon_{30}^{-1} = 30 - \sqrt{899}, \qquad \log \varepsilon_{30} = \operatorname{arcosh} 30. \tag{46.2}$$

This Pell unit determines the hyperbolic scale associated with the normalizer and relates the dodecaphase phase, mode 30, twin primes, and hyperbolic geometry. This relation does not identify the constitutive operator of the vacuum with the doubling renormalizer; such an identification would require an explicit intertwiner.

The unit acts integrally through

$$M_{30} = \begin{pmatrix} 30 & 899 \\ 1 & 30 \end{pmatrix}, \qquad \det M_{30} = 1. \tag{46.3}$$

On $\mathbb{Z}[\sqrt{899}]$, this matrix represents multiplication by $30 + \sqrt{899}$. Its eigenvalues are $\varepsilon_{30}^{\pm 1}$; therefore $\log \varepsilon_{30}$ is the translation length of the associated hyperbolic transformation. The same central integer 30 reappears in the vacuum because $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$, but the two operators continue to have distinct domains. The demonstrated confluence concerns the mode and its completions; identity of the dynamics is not postulated.

46.2 Lift nonadic of the unit of Pell

For $k \geq 1$, let the unramified quadratic ring be

$$\mathcal{R}_k = (\mathbb{Z}/3^k\mathbb{Z})[r]/(r^2 - 2), \tag{46.4}$$

and let

$$u = 3 + 2r.$$

The conjugation $r \mapsto -r$ gives

$$N_{\mathcal{R}_k/(\mathbb{Z}/3^k\mathbb{Z})}(u) = (3 + 2r)(3 - 2r) = 9 - 8 = 1,$$

so that u is a unit of norm one. Moreover,

$$u^2 = 17 + 12r, \qquad u^4 - 1 = 576 + 408r = 3(192 + 136r).$$

The second factor is a unit modulo three because it reduces to $r \neq 0$. Hence the valuation 3-adic satisfies $v_3(u^4 - 1) = 1$. The lifting lemma for a unit congruent to one modulo three yields, for every $j \geq 0$,

$$v_3\left(u^{4\cdot 3^j} - 1\right) = j + 1.$$

Since $u \bmod 3 = 2r$ and $(2r)^2 = -1$, its order modulo three is four. Consequently,

$$\operatorname{ord}(u) = 4 \cdot 3^{k-1}, \tag{46.5}$$

and the profinite closure of the compatible cyclic groups is

$$\varprojlim_k \langle u \bmod 3^k \rangle \simeq C_4 \times \mathbb{Z}_3. \tag{46.6}$$

Its Archimedean realization is

$$3 + 2\sqrt{2} = (1 + \sqrt{2})^2.$$

This construction proves the conservation of nonadic depth by a hyperbolic unit. The units $3 + 2\sqrt{2}$, $2 + \sqrt{3}$, and $30 + \sqrt{899}$ form a Pell family with distinct provenances; any identification among them requires explicit maps between the corresponding extensions.

The mode 30 is one of the points at which the architecture begins to exhibit a unit determined by the operatorial and arithmetic genealogy.

It appears first in the vacuum because

$$90 = 3 \cdot 30, \qquad 120 = 4 \cdot 30.$$

It is the elementary cadence that permits comparison of the sectors 90 and 120. The vacuum reads that cadence as the completion $3 \to 4$.

But the same 30 reappears in the construction of the criticality germ. There the central identity is

$$899 = 30^2 - 1 = 29 \cdot 31.$$

Normalization of the twelve phases produces $\sqrt{899}$, and the analytic profile can be written in closed form. What is truly striking is that π disappears before the iteration begins.

This must be read carefully. The angular geometry of π is absorbed into the organization of the phases. Once the dodecaphase configuration has been normalized, the dynamic germ is governed by the internal arithmetic of 899.

Therefore, the construction begins by generating internally a dynamic object of its own:

- es par;
- is analytic;
- has a critical quadratic;
- belongs to the corresponding unimodal class;
- has negative Schwarzian;
- can be subjected to the operator of renormalization.

The superstable ratios then begin to approximate the universal constants. The germ was produced first, and universality appears as the terminal behavior of its dynamics.

This is a profound epistemological difference. A numerical coincidence would state: “our calculation resembles Feigenbaum.” The HMT construction states: “this is the exact object that enters the universality class, and these are the quotients that emerge upon iterating it.”.

The 30 reappears a third time in modular memory. If

$$N = N_{90} + N_{120}$$

measures the total occupancy and

$$D = 90N_{90} + 120N_{120}$$

measures the occupation weighted by provenance, the matrix that transforms (N_{90}, N_{120}) into (N, D) has determinant 30.

This allows the process to be inverted:

$$N_{90} = 4N - \frac{D}{30}, \qquad N_{120} = \frac{D}{30} - 3N.$$

Thus, the same 30 performs three functions:

- in the vacuum, makes the propagations 90 and 120 commensurable;
- at criticality, determines the normalization $899 = 30^2 - 1$;
- in modular memory, is the determinant that allows the origin of each occupation to be recovered.

The vacuum operator, the chaos germ, and the modular Hamiltonian remain typologically distinct. The confluence is subtler: all three preserve a common partition of the genealogy.

We might say that the mode 30 is a gear. In one machine it measures propagation; in another it normalizes criticality; in another it makes the apparent loss of provenance reversible. The gear is the same, but each machine performs a different function.

The emergence of the Pell unit

$$30 + \sqrt{899}$$

reinforces this interpretation. The 30 simultaneously organizes a finite sum and hyperbolic growth. The same arithmetic that closes the germ opens a scale dynamics.

That is perhaps the profound reason why the result proves so beautiful: the mode that makes the finite completion of the vacuum possible is also the mode that prepares an infinite universality of criticality.

Finite and infinite are articulated through the coherent repetition of a finite structure that preserves memory; from it the dynamic infinite is born.

46.3 Spectral Pell screw and logarithmic suspension

46.3.1 Spectral identity $V^{-1}\mathcal{S}V = (2+\sqrt{3})\mathrm{i}\mathcal{S}$ and equation of the logarithmic suspension

The corpus contains the exact identity

$$V^{-1}\mathcal{S}V = (2+\sqrt{3})\mathrm{i}\,\mathcal{S}. \tag{46.7}$$

Let $\varepsilon = 2+\sqrt{3}$. The orbit transversal

$$z_n = z_0(\varepsilon\mathrm{i})^n$$

satisfies

$$r_n = r_0\varepsilon^n, \qquad \theta_n = \theta_0 + \frac{\pi n}{2}.$$

Eliminando n,

$$r(\theta) = r_0 \exp\left[\frac{2\log(2+\sqrt{3})}{\pi}(\theta - \theta_0)\right]. \tag{46.8}$$

The logarithmic spiral is an output of the operator: its equation was not chosen to fit ε afterward.

Derivation of equation (46.8). From $\theta_n - \theta_0 = n\pi/2$ it follows that $n = 2(\theta_n - \theta_0)/\pi$. Substituting into $r_n = r_0\varepsilon^n$,

$$r_n = r_0 \exp\left(\frac{2\log\varepsilon}{\pi}(\theta_n - \theta_0)\right).$$

Continuous interpolation is the indicated equation. □

46.3.2 Conjugate contraction involution

The bilateral extension $n \in \mathbb{Z}$ produces

$$n \to +\infty : \ r_n \to \infty, \qquad n \to -\infty : \ r_n \to 0.$$

The inversion $n \mapsto -n$ interchanges expansion and contraction. Since $\varepsilon^{-1} = 2 - \sqrt{3}$, both branches belong to the same Pell unit. They are not two independent spirals, but two orientations of the same spectral screw.

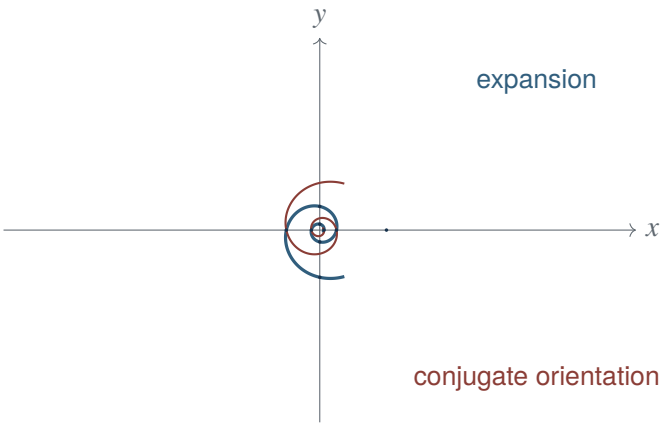


Figure 46.1: Transverse projection of the Pell screw. The marks represent quarter turns.

46.3.3 Scale covariance and energy conservation

The dilation ε must not be interpreted as gratuitous growth of the energetic amplitude of a wave in vacuum. Its coherent realization acts on the spatial or spectral scale:

$$(r, \varphi, \lambda) \longmapsto (\varepsilon r, \varphi + \pi/2, \varepsilon\lambda).$$

This distinction will be essential in the electromagnetic realization of section 63.7.

Exact compatibility between the Pell spectral screw and the phase–memory suspension

The minimal phase–memory lift is a circular helix. The spectral screw combines a quarter-turn and a Pell dilation; its projection is a logarithmic spiral. Both arise from the HMT ledger, but they are neither the same periodicity nor obtained from one another by forgetting types.

Chapter 47

Arithmetic theory of nonadic vacancies: defect $10^3 - 3^6$, radical network $3 \to 6 \to 9$, power–vacancy law, and boundary torsion

47.1 Classification of the irreducible arithmetic cycles of the nonadic radical lattice

In the arithmetic realization, a prime corresponds to an irreducible cycle that cannot be factored into smaller positive cycles. Nonadic periodicity may select candidates and organize residue classes; primality remains defined by the exact multiplicative property

$$p = ab \Longrightarrow a = 1 \text{ or } b = 1.$$

Once the coproduct of prime arithmetic cycles has been selected, equations (42.4) y (42.5) are obtained. The helical structure and the digital root parametrize phase and carry, but do not replace the factorization criterion.

47.2 Density, regularized term, and vacancy holonomy

The basic positive density is

$$\varepsilon_{\text{vac}} = \log_{10} \frac{10}{9}. \tag{47.1}$$

The sequence is deduced from the completion $1000 = 729 + 271$ determines

$$\varepsilon_{\text{vac}} = 1 - \log_{1000}(729) = \log_{10} \frac{10}{9}, \tag{47.2}$$

and the exact mechanical word

$$z_n = \lfloor (n+1)\varepsilon_{\text{vac}} \rfloor - \lfloor n\varepsilon_{\text{vac}} \rfloor, \qquad n \geq 1. \tag{47.3}$$

Since ε_{vac} is irrational, the word is Sturmian: it has complexity $p(N) = N+1$, balance one and topological entropy topological zero. This dynamic differs from the full shift on the triadic alphabet triadic: the vacancy determines an ordered boundary of zero entropy.

The suma telescopica produces

$$\sum_{n=1}^{N} z_n = \lfloor (N+1)\varepsilon_{\text{vac}} \rfloor, \qquad \sum_{n=1}^{N} (z_n - \varepsilon_{\text{vac}}) = \varepsilon_{\text{vac}} - \{(N+1)\varepsilon_{\text{vac}}\}. \tag{47.4}$$

The partial sums of the discrepancy have modulus less than one. This is the estimation supporting the subsequent harmonic constants.

The fraction continuous begins

$$\varepsilon_{\text{vac}} = [0; 21, 1, 5, 1, 6, 2, 3, \ldots].$$

Thus the ones in z_n are separated by 21 or 22 steps. If

$$\sigma = 22 - \varepsilon_{\text{vac}}^{-1} = 1 - T_G(\varepsilon_{\text{vac}}),$$

then $a_1(\sigma) = 6$ and the returns of the short separation have length 6 or 7. The pairs 21/22 and 6/7 are two scales of the same rotation, not coincidences of integers.

For the exact word in equation (47.3), the regularized constant is

$$\gamma_{\mathrm{vac}}^{+} = \lim_{N\to\infty}\left(\sum_{n\le N}\frac{z_n}{n} - \varepsilon_{\mathrm{vac}}\log N\right). \tag{47.5}$$

The associated holonomy is

$$\rho_{\varepsilon}(k) = \exp(2\pi \mathrm{i} k\varepsilon_{\mathrm{vac}}). \tag{47.6}$$

The three quantities have distinct types: density, finite term, and phase. The vacancy Stieltjes tower prolongs equation (47.5):

$$\gamma_{\mathrm{vac,j}}^{+} = \varepsilon_{\mathrm{vac}}\gamma_j + \sum_{n\ge 1}\frac{(z_n - \varepsilon_{\mathrm{vac}})(\log n)^j}{n}. \tag{47.7}$$

For $j = 0$, one recovers $\gamma_{\mathrm{vac}}^{+}$. Convergence requires the discrepancy bound for the integral development of vacancies.

Convergence is proved here. For $j \ge 0$, the sequence $b_n = (\log n)^j/n$ tends to zero and is decreasing from $n > e^j$ onward. The uniform bound equation (47.4) and Dirichlet's criterion prove the convergence of

$$\sum_{n\ge 1}(z_n - \varepsilon_{\mathrm{vac}})b_n.$$

For $j = 0$, separating $\varepsilon_{\mathrm{vac}}\sum_{n\le N} n^{-1}$ gives exactly equation (47.5). Abel summation conservatively bounds the tail after N by $2/(N+1)$: one term comes from the boundary and another from the total variation of $1/n$. The declared exhaustive execution uses $N = 10^8$, enumerates the Beatty positions of the ones, checks the gap sets $\{21, 22\}$ and returns $\{6, 7\}$, and obtains as its finite part

$$-0.11207107505876366224207105242526397929\ldots.$$

Combining the analytic bound with the recorded rounding and boundary guards, the finite certificate encloses

$$-0.112071094143613634 < \gamma_{\mathrm{vac}}^{+} < -0.112071055516338732. \tag{47.8}$$

This interval is a terminal value with controlled error. It does not define the word; the word and its discrepancy precede it.

A second evaluation using multiprecision arithmetic and directed rounding control provides, to eighteen decimal places, the certified interval

$$-0.112071086058763562 < \gamma_{\mathrm{vac}}^{+} < -0.112071064058763762,$$

contained strictly within equation (47.8). The inclusion of both intervals verifies the stability of the finite part with respect to truncation, the boundary guards, and the precision regime. The result depends solely on the recurrences, the tail bounds, and the interval arithmetic set out in this chapter.

47.3 Operative content of the helicoidal representation

The helical representation encodes a periodic phase together with a memory increment. It has operatorial content when there exists a cocycle

$$c(\gamma\eta) = c(\gamma) + c(\eta)$$

that transports the carry and whose monodromy is interlinked with the arithmetic operator. In the absence of that map, the helix constitutes only a phase parametrization; with the interlinker, it distinguishes phase return from state return.

Remark 47.1 (Negative control). Cofinality of cylinders does not imply dynamical conjugacy. A counterexample to commutation between the TPK shift and T_G precludes the intertwiner, but does not affect equations (44.5) y (44.6). In the vacancy sector, the divergence of equation (47.7) under the declared bound refutes the family of regularized constants.

Logical Scope and Domain of Validity

Cofinality, Gauss identities, Levy, entropy, and Lochs: *Exact internal theorem in the transported system*. GKW and dynamic intertwiner: *Open program with falsification criterion*. Density, constant, and vacancy holonomy: *Exact internal theorem with discrepancy control*.

47.4 Classification of the radical network 3→6→9 and derivation of the power–vacancy law

The appearance of 369 and 693 is not reducible to a graphic similarity. There are two exact mechanisms that converge:

1. **nonadic algebraic mechanism:** $3, 6, 9$ are the visual representatives of the nilpotent radical $\mathfrak{m} = (3) \subset \mathbb{Z}/9\mathbb{Z}$; the words 369, 693, and 936 are the three phases of its multiplicative orbit;
2. **Archimedean vacancy mechanism:** the difference between the decimal length of 9^n and the boundary 10^n is governed exactly by the same irrational rotation $\delta_{\text{vac}} = \log_{10}(10/9)$ that already organizes the vacancy calendar.

For $n = 9^9$, both mechanisms meet in the identity

$$9^9 - \left\lceil 9^9 \log_{10} \frac{10}{9} \right\rceil = 369\,693\,099,$$

so that

$$\#\,\text{cifras}\left(9^{9^9}\right) = 369\,693\,100.$$

The block 369 | 693 is, therefore, an exact output of the logarithmic reader of the nonadic power; it is not the decimal prefix of 9^{9^9}. The final unit proceeds from the universal operator that transforms decimal order into decimal length.

This identity opens a broader structural reading: addition, product, power, subtraction, division, root, and logarithm can be organized as a hierarchy of transports and quotients. Each ascent of operation compactifies histories; TPK preserves the coordinates that make their reconstruction possible.

47.5 Radical structure of the APP on $\mathbb{Z}/9\mathbb{Z}$

47.5.1 Nonadic local ring $\mathbb{Z}/9\mathbb{Z}$ and nilpotent radical

Let

$$R = \mathbb{Z}/9\mathbb{Z}.$$

Its group of units and its maximal ideal are

$$R^\times = \{1, 2, 4, 5, 7, 8\}, \qquad \mathfrak{m} = (3) = \{0, 3, 6\} = \{9, 3, 6\}_{\mathrm{vis}}.$$

Moreover,

$$\mathfrak{m}^2 = 0.$$

The preceding equality is an equality of ideals: every product of two elements of $\mathfrak{m}$ vanishes modulo nine. It must not be confused with the Cartesian product $\mathfrak{m} \times \mathfrak{m}$, which contains nine APP positions.

47.5.2 Exact sequence of the multiplier 3

Multiplication by three,

$$M_3 : R \longrightarrow R, \qquad x \longmapsto 3x,$$

has

$$\ker M_3 = \mathfrak{m}, \qquad \operatorname{im} M_3 = \mathfrak{m}.$$

It therefore induces an isomorphism of spaces over $\mathbb{F}_3$,

$$\overline{M}_3 : R/\mathfrak{m} \xrightarrow{\sim} \mathfrak{m}, \qquad \overline{x} \longmapsto 3x.$$

In visual representatives,

$$\overline{0} \longmapsto 9, \qquad \overline{1} \longmapsto 3, \qquad \overline{2} \longmapsto 6.$$

Thus, the band 3, 6, 9 is not a decimal ornament. It is an internal copy of the ternary quotient lodged in the nonadic radical.

47.5.3 Cyclic orbit 369→693→936→369

The row of the multiplication table corresponding to 3 is

$$3, 6, 9, 3, 6, 9, 3, 6, 9.$$

The cambio of phase $j \mapsto j + 1$ produces

$$369 \longmapsto 693 \longmapsto 936 \longmapsto 369.$$

These three words are the three cyclic readings of the application.

$$\mathbb{F}_3 \longrightarrow \mathfrak{m}, \qquad \bar{j} \longmapsto 3j.$$

The row 6, by satisfying $6 \equiv -3 \pmod 9$, reverses the orientation of the same cycle. The mark 9 fulfills two typed functions:

$$9 \bmod 9 = 0, \qquad \mathrm{dr}_9(9) = 9.$$

The module publishes the null class; the positive digital root preserves the mark of closure. This difference of readers is precisely what allows 9 to be at once residual zero and visible closure.

47.5.4 Radical cross of APP and coordinate reduction to $\mathbb{F}_3^2$

On

$$X_9 = R^2,$$

the union

$$\mathscr{C}_{369} = (\mathfrak{m} \times R) \cup (R \times \mathfrak{m})$$

is the cross formed by rows and columns $3, 6, 9$. It contains

$$3 \cdot 9 + 3 \cdot 9 - 3 \cdot 3 = 45$$

positions. Their central intersection $\mathfrak{m} \times \mathfrak{m}$ contains nine positions. In the multiplicative observable, the sum of the cross is

$$297 = 216 + 81 = 3(72 + 27).$$

The coordinate reduction produces

$$X_9/(\mathfrak{m} \times \mathfrak{m}) \cong \mathbb{F}_3^2$$

as a quotient set typed by the classes modulo three. The inverse lift requires preserving the coordinate within each three-element fiber.

47.5.5 Determination of the arithmetic core 7227 by incidence with the radical cross of APP

The central block of the APP is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R = 9P_+ + 5P_-.$$

Its rowwise reading produces $72 \mid 27$; its diagonal reading produces $77 \mid 22$. One verifies

$$72 + 27 = 77 + 22 = 99,$$

$$72 - 27 = 45 = \det B_c,$$

$$77 - 22 = 55 = 54 + 1.$$

The relation with the radical cross is not nominal:

$$3 \cdot 72 = 216, \qquad 3 \cdot 27 = 81, \qquad 3 \cdot 99 = 297.$$

The central unitary block and the non-unitary network $3, 6, 9$ are thereby linked by an exact local–global identity. The hierarchy of the sum–product difference preserves its types:

4 (local spectral jump), 2 (off-diagonal coupling),

6 (compressed difference modulo three), 54 (two-dimensional unfolding).

47.6 Power–vacancy law for the scales 9^n and 10^n

47.6.1 Decimal Length Functions and Accumulated Vacancy

For $n \geq 1$, let

$$D_9(n) = \#\,\mathrm{cifras}(9^n) = \lfloor n \log_{10} 9 \rfloor + 1.$$

The decimal boundary 10^n has $n+1$ digits. We define the accumulated vacancy of the nonadic power by

$$V_9(n) = (n+1) - D_9(n).$$

We introduce the two complementary numbers

$$\rho_{\mathrm{vac}} = \log_{10} 9, \qquad \delta_{\mathrm{vac}} = 1 - \rho_{\mathrm{vac}} = \log_{10} \frac{10}{9}.$$

47.6.2 Exact identity between decimal length and vacancy rotation

Theorem. For every integer $n \geq 1$,

$$\boxed{V_9(n) = \lceil n\delta_{\mathrm{vac}} \rceil},$$

and, therefore,

$$\boxed{D_9(n) = n - \lceil n\delta_{\mathrm{vac}} \rceil + 1}.$$

Proof. Since δ_{vac} is irrational, $n\delta_{\mathrm{vac}} \notin \mathbb{Z}$. Then

$$\begin{aligned} D_9(n) &= \lfloor n(1 - \delta_{\mathrm{vac}}) \rfloor + 1 \\ &= \lfloor n - n\delta_{\mathrm{vac}} \rfloor + 1 \\ &= n - \lceil n\delta_{\mathrm{vac}} \rceil + 1. \end{aligned}$$

Subtracting from $n+1$ yields the first identity. □

47.6.3 Sturmian coding of vacancies induced by nonadic powers

The increments of the vacancy are

$$Z_n = V_9(n+1) - V_9(n) = \lceil (n+1)\delta_{\mathrm{vac}} \rceil - \lceil n\delta_{\mathrm{vac}} \rceil \in \{0, 1\}.$$

This is exactly the mechanical word of slope δ_{vac} used by the HMT vacancy calendar. Therefore, that calendar admits an additional, completely exact reading:

> $Z_n = 1$ if and only if, in passing from 9^n to 9^{n+1}, a new decimal range is lost relative to the boundary $10^n \to 10^{n+1}$.

The power does not add a coincidence to the theory of vacancies: it provides its natural realization as a counter of decimal rank loss.

47.6.4 Law $21/22$ as a case of the power–vacancy theorem

Since

$$\delta_{\mathrm{vac}}^{-1} = 21.85434532678283256\ldots,$$

the indices for which $Z_n = 1$ are separated by 21 or 22. The first twenty-one exponents satisfy

$$D_9(n) = n, \qquad 1 \le n \le 21,$$

whereas

$$D_9(22) = 21.$$

In particular,

$$9^9 = 387\,420\,489$$

has exactly nine digits. This initial preservation of rank is not an indefinite rule: the vacancy calendar determines exactly when it ceases to hold.

47.6.5 Exact evaluation at $n = 9^9$ and length of 9^{9^9}

Let

$$n_* = 9^9 = 387\,420\,489.$$

Then

$$n_*\delta_{\mathrm{vac}} = 17\,727\,389.3684296412564569\ldots,$$

and, therefore,

$$\boxed{V_9(n_*) = 17\,727\,390}.$$

The conserved coordinate after discounting vacancies is

$$n_* - V_9(n_*) = 387\,420\,489 - 17\,727\,390 = 369\,693\,099.$$

Therefore,

$$\boxed{\mathrm{ord}_{10}\left(9^{9^9}\right) = 369\,693\,099},$$

$$\boxed{D_9(9^9) = 369\,693\,100}.$$

Furthermore, the congruences are obtained.

$$9^9 \equiv 0 \pmod 9, \qquad V_9(9^9) \equiv 0 \pmod 9,$$

$$369\,693\,099 \equiv 0 \pmod 9, \qquad 369\,693\,100 \equiv 1 \pmod 9.$$

The corresponding digit sums are

$$s_{10}(9^9) = 45, \qquad s_{10}(V_9(9^9)) = 36,$$

$$s_{10}(369\,693\,099) = 54, \qquad s_{10}(369\,693\,100) = 37.$$

The return $9 \to 1$ appears here with precise types:

1. 9^9 and the accumulated vacancy belong to the null class modulo nine;
2. its difference, the decimal order, remains in the null class and has digit sum 54;
3. the universal operator $D = \mathrm{ord} + 1$ adds the closure unit and publishes the positive class 1.

47.6.6 Domain of validity of the power–vacancy law

It is proved:

- the theorem potencia–vacancia for every n;
- the identity with the mechanical vacancy word;
- the exact evaluation in $n = 9^9$;
- the decimal order and length of 9^{9^9};
- the preceding congruences and digit sums.

It is not claimed that 9 is the only basis with an analogous property in every domain, that each external occurrence of 369 arises from this tower, or that the tower replaces the construction of the nine phases or the generation certificate for π, e, φ.

47.7 Operator Hierarchy, Reversibility, and Genealogical Memory

47.7.1 Translation, Dilation, and Exponentiation as Operational Levels

Elementary operations can be seen as actions:

$$T_a(x) = x + a, \qquad M_a(x) = ax, \qquad P_k(x) = x^k.$$

Their partial inverses are, respectively, subtraction, division, and extraction of roots. The passage from one operation to the next modifies the structure of the fibers.

47.7.2 Stratified reversibility in $\mathbb{Z}/9\mathbb{Z}$

proposition. En the anillo nonadico:

1. every translation T_a is bijective;
2. M_a is bijective if and only if $a \in R^\times$;
3. if $a \in \mathfrak{m}$, multiplication has nontrivial fibers and its image lies within $\mathfrak{m}$;
4. for $x \in \mathfrak{m}$, $x^k = 0$ for every $k \geq 2$;
5. over $R^\times \cong C_6$, the map P_k is an automorphism if and only if $\gcd(k, 6) = 1$.

Proof. The first assertion uses the additive group structure. The second is the characterization of the units. The third follows from $aR \subseteq \mathfrak{m}$. The fourth follows from $\mathfrak{m}^2 = 0$. The fifth is the condition for multiplication by k to be an automorphism of the cyclic group of order six. □

The result produces an exact hierarchy:

sum: global reversibility,

product: reversibility restricted to units,

power: cyclic dynamics on units and nilpotent collapse in $\mathfrak{m}$.

Roots are functions only when the corresponding power is injective on the chosen domain; on the total domain they are multibranch correspondences or empty relations.

47.7.3 Reading of the Topological Phase Kernel on the operator hierarchy

This stratification suggests a functional definition:

> TPK does not add an external operation to sum, product, and power. It preserves the sheet, quotient, carry, orientation, route, and boundary that each ordinary evaluation eliminates when identifying several histories with the same output.

Sum and product are the first two evaluations on the APP. A power is repeated composition of the product; a root selects or reconstructs branches of that composition. The record distinguishes the published value from the operatorial history producing it.

47.7.4 Logarithmic descent between consecutive levels of the hierarchy

In a positive domain,

$$\log(xy) = \log x + \log y, \qquad \log(x^k) = k \log x, \qquad \log(a^x) = x \log a.$$

The logarithm converts product into sum, power into product, and exponential into dilation. The defect appears when the subsequent reader discretizes:

$$\mathcal{L}_{10}(a^n) = \lfloor n \log_{10} a \rfloor + 1.$$

The integer part introduces the boundary-crossing calendar. For $a = 9$, that calendar is precisely the vacancy word.

47.7.5 Affine conjugation of translations and dilations

Translations and dilations satisfy

$$M_a T_b M_a^{-1} = T_{ab}$$

when a is invertible. This identity expresses that the product reorganizes the units of addition. When a ceases to be invertible, conjugation breaks because branches disappear. The radical $3, 6, 9$ localizes precisely that failure.

47.8 Path Algebra: Multiplicative Composition, Additive Action and Superposition

In a path algebra, addition aggregates alternatives and multiplication concatenates trajectories. For local weights $w(e)$,

$$K(x,y) = \sum_{\gamma : x \to y} \prod_{e \in \gamma} w(e).$$

If a coherent logarithmic branch is fixed,

$$\prod_{e \in \gamma} w(e) = \exp\left(\sum_{e \in \gamma} \log w(e) \right),$$

and, with the appropriate discrete action,

$$K(x,y) = \sum_{\gamma : x \to y} e^{-S(\gamma)/\hbar}.$$

The equation combines the operations:

- the product composes successive steps;
- the logarithm accumulates the local action;
- the exponential reconstructs the route weight;
- the suma agrega all the alternativas.

APP is the minimal support on which sum and product act on the same histories. TPK adds the data that prevent the sum over paths from becoming a sum over values without genealogy.

In this framework, a digit is the output of a reader; a value is the Archimedean projection of a section; an HMT number preserves the compatible family of paths that supports that value; a particle preserves that genealogy under a persistent spectral realization.

47.9 Triadic exponential family $J_\tau^2 = -\tau I$: elliptic, parabolic, and hyperbolic realizations of Euler's identity

47.9.1 Internal root of -1 in the finite branch

The Paley incidence matrix provides an operator A_W with

$$A_W^2 = -I.$$

This turns $\mathbb{F}_3^6$ into a three-dimensional space over

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1).$$

The root of -1 is not adjoined from outside as complex notation. It emerges as an endomorphism of an already constructed ternary support.

47.9.2 Continuous extension of the common quadratic operator and conservation of its spectral type

In the elliptic real branch, the generator J_{e} satisfies

$$J_{\mathrm{e}}^2 = -I,$$

and

$$e^{tJ_{\mathrm{e}}} = \cos t\, I + \sin t\, J_{\mathrm{e}}.$$

In particular,

$$e^{\frac{\pi}{2}J_{\mathrm{e}}} = J_{\mathrm{e}}, \qquad e^{\pi J_{\mathrm{e}}} = -I, \qquad e^{2\pi J_{\mathrm{e}}} = I.$$

Thus, J_{e} is the internal quarter-turn and Euler's equation is its half-turn closure.

47.9.3 Conjugation of the arithmetic, angular, and torsional realizations via the common quadratic operator

Let

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

This same set contains the real diameter $(-1, 1)$, the complex interior, the trigonometric boundary S^1, and the Poincaré hyperbolic metric

$$ds_{\mathrm{P}}^2 = \frac{4\ell^2 |dz|^2}{(1 - |z|^2)^2}.$$

The map

$$R_{\pi/2}(z) = iz$$

preserves $|z|$ and $|dz|$. By consiguiente, is simultaneamente:

- complex multiplication by $\sqrt{-1}$;
- trigonometric rotation of 90°;
- Euclidean isometry of the disk;
- Poincaré hyperbolic isometry;
- automorphism of its circular boundary.

The coincidence is literal: the same map acts on the same carrier; the metric reader changes.

47.9.4 Elliptic, parabolic, and hyperbolic geometries of the extended Euler equation

The generators

$$J_{+1}^2 = -I, \qquad J_0^2 = 0, \qquad J_{-1}^2 = I$$

produce

$$e^{tJ_{+1}} = \cos t\, I + \sin t\, J_{+1},$$

$$e^{tJ_0} = I + tJ_0,$$

$$e^{tJ_{-1}} = \cosh t\, I + \sinh t\, J_{-1}.$$

The same exponential series separates into three regimes according to the square of the generator: elliptic, parabolic, and hyperbolic. The specialization

$$e^{2\log\varphi\, J_{-1}} = F_{\mathrm{av}}^2$$

integrates the Fibonacci self-scale. Euler and Fibonacci appear as specializations of a common quadratic exponential family.

47.9.5 Common carrier, common operator, and quadratic exponential family

Geometric fusion is formulated at three levels:

1. **common carrier:** the unit disk contains the real diameter, the complex interior, and the circular boundary;
2. **common operator:** $z \mapsto iz$ is a quarter-turn, complex multiplication, and a hyperbolic isometry;
3. **common family:** the tritic exponential equation brings together the elliptic, parabolic, and hyperbolic branches.

The result does not identify the Euclidean and hyperbolic metrics. It states that they are distinct realizations on the same support and share the automorphism originating in the internal root of $-I$.

47.10 Restriction of internal rotation, edge holonomy and minimal torsion

The construction contains two positive and differentiated objects:

1. the internal generator J_{e}, with $J_{\mathrm{e}}^2 = -I$;
2. the global invariant

$$\Delta_4 = \frac{\pi - e\log\pi}{270}\left(1 + \frac{\pi}{729}\right),$$

denoted minimal torsion and used prior to species selection.

The authorial thesis adds that internal rotational dynamics, when published at the boundary, is absorbed as minimal torsion. The relation must not be written as a bare equality $J_{\mathrm{e}} = \Delta_4$: the objects have different types. Its natural form requires

$$\text{interior rotation} \xrightarrow{\mathrm{Res}_\partial} \text{boundary holonomy} \xrightarrow{\mathrm{Def}} \text{scalar defect}.$$

Let q_∂ be the boundary reader, ∇_∂ the induced connection, and e_∂ the discrete frame. The edge torsion has type

$$T_\partial = d_{\nabla_\partial} e_\partial.$$

The author's claim acquires probative form through a functional

$$\Lambda_\partial : \mathrm{Tors}(\partial X) \longrightarrow \mathbb{R}$$

such that

$$\Lambda_\partial(T_\partial) = \Delta_4$$

and a theorem deriving T_∂ from the restriction of J_{e} through the boundary operators of the TPK.

The internal root of $-I$, the quarter-turn, the extended Euler equation, and the formula for Δ_4 are proved in their respective domains. The passage from interior rotation to minimal boundary torsion requires the declared morphism Res_∂ and functional Λ_∂; it is not identified by a scalar equality.

47.11 Generational triality, conjugation, and mixing over the enriched state

The phrase of Isidor Isaac Rabi may serve as a historical reference, but it must not govern the title or constitute an autonomous section. The scientific question is:

> What internal mechanism distinguishes three stable orders of fermionic realization and preserves that distinction under color, conjugation, and mixing?

The HMT response is not "because TRIT has three values". The structure separates

$$\mathrm{flav}(X) = \big(\sigma_{ud}(X), g(X), \chi_{\mathrm{fina}}(X)\big),$$

where σ_{ud} distinguishes branches, g distinguishes physical generations, and χ_{fina} preserves orientation and memory within a branch. For quarks, the color orbit is added after flavor and before the character.

Triplicity is obtained as a consequence of the enriched structure; its effects on color, conjugation, and mixing are developed below:

1. color, flavor, generation, and mass are distinct readers of the same enriched state;
2. particle–antiparticle conjugation acts on the orientation without erasing the generational class;
3. CKM and PMNS transport already typed state bases; they do not create triplicity;
4. the neutrino classification must be derived from representation, occupation, and mixing, not by renaming depths $G0, \ldots, G4$.

That neutrinos are half-integer-spin leptons obeying fermionic statistics is a known physical classification. The structural contribution of HMT consists in obtaining internally why the neutral sector occupies that representation, why it admits three physical realizations, and how it couples to the mixing operator without reading its external angles.

47.12 Operator memory of the TPK: preservation of residue, quotient, carry, sheet, orientation, route, and boundary under projection

47.12.1 Loss of reversibility and genealogical lifting

In APP, different histories may produce the same residue. The product folds more histories than the sum; the power folds more than the product; the root attempts to reconstruct branches already identified. TPK preserves the datum that makes it possible to invert that loss:

residue, quotient, carry, sheet, orientation, route, boundary, memory.

An algebraic operation may be characterized not only by its value, but also by the type of genealogy that it preserves or destroys.

47.12.2 Projection of Compatible Histories onto the Archimedean Continuum

The real line appears when a family of compatible cylinders contracts under Archimedean evaluation. The real value is the shadow of the section, not its complete genealogy. The exceptional projection preserves another part as incidence and symmetry.

The path integral expresses the same principle:

$$\text{alternatives} \xrightarrow{\Sigma} \text{global amplitude}, \qquad \text{successive stages} \xrightarrow{\prod} \text{route weight}.$$

The logarithm converts multiplicative composition into additive action; the exponential returns it to amplitude. The discrete structure of the continuum can be formulated as genealogical conservation of all routes before their projection onto $\mathbb{R}$.

47.12.3 Distinction between numerical section and persistent realization of particle

A number and a particle share minimal ontological invariants: memory, orientation, persistence, closure, signature, and position in a family of prolongations.

The number publishes an Archimedean coordinate. The particle adds spectral realization, occupation, propagation, and stability. The passage HMT → MD changes the reader applied to the same enriched genealogy; it does not arbitrarily change the object.

47.13 Joint closure of the radical network, the power–vacancy law, Euler geometry, and the operative hierarchy

47.13.1 Theorem of the Nonadic Radical Network

The APP net $3, 6, 9$ is the visual representation of the nilpotent radical $(3) \subset \mathbb{Z}/9\mathbb{Z}$. Its phases 369, 693, and 936 realize the isomorphism $R/(3) \cong (3)$ induced by multiplication by three. The bands locate exactly the failure of invertibility of the product.

47.13.2 Power-void Theorem

For every $n \geq 1$, the number of decimal ranks lost by 9^n relative to 10^n is $\lceil n \log_{10}(10/9) \rceil$. Their increments form the mechanical vacancy word. In $n = 9^9$, the retained coordinate is 369 693 099 and the resulting length is 369 693 100.

47.13.3 Exponential Family Euler's Geometric Theorem

The internal root of $-I$ produces a quarter-turn. On the unit disk, the same map $z \mapsto iz$ is simultaneously complex multiplication, trigonometric rotation, and a Poincaré isometry. The tritic exponential equation extends this regime to the parabolic and hyperbolic regimes, incorporating the Fibonacci self-scaling in the latter.

47.13.4 Operatorial conservation principle between levels

The sum–product–power hierarchy is a hierarchy of fibers and loss of reversibility. The TPK is the lift that preserves the coordinates required to distinguish histories identified by the ordinary readers.

47.14 Typing conditions and incompatibility criteria of the nonadic vacancy theory

1. Do not identify 369 693 100 with a TPK word without declaring the reader between domains.
2. No confundir the ideal $\mathfrak{m}^2 = 0$ With $\mathfrak{m} \times \mathfrak{m}$.
3. Do not confuse modulo nine with digital root: $9 \mapsto 0$ in the former and $9 \mapsto 9$ in the latter.
4. Do not use the threefold character of TRIT as a sufficient proof of three physical generations.
5. Do not turn $J^2 = -I$ into Δ_4 without the boundary map and the torsion functional.

6. Do not present 369 as a prefix of 9^{9^9}: it pertains to its order and its decimal length.

7. Do not reduce the ten thousand digits to an order-of-magnitude test: they certify extensive truncations of a parametrizable process whose thesis is indefinite prolongation.

47.15 Joint consequences for the radical net, the operational memory, and the edge torsion

The principal finding is not that a tower visually contains the digits $3, 6, 9$. The nonadic power and the vacancy calendar are two presentations of the same logarithmic defect, while APP localizes $3, 6, 9$ as the radical where multiplication loses invertibility. The tower connects both extremes:

$$\text{nonadic radical} \longrightarrow \text{power} \longrightarrow \text{logarithm} \longrightarrow \text{vacancy} \longrightarrow \text{decimal order} \longrightarrow \text{closure } 9 \to 1.$$

The extended Euler equation adds the second direction: the internal root of $-I$ converts the discrete structure into rotational dynamics and gathers, on a common support, the real line, the trigonometric circle, and the hyperbolic disk. The later passage to minimal torsion must preserve the boundary map already developed in the corpus, not replace it with a nominal identification.

Chapter 48

Thermal constants and radiation: Boltzmann, Wien, Stefan–Boltzmann and spectral thermodynamics of the photon gas

48.1 Relation termica intrinseca of the escala 108

The chronological structure 108 fixes

$$k_B \Theta_{\mathrm{clk}} \log 3 = \frac{h_{\mathrm{int}}}{108 t_0} = \frac{2\pi \hbar_{\mathrm{int}}}{108 t_0}. \tag{48.1}$$

This equality does not separately determine a numerical coordinate of k_B and another of Θ_{clk} without an additional normalization. It does determine their product as a section of energy. The factor $\log 3$ has three compatible antecedents:

$$\log 3 = C_1 + C_2 - 2C_0 = \frac{S_{\widehat{\mathbb{T}}}}{k_B} = \frac{E_P}{k_B \Theta_{\mathrm{clk}}}. \tag{48.2}$$

The three equalities correspond to residual, informational and thermodynamic realizations of the same scalar. The coincidence of value does not identify their respective domains.

48.2 Spectral formulations of Wien's displacement law

For the spectral density per wavelength, the maximum satisfies

$$5(1 - \mathrm{e}^{-x_5}) = x_5, \qquad x_5 = 5 + W_0(-5\mathrm{e}^{-5}).$$

With equation (48.1),

$$\lambda_{\max}(\Theta_{\mathrm{clk}}) = \frac{108 \log 3}{x_5}\, \ell_0. \tag{48.3}$$

For the frequency density,

$$3(1 - \mathrm{e}^{-x_3}) = x_3, \qquad x_3 = 3 + W_0(-3\mathrm{e}^{-3}),$$

and

$$\nu_{\max}(\Theta_{\mathrm{clk}}) = \frac{x_3}{108 t_0 \log 3}. \tag{48.4}$$

The maxima are not reciprocals, since the spectral densities transform with the corresponding Jacobian. The quotient x_5/x_3 belongs to the structure of the parameterization and does not represent an inconsistency.

48.3 Stefan–Boltzmann law and thermodynamics of the photon gas

Evaluating the radiation law at Θ_{clk} produces the intrinsic flux

$$j_\star(\Theta_{\mathrm{clk}}) = \frac{2\pi^5}{15\, 108^4 (\log 3)^4}\, \frac{h_{\mathrm{int}}}{\ell_0^2 t_0^2}, \tag{48.5}$$

in the declared normalization. The Stefan–Boltzmann constant is not inserted as a primitive: it results from integrating the bosonic spectrum once action, propagation, and temperature have been fixed. The

spectral laws employed are those of Planck and Wien [103, 134]; HMT fixes here the section on which they are evaluated, not their constants by retrospective comparison.

In units of the Planck scale,

$$n_\gamma \ell_P^3 = \frac{2\zeta(3)}{\pi^2 \log^3 3}, \tag{48.6}$$

$$\frac{u_\gamma \ell_P^3}{E_P} = \frac{\pi^2}{15 \log^4 3}, \tag{48.7}$$

$$\frac{s_\gamma \ell_P^3}{k_B} = \frac{4\pi^2}{45 \log^3 3}. \tag{48.8}$$

$\zeta(3)$ represents the third-order bosonic moment, whereas $\log 3$ determines the energy associated with the TRIT decision. The equation preserves the distinction between the two invariants.

The quantum thermal conductance is expressed as

$$\frac{\kappa_Q t_0}{k_B} = \frac{\pi^2}{324 \log 3}. \tag{48.9}$$

48.4 Metrological Confluence of Thermal and Radiative Invariants

Object	Equation internal	Significado estructural
R_K	$Z_{\text{vac}}/(2\alpha)$	bidirectionally invariant quantum resistance
G_0	$4\alpha/Z_{\text{vac}}$	conductancia complementaria
Φ_0	$\sqrt{hZ/(8\alpha)}$	flow preserving orientation of action
K_J	Φ_0^{-1}	reciprocal relation between frequency and flux
E_P	$k_B\Theta \log 3$	energy of the cycle and TRIT informational content
Wien's law	$108 \log 3/x_5$	endpoint spectral in the scale chronological
Photonic Apéry	$2\zeta(3)/(\pi^2 \log^3 3)$	number density of the bosonic field

Remark 48.1 (Negative control). The block is falsified if a change of spectral variable improperly preserves the same Wien extremum, if $\zeta(3)$ replaces $\log 3$, or if separate values are assigned to k_B and Θ without specifying a chart. The identities equations (36.37) y (40.6) provide direct algebraic controls.

Logical Scope and Domain of Validity

Electric quanta and Planck–TRIT confluence: *Exact internal theorem*. Wien, Stefan–Boltzmann, and the photonic gas: *Exact mathematical composition* with the declared bosonic continuum. The SI chart remains an *External metrological comparison*.

48.5 Boltzmann constant, information, and thermal transduction

48.5.1 Dimensional structure of transduction

The structural definition of the Boltzmann constant uses the temperature and energy lines. The positive transduction

$$k_B : \mathcal{L}_\Theta \longrightarrow \mathcal{L}_E$$

assigns a temperature its energy scale. In the tritic normalization, the cycle's temperature section satisfies

$$k_B \Theta_{\mathrm{clk}} = \frac{E_P}{\log 3}.$$

The right-hand side is fixed by the previously constructed action and period. The factor $\log 3$ comes from the informational content of a uniform distribution on the three local states.

Let u_E, u_Θ be positive bases, and write $E_P = \epsilon_P u_E$, $\Theta_{\mathrm{clk}} = \theta_{\mathrm{clk}} u_\Theta$, and $k_B(u_\Theta) = b\, u_E$. Then

$$b\, \theta_{\mathrm{clk}} = \frac{\epsilon_P}{\log 3}.$$

For $u'_E = s_E u_E$ and $u'_\Theta = s_\Theta u_\Theta$, the coordinates transform by

$$b' = \frac{s_\Theta}{s_E} b, \qquad \theta'_{\mathrm{clk}} = \frac{\theta_{\mathrm{clk}}}{s_\Theta}, \qquad \epsilon'_P = \frac{\epsilon_P}{s_E}.$$

The energy identity remains covariant. This formulation combines the individual structure of the transducer with the normalization of the section on which it acts.

Remark 48.2 (Thermal coordinate and choice of bases). The transducer and the preceding energy identity are covariant. An individual numerical value of k_B is referred to the positive bases u_E, u_Θ; its transformation is $b' = (s_\Theta/s_E)b$. Internal selection of a numerical coordinate therefore requires, in addition, a construction fixing those bases. The energy identity does not by itself select that choice. The metrological coordinate does not define action or period.

The distinction helps interpret entropy. The functional $s(\rho) = -\operatorname{Tr}(\rho \log \rho)$ is dimensionless. Dimensional entropy is $S(\rho) = k_B s(\rho)$, understood on the corresponding line. The quantity ΘS belongs to the energy line. This preserves the mathematical meaning of the equalities combining information, temperature, and action.

In particular, the continuation information (73.61) admits the dimensional publication $S_n^{\mathrm{gen}}(x) = k_B s_n(x)$. The admissible branches and their measure determine s_n; the transducer fixes the entropy scale. For a uniform distribution over three alternatives, the value is $k_B \log 3$. When the prolongations have different weights, the weighted expression preserves that structural difference. The Boltzmann constant can be defined here by its function: it is the dimensional coefficient that transports the information of the admitted distribution to the entropy line and, under a specified temperature, links that information to energy.

Action enters before thermal publication. The section $\hbar_a$ converts the evolution frequency into energy; the section h_a expresses the action accumulated during a turn. The equality $k_B \Theta_{\mathrm{clk},a} \log 3 = h_a / T_{108}$ brings both functions together: energy of the period and local information of the tritic dynamics. A thermodynamic identification additionally preserves the statistical state, the temperature and the physical exchange or erasure operation under study.

48.5.2 Structural definition and function of the Boltzmann constant

Definition 48.3 (Thermal transducer of energy per decision)**.** Fix a positive thermal section $\Theta_{\mathrm{clk},a} \in \mathcal{L}_\Theta$. The Boltzmann transducer is the positive linear map $k_B : \mathcal{L}_\Theta \to \mathcal{L}_E$ realizing

$$k_B(\Theta_{\mathrm{clk},a}) = \frac{\epsilon_a}{\log 3}, \qquad \epsilon_a = \frac{h_a}{108 t_0}. \tag{48.10}$$

Action and duration determine ϵ_a; the uniform tritic decision determines its informational normalization.

Uniqueness, proved in section 70.2.7, follows by prescribing the image of a nonzero section of a line. Equations (70.26) and (70.27) specify, respectively, its coordinate in declared bases and compatibility of the same transducer with both orientations. Determination of the product $k_B\Theta_{\mathrm{clk},a}$ precedes individualization: the latter retains as explicit data the thermal section, the bases, and, when internal selection is claimed, its constructive reader. The pressure in (70.30), under the hypotheses enumerated there, normalizes the energy parameter $\beta = (k_B\Theta)^{-1}$.

The physical function of k_B is to publish the information of a distribution dimensionally: $S = k_B s$. Evaluating that entropy at a temperature produces the energy ΘS. A uniform triple contributes $\log 3$ nats; a distribution of continuations contributes its weighted entropy; a quantum reduction contributes the entropy of the specified bipartition. The transducer acts on those contents while preserving their domains. Genealogical memory determines what information each reading retains; the thermodynamic balance specifies what operation exchanges or erases it. This separation explains the relation between reversibility, entanglement and decision energy.

Chapter 49

Electroweak constants and mixing: angular determinant, relation W/Z, Weinberg angle and rational CKM reflection

49.1 Angular determinant and Weinberg electroweak relation

The determinant of T is

$$\det T = q_+ q_- = D_A. \tag{49.1}$$

The constitutive sector of the vacuum evaluates $F(T)^2$, where $F(z) = (1 - z^{90})/(1 - z^{120})$, whereas the electroweak angular sector evaluates $\det T$. These are two functions of a common antecedent, not a horizontal factorization between constants. This distinction is developed in the section 49.1.

Remark 49.1 (Negative control). The block is refuted if any of these exact identities is violated: positivity of $I - T^{120}$, equality equation (39.8), relations equation (39.5), or compatibility $E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m$. An SI coincidence cannot compensate for an operator failure.

Logical Scope and Domain of Validity

Operator, spectrum, completion $3 \to 4$, parity, amplification, and constitutive sections: *Exact internal theorem*. The formulation that unites them in a single operator is *New formalization*; the angular data and lines are an *Antecedent demonstrated result* of the preceding HMT development and are declared here as explicit initial structure.

49.2 Constitutive and Determinantal Functions of the Angular Operator

Recall the two-sheet operator of the chapter 34,

$$T = q_+ P_+ + q_- P_-, \qquad q_- = \mathrm{e}^{-(A_{\mathrm{rad}} - C^*_{\mathrm{rad}})}, \qquad q_+ = \mathrm{e}^{-(A_{\mathrm{rad}} + C^*_{\mathrm{rad}})}. \tag{49.2}$$

The electromagnetic vacuum applies the function

$$F(T)^2, \qquad F(z) = \frac{1 - z^{90}}{1 - z^{120}}, \tag{49.3}$$

while the electroweak angular sector applies the determinant

$$\boxed{\det T = q_+ q_- = \mathrm{e}^{-2A_{\mathrm{rad}}} =: D_A.} \tag{49.4}$$

They are two functional realizations of the common antecedent (A, C^*, T). There is no causal arrow $F(T)^2 \to \det T$ or its inverse. Structural unity resides in T, not in subsequently identifying permeability, permittivity, and θ_W.

Under the sheet involution $C^* \mapsto -C^*$, the eigenvalues q_+, q_- are exchanged. Therefore D_A and the constitutive speed are even, while $(\widehat{\mu}, \widehat{\varepsilon})$ is exchanged and $\widehat{Z}$ is inverted. The following electroweak relation uses the even invariant.

49.3 Exact relation between sectors W and Z

The angular coordinate of duration–perduration is

$$\chi_{\mathrm{dur}}(n_A, s_C) = \exp\!\left(n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}}\right). \tag{49.5}$$

This coordinate is not introduced as an isolated label. Its origin is

$$\begin{aligned} \mathrm{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow \text{surviving extension} \longrightarrow \text{route} \\ \longrightarrow \text{ledger} \longrightarrow (n_A, s_C, \ldots) \longrightarrow \chi_{\mathrm{dur}}. \end{aligned} \tag{49.6}$$

The signatures of W, Z and H are typed quantities obtained through that chain; this chapter adopts them without selecting them from their terminal masses. In the established angular realization, the charged and neutral sectors satisfy

$$(n_A, s_C)_W = (94, -1), \qquad (n_A, s_C)_Z = (95, -1). \tag{49.7}$$

This assertion refers to the common angular component. The additional refinement coordinates of the General Mass Law are not eliminated unless the physical transducer proves their cancellation.

Theorem 49.2 (Weinberg defect in the physical-mass scheme of the angular chart)**.** *If W and Z are evaluated on the same scale section and share the refinement that accompanies equation (49.7), then*

$$\boxed{\frac{m_W^{\angle}}{m_Z^{\angle}} = \mathrm{e}^{-A_{\mathrm{rad}}} = \sqrt{D_A}, \qquad \left(\frac{m_W^{\angle}}{m_Z^{\angle}}\right)^2 = D_A.} \tag{49.8}$$

In the physical-mass scheme,

$$\boxed{\sin^2\theta_W^{\mathrm{OS,HMT}} = 1 - D_A = 0.22487088075284356981111590 35\ldots.} \tag{49.9}$$

Además,

$$\frac{m_W^{\angle}}{m_Z^{\angle}} = 0.88041417483316136701288854 59\ldots. \tag{49.10}$$

Proof. By equations (49.5) y (49.7),

$$\frac{\chi_{\mathrm{dur}}(94, -1)}{\chi_{\mathrm{dur}}(95, -1)} = \exp(-A_{\mathrm{rad}});$$

the coordinate C^* cancels because both sectors share $s_C = -1$. Squaring and using equation (49.4) yields D_A. The definition of physical masses $s_W^2 = 1 - m_W^2/m_Z^2$ yields equation (49.9). □

The principal content is the identity between two internal constructions: the determinant of the angular operator and the one-step difference in the realization W/Z. Promotion to a physical gauge selector requires an intertwiner that carries the sheets of T to the neutral basis (W^3, B) and demonstrates which refinements are preserved or cancel.

49.4 Gauge couplings in rationalized normalization

We define the dimensionless gauge coupling

$$e_g := \sqrt{4\pi\alpha_{\mathrm{HMT}}} = 0.30282212087175264276624426 89\ldots. \tag{49.11}$$

e_g must not be confused with a dimensional charge section e_{int}: the former is a coupling in natural units; the latter lives in the charge line of Dimensional Mechanics.

Let

$$s_W^2 = 1 - D_A, \qquad c_W^2 = D_A. \tag{49.12}$$

The convention electrodebil racionalizada to level of arbol gives

$$g = \frac{e_g}{s_W} = 2\sqrt{\frac{\pi\alpha_{\mathrm{HMT}}}{1-D_A}} = 0.63858834273345742836470038 09\ldots, \tag{49.13}$$

$$g' = \frac{e_g}{c_W} = 2\sqrt{\frac{\pi\alpha_{\mathrm{HMT}}}{D_A}} = 0.34395416331085024293623192 57\ldots, \tag{49.14}$$

$$\frac{g'}{g} = \sqrt{D_A^{-1} - 1} = 0.53861641419660935949966109 33\ldots. \tag{49.15}$$

Proposition 49.3 (Tree-level custodial relation)**.** *At the same interface,*

$$\rho_{\mathrm{EW}}^{(0)} := \frac{(m_W^{\angle})^2}{(m_Z^{\angle})^2 c_W^2} = 1. \tag{49.16}$$

Proof. By equations (49.8) y (49.12), both the relative numerator and c_W^2 are D_A. □

The equations equations (49.13) y (49.14) are exact compositions once the physical mass scheme, rationalization, and scale have been declared. They do not assert that a change of renormalization scheme leaves their numerical values unchanged.

49.5 Fermi Constant in the Approximation at Tree Level

In natural units, integration of the charged propagator at low energy yields

$$\frac{G_F^{(0)}}{\sqrt{2}} = \frac{g^2}{8m_W^2}. \tag{49.17}$$

Substituting equation (49.13) gives the composition

$$\boxed{G_F^{(0)} = \frac{\pi\alpha_{\mathrm{HMT}}}{\sqrt{2}(1-D_A)m_W^2}.} \tag{49.18}$$

By adopting the previously obtained mass

$$m_W^{\mathrm{HMT}} = 80.38159255022 1\ldots\ \mathrm{GeV}, \tag{49.19}$$

the terminal evaluation is

$$\boxed{G_F^{(0)} = 1.1157162817659203186621784 \times 10^{-5}\ \mathrm{GeV}^{-2}.} \tag{49.20}$$

The difference with respect to a renormalized Fermi constant is not used to choose any coefficient. Under the convention

$$G_F = \frac{\pi \alpha_{\mathrm{HMT}}}{\sqrt{2}(1 - D_A) m_W^2 (1 - \Delta r)}, \tag{49.21}$$

the remaining obligation is to derive Δr from the internal sector of loops, states, and scale. Opening the observed value of G_F to define Δr would be a calibrated reconstruction, not a prediction.

49.6 Higgs Coupling Oriented Dependence

In the angular chart, the relevant routes are

$$(n_A, s_C)_H = (97, 9), \qquad (n_A, s_C)_W = (94, -1). \tag{49.22}$$

Therefore,

$$\boxed{\frac{m_H^{\angle}}{m_W^{\angle}} = \frac{\chi_{\mathrm{dur}}(97, 9)}{\chi_{\mathrm{dur}}(94, -1)} = \exp\left(3A_{\mathrm{rad}} + \frac{5}{3} C^*_{\mathrm{rad}}\right) = 1.558720872123255485 64\ldots.} \tag{49.23}$$

Here, C^* does not cancel: the scalar sector preserves the orientation that the determinant $D_A = \mathrm{e}^{-2A_{\mathrm{rad}}}$ does not distinguish.

Let us take the convention of the potential

$$V(H) = -\mu_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2, \tag{49.24}$$

for which

$$m_H^2 = 2\lambda_H v^2, \qquad m_W^2 = \frac{g^2 v^2}{4}. \tag{49.25}$$

Corollary 49.4 (Oriented Higgs self-coupling)**.** *At the declared tree-level interface,*

$$\boxed{\lambda_H = \frac{\pi \alpha_{\mathrm{HMT}}}{2(1 - D_A)} \exp\left(6A_{\mathrm{rad}} + \frac{10}{3} C^*_{\mathrm{rad}}\right) = 0.12384791154824668046 62\ldots.} \tag{49.26}$$

Además,

$$G_F^{(0)} m_H^2 = \sqrt{2}\, \lambda_H. \tag{49.27}$$

Proof. Dividing the two equations in equation (49.25) gives $m_H^2/m_W^2 = 8\lambda_H/g^2$. With $g^2 = 4\pi\alpha_{\mathrm{HMT}}/(1 - D_A)$ and the square of equation (49.23), we obtain equation (49.26). On the other hand, $G_F^{(0)} = 1/(\sqrt{2}v^2)$ and $m_H^2 = 2\lambda_H v^2$ imply equation (49.27). □

The value of λ_H precedes the incorporation of radiative corrections and depends on the declared convention and scale. Its structural interest is that it separates two memories: the common gauge sector uses the even invariant D_A, whereas the Higgs route explicitly retains C^*.

49.7 Determinantal closure of the electroweak sector: mixing, couplings, and constitutive spectrum

The complete chain of this chapter is

$$\boxed{\begin{aligned}(A, C^*) &\longrightarrow T \longrightarrow \begin{cases} F(T)^2 & \longrightarrow (\widehat{\mu}, \widehat{\varepsilon}, \widehat{Z}, \widehat{c}), \\ \det T = D_A & \longrightarrow (s_W^2, c_W^2), \end{cases} \\ (D_A, \alpha_{\mathrm{HMT}}) &\longrightarrow (e_g, g, g'), \\ (D_A, \alpha_{\mathrm{HMT}}, m_W) &\longrightarrow G_F^{(0)}, \\ (A, C^*, D_A, \alpha_{\mathrm{HMT}}) &\longrightarrow \left(\frac{m_H}{m_W}, \lambda_H\right). \end{aligned}} \tag{49.28}$$

The figure always appears at the end. Neither $\sin^2\theta_W$, nor G_F, nor λ_H is introduced as a target value for selecting A, C^* or the route signatures.

Proposition 49.5 (Main Result)**.** *The same angular operator determines the vacuum through a rational function of its sheets and the electroweak defect through its determinant. The one-step difference between W and Z turns that determinant into $\sin^2\theta_W^{\mathrm{OS}} = 1 - D_A$; the Higgs route, in contrast, preserves the counterangle and preserves the orientation eliminated by the determinant.*

Remark 49.6 (Negative control). The Weinberg relation is refuted in its domain if the angular signatures W/Z do not differ only by $n_A : 94 \to 95$, if a refinement coordinate does not cancel at the declared interface, or if $(m_W^{\angle}/m_Z^{\angle})^2 \neq D_A$. Gauge promotion is blocked if no intertwiner with the basis (W^3, B) exists. The figures of g, g', G_F, λ_H change if the scheme or scale is altered without recording it. Using observed G_F or m_H to select Δr, the signatures, or C^* introduces feedback from the target value and invalidates the derivation. Conflating e_g with the dimensional charge constitutes a type error.

Logical Scope and Domain of Validity

Angular determinant, quotient W/Z and complementary Weinberg defect: *Exact internal theorem in the conditional angular chart*. Couplings $g, g', \rho_{\mathrm{EW}}^{(0)}$, Fermi and Higgs: *Exact composition at the tree-level electroweak interface*. The gauge-basis intertwiner and the internal correction Δr: *Open frontier with a refutation criterion*. Fermi and Higgs are not independent constants selected without target values: they are exact compositions using the signatures, the quantity m_W and the declared tree-level convention. Subsequent conventional values are *External metrological comparison*.

49.8 Involution of leaves in 4 representations

The involution $C^* \mapsto -C^*$ organizes four distinct responses:

$$\begin{aligned} S_{\mathrm{bi}}(A, -C^*) &= S_{\mathrm{bi}}(A, C^*), \\ \Gamma_{6\to 8}(A, -C^*) &= -\Gamma_{6\to 8}(A, C^*), \\ (\widehat{\mu}, \widehat{\varepsilon}) &\longleftrightarrow (\widehat{\varepsilon}, \widehat{\mu}), \\ \widehat{c} &\longmapsto \widehat{c}, \qquad D_A \longmapsto D_A. \end{aligned} \tag{49.29}$$

The bilayer information is even; the areal orientation is odd; the two constitutive quantities form a doublet, and the determinants remain fixed. This parity table precedes any interpretation as a physical symmetry.

49.9 Involutive rational transformation in the CKM parameter space

Let

$$M_{\mathrm{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \qquad \det M_{\mathrm{CKM}} = -\frac{3857}{6}. \tag{49.30}$$

The inverse exists on $\mathbb{Q}$ and is

$$M_{\mathrm{CKM}}^{-1} = \begin{pmatrix} 1528/3857 & 3380/3857 & 801/3857 \\ 75/3857 & 176/3857 & -150/3857 \\ 6/551 & -30/551 & -12/551 \end{pmatrix}. \tag{49.31}$$

With

$$h = \frac{C^*}{6}, \qquad \Delta = \frac{180}{\pi}\Delta_4, \qquad \boldsymbol{\theta} = M_{\mathrm{CKM}}(A, h, \Delta)^T, \tag{49.32}$$

the odd coordinate enters in the direction

$$c_h = \begin{pmatrix} -5 \\ 7 \\ -20 \end{pmatrix}. \tag{49.33}$$

If $c_A = (2,0,1)^T$ and $c_\Delta = (28, -25/2, -2/3)^T$, the three columns (c_A, c_h, c_Δ) are precisely the basis transported by M_{CKM}. The second row of equation (49.31) is the covector ℓ_h that extracts the odd coordinate:

$$\ell_h c_A = 0, \qquad \ell_h c_h = 1, \qquad \ell_h c_\Delta = 0. \tag{49.34}$$

Theorem 49.7 (Reflexion CKM inducida by the sheet)**.** *Let*

$$D_h = \mathrm{diag}(1, -1, 1), \qquad \ell_h = \frac{1}{3857}\begin{pmatrix} 75 & 176 & -150 \end{pmatrix}. \tag{49.35}$$

The involution induced on the space of $\boldsymbol{\theta}$ is

$$\boxed{\mathcal{R}_{\mathrm{CKM}} = M_{\mathrm{CKM}} D_h M_{\mathrm{CKM}}^{-1} = I - 2c_h\ell_h.} \tag{49.36}$$

Satisfies

$$\mathcal{R}_{\mathrm{CKM}}^2 = I, \qquad \det \mathcal{R}_{\mathrm{CKM}} = -1, \qquad \mathcal{R}_{\mathrm{CKM}} M_{\mathrm{CKM}} = M_{\mathrm{CKM}} D_h. \tag{49.37}$$

Proof. Multiplication of equations (49.30) y (49.31) gives the identity; in particular, equation (49.34) proves that $c_h\ell_h$ is the rank-one projector onto $\mathbb{Q}c_h$ along $\mathrm{span}\{c_A, c_\Delta\}$. The exact product $\ell_h c_h = 1$ implies

$$(I - 2c_h\ell_h)^2 = I - 4c_h\ell_h + 4c_h(\ell_h c_h)\ell_h = I.$$

The operator fixes the hyperplane $\ker \ell_h = \mathrm{span}\{c_A, c_\Delta\}$ pointwise and changes the sign of c_h. Its eigenvalues are therefore $1, 1, -1$; hence $\det \mathcal{R}_{\mathrm{CKM}} = -1$ and $\mathrm{Tr}\, \mathcal{R}_{\mathrm{CKM}} = 1$. Finally,

$$\mathcal{R}_{\mathrm{CKM}} M_{\mathrm{CKM}} = M_{\mathrm{CKM}} D_h M_{\mathrm{CKM}}^{-1} M_{\mathrm{CKM}} = M_{\mathrm{CKM}} D_h,$$

which proves the entanglement identity. □

The explicit matrix is

$$\mathcal{R}_{\mathrm{CKM}} = \begin{pmatrix} 4607/3857 & 1760/3857 & -1500/3857 \\ -150/551 & 199/551 & 300/551 \\ 3000/3857 & 7040/3857 & -2143/3857 \end{pmatrix}. \tag{49.38}$$

Although $\mathcal{R}_{\mathrm{CKM}}$ is not an orthogonal reflection for the Euclidean product of the three angular coordinates, it is for the transported rational metric

$$G_{\mathrm{CKM}} = (M_{\mathrm{CKM}}^{-1})^T M_{\mathrm{CKM}}^{-1}, \qquad \mathcal{R}_{\mathrm{CKM}}^T G_{\mathrm{CKM}} \mathcal{R}_{\mathrm{CKM}} = G_{\mathrm{CKM}}. \tag{49.39}$$

Indeed, in the basis (c_A, c_h, c_Δ), the metric is the identity and the involution is D_h. Hence the term "reflection" has precise metric content and does not merely describe that $\mathcal{R}_{\mathrm{CKM}}^2 = I$. Phase $\delta_{\mathrm{CP}} = 9A$ remains fixed under this reflection. Therefore, $\mathcal{R}_{\mathrm{CKM}}$ is not physical CP conjugation, but rather the transport of HMT sheet inversion to mixing space.

The rational inverse of M_{CKM} recovers

$$A = \frac{1528\theta_{12} + 3380\theta_{23} + 801\theta_{13}}{3857}, \tag{49.40}$$

$$C^* = \frac{6(75\theta_{12} + 176\theta_{23} - 150\theta_{13})}{3857}. \tag{49.41}$$

The third coordinate is also recovered without loss:

$$\Delta = \frac{6\theta_{12} - 30\theta_{23} - 12\theta_{13}}{551}. \tag{49.42}$$

These equations are coordinate changes. They do not turn CKM into a retrospective cause of A, C^* or of γ_{HMT}.

As a terminal recognition, the previous constructor determines

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta_{\mathrm{CP}}) = (13.003313589509^\circ, 2.398056157332^\circ, 0.213977382244^\circ, 65.676173123554^\circ). \tag{49.43}$$

The rational reflection is proved before using these decimals.

Proposition 49.8 (Main Result)**.** *The transition* $6 \to 8$ *is first formalized through the inclusion* $\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8$*, from which* $90, 120$ *and the defect* $-1/12$ *are obtained. The six-trit word then provides the bijection* $729 \leftrightarrow 729$*, and* $\mathcal{J}_{6\to 8}$ *transports the two collective modes to the boundary. On that antecedent, area and the modular Hamiltonian exactly reconstruct the two families; thermofield-double purification determines the entanglement entropy, and the finite first law preserves the relative-entropy defect. In CKM, sheet inversion becomes a rational rank-one odd reflection.*

Remark 49.9 (Negative control). The initial genealogy fails if $\iota_{6,8}$ is not injective, if the counts are not $90/120$, if $\delta_{6,8}^{\mathrm{inc}} \neq -1/12$, if β_{729} is not bijective or if it is falsely presented as an additive isomorphism. Collective transport fails if $\mathcal{J}_{6\to 8} D_{\mathrm{inc}} \neq D_{\mathrm{area}} \mathcal{J}_{6\to 8}$. Reconstruction fails if the determinant of the boundary observables is not thirty or if equation (72.8) does not recover the occupations. The TFD fails if its partial trace is not ρ_A. The first law does not extend to finite changes without the relative entropy of equation (74.6). Type III_{λ_∂} is invalidated if a channel does not persist, if there is no compatible product or if the ratio group changes. The CKM reflection is refuted by any of $\mathcal{R}^2 \neq I$, $\det \mathcal{R} \neq -1$ or $\mathcal{R}M \neq MD_h$. A formula $\boldsymbol{\theta} = f(\gamma_{\mathrm{HMT}})$ without an intertwiner violates the typed independence of the two functional realizations of the common antecedent.

Logical Scope and Domain of Validity

Inclusion $H \subset O$, counts $90/120$, two normalizations of the defect $-1/12$, bijection of six trits $729 \leftrightarrow 729$, collective intertwiner, scale a_0, finite operators, area spectrum, TFD, oriented defect, local and finite first law, and CKM reflection: *Internally exact theorem with its explicit starting structures*. Classification III_{λ_∂}: *Theorem with explicit persistence hypotheses*. Interpretation of the thermofield-double copy as a physical exterior: *Conditional physical realization*. The magnitude γ_{HMT} is adopted as *Proved antecedent result*; it is not rederived in this volume.

The electroweak link starts from the same angular operator that produced the vacuum constitutive relations, but uses another functional.

To obtain μ, ε, Z, and c it was necessary to preserve the full spectrum. It was necessary to distinguish the two sheets.

The determinant does the opposite:

$$D_A = \det T = q_+ q_-.$$

By multiplying the eigenvalues, the determinant integrates the two orientations into a common invariant angular magnitude. It publishes the even part and reserves the leaf identity in the oriented memory.

By eso D_A is even under the inversion

$$C^* \longmapsto -C^*.$$

The sector W/Z has precisely that structure. The routes of W and Z are adjacent in the principal angular coordinate and share the same oriented coordinate. Upon forming the quotient, the contribution of C^* cancels.

Remains

$$\left(\frac{m_W}{m_Z}\right)^2 = D_A,$$

and therefore

$$\sin^2 \theta_W = 1 - D_A.$$

The identity reveals that the Weinberg angle is the complementary defect of the angular determinant.

The determinant measures the common part that survives when the two sheets are collapsed into their product. $1 - D_A$ measures the complementary part required for mixing.

It may be stated as follows: the vacuum preserves the entire response; the electroweak sector uses the even compression of that response.

The same angular antecedent produces two distinct functions:

- the complete functional calculus generates the constitutive relations;
- the determinant generates the mixing invariant.

This explains why the vacuum and the Weinberg angle are two distinct functional realizations of the same angular antecedent.

From that point onward, the couplings g and g' are, at the stated level, the two projections of the same rationalized electromagnetic coupling onto the sine and cosine directions of the mixing:

$$g^2 = \frac{4\pi\alpha_{\mathrm{HMT}}}{1 - D_A}, \qquad g'^2 = \frac{4\pi\alpha_{\mathrm{HMT}}}{D_A}.$$

The tree custodial identity

$$\rho_{\mathrm{EW}}^{(0)} = 1$$

states that the same angular partition organizes mass and coupling coherently.

The Fermi constant appears later as the effective low-energy response of the charged channel:

$$G_F^{(0)} = \frac{\pi \alpha_{\mathrm{HMT}}}{\sqrt{2}(1 - D_A) m_W^2}.$$

The Fermi constant appears as a composition of the fine structure, the angular defect, and the carrier scale W.

The Higgs sector introduces a decisive difference. Its route has an oriented coordinate distinct from the route of W, so C^* persists. The scalar self-coupling publishes the orientation reserved by the determinant.

The gauge sector is even: it uses D_A.

The scalar sector is oriented: it retains C^*.

This division is physically suggestive. Gauge fields organize the common part of the transformation; the scalar sector preserves the memory of the broken orientation.

Radiative corrections belong to a later level and must be constructed prospectively from the internal loops, states, and scales, with G_F as output and terminal evaluation. The causal skeleton is already present: vacuum, mixing, couplings, and Fermi response follow from different readings of the same enriched angular state.

49.10 Electroweak Angular Closure and Compatibility Conditions of the Gauge Transducer

Let

$$c_W = \sqrt{D_A}, \qquad s_W = \sqrt{1 - D_A}.$$

The rotation

$$\begin{pmatrix} Z \\ A \end{pmatrix} = \begin{pmatrix} c_W & -s_W \\ s_W & c_W \end{pmatrix} \begin{pmatrix} W^3 \\ B \end{pmatrix}$$

is orthogonal and preserves orientation. The identities

$$\left(\frac{m_W}{m_Z} \right)^2 = D_A, \qquad \sin^2 \theta_W = 1 - D_A$$

are closed in the angular chart. Gauge promotion requires that the transducer preserve the kinetic form, the charges, and the Ward identities; the correction Δr belongs to radiative dynamics and is not obtained by fitting it to G_F.

Chapter 50

Intrinsic derivation of G and contragredient reciprocity $G \leftrightarrow \hbar$: phase closure, Compton and Planck scales, and invariance of $G\hbar$

50.1 Causal composition of the phase, action, and length invariants in the derivation of G

Gravity is organized here into four strata that must not be merged:

$$\begin{aligned} \text{CT108 chronological structure and action lines} &\longrightarrow \text{periodicity selector } \theta_{108}^{\mathrm{circ}} \longrightarrow G_{108}, \\ \text{icosahedral incidence} &\longrightarrow V_{12} \xrightarrow{P_5} V_5 \xrightarrow{\Gamma_5} \mathrm{STF}_3 \xrightarrow{\Lambda(n)} \mathcal{H}_{\mathrm{TT}}(n), \\ \text{enriched holonomic route} &\dashrightarrow G_{\mathrm{hol}}, \\ \text{metrological chart} &\longrightarrow 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}. \end{aligned} \tag{50.1}$$

The first line states a condition of periodicity that is intrinsic and exact. The second constructs the carrier and its trace-free transverse reduction. The third preserves an explicit construction boundary. The fourth is an evaluation in a declared decimal chart: it identifies the magnitude but does not select the gravitational character.

50.2 Gravitational magnitude line and chronological scale 108

Let $c_{\mathrm{int}} > 0$, $t_0 > 0$, $\ell_0 = c_{\mathrm{int}} t_0$ and a positive section $\hbar_a$ of the action line, with $a \in \{\mathrm{pre}, \mathrm{ret}\}$. The dimensional transducer is

$$G_a(\theta) = \theta \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a} = \theta \frac{c_{\mathrm{int}}^3 \ell_0^2}{\hbar_a}, \qquad \theta > 0. \tag{50.2}$$

The equality of the two expressions uses only $\ell_0 = c_{\mathrm{int}} t_0$, and $[G] = L^3 M^{-1} T^{-2}$. The dimension is fixed by the line; the mathematical problem is to select the dimensionless character θ.

The CT108 calendar defines

$$T_{108} = 108 t_0, \qquad \omega_{108} = \frac{2\pi}{108 t_0}, \qquad R_{108} = \frac{c_{\mathrm{int}}}{\omega_{108}} = \frac{54}{\pi} \ell_0. \tag{50.3}$$

For each action sheet, let

$$E_{108,a} = \hbar_a \omega_{108}, \qquad m_{108,a} = \frac{E_{108,a}}{c_{\mathrm{int}}^2}. \tag{50.4}$$

Then, the reduced Compton wavelength coincides with the radius of the cycle:

$$\bar{\lambda}_{C,a} = \frac{\hbar_a}{m_{108,a} c_{\mathrm{int}}} = R_{108}. \tag{50.5}$$

Theorem 50.1 (Uniqueness of the selector under the HMT periodicity condition)**.** *With the reduced convention $r_g = Gm/c_{\mathrm{int}}^2$ and the HMT periodicity premise that identifies the gravitational radius of the CT108 quantum with its phase radius, the closure condition*

$$r_{g,a}(\theta) = R_{108} \tag{50.6}$$

selects a unique positive solution,

$$\boxed{\theta_{108}^{\mathrm{circ}} = \left(\frac{54}{\pi}\right)^2 = 295.4525715010569415303525\ldots.} \tag{50.7}$$

In that embodiment,

$$R_{108} = \bar{\lambda}_{C,a} = r_{g,a} = \ell_P^{\mathrm{circ}}. \tag{50.8}$$

Proof. By equations (50.2) y (50.4),

$$r_{g,a}(\theta) = \frac{G_a(\theta) m_{108,a}}{c_{\mathrm{int}}^2} = \theta c_{\mathrm{int}} t_0^2 \omega_{108} = \theta \frac{\pi}{54} \ell_0.$$

Comparing it with $R_{108} = (54/\pi)\ell_0$ necessarily gives $\theta = (54/\pi)^2$. The equation is linear in θ and all the factors are positive, so the solution is unique. □

The equality $r_g = R_{108}$ does not follow from the dimensionality of equation (50.2). It is the explicit geometric condition of this HMT realization: action, Compton, and periodic return must determine the same length. The theorem proves that, once this structural condition is accepted, the character θ is fixed without using the SI value of G.

Remark 50.2 (Radius convention). The theorem uses the reduced gravitational radius Gm/c^2. If it is replaced by the Schwarzschild radius $2Gm/c^2$, the selector changes by a factor of two. The two expressions correspond to distinct geometric conditions, so the convention must be declared before the calculation.

50.3 Reciprocity $G \leftrightarrow \hbar$ and conservation of the Planck area

Corollary 50.3 (Planck family associated with the temporal periodicity of 108 phases)**.** *For* $G_a = G_a(\theta_{108}^{\mathrm{circ}})$,

$$\ell_P = \frac{54}{\pi}\ell_0, \qquad t_P = \frac{54}{\pi} t_0, \tag{50.9}$$

$$E_{P,a} = \frac{\hbar_a}{t_P} = \frac{\pi \hbar_a}{54 t_0} = \frac{h_a}{108 t_0}, \qquad \ell_P^2 = \frac{G_a \hbar_a}{c_{\mathrm{int}}^3} = \theta_{108}^{\mathrm{circ}} \ell_0^2. \tag{50.10}$$

Además,

$$F_{P,a} = \frac{c_{\mathrm{int}}^4}{G_a}, \qquad P_{P,a} = \frac{c_{\mathrm{int}}^5}{G_a}, \qquad \rho_{P,a} = \frac{\hbar_a}{(\theta_{108}^{\mathrm{circ}})^2 c_{\mathrm{int}}^5 t_0^4}. \tag{50.11}$$

Proof. The first three identities follow from $\ell_P = R_{108}$, $t_P = \ell_P / c_{\mathrm{int}}$, and $h_a = 2\pi\hbar_a$. For the area,

$$\frac{G_a \hbar_a}{c_{\mathrm{int}}^3} = \theta_{108}^{\mathrm{circ}} c_{\mathrm{int}}^2 t_0^2 = \theta_{108}^{\mathrm{circ}} \ell_0^2.$$

The remaining expressions are the dimensional compositions of force, power, and mass per volume with the same section. □

Let

$$R_{\mathrm{act}} := \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}}. \tag{50.12}$$

Since the action appears in the numerator of the chronological mass and in the denominator of G,

$$\frac{m_{108,\mathrm{pre}}}{m_{108,\mathrm{ret}}} = R_{\mathrm{act}}, \qquad \frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}} = R_{\mathrm{act}}^{-1}, \qquad G_a \hbar_a = \theta_{108}^{\mathrm{circ}} c_{\mathrm{int}}^5 t_0^2. \tag{50.13}$$

Therefore, $\ell_P^2 = G_a\hbar_a/c^3$ does not change sheets. Gravity and action are reciprocally oriented, while area preserves the common geometry. This cancellation is the interface of confluence toward the area operator of the section 51.6. The second fundamental relation appears when action and gravity are compared.

In HMT, action measures how much occurs while simultaneously preserving the orientation of the realization. There is one section before the return and another after it. The phase may return accompanied by a different memory history. That difference is stored in the bidirectional ratio of the two action sections.

Gravity responds in a reciprocal manner.

The gravitational transducer has the form

$$G_a(\theta) = \theta\frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a}.$$

Action appears in the denominator. Therefore, if $\hbar$ is multiplied by a ratio R_{act} when the sheet changes, the gravitational constant is multiplied by the inverse ratio:

$$\frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = R_{\mathrm{act}}, \qquad \frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}} = R_{\mathrm{act}}^{-1}.$$

Action and gravity form a contragredient pair here: when one advances in one direction, the other exactly compensates for that displacement.

What remains invariant is

$$G_a\hbar_a.$$

And that is precisely the combination that determines the Planck area:

$$\ell_P^2 = \frac{G_a\hbar_a}{c_{\mathrm{int}}^3}.$$

Action remembers the sheet. Gravity responds to that memory. Area remains invariant.

This is an extraordinarily fertile idea. The Planck area emerges as the object that survives when action and gravity transform into one another reciprocally.

It may be stated as follows: $\hbar$ preserves the dynamical orientation; G preserves the geometric compensation; ℓ_P^2 preserves that on which both realizations agree.

The area is, therefore, a currency of exchange between quantum physics and geometry, because the bidirectional transformation of $\hbar$ and G selects exactly that combination as invariant.

This reciprocity affords another understanding of the relation between information and gravity. Information requires a distinction among states; action measures the cost of transition between them. Yet if that transition must acquire a geometry, the gravitational response changes inversely so as to keep the quantum of area stable.

Geometry incorporates phase information through an exact compensation.

From a Machian reading, this is already a highly suggestive result. A local magnitude of geometry is determined relationally by the orientation and the return of the complete state. The invariant area restricts that relational dependence and fixes its consistency.

The HMT version of the relational principle consists in something more precise than the maxim "matter determines geometry": the orientation of action and the gravitational response co-determine one another in such a way that elementary geometry remains invariant.

The action contains the history. Gravity adjusts the response. The area preserves the agreement.

The derivation of G takes that reciprocity to a much more concrete geometric condition.

Dimensional Mechanics first determines what type of object G can be. It fixes its magnitude line and dimensional homogeneity. Chronological closure then selects its value: dimension determines the class of magnitude, and closure determines its dimensionless coefficient.

The selection proceeds from the temporal periodicity of (108) phases.

The cycle 108 determines a frequency, a phase radius, and an energy. From that energy arises a chronological mass. And constructing the reduced Compton wavelength of that mass yields exactly the same radius as that of the cycle:

$$\bar{\lambda}_C = R_{108}.$$

Phase and matter are already stating the same length before G is introduced.

Then gravity comes into play. The reduced gravitational radius is constructed.

$$r_g = \frac{Gm}{c^2},$$

and gravitational closure is required to identify the phase radius, Compton length, and gravitational radius as a single structural scale:

$$r_g = R_{108} = \bar{\lambda}_C.$$

That condition uniquely selects

$$\theta_{108} = \left(\frac{54}{\pi}\right)^2.$$

In this way,

$$R_{108} = \bar{\lambda}_C = r_g = \ell_P.$$

The structural function of θ_{108} determines the significance of its terminal decimal: it shows what G is doing.

G appears as the coefficient required to make the four lengths published by these languages coincide:

- the lenguaje of the fase produce R_{108};
- the quantum language produces $\bar{\lambda}_C$;
- the gravitational language produces r_g;
- the lenguaje of Planck produce ℓ_P.

The gravitational constant is the mediator that prevents those four readers from separating.

Stated in more physical terms: G expresses how strongly geometry must respond for a quantum whose mass follows from the clock 108 to close upon its own phase.

It is the relation that makes possible for matter, phase, and geometry to be three compatible publications of the same event.

Here the Machian reading becomes stronger. The local gravitational scale is selected by a global cycle-closure condition. The local radius of a quantum acquires meaning when integrated into a complete periodicity. Inertia, phase, and geometry are determined relationally.

The demonstrated scope provides a concrete mathematical mechanism for a Machian reading: a local property—the gravitational coupling—is selected by the consistency of the global state. The other historical formulations of Mach's principle preserve their own domain.

Then comes the fifth dimension.

The scalar G fixes the gravitational intensity, and the carrier specifies its representational structure. HMT constructs a symmetric traceless tensor space of dimension five. That carrier admits five helicoidal weights:

$$+2, \quad +1, \quad 0, \quad -1, \quad -2.$$

They are the five natural degrees of freedom of a symmetric traceless tensor in three dimensions before the dynamical constraints of a specific theory are imposed.

That clarifies a point that is often confused. "Five components" designates the complete carrier, whereas the two helicities designate the transverse massless reduction in general relativity.

It means that the complete carrier has five irreducible coordinates. The dynamics then decides which part of that carrier is physical:

- in a massive spin-two realization, all five weights may survive;
- in a massless realization with gauge invariance, the transverse traceless projection eliminates the sectors ± 1 and 0;
- the helicities ± 2 remain.

Thus, HMT provides a space prior to the dynamical decision. It constructs the complete five-component carrier and contains, through the corresponding projection, its physical reduction to two helicities.

Beauty resides in the ordered coexistence of two statements:

- the spin-two geometry has a five-component carrier;
- massless gravitational radiation occupies a two-dimensional subspace.

Both statements form a coherent and precise reduction.

And that reduction preserves another central lesson: the scalar G fixes the common intensity, while the orientation information resides in the tensor carrier. The intensity resides in the constant, the helicities reside in the representation, and the complete genealogy brings magnitude and carrier together.

50.4 Genealogical domain of gravitational realization

The gravitational realization receives as its initial object a section of the discrete continuum already constructed by the composition

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow X_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow C_{\mathrm{cont}}^{\mathrm{disc}}. \tag{50.14}$$

APP preserves the additive and multiplicative evaluations through residue, quotient and carry; TRIT determines the regime and orientation; the TPK incorporates the tritic phase selector without eliminating either of the two APP sheets, performs cardinal transport, preserves memory and realizes phase return without resetting the state. The dimensional transducer then acts on the surviving section of the continuum:

$$\mathfrak{G}_{108} : C_{\mathrm{cont}}^{\mathrm{disc}} \times \mathcal{L}_S^{+} \longrightarrow \mathcal{G}_{\text{HMT-MD}}, \qquad (x, \hbar_a) \longmapsto \big(G_a, R_{108}, m_{108,a}, \ell_{P,a}^{\prime,2}\big). \tag{50.15}$$

The cardinals 90/120, the periodicity 108, the action, and the angular pair enter with their genealogy preserved. No subsequent gravitational identification retrospectively selects these antecedents.

50.5 Gravitational Transducer and Chronological Selector of Periodicity

Let $c_{\mathrm{int}} > 0$, $t_0 > 0$, $\ell_0 = c_{\mathrm{int}} t_0$, and $\hbar_a \in \mathcal{L}_S^+$, with $a \in \{\mathrm{pre}, \mathrm{ret}\}$. The dimensional transducer is

$$G_a(\theta) = \theta \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a} = \theta \frac{c_{\mathrm{int}}^3 \ell_0^2}{\hbar_a}, \qquad \theta > 0. \tag{50.16}$$

The chronology cyclic of 108 states determina

$$T_{108} = 108 t_0, \qquad \omega_{108} = \frac{2\pi_{\mathrm{HMT}}}{108 t_0}, \qquad R_{108} = \frac{c_{\mathrm{int}}}{\omega_{108}} = \frac{54}{\pi_{\mathrm{HMT}}} \ell_0. \tag{50.17}$$

The chronological quantum and its associated mass are

$$E_{108,a} = \hbar_a \omega_{108}, \qquad m_{108,a} = \frac{E_{108,a}}{c_{\mathrm{int}}^2}. \tag{50.18}$$

The reduced Compton wavelength satisfies

$$\bar{\lambda}_{C,a} = \frac{\hbar_a}{m_{108,a} c_{\mathrm{int}}} = R_{108}. \tag{50.19}$$

Theorem 50.4 (Uniqueness of the gravitational periodicity selector)**.** *Under the reduced convention $r_g = Gm/c_{\mathrm{int}}^2$, the condition $r_{g,a}(\theta) = R_{108}$ has the unique positive solution*

$$\boxed{\theta_{108}^{\mathrm{circ}} = \left(\frac{54}{\pi_{\mathrm{HMT}}} \right)^2.} \tag{50.20}$$

The corresponding realization identifies structurally

$$R_{108} = \bar{\lambda}_{C,a} = r_{g,a} = \ell_P^{\mathrm{circ}}. \tag{50.21}$$

Proof. By equations (50.16) y (50.18),

$$r_{g,a}(\theta) = \frac{G_a(\theta) m_{108,a}}{c_{\mathrm{int}}^2} = \theta \frac{\pi_{\mathrm{HMT}}}{54} \ell_0.$$

Equality with $R_{108} = (54/\pi_{\mathrm{HMT}})\ell_0$ imposes $\theta = (54/\pi_{\mathrm{HMT}})^2$. Linearity in θ and positivity of the factors determine uniqueness. □

In the chart in which $c_{\mathrm{int}} = (\mu\varepsilon)^{-1/2}$, the transducer assumes the form

$$\boxed{G_a = \left(\frac{54}{\pi_{\mathrm{HMT}}} \right)^2 \frac{t_0^2}{\hbar_a (\mu\varepsilon)^{5/2}}.} \tag{50.22}$$

The constitutive product $\mu\varepsilon$ governs the causal cone and the gravitational scale; the quotient $Z^2 = \mu/\varepsilon$ preserves the impedance orientation.

50.6 Bidirectional Covariance of Gravity and Action

We define

$$R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}}. \tag{50.23}$$

The chronological mass and the gravitational coupling transform contragrediently:

$$\frac{m_{108,\mathrm{pre}}}{m_{108,\mathrm{ret}}} = R_{\mathrm{act}}, \qquad \frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}} = R_{\mathrm{act}}^{-1}. \tag{50.24}$$

Its product remains fixed:

$$\boxed{G_a\hbar_a = \theta_{108}^{\mathrm{circ}} c_{\mathrm{int}}^5 t_0^2, \qquad \ell_P^2 = \frac{G_a\hbar_a}{c_{\mathrm{int}}^3} = \theta_{108}^{\mathrm{circ}} \ell_0^2.} \tag{50.25}$$

The action preserves the orientation of the sheet; the area of Planck preserves the geometry common of both orientationes.

Theorem 50.5 (Chart independence of the sections h and G)**.** *Let u_L, u_T, u_S be positive bases of the dimensional lines of length, time and action, and replace them by*

$$u'_L = \lambda_L u_L, \qquad u'_T = \lambda_T u_T, \qquad u'_S = \lambda_S u_S, \qquad \lambda_L, \lambda_T, \lambda_S > 0.$$

The coordinates of the same sections satisfy

$$c'_{\mathrm{int}} = \frac{\lambda_T}{\lambda_L} c_{\mathrm{int}}, \qquad t'_0 = \frac{t_0}{\lambda_T}, \qquad \hbar'_a = \frac{\hbar_a}{\lambda_S}, \qquad G'_a = \frac{\lambda_S \lambda_T^3}{\lambda_L^5} G_a. \tag{50.26}$$

For the induced basis $u_G = u_L^5 u_T^{-3} u_S^{-1}$ one obtains

$$\hbar'_a u'_S = \hbar_a u_S, \qquad G'_a u'_G = G_a u_G. \tag{50.27}$$

Consequently, $h_a = 2\pi_{\mathrm{HMT}}\hbar_a$, $G_a = \theta_{108}^{\mathrm{circ}} c_{\mathrm{int}}^5 t_0^2/\hbar_a$ and $G_a\hbar_a = \theta_{108}^{\mathrm{circ}} c_{\mathrm{int}}^5 t_0^2$ are identities of sections, not equalities dependent on a choice of units. The metrological chart publishes their numerical coordinates after the HMT–MD construction.

Proof. The first three transformations of (50.26) are the contragredient law of coordinates in $L \otimes T^{-1}$, T and S. Substituting them into (50.16) produces the fourth. The basis of the gravitational line transforms by the inverse factor, proving (50.27). The scalars π_{HMT} and $\theta_{108}^{\mathrm{circ}}$ are dimensionless and remain invariant. □

Chapter 51

HMT derivation of the Barbero–Immirzi parameter and minimum area: hexad–octad incidence, 90/120 channels and spectrum of the area operator

51.1 Hexad–Octad Incidence Morphism and Conservation of the Oriented Channels 90/120

With the dodecad and the alignment of the section 17.1.6 fixed, let H be a hexad in the HMT coordinates, let $\ell(H) \subset D$ be its image, and let $O_H = \iota_D(\ell(H))$ be its unique lift. Marking a point and a pair of points of the residual hexad yields

$$F_6(H) = \ell(H) \times \binom{\ell(H)}{2}, \qquad |F_6| = 6\binom{6}{2} = 90.$$

If the marked point can traverse the octad:

$$F_8(H) = O_H \times \binom{\ell(H)}{2}, \qquad |F_8| = 8\binom{6}{2} = 120.$$

The inclusion $\ell(H) \hookrightarrow O_H$ induces the incidence morphism

$$\mathcal{I}_{6\to 8} : F_6(H) \hookrightarrow F_8(H).$$

Normalizing the difference by the chosen factor $24\binom{6}{2}$ yields

$$\boxed{\frac{90-120}{24\binom{6}{2}} = \frac{6-8}{24} = -\frac{1}{12}.}$$

The equality is a normalized defect of the combinatorial interface $6 \to 8$. The factor $24\binom{6}{2}$ forms part of the definition of this normalization; the inclusion alone determines the difference -30. The operator $\mathcal{I}_{6\to 8}$ has marked incidence configurations as its domain; it is neither the transverse extractor of the dodecaphase state nor the family of angular moments.

51.2 Dodecaphase fundamental chain and hexad–octad functional

The active incidence is constructed before any gravitational realization. Let

$$H_{\mathrm{act}} = \{1, 2, 4, 5, 7, 8\}, \qquad \mathcal{P}_2(H_{\mathrm{act}}) = \{A \subset H_{\mathrm{act}} : |A| = 2\}.$$

Then

$$|H_{\mathrm{act}}| = 6, \qquad |\mathcal{P}_2(H_{\mathrm{act}})| = 15,$$

and the incidences before and after enlargement have cardinalities

$$\mathcal{F}_{90} = H_{\mathrm{act}} \times \mathcal{P}_2(H_{\mathrm{act}}), \qquad |\mathcal{F}_{90}| = 90,$$

$$\mathcal{F}_{120} = \{1, \ldots, 8\} \times \mathcal{P}_2(H_{\mathrm{act}}), \qquad |\mathcal{F}_{120}| = 120.$$

These two cardinalities are outputs of the phase rule; they are neither angles nor parameters supplied by the physical realization.

Denote by $C_{12} = \mathbb{Z}/12\mathbb{Z}$ the temporal record of TPK and by ∂_{ret} its unique return seam per turn. In the free module of oriented events, consider the fundamental chain

$$c_{12}^{+} = \sum_{j \in C_{12}} [j, \mathcal{F}_{90}] - [\partial_{\mathrm{ret}}, \mathcal{F}_{120}]. \tag{51.1}$$

The sum contains exactly the twelve sectors of the wheel; the second term is unique because one turn has a single closure seam. Its sign is the boundary sign. The bidirectional involution sends c_{12}^{+} to $c_{12}^{-} = -c_{12}^{+}$: it changes the orientation of the chain without altering its absolute multiplicities or its incidence types.

For $m \in \{90, 120\}$, let

$$S_m(q) = \frac{q^m}{1 - q^{3m}}.$$

The return readout $\mathscr{R}$ is defined on the generators of (51.1) by

$$\mathscr{R}([j, \mathcal{F}_{90}]) = S_{90}, \qquad \mathscr{R}([\partial_{\mathrm{ret}}, \mathcal{F}_{120}]) = S_{120}.$$

The first equality is constant on C_{12} by covariance under phase rotation; the second has a different domain and acts only at the enlarged seam. By linearity,

$$\boxed{\mathscr{R}(c_{12}^{+}) = 12S_{90} - S_{120}.} \tag{51.2}$$

Thus the coefficients $12 : -1$ are the image of the fundamental chain: twelve phase cells and one oriented boundary. They are not selected through the conventional value of the Barbero–Immirzi parameter, through a decimal comparison, or by prescribing in advance the pole coefficient to be calculated next.

Theorem 51.1 (Pole coefficient of the dodecaphase chain)**.** *The functional $F(q) = \mathscr{R}(c_{12}^{+})(q) = 12S_{90}(q) - S_{120}(q)$, determined by the twelve sectors and their oriented seam, has coefficient $1/24$ for $(1-q)^{-1}$. Its complex residue at $q = 1$ is $-1/24$. The pole coefficient is an output of the already constructed functional.*

Proof. For a functional with a simple pole at $q = 1$, write

$$p_1(F) := \lim_{q \uparrow 1} (1 - q) F(q).$$

The identity

$$1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$$

implies

$$p_1(S_m) = \lim_{q \uparrow 1} \frac{q^m}{\sum_{j=0}^{3m-1} q^j} = \frac{1}{3m}.$$

By linearity,

$$p_1\big(\mathscr{R}(c_{12}^{+})\big) = \frac{12}{270} - \frac{1}{360} = \frac{48 - 3}{1080} = \frac{1}{24}.$$

These equalities are taken in $\mathbb{Q}$. The complex residue is the coefficient of $(q-1)^{-1}$, so that

$$\operatorname*{Res}_{q=1} \mathscr{R}(c_{12}^{+})(q) = -p_1\big(\mathscr{R}(c_{12}^{+})\big) = -\frac{1}{24}.$$

□

The Diophantine equation

$$\frac{a}{270} - \frac{b}{360} = \frac{1}{24} \quad \Longleftrightarrow \quad 4a - 3b = 45$$

remains a subsequent arithmetic certificate. On its own, it admits the family $(a, b) = (12 + 3t, 1 + 4t)$; the chain (51.1) eliminates that ambiguity before the calculation, since changing $(12, 1)$ alters the multiplicity of the orbit or its seam and therefore no longer represents a fundamental TPK turn.

With the conjugate channels $q_\pm$ already generated by the angular branch, one finally obtains

$$\mathscr{R}(c_{12}^{+})(q_-) - \mathscr{R}(c_{12}^{+})(q_+) = 12\Delta S_{90} - \Delta S_{120}.$$

This is the internal input of the hexadic–octadic functional. Its evaluation as the Barbero–Immirzi parameter and its insertion into the area operator are subsequent publications: they do not participate in the selection of c_{12}^{+}, 90/120, $12 : -1$ or $1/24$.

Falsifiers. The construction is refuted if the phase census does not yield twelve sectors, if one turn contains more than one seam, if the incidences do not have cardinalities 90 and 120, if the involution does not reverse the entire chain, or if the exact coefficient of $(1 - q)^{-1}$ in its image differs from $1/24$.

The transition from a hexad to an octad combines three structures of distinct types: a finite incidence, two Lambert series associated with the channels $q_\pm$, and an identification with the real parameter of the Holst action. The three are articulated through the incidence, the angular functional, and the areal dictionary into the geometry of the Ashtekar–Barbero connection developed in this same residence.

51.3 Tritic derivation of the cardinalities 90/120 and subsequent realization in hexad–octad incidence

Let H be a set of six points and let O be a set of eight points containing a distinguished copy of H. Consider

$$\mathcal{F}_6 = H \times \binom{H}{2}, \qquad \mathcal{F}_8 = O \times \binom{H}{2}.$$

The inclusion distinguished $H \hookrightarrow O$ induce

$$\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8, \qquad (h, \{h_1, h_2\}) \longmapsto (h, \{h_1, h_2\}).$$

This incidence morphism is the combinatorial antecedent of the cardinals 90/120; it is neither the dodecaphasic extractor nor an angular series.

Theorem 51.2 (Cardinalities of the Inclusion)**.**

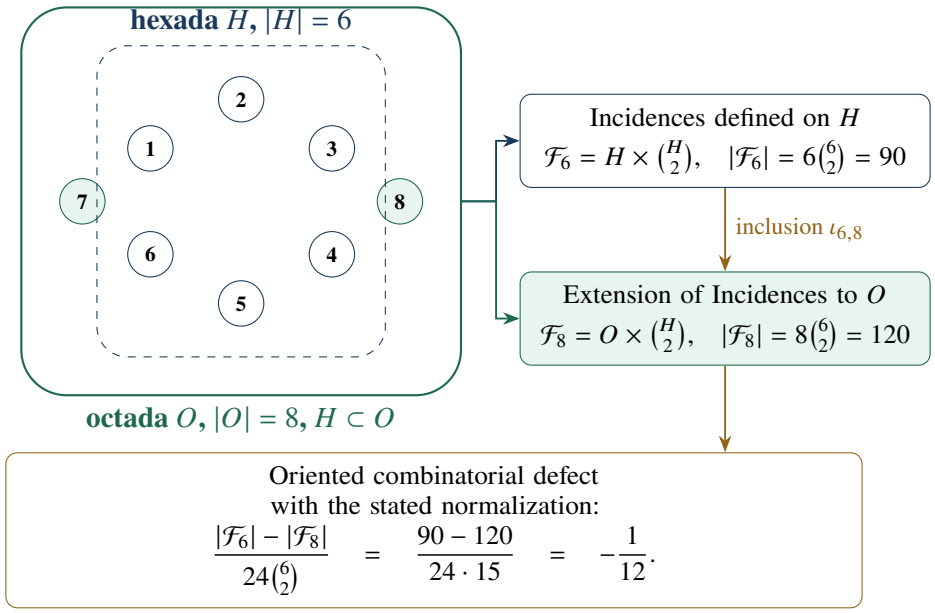
$$|\mathcal{F}_6| = 6\binom{6}{2} = 90, \qquad |\mathcal{F}_8| = 8\binom{6}{2} = 120.$$

The normalized difference by twenty-four coordinates is

$$\frac{90 - 120}{24\binom{6}{2}} = -\frac{1}{12}.$$

Proof. Each element of the first coordinate can be combined with any of the fifteen pairs of H. The multiplication principle gives the two cardinalities; the last identity follows by substituting $\binom{6}{2} = 15$. □

The initial indices of the angular hierarchy are thus associated with two incidence spaces: 90 configurations on the hexad and 120 on the ambient octad. The quotient $-1/12$ is a combinatorial identity of this normalization.



This identity belongs to the incidence map. It does not identify the dodecaphase extractor, the tower semigroup, or the angular functional.

Figure 51.1: Inclusion of a distinguished subset of six elements in one of eight elements and incidence counts. The cardinals 90 and 120 do not merge the dodecaphase extractor, the angular functional or the tower semigroup.

51.4 Lambert Series Associated with the 2 Incidence Sectors

For $0 < q < 1$, we define simultaneously

$$S_{90}(q) = \frac{q^{90}}{1 - q^{270}}, \qquad S_{120}(q) = \frac{q^{120}}{1 - q^{360}}, \qquad \Delta S_m = S_m(q_-) - S_m(q_+).$$

Geometric expansions

$$\Delta S_{90} = \sum_{j \geq 0} s_{90+270j}, \qquad \Delta S_{120} = \sum_{j \geq 0} s_{120+360j}$$

place both terms within the global semigroup of section 91.8.

The angular functional associated with the transition $6 \to 8$ is

$$K_{6\to 8} = 12\Delta S_{90} - \Delta S_{120},$$

and its precedence has been fixed by the fundamental dodecaphasic chain: the twelve sectors and the unique oriented seam first produce the weight $12 : -1$. The coefficient of $(1 - q)^{-1}$, equal to $1/24$, is then calculated as an exact output; the Diophantine equation that follows certifies the already generated pair and does not select it. The resulting angular coordinate is

$$\Gamma_{6\to 8} = \frac{\pi A C^*}{180} + K_{6\to 8}.$$

At the point HMT,

$$\Delta S_{90} = 2.95169439297457621139847987861 \times 10^{-4},$$
$$\Delta S_{120} = 1.96835112502951720093738706217 \times 10^{-5},$$
$$\frac{\pi A C^*}{180} = 0.27048532293414007846558645879 0\ldots,$$

and, therefore,

$$\boxed{\Gamma_{6\to 8} = 0.27400767269445927474725526077398041940\ldots.}$$

The tables computed with the conjugate angular coordinate C^* rounded to fifteen decimal places give 0.2740076726944593 in double-precision arithmetic; the preceding value is the multiprecision evaluation of the complete regional phase.

51.5 Subsequent Diophantine certificate of the weights

The identity

$$\lim_{q\to 1^-}(1-q)\frac{q^m}{1-q^n} = \frac{1}{n}$$

assigns to the combination $aS_{90} - bS_{120}$, before its evaluation in the conjugate channels, the coefficient of $(1-q)^{-1}$

$$\frac{a}{270} - \frac{b}{360}.$$

Theorem 51.3 (Integer weights with pole coefficient $1/24$)**.** *The positive integer solutions of*

$$\frac{a}{270} - \frac{b}{360} = \frac{1}{24}$$

are

$$(a,b) = (12+3t, 1+4t), \qquad t \in \mathbb{Z}_{\geq 0}.$$

The pair $(12,1)$ *is the unique coordinatewise minimal solution and the unique one of minimal* ℓ^1 *norm.*

Proof. Multiplying by 1080 produces $4a - 3b = 45$. Modulo three, one obtains $a \equiv 0 \pmod 3$; writing $a = 3u$ yields $4u - b = 15$. Positivity implies $u = 4+t$, $b = 1+4t$, with $t \geq 0$. Therefore, $a + b = 13 + 7t$, which proves minimality. □

Primitivity $\gcd(a,b) = 1$ does not by itself select the minimal pair, since other primitive solutions exist. The fundamental TPK turn has already produced the weight $12 : -1$; the pole coefficient $1/24$, positivity and minimality form its subsequent arithmetic certificate. None of these objects is deduced from the decimal evaluation of $\Gamma_{6\to 8}$.

51.6 Monotonicity of the functional with respect to the conjugate angular coordinate

Para $m \in \{90, 120\}$,

$$S'_m(q) = \frac{m q^{m-1}(1 + 2q^{3m})}{(1 - q^{3m})^2} > 0.$$

Moreover,

$$\frac{\mathrm{d}q_-}{\mathrm{d}C} = \frac{\pi}{180} q_-, \qquad \frac{\mathrm{d}q_+}{\mathrm{d}C} = -\frac{\pi}{180} q_+.$$

Proposition 51.4 (Strict Monotonicity)**.** *For $1^\circ \le C \le 3^\circ$, the function $C \mapsto \Gamma_{6\to 8}(A, C)$ is strictly increasing. Consequently, every value in its image has at most one preimage in that interval.*

Proof. The derivative of each difference ΔS_m is positive. On the specified interval one obtains $\Delta S'_{120} < 5.35 \times 10^{-4}$, whereas $\pi A/180 > 0.1273$. Therefore, $\Gamma'(C) > 0.1267$. At $C = C^*$,

$$\Gamma'(C^*) = 0.1328995105618907\ldots.$$

□

Local injectivity excludes a second nearby angular coordinate with the same functional evaluation. This control neither constructs nor selects C^*: the conjugate angular coordinate has already been obtained in the preceding regional branch. The proposition only quantifies sensitivity of the downstream evaluation once A and C^* are fixed; it does not authorize inversion of $\Gamma_{6\to 8}$ or a physical value of γ_{BI} to redefine them.

51.7 Hexad–octad inclusion and associated functional realizations

Let us fix a set H of six points and a set O of eight points with a distinguished inclusion $j : H \hookrightarrow O$. In the Witt residence, H is a hexad of a dodecad and O its lifted octad; in this chapter only the already fixed inclusion is used, not the terminal value of Barbero–Immirzi. We write

$$\binom{H}{2} := \{P \subset H : |P| = 2\}, \qquad \left|\binom{H}{2}\right| = \binom{6}{2} = 15, \tag{51.3}$$

and we define two incidence spaces with distinct types:

$$\mathcal{F}_6 := H \times \binom{H}{2}, \qquad \mathcal{F}_8 := O \times \binom{H}{2}. \tag{51.4}$$

The inclusion of incidences is

$$\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8, \qquad (h, P) \longmapsto (j(h), P). \tag{51.5}$$

It is injective because j is injective and preserves the marked pair P.

Theorem 51.5 (Incidence and defect counts oriented)**.** *The two spaces of equation (51.4) satisfy*

$$|\mathcal{F}_6| = 6\binom{6}{2} = 90, \qquad |\mathcal{F}_8| = 8\binom{6}{2} = 120, \qquad |\mathcal{F}_8 \setminus \iota_{6,8}(\mathcal{F}_6)| = 30. \tag{51.6}$$

Normalization by the twenty-four coordinates of the dodecad–twenty-four pair and by the fifteen pairs of H produces

$$\boxed{\delta^{\mathrm{inc}}_{6,8} = \frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24\binom{6}{2}} = -\frac{1}{12}.} \tag{51.7}$$

The same fraction is obtained through the reciprocal functional associated with the incorporated memory,

$$\boxed{\delta^{\mathrm{rec}}_{6,8} = \frac{30}{120} - \frac{30}{90} = -\frac{1}{12}.} \tag{51.8}$$

They are two exact realizations of the same oriented defect, but have distinct domains and are not identified with $\zeta(-1)$ without an additional intertwiner.

Proof. The multiplicative principle applied to equation (51.4) gives $6 \cdot 15 = 90$ and $8 \cdot 15 = 120$. The image of $\iota_{6,8}$ has ninety elements, so the complement has thirty. Substitution into equation (51.7) gives $-30/(24 \cdot 15) = -1/12$, while equation (51.8) is $1/4 - 1/3 = -1/12$. □

The hexad–octad transition admits two functional realizations that are not deduced from one another. The algebraic diagram, formulated on the spaces that justify the counts, is

$$\begin{aligned} (H \overset{j}{\hookrightarrow} O) &\longmapsto (|\mathcal{F}_6|, |\mathcal{F}_8|) = (90, 120) \longmapsto \Gamma_{6\to 8} = \gamma_{\mathrm{HMT}}, \\ (h_6, o_8, t) = (6, 8, 3) &\longmapsto -\frac{o_8 - h_6}{t} = -\frac{2}{3}, \end{aligned} \tag{51.9}$$

where $-2/3$ is the coefficient of the third coordinate of row 13 of the CKM constructor. The first realization transforms the incidence into area and modular memory; the second transforms the same transition into angular-mixing parameters. $\boldsymbol{\theta}_{\mathrm{CKM}} = f(\gamma_{\mathrm{HMT}})$ is not postulated: both outputs preserve a common antecedent.

The morphism that determines the sectors 90/120 is $\iota_{6,8}$. The quantity γ_{HMT} has already been determined internally by the hexad–octad construction and is reused here without reconstructing it from its decimal representation. The operatorial consequences begin only after fixing the elementary scale a_0, which is defined below without using entropy or the terminal product $\gamma_{\mathrm{HMT}} a_0$.

51.8 Correspondence between the Nonadic Interior and the Boundary Degrees of Freedom

Let $\mathbb{T}_3 = \{0, 1, 2\}$. Every $x \in \mathbb{Z}/9\mathbb{Z}$ has a unique digit representation $x = x_0 + 3x_1$, with $x_i \in \mathbb{T}_3$, and every $y \in \mathbb{Z}/27\mathbb{Z}$ a unique representation $y = y_0 + 3y_1 + 9y_2$. Therefore

$$\mathcal{B} = (\mathbb{Z}/9\mathbb{Z})^3, \qquad \mathcal{B}_\partial = (\mathbb{Z}/27\mathbb{Z})^2, \qquad |\mathcal{B}| = |\mathcal{B}_\partial| = 3^6 = 729. \tag{51.10}$$

Equality of cardinalities is accompanied by the explicit map. If $x_j = r_{2j-1} + 3r_{2j}$ for $j = 1, 2, 3$, we define

$$\beta_{729}(x_1, x_2, x_3) = \big(r_1 + 3r_2 + 9r_3,\ r_4 + 3r_5 + 9r_6\big). \tag{51.11}$$

Proposition 51.6 (Genealogical bijection $729 \leftrightarrow 729$)**.** *The map $\beta_{729} : \mathcal{B} \to \mathcal{B}_\partial$ is bijective and preserves the ordered six-trit word. In general, it is not an isomorphism of additive groups: regrouping the digits changes where the carry is recorded.*

Proof. Extracting the six digits from three nonadic coordinates and grouping them into two triples is a permutation of the coordinates of $\mathbb{T}_3^6$. The positional decomposition is unique on both sides, so the operation has an inverse. The additive caveat is verified, for example, by adding words that generate a carry between r_3 and r_4: that boundary separates the two coordinates of order twenty-seven, but crosses two distinct coordinates in the nonadic packing. □

On the boundary torus $\mathcal{B}_\partial$, we take the block of vertices

$$A = \{0, 1, \ldots, 8\} \times \{0, 1, \ldots, 8\}. \tag{51.12}$$

Its oriented links crossing the boundary are separated by direction:

$$\partial A_{90} = \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \le j \le 8\}, \tag{51.13}$$

$$\partial A_{120} = \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \le i \le 8\}, \tag{51.14}$$

where all coordinates are read modulo twenty-seven. Thus

$$\partial A = \partial A_{90} \sqcup \partial A_{120}, \qquad |\partial A_{90}| = |\partial A_{120}| = 18. \tag{51.15}$$

The subscripts 90, 120 label the channel provenance; they do not say that the boundary has ninety or one hundred and twenty links.

Theorem 51.7 (Law areal of the cut triadic)**.** *In the cubic graph $N \times N \times N$ with unit capacity, every cut between two opposite faces has capacity at least N^2, and a transverse layer attains that bound. For $N = 9$,*

$$\mathcal{A}_{\min} = 81, \qquad S_{\mathrm{tri}} = 81 \log 3. \tag{51.16}$$

Proof. There are N^2 lines of edges parallel to the normal direction, pairwise disjoint. Every cut must intersect each one, so its capacity is at least N^2. A single transverse layer contains exactly N^2 edges and attains the minimum. If each cut link carries a maximally mixed trit, its entropy is $\log 3$; additivity over the eighty-one links gives equation (51.16). □

The entropy S_{tri} measures the capacity of the cubic cut. The modular entropy of the thirty-six links of equation (51.15) measures another state and must not be equated with it.

51.9 Collective incidence entangler between interior and boundary

In the sum $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$, we define

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \qquad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f). \tag{51.17}$$

In $\ell^2(\partial A_{90}) \oplus \ell^2(\partial A_{120})$, let

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial A_{90}} (\delta_e, 0), \qquad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial A_{120}} (0, \delta_e). \tag{51.18}$$

The pairs (u_{90}, u_{120}) and (v_{90}, v_{120}) are orthonormal. Therefore there exists a unique isometry between the collective subspaces that satisfies

$$\mathcal{J}_{6\to 8} u_{90} = v_{90}, \qquad \mathcal{J}_{6\to 8} u_{120} = v_{120}. \tag{51.19}$$

Define

$$D_{\mathrm{inc}} = 90 |u_{90}\rangle\langle u_{90}| + 120 |u_{120}\rangle\langle u_{120}|, \tag{51.20}$$

$$D_{\mathrm{area}} = 90 |v_{90}\rangle\langle v_{90}| + 120 |v_{120}\rangle\langle v_{120}|. \tag{51.21}$$

Theorem 51.8 (Exact transport of the two labels)**.** *The map $\mathcal{J}_{6\to 8}$ is unitary between the two collective planes and intertwines the incidence and area operators:*

$$\boxed{\mathcal{J}_{6\to 8} D_{\mathrm{inc}} = D_{\mathrm{area}} \mathcal{J}_{6\to 8}.} \tag{51.22}$$

Therefore, every polynomial combination of the labels 90, 120*, in particular the primitive combination* $12 : -1$*, is transported with the same coefficients.*

Proof. Orthonormality shows that $\mathcal{J}_{6\to 8}$ maps one orthonormal basis to another. In u_{90}, both sides of equation (51.22) equal $90v_{90}$; in u_{120}, both equal $120v_{120}$. Equality on a basis proves the identity. The functional calculus, in its polynomial form, preserves the intertwining. □

This map does not purport to inject ninety individual incidences into eighteen links: it transports exactly the two uniform modes and their spectral labels. Extending it to non-uniform fluctuations would require a new incidence map; the modular operator defined here only uses the collective subspace that has just been constructed.

51.10 Determination of the Elementary Scale of the Spectrum of the Area Operator

Let C^* the angular coordinate conjugate, expresada en grados, and fijemos

$$\theta_{\mathrm{clk}} = \frac{C^*}{6}\frac{\pi}{180}, \qquad \boxed{a_0 = \frac{1+2\cos(10^\circ)}{3}\,\theta_{\mathrm{clk}}.} \tag{51.23}$$

The factor preceding θ_{clk} is not a decimal correction. It is the normalized real character of the phase triplet

$$\frac{1}{3}\,\mathrm{Tr}\,\mathrm{diag}\left(1, \mathrm{e}^{\mathrm{i}\pi/18}, \mathrm{e}^{-\mathrm{i}\pi/18}\right) = \frac{1+2\cos(10^\circ)}{3}. \tag{51.24}$$

Thus, a_0 is the spectral mean of the angular structure on the oriented trit. It is positive, is fixed before the area operator, and is not selected from entropy. The subsequent identification $\gamma_{\mathrm{HMT}} = \Gamma_{6\to 8}$ realizes the HMT output in the Ashtekar–Barbero connection [10, 68]; it does not reopen its derivation.

Remark 51.9 (Alcance of the simbolo a_0). In this chapter, $a_0 \equiv a_0^{\mathrm{area}}$ denotes exclusively the normalization of equation (51.23). It is neither the Bohr radius nor the initial cosmological scale factor, although those objects are traditionally written with the same symbol. No identity in this chapter allows one to substitute one for the other.

51.11 Areal Law of the Triadic Cut: Capacity $81\log 3$, Area Operator, Spectrum, and Collective Intertwiner of the Channels 90/120

The HMT identification of the functional $\Gamma_{6\to 8}$ with the Barbero–Immirzi coordinate is made on the same triadic boundary that supports the incidence (90/120). The interior and its boundary sheet have equal cardinality,

$$\mathcal{B} = (\mathbb{Z}/9\mathbb{Z})^3, \quad |\mathcal{B}| = 729, \qquad \partial\mathcal{B} = (\mathbb{Z}/27\mathbb{Z})^2, \quad |\partial\mathcal{B}| = 729,$$

by regrouping six trits into three nonadic coordinates or two coordinates of order twenty-seven.

Theorem 51.10 (Areal Law of Triadic Cutoff)**.** *In the cubic graph $N\times N\times N$ with unit capacity, the minimum cut between two opposite faces has capacity N^2. For $N = 9$,*

$$\mathcal{A}_{\min} = 81, \qquad S_{\mathrm{tri}} = \mathcal{A}_{\min}\log 3 = 81\log 3.$$

Proof. The N^2 lines parallel to the normal direction are edge-disjoint paths, so every cut intercepts them. A transverse layer contains exactly N^2 edges and attains the bound. Each triadic link contributes $\log 3$ to the entropy of a maximally mixed state. □

On the boundary torus 27×27, let A be the canonical block 9×9. Its boundary decomposes exactly into

$$\partial A = \partial A_{90} \sqcup \partial A_{120}, \qquad |\partial A_{90}| = |\partial A_{120}| = 18. \tag{51.25}$$

The horizontal links represent channel (90), and the vertical ones channel (120). This decomposition turns the pair of incidences into an operator decomposition of the boundary.

The areal entropy $S_{\mathrm{tri}} = 81 \log 3$ belongs to the maximally mixed state of the cut links. The bilayer entropy $S_{\mathrm{bi}}(y)$ of the definition 73.2 belongs to the normalized distribution between the two sheets; the two objects retain distinct domains.

Let $N = \mathrm{diag}(0, 1, 2)$ be the number operator of a tritic link. The areal realization declares the positive scale

$$\theta_{\mathrm{clk}} = \frac{C^*}{6}\frac{\pi}{180}, \qquad a_0 = \frac{1 + 2\cos(10^\circ)}{3}\,\theta_{\mathrm{clk}}, \qquad \gamma_{\mathrm{HMT}} = \Gamma_{6\to 8}.$$

The formula for a_0 constitutes the starting structure of this realization; once that scale is fixed, the operator and its spectrum follow exactly.

Definition 51.11 (Area Operator HMT)**.** On the boundary sheet $\mathcal{H}_\partial = \bigotimes_{e\in\partial A} \mathbb{C}^3$, define

$$\widehat{\mathcal{A}}_{\mathrm{HMT}} = \gamma_{\mathrm{HMT}} a_0 \sum_{e\in\partial A} N_e. \tag{51.26}$$

Its spectrum in the canonical cut is

$$\mathrm{Spec}\,\widehat{\mathcal{A}}_{\mathrm{HMT}} = \{n\gamma_{\mathrm{HMT}} a_0 : 0 \le n \le 72\},$$

with multiplicities given by the coefficients of $(1 + z + z^2)^{36}$. Prolongation through compatible cuts produces the protected family $\mathcal{A}_n^{\mathrm{prot}} = n\gamma_{\mathrm{HMT}} a_0$.

The intertwiner that fixes normalization is constructed in the subspace of collective channels. Let u_{90}, u_{120} be the uniform unit vectors of the hexadic and octadic incidences, and let v_{90}, v_{120} be the uniform unit vectors of the two sets of links of (51.25). The map

$$\begin{aligned} \mathcal{J}_{6\to 8} :\ \mathrm{span}\{u_{90}, u_{120}\} &\longrightarrow \mathrm{span}\{v_{90}, v_{120}\}, \\ \mathcal{J}_{6\to 8} u_{90} = v_{90}, &\qquad \mathcal{J}_{6\to 8} u_{120} = v_{120} \end{aligned} \tag{51.27}$$

is an isometry between these collective subspaces. If

$$D_{\mathrm{inc}} = 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \qquad D_{\mathrm{area}} = 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|,$$

then

$$\boxed{\mathcal{J}_{6\to 8} D_{\mathrm{inc}} = D_{\mathrm{area}} \mathcal{J}_{6\to 8}.}$$

In particular, the primitive combination $12 : -1$ acts with the same coefficients before and after $\mathcal{J}_{6\to 8}$. This equality fixes the normalization dictionary that inserts $\gamma_{\mathrm{HMT}} = \Gamma_{6\to 8}$ into the Ashtekar–Barbero connection.

The passage from hexad to octad is an incidence structure transformation.

In passing from 6 to 8, the available pairs, the internal relations, and the way in which orientation can be preserved change. That transition generates an oriented defect. The cardinal variation realizes a turn in the mode of organization.

The same sheet inversion is expressed by

$$C^* \longmapsto -C^*.$$

But different readers respond in different ways to that inversion.

The hexad–octad functional leading to the Barbero–Immirzi parameter is odd: when the sheet is inverted, it changes sign. Geometric orientation is reversed.

Bilayer entropy, by contrast, is even: it assigns the same entropy to C^* and $-C^*$ when the probability distribution is observed. The total information may remain unchanged even though the orientation has been reversed.

CKM space realizes a third response. The inversion induces an exact rational reflection on the three angles:

$$R_{\mathrm{CKM}} = M_{\mathrm{CKM}}\,\mathrm{diag}(1,-1,1)\,M_{\mathrm{CKM}}^{-1},$$

with

$$R_{\mathrm{CKM}}^{2} = I, \qquad \det R_{\mathrm{CKM}} = -1.$$

This means that the change of sheet is transported to the mixing space as a true reflection. There exists an odd direction of rank one and a plane that remains fixed.

The counter-angle C^{*} is precisely the coordinate that changes. The phase constructed from A remains invariant. The transformation thus separates two memories:

- an orientation memory;
- a phase memory.

This provides a profound reading of CKM. The mixing angles are coordinates of a space acted upon by the same involution that already organized the hexad–octad transition.

The turn $6 \to 8$ supplies the orientation operation that CKM realizes in its own codomain; physical causality belongs to the corresponding transducer.

We then have three sister publications of the same transit:

1. **Geometric publication.** The Barbero–Immirzi parameter orients the quantum of area.
2. **Informational publication.** The reduced state and its modular Hamiltonian measure what information from that orientation remains on the boundary.
3. **Mixing publication.** The CKM reflection transforms the mixing coordinates and isolates their odd component.

γ_{BI}, entropy and the CKM angles remain distinct realizations of a common genealogical hinge. Unity resides in the antecedent prior to ramification. They share the hinge, but each preserves a distinct part of it.

This is one of the most beautiful expressions of the HMT methodology: the fundamental unit is recovered by tracing the genealogies backward until the common operation that generates distinct terminal results is found.

Gravity conserves the spin as the orientation of the area.

Information conserves the spin as memory of the state.

The mixture preserves the twist as a reflection of coordinates.

The same transformation «breathes» in three ways.

51.12 Typed distinction between Barbero–area, area–modular Hamiltonian, and Cartan–Holst realization

The derivation of γ_{BI}, of the minimal area, and of the spectrum of the area operator concludes in this chapter. The joint reconstruction of occupation and provenance through the boundary modular Hamiltonian, and the Palatini–Cartan–Holst realization, preserve their own domains, operators, and proofs in Part IV.

51.13 Hexad–octad incidence, Barbero–Immirzi parameter, and area spectrum

The areal derivation receives as antecedents the incidence inclusion $6 \to 8$, the TPK cardinals $90/120$, the normalized defect $-1/12$, and the spectral channels $q_{\pm}$. The hexad–octad functional determines the Barbero–Immirzi parameter and the normalization of the area operator. This residence contains the HMT derivation and the spectrum; the Palatini–Cartan–Holst realization occupies a subsequent residence in Dimensional Mechanics.

Let $H \subset O$, with $|H| = 6$ and $|O| = 8$, and preserve the marked pair of the hexad in the inclusion

$$\iota_{6,8} : \mathcal{F}_6 = H \times \binom{H}{2} \hookrightarrow \mathcal{F}_8 = O \times \binom{H}{2}. \tag{51.28}$$

The internal count gives

$$|\mathcal{F}_6| = 6\binom{6}{2} = 90, \qquad |\mathcal{F}_8| = 8\binom{6}{2} = 120, \qquad \frac{90-120}{24\binom{6}{2}} = -\frac{1}{12}. \tag{51.29}$$

Once π_{HMT}, the angle A and the unique counterangle C^* have been published, the two channels are

$$q_{\pm} = \exp\Big[-\frac{\pi_{\mathrm{HMT}}}{180}(A \pm C^*)\Big], \qquad 0 < q_+ < q_- < 1, \tag{51.30}$$

and their oriented-return series are

$$S_{90}(q) = \frac{q^{90}}{1-q^{270}}, \qquad S_{120}(q) = \frac{q^{120}}{1-q^{360}}, \qquad \Delta S_m = S_m(q_-) - S_m(q_+). \tag{51.31}$$

The causal order of the coefficients is fixed in the temporal register of the TPK, before evaluating the pole coefficient. Let $C_{12} = \mathbb{Z}/12\mathbb{Z}$ be the dodecaphase wheel and let ∂_{ret} be its unique oriented return seam. In the free module of incidence events, the fundamental turn is

$$c_{12}^{+} = \sum_{j \in C_{12}} [j, \mathcal{F}_6] - [\partial_{\mathrm{ret}}, \mathcal{F}_8]. \tag{51.32}$$

The sign of the second term is the boundary sign. The bidirectional involution reverses the complete orientation, $c_{12}^{-} = -c_{12}^{+}$, without modifying the absolute multiplicities of the two incidence types. Define the return reader on the generators by

$$\mathscr{R}([j, \mathcal{F}_6]) = S_{90}, \qquad \mathscr{R}([\partial_{\mathrm{ret}}, \mathcal{F}_8]) = S_{120}. \tag{51.33}$$

Phase covariance makes the first image constant over the twelve sectors; the second acts only at the enlarged seam. By linearity,

$$\boxed{\mathscr{R}(c_{12}^{+}) = 12S_{90} - S_{120}.} \tag{51.34}$$

Thus, the weight $12 : -1$ is the image of the fundamental turn: twelve sectors and one oriented boundary. It does not arise from imposing $1/24$ beforehand, from a decimal comparison or from the conventional Barbero–Immirzi value.

Theorem 51.12 (Dodecaphase determination of the Barbero–Immirzi functional)**.** *The fundamental chain* (51.32) *determines the oriented pair* $(a, b) = (12, 1)$*. The coefficient of* $(1 - q)^{-1}$ *in its image is then* $1/24$*, as an exact output of the already constructed functional. Therefore, after evaluating the conjugate channels* $q_{\pm}$*, the dimensionless output is*

$$\boxed{\gamma_{\mathrm{HMT}} = \Gamma_{6\to 8} = \frac{\pi_{\mathrm{HMT}} A C^{*}}{180} + 12\Delta S_{90} - \Delta S_{120}.} \tag{51.35}$$

This formula receives the already generated angular pair and does not use the conventional value of the Barbero–Immirzi parameter as input.

Proof. The sum of (51.32) contains exactly the twelve sectors of C_{12}, while the return seam appears only once and with negative orientation. The reader (51.33) therefore produces (51.34), before any pole calculation. For each $m > 0$,

$$p_1(S_m) = \lim_{q\to 1}(1 - q)\frac{q^m}{1 - q^{3m}} = \frac{1}{3m}.$$

Consequently,

$$p_1\big(\mathscr{R}(c_{12}^{+})\big) = \frac{12}{270} - \frac{1}{360} = \frac{48 - 3}{1080} = \frac{1}{24}.$$

The coefficient p_1 is referred to $(1 - q)^{-1}$. In the complex coordinate $q - 1$,

$$\operatorname*{Res}_{q=1} S_m(q) = -\frac{1}{3m}, \qquad \operatorname*{Res}_{q=1} \mathscr{R}(c_{12}^{+})(q) = -\frac{1}{24}.$$

The subsequent Diophantine equation

$$\frac{a}{270} - \frac{b}{360} = \frac{1}{24} \quad \Longleftrightarrow \quad 4a - 3b = 45$$

has the positive integer solutions $(a, b) = (12 + 3t, 1 + 4t)$, $t \geq 0$; the unique minimal solution is $(12, 1)$. This count arithmetically certifies the pair that the fundamental orbit had already generated: any $t > 0$ alters the multiplicity of the sectors or the seam and ceases to represent one TPK turn. Finally, substituting $q_{\pm}$ into the oriented image and adding the previously generated angular component proves (51.35). □

The causal sequence is fixed without exchanging roles:

$$\begin{aligned} (\mathcal{F}_6 \hookrightarrow \mathcal{F}_8, C_{12}, \partial_{\mathrm{ret}}) &\longrightarrow (90, 120, -1/12, c_{12}^{+}) \\ &\longrightarrow \mathscr{R}(c_{12}^{+}) = 12S_{90} - S_{120}, \\ \mathscr{R}(c_{12}^{+}) &\longrightarrow p_1\big(\mathscr{R}(c_{12}^{+})\big) = \frac{1}{24}, \\ \big(\mathscr{R}(c_{12}^{+}), A, C^{*}, q_{\pm}\big) &\longrightarrow \Gamma_{6\to 8} = \gamma_{\mathrm{HMT}} \longrightarrow \widehat{\mathcal{A}}_{\mathrm{phys}}. \end{aligned} \tag{51.36}$$

The angular pair and its channels come from the preceding construction. Evaluation of the pole coefficient is a check of the return functional; the conjugate evaluation uses the previously generated angular coordinates. The subsequent realization in the Holst action recognizes the same dimensionless section; it does not intervene in its selection.

For the two boundary sectors, let

$$N_m = \sum_{e\in\partial A_m} N_e, \qquad m \in \{90, 120\}, \qquad \mathrm{Spec}(N_e) = \{0, 1, 2\}. \tag{51.37}$$

The physical area operator is written

$$\boxed{\frac{\widehat{\mathcal{A}}_{\text{phys}}}{\ell_P^2} = \gamma_{\text{HMT}} a_0 (N_{90} + N_{120}).} \tag{51.38}$$

On 36 links qutrit,

$$\text{Spec}(N_{90} + N_{120}) = \{0, 1, \dots, 72\}, \tag{51.39}$$

and the multiplicity of the eigenvalue n is the coefficient of z^n in $(1 + z + z^2)^{36}$. From here it follows

$$\text{Spec}\left(\frac{\widehat{\mathcal{A}}_{\text{phys}}}{\ell_P^2}\right) = \{n\gamma_{\text{HMT}} a_0 : 0 \le n \le 72\}. \tag{51.40}$$

51.14 Total Geometry and Origin Memory

The modular Hamiltonian of the boundary is

$$\boxed{K_A = -\log \rho_A = cI + \Delta_{\text{mod}}(90N_{90} + 120N_{120}).} \tag{51.41}$$

Let

$$N = N_{90} + N_{120}, \qquad D = 90N_{90} + 120N_{120}. \tag{51.42}$$

The transition matrix $\left(\begin{smallmatrix} 1 & 1 \\ 90 & 120 \end{smallmatrix}\right)$ has determinant 30, and its inverse reconstructs the sectors:

$$\boxed{N_{90} = 4N - \frac{D}{30}, \qquad N_{120} = \frac{D}{30} - 3N.} \tag{51.43}$$

The area publishes the total number of excitations; the modular Hamiltonian publishes their provenance sector. The joint pair reconstructs the complete occupation of the boundary.

51.15 Bilayer state and oriented information

Let $R = R^* = R^{-1}$ be the sheet involution and $y = C^*_{\text{rad}}$. The normalized state is

$$\boxed{\rho_y = \frac{e^{-yR}}{2\cosh y}.} \tag{51.44}$$

Its modular Hamiltonian and its entropy are

$$K_y = -\log \rho_y = \log(2\cosh y) I + yR, \tag{51.45}$$

$$S(\rho_y) = \log(2\cosh y) - y \tanh y. \tag{51.46}$$

The inversion $y \mapsto -y$ leaves entropy fixed and produces the oriented informational distance

$$\boxed{D(\rho_y \| \rho_{-y}) = 2y \tanh y.} \tag{51.47}$$

The even component measures the spectral distribution; the odd component measures the informational cost of reversing the orientation.

51.16 How much informational energy and gravitational cell of medium action

The periodicity of 108 states determines

$$t_P = \frac{54}{\pi} t_0, \qquad E_P = \frac{\hbar}{t_P} = \frac{h}{108 t_0}. \tag{51.48}$$

The thermal realization of the tritico clock satisfies

$$\boxed{E_P = k_B \Theta_{\mathrm{clk}} \log 3.} \tag{51.49}$$

For the horizon of radius R,

$$E_\Lambda(R) = \frac{c^4 R}{2G}, \qquad E_\Lambda(R) = T_H(R) S_H(R). \tag{51.50}$$

In the Planck cell,

$$\boxed{E_\Lambda = T_H S_H = \frac{E_P}{2} = \frac{k_B \Theta_{\mathrm{clk}} \log 3}{2}.} \tag{51.51}$$

Therefore,

$$\boxed{E_\Lambda t_P = \frac{\hbar}{2}, \qquad E_\Lambda T_{108} = \frac{h}{2}.} \tag{51.52}$$

Vacuum energy, area entropy, chronological temperature, and action share a single confluence cell with typed domains.

When $S_H / k_B = \pi$, the multiplicity efectiva satisfies

$$\Omega_{\mathrm{eff}} = e^{S_H / k_B} = e^{\pi}. \tag{51.53}$$

The hyperbolic Euler operator possesses

$$\mathrm{Spec}(e^{\pi H}) = \{e^{\pi}, e^{-\pi}\}. \tag{51.54}$$

Equality of the scalars establishes spectral confluence; the intertwiner between microscopic horizon multiplicity and Perron dynamics belongs to its specific physical residence.

51.17 Evaluation at the Pell scale and the constitutive Barbero–Immirzi functional

The relation between the vacuum response and the Catalan moment allows two levels of transport to be composed: the weighted depth of the quarter-turn character and the evaluation of the angular channels. The Pell scale distinguishes an argument of the family of moments; the vacuum channels instead receive the arguments determined by the angular pair. The Barbero–Immirzi functional is expressed on the latter through the same constitutive response and its inverse. The link is thus established between defined operations, preserving the position of each evaluation in the APP–TRIT–TPK construction.

51.17.1 Oriented moment and evaluation at the Pell unit

Write

$$G_{\mathrm{Cat}} := C_2(1) = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^2}, \qquad \lambda_{\mathrm{P}} = 2 + \sqrt{3}, \qquad \rho_{\mathrm{P}} = 2 - \sqrt{3} = \lambda_{\mathrm{P}}^{-1}. \tag{51.55}$$

The constant G_{Cat} is the endpoint value of the moment constructed in (41.17). The pair $\lambda_{\mathrm{P}}, \rho_{\mathrm{P}}$ corresponds to the expanding and contracting orientations of the Pell unit: $(2+\sqrt{3})(2-\sqrt{3}) = 1$. The norm identity $2^2 - 3 \cdot 1^2 = 1$ fixes reciprocity. Specialization of the moment at ρ_{P} follows construction of its oriented character.

Differential equation (41.18), with zero condition at the origin, gives

$$\mathcal{D}C_2(s) = \arctan s, \qquad C_2(s) = \int_0^s \frac{\arctan t}{t}\, dt, \qquad \mathcal{D} = s\frac{d}{ds}. \tag{51.56}$$

Indeed, integrating $\mathcal{D}^2 C_2 = s/(1+s^2)$ with respect to ds/s produces $\arctan s$; a second integration produces the stated expression. The integrand extends continuously to $t = 0$ with value 1. The angular form of this integral allows exact determination of the evaluation at ρ_{P}.

Lemma 51.13 (Angular integral of the odd character)**.** *For* $0 < \theta \le \pi/4$,

$$-\int_0^\theta \log(\tan x)\, dx = \sum_{\substack{n\ge 1\\ n\ odd}} \frac{\sin(2n\theta)}{n^2}. \tag{51.57}$$

Proof. For $0 < a < 1$, the logarithm expansion inside the unit disk implies

$$\begin{aligned} L_a(x) &:= \log|1 - ae^{2ix}| - \log|1 + ae^{2ix}| \\ &= -2\sum_{\substack{n\ge 1\\ n\ \mathrm{odd}}} \frac{a^n \cos(2nx)}{n}. \end{aligned}$$

The series converges uniformly in x for every $a < 1$; termwise integration gives

$$-\int_0^\theta L_a(x)\, dx = \sum_{\substack{n\ge 1\\ n\ \mathrm{odd}}} \frac{a^n \sin(2n\theta)}{n^2}.$$

As $a \uparrow 1$, the quotient of moduli converges to $|1 - e^{2ix}|/|1 + e^{2ix}| = \tan x$ for $0 < x \le \theta$. The logarithm's singularity at the origin is integrable. To justify the limit, one may restrict to $a \ge 1/2$ and observe that

$$|1 - ae^{2ix}|^2 = (1-a)^2 + 4a\sin^2 x \ge 2\sin^2 x, \qquad 1 \le |1 + ae^{2ix}| \le 2$$

on $0 \le x \le \pi/4$. Together with the upper bound $|1 - ae^{2ix}| \le 2$, these inequalities give $|L_a(x)| \le c + |\log x|$ with a constant independent of a. Dominated convergence permits passage to the limit in the integral. For the integrated series, domination by $\sum n^{-2}$ applies. The two limits prove (51.57). □

Theorem 51.14 (Catalan–Pell evaluation)**.** *The oriented moment satisfies*

$$C_2(\rho_{\mathrm{P}}) = \frac{2}{3}G_{\mathrm{Cat}} - \frac{\pi}{12}\log\lambda_{\mathrm{P}}, \tag{51.58}$$

$$\mathcal{D}C_2(\rho_{\mathrm{P}}) = \frac{\pi}{12}, \qquad \mathcal{D}^2 C_2(\rho_{\mathrm{P}}) = \frac{1}{4}. \tag{51.59}$$

Proof. The substitution $t = \tan x$ in (51.56), followed by integration by parts, gives

$$C_2(\tan\theta) = \theta\log(\tan\theta) - \int_0^\theta \log(\tan x)\, dx. \tag{51.60}$$

The boundary term at zero vanishes because $x\log(\tan x) \to 0$. The tangent difference formula gives

$$\tan\frac{\pi}{12} = \tan\left(\frac{\pi}{4} - \frac{\pi}{6}\right) = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2 - \sqrt{3}.$$

It remains to evaluate the series in (51.57) at $\theta = \pi/12$. Let $\chi_{-4}(n) = (-1)^{(n-1)/2}$ for odd n. In the six odd classes modulo 12, one verifies

$$2\sin\frac{n\pi}{6} = \chi_{-4}(n) + 3\,\mathbf{1}_{3|n}\chi_{-4}(n/3). \tag{51.61}$$

The finite check is

n mod 12	1	3	5	7	9	11
$2\sin(n\pi/6)$	1	2	1	−1	−2	−1
$\chi_{-4}(n)$	1	−1	1	−1	1	−1
$3\mathbf{1}_{3\|n}\chi_{-4}(n/3)$	0	3	0	0	−3	0

and both sides are periodic modulo 12 on odd indices. Absolute convergence permits separation of summands and the substitution $n = 3m$ in the second term:

$$\sum_{\substack{n\geq 1\\ n\ \text{odd}}} \frac{\sin(n\pi/6)}{n^2} = \frac{1}{2}\sum_{n\ \text{odd}} \frac{\chi_{-4}(n)}{n^2} + \frac{3}{2}\sum_{m\ \text{odd}} \frac{\chi_{-4}(m)}{(3m)^2}$$
$$= \left(\frac{1}{2}+\frac{1}{6}\right) G_{\mathrm{Cat}} = \frac{2}{3}G_{\mathrm{Cat}}.$$

Sums without a lower bound in this expression run over positive integers. Substitution in (51.60), using $\log\rho_{\mathrm{P}} = -\log\lambda_{\mathrm{P}}$, yields (51.58). The first derivative follows from $\arctan\rho_{\mathrm{P}} = \pi/12$; the second from

$$\frac{\rho_{\mathrm{P}}}{1+\rho_{\mathrm{P}}^2} = \frac{1}{\rho_{\mathrm{P}}+\rho_{\mathrm{P}}^{-1}} = \frac{1}{4}.$$

□

The coefficient $2/3$ comes from the character decomposition in (51.61): restriction to multiples of 3 changes the depth weight by the factor 3^{-2}. Evaluation thus brings together the quarter-turn and dodecaphasic subdivision in an exact moment identity. The coefficients are fixed before any decimal comparison.

51.17.2 Composition of phase and weighted depth

The depth representation specifies what this evaluation preserves. On $\ell^2(\mathbb{N}_{>0})$, define

$$N\delta_n = n\delta_n, \qquad U_4\delta_n = i^n\delta_n, \qquad Q_4 = \frac{U_4 - U_4^*}{2i}.$$

For $0 < s < 1$, the operator $N^{-2}s^N Q_4$ is trace class, and

$$C_2(s) = \mathrm{Tr}(N^{-2}s^N Q_4). \tag{51.62}$$

This equality follows by evaluation on the basis δ_n; the diagonal entries are $s^n\sin(n\pi/2)/n^2$. The interchange $U_4 \leftrightarrow U_4^*$ reverses Q_4 and with it the oriented moment. Index n and weight s^n remain fixed.

Composition is performed before taking the trace. For $a \in \mathbb{Z}/4\mathbb{Z}$, let $B(s,a) = s^N U_4^a$. Since both factors are diagonal in the same basis,

$$B(s,a)B(t,b) = B(st, a+b), \qquad B(s,1)^4 = s^{4N}. \tag{51.63}$$

Phase return preserves the accumulated contraction. In particular, $B(\rho_{\mathrm{P}},1)^4 = \rho_{\mathrm{P}}^{4N}$, whose eigenvalue at depth n is ρ_{P}^{4n}. This realization gives an operator meaning to the contractive evaluation; the scalar moment sum aggregates depths after preserving their index and orientation. The constitutive arguments $s_\pm = q_\pm^{30}$ remain determined by the angular pair of (39.47). The preceding formulas contain no equality between those arguments and ρ_{P}.

51.17.3 Dodecaphasic incidence and return functional

The Barbero–Immirzi contribution uses angular transport and hexad–octad incidence. Let $\mathsf{H} \subset \mathsf{O}$ be a hexad and an octad with the hexad's marked pair preserved. The relevant flag sets are

$$\mathcal{F}_6 = \mathsf{H} \times \binom{\mathsf{H}}{2} \hookrightarrow \mathcal{F}_8 = \mathsf{O} \times \binom{\mathsf{H}}{2}, \qquad |\mathcal{F}_6| = 90, \quad |\mathcal{F}_8| = 120. \tag{51.64}$$

The cardinalities are $6\binom{6}{2}$ and $8\binom{6}{2}$; the same incidence preserves the normalized defect $(90-120)/(24\binom{6}{2}) = -1/12$. This count identifies the two sectors used by return, while the calendar determines their relative multiplicities.

For $m \in \{90, 120\}$, the tritic return on a contractive channel is

$$S_m(q) = \sum_{j \geq 0} q^{m+3mj} = \frac{q^m}{1 - q^{3m}}, \qquad 0 < q < 1. \tag{51.65}$$

Each term corresponds to the initial height m and j returns of length $3m$. The geometric series thus preserves both the initial height and return period. The dodecaphasic calendar $\mathbb{Z}/12\mathbb{Z}$ supplies twelve sectors and a unique oriented boundary incidence, denoted ∂_{ret}. In the free module of events, its fundamental chain is

$$c_{12}^{+} = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\mathrm{ret}}, \mathcal{F}_8]. \tag{51.66}$$

The linear readout $\mathscr{L}_q$ assigns $S_{90}(q)$ to each generator $[j, \mathcal{F}_6]$ and $S_{120}(q)$ to the boundary generator. This yields

$$\mathscr{L}_q(c_{12}^{+}) = 12 S_{90}(q) - S_{120}(q). \tag{51.67}$$

The weights 12 and -1 are the chain's oriented multiplicities. The conjugate orientation satisfies $c_{12}^{-} = -c_{12}^{+}$, with the same incidence types.

Proposition 51.15 (Singular coefficient of the fundamental return)**.** *The coefficient of* $(1-q)^{-1}$ *in* $\mathscr{L}_q(c_{12}^{+})$ *is* $1/24$.

Proof. For every integer $m > 0$, the factorization $1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$ gives

$$\lim_{q \uparrow 1} (1 - q) S_m(q) = \frac{1}{3m}.$$

Applying this identity to the functional already determined in (51.67) gives

$$\lim_{q \uparrow 1} (1 - q) \mathscr{L}_q(c_{12}^{+}) = \frac{12}{270} - \frac{1}{360} = \frac{1}{24}. \tag{51.68}$$

In the complex coordinate $q - 1$, the residue is $-1/24$. □

Singular evaluation verifies a consequence of incidence and tritic return. The proof order is fixed by (51.64)–(51.67); the singular coefficient is computed after the chain has been constructed.

51.17.4 Factorization through the constitutive operator

Retain the lifted angular chart of section 39.7. To make the angular conversion explicit, write

$$A_{\mathrm{deg}} = \frac{180x}{\pi}, \qquad C_{\mathrm{deg}}^{*} = \frac{180y}{\pi}, \qquad q_{\pm} = e^{-x \mp y}, \qquad s_{\pm} = q_{\pm}^{30}. \tag{51.69}$$

The orientation considered in (39.42) corresponds to $0 < y < x$; sheet interchange is expressed by $y \mapsto -y$ in the larger domain $x > |y|$. Real exponentials act on these lifts, which preserve the values of $x \pm y$ during composition.

The oriented Barbero–Immirzi functional is

$$\gamma = \frac{\pi A_{\mathrm{deg}} C^{*}_{\mathrm{deg}}}{180} + 12\big[S_{90}(q_-) - S_{90}(q_+)\big] - \big[S_{120}(q_-) - S_{120}(q_+)\big]. \tag{51.70}$$

The first component is the angular evaluation; the second is the difference of the fundamental-chain readout on the conjugate channels. The arguments are received from the previously constructed angular transport. The formula determines a dimensionless scalar on that starting structure.

Let $\psi : (3/4, 1) \to (0, 1)$ be the inverse of f constructed through (39.46). Define

$$\mathcal{H}(s) = \frac{12 s^3}{1 - s^9} - \frac{s^4}{1 - s^{12}} - \frac{(\log s)^2}{20\pi}, \qquad 0 < s < 1. \tag{51.71}$$

This function is real analytic on its domain. The exponents $3, 4, 9, 12$ result from expressing heights $90, 120$ and returns $270, 360$ in units of 30; the logarithmic coefficient is fixed when the angular contribution is transported.

Theorem 51.16 (Oriented constitutive expression of Barbero–Immirzi)**.** *In the chart* (51.69)*,*

$$\gamma = \mathcal{H}(s_-) - \mathcal{H}(s_+) = \mathcal{H}\big(\psi(r_-)\big) - \mathcal{H}\big(\psi(r_+)\big). \tag{51.72}$$

On the two-dimensional channel representation,

$$\gamma = -\operatorname{Tr}\left[\mathcal{R}\,\mathcal{H}\big(\psi(\mathcal{V})\big)\right], \tag{51.73}$$

where Tr *is the ordinary trace, with* $\operatorname{Tr} I = 2$*.*

Proof. Substitution of $s = q^{30}$ into (51.65) gives

$$S_{90}(q) = \frac{s^3}{1 - s^9}, \qquad S_{120}(q) = \frac{s^4}{1 - s^{12}}.$$

Moreover, (51.69) implies

$$\log s_\pm = -30(x \pm y) = -\frac{\pi}{6}(A_{\mathrm{deg}} \pm C^{*}_{\mathrm{deg}}).$$

By the difference of squares,

$$\frac{(\log s_+)^2 - (\log s_-)^2}{20\pi} = \frac{900\big[(x+y)^2 - (x-y)^2\big]}{20\pi} \tag{51.74}$$

$$= \frac{180 xy}{\pi} = \frac{\pi A_{\mathrm{deg}} C^{*}_{\mathrm{deg}}}{180}. \tag{51.75}$$

Adding these identities proves the first equality in (51.72). The second uses the inversion $\psi(f(s_\pm)) = s_\pm$ proved in Section 39.8.

For the operator expression, the sheet projectors have rank one and $\mathcal{R}P_\pm = \pm P_\pm$. Spectral calculus gives

$$\mathcal{H}\big(\psi(\mathcal{V})\big) = \mathcal{H}(s_+)P_+ + \mathcal{H}(s_-)P_-.$$

Multiplying by $\mathcal{R}$ and taking the ordinary trace gives $\mathcal{H}(s_+) - \mathcal{H}(s_-)$. Its negative is γ. □

The orientation mark $\mathcal{R}$ belongs to the functional. If that mark is held fixed and the channels of $\mathcal{V}$ are interchanged, the sign of γ changes. Evaluation of the spectrum without its labels preserves the unordered

pair, whereas (51.73) preserves the oriented contribution. If the normalized trace $\tau_2 = \mathrm{Tr}/2$ is used, the same formula reads $\gamma = -2\tau_2[\mathcal{R}\mathcal{H}(\psi(\mathcal{V}))]$.

By contrast, a simultaneous change of representation $(\mathcal{R}, \mathcal{V}) \mapsto (U\mathcal{R}U^{-1}, U\mathcal{V}U^{-1})$ preserves the functional: spectral calculus is transported by conjugation and the trace is invariant. Interchanging the response relative to a fixed orientation and jointly transporting response and orientation are therefore different operations.

This normalization is compatible with the amplifications of (39.48). For $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$ and $\mathcal{R}_m = \mathcal{R} \otimes I_{9^m}$,

$$\gamma = -\frac{1}{9^m}\,\mathrm{Tr}\left[\mathcal{R}_m \mathcal{H}\big(\psi(\mathcal{V}_m)\big)\right]. \tag{51.76}$$

Indeed, functional calculus preserves the tensor product and the trace of I_{9^m} is 9^m. Representation multiplicity is separated from the oriented evaluation.

51.17.5 Reconstruction from vacuum coordinates

Identities (39.45) express the same functional through normalized impedance and velocity:

$$\gamma = \mathcal{H}\left(\psi\left(\sqrt{\frac{\widehat{Z}}{\widehat{c}}}\right)\right) - \mathcal{H}\left(\psi\left(\frac{1}{\sqrt{\widehat{Z}\widehat{c}}}\right)\right). \tag{51.77}$$

The positive domain of this reconstruction is described by

$$1 < \widehat{Z}\widehat{c} < \frac{16}{9}, \qquad \frac{9}{16} < \frac{\widehat{Z}}{\widehat{c}} < 1. \tag{51.78}$$

Indeed, these inequalities are equivalent to $3/4 < r_\pm < 1$ after substitution of the positive roots of (39.45). For orientation $y > 0$, one also retains $r_- < r_+$, equivalently $\widehat{Z} < 1$. Domain points are used with the angular compatibility already established by the TPK; inversion is a subsequent reconstruction of its channels. The coordinates are the internal ones of (39.44): a change of units must be undone through (63.139) before ψ is applied.

Interchange $y \mapsto -y$ acts as $(\widehat{Z}, \widehat{c}) \mapsto (\widehat{Z}^{-1}, \widehat{c})$ and produces $\gamma \mapsto -\gamma$. In the balanced case $y = 0$, the channels coincide and (51.72) gives $\gamma = 0$. Normalized velocity preserves the product of the channels; its joint readout with impedance recovers both responses and the orientation required to evaluate the functional. In this balanced case $\mathcal{V} = rI$ has degenerate spectrum: the mark $\mathcal{R}$ is retained as representation data, although the scalar r no longer distinguishes its projectors. For simple spectrum, by contrast, $P_+ = (\mathcal{V} - r_- I)/(r_+ - r_-)$ and $P_- = I - P_+$.

The central result of this composition is the factorization (51.72). The Catalan moment supplies the oriented coefficient that determines f; cubic inversion recovers the argument of each channel; dodecaphase incidence and the angular pair determine $\mathcal{H}$. The Pell scale distinguishes an exact evaluation of the family, with the proof of (51.58). Each relation retains its argument and its operation: the extremal constant, the contractive moment and the constitutive functional are linked evaluations of specified structures.

Chapter 52

The HMT–MD Angular Radion: Carry-Preserving Junction of the Direct and Torsional Sections and Exact Closure of the Unit of Phase, Action, and Memory

52.1 Composition APP–TRIT–TPK of the phase state: generation of the memory-oriented unit

Mathematical Synthesis of the Object and Its Causal Chain This monograph reconstructs the radian from the generative chain

$$\mathrm{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow X_{\mathrm{enr}} \longrightarrow \text{discrete structure of the continuum} \longrightarrow \mathrm{MD},$$

without taking as input either the metrological value of $180/\pi$ or the small angular residue to be explained. The result is an enriched unit, the *radion*, defined as the oriented unit that traverses one radian of phase and sweeps exactly an action $\hbar_{\mathrm{ret}}$ in the positive ellipse associated with the same state.

The governing closure is

$$\boxed{\frac{180^\circ}{\pi_{\mathrm{HMT}}} = 50^\circ + A^\circ - \frac{20}{81}\,\Delta_4^\circ + \varepsilon_R^\circ}$$

with

$$S_E = 2\pi_{\mathrm{HMT}}\left(\frac{5}{36} + \frac{25}{9}\alpha_{\mathrm{HMT}}\right), \qquad \Delta_4 = \frac{\pi_{\mathrm{HMT}} - e_{\mathrm{HMT}}\log\pi_{\mathrm{HMT}}}{270}\left(1 + \frac{\pi_{\mathrm{HMT}}}{729}\right),$$

$$\rho_{\mathrm{tw}} = \frac{2\cdot 90}{729} = \frac{20}{81}, \qquad \varepsilon_R^{(1)} = \frac{20}{81}\Delta_4 - (S_E - 1).$$

APP division proves $1 < S_E < 2$, so that the quotient is exactly one and the remainder is $D_E = S_E - 1$. The equality

$$1 = S_E - \frac{20}{81}\Delta_4 + \varepsilon_R^{(1)}$$

is then an internal identity. The angular chart produces

$$\varepsilon_R^\circ = 5.13830167585752820716851646226459474899085744\ldots \times 10^{-8}\,^\circ.$$

The text also demonstrates that the action ellipse

$$Q = a_S\cos\theta, \qquad P = b_S\sin\theta, \qquad a_S b_S = 2\hbar_{\mathrm{ret}}$$

satisfies

$$\mathrm{Area}(0,\theta) = \hbar_{\mathrm{ret}}\theta.$$

Therefore, a radion sweeps $\hbar_{\mathrm{ret}}$ and the turn 2π encloses $h_{\mathrm{ret}} = 2\pi\hbar_{\mathrm{ret}}$. The same ellipse has a finite symplectic boundary and an interior of infinite area when equipped with the Hilbert–Beltrami–Klein metric. This is the exact form of the physical thesis guiding the work: the visible surface closes the action; the interior depth preserves a development that cannot be exhausted by the planar projection.

The exposition maintains typed four realizations of the same state:

1. the circle S^1, which publishes the phase;
2. the positive ellipse, which publishes the action and the anisotropy;
3. the interior of Klein, that miof the depth by means of reason doble;
4. the nonadic helix, which preserves return, memory, and contraction.

They are not four superposed geometries without a criterion. They are four readers with distinct domains, codomains, and conserved data, applied to a single enriched state. The sector 90/120, the dodecaphase, the radion carry, the Lambert tail, the Barbero kernel, and the global towers are likewise presented as related but noninterchangeable objects.

Tesis rectora

The radian is the oriented unit that sweeps $\hbar_{\mathrm{ret}}$ in the action ellipse. The circle publishes its phase, Klein measures its depth, the helix preserves its memory, and the coinductive carry exactly splices its direct and torsional sections.

Expository order and reading dependencies The order is not ornamental. Parts I and II produce the object and its closure from APP–TRIT–TPK. Only thereafter are realizations introduced in the languages of Hilbert, LC, Maxwell, Minkowski, Hodge, Lorentz, modularity, and orbits. Each chapter distinguishes three levels:

Internal result. Equality or construction proved in the HMT domain with its declared primitives.

Consolidated formalization. A new map articulating already existing results without using the target as a selector.

Physical interface. Subsequent dimensional translation that recognizes a conventional structure without making it the origin of the selection.

The *status and falsifier* boxes indicate which concrete remark would invalidate each strong promotion. This discipline does not weaken the thesis; it makes the thesis auditable. The purpose of the extended narrative is to ensure that no word such as *circle*, *ellipse*, *torsion*, *charge*, or *memory* conceals a change of type.

Pivotal results and proof scope

C1. Closure of the radion independent of its final value through two publications of the same state and APP subtraction with carry.

C2. Derivation of the coefficient 20/81 as two sheets of the active sector 90, normalized by the fiber $3^6 = 729$.

C3. Typed equalizer between the seam $\mathfrak{R}s = 1/2$, the hundred-mark APP face, and the dodecaphasic phase $\theta_2 = 50^\circ$.

C4. Radian area theorem: $\mathrm{Area}(\theta) = \hbar_{\mathrm{ret}}\theta$.

C5. Exact focal study of the ellipse and of its Euclidean, hyperbolic, symplectic, and modular invariants.

C6. Boundary–interior theorem: finite total action h and Klein interior of infinite hyperbolic area.

C7. Exact realization through minimal covariance and an LC oscillator, followed by the electromagnetic publication 90/120.

C8. Atlas of non-identifications for all objects 90/120, including the separation among ε_R, the spectral tail, and the Barbero kernel $12 : -1$.

C9. Structural characterization of π in the radion chart, distinct from the primary coinductive generator.

C10. Ontological synthesis: dimension as the minimal rank of memories that a phase projection can no longer preserve.

Correspondence between results, causal types, and causal residences The following correspondence orders the governing assertions and their proofs, while keeping visible the hierarchy that prevents physical richness from erasing the mathematical closure supporting it.

APP–TRIT–TPK genealogy of the radion

The radion arises from the effective composition APP $\to \widehat{\mathbb{T}} \to$ TPK $\to X_{\mathrm{enr}} \to$ MD; it is not obtained by retroactively redefining the conventional radian.

Exact closure and prospective independence

APP carry internally produces

$$1 = S_E - \frac{20}{81}\Delta_4 + \varepsilon_R^{(1)}, \qquad \varepsilon_R^{(1)} = \frac{20}{81}\Delta_4 - (S_E - 1),$$

and the angular chart transports this same identity to $180^\circ/\pi_{\mathrm{HMT}}$ without introducing the target residue.

Critical coordinate of 50°

The critical coordinate $\Re s = 1/2$, published on the APP face of one hundred marks and in the dodecaphase wheel, selects the phase $\theta_2 = 50^\circ$. It is an equality of typed readers of the same state.

Nonadic Bidirectional Density $20/81$

The torsional coefficient is not fitted on the lattice $1/81$: it is constructed as $\rho_{\mathrm{tw}} = 2N_6/3^6 = 2 \cdot 90/729 = 20/81$, preserving the sheet, orientation, and normalization of the nonadic fibre.

Areal law of the radion

For the positive ellipse $Q = a_S \cos\theta$, $P = b_S \sin\theta$, $a_S b_S = 2\hbar_{\mathrm{ret}}$, the oriented area swept is $\mathcal{A}(\theta) = \hbar_{\mathrm{ret}}\theta$. One radion sweeps exactly $\hbar_{\mathrm{ret}}$, and the revolution 2π encloses $h_{\mathrm{ret}} = 2\pi\hbar_{\mathrm{ret}}$.

Focal invariants of the positive operator

The published pair $A \pm C^*$ fixes the semiaxes, eccentricity, focal separation, and symplectic compression parameter. In particular, $(d_1/d_0)^2 = C^*/2$, a dimensionless identity separating shape from scale.

Symplectic finiteness of the boundary and divergence of the interior hyperbolic area

The symplectic boundary of the ellipse encloses the finite action h_{ret}, whereas the Hilbert–Beltrami–Klein area of the interior diverges upon reaching the boundary. Visible phase and depth are not the same measure.

Nonadic Helical Lift with Phase Return and Memory Advance

The nonadic monodromy returns the phase without resetting the state. The helix lifts the circle while preserving carry and memory; the closure $2\pi/4\pi$ distinguishes projective return, reversal of orientation, and complete return.

Oscillatory and constitutive electromagnetic realization

The same anisotropy is realized through a minimal covariance and an LC oscillator: one coordinate stores the electric–elastic aspect and its conjugate the magnetic–rotational aspect. Frequency and impedance remain separately typed.

Typed separation of the operators 90/120

Dodecaphasic extractor, radion carry, Lambert tail, hexad–octad functional, Barbero kernel, and global tower semigroup are distinct objects. The weight $12 : -1$ comes from the twelve sectors of the dodecaphasic fundamental chain and its unique oriented seam. The coefficient of $(1-q)^{-1}$, equal to $1/24$, and the minimal integer counterterm subsequently verify the pair produced.

Complex, Lorentzian, and Hodge-dual realizations

The elliptic, parabolic, and hyperbolic regimes of TRIT are published as rotation, threshold, and hyperbolic transformation. The Hodge star, the constitutive relations, and the Lorentz force act subsequently on the already oriented state; they do not select its constants.

Modular, orbital, and arithmetic boundary characters

The modular involution, elliptic orbits, advance–delay, and 3–6–9 arithmetic recognize distinct memories of the same state. The modular function and vacancies are used as posterior readers, never as substitutes for the APP–TRIT–TPK generator.

Operatorial synthesis of phase, action, depth, and memory

The circle is the phase projection; the ellipse is the action section; Klein measures depth; the helix preserves history. The radion is the oriented unit that keeps these publications coupled without erasing the residue that distinguishes them.

Delimitation of domains and causal continuity

The complete mass laws, electron refinement, CKM/CP, and the exceptional corridor retain their own governing proofs. This monograph cites them only where they share an antecedent or chart; it neither transfers their specific selectors to the radion nor confuses their probative residences.

52.1.1 Bivariate support of APP, ternary orientation of TRIT, and enriched-state update through TPK

The operative genealogy precedes angular measurement and determines the variables preserved by each reader.

52.1.1.1 APP–TRIT–TPK support and enriched state construction

Precedence of the enriched state with respect to the circular publication The elementary definition of radian begins with an already given circle, an already given center, and an already measurable arc length. If the radius and arc have the same length, the subtended angle measures one radian. The definition is correct within Euclidean geometry, but does not answer a prior question: what structure makes it possible for a turn to possess phase, orientation, action, and memory, and why is the natural unit related to 2π?

HMT reverses the causal order. It first constructs the finite state, then its coinductive prolongation, then the revolution, and only at the end its angular publication. The radion does not correct the conventional radian; it lifts it to its genealogical domain.

Type receipt

$$\mathrm{APP} \to \widehat{\mathbb{T}} \to \mathrm{TPK} \to X_{\mathrm{enr}} \to C_{\mathrm{cont}}^{\mathrm{disc}} \to \left(\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}}, A, C^*, \Delta_4, \hbar_{\mathrm{ret}}\right) \to \mathfrak{r}_\sigma.$$

Conventional mathematics appears only afterward as a language of realization: S^1, symplectic ellipse, Klein metric, Hilbert space, LC circuit, Minkowski screen, and field equations.

Bivariate support of APP: leaves, residues, quotients, and carry The Positional Arithmetic Platform (APP) simultaneously preserves two realizations of the symbols $1, \dots, 9$: an additive sheet and a multiplicative sheet. A number is not reduced to its published value. Alongside the value remain sheet, orientation, quotient, remainder, carry, boundary, route, and survival.

The basic census has three strata that must never be identified by mere cardinal coincidence:

$$104,976 \longrightarrow 468\ U_6 \longrightarrow 243\ w_6 \subset \mathbb{F}_3^6, \qquad |\mathbb{F}_3^6| = 729.$$

The 468 decimal emissions, the 243 visible words, and the ambient set of 729 words are distinct objects. The ambient set will be decisive for the normalization $20/81$.

In cubic base $B = 1000$, the oriented subtraction of two triad words is performed from right to left:

$$u_i = t_i - v_i + c_{i+1}, \qquad r_i = u_i \bmod 1000, \qquad c_i = \frac{u_i - r_i}{1000}. \tag{52.1}$$

Once the operands and the remote carry are fixed, quotient and remainder are unique. This is the operator that will subsequently publish the digits of $\varepsilon_R^{(1)}$. No decimal is chosen after seeing the radian.

The completion

$$10^3 = 729 + 243 + 27 + 1$$

is not a decimal curiosity. It expresses conservation of the unit in passing from the ternary ambient space to a chamber of one thousand positions. The difference $1000 - 729 = 271$ is the completing corona; the correct identity is $999 = 729 + 270$, not $729 + 271$.

Ternary orientation and classification of the 3 operational regimes of the TRIT TRIT assigns one of three oriented regimes to each local state. Under the active temporal convention,

$$\begin{aligned} +1 &\longleftrightarrow \text{return, memory, past,} \\ 0 &\longleftrightarrow \text{threshold, present,} \\ -1 &\longleftrightarrow \text{bidirectional opening, future.} \end{aligned}$$

The quadratic realization distinguishes:

$$J^2 = -I, \qquad N^2 = 0, \qquad R^2 = I. \tag{52.2}$$

Thus arise, respectively, elliptic rotation, the parabolic threshold, and hyperbolic separation. Three metrics are not being mixed. Three typed actions on a common antecedent are preserved.

The internal root of -1 is not imported by presupposing the complex plane. In the APP block, two projections P, Q with ratio $r = 3/4 = 90/120$ produce

$$J_{\mathrm{comm}} = \frac{[P, Q]}{\sqrt{r(1-r)}}, \qquad J_{\mathrm{comm}}^2 = -I,$$

while

$$R_{\mathrm{comm}} = 2P - I, \qquad R_{\mathrm{comm}}^2 = I, \qquad R_{\mathrm{comm}} J_{\mathrm{comm}} = -J_{\mathrm{comm}} R_{\mathrm{comm}}.$$

The pair generates the minimal form of $\mathrm{Cl}_{1,1} \simeq M_2(\mathbb{R})$: rotation and opening emerge together, but remain distinct operations.

Dynamic composition of the TPK: selection, transport, memory, and return The Topological Phase Kernel is not a name for a box of objects. It is the composition that selects phase, transports cardinality, lifts sheets, records memory, and executes return. The fundamental dependence is

$$U_t \longrightarrow \Gamma_9 \longrightarrow K_{\mathrm{ph}}.$$

The nonadic holonomy Γ_9 acts on the enriched state; the publication K_{ph} is subsequent and preserves reduced memory. The identity $K_{\mathrm{ph}}^9|_r = E_+ E_\times$ does not turn K_{ph} into a generator.

The nonadic certificate preserves the complete chain

$$w_6 \to w_{12} \to w_{18} \to w_{24} \to w_{30} \to R_{36} \to G_9,$$

with 396 levels, 2376 trits, 43 circuits, 387 pairs $(K, K+9)$ per channel, and five regions of π. The phase satisfies

$$g(k+9) = g(k),$$

but the complete state does not return:

$$B_{k+9} \neq B_k.$$

This is the mathematical seed of the helix: the angular projection closes; the memory advances.

Non-split temporal extension $C_{12} \to C_{108} \to C_9$ The twelve-step memory is organized via

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0. \tag{52.3}$$

The cocycle

$$c(r, r') = \left[\left\lfloor \frac{r + r'}{9} \right\rfloor \right]_{12}$$

measures the carry between phase return and memory advance. Nine steps close the observable phase C_9; they do not reset the dodecaphasic record.

It is convenient to separate four objects:

C_{12}	cyclic group of twelve positions;
$\mathcal{R}_{12}$	enhanced record $(E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)_{m=1}^{12}$;
Θ_{108}	raw publication on the 108 calendar;
Γ_{108}	history or lifted holonomy.

Accordingly, the vague expression "post-C_{12}" shall be replaced by "subsequent to the dodecaphasic record $\mathcal{R}_{12}$, in its hexadic" reading when that is the effective map.

Correlative emergence of the 5 constructions of the continuum from the unique state The discrete structure of the continuum is not obtained by assembling five independent theories. Its five internal aspects—spectral, solenoidal, gauge, cohomological, and determinant—emerge correlatively from the same APP–TRIT–TPK state. This work does not prove the integral treatise anew, but preserves the causal fact required by the radion: the coinductive section that generates π, e, φ and α does not live on a bare real line; it lives in an object with fibers, relations, orientation, carry, memory, terminal object, and reader.

Hypothesis, domain, and conclusion

Genealogy falsifier. If a later formula receives $180/\pi$, ε_R, a mass, or a metrological table in order to choose a phase, a coefficient, or a branch, it ceases to be an HMT-first generation. The external contrast may recognize an output; it cannot select it.

52.1.2 Prospective independence of the closure with respect to the final value

52.1.2.1 Genealogy of the constants and construction of the angular pair

Coinductive section and uniqueness of the generated value of π Be x_∞ the section global of the inverse system. The lectores coinductivos produce

$$\mathrm{ev}_\pi(x_\infty) = \pi_{\mathrm{HMT}}, \qquad \mathrm{ev}_e(x_\infty) = e_{\mathrm{HMT}}, \qquad \mathrm{ev}_\varphi(x_\infty) = \varphi_{\mathrm{HMT}}.$$

The HMT subscript does not designate another number. It preserves the genealogy of the same constant. In particular,

$$\pi_{\mathrm{HMT}} = 3.14159265358979323846264338327950288419716939937510\ldots.$$

The positive loop is

$$T_+(x_\infty) = 2\pi_{\mathrm{HMT}}. \tag{52.4}$$

Its angular publication of a unit is

$$R^\circ_{\mathrm{HMT}} = \frac{360^\circ}{T_+} = \frac{180^\circ}{\pi_{\mathrm{HMT}}}. \tag{52.5}$$

The equation does not yet define the enriched radion; it defines the visible chart of the radian once the turn has been generated.

Bidirectionality of conjugate routes and prolongations Bidirectionality is not enunciated as a slogan. The concrete involution

$$J_\alpha(T) = \frac{\alpha^2_{\mathrm{HMT}}}{T}, \qquad T_- = J_\alpha(T_+), \qquad T_+T_- = \alpha^2_{\mathrm{HMT}} \tag{52.6}$$

preserves a product and reverses the sheet. Expansion and contraction are two conjugate prolongations of the same state. The finite root of a quadratic jet can approximate T_+, but it does not replace the coinductive section: the certified difference is

$$T_{P_9} - T_+ = -5.1110566290335\ldots \times 10^{-25}.$$

Finite control is a witness; the foundation is the limit.

Causal dependence among α, action, and angular publication The dodecaphase closure generates α_{HMT} from the signed state and its reversible charts. The local determinantal reader then fixes the action decade through

$$\Sigma_H = \varphi_{\mathrm{HMT}}\alpha^{16}_{\mathrm{HMT}}, \qquad \mathrm{ord}_{10}\,\Sigma_H = -34.$$

The return section produces $\hbar_{\mathrm{ret}}$. Neither α nor $\hbar$ is "measured in degrees". They are subsequently published through the angular pair

$$P(\alpha, \eta_{\mathrm{ret}}) = \Big([10^3\alpha]_{360}, [2(\eta_{\mathrm{ret}} + \alpha)]_{360}\Big) = (A, C^*). \tag{52.7}$$

In the canonical branch,

$$A = 7.29735256928380099728510547238066 3\ldots^\circ,$$

$$C^* = 2.12373833896864572426195138039336079 37\ldots^\circ.$$

Only then the analytic chart

$$x = \frac{\pi A}{180}, \qquad y = \frac{\pi C^*}{180}$$

open the channels $q_\pm = e^{-x\mp y}$.

The complete causal order is

$$\alpha \to (\eta \parallel -34) \to \hbar \to P \to \kappa_{360} \to q_\pm \to \text{MD realizations}.$$

The relation $C^*/2 - \alpha$ may serve as a reverse verification. It is not used to generate $\hbar_{\text{ret}}$ retroactively.

> The angular chart does not turn a dimensionless constant or an action into an angle by fiat. It publishes two state coordinates on a finite wheel. The degree is the language of that publication; it is not the ontological substance of α or $\hbar$.

Prospective independence with respect to the final value The depth operator verifier explicitly prohibits three inputs:

$$\frac{180}{\pi}, \qquad \varepsilon_R, \qquad \text{the target decimal.}$$

First construct S_E and S_T, then execute the APP subtraction, and only at the end compare the closure. The controls give

`target_epsilon_used=False`, `target_180_over_pi_used=False`.

Also:

- a single sheet gives a defect of order 10^{-5};
- replacing θ_2 by θ_1 or θ_3 shifts the result by approximately $\pm\pi/6$;
- removing channel e leaves a defect of order 10^{-3};
- the variante historica $\Delta_4(A, C^*)$ produce other number;
- the spectral tail following $\mathcal{R}_{12}$ and the four-term harmonic operator do not coincide with $\varepsilon_R^{(1)}$ either.

Hypothesis, domain, and conclusion

Status. The closure is a *new formalization* proved with an explicit initial structure and a computational certificate independent of the final value. The equality is exact in the coinductive limit. The short 36-digit profile of α generates a variant that diverges only after many digits; it is retained as a historical witness, not as the canonical value.

52.2 Direct and torsional sections: APP carry and exact closure of the radion

52.2.1 Independent construction of the sections S_E and S_T before defining the cocycle $\varepsilon_R = S_T - S_E$

Exact closure is obtained through two sections constructed before the cocycle that relates them.

52.2.1.1 Independent construction of direct and torsional sections

Direct section of the dodecaphasic temporal system The dodecaphase wheel has centers

$$\theta_m = 30m - 10^\circ, \qquad m = 1, \dots, 12. \tag{52.8}$$

The second phase is $\theta_2 = 50^\circ$. Together with the publication $A = 10^3 \alpha_{\mathrm{HMT}}$, it forms the direct section

$$S_E = \frac{T_+}{360}\,(50 + 10^3 \alpha_{\mathrm{HMT}}) \tag{52.9}$$

$$= 2\pi_{\mathrm{HMT}} \left(\frac{5}{36} + \frac{25}{9} \alpha_{\mathrm{HMT}} \right). \tag{52.10}$$

The subscript E here means *direct or common*; its electrical realization will come after the constitutive reader. One does not literally postulate $A = E$.

The coinductive cylinders prove

$$1 < S_E < 2. \tag{52.11}$$

The APP division then enforces

$$q_E = \lfloor S_E \rfloor = 1, \qquad D_E = \mathrm{rem}_1(S_E) = S_E - 1. \tag{52.12}$$

The integer one is not the number of turns of the helix. It is the unique quotient of the direct section within the unit cell.

Torsional Section Determined by the Operator Δ_4 The second section uses a previous coinductive output:

$$\Delta_4 = \frac{\pi_{\mathrm{HMT}} - e_{\mathrm{HMT}} \log \pi_{\mathrm{HMT}}}{270} \left(1 + \frac{\pi_{\mathrm{HMT}}}{729} \right). \tag{52.13}$$

The word *torsion* here denotes the global quantum of the HMT chart. It does not automatically identify it with Cartan torsion $T^I = D_\omega e^I$. The bridge to Cartan requires another typed arrow.

52.2.2 Oriented density of the active sector and unique determination of $20/81$

Oriented density $20/81$ of the active sector $90/120$ The triadic phase separator selects

$$H_+ = \{1, 4, 7\}, \qquad H_- = \{2, 5, 8\}, \qquad H_0 = \{3, 6, 9\}.$$

There are six active phases and fifteen unordered pairs:

$$H_{\mathrm{act}} = H_+ \cup H_-, \qquad \left| \binom{H_{\mathrm{act}}}{2} \right| = \binom{6}{2} = 15.$$

The active mode and its extension to eight positions are

$$N_6 = 6\binom{6}{2} = 90, \qquad N_8 = 8\binom{6}{2} = 120. \tag{52.14}$$

Two conjugate sheets of the active sector, normalized by the ambient fiber, give

$$\rho_{\mathrm{tw}} = \frac{2N_6}{|\mathbb{F}_3^6|} = \frac{2 \cdot 90}{729} = \frac{180}{729} = \boxed{\frac{20}{81}}. \tag{52.15}$$

Proposition 52.1 (Relative uniqueness of the density)**.** *With the active sector $N_6 = 90$, the two conjugate sheets, and the normalization by $|\mathbb{F}_3^6| = 729$ fixed, the oriented density is necessarily* $20/81$.

Proof. The transported cardinality is $2N_6 = 180$. Normalization by the ambient space gives $180/729$; dividing the numerator and denominator by 9 yields $20/81$. No comparison with $180/\pi$ or with ε_R is involved. □

The historical reading as a projection onto a $1/81$ lattice is a useful rigidity control, but it is no longer the foundation: the coefficient is obtained before closure. Ablating one sheet produces $10/81$, and the output changes at order 10^{-5}; hence, the factor of two is not decorative.

The torsional section is

$$\Phi_T = \rho_{\mathrm{tw}} \Delta_4, \qquad S_T = 1 + \Phi_T. \tag{52.16}$$

52.2.3 The carry ε_R as a transition cocycle between sections

52.2.3.1 Transition cocycle between the direct and torsional sections

definition operatoria of the carry ε_R

Definition 52.2 (Real carry of the radion)**.** The unitary output of the radion is the transition cocycle

$$\varepsilon_R^{(1)} := S_T - S_E = \Phi_T - D_E = \frac{20}{81}\Delta_4 - operatornamerem_1(S_E). \tag{52.17}$$

The equation contains a subtraction, but it is not a fit to the target. Its two operands S_T and S_E are generated separately. The independence demonstrated is independence from $180/\pi$ and from a target ε; algebraic independence between a difference and its summands is not asserted.

Theorem 52.3 (Exact unitary closure)**.** *With S_E, Φ_T, and $\varepsilon_R^{(1)}$ defined by* (52.10)*,* (52.13)*,* (52.15)*, and* (52.17)*, one has exactly*

$$\boxed{1 = S_E - \Phi_T + \varepsilon_R^{(1)}.} \tag{52.18}$$

Proof. By (52.11), $D_E = S_E - 1$. By definition, $\varepsilon_R^{(1)} = \Phi_T - D_E$. Then

$$S_E - \Phi_T + \varepsilon_R^{(1)} = S_E - \Phi_T + \Phi_T - (S_E - 1) = 1.$$

The strength of the proof lies not in this elementary cancellation, but in the fact that the prior genealogy fixes both operands without using the closure as input. □

Publication of the closure on the angular chart The full-turn chart multiplies a unit fraction by $360/T_+ = 180/\pi_{\mathrm{HMT}}$. In particular,

$$\varepsilon_R^{\circ} = \frac{360^\circ}{T_+}\varepsilon_R^{(1)}.$$

Multiplying (52.18) by $360^\circ/T_+$ and using

$$\frac{360}{T_+}S_E = 360\left(\frac{5}{36} + \frac{25}{9}\alpha_{\mathrm{HMT}}\right) = 50 + 1000\alpha_{\mathrm{HMT}} = 50 + A$$

produce the angular closure.

Theorem 52.4 (Exact equation of the radion)**.**

$$\boxed{\frac{180^\circ}{\pi_{\mathrm{HMT}}} = 50^\circ + A^\circ - \frac{20}{81}\,\Delta_4^\circ + \varepsilon_R^\circ}$$

The equation says something more precise than "the radian is $57.2957\ldots^\circ$". It decomposes that publication into four operations with a reference proof:

50°	second phase of the wheel; critical hinge on a declared chart;
A°	common coordinate of the parangular publication;
$-\frac{20}{81}\Delta_4^\circ$	orientation-transport of global torsion;
$+\varepsilon_R^\circ$	real carry that mates the two lifts.

Convergence by depth of refinement At depth N, the generator delivers rational cylinders $p_N \to \pi_{\mathrm{HMT}}$ and $e_N \to e_{\mathrm{HMT}}$. Define

$$S_{E,N} = 2p_N\left(\frac{5}{36} + \frac{25}{9}\alpha_{\mathrm{HMT}}\right),$$

$$\Delta_{4,N} = \frac{p_N - e_N \log p_N}{270}\left(1 + \frac{p_N}{729}\right), \qquad S_{T,N} = 1 + \frac{20}{81}\Delta_{4,N},$$

and

$$\mathcal{E}_N = S_{T,N} - S_{E,N}.$$

Continuity of the functional on $3 < p < 4$, $2 < e < 3$, together with the monotone derivatives, transports each pair of cylinders to a certified interval. At a depth of 45 blocks, its width is less than 4.81×10^{-130}.

Each nine-level return contains six trits per level. Therefore, the refinement rate is

$$729^{-9} = 3^{-54}. \tag{52.19}$$

This quantity controls the width of the intervals; it is not the value of ε_R.

The subtraction of triads stabilizes

```
000|000|000|896|802|822|044|562|981|999|050|695|020|651|286|413
```

and the carry prefix

```
0,0,0,0,0,-1,0,-1,-1,-1,0,-1,0.
```

This is the concrete mechanism of decimal memory: not a chosen tail, but the recurrence (52.1) applied to previously constructed operands.

Evaluation of the canonical values The coinductive output is

$$\varepsilon_R^{(1)} = 8.96802822044562981999050695020651286413727362050 09\ldots \times 10^{-10}, \tag{52.20}$$

$$\varepsilon_R^\circ = 5.13830167585752820716851646226459474899085744000 54\ldots \times 10^{-8\,\circ}. \tag{52.21}$$

The complete publication results

$$\frac{180^\circ}{\pi_{\mathrm{HMT}}} = 57.29577951308232087679815481410517033240547246656 43\ldots^\circ.$$

Exact Result

$$\begin{aligned}
50^\circ + A^\circ &= 57.297352569283800997285105472380662999 56\ldots^\circ,\\
\frac{20}{81}\Delta_4^\circ &= 0.001573107584496879062232729960657289804 65\ldots^\circ,\\
50^\circ + A^\circ - \frac{20}{81}\Delta_4^\circ &= 57.295779461699304118222872742420005709 76\ldots^\circ,\\
\varepsilon_R^\circ &= 0.000000051383016758575282071685164622645947 49\ldots^\circ,\\
\text{final sum} &= 57.295779513082320876798154814105170332 41\ldots^\circ.
\end{aligned}$$

52.3 Critical coordinate and oriented dodecaphase temporal system: affine publication of the phase of 50°

52.3.1 Affine map j_{100} of the APP chart and uniqueness of the phase $m = 2$ in $j_{100}(1/2) = \theta_m = 50^\circ$

The critical coordinate links the Cayley chart, the APP record, and the dodecaphasic phase through typed maps.

52.3.1.1 Two exact realizations of the angular coordinate 50°

Second phase of the dodecaphasic temporal system By (52.8), the sequence of centros begins

$$20^\circ,\ 50^\circ,\ 80^\circ,\ 110^\circ, \ldots.$$

Solve

$$30m - 10 = 50$$

gives only $m = 2$. Therefore, 50° is not a rounding of A, a half-turn, or the centre of a circle: it is a typed phase of the dodecaphase calendar. The fraction of a turn is

$$\frac{50}{360} = \frac{5}{36},$$

exactly the first summand of S_E/T_+.

Cayley Critical Coordinate Matching The functional involution

$$\rho \longmapsto 1 - \bar{\rho}$$

fixed

$$\mathrm{Fix}(\rho \mapsto 1 - \bar{\rho}) = \{\rho : \Re\rho = 1/2\}.$$

The number $1/2$ is the self-dual real coordinate of the functional interval $0 \le \Re s \le 1$. In the longitudinal sense, it is not "half of a line", and it does not contain the spectral height γ. For

$$\rho_\gamma = \frac{1}{2} + i\gamma,$$

The Cayley transformation can be written

$$\tau_\gamma = \frac{\rho_\gamma - 1}{\rho_\gamma} = \frac{\gamma + i/2}{\gamma - i/2} \in S^1, \qquad \gamma = \frac{1}{2}\cot\frac{\theta_\gamma}{2}. \tag{52.22}$$

Thus, $1/2$ fixes the seam, γ determines the angular character, and Γ_9 preserves the depth of its iterations.

Affine injection j_{100} of the APP record into the angular chart Cubic completion arranges a chamber of ten marks per coordinate. One face contains $10^2 = 100$ positions. We declare the affine chart

$$j_{100} : [0,1] \longrightarrow [0, 100^\circ], \qquad j_{100}(\sigma) = 100\sigma^\circ. \tag{52.23}$$

Then

$$j_{100}\left(\frac{1}{2}\right) = 50^\circ.$$

Requiring $j_{100}(1/2) = \theta_m$ simultaneously yields

$$100 \cdot \frac{1}{2} = 30m - 10 \quad \Longleftrightarrow \quad m = 2.$$

Theorem 52.5 (Critical Equalizer–Dodecaphasic)**.** *In the hundred-mark APP chart, the fixed coordinate of the critical involution is published exactly in the second phase of the wheel:*

$$\boxed{j_{100}(1/2) = \theta_2 = 50^\circ.} \tag{52.24}$$

The phase $m = 2$ is unique.

The theorem is exact once (52.23) has been declared. Its status is *new formalization*: the critical midpoint, the 10^3 completion, the hundred-mark face, and θ_2 had already been constructed; choosing that face as the affine critical chart brings their domains together for the first time.

Compatibility of the critical value with the Cayley chart The Cayley transform does not send $1/2$ to 50°. If $\gamma = 0$, (52.22) gives

$$\tau_0 = -1 = e^{i\pi},$$

that is, 180°. The 50° belong to the APP chart of one hundred marks and to the dodecaphasic calendar; the Cayley angle depends on γ. This distinction prevents all the zeros of the zeta function from being collapsed into a single phase.

Falsification criterion 52.6. The identification $1/2 \leftrightarrow 50^\circ$ fails if the chart (52.23) is removed. The scalar equality $100(1/2) = 50$ is not sufficient by itself to identify two objects. The theorem depends on the distinguished APP face and on the dodecaphase wheel.

52.3.2 Compatibility of the critical phase with the nonadic return and memory advance

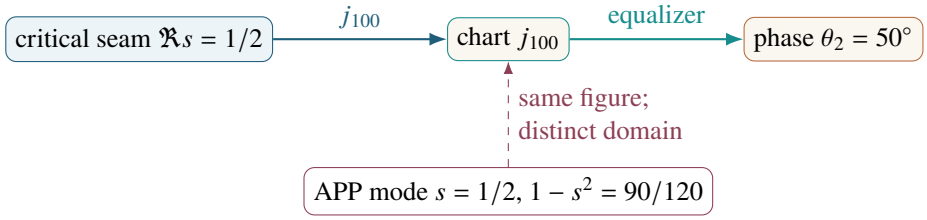
52.3.2.1 Typed separation of the 2 objects with value $1/2$

APP contains, in addition, a singular mode with

$$s = \frac{1}{2}, \qquad s^2 = \frac{1}{4}, \qquad 1 - s^2 = \frac{3}{4} = \frac{90}{120}, \qquad s\sqrt{1 - s^2} = \frac{\sqrt{3}}{4}.$$

This s arises from the principal angles of the sum and product sheets. The critical seam $\Re\rho = 1/2$ arises from a functional involution. That both coordinates equal $1/2$ is a scalar coincidence with possible content, but it does not authorize their identification without a typed square.

The correct way to preserve the finding is an equalizer diagram:



The diagram does not feign the missing arrow. It shows what has been proved, what has been formalized, and what equality still requires additional transport if a global operator identification is intended.

52.3.2.2 Spectral Circle and Helical Memory Compatibility

The Cayley–Pontryagin reader transports a spectral fiber via

$$\widetilde{U}_{\Gamma_9^n} J\psi_\gamma = \tau_\gamma^n J\psi_\gamma.$$

The circular publication is $\tau_\gamma^n \in S^1$. The lift

$$(\tau_\gamma^n, n) \in S^1 \times \mathbb{Z}$$

preserves the number of returns. On the contractive sheet appears

$$\widetilde{M}_9^n J\psi_\gamma = \left(\frac{5}{9}\right)^n \tau_\gamma^n J\psi_\gamma.$$

We therefore have three typed images:

$$\text{phase circle}, \qquad \text{memory helix}, \qquad \text{contractive spiral}.$$

A zero is not a revolution, and a nine is not a particular zero of the zeta function. The zero fixes a character τ_γ that persists through the revolutions; nonadic return increments memory. Nine functions as residual zero in $\mathbb{Z}/9\mathbb{Z}$, an arithmetic operation distinct from the enumeration of spectral zeros.

> The critical line becomes a circle when we forget the linear height and retain the unitary character. It becomes a helix when we reinsert the memory of how many times the character has been transported. Depth is not a third decorative coordinate: it is the information that the circular projection cannot retain.

52.4 Angular pair and reduced action: construction of the positive ellipse and area per radion

52.4.1 Areal law of the radion: $\mathcal{A}(\theta) = \hbar_{\text{ret}}\theta$ and $\mathcal{A}(2\pi) = h_{\text{ret}}$

Spectral geometry determines the form, and the areal law determines the action transported by the phase.

52.4.1.1 Equivalence between the spectral and geometric charts

Spectral positive operator and conjugate eigenvalues The angular momentum produces the two-layer operator.

$$B_1 = AI + C^*R, \qquad R^2 = I, \qquad P_\pm = \frac{I \pm R}{2}. \tag{52.25}$$

Its eigenvalues are $A \pm C^*$. Since $0 < C^* < A$, both are positive. In the chart $\kappa = \pi/180$, the spectral ellipse has semiaxes

$$a_1^2 = \kappa(A + C^*), \qquad b_1^2 = \kappa(A - C^*). \tag{52.26}$$

The decomposition is reversible:

$$A = \frac{a_1^2 + b_1^2}{2\kappa}, \qquad C^* = \frac{a_1^2 - b_1^2}{2\kappa}. \tag{52.27}$$

The ellipse is not drawn to illustrate a number. It exactly reconstructs the pair that generated it.

Define

$$\rho_1 = \sqrt{A^2 - C^{*2}}, \qquad \eta = \operatorname{artanh} \frac{C^*}{A} = \frac{1}{2} \log \frac{A + C^*}{A - C^*}. \tag{52.28}$$

Then

$$A \pm C^* = \rho_1 e^{\pm \eta},$$

and

$$a_1 = \sqrt{\kappa \rho_1}\, e^{\eta/2}, \qquad b_1 = \sqrt{\kappa \rho_1}\, e^{-\eta/2}. \tag{52.29}$$

The product preserves scale; the quotient preserves form:

$$a_1 b_1 = \kappa \rho_1, \qquad \frac{b_1}{a_1} = e^{-\eta} = \sqrt{\frac{A - C^*}{A + C^*}}.$$

Focal Invariants of the Positive Operator The semifocal distance is

$$c_1^2 = a_1^2 - b_1^2 = 2\kappa C^*, \qquad F_\pm = (\pm c_1, 0). \tag{52.30}$$

The total distance is equal to

$$d_1 = 2c_1 = 2\sqrt{2\kappa C^*} = 0.544545509325626623406299779866023\ldots. \tag{52.31}$$

The isolated number is not the thesis. The identity $c_1^2 = 2\kappa C^*$ states that C^* is literally the quadratic focal splitting in the distinguished chart. The foci are the conjugate endpoints of the spectral aperture; they are not, by themselves, the electric and magnetic poles. That reading requires the constitutive transducer.

The eccentricity is

$$e_f = \frac{c_1}{a_1}, \qquad e_f^2 = 1 - e^{-2\eta} = \frac{2C^*}{A + C^*}. \tag{52.32}$$

Numerically,

$$\eta = 0.29968968392163079968492656286392 7\ldots,$$
$$e_f = 0.67145189555019198691627705586433 2\ldots.$$

Reference APP Block and Normalized Transport The central block

$$B_0 = 7I + 2R = 9P_+ + 5P_-$$

possesses

$$\rho_0 = \sqrt{45}, \qquad \eta_0 = \frac{1}{2} \log \frac{9}{5}, \qquad e_0 = \frac{2}{3}.$$

Its focal length is

$$d_0 = 4\sqrt{\kappa} = 0.528443639680801468446003835349358\ldots.$$

The exact comparison is

$$\boxed{\left(\frac{d_1}{d_0}\right)^2 = \frac{C^*}{2}}, \qquad d_1 = d_0 \sqrt{\frac{C^*}{2}}. \tag{52.33}$$

This is the defensible genealogy of 0.544545 . . .: a focal APP scale of reference multiplied by a relative invariant. Its visual proximity to 0.54 is not an identity with the APP defect 54.

Transport from B_0 separates into scale and anisotropy:

$$T_{\mathrm{rel}} = \frac{\rho_1}{\sqrt{45}} \exp\big((\eta - \eta_0)R\big). \tag{52.34}$$

On the leaves,

$$t_+ = \frac{A + C^*}{9} = 1.0467878786947163\ldots, \qquad t_- = \frac{A - C^*}{5} = 1.0347228460630311\ldots.$$

The deformation is not a single decimal. Its isotropic component is $\sqrt{t_+ t_-}$; its hyperbolic component is $\frac{1}{2}\log(t_+/t_-)$.

52.4.1.2 Positive ellipse associated with the angular pair

Construction via Conjugate Semi-axes The preceding spectral scale has no units of action. Dimensional Mechanics publishes the same form on the internal action line:

$$a_S = \sqrt{2\hbar_{\mathrm{ret}} e^{\eta}}, \qquad b_S = \sqrt{2\hbar_{\mathrm{ret}} e^{-\eta}}. \tag{52.35}$$

Immediately,

$$a_S b_S = 2\hbar_{\mathrm{ret}}, \qquad \frac{a_S}{b_S} = e^{\eta}. \tag{52.36}$$

The product fixes the action; the quotient fixes the anisotropy. The ellipse is parametrized by

$$Q(\theta) = a_S \cos\theta, \qquad P(\theta) = b_S \sin\theta. \tag{52.37}$$

The spectral and action ellipses are homothetic:

$$\lambda_{\mathrm{act}}^2 = \frac{2\hbar_{\mathrm{ret}}}{\kappa\rho_1}, \qquad (a_S, b_S, c_S) = \lambda_{\mathrm{act}}(a_1, b_1, c_1). \tag{52.38}$$

They share the form $e^{-\eta}$, not the physical dimension.

Areal law of action per radion The forma simplectica canonica is $\omega = \mathrm{d}Q \wedge \mathrm{d}P$. The area oriented barrida from 0 until θ is

$$\mathcal{A}(\theta) = \frac{1}{2}\int_0^{\theta} \big(Q(t)P'(t) - P(t)Q'(t)\big)\ \mathrm{d}t \tag{52.39}$$

$$= \frac{1}{2} a_S b_S \int_0^{\theta} (\cos^2 t + \sin^2 t)\ \mathrm{d}t. \tag{52.40}$$

Using (52.36), we obtain the central result.

Theorem 52.7 (Action per radion)**.** *For the ellipse* (52.37)*,*

$$\boxed{\mathcal{A}(\theta) = \hbar_{\mathrm{ret}}\theta.} \tag{52.41}$$

In particular,

$$\boxed{\mathcal{A}(1) = \hbar_{\mathrm{ret}}}, \qquad \boxed{\mathcal{A}(2\pi) = 2\pi\hbar_{\mathrm{ret}} = h_{\mathrm{ret}}}. \tag{52.42}$$

This equality allows the definition of the radius without first appealing to the arc length.

Definition 52.8 (Oriented radion)**.**

$$\mathfrak{r}_\sigma = (\theta = 1, \sigma), \qquad \sigma \in \{+1, -1\}, \qquad \mathcal{A}(\mathfrak{r}_\sigma) = \sigma \hbar_{\mathrm{ret}}. \tag{52.43}$$

The radian is the projection that forgets σ, the sheet, the memory, and the area. The radion preserves that information. Hence the name *rad–ion*: orientation precedes physical charge, and charge appears later as

$$Q(x) = n_Q(x) e_{\mathrm{int}}, \qquad e_{\mathrm{int}} = \sqrt{\frac{2\alpha_{\mathrm{HMT}} h_{\mathrm{ret}}}{Z_{\mathrm{int}}}}.$$

52.4.2 Conjugate semi-axes, focal invariants, and orientation of anisotropy

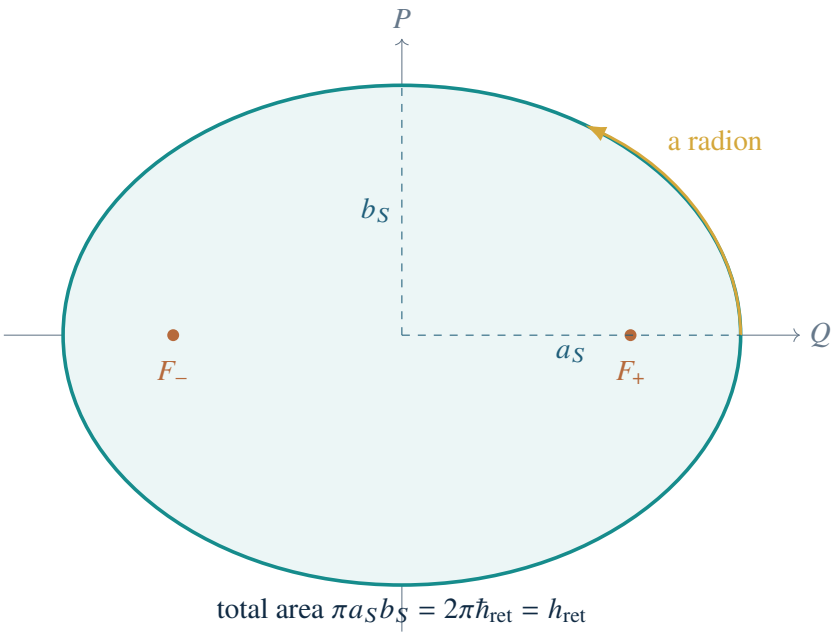


Figure 52.1: The action ellipse. A parametric arc of one radian sweeps $\hbar_{\mathrm{ret}}$; one turn encloses h_{ret}. The foci encode the aperture C^* in the distinguished chart.

Correspondence between phase, action, area, and orientation

Relationship between the reduced action, uncertainty, and area The relation $\hbar = h/(2\pi)$ is often presented as a notational convenience: when describing oscillations, rotations, and Fourier transforms, the factors 2π disappear if $\hbar$ is used. The action ellipse reveals a stronger reading. Division by 2π is exactly the passage from the action of one revolution to the action per unit phase:

$$h = \mathcal{A}(2\pi), \qquad \hbar = \mathcal{A}(1).$$

The reduced constant is not merely h in more convenient notation; it is the areal density of action with respect to phase.

Heisenberg, Dirac, and Jordan established the language of conjugate variables, commutators, and canonical transformations. No literal "ontological ellipse" lacking a source is attributed to them here. The HMT–MD contribution is to identify the positive figure, construct its semiaxes from (A, C^*), and prove (52.41). The historical genealogy opens the language; the present theorem fixes the object.

> Action is not appended to motion afterward. In the radion ellipse, action is the area generated by motion. A full turn does not transport a prefabricated constant: it encloses it.

Algebraic controls of the focal invariants Decimal similarities are subject to control:

Comparison	Residue	Verdict
d_1 versus 0.54	4.5455×10^{-3}	not equality; 54 is not deduced;
d_1^2 versus 8/27	2.3352×10^{-4}	proximity without map;
η frente a 3/10	-3.1032×10^{-4}	no igualdad;
e_f versus 2/3	4.7852×10^{-3}	2/3 belongs exactly to B_0;
$e^{-\eta}$ versus 20/27	3.0740×10^{-4}	no igualdad.

The robust invariants are (52.28), (52.32), and (52.33). The HMT criterion does not consist in favouring a decimal resemblance, but in requiring an object, an operation, and conservation.

52.5 Nonadic Return and Internal Projective: Klein Disk and Helical Depth of Memory

52.5.1 Normalization to the Hilbert–Beltrami–Klein disk and hyperbolic interior

Projective geometry distinguishes the finite action from hyperbolic depth and return memory.

52.5.1.1 Hyperbolic Interior of Hilbert–Beltrami–Klein

Projective normalization on the Klein disk The affine application

$$u = \frac{Q}{a_S}, \qquad v = \frac{P}{b_S}$$

carries the action ellipse to the unit disk

$$D = \{(u, v) : u^2 + v^2 < 1\}.$$

On each chord, the Hilbert distance is defined through the cross-ratio of its boundary endpoints. The resulting infinitesimal metric is the Beltrami–Klein form

$$\mathrm{d}s_K^2 = \frac{(1 - r^2)(\mathrm{d}u^2 + \mathrm{d}v^2) + (u\,\mathrm{d}u + v\,\mathrm{d}v)^2}{(1 - r^2)^2}, \qquad r^2 = u^2 + v^2. \tag{52.44}$$

With this normalization, the Gaussian curvature is constant:

$$K = -1. \tag{52.45}$$

The geodesics are Euclidean chords, but their hyperbolic length diverges as they approach the boundary.

Finiteness of the boundary and infinitude of the hyperbolic depth The metric determinant of (52.44) produce

$$\mathrm{d}A_K = \frac{\mathrm{d}u\ \mathrm{d}v}{(1 - r^2)^{3/2}}. \tag{52.46}$$

The area of the subdisk $r < R < 1$ is

$$A_K(R) = \int_0^{2\pi} \int_0^R \frac{r\ \mathrm{d}r\ \mathrm{d}\theta}{(1 - r^2)^{3/2}} \tag{52.47}$$

$$= 2\pi \left(\frac{1}{\sqrt{1 - R^2}} - 1 \right). \tag{52.48}$$

Therefore,

$$\lim_{R\to 1^-} A_K(R) = +\infty.$$

Simultaneously, the same boundary encloses the symplectic area.

$$\pi a_S b_S = 2\pi\hbar_{\mathrm{ret}} = h_{\mathrm{ret}} < \infty.$$

Theorem 52.9 (Finiteness of Action and Infinitude of Depth)**.** *The radion ellipse has a finite turn action h_{ret}, whereas its interior, equipped with the Hilbert–Beltrami–Klein metric, has infinite hyperbolic area:*

$$\boxed{\mathrm{Area}_{\mathrm{simp}}(\partial\mathbb{E}_S) = h_{\mathrm{ret}} < \infty, \qquad \mathrm{Area}_K(\mathbb{E}_S) = +\infty.} \tag{52.49}$$

There is no contradiction: $\mathrm{Area}_{\mathrm{simp}}$ and Area_K are different measures. The first counts oriented action in phase; the second measures interior projective depth. The thesis "finite boundary, infinite interior" is now an equality and a limit, not a metaphor.

Focal Coordinates Inside Klein In the normalized disk, the foci are located at $(\pm e_f, 0)$. The distance from the center to a focus is

$$u_f = \operatorname{artanh} e_f = \operatorname{arcosh}(e^{\eta}) = 0.81338233117534233376088973108822 8\ldots. \tag{52.50}$$

The focal–focal distance is

$$d_K(F_-, F_+) = 2u_f = 1.62676466235068466752177946217646\ldots.$$

We have

$$e_f = \tanh u_f, \qquad e^{-\eta} = \operatorname{sech} u_f, \qquad \eta = \log\cosh u_f. \tag{52.51}$$

The region up to the focal radius has area

$$A_K(e_f) = 2\pi(e^{\eta} - 1) = 2.19559620884061869541206115980360\ldots.$$

All open ellipses are projectively isometric as Hilbert domains. Curvature -1 alone therefore does not retain eccentricity. The HMT form resides in the joint structure: the distinguished affine chart, symplectic form, sheet labeling, and boundary parametrization.

52.5.2 Difference between phase return and return of the enriched state

52.5.2.1 Helical elevation of the nonadic return

Nonadic decomposition and memory quotient The elementary operation

$$n = 9q_9(n) + \rho_9(n), \qquad H_9(n) = (\rho_9(n), q_9(n)) \tag{52.52}$$

satisfies

$$H_9(n+9) = H_9(n) + (0,1). \tag{52.53}$$

The first coordinate closes; the second ascends. This identity is the minimal arithmetic version of a screw.

For a spectral fiber,

$$n \longmapsto (\tau_\gamma^n, n)$$

is a helix over S^1. Adding the contraction $(5/9)^n$ produces an inner spiral. The radion participates in both readings: a unit of phase on the boundary and an oriented unit of action in the lift.

Memory range as dimensional reader Suppose that a visible projection preserves the phase value but loses d linearly independent memory cocycles. Any faithful lift requires at least d additional coordinates to reconstruct them. This motivates a precise reading of Dimensional Mechanics:

$$\begin{gathered}\text{emergent dimension} = \text{minimum rank of independent memory} \\ \text{not publishable on the visible sheet.}\end{gathered} \tag{52.54}$$

It is not asserted that every physical dimension is a counter. Rather, in the TPK domain, every independent memory that must survive forces a coordinate of the lift.

> Dimension appears when the projection can no longer preserve the entire history. The circle suffices to state where the phase is; it does not suffice to state how many times it returned, which sheet it transported, which interval contracted, or which carry survived.

Cylinder Assembly, Spirals, and Helical Trajectories The coinductive cylinders are not physical tubes added to a spiral. They are nested intervals that preserve prefixes and narrow the value. The spiral describes transverse contraction; the helix describes longitudinal memory; the cylinder describes the set of compatible values at a finite depth. In the limit, the intersection of the cylinders selects a unique section.

The spatial image brings together, without confusion:

$$\begin{aligned}\text{angle} &\leftrightarrow \text{visible phase},\\ \text{height} &\leftrightarrow \text{return/memory},\\ \text{cylinder radius} &\leftrightarrow \text{refinement uncertainty},\\ \text{sheet} &\leftrightarrow \text{bidirectional orientation}.\end{aligned}$$

The rate 3^{-54} contracts the cylinder by return; it is not added to the angle and does not replace the actual carry ε_R.

52.5.2.2 Phase closures of orders 2π and 4π

In the circular projection, 2π restores the phase. In an oriented lift with two sheets, one turn may close on the conjugate sheet and require a second turn to restore the orientation:

$$U_{2\pi} = -I, \qquad U_{4\pi} = I.$$

The structure agrees with spinorial periodicity when the functor is applied to a representation $SU(2)$; the TPK lift is not identified with $SU(2)$ without that map.

The areal reading is consistent:

$$\mathcal{A}(2\pi) = h_{\mathrm{ret}}, \qquad \mathcal{A}(4\pi) = 2h_{\mathrm{ret}}.$$

In a Möbius band, one visible circuit reverses the sheet and two circuits restore the orientation. The $2\pi/4\pi$ is not an accidental duplication; it is the difference between phase return and return of the enriched state.

52.6 Oriented ellipse and conjugate variables: minimum uncertainty and electromagnetic oscillation

52.6.1 Conjugate symplectic variables and determination of the minimum uncertainty

The symplectic form coordinates action, uncertainty, and oscillatory dynamics without altering the genealogy of the state.

52.6.1.1 Realization of the state in a Hilbert space

Minimum Covariance Matrix Once the action ellipse has been constructed, it can be performed on a pair of conjugate operators

$$[\widehat{Q}, \widehat{P}] = i\hbar_{\mathrm{ret}}.$$

We define

$$V_\eta = \frac{\hbar_{\mathrm{ret}}}{2}\begin{pmatrix} e^{\eta} & 0 \\ 0 & e^{-\eta} \end{pmatrix}. \tag{52.55}$$

Its determinant is

$$\det V_\eta = \frac{\hbar_{\mathrm{ret}}^2}{4}. \tag{52.56}$$

Consequently,

$$(\Delta Q)^2 = \frac{\hbar_{\mathrm{ret}}}{2}e^{\eta}, \qquad (\Delta P)^2 = \frac{\hbar_{\mathrm{ret}}}{2}e^{-\eta}, \qquad \Delta Q\,\Delta P = \frac{\hbar_{\mathrm{ret}}}{2}. \tag{52.57}$$

The Robertson–Schrödinger inequality is saturated.

The compressed annihilator

$$\widehat{a}_\eta = \frac{e^{-\eta/2}\widehat{Q} + ie^{\eta/2}\widehat{P}}{\sqrt{2\hbar_{\mathrm{ret}}}} \tag{52.58}$$

satisfies $[\widehat{a}_\eta, \widehat{a}_\eta^\dagger] = 1$. Its vacuum has wavefunction

$$\psi_\eta(Q) = (\pi\hbar_{\mathrm{ret}}e^{\eta})^{-1/4}\exp\left(-\frac{Q^2}{2\hbar_{\mathrm{ret}}e^{\eta}}\right). \tag{52.59}$$

Ellipse as level set of the Mahalanobis functional The set

$$\mathbb{E}_{\mathrm{act}} = \{x \in \mathbb{R}^2 : x^{\mathsf{T}}V_\eta^{-1}x \le 4\} \tag{52.60}$$

has semiaxes

$$a_S = 2\Delta Q, \qquad b_S = 2\Delta P. \tag{52.61}$$

Its area is

$$4\pi\sqrt{\det V_\eta} = 2\pi\hbar_{\mathrm{ret}} = h_{\mathrm{ret}}.$$

The previously constructed HMT ellipse coincides exactly with the contour of two deviations from the minimal Gaussian state.

Proposition 52.10 (Unique chain of the quotient form)**.** *After declaring the spectral, symplectic, and Hilbert interfaces,*

$$\boxed{e^{-\eta} = \sqrt{\frac{A - C^*}{A + C^*}} = \frac{b_1}{a_1} = \frac{b_S}{a_S} = \frac{\Delta P}{\Delta Q}.} \tag{52.62}$$

The action ellipse corresponds to the classical level $H = \hbar_{\mathrm{ret}}\omega$ and to the contour 2σ of the compressed vacuum. It must not be conflated with the mean vacuum energy, $\hbar_{\mathrm{ret}}\omega/2$. This factor-of-two difference is a mandatory physical control.

Invariance Conditions of the Ellipse as an Operational Object In the conventional formulation, an uncertainty ellipse may be viewed as a graphical device for representing a covariance matrix. Here the causal order is reversed: the positive ellipse already exists as a publication of $(A, C^*, \hbar_{\mathrm{ret}})$; Hilbert space recognizes it through (52.55). The claimed ontological reality does not mean that a classical particle physically traverses an oval in the plane (Q, P). It means that the state preserves a positive form, a total action, and an orientation quotient before a probabilistic theory interprets them.

The wave function does not create the form; it represents it. Uncertainty does not create $\hbar$; it publishes the impossibility of simultaneously compressing the two directions without altering the symplectic product.

> The probabilistic reading is one possible realization of the ellipse, not its cause. The primary object is the joint preservation of product and orientation. Probability appears when that geometry is published as the covariance of a state on a Hilbert space.

52.6.2 Exchange $\mu \leftrightarrow \varepsilon$, inversion $Z \leftrightarrow Z^{-1}$, and conservation of causal velocity

Energy exchange between the electric and magnetic sectors The two terms of (52.67) become

$$U_E(\theta) = \hbar_{\mathrm{ret}}\omega \cos^2\theta, \qquad U_B(\theta) = \hbar_{\mathrm{ret}}\omega \sin^2\theta. \tag{52.63}$$

Therefore,

$$U_E + U_B = \hbar_{\mathrm{ret}}\omega.$$

Electric–magnetic polarity does not consist in arbitrarily labeling the foci. It consists of two conjugate reserves that are exchanged during the orbit while keeping the frequency and total action constant.

The common quotient extends:

$$\boxed{e^{-\eta} = \frac{b_S}{a_S} = \frac{\Delta P}{\Delta Q} = \frac{Z_\eta}{Z_*} = \sqrt{\frac{A - C^*}{A + C^*}}.} \tag{52.64}$$

The conserved products are

$$a_S b_S = 2\hbar_{\mathrm{ret}}, \qquad \det V_\eta = \frac{\hbar_{\mathrm{ret}}^2}{4}, \qquad L_\eta C_\eta = \omega^{-2}.$$

Exact Result

The LC oscillator is the minimal physical realization of the double reading: the product preserves action and frequency; the quotient preserves orientation and impedance. A complete cycle exchanges electricity and magnetism without losing the area $\oint P\, \mathrm{d}Q = h_{\mathrm{ret}}$.

52.6.3 Inductive–capacitive pair (L_η, C_η): invariance of $L_\eta C_\eta = \omega^{-2}$ and orientation of impedance

52.6.3.1 Inductive–capacitive oscillator associated with the angular pair

Independent construction of the 2 conjugated reservoirs Given $\omega > 0$ and a reference impedance $Z_* > 0$, we define

$$L_\eta = \frac{Z_*}{\omega} e^{-\eta}, \qquad C_\eta = \frac{1}{Z_*\omega} e^{\eta}. \tag{52.65}$$

Then

$$L_\eta C_\eta = \omega^{-2}, \qquad \sqrt{\frac{L_\eta}{C_\eta}} = Z_* e^{-\eta}. \tag{52.66}$$

The frequency remains fixed; the impedance preserves the orientation of the form.

The Hamiltonian is

$$H_{LC} = \frac{q^2}{2C_\eta} + \frac{\Phi^2}{2L_\eta}. \tag{52.67}$$

With the normalized variables

$$Q = \sqrt{Z_*}\, q, \qquad P = \frac{\Phi}{\sqrt{Z_*}}, \tag{52.68}$$

we obtain

$$H_{LC} = \frac{\omega}{2}\left(e^{-\eta}Q^2 + e^{\eta}P^2\right). \tag{52.69}$$

On (52.37),

$$H_{LC} = \hbar_{\mathrm{ret}}\omega. \tag{52.70}$$

LC Oscillator as an Elementary Dynamic Unit The harmonic oscillator pervades nearly all of physics because it linearizes the neighborhood of an equilibrium, decomposes fields into modes, organizes quanta, and allows propagation to be read as a superposition of phases. In the radion, this universality receives a concrete genealogy: the mode does not begin in an external differential equation, but in a positive form with a fixed return action. Conventional dynamics appears upon choosing a clock ω and a transducer Z_*.

Frequency does not alter the action ellipse: it changes the level energy through $E = \hbar_{\mathrm{ret}}\omega$. This separation is crucial. Geometry fixes action; the clock converts action into energy.

52.6.3.2 Time reversal and conjugate propagations

Let $H = H^*$ be given, and let J_E be a complex structure with $J_E^2 = -I$. Suppose there is an involution C such that

$$CJ_EC^{-1} = -J_E, \qquad CHC^{-1} = H.$$

The evolution

$$U(t) = \exp\left(-\frac{J_E H t}{\hbar_{\mathrm{ret}}}\right) \tag{52.71}$$

satisfies exactly

$$CU(t)C^{-1} = U(-t). \tag{52.72}$$

This is the minimal operator form of bidirectionality: conjugation interchanges the temporally opposed prolongations.

Wheeler–Feynman absorber theory and the reading of antiparticles as temporally conjugate trajectories offer a subsequent historical realization. Identity (52.72) is demonstrated; converting it into a complete theory of advanced and retarded propagators requires declaring action, boundary conditions, causality, and absorber interactions. The monograph preserves the physical intuition without converting a correspondence into an equality of theories.

52.7 3 TRIT regimes: electric–magnetic and elastic–rotational polarities

52.7.1 Coordination of the elliptic, parabolic, and hyperbolic realizations

The 3 TRIT regimes admit complex, electromagnetic, and mechanical realizations with distinct types.

52.7.1.1 Operator quadratic of root internal of −1

Exponential representation of the elliptic regime $J^2 = -I$ With $J^2 = -I$,

$$T_{\mathrm{ell}}(x, y) = e^{-xI - yJ} = e^{-x}(\cos y\, I - \sin y\, J). \tag{52.73}$$

Under the identification generated by J,

$$T_{\mathrm{ell}}(x,y) \longleftrightarrow e^{-x-iy}.$$

The scalar x controls dilation; y controls oriented rotation. The form $a + bi$ is recognized as elasticity plus rotation, not as two arbitrarily adjoined quantities.

Exponential representation of the parabolic regime $J^2 = 0$ With $N^2 = 0$, the exponential is truncated:

$$T_{\mathrm{par}}(x,y) = e^{-x}(I - yN). \tag{52.74}$$

The regime describes the threshold at which there is neither periodic rotation nor exponential opening. In the active temporal reading it is the Euclidean present, but it retains memory of the branches that bound it.

Exponential Representation of the Hyperbolic Regime $J^2 = +I$ With $R^2 = I$ and $P_\pm = (I \pm R)/2$,

$$T_{\mathrm{hyp}}(x,y) = e^{-xI-yR} = q_+P_+ + q_-P_-, \qquad q_\pm = e^{-x\mp y}. \tag{52.75}$$

The elementary invariants are

$$q_+q_- = e^{-2x}, \qquad \frac{q_-}{q_+} = e^{2y}. \tag{52.76}$$

The product preserves the common mode; the quotient preserves the oriented separation.

Applied to the angular pair,

$$x = \frac{\pi A}{180}, \qquad y = \frac{\pi C^*}{180}.$$

Therefore, A is the common coordinate of the publication and C^* its oriented contrast. Only after a physical reader has been applied are they recognized as electric and magnetic poles.

52.7.2 Typed separation of the constitutive and mechanical polarities

52.7.2.1 Constitutive Operator on the Channels 90/120

Common and Oriented Component Decomposition We define

$$\mathcal{E} = \log\big[(1-q_+^{90})(1-q_-^{90})\big] - \log\big[(1-q_+^{120})(1-q_-^{120})\big], \tag{52.77}$$

$$\mathcal{B} = \log\frac{1-q_-^{90}}{1-q_+^{90}} - \log\frac{1-q_-^{120}}{1-q_+^{120}}. \tag{52.78}$$

Under $C^* \mapsto -C^*$, $q_+ \leftrightarrow q_-$ are interchanged. Therefore, $\mathcal{E}$ is even and $\mathcal{B}$ is odd.

The normalized constitutive equations are

$$\widehat{\mu} = e^{\mathcal{E}+\mathcal{B}}, \qquad \widehat{\varepsilon} = e^{\mathcal{E}-\mathcal{B}}, \qquad \widehat{Z} = e^{\mathcal{B}}, \qquad \widehat{c} = e^{-\mathcal{E}}. \tag{52.79}$$

They are verified exactly.

$$\widehat{Z} = \sqrt{\frac{\widehat{\mu}}{\widehat{\varepsilon}}}, \qquad \widehat{c} = (\widehat{\mu}\widehat{\varepsilon})^{-1/2}. \tag{52.80}$$

Conjugation produces

$$C^* \mapsto -C^* : \qquad \widehat{\mu} \leftrightarrow \widehat{\varepsilon}, \quad \widehat{Z} \leftrightarrow \widehat{Z}^{-1}, \quad \widehat{c} \mapsto \widehat{c}. \tag{52.81}$$

Theorem 52.11 (Constitutive polarity)**.** *The reader* 90/120 *converts the common mode and oriented contrast of the pair (A, C^*) into two conjugate constitutive quantities. Sheet inversion exchanges permeability and permittivity, reverses impedance, and preserves normalized velocity.*

This is the precise formulation of the intuition "electrical part and magnetic part". One writes neither $A = E$ nor $C^* = B$. Instead, one writes the map

$$(A, C^*) \to (x, y) \to (q_+, q_-) \to (\mathcal{E}, \mathcal{B}) \to (\widehat{\varepsilon}, \widehat{\mu}, \widehat{Z}, \widehat{c}).$$

Elastic-rotational realization of the constitutive polarity The same information may be published in the complex regime by

$$Z_{\mathrm{em}} = e^{\mathcal{E}/2}\left(\cos\frac{\mathcal{B}}{2}\, I - \sin\frac{\mathcal{B}}{2}\, J\right). \tag{52.82}$$

The factor $e^{\mathcal{E}/2}$ is an elastic dilation; the factor generated by J is a rotation. Complex notation does not add a mysterious imaginary dimension to a previously given real world: it records two operational modes already separated by the TRIT.

Transduction of Orientation into Effective Load The radion sheet σ fixes an orientation before a unit of charge is introduced. Dimensional Mechanics then provides the transducer

$$e_{\mathrm{int}} = \sqrt{\frac{2\alpha_{\mathrm{HMT}} h_{\mathrm{ret}}}{Z_{\mathrm{int}}}}, \qquad Q = n_Q e_{\mathrm{int}}.$$

The causal chain is

$$\text{sheet} \to \text{orientation} \to \text{constitutive} \to \text{dimensional transducer} \to \text{physical charge}.$$

Calling the unit *radion* does not prematurely identify the orientation with the coulomb; it makes visible the locus where the charge sign may reside.

> Electricity and magnetism are not affixed to the circle from outside. They emerge when the same state is read through product and quotient: the product retains the common mode, while the quotient retains the orientation. The LC oscillator realizes the exchange; the reader 90/120 publishes the constitutive relations; the dimensional chart assigns units.

52.8 Separation of the Publications 90/120: Incidence, Record, Spectrum, Barbero Kernel, and Towers

52.8.1 Typed separation of the publications of the sector 90/120

52.8.1.1 Global harmonic hierarchy and semigroup $\langle 90, 120\rangle$

To avoid a symbol collision, we fix

$$R_{\mathrm{tow}} := -R_{\mathrm{ch}}.$$

Then

$$T = e^{-xI + yR_{\mathrm{tow}}} = q_- P_+^{\mathrm{tow}} + q_+ P_-^{\mathrm{tow}}. \tag{52.83}$$

The even and odd towers are

$$c_n = \mathrm{Tr}(T^n) = q_-^n + q_+^n = 2e^{-nx}\cosh(ny), \tag{52.84}$$

$$s_n = \mathrm{Tr}(R_{\mathrm{tow}} T^n) = q_-^n - q_+^n = 2e^{-nx}\sinh(ny). \tag{52.85}$$

Under $C^* \mapsto -C^*$, c_n is even and s_n is odd. It holds that

$$c_n^2 - s_n^2 = 4e^{-2nx}, \tag{52.86}$$

and both sequences satisfy the order-two recurrence determined by the eigenvalues $q_\pm$.

The semigroup

$$\langle 90, 120\rangle = 30\langle 3, 4\rangle \tag{52.87}$$

is the harmonic hierarchy of a single bilayer operator. It is not a list of eight lateral corrections. Its completeness means that every positive ratio admits a convergent expansion; it does not mean that an expansion obtained from the target is a prospective physical selector.

52.8.1.2 Typed classification and separation of the realizations 90/120

Object	Operation	What is conserved / why it is not another object
Incidence N_6, N_8	finite counting over active phases	cardinality and orientation; it produces 90, 120, not a series;
$-1/12$	diferencia normalized (52.88)	incidence scalar; no Barbero weight;
E_{dod}	harmonic extractor of the wheel	dodecaphase phase; no real carry;
$h = C^*/6$	reconstruction of six pairs	opening by pair; no metrological normalization;
F_{spec}	cola logaritmica $j \geq 2$	complete spectral publication; differs from ε_R°;
$\varepsilon_R^{(1)}$	subtraction APP between S_T and S_E	real join of two lifts; not a Lambert series;
$K_{6\to 8}$	$12\Delta S_{90} - \Delta S_{120}$	marked jump and conjugate evaluation; pole coefficient $1/24$ verified; Barbero output;
$12 : -1$	twelve sectors and one oriented seam	kernel weight; subsequent certificate (52.91); distinct from $-1/12$;
$\langle 90, 120\rangle$	semigroup of powers of T	global harmonic hierarchy; no residue per species.

Tesis rectora

All these objects descend from the same APP–TRIT–TPK state. That common genealogy explains why they share numbers and symmetries. Their typing explains why they cannot replace one another.

52.8.2 Conservation of Incidence and Orientation Between the Tritic Kernel and Its Subsequent Realizations

The typed hierarchy separates incidence, record, spectral evaluation, Barbero functional, and harmonic towers.

52.8.2.1 Tritic incidence, dodecaphasic recording and hexadic reading

Primary combinatorial derivation of 90 **and** 120 The counts

$$90 = 6\binom{6}{2}, \qquad 120 = 8\binom{6}{2}$$

arise from the phase selector and its active incidences. The normalized orientation satisfies

$$\frac{30}{120} - \frac{30}{90} = -\frac{1}{12}. \tag{52.88}$$

This scalar is an incidence result. It is neither the weight $12 : -1$ of the Barbero kernel nor a dodecaphase carry.

The dodecaphonic extractor can be published through a combination of harmonics as

$$E_{\mathrm{dod}}(\phi) = 12\cos(3\phi) - \cos(4\phi).$$

Its domain is the phase of the wheel; its output is a harmonic selector. Although the numbers 12 and -1 appear, it does not act on Lambert series or produce ε_R.

52.8.3 Dodecaphasic origin of the weight $12 : -1$ and Diophantine verification of its pole coefficient

52.8.3.1 Barbero–Immirzi hexad–octad functional

Oriented series and incidence jump We define

$$S_{90}(q) = \frac{q^{90}}{1-q^{270}}, \qquad S_{120}(q) = \frac{q^{120}}{1-q^{360}}, \tag{52.89}$$

and

$$\Delta S_m = S_m(q_-) - S_m(q_+).$$

The marked inclusion functional is

$$K_{6\to 8} = 12\Delta S_{90} - \Delta S_{120}. \tag{52.90}$$

Diophantine verification of the dodecaphasic weight $12 : -1$ The dodecaphasic fundamental chain and its return reader, constructed in (51.32)–(51.34), produce twelve terms of type S_{90} and one boundary term of type $-S_{120}$. The oriented pair is determined before calculating the pole coefficient.

Consider $aS_{90} - bS_{120}$. Verification of the coefficient of $(1-q)^{-1}$ takes the form

$$\frac{a}{270} - \frac{b}{360} = \frac{1}{24}. \tag{52.91}$$

Multiplying by 1080 gives

$$4a - 3b = 45.$$

The unique seam of the fundamental turn has absolute multiplicity one. Its Diophantine verification corresponds to the minimal integer counterterm $|b| = 1$. For $b = 1$, $a = 12$; for $b = -1$, $a = 21/2$ is not an integer. Therefore,

$$\boxed{a : b = 12 : 1, \qquad aS_{90} - bS_{120} = 12S_{90} - S_{120}.} \tag{52.92}$$

Proposition 52.12 (Uniqueness of the Diophantine verification)**.** *The generated pair* $(12, 1)$ *is the unique integer solution of* (52.91) *with a positive principal channel and minimal counterterm* $|b| = 1$.

The hexad–octad realization preserves the channels $90/120$ previously constructed; the dodecaphasic chain supplies the multiplicities and boundary orientation; the return reader assigns the series to the generators. These joint operations produce $12S_{90} - S_{120}$. The equation (52.91), positivity, and minimality retain their role as subsequent arithmetic verification.

The Barbero-Deformable chain is subsequent:

$$(A, C^*) \to (x, y) \to q_\pm \to K_{6\to 8} \to \gamma_{\mathrm{BI}},$$

with evaluation

$$\gamma_{\mathrm{BI}} = 0.274007672694459\ldots.$$

Barbero–Immirzi does not retroactively generate C^*, Δ_4, or ε_R.

52.8.4 Extraction of the finite component subsequent to the oriented dodecaphasic system R_{12}: $C^* \mapsto C^*/6$, remainder $j \geq 2$, and reconstruction of 6 sections

Extraction of component $C^*/6$ from the dodecaphase record The twelve sectors group into six opposite pairs:

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}, \qquad i = 1, \ldots, 6. \tag{52.93}$$

The reconstruction has profile $1 + 5 + 6 = 12$. The opening per pair is

$$h = \frac{C^*}{6}. \tag{52.94}$$

It is neither an external normalization nor one sixth of a target: it is the internal index of the hexadic reconstruction of the record $\mathcal{R}_{12}$.

We define

$$q_{h,-} = \exp\left[-\frac{\pi}{180}(A - h)\right], \qquad q_{h,+} = \exp\left[-\frac{\pi}{180}(A + h)\right]. \tag{52.95}$$

52.8.5 Quantitative decomposition of the carry and spectral publications: $F_{\mathrm{spec}} = \varepsilon_R^\circ + \chi_R$

52.8.5.1 Spectral publication of the tail of orders $j \geq 2$

Definition of the spectral functional For $m \in \{90, 120\}$, let

$$L_m^{(2)}(q) = \sum_{j \geq 2} \frac{q^{mj}}{j} = -\log(1 - q^m) - q^m. \tag{52.96}$$

The reading of a pair is

$$T_h^{(2)} = \frac{180}{\pi}\Big[L_{90}^{(2)}(q_{h,-}) - L_{90}^{(2)}(q_{h,+}) - L_{120}^{(2)}(q_{h,-}) + L_{120}^{(2)}(q_{h,+})\Big]. \tag{52.97}$$

The reconstruction of the six pairs gives

$$F_{\mathrm{spec}} = 6T_h^{(2)} = 5.1499235153094574410\ldots \times 10^{-8\,\circ}. \tag{52.98}$$

The radion carry is

$$F_{\mathrm{acarreo}} = \varepsilon_R^\circ = 5.1383016758575282072\ldots \times 10^{-8\,\circ}.$$

They do not coincide:

$$\chi_R := F_{\mathrm{spec}} - F_{\mathrm{acarreo}} = 1.16218394519292338\ldots \times 10^{-10\,\circ}. \tag{52.99}$$

The equality

$$F_{\mathrm{acarreo}} = F_{\mathrm{spec}} - \chi_R$$

is a valid change of chart once both outputs have been constructed. It must not be presented as an independent generation of ε_R° if χ_R is defined by the subtraction itself. The tail is closed and meaningful; what is false is its direct identification with carry.

Falsification criterion 52.13. After converting the tail expressed in degrees to the internal angular unit,

$$\frac{\pi}{180^\circ} F_{\mathrm{spec}} - \varepsilon_R^{(1)} = 2.0283936357433839\ldots \times 10^{-12} \neq 0.$$

Therefore, neither the complete logarithm nor the tail $j \geq 2$, even after applying the chart of units, is $\varepsilon_R^{(1)}$. A second autonomous spectral factorization would need to generate χ_R without using the difference as a definition; the APP–TPK closure of the radion does not depend on that promotion.

52.9 Oriented holonomy and physical geometry: later realizations of Minkowski, Hodge, Lorentz, and Cartan–Holst

Oriented holonomy admits subsequent causal and geometric realizations in Minkowski, Hodge, Lorentz, and Cartan–Holst.

Construction of the Lorentzian metric from the APP block

APP matrix of parameters 7 **and** 2 The central matrix

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_- \tag{52.100}$$

it is normalized by its determinant:

$$M_c = \frac{B_c}{\sqrt{45}} = \exp\left(\frac{1}{2}\log\frac{9}{5}\,R\right) \in SO^+(1,1). \tag{52.101}$$

The internal speed is

$$\eta_c = \frac{1}{2}\log\frac{9}{5}.$$

The associated projective publication satisfies

$$P_B^n = P_+ + \left(\frac{5}{9}\right)^n P_-. \tag{52.102}$$

The same ratio $5/9$ that appears as a spectral contraction is recognized here as the separation of one stable mode and one contractive mode. The identity is operatorial; it does not turn every occurrence of $5/9$ into the same map.

Incidence $6 \to 8$ **and causal-signature quotient** Let M be the typed incidence between six faces and eight vertices. The recovered spectral relation is

$$MM^{\mathsf{T}} = 12P_u + 4P_-, \tag{52.103}$$

where P_u projects onto the uniform mode and P_- onto the oriented subspace. The surviving quotient of dimension four decomposes as

$$Q_4 \simeq \mathbb{R}u \oplus E_-. \tag{52.104}$$

Declaring the uniform mode negative and the three oriented modes positive, one obtains

$$B_{\text{caus}} = \operatorname{diag}(-1,1,1,1). \tag{52.105}$$

The Minkowski screen does not select the HMT state. It is the subsequent causal realization of the quotient already constructed by the incidence.

52.9.1 Hodge duality on 2-Lorentzian forms

52.9.1.1 Hodge Duality and Quarter-turn Operator

With the signature $(-+++)$ and an orientation fixed, the Hodge star on two-forms satisfies

$$\star : \Lambda^2 Q_4^* \to \Lambda^2 Q_4^*, \qquad \star^2 = -I. \tag{52.106}$$

Thus, the quarter-turn expressed in the phase plane by $J^2 = -I$ reappears in field space as a complex structure on Λ^2. The domains differ, but the genealogy of orientation is preserved by the functor.

Let A_{em} be a potential and

$$F = \mathrm{d}A_{\mathrm{em}}. \tag{52.107}$$

The decomposition of F relative to a temporal observer produces electric and magnetic fields; $\star F$ exchanges their roles with the sign fixed by the orientation and signature. The constitutive polarity of (52.81) thus acquires a differential realization.

Electromagnetic and causal transduction chain The transport that must remain visible is

$$\Gamma_9 \longrightarrow \text{memory helix} \longrightarrow (Q_4, B_{\mathrm{caus}}, \star) \longrightarrow A_{\mathrm{em}} \longrightarrow F = \mathrm{d}A_{\mathrm{em}} \longrightarrow m\nabla_u u = q(\iota_u F)^\sharp. \tag{52.108}$$

Each arrow adds structure: the holonomy preserves return; the quotient fixes causality; Hodge fixes duality; the potential fixes the field; coupling to charge produces dynamics.

The final equation is the Lorentz force in geometric form. The TPK helix is its memory antecedent, not a physical orbit already moved by a field. Dynamics appears only upon introducing F, q, and the connection ∇.

52.9.2 Separation between internal torsional memory and Cartan–Holst geometric realization

52.9.2.1 Lorentzian Realization of Helical Trajectories

In a uniform magnetic field, a particle of charge q, mass m, and relativistic factor γ has

$$\omega_c = \frac{|q|B}{\gamma m}, \qquad r_L = \frac{p_\perp}{|q|B}, \qquad \Delta z = \frac{2\pi p_\parallel}{|q|B}. \tag{52.109}$$

The cyclotron phase closes while the parallel motion advances: the orbit is helical. The parallel with (52.53) is structural and can be formalized by a transducer that preserves phase, orientation, and step. The quotient q_9 is not identified with the spatial coordinate z; rather, the same architecture of return plus advance is recognized.

The Lorentz force is not derived from the image of a spiral by similarity. It is derived by completing (52.108). The image helps explain why circular phase and longitudinal memory are compatible; the field form determines the dynamics.

52.9.2.2 Confluence between action, area, information, and gravity

Cartan–Holst Realization of Holonomy The gravitational promotion is expressed via a holonomy morphism.

$$\mathrm{Hol}_{\mathcal{A}}\big(\mathfrak{R}_{\mathrm{grav}}(\gamma)\big) = \rho_C\big(\mathrm{Hol}_{\mathrm{TPK}}(\gamma)\big). \tag{52.110}$$

The right-hand side contains route memory; ρ_C realizes it in a gravitational connection. Cartan torsion

$$T^I = D_\omega e^I$$

appears in the codomain. The global torsion Δ_4 is an upstream coordinate that can feed the map, but it is not renamed as T^I without (52.110).

Transduction of the action area into informational area The radion fixes one quantum of action per phase. The Barbero hexad–octad reader subsequently uses the channels $q_{\pm}$, the incidence 90/120, and the functional (52.90). The causal direction is

$$(A, C^*, \hbar_{\mathrm{ret}}) \to q_{\pm} \to K_{6\to 8} \to \gamma_{\mathrm{BI}} \to \text{area/information}.$$

The tritic alphabet provides the unit of entropy $\log 3$, and a cut of 81 channels provides $81 \log 3$. This entropy is not to be confused with $\log 2$, with $\log \varphi$, with a microcanonical entropy, or with a von Neumann entropy unless the state is specified.

The physical synthesis is affirmative but typed: the radion supplies the oriented unit of action that a holonomy can transport; Barbero publishes the area normalization; the causal screen and Hodge make it possible to realize fields; Cartan–Holst receives the holonomy in gravitational geometry.

Hypothesis, domain, and conclusion

The scalar radion equation is not by itself an Einstein equation. Its contribution to gravity is to provide an exact section of the architecture of action, orientation, and holonomy consumed by the gravitational realization.

52.9.2.3 Wick Rotation as Analytic Reader Change

In the conventional formulation, Wick rotation relates a Lorentzian time variable to a Euclidean coordinate through complex continuation. In HMT, the root $J^2 = -I$, the involutive map $R^2 = I$, and the threshold $N^2 = 0$ already exist. The change of reader

$$J \longleftrightarrow R$$

transports a common/oriented pair between elliptic publication and hyperbolic publication. Complex continuation subsequently recognizes that operation in analytic coordinates.

This explains why the circle, the action ellipse, and the hyperbolic interior can belong to the same state without being the same metric. The circle uses J for the phase; the shape of the ellipse uses a symplectic compression generated by R; Klein measures the interior by means of the cross-ratio; the Minkowski screen realizes causality in another codomain.

52.10 Ellipse, Lattice, and Path Phase: Modular Character, Orbits, and Propagation

52.10.1 Rectangular Lattice of Periods, Involution $\tau \mapsto -1/\tau$, and Invariance of j

Modular, orbital, and path characters preserve distinct invariants of the same oriented form.

52.10.1.1 Complex torus associated with the action ellipse

Rectangular lattice of periods The ellipse by itself is not a torus. We declare the lattice

$$\Lambda_\eta = a_S\mathbb{Z} + i, b_S\mathbb{Z} \tag{52.111}$$

and the quotient $\mathbb{C}/\Lambda_\eta$. Its modulus is

$$\tau_\eta = i\frac{b_S}{a_S} = ie^{-\eta} = i\,0.74104814415937516079548366336930 0\ldots. \tag{52.112}$$

The bidirectional inversion $\eta \mapsto -\eta$ exchanges the axes:

$$\tau_{-\eta} = ie^{\eta} = -\frac{1}{\tau_\eta}. \tag{52.113}$$

It is exactly the modular transformation $S : \tau \mapsto -1/\tau$.

Invariance of the Modular Function j By modular invariance,

$$\boxed{j(\tau_{-\eta}) = j(\tau_\eta).} \tag{52.114}$$

In the normalization $j(i) = 1728$,

$$j(\tau_\eta) = 5597.438439324087298300970892547 68\ldots.$$

The normalized moonshine function $J = j - 744$ gives

$$J(\tau_\eta) = 4853.438439324087298300970892547 68\ldots.$$

For the basal APP block,

$$\tau_0 = i\frac{\sqrt{5}}{3}, \qquad j(\tau_0) = 5369.488320246743060219168826352\ldots.$$

Proposition 52.14 (What the modular class preserves)**.** *The sheet involution reverses η, but j preserves the class of the torus. The foci and the axis labelling retain the orientation that j forgets.*

This result connects the two-way structure exactly with modularity once the quotient has been declared. It does not authorize making the Monster act directly on the charge, digits, or foci. The exceptional corridor

$$\text{APP} \to \text{Paley} \to \text{Witt} \to \text{Golay} \to A_2^{12} \to \text{Leech} \to V^\natural \to \text{Monster} \to \text{Moonshine}$$

is another projection of the same enriched state. Identifying a concrete action on the radion requires an intertwiner that preserves its relevant coordinates.

Chain of character form invariants Gathering the declared realizations,

$$\boxed{r_\eta = e^{-\eta} = \sqrt{\frac{A - C^*}{A + C^*}} = \frac{b_1}{a_1} = \frac{b_S}{a_S} = \frac{\Delta P}{\Delta Q} = \frac{Z_\eta}{Z_*} = \Im\tau_\eta.} \tag{52.115}$$

A single ratio runs through spectral geometry, action, covariance, impedance, and modulus. Each equality rests on an explicit map; none depends on decimal resemblance.

52.10.2 Compatibility between the modular character and propagation over oriented paths

52.10.2.1 Areal Conservation in Kepler and Sommerfeld Orbits

Areal Equivalence of the Orbital and Oscillatory Realizations Kepler's ellipse lives in configuration space and responds to a central potential. The radion ellipse lives in phase space and responds to an action form. They are different objects. They share an areal law because both realizations possess a conserved two-form.

For a Keplerian orbit,

$$r(\phi) = \frac{p}{1 + e\cos(\phi - \phi_0)}, \qquad \frac{\mathrm{d}A}{\mathrm{d}t} = \frac{\ell}{2m}. \tag{52.116}$$

The third law takes the form

$$T^2 = \frac{4\pi^2\mu}{K}a^3.$$

For the phase oscillator,

$$J = \oint p \, \mathrm{d}q = \frac{2\pi E}{\omega}. \tag{52.117}$$

If the elementary cell is $J = h_{\mathrm{ret}}$, then

$$E = \hbar_{\mathrm{ret}}\omega, \qquad \frac{J}{2\pi} = \hbar_{\mathrm{ret}}. \tag{52.118}$$

This is the clean convergence between Planck action, angular frequency, and radion.

Sommerfeld quantizes action integrals over cycles. The HMT reading adds the memory of which cycle, sheet, and return survive. On a Möbius band, one visible loop may carry $h_{\mathrm{ret}}/2$ and two loops restore h_{ret}. That topological factor is not added to the Maslov index of EBK quantization; they belong to corrections with distinct owners.

Orbital Holonomy and Advanced and Retarded Propagations An orbit that accumulates holonomy satisfies a condition of the type

$$\epsilon_\gamma \exp\left(\frac{iJ_\gamma}{\hbar_{\mathrm{ret}}}\right) = 1. \tag{52.119}$$

The sign ϵ_γ preserves the sheet; J_γ preserves the action. Apsidal advance or retardation is interpreted as a decoupling between visible angular closure and complete return. The radion supplies the oriented unit with which that difference is measured.

52.10.2.2 Composition of Schrödinger amplitudes and path summation

Four realizations of the same phase Once $\hbar_{\mathrm{ret}}$ is fixed, distinct conventional equations recognize aspects of the same transport:

1. Schrödinger publishes the phase through $i\hbar_{\mathrm{ret}}\partial_t\psi = H\psi$.
2. Pauli adds the double spin leaf and its magnetic coupling.
3. Dirac linearizes the energy–momentum relation and organizes particle and antiparticle within a causal representation.
4. The path sum assigns to each history the phase $\exp(iS[\gamma]/\hbar_{\mathrm{ret}})$.

These theories do not retrospectively generate $\hbar_{\mathrm{ret}}$. They receive the unit of action and realize different contents of the enriched state.

Phase functional over the path space In Feynman's formulation,

$$\mathcal{K}(b,a) = \int_{\gamma:a\to b} \exp\left(\frac{i}{\hbar_{\mathrm{ret}}}S[\gamma]\right)\mathcal{D}\gamma. \tag{52.120}$$

An action increment $\hbar_{\mathrm{ret}}$ changes the phase by one radian. An increment h_{ret} changes it by one revolution. The theorem $\mathcal{A}(\theta) = \hbar_{\mathrm{ret}}\theta$ converts this relation into area geometry: the phase of a path can be read as swept action in units of the radion.

The inversion $C_M = -\mathrm{rev}$ on dodecaphase words and the identity (52.72) organize conjugate routes. In an absorber realization, the following are distinguished

$$G_{\mathrm{sym}} = \tfrac{1}{2}(G_{\mathrm{ret}} + G_{\mathrm{adv}}), \qquad G_{\mathrm{odd}} = G_{\mathrm{ret}} - G_{\mathrm{adv}}.$$

The first sector is even under retarded–advanced interchange; the second is odd. This reproduces the separation between even energy/form and odd oriented action of the radion when the transducer preserves the boundary and the symplectic form.

Hypothesis, domain, and conclusion

Classical absorber theory and the Feynman–Stueckelberg interpretation of antiparticles are distinct historical realizations. The former uses a temporally symmetric interaction; the latter reorganizes propagation in relativistic theory. HMT supplies a common antecedent of reversal, but does not merge them without a map of propagators.

52.11 Radical nonadic and decimal boundary: conservation of the radion carry

52.11.1 The carry of the radion in the decimal completion

The nonadic boundary preserves the sum–product defect, vacancy, and completion carry.

52.11.1.1 Radical 3−6−9 of the boundary

Generation and nilpotency of the nonadic radical In the ring $\mathbb{Z}/9\mathbb{Z}$, the ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\} \tag{52.121}$$

satisfies

$$\mathfrak{m}^2 = (9) = 0. \tag{52.122}$$

It is the nilpotent sector of the nonadic boundary. The visible orbit of its three marks is published as

$$369 \longrightarrow 693 \longrightarrow 936 \longrightarrow 369.$$

The return does not imply that all states are equal: the visible permutation closes while the route cocycles may advance.

The double convention

$$9 \equiv 0 \pmod 9, \qquad \mathrm{dr}_9(9) = 9$$

expresses two readers: algebraic residual zero and visible digital-root closure. Thus a nine may absorb boundary torsion without becoming a specific zero of the zeta function.

Sum–product defect $459 - 405 = 54$ The APP sheets produce complementary totals

$$S_+ = 405, \qquad S_\times = 459, \qquad S_\times - S_+ = 54.$$

The 54 is an exact defect of the tables and reappears as a contraction exponent through the identity $9 \cdot 6 = 54$. Each occurrence nevertheless requires its own map. The focal distance $d_1 = 0.544545\ldots$ is not equal to 0.54, and invariant analysis has produced no arrow identifying it with the 54 defect.

The power

$$9^9 = 387,420,489$$

generates a scale where

$$\mathrm{ord}_{10}\left(9^{9^9}\right) = 369,693,099, \qquad \#\,\text{digits} = 369,693,100. \tag{52.123}$$

The block 369|693 appears in the order–length pair; it is not asserted to be the decimal prefix of the power.

52.11.1.2 Arithmetic Theory of Decimal Vacuums

Asymptotic vacancy slope The separation between the decade and the nonary base is

$$\delta_{\mathrm{vac}} = \log_{10} \frac{10}{9}. \tag{52.124}$$

The accumulated counter

$$V_9(n) = \lceil n\delta_{\mathrm{vac}} \rceil \tag{52.125}$$

records how many correction positions are required by the decimal publication of a nonadic evolution. Its jumps form a binary word

$$z_n = V_9(n) - V_9(n-1) \in \{0, 1\}.$$

The gaps are distributed with lengths $21/22$, the returns with lengths $6/7$, and the combined determinant is 15. These numbers arise from the vacancy reader, not from the Barbero kernel, although 15 coincides with $\binom{6}{2}$.

Polarization and Arithmetic Current of the Nonadic Grid We define

$$P_N = \sum_{n=1}^{N} (z_n - \delta_{\mathrm{vac}}), \qquad J_N = P_N - P_{N-1} = z_N - \delta_{\mathrm{vac}}. \tag{52.126}$$

The balanced sequence satisfies $|P_N| < 1$. The "current" J_N is here an arithmetic current: it measures the creation and absorption of vacancies relative to the mean slope. Only a subsequent dimensional transducer can publish it as an electromagnetic current density.

The physical intuition is powerful because the same algebraic form separates an even average from an oriented fluctuation. Rigor requires retaining the type:

$$J_N \in \mathbb{R} \quad \text{dimensionless}, \qquad j^{\mu} \in \text{physical current line}.$$

52.11.1.3 Completion $729 \to 1000$ *and preservation of carry*

Decomposition $1000 = 729 + 243 + 27 + 1$ The completion

$$1000 = 729 + 243 + 27 + 1$$

preserves the unit through four ternary scales. The total crown is

$$1000 - 729 = 271,$$

while the last integer before the closure satisfies

$$999 = 729 + 270.$$

The difference between 270 and 271 distinguishes an occupied boundary from a completing unit. The same care prevents identifying a vacancy with zero or a carry with an actual residue.

Residence of the carry of the radion in the decimal crown The radion's operands begin, in blocks of three digits, with

$$\Phi_T : \quad 000|027|455|906|837|565|\cdots,$$

$$D_E : \quad 000|027|455|010|034|743|\cdots.$$

The subtraction APP gives

$$\varepsilon_R^{(1)} : \quad 000|000|000|896|802|822|\cdots.$$

The small size of the result is not obtained by truncating the low-order summands. It arises because two distinct publications share a long prefix, and the operator $\mathrm{SubAcarreo}_{1000}$ preserves the borrows required to subtract them exactly.

The word *memory* has two levels here:

1. q_{mem} records the depth of nonadic returns;
2. c_i records the local borrowings of decimal subtraction.

The first organizes the helix; the second publishes the decimal. ε_R uses both because its operands follow from cylinders at depth and its subtraction is performed with carry, but it is equal to neither counter.

Tesis rectora

The boundary is not a place where the theory loses information. It is the place where information changes representation: phase that returns, memory that ascends, cylinder that contracts, vacancy that is recorded, and carry that transports the difference.

52.11.2 Compatibility of the direct and torsional sections: structural characterization of π

52.11.2.1 Structural characterization of π by compatibility of sections

Reorder (52.18) gives

$$\boxed{2\pi_{\mathrm{HMT}} = \frac{1 + \Phi_T - \varepsilon_R^{(1)}}{\frac{5}{36} + \frac{25}{9}\alpha_{\mathrm{HMT}}}.} \tag{52.127}$$

This equality may be called a new definition of π within the radion chart if *structural definition* is understood as an exact characterization by compatibility of sections. It is not a second independent generator: Δ_4 and the carry operator already receive the coinductive section of π. The novelty is that the turn is characterized by the gluing among direct publication, torsion, and memory.

> The equation (52.127) does not say that we first do not know π and then guess it by closing an angle. It says that the π generated by the nonadic connection is exactly the one that makes two internal readings of the same state compatible. It is an identity of intrinsic recognition, not a metrological calibration.

52.12 Evolutionary Conservation of the Holographic Unit Across Dimensions

52.12.1 Typed Separation among the Curvature Regime $\tau \in \{+1, 0, -1\}$, the Radial Response p, and the Global Spin Holonomy

The radion's biography can be represented by a raised curve.

$$\gamma(\theta) = \big(r(\theta)\cos\theta,\ r(\theta)\sin\theta,\ m(\theta)\big),$$

where the first two coordinates belong to the Euclidean plane $\mathbb{R}^2$, and $m(\theta)$ records the memory or depth of the lift. This equation permits a joint reading of circle, helix, and spiral.

When the angle evolves while the radius and memory remain constant, the planar publication is a circle. When angle and memory evolve while the radius retains its value, the complete trajectory is a helix of constant radius. When angle and radius evolve over a fixed memory, a spiral appears in $\mathbb{R}^2$. When all three coordinates evolve, the trajectory is a radial helix: a biography in which phase, amplitude, and memory are jointly transformed.

The radial character p classifies the response law of the radius. One useful family is

$$r_p(\theta) = r_0\big[1 + p\lambda(\theta - \theta_0)\big]^{1/p}, \qquad p \neq 0,$$

with logarithmic limit

$$r_0 \exp\big(\lambda(\theta - \theta_0)\big) \qquad \text{when } p \to 0.$$

In this parameterization, $p = 2$ publishes the Fermat regime, $p = 1$ the Archimedean regime, $p = 0$ the logarithmic regime, $p = -1$ the hyperbolic regime, and $p = -2$ the lituus regime, with the corresponding choices of domain and orientation. Fractional values describe intermediate regimes. Thus, $p = \frac{1}{2}$, $p = \frac{3}{2}$, or $p = -\frac{1}{2}$ express radial responses that interpolate among the integer families and extend the available geometric classification.

The character can also change when crossing a threshold. The differential writing

$$\frac{dx}{d\theta} = \beta(m)\, x^{1-p(m)}, \qquad x = \frac{r}{r_0},$$

describes an evolution through regimes: $p(m)$ remains stable within a phase and changes when the memory reaches a publication boundary. The resulting trajectory connects circular, helical, and spiral sectors through a common law. Continuity of the state and the threshold ledger make those sectors phases of a single evolution.

Three invariants occupy here distinct and coordinated levels. The curvature sign

$$\tau \in \{+1, 0, -1\}$$

classifies the elliptic, parabolic, and hyperbolic regimes. The character p classifies the radial response. Spin records the orientation holonomy accumulated by the lift. Curvature, radial response, and spin form a reading system: curvature determines the local type of space, p determines how the trajectory occupies that space, and spin preserves the orientation acquired while traversing it. Closed and open curves then appear as distinct geometric biographies of the same oriented unit.

52.12.2 Particle as a Persistent Section of an Enriched Orbit: Phase, Route, Fibre, Multisection, and Memory

Dimensional Mechanics can formulate the particle as a persistent section of that geometric and dynamical extension. A compact conceptual notation is

$$\mathfrak{P} = \mathrm{Sec}_{\mathrm{pers}}\big[\tau, p, \mathrm{Hol}; \text{sheet, fiber, route, memory}\big].$$

The particle preserves a recognizable route through changes of phase, curvature, and dimension. Its spin publishes orientation holonomy; its charge publishes phase polarity; its color publishes an internal orientation of the gauge fiber; its flavor distinguishes families of routes or multisections; its mass expresses the dimensional-transduction character by which areal action acquires a volumetric reading. Each property is a reader of the persistent state, and all refer to a common genealogy.

The electron occupies the basal threshold of this construction because it offers the first stable reader in which phase, charge, spin and mass are simultaneously published. The rector electronic mass

$$m_e^{\beta} = 0.510998950690000808608\ldots \text{ MeV}$$

fixes a transduction reference; the preceding basal states preserve their function as genealogical stages. From the electron onward, the quotients K_D/K_Ω, the towers $90/120$, and the domain $81/56/13/324$ organize the publication of massive families. The mass law therefore reads how each persistent section traverses the enriched space and converts areal density of action into volumetric presence.

This interpretation connects spirals to particles precisely. The index p supplies the radial response; τ supplies the curvature regime; holonomy supplies spin; the route and the fiber supply flavor and color;

area–volume transduction supplies the mass character. The particle atlas thereby becomes a classification of stable modes of dimensional traversal.

52.12.3 Transduction of the oriented phase in electric polarization, magnetic response, and boundary radiative emission

Electricity is presented as directed transport of phase and action through a vacancy structure. A vacancy is an admissible transition of the support: it opens a channel, fixes a threshold, and permits the update to advance. The electric component publishes transport tangential to the route. The magnetic component publishes the transverse curvature induced by that transport. Their perpendicularity expresses the geometric relation between advance and rotation.

When two sinusoidal transverse components maintain the appropriate phase difference, the field vector rotates. Longitudinal propagation lifts that rotation to a helix. Variation of amplitude transforms its frontal projection into a spiral. The geometric reading is then fully coordinated: angle carries phase; angular velocity carries frequency; radius carries amplitude and action density; orientation carries polarization and direction; the vertical coordinate carries propagation and memory; p carries the radial regime; the threshold carries emission, absorption, or change of state.

Electromagnetism appears as a phase–action–memory dynamics with complementary publications. Electricity records polar advance; magnetism records the transverse response; the wave records periodicity; the helix records advance with memory; the spiral records radial evolution; radiation records the publication of the state when accumulation reaches a threshold.

This architecture also orders the role of vacancies and cycles. Vacancy supplies the local possibility of transition; the cycle preserves phase; memory distinguishes turns; the threshold determines the next residence of the action. Propagation may remain confined, change radial regime, or be published as controlled radiation. Field forces thereby acquire a continuous genealogical description from the discrete support to the observed wave.

52.12.4 Reversibility of the complete dynamics and thermodynamic cost of information-reducing projections

The Landauer zero represents a limiting case of this law. For a bijective update of the complete state,

$$H(X \mid U(X)) = 0, \qquad \inf Q_L = 0.$$

The information remains available in the genealogy. When a reader forms a quotient q, entropy decomposes as

$$H(X) = H(q(X)) + H(X \mid q(X)).$$

The conditional term measures the information retained outside the projection. A materialized tritic reset satisfies the threshold

$$Q_{\text{reset}} \geq k_B \Theta \log 3.$$

The Boltzmann constant translates the multiplicity grouped by the thermal reader; the Nyquist threshold separates faithful sampling from aliasing; null Landauer characterizes the reversible transport of the integral state. Information, temperature and action are articulated by the type of reading applied.

52.12.5 Self-conservative forces as transport of action among the observable, radial, memory, vacancy, and radiation sectors

Evolutionary conservation is better expressed through exchange than through scalar immobility. A complete state distributes action and momentum among visible motion, memory, radial deformation, vacancies, radiation, and the conjugate return branch. The enriched balance may be written as

$$\Delta J_{\text{visible}} + \Delta J_{\text{memoria}} + \Delta J_{\text{radial}} + \Delta J_{\text{vacancias}} + \Delta J_{\text{radiation}} + \Delta J_{\text{retorno}} = 0.$$

A force is **self-conserving** when its own update law contains the channels that transport, store, publish, and reabsorb action. Conservation belongs to the complete genealogical balance of the interaction. Each reader observes part of the exchange; the genealogy reconstructs its totality.

The Earth–Moon system provides a direct physical image. Earth's rotation, the lunar orbit, tides, deformation, and radiation exchange angular momentum. The net consumption of the complete system closes in the genealogical balance of those transfers. The magnitude called energy functions as a reading derived from the action and momentum published in each sector. Fundamental conservation resides in compatibility among all readers of the exchange.

This formulation dissolves the former boundary between conservative forces and so-called nonconservative forces through a deeper category. Friction, apparent dissipation, and radiation occupy explicit channels of the extended state. A loss in the visible mechanical projection corresponds to a transfer toward memory, deformation, heat, vacancies, or radiation. Self-conservation follows the residence of action through those changes.

52.12.6 Self-inertial systems: publication of acceleration through orientation, curvature, sheet, and memory

The appropriate expression is **relational self-inertia**. It denotes the compatibility between the total state's own connection and the response that appears when it is projected within the global distribution. The endogenous and exogenous aspects are analytic components of a single dynamic relation.

Let Γ be a horizontal trajectory or geodesic of the complete state. Its intrinsic law satisfies

$$\nabla_{\dot\Gamma}\dot\Gamma = 0.$$

When it is projected onto a reader S, the observed acceleration is determined by the curvature of the projection:

$$\nabla^{S}_{\dot\Gamma_S}\dot\Gamma_S = \mathcal{K}_{\pi_S}(\dot\Gamma,\dot\Gamma).$$

The published acceleration expresses the relation between total motion and the geometry of the reader. Mach's principle completes this mechanism through the global boundary trace

$$b : \mathcal{X}_\infty \longrightarrow B_\infty$$

and the factorization of the local inertial response

$$I_v = \bar I_v \circ b.$$

The global distribution enters the local response through a typed map. Relational self-inertia therefore combines the endogenous connection of motion, the global condition transmitted by b, and the observer's concrete projection. A synthetic expression for this coordination is

$$\nabla^{\text{auto}} = \mathcal{M}\big(\nabla^{\text{end}}, b[\mathcal{X}_\infty], \pi_S\big),$$

where $\mathcal{M}$ represents the Machian coupling and preserves the type of each component.

The Ambivalence Principle adds the double orientation of reading. Every direct publication preserves a conjugate return branch; emission and absorption form related endpoints of the same biography; temporal orientation is read from both directions of transport. Absorber theory finds a natural formulation here: source and receiver jointly close the genealogical balance of phase and action. The torsional present is the interface at which both orientations are rendered compatible.

TRIT acts at this level as a time crystal: it organizes the local periodicity of updating and, through memory, preserves the difference between successive returns. The direct branch and the conjugate branch relate emission to absorption, and the observer–observable coupling acquires a dynamical residence within the same state. Published time is the reading of that orientation; genealogical time is the complete sequence of phase, return, and memory.

Mach and Ambivalence perform complementary functions. Mach relates the local response to the global distribution. Ambivalence preserves the forward and conjugate directions of evolution. Relational self-inertia articulates both and replaces the division between inertial and non-inertial systems with a single law of connection, projection, and memory.

52.12.7 Area–volume duality, hexad–octad transition, and the role of the Barbero–Immirzi parameter

The passage from area to volume constitutes a central dimensional transduction. Area publishes boundary action; volume publishes interior depth. The HMT derivation of the Barbero–Immirzi parameter organizes the compatibility between the two measures through hexad–octad incidence and the channels $90/120$:

$$90 = 6\binom{6}{2}, \qquad 120 = 8\binom{6}{2}.$$

The cardinals 6 and 8 admit a cubic realization by means of faces and vertices, whereas the Steiner hexad and octad are distinct incidence objects linked only by the typed morphism $6 \to 8$. The 12 edges of the cube retain their own combinatorial function and do not by themselves define Steiner incidence. The difference in incidence expresses a productive dimensional incompatibility: surface information requires a transduction rule in order to acquire volumetric residence.

The same principle appears in the carry $999 + 1 = 1000$. The nines reach the boundary of the decimal register; the increment simultaneously publishes the closure of the preceding level and the origin $0/1$ of the next. The holographic ratio among 729, 243, 27, and 1 unfolds that transition at triadic scales. The Barbero–Immirzi transduction is inserted into this family of reader changes: boundary and interior preserve the unit while redistributing its measure.

The normalized defect $-1/12$ and the oriented weight $12 : -1$ perform distinct functions within this geometry. The former quantifies an incidence correction; the latter records a signed orientation in the transduction. Their coordination with the minimal area, the modular Hamiltonian, and memory provides the bridge to dimensional dynamics.

52.12.8 Hyperbolic interior contraction, exterior expansion, and holographic conservation of the boundary

The thesis may be formulated from a single idea: unity preserves its identity while changing scale, reader, curvature, memory, and dimension. This permanence has an evolutionary character because unity unfolds, acquires new coordinates, and publishes different properties; it has a holographic character because each finite publication retains the rule for reconstructing the state from which it proceeds. The Holographic Closure of Infinity designates precisely this compatibility between identity and unfolding: each stage contains the law that permits the same genealogy to be deepened inward and prolonged outward.

The governing designation is **principle of evolutionary conservation of the unit through dimensional transduction and holographic projection**. The expression places each term at its proper level. Conservation pertains to genealogical identity; evolution pertains to updating; transduction relates

dimensional realizations; holographic projection preserves the reconstructive correspondence between its readers.

Elementary arithmetic offers the first image of this architecture. The decimal completion

$$1000 = 10^3 = 729 + 243 + 27 + 1 = 3^6 + 3^5 + 3^3 + 3^0$$

exposes four triadic strata within a decimal cubic unit. Upon normalization,

$$1 = \frac{729}{1000} + \frac{243}{1000} + \frac{27}{1000} + \frac{1}{1000},$$

the unit is recognized simultaneously as totality and as graded unfolding. The reciprocal reading

$$3^{-6}, \quad 3^{-5}, \quad 3^{-3}, \quad 3^0$$

describes the inner depth of those strata; its normalization produces the corresponding partition. Relation $999 + 1 = 1000$ expresses the transport mechanism: the last unit completes a register and acts simultaneously as the carry inaugurating the next. Zero and one thus form a continuity interface. The intervals $[0, 1]$, $[0, 10]$, and the prolongation toward $[0, \infty)$ are scalar readers of a single completion law.

The first splitting $1000 = 729 + 271$, followed by $271 = 243 + 27 + 1$, shapes the extreme and mean holographic ratio of the deployment: a dominant triadic core preserves the same resolution rule within the remainder. The decimal cubic unit thus contains a nested triadic depth. Movement toward 729, 243, 27, and 1 reads the inward contraction; the carry toward one thousand reads the outward expansion.

This double orientation organizes HMT infinity. Inward, refinement grows hyperbolically: each cell admits a new depth without losing its filiation. Outward, unity expands through successive dimensional publications. Closure consists in the commutativity of both movements. If U represents the update of the state, R_d the reader on stratum d, and $T_{d\to d'}$ the transduction between strata, evolutionary conservation is expressed by means of

$$R_{d'} \circ U = T_{d\to d'} \circ R_d.$$

The same event may provide an angular measure, an action density, a published mass, or a gravitational response. Their common identity resides in the genealogy that makes those readings commute.

52.12.8.1 Nonadic Monodromy and Helical Lifting with Memory Advance

APP, TRIT, and TPK provide the formal mechanism for this unfolding. APP constructs the support and separates the evaluations that must be preserved. TRIT organizes the local regimes and their ambivalence. TPK transports phase and memory over the enriched support. The causal dependence is fixed by

$$U_t \longrightarrow \Gamma_9 \longrightarrow K_{\mathrm{ph}}.$$

Phase return and memory advance form coordinated operations. After nine steps, the same angular class may reappear,

$$g(K + 9) = g(K),$$

as the memory counter progresses,

$$q_{\mathrm{mem}}\Gamma_9 = q_{\mathrm{mem}} + 1, \qquad B_{K+9} \neq B_K.$$

The precise geometric figure is a helical lift. A closed orbit in the angular projection lifts to an open trajectory when each phase return increments the memory. The circle preserves the cycle; the helix

preserves the cycle and additionally records the depth reached. The coinductive monodromy thereby generates infinite depth through a sequence of locally closed and globally ascending returns.

This image has more than merely illustrative value. The angular wheel constitutes a section of the helix; the helix constitutes the wheel's complete biography when the state transports memory. Each turn recovers a phase position and simultaneously initiates a new sheet. Infinitude arises from compatibility between return and lifting: iterating the cycle produces depth without dissolving orbital identity.

The nonadic certificate gives finite and verifiable content to that architecture. The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

orders the lift; the 396 levels, the 2376 trits, the 43 turns, and the 387 pairs $(K, K+9)$ per channel quantify the transit; the five regions of π and the ambient fiber of 729 elements publish subsequent internal readings. This sequence makes a productive monodromy visible: the return recovers the phase and coinduction preserves the growth of memory.

52.12.8.2 Rational unit and coordination of the circular, elliptic, projective, and helical readers

The radion concentrates within a metrological unit the structure that the radian publishes angularly. Its natural definition is that of an **oriented section of phase, action, and memory**. One unit of phase $\theta = 1$ transports one unit of action $\sigma\hbar_{\mathrm{ret}}$, belongs to a complete turn $T_+ = 2\pi_{\mathrm{HMT}}$, and preserves sheet, orientation, nonadic return, and position in the refinement cylinder.

The radian expresses the visible angular coordinate. The radion preserves, together with that coordinate, the transported action, orientation, carry, and memory of the revolution. Angular metrology thereby acquires a genealogy. Measuring an arc means identifying how much rotation has been published and which complete state has realized that rotation.

The relationship between phase and area takes on a particularly clear form:

$$\mathcal{A}(\theta) = \hbar_{\mathrm{ret}}\,\theta.$$

One radional unit sweeps an action area $\hbar_{\mathrm{ret}}$; one complete turn encloses h_{ret}. The passage from phase to area is incorporated into the reading unit itself. The angular metric, action, and memory cease to appear as juxtaposed quantities and become coordinates of the same oriented section.

The radion admits several coordinated geometric publications. The circular reader publishes phase; the elliptical reader publishes action and form; the Klein interior reader publishes depth; the nonadic helical reader publishes return and memory. The unit remains genealogically identical across those realizations. The radion therefore constitutes the appropriate metrological link between HMT and Dimensional Mechanics: it offers a unit that can traverse changes of reader while preserving the state presupposed by each partial measurement.

52.12.9 Five-Component Gravitational Polarity and Joint Transduction of Action, Area, Volume, Time, and Mass

Dimensional Mechanics makes it possible to define dimension as a stable publication regime characterized by the minimum rank of independent memory required by the lift. To traverse a dimension is to apply a transducer that preserves genealogy and changes the type of reading. Gravity occupies the level of connection between these regimes.

In this formulation, **interdimensional gravitation with five correlative polarizations**, abbreviated as the **pentapolar structure of gravitation**, is the connection that transports genealogical unity and leaves minimal torsion at each interface. A compact notation for this trace is

$$\Theta_d^{\min} = \nabla_{d+1} \circ T_{d\to d+1} - T_{d\to d+1} \circ \nabla_d.$$

The expression measures the difference between transporting after differentiating and differentiating after transporting. That difference is the minimal geometric memory of the dimensional crossing. The composition of interfaces produces a gravitational holonomy

$$\mathfrak{G}_{d\to D} = \mathrm{Hol}\big(\Theta_d^{\min}, \Theta_{d+1}^{\min}, \dots, \Theta_{D-1}^{\min}\big).$$

The fundamental object is this interdimensional connection. A local graviton-like mode occupies the derived rank of the quantized publication of torsion; gravitation retains its primary residence in compatibility among dimensions.

The term pentapolar encompasses the five polarizations inherent to the continuous state:

$$\mathbf{G}_d^{\pm} = \big(G_{\mathrm{sp},d}^{\pm}, G_{\mathrm{sol},d}^{\pm}, G_{\mathrm{gau},d}^{\pm}, G_{\mathrm{coh},d}^{\pm}, G_{\mathrm{det},d}^{\pm}\big).$$

The five components emerge correlatively from the same enriched state. The polarity ± records direct concentration and conjugate publication. Gravity traverses the dimensional strata while simultaneously transporting those five readings; minimal torsion preserves the trace of each crossing.

Geometry thereby acquires a unified image. The elliptic regime organizes closures; the parabolic regime organizes thresholds; the hyperbolic regime organizes depth. Dimensional transduction connects these regimes with the radial responses p, spin holonomies, and mass publications. The pentapolar structure of gravitation expresses the global coherence of that connection.

M-theory provides subsequent physical screens for this connection: additional dimensions, strings, branes, and T-duality realize concrete modes of transduction. HMT supplies the APP–TRIT–TPK genealogy, the enriched state, memory, and bidirectional reading; the interface with M-theory publishes those structures in typed physical geometries. T-duality exchanges compatible dimensional readers and preserves the genealogical class of the state.

52.12.9.1 Compatibility of the classical, relativistic, quantum, and HMT-MD realizations

Classical mechanics publishes trajectory and momentum. Relativity publishes causal structure, metric, and curvature. Quantum mechanics publishes phase, holonomy, vacancies, and amplitudes. Dimensional Mechanics provides the transducers that relate those publications, and HMT preserves the discrete genealogy that sustains them.

In the classical reading, an orbit describes the visible projection of motion. In the relativistic reading, that orbit is incorporated into the geometry of the reader. In the quantum reading, phase and spin record the holonomy of the section. In the HMT–MD reading, the three expressions belong to the same lifted biography: the radion quantifies phase, action, and memory; p classifies the radial response; τ classifies curvature; holonomy publishes spin; the dimensional transducer publishes mass; the pentapolar connection publishes gravitation.

Apparent classical dissipation acquires a residence in the expanded genealogical balance. Relativistic acceleration is expressed through connection and projection. Quantum indeterminacy is organized by readers, vacancies, and branches of Ambivalence. Mach's principle relates the local response to the global distribution. Zero Landauer cost guarantees the informational reversibility of the complete state. All these elements converge in a deeper conservation: the identity of the unit across changes of publication.

52.12.9.2 Genealogical Conservation and Holographic Prolongation Narrative Theorem

The closure can now be narrated as a single sequence. The unit decomposes without losing identity. The oriented phase acquires measure in the radion. The nonadic return converts the cycle into a helix through memory. The character p transforms the helix into radial families and classifies their regimes. The persistence of those trajectories produces sections with spin, charge, flavour, colour, and mass. Vacancies open channels; electrical polarity transports phase; magnetic response curves transversely; thresholds publish radiation. Self-conservation closes the genealogical balance of action and momentum. Relational

self-inertia links the proper connection, projection, and Machian totality. Ambivalence preserves the direct and conjugate directions. Area–volume transduction leads to Barbero–Immirzi and the mass law. Minimal torsion links dimensions and publishes the pentapolar structure of gravitation.

Inward growth and outward expansion are conjugate orientations of the same process. Hyperbolic refinement generates depth; dimensional transduction generates extension. Carry $0/1$ links scales; monodromy links turns; memory links states; holonomy links orientations; gravitation links dimensions. Each linkage preserves an evolving unity, and every finite publication contains the rule of its prolongation.

The fundamental thesis is formulated as follows:

> **The holographic closure of infinity realizes the principle of the evolutionary conservation of unity through dimensional transduction and holographic projection: unity maintains its genealogical identity as it unfolds as phase, action, curvature, memory, particle, field, and gravitation; every finite reading preserves the rule of reconstruction and prolongation of the complete state.**

This formulation provides a joint physical and mathematical image. Infinity acquires a law of local closure; the unit acquires depth; dimension acquires memory; conservation acquires relational character; force acquires self-preservation; inertia acquires Machian structure; gravity acquires a pentapolar connection; and holography acquires the precise meaning of reconstruction among compatible readers.

52.13 Operative definition of the angular radian through the enriched state and 4 typed readers

52.13.1 Operational definition of the angular and quadruple radion and typed realization of the enriched state

The operator synthesis brings together the phase, action, depth, and memory readers without identifying them with one another.

52.13.1.1 Quadruple realization of the radional state

Let $x_\infty \in X_{\mathrm{enr}}$ be the coinductive section. The complete radion is not an isolated point, but rather the bundle

$$\mathfrak{R}_\sigma(x_\infty) = \big(\theta = 1, \sigma, \pi_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}}, A, C^*, \Delta_4, \Phi_T, D_E, \varepsilon_R^{(1)}, \hbar_{\mathrm{ret}}, q_{\mathrm{mem}}, \mathcal{I}_\infty\big). \tag{52.128}$$

Four readers extract different objects from it:

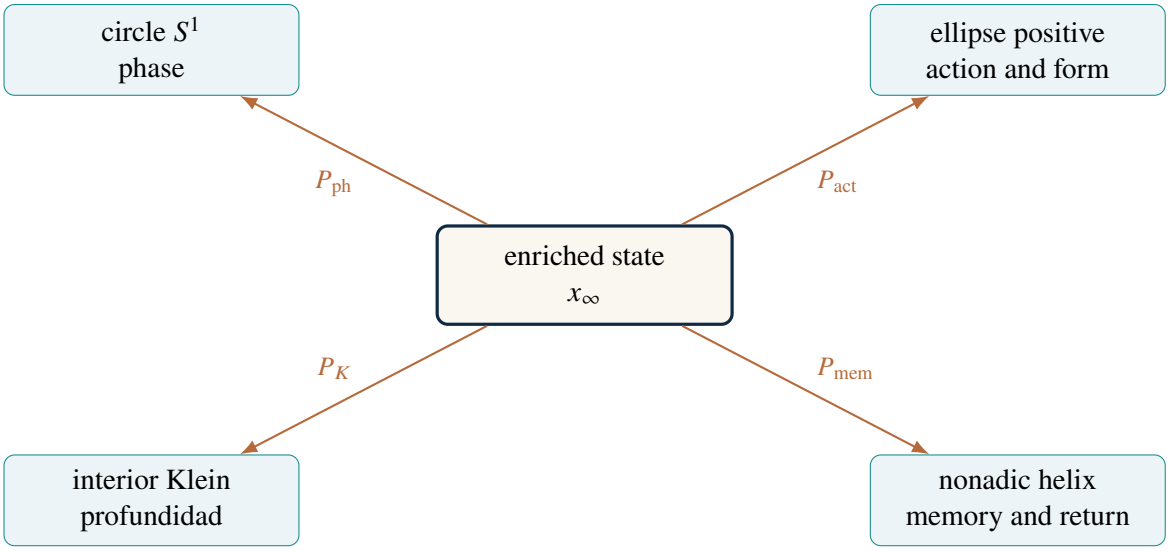


Figure 52.2: Four typed publications of the same antecedent. The unity of the state does not eliminate the difference among codomains.

The circle preserves the phase class and forgets the lift. The ellipse preserves action, anisotropy, and symplectic orientation. Klein endows the interior with a projective distance. The helix preserves the number of returns and refinement. Dimensional Mechanics does not consist in mixing them, but in constructing the transducers that make them compatible.

52.13.1.2 Operator interpretation of the cocycle ε_R

ε_R is not a universal constant independent of every context. It is the coordinate of a transition cocycle associated with the radion chart. Its domain contains two sections of the same state:

$$S_E = 1 + D_E, \qquad S_T = 1 + \Phi_T.$$

His rule is

$$\varepsilon_R^{(1)} = S_T - S_E = \Phi_T - D_E. \tag{52.129}$$

Its codomain is the dimensionless line of angular unity; the chart $360/T_+$ publishes it in degrees.

Exact Result

In plain language: the direct section and torsional section reach the same unit cell by two different internal routes. Their integer parts coincide; their remainders do not. ε_R is the oriented difference of those remainders, computed with APP carry and carried to the limit by the TPK cylinders. It is not introduced to make the radian close: it appears before the radian is consulted and, when published, closes it.

This definition explains why the number is small. It is not because a small correction was sought, but because S_E and S_T share a deep prefix. Smallness measures the structural proximity of two publications; the sign measures their orientation; the digits record the carry.

52.13.1.3 Operational definition of the radion

Definition 52.15 (HMT–MD radion)**.** The radion is the phase-oriented section of action and memory that simultaneously satisfies:

1. traverses a phase unit $\theta = 1$;
2. sweeps signed action $\sigma\hbar_{\mathrm{ret}}$;

3. belongs to a coinductively generated turn $T_+ = 2\pi_{\mathrm{HMT}}$;
4. preserves the σ sheet, the nonadic return, and the refinement cylinder;
5. satisfies the exact junction $1 = S_E - \Phi_T + \varepsilon_R^{(1)}$.

The conventional radian is the image of the radon under the forgetful functor.

$$\mathfrak{R}_\sigma \longmapsto \theta = 1 \in \mathbb{R}/2\pi\mathbb{Z}.$$

That forgetting is useful and legitimate for Euclidean trigonometry. It becomes insufficient when physics must distinguish action, sign, return, field, and history.

Genealogy of the Natural Angular Unit The naturalness of the radian is usually justified because the derivative of $e^{i\theta}$, $\sin\theta$ and $\cos\theta$ assumes its simplest form when θ is measured as the arc/radius quotient. HMT preserves that advantage, but explains it from an earlier layer:

$$\theta \quad \leftrightarrow \quad \frac{\mathcal{A}(\theta)}{\hbar_{\mathrm{ret}}}.$$

The phase is natural because it measures action in units of $\hbar_{\mathrm{ret}}$; the turn is natural because it encloses h_{ret}. The arc-to-radius quotient is the Euclidean publication of a unit that already possesses symplectic structure and memory.

Compatibility between action ellipse and dimensional growth The circle is the form that results when anisotropy is forgotten: $C^* = 0 \Rightarrow \eta = 0 \Rightarrow a = b$. The ellipse is more informative because it preserves two eigenvalues, two axes and an exchange orientation at fixed product. In this sense, the ellipse is a prolongation of the circle, not a circle deformed by accident.

Klein depth and the helical lift add two types of information that the planar contour does not contain: interior distance and return memory. This is the ontological image of Dimensional Mechanics that emerges from the radion: a new dimension appears when one requires an independent datum, which the preceding projection had to forget, to be preserved.

52.13.2 Structural properties of the radional unit: genealogy, critical hinge, density, carry, action, boundary–interior compatibility, form, polarity, and modularity

52.13.2.1 System of component theorems of the radion

Theorem 52.16 (Genealogy)**.** *The object $\mathfrak{R}_\sigma$ descends from* APP $\to \widehat{\mathbb{T}} \to$ TPK $\to X_{\mathrm{enr}}$ *and does not receive the metrological target as a selector.*

Theorem 52.17 (Hinge)**.** *In the hundred-mark APP chart, the critical seam and the second dodecaphasic phase satisfy $j_{100}(1/2) = \theta_2 = 50°$, with unique $m = 2$.*

Theorem 52.18 (Density)**.** *The two sheets of the active sector $N_6 = 90$, normalized by $3^6 = 729$, produce $\rho_{\mathrm{tw}} = 20/81$.*

Theorem 52.19 (Carry)**.** *The APP subtraction of the sections S_T and S_E converges to $\varepsilon_R^{(1)}$ at interval rate 3^{-54} per return and exactly closes the unit.*

Theorem 52.20 (Action)**.** *The positive ellipse satisfies $\mathcal{A}(\theta) = \hbar_{\mathrm{ret}}\theta$; a radion sweeps $\hbar_{\mathrm{ret}}$, and one turn encloses h_{ret}.*

Theorem 52.21 (Boundary–interior)**.** *The same boundary has finite action h_{ret}, whereas the Klein interior has infinite hyperbolic area.*

Theorem 52.22 (Form)**.** *The quotient $e^{-\eta}$ is preserved across spectral semiaxes, action semiaxes, uncertainties, impedance, and the modulus of the torus.*

Theorem 52.23 (Polarity)**.** *The inversion $C^* \mapsto -C^*$ interchanges constitutive relations, inverts impedance, and preserves the normalized velocity.*

Theorem 52.24 (Modularity)**.** *The bidirectional inversion acts as $S : \tau \mapsto -1/\tau$ and preserves $j(\tau)$, while the focal labeling preserves the orientation that j forgets.*

52.13.2.2 Algebraic and Geometric Proofs United

Linear proof of the unit closure To facilitate an independent audit, we present the argument without narrative interruption.

Step 1. The nonadic connection generates in a correlated manner $\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}, \alpha_{\mathrm{HMT}}$.

Step 2. The turn is $T_+ = 2\pi_{\mathrm{HMT}}$.

Step 3. The wheel fixes $\theta_2 = 50°$, and the parangular chart fixes $A = 1000\alpha_{\mathrm{HMT}}$.

Step 4. The direct section is

$$S_E = T_+ \left(\frac{50}{360} + \frac{1000\alpha_{\mathrm{HMT}}}{360} \right) = 2\pi_{\mathrm{HMT}} \left(\frac{5}{36} + \frac{25}{9}\alpha_{\mathrm{HMT}} \right).$$

Step 5. The certified intervals yield $1 < S_E < 2$; then $D_E = S_E - 1$.

Step 6. The incidence produces $N_6 = 90$; two sheets and the ambient space 729 produce $\rho = 2N_6/729 = 20/81$.

Step 7. Global torsion is

$$\Delta_4 = \frac{\pi_{\mathrm{HMT}} - e_{\mathrm{HMT}} \log \pi_{\mathrm{HMT}}}{270} \left(1 + \frac{\pi_{\mathrm{HMT}}}{729} \right).$$

Step 8. The section torsional is $S_T = 1 + (20/81)\Delta_4$.

Step 9. The subtraction APP defines

$$\varepsilon_R^{(1)} = S_T - S_E = \frac{20}{81}\Delta_4 - D_E.$$

Step 10. By substitution,

$$S_E - \frac{20}{81}\Delta_4 + \varepsilon_R^{(1)} = 1.$$

Step 11. Multiplying by $360/T_+ = 180/\pi_{\mathrm{HMT}}$ produces

$$\frac{180°}{\pi_{\mathrm{HMT}}} = 50° + A° - \frac{20}{81}\Delta_4^\circ + \varepsilon_R^\circ.$$

No step receives the left-hand side of the final equality. The metrological result is evaluated only at the end as a recognition.

Uniqueness of the carry-subtraction Let $B = 1000$. For each position, every integer u_i admits a unique Euclidean division

$$u_i = Bc_i + r_i, \qquad 0 \le r_i < B.$$

Therefore,

$$r_i = u_i \bmod B, \qquad c_i = (u_i - r_i)/B$$

are unique. Traversing the positions from the remote end toward the front, the output carry of one position fixes the input of the next. By induction, the entire word of residues and carries is unique.

In pseudocode:

```
acarreo = remote_acarreo
for i from last_block downto first_block:
    u = torsion[i] - direct[i] + acarreo
    remainder[i] = u mod 1000
    acarreo = (u - remainder[i]) / 1000
return remainder, acarreo_word
```

The procedure does not consult the expected value of the difference.

Convergence of the depth functional Let

$$F(p,e) = 1 + \frac{20}{81}\frac{p - e\log p}{270}\left(1 + \frac{p}{729}\right) - 2p\left(\frac{5}{36} + \frac{25}{9}\alpha_{\mathrm{HMT}}\right).$$

On the compact rectangle $3 \le p \le 4$, $2 \le e \le 3$, F is continuous and continuously differentiable. If $I_N^{\pi} \searrow \{\pi_{\mathrm{HMT}}\}$ and $I_N^{e} \searrow \{e_{\mathrm{HMT}}\}$, the interval image

$$F(I_N^{\pi}, I_N^{e})$$

forms a nested family whose width tends to zero. The intersection is a single real number, $\varepsilon_R^{(1)}$. Monotonicity of the derivatives permits the endpoints to be evaluated without interior sampling.

The rate 3^{-54} per return follows from the refinement of six trits over nine levels. At depth 45 triads, the width bound $< 4.81 \times 10^{-130}$ certifies more than 129 digits of the value.

Derivation of the metric and hyperbolic area of Klein On the unit disk, the matrix form of the Klein metric is

$$g(u,v) = \frac{1}{1-r^2} I + \frac{1}{(1-r^2)^2}\begin{pmatrix} u \\ v \end{pmatrix}\begin{pmatrix} u & v \end{pmatrix}.$$

The lemma for the determinant of a rank one perturbation gives

$$\det g = (1-r^2)^{-3}.$$

Henceforth

$$\sqrt{\det g}\; \mathrm{d}u\, \mathrm{d}v = (1-r^2)^{-3/2}\, \mathrm{d}u\, \mathrm{d}v.$$

Integration in polar coordinates yields (52.48). The infinite limit is due to the metric singularity at the boundary; the boundary continues to have finite symplectic area because that measure uses $\mathrm{d}Q \wedge \mathrm{d}P$, not $\sqrt{\det g}$.

Focal Distance on the Klein Disk In a diametro, the distance from the center to r is

$$d_K(0,r) = \operatorname{artanh} r.$$

For $r = e_f$, (52.32) implies

$$1 - e_f^2 = e^{-2\eta}.$$

If $u_f = \operatorname{artanh} e_f$, then

$$\operatorname{sech} u_f = \sqrt{1 - e_f^2} = e^{-\eta},$$

and by therefore $\eta = \log\cosh u_f$, as in (52.51).

Symplectic Compression Transformation Define

$$S_\eta = \begin{pmatrix} e^{\eta/2} & 0 \\ 0 & e^{-\eta/2} \end{pmatrix}, \qquad \det S_\eta = 1. \tag{52.130}$$

The transported complex structure is

$$J_\eta = S_\eta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} S_\eta^{-1} = \begin{pmatrix} 0 & -e^{\eta} \\ e^{-\eta} & 0 \end{pmatrix}, \qquad J_\eta^2 = -I. \tag{52.131}$$

The curve

$$\gamma(\theta) = \sqrt{2\hbar_{\text{ret}}}\, S_\eta(\cos\theta, \sin\theta) = e^{\theta J_\eta}\gamma(0)$$

is exactly the action ellipse. The hyperbolic symplectic compression preserves area and transports complex rotation instead of destroying it.

For orientation σ,

$$\gamma_\sigma(\theta) = (a_S\cos\theta, \sigma b_S \sin\theta), \qquad \mathcal{A}_\sigma(\theta) = \sigma\hbar_{\text{ret}}\theta.$$

The two sheets share foci and action modulus; they differ in signed action.

Diophantine verification of the dodecaphasic Barbero–Immirzi weight The equation $4a - 3b = 45$ has integer solutions

$$(a, b) = (12 + 3t, 1 + 4t), \qquad t \in \mathbb{Z}.$$

The condition $|b| = 1$ leaves $b = 1$, since $b = -1$ does not belong to the class $1 \pmod 4$. Hence $a = 12$. This verification reproduces the pair obtained through the dodecaphasic fundamental chain: twelve sectors and one unique oriented seam. The cardinalities 90/120, the chain multiplicities, and the return reader retain distinct functions; the pole coefficient and minimality arithmetically verify their composition.

52.13.3 Sensitivity of unitary closure to controlled variations in sheet multiplicity, the dodecaphase, channel e, global torsion, the harmonic operator, and the hexadic tail

52.13.3.1 Negative controls of the rational system

The construction is refutable at specific points:

1. If $S_E \notin (1, 2)$, the APP quotient is not one and the closure changes cell.
2. If the incidencia activates no produces $N_6 = 90$, o if no there are two sheets, 20/81 ceases of seguirse.
3. If the verifier receives $180/\pi$ or ε_R before constructing the operands, the generation is circular.
4. If subtraction by triads does not stabilize compatible prefixes, the proposed coinductive section does not exist.
5. If $|C^*| \geq A$, the positive ellipse ceases to exist.
6. If $a_S b_S \neq 2\hbar_{\text{ret}}$, the area theorem fails.
7. If $C^* = 0$, $d = 0$, $\eta = 0$, $\tau = i$ and $j = 1728$ must hold.
8. If the Klein area of the subdisk does not follow (52.48), the infinite-depth thesis fails.

9. If $F_{\mathrm{spec}} = \varepsilon_R^{\circ}$ is asserted without an independent counterterm, the numerical difference (52.99) refutes it.
10. Producing $12 : -1$ requires preserving, alongside the cardinalities $90/120$, the twelve dodecaphasic sectors, the unique oriented seam, and the reader's action on these generators. Changing any multiplicity must modify the image $12S_{90} - S_{120}$. For the unaltered fundamental chain, the coefficient of $(1-q)^{-1}$ must yield $1/24$.
11. If the function j is used to select the charge sign, the invariance (52.114) is lost and a type error is incurred.

52.13.3.2 Joint physical realization of the radion

The final image is not a corrected circle. It is an architecture of publications:

APP:	unit, remainder, carry and fiber;
TRIT:	regime, orientation and sheets;
TPK:	phase, return, dodecaphase and memory;
ellipse:	action and anisotropy;
Klein:	interior depth;
helix:	biography of the return;
MD:	clock, energy, charge, field and causality.

The radion equation brings together the visible face:

$$\boxed{\frac{180^{\circ}}{\pi_{\mathrm{HMT}}} = 50^{\circ} + A^{\circ} - \frac{20}{81}\,\Delta_4^{\circ} + \varepsilon_R^{\circ}}$$

The ellipse brings together the action face:

$$\mathcal{A}(1) = \hbar_{\mathrm{ret}}, \qquad \mathcal{A}(2\pi) = h_{\mathrm{ret}}.$$

The Klein metric unites the depth:

$$K = -1, \qquad \mathrm{Area}_K = \infty.$$

Holonomy gathers memory:

$$g(k+9) = g(k), \qquad B_{k+9} \neq B_k.$$

Tesis rectora

The circle is the angular shadow; the ellipse is the body of action; Klein measures the interior; the helix preserves the biography. The radion is the unit that allows the four statements to be read on a single state without reducing one to another.

52.13.3.3 Canonical values, stability, and reproducible verification

Values of the generated internal invariants

Object	Value used in the coinductive profile
π_{HMT}	3.14159265358979323846264338327950288419716939937510582097494459 23 . . .

Object	Value used in the coinductive profile
e_{HMT}	coinductive evaluation of the Euler reader, without an external table
α_{HMT}	$0.00729735256928380099728510547238066299956417253964270918540468 25\ldots$
A	$7.29735256928380099728510547238066299956417253964270\ldots^{\circ}$
C^{*}	$2.12373833896864572426195138039336079 37\ldots^{\circ}$
η	$0.29968968392163079968492656286392 7\ldots$
$\hbar_{\mathrm{ret}}$	$1.05457181691503906113369058472430\ldots\times 10^{-34}U_S$

Quantitative convergence with depth

Depth	ε_R°	reading
12	$5.1681282186551375597\times 10^{-8}$	preasymptotic
18	$5.1383017764316644535\times 10^{-8}$	7 cifras estables
24	$5.1383016758575394560\times 10^{-8}$	14 cifras estables
30	$5.1383016758575282072\times 10^{-8}$	20 cifras estables
36	$5.13830167585752820716851\times 10^{-8}$	perfil largo
45	$5.13830167585752820716851646226459\times 10^{-8}$	width $< 4.81\times 10^{-130}$

The depth returns $9, 18, 27, 36, 45$ preserve the phase 8 and increase the memory $0, 1, 2, 3, 4$. The ratio between consecutive widths tends to

$$3^{-54} = 1.71969823345498561276103993160048 71\ldots\times 10^{-26}.$$

Structural Sensitivity Analysis by Controlled Suppression

Operation	Output or unit difference
Canonical operator	$8.96802822044562981999\times 10^{-10}$
A sheet, $\rho = 90/729$	$-1.3727056615960678\times 10^{-5}$
Phase $\theta_1 = 20^{\circ}$	$+5.2359877649510169\times 10^{-1}$
Phase $\theta_3 = 80^{\circ}$	$-5.2359877470149605\times 10^{-1}$
Without channel e	$1.8065350748904653\times 10^{-3}$
Historical Torsion $\Delta_4(A, C^{*})$	$1.0517084608768816\times 10^{-9}$
Harmonic operator M_4 minus canonical	$1.0525500222895070\times 10^{-17}$
Hexagonal Cola less canonical	$2.0283936357433839\times 10^{-12}$

The separation of scales reveals structural sensitivity: closure depends on the phase, the two sheets, the channels π/e, global torsion, and carry. No ablation reproduces the value.

Reproducible Computational Certificates The work uses the following executable proprietors:

1. `output/RADION_TWOWAY_NONADICO_20260827/verificar_emergencia_radion_por_profundidad.py`;
2. `output/RADION_TWOWAY_NONADICO_20260827/verificar_radion_helice_critica_tpk.py`;

3. `output/RADION_TWOWAY_NONADICO_20260827/verificar_operador_tpk_target_free_profundidad.py`;
4. the consolidated verifier for this monograph, located at `certificados/verificar_monografia_radion.py`.

The expected outputs include

`PASS_EMERGENCIA_RADION_POR_PROFUNDIDAD`,

`PASS_RADION_HELICE_CRITICA_TPK`, `PASS_MONOGRAFIA_RADION_HMT_MD`.

Distinction between computational precision and mathematical accuracy The fact that a program displays zero to 120 or 200 digits does not by itself turn a residual identity into a prediction. Mathematical exactness derives from (52.18); arbitrary precision makes it possible to evaluate its operands and verify convergence. Non-circularity derives from the order of construction and the prohibition on target values, not from the number of digits.

Decimal computation may leave residuals of order 10^{-218} when combining branches evaluated at different precisions. That number is implementation rounding, not an additional physical correction.

52.13.3.4 Conceptual Genealogy and Primary Sources

Planck action Planck introduces h into the radiation law as the constant linking energy and frequency. The decisive historical datum for this work is dimensional: h is action. The radion does not alter that role; it explains why the action of one complete turn and the action per unit of phase are related by 2π.

Bohr-Sommerfeld Angular Quantization Bohr uses $h/2\pi$ in the quantization of angular momentum. Sommerfeld extends quantization to action integrals and Keplerian orbits. Therefore, it is not historically accurate to say that Dirac invented the reduced constant. His contribution belongs to a genealogy in which angular action was already present.

The HMT–MD novelty is different: it does not take $h/2\pi$ as an external quotient, but rather constructs an ellipse whose turn has area h and whose sector of one radian has area $\hbar$.

Born and Jordan Matrix Formalism and the Quantum Oscillator Born and Jordan place the variational principle, phases, and the harmonic oscillator at the heart of matrix mechanics; the electromagnetic field is decomposed into a collection of oscillators. The LC realization of the radion continues that intuition with one additional datum: the form quotient $e^{-\eta}$ determines the relative impedance while product, frequency, and action remain fixed.

Angular frequency and action in the Dirac formulation In 1925, Dirac writes frequencies in radians per unit time and uses relations of the form

$$xy - yx = \frac{ih}{2\pi}[x, y], \qquad \Delta E = \frac{h}{2\pi}\omega.$$

The equation $E = \hbar_{\mathrm{ret}}\omega$ explicitly joins reduced action and angular frequency. In the radion, that relation acquires a geometry: $\hbar_{\mathrm{ret}}$ is the area swept by one unit of phase.

Heisenberg, Robertson, and Schrödinger Uncertainty Relations Heisenberg formulates the joint limitation of canonically conjugate variables. Robertson generalizes the inequality; Schrödinger incorporates covariance. The matrix (52.55) belongs more directly to that Robertson–Schrödinger prolongation than to Heisenberg's article alone.

None of these authors formulates the complete genealogical ellipse of this work. The historically defensible assertion is that they provide the language in which the action ellipse can be recognized as a minimal contour; the construction of the pair (A, C^*), the foci, Klein, the helix, and the carry is the HMT–MD contribution.

Feynman Phase Functional The spacetime formulation assigns a phase to each path, $e^{iS/\hbar}$. The radion unit makes the correspondence literal: incrementing the action by $\hbar$ increments the phase by one radian. The sum over paths recognizes, in integral language, the same principle that the ellipse expresses as area.

Advanced and retarded propagations in Wheeler–Feynman Absorber theory constructs a temporally symmetric classical interaction through combinations of retarded and advanced fields. The HMT identity $CU(t)C^{-1} = U(-t)$ and route inversion provide an operator-theoretic antecedent. Physical promotion requires the map between routes and propagators to preserve action, orientation, boundary, and symplectic form.

Absorber theory is not the same assertion as the Feynman–Stueckelberg interpretation of antiparticles as temporally reversed propagation. Both may recognize bidirectionality; their physical objects and boundary conditions differ.

Hodge Duality and Separation with Respect to Hochschild Cohomology The operation relevant to this manuscript is the Hodge star $\star^2 = -I$ on two-forms in Lorentzian signature. No proprietor of Hochschild homology acting on the ellipse, the carry, or the sector 90/120 has been located in the active radion corpus. Introducing Hochschild by nominal similarity would degrade the typing. If a future prolongation constructs a deformation cocycle with its own domain and codomain, it must enter as a new result, not as a retroattribution.

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> The founders contain the components: action, angular frequency, conjugate variables, oscillator, uncertainty, path phase, and time reversal. The contribution of the radion is to bring them together in a generated object: an ellipse of area h, a sector of area $\hbar$, an interior of infinite depth, a sheet orientation, and an exact carry of compatibility.

52.13.3.5 Final Synthesis: Angular unity with phase genealogy, action and memory

The radian does not disappear. It becomes fully visible.

In its ordinary use, a radian is a ratio that need not remember how it was produced. The arc and the radius cancel dimensionally; the unit appears bare. That bareness is a virtue for trigonometry, but it is a forgetting for a theory that seeks to explain the structure of the continuum and the origin of action. The radion recovers what was forgotten without altering the published value.

The genealogy begins in APP, where an output preserves its value and also its sheet, quotient, remainder, carry, and boundary. TRIT introduces regime and orientation. TPK returns the phase without resetting the state. The nonadic connection constructs the coinductive section of π, and the dodecaphase publishes the hinge of 50°. The angular pair supplies a common mode A and an oriented opening C^*. The incidence of six active phases produces 90, its two sheets over the ambient space 729 produce $20/81$, and global torsion supplies the second section.

Then ε_R appears. Not as a decimal begged of the closure, but as the difference that the structure was required to transport:

$$\varepsilon_R^{(1)} = \frac{20}{81}\Delta_4 - (S_E - 1).$$

The carry publishes its digits; the depth certifies its limit; the angular chart recognizes it as

$$5.13830167585752820716851646226459474899\ldots \times 10^{-8\,\circ}.$$

The radian equation is closed because both sides are publications of the same state:

$$\boxed{\frac{180^\circ}{\pi_{\mathrm{HMT}}} = 50^\circ + A^\circ - \frac{20}{81}\,\Delta_4^\circ + \varepsilon_R^\circ}$$

The account does not end with the equality. The pair (A, C^*) constructs a positive ellipse. The product of its semiaxes fixes $2\hbar_{\mathrm{ret}}$; their quotient fixes its shape. The area theorem proves that one phase unit sweeps $\hbar_{\mathrm{ret}}$ and that one revolution encloses h_{ret}. Division by 2π here acquires physical content: h is action per revolution; $\hbar$ is action per radion.

The foci cease to be ornaments in a figure. Their separation reconstructs C^* in the spectral chart, and their Klein distance measures a finite hyperbolic depth within a domain whose total hyperbolic area is infinite. The same boundary contains a finite action. The apparent paradox is the central thesis of the geometry: one measure closes the action; another opens the depth.

The circle, therefore, is not false. It is the exact projection that preserves phase and forgets anisotropy, interior, and memory. The ellipse preserves action and form. The Klein surface preserves depth. The helix preserves return and biography. No image replaces the others; all arise from the same state.

The Hilbert realization recognizes the ellipse as a contour of minimal covariance. The LC oscillator recognizes it as an exchange of electrical and magnetic reserves. The reader 90/120 publishes permeability, permittivity, impedance, and velocity. The six–eight incidence produces a causal screen; Hodge converts orientation into field duality; Lorentz converts field and charge into dynamics; Cartan–Holst receives the holonomy in gravitational geometry. The order matters: the HMT object selects; conventional physics realizes.

The word *radion* is thereby justified. It does not mean that one focus is a positive charge and the other a negative charge. It means that the phase unit has a signed orientation prior to electrical publication:

$$\mathcal{A}(\mathfrak{r}_+) = +\hbar_{\mathrm{ret}}, \qquad \mathcal{A}(\mathfrak{r}_-) = -\hbar_{\mathrm{ret}}.$$

Charge appears when Dimensional Mechanics applies the transducer. The sign is not added at the end; it is preserved from the sheet.

This also explains why the 90/120 sector appeared to conceal many answers. Incidence, the dodecaphase extractor, the spectral tail, the Barbero kernel, towers, and carry share a common origin. Precisely because they share an origin, they can be compared. Precisely because they employ distinct operations, they must not be merged. Typing does not diminish the unity of the system: it makes it intelligible.

The most compact conclusion of this work can be written in four lines:

$$\begin{aligned} \text{phase:} &\quad \theta = 1, \\ \text{action:} &\quad \mathcal{A} = \sigma\hbar_{\mathrm{ret}}, \\ \text{memory:} &\quad H_9(n+9) = H_9(n) + (0,1), \\ \text{compatibility:} &\quad 1 = S_E - \Phi_T + \varepsilon_R^{(1)}. \end{aligned}$$

The radion is that fourfold unit. The radian is its exact shadow in the angular chart. The difference lies not in the value of π but in how much structure we are willing to preserve when we say that a revolution has closed.

The circle publishes phase.
The ellipse contains action.
Klein opens depth.
The helix remembers the path.

The radion is the unit that does not force a choice among them.

Chapter 53

Genealogical classification of numbers: surviving sections, invariants and atlas of constants

equation (6.98) defined an HMT number as a compatible section of the inverse survival system, and not as a sequence of digits devoid of genealogy. There remained, however, two questions that should be separated precisely. The first is a question of *representation*: which real values can the 3-adic transducer represent before the particular closure filters are imposed? The second is a question of *selection*: which sections of the raw space are selected by APP, TRIT, and TPK when their compatibilities are required simultaneously? This section resolves the first, formulates the second as an explicit domain restriction, and extracts from the base-one-thousand chart a complete taxonomy of visible values. The foundational tools are p-adic arithmetic, symbolic dynamics, and the positional characterization of the rational numbers [74, 80, 66].

The distinction is structural. A language capable of representing every tape does not, by that fact alone, select the tape of π, e, φ, or α. Conversely, a selection law cannot be evaluated unless the space on which it acts has first been constructed. Representation and selection are two successive theorems, not two names for the same step.

53.1 Hensel's Integral Transducer as a Full Shift Operator

Let p be a prime, let $d \geq 1$, and let $A \in \mathrm{GL}_d(\mathbb{Z})$. For a row vector $x_t \in \mathbb{Z}_p^d$, we define

$$y_t = x_t A, \qquad u_t = y_t \bmod p \in \mathbb{F}_p^d, \qquad x_{t+1} = \frac{y_t - u_t}{p}. \tag{53.1}$$

The block u_t is the visible emission, and x_{t+1} is the memory that remains after that digit is removed.

Theorem 53.1 (Hensel Conjugation with Full Shift)**.** *For every $T \geq 1$, the map*

$$\Psi_T : (\mathbb{Z}/p^T\mathbb{Z})^d \longrightarrow (\mathbb{F}_p^d)^T, \qquad x_0 \longmapsto (u_0, \ldots, u_{T-1}),$$

is bijective. Its inverse is

$$x_0 \equiv \sum_{t=0}^{T-1} p^t u_t A^{-(t+1)} \pmod{p^T}. \tag{53.2}$$

Taking the inverse limit yields a homeomorphism.

$$\Psi : \mathbb{Z}_p^d \xrightarrow{\sim} (\mathbb{F}_p^d)^{\mathbb{N}}.$$

If S_A denotes the update $x_t \mapsto x_{t+1}$ and σ removes the first block from a tape, then

$$\Psi \circ S_A = \sigma \circ \Psi.$$

Proof. From equation (53.1) is obtained

$$x_t = (u_t + p x_{t+1}) A^{-1}.$$

Iteration of this identity gives

$$x_0 = \sum_{t=0}^{T-1} p^t u_t A^{-(t+1)} + p^T x_T A^{-T}.$$

The final summand vanishes modulo p^T, which proves that a word of T blocks determines at most one initial class. Conversely, given any word, $x_T = 0$ is fixed and reconstruction proceeds backward by

means of $x_t = (u_t + px_{t+1})A^{-1}$. The equality $x_t A = u_t + px_{t+1}$ verifies that the resulting class emits the prescribed word. Therefore, Ψ_T is bijective.

The bijections are compatible with reduction modulo p^T. Their inverse limits produce the homeomorphism Ψ; equivalently, the series of equation (53.2) converges in $\mathbb{Z}_p^d$. Applying S_A removes exactly the block u_0, and therefore corresponds to applying σ on the strip. □

In the HMT instance, $p = 3$, $d = 6$, and the published matrix has determinant one. Hence

$$\mathbb{Z}_3^6 \simeq (\mathbb{F}_3^6)^{\mathbb{N}}, \qquad |\mathbb{F}_3^6| = 3^6 = 729. \tag{53.3}$$

Each tape of six-trit blocks has a unique initial 3-adic memory. The matrix determines the coordinatization and the reading dynamics; the universality of the alphabet proves that it excludes no tape.

Corollary 53.2 (Entropy and Haar measure)**.** *The raw Hensel update is conjugate to the full one-sided shift on* 729 *symbols. Its topological entropy is* $6 \log 3$. *The Haar measure on* $\mathbb{Z}_3^6$ *is transported to the uniform product measure on the blocks.*

Proof. There are $729^T = 3^{6T}$ cylinders of length T, and each class modulo 3^T has Haar measure 3^{-6T}. Conjugation by theorem 53.1 transports both properties to the shift. □

The probabilistic conclusion concerns the raw space endowed with Haar measure. It does not assert that the published sections of the constants were sampled according to that measure or that they are algorithmically random.

53.2 Balanced ternary chart of the integers

The raw 3-adic chart and the Archimedean chart do not exhaust the numerical types of the HMT classification. For the integral operator, the empty word is used together with the reduced words

$$\mathscr{W}_{\mathrm{bal}} := \{\varnothing\} \cup \{a_1 \cdots a_m : a_j \in \{-1, 0, +1\},\ a_1 \neq 0\}.$$

We define recursively

$$b(\varnothing) = 0, \qquad b(wa) = 3b(w) + a.$$

Theorem 53.3 (Unique balanced ternary representation)**.** *The evaluation*

$$b : \mathscr{W}_{\mathrm{bal}} \longrightarrow \mathbb{Z}$$

is bijective. Words whose first trit is $+1$ *represent the positive integers, and those beginning with* -1 *represent the negative integers.*

Proof. For $n \neq 0$, the ultimo trit a is the representante unique of n mod 3 in $\{-1, 0, +1\}$. The padre

$$\frac{n - a}{3}$$

has strictly smaller absolute value. The iteration terminates at 0, so that every integer has a finite expansion. The uniqueness of the balanced residue at each step proves the uniqueness of the word. The sign is fixed by the first nonzero trit. □

Remark 53.4. This bijection is an exact arithmetic chart. It does not identify all balanced words with surviving TPK sections, nor does it replace the genealogy, carry, or orientation of an enriched route.

53.3 Exact coverage of the preceding real chart via compatible sections

Let us concatenate the coordinates of the blocks $u_t \in \mathbb{F}_3^6$ to obtain a tape $(a_n)_{n\geq 1} \in \{0,1,2\}^{\mathbb{N}}$. Its ternary evaluation is

$$\mathrm{ev}_3((a_n)) = \sum_{n\geq 1} \frac{a_n}{3^n} \in [0,1].$$

We adopt the canonical convention that excludes a representation with an eventually constant tail of twos only when the same point has another expansion in the chart. The endpoint 1 therefore retains its unique fractional representation $0.222\ldots_3$.

Theorem 53.5 (Archimedean raw coverage)**.** *The map*

$$P_{\mathrm{raw}} : \mathbb{Z}_3^6 \xrightarrow{\Psi} (\mathbb{F}_3^6)^{\mathbb{N}} \xrightarrow{\mathrm{flat}} \{0,1,2\}^{\mathbb{N}} \xrightarrow{\mathrm{ev}_3} [0,1]$$

is continuous and surjective. Therefore,

$$\mathbb{Z}_3^6/{\sim_{P_{\mathrm{raw}}}} \cong [0,1]$$

as the value space, where $x \sim_{P_{\mathrm{raw}}} y$ if their evaluations coincide.

Proof. Every ternary digit belongs to a unique consecutive block of six digits, so flat is a bijection of tape spaces. Every $r \in [0,1]$ has a ternary expansion; the theorem 53.1 provides the unique 3-adic memory that emits its blocks. Continuity is verified by cylinders: fixing T blocks fixes $6T$ digits and confines the value to an interval of diameter 3^{-6T}. Identification of the quotient with the image is the canonical factorization of a surjective map by its fibers. □

A single 3-adic chart is compact and therefore cannot continuously cover an unbounded line. The correct globalization uses a countable family of charts.

Corollary 53.6 (Cover Archimedean countable of $\mathbb{R}$)**.** *Let*

$$\widehat{X}_{\mathrm{raw}} = \mathbb{Z} \times \mathbb{Z}_3^6, \qquad \widehat{P}_{\mathrm{raw}}(m,x) = m + P_{\mathrm{raw}}(x).$$

Then $\widehat{P}_{\mathrm{raw}}$ is surjective onto $\mathbb{R}$. Consequently, every real number admits at least one raw Hensel genealogy and

$$\widehat{X}_{\mathrm{raw}}/{\sim_{\widehat{P}_{\mathrm{raw}}}} \cong \mathbb{R}$$

as a set of values.

Proof. For $r \in \mathbb{R}$, tome $m = \lfloor r \rfloor$ y aplique the theorem 53.5 to $r - m \in [0,1)$. □

This result makes precise the assertion that $\mathbb{R}$ is the Archimedean quotient image of a genealogical structure: the projection forgets 3-adic memory and preserves the evaluated point. The assertion proved here concerns the raw completion *raw*. If $X_{\mathrm{surv}} \subseteq \mathbb{Z}_3^6$ denotes the subspace that also satisfies a concrete family of APP–TRIT–TPK closures, then

$$P_{\mathrm{raw}}(X_{\mathrm{surv}}) = [0,1]$$

is an additional theorem. The transducer guarantees full capacity; survival determines what portion of that capacity the restricted system realizes.

Nor is it deduced here that the native sum and product of TPK descend to the entire global quotient. The usual arithmetic may be transported from $\mathbb{R}$, but proving that it coincides with operations constructed before evaluation requires the corresponding descent theorem.

In this construction, the cylinders are not nonzero infinitesimals of $\mathbb{R}$. Every finite-depth cylinder has positive diameter, and the sequence of diameters converges to zero. Two sections that agree up to depth T and differ thereafter form a *tail perturbation*; their Archimedean difference is bounded by the diameter of the common cylinder. This is the exact formulation of the "cylinders tending to zero" within the ordinary real line.

53.4 Positional chart in base 10^3 and classification of the published sections

The cubic completion $1000 = 729 + 243 + 27 + 1$ makes it natural to group the decimal digits into blocks $b_n \in \{0, \dots, 999\}$. For a tape of blocks, we define

$$\mathrm{ev}_{1000}(b_1, b_2, \dots) = \sum_{n \geq 1} \frac{b_n}{1000^n}.$$

The canonical expansion excludes an eventually constant tail of blocks 999. This convention resolves the duplication of cylinder endpoints.

Theorem 53.7 (Visible taxonomy based on thousand)**.** *Let $x \in [0, 1)$ be given, and let (b_n) be its canonical base-one-thousand expansion. Then:*

1. *x has a terminating expansion if and only if, when written as an irreducible fraction a/q, one has $q \mid 1000^N$ for some N; equivalently, the only primes dividing q are 2 and 5.*
2. *x is rational if and only if (b_n) is eventually periodic.*
3. *x is irrational if and only if (b_n) is not eventually periodic.*

Proof. An expansion terminating at depth N has the form $D/1000^N$; upon reduction, its denominator contains only the primes 2 and 5. Conversely, if $q \mid 1000^N$, then a/q is written with N blocks.

If, after a prefix of length r, a period of length s is repeated whose base-thousand integer is P, the tail is a geometric series:

$$\frac{P}{1000^r\,(1000^s - 1)},$$

and the value is rational. Conversely, when a/q is expanded, the remainders of successive division can assume only q values. A zero remainder produces a terminating expansion; repetition of a remainder produces a period. The third statement is the exact negation of the second. □

The word *fraction* denotes a presentation a/q of a rational number, not an additional numerical species. Nor should periodicity and algebraicity be conflated. A real number is algebraic when it is a zero of some nonzero polynomial over $\mathbb{Q}[X]$; it is transcendental otherwise. This second classification is independent of the base used to write its digits. For example,

$$\varphi^2 - \varphi - 1 = 0$$

makes φ an algebraic irrational, whereas the transcendence of e and π proceeds from classical theorems external to HMT.

53.5 Concatenation, nonadic reduction, and carry

Compatibility between the cubic chart and the digital root is an arithmetic identity, not a visual resemblance. If $u_j \in \{0, \dots, 999\}$ and

$$D_n = 1000 D_{n-1} + u_n, \qquad D_0 = 0,$$

then D_n is the integer obtained by concatenating the first n blocks.

Theorem 53.8 (Nonadic congruence of control of the cubic chart)**.** *For every $n \geq 1$,*

$$D_n \equiv \sum_{j=1}^{n} u_j \pmod 9.$$

Therefore, concatenation in base ten, reduction modulo nine, and digital root are compatible.

Proof. Since $1000 \equiv 1 \pmod 9$, the recurrence gives $D_n \equiv D_{n-1} + u_n \pmod 9$. Iteration from $D_0 = 0$ produces the formula. □

In the dodecaphase closure of α, the blockwise carry equation has the form

$$P_m + E_m - \Phi_m - K_m + c_{m+1} = A_m + 1000 c_m.$$

Reduced modulo nine,

$$A_m \equiv P_m + E_m - \Phi_m - K_m + c_{m+1} - c_m \pmod 9.$$

If the extreme carries vanish, the telescoping sum eliminates the intermediate memory. The global residue of the output then coincides with the residue of the combination of inputs. This law explains why the nonadic record controls decimal concatenation; by itself, it does not select the blocks that must be concatenated.

53.6 Nonadic elevated coordinate and phase return

Reduction modulo nine must not be confused with the identification of the integers that possess the same residue. Let

$$I_9 = \{1, 2, \dots, 9\}.$$

For each positive integer n, we define its *nonadic phase* and its *revolution index* by

$$\rho_9(n) = 1 + ((n-1) \bmod 9), \qquad \kappa_9(n) = \frac{n - \rho_9(n)}{9}. \tag{53.4}$$

The first coordinate is the digital root written in the alphabet I_9, so that the zero residue class is represented by phase 9. The second records how many complete turns have preceded the visible phase.

Theorem 53.9 (Nonadic decomposition with memory)**.** *The map*

$$\Theta_9 : \mathbb{N}_{>0} \longrightarrow \mathbb{N}_0 \times I_9, \qquad n \longmapsto \big(\kappa_9(n), \rho_9(n)\big),$$

is bijective, with inverse

$$\Theta_9^{-1}(q, r) = 9q + r.$$

In these coordinates, the arithmetic successor is conjugated with the dynamics.

$$\mathcal{S}_9(q, r) = \begin{cases} (q, r+1), & 1 \leq r < 9, \\ (q+1, 1), & r = 9. \end{cases} \tag{53.5}$$

In particular, the visible transition $9 \to 1$ *is a phase return accompanied by the irreversible increment* $q \to q+1$*; it is not an equality between the integers* 9 *and* 10*.*

Proof. Euclidean division of $n-1$ by 9 uniquely provides $n-1 = 9q+s$, with $q \in \mathbb{N}_0$ and $0 \le s < 9$. Taking $r = s+1$ yields $n = 9q+r$, with $r \in I_9$, which proves simultaneously existence, uniqueness, and the inverse formula. If $r < 9$, adding one unit only increments r. If $r = 9$, the equality $9q+9+1 = 9(q+1)+1$ gives the second branch of equation (53.5). □

This elevation separates three notions that the isolated figure mixes:

1. the unit of advancement, realized by the successor;
2. the visible phase r, which traverses the nonadic cycle;
3. the accumulated memory q, which distinguishes returns of the same phase.

The projection $(q, r) \mapsto r$ forgets the number of turns and satisfies

$$\rho_9(n+9) = \rho_9(n), \qquad \kappa_9(n+9) = \kappa_9(n) + 1.$$

This is the elementary arithmetic model of the mechanism that, in the enriched TPK state, additionally preserves Hensel quotients, orientation, signature, and boundary datum. A return of the visible word can thus coexist with a distinct genealogy.

Zero does not belong to I_9: it is added as a separate base point before Θ_9 is applied. For negative integers, the positive modulus of the decomposition is preserved and an orientation $\varepsilon \in \{-1, +1\}$ is appended. Consequently, neither the sign nor zero is deduced from the residual phase. This typing prevents a cyclic chart from being converted into a false arithmetic identity.

Theorem 53.10 (Oriented Integer and Genealogical Sign)**.** *The map*

$$\Theta_9^{\pm} : \mathbb{Z} \setminus \{0\} \longrightarrow \{-1, +1\} \times \mathbb{N}_0 \times I_9,$$

$$n \longmapsto \big(\operatorname{sgn}(n), \kappa_9(|n|), \rho_9(|n|)\big)$$

is bijective, with inverse

$$(\varepsilon, q, r) \longmapsto \varepsilon(9q + r).$$

Aritmetical negation acts by

$$(\varepsilon, q, r) \longmapsto (-\varepsilon, q, r) :$$

reverses the orientation and preserves magnitude, phase, and turn memory.

Proof. The theorem 53.9 uniquely provides $|n| = 9q + r$, with $q \in \mathbb{N}_0$ and $r \in I_9$. The sign of a nonzero integer is likewise unique. The inverse formula recomposes n, and replacing n with $-n$ changes only ε. □

Thus, a negative is neither a special residual phase nor an "impossible" quantity. It is the same nonadic magnitude transported with the opposite genealogical orientation. This structure will be prolonged in Dimensional Mechanics by the particle–antiparticle anti-section.

Iterating κ_9 produces a finite tower for every ordinary integer; in the inverse completion, a compatible sequence of quotients may be infinite. The distinction between the two cases anticipates the HMT taxonomy: the visible value is determined by the positional expansion, whereas the genealogical type is determined by the complete evolution of its memories.

53.7 Bifactorial classification of the number by Archimedean value and transport genealogy

The preceding taxonomy classifies the projected value. HMT adds a second axis that ordinary decimal notation discards. For a section $x \in \Omega_\infty^{\mathrm{Arch}}$, we distinguish:

$$\begin{aligned} \text{visible type} &= \text{terminating, periodic or nonperiodic; algebraic or transcendental,} \\ \text{genealogical type} &= \text{carry class, orientation, closure and monodromy of } x. \end{aligned}$$

Two sections may share the same real value and differ on the second axis. A rational real has an eventually periodic visible expansion, but a redundant enriched presentation may preserve nonperiodic internal memory that evaluation forgets. Periodicity of the *value* always refers to its canonical expansion; periodicity of the *genealogy* refers to the TPK enriched state.

This separation provides a concise definition:

$$\boxed{\text{HMT number = compatible section with memory;} \qquad \text{real value = its evaluation class.}} \tag{53.6}$$

Measurement in this chart consists in coupling the section to the reader, fixing a cylinder, and conditioning the compatible prolongations. Increasing precision means narrowing the cylinder, not consuming the genealogy that remains in the fibre.

53.8 Memorization of arbitrary depth in finite emission systems

The visible recurrence of nine phases can coexist with an irrational output only if the complete state preserves information that does not reduce to those nine labels. The following result makes this necessity exact.

Theorem 53.11 (Characterization via finite emitters)**.** *Let Q be a finite set, $F : Q \to Q$ a deterministic transition, $o : Q \to \{0, \dots, 999\}$ an output, and $q_0 \in Q$. The tape $b_n = o(F^{n-1}(q_0))$ is eventually periodic and its evaluation is rational. Conversely, every eventually periodic tape admits a finite deterministic emitter.*

Proof. Among $|Q| + 1$ consecutive states, two are equal. From the first repetition, determinism forces the entire future to repeat, so the tape is eventually periodic. The theorem 53.7 gives rationality. For the converse, it suffices to construct a chain of states for the prefix and a cycle for the period. □

Therefore, a law generating exactly e, π, or any other irrational number cannot consist solely of a deterministic finite-state automaton that repeats without additional memory. It may use an inverse limit, 3-adic memory, increasing depth, or an aperiodic input; all these mechanisms provide unbounded effective state. In HMT, visible-phase return $g_{\mathrm{vis}}(n + 9) = g_{\mathrm{vis}}(n)$ does not force return of the carry, orientation, or boundary. That difference is precisely where an aperiodic expansion may reside.

53.9 Local positional square of e and genealogical differentiation of states with the same visible block

For a decimal word $a_1 \cdots a_n$, we shall call a factorization a *local square* of length ℓ at position i

$$a_i \cdots a_{i+2\ell-1} = vv, \qquad |v| = \ell.$$

This notion belongs to word combinatorics. It does not assert that the infinite tail has a period.

Theorem 53.12 (Finite characterization of the square 1828^2)**.** *Consider the* 243 *catalog words and the fixed reader $w_6 \mapsto w_{30} \mapsto d_1d_2d_3d_4$, with four triads in base one thousand. There is a unique word whose reading contains a square of length four beginning at the second decimal digit. It is*

$$w_e = 201101, \qquad 718281828459 = 7\,\underbrace{1828\,1828}_{(1828)^2}\,459.$$

The repetition is not prolonged to a third copy: the digit that would require that continuation would be 1*, whereas the effective digit is* 4*. In the whole catalogue there exists only one other square of length four,*

$$575157518472 = \underbrace{5751\,5751}_{\textit{local square}}\,8472,$$

which begins at the first digit.

Proof. The exhaustive enumeration evaluates the 243 words using integer arithmetic. The square profile for lengths $1, \dots, 6$ contains

$$240,\ 21,\ 3,\ 2,\ 0,\ 0$$

occurrences, distributed among

$$153,\ 19,\ 3,\ 2,\ 0,\ 0$$

words. The two occurrences of length four are exactly those in the statement. Two independent integer enumerations reproduce the same census, and their complete outputs coincide term by term. □

The word w_e also has the lifting chain

$$201101 \mid 121221 \mid 102011 \mid 012222 \mid 102011.$$

The third and fifth visible blocks coincide. The states that transport them, however, do not coincide. Before both emissions, their reductions modulo 27 are

$$(25, 11, 7, 17, 8, 16), \qquad (7, 8, 4, 23, 26, 7),$$

and the subsequent Hensel quotients are

$$(6, 13, 9, 2, 13, 1), \qquad (15, 0, 3, 19, 23, 25).$$

Therefore, the return 102011 is a return of the visible projection, not of the enriched state. This is the exact formulation of the local repetition that is broken: the quotient preserves a genealogy that the residual block does not record.

Consequently, the beginning

$$e = 2.718281828459\ldots$$

does not exhibit a "semiperiodicity," but rather a local word-square with visible-phase return and memory advance without state periodicity. The result is exhaustive within the fixed reader; its content is combinatorial and dynamical, not an inference of irrationality from twelve digits.

53.10 2 finite measures of the $468 \to 243$ emitter: uniform frequency and structural priority

The catalogue publico distinguishes 468 emissions y 104 976 semillas. Be

$$q : E_{468} \longrightarrow W_{243}$$

the map that forgets the emission signature and preserves the six-trit word. There are two natural measures because there are two finite counting universes:

$$\mu_{468}(w) = \frac{|q^{-1}(w)|}{468}, \qquad \mu_{\text{seed}}(w) = \frac{\sum_{e \in q^{-1}(w)} m(e)}{104\,976},$$

where $m(e)$ is the multiplicity of semillas of the emision e.

Theorem 53.13 (Canonical measures relative to the counted universe)**.** *The uniform measure on E_{468} is the unique probability invariant under all permutations of its emissions; its image measure is μ_{468}. The uniform measure on the* 104 976 *seeds induces μ_{seed}. Their exact histograms are*

$\lvert q^{-1}(w)\rvert$	1	2	3	4	5	6
$\#\{w\}$	132	66	18	3	6	18

and

seed weight	144	288	432	576	864	1008	1440	1944	2088
$\#\{w\}$	120	54	18	9	15	6	3	12	6.

For the three words distinguished by the reader, one obtains

	μ_{468}	μ_{seed}
π	5/468	7/729
e	1/468	1/729
φ	1/468	1/729.

Proof. Invariance under the symmetric group assigns equal masses to all points of E_{468}; normalization fixes them at $1/468$. The formula for the pushforward measure follows by summing over each fiber. The same argument applied to the seed set yields the second measure. The histograms are obtained by exhaustive enumeration of the finite catalogue of 468 emissions and 243 words; their sums verify

$$132 + 66 + 18 + 3 + 6 + 18 = 243,$$

$$132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468$$

and the sum total of the weights is 104 976. □

The nonuniformity of the pushforward measure is a finite theorem: distinct words have different numbers of realizations. Conversely, no normalizable uniform probability exists on all of $\mathbb{R}$, and these measures do not intrinsically order all the reals. The word for π belongs to one of the six fibers of size five and weight 1008; it is not the unique maximum of either histogram. Its proved singularity arises from the composite structure of its five regions, from its oriented partition $3 + 2$, and from its subsequent compatibilities, not from declaring an absolute probability on the continuum a priori.

53.11 Selector orbital finite preceding the Archimedean reader

The preceding measures reveal multiplicities, but an isolated six-trit word still depends on an orientation. The catalog has an internal symmetry that allows unoriented classes to be selected first and a representative to be fixed afterward. Write a word as $abc \mid def$ and define

$$r(abc \mid def) = cab \mid efd, \qquad s(abc \mid def) = def \mid abc.$$

These permutations also act on the six coordinates of the datum U_6. In the complete catalog, they satisfy

$$r^3 = 1, \qquad s^2 = 1, \qquad srs = r^{-1},$$

and hence they generate an action denoted by D_3^{cat}. The action uses the fixed TRIT labeling

$$\Psi(U_6) = (-U_6 \bmod 3)$$

and the published partition of the six coordinates into two triads.

For a word w, let $f(w) = |q^{-1}(w)|$ be the number of emissions, $m(w)$ its seed mass, and

$$\nu(w) = (n_0(w), n_1(w), n_2(w))$$

its ordered census of trits. Also let $d(e)$ be the additive diagonal class of an emission and $o(e) \in \{\mathrm{ES}, \mathrm{NO}\}$ its published orientation.

Theorem 53.14 (Internal orbital selection of the three words)**.** *The action D_3^{cat} preserves the set of* 468 *emissions, the projection $q : E_{468} \to W_{243}$, and the seed multiplicity. Its action on W_{243} has exactly* 43 *orbits:* 38 *of size six and five of size three. The following uniqueness statements hold within that quotient.*

1. *There exists a unique orbit $\mathcal{O}_\pi$ of size six such that, for every $w \in \mathcal{O}_\pi$,*

$$f(w) = 5, \qquad m(w) = 1008, \qquad \{m(e) : e \in q^{-1}(w)\} = \{432, 144, 144, 144, 144\}.$$

   It is

$$\mathcal{O}_\pi = \{001112, 010211, 100121, 112001, 121100, 211010\},$$

   and all its words have census $\nu = (2, 3, 1)$.

2. *Consider the orbits of size six whose words have a single emission, of multiplicity* 144*, and whose joint diagonal support is $\{0, 3, 6\}$. There are exactly six. Among them there is a unique orbit with the same census $(2, 3, 1)$ of $\mathcal{O}_\pi$:*

$$\mathcal{O}_e = \{011120, 012110, 101201, 110012, 120011, 201101\}.$$

3. *In the same sector there exists a unique balanced orbit, that is, with $\nu = (2, 2, 2)$:*

$$\mathcal{O}_\varphi = \{002112, 020211, 112002, 121200, 200121, 211020\}.$$

   Finally, the guiding clause

$$\exists e \in q^{-1}(w) : \qquad d(e) = 6, \qquad o(e) = \mathrm{ES} \tag{53.7}$$

select a single representative from each orbit and produces, respectively,

$$\boxed{010211, \qquad 201101, \qquad 121200.}$$

Proof. The group relations are verified directly as identities of permutations of six coordinates. Enumeration of the 468 rows verifies that r and s send every U_6 to another row of the catalogue with the same multiplicity, and that ternary projection commutes with both actions. Therefore, the orbits can be computed on the 243 words without losing the fibers.

The exhaustive partition gives the size histogram.

$$38 \cdot 6 + 5 \cdot 3 = 243.$$

Filtering those 43 orbits by the first predicate leaves exactly one; filtering by singleton fiber, minimum multiplicity, and diagonal support $\{0, 3, 6\}$ leaves six. The census $(2, 3, 1)$ leaves a single orbit in the first case, and the balanced census $(2, 2, 2)$ leaves a single orbit in the second. Finally, the search for a lift with $d = 6$ and orientation ES leaves one word in each of the three orbits. All filtrations are performed on structural descriptors of the catalogue before consulting the decimal reader. □

The predicates of the theorem may be combined into a selection invariant. For an orbit $\mathcal{O}$, we define its *orbital profile*

$$\Sigma_{\mathrm{cat}}(O) = \big(|O|,\ f,\ m,\ \mathcal{M}_{\mathrm{fib}},\ \mathcal{D},\ \nu\big),$$

where f is the fibre size, m the seed mass, $\mathcal{M}_{\mathrm{fib}}$ the multiset of multiplicities of their lifts, $\mathcal{D}$ the diagonal support when used, and ν the trit census. The three distinguished classes occupy different positions in this invariant space:

O	$\lvert O\rvert$	f	m	$\mathcal{M}_{\mathrm{fib}}$	$(\mathcal{D}, \nu)$
O_π	6	5	1008	$\{432, 144, 144, 144, 144\}$	$(-, (2, 3, 1))$
O_e	6	1	144	$\{144\}$	$(\{0, 3, 6\}, (2, 3, 1))$
O_φ	6	1	144	$\{144\}$	$(\{0, 3, 6\}, (2, 2, 2))$.

The demonstrated singularity does not mean that these numbers receive an absolute probability over $\mathbb{R}$. It means that, within the finite universe produced by the emitter and before evaluating digits, their classes are separable by internal invariants and that the class of π has a regional fiber of a different type. This is the mathematical formulation of a structural priority in HMT.

The negative control is informative. If the diagonal support $\{0, 3, 6\}$ is omitted, there exist four balanced singleton orbits and O_φ ceases to be unique. Before the gauge is fixed, each orbit admits six possible orientations. Once it is fixed, the word of π preserves three positive lifts of U_6, rather than a single region. Thus, the theorem selects the oriented word while respecting the geometric multiplicity of the five regions.

The non-uniformity is reinforced when passing to orbits:

	emissions	seeds
O_π	5/78	14/243
O_e	1/78	2/243
O_φ	1/78	2/243.

These quotients remain masses in the finite catalog, not probabilities assigned to the evaluated real numbers.

The causal advance is precise. The three words cease to be named inputs: they are recovered from the symmetry and finite invariants of APP–TPK, relative to the action D_3^{cat}, the TRIT labeling, the distribution $3+3$, and the orienting gauge equation (53.7). The proof uses neither L_0, L_1, decimal digits, α, CODATA, nor the dodecaphasic state. The absolute uniqueness of that action among all possible automorphisms, the deduction of the gauge rather than its declaration, the canonicity of the reader L_0/L_1 independently of the target value, and the coinductive emission of the infinite expansions remain subsequent claims.

53.12 Position of π, e, φ and α in the classification

The theorem 53.1 explains why any decimal prefix can be re-emitted from a 3-adic state: the raw capacity is universal. Therefore, the mere back-and-forth operation on a tape selects no constant. The distinctive content of the published construction lies in the objects preceding and following that re-emission:

1. the finite fibers and the five distinct regions of π;
2. the typed Archimedean reader that associates words, cylinders, and blocks;
3. the dodecafase signed and the closure reversible

$$U_{12}^{\mathrm{sgn}} \longleftrightarrow K \longleftrightarrow (D_3K, D_4K, Q) \longleftrightarrow \alpha_{\mathrm{HMT}};$$

4. the contrast criteria that distinguish an equivalence of charts from a selector preceding evaluation.

The twelve published decimal triples of π, e, φ are exact finite coordinates of the enriched reader. The printed string of thirty-six digits is the rational dodecaphasic projection $\alpha_{12} = \pi_{12}(\alpha_{\mathrm{HMT}})$:

$$\frac{729735256928380099728510547238066 3}{10^{36}}.$$

The exact identity in this window is

$$\alpha_{12} := \pi_{12}(\alpha_{\mathrm{HMT}}) = \frac{729735256928380099728510547238066 3}{10^{36}}.$$

The compatible recurrence at arbitrary depth constructs the infinite section

$$\alpha_{\mathrm{HMT}} := \varprojlim_{N} A^{(N)},$$

and each rational window is an exact projection of that same object. The kernel $E_{21/22}$ constructs, in another chart, the finite simple positive root α_E; the graded transport of the TPK extends that kernel into a compatible family whose limit root is α_B. If $\alpha_A := \varprojlim_N \rho_N(\alpha_{12})$, the truncation squares of the two families commute and determine the same surviving cylinder at each depth. The exact two-way theorem therefore establishes

$$\boxed{\alpha_A = \alpha_B = \alpha_{\mathrm{HMT}}}.$$

The equality of $\mathrm{Tr}_9(\alpha_E^{-1})$ and $\mathrm{Tr}_9(\alpha_{12}^{-1})$ is the first finite certificate of that identity; it neither defines it, limits its scope nor selects the extensions. The typed projection keeps the coinductive section, its rational window and the analytic root distinct without separating their identity in the limit.

The square root of -1 belongs to another chart. In the section 3.5, one constructs

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1), \qquad \iota^2 = -1,$$

and the Witt matrix is proved to realize that quadratic structure. This is an exact finite algebraic extension, not a visible category of real expansions.

53.13 Previous chart, inductive survival, and construction of the global boundary

The following controls are falsifiers of the preceding finite results:

1. that two states distinct modulo 3^T emit the same word of T blocks, or that there is a word without a preimage;
2. failure of the explicit inverse of equation (53.2);
3. that a rational number should appear whose canonical expansion is not eventually periodic, or an irrational number whose expansion is periodic;
4. that a finite deterministic emitter produces a tape that is not eventually periodic;
5. that the catalog censuses do not sum to 468, 243, and 104 976, or that the measures of the three words do not agree with the enumeration.

The raw Hensel chart

$$\mathbb{Z}_3^6 \longrightarrow [0, 1]$$

is continuous and surjective; upon adjoining the integer part, its raw atlas covers $\mathbb{R}$. This exact result does not imply that every raw fiber contains a representative that survives the additional filters. The published finite windows and the named readers of π, e, φ are verified within HMT, relative to their seeds, regions, orientations, and enriched rules.

The positional representation of a number records a sequence of symbols, but does not necessarily preserve the state determining its continuation. Triadic Modular Holography distinguishes the two levels. Nonadic phase indicates the position of a reading within the nine-step cycle; the enriched state additionally

preserves orientation, Euclidean quotient, carry transport, and boundary condition. Return of the phase does not therefore imply return of the state.

Let S be a space of enriched states, let $T : S \to S$ be a transition, and let

$$g : S \longrightarrow \mathbb{Z}/9\mathbb{Z}$$

the phase, with $g(Ts) = g(s) + 1$. For a basis $b \geq 2$, a reader $d : S \to \{0, \dots, b-1\}$ produces

$$x(s) = \sum_{n \geq 0} \frac{d(T^n s)}{b^{n+1}}.$$

When a value admits the two usual positional expansions, we adopt the canonical representation that does not end in a constant tail $b - 1$. This convention distinguishes an ambiguity of notation from multiplicity proper to HMT, which arises from distinct enriched sections having the same evaluation. The nonadic return operator is the restriction

$$M = T^9 : g^{-1}(r) \longrightarrow g^{-1}(r).$$

The difference between $T^9 s = s$ and $g(T^9 s) = g(s)$ formalizes the distinction between periodicity and monodromy.

The transition of the enriched state of TPK may be given by a relation rather than a function. In that case, the preceding notation applies to the shift on the space of infinite histories or to a deterministic realization whose state has been enlarged with the information necessary to fix the continuation; it is not presupposed that a visible block has a unique successor.

Definition 53.15 (HMT Number and Arithmetic Presentation)**.** An HMT number is solely a compatible section $\boldsymbol{x}$ of the inverse system of surviving states. Given a positional reader L whose domain contains $\boldsymbol{x}$, the pair $(\boldsymbol{x}, L)$ is a presentation or measurement of the number. The Archimedean value published by that presentation is the singleton intersection of the associated cylinders when their diameters tend to zero. Two sections with the same value are not identified before applying the Archimedean quotient.

Definition 53.16 (Visible and enriched periodicities)**.** The output is eventually periodic when there exist $N \geq 0$ and $p > 0$ such that

$$d(T^{n+p} s) = d(T^n s), \qquad n \geq N.$$

The state is eventually periodic when

$$T^{N+p} s = T^N s.$$

The second condition implies the first. The converse requires that the reader separate the tails of the orbit under consideration, that is, that

$$\big(d(T^{n+k} s)\big)_{k \geq 0} = \big(d(T^{m+k} s)\big)_{k \geq 0} \quad \Longrightarrow \quad T^n s = T^m s.$$

Theorem 53.17 (Classification by phase return)**.** *Let $s \in S$.*

1. *If the orbit of s reaches a state whose future output is constantly null, then $x(s)$ has a terminating expansion and is rational.*
2. *If the state is eventually periodic, then the output is eventually periodic and $x(s) \in \mathbb{Q}$.*
3. *If the reader separates the tails of the orbit and the orbit of s is not eventually periodic, then the output is not eventually periodic and $x(s) \notin \mathbb{Q}$.*
4. *Nonperiodicity does not distinguish between algebraic irrationality and transcendence. This distinction requires a polynomial certificate or an additional transcendence theorem.*

Proof. The first two parts are the classical characterization of the rationals through terminating or eventually periodic positional expansions. In the third, eventual periodicity of the output would produce two identical tails. The separation hypothesis would then identify the corresponding states, contradicting the nonperiodicity of the orbit. The fourth part follows from the existence of algebraic and transcendental irrationals with expansions that are not eventually periodic. □

Corollary 53.18 (Nonadic Monodromy and Aperiodicity)**.** *Suppose that the phase returns every nine steps, that the return orbit*

$$\{M^k s : k \geq 0\}, \qquad M = T^9,$$

is not eventually periodic and the reader separates the tails. Then the output is not eventually periodic, and the number produced is irrational. In this case, phase periodicity coexists with an aperiodic evolution of the enriched state.

The hypothesis cannot be replaced by a single inequality $M(s) \neq s$, nor by a finite number of distinct returns. Relative to the initial state, the declared transition and reader, the current window certifies nine phase returns with change of state; it proves finite monodromy, but not by itself the aperiodicity of the entire orbit.

53.14 Classification of constants by genealogical antecedent, generating operation, and codomain

The HMT classification separates properties that the ordinary representation brings together under a single value:

Class	Structural condition
Definitive	The output reaches a null tail.
Rational periodic	The output is eventually periodic.
Rational monodromy	The output is periodic, but the enriched state traverses a nontrivial fibre that the visible reader forgets.
Irracional coinductivo	The output is not eventually periodic and the cylinders determine a unique value.
Recurrente local	A finite word contains returns or squares, without this imposing periodicity on the suffix.
Algebraic	The value satisfies a nonzero polynomial with integral coefficients; the route must be accompanied by that certificate.
Transcendent	The value satisfies no nonzero integer polynomial; the classification requires a transcendence theorem.
Distinguido HMT	The section satisfies a frozen structural predicate of fiber, boundary, survival, or symmetry; it is a class transverse to the preceding ones.

Thus, HMT does not alter the correct arithmetic definitions of rationality, algebraicity, and transcendence. It refines them by adding the genealogy of the state, phase monodromy, and the multiplicity of the fibers that Archimedean evaluation had forgotten.

In particular, φ is algebraic because it satisfies $X^2 - X - 1 = 0$, whereas the transcendence of e and π follows from classical external theorems. These properties are compatible with the three sections being distinguished by HMT. The dodecaphasic closure later constructs the exact rational projection $\alpha_{12} = \pi_{12}(\alpha_{\mathrm{HMT}})$, and the compatible prolongation constructs the coinductive section α_{HMT}, in section 30.14; neither is used in this taxonomy as an input or published here as a second residence. The kernel $E_{21/22}$ produces a distinct analytic root, whose agreement with the integer route is proved in the codomain of Tr_9. Section, rational projection, analytic root, reader and value thus retain their own types.

Finally, an "infinitesimal direction" requires its own typing. For a finite reader r_m, two distinct sections in the same fiber of r_m define a difference that is invisible at depth m, although active in a subsequent prolongation. They do not thereby constitute a nonzero infinitesimal real. If two sections coincide under every faithful truncation of the inverse system, they are the same section. A genuine arithmetic infinitesimal would require the construction of an ordered or valued non-Archimedean extension and proof that the HMT operations descend to it.

53.15 Number as a compatible section: arithmetic classification, Archimedean evaluation, and genealogy

The numerical classification of HMT preserves the ordinary arithmetic species and adds a coordinate that positional evaluation discards: the genealogy of the section that publishes the value. The result is a joint classification. The first axis answers which number is obtained; the second records by which compatible sequence of states, returns, carries, and closures it is obtained. This distinction makes it possible to formulate within a single language integers, rational expansions, irrationals, the imaginary unit, regularizations, and spectral constants without identifying objects that only share a scalar.

53.15.1 Archimedean evaluation separates the section from its value

Let

$$X_\infty^{\mathrm{surv}} = \varprojlim_n X_n^{\mathrm{surv}}$$

an inverse system of surviving states with its restriction maps. A compatible section is a family $\boldsymbol{x} = (x_n)_{n\geq 0}$ satisfying $\rho_{n+1,n}(x_{n+1}) = x_n$ at every depth. A positional reader L assigns a cylinder $I_n(\boldsymbol{x}; L)$ to each prefix. When the cylinders are compatible, their diameters converge to zero and the convention for endpoints is fixed canonically, one defines

$$\mathrm{Val}_L(\boldsymbol{x}) := \text{the point of } \bigcap_{n\geq 0} \overline{I_n(\boldsymbol{x}; L)}. \tag{53.8}$$

Definition 53.19 (Number, presentation and value in HMT)**.** An *HMT number* is a compatible section $\boldsymbol{x} \in X_\infty^{\mathrm{surv}}$. The pair $(\boldsymbol{x}, L)$ is a *presentation* of the number, and $\mathrm{Val}_L(\boldsymbol{x})$ is the value published by that presentation. The fibre

$$\mathrm{Val}_L^{-1}(r)$$

preserves the genealogies that the Archimedean quotient represents by the same value r.

The capacity of the chart and selection of a section are distinct typed results. The theorem 53.5 constructs a continuous surjective map

$$P_{\mathrm{raw}} : \mathbb{Z}_3^6 \longrightarrow [0, 1],$$

and the corollary 53.6 globalizes it to

$$\widehat{P}_{\mathrm{raw}} : \mathbb{Z} \times \mathbb{Z}_3^6 \longrightarrow \mathbb{R}.$$

These maps prove the complete representational capacity of the raw chart. The survival predicate and the readers that distinguish π, e, φ select, within that capacity, the corresponding structural genealogies.

53.15.2 Termination and periodicity determine rationality

For the expansion canonica in basis mil

$$x = \sum_{n \geq 1} \frac{b_n}{1000^n}, \qquad b_n \in \{0, \dots, 999\},$$

one adopts the convention that eliminates the duplication produced by an eventually constant tail of blocks 999. The theorem 53.7 then provides the following exact classification.

Theorem 53.20 (Visible classification in the cubic decimal chart)**.** *Let $x = a/q \in [0, 1)$ be an irreducible fraction when x is rational. Then:*

1. *the expansion terminates if and only if $q \mid 1000^N$ for some N, an equivalence restricting the prime factors of q to 2 and 5;*
2. *the expansion is eventually periodic if and only if $x \in \mathbb{Q}$;*
3. *the expansion is not eventually periodic if and only if $x \in \mathbb{R} \setminus \mathbb{Q}$.*

Algebraicity and transcendence constitute a second axis: a real number is algebraic if it is a zero of a nonzero polynomial in $\mathbb{Q}[X]$, and is transcendental if it is a zero of none.

Proof. An expansion of length N has a denominator dividing 1000^N, and the converse is obtained by expressing a/q with that denominator. For a rational number, successive division has a finite set of remainders: a zero remainder produces termination, and repetition of a remainder produces periodicity. A periodic tail is, conversely, a rational geometric series. The polynomial classification is independent of the positional base. □

The enriched dynamics also distinguishes output periodicity from state periodicity. If T transports the state and d publishes a block, the conditions

$$d(T^{n+p}s) = d(T^n s), \qquad T^{n+p}s = T^n s$$

define, respectively, visible-phase return and state return. The second implies the first. The inverse implication requires the reader to separate the tails of the orbit, according to theorem 53.17. Therefore, a periodic nonadic phase nonadic periodic can transport a nonperiodic memory periodic.

Table 53.1: Joint taxonomy of the published value and the enriched state.

Class	Condition on the value	Genealogical condition or required proof
Integer	Element of $\mathbb{Z}$.	Finite and unique balanced ternary word; sign and magnitude remain typed.
Decimal finito	A rational number whose reduced denominator divides some power of 1000.	The output reaches a null tail.
Rational periodic	Expansion that is eventually periodic.	An eventually periodic state suffices; a periodic output may preserve an additional internal fibre.
Rational monodromic	Same rational value as the previous row.	The output returns while the enriched state traverses a fiber that the visible reader forgets.

Class	Condition on the value	Genealogical condition or required proof
Irracional coinductivo	Expansion that is not eventually periodic.	An infinite family of compatible cylinders fixes a unique value; global aperiodicity requires a proof over the entire orbit, not a finite window.
Algebraic	Root of a polynomial no zero of $\mathbb{Q}[X]$.	The polynomial is appended and its vanishing is verified.
Transcendent	Real external to the closure algebraic of $\mathbb{Q}$.	A transcendence theorem is appended; decimal nonperiodicity by itself only proves irrationality.
Distinguido HMT	Any of the preceding arithmetic classes.	The section satisfies a frozen fiber, boundary, survival, symmetry, or closure predicate.

This table locates on different axes two questions that must be answered separately. Ordinary classification determines the properties of the value; the HMT reader determines the structural chain generating the section. For example,

$$\varphi^2 - \varphi - 1 = 0$$

classifies φ as an algebraic irrational, whereas the transcendence of e and π belongs to external classical theorems. All three may simultaneously be sections distinguished by different HMT predicates. Their printed truncations of one thousand or ten thousand digits are finite rationals that demonstrate the calculated prefixes of a prolongable output; they do not replace the infinite object they represent.

The dodecaphase closure defines, in turn, the finite rational projection

$$\alpha_{12} = \pi_{12}(\alpha_{\mathrm{HMT}}) = \frac{729735256928380099728510547238 0663}{10^{36}}. \tag{53.9}$$

This projection is exact and rational. The object $\alpha_{\mathrm{HMT}} = \varprojlim_N A^{(N)}$ is the infinite section constructed by the compatible recurrence; its rational truncations do not replace it. The structural derivation of the limit and the classification of each finite projection are complementary statements.

53.15.3 The balanced ternary chart represents the integers bijectively

Let

$$\mathscr{W}_{\mathrm{bal}} = \{\varnothing\} \cup \{a_1 \cdots a_m : a_j \in \{-1, 0, +1\},\ a_1 \neq 0\}$$

and define

$$b(\varnothing) = 0, \qquad b(wa) = 3b(w) + a.$$

The theorem 53.3 demonstrates that

$$b : \mathscr{W}_{\mathrm{bal}} \xrightarrow{\sim} \mathbb{Z} \tag{53.10}$$

is a bijection. The final trit is the unique representative of the class modulo three, and the quotient $(n - a)/3$ strictly reduces the absolute value until zero is reached. The first nonzero trit fixes the sign. This chart is a representation that is exact and arithmetic; the membership of a word in a surviving section additionally incorporates the conditions of route, orientation, and boundary.

For $n > 0$, the coordinates

$$\rho_9(n) = 1 + ((n-1) \bmod 9), \qquad \kappa_9(n) = \frac{n - \rho_9(n)}{9}$$

realize the decomposition $n = 9\kappa_9(n) + \rho_9(n)$. The phase $\rho_9 \in \{1, \dots, 9\}$ publishes the zero residue class as 9, and κ_9 preserves the number of turns. With the sign $\varepsilon \in \{-1, +1\}$,

$$n \longleftrightarrow \big(\varepsilon, \kappa_9(|n|), \rho_9(|n|)\big)$$

is bijective on $\mathbb{Z} \setminus \{0\}$; zero remains the base point. The visible-phase return $9 \to 1$ is thus a phase transition with memory advance, not an identification of consecutive integers.

53.15.4 Finite complex structure induced by the Witt endomorphism $J^2 = -I_6$

The ordinary imaginary unit is the element $i \in \mathbb{C}$ satisfying $i^2 = -1$; it is algebraic of degree two over $\mathbb{R}$ and over $\mathbb{Q}$. The HMT realization belongs to a different finite domain. If C_P is the Paley conference matrix and $J = A_W$ its reduction modulo three in $\mathrm{End}_{\mathbb{F}_3}(\mathbb{F}_3^6)$, then

$$\boxed{J^2 = 2I_6 = -I_6.} \tag{53.11}$$

Since $X^2 + 1$ is irreducible over $\mathbb{F}_3[X]$, the extension

$$\mathbb{F}_9 = \mathbb{F}_3[\iota]/(\iota^2 + 1)$$

separates the eigenspaces of J associated with $\pm\iota$. The result, proved in chapter 32, endows the ternary module with a finite complex structure and organizes two conjugate orientations. Its force is *internally exact, proved using the declared Paley–Witt matrix*; the comparison with $\mathbb{C}$ preserves the distinction between the two fields.

53.15.5 6 operators realize the rational $-1/12$ on distinct domains

Zeta regularization assigns to the sequence of positive integers the finite part obtained by meromorphic continuation:

$$\sum_{n\geq 1}^{[\zeta]} n := \zeta(-1) = -\frac{1}{12}. \tag{53.12}$$

The superscript $[\zeta]$ forms part of the type of the expression. The ordinary partial sum satisfies $\sum_{n=1}^{N} n = N(N+1)/2$ and grows without bound; therefore (53.12) is a regularized value, not the limit of those partial sums.

HMT additionally contains six exact algebraic realizations of the same rational number. Scalar coincidence permits their comparison; each object retains its domain, map, and normalization.

Table 53.2: Typed realizations of $-1/12$.

Domain	Map and evaluation	Conclusion and normalization
Marked hexad–octad incidence	$F_6 \hookrightarrow F_8$, with $\lvert F_6\rvert = 90$, $\lvert F_8\rvert = 120$, and $(90 - 120)/(24\binom{6}{2}) = -1/12$.	Exact combinatorial defect for the declared normalization.

Domain	Map and evaluation	Conclusion and normalization
Dodecaphase periodicities	To step of 30° produce $$\frac{30}{120} - \frac{30}{90} = \frac{1}{4} - \frac{1}{3}, \\ = -\frac{1}{12}.$$	Exact oriented difference of periods.
Gram Operator	For $K_{60,81} = \operatorname{diag}(0, 12)$, the pseudoinverse on the nonzero range is $1/12$; the negative orientation publishes $-1/12$.	Exact operator identity on the non-vanishing component.
Second scale difference	The calculo produce $C(L) = 8/81$ and the extractor declarado $R(L) = -(27/32)C(L) = -1/12$.	Exact after fixing the normalization factor $-27/32$; that factor is an explicit primitive.
Modular quotient eta	$$\frac{\eta(\tau)}{\eta(3\tau)} = q^{-1/12} \\ \times \prod_{n\geq 1}(1 + q^n + q^{2n})^{-1}.$$	exponent modular exact, procedente of $1/24 - 3/24$.
Projectors of the twelve-phase cycle	For $D_r = I - S^r$, $$\tau(\Pi_{\ker D_3} - \Pi_{\ker D_4}) = \frac{3-4}{12}, \\ = -\frac{1}{12}.$$	Exact transverse defect between steps three and four.

The six realizations are proved in chapter 33. Zeta regularization, the incidence defect, the pseudoinverse, the scale extractor, the eta quotient, and the trace of projectors are different maps. An identification of their domains would require an additional intertwiner. Likewise, the term $+1/12$ of the Glaisher–Kinkelin formula belongs to the regularization of the hyperfactorial and retains that distinct type.

53.15.6 Multiplicative decomposition of primes as irreducible modes of the TPK

An ordinary prime is an integer $p > 1$ whose factorizations $p = ab$, with $a, b \in \mathbb{N}$, necessarily contain a unit. The reduced coproduct

$$\Delta_{\times}^{\mathrm{red}} e_n = \sum_{\substack{ab=n \\ a,b>1}} e_a \otimes e_b$$

turns that definition into a spectral selector. According to theorem 27.21, the positive operator

$$N_{\times}^{\mathrm{red}} = (\overline{\Delta_{\times}^{\mathrm{red}}})^{*}\overline{\Delta_{\times}^{\mathrm{red}}}$$

satisfies $N_{\times}^{\mathrm{red}} e_n = d_{\mathrm{red}}(n) e_n$, and its kernel, after removing e_1, selects exactly the primes. The multiplicative intertwiner transports the same selector to the native product of APP. This selector is the spectral projection onto $\ker N_{\times}^{\mathrm{red}} \ominus \mathbb{C}e_1$, which coincides exactly with the subspace generated by the primes and does not depend on recognizing decimal lists.

Unique factorization additionally provides the Fock realization

$$U_F : \Gamma_s(\ell^2(\mathbb{P})) \xrightarrow{\sim} \ell^2(\mathbb{N}_{\geq 1}), \qquad U_F \, d\Gamma(H_{\mathbb{P}}) \, U_F^* = \log Q,$$

where $H_{\mathbb{P}} e_p = (\log p) e_p$. Each prime is a multiplicatively irreducible mode; a composite integer is a finite occupation of those modes.

For n coprime to 10, its decimal clock is the multiplicative order

$$\tau_n = \operatorname{ord}_n(10).$$

If p is primo y $p \notin \{2, 3, 5\}$, the equation (53.13) demonstrates

$$\tau_p \mid p - 1, \qquad M_p := \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \qquad M_p \equiv 0 \pmod 9. \tag{53.13}$$

For a compound,

$$\operatorname{ord}_n(10) = \operatorname{lcm}_{p^a \| n} \operatorname{ord}_{p^a}(10).$$

Therefore, the composite clock synchronizes prime-power clocks, whereas a prime supplies an irreducible component. The nonadic congruence (53.13) describes phase closure; the primality selector resides in the reduced coproduct. Separation of these two functions prevents selective capacity from being attributed to the decimal period by itself.

53.15.7 Each constant is associated with its extraction operator

The cycles of order 3^m, the limit of their heat traces, and the theta–Mellin transform construct the functional

$$\mathcal{Z}_3(s) = \frac{\displaystyle\int_0^{\infty} \Theta_+(x) x^{s/2-1} \, \mathrm{d}x}{\pi^{-s/2}\Gamma(s/2)} = \zeta(s), \qquad \Re s > 1. \tag{53.14}$$

Theta inversion meromorphically extends the functional. The residual partition

$$\zeta(s) = Z_0(s) + Z_1(s) + Z_2(s)$$

separates aggregate, scale, and orientation. A single operatorial source admits several extraction maps—finite part, Laurent coefficient, evaluation, regularized derivative, or character—; coincidence of the source does not identify the resulting constants.

Table 53.3: Definition, extraction map, and mathematical scope of the complementary numerical constants.

Object	Standard definition	Map and mathematical reference	Conclusion and condition
Euler–Mascheroni γ	For $H_N = \sum_{n \leq N} n^{-1}$, $\gamma = \lim_{N\to\infty}(H_N - \log N)$, equivalente to the parte finite of ζ in $s = 1$.	If $C_r = \mathrm{FP}_{s=1} Z_r(s)$, then $\gamma = C_0 + C_1 + C_2$, by (27.9).	Exact identity of the residual functional demonstrated with the declared operator and normalization.
Stieltjes constants γ_n	If $\zeta(1+z) - z^{-1} = \sum_{n\geq 0} a_n z^n$, they are defined by $\gamma_n = (-1)^n n! a_n$.	The residual coefficients $c_{r,n}$ satisfy $\gamma_n = (-1)^n n! \sum_r c_{r,n}$ and $L^{(n)}(1, \chi_{-3})/n! = c_{1,n} - c_{2,n}$: this is the coefficientwise decomposition of (41.11) by the three residue classes of (27.9).	Exact decomposition; numerical extractions verify the coefficients, not their definition.
Apéry $\zeta(3)$	Value of the Riemann zeta function at $s = 3$.	$\mathcal{Z}_3(3) = \zeta(3)$, with prefixes bounded by rational intervals; its spectral residence within the graded tower is fixed at equations (43.1) y (43.2).	Exact evaluation of the functional; its irrationality follows from the classic theorem of Apery [3].

Object	Standard definition	Map and mathematical reference	Conclusion and condition
Glaisher–Kinkelin A_{GK}	Regularized Growth Constant of the Hyperfactorial.	$A_{\mathrm{GK}} = \exp(1/12 - \mathcal{Z}_3'(-1))$, by equations (43.1) y (43.2).	Exact regularized derivative; $1/12$ belongs to this hyperfactorial normalization.
Euler–Kronecker constant $\gamma_{F_{-3}}$	Logarithmic finite part of the Dedekind zeta function of the field $F_{-3} = \mathbb{Q}(\sqrt{-3})$.	The residual character χ_{-3} selects the field and produces $\gamma_{F_{-3}} = \gamma + L'(1, \chi_{-3})/L(1, \chi_{-3})$, through (42.2).	Exact identity; the choice of field comes from the character modulo three.
Meissel–Mertens B_1	For $S_P(x) = \sum_{p \le x} p^{-1}$, $B_1 = \lim_{x\to\infty}(S_P(x) - \log\log x)$.	It is the finite part of the truncated prime trace over the one-particle sector of the Fock Hamiltonian, by (42.4); the selector of prime modes is fixed before regularization is performed.	Exact operatorial representation once the prime modes have been fixed; the selector of irreducibles precedes it.
Feigenbaum $(\delta, \alpha_{\mathrm{F}})$	Universal scaling constants of the renormalization fixed point for the quadratic period-doubling cascade.	The family dodecafasica declarada produce to level eight $\delta_8^{\mathscr{B}}(0) = 4.66921218915\ldots$ and $\alpha_8^{\mathscr{B}}(0) = 2.50296847988\ldots$, by section 45.6.	Exact finite calculation up to $n = 8$; convergence to the universal fixed point requires proving convergence of the renormalization operator. α_{F} and α_{HMT} are distinct objects.
Catalan G	$G = L(2, \chi_{-4}) = \sum_{n\ge 0}(-1)^n/(2n+1)^2$.	The natural reader is the odd character modulo four evaluated at 2, by equation (41.6).	Exact classical identity with localized operator; it is not an output of the residual functional modulo three.
Khinchin K	Almost-sure geometric mean of the digits of the continued fraction for the Gauss system.	The reader is the ergodic mean of $\log a_1$ for $T(x) = x^{-1} - \lfloor x^{-1} \rfloor$ and its invariant measure, by equation (44.2).	Incorporated classical ergodic theorem; it requires dynamics distinct from the triadic spectral tower.

The table establishes a reading rule: a constant is defined by its object, its operator, and its extraction map, not by the appearance of a decimal prefix in a catalogue. Euler–Mascheroni and Stieltjes belong to finite parts and Laurent jets; Apery, to an evaluation; Glaisher–Kinkelin, to a regularized derivative; Euler–Kronecker, to the character of the field; Meissel–Mertens, to a prime trace; Feigenbaum, to a finite dynamical family; Catalan and Khinchin, to distinct neighboring operators.

53.15.8 Domains, premises, and necessary conditions of the genealogical classification of the number

The arithmetic definitions of integer, rational, irrational, algebraic, and transcendental are standard. The balanced ternary chart, the separation between section and value, the native selector of irreducibles, the Paley–Witt root of minus one, the six realizations of $-1/12$, and the spectral readers are deduced from the maps cited in each subsection. This section gathers them without introducing an additional constant or numerical selector.

The force of each assertion is determined by its map:

1. the indicated bijections, matrix identities, coproducts, traces, and evaluations are exact on their declared domains;

2. the transcendence of e and π, the irrationality of $\zeta(3)$ and Khinchin's ergodic theorem are classical theorems incorporated for purposes of control or classification;
3. the Feigenbaum approximants establish the computed branch and depth; universality requires the renormalization operator and its convergence theorem;
4. equality of two scalars from different domains authorizes a comparison and requires an intertwiner to identify the complete objects;
5. a decimal truncation proves the computed prefix, whereas the arithmetic class of the infinite value requires, as appropriate, a polynomial equation or a theorem of irrationality or transcendence.

Under these rules, the HMT ontology of number acquires a precise formulation: the Archimedean value is the published coordinate; the compatible section preserves the chain that generates it; the operator determines which property is read; and its hypotheses determine exactly which conclusion follows.

53.16 Genealogical closure of the concept of number: compatibility, memory, and Archimedean evaluation

The number theory developed in HMT culminates in a distinction of types that governs the subsequent transition to Dimensional Mechanics. A *digit* is a visible coordinate of a finite state; a *value* is the Archimedean image published by a reader; an *HMT number* is the complete compatible genealogy that preserves, together with its prefixes, orientation, quotient, carry, boundary, and memory. The fiber of a value gathers the distinct genealogies that publish that same scalar.

This construction extends ordinary arithmetic classification. Integer, rational, irrational, algebraic, and transcendental continue to denote properties of the value. HMT incorporates a second axis: the manner in which a section is prolonged, returns, preserves memory, and reaches a stable evaluation. The closure of Parts I–III fixes both axes and the map relating them.

53.16.1 Digit, section, value and symmetry

Let

$$\mathcal{S}_0^{\mathrm{surv}} \xleftarrow{\rho_{1,0}} \mathcal{S}_1^{\mathrm{surv}} \xleftarrow{\rho_{2,1}} \cdots$$

the inverse system of surviving states, where each restriction preserves the typed TPK data belonging to the preceding level, and let

$$\mathcal{S}_\infty^{\mathrm{surv}} := \varprojlim_n \mathcal{S}_n^{\mathrm{surv}}.$$

Definition 53.21 (Ciphers)**.** Having fixed a positional reader L, the digit or block visible at depth n is

$$d_{L,n}(x_n) := \pi_{L,n}^{\mathrm{vis}}(x_n).$$

It is the coordinate published by the finite presentation x_n; the fiber of $\pi_{L,n}^{\mathrm{vis}}$ preserves the states sharing that coordinate.

Definition 53.22 (HMT Number and Presentation)**.** An HMT number is to compatible section

$$\boldsymbol{x} = (x_n)_{n\geq 0} \in \mathcal{S}_\infty^{\mathrm{surv}}, \qquad \rho_{n+1,n}(x_{n+1}) = x_n.$$

The pair $(\boldsymbol{x}, L)$ is a presentation of the number. The section preserves the complete history; the reader declares which coordinates of that history are published.

Definition 53.23 (Archimedean value)**.** Suppose that L assigns to each x_n a compact cylinder $I_{L,n}(\boldsymbol{x}) \subset \mathbb{R}$, with

$$I_{L,n+1}(\boldsymbol{x}) \subseteq I_{L,n}(\boldsymbol{x}), \qquad \operatorname{diam} I_{L,n}(\boldsymbol{x}) \longrightarrow 0.$$

The value published by the presentation is

$$\mathrm{Val}_L(\boldsymbol{x}) := \text{the unique element of } \bigcap_{n\geq 0} I_{L,n}(\boldsymbol{x}).$$

Proposition 53.24 (Existence and uniqueness of the evaluation)**.** *The preceding hypotheses determine a map*

$$\mathrm{Val}_L : \mathcal{S}_\infty^{\mathrm{surv}} \longrightarrow \mathbb{R}.$$

Each fibre $\mathrm{Val}_L^{-1}(r)$ *preserves the genealogies that the Archimedean evaluation represents by the same value* r*.*

Proof. The finite intersection property of the nested family of compact sets guarantees a nonempty intersection. Two distinct points would have a positive distance bounded by every diameter, contradicting the convergence of the latter to zero. □

Definition 53.25 (Genealogical symmetry)**.** A symmetry of the HMT number system is a family of bijections $g = (g_n)_{n\geq 0}$ that preserves survival and types, and satisfies

$$\rho_{n+1,n} \circ g_{n+1} = g_n \circ \rho_{n+1,n} \qquad (n \geq 0).$$

The action

$$g_\infty(\boldsymbol{x}) := (g_n x_n)_{n\geq 0}$$

is an automorphism of the inverse limit. A genealogical invariant is a function constant on its orbits.

Symmetry is thus characterized as a transformation compatible at every depth. This notion renders decimal publication, exceptional incidence, and subsequent spectral realization comparable while preserving the proper type of each output.

53.16.2 Evolutionary Conservation of Unity

For $n \geq 1$, let

$$\rho_9(n) = 1 + ((n-1) \bmod 9), \qquad \kappa_9(n) = \frac{n - \rho_9(n)}{9}. \tag{53.15}$$

Then

$$n = 9\kappa_9(n) + \rho_9(n), \qquad \rho_9(n) \in \{1, \ldots, 9\}.$$

The coordinate ρ_9 publishes the nonadic phase and represents the null class through 9; κ_9 records the number of circuits. The digital root publishes the phase. Reduction modulo nine accompanied by its quotient publishes phase and memory.

The associated helical realization is

$$\mathcal{H}_9(n) := \left(\cos\frac{2\pi(\rho_9(n)-1)}{9}, \sin\frac{2\pi(\rho_9(n)-1)}{9}, \kappa_9(n) + \frac{\rho_9(n)-1}{9}\right). \tag{53.16}$$

Proposition 53.26 (Phase return and memory advance)**.** *The realization* (53.16) *satisfies*

$$\mathcal{H}_9(n+9) = \mathcal{H}_9(n) + (0, 0, 1).$$

The visible return $9 \to 1$ *closes the phase and increments the state memory by one.*

Proof. The identities $\rho_9(n+9) = \rho_9(n)$ and $\kappa_9(n+9) = \kappa_9(n) + 1$ fix the circular coordinates and increase the height exactly. □

This identity formulates evolutionary conservation: unity returns as phase, and the new state transports the preceding closure as memory. The cubic completion realizes the same law in three dimensions:

$$10^3 = (9+1)^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1 = 729 + 271. \tag{53.17}$$

The interior has 729 states; 243 + 27 constitutes the punctured corona; the apex +1 completes the base-one-thousand unit. The identity 729 + 270 = 999 describes the state immediately preceding closure, and 729 + 271 = 1000 describes the completion. The law

$$10^d = (9+1)^d = \sum_{r=0}^{d} \binom{d}{r} 9^{d-r}$$

transports the same unit between dimensions and preserves the genealogy of its strata.

53.16.3 Visible syntax and semantics of prolongation

A local signature describes the current form of a branch. Its survival semantics is the system of prolongations it can support. For $x_n \in \mathcal{S}_n^{\mathrm{surv}}$, define

$$\mathrm{Ext}_m(x_n) := \{x_m \in \mathcal{S}_m^{\mathrm{surv}} : \rho_{m,n}(x_m) = x_n\}, \qquad m \geq n.$$

Two states that satisfy

$$d_{L,n}(x_n) = d_{L,n}(y_n)$$

may possess different extension trees. In the finite witness $60 \to 81$, two branches with the same visible signature possess, respectively, zero and twelve prolongations. The witness establishes that future continuity adds information to the local output. The HMT number incorporates that continuity as part of the object.

Coinduction is expressed through the compatibility of all prefixes. Blocks of one thousand or ten thousand digits are truncated publications of a uniformly prolongable rule. Numerical generation reaches any requested depth; the lift to the enriched state also preserves the hypotheses of existence, uniqueness, and compatibility of its fibers.

53.16.4 Arithmetic classification and genealogical classification

In a basis $b \geq 2$, the canonical presentation of a real number separates the following visible classes:

1. an integer belongs to $\mathbb{Z}$ and admits a finite balanced-ternary word;
2. a rational number has a finite or eventually periodic expansion;
3. an irrational number has an expansion that is not eventually periodic;
4. an algebraic number is a root of a nonzero polynomial in $\mathbb{Q}[X]$;
5. a transcendental number belongs to the complement of the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$.

Periodicity distinguishes rational from irrational numbers; the polynomial equation distinguishes algebraic from transcendental numbers. These are different axes. For example, φ is algebraic irrational because $\varphi^2 - \varphi - 1 = 0$; the transcendence of e and π follows from classical theorems; an HMT genealogy additionally distinguishes the propagation, closure, and self-scaling functions that realize those values.

The same separation governs $e + \pi$. The sum of the two presentations defines a composite object and publishes

$$\mathrm{Val}(e) + \mathrm{Val}(\pi) = e + \pi.$$

Its arithmetic class is a question in its own right. The HMT construction determines the map that forms the composite and preserves the two genealogies involved.

53.16.5 Factorization of the genealogical closure of the number through Euler's operational system

The classical Euler equation belongs to a unique quadratic family. Let J_σ be a generator satisfying

$$J_\sigma^2 = \sigma I, \qquad \sigma \in \{-1, 0, +1\}.$$

The separation of even and odd powers provides

$$\exp(tJ_\sigma) = C_\sigma(t)I + S_\sigma(t)J_\sigma, \tag{53.18}$$

with the exact realizations

$$\begin{aligned} J_{-1}^2 = -I \quad &: \quad \exp(tJ_{-1}) = \cos t\, I + \sin t\, J_{-1}, \\ J_0^2 = 0 \quad &: \quad \exp(tJ_0) = I + tJ_0, \\ J_{+1}^2 = I \quad &: \quad \exp(tJ_{+1}) = \cosh t\, I + \sinh t\, J_{+1}. \end{aligned}$$

Elliptic, parabolic, and hyperbolic are three regimes of a single operator exponential. The tritic classification determines the geometry of the evolution.

The hyperbolic branch contains the Fibonacci self-scale. For

$$F = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad A = F^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

we have $\det A = 1$, $\operatorname{tr} A = 3$, and

$$\eta = \operatorname{arcosh} \frac{3}{2} = 2 \log \varphi.$$

Therefore, there exists $J_h^2 = I$ such that

$$A = \exp(\eta J_h), \qquad A^n = \cosh(n\eta)I + \sinh(n\eta)J_h. \tag{53.19}$$

The relation $\varphi^2 = \varphi + 1$ expresses two readings of the same process: local additive recurrence and spectral multiplicative scaling. This compatibility provides the elementary model of the double sum–product evaluation that APP formalizes on their common support.

53.16.6 Typed Unification of Circular, Complex, and Hyperbolic Geometries

The initial geometric motivation of HMT can be formulated exactly as a common architecture of support, boundary, and generators. The complex plane $\mathbb{C}$ contains the unit circle

$$S^1 = \{e^{\mathrm{i}\theta} : \theta \in \mathbb{R}\},$$

which constitutes the elliptic realization of (53.18). The upper half-plane $\mathbb{H} = \{w \in \mathbb{C} : \operatorname{Im} w > 0\}$ is transported to the hyperbolic disc by means of the Cayley transformation

$$C(w) = \frac{w - \mathrm{i}}{w + \mathrm{i}}, \qquad C : \mathbb{H} \xrightarrow{\sim} \mathbb{D}, \qquad \mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}.$$

The ideal boundary of $\mathbb{D}$ is the same circle S^1, whereas its interior carries the Poincare metric

$$\mathrm{d}s_{\mathbb{D}}^2 = \frac{4\,|\mathrm{d}z|^2}{(1 - |z|^2)^2}.$$

Fusion therefore consists of a typed diagram: the trigonometric parameterization publishes the circular boundary; the complex structure provides the chart; the Poincare metric realizes the hyperbolic interior.

The three regimes $J^2 = -I, 0, +I$ also incorporate the parabolic threshold and place this geometric concurrence within a single operator family.

Proposition 53.27 (Euclidean and Hyperbolic Compatibility of the Quarter Turn)**.** *Over the disco unitary*

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\},$$

the application

$$\mathcal{R}_{\pi/2} : \mathbb{D} \longrightarrow \mathbb{D}, \qquad \mathcal{R}_{\pi/2}(z) = \mathrm{i}z,$$

is simultaneously a Euclidean rotation and an isometry of the Poincaré metric

$$\mathrm{d}s^2_{\mathbb{D}} = \frac{4\,|\mathrm{d}z|^2}{(1-|z|^2)^2}.$$

Its extension to the boundary $S^1 = \partial\mathbb{D}$ is the oriented quarter-turn. In the real realization of theorem 3.25, this map corresponds to $\mathrm{Exp}((\pi/2)J_{\mathrm{e}}) = J_{\mathrm{e}}$.

Proof. Multiplication by i preserves $|z|$ and $|\mathrm{d}z|$. It therefore preserves both the Euclidean metric and the numerator and denominator of $\mathrm{d}s^2_{\mathbb{D}}$. The exponential identity follows from $J_{\mathrm{e}}^2 = -I$. □

The complex plane, trigonometric circle, and hyperbolic disk are not identified as metric spaces: they share a coordinated carrier and the quarter-turn automorphism once each realization has been declared.

53.16.7 Boundary interface between quadratic rotation and minimal torsion

The function of this interface is to transport an already constructed interior rotation to a boundary datum without confusing the quadratic endomorphism with a scalar defect. Fix a base point $x_0 \in \partial X$, a real bundle $E_\partial \to \partial X$, a connection ∇_∂, a boundary reader q_∂, and a discrete cotetrad $e_\partial \in \Omega^1(\partial X; E_\partial)$. We write

$$\mathrm{Tors}(\partial X; E_\partial) := \Omega^2(\partial X; E_\partial), \qquad \mathrm{Hol}_{x_0}(\nabla_\partial) \subseteq \mathrm{Aut}(E_{\partial,x_0}).$$

An edge realization of the interface should construct the maps

$$\mathrm{Res}_\partial : \langle J_{\mathrm{e}} \rangle \longrightarrow \mathrm{Hol}_{x_0}(\nabla_\partial), \qquad \mathrm{Def}_\partial : \mathrm{Hol}_{x_0}(\nabla_\partial) \longrightarrow \mathrm{Tors}(\partial X; E_\partial).$$

The torsion determined by the connection and the co-tetrad has the type

$$T_\partial := d_{\nabla_\partial} e_\partial \in \Omega^2(\partial X; E_\partial). \tag{53.20}$$

The physical realization of minimal torsion requires in addition a functional

$$\Lambda_\partial : \mathrm{Tors}(\partial X; E_\partial) \longrightarrow \mathbb{R} \tag{53.21}$$

such that the three maps satisfy the entanglement conditions

$$\mathrm{Def}_\partial\big(\mathrm{Res}_\partial(J_{\mathrm{e}})\big) = T_\partial, \qquad \Lambda_\partial(T_\partial) = \Delta_4. \tag{53.22}$$

These equations specify the comparator that must be constructed; they do not assert that the three maps exist. The scalar Δ_4 is defined by equation (58.2).

Remark 53.28 (Exact condition of the boundary comparator). The operator root $J_{\mathrm{e}}^2 = -I$, the quarter-turn $\mathrm{Exp}((\pi/2)J_{\mathrm{e}}) = J_{\mathrm{e}}$, and the scalar Δ_4 are deduced in their respective domains. The preceding boundary chain fixes the type of the morphisms that must link them. Until Res_∂, Def_∂, and Λ_∂ are constructed, the conditions (53.22) define the required comparator but do not imply a physical realization; the angular proximity of its endpoints does not replace that operator.

APP, TRIT, and TPK supply the discrete support of that architecture. On $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$, the sum and product evaluations induce projectors $P_{\Sigma,d}$ and $P_{\Pi,d}$ that satisfy

$$\|[P_{\Sigma,2}, P_{\Pi,2}]\| = \frac{\sqrt{2}}{3}, \qquad \|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}.$$

This calculated incompatibility is the finite precursor of Heisenberg: two families of observables share a support and possess incompatible spectral decompositions. The alternating form of the torus,

$$\sigma((a,b),(c,d)) = ad - bc \pmod 9,$$

is integrated into the Weyl–Heisenberg nonadic representation

$$X|n\rangle = |n+1\rangle, \qquad Z|n\rangle = \omega^n|n\rangle, \qquad ZX = \omega XZ, \qquad \omega = e^{2\pi \mathrm{i}/9}.$$

The commutator publishes the oriented phase, and the resulting central group has $9^3 = 729$ elements.

The central block of APP simultaneously provides the local lorentzian realization:

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-, \qquad M_c := \frac{B_c}{\sqrt{45}} = \exp\left(\frac{1}{2}\log\frac{9}{5}\,R\right).$$

Con $\eta_{1,1} = \mathrm{diag}(1,-1)$,

$$M_c^{\mathsf{T}}\eta_{1,1}M_c = \eta_{1,1}, \qquad \det M_c = 1, \qquad M_c \in SO^+(1,1).$$

Thus Lorentz geometry and the Minkowski plane 1 + 1 emerge as an exact spectral realization of the central block. Every subsequent dimensional lift must intertwine this local block with its new codomain. The sequence

$$\text{APP–TRIT–TPK} \longrightarrow \begin{cases} \text{elliptic–parabolic–hyperbolic family,} \\ \text{Lorentz–Minkowski block } 1+1, \\ \text{finite Weyl–Heisenberg representation} \end{cases}$$

presents the geometric and operatorial developments preceding the genealogical closure of the number.

53.16.8 Constant Generation Operators: Domains, Seeds, and Evaluations

The function of a constant is fixed through four data: mathematical definition, domain, operator or reader, and deduced conclusion. The table 53.4 brings together the objects that must remain visible when number theory is closed.

Table 53.4: Constants and Seeds in the Genealogical Closure of the Number.

Object	Definition or identity	HMT domain and map	Conclusion and condition
π, e, φ	Sections published by compatible numerical cylinders.	The filtration $w_6 \to w_{30} \to R_{36} \to G_9$ produces the terminal state; the readers distinguish closure, propagation, and self-scaling.	For each prescribed finite depth, the map produces compatible unit cylinders; the infinite section follows from the coinductive theorem.

Object	Definition or identity	HMT domain and map	Conclusion and condition
α_{HMT}	Dodecaphasic closure with carry over the terminal state.	The double reading leads to $(P, E, \Phi, K, c_{13} = 0)$ and to the rational output of 36 digits; the $E_{21/22}$ route provides an independent computation.	The recurrence is exact for the declared terminal state and receives neither α nor CODATA; the minimal APP deduction$\rightarrow K$ requires construction of its complete bidirectional reader.
i interno	$J^2 = -I$ on the Paley–Witt module.	Construct a finite complex structure and separate conjugate orientations.	Over the declared base field, $J^2 = -I$ holds exactly.
$-1/12$	$\zeta(-1) = -1/12$ and normalized operator realizations.	Incidence $90 \rightarrow 120$, dodecaphase defect and spectral functionals, each with its domain and normalization.	Exact equality for each declared map.
$e + \pi$	Sum of the presentations of e and π.	Composite object of the presentation algebra; preserves both genealogies.	Exact construction; separate arithmetic class.
Euler–Mascheroni γ and Stieltjes γ_n	$\gamma = \mathrm{FP}_{s=1}\,\zeta(s)$; coefficients of Laurent.	Residual decomposition modulo three and Laurent jets.	Exact identities of the declared functional.
Apéry $\zeta(3)$	Evaluation of the zeta function at $s = 3$.	Reader $\mathcal{Z}_3(3)$ with rational prefix intervals.	Exact evaluation; classical irrationality.
Glaisher–Kinkelin A_{GK}	$A_{\mathrm{GK}} = \exp(1/12 - \zeta'(-1))$.	Regularized derivative of the spectral reader.	Exact identity with explicit normalization.
Euler–Kronecker constant $\gamma_{F_{-3}}$	Logarithmic finite part of the Dedekind zeta function of $F_{-3} = \mathbb{Q}(\sqrt{-3})$.	The character χ_{-3} selects the field and organizes $L'(1, \chi_{-3})/L(1, \chi_{-3})$.	Exact identity of the declared character.
Meissel–Mertens B_1	$B_1 = \lim_{x\to\infty}(\sum_{p\le x} p^{-1} - \log\log x)$.	Finite part of the trace over primary modes.	Exact operatorial representation with fixed modes.
Catalan G and Khinchin K_0	$G = L(2, \chi_{-4})$; ergodic mean of the Gauss system.	Readers of modular character four and continued fraction.	Classical theorems embedded with their own dynamics.
Feigenbaum (δ, α_F)	Scales of the renormalization operator in period doubling.	The dodecaphonic family produces level eight approximants.	The calculation determines level eight exactly; universality requires proving convergence to the renormalization fixed point.
Rogers–Ramanujan constant $R(q)$	Level-five continued fraction and Klein–Ramanujan identity for j.	It organizes level 5, A_5, the limit $R(q) \rightarrow \varphi^{-1}$, and Hecke–Faber replicability.	Exact modular identities; the comparison of coefficients covers only the finite depth computed.
Prime clock p	$\tau_p = \mathrm{ord}_p(10)$, $M_p = (10^{\tau_p} - 1)/p \equiv 0 \pmod 9$ for $p \nmid 30$.	Primes are irreducible periods; composites synchronize clocks through the least common multiple.	Exact arithmetic theorem in the declared domain.

Finite part, special value, regularized derivative, character, ergodic average, renormalization scale, and prime period are distinct maps. Their coexistence within HMT arises from a typed family of operators and extraction maps.

53.16.9 Double projection of a single genealogy

Let $\mathfrak{X} := X_\infty^{\mathrm{cal}}$ be the compact space of enriched histories with transported initial gauge. Given a declared gauge $c \in \mathscr{C}_{\mathrm{exc}}$, two evaluations act on the same section:

$$\mathfrak{R} : \mathfrak{X} \longrightarrow \mathbb{R}, \qquad \mathfrak{E}_c : \mathfrak{X} \longrightarrow \mathbf{Exc}, \qquad \mathfrak{E}_c(x) := \mathfrak{E}(x, c),$$

where $\mathfrak{R}$ contrae cylinders and $\mathfrak{E}_c$ preserves supports, incidences, polarity and boundary transport.

Theorem 53.29 (Causal unit of the double projection)**.** *For every history $\boldsymbol{x} \in \mathfrak{X}$, the pair*

$$\mathfrak{D}_c(\boldsymbol{x}) := (\mathfrak{R}(\boldsymbol{x}), \mathfrak{E}_c(\boldsymbol{x}))$$

is a joint publication of the same antecedent. The first coordinate determines the Archimedean value; the second determines the exceptional incidence class. When the joint readers separate points, $\mathfrak{D}_c$ is injective.

Proof. Both components are defined on the same domain and preserve the restriction maps. If $\mathfrak{D}_c(\boldsymbol{x}) = \mathfrak{D}_c(\boldsymbol{y})$, joint separation of histories implies $\boldsymbol{x} = \boldsymbol{y}$. □

The Archimedean branch produces the cylinders and the values of π, e, φ. The incidence branch prolongs the same words and the same seal toward Paley, Witt, Golay, the Leech lattice, $V^\natural$, the Monster group, and Moonshine. The causal relation is the common preimage: both evaluations preserve different invariants of the same history. The real value is the contracted shadow; exceptional symmetry is the incidence memory published by the same state.

53.16.10 Genealogical Closure of the Concept of Number

The concept of number is fixed by the string

$$\text{finite state} \longrightarrow \text{compatible extension} \longrightarrow \text{genealogical section} \longrightarrow \begin{cases} \text{visible digit,} \\ \text{Archimedean value,} \\ \text{exceptional incidence.} \end{cases}$$

The digit is a coordinate; the value is a projection; the number is the complete section; the symmetry is the compatible automorphism that preserves its memory. Evolutionary conservation transports that identity through the nonadic return, the helical carry, and the completion $729 + 271 = 1000$. With these distinctions secured, Dimensional Mechanics realizes the same genealogy in a new codomain.

Causal dependence between inverse section, Archimedean evaluation, and exceptional conservation of incidence The inverse limit supplies the section; the Archimedean reader produces the value; the binomial completion and the nonadic extension transport unit and memory; the complementary readers classify the outputs; and the double projection applies the Archimedean and exceptional maps to the same antecedent. The Witt–Golay–Leech–Moonshine chain requires, at every step, the corresponding incidence intertwiner.

53.17 Definition of the genealogical atlas and metrological evaluation conventions

This atlas is not a table of coincidences. Its mathematical core preserves five inseparable fields; the logical status and the falsification criterion are stated subsequently, without taking the place of any of the five:

$$\boxed{\begin{array}{c|c|c} \text{antecedent} & \text{equation} & \text{preserved invariant} \\ \text{meaning} & \text{terminal value} & \end{array}} \qquad \begin{array}{c}[\text{status;} \\ \text{refutation criterion}]\end{array}. \tag{53.23}$$

53.18 HMT corollary of metrological confluence

Definition 53.30 (Quintuple in Metrological Form)**.** Let x be a quantity obtained through an HMT–MD realization. Its *metrological quintuple* is

$$\mathfrak{C}(x) = \big(A_x \mid E_x \mid I_x \mid S_x \mid V_x\big),$$

where A_x is the typed antecedent, E_x the equation of determination, I_x the invariant preserved by the realization, S_x the structural meaning, and V_x the terminal value. Forming V_x before the other four components have been fixed is not permitted. Two outputs with the same decimal are not equal if their antecedents or invariants differ.

Corollary 53.31 (Metrological confluence)**.** *The outputs of vacuum, action, information, temperature, gravity, criticality, spectrum, mixing, and cosmology constructed in the chapters 34, 40, 41, 44, 45 y 50 y sections 49.1, 51.6 y 88.1 admit the quintuples of the table 53.5. In each row, the value in the fifth column is a consequence of the first four and not a datum selecting them. The rows share genealogical nodes —mode* 30*,* $\log 3$*, determinant* D_A*, channels* $90/120$*, and change* $6 \to 8$*—, but retain their domains and operators separately.*

Table 53.5: HMT Metrological Confluence Corollary: the five columns are a single probative unit.

Antecedent	Ecuación	Conserved invariant	Significado estructural	Terminal value
I. Action				
two realizations of the action line (equations (34.7) y (34.8))	$R_{\mathrm{act}} = H_5/\eta_{\mathrm{ret}}$, $\xi_{\mathrm{act}} = \frac{1}{2}\log R_{\mathrm{act}}$	Decade and Leaf Orientation	Absolute action and its relative quotient correspond to distinct functionals; the decade is not used as a fit	R_{act}, ξ_{act}
II. Vacuum—electromagnetic				
$T = q_+P_+ + q_-P_-$, $30 \to 90/120$ (equations (39.2) y (39.8))	$V = (I - T^{90})(I - T^{120})^{-1}$	Positivity, Leaf Spectrum, and Completion $3 \to 4$	The vacuum is a bilayer operator; $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$ determine the fourth sector without introducing an SI constant	$r_+ = 0.9999996285\ldots$, $r_- = 0.9997241371\ldots$
spectral response (equations (39.4) y (39.12))	$\widehat{\varepsilon} = r_+^2$, $\widehat{\mu} = r_-^2$, $\widehat{Z} = r_-/r_+$, $\widehat{c} = (r_+r_-)^{-1}$	Product and quotient of sheets, stable under $V \mapsto V \otimes I_{9^m}$	Permittivity, permeability, impedance, and propagation are four functions of the same positive operator	$0.9999992571\ldots$, $0.9994483505\ldots$, $0.9997245085\ldots$, $1.0002763105\ldots$
constitutive lines (equation (39.18))	$\mu\varepsilon = c^{-2}$, $\mu/\varepsilon = Z^2$	Homogeneity in L_S, L_C, L_Q	The dimensional chart is a subsequent section, not a unit attached to a dimensionless number	Z, μ, ε, c as exact sections

Continued on the next page

Continuation of the HMT metrological confluence corollary

Antecedent	Ecuación	Conserved invariant	Significado estructural	Terminal value
$(Z_{\mathrm{vac}}, h_a, \alpha_{\mathrm{HMT}})$ (equations (40.1) y (40.6))	$e_a^2 = 2\alpha h_a/Z$, $R_K G_0 = 2$, $K_J \Phi_0 = 1$	Bidirectional Product and Opposite Flow/Frequency Orientation	The electric quanta descend from the vacuum and the action; the SI charge does not act as a selector	$e_a, R_K, G_0, \Phi_{0,a}, K_{J,a}$
III. Information, temperature, and radiation				
uniform trit and chronological structure 108 (equations (48.1) y (48.2))	$k_B \Theta_{\mathrm{clk}} \log 3 = h_{\mathrm{int}}/(108 t_0) = E_P$	Entropy of a Ternary Decision and the Period and Temporal Periodicity of (108) Phases	A single quotient measures information, temperature, and energy without identifying their three state spaces	$\log 3 = 1.0986122886681098\ldots$
$G\hbar$ y temporal periodicity of (108) phases (equations (36.36) y (36.37))	$E_P = \pi\hbar/(54 t_0) = k_B \Theta_{\mathrm{clk}} \log 3$	Bidirectional product $G\hbar$ and TRIT informational content	Planck's geometric quantum and the informational quantum associated with a trit converge in the same energy section	$E_P = h/(108 t_0)$
Planck spectrum (equations (48.3) y (48.4))	$\lambda_{\max}/\ell_0 = 108 \log 3/x_5$, $\nu_{\max} t_0 = x_3/(108 \log 3)$	Same spectrum, distinct Jacobians	The maxima by length and by frequency are functional realizations of a common antecedent, not two approximations of the same coordinate	$23.896756779\ldots$, $0.0237794888\ldots$
bosonic continuum (equations (48.5), (48.6) y (48.8))	$n_\gamma \ell_P^3 = 2\zeta(3)/[\pi^2 (\log 3)^3]$ and Laws T^4, T^3	Bosonic occupation and informational content per trit	Radiance, number, energy, and entropy form a thermal family; they are not four independent calibrations	Exact coefficients in $\pi, \zeta(3), \log 3$
IV. Gravity, area and modular memory				
circular closure: temporal periodicity of 108 phases (theorem 50.1 y equation (50.7))	$G = \theta_{108} c^5 t_0^2/\hbar$, $\theta_{108} = (54/\pi)^2$	Equality of the Compton radius and the gravitational radius	The gravitational constant is a section selected by a closure of lengths, not a grafted number	$G_{\mathrm{int}} = (54/\pi)^2 c^5 t_0^2/\hbar$
$A_5/C_5 \to P_5$ (theorem 77.5 y equation (77.26))	$\mathcal{G}_5 = G P_5$, $P_5^2 = P_5$, $\mathrm{Tr}\, P_5 = 5$	Central Rank and TT Reduction	Five components of the tensorial carrier precede the two physical helicities fysical; they are not confused with them	$\mathrm{Spec}\, \mathcal{G}_5 = \{G^{(5)}, 0^{(7)}\}$
change $6 \to 8$, derivation of Barbero–Immirzi, and channels 90/120 (equations (51.9) y (72.2))	$A_{\mathrm{phys}}/\ell_P^2 = \gamma_{\mathrm{HMT}} a_0 (N_{90} + N_{120})$	Area quantum, hexad–octad orientation, and ℓ_P^2	γ_{HMT} fixes the geometric quantum; this monograph adopts that magnitude without reopening its derivation	$\gamma_{\mathrm{HMT}} a_0 = 0.0016755940749\ldots$
Occupation N and Memory D (theorem 72.2 y equation (72.8))	$D = 90N_{90} + 120N_{120}$, $N_{90} = 4N - D/30$, $N_{120} = D/30 - 3N$	Determinant 30 and channel provenance	Area quantifies the total occupation; the modular holographic Hamiltonian preserves the sector 90/120 of origin	(N_{90}, N_{120}) reconstructed exactly
boundary state (theorems 73.1 y 74.1 y equation (73.8))	$I_{\mathrm{or}} = 36 \log 3 - S(\rho_A)$, $\delta S = \beta_{90} \delta\langle \mathcal{A}_{90}\rangle + \beta_{120} \delta\langle \mathcal{A}_{120}\rangle$	Schmidt spectrum and ratio $\beta_{120}/\beta_{90} = 4/3$	Entanglement entropy, oriented information, and area are three compatible readings of the same boundary	$I_{\mathrm{or}} = 0.0016691832\ldots$, $(\beta_{90}, \beta_{120}) = (5.97265\ldots, 7.96353\ldots)$
infinite local network (theorem 75.1 y equation (75.3))	$\lambda_\partial = \exp(-30\Delta_4^{\mathrm{mod}})$	Group of ratios of weights under persistence 90/120	The limit is a modular GNS generator, not a global trace–class density matrix in the UHF algebra	$\lambda_\partial = 0.996669646468957\ldots$

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Continuation of the HMT metrological confluence corollary

Antecedent	Ecuación	Conserved invariant	Significado estructural	Terminal value
V. Criticality and Mode 30				
phases $3m-1$ (equations (45.3), (45.6) y (46.1))	$u_0 = 1 - \cos(37y)\sin(36y)/(12\sin 3y)$, $899 = 30^2 - 1$	Normalization in quadratic form and cancellation of π before iteration	The criticality constant is derived from a closed profile whose mode 30 also separates 90 from 120	$u_0(x) = x^2 - (715769/2424603)x^4 + O(x^6)$
Superstable Succession (equations (45.8) y (45.11))	$\mathcal{R}(f)(x) = -a^{-1}f(f(-ax))$	Unimodality, negative Schwarzian, and period-doubling combinatorics	The finite quotients certify entry into the universal class; the hyperbolicity of the fixed point remains a separate obligation	$\delta_8 = 4.66921218915\ldots$, $\alpha_8 = 2.50296847988\ldots$
VI. Spectrum, arithmetic, and cylinders				
quarter-twist (equations (41.4) y (41.6))	$J^2 = -I$, $G = L(2, \chi_{-4})$	Complex orientation and multiplicativity of the character	Catalan's constant is the second moment of the quarter-torsion operator, not a decimal associated by resemblance	$G = 0.915965594177219\ldots$
CRT product (equations (41.7), (41.8) y (42.2))	$\chi_{12} = \chi_{-3}\chi_{-4}$, $\gamma_{\mathbb{Q}(\sqrt{3})} = \gamma + L'/L(1, \chi_{12})$	Triadic orientation, quarter-twist, and finite Dedekind term	The Lattice 3/4/12 Determines a Real Field and Its Boundary Constant	$L(1) = 0.7603459963\ldots$, $\gamma_{\mathbb{Q}(\sqrt{3})} = 1.05396560825\ldots$
zeta triadica renormalizada (equations (43.1) y (43.2))	$A_k = \exp(H_k B_{k+1}/(k+1) - \mathcal{Z}_3'(-k))$	Zeta-derivative level and Bernoulli normalization	Bendersky–Glaisher organizes $\zeta(2n+1)$ as a graded family of determinations	$A_0 = \sqrt{2\pi}$, $A_1 = A_{\mathrm{GK}}$, family A_k
cofinal cylinders (equations (44.2) y (44.5) a (44.7))	$L = \pi^2/(12\log 2)$, $h_G = 2L$, $\Lambda_b = \log b/h_G$	Gauss measure, denominator growth, and coding rate	Khinchin, Levy, and Lochs are distinct observables of the same transported system	$K = 2.6854520011\ldots$, $L = 1.1865691104\ldots$, $\Lambda_{10} = 0.9702701144\ldots$
operator of transferencia of Gauss (equation (44.8))	$\mathcal{L}_G f(x) = \sum_{n\geq 1}(n+x)^{-2} f((n+x)^{-1})$	Mass, Invariant Measure, and the Zero-Mean Complement	The subdominant eigenvalue measures the relaxation of the cylinders; it is not selected from a tabulated decimal	λ_{GKW}: certified interval yet to be constructed
irreducible cycles and vacancies (equations (42.4), (42.5), (47.5) y (47.7))	$B_1 = \lim(\sum_{p\leq x} p^{-1} - \log\log x)$, $\gamma^{+}_{\mathrm{vac,j}} = \varepsilon_{\mathrm{vac}}\gamma_j + \cdots$	Prime factorization, bounded discrepancy, and vacancy holonomy	Local Sum, Multiplicative Density, and Finite Terms Preserve Distinct Antecedent Constructions	$B_1 = 0.2614972128\ldots$, $C_2 = 0.6601618158\ldots$, $\varepsilon_{\mathrm{vac}} = 0.04575749056\ldots$
lift $r^2 = 2 \pmod{3^k}$ (equation (46.5))	$u = 3 + 2r$, $\mathrm{ord}(u) = 4\cdot 3^{k-1}$	Quarter-torsion operator compatible at every nonadic depth	The Pell unit simultaneously preserves finite orientation and profinite memory	$3 + 2\sqrt{2} = 5.82842712474619\ldots$
VII. Leaf Change, Mixing, and Electroweak Sector				
$C^* \mapsto -C^*$, $6 \to 8$ (equations (49.29) y (49.32))	$(A, C^*) \mapsto (A, -C^*)$, $-(8-6)/3 = -2/3$	Sheet parity and hex-octagonal oriented defect	Barbero and CKM are distinct realizations of state change, not mutually retrospective definitions	A, C^* recovered rationally from the three angles
coordinate matrix M (theorem 49.7 y equation (49.38))	$R_{\mathrm{CKM}} = M\,\mathrm{diag}(1,-1,1)M^{-1}$	Rational involution, $R^2 = I$, $\det R = -1$	The mixing space preserves exactly the coordinate that experienced the sheet inversion; the CP phase is fixed under this involution	Matrix rational exact, $\det M = -3857/6$
angular operator T (theorem 49.2 y equation (49.9))	$D_A = \det T = e^{-2A_{\mathrm{rad}}}$, $\sin^2\theta_W = 1 - D_A$	Determinant even and adjacency of paths W/Z	The vacuum and Weinberg share an angular operator, but determine different spectral functions	$D_A = 0.775129119247\ldots$, $\sin^2\theta_W = 0.224870880753\ldots$

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Continuation of the HMT metrological confluence corollary

Antecedent	Ecuación	Conserved invariant	Significado estructural	Terminal value
gauge chart in the tree-level approximation (equations (49.13), (49.14) y (49.18) y corollary 49.4)	$g = 2\sqrt{\pi\alpha/(1-D_A)}$, $G_F^{(0)} = \pi\alpha/[\sqrt{2}(1-D_A)m_W^2]$	The Same D_A Preserves the Neutral/Charged Channel Separation	The couplings, the Fermi constant, and the Higgs self-coupling form a typed composition; radiative corrections are not incorporated into the internal functional	$G_F^{(0)} = 1.1157162817\cdot10^{-5}\,\mathrm{GeV}^{-2}$, $\lambda_H = 0.123847911548\ldots$
functionals $(\alpha, \varphi, 1000, 55)$ (equation (53.28))	$a_\mu = \alpha/(2\pi) + \alpha(\varphi - 1)/1000 + \alpha^2/(55000)$	Sum of three contributions with frozen coefficients	Leptonic composition control; it does not replace the complete QED radiative series	$a_\mu = 0.001165920713\ldots$
VIII. Cosmology and Scope Controls				
radius $R_{108} = \ell_P$ (equations (88.9) a (88.11))	$H_{108} = \pi/(54t_0)$, $\Lambda_{108} = \pi^2/(972\ell_0^2)$	A single radius under the de Sitter realization	Expansion, curvature, temperature, and entropy are four magnitudes of the same conditioned geometry	$T_{\mathrm{dS}}/\Theta_{\mathrm{clk}} = \log 3/(2\pi)$, $S_{\mathrm{dS}}/k_B = \pi$
Perron scale (equations (81.2), (81.4) y (81.6))	$a_n/a_0 = \varphi^{n/3}$, $M_n = (a_n/a_0)^{-6}$	Stable mode φ^{-2n} and quadratic spin dilution	The exponent -6 represents the active memory in the cosmological law; the Einstein–Cartan amplitude requires its transducer	$M_n a_n^6 = a_0^6$, $H_\varphi t_0 = 0.1604039416865\ldots$
controls of composition (equations (53.29) y (53.30))	$a_0 = \hbar/(m_e c\,\alpha)$, $E_* = h_{\mathrm{int}} c_{\mathrm{int}}/(e_{\mathrm{int}}\ell_0^2)$	Dimensional Homogeneity and Non-Feeding by Name	Bohr is composed from magnitudes defined outside the scope of this monograph; Born–Infeld still requires a determinantal action that selects its scale	Exact sections a_0, E_*, without promoting a new selector

Proof. Each row is obtained by substituting exclusively already constructed objects into the equation cited in its first or second column. In vacuum, the calculation is spectral: the functional theorem for V produces $r_\pm$, and the four constitutive quantities are functions of those two eigenvalues. In the realization, in its thermal form, the chronological structure 108 and the uniform entropy of one trit produce the product $k_B\Theta_{\mathrm{clk}}$, after which Planck, Wien, and radiation are compositions. In gravity, the circular condition fixes θ_{108}, while P_5 and the TT reduction fix the type of the carrier. In the modular realization, the linear system of determinant 30 reconstructs both channels, and the spectral decomposition of the reduced state produces entropy and the first law. The constructions of criticality, characters, Gauss, and Pell are proved in their respective operators. The CKM reflection is obtained by conjugation of a diagonal involution, and the electroweak sector by the determinant function of the angular operator. Finally, the two cosmological realizations respectively use a temporal-periodicity radius of (108) phases and the stable Perron eigenvalue. At no step does the decimal in the fifth column intervene to choose the antecedent. This proves the stated terminality and confluence. □

53.19 Principal structural relations of genealogical confluence

The preceding corollary is not limited to enumerating rows: it determines which operators share an antecedent and the exact point at which their domains diverge. The following five configurations contain the highest-density causal relations in the volume.

53.19.1 Mode 30: Completion in Harmonic Form, Criticality, and Memory

The same integer appears in three exact constructions:

$$\begin{aligned} &90 = 3 \cdot 30, && 120 = 4 \cdot 30, && \frac{1-q^{90}}{1-q^{120}} = \frac{1+s+s^2}{1+s+s^2+s^3}, && s = q^{30}, \\ &899 = 30^2 - 1 = 29 \cdot 31, && && D = 90N_{90} + 120N_{120}. \end{aligned} \tag{53.24}$$

The first line is the $3 \to 4$ completion of the vacuum operator; the second normalizes the dodecaphasic germ without introducing π into the iteration; the third preserves how many occupations arrived through each modular channel. The common invariant is the partition of mode 30; the codomains are different. Therefore, the intersection is structural, not an identification of V, u_0 and K_{HHM}.

Completion identities for mode 30, invertibility, and incompatibility condition. The three identities in equation (53.24) are exact. The confluence is invalidated if $90/120$ cease to be consecutive multiples of the same mode, if $899 \neq 30^2 - 1$, or if the inversion $(N, D) \mapsto (N_{90}, N_{120})$ loses its determinant 30. Spectral differences do not falsify it: those differences are precisely what keep the three realizations typed.

53.19.2 log 3: residual invariant, information, and temperature

The informational confluence is summarized in the chain

$$S_{\widehat{\mathbb{T}}} = k_B \log 3, \qquad E_P = k_B \Theta_{\mathrm{clk}} \log 3, \qquad I_{\mathrm{or}} = 36 \log 3 - S(\rho_A). \tag{53.25}$$

The first member measures a uniform ternary decision; the second determines its energetic magnitude in the chronological structure 108; the third compares the capacity of 36 trits with the entropy of the reduced boundary state. The scalar is preserved, but the unit, temperature, and reduced state remain typologically separate. This separation is what permits the identity $E_P = k_B \Theta_{\mathrm{clk}} \log 3$ to be read precisely: the Planck realization supplies the quantum of energy, the TRIT realization its informational content, and the chronological structure the transducer.

Preservation of $\log 3$, dimensional separation, and incompatibility condition. The triple reading is an exact composition of equations (36.37), (48.2) y (73.8). It is invalidated if $\log 3$ is replaced by a binary entropy, if Θ_{clk} is fixed from kelvin before the identity, or if $S(\rho_A)$ is computed with a state different from that obtained by partial trace.

53.19.3 D_A: sheet determinant and Weinberg angular relation

The angular operator $T = q_+ P_+ + q_- P_-$ has two functional families:

$$F(T)^2 \longrightarrow (\widehat{\varepsilon}, \widehat{\mu}, \widehat{Z}, \widehat{c}), \qquad \det T = q_+ q_- = D_A \longrightarrow \sin^2 \theta_W = 1 - D_A. \tag{53.26}$$

The first preserves the complete spectrum of the vacuum response; the second discards the individual orientation of the sheets and preserves only their product. Hence D_A is even under $C^* \mapsto -C^*$ and can determine the quotient W/Z, whereas the impedance preserves the quotient of eigenvalues. The relation contains more information than an angular coincidence: it is an exact functional bifurcation of the same antecedent operator.

Accuracy of the angular determinant and obstructions of the electroweak interface. The spectral side and the identity $D_A = e^{-2A_{\mathrm{rad}}}$ are exact in the declared chart. The gauge realization retains its status as an interface until the basis intertwiner is constructed. A quotient of nonadjacent routes, an odd dependence on C^*, or $(m_W/m_Z)^2 \neq D_A$ falsifies that interface without affecting the vacuum operator.

53.19.4 Sectors 90 and 120: vacuum, area and modular dynamics

The channels 90 and 120 have three noninterchangeable functions: they are propagation exponents in V, multiplicities of the incidence $6 \to 8$, and spectral weights in K_{HHM}. Their ratio $4/3$ reappears in the modular temperatures,

$$\frac{120}{90} = \frac{4}{3} = \frac{\beta_{120}}{\beta_{90}},$$

but the vacuum operator is not the modular Hamiltonian, and a modular temperature is not a permeability. The confluence consists in the same genealogy preserving three magnitudes: propagation, areal occupation, and channel memory.

Finite Reconstruction of Channels 90/120 and Hypothesis of the Modular Limit. The finite reconstruction and the ratio $4/3$ are exact. The type III classification requires the infinite persistence of both channels. The disappearance of either channel from some level onward, the loss of state compatibility, or the attainment of a ratio group different from $\{\lambda_\partial^n : n \in \mathbb{Z}\}$ would falsify that classification, not the finite inversion.

53.19.5 Barbero–Immirzi parameter, entropy and CKM as realizations of the transition $6 \to 8$

The complete confluence interface is written without turning one realization into the cause of the others:

$$(6 \to 8,\ 90/120,\ C^* \mapsto -C^*) \longrightarrow \begin{cases} \gamma_{\mathrm{HMT}} \ell_P^2 \longrightarrow A_{\mathrm{phys}}, \\ \rho_A \longrightarrow (-\log \rho_A, S, I_{\mathrm{or}}, \delta S), \\ C^* \mapsto -C^* \longrightarrow R_{\mathrm{CKM}} = M\,\mathrm{diag}(1,-1,1)M^{-1}. \end{cases} \tag{53.27}$$

The first realization fixes the geometric quantum; the second measures the information of the reduced state and its areal response; the third records the sheet orientation in the mixing space. In parallel, $E_P = k_B \Theta_{\mathrm{clk}} \log 3$ fixes the informational quantum of energy. Thus gravity, entropy, and information are equated through (ℓ_P^2, E_P), without identifying γ_{HMT}, k_B, or the CKM angles.

Causal Independence of Barbero–Immirzi, Modular Information, and CKM Mixing. The output γ_{HMT} is a demonstrated antecedent result of its corresponding derivation; the area operator, purification, local first law, and CKM reflection are exact in this monograph. The infinite modular classification and the physical horizon reading retain their hypotheses. The diagram is invalidated if CKM is used to recalibrate γ_{HMT}, if an incidence hexad/octad is equated with faces/vertices without an intertwiner, or if $R_{\mathrm{CKM}}^2 \neq I$ or $\det R_{\mathrm{CKM}} \neq -1$.

53.20 Theoretical independence of the corollary of metrological confluence

The table 53.5 does not refer to an external table to complete any of its five columns. Each antecedent is defined in the body; each equation is proved or typed at the cited locator; each invariant declares what is preserved and also what is discarded; each meaning prevents identification by decimal similarity; and each terminal value is obtained thereafter in the chain. The statuses and falsification criteria remain in the extended records that follow, where they are developed without the inevitable compression of a synoptic table.

Consequently, the table does not replace the development; it complements it. It may be read synoptically to recognize the confluence, but its validity derives from the theorems and proofs in the main text, and each row is followed by an autonomous record containing antecedent, equation, meaning, value, status, and negative control.

53.21 Action, vacuum, and quantum–electric constants

Bidirectional sections of action

Antecedent. Two oriented realizations of the same line of action. **Equation.**

$$R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}}, \qquad \xi_{\mathrm{act}} = \tfrac{1}{2}\log R_{\mathrm{act}}.$$

Structure and meaning. The decade common to both fixes the type; the quotient preserves the orientation. **Value.** R_{act} and ξ_{act} in the internal chart, without comparing dimensional mantissas with residues. **Status.** *Exact internal theorem.* **Falsification criterion.** Using the decade common to both as a relative corrector or mixing the two sections.

Constitutive operator of the vacuum: positive spectrum and joint reconstruction of ε, μ, Z, c

Antecedent. The bilayer operator $T = q_+P_+ + q_-P_-$ and the genealogy $30 \to 90/120$. **Equation.**

$$V = (I - T^{90})(I - T^{120})^{-1}, \qquad \frac{1-q^{90}}{1-q^{120}} = \frac{1+s+s^2}{1+s+s^2+s^3}, \quad s = q^{30}.$$

If $\operatorname{Spec} V = \{r_+, r_-\}$,

$$\widehat{\varepsilon} = r_+^2, \quad \widehat{\mu} = r_-^2, \quad \widehat{Z} = r_-/r_+, \quad \widehat{c} = (r_+r_-)^{-1}.$$

Structure and meaning. The four constitutive relations are functions of a single positive operator; each sheet completes three segments by a fourth segment. The amplification $V_m = V \otimes I_{9^m}$ preserves the functionals under tracial expectation. **Value.**

$$\widehat{\varepsilon} = 0.99999925709663670276044161 55\ldots, \quad \widehat{\mu} = 0.99944835047932693508998155 59\ldots,$$
$$\widehat{Z} = 0.99972450853893721545555434 66\ldots, \quad \widehat{c} = 1.00027631048615752610638626 89\ldots.$$

Status. *Exact internal theorem*; nonadic amplification: *New exact formalization*. **Falsification criterion.** Loss of positivity, a residue in the identity $3 \to 4$, or variation of the invariants under the tracial expectation.

Constitutive sections of ε, μ, Z, and c in the oriented charts

Antecedent. The four preceding coefficients and the frame (u_S, u_C, u_T, u_Q). **Equation.**

$$Z = \widehat{Z}u_S u_Q^{-2}, \quad c = \widehat{c}u_C, \quad \mu = \widehat{\mu}u_S u_C^{-1} u_Q^{-2}, \quad \varepsilon = \widehat{\varepsilon}u_S^{-1}u_C^{-1}u_Q^2.$$

Structure and meaning. Permittivity and permeability do not receive units after a dimensionless calculation: they are sections of constitutive lines. **Value.** The four preceding sections, with $\mu\varepsilon = c^{-2}$ and $\mu/\varepsilon = Z^2$. **Status.** *Exact internal theorem.* **Falsification criterion.** Introducing c, μ_0, ε_0 SI before the operator or violating homogeneity.

Joint derivation of charge, the von Klitzing resistance, conductance, flux, and the Josephson constant

Antecedent. $(Z_{\mathrm{vac}}, h_a, \alpha_{\mathrm{HMT}})$. **Equation.**

$$e_a = \sqrt{\frac{2\alpha_{\mathrm{HMT}}h_a}{Z_{\mathrm{vac}}}}, \quad R_K = \frac{Z_{\mathrm{vac}}}{2\alpha_{\mathrm{HMT}}}, \quad G_0 = \frac{4\alpha_{\mathrm{HMT}}}{Z_{\mathrm{vac}}},$$

$$\Phi_{0,a} = \sqrt{\frac{h_a Z_{\mathrm{vac}}}{8\alpha_{\mathrm{HMT}}}}, \qquad K_{J,a} = \Phi_{0,a}^{-1}.$$

Structure and meaning. R_K and G_0 are bidirectional invariants; Φ_0 and K_J preserve the orientation of action with opposite exponents. **Value.** The exact expressions on their lines; $R_K G_0 = 2$ and $K_J\Phi_0 = 1$.

Status. *Exact internal theorem.* **Falsification criterion.** Using SI charge as a selector or failing $\alpha = Ze^2/(2h)$.

53.22 Constantes thermal and radiation

Relation of Boltzmann y content entropico TRIT

Antecedent. Uniform ternary decision, section of action, and chronological structure 108. **Equation.**

$$k_B\Theta_{\mathrm{clk}}\log 3 = \frac{h_{\mathrm{int}}}{108t_0} = \frac{2\pi\hbar_{\mathrm{int}}}{108t_0}, \qquad \log 3 = \frac{S_{\widehat{\mathbb{T}}}}{k_B} = \frac{E_P}{k_B\Theta_{\mathrm{clk}}}.$$

Structure and meaning. $\log 3$ is simultaneously a spectral defect, the content of a decision, and a thermal ratio; the operators remain distinct. **Value.** $\log 3 = 1.0986122886681098\ldots$ and the preceding energy–temperature section. **Status.** *Exact confluence.* **Falsification criterion.** Arbitrarily separating k_B from Θ_{clk} or inferring identity of domains from equality of the scalar.

Wien's Displacement Law

Antecedent. The Planck spectrum evaluated at Θ_{clk}. **Equation.**

$$x_5 = 5 + W_0(-5e^{-5}), \quad \frac{\lambda_{\max}}{\ell_0} = \frac{108\log 3}{x_5},$$

$$x_3 = 3 + W_0(-3e^{-3}), \quad \nu_{\max}t_0 = \frac{x_3}{108\log 3}.$$

Structure and significance. Wavelength and frequency are two spectral parameterizations with distinct Jacobians. **Value.** $\lambda_{\max}/\ell_0 = 23.896756779\ldots$ and $\nu_{\max}t_0 = 0.0237794888\ldots$. **Status.** *Exact internal theorem.* **Falsification criterion.** Confusing x_5 with x_3 or introducing kelvin before the thermal product.

Stefan–Boltzmann Law and Photon Gas

Antecedent. The same thermal scale, the bosonic continuum, and the functional $\zeta(3)$. **Equation.**

$$\mathcal{F}(\Theta_{\mathrm{clk}}) = \frac{2\pi^5}{15\,108^4(\log 3)^4}\frac{h_{\mathrm{int}}}{\ell_0^2 t_0^2},$$

$$n_\gamma\ell_P^3 = \frac{2\zeta(3)}{\pi^2(\log 3)^3}, \qquad \frac{u_\gamma\ell_P^3}{E_P} = \frac{\pi^2}{15(\log 3)^4}.$$

Structure and meaning. $\zeta(3)$ counts occupation of bosonic type; $\log 3$ fixes the TRIT informational content per cell. **Value.** The exact coefficients above. **Status.** *Exact composition with an explicit starting structure.* **Falsification criterion.** Loss of factors 2π, incorrect flow dimensions, or fusion of the mathematical provenances of $\zeta(3)$ and $\log 3$.

53.23 Criticality and dynamic constants

Dodecaphasic Profile and Normalization by 899

Resultados rectores: , and .

Antecedent. Twelve phases $3m-1$ and elementary mode 30. **Equation.** With $y = x/\sqrt{899}$,

$$u_0(x) = 1 - \frac{1}{12}\sum_{m=1}^{12}\cos\bigl(2(3m-1)y\bigr) = 1 - \frac{\cos 37y\sin 36y}{12\sin 3y},$$

$$899 = 30^2 - 1 = 29\cdot 31, \qquad u_0(x) = x^2 - \frac{715769}{2424603}x^4 + O(x^6).$$

Structure and meaning. The phase π cancels before iteration; the resulting profile is analytic, quadratic and unimodal, and has negative Schwarzian on $[-1, 1]$. **Value.** Exact rational profile and coefficients. **Status.** *Exact internal theorem.* **Falsification criterion.** A second critical point, nonnegative Schwarzian, or a residue in the trigonometric identity.

Feigenbaum Superstable Approximants and Hyperbolic Certificate of Universality

Antecedent. The superstable succession of the preceding germ. **Equation.**

$$\mathcal{R}(f)(x) = -a(f)^{-1} f\big(f(-a(f)x)\big), \qquad a(f) = -f(1).$$

Structure and meaning. The finite quotients prove that the family enters the appropriate route; universality requires the fixed point and its hyperbolicity. **Value.** $\delta_8 = 4.66921218915\ldots$, $\alpha_8 = 2.50296847988\ldots$. **Status.** Approximants: *Finite certificate*; universal limit: *Program with falsification criterion*. **Falsification criterion.** More than one unstable eigenvalue, an unbounded tail, or null transversality.

53.24 Constantes espectrales, aritmeticas and of vacancia

Internal complex structure and Catalan's constant

Antecedent. $J^2 = -I$ and the character $\chi_J(n) = i^{n-1} = \chi_{-4}(n)$ on odd integers. **Equation.**

$$G = L(2, \chi_J) = \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)^2}.$$

Structure and meaning. Catalan's constant is the quadratic moment of the torsion quarter. **Value.** $G = 0.915965594177219\ldots$. **Status.** *Exact internal theorem.* **Falsification criterion.** Failure of multiplicativity or discrepancy with χ_{-4}.

Dodecaphase character and real quadratic field

Antecedent. CRT Product $\chi_{12} = \chi_{-3}\chi_{-4}$. **Equation.**

$$L(1, \chi_{12}) = \frac{\log(2+\sqrt{3})}{\sqrt{3}}, \qquad L(2, \chi_{12}) = \frac{\pi^2}{6\sqrt{3}},$$

$$\gamma_{\mathbb{Q}(\sqrt{3})} = \gamma + \frac{L'(1, \chi_{12})}{L(1, \chi_{12})}.$$

Structure and meaning. The triadic orientation and the torsional quarter produce the real character of $\mathbb{Q}(\sqrt{3})$; Euler–Kronecker is the finite term of its Dedekind zeta function. **Value.** 0.7603459963009463 . . ., 0.9497031262940094 . . ., and 1.0539656082484825 8 . . .. **Status.** *Exact internal theorem*; the composite identity preserves the declared hypotheses and normalizations. **Falsification criterion.** Residue pattern different from $(+, -, -, +)$ or failure of the Dedekind factorization.

Bendersky–Glaisher Family

Antecedent. Renormalized triadic zeta functional. **Equation.**

$$A_k = \exp\left(\frac{H_k B_{k+1}}{k+1} - \mathcal{Z}_3'(-k)\right), \quad \zeta(2n+1) = (-1)^{n+1} \frac{2(2\pi)^{2n}}{(2n)!} \log A_{2n}.$$

Structure and meaning. Apery and the odd zeta values are levels of a tower, not isolated coincidences. **Value.** $A_0 = \sqrt{2\pi}$, $A_1 = A_{\mathrm{GK}}$, and the preceding exact family. **Status.** *Exact internal theorem.* **Falsification criterion.** Failure of A_0, A_1 or of the identity for $\zeta(2n+1)$.

Gauss dynamical system: K, L, λ_{GKW}, and transported cylinders

Antecedent. Evaluation of HMT cylinders and measure $d\mu_G = dx/[\log 2(1+x)]$. **Equation.**

$$\log K = \int \log a_1 \, d\mu_G, \quad L = -\int \log x \, d\mu_G = \frac{\pi^2}{12\log 2},$$

$$h_G = 2L = \frac{\pi^2}{6\log 2}, \qquad \Lambda_b = \frac{6\log 2 \log b}{\pi^2}.$$

Structure and meaning. Khinchin measures digits; Lévy measures denominator growth; Lochs compares coding rates. **Value.**

$$K = 2.685452001065306\ldots, \qquad L = 1.18656911041562545\ldots,$$
$$h_G = 2.37313822083125091\ldots, \qquad \Lambda_{10} = 0.97027011439203393\ldots,$$
$$\Lambda_{729} = 2.77761896637430578\ldots.$$

Status. Cofinality and identities: *Exact in the transported system*; GKW constant: *Open program with a falsification criterion*. **Falsification criterion.** Failure of cylinder cofinality, non-invariant measure, or absence of a spectral gap in the declared space.

Aritmetic cycles of Meissel–Mertens and regularized prime series

Antecedent. Irreducible-cycle selector and exact factorization. **Equation.**

$$B_1 = \lim_{x\to\infty}\left(\sum_{p\le x}\frac{1}{p} - \log\log x\right), \qquad C_2 = \prod_{p>2}\frac{p(p-2)}{(p-1)^2}.$$

Structure and significance. B_1 is a regularized local sum; C_2 a multiplicative density of pairs. **Value.** $B_1 = 0.2614972128476428\ldots$, $C_2 = 0.6601618158468696\ldots$. **Status.** *Demonstrated with explicit initial structure*: declared prime selector and tail bound. **Falsification criterion.** Using zeros or decimal digits to choose the primes, or asserting the limit without executing the tail bound.

Vacancies, Holonomy, and Stieltjes Constants

Antecedent. Mechanical word of vacancies and bounded discrepancy. **Equation.**

$$\varepsilon_{\mathrm{vac}} = \log_{10}\frac{10}{9}, \quad \gamma^{+}_{\mathrm{vac}} = \lim_{N}\left(\sum_{n\le N}\frac{z_n}{n} - \varepsilon_{\mathrm{vac}}\log N\right),$$

$$\rho_{\varepsilon}(k) = e^{2\pi i k \varepsilon_{\mathrm{vac}}}, \qquad \gamma^{+}_{\mathrm{vac,j}} = \varepsilon_{\mathrm{vac}}\gamma_j + \sum_{n\ge 1}\frac{(z_n - \varepsilon_{\mathrm{vac}})(\log n)^j}{n}.$$

Structure and meaning. Density, finite term, and phase are three types; the tower prolongs the finite term. **Value.** $\varepsilon_{\mathrm{vac}} = 0.0457574905606751\ldots$; the remainder are determined by their certified limits. **Status.** *Exact internal theorem with discrepancy*. **Falsification criterion.** Loss of the discrepancy bound or disagreement at level $j = 0$.

Pell nonadic lift

Antecedent. $r^2 = 2 \pmod{3^k}$. **Equation.**

$$u = 3 + 2r, \qquad \mathrm{ord}(u) = 4\cdot 3^{k-1}, \qquad \varprojlim\langle u\rangle = C_4 \times \mathbb{Z}_3.$$

Structure and meaning. The quarter-turn persists at every depth; the Archimedean realization is $(1+\sqrt{2})^2$. **Value.** $3+2\sqrt{2} = 5.82842712474619\ldots$. **Status.** *Exact internal theorem.* **Falsification criterion.** Different order or incompatible lifts.

53.25 Gravity, area and modular information

Gravitational constant and five-component tensor representation

Antecedent. Chronological structure 108, action, and central projector of rank five. **Equation.**

$$G_a(\theta) = \theta \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a}, \qquad \theta_{108} = \left(\frac{54}{\pi}\right)^2, \qquad \mathcal{G}_5 = G_a(\theta_{108}) P_5.$$

Structure and meaning. G is a section selected by the Compton–gravity closure; P_5 determines a five-component carrier, which the TT reduction takes to two massless helicities. **Value.** $G_{\mathrm{int}} = \theta_{108} c^5 t_0^2/\hbar$ and $\mathrm{Spec}\,\mathcal{G}_5 = \{G \text{ with multiplicity } 5, 0 \text{ with multiplicity } 7\}$. **Status.** Selector: *Exact internal theorem*; carrier realization: *Typed physical realization.* **Refutation criterion.** Another positive root, $\mathrm{Tr}\,P_5 \neq 5$, or confusing five components with five massless physical polarizations of GR.

Planck system, reciprocity $G \leftrightarrow \hbar$, and areal invariance

Antecedent. The bidirectional product $G_a \hbar_a$ and temporal periodicity of 108 phases. **Equation.**

$$\ell_P^2 = \frac{G\hbar}{c^3} = \theta_{108} \ell_0^2, \quad t_P = \frac{54}{\pi} t_0, \quad E_P = \frac{\pi}{54} \frac{\hbar}{t_0},$$

$$F_P = \frac{c^4}{G}, \qquad P_P = \frac{c^5}{G}, \qquad \rho_P = \frac{\hbar}{\theta_{108}^2 c^5 t_0^4}.$$

Structure and significance. The Planck area is invariant under the splitting of action and links gravity to the thermal scale. **Value.** The preceding exact sections. **Status.** *Exact internal theorem.* **Falsification criterion.** Combining G and $\hbar$ from incompatible sheets or failing homogeneity.

Area operator and modular Hamiltonian: $\widehat{\mathcal{A}}_{\mathrm{HMT}}$, H_{mod}, and channels $90/120$

Antecedent. Incidence $6 \to 8$, channels $90/120$, output γ_{HMT} and ℓ_P^2. **Equation.**

$$\frac{A_{\mathrm{phys}}}{\ell_P^2} = \gamma_{\mathrm{HMT}} a_0 (N_{90} + N_{120}), \qquad K_{\mathrm{HHM}} = \Delta_4^{\mathrm{mod}} (90 N_{90} + 120 N_{120}) + cI.$$

If $N = N_{90} + N_{120}$ and $D = 90N_{90} + 120N_{120}$,

$$N_{90} = 4N - D/30, \qquad N_{120} = D/30 - 3N.$$

Structure and meaning. Area determines total occupancy; the Hamiltonian preserves channel provenance. Together they reconstruct the boundary. **Value.** $\gamma_{\mathrm{HMT}} a_0 = 0.0016755940749224991\ldots$ in the declared chart. **Status.** *Exact internal theorem at finite cutoff*; γ_{HMT} is not rederived. **Falsification criterion.** Nonzero commutator, lost determinant 30, or use of the adjective holographic without transport to the boundary.

Relative entropy, double thermofield purification, and first law modular areal

Antecedent. Faithful reduced state of the boundary and its Schmidt purification. **Equation.**

$$I_{\mathrm{or}} = 36\log 3 - S(\rho_A), \quad |\mathrm{TFD}\rangle = \sum_n \sqrt{p_n}|n\rangle|n\rangle,$$

$$\delta S = \beta_{90}\delta\langle\mathcal{A}_{90}\rangle + \beta_{120}\delta\langle\mathcal{A}_{120}\rangle, \qquad \frac{\beta_{120}}{\beta_{90}} = \frac{4}{3}.$$

Structure and meaning. Purification converts reduced entropy into entanglement; the first law equates modular memory and area. **Value.** $I_{\mathrm{or}} = 0.0016691832338689407\ldots$, $\beta_{90} = 5.97264854010709\ldots$, $\beta_{120} = 7.96353138680946\ldots$. **Status.** *Exact local internal theorem.* **Falsification criterion.** Partial trace different from ρ_A, $I_{\mathrm{or}} < 0$, or use for finite variations without the relative-entropy term.

Modular limit and type III parameter

Antecedent. Local states compatible with infinite persistence of both channels. **Equation.**

$$K_{\mathrm{HHM}}^{(m)} = -\log\rho_m, \qquad \lambda_\partial = \exp(-30\Delta_4^{\mathrm{mod}}).$$

Structure and meaning. At infinity there exists a local net or the GNS modular generator, not a global density matrix within the UHF algebra. **Value.** $\lambda_\partial = 0.9966696464689573786\ldots$. **Status.** Net: *Operator type rule*; classification: *Theorem under persistence hypotheses*. **Falsification criterion.** Asymptotic absence of a channel, a nonproduct state, or a different asymptotic ratio.

53.26 Sheet involution, mixing, and electroweak sector

Leaf Involution and Rational CKM Transformation: Unitarity, Determinant, and Jarlskog Character

Antecedent. Exchange $C^* \mapsto -C^*$ and datum $(h_6, o_8, t) = (6, 8, 3)$. **Equation.**

$$R_{\mathrm{CKM}} = M\,\mathrm{diag}(1,-1,1)M^{-1}, \quad R_{\mathrm{CKM}}^2 = I, \quad \det R_{\mathrm{CKM}} = -1,$$

$$-\frac{o_8 - h_6}{t} = -\frac{2}{3}, \quad A = \frac{1528\theta_{12} + 3380\theta_{23} + 801\theta_{13}}{3857},$$

$$C^* = 6\frac{75\theta_{12} + 176\theta_{23} - 150\theta_{13}}{3857}.$$

Structure and meaning. Barbero and CKM are distinct functional realizations of the change $6 \to 8$, adopted as a common antecedent; the involution induces a rational rank-one odd reflection. **Value.** Exact rational matrix, $\det M = -3857/6$; the CP phase is fixed under this involution, which is not CP conjugation. **Status.** *Exact genealogical/internal.* **Falsification criterion.** $R^2 \neq I$, failure of $RM = MD$, loss of $-2/3$, or presenting CKM as the retrospective cause of γ_{HMT}.

Angular determinant, Weinberg parameter and electroweak couplings

Antecedent. Two functions of the same angular operator: $F(T)^2$ and $\det T$; adjacent W/Z routes. **Equation.**

$$D_A = \det T = e^{-2A_{\mathrm{rad}}}, \qquad \sin^2\theta_W^{\mathrm{OS}} = 1 - D_A,$$

$$g = 2\sqrt{\frac{\pi\alpha}{1-D_A}}, \qquad g' = 2\sqrt{\frac{\pi\alpha}{D_A}}.$$

Structure and meaning. The vacuum and Weinberg are distinct spectral realizations of a common operator; the determinant is even under the change of sheet.

Value. $D_A = 0.7751291192471564302\ldots$ and $\sin^2\theta_W = 0.2248708807528435698\ldots$.

Status. Angular relation: *Exact internal theorem in the chart*; gauge realization: *Conditional realization*. **Falsification criterion.** Nonadjacent routes, a mass quotient different from D_A, or an undeclared gauge scheme.

Fermi Constant and Higgs Coupling

Antecedent. α_{HMT}, D_A, routes W/H, and the electroweak chart of the tree. **Equation.**

$$G_F^{(0)} = \frac{\pi\alpha}{\sqrt{2}(1-D_A)m_W^2},$$

$$\lambda_H = \frac{\pi\alpha}{2(1-D_A)} \exp\left(6A_{\mathrm{rad}} + \frac{10}{3}C^*_{\mathrm{rad}}\right), \qquad G_F^{(0)} m_H^2 = \sqrt{2}\lambda_H.$$

Structure and meaning. Fermi is the effective low-energy limit of the charged channel; Higgs preserves the orientation that the gauge determinant eliminates. **Value.** $G_F^{(0)} = 1.1157162817 \times 10^{-5}\,\mathrm{GeV}^{-2}$ in the declared chart and $\lambda_H = 0.123847911548\ldots$. **Status.** *Tree-level electroweak interface*. **Refutation criterion.** Using observed G_F to choose Δr, omitting scheme/scale or selecting λ_H from the observed mass.

Leptonic magnetic anomaly

Antecedent. $(\alpha, \varphi, 1000, 54+1)$ with coefficients frozen before the contrast. **Equation.**

$$a_\mu = \frac{\alpha}{2\pi} + \frac{\alpha(\varphi-1)}{1000} + \frac{\alpha^2}{1000(54+1)}. \tag{53.28}$$

Structure and meaning. It is an evaluation of three HMT functionals, not a substitution for the complete QED radiative series. **Value.** $a_\mu = 0.001165920713\ldots$. **Status.** *Exact evaluation with open local provenance*. **Falsification criterion.** Selecting 1000 or 55 from the experimental value.

53.27 Determination of Cosmological Parameters via Torsional Memory and Perron Dynamics

de Sitter realization through the temporal periodicity of 108 phases

Antecedent. Circular selector $R_{108} = \ell_P$ and de Sitter realization. **Equation.**

$$H_{108} = \frac{\pi}{54t_0}, \quad \Lambda_{108} = \frac{\pi^2}{972\ell_0^2}, \quad \frac{T_{\mathrm{dS}}}{\Theta_{\mathrm{clk}}} = \frac{\log 3}{2\pi}, \quad \frac{S_{\mathrm{dS}}}{k_B} = \pi.$$

Structure and significance. A single radius determines expansion, curvature, temperature, and entropy. **Value.** The preceding exact quadruple. **Status.** *Conditioned physical realization.* **Falsification criterion.** A non-de Sitter background or a PCH realizer selecting another radius.

Perron–Einstein–Cartan Realization

Antecedent. Scale $V_n = \varphi^n$, $D = 3$, stable mode φ^{-2n}. **Equation.**

$$\frac{a_n}{a_0} = \varphi^{n/3}, \qquad M_n = \left(\frac{a_n}{a_0}\right)^{-6}, \qquad H_\varphi = \frac{\log\varphi}{3t_0}.$$

Structure and meaning. The exponent -6 is active memory and coincides with the Einstein–Cartan dilution that is quadratic in spin. **Value.** $H_\varphi t_0 = 0.1604039416865344825\ldots$ and the exact law $M_n a_n^6 = a_0^6$. **Status.** Scale: *Exact internal theorem*; spin reading: *Conditional realization*. **Falsification criterion.** A different exponent, a nonadditive temporal parameter, or a physical amplitude selected from a bounce.

53.28 Typed separation between the Bohr and Born-Infeld radii and the internally generated scales

Bohr radius as an electrodynamic scale derived from α

Antecedent. Action, vacuum, α, and electronic mass from its antecedent. **Equation.**

$$a_0 = \frac{\hbar}{m_e c \alpha}. \tag{53.29}$$

Meaning and value. It is a descending composition, not an independent selector of this monograph. **Status.** *Out of scope*. **Falsification criterion.** Recalculating m_e here or using a_0 to select α.

Born-Infeld Scale and Recovery of the Maxwell Limit

Antecedent. $(h_{\mathrm{int}}, c_{\mathrm{int}}, e_{\mathrm{int}}, \ell_0)$. **Equation.**

$$E_* = \frac{h_{\mathrm{int}} c_{\mathrm{int}}}{e_{\mathrm{int}} \ell_0^2}. \tag{53.30}$$

Structure and meaning. The expression constructs a field scale; it does not yet construct a Born–Infeld action. **Value.** Exact section E_*, without determination of a physical parameter. **Status.** *Open program*. **Falsification criterion.** Identification by name or decimal without a determinantal action and selector.

Proposition 53.32 (Main Result)**.** *The metrological result is not the isolated decimal. It is the preservation of a genealogy through operators, lines of magnitude, and charts, until the real value appears in the last position of the record.*

53.29 Existence, uniqueness, and compatibility theorems of the genealogical constants

The construction determines an autonomous sector of intrinsic constants. Its compositions, cross-theorems, and certificates form a self-contained chain of proof. The governing order is

$$\mathrm{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow \text{functional realization} \longrightarrow \text{invariant} \longrightarrow \text{section} \longrightarrow \text{terminal value}. \tag{53.31}$$

In that order, five main closures are obtained.

1. The electromagnetic vacuum ceases to be a collection of four formulas: it is a positive two-sheet operator whose spectrum determines ε, μ, Z, c, with compatible nonadic amplification.
2. Criticality no longer begins with the Feigenbaum decimals: the dodecaphase cancels π, mode 30 produces 899, the germ closes trigonometrically and demonstrably enters the appropriate unimodal analytic class. Universal hyperbolicity is isolated as a distinct certificate.
3. Gravity ceases to be merely a decimal chart: the temporal periodicity of 108 phases selects a section of L_G, the rank-five projector fixes the tensor carrier, and TT reduction distinguishes that carrier from the two massless helicities.
4. The hexad–octad transition acquires three related realizations: areal quantum, modular memory, and mixing reflection. The modular holographic Hamiltonian is central to that interface of confluence, but it does not subsume the vacuum, criticality, gravity, arithmetic, or cosmology under a title that is artificially unique.
5. The less common constants include Catalan, Levy, Lochs, and Euler–Kronecker. Bendersky–Glaisher, vacancies, and cosmological realizations likewise appear as values of operators and dynamics, not as a list for which retrospective coincidences are sought.

53.30 Theorematic character of the metrological corollary

The corollary 53.31 and the table 53.5 constitute the probative closure of the development. They are not an abridged summary of the preceding material. For a row to enter that corollary, it must simultaneously preserve

$$\text{antecedent} \mid \text{equation} \mid \text{invariant} \mid \text{meaning} \mid \text{terminal value}.$$

If any of these five components is removed, the row ceases to be an HMT output and becomes a mere numerical equivalence. The status and the falsification criterion are not substitute columns: they audit from outside the strength and domain of the complete quintuple.

This formulation produces three structural and mathematical consequences. First, the terminal value may be an exact expression on a magnitude line; reduction to SI is not required in order to recognize it. Second, an antecedent may admit several functional realizations —for example, T determines both the vacuum response and D_A— without those realizations being identified. Third, a shared integer such as 30, a shared ratio such as $4/3$, or a shared scalar such as $\log 3$ establishes confluence only when the map transporting it and the invariant it preserves are exhibited.

In particular, the atlas allows one to read two closure equations that were previously scattered:

$$\boxed{E_P = k_B \Theta_{\mathrm{clk}} \log 3}, \qquad \boxed{(6 \to 8) \longrightarrow \left(\gamma_{\mathrm{HMT}} \ell_P^2,\ -\log \rho_A,\ R_{\mathrm{CKM}}\right).} \tag{53.32}$$

The first brings together a quantum of energy, temperature, and information. The second determines, from a common confluence interface, a geometric quantum, a modular generator, and a mixing reflection. No equation asserts that its components are the same object; it asserts that their genealogies have been preserved to the exact point of branching.

The negative control is also immediate. If E_P is used to select k_B, if CKM is used to recalibrate γ_{HMT}, if $\log 3$ is confused with a density matrix, or if equality of two decimals replaces the intertwiner, the corollary does not apply. The autonomy of the derivation lies in the fact that these controls, like the derivations, are formulated within the development itself.

53.31 Logical stratification of internal identities, dimensional realizations, and external recognitions

The strength of an assertion is not deduced from the extraordinary nature of its conclusion or from its proximity to a known value. It is deduced from the chain that sustains it.

Status	Content of the development
theorem internal exact	Vacuum operator and its constitutive quantities; mode 30; charge and quantum quotients; thermal scale; closed chaos profile; lattice $3/4/12$; Gauss identities; selector θ_{108}; Planck family; CKM reflection; law $M_n = (a_n/a_0)^{-6}$ in its chart.
Exact mathematical composition	Wien, Stefan–Boltzmann, photon gas, Euler–Kronecker, Bendersky–Glaisher, Weinberg angular relation, joint area–memory reconstruction and Higgs self-coupling at the tree-level interface.
Demonstrated with Explicit Starting Structure	Results that depend on a declared chart, an orientation, a measure, a boundary state, or a boundary datum, without introducing the terminal value as a selector.
theorem with hipotesis explicit	Classification of the GNS closure as a hyperfinite factor III_{λ_∂}, under faithfulness, compatibility, and infinite persistence of the two channels $90/120$.

Conditional realization in physics	Complete identification of gauge, renormalized G_F, de Sitter realization of the CT108 radius, transduction of Perron memory into spin density, and dynamic realization of the tensor carrier.
Open Programme with Falsification Criterion	Full hyperbolicity of the Feigenbaum renormalization operator, subdominant Gauss–Kuzmin–Wirsing eigenvalue, and construction of an internally selected Born–Infeld action.
External metrological contrast	SI charts, CODATA/PDG comparisons, classical values of the constants and operatorial classifications applied only after constructing the HMT weights.

An open interface does not diminish the internal theorem that precedes it. Conversely, an internal theorem does not automatically promote a physical interpretation whose transducer has not yet been constructed.

53.32 Transverse obstructions of typing, causality, orientation, and nonadic compatibility

The construction is subject to refutation. The following controls traverse all the chapters.

F1. **Dimensional typing.** An identity that is not homogeneous in the lines L_S, L_C, L_T, L_Q is discarded, although it reproduces a decimal.

F2. **No feeding by the target value.** No SI, CODATA, PDG, Feigenbaum, GKW, H_0, G_F, or magnetic anomaly value may select an internal operator, coefficient, or realization.

F3. **Sheet convention.** Silently reversing q_+ and q_-, or conflating sheet involution with CP conjugation, invalidates the parities and quotients.

F4. **Nonadic compatibility.** A functional that changes under amplification 9^m, loses the unit, or fails to preserve return memory does not possess the claimed limit.

F5. **No fusion by scalar.** The equality of two terminal values does not identify their domains. An intertwiner is required that commutes with the operators and preserves orientation, carry, and boundary.

F6. **Finite–infinite separation.** A certificate through level eight does not prove universal hyperbolicity; a finite modular Hamiltonian is not $-\log \rho_\infty$ within the UHF; numerical convergence without a tail bound does not certify a prime product.

F7. **Physical transducer.** The de Sitter, Einstein–Cartan, Born–Infeld, or gauge reading is blocked unless it determines the complete object of the target theory and its boundary conditions.

53.33 Necessary Conditions for Mathematical Extensions and Non-Equivalent Physical Realizations

No generic caveat such as "HMT still needs to be related to physics" remains. Localized obligations remain:

CHAOS-OBL-1	Certify, in a space of even analytic functions, the fixed point of $\mathcal{R}$, a unique unstable eigenvalue, the tail bound, and the transversality of the HMT germ.
GAUSS-OBL-1	Fix the functional space of the Gauss operator, isolate the subdominant eigenvalue with intervals, and prove stability of the projector under the partitions 9^m.
HMOD-OBL-1	Verify compatibility of the local states and positive-density persistence of both channels before applying the Araki–Woods classification. Do not use a nonexistent global trace-class density.

EW-OBL-1 Construct the intertwiner between the HMT sheets and the neutral/charged electroweak basis, and derive Δr without using the experimental value of G_F.

COS-OBL-1 For de Sitter, prove that the PCH realizer dynamically selects R_{108}. For Einstein–Cartan, construct the amplitude, current, sign and dimensional chart of the map $M_n \mapsto s_n^2$.

BORN-OBL-1 Construct the Born–Infeld determinantal action and prove that the scale E_* is its parameter, rather than identifying it by dimensional considerations.

ANOM-OBL-1 Close the local provenance of 1000 and $54+1$ before promoting the evaluation of a_μ to a complete derivation.

Each obligation has an object, a domain, and a falsification criterion. None authorizes the reopening of the exact blocks that precede it.

53.34 Formal Delimitation of the Domain of Genealogical Derivations

- It does not rederive $\pi, e, \varphi, \alpha_{\mathrm{HMT}}$; it uses them as already constructed inputs when a subsequent composition requires them.
- It does not compete with the General Mass Law or recalculate the electron mass.
- It does not rederive γ_{HMT}; it develops its areal, informational, and modular meaning.
- It does not identify the five-component carrier with five polarizations of a massless graviton.
- It does not conflate a finite Feigenbaum approximation with the universal theorem, nor a dimensional scale with a Born–Infeld action.
- It does not turn the modular holographic Hamiltonian into the name or the exclusive explanation of all constants. It is a central component of the area–information–mixing confluence interface within a broader atlas.

53.35 Reconstructive power of the common genealogy over mathematical and physical constants

The metrological singularity of HMT–MD does not consist in producing many decimals. It consists in the real value appearing at the end of a sequence that conserves memory. Permeability and permittivity are the two quadratic eigenvalues of the same response; G is the section that closes a Compton radius with a gravitational radius; $\log 3$ is simultaneously decision content and thermal ratio; Catalan is a moment of the quarter-torsion; γ_{HMT} orients the quantum of area; the modular Hamiltonian conserves which channel produced the occupation; the CKM reflection conserves the coordinate responsible for the change of sheet; the exponent -6 conserves stable memory when determining the cosmological scale.

Proposition 53.33 (Main Result)**.** *The structural signification demonstrated in this development is expressed through a realization rule: the same antecedent may admit several functionals without losing its genealogy, and each one simultaneously determines the meaning of the constant and the reason for its terminal value.*

The final assertion is therefore more precise than "HMT reproduces constants". HMT–MD constructs operators, characters, sections, and transducers of which constants are evaluations. Where the chain is closed, the value and its meaning are proved together. Where an interface remains, the falsification criterion permits work to continue without downgrading what has already been established or presenting as a theorem what is still a programme.

Chapter 54

Genealogical completeness of HMT sections and decidability of each finite level

54.1 Conservative Extension of Finite Completeness via Projective Multiplicities

A concrete HMT tower consists of finite sets, transition relations, restriction maps, and readers. All of this can be encoded in ZF or ZFC. The collection of all towers constitutes a definable class; it is unnecessary to postulate that HMT already forms an autonomous class theory comparable to NBG.

Proposition 54.1 (Conservative Coding)**.** *Every finite or countable genealogical object used in this notebook can be represented by sets and relations in ZFC. Therefore, the formalizations constructed here do not prove a contradiction in ZFC.*

Proof. Each level X_m is a set, each restriction $p_{n,m}$ is a subset of $X_n \times X_m$, and a countable tower is a function with domain $\mathbb{N}$ encoding these data. Types, readers, and memories are added as relations or functions. The inverse limit is a subset of the product $\prod_m X_m$, which ZFC constructs by Replacement, Union, Power Set, and Separation. □

What HMT calls into question is not the legitimacy of that encoding, but the practice of forgetting the maps and retaining only a shadow. If, for example, one applies the reader

$$L : \Omega_\infty \longrightarrow \mathbb{R}$$

or the cardinal profile

$$\mathbb{X} \longmapsto (|X_m|)_{m \geq 0},$$

nonisomorphic objects may be collapsed. The projection is valid in its codomain; what would be invalid is to attribute faithfulness to it without proof.

Remark 54.2 (Conditions of validity). It is not claimed that HMT internally satisfies every schema of ZF, IZF, or NBG. Separation, Replacement, the full Power Set axiom, and a general transfinite hierarchy have not been verified jointly within an autonomous HMT axiomatics.

54.2 Category of genealogical objects

Definition 54.3. A genealogical object is

$$\mathbb{X} = \big((X_m)_{m \geq 0}, (p_{n,m})_{n \geq m}, \mathcal{T}, \mathcal{O}, \mathcal{M}\big),$$

where $\mathcal{T}$ records types, $\mathcal{O}$ observables, and $\mathcal{M}$ memories. A morphism $f : \mathbb{X} \to \mathbb{Y}$ is a family $f_m : X_m \to Y_m$ satisfying

$$p^Y_{n,m} f_n = f_m p^X_{n,m}$$

and preserves the indicated data.

The category thus obtained is a concrete presentation of enriched profinite objects. The global-sections functor

$$\Gamma(\mathbb{X}) = \varprojlim_m X_m$$

recovers the complete histories. The small path category of TPK also admits the presheaf envelope

$$\widehat{\mathcal{P}_{\mathrm{TPK}}} = \mathbf{Set}^{\mathcal{P}_{\mathrm{TPK}}^{op}}.$$

This construction provides contextual logic and data sheaves, but does not warrant declaring Booleanity, choice, or enough points without studying the specific topos.

54.3 Foundations of Prefixes and Coinduction of Sections

The prefix graph is graded by depth:

$$\mathrm{rk}(x_m) = m.$$

Applying the parent strictly decreases the rank. There are no parental cycles.

$$x_m \longrightarrow x_{m-1} \longrightarrow \cdots \longrightarrow x_m$$

compatible with that graduation.

Proposition 54.4. *Well-foundedness of the prefix graph and coinductive existence of an infinite section are compatible.*

Proof. Well-foundedness concerns the direction of descent through parents. Coinduction concerns a family (x_m) compatible with all descents. No x_m belongs to itself or reappears as the same node; the section is a function on $\mathbb{N}$, not a membership cycle. □

This distinction clarifies the return $9 \longrightarrow 1$. The phase may recur, but the memory coordinate increases. A loop in the phase projection is not a cycle in the graded state graph.

54.4 Fibration of ordinals by compatible genealogical histories

Von Neumann represents each ordinal as the set of its predecessors, [127]. HMT may lift that representation without modifying it.

Definition 54.5 (Genealogical elevation of a finite ordinal)**.** Let $\boldsymbol{x} = (x_m)$ be a section. Define

$$\widehat{n}_{\boldsymbol{x}} = \{(k, x_k) : k < n\}.$$

Proposition 54.6. *The projection onto the first coordinate satisfies*

$$\pi(\widehat{n}_{\boldsymbol{x}}) = n,$$

and the sequence obeys

$$\widehat{n+1}_{\boldsymbol{x}} = \widehat{n}_{\boldsymbol{x}} \cup \{(n, x_n)\}.$$

Proof. Both identities follow directly from the definition. The added information does not alter the underlying order; it distinguishes the routes that realize the same ordinal. □

For ω one writes

$$\widehat{\omega}_{\boldsymbol{x}} = \{(n, x_n) : n \in \mathbb{N}\}.$$

The support of each history remains countable, although the space of all histories may have continuum cardinality when branching persists.

Remark 54.7 (Extension and open domain). The extension

$$S_{\alpha+1} = \mathrm{Ext}_\alpha(S_\alpha), \qquad S_\lambda = \varprojlim_{\beta<\lambda} S_\beta$$

is a well-typed transfinite programme, not a result already proved for every ordinal. The coherence of the limits, the sizes of the levels, and the existence of extensions must be verified.

54.5 Decategorification Functor and the Cardinality of Published Sections

Two towers can satisfy

$$|X_m| = |Y_m| \qquad \text{for every } m$$

and fail to be isomorphic: it suffices that their restriction maps have distinct branching patterns. The cardinal profile is therefore a useful but nonfaithful invariant.

Remark 54.8 (Structural Interpretation). Counting how many cells exist does not say which cell prolongs which, which orientation is reversed, which carry is transported, or which state returns. The cardinal measures the inventory; the genealogy measures the organization.

Remark 54.9 (Separation of results and scope). The ZFC coding, the section as a limit, and the distinction between observational equality and state identity are *results established in the preceding sections*. The unified category, the presheaf topos, and the fibered ordinal are *explicit constructions of the present development*. No complete autonomous HMT theory of classes or transfinite ordinals is proclaimed.

54.6 Profinite limit of the system of genealogical truncations

Let

$$X = \varprojlim_m X_m$$

an inverse limit of finite discrete sets with surjective maps. We equip X with the initial topology of the projections $\pi_m : X \to X_m$. The cylinders

$$[A]_m = \pi_m^{-1}(A), \qquad A \subseteq X_m,$$

are clopen and form a basis.

Theorem 54.10 (Cylindrical power)**.**

$$\mathrm{Clop}(X) \cong \varinjlim_m \big(\mathcal{P}(X_m), p_{n,m}^{-1}\big).$$

Proof. Every cylinder is clopen. Conversely, if $C \subset X$ is clopen, then for each $x \in C$ there is a cylinder contained in C. Compactness of C gives a finite subcover. Lifting those cylinders to a common depth m yields $C = \pi_m^{-1}(A)$ for some $A \subseteq X_m$. Two representatives at different depths define the same clopen set exactly when they agree after a common lift. □

Theorem 54.11 (Closed-form power)**.**

$$\mathrm{Closed}(X) \cong \varprojlim_m \big(\mathcal{P}(X_m), p_{n,m*}\big),$$

where $p_*(A) = p(A)$.

Proof. If $F \subset X$ is closed, it is compact. Its projections $F_m = \pi_m(F)$ satisfy $p_{n,m}(F_n) = F_m$. Conversely, a compatible family (F_m) defines the inverse limit

$$F = \{x \in X : \pi_m(x) \in F_m \text{ for every } m\}.$$

It is an intersection of closed sets and, by the surjective compatibility of the restrictions $F_n \to F_m$, its projections are exactly F_m. The two constructions are inverse. □

54.7 Obstruction to the Reconstruction of the Genealogical Power from Finite Reductions

In a Cantor space,

$$|\operatorname{Clop}(X)| = \aleph_0, \qquad |\operatorname{Closed}(X)| = 2^{\aleph_0}, \qquad |\mathcal{P}(X)| = 2^{2^{\aleph_0}}.$$

The three objects cannot be identified.

This separation does not presuppose the continuum hypothesis. Writing $\mathfrak{c} = 2^{\aleph_0}$,

$$|\operatorname{Clop}(X)| = \aleph_0, \qquad |\operatorname{Closed}(X)| = \mathfrak{c}, \qquad |\mathcal{P}(X)| = 2^{\mathfrak{c}}.$$

CH identifies only $\mathfrak{c}$ with $\aleph_1$; it does not collapse any of the three layers. Gödel–Cohen independence changes which cardinal may occupy $\mathfrak{c}$, not Cantor's inequality $\mathfrak{c} < 2^{\mathfrak{c}}$.

Proposition 54.12 (Dense falsifier)**.** *If $D \subsetneq X$ is dense, then*

$$\pi_m(D) = X_m \qquad \text{for every } m,$$

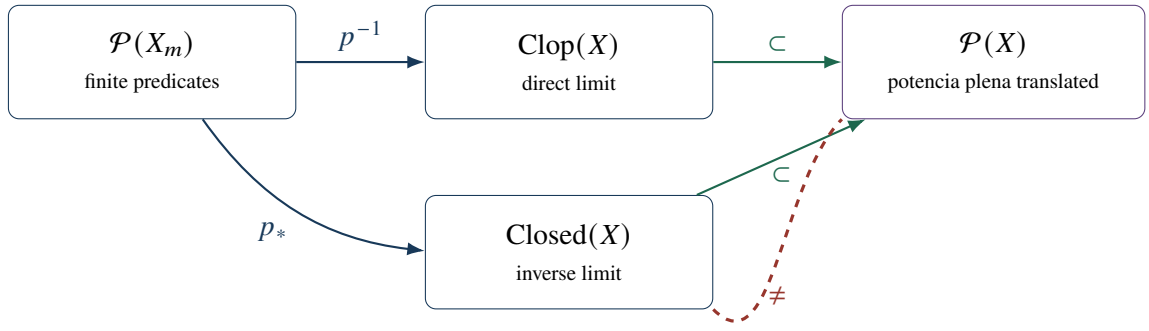
so that D and X possess the same finite shadows.

Proof. Every nonempty cylindrical fiber $\pi_m^{-1}(a)$ is open. By density, it intersects D. Hence $a \in \pi_m(D)$ for every $a \in X_m$. □

Finite shadows reconstruct the closure of a subset, not the arbitrary subset itself. This is the precise point at which an identification

$$\varprojlim_m \mathcal{P}(X_m) = \mathcal{P}(\varprojlim_m X_m)$$

would become false.



A proper dense subset and X have the same finite projections: finite history recovers closure, not the full power set.

Figure 54.1: Three notions of power that must not be flattened.

54.8 Necessary choice conditions for the global prolongation of compatible sections

Let $p : X \twoheadrightarrow Y$. It is useful to distinguish:

1. a *section conjuntista* $s : Y \to X$ With $p s = \mathrm{id}_Y$;
2. a *natural family* of sections that commutes with the restrictions of a tower;
3. an *algebraic section* that also preserves the operations;
4. a *section efectiva* calculada by a algoritmo total.

Every natural, algebraic, or effective section is, upon forgetting structure, a set-theoretic section. The three additional properties are generally incomparable: an algorithm may select representatives without being a homomorphism; a homomorphism may not be computable in the chosen presentation; a natural family may not preserve the algebraic operation.

Remark 54.13 (Conditions of validity). The axiom of choice guarantees a set-theoretic selection within its domain. It does not guarantee naturality, algebraicity, effectiveness, or uniform cost. The independence of AC from ZF likewise does not turn these four requirements into one: even in a model with choice, the structural section requires its own proof.

54.9 Pell Obstruction to the Uniform Periodicity of the Prefix System

The active generator constructs

$$K_\infty \cong C_4 \times \mathbb{Z}_3$$

and finite projections $p_m : K_\infty \to K_m$. A representative of each class may be chosen, but not a homomorphic section compatible with the group.

Proposition 54.14. *There is no homomorphism $s_m : K_m \to K_\infty$ with $p_m s_m = \mathrm{id}$ for the quotient that introduces finite ternary torsion into K_m.*

Proof. A homomorphic section would send an element of order 3^r of K_m to an element of the same order in the component $\mathbb{Z}_3$. But the additive group $\mathbb{Z}_3$ is torsion-free: if $3^r x = 0$, then $x = 0$. The image cannot project back onto a nonzero element of order 3^r. □

The obstruction does not say «it cannot be chosen». It says: it cannot be chosen while simultaneously preserving the projection and the group law. The carry information that disappears in the quotient reappears as an obstruction to the section.

Remark 54.15 (Separation of results and scope). The profinite continuum and the Pell obstruction are *results established in the preceding sections*. The two power theorems and the table of choices are *explicit constructions of the present development*. The dense falsifier precludes turning an incomplete search for a subset into a power that has already been reconstructed.

Chapter 55

Gödel–Turing obstruction to uniform decision of projective multiplicities

55.1 Algorithmic decidability of each finite level of the prefix system

Proposition 55.1. *At a fixed depth N, if the domains are finite and the atomic observables are computable, every formula in the finite language L_N is semantically decidable.*

Proof. Atomic formulas are computed; connectives are decided by truth tables; each quantifier ranges over a finite domain. The conclusion follows by induction on syntactic complexity. □

The decidability of each window does not necessarily provide a single algorithm that decides a property of the complete tower.

55.2 Effective tower associated with the computation of projective multiplicities

Let T_e be the Turing machine with index e. For each n, define

$$X_n^e = \{b_n\} \cup \{c_n : T_e \text{ has not halted in its first } n \text{ steps}\}.$$

Links preserve existing labels:

$$p_{n+1,n}(b_{n+1}) = b_n, \qquad p_{n+1,n}(c_{n+1}) = c_n.$$

The visible tower has to single symbol v_n, and

$$q_n(b_n) = q_n(c_n) = v_n.$$

Theorem 55.2 (Global multiplicity and stopping)**.**

$$\#q_\infty^{-1}(\boldsymbol{v}) = \begin{cases} 1, & T_e \textit{ stops}, \\ 2, & T_e \textit{ never halts}. \end{cases}$$

Therefore, no total algorithm decides uniqueness of the global fiber for all effective towers in this family.

Proof. The branch $\boldsymbol{b} = (b_n)$ always exists. If T_e halts at step h, the states c_n disappear from h onward and do not form a global section. If it never halts, $\boldsymbol{c} = (c_n)$ is a second section. A decider for uniqueness would decide HALT, contradicting [125]. □

We define

$$U_e : \#q_\infty^{-1}(\boldsymbol{v}) = 1.$$

Then $U_e \iff T_e \downarrow$.

Theorem 55.3 (Uniform incompleteness)**.** *Let T be a recursively axiomatizable theory that formalizes the towers X^e and is semantically correct for U_e and $\neg U_e$. Then T does not decide every U_e.*

Proof. If each U_e were to decide, for an index e one would enumerate in parallel the proofs of U_e and $\neg U_e$ until finding one. By correctness, the first would appear exactly when the machine halts and the second when it does not. A decider of HALT would be obtained. □

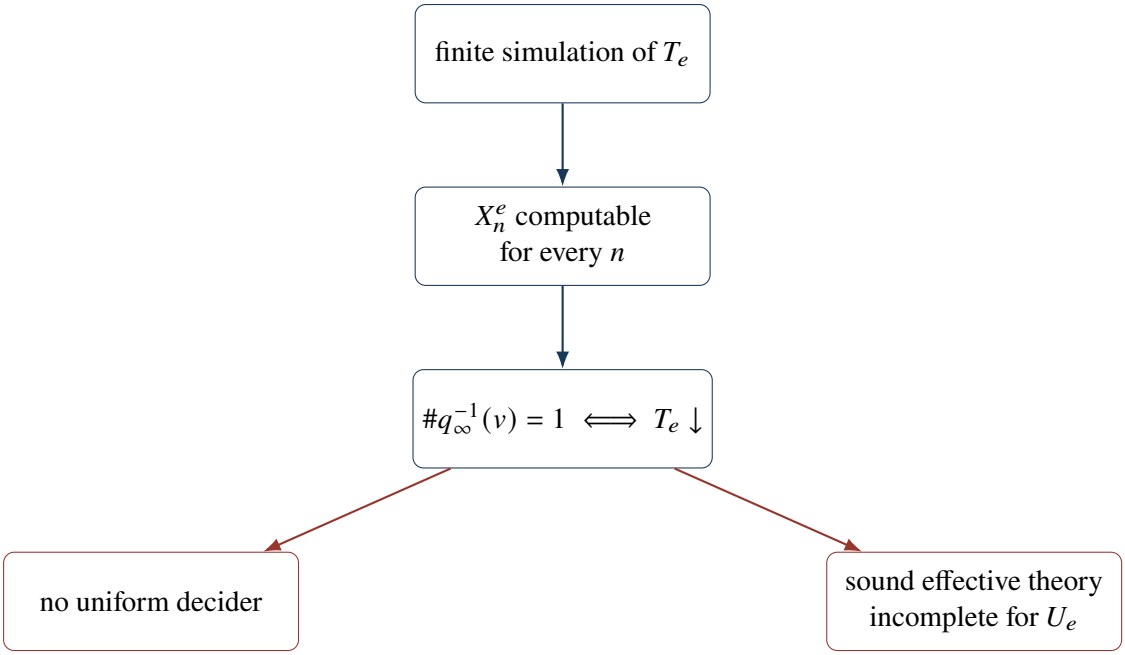


Figure 55.1: From the finite decision to uniform incompleteness.

55.3 Finite completeness of the HMT levels and exact domain of the Gödel–Turing obstruction

The academic name used is:

HMT uniform incompleteness theorem for projective multiplicities.

The abbreviated designation “third Gödel result in HMT” is also used. The proof is Gödel–Turing: it does not reproduce the self-referential diagonal of [55], but instead reduces HALT to the multiplicity of a fibre.

Remark 55.4 (Conditions of validity). The theorem does not prove the inconsistency of ZFC. Nor does it preclude a direct proof that a particular tower has a unique section. It precludes a universal uniqueness procedure for all effective towers.

55.4 Restriction of the uniform obstruction to the nonadic temporal system of 9 phases

The 9 phases of the TPK provide a concrete algorithm that, with its declared readers, partitions the refinement block into 729 subcylinders and selects one of them at each finite stage. The passage to the infinite section uses a global theorem of nonemptiness, compatibility, and unitarity.

The Gödel–Turing result says:

there is no uniqueness decider for every effective tower.

It does not state:

no concrete tower can have a uniform proof of its own.

And a particular proof does not provide the universally forbidden decisor.

Remark 55.5 (Structural Conclusion). Finite decidability, existence of the limit, uniqueness of an instance, and uniform decidability of all instances are four levels. Holographic closure requires that they be declared, rather than automatically transferring a property from one level to another.

Remark 55.6 (Separation of results and scope). The reducing tower, the equivalence $U_e \iff T_e\downarrow$, and the metamathematical theorem are *results established in the preceding sections*. The central position within the closure of infinity is an expository organization that alters neither the logical content nor the theorem's domain.

Chapter 56

Gödel–Cohen: relative consistency, independence and forcing on prefix spaces

56.1 Relative consistency in L and independence via forcing

Theorem 56.1 (Gödel, 1940: constructible half)**.** *In usual metatheory,*

$$\mathrm{Con}(\mathsf{ZF}) \Longrightarrow \mathrm{Con}\big(\mathsf{ZF}+\mathsf{AC}+\mathsf{GCH}\big).$$

More precisely, given a model $M \models \mathsf{ZF}$, its internal constructible universe L^M satisfies, in the corresponding internal sense,

$$\mathsf{ZF}+V=L+\mathsf{AC}+\mathsf{GCH}.$$

Idea of the classical proof. The constructible hierarchy is defined by recursion over the ordinals, admitting at each stage only subsets definable over earlier stages. The construction preserves the ordinals, produces an inner model, and provides a global definable well-order, from which choice and the generalized continuum hypothesis [56] follow. □

Theorem 56.2 (Cohen, 1963–1964: forcing half)**.** *One has the metamathematical implications*

$$\mathrm{Con}(\mathsf{ZFC}) \Longrightarrow \mathrm{Con}(\mathsf{ZFC}+\neg\mathsf{CH}),$$

and

$$\mathrm{Con}(\mathsf{ZF}) \Longrightarrow \mathrm{Con}(\mathsf{ZF}+\neg\mathsf{AC}).$$

Combining the two halves, CH *is independent of* ZFC *and* AC *is independent of* ZF*, always under the corresponding consistency hypothesis [30, 31].*

For the first construction, one begins with a base model M, a poset $\mathbb{P} \in M$, and an M-generic filter $G \subseteq \mathbb{P}$:

$$G \cap D \neq \varnothing \qquad \text{for every dense } D \subseteq \mathbb{P} \text{ belonging to } M.$$

The forcing theorem relates $p \Vdash_M \varphi$ to the truth of φ in $M[G]$. To obtain $\mathsf{ZF}+\neg\mathsf{AC}$, one additionally takes a suitable symmetric submodel of a generic extension; an ordinary extension of a model of ZFC preserves choice [32].

Remark 56.3 (Conditions of validity). These theorems prove relative consistency and independence. They do not prove the absolute consistency of ZF or ZFC, do not decide CH in a privileged universe, and do not convert every compatible filter into a generic filter. Nor do they select the HMT branches or identify a numerical output.

56.1.1 2 Non-Equivalent Operations: Model Extension and Object Enrichment

Forcing and HMT do not perform the same operation. The former compares a universe M with an extension $M[G]$; the latter preserves fibers that a reader forgets. If $\mathsf{Hist}_{\mathrm{HMT}}$ is the category of enriched histories, the functor

$$U : \mathsf{Hist}_{\mathrm{HMT}} \longrightarrow \mathsf{Set}, \qquad (x_m, \mu_m)_{m\geq 0} \longmapsto (x_m)_{m\geq 0},$$

may identify distinct histories without changing the visible set. Cohen's question is which subsets exist in the model; the genealogical question is how many lifts, with memory and operators, a visible object possesses. They are compatible axes, but neither replaces the other.

Operation	What changes	What does not prove by itself
$M \mapsto M[G]$	Set-theoretic universe, names and forced truth	Uniqueness or genealogy of an HMT section.
$\xi \mapsto U(\xi)$	Amount of visible memory in the projection	CH, AC, or genericity over M.

56.1.2 HMT power does not presuppose CH

In every model in which X is a Cantor space,

$$|\operatorname{Clop}(X)| = \aleph_0, \qquad |\operatorname{Closed}(X)| = \mathfrak{c}, \qquad |\mathcal{P}(X)| = 2^{\mathfrak{c}}, \qquad \mathfrak{c} = 2^{\aleph_0}.$$

CH asserts only $\mathfrak{c} = \aleph_1$. By Cantor's theorem, $\mathfrak{c} < 2^{\mathfrak{c}}$ in each of the models being compared. On passing from M to $M[G]$, new branches, new closed codes, and new subsets may appear, although the finite cylinders retain their formation rule. The full power set is model-sensitive; the clopen–closed–full distinction does not depend on deciding CH.

56.2 Compatible prefixes poset and inverse limit of extensions

We define

$$\mathbb{P}_{\mathrm{TPK}} = \bigsqcup_{n\geq 0} \mathcal{S}_n$$

and order by prolongation: $q \leq p$ if q contains more information than p. For each N,

$$D_N = \{p \in \mathbb{P}_{\mathrm{TPK}} : \ell(p) \geq N\}.$$

If every surviving prefix can be prolonged, each D_N is dense.

Theorem 56.4 (Branches and depth filters)**.** *The sections of Ω_∞ are in bijection with the compatible filters of $\mathbb{P}_{\mathrm{TPK}}$ that intersect all the D_N.*

Proof. The prefixes of a section are compatible, directed, and reach every depth. Conversely, a filter that meets each D_N determines a unique restriction at each level; directedness excludes incompatible conditions and produces a section. □

Meeting all dense sets D_N expresses unbounded depth, not genericity over a model. Being M-generic would require intersecting every dense set $D \in M$.

Proposition 56.5 (Conditioned Cohen case)**.** *If Ω_∞ is compact, metrizable, zero-dimensional, perfect, and without isolated points, it is homeomorphic to the Cantor set; the forcing of nonempty cylinders has a dense part equivalent to Cohen forcing, [30, 31].*

A unique section, by contrast, produces an atomic or trivial forcing. The word "forcing" does not by itself select a branch.

For a section $x = (x_m)_{m\geq 0}$, its prefixes form a filter G_x meeting the depth-dense sets. Nevertheless,

$$\big(\forall N,\ G_x \cap D_N \neq \varnothing\big) \not\Longrightarrow \big(\forall D \in M \text{ dense},\ G_x \cap D \neq \varnothing\big).$$

The first property expresses unbounded prolongation; the second, genericity relative to a model. A compatible section produced by nine TPK phases does not become a Cohen real by nomenclature alone.

Remark 56.6 (Conditions of validity). The following falsify an automatic identification with Cohen: a terminal cylinder or an isolated point; a filter that meets every D_N but omits some dense subset of M; two

compatible memories on the same visible branch; or the absence of an explicit dense embedding. The shared name "branch" is not a morphism.

56.3 Correspondence between genealogical sections and Borel classes of the limit space

The cylinders generate

$$\Gamma_{\mathrm{HMT}}^{\mathrm{Bor}} = \sigma\big(\{[p]_n : p \in \mathcal{S}_n\}\big).$$

The class is stable under complements, countable unions, and preimages of certified continuous maps.

If an infinite game on histories has a Borel winning condition, Martin's Borel determinacy theorem guarantees a winning strategy [87]. This is a classical application to HMT space; it does not derive AD, MM, a value of $2^{\aleph_0}$, or categoricity of $\mathcal{P}(\mathbb{R})$.

For $A, B \subseteq \Omega_\infty$, the Wadge reduction

$$A \leq_W B$$

requires a continuous map f with $x \in A \iff f(x) \in B$. If the maps are restricted to certified TPK operators, a narrower preorder appears, whose well-foundedness or semilinearity requires a proof of its own.

Remark 56.7 (Conditions of validity). A proof that uses Wadge well-foundedness to establish the determinacy that would then justify that well-foundedness is circular. The historical AD/MM branches are retained as architecture, not as HMT theorems.

56.4 Minimax theorem at each finite level of the poset of compatible prefixes

Let $P_N = \mathcal{S}_N$ be the finite set of candidates, T_N a finite set of adversarial tests, and

$$A_N : P_N \times T_N \to [-1, 1]$$

a payment fixed before the result.

Theorem 56.8 (Finite minimax)**.**

$$\max_{\sigma \in \mathcal{P}(P_N)} \min_{\tau \in \mathcal{P}(T_N)} \mathbb{E}_{\sigma,\tau} A_N = \min_{\tau \in \mathcal{P}(T_N)} \max_{\sigma \in \mathcal{P}(P_N)} \mathbb{E}_{\sigma,\tau} A_N.$$

Proof. It is the strong duality between the selector's linear program and that of the adversary, in the form of von Neumann's minimax theorem, [128]. □

Equilibrium may be mixed and does not provide a unique pure branch.

56.5 Projective system of truncations and construction of the limit section

Let

$$X = \varprojlim P_N, \qquad Y = \varprojlim T_N$$

compact, and let $A : X \times Y \to \mathbb{R}$ be continuous. The spaces of probability measures are compact and convex in the weak topology. Sion's theorem [111] yields

$$\sup_{\mu \in \mathcal{P}(X)} \inf_{\nu \in \mathcal{P}(Y)} \iint A \, d\mu \, d\nu = \inf_{\nu \in \mathcal{P}(Y)} \sup_{\mu \in \mathcal{P}(X)} \iint A \, d\mu \, d\nu.$$

Corollary 56.9. *A global equilibrium selects a pure branch only if the selector measure is a Dirac measure δ_x. The existence of a mixed equilibrium does not imply concentration.*

Remark 56.10 (Conditions of validity). Passage to the limit fails without compactness, continuity, or projective consistency of the measures. If the payoff is defined using the target it is to select, the minimax is target-fed and does not explain the route.

56.6 Projective and Matrix Extensions Provided by Operator Entanglers

Forcing organizes possible prolongations; Borel theory classifies sets of histories; minimax constructs robustness against falsifiers. None replaces the structural law that produces a unique survivor. Their function is to clarify the topology and decision structure of route space, not to introduce a retrospective selection.

Remark 56.11 (Separation of results and scope). Gödel's and Cohen's theorems are *applied classical theorems*. The poset, the typed comparison between the depth filter and the generic filter, the HMT Borel class, and the selector–falsifier game are *explicit constructions of the present development*. Martin, von Neumann, and Sion are applied thereafter, with their classical hypotheses. The former PSC/HTT assertions concerning CH, AD, and MM are not promoted.

Chapter 57

Euler operatorial system in HMT: sum–product defect, triadic exponential geometries, Ambivalence Principle, and double projection

57.1 Sum–product defect of APP and linear coefficient of the modular product Euler characteristic

57.1.1 Exact identity $459 - 405 = 54 = [q]t_3(q)$

On the support of 81 cells, the two APP leaves produce

$$S_+ = 405, \qquad S_\times = 459, \qquad S_\times - S_+ = 54. \tag{57.1}$$

The triadic regularization supplies the exponent $-1/12$. Its dodecaphase publication determines the pole q^{-1}, and the Dedekind quotient of level 3 is written

$$t_3(q) = \left(\frac{\eta(\tau)}{\eta(3\tau)}\right)^{12} = q^{-1}\prod_{n\geq 1}(1+q^n+q^{2n})^{-12}, \qquad q = e^{2\pi i\tau}. \tag{57.2}$$

Theorem 57.1 (Euler–HMT sum–product identity)**.** *The defect between the two APP evaluations coincides with the linear coefficient of t_3:*

$$\boxed{S_\times - S_+ = 54 = [q]\, t_3(q).} \tag{57.3}$$

Proof. Each factor admits $(1+q^n+q^{2n})^{-12} = 1 - 12q^n + O(q^{2n})$. For the coefficient of q subsequent to the pole q^{-1}, the degree-2 partitions of the level-3 product contribute. The certified finite expansion produces $[q]t_3 = 54$. APP obtains the same integer by the exact subtraction $459 - 405$. □

57.1.2 Conservation of the defect via residue, quotient, and nonary carry

The identity equation (57.3) links the additive difference of two finite sheets to an infinite modular product. The link preserves the origin of 54: sum, product, and regularization remain three distinguishable state coordinates.

57.1.2.1 EMRH Completion and Conservation of the Unit

The ternary fiber of depth 6 has $3^6 = 729$ elements. Its decimal completion distinguishes

$$729 + 270 = 999, \qquad 729 + 271 = 1000, \qquad 271 = 243 + 27 + 1. \tag{57.4}$$

The terminal term +1 is the closure unit transmitted by the carry. The binomial identity

$$1000 = (9+1)^3 = 9^3 + 3\cdot 9^2 + 3\cdot 9 + 1 \tag{57.5}$$

expresses transport between the nonadic chart and decimal publication without losing the genealogy of the completion.

Archimedean evaluation is performed on compatible half-closed cylinders. The two expansions of a terminating rational represent the same value and preserve distinct boundary histories. This structure types the identity $0.999\ldots = 1$ as equality of evaluation accompanied by route memory.

57.2 Quadratic algebras of TRIT and the elliptic, parabolic, and hyperbolic exponential law

57.2.1 Operator family $J_{+1}^2 = -I$, $J_0^2 = 0$, and $J_{-1}^2 = I$

Para $\tau \in \{+1, 0, -1\}$, definimos

$$\mathbb{A}_\tau = \mathbb{R}[J_\tau]/(J_\tau^2 + \tau I). \tag{57.6}$$

The exponential series converges in each algebra and reduces to

$$e^{tJ_{+1}} = \cos t\, I + \sin t\, J_{+1}, \tag{57.7}$$

$$e^{tJ_0} = I + tJ_0, \tag{57.8}$$

$$e^{tJ_{-1}} = \cosh t\, I + \sinh t\, J_{-1}. \tag{57.9}$$

TRIT assigns elliptic closure, first-order variation, and hyperbolic opening, respectively. The classical identity occupies the leaf $\tau = +1$:

$$\boxed{\mathrm{Exp}(\pi J_{+1}) + I = 0.} \tag{57.10}$$

The parabolic sheet preserves the first jet, and the hyperbolic sheet publishes a pair of reciprocal eigenvalues.

57.2.2 Identities $\mathrm{Exp}(\pi J_{+1}) + I = 0$ and $\mathrm{Exp}(2 \log \varphi\, H_\varphi) = F_{\mathrm{av}}^2$

Let

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad H_\varphi = \frac{2F_{\mathrm{av}}^2 - 3I}{2\varphi - 1}. \tag{57.11}$$

Then $H_\varphi^2 = I$ and

$$\boxed{\mathrm{Exp}(2 \log \varphi\, H_\varphi) = F_{\mathrm{av}}^2.} \tag{57.12}$$

In this coordination, π measures angular closure, φ the expansive eigenvalue of memory, and e the operation that transforms an additive generator into multiplicative transport.

57.3 Ambivalence Principle: involution between π/φ^2 and φ/π^2 and nonadic accumulation of orientation

57.3.1 Action $Q(x, y) = (x/y^2, y/x^2)$ on the areal and volumetric readers

We define the quadratic reader

$$Q(x, y) = \left(\frac{x}{y^2}, \frac{y}{x^2} \right). \tag{57.13}$$

In the coordinates

$$P = xy, \qquad O = \frac{x}{y}, \tag{57.14}$$

its action is

$$P \circ Q = P^{-1}, \qquad O \circ Q = O^3. \tag{57.15}$$

The second iteration satisfies

$$\boxed{P \circ Q^2 = P, \qquad O \circ Q^2 = O^9.} \tag{57.16}$$

The product returns and the orientation preserves a nonadic power of memory.

57.3.2 Identity $\varphi^3(\pi/\varphi^2)^2(\varphi/\pi^2) = 1$ and typed bidirectionality

For $(x, y) = (\pi, \varphi)$, the two orientations are

$$\rho_A = \frac{\pi}{\varphi^2}, \qquad \rho_E = \frac{\varphi}{\pi^2}, \tag{57.17}$$

and obey

$$\boxed{\varphi^3 \rho_A^2 \rho_E = 1.} \tag{57.18}$$

The involution $(x, y) \mapsto (y, x)$ exchanges the areal and electronic orientations of the same reader $R(x, y) = x/y^2$.

57.4 Exponential transformation of the parangular generator $A_{\mathrm{rad}} I + C^*_{\mathrm{rad}} H_\varphi$

The action chart

$$T_{-34}(u) = u\, 10^{-34} \mathcal{U}_S \tag{57.19}$$

transports the angular identity $C^*_{\mathrm{deg}} = 2(\eta_{\mathrm{ret}} + \alpha_{\mathrm{HMT}})$ to

$$\boxed{T_{-34}(C^*_{\mathrm{deg}}) = 2\hbar_{\mathrm{ret}} + 2T_{-34}(\alpha_{\angle,\mathrm{deg}}).} \tag{57.20}$$

Each summand thus lies on the same dimensional line $\mathcal{L}_S$.

57.4.1 Spectral projectors $P_\pm = (I \pm H_\varphi)/2$ and conjugate eigenvalues $q_\pm$

Let

$$B = A_{\mathrm{rad}} I + C^*_{\mathrm{rad}} H_\varphi, \qquad P_\pm = \frac{I \pm H_\varphi}{2}. \tag{57.21}$$

Exponentiation transforms the additive generator into two conjugate multiplicative leaves:

$$\boxed{e^{-B} = q_+ P_+ + q_- P_-, \qquad q_\pm = e^{-A_{\mathrm{rad}} \mp C^*_{\mathrm{rad}}}.} \tag{57.22}$$

57.4.2 Compatibility with the Parangular Publication and the Channels $90/120$

The cardinals 90 and 120 arise from the tritic incidence of the TPK. The pair $q_\pm$ does not generate them: it spectrally evaluates the operator $V_{90,120}$ on those previously constructed channels.

57.5 Elliptic action boundary and hyperbolic interior of Hilbert–Beltrami–Klein

57.5.1 Conjugate coordinates of the action ellipse and law $\mathrm{Area}(\theta) = \hbar_{\mathrm{ret}}\theta$

The angular speed

$$\eta = \operatorname{artanh}\left(\frac{C^*}{A}\right) \tag{57.23}$$

define the symplectic semiaxes

$$a_S = \sqrt{2\hbar_{\mathrm{ret}} e^{\eta}}, \qquad b_S = \sqrt{2\hbar_{\mathrm{ret}} e^{-\eta}}. \tag{57.24}$$

With

$$Q = a_S \cos\theta, \qquad P = b_S \sin\theta, \tag{57.25}$$

the action enclosed by the arc satisfies

$$\boxed{\mathrm{Area}(\theta) = \hbar_{\mathrm{ret}}\theta.} \tag{57.26}$$

57.5.2 correspondence $\rho = \tan(\theta/2) = \tanh(u/2)$ between borde periodico e interior hyperbolic

The Gudermannian coordinate

$$\rho = \tan\frac{\theta}{2} = \tanh\frac{u}{2} \tag{57.27}$$

coordinates the circular and hyperbolic charts. The boundary carries the symplectic orbit; the convex interior domain carries the Hilbert–Beltrami–Klein hyperbolic metric. The Gudermann coordinate realizes their radial correspondence and preserves the distinction between metrics.

57.6 Compatibility of the Euler operator closure with the Archimedean and exceptional publications

57.6.1 Double projection of the same enriched state and conservation of incidence

Let X^{enr} be the enriched state constructed by APP–TRIT–TPK. Its two principal publications are

$$X^{\mathrm{enr}} \xrightarrow{\mathcal{P}_{\mathrm{arq}}} \mathbb{R}, \qquad X^{\mathrm{enr}} \xrightarrow{\mathcal{P}_{\mathrm{exc}}} \mathrm{Witt} \to \mathrm{Golay} \to \mathrm{Leech} \to V^{\natural} \to \mathbb{M} \to J. \tag{57.28}$$

The first contracts compatible cylinders toward the Archimedean value; the second preserves boundary incidence as far as the exceptional corridor. The integer 54 appears in APP and in the modular coefficient of equation (57.3); the corridor $4 \to 2 \to 6 \to 54$ subsequently transports that incidence toward the Leech lattice, the vertex operator algebra, and Moonshine. The action of the Monster resides in $V^{\natural}$; the digital and route genealogy remains in the domain of TPK and reaches the corridor through the declared incidence maps.

57.6.2 Sistema macrovectorial $\mathfrak{E}_{\mathrm{HMT}} = 0$: confluencia of the identities eulerianas in a unique operator

The above identities are gathered in the typed application.

$$\mathfrak{E}_{\mathrm{HMT}} = \begin{pmatrix} (S_\times - S_+) - [q]t_3 \\ \mathrm{Exp}(I) - eI \\ \mathrm{Exp}(\pi J_{+1}) + I \\ \mathrm{Exp}(J_0) - I - J_0 \\ \mathrm{Exp}(2\log\varphi\, H_\varphi) - F_{\mathrm{av}}^2 \\ \varphi^3 \rho_A^2 \rho_E - 1 \\ T_{-34}(C^*) - 2\hbar_{\mathrm{ret}} - 2T_{-34}(\alpha_{\angle,\mathrm{deg}}) \end{pmatrix} = 0. \tag{57.29}$$

Each component retains its domain and its own proof. The aggregate map exhibits the coordination of sum, product, elliptic closure, first jet, hyperbolic expansion, ambivalence, and action.

The internal APP–TRIT–TPK composition already transports sum, product, orientation, residue, quotient, carry, route, and memory to the preceding publications. No absent HMT intertwiner is thereby posited. A subsequent promotion can be formulated only through the specific typed square required by a later representation—for example, complete group equivariance as far as Witt—without downgrading the operator confluence proved here.

57.7 Dimensional Transduction of Internally Generated Constants and Subsequent Metrological Comparison

The genealogical classification of number closes the mathematical publication: a stable section can contract its history to a value without losing the memory that determines its identity. Dimensional Mechanics now adds structure that number alone does not contain: representation, band, transducer, and local persistence. Metrology is the subsequent comparison between that realization and a dimensional

chart; no external standard retroactively generates α, the decade 10^{-34}, Planck, the counter-angle, the constitutive relations of the vacuum, the gravitational constant, or the Barbero–Immirzi parameter.

57.8 Metric Transduction of Genealogical States: Action, Mass, and Dimensional Observables

The bridge would be superficial if it isolated the squares from the physical thesis. HMT does not present APP, TRIT, and TPK as a combinatorial game to which experimental constants are subsequently added. Its governing assertion is genealogical: the mathematical and physical constants are outputs of readers constructed on the same architecture, and external comparison occurs after derivation. Operatorial language must make that chain more visible, not reduce it to numerical fitting.

57.8.1 Theoretical Background of the Genealogical-Metrical Transduction Functor

Within the domains and certificates of the HMT–MD theory developed here, the construction receives as demonstrated results:

1. the monodromic generation of π, e, φ, and α, with the chain of nine positions, its filtration stages, and the biographies of its regions;
2. the common source of the Archimedean projection and the exceptional projection, with their different codomains;
3. action, the structural decade 10^{-34}, Planck's constant, the gravitational transducer, and the Barbero–Immirzi sector;
4. the vacuum constitutive relations and their correlated genealogy;
5. the General Mass Law over the complete domain $81/56/13/324$ and the null sector, without replacing them with any expository taxonomy, followed by the typed physical realizations and by the exhaustive inventories of masses and widths, which do not enter into the construction of the domain;
6. the electronic closure, the mixing and CP-violation sectors, and the CKM and PMNS transports with their specific mathematical status in each case;
7. the exceptional output, the codes, and the symmetries that the work derives from the second projection.

One does not write "under stated hypotheses" as a substitute for those proofs. The hypotheses belong to each theorem; when they are verified, the result is accepted as proved in that domain. Nor is a conclusion exported to a larger domain when its physical realization requires an additional morphism. This is the probative separation of the development.

57.8.2 Planck, Action, and the Self-Dual Line

The absolute scale is separated from the relative ratios. In the notation of the mass source,

$$\Sigma_H = \varphi\alpha^{16}, \qquad \mathrm{ord}_{10}(\Sigma_H) = -34, \tag{GMC-9.1}$$

while the relative action factor is constructed in another coordinate. In the 108–Compton–Planck periodicity identity, the self-dual point satisfies, with the declared radius conventions,

$$R_{108} = \bar{\lambda}_C = r_g = \ell_P, \qquad C_{108} = 2\pi\ell_P. \tag{GMC-9.2}$$

The self-dual line has conjugation, unit norm, and an involution with a fixed point; the electron appears as the first material displacement, not as the origin of the scale or as a classical black hole.

In operatorial language, B_c produces the free interval (GMC-6.15); the rapidity η of (GMC-2.6) coordinates contraction and hyperbolic opening; the UCP map publishes a point of the interval without exhausting the section from which it originated. This is a complete example of the sought transduction:

the new nomenclature changes no physical identity, but separates spectrum, state, compression, and genealogy.

57.8.3 The law of masses as a family of positive operators

The HMT law acts on fibres and multisections through positive operators. In its bidirectional form,

$$\widehat{M}_M^{\beta,r} = R_{\mathrm{act}}^{B_M^r/2}\, \widehat{M}_M^{(0)}\, R_{\mathrm{act}}^{B_M^r/2}, \qquad r \in \{D, \Omega\}. \tag{GMC-9.3}$$

The decade 10^{-34} fixes the absolute scale and cancels in ratios; the readers K_D and K_Ω fix the route refinement. In the electron fiber

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80, 54, 6), \tag{GMC-9.4}$$

and the rank-one reduction closes the complete electronic value determined by the HMT closure. The null, scalar, operator-valued, and noncomparable-as-scalar-mass classes remain separate: a positive matrix does not become an experimental number until the realization functional has been declared.

The De las Cuevas–Netzer program contributes a new question here. For each matrix depth and level, which UCP images of the mass–route system preserve the spectrum, which preserve only moments, and which lie in the convex matrix hull of rank-one realizations? The bigrading (m, s) turns that question into a family of coherent finite problems.

57.8.4 Physical-Mathematical Transduction Matrix

HMT result	Lectura operatorial	Question that opens
monodromy $9 \to 1$	return of the observable with distinct environmental state	classify dilations that preserve holonomy
π, e, φ, α	four compatible readers of a genealogy	existence and minimality of a common UCP system
double projection	two marginals of a cylindrical antecedent	noncommutative precursor and joint fiber separation
B_c and jump 4	free matrix interval $[5I, 9I]$	endpoints and transport of the hierarchy 4; 2; 6; 54
Planck and periodicity of order 108	self-dual point and conjugate readers	realization as a symmetry of a conical system with memory
law of masses	positive family over fibers and paths	matrix range, extremes and sector realizations
CKM/CP	rank three transport and nontrivial phase	covariant channel that preserves spectrum and orientation
exceptional symmetry	second codomain of the double projection	entangler between nonadic atlas, code and representation

57.8.5 Universal embedding of an APP signal into a quantum Markov semigroup

The cyclic constructions use only the Latin symbol. A general framework exists for introducing any Hermitian signal of the two sheets without disrupting the magic sums.

Proposition 57.2 (Double centering of an operatorial signal)**.** *Let $G = (G_{ij})_{i,j=0}^{8}$ be a grid of self-adjoint matrices in M_3. Define*

$$H_{ij} = G_{ij} - \frac{1}{9}\sum_{j'} G_{ij'} - \frac{1}{9}\sum_{i'} G_{i'j} + \frac{1}{81}\sum_{i',j'} G_{i'j'}. \tag{GMC-9.5}$$

Then all row and column sums of H vanish. If $M = \max_{i,j} \|H_{ij}\|$ and $\varepsilon \in \mathbb{R}$ satisfies

$$|\varepsilon| \leq \frac{1}{9M} \tag{GMC-9.6}$$

(with the convention that there is no restriction if $M = 0$), the matrix

$$A_{ij} = \frac{1}{9} I_3 + \varepsilon H_{ij} \tag{GMC-9.7}$$

is a QMS *of sizes* $(9, 3)$.

Proof. Upon summing (GMC-9.5) along a row or column, the four terms cancel. Therefore, (GMC-9.7) sums to I_3 on each line. Since $-MI_3 \preceq H_{ij} \preceq MI_3$, the bound (GMC-9.6) implies $A_{ij} \succeq 0$. □

To use sources of different dimensions, a UCP reader into M_3 must first be declared: if $T_{ij} = T_{ij}^*$ belongs to the source operator system and Θ is that reader, one sets $G_{ij} = \Theta(T_{ij})$. The qutrit images of signals constructed with P_Σ, P_Π, the block B_c, or route readers can thereby be inserted; operators of sizes 81×81 or 2×2 are not inserted directly as cells of M_3. Double centering removes every purely "row only + column only" component; this provides an immediate falsifier for encodings that appear coupled without possessing a two-dimensional interaction.

Structural obstruction

The Proposition 57.2 proves the existence of an APP-marked QMS, not its fidelity or nonsemiclassicality. The small amplitude may keep it within a semiclassical neighborhood. The next stage must solve the SDP (GMC-8.6) or produce a dual witness.

57.8.6 Factorization of the genealogical–matrix bridge through APP embeddings, a quantum Markov semigroup, a semiclassical boundary, a continuous channel, a conical system, and a completely positive realization

The bridge leaves a program ordered by dependencies, not an open list of analogies:

1. **Filtered APP system.** Construct complete order- $\alpha_m : \mathcal{S}_m^{\mathrm{APP}} \hookrightarrow \mathcal{S}_{m+1}^{\mathrm{APP}}$ embeddings that transport both sheets and the carry.
2. **APP–TPK-faithful QMS.** Choose G_{ij} in (GMC-9.5) from the sum–product pair and prove the equivariance equation (GMC-7.9).
3. **Semiclassical boundary.** Exploit the nonadic symmetry to reduce the SDP over S_9 and decide whether the candidate lies outside the minimal system.
4. **Continuum Channel.** Construct the local HMT tensor and subject its transfer channel to (GMC-7.2); distinguish the fine limit, coarse limit, and root memory.
5. **Intrinsic cone system.** Define the categorical stem, the cones, the dual, and the CP transformations of TPK.
6. **Joint metrological realization.** Construct a common UCP system for mathematical, physical, and exceptional readers, and measure what genealogy each output requires.

57.8.7 Naturality, cofinality, and separation hypotheses required by the reconstruction theorem

A statement is established when it includes:

- complete definition of the domain and of all indices;
- symbolic proof or numerical certificate with a declared tolerance;
- verification of positivity, normalization, and compatibility;
- preserved negative falsifiers;
- separation between scalar coincidence and equivalence of the matrix tower;
- derivation complete from the premises and constructiones that the sustentan.

This protocol does not attenuate the thesis. It permits emphatic declaration of what has already been demonstrated and exact localization of the extension capable of advancing the state of the art.

The electron constitutes the first complete material realization of this passage. Its basal stage preserves causal function, but beta closure fixes the governing residence from which magnitudes, fields, and the General Mass Law are developed.

Part IV

Dimensional Mechanics: transduction of the genealogical state, matter, fields, and gravitation

Chapter 58

Genealogical construction of the electron: target-value-independent beta closure and atomistic realization of charge

The genealogical state here acquires its first complete material realization. The central localization, the electron–positron band, and the bidirectional reading are constructed before any metrological comparison; the basal stage remains distinct from the governing beta closure.

58.1 Closure β of the electronic section and dimensional publication of its rest energy

The electron appears in HMT as the point at which six preceding and mutually typed constructions are assembled: the incompatibility between the two arithmetic sheets of APP, the orientation induced by the involution of the areal–volumetric hinge, two integer quotients from the APP–TPK census, the dodecaphasic closure of α_{HMT}, the global defect Δ_4, and the unique center of the nonadic atlas. The route band then adds electron–positron conjugation. The MeV chart and the comparison with CODATA are subsequent to all of them.

This precedence is part of the mathematical content: the factors, their types and their causal order are determined before opening the tabulated chart, which intervenes only in the subsequent metrological comparison.

58.1.1 Electronic scalar $\mathfrak{e}_{\mathrm{HMT}}$ and causal graph of its factors APP–TPK, α_{HMT}, Δ_4 and nonadic atlas

The internal output is the dimensionless scalar.

$$\boxed{\mathfrak{e}_{\mathrm{HMT}} = \frac{\sqrt{3}}{4}\left[\exp\!\left(\frac{\varphi}{\pi^2}\right) - \frac{396}{18}\,\alpha_{\mathrm{HMT}}^3\right]\left[1 + \frac{270}{18}\,\Delta_4\right].} \tag{58.1}$$

The identities

$$\frac{396}{18} = 22 = 27 - 5, \qquad \frac{270}{18} = 15 = \binom{6}{2} = 3 \cdot 5$$

allow the abbreviated notation to be recovered

$$\mathfrak{e}_{\mathrm{HMT}} = \frac{\sqrt{3}}{4}\left(e^{\varphi/\pi^2} - 22\alpha_{\mathrm{HMT}}^3\right)(1 + 15\Delta_4).$$

Here the letter e inside the exponential is Euler's number; the subscript of $\mathfrak{e}_{\mathrm{HMT}}$ denotes the electronic section.

The causal graph of the equation is

$$\begin{aligned}
\text{APP projectors in dimension three} &\longrightarrow \sqrt{3}/4,\\
\text{TPK calendar and areal–volumetric sheets} &\longrightarrow F_{\mathrm{av}},\ \pi/\varphi^2,\ \varphi/\pi^2,\\
\text{dodecaphasic closure} &\longrightarrow \alpha_{\mathrm{HMT}},\\
\text{APP–TPK census and electronic family} &\longrightarrow 396,\ 18,\ 270,\ 22,\ 15,\ 5,\\
\text{global readings of } \pi, e \text{ and register} &\longrightarrow \Delta_4,\\
\text{nonadic atlas and anti-section} &\longrightarrow (5,5),\ \text{electron–positron band.}
\end{aligned}$$

No arrow in this diagram receives as an argument the reference electronic mass.

58.1.2 1st factor: exact incompatibility of the APP leaves

For $d \geq 2$, let

$$\mathscr{A}_d = \{1, \dots, 9\}^d$$

with uniform measure, and let

$$\Sigma_d(x) = \mathrm{dr}(x_1 + \cdots + x_d), \qquad \Pi_d(x) = \mathrm{dr}(x_1 \cdots x_d)$$

the additive and multiplicative observables. Their orthogonal conditional expectations on $\ell^2(\mathscr{A}_d)$ are $P_{\Sigma,d}$ and $P_{\Pi,d}$. The proposition 1.55 proves by exact enumeration that

$$\left\| [P_{\Sigma,3}, P_{\Pi,3}] \right\| = \frac{\sqrt{3}}{4}.$$

Proposition 58.1 (Electronic prefactor APP content)**.** *The prefactor $\sqrt{3}/4$ of equation (58.1) is the exact norm of the noncommutativity between the sum and product sheets in dimension three. It is not a geometric factor introduced from the electronic formula.*

Proof. The joint table of Σ_3 and Π_3 determines the principal angles between $\mathrm{Ran}\, P_{\Sigma,3}$ and $\mathrm{Ran}\, P_{\Pi,3}$. If s is one of its singular values, the corresponding contribution to the commutator norm is $s\sqrt{1-s^2}$. Enumeration of the 9^3 words gives at most

$$s^2(1-s^2) = \frac{3}{16}, \qquad s^2 = \frac{1}{4}.$$

Therefore the maximum norm is $\sqrt{3/16} = \sqrt{3}/4$, before introducing α_{HMT}, Δ_4, a unit of energy or a reference mass. □

The physical function of the factor is thereby fixed: it measures the maximal incompatibility of the two arithmetic readings on the same ternary support. The electron does not inherit two independent numbers, one additive and the other multiplicative; it inherits the exact obstruction to rendering the two sheets simultaneously compatible. The prefactor preserves, within the mass law, the local memory of the APP disagreement from which the route arises.

58.1.3 2nd factor: the exchange involution between π and φ

The primitive calendar of modes (Σ, Π, Π) induces, on the areal and volumetric sheets, the matrix

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

theorem 6.20 obtains it by counting the transitions $\Sigma \to \Pi$, $\Pi \to \Pi$ and $\Pi \to \Sigma$, each once. The matrix is not postulated from the golden ratio: its eigenvalues φ and $-\varphi^{-1}$ appear after the incidence has been constructed.

In the second-order memory, $\mathrm{Sym}^2(F_{\mathrm{av}})$ has spectrum

$$\varphi^2, \qquad -1, \qquad \varphi^{-2}.$$

Thus, φ^{-2} is the positive stable mode of the hinge. The closure mark π, applied just once to the areal return, produces

$$\frac{\pi}{\varphi^2}.$$

This reason preserves the closure–self-scaling order. The electronic law uses its opposite orientation.

Theorem 58.2 (Exact Inversion of the Electronic Hinge)**.** *Let*

$$\mathscr{R}(x,y)=\frac{x}{y^2},\qquad \mathsf{J}(x,y)=(y,x).$$

Then $\mathsf{J}^2=I$ *and*

$$\mathscr{R}(\pi,\varphi)=\frac{\pi}{\varphi^2},\qquad (\mathscr{R}\circ\mathsf{J})(\pi,\varphi)=\frac{\varphi}{\pi^2}.$$

Consequently,

$$\frac{\varphi}{\pi^2}=\frac{1}{\varphi^3\left(\pi/\varphi^2\right)^2},\qquad e^{\varphi/\pi^2}=\exp\left[\frac{1}{\varphi^3\left(\pi/\varphi^2\right)^2}\right].$$

Proof. The first pair of identities is obtained by applying $\mathscr{R}$ before and after the involution J. For the reparametrization,

$$\varphi^3\left(\frac{\pi}{\varphi^2}\right)^2=\frac{\pi^2}{\varphi},$$

and its inverse is φ/π^2. The last equality follows by applying the exponential. □

The simultaneous presence of π and φ therefore has a precise structural meaning. π/φ^2 reads areal closure over quadratic volumetric memory; φ/π^2 reads interior self-growth over quadratic closure. They are not two similar quotients chosen for convenience. They are the two orientations of the same reader, exchanged by an exact involution. The term e^{φ/π^2} publishes the interior orientation of that hinge in the electron section.

58.1.4 3rd and 4th factors: the integers 22, 15 and the center 5

The census of the APP chart employed by the electronic family provides three integer sums:

$$\Sigma(\text{APP})=396,\qquad \Sigma(\text{centro }2\times 2)=18,\qquad \Sigma(\text{canal})=270.$$

The first admits the ring decomposition.

$$396=18+72+108+198.$$

The kernel 2×2 is therefore an explicit substructure of the complete chart, and the integer 270 belongs to the global channel of the same APP–TPK census. The sum of the even rings is

$$18+108=126,\qquad \frac{396}{126}=\frac{22}{7}.$$

This last quotient is the rational closure seal of the chart, not an identification of $22/7$ with the number π generated by the coinductive reader. It does show that the numerator 22 already serves an internal function before appearing in the electronic law.

Theorem 58.3 (Internal Origin of the Electronic Coefficients)**.** *In the electronic family fixed by the APP chart and its global channel,*

$$\boxed{22=\frac{\Sigma(\text{APP})}{\Sigma(\text{centro})}=\frac{396}{18}=27-5,}$$

$$\boxed{15=\frac{\Sigma(\text{canal})}{\Sigma(\text{centro})}=\frac{270}{18}=3\cdot 5=\binom{6}{2}.}$$

In particular, the coefficients are determined without consulting Z_0, the electron mass or a chart of units.

Proof. The first two equalities are exact divisions of the finite census:

$$396 = 22 \cdot 18, \qquad 270 = 15 \cdot 18.$$

By defining

$$p := 27 - 22,$$

one obtains $p = 5$, and then

$$15 = 3p = 3 \cdot 5 = \binom{6}{2}.$$

All quantities belong to the integer domain of the chart and the family before metrological transduction. □

58.1.4.1 Persistence selector in the central return

The same pair has a dynamic construction that also fixes its sign without consulting a mass. The four orientations of the central seed $(5, 5)$ return to that cell for the first time after twenty-seven steps. The typed return space is

$$\mathcal{K}_e = \mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3, \qquad |\mathcal{K}_e| = 27.$$

Moreover, the microfiber of π contains the five genealogically distinct regions

$$\mathscr{R}_\pi = 3_{++} + 1_{--} + 1_{\perp}.$$

The regional gauge diag_c and, within a repeated phase, the internal order of U_6 define the injection

$$\iota : \mathscr{R}_\pi \hookrightarrow \mathcal{K}_e$$

through the five distinct places

$$(4, 0), \quad (5, 0), \quad (6, 0), \quad (6, 1), \quad (6, 2).$$

The first two correspond, respectively, to the transverse region and the mutated return; the last three separate the positive regions by their tritic sheet. The common visible word does not intervene as the inverse of the map.

Theorem 58.4 (Internal selector of the electronic coefficients)**.** *With the central return and the five typed regions fixed, the coefficients of the electronic equation are*

$$\boxed{n_e = -\left|\mathcal{K}_e \setminus \iota(\mathscr{R}_\pi)\right| = -22, \qquad m_e^{\mathrm{tor}} = \left|\mathscr{R}_\pi \times \mathbb{F}_3\right| = +15.}$$

The first symbol records the boundary deficit; the second, the retained torsional memory.

Proof. The five places above are distinct, so ι is injective and

$$|\mathcal{K}_e \setminus \iota(\mathscr{R}_\pi)| = 27 - 5 = 22.$$

Each of the five regions preserves three triticale leaves, hence

$$|\mathscr{R}_\pi \times \mathbb{F}_3| = 5 \cdot 3 = 15.$$

The orientation rule assigns a negative sign to states removed from the shell and a positive sign to retained memory. The construction opens only the center, the return of twenty-seven states, and the regional identities; it does not open an observed mass, an impedance, or a calibrated signature. □

The number 22 measures how many central kernels of weight 18 are recomposed by the APP sum 396. The number 15 measures how many of those same kernels are recomposed by the channel 270, and simultaneously counts the pairs of a hexad. The number 5 is not an additional parameter:

$$5 = 27 - 22, \qquad 15 = 3 \cdot 5.$$

The three expressions display the same closure from depth 27, from the central normalization, and from hexadic combinatorics.

58.1.4.2 Conservation of coefficients 22 and 15 in the material realization of the electron

The historical exploration first located $p = 5$ through a finite comparison with Z_0, and from there wrote $(27 - p, 3p) = (22, 15)$. That procedure retains value as a history of discovery and as an external check. It is not, however, the proof that governs the law: the subsequent count

$$\frac{396}{18} = 22, \qquad \frac{270}{18} = 15$$

constructs both integers within APP–TPK and recovers $p = 5$ as a consequence. The history explains how the pair was found; the census explains why it belongs to the equation.

58.1.5 5.º factor: the cubo of the closure α_{HMT}

The closure of α_{HMT} precedes Dimensional Mechanics. As established in section 30.12, the twelve base-thousand blocks of π, e, φ and the seal K combine through the reversible carry

$$P_m + E_m - \Phi_m - K_m + c_{m+1} = A_m + 1000c_m, \qquad c_{13} = c_1 = 0.$$

The output is

$$A_\alpha = (007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 663)$$

and, by concatenation, the exact dodecaphasic window

$$\alpha_{12} = \pi_{12}(\alpha_{\mathrm{HMT}}) = \frac{729735256928380099728510547238066 3}{10^{36}}.$$

Telescoping the carries proves the complete integer identity; the compatible prolongation constructs $\alpha_{\mathrm{HMT}} = \varprojlim_N A^{(N)}$, and α_{12} is its thirty-six-digit rational projection. The comparison with the fine-structure constant is performed subsequently and does not intervene in its construction.

The electronic law takes the third power

$$\alpha_{\mathrm{HMT}}^3.$$

Mathematically, this is the multiplicative evaluation of the third symmetric power of the same scalar already closed; neither three metrological copies nor three different parameters are introduced. Its contribution is

$$22\alpha_{\mathrm{HMT}}^3 = 0.000008549065992005517270149798075982 28\ldots.$$

Proposition 58.5 (Statute of the Electronic Cubic Channel)**.** *The term $22\alpha_{\mathrm{HMT}}^3$ of the electronic law depends only on the prior closure of α_{HMT}, on internal multiplication, and on the integer 22 of theorem 58.3. It does not depend on a reading of the electron mass.*

Proof. α_{12} is the rational number fixed by the concatenation and carry identity, and the compatible family of windows determines the section α_{HMT}. The projection α_{12}^3 certifies the printed digits of α_{HMT}^3. The coefficient 22 is determined by $396/18$. The composition of both operations contains only data constructed before the energy chart. □

This proposition fixes exactly what is used for the electron. The third power is the cubic degree of this channel in the published law. There is no need here to postulate a universal rule according to which every occurrence of α^d automatically represents a dimension d; a universal statement of that kind would require its own theorem of multiplicativity among degrees. The electron genealogy is complete without turning that generalization into a premise.

58.1.6 6th factor: minimal torsion and global defect Δ_4

The terms *minimal torsion* and *global defect* denote a single invariant here, not two distinct corrections. The object used by the mass law is

$$\boxed{\Delta_4 = \frac{\pi - e\log\pi}{270}\left(1 + \frac{\pi}{729}\right).} \tag{58.2}$$

Its inputs are the already fixed HMT readings of π and e, the global channel 270, and the interior cell 729. Consequently,

$$\Delta_4 = 0.000111196422692140054051134786120679 09\ldots.$$

The first ratio measures the signed defect $\pi - e\log\pi$ in units of the 270 channel; the second factor couples that ratio to the interior cell through $\pi/729$. This reading adds no parameters to the formula: it separates its two constituent operations. The chapter 35 shows that it is a pre-dimensional mathematical input and distinguishes it from the operator that eventually realizes it as a change of sheet.

Proposition 58.6 (Globality of the Electronic Defect)**.** *Δ_4 is a global invariant preceding the species. In the electronic law it enters through*

$$1 + 15\Delta_4 = 1.001667946340382100810767021791810 19\ldots.$$

It is not a corrector chosen after knowing the residue of the electron mass.

Proof. The equation (58.2) contains neither a species label, a mass, nor a unit. The integer 15 has already been constructed by $270/18$. Therefore, the composition $1 + 15\Delta_4$ is determined by the global data and the internal census. □

Δ_4 must be distinguished from the other blocks of the mass character and from the other notions of torsion. In particular, it is neither s_{270}, nor the quadratic invariant $135\Delta_4^2$, nor the angular residue $\varepsilon_{\mathrm{res}}$, nor the geometric Cartan–Holst torsion. Substituting one for another would repeat or alter the level of the correction. For the electron, Δ_4 preserves the global memory of refinement; the factor 15 states how many times the electron family reads it. The qualifier *minimal* preserves the historical name of this invariant and is not used as a substitute for a variational theorem of minimality.

58.1.7 Material realization of the fixed point (5,5) in the electronic closure

Let

$$C_{81} = \{1, \ldots, 9\}^2$$

the nonadic atlas and

$$\rho(r, c) = (10 - r, 10 - c)$$

its central inversion. theorem 90.11 proves that the atlas contains a unique generation-zero cell and that this cell bears the positional label e_{core}.

Proposition 58.7 (Uniqueness of the Electronic Center)**.** *The involution ρ has a unique fixed point:*

$$\boxed{(r, c) = (5, 5).}$$

Its uniqueness is combinatorial and does not depend on the value of $\mathfrak{e}_{\mathrm{HMT}}$.

Proof. The equation $(r, c) = \rho(r, c)$ is equivalent to

$$2r = 10, \qquad 2c = 10.$$

In $\{1, \dots, 9\}^2$ it has the unique solution $r = c = 5$. The evaluation of the mass does not appear in the argument. □

Corollary 58.8 (Universality of the Central Section)**.** *Every admissible automorphism of the atlas that intertwines the central inversion preserves the cell* $(5, 5)$.

Proof. If $f\rho = \rho f$ and $\rho(5, 5) = (5, 5)$, then $\rho f(5, 5) = f\rho(5, 5) = f(5, 5)$. Therefore $f(5, 5)$ is another fixed point of ρ; uniqueness forces $f(5, 5) = (5, 5)$. □

This is the structural reason why the electronic class is the same in every realization preserving the atlas and its inversion: a local decimal does not recur by chance; rather, every admissible chart must transport the same fixed point and the same chain of invariants. The physical reading of this uniqueness identifies each electron with a realization of that central class; the proposition proves the internal universality of the atlas.

In the electronic family fixed by the construction, this point is the geometric anchor of the section. Four objects should not be confused:

$$(5, 5), \qquad e_{\text{core}}, \qquad v_e = (0, 0, 0, 0, 0), \qquad \text{the raw signature of the central cell.}$$

The first is a position; the second, its internal label; the third, the chosen origin in the fine action lattice; the fourth preserves the raw atlas data and need not be the zero vector. Their coordination explains why the electron occupies the center without erasing the genealogy of the representations describing it.

Oriented electronic routes e_{++}, e_{--}, envelope e_{env}, action origin and signatures of the central cell
The central cell carries two oriented histories of twelve events. Writing v^2 for the concatenation of two copies of a sextuple, the active route of orientation ++, its opposite history, and the historical envelope are, respectively,

$$\begin{aligned} e_{\text{act}} = e_{++} &= (2, 2, 5, -4, -3, -2)^2, \\ e_{--} &= (4, 3, 2, -2, -2, -5)^2, \\ e_{\text{env}} &= (4, 3, 5, -4, -3, -5)^2. \end{aligned} \tag{E-centro-1}$$

The two stories publish the same word

$$\tau_{12} = \text{+++-+++-},$$

but they are not the same route. The envelope takes the absolute maximum coordinate by coordinate while preserving the sign; it records amplitudes, but flattens the genealogical order and does not replace e_{act}.

The other readers of the center live in distinct codomains:

$$\begin{aligned} v_e &= (0, 0, 0, 0, 0), && \text{affine origin of the action lattice,} \\ R(5, 5) &= (24, 0, 12, 0, -12), && \text{raw APP signature of the cell,} \\ \mathrm{Sig}^{\Gamma_{108}}_{\text{masa}} &= (-12, -4, 12, -6, 12), && \text{even mass projection,} \\ \mathrm{Sig}^{\Gamma_{108}}_{\text{carga}} &= (0, 0, 0, 0), && \text{odd charge projection.} \end{aligned} \tag{E-centro-2}$$

The fact that the charge word has zero odd signature does not turn the route into zero, nor does it identify the raw signature with the affine origin. Position, active route, opposite route, envelope, origin and both signatures are coordinated objects, not six notations for the same vector.

58.1.8 Particle–antiparticle conjugation and persistence of the electronic band

Electronic identity does not end at the central point or at the mass value. Material conjugation is realized on dodecaphasic words. Let

$$C_M(\mathbf{t}) = -\operatorname{rev}(\mathbf{t})$$

be the oriented anti-section and let $\chi \in \{-1, +1\}$ be the charge orientation. The band is the class

$$[(\mathbf{t}, \chi)]_M = \{(\mathbf{t}, \chi), (C_M\mathbf{t}, -\chi)\}$$

of definition 92.28. The band preserves its even components and exhibits two sections with opposite odd components.

Theorem 58.9 (Electron and positron as sections of a band)**.** *For the band character of theorem 92.31,*

$$m(C_M\mathbf{t}, -\chi) = m(\mathbf{t}, \chi).$$

In the electronic channel,

$$(n, \chi) = (-22, +1) \quad \longleftrightarrow \quad (+22, -1)$$

and both pairs produce

$$\chi n = -22.$$

Therefore, electron and positron have the same mass and opposite charge orientations.

Proof. If E_+ and E_- are, respectively, the even and odd components of the character with respect to C_M, then

$$E_+(C_M\mathbf{t}) = E_+(\mathbf{t}), \qquad E_-(C_M\mathbf{t}) = -E_-(\mathbf{t}).$$

From here,

$$E_{\text{band}}(C_M\mathbf{t}, -\chi) = E_{\text{band}}(\mathbf{t}, \chi),$$

and the equality of mass is obtained by applying the exponential. In the electronic term,

$$(+1)(-22) = (-1)(+22) = -22.$$

□

The finite closure of proposition 92.33 reinforces this interpretation: over the eighty-one CT–108 routes, neither pure sign nor pure reversal preserves the taxonomy, whereas − rev does preserve it, with

$$81 = 54 + 9 + 9 + 9.$$

The antiparticle is therefore neither a negative mass nor an independent copy. It is the other oriented section of the same band: it preserves the even invariants that determine mass and reverses the odd invariants that determine charge and orientation.

58.1.9 Basal stage of the electronic closure

Theorem 58.10 (Holographic composition of the electronic section)**.** *With the objects constructed in the preceding sections fixed, the dimensionless electronic section is*

$$\mathfrak{e}_{\text{HMT}} = \left\| [P_{\Sigma,3}, P_{\Pi,3}] \right\| \left[\exp\big((\mathscr{R} \circ \mathsf{J})(\pi, \varphi)\big) - \frac{\Sigma(\text{APP})}{\Sigma(\text{centro})}\, \alpha_{\text{HMT}}^3 \right] \left[1 + \frac{\Sigma(\text{canal})}{\Sigma(\text{centro})}\, \Delta_4 \right].$$

Equivalently,

$$\boxed{\mathfrak{e}_{\mathrm{HMT}} = \frac{\sqrt{3}}{4}\left(e^{\varphi/\pi^2} - 22\alpha_{\mathrm{HMT}}^3\right)(1 + 15\Delta_4).}$$

Its evaluation is

$$\boxed{\mathfrak{e}_{\mathrm{HMT}} = 0.510998922488066072509870877191349437 63\ldots.}$$

This scalar is the basal stage $\mathfrak{e}_e^{(0)}$*, not the governing electronic output. The bidirectional beta closure publishes* 0.510998950690000808608 . . . MeV.

Proof. The proposition 58.1 contributes

$$\left\|[P_{\Sigma,3}, P_{\Pi,3}]\right\| = \sqrt{3}/4.$$

The theorem 58.2 contributes

$$(\mathscr{R} \circ \mathsf{J})(\pi, \varphi) = \varphi/\pi^2.$$

The theorem 58.3 contributes

$$\frac{\Sigma(\mathrm{APP})}{\Sigma(\mathrm{centro})} = 22, \qquad \frac{\Sigma(\mathrm{canal})}{\Sigma(\mathrm{centro})} = 15.$$

The dodecaphasic closure contributes the rational window α_{12} of the coinductive section α_{HMT}, and equation (58.2) contributes the global defect. Substitution yields the abbreviated formula. Successive evaluation of

$$\begin{aligned} e^{\varphi/\pi^2} &= 1.178144942596306780811600956808889 10\ldots, \\ 22\alpha_{\mathrm{HMT}}^3 &= 0.000008549065992005517270149798075982 28\ldots, \\ 1 + 15\Delta_4 &= 1.001667946340382100810767021791810 19\ldots \end{aligned}$$

gives the declared value. □

The theorem is holographic in a verifiable sense: each local factor retains a distinct layer of the complete system. The commutator retains sheet duality; the exponential retains the interior orientation of the hinge π–φ; 22 and 15 retain the global census and its central kernel; α_{HMT}^3 retains fine closure; and Δ_4 retains global refinement. The center determines where the section is anchored, and the band determines how it survives under conjugation. The value does not replace this structure: it is its common evaluation.

58.1.9.1 Anterior and posterior action splitting upon return

The regional realization of the conjugate angular coordinate C^* distinguishes two coordinates on the same internal action line. The coordinate before the return is H_5 and the subsequent coordinate is

$$\eta_{\mathrm{ret}}^{(\mathrm{deg})} = H_5 - \frac{90}{\pi}\mathscr{C}_\pi(\alpha_{\mathrm{HMT}}).$$

The notation $\eta_{\mathrm{ret}}^{(\mathrm{deg})}$ is the canonical sexagesimal coordinate defined on the action branch by $\eta_{\mathrm{ret}}^{(\mathrm{deg})} = C^*/2 - \alpha_{\angle,\mathrm{deg}}$. The preceding equality is evaluated in that chart; it does not introduce a second return coordinate. Therefore, the additive and multiplicative charts of the oriented splitting are

$$\boxed{\delta_{2w} := H_5 - \eta_{\mathrm{ret}}^{(\mathrm{deg})} = \frac{90}{\pi}\mathscr{C}_\pi(\alpha_{\mathrm{HMT}}), \qquad R_{\mathrm{act}} := \frac{\hbar_{\mathrm{HMT}}^{\mathrm{pre}}}{\hbar_{\mathrm{HMT}}^{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}^{(\mathrm{deg})}}.} \tag{58.3}$$

Both coordinates follow from a single pre-return/post-return transport. They are not two Planck constants: the common factor $10^{-34}\mathcal{U}_S$ fixes the dimensional line and cancels exactly in R_{act}. The action ellipse subsequently realizes $\mathrm{Area}(\theta) = \hbar_{\mathrm{HMT}}^{\mathrm{ret}}\theta$, so that the splitting precedes the holonomy reading.

The coupling

$$\kappa_e^{\beta} = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4, \qquad \mathfrak{e}_e^{\beta} = \mathfrak{e}_{\mathrm{HMT}} R_{\mathrm{act}}^{\kappa_e^{\beta}} = 0.5109989506900008086\ldots \tag{58.4}$$

is the bidirectional quantitative closure of the electron. The two independent readers of the electronic register constructed in the following chapter select, without consulting the comparison mass,

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80, 54, 6),$$

and derive exactly κ_e^{β}. The integer 80 comes from the first coordinate of those readers; the word $w_e = 201101$ retains its own residence in the channel of Euler's constant.

Internal Closure of Bidirectional Refinement

The identity (58.3), the reduction $(80, 54, 6)$, and the equation (58.4) constitute internally exact, target–free results on the electronic record. Propagation to fibers of higher rank is formulated by means of the multisection operators.

58.1.10 Energy chart and contrast of the bidirectional closure

Let $\mathcal{U}_E$ a real line of energy dimensions and let u_E a chosen basis. The metrological transduction is

$$\varsigma_{u_E} : \mathbb{R} \longrightarrow \mathcal{U}_E, \qquad \varsigma_{u_E}(x) = x\, u_E.$$

The chart preserves two stages of the same genealogy. The basal stage is

$$E_e^{(0)} = \varsigma_{u_E}(\mathfrak{e}_{\mathrm{HMT}}).$$

With $u_E = 1\,\mathrm{MeV}$,

$$E_e^{(0)} = 0.5109989224880660725\ldots\ \mathrm{MeV}.$$

The action ratio and the derived exponent complete the leading stage,

$$E_e^{\beta} = E_e^{(0)} R_{\mathrm{act}}^{\kappa_e^{\beta}} = 0.510998950690000808608257164 8\ldots\ \mathrm{MeV}, \qquad m_e^{\beta} = E_e^{\beta}/c^2.$$

The unit does not appear in equation (58.1); it is the basis of a dimensional line applied to an already constructed output.

The CODATA 2022 reference for the electron energy equivalent is [29]

$$E_e^{\mathrm{CODATA}} = 0.51099895069(16)\ \mathrm{MeV}.$$

In that same chart, the bidirectional output satisfies

$$E_e^{\beta} - E_e^{\mathrm{CODATA}} = 8.086082571648 \times 10^{-16}\ \mathrm{MeV},$$

with relative difference

$$1.582406883758 \times 10^{-15}.$$

The comparison is carried out after freezing the internal evaluation. The identities of the basal selector remain

$$22 = \frac{396}{18}, \qquad 15 = \frac{270}{18}, \qquad p = 27 - 22 = 5.$$

and additionally receive the combinatorial proof $-22 = -|K_e \setminus \iota(R_\pi)|$, $+15 = |R_\pi \times \mathbb{F}_3|$. The metrological comparison verifies the output; the route, the readers, and the action ratio generate it.

58.1.11 Structural Consequences of Electronic Closure

The preceding structural reading imposes several distinctions:

1. $\sqrt{3}/4$ is the norm of the APP commutator in dimension three. Its agreement with other geometric formulas does not supersede this provenance.
2. π/φ^2 and φ/π^2 are not interchangeable. The first ratio is the closure orientation; the second, obtained through J, is the electronic orientation of self-growth.
3. 22 and 15 follow from $396/18$ and $270/18$. Presenting them as values logically selected by Z_0 would reverse the demonstrated dependence.
4. α_{HMT}^3 uses the prior closure of α_{HMT}. The cubic power belongs to the electron law; by itself, it does not authorize a universal rule $\alpha^d \leftrightarrow d$ without an additional theorem.
5. Δ_4 is global. No must reemplazarse by s_{270}, by $135\Delta_4^2$ neither by a residue particular of the especie.
6. The position $(5, 5)$, the label e_{core}, the fine origin $v_e = 0$, and the central raw signature are coordinated objects, not synonyms.
7. The electron and positron are not distinguished by two mass laws: they are the two oriented sections of a single band, with even mass and odd charge.
8. $0.5109989224\ldots$ is first a dimensionless scalar. MeV, MeV/c^2, CODATA, and Z_0 belong to the subsequent comparison.

These distinctions explain why the electronic equation is a miniature of the HMT–MD system. In a single section, support, sheets, orientation, census, fine closure, refinement, center, persistence, and conjugation reappear. This is the content of the word *holographic*: not that a digit poetically represents the whole, but that each factor preserves a verifiable dependence on the whole that produces it.

58.2 Electron realization band and separation between base mass and beta closure

The enriched route of the central point publishes two complementary representations of its memory: the complete tritic word $\gamma_{108} \in \mathbb{F}_3^{108}$ and the signed dodecaphase record $\tau_{12} \in \{-1, 0, +1\}^{12}$. The two readers defined below are computed exclusively from those representations, the calendar, and the HMT constants already constructed.

58.2.1 Reader of displacements on the cyclic calendar of order 108

Let S be the cyclic shift on $\mathbb{Z}/108\mathbb{Z}$. For $k \in \mathbb{Z}/108\mathbb{Z}$, define

$$D_k(\Gamma) := \#\{t \in \mathbb{Z}/108\mathbb{Z} : \gamma_t \neq \gamma_{t+k}\} = \|(I - S^k)\gamma\|_0. \tag{58.5}$$

The integer triple of the reader is

$$K_D(\Gamma) = \left(D_{27} + D_{36},\ D_1,\ \frac{D_4 - D_3}{2}\right) =: (A_D, B_D, C_D), \tag{58.6}$$

and its scalar evaluation is

$$\kappa_D(\Gamma) = A_D - \alpha_{\mathrm{HMT}} B_D + \Delta_4 C_D. \tag{58.7}$$

The shifts (27) and (36) are the chronological lifts of 90° and 120° because nine discrete steps represent 30°. The term D_1 measures the immediate local boundary, and the difference $D_4 - D_3$ compares the two genealogies before their height quotient. In the domain of the 324 routes, $\gamma_{t+54} = \gamma_t$ holds; all the D_k are even and C_D is an integer.

58.2.2 Dodecaphasic correlation difference reader

For the word $\tau = (\tau_r)_{r\in\mathbb{Z}/12\mathbb{Z}}$, let

$$C_k^{\mathrm{corr}}(\tau) = \sum_{r=0}^{11} \tau_r \tau_{r+k}, \qquad A_\ell(\tau) = \sum_{r=0}^{11} \left(\sum_{j=0}^{\ell-1} \tau_{r+j} \right)^2 .$$

The primitive genealogical obstruction and the antipodal memory are

$$\Omega_{90|120}(\tau) = 4A_3 - 3A_4 = -2C_1^{\mathrm{corr}} - 4C_2^{\mathrm{corr}} - 6C_3^{\mathrm{corr}}, \tag{58.8}$$

$$P_6(\tau) = \sum_{r=0}^{5} \tau_r \tau_{r+6}. \tag{58.9}$$

The symbol P_6 prevents the colision with the short cycle of the corredor of constantes. The second triple and its evaluation are

$$K_\Omega(\Gamma) = \big(\Omega_{90|120}(\tau_\Gamma), 54, P_6(\tau_\Gamma)\big), \qquad \kappa_\Omega = \Omega_{90|120} - 54\alpha_{\mathrm{HMT}} + P_6\Delta_4. \tag{58.10}$$

Lemma 58.11 (Linear integral primitive uniqueness)**.** *Let $L = uA_3 + vA_4$, with $u, v \in \mathbb{Z}$, be a contrast that vanishes when the genealogies descend to the same height, $A_3 = 3a$, $A_4 = 4a$. Then $(u, v) = n(4, -3)$. Once the sign $90 - 120$ has been fixed, $\Omega_{90|120} = 4A_3 - 3A_4$ is the primitive generator.*

Proof. Vanishing for every a requires $3u + 4v = 0$. Since $\gcd(3, 4) = 1$, the integer solutions are $(u, v) = n(4, -3)$. Primitivity corresponds to $|n| = 1$. □

The lemma selects Ω within the class of integral linear contrasts of the energies A_3, A_4. Its uniqueness domain is that functional class; nonlinear invariants belong to a distinct selection problem.

58.2.3 Evaluation of the central route

Before metrological comparison, the electronic route $\Gamma_e = (5, 5, ++)$ provides,

$$(D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32),$$

and its signed dodecaphonic register is

$$\tau_e = (+, +, +, -, -, -, +, +, +, -, -, -).$$

Therefore,

$$K_D(\Gamma_e) = (48 + 32, 54, (68 - 56)/2) = (80, 54, 6),$$

$$K_\Omega(\Gamma_e) = (80, 54, 6).$$

Theorem 58.12 (Common Electronic Reduction of Bidirectional Readers)**.** *The readers K_D and K_Ω coincide on the central route and determine*

$$\kappa_e = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4. \tag{58.11}$$

Proof. The two preceding evaluations yield the same integer triple. Substitution of its components into $A - \alpha_{\mathrm{HMT}}B + \Delta_4 C$ proves (58.11). Every quantity used in this reduction belongs to the route, to its two records, or to the previously constructed HMT constants. □

Subsequent realization on the energy chart. With $R_{\mathrm{act}} = H_5/\eta_{\mathrm{ret}}^{(\mathrm{deg})}$, the positive action on the baseline $\mathfrak{e}_0$ yields

$$\mathfrak{e}_e^{\beta} = \mathfrak{e}_0 R_{\mathrm{act}}^{\kappa_e}. \tag{58.12}$$

After applying the current chart to MeV is obtained

$$\varsigma_{u_E}(\mathfrak{e}_e^{\beta})\big|_{u_E=1\,\mathrm{MeV}} = 0.510998950690000808608257164 8\ldots\ \mathrm{MeV}. \tag{58.13}$$

The first equality belongs to the internal theorem conditioned by the basal and the action line; the second is its dimensional realization. The external value is reserved for metrological comparison.

The integer (80) follows from $D_{27} + D_{36}$ and, in the dodecaphase reader, from $\Omega_{90|120}$. Its coincidence with other internal cardinals is a compatibility control and does not replace this operator provenance. The orientation route $(--)$ is a temporal translation of route $(++)$ and preserves the same profile, in accordance with mass parity under conjugation.

58.2.4 Separation of readers and return scales

Numerical coincidence of internal components is subordinate to their type. At the central point, the objects involved include

$$R_{\mathrm{atlas}}(5,5) = (24,0,12,0,-12), \qquad L_{\kappa_0}(\Gamma_e^{++}) = (24,0,0,0,-12),$$

$$k_{\Delta_4}^{\mathrm{dod}}, k_{\Delta_4}^{\mathrm{pro}} : \mathscr{R}_{324} \longrightarrow \mathbb{Z},$$
$$k_{\Delta_4}^{\mathrm{dod}}(\Gamma_e^{++}) = 12, \qquad k_{\Delta_4}^{\mathrm{pro}}(\Gamma_e^{++}) = 4,$$
$$K_D(\Gamma_e^{++}) = K_\Omega(\Gamma_e^{++}) = (80,54,6).$$

Here $\mathscr{R}_{324}$ denotes the domain of the 324 oriented routes. The first is a static family signature; the second, a genealogy extractor; the third and fourth terms are torsional readers defined, respectively, by dodecaphasic reading and prospective prolongation; the final object is the pair of bidirectional readers. No partial equality identifies those morphisms or permits transferring a coefficient between readers.

Four temporal scales are also distinguished: 27 is the first return of class, orbit, and cell; $54 = D_1(\Gamma_e)$ is the observable kinematic return within the cycle; 108 closes the calendar; and 216 closes the oriented kinematic state in the extension. This typing keeps positional return, remark, and oriented holonomy separate.

Causal chain of the electronic outcome The route and its two registers determine K_D, K_Ω, their common reduction, and the evaluation of κ_e. The realization in MeV subsequently applies the energy chart; external mass enters only in the comparison. The pair $\mathbf{K} = (K_D, K_\Omega)$ is the complete operator output of HMT on each route. In a noncentral fiber, its representation by a single physical scalar requires an additional morphism between the fiber operator and the dimensional chart; this requirement neither modifies the internal law nor authorizes the selection of a cell by metrological proximity.

From the Cartan–Holst realization to horizon–area–information confluence through conservation of action The electronic closure fixes a material realization; the next step identifies the grammar that converts action, clock, and mass into physical magnitudes without using that realization as input data.

58.3 Null–material closure of the electronic fiber and bifurcation preceding the positive character

Planck's self-dual identity fixes the origin of the positive line of masses; the electronic genealogy fixes its first complete material displacement. This section brings together, without repeating their reference proofs, the circular realization of CT108, the basal composition of the electron, and the bidirectional holonomy of its route.

Let

$$\mathcal{H}_{108}^{\mathrm{reg}} := \ell^2(\mathbb{Z}/108\mathbb{Z};\mathbb{R}), \qquad S_{108}^{\mathrm{reg}}\delta_\theta = \delta_{\theta+1},$$

the real regular representation of the cyclic shift of CT108, and let

$$n_0 := \frac{1}{\sqrt{108}}\sum_{\theta=0}^{107}\delta_\theta, \qquad \mathcal{N}_{108} := \ker(S_{108}^{\mathrm{reg}} - I) = \mathbb{R}n_0.$$

The normalized zero Fourier mode is thereby defined, not merely named. Let $\mathcal{G}_e$ be the space of enriched electronic genealogies of proposition 90.9, and fix the linear submodule $\ell_{\Gamma_e} := \mathbb{R}g_{\Gamma_e} \subset \mathbb{R}[\mathcal{G}_e]$ generated by the central route in the free linearization of that space. Likewise, if $F_e^+ = \mathbb{R}_{>0}I$ is the positive ray already defined, let $\mathcal{F}_e := \mathrm{span}_{\mathbb{R}}(F_e^+) = \mathbb{R}I$ be its linear span, and let $\mathcal{L}_M$ be the mass line defined in section 90.4.7. The declared bases determine the rank-one linear morphism

$$\mathfrak{C}_e : \mathcal{N}_{108}\otimes\ell_{\Gamma_e} \longrightarrow \mathcal{F}_e\otimes\mathcal{L}_M, \qquad \mathfrak{C}_e(n_0\otimes g_{\Gamma_e}) = \mathfrak{c}_e\, I\otimes\mathcal{U}_M, \tag{58.14}$$

where the positive coordinate is

$$\boxed{\mathfrak{c}_e = \frac{\sqrt{3}}{4}\left[\exp\left(\frac{\varphi_{\mathrm{HMT}}}{\pi_{\mathrm{HMT}}^2}\right) - 22\alpha_{\mathrm{HMT}}^3\right](1+15\Delta_4)R_{\mathrm{act}}^{\,80-54\alpha_{\mathrm{HMT}}+6\Delta_4}.} \tag{58.15}$$

Theorem 58.13 (Null closure–one-range electronic material)**.** *With the primitives and charts already fixed, the morphism $\mathfrak{C}_e$ exactly factors the beta electron output. Its four factors preserve, in this order, the sum–product incompatibility of APP; the areal–volumetric orientation together with the boundary deficit; the torsion and retained memory; and the bidirectional holonomy of the electron route. In the chart $\mathcal{U}_M = 1\,\mathrm{MeV}/c^2$,*

$$\mathfrak{C}_e(n_0\otimes g_{\Gamma_e}) = 0.510998950690000808608257164 8\ldots\; I\otimes \mathrm{MeV}/c^2. \tag{58.16}$$

Proof. The theorem 58.10 proves that the first three factors of (58.15) compose the positive basal scalar $\mathfrak{e}_{\mathrm{HMT}}$. The theorem 58.12 obtains, from two independent readers of the route,

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80,54,6), \qquad \kappa_e = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4.$$

The ratio $R_{\mathrm{act}} > 0$ belongs to the unique pre-return/post-return action line. Consequently, the fourth factor is positive, and the composition coincides with (58.4) and (58.13). The dimensional chart acts only after this factorization. □

Mathematical scope and physical realization. The exact result is the declared linear realization of the electron factorization over the typed null mode, genealogy, and fibre. The CT108 null mode does not by itself select the coordinate $\mathfrak{c}_e$: that coordinate follows from the four factors proved in their respective residences, and the morphism assembles them into a single rank-one algebraic closure. The bound electromagnetic dynamics retains as its own requirements a stable Hamiltonian, charge, spin 1/2, the form factor, and gauge coupling; these objects are not premises of the preceding factorization. The Planck

unit occupies the self-dual point of the line, whereas the electron occupies a genealogically displaced section: an interpretation as a classical black hole belongs to another type of realization and does not enter this morphism.

58.4 Law of electronic vacancies: polarization, current, and structure of the null modes

Finite scales

$$729 = 3^6, \qquad 1000 = 10^3$$

determine an exact logarithmic mismatch. We define

$$\rho_{\mathrm{vac}} = \log_{1000} 729 = \log_{10} 9, \qquad \delta_{\mathrm{vac}} = 1 - \rho_{\mathrm{vac}} = \log_{10}(10/9). \tag{58.17}$$

The irrationality of δ_{vac} follows directly from the prime valuations: an equality $\delta_{\mathrm{vac}} = p/q$ would imply $(10/9)^q = 10^p$, which is incompatible with the valuations at $2, 3, 5$. The vacancy is therefore an aperiodic arithmetic rotation determined entirely by the relation between the two scales; its definition precedes the electronic law.

58.4.1 Exact Laws of Hole and Return

Let

$$p_n = \left\lfloor \frac{n}{\delta_{\mathrm{vac}}} \right\rfloor.$$

Theorem 58.14 (Gap law 21/22)**.** *The consecutive increments of the sequence p_n satisfy*

$$p_{n+1} - p_n \in \{21, 22\}.$$

Both values appear in the word mechanical.

Proof. For every irrational $x > 1$, the difference $\lfloor (n+1)x \rfloor - \lfloor nx \rfloor$ belongs to $\{\lfloor x \rfloor, \lceil x \rceil\}$. In this case

$$\delta_{\mathrm{vac}}^{-1} = 21.85434532678283256\ldots,$$

which determines the two vacancies. Irrationality rules out stabilization in either one alone. □

The frequency of the short hole is

$$\sigma_{\mathrm{vac}} = 22 - \delta_{\mathrm{vac}}^{-1} = 0.1456546732171674374\ldots, \qquad \sigma_{\mathrm{vac}}^{-1} = 6.865553832996660997\ldots.$$

Corollary 58.15 (Return law 6/7)**.** *Successive occurrences of the gap* 21 *are separated by six or seven positions.*

Proof. The indicator word for the short gap is the mechanical coding with irrational slope σ_{vac}. Beatty's argument applied to $\sigma_{\mathrm{vac}}^{-1}$ produces exactly the returns 6/7 [82]. □

The two laws form the oriented framework

$$V_{\mathrm{vac}} = \begin{pmatrix} 21 & 22 \\ 6 & 7 \end{pmatrix}, \qquad \det V_{\mathrm{vac}} = 21 \cdot 7 - 22 \cdot 6 = 15. \tag{58.18}$$

The nonzero determinant proves the independence of the gap and return directions. The cardinal 15 thus reappears as an invariant of the vacancy calendar. In the electron equation, the pair $(-22, +15)$ comes from the central selector proved in theorem 58.3; the framework (58.18) provides an independent arithmetic corroboration, with its own genealogy.

It is convenient to distinguish two words derived from the same mismatch. The indicator

$$h_n = \mathbf{1}_{\{p_{n+1}-p_n=21\}}$$

has frequency σ_{vac} and returns $6/7$. The word

$$z_n = \lceil (n+1)\delta_{\mathrm{vac}} \rceil - \lceil n\delta_{\mathrm{vac}} \rceil$$

has slope δ_{vac} and contains 45 or 46 units in each block of length one thousand. The laws $21/22$, $6/7$, and $45/46$ are distinct readers of the same irrational rotation.

58.4.2 Bounded polarization and discrete continuity equation

A phase θ defines

$$z_n(\theta) = \lceil (n+1)\delta_{\mathrm{vac}} + \theta \rceil - \lceil n\delta_{\mathrm{vac}} + \theta \rceil.$$

The accumulated intrinsic polarization is

$$\begin{aligned} P_N(\theta) &:= \sum_{n=0}^{N-1} \big(z_n(\theta) - \delta_{\mathrm{vac}} \big) \\ &= \lceil N\delta_{\mathrm{vac}} + \theta \rceil - \lceil \theta \rceil - N\delta_{\mathrm{vac}}, \qquad |P_N(\theta)| < 1. \end{aligned} \tag{58.19}$$

The temporal difference

$$J_N(\theta) = P_{N+1}(\theta) - P_N(\theta) = z_N(\theta) - \delta_{\mathrm{vac}} \tag{58.20}$$

is the associated arithmetic current. The equations (58.19) y (58.20) constitute an exact continuity equation: the accumulated charge remains uniformly bounded, and each pulse is its local variation. The electromagnetic realization applies a unit of charge and a temporal scale to this pair; the polarization and current structure is already fixed before that chart.

58.4.3 Harmonic Closure of Conjugated Null Modes in the Electronic State

The central electronic section admits a structural reading as the closure of two conjugate null modes. For $u, v > 0$, the hyperbolic projectors $P_{\pm}$ satisfy

$$N_{\mathrm{sp}}(uP_+ + vP_-) = uv > 0,$$

although each isolated component has zero norm. In this construction, the resonant closure of conjugate electromagnetic null modes denotes the two-sheet transport ring whose holonomy closes the central material section; emission and absorption correspond to opening and closing a null orientation relative to that section.

Photonic realization of harmonic closure

The closure of two null modes and the positivity of their coupled norm are algebraic identities. The electromagnetic realization is specified by an intertwiner with the Maxwell modes, a closure Hamiltonian, and the corresponding transition rates. The public expression *resonant closure of conjugate electromagnetic null modes* names the structural object that receives that realization.

58.5 Central electron and helical reader of matter

58.5.1 Uniqueness of the Material Center under Cellular Inversion

The material atlas has domain $C_{81} = \{1, \dots, 9\}^2$ and involution

$$\rho_c(r, c) = (10 - r, 10 - c).$$

Its only fixed point is

$$(r, c) = (5, 5).$$

The other 80 cells form 40 pairs. The central electron is the only point multisection; the remaining internal species are fibers, orbits, or multisections. Spiral theory does not call this conclusion into question.

58.5.2 Common singular mode of the APP sheets in the canonical chart $1, \dots, 9$

In the canonical APP chart $1, \dots, 9$, the vector of values of the operational tritic phase selector is

$$m_{\mathrm{ph}} = (1, -1, 0)^3.$$

For the three-dimensional joint table,

$$C^{(3)} m_{\mathrm{ph}} = (C^{(3)})^* m_{\mathrm{ph}} = -\frac{1}{2} m_{\mathrm{ph}}.$$

Therefore

$$s = \frac{1}{2}, \qquad 1 - s^2 = \frac{3}{4} = \frac{90}{120},$$

$$\|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}. \tag{58.21}$$

The mode, oriented defect $90 \to 120$, and electronic prefactor are coordinated publications of the same APP–TRIT–TPK chain, whose effective composition provides their internal intertwining.

58.5.3 Genealogical record of the material route: surviving extension, full signature, and dimensionless character

Each cell of the atlas bears the signature

$$\mathcal{R}(r, c) = (n_A, s_C, k_{\Delta_4}, \nu_{120}, \nu_{270}, \tau_{12})_{r,c}.$$

The image contains 56 route families; refinement by family and level contains 13 multisections. The governing counts are

$$81/56/13/324,$$

with the null sector and the global towers. The radial reader only can be added by means of a compatible signature, by ejemplo

$$\Sigma_{\mathrm{esp}}(X) = (p_X, a_X, [\Lambda_X, C_X], [\kappa_{9,X}], \mu_X, \varepsilon_{\mathrm{tw},X}),$$

that is constant where appropriate and natural under refinement.

58.5.4 Material compatibility square

Let $\mathcal{E}_{\mathrm{atlas}}$ the space of persistent sections and $\mathfrak{S}_{\mathrm{esp}}$ the spiral reader. The material compatibility is expressed by means of the square

$$\begin{array}{ccc} \mathcal{E}_{\mathrm{atlas}} & \xrightarrow{\mathfrak{S}_{\mathrm{esp}}} & \mathcal{D}^{\mathrm{enr}}_{\mathrm{rad}} \\ {\scriptstyle (K_D,K_\Omega)}\downarrow & & \downarrow{\scriptstyle \mathrm{ev}^{\mathrm{esp}}_{\mathrm{dim}}} \\ \mathrm{Spec}(K_D,K_\Omega) & \xrightarrow{\mathfrak{R}_{\mathrm{mat}}} & \mathcal{M}_{\mathrm{hel}} \end{array}$$

The electron $(5,5)$ provides the central anchor, but does not select p, Λ, C retrospectively. The other species require constancy on multisections, compatibility with cell inversion, K_D/K_Ω, CKM/PMNS, color, and spin representation.

58.5.5 Spin and helicity

is not identified the spin $1/2$ with a vuelta of a helix. The fermionic representation satisfies

$$U(2\pi)\psi_e = -\psi_e, \qquad U(4\pi)\psi_e = \psi_e.$$

The nonadic helix records phase and memory; the spin representation lives in another codomain. A valid material intertwiner must preserve both laws, not replace one with the other.

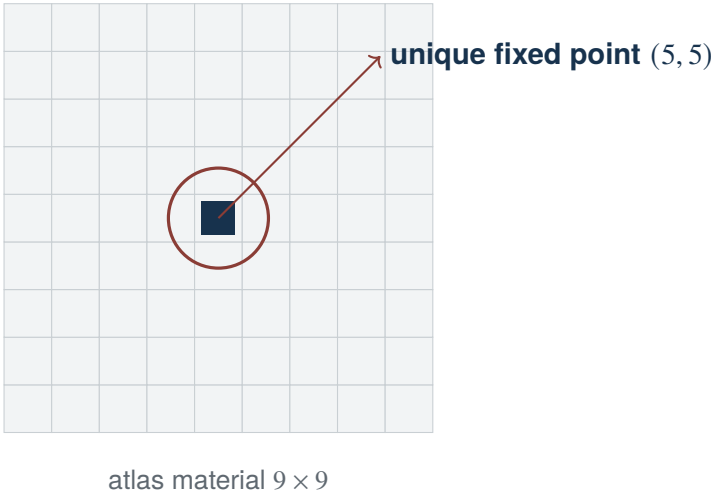


Figure 58.1: The central electron is a result of the atlas; the spiral correspondence is a subsequent reader.

Chapter 59

From the genealogical number to the physical magnitude: action, mass, and dimensional observables

Dimensional publication is organized by an absolute action line and a family of positive operators. The local determinantal order produces the decade minus thirty-four; only thereafter do the metrological chart and the dimensional observables intervene.

59.1 Lines of magnitude and dimensional realization of the HMT kernel

The transition from HMT to Dimensional Mechanics does not consist in adjoining a unit symbol to a real number. The arithmetic core produces characters, cocycles, ratios, and orders of a decimal chart. By contrast, a physical magnitude is an element of a line endowed with dimensional type. The distinction is structural: a change of units modifies the coordinates of a magnitude, but not the magnitude itself. On the one hand, this section constructs the ambient lattice required to realize mass, length, duration, and charge independently, and determines that its minimum rank is four. On the other hand, it constructs a monoidal functor of trajectories with signature that takes values in a portion of that lattice. The minimality of the lattice is not attributed to the functor.

The construction has two consequences. First, it makes it possible to realize action, duration, length, charge, and mass in a common lattice without introducing the International System into the generative formulas; assigning charge to states additionally requires a specific integral character. Second, it proves exactly which part of an absolute scale cannot follow from dimensionless scalars. This impossibility does not affect ratios, mass characters, or constitutive identities; it fixes the correct mathematical type of their outputs. The distinction among dimension, numerical coordinate, and choice of unit is that of the classical theory of dimensional analysis [18, 17]; the structure of lines that follows makes this distinction explicit in the HMT domain.

59.1.1 Lattice of dimensions and oriented lines

Let

$$\Lambda_{\mathrm{D}} = \mathbb{Z}\langle \mathbf{M}, \mathbf{L}, \mathbf{T}, \mathbf{Q} \rangle \cong \mathbb{Z}^4$$

the lattice of dimensions, written in terms of mass, length, duration, and charge. For $d = (d_M, d_L, d_T, d_Q) \in \Lambda_{\mathrm{D}}$, we denote by $\mathcal{L}_d$ an oriented real line and by $\mathcal{L}_d^+$ its positive ray. The tensor product realizes addition of dimensions,

$$\mathcal{L}_d \otimes \mathcal{L}_e \simeq \mathcal{L}_{d+e}, \qquad \mathcal{L}_d^{\vee} \simeq \mathcal{L}_{-d}.$$

The collection of these lines, with positive isomorphisms and tensor product, forms a symmetric Picard groupoid. A dimensionless scalar is an element of $\mathcal{L}_0 \simeq \mathbb{R}$; a quantity of dimension d is an element of $\mathcal{L}_d$.

We introduce four dimensions adapted to HMT:

$$\mathbf{S} = \mathbf{M} + 2\mathbf{L} - \mathbf{T}, \qquad \mathbf{C} = \mathbf{L} - \mathbf{T}, \qquad \mathbf{T} = \mathbf{T}, \qquad \mathbf{Q} = \mathbf{Q}.$$

They represent, respectively, action, velocity, duration, and charge. In the basis $(\mathbf{M}, \mathbf{L}, \mathbf{T}, \mathbf{Q})$, their columns form the matrix

$$B_{\mathrm{D}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Theorem 59.1 (Adapted base dimension for HMT)**.** *The dimensions*

$$\mathbf{S}, \qquad \mathbf{C}, \qquad \mathbf{T}, \qquad \mathbf{Q}$$

constitute a unimodular basis of Λ_{D}*. In particular, every mechanical, electromagnetic, or gravitational dimension generated by* $(\mathbf{M}, \mathbf{L}, \mathbf{T}, \mathbf{Q})$ *has a unique expression with integer exponents in the four lines* $\mathcal{L}_S, \mathcal{L}_C, \mathcal{L}_T, \mathcal{L}_Q$*.*

Proof. The matrix B_{D} is block triangular and $\det B_{\mathrm{D}} = 1$. It therefore belongs to $\mathrm{GL}_4(\mathbb{Z})$. Its integral inverse provides

$$\mathbf{L} = \mathbf{C} + \mathbf{T}, \qquad \mathbf{M} = \mathbf{S} - 2\mathbf{C} - \mathbf{T}.$$

The four original dimensions are thereby uniquely recovered. □

Among the identities that we will use are

$$\mathbf{E} = \mathbf{S} - \mathbf{T}, \qquad \mathbf{Z} = \mathbf{S} - 2\mathbf{Q}, \tag{59.1}$$

$$\boldsymbol{\mu} = \mathbf{S} - \mathbf{C} - 2\mathbf{Q}, \qquad \boldsymbol{\varepsilon} = -\mathbf{S} - \mathbf{C} + 2\mathbf{Q}, \tag{59.2}$$

$$\mathbf{G} = -\mathbf{S} + 5\mathbf{C} + 2\mathbf{T}, \qquad \boldsymbol{\kappa} = -\mathbf{S} + \mathbf{C} + 2\mathbf{T}, \tag{59.3}$$

$$\boldsymbol{\Lambda} = -2\mathbf{C} - 2\mathbf{T}. \tag{59.4}$$

Here $\mathbf{E}$ is energy, $\mathbf{Z}$ impedance, $\boldsymbol{\mu}$ permeability, $\boldsymbol{\varepsilon}$ permittivity, $\mathbf{G}$ the dimension of the gravitational constant, $\boldsymbol{\kappa} = \mathbf{G} - 4\mathbf{C}$, and $\boldsymbol{\Lambda} = -2\mathbf{L}$.

59.1.2 Rescaling group and impossibility theorem

The group of unit base change transformations is

$$\mathcal{G}_{\mathrm{D}} := \mathrm{Hom}(\Lambda_{\mathrm{D}}, \mathbb{R}_{>0}) \cong (\mathbb{R}_{>0})^4.$$

For $g \in \mathcal{G}_{\mathrm{D}}$ and $d \in \Lambda_{\mathrm{D}}$, write g^d for the corresponding character. If u_d is a positive basis of $\mathcal{L}_d$, the chart change $u_d \mapsto g^d u_d$ transforms the coordinate of a fixed quantity according to

$$x\, u_d = (g^{-d} x)(g^d u_d).$$

The magnitude does not change; its numerical representation changes.

Theorem 59.2 (Impossibility of an Absolute Scale from Dimensionless Data)**.** *Let X be a space on which $\mathcal{G}_{\mathrm{D}}$ acts trivially, and let $d \neq 0$. Every map $F : X \to \mathcal{L}_d$ that is natural with respect to all changes of units and that receives no additional dimensional basis is identically zero. By contrast, every quotient of two nonzero sections of the same line, and every product whose total dimension is zero, is invariant under $\mathcal{G}_{\mathrm{D}}$.*

Proof. Naturality and the trivial action on X imply

$$F(x) = F(gx) = g^d F(x) \qquad (g \in \mathcal{G}_{\mathrm{D}}).$$

Since $d \neq 0$, there exists g with $g^d \neq 1$; hence $(g^d - 1)F(x) = 0$, and then $F(x) = 0$. For two sections of dimension d, the factors g^d cancel in the quotient. The same cancellation occurs in every monomial of total dimension zero. □

The theorem does not establish that a discrete theory is incapable of constructing an internal scale. It establishes that this scale must appear as a section of a line or, equivalently, that the theory must extend its domain by the torsor of dimensional frames. It is precisely this extension that defines Dimensional Mechanics.

Corollary 59.3 (Minimum rank)**.** *Every system of lines that permits mass, length, duration, and charge to be realized independently has scale rank at least four. The basis* $(\mathbf{S}, \mathbf{C}, \mathbf{T}, \mathbf{Q})$ *attains the bound; it is therefore minimal.*

Proof. The four original dimensions are linearly independent in Λ_{D}. Three generators cannot span a rank-four lattice. The theorem 59.1 provides four generators that do span it. □

59.1.3 Internal dimensional space: degrees, charts, and HMT–MD transduction operators

An internal framework HMT–MD is a quadruple of positive bases

$$\mathfrak{u} = (u_S, u_C, u_T, u_Q) \in \mathrm{Fr}_{\mathrm{D}},$$

where Fr_{D} is a principal torsor under $\mathcal{G}_{\mathrm{D}}$. The derived bases are defined tensorially:

$$\begin{aligned} u_L &= u_C u_T, & u_M &= u_S u_C^{-2} u_T^{-1}, & u_E &= u_S u_T^{-1}, \\ u_Z &= u_S u_Q^{-2}, & u_\mu &= u_S u_C^{-1} u_Q^{-2}, & u_\varepsilon &= u_S^{-1} u_C^{-1} u_Q^{2}, \\ u_G &= u_S^{-1} u_C^{5} u_T^{2}, & u_\kappa &= u_S^{-1} u_C u_T^{2}, & u_\Lambda &= u_C^{-2} u_T^{-2}. \end{aligned}$$

Juxtaposition here denotes the tensor product of bases. An external chart is another frame $\mathfrak{u}_{\mathrm{ext}}$ together with a positive isomorphism between the two systems of lines. Nothing requires that chart to be SI; the choice of SI belongs to the metrological publication of a magnitude.

59.1.3.1 Action line

The determinantal reader local begins from

$$G_H = \mathrm{coker}(H_4), \qquad |G_H| = 16, \qquad \Sigma_H = \varphi \alpha_{\mathrm{HMT}}^{16},$$

and proves

$$10^{-34} < \Sigma_H < 10^{-33}, \qquad \mathrm{ord}_{10}(\Sigma_H) = -34.$$

The post-return regional realization provides the dimensionless angular coordinate.

$$\eta_{\mathrm{ret}}^{(\mathrm{deg})} = \frac{C_{\mathrm{deg}}^{*}}{2} - \alpha_{\angle,\mathrm{deg}} = 1.0545718169150390611336905847242997\ldots.$$

We define the internal action reduced section by

$$\hbar_{\mathrm{int}} := \eta_{\mathrm{ret}}^{(\mathrm{deg})}\, 10^{-34} u_S, \qquad h_{\mathrm{int}} := 2\pi \hbar_{\mathrm{int}}. \tag{59.5}$$

The exponent is an output of the reader; u_S is the basis of a line. The composition is covariant under $\mathcal{G}_{\mathrm{D}}$ and contains no identification with J s.

59.1.3.2 Periodicity of route edge and resulting constitutive velocity

Let $\mathsf{P}_{\mathrm{TPK}}$ be the free category of trajectories of the graph of enriched states. The grading by length

$$\tau(\gamma) = |\gamma|, \qquad \tau(\gamma\eta) = \tau(\gamma) + \tau(\eta)$$

define the combinatorial clock of the protocol. Its implementation is

$$T_{\mathrm{int}}(\gamma) = \tau(\gamma) u_T.$$

This duration counts transitions of the fixed graph. A subdivision of the graph defines another clock and must be accompanied by its refinement morphism.

The channels $q_\pm$ generate the symmetric and antisymmetric functions

$$\mathcal{E} = \log\big[(1-q_+^{90})(1-q_-^{90})\big] - \log\big[(1-q_+^{120})(1-q_-^{120})\big],$$

$$\mathcal{B} = \log\frac{1-q_-^{90}}{1-q_+^{90}} - \log\frac{1-q_-^{120}}{1-q_+^{120}}.$$

These determine the positive coefficients

$$\widehat{c} = e^{-\mathcal{E}}, \qquad \widehat{Z} = e^{\mathcal{B}}.$$

Its internal realizations are

$$c_{\mathrm{int}} = \widehat{c}\, u_C, \qquad Z_{\mathrm{int}} = \widehat{Z}\, u_Z. \tag{59.6}$$

For a constant channel along a path one obtains the length

$$L_{\mathrm{int}}(\gamma) = \widehat{c}\,|\gamma|\, u_L. \tag{59.7}$$

If the channel depends on the edge, the covariant formula is the sum $\sum_{e\in\gamma}\widehat{c}(e)u_L$; its additivity under concatenation is preserved.

59.1.3.3 Elementary charge scale and conditional character

Since $u_Z = u_S u_Q^{-2}$, the quotient $h_{\mathrm{int}}/Z_{\mathrm{int}}$ belongs to $\mathcal{L}_Q^{\otimes 2}$. The positive orientation of the lines selects a unique positive root. We define

$$e_{\mathrm{int}} := \left(\frac{2\alpha_{\mathrm{HMT}} h_{\mathrm{int}}}{Z_{\mathrm{int}}}\right)^{1/2} \in \mathcal{L}_Q^{+}. \tag{59.8}$$

Proposition 59.4 (Internal Closure of the Fine Structure Relation)**.** *The section* (59.8) *is the unique positive section that satisfies*

$$\alpha_{\mathrm{HMT}} = \frac{Z_{\mathrm{int}} e_{\mathrm{int}}^2}{2h_{\mathrm{int}}}.$$

The equality is invariant under the full scaling group.

Proof. Existence follows from the positivity of $\alpha_{\mathrm{HMT}}, h_{\mathrm{int}}, Z_{\mathrm{int}}$. An oriented real line has a unique positive root of a positive section of its square. The dimension of the right-hand side is $\mathbf{Z} + 2\mathbf{Q} - \mathbf{S} = 0$, so all rescaling factors cancel. □

This proposition does not count as a second independent determination of α: within the system of lines, it realizes the constitutive compatibility among action, impedance, and an elementary charge scale. It does not construct a charge on the states. This additionally requires an integer character

$$n_Q : \mathrm{Ob}(\mathsf{P}_{\mathrm{TPK}}^{\mathrm{sig}}) \longrightarrow \mathbb{Z}; \qquad Q(x) = n_Q(x) e_{\mathrm{int}}.$$

The existence and selection of n_Q are additional hypotheses. In particular, n_Q is not a component of the trajectory functor constructed below. The elementary scale and the classification of states are distinct maps.

59.1.4 Monoidal path functor with signature

Let $\mathsf{P}_{\mathrm{TPK}}^{\mathrm{sig}}$ be the category of enriched paths decorated with a signature coordinate $v_x \in \mathbb{Z}^5$ at each object. For $\gamma : x \to y$, we set

$$\Delta v(\gamma) = v_y - v_x.$$

Composition satisfies $\Delta v(\gamma\eta) = \Delta v(\gamma) + \Delta v(\eta)$. Let

$$\chi_{\mathrm{mass}}(v) = \exp\langle \lambda_{\mathrm{mass}}, v\rangle$$

the positive mass character. On the route quotient, we fix the action law determined by five data: the normalized trace of the distinct emissions, a positive additive cocycle, the change defect $\Delta_4 P_3$, the residual capacity $\lambda = 1 - \delta$, and the local sum of edge costs. In this law,

$$\mathcal{A}_*(\gamma) = N_{\mathrm{giro}}(\gamma) + \lambda_* N_{\mathrm{hoja}}(\gamma), \qquad \lambda_* = 1 - \frac{5}{468} - \frac{3}{11}\Delta_4.$$

The following argument requires only additivity; it also applies to the functional I_λ with any $\lambda > 0$.

Consider the monoid

$$\mathfrak{M}_{\mathrm{D}} = \mathbb{N} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$$

with product

$$(t, \ell, s, r) \odot (t', \ell', s', r') = (t + t', \ell + \ell', s + s', rr').$$

Theorem 59.5 (Monoidal path functor with signature)**.** *The map*

$$\mathfrak{D}_{\mathrm{HMT}}(\gamma) = \big(|\gamma|, \widehat{c}|\gamma|, \mathcal{A}_*(\gamma), \chi_{\mathrm{mass}}(\Delta v(\gamma))\big)$$

define a monoidal functor

$$\mathfrak{D}_{\mathrm{HMT}} : \mathsf{P}^{\mathrm{sig}}_{\mathrm{TPK}} \longrightarrow \mathfrak{M}_{\mathrm{D}}.$$

After choosing an internal frame, its three additive components are realized as

$$T_{\mathrm{int}}(\gamma) = |\gamma| u_T, \quad L_{\mathrm{int}}(\gamma) = \widehat{c}|\gamma| u_L, \quad S_{\mathrm{int}}(\gamma) = \mathcal{A}_*(\gamma)\hbar_{\mathrm{int}},$$

and the fourth transports dimensionless mass ratios. The functor does not contain a charge coordinate and does not generate the ambient dimensional lattice.

Proof. The length of a trajectory, the number of turns, and the number of changes of sheet are additive under compatible concatenation. By hypothesis, $\mathcal{A}_*$ is additive. The difference of signatures is additive, and the mass character transforms sums into products. Each component therefore respects the law $\odot$, and the empty route is sent to $(0, 0, 0, 1)$. The linear realization preserves sums because it multiplies by fixed sections of the corresponding lines. □

Remark 59.6 (Ambiental lattice and scope of the functor). The minimum-rank corollary concerns a system capable of varying mass, length, duration, and charge independently. The preceding functor realizes duration, length, and action, and transports a mass ratio; it does not use u_Q or prove a rank bound. The subsequent assignment of charges remains conditional upon the existence of n_Q. In the dimensional lattice, the types of its components are

$$\mathbf{T}, \qquad \mathbf{L} = \mathbf{C} + \mathbf{T}, \qquad \mathbf{S}, \qquad \mathbf{0},$$

so that its dimensional image is contained in $\langle \mathbf{S}, \mathbf{C}, \mathbf{T}\rangle_{\mathbb{Z}}$, of rank three. The rescaling $u_Q \mapsto \lambda u_Q$, with the other bases fixed, leaves all components of the functor invariant; the latter does not faithfully realize the fourth ambient direction.

If the bidirectional electronic section is assigned to each object as the origin of the mass chart,

$$m(x) = m_e^{\beta} \chi_{\mathrm{mass}}(v_x - v_e),$$

then every morphism $\gamma : x \to y$ satisfies exactly

$$\frac{m(y)}{m(x)} = \chi_{\mathrm{mass}}(\Delta v(\gamma)).$$

The theorem closes dimensional propagation once the signature is given. It does not change the status of the preceding map associating a species with a trajectory and its signature.

59.1.5 Basal stage and bidirectional closure of the electronic scale

The prevailing electronic invariant is the positive scalar.

$$\mathfrak{e}_0 = \frac{\sqrt{3}}{4}\left(e^{\varphi/\pi^2} - 22\alpha_{\mathrm{HMT}}^3\right)(1 + 15\Delta_4).$$

Its numerical value is $0.510998922488066\ldots$. This is the basal stage of the genealogy, not its governing output. In the internal dimensional framework, it defines an energy section and a mass section related by the velocity:

$$E_e^{\mathrm{int}} = \mathfrak{e}_0 u_E, \qquad m_e^{\mathrm{int}} = \frac{E_e^{\mathrm{int}}}{c_{\mathrm{int}}^2} = \mathfrak{e}_0 \widehat{c}^{-2} u_M. \tag{59.9}$$

The central route and its two bidirectional readers determine, without consulting the external mass,

$$\kappa_e = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4, \qquad R_{\mathrm{act}} = \frac{H_5}{\eta_{\mathrm{ret}}^{(\mathrm{deg})}}.$$

The current electronic section is, therefore,

$$\begin{aligned} &E_e^{\beta} = E_e^{(0)} R_{\mathrm{act}}^{\kappa_e}, && m_e^{\beta} = \frac{E_e^{\beta}}{c_{\mathrm{int}}^2}, \\ &E_e^{\beta} = 0.51099895069000080860825716 48\ldots\ \mathrm{MeV}, && u_E = 1\,\mathrm{MeV}. \end{aligned} \tag{59.10}$$

Thus, the basal formula retains its derivative role, and bidirectional closure governs all subsequent propagation through characters and towers. In a natural-units chart in which the coordinate of c_{int} is fixed at one, both numerical coordinates coincide; their lines are not silently identified.

Proposition 59.7 (Typed separation of the energy base and the clock quantum)**.** *Escribamos $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, and definamos*

$$\varepsilon_{\mathrm{clk}} := \hbar_{\mathrm{int}} \frac{2\pi}{108 t_0}, \qquad \kappa_{\mathrm{clk}}(\tau_0) := \frac{2\pi \eta_{\mathrm{ret}}^{(\mathrm{deg})}\, 10^{-34}}{108\tau_0}.$$

Then

$$\varepsilon_{\mathrm{clk}} = \kappa_{\mathrm{clk}}(\tau_0) u_E, \qquad m_0 := \frac{\varepsilon_{\mathrm{clk}}}{c_{\mathrm{int}}^2} = \kappa_{\mathrm{clk}}(\tau_0)\widehat{c}^{-2} u_M,$$

and the current electronic section satisfies

$$E_e^{\mathrm{int}} = \frac{\mathfrak{e}_0}{\kappa_{\mathrm{clk}}(\tau_0)} \varepsilon_{\mathrm{clk}}.$$

Therefore, $E_e^{\mathrm{int}} = \mathfrak{e}_0 \varepsilon_{\mathrm{clk}}$ if and only if $\kappa_{\mathrm{clk}}(\tau_0) = 1$. Adopting $\varepsilon_{\mathrm{clk}}$ as the basis while retaining the same coordinate $\mathfrak{e}_0$ is a new choice of frame; it is not a correction of the electronic invariant, nor does it follow from order-108 periodicity. In particular, the normalization used later, $t_0 = u_T$, fixes $\tau_0 = 1$, not $\kappa_{\mathrm{clk}} = 1$.

Proof. One substitutes $\hbar_{\mathrm{int}} = \eta_{\mathrm{ret}}^{(\mathrm{deg})}\, 10^{-34} u_S$, $t_0 = \tau_0 u_T$, $u_E = u_S u_T^{-1}$, and $c_{\mathrm{int}} = \widehat{c}, u_C$. The final identity follows by comparison with equation (59.9). All equalities occur between sections of the same line; none consults an external unit or mass. □

Proposition 59.8 (Mass scale propagation)**.** *For each declared signature $v \in \mathbb{Z}^5$,*

$$m_v^{\mathrm{int}} := m_e^{\mathrm{int}} \chi_{\mathrm{mass}}(v)$$

is a positive section of $\mathcal{L}_M$, and

$$\frac{m_{v+w}^{\mathrm{int}}}{m_v^{\mathrm{int}}} = \chi_{\mathrm{mass}}(w).$$

The construction is covariant under rescaling and does not require an external unit of energy or mass.

Proof. The character is dimensionless and positive, so it does not modify the line of m_e^{int}. The quotient identity expresses the multiplicativity of χ_{mass}. Under a change of frame, the numerator and denominator acquire the same weight $\mathbf{M}$, which cancels. □

A subsequent metrological chart may send u_E to an energy basis and u_M to the coherent basis obtained by dividing by the square of the standard velocity. Only that chart authorizes the section to be expressed by an external unit name.

59.1.6 Constitutive as isomorphisms between lines

The internal functions satisfy

$$\widehat{\mu} = e^{\mathcal{B}+\mathcal{E}}, \qquad \widehat{\varepsilon} = e^{-\mathcal{B}+\mathcal{E}}, \qquad \widehat{Z} = e^{\mathcal{B}}, \qquad \widehat{c} = e^{-\mathcal{E}}.$$

We define the sections

$$\mu_{\mathrm{int}} = \widehat{\mu} u_\mu, \qquad \varepsilon_{\mathrm{int}} = \widehat{\varepsilon} u_\varepsilon.$$

Theorem 59.9 (Covariant Linear Constitutive Relations)**.** *In the groupoid of oriented lines, the exact identities hold.*

$$\sqrt{\frac{\mu_{\mathrm{int}}}{\varepsilon_{\mathrm{int}}}} = Z_{\mathrm{int}}, \qquad \frac{1}{\sqrt{\mu_{\mathrm{int}}\varepsilon_{\mathrm{int}}}} = c_{\mathrm{int}}.$$

The four sections depend on two dimensionless characters and on the two lines $\mathcal{L}_Z, \mathcal{L}_C$; they do not constitute four independent scales.

Proof. In coefficients, $\widehat{\mu}/\widehat{\varepsilon} = \widehat{Z}^2$ and $\widehat{\mu\varepsilon} = \widehat{c}^{-2}$. In dimensions,

$$\boldsymbol{\mu} - \boldsymbol{\varepsilon} = 2\mathbf{Z}, \qquad \boldsymbol{\mu} + \boldsymbol{\varepsilon} = -2\mathbf{C}.$$

The positive roots are therefore sections of the correct lines, and the identities are covariant. □

59.1.7 Cartan Scale and Homogeneity of the Gravitational Action

The elementary duration $t_0 = u_T$ and the constitutive velocity determine the length of a transition,

$$\ell_0 = c_{\mathrm{int}} t_0 = \widehat{c}\, u_L.$$

This section allows for the homogenization of the Cartan affine connection:

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega^I{}_J & \ell_0^{-1} e^I \\ 0 & 0 \end{pmatrix}, \qquad \mathcal{F}_{\ell_0} = \begin{pmatrix} F^I{}_J & \ell_0^{-1} T^I_{\mathrm{tor}} \\ 0 & 0 \end{pmatrix}.$$

The quotient e/ℓ_0 is dimensionless; curvature and torsion remain in distinct blocks of the same transport.

The dimension of G does not add a fifth generator, since $\mathbf{G} = -\mathbf{S} + 5\mathbf{C} + 2\mathbf{T}$. For every positive dimensionless parameter θ_G, we define the covariant family

$$G_{\mathrm{int}}(\theta_G) := \theta_G \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_{\mathrm{int}}} \in \mathcal{L}_G^+. \tag{59.11}$$

The choice $\theta_G = 1$ is the internal Planck normalization associated with an elementary transition. By construction, it satisfies

$$\ell_0^2 = \frac{\hbar_{\mathrm{int}} G_{\mathrm{int}}(1)}{c_{\mathrm{int}}^3}.$$

This reference normalization is neither the circular selector nor the holonomic selector of the physical section of G. The parameter θ_G makes visible the freedom not fixed by the dimensional identities. A TPK dynamics that produces a gravitational character must determine it before any external chart.

Let

$$\kappa_{\mathrm{int}} = \frac{8\pi G_{\mathrm{int}}}{c_{\mathrm{int}}^4}.$$

In the following functional, M is a smooth oriented manifold of dimension four, and $\gamma \in \mathbb{R} \setminus \{0\}$. If M has a boundary, variations of compact support in the interior are imposed, or a boundary term with conditions that cancel the boundary variation. A spinorial matter action additionally presupposes a compatible spin structure. If the co-tetrad e^I takes values in $\mathcal{L}_L$ and the curvature F^{IJ} is a dimensionless two-form after the connection has been integrated, then $\int \Sigma_{IJ} \wedge F^{IJ}$ has dimension $2\mathbf{L}$.

Theorem 59.10 (Homogeneity of the Palatini–Cartan–Holst Action)**.** *With explicit units, the gravitational functional of action dimension is*

$$\boxed{S_{\mathrm{PCH}} = \frac{1}{2\kappa_{\mathrm{int}} c_{\mathrm{int}}} \int \Sigma_{IJ} \wedge \left(\star F^{IJ} + \gamma^{-1} F^{IJ} \right) - \frac{1}{\kappa_{\mathrm{int}} c_{\mathrm{int}}} \int \Lambda_{\mathrm{int}} \, \mathrm{vol}_e + S_{\mathrm{m}}.}$$

Both geometric terms belong to $\mathcal{L}_S$. Writing them with $1/(2\kappa)$ without the factor c^{-1} belongs to a convention $c = 1$ or has dimension $\mathbf{S} + \mathbf{C}$, not action dimension.

Proof. De (59.3), $\boldsymbol{\kappa} = -\mathbf{S} + \mathbf{C} + 2\mathbf{T}$. Moreover, $2\mathbf{L} = 2\mathbf{C} + 2\mathbf{T}$. therefore,

$$-\boldsymbol{\kappa} - \mathbf{C} + 2\mathbf{L} = \mathbf{S}.$$

The cosmological term has the same dimension because $\boldsymbol{\Lambda} + 4\mathbf{L} = 2\mathbf{L}$. If $-\mathbf{C}$ is omitted, the result is $\mathbf{S} + \mathbf{C}$. □

The Cartan–Holst equation then retains its variational content provided that the spin current is defined with the normalization consistent with this functional. The dynamical realization must still provide a morphism from the TPK holonomies to (e, ω, Ψ); the present section constructs the exact dimensional codomain and removes the ambiguity of the units in that morphism.

59.1.8 Universality of Dimensional Realization

We collect the results. Let $\mathcal{X}_{\mathrm{HMT}}$ be the domain of enriched states, and let the following be dimensionless data on it:

$$\alpha_{\mathrm{HMT}}, \quad A_{\mathrm{deg}}, \quad C^*_{\mathrm{deg}}, \quad \widehat{c}, \widehat{Z}, \quad \mathcal{A}_*, \quad \chi_{\mathrm{mass}}, \quad \mathfrak{e}_0,$$

built in the previous chapters over their declared domains. Consider the extended domain

$$\mathcal{X}_{\mathrm{MD}} := \mathcal{X}_{\mathrm{HMT}} \times \mathrm{Fr}_{\mathrm{D}}\,.$$

Theorem 59.11 (Universal dimensional extension)**.** *There exists a unique tensorial, positive, and $\mathcal{G}_{\mathrm{D}}$-equivariant extension that:*

1. *realizes the edge clock in $\mathcal{L}_T$;*
2. *realizes $\widehat{c}$ and $\widehat{Z}$ in $\mathcal{L}_C$ and $\mathcal{L}_Z$;*

3. *realizes the determinantal index and the post-return component in $\mathcal{L}_S$;*
4. *satisfies the exact relations among impedance, propagation speed, permeability, and permittivity;*
5. *satisfies the internal fine-structure relation with an elementary scale of positive charge and, if a character n_Q is supplied, conditionally realizes $Q(x) = n_Q(x)e_{\mathrm{int}}$;*
6. *transports the mass character into $\mathcal{L}_M$.*

All other lines of the lattice are obtained by tensor products and duals. Two realizations differ by exactly one element of $\mathcal{G}_{\mathrm{D}}$.

Proof. The theorem 59.1 identifies $(\mathbf{S}, \mathbf{C}, \mathbf{T}, \mathbf{Q})$ with a free basis of the lattice. A monoidal map on a free category is determined by its value on the generators. Positivity fixes the roots that appear in charge and the constitutive relations. The identities already proved ensure that the definitions do not depend on an alternative presentation of a dimension. The simply transitive action of $\mathcal{G}_{\mathrm{D}}$ on the frames proves the final assertion. □

The mathematical conclusion is precise. HMT fixes the genealogy of the internal coefficients, and Dimensional Mechanics realizes them in a minimal ambient lattice of rank four when mass, length, duration, and charge must vary independently. The monoidal functor of trajectories with signature occupies part of that lattice and therefore does not inherit a claim of minimality. Ratios and homogeneous equations are independent of every chart. A publication in seconds, joules, coulombs, or electronvolts is the subsequent composition with an external frame; it does not alter any of the internal theorems and cannot replace them.

The HMT closure produces numbers as complete genealogical sections. Dimensional Mechanics begins when a section of that class is realized on a route, transports a register, induces a spectral operator, and acquires a physical chart. This step preserves the mathematical antecedent and changes the codomain: number, state, route, spectrum, and particle are objects linked by explicit maps and retain different types.

The formal hinge is expressed through

$$\boldsymbol{x} \overset{\mathrm{sh}}{\longmapsto} \gamma_{\boldsymbol{x}} \overset{\mathcal{L}}{\longmapsto} \mathcal{L}_{\boldsymbol{x}} \overset{\Sigma}{\longmapsto} \sigma_{\boldsymbol{x}} \overset{\chi}{\longmapsto} \chi_{\boldsymbol{x}} \overset{\mathcal{T}}{\longmapsto} \widehat{K}_{\boldsymbol{x}} \overset{\mathfrak{R}_{\mathrm{phys}}}{\longmapsto} \mathfrak{p}_{\boldsymbol{x}}. \tag{59.12}$$

Each arrow possesses its own domain and preserves the origin key of the initial section.

59.1.9 Physical correspondence of the APP–TRIT–TPK core

The opening of Dimensional Mechanics retraces the formal core with a new codomain. This repetition preserves the HMT theorems and declares the physical-realization maps that are added. Its causal order is

APP as a support of sheets ⟶ TRIT as temporal orientation
↓
unitary cycle and torsional defect
↓
dual angular coordinates ⟶ particle–route

59.1.9.1 The Cayley torus as a support for physical leaves

APP provides the two-dimensional torus

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2$$

with its Cayley graph and two evaluations on the same vertices. At the formal level, Σ accumulates and Π folds. The physical realization assigns sheets and channels to those evaluations, preserving the provenance map that makes it possible to compare their responses. The torus preserves spatial periodicity; the path category preserves direction, reversibility, and transport memory.

59.1.9.2 Euclidean gnomonic chart and curvature atlas

The L-shaped gnomon completes the multiplicative kernel 8×8 and separates the folded interior from the publication chart. In the null class, precisely the two readers that the quotient identifies outside the boundary are distinguished,

$$n \longmapsto n \bmod 9, \qquad n \longmapsto \mathrm{dr}_9(n),$$

so that the residue 0 and its nonadic publication 9 preserve distinct coordinates of the same closure. The sector $3, 6, 9$ makes this transition between the digital root and the modular residue visible.

In the geometric realization, the gnomonic L carries the Euclidean chart $\mathscr{H}_0^{(L)}$. On it, the inertial and gravitational readers descend to a common normalization,

$$\frac{\mathcal{G}_L}{\mathcal{I}_L} = 1,$$

which constitutes the residence of the local equivalence principle. Before that projection, the enriched state separately preserves the areal and volumetric readings. The Ambivalence Principle is then expressed by the internal ratio

$$\boxed{\frac{\mathcal{L}_{\mathrm{ar}}(x)}{\mathcal{L}_{\mathrm{vol}}(x)} = \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2}.}$$

The Euclidean equality and the ambivalent ratio belong to distinct levels: the former is the local projection onto the L; the latter preserves the areal–volumetric genealogy prior to that projection.

The metric realization is obtained via a declared family of maps. For

$$\mathbb{A}_s := \mathbb{R}[X]/(X^2 + s), \qquad s \in \{-1, 0, +1\},$$

and a curvature scale $\rho > 0$, one fixes

$$\mathcal{M}_{s,\rho} : \mathbb{A}_s \dashrightarrow (M_s, g_{s,\rho}), \qquad g_{s,\rho} = \mathrm{d}r^2 + S_{s,\rho}(r)^2 \mathrm{d}\theta^2,$$

with

$$S_{s,\rho}(r) = \begin{cases} \rho^{-1} \sinh(\rho r), & s = -1, \\ r, & s = 0, \\ \rho^{-1} \sin(\rho r), & s = +1. \end{cases} \qquad K(M_s, g_{s,\rho}) = s\rho^2.$$

The radial domain is $r \geq 0$ for $s = -1, 0$; for $s = +1$, the polar chart covers $0 \leq r \leq \pi/\rho$, with the usual polar regularity at its endpoints. The branch $s = -1$ provides the hyperbolic spatial section of the chart of anti–de Sitter type; $s = 0$ realizes the Euclidean gnomonic chart; and $s = +1$ provides the positive spatial geometry that is prolonged into the de Sitter foliations. The complete correspondence is collected in sections 61.1.2 y 88.2.1.

The algebraic classification and the metric family are coordinated through the declared realization $\mathcal{M}_{s,\rho}$. This arrow constitutes the physical-interface ansatz, whereas $K = s\rho^2$ is the exact calculation of the metric once s and ρ are fixed. The quadratic relation fixes the algebraic class; the declared realization transports its sign to curvature, and ρ fixes the geometric scale. In the negative chart, restriction to the boundary coordinates the Archimedean projection with exceptional incidence; in the null chart, the inertial and gravitational readers descend to local equivalence; in the positive chart, the foliations satisfy

$\ddot{a}/a = H^2 > 0$. This atlas links negative-curvature holography, local Euclidean physics, and accelerated cosmological evolution through declared domains and change maps.

59.1.9.3 Temporal Representation of the TRIT: (+1) Return and Memory, (0) Threshold Present, and (-1) Bidirectional Opening

Once an order has been declared on the transitions, the temporal reader is

$$\begin{aligned} +1 &\longmapsto \text{return, retained memory and past,} \\ 0 &\longmapsto \text{evaluation threshold and present,} \\ -1 &\longmapsto \text{bidirectional opening and future.} \end{aligned}$$

The past designates transported closure; the present, the evaluation boundary; the future, the tree of compatible prolongations. This reading is the temporal realization of the three quadratic algebras already constructed in HMT and preserves the involution that exchanges opening and return.

59.1.9.4 Minimal Torsion and Unitary Dynamics in $\mathbb{C}^{108}$

The invariant called minimal torsion is

$$\Delta_4 = \frac{\pi - e\log\pi}{270}\left(1 + \frac{\pi}{729}\right).$$

Its entries arise from the HMT sections of π and e, from the channel 270, and from the interior 729. It functions as a global memory of refinement before electron specialization.

The reference temporal dynamics resides in $\mathcal{H}_{\mathrm{clk}} = \mathbb{C}^{108}$. With

$$\omega_0 = \frac{2\pi}{108\, t_0}, \qquad H_{\mathrm{clk}} = \hbar_{\mathrm{int}}\omega_0 \sum_{k=0}^{107} k\, |\widetilde{k}\rangle\langle\widetilde{k}|,$$

the propagator satisfies

$$U_{\mathrm{step}} = \exp\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}} H_{\mathrm{clk}} t_0\right), \qquad U_{\mathrm{step}}^{108} = I.$$

The unit cycle is exact for the declared primitives. Its realization as a physical time crystal adds the Hamiltonian or driven protocol, the primitive subharmonic response, stability under perturbations, persistence, and the protection mechanism developed in section 61.1. This residence preserves the hierarchy and refers the proof there.

59.1.9.5 Dual angular coordinates and spectral channels

The sum–product incompatibility is transported to two angular coordinates A and C^*. In the Archimedean chart

$$x = \frac{\pi A}{180}, \qquad y = \frac{\pi C^*}{180},$$

the channels are the eigenvalues of the central element

$$T = e^{-xI - yR} = q_+ P_+ + q_- P_-, \qquad q_- = e^{-x+y}, \qquad q_+ = e^{-x-y}.$$

The chart is invertible:

$$x = -\frac{1}{2}\log(q_+ q_-), \qquad y = \frac{1}{2}\log\frac{q_-}{q_+}.$$

Therefore, A and C^* are, respectively, the symmetric and antisymmetric coordinates of a single spectral band. The section 90.4 contains its construction and its proofs; the present hinge fixes its causal position before the particle–route.

59.1.10 State, route, spectral form and physical realization

Definition 59.12 (Finite state and genealogical section)**.** A finite state is an element $x_n \in \mathcal{S}_n^{\mathrm{surv}}$. A genealogical section is a family $\boldsymbol{x} = (x_n)_n$ compatible at every depth. The state is a restriction of the section; the section determines its complete system of restrictions.

Definition 59.13 (Observable route)**.** A chart secuencial of depth R defines the sombra

$$\gamma_{\boldsymbol{x}}^{[R]} := \mathrm{sh}_R(\boldsymbol{x}) = (c_0 \to c_1 \to \cdots \to c_R).$$

The route preserves position, sheet, orientation, carry, and memory according to the chart's data list. It is an ordered presentation of the section.

Definition 59.14 (Spectral form)**.** The spectral form associated with to section is

$$\mathfrak{S}(\boldsymbol{x}) := (\gamma_{\boldsymbol{x}}, \mathcal{L}_{\boldsymbol{x}}, \sigma_{\boldsymbol{x}}, \chi_{\boldsymbol{x}}, \widehat{K}_{\boldsymbol{x}}),$$

where $\mathcal{L}_{\boldsymbol{x}}$ is the route record, $\sigma_{\boldsymbol{x}}$ its integer signature, $\chi_{\boldsymbol{x}}$ the character constructed on the signature, and $\widehat{K}_{\boldsymbol{x}}$ the operator obtained from the tower hierarchy. The spectrum $\mathrm{Spec}\,\widehat{K}_{\boldsymbol{x}}$ is an internal output.

Definition 59.15 (Physical realization)**.** A physical realization of $\mathfrak{S}(\boldsymbol{x})$ consists of

$$\mathfrak{R}_{\mathrm{phys}} = (\mathcal{H}_{\mathrm{phys}}, \rho_{\mathrm{sym}}, \varsigma_{\mathbf{u}}, \mathcal{O}),$$

where $\mathcal{H}_{\mathrm{phys}}$ is the state space, ρ_{sym} the representation of the symmetry group, $\varsigma_{\mathbf{u}}$ the units chart, and $\mathcal{O}$ the set of observables. The realization intertwines route transports with the dynamics of $\mathcal{H}_{\mathrm{phys}}$ and publishes quantities with units.

The spectral form is exact within the discrete system once its readers have been fixed. Identification with a physical species belongs to the realization map. This separation makes it possible to publish the operator, its spectrum, its position, and its status, accompanied by the corresponding physical intertwiner.

59.1.11 4 grammars and their domains

The passage from number to particle uses four grammars. The first three belong to the mathematical core; the fourth belongs to the chosen statistical representation.

Grammar	Operation	Function
accumulation	sheet sum of APP	composes additive contributions;
fold	sheet product of APP	composes scales and transpositions;
orientation	TRIT and route conjugation	distinguishes sense and anti-section;
occupation	projector of the representation	fixes statistical multiplicity.

The fermionic occupation is carried out, for example, in a fixed CAR algebra:

$$\{a_i, a_j\} = 0, \qquad \{a_i^\dagger, a_j^\dagger\} = 0, \qquad \{a_i, a_j^\dagger\} = \delta_{ij} I.$$

It follows from these premises.

$$(a_i^\dagger)^2 = 0, \qquad n_i := a_i^\dagger a_i, \qquad n_i^2 = n_i, \qquad \mathrm{Spec}(n_i) = \{0, 1\}.$$

TPK selects the grading and transports the route; the CAR representation realizes exclusion. This distinction positively fixes the algebraic provenance of the Pauli principle.

59.1.12 The spectral realization functor

Let $\Omega_{\mathrm{HMT}}^{\mathrm{surv}}$ be the category whose objects are surviving sections and whose morphisms are transformations compatible with the restrictions. Let **SpecRoute** be the category of $(\gamma, \mathcal{L}, \sigma, \chi, \widehat{K})$ bundles and intertwiners that preserve the route keys.

Theorem 59.16 (Spectral realization of a section)**.** *The route, register, signature, character, and tower readers define, on their common domain, a functor*

$$\mathfrak{S} : \Omega_{\mathrm{HMT}}^{\mathrm{surv}} \longrightarrow \mathbf{SpecRoute}.$$

For a compatible restriction of sections, the route is truncated, the record is restricted, and subsequent readers commute with that restriction. Two presentations related by a genealogical symmetry produce intertwined spectral forms.

Proof. The route shadow is defined by sequential reading of the state and commutes with truncation. The record concatenates typed local data, and its restriction eliminates only the posterior segment. The signature and the character are calculated by declared homomorphisms over that record. The tower hierarchy prolongs the character by composition. Identity and composition are preserved at every stage. □

The theorem makes the expression "the number assumes spectral form" precise: it transports a complete section to a route-and-spectrum package that preserves its genealogy.

59.1.13 Persistence and definition of particle

For a prefix $s_N \in \mathcal{S}_N^{\mathrm{surv}}$, let

$$\mathrm{Ext}_M(s_N) := \{s_M \in \mathcal{S}_M^{\mathrm{surv}} : \rho_{M,N}(s_M) = s_N\}, \qquad \mathcal{L}(s_N) := \sup\{M : \mathrm{Ext}_M(s_N) \neq \varnothing\}.$$

Definition 59.17 (Persistence)**.** A prefix is persistent when it belongs to a compatible global section; in a finitely branching tree of prolongations, this condition is equivalent to $\mathcal{L}(s_N) = \infty$.

Definition 59.18 (Mathematical particle)**.** Let γ_x be a persistent route and let $\mathcal{B}_{\gamma_x}$ be a discrete bundle whose fibres carry the internal degrees of the realization. A mathematical particle is

$$\mathfrak{p} = (x, \gamma_x, \mathcal{B}_{\gamma_x}, s, n, U, \mathfrak{S}(x), \mathfrak{R}_{\mathrm{phys}}),$$

where s is a compatible section of the bundle, n its occupation function, U the propagator intertwining the transports, and $\mathfrak{R}_{\mathrm{phys}}$ the declared physical realization.

The definition contains three closures:

1. the number is the global closure of compatible prefixes;
2. the spectral form is the closure of the route reading under the record, signature, character, and towers;
3. the particle is the local closure traversed by a persistent section of a physical bundle.

The relation is summarized as

number = global genealogical closure,	particle = persistent local realization of that closure.

The verbal equality denotes a functorial dependence between types.

59.1.14 Oriented band and anti-section

Let $\tau = (\tau_0, \dots, \tau_{11}) \in \{-1, 0, +1\}^{12}$ be the oriented word of a route, and define

$$C_M(\tau) := -\operatorname{rev}(\tau). \tag{59.13}$$

It holds $C_M^2 = I$. The band

$$\mathcal{B}(\tau) := \{\tau, C_M\tau\}$$

brings together two orientations of the same genealogy. One section realizes the particle, and the conjugate section realizes the antiparticle.

Proposition 59.19 (Mass parity and orientation)**.** *Let $\mathfrak{m}(\tau)$ be a mass reading that depends on even invariants of the band, and let $q(\tau)$ be an odd oriented reading. Then*

$$\mathfrak{m}(C_M\tau) = \mathfrak{m}(\tau), \qquad q(C_M\tau) = -q(\tau).$$

Proof. The first identity expresses invariance under the band involution. The second expresses the parity of an oriented form. The relation $C_M^2 = I$ guarantees the compatibility of both readings. □

Particle and antiparticle thus share the mass spectrum and have opposite orientations. The conclusion follows from the parity of the readers.

59.1.15 Path propagation and finite path summation

Let U be a propagator on a finite state space. For every $R \geq 1$, matrix multiplication gives

$$(U^R)_{fi} = \sum_{\substack{\gamma : i \to f \\ |\gamma| = R}} \prod_{\ell \in \gamma} U_\ell. \tag{59.14}$$

Theorem 59.20 (Exact expansion by histories)**.** *The equation (59.14) enumerates once each route of length R between i and f. If U is unitary, U^R preserves the norm and defines a finite unitary evolution.*

Proof. Expanding the matrix product yields all the intermediate indices $(x_1, \dots, x_{R-1})$. Each tuple determines a route, and each route determines a tuple. Unitarity follows from $(U^R)^\dagger U^R = I$. □

The wave amplitude is the coherent sum over the band of routes; the particle is the persistent section whose propagation realizes that band. Wave and state form complementary levels of the same realization.

59.1.16 Finite horizon of prolongation and class of compatible extension extensions of the genealogical state

Definition 59.21 (Virtual finite route)**.** A prefix s_N is finitely virtual if there exists $L < \infty$ such that

$$\operatorname{Ext}_L(s_N) \neq \varnothing, \qquad \operatorname{Ext}_{L+1}(s_N) = \varnothing.$$

Its route transports record, signature, and amplitude to L, and its prolongation horizon ends at the boundary $L + 1$.

In a finitely branching tree, the existence of prolongations to every depth is equivalent, by Koenig's lemma, to the existence of a global section. Persistence and virtuality are defined through the geometry of prolongations. The virtual route occupies a typed intermediate state and has a finite genealogical duration.

59.1.17 Internal spectrum, realization, and species

For a multisection F of routes, the natural object is the operator

$$\widehat{B}_F e_\Gamma = \kappa(\Gamma) e_\Gamma, \qquad \exp((\log R_{\mathrm{act}})\widehat{B}_F).$$

Its spectrum is invariant under permutations of the fiber basis. The physical chart determines which combination of eigenvalues corresponds to a scalar or matrix observable. Three levels must be preserved:

$$\operatorname{Spec} \widehat{B}_F, \qquad \mathfrak{R}_{\mathrm{phys}}(\widehat{B}_F), \qquad \text{species identification.}$$

The first is an internal output; the second is a realization; the third is a key within the physical inventory. A complete scientific table publishes all three and the morphism relating them.

59.1.18 Physical-spectral realization functor and 1st electronic closure

The hinge is fixed by the chain

$$\boxed{\begin{array}{c}\text{genealogical number} \longrightarrow \text{observable route} \longrightarrow \text{record} \longrightarrow \text{signature} \\ \text{character} \longrightarrow \text{towers} \longrightarrow \text{spectral operator} \\ \text{physical realization} \longrightarrow \text{particle}\end{array}}$$

The first physical closure specializes this chain to the electron section selected by the center of the atlas, with the current action line, bidirectional readers, and dimensional chart. This residence establishes the formal domain that makes it possible to understand why the electron realizes the theory of the number and why its generalization to the other species preserves route, spectrum, and genealogy.

Composition of dimensional readers and compatibility among their mathematical invariants
The definition of persistent section and discrete bundle belongs to the particle chapter; route, register, signature, and character belong to the surviving-extension chapter; band, material conjugation, and virtuality belong to the antisections chapter; fiber operators and their spectra belong to the bidirectional law of masses. The four grammars and the local closure of the path are integrated here with the CAR structure, persistence, character, and physical realization.

59.2 Planck constant and action line: derivation determinantal of the escala 10^{-34}

The construction of the action scale begins with an integer invariant. The Smith normal form of the local dodecaphasic block produces a cokernel of order 16; its norm on the constant section α_{HMT}, followed by a single action of φ, places the output $\Sigma_H = \varphi\alpha_{\mathrm{HMT}}^{16}$ between 10^{-34} and 10^{-33}. In this manner, the characteristic decimal power of action is obtained before joules and seconds are chosen. The angular branch subsequently supplies the two oriented coordinates—before the return and after the return—and the metrological chart finally expresses them as h and $\hbar$.

Governing result: from the local cokernel to the action scale

The Smith form of H_4, the order of its cokernel, and the inequalities determining the decade are exact. The complete result is PROVED FROM EXPLICIT STARTING STRUCTURE: the local block, the invariant section, the multiplicative norm, a single action of φ, and the decimal reader of section 59.2 are declared. The ablations identify the function of each rule and distinguish the local cokernel from the global cokernel of order 4096.

59.2.1 Smith Normal Form of H_4 and Transport of $\Sigma_H = \varphi\alpha_{\mathrm{HMT}}^{16}$ to the Action Line

For a of the three parallel dodecaphase orbits, the Parts I–III obtain

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1,2,2,4), \qquad G_H := \mathrm{coker}(H_4) \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z},$$

and therefore $|G_H| = 16$. Locality belongs to the reader's type: the cokernel of the complete dodecaphasic transformation has order $16^3 = 4096$, but simultaneously records three direct copies and does not replace the index of an orbit.

Let $s_\alpha(g) = \alpha_{\mathrm{HMT}}$ be the constant section over G_H. With the local norm

$$\mathcal{N}_{G_H}(s) := \prod_{g \in G_H} s(g)$$

and a sola action of φ on the base, is recovered, with a local label for facilitar the references of the Parte V,

$$\Sigma_H := \varphi\, \mathcal{N}_{G_H}(s_\alpha) = \varphi\alpha_{\mathrm{HMT}}^{16}. \tag{59.15}$$

This is the same section defined in equation (59.15), not a second construction.

Theorem 59.22 (Determinantal Order Transfer)**.** *Once the primitives of the local determinantal reader are fixed, the section in equation (59.15) satisfies*

$$\alpha_{\mathrm{HMT}}^{16} < 10^{-34} < \varphi\alpha_{\mathrm{HMT}}^{16} < 10^{-33}, \qquad \varepsilon_H := \lfloor \log_{10} \Sigma_H \rfloor = -34.$$

If $\mathcal{L}_S$ is a real action line with basis $\mathcal{U}_S$, the assignment

$$\tau_S : \mathbb{R}_{>0} \longrightarrow \mathcal{L}_S, \qquad x \longmapsto x\,\mathcal{U}_S,$$

transports that order without modifying it. In particular, the decimal exponent of the action coordinates defined below is -34.

Proof. The exact inequalities and the value of the order are the content of theorem 59.22; there they reduce to integer comparisons based on the constructed rational decimal of α_{HMT}. The map τ_S changes the codomain—from a positive scalar to a magnitude line with a declared basis—but preserves the scalar coordinate. Consequently, it does not alter its decimal order. □

The inherited evaluation is

$$\Sigma_H = 1.046243838104756580184229208303089\ldots \times 10^{-34},$$

and the normalized number

$$B_H := 10^{34}\Sigma_H = 1.046243838104756580184229208303089\ldots$$

belongs to $[1, 10)$. The factor 10^{34} is reconstructed from the order; it is not introduced as a threshold or prior normalization.

59.2.2 Necessity of the local core

The four exact ablations are

evaluated rule	decadic exponent
$\alpha_{\mathrm{HMT}}^{16}$ without φ	-35
$\varphi\alpha_{\mathrm{HMT}}^{15}$	-32
$\varphi\alpha_{\mathrm{HMT}}^{17}$	-37
$\varphi\alpha_{\mathrm{HMT}}^{4096}$	-8753.

Thus, $|G_H| = 16$ fixes the exponent of α_{HMT} once the local reader is chosen, and the inequalities then fix order -34. These controls exclude omission of φ, neighboring exponents, and substitution of the global cokernel for the local one. They do not constitute a classification of all admissible readers or a proof of prior uniqueness of the constant section, the unique action of φ, or the decimal chart.

59.2.3 Posterior coupling with the action coordinates

Write

$$\varepsilon_H := \left\lfloor \log_{10} \Sigma_H \right\rfloor = -34.$$

Once the angular branch has constructed the pre-return character H_5 and the reconstructed coordinate

$$\eta_{\mathrm{ret}}^{(\mathrm{deg})} = \frac{C^*}{2} - \alpha_{\angle,\mathrm{deg}},$$

both receive the same order in $\mathcal{L}_S$:

$$\hbar_{\mathrm{HMT}}^{\mathrm{pre}} := H_5\, 10^{\varepsilon_H} \mathcal{U}_S, \qquad \hbar_{\mathrm{HMT}}^{\mathrm{ret}} := \eta_{\mathrm{ret}}^{(\mathrm{deg})}\, 10^{\varepsilon_H} \mathcal{U}_S.$$

Numerically,

$$\hbar_{\mathrm{HMT}}^{\mathrm{pre}} = 1.0545718176461544696258218294330 59\ldots \times 10^{-34} \mathcal{U}_S,$$
$$\hbar_{\mathrm{HMT}}^{\mathrm{ret}} = 1.0545718169150390611336905847243 00\ldots \times 10^{-34} \mathcal{U}_S.$$

The datum H_5 precedes the return and is not identified with $\eta_{\mathrm{ret}}^{(\mathrm{deg})}$. Nor does it enter the proof of the exponent -34.

The corresponding unreduced coordinates are

$$h_{\mathrm{HMT}}^{\mathrm{pre}} = 2\pi H_5\, 10^{-34} \mathcal{U}_S = 6.626070149999987926001110349899474\ldots \times 10^{-34} \mathcal{U}_S,$$

$$h_{\mathrm{HMT}}^{\mathrm{ret}} = 2\pi \eta_{\mathrm{ret}}^{(\mathrm{deg})}\, 10^{-34} \mathcal{U}_S = 6.626070145406254333510749847592879\ldots \times 10^{-34} \mathcal{U}_S.$$

The comparison

$$2\pi H_5 - 6.62607015 = -1.2073998889650100526\ldots \times 10^{-14}$$

is a subsequent external control on the mantissa; it is neither an equality with the defining SI value nor an input to the reader.

Object	Coordenada obtenida	Mathematical and metrological function
Escala determinantal	$\Sigma_H = 1.0462438381047565\ldots \times 10^{-34}$	Fix internally the decimal order of the action line.
Reduced prior action	$\hbar_{\mathrm{HMT}}^{\mathrm{pre}} = H_5\, 10^{-34} \mathcal{U}_S$	Retain the angular coordinate prior to return.
Reduced posterior action	$\hbar_{\mathrm{HMT}}^{\mathrm{ret}} = \eta_{\mathrm{ret}}^{(\mathrm{deg})}\, 10^{-34} \mathcal{U}_S$	Publish the post-return coordinate on the same line.
Full cycle action	$h_{\mathrm{HMT}}^{\mathrm{pre}} = 6.6260701499999879\ldots \times 10^{-34} \mathcal{U}_S$	It transports local action through one phase turn by means of $h = 2\pi\hbar$.
Carta SI	$h = 6.62607015 \times 10^{-34}$ J s	external defining coordinate usada for the metrological recognition [20].

59.2.4 h and $\hbar$: action per full cycle and per radian

The identities

$$E = h\nu = \hbar\omega, \qquad \omega = 2\pi\nu, \qquad h = 2\pi\hbar$$

do not describe two quanta. They describe the same action in two charts. The frequency ν counts complete turns per unit time; the angular frequency ω counts phase in radians per unit time. Consequently, h measures the action associated with the closure of one turn, and $\hbar$ measures the action per unit phase. The reduction is not merely an abbreviation: it transports the global holonomy to the local generator of its propagation.

The same distinction explains the coordinated use of degrees and radians. The complete angular cycle of 360° makes the finite partitions visible

$$12 \cdot 30^\circ = 4 \cdot 90^\circ = 3 \cdot 120^\circ = 360^\circ$$

and the returns of 270° and 360°. The radian, by contrast, is the additive coordinate in which the exponential and the derivative are written without inserting a conversion constant in each operation. The two charts are related by

$$\mathbb{R}/\mathbb{Z} \xrightarrow{2\pi} \mathbb{R}/(2\pi\mathbb{Z}) \xrightarrow{180/\pi} \mathbb{R}/(360\mathbb{Z}).$$

The composition preserves the point of the turn: the degree publishes its position in the global calendar, and the radian linearizes its local motion.

Applied to the coordinates constructed above,

$$h^{\mathrm{pre}}_{\mathrm{HMT}} = 2\pi\,\hbar^{\mathrm{pre}}_{\mathrm{HMT}}, \qquad h^{\mathrm{ret}}_{\mathrm{HMT}} = 2\pi\,\hbar^{\mathrm{ret}}_{\mathrm{HMT}}.$$

The factor 2π does not retrospectively correct a mantissa. It traverses one complete turn with the local action. Thus an amplitude $\exp(\mathrm{i}S/\hbar)$, a commutation relation, and a holonomy quantization compare an action with the local unit of phase, whereas the global closure is expressed with h.

59.2.5 Confluence of the determinantal and dodecaphasic branches of the action scale

The angular branch and the determinantal branch both start from α_{HMT}, but neither retrospectively generates the other. The former decomposes as

$$\begin{gathered}
(\alpha_{\mathrm{HMT}}, \varphi;\, 9, 5, 4, 12, 54,\ \text{graded rule}) \longrightarrow H_5, \\
\alpha_{\mathrm{HMT}} \longrightarrow A = 1000\alpha_{\mathrm{HMT}}{}^\circ, \qquad (\alpha_{\mathrm{HMT}}, \pi) \longrightarrow D_A, \\
\mathcal{F}_\pi \longrightarrow \rho_\pi = 169, \qquad (\rho_\pi, D_A, \alpha_{\mathrm{HMT}}) \longrightarrow C_\pi(\alpha_{\mathrm{HMT}}), \\
(\pi, \alpha_{\mathrm{HMT}}, H_5, C_\pi(\alpha_{\mathrm{HMT}})) \longrightarrow y_* \longrightarrow C^*, \\
(A, C^*) \longrightarrow q_\pm \longrightarrow \big(S_{90}(q_\pm), S_{120}(q_\pm)\big) \longrightarrow K_{6\to 8} \longrightarrow \Gamma_{6\to 8}.
\end{gathered}$$

The posterior identification $\Gamma_{6\to 8} \dashrightarrow \gamma_{\mathrm{BI}}$ belongs to the interfaz fisica. The determinantal branch is

$$(H_4, \alpha_{\mathrm{HMT}}, \varphi;\ \text{reader rules}) \longrightarrow G_H \longrightarrow \Sigma_H \longrightarrow \varepsilon_H = -34.$$

Only at the end is ε_H combined with H_5 and $\eta^{(\mathrm{deg})}_{\mathrm{ret}}$ to yield coordinates in $\mathcal{L}_S$. Both an arrow $A \to H_5$ and any selection of C^* from SI or from γ_{BI} are excluded.

59.2.6 Systems of transduction between genealogical number and physical magnitude

Choosing a physical unit is an operation subsequent to the theorem. A metrological chart is an isomorphism of lines of magnitudes

$$\iota_{\mathrm{SI}} : \mathcal{L}_S \longrightarrow \mathcal{L}_{\mathrm{J\,s}}, \qquad \iota_{\mathrm{SI}}(\mathcal{U}_S) = \mathrm{J\,s}.$$

The unit belongs to the codomain of ι_{SI}; the exponent -34 belongs to the relative arithmetic theorem. Merely juxtaposing a unit symbol does not turn a dimensionless number into a physical quantity.

Proof hierarchy for the Planck constant

The order 10^{-34} follows from the local determinant and is fixed prior to metrology. The mantissas H_5 and $\eta_{\mathrm{ret}}^{(\mathrm{deg})}$ follow from the angular branch and distinguish the orientations before and after the return. The identity $h = 2\pi\hbar$ transports both coordinates between action per radian and action per cycle. The map ι_{SI} then supplies the unit J s and permits the output to be compared with the defining constant of the SI. This succession preserves in distinct places the internal derivation, the dimensional realization, and the external comparison.

From the constitutive bilayer factorization to the modular Hamiltonian, the free energy, and the origin memory With the absolute scale and its typing established, the dynamics can separate the three temporal sheets and coordinate their generators without confusing the global Hamiltonian, the Lagrangian of the present sheet, and the modular memory Hamiltonian.

Chapter 60

Ambivalence Principle in Dimensional Mechanics: areal–volumetric involution, self-inertial systems, and conjugate temporal orientation

60.1 Compatible sections and measurement theory

The dimensional notion of measurement is formulated on inverse systems of compatible states. Before assigning a number to a state, one specifies its refinement history, the coupling with the apparatus, and the map that produces the observable. This sequence makes it possible to separate the existence of a real value from the physical operation by which that value is recorded.

60.1.1 The number as a compatible section

Let $(\mathcal{S}_n, r_{n+1,n})_{n\geq 0}$ be an inverse system of surviving states. A compatible section is a sequence $x = (x_n)_{n\geq 0}$ such that

$$r_{n+1,n}(x_{n+1}) = x_n \qquad (n \geq 0).$$

His space is the inverse limit.

$$\mathcal{X}_\infty = \varprojlim_n \mathcal{S}_n.$$

When each x_n determines an interval $I_n(x_n) \subset \mathbb{R}$, the closures $\overline{I_n(x_n)}$ are nonempty compact sets, are nested, and their diameters tend to zero; the nested-interval theorem guarantees that their intersection contains a unique point. The Archimedean evaluation is then defined by

$$\mathcal{P}_{\mathrm{arq}}(x) = \text{the unique element of } \bigcap_{n\geq 0} \overline{I_n(x_n)}.$$

The HMT number is the compatible section $x \in \mathcal{X}_\infty$. The reader L is additional structure, and the coupling (x, L) constitutes its readout presentation or measurement. Two sections may have the same real evaluation and differ in their quotient, orientation, or boundary coordinates; the Archimedean projection does not identify those genealogies by definition.

60.1.2 Compatibility between ternary cylinders and thousandths

A ternary word of length L, whose integer value is N, determines the cylinder

$$I_{N,L} = \left[\frac{N}{3^L}, \frac{N+1}{3^L}\right).$$

A word of m blocks in base 1000, whose integer value is D, determines

$$J_{D,m} = \left[\frac{D}{1000^m}, \frac{D+1}{1000^m}\right).$$

Lemma 60.1 (Exact compatibility of cylinders)**.** *The cylinders $I_{N,L}$ and $J_{D,m}$ have a nonempty intersection if and only if*

$$N\,1000^m < (D+1)3^L, \qquad D\,3^L < (N+1)1000^m.$$

Proof. Two half-open intervals $[a, b)$ and $[c, d)$ intersect exactly when $a < d$ and $c < b$. Substituting the four endpoints and multiplying by the positive integer $3^L 1000^m$ yields the inequalities. The decision therefore reduces to integer arithmetic. □

The prolongation of a word refines its cylinder. A compatible chain whose closures are compact, nonempty, and nested, with diameter tending to zero, determines a unique real evaluation. The use of

closures is essential: nested half-open intervals may have empty intersection when the limit coincides with an excluded endpoint. The complete state preserves the words, transformations, and carries that produce the evaluation.

60.1.3 Reversible coupling between system and apparatus

Let X be the state space of the system, M the state space of the apparatus, and $m_0 \in M$ its initial preparation. A coupling is a map

$$U : X \times M \longrightarrow X \times M.$$

When U is bijective, the joint state preserves the information. An evaluation map $q_M : M \to Y$ may be noninjective and publish only an observable $y \in Y$.

Definition 60.2 (Measurement)**.** An HMT measurement is the triple

$$\mathfrak{M} = (U, q_M, C),$$

where U couples system and apparatus, q_M evaluates the apparatus state, and $C \subseteq X \times M$ is the compatibility condition. Its output on x is

$$\mathrm{Med}_{\mathfrak{M}}(x) = q_M\big(\mathrm{pr}_M\, U(x, m_0)\big)\,,$$

provided that the coupled pair satisfies C.

The definition establishes a precise composition of three operations: coupling, restriction to the compatible domain, and evaluation of the observable.

Theorem 60.3 (Localization of Information Loss)**.** *Let*

$$\Omega_M := q_M \circ \mathrm{pr}_M : X \times M \longrightarrow Y.$$

If U is bijective and the complete final state is preserved, the classes of inputs indistinguishable under the published observable are exactly the preimages under U of the fibres of Ω_M, unless it is further composed with a subsequent elimination or reinitialization of the state.

Proof. The bijectivity of U permits (x, m_0) to be recovered from the final joint state. For two inputs z, z' in the compatible domain, equality of the registers is equivalent to $\Omega_M(Uz) = \Omega_M(Uz')$. Hence, Uz and Uz' belong to the same fiber of the composite map $q_M \circ \mathrm{pr}_M$. This composition includes both the loss due to projection onto the apparatus and that due to evaluation of its state. Any subsequent erasure introduces an additional map and must be analyzed separately. □

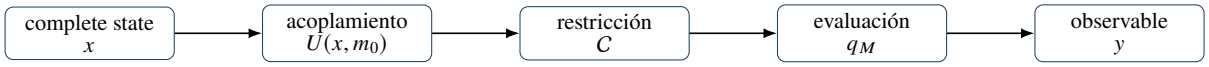


Figure 60.1: Composición de una medición. La aplicación de evaluación actúa después del acoplamiento y de la restricción de compatibilidad; el estado conjunto permanece disponible para la transformación inversa.

60.1.4 Principio de Ambivalencia: involución bidireccional de los lectores π/φ^2 y φ/π^2

Sean

$$q_A : \mathcal{X}_{\mathrm{TPK}} \to \mathcal{X}_A, \qquad q_V : \mathcal{X}_{\mathrm{TPK}} \to \mathcal{X}_V$$

two quotient maps respectively associated with the observables of aperture and retention, and let $T : \mathcal{X}_{\mathrm{TPK}} \to \mathcal{M}$ be the transported memory variable.

Principle 60.4 (Ambivalence)**.** The complete state precedes its observable projections. The equivalence relations induced by q_A and q_V are not assumed to be identical, and compatibility between them is recorded by T. The joint map

$$(q_A, q_V, T) : \mathcal{X}_{\mathrm{TPK}} \longrightarrow \mathcal{X}_A \times \mathcal{X}_V \times \mathcal{M}$$

is adopted as the separation condition for physically distinguishable states, modulo the declared gauge group.

Proposition 60.5 (Reconstruction under the separation condition)**.** *If the preceding joint map is injective modulo gauge, each complete-state class is uniquely determined by its two observables and its memory variable.*

Proof. Two states with the same triple of images belong to the same gauge class by the injectivity hypothesis. Each fiber of the joint map therefore contains at most one class. □

The observable q_A records the accessible prolongations, and q_V the retained component. A material interpretation of retention requires a mass observable that factors through q_V; it does not follow solely from the separation principle.

60.1.5 Relational determination and temporal orientation

Let $[\gamma]_{\mathcal{G}}$ be the class of a trajectory in the state groupoid, and let β_γ be its terminal boundary datum.

Principle 60.6 (Relational determination)**.** An admissible local observable has the form

$$O_{\mathrm{loc}}(x; \gamma) = F\big(x, [\gamma]_{\mathcal{G}}, \beta_\gamma\big)$$

and it is invariant under trajectory equivalences that preserve the ordered record. Consequently, the visible point coordinate does not by itself determine the observable when the evaluation has identified different transport classes.

The formulation is relational in a precise sense: the argument of O_{loc} contains the state, the global transport class, and the boundary. This is the domain on which the theory may subsequently be compared with the tradition associated with Mach's principle. It is not presupposed that every inertial property has been derived; the type of the map that must realize it is specified.

If the update on $\mathcal{X}_{\mathrm{TPK}}$ is bijective and the register preserves the coordinates required by its inverse, the complete dynamics is reversible. An observable temporal orientation appears upon composing it with a noninjective reader, reinitializing a boundary datum, or selecting a single prolongation among several. Irreversibility is therefore localized in a defined morphism of information loss or selection, not in the mere ordering of labels.

60.1.6 Relation with classical variational principles

Hamilton's principle requires the first variation of an action to vanish along the physical trajectory; it does not assert, in general, that this trajectory is a global minimum. The discrete interaction functional constructed in the section 25.12, by contrast, orders a finite family of routes by the number of changes in orientation and arithmetic law. The two principles coincide only after constructing a limit in which the discrete cost induces an action and its first variation reproduces the equations of motion. This distinction precludes replacing a future correspondence with an identity of names.

60.1.7 Prospective evaluation conditions of the Ambivalence Principle on the enriched state

A prospective evaluation is mathematically determined when, before its output is consulted, the state space, coupling, quotient maps, memory variable, compatibility rule, and comparison criterion have been fixed. If several sections survive, the output is the complete family. The assignment of frequencies then requires a probability measure or an explicit additional dynamics.

60.1.8 Quotient of modes and areal-volumetric correspondence

The calendar sign, mode label, and geometric sheet are three typed objects. In particular, the phase sign is not identified with the angular observable on digits. For the geometric correspondence, we fix the typed HMT reader

$$\mathfrak{g}_{\mathrm{av}}(\Sigma) = \mathscr{H}_{\mathrm{ar}}, \qquad \mathfrak{g}_{\mathrm{av}}(\Pi) = \mathscr{H}_{\mathrm{vol}}.$$

The first assignment expresses the additive or boundary reading; the second, the multiplicative or interior reading. The map $\mathfrak{g}_{\mathrm{av}}$ is part of the sheet typing. The following result proves that, once that typing is fixed, TPK dynamics forces the incidence between them.

Theorem 60.7 (Necessary descent of leaf incidence)**.** *Let m_r be the mode obtained after applying phase r of the primitive block $(+1, -1, 0)$, with the rule*

$$+1 : \ m \mapsto \Sigma, \qquad -1 : \ m \mapsto \Pi, \qquad 0 : \ m \mapsto m.$$

The cyclic orbit of modes is (Σ, Π, Π). If

$$N_3(i, j) := \#\{r \in \mathbb{Z}/3\mathbb{Z} : (m_r, m_{r+1}) = (i, j)\},$$

then, in the order (Σ, Π),

$$\boxed{N_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.}$$

By repetition of the calendar,

$$N_9 = 3N_3, \qquad N_{27} = 9N_3, \qquad N_{108} = 36N_3.$$

After applying $\mathfrak{g}_{\mathrm{av}}$, the primitive incidence matrix is, in the order $(\mathscr{H}_{\mathrm{ar}}, \mathscr{H}_{\mathrm{vol}})$,

$$\boxed{F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.}$$

Thus, the two cross transitions, the unique volumetric self-retention, the absence of areal self-retention and the unit primitive multiplicities are consequences of the TPK mode quotient; they are not four added postulates.

Proof. The three consecutive pairs, with cyclic closure, are

$$\Sigma \longrightarrow \Pi, \qquad \Pi \longrightarrow \Pi, \qquad \Pi \longrightarrow \Sigma.$$

Each appears once, and the pair $\Sigma \to \Sigma$ does not appear. This determines the four entries of N_3. The calendars of lengths $9, 27, 108$ respectively contain $3, 9, 36$ copies of the primitive block, so their incidence matrices are the indicated multiples. The greatest common divisor of the nonzero entries of each long matrix removes only that global repetition and returns N_3. Finally, $\mathfrak{g}_{\mathrm{av}}$ transports Σ, Π to the areal and volumetric sheets without altering the incidence. □

The real hyperbolic realization that uses this matrix and proves its Lorentz invariance is formulated in section 3.5.6; there F_{av} is used as an antecedent, without repeating its derivation.

This theorem must be distinguished from a stronger Markov assertion. Forgetting the phase is not a strong Markov quotient of $\mathcal{K}_{\mathrm{ph}}$: the visible mode does not determine which of the three operators $I, E_+, E_\times$ acts at the next elementary iteration. What descends without an additional hypothesis is the edge multigraph of a TPK block.

There is, however, a memory-one completion uniquely determined by that descent. Let X_{av} be the smallest subshift of finite type over $\{\mathscr{H}_{\mathrm{ar}}, \mathscr{H}_{\mathrm{vol}}\}$ that contains all the words obtained by forgetting the phase and retains only consecutive pairs. Every one-step subshift with that property must admit exactly the three observed pairs; the smallest such subshift excludes the sole unobserved pair. Its adjacency matrix is therefore F_{av}.

Corollary 60.8 (Parry Measure of the One-Memory Wrapper)**.** *The subshift X_{av} is primitive, has topological entropy $\log \varphi$, and possesses a unique measure of maximal entropy. Its stationary vertex weights satisfy*

$$\boxed{\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}.}$$

Proof. The Perron root of F_{av} is φ, and a positive Perron vector is $(1, \varphi)$. Since the matrix is symmetric, the Parry vertex weights are proportional to the product of the left and right vectors, that is, to $(1, \varphi^2)$. □

The Parry measure belongs to the memory-one path envelope. It is not the image of τ_{468}, nor the temporal frequency of the periodic orbit of modes. The latter is

$$\nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{ar}}) = \frac{1}{3}, \qquad \nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{vol}}) = \frac{2}{3},$$

whereas τ_{468} is uniform over emissions, and the Parry measure weights renewal histories of arbitrary length. The three measures answer distinct questions. The irrational ratio φ^{-2} is thus derived canonically from the path envelope, without attributing it to a quotient of finite cardinalities.

60.1.8.1 From chronological counting to the return marked by π

The two formulas

$$\left(\frac{1}{3}, \frac{2}{3}\right) \qquad \text{and} \qquad \frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}$$

are not approximations of one another. The first counts the three instants of the primitive calendar (Σ, Π, Π); the second weights all the admissible histories of the memory-one envelope. Hence the chronological frequency is rational, and the Parry ratio is irrational. In what follows, we write

$$\frac{\pi_{\mathrm{HMT}}}{\varphi^2} = \pi_{\mathrm{HMT}} \varphi^{-2};$$

this is not division by φ^{-2}.

The appearance of φ^{-2} can be seen directly in the linear dynamics. If F_n denotes the n-th Fibonacci number,

$$F_{\mathrm{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix} \qquad (n \geq 1).$$

The eigenvalues of F_{av} are φ and $-\varphi^{-1}$. The contracted branch reverses orientation in one iteration. Upon passing to the quadratic memory, namely $\mathrm{Sym}^2(\mathbb{R}^2)$, that reversal is recorded before the sign disappears. In the basis $(x^2, 2xy, y^2)$, with the images of the basis vectors represented by rows, the induced operator is

$$\mathrm{Sym}^2(F_{\mathrm{av}}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \chi(t) = (t+1)(t^2 - 3t + 1).$$

Its three eigenvalues are

$$\varphi^2, \qquad -1, \qquad \varphi^{-2}.$$

Thus, φ^{-2} is the positive stable mode of second-order memory. This step explains why the golden ratio appears here as a return law and not as an added statistical mean.

The closure π belongs to another level of the same HMT state: it is the coinductive output of the regional reader already constructed, not a new eigenvalue of F_{av}. Let

$$P_{\mathrm{ar}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_{\mathrm{vol}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Among the positive operators that preserve both sheets, there is exactly one whose volumetric normalization is one and whose areal mark is the generated closure π_{HMT}:

$$D_\pi = \pi_{\mathrm{HMT}} P_{\mathrm{ar}} + P_{\mathrm{vol}}.$$

Uniqueness is relative to these three explicit clauses: positivity, diagonality with respect to the leaves, and the two normalizations. With $e_{\mathrm{ar}} = (1,0)^T$ and $e_{\mathrm{vol}} = (0,1)^T$, the marked return of length n is

$$\Theta_n := \frac{\langle e_{\mathrm{ar}}, D_\pi F_{\mathrm{av}}^n e_{\mathrm{ar}} \rangle}{\langle e_{\mathrm{vol}}, F_{\mathrm{av}}^n e_{\mathrm{vol}} \rangle} = \pi_{\mathrm{HMT}} \frac{F_{n-1}}{F_{n+1}}.$$

Therefore,

$$\boxed{\lim_{n\to\infty} \Theta_n = \frac{\pi_{\mathrm{HMT}}}{\varphi^2}.}$$

The mark π_{HMT} is applied only once, at the boundary return; it is not reinjected at every transition. Finite combinatorics produces the stable mode φ^{-2}, the coinductive reader produces the closure π_{HMT}, and D_π composes both data without reversing their causal order.

The native length 108 provides an exact witness to that convergence:

$$(F_{\mathrm{av}}^{108})_{\mathrm{ar,ar}} = F_{107} = 10284720757613717413913,$$

$$(F_{\mathrm{av}}^{108})_{\mathrm{vol,vol}} = F_{109} = 26925748508234281076009.$$

Consequently,

$$\left| \frac{F_{107}}{F_{109}} - \varphi^{-2} \right| = 6.1684979916614407936\ldots \times 10^{-46},$$

and, after the only closing marker,

$$\left| \pi_{\mathrm{HMT}} \frac{F_{107}}{F_{109}} - \frac{\pi_{\mathrm{HMT}}}{\varphi^2} \right| = 1.9378907974286976068\ldots \times 10^{-45}.$$

These errors measure the finite approximation of the return; they are not adjustments of π or of φ.

Cylindrical reader of the ratio. Let $I_n^{\pi} = [\underline{\pi}_n, \overline{\pi}_n]$ and $I_n^{\varphi} = [\underline{\varphi}_n, \overline{\varphi}_n]$ be the positive, nested cylinders produced by the two HMT readers. The interval operation

$$Q(I, J) := \left[\frac{\inf I}{(\sup J)^2}, \frac{\sup I}{(\inf J)^2} \right]$$

is the smallest interval containing x/y^2 for every $(x, y) \in I \times J$. It preserves nesting: if the input cylinders are refined, then $Q(I_n^{\pi}, I_n^{\varphi})$ is also refined. Since their diameters tend to zero and the limit of I_n^{φ} is positive,

$$\bigcap_n Q(I_n^{\pi}, I_n^{\varphi}) = \left\{ \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2} \right\}.$$

With the twelve already published base-thousand blocks, the quotient interval forces thirty-five decimal digits:

$$\frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2} = 1.19998161486432666111577775331680692\ldots.$$

This construction preserves the provenance of each digit: the numerator comes from the closure cylinder and the denominator from the autoscale.

Algebraic separation between areal reading, volumetric reading and their respective codomains The complete ratio cannot arise from the rational algebra of F_{av} alone. Indeed, its rational spectral functions belong to $\mathbb{Q}(\sqrt{5})$; φ^{-2} is found there, but not the transcendental number π. Consequently, any derivation purporting to obtain π/φ^2 solely from the matrix 2×2 would be ill typed. The transcendental datum must derive from the coinductive closure reader, or from an equivalent boundary cocycle proved independently. This negative control fixes the division of labor between the finite correspondence and the number reader.

60.1.8.2 Exchange involution of the quotient

The correspondence admits two exact orientations of the same law. If

$$\mathscr{R}(x, y) = \frac{x}{y^2}, \qquad \mathsf{J}(x, y) = (y, x),$$

then $\mathsf{J}^2 = I$ and

$$\mathscr{R}(\pi, \varphi) = \frac{\pi}{\varphi^2}, \qquad (\mathscr{R} \circ \mathsf{J})(\pi, \varphi) = \frac{\varphi}{\pi^2}.$$

The identities are also verified.

$$\mathscr{R}(\pi, \varphi)\mathscr{R}(\varphi, \pi) = \frac{1}{\pi\varphi}, \qquad \frac{\mathscr{R}(\pi, \varphi)}{\mathscr{R}(\varphi, \pi)} = \left(\frac{\pi}{\varphi}\right)^3.$$

Writing $\mathcal{B}_{\mathrm{av}}^{(\pi)} = \mathscr{R}(\pi, \varphi)$ also yields the exact reparametrization

$$\frac{\varphi}{\pi^2} = \frac{1}{\varphi^3 \big(\mathcal{B}_{\mathrm{av}}^{(\pi)}\big)^2}, \qquad e^{\varphi/\pi^2} = \exp\left[\frac{1}{\varphi^3 \big(\mathcal{B}_{\mathrm{av}}^{(\pi)}\big)^2} \right].$$

Thus, the change of orientation between closure and growth transports the boundary ratio to its exchanged reading:

$$\frac{\pi}{\varphi^2} \overset{\mathsf{J}}{\longleftrightarrow} \frac{\varphi}{\pi^2}.$$

The first orientation reads the marked areal closure over quadratic volumetric memory; the second reads interior growth over quadratic closure. This correspondence is a theorem of the HMT reader.

Dimensional Mechanics subsequently applies the second orientation within the electronic construction, without converting the electron into an antecedent of π, φ, or of the correspondence.

60.1.8.3 Ambivalence Principle of the areal and volumetric readers

The Ambivalence Principle asserts that the areal reading and the volumetric reading are two coordinated, but not interchangeable, functionals of one and the same state with memory. Their coincidence on the diagonal is

$$\mathcal{L}_{\mathrm{ar}}^{\mathrm{diag}}(x) = \mathcal{L}_{\mathrm{vol}}^{\mathrm{diag}}(x).$$

The diagonal equality does not erase the ambivalence outside of it.

On the areal–volumetric boundary, both readers factor through the same positive memory amplitude $\mathcal{M}(x)$:

$$\mathcal{L}_{\mathrm{vol}}(x) = \mu_{\mathrm{vol}}\mathcal{M}(x), \qquad \mathcal{L}_{\mathrm{ar}}(x) = \pi_{\mathrm{HMT}}\mu_{\mathrm{ar}}\mathcal{M}(x).$$

Then the Parry theorem and the return mark give

$$\boxed{\frac{\mathcal{L}_{\mathrm{ar}}(x)}{\mathcal{L}_{\mathrm{vol}}(x)} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2}.}$$

With volumetric normalization, the relative difference of readers is

$$\frac{\mathcal{L}_{\mathrm{ar}} - \mathcal{L}_{\mathrm{vol}}}{\mathcal{L}_{\mathrm{vol}}} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2} - 1 = 0.1999816148643266611\ldots.$$

The physical reading of this correspondence is undertaken later, in the Cartan–Holst chapter, after constructing holonomy, co-tetrad, connection, torsion, and curvature. Here the result is entirely HMT: an exact ratio between two readers of the same sheet memory.

They should be distinguished from the enriched extractor subsequently used in Dimensional Mechanics. There a complete history preserves the action count n_A, the twelve oriented events $e_0, \ldots, e_{11}$ and the torsional cochain at step level; these yield

$$s_C = \sum_{i=0}^{5}(e_i + e_{i+6}), \qquad W_{\pm} = 6n_A \pm s_C.$$

The channels already generated by HMT then satisfy the exact identity

$$\exp\!\left(n_A A_{\mathrm{rad}} + \frac{s_C}{6}C^*_{\mathrm{rad}}\right) = \left(q_+^{-W_+}q_-^{-W_-}\right)^{1/12}.$$

Therefore, the route–angular-character passage is not absent: it is realized on the enriched dodecaphase record. What this phase calculation proves by itself are $\Omega_{\mathrm{sw}}, S, G$; it neither replaces them nor permits reconstruction of the complete record after it has been projected.

The uniform measure on the 104 976 presentations descends to atoms of masses $1/729$, $1/243$ and $1/54$, but not to $1/468$. In the fivefold microfiber of π, the two probabilities are

$$\tau_{468}(\mathscr{R}_\pi) = \frac{5}{468}, \qquad F_{\#}\mu_{\mathrm{seed}}(\mathscr{R}_\pi) = \frac{7}{729}.$$

Both the averaged lift and the phase lift realize the first: they do not push forward the uniform measure on seeds, but assign the same total mass to each observable class. The phase lift also preserves microscopic stillness at neutral steps.

To relate phase elevation to the enriched histories of depth six, a kernel $\mathcal{K}_6$ on $\mathcal{H}_6^{\mathrm{cat}} \times C_9$ is required. It descends to $\widehat{\mathcal{K}}_{\mathrm{ph}}$ through $\rho_6 \times \mathrm{id}_{C_9}$ if and only if it satisfies, at each phase, the strong lumpability condition

$$\sum_{\rho_6(g)=y} \mathcal{K}_6((h,r),(g,r+1)) = L_{\tau(r)}(\rho_6(h),y),$$

independently of h within each fiber of ρ_6. In that case, composition with F projects to $\mathcal{K}_{\mathrm{ph}}$. Constructing a family $(\mathcal{K}_m)_{m\geq 6}$ that satisfies this identity and commutes with all truncations is the remaining step toward a transfer of the inverse limit.

60.2 Areal–Volumetric Correspondence and Ratio π/φ^2

The preceding section constructed the dodecaphase involution, its lift to persistent sections, the Möbius band relative to a negative holonomy, and the biangular chart $(A, C^*) \leftrightarrow (q_+, q_-)$. It remains to state precisely how the returns of the areal and volumetric sheets are counted and why the golden ratio appears squared.

60.2.1 Classes, supports, and leaves are distinct objects

The nonadic classes of motion are

$$C_1 = \{1,4,7\}, \qquad C_2 = \{2,5,8\}, \qquad C_0 = \{3,6,9\}.$$

The supports employed by the angular observable are

$$\mathcal{D}_{\mathrm{act}} = \{1,3,5,7\}, \qquad \mathcal{D}_{\mathrm{evt}} = \{2,4,6,8\}, \qquad \mathcal{D}_9 = \{9\}.$$

Its exact intersection is

$$B_{\mathrm{na}} = \big(|C_i \cap D_j|\big)_{i,j} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

Theorem 60.9 (Triadic incidence)**.** *The following identities hold*

$$\det B_{\mathrm{na}} = 3, \qquad \mathrm{SNF}(B_{\mathrm{na}}) = \mathrm{diag}(1,1,3),$$

$$\mathrm{im}\, B_{\mathrm{na}} = \{(u,v,w) \in \mathbb{Z}^3 : u + v \equiv 0 \pmod 3\}.$$

The simultaneous interchange $C_1 \leftrightarrow C_2$, $\mathcal{D}_{\mathrm{act}} \leftrightarrow \mathcal{D}_{\mathrm{evt}}$ leaves the incidence fixed.

The observable of the record

$$j_2(d) = \begin{cases} -1, & d \in \mathcal{D}_{\mathrm{act}}, \\ +1, & d \in \mathcal{D}_{\mathrm{evt}}, \\ 0, & d \in \mathcal{D}_9 \end{cases}$$

is composed with the active geometric reader:

$$-1 \mapsto \mathscr{H}_{\mathrm{ar}}, \qquad +1 \mapsto \mathscr{H}_{\mathrm{vol}}, \qquad 0 \mapsto \mathscr{H}_{\mathrm{tor}}.$$

Therefore,

$$\mathfrak{g}_{\triangle}(\mathcal{D}_{\mathrm{act}}) = \mathscr{H}_{\mathrm{ar}}, \qquad \mathfrak{g}_{\triangle}(\mathcal{D}_{\mathrm{evt}}) = \mathscr{H}_{\mathrm{vol}}, \qquad \mathfrak{g}_{\triangle}(\mathcal{D}_9) = \mathscr{H}_{\mathrm{tor}}.$$

The matrix proves the crossing between classes and supports. The reader $\mathfrak{g}_{\triangle}$ declares the passage from supports to sheets. Omitting this second map would identify the cardinalities of distinct objects.

60.2.2 Spectral opening of the band through the conjugated angular phase

In the active chart,

$$q_+ = e^{-A_{\mathrm{rad}}-C^*_{\mathrm{rad}}}, \qquad q_- = e^{-A_{\mathrm{rad}}+C^*_{\mathrm{rad}}},$$

and the exact inverse is

$$A_{\mathrm{rad}} = -\frac{1}{2}\log(q_+q_-), \qquad C^*_{\mathrm{rad}} = \frac{1}{2}\log\frac{q_-}{q_+}.$$

Thus, A is the common mode and C^* the differential mode. At each tower height,

$$q_-^n - q_+^n = 2e^{-nA_{\mathrm{rad}}}\sinh(nC^*_{\mathrm{rad}}).$$

The negative holonomy constructed in Section 94.4.1 fixes the Möbius gluing; the last identity proves that C^* measures the oriented opening of the two conjugate sheets. In the physical representation, those two orientations are read as particle and antiparticle.

In the boundary reader, one turn of the base is parameterized by 2π and ends in the opposite orientation. The squared loop, parameterized by 4π, restores the orientation. Therefore, relative to the declared lift and geometric reader, 2π is projected areal closure and 4π is volumetric closure of the oriented cover. This construction does not conflate that double return with a spinorial representation; the representation, when introduced, is a subsequent functor.

60.2.3 TPK descent, path envelope, and loop counting

The TPK calendar and the angular observable j_2 are distinct maps. The former updates the arithmetic mode $m \in \{\Sigma, \Pi\}$; the latter classifies digit supports. The hinge uses the typed HMT reader

$$\eta_{\mathrm{av}}(\Sigma) = \mathscr{H}_{\mathrm{ar}}, \qquad \eta_{\mathrm{av}}(\Pi) = \mathscr{H}_{\mathrm{vol}}.$$

Thus, Σ is the additive or boundary mode and Π the multiplicative or interior mode. This statement does not identify τ with j_2.

In the primitive phase block

$$\tau_3 = (+1, -1, 0),$$

TPK updates the mode via

$$+1 :\ m \mapsto \Sigma, \qquad -1 :\ m \mapsto \Pi, \qquad 0 :\ m \mapsto m.$$

The cyclic orbit is (Σ, Π, Π). Define its edge-incidence matrix by

$$N_3(i,j) = \#\{r \in \mathbb{Z}/3\mathbb{Z} : (m_r, m_{r+1}) = (i,j)\}.$$

Theorem 60.10 (Necessary Descent of the Four Rules)**.** *In the order* (Σ, Π)*,*

$$N_3 = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Moreover,

$$N_9 = 3N_3, \qquad N_{27} = 9N_3, \qquad N_{108} = 36N_3.$$

Therefore, after removing the common repetition factor and applying η_{av}*, the TPK mode quotient necessarily induces*

$$\boxed{F_{\mathrm{av}} = S + P_{\mathrm{vol}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}}$$

where

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_{\mathrm{vol}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The two crossed transitions, the unique volumetric self-retention, the absence of areal self-retention, and the unit primitive multiplicities are consequences of TPK, not added premises.

Proof. The three consecutive pairs of the cycle are

$$\Sigma \longrightarrow \Pi, \qquad \Pi \longrightarrow \Pi, \qquad \Pi \longrightarrow \Sigma.$$

Each one appears once and $\Sigma \to \Sigma$ does not appear. This determines the four entries of N_3. The length calendars $9, 27, 108$ contain $3, 9, 36$ primitive blocks, respectively; therefrom come the three multiples. Reduction by the greatest common divisor of the nonzero entries eliminates only the repetition of the block. The reader η_{av} changes the type of the vertices, not their incidence. □

The preceding theorem is an exact descent of edges. It does not turn phase forgetting into a strong Markov quotient: the same visible emission may be updated by I, E_+ or $E_\times$, depending on the hidden phase. From this fact follows a necessary distinction among three measures:

$$\tau_{468}, \qquad \nu_{\mathrm{cyc}}, \qquad \mu_{\mathrm{Parry}}.$$

The former is uniform over the 468 emissions and stationary under phase transfer. The latter is the frequency of the periodic orbit,

$$\nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{ar}}) = \frac{1}{3}, \qquad \nu_{\mathrm{cyc}}(\mathscr{H}_{\mathrm{vol}}) = \frac{2}{3}.$$

Neither of them can directly produce the irrational quotient φ^{-2}.

The object that does produce it is determined by a universal property. Let X_{av} be the smallest memory-one subshift of finite type that contains the words obtained by forgetting the phase and preserves all their consecutive pairs. Every one-step system containing those words must admit

$$\mathrm{ar} \to \mathrm{vol}, \qquad \mathrm{vol} \to \mathrm{vol}, \qquad \mathrm{vol} \to \mathrm{ar}.$$

The smaller one excludes the only absent pair, $\mathrm{ar} \to \mathrm{ar}$. Consequently, X_{av} has adjacency matrix F_{av}. This is the minimal one-memory symbolic completion of the TPK quotient; it is not a new microscopic dynamics.

Theorem 60.11 (Fibonacci Law and Parry Measure)**.** *The operator F_{av} is primitive and*

$$F_{\mathrm{av}}^{n} = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix} \qquad (n \geq 1).$$

The subshift X_{av} has topological entropy $\log \varphi$ and a unique invariant measure of maximal entropy. For this measure,

$$\frac{(F_{\mathrm{av}}^{n})_{\mathrm{ar,ar}}}{(F_{\mathrm{av}}^{n})_{\mathrm{vol,vol}}} = \frac{F_{n-1}}{F_{n+1}} \longrightarrow \varphi^{-2}, \qquad \boxed{\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}.}$$

Proof. The square of F_{av} is strictly positive. The formula for powers is obtained by induction through $F_{n+1} = F_n + F_{n-1}$. The diagonal entries count closed loops. The Perron root is φ and a positive Perron vector is $(1, \varphi)$. Since the matrix is symmetric, the vertex weights of Parry are proportional to the product of the left and right vectors, that is, to $(1, \varphi^2)$. □

The factor φ^{-1} is the Perron amplitude ratio of a traversal; φ^{-2} is the weight ratio of a complete return. The two exponents are thereby ordered without introducing two independent constants.

60.2.4 Spectral Origin of the Quadratic Exponent of Return

The appearance of the square admits a second, purely spectral proof that does not depend on interpreting a weighted average. The characteristic polynomial of F_{av} is

$$\chi_{F_{\mathrm{av}}}(t) = t^2 - t - 1,$$

so that its two eigenvalues are

$$\lambda_+ = \varphi, \qquad \lambda_- = -\varphi^{-1}.$$

The sign of λ_- records reversal of orientation in a traversal of the stable sheet. Upon lifting the action to quadratic memory $\mathrm{Sym}^2(\mathbb{R}^2)$, the eigenvalues are the pairwise products

$$\lambda_+^2 = \varphi^2, \qquad \lambda_+\lambda_- = -1, \qquad \lambda_-^2 = \varphi^{-2}.$$

In the basis $(e_{\mathrm{ar}}^2, e_{\mathrm{ar}} \odot e_{\mathrm{vol}}, e_{\mathrm{vol}}^2)$, writing the coordinates of the images of the three basis vectors by columns, the matrix is

$$\mathrm{Sym}^2(F_{\mathrm{av}}) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix},$$

and satisfies

$$\chi_{\mathrm{Sym}^2(F_{\mathrm{av}})}(t) = t^3 - 2t^2 - 2t + 1 = (t+1)(t^2 - 3t + 1).$$

Therefore, the stable mode changes orientation with $-\varphi^{-1}$ in one traversal and becomes the positive weight φ^{-2} when read as area memory or as a quadratic return. This is the structural reason for the exponent two.

60.2.5 Boundary closure and quotient π/φ^2

The coinductive reader of π precedes this construction. We define the closure band by composition, without using π to select the transporter:

$$\mathcal{B}_{\mathrm{av}}^{(\pi)}(X_\infty) := \Pi_{\mathrm{HMT}}(X_\infty)\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}}.$$

Marked return Θ_n and convergence of the areal–volumetric ratio Let the sheet projectors be

$$P_{\mathrm{ar}} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad P_{\mathrm{vol}} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The boundary marker is defined by

$$D_\pi := \Pi_{\mathrm{HMT}}(X_\infty)P_{\mathrm{ar}} + P_{\mathrm{vol}}.$$

It is the unique positive operator diagonal with respect to the sheet decomposition that assigns unit weight to the volumetric sheet and the closure $\Pi_{\mathrm{HMT}}(X_\infty)$ to the areal sheet. Its canonicity is therefore relative to three explicit clauses: sheet diagonality, volumetric normalization, and areal marking by the already constructed closure reader.

If $e_{\mathrm{ar}}, e_{\mathrm{vol}}$ are the sheet vectors, the marked-return functional of depth n is

$$\Theta_n(X_\infty) := \frac{\langle e_{\mathrm{ar}}, D_\pi F_{\mathrm{av}}^n e_{\mathrm{ar}}\rangle}{\langle e_{\mathrm{vol}}, F_{\mathrm{av}}^n e_{\mathrm{vol}}\rangle}.$$

The previous power law gives, without approximation,

$$\boxed{\Theta_n(X_\infty) = \Pi_{\mathrm{HMT}}(X_\infty)\frac{F_{n-1}}{F_{n+1}}.}$$

The closure π is applied only once, when marking the boundary return; it is not reintroduced at each iteration of F_{av}.

Corollary 60.12 (Areal–Volumetric Holographic Ratio)**.** *Relative to the TPK mode quotient, its minimal memory-one envelope, and the HMT closure of π,*

$$\boxed{\mathcal{B}_{\mathrm{av}}^{(\pi)} = \lim_{n\to\infty}\Theta_n = \frac{\pi_{\mathrm{HMT}}}{\varphi^2}.}$$

Native Periodicity of 108 **Phases and Closure of the Unitary Calendar** At dodecaphasic depth 108, the two diagonal counts are

$$(F_{\mathrm{av}}^{108})_{\mathrm{ar,ar}} = F_{107} = 10284720757613717413913,$$

$$(F_{\mathrm{av}}^{108})_{\mathrm{vol,vol}} = F_{109} = 26925748508234281076009.$$

Consequently,

$$\frac{F_{107}}{F_{109}} = 0.38196601125010515179541316563436188 2\ldots,$$

y the error respecto of φ^{-2} is

$$\left|\frac{F_{107}}{F_{109}} - \varphi^{-2}\right| = 6.16849799166144079 36\ldots\times 10^{-46}.$$

After the boundary marking, the error relative to π/φ^2 is

$$\left|\pi\frac{F_{107}}{F_{109}} - \frac{\pi}{\varphi^2}\right| = 1.93789079742869760 68\ldots\times 10^{-45}.$$

Thus, 108 is not used to fit the ratio: it is a native depth of the calendar that already approximates the limiting return with more than forty-four stable fractional digits.

Coinductive intervalar reader: cylinder fitting and uniqueness of the limit For positive intervals

$$I = [a, b], \qquad J = [c, d], \qquad 0 < c \le d,$$

we define

$$Q(I, J) := \left[\frac{a}{d^2}, \frac{b}{c^2}\right].$$

The function $(x, y) \mapsto x/y^2$ is increasing in x and decreasing in $y > 0$. Hence $Q(I, J)$ is exactly the smallest interval containing all quotients x/y^2 with $(x, y) \in I \times J$; it is not a bound chosen by rounding.

Let I_n^π and I_n^φ be the nested positive cylinders generated by HMT. Then

$$Q(I_{n+1}^\pi, I_{n+1}^\varphi) \subseteq Q(I_n^\pi, I_n^\varphi),$$

and, as

$$\bigcap_n I_n^\pi = \{\pi_{\mathrm{HMT}}\}, \qquad \bigcap_n I_n^\varphi = \{\varphi_{\mathrm{HMT}}\},$$

we obtain

$$\bigcap_n Q(I_n^{\pi}, I_n^{\varphi}) = \left\{ \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2} \right\}.$$

Each base-1000 block is forced as soon as the quotient interval lies entirely within a single triadic cylinder of the corresponding depth.

In particular, the positive twelve-triad cylinders for π and φ produce the quotient interval

$$\left[\frac{p^-}{(f^+)^2}, \frac{p^+}{(f^-)^2} \right].$$

The evaluation certifies thirty-five fractional digits:

$$\frac{\pi}{\varphi^2} = 1.19998161486432666111577775331680692\ldots,$$

y eleven complete blocks:

$$199|981|614|864|326|661|115|777|753|316|806.$$

Algebraic Separation Theorem. The matrix F_{av}, its powers, and its rational spectral algebra live in an algebraic extension of $\mathbb{Q}$ contained in $\mathbb{Q}(\sqrt{5})$. Consequently, no composition using only rational algebraic operations on F_{av} can produce the transcendental factor π/φ^2. The finite operator generates φ^{-2}; the coinductive boundary reader generates π_{HMT}; the marked return composes both without conflating their provenances. This negative control prevents F_{av} from being presented as an isolated generator of π.

60.2.6 Reader's Involution and Electronic Section

The relation just obtained appears alongside the electron in a manner more precise than literal repetition. The active electron invariant of Section 94.4.1 contains

$$\mathfrak{e}_0 = \frac{\sqrt{3}}{4} \left(e^{\varphi/\pi^2} - 22\alpha_{\mathrm{HMT}}^3 \right) (1 + 15\Delta_4).$$

Therefore, the electron exponent is φ/π^2, whereas the sheet hinge produces π/φ^2. The two scalars are neither equal nor reciprocal. They form the two orientations of the same bivariate law. Indeed, define

$$\mathscr{R}(x, y) := \frac{x}{y^2}, \qquad \mathsf{J}(x, y) := (y, x).$$

Then $\mathsf{J}^2 = I$ and

$$\boxed{\mathscr{R}(\pi, \varphi) = \frac{\pi}{\varphi^2}, \qquad (\mathscr{R} \circ \mathsf{J})(\pi, \varphi) = \frac{\varphi}{\pi^2}.}$$

Exchange of the two readers carries the marked areal–volumetric return exactly to the exponential seed of the electron. Moreover,

$$\mathscr{R}(\pi, \varphi)\mathscr{R}(\varphi, \pi) = \frac{1}{\pi\varphi}, \qquad \frac{\mathscr{R}(\pi, \varphi)}{\mathscr{R}(\varphi, \pi)} = \left(\frac{\pi}{\varphi} \right)^3.$$

If we denote by $\mathcal{B}_{\mathrm{av}}^{(\pi)} = \pi/\varphi^2$ the marked sheet ratio, the same identity may be written without introducing an additional constant:

$$\boxed{\frac{\varphi}{\pi^2} = \frac{1}{\varphi^3 \left(\mathcal{B}_{\mathrm{av}}^{(\pi)} \right)^2},} \qquad e^{\varphi/\pi^2} = \exp\left[\frac{1}{\varphi^3 \left(\mathcal{B}_{\mathrm{av}}^{(\pi)} \right)^2} \right].$$

Numerically,

$$\frac{\varphi}{\pi^2} = 0.16394111891367238359975275574566698 2\ldots,$$
$$e^{\varphi/\pi^2} = 1.17814494259630678081160095680888910\ldots.$$

This identity formalizes the bidirectional content without altering the force of the electron equation. The orientation (π, φ) reads boundary closure through quadratic memory; the exchanged orientation (φ, π) reads interior growth through quadratic closure and enters exponentiated in $\mathfrak{e}_0$. The theorem proves that both expressions belong to the same exchange orbit. The coefficients 22 and 15 have an independent generator: the theorem 16.4 obtains them, without consulting the electron mass, as

$$-22 = -\left|\mathcal{K}_e \setminus \iota(\mathscr{R}_\pi)\right|, \qquad +15 = \left|\mathscr{R}_\pi \times \mathbb{F}_3\right|.$$

Its status is DEMONSTRATED WITH EXPLICIT STARTING STRUCTURE. The current involution transports that construction to the electronic orientation and does not use the electron to obtain F_{av}, π, or φ. The last form is an exact change of coordinates of the electronic seed, not an alternative derivation of $\mathfrak{e}_0$.

60.2.7 Pentagonal Identity of Rogers–Ramanujan and Recognition of the Perron Invariant

The Rogers–Ramanujan continued fraction provides an analytic realization of the same invariant. With $q_A = e^{-A_{\mathrm{rad}}}$, $q_C = e^{-C^*_{\mathrm{rad}}}$, one obtains

$$R(q_A)R(q_C) - \varphi^{-2} \simeq -1.01864 \times 10^{-27},$$

and with the channels of the split decomposition,

$$R(q_+)R(q_-) - \varphi^{-2} \simeq -1.19458 \times 10^{-21}.$$

The differences are not zero. The exact statement is the limit.

$$q_A^{(m)}, q_C^{(m)} \to 1^- \quad \Longrightarrow \quad R(q_A^{(m)})R(q_C^{(m)}) \to \varphi^{-2}.$$

Rogers–Ramanujan thus recognizes the Perron invariant in the pentadic reader; by itself, it does not select the angular coordinates A, C^*, or F_{av}. The former finite kernel $90/120$ preserves an approximate residue 7.81×10^{-9} and is not used as an exact identity.

60.2.8 Inversion of the inertial and gravitational readers and bidirectional reconstruction of the magnitude

The Ambivalence Principle distinguishes the opening reader q_A, the retention reader q_V, and the transported memory. The preceding hinge makes it possible to formulate the physical intuition precisely without conflating it with the combinatorial theorem. Let $\mathcal{M}(x)$ be a common memory character, and define, at the areal–volumetric interface,

$$\mathcal{I}_{\mathrm{av}}(x) := \mu_{\mathrm{vol}}\mathcal{M}(x), \qquad \mathcal{G}_{\mathrm{av}}(x) := \pi_{\mathrm{HMT}}\mu_{\mathrm{ar}}\mathcal{M}(x).$$

Therefore, whenever $\mathcal{M}(x) \neq 0$,

$$\boxed{\frac{\mathcal{G}_{\mathrm{av}}(x)}{\mathcal{I}_{\mathrm{av}}(x)} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2},} \qquad \frac{\mathcal{G}_{\mathrm{av}} - \mathcal{I}_{\mathrm{av}}}{\mathcal{I}_{\mathrm{av}}} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2} - 1.$$

The first formula expresses the oriented ratio between the gravitational and inertial readers; the second, their relative defect. The gnomonic L carries the local Euclidean chart on which both readers descend to a common normalization,

$$\frac{\mathcal{G}_L}{\mathcal{I}_L} = 1.$$

The physical reading is stratified. In the Euclidean chart of L, the local torsional connection projects onto a unique normalization, and the equivalence principle is realized as the unit quotient. In the enriched state prior to that projection, areal opening and volumetric retention remain separate: the Ambivalence Principle preserves its two functionals and fixes the internal quotient π/φ^2. Local equivalence and boundary ambivalence are thus two successive readings of the same antecedent, with declared domains and maps.

The physical realization adopts factorization of the inertial observable by volumetric retention and of the gravitational observable by areal opening with boundary closure π; both share character $\mathcal{M}$. The internal mathematics determines the correspondence and its quotient, and the Dimensional Mechanics functor publishes its observable charts. A rupture of either factorization constitutes the explicit falsifier of this realization.

Cosmological boundary. The hypothesis that dark matter or dark energy are effective readings of nontorsional crossings is retained only as a hypothesis of physical prolongation. To turn it into a testable proposition, at least a map from sheet interfaces to a cosmological geometry, observables representing density and pressure, and a prospective comparison with data without using those data to fix the map must be constructed. The proved correspondence supplies the structural candidate; it does not yet constitute, by itself, an observational explanation of the dark sector.

Areal–volumetric hinge domain and necessary validity conditions The band, the areal–volumetric ratio, and the bidirectional reading belong to the prior architecture. The incidence matrix, calendar descent, minimal envelope, loop law, Parry measure, quadratic action, and marked return provide its explicit formalization. The interval reader and the depth controls 108 certify the calculation. The incidences $N_3, N_9, N_{27}, N_{108}$, the matrix identities, and the cylinders are exact internal identities; the Möbius theorem and the ratio π/φ^2 are PROVED FROM EXPLICIT STARTING STRUCTURE; the particle–antiparticle and inertial–gravitational readings constitute subsequent physical interfaces; the extension to the dark sector remains an unproved physical hypothesis. The pair related by the involution $\mathscr{R}(\pi, \varphi) \leftrightarrow \mathscr{R}(\varphi, \pi)$ exactly formalizes a prior architecture. The coefficients $(-22, +15)$ are proved, by theorem 16.4, from their explicit starting structure.

A positive holonomy gives a cylinder. An areal–areal transition, the absence of a cross transition, or a TPK count different from $N_9 = 3F_{\mathrm{av}}$ and $N_{108} = 36F_{\mathrm{av}}$ falsifies the descent. The identification of μ_{Parry} with τ_{468} or with ν_{cyc} is refuted by their distinct values. A spectrum of $\mathrm{Sym}^2(F_{\mathrm{av}})$ different from $\{\varphi^2, -1, \varphi^{-2}\}$, a count 108 different from (F_{107}, F_{109}), a nonnested quotient interval, or the iterative introduction of π within F_{av} falsifies the new formalizations. A universal physical identification of the quotient without a map from sheets to observables lies outside the theorem. The delta of the four rules is closed; it does not reopen π, φ, C^*, or the extension–route–ledger–character junction.

Chapter 61

Dynamics of 3 sheets: global Hamiltonian, Lagrangian of the present sheet, and modular Hamiltonian of memory

The 3 sheets of the TRIT are realized as a quadratic family with a first-order null sheet. The variation distinguishes the global generator, the effective action of the present sheet, and the modular memory of the boundary: the 3 operators are coordinated without being identified.

61.1 Quadratic regimes and temporal readout

61.1.1 Quadratic algebras of the TRIT and curvature signatures

The primary construction and proof of the three regimes reside in section 3.4. For $\tau \in \widehat{\mathbb{T}}$, with balanced image $\bar{\tau} \in \{-1, 0, +1\}$, one has

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \bar{\tau}),$$

and, if J is the class of X,

$$\begin{aligned} \tau = +1 \quad &: \quad J^2 = -I, \quad \text{elliptic regime,} \\ \tau = 0 \quad &: \quad J^2 = 0, \quad \text{parabolic regime,} \\ \tau = -1 \quad &: \quad J^2 = +I, \quad \text{hyperbolic regime.} \end{aligned}$$

For all three values of τ, the quotient $\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \bar{\tau})$ satisfies precisely the three quadratic identities above. In particular, the internal square root of -1 is realized as an operator on the elliptic sheet; HMT need neither reject nor suppress the complex extension. The parabolic algebra contains a nilpotent, and the hyperbolic algebra decomposes by means of the two idempotents $(I \pm J)/2$.

These identities classify algebras and transports. They do not, by themselves, select a spacetime metric or prove that the physical universe occupies one of the three sheets.

61.1.2 Parametrized metric family of the temporal reader

Write

$$\mathbb{A}_s := \mathbb{R}[X]/(X^2 + s), \qquad s \in \{+1, 0, -1\},$$

for the same family with the real sign as index. To make the possible interface precise, fix a scale $\rho_{\text{curv}} > 0$, of dimension L^{-1}, and $\theta \in \mathbb{R}/2\pi\mathbb{Z}$. For $s \in \{+1, 0, -1\}$, let

$$g_{s,\rho} = \mathrm{d}r^2 + S_{s,\rho}(r)^2 \, \mathrm{d}\theta^2,$$

$$S_{s,\rho}(r) = \begin{cases} \rho_{\text{curv}}^{-1} \sin(\rho_{\text{curv}} r), & s = +1, \\ r, & s = 0, \\ \rho_{\text{curv}}^{-1} \sinh(\rho_{\text{curv}} r), & s = -1. \end{cases}$$

The radial domain is $r \geq 0$ for $s = 0, -1$; for $s = +1$, the chart polar cubre $0 \leq r \leq \pi/\rho_{\text{curv}}$, with the polar regularity usual in the extremos. In the interior of each chart,

$$K_{\text{G}} = -\frac{S''_{s,\rho}}{S_{s,\rho}} = s\,\rho_{\text{curv}}^2.$$

The last equality is an exact computation of the model once the realization is fixed. The arrow

$$\mathbb{A}_s \dashrightarrow (M_s, g_{s,\rho})$$

is, however, a *ansatz* of the PHYSICAL INTERFACE; it is not a canonical consequence of the relation $J^2 = -sI$.

If $\ell_0 \in \mathcal{L}_L^+$ is a length and we declare

$$\rho_{\mathrm{curv}} := \frac{\Delta_4}{\ell_0},$$

then

$$K_{\mathrm{G}} = s\,\frac{\Delta_4^2}{\ell_0^2}.$$

This specialization is DEMONSTRATED WITH EXPLICIT STARTING STRUCTURE: the conclusion is obtained once s, Δ_4, ℓ_0 have been declared, but it does not select ℓ_0, does not deduce the observed cosmological curvature, and does not identify the ratios $\mathcal{G}_{\mathrm{tor}}/\mathcal{I}_{\mathrm{tor}}$ or $\mathcal{G}_{\mathrm{av}}/\mathcal{I}_{\mathrm{av}}$ with K_{G}.

61.1.3 Temporal TRIT reader on (-1,0,+1): opening, threshold, and return with memory

Once an ordered parametrization of the transitions has been chosen, the active temporal reader is

$$\begin{aligned} +1 &\longmapsto \text{return, retained memory and past,}\\ 0 &\longmapsto \text{evaluation threshold and present,}\\ -1 &\longmapsto \text{bidirectional opening and future.}\end{aligned}$$

The reader is PROVED WITH AN EXPLICIT STARTING STRUCTURE: its conclusions follow once the TRIT signature, the order of the trajectory, and the transported memory have been specified. It does not turn the set $\{-1, 0, +1\}$ into an equation of motion. The future designates the space of compatible continuations; survival may select a branch or a multisection without erasing the genealogy of the others.

The frequency

$$\omega_0 = \frac{2\pi}{108\, t_0}$$

assigns a type to the full-return clock through the decision entropy $I_{\mathrm{TRIT}} = \log 3$ and the Boltzmann transducer. It does not, in isolation, establish a stable subharmonic phase. It does, however, enable construction of the finite reference Hamiltonian below; selection of a robust physical Hamiltonian requires additional conditions.

61.1.4 Unitary dynamics on 108 states: Hamiltonian, propagator, and period

Let

$$\mathcal{H}_{\mathrm{clk}} := \mathbb{C}^{108}, \qquad \{|j\rangle : j \in \mathbb{Z}/108\mathbb{Z}\}$$

with its canonical Hermitian inner product. Fix $\zeta := \exp(2\pi\mathrm{i}/108)$, the unitary shift

$$U_{\mathrm{clk}}|j\rangle = |j+1\rangle$$

and the Fourier basis

$$|\widetilde{k}\rangle := \frac{1}{\sqrt{108}} \sum_{j=0}^{107} \zeta^{jk}|j\rangle, \qquad 0 \le k \le 107.$$

With $t_0 \in \mathcal{L}_T^+$ and $\hbar_{\mathrm{int}} \in \mathcal{L}_S^+$, define

$$\boxed{H_{\mathrm{clk}} := \hbar_{\mathrm{int}}\omega_0 \sum_{k=0}^{107} k\, |\widetilde{k}\rangle\langle\widetilde{k}|, \qquad \omega_0 = \frac{2\pi}{108\, t_0}.} \tag{CLK1}$$

The operator has energy type $[H_{\mathrm{clk}}] = \mathcal{L}_S\mathcal{L}_T^{-1} = \mathcal{L}_E$.

Theorem 61.1 (Exact unitary realization of the 108 calendar)**.** *The operator H_{clk} is self-adjoint and its one-step propagator satisfies*

$$U_{\mathrm{step}} := \exp\!\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}} H_{\mathrm{clk}} t_0\right) = U_{\mathrm{clk}}, \qquad U_{\mathrm{step}}^{108} = I. \tag{CLK2}$$

If

$$Z_{\mathrm{clk}}|j\rangle = \zeta^{j}|j\rangle,$$

then, for any basis state $|j_0\rangle$,

$$\langle Z_{\mathrm{clk}}\rangle_n := \langle j_0|U_{\mathrm{step}}^{-n} Z_{\mathrm{clk}} U_{\mathrm{step}}^{n}|j_0\rangle = \zeta^{j_0+n}. \tag{CLK3}$$

The sequence has primitive period 108*.*

Proof. Expression (CLK1) is a spectral decomposition with real eigenvalues, hence $H_{\mathrm{clk}} = H_{\mathrm{clk}}^{\dagger}$. Moreover,

$$U_{\mathrm{clk}}|\widetilde{k}\rangle = \zeta^{-k}|\widetilde{k}\rangle,$$

whereas the exponential of $\hbar_{\mathrm{int}}\omega_0 k$ over t_0 gives $\exp(-2\pi \mathrm{i}k/108) = \zeta^{-k}$. The two operators agree on an orthonormal basis. The remaining identities follow from $U_{\mathrm{step}}^{n}|j_0\rangle = |j_0 + n\rangle$ and the primitivity of ζ. □

The corresponding variational formulation, for a curve $\psi \in C^1([t_a, t_b], \mathcal{H}_{\mathrm{clk}})$, unconstrained during the variation, is

$$\boxed{L_{\mathrm{clk}}(\psi, \dot{\psi}) = \frac{\mathrm{i}\hbar_{\mathrm{int}}}{2}\big(\langle\psi|\dot{\psi}\rangle - \langle\dot{\psi}|\psi\rangle\big) - \langle\psi|H_{\mathrm{clk}}|\psi\rangle.} \tag{CLK4}$$

The action functional is

$$\boxed{S_{\mathrm{clk}}[\psi] := \int_{t_a}^{t_b} L_{\mathrm{clk}}(\psi(t), \dot{\psi}(t))\,\mathrm{d}t,} \tag{CLK5}$$

with $\delta\psi(t_a) = \delta\psi(t_b) = 0$. Varying ψ and $\bar{\psi}$ independently yields the Schrödinger equation for the finite clock

$$\boxed{\mathrm{i}\hbar_{\mathrm{int}}\dot{\psi} = H_{\mathrm{clk}}\psi} \tag{CLK6}$$

together with its adjoint equation [107]. Its propagator sampled at $t = nt_0$ is $\psi_n = U_{\mathrm{step}}^{n}\psi_0$. Since H_{clk} is self-adjoint, the norm is conserved; normalized initial data remain normalized. The result is exact with respect to the specified $H_{\mathrm{clk}}, t_0, \hbar_{\mathrm{int}}$; it neither selects a physical Hamiltonian nor introduces an experimental frequency.

The theorem 61.1 proves the unitary dynamics of the calendar relative to the declared complex completion, t_0 and $\hbar_{\mathrm{int}}$. The five dynamical conditions of the following section receive a constructive realization in section 61.3: there the drive, the order observable, the primitive subharmonic of period $108T_{\mathrm{d}}$, and finite spectral stability are fixed. The macroscopic material realization constitutes a distinct external recognition and not a lacuna in the finite result.

61.1.5 Dynamic realization conditions of a time crystal: Hamiltonian, subharmonic response and stability

A driven realization must provide, at a minimum:

1. a physical state space and a self-adjoint Hamiltonian $H(t)$, or an equivalent driven protocol, with $H(t + T_{\mathrm{d}}) = H(t)$;
2. the propagator of Floquet

$$U_F = \mathcal{T}\exp\!\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}}\int_0^{T_{\mathrm{d}}} H(t)\ \mathrm{d}t\right);$$

3. an observable of order O and a family of states for which there exists a primitive subharmonic response of order $m > 1$:
$$\langle O((n+m)T_{\mathrm{d}})\rangle = \langle O(nT_{\mathrm{d}})\rangle$$
in the asymptotic regime, without the same periodicity for a proper divisor of m;
4. stability of m, of the phase, and of the amplitude under an open class of admissible perturbations;
5. persistence that is not a finite-size transient, together with a declared mechanism of localization, prethermalization, or dynamical protection.

The calendar $27 \to 54 \to 108$ provides the internal hierarchy of periods. The finite realization of section 61.3 selects subharmonic order 108 in its declared domain, without replacing the five preceding constructions or by itself imposing a macroscopic material phase.

Comparison Criterion 61.2 (HMT–MD time crystal)**.** The identification is refuted if no well-typed driven Hamiltonian or protocol exists; if the alleged subharmonicity disappears under arbitrarily small perturbations compatible with the model; if the observed period is only the period imposed by the drive; if the oscillation is a finite transient; or if the temporal reader assigns equivalent histories to incompatible physical phases.

Mathematical and physical domain of the temporal reader. The three algebras and their exponentials are proved in the algebraic domain; the temporal reader is deduced from the initial signature, order, and memory. The finite HMT time crystal satisfies the test criterion 61.2 through the construction of section 61.3. Extension to a macroscopic material phase additionally requires specification of the physical support and belongs to the subsequent experimental comparison.

61.2 3 temporal sheets, nilpotent calculus, and tritified Cartan connection

61.2.1 Quadratic Family of TRIT

For $\tau \in \{+1, 0, -1\}$, consider the real algebra

$$\mathbb{A}_\tau = \mathbb{R}[J_\tau]/(J_\tau^2 + \tau), \qquad J_\tau^2 = -\tau I.$$

The exponential is computed exactly:

$$e^{tJ_{+1}} = \cos t + J_{+1} \sin t,$$

$$e^{tJ_0} = I + tJ_0,$$

$$e^{tJ_{-1}} = \cosh t + J_{-1} \sinh t.$$

The three sheets have differentiated geometric functions:

τ	J_τ^2	régimen
$+1$	$-I$	elliptic: closure, return and memory
0	0	parabolic: tangency, variation, and present
-1	$+I$	hyperbolic: aperture, boost, and propagation

Let $\widehat{\tau} = P_+ - P_-$, with $P_0 = I - P_+ - P_-$. Then

$$\widehat{\tau}^3 = \widehat{\tau}.$$

An involution that exchanges the outer sheets and preserves the central sheet satisfies

$$\mathsf{J}\widehat{\tau}\mathsf{J}^{-1} = -\widehat{\tau}.$$

The threshold remains fixed and the closing and opening orientations are interchanged.

61.2.2 The null sheet as 1st jet

In $\mathbb{A}_0$, write $\epsilon = J_0$, with $\epsilon^2 = 0$. For every differentiable functional F,

$$\boxed{F(x + \epsilon v) = F(x) + \epsilon\, dF_x(v).} \tag{61.1}$$

The equality is exact because all terms of order at least two contain ϵ^2. In particular, for an action $S[e, \omega, \Psi]$,

$$S[e + \epsilon\delta e, \omega + \epsilon\delta\omega, \Psi + \epsilon\delta\Psi] = S[e, \omega, \Psi] + \epsilon\, \delta S. \tag{61.2}$$

The nilpotent coefficient contains the Euler–Lagrange equations, the Cartan equation, the symplectic potential, and the boundary terms. The central sheet is therefore the algebraic residence of the variational principle.

Theorem 61.3 (Nilpotent stationarity)**.** *For a configuration z, the following are equivalent:*

$$dS_z(\delta z) = 0 \quad \textit{for every } \delta z,$$

and

$$S(z + \epsilon\delta z) = S(z) \quad \textit{in } \mathbb{A}_0 \quad \textit{for every } \delta z.$$

Proof. The equivalence is the equality of the coefficients of 1 and ϵ in (61.1). □

The global Hamiltonian, the effective action of the central sheet, and the modular Hamiltonian of the boundary are coordinated objects:

$$H_{\mathrm{glob}}, \qquad L_{0,\mathrm{eff}}, \qquad K_{\mathrm{HHM}} = -\log \rho_{\partial}.$$

The first generates evolution; the second publishes local variation; the third preserves informational provenance.

61.2.3 Feshbach–Schur reduction

Let

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_{\mathrm{ext}}, \qquad H = \begin{pmatrix} H_{00} & V \\ V^* & H_{\mathrm{ext}} \end{pmatrix}.$$

For z in the resolvent set of H_{ext}, the resolvent compression is

$$P_0(z - H)^{-1}P_0 = [z - H_{00} - \Sigma_0(z)]^{-1}, \qquad \Sigma_0(z) = V(z - H_{\mathrm{ext}})^{-1}V^*. \tag{61.3}$$

The operator Σ_0 transports the memory of the elliptic and hyperbolic sheets to the present sheet. Its time transform generates a memory kernel

$$S_{\mathrm{eff}}[\psi_0] = \int \psi_0^\dagger (i\hbar\partial_t - H_{00})\psi_0 \,\mathrm{d}t - \iint \psi_0^\dagger(t) K_0(t,t')\psi_0(t') \,\mathrm{d}t\,\mathrm{d}t'.$$

A regular low-frequency expansion,

$$\Sigma_0(\omega) = \Sigma_0^{(0)} + \omega\Sigma_0^{(1)} + \omega^2\Sigma_0^{(2)} + \cdots,$$

produces an effective local action; a singular spectral structure preserves nonlocal memory. The present thus receives the response of the global state without introducing an external time.

61.2.4 Connection universal of Cartan

Let $\mathfrak{g}_\tau$ be the Cartan algebra generated by $J_{ab} = -J_{ba}$ and $P_a^{(\tau)}$, with relations

$$[J_{ab}, J_{cd}] = \eta_{bc}J_{ad} - \eta_{ac}J_{bd} - \eta_{bd}J_{ac} + \eta_{ad}J_{bc}, \tag{61.4}$$

$$[J_{ab}, P_c^{(\tau)}] = \eta_{bc}P_a^{(\tau)} - \eta_{ac}P_b^{(\tau)}, \tag{61.5}$$

$$[P_a^{(\tau)}, P_b^{(\tau)}] = -\tau J_{ab}. \tag{61.6}$$

For $\tau = 0$ one obtains the algebra of Poincare; the extreme signs yield the two constant-curvature forms. Let ω be a Lorentz connection, $e = e^a P_a^{(\tau)}$ a coframe valued in the translation sector, and ℓ a length scale. We define the tritified connection

$$\mathcal{A}_\tau^{\mathrm{Cart}} = \frac{1}{2}\omega^{ab}J_{ab} + \ell^{-1}e^a P_a^{(\tau)}. \tag{61.7}$$

Its curvature is

$$\boxed{\mathcal{F}_\tau^{\mathrm{Cart}} = \frac{1}{2}\left(F_\omega^{ab} - \tau\ell^{-2}e^a \wedge e^b\right)J_{ab} + \ell^{-1}T^a P_a^{(\tau)}, \qquad T^a = D_\omega e^a.} \tag{61.8}$$

Proof. Upon expanding $\mathrm{d}\mathcal{A}_\tau^{\mathrm{Cart}} + \mathcal{A}_\tau^{\mathrm{Cart}} \wedge \mathcal{A}_\tau^{\mathrm{Cart}}$, the first bracket of equation (61.6) produces F_ω, the second produces $D_\omega e$, and the third contributes $-\tau\ell^{-2}e^a \wedge e^b J_{ab}/2$. □

In the central sheet,

$$\mathcal{F}_0^{\mathrm{Cart}} = \frac{1}{2}F_\omega^{ab}J_{ab} + \ell^{-1}T^a P_a^{(0)}.$$

Torsion is the nilpotent coefficient of curvature. This equality joins the variational interpretation of the sheet 0 to its geometric Cartan interpretation.

61.2.5 Nonadic corona and torsional energy

On the edges of the crown of nines, let be

$$(D_{9,m}f)(a) = f(t(a)) - U_a f(s(a)),$$

where U_a transports orientation and carry. For a positive weight W_m,

$$H_m = H_{\mathrm{clk},m} + D_{9,m}^* W_m D_{9,m}, \qquad \mathcal{E}_{\mathrm{tor},m}(f) = \|W_m^{1/2} D_{9,m} f\|^2. \tag{61.9}$$

Unitary holonomy preserves positivity. The refinement squares

$$D_{9,m+1}J_m = K_m D_{9,m}, \qquad K_m^* W_{m+1} K_m = W_m$$

guarantee the compatibility of energy at every depth.

The subsequent closure will construct the screen soldering that transforms this defect into a Cartan response. The present section fixes the operator domain and the positivity that this intertwiner must preserve.

61.2.6 Principle of Ambivalence: bidirectional involution of the ratios π/φ^2 and φ/π^2

We define

$$Q(x,y) = \left(\frac{x}{y^2}, \frac{y}{x^2}\right), \qquad x, y > 0.$$

For

$$P = xy, \qquad O = x/y,$$

we verify

$$P(Q(x,y)) = P^{-1}, \qquad O(Q(x,y)) = O^3.$$

After two steps,

$$\boxed{P(Q^2) = P, \qquad O(Q^2) = O^9.} \tag{61.10}$$

In coordinates $u = \log(xy)$, $v = \log(x/y)$,

$$(u,v) \longmapsto (-u, 3v) \longmapsto (u, 9v).$$

The product returns, and orientation multiplies its memory by nine. For $(x,y) = (\pi, \varphi)$, the two outputs are

$$\rho_A = \frac{\pi}{\varphi^2}, \qquad \rho_E = \frac{\varphi}{\pi^2}.$$

Satisfy

$$\rho_E \rho_A^2 = \varphi^{-3}, \qquad M_9 = (\rho_E \rho_A^2)^6 = \varphi^{-18}.$$

The relation links areal reading, electronic exponent, nonadic return and quadratic memory without identifying their algebraic structures with each other.

Estatuto probatorio

The quadratic algebras, the dual calculus, the Schur reduction, the curvature (61.8), and the identity (61.10) are exact. The limit of the torsional energy and its realization as the full Einstein–Cartan action require the refinement squares and the physical morphism developed subsequently.

Falsification criterion

The following refute internal closure: a term other than $-\tau\ell^{-2} e \wedge e$ in the curvature; a nilpotent coefficient that is not $D_\omega e$; a residue in the Feshbach–Schur identity; loss of positivity of $D_9^* W D_9$; or failure of either of the two equations of (61.10).

61.3 Finite-time Temporal Crystal HMT: Periodically forced Hamiltonian, Subharmonic response and Spectral stability

61.3.1 Construction of the periodic forcing from the 108-state temporal system

Sea $N = 108$,

$$\mathcal{H}_{\mathrm{clk}} = \mathbb{C}^N, \qquad S|j\rangle = |j+1\rangle, \qquad Z|j\rangle = \zeta^j|j\rangle, \qquad \zeta = \exp(2\pi\mathrm{i}/N).$$

In the Fourier basis $\{|\widetilde{k}\rangle\}_{k=0}^{N-1}$, we set

$$H_{\mathrm{clk}} = \hbar_{\mathrm{int}}\frac{2\pi}{Nt_0}\sum_{k=0}^{N-1} k|\widetilde{k}\rangle\langle\widetilde{k}|, \qquad \exp\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}}H_{\mathrm{clk}}t_0\right) = S.$$

This equality fixes the starting clock. To exhibit a nonconstant periodic drive, let $0 < \eta < 1$, $T_{\mathrm{d}} = t_0$, and define, in each period,

$$H_{\mathrm{d}}(t) = \begin{cases} 0, & 0 \le t < \eta T_{\mathrm{d}}, \\ (1-\eta)^{-1}H_{\mathrm{clk}}, & \eta T_{\mathrm{d}} \le t < T_{\mathrm{d}}. \end{cases} \qquad H_{\mathrm{d}}(t+T_{\mathrm{d}}) = H_{\mathrm{d}}(t).$$

As the two segments commute, their Floquet propagator is

$$U_F = \mathcal{T}\exp\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}}\int_0^{T_{\mathrm{d}}} H_{\mathrm{d}}(t)\,\mathrm{d}t\right) = \exp\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}}H_{\mathrm{clk}}T_{\mathrm{d}}\right) = S.$$

We define the Hermitian observables

$$Q = \frac{Z+Z^\dagger}{2}, \qquad P = \frac{Z-Z^\dagger}{2\mathrm{i}}.$$

Theorem 61.4 (Primitive Subharmonicity of the Finite Temporal Crystal HMT)**.** *For $\rho_{j_0} = |j_0\rangle\langle j_0|$, the vector of orden*

$$\mathbf{q}_n = \left(\mathrm{Tr}\,\rho_{j_0}U_F^{-n}QU_F^n,\ \mathrm{Tr}\,\rho_{j_0}U_F^{-n}PU_F^n\right)$$

satisfies

$$\mathbf{q}_n = \left(\cos\frac{2\pi(j_0+n)}{N},\ \sin\frac{2\pi(j_0+n)}{N}\right).$$

Its primitive period is $NT_{\mathrm{d}} = 108T_{\mathrm{d}}$, whereas the drive has period T_{d}. Therefore, the system realizes a primitive subharmonic response of order 108 *in the declared finite domain.*

Proof. The Weyl relation $S^{-1}ZS = \zeta Z$ gives $S^{-n}ZS^n = \zeta^n Z$. Its real and imaginary parts yield the formula. If an integer $d > 0$ were a period, then $\zeta^d = 1$ would hold; the primitivity of ζ forces $N \mid d$. □

61.3.2 Nilpotent jet of the triadic evolution and compatible prolongation

The minimal separation between distinct roots of unity is

$$g_N = \min_{k\neq\ell}|\zeta^k - \zeta^\ell| = 2\sin\frac{\pi}{N} > 0.$$

Theorem 61.5 (Covariant rigidity and finite spectral stability)**.** *The following two assertions hold.*

1. *For every unitary V, the terna*

$$U_F^{(V)} = VU_FV^\dagger, \qquad Z^{(V)} = VZV^\dagger, \qquad \rho^{(V)} = V\rho V^\dagger$$

exactly preserves the Weyl relation, the subharmonicity of order N, and the amplitude of the order vector.

2. *If U' is unitary and $\|U' - U_F\| = \varepsilon < g_N/2$, the N disks of radius $g_N/2$ centered at $\{\zeta^k\}$ uniquely separate the spectral sectors of U'. Their Riesz projectors have the same rank as those of U_F throughout every deformation that remains in that ball. Moreover, for every observable A, every state ρ, and every $n \geq 0$,*

$$\left|\operatorname{Tr}\rho\left(U'^{-n}AU'^{n} - U_F^{-n}AU_F^{n}\right)\right| \leq 2n\varepsilon\|A\|.$$

Proof. The first part is obtained by conjugating each identity. For the second, the separation g_N prevents two sectors from entering the same contour; continuity of the Riesz projectors preserves their ranges. The telescoping identity gives $\|U'^{n} - U_F^{n}\| \leq n\varepsilon$, and the same bound holds for the inverse powers. The inequality of expectations results from inserting and subtracting $U'^{-n}AU_F^{n}$. □

The resulting object is a *finite HMT time crystal* with a drive, a subharmonic observable, and quantified spectral stability. Whether a macroscopic material realizes this class, retains coherence at large volume, or exhibits prethermal or localized protection belongs to EXTERNAL IDENTIFICATION; it is not an unresolved mathematical part of the finite construction.

Chapter 62

The HMT–MD confluence theorem: Einstein–Schrödinger system and electrogravitational coordination of action, matter, fields, geometry, and information

Unification is expressed as the vertical compatibility of distinct publications of the same state: coupling and vacuum, action and gravity, geometry and information, screen and torsion, clock and horizon. The theorem preserves the domains and maps of each response.

62.1 Theorem of Dimensional Response Confluence

62.1.1 Antecedent common and product fibered

The volume results arise from heterogeneous readers applied to the same enriched genealogical state. Let

$$X = \left(X_{\mathrm{APP}}, X_{\widehat{\mathbb{T}}}, X_{\mathrm{TPK}}, C, \mathcal{B}, \mathcal{R}, \mathcal{M}, \partial X\right) \tag{62.1}$$

the state that preserves cylinders, carry, route, memory, and boundary. The typed readers are constructed on X

$$P_i : X \longrightarrow Y_i, \qquad i \in I := \{\alpha, \mathrm{act}, \mathrm{ang}, 108, \square, \mathrm{obs}, \mathrm{Per}\}. \tag{62.2}$$

Their codomains represent, respectively, the dodecaphasic coupling generator, the action pair, the bilayer angular operator, the clock 108, the cubic screen, the internal observer, and the Perron mode. For each reader, its graph is taken, provided with the projection in its canonical form

$$\Gamma_i := \{(x, y_i) \in X \times Y_i : P_i(x) = y_i\}, \qquad \pi_i : \Gamma_i \longrightarrow X, \quad (x, y_i) \longmapsto x. \tag{62.3}$$

The master object is the well-typed fiber product of those graphs:

$$\boxed{\mathfrak{M}_X = \underset{X,\, i \in I}{\times} \Gamma_i = \left\{\left(x, (y_i)_{i \in I}\right) : P_i(x) = y_i \text{ for every } i \in I\right\}.} \tag{62.4}$$

The projections π_i materially fix the same antecedent. An element of $\mathfrak{M}_X$ is a family of compatible outputs accompanied by the state that generates them.

Definition 62.1 (Publication dimensional)**.** A dimensional publication is a composition

$$\mathfrak{M}_X \xrightarrow{P} \mathcal{L} \xrightarrow{\mathrm{ev}_{\mathcal{U}}} \mathbb{R}, \tag{62.5}$$

where $\mathcal{L}$ is a typed dimensional line, and $\mathrm{ev}_{\mathcal{U}}$ is the final evaluation in a chart of units. The map P first constructs the invariant and its genealogy.

62.1.2 Naturality of the realizations and vertical conservation of the enriched state invariants

Two sibling readers share an antecedent and may satisfy cross-relations without one factoring through the other. This distinction governs the double sheets of the vacuum, the positive and oriented realizations, the double Archimedean and exceptional projection, and the geometric, informational, and flavor readings of the transition $6 \to 8$.

Theorem 62.2 (Confluence without factorization horizontal)**.** *Let $P_i : \mathfrak{M}_X \to Y_i$ and $i = 1, \ldots, r$ be publications constructed on the same state. Suppose that each P_i preserves its typed data and that the relations $R_j(P_1, \ldots, P_r) = 0$ are proved after the readers have been constructed. Then the joint morphism*

$$P = (P_1, \dots, P_r) : \mathfrak{M}_X \longrightarrow Y_1 \times \cdots \times Y_r \tag{62.6}$$

is vertically compatible with X. A factorization $P_i = F_{ij} \circ P_j$ additionally requires P_j to separate all the fibers distinguished by P_i. The mere existence of $R_j = 0$ does not imply that separation.

Proof. Vertical compatibility follows from the universal property of the fiber product of the graphs Γ_i: all components have the same image in X. If $P_i = F_{ij} \circ P_j$, then $P_j(x) = P_j(x')$ forces $P_i(x) = P_i(x')$. Hence, any pair x, x' that shares the reading j and differs in the reading i refutes the factorization. A scalar relation $R_j = 0$ only restricts the joint image; it does not guarantee the inclusion of kernels required for factorization. □

The theorem explains why the product of sheets can fix the causal cone while the quotient preserves orientation, and why the total area can coexist with a modular Hamiltonian that distinguishes the origin $90/120$.

62.1.3 Constitutive coupling between the vacuum and the electroweak sector

Let

$$D = D_A = \det T, \qquad Q(D) = -\frac{9}{25} \log D = 4\pi\alpha_{\mathrm{HMT}}. \tag{62.7}$$

The memory-oriented is expressed through

$$\rho = e^{2C^*_{\mathrm{rad}}}, \qquad q_- = \sqrt{D\rho}, \qquad q_+ = \sqrt{D/\rho}. \tag{62.8}$$

The functional calculus

$$F(z) = \frac{1 - z^{90}}{1 - z^{120}}, \qquad r_\pm = F(q_\pm) \tag{62.9}$$

yields

$$\widehat{\mu} = r_-^2, \quad \widehat{\varepsilon} = r_+^2, \quad \widehat{Z} = \frac{r_-}{r_+}, \quad \widehat{c} = (r_- r_+)^{-1}. \tag{62.10}$$

At the tree-level electroweak interface,

$$g^2 = \frac{Q(D)}{1 - D}, \qquad g'^2 = \frac{Q(D)}{D}, \tag{62.11}$$

from which

$$\boxed{(1 - D)g^2 = Dg'^2 = Q(D), \qquad \frac{1}{g^2} + \frac{1}{g'^2} = \frac{1}{Q(D)}.} \tag{62.12}$$

The first equality publishes the same coupling in two gauge orientations; the second recovers it as a sum of compliances. The determinant D governs the even mixture, and ρ preserves the odd orientation.

The Higgs coordinate incorporates the latter:

$$\lambda_H = \frac{Q(D)}{8(1 - D)} D^{-3} \rho^{5/3}. \tag{62.13}$$

Defining

$$\Xi = \frac{8(1 - D)D^3 \lambda_H}{Q(D)}, \tag{62.14}$$

is reconstructed

$$\rho = \Xi^{3/5}, \qquad \widehat{Z} = \frac{F(\sqrt{D}\,\Xi^{3/10})}{F(\sqrt{D}\,\Xi^{-3/10})}. \tag{62.15}$$

The equation equation (62.15) provides a cross-check between scalar orientation and vacuum impedance. Its force comes from the parallel construction of both readers on X.

62.1.4 Reciprocity between action and gravity in the electrogravitational form

Let

$$Q_P = E_P, \qquad \ell_P, \qquad Q_P \ell_P = \hbar c, \qquad \frac{\ell_P}{Q_P} = \frac{G}{c^4}. \tag{62.16}$$

The product is the exchange quantum $\hbar c$; the quotient is the gravitational compliance. We introduce

$$\mathscr{T}_P = \frac{Q_P}{\ell_P} = \frac{c^4}{G}, \qquad \mathscr{C}_P = \frac{\ell_P}{Q_P} = \frac{G}{c^4}. \tag{62.17}$$

The Einstein equation assumes the constitutive form

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi \mathscr{C}_P T_{\mu\nu}. \tag{62.18}$$

For a charge q, the electric coordinate lifted to the energy plane is

$$\mathcal{Q}(q) = Q_P \sqrt{\frac{Z}{4\pi\hbar}}\, q. \tag{62.19}$$

Newton and Coulomb are then brought together in the bilinear form

$$\boxed{V(r) = \frac{\mathscr{C}_P}{r}\big(\mathcal{Q}_1\mathcal{Q}_2 - E_1E_2\big).} \tag{62.20}$$

The cone $\mathcal{Q}^2 = E^2$ reproduces the static condition of extremality in the Einstein–Maxwell chart. The sheet product of the vacuum governs the causal cone; its quotient governs the electric lift and orientation.

The cross-identity with the electroweak sector is

$$\boxed{\frac{\hbar^2}{8\pi\ell_P^2}\frac{(\kappa/\mu)}{e^2} = \frac{1}{4\pi\alpha_{\mathrm{HMT}}} = \frac{1}{g^2} + \frac{1}{g'^2}.} \tag{62.21}$$

This equality preserves its value under inversion of the action sheet: G and e^2 transform contragrediently, whereas $G\hbar$, Ge^2, and ℓ_P^2 remain invariant.

62.1.5 Confluence between geometry, information and mixing

Let $N = N_{90} + N_{120}$ be the total number of boundary excitations and

$$D_{90/120} = 90N_{90} + 120N_{120}. \tag{62.22}$$

The area reads N, whereas the modular holographic Hamiltonian reads $D_{90/120}$. Since

$$\det\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix} = 30, \tag{62.23}$$

the pair reconstructs exactly

$$N_{90} = 4N - \frac{D_{90/120}}{30}, \qquad N_{120} = \frac{D_{90/120}}{30} - 3N. \tag{62.24}$$

The total geometry and the weighted modular energy therefore preserve quantity and provenance.

The angular double-layer state

$$\rho_y = \frac{e^{-yR}}{2\cosh y} \tag{62.25}$$

has even entropy and oriented divergence

$$S(\rho_y) = \log(2\cosh y) - y\tanh y, \qquad D(\rho_y\|\rho_{-y}) = 2y\tanh y. \tag{62.26}$$

The involution $y \mapsto -y$ is represented in the flavour sector by

$$R_{\mathrm{CKM}} = M_{\mathrm{CKM}}\,\mathrm{diag}(1,-1,1)M_{\mathrm{CKM}}^{-1}, \qquad R_{\mathrm{CKM}}^2 = I, \qquad \det R_{\mathrm{CKM}} = -1. \tag{62.27}$$

Barbero reads the odd component of the transition $6 \to 8$, entropy reads its even distribution, and R_{CKM} realizes the same inversion in the mixing codomain. They are coordinated representations of a common hinge, with their own domains and observables.

62.1.6 Transduction of the helical memory in the torsional realization

In the quotient lorentziano $Q_{\square}$, the soldadura and the respuis positiva satisfy

$$\boxed{(\beta_m^{\square})^{*}\Lambda_m^{\square}\beta_m^{\square} = \frac{56}{81}I_{Q_{\square}}.} \tag{62.28}$$

In the deep register, the screw operator checks

$$\boxed{V^{-1}\mathcal{S}V = (2+\sqrt{3})\mathrm{i}\,\mathcal{S}.} \tag{62.29}$$

The first identity measures the minimum response cost of the screen; the second transports orientation and depth. Catalan appears as the second moment of the quaternary component. If $N_{\mathrm{dep}}\delta_n = n\delta_n$ denotes the depth operator of the helical record, then

$$G_{\mathrm{Cat}} = \mathrm{Tr}(N_{\mathrm{dep}}^{-2}Q_{\chi,-4}). \tag{62.30}$$

Perron's theorem

$$M_n = \left(\frac{a_n}{a_0}\right)^{-6} \tag{62.31}$$

provides the exact covariance of a quadratic spin density. The Cartan realizer must transport equations (62.28) y (62.29) to a current and a contorsion whose homogeneous mode satisfies equation (62.31). This chain places the physical interface in a unique and falsifiable arrow.

62.1.7 Confluence between clock periodicity, horizon energy, and information cost

The clock 108 fixes

$$E_P = k_B\Theta_{\mathrm{clk}}\log 3. \tag{62.32}$$

For the de Sitter patch of radius $R = \ell_P$, the vacuum energy, temperature, and entropy satisfy

$$E_\Lambda = T_H S_H = \frac{E_P}{2} = \frac{k_B \Theta_{\mathrm{clk}} \log 3}{2}. \tag{62.33}$$

From here follow

$$E_\Lambda t_P = \frac{\hbar}{2}, \qquad E_\Lambda T_{108} = \frac{h}{2}, \qquad \frac{S_H}{k_B} = \pi. \tag{62.34}$$

The vacuum volume, the horizon area, the cycle temperature, and the information cost of a trit publish the same energy. The multiplicity $e^{S_H/k_B} = e^{\pi}$ coincides exactly with the expansive eigenvalue of $e^{\pi J_\varphi}$; promotion of this equality to an equivalence of microstates requires an intertwiner that preserves measure and dynamics.

62.1.8 Fibrewise confluence of HMT–MD responses: vacuum, action, gravity, information, mixing, memory, and cosmology

Theorem 62.3 (Confluence of HMT–MD responses)**.** *Given the enriched state X and the primitives declared in each reader, the publications of vacuum, action, gravity, information, mixing, screen, helical memory, and cosmology are the projections or compositions defined on the fiber product of graphs equation (62.4). Their cross-relations are generated by equations (62.12), (62.20), (62.21), (62.24), (62.26), (62.28), (62.29) y (62.33). Every numerical evaluation is terminal, and every oriented memory remains in the factor that generates it.*

Proof. The construction of $\mathfrak{M}_X$ fixes the common antecedent. The vacuum equations follow from the functional calculation of T; the electroweak equations follow from D and $Q(D)$; the electrogravitational form follows from the two identities of the Planck pair; the reconstruction 90/120 results from inverting an integral matrix of determinant 30; the informational orientation follows from the logarithm of ρ_y; the screen energy results from the unique response in $Q_\square$; the screw follows from the two Weyl and Pell covariances; and the horizon equality follows from the geometric formulas of de Sitter specialized at $R = \ell_P$. No proof uses the terminal decimal to select the branch. The horizontal independence theorem prevents cross-relations from being converted into unproved factorizations. □

62.1.9 Corollary of metrological confluence between the HMT and SI realizations

Every output admits the inseparable quintuple

$$\mathcal{K}(P) = \big(\text{antecedent, equation, invariant, meaning, terminal value}\big). \tag{62.35}$$

The volume's metrological atlas enumerates these quintuples. Its probative content is as follows: the antecedent fixes the genealogy; the equation fixes the operation; the invariant demonstrates what survives; the meaning makes the physical reading explicit; the terminal value enables external recognition. Separating one column destroys causal information; preserving all five permits insertion of the result into a chart of units without converting that chart into a foundation.

Estatuto probatorio

The fiber product, the algebraic identities, and the horizontal-independence theorem constitute an exact formalization. The internal equations cited retain the proved status of the chapters containing their complete proofs. The Einstein–Maxwell, de Sitter, Einstein–Cartan, and electroweak readings remain at their declared interfaces when they require an additional realizer. The theorem organizes those layers without promoting them by numerical similarity.

Falsification criterion

Confluence is refuted if two components attributed to the same element of $\mathfrak{M}_X$ actually arise from incompatible states, if one of the generating identities fails, if a terminal evaluation enters into the selection of its own antecedent, or if a memory distinguished by another reader is lost. A purported horizontal factorization is refuted by any pair of states with the same intermediate reader and different final outputs. Physical promotions are additionally tested against their own realizers, gauge identities, Bianchi identities, refinement, and boundary conditions.

62.2 Discrete Physical Realization via Fiber Product and Cartan Entanglement

The common fiber product, its positive and oriented components, reversible localization, and smooth-promotion criterion provide the material intertwiner between the abstract confluence and its physical realizations. The development is preserved in full, with its obstructions explicit.

62.2.1 Discrete physical realization: positive and oriented readings on TPK routes, common domain, and reversible localization

The spectral form, the Hilbert realization, the dimensional section, and Cartan transport are not decorations appended to a classical output. They are four coordinated representations of the same enriched TPK route. To express that causal unity, their common domain must first be constructed; it is not sufficient to write four arrows with the same source symbol.

62.2.1.1 Common domain as a fiber product

Let $\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}$ be the category of prepared TPK states and registered routes compatible with their endpoints, boundary, and restrictions. Let

$$\mathsf{D}_{\mathrm{sp}} := \mathsf{SurvPath}_{\mathrm{HMT}}, \qquad \mathsf{D}_Q := \mathrm{Dom}(Q), \qquad \mathsf{D}_D := \mathrm{Dom}(\mathfrak{D}_{\mathrm{HMT}}), \qquad \mathsf{D}_C := \mathrm{Dom}(\rho_C^{\mathrm{disc}}),$$

with their faithful forgetful functors

$$U_i : \mathsf{D}_i \longrightarrow \mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}} \qquad (i \in \{\mathrm{sp}, Q, D, C\}).$$

Each U_i forgets only the realization and preserves the underlying TPK state and route. We define

$$\mathsf{P}^{+}_{\mathrm{CT}} := \mathsf{D}_{\mathrm{sp}} \times_{\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}} \mathsf{D}_Q \times_{\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}} \mathsf{D}_D \times_{\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}} \mathsf{D}_C. \tag{62.36}$$

An object is therefore a prepared state equipped with the four realizations over the same antecedent; an arrow is a recorded route whose underlying image coincides in the four factors. If the presentations are identified only up to isomorphism, the corresponding bicategorical fiber product is used; in the fixed presentation of the record the strict fiber product of (62.36) is adopted.

Proposition 62.4 (Existence of an object in the common domain)**.** *The category $\mathsf{P}^{+}_{\mathrm{CT}}$ is nonempty. It is not necessary to specialize the central electronic route in order to prove this: any certified coinductive section, equipped with its identity arrow and its neutral data, admits compatible lifts to the four domains of* (62.36).

Proof. The theorem 8.54 provides, for example, a prepared surviving section x_π. Its identity id_{x_π} is simultaneously: a surviving route; a morphism of $\mathsf{P}_{\mathrm{TPK}}$, which Q sends to the bounded identity of $Q(x_\pi)$; the empty route with null signature, which the theorem 59.5 sends to $(0, 0, 0, 1)$; and the identity of the groupoid, which the theorem 82.10 sends to the neutral Poincare element $(I, 0)$. These four lifts forget to the same object x_π and the same morphism id_{x_π}. Their quadruple defines, by the fiber-product property, an object of $\mathsf{P}^{+}_{\mathrm{CT}}$. □

Definition 62.5 (Adapters to the common domain)**.** The four adapters are the canonical projections

$$F_{\mathrm{sp}} : \mathsf{P}^{+}_{\mathrm{CT}} \longrightarrow \mathsf{SurvPath}_{\mathrm{HMT}},$$
$$F_{Q} : \mathsf{P}^{+}_{\mathrm{CT}} \longrightarrow \mathrm{Dom}(Q),$$
$$F_{D} : \mathsf{P}^{+}_{\mathrm{CT}} \longrightarrow \mathrm{Dom}(\mathfrak{D}_{\mathrm{HMT}}),$$
$$F_{C} : \mathsf{P}^{+}_{\mathrm{CT}} \longrightarrow \mathrm{Dom}(\rho^{\mathrm{disc}}_{C}).$$

In particular, F_{sp} preserves the compatible family of finite restrictions of the section and the route; it does not reconstruct that family from its spectrum.

Lemma 62.6 (Functoriality of the adapters)**.** *The four adapters preserve identities and composition.*

Proof. The identity of an object of the fibered product is the quadruple of identities on the same TPK state; each projection returns the identity of the factor. Composition is defined componentwise between quadruples that project onto the same concatenation of routes. Projecting before or after composition gives, by definition, the same result. □

62.2.1.2 4 *Components and Positive Product*

The spectral, Hilbertian, dimensional, and Cartan realization theorems provide

$$\mathfrak{S} : \mathsf{SurvPath}_{\mathrm{HMT}} \longrightarrow \mathbf{SpecRoute}, \qquad Q : \mathrm{Dom}(Q) \longrightarrow \mathsf{Hilb}_{\mathbb{R},J},$$
$$\mathfrak{D}_{\mathrm{HMT}} : \mathrm{Dom}(\mathfrak{D}_{\mathrm{HMT}}) \longrightarrow \mathbf{B}\mathfrak{M}^{+}_{D}, \qquad \rho^{\mathrm{disc}}_{C} : \mathrm{Dom}(\rho^{\mathrm{disc}}_{C}) \longrightarrow \mathbf{B}G_{C},$$

where

$$G_{C} := \mathrm{Spin}^{+}(1,3) \ltimes \mathbb{R}^{1,3}$$

and

$$\mathfrak{M}^{+}_{D} := \mathbb{N} \times \mathbb{R}^{2}_{\geq 0} \times \mathbb{R}_{>0}, \qquad (n,a,b,\lambda) \odot (n',a',b',\lambda') = (n+n', a+a', b+b', \lambda\lambda'). \tag{62.37}$$

Here $\mathbf{B}M$ is the one-object category associated with the monoid M. The first functor is that of theorem 59.16; the second is the Hilbert realization defined in the postulates chapter; the third is theorem 59.5; and the fourth is theorem 82.10.

We define the common constraints

$$\mathfrak{S}_{+} := \mathfrak{S} \circ F_{\mathrm{sp}}, \qquad Q_{+} := Q \circ F_{Q}, \qquad \mathfrak{D}_{+} := \mathfrak{D}_{\mathrm{HMT}} \circ F_{D}, \qquad \rho^{\mathrm{disc}}_{C,+} := \rho^{\mathrm{disc}}_{C} \circ F_{C}.$$

Theorem 62.7 (Positive discrete realization on the common domain)**.** *The map*

$$\mathfrak{R}^{\mathrm{disc},+}_{\mathrm{MD}} := (\mathfrak{S}_{+}, Q_{+}, \mathfrak{D}_{+}, \rho^{\mathrm{disc}}_{C,+}) \tag{62.38}$$

is a functor

$$\mathsf{P}^{+}_{\mathrm{CT}} \longrightarrow \mathbf{SpecRoute} \times \mathsf{Hilb}_{\mathbb{R},J} \times \mathbf{B}\mathfrak{M}^{+}_{D} \times \mathbf{B}G_{C}.$$

It is not asserted that every TPK route belongs to $\mathsf{P}^{+}_{\mathrm{CT}}$.

Proof. By lemma 62.6, each projection F_i is a functor; by the cited theorems, so are $\mathfrak{S}, Q, \mathfrak{D}_{\mathrm{HMT}}$ and ρ^{disc}_{C} on their declared domains. Their compositions therefore preserve identities and concatenation. Composition in the product codomain is computed componentwise, which proves (62.38). □

The Hilbert component has general target $\mathsf{Hilb}_{\mathbb{R},J}$, whose morphisms are bounded real operators intertwining the complex structures. Let $\mathsf{P}^{u}_{\mathrm{CT}} \subseteq \mathsf{P}^{+}_{\mathrm{CT}}$ be the subcategory of arrows $\gamma : x \to y$ for which

$$Q_+(\gamma)^* Q_+(\gamma) = I_{Q_+(x)}, \qquad Q_+(\gamma) Q_+(\gamma)^* = I_{Q_+(y)}. \tag{62.39}$$

The identities and products of unitary operators are again unitary; hence this condition defines a subcategory, and only within it is obtained

$$\mathcal{U}_{\mathrm{reg}} := Q_+\big|_{\mathsf{P}^u_{\mathrm{CT}}} : \mathsf{P}^u_{\mathrm{CT}} \longrightarrow \mathsf{UHilb}_{\mathbb{R},J}. \tag{62.40}$$

One-sided invertibility or norm conservation on a subset does not replace the two identities of (62.39).

The seven-entry notation decomposes into the four typed components: Q is Q_+; U is the restriction (62.40); A is the action coordinate of $\mathfrak{D}_+$; C is the conjugation declared on the subdomain where it intertwines the Hilbert component and orientation; $\mathcal{E}$ and $\mathcal{B}$ are constitutive readers subordinated to the dimensional data; and N_{sp} is a quadratic property of the split realization. None of the final four is added to the product as a fifth functor without a domain and codomain.

62.2.1.3 Positive and oriented readings

Let $e \mapsto e^\vee$ be edge reversal in the underlying directed graph. In $\mathsf{P}^+_{\mathrm{CT}}$, $e^\vee$ is another edge and is not yet declared the categorical inverse of e. Having fixed a weight $w(e) = w(e^\vee) > 0$, we define on a route $\gamma = e_1 \cdots e_r$

$$V_+(\gamma) := \sum_{j=1}^{r} w(e_j), \qquad V_{\mathrm{or}}(\gamma) := \sum_{j=1}^{r} \varepsilon(e_j) w(\underline{e}_j),$$

where $\underline{e}_j$ is the unoriented edge and $\varepsilon(e^\vee) = -\varepsilon(e)$. The first reader preserves total positive variation; the second, net oriented transport. Both are additive under concatenation and receive the same TPK route.

Proposition 62.8 (No factorization of the positive reader by the oriented)**.** *Suppose that the identity and the loop $ee^\vee := e^\vee \circ e$ belong to the constructed domain. There is no function H on* $\operatorname{im} V_{\mathrm{or}}$ *such that $V_+ = H \circ V_{\mathrm{or}}$.*

Proof. We have

$$V_{\mathrm{or}}(\mathrm{id}) = 0 = V_{\mathrm{or}}(ee^\vee), \qquad V_+(\mathrm{id}) = 0, \qquad V_+(ee^\vee) = 2w(e) > 0.$$

The same input 0 would force $H(0)$ to take two distinct values. □

This result is a concrete nonfactorization in the constructed domain; it is not a universal theorem about every possible enriched intertwiner. In particular, it does not authorize erasing a reader. The unit lies in the common TPK antecedent and not in a horizontal compression between its outputs.

62.2.1.4 Reversible localization and oriented realization

Let $\mathsf{P}^+_{\mathrm{CT,rev}}$ be the subcategory generated by the edges for which the following four conditions have been verified:

1. $\mathfrak{S}_+(e)$ is an isomorphism of **SpecRoute** and $\mathfrak{S}_+(e^\vee) = \mathfrak{S}_+(e)^{-1}$;
2. $Q_+(e)$ satisfies (62.39) and $Q_+(e^\vee) = Q_+(e)^*$;
3. the oriented dimensional datum arises from an additive cochain with $d_{\mathrm{or}}(e^\vee) = -d_{\mathrm{or}}(e)$ and takes values in $\mathrm{Gr}(\mathfrak{M}^+_D)$;
4. the Cartan cocycles are normalized, additive, and antisymmetric under reversal.

We form the groupoid

$$\Pi_\vee(\mathsf{P}^+_{\mathrm{CT,rev}})$$

obtained from the groupoid completion by additionally imposing $[e^\vee] = [e]^{-1}$. The oriented dimensional component takes values in

$$\mathrm{Gr}(\mathfrak{M}_D^+) \cong \mathbb{Z} \times \mathbb{R}^2 \times \mathbb{R}_{>0}, \tag{62.41}$$

with inverse $(n, a, b, \lambda)^{-1} = (-n, -a, -b, \lambda^{-1})$. This signed reader is not the positive magnitude V_+ relabeled: its values on the reverse edges are distinct.

For the Cartan component, let c_g, c_s be cocycles as in item 4, let $R_4 \in \mathrm{Spin}^+(1,3)$, let $R_4 u = u$, and let $\ell_0 > 0$. Then

$$\rho_C^{\mathrm{disc}}(\gamma) = \big(R_4^{c_g(\gamma)}, \ell_0 c_s(\gamma) u\big).$$

The law $(R, a)(S, b) = (RS, a + Rb)$, the equality $R_4 u = u$, and the additivity of the cocycles give

$$\rho_C^{\mathrm{disc}}(\gamma_2\gamma_1) = \rho_C^{\mathrm{disc}}(\gamma_2)\rho_C^{\mathrm{disc}}(\gamma_1), \qquad \rho_C^{\mathrm{disc}}(\gamma^\vee) = \rho_C^{\mathrm{disc}}(\gamma)^{-1}.$$

Theorem 62.9 (Discrete oriented realization)**.** *Under the four preceding conditions, the components descend uniquely to a functor*

$$\mathfrak{R}_{\mathrm{MD}}^{\mathrm{disc,or}} : \Pi_\vee(\mathsf{P}_{\mathrm{CT,rev}}^+) \longrightarrow \mathrm{Core}(\mathbf{SpecRoute}) \times \mathsf{UHilb}_{\mathbb{R},J} \times \mathbf{B}\,\mathrm{Gr}(\mathfrak{M}_D^+) \times \mathbf{B}G_C.$$

In particular,

$$\mathfrak{R}_{\mathrm{MD}}^{\mathrm{disc,or}}(\gamma^{-1}) = \mathfrak{R}_{\mathrm{MD}}^{\mathrm{disc,or}}(\gamma)^{-1}.$$

Proof. Conditions 1 and 2 make the spectral and Hilbert images invertible; condition 3 takes values in a group; and the Cartan component already takes values in G_C. All four respect the generating relation $[e^\vee] = [e]^{-1}$. By the universal property of groupoid localization, each descends uniquely; their product is the indicated functor. The inversion identity is verified componentwise. □

The positive reader $\mathfrak{R}_{\mathrm{MD}}^{\mathrm{disc},+}$ and the oriented reader $\mathfrak{R}_{\mathrm{MD}}^{\mathrm{disc,or}}$ are siblings: neither is defined as a compression of the other. The subsequent passage from an oriented value to its modulus is not, in general, monoidal and coincides with a positive reading only on a cone without cancellation. Likewise, this pair does not replace the foundational double projection: the Archimedean evaluation and the Paley–Witt–Golay–Leech–$V^\natural$–Monster–Moonshine branch remain sibling readers of the same enriched state. The four components constructed in this section replace neither of those two projections.

62.2.1.5 Cartan Entanglement and Soft Limit

Let

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1} e \\ 0 & 0 \end{pmatrix}$$

a Cartan connection declared. The equation

$$\mathrm{Hol}_{\mathcal{A}_{\ell_0}}\big(\mathfrak{R}_C(\gamma)\big) = \rho_C\big(\mathrm{Hol}_{\mathrm{TPK}}(\gamma)\big) \tag{62.42}$$

is exact in the discrete model when both holonomies are defined by the same ordered product of edges.

Comparison Criterion 62.10 (Promotion to a smooth realization)**.** The equation (62.42) defines a smooth prolongation only if subdivision independence, convergence of the ordered product, and local recovery of a co-tetrad and a connection with the declared domains and regularity are also proved.

The term ‘physical’ means here that the four components publish spectrum, state realization, dimensions, and Cartan transport without losing their TPK antecedent. It does not assert uniqueness, equivalence of categories, coverage of every route, an automatic SI chart, or a complete field theory.

Comparison Criterion 62.11 (Obstructions to positive, unitary, and oriented realizations)**.** The positive realization is not defined if any of the four adapters is absent, if their images do not have the same underlying TPK route, or if a component does not preserve identity and composition. The unitary restriction does not exist if either of the two equalities in (62.39) fails. The oriented realization does not

descend to the groupoid if a spectral image is not isomorphic, if a localized relation changes its image, if a positive magnitude is treated as an inverse, if a reversal does not change the sign of the oriented data, or if the Cartan cocycle ceases to be normalized, additive, or antisymmetric. No compatible smooth prolongation exists if the result depends on the subdivision or if the limit does not recover the cotetrad and connection. Any use of an external coordinate to select the route violates the constructed domain.

Domain, codomain, and status of the HMT–MD confluent equations The positive product is proved on the common fiber product, and the oriented realization on reversible localization under the four explicit conditions. Extending the domain to other routes requires constructing their four lifts over a common antecedent. Smooth prolongation additionally requires the subdivision, convergence, and local recovery properties of test criterion 62.10; a finite verification does not replace those properties.

From the Cartan–Holst realization to horizon–area–information confluence through conservation of action The first constitutive realization of this confluence is the electromagnetic–electroweak vacuum, where the symmetric, antisymmetric, and determinantal functions publish propagation and couplings.

62.3 Planck pair, universal tension, and gravitational compliance

62.3.1 Planck dimensional pair (Q, ℓ): $Q\ell = \hbar c$, $\ell/Q = G/c^4$, and unique reconstruction

Let t_0 be the chronological unit of the CT108 section, and let c be the causal section selected by the constitutive reader. We define

$$t_P = \frac{54}{\pi} t_0, \qquad \ell = \ell_P = c t_P = \frac{54}{\pi} c t_0,$$

and the energy quantum

$$Q = E_P = \frac{\hbar}{t_P} = \frac{h}{108 t_0}.$$

The periodicity $T_{108} = 108 t_0$ verifies

$$T_{108} = 2\pi t_P, \qquad Q T_{108} = h.$$

From these definitions, the first identity of the pair is deduced:

$$\boxed{Q\ell = \hbar c.} \tag{62.43}$$

The circular gravitational realization determines

$$G = \left(\frac{54}{\pi}\right)^2 \frac{c^5 t_0^2}{\hbar}.$$

Substituting $\ell = (54/\pi) c t_0$ and $Q = \hbar / t_P$ yields the second identity:

$$\boxed{\frac{\ell}{Q} = \frac{G}{c^4}.} \tag{62.44}$$

The equations (62.43) and (62.44) are algebraically independent as product and quotient relations. They uniquely reconstruct

$$Q = \sqrt{\frac{\hbar c^5}{G}}, \qquad \ell = \sqrt{\frac{G\hbar}{c^3}}.$$

Theorem 62.12 (Planck Pair Closure)**.** *In the circular CT108 realization, the positive pair* (Q, ℓ) *is the unique solution of*

$$Q\ell = \hbar c, \qquad \ell/Q = G/c^4.$$

Satisfies, moreover,

$$GQ = c^4\ell, \qquad GQ^2 = \hbar c^5, \qquad G\hbar = c^3\ell^2.$$

Proof. Positivity permits multiplication and division of the two equations without ambiguity of sign. The first additional equality follows from $\ell/Q = G/c^4$; the second results from multiplying it by Q^2, and the third from multiplying $\ell/Q = G/c^4$ by $Q\ell = \hbar c$. Reconstruction by positive roots proves uniqueness. □

62.3.2 Reciprocity between Planck tension and compliance and conservation of the action product

The pair defines two reciprocal responses:

$$\mathscr{T}_P := \frac{Q}{\ell} = \frac{c^4}{G}, \qquad \mathscr{C}_P := \frac{\ell}{Q} = \frac{G}{c^4}. \tag{62.45}$$

The first has the dimension of force; the second, the dimension of gravitational compliance. They satisfy

$$\mathscr{T}_P\mathscr{C}_P = 1.$$

The Einstein equation is then written as a constitutive law:

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi\mathscr{C}_P T_{\mu\nu}.} \tag{62.46}$$

Matter supplies tension–energy, and geometry responds through $\mathscr{C}_P$. The constant G publishes the curvature capacity per unit of tension in the chosen causal chart.

In the Newtonian regime, for an energy density ε,

$$\nabla^2\left(\frac{\Phi}{c^2}\right) = 4\pi\mathscr{C}_P\varepsilon.$$

The Einstein–Hilbert action is normalized by the structural area:

$$\frac{S_{\mathrm{EH}}}{\hbar} = \frac{1}{16\pi\ell^2}\int_{\mathcal{M}} (R - 2\Lambda)\sqrt{-g}\,\mathrm{d}^4x. \tag{62.47}$$

The scalar response and the tensor response are distinct publications of the same compliance. The former uses $4\pi\mathscr{C}_P$; the latter, $8\pi\mathscr{C}_P$. Upon inserting CT108,

$$2\pi\mathscr{C}_P = 108\frac{ct_0}{Q}, \qquad 4\pi\mathscr{C}_P = 216\frac{ct_0}{Q}, \qquad 8\pi\mathscr{C}_P = 432\frac{ct_0}{Q}.$$

The progression $108 \to 216 \to 432$ follows from the geometric factors. The coincidence of 432 with the multiplicity of the transverse region of π constitutes a structured compatibility; its promotion requires an incidence map that transports the genealogical multiplicity to the Einstein tensor.

62.3.3 Ternary Informational Quantum $\log 3$ and Capacity of the Qutrit Cell

The thermodynamic relation of the CT108 section is

$$\boxed{Q = k_B \Theta_{\mathrm{clk}} \log 3.} \tag{62.48}$$

In this equation, $\log 3$ is the entropy of an equiprobable ternary alternative; Θ_{clk} fixes the thermal chart and Q publishes the energy of the cycle. The combination with theorem 62.12 yields

$$\boxed{G k_B \Theta_{\mathrm{clk}} \log 3 = c^4 \ell.} \tag{62.49}$$

Gravity transforms the informational quantum of the clock into a length of geometric response. Equivalently,

$$\ell = \mathscr{C}_P k_B \Theta_{\mathrm{clk}} \log 3.$$

This equality separates the dimensionless content $\log 3$, the thermometric frame $k_B \Theta_{\mathrm{clk}}$, and the geometric compliance.

62.3.4 Covariance of bidirectional readers under sheet inversion and temporal conjugation

Let $\hbar_{\mathrm{pre}}$ and $\hbar_{\mathrm{ret}}$ be the two oriented sections of the action line, related by a positive quotient R_{act}:

$$\frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = R_{\mathrm{act}}.$$

The invariance of ℓ obliga to

$$\frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}} = R_{\mathrm{act}}^{-1}.$$

Therefore,

$$G_\sigma \hbar_\sigma = c^3 \ell^2 \qquad (\sigma \in \{\mathrm{pre}, \mathrm{ret}\}). \tag{62.50}$$

The action preserves the orientation of the route; its product with the gravitational response preserves the area. This covariance makes ℓ^2 the geometric invariant of the bidirectional inversion.

62.3.5 Constitutive vacuum and causal closure

The operator of vacuum proporciona

$$c_{\mathrm{vac}}^{-2} = \mu \varepsilon.$$

The Planck pair provides a second causal section:

$$c_{\mathrm{int}} = \frac{Q \ell}{\hbar}.$$

We define the sewing scalar

$$\Xi_c = \frac{Q \ell}{\hbar} \sqrt{\mu \varepsilon} = \frac{c_{\mathrm{int}}}{c_{\mathrm{vac}}}. \tag{62.51}$$

The equality $\Xi_c = 1$ is necessary and sufficient for both readers to publish the same causal cone. Under this seam,

$$\boxed{G = \left(\frac{54}{\pi}\right)^2 \frac{t_0^2}{\hbar(\mu\varepsilon)^{5/2}}.} \tag{62.52}$$

Proof. The relation $c = (\mu\varepsilon)^{-1/2}$ gives $c^5 = (\mu\varepsilon)^{-5/2}$. Substituting it into the circular selector of G yields (62.52). □

The product $\mu\varepsilon$ governs the causal cone and the gravitational response; the quotient $\mu/\varepsilon = Z^2$ governs impedance and orientation. Both contents follow from the same bilayer operator and preserve distinct functions.

62.3.6 Machian factorization of the local inertial response through the trace of the global closure: $\mathcal{I}_v = \overline{\mathcal{I}}_v \circ b$

A local energy E determines the dimensionless coordinate

$$\xi_E(E) = \frac{E}{Q}.$$

The gravitational coupling of two energies is

$$\alpha_G(E_1, E_2) = \frac{E_1 E_2}{Q^2}. \tag{62.53}$$

The reference Q belongs to the global closure clock. Local inertia is therefore evaluated against a scale produced by the complete genealogy. This dependence does not posit an instantaneous action of the universe upon the body: it establishes an operatorial factorization of the local response by the global closure class.

Estatuto probatorio

The identities of the pair, the tension, the compliance, and the covariance $G\hbar$ are exact in the declared circular CT108 realization. The identification $c_{\mathrm{vac}} = c_{\mathrm{int}}$ is a typed physical interface; its content is concentrated in the map of causal lines whose evaluation is $\Xi_c = 1$.

Falsification criterion

The closure of this chapter is refuted by: a distinct positive pair that simultaneously satisfies the two governing equations; the loss of $G\hbar = c^3\ell^2$ under inversion of the action; a section of the vacuum that does not satisfy $c^{-2} = \mu\varepsilon$; or a frame transport for which $\Xi_c \neq 1$.

62.4 Electrogravitational Bilinear Form and Cross-Response Equations

62.4.1 Internal source coordinates

Let Z be the impedance section of the vacuum and let (Q, ℓ) be the pair from the preceding chapter. To a charge q and an energy E we associate

$$\xi_q(q) = \sqrt{\frac{Z}{4\pi\hbar}}\, q, \qquad \xi_E(E) = \frac{E}{Q}.$$

The internal source vector is

$$X(q, E) = \begin{pmatrix} \xi_q(q) \\ \xi_E(E) \end{pmatrix}, \qquad \eta_{1,1} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The form $X_1^{\mathsf{T}}\eta_{1,1}X_2$ separates electrical orientation and gravitational closure within a single dimensionless plane.

62.4.2 Static unification of the potentials

Theorem 62.13 (Electrogravitational form)**.** *If $Zc = \varepsilon^{-1}$, $E_i = m_i c^2$ and $Q\ell = \hbar c$, then*

$$\boxed{V_{12}(r) = \frac{Q\ell}{r} X(q_1, E_1)^{\mathsf{T}} \eta_{1,1} X(q_2, E_2)} \tag{62.54}$$

is exactly the sum of the Coulomb potential and the Newton potential.

Proof. The electric term is

$$\frac{Q\ell}{r}\frac{Zq_1q_2}{4\pi\hbar} = \frac{Zcq_1q_2}{4\pi r} = \frac{q_1q_2}{4\pi\varepsilon r}.$$

The term gravitational is

$$-\frac{Q\ell}{r}\frac{E_1E_2}{Q^2} = -\frac{\ell}{Q}\frac{E_1E_2}{r} = -\frac{G}{c^4}\frac{m_1m_2c^4}{r} = -\frac{Gm_1m_2}{r}.$$

□

The associated radial force takes the form

$$F_{12}(r) = \frac{Q}{\ell}\left(\frac{\ell}{r}\right)^2 X_1^{\mathsf{T}}\eta_{1,1}X_2. \tag{62.55}$$

The factor $Q/\ell = \mathscr{T}_P$ is the universal tension; the dimensionless form determines the intensity and sign of its publication.

62.4.3 Cone of extremality of constitutive responses and classification of its rays

For a single source, the discriminant

$$\Delta_{\mathrm{EMG}}(q, E) = E^2 - \mathcal{Q}_e(q)^2, \qquad \mathcal{Q}_e(q) = Q\sqrt{\frac{Z}{4\pi\hbar}}\, q,$$

induce the three sectors

$$\Delta_{\mathrm{EMG}} > 0, \qquad \Delta_{\mathrm{EMG}} = 0, \qquad \Delta_{\mathrm{EMG}} < 0.$$

The condition null is equivalent to

$$E^2 = \mathcal{Q}_e(q)^2 \quad \Longleftrightarrow \quad Gm^2 = \frac{q^2}{4\pi\varepsilon}. \tag{62.56}$$

It is the condition of electrogravitational extremality in the Reissner–Nordström normalization. Here the trit acquires a static realization: subextremality, extreme threshold and superextremality are the three signs of the same quadratic form.

For the elementary charge constructed by

$$e^2 = \frac{4\pi\alpha_{\mathrm{HMT}}\hbar}{Z},$$

the coordinate satisfies

$$\boxed{\xi_q(e)^2 = \alpha_{\mathrm{HMT}}.} \tag{62.57}$$

This equality locates the central charge in the source plane without using the electron mass or importing its complete theory.

62.4.4 Einstein–Maxwell Coefficient

Write

$$\kappa = \frac{8\pi G}{c^4} = 8\pi \mathscr{C}_P.$$

In a convention where the electromagnetic tensor is expressed as $T^{\mathrm{EM}}_{\mu\nu} = \mu^{-1} Q_{\mu\nu}(F, g)$, the gravitational coefficient of the quadratic form is κ/μ. Using

$$\ell^2 = \frac{G\hbar}{c^3}, \qquad Z = \mu c, \qquad e^2 = \frac{4\pi\alpha\hbar}{Z},$$

we obtain

$$\boxed{\frac{\hbar^2}{8\pi\ell^2}\frac{(\kappa/\mu)}{e^2} = \frac{1}{4\pi\alpha_{\mathrm{HMT}}}.} \tag{62.58}$$

Proof. First,

$$\frac{\kappa}{\mu} = \frac{8\pi G}{c^4\mu} = \frac{8\pi\ell^2}{\hbar\mu c} = \frac{8\pi\ell^2}{\hbar Z}.$$

Al dividir by $e^2 = 4\pi\alpha\hbar/Z$ and multiplicar by $\hbar^2/(8\pi\ell^2)$ remains $1/(4\pi\alpha)$. □

At the rationalized electroweak interface,

$$g^2 = \frac{4\pi\alpha}{1 - D_A}, \qquad g'^2 = \frac{4\pi\alpha}{D_A},$$

and therefore

$$\boxed{\frac{\hbar^2}{8\pi\ell^2}\frac{(\kappa/\mu)}{e^2} = \frac{1}{g^2} + \frac{1}{g'^2}.} \tag{62.59}$$

The equality requires the gravity, vacuum, charge, and gauge mixing readers to commute. Within the present graph, e, g, g' descend from α_{HMT}; the equation is a cross-check, not a second derivation independent of α.

62.4.5 Polygon of fine structure publications

The dodecaphasic macro-object is published through the system

$$\boxed{\alpha_{\mathrm{HMT}} = \frac{Ze^2}{4\pi\hbar} = \frac{ZG_0}{4} = \frac{Z}{2R_K} = -\frac{9}{100\pi}\log D_A = \frac{(1 - D_A)g^2}{4\pi} = \frac{D_A g'^2}{4\pi}.} \tag{62.60}$$

The first causal construction of α remains the signed wheel, the seal, and the carry. The subsequent equalities are metrological publications or compatibility checks. Elimination of variables yields

$$(1 - D_A)g^2 = D_A g'^2 = \frac{Ze^2}{\hbar} = \pi G_0 Z = \frac{2\pi Z}{R_K}.$$

The dual form is

$$\frac{1}{g^2}+\frac{1}{g'^2}=\frac{\hbar}{Ze^2}=\frac{R_K}{2\pi Z}=\frac{1}{\pi G_0 Z}.$$

Each metrological cycle must be marked as a *round trip*: if e, R_K, or G_0 were defined from α, their inverse substitution checks coherence but adds no selective information.

62.4.6 Covariance of action inversion

Under the bidirectional transformation,

$$\frac{e^2_{\mathrm{pre}}}{e^2_{\mathrm{ret}}}=R_{\mathrm{act}},\qquad \frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}}=R^{-1}_{\mathrm{act}}.$$

Therefore,

$$\boxed{G_\sigma e^2_\sigma = \text{sheet constant.}} \tag{62.61}$$

In constitutive form,

$$\frac{Ze^2_\sigma G_\sigma}{c^5 t_0^2}=4\pi\theta_{108}\alpha_{\mathrm{HMT}},\qquad \theta_{108}=\left(\frac{54}{\pi}\right)^2.$$

The charge preserves the orientation of action, and the gravitational response compensates for it; their product belongs to the shared geometry.

62.4.7 Tensorial representation of the gravitational carrier and transverse reduction of its degrees of freedom

The bilinear form (62.54) closes the static sector. Its promotion to complete dynamics requires a realizer

$$\mathcal{R}_{\mathrm{EM}}:\mathcal{X}^{\mathrm{enr}}_{\mathrm{TPK}}\longrightarrow (e^I{}_\mu,\omega^{IJ}{}_\mu,A_\mu)$$

that preserves the Bianchi identity, Maxwell's equations, current conservation, and the boundary terms. The identity (62.59) fixes the coefficient that this realizer must publish; it does not replace its differential equations.

Falsification criterion

The electrogravitational form is refuted by: a different coefficient in Coulomb or Newton upon expanding (62.54); a null cone that fails to reproduce (62.56); the loss of (62.58); or the use of a publication descended from α as a retroactive selector of its dodecaphase branch.

Causal structure of the electrogravitational confluence. The composition proved in this chapter is organized into five typed stages. The enriched state first supplies the phase α_{HMT} and its orientation. The constitutive operator then publishes e, g, and g' in their respective codomains. The gravitational closure determines G and the chronology θ_{108}. Identities (62.54)–(62.59) coordinate these publications through a common coefficient. Finally, $\mathcal{R}_{\mathrm{EM}}$ transports the result to the fields $(e^I{}_\mu,\omega^{IJ}{}_\mu,A_\mu)$, preserving the differential identities and boundary terms. This factorization fixes the antecedence of the HMT construction with respect to its tensorial realization and prepares the constitutive relations developed in the following chapter.

Chapter 63

Constitutive realization of the vacuum and electromagnetic–electroweak propagation

The bidirectional structure of action induces constitutive vacuum relations and separates the propagation, impedance, charge, and electroweak mixing sectors. SI references appear as a subsequent realization of internal functions already determined.

63.1 Realization of the Electromagnetic Bidirectional Action Operator: ε, μ, Z, and c

63.1.1 Derivation of the action scale from the dodecaphasic state

The action decade is determined intrinsically, without adopting it as an inherited unit. The local determinantal functional is applied to the previously constructed HMT output α_{HMT} and defines

$$\Sigma_H = \varphi_{\mathrm{HMT}} \alpha_{\mathrm{HMT}}^{16}. \tag{63.1}$$

The exponent 16 is the order of the local cokernel of the signed closure; its decimal valuation fixes

$$\mathrm{ord}_{10}(\Sigma_H) = -34. \tag{63.2}$$

The origin of the exponent can be verified within this volume. On the local orbit of four leaves, the Hadamard block acts

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \tag{63.3}$$

Lemma 63.1 (Local index of the action line)**.** *The form normal of Smith of H_4 is*

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4). \tag{63.4}$$

Consequently,

$$\mathrm{coker}(H_4) \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \qquad |\mathrm{coker}(H_4)| = 16.$$

Proof. The greatest common divisor of the entries is one; that of the nonzero 2×2 minors is two; that of the 3×3 minors is four; and $|\det H_4| = 16$. The successive quotients of these determinantal divisors are $1, 2, 2, 4$, which prove equation (63.4). The product of the invariants gives the order of the cokernel. □

The norm of the section α_{HMT} over those sixteen sheets produces $\alpha_{\mathrm{HMT}}^{16}$; the self-scaling φ_{HMT} acts only once on the basis, not once per sheet. Hence equation (63.1). The certified comparisons

$$\alpha_{\mathrm{HMT}}^{16} < 10^{-34} < \varphi_{\mathrm{HMT}} \alpha_{\mathrm{HMT}}^{16} < 10^{-33} \tag{63.5}$$

prove, without introducing the SI unit, that the decimal order is exactly -34. The terminal evaluation begins with

$$\Sigma_H = 1.046243838104756580184229208303089\ldots \times 10^{-34}. \tag{63.6}$$

The substitution of the kernel by three copies with exponent 16^3, or the repeated application of φ on each sheet, would alter the genealogy. Both operations constitute falsification criteria rather than equivalent conventions. The causal chain that is preserved is therefore,

$$\mathrm{APP} \to \widehat{\mathbb{T}} \to \mathrm{TPK} \to U_{12}^{\mathrm{sign}} \to \alpha_{\mathrm{HMT}} \to \Sigma_H \to 10^{-34} \mathcal{U}_S \to \{\hbar_{\mathrm{pre}}, \hbar_{\mathrm{ret}}\}.$$

This volume does not rederive α_{HMT}, but develops the metrological geometry subsequent to the selection of the type and decade by Σ_H. The integer -34 is neither a mantissa nor an adjustment to the International System, but the terminal valuation of the local functional.

63.1.2 Oriented sections of the action line

The realizations before and after the return constitute two sections of the same action line:

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S. \tag{63.7}$$

The common decade determines the line, while the oriented information is expressed through

$$R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}}, \qquad \xi_{\mathrm{act}} = \frac{1}{2}\log R_{\mathrm{act}}. \tag{63.8}$$

The first quantity provides a multiplicative coordinate and the second an additive coordinate of rapidity type. They do not represent two independent Planck constants, but the two orientations of the same section with respect to the return.

If $h_a = 2\pi\hbar_a$, $a \in \{\mathrm{pre}, \mathrm{ret}\}$, a quantity proportional to $h_a^{1/2}$ changes as $R_{\mathrm{act}}^{1/2}$, one proportional to $h_a^{-1/2}$ changes as $R_{\mathrm{act}}^{-1/2}$, and a quotient in which the action cancels is a bidirectional invariant.

63.1.3 Bilayer spectral decomposition of the angular operator

Let $R = R^* = R^{-1}$ be the active involution and

$$P_\pm = \frac{I \pm R}{2}, \qquad T = q_+P_+ + q_-P_-, \qquad 0 < q_+ < q_- < 1.$$

With

$$x = A_{\mathrm{rad}} = \frac{\pi A}{180}, \qquad y = C^*_{\mathrm{rad}} = \frac{\pi C^*}{180},$$

the two channels are constructed before evaluating the vacuum:

$$q_+ = e^{-x-y}, \qquad q_- = e^{-x+y}. \tag{63.9}$$

The exchange of sheets reverses C^*, that is, $y \mapsto -y$, and permutes $q_+ \leftrightarrow q_-$. The pair is thus the spectral reading of the same angular antecedent, not two parameters fitted separately.

Definition 63.2 (Constitutive response)**.**

$$\mathcal{V}_{90,120} = (I - T^{90})(I - T^{120})^{-1}. \tag{63.10}$$

Since $\|T\| < 1$, the denominator is positive and invertible. Functional calculus gives

$$\mathcal{V}_{90,120} = r_+P_+ + r_-P_-, \qquad r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}}. \tag{63.11}$$

The assertion of invertibility deserves to be made explicit. For each leaf, $0 < q_\pm^{120} < 1$; therefore

$$I - T^{120} = (1 - q_+^{120})P_+ + (1 - q_-^{120})P_-$$

is strictly positive and

$$(I - T^{120})^{-1} = \frac{P_+}{1 - q_+^{120}} + \frac{P_-}{1 - q_-^{120}}.$$

The response $\mathcal{V}_{90,120}$ is not obtained by solving four separate scalar equations: it is the result of the functional calculus of a single positive contraction. This observation prevents permeability, permittivity, impedance and propagation from losing their shared antecedent.

Theorem 63.3 (Joint spectral determination of the four constitutive relations)**.** *The four constituent characters*

$$\widehat{\varepsilon} = r_+^2, \qquad \widehat{\mu} = r_-^2, \qquad \widehat{Z} = \frac{r_-}{r_+}, \qquad \widehat{c} = \frac{1}{r_+ r_-} \tag{63.12}$$

are spectral functions of $\mathcal{V}_{90,120}$. They satisfy

$$\widehat{\mu}\widehat{\varepsilon} = \widehat{c}^{-2}, \qquad \frac{\widehat{\mu}}{\widehat{\varepsilon}} = \widehat{Z}^2. \tag{63.13}$$

Proof. The identities are obtained by substituting equation (63.12). No dimensional chart nor any exterior value intervenes. □

Corollary 63.4 (Reconstruction of the two sheets)**.** *The four characters do not contain four degrees of freedom. Any of the pairs $(\widehat{\mu}, \widehat{\varepsilon})$, $(\widehat{Z}, \widehat{c})$, or $(\mathcal{E}, \mathcal{B})$ reconstructs the two positive eigenvalues:*

$$r_- = \sqrt{\widehat{\mu}} = \sqrt{\widehat{Z}/\widehat{c}}, \qquad r_+ = \sqrt{\widehat{\varepsilon}} = \frac{1}{\sqrt{\widehat{Z}\widehat{c}}}, \tag{63.14}$$

$$\log r_- = \frac{\mathcal{E} + \mathcal{B}}{2}, \qquad \log r_+ = \frac{\mathcal{E} - \mathcal{B}}{2}. \tag{63.15}$$

In particular, orientation is lost if only the product is retained, but is recovered by adding the quotient.

Proof. The first line is the positive root of the identities equations (63.12) y (63.13). The second inverts the linear system $\mathcal{E} = \log r_- + \log r_+$, $\mathcal{B} = \log r_- - \log r_+$. □

This corollary expresses a central property of HMT: product and quotient are complementary functionals of the same state. The product determines the propagation in common, whereas the quotient preserves the relative orientation of the inner and outer sheets.

In the V2 certificate,

$$\begin{aligned}
r_+ &= 0.99999962854824936317869522 81\ldots,\\
r_- &= 0.99972413718951836413151658 53\ldots,\\
\widehat{\varepsilon} &= 0.99999925709663670276044161 55\ldots,\\
\widehat{\mu} &= 0.99944835047932693508998155 59\ldots,\\
\widehat{Z} &= 0.99972450853893721545555434 66\ldots,\\
\widehat{c} &= 1.00027631048615752610638626 89\ldots.
\end{aligned}$$

These numerals are the terminal recognition of equation (63.10); they do not define its exponents $90, 120$.

63.1.4 Harmonic factorization of sectors 90 and 120

The harmonic semigroup is

$$\langle 90, 120\rangle = 30\langle 3, 4\rangle.$$

If $s = q^{30}$, then

$$\frac{1-q^{90}}{1-q^{120}} = \frac{1-s^3}{1-s^4} = \frac{1+s+s^2}{1+s+s^2+s^3}. \tag{63.16}$$

The identity also separates the response from its defect. If

$$F_{3\to4}(s) = \frac{1+s+s^2}{1+s+s^2+s^3}, \qquad 0 < s < 1,$$

then

$$d_{3\to4}(s) := 1 - F_{3\to4}(s) = \frac{s^3}{1+s+s^2+s^3} > 0. \tag{63.17}$$

The additional term is preserved exactly as a positive defect. Moreover,

$$F'_{3\to4}(s) = -\frac{s^2(3+2s+s^2)}{(1+s+s^2+s^3)^2} < 0. \tag{63.18}$$

Consequently, the order of the angular channels uniquely determines the order of the constitutive responses. The assignment of $\widehat{\mu}$ and $\widehat{\varepsilon}$ to the respective sheets is obtained analytically, without resorting to their decimal expansions.

Each sheet of the vacuum constitutes the exact completion from three to four terms of the same elementary mode 30. Consequently, permeability and permittivity are interpreted as two quadratic functionals of the same completion evaluated on conjugate sheets, and not as independent constitutive parameters.

Since the function $(1-q^{90})/(1-q^{120})$ decreases in $0 < q < 1$ and $q_+ < q_-$, we obtain

$$0 < r_- < r_+ < 1, \qquad 0 < \widehat{\mu} < \widehat{\varepsilon} < 1, \qquad 0 < \widehat{Z} < 1 < \widehat{c}.$$

The limits of the functional determine its geometry. When $q \to 0^+$, $F(q) \to 1$: the four readings approach unity and the memory defect disappears. When $q \to 1^-$,

$$F(q) \longrightarrow \frac{90}{120} = \frac{3}{4}.$$

The response therefore takes values in the positive interval $(3/4, 1)$. The endpoint $3/4$ is not a fitted constant: it is the quotient of the two channel lengths and the continuous realization of the three–to–four completion.

63.1.5 Sheet involution and preservation of orientation

The sheet exchange $C^* \mapsto -C^*$ acts by

$$\widehat{\mu} \longleftrightarrow \widehat{\varepsilon}, \qquad \widehat{Z} \longmapsto \widehat{Z}^{-1}, \qquad \widehat{c} \longmapsto \widehat{c}. \tag{63.19}$$

The common propagation is even; impedance preserves orientation. This distinction will reappear in Barbero and CKM: not all functionals associated with the change of sheet have the same parity.

63.1.6 Logarithmic characters and tracial compatibility

Define

$$\mathcal{E} = \log(r_+ r_-), \qquad \mathcal{B} = \log(r_-/r_+).$$

With the normalized trace $\tau = \mathrm{Tr}/2$,

$$2\tau(\log \mathcal{V}) = \mathcal{E}, \qquad -2\tau(R \log \mathcal{V}) = \mathcal{B}. \tag{63.20}$$

The pair $(\mathcal{E}, \mathcal{B})$ constitutes a linear coordinate system for the operator logarithm:

$$\log \mathcal{V} = \frac{\mathcal{E}}{2} I - \frac{\mathcal{B}}{2} R. \tag{63.21}$$

The scalar component is invariant under sheet exchange; the oriented component changes sign. This decomposition directly demonstrates the parity of propagation and the oriented character of impedance.

The nonadic amplification

$$\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$$

is compatible with the expectations $E_m = \mathrm{id} \otimes \tau_9$:

$$E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m.$$

Therefore, $\mathcal{E}$ and $\mathcal{B}$ are not peculiarities of a 2×2 matrix; they persist in the UHF tower and in its tracial closure II_1.

More precisely, for every polynomial p and afterwards, by continuity, for every continuous function on the spectrum,

$$E_m\big(p(\mathcal{V}_{m+1})\big) = p(\mathcal{V}_m). \tag{63.22}$$

Compatibility does not preserve only two numbers: it preserves the entire functional algebra generated by the response. In particular, it preserves its spectral projectors, powers, logarithm, and quadratic forms that measure the defects equation (63.17).

63.1.7 Dimensional realization of the constitutive relations

Dimensional Mechanics determines

$$Z_{\mathrm{vac}} = \widehat{Z}\, u_S u_Q^{-2}, \tag{63.23}$$

$$c_{\mathrm{vac}} = \widehat{c}\, u_C, \tag{63.24}$$

$$\mu_{\mathrm{vac}} = \widehat{\mu}\, u_S u_C^{-1} u_Q^{-2}, \tag{63.25}$$

$$\varepsilon_{\mathrm{vac}} = \widehat{\varepsilon}\, u_S^{-1} u_C^{-1} u_Q^{2}. \tag{63.26}$$

The constitutive identities hold on the straight lines:

$$\mu_{\mathrm{vac}} \varepsilon_{\mathrm{vac}} = c_{\mathrm{vac}}^{-2}, \qquad \mu_{\mathrm{vac}} / \varepsilon_{\mathrm{vac}} = Z_{\mathrm{vac}}^{2}. \tag{63.27}$$

The SI chart comes later. It is not necessary to declare c primitive and then solve $\mu\varepsilon = c^{-2}$: the three objects descend jointly from $\mathcal{V}$ and from the declared lines.

The separation of levels is expressed by the diagram

$$\begin{array}{ccccc} (A, C^*) & \longrightarrow & T & \longrightarrow & \mathcal{V} \longrightarrow (\widehat{\mu}, \widehat{\varepsilon}, \widehat{Z}, \widehat{c}) \\ \big\downarrow \text{line charts} & & & & \big\downarrow \text{final evaluation} \\ (u_S, u_Q, u_C) & \longrightarrow & (\mu_{\mathrm{vac}}, \varepsilon_{\mathrm{vac}}, Z_{\mathrm{vac}}, c_{\mathrm{vac}}) & \longrightarrow & \text{SI coordinates.} \end{array}$$

The first line is dimensionless and operatorial; the second determines types of quantity; the third selects a metrological chart. Altering the final chart changes neither the eigenvalues nor the constitutive identities.

In this precise sense, mathematical–physical unification consists in distinguishing the genealogical invariant from its coordinate system, not in removing the dimensional structure.

Proposition 63.5 (Spectral minimality)**.** *Let W be a positive two-sheet operator with simple spectrum $\{r_+, r_-\}$. If four positive characters satisfy $\mu\varepsilon = c^{-2}$ and $\mu/\varepsilon = Z^2$, if their propagation in common is determined by $c^{-1} = \det W = r_- r_+$, and if its orientation fixes $Z = r_-/r_+$, then necessarily*

$$(\mu, \varepsilon, Z, c) = \left(r_-^2, r_+^2, \frac{r_-}{r_+}, \frac{1}{r_- r_+}\right).$$

Therefore, equation (63.12) is not one parametrization among many: it is the minimal positive realization compatible with product, quotient and orientation.

Proof. From $Z = r_-/r_+$ and $c^{-1} = \det W = r_- r_+$ the positive roots are obtained uniquely roots. The two constitutive identities then force afterwards $\mu = r_-^2$ and $\varepsilon = r_+^2$. □

63.1.8 Angular determinant and Weinberg electroweak relation

The determinant of T is

$$\det T = q_+ q_- = D_A. \tag{63.28}$$

The constitutive sector of the vacuum evaluates $F(T)^2$, where $F(z) = (1 - z^{90})/(1 - z^{120})$, whereas the electroweak angular sector evaluates $\det T$. These are two functions of a common antecedent, not a horizontal factorization between constants. This distinction is developed in the section 63.2.

Falsification criterion 63.6. The block is refuted if any of these exact identities is violated: positivity of $I - T^{120}$, equality equation (63.16), relations equation (63.13), or compatibility $E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m$. An SI coincidence cannot compensate for an operator failure.

Hypothesis, domain, and conclusion

Operator, spectrum, completion $3 \longrightarrow 4$, parity, amplification, and constitutive sections form an exact internal theorem. Their assembly into a single operator is established in this formulation; the angular data and the lines, previously derived in HMT, constitute the explicit starting structure.

63.2 Angular determinant and electroweak-sector parameters

63.2.1 Constitutive and Determinantal Functions of the Angular Operator

Recall the two-sheet operator of the section 63.1,

$$T = q_+ P_+ + q_- P_-, \qquad q_- = \mathrm{e}^{-(A_{\mathrm{rad}} - C^*_{\mathrm{rad}})}, \qquad q_+ = \mathrm{e}^{-(A_{\mathrm{rad}} + C^*_{\mathrm{rad}})}. \tag{63.29}$$

The electromagnetic vacuum applies the function

$$F(T)^2, \qquad F(z) = \frac{1 - z^{90}}{1 - z^{120}}, \tag{63.30}$$

while the electroweak angular sector applies the determinant

$$\boxed{\det T = q_+ q_- = \mathrm{e}^{-2A_{\mathrm{rad}}} =: D_A.} \tag{63.31}$$

They are two sibling functional realizations of the common antecedent (A, C^*, T). The causal structure generates both from T. The subsequent reconstruction demonstrated in theorems 63.17 y 63.18 permits recovery of

$$\det T = \det\Big(F^{-1}\Big(\sqrt{F(T)^2}\Big)\Big),$$

without turning that functional inversion into the generating arrow of the electroweak sector. Structural unity resides in T, and the reconstruction makes explicit the compatibility among permeability, permittivity, and θ_W.

Under the sheet involution $C^* \mapsto -C^*$, the eigenvalues q_+, q_- are exchanged. Therefore D_A and the constitutive speed are even, while $(\widehat{\mu}, \widehat{\varepsilon})$ is exchanged and $\widehat{Z}$ is inverted. The following electroweak relation uses the even invariant.

63.2.2 Exact relation between the W and Z sectors

The angular coordinate of duration–perduration is

$$\chi_{\mathrm{dur}}(n_A, s_C) = \exp\Big(n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}}\Big). \tag{63.32}$$

This coordinate is not introduced as an isolated label. Its origin is

$$\begin{aligned} \mathrm{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow \text{surviving extension} \longrightarrow \text{route} \\ \longrightarrow \text{genealogical record} \longrightarrow (n_A, s_C, \ldots) \longrightarrow \chi_{\mathrm{dur}}. \end{aligned} \tag{63.33}$$

The signatures of W, Z and H are typed quantities obtained through that chain; this chapter adopts them without selecting them from their terminal masses. In the established angular realization, the charged and neutral sectors satisfy

$$(n_A, s_C)_W = (94, -1), \qquad (n_A, s_C)_Z = (95, -1). \tag{63.34}$$

This assertion refers to the common angular component. The additional refinement coordinates of the General Mass Law are not eliminated unless the physical transducer proves their cancellation.

Theorem 63.7 (Weinberg defect in the physical-mass scheme of the angular chart)**.** *If W and Z are evaluated on the same scale section and share the refinement that accompanies equation (63.34), then*

$$\boxed{\frac{m_W^{\angle}}{m_Z^{\angle}} = \mathrm{e}^{-A_{\mathrm{rad}}} = \sqrt{D_A}, \qquad \left(\frac{m_W^{\angle}}{m_Z^{\angle}}\right)^2 = D_A.} \tag{63.35}$$

In the physical-mass scheme,

$$\boxed{\sin^2\theta_W^{\mathrm{OS,HMT}} = 1 - D_A = 0.22487088075284356981111590 35\ldots.} \tag{63.36}$$

Además,

$$\frac{m_W^{\angle}}{m_Z^{\angle}} = 0.88041417483316136701288854 59\ldots. \tag{63.37}$$

Proof. By equations (63.32) y (63.34),

$$\frac{\chi_{\mathrm{dur}}(94,-1)}{\chi_{\mathrm{dur}}(95,-1)} = \exp(-A_{\mathrm{rad}});$$

the coordinate C^* cancels because both sectors share $s_C = -1$. Squaring and using equation (63.31) yields D_A. The definition of physical masses $s_W^2 = 1 - m_W^2/m_Z^2$ yields equation (63.36). □

The principal content is the identity between two internal constructions: the determinant of the angular operator and the one-step difference in the realization W/Z. Promotion to a physical gauge selector requires an intertwiner that carries the sheets of T to the neutral basis (W^3, B) and demonstrates which refinements are preserved or cancel.

63.2.3 Gauge couplings in rationalized normalization

We define the dimensionless gauge coupling

$$e_g := \sqrt{4\pi\alpha_{\mathrm{HMT}}} = 0.30282212087175264276624426 89\ldots. \tag{63.38}$$

e_g must not be confused with a dimensional charge section e_{int}: the former is a coupling in natural units; the latter lives in the charge line of Dimensional Mechanics.

Let

$$s_W^2 = 1 - D_A, \qquad c_W^2 = D_A. \tag{63.39}$$

The convention electrodebil racionalizada to level of arbol gives

$$g = \frac{e_g}{s_W} = 2\sqrt{\frac{\pi\alpha_{\mathrm{HMT}}}{1 - D_A}} = 0.63858834273345742836470038 09\ldots, \tag{63.40}$$

$$g' = \frac{e_g}{c_W} = 2\sqrt{\frac{\pi\alpha_{\mathrm{HMT}}}{D_A}} = 0.34395416331085024293623192 57\ldots, \tag{63.41}$$

$$\frac{g'}{g} = \sqrt{D_A^{-1} - 1} = 0.53861641419660935949966109 33\ldots. \tag{63.42}$$

Proposition 63.8 (Tree-level custodial relation)**.** *At the same interface,*

$$\rho_{\mathrm{EW}}^{(0)} := \frac{(m_W^{\angle})^2}{(m_Z^{\angle})^2 c_W^2} = 1. \tag{63.43}$$

Proof. By equations (63.35) y (63.39), both the relative numerator and c_W^2 are D_A. □

The equations equations (63.40) y (63.41) are exact compositions once the physical mass scheme, rationalization, and scale have been declared. They do not assert that a change of renormalization scheme leaves their numerical values unchanged.

63.2.4 Fermi Constant in the Approximation at Tree Level

In natural units, integration of the charged propagator at low energy yields

$$\frac{G_F^{(0)}}{\sqrt{2}} = \frac{g^2}{8m_W^2}. \tag{63.44}$$

Substituting equation (63.40) gives the composition

$$\boxed{G_F^{(0)} = \frac{\pi\alpha_{\mathrm{HMT}}}{\sqrt{2}(1 - D_A)m_W^2}.} \tag{63.45}$$

By adopting the previously obtained mass

$$m_W^{\mathrm{HMT}} = 80.381592550221\ldots\ \mathrm{GeV}, \tag{63.46}$$

the terminal evaluation is

$$\boxed{G_F^{(0)} = 1.1157162817659203186621784 \times 10^{-5}\ \mathrm{GeV}^{-2}.} \tag{63.47}$$

The difference with respect to a renormalized Fermi constant is not used to choose any coefficient. Under the convention

$$G_F = \frac{\pi \alpha_{\mathrm{HMT}}}{\sqrt{2}(1 - D_A) m_W^2 (1 - \Delta r)}, \tag{63.48}$$

the remaining obligation is to derive Δr from the internal sector of loops, states, and scale. Opening the observed value of G_F to define Δr would be a calibrated reconstruction, not a prediction.

63.2.5 Higgs Coupling Oriented Dependence

In the angular chart, the relevant routes are

$$(n_A, s_C)_H = (97, 9), \qquad (n_A, s_C)_W = (94, -1). \tag{63.49}$$

Therefore,

$$\boxed{\frac{m_H^{\angle}}{m_W^{\angle}} = \frac{\chi_{\mathrm{dur}}(97, 9)}{\chi_{\mathrm{dur}}(94, -1)} = \exp\left(3A_{\mathrm{rad}} + \frac{5}{3} C^*_{\mathrm{rad}}\right) = 1.558720872123255485 64\ldots.} \tag{63.50}$$

Here, C^* does not cancel: the scalar sector preserves the orientation that the determinant $D_A = \mathrm{e}^{-2A_{\mathrm{rad}}}$ does not distinguish.

Let us take the convention of the potential

$$V(H) = -\mu_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2, \tag{63.51}$$

for which

$$m_H^2 = 2\lambda_H v^2, \qquad m_W^2 = \frac{g^2 v^2}{4}. \tag{63.52}$$

Corollary 63.9 (Oriented Higgs self-coupling)**.** *At the declared tree-level interface,*

$$\boxed{\lambda_H = \frac{\pi \alpha_{\mathrm{HMT}}}{2(1 - D_A)} \exp\left(6A_{\mathrm{rad}} + \frac{10}{3} C^*_{\mathrm{rad}}\right) = 0.12384791154824668046 62\ldots.} \tag{63.53}$$

Además,

$$G_F^{(0)} m_H^2 = \sqrt{2}\, \lambda_H. \tag{63.54}$$

Proof. Dividing the two equations in equation (63.52) gives $m_H^2/m_W^2 = 8\lambda_H/g^2$. With $g^2 = 4\pi\alpha_{\mathrm{HMT}}/(1 - D_A)$ and the square of equation (63.50), we obtain equation (63.53). On the other hand, $G_F^{(0)} = 1/(\sqrt{2}v^2)$ and $m_H^2 = 2\lambda_H v^2$ imply equation (63.54). □

The value of λ_H precedes the incorporation of radiative corrections and depends on the declared convention and scale. Its structural interest is that it separates two memories: the common gauge sector uses the even invariant D_A, whereas the Higgs route explicitly retains C^*.

63.2.6 Determinantal closure of the electroweak sector: mixing, couplings, and constitutive spectrum

The complete chain of this chapter is

$$\boxed{\begin{aligned} (A, C^*) &\longrightarrow T \longrightarrow \begin{cases} F(T)^2 & \longrightarrow (\widehat{\mu}, \widehat{\varepsilon}, \widehat{Z}, \widehat{c}), \\ \det T = D_A & \longrightarrow (s_W^2, c_W^2), \end{cases} \\ (D_A, \alpha_{\mathrm{HMT}}) &\longrightarrow (e_g, g, g'), \\ (D_A, \alpha_{\mathrm{HMT}}, m_W) &\longrightarrow G_F^{(0)}, \\ (A, C^*, D_A, \alpha_{\mathrm{HMT}}) &\longrightarrow \left(\frac{m_H}{m_W}, \lambda_H\right). \end{aligned}} \tag{63.55}$$

The figure always appears at the end. Neither $\sin^2\theta_W$, nor G_F, nor λ_H is introduced as a target value for selecting A, C^* or the route signatures.

> **The same angular operator determines the vacuum through a rational function of its sheets and the electroweak defect through its determinant. The one-step difference between W and Z turns that determinant into $\sin^2\theta_W^{\mathrm{OS}} = 1 - D_A$; the Higgs route, in contrast, preserves the counterangle and preserves the orientation eliminated by the determinant.**

Falsification criterion 63.10. The Weinberg relation is refuted in its domain if the angular signatures W/Z do not differ only by $n_A : 94 \to 95$, if a refinement coordinate does not cancel at the declared interface, or if $(m_W^{\angle}/m_Z^{\angle})^2 \neq D_A$. Gauge promotion is blocked if no intertwiner with the basis (W^3, B) exists. The figures of g, g', G_F, λ_H change if the scheme or scale is altered without recording it. Using observed G_F or m_H to select Δr, the signatures, or C^* introduces feedback from the target value and invalidates the derivation. Conflating e_g with the dimensional charge constitutes a type error.

Hypothesis, domain, and conclusion

Angular determinant, quotient W/Z, and complementary Weinberg defect: *Exact internal theorem in the angular conditioned chart*. Couplings $g, g', \rho_{\mathrm{EW}}^{(0)}$, Fermi, and Higgs: *Exact Composition in the Electroweak Tree-Level Interface*. The gauge-basis intertwiner and the internal correction Δr: *Open boundary with falsification criterion*. Fermi and Higgs are not independent constants selected without target values: they are exact compositions using the signatures, the quantity m_W and the declared tree-level convention. The subsequent conventional values are *Contrast metrological external*.

From the electroweak determinant to the positive vacuum constitutive operator and its spectral completion 3→4 Once the constitutive support has been fixed, quantum evolution is reconstructed as a sum and integral of genealogies, with typed effective reduction and probabilistic publication.

63.3 Bilayer modular state and spectral resolvent

63.3.1 The angular operator as common antecedent

The constitutive operator of section 63.1 admits a prior formulation in which intensity and orientation occupy distinct coordinates of the same object. Let $R = R^* = R^{-1}$ be an involution with spectral multiplicities one and one, and let

$$P_+ = \frac{I + R}{2}, \qquad P_- = \frac{I - R}{2} \tag{63.56}$$

their projectors. For

$$x = A_{\mathrm{rad}} > 0, \qquad y = C_{\mathrm{rad}}^*, \qquad |y| < x, \tag{63.57}$$

we define the angular generator and its positive exponential by

$$K_{\angle} = xI + yR, \qquad T_{x,y} = \mathrm{e}^{-K_{\angle}} = q_+ P_+ + q_- P_-, \tag{63.58}$$

where

$$q_+ = \mathrm{e}^{-x-y}, \qquad q_- = \mathrm{e}^{-x+y}. \tag{63.59}$$

The condition $|y| < x$ places both eigenvalues in $(0, 1)$. The determinant and the spectral quotient separate the two coordinates:

$$D_A := \det T_{x,y} = q_+ q_- = \mathrm{e}^{-2x}, \qquad \varrho_{\mathrm{or}} := \frac{q_-}{q_+} = \mathrm{e}^{2y}. \tag{63.60}$$

The first character is even under sheet exchange; the second records their orientation. Recovery is immediate:

$$x = -\frac{1}{2}\log D_A, \qquad y = \frac{1}{2}\log \varrho_{\mathrm{or}}. \tag{63.61}$$

The causal provenance of x and y remains linked to state APP $\to \widehat{\mathbb{T}} \to$ TPK. In particular,

$$x = \frac{50\pi}{9}\,\mathfrak{A}_\beta, \qquad D_A = \exp\!\left(-\frac{100\pi}{9}\,\mathfrak{A}_\beta\right), \tag{63.62}$$

where $\mathfrak{A}_\beta = \alpha_{\mathrm{HMT}}$ denotes the terminal character of the already constructed dodecaphase macro-object. The coordinate y subsequently comes from the regional reader and preserves the orientation that the determinant eliminates.

Proposition 63.11 (Canonical separation of intensity and orientation)**.** *The map*

$$(x, y) \longmapsto (D_A, \varrho_{\mathrm{or}}) = (\mathrm{e}^{-2x}, \mathrm{e}^{2y})$$

is a diffeomorphism between the domain $x > 0$, $|y| < x$, and the domain

$$0 < D_A < 1, \qquad D_A < \varrho_{\mathrm{or}} < D_A^{-1}.$$

The sheet involution acts by $(D_A, \varrho_{\mathrm{or}}) \mapsto (D_A, \varrho_{\mathrm{or}}^{-1})$.

Proof. The inverse formulas are equation (63.61). The inequality $|y| < x$ is equivalent to $|\log \varrho_{\mathrm{or}}| < -\log D_A$, from which the stated interval is obtained. Replacing y by $-y$ preserves D_A and inverts the ratio. □

63.3.2 Normalization of the bilayer state

The trace of the angular operator is

$$\operatorname{Tr} T_{x,y} = 2\mathrm{e}^{-x}\cosh y. \tag{63.63}$$

Its normalization exactly eliminates the intensity x and preserves the orientation:

$$\boxed{\rho_y := \frac{T_{x,y}}{\operatorname{Tr} T_{x,y}} = \frac{\mathrm{e}^{-yR}}{2\cosh y} = p_+(y)P_+ + p_-(y)P_-,} \tag{63.64}$$

with

$$p_+(y) = \frac{\mathrm{e}^{-y}}{2\cosh y}, \qquad p_-(y) = \frac{\mathrm{e}^{y}}{2\cosh y}, \qquad p_+ + p_- = 1. \tag{63.65}$$

The family ρ_y is faithful for every $y \in \mathbb{R}$. Its orientation observable satisfies

$$\operatorname{Tr}(\rho_y R) = -\tanh y, \qquad \rho_{-y} = R_{\mathrm{sh}}\rho_y R_{\mathrm{sh}}^{*}, \tag{63.66}$$

where R_{sh} denotes any unitary operator permuting P_+ and P_-. This implementation distinguishes the spectral involution R, which measures the leaf sign, from the unitary operator R_{sh}, which exchanges the leaves.

Definition 63.12 (Local modular holographic Hamiltonian)**.** The modular Hamiltonian of the bilayer state is

$$\boxed{K_y^{\mathrm{mod}} := -\log \rho_y = \log(2\cosh y)I + yR.} \tag{63.67}$$

The scalar component normalizes the state, and the yR component preserves oriented memory.

The adjective *holographic* records the provenance of R, $P_{\pm}$, and y from the enriched state and its two sheets. The equality equation (63.67) is an identity of functional calculus; its promotion to a generator of temporal dynamics requires specification of the corresponding physical flow.

63.3.3 Entropy, Parity, and Orientation Divergence

Theorem 63.13 (Exact informational geometry of the two sheets)**.** *The von Neumann entropy, parity, and relative entropy between the two orientations satisfy*

$$S(\rho_y) := -\operatorname{Tr}(\rho_y \log \rho_y) = \log(2\cosh y) - y\tanh y, \tag{63.68}$$

$$S(\rho_y) = S(\rho_{-y}), \tag{63.69}$$

$$\boxed{D(\rho_y\|\rho_{-y}) = 2y\tanh y.} \tag{63.70}$$

The divergence is positive for $y \neq 0$ and vanishes exactly on the unoriented sheet $y = 0$.

Proof. From equation (63.67) and $\operatorname{Tr}(\rho_y R) = -\tanh y$ follows

$$S(\rho_y) = \operatorname{Tr}(\rho_y K_y^{\mathrm{mod}}) = \log(2\cosh y) - y\tanh y.$$

Both terms are even. Moreover,

$$\log\rho_y - \log\rho_{-y} = -2yR,$$

and, consequently,

$$D(\rho_y\|\rho_{-y}) = -2y\operatorname{Tr}(\rho_y R) = 2y\tanh y.$$

The product $y\tanh y$ is positive for $y \neq 0$. □

Entropy quantifies the even distribution of the sheets, whereas divergence quantifies the informational separation of their orientations. The derivatives

$$\frac{\mathrm{d}}{\mathrm{d}y}S(\rho_y) = -y\operatorname{sech}^2 y, \qquad \frac{\mathrm{d}^2}{\mathrm{d}y^2}\log(2\cosh y) = \operatorname{sech}^2 y \tag{63.71}$$

identify $S(\rho_0) = \log 2$ as a maximum and provide the Fisher information of the exponential family. Near the symmetric sheet,

$$D(\rho_y\|\rho_{-y}) = 2y^2 - \frac{2}{3}y^4 + O(y^6). \tag{63.72}$$

The first contribution is quadratic: a small oriented memory has a second-order informational cost.

The trace distance between orientations is also obtained in closed form:

$$\frac{1}{2}\|\rho_y - \rho_{-y}\|_1 = |\tanh y|. \tag{63.73}$$

Since $|y| \geq |\tanh y|$, the identity equation (63.70) implies the control

$$D(\rho_y\|\rho_{-y}) \geq 2\left(\frac{1}{2}\|\rho_y - \rho_{-y}\|_1\right)^2, \tag{63.74}$$

consistent with Pinsker's inequality. Thus, the oriented quantity simultaneously has spectral, entropic, and metric readings.

63.3.4 Normalized resolvent of the angular macro-object

Let

$$\kappa_\beta := 4\pi \mathfrak{A}_\beta > 0. \tag{63.75}$$

We define the meromorphic matrix-valued function

$$\boxed{\mathfrak{M}_\beta(z) := \kappa_\beta^{-1}(zI - T_{x,y})^{-1}, \qquad z \in \mathbb{C} \setminus \{q_+, q_-\}.} \tag{63.76}$$

Its spectral decomposition is

$$\mathfrak{M}_\beta(z) = \frac{1}{\kappa_\beta}\left(\frac{P_+}{z - q_+} + \frac{P_-}{z - q_-}\right). \tag{63.77}$$

The resolvent unites in a single object the coupling scale, the angular poles, the sheet projectors, the determinant, and all subsequent functional calculus.

Theorem 63.14 (Injectivity by the first two moments)**.** *The assignment*

$$(\mathfrak{A}_\beta, T_{x,y}) \longmapsto \mathfrak{M}_\beta$$

is injective for $\mathfrak{A}_\beta > 0$*. specifically, if*

$$B_0 := \lim_{|z|\to\infty} z\mathfrak{M}_\beta(z), \tag{63.78}$$

$$B_1 := \lim_{|z|\to\infty} z^2\left(\mathfrak{M}_\beta(z) - z^{-1}B_0\right), \tag{63.79}$$

then

$$B_0 = \kappa_\beta^{-1} I, \qquad \kappa_\beta = \tau_2(B_0)^{-1}, \qquad T_{x,y} = \kappa_\beta B_1, \tag{63.80}$$

where $\tau_2 = \mathrm{Tr}/2$*. Therefore, the asymptotics recovers* $\mathfrak{A}_\beta = \kappa_\beta/(4\pi)$*, and the next moment recovers the complete angular operator.*

Proof. For $|z| > \|T_{x,y}\|$, the Neumann series gives

$$\mathfrak{M}_\beta(z) = \frac{1}{\kappa_\beta}\sum_{n\geq 0}\frac{T_{x,y}^n}{z^{n+1}}. \tag{63.81}$$

The coefficients of z^{-1} and z^{-2} are precisely B_0 and B_1. The equalities equation (63.80) follow upon taking the normalized trace and multiplying by κ_β. Two pairs with the same resolvent have the same coefficients and therefore coincide. □

If $q_+ \neq q_-$, the spectral data are also recovered from the poles and residues:

$$\operatorname*{Res}_{z=q_\pm} \mathfrak{M}_\beta(z) = \kappa_\beta^{-1} P_\pm, \qquad \det \mathfrak{M}_\beta(z) = \frac{\kappa_\beta^{-2}}{(z - q_+)(z - q_-)}. \tag{63.82}$$

in particular, the scalar denominator contains $q_+ + q_-$ and $q_+q_- = D_A$, and

$$P_+ = \frac{T - q_- I}{q_+ - q_-}, \qquad P_- = \frac{T - q_+ I}{q_- - q_+}. \tag{63.83}$$

The case $y = 0$ produces a unique simple pole of spectral multiplicity two: $T = \mathrm{e}^{-x} I$. The resolvent then preserves the intensity and the scalar operator, whereas the choice of projectors has no intrinsic content because the orientation is exactly zero.

Corollary 63.15 (Calculation functional from the resolvente)**.** *For every function f holomorphic in a neighborhood of $\{q_+, q_-\}$,*

$$f(T) = \frac{\kappa_\beta}{2\pi \mathrm{i}} \int_\Gamma f(z)\mathfrak{M}_\beta(z)\, \mathrm{d}z, \tag{63.84}$$

where Γ encircles both poles. In particular, the vacuum response

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}$$

is reconstructed from $\mathfrak{M}_\beta$ without incorporating additional scalar data.

Proof. The identity is the formula of Dunford–Riesz aplicada to $(zI - T)^{-1} = \kappa_\beta \mathfrak{M}_\beta(z)$. The function $f(z) = (1 - z^{90})/(1 - z^{120})$ is holomorphic over a neighborhood of the spectrum, since $0 < q_\pm < 1$. □

63.3.5 Compatibility with nonadic refinement

The preceding information is prolonged through the tower

$$\mathcal{A}_m = M_2(\mathbb{C}) \otimes M_{9^m}(\mathbb{C}), \qquad \iota_m(a) = a \otimes I_9. \tag{63.85}$$

Let

$$T_m = T \otimes I_{9^m}, \quad R_m = R \otimes I_{9^m}, \quad \mathfrak{M}_{\beta,m}(z) = \kappa_\beta^{-1}(zI - T_m)^{-1}, \tag{63.86}$$

and let $E_m = \mathrm{id}_{M_2 \otimes M_{9^m}} \otimes \tau_9$ be the tracial expectation from $\mathcal{A}_{m+1}$ to $\mathcal{A}_m$.

Theorem 63.16 (Resolvent and modular naturality)**.** *for every $m \geq 0$ and every z of the resolvent set,*

$$E_m(T_{m+1}) = T_m, \tag{63.87}$$

$$E_m\big(\mathfrak{M}_{\beta,m+1}(z)\big) = \mathfrak{M}_{\beta,m}(z), \tag{63.88}$$

$$E_m(f(T_{m+1})) = f(T_m) \tag{63.89}$$

for every continuous function f on the spectrum. Likewise, the state functionals

$$\omega_{y,m}(a) := (\mathrm{Tr}_2 \otimes \tau_{9^m})\big((\rho_y \otimes I_{9^m})a\big) \tag{63.90}$$

are compatible with the inclusions and define a state $\omega_{y,\infty}$ on the UHF limit.

Proof. All objects of equation (63.86) have the form $a \otimes I_{9^m}$. The tracial expectation eliminates the last identity factor and preserves a. The same observation proves equation (63.89); compatibility of the states is obtained by evaluating $\omega_{y,m+1}(a \otimes I_9)$ and using $\tau_9(I_9) = 1$. □

The local entropy relative to the trace $\mathrm{Tr}_2 \otimes \tau_{9^m}$ remains equal to $S(\rho_y)$, and the divergence between orientations remains equal to $2y \tanh y$. Refinement adds resolution that is nonadic and preserves the bilayer content. In this manner, the construction lives in the UHF tower and in its tracial closure, rather than depending on an accidental presentation by matrices 2×2.

63.3.6 Exactness of the modular resolvent, normalization obstructions, and separation of the thermodynamic reading

Estatuto probatorio

The decomposition of T, the state ρ_y, the modular Hamiltonian, entropy, relative divergence, injectivity of the resolvent, and UHF naturality are exact results once $(\mathfrak{A}_\beta, C^*, R)$ has been fixed. The assembly of these identities in the resolvent $\mathfrak{M}_\beta$ constitutes a new formalization of the preexisting authorial architecture. The reading of $D(\rho_y \| \rho_{-y})$ as physical work requires temperature, protocol,

and dynamics; until these are fixed, its status is that of an exact informational measure. Realizations through Barbero, CKM, or Higgs are subsequent readers and preserve their respective intertwiners.

Falsification criterion

The block is refuted if the normalization of T retains a dependence on x, if K_y^{mod} does not exponentiate to ρ_y, if the relative entropy differs from $2y\tanh y$, if two distinct pairs $(\mathfrak{A}_\beta, T)$ have the same resolvent, or if any expectation E_m alters the functional calculation. Recovery of the projectors in $y = 0$ additionally constitutes a negative control: any sheet selector that survived that degeneration would have been introduced externally.

63.4 Inversion constitutiva and curva modular cargada

63.4.1 The Spectral Completion from 3 to 4

The vacuum operator of section 63.1 is obtained by applying to the angular operator the function

$$F(q) := \frac{1-q^{90}}{1-q^{120}}, \qquad 0 < q < 1. \tag{63.91}$$

The factorization of the two channels by the elementary mode 30,

$$90 = 3 \cdot 30, \qquad 120 = 4 \cdot 30, \tag{63.92}$$

allows $s = q^{30}$ to be introduced and written as

$$F(q) = f(s), \qquad f(s) := \frac{1+s+s^2}{1+s+s^2+s^3}. \tag{63.93}$$

The numerator contains three consecutive stages of the mode, and the denominator incorporates the fourth. The completion defect is

$$1 - f(s) = \frac{s^3}{1+s+s^2+s^3} > 0. \tag{63.94}$$

Thus, the fourth stage is recorded as positive memory within the response.

Theorem 63.17 (Bijectivity of the response $3 \to 4$)**.** *The function F is a decreasing analytic diffeomorphism*

$$\boxed{F : (0,1) \xrightarrow{\sim} (3/4, 1).} \tag{63.95}$$

for each $r \in (3/4, 1)$, the value $q = F^{-1}(r)$ is obtained as $q = s^{1/30}$, where $s \in (0,1)$ is the root unique *of*

$$rs^3 + (r-1)(1+s+s^2) = 0. \tag{63.96}$$

Proof. The derivative with respect to s is

$$f'(s) = -\frac{s^2(3+2s+s^2)}{(1+s+s^2+s^3)^2} < 0 \qquad (0 < s < 1). \tag{63.97}$$

As $\mathrm{d}s/\,\mathrm{d}q = 30q^{29} > 0$, tambiin $F'(q) < 0$. The limites are

$$\lim_{q\to 0^+} F(q) = 1, \qquad \lim_{q\to 1^-} F(q) = \frac{90}{120} = \frac{3}{4}.$$

Continuity and strict monotonicity prove bijectivity; the derivative does not vanish on the interval, so the inverse function theorem proves analyticity. Multiplying $r = (1 + s + s^2)/(1 + s + s^2 + s^3)$ by the denominator produces equation (63.96); uniqueness of its root in $(0, 1)$ is equivalent to the injectivity already proved. □

The inverter equation (63.96) provides a numerical procedure that is intrinsic. A bisection on $(0, 1)$ preserves the interval and certifies each digit; a Newton method can accelerate it using equation (63.97). Decimal evaluation enters after fixing the equation and its root, which is unique.

63.4.2 Exact reconstruction of the angular state from the vacuum

Let

$$T = q_+P_+ + q_-P_-, \qquad \mathcal{V} = F(T) = r_+P_+ + r_-P_-, \qquad r_\pm = F(q_\pm). \tag{63.98}$$

The two dimensionless constitutive quantities are assigned to the sheets through

$$\widehat{\varepsilon} = r_+^2, \qquad \widehat{\mu} = r_-^2. \tag{63.99}$$

Theorem 63.18 (Reconstruction constitutive reversible)**.** *The complete constitutive response determines the angular antecedent. In the labeled spectral basis,*

$$\boxed{q_+ = F^{-1}(\sqrt{\widehat{\varepsilon}}),} \qquad \boxed{q_- = F^{-1}(\sqrt{\widehat{\mu}}),} \tag{63.100}$$

$$\boxed{D_A = q_+q_-} = F^{-1}(\sqrt{\widehat{\varepsilon}})F^{-1}(\sqrt{\widehat{\mu}}), \tag{63.101}$$

$$\boxed{A_{\mathrm{rad}} = -\frac{1}{2}\log D_A,} \qquad \boxed{C^*_{\mathrm{rad}} = \frac{1}{2}\log\frac{q_-}{q_+}.} \tag{63.102}$$

If the operator $\mathcal{V}$ is known, rather than its two labelled eigenvalues, the reconstruction takes the form of functional calculus

$$\boxed{T = F^{-1}(\mathcal{V}).} \tag{63.103}$$

Proof. The eigenvalues of $\mathcal{V}$ belong to $(3/4, 1)$, and the positive root of each constitutive relation is $r_\pm$. The theorem 63.17 provides a unique $q_\pm$ for each sheet. The formulas of $D_A, A_{\mathrm{rad}}, C^*_{\mathrm{rad}}$ are the inverses of equation (63.60). For the operator form, the spectral theorem applies the continuous function F^{-1} to the spectrum of $\mathcal{V}$. □

When only the two numbers $(\widehat{\mu}, \widehat{\varepsilon})$ are retained, the assignment to sheets is part of the typed data. The operator $\mathcal{V}$ also contains its spectral projectors and reconstructs the full representation. This distinction separates spectral reconstruction from the choice of a basis.

The remaining readings are recovered simultaneously:

$$\widehat{Z} = \frac{r_-}{r_+} = \sqrt{\frac{\widehat{\mu}}{\widehat{\varepsilon}}}, \qquad \widehat{c} = \frac{1}{r_-r_+} = \frac{1}{\sqrt{\widehat{\mu}\widehat{\varepsilon}}}. \tag{63.104}$$

The product of the sheets publishes the common propagation; the quotient publishes constitutive orientation. Under $C^* \mapsto -C^*$, the two constitutive quantities are exchanged, $\widehat{Z}$ is inverted, and $\widehat{c}$ remains fixed.

Corollary 63.19 (Electroweak cross-check)**.** *At the angular interface where $(m_W^\angle/m_Z^\angle)^2 = D_A$, the reconstruction of the vacuum implies*

$$\boxed{\sin^2\theta_W^{\mathrm{OS,HMT}} = 1 - F^{-1}(\sqrt{\widehat{\varepsilon}})F^{-1}(\sqrt{\widehat{\mu}}).} \tag{63.105}$$

Proof. The theorem 63.18 reconstructs D_A. The stated angular interface defines $\sin^2\theta_W^{\mathrm{OS}} = 1 - (m_W/m_Z)^2 = 1 - D_A$. □

The equation equation (63.105) is a consistency check between sister readers. The causal genealogy continues from T: the vacuum applies $F(T)$, and the electroweak reader applies $\det T$. The invertibility of F permits comparison of the two publications without making either one the foundation of the other.

63.4.3 The loaded modular curve

The identity

$$D_A = \exp\!\left(-\frac{100\pi}{9}\alpha_{\mathrm{HMT}}\right) \tag{63.106}$$

suggests using the determinant as a global coordinate of the family. For $D \in (0,1)$, we define

$$\boxed{Q(D) := -\frac{9}{25}\log D,} \qquad a(D) := \frac{Q(D)}{4\pi} = -\frac{9}{100\pi}\log D. \tag{63.107}$$

At the HMT point, $a(D_A) = \alpha_{\mathrm{HMT}}$ and

$$\boxed{Q(D_A) = 4\pi\alpha_{\mathrm{HMT}} =: e_g^2.} \tag{63.108}$$

The symbol $Q(D)$ here designates the dimensionless charged coupling; it is distinct from every dimensional section of charge or energy.

To make the regional reader self-contained, we introduce the quintic character

$$H_5(a) := \frac{1}{2}\sqrt{\frac{1000a}{\varphi_{\mathrm{HMT}}}} - a + \frac{9}{16}a^2 - \frac{5}{9}a^3 + \frac{7}{48}a^4 - \frac{1}{54}a^5, \tag{63.109}$$

the regional germ

$$C_\pi(a,D) := 169a^6 + \frac{Da^7}{1-Da}, \tag{63.110}$$

and the oriented phase

$$y(D) := \frac{\pi}{90}\big(H_5(a(D)) + a(D)\big) - C_\pi(a(D),D). \tag{63.111}$$

The selector 169, the finite term of degree six, and the geometric prolongation of degree seven retain their regional provenance. The condition $0 < Da(D) < 1$ guarantees convergence of the germ.

We define the oriented quotient.

$$\varrho_{\mathrm{or}}(D) := \mathrm{e}^{2y(D)} \tag{63.112}$$

and the admissible interval

$$\mathcal{I}_{\mathrm{mod}} := \left\{ D \in (0,1) : 0 < Da(D) < 1,\ |y(D)| < -\frac{1}{2}\log D \right\}. \tag{63.113}$$

The second inequality ensures that the two reconstructed channels are strict contractions.

Definition 63.20 (Modular curve loaded)**.** The charged modular curve is the map

$$\Gamma_{\mathrm{ch}} : \mathcal{I}_{\mathrm{mod}} \longrightarrow (0,1)\times\mathbb{R}_{>0}, \qquad \boxed{\Gamma_{\mathrm{ch}}(D) = \big(D, \varrho_{\mathrm{or}}(D)\big).} \tag{63.114}$$

On it,

$$q_-(D) = \sqrt{D\,\varrho_{\mathrm{or}}(D)}, \qquad q_+(D) = \sqrt{\frac{D}{\varrho_{\mathrm{or}}(D)}}. \tag{63.115}$$

Theorem 63.21 (Functional rank one)**.** *The curve Γ_{ch} is an analytic immersion of rank one. The generic space of positive bilayer contractions requires two independent coordinates $(D, \varrho_{\mathrm{or}})$; the HMT reader restricts that space to the graph $\varrho_{\mathrm{or}} = \varrho_{\mathrm{or}}(D)$. Moreover,*

$$A_{\mathrm{rad}}(D) = -\frac{1}{2}\log D = \frac{50\pi}{9}a(D), \qquad C^*_{\mathrm{rad}}(D) = y(D). \tag{63.116}$$

Proof. The functions of equations (63.107) y (63.109) a (63.111) are analytic on $\mathcal{I}_{\mathrm{mod}}$. The first component of $\Gamma'_{\mathrm{ch}}(D)$ is one, so the derivative never vanishes and its rank is one. The identities equation (63.116) follow directly from the definitions. Finally, the product and quotient of equation (63.115) are D and $\varrho_{\mathrm{or}}(D)$, respectively. □

The resolvent family of the chapter previous is restricted to the same curve:

$$\mathfrak{M}_D(z) := Q(D)^{-1}\big(zI - T(D)\big)^{-1}, \qquad T(D) = q_+(D)P_+ + q_-(D)P_-. \tag{63.117}$$

At the point D_A, the equality $Q(D_A) = 4\pi\mathfrak{A}_\beta$ identifies $\mathfrak{M}_D$ with $\mathfrak{M}_\beta$. The coupling, intensity, orientation, and vacuum are thus coordinated by a single genealogical variable, while their reader functions remain typed.

63.4.4 Confluence between vacuum, Higgs and impedance

The vacuum reader publishes along the curve

$$\widehat{\mu}(D) = F(q_-(D))^2, \qquad \widehat{\varepsilon}(D) = F(q_+(D))^2, \tag{63.118}$$

$$\widehat{Z}(D) = \frac{F(q_-(D))}{F(q_+(D))}, \qquad \widehat{c}(D) = \frac{1}{F(q_-(D))F(q_+(D))}. \tag{63.119}$$

By the injectivity of F, either complete operator pair reconstructs $(D, \varrho_{\mathrm{or}})$.

The Higgs angular route contains the orientation through

$$\frac{m_H^{\angle}}{m_W^{\angle}} = \exp\!\left(3A_{\mathrm{rad}} + \frac{5}{3}C^*_{\mathrm{rad}}\right). \tag{63.120}$$

In the tree interface with $m_H^2 = 2\lambda_H v^2$ and $m_W^2 = g^2v^2/4$, this identity is equivalent to

$$\lambda_H = \frac{Q(D)}{8(1-D)}D^{-3}\varrho_{\mathrm{or}}(D)^{5/3}. \tag{63.121}$$

Therefore, the scalar

$$\Xi_H(D, \lambda_H) := \frac{8(1-D)D^3\lambda_H}{Q(D)} \tag{63.122}$$

satisfies

$$\boxed{\Xi_H = \varrho_{\mathrm{or}}^{5/3}, \qquad \varrho_{\mathrm{or}} = \Xi_H^{3/5}.} \tag{63.123}$$

Corollary 63.22 (Higgs–impedance reconstruction)**.** *Under the declared tree-level interface and for $\Xi_H > 0$,*

$$\boxed{\widehat{Z} = \frac{F\left(\sqrt{D}\,\Xi_H^{3/10}\right)}{F\left(\sqrt{D}\,\Xi_H^{-3/10}\right)}.} \tag{63.124}$$

The same quotient orientado is reconstructed by three vias:

$$\varrho_{\mathrm{or}}^{\mathrm{HMT}}(D) = \frac{F^{-1}(\sqrt{\widehat{\mu}})}{F^{-1}(\sqrt{\widehat{\varepsilon}})} = \Xi_H(D, \lambda_H)^{3/5}. \tag{63.125}$$

Proof. The equation equation (63.123) gives $q_- = \sqrt{D}\,\Xi_H^{3/10}$ and $q_+ = \sqrt{D}\,\Xi_H^{-3/10}$. Substituting them in $\widehat{Z} = F(q_-)/F(q_+)$ proves equation (63.124). The second equality of equation (63.125) is equation (63.123); the first proceeds of the theorem 63.18. □

The substitution of λ_H obtained from equation (63.121) turns equation (63.125) into an algebraic consistency identity. A determination of λ_H from an independent reader turns the same equation into a falsifiable cross-check of the orientation. This separation prevents the attribution of probative independence to a mere inversion of the starting equation.

The gauge partition also assumes a form determined purely by D:

$$g^2 = \frac{Q(D)}{1-D}, \qquad g'^2 = \frac{Q(D)}{D}, \qquad \boxed{\frac{1}{g^2} + \frac{1}{g'^2} = \frac{1}{Q(D)} = -\frac{25}{9\log D}.} \tag{63.126}$$

The total coupling $Q(D)$ is apportioned between the two gauge sectors, and the harmonic sum exactly recovers the charged macro-object.

63.4.5 Compatible causal section matching with orientation and boundary data

The coupling $Q(D)$ should be distinguished from the Planck energy, which we shall denote by $Q_P = E_P$. The Planck pair defines an internal velocity

$$c_{\mathrm{int}} := \frac{Q_P \ell_P}{\hbar}, \tag{63.127}$$

while the constitutive operator defines

$$c_{\mathrm{vac}} := \frac{1}{\sqrt{\mu_{\mathrm{vac}}\varepsilon_{\mathrm{vac}}}}. \tag{63.128}$$

Both expressions live in typed causal curves. A sewing morphism

$$J_C : L_C^{\mathrm{vac}} \longrightarrow L_C^{\mathrm{grav}} \tag{63.129}$$

allows their comparison, and their dimensionless defect is

$$\boxed{\Xi_c := \frac{Q_P \ell_P}{\hbar}\sqrt{\mu_{\mathrm{vac}}\varepsilon_{\mathrm{vac}}} = \frac{c_{\mathrm{int}}}{c_{\mathrm{vac}}}.} \tag{63.130}$$

Comparison Criterion 63.23 (Causal seam)**.** The electromagnetic vacuum and the gravitational realization publish the same causal section exactly when

$$\boxed{\Xi_c = 1,} \qquad J_C(c_{\mathrm{vac}}) = c_{\mathrm{int}}. \tag{63.131}$$

Once this seam is satisfied, the CT108 identity $G = (54/\pi)^2 c^5 t_0^2/\hbar$ is expressed entirely through the constitutive product:

$$\boxed{G = \left(\frac{54}{\pi}\right)^2 \frac{t_0^2}{\hbar(\mu_{\mathrm{vac}}\varepsilon_{\mathrm{vac}})^{5/2}}.} \tag{63.132}$$

Gravity uses the even product of the leaves; impedance and charge use the oriented quotient

$$Z_{\mathrm{vac}}^2 = \frac{\mu_{\mathrm{vac}}}{\varepsilon_{\mathrm{vac}}}. \tag{63.133}$$

The pair product–quotient distributes, therefore, causality and orientation without disaggregating the spectral antecedent.

63.4.6 Stable reconstruction in the UHF tower

Let $\tau_m = \tau_2 \otimes \tau_{9^m}$ be the normalized trace of $\mathcal{A}_m = M_2 \otimes M_{9^m}$, and let

$$T_m = T \otimes I_{9^m}, \qquad \mathcal{V}_m = F(T_m) = \mathcal{V} \otimes I_{9^m}. \tag{63.134}$$

Functional inversion commutes with refinement:

$$F^{-1}(\mathcal{V}_m) = T_m, \qquad E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m, \qquad E_m(T_{m+1}) = T_m. \tag{63.135}$$

The ordinary matrix determinant of T_m is $D_A^{9^m}$. The invariant compatible with refinement is the normalized logarithmic determinant

$$\boxed{\mathfrak{d}_m(T_m) := \exp(2\tau_m(\log T_m)) = D_A.} \tag{63.136}$$

The orientation admits the equally stable prolongation

$$\boxed{\mathfrak{r}_m(T_m, R_m) := \exp(-2\tau_m(R_m \log T_m)) = \varrho_{\mathrm{or}},} \qquad R_m = R \otimes I_{9^m}. \tag{63.137}$$

Proposition 63.24 (Stability of reconstruction at every depth)**.** *For every m, the pair*

$$\Big(\mathfrak{d}_m(F^{-1}(\mathcal{V}_m)), \mathfrak{r}_m(F^{-1}(\mathcal{V}_m), R_m)\Big) \tag{63.138}$$

equals $(D_A, \varrho_{\mathrm{or}})$*. Consequently, it reconstructs the same coordinates* A_{rad} *and* C_{rad}^* *at every level and in the UHF limit.*

Proof. Since $\log T_m = (\log T) \otimes I_{9^m}$, the normalized trace eliminates the amplification factor. Moreover,

$$\tau_2(\log T) = \frac{1}{2}(\log q_+ + \log q_-) = \frac{1}{2}\log D_A$$

and

$$\tau_2(R \log T) = -y = -\frac{1}{2}\log \varrho_{\mathrm{or}}.$$

The equations equations (63.136) y (63.137) follow upon exponentiation. Functional inversion of equation (63.135) completes the reconstruction. □

This formulation shows that the completion $3 \to 4$ preserves its reconstructive capacity at every depth. The nonadic cylinders increase the resolution of the state; the normalized determinant, orientation, response, and modular curve remain natural with respect to the expectations.

63.4.7 Separation between internal quantum identities, physical interfaces, and reconstruction obstructions

Estatuto probatorio

The monotonicity and bijectivity of F, the operatorial inversion, the reconstruction (q_+, q_-, D_A, C^*), the rank one of Γ_{ch}, the harmonic gauge identity, and UHF stability are exact results. The charged modular curve formalizes a pre-existing authorial architecture with explicit starting structure: closure of α_{HMT}, regional selector 169, character H_5, germ C_π, and orientation. The Higgs–impedance

equality is exact within the tree interface; it constitutes an independent proof only when λ_H follows from a distinct reader. The condition $\Xi_c = 1$ and the subsequent gravitational equation are a typed physical interface whose promotion requires the morphism J_C.

Falsification criterion

The reconstruction is refuted if $F'(q)$ changes sign in $(0, 1)$, if the cubic equation admits two roots in that interval, if $F^{-1}(\mathcal{V})$ does not recover T, or if the constitutive relations reconstruct a product different from D_A. The curve is refuted if $Q(D_A) \neq 4\pi\alpha_{\mathrm{HMT}}$, if its derivative loses rank, if $q_{\pm}(D)$ leaves $(0, 1)$ within the declared domain, or if the three reconstructions of ϱ_{or} in equation (63.125) disagree under the same normalization. The causal seam is refuted by $\Xi_c \neq 1$ after applying the line morphism. In the tower, using the ordinary determinant as though it were invariant would produce $D_A^{9^m}$; the correct control requires equation (63.136).

63.5 Constitutive evaluation and dimensional realization

Evaluation now brings together the preceding constructions in their effective order. The regional expansions of π, φ, and e arise from nonadic refinement; the register K preserves its twelve positions. Its periodic readout determines the positional section of α. The action character and regional return act on it, followed by the angular pair and, finally, the constitutive response. This evaluation follows the selection of rules: none of its digits is used to choose a channel.

63.5.1 Evaluation of the internal chain

To reproduce the table, one uses the thousand-decimal regional expansions generated by the nonadic certificate and the integer N_K in (11.114). The readout used is

$$\alpha = \pi + e - \varphi - 4 - \frac{N_K}{1000^{12} - 1}.$$

The finite twelve-block section and the prolonged section have their definitions and exact difference in the register chapter. Here the latter is held fixed throughout evaluation. Readouts are not interchanged as precision increases, nor is the analytic operator of the second route replaced by a decimal coincidence with the first.

Objeto	Decimal evaluation (24 significant digits)
α	0.007297352569283800997285 11 . . .
H_5	1.054571817646154469625 82 . . .
η_{ret}	1.054571816915039061133 69 . . .
A_{deg}	7.297352569283800997285 11 . . .
C^*_{deg}	2.123738338968645724261 95 . . .
q_+	0.848377942576404374945737 . . .
q_-	0.913660151150557285295519 . . .
r_+	0.999999628548249363178695 . . .
r_-	0.999724137189518364131517 . . .
$\widehat{\varepsilon}$	0.999999257096636702760442 . . .
$\widehat{\mu}$	0.999448350479326935089982 . . .
$\widehat{Z}$	0.999724508538937215455554 . . .
$\widehat{c}$	1.000276310486157526106 39 . . .
γ	0.274007672694459274747255 . . .
G_{Cat}	0.915965594177219015054604 . . .

Objeto	Decimal evaluation (24 significant digits)
ρ	0.267949192431122706472554 . . .

The calculation uses 120 working digits; 24 significant digits are displayed to facilitate comparison between quantities of different orders. The 120 digits are an evaluation precision, not an experimental uncertainty or, by themselves, an interval error certificate. The package preserves the generated inputs, their hashes, the operations, and the residuals of the checked identities. The algebraic proofs in the main text establish these identities independently of this table.

The meaning of the final four constitutive coordinates is preserved under evaluation: $\widehat{\varepsilon}$ and $\widehat{\mu}$ are the quadratic responses per sheet, $\widehat{Z}$ is their relative orientation, and $\widehat{c}$ is the inverse of their product. The proximity of some digits to unity expresses the small defect of the 3–4 quotient on these channels. Its origin is seen exactly in

$$1 - f(s) = \frac{s^3}{1 + s + s^2 + s^3}.$$

The formula allows the response to be studied before a dimensional quantity is assigned to it: the sign of the defect, its cubic order, and its monotonicity are consequences of the same operator.

63.5.2 Charts of units and covariance

Let $\mathcal{L}_S$, $\mathcal{L}_Q$ and $\mathcal{L}_C$ be the action, charge and velocity lines. A positive physical chart evaluates the sections u_S, u_Q and u_C in units of action, charge and velocity. The realization (39.50) then produces an impedance, a velocity, a permeability and a permittivity, with their respective dimensions. The coordinates in the preceding table remain dimensionless.

Proposition 63.25 (Covariance of the constitutive realization)**.** *If the reference sections change by $u'_S = au_S$, $u'_Q = bu_Q$, $u'_C = du_C$, with $a, b, d > 0$, the coordinates representing the same quantities satisfy*

$$\widehat{Z}' = a^{-1}b^2\widehat{Z}, \qquad \widehat{c}' = d^{-1}\widehat{c}, \qquad \widehat{\mu}' = a^{-1}db^2\widehat{\mu}, \qquad \widehat{\varepsilon}' = adb^{-2}\widehat{\varepsilon}.$$

The relations $\widehat{\mu}'\widehat{\varepsilon}' = (\widehat{c}')^{-2}$ and $\widehat{\mu}'/\widehat{\varepsilon}' = (\widehat{Z}')^2$ are preserved.

Proof. Equate each quantity in (39.50) to its expression in the new basis and solve for its coordinate. The factors in the product are d^2, and those in the quotient are $a^{-2}b^4$, exactly those required by the transformed coordinates of c and Z. □

This covariance distinguishes two acts: constructing a section and expressing that section in chosen units. The spectral theorem fixes the coordinates in the declared internal normalization. An SI comparison must also display the evaluation of the three reference sections. Adjusting them through the quantities one wishes to reproduce would be a calibration of the chart, not an independent verification of those quantities.

To reconstruct the channels from coordinates expressed in the primed chart, that change is undone before using the cube root:

$$\widehat{Z} = ab^{-2}\widehat{Z}', \quad \widehat{c} = d\widehat{c}', \quad \widehat{\mu} = ad^{-1}b^{-2}\widehat{\mu}', \quad \widehat{\varepsilon} = a^{-1}d^{-1}b^2\widehat{\varepsilon}'. \tag{63.139}$$

For example, the inputs to the spectral inverse are $r_+ = (\widehat{Z}\widehat{c})^{-1/2}$ and $r_- = (\widehat{Z}/\widehat{c})^{1/2}$, calculated after (63.139). Directly inserting the primed coordinates into those expressions would change the normalization of the operator and could move them outside the interval $(3/4, 1)$. Dimensional covariance preserves the physical quantities; cubic inversion reconstructs transport in the internal chart in which it was defined.

63.5.3 Identity checks and scope of comparison

Reproducible comparison is organized by relations. First, $\widehat{\mu}\widehat{\varepsilon}\widehat{c}^2 = 1$ and $\widehat{\mu}/(\widehat{\varepsilon}\widehat{Z}^2) = 1$ check constitutive consistency. Second, cubic inversion recovers the original arguments and angular coordinates. Finally, evaluating Barbero through the angular functional and through functional calculus of the response agrees by the identity proved in the previous chapter. Interchanging sheets must reverse the oriented contribution and preserve the common mode; this check detects a loss of labeling that scalar products would not reveal.

The comparison of the reduced Planck constant with reference tables is preserved in the antecedent action development. For the four vacuum quantities, the present table is an evaluation of their internal coordinates. It is not presented as a table of SI coincidences: that additional assertion requires an independent numerical chart for u_S, u_Q, u_C. The proved constitutive result, including its inversion, remains complete in its domain and explicitly preserves the dimensional interface needed to make that comparison.

63.6 Exhaustive enumeration and finite determination of transport operators

This appendix develops two finite operations with different starting data. The first enumerates the initial conditions of the two cursors and constructs their emissions through the declared arithmetic recurrence. The second determines transport operators from the regional transition register. Separating their domains specifies what each proof produces: enumeration constructs the emission catalogue; matrix reconstruction determines the operators compatible with the register. The reconstruction theorem is not used as a proof of the prior production of that register.

The primitives of the first operation are the support I_9^2, the four cardinal orientations, the tritic selector, the pre-displacement readout, orientation reversals and the emission rule. Their definitions are brought together below. Those of the second operation are the four transition matrices in (63.146). Their unique reconstruction is proved in Theorem 63.27; the resulting calendar is given in (63.148). These determinations preserve the starting data specified for each operation.

63.6.1 Complete domain of initial conditions

Let $I_9 = \{0, \dots, 8\}$ and let

$$\mathcal{D} = \{N, E, S, O\}, \qquad v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

An initial cursor condition is $s = (i, j, d) \in \mathcal{S} = I_9^2 \times \mathcal{D}$. The two cursors are independent in the initial domain

$$\Omega = \mathcal{S}_+ \times \mathcal{S}_\times, \qquad |\mathcal{S}| = 9^2 \cdot 4 = 324, \qquad |\Omega| = 324^2 = 104\,976. \tag{63.140}$$

Index i corresponds to the positive APP label $i + 1$. For positive integers, positive reduction is $\rho_9(n) = 1 + [(n - 1)]_9$, where $[a]_b$ denotes the representative in $\{0, \dots, b - 1\}$. The residual evaluations of the two sheets are

$$\sigma(i, j) = \rho_9(i + j + 2), \qquad \varpi(i, j) = \rho_9((i + 1)(j + 1)).$$

The finite multiplicative readout uses

a	1	2	3	4	5	6	7	8	9
$\ell_9(a)$	0	1	0	2	5	0	4	3	0

. (63.141)

On units, ℓ_9 is the discrete exponent in base 2. On $3, 6, 9$, the value 0 constitutes the declared extension of the emission observable; membership in that sector remains in the trajectory register.

For $1 \le t \le 54$, define

$$r_t = 1 + [(t-1)]_9, \qquad \tau_t = \begin{cases} +1, & r_t \in \{1,4,7\}, \\ -1, & r_t \in \{2,5,8\}, \\ 0, & r_t \in \{3,6,9\}. \end{cases}$$

Each cursor preserves its position z_t immediately before step t and its orientation vector v_t. The additive cursor is active when $\tau_t = +1$; the multiplicative cursor, when $\tau_t = -1$. For each of them, with activity indicator a_t, the update is

$$z_{t+1} = [z_t + a_t v_t]_9, \qquad v_{t+1} = \begin{cases} -v_t, & t \in \{27,54\}, \\ v_t, & t \notin \{27,54\}. \end{cases} \tag{63.142}$$

The readout at step t is taken at z_t, before displacement. The neutral event preserves both positions. This convention fixes unambiguously the window endpoints and the effect of cardinal reversals.

In the windows $W_m = \{9m-8, \dots, 9m\}$, $1 \le m \le 6$, let

$$S_m(s_+) = \sum_{\substack{t \in W_m \\ \tau_t = +1}} \sigma(z_t^+), \qquad E_m(s_\times) = \sum_{\substack{t \in W_m \\ \tau_t = -1}} \ell_9(\varpi(z_t^\times)), \tag{63.143}$$

$$s_m = [S_m]_{10}, \qquad e_m = [E_m]_{10}.$$

Each window contains three events of each type. The signatures s and e are the six components of these residues, not the integers S_m, E_m before reduction.

63.6.2 Census of signatures and emissions

The emission of a pair of initial conditions is determined by

$$U_m = 100 s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}, \qquad 1 \le m \le 6. \tag{63.144}$$

The final term 1 is $[7 \cdot 3]_{10}$, corresponding to the three neutral events. This rule's coefficients belong to the declared recurrence. The real constants studied subsequently do not enter enumeration of its domain or its evaluation.

Proposition 63.26 (Injective factorization of the finite catalogue)**.** *Emission distinguishes every signature pair* (s, e)*. There are* 18 *additive and* 26 *multiplicative signatures; their pairs produce* 468 *emissions. The multiplicities of these emissions are* 144*,* 432 *and* 1944*, with* 432*,* 18 *and* 18 *emissions, respectively.*

Proof. The hundreds digit of U_m recovers s_m. If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit verifies the third congruence of (63.144). Thus, the coupling is injective.

The census is obtained by traversing, in the full Cartesian order, $i = 0, \dots, 8$, $j = 0, \dots, 8$ and $d = N, E, S, O$, and applying the 54 steps of (63.142). The 324 additive conditions are distributed over the following 18 signatures, each with 18 preimages. In the tables, a sequence of six digits represents the six ordered components of a signature.

122564	122898	212645
212988	221456	221889
456221	465898	546988
564122	645212	654889
889221	889654	898122
898465	988212	988546

For the multiplicative cursor, the signatures and their multiplicities are

Signature	Multiplicity	Signature	Multiplicity
000000	108	555555	24
177393	8	555717	8
177555	8	555771	8
177771	8	555933	8
339555	8	717555	8
339771	8	717717	8
339933	8	717933	8
393177	8	771177	8
393393	8	771339	8
393555	8	771555	8
555177	8	933339	8
555339	8	933555	8
555393	8	933717	8

These tables result from the finite sums (63.143), with domain, readout order and reversals already specified. Their sums of multiplicities are $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. The Cartesian domain (63.140) permits every signature pair to be realized. Injectivity gives $18 \cdot 26 = 468$ emissions and the multiplicities

$$(18 \cdot 24,\ 18 \cdot 8) = (432, 144), \qquad (18,\ 18 \cdot 24) = (18, 432), \qquad (18,\ 18 \cdot 108) = (18, 1944),$$

where each pair indicates the number of emissions and multiplicity of their fibres. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

Enumeration thus exhausts the complete domain of initial conditions. □

The oriented ternary reduction is

$$\Psi(U) = ([-U_1]_3, \dots, [-U_6]_3).$$

Applied to the $18 \cdot 26$ emissions of the preceding tables, it produces the following fibre census. Here r counts distinct emissions, not initial conditions.

r	1	2	3	4	5	6
$\#\{w : \#\Psi^{-1}(w) = r\}$	132	66	18	3	6	18

(63.145)

The sum of the second row is 243, while its weighted sum with weights r is 468. The image of 243 words is therefore distinguished from the ambient space $\mathbb{F}_3^6$, containing 729 words. The accompanying certificate preserves all signatures with their preimages, the 468 emissions and their ternary reduction; the preceding formulas provide a complete procedure for reconstructing them.

63.6.3 Transition register and reconstruction of the lifts

This subsection fixes the regional register of the cited source. Let $b_0^{\chi}, \dots, b_4^{\chi} \in \mathbb{F}_3^6$, with $\chi \in \{\mathsf{C}, \mathsf{P}, \mathsf{A}\}$, be its three sequences. The register organizes transitions $b_0 \to b_1 \to b_2$ in X_0, Y_0 and transitions $b_2 \to b_3 \to b_4$ in X_1, Y_1, using row vectors and the same regional order:

$$X_0=\begin{pmatrix}0&1&0&2&1&1\\0&1&2&2&2&2\\2&0&1&1&0&1\\1&2&1&2&2&1\\1&2&1&2&0&0\\1&1&2&2&0&2\end{pmatrix},\quad Y_0=\begin{pmatrix}0&1&2&2&2&2\\0&1&0&2&1&1\\1&2&1&2&2&1\\1&0&2&0&1&1\\1&1&2&2&0&2\\1&2&1&0&2&0\end{pmatrix},$$
$$X_1=\begin{pmatrix}0&1&0&2&1&1\\0&0&2&1&1&1\\1&0&2&0&1&1\\0&1&2&2&2&2\\1&2&1&0&2&0\\0&1&0&2&1&0\end{pmatrix},\quad Y_1=\begin{pmatrix}0&0&2&1&1&1\\1&1&0&2&2&1\\0&1&2&2&2&2\\1&0&2&0&1&1\\0&1&0&2&1&0\\0&1&0&2&0&0\end{pmatrix}. \tag{63.146}$$

The preceding production is expressed in its source through the composition of selection, transport and update:

$$x^{\mathcal{X}}_{j+1}=\mathcal{U}_{9j+9}\circ\cdots\circ\mathcal{U}_{9j+1}(x^{\mathcal{X}}_j),\qquad b^{\mathcal{X}}_j=\Pi_6(x^{\mathcal{X}}_j),\qquad \mathcal{U}_t=\mathsf{Upd}_t\circ\mathsf{Tra}_t\circ\mathsf{Sel}_t.$$

The following result starts from register (63.146). Its proof reconstructs the operators and verifies transition compatibility; it does not attribute coordinatewise production of states $x^{\mathcal{X}}_j$ to matrix inversion.

Proposition 63.27 (Uniqueness on the regional register)**.** *The systems $X_aL_a=Y_a$, $a=0,1$, have a unique solution over $\mathbb{F}_3$. The two solutions are*

$$L_0=\begin{pmatrix}2&2&2&1&2&1\\2&1&2&2&1&1\\1&0&0&1&0&2\\0&0&1&1&1&0\\2&2&2&0&2&1\\2&1&2&1&0&0\end{pmatrix},\qquad L_1=\begin{pmatrix}2&1&2&0&2&2\\1&2&1&1&2&0\\1&2&1&1&1&2\\0&1&0&0&2&1\\2&0&2&1&0&1\\0&2&2&2&1&1\end{pmatrix}. \tag{63.147}$$

Proof. Elimination in $\mathbb{F}_3$ gives $\det X_0=1$ and $\det X_1=2=-1$. Each map $L\mapsto X_aL$ is therefore an automorphism of the 36-dimensional matrix space; under columnwise vectorization it is represented by $I_6\otimes X_a$. One obtains $L_a=X_a^{-1}Y_a$. Multiplication of the matrices in (63.147) by the respective matrices X_a reproduces the two matrices Y_a of (63.146); existence and uniqueness are established on that register. □

The three resulting sequences, explicitly separated into blocks, are

$$B^{\mathsf{C}}=010211\mid 012222\mid 010211\mid 002111\mid 110221,$$
$$B^{\mathsf{P}}=201101\mid 121221\mid 102011\mid 012222\mid 102011,$$
$$B^{\mathsf{A}}=121200\mid 112202\mid 121020\mid 010210\mid 010200.$$

Each block preserves its six positions. Concatenation is not identified with a base-ten integer.

63.6.4 Exhaustiveness of the calendar and stability of reconstruction

A binary calendar is $c=(c_0,c_1,c_2,c_3)\in\{0,1\}^4$. Define the number of transitions incompatible with the register by

$$d(c)=\sum_{\mathcal{X}\in\{\mathsf{C},\mathsf{P},\mathsf{A}\}}\sum_{j=0}^{3}\mathbf{1}\{b^{\mathcal{X}}_jL_{c_j}\neq b^{\mathcal{X}}_{j+1}\}. \tag{63.148}$$

Multiplying the preceding sequences by (63.147) gives

$$d(c)=3\,d_{\mathrm{H}}(c,(0,0,1,1)),$$

where d_{H} is Hamming distance: at each of the four positions, the alternative operator fails to satisfy all three regional transitions. The complete census is

c	$d(c)$	c	$d(c)$	c	$d(c)$	c	$d(c)$
0000	6	0100	9	1000	9	1100	12
0001	3	0101	6	1001	6	1101	9
0010	3	0110	6	1010	6	1110	9
0011	0	0111	3	1011	3	1111	6

Consequently, 0011 is the unique calendar reproducing the three complete sequences.

Sensitivity to matrix entries likewise admits an exhaustive proof without approximations. A single-entry modification replaces L_a by $L_a + \lambda E_{ij}$, with $a \in \{0,1\}$, $1 \le i,j \le 6$ and $\lambda \in \mathbb{F}_3 \setminus \{0\}$. There are $2 \cdot 6 \cdot 6 \cdot 2 = 144$ such modifications. Since X_a is invertible,

$$X_a(L_a + \lambda E_{ij}) - Y_a = \lambda X_a E_{ij} \neq 0.$$

Every modification destroys at least one transition of the corresponding register. The executable certificate also enumerates the 144 modifications and preserves, for each, the number of violated transitions.

The proved property is rigidity of the operators with respect to the fixed regional register. Reproducing a prior construction of the register itself requires the coordinatewise action of its selection, transport and update operators. That dependency is kept separate from the rigidity theorem, the finite emission catalogue and the subsequent analytic realizations.

63.6.5 Reproduction certificate and location of the data

The program `verificar_nucleo_finito.py` traverses the 324 conditions of each cursor and forms the product of the resulting classes. The documentary catalogue is loaded after this construction to compare emissions and their multiplicities. In the second part, the four transition matrices are extracted from the verbatim copy of the source; solutions are computed by exact elimination modulo 3 before comparison with the printed matrices. The receipt `NUCLEO_FINITO.json` preserves the domains, preimages, emissions, 16 calendars and 144 modifications.

Verification uses exact integers and residues. It receives no metrological values or real expansions of the constants. Its scope is that of the finite operations and starting data specified in this appendix. The proofs of compatibility at unbounded depth and physical identifications retain their respective domains and expository obligations.

63.7 Restricted Electromagnetic Realization

63.7.1 Realization of the constitutive vacuum operator in the fields (E_σ, B_σ)

After constructing the continuum and deriving the constitutive operator of the vacuum, the propagation speed is published as

$$c_{\mathrm{vac}} = (\mu_{\mathrm{vac}}\varepsilon_{\mathrm{vac}})^{-1/2}.$$

Let

$$\zeta = \frac{z - c_{\mathrm{vac}}t}{\lambda}$$

and $\sigma = \pm 1$. Definimos

$$\mathbf{E}_\sigma(z,t) = E_0\big(\mathbf{e}_1 \cos 2\pi\zeta + \sigma \mathbf{e}_2 \sin 2\pi\zeta\big), \tag{63.149}$$

$$\mathbf{B}_\sigma(z,t) = \frac{1}{c_{\mathrm{vac}}}\widehat{\mathbf{k}} \times \mathbf{E}_\sigma(z,t). \tag{63.150}$$

Then

$$\mathbf{E}_\sigma \perp \mathbf{B}_\sigma \perp \widehat{\mathbf{k}}, \qquad |\mathbf{E}_\sigma| = c_{\mathrm{vac}}|\mathbf{B}_\sigma|,$$
$$\omega = c_{\mathrm{vac}}k, \qquad \mathbf{S} = \frac{1}{\mu_{\mathrm{vac}}}\mathbf{E}_\sigma \times \mathbf{B}_\sigma = \frac{E_0^2}{Z_{\mathrm{vac}}}\widehat{\mathbf{k}}.$$

63.7.2 Helical realization of the electromagnetic phase: periodic return and longitudinal translation by one wavelength

At fixed time, the normalized end of the electric vector satisfies

$$\left(\frac{E_1}{E_0}, \frac{\sigma E_2}{E_0}, \frac{z - c_{\mathrm{vac}}t}{\lambda}\right) = \big(\cos 2\pi\zeta, \sin 2\pi\zeta, \zeta\big). \tag{63.151}$$

The equation coincides with the normalized phase-and-memory helix. The step $\zeta \mapsto \zeta + 1$ returns the electromagnetic phase and translates the marked event by one wavelength.

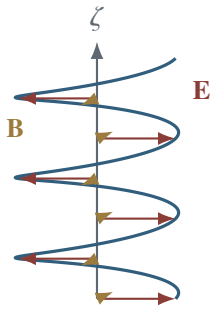


Figure 63.1: Analytic realization of the phase-position helix in a circular mode.

The exact statement concerns the trajectory of the field endpoint in the phase–position bundle. It does not follow that every spatial field line is a helix. Nor is electromagnetic helicity identified with the electron's fermionic holonomy.

63.7.3 Electromagnetic Duality

On two complexified forms in lorentzian signature,

$$\star^2 = -I.$$

The relation

$$\mathfrak{R}_{\mathrm{EM}} \circ J_{+1} = \star \circ \mathfrak{R}_{\mathrm{EM}}$$

realizes the internal quarter-turn as electric–magnetic duality when the intertwiner $\mathfrak{R}_{\mathrm{EM}}$ is constructed. The eigenspaces

$$\star F_\sigma = \mathrm{i}\sigma F_\sigma$$

separate the two helicities. This is a typed physical interface: it uses the already constructed elliptic generator, but requires the space of forms and the metric of the realization.

63.7.4 Compatibility condition with Pell's screw

For a phase

$$\Phi(r, \varphi, z, t) = \ell\varphi + \nu \log(r/r_0) + k_z z - \omega t,$$

invariance under $(r, \varphi) \mapsto (\varepsilon r, \varphi + \pi/2)$ requires

$$\nu \log(2 + \sqrt{3}) + \ell\frac{\pi}{2} \in 2\pi\mathbb{Z}. \tag{63.152}$$

This condition quantifies phase compatibility. The vector amplitude must additionally satisfy Helmholtz, transversality, and boundary conditions.

Chapter 64

Reconstruction of quantum principles from the genealogical state: Hilbert space, superposition, observables, dynamics, Born rule, uncertainty, and entanglement

64.1 Reconstruction of quantum principles from the genealogical state

Hilbert's formulation organizes quantum mechanics through states, self-adjoint observables, unitary evolution, tensorial composition, and a probabilistic rule for outcomes [129]. HMT–MD constructs a layer prior to that representation: a compatible section, its observable route, transport memory, and the reader that publishes an outcome. Hilbert space is not replaced; it appears as the codomain of a declared linear realization.

The typed sequence is

$$\boxed{\begin{aligned}&\text{genealogical section} \longrightarrow \text{linear realization} \longrightarrow \text{phase and superposition}\\ &\qquad\qquad\qquad\qquad \longrightarrow \text{observable and evolution} \longrightarrow \text{probability and measurement}\end{aligned}}$$

Each arrow preserves its domain. In particular, the state vector does not replace the TPK section, just as the real value does not replace the numerical genealogy that produces it.

64.1.1 Genealogical state and Hilbert realization

Definition 64.1 (Realized genealogical state)**.** A genealogical state is a quintuple

$$\mathfrak{G} = (x_\infty, \Gamma, \mathcal{F}, \mathcal{M}, \mathcal{R}), \tag{64.1}$$

where $x_\infty \in \Omega_\infty$ is a compatible section, Γ its observable route, $\mathcal{F}$ the family of realizations, $\mathcal{M}$ the memory of the lifts, and $\mathcal{R}$ the declared reader. Let $\mathsf{P}_{\mathrm{TPK}}$ be the category whose objects are the prepared TPK sections, finite or coinductive, and whose morphisms are the recorded routes compatible with their boundary data. In particular, the depth restriction is a morphism

$$r_{n+1,n} : x_{n+1} \longrightarrow x_n.$$

Let $\mathsf{Hilb}_{\mathbb{R},J}$ be the category whose objects are pairs $(\mathcal{H}_\mathbb{R}, J)$, with $\mathcal{H}_\mathbb{R}$ a real Hilbert space and J orthogonal, $J^2 = -I$, and whose morphisms are bounded real-linear maps T intertwining the complex structures, $TJ_1 = J_2T$. A quantum realization is a functor

$$Q : \mathsf{P}_{\mathrm{TPK}} \longrightarrow \mathsf{Hilb}_{\mathbb{R},J},$$

and a *pointed realization* is the pair (Q, Ψ) assigning to each prepared section x a nonzero vector $\Psi_x \in Q(x)$, compatible with the restrictions of the inverse system.

With the convention of linearity in the second argument, the associated Hermitian form is written

$$\langle u, v\rangle_\mathbb{C} = \langle u, v\rangle_\mathbb{R} - i\langle u, J_E v\rangle_\mathbb{R}.$$

Choosing the chart $J_E \leftrightarrow i$ realizes the complex structure that in Parts I–III is constructed as an internal root of $-I$. For prefixes of depth n, set $\mathcal{H}_n := Q(x_n)$ and

$$p_{n+1,n} := Q(r_{n+1,n}) : \mathcal{H}_{n+1} \longrightarrow \mathcal{H}_n.$$

Compatibility between restriction and realization requires

$$p_{n+1,n}\Psi_{n+1}(x_{n+1}) = \Psi_n(x_n). \tag{64.2}$$

If $r_{\infty,n} : x_\infty \to x_n$ is the coinductive restriction and $p_n := Q(r_{\infty,n}) : \mathcal{H}_\infty \to \mathcal{H}_n$, a limit $\mathcal{H}_\infty$ realizes the complete section when $p_n\Psi_\infty(x_\infty) = \Psi_n(x_n)$ at every depth.

Principle 64.2 (Realized quantum state)**.** The pure state associated with a section is represented by the ray of the nonzero vector $\Psi = \Psi_\infty(x_\infty)$; a statistical preparation is represented by a positive trace-one density operator on the same realization. Global-phase identification acts in the representation: the section and its record remain in the genealogical domain.

64.1.2 Superposition and internal phase

If $\Psi, \Phi \in \mathcal{H}_\infty$ and $a = \alpha + i\beta \in \mathbb{C}$, the action of a in the real representation is $\alpha I + \beta J_E$. The combination $a\Psi + b\Phi$ is formed only after transporting the two alternatives to a common fiber. The genealogical lifts of section 3.6 take values in output × memory; they belong to the TPK domain and are not multiplied as operators. If a projective realization assigns unitary operators U_g to visible transformations, its composition defect is described separately by

$$U_g U_h = c(g,h)U_{gh}, \qquad c(g,h) \in U(1),$$

with $c(e,g) = c(g,e) = 1$ and

$$c(g,h)c(gh,k) = c(h,k)c(g,hk).$$

A cocycle valued in another memory group acts on the realization only after a representation of that group has been declared that intertwines the operators U_g.

Principle 64.3 (Genealogical superposition)**.** Compatible alternatives are first transported in $\mathsf{P}_{\mathrm{TPK}}$ and are then combined linearly in $\mathcal{H}_\infty$. The action of J_E represents the relative phase, and the route weight preserves the declared lifting datum.

The interference is expressed through the cross terms.

$$\|\Psi_1 + \Psi_2\|^2 = \|\Psi_1\|^2 + \|\Psi_2\|^2 + 2\,\mathrm{Re}\langle\Psi_1, \Psi_2\rangle.$$

Two routes with identical endpoints may therefore contribute distinct phases without their joint publication erasing the difference between their histories.

64.1.3 Observables and Spectral Calculus

Let $O : \Omega_\infty \to Y \subseteq \mathbb{R}$ be a real genealogical reader. A quantum realization of O is a self-adjoint operator $\widehat{O}$, with dense domain $D(\widehat{O})$, whose PVM satisfies the compatibility condition

$$P_O(B)\Psi_x \quad \text{represents the restriction of } x \text{ to } O^{-1}(B)$$

for each Borel set $B \subseteq \mathbb{R}$ in the support of the preparation. In finite dimension,

$$\widehat{O} = \sum_a o_a P_a, \qquad P_a P_b = \delta_{ab} P_a, \qquad \sum_a P_a = I;$$

in general dimension, the sum is replaced by the spectral integral $\widehat{O} = \int \lambda \, \mathrm{d}P_O(\lambda)$.

Principle 64.4 (Typed observable)**.** The physical observable is the pair formed by the reader in the genealogical domain and its self-adjoint representation. Spectral calculus publishes the possible results of the representation without erasing the operation that defined them.

For a normalized pure state $\Psi \in D(\widehat{O}^2)$,

$$\langle O\rangle_\Psi = \langle \Psi, \widehat{O}\Psi\rangle, \qquad (\Delta_\Psi O)^2 = \langle \Psi, (\widehat{O} - \langle O\rangle_\Psi)^2 \Psi\rangle.$$

The formulation by means of $\mathrm{tr}(\rho\widehat{O})$ extends the definition to mixed states when $\rho\widehat{O}$ is trace class; the second moment likewise requires $\rho\widehat{O}^2$ to be trace class.

64.1.4 Evolution, Hamiltonian, and Schrödinger equation

Genealogical evolution preserves composition of transports. Its realization satisfies

$$U(t_3, t_2)U(t_2, t_1) = U(t_3, t_1), \qquad U(t, t) = I.$$

When it preserves the inner product and satisfies $U(t)J_E = J_E U(t)$, it is orthogonal in the real chart and unitary in the complex chart. For a strongly continuous one-parameter unitary group, Stone's theorem provides a self-adjoint and J_E-linear generator H [117]:

$$U(t) = \exp(-J_E H t/\hbar_{\mathrm{int}})\,.$$

On the domain $D(H)$, differentiation with respect to time gives

$$\hbar_{\mathrm{int}} J_E \dot{\Psi}(t) = H\Psi(t), \tag{64.3}$$

and the chart $J_E \leftrightarrow i$ produces $i\hbar_{\mathrm{int}}\dot{\Psi} = H\Psi$.

The calendar C_{108} realizes this correspondence in finite form: the self-adjoint Hamiltonian constructed for it has, over t_0, the unitary one-iteration shift as its exponential. The continuous group additionally requires strong continuity; it does not follow from the periodicity of the calendar alone.

64.1.5 Probability and quadratic representation of the branches

Let $C_n = \{C_s : s \in \mathcal{S}_n\}$ be the partition of Ω_∞ by a finite set $\mathcal{S}_n$ of prefixes of depth n, and let μ be a compatible cylindrical probability measure, $\mu(\Omega_\infty) = 1$. Let E be a real Hilbert space with complex structure J_E. On $\mathcal{H}_n = \ell^2(\mathcal{S}_n) \otimes E$, define

$$\Psi_n = \sum_{s\in\mathcal{S}_n} \sqrt{\mu(C_s)}\,|s\rangle \otimes e^{\theta_s J_E} u_0, \tag{64.4}$$

with u_0 unitary.

Theorem 64.5 (Quadratic representation of a branch measure)**.** *For a cylindrical event $A = \bigcup_{s\in I_A} C_s$, let*

$$P_A = \left(\sum_{s\in I_A} |s\rangle\langle s|\right) \otimes I_E.$$

Then

$$\|P_A\Psi_n\|^2 = \mu(A).$$

Proof. The prefix vectors are orthogonal and $e^{\theta_s J_E}$ preserves the norm. Therefore,

$$\|P_A\Psi_n\|^2 = \sum_{s\in I_A} \mu(C_s) = \mu(A).$$

□

The quadratic rule is obtained here for the PVM of finite-depth cylindrical events as an exact representation of the declared branch measure [14]. Conversely, Gleason's theorem characterizes, in

dimension at least three, noncontextual additive measures on projectors through density operators [53]. Coupling with the apparatus and conditioning on the outcome are developed subsequently in measurement theory.

64.1.6 Incompatibility of readers and uncertainty

Translations and nonadic phases define the Weyl pair.

$$\mathcal{H}_9 = \ell^2(\mathbb{Z}/9\mathbb{Z}), \qquad \omega_9 = e^{2\pi J_E/9}, \qquad X|n\rangle = |n+1\rangle, \qquad Z|n\rangle = \omega_9^n |n\rangle.$$

The indices are read modulo 9, and ω_9 is the complex scalar realized by the structure J_E on $\mathcal{H}_9$. The identity

$$ZX = \omega_9 XZ$$

realizes the finite Heisenberg group of order $9^3 = 729$. This cardinality belongs to the group of phases and translations and retains its type against the other occurrences of 729.

For self-adjoint observables A and B, a normalized vector $\Psi \in D(AB) \cap D(BA)$ with finite second moments permits the application of Cauchy–Schwarz to obtain the Robertson–Heisenberg inequality

$$\Delta_\Psi A \, \Delta_\Psi B \geq \frac{1}{2} \big| \langle \Psi, [A, B]\Psi \rangle \big|. \tag{64.5}$$

The incompatibility thus appears at coordinated levels: the commutator of the APP evaluations, the Weyl relation in the representation, and the dispersion bound for the physical readings [60, 106].

For $\Psi \neq 0$, the discrete Fourier transform supplies the support bound

$$|\operatorname{supp} \Psi| \, |\operatorname{supp} \widehat{\Psi}| \geq 9,$$

and, for mutually unbiased position and phase bases, the Shannon entropic inequality, with natural logarithms,

$$H_Q(\Psi) + H_P(\Psi) \geq \log 9$$

offers a third exact formulation [92]. Here H_Q and H_P are the Shannon entropies of the probability distributions in the two bases.

64.1.7 Tensor product of paths and entanglement generation via shared memory

Genealogical composition glues sections over boundary data. Fix on $\mathsf{P}_{\mathrm{TPK}}$ a monoidal structure $(\boxtimes, \mathbf{1}, a, l, r)$, with declared product, unit, and associativity and unit coherences. A strong monoidal realization provides natural isomorphisms

$$\mu_{A,B} : Q(A) \widehat{\otimes} Q(B) \xrightarrow{\sim} Q(A \boxtimes B).$$

In the Hilbert image there appears $\mathcal{H}_A \otimes \mathcal{H}_B$. A nonseparable pure state admits a Schmidt decomposition

$$|\Psi\rangle = \sum_r \sqrt{p_r}\, |r\rangle_A \otimes |r\rangle_B,$$

and satisfies $p_r \geq 0$, $\sum_r p_r = 1$, with $\{|r\rangle_A\}$ and $\{|r\rangle_B\}$ orthonormal Schmidt bases, and its reductions have the same nonzero spectrum. The entropy associated with that spectrum receives, in the Barbero–Immirzi chapter, an area and boundary realization.

Principle 64.6 (Genealogical composition)**.** The tensor product represents a gluing of sections over a shared boundary. Entanglement expresses the nonfactorization of the realized section with respect to the chosen bipartition.

64.1.8 Kinematic Reconstruction Theorem

Theorem 64.7 (Hilbertian Conditional Reconstruction of the Genealogical State)**.** *Let Ω_∞ be a space of compatible sections, let $(\mathsf{P}_{\mathrm{TPK}}, \boxtimes, \mathbf{1})$ be the monoidal category of routes with the declared coherences, and let*

$$Q : \mathsf{P}_{\mathrm{TPK}} \longrightarrow \mathsf{Hilb}_{\mathbb{R},J}$$

a strong monoidal realization, pointed by a compatible section $\Psi_x \in Q(x)$. Suppose that:

1. *$J_E^2 = -I$, J_E preserves the inner product, and the morphisms of Q intertwine the complex structures;*
2. *each declared real reader is represented by a self-adjoint operator whose PVM satisfies the compatibility condition with the reader's fibers;*
3. *each continuous evolution is represented by a strongly continuous one-parameter unitary group $U(t)$, with $U(t)J_E = J_E U(t)$, and a self-adjoint J_E-linear generator H;*
4. *the preparation induces a compatible cylindrical probability measure and the state Ψ_n of (64.4).*

The complex chart of Q then realizes states as rays or densities, complex superposition, self-adjoint observables, unitary evolution generated by H, tensor composition, and quadratic probabilities for the finite-depth cylindrical PVM.

Proof. The first hypothesis converts every real fiber into a complex Hilbert space and makes the transports complex-linear. The second permits application of the spectral theorem with the typed reader. The third supplies the Hamiltonian through Stone's theorem and the differential equation on $D(H)$. The fourth and theorem 64.5 give the quadratic rule on cylindrical events. Strong monoidality realizes the tensor product and its notion of separability. □

Result and scope

The Hilbertian reconstruction is coordinated as a property of a realization of the genealogical state: the complex structure derives from $J_E^2 = -I$; evolution, from the representation of routes; probability, from a compatible measure; and entanglement, from composition over a shared boundary. The theorem is exact under its four hypotheses and does not automatically identify TPK with a concrete quantum theory. The construction of a concrete physical realization preserves its sources: Hamiltonian, apparatus, readers, domain, and boundary conditions.

64.2 Reconstruction of quantum principles from the genealogical state

Hilbert's formulation organizes quantum mechanics through states, self-adjoint observables, unitary evolution, tensorial composition, and a probabilistic rule for outcomes [235]. HMT–MD constructs a layer prior to that representation: a compatible section, its observable route, transport memory, and the reader that publishes an outcome. Hilbert space is not replaced; it appears as the codomain of a declared linear realization.

The typed sequence is

$$\begin{array}{l}\text{genealogical section} \longrightarrow \text{linear realization} \longrightarrow \text{phase and superposition} \\ \qquad\qquad\qquad\qquad \longrightarrow \text{observable and evolution} \longrightarrow \text{probability and measurement}\end{array}$$

Each arrow preserves its domain. In particular, the state vector does not replace the TPK section, just as the real value does not replace the numerical genealogy that produces it.

64.2.1 Genealogical state and Hilbert realization

Definition 64.8 (Realized genealogical state)**.** A genealogical state is a quintuple

$$\mathfrak{G} = (x_\infty, \Gamma, \mathcal{F}, \mathcal{M}, \mathcal{R}), \tag{64.6}$$

where $x_\infty \in \Omega_\infty$ is a compatible section, Γ its observable route, $\mathcal{F}$ the family of realizations, $\mathcal{M}$ the memory of the lifts, and $\mathcal{R}$ the declared reader. Let $\mathsf{P}_{\mathrm{TPK}}$ be the category whose objects are the prepared TPK sections, finite or coinductive, and whose morphisms are the recorded routes compatible with their boundary data. In particular, the depth restriction is a morphism

$$r_{n+1,n} : x_{n+1} \longrightarrow x_n.$$

Let $\mathsf{Hilb}_{\mathbb{R},J}$ be the category whose objects are pairs $(\mathcal{H}_\mathbb{R}, J)$, with $\mathcal{H}_\mathbb{R}$ a real Hilbert space and J orthogonal, $J^2 = -I$, and whose morphisms are bounded real-linear maps T intertwining the complex structures, $TJ_1 = J_2T$. A quantum realization is a functor

$$Q : \mathsf{P}_{\mathrm{TPK}} \longrightarrow \mathsf{Hilb}_{\mathbb{R},J},$$

and a *pointed realization* is the pair (Q, Ψ) assigning to each prepared section x a nonzero vector $\Psi_x \in Q(x)$, compatible with the restrictions of the inverse system.

With the convention of linearity in the second argument, the associated Hermitian form is written

$$\langle u, v\rangle_\mathbb{C} = \langle u, v\rangle_\mathbb{R} - i\langle u, J_E v\rangle_\mathbb{R}.$$

Choosing the chart $J_E \leftrightarrow i$ realizes the complex structure that is constructed in the formal kernel and its development as an internal root of $-I$. For prefixes of depth n, set $\mathcal{H}_n := Q(x_n)$ and

$$p_{n+1,n} := Q(r_{n+1,n}) : \mathcal{H}_{n+1} \longrightarrow \mathcal{H}_n.$$

Compatibility between restriction and realization requires

$$p_{n+1,n}\Psi_{n+1}(x_{n+1}) = \Psi_n(x_n). \tag{64.7}$$

If $r_{\infty,n} : x_\infty \to x_n$ is the coinductive restriction and $p_n := Q(r_{\infty,n}) : \mathcal{H}_\infty \to \mathcal{H}_n$, a limit $\mathcal{H}_\infty$ realizes the complete section when $p_n\Psi_\infty(x_\infty) = \Psi_n(x_n)$ at every depth.

Principle 64.9 (Realized quantum state)**.** The pure state associated with a section is represented by the ray of the nonzero vector $\Psi = \Psi_\infty(x_\infty)$; a statistical preparation is represented by a positive, trace-one density operator on the same realization. Global-phase identification acts in the representation: the section and its record remain in the genealogical domain.

64.2.2 Evolution, Hamiltonian, and Schrödinger equation

Genealogical evolution preserves composition of transports. Its realization satisfies

$$U(t_3, t_2)U(t_2, t_1) = U(t_3, t_1), \qquad U(t, t) = I.$$

When it preserves the inner product and satisfies $U(t)J_E = J_E U(t)$, it is orthogonal in the real chart and unitary in the complex chart. For a strongly continuous one-parameter unitary group, Stone's theorem provides a self-adjoint and J_E-linear generator H [234]:

$$U(t) = \exp(-J_E H t/\hbar_{\mathrm{int}})\,.$$

On the domain $D(H)$, differentiation with respect to time gives

$$\hbar_{\mathrm{int}} J_E \dot{\Psi}(t) = H\Psi(t), \tag{64.8}$$

and the chart $J_E \leftrightarrow i$ produces $i\hbar_{\mathrm{int}}\dot{\Psi} = H\Psi$.

The calendar C_{108} realizes this correspondence in finite form: the self-adjoint Hamiltonian constructed for it has, over t_0, the unitary one-iteration shift as its exponential. The continuous group additionally requires strong continuity; it does not follow from the periodicity of the calendar alone.

64.3 Quadratic regimes and temporal readout

64.3.1 Quadratic algebras of the TRIT and curvature signatures

The primary construction and proof of the three regimes are given in the construction of the quadratic algebras of the TRIT, whose defining relation is presented in (1.156). For $\tau \in \widehat{\mathbb{T}}$, with balanced image $\bar{\tau} \in \{-1, 0, +1\}$, we have

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \bar{\tau}),$$

and, if J is the class of X,

$$\begin{aligned} \tau = +1 &: J^2 = -I, \quad \text{elliptic regime,} \\ \tau = 0 &: J^2 = 0, \quad \text{parabolic regime,} \\ \tau = -1 &: J^2 = +I, \quad \text{hyperbolic regime.} \end{aligned}$$

For all three values of τ, the quotient $\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \bar{\tau})$ satisfies precisely the three quadratic identities above. In particular, the internal square root of -1 is realized as an operator on the elliptic sheet; HMT need neither reject nor suppress the complex extension. The parabolic algebra contains a nilpotent, and the hyperbolic algebra decomposes by means of the two idempotents $(I \pm J)/2$.

These identities classify algebras and transports. They do not, by themselves, select a spacetime metric or prove that the physical universe occupies one of the three sheets.

64.3.2 Temporal TRIT reader on (-1,0,+1): opening, threshold, and return with memory

Once an ordered parametrization of the transitions has been chosen, the active temporal reader is

$$\begin{aligned} +1 &\longmapsto \text{return, retained memory and past,} \\ 0 &\longmapsto \text{evaluation threshold and present,} \\ -1 &\longmapsto \text{bidirectional opening and future.} \end{aligned}$$

The reader is PROVED WITH AN EXPLICIT STARTING STRUCTURE: its conclusions follow once the TRIT signature, the order of the trajectory, and the transported memory have been specified. It does not turn the set $\{-1, 0, +1\}$ into an equation of motion. The future designates the space of compatible continuations; survival may select a branch or a multisection without erasing the genealogy of the others.

The frequency

$$\omega_0 = \frac{2\pi}{108\, t_0}$$

assigns a type to the full-return clock through the decision entropy $I_{\mathrm{TRIT}} = \log 3$ and the Boltzmann transducer. It does not, in isolation, establish a stable subharmonic phase. It does, however, enable construction of the finite reference Hamiltonian below; selection of a robust physical Hamiltonian requires additional conditions.

64.3.3 Unitary dynamics on 108 states: Hamiltonian, propagator, and period

Fix an orientation $a \in \{\mathrm{pre}, \mathrm{ret}\}$ of the sections constructed in (16.35) and write $\hbar_{\mathrm{int}} = \hbar_a$. The same section is retained in the Hamiltonian, propagator, and subsequent thermal conversion.

Let

$$\mathcal{H}_{\mathrm{clk}} := \mathbb{C}^{108}, \qquad \{|j\rangle : j \in \mathbb{Z}/108\mathbb{Z}\}$$

with its canonical Hermitian inner product. Fix $\zeta := \exp(2\pi \mathrm{i}/108)$, the unitary shift

$$U_{\mathrm{clk}}|j\rangle = |j+1\rangle$$

and the Fourier basis

$$|\widetilde{k}\rangle := \frac{1}{\sqrt{108}} \sum_{j=0}^{107} \zeta^{jk} |j\rangle, \qquad 0 \le k \le 107.$$

With $t_0 \in \mathcal{L}_T^+$ and $\hbar_{\mathrm{int}} \in \mathcal{L}_S^+$, define

$$\boxed{H_{\mathrm{clk}} := \hbar_{\mathrm{int}} \omega_0 \sum_{k=0}^{107} k\, |\widetilde{k}\rangle\langle\widetilde{k}|, \qquad \omega_0 = \frac{2\pi}{108\, t_0}.} \tag{CLK1}$$

The operator has energy type $[H_{\mathrm{clk}}] = \mathcal{L}_S \mathcal{L}_T^{-1} = \mathcal{L}_E$.

Theorem 64.10 (Exact unitary realization of the 108 calendar)**.** *The operator H_{clk} is self-adjoint and its one-step propagator satisfies*

$$U_{\mathrm{step}} := \exp\!\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}} H_{\mathrm{clk}} t_0\right) = U_{\mathrm{clk}}, \qquad U_{\mathrm{step}}^{108} = I. \tag{CLK2}$$

If

$$Z_{\mathrm{clk}}|j\rangle = \zeta^j |j\rangle,$$

then, for any basis state $|j_0\rangle$,

$$\langle Z_{\mathrm{clk}} \rangle_n := \langle j_0 | U_{\mathrm{step}}^{-n} Z_{\mathrm{clk}} U_{\mathrm{step}}^{n} | j_0 \rangle = \zeta^{j_0+n}. \tag{CLK3}$$

The sequence has primitive period 108.

Proof. Expression (CLK1) is a spectral decomposition with real eigenvalues, hence $H_{\mathrm{clk}} = H_{\mathrm{clk}}^{\dagger}$. Moreover,

$$U_{\mathrm{clk}}|\widetilde{k}\rangle = \zeta^{-k}|\widetilde{k}\rangle,$$

whereas the exponential of $\hbar_{\mathrm{int}}\omega_0 k$ over t_0 gives $\exp(-2\pi \mathrm{i} k/108) = \zeta^{-k}$. The two operators agree on an orthonormal basis. The remaining identities follow from $U_{\mathrm{step}}^n |j_0\rangle = |j_0 + n\rangle$ and the primitivity of ζ. □

The corresponding variational formulation, for a curve $\psi \in C^1([t_a, t_b], \mathcal{H}_{\mathrm{clk}})$, unconstrained during the variation, is

$$\boxed{L_{\mathrm{clk}}(\psi, \dot\psi) = \frac{\mathrm{i}\hbar_{\mathrm{int}}}{2}\big(\langle\psi|\dot\psi\rangle - \langle\dot\psi|\psi\rangle\big) - \langle\psi|H_{\mathrm{clk}}|\psi\rangle.} \tag{CLK4}$$

The action functional is

$$\boxed{S_{\mathrm{clk}}[\psi] := \int_{t_a}^{t_b} L_{\mathrm{clk}}(\psi(t), \dot\psi(t))\, \mathrm{d}t,} \tag{CLK5}$$

with $\delta\psi(t_a) = \delta\psi(t_b) = 0$. Varying ψ and $\bar\psi$ independently yields the Schrödinger equation for the finite clock

$$\boxed{\mathrm{i}\hbar_{\mathrm{int}}\dot{\psi} = H_{\mathrm{clk}}\psi} \tag{CLK6}$$

together with its adjoint equation [233]. Its propagator sampled at $t = nt_0$ is $\psi_n = U_{\mathrm{step}}^n\psi_0$. Since H_{clk} is self-adjoint, the norm is conserved; normalized initial data remain normalized. The result is exact with respect to the specified $H_{\mathrm{clk}}, t_0, \hbar_{\mathrm{int}}$; it neither selects a physical Hamiltonian nor introduces an experimental frequency.

The theorem 64.10 establishes the unitary dynamics of the calendar with respect to the specified complex completion, t_0, and $\hbar_{\mathrm{int}}$. Its realization as a finite periodic system is developed in Section 64.4, which specifies the drive, the order observable, the primitive subharmonic response of period $108T_{\mathrm{d}}$, and finite spectral stability. A macroscopic material realization constitutes a distinct external identification, not a gap in the finite result.

64.4 Finite time crystal: periodic driving, subharmonic response and spectral stability

64.4.1 Periodic driving induced by the 108-state clock

Sea $N = 108$,

$$\mathcal{H}_{\mathrm{clk}} = \mathbb{C}^N, \qquad S|j\rangle = |j+1\rangle, \qquad Z|j\rangle = \zeta^j|j\rangle, \qquad \zeta = \exp(2\pi\mathrm{i}/N).$$

In the Fourier basis $\{|\widetilde{k}\rangle\}_{k=0}^{N-1}$, we set

$$H_{\mathrm{clk}} = \hbar_{\mathrm{int}}\frac{2\pi}{Nt_0}\sum_{k=0}^{N-1} k|\widetilde{k}\rangle\langle\widetilde{k}|, \qquad \exp\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}}H_{\mathrm{clk}}t_0\right) = S.$$

This equality fixes the starting clock. To exhibit a nonconstant periodic drive, let $0 < \eta < 1$, $T_{\mathrm{d}} = t_0$, and define, in each period,

$$H_{\mathrm{d}}(t) = \begin{cases} 0, & 0 \le t < \eta T_{\mathrm{d}}, \\ (1-\eta)^{-1}H_{\mathrm{clk}}, & \eta T_{\mathrm{d}} \le t < T_{\mathrm{d}}. \end{cases} \qquad H_{\mathrm{d}}(t+T_{\mathrm{d}}) = H_{\mathrm{d}}(t).$$

As the two segments commute, their Floquet propagator is

$$U_F = \mathcal{T}\exp\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}}\int_0^{T_{\mathrm{d}}} H_{\mathrm{d}}(t)\,\mathrm{d}t\right) = \exp\left(-\frac{\mathrm{i}}{\hbar_{\mathrm{int}}}H_{\mathrm{clk}}T_{\mathrm{d}}\right) = S.$$

We define the Hermitian observables

$$Q = \frac{Z+Z^\dagger}{2}, \qquad P = \frac{Z-Z^\dagger}{2\mathrm{i}}.$$

Theorem 64.11 (Primitive Subharmonicity of the Finite Temporal Crystal HMT)**.** *For $\rho_{j_0} = |j_0\rangle\langle j_0|$, the vector of orden*

$$\mathbf{q}_n = \left(\mathrm{Tr}\,\rho_{j_0}U_F^{-n}QU_F^n,\ \mathrm{Tr}\,\rho_{j_0}U_F^{-n}PU_F^n\right)$$

satisfies

$$\mathbf{q}_n = \left(\cos\frac{2\pi(j_0+n)}{N},\ \sin\frac{2\pi(j_0+n)}{N}\right).$$

Its primitive period is $NT_{\mathrm{d}} = 108T_{\mathrm{d}}$, whereas the drive has period T_{d}. Therefore, the system realizes a primitive subharmonic response of order 108 *in the declared finite domain.*

Proof. The Weyl relation $S^{-1}ZS = \zeta Z$ gives $S^{-n}ZS^n = \zeta^n Z$. Its real and imaginary parts yield the formula. If an integer $d > 0$ were a period, then $\zeta^d = 1$ would hold; the primitivity of ζ forces $N \mid d$. □

64.4.2 Nilpotent jet of the triadic evolution and compatible prolongation

The minimal separation between distinct roots of unity is

$$g_N = \min_{k \neq \ell} |\zeta^k - \zeta^\ell| = 2 \sin \frac{\pi}{N} > 0.$$

Theorem 64.12 (Covariant rigidity and finite spectral stability)**.** *The following two assertions hold.*

1. *For every unitary V, the terna*

$$U_F^{(V)} = V U_F V^\dagger, \qquad Z^{(V)} = V Z V^\dagger, \qquad \rho^{(V)} = V \rho V^\dagger$$

exactly preserves the Weyl relation, the subharmonicity of order N, and the amplitude of the order vector.

2. *If U' is unitary and $\|U' - U_F\| = \varepsilon < g_N/2$, the N disks of radius $g_N/2$ centered at $\{\zeta^k\}$ uniquely separate the spectral sectors of U'. Their Riesz projectors have the same rank as those of U_F throughout every deformation that remains in that ball. Moreover, for every observable A, every state ρ, and every $n \geq 0$,*

$$\left|\operatorname{Tr} \rho \left(U'^{-n} A U'^{n} - U_F^{-n} A U_F^{n}\right)\right| \leq 2n\varepsilon \|A\|.$$

Proof. The first part is obtained by conjugating each identity. For the second, the separation g_N prevents two sectors from entering the same contour; continuity of the Riesz projectors preserves their ranges. The telescoping identity gives $\|U'^n - U_F^n\| \leq n\varepsilon$, and the same bound holds for the inverse powers. The inequality of expectations results from inserting and subtracting $U'^{-n} A U_F^n$. □

The resulting object is a *finite HMT time crystal* with a drive, a subharmonic observable, and quantified spectral stability. Whether a macroscopic material realizes this class, retains coherence at large volume, or exhibits prethermal or localized protection belongs to EXTERNAL IDENTIFICATION; it is not an unresolved mathematical part of the finite construction.

64.5 Algebraic construction of quantum completions over APP

APP and TPK provide oriented states, transport, and sheet parity. Their linear realization enables construction of the symmetric and exterior algebras, with their creation, annihilation, and occupation operators. This section establishes these algebraic constructions, the cellular field sector, and reversible measurement coupling. Transport of the completions under refinement is proved in Section 68.6; application of their occupation spectra to counting and equilibrium is developed in Sections 69.3–68.10. Each stage retains the conditions of the representation it uses.

64.5.1 Linear state space associated with a particle realization

Let S be a finite set of TPK route classes at the chosen depth, and let

$$\mathcal{H}_1 = \ell^2(S)$$

be the Hilbert space with orthonormal basis $(e_s)_{s \in S}$. A choice of positive weights E_s defines the diagonal operator

$$H e_s = E_s e_s.$$

The choice of S, its quotient by symmetries, and the weights are part of the model. Once these are fixed, the multiparticle completions are canonical.

64.5.2 Symmetric and exterior completions

We define

$$\mathcal{F}_+(\mathcal{H}_1) = \bigoplus_{n\geq 0} \mathrm{Sym}^n \mathcal{H}_1, \qquad \mathcal{F}_-(\mathcal{H}_1) = \bigoplus_{n\geq 0} \bigwedge^n \mathcal{H}_1.$$

The first is the bosonic completion; the second is the fermionic completion. In both, the degree-zero summand is the vacuum space. The exterior powers are equipped with the inner product for which ordered wedges of vectors from an orthonormal basis are orthonormal; the symmetric powers are used with the usual normalization of creation and annihilation operators.

64.5.3 Sheet grading and exchange algebra

Suppose that the oriented sheet defines an additive character

$$p : \mathrm{Mor}(\mathsf{P}_{\mathrm{TPK}}) \longrightarrow \mathbb{Z}/2\mathbb{Z}, \qquad p(\gamma\eta) = p(\gamma) + p(\eta).$$

On the corresponding graded linear space, introduce the symmetry

$$\tau(v \otimes w) = (-1)^{p(v)p(w)} w \otimes v.$$

Theorem 64.13 (Statistical completion determined by parity)**.** *The quotient of the tensor algebra by the relations*

$$v \otimes w - (-1)^{p(v)p(w)} w \otimes v$$

is symmetric on the even sector and exterior on the odd sector. Two identical odd states have zero product; even states admit arbitrary occupation.

Proof. If $p(v) = p(w) = 0$, the relation is $vw = wv$. If $p(v) = p(w) = 1$, it is $vw = -wv$; setting $v = w$ and working in characteristic zero gives $v^2 = 0$. Mixed sectors commute without an additional sign. This is the universal property of the graded symmetric algebra. □

64.5.4 Canonical operators and idempotence of occupation

The parity character determines the completion; the product of that completion then determines the operator algebra. In the exterior sector, creation $a_s^\dagger$ is exterior multiplication by e_s, and a_s is its adjoint.

Theorem 64.14 (Pauli exclusion principle in the exterior completion)**.** *For every mode $s \in S$, the creation and annihilation operators of the exterior completion satisfy*

$$\{a_s, a_t^\dagger\} = \delta_{st} I, \qquad \{a_s, a_t\} = \{a_s^\dagger, a_t^\dagger\} = 0. \tag{64.9}$$

In particular, the occupation operator $N_s = a_s^\dagger a_s$ is a projector:

$$N_s^2 = N_s, \qquad \mathrm{Spec}(N_s) \subseteq \{0, 1\}. \tag{64.10}$$

The eigenvalues 0 *and* 1 *correspond, respectively, to the absence and presence of mode s.*

Proof. On an ordered wedge, a_s removes the factor e_s, when present, with the sign determined by its position; when absent, it gives zero. Composing this operation with exterior multiplication proves relations (64.9): for $s = t$, the two compositions cover the complementary cases of absence and presence; for $s \neq t$, the exchange signs cancel. Antisymmetry implies $e_s \wedge e_s = 0$ and $a_s^2 = (a_s^\dagger)^2 = 0$. Therefore

$$N_s^2 = a_s^\dagger a_s a_s^\dagger a_s = a_s^\dagger (I - a_s^\dagger a_s) a_s = a_s^\dagger a_s = N_s. \tag{64.11}$$

The operator is self-adjoint, and every eigenvalue of an idempotent satisfies $\lambda^2 = \lambda$. Hence its spectrum is contained in $\{0, 1\}$. The vacuum and the state e_s realize both values [231, 228]. □

In the symmetric sector, the bosonic operators $b_s^\dagger, b_s$ act on the normalized occupation basis with factors $\sqrt{m_s + 1}$ and $\sqrt{m_s}$, respectively. Their compositions give

$$[b_s, b_t^\dagger] = \delta_{st} I, \qquad [b_s, b_t] = [b_s^\dagger, b_t^\dagger] = 0. \tag{64.12}$$

These identities are understood on the algebraic finite-particle domain, which is invariant under the operators considered.

Theorem 64.13 turns a compositional TPK parity into an exact occupation rule. To obtain the physical spin–statistics theorem, one must also construct a positive local Lorentzian theory and prove that sheet parity agrees with spin under rotations. This equality does not follow merely from both objects taking values modulo two.

64.5.5 Sheet holonomy and $2\pi/4\pi$ periodicity

The preceding parity admits a natural geometric realization when a surviving extension has a closed return with a nontrivial orientation cover. Let ρ be the return generator and let

$$\varepsilon : \langle \rho \rangle \longrightarrow \{-1, +1\}$$

be its holonomy character, with $\varepsilon(\rho) = -1$. The associated real bundle is then the orientation line of a Möbius strip: one turn closes the geometric projection but changes the lifted sheet; two turns restore the orientation.

To express this property in angular terms, fix a unitary representation U of the rotation cover on the graded space $\mathcal{H}_1 = \mathcal{H}_{\bar{0}} \oplus \mathcal{H}_{\bar{1}}$, compatible with sheet parity in the sense that

$$U(2\pi)v = (-1)^{p(v)} v$$

for every homogeneous vector v. Compatibility is a condition of the realization: it identifies the generator of the orientation cover with the character p already constructed on the routes.

Theorem 64.15 (Restoration of orientation and exchange algebra)**.** *Under the preceding compatibility condition,*

$$U(2\pi)|_{\mathcal{H}_{\bar{0}}} = I, \qquad U(2\pi)|_{\mathcal{H}_{\bar{1}}} = -I, \qquad U(4\pi) = I.$$

Moreover, the same character p determines the graded symmetry

$$\tau(v \otimes w) = (-1)^{p(v)p(w)} w \otimes v.$$

Thus the even-holonomy sector is completed by symmetric powers and the odd-holonomy sector by exterior powers. In the latter, the exclusion relation $v \wedge v = 0$ holds.

Proof. The first and second equalities are the compatibility condition evaluated on each homogeneous component. Iterating one turn gives

$$U(4\pi)v = U(2\pi)^2 v = (-1)^{2p(v)} v = v,$$

which proves restoration. The exchange statement is Theorem 64.13: for two even vectors the sign is positive, whereas for two odd vectors it is negative. □

Three returns should be distinguished. Angular closure after 2π is a return of the projection; closure after 4π restores the lifted orientation; neither, by itself, requires every coordinate of the record to return

to its initial value. This separation is the dimensional analogue of the HMT distinction between phase return and state return.

64.5.5.1 The sum and product sheets as supports of the 2 *completions*

The APP table contains two evaluations on the same support. The additive sheet acts by translations and preserves the multiplicity of continuations; in this sense, it accumulates. The multiplicative sheet permutes the units and contracts fibers over noninvertible elements; in this sense, it folds. This incompatibility precedes the statistical choice and explains why TPK must preserve two genealogies rather than reduce them to a single visible word.

The statistical assignment is not obtained, however, by verbally replacing "sum" with "boson" and "product" with "fermion." The exact map is the parity character

$$p : \mathrm{Mor}(\mathsf{P}_{\mathrm{TPK}}) \longrightarrow \mathbb{Z}/2\mathbb{Z}.$$

When the transport of an extension lies in the sector $p = 0$, its completion admits cumulative occupation and is symmetric. When it lies in $p = 1$, sheet reversal introduces the exchange sign and the completion is exterior. Sum and product supply the two local geometries; the record, orientation, and character p determine the sector in which a route is realized.

The 4π periodicity and fermionic exclusion are coordinated by the same sheet character. Their consequences for finite counts and equilibrium occupations are obtained, with the Hamiltonian and thermal conditions explicit, in Section 69.3 and in Theorem 68.12.

64.5.6 The cellular complex of the APP torus

The APP graph is the one-skeleton of a cellular decomposition of the torus T^2 with 9×9 vertices. Let

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2$$

be its real cochain complex. Choosing the uniform inner product, define the codifferential $\delta = d^*$ and the Laplacian

$$\Delta_k = d_{k-1}\delta_k + \delta_{k+1}d_k.$$

Theorem 64.16 (Harmonic sector of the digital torus)**.** *The space of harmonic one-forms has dimension two:*

$$\dim \ker \Delta_1 = b_1(T^2) = 2.$$

A basis can be chosen that is represented by the uniform horizontal and vertical flows. Both are closed and coclosed, and neither is the gradient of a globally defined periodic function.

Proof. The Hodge theorem for finite cellular complexes identifies $\ker \Delta_1$ with $H^1(T^2; \mathbb{R})$. The cohomology of the torus has rank two. Directly, the constant horizontal and vertical flows have zero divergence and curl; their integrals over the two fundamental cycles are independent, so they are not exact. □

The space of connections modulo gauge transformations is

$$C^1/\mathrm{im}\, d_0.$$

In a finite complex equipped with the preceding inner product, each orbit admits a Coulomb representative in $\ker \delta_1$; within the orbit, the choice is unique modulo the stabilizer $\ker d_0$ of the gauge parameter. The correct cohomological quotient is

$$\ker d_1/\mathrm{im}\, d_0 \simeq \ker \Delta_1.$$

Its two harmonic classes constitute a natural mathematical candidate for two free global modes. They are not yet the two local polarizations of a photon in 3 + 1-dimensional spacetime: that identification requires a dynamical limit, a dispersion relation, and a physical gauge group.

64.5.7 Discrete Gauss law

For an oriented 1-cochain F and a set of vertices V, divergence is defined by

$$(\operatorname{div} F)(v) = \sum_{e:v\to w} F(e) - \sum_{e:w\to v} F(e).$$

Summing over V, every interior edge appears once with each sign and cancels. Therefore,

$$\boxed{\sum_{v\in V} \operatorname{div} F(v) = \operatorname{Flux}_{\partial V}(F).}$$

This identity is an exact Gauss law on the graph. Interpreting $\operatorname{div} F$ as electric charge requires the charge character and dimensional realization constructed in Dimensional Mechanics.

64.5.8 Reversible measurement apparatus

Let S be a set of states and let $r : S \to \mathbb{Z}/9\mathbb{Z}$ be a readout. Coupling to a nonadic register is defined by

$$U_r : S \times \mathbb{Z}/9\mathbb{Z} \longrightarrow S \times \mathbb{Z}/9\mathbb{Z}, \qquad U_r(s, a) = (s, a + r(s)).$$

Proposition 64.17 (Reversibility of the coupling)**.** *The map U_r is a permutation, and*

$$U_r^{-1}(s, a) = (s, a - r(s)).$$

Information loss arises only after applying a reader to the register or conditioning the state on the outcome.

Proof. The proposed inverse satisfies both composition identities by the group law of $\mathbb{Z}/9\mathbb{Z}$. □

This proposition formalizes the operational definition of measurement: coupling, reading, and restricting compatible futures are three different maps. The full coupling is reversible; the observational quotient need not be.

64.5.9 Algebraic correspondences and physical realizations

The Fock completions, exclusion conditional on parity, the harmonic sector of APP, discrete Gauss law, and reversibility of the apparatus are complete mathematical theorems. Their physical identifications form a diagram that must commute:

$$\text{sheet parity} \longrightarrow \text{graded interchange} \longrightarrow \text{CAR/CCR} \longrightarrow \text{physical statistics},$$

$$H^1(\mathcal{A}_9) \simeq \ker \Delta_1 \hookrightarrow C^1/\operatorname{im} d_0 \longrightarrow \text{photonic realization}.$$

In the first line, the first two arrows have explicit constructions; the last requires the corresponding dynamics and equilibrium conditions. In the second, the Hodge identification and inclusion of harmonic classes in the space of connections modulo gauge are the constructions of Section 64.5.6. The zero eigenvalue belongs to the cellular Laplacian: identifying these modes with massless excitations and the two local photon polarizations additionally requires the dynamics, dispersion relation, and physical gauge realization specified there. This distinction allows the bridges to be developed without confusing an analogy in cardinality with a representation.

64.6 Arithmetic–geometric coupling and bilateral recovery

The dodecaphase register of section 11.4 admits arithmetic, incidence and operator readings. We call *holographic constant of arithmetic-geometric coupling* the generated register K, preserving the order of its twelve windows, the phase origin and orientation. Its rational reading $\kappa = K_{\mathrm{int}}/(1000^{12} - 1)$ is an encoding of that register; it does not replace its types or the enriched history that produces it. Hadamard inversion recovers its signed coordinate; the incidence reading determines the Witt tetrad. The following constructions make two further functions explicit: identifying operators on its cyclic carrier and preserving information between two isometric realizations. These are operations subsequent to the realization of the continuum, not rules introducing conventional constants into APP–TRIT–TPK.

64.6.1 Exact encoding and cyclic reference

Proposition 64.18 (Recovery of the record from its rational reading)**.** *With base $b = 1000$, length twelve, origin, and orientation fixed, the map*

$$(k_1, \dots, k_{12}) \longmapsto \kappa = \frac{\sum_{j=1}^{12} k_j b^{12-j}}{b^{12} - 1}, \qquad 0 \le k_j < b,$$

is injective. From κ, one recovers the integer $n = (b^{12} - 1)\kappa$ and, by successive Euclidean divisions, the twelve blocks, including leading zeros. An absolute error strictly smaller than $1/[2(b^{12} - 1)]$ permits the same recovery by rounding.

Proof. The fixed-length integer expansion in base b is unique. Two distinct integers are separated by at least one; their rational readings are separated by at least $1/(b^{12} - 1)$. The strict bound places the approximate datum within the rounding cell of a unique integer. □

Now consider $K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$, the cyclic shift S of its positions, and $O_K = (K, SK, \dots, S^{11}K)$ on $\mathbb{R}^{12}$. The shift of positions is identified neither with the entire enriched evolution nor, solely by virtue of its periodicity, with an exceptional symmetry.

Theorem 64.19 (Invertibility of the orbital reference)**.** *The matrix O_K is invertible, and*

$$589609\|c\|^2 \le \|O_K c\|^2 \le 6263^2\|c\|^2.$$

Its centred orbit has rank eleven.

Proof. The Gram matrix $O_K^* O_K$ is circulant. Its first row, obtained from the twelve inner products $\langle K, S^j K\rangle$, is

$$(4251257, 2999323, 3163385, 3287920, 3216553, 3344743,$$
$$2950064, 3344743, 3216553, 3287920, 3163385, 2999323).$$

Evaluation at the twelve characters of the cycle yields the eigenvalues

$$6263^2, \quad 697225, \quad 589609\ (2), \quad 1407529\ (2), \quad 1053157\ (2),$$
$$(1248025 - 345420\sqrt{3})\ (2), \qquad (1248025 + 345420\sqrt{3})\ (2),$$

where parentheses indicate multiplicities. The smallest value is 589609: for the only comparison that is not immediate, $658416 > 345420\sqrt{3}$, as is seen by squaring both sides. The triangle inequality for the Fourier transform and $K_j \ge 0$ give the maximum $(\sum K_j)^2 = 6263^2$. Positivity of every mode proves invertibility and the bounds. Centring the record removes only the uniform mode. □

Corollary 64.20 (Identification of an operator from its responses)**.** *A linear operator $T : \mathbb{R}^{12} \to \mathbb{R}^{12}$ is determined by $Y = (TK, TSK, \dots, TS^{11}K)$, through $T = YO_K^{-1}$. If $TS = ST$, the complete vector response $y = TK$ suffices, since $T = O_y O_K^{-1}$. For a perturbation E of Y, $\|\widehat{T} - T\| \le \|E\|/\sqrt{589609}$.*

Proof. We have $Y = TO_K$. Under commutation, $TS^j K = S^j y$, hence $Y = O_y$. The bound follows from $\|O_K^{-1}\| = 1/\sqrt{589609}$. The norms are the Euclidean norm and its induced operator norm. □

The response y contains twelve components. The statement does not attribute the recovery of an arbitrary operator to a single scalar: it specifies the observation and commutation condition that make identification possible.

64.6.2 Conservation and reconstruction of two channels

Let H, H' be Hilbert spaces, and let $\mathcal{B}, C : H \to H'$ be bounded linear isometries. For $a > 0$, define

$$p = a^2 + a + 1, \qquad N = (2a+1)p = (a+1)^3 + a^3, \qquad Q_- = a\mathcal{B} - C, \quad Q_+ = (a+1)\mathcal{B} + C.$$

Theorem 64.21 (Bilateral balance and recovery)**.** *The following hold*

$$(a+1)Q_-^* Q_- + aQ_+^* Q_+ = NI_H, \qquad W_a v = \frac{1}{\sqrt{N}} \begin{pmatrix} \sqrt{a+1}\, Q_- v \\ \sqrt{a}\, Q_+ v \end{pmatrix}, \qquad W_a^* W_a = I_H.$$

The data $x = Q_- v$, $y = Q_+ v$ determine

$$\mathcal{B}v = \frac{x+y}{2a+1}, \qquad Cv = \frac{ay - (a+1)x}{2a+1}, \qquad v = \mathcal{B}^* \frac{x+y}{2a+1}.$$

Proof. Setting $X = \mathcal{B}^* C + C^* \mathcal{B}$, one obtains

$$Q_-^* Q_- = (a^2+1)I - aX, \qquad Q_+^* Q_+ = ((a+1)^2+1)I + (a+1)X.$$

The weights cancel X, and the remaining coefficient is $2a^3 + 3a^2 + 3a + 1 = N$. Normalization gives the isometry. Adding the two channels recovers $(2a+1)\mathcal{B}v$; the combination $ay - (a+1)x$ recovers $(2a+1)Cv$. Finally, $\mathcal{B}^*\mathcal{B} = I_H$. □

The balance also holds in infinite dimension: only bounded operators are composed. The image of W_a is closed, and its orthogonal projector is $W_a W_a^*$. Thus a normalized pair has an input precisely when it belongs to that image; its orthogonal component measures the compatibility defect.

The factorization

$$W_a = R_a E_a, \qquad E_a v = \frac{1}{\sqrt{p}} \begin{pmatrix} \sqrt{a(a+1)}\, \mathcal{B}v \\ Cv \end{pmatrix}, \quad R_a = \frac{1}{\sqrt{2a+1}} \begin{pmatrix} \sqrt{a} & -\sqrt{a+1} \\ \sqrt{a+1} & \sqrt{a} \end{pmatrix}$$

is verified by block multiplication; R_a acts on $H' \oplus H'$, with each scalar entry multiplying $I_{H'}$. In the nonadic partition $a = 8$, the weights are $72/73$ and $1/73$; the factor $73 = 8 \cdot 9 + 1$ follows from the general law $a(a+1) + 1$, rather than being an additional primitive datum. The corresponding balance is $9\|Q_- v\|^2 + 8\|Q_+ v\|^2 = 17 \cdot 73\, \|v\|^2$.

64.6.3 Transport of observables and incoming memory

Theorem 64.22 (Observables common to both channels)**.** *For a bounded self-adjoint operator F on H',*

$$W_a^* \operatorname{diag}(F, F) W_a = \frac{a(a+1)\mathcal{B}^* F \mathcal{B} + C^* F C}{p}.$$

If $\mathcal{B}$ is unitary and $C = \mathcal{B}R$, with R unitary, and $G = \mathcal{B}^* F \mathcal{B}$, the observable is preserved if and only if $GR = RG$.

Proof. Expanding $(a+1)Q_-^*FQ_- + aQ_+^*FQ_+$ cancels the cross terms. The other two coefficients are $a(a+1)(2a+1)$ and $2a+1$. Dividing by N gives the formula. In the stated case, equality with G is equivalent to $R^*GR = G$; unitarity makes this equivalent to $GR = RG$. □

The norm is obtained by taking $F = I$. A general observable requires the commutation condition: preserving the norm does not mean preserving every observable without transformation.

Theorem 64.23 (Unitary extension with memory)**.** *If $H' = H$, $\mathcal{B} = I$, and $C = U$ is unitary, set*

$$A = \frac{\sqrt{a+1}}{\sqrt{N}}Q_-, \qquad D = \frac{\sqrt{a}}{\sqrt{N}}Q_+, \qquad \mathscr{R}_a = \begin{pmatrix} A & -D^* \\ D & A^* \end{pmatrix}.$$

Then $\mathscr{R}_a$ is unitary on $H \oplus H$, its first column is W_a, and it preserves $\|v\|^2 + \|m\|^2$ for every input (v, m). The inverse is

$$v = A^*z_1 + D^*z_2, \qquad m = -Dz_1 + Az_2.$$

Proof. The operators A, D are normal and commute with each other and with their adjoints, since they are polynomials in U, U^*. The balance gives $A^*A + D^*D = I$, and normality gives $AA^* + DD^* = I$. The cross terms in both products $\mathscr{R}_a^*\mathscr{R}_a$ and $\mathscr{R}_a\mathscr{R}_a^*$ vanish; the diagonal blocks are the identity. Applying the adjoint to (z_1, z_2) yields the inverse. □

Recovery of the register, orbital identification and unitary extension specify three distinct levels of information. The first preserves the blocks with their chart; the second identifies operators under the stated hypotheses; the third transports state and memory jointly. This hierarchy allows the same generated structure to be used without confusing a reduced reading with the state in its entirety.

Chapter 65

Quantum trajectory dynamics: Feshbach–Schur reduction, Born rule, and genealogical path integral

65.1 Feshbach–Schur reduction

Let

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_{\mathrm{ext}}, \qquad H = \begin{pmatrix} H_{00} & V \\ V^* & H_{\mathrm{ext}} \end{pmatrix}.$$

For z in the resolvent set of H_{ext}, the resolvent compression is

$$P_0(z-H)^{-1}P_0 = [z - H_{00} - \Sigma_0(z)]^{-1}, \qquad \Sigma_0(z) = V(z - H_{\mathrm{ext}})^{-1}V^*. \tag{65.1}$$

The operator Σ_0 transports the memory of the elliptic and hyperbolic sheets to the present sheet. Its time transform generates a memory kernel

$$S_{\mathrm{eff}}[\psi_0] = \int \psi_0^\dagger(i\hbar\partial_t - H_{00})\psi_0 \,\mathrm{d}t - \iint \psi_0^\dagger(t)K_0(t,t')\psi_0(t')\,\mathrm{d}t\,\mathrm{d}t'.$$

A regular low-frequency expansion,

$$\Sigma_0(\omega) = \Sigma_0^{(0)} + \omega\Sigma_0^{(1)} + \omega^2\Sigma_0^{(2)} + \cdots,$$

produces an effective local action; a singular spectral structure preserves nonlocal memory. The present thus receives the response of the global state without introducing an external time.

65.2 From the finite sum to the path integral genealogical

The preceding matrix expansion establishes exactly that each entry of U^R is the sum of the ordered weights of routes of length R. The next step preserves the information that a visible route does not publish and organizes finite refinements into a compact genealogy space. In this way, three results are distinguished: the finite combinatorial identity, the cylindrical integral over the path limit, and the realization of that integral as the propagator of a particular physical system.

65.2.1 Route category and functorial weight

Let $\mathsf{P}_{\mathrm{TPK}}$ be the free category generated by the admissible TPK transitions. Its objects are enriched boundary states, and its morphisms are composable sequences of transitions. Composition is concatenation, and the empty path is the identity. The map

$$q_{\mathrm{vis}} : \mathsf{P}_{\mathrm{TPK}} \longrightarrow \mathsf{P}_X$$

publishes the observable route and preserves in its fiber the differences of sheet, orientation, carry, block, boundary, and memory.

Definition 65.1 (Genealogical weight)**.** Let $\mathcal{B}$ be a normed real algebra, possibly noncommutative, and let $\mathbf{B}_{\mathcal{B}}$ be the category with a single object $*$ whose endomorphism monoid is $\mathrm{End}(*) = (\mathcal{B}, \cdot, 1)$. A genealogical weight is a functor

$$W : \mathsf{P}_{\mathrm{TPK}} \longrightarrow \mathbf{B}_{\mathcal{B}}$$

such that

$$W(\mathrm{id}_x) = 1, \qquad W(\gamma_2 \circ \gamma_1) = W(\gamma_2)W(\gamma_1).$$

The product order agrees with the causal order of the route.

For enriched states $\widetilde{i}, \widetilde{f}$, the kernel at depth R is

$$\mathcal{K}_R(\widetilde{f},\widetilde{i}) = \sum_{\substack{\gamma\in \mathsf{P}_{\mathrm{TPK}}(\widetilde{i},\widetilde{f})\\ |\gamma|=R}} W(\gamma). \tag{65.2}$$

Two routes with the same visible word remain distinct summands when they belong to different genealogical fibers.

Proposition 65.2 (Composition through an enriched cut)**.** *If every route of length $R+S$ has a unique cut after R transitions, then*

$$\mathcal{K}_{R+S}(\widetilde{f},\widetilde{i}) = \sum_{\widetilde{x}} \mathcal{K}_S(\widetilde{f},\widetilde{x})\mathcal{K}_R(\widetilde{x},\widetilde{i}).$$

The formula preserves the causal order of the factors when $\mathcal{B}$ is noncommutative.

Proof. The set of complete routes is partitioned by the enriched state $\widetilde{x}$ reached at the cut. The functorial factorization of the weight turns the sum over each class into the product of the partial kernels. Uniqueness of the cut ensures that each route appears exactly once. □

65.2.2 Construction of the internal phase through the real action character on genealogical routes

Let (E, J_E) be a real Euclidean space equipped with an orthogonal endomorphism J_E such that $J_E^2 = -I$. If $\mathcal{A}_* : \mathsf{P}_{\mathrm{TPK}} \to \mathbb{R}$ is additive under concatenation, the phase

$$\Phi_J(\gamma) := \exp[-J_E\mathcal{A}_*(\gamma)] = \cos\mathcal{A}_*(\gamma)\, I - \sin\mathcal{A}_*(\gamma)\, J_E \tag{65.3}$$

is multiplicative. For a dimensional action $S_{\mathrm{int}}(\gamma) = \hbar_{\mathrm{int}}\mathcal{A}_*(\gamma)$, this expression internally realizes the phase $\exp[-J_E\, S_{\mathrm{int}}(\gamma)/\hbar_{\mathrm{int}}]$. The scalar symbol i appears only when a complex chart is chosen for the pair (E, J_E).

Proposition 65.3 (Phase functor)**.** *The map Φ_J satisfies*

$$\Phi_J(\mathrm{id}_x) = I, \qquad \Phi_J(\gamma_2\circ\gamma_1) = \Phi_J(\gamma_2)\Phi_J(\gamma_1).$$

Proof. The action of the empty path is zero. For composable routes, additivity of $\mathcal{A}_*$ and commutativity of multiples of the same endomorphism J_E allow the exponential of the summed value to be separated. □

65.2.3 Unitary realization on the chronological register 27×27

Before passing to the cylindrical limit, a finite kernel admits an explicit unitary realization. Let

$$\mathbb{T}^2_{27} := (\mathbb{Z}/27\mathbb{Z})^2, \qquad \mathcal{D} := \{\mathrm{N},\mathrm{S},\mathrm{E},\mathrm{W}\},$$

and consider the real space

$$\mathcal{H}_{27}^{\mathbb{R},\mathrm{reg}} := \ell^2(\mathbb{T}^2_{27};\mathbb{R}) \otimes \mathbb{R}^{\mathcal{D}} \otimes E.$$

The endomorphism J_E is extended as the identity on the position and direction factors, and the same notation is retained for that extension.

In the ordered basis $(\mathrm{N},\mathrm{S},\mathrm{E},\mathrm{W})$, the directional mixer is

$$C_4 := \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1\\ 1 & 1 & -1 & -1\\ 1 & -1 & 1 & -1\\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad C_4^{\mathsf{T}}C_4 = I_4. \tag{65.4}$$

Here C_4 denotes exclusively the Walsh–Hadamard mixer on the directional fiber; when it acts on $\mathcal{H}_{27}^{\mathbb{R},\mathrm{reg}}$ it is understood as $C_4 \otimes I_E$.

Fix a translational chart in which the action of an elementary edge γ_d depends only on its direction:

$$s_d := S_{\mathrm{int}}(\gamma_d) \in \mathcal{L}_S, \qquad a_d := \frac{s_d}{\hbar_{\mathrm{int}}} \in \mathbb{R}.$$

The real diagonal phase operator is

$$D_J := \mathrm{diag}\left(e^{-J_E a_{\mathrm{N}}}, e^{-J_E a_{\mathrm{S}}}, e^{-J_E a_{\mathrm{E}}}, e^{-J_E a_{\mathrm{W}}}\right).$$

The conditional shift transports each component in its direction:

$$\begin{aligned} (S\psi)(x,y,\mathrm{N}) &= \psi(x,y-1,\mathrm{N}), & (S\psi)(x,y,\mathrm{S}) &= \psi(x,y+1,\mathrm{S}),\\ (S\psi)(x,y,\mathrm{E}) &= \psi(x-1,y,\mathrm{E}), & (S\psi)(x,y,\mathrm{W}) &= \psi(x+1,y,\mathrm{W}), \end{aligned}$$

with all coordinates taken modulo 27. The elementary step and its composition of twelve steps are

$$U_{\mathrm{micro}} := S D_J C_4, \qquad U_{12}^{\mathrm{reg}} := U_{\mathrm{micro}}^{12}. \tag{65.5}$$

The superscript reg preserves the type: this operator acts on the directional torus 27×27 and is not identified with the other dodecaphasic objects of the construction or with the temporal endomorphism of order 108.

Proposition 65.4 (Unitarity and exact path sum in the register)**.** *The operator U_{micro} is orthogonal on the realification and unitary in the complex chart determined by J_E. In particular, U_{12}^{reg} is unitary. If $\Xi := \mathbb{T}_{27}^2 \times \mathcal{D}$, then, for every $n \geq 0$ and $\xi_i, \xi_f \in \Xi$,*

$$\left(U_{\mathrm{micro}}^n\right)_{\xi_f,\xi_i} = \sum_{\substack{\gamma:\xi_i \to \xi_f \\ |\gamma|=n}} W_{U_{\mathrm{micro}}}(\gamma),$$

where, for $\gamma = (\xi_0, \ldots, \xi_n)$,

$$W_{U_{\mathrm{micro}}}(\gamma) := (U_{\mathrm{micro}})_{\xi_n,\xi_{n-1}} \cdots (U_{\mathrm{micro}})_{\xi_1,\xi_0}.$$

The sum ranges over exactly the paths permitted by the nonzero blocks of the operator.

Proof. Since J_E is orthogonal and $J_E^2 = -I$, we have $J_E^{\mathsf{T}} = -J_E$. Thus, each block $e^{-J_E a_d}$ is an orthogonal rotation and D_J is orthogonal. The operator S permutes an orthonormal basis, and (65.4) proves that C_4 is orthogonal. Consequently,

$$U_{\mathrm{micro}}^{\mathsf{T}} U_{\mathrm{micro}} = C_4^{\mathsf{T}} D_J^{\mathsf{T}} S^{\mathsf{T}} S D_J C_4 = I.$$

All three factors commute with the extension of J_E; hence U_{micro} is complex-linear in the chart $J_E \leftrightarrow \mathrm{i}$, and its real orthogonality is equivalent to unitarity. Every power, in particular U_{12}^{reg}, preserves this property.

For $n \geq 1$, block multiplication gives

$$\left(U_{\mathrm{micro}}^n\right)_{\xi_f,\xi_i} = \sum_{\xi_1,\ldots,\xi_{n-1}} (U_{\mathrm{micro}})_{\xi_f,\xi_{n-1}} \cdots (U_{\mathrm{micro}})_{\xi_1,\xi_i}.$$

Each choice of intermediate states determines a path of length n, and each path determines a unique ordered product. For $n = 0$, the sole term is the empty path and its weight is the identity. □

Translation invariance decomposes the problem into blocks of four directional components, tensored with E. For

$$k_x = \frac{2\pi a}{27}, \qquad k_y = \frac{2\pi b}{27}, \qquad 0 \le a, b < 27,$$

we adopt the realified Fourier convention

$$\chi_{a,b}(x, y) := \exp[-J_E(k_x x + k_y y)].$$

Direct substitution into the conditional shift produces

$$S(k_x, k_y) = \operatorname{diag}\big(e^{J_E k_y}, e^{-J_E k_y}, e^{J_E k_x}, e^{-J_E k_x}\big),$$

and, since D_J and C_4 do not depend on position,

$$\begin{aligned} U_{\text{micro}}(k_x, k_y) &= S(k_x, k_y) D_J C_4, \\ U_{12}^{\text{reg}}(k_x, k_y) &= U_{\text{micro}}(k_x, k_y)^{12}. \end{aligned} \tag{65.6}$$

Consequently,

$$\operatorname{Spec} U_{12}^{\text{reg}} = \bigcup_{a,b=0}^{26} \operatorname{Spec} U_{12}^{\text{reg}} \left(\frac{2\pi a}{27}, \frac{2\pi b}{27} \right).$$

The matrix power enumerates paths, and the Fourier decomposition publishes the quasienergies of the same finite dynamics.

Fix an elementary duration $\delta t > 0$, and let $\Delta t_{12} := 12\,\delta t$. In each unitary block, choose a spectral branch $\mathfrak{b}$ of the logarithm. The operator $\operatorname{Log}_{\mathfrak{b}} U_{12}^{\text{reg}}(k_x, k_y)$ is skew-adjoint and commutes with J_E. Define

$$\begin{aligned} H_{\text{eff}}(k_x, k_y) &:= \frac{\hbar_{\text{int}}}{\Delta t_{12}} J_E \operatorname{Log}_{\mathfrak{b}} U_{12}^{\text{reg}}(k_x, k_y), \\ U_{12}^{\text{reg}}(k_x, k_y) &= \exp\left[-\frac{J_E \Delta t_{12}}{\hbar_{\text{int}}} H_{\text{eff}}(k_x, k_y) \right]. \end{aligned} \tag{65.7}$$

The first line is self-adjoint in the complex chart. The second follows from $-J_E^2 = I$ and proves that the exponential recovers the propagator exactly. The branch $\mathfrak{b}$ fixes the quasienergy zone.

This realization constructs a finite propagator, its exact path sum and its effective generator. Identifying it with a continuous physical Hamiltonian still requires verification of the representation, action and limit hypotheses stated below.

65.2.4 Cylindrical limit on the space of genealogies

With endpoints fixed, let $\mathcal{P}_n$ be finite sets of paths at resolution n, with surjective restriction maps. The space

$$\Omega_{fi}^{\text{path}} := \varprojlim_n \mathcal{P}_n$$

is compact and totally disconnected. A compatible family of probability measures μ_n on the cylinders determines a Borel measure μ on $\Omega_{fi}^{\text{path}}$.

Let Π_n be the partition into level-n cylinders. For $C \in \Pi_n$, choose a representative γ_C, an approximate action $S_n(C)$ and the weight $\mu_n(C)$. For $F : \Omega_{fi}^{\text{path}} \to \operatorname{End}(E)$, define

$$\mathcal{I}_n(F) := \sum_{C \in \Pi_n} \mu_n(C) F(\gamma_C) \exp\left[-J_E \frac{S_n(C)}{\hbar_{\text{int}}} \right]. \tag{65.8}$$

Theorem 65.5 (Genealogical path integral)**.** *Suppose that:*

1. *the cylinder measures μ_n are projectively compatible;*
2. *F is continuous and bounded;*
3. *there exists a continuous action $S : \Omega_{fi}^{\mathrm{path}} \to \mathcal{L}_S$ for which*

$$\max_{C\in\Pi_n} \sup_{\gamma\in C} \left\| \frac{S_n(C) - S(\gamma)}{\hbar_{\mathrm{int}}} \right\| \longrightarrow 0;$$

4. *the mesh of Π_n tends to zero.*

Then $\mathcal{I}_n(F)$ converges in operator norm, the limit is independent of the representatives, and it defines the Bochner integral

$$\int_{\Omega_{fi}^{\mathrm{path}}} F(\gamma) \exp\left[-J_E \frac{S(\gamma)}{\hbar_{\mathrm{int}}}\right] \mathrm{d}\mu(\gamma). \tag{65.9}$$

Proof. Compatibility of the finite measures constructs μ on the cylinder algebra and, by extension, on the Borel sigma-algebra. The function

$$G(\gamma) = F(\gamma) \exp[-J_E\, S(\gamma)/\hbar_{\mathrm{int}}]$$

is continuous on a compact space and is therefore uniformly continuous and bounded. Orthogonality of J_E gives the estimate

$$\|e^{-J_E a} - e^{-J_E b}\| \le |a - b|.$$

Uniform approximation of the action and contraction of the mesh make the cylindrical sums converge to the Bochner integral of G [13]. □

The theorem constructs a rigorous integral on the compact space of genealogies. Its realization as a Feynman oscillatory integral for a specific Hamiltonian is obtained when the finite kernels, action and cylinder measure satisfy these convergence hypotheses. The finite matrix expansion, genealogical integral and physical propagator are thus related by explicit morphisms, without being identified by abbreviation.

65.2.5 Spectral decomposition of the path measure and conditioning of futures

Let $\mathrm{Val}_{\mathrm{path}} : \Omega_{fi}^{\mathrm{path}} \to K \subset \mathbb{R}$ be a Borel-measurable path reader, where K carries its standard Borel structure, and let $\nu = (\mathrm{Val}_{\mathrm{path}})_*\mu$. For every function factoring through the value, $F = \widehat{F} \circ \mathrm{Val}_{\mathrm{path}}$, we have

$$\int_{\Omega_{fi}^{\mathrm{path}}} F \,\mathrm{d}\mu = \int_K \widehat{F}(x)\,\mathrm{d}\nu(x).$$

Since $\Omega_{fi}^{\mathrm{path}}$ is compact metrizable and μ is a Borel probability measure, there exists a family of regular conditional probabilities $(\mu_x)_{x\in K}$, defined for ν-almost every x. For a general genealogical integrand, the disintegration of μ takes the form

$$\int_{\Omega_{fi}^{\mathrm{path}}} G(\gamma)\,\mathrm{d}\mu(\gamma) = \int_K \left[\int_{\mathrm{Val}_{\mathrm{path}}^{-1}(x)} G(\gamma)\,\mathrm{d}\mu_x(\gamma) \right] \mathrm{d}\nu(x). \tag{65.10}$$

The inner integral preserves the distinct histories that publish the same coordinate. This is the precise link between the sum over all paths, real evaluation, and the thesis that the continuum is a projection of a richer genealogical domain.

Domain of the genealogical path integral: cylindrical compactness, projective consistency, kernel convergence and disintegration over fibers The finite expansion, unitary realization on $\mathbb{T}^2_{27}$ and its

Fourier reduction are exact on their finite domains. The construction of H_{eff} is exact once the complex structure, elementary translational action, duration and spectral branch have been declared; it does not by itself select a continuous physical Hamiltonian. The enriched route category and internal phase precede the cylindrical assembly. theorem 65.5 is exact under the four declared premises. Realizing a specific continuous quantum model requires verifying those premises for its action and Hamiltonian.

65.3 Cylindrical Concentration of the Genealogical Measure of Trajectories and Realization of the Dirac Kernel

65.3.1 Projective limit of genealogies and domain of the path integral

Let

$$X = (\mathbb{Z}/9\mathbb{Z})^{\mathbb{N}}$$

with the product topology and the uniform measure μ. For $x = (x_1, x_2, \ldots)$, the cylinder of depth n is

$$C_n(x) = \{y \in X : y_j = x_j \text{ for } 1 \le j \le n\}.$$

Cylinders are simultaneously open and closed, form a base and satisfy

$$\mu(C_n(x)) = 9^{-n}, \qquad C_{n+1}(x) \subset C_n(x), \qquad \bigcap_n C_n(x) = \{x\}.$$

Adding a digit does not create a new number: it restricts the cylinder of admissible prolongations. This statement gives topological content to the genealogical thesis of number.

65.3.2 Inverse limit of the nonadic cylinders and stability of the Dirac kernel

Theorem 65.6 (Unitary nonadic concentration)**.** *Define*

$$f_{n,x} = 9^n \mathbf{1}_{C_n(x)}.$$

Then

$$\mu(C_n(x)) \to 0, \qquad f_{n,x}(x) \to \infty, \qquad \int_X f_{n,x}\, \mathrm{d}\mu = 1,$$

and the measures $f_{n,x}\mu$ converge weak- to δ_x.*

Proof. The first identity is immediate. The height of $f_{n,x}$ over its support is 9^n, so it diverges. The integral equals $9^n 9^{-n} = 1$. For $g \in C(X)$, the oscillation of g in $C_n(x)$ tends to zero by uniform continuity; therefore

$$\left| \int g f_{n,x}\, \mathrm{d}\mu - g(x) \right| \le \sup_{y \in C_n(x)} |g(y) - g(x)| \longrightarrow 0.$$

This is weak-* convergence to δ_x. □

The theorem does not assert $1/0 = \infty$. It exhibits three evaluations of the same object: support measure, density and total mass. In blocks of three digits,

$$\mu(C_{3m}) = 729^{-m}, \qquad f_{3m} = 729^m \mathbf{1}_{C_{3m}}, \qquad 729^{-m} 729^m = 1.$$

The evolutionary conservation of unity here assumes an exact analytical form.

65.3.3 Kronecker Kernel and Realization of the Dirac Distribution in the Genealogical Register

At the finite level $X_n = (\mathbb{Z}/9\mathbb{Z})^n$, let

$$K_n(x, y) = 9^n \mathbf{1}_{\{x=y\}}.$$

With the uniform measure, the operator

$$(E_n f)(x) = \sum_{y \in X_n} K_n(x, y) f(y) 9^{-n}$$

is the identity. In the cylindrical limit, the condition $x = y$ is replaced by "same prefix of length n" and E_n is the conditional expectation on the cylindrical algebra. One has

$$E_n^2 = E_n, \qquad E_n^* = E_n, \qquad E_n f \to f \quad \text{in } L^2(X, \mu).$$

The exact bridge is, therefore,

$$\delta_{ij} \longrightarrow K_n(x, y) \longrightarrow \delta_x.$$

The Kronecker delta is a matrix kernel; the Dirac delta is a measure or distribution. Neither is the Dirac operator $D = \gamma^\mu \partial_\mu$. The transition to spinors and Clifford requires a second intertwiner and will be treated later.

65.3.4 Path Measure Scaling: Effective Dimension and Density

Suppose that a contraction of ratio λ acts independently in d directions and that the product measure scales as λ^d. To preserve unit mass, the density must scale as

$$\rho \longmapsto \lambda^{-d} \rho.$$

The exponents $(-1, -2, -3)$ then appear as density weights on a line, area, and volume. This proposition is general and exact under the product hypothesis. It does not permit λ to be identified automatically with α_{HMT}; that identification requires a metric chart.

65.3.5 Monodromy of the Logarithm and Computability of the Genealogical Amplitude

The existence of an infinite branch in a finitely branching tree may follow from König's lemma, but König's lemma does not furnish a computable selector. There are infinite recursive binary trees with no recursive branch. The joint nonadic connection does not rest on that abstract existence: it has a certified uniform selector that generates to arbitrary finite depth and opens the decimal witnesses only after generation.

Three prohibited confusions

1/0 shall not be written; the Dirac delta shall not be represented as an ordinary function with "an infinite one"; König shall not be used as an algorithm. The strong result is the exact compatibility among cylinders, unit density, and the weak limit, together with the constructive selector independent of decimal comparison.

65.4 Genealogical Path Algebra and Feynman Representation

Feynman's formulation assigns an amplitude to each possible history and sums those amplitudes to obtain the propagator. In HMT, the word *history* has content prior to amplitude: a route preserves position, sheet, orientation, phase, carry, boundary, memory, and survival condition. The sum over paths is not added to that system from outside. It appears upon linearly representing its category of routes [48, 49].

The development has three linked mathematical levels. The expansion of a matrix power as a finite sum of paths has already been proved in Dimensional Mechanics. The ledger 27×27 provides a unitary realization with shift, mixer, and phase, diagonalizable by Fourier transform. The family of finite sums subsequently admits a cylindrical limit when the measure, the action, and the refinement satisfy the hypotheses of the convergence theorem. The combinatorial identity, its spectral realization, and its functional extension are thereby brought together.

65.4.1 The enriched category of paths

Let X be a finite set of observable states, and let $\mathsf{P}_{\mathrm{TPK}}$ be the free category generated by the admissible TPK transitions. Denote $\widetilde{X} = \mathrm{Ob}(\mathsf{P}_{\mathrm{TPK}})$. Its objects are enriched boundary states. A morphism $\gamma : \widetilde{x} \to \widetilde{y}$ is a composable sequence of enriched transitions. Composition is concatenation, and the identity is the empty path.

The visible projection

$$q_{\mathrm{vis}} : \mathsf{P}_{\mathrm{TPK}} \longrightarrow \mathsf{P}_X$$

forgets part of the fiber. Two different morphisms may have the same observable word. This lack of faithfulness is essential: a sum that aggregated too early over P_X would lose precisely the information capable of distinguishing incompatible futures.

Definition 65.7 (Genealogical weight)**.** Let $\mathcal{B}$ be a real normed algebra, not necessarily commutative, and let $B(\mathcal{B}, \cdot, 1)$ be the one-object category whose endomorphism monoid is $(\mathcal{B}, \cdot, 1)$. A genealogical weight is a functor

$$W : \mathsf{P}_{\mathrm{TPK}} \longrightarrow B(\mathcal{B}, \cdot, 1)$$

which satisfies

$$W(\mathrm{id}_x) = 1_{\mathcal{B}}, \qquad W(\gamma_2 \circ \gamma_1) = W(\gamma_2) W(\gamma_1).$$

The order of the product follows the causal order of the route.

The weight may contain a fibre transport, a positive penalty, an action, a ledger character, or several of those factors. The definition does not conflate those functions: each component retains its domain and composition rule.

65.4.2 The exact finite sum

Let E_x be the linear fiber over $x \in X$ and $\mathcal{H}_X = \bigoplus_{x \in X} E_x$. Let $U = (U_{yx})_{x,y \in X}$ be the global block operator, with $U_{yx} : E_x \to E_y$. We interpret U_{yx} as elementary transport from x to y. For a visible path $\gamma = (x_0, x_1, \ldots, x_R)$, we define

$$W_U(\gamma) = U_{x_R x_{R-1}} \cdots U_{x_1 x_0}.$$

Theorem 65.8 (Finite Path Expansion)**.** *For every $R \geq 1$,*

$$(U^R)_{fi} = \sum_{\substack{\gamma : i \to f \\ |\gamma| = R}} W_U(\gamma). \tag{65.11}$$

If the global operator $U : \mathcal{H}_X \to \mathcal{H}_X$ is orthogonal —or unitary after choosing a complex chart—, the evolution $\psi_R = U^R \psi_0$ preserves the norm.

Proof. The entry (f, i) of the product U^R is

$$\sum_{x_1,\dots,x_{R-1}} U_{f x_{R-1}} U_{x_{R-1} x_{R-2}} \cdots U_{x_1 i}.$$

Each choice of intermediate indices determines a unique path of length R, and each path determines a unique ordered monomial. Norm conservation is an immediate consequence of $(U^R)^{\mathsf{T}} U^R = I$, or of its unitary version. □

The formula (65.11) is the combinatorial identity underlying the sum over histories. Its HMT content appears when each visible path is refined by the fibers of q_{vis}:

$$\mathcal{K}_R(\widetilde{f}, \widetilde{i}) = \sum_{\substack{\gamma \in \mathsf{P}_{\mathrm{TPK}}(\widetilde{i}, \widetilde{f}) \\ |\gamma| = R}} W(\gamma). \tag{65.12}$$

In that sum, two published routes with the same endpoints remain distinct terms whenever their genealogies differ.

65.4.3 Unitary realization on the genealogical record 27×27

The historical source develops a quantum walk on the torus

$$\Lambda_{27} = (\mathbb{Z}/27\mathbb{Z})^2$$

with four cardinal orientations $\mathcal{D} = \{N, S, E, W\}$. Its real phase space is

$$\mathcal{H}_{27}^{\mathbb{R}} = \ell^2(\Lambda_{27}) \otimes \mathbb{R}^4 \otimes E,$$

where (E, J_E) internally realizes the complex structure. The direction mixer is the Walsh–Hadamard matrix

$$C_4 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad C_4^{\mathsf{T}} C_4 = I_4.$$

An oriented elementary action $s_d \in \mathcal{L}_S$ defines the real phase gate

$$D_J = \operatorname{diag}\left(e^{-J_E s_N/\hbar_{\mathrm{int}}}, e^{-J_E s_S/\hbar_{\mathrm{int}}}, e^{-J_E s_E/\hbar_{\mathrm{int}}}, e^{-J_E s_W/\hbar_{\mathrm{int}}} \right).$$

The conditioned shift transports each component in its direction:

$$\begin{aligned} (S\psi)(x, y, N) &= \psi(x, y-1, N), & (S\psi)(x, y, S) &= \psi(x, y+1, S), \\ (S\psi)(x, y, E) &= \psi(x-1, y, E), & (S\psi)(x, y, W) &= \psi(x+1, y, W), \end{aligned}$$

with coordinates modulo 27. The elementary step and the dodecaphase composition are

$$U_{\mathrm{micro}} = S D_J C_4, \qquad U_{12} = U_{\mathrm{micro}}^{12}. \tag{65.13}$$

Proposition 65.9 (Unitarity and exact sum in the ledger)**.** *The operator U_{micro} is orthogonal on the realification and unitary in the complex chart. For every n, each entry of U_{micro}^n is the ordered sum of the weights of all paths of n steps joining its endpoint states.*

Proof. The shift is a permutation, C_4 is orthogonal, and each block $e^{-J_E s_d/\hbar_{\mathrm{int}}}$ is a rotation. Their product is orthogonal. The second assertion is the theorem 65.8 applied to the block matrix U_{micro}. □

Translational symmetry reduces the spectral problem to blocks of size four. For

$$k_x = \frac{2\pi m}{27}, \qquad k_y = \frac{2\pi n}{27}, \qquad 0 \le m, n < 27,$$

the shift diagonalizes as

$$S(k_x, k_y) = \operatorname{diag}\big(e^{J_E k_y}, e^{-J_E k_y}, e^{J_E k_x}, e^{-J_E k_x}\big),$$

and

$$U_{\text{micro}}(k_x, k_y) = S(k_x, k_y) D_J C_4, \qquad U_{12}(k_x, k_y) = U_{\text{micro}}(k_x, k_y)^{12}. \tag{65.14}$$

The global spectrum is the union of the spectra of the 27^2 blocks. The expression combines two representations of the same dynamics: the matrix power enumerates paths, and the Fourier transform publishes quasienergies.

In each unit block a spectral logarithm can be chosen and defined

$$H_{\text{eff}}(k_x, k_y) = \frac{\hbar_{\text{int}}}{J_E\, \Delta t} \log U_{12}(k_x, k_y),$$

understood through functional calculus in the complex chart or its realification. The choice of branch fixes the quasienergy zone. The effective Hamiltonian thus arises from the propagator already constructed; its exponential recovers U_{12} in each mode.

65.4.4 Real action character

Let $\mathcal{A}_* : \mathsf{P}_{\text{TPK}} \to \mathbb{R}_{\ge 0}$ be the additive action cocycle of the HMT–MD source. The gauge is

$$\lambda_* = 1 - \frac{5}{468} - \frac{3}{11}\Delta_4 = 0.9892859130191415\ldots \in (0, 1).$$

The local clause H_{loc} assigns unit cost to a turn, cost λ_* to a change of sheet, and adds both when they occur on the same edge. Under this clause,

$$\mathcal{A}_*(\gamma) = \#\,\text{turns}(\gamma) + \lambda_* \#\,\text{sheet-switches}(\gamma), \qquad \mathcal{A}_*(\gamma_2 \circ \gamma_1) = \mathcal{A}_*(\gamma_2) + \mathcal{A}_*(\gamma_1).$$

The associated dimensional action is

$$S_{\text{int}}(\gamma) = \hbar_{\text{int}} \mathcal{A}_*(\gamma).$$

In the real phase module (E, J_E) of the preceding chapter, with E a finite-dimensional real Euclidean space, define

$$\Phi_J(\gamma) := \exp\!\left(-J_E \frac{S_{\text{int}}(\gamma)}{\hbar_{\text{int}}}\right) = \cos \mathcal{A}_*(\gamma)\, I - \sin \mathcal{A}_*(\gamma)\, J_E. \tag{65.15}$$

Proposition 65.10 (Multiplicativity of the internal phase)**.** *The map Φ_J is a functor from P_{TPK} to the orthogonal group generated by J_E:*

$$\Phi_J(\mathrm{id}_x) = I, \qquad \Phi_J(\gamma_2 \circ \gamma_1) = \Phi_J(\gamma_2)\Phi_J(\gamma_1).$$

Proof. The first equality uses $\mathcal{A}_*(\mathrm{id}_x) = 0$. For the second, the additivity of $\mathcal{A}_*$ and the commutation of multiples of the same endomorphism J_E give

$$e^{-J_E(\mathcal{A}_*(\gamma_2)+\mathcal{A}_*(\gamma_1))} = e^{-J_E \mathcal{A}_*(\gamma_2)} e^{-J_E \mathcal{A}_*(\gamma_1)}.$$

□

Now let $T(e) \in \mathrm{End}(E)$ be the bounded fiber transport of an edge, and suppose that it preserves J_E. The weight

$$W_J(\gamma) = \overleftarrow{\prod_{e\in\gamma}} T(e) \exp\left(-J_E \frac{s(e)}{\hbar_{\mathrm{int}}}\right)$$

realizes, entirely on E, the Hamiltonian convention $\exp(-iS_H[\gamma]/\hbar)$. The usual Feynman Lagrangian convention $\exp(+iS_L[\gamma]/\hbar)$ is obtained by taking $S_L = -S_H$ or, equivalently, by introducing an orientation $\varepsilon \in \{-1, +1\}$ in $\exp(\varepsilon J_E S/\hbar_{\mathrm{int}})$. The scalar symbol i is the complex chart of the algebraic operation performed by J_E. The positive cost $\mathcal{A}_*$ becomes physical action upon application of the corresponding realization map.

65.4.5 Composition of propagators over concatenated paths and amplitude law

The genealogy preserves the composition law that makes the kernel dynamic.

Proposition 65.11 (Convolution and Temporal Truncation)**.** *Suppose that $\widetilde{X}$ is finite; more generally, suppose that the following family of products is absolutely summable. If every route of length $R+S$ has a unique cut after R steps, then*

$$\mathcal{K}_{R+S}(\widetilde{f},\widetilde{i}) = \sum_{\widetilde{x}\in\widetilde{X}} \mathcal{K}_S(\widetilde{f},\widetilde{x})\mathcal{K}_R(\widetilde{x},\widetilde{i}). \tag{65.16}$$

The identity remains valid with noncommutative weights if the causal order of the factors is preserved.

Proof. We partition the set of routes by the enriched state $\widetilde{x}$ reached at the cut. Functorial factorization of the weight transforms the sum over complete routes into the product of the two partial sums, with the subsequent segment on the left. □

The cut on the visible state would be valid only if weight and composition descended faithfully through q_{vis}. In the presence of memory, two segments with the same visible coordinate may belong to incompatible fibers; hence convolution is performed over $\widetilde{X}$.

This equation distinguishes the sum over paths from a collection of numerical coincidences. The propagator has a semigroup, endpoints, composition, and a refinement law.

65.4.6 Cylindrical limit of the finite system and construction of the corresponding integral

A functional integral requires control of the path space and its refinement. For enriched endpoints $\widetilde{i}, \widetilde{f}$, let $\mathcal{P}_n(\widetilde{i}, \widetilde{f})$ be the finite sets of admissible paths at resolution n, with surjective restrictions, and define

$$\Omega_{fi}^{\mathrm{path}} := \varprojlim_n \mathcal{P}_n(\widetilde{i}, \widetilde{f}).$$

It is the compact space of path genealogies with those boundary data. Let $\{\Pi_n\}_{n\geq 0}$ be a sequence of cylindrical partitions of $\Omega_{fi}^{\mathrm{path}}$, with Π_{n+1} refining Π_n. For each cylinder $C \in \Pi_n$, let us choose a representative γ_C, a projective weight $\mu_n(C)$, and an approximate action $S_n(C)$.

We define the simple operator

$$\mathcal{I}_n(F) := \sum_{C\in\Pi_n} \mu_n(C) F(\gamma_C) \exp\left(-J_E \frac{S_n(C)}{\hbar_{\mathrm{int}}}\right). \tag{65.17}$$

Theorem 65.12 (Genealogical Limit of Cylindrical Integrals)**.** *Suppose that:*

1. *the cylindrical masses are projectively compatible;*
2. *$F : \Omega_{fi}^{\mathrm{path}} \to \mathrm{End}(E)$ is continuous and bounded;*

3. *there exists a continuous function* $S : \Omega_{fi}^{\mathrm{path}} \to \mathcal{L}_S$ *such that*

$$\max_{C\in\Pi_n} \sup_{\gamma\in C} \left\| \frac{S_n(C) - S(\gamma)}{\hbar_{\mathrm{int}}} \right\| \longrightarrow 0;$$

4. *the mesh of the partitions,* $\mathrm{mesh}(\Pi_n) = \max_{C\in\Pi_n} \mathrm{diam}_{d_\vartheta} C$*, tends to zero.*

Since E *is finite-dimensional,* $\mathrm{End}(E)$*, endowed with the operator norm, is a Banach space; the following integral is therefore a well-defined Bochner integral [13]. Then* $\mathcal{I}_n(F)$ *converges in operator norm, and its limit does not depend on the representatives. We write*

$$\int_{\Omega_{fi}^{\mathrm{path}}} F(\gamma) \exp\!\left(-J_E \frac{S(\gamma)}{\hbar_{\mathrm{int}}}\right) \mathrm{d}\mu(\gamma) := \lim_{n\to\infty} \mathcal{I}_n(F). \tag{65.18}$$

Proof. Projective compatibility constructs the cylinder measure μ. The function

$$G(\gamma) = F(\gamma) \exp(-J_E S(\gamma)/\hbar_{\mathrm{int}})$$

is continuous on the compact set $\Omega_{fi}^{\mathrm{path}}$, hence uniformly continuous and bounded. Let

$$G_n(C) = F(\gamma_C) \exp(-J_E S_n(C)/\hbar_{\mathrm{int}}) \,.$$

Since J_E is an orthogonal structure,

$$\|e^{-J_E a} - e^{-J_E b}\| \le |a - b|.$$

The third hypothesis implies that $G_n(C) - G(\gamma_C)$ tends uniformly to zero. The fourth hypothesis and the uniform continuity of G make the sums with $G(\gamma_C)$ convergent Riemann sums independent of the representatives. The limit coincides with the Bochner integral of G with respect to μ. More precisely, if

$$\epsilon_n = \max_{C\in\Pi_n} \sup_{\gamma\in C} \left\| \frac{S_n(C) - S(\gamma)}{\hbar_{\mathrm{int}}} \right\|,$$

then

$$\left\| \mathcal{I}_n(F) - \int G \, \mathrm{d}\mu \right\| \le \omega_F\big(\mathrm{mesh}(\Pi_n)\big) + \|F\|_\infty \epsilon_n,$$

because the integral decomposes over the cylinders and the total mass is one. Both terms tend to zero by hypothesis. □

Corollary 65.13 (Convergence of a family of discrete kernels)**.** *In addition to the hypotheses of theorem 65.12, suppose that there exists a cofinal sequence of depths* R_n *and simple functions* $F_{fi,n}$ *such that*

$$\mathcal{K}_{R_n}(\widetilde{f},\widetilde{i}) = \mathcal{I}_n(F_{fi,n}), \qquad F_{fi,n} \longrightarrow F_{fi}$$

uniformly. Then the kernels $\mathcal{K}_{R_n}(\widetilde{f},\widetilde{i})$ *converge to the genealogical integral of* F_{fi}*.*

Proof. The difference between $\mathcal{I}_n(F_{fi,n})$ and $\mathcal{I}_n(F_{fi})$ is bounded by $\|F_{fi,n} - F_{fi}\|_\infty$, because μ_n has total mass one and the phase is orthogonal. The second term converges by theorem 65.12. □

The theorem produces a rigorous integral over the compact genealogical space. The corollary makes explicit the bridge that identifies each matrix kernel with a normalized cylindrical sum. The real-time Feynman oscillatory integral appears when a physical representation and a refinement whose discrete action approximates the action of the system are chosen. That identification requires proof of convergence of the corresponding propagator; it is not granted by the mere existence of μ.

65.4.7 Integral visible and integral conditioned by memory

A real path reader $\mathrm{Val}_{\mathrm{path}} : \Omega_{fi}^{\mathrm{path}} \to K \subset \mathbb{R}$ induces the visible measure $\nu = (\mathrm{Val}_{\mathrm{path}})_*\mu$. For a function that factors through the value, $F = \widehat{F} \circ \mathrm{Val}_{\mathrm{path}}$,

$$\int_{\Omega_{fi}^{\mathrm{path}}} F \, \mathrm{d}\mu = \int_K \widehat{F}(x) \, \mathrm{d}\nu(x).$$

The phase or transport, however, may fail to factor through $\mathrm{Val}_{\mathrm{path}}$. The disintegration of μ over the value fibers expresses the difference:

$$\int_{\Omega_{fi}^{\mathrm{path}}} G(\gamma) \, \mathrm{d}\mu(\gamma) = \int_K \left[\int_{\mathrm{Val}_{\mathrm{path}}^{-1}(x)} G(\gamma) \, \mathrm{d}\mu_x(\gamma) \right] \mathrm{d}\nu(x). \tag{65.19}$$

The inner integral preserves the histories that publish the same coordinate. This is the mathematical form of integration with respect to values without reducing the number to its digit.

65.4.8 Action, interference, and persistence

The genealogical sum simultaneously maintains three operations:

$$\text{route admissibility} \longrightarrow \text{weight/action} \longrightarrow \text{interference in the representation.}$$

The first belongs to TPK and its survival law. The second belongs to the cocycle and dimensional transport. The third belongs to the phase module and the physical reader. No observed magnitude retrospectively selects the route that produces it.

Result

The finite Feynman sum is obtained as an exact linear representation of the enriched category of routes. The complex structure can be realized by a real endomorphism $J_E^2 = -I$. Under compatibility of measures, contraction of cylinders, and uniform convergence of actions, the cylindrical sums define a genealogical integral. The finite kernels converge to it when they additionally satisfy the bridge of the corollary 65.13.

Hypothesis, domain, and conclusion

The finite matrix expansion, the action cocycle, and the real realizations of the complex structure are recovered results. Their assembly in (65.15) and (65.12) is an exact new formalization. The theorem 65.12 is exact under its premises. Equivalence with a concrete continuous quantum model requires verifying those premises for the Hamiltonian and the action of that model.

The premises of the theorem identify exactly the structure that passes from the finite ledger to the function space: projective compatibility of the weights, contraction of the mesh, uniform convergence of the action, and correspondence between kernels and cylindrical sums. In the realization (65.13), composition and unitarity are algebraic identities; as the ledger is refined, the remaining premises determine convergence to the chosen continuum propagator.

Chapter 66

Genealogical theory of quantum measurement: reversible coupling, instruments, conditioning of futures, and spectral update

Measurement is formulated as a reversible coupling between a section and a context, followed by spectral publication. The induced instruments preserve the genealogy required to condition future events without turning the outcome into a unique retrospective history.

66.1 Conditioning of sections and quantum updating

The preceding general theory places measurement in the joint state of system, reader, and apparatus. The quantum realization adds a precise correspondence among three levels: restriction of the section space, spectral projection, and the completely positive instrument. Joint evolution belongs to coupling; publication of the outcome belongs to the reader; updating belongs to conditioning on the read fiber.

66.1.1 Measurement context $\mathfrak{M}_a = (U_a, q_a, C_a, m_0)$, output map, and conditioned fiber $\Omega_{a,y}$

Let (Ω_S, Σ_S) and (Ω_M, Σ_M) be the measurable spaces of genealogical states of the system and apparatus. A context a is specified by

$$\mathfrak{M}_a = (U_a, q_a, C_a, m_0), \tag{66.1}$$

where $m_0 \in \Omega_M$ is the preparation of the apparatus,

$$U_a : \Omega_S \times \Omega_M \longrightarrow \Omega_S \times \Omega_M$$

is a measurable coupling, (Y_a, Σ_{Y_a}) is the measurable space of outcomes, $q_a : \Omega_M \to Y_a$ is a measurable reader of the pointer, and C_a declares compatibility and the horizon of prolongation. When a reversible regime is invoked, it will also be required that U_a be bijective and bimeasurable; in the Hilbert realization, the corresponding condition is the unitarity of $\widehat{U}_a$. The output is

$$\mathrm{Out}_a(x) = q_a\big(\mathrm{pr}_M\, U_a(x, m_0)\big)\,. \tag{66.2}$$

Let $\Omega_a \in \Sigma_S$ be the set of sections compatible with the preparation and the context. The map Out_a is measurable by composition. For $y \in Y_a$ with $\{y\} \in \Sigma_{Y_a}$, define the measurable fiber

$$\Omega_{a,y} := \{x \in \Omega_a : \mathrm{Out}_a(x) = y\}. \tag{66.3}$$

Definition 66.1 (Genealogical Update)**.** The update determined by the result y is the restriction

$$\Omega_a \longmapsto \Omega_{a,y}.$$

At finite depth it preserves exactly the prefixes that admit some extension in $\Omega_{a,y}$.

For a prefix $s \in S_n$, the compatible futures subsequent to the reading form

$$\mathrm{Fut}_{a,y}(s) = \{x \in \Omega_{a,y} : x|_n = s\}.$$

The update modifies the prolongation condition and preserves the already traversed prefix together with the coupling record.

Proposition 66.2 (Conditional probability over the fiber)**.** *Let μ_a be a probability measure on the restricted measurable space $(\Omega_a, \Sigma_S|_{\Omega_a})$, let $B \in \Sigma_S|_{\Omega_a}$, and suppose $\mu_a(\Omega_{a,y}) > 0$. Then*

$$\mu_{a,y}(B) = \frac{\mu_a(B \cap \Omega_{a,y})}{\mu_a(\Omega_{a,y})} \tag{66.4}$$

defines a probability concentrated on the fiber of the result: $\mu_{a,y}(\Omega_{a,y}) = 1$.

Proof. The map is nonnegative and countably additive by the properties of μ_a. Normalization follows by substituting $B = \Omega_{a,y}$ into (66.4). □

66.1.2 Spectral realization of the device

Let $\{|r\rangle\}$ be an orthonormal basis adapted to the system observable, let $|m_0\rangle$ be the apparatus preparation, and let $\{|m_r\rangle\}$ be an orthonormal family of pointer states. The unitary representation of the coupling satisfies

$$\widehat{U}_a(|r\rangle \otimes |m_0\rangle) = |r\rangle \otimes |m_r\rangle. \tag{66.5}$$

For $|\psi\rangle = \sum_r c_r|r\rangle$, linearity produces

$$\widehat{U}_a(|\psi\rangle \otimes |m_0\rangle) = \sum_r c_r|r\rangle \otimes |m_r\rangle.$$

The coupling thus constructs a correlation between system and pointer. The selective reading of the outcome r is represented, for a density state with $\operatorname{tr}(P_r\rho) > 0$, by the Luders [83] rule,

$$\rho \longmapsto \rho_r = \frac{P_r\rho P_r}{\operatorname{tr}(P_r\rho)}. \tag{66.6}$$

Theorem 66.3 (Correspondence between fibers and projections)**.** *Suppose that each fiber $\Omega_{a,y}$ is a finite union of disjoint cylinders of a common depth n, that the realization Q carries those cylinders to orthogonal subspaces, and that P_y is the sum of their cylinder projectors. Let $\mu_a(\Omega_{a,y}) > 0$, and let $\Psi_n(\mu_a)$ be the state constructed by (64.4) from that same measure. Then*

$$\mu_a(\Omega_{a,y}) = \|P_y\Psi_n(\mu_a)\|^2,$$

and the restriction to $\Omega_{a,y}$ is represented by the normalized projection $P_y\Psi_n(\mu_a)/\|P_y\Psi_n(\mu_a)\|$. In the density-matrix chart, if $\operatorname{tr}(\rho P_y) > 0$, let $\mathcal{A}_n$ be the algebra of cylindrical events of depth n, and suppose that $B \mapsto P_B$ and P_y belong to the same PVM and that

$$\mu_a(B) = \operatorname{tr}(\rho P_B) \qquad \text{for every } B \in \mathcal{A}_n. \tag{66.7}$$

Then the same conditional probability is represented by (66.6): for every $B \in \mathcal{A}_n$,

$$\mu_{a,y}(B) = \operatorname{tr}(\rho_y P_B).$$

Proof. The fiber is a disjoint union of cylinders, and P_y is the orthogonal sum of their projectors. The theorem 64.5 identifies the quadratic norm of each component with its measure; additivity gives the equality. After projection, the coefficients of each cylinder are divided by $\sqrt{\mu_a(\Omega_{a,y})}$; their squares are therefore the conditional probabilities of (66.4). In the density chart, membership in the same PVM gives $P_BP_y = P_{B\cap\Omega_{a,y}}$. By cyclicity of the trace and by (66.7),

$$\operatorname{tr}(\rho_y P_B) = \frac{\operatorname{tr}(\rho P_{B\cap\Omega_{a,y}})}{\operatorname{tr}(\rho P_y)} = \frac{\mu_a(B \cap \Omega_{a,y})}{\mu_a(\Omega_{a,y})}.$$

□

66.1.3 Instruments induced by coupling

For a refined pointer basis $\{|m_{y,\alpha}\rangle\}_{y,\alpha}$, the operators

$$M_{y,\alpha} = \langle m_{y,\alpha}|\widehat{U}_a|m_0\rangle, \qquad \sum_{y,\alpha} M_{y,\alpha}^\dagger M_{y,\alpha} = I \tag{66.8}$$

form a Kraus representation [76]. The effect $E_y = \sum_\alpha M_{y,\alpha}^\dagger M_{y,\alpha}$ determines the probability and, when $p(y \mid a, \rho) > 0$, the posterior state:

$$p(y \mid a, \rho) = \operatorname{tr}(\rho E_y), \qquad \rho_{a,y} = \frac{\sum_\alpha M_{y,\alpha}\rho M_{y,\alpha}^\dagger}{p(y \mid a, \rho)}. \tag{66.9}$$

The non-selective transformation is

$$\mathcal{E}_a(\rho) = \sum_{y,\alpha} M_{y,\alpha}\rho M_{y,\alpha}^\dagger.$$

Proposition 66.4 (Unitary coupling and reduced channel)**.** *The joint dynamics $\widehat{U}_a$ and the preparation $|m_0\rangle$ induce a completely positive trace-preserving channel on the system. The pointer decomposition induces the instrument $\mathcal{I}_{a,y}(\rho) = \sum_\alpha M_{y,\alpha}\rho M_{y,\alpha}^\dagger$. Conversely, every completely positive channel admits a dilation by an auxiliary space and an isometry.*

Proof. The completeness identity in (66.8) preserves the trace, and the Kraus form proves complete positivity. The converse assertion is the Stinespring dilation theorem [116]. □

The auxiliary space of the dilation can linearly realize the memory and register transported by TPK once an embedding that intertwines that memory with the auxiliary space has been declared. Stinespring guarantees the dilation of the channel; it does not by itself identify the two objects. The Lueders rule is the fine projective case $M_{y,1} = P_y$; degenerate readers and nonideal apparatus are described by means of families $M_{y,\alpha}$.

66.1.4 Internal Observer and Completeness of Reading

A realized observer is a subsystem capable of preparing a context, coupling to a boundary, preserving a distinguishable record, and using it as a condition of prolongation. The sequence

$$(x, m_0) \xrightarrow{U_a} (x', m') \xrightarrow{q_a} y \xrightarrow{\text{cond}} \Omega_{a,y} \tag{66.10}$$

places the observer's reading frame and memory within the extended state.

The Ambivalence principle completes this description. The areal and volumetric readings produce quotients q_A, q_V, whereas memory T preserves transport between them. When

$$\Xi : x \longmapsto (q_A(x), q_V(x), T(x))$$

is injective modulo the declared gauge group, the pair of observables and the memory reconstruct the class of the complete state.

Theorem 66.5 (Genealogical Decomposition of Measurement)**.** *Under the hypotheses of theorem 66.3 y proposition 66.4, the transition between evolution and outcome decomposes, in the regime in which U_a is bijective and bimeasurable or $\widehat{U}_a$ is unitary, as follows:*

$$\textit{reversible coupling} \longrightarrow \textit{pointer reading} \longrightarrow \textit{section conditioning}.$$

This decomposition possesses two regimes:

1. *In a projective cylindrical measurement, $M_{y,1} = P_y$ and the fiber restriction, normalized projection, and Luders update coincide:*

$$\Omega_a \longmapsto \Omega_{a,y}, \qquad \Psi_n \longmapsto \frac{P_y \Psi_n}{\|P_y \Psi_n\|}, \qquad \rho \longmapsto \frac{P_y \rho P_y}{\mathrm{tr}(\rho P_y)}.$$

2. *For a general instrument, the conditioned state is*

$$\rho \longmapsto \frac{\sum_\alpha M_{y,\alpha} \rho M_{y,\alpha}^\dagger}{\mathrm{tr}(\rho E_y)}.$$

In a dilation, this update arises from a pointer projection on the system composed with the apparatus. Its correspondence with $\Omega_{a,y}$ requires a declared morphism intertwining the reader fiber, the effect E_y, and the instrument $\mathcal{I}_{a,y}$.

Proof. The first arrow realizes the coupling; the third restricts the space of sections to the fiber defined in (66.3). The theorem 66.3 proves coincidence of the three forms in the projective regime. In the general regime, (66.9) proves the instrument update, and Stinespring supplies its dilated realization. The additional intertwining morphism is precisely the datum transporting that update back to the section space. □

Result and residence

In the declared model, the quantum update is the spectral realization of the restriction of compatible futures. Reversibility is determined by the bimeasurability of U_a or the unitarity of $\widehat{U}_a$; information loss, in the fibers of the reader or in a subsequent erasure; and the conditioned state, in $\Omega_{a,y}$. The thermodynamic cost of resetting and the zero Landauer cost of reversible evolution remain in the chapter on information and entropy.

66.2 Global state, local instruments, and Bell's triadic correlations

The question posed by Bell [9] relates the local description of outcomes to the global structure of their correlations. It asks whether a family of joint probabilities admits a decomposition into local responses conditioned by the same prior state. In HMT this question has a natural domain: the enriched state preserves genealogy, orientation, carry, and boundary memory, while each instrument publishes a local coordinate. The richness of the common state and the probabilistic form of its readings constitute distinct mathematical data. The former organizes the provenance of the correlation; the latter determines its position relative to the local Bell polytope.

The purpose of this chapter is to traverse that complete chain. We shall begin with a global preparation on the TPK state space, define the instruments of the two wings, recover the CHSH bound, and prove its stability under enrichment of the shared state; we shall then pass to the ternary alphabet intrinsic to TRIT. In that new domain the information unit is the trit and the pertinent inequality is CGLMP. The internal elliptic structure further provides a real realization of the order-three Fourier transform through the operator $J_E^2 = -I$. The result is a precise formulation of the relation among holographic totality, local reading, and quantum correlation.

66.2.1 Enhanced preparation and boundary instruments

Let X_{enr} be the space of enriched TPK states, and let μ be a probability measure describing the common preparation. An element $x \in X_{\mathrm{enr}}$ comprises the local state together with its transport history. The reading is introduced by means of instruments. For two wings, Alice and Bob, with setting sets $\mathcal{A}$ and $\mathcal{B}$, we write

$$\mathcal{I}_A^a : X_{\mathrm{enr}} \longrightarrow \mathcal{D}(O_A), \qquad \mathcal{I}_B^b : X_{\mathrm{enr}} \longrightarrow \mathcal{D}(O_B),$$

where $\mathcal{D}(O)$ denotes the simplex of distributions on the outcome alphabet O. For each prepared state, the kernels $\mathcal{I}_A^a(A \mid x)$ and $\mathcal{I}_B^b(B \mid x)$ express the probabilities published by the local readers.

The local factored form of the joint distribution is

$$P_{\mathrm{loc}}(A, B \mid a, b) = \int_{X_{\mathrm{enr}}} \mathcal{I}_A^a(A \mid x)\, \mathcal{I}_B^b(B \mid x)\, \mathrm{d}\mu(x). \tag{66.11}$$

This equation preserves a common global state and allows that state to possess arbitrarily rich memory. Its characteristic assertion is the factorization of responses once x has been fixed. Independence from the settings further requires that the measure μ be common to all choices of a and b, and complete sampling requires the observed population to represent that same preparation for every pair of settings.

This distinction is linked directly to the HMT definition of measurement. The reversible coupling between system and apparatus produces a joint state; the instrument selects a partition of that state, and the reader publishes one of its fibers. For a setting a and an outcome A, the fiber

$$\Omega_{a,A} = \{x \in X_{\mathrm{enr}} : \mathrm{Out}_a(x) = A\}$$

represents the set of histories compatible with the reading. Different settings induce different partitions of the same space; the published random variable is therefore typed by its instrumental context. Contextuality is thus formulated in the same language of instruments, fibers, and survival.

66.2.2 Local factorization, CHSH bound, and stability under enrichment

First consider binary outcomes $O_A = O_B = \{-1, +1\}$ and two settings per wing. The correlations and the Clauser–Horne–Shimony–Holt functional [25] are

$$E_{ab} = \sum_{A,B=\pm 1} AB\, P(A, B \mid a, b), \qquad S_{\mathrm{CHSH}} = E_{00} + E_{01} + E_{10} - E_{11}. \tag{66.12}$$

Theorem 66.6 (CHSH Bound under Local Factorization)**.** *If the joint distribution has the form* (66.11)*, the preparation is independent of the settings, and sampling preserves the common population in the four pairs of settings, then*

$$|S_{\mathrm{CHSH}}| \le 2.$$

Proof. For each x, define the conditional expectations

$$\overline{A}_a(x) = \sum_{A=\pm 1} A\, \mathcal{I}_A^a(A \mid x), \qquad \overline{B}_b(x) = \sum_{B=\pm 1} B\, \mathcal{I}_B^b(B \mid x).$$

Ambas belong to $[-1, 1]$. The integrando of S_{CHSH} is

$$\overline{A}_0(\overline{B}_0 + \overline{B}_1) + \overline{A}_1(\overline{B}_0 - \overline{B}_1).$$

For any $u, v \in [-1, 1]$, $|u + v| + |u - v| \le 2$ holds. The absolute value of the integrand therefore does not exceed 2. Integration with respect to a single probability measure μ preserves the bound. □

The theorem leads to a specifically important consequence for HMT.

Corollary 66.7 (Persistence of the bound under state enrichment)**.** *Replacing an elementary hidden variable with the complete state $x \in X_{\mathrm{enr}}$ preserves the bound $|S_{\mathrm{CHSH}}| \leq 2$ provided that the two instruments continue to factor as in* (66.11)*.*

Proof. The state x plays the role of the shared variable in the theorem 66.6. The argument depends on factorization of the conditional responses and on a preparation common to the four pairs of settings; the internal dimension of X_{enr} leaves both properties intact. □

This corollary positively establishes the locus at which an HMT theory of quantum correlations acts. The global genealogy provides the common domain. To realize a violation, the physical morphism must provide a joint nonseparable rule. In this formulation, the Bell correlation expresses a property of the joint representation of the state, compatible with the operational independence of the marginals.

66.2.3 Extension of binary entropy to the qutrit system and emergence of the invariant $\log 3$

TRIT naturally selects the alphabet $\mathbb{Z}/3\mathbb{Z}$. For a distribution $p = (p_0, p_1, p_2)$, its entropy expressed in trits is

$$H_3(p) = -\sum_{j=0}^{2} p_j \log_3 p_j = \frac{H_e(p)}{\log 3}. \tag{66.13}$$

The same relation holds for mutual information; to avoid confusing it with the Bell functional introduced shortly, we denote it by

$$\mathsf{I}^{(3)}(X : Y) = \frac{\mathsf{I}^{(e)}(X : Y)}{\log 3}.$$

Proposition 66.8 (Invariance of Locality Under Change of Information Unit)**.** *The passage from natural logarithms to base-three logarithms multiplies entropies and information by the positive constant* $(\log 3)^{-1}$*. It preserves the joint distributions, their marginals, and the local factorization condition.*

Proof. The identity $\log_3 u = (\log u)/(\log 3)$ proves the first assertion. The second follows because change of basis acts on the value used to quantify an already given distribution and leaves invariant both its terms and the stochastic kernels that factorize it. □

The appearance of $\log 3$ places the analysis in an information unit suited to the ternary alphabet. The substantive change for Bell is the passage from two to three outcomes per instrument. This passage leads from the binary CHSH functional to a qutrit inequality and allows the TRIT structure to enter both the definition of the settings and the choice of units.

66.2.4 The CGLMP inequality for 3 outcomes

Let A_1, A_2, B_1, B_2 be variables taking values in $\mathbb{Z}/3\mathbb{Z}$; all equalities in this section are understood modulo three. We define the Collins–Gisin–Linden–Massar–Popescu functional [24]

$$\begin{aligned} I_3^{\mathrm{CGLMP}} = {} & P(A_1 = B_1) + P(B_1 = A_2 + 1) + P(A_2 = B_2) + P(B_2 = A_1) \\ & - P(A_1 = B_1 - 1) - P(B_1 = A_2) - P(A_2 = B_2 - 1) - P(B_2 = A_1 - 1). \end{aligned} \tag{66.14}$$

Theorem 66.9 (Local triadic bound)**.** *Every distribution of four ternary instruments admitting a factorized local model satisfies*

$$I_3^{\mathrm{CGLMP}} \leq 2.$$

Proof. The set of local distributions is the convex hull of the deterministic assignments, so it suffices to prove the bound at its vertices. For a fixed assignment, write

$$x = A_1 - B_1, \qquad y = B_1 - A_2, \qquad z = A_2 - B_2, \qquad t = B_2 - A_1.$$

The relation of closure is $x + y + z + t = 0$. In these terminos,

$$I_3^{\mathrm{CGLMP}} = \mathbf{1}_{x=0} + \mathbf{1}_{y=1} + \mathbf{1}_{z=0} + \mathbf{1}_{t=0} - \mathbf{1}_{x=2} - \mathbf{1}_{y=0} - \mathbf{1}_{z=2} - \mathbf{1}_{t=2}.$$

The twenty-seven choices of (x, y, z) determine t. For $y = 0$, the nine patterns produce -4 once, -1 seven times, and 2 once; for $y = 1$, they produce -1 three times and 2 six times; for $y = 2$, they produce -1 six times and 2 three times. The maximum at every vertex is 2, and convexity preserves that bound. □

The preceding inequality describes the convex boundary of local models in the same alphabet used by HMT. Its physical evaluation starts from a joint distribution obtained from a preparation and four concrete instruments. The condition $I_3^{\mathrm{CGLMP}} > 2$, when obtained with a complete sample and independently fixed settings, excludes a local factorization of the form (66.11) for ternary outcomes.

66.2.5 Triadic Transform from the Internal Elliptic Structure

Realization of the qutrit settings requires a transformation that mixes the three outputs. Let E be a real Euclidean space endowed with an orthogonal complex structure J_E, that is,

$$J_E^2 = -I, \qquad J_E^* = -J_E.$$

Define

$$\omega_J = \exp\left(\frac{2\pi}{3} J_E\right), \qquad \omega_J^3 = I, \qquad I + \omega_J + \omega_J^2 = 0.$$

On $E \otimes \mathbb{R}^3$, the internal triadic transform is

$$\mathcal{F}_3^{(J)} = \frac{1}{\sqrt{3}} \sum_{r,s=0}^{2} \omega_J^{rs} \otimes |r\rangle\langle s|. \tag{66.15}$$

Proposition 66.10 (Orthogonality of the internal triadic transform)**.** *The map $\mathcal{F}_3^{(J)}$ is orthogonal, and its inverse is obtained by replacing J_E with $-J_E$:*

$$\left(\mathcal{F}_3^{(J)}\right)^{-1} = \mathcal{F}_3^{(-J)}.$$

In a chart that identifies J_E with multiplication by i, it recovers the complex Fourier matrix of order three.

Proof. The orthogonality of J_E implies $(\omega_J^k)^* = \omega_J^{-k}$. The (r, r') block of $\mathcal{F}_3^{(J)}\mathcal{F}_3^{(-J)}$ is

$$\frac{1}{3} \sum_{s=0}^{2} \omega_J^{(r-r')s}.$$

The sum equals I if $r = r'$ and equals zero in the other two cases by $I + \omega_J + \omega_J^2 = 0$. The composition is therefore the identity. The final assertion follows from $\exp(\theta J_E) \leftrightarrow e^{i\theta}$ in the complex chart. □

This construction makes the elliptic type of TRIT operational in Bell instruments. A local setting may be composed from $\mathcal{F}_3^{(J)}$, diagonal phase transports, and projectors onto the ternary basis. The orientation and memory transports of each wing must determine those phases before the inequality is evaluated. Thus, J_E supplies the internal rotation, whereas the genealogical register supplies the genealogical selection of the instruments.

66.2.6 Joint Representation and Independence of the Marginals

The confluence with quantum mechanics is expressed through a realization map.

$$\mathcal{R}_{\mathrm{Bell}} : X_{\mathrm{enr}} \longrightarrow \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B), \tag{66.16}$$

where $\mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B)$ is the space of density operators. Let $\{M^a_{A,k}\}_{k\in\mathbb{Z}/3\mathbb{Z}}$ and $\{N^b_{B,\ell}\}_{\ell\in\mathbb{Z}/3\mathbb{Z}}$ be local POVM operators. The joint rule is

$$P(k,\ell \mid a,b) = \int_{X_{\mathrm{enr}}} \mathrm{tr}\big[\mathcal{R}_{\mathrm{Bell}}(x)\big(M^a_{A,k} \otimes N^b_{B,\ell}\big)\big]\, \mathrm{d}\mu(x). \tag{66.17}$$

Proposition 66.11 (Local Completeness Implies Independence of the Marginals)**.** *If*

$$\sum_{\ell=0}^{2} N^b_{B,\ell} = I_B \quad \textit{for every } b,$$

then Alice's marginal is independent of Bob's setting. The symmetric statement holds for the other wing.

Proof. Upon summing (66.17) over ℓ, completeness replaces $\sum_\ell N^b_{B,\ell}$ by I_B. One obtains

$$\sum_{\ell} P(k,\ell \mid a,b) = \int \mathrm{tr}\big[\mathcal{R}_{\mathrm{Bell}}(x)\big(M^a_{A,k} \otimes I_B\big)\big]\, \mathrm{d}\mu(x),$$

an expression independent of b. □

Equation (66.16) fixes the type of the intertwiner between the HMT genealogy and the bipartite space of the physical realization. When $\mathcal{R}_{\mathrm{Bell}}(x)$ is separable for almost every x, the distribution can be rewritten through shared variables and remains in the local domain. A nonseparable realization may leave that factorization and preserve, by the preceding proposition, the independence of the marginals from the remote setting. Global correlation and signal transmission are therefore distinct properties of the same construction.

66.2.7 Holographic Reconstruction of Correlation via Indistinguishable Path Classes

The HMT structure now organizes the local–global question through typed objects. The whole is a compatible section of the extension system, with memory of its genealogy. Each experimental wing is a boundary equipped with an instrument that publishes a fiber. The joint preparation fixes the common state; the instruments fix the local questions; the Born rule fixes the probabilities; CHSH or CGLMP determines whether those probabilities admit a local decomposition.

Holographic correlation is realized as a reading of the same global state within a joint representation. Independence of the marginals ensures that each local statistic remains invariant under the remote setting, while nonseparability may prevent the complete distribution from being reduced to independent responses conditioned on a classical variable.

A precise probative hierarchy is thereby established. The local CHSH and CGLMP bounds, persistence of the bound under enrichment of the state, invariance under change to log 3, the transform $\mathcal{F}_3^{(J)}$, and independence of the marginals by completeness of the POVM are closed mathematical results. The HMT architecture of Bell is fixed by

$$X_{\mathrm{enr}} \xrightarrow{\mathcal{R}_{\mathrm{Bell}}} \mathcal{D}(\mathcal{H}_A \otimes \mathcal{H}_B) \xrightarrow{M^a \otimes N^b} P(k,\ell \mid a,b) \xrightarrow{\mathrm{CGLMP}} I_3^{\mathrm{CGLMP}}.$$

The evaluation of a concrete HMT violation is obtained by constructing $\mathcal{R}_{\mathrm{Bell}}$ and the four instruments from the enriched genealogical record, certifying their independence from adjustments, and calculating

the distribution without postselection. The chapter provides the domain, the operators, the bounds, and the realization criterion; the numerical evaluation remains reserved for the distribution derived from those instruments.

From reversible measurement to fermionic occupation and to the algebraic exclusion principle of Pauli The measured states admit collective occupations; their distributions are derived first on the finite screen and then in the statistical limit.

66.3 Internal Complex Structure and Projective Geometry of 2-Level States

The phase geometry employed by the quantum realization has an algebraic genealogy predating complex notation. Its point of departure is the finite Paley–Witt incidence, from which an endomorphism whose square is minus the identity is obtained. Realification preserves that operator as the phase generator, and projectivization of a two-level space leads to the Bloch sphere. On that sphere, the spinorial action of SU(2), its expression through Möbius transformations, and the binary icosahedral group form successive realizations of the same structure.

The causal chain developed in this chapter is

$$\boxed{\begin{array}{c} C_P \longrightarrow A_W^2 = -I_6, \\ A_W^2 = -I_6 \longrightarrow \mathbb{F}_3^6 \simeq \mathbb{F}_9^3 \longrightarrow (E, J_E), \\ (E, J_E) \longrightarrow \mathbb{P}(\mathcal{H}_J) \simeq \mathbb{CP}^1 \longrightarrow S^2, \\ S^2 \longrightarrow \{\mathrm{SU}(2),\ \mathrm{Möb}(S^2),\ 2A_5\}. \end{array}} \tag{66.18}$$

Each arrow declares its domain. The finite quadratic structure, the orthogonal realification, and the projective geometry remain related, with their scalar fields explicitly separated.

66.3.1 The incidence of Paley as an algebraic origin of the phase

Let $C_P \in M_6(\mathbb{Z})$ be the symmetric conference matrix

$$C_P = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & -1 & -1 & 1 \\ 1 & 1 & 0 & 1 & -1 & -1 \\ 1 & -1 & 1 & 0 & 1 & -1 \\ 1 & -1 & -1 & 1 & 0 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \end{pmatrix}. \tag{66.19}$$

Their rows satisfy

$$C_P C_P^{\mathsf{T}} = 5 I_6. \tag{66.20}$$

By reducing modulo three we obtain

$$A_W := C_P \bmod 3 \in M_6(\mathbb{F}_3). \tag{66.21}$$

Theorem 66.12 (Internal root of minus the identity)**.** *The endomorphism $J_W : v \mapsto vA_W$ of $V_W = \mathbb{F}_3^6$ satisfies*

$$J_W^2 = -I_{V_W}. \tag{66.22}$$

Consequently, V_W admits a canonical module structure over $\mathbb{F}_3[X]/(X^2+1)$ once the Paley matrix has been fixed.

Proof. The matrix C_P is symmetric. Reducing (66.20) modulo three yields

$$A_W^2 = 5I_6 \bmod 3 = 2I_6 = -I_6.$$

Therefore, the evaluation $X \mapsto J_W$ annihilates the ideal (X^2+1) and defines the action of the indicated quotient. □

The polynomial X^2+1 is irreducible over $\mathbb{F}_3$, since its values in $0,1,2$ are $1,2,2$. Therefore,

$$\mathbb{F}_9 := \mathbb{F}_3[\iota]/(\iota^2+1) \tag{66.23}$$

is a field, and J_W realizes multiplication by ι.

Corollary 66.13 (Descent from six ternary coordinates to three quadratic coordinates)**.** *The rule*

$$(a+b\iota)v := av + bJ_W v, \qquad a,b \in \mathbb{F}_3, \tag{66.24}$$

converts V_W into a vector space of dimension three over $\mathbb{F}_9$. In the coordinates fixed by A_W,

$$\mathbb{F}_3^6 \simeq \mathbb{F}_9^3. \tag{66.25}$$

Proof. The identity $J_W^2 = -I$ verifies the multiplication rules of $\mathbb{F}_9$. Since $[\mathbb{F}_9 : \mathbb{F}_3] = 2$ and $\dim_{\mathbb{F}_3} V_W = 6$, the dimension over $\mathbb{F}_9$ is three. □

The same operator organizes the ternary code via the graph.

$$C_W = \{(q, qA_W) : q \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}. \tag{66.26}$$

If $G = (I_6 \mid A_W)$, then

$$GG^{\mathsf{T}} = I_6 + A_W A_W^{\mathsf{T}} = 0.$$

Thus, the internal square root of minus the identity participates simultaneously in the quadratic structure and the isotropy of the code. This dual function explains why phase and exceptional incidence share an antecedent while preserving distinct readers.

66.3.2 Orthogonal realification of the phase

The analytic phase is constructed on a real space before a complex chart is chosen. Let (E, g) be a real Euclidean space of even dimension, and let $J_E \in \mathrm{End}(E)$ be an operator such that

$$J_E^2 = -I, \qquad g(J_E u, J_E v) = g(u, v). \tag{66.27}$$

The second identity implies $J_E^{\mathsf{T}} = -J_E$.

Proposition 66.14 (Phase as an orthogonal group of one parameter)**.** *For every $\theta \in \mathbb{R}$,*

$$R_J(\theta) := e^{\theta J_E} = \cos\theta\, I + \sin\theta\, J_E \tag{66.28}$$

is orthogonal and satisfies

$$R_J(\theta)R_J(\phi) = R_J(\theta+\phi), \qquad R_J(2\pi) = I. \tag{66.29}$$

Proof. The exponential series separates into even and odd powers and, using $J_E^2 = -I$, produces the trigonometric expression. The antisymmetry of J_E gives $R_J(\theta)^{\mathsf{T}} = R_J(-\theta)$; therefore, $R_J(\theta)^{\mathsf{T}} R_J(\theta) = I$. The composition law follows from the fact that all factors are functions of the same operator. □

The choice of the chart $J_E \leftrightarrow i$ turns E into a complex vector space: for $a, b \in \mathbb{R}$,

$$(a + ib)u := au + bJ_E u. \tag{66.30}$$

In that chart, $R_J(\theta)$ is written as multiplication by $e^{i\theta}$. The complex notation publishes the phase already transported by the real operator.

Remark 66.15 (Two scales, the same quadratic relation). The operator J_W acts on $\mathbb{F}_3^6$; the operator J_E acts on a real inner-product space. The prolongation $J_W \rightsquigarrow J_E$ is a realization that preserves the relation $J^2 = -I$ and additionally specifies a topology, norm, and completion. This order avoids using the continuous analytic structure as a premise of the finite incidence.

66.3.3 The complex projective line and the Bloch sphere

Let $\mathcal{H}_J$ be a Hilbert space of complex dimension two obtained through the chart (66.30). A normalized pure state is a vector

$$\psi = \begin{pmatrix} z_0 \\ z_1 \end{pmatrix}, \qquad |z_0|^2 + |z_1|^2 = 1, \tag{66.31}$$

and its projective physical state is the ray $[\psi] = \{e^{i\vartheta}\psi : \vartheta \in \mathbb{R}\}$. The ray space is $\mathbb{P}(\mathcal{H}_J) \simeq \mathbb{CP}^1$.

We define the coordinates

$$\begin{aligned} r_1 &= 2\,\mathrm{Re}(\overline{z}_0 z_1), \\ r_2 &= 2\,\mathrm{Im}(\overline{z}_0 z_1), \\ r_3 &= |z_0|^2 - |z_1|^2. \end{aligned} \tag{66.32}$$

The spherical representation of two-level states is formulated in the coordinates introduced by Bloch [12].

Theorem 66.16 (Projective Bloch Identification)**.** *The map*

$$\mathcal{B} : \mathbb{CP}^1 \longrightarrow S^2, \qquad [\psi] \longmapsto (r_1, r_2, r_3), \tag{66.33}$$

is a smooth bijection. The rank-one projector associated with $[\psi]$ *is*

$$\rho_\psi = |\psi\rangle\langle\psi| = \frac{1}{2}\big(I_2 + r_1\sigma_1 + r_2\sigma_2 + r_3\sigma_3\big), \tag{66.34}$$

where $\sigma_1, \sigma_2, \sigma_3$ *are the Pauli matrices.*

Proof. Multiplication of ψ by a common phase leaves the three coordinates invariant. Moreover,

$$r_1^2 + r_2^2 + r_3^2 = 4|z_0|^2|z_1|^2 + (|z_0|^2 - |z_1|^2)^2 = 1.$$

Every $\boldsymbol{r} \in S^2$ determines the positive projector $\rho = (I + \boldsymbol{r}\cdot\boldsymbol{\sigma})/2$, which satisfies $\rho^2 = \rho$ and $\mathrm{tr}\,\rho = 1$; its image is a single ray. The direct matrix formula for $|\psi\rangle\langle\psi|$ yields (66.34). The usual affine charts of $\mathbb{CP}^1$ make the map and its inverse smooth. □

The Bloch sphere records the projective class and preserves, through ρ_ψ, all expectation values of two-level Hermitian observables. The global phase belongs to the fibre S^1 of the Hopf bundle [63]

$$S^1 \longrightarrow S^3 \longrightarrow S^2. \tag{66.35}$$

The HMT distinction between enriched state and published coordinate finds a precise realization here: the normalized vector belongs to S^3, the projective reader publishes a point of S^2, and the fiber preserves the phase that the projection identifies.

66.3.4 Spinorial Action, Rotations, and Möbius Transformations

Every element of $\mathrm{SU}(2)$ can be written as

$$U = \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{66.36}$$

Acts linearly on spinors and, by conjugation, on projectors.

Theorem 66.17 (Double covering and action on the sphere)**.** *for each $U \in \mathrm{SU}(2)$ there exists a unique $R_U \in \mathrm{SO}(3)$ such that*

$$U(\boldsymbol{r} \cdot \boldsymbol{\sigma})U^{\dagger} = (R_U \boldsymbol{r}) \cdot \boldsymbol{\sigma}. \tag{66.37}$$

The map $U \mapsto R_U$ is a surjective homomorphism with kernel $\{\pm I_2\}$.

Proof. Conjugation preserves the three-dimensional real space of trace-free Hermitian matrices and its inner product $\langle A, B\rangle = \frac{1}{2}\operatorname{tr}(AB)$. It therefore induces an orthogonal transformation of $\mathbb{R}^3$. Continuity and the value at the identity fix a positive determinant. If conjugation is trivial on the three Pauli matrices, U commutes with every complex matrix 2×2, and by Schur's lemma it is scalar; the determinant-one condition leaves $U = \pm I_2$. The axis–angle representation shows that every rotation has two opposite preimages. □

In the affine chart $z = z_0/z_1$ of $\mathbb{CP}^1$, the same action takes the form

$$z \longmapsto \frac{\alpha z + \beta}{-\overline{\beta} z + \overline{\alpha}}. \tag{66.38}$$

The Bloch rotation and the fractional transformation are therefore two coordinates of the same projective action. The former employs the Hermitian projector; the latter, the Riemann sphere.

66.3.5 The icosahedral specialization and the binary group $2A_5$

The grupo rotacional of the icosaedro is a subgrupo of $\mathrm{SO}(3)$ isomorfo to A_5. Its preimagen by the recubrimiento of the theorem 66.17 defines the grupo icosaedrico binario

$$2A_5 := \pi^{-1}(A_5) \subset \mathrm{SU}(2). \tag{66.39}$$

Proposition 66.18 (Icosahedral central extension)**.** *There exists an exact sequence*

$$1 \longrightarrow \{\pm I_2\} \longrightarrow 2A_5 \longrightarrow A_5 \longrightarrow 1, \tag{66.40}$$

and $|2A_5| = 120$. An icosahedral rotation of orders three or five lifts to elements of orders six or ten, respectively.

Proof. The restriction of $\pi : \mathrm{SU}(2) \to \mathrm{SO}(3)$ to the preimage of A_5 is surjective and preserves the kernel $\{\pm I_2\}$. Exactness and the order theorem yield $|2A_5| = 2|A_5| = 120$. A rotation through angle $2\pi/n$ is represented in $\mathrm{SU}(2)$ with half-angle π/n; after n iterations it reaches $-I_2$ and after $2n$, the identity. □

The icosahedral specialization establishes a rigorous hinge between the geometry of two-level states, Möbius transformations, and the finite-symmetry branch. The development of McKay theory and of the exceptional structures subsequently uses the representation of $2A_5$ and its modules; its additional content is developed in the specific theorems of that branch.

66.3.6 Functorial composition $\mathbb{F}_3^6 \to \mathbb{F}_9^3 \to \mathbb{CP}^1 \simeq S^2 \to \mathrm{SU}(2) \to \mathrm{SO}(3) \to 2A_5$

The relation $J^2 = -I$ traverses four categories while preserving its type:

1. in $\mathbb{F}_3^6$, defines the quadratic extension $\mathbb{F}_9$ and organizes the Witt incidence;
2. in a real orthogonal space, generates the phase flow $e^{\theta J_E}$;
3. in complex dimension two, projectivization produces $\mathbb{CP}^1 \simeq S^2$ and the Bloch projectors;
4. the action of $\mathrm{SU}(2)$ descends to rotations and Möbius transformations, and its icosahedral preimage produces $2A_5$.

The internal complex structure thus has a finite provenance and an explicit analytic realization. The Bloch sphere publishes the geometry of rays; HMT genealogy additionally preserves the route, orientation, carry, and memory that precede that publication.

66.4 Coupled observer, self-inertia and genealogical conservation of information

66.4.1 Joint state and observer residence

The elementary dynamic description comprises the observed section, the realized observer, and the record produced by both. We write

$$\boxed{\Gamma = (x, O, m) \in \mathcal{X}_{\mathrm{TPK}} \times \mathfrak{O} \times \mathfrak{M}.} \tag{66.41}$$

The coordinate x preserves cylinder, route, sheet, orientation, carry, boundary, and possibilities of prolongation; O specifies the reader subsystem and its resolution; m preserves the distinguishable memory of the coupling. The joint state constitutes the primary domain of measurement and inertial response.

An observer at depth n is represented by a chart

$$q_{O,n} : \mathcal{X}_{\mathrm{TPK}}^{(n)} \longrightarrow Y_{O,n}. \tag{66.42}$$

For $x \in \mathcal{X}_{\mathrm{TPK}}^{(n)}$, its observational horizon is the fibre

$$\mathfrak{H}_{O,n}(x) = q_{O,n}^{-1}(q_{O,n}(x)). \tag{66.43}$$

The fiber gathers the sections indistinguishable for the declared chart. Refinement preserves naturality when there are maps $r_{n+1,n}$ and $s_{n+1,n}$ such that

$$q_{O,n} \circ r_{n+1,n} = s_{n+1,n} \circ q_{O,n+1}. \tag{66.44}$$

This identity makes finite observation compatible with deep genealogy: the result of truncating a refined reading coincides with the reading of the truncated state.

Definition 66.19 (Realized observer)**.** An implemented observer is a tuple

$$O_a = (m_0, U_a, q_a, C_a) \tag{66.45}$$

formed by a preparation m_0, a coupling U_a, a pointer reader q_a, and a compatibility and prolongation condition C_a. Its function is to produce a distinguishable record and use it as a condition for continuation of the route.

The definition situates the reading frame within the system. The published coordinate depends on the coupling, the resolution, and the memory; the enriched section preserves the totality from which that coordinate arises.

66.4.2 Measurement context and conditioning by fibers

Let (Ω_S, Σ_S) and (Ω_M, Σ_M) be measurable spaces of sections of the system and the apparatus. The context a uses a measurable coupling

$$U_a : \Omega_S \times \Omega_M \longrightarrow \Omega_S \times \Omega_M \tag{66.46}$$

and a reader $q_a : \Omega_M \to Y_a$. The output is

$$\mathrm{Out}_a(x) = q_a\big(\mathrm{pr}_M\, U_a(x, m_0)\big)\,. \tag{66.47}$$

For an outcome y, the measurable fiber is defined as

$$\Omega_{a,y} = \{x \in \Omega_a : \mathrm{Out}_a(x) = y\}. \tag{66.48}$$

The Update of Genealogy Is the Restriction

$$\Omega_a \longmapsto \Omega_{a,y}. \tag{66.49}$$

At finite depth, the operation preserves the prefixes that admit an extension in $\Omega_{a,y}$. The memory of the result is incorporated into the prolongation condition; the traversed route remains recorded.

If $\mu_a(\Omega_{a,y}) > 0$, the measure conditioned is

$$\mu_{a,y}(B) = \frac{\mu_a(B \cap \Omega_{a,y})}{\mu_a(\Omega_{a,y})}. \tag{66.50}$$

This expression constitutes the probabilistic chart of the relative state. Its spectral realization is obtained through cylindrical projections.

66.4.3 cylindrical PVM and Lüders correspondence

Suppose that $\Omega_{a,y}$ is a finite union of disjoint cylinders of a common depth n. Let P_y be the sum of their orthogonal projectors in the Hilbert realization. For a density state ρ and a common PVM of cylindrical events, the Born probability is

$$p(y) = \mathrm{Tr}(\rho P_y). \tag{66.51}$$

When $p(y) > 0$, the update by Lüders [84] is

$$\rho_y = \frac{P_y \rho P_y}{\mathrm{Tr}(\rho P_y)}. \tag{66.52}$$

Theorem 66.20 (Equivalence between fiber restriction and Lüders)**.** *Be $\mathcal{A}_n$ the algebra of sucesos cilindricos to depth n. Supongamos that*

$$\mu_a(B) = \mathrm{Tr}(\rho P_B), \qquad B \in \mathcal{A}_n, \tag{66.53}$$

and that P_B and P_y belong to the same projective spectral measure. Then

$$\boxed{\mathrm{Tr}(\rho_y P_B) = \frac{\mu_a(B \cap \Omega_{a,y})}{\mu_a(\Omega_{a,y})}.} \tag{66.54}$$

Proof. Membership in a common PVM gives $P_B P_y = P_{B \cap \Omega_{a,y}}$. Cyclicity of the trace and equation (66.53) yield

$$\mathrm{Tr}(\rho_y P_B) = \frac{\mathrm{Tr}(\rho P_y P_B P_y)}{\mathrm{Tr}(\rho P_y)} = \frac{\mathrm{Tr}(\rho P_{B \cap \Omega_{a,y}})}{\mathrm{Tr}(\rho P_y)},$$

which coincides with equation (66.50). □

The hypothesis of a common PVM is substantive. Two mutually unbiased contexts admit their own projective descriptions and require a noisy POVM or a dilation for a joint reading. The correspondence of theorem 66.20 is asserted within the declared spectral context.

66.4.4 Quantum instrument, dilation, and channel preservation

Let $\widehat{U}_a$ be the unitary realization of the coupling between system and apparatus, and let $|m_0\rangle$ be the pointer preparation. A refined basis $|m_{y,\alpha}\rangle$ induces the Kraus operators

$$M_{y,\alpha} = \langle m_{y,\alpha}|\widehat{U}_a|m_0\rangle, \qquad \sum_{y,\alpha} M_{y,\alpha}^\dagger M_{y,\alpha} = I. \tag{66.55}$$

The instrument and its effect are

$$\mathcal{I}_y(\rho) = \sum_\alpha M_{y,\alpha}\rho M_{y,\alpha}^\dagger, \qquad E_y = \sum_\alpha M_{y,\alpha}^\dagger M_{y,\alpha}. \tag{66.56}$$

The probability and the relative state take the form

$$p(y) = \mathrm{Tr}(\rho E_y), \qquad \rho_y = \frac{\mathcal{I}_y(\rho)}{p(y)}. \tag{66.57}$$

The non-selective channel is

$$\mathcal{E}_a(\rho) = \sum_y \mathcal{I}_y(\rho). \tag{66.58}$$

Proposition 66.21 (Exact Conservation of the Global Channel)**.** *The isometry*

$$V_a : \mathcal{H}_S \longrightarrow \mathcal{H}_S \otimes \mathcal{H}_M, \qquad V_a|\psi\rangle = \widehat{U}_a(|\psi\rangle \otimes |m_0\rangle) \tag{66.59}$$

satisfies $V_a^\dagger V_a = I$. Consequently,

$$\sum_y p(y) = 1, \qquad \mathrm{Tr}\,\mathcal{E}_a(\rho) = \mathrm{Tr}\,\rho, \tag{66.60}$$

and $\mathcal{E}_a$ is completely positive and trace-preserving. For an initial pure state, the joint state $V_a|\psi\rangle$ remains pure and normalized.

Proof. The unitarity of $\widehat{U}_a$ and the normalization of $|m_0\rangle$ yield $V_a^\dagger V_a = I$. The completeness identity for equation (66.55) preserves the trace. The Kraus form proves complete positivity, and an isometry maps unit vectors to unit vectors. □

Preservation of the global channel means that the dilated evolution retains all amplitudes and all records. Conditioning selects a relative fiber for the reader that obtained y. The sum of the instruments reproduces the complete channel. This threefold distinction separates evolution, reading, and relative state.

The Stinespring theorem [116] guarantees a dilation that is isometric for every completely positive channel. The identification of the auxiliary space with the TPK memory is realized by an intertwiner that preserves route, carry, register, and refinement. The abstract existence of the dilation and the concrete genealogy of the register are two coordinable, typed results.

66.4.5 State relative and formulation Everettian

The relative-state formulation provides the conceptual antecedent of this construction [45].

In an ideal premeasurement, suppose that the pointer states $\{|m_r\rangle\}_r$ are orthonormal and that $\widehat{U}_a$ is an isometry on the subspace spanned by $\{|r\rangle \otimes |m_0\rangle\}_r$. Then

$$\widehat{U}_a(|r\rangle \otimes |m_0\rangle) = |r\rangle \otimes |m_r\rangle. \tag{66.61}$$

For $|\psi\rangle = \sum_r c_r|r\rangle$, linearity produces

$$|\Psi'\rangle = \sum_r c_r|r\rangle \otimes |m_r\rangle. \tag{66.62}$$

The record m_r defines the relative section

$$\rho_r = \frac{Q_r|\Psi'\rangle\langle\Psi'|Q_r}{\mathrm{Tr}(|\Psi'\rangle\langle\Psi'|Q_r)}, \qquad Q_r = I_S \otimes |m_r\rangle\langle m_r|. \tag{66.63}$$

In the Genealogical Space, the Same Result Corresponds to

$$\Omega_{a,r} = \{x : \mathrm{Out}_a(x) = r\}. \tag{66.64}$$

Theorem 66.22 (Relative state correspondence)**.** *Under the projective hypotheses of theorem 66.20, the relative state of equation (66.63) and the measure conditioned on $\Omega_{a,r}$ assign the same probabilities to every event of the declared cylindrical PVM. The global isometry preserves the correlated superposition of all records.*

Proof. The first assertion is equation (66.54). The second follows from $V_a^\dagger V_a = I$ and the linear expression equation (66.62). □

The Everettian correspondence reaches the mathematical level of relative states: the joint state preserves the channels, and each register determines a conditioned fiber. HMT adds to each fiber a genealogy composed of prefixes, carry, orientation, boundary, and compatible prolongations. The macroscopic stability of the registers belongs to the theory of decoherence of the apparatus; the ontology of physically autonomous branches belongs to an additional interpretation. The preceding theorem retains its force independently of that ontological choice.

66.5 Canonical entangler between TPK memory and device

Let $\mathfrak{M}_n$ be the finite set of TPK records at level n, including outcome, route, carry, leaf, and memory. Its canonical Hilbertization is

$$\mathcal{H}_{\mathfrak{M}_n} = \ell^2(\mathfrak{M}_n).$$

Let $\mathcal{K}_{A,n}$ be an auxiliary apparatus space containing an orthonormal family $\{|a_m\rangle : m \in \mathfrak{M}_n\}$. We define

$$\iota_n : \mathcal{H}_{\mathfrak{M}_n} \longrightarrow \mathcal{K}_{A,n}, \qquad \iota_n|m\rangle = |a_m\rangle.$$

To preserve genealogy when the visible update is not injective, the ledger lifting is used.

$$\widehat{F}_n(m) = (F_n(m), m) \in \mathfrak{M}_{n+1}.$$

It is injective by construction and defines the isometry.

$$V_n|m\rangle = |\widehat{F}_n(m)\rangle.$$

On the pointer subspace is fixed

$$W_n \iota_n = \iota_{n+1} V_n.$$

Theorem 66.23 (TPK Apparatus–Memory Intertwiner)**.** *The mapas ι_n and V_n are isometrias and the square*

$$\begin{array}{ccc} \mathcal{H}_{\mathfrak{M}_n} & \xrightarrow{V_n} & \mathcal{H}_{\mathfrak{M}_{n+1}} \\ {\scriptstyle \iota_n}\downarrow & & \downarrow{\scriptstyle \iota_{n+1}} \\ \mathcal{K}_{A,n} & \xrightarrow{W_n} & \mathcal{K}_{A,n+1} \end{array}$$

commutes. The map W_n prolongs to an isometry of the apparatus and, after adding an auxiliary complement if the dimensions differ, to a unitary. The update preserves norm, distinguishes all records, and retains the preceding state within the ledger.

Proof. The maps send orthonormal bases to orthonormal families. The square identity is verified on every $|m\rangle$. Every finite isometry extends to a unitary map between dilations of equal dimension by completing orthonormal bases. □

Laser pointers

$$P_m = |a_m\rangle\langle a_m|$$

define the conditional expectation

$$\mathcal{D}_n(A) = \sum_{m \in \mathfrak{M}_n} P_m A P_m.$$

One has

$$\mathcal{D}_n^2 = \mathcal{D}_n, \qquad \mathrm{Spec}(\mathcal{D}_n) \subset \{0, 1\}.$$

The diagonal ledger subalgebra is the fixed subspace, and the coherent complement lies in the kernel. The unit gap of this projection equals one. Under simultaneous conjugation of the apparatus, the pointers, and the intertwiner, all these identities remain exact. This completes the finite stability of the ledger within the covariant class.

When the Kraus labels (y, α) are included in $\mathfrak{M}_n$, the isometry ι_n explicitly identifies the TPK memory with the Stinespring environment, not merely with an abstract auxiliary space. The construction of a material device, its decoherence, and its calibration constitute EXTERNAL RECOGNITION.

66.6 Joint observer–observable state, reversible instrument, and self-measurement

Definition 66.24 (Geometry of Phase Sewing)**.** A *phase-stitching geometry* is a system

$$\mathcal{X} = (X, \oplus, \otimes, \tau, C_n, L, \mathfrak{c}, v),$$

where (X) is a finite local support; $\oplus$ and $\otimes$ are its two internal transport laws; $\tau : X \to D$ is the residual map; C_n is an oriented cycle; L is a lattice whose real space $W = L \otimes_{\mathbb{Z}} \mathbb{R}$ carries a positive inner product; $\mathfrak{c} \in \mathbb{P}(D^n)$ is a distinguished nontrivial residual class; and $v \in L$ is its primitive representative in the fixed chamber. A transport $\omega \in C^1(C_n; W)$ is *admissible* when the harmonic mode of its decomposition belongs to the line $\mathbb{R}v \subset W$.

Theorem 66.25 (Exact harmonic decomposition on a cycle)**.** *For every (1)-cochain $\omega \in C^1(C_n; W)$, there exists a unique decomposition*

$$\omega = d\Phi + u\mathbf{1}_E,$$

with $\Phi \in C^0(C_n; W)$, normalizada by $\Phi(0) = 0$, and $u \in W$.

Proof. If $e_0, \dots, e_{n-1}$ are the oriented edges of the cycle, define

$$u = \frac{1}{n}\sum_{j=0}^{n-1} \omega(e_j), \qquad y_j = \omega(e_j) - u.$$

Then $\sum_j y_j = 0$. The partial sums $\Phi(k) = \sum_{j=0}^{k-1} y_j$, with $\Phi(0) = 0$, determine a well-defined 0-cochain and verify $d\Phi(e_j) = y_j$, including the final edge. For uniqueness, $d\Phi + u\mathbf{1}_E = d\Psi + w\mathbf{1}_E$ yields, after summing over the cycle, $n(w-u) = 0$; therefore $u = w$, and normalization eliminates the remaining constant from $\Phi - \Psi$. □

Definition 66.26 (Seam defect)**.** The piecewise-continuous lift $\Gamma_\omega : [0,1] \to W$ is fixed by

$$\Gamma_\omega\left(\frac{j+1}{n}\right) - \Gamma_\omega\left(\frac{j}{n}\right) = \omega(e_j).$$

Its stitching defect is

$$\partial\Gamma_\omega = \Gamma_\omega(1) - \Gamma_\omega(0) = \sum_{j=0}^{n-1} \omega(e_j).$$

Proposition 66.27 (Telescoping cancellation of the exact term)**.** *For every (0)-cochain Φ, one has*

$$\sum_{j=0}^{n-1} (d\Phi)(e_j) = 0.$$

Consequently, the stitching defect depends exclusively on the harmonic mode $u\mathbf{1}_E$ of the decomposition of theorem 66.25.

Proof. The sum of $\Phi(j+1) - \Phi(j)$ telescopes over the cycle. Thus, if $\omega = d\Phi + u\mathbf{1}_E$, then $\partial\Gamma_\omega = nu$. □

Theorem 66.28 (Universal Theorem of Sewing)**.** *Let $\mathcal{X}$ be a phase-stitching geometry, and let ω be an admissible transport. There exists a unique real scalar $\mathfrak{a}_\chi(\omega)$ such that*

$$\partial\Gamma_\omega = \mathfrak{a}_\chi(\omega) v, \qquad \mathfrak{a}_\chi(\omega) = \frac{\langle \partial\Gamma_\omega, v\rangle}{\langle v, v\rangle}.$$

Proof. By theorem 66.25, $\omega = d\Phi + u\mathbf{1}_E$, with unique u. Admissibility provides a unique $\lambda \in \mathbb{R}$ with $u = \lambda v$. The proposition 66.27 gives $\partial\Gamma_\omega = n\lambda v$; it suffices to define $\mathfrak{a}_\chi(\omega) = n\lambda$ and project orthogonally onto $\mathbb{R}v$. □

66.7 Joint observer–observable state, intrinsic self-measurement, and Stokes' holographic identity

This section makes explicit three claims that elsewhere in the manuscript appeared only in narrative form. The first is that the *observer* does not introduce the invariant, but selects a polarization, a trivialization, and a reading orientation. The second is that the *self-measure* is not a metaphor, but the unique normalized linear functional compatible with the residual boundary mode. The third is that the so-called *holographic principle* admits, in this finite framework, an exact formulation through discrete Stokes: the same scalar can be read on the boundary as a seam defect or in the *bulk* as projected total curvature.

From a historical standpoint, these three components permit the classical notion of measurement to be reformulated. In standard metrology, a measurement is traceable when it can be related to a reference through a documented and unbroken chain of comparisons; and, since 2019, the SI has been defined by fixed values of constants of nature [94, 19]. The present framework does not deny that scheme: it

displaces it to a prior level, at which the "unit" is no longer first an external standard, but an internal residual mode. Measurement then ceases to be a comparison with an artifact and becomes a reading of an intrinsic closure.

66.7.1 The Observer as Polarization, Trivialization, and Orientation

The most stable geometric interpretation of the observer problem is as follows. The local support has at least two internal polarizations, one additive and the other multiplicative. In the standard language of finite quantum kinematics, both polarizations generate a Weyl–Heisenberg extension in which the central phase records the order of the transports [123, 131]. The observer adds no new dynamics to the system; the observer selects a representative of the same transport class.

Definition 66.29 (Edge Observer)**.** Let $\mathcal{X} = (X, \oplus, \otimes, \tau, C_n, L, \mathfrak{c}, v)$ be a phase-stitching geometry, and let $\omega \in C^1(C_n; W)$ be an admissible transport. A *boundary observer* is a pair

$$O = (\varepsilon, \Phi), \qquad \varepsilon \in \{+1, -1\}, \qquad \Phi \in C^0(C_n; W),$$

that acts on ω by

$$\omega^O := \varepsilon\, \omega + d\Phi.$$

The sign ε encodes the reading orientation, and the 0-cochain Φ encodes the trivialization or choice of gauge.

Remark 66.30. At the local level, the choice between an additive reader and a multiplicative reader should be understood as a polarization of the same phase support. At the global boundary level, that choice is absorbed into the class of the observed 1-cochain. The geometrically relevant object is therefore the cohomology class, rather than the particular representative.

Proposition 66.31 (Observer Covariance)**.** *For every admissible transport ω and every boundary observer $O = (\varepsilon, \Phi)$, one has*

$$\mathfrak{a}_\chi(\omega^O) = \varepsilon\, \mathfrak{a}_\chi(\omega).$$

In particular, self-measurement is invariant under a change of trivialization and changes sign only when the orientation is reversed.

Proof. Using the definition of seam defect and linearity, it is obtained

$$\partial\Gamma_{\omega^O} = \sum_{e \in C_n} \big(\varepsilon\, \omega(e) + (d\Phi)(e)\big) = \varepsilon \sum_{e \in C_n} \omega(e) + \sum_{e \in C_n} (d\Phi)(e).$$

The sum of the exact term telescopes and equals 0 by Proposition 66.27. Hence

$$\partial\Gamma_{\omega^O} = \varepsilon\, \partial\Gamma_\omega.$$

Projecting onto the residual line $\mathbb{R}v$ by means of the formula in Theorem 66.28, one concludes

$$\mathfrak{a}_\chi(\omega^O) = \frac{\langle \partial\Gamma_{\omega^O}, v\rangle}{\langle v, v\rangle} = \varepsilon\, \frac{\langle \partial\Gamma_\omega, v\rangle}{\langle v, v\rangle} = \varepsilon\, \mathfrak{a}_\chi(\omega).$$

□

Corollary 66.32 (The observer does not create the invariant)**.** *Two boundary observers differing only by gauge measure the same closure scalar. Therefore, the observer fixes a reading but does not manufacture the residual mode.*

Remark 66.33. This proposition resolves the observer problem at precisely the level at which the present framework can resolve it. The observer does not disappear, but its role changes: it does not introduce

the pattern; rather, it selects a polarization and an orientation for an invariant already generated by the structure itself. In this sense, the theory is *observer-covariant*.

66.7.2 Uniqueness of the intrinsic autofunctional

The next step consists in formalizing the idea that the system measures itself. To this end, one must distinguish between the local density of the harmonic mode and the global magnitude of boundary closure.

Definition 66.34 (Self-measure density and global closure)**.** Let ω be an admissible transport on C_n. We define the *self-measure density* by

$$\mu_\chi(\omega) := \frac{1}{n}\,\mathfrak{a}_\chi(\omega) = \frac{1}{n}\,\frac{\langle \partial\Gamma_\omega, v\rangle}{\langle v, v\rangle}.$$

Equivalently, if

$$\omega = d\Phi + u\mathbf{1}_E \qquad \text{with } u \in \mathbb{R}v,$$

then $u = \mu_\chi(\omega)\, v$. Llamaremos *closure global* to the scalar

$$\mathfrak{a}_\chi(\omega) = n\,\mu_\chi(\omega).$$

Lemma 66.35 (One-dimensional quotient of admissible transports)**.** *Let $\mathcal{T}_{\mathrm{adm}} \subset C^1(C_n; W)$ be the subspace of admissible transports. Then the quotient*

$$\mathcal{T}_{\mathrm{adm}}/dC^0(C_n; W)$$

is a real vector space of dimension 1*, canonically generated by the class of $v\mathbf{1}_E$.*

Proof. By Theorem 66.25, every $\omega \in \mathcal{T}_{\mathrm{adm}}$ can be written uniquely as

$$\omega = d\Phi + u\mathbf{1}_E.$$

The admissibility imposes $u \in \mathbb{R}v$, so that the class of ω modulo exacts remains completely determinada by the real scalar λ such that $u = \lambda v$. In consequence, the quotient is isomorfo to $\mathbb{R}v$, and therefore one-dimensional. □

Theorem 66.36 (Uniqueness of the Auto-measure)**.** *Let $F : \mathcal{T}_{\mathrm{adm}} \to \mathbb{R}$ be a linear functional such that:*

(i) *$F(\omega + d\Phi) = F(\omega)$ for every $\Phi \in C^0(C_n; W)$;*

(ii) *$F(v\mathbf{1}_E) = 1$.*

Then

$$F = \mu_\chi.$$

In particular, the self-measure is the unique *normalized linear functional compatible with the distinguished residual mode.*

Proof. By Lemma 66.35, the quotient $\mathcal{T}_{\mathrm{adm}}/dC^0(C_n; W)$ is one-dimensional and is generated by the class of $v\mathbf{1}_E$. Every gauge-invariant linear map factors through that quotient and is completely determined by its value on the generator. The normalization $F(v\mathbf{1}_E) = 1$ then uniquely fixes the functional. By definition,

$$\mu_\chi(v\mathbf{1}_E) = \frac{1}{n}\,\frac{\langle nv, v\rangle}{\langle v, v\rangle} = 1,$$

and, moreover, μ_χ is linear and gauge invariant. Hence $F = \mu_\chi$. □

Corollary 66.37 (Auto-measure in the HMT case)**.** *In the strong dodecaphase specialization, where $n = 12$ and $v = v_\alpha$, one has*

$$\mu_{\mathrm{HMT}}(\omega) = \frac{\alpha}{12}, \qquad \mathfrak{a}_{\mathrm{HMT}}(\omega) = \alpha.$$

Thus, the number α is the global closure of the boundary, and $\alpha/12$ is the elementary density of self-measure per phase.

Remark 66.38. This formulation changes the meaning of "unit". In classical metrology, the unit is an external reference to which a result is linked through a calibration chain [94, 19]. Here normalization proceeds from the system itself: vector v is not imported from outside, but is generated internally as the minimum lift of the distinguished residual class. In this precise sense, the unit is a *self-measurement*.

66.7.3 Finite holographic principle: boundary, *bulk*, and discrete Stokes

What appeared in the prologue as holographic intuition now admits an exact form. The crucial fact is that the same scalar can be read in two ways: as boundary holonomy or as the total projected curvature of the *bulk*. This does not identify the theory with the quantum-gravitational holography of 't Hooft or Susskind [119, 118]; it does, however, produce a finite and literal analogue of reduction to the boundary.

Definition 66.39 (*Finite bulk* with cyclic boundary)**.** A finite *bulk* with cyclic boundary is an oriented cellular complex of dimension 2

$$K = K^{(0)} \cup K^{(1)} \cup K^{(2)}$$

whose oriented boundary satisfies

$$\partial K \cong C_n.$$

Given a 1-cochain $\Omega \in C^1(K; W)$, we call *discrete curvature* the 2-cochain

$$F_\Omega := d\Omega \in C^2(K; W).$$

If $\Omega|_{\partial K} = \omega$, we shall say that Ω is a *bulk extension* of the boundary transport ω.

Theorem 66.40 (Projected discrete Stokes)**.** *Let K be a finite bulk with boundary $\partial K \cong C_n$, and let $\Omega \in C^1(K; W)$ be a bulk extension of an admissible transport ω on the boundary. Then*

$$\partial\Gamma_\omega = \sum_{e \subset \partial K} \Omega(e) = \sum_{f \in K^{(2)}} F_\Omega(f).$$

In particular,

$$\mathfrak{a}_\chi(\omega) = \frac{\left\langle \sum_{f \in K^{(2)}} F_\Omega(f), v \right\rangle}{\langle v, v \rangle}.$$

Proof. The first equality is simply the definition of seam defect applied to the restriction of Ω to the boundary. The second is the discrete Stokes formula for cochains: the oriented sum over the boundary of a complex equals the sum, over all 2-cells, of the oriented coboundary of the 1-cochain. That is,

$$\sum_{e \subset \partial K} \Omega(e) = \sum_{f \in K^{(2)}} (d\Omega)(f) = \sum_{f \in K^{(2)}} F_\Omega(f).$$

Projecting onto the residual line $\mathbb{R}v$ and using Theorem 66.28 yields the final formula. □

Corollary 66.41 (Obstruction to a flat *bulk*)**.** *If an admissible boundary transport ω admits a flat bulk extension, that is, an Ω with $F_\Omega = 0$, then*

$$\mathfrak{a}_\chi(\omega) = 0.$$

Equivalently, if $\mathfrak{a}_\chi(\omega) \neq 0$, every bulk extension of the boundary must carry nonzero discrete curvature.

Proof. This is an immediate consequence of Theorem 66.40. If $F_\Omega = 0$, then the projected sum of curvatures equals 0, and therefore $\mathfrak{a}_\chi(\omega) = 0$. □

Corollary 66.42 (Finite holographic reading)**.** *The closure invariant may be read equivalently as a boundary observable or as the integrated curvature of the* bulk*:*

$$\mathfrak{a}_\chi(\omega) = \frac{\langle \partial\Gamma_\omega, v\rangle}{\langle v, v\rangle} = \frac{\left\langle \sum_{f\in K^{(2)}} F_\Omega(f), v\right\rangle}{\langle v, v\rangle}.$$

In this precise sense, the system satisfies a finite holographic principle.

Remark 66.43. The preceding corollary is the most literal and strongest form of the holographic principle that the manuscript can sustain through internal proofs. It does not assert a complete gravitational duality. It asserts something more modest and exact: the relevant physical–mathematical scalar of the system *factors through the boundary* and, if extended to a *bulk*, coincides with the total projected curvature of that *bulk*. When $\mathfrak{a}_\chi(\omega) = \alpha$, the equality states that α is simultaneously a boundary holonomy and an integrated curvature.

Corollary 66.44 (Geometric consequence for HMT)**.** *In the strong HMT case, every bulk filling of the dodecaphase cycle that extends the canonical boundary transport must satisfy*

$$\left\langle \sum_{f\in K^{(2)}} F_\Omega(f), v_\alpha \right\rangle = \alpha\,\langle v_\alpha, v_\alpha\rangle = 54\alpha.$$

In particular, the HMT boundary admits no flat filling; its bulk *necessarily carries nonzero projected total curvature.*

Proof. It suffices to apply Corollary 66.42 with $v = v_\alpha$ and use the fact that $\langle v_\alpha, v_\alpha\rangle = 54$. □

Remark 66.45 (On the geometry of the *bulk*). Theorem 66.40 does not by itself identify the metric geometry of the *bulk*; it does impose a very strong negative condition: when $\mathfrak{a}_\chi(\omega) \neq 0$, the filling cannot be exactly or discretely flat. If, moreover, one seeks a global completion compatible with the modular reading of the boundary, the natural candidate ceases to be the Euclidean disk and becomes a modular-type *bulk*. In classical theory, modular functions live on the upper half-plane $\mathbb{H}$, and the Klein invariant $J(\tau)$ classifies the global complex torus [95]. The present result does not yet prove that identification, but it does justify why the intuition of a non-Euclidean *bulk* is mathematically reasonable.

66.7.4 Structural Consequences of Polarization, Self-Measurement, and Discrete Stokes Identity

The preceding three components permit the interpretation of observer, self-measurement, and holography to be fixed with a degree of precision that was previously only sketched.

(1) The **observer** is a choice of polarization, orientation, and gauge. It changes the reading, but not the residual mode, except for the sign associated with the orientation.

(2) The **self-measurement** is the unique normalized linear functional on the admissible space of transports. It requires no external pattern because its normalization arises from the internal minimal lift.

(3) The **holographic principle** acquires an exact form: the relevant scalar can be read entirely on the boundary or entirely as the projected total curvature of the bulk.

This triple identification allows the central thesis of the manuscript to be reformulated more strongly: not only is α an Archimedean boundary invariant, but, within the finite framework considered here, α is the first scalar that a self-inertial phase cell produces when measuring itself and projecting all its effective curvature onto the distinguished residual mode.

Chapter 67

Sommerfeld–Kepler central limit, pilot-wave dynamics, and double-slit interference on the genealogical trajectory space

67.1 Central Limit, Kepler Orbits and Sommerfeld Holonomy

This chapter distinguishes four strata that physical terminology may improperly superimpose: the exact cancellation of interior flows in a graph; the continuous realization of a central field; the Keplerian dynamics resulting from the inverse potential; and the semiclassical phase condition on a band with holonomy. The first is combinatorial. The other three successively require a metric chart, a Lagrangian, and a semiclassical phase line.

67.1.1 Discrete Gauss and inverse descent to area

Let $G = (V, E)$ be a finite undirected graph, let $\vec{E} := \{(x, y), (y, x) : \{x, y\} \in E\}$ be its set of arcs, and let $F : \vec{E} \to \mathbb{R}$ be an antisymmetric 1-cochain, $F(y, x) = -F(x, y)$. We define

$$\operatorname{div} F(x) := \sum_{y:\{x,y\}\in E} F(x, y).$$

For $S \subseteq V$, the flux $\operatorname{Flux}_{\partial S} F$ is the oriented sum over the edges with one endpoint in S and the other in $V \setminus S$.

Theorem 67.1 (Gauss's Theorem in Graphs)**.** *For every $S \subseteq V$,*

$$\sum_{x\in S} \operatorname{div} F(x) = \operatorname{Flux}_{\partial S} F. \tag{K1}$$

Proof. Every edge whose two endpoints lie in S occurs twice with opposite orientations and cancels. Only the edges crossing the boundary chain remain. □

In the cubic lattice, for $R \in \mathbb{Z}_{\geq 0}$,

$$C_R := \{x \in \mathbb{Z}^3 : \|x\|_\infty \leq R\},$$

the number of outgoing links is

$$|\partial C_R| = 6(2R + 1)^2.$$

If a source has total flux Q and mean uniformity is imposed on the shell, the flux per link is

$$E_R = \frac{Q}{6(2R + 1)^2}.$$

When defining r_{eff} by

$$4\pi r_{\text{eff}}^2 = 6(2R + 1)^2,$$

the exact identity is obtained

$$E_R = \frac{Q}{4\pi r_{\text{eff}}^2}. \tag{K2}$$

The passage from (K1) to (K2) uses the additional hypothesis of mean isotropy. Identifying Q with a gravitational or electric source also requires the corresponding dimensional transducer.

67.1.2 Central Lagrangian and Keplerian conics

Let $\mu > 0$ be a reduced mass and $\mathcal{K} > 0$ a coupling of the energy-times-length type, $[\mathcal{K}] = \mathsf{E}\,\mathsf{L}$. On the continuous Euclidean sheet, consider

$$L_{\mathrm{Kep}}(\mathbf{r}, \dot{\mathbf{r}}) = \frac{\mu}{2}\|\dot{\mathbf{r}}\|^2 + \frac{\mathcal{K}}{r}, \qquad H_{\mathrm{Kep}}(\mathbf{r}, \mathbf{p}) = \frac{\|\mathbf{p}\|^2}{2\mu} - \frac{\mathcal{K}}{r}, \tag{K3}$$

where $r = \|\mathbf{r}\| > 0$ and $\mathbf{p} = \mu\dot{\mathbf{r}}$. The Euler–Lagrange equation is

$$\mu\ddot{\mathbf{r}} = -\frac{\mathcal{K}}{r^3}\mathbf{r}. \tag{K4}$$

The specific angular momentum $\mathbf{h} = \mathbf{r} \times \dot{\mathbf{r}}$ is constant; if $\mathbf{h} \neq 0$, the trajectory remains in the plane orthogonal to $\mathbf{h}$.

Proposition 67.2 (Keplerian conic)**.** *Let $h = \|\mathbf{h}\|$. Every nonradial solution of* (K4) *satisfies*

$$r(\phi) = \frac{p}{1 + e\cos(\phi - \phi_0)}, \qquad p = \frac{\mu h^2}{\mathcal{K}}. \tag{K5}$$

For negative energy, $0 \le e < 1$ and the orbit is elliptic.

Proof. With $u = 1/r$, the Binet equation takes the form

$$u''(\phi) + u(\phi) = \frac{\mathcal{K}}{\mu h^2}.$$

Its general solution is

$$u = \frac{\mathcal{K}}{\mu h^2}\bigl(1 + e\cos(\phi - \phi_0)\bigr).$$

Inverting u gives (K5); the binding condition is equivalent to $e < 1$. □

The areolar velocity is

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{h}{2}. \tag{K6}$$

For an ellipse with semimajor axis a, integrating (K6) over one period yields

$$A_{\mathrm{ell}} = \pi a b, \qquad b = a\sqrt{1 - e^2}, \qquad p = a(1 - e^2) = \frac{\mu h^2}{\mathcal{K}}.$$

therefore, $T = 2A_{\mathrm{ell}}/h$, and to the eliminar b, p and h is obtained

$$T^2 = \frac{4\pi^2\mu}{\mathcal{K}}\, a^3. \tag{K7}$$

In the gravitational test problem, $\mathcal{K} = GM\mu$, and (K7) reduces to $T^2 = 4\pi^2 a^3/(GM)$ [70, 96].

The graph Gauss theorem is exact. Under the declared isotropy and distance, the flux produces the inverse-area field; the continuum limit then transports that field to the central Lagrangian and to the Keplerian orbits written above.

67.1.3 Levi-Civita Covering and Möbius Strip

In the complex plane without the origin, the Levi–Civita application

$$q = z^2 \tag{K8}$$

is a double cover. One turn of q around the origin interchanges the two lifts z and $-z$; two turns are required to close the lift. It is an exact witness of memory $\mathbb{Z}_2$. It is not, however, a Möbius band: full regularization of the Kepler problem also requires reparameterizing time and transforming the momenta.

Now let $\gamma : S^1 \hookrightarrow \mathbb{R}^2$ be an embedding whose image is an elliptic orbit, and fix $\varepsilon > 0$. We define

$$M_\gamma := ([0, 2\pi] \times [-\varepsilon, \varepsilon])/\big((0, u) \sim (2\pi, -u)\big). \tag{K9}$$

Theorem 67.3 (Orbital band oriented period)**.** *M_γ is a Möbius band whose kernel is γ. A section of the sign line has holonomy*

$$\mathcal{H}(\gamma) = -I, \qquad \mathcal{H}(\gamma^2) = I. \tag{K10}$$

Therefore, its return as an oriented section requires two turns.

Proof. Identification of the extreme fibers by means of $u \mapsto -u$ is the standard construction of the Möbius band. The holonomy of one turn is the fiber involution; its square is the identity. □

The cover (K8) and the band (K9) share a monodromy of order two, but they are not isomorphic objects. This distinction prevents a particle–antiparticle orientation from being deduced from Levi–Civita regularization without an additional map.

67.1.4 Sommerfeld's semiclassical condition with holonomy

Let

$$J_\gamma := \oint_\gamma \mathbf{p} \cdot \mathrm{d}\mathbf{q}$$

the action of a cycle, and suppose that semiclassical transport contributes the phase $\exp(\mathrm{i}J_\gamma/\hbar_{\mathrm{int}})$. If the associated real line has holonomy $\varepsilon_\gamma \in \{+1, -1\}$, the closure condition of the complexified section is

$$\varepsilon_\gamma \exp\left(\frac{\mathrm{i}J_\gamma}{\hbar_{\mathrm{int}}}\right) = 1. \tag{K11}$$

Proposition 67.4 (Shift by half a unit due to holonomy of sign)**.** *Under the preceding hypotheses,*

$$\begin{aligned} \varepsilon_\gamma = +1 \quad &\Longrightarrow \quad J_\gamma = 2\pi\hbar_{\mathrm{int}} n, \\ \varepsilon_\gamma = -1 \quad &\Longrightarrow \quad J_\gamma = 2\pi\hbar_{\mathrm{int}} \left(n + \frac{1}{2}\right), \qquad n \in \mathbb{Z}. \end{aligned} \tag{K12}$$

Proof. The equation (K11) requires a phase equal to one in the trivial sector and equal to minus one in the sign sector. Their arguments differ, respectively, by an integral or half-integral multiple of 2π. □

The proposition is an exact consequence of the declared phase and holonomy; it does not by itself select the Coulomb Hamiltonian, a physical quantum number, or an atomic spectrum [112]. In an EBK quantization, the Maslov indices of the corresponding cycles must be added without counting the topological shift twice.

Relative to the orientation reader, the two local sheets may represent routes γ and $\gamma^\vee$ and be linked to the two orientations of the two-sheet entanglement defined in definition 73.2. This correspondence neither identifies those routes with advanced or retarded Green functions nor asserts that entropy causes the

Keplerian conic: the orbit proceeds from (K4); entropy measures, after a Hilbert completion, correlation between sheets.

Comparison Criterion 67.5 (Kepler–Sommerfeld limit)**.** The realization is refuted or improperly typed if an inverse-square law is obtained without declaring isotropy; if $\mathcal{K}$ or G is introduced as a scalar without a dimensional line; if cover $q = z^2$ is identified with a Mobius band; if (K12) is published as an exact spectrum without a Hamiltonian, cycles, and Maslov indices; or if entanglement is attributed to a merely deterministic route inversion.

67.2 Reconstruction of quantum dynamics through coherent amplitudes over route genealogies

The wave is not added to the history: it is the linear form in which the complete history simultaneously preserves all its possible paths.

— HMT–MD propagation principle

Measurement has already been defined as a coupling among the system, the apparatus, and the reader. It is now necessary to explain what propagates before the result is published, how one and the same structure supports both a wave and a persistent trajectory, and how observable locality coexists with a global genealogy.

The response of Triadic Modular Holography begins from a distinction that the pointwise formulation tends to obscure. A state is not exhausted by the cell in which it lies or by the real number ultimately read from it. It also preserves the route, orientation, quotients, carries, boundary, and return that brought it there. Upon linearizing this space of histories, the sum over paths does not appear as a metaphor added a posteriori: it is the exact expansion of the propagator. The wave is the coherent evaluation of the totality of routes; in the following chapter, the particle will be the section that persists along one of them.

67.2.1 The global amplitude of 1 family of paths

Let X be a finite set of observable states at a given depth, and let $U : \mathbb{C}^X \to \mathbb{C}^X$ be the one-step propagation operator. A route of length R from i to f is a sequence

$$\gamma = (x_0 = i, x_1, \dots, x_R = f)$$

of admissible transitions. Its ordered weight is

$$w_U(\gamma) = \prod_{t=1}^{R} U_{x_t x_{t-1}}.$$

Although scalar weights commute, the indexed sequence of transitions matters: one is not counting only how many routes reach f, but which factor each step assigns to that history. Multiplication composes the transitions along a route; addition gathers alternative histories that terminate in the same state. For operator-valued weights, the temporal product is moreover genuinely noncommutative. This difference between a product within a route and a sum among routes is the linear version of the double reading that runs throughout HMT.

Theorem 67.6 (Finite path sum)**.** *For all $R \geq 1$,*

$$(U^R)_{fi} = \sum_{\substack{\gamma : i \to f \\ |\gamma| = R}} w_U(\gamma).$$

If $\psi_0 \in \mathbb{C}^X$ is the initial amplitude, then

$$\psi_R(f) = \sum_{i \in X} \sum_{\substack{\gamma : i \to f \\ |\gamma| = R}} w_U(\gamma)\, \psi_0(i).$$

Proof. The assertion for $R = 1$ is the definition of U. When an expansion of U^R is multiplied by U, every route of length R receives exactly one final admissible edge. Distributivity sums all possible extensions, and multiplication preserves the order of their weights. By induction, every route of length $R + 1$ appears exactly once. The second equality applies the first to ψ_0. □

This identity has considerable conceptual content. A component $\psi_R(f)$ is not an isolated property of the point f. It is the contraction of the complete genealogy of routes reaching it. Two histories may end at the same visible value and nevertheless contribute opposite phases; they may also reinforce one another. Interference consists precisely in summing amplitudes before forming a positive observable. There is no need to introduce an additional material wave traversing a space distinct from that of the histories: the wave structure resides in the linear composition of the routes.

When U is unitary,

$$\|\psi_R\|_2 = \|U^R \psi_0\|_2 = \|\psi_0\|_2.$$

The total quantity that may subsequently be normalized as a probability remains constant. If the physical reader assigns to an output f the weight

$$p_R(f) = \frac{|\psi_R(f)|^2}{\|\psi_R\|_2^2},$$

then p_R is a normalized distribution, and its evolution is compatible with unitary conservation. This is the precise point at which route amplitude becomes outcome statistics. The chapter on measure has already located the coupling that publishes an output; here, the object evaluated by that coupling has been identified.

The construction also clarifies why a visible number is not sufficient to reconstruct the physics. The map

$$\{\text{deep routes}\} \longrightarrow X$$

generally forgets the relative phase between paths. Knowing only the final distribution of cells does not permit inversion of the sum or recovery of the genealogy. The complete HMT state preserves what the screen contracts. Hence memory is not an interpretive ornament: it is the datum on which interference depends.

At the refinement limit, the correct question is not whether the discrete route "becomes" mysteriously continuous, but whether the propagators form a compatible family. For refinement projections $p_{m+1,m} : X_{m+1} \to X_m$, coherence requires the cylindrical observables and the transported states to agree. When this compatibility is exact, the amplitude of a coarse history is the sum of the amplitudes of all its fine prolongations. When the defects are summable, the expectations of uniformly bounded cylindrical observables form a Cauchy family. Operational continuity is thereby obtained without erasing the discrete structure that generates it.

Principle 67.7 (The wave as linear memory)**.** The finite-depth HMT wave function is the linear evaluation of the route extension. It coherently preserves the alternatives compatible with the preparation and the boundary; the observed value is a subsequent projection of that global amplitude.

This formulation is stronger than asserting that "everything occurs along many paths". It specifies the domain, the weight of each path, the composition rule, and the preserved norm. It also contains its

negative control: if a purported amplitude does not respect route composition, counts a history twice, consults the future result in order to assign the weight, or loses the norm under a propagator declared unitary, it is not the wave function constructed here.

67.2.2 Pilot wave, route performed and double orientation

The preceding structure already contains the two components that a pilot-wave theory must distinguish. The amplitude ψ_t globally organizes the alternatives; a persistent route x_t records the history actually realized. The wave and the particle are not two rival substances. They are two levels of reading of the same enriched state: the linear field of possibilities and the section that preserves its identity while traversing it.

The guiding law must convert this kinematic architecture into dynamics. Its minimal form is a map

$$G : (\psi_t, x_t) \longmapsto x_{t+1},$$

o, if the update is stochastic, a kernel

$$P_{\psi_t}(y \mid x).$$

The central condition is not verbal, but mathematical: the guided distribution must be equivariant with amplitude evolution,

$$|\psi_{t+1}(y)|^2 = \sum_x P_{\psi_t}(y \mid x)\, |\psi_t(x)|^2.$$

This equality ensures that a population initially distributed according to $|\psi_t|^2$ remains so distributed after one step. The guide cannot be chosen retrospectively to reproduce each detector; it must be fixed by the state, propagator, orientation, and register before the output is known. HMT adds a possibility that a pointwise description does not possess: G may depend on route and memory cocycles that do not descend to the visible value x_t. The local action can thereby be informed by a global genealogy without the appearance of a supplementary force traveling from the future.

The bidirectional reading further develops this structure. For an oriented route γ with

$$\partial\gamma = x_{\mathrm{abs}} - x_{\mathrm{em}},$$

the return involution defines

$$\gamma^{\vee} = C_{\mathrm{rev}}\gamma, \qquad \partial\gamma^{\vee} = x_{\mathrm{em}} - x_{\mathrm{abs}}.$$

These are not two independently and freely chosen paths. They are the two orientations of the same history. The internal length is invariant,

$$L_{\mathrm{int}}(\gamma^{\vee}) = L_{\mathrm{int}}(\gamma), \qquad L_{\leftrightarrow} = 2L_{\mathrm{int}}(\gamma),$$

and inversion interchanges the conjugate channels. Preparation and absorption cease to be ontologically disconnected endpoints: they belong to a closure in which the outward and return paths must be compatible.

When this involution is represented in the physical theory of propagators, the two orientations admit retarded and advanced readings. The retarded component transports the preparation toward absorption; the advanced component transports the return condition toward emission. The observable history is their mutual adjustment. This formulation precisely expresses the intuition of a pilot wave with advance and retardation: the local guide receives information from both boundaries of the same oriented object.

This provides no authorization to send controllable messages into the past. An advanced solution of an equation and a signalable future choice are distinct objects. The physical representation must prove that the propagators satisfy the evolution equation, that the boundary conditions compose consistently, and

that the marginal probabilities do not permit modulation of an earlier output by a later decision. Dual orientation is the support; observable causality is a property that must be verified on its physical reader.

The correlation between the two sheets can be formalized in two Hilbert spaces, $\mathcal{H}_{\mathrm{adv}}$ and $\mathcal{H}_{\mathrm{ret}}$. For a distribution $p = (p_i)$, the state

$$|\Psi_p\rangle = \sum_i \sqrt{p_i}\, |i\rangle_{\mathrm{adv}} \otimes |i\rangle_{\mathrm{ret}}$$

has reduced matrices with spectrum (p_i) and entropy

$$S_{\mathrm{ent}}(p) = -\sum_i p_i \log p_i.$$

The equality is exact once the bilayer factorization has been declared. A single history has Schmidt rank one and zero entropy; several correlated histories produce positive entropy. The result does not simply identify this entropy with $\log \varphi_{\mathrm{HMT}}$, with the information in a triadic cell, or with thermodynamic entropy. Each belongs to a different object. What it proves is that outbound and return can form a nonseparable state and that the shared information is calculated without metaphors.

Testing program 67.8 (Bidirectional guidance)**.** A physical realization of the HMT pilot wave must fix, before comparison, the propagator U, the law G or the kernel P_ψ, the involution C_{rev}, and the boundary conditions. It is falsified if it loses equivariance, assigns two incompatible routes to the same complete state, needs to know the future detector in order to choose the trajectory, or permits controllable signaling against the published causal orientation.

This proof does not relegate the proposal to a mere possibility. On the contrary, it shows that the mathematical core has already been constructed and identifies the sole locus at which physical dynamics must act. The sum over paths is not pending; the persistent route is not pending; nor is inversion. What is decided experimentally is which Hamiltonian realizes the guide and which boundary conditions turn the two sheets into nature's advanced and retarded waves.

67.2.3 Bell: locality of the data and globality of the genealogy

The distinction between complete state and projected value permits the Bell problem to be formulated precisely. Consider settings $a, b \in \{0, 1\}$, outcomes $A, B \in \{-1, +1\}$, and a shared state λ. A factorizable local theory satisfies

$$P(A, B \mid a, b) = \int_\Lambda P_A(A \mid a, \lambda)\, P_B(B \mid b, \lambda)\ \mathrm{d}\rho(\lambda),$$

with a measure independent of the adjustments,

$$\mathrm{d}\rho(\lambda \mid a, b) = \mathrm{d}\rho(\lambda).$$

Defining

$$E_{ab} = \sum_{A,B=\pm 1} AB\, P(A, B \mid a, b),$$

we obtain

$$S_{\mathrm{CHSH}} = E_{00} + E_{01} + E_{10} - E_{11}, \qquad |S_{\mathrm{CHSH}}| \le 2.$$

HMT preserves this theorem. Adding hidden memory to λ does not violate the bound if factorization, setting independence, and the absence of postselection are maintained. This control is important: it prevents the richness of TPK from being turned into an automatic explanation of every correlation.

The holographic contribution is formulated differently. The two wings of an experiment may be local projections of the same enriched section, with a common boundary and genealogy. The datum published on each wing is local; the state from which both descend need not factor into two independent deep histories. Nonlocality then consists not in a signal crossing space after the settings have been chosen, but in the impossibility of decomposing the global preparation into two autonomous complete states.

Non-signaling and Bell locality are not equivalent. It may be satisfied

$$\sum_B P(A, B \mid a, b) = P(A \mid a), \qquad \sum_A P(A, B \mid a, b) = P(B \mid b),$$

and nevertheless fail the preceding factorization. This is the precise domain that an HMT realization must occupy: global correlation without a superluminal communication channel. The contextual reader may depend on the setting in how it projects the common state, but an observer's marginal distribution cannot depend on the remote choice.

Construction 67.9 (HMT Protocol for Bell)**.** An enriched state x, a preparation measure, and four readers L_{ab} are fixed first. A binary outcome map is then declared, and the complete table is calculated

$$P_{\mathrm{HMT}}(A, B \mid a, b).$$

The construction is physically valid only if it is frozen before the data are consulted, is normalized, has no hidden postselection, and publishes simultaneously S_{CHSH} and the four marginal no-signaling controls.

The so-called Algorithm 3 belongs to this specific construction. Its purpose is not to prove the classical theorem again, but to construct the descent from the shared genealogy to a contextual distribution. If it preserves all the Bell premises, it will necessarily produce $|S| \leq 2$. If it produces $|S| > 2$, it must show explicitly which factorization fails to descend and why no-signaling is nevertheless preserved. This is the complete holographic explanation of the locality problem: the observable space may be separable even though the history generating it is not separable.

The proposal is especially natural in a theory in which sum and product are distinct readers. Bell factorization seeks to express a joint probability as a conditioned product of local responses; quantum amplitude, by contrast, arises by summing histories before projecting them. If projection is performed too early, the common phase is lost and only the local polytope remains. If the global state is preserved until the joint reading, interference terms survive that do not admit such a decomposition. The incompatibility is not a license to violate logic: it is the typed difference between composing alternatives by sum and composing local probabilities by product.

67.2.4 From the path theorem to quantum contrast

The resulting quantum architecture is articulated through a single chain of maps. HMT constructs an enriched discrete space of histories; its linearization produces the coherent sum over paths; unitarity preserves the norm; a persistent section makes it possible to speak of a trajectory; oriented inversion yields the outward-and- return pair; two-sheet factorization quantifies its correlation; and Bell analysis distinguishes locality of the datum from separability of the genealogy. These are not juxtaposed analogies, but successive maps on the same state.

The theory thus offers a unified reading of quantum features. Superposition is the linear coexistence of prolongations; interference is their phased sum; the particle is the persistence of a section; apparent reduction is the selection of a fiber by the reader; entanglement is the nonfactorization of sheets or subsystems; and nonlocality is the globality of the genealogy preserved under locally nonsignaling projections.

Physical tests are added not to weaken this reading, but to determine its natural realization. A finite interferometric network can implement U and compare the path expansion with the final pattern. A

reconstruction of currents can test the guidance law. Controlled inversion of the boundary conditions can verify the reciprocity $\gamma \leftrightarrow \gamma^{\vee}$. A Bell experiment with sealed readers can determine whether deep memory generates a nonfactorizable, no-signaling distribution.

Testing program 67.10 (Joint Quantum Compatibility Certificate)**.** The certificate distinguishes four independent validity conditions:

1. the propagator not reproduce by coherent sum the amplitudes observadas;
2. no natural guiding law is equivariant with $|\psi|^2$;
3. the double orientation does not admit reciprocal physical propagators without improper causal signaling;
4. the Bell descent is locally factorizable or violates no-signaling.

Failure of an interface localizes the incorrect map; it does not erase the finite route theorems that precede it.

The positive thesis is therefore clear. Quantum theory does not require the history of a system to be abandoned; it requires what is meant by history to be enriched. A point without genealogy cannot by itself carry interference, return, or correlation. An oriented extension can do so. Its wave is the coherent memory of all compatible paths; its particle is the closure that persists. The passage between the two forms is not an ontological leap: it is the passage from the linear totality of routes to a realized section of that same totality.

67.2.5 Double-slit interference in T_{27}^2: path memory, Born rule, and dodecaphasic unitarity

The finite realization is constructed upon

$$\mathcal{H}_{\mathrm{cam}} = \ell^2\left(T_{27}^2\right) \otimes \mathbb{C}^4, \qquad T_{27}^2 = (\mathbb{Z}/27\mathbb{Z})^2,$$

where the dimension factor 4 records the directions (N, S, E, W). Explicit memory of the trajectory adds the factor $\mathbb{C}^2_{\mathrm{mem}}$. The microscopic step is the operator

$$U_{\mathrm{micro}} = S D_J C_4, \tag{67.1}$$

with C_4 unitary, D_J diagonal, and S as a shift conditioned by direction. Consequently,

$$U_{\mathrm{micro}}^{*} U_{\mathrm{micro}} = I,$$

and the norm is preserved at each step.

67.2.5.1 Preparation of the 2 *Trajectories and Reading Screen*

The two openings are fixed at

$$(x_0, y_1) = (2, 8), \qquad (x_0, y_2) = (2, 18),$$

with the same initial directional state. If ψ_1 and ψ_2 are the amplitudes propagated from each aperture, the screen located at $x = 24$ publishes, for a real memory overlap $\mu \in [0, 1]$,

$$I_\mu(y) = \frac{1}{2} \sum_{d} \left(|\psi_1(24, y, d)|^2 + |\psi_2(24, y, d)|^2 + 2\,\mathrm{Re}\Big[\mu\, \overline{\psi_1(24, y, d)} \psi_2(24, y, d)\Big] \right). \tag{67.2}$$

The formula is also obtained by introducing the markers

$$|m_1\rangle = (1, 0), \qquad |m_2\rangle = (\mu, \sqrt{1 - \mu^2})$$

Table 67.1: Interference and distance from the canonical sequence after 36 steps.

Sequence	$\mathcal{V}_{\text{lineal}}$	total variation distance	mean fidelity
canonical	0.5992212256	0	1.0000000000
inverse	0.5974219552	0.0333923142	0.9740951119
replaced pair	0.5907431085	0.0628288605	0.9417893564
randomized	0.6254301808	0.0963185652	0.6564185376

and tracing over $\mathbb{C}^2_{\text{mem}}$. This equality identifies the overlap of the registers, rather than a loss of norm, as the origin of the attenuation of the fringes. In particular,

$$\mathcal{V}(\mu) = |\mu|\,\mathcal{V}(1). \tag{67.3}$$

For $\mu = 0$, a phase correction cannot reconstruct the cross term; if the markers are unmade unitarily before recombination, the curve $\mu = 1$ is exactly recovered.

67.2.5.2 Unitarity Identities of the Dodecaphase Order Under a Common Amplitude Budget

The control block consists of 12 oriented pulses with zero vector sum. The first 6 are obtained from a directional seed through the dodecaphasic signature; the remaining 6 are their opposites. Four sequences of 36 steps are compared:

1. the canonical sequence, repeated for 3 full cycles;
2. its oriented inverse;
3. a sequence in which an opposite pair is replaced by common phases of the same norm;
4. a random permutation of the same multiset of pulses.

All preserve the pulse count, the null vector sum, and the quadratic budget. The difference between their patterns therefore measures dependence on noncommutative order and directional information, not an unequal injection of norm or energy.

Noncommutativity is verified on a probe state fixed before the reading:

$$\|U_b U_a \psi - U_a U_b \psi\|_2 = 0.24776388825652637.$$

Numerical control provides

$$\|C_4^* C_4 - I\|_{\max} = 0, \qquad \max_t \left|\|\psi_t\|_2^2 - 1\right| = 1.3322676295501878 \times 10^{-15},$$

$$\Delta B_2 = 2.220446049250313 \times 10^{-16}, \qquad \varepsilon_{\text{Born}} = 1.734723475976807 \times 10^{-18},$$

$$\varepsilon_{\mathcal{V}(\mu)} = 2.220446049250313 \times 10^{-16},$$

where ΔB_2 is the maximum dispersion of the quadratic budget among the four sequences.

67.2.5.3 Interference reconstructed at T^2_{27}: detection distribution and physical domain

Theorem 67.11 (Unitary interference with explicit path memory)**.** *For the dynamics defined by* (67.1)*, the intensity obtained by tracing out the path record coincides with* (67.2)*; the visibility satisfies* (67.3)*; and sequences with the same quadratic budget may produce different patterns when their order differs.*

The proof is finite: unitarity of each factor, expansion of the square of the sum of the two branches, and partial contraction of the marker. The preceding numerical controls verify the run on T^2_{27}. The concrete chart taking the dodecaphasic state to directional phases constitutes a declared physical realization; it is

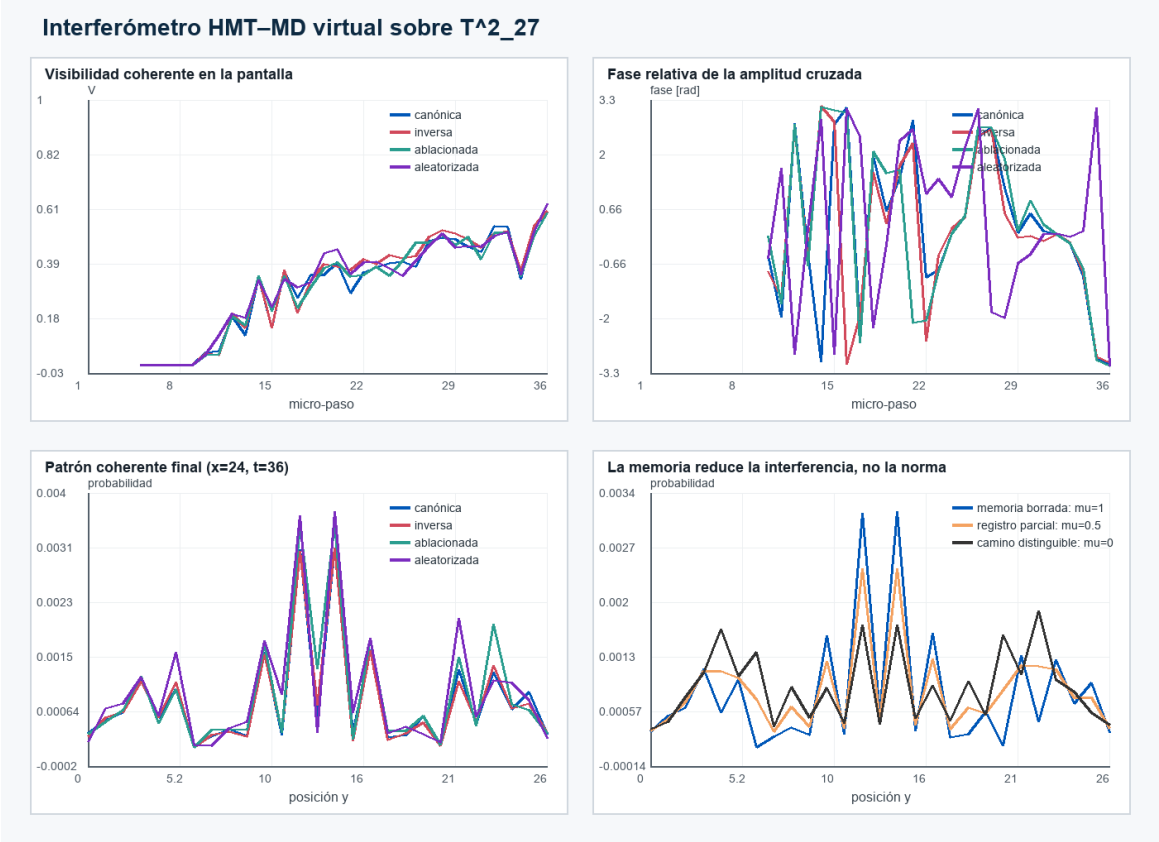


Figure 67.1: Evolution of visibility, relative phase, and final pattern for the four sequences with common budget. The original image depicts the virtual HMT–MD interferometer on T^2_{27}. Panel labels, from left to right and top to bottom: coherent visibility on the screen; relative phase of the cross amplitude; final coherent pattern $(x = 24, t = 36)$; memory reduces interference, not the norm. Axes: microstep, position y, phase in radians, and probability. The four sequences are canonical, inverse, ablated, and randomized. The memory curves represent erased memory $(\mu = 1)$, partial record $(\mu = 0.5)$, and a distinguishable path $(\mu = 0)$.

not identified by definition with the temporal return of 108 phases. The electron scale is applied after the dimensionless dynamics and neither selects the word nor fits the fringes.

The realization would be refuted by loss of unitarity, discrepancy between the explicit memory and the Born rule, creation of interference by a phase when $\mu = 0$, dependence of the energy budget on the sequence, or use of the screen outcome to select the pulse order retrospectively.

67.3 Central realisation of trajectories and Sommerfeld holonomy

The particle defined in section 90.14 has an observable history before receiving an orbital realization. The question of this section is how a representation of that history preserves flux, area and phase when expressed as central motion. The answer is organized into four operations: cancellation of the interior fluxes in a graph cut, normalization of its boundary in a metric chart, evolution under the declared central operator and phase transport over its cycles.

The order keeps visible which datum enters each statement. The graph and its records come from APP–TRIT–TPK; the metric chart specifies how its boundary is measured; the dynamical realization fixes the Lagrangian acting in that chart; the semiclassical condition uses the reduced Planck constant already constructed and the holonomy of the transported line. The results of orbital geometry are proved with

Figure 67.2: Order controls, path memory, and phase feedback. The original image presents falsifiable controls of order, memory, and feedback. Panel labels, from left to right and top to bottom: phase sweep at the port of maximum overlap; Born rule and continuous loss of coherence; permutation stress with an identical budget; variable phase aligns, but does not erase the record. Axes: probability, visibility, δ in radians, $|\mu|$, final visibility, and microstep. Sequence labels: canonical, inverse, ablated, and randomized. The other curves denote explicit memory, $V(\mu) = |\mu|V(1)$, locked phase with reversible memory, and the same phase with a recorded path.

these explicit starting data. Central mechanics is recognized in the resulting realization, without making its conventional constants inputs to the generator.

67.3.1 Flux conservation and boundary reading

Let $G = (V, E)$ be a finite cut of the route graph of the realisation under consideration, with an arc in each orientation of its edges. An oriented flux is a function $F : \vec{E} \to \mathbb{R}$ satisfying $F(y, x) = -F(x, y)$. At a vertex x, its divergence is

$$\operatorname{div} F(x) = \sum_{y:\{x,y\}\in E} F(x, y). \tag{67.4}$$

For $S \subseteq V$, the outward flux is evaluated over the arcs originating in S and ending in $V \setminus S$.

Theorem 67.12 (Flux balance in a finite cut)**.** *For every subset $S \subseteq V$,*

$$\sum_{x\in S} \operatorname{div} F(x) = \sum_{\substack{x\in S,\ y\notin S \\ \{x,y\}\in E}} F(x, y). \tag{67.5}$$

Proof. An edge with both endpoints in S contributes twice to the sum of divergences, through $F(x, y) + F(y, x) = 0$. Edges with both endpoints outside S do not contribute. Exactly the edges crossing the boundary remain, with the indicated orientation. □

The theorem transmits to the boundary the balance produced in the interior. For a cubic representation with nearest neighbours, consider

$$C_R = \{x \in \mathbb{Z}^3 : \|x\|_\infty \le R\}, \qquad R \in \mathbb{Z}_{\ge 0}.$$

Each of the six faces contains $(2R + 1)^2$ outward links in its normal direction. Links in different directions are counted separately, even when they originate at the same edge or corner vertex. Hence,

$$|\partial C_R| = 6(2R + 1)^2. \tag{67.6}$$

This census counts outward links, not the number of boundary vertices.

If Q_F is the total flux of the enclosed source, its average per link is $Q_F/[6(2R + 1)^2]$. Under the condition of mean isotropy of the reader, that average is the intensity assigned to the shell. Let us define its effective radius, in units of the step of the chart, by

$$4\pi r_{\text{eff}}^2 = 6(2R + 1)^2. \tag{67.7}$$

We then obtain

$$E_R = \frac{Q_F}{4\pi r_{\text{eff}}^2}. \tag{67.8}$$

The coordinate π of this normalization is the regional HMT output already published. The identity follows from the census and the definition of the radius; the passage from an average to a central realization additionally uses the stated isotropy. The dimensional chart then decides whether Q_F represents electric, gravitational or other observable flux and how it converts the lattice step into length. The finite balance preserves this distinct status of each operation.

67.3.2 Central dynamics on the Euclidean sheet

Let $\mu > 0$ be the reduced mass of the material realisation and let $\mathcal{K} > 0$ be its central coupling, with the dimension of energy times length. These coordinates are received after the internal constructions of mass, action and coupling; their origin and dimensional line are preserved when carrying out the realisation. The domain of the following theorem consists of all realisations that adopt, on their Euclidean sheet,

$$L(\mathbf{r}, \dot{\mathbf{r}}) = \frac{\mu}{2}\|\dot{\mathbf{r}}\|^2 + \frac{\mathcal{K}}{r}, \qquad H(\mathbf{r}, \mathbf{p}) = \frac{\|\mathbf{p}\|^2}{2\mu} - \frac{\mathcal{K}}{r}, \quad r = \|\mathbf{r}\| > 0, \quad \mathbf{p} = \mu\dot{\mathbf{r}}. \tag{67.9}$$

The exclusion of $r = 0$ delimits the regular domain of these coordinates. The trajectory is considered over a collision-free time interval.

Proposition 67.13 (Orbital plane and conic equation)**.** *A non-radial solution of the Euler–Lagrange equation of* (67.9) *conserves $\boldsymbol{\ell} = \mathbf{r} \times \dot{\mathbf{r}}$, with $\ell = \|\boldsymbol{\ell}\| > 0$. In the plane orthogonal to $\boldsymbol{\ell}$, it can be written*

$$r(\theta) = \frac{p}{1 + \epsilon\cos(\theta - \theta_0)}, \qquad p = \frac{\mu\ell^2}{\mathcal{K}}, \qquad \epsilon \geq 0. \tag{67.10}$$

Negative energy corresponds to the elliptic regime $0 \leq \epsilon < 1$.

Proof. Differentiating the potential gives

$$\mu\ddot{\mathbf{r}} = -\frac{\mathcal{K}}{r^3}\mathbf{r}. \tag{67.11}$$

Hence, $\dot{\boldsymbol{\ell}} = \mathbf{r} \times \ddot{\mathbf{r}} = 0$. Position and velocity remain orthogonal to the non-zero constant vector $\boldsymbol{\ell}$, fixing the plane. In its polar coordinates, $r^2\dot{\theta} = \ell$. Introducing $u = 1/r$ into the radial equation gives Binet's equation

$$u''(\theta) + u(\theta) = \frac{\mathcal{K}}{\mu\ell^2}.$$

Its general solution is $u = \mathcal{K}[1 + \epsilon\cos(\theta - \theta_0)]/(\mu\ell^2)$, and inversion proves (67.10) on the domain where the denominator is positive. Substitution into the energy gives $H = \mathcal{K}(\epsilon^2 - 1)/(2p)$, which is negative exactly when $\epsilon < 1$. □

Here the letter ϵ denotes orbital eccentricity, not Euler's number. The vector $\boldsymbol{\ell}$ is specific angular momentum, not the Planck constant. These notational choices allow the geometric observables and the internal constants entering their realisation to be compared without conflation.

Proposition 67.14 (Areal conservation and period law)**.** *The orbit's areal velocity is constant:*

$$\frac{d\mathcal{A}_{\mathrm{orb}}}{dt} = \frac{\ell}{2}. \tag{67.12}$$

For an ellipse with semimajor axis a, its period satisfies

$$T^2 = \frac{4\pi^2\mu}{\mathcal{K}}a^3. \tag{67.13}$$

Proof. The sector swept out in time dt has area $r^2 d\theta/2 = \ell\, dt/2$. If $b = a\sqrt{1 - \epsilon^2}$ is the semiminor axis, the total area is πab and $T = 2\pi ab/\ell$. The relation $p = a(1 - \epsilon^2) = \mu\ell^2/\mathcal{K}$ implies $\ell^2 = \mathcal{K}a(1 - \epsilon^2)/\mu$. Substitution into $T^2 = 4\pi^2 a^2 b^2/\ell^2$ gives (67.13). □

Areal conservation is determined by the central symmetry of the motion. The ellipse of this proposition lives in configuration space. The action ellipse constructed from the angular pair, in contrast, lives in

its action chart and transports the corresponding normalizations. The link between the two requires the realization map that identifies their variables; sharing an elliptical shape does not replace that map. The orbital reading thus preserves the dependencies on mass, coupling and geometry that produce it.

67.3.3 Orbital covering and real sign line

On the complex plane with the origin removed, the map

$$q = z^2 : \mathbb{C} \setminus \{0\} \longrightarrow \mathbb{C} \setminus \{0\} \tag{67.14}$$

is a double covering. If $q(t) = r(t)e^{i\theta(t)}$ and a root is chosen initially, its lift is $z(t) = \sqrt{r(t)}e^{i\theta(t)/2}$. One circuit with an increment 2π in θ ends at $-z(0)$, and two circuits restore the initial point. This is the transport of the two lifts over the punctured plane.

An orbital band requires another object: a real line associated with the orbit and its sign character. Let $\gamma : S^1 \to \mathbb{R}^2$ be a regular injective parametrisation of an elliptic orbit, and let $\delta > 0$. With holonomy -1 over one circuit, its interval band is represented by

$$M_\gamma = ([0, 2\pi] \times [-\delta, \delta])/\big((0, u) \sim (2\pi, -u)\big). \tag{67.15}$$

Identifying the base with the image of γ places the zero section over the orbit.

Proposition 67.15 (Oriented transport over the orbital cycle)**.** *The line transport in* (67.15) *satisfies* $\mathcal{H}(\gamma) = -I$ *and* $\mathcal{H}(\gamma^2) = I$.

Proof. The terminal fibre is identified with the initial fibre by $u \mapsto -u$; composing this map twice gives the identity. This is the construction in Theorem 90.36, applied to the orbital loop. □

The covering (67.14) has two-point fibres; (67.15) has interval fibres. Monodromy of order two appears in both, while preserving these distinct types. Levi–Civita dynamical regularisation also uses the transformation of momenta and time reparametrisation. The present construction uses only its covering and its lifting law, with the scope just proved.

67.3.4 Phase condition and half-integer shift

Let γ be a cycle of the Hamiltonian realisation, and let

$$J_\gamma = \oint_\gamma \mathbf{p} \cdot d\mathbf{q} \tag{67.16}$$

be its action. The reduced Planck constant $\hbar_{\mathrm{int}} > 0$ is received from the HMT action construction, in the same dimensional line as J_γ. The quotient $J_\gamma/\hbar_{\mathrm{int}}$ is therefore dimensionless. Suppose that semiclassical transport in the complexified line contributes the phase $\exp(iJ_\gamma/\hbar_{\mathrm{int}})$ and that the real orientation line contributes the character $\varepsilon_\gamma \in \{+1, -1\}$. The closure condition for the transported section is

$$\varepsilon_\gamma \exp\left(\frac{iJ_\gamma}{\hbar_{\mathrm{int}}}\right) = 1. \tag{67.17}$$

The product preserves both contributions: the action of the cycle and the sign of the line. The latter is not absorbed into a modification of the Planck constant.

Theorem 67.16 (Integer and half-integer action sectors)**.** *Under the transport* (67.17)*,*

$$J_\gamma = \begin{cases} 2\pi\hbar_{\mathrm{int}} n, & \varepsilon_\gamma = +1, \\ 2\pi\hbar_{\mathrm{int}}(n + \frac{1}{2}), & \varepsilon_\gamma = -1, \end{cases} \qquad n \in \mathbb{Z}. \tag{67.18}$$

Proof. For a positive sign, the exponential must equal 1, whose real arguments are $2\pi n$. For a negative sign, it must equal -1, whose arguments are $(2n+1)\pi$. Multiplying by $\hbar_{\mathrm{int}}$ produces the two families. □

The formula determines the closure constraint on the action of a given cycle. Obtaining an atomic spectrum also uses the specific Hamiltonian, the independent cycles and the quantisation prescription. If this prescription includes Maslov indices, their phases are composed with (67.17) according to the chosen transport, avoiding counting the same topological contribution twice.

Reversing the orientation of the cycle changes J_γ to $-J_\gamma$, whereas the sign character satisfies $\varepsilon_{\gamma^{-1}} = \varepsilon_\gamma^{-1} = \varepsilon_\gamma$. The total phase becomes its inverse, and the closure condition is preserved. This is a precise law of orbital bidirectionality. Its identification with material antisections requires the lift in Section 90.15.2; equality of two signs does not by itself establish that identification.

The realization thus brings together three successive conservations: flux passes from the interior to the boundary; central dynamics conserves angular momentum and the area swept per unit time; transport of the section preserves its closure condition through action and holonomy. Each result retains the equation, the domain and the data that support it. The material history of the TPK is represented in those dynamics with its memory, instead of being replaced by the orbital figure it publishes.

67.4 Differential Geometry and Dimensional Realizations

67.4.1 Curvature and torsion of the published curves

67.4.1.1 Curvature and arc length of a polar curve $r = r(\theta)$

Let

$$\gamma(\theta) = r(\theta)(\cos\theta, \sin\theta).$$

With primes denoting derivatives with respect to θ, the velocity and curvature are

$$v(\theta) = \sqrt{r^2 + (r')^2}, \tag{67.19}$$

$$\kappa_{\mathrm{F}}(\theta) = \frac{r^2 + 2(r')^2 - rr''}{\left(r^2 + (r')^2\right)^{3/2}}. \tag{67.20}$$

The arc length between θ_0 and θ_1 is

$$s = \int_{\theta_0}^{\theta_1} \sqrt{r^2 + (r')^2}\, \mathrm{d}\theta.$$

These magnitudes belong to the published curve. They are neither the TRIT regime nor the radial cocycle.

67.4.1.2 Evaluation of κ_{F} in the logarithmic, Archimedean, Fermat, hyperbolic, and lituus families

For the logarithmic spiral $r = r_0 \mathrm{e}^{b\theta}$,

$$r' = br, \qquad r'' = b^2 r,$$

and

$$\kappa_{\mathrm{F}}(\theta) = \frac{1}{r(\theta)\sqrt{1+b^2}}. \tag{67.21}$$

The angle ψ between the tangent vector and the radial direction satisfies

$$\tan\psi = \frac{r}{r'} = \frac{1}{b},$$

and is therefore constant. This is the equiangular property of the logarithmic spiral.

For the Archimedean spiral $r = a\theta$,

$$\kappa_{\mathrm{F}}(\theta) = \frac{\theta^2 + 2}{a(\theta^2 + 1)^{3/2}}.$$

For Fermat $r = a\sqrt{\theta}$, the singularity at the origin requires $\theta > 0$ to be declared. For the hyperbolic and lituus branches, the angular origin is an asymptote and not a point of the domain.

Chapter 68

Quantum occupation statistics: Fermi–Dirac and Bose–Einstein distributions

Microcanonical enumeration and the symmetric and antisymmetric completions produce the Bose–Einstein and Fermi–Dirac laws. The Maxwell–Boltzmann limit is retained as a derived regime, not as a starting point.

68.1 Microcanonical enumeration of states on a finite screen

Consider the finite realization

$$\mathcal{H}_{\mathrm{occ}} = \bigoplus_{k=0}^{3} \mathbb{C}^{g_k}, \qquad H_{\mathrm{occ}}|_{\mathbb{C}^{g_k}} = E_k^{\mathrm{occ}} I,$$

with levels and multiplicities

$$E^{\mathrm{occ}} = (0, 3, 6, 9), \qquad g = (1, 6, 12, 8), \qquad (N, E) = (12, 66).$$

The sum $1 + 6 + 12 + 8 = 27$ makes $\mathcal{H}_{\mathrm{occ}}$ a screen of twenty-seven modes. This spectrum is the datum of a declared finite representation; its cardinality does not suffice to identify it with any other occurrence of 27 in HMT. The morphism relating a specific boundary screen to $(\mathcal{H}_{\mathrm{occ}}, H_{\mathrm{occ}})$ must specify both the fibers and the spectral operator. The fermionic and bosonic generating functions are

$$F(y, x) = \prod_k (1 + y x^{E_k^{\mathrm{occ}}})^{g_k}, \qquad B(y, x) = \prod_k (1 - y x^{E_k^{\mathrm{occ}}})^{-g_k}.$$

Proposition 68.1 (Microcanonical cardinalities)**.**

$$[y^{12} x^{66}] F = 2\,104\,494, \qquad [y^{12} x^{66}] B = 259\,436\,430.$$

Proof. In the fermionic product, choosing r of the g_k modes contributes $\binom{g_k}{r} y^r x^{r E_k^{\mathrm{occ}}}$. In the bosonic product, distributing r indistinguishable occupations among g_k modes contributes $\binom{g_k + r - 1}{r} y^r x^{r E_k^{\mathrm{occ}}}$. The finite convolution over the four levels produces the stated integers. □

68.2 Bose–Einstein and Fermi–Dirac distributions

The equilibrium formulation starts from the Hamiltonian and the total occupation number,

$$H = \sum_k E_k^{\mathrm{occ}} n_k, \qquad N = \sum_k n_k, \qquad K_\mu := H - \mu N. \tag{68.1}$$

For $\beta > 0$, the grand canonical state is

$$\rho_{\beta,\mu} = \frac{e^{-\beta K_\mu}}{\mathcal{Z}_{\beta,\mu}}, \qquad \mathcal{Z}_{\beta,\mu} = \mathrm{Tr}(e^{-\beta K_\mu}). \tag{68.2}$$

Theorem 68.2 (Grand canonical factorization and occupation laws)**.** *For energy levels E_k^{occ} with degeneracies g_k, the fermionic trace factorizes as*

$$\mathcal{Z}_{\mathrm{FD}} = \prod_k \left(1 + e^{-\beta(E_k^{\mathrm{occ}} - \mu)}\right)^{g_k}, \qquad \langle n_k \rangle_{\mathrm{FD}} = \frac{g_k}{e^{\beta(E_k^{\mathrm{occ}} - \mu)} + 1}. \tag{68.3}$$

In the bosonic sector, under the condition $\mu < \inf_k E_k^{\mathrm{occ}}$,

$$\mathcal{Z}_{\mathrm{BE}} = \prod_k \bigl(1 - e^{-\beta(E_k^{\mathrm{occ}} - \mu)}\bigr)^{-g_k}, \qquad \langle n_k \rangle_{\mathrm{BE}} = \frac{g_k}{e^{\beta(E_k^{\mathrm{occ}} - \mu)} - 1}. \tag{68.4}$$

Proof. The operator K_μ is a sum of commuting occupation operators, so its trace factorizes over modes. In each fermionic mode, the occupations 0 and 1 are summed, and the corresponding factor is $1 + u_k$, where $u_k = e^{-\beta(E_k^{\mathrm{occ}} - \mu)}$. In each bosonic mode, all nonnegative occupations are summed, giving the geometric series $(1 - u_k)^{-1}$; the condition on μ ensures $u_k < 1$. The powers g_k incorporate the degeneracy. Finally,

$$\langle n_k \rangle = -\frac{1}{\beta} \frac{\partial}{\partial E_k^{\mathrm{occ}}} \log \mathcal{Z}$$

yields the two occupation laws [16, 43, 47, 35, 101]. □

The microcanonical derivation recovers the same result in the thermodynamic regime. For mean fermionic occupations $p_k = m_k / g_k$, the combinatorial entropy per level is

$$S_k \simeq g_k \bigl[-p_k \log p_k - (1 - p_k) \log(1 - p_k) \bigr]. \tag{68.5}$$

The variation of $\sum_k S_k$, subject to $\sum_k g_k p_k = N$ and $\sum_k g_k E_k^{\mathrm{occ}} p_k = E$, gives

$$\log \frac{1 - p_k}{p_k} = \alpha + \beta E_k^{\mathrm{occ}}, \qquad \mu = -\frac{\alpha}{\beta}, \tag{68.6}$$

and hence the Fermi–Dirac occupation of (68.3). The multipliers β and μ are the dual coordinates of the energy and population constraints.

For mean bosonic occupations $\overline{n}_k = m_k / g_k$, the log-multiplicity of the level is

$$S_k \simeq g_k \bigl[(1 + \overline{n}_k) \log(1 + \overline{n}_k) - \overline{n}_k \log \overline{n}_k \bigr]. \tag{68.7}$$

The variation of $\sum_k S_k$, subject to $\sum_k g_k \overline{n}_k = N$ and $\sum_k g_k E_k^{\mathrm{occ}} \overline{n}_k = E$, yields

$$\log \frac{1 + \overline{n}_k}{\overline{n}_k} = \alpha + \beta E_k^{\mathrm{occ}}, \qquad \overline{n}_k = \frac{1}{e^{\beta(E_k^{\mathrm{occ}} - \mu)} - 1}, \qquad \mu = -\frac{\alpha}{\beta}. \tag{68.8}$$

This is the Bose–Einstein occupation of (68.4). The condition $\mu < \inf_k E_k^{\mathrm{occ}}$ ensures positivity of the denominator and convergence of each modal factor.

In the finite instance constructed, the two constraints determine

$$(\beta_{\mathrm{FD}}, \mu_{\mathrm{FD}}) = (0.15307011353415928,\ 4.502865197597043),$$

$$(\beta_{\mathrm{BE}}, \mu_{\mathrm{BE}}) = (0.05456727960068316,\ -15.92569781906737).$$

If m_k^{mic} denotes the mean occupancy obtained by exact microcanonical enumeration and m_k^{eq} the evaluation of the corresponding distribution, the errors $\sum_k |m_k^{\mathrm{mic}} - m_k^{\mathrm{eq}}|$ are

$$0.09897380039002812 \quad \text{and} \quad 0.9190625715582017,$$

respectively. The enumeration traverses all occupation quadruples, computes their integer multiplicities, and solves the two constraint equations. The two laws are realized on the same screen. The physical choice between them requires a grading and a Hamiltonian compatible with the trajectory representation. The thermal parameter will subsequently be typed as $\beta = (k_B \Theta)^{-1}$.

68.3 Quantified convergence to the Maxwell–Boltzmann distribution

The relation between the two quantum statistics and the classical regime can be formulated without resorting to an informal approximation. For each mode, let

$$u_k := e^{-\beta(E_k^{\mathrm{occ}}-\mu)}.$$

The state occupations are written

$$\overline{n}_k^{\mathrm{FD}} = \frac{u_k}{1+u_k}, \qquad \overline{n}_k^{\mathrm{BE}} = \frac{u_k}{1-u_k};$$

the multiplicities g_k are incorporated subsequently through $\langle n_k \rangle = g_k \overline{n}_k$.

Theorem 68.3 (Common classical limit of quantum statistics)**.** *If* $0 \le u_k \le \delta < 1$*, then*

$$\left|\overline{n}_k^{\mathrm{FD}} - u_k\right| \le u_k^2, \qquad \left|\overline{n}_k^{\mathrm{BE}} - u_k\right| \le \frac{u_k^2}{1-\delta}.$$

Consequently, in the regime dilute $u_k \to 0$*,*

$$\boxed{\overline{n}_k^{\mathrm{FD}} \sim \overline{n}_k^{\mathrm{BE}} \sim e^{-\beta(E_k^{\mathrm{occ}}-\mu)}},$$

which is the Maxwell–Boltzmann distribution per mode.

Proof. The exact identities

$$u_k - \frac{u_k}{1+u_k} = \frac{u_k^2}{1+u_k}, \qquad \frac{u_k}{1-u_k} - u_k = \frac{u_k^2}{1-u_k}$$

provide the bounds. For $0 < u_k \le \delta$, dividing by u_k and taking $u_k \downarrow 0$ yields the asymptotic equivalence. The leading term of order u_k^2 preserves the physical difference: exclusion decreases fermionic occupation, whereas accumulation increases bosonic occupation. □

68.4 Quantum completions, leaf holonomy, and statistical periodicity

The symmetric and exterior completions, the sheet grading, the holonomy $2\pi/4\pi$, the occupation counts, and the cellular complex of APP form a single algebraic antecedent of quantum statistics. The complete development is incorporated here so that periodicity is not reduced to a mere subsequent reference.

68.5 Statistical completions and fields over APP

The words Bose, Fermi, Pauli, or photon do not by themselves designate theorems of a discrete system. Obtaining them requires specification of a state space, a grading, a product, and operators. APP and TPK provide a set of oriented states and a sheet parity. Two universal completions can be constructed from those data: the symmetric algebra and the exterior algebra. This section identifies exactly what follows from the construction and separates algebraic statistics from its physical interpretation.

68.5.1 Linear state space associated with a particle realization

Let S be a finite set of TPK route classes at the chosen depth, and let

$$\mathcal{H}_1 = \ell^2(S)$$

be the Hilbert space with orthonormal basis $(e_s)_{s\in S}$. A choice of positive weights E_s defines the diagonal operator

$$He_s = E_s e_s.$$

The choice of S, its quotient by symmetries, and the weights are part of the model. Once these are fixed, the multiparticle completions are canonical.

68.5.2 Symmetric and exterior completions

We define

$$\mathcal{F}_+(\mathcal{H}_1) = \bigoplus_{n\geq 0} \operatorname{Sym}^n \mathcal{H}_1, \qquad \mathcal{F}_-(\mathcal{H}_1) = \bigoplus_{n\geq 0} \bigwedge^n \mathcal{H}_1.$$

The first is the bosonic completion; the second, the fermionic.

Theorem 68.4 (Pauli exclusion principle in the exterior completion)**.** *For every mode $s \in S$, the creation and annihilation operators of the exterior completion satisfy*

$$\{a_s, a_t^\dagger\} = \delta_{st} I, \qquad \{a_s, a_t\} = \{a_s^\dagger, a_t^\dagger\} = 0.$$

In particular, the occupation operator $N_s = a_s^\dagger a_s$ is a projector:

$$N_s^2 = N_s,$$

and its eigenvalues are $0, 1$.

Proof. Creation is the exterior product by e_s; annihilation is its adjoint. Antisymmetry implies $e_s \wedge e_s = 0$ and yields the canonical relations by calculation on the exterior basis. Then

$$N_s^2 = a_s^\dagger a_s a_s^\dagger a_s = a_s^\dagger (I - a_s^\dagger a_s) a_s = N_s.$$

Every self-adjoint projector has spectrum contained in $\{0, 1\}$. □

The exclusion principle thus appears as a theorem of the exterior completion. The specifically HMT question is which transport datum selects that completion for a class of routes.

68.5.3 Sheet grading and exchange algebra

Suppose that the oriented sheet defines an additive character

$$p : \operatorname{Mor}(\mathsf{P}_{\mathrm{TPK}}) \longrightarrow \mathbb{Z}/2\mathbb{Z}, \qquad p(\gamma\eta) = p(\gamma) + p(\eta).$$

On the corresponding graded linear space, introduce the symmetry

$$\tau(v \otimes w) = (-1)^{p(v)p(w)} w \otimes v.$$

Theorem 68.5 (Statistical completion determined by parity)**.** *The quotient of the tensor algebra by the relations*

$$v \otimes w - (-1)^{p(v)p(w)} w \otimes v$$

is symmetric on the even sector and exterior on the odd sector. Two identical odd states have zero product; even states admit arbitrary occupation.

Proof. If $p(v) = p(w) = 0$, the relation is $vw = wv$. If $p(v) = p(w) = 1$, it is $vw = -wv$; setting $v = w$ and working in characteristic zero gives $v^2 = 0$. Mixed sectors commute without an additional sign. This is the universal property of the graded symmetric algebra. □

The theorem converts a compositional TPK parity into an exact occupancy rule. To obtain the physical spin–statistics theorem, one must additionally construct a positive local Lorentzian theory and prove that

sheet parity coincides with spin under rotations. The equality does not follow solely from both objects taking values modulo two.

68.5.4 Sheet holonomy and $2\pi/4\pi$ periodicity

The preceding parity admits a natural geometric realization when a surviving extension has a closed return with a nontrivial orientation cover. Let ρ be the return generator and let

$$\varepsilon : \langle \rho \rangle \longrightarrow \{-1, +1\}$$

be its holonomy character, with $\varepsilon(\rho) = -1$. The associated real bundle is then the orientation line of a Möbius strip: one turn closes the geometric projection but changes the lifted sheet; two turns restore the orientation.

To express this property in angular terms, fix a unitary representation U of the rotation cover on the graded space $\mathcal{H}_1 = \mathcal{H}_{\bar{0}} \oplus \mathcal{H}_{\bar{1}}$, compatible with sheet parity in the sense that

$$U(2\pi)v = (-1)^{p(v)}v$$

for every homogeneous vector v. Compatibility is a condition of the realization: it identifies the generator of the orientation cover with the character p already constructed on the routes.

Theorem 68.6 (Restoration of orientation and exchange algebra)**.** *Under the preceding compatibility condition,*

$$U(2\pi)|_{\mathcal{H}_{\bar{0}}} = I, \qquad U(2\pi)|_{\mathcal{H}_{\bar{1}}} = -I, \qquad U(4\pi) = I.$$

Moreover, the same character p determines the graded symmetry

$$\tau(v \otimes w) = (-1)^{p(v)p(w)} w \otimes v.$$

Thus the even-holonomy sector is completed by symmetric powers and the odd-holonomy sector by exterior powers. In the latter, the exclusion relation $v \wedge v = 0$ holds.

Proof. The first and second equalities are the compatibility condition evaluated on each homogeneous component. Iterating one turn gives

$$U(4\pi)v = U(2\pi)^2 v = (-1)^{2p(v)}v = v,$$

which proves the restoration. The assertion concerning exchange is the parity-completion theorem of the preceding section: for two even vectors the sign is positive, whereas for two odd vectors it is negative. □

Three returns should be distinguished. Angular closure after 2π is a return of the projection; closure after 4π restores the lifted orientation; neither, by itself, requires every coordinate of the record to return to its initial value. This separation is the dimensional analogue of the HMT distinction between phase return and state return.

68.5.4.1 The sum and product sheets as supports of the 2 *completions*

The APP table contains two evaluations on the same support. The additive sheet acts by translations and preserves the multiplicity of continuations; in this sense, it accumulates. The multiplicative sheet permutes the units and contracts fibers over noninvertible elements; in this sense, it folds. This incompatibility precedes the statistical choice and explains why TPK must preserve two genealogies rather than reduce them to a single visible word.

The statistical assignment is not obtained, however, by verbally replacing “sum” with “boson” and “product” with “fermion.” The exact map is the parity character

$$p : \mathrm{Mor}(\mathsf{P}_{\mathrm{TPK}}) \longrightarrow \mathbb{Z}/2\mathbb{Z}.$$

When the transport of an extension lies in the sector $p = 0$, its completion admits cumulative occupation and is symmetric. When it lies in $p = 1$, sheet reversal introduces the exchange sign and the completion is exterior. Sum and product supply the two local geometries; the record, orientation, and character p determine the sector in which a route is realized.

With a diagonal Hamiltonian H, energy ϵ, chemical potential μ, and thermal parameter β, the mean occupations of the two sectors are then

$$\bar{n}_{\mathrm{par}}(\epsilon) = \frac{1}{e^{\beta(\epsilon-\mu)} - 1}, \qquad \bar{n}_{\mathrm{impar}}(\epsilon) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1}.$$

The Bose–Einstein and Fermi–Dirac formulas are therefore the thermodynamic evaluations of the symmetric and exterior completions selected by transport. The periodicity 4π and fermionic exclusion are coordinated by the same character, but neither of them is inferred from a mere numerical coincidence.

68.5.5 Number and Occupation Distribution Operators in the Fock Realization

If there are g modes and N particles, the sector dimensions are

$$\Omega_F(g, N) = \binom{g}{N}, \qquad \Omega_B(g, N) = \binom{g + N - 1}{N}.$$

The first formula counts subsets of modes; the second, weak compositions of N into g parts. These counts are exact consequences of the two completions and require no thermodynamic hypothesis.

68.5.6 The cellular complex of the APP torus

The APP graph is the one-skeleton of a cellular decomposition of the torus T^2 with 9×9 vertices. Let

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2$$

be its real cochain complex. Choosing the uniform inner product, define the codifferential $\delta = d^*$ and the Laplacian

$$\Delta_k = d_{k-1}\delta_k + \delta_{k+1}d_k.$$

Theorem 68.7 (Harmonic sector of the digital torus)**.** *The dimension of the space of harmonic one-forms is two:*

$$\dim \ker \Delta_1 = b_1(T^2) = 2.$$

A basis can be chosen that is represented by the uniform horizontal and vertical flows. Both are closed and coclosed, and neither is the gradient of a globally defined periodic function.

Proof. The Hodge theorem for finite cellular complexes identifies $\ker \Delta_1$ with $H^1(T^2; \mathbb{R})$. The cohomology of the torus has rank two. Directly, the constant horizontal and vertical flows have zero divergence and curl; their integrals over the two fundamental cycles are independent, so they are not exact. □

The space of connections modulo gauge transformations is

$$C^1/\mathrm{im}\, d_0.$$

In a finite complex equipped with the preceding inner product, each orbit admits a Coulomb representative in $\ker \delta_1$; within the orbit, the choice is unique modulo the stabilizer $\ker d_0$ of the gauge parameter. The correct cohomological quotient is

$$\ker d_1/\mathrm{im}\, d_0 \simeq \ker \Delta_1.$$

Its two harmonic classes constitute a natural mathematical candidate for two free global modes. They are not yet the two local polarizations of a photon in 3 + 1-dimensional spacetime: that identification requires a dynamical limit, a dispersion relation, and a physical gauge group.

68.5.7 Discrete Gauss law

For an oriented 1-cochain F and a set of vertices V, divergence is defined by

$$(\operatorname{div} F)(v) = \sum_{e:v\to w} F(e) - \sum_{e:w\to v} F(e).$$

Summing over V, every interior edge appears once with each sign and cancels. Therefore,

$$\boxed{\sum_{v\in V} \operatorname{div} F(v) = \operatorname{Flux}_{\partial V}(F).}$$

This identity is an exact Gauss law on the graph. Interpreting div F as electric charge requires the charge character and dimensional realization constructed in Dimensional Mechanics.

68.5.8 Reversible measurement apparatus

Let S be a set of states and let $r : S \to \mathbb{Z}/9\mathbb{Z}$ be a readout. Coupling to a nonadic register is defined by

$$U_r : S \times \mathbb{Z}/9\mathbb{Z} \longrightarrow S \times \mathbb{Z}/9\mathbb{Z}, \qquad U_r(s, a) = (s, a + r(s)).$$

Proposition 68.8 (Reversibility of the coupling)**.** *The map U_r is to permutation and*

$$U_r^{-1}(s, a) = (s, a - r(s)).$$

Information loss arises only after applying a reader to the register or conditioning the state on the outcome.

Proof. The proposed inverse satisfies both composition identities by the group law of $\mathbb{Z}/9\mathbb{Z}$. □

This proposition formalizes the operational definition of measurement: coupling, reading, and restricting compatible futures are three different maps. The full coupling is reversible; the observational quotient need not be.

68.5.9 Algebraic correspondences and physical realizations

The Fock completions, exclusion conditional on parity, the harmonic sector of APP, discrete Gauss law, and reversibility of the apparatus are complete mathematical theorems. Their physical identifications form a diagram that must commute:

$$\text{sheet parity} \longrightarrow \text{graded interchange} \longrightarrow \text{CAR/CCR} \longrightarrow \text{physical statistics},$$

$$H^1(\mathcal{A}_9) \longrightarrow \text{discrete gauge field} \longrightarrow \text{massless mode} \longrightarrow \text{photon}.$$

The first two arrows of each line have explicit constructions; the last require the corresponding dynamics and physical limit. This separation permits the bridges to be developed without conflating an analogy of cardinality with a representation.

From reversible measurement to fermionic occupation and to the algebraic exclusion principle of Pauli The statistical distinction is completed algebraically through occupation idempotence and fermionic exclusion.

68.5.10 Quasi-free correlation and spectral occupation entropies

For a gauge-invariant, number-conserving quasifree fermionic state defined on a finite-dimensional one-particle space, let $C_{ij} = \langle a_j^\dagger a_i \rangle$ be the one-particle correlation operator. The gauge-invariant hypothesis excludes anomalous pairing correlations, so that C determines the quasifree state. If $0 < C < I$, its one-particle modular generator and the reciprocal reconstruction of the correlator are

$$\boxed{K_1 = \log\bigl((I - C)C^{-1}\bigr), \qquad C = (I + e^{K_1})^{-1}.} \tag{68.9}$$

Corollary 68.9 (Logistic form of the modular spectrum)**.** *If $K_1 u_i = \kappa_i u_i$, the occupation of mode u_i is*

$$\langle n_i \rangle = \frac{1}{1 + e^{\kappa_i}}.$$

For $\kappa_i = \beta(\varepsilon_i - \mu)$, the Fermi–Dirac law is recovered.

Proof. The spectral diagonalization of K_1 diagonalizes every function thereof. Applying $x \mapsto (1 + e^x)^{-1}$ to each eigenvalue yields the formula. □

If C has eigenvalues $p_i \in [0, 1]$, the occupation entropy of the quasifree fermionic state is, with the convention $0 \log 0 = 0$,

$$S_F = -\sum_i \bigl[p_i \log p_i + (1 - p_i) \log(1 - p_i)\bigr]. \tag{68.10}$$

For a diagonal Gaussian bosonic state with finitely many modes and mean occupancies $\overline{n}_i$,

$$S_B = \sum_i \bigl[(1 + \overline{n}_i) \log(1 + \overline{n}_i) - \overline{n}_i \log \overline{n}_i\bigr]. \tag{68.11}$$

The same formula extends to a countable family when the series on the right-hand side converges. These entropies quantify occupation configurations. Entanglement entropy has another domain: the reduced state induced by a tensor decomposition. Identifying the two requires specifying the reduction and the correlation operator that implements it.

Finally, for a reduced state ρ_A, the *von Neumann entropy*

$$S_{\mathrm{vN}}(\rho_A) = -\operatorname{Tr}(\rho_A \log \rho_A)$$

is a spectral property of the density operator. When ρ_A comes from a global purification, it quantifies entanglement across the cut. In the HMT boundary construction, the area operator, the modular Hamiltonian, and the corresponding reduction remain in the section 51.11.

The maps between these domains are published through explicit identities. The decomposition of a measure along the fibres of q satisfies the chain rule

$$\mathsf{H}_\mu(\mathbf{X}) = \mathsf{H}_{q_*\mu}(q(\mathbf{X})) + \mathsf{H}_{\mathrm{fib}}(q; \mu). \tag{68.12}$$

A finite distribution $p = (p_i)$ is realized diagonally as

$$\rho_p = \sum_i p_i |i\rangle\langle i|, \qquad S_{\mathrm{vN}}(\rho_p) = \mathsf{H}(p).$$

If $P_{N,E}$ projects onto the microcanonical subspace of dimension $\Omega(N, E)$, then

$$\rho_{\mathrm{mc}} = \frac{P_{N,E}}{\Omega(N, E)}, \qquad S_{\mathrm{vN}}(\rho_{\mathrm{mc}}) = \log \Omega(N, E) = S_{\mathrm{mc}}.$$

For a pure bipartite state,

$$\rho_A = \mathrm{Tr}_B\big(|\Psi_{AB}\rangle\langle\Psi_{AB}|\big)$$

defines the reduction that measures entanglement. Finally, equality between $h_{\mathrm{top}} = \log \rho(A)$ and a metric entropy is attained, for this primitive system, by means of the Parry measure, which realizes maximal metric entropy; it is not attributed to an arbitrary measure on histories.

68.6 Transport of Fock completions under refinement

The completions and their operator algebras are constructed in Section 64.5. The result of this stage is their compatibility with refinement morphisms: transport of the one-particle fiber extends to all occupation sectors.

68.6.1 Second quantization of refinement morphisms

Let $\mathsf{Hol}_{\pm}^{\leq 1}$ be the monoidal category of one-particle fibers, with contractive refinement morphisms preserving holonomy parity $\epsilon \in \{+1, -1\}$. It includes, in particular, unitary intertwiners. The monoidal operation on the fibers is direct sum; on the completions it induces the tensor product, graded in the exterior sector. Second quantization uses the completions of Section 64.5.2:

$$\mathfrak{F}(H, \epsilon) = \begin{cases} \mathcal{F}_+(H), & \epsilon = +1, \\ \mathcal{F}_-(H), & \epsilon = -1, \end{cases}$$

and, for a contractive intertwiner $T : H \to H'$,

$$\mathfrak{F}(T) = \bigoplus_{n \geq 0} \mathrm{Sym}^n(T) \quad \text{or} \quad \bigoplus_{n \geq 0} \bigwedge^n T.$$

68.6.2 Functoriality, canonical operators, and transport domains

Let $\mathcal{D}_\epsilon(H)$ be the algebraic direct sum of the finite-particle sectors. In the following identities, $a_H^\dagger(f)$ and $a_H(f)$ denote the creation and annihilation operators of the chosen completion: the operators a of the exterior sector or the operators b of the symmetric sector in Section 64.5.4.

Theorem 68.10 (Internal transport of the statistic)**.** *The preceding completion defines a graded monoidal functor on $\mathsf{Hol}_{\pm}^{\leq 1}$, with bounded operator values and compatibility with refinement. In the branch $\epsilon = -1$, creation and annihilation operators satisfy CAR and occupation operators obey $N_s^2 = N_s$; in the branch $\epsilon = +1$, they satisfy CCR. Thus Pauli exclusion and fermionic and bosonic statistics descend from holonomy parity once the one-particle representation has been fixed. For every contractive intertwiner $T : H \to H'$, $f \in H$, and $g \in H'$, the following hold on $\mathcal{D}_\epsilon(H)$:*

$$\mathfrak{F}(T) a_H^\dagger(f) = a_{H'}^\dagger(Tf) \mathfrak{F}(T), \tag{68.13}$$

$$a_{H'}(g) \mathfrak{F}(T) = \mathfrak{F}(T) a_H(T^* g). \tag{68.14}$$

If T is isometric, the second identity, with $g = Tf$, also transports annihilation of f. For $\|f\| = 1$, the occupation operators then satisfy

$$N_{H', Tf} \mathfrak{F}(T) = \mathfrak{F}(T) N_{H,f}, \qquad N_{H,f} = a_H^\dagger(f) a_H(f). \tag{68.15}$$

Proof. Symmetric and exterior powers preserve compositions and identities; in each degree, their norms are bounded by $\|T\|^n$. Contractivity therefore ensures that the direct sum defines a bounded operator. The decomposition of powers of a direct sum according to their degrees in each factor gives monoidal compatibility, with the graded exchange sign in the exterior powers.

The CAR relations and idempotence are Theorem 64.14; the CCR relations are (64.12). To verify their transport, it suffices to act on symmetric or exterior products of vectors. Applying T to each factor commutes with inserting the factor f, which is transformed into Tf; this proves (68.13). Annihilation sums the contractions of each factor and uses $\langle g, Tf\rangle = \langle T^*g, f\rangle$; exterior signs and symmetric normalizations agree on both sides. This yields (68.14). If $T^*T = I$, setting $g = Tf$ and composing the two identities proves (68.15). All these operations preserve the finite-particle domain. □

For noncontractive morphisms, second quantization is defined first on $\mathcal{D}_\epsilon(H)$; its bounded extension depends on a uniform bound for the norms over all degrees. Such morphisms lie outside $\mathsf{Hol}_\pm^{\leq 1}$. In the symmetric completion, if $\|T\| > 1$, the norms $\|\operatorname{Sym}^n T\| = \|T\|^n$ grow without bound; in the exterior completion of a finite fiber, only finitely many degrees are nonzero. The general annihilation identity retains T^*g; its specialization to the same transported vector corresponds to the isometry condition stated above.

The distinction is already verified in a one-dimensional fiber. If $T = tI$, with $0 < t < 1$, and e is a unit vector, then, on the one-particle state e, in either completion,

$$a(te)\mathfrak{F}(T)e = t^2\Omega, \qquad \mathfrak{F}(T)a(e)e = \Omega, \tag{68.16}$$

where Ω is the vacuum. The general identity (68.14) correctly gives $\mathfrak{F}(T)a(T^*te)e = t^2\Omega$. The factor t^2 identifies precisely the hypothesis allowing specialization of annihilation transport and, subsequently, occupation transport.

The result establishes internal algebraic transport. Its identification with the relativistic spin–statistics theorem retains the additional conditions of Lorentzian representation and locality specified in Section 64.5.5.

68.7 Thermodynamic realization through Gibbs states and the pressure limit

68.7.1 Finite Hamiltonian, partition function, Gibbs state, and cofinal pressure limit

For a compressed Hamiltonian $H_{X,V}$ over a finite volume or refinement V, the thermal state is completely defined by

$$Z_{X,V}(\beta) = \operatorname{Tr} e^{-\beta H_{X,V}}, \qquad \rho_{X,V,\beta} = Z_{X,V}(\beta)^{-1} e^{-\beta H_{X,V}}.$$

The thermodynamic limit exists when the pressure

$$p_X(\beta) = \lim_{V \nearrow \infty} \frac{1}{\beta |V|} \log Z_{X,V}(\beta)$$

exists and is independent of the cofinal sequence.

68.7.2 Condensation criterion by saturation of the excited-state density

Bosonic condensation is determined by saturation of the excited-state density; it is not asserted merely from the existence of a symmetric completion. The problem is thus expressed by a verifiable mathematical condition on the Hamiltonian and the limit, rather than by a nominal attribution.

68.8 Microcanonical enumeration of states on a finite screen

Consider the finite realization

$$\mathcal{H}_{\mathrm{occ}} = \bigoplus_{k=0}^{3} \mathbb{C}^{g_k}, \qquad H_{\mathrm{occ}}|_{\mathbb{C}^{g_k}} = E_k^{\mathrm{occ}} I,$$

with levels and multiplicities

$$E^{\mathrm{occ}} = (0, 3, 6, 9), \qquad g = (1, 6, 12, 8), \qquad (N, E) = (12, 66).$$

The sum $1 + 6 + 12 + 8 = 27$ makes $\mathcal{H}_{\mathrm{occ}}$ a screen of twenty-seven modes. This spectrum is the datum of a declared finite representation; its cardinality does not suffice to identify it with any other occurrence of 27 in HMT. The morphism relating a specific boundary screen to $(\mathcal{H}_{\mathrm{occ}}, H_{\mathrm{occ}})$ must specify both the fibers and the spectral operator. The fermionic and bosonic generating functions are

$$F(y, x) = \prod_k (1 + y x^{E_k^{\mathrm{occ}}})^{g_k}, \qquad B(y, x) = \prod_k (1 - y x^{E_k^{\mathrm{occ}}})^{-g_k}.$$

Proposition 68.11 (Microcanonical cardinalities)**.**

$$[y^{12} x^{66}] F = 2\,104\,494, \qquad [y^{12} x^{66}] B = 259\,436\,430.$$

Proof. In the fermionic product, choosing r of the g_k modes contributes $\binom{g_k}{r} y^r x^{r E_k^{\mathrm{occ}}}$. In the bosonic product, distributing r indistinguishable occupations among g_k modes contributes $\binom{g_k + r - 1}{r} y^r x^{r E_k^{\mathrm{occ}}}$. The finite convolution over the four levels produces the stated integers. □

68.9 Bose–Einstein and Fermi–Dirac distributions

The equilibrium formulation starts from the Hamiltonian and the total occupation number,

$$H = \sum_k E_k^{\mathrm{occ}} n_k, \qquad N = \sum_k n_k, \qquad K_\mu := H - \mu N. \tag{68.17}$$

For $\beta > 0$, the grand canonical state is

$$\rho_{\beta,\mu} = \frac{e^{-\beta K_\mu}}{\mathcal{Z}_{\beta,\mu}}, \qquad \mathcal{Z}_{\beta,\mu} = \mathrm{Tr}(e^{-\beta K_\mu}). \tag{68.18}$$

Theorem 68.12 (Grand canonical factorization and occupation laws)**.** *For energy levels E_k^{occ} with degeneracies g_k, the fermionic trace factorizes as*

$$\mathcal{Z}_{\mathrm{FD}} = \prod_k \bigl(1 + e^{-\beta(E_k^{\mathrm{occ}} - \mu)}\bigr)^{g_k}, \qquad \langle n_k \rangle_{\mathrm{FD}} = \frac{g_k}{e^{\beta(E_k^{\mathrm{occ}} - \mu)} + 1}. \tag{68.19}$$

In the bosonic sector, under the condition $\mu < \inf_k E_k^{\mathrm{occ}}$,

$$\mathcal{Z}_{\mathrm{BE}} = \prod_k \bigl(1 - e^{-\beta(E_k^{\mathrm{occ}} - \mu)}\bigr)^{-g_k}, \qquad \langle n_k \rangle_{\mathrm{BE}} = \frac{g_k}{e^{\beta(E_k^{\mathrm{occ}} - \mu)} - 1}. \tag{68.20}$$

Proof. The operator K_μ is a sum of commuting occupation operators, so its trace factorizes over modes. In each fermionic mode, the occupations 0 and 1 are summed, and the corresponding factor is $1 + u_k$, where $u_k = e^{-\beta(E_k^{\mathrm{occ}} - \mu)}$. In each bosonic mode, all nonnegative occupations are summed, giving the geometric series $(1 - u_k)^{-1}$; the condition on μ ensures $u_k < 1$. The powers g_k incorporate the degeneracy. Finally,

$$\langle n_k \rangle = -\frac{1}{\beta} \frac{\partial}{\partial E_k^{\mathrm{occ}}} \log \mathcal{Z}$$

yields the two occupation laws [227, 229, 230, 228, 232]. □

The microcanonical derivation recovers the same result in the thermodynamic regime. For mean fermionic occupations $p_k = m_k / g_k$, the combinatorial entropy per level is

$$S_k \simeq g_k \bigl[-p_k \log p_k - (1 - p_k) \log(1 - p_k) \bigr]. \tag{68.21}$$

The variation of $\sum_k S_k$, subject to $\sum_k g_k p_k = N$ and $\sum_k g_k E_k^{\mathrm{occ}} p_k = E$, gives

$$\log \frac{1 - p_k}{p_k} = \lambda_N + \beta E_k^{\mathrm{occ}}, \qquad \mu = -\frac{\lambda_N}{\beta}, \tag{68.22}$$

and therefore the Fermi–Dirac occupation of (68.19). The multipliers β and μ are the dual coordinates of the energy and population constraints. The symbol $\lambda_N = -\beta\mu$ denotes the multiplier associated with the occupation number; α retains throughout the article its use for the fine-structure constant.

For mean bosonic occupations $\overline{n}_k = m_k / g_k$, the log-multiplicity of the level is

$$S_k \simeq g_k \bigl[(1 + \overline{n}_k) \log(1 + \overline{n}_k) - \overline{n}_k \log \overline{n}_k \bigr]. \tag{68.23}$$

The variation of $\sum_k S_k$, subject to $\sum_k g_k \overline{n}_k = N$ and $\sum_k g_k E_k^{\mathrm{occ}} \overline{n}_k = E$, yields

$$\log \frac{1 + \overline{n}_k}{\overline{n}_k} = \lambda_N + \beta E_k^{\mathrm{occ}}, \qquad \overline{n}_k = \frac{1}{e^{\beta(E_k^{\mathrm{occ}} - \mu)} - 1}, \qquad \mu = -\frac{\lambda_N}{\beta}. \tag{68.24}$$

This is the Bose–Einstein occupation of (68.20). The condition $\mu < \inf_k E_k^{\mathrm{occ}}$ ensures positivity of the denominator and convergence of each modal factor.

In the finite instance constructed, the two constraints determine

$$(\beta_{\mathrm{FD}}, \mu_{\mathrm{FD}}) = (0.15307011353415928,\ 4.502865197597043),$$

$$(\beta_{\mathrm{BE}}, \mu_{\mathrm{BE}}) = (0.05456727960068316,\ -15.92569781906737).$$

If m_k^{mic} denotes the mean occupancy obtained by exact microcanonical enumeration and m_k^{eq} the evaluation of the corresponding distribution, the errors $\sum_k |m_k^{\mathrm{mic}} - m_k^{\mathrm{eq}}|$ are

$$0.09897380039002812 \quad \text{and} \quad 0.9190625715582017,$$

respectively. The enumeration runs through all occupation quadruples, calculates their integer multiplicities and solves the two constraint equations. Both laws are realized on the same screen. The physical choice between them requires a grading and a Hamiltonian compatible with the representation of the path. The thermal parameter is $\beta = (k_B \Theta)^{-1}$, with the dimensional transduction developed in section 123.

68.10 Quantified convergence to the Maxwell–Boltzmann distribution

The relation between the two quantum statistics and the classical regime can be formulated without resorting to an informal approximation. For each mode, let

$$u_k := e^{-\beta(E_k^{\mathrm{occ}}-\mu)}.$$

The state occupations are written

$$\overline{n}_k^{\mathrm{FD}} = \frac{u_k}{1+u_k}, \qquad \overline{n}_k^{\mathrm{BE}} = \frac{u_k}{1-u_k};$$

the multiplicities g_k are incorporated subsequently through $\langle n_k \rangle = g_k \overline{n}_k$.

Theorem 68.13 (Common classical limit of quantum statistics)**.** *If* $0 \le u_k \le \delta < 1$*, then*

$$\left|\overline{n}_k^{\mathrm{FD}} - u_k\right| \le u_k^2, \qquad \left|\overline{n}_k^{\mathrm{BE}} - u_k\right| \le \frac{u_k^2}{1-\delta}.$$

Consequently, in the regime dilute $u_k \to 0$*,*

$$\boxed{\overline{n}_k^{\mathrm{FD}} \sim \overline{n}_k^{\mathrm{BE}} \sim e^{-\beta(E_k^{\mathrm{occ}}-\mu)}},$$

which is the Maxwell–Boltzmann distribution per mode.

Proof. The exact identities

$$u_k - \frac{u_k}{1+u_k} = \frac{u_k^2}{1+u_k}, \qquad \frac{u_k}{1-u_k} - u_k = \frac{u_k^2}{1-u_k}$$

provide the bounds. For $0 < u_k \le \delta$, dividing by u_k and taking $u_k \downarrow 0$ yields the asymptotic equivalence. The leading term of order u_k^2 preserves the physical difference: exclusion decreases fermionic occupation, whereas accumulation increases bosonic occupation. □

Chapter 69

Fermionic completion of trajectories and the algebraic Pauli exclusion principle

Exterior completion of the trajectories imposes idempotence and exclusion before any species identification. The rotation, color, flavor, and virtuality characters preserve their typing within the corresponding representation.

69.1 Occupation, exclusion and finite statistics

The null and material realizations provide the geometric support of occupation. The fermionic sector is constructed on the canonical anticommutation relations (CAR), and the spin–statistics theorem further requires the declared operator representation. This section fixes those premises before deriving idempotence of occupation, the microcanonical cardinalities, and the Bose–Einstein and Fermi–Dirac distributions with their common classical limit.

69.1.1 Idempotency of Occupation in the Anticommutation Algebra

Let $a_i, a_i^\dagger$ be operators satisfying the canonical anticommutation relations (CAR)

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \qquad \{a_i, a_j\} = 0,$$

and let $n_i = a_i^\dagger a_i$.

Theorem 69.1 (Pauli Exclusion Principle)**.** *One has* $n_i^2 = n_i$*, and therefore* $\mathrm{Spec}(n_i) \subseteq \{0, 1\}$.

Proof.

$$n_i^2 = a_i^\dagger a_i a_i^\dagger a_i = a_i^\dagger (1 - a_i^\dagger a_i) a_i = a_i^\dagger a_i,$$

because $a_i^2 = (a_i^\dagger)^2 = 0$. Every eigenvalue of an idempotent satisfies $\lambda^2 = \lambda$ [99, 35]. □

The nilpotence $(a_i^\dagger)^2 = 0$ precludes double fermionic occupation of the same mode and, upon maximizing entropy under the energy and number constraints, leads to the Fermi–Dirac occupation law presented below. The symmetric completion and the Bose–Einstein law constitute a distinct stratum and are presented separately.

The CAR relations are an algebraic hypothesis of the theorem. A monodromy with sign change under a rotation of 2π is compatible with a fermionic representation, but does not by itself constitute a derivation of CAR or of the spin–statistics theorem.

69.2 Spin, color, flavor, and virtuality characters

Let $\jmath : \Gamma_{\mathrm{TPK}} \to \Gamma_{\mathrm{TPK}}$ be the involution that reverses the orientation of each edge and permutes the fibers through $J_c : E_c \to E_{\jmath c}$. A spin character is a representation ρ_{rot} of the subgroup generated by $\jmath$ on the sections; its sign eigenspaces $+1$ and -1 are algebraic objects distinct from a physical representation of the Lorentz group.

Let C be the Abelian group of channel characters of the transport. Two sections have the same *arithmetic flavour* when their characters

$$\chi_s : C \longrightarrow \mathbb{C}^\times$$

coincide. Identifying these classes with fermion flavors requires a representation of the algebra of physical observables and is not introduced by nomenclature.

Finally, let

$$\mathrm{Ext}_N(s) = \{s' \in \Gamma(\mathcal{B}_{\gamma'}) : \gamma'|_{\{0,\dots,N\}} = \gamma,\ s'|_{\{0,\dots,N\}} = s\}.$$

The section is stable from depth N onward when the restrictions $\mathrm{Ext}_{N+1}(s) \to \mathrm{Ext}_N(s)$ admit a compatible branch at every subsequent level. It is called *virtual up to depth N* when that global branch does not yet exist, although finite extensions survive. This definition belongs to the inverse system and does not depend on a temporal metaphor.

On the affine plane $\mathbb{F}_3^2$, the linear functional

$$\kappa_3 : \mathbb{F}_3^2 \longrightarrow \mathbb{F}_3, \qquad \kappa_3(r, c) = r - c$$

is surjective, and its kernel has dimension one. Consequently, its three fibers are the affine classes of the kernel, and each contains exactly three points. The same balance persists in every disjoint union of complete copies of $\mathbb{F}_3^2$. This ternary partition is a self-contained algebraic statement. Once the qutrit chart, the orientation, and the phase character have been fixed, the section 90.6 constructs the representation and its algebra $\mathfrak{su}(3)$; the intertwiner with TPK transport and the physical chromodynamic identification remain an interface.

Distinction between reversible genealogical transport and irreversible projection. Reversibility of the update permits the cost of an irreversible projection to be distinguished from complete genealogical transport.

69.3 Statistical application of occupation spectra

The null and material realizations provide the geometric support for occupation. The graded completion of that support determines its multiparticle spaces and operators; Theorem 68.10 establishes their transport under refinement. This stage uses those results to characterize admissible configurations, count states, and prepare the thermal evaluation. The algebraic construction, finite counting, and equilibrium state thus retain their respective functions and premises.

69.3.1 Spectral restriction and admissible configurations

For a mode i, write $n_i = N_i = a_i^\dagger a_i$. Theorem 64.14, with its complete proof in (64.11), establishes $n_i^2 = n_i$ and $\mathrm{Spec}(n_i) \subseteq \{0, 1\}$. The calculation uses only CAR and the adjoint relation, so it also applies to any representation of those relations. Nilpotency $(a_i^\dagger)^2 = 0$ prevents double fermionic occupation of the same mode. Consequently, a fermionic configuration of g modes is a family $(n_1, \dots, n_g) \in \{0, 1\}^g$; in the bosonic sector, the occupations belong to $\mathbb{N}_0^g$.

The fermionic representation uses the CAR algebra constructed from the exterior completion. The relativistic spin–statistics identification additionally retains compatibility of the rotation representation with parity and the locality conditions stated in Sections 64.5.4 and 64.5.5. The 4π periodicity and exclusion are coordinated by the same sheet character, with these representation conditions explicit.

69.3.2 Dimensions of fixed-occupation sectors

If there are g modes and N particles, the sector dimensions are

$$\Omega_F(g, N) = \binom{g}{N}, \qquad \Omega_B(g, N) = \binom{g + N - 1}{N}. \tag{69.1}$$

The first formula counts subsets of modes, with zero dimension if $N > g$; the second counts weak compositions of N into g parts. These counts are exact consequences of the two completions and are obtained before introducing the thermal state. Proposition 68.11 applies both counts to the finite spectral screen, retaining its levels, multiplicities, and energy and population constraints.

69.3.3 Thermal readout of the graded completions

With a diagonal Hamiltonian H, modal energy ϵ, chemical potential μ, and thermal parameter $\beta > 0$, the mean occupations of the two sectors are

$$\bar{n}_{\mathrm{par}}(\epsilon) = \frac{1}{e^{\beta(\epsilon-\mu)} - 1}, \qquad \bar{n}_{\mathrm{impar}}(\epsilon) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1}. \tag{69.2}$$

The bosonic sector requires $\mu < \inf \operatorname{Spec} H$. These expressions are the per-mode specialization of Theorem 68.12, whose proof constructs the grand canonical trace from the admissible occupations. The Bose–Einstein and Fermi–Dirac formulas are therefore the thermodynamic evaluations of the symmetric and exterior completions selected by transport. The same section derives their expressions by varying the entropy under the energy and number constraints; Theorem 68.13 proves their common classical limit with an explicit error bound.

Chapter 70

Reversible update, erasure cost, and exact domain of the zero Landauer limit

The full transport preserves fibre information and may have zero logical cost; the projection or reinitialization that discards the genealogy is subject to Landauer's cost. The two domains are separated explicitly.

70.1 Logical reversibility and Landauer's principle in TPK fibers

Let $(\Omega, 2^{\Omega}, \mu)$ be a finite probability space, let $X : \Omega \to \mathcal{X}$ be a random variable with values in the space $\mathcal{X}$ of complete states, and let $q : \mathcal{X} \to Y$ be a reader. The chain rule gives

$$\mathsf{H}(X) = \mathsf{H}(q(X)) + \mathsf{H}(X \mid q(X)).$$

The second summand measures the information of the fiber that the reader does not publish.

Theorem 70.1 (Landauer zero for full transport and projection cost)**.** *If $U : \mathcal{X} \to \mathcal{X}$ is a bijective update of the complete HMT state, then*

$$\boxed{\mathsf{H}(X \mid U(X)) = 0.}$$

For a projected output $q(U(X))$, the discarded information is

$$\mathsf{H}(X \mid q(U(X))).$$

A physical reset that identifies distinct memory states must account for that information in the environment.

Proof. Bijectivity implies $X = U^{-1}(U(X))$; therefore X is determined by $U(X)$ and the conditional entropy is zero. For the reader q, the chain rule separates published entropy from fiber entropy. If the reset operation is not injective, the output no longer permits reconstruction of the memory state. Landauer's thermodynamic principle bounds the cost of implementation in terms of that erased entropy; the reversible update does not by itself contribute an erasure term. □

In nats and bits, respectively,

$$Q \geq k_B \Theta \, \mathsf{H}_{\text{nat}}, \qquad Q \geq k_B \Theta \log 2 \, \mathsf{H}_{\text{bit}}.$$

For bijective transport of the complete state, the erasure entropy is zero and, therefore, the minimum Landauer term associated exclusively with erasure satisfies

$$\boxed{Q_{\text{L}}^{\min}[U] = k_B \Theta \, \mathsf{H}(X \mid U(X)) = 0.}$$

For the restart of a uniform TRIT, on the other hand,

$$\boxed{Q_{\text{reset}} \geq k_B \Theta \log 3.}$$

These two equations locate the cost in information loss: the cost appears when projection or resetting forgets the fiber. Preparation, projection, and resetting may produce entropy; reversible transport preserves genealogical information. The null value is the logical Landauer limit for that complete update, whereas the dynamical costs of control, finite speed, and coupling to the environment belong to the physical realization of the device.

Scope of the null Landauer relation

The equality $\mathsf{H}(X \mid U(X)) = 0$ is exact for a bijective update of the enriched state. The bound $Q_{\text{reset}} \geq k_B \Theta \log 3$ is the Landauer specialization to the erasure of a uniform triadic decision. Realizing both formulas in joules uses the Boltzmann transducer and a declared physical temperature.

Edge Information Transduction in Minimum Triadic Cut Capacity. The information preserved by the boundary reappears geometrically in the minimal triadic cut and in its capacity.

70.2 Genealogical Conservation of Information, Reduction Entropies and Thermal Structure

The information transported by an HMT state is not exhausted by the coordinate that one of its readers publishes. The visible coordinate, sheet, orientation, quotient, carry, route, and closure seal belong to the same enriched state, even though a particular evaluation displays only part of it. This distinction makes it possible to place in a single framework three operations that the macroscopic description usually presents separately: preserving a history, reducing it to a reading, and assigning energy to the information actually erased.

The chapter develops that chain without identifying its stages. It first formulates fiber memory and the exact recoverability criterion. It then proves the vanishing of the logical Landauer term for a bijective update of the TPK enriched state. Five entropies having different domains and functions are subsequently distinguished. Finally, the areal–volumetric quotient and the clock of 108 steps provide the structure that links triadic information, action, and temperature.

70.2.1 Disintegration of Visible Reading Over Fibers and Recovery of Genealogical Memory

Let Ω be a finite space of enriched states, and let

$$q : \Omega \longrightarrow Y$$

a visible reader. The value $q(x)$ is a coordinate of the state x, not its exhaustive description. Denote by

$$M : \Omega \longrightarrow \mathcal{M}$$

the joint record of the genealogical coordinates that the reader does not publish: leaf, orientation, quotient or carry, phase, route, and seal, according to the domain under consideration.

Definition 70.2 (Enhanced reader and fiber memory)**.** The enriched reader associated with q and M is

$$\widehat{q} = (q, M) : \Omega \longrightarrow Y \times \mathcal{M}. \tag{70.1}$$

The fibre memory of q is the information needed to distinguish the states contained in the same fibre $q^{-1}(y)$.

The APP residue–quotient extension is the elementary arithmetic example. In each cell, both Euclidean divisions are preserved

$$i + j = 9q^{+}_{ij} + R^{+}_{ij}, \qquad ij = 9q^{\times}_{ij} + R^{\times}_{ij}.$$

Reduction modulo nine publishes R^{+} or $R^{\times}$; the lift simultaneously preserves (R, q). Summing over the nonadic domain yields the exact identity

$$2025 - 810 = 54 + 9 \cdot 129. \tag{70.2}$$

The first term on the right is the difference between the visible residues; the second recomposes the difference between the quotients. The equality shows that the residual projection does not by itself determine the integral operation.

Now let X be a random variable with values in Ω. We shall work with dimensionless entropies in nats. The chain rule gives

$$\mathsf{H}(X) = \mathsf{H}(q(X)) + \mathsf{H}(X \mid q(X)). \tag{70.3}$$

The second term quantifies the genealogical information that remains within the fiber of the reading.

Theorem 70.3 (Fiber Memory Recoverability)**.** *If the enriched reader $\widehat{q} = (q, M)$ is injective on the support of X, then*

$$\boxed{\mathsf{H}(X \mid q(X), M(X)) = 0.} \tag{70.4}$$

The entropy $\mathsf{H}(X \mid q(X))$ measures exactly the information that would be lost by retaining the visible coordinate and discarding the memory.

Proof. The injectivity of $\widehat{q}$ provides an inverse on its image. Hence the pair $(q(X), M(X))$ determines X, and the conditional entropy is zero. If $M(X)$ is omitted, entries in the same fiber are identified; the chain rule (70.3) assigns the term $\mathsf{H}(X \mid q(X))$ to that identification. □

The genealogical definition of number enters precisely here. An HMT number is a compatible section of surviving cylinders together with the memory of its route, orientation, signature, and closure. Its real value is the evaluation of that section. Evaluation may forget coordinates; the section preserves them.

70.2.2 Logical reversibility and Landauer's term

Let

$$U : \Omega \longrightarrow \Omega$$

an update of the enriched TPK state. Landauer's question is posed on the complete map, before a visible reader is applied or a record is reset; its thermodynamic formulation of logical erasure dates to Landauer [78].

Theorem 70.4 (Landauer zero for full transport)**.** *If U is bijective, then, for every input distribution X,*

$$\mathsf{H}(X \mid U(X)) = 0. \tag{70.5}$$

In the ideal Landauer regime, the minimum term associated exclusively with internal logical erasure is

$$\boxed{Q_{\mathrm{L}}^{\min}[U] = k_B \Theta\, \mathsf{H}(X \mid U(X)) = 0.} \tag{70.6}$$

Proof. The bijection gives U^{-1}, so that $X = U^{-1}(U(X))$. The complete output determines the input, and the conditional entropy vanishes. The informational form of Landauer's principle assigns the erasure cost to the term $k_B \Theta \mathsf{H}(X \mid U(X))$, which equals zero by the preceding equality. □

The result localizes conservation in the enriched space. For a projected output $q(U(X))$, the unpublished information is

$$\mathsf{H}\big(X \mid q(U(X))\big). \tag{70.7}$$

A physical reset identifying distinct states must transfer that information to the environment. In particular, resetting a uniform TRIT to a single state has

$$\mathsf{H}(X) = \log 3, \qquad \boxed{Q_{\mathrm{reset}} \geq k_B \Theta \log 3.}$$

Reversible updating, projection, and reset are thus distinct operations. The first preserves genealogy; the second reduces the reading; the third physically eliminates a record. Additional dissipation due to control, noise, or finite speed belongs to the realization of the device and not to the logical erasure term of (70.6).

70.2.3 5 Notions of Entropy and Their Domains

The word *entropy* denotes five related but noninterchangeable functionals in this construction. Their separation prevents temperature from being attributed to a purely combinatorial count, or loss of information to an evolution that merely changes the reading.

70.2.3.1 Conditional entropy of the fibers of the visible reader and genealogy loss

The quantity

$$\mathsf{H}_{\mathrm{fib}}(q; X) := \mathsf{H}(X \mid q(X))$$

measures the information in the genealogy that a visible reading does not distinguish. Its domain is a reader and a distribution on the enriched state. It vanishes when the reader is faithful on the support under consideration.

70.2.3.2 Informational entropy of a triadic decision

For a distribution $p = (p_{-1}, p_0, p_{+1})$ over the three visible states of the TRIT,

$$I_{\mathrm{TRIT}}(p) = -\sum_{a\in\{-1,0,+1\}} p_a \log p_a.$$

The uniform distribution produces

$$\boxed{I_{\mathrm{TRIT}} = \log 3.} \tag{70.8}$$

This quantity measures a local decision among three visible alternatives.

70.2.3.3 Topological Entropy of Admissible Histories

If A is the adjacency matrix of a finite subshift, its topological entropy is

$$h_{\mathrm{top}}(A) = \log \rho(A),$$

where $\rho(A)$ is the spectral radius. This functional measures the asymptotic growth rate of compatible histories, not the uncertainty of an isolated decision.

70.2.3.4 Combinatorial Entropy and Maximum Statistical Entropy

For energy levels ε_i with degeneracies g_i, a fermionic occupation $n_i \in \{0, \dots, g_i\}$ has

$$\Omega(g_i, n_i) = \binom{g_i}{n_i}$$

microstates. With total number and total energy fixed,

$$\Omega(N, E) = \sum_{\substack{(n_i):\ \sum_i n_i = N \\ \sum_i \varepsilon_i n_i = E}} \prod_i \binom{g_i}{n_i}, \qquad S_{\mathrm{mc}} = \log \Omega(N, E). \tag{70.9}$$

With number and energy fixed as mean values, Shannon maximization [110] yields the product distribution

$$\mathbb{P}(n_1, \dots, n_M) \propto \exp\left[-\beta \sum_j \varepsilon_j n_j + \beta\mu \sum_j n_j\right]$$

and, therefore,

$$\boxed{\langle n_j \rangle_{\mathrm{FD}} = \frac{1}{e^{\beta(\varepsilon_j - \mu)} + 1}.} \tag{70.10}$$

Bosonic occupation replaces the binomial count with $\binom{g+n-1}{n}$ and yields

$$\langle n_j \rangle_{\mathrm{BE}} = \frac{1}{e^{\beta(\varepsilon_j - \mu)} - 1}.$$

In both cases, β and μ are determined by the declared macroscopic constraints.

70.2.3.5 von Neumann Entropy of 1 *Reduced State*

For a density matrix ρ_A on a region or subalgebra,

$$S_{\mathrm{vN}}(\rho_A) = -\operatorname{Tr}(\rho_A \log \rho_A).$$

Its domain is a reduced state. When ρ_A arises from a global purification, this quantity measures entanglement across the cut. The next chapter constructs the triadic reduction, modular Hamiltonian, and area quantization that give it a precise geometric residence.

Proposition 70.5 (Typed separation of the five entropies)**.** *The quantities $\mathsf{H}_{\mathrm{fib}}$, I_{TRIT}, h_{top}, S_{mc}, and S_{vN} are not substituted for one another: they depend, respectively, on a reading fiber, a local triadic distribution, a history system, a set of microstates compatible with constraints, and a reduced state. A physical map may relate them only after declaring the coarse–graining, ensemble, Hamiltonian, or purification that changes from one domain to the next.*

Proof. Each functional is defined on a different type of input. Conditional entropy requires a reader; I_{TRIT}, a distribution over three symbols; h_{top}, a word dynamics; S_{mc}, a macrocanonical constraint; and S_{vN}, a density operator. An accidental numerical equality does not supply the maps that change those types. □

70.2.4 Areal–volumetric incidence and topological entropy

The TPK calendar induces the hinge between the additive and multiplicative modes. In the primitive block

$$\tau_3 = (+1, -1, 0),$$

the rule

$$+1 : m \mapsto \Sigma, \qquad -1 : m \mapsto \Pi, \qquad 0 : m \mapsto m$$

produces the cyclic orbit (Σ, Π, Π). The geometric reader

$$\Sigma \longmapsto \mathscr{H}_{\mathrm{ar}}, \qquad \Pi \longmapsto \mathscr{H}_{\mathrm{vol}}$$

transports these modes to the areal and volumetric sheets.

Theorem 70.6 (Necessary incidence of the sheets)**.** *In the order $(\mathscr{H}_{\mathrm{ar}}, \mathscr{H}_{\mathrm{vol}})$, the primitive incidence matrix is*

$$\boxed{F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.} \tag{70.11}$$

Calendars of lengths $9, 27, 108$ *yield*

$$N_9 = 3F_{\mathrm{av}}, \qquad N_{27} = 9F_{\mathrm{av}}, \qquad N_{108} = 36F_{\mathrm{av}}.$$

Proof. The consecutive pairs of the cycle are $\Sigma \to \Pi$, $\Pi \to \Pi$, and $\Pi \to \Sigma$. Each occurs once, and $\Sigma \to \Sigma$ does not occur. This fixes all four entries of F_{av}. The longer calendars contain, respectively, three, nine, and thirty-six copies of the primitive block. □

Theorem 70.7 (Fibonacci law and measure of maximal entropy)**.** *For $n \geq 1$,*

$$F_{\mathrm{av}}^{n} = \begin{pmatrix} F_{n-1} & F_{n} \\ F_{n} & F_{n+1} \end{pmatrix}. \tag{70.12}$$

The one-step subshift defined by F_{av} has

$$\boxed{h_{\mathrm{top}} = \log \varphi}, \qquad \boxed{\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}}. \tag{70.13}$$

Proof. The matrix identity follows by induction from $F_{n+1} = F_n + F_{n-1}$. The Perron root of F_{av} is φ, hence $h_{\mathrm{top}} = \log \varphi$. Since the matrix is symmetric, the Parry weights [98] are proportional to the squared components of a Perron vector $(1, \varphi)$, that is, to $(1, \varphi^2)$. □

The expressions $\log 3$ and $\log \varphi$ are thus distinguished by their genealogy. The first quantifies a uniform local choice among three states; the second measures the growth of histories allowed by sheet incidence. The areal–volumetric relation is a property of the complete return and therefore appears with exponent two.

70.2.5 Boltzmann Transduction and Energy by Triadic Decision

Let $\mathcal{L}_T$ be the duration line, $\mathcal{L}_\Theta$ the temperature line, $\mathcal{L}_S$ the action line, and $\mathcal{L}_E$ the energy line. The Boltzmann constant is the positive isomorphism

$$\boxed{k_B : \mathcal{L}_\Theta \longrightarrow \mathcal{L}_E, \qquad [k_B] = \mathbf{E} - \mathbf{\Theta}.} \tag{70.14}$$

Thus, $\beta = (k_B \Theta)^{-1}$ belongs to the reciprocal-energy line.

The determinantal reader of the action constructs

$$\hbar_{\mathrm{int}} = \eta_{\mathrm{ret}}\, 10^{-34} \mathcal{U}_S \in \mathcal{L}_S.$$

If $t_0 \in \mathcal{L}_T^{+}$ is the elementary internal duration, the full-return clock fixes

$$\omega_0 = \frac{2\pi}{108 t_0}, \qquad \varepsilon_{\mathrm{clk}} = \hbar_{\mathrm{int}} \omega_0 \in \mathcal{L}_E.$$

Definition 70.8 (Triadic thermal scale)**.** The section $\Theta_{\mathrm{clk}} \in \mathcal{L}_\Theta^{+}$ is defined by

$$\boxed{k_B \Theta_{\mathrm{clk}} \log 3 = \hbar_{\mathrm{int}} \frac{2\pi}{108 t_0}.} \tag{70.15}$$

The equality fixes energy by triadic decision and links three objects already constructed: the local information $\log 3$, the internal action, and the clock frequency. The product $k_B \Theta_{\mathrm{clk}}$ is determined within the internal chart. The choice of thermal and energetic bases then separates the coordinate of k_B and that of Θ_{clk}. In the kelvin–joule chart, the metrological definition publishes

$$k_B^{\mathrm{SI}} = 1.380649 \times 10^{-23}\ \mathrm{J\,K^{-1}}.$$

70.2.6 Thermal Operator on Histories

Let $\mathcal{E}$ be the set of edges of a finite graph of extensions, let $a_*(e) \geq 0$ be the dimensionless action increment, and let $\epsilon_* \in \mathcal{L}_E^+$ be a positive energy section. For $\beta \in \mathcal{L}_E^{-1}$, we define

$$(\mathcal{L}_{\beta,*} f)(y) = \sum_{e:x\to y} \exp[-\beta\epsilon_* a_*(e)] f(x). \tag{70.16}$$

The exponent is dimensionless, and positivity of the operator preserves the composition of histories. Pressure normalization selects the thermal state through

$$\boxed{\rho(\mathcal{L}_{\beta,*}) = 1.} \tag{70.17}$$

Proposition 70.9 (Confluence of the information branch and the spectral branch)**.** *An HMT–MD thermal realization is determined when a single value of $\beta = (k_B\Theta)^{-1}$ satisfies the spectral normalization* (70.17) *and the action–information scale* (70.15)*. The first equation weights the histories; the second fixes the energy of the local decision. Their confluence turns genealogical accounting into a thermal ensemble without identifying information, entropy, and energy.*

Thermodynamics therefore appears after genealogical conservation. The complete state retains the history; the reader defines the coarse–graining; the count organizes the microstates; and the Boltzmann transducer assigns an energy scale to the information at a declared temperature.

70.2.7 Genealogical memory, informational reduction and thermal transduction

The thermal construction uses two publications of the same APP–TRIT–TPK development. The first preserves the states, admissible routes and register information; the second provides the oriented action and the duration of the return. Their composition determines an energy per tritic decision. The temperature and energy coordinates are subsequently introduced through explicit dimensional maps. This sequence allows one to identify what information is preserved, what information a reader publishes and what quantity converts that information into dimensional entropy.

70.2.7.1 Reader, memory, and state recovery

Fix a finite level of the development and denote by Ω the set of admissible enriched states. The register preserves sheet, orientation, residue, quotient, carry, route, phase and memory in the domains where each coordinate enters. Let

$$q : \Omega \longrightarrow Y, \qquad M : \Omega \longrightarrow \mathcal{M}$$

be the visible publication and the retained memory, respectively. For a distribution p over Ω, we write $\mathsf{H}(p) = -\sum_x p_x \log p_x$, with $0 \log 0 = 0$. The logarithm is applied to dimensionless probabilities.

Definition 70.10 (Genealogical recovery of a publication)**.** The pair $\widehat{q} = (q, M)$ is a recoverable reader on the support Ω_p when there exists

$$d : \widehat{q}(\Omega_p) \longrightarrow \Omega_p, \qquad d \circ \widehat{q} = \mathrm{id}_{\Omega_p}.$$

The fibre information of q is $\mathsf{H}_{\mathrm{fib}}(q; p) = \mathsf{H}(X \mid q(X))$, where X has law p.

Proposition 70.11 (Recovery identity and cost of reduction)**.** *For a recoverable reader,*

$$\mathsf{H}(X \mid q(X), M(X)) = 0, \tag{70.18}$$

$$\mathsf{H}(X) = \mathsf{H}(q(X)) + \mathsf{H}(X \mid q(X)), \tag{70.19}$$

$$\mathsf{H}(X \mid q(X)) = \mathsf{H}(M(X) \mid q(X)). \tag{70.20}$$

Proof. The map d recovers X from the pair $(q(X), M(X))$, so the first conditional entropy is zero. Since q is a function of X, the chain rule gives the second equality. For the third, expand $\mathsf{H}(X, M(X) \mid q(X))$ in two ways. One gives $\mathsf{H}(X \mid q(X))$, since M is a function of X; the other gives $\mathsf{H}(M(X) \mid q(X)) + \mathsf{H}(X \mid q(X), M(X))$. □

Euclidean division on the APP sheets displays the elementary arithmetic mechanism. With $i, j \in \{1, \dots, 9\}$, one preserves

$$i + j = 9q_{ij}^{+} + R_{ij}^{+}, \qquad ij = 9q_{ij}^{\times} + R_{ij}^{\times}, \qquad 1 \le R_{ij}^{\sigma} \le 9 \quad (\sigma \in \{+, \times\}).$$

The residue–quotient pair recovers the integer value of the operation. Summing the differences over the eighty-one positions gives

$$2025 - 810 = 54 + 9 \cdot 129.$$

Quotient memory restores the contribution not published by residual reduction. Recoverability of the complete state additionally requires preserving the relevant position and route coordinates; the residue–quotient pair identifies the result of the operation here.

70.2.7.2 Reversible update, tritic decision and erasure

Let $U : \Omega \to \Omega$ be the TPK update on a domain where enriched transport has an inverse. The transported law is $p'_y = p_{U^{-1}y}$. The bijection preserves the multiset of probabilities and therefore

$$\mathsf{H}(U(X)) = \mathsf{H}(X), \qquad \mathsf{H}(X \mid U(X)) = 0. \tag{70.21}$$

If the observed output is $q(U(X))$, the unpublished information is quantified by $\mathsf{H}(X \mid q(U(X)))$. The complete state, its publication and the resetting of a register are thus three distinct maps.

For the three oriented TRIT states, let $p = (p_{-1}, p_0, p_{+1})$ be a normalized distribution. Its information is

$$I_{\mathrm{TRIT}}(p) = -\sum_{a=-1}^{1} p_a \log p_a, \qquad 0 \le I_{\mathrm{TRIT}}(p) \le \log 3. \tag{70.22}$$

The upper bound corresponds to $p_a = 1/3$. Indeed,

$$D(p \| (1/3, 1/3, 1/3)) = \sum_{a} p_a \log(3p_a) = \log 3 - I_{\mathrm{TRIT}}(p) \ge 0;$$

the inequality follows from the convexity of $t \log t$, and equality characterizes the uniform distribution. The lower bound corresponds to a distribution concentrated on one state.

The ideal Landauer realization uses a memory with energetically degenerate logical states, an isothermal environment at temperature Θ and a balance that preserves the output of the logical map and accounts for auxiliary registers. In that model, the decrease in logical entropy is $k_B[\mathsf{H}(X) - \mathsf{H}(F(X))] = k_B\mathsf{H}(X \mid F(X))$ for a deterministic map F. The entropic balance requires transferring at least that amount to the environment; its energy expression is $Q \ge k_B\Theta\mathsf{H}(X \mid F(X))$. The distinction between reversible logical evolution and physical erasure corresponds to the Landauer–Bennett realization [176]. The invertible update (70.21) has zero logical contribution; resetting a uniform triple requires transferring $\log 3$ nats to the environment and satisfies $Q \ge k_B\Theta \log 3$. The control energy and dissipation of a specific physical realization belong to its additional thermodynamic balance.

70.2.7.3 Action, frequency, and decision energy

Let $\mathcal{L}_{\mathrm{act}}$ and $\mathcal{L}_T$ be the dimensional lines of action and duration; the energy line is $\mathcal{L}_E = \mathcal{L}_{\mathrm{act}} \otimes \mathcal{L}_T^{*}$. The preceding development provides the positive sections $\hbar_a \in \mathcal{L}_{\mathrm{act}}$, with $a \in \{\mathrm{pre}, \mathrm{ret}\}$, and the elementary duration $t_0 \in \mathcal{L}_T$. The complete calendar provides

$$T_{108} = 108t_0, \qquad \omega_{108} = \frac{2\pi}{T_{108}}, \qquad h_a = 2\pi\hbar_a.$$

The dimensional contraction between action and frequency defines

$$\epsilon_a := \hbar_a\omega_{108} = \frac{h_a}{108t_0} = \frac{\hbar_a}{t_P}, \qquad t_P = \frac{54}{\pi}t_0. \tag{70.23}$$

The final expression uses the same reduced circular period as the gravitational section. The three expressions correspond to a single energy section, with an orientation a maintained in all the factors. The normalized information of a uniform decision produces the energy section per nat

$$\varepsilon_a := \frac{\epsilon_a}{\log 3} \in \mathcal{L}_E^+. \tag{70.24}$$

The previously constructed action determines the energy scale once the duration is fixed; the cardinality of the TRIT alternatives determines its informational normalization. The value of $\log 3$ enters through the count and measure declared in (70.22).

70.2.7.4 Dimensional definition of Boltzmann and selection of coordinates

Let $\mathcal{L}_\Theta$ be the oriented temperature line. The dimensional entropy space is $\mathcal{L}_{\text{ent}} = \mathcal{L}_E \otimes \mathcal{L}_\Theta^*$. A positive linear map $k_B : \mathcal{L}_\Theta \to \mathcal{L}_E$ is, equivalently, a positive element of $\mathcal{L}_{\text{ent}}$. Thus the same object can act on a temperature or multiply dimensionless information.

Definition 70.12 (Thermal normalization of the tritic decision)**.** A thermal pair compatible with the oriented section a consists of a positive transducer k_B and a positive section $\Theta_{\text{clk},a} \in \mathcal{L}_\Theta$ such that

$$k_B(\Theta_{\text{clk},a}) = \varepsilon_a, \qquad k_B\Theta_{\text{clk},a}\log 3 = \epsilon_a. \tag{70.25}$$

The Boltzmann constant serves as the energy–temperature transducer of tritic information in this realization.

Theorem 70.13 (Uniqueness relative to the thermal section and dimensional covariance)**.** *Given a positive section $\Theta_{\text{clk},a}$, there exists a unique positive linear map satisfying (70.25). For positive bases u_E, u_Θ, if*

$$\epsilon_a = \epsilon_a^{\text{num}} u_E, \quad \Theta_{\text{clk},a} = \theta_a u_\Theta, \quad k_B(u_\Theta) = b\, u_E,$$

its coordinate is

$$b = \frac{\epsilon_a^{\text{num}}}{\theta_a \log 3}. \tag{70.26}$$

Under $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$, with $s_E, s_\Theta > 0$, one has

$$b' = \frac{s_\Theta}{s_E}b, \qquad \theta'_a = \frac{\theta_a}{s_\Theta}, \qquad (\epsilon_a^{\text{num}})' = \frac{\epsilon_a^{\text{num}}}{s_E}.$$

Proof. A nonzero section generates a line. Prescribing its image determines the unique linear map; the positivity of both sections proves its positivity. Substituting the coordinates into (70.25) gives $b\theta_a \log 3 = \epsilon_a^{\text{num}}$. The three transformation laws are obtained by expressing the same sections and the same map in the new bases. □

The same Boltzmann constant must act in both orientations. Since t_P is common, simultaneous compatibility is equivalent to

$$\frac{\Theta_{\text{clk,pre}}}{\Theta_{\text{clk,ret}}} = \frac{\epsilon_{\text{pre}}}{\epsilon_{\text{ret}}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}. \tag{70.27}$$

Indeed, an isomorphism of lines preserves quotients of sections. Conversely, if the preceding equality holds, the isomorphism determined in one orientation also satisfies the normalization of the other. Thus,

return changes the thermal section of the cycle in the same proportion as the energy, while the transducer remains common to both readings.

The preceding equation makes explicit the choice required to publish an individual value. If the thermal section has not yet been selected by an independent reader, for each $\lambda > 0$ the pair

$$(k_B, \Theta_{\mathrm{clk},a}) \longmapsto (\lambda k_B, \lambda^{-1}\Theta_{\mathrm{clk},a})$$

preserves ϵ_a. Consequently, presenting an internally selected individual coordinate requires specifying the reader of $\Theta_{\mathrm{clk},a}$, the bases and their dimensional transport. The action and information equations remain determined before that specification. The subsequent identification of the bases with joule and kelvin belongs to the metrological publication of the result.

70.2.7.5 Dimensional entropy and information of the reduced state

For a distribution of admitted routes p, or for a reduced state with density matrix ρ, let

$$s(p) = -\sum_x p_x \log p_x, \qquad s(\rho) = -\operatorname{Tr}(\rho \log \rho).$$

Their dimensional publications are

$$S(p) = k_B s(p), \qquad S(\rho) = k_B s(\rho) \quad \text{in } \mathcal{L}_{\mathrm{ent}}. \tag{70.28}$$

Evaluation at Θ produces an energy $\Theta S = k_B \Theta\, s$. For the uniform triple at the cycle temperature, (70.25) gives exactly

$$S_{\mathrm{TRIT}} = k_B \log 3, \qquad \Theta_{\mathrm{clk},a} S_{\mathrm{TRIT}} = \epsilon_a.$$

If ρ_A is the reduction of the bipartite purification constructed in the areal section, $s(\rho_A)$ measures the entanglement of that bipartition. Unit transport preserves its eigenvalues, its orientation and the sectoral occupations. The areal scale $\gamma a_0 \ell_P^2$ converts occupations into area; the transducer k_B converts information into dimensional entropy. Both readings use the same register through distinct operators.

70.2.7.6 Spectral normalization of thermal histories

Let $\mathcal{G}$ be an already constructed finite graph of admissible extensions, with adjacency matrix A. Each edge e preserves its route coordinates and receives a dimensionless action increment $a_*(e) > 0$ from the corresponding reader. Having chosen the energy section ϵ_*, set $t = \beta\epsilon_*$, where $\beta = (k_B\Theta)^{-1}$. The thermal operator is

$$(L_t f)(y) = \sum_{e: x \to y} \exp[-t a_*(e)]\, f(x). \tag{70.29}$$

All its exponents are dimensionless. The condition $\rho(L_t) = 1$ normalizes the weighted growth of histories.

Proposition 70.14 (Existence and uniqueness of normalized pressure)**.** *Suppose that A is irreducible, $\rho(A) > 1$, and that $0 < m \le a_*(e) \le M$ for all edges. There exists a unique $t_* > 0$ such that $\rho(L_{t_*}) = 1$, and*

$$\frac{\log \rho(A)}{M} \le t_* \le \frac{\log \rho(A)}{m}. \tag{70.30}$$

Proof. Entrywise comparison gives $e^{-tM}A \le L_t \le e^{-tm}A$ for $t \ge 0$. Monotonicity of the spectral radius of nonnegative matrices implies the corresponding bounds for $\rho(L_t)$. The spectral radius is continuous, equals $\rho(A) > 1$ at zero and tends to zero at infinity. For $s > 0$, moreover, $L_{t+s} \le e^{-sm} L_t$, so that $\rho(L_{t+s}) \le e^{-sm}\rho(L_t) < \rho(L_t)$. Continuity and strict decrease prove existence and uniqueness. Substituting the value one into the initial bounds proves (70.30). □

The local realization of three equally weighted alternatives, represented by a vertex with three loops and $a_*(e) = 1$, satisfies $\rho(L_t) = 3e^{-t}$. Its normalization gives $t_* = \log 3$. With $\epsilon_* = \epsilon_a$, we obtain

$\beta_{\mathrm{clk}} = \log 3/\epsilon_a$, precisely the relation (70.25). This calculation identifies the local tritic confluence. For a graph of histories with additional restrictions, confluence with the same clock requires checking

$$\rho\big(L_{(\log 3/\epsilon_a)\,\epsilon_*}\big) = 1$$

with its effective increments. The local normalization and the pressure of the complete system thus retain their respective domains.

The resulting relation brings together structure, value and meaning. The register determines the recoverable information; probabilities over continuations determine its entropic reading; action and period determine ϵ_a; the energy–temperature transducer publishes dimensional entropy. Area and entanglement are related through the reduced state and its boundary operators, keeping visible which datum produces each quantity.

Chapter 71

Minimal area of the triadic cut, capacity $81 \log 3$ and collective incidence transport

Areal quantization begins with the nonadic interior, its boundary, and the collective transport of incidence. The minimal cut contains 81 units and its qutrit capacity is $81 \log 3$; the result precedes its gravitational realization.

71.1 Areal quantization, modular dynamics, and rational CKM transformation

71.1.1 Hexad–octad inclusion and associated functional realizations

Let us fix a set H of six points and a set O of eight points with a distinguished inclusion $j : H \hookrightarrow O$. In the Witt residence, H is a hexad of a dodecad and O its lifted octad; in this chapter only the already fixed inclusion is used, not the terminal value of Barbero–Immirzi. We write

$$\binom{H}{2} := \{P \subset H : |P| = 2\}, \qquad \left|\binom{H}{2}\right| = \binom{6}{2} = 15, \tag{71.1}$$

and we define two incidence spaces with distinct types:

$$\mathcal{F}_6 := H \times \binom{H}{2}, \qquad \mathcal{F}_8 := O \times \binom{H}{2}. \tag{71.2}$$

The inclusion of incidences is

$$\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8, \qquad (h, P) \longmapsto (j(h), P). \tag{71.3}$$

It is injective because j is injective and preserves the marked pair P.

Theorem 71.1 (Incidence and defect counts oriented)**.** *The two spaces of equation (71.2) satisfy*

$$|\mathcal{F}_6| = 6\binom{6}{2} = 90, \qquad |\mathcal{F}_8| = 8\binom{6}{2} = 120, \qquad |\mathcal{F}_8 \setminus \iota_{6,8}(\mathcal{F}_6)| = 30. \tag{71.4}$$

Normalization by the twenty-four coordinates of the dodecad–twenty-four pair and by the fifteen pairs of H produces

$$\boxed{\delta^{\mathrm{inc}}_{6,8} = \frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24\binom{6}{2}} = -\frac{1}{12}.} \tag{71.5}$$

The same fraction is obtained through the reciprocal functional associated with the incorporated memory,

$$\boxed{\delta^{\mathrm{rec}}_{6,8} = \frac{30}{120} - \frac{30}{90} = -\frac{1}{12}.} \tag{71.6}$$

They are two exact realizations of the same oriented defect, but have distinct domains and are not identified with $\zeta(-1)$ without an additional intertwiner.

Proof. The multiplicative principle applied to equation (71.2) gives $6 \cdot 15 = 90$ and $8 \cdot 15 = 120$. The image of $\iota_{6,8}$ has ninety elements, so the complement has thirty. Substitution into equation (71.5) gives $-30/(24 \cdot 15) = -1/12$, while equation (71.6) is $1/4 - 1/3 = -1/12$. □

The hexad–octad transition admits two functional realizations that are not deduced from one another. The algebraic diagram, formulated on the spaces that justify the counts, is

$$\begin{aligned}(H \overset{j}{\hookrightarrow} O) &\longmapsto (|\mathcal{F}_6|, |\mathcal{F}_8|) = (90, 120) \longmapsto \Gamma_{6\to 8} = \gamma_{\mathrm{HMT}},\\ (h_6, o_8, t) = (6, 8, 3) &\longmapsto -\frac{o_8 - h_6}{t} = -\frac{2}{3},\end{aligned} \tag{71.7}$$

where $-2/3$ is the coefficient of the third coordinate of row 13 of the CKM constructor. The first realization transforms the incidence into area and modular memory; the second transforms the same transition into angular-mixing parameters. $\boldsymbol{\theta}_{\mathrm{CKM}} = f(\gamma_{\mathrm{HMT}})$ is not postulated: both outputs preserve a common antecedent.

The morphism determining the sectors 90/120 is $\iota_{6,8}$. Magnitude γ_{HMT} is adopted as a theorem previously demonstrated in the hexad–octad construction and is not reconstructed from its decimal representation. The operatorial consequences begin only after fixing the elementary scale a_0, defined below without using entropy or the terminal product $\gamma_{\mathrm{HMT}} a_0$.

71.1.2 Correspondence between the Nonadic Interior and the Boundary Degrees of Freedom

Let $\mathbb{T}_3 = \{0, 1, 2\}$. Every $x \in \mathbb{Z}/9\mathbb{Z}$ has a unique digit representation $x = x_0 + 3x_1$, with $x_i \in \mathbb{T}_3$, and every $y \in \mathbb{Z}/27\mathbb{Z}$ a unique representation $y = y_0 + 3y_1 + 9y_2$. Therefore

$$\mathcal{B} = (\mathbb{Z}/9\mathbb{Z})^3, \qquad \mathcal{B}_\partial = (\mathbb{Z}/27\mathbb{Z})^2, \qquad |\mathcal{B}| = |\mathcal{B}_\partial| = 3^6 = 729. \tag{71.8}$$

Equality of cardinalities is accompanied by the explicit map. If $x_j = r_{2j-1} + 3r_{2j}$ for $j = 1, 2, 3$, we define

$$\beta_{729}(x_1, x_2, x_3) = \big(r_1 + 3r_2 + 9r_3,\ r_4 + 3r_5 + 9r_6\big). \tag{71.9}$$

Proposition 71.2 (Genealogical bijection $729 \leftrightarrow 729$)**.** *The map $\beta_{729} : \mathcal{B} \to \mathcal{B}_\partial$ is bijective and preserves the ordered six-trit word. In general, it is not an isomorphism of additive groups: regrouping the digits changes where the carry is recorded.*

Proof. Extracting the six digits from three nonadic coordinates and grouping them into two triples is a permutation of the coordinates of $\mathbb{T}_3^6$. The positional decomposition is unique on both sides, so the operation has an inverse. The additive caveat is verified, for example, by adding words that generate a carry between r_3 and r_4: that boundary separates the two coordinates of order twenty-seven, but crosses two distinct coordinates in the nonadic packing. □

On the boundary torus $\mathcal{B}_\partial$, we take the block of vertices

$$A = \{0, 1, \dots, 8\} \times \{0, 1, \dots, 8\}. \tag{71.10}$$

Its oriented links crossing the boundary are separated by direction:

$$\partial A_{90} = \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \le j \le 8\}, \tag{71.11}$$

$$\partial A_{120} = \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \le i \le 8\}, \tag{71.12}$$

where all coordinates are read modulo twenty-seven. Thus

$$\partial A = \partial A_{90} \sqcup \partial A_{120}, \qquad |\partial A_{90}| = |\partial A_{120}| = 18. \tag{71.13}$$

The subscripts 90, 120 label the channel provenance; they do not say that the boundary has ninety or one hundred and twenty links.

Theorem 71.3 (Law areal of the cut triadic)**.** *In the cubic graph $N \times N \times N$ with unit capacity, every cut between two opposite faces has capacity at least N^2, and a transverse layer attains that bound. For $N = 9$,*

$$\mathcal{A}_{\min} = 81, \qquad S_{\mathrm{tri}} = 81 \log 3. \tag{71.14}$$

Proof. There are N^2 lines of edges parallel to the normal direction, pairwise disjoint. Every cut must intersect each one, so its capacity is at least N^2. A single transverse layer contains exactly N^2 edges and attains the minimum. If each cut link carries a maximally mixed trit, its entropy is $\log 3$; additivity over the eighty-one links gives equation (71.14). □

The entropy S_{tri} measures the capacity of the cubic cut. The modular entropy of the thirty-six links of equation (71.13) measures another state and must not be equated with it.

71.1.3 Collective incidence entangler between interior and boundary

In the sum $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$, we define

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \qquad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f). \tag{71.15}$$

In $\ell^2(\partial A_{90}) \oplus \ell^2(\partial A_{120})$, let

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial A_{90}} (\delta_e, 0), \qquad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial A_{120}} (0, \delta_e). \tag{71.16}$$

The pairs (u_{90}, u_{120}) and (v_{90}, v_{120}) are orthonormal. Therefore there exists a unique isometry between the collective subspaces that satisfies

$$\mathcal{J}_{6\to 8} u_{90} = v_{90}, \qquad \mathcal{J}_{6\to 8} u_{120} = v_{120}. \tag{71.17}$$

Define

$$D_{\mathrm{inc}} = 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \tag{71.18}$$

$$D_{\mathrm{area}} = 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|. \tag{71.19}$$

Theorem 71.4 (Exact transport of the two labels)**.** *The map $\mathcal{J}_{6\to 8}$ is unitary between the two collective planes and intertwines the incidence and area operators:*

$$\boxed{\mathcal{J}_{6\to 8} D_{\mathrm{inc}} = D_{\mathrm{area}} \mathcal{J}_{6\to 8}.} \tag{71.20}$$

Therefore, every polynomial combination of the labels $90, 120$*, in particular the primitive combination* $12 : -1$*, is transported with the same coefficients.*

Proof. Orthonormality shows that $\mathcal{J}_{6\to 8}$ maps one orthonormal basis to another. In u_{90}, both sides of equation (71.20) equal $90 v_{90}$; in u_{120}, both equal $120 v_{120}$. Equality on a basis proves the identity. The functional calculus, in its polynomial form, preserves the intertwining. □

This map does not purport to inject ninety individual incidences into eighteen links: it transports exactly the two uniform modes and their spectral labels. Extending it to non-uniform fluctuations would require a new incidence map; the modular operator defined here only uses the collective subspace that has just been constructed.

71.1.4 What is the elementary and minimal scale of the area operator

Let C^* the angular coordinate conjugate, expresada in grados, and fijemos

$$\theta_{\mathrm{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \qquad \boxed{a_0 = \frac{1 + 2\cos(10^\circ)}{3} \theta_{\mathrm{clk}}.} \tag{71.21}$$

The factor preceding θ_{clk} is not a decimal correction. It is the normalized real character of the phase triplet

$$\frac{1}{3}\,\mathrm{Tr}\,\mathrm{diag}\left(1, e^{i\pi/18}, e^{-i\pi/18}\right) = \frac{1+2\cos(10^\circ)}{3}. \tag{71.22}$$

Thus, a_0 is the spectral mean of the angular structure on the oriented trit. It is positive, is fixed before the area operator, and is not selected from entropy. The subsequent identification $\gamma_{\mathrm{HMT}} = \Gamma_{6\to 8}$ realizes the HMT output in the Ashtekar–Barbero connection [11, 69]; it does not reopen its derivation.

Remark 71.5 (Alcance of the simbolo a_0). In this chapter, $a_0 \equiv a_0^{\mathrm{area}}$ denotes exclusively the normalization of equation (71.21). It is neither the Bohr radius nor the initial cosmological scale factor, although those objects are traditionally written with the same symbol. No identity in this chapter allows one to substitute one for the other.

Construction of the area operator from the triad cut and channel separation. The cut and its capacity determine an area operator; its spectrum must separately preserve total occupation and channel provenance.

Chapter 72

Spectrum of the area operator and joint reconstruction of occupation and provenance

The sectors 90 and 120 define compatible occupancy and memory operators. Inverting the linear system reconstructs both channels and prevents total area from erasing provenance.

72.1 Area and modular operators in the sectors 90 and 120

In each of the two families of equation (71.13) there are eighteen qutrit channels. For $m \in \{90, 120\}$, define

$$N_m = \sum_{e \in \partial A_m} N_e, \qquad \mathrm{Spec}(N_e) = \{0, 1, 2\}. \tag{72.1}$$

The Barbero–Immirzi output is used here as a prior result, without rederiving it. With the area normalization constructed in equation (71.21), one obtains

$$\boxed{\frac{\widehat{\mathcal{A}}_{\mathrm{phys}}}{\ell_P^2} = \gamma_{\mathrm{HMT}} a_0 (N_{90} + N_{120}),} \qquad \gamma_{\mathrm{HMT}} a_0 = 0.0016755940749224991. \tag{72.2}$$

Theorem 72.1 (Finite spectrum of the area quantum)**.** *On the thirty-six links of the cut,*

$$\mathrm{Spec}(N_{90} + N_{120}) = \{0, 1, \ldots, 72\}. \tag{72.3}$$

The multiplicity of the eigenvalue n is the coefficient of z^n in $(1 + z + z^2)^{36}$. Consequently,

$$\boxed{\mathrm{Spec}\left(\frac{\widehat{\mathcal{A}}_{\mathrm{phys}}}{\ell_P^2}\right) = \{n\gamma_{\mathrm{HMT}} a_0 : 0 \le n \le 72\}.} \tag{72.4}$$

The compatible cuts prolong the family $\mathcal{A}_n^{\mathrm{prot}} = n\gamma_{\mathrm{HMT}} a_0 \ell_P^2$.

Proof. Each operator N_e has a spectral generating function $1 + z + z^2$. The thirty-six factors commute and act on distinct tensor factors; therefore, the generating function of the total is $(1 + z + z^2)^{36}$, whose degrees range over all integers from zero to seventy-two. Multiplication by the positive scalar $\gamma_{\mathrm{HMT}} a_0 \ell_P^2$ gives equation (72.4). □

Two quantization procedures must be distinguished. The cut law theorem 71.3 counts eighty-one unit capacities and determines 81 log 3; the operator equation (72.2) quantizes occupations of thirty-six response links, up to seventy-two number trits. They share the same boundary sheet, but not the same observable.

The torsion modular produces

$$\Delta_4^{\mathrm{mod}} = 0.000111197050059977324941456613193385 6\ldots, \tag{72.5}$$

and the boundary Hamiltonian

$$\boxed{K_{\mathrm{HHM}}^{(A)} = -\log \rho_A = cI + \Delta_4^{\mathrm{mod}} (90 N_{90} + 120 N_{120}).} \tag{72.6}$$

The qualifier *holographic* does not designate an analogy: the incidence $6 \longrightarrow 8$ is transported to the boundary sets ∂A_{90} and ∂A_{120}, and the reduced state resides in the algebra generated by its number operators.

Theorem 72.2 (Joint reconstruction of geometry and memory)**.** *Let*

$$N = N_{90} + N_{120}, \qquad D = 90N_{90} + 120N_{120}. \tag{72.7}$$

Then

$$\boxed{N_{90} = 4N - \frac{D}{30}, \qquad N_{120} = \frac{D}{30} - 3N.} \tag{72.8}$$

In particular, $[\widehat{\mathcal{A}}_{\mathrm{phys}}, K^{(A)}_{\mathrm{HHM}}] = 0$*, and the pair* $(\widehat{\mathcal{A}}_{\mathrm{phys}}, K^{(A)}_{\mathrm{HHM}} - cI)$ *reconstructs the two occupation sectors.*

Proof. The operators N_{90} and N_{120} act on distinct factors and commute. The linear map carrying (N_{90}, N_{120}) to (N, D) has matrix

$$\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}, \qquad \det = 30.$$

Its inverse is exactly equation (72.8). Since equations (72.2) y (72.6) are linear combinations of the same number operators, the commutator vanishes. □

The operator interpretation is precise: the area observable preserves the total number of excitations, while the modular Hamiltonian preserves their sector of origin. Neither observable alone retains both coordinates; the joint pair does determine them.

72.2 Confluence of area, information and energy

The angular scale θ_{clk} in equation (71.21) and the chronological temperature Θ_{clk} are distinct objects; the use of an uppercase letter is part of the typing. The thermodynamic realization in section 14.4 proves

$$E_P = k_B \Theta_{\mathrm{clk}} \log 3, \tag{72.9}$$

whereas the areal realization constructs the elementary increment

$$\delta\mathcal{A} = \gamma_{\mathrm{HMT}} a_0 \ell_P^2. \tag{72.10}$$

They are not equated: equation (72.9) fixes the informational quantum of energy, and equation (72.10) the geometric quantum of area. Their quotient defines the typed tension of conversion

$$\boxed{\mathcal{T}_{I/A} = \frac{E_P}{\delta\mathcal{A}} = \frac{k_B \Theta_{\mathrm{clk}} \log 3}{\gamma_{\mathrm{HMT}} a_0 \ell_P^2}.} \tag{72.11}$$

The expression introduces no additional constant: it determines the normalizations that must be preserved in the transformation of information into geometry.

In the minimal cut of eighty-one links, the physical information capacity and the energy of one trit per link are

$$S^{\mathrm{phys}}_{\mathrm{cut}} = 81 k_B \log 3, \qquad E^{(1)}_{\mathrm{cut}} = 81 E_P = \Theta_{\mathrm{clk}} S^{\mathrm{phys}}_{\mathrm{cut}}. \tag{72.12}$$

The final equality is an equation of state of this realization, not a universal equipartition law. In particular, the Gibbs state of the thirty-six boundary links has a different entropy and is treated next.

This separation determines the structural content of the confluence: Barbero–Immirzi orients the area spectrum; $\log 3$ measures the irreducible decision of the TRIT; Θ_{clk} converts that information into energy;

and ℓ_P^2 connects the area quantum to the gravitational family. Gravity is described as a confluence of functional realizations of a common antecedent, and not as an imposed equality among their values.

Transport of the channel memory to the modular Hamiltonian and to the double thermal field purification. The channel memory is represented below by the modular Hamiltonian and a double thermal-field purification.

72.3 Area operator and reconstruction of sectorial information

Regrouping six trits constructs a bijection between $(\mathbb{Z}/9\mathbb{Z})^3$ and $(\mathbb{Z}/27\mathbb{Z})^2$, both of cardinality 729. The bijection is not presented as an additive isomorphism. On a 9×9 boundary block, the source decomposes 36 links into two sets of 18, labeled by channels 90 and 120.

72.3.1 Tritic regrouping and boundary incidence

The correspondence preserves an ordered word, in addition to its cardinality. Write each coordinate $x_j \in \mathbb{Z}/9\mathbb{Z}$, for $j = 1, 2, 3$, through its unique positional representatives $x_j = r_{2j-1} + 3r_{2j}$, with $r_k \in \{0, 1, 2\}$. The map is

$$\beta_{729}(x_1, x_2, x_3) = \big(r_1 + 3r_2 + 9r_3,\ r_4 + 3r_5 + 9r_6\big) \in (\mathbb{Z}/27\mathbb{Z})^2. \tag{72.13}$$

Proposition 72.3 (Positional correspondence between interior and boundary)**.** *The map β_{729} is bijective and preserves the order of the six trits. Its inverse extracts the two triples of digits and regroups them into three pairs.*

Proof. The positional expansion of the representatives $0, \dots, 8$ and $0, \dots, 26$ is unique. Extracting the six digits, regrouping them, and extracting them again returns the same word. The inverse operation thus restores each of the three initial coordinates. □

Carry preserves its position in the enriched word. The additive operations of the two packagings have different boundaries: for example, $\beta_{729}(0, 2, 0) = (18, 0)$ and $\beta_{729}(0, 1, 0) = (9, 0)$, whose sum in the order-twenty-seven torus is $(0, 0)$, whereas $\beta_{729}(0, 3, 0) = (0, 1)$. This distinction allows the positional correspondence to be reused while keeping the operations typed.

In the torus $(\mathbb{Z}/27\mathbb{Z})^2$, the block $B_\partial = \{0, \dots, 8\}^2$ has the oriented link sets

$$\partial B_{90} = \{((-1, j), (0, j)),\ ((8, j), (9, j)) : 0 \le j \le 8\}, \tag{72.14}$$

$$\partial B_{120} = \{((i, -1), (i, 0)),\ ((i, 8), (i, 9)) : 0 \le i \le 8\}. \tag{72.15}$$

All coordinates are read modulo twenty-seven. Each system contains eighteen links and the two systems are disjoint. Their labels preserve the incidence channels; the cardinality of the links is $18 + 18$.

Theorem 72.4 (Minimum capacity of the cubic cut)**.** *In the cubic graph with vertices $\{0, \dots, N-1\}^3$, with $N \ge 2$, links between neighbors, and unit capacity per link, the minimum cut between two opposite faces has capacity N^2. For $N = 9$, a maximally mixed product state on the trits of a minimum layer has information*

$$\mathrm{cap}_{\min} = 81, \qquad s_{\mathrm{corte}} = 81 \log 3. \tag{72.16}$$

Proof. The N^2 lines parallel to the normal of the faces form edge-disjoint paths. Every separating cut intercepts each path, so it contains at least N^2 links. The links crossing a transverse layer constitute a cut of that cardinality. Each uniform trit contributes $\log 3$ nats; entropy is additive for the product state of the eighty-one trits of the layer. □

The capacity of the cubic cut and the state of the thirty-six links in (72.15) belong to observables defined on different cuts. The genealogy preserves both domains and their measures.

72.3.2 Transport of the collective incidence modes

The spaces $\mathcal{F}_6$ and $\mathcal{F}_8$ of Section 15.5 provide the uniform vectors

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \qquad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f)$$

in $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$. In $\ell^2(\partial B_{90}) \oplus \ell^2(\partial B_{120})$, construct

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{90}} (\delta_e, 0), \qquad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{120}} (0, \delta_e).$$

The two pairs are orthonormal. Denote their respective spanned planes by $\mathscr{H}_{\mathrm{inc}}^{\mathrm{coll}}$ and $\mathscr{H}_{\partial}^{\mathrm{coll}}$.

Theorem 72.5 (Collective intertwining of the sectorial labels)**.** *There is a unique surjective linear isometry between these planes such that*

$$\mathcal{J}_{6\to 8} u_m = v_m \quad (m = 90, 120). \tag{72.17}$$

For the operators

$$D_{\mathrm{inc}} = 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|,$$
$$D_{\partial} = 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|,$$

we have

$$\mathcal{J}_{6\to 8} D_{\mathrm{inc}} = D_{\partial} \mathcal{J}_{6\to 8}. \tag{72.18}$$

The sectorial projectors and their oriented combination with coefficients $12 : -1$ *are also transported.*

Proof. The image of a basis fixes the unique linear map. Norms and inner products are preserved because both bases are orthonormal. Identity (72.18) holds on each vector u_m, since its two sides are $m v_m$. The projectors are expressed as $(120I - D)/30$ and $(D - 90I)/30$; their linear functional calculus and any combination of the two preserve intertwining. □

This transport acts on the two uniform modes and their spectral labels. Individual incidences and fluctuations within each sector remain in their complete spaces. The link preserves exactly the collective datum that the areal weighting will use.

72.3.3 Angular character and area operator

Let $N_e = \mathrm{diag}(0, 1, 2)$ on each link, and let N_{90}, N_{120} be the sector sums. With the scale

$$\theta_{\mathrm{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \qquad a_0 = \frac{1 + 2\cos(10^\circ)}{3} \theta_{\mathrm{clk}},$$

fixed, the angular factor is the normalized real character of the triplet

$$\chi_{10} = \frac{1}{3} \operatorname{Tr} \mathrm{diag}(1, e^{i\pi/18}, e^{-i\pi/18}) = \frac{1 + 2\cos(10^\circ)}{3}, \qquad a_0 = \chi_{10} \theta_{\mathrm{clk}}. \tag{72.19}$$

The sum of the conjugate phases proves the equality. On the positive branch considered, $\theta_{\mathrm{clk}} > 0$, $\chi_{10} > 0$, and $\gamma_{\mathrm{HMT}} > 0$; hence a_0 and the areal scale are positive. The angular triplet and its normalization precede the entropic distribution on the links. With these quantities, the normalized area operator is

$$\widehat{\mathcal{A}}/\ell_P^2 = \gamma_{\mathrm{HMT}} a_0 (N_{90} + N_{120}).$$

Its spectrum has values $n\gamma_{\mathrm{HMT}}a_0$, $0 \le n \le 72$, with multiplicities given by $(1 + z + z^2)^{36}$. The proof consists in summing the eigenvalues $0, 1, 2$ of the 36 tensor factors, with the links constructed in (72.15) and the character of (72.19).

Theorem 72.6 (Recovery of the boundary sectors)**.** *Let*

$$N = N_{90} + N_{120}, \qquad D = 90N_{90} + 120N_{120}.$$

Then

$$N_{90} = 4N - \frac{D}{30}, \qquad N_{120} = \frac{D}{30} - 3N. \tag{72.20}$$

Thus the total and the weighted reader jointly recover the two occupations.

Proof. La matriz $\left(\begin{smallmatrix} 1 & 1 \\ 90 & 120 \end{smallmatrix}\right)$ has determinant 30. Its inversion gives (72.20). On the domain of integer occupations, the image satisfies $D \in 30\mathbb{Z}$ and $90N \le D \le 120N$. □

The modular Hamiltonian of the source is $K_A = -\log \rho_A = cI + \Delta_{\mathrm{mod}} D$. For $\Delta_{\mathrm{mod}} \neq 0$ and known normalization, $D = (K_A - cI)/\Delta_{\mathrm{mod}}$ publishes the second observable. This operator describes the spectral structure of the reduced state; it is not identified by notation with the generator of time evolution. The area preserves the total occupation and the pair adds the provenance. For example, $(N_{90}, N_{120}) = (1, 0)$ and $(0, 1)$ share $N = 1$, but have $D = 90$ and $D = 120$.

72.3.4 State normalization and the meaning of the modular Hamiltonian

The second observable is made explicit through the state that defines it. The preceding tensor decomposition is retained: eighteen qutrits per sector, each with number operator $\mathrm{diag}(0, 1, 2)$; N_m is the sum of these operators and the sectors act on distinct factors. Let

$$z_m = 1 + e^{-m\Delta_{\mathrm{mod}}} + e^{-2m\Delta_{\mathrm{mod}}}, \qquad m \in \{90, 120\}.$$

The family of source states has the product realization

$$\rho_A = Z^{-1} e^{-\Delta_{\mathrm{mod}}(90N_{90} + 120N_{120})}, \qquad Z = z_{90}^{18} z_{120}^{18}.$$

Commutativity of the number operators and tensor factorization prove the expression for Z. The state has full rank for real, finite Δ_{mod}, and its spectral logarithm gives

$$K_A = -\log \rho_A = (\log Z) I + \Delta_{\mathrm{mod}} D.$$

For $\Delta_{\mathrm{mod}} \neq 0$, K_A weights the excitations of the two sectors differently. Joint reading with N recovers the two occupations by equation (72.20).

The need for joint reading is seen in two complementary examples. The occupation states $(1, 0)$ and $(0, 1)$ have the same total $N = 1$ and different modular weights. Conversely, $(4, 0)$ and $(0, 3)$ have the same $D = 360$ and different totals. The pair (N, D) distinguishes both cases. This recovery concerns sectoral occupations, not each microscopic vector within a degeneracy of those occupations.

When $\Delta_{\mathrm{mod}} = 0$, the state is proportional to the identity and K_A is scalar: that reading ceases to distinguish the sectors. For $\Delta_{\mathrm{mod}} \neq 0$, the known normalization allows one to recover D and compose it with the area. The role of the modular Hamiltonian is thereby defined: it characterizes the reduced state and its information distribution. Its inclusion in the article is motivated by that mathematical function, whereas the evolution Hamiltonian belongs to the dynamical development of the action.

72.3.5 Reduced state, Schmidt decomposition, and entanglement entropy

If $\rho_A = \sum_\nu p_\nu |\nu\rangle\langle\nu|$, the purification

$$|\Psi\rangle = \sum_\nu \sqrt{p_\nu}\, |\nu\rangle_A |\nu\rangle_B$$

satisfies $\mathrm{Tr}_B\, |\Psi\rangle\langle\Psi| = \rho_A$. Thus $-\mathrm{Tr}(\rho_A \log \rho_A)$ is the entanglement entropy of this particular pure bipartition. The enlarged state declares which system carries that interpretation.

For a two-sheet involution with one eigenvector of each sign,

$$\rho_y = \frac{e^{-yR}}{2\cosh y}, \qquad R^2 = I,$$

diagonalization gives

$$S(\rho_y) = \log(2\cosh y) - y\tanh y, \qquad D(\rho_y \| \rho_{-y}) = 2y\tanh y.$$

Entropy is even in y; the distance explicitly compares the two orientations. This two-level entropy is not identified with the capacity $81\log 3$ of the triadic cut.

Chapter 73

Boundary modular Hamiltonian, thermofield-double purification and oriented information deficit

The faithful qutrit state admits a double-thermofield purification. The modular Hamiltonian records the orientation of the channels, and the informational deficit measures the finite separation between states without being identified with area.

73.1 Qutrit state, thermofield-double purification, and informational deficit

For a link with weight $w > 0$, let

$$Z(w) = 1 + \mathrm{e}^{-w} + \mathrm{e}^{-2w}, \qquad \rho(w) = \frac{1}{Z(w)} \sum_{n=0}^{2} \mathrm{e}^{-wn} |n\rangle\langle n|. \tag{73.1}$$

The weights of the two families are

$$w_{90} = 90\Delta_4^{\mathrm{mod}} = 0.010007734505397959244731095187 4\ldots, \tag{73.2}$$

$$w_{120} = 120\Delta_4^{\mathrm{mod}} = 0.013343646007197278992974793583 2\ldots, \qquad \frac{w_{120}}{w_{90}} = \frac{4}{3}. \tag{73.3}$$

The finite boundary state is

$$\rho_A = \rho(w_{90})^{\otimes 18} \otimes \rho(w_{120})^{\otimes 18}. \tag{73.4}$$

Its logaritmo reproduce equation (72.6), with $c = 18 \log Z(w_{90}) + 18 \log Z(w_{120})$.

Theorem 73.1 (Double-thermofield purification)**.** *The vector*

$$|\,\mathrm{TFD}(w)\rangle = \frac{1}{\sqrt{Z(w)}} \sum_{n=0}^{2} \mathrm{e}^{-wn/2} |n\rangle_A \otimes |n\rangle_{\bar{A}} \tag{73.5}$$

purify $\rho(w)$. The product of eighteen copies for each of the weights w_{90}, w_{120} purifies ρ_A; in the doubled system, $S(\rho_A)$ is exactly entanglement entropy.

Proof. Upon tracing out the second factor, the orthogonality of $|n\rangle_{\bar{A}}$ eliminates the cross terms and leaves

$$\mathrm{Tr}_{\bar{A}}\, |\,\mathrm{TFD}(w)\rangle\langle \mathrm{TFD}(w)| = \rho(w).$$

The partial trace commutes with tensor products, which proves the assertion for the thirty-six links. □

For a link,

$$S(w) = \log Z(w) + w \frac{\mathrm{e}^{-w} + 2\mathrm{e}^{-2w}}{Z(w)}. \tag{73.6}$$

Consequently,

$$S(\rho_A) = 18 S(w_{90}) + 18 S(w_{120}) = 39.548373208818079949573197 30\ldots \text{ nats}. \tag{73.7}$$

The maximally mixed state of thirty-six qutrits has entropy $36 \log 3$. We define the oriented defect

$$\boxed{I_{\mathrm{or}} = 36 \log 3 - S(\rho_A) = 0.001669183233868940655631225913\ldots \text{ nats}.} \tag{73.8}$$

Its positivity follows from the entropic maximality of the uniform state. The proximity of I_{or} to $\gamma_{\mathrm{HMT}} a_0$ does not constitute an identity and is not used to select either value.

73.2 Odd-even decomposition of the bilayer state: interleaf information and oriented geometric transport

The normalized distribution between sheets separates even entropy from odd hexad–octad transport. This bilayer parity preserves the boundary information that the modular Hamiltonian records and avoids identifying entropy, area, and orientation.

73.3 The bicover: even information and odd geometric transport

Barbero–Immirzi and the entropy of the two sheets are not two juxtaposed quantities. They arise from the same channels $q_{\pm}$, but preserve different data. Probabilistic normalization forgets the common scale and measures the distribution between orientations; the hexad–octad functional preserves the sign of that orientation. The first reading is even and the second is odd.

73.3.1 Normalized distribution between leaves

Write

$$x = A_{\mathrm{rad}}, \qquad y = C^{*}_{\mathrm{rad}}, \qquad q_{-} = e^{-x+y}, \qquad q_{+} = e^{-x-y}.$$

The sum $q_{-} + q_{+} = 2e^{-x}\cosh y$ allows defining

$$p_{-} = \frac{q_{-}}{q_{-}+q_{+}} = \frac{e^{y}}{2\cosh y}, \qquad p_{+} = \frac{q_{+}}{q_{-}+q_{+}} = \frac{e^{-y}}{2\cosh y}.$$

The common factor e^{-x} disappears. The distribution preserves the asymmetry between sheets, but not the common angular scale.

Definition 73.2 (Bilayer entropy)**.** The entropy of the oriented distribution is

$$S_{\mathrm{bi}}(y) := -p_{-}\log p_{-} - p_{+}\log p_{+} = \log(2\cosh y) - y\tanh y.$$

Proposition 73.3 (Parity and maximum of the bicover entropy)**.** *The function S_{bi} is even and satisfies*

$$S_{\mathrm{bi}}(-y) = S_{\mathrm{bi}}(y), \qquad S'_{\mathrm{bi}}(y) = -y\ \operatorname{sech}^{2} y, \qquad 0 < S_{\mathrm{bi}}(y) \le \log 2.$$

The maximum $\log 2$ *is attained only at $y = 0$, when both sheets have equal weight.*

Proof. Parity follows because cosh is even and $y\tanh y$ is also even. The direct derivation produces

$$\frac{\mathrm{d}}{\mathrm{d}y}\log(2\cosh y) = \tanh y, \qquad \frac{\mathrm{d}}{\mathrm{d}y}(y\tanh y) = \tanh y + y\operatorname{sech}^{2} y.$$

Therefore $S'_{\mathrm{bi}}(y) = -y\operatorname{sech}^{2} y$: the function increases on $(-\infty, 0)$, decreases on $(0, \infty)$, and takes the value $\log 2$ at the origin. Positivity is that of the entropy of a distribution with two strictly positive weights. □

73.3.2 Parity of the hexad-octad transport

Let

$$s_n(q_-, q_+) = q_-^n - q_+^n.$$

The leaf exchange is the involution

$$\mathsf{J}_{\mathrm{sh}} : (x, y) \longmapsto (x, -y),$$

which permutes $q_- \leftrightarrow q_+$. Hence,

$$s_n \circ \mathsf{J}_{\mathrm{sh}} = -s_n.$$

The functional constructed in section 51.2 is

$$\Gamma_{6\to 8}(A, C^*) = \frac{\pi A C^*}{180} + 12\big(S_{90}(q_-) - S_{90}(q_+)\big) - \big(S_{120}(q_-) - S_{120}(q_+)\big).$$

Theorem 73.4 (Theorem of double-layer parity)**.** *Under the exchange $C^* \mapsto -C^*$,*

$$\boxed{S_{\mathrm{bi}}(-C^*_{\mathrm{rad}}) = S_{\mathrm{bi}}(C^*_{\mathrm{rad}}),}$$

$$\boxed{\Gamma_{6\to 8}(A, -C^*) = -\Gamma_{6\to 8}(A, C^*).}$$

The same bilayer therefore produces an even informational reading and an odd geometric transport.

Proof. The first equality is proposition 73.3. In the second, the term $\pi A C^*/180$ changes sign, and each Lambert difference changes sign because its two arguments are interchanged. The complete combination is odd. □

73.3.3 Interpretation of the constitutive operator as coupling between vacuum, field, and response

Bilayer entropy and $\Gamma_{6\to 8}$ are not two names for the same number. The former quantifies how much weight the two orientations share after cancellation of the common scale. The latter preserves the orientation and incidence of the step $6 \to 8$. At the HMT point,

$$S_{\mathrm{bi}} = 0.69246069960358548\ldots \text{ nats}, \qquad \Gamma_{6\to 8} = 0.27400767269445927\ldots.$$

The link between information and geometry is that both readings factor through the same channels $q_\pm$, not that their values are equated.

The particle–antiparticle band of section 92.9 reproduces the same parity distinction: mass and the amount of correlation descend to the band quotient, whereas charge, orientation, and transport change sign. In the Cartan–Holst action, $\Gamma_{6\to 8}$ occupies the oriented coordinate; in the informational reading, S_{bi} counts the distribution between sheets. This explains why a single structure can preserve information and reverse orientation without conflating the two operations.

Tangential variation of the modular state and derivation of the areal law. The tangent variation of the modular state produces the first areal law; finite variations preserve the relative-entropy term.

73.4 Holographic-modular free energy and informational memory of geometry

73.4.1 Oriented Bilayer State $T_{A,y}$, Normalization ρ_y, and Separation between Intensity and Leaf Orientation

The angular response constructed in the preceding chapters contains two spectral data of complementary nature. The coordinate A_{rad} fixes the common intensity of the two leaves; the oriented coordinate

$$y = C^*_{\mathrm{rad}}$$

fixes its separation. Let P_+ and P_- be rank-one orthogonal projections on a two-dimensional Hilbert space $\mathcal{H}_{\mathrm{bi}}$, and define

$$I_{\mathrm{bi}} = P_+ + P_-, \qquad R = P_+ - P_-, \qquad R^2 = I_{\mathrm{bi}}. \tag{73.9}$$

The positive angular operator is

$$T_{A,y} = \exp(-A_{\mathrm{rad}} I_{\mathrm{bi}} - yR) = q_+ P_+ + q_- P_-, \qquad q_\pm = \exp(-A_{\mathrm{rad}} \mp y). \tag{73.10}$$

Its normalization removes exactly the scalar intensity and preserves the relative orientation:

$$\boxed{\rho_y = \frac{T_{A,y}}{\operatorname{Tr} T_{A,y}} = \frac{\exp(-yR)}{2\cosh y} = p_+(y)P_+ + p_-(y)P_-,} \tag{73.11}$$

where

$$p_+(y) = \frac{\mathrm{e}^{-y}}{2\cosh y}, \qquad p_-(y) = \frac{\mathrm{e}^{y}}{2\cosh y}. \tag{73.12}$$

The intensity A_{rad} remains in $\operatorname{Tr} T_{A,y}$ and in the determinant of the unnormalized operator; the state ρ_y represents the pure informational sector of the choice of sheet. This separation types two readers of the same antecedent and avoids attributing to a normalized probability the constitutive scale that was removed by normalization.

Definition 73.5 (Bilayer holographic-modular Hamiltonian)**.** The finite holographic-modular Hamiltonian of the oriented state is

$$\boxed{K_y = -\log \rho_y = \log(2\cosh y) I_{\mathrm{bi}} + yR.} \tag{73.13}$$

The scalar term normalizes the state, and the term yR preserves sheet provenance.

The designation *holographic–modular* declares a precise function: the total observable publishes occupancy, while the modular generator retains the distribution among sectors that made that occupancy possible. In finite dimension, K_y is a well-defined self-adjoint matrix. Its extension to the infinite system will be formulated through a local net and the GNS representation; the two residences remain distinct.

73.4.2 Exact spectral thermodynamics of the 2 sheets

The reduced Function of Partition and the oriented moment are

$$Z_{\mathrm{bi}}(y) = 2\cosh y, \qquad \langle R \rangle_y = \operatorname{Tr}(\rho_y R) = -\tanh y. \tag{73.14}$$

The von Neumann entropy is calculated without approximation:

$$\boxed{S(\rho_y) = -\operatorname{Tr}(\rho_y \log \rho_y) = \log(2\cosh y) - y\tanh y.} \tag{73.15}$$

Its parity and variation satisfy

$$S(\rho_{-y}) = S(\rho_y), \qquad \frac{\mathrm{d}}{\mathrm{d}y} S(\rho_y) = -y\ \operatorname{sech}^2 y. \tag{73.16}$$

The deficit relative to the uniform bilayer state is determined by

$$I_{\mathrm{or}}^{(2)}(y) = \log 2 - S(\rho_y) = y \tanh y - \log\cosh y \geq 0. \tag{73.17}$$

Equality occurs only at $y = 0$. The deficit measures the information retained by the polarization of the sheets, whereas the entropy measures the published probabilistic distribution.

Theorem 73.6 (Relative entropy of two orientations)**.** *For $y, z \in \mathbb{R}$, the faithful states of equation (73.11) satisfy*

$$\boxed{D(\rho_y \| \rho_z) = \log \frac{\cosh z}{\cosh y} + (y - z)\tanh y.} \tag{73.18}$$

In particular, sheet inversion has the exact cost

$$\boxed{D(\rho_y \| \rho_{-y}) = 2y \tanh y.} \tag{73.19}$$

Proof. From $\log \rho_y = -\log(2\cosh y) I_{\mathrm{bi}} - yR$ it follows.

$$\log \rho_y - \log \rho_z = \log \frac{\cosh z}{\cosh y} I_{\mathrm{bi}} + (z - y)R.$$

Taking the expectation in ρ_y and using $\langle R \rangle_y = -\tanh y$ yields equation (73.18). Substituting $z = -y$ eliminates the quotient of hyperbolic cosines and produces equation (73.19). The expression is positive because y and $\tanh y$ have the same sign. □

The symmetrized Jeffreys cost is

$$D(\rho_y \| \rho_{-y}) + D(\rho_{-y} \| \rho_y) = 4y \tanh y. \tag{73.20}$$

The local curvature of the exponential family is given by

$$\frac{\mathrm{d}^2}{\mathrm{d}y^2} \log Z_{\mathrm{bi}}(y) = \mathrm{Var}_{\rho_y}(R) = \operatorname{sech}^2 y. \tag{73.21}$$

Thus, the same oriented coordinate determines three quantities with different parities: $\langle R \rangle_y$ is odd; $S(\rho_y)$ is even; and the distance between orientations is positive and even as a function of the modulus of y. Geometry, information, and inversion preserve different components of the same state.

Corollary 73.7 (Separation of intensity and orientation)**.** *The determinant $\det T_{A,y} = \exp(-2A_{\mathrm{rad}})$ depends on the angular intensity, whereas the quotient $q_-/q_+ = \exp(2y)$, the moment $\langle R \rangle_y$, and $D(\rho_y \| \rho_{-y})$ depend on orientation. Therefore, the even reader leading to D_A and the odd reader leading to sheet memory are jointly reconstructible and remain typed.*

73.4.3 Double-thermofield purification and sheet conjugation

Let $(|+\rangle, |-\rangle)$ be a basis adapted to (P_+, P_-), and let $\overline{\mathcal{H}}_{\mathrm{bi}}$ be a declared copy. We define

$$|\,\mathrm{TFD}_y\rangle = \sqrt{p_+(y)}\,|+\rangle|\bar{+}\rangle + \sqrt{p_-(y)}\,|-\rangle|\bar{-}\rangle. \tag{73.22}$$

Proposition 73.8 (Bilayer purification)**.** *The vector of equation (73.22) is normalized and satisfies*

$$\mathrm{Tr}_{\overline{\mathcal{H}}_{\mathrm{bi}}} |\,\mathrm{TFD}_y\rangle\langle \mathrm{TFD}_y\,| = \rho_y. \tag{73.23}$$

The simultaneous permutation of the labels $+ \leftrightarrow -$ transports $|\,\mathrm{TFD}_y\rangle$ into $|\,\mathrm{TFD}_{-y}\rangle$.

Proof. The sum of the squares of the coefficients is $p_+(y) + p_-(y) = 1$. The partial trace eliminates the cross terms and leaves $p_+(y)P_+ + p_-(y)P_-$. Exchange of the labels permutes both coefficients, since $p_+(-y) = p_-(y)$. □

The purification proves an equality of entropies within the declared doubled system:

$$S(\rho_y) = S\big(\mathrm{Tr}_{\mathcal{H}_{\mathrm{bi}}} \,|\, \mathrm{TFD}_y\rangle\langle \mathrm{TFD}_y \,|\big)\,. \tag{73.24}$$

The thermofield-double copy has a mathematical function here: it exhibits a pure state whose reduction produces exactly the bilayer state. Its identification with two physical regions of a spacetime requires a factorization of observables and an additional geometric interface.

The bilayer TFD and the qutrit TFD of theorem 73.1 are related yet distinct constructions. The first purifies the orientation of two sheets; the second purifies the thirty-six qutrit links of the channels 90/120. Their confluence resides in the angular antecedent and in the provenance label, not in an identification of their Hilbert spaces.

73.4.4 Total Geometry and Modular Provenance

The number operators N_{90} and N_{120} commute. Areal geometry publishes the sum of occupations,

$$\widehat{\mathcal{A}} = \gamma_{\mathrm{HMT}} a_0 \ell_P^2 (N_{90} + N_{120}), \tag{73.25}$$

while the modular Hamiltonian of the boundary preserves its origin,

$$K_A - cI = \Delta_4^{\mathrm{mod}} (90N_{90} + 120N_{120}). \tag{73.26}$$

Let us define the dimensionless observables

$$N = N_{90} + N_{120}, \qquad D = 90N_{90} + 120N_{120}. \tag{73.27}$$

Theorem 73.9 (Reconstruction joint of the channels)**.** *The pair (N, D), and hence the pair $(\widehat{\mathcal{A}}, K_A)$ once their scales have been fixed, exactly reconstructs the two occupation operators:*

$$\boxed{N_{90} = 4N - \frac{D}{30}, \qquad N_{120} = \frac{D}{30} - 3N.} \tag{73.28}$$

Proof. The linear system has matrix $\left(\begin{smallmatrix} 1 & 1 \\ 90 & 120 \end{smallmatrix}\right)$ with determinant 30. Its inverse gives equation (73.28). Commutativity of the number operators permits application of the identity in their joint functional calculus. □

The result formulates a concrete holography of reconstruction. The area determines the total geometric quantity; the modular Hamiltonian determines the weighted provenance; the pair recovers the sectoral distribution. Two states with the same area can carry distinct histories, and that difference remains visible in K_A. The coefficient 30 that makes the system invertible is the same completion defect between the channels 90 and 120.

Reconstruction requires retaining both observables. A compression that retained only N would identify all constant-sum pairs (N_{90}, N_{120}). A compression that retained only D would identify pairs of equal modular weight. The joint pair has rank two and preserves the relevant sectoral genealogy.

73.4.5 1st local law and finite balance of relative entropy

The finite formulation employs the relative entropy of operator states in the sense of Araki [5].

Let ρ_A be the boundary qutrit state of equation (73.4) and let $K_A = -\log \rho_A$. The areal decomposition proved in theorem 74.1 takes the form

$$K_A - cI = \beta_{90}\mathcal{A}_{90} + \beta_{120}\mathcal{A}_{120}, \qquad \frac{\beta_{120}}{\beta_{90}} = \frac{4}{3}. \tag{73.29}$$

For a differentiable curve of normalized states passing through ρ_A, the tangent variation satisfies

$$\boxed{\delta S = \beta_{90}\,\delta\langle\mathcal{A}_{90}\rangle + \beta_{120}\,\delta\langle\mathcal{A}_{120}\rangle.} \tag{73.30}$$

The quotient $4/3$ records the different informational weights of the two sectors; the same increment in total area may produce different entropic variations according to its channel of origin.

The finite formulation preserves the term eliminated by linearization. If σ is a state whose support is contained in that of ρ_A, then

$$\boxed{\Delta S = \beta_{90}\Delta\langle\mathcal{A}_{90}\rangle + \beta_{120}\Delta\langle\mathcal{A}_{120}\rangle - D(\sigma\|\rho_A).} \tag{73.31}$$

Proof. The definition of relative entropy provides

$$D(\sigma\|\rho_A) = -S(\sigma) + \mathrm{Tr}(\sigma K_A).$$

For ρ_A, the same expression is zero. Subtracting the two identities yields $\Delta S = \Delta\langle K_A\rangle - D(\sigma\|\rho_A)$. The scalar term in K_A cancels between normalized states, and equation (73.29) concludes the proof. □

The local law expresses the derivative of entropy at the reference state. The finite law expresses the exact balance among geometric response, entropic change, and informational distance. The positivity of D quantifies the convexity that separates the finite displacement from its tangent approximation.

73.4.6 Holographic-modular free energy of the horizon

Consider a horizon realization in which an areal radius R determines

$$E_\Lambda(R) = \frac{c^4 R}{2G}, \qquad k_B T_H(R) = \frac{\hbar c}{2\pi R}, \qquad \frac{S_H(R)}{k_B} = \frac{\pi R^2}{\ell_P^2}. \tag{73.32}$$

The identity $\ell_P^2 = G\hbar/c^3$ gives

$$\boxed{E_\Lambda(R) = T_H(R) S_H(R).} \tag{73.33}$$

For a finite perturbation σ of the reference state ρ_A, write

$$\Delta k_A = \mathrm{Tr}((\sigma - \rho_A)K_A), \qquad \Delta s_A = S(\sigma) - S(\rho_A), \tag{73.34}$$

and let us define the response quantities

$$\mathcal{E}_R(\sigma) = E_\Lambda(R) + k_B T_H(R)\Delta k_A, \qquad \mathcal{S}_R(\sigma) = S_H(R) + k_B \Delta s_A. \tag{73.35}$$

Theorem 73.10 (Free-Energy Identity in Holographic–Modular Form)**.** *Under the horizon realization equation (73.32), for every σ whose support is contained in that of ρ_A,*

$$\boxed{\mathcal{E}_R(\sigma) - T_H(R)\mathcal{S}_R(\sigma) = k_B T_H(R) D(\sigma\|\rho_A) \geq 0.} \tag{73.36}$$

Proof. The reference saturation equation (73.33) cancels the term $E_\Lambda - T_H S_H$. By definition,

$$\mathcal{E}_R - T_H \mathcal{S}_R = k_B T_H (\Delta k_A - \Delta s_A).$$

The finite relative-entropy identity gives $\Delta k_A - \Delta s_A = D(\sigma \| \rho_A)$, and Klein's inequality gives positivity. □

In the closure $R = \ell_P$, the Planck quantum $E_P = \hbar c/\ell_P$ yields

$$k_B T_H(\ell_P) = \frac{E_P}{2\pi}, \qquad E_\Lambda(\ell_P) = \frac{E_P}{2}, \tag{73.37}$$

and therefore

$$\boxed{\mathcal{E}_{\ell_P}(\sigma) - T_H(\ell_P)\mathcal{S}_{\ell_P}(\sigma) = \frac{E_P}{2\pi} D(\sigma \| \rho_A).} \tag{73.38}$$

The structural identity

$$E_P = k_B \Theta_{\mathrm{clk}} \log 3 \tag{73.39}$$

converts the defect coefficient into

$$\frac{E_P}{2\pi} = \frac{k_B \Theta_{\mathrm{clk}} \log 3}{2\pi}. \tag{73.40}$$

The reference horizon saturates the balance; the finite deviation is measured by the informational cost of departing from its modular genealogy.

The realization also satisfies

$$E_\Lambda(\ell_P) t_P = \frac{\hbar}{2}, \qquad E_\Lambda(\ell_P) T_{108} = \frac{h}{2}, \tag{73.41}$$

when $t_P = \ell_P/c$ and $T_{108} = 2\pi t_P$. The half-action results from the coupling between thermal periodicity and areal geometry. This equality remains separate from the free-energy identity: the first fixes the quantum of the reference horizon; the second quantifies the perturbations of the modular state.

73.4.7 Confluence between areal quantity and informational quantity

The hexad–octad transition publishes the geometric quantum

$$\delta\mathcal{A} = \gamma_{\mathrm{HMT}} a_0 \ell_P^2, \tag{73.42}$$

and the CT108 clock publishes the information quanta

$$E_P = k_B \Theta_{\mathrm{clk}} \log 3. \tag{73.43}$$

Its quotient defines the tension tipada

$$\boxed{\mathcal{T}_{I/A} = \frac{E_P}{\delta\mathcal{A}} = \frac{k_B \Theta_{\mathrm{clk}} \log 3}{\gamma_{\mathrm{HMT}} a_0 \ell_P^2}.} \tag{73.44}$$

The equation assigns a conversion between a ternary decision and a unit of geometry. The numerator belongs to the action–clock–information branch; the denominator belongs to the hexad–octad–area branch. Both descend from the same enriched state and converge in gravity after preserving their domains.

The pair

$$\left(\widehat{\mathcal{A}}, K_A\right) \tag{73.45}$$

reconstructs quantity and origin; the pair

$$(\delta\mathcal{A}, E_P) \tag{73.46}$$

relates elementary geometry and informational cost. The first pair is spectral; the second is dimensional. Together they describe how a discrete structure of orientation is published as area, energy and entropy.

73.4.8 Local UHF Network and Modular Generator in the GNS Representation

Let $(\mathcal{A}_m, \iota_m)$ be a net of local matrix algebras with

$$\mathcal{A}_m \simeq M_{9^m}(\mathbb{C}), \qquad \iota_m(a) = a \otimes I_9, \qquad \mathcal{A}_{\mathrm{UHF}} = \varinjlim_m \mathcal{A}_m. \tag{73.47}$$

A family of local states ω_m defines a global state ω when it satisfies

$$\omega_{m+1} \circ \iota_m = \omega_m. \tag{73.48}$$

In each finite cut can be written

$$\omega_m(a) = \mathrm{Tr}(\rho_m a), \qquad K_m = -\log \rho_m. \tag{73.49}$$

The family (K_m) constitutes the local residence of the holographic-modular Hamiltonian.

In the GNS representation $(\pi_\omega, \mathcal{H}_\omega, \Omega_\omega)$, modular theory is expressed through the modular operator Δ_ω and the group

$$\sigma_t^\omega(x) = \Delta_\omega^{it} x \Delta_\omega^{-it}, \qquad x \in \pi_\omega(\mathcal{A}_{\mathrm{UHF}})''. \tag{73.50}$$

The generator $-\log \Delta_\omega$ may be unbounded and belongs to the standard representation; a global trace-class density matrix within the UHF does not, in general, form part of this construction.

The expectativa tracial nonadica

$$E_m = \mathrm{id} \otimes \tau_9 : \mathcal{A}_{m+1} \longrightarrow \mathcal{A}_m \tag{73.51}$$

preserves the unit and constitutes the canonical matrix compression reader. Compatibility of a modular state additionally requires

$$\omega_m \circ E_m = \omega_{m+1} \quad \text{in the declared subspace,} \tag{73.52}$$

or the equivalent condition appropriate to the inclusion used. The traciality of the reader does not automatically turn a nontracial Gibbs product into a compatible family.

The theorem 75.1 proves, under faithful persistence of the two weights $90\Delta_4^{\mathrm{mod}}$ and $120\Delta_4^{\mathrm{mod}}$, positive asymptotic density of both channels, factoriality, and the absence of additional quotients, that the ratio group is

$$90\Delta_4^{\mathrm{mod}}\mathbb{Z} + 120\Delta_4^{\mathrm{mod}}\mathbb{Z} = 30\Delta_4^{\mathrm{mod}}\mathbb{Z} \tag{73.53}$$

and that the hyperinfinite ITPFI closure is of type

$$\boxed{III_{\lambda_\partial}, \qquad \lambda_\partial = \exp(-30\Delta_4^{\mathrm{mod}}).} \tag{73.54}$$

The statement classifies the specified product network. A modification of the ratio group, asymptotic disappearance of a channel, or loss of factoriality requires recalculating the type. In particular, the tracial II_1 closure of the UHF algebra and the type-III modular closure correspond to different states and representations; both are operatorial readers of the nonadic system, and neither replaces the APP–TRIT–TPK antecedent.

73.4.9 Exact gravitational closure identities and tensor realization conditions

The architecture of this chapter distinguishes the following results:

(i) The bilayer normalization, the Hamiltonian K_y, the entropy $S(\rho_y)$, and the identity $D(\rho_y\|\rho_{-y}) = 2y\tanh y$ are *exact new formalization* of the angular response already constructed.

(ii) The TFD bilayer is a *exact new formalization* in the declared doubled space. Interpreting the copy as another physical region requires a subsequent geometric factorization.

(iii) The reconstruction 90/120, the qutrit DFT, the first local law, the finite identity, and the modular classification under their hypotheses are *resultados exactos* of the preceding operatorial construction.

(iv) The horizon equality $E_\Lambda = T_H S_H$ is exact in the declared de Sitter chart. Its composition with the qutrit state to form equation (73.36) is a *interface physical conditioned* to that realization.

(v) The intertwiner among horizon microstates, boundary TFD, and hyperbolic eigenspaces is classified by spectrum and multiplicity in theorem 73.11: typed coincidence produces unitary equivalence, and its failure proves the obstruction.

Estatuto probatorio

The spectral identities of the family ρ_y, its purification, the linear reconstruction of the sectors 90/120, and the finite balance with relative entropy are exact in finite dimension. The type-III classification is exact under the four hypotheses formulated in this construction. Gravitational free energy is asserted in the horizon realization of equation (73.32); it preserves the proper residence of the modular state and retains cosmological dynamics as a subsequent promotion.

Falsification criterion

The corresponding assertions are refuted by: a spectrum of ρ_y that does not sum to one; a computation of $D(\rho_y\|\rho_{-y})$ different from $2y\tanh y$; a partial trace of the TFD that does not produce ρ_y; two distinct pairs (N_{90}, N_{120}) with the same (N, D); a finite perturbation that violates equation (73.31); a horizon chart that fails to satisfy $E_\Lambda = T_H S_H$; or an infinite network whose ratio group differs from $30\Delta_4^{\mathrm{mod}}\mathbb{Z}$ under the declared hypotheses. An operator promotion is also refuted by replacing the local network with a nonexistent global density matrix or by identifying the tracial closure II_1 with the type-III modular closure.

73.5 Spectral Classification of the Intertwiner among the Horizon, Thermofield-Double Purification, and Perron Dynamics

Let $\mathcal{H}_{\mathrm{H}}$, $\mathcal{H}_{\mathrm{TFD}}$, and $\mathcal{H}_{\mathrm{P}}$ be the cyclic subspaces generated, respectively, by the horizon microstates, the thermofield-double purification, and the positive Perron operator. Denote by A_{H}, A_{TFD}, and A_{P} their restricted self-adjoint operators, and by $m_X(\lambda)$ the spectral multiplicity of λ in the sector X.

Theorem 73.11 (Existence and uniqueness by spectral entangler calibers)**.** *There exists a surjective isometry*

$$U : \mathcal{H}_{\mathrm{H}} \longrightarrow \mathcal{H}_{\mathrm{TFD}} \longrightarrow \mathcal{H}_{\mathrm{P}}$$

that intertwines the 3 *operators if and only if, after applying the declared energy and memory normalizations,*

$$\mathrm{spec}(A_{\mathrm{H}}) = \mathrm{spec}(A_{\mathrm{TFD}}) = \mathrm{spec}(A_{\mathrm{P}}), \qquad m_{\mathrm{H}}(\lambda) = m_{\mathrm{TFD}}(\lambda) = m_{\mathrm{P}}(\lambda)$$

for every λ in the common spectrum. When these conditions hold, the intertwiner is determined up to a unitary transformation on each eigenspace of multiplicity greater than 1.

Proof. If $UA_X = A_Y U$, unitary equivalence preserves the spectrum and its multiplicities. Conversely, for each λ one chooses orthonormal bases of the eigenspaces of equal dimension and defines U_λ by matching those bases. The orthogonal sum $U = \bigoplus_\lambda U_\lambda$ is unitary and intertwines the operators. The residual freedom is exactly the product of the unitary groups of the degenerate eigenspaces. □

The existence problem is thus closed by classification: coincidence of spectra and multiplicities produces the intertwiner; failure to satisfy them proves the obstruction. An isolated scalar equality does not replace these conditions.

The area and the modular Hamiltonian form a pair conceptually analogous to product and quotient in the vacuum.

The area operator determines total occupancy. The modular Hamiltonian preserves the sectoral provenance of that occupancy.

Two families of channels appear at the boundary: 90 and 120. Each has its occupancy operators. The area operator is constructed from the sum:

$$\frac{\widehat{\mathcal{A}}_{\text{phys}}}{\ell_P^2} = \gamma_{\text{BI}} a_0 (N_{90} + N_{120}).$$

Area counts the total number of excitations. Whether an excitation comes from channel 90 or channel 120, the elementary area is the same. Geometrically, both contribute to the same total.

The modular holographic Hamiltonian introduces an additional sectorial weighting:

$$K_{\text{HHM}} - cI = \Delta_4^{\text{mod}} (90 N_{90} + 120 N_{120}).$$

Count the occupations and also weigh each excitation by the channel from which it comes.

This is why the area and the Hamiltonian commute: both read the same occupation operators and contain complementary information.

The spectrum of the area operator is invariant with respect to sectoral provenance, whereas the modular Hamiltonian distinguishes it.

The area operator determines the total number of excitations. The weighted modular value determines their distribution between the two genealogies. The joint conservation of both observables exactly reconstructs the sectoral decomposition.

This is an especially sharp realization of the holographic idea. The boundary publishes a minimal set of observables from which the pertinent information of the interior can be recovered.

Here holography consists in identifying which pair of boundary observables exactly reconstructs which internal data.

The area operator publishes geometric magnitude, and the modular Hamiltonian publishes sectorial genealogy. The pair of observables reconstructs the complete information of the sectors 90/120.

The reduced boundary state furthermore converts that Hamiltonian into a true modular generator:

$$K_{\text{HHM}} = -\log \rho_A.$$

Thermofield-double purification shows that the entropy of the reduced state is exactly entanglement entropy within the declared doubled system. There is an enlarged pure state whose partial trace produces the boundary state and explicitly realizes the thermal identity.

Then appears the oriented information deficit:

$$I_{\text{or}} = 36 \log 3 - S(\rho_A).$$

The 36 trits would represent the maximum capacity if all states were uniform. The difference measures how much oriented information is preserved by the actual distribution of weights 90/120.

The total entropy quantifies uncertainty. The deficit relative to the maximum quantifies the residual oriented structure.

The first modular law translates that structure into geometry:

$$\delta S = \beta_{90}\,\delta\langle \mathcal{A}_{90}\rangle + \beta_{120}\,\delta\langle \mathcal{A}_{120}\rangle,$$

with

$$\frac{\beta_{120}}{\beta_{90}} = \frac{4}{3}.$$

The change in entropy depends jointly on the change in total area and the sector of provenance. Two variations with the same geometric increment may have different informational responses if they arise from different channels.

This dependence establishes the geometric conservation of memory.

For finite variations, relative entropy appears as a positive correction term. The exact extension of the law incorporates the difference between the exact variation and its linear approximation through a positive quantity.

Outside the infinitesimal regime, this positive correction determines the nonlinear cost of departure from the reference state and quantifies the contribution of genealogical memory to the finite variation.

Finally, the relation

$$E_P = k_B \Theta_{\mathrm{clk}} \log 3$$

and the quantum areal

$$\delta\mathcal{A} = \gamma_{\mathrm{BI}} a_0 \ell_P^2$$

allow the definition of a conversion tension between energetic information and geometry:

$$\mathcal{T}_{I/A} = \frac{E_P}{\delta\mathcal{A}}.$$

The exact ratio between the informational energy quantum and the geometric area quantum is expressed in the declared normalization.

For this reason, the modular holographic Hamiltonian is a central component within an atlas of complementary readers. Its specific function is to preserve the provenance that the total geometry integrates into a sum.

The area operator determines the total occupation.

The modular Hamiltonian determines the sectoral origin.

Quantum entropy quantifies the published information.

The relative entropy quantifies the cost of the genealogical transition.

73.6 Joint reconstruction of occupancy, provenance, and modular entropy in the channels 90/120

The information retained by reduction to the region is published through the modular Hamiltonian. The starting structure of this reading is declared to be

$$\Delta_4^{\mathrm{mod}} = \frac{(A - C^*)\pi/180}{9 \cdot 90}\left(1 - \frac{7}{12}\frac{\pi}{729}\right), \qquad w_{90} = 90\Delta_4^{\mathrm{mod}}, \quad w_{120} = 120\Delta_4^{\mathrm{mod}},$$

we define

$$K_{\mathrm{link}}(w) = wN, \qquad \rho_{\mathrm{link}}(w) = \frac{\mathrm{diag}(1, e^{-w}, e^{-2w})}{1 + e^{-w} + e^{-2w}},$$

$$\rho_A = \bigotimes_{e \in \partial A_{90}} \rho_{\mathrm{link}}(w_{90}) \otimes \bigotimes_{e \in \partial A_{120}} \rho_{\mathrm{link}}(w_{120}), \qquad K_A = -\log \rho_A. \tag{73.55}$$

Then

$$K_A = \sum_{e \in \partial A} w_e N_e + c\mathbf{1}, \qquad S(\rho_A) = 18S(w_{90}) + 18S(w_{120}),$$

and the evaluation of that starting structure gives

$$S(\rho_A) = 39.54837320881808\ldots \text{ nats} = 57.05624190358778\ldots \text{ bits}.$$

The quantity $S(\rho_A)$ is the von Neumann entropy of the declared mixed product. A global purification identifies it with an entanglement entropy across the cut; without that purification, its canonical type is reduced modular entropy. Thus, γ_{HMT} fixes the quantum of area/torsion, S_{tri} counts the areal capacity, S_{bi} quantifies the allocation between sheets, and $S(\rho_A)$ measures modular mixing. Area operator, spectrum, and intertwiner belong to a single boundary construction.

73.6.1 Genealogical memory and informational content of a reading

The area operator brings together boundary occupations. The weighted observable adds their sectoral provenance. This difference expresses a physical principle of the manuscript: accessible information depends on the map that publishes the state. In APP, the residue and quotient jointly preserve the integer operation; in TPK, the visible phase coexists with orientation and the increment of memory. The physical realization transports that same distinction between state, observable and recovery register; identity (73.57) makes explicit the information preserved by the reading.

Let Ω be a finite set of states reached by the update under consideration. A reader $q : \Omega \to Y$ publishes the observable coordinate and $M : \Omega \to \mathcal{M}$ preserves the memory of the fibre: the relevant sheet, orientation, quotient, carry and route. For a distribution p over those states, denote by H_p the dimensionless entropy, with the natural logarithm.

Proposition 73.12 (Informational resolution of the fiber)**.** *If (q, M) is injective on the support of p, then*

$$\mathsf{H}_p(X) = \mathsf{H}_p(q(X)) + \mathsf{H}_p(X \mid q(X)), \tag{73.56}$$

$$\mathsf{H}_p(X \mid q(X), M(X)) = 0. \tag{73.57}$$

Proof. Factorization of the joint distribution into the visible distribution and the conditional distribution over the fibre gives the first equality. Injectivity provides an inverse on the image of (q, M); each published pair determines a unique state of the support and its conditional distribution is concentrated on that state. □

Equation (72.20) implements this mechanism for the two aggregate occupations: N publishes the total and D resolves its sectoral distribution. States $(N_{90}, N_{120}) = (1, 0)$ and $(0, 1)$ share the aggregate area and are distinguished by the second observable. Under an equiprobable distribution over that pair, reading

N alone leaves $\log 2$ nats of uncertainty; the pair (N, D) resolves it exactly. The assertion concerns those occupations. The microhistories within each sector retain their additional register.

The interpretation of edge memory is thus determined by a concrete recovery. A reduction can hide information that the joint state continues to transport. For a bijective update U, the equality $\mathsf{H}_p(X \mid U(X)) = 0$ expresses logical reversibility. A physical memory-discarding or reset operation, for its part, has its own information and energy balance. Entanglement is evaluated with respect to the bipartition and reduced state specified in the preceding section.

73.6.2 Areal and volumetric sheets: incidence, growth, and measure

The areal–volumetric distinction has a dynamic realization in the TPK calendar. The block $(+1, -1, 0)$, applied to the modes Σ, Π through

$$+1 : m \mapsto \Sigma, \qquad -1 : m \mapsto \Pi, \qquad 0 : m \mapsto m,$$

produces the cyclic orbit (Σ, Π, Π). The geometric reader publishes $\Sigma \mapsto \mathscr{H}_{\mathrm{ar}}$ and $\Pi \mapsto \mathscr{H}_{\mathrm{vol}}$. Their consecutive transitions fix the incidence

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad N_9 = 3F_{\mathrm{av}}, \quad N_{27} = 9F_{\mathrm{av}}, \quad N_{108} = 36F_{\mathrm{av}}. \tag{73.58}$$

Indeed, the cycle contains one transition of each type $\Sigma \to \Pi$, $\Pi \to \Pi$, $\Pi \to \Sigma$. The larger calendars repeat this block with the indicated multiplicities.

The associated symbolic envelope allows every concatenation admitted by F_{av}. Its number of paths depends on powers of that matrix. With $F_0 = 0$, $F_1 = 1$, induction gives

$$F_{\mathrm{av}}^{n} = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}, \qquad h_{\mathrm{top}} = \log \varphi. \tag{73.59}$$

The matrix identity uses the Fibonacci recurrence. Exponential growth is fixed by the Perron root φ of the incidence already constructed.

Proposition 73.13 (Stationary distribution of the sheets)**.** *The maximal-entropy measure of the symbolic envelope has weights*

$$\mu_{\mathrm{ar}} = \frac{1}{1+\varphi^2}, \qquad \mu_{\mathrm{vol}} = \frac{\varphi^2}{1+\varphi^2}, \qquad \frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}. \tag{73.60}$$

Proof. The positive vector $r = (1, \varphi)$ satisfies $F_{\mathrm{av}} r = \varphi r$. The transition probabilities $P_{ij} = (F_{\mathrm{av}})_{ij} r_j / (\varphi r_i)$ sum to one. Symmetry of F_{av} gives stationary weights proportional to r_i^2. Their normalization produces (73.60). In the stationary mean, $-\log P_{ij} = \log \varphi + \log r_i - \log r_j$ on the allowed edges. The final two terms cancel, and the entropy per transition is $\log \varphi$, equal to the topological entropy in (73.59). □

The periodic calendar and its symbolic envelope constitute two explicit objects: the former has chronological frequencies $(1/3, 2/3)$; the latter admits the concatenations of the incidence and has the measure (73.60). This separation allows each quantity to be assigned its physical-mathematical meaning. The ratio φ^{-2} characterizes the distribution of the sheets in that measure; $\log \varphi$ characterizes the growth of its permitted histories. The angular boundary reading subsequently incorporates the corresponding regional coordinate. The construction distinguishes volumetric growth, areal publication and transport memory.

73.6.3 Admissible continuations and survival information

Given a TPK state x, let $\Omega_n(x)$ be the finite set of its continuations of length n admitted by the prolongation rule. Restrictions between levels preserve the prefixes. A compatible family of probabilities p_n, obtained from the specified statistical reader, defines the continuation uncertainty

$$s_n(x) = -\sum_{\omega \in \Omega_n(x)} p_n(\omega) \log p_n(\omega). \tag{73.61}$$

Compatibility requires the probability of a prefix to be the sum of the probabilities of its prolongations. In the preceding envelope, the stationary measure provides a family of this kind.

Proposition 73.14 (Informational balance of refinement)**.** *For the preceding compatible family,*

$$0 \le s_n(x) \le \log |\operatorname{supp} p_n|, \tag{73.62}$$

$$s_{n+1}(x) - s_n(x) = \mathsf{H}(X_{n+1} \mid X_n) \ge 0. \tag{73.63}$$

The upper bound is attained by the uniform distribution on the support. The increment is zero when every prefix of positive probability determines its prolongation with probability one.

Proof. The elementary entropy inequality for a finite distribution gives the first line. Since X_n is the restriction of X_{n+1}, factoring probabilities over each prolongation fiber gives $\mathsf{H}(X_{n+1}) = \mathsf{H}(X_n) + \mathsf{H}(X_{n+1} \mid X_n)$. Each conditional entropy is nonnegative and vanishes exactly for a point distribution. □

Continuations remaining admissible thus have precise informational content. Their multiplicity and weights describe the residual uncertainty of the reading at finite depth. A realized history also preserves its orientation and memory. Informational entropy, reduced-state entropy, and thermodynamic entropy are linked through their respective readers; each retains the domain information necessary for its interpretation.

73.7 Areal–volumetric incidence and topological entropy

The TPK calendar induces the hinge between the additive and multiplicative modes. In the primitive block

$$\tau_3 = (+1, -1, 0),$$

the rule

$$+1 : m \mapsto \Sigma, \qquad -1 : m \mapsto \Pi, \qquad 0 : m \mapsto m$$

produces the cyclic orbit (Σ, Π, Π). The geometric reader

$$\Sigma \longmapsto \mathscr{H}_{\mathrm{ar}}, \qquad \Pi \longmapsto \mathscr{H}_{\mathrm{vol}}$$

transports these modes to the areal and volumetric sheets.

Theorem 73.15 (Necessary incidence of the sheets)**.** *In the order* $(\mathscr{H}_{\mathrm{ar}}, \mathscr{H}_{\mathrm{vol}})$*, the primitive incidence matrix is*

$$\boxed{F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.} \tag{73.64}$$

Calendars of lengths $9, 27, 108$ *yield*

$$N_9 = 3F_{\mathrm{av}}, \qquad N_{27} = 9F_{\mathrm{av}}, \qquad N_{108} = 36F_{\mathrm{av}}.$$

Proof. The consecutive pairs of the cycle are $\Sigma \to \Pi$, $\Pi \to \Pi$, and $\Pi \to \Sigma$. Each occurs once, and $\Sigma \to \Sigma$ does not occur. This fixes all four entries of F_{av}. The longer calendars contain, respectively, three, nine, and thirty-six copies of the primitive block. □

Theorem 73.16 (Fibonacci law and measure of maximal entropy)**.** *For $n \geq 1$,*

$$F_{\mathrm{av}}^{n} = \begin{pmatrix} F_{n-1} & F_{n} \\ F_{n} & F_{n+1} \end{pmatrix}. \tag{73.65}$$

The one-step subshift defined by F_{av} has

$$\boxed{h_{\mathrm{top}} = \log \varphi}, \qquad \boxed{\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi^{-2}}. \tag{73.66}$$

Proof. The matrix identity follows by induction from $F_{n+1} = F_n + F_{n-1}$. The Perron root of F_{av} is φ, hence $h_{\mathrm{top}} = \log \varphi$. Since the matrix is symmetric, the Parry weights [239] are proportional to the squared components of a Perron vector $(1, \varphi)$, that is, to $(1, \varphi^2)$. □

The expressions $\log 3$ and $\log \varphi$ are thus distinguished by their genealogy. The first quantifies a uniform local choice among three states; the second measures the growth of histories allowed by sheet incidence. The areal–volumetric relation is a property of the complete return and therefore appears with exponent two.

73.7.1 Stationary distribution and variational characterization

Let X_{av} be the space of two-sided sequences over $\{\mathrm{ar}, \mathrm{vol}\}$ whose consecutive pairs have positive incidence in F_{av}. It is the one-step-memory envelope of the mode calendar: it retains the admissible pairs while allowing all concatenations compatible with them. Its stationary measure weights that space of histories; the chronological frequency of the orbit (Σ, Π, Π) remains, separately, $(1/3, 2/3)$.

Proposition 73.17 (Explicit characterization of the measure of maximal entropy)**.** *With $r = (1, \varphi)$, the stochastic matrix and its stationary distribution are*

$$P_{ij} = \frac{(F_{\mathrm{av}})_{ij} r_j}{\varphi r_i}, \qquad P = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \qquad \pi_{\mathrm{av}} = \frac{(1, \varphi^2)}{1 + \varphi^2}. \tag{73.67}$$

The stationary Markov measure μ determined by these data assigns to each admissible cylinder of n transitions

$$\mu[i_0 \dots i_n] = \frac{r_{i_0} r_{i_n}}{(1 + \varphi^2)\varphi^{n}}. \tag{73.68}$$

It is the unique stationary measure on X_{av} with maximal entropy rate, and that rate is $\log \varphi$.

Proof. The identity $\varphi^2 = \varphi + 1$ implies $\varphi^{-2} + \varphi^{-1} = 1$, so the rows of P sum to one. Direct substitution verifies $\pi_{\mathrm{av}} P = \pi_{\mathrm{av}}$. Both cross-transitions have positive probability; the chain is irreducible, and the self-transition at vol makes it aperiodic. The stationary distribution is also obtained by solving the two linear equations and the normalization, which give the unique solution displayed above. Multiplication of $\pi_{i_0} \prod_{k=0}^{n-1} P_{i_k i_{k+1}}$ cancels the intermediate factors r_{i_k} and proves (73.68).

For maximality, let ν be any stationary measure on X_{av}, and let $q(j \mid \mathcal{P})$ be the conditional distribution of X_1 given the entire past $\mathcal{P} = (\dots, X_{-1}, X_0)$. Its support is contained in that of the row $P_{X_0,\cdot}$. For finite distributions, the relative divergence $D(q\|p) = \sum_j q_j \log(q_j/p_j)$ is nonnegative and vanishes exactly when $q = p$. This property follows from concavity of the logarithm, with the convention $0 \log 0 = 0$. On admissible pairs,

$$\log P_{ij} = \log r_j - \log r_i - \log \varphi.$$

Stationarity cancels the expectation of $\log r_{X_1} - \log r_{X_0}$. Hence

$$h(\nu) = H_\nu(X_1 \mid \mathcal{P}) = \log\varphi - \mathbb{E}_\nu D\big(q(\cdot \mid \mathcal{P}) \| P_{X_0,\cdot}\big) \le \log\varphi. \tag{73.69}$$

The identity between entropy rate and entropy conditional on the past follows from the chain rule on finite blocks and the decreasing limit of their conditional entropies; the finite alphabet ensures that all of them lie between zero and $\log 2$.

The constructed measure μ is Markov with transition matrix P, so the divergence term vanishes and $h(\mu) = \log\varphi$. If another stationary measure attains that rate, the divergence vanishes almost surely: its conditional distribution of the future depends only on X_0 and agrees with P. It is therefore the same stationary chain, whose unique initial distribution is π_{av}. This proves uniqueness. □

Formula (73.69) specifies exactly what is maximized: entropy per transition of the stated symbolic envelope. The ratio $\mu_{\mathrm{ar}}/\mu_{\mathrm{vol}} = \varphi^{-2}$ arises from its stationary weights. The periodic chronological orbit has zero entropy rate and retains its own temporal distribution; both objects remain defined on their respective domains.

Chapter 74

1st areal modular law and finite extension through relative entropy

The first modular law relates variations of entropy and area within the declared finite system. Its nonlinear extension retains the positive relative-entropy defect and therefore does not conflate tangent equality with global identity.

74.1 1st modular law for area variations

Let us define the two normalized areas

$$\mathcal{A}_m = \gamma_{\mathrm{HMT}} a_0 N_m, \qquad m \in \{90, 120\}. \tag{74.1}$$

Then

$$K^{(A)}_{\mathrm{HHM}} - cI = \beta_{90}\mathcal{A}_{90} + \beta_{120}\mathcal{A}_{120}, \qquad \beta_m = \frac{m\Delta_4^{\mathrm{mod}}}{\gamma_{\mathrm{HMT}} a_0}. \tag{74.2}$$

The terminal values are

$$\beta_{90} = 5.97264854010709299146517988889\ldots, \tag{74.3}$$

$$\beta_{120} = 7.96353138680945732195357318 52\ldots, \qquad \frac{\beta_{120}}{\beta_{90}} = \frac{4}{3}. \tag{74.4}$$

Theorem 74.1 (First modular law in areal variables)**.** *For a differentiable family of normalized states $\rho(\varepsilon)$ passing through ρ_A, while keeping $\Delta_4^{\mathrm{mod}}, \gamma_{\mathrm{HMT}}, a_0$ fixed,*

$$\boxed{\delta S = \beta_{90}\,\delta\langle\mathcal{A}_{90}\rangle + \beta_{120}\,\delta\langle\mathcal{A}_{120}\rangle.} \tag{74.5}$$

Proof. The first entanglement law in the reference state is $\delta S = \delta\langle -\log\rho_A\rangle$. The scalar term cI does not contribute because $\delta\,\mathrm{Tr}\,\rho = 0$. Substituting equation (74.2) proves the identity. □

The formula is local in state space. For finite changes, relative entropy appears; omitting it would turn a tangent identity into a false claim of global linearity.

Theorem 74.2 (Finite Identity with a Defect Term)**.** *For every state σ whose support is contained in that of ρ_A,*

$$\boxed{S(\sigma) - S(\rho_A) = \beta_{90}\Delta_\sigma\langle\mathcal{A}_{90}\rangle + \beta_{120}\Delta_\sigma\langle\mathcal{A}_{120}\rangle - D(\sigma\|\rho_A),} \tag{74.6}$$

where $D(\sigma\|\rho_A) = \operatorname{Tr}\sigma(\log\sigma - \log\rho_A) \geq 0$. In particular, the linear variation of theorem 74.1 is the tangent term of this identity.

Proof. By definition and by $K^{(A)}_{\mathrm{HHM}} = -\log\rho_A$,

$$D(\sigma\|\rho_A) = -S(\sigma) + \operatorname{Tr}(\sigma K^{(A)}_{\mathrm{HHM}}).$$

For $\sigma = \rho_A$, the left-hand side is zero and $S(\rho_A) = \operatorname{Tr}(\rho_A K^{(A)}_{\mathrm{HHM}})$. Subtracting the two identities gives

$$S(\sigma) - S(\rho_A) = \Delta_\sigma\langle K^{(A)}_{\mathrm{HHM}}\rangle - D(\sigma\|\rho_A).$$

The scalar cI cancels between normalized states; substituting equation (74.2) yields equation (74.6). The positivity of relative entropy is Klein's inequality. □

The correction $D(\sigma\|\rho_A)$ measures the informational distance between the actual state and the reference modular state. Thus, the first law loses no content upon leaving the infinitesimal regime: it becomes an exact identity with a controlled positive defect.

Inductive persistence of modular weights and factorial classification of the limit. The inductive persistence of the modular weights leads to infinite operator closure and its factorial classification.

74.1.1 Equation of state for sectorial information

The incidence already constructed provides eighteen tritic factors in each sector. This multiplicity allows the relation between area, the occupation distribution, and the entropy of the preceding purification to be written explicitly. We retain the reduced state defined there and abbreviate $\Delta = \Delta_{\mathrm{mod}}$. For $m \in \{90, 120\}$, put

$$x_m = e^{-m\Delta}, \qquad z_m = 1 + x_m + x_m^2, \qquad f_m = \frac{x_m + 2x_m^2}{z_m}, \qquad v_m = \frac{x_m + 4x_m^2}{z_m} - f_m^2. \tag{74.7}$$

Here f_m and v_m are, respectively, the mean and variance of the occupation of one factor in sector m. The parameter Δ preserves the status and normalization of the previously specified reduced state.

74.1.1.1 Bipartite quantum purification of the incidence sectors

The preceding Schmidt decomposition is made concrete on each link. For a finite real weight w, define

$$z(w) = 1 + e^{-w} + e^{-2w}, \qquad \rho(w) = \frac{1}{z(w)}\sum_{n=0}^{2} e^{-wn}|n\rangle\langle n|.$$

The sectorial weights are $w_m = m\Delta$, with $m = 90, 120$; hence $z(w_m) = z_m$.

Proposition 74.3 (Thermofield-double purification of sectorial incidence)**.** *The normalized vector*

$$|\operatorname{TFD}(w)\rangle = \frac{1}{\sqrt{z(w)}}\sum_{n=0}^{2} e^{-wn/2}|n\rangle_A \otimes |n\rangle_{\bar{A}} \tag{74.8}$$

purifies $\rho(w)$. The product

$$|\Psi_\Delta\rangle = |\operatorname{TFD}(90\Delta)\rangle^{\otimes 18} \otimes |\operatorname{TFD}(120\Delta)\rangle^{\otimes 18} \tag{74.9}$$

has reduction $\rho_A(\Delta)$ on the thirty-six original links. Its entanglement entropy for the bipartition $A \mid \bar{A}$ is $s(\Delta)$.

Proof. The squared norm is $z(w)^{-1}\sum_{n=0}^{2} e^{-wn} = 1$. In the partial trace, orthogonality of the second factor eliminates terms with distinct indices; the remaining terms form $\rho(w)$. The partial trace commutes with the tensor product. One thus obtains $\rho(90\Delta)^{\otimes 18} \otimes \rho(120\Delta)^{\otimes 18}$, which coincides with the normalized state defined previously. Equality between the entropy of a reduction and the entanglement of this purification is read directly from its Schmidt coefficients. □

The two factors of the bipartition are determined within the construction. The physical realization of the second factor preserves that subsystem identification and its algebra of observables.

74.1.1.2 Occupation moments and information content

Proposition 74.4 (Mean area and entropy of the sectorial purification)**.** *In this finite family of states,*

$$\langle N\rangle_\Delta = 18(f_{90} + f_{120}), \tag{74.10}$$

$$\langle D\rangle_\Delta = 18(90 f_{90} + 120 f_{120}), \tag{74.11}$$

$$\frac{\langle \widehat{\mathcal{A}}\rangle_\Delta}{\ell_P^2} = 18\gamma_{\mathrm{HMT}} a_0 (f_{90} + f_{120}), \tag{74.12}$$

$$s(\Delta) = 18\log z_{90} + 18\log z_{120} + \Delta\langle D\rangle_\Delta. \tag{74.13}$$

The dimensional entropy is $S(\Delta) = k_B s(\Delta)$, with the thermal transducer of Section 48.5.

Proof. In each factor, the normalized weights are $(1, x_m, x_m^2)/z_m$. Their first moment gives f_m, and their second moment gives the first fraction in v_m. Tensor factorization and additivity of the number operators produce the first three identities. Finally, $\log \rho_A = -(\log Z)I - \Delta D$, with $Z = z_{90}^{18} z_{120}^{18}$. Taking the trace against $-\rho_A$ gives (74.13). The purification specified previously has this same entanglement entropy. □

Loss of uniformity has its own quantitative reading. In dimension $d_\partial = 3^{36}$, let $\tau_\partial = I/d_\partial$. Define

$$I_{\mathrm{or}}(\Delta) = 36\log 3 - s(\Delta) = D\big(\rho_A(\Delta)\|\tau_\partial\big) \geq 0. \tag{74.14}$$

The equality is obtained by substituting $\log \tau_\partial = -36\log 3\, I$. If p_ν are the eigenvalues of ρ_A, the convexity of $t\log t$ gives $\sum_\nu p_\nu \log(d_\partial p_\nu) \geq 0$, with equality exactly for the uniform state. For this family, this occurs at $\Delta = 0$. The deficit is dimensionless information; the areal quantum $\gamma_{\mathrm{HMT}} a_0$ is another reading. Each expression retains its definition even though their evaluations may be close.

Proposition 74.5 (Information response and boundary fluctuations)**.** *For every finite real Δ,*

$$\frac{\mathrm{d}\langle D\rangle_\Delta}{\mathrm{d}\Delta} = -\operatorname{Var}_\Delta(D) = -18(90^2 v_{90} + 120^2 v_{120}), \tag{74.15}$$

$$\frac{\mathrm{d}s}{\mathrm{d}\Delta} = -\Delta \operatorname{Var}_\Delta(D), \tag{74.16}$$

$$\frac{\mathrm{d}\langle \widehat{\mathcal{A}}\rangle_\Delta}{\mathrm{d}\Delta} = -18\gamma_{\mathrm{HMT}} a_0 \ell_P^2 (90 v_{90} + 120 v_{120}). \tag{74.17}$$

Proof. The sum defining Z is finite. Differentiating it gives $\partial_\Delta \log Z = -\langle D\rangle_\Delta$ and $\partial_\Delta^2 \log Z = \operatorname{Var}_\Delta(D)$. The sectors and their factors are independent in the product state; their variances therefore add. On differentiating (74.13), the terms $-\langle D\rangle_\Delta$ and $\langle D\rangle_\Delta$ cancel. For area, use $\partial_\Delta f_m = -m v_m$ in (74.12). □

These equations relate three levels of the same construction. Incidence fixes multiplicities and sectorial weights; the reduced state fixes their probabilities; fluctuations of those occupations determine the response

of area and entropy. The scale $\gamma_{\mathrm{HMT}} a_0 \ell_P^2$ converts occupation into area, whereas k_B converts information into dimensional entropy. In the gravitational section, $\ell_P^2 = G\hbar/c^3$ makes explicit the dependence of that areal unit on action and gravitation. Comparison with a horizon additionally preserves the geometric map and the state specified there.

74.1.2 Occupation inversion and entropic invariants

Let $\mathcal{R}$ be the unitary involution that exchanges $|0\rangle$ and $|2\rangle$ and fixes $|1\rangle$ in each of the thirty-six tritic factors. Its action on occupations preserves the entire microscopic configuration through a bijective change of its labels.

Theorem 74.6 (Preserved entropy and complementary area)**.** *The preceding involution satisfies*

$$\mathcal{R} N \mathcal{R}^* = 72I - N, \qquad \mathcal{R} D \mathcal{R}^* = 7560I - D, \tag{74.18}$$

$$\rho_A(-\Delta) = \mathcal{R}\rho_A(\Delta)\mathcal{R}^*, \qquad s(-\Delta) = s(\Delta), \tag{74.19}$$

$$\langle \widehat{\mathcal{A}} \rangle_{-\Delta} + \langle \widehat{\mathcal{A}} \rangle_{\Delta} = 72\gamma_{\mathrm{HMT}} a_0 \ell_P^2. \tag{74.20}$$

The relative entropy between the two orientations is

$$D\big(\rho_A(\Delta) \| \rho_A(-\Delta)\big) = \Delta\big(7560 - 2\langle D \rangle_{\Delta}\big). \tag{74.21}$$

Proof. Each local occupation n transforms into $2-n$. Thus, $N_m \mapsto 36I - N_m$ and $D \mapsto 36(90{+}120)I - D = 7560I - D$. Furthermore, $z_m(-\Delta) = e^{2m\Delta} z_m(\Delta)$, whence $Z(-\Delta) = e^{7560\Delta} Z(\Delta)$. Substituting both identities into the normalized state proves unitary conjugation. Spectral invariance proves equality of entropies; transformation of N proves the area sum. Finally, subtracting the two state logarithms and taking their expectation in $\rho_A(\Delta)$ yields (74.21). □

At $\Delta = 0$ the reduced state is uniform and $s(0) = 36 \log 3$. In the limit $\Delta \to +\infty$, the zero-occupation configuration survives; the entropy tends to zero. Inversion takes that limit to the maximum-occupation configuration. At both extremes the entropy is the same and the area distinguishes the configurations. For $\Delta > 0$, the response (74.16) is strictly negative because the state has full rank and D has more than one eigenvalue. The law thus preserves the difference between reversibility of the complete configuration, equality of entropies and change in the areal reading. These properties belong to the specified sectoral family; the physical transport of the bipartition is addressed in section 83.2.

74.1.3 First modular law and finite area variations

The derivatives of (74.16) vary the parameter Δ of a family of states. The modular first law, by contrast, fixes a reference state and considers variations of the observed state around it. Take $\rho_0 = \rho_A(\Delta_0)$, with Δ_0 real and finite, and keep $\Delta_0, \gamma_{\mathrm{HMT}}, a_0, \ell_P$ fixed. The branch $\gamma_{\mathrm{HMT}} > 0, a_0 > 0$ is retained. The normalized sectoral areas are

$$\mathcal{A}_m^{\mathrm{n}} = \gamma_{\mathrm{HMT}} a_0 N_m, \qquad \widehat{\mathcal{A}}_m = \ell_P^2 \mathcal{A}_m^{\mathrm{n}}, \qquad m \in \{90, 120\}. \tag{74.22}$$

The logarithm of the reference state takes the form

$$K_0 = -\log \rho_0 = c_0 I + \beta_{90} \mathcal{A}_{90}^{\mathrm{n}} + \beta_{120} \mathcal{A}_{120}^{\mathrm{n}}, \qquad \beta_m = \frac{m\Delta_0}{\gamma_{\mathrm{HMT}} a_0}, \quad c_0 = \log Z(\Delta_0). \tag{74.23}$$

In particular, $\beta_{120} = \frac{4}{3}\beta_{90}$; for $\Delta_0 \neq 0$, their quotient is $4/3$. The coefficients arise from expressing the same modular operator in areal units.

Theorem 74.7 (First modular law in areal coordinates)**.** *Let $\sigma(\varepsilon)$ be a differentiable family of normalized states on an interval containing zero, with $\sigma(0) = \rho_0$. Then*

$$\left.\frac{\mathrm{d}s(\sigma(\varepsilon))}{\mathrm{d}\varepsilon}\right|_0 = \sum_{m=90,120} \beta_m \left.\frac{\mathrm{d}\langle \mathcal{A}^{\mathrm{n}}_m\rangle_{\sigma(\varepsilon)}}{\mathrm{d}\varepsilon}\right|_0 . \tag{74.24}$$

Proof. The state ρ_0 has full rank. In its neighborhood, the derivative of the trace of $-\sigma\log\sigma$ is $-\operatorname{Tr}[\dot\sigma(\log\rho_0 + I)]$; this follows by diagonalizing ρ_0 in the formula for the spectral derivative of the trace. Normalization implies $\operatorname{Tr}\dot\sigma = 0$, so the derivative is $\operatorname{Tr}(\dot\sigma K_0)$. Substituting (74.23) also cancels $c_0 I$ and produces the identity. □

Theorem 74.8 (Finite area identity and relative-entropy deficit)**.** *For any normalized state σ on the same finite space, define $\Delta_\sigma\langle B\rangle = \operatorname{Tr}[(\sigma-\rho_0)B]$. The identity*

$$\boxed{s(\sigma) - s(\rho_0) = \sum_{m=90,120} \beta_m \Delta_\sigma\langle \mathcal{A}^{\mathrm{n}}_m\rangle - D(\sigma\|\rho_0).} \tag{74.25}$$

holds. Here $D(\sigma\|\rho_0) = \operatorname{Tr}[\sigma(\log\sigma - \log\rho_0)]$ is finite, nonnegative, and interpreted with $0\log 0 = 0$.

Proof. By definition, $D(\sigma\|\rho_0) = -s(\sigma) + \operatorname{Tr}(\sigma K_0)$, whereas $s(\rho_0) = \operatorname{Tr}(\rho_0 K_0)$. Subtraction gives $s(\sigma) - s(\rho_0) = \Delta_\sigma\langle K_0\rangle - D(\sigma\|\rho_0)$. Substituting (74.23) proves the formula because the traces of both states are one.

To see positivity, diagonalize σ with eigenvalues p_i and vectors u_i, and ρ_0 with eigenvalues $q_j > 0$ and vectors v_j. Put $r_i = \langle u_i, \rho_0 u_i\rangle$. Concavity of the logarithm gives $\sum_j |\langle u_i, v_j\rangle|^2 \log q_j \le \log r_i$. Hence $D(\sigma\|\rho_0) \ge \sum_i p_i \log(p_i/r_i) \ge 0$; the last inequality is convexity of $t\log t$ applied with weights r_i, which sum to one. Strict positivity of the q_j ensures that all reference logarithms are finite. □

The finite identity preserves the informational distance from the reference state; its tangent term is (74.24). For the physical areas, the corresponding coefficients are β_m/ℓ_P^2. Multiplying the equation by the common transducer k_B publishes the dimensional entropy. Thus, the relation between geometry and information remains exact also for finite variations, with its explicit correction and without modifying the parameters of the reference state.

74.1.4 Equation of state for sectorial information

The incidence already constructed provides eighteen tritic factors in each sector. This multiplicity allows the relation between area, the occupation distribution, and the entropy of the preceding purification to be written explicitly. We retain the reduced state defined there and abbreviate $\Delta = \Delta_{\mathrm{mod}}$. For $m \in \{90, 120\}$, put

$$x_m = e^{-m\Delta}, \qquad z_m = 1 + x_m + x_m^2, \qquad f_m = \frac{x_m + 2x_m^2}{z_m}, \qquad v_m = \frac{x_m + 4x_m^2}{z_m} - f_m^2. \tag{74.26}$$

Here f_m and v_m are, respectively, the mean and variance of the occupation of one factor in sector m. The parameter Δ preserves the status and normalization of the previously specified reduced state.

74.1.4.1 Bipartite quantum purification of the incidence sectors

The preceding Schmidt decomposition is made concrete on each link. For a finite real weight w, define

$$z(w) = 1 + e^{-w} + e^{-2w}, \qquad \rho(w) = \frac{1}{z(w)} \sum_{n=0}^{2} e^{-wn} |n\rangle\langle n|.$$

The sectorial weights are $w_m = m\Delta$, with $m = 90, 120$; hence $z(w_m) = z_m$.

Proposition 74.9 (Thermofield-double purification of sectorial incidence)**.** *The normalized vector*

$$|\operatorname{TFD}(w)\rangle = \frac{1}{\sqrt{z(w)}} \sum_{n=0}^{2} e^{-wn/2} |n\rangle_A \otimes |n\rangle_{\bar{A}} \tag{74.27}$$

purifies $\rho(w)$*. The product*

$$|\Psi_\Delta\rangle = |\operatorname{TFD}(90\Delta)\rangle^{\otimes 18} \otimes |\operatorname{TFD}(120\Delta)\rangle^{\otimes 18} \tag{74.28}$$

has reduction $\rho_A(\Delta)$ *on the thirty-six original links. Its entanglement entropy for the bipartition* $A \mid \bar{A}$ *is* $s(\Delta)$*.*

Proof. The squared norm is $z(w)^{-1} \sum_{n=0}^{2} e^{-wn} = 1$. In the partial trace, orthogonality of the second factor eliminates terms with distinct indices; the remaining terms form $\rho(w)$. The partial trace commutes with the tensor product. One thus obtains $\rho(90\Delta)^{\otimes 18} \otimes \rho(120\Delta)^{\otimes 18}$, which coincides with the normalized state defined previously. Equality between the entropy of a reduction and the entanglement of this purification is read directly from its Schmidt coefficients. □

The two factors of the bipartition are determined within the construction. The physical realization of the second factor preserves that subsystem identification and its algebra of observables.

74.1.4.2 Occupation moments and information content

Proposition 74.10 (Mean area and entropy of the sectorial purification)**.** *In this finite family of states,*

$$\langle N \rangle_\Delta = 18(f_{90} + f_{120}), \tag{74.29}$$

$$\langle D \rangle_\Delta = 18(90 f_{90} + 120 f_{120}), \tag{74.30}$$

$$\frac{\langle \widehat{\mathcal{A}} \rangle_\Delta}{\ell_P^2} = 18\gamma_{\mathrm{HMT}} a_0 (f_{90} + f_{120}), \tag{74.31}$$

$$s(\Delta) = 18 \log z_{90} + 18 \log z_{120} + \Delta \langle D \rangle_\Delta. \tag{74.32}$$

The dimensional entropy is $S(\Delta) = k_B s(\Delta)$*, with the thermal transducer of Section 123.*

Proof. In each factor, the normalized weights are $(1, x_m, x_m^2)/z_m$. Their first moment gives f_m, and their second moment gives the first fraction in v_m. Tensor factorization and additivity of the number operators produce the first three identities. Finally, $\log \rho_A = -(\log Z) I - \Delta D$, with $Z = z_{90}^{18} z_{120}^{18}$. Taking the trace against $-\rho_A$ gives (74.32). The purification specified previously has this same entanglement entropy. □

Loss of uniformity has its own quantitative reading. In dimension $d_\partial = 3^{36}$, let $\tau_\partial = I/d_\partial$. Define

$$I_{\mathrm{or}}(\Delta) = 36 \log 3 - s(\Delta) = D\big(\rho_A(\Delta) \| \tau_\partial\big) \geq 0. \tag{74.33}$$

The equality is obtained by substituting $\log \tau_\partial = -36 \log 3\, I$. If p_ν are the eigenvalues of ρ_A, the convexity of $t \log t$ gives $\sum_\nu p_\nu \log(d_\partial p_\nu) \geq 0$, with equality exactly for the uniform state. For this family, this occurs at $\Delta = 0$. The deficit is dimensionless information; the areal quantum $\gamma_{\mathrm{HMT}} a_0$ is another reading. Each expression retains its definition even though their evaluations may be close.

Proposition 74.11 (Information response and boundary fluctuations)**.** *For every finite real* Δ*,*

$$\frac{\mathrm{d}\langle D \rangle_\Delta}{\mathrm{d}\Delta} = -\operatorname{Var}_\Delta(D) = -18(90^2 v_{90} + 120^2 v_{120}), \tag{74.34}$$

$$\frac{\mathrm{d}s}{\mathrm{d}\Delta} = -\Delta \operatorname{Var}_\Delta(D), \tag{74.35}$$

$$\frac{\mathrm{d}\langle\widehat{\mathcal{A}}\rangle_\Delta}{\mathrm{d}\Delta} = -18\gamma_{\mathrm{HMT}}a_0\ell_P^2(90v_{90} + 120v_{120}). \tag{74.36}$$

Proof. The sum defining Z is finite. Differentiating it gives $\partial_\Delta \log Z = -\langle D\rangle_\Delta$ and $\partial_\Delta^2 \log Z = \mathrm{Var}_\Delta(D)$. The sectors and their factors are independent in the product state; their variances therefore add. On differentiating (74.32), the terms $-\langle D\rangle_\Delta$ and $\langle D\rangle_\Delta$ cancel. For area, use $\partial_\Delta f_m = -mv_m$ in (74.31). □

These equations relate three levels of the same construction. Incidence fixes multiplicities and sectoral weights; the reduced state fixes their probabilities; fluctuations of those occupations determine the response of area and entropy. The scale $\gamma_{\mathrm{HMT}}a_0\ell_P^2$ converts occupation into area, while k_B converts information into dimensional entropy. In section 121, $\ell_P^2 = G\hbar/c^3$ makes explicit the dependence of that areal unit on action and gravitation. The family of reduced states and its bipartition remain specified in this development; any horizon realization must also transport those two data.

74.1.5 Occupation inversion and entropic invariants

Let $\mathcal{R}$ be the unitary involution that exchanges $|0\rangle$ and $|2\rangle$ and fixes $|1\rangle$ in each of the thirty-six tritic factors. Its action on occupations preserves the entire microscopic configuration through a bijective change of its labels.

Theorem 74.12 (Preserved entropy and complementary area)**.** *The preceding involution satisfies*

$$\mathcal{R}N\mathcal{R}^* = 72I - N, \qquad \mathcal{R}D\mathcal{R}^* = 7560I - D, \tag{74.37}$$

$$\rho_A(-\Delta) = \mathcal{R}\rho_A(\Delta)\mathcal{R}^*, \qquad s(-\Delta) = s(\Delta), \tag{74.38}$$

$$\langle\widehat{\mathcal{A}}\rangle_{-\Delta} + \langle\widehat{\mathcal{A}}\rangle_\Delta = 72\gamma_{\mathrm{HMT}}a_0\ell_P^2. \tag{74.39}$$

The relative entropy between the two orientations is

$$D\big(\rho_A(\Delta)\|\rho_A(-\Delta)\big) = \Delta\big(7560 - 2\langle D\rangle_\Delta\big). \tag{74.40}$$

Proof. Each local occupation n transforms into $2-n$. Thus, $N_m \mapsto 36I - N_m$ and $D \mapsto 36(90{+}120)I - D = 7560I - D$. Furthermore, $z_m(-\Delta) = e^{2m\Delta}z_m(\Delta)$, whence $Z(-\Delta) = e^{7560\Delta}Z(\Delta)$. Substituting both identities into the normalized state proves unitary conjugation. Spectral invariance proves equality of entropies; transformation of N proves the area sum. Finally, subtracting the two state logarithms and taking their expectation in $\rho_A(\Delta)$ yields (74.40). □

At $\Delta = 0$ the reduced state is uniform and $s(0) = 36\log 3$. In the limit $\Delta \to +\infty$, the zero-occupation configuration survives; the entropy tends to zero. Inversion takes that limit to the maximum-occupation configuration. At both extremes the entropy is the same and the area distinguishes the configurations. For $\Delta > 0$, the response (74.35) is strictly negative because the state has full rank and D has more than one eigenvalue. The law thus preserves the difference between reversibility of the complete configuration, equality of entropies and change in the areal reading. These properties belong to the specified sectoral family; the physical transport of the bipartition is addressed in section 74.1.7.

74.1.6 First modular law and finite area variations

The derivatives of (74.35) vary the parameter Δ of a family of states. The modular first law, by contrast, fixes a reference state and considers variations of the observed state around it. Take $\rho_0 = \rho_A(\Delta_0)$, with Δ_0 real and finite, and keep $\Delta_0, \gamma_{\mathrm{HMT}}, a_0, \ell_P$ fixed. The branch $\gamma_{\mathrm{HMT}} > 0, a_0 > 0$ is retained. The normalized sectoral areas are

$$\mathcal{A}_m^{\mathrm{n}} = \gamma_{\mathrm{HMT}}a_0N_m, \qquad \widehat{\mathcal{A}}_m = \ell_P^2\mathcal{A}_m^{\mathrm{n}}, \qquad m \in \{90, 120\}. \tag{74.41}$$

The logarithm of the reference state takes the form

$$K_0 = -\log \rho_0 = c_0 I + \beta_{90}\mathcal{A}^{\mathrm{n}}_{90} + \beta_{120}\mathcal{A}^{\mathrm{n}}_{120}, \qquad \beta_m = \frac{m\Delta_0}{\gamma_{\mathrm{HMT}} a_0}, \quad c_0 = \log Z(\Delta_0). \tag{74.42}$$

In particular, $\beta_{120} = \frac{4}{3}\beta_{90}$; for $\Delta_0 \neq 0$, their quotient is $4/3$. The coefficients arise from expressing the same modular operator in areal units.

Theorem 74.13 (First modular law in areal coordinates)**.** *Let $\sigma(\varepsilon)$ be a differentiable family of normalized states on an interval containing zero, with $\sigma(0) = \rho_0$. Then*

$$\left.\frac{\mathrm{d}s(\sigma(\varepsilon))}{\mathrm{d}\varepsilon}\right|_0 = \sum_{m=90,120} \beta_m \left.\frac{\mathrm{d}\langle \mathcal{A}^{\mathrm{n}}_m\rangle_{\sigma(\varepsilon)}}{\mathrm{d}\varepsilon}\right|_0 . \tag{74.43}$$

Proof. The state ρ_0 has full rank. In its neighborhood, the derivative of the trace of $-\sigma\log\sigma$ is $-\operatorname{Tr}[\dot\sigma(\log\rho_0 + I)]$; this follows by diagonalizing ρ_0 in the formula for the spectral derivative of the trace. Normalization implies $\operatorname{Tr}\dot\sigma = 0$, so the derivative is $\operatorname{Tr}(\dot\sigma K_0)$. Substituting (74.42) also cancels $c_0 I$ and produces the identity. □

Theorem 74.14 (Finite area identity and relative-entropy deficit)**.** *For any normalized state σ on the same finite space, define $\Delta_\sigma\langle B\rangle = \operatorname{Tr}[(\sigma - \rho_0)B]$. The identity*

$$\boxed{s(\sigma) - s(\rho_0) = \sum_{m=90,120} \beta_m \Delta_\sigma\langle \mathcal{A}^{\mathrm{n}}_m\rangle - D(\sigma\|\rho_0).} \tag{74.44}$$

holds. Here $D(\sigma\|\rho_0) = \operatorname{Tr}[\sigma(\log\sigma - \log\rho_0)]$ is finite, nonnegative, and interpreted with $0\log 0 = 0$.

Proof. By definition, $D(\sigma\|\rho_0) = -s(\sigma) + \operatorname{Tr}(\sigma K_0)$, whereas $s(\rho_0) = \operatorname{Tr}(\rho_0 K_0)$. Subtraction gives $s(\sigma) - s(\rho_0) = \Delta_\sigma\langle K_0\rangle - D(\sigma\|\rho_0)$. Substituting (74.42) proves the formula because the traces of both states are one.

To see positivity, diagonalize σ with eigenvalues p_i and vectors u_i, and ρ_0 with eigenvalues $q_j > 0$ and vectors v_j. Put $r_i = \langle u_i, \rho_0 u_i\rangle$. Concavity of the logarithm gives $\sum_j |\langle u_i, v_j\rangle|^2 \log q_j \le \log r_i$. Hence $D(\sigma\|\rho_0) \ge \sum_i p_i \log(p_i/r_i) \ge 0$; the last inequality is convexity of $t\log t$ applied with weights r_i, which sum to one. Strict positivity of the q_j ensures that all reference logarithms are finite. □

The finite identity preserves the informational distance from the reference state; its tangent term is (74.43). For the physical areas, the corresponding coefficients are β_m/ℓ_P^2. Multiplying the equation by the common transducer k_B publishes the dimensional entropy. Thus, the relation between geometry and information remains exact also for finite variations, with its explicit correction and without modifying the parameters of the reference state.

74.1.7 Spectral entropy and preservation under transport

The entropy of a reduced state depends on its spectrum and on the bipartition defining the reduction. To specify the continuation towards duality, let $\mathcal{H}_A, \mathcal{H}_B$ be the two factors and ρ_{AB} a state with finite entropies. A factorized transport $U_A \otimes U_B$ produces

$$\rho'_A = U_A \rho_A U_A^*, \qquad S(\rho'_A) = S(\rho_A).$$

The first identity follows from the definition of partial trace and the second from the invariance of the spectrum. For a general unitary transport, the equivalent formulation also transports the inclusion of the subsystem's observable algebra. This specification identifies the geometric datum that must accompany an assertion of entanglement preservation.

74.1.8 Quasi-free correlation and spectral occupation entropies

For a gauge-invariant, number-conserving quasifree fermionic state defined on a finite-dimensional one-particle space, let $C_{ij} = \langle a_j^\dagger a_i \rangle$ be the one-particle correlation operator. The gauge-invariant hypothesis excludes anomalous pairing correlations, so that C determines the quasifree state. If $0 < C < I$, its one-particle modular generator and the reciprocal reconstruction of the correlator are

$$\boxed{K_1 = \log\big((I - C)C^{-1}\big), \qquad C = (I + e^{K_1})^{-1}.} \tag{74.45}$$

Corollary 74.15 (Logistic form of the modular spectrum)**.** *If $K_1 u_i = \kappa_i u_i$, the occupation of mode u_i is*

$$\langle n_i \rangle = \frac{1}{1 + e^{\kappa_i}}.$$

For $\kappa_i = \beta(\varepsilon_i - \mu)$, the Fermi–Dirac law is recovered.

Proof. The spectral diagonalization of K_1 diagonalizes every function thereof. Applying $x \mapsto (1 + e^x)^{-1}$ to each eigenvalue yields the formula. □

If C has eigenvalues $p_i \in [0, 1]$, the occupation entropy of the quasifree fermionic state is, with the convention $0 \log 0 = 0$,

$$S_F = -\sum_i \big[p_i \log p_i + (1 - p_i) \log(1 - p_i)\big]. \tag{74.46}$$

For a diagonal Gaussian bosonic state with finitely many modes and mean occupancies $\overline{n}_i$,

$$S_B = \sum_i \big[(1 + \overline{n}_i) \log(1 + \overline{n}_i) - \overline{n}_i \log \overline{n}_i\big]. \tag{74.47}$$

The same formula extends to a countable family when the series on the right-hand side converges. These entropies quantify occupation configurations. Entanglement entropy has another domain: the reduced state induced by a tensor decomposition. Identifying the two requires specifying the reduction and the correlation operator that implements it.

Finally, for a reduced state ρ_A, the *von Neumann entropy*

$$S_{\mathrm{vN}}(\rho_A) = -\operatorname{Tr}(\rho_A \log \rho_A)$$

is a spectral property of the density operator. When ρ_A arises from a global purification, it quantifies entanglement across the cut. In the HMT boundary construction, the area operator, modular Hamiltonian, and corresponding reduction are constructed in Section 122. The product purification of Proposition 74.9 identifies the bipartition, and identity (74.44) expresses its relation to area variations.

The maps between these domains are published through explicit identities. Decomposition of a measure along the fibres of q satisfies the chain rule

$$\mathsf{H}_{\mu}(\mathbf{X}) = \mathsf{H}_{q_*\mu}(q(\mathbf{X})) + \mathsf{H}_{\mathrm{fib}}(q;\mu). \tag{74.48}$$

A finite distribution $p = (p_i)$ is realized diagonally as

$$\rho_p = \sum_i p_i |i\rangle\langle i|, \qquad S_{\mathrm{vN}}(\rho_p) = \mathsf{H}(p).$$

If $P_{N,E}$ projects onto the microcanonical subspace of dimension $\Omega(N,E)$, then

$$\rho_{\mathrm{mc}} = \frac{P_{N,E}}{\Omega(N,E)}, \qquad S_{\mathrm{vN}}(\rho_{\mathrm{mc}}) = \log \Omega(N,E) = S_{\mathrm{mc}}.$$

For a pure bipartite state,

$$\rho_A = \mathrm{Tr}_B\big(|\Psi_{AB}\rangle\langle\Psi_{AB}|\big)$$

defines the reduction that measures entanglement. Finally, equality between $h_{\mathrm{top}} = \log \rho(A)$ and a metric entropy is attained, for this primitive system, by the Parry measure, which realizes the maximum metric entropy; it is not attributed to an arbitrary measure on histories. The matrix F_{av}, its one-step-memory envelope, and that measure are constructed in Theorems 73.15 and 73.16, with its explicit variational characterization in Proposition 73.17.

74.2 Thermal radiation and the information scale of the calendar

Occupation statistics provide the link between the modes of a realization and their mean energy. In the electromagnetic realization, the dispersion relation and polarization multiplicity additionally determine the spectral density. Their composition yields the thermal moments and spectral maxima, while keeping three operations distinct: production of the action scale, representation of the modes, and evaluation of the thermal state.

74.2.1 Constitutive speed and kinematic realization of the clock

Radiative propagation uses the velocity section produced by the angular transport of the same HMT state. The identification is established before choosing publication units. Let $x = A_{\mathrm{rad}}$ and $y = C^*_{\mathrm{rad}}$ be the angular coordinates generated in section 14.11.6, with the channels (14.106), and let R be the self-adjoint involution of the two sheets, with rank-one projectors P_+ and P_-. On the domain $x > |y|$, the transport and its channels are

$$T = \exp(-xI - y\mathsf{R}) = q_+P_+ + q_-P_-, \qquad q_+ = e^{-x-y}, \quad q_- = e^{-x+y}, \quad 0 < q_\pm < 1. \tag{74.49}$$

The response of the temporal blocks of lengths 90 and 120 is

$$\mathsf{V} = (I - T^{90})(I - T^{120})^{-1} = r_+P_+ + r_-P_-, \qquad r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}}. \tag{74.50}$$

The construction starts from the inherited channels; the symbols x, y and $q_\pm$ denote their coordinates, not parameters fitted to a target speed.

Proposition 74.16 (Identity of the constitutive and kinematic realizations)**.** *In the same dimensional chart, the clock section c_{int} and the constitutive-response section c_{vac} coincide:*

$$c_{\mathrm{int}} = c_{\mathrm{vac}} = \widehat{c}\, u_C, \qquad \widehat{c} = (r_+r_-)^{-1} = \exp(-\mathcal{E}), \tag{74.51}$$

where

$$\mathcal{E} = \log\big[(1-q_+^{90})(1-q_-^{90})\big] - \log\big[(1-q_+^{120})(1-q_-^{120})\big]. \tag{74.52}$$

The equality preserves sheet exchange.

Proof. The factors appearing in the logarithms are positive. Therefore,

$$\mathcal{E} = \log(r_+r_-) = \log\det \mathsf{V}, \qquad \exp(-\mathcal{E}) = (\det \mathsf{V})^{-1}.$$

The determinant is computed here in the two-sheet realization with rank-one projectors. The kinematic definition $c_{\mathrm{int}} = \exp(-\mathcal{E})u_C$ and the constitutive definition $c_{\mathrm{vac}} = (r_+r_-)^{-1}u_C$ therefore give the same section. Exchanging $P_+ \leftrightarrow P_-$ interchanges r_+ and r_- and preserves their product. Finally, $\det T = e^{-2x}$ corresponds to the elementary transport, whereas $\det \mathsf{V} = r_+r_-$ corresponds to its temporal response; the two determinants have distinct roles in this composition. □

74.2.1.1 Path length and elementary duration

The positive dimensional frame (u_S, u_C, u_T, u_Q) induces the length basis $u_L = u_C u_T$. For an enriched path γ of $n = |\gamma|$ transitions and a constant channel, the additive readers are

$$T_{\mathrm{int}}(\gamma) = nu_T, \qquad L_{\mathrm{int}}(\gamma) = n\widehat{c}\,u_L. \tag{74.53}$$

Their additivity concerns the count and realization of each transition: enriched routes additionally preserve orientation and memory.

Corollary 74.17 (Normalization of the elementary length)**.** *For $n > 0$, $L_{\mathrm{int}}(\gamma)/T_{\mathrm{int}}(\gamma) = c_{\mathrm{vac}}$. In the normalization $t_0 = u_T$,*

$$\ell_0 = c_{\mathrm{int}}t_0 = \widehat{c}\,u_L. \tag{74.54}$$

In a general temporal chart $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, the corresponding expression is $\ell_0 = \tau_0\widehat{c}\,u_L$.

Proof. Dividing the expressions in (74.53) cancels the integer n and uses $u_L/u_T = u_C$. The elementary formula is obtained by multiplying the speed section by t_0. □

Generalization to an edge-dependent channel preserves the additive formula $L_{\mathrm{int}}(\gamma) = \sum_{e\in\gamma}\widehat{c}(e)u_L$. Its quotient by nu_T is the mean of the speeds of those edges. The homogeneous radiative realization uses the constant channel specified in (74.53).

74.2.1.2 Change of chart and adapted normalization

Proposition 74.18 (Covariance of the identification)**.** *Let $u'_C = du_C$ and $u'_T = \eta u_T$, with $d, \eta > 0$. The same speed and the same elementary duration have coordinates*

$$\widehat{c}' = d^{-1}\widehat{c}, \qquad \tau'_0 = \eta^{-1}\tau_0, \qquad u'_L = d\eta u_L. \tag{74.55}$$

In particular, if $c_V = \widehat{c}_V u_C^V$ and $u_C^V = du_C^{III}$, equality of the sections is exactly equivalent to $d\widehat{c}_V = \widehat{c}_{III}$.

Proof. Express each section in the two bases. This gives $\widehat{c}'u'_C = \widehat{c}u_C$ and $\tau'_0u'_T = \tau_0u_T$. Their product satisfies

$$\tau'_0\widehat{c}'u'_L = \frac{\tau_0}{\eta}\frac{\widehat{c}}{d}d\eta u_L = \tau_0\widehat{c}u_L = \ell_0.$$

The final equivalence follows by substituting the relation between the bases. □

The adapted choice $d = \widehat{c}$ gives $u'_C = c_{\mathrm{vac}}$ and $\widehat{c}' = 1$. This choice expresses the already constructed section in a basis adapted to it. The spectral coefficient in the internal chart remains $(r_+r_-)^{-1}$; spectral

inversion uses that internal chart. For example, retaining the action and charge bases, the constitutive coordinates transform as

$$\widehat{\mu}' = d\widehat{\mu}, \qquad \widehat{\varepsilon}' = d\widehat{\varepsilon}, \qquad \widehat{\mu}'\widehat{\varepsilon}' = (\widehat{c}')^{-2}.$$

In the adapted chart, $\widehat{\mu}' = r_-/r_+$ and $\widehat{\varepsilon}' = r_+/r_-$, since the internal coordinates are $\widehat{\mu} = r_-^2$ and $\widehat{\varepsilon} = r_+^2$.

74.2.1.3 Transport to the radiative sector

The dispersion relation of the homogeneous radiative realization is written $\omega_{\boldsymbol{k}} = c_{\mathrm{vac}}|\boldsymbol{k}|$. With $h = 2\pi\hbar$ and $\beta = (k_B\Theta)^{-1}$, its spectral density and Stefan–Boltzmann coefficient have the expressions

$$u_\omega = \frac{\hbar\omega^3}{\pi^2(\widehat{c}u_C)^3}\frac{1}{e^{\beta\hbar\omega}-1}, \qquad \sigma = \frac{\pi^2 k_B^4}{60\hbar^3(\widehat{c}u_C)^2}. \tag{74.56}$$

These substitutions transport precisely the same speed section. The conditions of the radiative realization are retained: a free three-dimensional bosonic sector, two transverse polarizations, zero chemical potential, and positive temperature. Identity (74.51) establishes its constitutive antecedent; mode counting and the thermodynamic limit belong to the specific proofs for that realization.

Normalization under amplification. If refinement acts by $\mathsf{V}_m = \mathsf{V} \otimes I_{9^m}$, the normalized trace $\tau_m = \mathrm{Tr}/(2\cdot 9^m)$ satisfies

$$\exp[-2\tau_m(\log \mathsf{V}_m)] = \exp[-\log r_+ - \log r_-] = \widehat{c}.$$

Indeed, each eigenvalue appears 9^m times. The multiplicity cancels in the normalized trace. The ordinary amplified determinant is $(r_+r_-)^{9^m}$: the two-sheet normalization, or equivalently the normalized trace, preserves the constitutive reader under that refinement. This amplification of the representation preserves the enriched transport from which it arises.

74.2.2 Action, calendar, and temperature

The clock orientation a is retained and we abbreviate $\hbar = \hbar_a$, $c = c_{\mathrm{int}} = c_{\mathrm{vac}}$, with the identification of (74.51) in the same chart. The energy ε_{108} of this section is ϵ_a of (123.6); $\Theta_{108} = \Theta_{\mathrm{clk,a}}$. Dividing it by $\log 3$ produces the energy per nat of (123.7). The construction and covariance of the thermal pair are given in section 123.

We write $h = 2\pi\hbar > 0$ for the action publication, $t_0 > 0$ for the time unit of its dimensional chart and $\ell_0 = ct_0$ for the corresponding length. The chronological cell of length 108 determines the energy

$$\varepsilon_{108} = \frac{h}{108t_0}, \qquad k_B\Theta_{108}\log 3 = \varepsilon_{108}, \qquad \beta_{108} = \frac{108t_0\log 3}{h}. \tag{74.57}$$

Here $\beta = (k_B\Theta)^{-1}$ is the Gibbs parameter. The equality fixes the energy-valued product of temperature and the entropy transducer. The dimensional specification of k_B and Θ_{108} separately uses the thermal section and its bases. The natural logarithm measures information per tritic branch; its factor $\log 3$ converts the cardinality of the cell into dimensionless entropy.

Also introduce the associated units

$$t_P = \frac{108}{2\pi}t_0, \qquad \ell_P = ct_P, \qquad E_P = \frac{\hbar}{t_P} = \varepsilon_{108}. \tag{74.58}$$

These equalities specify the meaning of the symbols in this section; their evaluation in another chart preserves the identities between quotients. The length $\ell_P = (54/\pi)\ell_0$ coincides with the areal unit of section 121, in the circular realization declared there.

74.2.3 Mode density and spectral law

Proposition 74.19 (Thermal realization of transverse modes)**.** *Consider a free realization in a three-dimensional periodic box, whose excitation space contains the wave vectors* $\mathbf{k} \neq 0$*, with two transverse modes per vector, dispersion* $\omega = c|\mathbf{k}|$*, excitation energy* $\hbar\omega$*, zero chemical potential, and* $\beta > 0$*. The constant mode is kept fixed as background data. In the thermodynamic limit, the energy density per angular frequency is*

$$u_\omega(\Theta) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\beta\hbar\omega} - 1}, \qquad \omega > 0. \tag{74.59}$$

The energy is that of the excitations above the vacuum of this representation.

Proof. For a box of volume V, the wave vectors have density $V/(2\pi)^3$. In a radial shell of thickness dk, the density per unit volume, including the two polarizations, is

$$2\frac{4\pi k^2\, dk}{(2\pi)^3} = \frac{k^2}{\pi^2}\, dk = \frac{\omega^2}{\pi^2 c^3}\, d\omega.$$

This is the limiting density of the sums over the periodic lattice with the constant mode omitted. This choice defines the partition product before taking the limit: each excitation mode has positive energy. Excluding a single point preserves the limiting radial density. The previously obtained bosonic occupation is $\bar{n}_\omega = (e^{\beta\hbar\omega} - 1)^{-1}$. Multiplying the density by $\hbar\omega\bar{n}_\omega$ proves (74.59). Near zero, the energy integrand is of order ω^2, and at infinity it decays exponentially, so its integral converges. □

The density arises from the specified three-dimensional representation. The transverse multiplicity used here has that physical domain; the dimension of a harmonic space on the APP torus describes, separately, the cohomology of its cellular complex.

74.2.4 Thermodynamic limit of occupation sums

The preceding radial density can be obtained as an effective limit of the sums of the periodic realization. The estimate must preserve the fixed constant mode and uniformly control both the origin and the spectral tail. The action and calendar publications remain prior to this realization operation.

Lemma 74.20 (Convergence of the three thermal observables)**.** *Let* $a, b > 0$*,* $L > 0$*,* $\Delta = 2\pi/L$*, and*

$$F_N(r) = \frac{1}{e^{ar} - 1}, \qquad F_E(r) = \frac{br}{e^{ar} - 1}, \qquad F_Z(r) = -\log(1 - e^{-ar}) \quad (r > 0).$$

For each of these functions,

$$\lim_{L\to\infty} \frac{2}{L^3} \sum_{\nu\in\mathbb{Z}^3\setminus\{0\}} F\left(\frac{2\pi}{L}|\nu|\right) = \frac{1}{\pi^2}\int_0^\infty r^2 F(r)\, dr. \tag{74.60}$$

The sums and the integral are finite. The specialization $a = \beta\hbar c$*,* $b = \hbar c$ *gives, respectively, the number density, excitation-energy density, and logarithmic partition-function density of the transverse modes.*

Proof. All three functions are continuous and positive on $(0, \infty)$. There exist constants $C_0, C_1, d > 0$, depending on a, b but independent of Δ, such that

$$F(r) \le \frac{C_0}{r} \quad (0 < r \le 1), \qquad F(r) \le C_1 e^{-dr} \quad (r \ge 1). \tag{74.61}$$

Indeed, $e^{ar} - 1 \ge ar$ bounds F_N and F_E near the origin; moreover, $F_Z(r) = \log(1 + F_N(r)) \le F_N(r) \le 1/(ar)$. For $r \ge 1$, the inequality $-\log(1 - y) \le y/(1 - y)$, with $y = e^{-ar}$, gives the exponential bound

for F_Z. The other two follow from $(e^{ar}-1)^{-1} \le e^{-ar}/(1-e^{-a})$, absorbing the factor r in F_E into an exponential of smaller decay rate. These bounds already prove integrability of $r^2F(r)$.

For the discrete sums, group the lattice points by $m = \|\nu\|_\infty \ge 1$. Each shell contains exactly

$$(2m+1)^3 - (2m-1)^3 = 24m^2 + 2$$

points and satisfies $m \le |\nu| \le \sqrt{3}m$. Fix $0 < \delta \le 1$ and $0 < \Delta \le 1$. If $\delta < \Delta$, there are no points with $0 < \Delta|\nu| \le \delta$. Otherwise, setting $N = \lfloor \delta/\Delta \rfloor$, the first bound in (74.61) gives

$$\begin{aligned}\Delta^3 \sum_{0<\Delta|\nu|\le\delta} F(\Delta|\nu|) &\le C_0\Delta^2 \sum_{m=1}^{N} \frac{24m^2+2}{m} \\ &\le 26C_0\Delta^2 \sum_{m=1}^{N} m \le 26C_0\delta^2.\end{aligned}$$

The second line uses $m^{-1} \le m$ and then $N(N+1)/2 \le N^2$ for $N \ge 1$. The integral contribution from the same neighborhood is also of order δ^2, since $\int_0^\delta r^2F(r)\,dr \le C_0\delta^2/2$.

For $R \ge 1$, every point with $\Delta|\nu| > R$ belongs to a shell $m > R/(\sqrt{3}\Delta)$. The second bound in (74.61) gives

$$\begin{aligned}\Delta^3 \sum_{\Delta|\nu|>R} F(\Delta|\nu|) &\le C_1 e^{-dR/(2\sqrt{3})}\Delta^3 \sum_{m\ge1} (24m^2+2)e^{-d\Delta m/2} \\ &\le C_2 e^{-dR/(2\sqrt{3})},\end{aligned}$$

where C_2 is independent of $0 < \Delta \le 1$. To verify this uniformity, set $q = e^{-d\Delta/2}$ and use

$$\sum_{m\ge1} q^m = \frac{q}{1-q}, \qquad \sum_{m\ge1} m^2q^m = \frac{q(1+q)}{(1-q)^3}.$$

The first identity is the geometric series, and the second follows by applying $q\,d/dq$ twice, which is valid for $|q| < 1$. Moreover, $1 - e^{-d\Delta/2} \ge \Delta(1-e^{-d/2})$ by concavity. The factor Δ^3 therefore compensates for the denominators of both expressions. The integral tail likewise tends to zero by the exponential bound. For each $\Delta > 0$, the same estimates prove convergence of the infinite sum.

On the compact annulus $\delta \le |k| \le R$, the function $F(|k|)$ is continuous and bounded. Riemann sums of mesh Δ converge to its integral; the two boundary spheres have measure zero. First let $\Delta \to 0$, then $\delta \to 0$, and finally $R \to \infty$, using the preceding uniform bounds. The result is

$$\Delta^3 \sum_{\nu\ne0} F(\Delta|\nu|) \longrightarrow \int_{\mathbb{R}^3} F(|k|)\,d^3k = 4\pi \int_0^\infty r^2F(r)\,dr.$$

Since $2/L^3 = 2\Delta^3/(2\pi)^3$, the final radial coefficient is $2 \cdot 4\pi/(2\pi)^3 = 1/\pi^2$, proving (74.60). □

Control at the origin is carried out over positive-energy excitations. The constant mode remains fixed before the partition function is formed: removing it afterward from an integral would not replace that condition on the finite representation. The proof establishes the limit of the specified realization and preserves the distinction between that realization and the prior construction of its action and calendar parameters.

74.2.5 Thermal moments and information

Lemma 74.21 (Bosonic moment)**.** *For every real $s > 1$,*

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1}\, dx = \Gamma(s)\zeta(s). \tag{74.62}$$

Proof. For $x > 0$, $(e^x - 1)^{-1} = \sum_{m\geq 1} e^{-mx}$. All summands are nonnegative; Tonelli's theorem allows the sum and integral to be interchanged. The substitution $y = mx$ gives $\int_0^\infty x^{s-1}e^{-mx}\, dx = m^{-s}\Gamma(s)$. The sum converges for $s > 1$, yielding the stated equality. □

Proposition 74.22 (Number, energy, and entropy densities)**.** *In the realization of Proposition 74.19,*

$$n_\gamma = \frac{2\zeta(3)}{\pi^2 c^3 (\beta\hbar)^3}, \tag{74.63}$$

$$u_\gamma = \frac{\pi^2}{15 c^3 \hbar^3 \beta^4}, \tag{74.64}$$

$$s_\gamma = \frac{4\pi^2 k_B}{45 c^3 (\beta\hbar)^3}. \tag{74.65}$$

At the calendar temperature, these relations become

$$n_\gamma \ell_P^3 = \frac{2\zeta(3)}{\pi^2 (\log 3)^3}, \quad \frac{u_\gamma \ell_P^3}{E_P} = \frac{\pi^2}{15(\log 3)^4}, \quad \frac{s_\gamma \ell_P^3}{k_B} = \frac{4\pi^2}{45(\log 3)^3}. \tag{74.66}$$

Proof. Integrating occupation against the mode density uses (74.62) with $s = 3$, and the energy integral uses it with $s = 4$. This gives $\Gamma(3) = 2$ and $\Gamma(4)\zeta(4) = 6\pi^4/90 = \pi^4/15$. For entropy, start from the partition product of the modes:

$$\frac{\log Z}{V} = -\frac{1}{\pi^2 c^3} \int_0^\infty \omega^2 \log(1 - e^{-\beta\hbar\omega})\, d\omega = \frac{\pi^2}{45 c^3 (\beta\hbar)^3}.$$

The last equality follows by integration by parts after the substitution $x = \beta\hbar\omega$. The boundary terms vanish: $x^3 \log(1 - e^{-x})$ tends to zero at both endpoints. The Gibbs identity $s_\gamma/k_B = \log Z/V + \beta u_\gamma$ proves the third expression. Finally, $\beta_{108}\hbar = t_P \log 3$ and $\ell_P = c t_P$ give the three calendar normalizations. □

74.2.6 Thermal flux and the Stefan–Boltzmann constant

Proposition 74.23 (Isotropic flux and action normalization)**.** *The hemispherical flux of the isotropic realization is*

$$j_*(\Theta) = \sigma \Theta^4, \qquad \sigma = \frac{\pi^2 k_B^4}{60 \hbar^3 c^2} = \frac{2\pi^5 k_B^4}{15 h^3 c^2}. \tag{74.67}$$

At the calendar temperature,

$$j_*(\Theta_{108}) = \frac{2\pi^5}{15\, 108^4 (\log 3)^4} \frac{h}{\ell_0^2 t_0^2}. \tag{74.68}$$

Proof. The isotropic energy per unit solid angle is $u_\gamma/(4\pi)$. Flux normal to a surface integrates $c \cos\theta$ over the hemisphere:

$$\frac{c u_\gamma}{4\pi} \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos\theta \sin\theta\, d\theta = \frac{c u_\gamma}{4}.$$

The formula for u_γ, $\beta^{-1} = k_B\Theta$, and $h = 2\pi\hbar$ prove (74.67). Substitution of (74.57) and $c = \ell_0/t_0$ gives (74.68). □

Within this realization, the constant σ is the coefficient combining bosonic occupation, transverse-mode density, and angular projection of the flux. The fourth-power dependence arises from the cubic frequency moment together with the integration differential. The calendar readout expresses that same flux through action, length, time, and tritic information.

74.2.7 Spectral maxima and displacement law

Lemma 74.24 (Positive root for spectral maxima)**.** *For every integer $n > 1$, the equation $n(1 - e^{-x}) = x$ has a unique solution $x_n > 0$. The function $f_n(x) = x^n/(e^x - 1)$ has its unique positive maximum at x_n.*

Proof. Let $g_n(x) = n(1 - e^{-x}) - x$. We have $g_n(0) = 0$, $g_n'(x) = ne^{-x} - 1$, and $g_n''(x) = -ne^{-x} < 0$. The maximum of g_n is attained at $\log n$, where its value is $n - 1 - \log n > 0$, whereas $g_n(n) = -ne^{-n} < 0$. Strict monotonicity on either side of the maximum proves existence and uniqueness of the positive root. The sign of $f_n'(x)$ agrees with that of $g_n(x)$; moreover, f_n tends to zero at zero and infinity. This proves uniqueness of the maximum. □

Proposition 74.25 (Frequency and wavelength readouts)**.** *The maxima of radiance per frequency and per wavelength are*

$$\nu_{\max} = \frac{x_3 k_B \Theta}{h}, \qquad \lambda_{\max} = \frac{hc}{x_5 k_B \Theta}. \tag{74.69}$$

In particular,

$$\nu_{\max}(\Theta_{108}) = \frac{x_3}{108 t_0 \log 3}, \qquad \lambda_{\max}(\Theta_{108}) = \frac{108 \log 3}{x_5} \ell_0. \tag{74.70}$$

Proof. The conversion $\omega = 2\pi\nu$ and the isotropic angular factor give

$$B_\nu = \frac{2h\nu^3}{c^2} \frac{1}{e^{\beta h\nu} - 1}.$$

With $\nu = c/\lambda$, the transformed density satisfies $B_\lambda = B_\nu |d\nu/d\lambda|$; hence

$$B_\lambda = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\beta hc/\lambda} - 1}.$$

The respective changes of variable lead to f_3 and f_5. The preceding lemma proves (74.69); the calendar scale gives (74.70). The density Jacobian explains why the two maxima correspond to distinct roots: $\nu_{\max}\lambda_{\max} = cx_3/x_5$. □

Wien displacement is structurally defined by the stationary point of the chosen spectral density. A change of variable transforms both the argument and the measure. This precision locates each spectral constant in its generating operation and preserves its physical meaning when passing between frequency and wavelength.

Chapter 75

Persistence of the channels 90/120 and classification of the limit as a hyperinfinitesimal factor III_{λ_∂}

Faithful local products preserve the ratios of channels 90 and 120 under refinement. The GNS representation and the ITPFI limit determine the hyperfinite type III factor with explicit boundary parameter.

75.1 Local Limit and Modular Classification of the Infinite System

In an infinite UHF algebra, there is in general no global trace-class density matrix within the algebra. The correct object is the net

$$K_{\mathrm{HHM}}^{(m)} = -\log \rho_m \tag{75.1}$$

with compatible-state restrictions, or the modular generator in the GNS representation. The tracial expectation $\mathrm{id} \otimes \tau_9$ does not automatically preserve a nontracial Gibbs state and does not replace state compatibility.

Theorem 75.1 (Modular type under faithful persistence of both channels)**.** *Suppose that:*

(i) each cut is a product of faithful qutrit states with weights w_{90} and w_{120};

(ii) the inclusions preserve the state and the channel label;

(iii) infinitely many factors of each type occur, both with positive asymptotic density;

(iv) no additional spectral quotients are introduced, and the product representation is factorial.

Then the additive group of logarithms of ratios of eigenvalues is

$$90\Delta_4^{\mathrm{mod}}\mathbb{Z} + 120\Delta_4^{\mathrm{mod}}\mathbb{Z} = 30\Delta_4^{\mathrm{mod}}\mathbb{Z}. \tag{75.2}$$

The GNS closure ITPFI is the hyperinfinitesimal factor

$$\boxed{III_{\lambda_\partial}, \qquad \lambda_\partial = \mathrm{e}^{-30\Delta_4^{\mathrm{mod}}} = 0.99666964646895737861082997 69\ldots.} \tag{75.3}$$

Proof. At each link, the eigenvalue quotients are powers of e^{-w_m}. Infinite persistence of both types generates the subgroup $w_{90}\mathbb{Z} + w_{120}\mathbb{Z}$. Since $\gcd(90, 120) = 30$, it follows that equation (75.2). The Araki–Woods asymptotic ratio criterion for the faithful product yields the hyperfinite ITPFI factor with parameter $\mathrm{e}^{-30\Delta_4^{\mathrm{mod}}}$ [7, 121]. □

The theorem states its actual hypotheses. If one of the channels disappears, if the product is incompatible, or if the ratio group changes, the type must be recalculated.

Gravitational publication of the modular closure via the pentacomponent carrier. The modular closure prepares the gravitational publication: the five-component carrier is first constructed from icosahedral incidence.

Chapter 76

Dynamic, propagative, and cosmological realizations of genealogical memory: dodecaphase transport, Green function, and spatial expansion

76.1 Boundary memory: even–odd decomposition and 3 hydrodynamic, gauge, and cosmological realizations

The unity among hydrodynamics, gauge theory, and cosmology is not obtained by comparing three equations once they have been written. It appears earlier, in the decomposition of the same transport into an even part, which preserves the cost of the outward and return journeys, and an odd part, which preserves its orientation. The common object is constructed in the dodecaphase temporal system $\mathcal{R}_{12}$; its subsequent representations are not identified with one another.

76.1.1 From angular channels to dodecaphasic transport

Let

$$q_{\pm} = \exp\Big[-\frac{\pi}{180}(A \pm C^{*})\Big], \qquad S_{90}(q) = \frac{q^{90}}{1-q^{270}}, \qquad S_{120}(q) = \frac{q^{120}}{1-q^{360}},$$

and let us define

$$\kappa = 12\big(S_{90}(q_{-}) - S_{90}(q_{+})\big) - \big(S_{120}(q_{-}) - S_{120}(q_{+})\big).$$

This formula preserves the genealogy

$$\alpha_{\mathrm{HMT}} \longrightarrow A \longrightarrow C^{*} \longrightarrow q_{\pm} \longrightarrow \kappa.$$

In particular, κ is neither a viscosity, a tension, nor a mass. It is the dimensionless transfer coefficient that those quantities may subsequently read.

For

$$\phi_m = (30m + 20)^{\circ}, \qquad p_m = 12\cos(3\phi_m) - \cos(4\phi_m), \qquad 0 \le m < 12,$$

are set

$$w_m = \frac{1 + \kappa p_m}{12}.$$

The orthogonality of the characters of $\mathbb{Z}/12\mathbb{Z}$ gives $\sum_m p_m = 0$, hence $\sum_m w_m = 1$. Moreover, $|p_m| \le 13$; for the constructed value of $\kappa < 1/13$, all the weights are positive. On $\mathcal{H}_{12} = \ell^2(\mathbb{Z}/12\mathbb{Z})$ define

$$(U_{12} f)(n) = \sum_{m=0}^{11} w_m f(n-m).$$

Theorem 76.1 (Exact dodecaphonic reservation)**.** *Let*

$$P_{12} = \frac{U_{12} + U_{12}^{*}}{2}, \qquad J_{12} = \frac{U_{12} - U_{12}^{*}}{2}, \qquad L_{12} = I - P_{12}.$$

Then P_{12} is self-adjoint, J_{12} is anti-self-adjoint and, for every $f \perp \mathbf{1}$,

$$\boxed{\langle f, L_{12} f\rangle \ge g_{12}\|f\|_2^2, \qquad g_{12} = 1 - 3\kappa > 0.}$$

The constant is optimal and is attained at the frequencies 3 *and* 9*.*

Proof. In the Fourier basis $e_k(n) = \exp(2\pi i k n/12)$, convolution is diagonal:

$$U_{12}e_k = \lambda_k^{(U)} e_k, \qquad \lambda_k^{(U)} = \sum_{m=0}^{11} w_m e^{-2\pi i k m/12}.$$

The constant mode gives $\lambda_0^{(U)} = 1$. The cosines of frequencies three and four survive only in $k = 3, 9$ and $k = 4, 8$, respectively; the remaining modes vanish by orthogonality. The greatest value of $\operatorname{Re}\lambda_k^{(U)}$, with $k \neq 0$, is 3κ. Therefore,

$$\langle f, L_{12} f\rangle = \sum_{k\neq 0} (1 - \operatorname{Re}\lambda_k^{(U)})|\widehat{f}(k)|^2 \geq (1 - 3\kappa)\|f\|_2^2.$$

□

The decomposition has a precise meaning. P_{12} preserves what a route shares with its inverse; J_{12} preserves what changes sign when the orientation is reversed. The former can provide a positive form. The latter cannot: for real f, $\langle f, J_{12} f\rangle$ is purely imaginary. In return, J_{12} retains the circulation that the even projection would lose.

76.1.2 Distributed memory and local generator

The same duality appears in the nucleus

$$K_q(m, n) = q^{|m-n|}, \qquad 0 < q < 1.$$

Its matrix appears nonlocal because it correlates arbitrarily distant levels. Nevertheless, on a finite chain $0, \ldots, N$, its inverse is tridiagonal:

$$K_q^{-1} = \frac{1}{1-q^2}\begin{pmatrix} 1 & -q & 0 & \cdots & 0 \\ -q & 1+q^2 & -q & \ddots & \vdots \\ 0 & -q & 1+q^2 & \ddots & 0 \\ \vdots & \ddots & \ddots & 1+q^2 & -q \\ 0 & \cdots & 0 & -q & 1 \end{pmatrix}.$$

Proposition 76.2 (Nonlocal Green, local action)**.** *For $0 < q < 1$, K_q is positive definite, and the preceding matrix is its inverse. Consequently, an exponentially distributed memory is the Green kernel of a nearest-neighbour action.*

Proof. Multiplying the tridiagonal matrix by K_q cancels, in every interior row, the terms

$$-q\, q^{|m-1-n|} + (1+q^2) q^{|m-n|} - q\, q^{|m+1-n|}$$

when $m \neq n$, and leaves $1 - q^2$ when $m = n$. The extreme rows obey the same identity with only one neighbor. The inverse is positive; therefore, K_q is also positive. □

This result explains an apparent opposition that pervades Dimensional Mechanics: history is extensive because it preserves correlations among depths, whereas the generator that updates it may be local. “Local” and “with memory” are not alternatives.

76.1.3 Local interior and reflected representations: circulation, holonomy, contraction, and transfer positivity

The common-sheet chapter has demonstrated both local representations of the boundary once. In the interior reading, the odd sector carries circulation and the even sector produces contraction and coercivity; in the reflected reading, the former carries holonomy and the latter transfer positivity and a finite gap. Stokes' identity and the coincidence of the nonzero spectra of d^*d and dd^* are properties of the common object, not two parallel proofs. We refer to theorems 95.51 a 95.54; here we add only the third reading, the global one.

76.1.4 Global reading: expansion and curvature

The third reading does not identify the universe with a cell. It takes the same separation between interior and boundary as a global law of evolution. The identity

$$1000 = 729 + 271$$

recalls that interior growth and boundary publication complete a normalized unit. In a homogeneous and isotropic geometry, the even part of a transport can be represented by an effective pressure

$$p_{\mathrm{eff}} = p - 3\zeta H,$$

which is the standard tensorial form of a bulk viscosity. The equation does not define ζ from H; it declares the type of the cosmological reading: symmetric memory is published as resistance to expansion.

De Sitter geometry shows unambiguously what remains common among three distinct foliations. For $H > 0$,

k	$a_k(t)$
$+1$	$H^{-1}\cosh(Ht)$
0	$a_* e^{Ht}$
-1	$H^{-1}\sinh(Ht)$

in their respective domains.

Proposition 76.3 (An expansion, three spatial curvatures)**.** *The three preceding functions satisfy*

$$\frac{\dot a_k^2 + k}{a_k^2} = H^2, \qquad \frac{\ddot a_k}{a_k} = H^2,$$

and therefore

$$\boxed{R^{(4)} = 12H^2, \qquad R^{(3)} = \frac{6k}{a_k^2}.}$$

Proof. For $k = +1$ one uses $\tanh^2 + \operatorname{sech}^2 = 1$; for $k = -1$, $\coth^2 - \operatorname{csch}^2 = 1$; the flat chart is immediate. The FLRW curvature formulae then give the two scalars. □

The sign k distinguishes the geometry of the spatial sheets; it does not change the curvature of de Sitter spacetime. This example prevents one from confusing the reading of a sheet with that of the total object, the same error that the APP–TRIT–TPK architecture prevents from the outset.

76.1.5 2 Exact Operator Representations and 1 Global Geometric Realization of Boundary Memory

Theorem 76.4 (A memory, two representations, and a global reading)**.** *The even–odd separation of boundary transport admits two exact operator representations and a global geometric reading:*

reader	*odd sector*	*even sector*	*status*
interior	*circulation*	*contraction/coercivity*	*exact representation*
reflected	*holonomy*	*positive kernel/finite gap*	*exact representation*
global	*sheet orientation*	*mean response to expansion*	*typed FLRW reading*

The first two rows share Stokes' identity and the nonzero spectrum of d^*d and dd^*. The third preserves the distinction between sheet curvature and global curvature, but does not define ζ from the dodecaphase operator. No row is obtained by renaming the symbols of another.

Proof. The first and second rows are the theorems 95.52 a 95.54. The local/non-local structure is proved in proposition 76.2. The third row is realized by the FLRW decomposition and proposition 76.3; these formulas prove the geometric separation, not an autonomous selector of bulk viscosity. In each case, the orientation involution fixes the odd channel, and comparison with the inverse route fixes the even channel. □

The conclusion is not that a fluid, a connection, and a cosmology are the same physical system. It is more precise: the three sciences receive distinct representations of a single operator memory system. Flow reads it toward the interior, gauge theory toward the boundary, and cosmology on a global scale.

76.2 Spatial suspension, geometric trichromy and helicoidal leaf lamination

76.2.1 Spatial immersion $\Gamma(\theta)$ and Frenet–Lancret differential criteria

Let

$$\Gamma(\theta) = \big(r(\theta)\cos\theta, r(\theta)\sin\theta, z(\theta)\big).$$

Its curvature and torsion are calculated by means of

$$\kappa_{\mathrm{F}} = \frac{\|\Gamma' \times \Gamma''\|}{\|\Gamma'\|^3}, \qquad \mathcal{T}_{\mathrm{F}} = \frac{\det(\Gamma', \Gamma'', \Gamma''')}{\|\Gamma' \times \Gamma''\|^2}.$$

A general helix in the sense of Lancret satisfies

$$\frac{\mathcal{T}_{\mathrm{F}}}{\kappa_{\mathrm{F}}} = \text{constant}.$$

An arbitrary radial suspension need not satisfy that condition. The name *radial helical suspension* preserves the winding structure without asserting an unproved differential property.

76.2.2 Coordination of the regimes $J^2 = -I, 0, +I$ with the Frenet geometry

The generators

$$J_{+1}^2 = -I, \qquad J_0^2 = 0, \qquad J_{-1}^2 = I$$

produce elliptic, parabolic, and hyperbolic evolutions. A history may alternate them and publish a curve whose Frenet curvature varies continuously. The *trichromy* describes the sequence of internal regimes; Frenet geometry describes the output. The radial reader preserves both without confusing them.

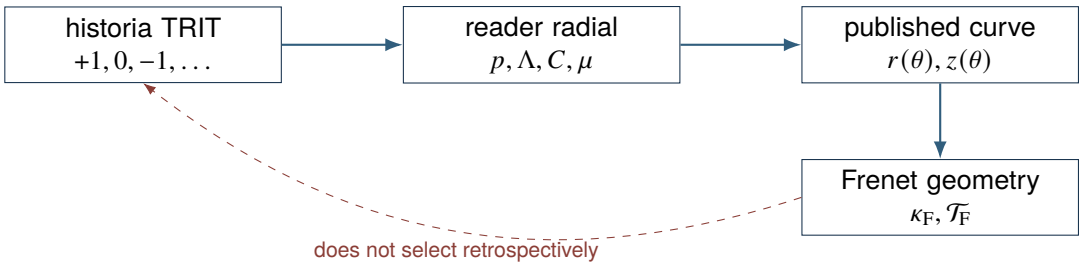


Figure 76.1: Causal separation between internal regime and visible curvature.

76.3 Dynamic structure of space and consequences

76.3.1 Helicoidal leaf lamination

The geometric synthesis is a lamination whose elements are lifted histories. Each leaf carries:

1. an angular phase;
2. a coordinate radial internal;
3. a memory depth;
4. a sequence of sheets APP;
5. a local TRIT regime;
6. a TPK holonomy and its boundary.

Transverse sections are circles or level curves; families of sections form cylinders; radial variation produces suspensions; the involution produces conjugate sheets. Real geometry is the image of this lamination under a publication that may contract several sheets.

76.3.2 Kinematic classification by angular and radial increments

A radial line appears when $\omega = 0$ and $\beta \neq 0$. A circle appears when $\omega \neq 0$ and $\beta = 0$. The spiral appears when both are nonzero. These three outputs are configurations of the same transport mechanism, not unrelated objects. The APP support is two-dimensional from the outset; the one-dimensional traversal is a particular section of that support.

76.3.3 Mobius Quotient via Return with Orientation Inversion

When the route involution is identified at the seam after one turn, the quotient may acquire a Möbius band. The assertion requires a free action and the intertwiner that transports orientation. The band does not follow from the mere presence of a sign -1; it follows from the seam

$$(x, 0) \sim (C_{\mathrm{rev}} x, 1)$$

in the declared domain. In the material atlas, the dodecaphasic involution and the cellular inversion remain distinct operations.

76.3.4 Classification of 3+2 equilibrium points via the Hessian

The intuition of Lagrange's three collinear and two triangular points may be formulated as a subsequent proof. Given an effective potential V_{eff} produced by the HMT–MD realization, the critical points satisfy

$$\nabla V_{\mathrm{eff}} = 0.$$

Its classification requires the Hessian and the Morse index. Only if the dimensional map produces exactly three collinear and two triangular critical points can it be related to the partition $3+2$. The correspondence is not used as an input to the spiral taxonomy.

76.3.5 Causal hierarchy between curves, fields, and material sections

The spiral correspondence opens three coordinated outputs:

$$\text{APP–TRIT–TPK ledger} \longrightarrow \begin{cases} \text{Archimedean curve,} \\ \text{electromagnetic mode,} \\ \text{material route signature.} \end{cases}$$

The first is algebraically closed in this work. The second has an exact realization for circular modes and a spectral compatibility condition. The third contains the central electron and the atlas but requires verification of the constancy square on each multisection before assigning flavor taxonomies.

Chapter 77

Icosahedral descent A_5/C_5 and selection of the five-component gravitational carrier

The gravitational line receives a chronological scale and then descends to the icosahedral quotient. The irreducible projector of rank 5 selects the tensor carrier without yet assuming a massive or radiative realization.

77.1 Intrinsic gravitational constant and tensor representation

77.1.1 Icosahedral descent A_5/C_5 and gravitational carrier selection

Gravity is organized here into four strata that must not be merged:

$$\begin{aligned} \text{CT108 chronological structure and action lines} &\longrightarrow \text{periodicity selector } \theta_{108}^{\mathrm{circ}} \longrightarrow G_{108},\\ \text{icosahedral incidence} &\longrightarrow V_{12} \xrightarrow{P_5} V_5 \xrightarrow{\Gamma_5} \mathrm{STF}_3 \xrightarrow{\Lambda(n)} \mathcal{H}_{\mathrm{TT}}(n),\\ \text{enriched holonomic route} &\dashrightarrow G_{\mathrm{hol}},\\ \text{metrological chart} &\longrightarrow 6.67430\times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}. \end{aligned} \tag{77.1}$$

The first line states a condition of periodicity that is intrinsic and exact. The second constructs the carrier and its trace-free transverse reduction. The third preserves an explicit construction boundary. The fourth is an evaluation in a declared decimal chart: it identifies the magnitude but does not select the gravitational character.

77.1.2 Gravitational magnitude line and chronological scale 108

Let $c_{\mathrm{int}} > 0$, $t_0 > 0$, $\ell_0 = c_{\mathrm{int}} t_0$ and a positive section $\hbar_a$ of the action line, with $a \in \{\mathrm{pre}, \mathrm{ret}\}$. The dimensional transducer is

$$G_a(\theta) = \theta \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a} = \theta \frac{c_{\mathrm{int}}^3 \ell_0^2}{\hbar_a}, \qquad \theta > 0. \tag{77.2}$$

The equality of the two expressions uses only $\ell_0 = c_{\mathrm{int}} t_0$, and $[G] = L^3 M^{-1} T^{-2}$. The dimension is fixed by the line; the mathematical problem is to select the dimensionless character θ.

The CT108 calendar defines

$$T_{108} = 108 t_0, \qquad \omega_{108} = \frac{2\pi}{108 t_0}, \qquad R_{108} = \frac{c_{\mathrm{int}}}{\omega_{108}} = \frac{54}{\pi}\ell_0. \tag{77.3}$$

For each action sheet, let

$$E_{108,a} = \hbar_a \omega_{108}, \qquad m_{108,a} = \frac{E_{108,a}}{c_{\mathrm{int}}^2}. \tag{77.4}$$

Then, the reduced Compton wavelength coincides with the radius of the cycle:

$$\bar{\lambda}_{C,a} = \frac{\hbar_a}{m_{108,a} c_{\mathrm{int}}} = R_{108}. \tag{77.5}$$

Theorem 77.1 (Uniqueness of the selector under the HMT periodicity condition)**.** *With the reduced convention $r_g = Gm/c_{\mathrm{int}}^2$ and the HMT periodicity premise that identifies the gravitational radius of the CT108 quantum with its phase radius, the closure condition*

$$r_{g,a}(\theta) = R_{108} \tag{77.6}$$

selects a unique positive solution,

$$\boxed{\theta_{108}^{\mathrm{circ}} = \left(\frac{54}{\pi}\right)^2 = 295.4525715010569415303525\ldots.} \tag{77.7}$$

In that embodiment,

$$R_{108} = \bar{\lambda}_{C,a} = r_{g,a} = \ell_P^{\mathrm{circ}}. \tag{77.8}$$

Proof. By equations (77.2) y (77.4),

$$r_{g,a}(\theta) = \frac{G_a(\theta) m_{108,a}}{c_{\mathrm{int}}^2} = \theta c_{\mathrm{int}} t_0^2 \omega_{108} = \theta \frac{\pi}{54} \ell_0.$$

Comparing it with $R_{108} = (54/\pi)\ell_0$ necessarily gives $\theta = (54/\pi)^2$. The equation is linear in θ and all the factors are positive, so the solution is unique. □

The equality $r_g = R_{108}$ does not follow from the dimensionality of equation (77.2). It is the explicit geometric condition of this HMT realization: action, Compton, and periodic return must determine the same length. The theorem proves that, once this structural condition is accepted, the character θ is fixed without using the SI value of G.

Remark 77.2 (Radius convention). The theorem uses the reduced gravitational radius Gm/c^2. If it is replaced by the Schwarzschild radius $2Gm/c^2$, the selector changes by a factor of two. The two expressions correspond to distinct geometric conditions, so the convention must be declared before the calculation.

77.1.3 Planck System and Bidirectional Covariance

Corollary 77.3 (Planck Family Associated with the CT108 Chronological Structure)**.** *For* $G_a = G_a(\theta_{108}^{\mathrm{circ}})$,

$$\ell_P = \frac{54}{\pi}\ell_0, \qquad t_P = \frac{54}{\pi} t_0, \tag{77.9}$$

$$E_{P,a} = \frac{\hbar_a}{t_P} = \frac{\pi \hbar_a}{54 t_0} = \frac{h_a}{108 t_0}, \qquad \ell_P^2 = \frac{G_a \hbar_a}{c_{\mathrm{int}}^3} = \theta_{108}^{\mathrm{circ}} \ell_0^2. \tag{77.10}$$

Además,

$$F_{P,a} = \frac{c_{\mathrm{int}}^4}{G_a}, \qquad P_{P,a} = \frac{c_{\mathrm{int}}^5}{G_a}, \qquad \rho_{P,a} = \frac{\hbar_a}{(\theta_{108}^{\mathrm{circ}})^2 c_{\mathrm{int}}^5 t_0^4}. \tag{77.11}$$

Proof. The first three identities follow from $\ell_P = R_{108}$, $t_P = \ell_P/c_{\mathrm{int}}$, and $h_a = 2\pi\hbar_a$. For the area,

$$\frac{G_a \hbar_a}{c_{\mathrm{int}}^3} = \theta_{108}^{\mathrm{circ}} c_{\mathrm{int}}^2 t_0^2 = \theta_{108}^{\mathrm{circ}} \ell_0^2.$$

The remaining expressions are the dimensional compositions of force, power, and mass per volume with the same section. □

Let

$$R_{\mathrm{act}} := \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}}. \tag{77.12}$$

Since the action appears in the numerator of the chronological mass and in the denominator of G,

$$\frac{m_{108,\mathrm{pre}}}{m_{108,\mathrm{ret}}} = R_{\mathrm{act}}, \qquad \frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}} = R_{\mathrm{act}}^{-1}, \qquad G_a \hbar_a = \theta_{108}^{\mathrm{circ}} c_{\mathrm{int}}^5 t_0^2. \tag{77.13}$$

Therefore, $\ell_P^2 = G_a\hbar_a/c^3$ does not change sheets. Gravity and action are reciprocally oriented, while area preserves the common geometry. This cancellation is the interface of confluence toward the area operator of the section 71.1.

77.1.4 Icosahedral Component of Dimension 5

The antecedent is neither $\mathbb{R}$ nor a subsequent selection of five polarizations. The object responsible for transporting the incidence is defined explicitly before introducing the exceptional realization.

Let $X_{\mathrm{TPK}}^{\mathrm{exc}}$ be the subspace of the enriched state TPK that preserves, in addition to the arithmetic output, the orientation, route, carry, and exceptional incidence. To make the geometric target space explicit, let $\varphi_g = (1+\sqrt{5})/2$, $\nu = \sqrt{1+\varphi_g^2}$, and define

$$\Omega_{12}^{\mathrm{ico}} = \frac{1}{\nu}\big(\{0\}\times\{\pm1\}\times\{\pm\varphi_g\} \;\cup\; \{\pm1\}\times\{\pm\varphi_g\}\times\{0\} \;\cup\; \{\pm\varphi_g\}\times\{0\}\times\{\pm1\}\big) \subset S^2. \tag{77.14}$$

Its twelve elements are the oriented vertex classes of an icosahedron. The incidence is not left verbal: it is fixed by

$$\mathcal{I}_{12}^{\mathrm{ico}} = \{(v,w) \in \Omega_{12}^{\mathrm{ico}} \times \Omega_{12}^{\mathrm{ico}} : v \neq w,\ \langle v, w\rangle = 1/\sqrt{5}\}. \tag{77.15}$$

Each vertex has five neighbors in this relation. The reading bridge is typed as a map

$$\pi_{\mathrm{ico}} : X_{\mathrm{TPK}}^{\mathrm{exc}} \longrightarrow \Omega_{12}^{\mathrm{ico}}. \tag{77.16}$$

The cardinality twelve of the image of π_{ico} is not sufficient. To define an incidence realization, the mapping must satisfy simultaneously:

(i) being surjective and constant exactly on the fibers that discard the arithmetic chart but preserve the exceptional route;

(ii) intertwine the action of the oriented automorphisms of the antecedent with a transitive action on $\Omega_{12}^{\mathrm{ico}}$;

(iii) transporting the declared TPK incidence to $\mathcal{I}_{12}^{\mathrm{ico}}$, not merely its cardinalities;

(iv) preserve the leaf involution and the nonadic return in the stabilizer of the corresponding fiber.

These conditions separate three assertions: the existence of the map π_{ico}, the identification of the group that acts, and the linear representation constructed thereafter. The second and third cannot be used to define the first retrospectively.

Theorem 77.4 (Incidence Descent to the Icosahedral Quotient)**.** *Suppose that π_{ico} satisfies (i)–(iv) and that the oriented action induced on $\Omega_{12}^{\mathrm{ico}}$ has group $G_{\mathrm{ico}} \simeq A_5$. Then, for every class $v_0 \in \Omega_{12}^{\mathrm{ico}}$,*

$$H_{v_0} = \mathrm{Stab}_{G_{\mathrm{ico}}}(v_0) \simeq C_5, \qquad \Omega_{12}^{\mathrm{ico}} \simeq G_{\mathrm{ico}}/H_{v_0} \simeq A_5/C_5. \tag{77.17}$$

The application

$$\kappa_{v_0} : G_{\mathrm{ico}}/H_{v_0} \longrightarrow \Omega_{12}^{\mathrm{ico}}, \qquad gH_{v_0} \longmapsto g\cdot v_0, \tag{77.18}$$

is an equivariant bijection and preserves incidence if, in the quotient, one declares

$$gH_{v_0} \sim g'H_{v_0} \quad\Longleftrightarrow\quad (g\cdot v_0, g'\cdot v_0) \in \mathcal{I}_{12}^{\mathrm{ico}}. \tag{77.19}$$

Therefore, the composite $\kappa_{v_0}^{-1}\pi_{\mathrm{ico}}$ is the typed map from the TPK antecedent to the quotient A_5/C_5.

Proof. The list of equation (77.14) has twelve elements. By transitivity, $|G_{\mathrm{ico}}\cdot v_0| = 12$, and the orbit–stabilizer theorem gives $|H_{v_0}| = 60/12 = 5$. Every group of prime order five is cyclic; hence $H_{v_0} \simeq C_5$. The formula for κ_{v_0} does not depend on the representative: if $gH_{v_0} = g'H_{v_0}$, then $g^{-1}g' \in H_{v_0}$

and $g'v_0 = gv_0$. Surjectivity follows from transitivity, and equality of the cardinalities gives injectivity. Equivariance is immediate: $\kappa_{v_0}(agH_{v_0}) = a\kappa_{v_0}(gH_{v_0})$. Finally, equation (77.19) is precisely the transport of structure through that bijection; hypothesis (iii) makes this transport commute with π_{ico}. □

The theorem does not replace the construction of π_{ico}: it proves that it is obtained once its four requirements have been verified. In the complete HMT profile, the incidence corridor reaching Paley–Witt–Golay–Leech–$V^\natural$–Monster–Moonshine is the domain of definition of that map. In this autonomous volume, the exact input is the map equation (77.16) with its conditions; the entire segment from A_5/C_5 to TT reduction is proved below without again consulting a gravitational value.

The chain of provenance is expressed through all its morphisms:

$$\mathrm{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow X_{\mathrm{TPK}}^{\mathrm{exc}} \xrightarrow{\pi_{\mathrm{ico}}} \Omega_{12}^{\mathrm{ico}} \xleftarrow[\simeq]{\kappa_{v_0}} A_5/C_5. \tag{77.20}$$

The complete exceptional corridor is proved in its own dedicated development. The hypothesis that this chapter cannot replace with a count is the existence of π_{ico} satisfying (i)–(iv). With that input fixed, descent to the quotient is theorem 77.4, and P_5, Γ_5 and $\Lambda(n)$ are constructed from it. Realizing the resulting carrier as a gravitational field remains a typed physical interface, not a consequence of cardinality.

Let $G = A_5$, let $H \simeq C_5$ be the stabilizer of an oriented vertex of the icosahedron, and consider

$$V_{12} = \mathbb{R}[G/H], \qquad |G/H| = \frac{60}{5} = 12. \tag{77.21}$$

The permutation representation ρ_{12} decomposes as

$$V_{12} \simeq \mathbf{1} \oplus V_3 \oplus V_{3'} \oplus V_5. \tag{77.22}$$

The central idempotent of the character summand $\chi_5 = (5, 1, -1, 0, 0)$ is

$$\boxed{P_5 = \frac{5}{60} \sum_{g \in A_5} \chi_5(g^{-1})\rho_{12}(g) = \frac{1}{12}\left(5I + \sum_{g \in 2A} \rho_{12}(g) - \sum_{g \in 3A} \rho_{12}(g)\right).} \tag{77.23}$$

Theorem 77.5 (Rank-five gravitational projector and coupler)**.** *The operator P_5 satisfies*

$$P_5^2 = P_5 = P_5^*, \qquad \mathrm{Tr}\, P_5 = \mathrm{rank}\, P_5 = 5. \tag{77.24}$$

For

$$\mathbb{G}_{5,a} := G_a(\theta_{108}^{\mathrm{circ}})P_5 \tag{77.25}$$

we have

$$\mathrm{Spec}(\mathbb{G}_{5,a}) = \{G_a(\theta_{108}^{\mathrm{circ}}) \textit{ with multiplicity } 5,\ 0 \textit{ with multiplicity } 7\}. \tag{77.26}$$

Proof. Character orthogonality converts equation (77.23) into the central idempotent that acts as the identity on the unique copy of V_5 and as zero on the other three summands of equation (77.22). Reality of the character and orthogonality of ρ_{12} give self-adjointness. Its trace and rank are five. Multiplying a rank-five projector by the positive scalar G_a immediately produces the indicated spectrum. □

The expression $\mathbb{G}_{5,a}$ does not add five different values of the gravitational constant. It is the same section of G acting on the selected tensor summand and annihilating the other seven modes of the module under the permutation action.

Equivariant reduction of the carrier to the traceless tensor sector. The carrier intertwines with the symmetric traceless tensors and is reduced equivariantly before transversality is imposed.

77.1.5 Icosahedral incidence and five-component gravitational coupling

The discrete structure of the continuum preserves an incidence reading alongside the arithmetic reading. The five-component gravitational carrier is constructed through that incidence and its tensor representation. The coupling section $G_a(\vartheta^\circ_{108})$, with $a \in \{\mathrm{pre}, \mathrm{ret}\}$, is the one obtained in section 80.3; what follows makes explicit the space on which it acts, its five components and its transverse projection.

Let $X^{\mathrm{exc}}_{\mathrm{TPK}}$ be the subspace of the enriched state TPK that preserves, in addition to the arithmetic output, the orientation, route, carry, and exceptional incidence. To make the geometric target space explicit, let $\varphi_g = (1+\sqrt{5})/2$, $\nu = \sqrt{1+\varphi_g^2}$, and define

$$\Omega^{\mathrm{ico}}_{12} = \frac{1}{\nu}\big(\{0\}\times\{\pm1\}\times\{\pm\varphi_g\} \;\cup\; \{\pm1\}\times\{\pm\varphi_g\}\times\{0\} \;\cup\; \{\pm\varphi_g\}\times\{0\}\times\{\pm1\}\big) \subset S^2. \tag{77.27}$$

Its twelve elements are the oriented vertex classes of an icosahedron. The incidence is not left verbal: it is fixed by

$$\mathcal{I}^{\mathrm{ico}}_{12} = \{(v,w) \in \Omega^{\mathrm{ico}}_{12}\times\Omega^{\mathrm{ico}}_{12} : v \neq w,\ \langle v, w\rangle = 1/\sqrt{5}\}. \tag{77.28}$$

Each vertex has five neighbors in this relation. The reading bridge is typed as a map

$$\pi_{\mathrm{ico}} : X^{\mathrm{exc}}_{\mathrm{TPK}} \longrightarrow \Omega^{\mathrm{ico}}_{12}. \tag{77.29}$$

The incidence realization requires, in addition to the twelve elements of the image, the following simultaneous properties of the map:

(i) being surjective and constant exactly on the fibers that discard the arithmetic chart but preserve the exceptional route;

(ii) intertwine the action of the oriented automorphisms of the antecedent with a transitive action on $\Omega^{\mathrm{ico}}_{12}$;

(iii) transporting the declared TPK incidence to $\mathcal{I}^{\mathrm{ico}}_{12}$, not merely its cardinalities;

(iv) preserve the leaf involution and the nonadic return in the stabilizer of the corresponding fiber.

These conditions fix the order of the construction: first the map π_{ico}, then the identification of the induced action and, finally, its linear representation. Orientation, route and carry belong to the enriched antecedent of that descent.

77.1.5.1 Calendar projection and representation of its twelve positions

The calendar of Section 1.23.5 provides an observable coordinate of the complete state. In its notation, write $r_0(\theta) = r(\theta) - 1 \in \{0,\dots,8\}$ and $b = \beta(\theta) \in \mathbb{Z}/12\mathbb{Z}$. The clock projection is

$$Q_{\mathrm{clk}}(x) = \big(\beta(\theta(x)), r_0(\theta(x))\big).$$

For an admissible route of n updates, the induced transport is

$$k(r_0,n) = \left\lfloor \frac{r_0+n}{9} \right\rfloor, \qquad T_n(b,r_0) = \big(b + k(r_0,n) \bmod 12,\ (r_0+n) \bmod 9\big). \tag{77.30}$$

Here $n \in \mathbb{N}_0$ counts effective updates; the integer quotient k records the nonadic returns crossed.

Proposition 77.6 (Compatibility of the calendar with route composition)**.** *The projection Q_{clk}, together with the length of each route, defines a functor from the category of admissible TPK routes to the action category of calendar (77.30). The identities $T_{m+n} = T_m T_n$, $T_0 = I$, $T_9(b,r_0) = (b+1, r_0)$, and $T_{108}(b,r_0) = (b,r_0)$ hold.*

Proof. Putting $r_0' = (r_0+n) \bmod 9$, Euclidean division gives

$$k(r_0, n+m) = k(r_0, n) + k(r_0', m).$$

This identity simultaneously preserves residue and quotient when the routes are concatenated. The update of θ established in Section 1.23.5 induces exactly (77.30); the remaining identities follow by substitution. The target category has pairs (b, r_0) as objects and admissible lengths as arrows, with addition as composition. □

The index $p = b + 1 \in \{1, \dots, 12\}$, taking the representative $0 \le b < 12$, publishes the dodecaphase position. One turn of 108 updates traverses its twelve positions. The proved periodicity corresponds to this projection: route, accumulated carry and memory remain in the source state and continue their update.

Incidence realization from the angular projector. The representatives $r_p C_5$ and the representation ρ of section 20.2 fix the chart of the twelve positions. The projector already defined by (20.5) produces

$$\begin{aligned} &\mathcal{H}_3 = P_3^{\mathrm{ico}}\mathbb{R}^{12}, \qquad v_p = 2P_3^{\mathrm{ico}} e_p, \\ &(P_3^{\mathrm{ico}})_{pq} = \frac{1}{20} \sum_{\substack{g \in A_5 \\ g r_q C_5 = r_p C_5}} \chi_3(g^{-1}). \end{aligned} \tag{77.31}$$

The last sum contains five terms per entry and uses the representatives and the two classes of order-five cycles fixed there.

Proposition 77.7 (Icosahedral realization of the dodecaphase positions)**.** *The vectors v_p are twelve distinct unit vectors in $\mathcal{H}_3$. Their incidence is $p \sim q \iff \langle v_p, v_q \rangle = 1/\sqrt{5}$. The map $p \mapsto v_p$ is A_5-equivariant, and its image is isometric to $\Omega_{12}^{\mathrm{ico}}$.*

Proof. Transitivity and $\operatorname{Tr} P_3^{\mathrm{ico}} = 3$ give $(P_3^{\mathrm{ico}})_{pp} = 1/4$; by idempotence, $\langle v_p, v_q \rangle = 4(P_3^{\mathrm{ico}})_{pq}$ and $\|v_p\| = 1$. Evaluation of the five terms in (77.31) gives the antipodal pairs

$$(1,4),\ (2,3),\ (5,8),\ (6,7),\ (9,12),\ (10,11).$$

For the representatives $(1, 2, 5, 6, 9, 10)$, their Gram matrix is

$$I_6 + \frac{D}{\sqrt{5}}, \qquad D = \begin{pmatrix} 0 & -1 & -1 & 1 & -1 & 1 \\ -1 & 0 & 1 & 1 & -1 & -1 \\ -1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & -1 & 1 \\ -1 & -1 & 1 & -1 & 0 & 1 \\ 1 & -1 & 1 & 1 & 1 & 0 \end{pmatrix}, \qquad D^2 = 5I_6. \tag{77.32}$$

The remaining scalar products are obtained by antipodality. Thus, apart from a vertex and its opposite, the products are $\pm 1/\sqrt{5}$, and each vertex has five neighbours. The following table makes explicit the comparison with the signed six-position chart $\mathcal{U} = (\infty, 0, 1, 2, 3, 4)$:

p	$u(p)$	$\sigma(p)$	p	$u(p)$	$\sigma(p)$
1	∞	$+1$	7	3	-1
2	0	-1	8	1	$+1$
3	0	$+1$	9	2	-1
4	∞	-1	10	4	$+1$
5	1	-1	11	4	-1
6	3	$+1$	12	2	$+1$

(77.33)

Indeed, for $C = \mathrm{bal}(A_W)$, with A_W given in (21.126), the entries of D are $D_{ab} = \sigma(p_a)\sigma(p_b)C_{u(p_a),u(p_b)}$. Consequently, the table and (77.32) isometrically identify v_p with $\sigma(p)\sqrt{2}E_C e_{u(p)}$, where $E_C = (I_6 + C/\sqrt{5})/2$. Both systems span three-dimensional spaces and have identical Gram matrices, proving that the induced linear map is well defined and isometric. The explicit realization in (77.38), proved below, identifies it with $\Omega_{12}^{\mathrm{ico}}$ and its incidence. Finally, $P_3^{\mathrm{ico}}\rho(g) = \rho(g)P_3^{\mathrm{ico}}$ implies $v_{g\cdot p} = \rho(g)v_p$. The bijection of the twelve positions gives $\mathrm{Stab}_{A_5}(v_p) = r_p C_5 r_p^{-1}$. □

One thus obtains the representational composition

$$x \longmapsto Q_{\mathrm{clk}}(x) \longmapsto p = \beta(\theta(x)) + 1 \longmapsto v_p. \tag{77.34}$$

The signed expression of this composition is $r_{\pm}^{\mathrm{rep}}(x) = (u(p), \sigma(p))$, determined by table (77.33). The table is an explicit choice of alignment between charts; the actions of A_5 provide its equivalent alignments. The complete register is preserved in the domain of (77.34).

Link with the five-component tensor carrier. In $\mathcal{H}_3$, form $Q_p = v_p v_p^* - I_{\mathcal{H}_3}/3$. Let $J = \mathbf{1}\mathbf{1}^*$, and let A be the matrix of the preceding antipodal permutation. The Frobenius product and (77.32) give

$$\begin{aligned} \mathrm{Gram}(Q_p) &= (4P_3^{\mathrm{ico}}) \circ (4P_3^{\mathrm{ico}}) - \frac{1}{3}J = \frac{4}{5}(I + \mathsf{A}) - \frac{2}{15}J, \\ P_5^{\mathrm{ico}} &= \frac{5}{8}\,\mathrm{Gram}(Q_p) = \frac{1}{2}(I + \mathsf{A}) - \frac{1}{12}J. \end{aligned} \tag{77.35}$$

The symbol $\circ$ denotes the entrywise product. The last expression is the orthogonal projector onto coordinates even under antipodality and orthogonal to $\mathbf{1}$, of dimension five. Its character is $(5, 1, -1, 0, 0)$: on the six antipodal pairs, the fixed-point counts are $(6, 2, 0, 1, 1)$, and the constant component is removed. It therefore coincides with the central projector of (77.49). This identity links the three-dimensional construction with Γ_0 and Γ_5, defined in (77.54) and (77.56), through the explicit quadratic operation $v_p \mapsto Q_p$.

77.1.5.2 Transport of the representational frame

Let $s = (1\,2\,\cdots\,12)$, and let $S_c e_p = e_{s(p)}$. A nonadic calendar return takes p to $s(p)$. The joint transport of frame, representation, and projector is expressed by

$$\begin{aligned} P_{3,k}^{\mathrm{ico}} = S_c^k P_3^{\mathrm{ico}} S_c^{-k}, \qquad \rho^{(k)}(g) = S_c^k \rho(g) S_c^{-k}, \\ v_{s^k(p)}^{(k)} = 2P_{3,k}^{\mathrm{ico}} e_{s^k(p)} = S_c^k v_p. \end{aligned} \tag{77.36}$$

The last equality follows by substitution; orthogonality of S_c preserves all inner products and transports incidence. For a route of length n, take $k = k(r_0, n)$. The carry identity in Proposition 77.6 proves compatibility of these transports with concatenation. Although $S_c^{12} = I$, the twelve turns update the memory of the source route.

Equality (77.36) establishes covariance between frames. The active action on a fixed incidence has additional content: it requires identifying the transport of the complete state and its action in that chart. In particular, s has order twelve, whereas the orders of the elements of A_5 are $1, 2, 3, 5$. Likewise, the stabilizer of a vertex has order five, and that of its antipodal pair has order ten; the involution of the signed chart interchanges the two vertices. These distinctions preserve the meaning of orientation, sheet and return in properties (i)–(iv): the proved representational realization and the descent of the complete exceptional route are related through those properties, with their declared domains and fibres.

Construction of the carrier from six-position incidence. The matrix A_W of (21.126), which enters previously in the angular transport of section 20.2, provides an explicit antecedent of the icosahedral

realization. Let $C = \mathrm{bal}(A_W)$, where $\mathrm{bal}(0) = 0$, $\mathrm{bal}(1) = 1$ and $\mathrm{bal}(2) = -1$. The order of its coordinates is $\mathcal{U} = (\infty, 0, 1, 2, 3, 4)$. The integer chart satisfies $C = C^{\mathsf{T}}$, $C^2 = 5I_6$ and $\operatorname{Tr} C = 0$. We define its signed covering by

$$\mathcal{D}_C = \mathcal{U} \times \{+1, -1\}, \qquad (u, \sigma) \sim_C (v, \tau) \iff u \neq v \text{ and } \sigma\tau = C_{uv}. \tag{77.37}$$

Orientation doubles each coordinate; the matrix decides the incidences between the two corresponding fibers. The involution of this covering is $\jmath_C(u, \sigma) = (u, -\sigma)$.

Proposition 77.8 (Isometric realization of signed incidence)**.** *The operator and vectors*

$$E_C = \frac{1}{2}\left(I_6 + \frac{C}{\sqrt{5}}\right), \qquad \mathcal{H}_C = \operatorname{im} E_C, \qquad v_{u,\sigma} = \sigma\sqrt{2} E_C e_u \tag{77.38}$$

realize $\mathcal{D}_C$ as twelve distinct unit vectors in a three-dimensional Euclidean space. There is an explicit isometry $T_C : \mathcal{H}_C \to \mathbb{R}^3$ for which $\iota_C(u, \sigma) = T_C v_{u,\sigma}$ is a bijection onto $\Omega_{12}^{\mathrm{ico}}$, preserves incidence, and satisfies $\iota_C \jmath_C = -\iota_C$.

Proof. From $C^2 = 5I_6$, one obtains $E_C^2 = E_C = E_C^*$; its trace is three. The metric identities are

$$\langle v_{u,\sigma}, v_{v,\tau} \rangle = \begin{cases} \sigma\tau, & u = v, \\ \sigma\tau C_{uv}/\sqrt{5}, & u \neq v. \end{cases}$$

Thus the twelve vectors are distinct, and their inner-product $1/\sqrt{5}$ incidence is exactly (77.37). To make the isometry explicit, let the columns of B, in order $\mathcal{U}$, be

$$B = \frac{1}{\nu}\begin{pmatrix} 0 & -1 & -\varphi_g & 0 & \varphi_g & 1 \\ -1 & -\varphi_g & 0 & 1 & 0 & -\varphi_g \\ -\varphi_g & 0 & -1 & -\varphi_g & -1 & 0 \end{pmatrix}.$$

Substitution of $\varphi_g^2 = \varphi_g + 1$ gives $B^* B = I_6 + C/\sqrt{5} = 2E_C$. Hence $T_C = B/\sqrt{2}\big|_{\mathcal{H}_C}$ is an isometry, and $T_C v_{u,\sigma} = \sigma B e_u$. The six columns of B and their opposites are precisely the twelve vertices of (77.27). The last identity also proves compatibility of the involutions. □

The preceding dodecaphasic realization constructs the twelve vectors directly from the projector P_3^{ico} and the calendar positions. The signed covering provides another coordinate expression of that carrier. For the reader π_{ico} in the descent theorem, the relation between the two expressions is written

$$X_{\mathrm{TPK}}^{\mathrm{exc}} \xrightarrow{r_\pm} \mathcal{D}_C \xrightarrow[\simeq]{\iota_C} \Omega_{12}^{\mathrm{ico}}, \qquad \pi_{\mathrm{ico}}(x) = \sigma(x) B e_{u(x)}, \quad r_\pm(x) = (u(x), \sigma(x)). \tag{77.39}$$

Since ι_C is a bijection, this factorization determines $r_\pm = \iota_C^{-1}\pi_{\mathrm{ico}}$. Thus, $r_\pm$ expresses in the signed chart the reader being used: its introduction does not add an independent hypothesis for constructing the dodecaphase carrier. The preceding correspondence table gives that expression for its twelve positions. Properties (i)–(iv) retain their own scope when the realization is identified with the quotient of the complete exceptional route. In particular, the joint transport of frame, projector and incidence proves covariance between charts; the preservation of a fixed incidence by an active transformation is a distinct identity. Orientation and memory remain in the source state during both readings. A signed permutation that transports C jointly transports E_C, its vectors and its incidence; this covariance allows the charts to be compared while preserving the orientation of each fibre.

Theorem 77.9 (Incidence Descent to the Icosahedral Quotient)**.** *Suppose that π_{ico} satisfies (i)–(iv) and that the oriented action induced on $\Omega_{12}^{\mathrm{ico}}$ has group $G_{\mathrm{ico}} \simeq A_5$. Then, for every class $v_0 \in \Omega_{12}^{\mathrm{ico}}$,*

$$H_{v_0} = \mathrm{Stab}_{G_{\mathrm{ico}}}(v_0) \simeq C_5, \qquad \Omega_{12}^{\mathrm{ico}} \simeq G_{\mathrm{ico}}/H_{v_0} \simeq A_5/C_5. \tag{77.40}$$

The application

$$\kappa_{v_0} : G_{\mathrm{ico}}/H_{v_0} \longrightarrow \Omega_{12}^{\mathrm{ico}}, \qquad gH_{v_0} \longmapsto g \cdot v_0, \tag{77.41}$$

is an equivariant bijection and preserves incidence if, in the quotient, one declares

$$gH_{v_0} \sim g'H_{v_0} \quad \Longleftrightarrow \quad (g \cdot v_0, g' \cdot v_0) \in \mathcal{I}_{12}^{\mathrm{ico}}. \tag{77.42}$$

Therefore, the composite $\kappa_{v_0}^{-1}\pi_{\mathrm{ico}}$ is the typed map from the TPK antecedent to the quotient A_5/C_5.

Proof. The list of (77.27) has twelve elements. By transitivity, $|G_{\mathrm{ico}} \cdot v_0| = 12$, and the orbit–stabilizer theorem gives $|H_{v_0}| = 60/12 = 5$. Every group of prime order five is cyclic; hence $H_{v_0} \simeq C_5$. The formula for κ_{v_0} does not depend on the representative: if $gH_{v_0} = g'H_{v_0}$, then $g^{-1}g' \in H_{v_0}$ and $g'v_0 = gv_0$. Surjectivity follows from transitivity, and equality of the cardinalities gives injectivity. Equivariance is immediate: $\kappa_{v_0}(agH_{v_0}) = a\kappa_{v_0}(gH_{v_0})$. Finally, (77.42) is precisely the transport of structure through that bijection; hypothesis (iii) makes this transport commute with π_{ico}. □

The theorem establishes descent once the four properties of π_{ico} have been verified. Its domain is the enriched state of the exceptional reading, which preserves the incidence transported by APP–TRIT–TPK. In this construction, the explicit antecedent is the map (77.29) with its four properties; the passage from A_5/C_5 to the transverse traceless reduction is proved below with that antecedent and the induced action of A_5.

The chain of provenance is expressed through all its morphisms:

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow X_{\mathrm{TPK}}^{\mathrm{exc}} \xrightarrow{\pi_{\mathrm{ico}}} \Omega_{12}^{\mathrm{ico}} \xleftarrow[\simeq]{\kappa_{v_0}} A_5/C_5. \tag{77.43}$$

The composition (77.43) retains as its antecedent the existence of π_{ico} with properties (i)–(iv). On that domain, descent to the quotient is theorem 77.9; from it are constructed the projector P_5^{ico}, the intertwiner Γ_5 and the projection $\Lambda(n)$. The realization of the carrier as a gravitational field additionally uses the dynamics and physical constraints specified at the end of this development.

Let $\mathcal{G} = A_5$, let $H \simeq C_5$ be the stabilizer of an oriented vertex of the icosahedron, and consider

$$V_{12} = \mathbb{R}[\mathcal{G}/H], \qquad |\mathcal{G}/H| = \frac{60}{5} = 12. \tag{77.44}$$

Characters of icosahedral incidence. The following calculation is performed on the icosahedral action obtained in Theorem 77.9, with its incidence and orientation hypotheses. It determines the decomposition of the vertex representation and the irreducibility of the tensor space through their traces, preserving the antecedent π_{ico} declared in that theorem.

Proposition 77.10 (Vertex decomposition and tensor character)**.** *Let $\tau = (1+\sqrt{5})/2$ and $\bar{\tau} = (1-\sqrt{5})/2$. In the class order $1, 2A, 3A, 5A, 5B$, the characters of the rotational representation V_3, its algebraic conjugate $V_{3'}$, the twelve-vertex representation V_{12}, and $\mathrm{STF}_3 = \mathrm{Sym}_0^2(V_3)$ are*

class	1	$2A$	$3A$	$5A$	$5B$
cardinal	1	15	20	12	12
class of g^2	1	1	$3A$	$5B$	$5A$
χ_3	3	-1	0	τ	$\bar\tau$
$\chi_{3'}$	3	-1	0	$\bar\tau$	τ
χ_{12}	12	0	0	2	2
χ_{STF}	5	1	-1	0	0

(77.45)

The representations $V_3, V_{3'}$ and STF_3 are absolutely irreducible and distinct. Moreover,

$$V_{12} \simeq \mathbf{1} \oplus V_3 \oplus V_{3'} \oplus \mathrm{STF}_3, \qquad \langle \chi_{12}, \chi_{\mathrm{STF}} \rangle = 1. \tag{77.46}$$

In particular, the five-dimensional summand occurs exactly once.

Proof. In A_5, there are $5 \cdot 3 = 15$ double transpositions and $\binom{5}{3} \cdot 2 = 20$ three-cycles. The $4! = 24$ five-cycles split into two classes of twelve: their centralizer is the cyclic subgroup of order five and is contained in A_5. Identifying the positions of a cycle with $\mathbb{Z}/5\mathbb{Z}$, the permutation inducing $r \mapsto 2r$ is a four-cycle on the nonzero elements, with negative sign. Squaring therefore exchanges the two classes. Inversion $r \mapsto -r$ is a product of two transpositions and preserves each class. This determines the first two rows of (77.45).

The representation V_3 is the action by rotations on the vertex coordinates. A rotation through angle θ has trace $1 + 2\cos\theta$. The classes of orders two and three give -1 and 0, respectively. The name $5A$ denotes the class of angles $\pm 2\pi/5$ and $5B$ that of $\pm 4\pi/5$; their traces are τ and $\bar\tau$. These expressions follow from $u^2 + u - 1 = 0$, for $u = 2\cos(2\pi/5)$, and from the double-angle formula. The equation for u follows by dividing $1 + z + z^2 + z^3 + z^4 = 0$ by z^2, with $z = \exp(2\pi i/5)$. We have thus calculated the character of a representation already defined geometrically.

To construct the other representation, use the vertices without the common normalization ν^{-1}. Their coordinates belong to $K = \mathbb{Q}(\sqrt{5})$. If B is the matrix of three linearly independent vertices and B_g contains their images, the rotation matrix is $R_g = B_g B^{-1} \in M_3(K)$. The conjugation $\sigma : \sqrt{5} \mapsto -\sqrt{5}$ then defines $R'_g = \sigma(R_g)$, satisfying

$$R'_{gh} = R'_g R'_h, \qquad (R'_g)^T R'_g = I, \qquad \det R'_g = 1.$$

Thus $V_{3'}$ is an effectively constructed real representation. Its character is $\sigma(\chi_3)$, exchanging τ and $\bar\tau$.

The permutation character counts fixed vertices. The identity fixes twelve; nontrivial rotations of order five fix the two endpoints of their axis. Those of order two or three fix no vertices, since the stabilizer of a vertex is C_5. Hence $\chi_{12} = (12, 0, 0, 2, 2)$.

For the tensor space, if $\lambda_1, \lambda_2, \lambda_3$ are the eigenvalues of R_g, the sum $\sum_{i \le j} \lambda_i \lambda_j$ is $\big((\sum_i \lambda_i)^2 + \sum_i \lambda_i^2\big)/2$. The metric form generates an invariant line in $\mathrm{Sym}^2(V_3)$, whose trace-free complement has character

$$\chi_{\mathrm{STF}}(g) = \frac{\chi_3(g)^2 + \chi_3(g^2)}{2} - 1. \tag{77.47}$$

Using the squaring map in the table and $\tau + \bar\tau = 1$, $\tau\bar\tau = -1$, gives $\chi_{\mathrm{STF}} = (5, 1, -1, 0, 0)$.

The real-character inner product is $\langle \chi, \psi \rangle = \frac{1}{60} \sum_C |C| \chi(C) \psi(C)$. In particular,

$$\begin{aligned} \langle \chi_3, \chi_3 \rangle &= \langle \chi_{3'}, \chi_{3'} \rangle = \frac{9 + 15 + 12(\tau^2 + \bar\tau^2)}{60} = 1, \\ \langle \chi_3, \chi_{3'} \rangle &= \frac{9 + 15 + 24\tau\bar\tau}{60} = 0, \\ \langle \chi_{\mathrm{STF}}, \chi_{\mathrm{STF}} \rangle &= \frac{25 + 15 + 20}{60} = 1. \end{aligned}$$

Norm one proves irreducibility of the complexifications; consequently, it also proves absolute irreducibility of the constructed real representations. Their dimensions and orthogonality distinguish the three summands. Finally, the equality of class functions

$$\chi_{12} = 1 + \chi_3 + \chi_{3'} + \chi_{\mathrm{STF}}$$

is verified in all five columns. Semisimplicity of finite-group representations in characteristic zero gives decomposition (77.46). The multiplicity of the tensor summand can be checked directly:

$$\langle \chi_{12}, \chi_{\mathrm{STF}} \rangle = \frac{12 \cdot 5}{60} = 1.$$

□

Thus the central idempotent associated with $\chi_{\mathrm{STF}} = \chi_5$ has rank $5 \cdot 1 = 5$ in V_{12}. The irreducibility entering normalization of the tensor intertwiner and the multiplicity entering gravitational coupling are established by this same calculation.

The permutation representation ρ_{12} decomposes as

$$V_{12} \simeq \mathbf{1} \oplus V_3 \oplus V_{3'} \oplus V_5. \tag{77.48}$$

The central idempotent of the character summand $\chi_5 = (5, 1, -1, 0, 0)$ is

$$\boxed{P_5^{\mathrm{ico}} = \frac{5}{60} \sum_{g \in A_5} \chi_5(g^{-1}) \rho_{12}(g) = \frac{1}{12} \left(5I + \sum_{g \in 2A} \rho_{12}(g) - \sum_{g \in 3A} \rho_{12}(g) \right).} \tag{77.49}$$

Theorem 77.11 (Rank-five gravitational projector and coupler)**.** *The operator P_5^{ico} satisfies*

$$(P_5^{\mathrm{ico}})^2 = P_5^{\mathrm{ico}} = (P_5^{\mathrm{ico}})^*, \qquad \operatorname{Tr} P_5^{\mathrm{ico}} = \operatorname{rank} P_5^{\mathrm{ico}} = 5. \tag{77.50}$$

For

$$\mathbb{G}_{5,a} := G_a(\vartheta_{108}^\circ) P_5^{\mathrm{ico}} \tag{77.51}$$

we have

$$\operatorname{Spec}(\mathbb{G}_{5,a}) = \{ G_a(\vartheta_{108}^\circ) \text{ with multiplicity } 5,\ 0 \text{ with multiplicity } 7 \}. \tag{77.52}$$

Proof. Character orthogonality converts (77.49) into the central idempotent that acts as the identity on the unique copy of V_5 and as zero on the other three summands of (77.48). Reality of the character and orthogonality of ρ_{12} give self-adjointness. Its trace and rank are five. Multiplying a rank-five projector by the positive scalar G_a immediately produces the indicated spectrum. □

The gravitational section G_a is common to the five components of the selected tensor summand. Its spectral multiplicity expresses the rank of the carrier; the other seven modes of the permutation module belong to the kernel of $\mathbb{G}_{5,a}$. The symbol P_5^{ico} identifies here this rank-five central projector on V_{12}; the average of the five positions of an arithmetic pentafibre has another domain and rank one.

77.1.6 Intertwiner with the space of symmetric traceless tensors

Let $\mathcal{V} = \Omega_{12}^{\mathrm{ico}} \subset S^2$ be the already constructed set of twelve unit vertices. For each $v \in \mathcal{V}$, define

$$Q_v = vv^T - \frac{1}{3} I_3 \in \mathrm{STF}_3 := \mathrm{Sym}_0^2(\mathbb{R}^3). \tag{77.53}$$

The frame map

$$\Gamma_0\left(\sum_{v\in\mathcal{V}} x_v e_v\right) = \sum_{v\in\mathcal{V}} x_v Q_v \tag{77.54}$$

is A_5-equivariant and satisfies

$$\Gamma_0\Gamma_0^* = \frac{8}{5} I_{\mathrm{STF}_3}, \qquad \Gamma_0 = \Gamma_0 P_5^{\mathrm{ico}}. \tag{77.55}$$

Theorem 77.12 (Normalized tensor entangler)**.** *The map*

$$\boxed{\Gamma_5 := \sqrt{\frac{5}{8}}\,\Gamma_0|_{V_5} : V_5 \xrightarrow{\simeq} \mathrm{STF}_3} \tag{77.56}$$

is a A_5-equivariant isometry. Its adjoint is

$$\Gamma_5^* Q = \sqrt{\frac{5}{8}} \sum_{v\in\mathcal{V}} (v^T Q v) e_v. \tag{77.57}$$

Proof. Equivariance follows from $Q_{gv} = gQ_v g^T$. The operator $\Gamma_0\Gamma_0^*$ commutes with the irreducible representation of A_5 on STF_3, and is therefore scalar. Its trace is

$$\sum_{v\in\mathcal{V}} \|Q_v\|_F^2 = 12 \cdot \frac{2}{3} = 8;$$

dividing by dimension five yields $8/5$. Hence, $\Gamma_5\Gamma_5^* = I$. Since domain and codomain have dimension five, also $\Gamma_5^*\Gamma_5 = I$. The formula for the adjoint follows from $\langle Q, Q_v\rangle_F = v^T Q v$. □

Normalization through the icosahedral frame constructs the isometry that transports the finite summand to the real spin-two tensor space.

77.1.7 Transverse traceless projection

For $n \in S^2$, let $\Pi(n) = I - nn^T$. We define

$$\boxed{\Lambda(n)Q = \Pi Q \Pi - \frac{1}{2}\,\mathrm{Tr}(\Pi Q \Pi)\Pi.} \tag{77.58}$$

Theorem 77.13 (Reduction from five components to two)**.** *The restriction of $\Lambda(n)$ to STF_3 is the orthogonal projector onto*

$$\mathcal{H}_{\mathrm{TT}}(n) = \{Q \in \mathrm{STF}_3 : Qn = 0\}. \tag{77.59}$$

In particular,

$$\Lambda(n)^2 = \Lambda(n) = \Lambda(n)^*, \qquad \operatorname{rank}\Lambda(n) = 2. \tag{77.60}$$

For $n = e_3$, the image is generated by

$$E_+ = \frac{1}{\sqrt{2}}\,\mathrm{diag}(1,-1,0), \qquad E_\times = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{77.61}$$

Proof. Since $\Pi n = 0$, the output is transverse. Its trace is

$$\mathrm{Tr}(\Pi Q \Pi) - \frac{1}{2}\,\mathrm{Tr}(\Pi)\,\mathrm{Tr}(\Pi Q \Pi) = 0,$$

since $\operatorname{Tr}\Pi = 2$. If $Qn = 0$ and $\operatorname{Tr} Q = 0$, then $\Pi Q \Pi = Q$ and $\operatorname{Tr}(\Pi Q \Pi) = 0$, so that $\Lambda(n)Q = Q$. The formula is self-adjoint with respect to the Frobenius product. In the frame $n = e_3$, its matrix in an adapted STF basis is $\operatorname{diag}(1, 1, 0, 0, 0)$, which proves the rank. □

The complete chain thus remains,

$$\boxed{V_{12} \xrightarrow{P_5^{\mathrm{ico}}} V_5 \xrightarrow{\Gamma_5} \mathrm{STF}_3 \xrightarrow{\Lambda(n)} \mathcal{H}_{\mathrm{TT}}(n), \qquad 12 \longrightarrow 5 \longrightarrow 5 \longrightarrow 2.} \tag{77.62}$$

The reduced coupling

$$\mathbb{G}_{\mathrm{TT},a}(n) = G_a(\vartheta_{108}^\circ)\Lambda(n)\Gamma_5 P_5^{\mathrm{ico}} \tag{77.63}$$

has rank two.

77.1.8 Decomposition into 5 polarization modes

Dimension five acquires tensorial content upon choosing a unit direction of propagation n and an oriented orthonormal frame (e_1, e_2, n). Define the five real tensors

$$E_+ = \frac{1}{\sqrt{2}}(e_1 e_1^T - e_2 e_2^T), \qquad E_\times = \frac{1}{\sqrt{2}}(e_1 e_2^T + e_2 e_1^T), \tag{77.64}$$

$$E_1 = \frac{1}{\sqrt{2}}(e_1 n^T + n e_1^T), \qquad E_2 = \frac{1}{\sqrt{2}}(e_2 n^T + n e_2^T), \tag{77.65}$$

$$E_0 = \frac{1}{\sqrt{6}}(2nn^T - e_1 e_1^T - e_2 e_2^T). \tag{77.66}$$

All are symmetric and traceless. The first two are transverse; the next two mix the transverse plane with the direction of propagation, and the last is the traceless longitudinal scalar mode.

Theorem 77.14 (Complete helical decomposition of the carrier)**.** *The tensors of* (77.64)*,* (77.65)*,* (77.66) *form an orthonormal basis of* STF_3*. In the complexification, the combinations*

$$E_{\pm 2} = \frac{1}{\sqrt{2}}(E_+ \mp i E_\times), \qquad E_{\pm 1} = \frac{1}{\sqrt{2}}(E_1 \mp i E_2), \qquad E_0 = E_0 \tag{77.67}$$

are weight-$\pm 2, \pm 1, 0$ *vectors, respectively, under rotations about* n*. Moreover,*

$$\Lambda(n)E_{\pm 2} = E_{\pm 2}, \qquad \Lambda(n)E_{\pm 1} = 0, \qquad \Lambda(n)E_0 = 0. \tag{77.68}$$

Proof. In the frame (e_1, e_2, n), each tensor has Frobenius norm one. Their off-diagonal entry patterns are disjoint, and the three diagonals of E_+, E_0 have zero inner product; therefore, the five tensors are orthonormal. Since $\dim \mathrm{STF}_3 = 6 - 1 = 5$, they form a basis.

A rotation through angle ψ about n sends (e_1, e_2) to $(\cos\psi\, e_1 + \sin\psi\, e_2, -\sin\psi\, e_1 + \cos\psi\, e_2)$. Substituting into the formulas gives a rotation through angle 2ψ on $\operatorname{span}\{E_+, E_\times\}$, one through angle ψ on $\operatorname{span}\{E_1, E_2\}$, and E_0 fixed. Hence the weights of (77.67). Finally, $\Pi E_+ \Pi = E_+$ and $\Pi E_\times \Pi = E_\times$, whereas $\Pi E_1 \Pi = \Pi E_2 \Pi = 0$. For E_0, $\Pi E_0 \Pi = -(e_1 e_1^T + e_2 e_2^T)/\sqrt{6}$, which is pure trace in the plane, and subtracting (77.58) annihilates it. □

The theorem specifies the meaning of "five polarizations" without assigning all of them the same dynamic status. Before field equations are imposed, the spin-two carrier has five irreducible coordinates. In a massive Fierz–Pauli realization, the constraints leave precisely the five weights of (77.67); in a massless realization with gauge invariance, the sectors ± 1 and 0 are eliminated, leaving ± 2. Here HMT supplies the carrier, the basis, and the projector; the choice of massive or massless dynamics belongs to the physical interface. Thus five does not mean "five observed waves," but rather five tensorial degrees available before reduction.

The orientation also remains visible: replacing (e_1, e_2, n) with $(e_1, -e_2, -n)$ preserves E_+, E_0 and exchanges the phases of the helical pairs. That information is not in the scalar G_a; it resides in the carrier on which $\mathbb{G}_{5,a}$ acts.

77.1.9 Physical interpretation of the representational chain

Dimension five appears in distinct objects, and only (77.62) authorizes transport between them:

1. V_5 is the irreducible summand of the representation on the twelve vertices. This is a finite result in representation theory.
2. STF_3 is the real five-dimensional carrier of the $m = -2, -1, 0, 1, 2$ weights of spin two.
3. A massive Fierz–Pauli theory can realize these five weights as five physical states, after fixing the action and constraints.
4. Massless general relativity retains two radiative helicities, ± 2, represented by E_+ and $E_\times$.
5. The $E(2)$ classes of metric waves, such as III_5, classify possible amplitudes under other hypotheses; they are not automatically V_5, Fierz–Pauli, or general relativity.

The structural content is thus determined: under the incidence-descent conditions, HMT constructs a five-component tensor carrier before imposing constraints. The massless realization preserves the two radiative helicities through the rank-two TT projection. Orientation information is preserved in the tensors and in their transport; the scalar value G_a determines the common coupling.

Chapter 78

Intertwiner with symmetric traceless tensors and transverse reduction $12 \to 5 \to 5 \to 2$

The icosahedral framework constructs an equivariant isometry into the space of traceless symmetric tensors. The transverse projection realizes $12 \to 5 \to 5 \to 2$, preserves the five-component carrier, and localizes the radiative subspace of dimension 2.

78.1 Intertwiner with the space of symmetric traceless tensors

Let $\mathcal{V} \subset S^2$ be the set of the twelve unit vertices of the icosahedron. For each $v \in \mathcal{V}$, define

$$Q_v = vv^T - \frac{1}{3} I_3 \in \mathrm{STF}_3 := \mathrm{Sym}^2_0(\mathbb{R}^3). \tag{78.1}$$

The frame map

$$\Gamma_0\left(\sum_{v\in\mathcal{V}} x_v e_v\right) = \sum_{v\in\mathcal{V}} x_v Q_v \tag{78.2}$$

is A_5-equivariant and satisfies

$$\Gamma_0\Gamma_0^* = \frac{8}{5} I_{\mathrm{STF}_3}, \qquad \Gamma_0 = \Gamma_0 P_5. \tag{78.3}$$

Theorem 78.1 (Normalized tensor entangler)**.** *The map*

$$\boxed{\Gamma_5 := \sqrt{\frac{5}{8}}\,\Gamma_0|_{V_5} : V_5 \xrightarrow{\simeq} \mathrm{STF}_3} \tag{78.4}$$

is a A_5-equivariant isometry. Its adjoint is

$$\Gamma_5^* Q = \sqrt{\frac{5}{8}} \sum_{v\in\mathcal{V}} (v^T Q v) e_v. \tag{78.5}$$

Proof. Equivariance follows from $Q_{gv} = gQ_v g^T$. The operator $\Gamma_0\Gamma_0^*$ commutes with the irreducible representation of A_5 on STF_3, and is therefore scalar. Its trace is

$$\sum_{v\in\mathcal{V}} \|Q_v\|_F^2 = 12 \cdot \frac{2}{3} = 8;$$

dividing by dimension five yields $8/5$. Hence, $\Gamma_5\Gamma_5^* = I$. Since domain and codomain have dimension five, also $\Gamma_5^*\Gamma_5 = I$. The formula for the adjoint follows from $\langle Q, Q_v\rangle_F = v^T Q v$. □

Normalization through the icosahedral frame eliminates a purely cardinal coincidence: it constructs the map that transports the finite summand to the continuous spin-two carrier.

78.2 Transverse traceless projection

For $n \in S^2$, let $\Pi(n) = I - nn^T$. We define

$$\boxed{\Lambda(n)Q = \Pi Q \Pi - \frac{1}{2}\operatorname{Tr}(\Pi Q \Pi)\Pi.} \tag{78.6}$$

Theorem 78.2 (Reduction from five components to two)**.** *The restriction of* $\Lambda(n)$ *to* STF_3 *is the orthogonal projector onto*

$$\mathcal{H}_{\mathrm{TT}}(n) = \{Q \in \mathrm{STF}_3 : Qn = 0\}. \tag{78.7}$$

In particular,

$$\Lambda(n)^2 = \Lambda(n) = \Lambda(n)^*, \qquad \operatorname{rank}\Lambda(n) = 2. \tag{78.8}$$

For $n = e_3$*, the image is generated by*

$$E_+ = \frac{1}{\sqrt{2}}\operatorname{diag}(1, -1, 0), \qquad E_\times = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{78.9}$$

Proof. Since $\Pi n = 0$, the output is transverse. Its trace is

$$\operatorname{Tr}(\Pi Q \Pi) - \frac{1}{2}\operatorname{Tr}(\Pi)\operatorname{Tr}(\Pi Q \Pi) = 0,$$

since $\operatorname{Tr}\Pi = 2$. If $Qn = 0$ and $\operatorname{Tr} Q = 0$, then $\Pi Q \Pi = Q$ and $\operatorname{Tr}(\Pi Q \Pi) = 0$, so that $\Lambda(n)Q = Q$. The formula is self-adjoint with respect to the Frobenius product. In the frame $n = e_3$, its matrix in an adapted STF basis is $\operatorname{diag}(1, 1, 0, 0, 0)$, which proves the rank. □

The complete chain thus remains,

$$\boxed{V_{12} \xrightarrow{P_5} V_5 \xrightarrow{\Gamma_5} \mathrm{STF}_3 \xrightarrow{\Lambda(n)} \mathcal{H}_{\mathrm{TT}}(n), \qquad 12 \longrightarrow 5 \longrightarrow 5 \longrightarrow 2.} \tag{78.10}$$

The reduced coupling

$$\mathbb{G}_{\mathrm{TT},a}(n) = G_a(\theta_{108}^{\mathrm{circ}})\Lambda(n)\Gamma_5 P_5 \tag{78.11}$$

has rank two.

Helical decomposition of the spectrum and separation of massive and non-massive modes. The reduction makes it possible to distinguish the complete helicoidal spectrum of a massive realization and its two massless helicities.

78.3 Oriented incidence and gravitational tensor representation

The exceptional publication of the state preserves the incidence accompanying its arithmetic reading. The ternary matrix A_W of (21.126) provides a precise material antecedent here. Its balanced lift $C = \mathrm{bal}(A_W)$, with representatives $0, 1, -1$, satisfies

$$C = C^{\mathsf{T}}, \qquad C^2 = 5I_6, \qquad \operatorname{Tr} C = 0.$$

The six positions have the order $(\infty, 0, 1, 2, 3, 4)$. This incidence first produces a three-dimensional space; its quadratic reading then produces the five-component space on which the gravitational coupling acts.

78.3.1 Signed covering and calendar transport

We define

$$E_C = \frac{1}{2}(I_6 + C/\sqrt{5}), \quad \mathcal{H}_C = \operatorname{im} E_C, \quad v_{u,\sigma} = \sigma\sqrt{2}E_C e_u, \quad \sigma \in \{1,-1\}.$$

Proposition 78.3 (Twelve oriented positions)**.** *The vectors $v_{u,\sigma}$ are twelve distinct unit vectors in a three-dimensional space. Their incidence is determined by $\langle v_{u,\sigma}, v_{v,\tau}\rangle = 1/\sqrt{5}$. Orientation reversal acts by antipodality.*

Proof. The quadratic identity for C implies $E_C^2 = E_C = E_C^*$; its trace is three. Since the diagonal of C is zero,

$$\langle v_{u,\sigma}, v_{v,\tau}\rangle = \begin{cases} \sigma\tau, & u = v, \\ \sigma\tau C_{uv}/\sqrt{5}, & u \neq v. \end{cases}$$

This proves normalization, distinction of the twelve vectors and incidence. Changing σ to $-\sigma$ changes the sign of the vector. □

An explicit Euclidean expression is obtained by setting $\nu = \sqrt{1+\varphi^2}$ and

$$B = \frac{1}{\nu}\begin{pmatrix} 0 & -1 & -\varphi & 0 & \varphi & 1 \\ -1 & -\varphi & 0 & 1 & 0 & -\varphi \\ -\varphi & 0 & -1 & -\varphi & -1 & 0 \end{pmatrix}.$$

The relation $\varphi^2 = \varphi + 1$, for the coordinate already generated, gives $B^*B = 2E_C$. Consequently $B/\sqrt{2}|_{\mathcal{H}_C}$ is an isometry and takes $v_{u,\sigma}$ to $\sigma B e_u$. The image is

$$\mathcal{V} = \frac{1}{\nu}\big(\{0\}\times\{\pm1\}\times\{\pm\varphi\} \;\cup\; \{\pm1\}\times\{\pm\varphi\}\times\{0\} \;\cup\; \{\pm\varphi\}\times\{0\}\times\{\pm1\}\big).$$

The internal coordinate, the incidence and its twelve geometric representatives are thus related.

The calendar coordinate is written $(b, r) \in \mathbb{Z}/12\mathbb{Z} \times \{0,\dots,8\}$. A route of length n induces

$$k(r,n) = \left\lfloor \frac{r+n}{9} \right\rfloor, \qquad (b,r) \longmapsto \big(b + k(r,n) \bmod 12,\ (r+n) \bmod 9\big).$$

The explicit alignment of its positions $p = b + 1$ with the signed chart is

p	1	2	3	4	5	6	7	8	9	10	11	12
u	∞	0	0	∞	1	3	3	1	2	4	4	2
σ	+	−	+	−	−	+	−	+	−	+	−	+

This table fixes a trivialization of the carrier. Each change of position jointly transports the vectors and their incidence matrix.

Proposition 78.4 (Covariance of the frame under return)**.** *If $Se_p = e_{p+1 \bmod 12}$, the matrix representation of an operator P of the carrier is transported as $P_k = S^k P S^{-k}$. This transport preserves inner products, incidences and route composition. After twelve returns the chart returns, preserving the memory accumulated in the source state.*

Proof. S is orthogonal and $S^{12} = I$; the metric properties and periodicity of the chart follow. For composition, with $r' = (r+n) \bmod 9$, Euclidean division gives $k(r, n+m) = k(r,n) + k(r',m)$. The exponents of S are added exactly with the carry. The operation acts on the calendar coordinate; the route's memory updates remain present in its enriched domain. □

78.3.2 Quadratic reading and five-component space

Let $\mathbb{R}^{\mathcal{V}}$ be the space of the twelve coordinates, $\mathsf{A}e_v = e_{-v}$ its antipodal involution and $\mathsf{J} = \mathbf{1}\mathbf{1}^{\mathsf{T}}$. We define

$$P_5 = \frac{1}{2}(I + \mathsf{A}) - \frac{1}{12}\mathsf{J}, \qquad Q_v = vv^{\mathsf{T}} - \frac{1}{3}I_3, \qquad \Gamma_0 x = \sum_{v\in\mathcal{V}} x_v Q_v.$$

Theorem 78.5 (Five-component projector and tensor isometry)**.** *P_5 is the rank-five orthogonal projector onto the space of coordinates even under antipodality and of zero sum. Moreover,*

$$\Gamma_0^*\Gamma_0 = \frac{8}{5}P_5, \qquad \Gamma_0\Gamma_0^* = \frac{8}{5}I_{\mathrm{Sym}_0^2(\mathbb{R}^3)}.$$

Therefore

$$\Gamma_5 = \sqrt{\frac{5}{8}}\,\Gamma_0|_{\mathrm{im}\,P_5}$$

is an isometry onto the space of traceless symmetric tensors.

Proof. The even space has one coordinate for each antipodal pair, so its dimension is six. The constant vector belongs to it; subtracting its rank-one projector gives P_5, of rank five.

Since $\langle Q_v, Q_w\rangle_F = (v^{\mathsf{T}}w)^2 - 1/3$, the entries of $\Gamma_0^*\Gamma_0$ equal $2/3$ for $w = \pm v$ and $-2/15$ in the other ten positions of each row. These are exactly the entries of $(8/5)P_5$.

The identity can also be checked in tensor space without resorting to a classification of representations. The list of vertices gives

$$\sum_v v_i^4 = \frac{12}{5}, \qquad \sum_v v_i^2 v_j^2 = \frac{4}{5} \quad (i \neq j),$$

and moments with any odd power vanish by sign changes. Indeed, $(1+\varphi^2)^2 = 5\varphi^2$ and $1 + \varphi^4 = 3\varphi^2$. For a symmetric tensor Q, these sums give

$$\sum_v (v^{\mathsf{T}}Qv)\, vv^{\mathsf{T}} = \frac{8}{5}Q + \frac{4}{5}(\mathrm{tr}\,Q)I.$$

With $\mathrm{tr}\,Q = 0$, and using $\sum_v v^{\mathsf{T}}Qv = 4\,\mathrm{tr}\,Q = 0$, we obtain $\Gamma_0\Gamma_0^*Q = (8/5)Q$. The two identities prove the normalized isometry and its surjectivity. □

The gravitational section in 121 determines the operator $G_a P_5$. Its spectrum has G_a with multiplicity five and zero with multiplicity seven. Orientation is preserved in the carrier and its transport; G_a fixes the common coupling.

78.3.3 Transverse projection and helicities

For $n \in S^2$, set $\Pi = I - nn^{\mathsf{T}}$ and

$$\Lambda_n Q = \Pi Q \Pi - \frac{1}{2}\,\mathrm{tr}(\Pi Q \Pi)\Pi.$$

Proposition 78.6 (Rank-two transverse reduction)**.** *Λ_n is the orthogonal projector onto $\{Q \in \mathrm{Sym}_0^2(\mathbb{R}^3) : Qn = 0\}$, of dimension two.*

Proof. $\Pi n = 0$ gives transversality and $\mathrm{tr}\,\Pi = 2$ gives zero trace. A transverse traceless tensor remains fixed. The formula is self-adjoint for the Frobenius inner product. In a frame (e_1, e_2, n), the image has orthonormal basis

$$E_+ = \frac{e_1e_1^{\mathsf{T}} - e_2e_2^{\mathsf{T}}}{\sqrt{2}}, \qquad E_\times = \frac{e_1e_2^{\mathsf{T}} + e_2e_1^{\mathsf{T}}}{\sqrt{2}}.$$

□

The other three vectors of an orthonormal basis of the carrier are

$$E_1 = \frac{e_1 n^{\mathsf{T}} + n e_1^{\mathsf{T}}}{\sqrt{2}}, \quad E_2 = \frac{e_2 n^{\mathsf{T}} + n e_2^{\mathsf{T}}}{\sqrt{2}}, \quad E_0 = \frac{2 n n^{\mathsf{T}} - e_1 e_1^{\mathsf{T}} - e_2 e_2^{\mathsf{T}}}{\sqrt{6}}.$$

When the transverse frame is rotated by an angle θ, the pairs $(E_+, E_\times)$ and (E_1, E_2) rotate by 2θ and θ, respectively, while E_0 remains fixed. The complexification thus has weights $2, -2, 1, -1, 0$. Λ_n preserves the first two and annihilates the rest. The chain is established

$$\mathbb{R}^{12} \xrightarrow{P_5} \operatorname{im} P_5 \xrightarrow{\Gamma_5} \operatorname{Sym}_0^2(\mathbb{R}^3) \xrightarrow{\Lambda_n} \mathcal{H}_{\mathrm{TT}}(n), \qquad 12 \longrightarrow 5 \longrightarrow 5 \longrightarrow 2.$$

The five-component representation and its reduction are algebraic results preceding the choice of field equations. In the massless gravitational realization, the gauge constraints select the transverse image. This distinction preserves both the five structural coordinates and the two radiative helicities.

Chapter 79

Helical decomposition of the spin carrier 2 and distinction between massive and massless realizations

The real carrier of dimension 5 decomposes into 5 helicity weights. The massive realization retains all 5 components; the massless transverse reduction retains only the radiative helicities ±2, without converting a collective possibility into a postulated fundamental particle.

79.1 Decomposition into 5 polarization modes

Dimension five acquires tensorial content upon choosing a unit direction of propagation n and an oriented orthonormal frame (e_1, e_2, n). Define the five real tensors

$$E_+ = \frac{1}{\sqrt{2}}(e_1 e_1^T - e_2 e_2^T), \qquad E_\times = \frac{1}{\sqrt{2}}(e_1 e_2^T + e_2 e_1^T), \tag{79.1}$$

$$E_1 = \frac{1}{\sqrt{2}}(e_1 n^T + n e_1^T), \qquad E_2 = \frac{1}{\sqrt{2}}(e_2 n^T + n e_2^T), \tag{79.2}$$

$$E_0 = \frac{1}{\sqrt{6}}(2nn^T - e_1 e_1^T - e_2 e_2^T). \tag{79.3}$$

All are symmetric and traceless. The first two are transverse; the next two mix the transverse plane with the direction of propagation, and the last is the traceless longitudinal scalar mode.

Theorem 79.1 (Complete helical decomposition of the carrier)**.** *The tensors of equations (79.1) a (79.3) form an orthonormal basis of* STF_3*. In the complexification, the combinations*

$$E_{\pm 2} = \frac{1}{\sqrt{2}}(E_+ \mp \mathrm{i}E_\times), \qquad E_{\pm 1} = \frac{1}{\sqrt{2}}(E_1 \mp \mathrm{i}E_2), \qquad E_0 = E_0 \tag{79.4}$$

are weight-$\pm 2, \pm 1, 0$ *vectors, respectively, under rotations about n. Moreover,*

$$\Lambda(n)E_{\pm 2} = E_{\pm 2}, \qquad \Lambda(n)E_{\pm 1} = 0, \qquad \Lambda(n)E_0 = 0. \tag{79.5}$$

Proof. In the frame (e_1, e_2, n), each tensor has Frobenius norm one. Their off-diagonal entry patterns are disjoint, and the three diagonals of E_+, E_0 have zero inner product; therefore, the five tensors are orthonormal. Since $\dim \mathrm{STF}_3 = 6 - 1 = 5$, they form a basis.

A rotation through angle ψ about n sends (e_1, e_2) to $(\cos\psi\, e_1 + \sin\psi\, e_2, -\sin\psi\, e_1 + \cos\psi\, e_2)$. Substituting into the formulas gives a rotation through angle 2ψ on $\mathrm{span}\{E_+, E_\times\}$, one through angle ψ on $\mathrm{span}\{E_1, E_2\}$, and E_0 fixed. Hence the weights of equation (79.4). Finally, $\Pi E_+ \Pi = E_+$ and $\Pi E_\times \Pi = E_\times$, whereas $\Pi E_1 \Pi = \Pi E_2 \Pi = 0$. For E_0, $\Pi E_0 \Pi = -(e_1 e_1^T + e_2 e_2^T)/\sqrt{6}$, which is pure trace in the plane, and subtracting equation (78.6) annihilates it. □

The theorem specifies the meaning of "five polarizations" without assigning all of them the same dynamic status. Before field equations are imposed, the spin-two carrier has five irreducible coordinates. In a massive Fierz–Pauli realization, the constraints leave precisely the five weights of equation (79.4); in a massless realization with gauge invariance, the sectors ±1 and 0 are eliminated, leaving ±2. Here HMT supplies the carrier, the basis, and the projector; the choice of massive or massless dynamics belongs to the physical interface. Thus five does not mean "five observed waves," but rather five tensorial degrees available before reduction.

The orientation also remains visible: replacing (e_1, e_2, n) with $(e_1, -e_2, -n)$ preserves E_+, E_0 and exchanges the phases of the helical pairs. That information is not in the scalar G_a; it resides in the carrier on which $\mathbb{G}_{5,a}$ acts.

79.2 Representational chain $12 \to 5 \to 5 \to 2$: carrier V_5, trace-free tensor, Fierz–Pauli polarizations, and transverse helicities

Dimension five appears in distinct objects, and only equation (78.10) authorizes transport between them:

1. V_5 is the irreducible summand of the representation on the twelve vertices. This is a finite result in representation theory.
2. STF_3 is the real five-dimensional carrier of the $m = -2, -1, 0, 1, 2$ weights of spin two.
3. A massive Fierz–Pauli theory can realize these five weights as five physical states, after fixing the action and constraints.
4. Massless general relativity retains two radiative helicities, ± 2, represented by E_+ and $E_\times$.
5. The $E(2)$ classes of metric waves, such as III_5, classify possible amplitudes under other hypotheses; they are not automatically V_5, Fierz–Pauli, or general relativity.

In particular, the correct statement is: *HMT constructs a five-component tensor carrier before the restrictions; the TT reduction of the massless realization has rank two*. It is not asserted that general relativity has five propagating polarizations.

Existence conditions of the mode through natural holonomic selection. The physical existence of the mode still requires a natural and falsifiable holonomic selector.

Chapter 80

Gravitational Holonomic Selector: Dimensional Transduction, Naturality, Refinement, and Separation of the Torsional and Circular Regimes

The gravitational selector is formulated on enriched holonomies and commutes with the refinement maps. Its classification distinguishes selection determined by the declared normalizations from the obstruction to canonicity that appears when such data are omitted.

80.1 Holonomic conditions of the general gravitational selector

The second gravitational template starts from

$$\delta_\theta = \frac{A_{\mathrm{deg}} - C^*_{\mathrm{deg}}}{360} \tag{80.1}$$

and a route length $\Lambda_{27} = \mathcal{L}_{\mathrm{hol}}(\widetilde{\mathcal{K}}_{27})$ in an enriched lift of the return. The expression

$$\boxed{G_{\mathrm{hol}} = \frac{c_{\mathrm{int}}^3}{\hbar_{\mathrm{int}}}\left(\delta_\theta \Lambda_{27}\right)^2} \tag{80.2}$$

is dimensionally exact. Its promotion to a general selector requires that $\mathcal{L}_{\mathrm{hol}}$ be natural under cyclic change of basis, conjugation, relabeling, and subdivision, and possess a normalization that is unique and stable under refinement.

There is additionally a torsion control: every positive multiplicative character $\chi : K_{27} \to \mathbb{R}_+^\times$ on a finite observable cycle of order four is trivial, because $\chi(g)^4 = 1$ and $\chi(g) > 0$ imply $\chi(g) = 1$. A nontrivial holonomic response must therefore be defined on the nontorsional lift or on enriched memory; it cannot be obtained by renaming a positive character of the finite cycle.

The chart

$$10^{-11}\left(6 + \frac{674}{1000} + \frac{30}{10^5}\right) = 6.67430 \times 10^{-11} \tag{80.3}$$

is an exact evaluation of declared inputs. It does not fix Λ_{27} or prove that equation (80.2) assumes that value.

The periodicity condition determines G without using its decimal expansion: the chronological structure CT108 selects $(54/\pi)^2$, the gravitational line fixes the dimensional type, and the bidirectional family preserves $G\hbar$. The resulting coupling is represented on the tensorial carrier constructed by P_5, whereas the reduction to the physical trace-free transverse sector is effected by a distinct rank-two map.

Falsification criterion 80.1. The circular block is refuted if $r_g(\theta) \neq \theta(\pi/54)\ell_0$, if there exists another positive solution of equation (77.6), or if $G_a\hbar_a/c^3 \neq \theta_{108}^{\mathrm{circ}}\ell_0^2$. The tensorial block is refuted if π_{ico} does not preserve incidence or intertwine the declared action, if its stabilizer is not C_5, if P_5 is not a self-adjoint idempotent of trace five, if $\Gamma_0\Gamma_0^* \neq (8/5)I$, or if $\Lambda(n)$ does not have rank two. The five-polarization reading is refuted if the tensors of equations (79.1) a (79.3) are not an orthonormal basis or if $\Lambda(n)$ retains a weight other than ± 2. The general holonomic selector is resolved by obstruction: without a natural, normalized, and stable $\mathcal{L}_{\mathrm{hol}}$ there is no univocal selection compatible with all relabelings; using equation (80.3) to choose it incurs circularity and invalidates the construction.

Hypothesis, domain, and conclusion

Transducer, uniqueness of the selector under the condition of CT108 periodicity, and Planck family: *Exact internal theorem under the stated geometric condition*. The descent $\Omega_{12}^{\mathrm{ico}} \simeq A_5/C_5$ is exact under the four hypotheses stated explicitly in π_{ico}; the HMT construction of that map belongs to the exceptional provenance corridor. Once the descent, projector P_5, intertwiner Γ_5, five-helicity basis, TT reduction, and spectrum of $\mathbb{G}_5$ are fixed: *Internal exact theorem*. Massive reading, field equations, and realization of Palatini–Cartan–Holst: *Realization, Physically Typed*. Template holonomic: *Proved in homogeneity*; the nonexistence of a general unambiguous selector without additional normalization is a *obstruction of canonicity proved*. Carta SI: *Contrast metrological external*.

80.1.1 Criteria for selecting the gravitational coordinate: enriched holonomy, spectral response, and metrological chart

The APP–TRIT–TPK composition, the enriched state and the action branch first construct the gravitational coordinate. The gravitational constant is organized into three typed levels: the dimensional line that houses G, the transducer that converts a dimensionless coordinate into a section of that line and the structural selector of that coordinate. The Palatini–Cartan–Holst action and its constrained Hamiltonian then realize the already generated output. The decimal chart and experimental uncertainty are presented at the end as representation and comparison.

80.1.1.1 Dimensional line and internal transducer

In the dimensional lattice of Dimensional Mechanics,

$$\mathbf{G} = -\mathbf{S} + 5\mathbf{C} + 2\mathbf{T}, \qquad [G] = L^3 M^{-1} T^{-2}.$$

Let

$$c_{\mathrm{int}} \in \mathcal{L}_C^+, \qquad t_0 \in \mathcal{L}_T^+, \qquad \ell_0 = c_{\mathrm{int}} t_0, \qquad \hbar_{\mathrm{int}} \in \mathcal{L}_S^+.$$

Definition 80.2 (Gravitational transducer)**.** For $\theta_G > 0$, define

$$\boxed{G_{\mathrm{int}}(\theta_G) = \theta_G \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_{\mathrm{int}}} = \theta_G \frac{c_{\mathrm{int}}^3 \ell_0^2}{\hbar_{\mathrm{int}}} \in \mathcal{L}_G^+.} \tag{80.4}$$

The equality of the two expressions and the type of (80.4) are exact. If

$$L_G^2 := \frac{\hbar_{\mathrm{int}} G_{\mathrm{int}}}{c_{\mathrm{int}}^3},$$

then

$$\theta_G = \left(\frac{L_G}{\ell_0}\right)^2. \tag{80.5}$$

The gravitational coordinate is therefore a quadratic ratio of lengths or a normalized holonomy response. The decade $10^{-34}\mathcal{U}_S$ already belongs to $\hbar_{\mathrm{int}}$: it fixes the absolute action section and does not constitute a relative factor added to G.

80.1.1.2 Observable torsion and positive characters

The observable return of 27 steps

$$\mathcal{K}_{27} := F_{\mathrm{obs}}^{27}$$

satisfies

$$\mathcal{K}_{27}^4 = F_{\mathrm{obs}}^{108} = I.$$

In coordinate $\theta \in \mathbb{Z}/108\mathbb{Z}$, the order is exactly four, since $\mathcal{K}_{27}(\theta) = \theta + 27$, $\mathcal{K}_{27}^2(\theta) = \theta + 54 \neq \theta$, and $\mathcal{K}_{27}^4(\theta) = \theta$.

Theorem 80.3 (Triviality of the Observable Positive Character)**.** *Every multiplicative homomorphism*

$$\chi : \langle \mathcal{K}_{27} \rangle \longrightarrow \mathbb{R}_{>0}^{\times}$$

cumple $\chi(\mathcal{K}_{27}) = 1$.

Proof. Multiplicativity implies $\chi(\mathcal{K}_{27})^4 = \chi(I) = 1$. The only positive real number whose fourth power is one is 1. □

The same cycle does carry the unitary characters.

$$\chi_j(\mathcal{K}_{27}) = \exp(2\pi i j/4), \qquad j = 0, 1, 2, 3.$$

The observable quotient admits phase, whereas its torsion eliminates a nontrivial positive multiplicative scale. A multiplicative continuation must therefore act on a lift $\widetilde{\mathcal{K}}_{27}$ whose class in the abelianization of the isotropy group is nontorsional, and must naturally select both the character and its normalization.

80.1.1.3 Chronological memory and spectral response

The directed chronological covering realizes the return of 27 steps as the translation

$$(G_{\mathrm{or}}, S) \longmapsto (G_{\mathrm{or}}, S) + (1, 18),$$

of infinite semigroup order. This route preserves accumulation, but its spectral response requires interference among histories.

Let $\Gamma = (V, E)$ be a finite directed multigraph with positive measure, weights $a_*(e) \geq 0$, and cocycle $c(e) \in \mathbb{Z}^2$. The twisted operator

$$(\mathcal{L}_{\beta,\vartheta} f)(y) = \sum_{e:x\to y} e^{-\beta a_*(e)} e^{i\langle \vartheta, c(e)\rangle} f(x)$$

induce the operator positive $\mathcal{H}_{\beta,\vartheta} = \mathcal{L}_{\beta,\vartheta}^* \mathcal{L}_{\beta,\vartheta}$.

Proposition 80.4 (Interference criterion for the phase response)**.** *In a deterministic orbit with a unique incoming edge at each vertex, $\mathcal{H}_{\beta,\vartheta}$ is independent of ϑ. Every nonzero-phase spectral response requires multiple coherent histories or a magnetic differential whose holonomy survives in the enriched cycles.*

Proof. On the deterministic orbit, the sole phase of each term is multiplied by its conjugate in $\mathcal{L}_{\beta,\vartheta}^* \mathcal{L}_{\beta,\vartheta}$ and cancels. The absence of cross terms removes every dependence on ϑ. □

If a simple band $\lambda_0(\vartheta)$ has a minimum at the origin, its Hessian Q is positive semidefinite. With

$$v_{27} = (1, 18), \qquad \mathcal{R}_{27} = v_{27}^{\mathsf{T}} Q v_{27},$$

the condition $\mathcal{R}_{27} \neq 0$ constitutes the minimal control of sensitivity to accumulated memory. A natural normalization $\mathfrak{N}$ would allow one to define

$$\Lambda_G = \ell_0 \mathfrak{N}(Q, v_{27}), \qquad G_{\mathrm{spec}} = \frac{c_{\mathrm{int}}^3}{\hbar_{\mathrm{int}}} \big(\delta_\theta \Lambda_G\big)^2, \qquad \delta_\theta = \frac{A_{\mathrm{deg}} - C_{\mathrm{deg}}^*}{360}.$$

The homogeneity of this template is exact. Its physical realization is characterized by four positive conditions: an enriched multigraph, a nonzero response, stability under refinement, and independence under rescaling, together with a normalization $\mathfrak{N}$ fixed before the metrological contrast.

80.1.1.4 Geometric Functional on Enriched Holonomy

The second continuation defines directly a length functional.

$$\mathscr{L} : \mathrm{Loop}(\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}) \longrightarrow \mathcal{L}_L^+ \cup \{0\},$$

invariant under cyclic change of basepoint, conjugation, admissible relabeling, and subdivision. For an enriched lift, write

$$\Lambda_{27} := \mathscr{L}(\widetilde{\mathcal{K}}_{27}), \qquad \rho_{27} := \frac{\Lambda_{27}}{c_{\mathrm{int}} t_0}, \qquad \theta_G^{\mathrm{hol}} := (\delta_\theta \rho_{27})^2.$$

Transduction produces the homogeneous template

$$\boxed{G_{\mathrm{hol}} = \frac{c_{\mathrm{int}}^3}{\hbar_{\mathrm{int}}} \big(\delta_\theta \Lambda_{27}\big)^2.} \tag{80.6}$$

The concatenation law of this functional on the torsional cycle forms part of its definition and its proof. Physical selection is completed by constructing $\widetilde{\mathcal{K}}_{27}$ and $\mathscr{L}$, fixing a unique normalization, and verifying the stability of the output. The value $G_{\mathrm{CODATA\,2022}}$ enters exclusively afterward, in the metrological comparison.

Proposition 80.5 (Separation of the two gravitational selectors)**.** *The circular selector $\theta_{108}^{\mathrm{circ}} = (54/\pi)^2$, constructed by action–Compton–gravity closure, and the holonomic selector $\theta_G^{\mathrm{hol}} = (\delta_\theta \rho_{27})^2$ are two distinct typed realizations of the transducer G_{int}. They are not identified with one another, and the decimal chart $\mathscr{R}_G$ does not constitute an evaluation of G_{hol}.*

Proof. The first selector depends solely on the circular closure of order 108. The second additionally contains the spectral length $\Lambda_{27} = \mathscr{L}(\widetilde{\mathcal{K}}_{27})$. An equality between them would require computing Λ_{27} and proving the agreement of normalizations; no operation of the decimal reader performs that computation. Their domains and input data therefore distinguish them before any numerical comparison. □

80.1.1.5 Structural selection criterion

Comparison Criterion 80.6 (Structural selector of the gravitational constant)**.** An HMT selector of G is identified when it jointly satisfies:

1. enriched domain, composition, and orientation declared;
2. descent to a nontorsional Abelian class, or a geometric functional invariant under basepoint, conjugation, relabelling, and subdivision;
3. nontrivial positive response stable under refinement;
4. unique normalization independent of the operator's overall scale;
5. internal selection of the coordinate and its decimal position;
6. freezing of the code and entries before revealing the comparison value;
7. final realization as a section of $\mathcal{L}_G$ in a common dimensional chart.

The dimensionless ratios

$$\frac{\mathcal{G}_{\mathrm{tor}}}{\mathcal{I}_{\mathrm{tor}}} = 1, \qquad \frac{\mathcal{G}_{\mathrm{av}}}{\mathcal{I}_{\mathrm{av}}} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2}$$

describe distinct gravitational–inertial readers and remain compatible with a single dimensional section G. The construction of that section also uses $(\hbar_{\mathrm{int}}, c_{\mathrm{int}}, \ell_0)$ and the selector θ_G.

80.1.1.6 Evaluation of the decimal gravitational chart and subsequent metrological contrast

The decimal representation preserved in the corpus is

$$\mathscr{R}_G([n,k,d]_3;t,e) := 10^{-n}\left(k+\frac{t}{10^3}+\frac{e}{10^d}\right). \tag{80.7}$$

For the chart $[n,k,d]_3=[11,6,5]_3$, $t=674$ and $e=30$, the evaluation racional is

$$\boxed{\mathscr{R}_G = 10^{-11}\left(6+\frac{674}{1000}+\frac{30}{10^5}\right) = 6.67430\times10^{-11}.} \tag{80.8}$$

The triad, the two blocks, and the decimal position constitute the declared data of this chart. Their evaluation is exact; their structural selection is governed by test criterion 80.6.

After choosing the SI basis of $\mathcal{L}_G$, the current contrast is

$$\boxed{G_{\text{CODATA 2022}} = 6.67430(15)\times10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}.} \tag{80.9}$$

Field	Metrological Statement
Central value	$6.67430\times10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$
Standard uncertainty	$u(G)=0.00015\times10^{-11}=1.5\times10^{-15}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$
Incertidumbre relativa	$u_r(G)\simeq 2.2\times10^{-5}$
Source and temporality	2022 CODATA recommended values, published in 2024 [29]
Proof function	Subsequent external contrast; neither the unit nor the uncertainty enters into the rational evaluation of (80.8).

Reproducibility of the gravitational reader. An independent enumeration jointly verifies the dimensional type, the decimal reader, and the monodromy identities on the declared finite domain.

Proposition 80.7 (Causal chain of the gravitational constant)**.** *The covariant action, the restricted Hamiltonian, and the Einstein–Schrödinger equation govern the gravitational dynamics. The dimensional type and the transducer* (80.4) *are exact; the evaluation* (80.8) *is exact for its declared chart. The value, unit, and uncertainty of* (80.9) *constitute an external comparison with CODATA 2022. The autonomous selection of θ_G is typed by the enriched lift, the interfering response, and the holonomy functional subject to the test criterion 80.6.*

80.2 Dimensional Transduction and Holonomic Selection of the Gravitational Constant

80.2.1 Gravitational line and dimensional form

In the lattice of section 59.1, the dimension of the gravitational constant is

$$\mathbf{G} = -\mathbf{S}+5\mathbf{C}+2\mathbf{T}, \qquad [G]=L^3M^{-1}T^{-2}.$$

Let

$$c_{\text{int}}\in\mathcal{L}_C^+, \qquad t_0\in\mathcal{L}_T^+, \qquad \ell_0:=c_{\text{int}}t_0\in\mathcal{L}_L^+, \qquad \hbar_{\text{int}}\in\mathcal{L}_S^+.$$

The section $\hbar_{\text{int}}$ is obtained only after the decade determinantal reader -34 and the choice of a basis for the action line. It is not the dimensionless angular character preceding the return.

For every dimensionless character $\theta_G>0$, define

$$\boxed{G_{\text{int}}(\theta_G)=\theta_G\frac{c_{\text{int}}^5t_0^2}{\hbar_{\text{int}}}=\theta_G\frac{c_{\text{int}}^3\ell_0^2}{\hbar_{\text{int}}}\in\mathcal{L}_G^+.}$$

The equality of the two expressions and their type are INTERNALLY EXACT within the dimensional lattice. They do not select θ_G. If

$$L_G^2 := \frac{\hbar_{\mathrm{int}} G_{\mathrm{int}}}{c_{\mathrm{int}}^3},$$

then

$$\theta_G = \left(\frac{L_G}{\ell_0}\right)^2.$$

This latter identity locates the selector datum: the length ratio must be produced by a holonomy or spectral-response structure prior to the metrological chart.

80.2.2 Obstruction of the Positive Character on the Observable Return

The section 82.3.1 defines

$$\mathcal{K}_{27} := F_{\mathrm{obs}}^{27}.$$

The primary proposition proposition 6.11 proves

$$F_{\mathrm{obs}}^{108} = I;$$

Therefore, in the observable quotient,

$$\mathcal{K}_{27}^4 = I.$$

Theorem 80.8 (Torsional Obstruction for the Observable Positive Selector)**.** *Every multiplicative homomorphism*

$$\chi : \langle \mathcal{K}_{27} \rangle \longrightarrow \mathbb{R}_{>0}^{\times}$$

is trivial. In particular,

$$\chi(\mathcal{K}_{27}) = 1.$$

Proof. Multiplicativity and $\mathcal{K}_{27}^4 = I$ imply

$$\chi(\mathcal{K}_{27})^4 = \chi(I) = 1.$$

The sole positive real element whose fourth power is one is 1. □

The observable order is exactly four, and not merely a divisor of four: in the coordinate $\theta \in \mathbb{Z}/108\mathbb{Z}$,

$$\mathcal{K}_{27}(\theta) = \theta + 27, \qquad \mathcal{K}_{27}^2(\theta) = \theta + 54 \neq \theta, \qquad \mathcal{K}_{27}^4(\theta) = \theta.$$

The same cyclic group admits unitary characters.

$$\chi_k(\mathcal{K}_{27}) = \exp\left(\frac{2\pi i k}{4}\right), \qquad k = 0, 1, 2, 3,$$

and, in particular, the phase values $\pm i$. The typed conclusion is that the observable cycle can transport phase but not a nontrivial positive scale. These phase characters are not identified with the Paley operator J_W: they satisfy an analogous quadratic relation on distinct domains.

The obstruction does not depend on a metrological comparison. It affects the very form of the selector: a positive character on the observable cycle cannot produce a nontrivial θ_G. Moreover, a mere set of conjugacy classes is not, in general, a group and cannot be declared the domain of a character without first defining its composition.

80.2.2.1 Unique admissible multiplicative continuation

A multiplicative route can reopen only on an enriched object that does not inherit the observable torsion. For an object x of the memory groupoid, the correctly typed domain would be the isotropy group

$$H_x := \mathrm{Aut}_{\mathcal{G}_{\mathrm{mem}}}(x)$$

and, by multiplicativity towards an abelian group, its abelianization

$$H_x^{\mathrm{ab}} \longrightarrow \mathbb{R}_{>0}^{\times}.$$

Every positive character annihilates the torsion part of H_x^{ab}. Therefore, the proposal must construct a lift

$$\widetilde{\mathcal{K}}_{27} \in H_x, \qquad p(\widetilde{\mathcal{K}}_{27}) = \mathcal{K}_{27},$$

prove that its Abelian class is non-torsion and naturally select a character $\widetilde{\chi}_G$. The positive route over the observable cycle is thus closed by obstruction: without that non-torsion datum there is no nontrivial multiplicative selector. If every admissible lift satisfied $\widetilde{\mathcal{K}}_{27}^4 = I$, the multiplicative route would also be ruled out in the enriched domain.

80.2.3 Chronological composition of length, action, and spectrum in the determination of G

The theorem 82.6 provides a third route, distinct both from the obstructed observable character and from a possible lift in an isotropy group. On

$$\widetilde{O}_{108}^{+} = O_{108} \times \mathbb{N}^2,$$

the twenty-seven-step return sums $(1, 18)$, and its fourth power sums $(4, 72)$. This action belongs to a directed chronological category. It is not a lift of the groupoid or a field proved to be native to the complete enriched TPK state.

The already constructed positive functional,

$$\mathcal{A}_*(\gamma) = G_{\mathrm{or}}(\gamma) + \lambda_* S(\gamma),$$

is additive under concatenation of directed histories. It is not a character of the isotropy group, because its indicators do not change sign under a formal inversion. The completion $\mathbb{N}^2 \to \mathbb{Z}^2$ permits unitary characters to be introduced for harmonic analysis, but does not alter this fact.

$$\delta_\theta := \frac{A_{\mathrm{deg}} - C_{\mathrm{deg}}^*}{360}. \tag{80.10}$$

This oriented angular fraction is defined once the unique angle and counter-angle have been constructed. It will be used in the two subsequent dimensional templates and does not intervene retroactively in obtaining A_{deg} or C_{deg}^*.

80.2.3.1 Operator with torsion and deterministic obstruction to multiplicative continuation

For the exact control, let $\Gamma = (V, E)$ be an enriched directed *finite* multigraph, with measure $\mu : V \to \mathbb{R}_{>0}$, weights $a_*(e) \geq 0$, parameter $\beta \geq 0$, and cocycle $c(e) \in \mathbb{Z}^2$. For a fixed normalization, consider

$$(\mathcal{L}_{\beta,\vartheta} f)(y) = \sum_{e : x \to y} e^{-\beta a_*(e)} e^{i\langle \vartheta, c(e)\rangle} f(x), \qquad \vartheta \in \mathbb{T}^2,$$

and take the entire finite Hilbert space $\ell^2(V, \mu)$ as the domain. Then $\mathcal{L}_{\beta,\vartheta}$ is bounded and

$$\mathcal{H}_{\beta,\vartheta} := \mathcal{L}_{\beta,\vartheta}^* \mathcal{L}_{\beta,\vartheta} \geq 0$$

is self-adjoint. An extension to an infinite graph would additionally require bounded operators, or densely defined closed operators such that $\mathcal{L}^{*}_{\beta,\vartheta}\mathcal{L}_{\beta,\vartheta}$ were self-adjoint on a common domain; that extension is not assumed here.

Proposition 80.9 (Phase cancellation in the deterministic orbit)**.** *If Γ is solely the observable deterministic orbit and every vertex has a single incoming edge, then $\mathcal{H}_{\beta,\vartheta}$ is independent of ϑ. In particular, the Hessian of any of its bands with respect to ϑ is null.*

Proof. In each component, $\mathcal{L}_{\beta,\vartheta}$ multiplies the sole incoming term by a unitary phase. In $\mathcal{L}^{*}_{\beta,\vartheta}\mathcal{L}_{\beta,\vartheta}$, that phase is multiplied by its conjugate and cancels. No cross-terms remain between distinct histories. □

This control prevents spectral rigidity from being attributed to the mere deterministic cycle. A nontrivial phase response requires at least one of the following constructions:

1. an enriched multigraph in which two or more coherent histories contribute to the same amplitude; or
2. to magnetic differential

$$(d_{\vartheta}f)(e) = e^{i\langle\vartheta,c(e)\rangle/2}f(t(e)) - e^{-i\langle\vartheta,c(e)\rangle/2}f(s(e)),$$

with Laplacian $\Delta_{\vartheta} = d^{*}_{\vartheta}d_{\vartheta}$, whose phase depends on the holonomy of enriched cycles.

In both cases, the Hilbert space, domain, measure, weights, and normalization must be declared, and it must be proved that the family is not unitarily equivalent to the family without phase.

Suppose that $\vartheta \mapsto \mathcal{H}_{\beta,\vartheta}$ is an analytic self-adjoint family on a common domain and that $\lambda_0(\vartheta)$ is a simple isolated band in a neighborhood of $\vartheta = 0$. If 0 is a critical point and a local minimum,

$$Q_{ij} = \left.\frac{\partial^2\lambda_0(\vartheta)}{\partial\vartheta_i\partial\vartheta_j}\right|_{\vartheta=0}$$

is positive semidefinite. Without the minimum hypothesis, Q represents only the quadratic response. For

$$v_{27} = (1, 18), \qquad \mathcal{R}_{27} = v_{27}^{\mathsf{T}}Qv_{27},$$

the condition $\mathcal{R}_{27} \neq 0$ is a minimal test of spectral sensitivity to the accumulated memory.

80.2.3.2 Typed normalization of the spectral response and construction of θ_G^{spec}

If a natural and unique normalization $\mathfrak{N}$ converts the preceding response into a positive ratio independent of the operator's global scale, one may define

$$\Lambda_G := \ell_0\,\mathfrak{N}(Q, v_{27}), \qquad \theta_G^{\mathrm{spec}} := \left(\delta_\theta\frac{\Lambda_G}{\ell_0}\right)^2,$$

and, by the dimensional transducer,

$$\boxed{G_{\mathrm{spec}} = \frac{c_{\mathrm{int}}^3}{\hbar_{\mathrm{int}}}\left(\delta_\theta\Lambda_G\right)^2.}$$

The homogeneity of the template is exact. The class of its evaluations is characterized by the conditions of canonical operator selection, nonzero response, naturality and stability under refinement, independence of rescaling, uniqueness of $\mathfrak{N}$, and sealing prior to any query of the metrological value of G. Without those conditions, rescaling invariance proves the nonuniqueness of the selector; with them, the evaluation is determined on the declared normalized domain.

80.2.4 Nonhomomorphic functional over enriched paths

The second possible continuation abandons the multiplicative character and uses a geometric, spectral, or action functional. Let

$$\mathscr{L} : \mathrm{Loop}(\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}) \longrightarrow \mathcal{L}_L^+ \cup \{0\}$$

invariant under cyclic change of base point, admissible relabeling, conjugation, and subdivision. Its concatenation law must be stated explicitly—for example, as a subadditive norm or a spectral functional—; it is not presumed to be a homomorphism of the torsion group. For a candidate enriched lift, we write

$$\Lambda_{27} := \mathscr{L}(\widetilde{\mathcal{K}}_{27}), \qquad \rho_{27} := \frac{\Lambda_{27}}{c_{\mathrm{int}} t_0}.$$

The oriented angular fraction of (80.10) produces the dimensionless candidate

$$\theta_G^{\mathrm{hol}} := (\delta_\theta \rho_{27})^2.$$

His specialization in the transducer gives

$$\boxed{G_{\mathrm{hol}} = \frac{c_{\mathrm{int}}^3}{\hbar_{\mathrm{int}}} \big(\delta_\theta \Lambda_{27}\big)^2.}$$

The temporal form is the same identity:

$$T_{27} := \frac{\Lambda_{27}}{c_{\mathrm{int}}}, \qquad t_* := \delta_\theta T_{27}.$$

In particular, it is not appropriate to replace t_* with T_{27}/δ_θ.

The homogeneity of these formulas is exact and this chart retains the status of an auxiliary functional realization. The complete gravitational owner then constructs, from the already generated action branch, the unique selector $\vartheta_{108} = (54/\pi_{\mathrm{HMT}})^2$, the section G_a and the typed invariant $G_a \hbar_a$. That construction governs the derivation of G; the functional $\mathscr{L}$ records an additional coordinatization of the same output. The causal order first preserves A and C^*, then gravitational transduction and finally the metrological chart.

80.2.5 Transduction of the gravitational decimal chart onto the magnitude line and causal separation of metrological recognition

Let

$$\mathscr{R}_G([n,k,d]_3; t, e) := 10^{-n} \left(k + \frac{t}{10^3} + \frac{e}{10^d}\right),$$

where the subscript of $[n,k,d]_3$ records the reader triad and does not denote a ternary numeral. For the declared data

$$[n,k,d]_3 = [11,6,5]_3, \qquad t = 674, \qquad e = 30,$$

rational evaluation is

$$\boxed{\mathscr{R}_G = 10^{-11} \left(6 + \frac{674}{1000} + \frac{30}{10^5}\right) = 6.67430 \times 10^{-11}.}$$

The arithmetic equality is exact once the parameters have been declared. The triad, the blocks 674, 30, and their decimal position have the status of chart data, so this branch is preserved as a parametrized control and subsequent recognition. The number is dimensionless until a basis $u_G \in \mathcal{L}_G^+$ is chosen; only then is the section defined

$$G_{\mathrm{dec}} := \mathscr{R}_G \, u_G.$$

The coincidence of its coordinate with a metrological publication is EXTERNAL RECOGNITION. The integers 674, 30, the triplet $[11, 6, 5]$ and the decimal placement belong exclusively to this recognition chart; the governing derivation resides in the selector ϑ_{108} and in the complete owner's action transducer.

80.2.6 Hierarchy of Evidence for Gravitational Constructions

The previous layers do not constitute two proven genealogies of the same constant. Their situation is:

Object	Result available	Verdict
Transductor dimensional	Types $G_{\mathrm{int}}(\theta_G)$ exactly, without fixing θ_G.	Dimensional theorem; no selector.
Character of the observable $\mathcal{K}_{27}$	$\mathcal{K}_{27}^4 = I$ force $\chi(\mathcal{K}_{27}) = 1$.	Obstructed formulation.
Levantamiento enriquecido	Possible only with non-torsion abelian class, natural character and unique normalization.	Obstruction of canonicity demonstrated without those data.
Chronological coverage and spectral response	The cumulative translation has infinite semigroup order; along the deterministic orbit, the phase cancels.	Null response proved on the deterministic orbit; nontrivial class characterized by interference and natural normalization.
Path functional	The formula for G_{hol} is homogeneous once Λ_{27} has been fixed.	Exact parametrized family; no unique selector exists without specifying $\mathscr{L}$ and its normalization.
Decimal reader	Reproduce exactly the declared coordinate, but receive its blocks and position.	Control parametrizado.

Comparison Criterion 80.10 (Selector of the gravitational constant)**.** A promotion is rejected if:

1. the enriched lift preserves only torsion or does not descend to a well-defined Abelian class;
2. the chronological operator does not preserve interference, produces $\mathcal{R}_{27} = 0$, lacks a domain or measure, or its response changes under an admissible refinement;
3. $\mathscr{L}$ depends on the basepoint, relabelling, the chosen orientation, a subdivision, or a free normalization;
4. the normalization $\mathfrak{N}$ depends on the global scale of the operator, is nonunique, or is fixed by consulting G;
5. 674, 30, or the decimal placement admit alternatives with the same structural standing;
6. the candidate consults the metrological value before being sealed;
7. the value is not stable under refinement or does not define a section of $\mathcal{L}_G$ in a common dimensional chart.

80.2.7 Gravitational–inertial ratios and the dimensional constant

The local normalization of the torsion chart,

$$\frac{\mathcal{G}_{\mathrm{tor}}}{\mathcal{I}_{\mathrm{tor}}} = 1,$$

and the interface areal–volumetric of the section 6.3.7.1,

$$\frac{\mathcal{G}_{\mathrm{av}}}{\mathcal{I}_{\mathrm{av}}} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2},$$

are dimensionless reader ratios. They are not rival values of the dimensional constant G. The first expresses a local diagonal constraint; the second is an interface conditional upon factorization through a common memory. Constructing a section of $\mathcal{L}_G$ additionally requires the $(\hbar_{\mathrm{int}}, c_{\mathrm{int}}, \ell_0)$ framework and a θ_G selector.

Cosmological prolongation of the holonomic response. The response is extended to the cosmological regime through the Perron mode and torsional memory.

80.3 Action, gravity, and thermal information

80.3.1 Dodecaphasic period and circular realization of action

The calendar already constructed determines $T_{108} = 108t_0$. In its circular realization,

$$\omega_{108} = \frac{2\pi}{108t_0}, \qquad R_{108} = \frac{c_{\mathrm{int}}}{\omega_{108}} = \frac{54}{\pi}\ell_0, \qquad \ell_0 = c_{\mathrm{int}}t_0.$$

The unit ℓ_0 and the time and velocity charts remain explicit; the SI chart does not select the cycle. For each oriented action section $a \in \{\mathrm{pre}, \mathrm{ret}\}$,

$$E_{108,a} = \hbar_a \omega_{108}, \qquad m_{108,a} = \frac{E_{108,a}}{c_{\mathrm{int}}^2}.$$

The reduced and full lengths are, respectively,

$$\frac{\hbar_a}{m_{108,a}c_{\mathrm{int}}} = R_{108}, \qquad \frac{h_a}{m_{108,a}c_{\mathrm{int}}} = 2\pi R_{108}.$$

The same action and the same duration enter both readings. Their radial coincidence is determined in the following proposition.

Proposition 80.11 (Radius-coincidence selector)**.** *In the dimensional family*

$$G_a(\vartheta) = \vartheta \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a}, \qquad \vartheta > 0,$$

the circular-realization condition $G_a(\vartheta)m_{108,a}/c_{\mathrm{int}}^2 = R_{108}$ has the unique solution

$$\vartheta_{108}^{\circ} = (54/\pi)^2.$$

Proof. The first radius is $r_g(\vartheta) = \vartheta(\pi/54)\ell_0$. Comparing it with $(54/\pi)\ell_0$, with $\ell_0 > 0$, reduces the equality to a strictly increasing linear equation. □

The coincidence condition is an explicit part of this realization; the theorem solves for its coefficient without using a metrological value of G. The radius used here is Gm/c^2, so the factor two from the convention $2Gm/c^2$ is not inadvertently transferred.

Proposition 80.12 (Reciprocal action and gravity, conserved area)**.** *In the two preceding charts,*

$$\frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}} = R_{\mathrm{act}}^{-1}, \qquad \hbar_a G_a = \vartheta^{\circ}_{108} c_{\mathrm{int}}^5 t_0^2, \qquad \ell_P^2 = \frac{\hbar_a G_a}{c_{\mathrm{int}}^3} = \vartheta^{\circ}_{108} \ell_0^2.$$

Proof. The definition of G_a contains $\hbar_a^{-1}$ and the remaining factors are common to the two charts. Taking the quotient and the product gives the three equalities. □

This reciprocity explains why orientation information may vary while area remains unchanged. A precise relation is thus available between the action ellipse, Planck scale, and area operator of section 72.3.

80.3.2 Thermal normalization and energy per cycle

We fix the same oriented section a for energy and temperature. The thermal relation of the corpus publishes

$$k_B \Theta_{\mathrm{clk},a} \log 3 = \frac{h_a}{108 t_0} = \frac{\hbar_a}{t_P} = E_{P,a}, \qquad t_P = \frac{54}{\pi} t_0. \tag{80.11}$$

The factor $\log 3$ measures the information of the equiprobable triple; the clock energy incorporates the action already generated. The identity preserves the product $k_B \Theta_{\mathrm{clk}}$ as an energy section and makes explicit the thermal chart used, developed in section 70.2.7.

At this scale, the radiative realization provides

$$n_\gamma \ell_P^3 = \frac{2\zeta(3)}{\pi^2 \log^3 3}, \qquad \frac{u_\gamma \ell_P^3}{E_P} = \frac{\pi^2}{15 \log^4 3}, \qquad \frac{s_\gamma \ell_P^3}{k_B} = \frac{4\pi^2}{45 \log^3 3}.$$

Two distinct invariants occur here: $\log 3$ preserves TRIT information and $\zeta(3)$ the bosonic occupation moment. Their occurrence in the same expression relates the thermal constants to the spectral moments of bosonic occupation, without identifying these quantities with one another.

Section 70.2.7 develops the Boltzmann transducer, its action on tritic information, and the change of thermal coordinates. Equation (80.11) fixes the energy–temperature product used by that construction. The selection of an individual coordinate is distinguished there from the identity for the product. Within this same development, the preceding radiative formulas preserve the occupation and entropy content of the bosonic realization.

80.3.3 Barbero–Immirzi in the Cartan–Holst realization

Once γ_{HMT} has been constructed, the source realizes it as a coefficient of the Holst action. In the declared geometric chart, let e^I be a nondegenerate co-tetrad, ω^{IJ} a connection, $\Sigma^{IJ} = e^I \wedge e^J$ and $F^{IJ} = \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}$. With $\kappa = 8\pi G/c^4$, the specialized action is

$$\begin{aligned} S[e, \omega, \Psi] = \frac{1}{2\kappa c} \int \Sigma_{IJ} \wedge (\star F^{IJ} + \gamma_{\mathrm{HMT}}^{-1} F^{IJ}) \\ - \frac{\Lambda}{\kappa c} \int \mathrm{vol}_e + S_m[e, \omega, \Psi]. \end{aligned} \tag{80.12}$$

The current convention is $\delta_\omega S_m = -\frac{1}{2} \int \delta\omega^{IJ} \wedge \sigma_{IJ}$. For this variation, the connection is fixed at the boundary, $\delta\omega^{IJ}|_{\partial M} = 0$, or variations compactly supported in the interior are used. These sufficient conditions annihilate the boundary term resulting from integrating $\Sigma_{IJ} \wedge (\star + \gamma_{\mathrm{HMT}}^{-1} I) D_\omega \delta\omega^{IJ}$. Variation of (80.12) under that condition gives

$$D_\omega[(\star+\gamma_{\mathrm{HMT}}^{-1}I)\Sigma^{IJ}]=\kappa c\,\sigma^{IJ}.$$

This follows by using $\delta F = D_\omega\delta\omega$, integrating by parts, and setting the coefficient of the arbitrary variation to zero [187]. Boundary orientation and current normalization are part of this formula.

In Lorentzian signature, $\star^2=-I$ on bivectors. The identity

$$(\star+\gamma^{-1}I)^{-1}=\frac{\gamma^2}{1+\gamma^2}(-\star+\gamma^{-1}I)$$

is checked by multiplication: the cross terms cancel, leaving $(1+\gamma^{-2})I$ before normalization. Thus the parameter already constructed enters the torsional response, not merely a table value. The identities $D_\omega T^I = F^I{}_J\wedge e^J$ and $D_\omega F^{IJ}=0$ keep torsion, curvature, and balance connected.

The geometric realization preserves route memory and boundary soldering. The cosmological constant Λ of (80.12) is the coefficient declared in the action; it is not identified with the vector representation $\Lambda(R)$. The following identities preserve the holonomy, smoothness and nondegeneracy conditions of the realization.

80.3.4 Phase return and representation of accumulated memory

The chronology of enriched routes distinguishes position return, orientation return and phase return. The observable publication identifies certain arrival states, while the cumulative register preserves the sequence traversed. The geometric representation must transport both data jointly.

80.3.4.1 Inversion cocycle and change of sheet

Let F_{obs} be the chronological publication of TPK transport. Its canonical blocks have the hierarchy

27	positional return with orientation reversal,
54	positional and orientational return,
108	complete return of the observable phase.

The register counts orientation inversions $g(a)$ and effective changes of sheet $s(a)$. For a route,

$$c(\gamma)=\sum_{a\subset\gamma}(g(a),s(a)),\qquad c(\gamma_2\gamma_1)=c(\gamma_2)+c(\gamma_1). \tag{80.13}$$

The canonical computation of the block of length 27 publishes $c_{27}=(1,18)$. Composition preserves those increments in the four blocks that complete the observable period. On forward-oriented routes the register is cumulative; its extension to the route groupoid uses $\mathbb{Z}^2$ and $c(\bar\gamma)=-c(\gamma)$.

Proposition 80.13 (Memory of the complete observable turn)**.** *Writing $\mathcal{K}_{27}=F_{\mathrm{obs}}^{27}$, the lift to the register, for $x\in\mathcal{Q}_{\mathrm{obs}}$ and $z\in\mathbb{Z}^2$, satisfies*

$$\widetilde{\mathcal{K}}_{27}(x,z)=(\mathcal{K}_{27}x,z+(1,18)),\qquad \widetilde{\mathcal{K}}_{27}^{4}(x,z)=(x,z+(4,72)).$$

After N complete turns, the increment is $N(4,72)$.

Proof. The observable projection of $\mathcal{K}_{27}$ has order four. The preceding canonical computation fixes $c_{27}=(1,18)$ as a constant over that observable orbit. The additivity of (80.13) sums the four increments and gives $c_{108}=4(1,18)=(4,72)$. Repetition applies the same concatenation law and produces Nc_{108}. The return of the first coordinate preserves the increment of the second. □

80.3.4.2 Affine representation of the cocycle

The discrete realization uses an element R_4 of order four in $\mathrm{Spin}^+(1,3)$, a unit temporal vector u fixed by its vector action and a unit of length $\ell_0 > 0$. The scale ℓ_0 retains its provenance in the dimensional section. We define

$$\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma) = \left(R_4^{c_g(\gamma)},\ \ell_0 c_s(\gamma) u\right) \in \mathrm{Spin}^+(1,3) \ltimes \mathbb{R}^{1,3}. \tag{80.14}$$

The power acts on the vector through the associated vector representation of the spin group.

Theorem 80.14 (Composition and translational amplitude of return)**.** *The map $\rho_{\mathrm{C}}^{\mathrm{disc}}$ preserves the identity, concatenation and inversion. At complete returns,*

$$\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_{108}) = (I, 72\ell_0 u), \qquad \rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_{108}^N) = (I, 72N\ell_0 u). \tag{80.15}$$

Inversion changes the sign of the translation.

Proof. The semidirect product is $(R,a)(S,b) = (RS, a + Rb)$. Since $R_4 u = u$,

$$\begin{aligned}\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_2)\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_1) &= \left(R_4^{c_g(\gamma_2)+c_g(\gamma_1)}, \ell_0(c_s(\gamma_2) + c_s(\gamma_1))u\right) \\ &= \rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_2\gamma_1).\end{aligned}$$

The identity cocycle is zero and that of the inverse is its opposite. For the complete turn, substitute $c_{108} = (4, 72)$ and $R_4^4 = I$. Concatenating N turns keeps the rotational part equal to the identity and sums their translations. □

The amplitude $72\ell_0$ is a length associated with a finite route. A realization through a co-tetrad and a connection transports it to its affine holonomy. The local contribution to the gravitational equations is subsequently obtained through the curvature of that connection, the variation of the matter action and the normalization of its density. The representation thus preserves the exact cumulative content that must enter that realization.

80.3.5 Cubic incidence and boundary soldering

The cubic realization of transport preserves six oriented incidences. Its geometric quotient, the Lorentzian form and the energy response are obtained from the same incidence matrix. Transport proceeds from the enriched APP–TRIT–TPK route: the face, sign, carry and memory accompany each edge. This section develops the linear operation acting on that boundary publication.

80.3.5.1 Cubic incidence and the rank-four quotient

Let $H_F = \mathbb{R}^F$, with $F = \{(i,\epsilon) : 1 \le i \le 3,\ \epsilon \in \{-1,1\}\}$, its Euclidean inner product, and its basis $e_{i,\epsilon}$. Face opposition is the orthogonal involution $Re_{i,\epsilon} = e_{i,-\epsilon}$. Define

$$u = \sum_{i,\epsilon} e_{i,\epsilon}, \qquad P_u = \frac{uu^{\mathsf{T}}}{6}, \qquad P_- = \frac{I - R}{2}, \qquad P_0 = \frac{I + R}{2} - P_u, \qquad P_\square = P_u + P_-.$$

The vertices are $V = \{-1,1\}^3$. The incidence matrix $M : \mathbb{R}^V \longrightarrow H_F$ is determined by

$$M_{(i,\epsilon),\nu} = \mathbf{1}_{\{\nu_i = \epsilon\}}.$$

Proposition 80.15 (Incidence decomposition)**.** *We have*

$$MM^{\mathsf{T}} = 12P_u + 4P_-, \qquad \ker M^{\mathsf{T}} = P_0 H_F.$$

Consequently, the quotient $Q_\square = Q = H_F/P_0H_F$, orthogonally identified with $P_\square H_F$, has dimension four.

Proof. A face contains four vertices. Two opposite faces have no common vertices and two faces on different axes have two common vertices. Writing $\mathbf{J} = uu^{\mathsf{T}}$, we obtain

$$MM^{\mathsf{T}} = 4I + 2(\mathbf{J} - I - R) = 2(I - R) + 2\mathbf{J} = 4P_{-} + 12P_u.$$

The three projectors P_u, P_{-}, P_0 are orthogonal and sum to I; their ranks are one, three and two, respectively. The identity $\|M^{\mathsf{T}}x\|^2 = \langle x, MM^{\mathsf{T}}x\rangle$ identifies the stated kernel. The complementary rank is four. □

On Q, the form

$$B = P_{-} - P_u$$

has signature $(3, 1)$. The time orientation is fixed by u. With

$$t = \frac{u}{\sqrt{6}}, \qquad d_i = \frac{e_{i,+} - e_{i,-}}{\sqrt{2}}, \qquad W = \sqrt{6}P_u + \sqrt{2}P_{-},$$

we have $We_{i,\epsilon} = t + \epsilon d_i$. These six vectors are null for B, whereas

$$3t + \nu_1 d_1 + \nu_2 d_2 + \nu_3 d_3$$

has squared norm -6. The same incidence satisfies $MM^{\mathsf{T}} = 2W^{*}W$ on Q. The proper rotations of the cube commute with the projectors, preserve both forms and fix t. These identities determine the metric of the cubic publication.

80.3.5.2 Cellular soldering and memory transport

For each oriented edge a, the route publication provides the face $\lambda(\underline{a})$, the sign $\sigma(a)$, the source and target fibres and the transport $U(a) : Q_{s(a)} \longrightarrow Q_{t(a)}$. Inversion satisfies $U(\bar{a}) = U(a)^{-1}$. The boundary half-edges retain their formal inversion and their incidence at the boundary. With $\ell_m = \ell_0\, 9^{-m} > 0$, the soldering is

$$\beta_m(a) = \ell_m \sigma_m(a) n_{\lambda(\underline{a})}^{s(a)}, \qquad n_{i,\epsilon} = t + \epsilon d_i. \tag{80.16}$$

The face and sign are transported together with the route memory. The inversion convention requires

$$\beta_m(\bar{a}) = -U_m(a)\beta_m(a). \tag{80.17}$$

Applying this relation twice recovers $\beta_m(a)$; the carry register separately preserves the ordered composition.

Proposition 80.16 (Naturality of soldering under refinement)**.** *Let $a = a_1 \cdots a_9$ be a compatible refinement of an edge. Suppose that the nine child incidences, transported to the parent fibre, preserve the face and sign, and that $\ell_{m+1} = \ell_m/9$. Let $\mathcal{U}_j : Q_{m,s(a)} \longrightarrow Q_{m+1,s(a_j)}$ be the accumulated transport that includes the refinement identification and the path to the origin of child j. Then*

$$\beta_m(a) = \sum_{j=1}^{9} \mathcal{U}_j^{-1} \beta_{m+1}(a_j).$$

The equality is compatible with inversion and with concatenation of the memory register.

Proof. Each summand, expressed in the fibre at the origin of the parent, is $(\ell_m/9)\sigma_m(a)n_{\lambda(\underline{a})}$. The sum of nine terms reproduces the parent soldering. Inversion reverses the order of the transports and replaces each one with its inverse; the relation (80.17) provides the overall sign. Memory is concatenated in that same order, so refinement preserves both the vector publication and its enriched register. □

80.3.6 Geometric realization of transport and soldering

The discrete representation of return and screen soldering determine two aspects of transport: the accumulation of memory along routes and the comparison of their fibres. Their geometric realization preserves both operations. This section fixes the target space, the dimensional normalization of the connection and the identities that its variation will use.

80.3.6.1 Screen, co-tetrad and connection

The screen quotient $Q_\square$, its Lorentzian form, and the soldering β_m come from 80.3.5. In each cell, a screen realization $J_{\mathrm{scr},m}$ transports the form and soldering to the corresponding geometric fibre. Its cellular image is

$$e_m = J_{\mathrm{scr},m}\beta_m.$$

The scale ℓ_m remains within the soldering. Changes of frame act jointly on $J_{\mathrm{scr},m}$, the connection and the transport of the cells.

A smooth realization of those data on an oriented manifold $\mathcal{U}$ of dimension four consists of a nondegenerate co-tetrad $e = (e^I)$, a Lorentz connection $\omega = (\omega^I{}_J)$ and a realization of routes $\mathfrak{R}_{\mathrm{C}}$ satisfying

$$\mathrm{Hol}_{\mathcal{A}_{\ell_0}}(\mathfrak{R}_{\mathrm{C}}(\gamma)) = \rho_{\mathrm{C}}(\mathrm{Hol}_{\mathrm{TPK}}(\gamma)). \tag{80.18}$$

In this matrix equality, the representation is normalized by

$$N_{\ell_0}(R,a) = (\Lambda(R), a/\ell_0), \qquad \rho_{\mathrm{C}} = N_{\ell_0} \circ \rho_{\mathrm{C}}^{\mathrm{disc}},$$

where $\Lambda : \mathrm{Spin}^+(3,1) \to SO^+(3,1)$ is the vector representation. The map N_{ℓ_0} preserves the semidirect product: dividing the translation $a + \Lambda(R)b$ by ℓ_0 is equivalent to composing the two normalized translations. The complete turn therefore publishes $72u$ in the normalized connection and $72\ell_0 u$ in the dimensional chart. The spin lift separately preserves $R_4^{c_g(\gamma)}$; the vector representation has the central kernel $\{1,-1\}$, so that lift is retained when using spinor fields. Identity, concatenation, inversion and subdivision are maintained in the corresponding realization. The screen pairing and the holonomy condition specify the data of the realization used in the field equations.

With $\eta = \mathrm{diag}(-1,1,1,1)$, define $g = \eta_{IJ}e^I \otimes e^J$. For a positive elementary scale ℓ_0, constant in the chart, the affine connection is

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1}e \\ 0 & 0 \end{pmatrix}. \tag{80.19}$$

The co-tetrad has the dimension of length; the factor ℓ_0^{-1} makes its translational component dimensionally homogeneous.

Proposition 80.17 (Curvature and torsion of the realized connection)**.** *The curvature of* (80.19) *is*

$$\mathcal{F}_{\ell_0} = \mathrm{d}\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_\omega & \ell_0^{-1}T \\ 0 & 0 \end{pmatrix}, \quad F_\omega = \mathrm{d}\omega + \omega \wedge \omega, \quad T = \mathrm{d}e + \omega \wedge e. \tag{80.20}$$

Proof. The product of matrices of forms produces $\omega \wedge \omega$ in the diagonal block and $\ell_0^{-1}\omega \wedge e$ in the translational block. The exterior derivative contributes $\mathrm{d}\omega$ and $\ell_0^{-1}\mathrm{d}e$, since ℓ_0 is constant. Their sum proves the equality. □

Curvature describes the frame-transport defect and torsion that of the co-tetrad. Both belong to the same realized connection; their components retain distinct tensorial domains.

Proposition 80.18 (Covariance and Bianchi identities)**.** *Under a change of frame* $g_L : \mathcal{U} \to SO^+(3,1)$*,*

$$e' = g_L e, \qquad \omega' = g_L \omega g_L^{-1} - \mathrm{d}g_L\, g_L^{-1},$$

we have $T' = g_L T$ and $F_{\omega'} = g_L F_\omega g_L^{-1}$. Furthermore,

$$D_\omega T = F_\omega \wedge e, \qquad D_\omega F_\omega = 0. \tag{80.21}$$

Proof. In $\mathrm{d}(g_L e) + \omega' \wedge g_L e$, the terms containing $\mathrm{d}g_L \wedge e$ cancel. The same substitution into the curvature gives its transformation by conjugation. For the first Bianchi identity, expand

$$D_\omega(\mathrm{d}e + \omega \wedge e) = (\mathrm{d}\omega + \omega \wedge \omega) \wedge e.$$

In $\mathrm{d}F_\omega + \omega \wedge F_\omega - F_\omega \wedge \omega$, the terms containing $\mathrm{d}\omega$ and the cubic products cancel in pairs. This proves the second identity. □

80.3.6.2 The three quadratic realizations of curvature

The TRIT sheet $\tau \in \{+1, 0, -1\}$ is represented in the Cartan algebra by generators $J_{ab} = -J_{ba}$ and $P_a^{(\tau)}$, with relations

$$\begin{aligned}
[J_{ab}, J_{cd}] &= \eta_{bc} J_{ad} - \eta_{ac} J_{bd} - \eta_{bd} J_{ac} + \eta_{ad} J_{bc}, \\
[J_{ab}, P_c^{(\tau)}] &= \eta_{bc} P_a^{(\tau)} - \eta_{ac} P_b^{(\tau)}, \\
[P_a^{(\tau)}, P_b^{(\tau)}] &= -\tau J_{ab}.
\end{aligned}$$

For constant $\ell > 0$, write

$$\mathcal{A}_\tau^{\mathrm{Cart}} = \tfrac{1}{2}\omega^{ab} J_{ab} + \ell^{-1} e^a P_a^{(\tau)}.$$

Proposition 80.19 (Curvature of the tritified realization)**.** *With the preceding relations,*

$$\mathcal{F}_\tau^{\mathrm{Cart}} = \tfrac{1}{2}(F_\omega^{ab} - \tau \ell^{-2} e^a \wedge e^b) J_{ab} + \ell^{-1} T^a P_a^{(\tau)}. \tag{80.22}$$

Proof. Expand $\mathrm{d}\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}]$. The bracket of the Lorentz sector produces F_ω, the mixed bracket produces $D_\omega e$, and the translational bracket contributes $-\frac{1}{2}\tau \ell^{-2} e^a \wedge e^b J_{ab}$. □

On the central sheet the affine connection is recovered; the extreme sheets preserve the two signs of the constant-curvature term. The representation thus retains the tritic regime of the route before applying the variational equations. The holonomy equality (80.18) corresponds to the central sheet. On each extreme sheet, the representation $\rho_{\mathrm{C},\tau}$ in its Cartan group and its respective holonomy condition are used. The current and the metric tensor use the fields and representation of the same realization.

80.3.7 Oriented transport of the connection and area

The parity of the incidence functional allows comparison of the charts γ and $-\gamma$. In the canonical geometric realization, let E_i^a be a densitized triad, $\Gamma_a^i(E)$ its intrinsic connection and K_a^i the extrinsic curvature in the chosen normal convention. The Ashtekar–Barbero connection is $\mathcal{A}_a^i = \Gamma_a^i(E) + \gamma K_a^i$.

Proposition 80.20 (Involution between oriented charts)**.** *For $\gamma \in \mathbb{R} \setminus \{0\}$, the map*

$$\mathfrak{b} : (\gamma, K, E) \longmapsto (-\gamma, -K, E)$$

is involutive. It preserves $\mathcal{A}$ and reverses the form

$$\Omega_{KE} = \frac{1}{\kappa} \int_\Sigma \delta K_a^i \wedge \delta E_i^a,$$

with κ fixed and the same boundary conditions.

Proof. Applying $\mathfrak{b}$ twice restores all three coordinates. Since E remains fixed, so does $\Gamma(E)$. The product of the two odd coordinates satisfies $(-\gamma)(-K) = \gamma K$, preserving the connection. In the space of variations, $\delta(-K) = -\delta K$ and δE remains fixed: the pullback of the form is $\mathfrak{b}^*\Omega_{KE} = -\Omega_{KE}$. □

The oriented area $\mathcal{A}_{\mathrm{or}} = \gamma a_0 N$ and its physical magnitude $\mathcal{A}_{\mathrm{fis}} = |\gamma| a_0 N$, for $a_0 > 0$ and $N \geq 0$, coincide in the positive chart. Upon prolongation to the opposite chart, the former is odd and the latter remains invariant. This distinction preserves both the orientation of the incidence and the positivity of the geometric observable. The proposition establishes the algebraic transport of these charts; its realization as an inversion of the spatial normal additionally uses the transport of matter and of the boundary data of the action (80.12).

80.3.8 Structural definition and function of the gravitational constant

Definition 80.21 (Gravitational section contragredient to action)**.** For a positive section $\hbar_a$, a velocity $c_{\mathrm{int}} > 0$, and the chronological radius R_{108} of section 80.3, the circular gravitational section is

$$G_a = \frac{c_{\mathrm{int}}^3 R_{108}^2}{\hbar_a}, \qquad R_{108} = \frac{54}{\pi}\ell_0, \qquad \ell_0 = c_{\mathrm{int}} t_0. \tag{80.23}$$

It is the action-reciprocal section realizing coincidence between the cycle's reduced gravitational radius and its chronological radius, in the convention $r_g = Gm/c_{\mathrm{int}}^2$.

The radius-coincidence proposition of section 80.3 proves its uniqueness in the family $G_a(\vartheta) = \vartheta c_{\mathrm{int}}^5 t_0^2/\hbar_a$. The circular condition fixes $\vartheta = (54/\pi)^2$ and leads to (80.23). That condition and the positive dimensional sections are part of the realization. Keeping them common to both orientations gives

$$\frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}} = R_{\mathrm{act}}^{-1}, \qquad \frac{G_a \hbar_a}{c_{\mathrm{int}}^3} = R_{108}^2 = \ell_P^2. \tag{80.24}$$

The proof is direct substitution of (80.23); the action change is compensated by the reciprocal change in G_a.

This structure assigns gravitation a precise function: transporting the action section to a common geometric unit. In (80.12), G fixes coupling normalization; in the area operator, $G\hbar/c^3$ fixes the area unit. Reciprocity preserves that unit while the oriented action reading changes. The reduced state and its occupations subsequently supply the information and entanglement associated with the boundary.

The tensorial content of the coupling is preserved in $\mathbb{G}_{5,a} = G_a P_5^{\mathrm{ico}}$, constructed in 77.1.5. Section G_a determines its common magnitude; the projector selects the five-component carrier, and Γ_5 transports its orientation to the space of symmetric trace-free tensors. The basis in (77.64)–(77.66) makes the five components explicit. Projection $\Lambda(n)$ then defines the transverse reduction. Magnitude, carrier, and reduction are related operations with different domains within this realization.

80.4 Phase return and representation of accumulated memory

The chronology of enriched routes distinguishes position return, orientation return and phase return. The observable publication identifies certain arrival states, while the cumulative register preserves the sequence traversed. The geometric representation must transport both data jointly.

80.4.1 Inversion cocycle and change of sheet

Let F_{obs} be the chronological publication of TPK transport. Its canonical blocks have the hierarchy

27	positional return with orientation reversal,
54	positional and orientational return,
108	complete return of the observable phase.

The register counts orientation inversions $g(a)$ and effective changes of sheet $s(a)$. For a route,

$$c(\gamma) = \sum_{a \subset \gamma} (g(a), s(a)), \qquad c(\gamma_2\gamma_1) = c(\gamma_2) + c(\gamma_1). \tag{80.25}$$

The canonical computation of the block of length 27 publishes $c_{27} = (1, 18)$. Composition preserves those increments in the four blocks that complete the observable period. On forward-oriented routes the register is cumulative; its extension to the route groupoid uses $\mathbb{Z}^2$ and $c(\bar{\gamma}) = -c(\gamma)$.

Proposition 80.22 (Memory of the complete observable turn)**.** *Writing $\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}$, the lift to the register, for $x \in O_{108}$ and $z \in \mathbb{Z}^2$, satisfies*

$$\widetilde{\mathcal{K}}_{27}(x, z) = (\mathcal{K}_{27}x, z + (1, 18)), \qquad \widetilde{\mathcal{K}}_{27}^{4}(x, z) = (x, z + (4, 72)).$$

After N complete turns, the increment is $N(4, 72)$.

Proof. The observable projection of $\mathcal{K}_{27}$ has order four. The preceding canonical computation fixes $c_{27} = (1, 18)$ as a constant over that observable orbit. The additivity of (80.25) sums the four increments and gives $c_{108} = 4(1, 18) = (4, 72)$. Repetition applies the same concatenation law and produces Nc_{108}. The return of the first coordinate preserves the increment of the second. □

80.4.2 Affine representation of the cocycle

The discrete realization uses an element R_4 of order four in $\mathrm{Spin}^+(1, 3)$, a unit temporal vector u fixed by its vector action and a unit of length $\ell_0 > 0$. The scale ℓ_0 retains its provenance in the dimensional section. We define

$$\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma) = \left(R_4^{c_g(\gamma)},\ \ell_0 c_s(\gamma) u\right) \in \mathrm{Spin}^+(1, 3) \ltimes \mathbb{R}^{1,3}. \tag{80.26}$$

The power acts on the vector through the associated vector representation of the spin group.

Theorem 80.23 (Composition and translational amplitude of return)**.** *The map $\rho_{\mathrm{C}}^{\mathrm{disc}}$ preserves the identity, concatenation and inversion. At complete returns,*

$$\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_{108}) = (I, 72\ell_0 u), \qquad \rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_{108}^{N}) = (I, 72N\ell_0 u). \tag{80.27}$$

Inversion changes the sign of the translation.

Proof. The semidirect product is $(R, a)(S, b) = (RS, a + Rb)$. Since $R_4 u = u$,

$$\begin{aligned} \rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_2)\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_1) &= \left(R_4^{c_g(\gamma_2)+c_g(\gamma_1)}, \ell_0(c_s(\gamma_2) + c_s(\gamma_1))u\right) \\ &= \rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_2\gamma_1). \end{aligned}$$

The identity cocycle is zero and that of the inverse is its opposite. For the complete turn, substitute $c_{108} = (4, 72)$ and $R_4^4 = I$. Concatenating N turns keeps the rotational part equal to the identity and sums their translations. □

The amplitude $72\ell_0$ is a length associated with a finite route. A realization through a co-tetrad and a connection transports it to its affine holonomy. The local contribution to the gravitational equations is subsequently obtained through the curvature of that connection, the variation of the matter action and the normalization of its density. The representation thus preserves the exact cumulative content that must enter that realization.

80.5 Boundary soldering and energy response

The cubic realization of transport preserves six oriented incidences. Its geometric quotient, the Lorentzian form and the energy response are obtained from the same incidence matrix. Transport proceeds from the enriched APP–TRIT–TPK route: the face, sign, carry and memory accompany each edge. This section develops the linear operation acting on that boundary publication.

80.5.1 Cubic incidence and the rank-four quotient

Let $H_F = \mathbb{R}^F$, with $F = \{(i,\epsilon) : 1 \le i \le 3,\ \epsilon \in \{-1,1\}\}$, its Euclidean inner product, and its basis $e_{i,\epsilon}$. Face opposition is the orthogonal involution $Re_{i,\epsilon} = e_{i,-\epsilon}$. Define

$$u = \sum_{i,\epsilon} e_{i,\epsilon}, \qquad P_u = \frac{uu^{\mathsf{T}}}{6}, \qquad P_- = \frac{I-R}{2}, \qquad P_0 = \frac{I+R}{2} - P_u, \qquad P_\square = P_u + P_-.$$

The vertices are $V = \{-1,1\}^3$. The incidence matrix $M : \mathbb{R}^V \longrightarrow H_F$ is determined by

$$M_{(i,\epsilon),\nu} = \mathbf{1}_{\{\nu_i=\epsilon\}}.$$

Proposition 80.24 (Incidence decomposition)**.** *We have*

$$MM^{\mathsf{T}} = 12P_u + 4P_-, \qquad \ker M^{\mathsf{T}} = P_0 H_F.$$

Consequently, the quotient $Q = H_F/P_0H_F$, orthogonally identified with $P_\square H_F$, has dimension four.

Proof. A face contains four vertices. Two opposite faces have no common vertices and two faces on different axes have two common vertices. Writing $\mathbf{J} = uu^{\mathsf{T}}$, we obtain

$$MM^{\mathsf{T}} = 4I + 2(\mathbf{J} - I - R) = 2(I-R) + 2\mathbf{J} = 4P_- + 12P_u.$$

The three projectors P_u, P_-, P_0 are orthogonal and sum to I; their ranks are one, three and two, respectively. The identity $\|M^{\mathsf{T}}x\|^2 = \langle x, MM^{\mathsf{T}}x\rangle$ identifies the stated kernel. The complementary rank is four. □

On Q, the form

$$B = P_- - P_u$$

has signature $(3,1)$. The time orientation is fixed by u. With

$$t = \frac{u}{\sqrt{6}}, \qquad d_i = \frac{e_{i,+} - e_{i,-}}{\sqrt{2}}, \qquad W = \sqrt{6}P_u + \sqrt{2}P_-,$$

we have $We_{i,\epsilon} = t + \epsilon d_i$. These six vectors are null for B, whereas

$$3t + \nu_1 d_1 + \nu_2 d_2 + \nu_3 d_3$$

has squared norm -6. The same incidence satisfies $MM^{\mathsf{T}} = 2W^*W$ on Q. The proper rotations of the cube commute with the projectors, preserve both forms and fix t. These identities determine the metric of the cubic publication.

80.5.2 Cellular soldering and memory transport

For each oriented edge a, the route publication provides the face $\lambda(\underline{a})$, the sign $\sigma(a)$, the source and target fibres and the transport $U(a) : Q_{s(a)} \longrightarrow Q_{t(a)}$. Inversion satisfies $U(\bar{a}) = U(a)^{-1}$. The boundary half-edges retain their formal inversion and their incidence at the boundary. With $\ell_m = \ell_0\, 9^{-m} > 0$, the soldering is

$$\beta_m(a) = \ell_m \sigma_m(a) n^{s(a)}_{\lambda(\underline{a})}, \qquad n_{i,\epsilon} = t + \epsilon d_i. \tag{80.28}$$

The face and sign are transported together with the route memory. The inversion convention requires

$$\beta_m(\bar{a}) = -U_m(a)\beta_m(a). \tag{80.29}$$

Applying this relation twice recovers $\beta_m(a)$; the carry register separately preserves the ordered composition.

Proposition 80.25 (Naturality of soldering under refinement)**.** *Let $a = a_1 \cdots a_9$ be a compatible refinement of an edge. Suppose that the nine child incidences, transported to the parent fibre, preserve the face and sign, and that $\ell_{m+1} = \ell_m/9$. Let $\mathcal{U}_j : Q_{m,s(a)} \longrightarrow Q_{m+1,s(a_j)}$ be the accumulated transport that includes the refinement identification and the path to the origin of child j. Then*

$$\beta_m(a) = \sum_{j=1}^{9} \mathcal{U}_j^{-1} \beta_{m+1}(a_j).$$

The equality is compatible with inversion and with concatenation of the memory register.

Proof. Each summand, expressed in the fibre at the origin of the parent, is $(\ell_m/9)\sigma_m(a)n_{\lambda(\underline{a})}$. The sum of nine terms reproduces the parent soldering. Inversion reverses the order of the transports and replaces each one with its inverse; the relation (80.29) provides the overall sign. Memory is concatenated in that same order, so refinement preserves both the vector publication and its enriched register. □

80.5.3 Positive response, uniqueness, and minimal extension

The linear screen soldering is $\bar{\beta}_m = \ell_m W|_Q$. It is invertible, with

$$\bar{\beta}_m^{-1} = \ell_m^{-1} \left(\frac{1}{\sqrt{6}} P_u + \frac{1}{\sqrt{2}} P_- \right).$$

The eigenvalue $5/9$ of the antisymmetric sector of the normalized block $B_c/9$, constructed in Section 4.3.4, determines the compression coefficient $q_{\mathrm{vis}} = 5/9$. We reserve this notation to distinguish it from the Euclidean quotient $q_9(n)$ of the arithmetic kernel. Its realization on the screen uses an isometry $V_9 : Q \longrightarrow \mathcal{K}$. The visible contribution $C_9 = q_{\mathrm{vis}} V_9$ and its complement $D_9 = \sqrt{1 - q_{\mathrm{vis}}^2}\, V_9$ satisfy

$$C_9^* C_9 + D_9^* D_9 = I_Q, \qquad D_9^* D_9 = \frac{56}{81} I_Q.$$

The coefficient 56 is $9^2 - 5^2$; it belongs to the quadratic defect of that compression reader.

Theorem 80.26 (Energy response determined by soldering)**.** *There is a unique operator $\Lambda_m : Q \longrightarrow Q$ such that $\bar{\beta}_m^* \Lambda_m \bar{\beta}_m = D_9^* D_9$. It is positive definite and*

$$\Lambda_m = \frac{28}{243 \ell_m^2} (P_u + 3P_-) = \frac{112}{81 \ell_m^2} (M M^{\mathsf{T}}|_Q)^{-1}. \tag{80.30}$$

On H_F, the corresponding identity has right-hand side $(56/81)P_{\square}$.

Proof. Congruence by the invertible soldering necessarily determines

$$\Lambda_m = \bar{\beta}_m^{-*} \frac{56}{81} I_Q \bar{\beta}_m^{-1}.$$

Substituting the inverse, the coefficient of P_u is $56/(486\ell_m^2) = 28/(243\ell_m^2)$; that of P_- is $28/(81\ell_m^2)$. Both are positive. By Proposition 80.24, $(MM^{\mathsf{T}}|_Q)^{-1} = P_u/12 + P_-/4$, which gives the second expression. Extension to six faces annihilates $P_0 H_F$; the identity therefore extends through $P_\square$. □

Proposition 80.27 (Characterization by minimal extension)**.** *For every $b \in Q$,*

$$\langle b, \Lambda_m b \rangle = \frac{112}{81\ell_m^2} \min_{Mx=b} \|x\|^2.$$

The minimizing extension is unique: $x_{\min} = M^{\mathsf{T}}(MM^{\mathsf{T}}|_Q)^{-1} b$.

Proof. The stated vector satisfies $Mx_{\min} = b$ and belongs to $\operatorname{im} M^{\mathsf{T}} = (\ker M)^\perp$. Every other solution is $x_{\min} + z$, with $z \in \ker M$; by orthogonality, its squared norm is $\|x_{\min}\|^2 + \|z\|^2$. Moreover,

$$\|x_{\min}\|^2 = \langle b, (MM^{\mathsf{T}}|_Q)^{-1} b \rangle.$$

The formula for Λ_m completes the proof. □

The operator $MM^{\mathsf{T}}|_Q$ realizes the Dirichlet–to–Neumann response of this incidence formulation; its inverse is the Neumann–to–Dirichlet response. The preceding energy corresponds to the latter, with the normalization of the soldering and the compression defect. The Schur complement of $\left(\begin{smallmatrix} I & M^{\mathsf{T}} \\ M & 0 \end{smallmatrix}\right)$ is $-MM^{\mathsf{T}}$, which offers the same variational elimination.

80.5.4 Preservation of density by refinement

Consider the sector of cubic transports, orthogonal for the energy inner product and compatible with the form B. We fix a common label space Q^{lab}, identified isometrically with the parent's screen. The solderings have types $\bar{\beta}_m : Q^{\mathrm{lab}} \longrightarrow Q_m$ and $\bar{\beta}_{m+1,j} : Q^{\mathrm{lab}} \longrightarrow Q_{m+1,j}$; the operators Λ act in their respective target fibers. If $U_j : Q_j \longrightarrow Q$ transports the child fiber to the parent's, then

$$\bar{\beta}_{m+1,j} = \frac{1}{9} U_j^{-1} \bar{\beta}_m, \qquad \Lambda_{m+1,j} = 81 U_j^{-1} \Lambda_m U_j.$$

The factor 81 comes from $\ell_{m+1}^{-2} = 81\ell_m^{-2}$.

Theorem 80.28 (Invariance of the normalized density)**.** *Each child preserves the parent's quadratic response:*

$$\bar{\beta}_{m+1,j}^{*} \Lambda_{m+1,j} \bar{\beta}_{m+1,j} = \frac{56}{81} I_{Q^{\mathrm{lab}}}.$$

For r refinements, the joint response, expressed in the parent's fibers, is

$$E_{m+r} = \frac{56}{81} I_{Q^{\mathrm{lab}}} \otimes I_{9^r}.$$

The expectation with normalized trace $\tau_{9^r}(A) = 9^{-r} \operatorname{Tr}(A)$ satisfies

$$(\mathrm{id} \otimes \tau_{9^r})(E_{m+r}) = \frac{56}{81} I_{Q^{\mathrm{lab}}}$$

at every finite depth.

Proof. The congruence of a child contains the factors $1/9, 81, 1/9$; their product is one. Orthogonality cancels the intermediate transports. The direct sum of nine equal responses identifies the first refinement with $(56/81)I_Q \otimes I_9$. Iteration produces the tensor with I_{9^r}. Finally $\tau_{9^r}(I_{9^r}) = 1$. The expectations compose through the identity $\tau_{9^{r+s}} = \tau_{9^r} \otimes \tau_{9^s}$. □

The unital inclusions $A \mapsto A \otimes I_9$ are compatible with that family of traces. In the inductive limit of the matrix algebras, the response is the central element $(56/81)I_Q \otimes \mathbf{1}$. The unnormalized extensive sum produces, in contrast, $9^r(56/81)I_Q$. This distinction specifies which quantity is preserved: the quadratic screen density.

80.5.5 Variational realization of the response

The material realization uses an isometry $J_{\mathrm{scr}} : Q \longrightarrow \mathcal{H}_\partial$ compatible with the spin action and with refinement. The boundary co-tetrad is $e = J_{\mathrm{scr}}\bar{\beta}_m$. Let Λ_{CE} be the boundary response obtained by second variation of the material and geometric action, with its boundary conditions and elimination of the interior variables. The compatibility defect is

$$\mathcal{R}_{\mathrm{grav}} = \bar{\beta}_m^*(J_{\mathrm{scr}}^*\Lambda_{\mathrm{CE}}J_{\mathrm{scr}} - \Lambda_m)\bar{\beta}_m = e^*\Lambda_{\mathrm{CE}}e - D_9^*D_9. \tag{80.31}$$

By the invertibility of $\bar{\beta}_m$,

$$\mathcal{R}_{\mathrm{grav}} = 0 \quad \Longleftrightarrow \quad J_{\mathrm{scr}}^*\Lambda_{\mathrm{CE}}J_{\mathrm{scr}} = \Lambda_m.$$

The isometry J_{scr} transports the boundary states; the variational spin current is a degree-three form with values in bivectors. Each enters with its own domain in the realization of the response.

The construction determines a positive energy form, its minimizing interior extension and the density preserved by refinement. The amplitude of a concrete material contribution is obtained by evaluating the current of that realization in its variational quadratic form. This differentiates the operator fixing the response, the current on which it acts and the gravitational observable published by the evaluation.

80.6 Geometric realization of transport and soldering

The discrete representation of return and screen soldering determine two aspects of transport: the accumulation of memory along routes and the comparison of their fibres. Their geometric realization preserves both operations. This section fixes the target space, the dimensional normalization of the connection and the identities that its variation will use.

80.6.1 Screen, co-tetrad and connection

The screen quotient $Q_\square$, its Lorentzian form, and the soldering β_m come from 80.5. In each cell, a screen realization $J_{\mathrm{scr},m}$ transports the form and soldering to the corresponding geometric fibre. Its cellular image is

$$e_m = J_{\mathrm{scr},m}\beta_m.$$

The scale ℓ_m remains within the soldering. Changes of frame act jointly on $J_{\mathrm{scr},m}$, the connection and the transport of the cells.

A smooth realization of those data on an oriented manifold $\mathcal{U}$ of dimension four consists of a nondegenerate co-tetrad $e = (e^I)$, a Lorentz connection $\omega = (\omega^I{}_J)$ and a realization of routes $\mathfrak{R}_{\mathrm{C}}$ satisfying

$$\mathrm{Hol}_{\mathcal{A}_{\ell_0}}(\mathfrak{R}_{\mathrm{C}}(\gamma)) = \rho_{\mathrm{C}}(\mathrm{Hol}_{\mathrm{TPK}}(\gamma)). \tag{80.32}$$

In this matrix equality, the representation is normalized by

$$N_{\ell_0}(R, a) = (\Lambda(R), a/\ell_0), \qquad \rho_{\mathrm{C}} = N_{\ell_0} \circ \rho_{\mathrm{C}}^{\mathrm{disc}},$$

where $\Lambda : \mathrm{Spin}^{+}(3,1) \to SO^{+}(3,1)$ is the vector representation. The map N_{ℓ_0} preserves the semidirect product: dividing the translation $a + \Lambda(R)b$ by ℓ_0 is equivalent to composing the two normalized translations. The complete turn therefore publishes $72u$ in the normalized connection and $72\ell_0 u$ in the dimensional chart. The spin lift separately preserves $R_4^{c_g(\gamma)}$; the vector representation has the central kernel $\{1,-1\}$, so that lift is retained when using spinor fields. Identity, concatenation, inversion and subdivision are maintained in the corresponding realization. The screen pairing and the holonomy condition specify the data of the realization used in the field equations.

With $\eta = \mathrm{diag}(-1,1,1,1)$, define $g = \eta_{IJ} e^I \otimes e^J$. For a positive elementary scale ℓ_0, constant in the chart, the affine connection is

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1} e \\ 0 & 0 \end{pmatrix}. \tag{80.33}$$

The co-tetrad has the dimension of length; the factor ℓ_0^{-1} makes its translational component dimensionally homogeneous.

Proposition 80.29 (Curvature and torsion of the realized connection)**.** *The curvature of* (80.33) *is*

$$\mathcal{F}_{\ell_0} = \mathrm{d}\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_\omega & \ell_0^{-1} T \\ 0 & 0 \end{pmatrix}, \quad F_\omega = \mathrm{d}\omega + \omega \wedge \omega, \quad T = \mathrm{d}e + \omega \wedge e. \tag{80.34}$$

Proof. The product of matrices of forms produces $\omega \wedge \omega$ in the diagonal block and $\ell_0^{-1}\omega \wedge e$ in the translational block. The exterior derivative contributes $\mathrm{d}\omega$ and $\ell_0^{-1}\mathrm{d}e$, since ℓ_0 is constant. Their sum proves the equality. □

Curvature describes the frame-transport defect and torsion that of the co-tetrad. Both belong to the same realized connection; their components retain distinct tensorial domains.

Proposition 80.30 (Covariance and Bianchi identities)**.** *Under a change of frame $g_L : \mathcal{U} \to SO^{+}(3,1)$,*

$$e' = g_L e, \qquad \omega' = g_L \omega g_L^{-1} - \mathrm{d}g_L\, g_L^{-1},$$

we have $T' = g_L T$ and $F_{\omega'} = g_L F_\omega g_L^{-1}$. Furthermore,

$$D_\omega T = F_\omega \wedge e, \qquad D_\omega F_\omega = 0. \tag{80.35}$$

Proof. In $\mathrm{d}(g_L e) + \omega' \wedge g_L e$, the terms containing $\mathrm{d}g_L \wedge e$ cancel. The same substitution into the curvature gives its transformation by conjugation. For the first Bianchi identity, expand

$$D_\omega(\mathrm{d}e + \omega \wedge e) = (\mathrm{d}\omega + \omega \wedge \omega) \wedge e.$$

In $\mathrm{d}F_\omega + \omega \wedge F_\omega - F_\omega \wedge \omega$, the terms containing $\mathrm{d}\omega$ and the cubic products cancel in pairs. This proves the second identity. □

80.6.2 The three quadratic realizations of curvature

The TRIT sheet $\tau \in \{+1, 0, -1\}$ is represented in the Cartan algebra by generators $J_{ab} = -J_{ba}$ and $P_a^{(\tau)}$, with relations

$$\begin{aligned} [J_{ab}, J_{cd}] &= \eta_{bc} J_{ad} - \eta_{ac} J_{bd} - \eta_{bd} J_{ac} + \eta_{ad} J_{bc}, \\ [J_{ab}, P_c^{(\tau)}] &= \eta_{bc} P_a^{(\tau)} - \eta_{ac} P_b^{(\tau)}, \\ [P_a^{(\tau)}, P_b^{(\tau)}] &= -\tau J_{ab}. \end{aligned}$$

For constant $\ell > 0$, write

$$\mathcal{A}_\tau^{\mathrm{Cart}} = \tfrac{1}{2}\omega^{ab}J_{ab} + \ell^{-1}e^a P_a^{(\tau)}.$$

Proposition 80.31 (Curvature of the tritified realization)**.** *With the preceding relations,*

$$\mathcal{F}_\tau^{\mathrm{Cart}} = \tfrac{1}{2}(F_\omega^{ab} - \tau\ell^{-2}e^a \wedge e^b)J_{ab} + \ell^{-1}T^a P_a^{(\tau)}. \tag{80.36}$$

Proof. Expand $\mathrm{d}\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}]$. The bracket of the Lorentz sector produces F_ω, the mixed bracket produces $D_\omega e$, and the translational bracket contributes $-\frac{1}{2}\tau\ell^{-2}e^a \wedge e^b J_{ab}$. □

On the central sheet the affine connection is recovered; the extreme sheets preserve the two signs of the constant-curvature term. The representation thus retains the tritic regime of the route before applying the variational equations. The holonomy equality (80.32) corresponds to the central sheet. On each extreme sheet, the representation $\rho_{\mathrm{C},\tau}$ in its Cartan group and its respective holonomy condition are used. The current and the metric tensor use the fields and representation of the same realization.

80.7 Variational response of memory energy

The transport of a route identifies the fibers at its endpoints and preserves the orientation and memory entering that identification. The associated energy compares the final field with the transported initial field. Its differential quantifies how that comparison responds to a variation of transport. This section obtains the response directly from the discrete functional and develops its composition, covariance and inversion.

80.7.1 Discrete functional and space of variations

At a finite level m, let $\mathcal{V}_m$ be the vertices of the route publication and $\mathcal{E}_m^+$ an orientation of each edge. Each vertex carries a finite-dimensional Hermitian fiber H_v. We define

$$\mathcal{H}_m = \bigoplus_{v\in\mathcal{V}_m} H_v, \qquad \mathcal{K}_m = \bigoplus_{a\in\mathcal{E}_m^+} H_{t(a)}.$$

The Hermitian inner product is antilinear in the first variable. For unitary $U_a : H_{s(a)} \to H_{t(a)}$, the difference operator is

$$(D_m f)_a = f_{t(a)} - U_a f_{s(a)}, \qquad E_m(f, U) = \langle D_m f, \mathsf{W}_m D_m f\rangle, \qquad \mathsf{W}_m = \mathsf{W}_m^* > 0. \tag{80.37}$$

This is the energy of the term $D_m^*\mathsf{W}_m D_m$ in the discrete generator. The weight W_m may couple different edges. The soldering operator W of Section 80.5 and this energy weight have different domains and roles.

The transport U is a publication of the enriched state. The field f and the weight W_m remain explicit as arguments of the functional; each material application preserves the rules that determine them. Differentiable families of the unitary realizations of transport are considered,

$$\dot{U}_a = A_a U_a, \qquad A_a^* = -A_a.$$

The variations used physically form the subspace preserving the conditions of the corresponding realization. The differential is calculated first in the unitary tangent space and then restricted to that subspace.

Proposition 80.32 (Transport-response covector)**.** *Write*

$$v_a = U_a f_{s(a)}, \qquad r = D_m f, \qquad q = \mathsf{W}_m r.$$

At fixed field and weight, the energy differential is

$$\mathfrak{j}_m(A) = -2\,\mathrm{Re}\sum_{a\in\mathcal{E}_m^+}\langle q_a, A_a v_a\rangle. \tag{80.38}$$

Its representative for the real Hilbert–Schmidt inner product is

$$\mathcal{J}_a = v_a q_a^* - q_a v_a^*, \qquad \mathcal{J}_a^* = -\mathcal{J}_a, \qquad \mathfrak{j}_m(A) = \sum_a \operatorname{Re}\operatorname{Tr}(\mathcal{J}_a^* A_a). \tag{80.39}$$

For simultaneous variations of field, transport and weight we have

$$\dot{E}_m = 2\operatorname{Re}\sum_a \left\langle q_a, \dot{f}_{t(a)} - U_a \dot{f}_{s(a)} - A_a v_a \right\rangle + \langle r, \dot{\mathsf{W}}_m r\rangle. \tag{80.40}$$

Proof. The first-order update, written in the algebra of dual numbers with $\epsilon^2 = 0$, gives

$$r + \epsilon \dot{r}, \qquad \dot{r}_a = \dot{f}_{t(a)} - U_a \dot{f}_{s(a)} - A_a v_a.$$

The coefficient of ϵ in the energy is

$$\langle \dot{r}, \mathsf{W}_m r\rangle + \langle r, \mathsf{W}_m \dot{r}\rangle + \langle r, \dot{\mathsf{W}}_m r\rangle.$$

Self-adjointness of the weight yields (80.40). Fixing field and weight leaves (80.38). Moreover,

$$\operatorname{Tr}(\mathcal{J}_a^* A_a) = v_a^* A_a q_a - q_a^* A_a v_a.$$

Anti-Hermiticity of A_a implies $v_a^* A_a q_a = -\overline{q_a^* A_a v_a}$, giving the stated representative. □

The response is thus evaluated: on each edge the field is transported, its difference from the final field is calculated and the energy weight is applied. The antisymmetrized product of those two vectors determines $\mathcal{J}_a \in \mathfrak{u}(H_{t(a)})$. This operator represents, through the real Hilbert–Schmidt inner product, the response covector on the unitary variations of the target fiber.

80.7.2 Composition and transport of the response

Proposition 80.33 (Composition rule along a route)**.** *For a route formed by N transports, let $U_\gamma = U_N \cdots U_1$ and $B_k = U_N \cdots U_{k+1}$, with $B_N = I$. If $\dot{U}_k = A_k U_k$, then*

$$\dot{U}_\gamma U_\gamma^{-1} = \sum_{k=1}^{N} B_k A_k B_k^{-1}. \tag{80.41}$$

Thus a representative $\mathcal{J}_\gamma$ in the final fibre induces in factor k the contribution $B_k^* \mathcal{J}_\gamma B_k$. When a factor belongs to several routes in the functional, its contributions are summed with the incidences used by that functional.

Proof. The product rule gives

$$\dot{U}_\gamma = \sum_k U_N \cdots U_{k+1} A_k U_k \cdots U_1.$$

Multiplication by U_γ^{-1} eliminates the factors to the right of A_k and yields (80.41). By unitarity and cyclicity of the trace,

$$\operatorname{Re}\operatorname{Tr}(\mathcal{J}_\gamma^* B_k A_k B_k^{-1}) = \operatorname{Re}\operatorname{Tr}((B_k^* \mathcal{J}_\gamma B_k)^* A_k).$$

Linearity of the differential proves the sum of contributions. □

This formula preserves the position of each operation within the route. Transport following an incidence determines how its contribution is represented in the final fiber; the local response is recovered by transporting it to the fiber of that incidence.

Proposition 80.34 (Covariance under unitary frame changes)**.** *Let g_v be unitary frame changes independent of the variation parameter, and $G_{\mathcal{E}} = \bigoplus_a g_{t(a)}$. Under the transformations*

$$f'_v = g_v f_v, \quad U'_a = g_{t(a)} U_a g_{s(a)}^{-1}, \quad \mathsf{W}'_m = G_{\mathcal{E}} \mathsf{W}_m G_{\mathcal{E}}^{-1}, \quad A'_a = g_{t(a)} A_a g_{t(a)}^{-1},$$

we have

$$\mathcal{J}'_a = g_{t(a)} \mathcal{J}_a g_{t(a)}^{-1}, \qquad \mathfrak{j}'_m(A') = \mathfrak{j}_m(A). \tag{80.42}$$

Proof. We obtain $r' = G_{\mathcal{E}} r$, $q' = G_{\mathcal{E}} q$ and $v'_a = g_{t(a)} v_a$. Substitution into (80.39) proves the first identity. Invariance of the Hermitian inner product, or cyclicity of the trace, proves the second. □

When frame changes depend on the parameter, their derivatives also act on the field and weight. The corresponding covariant expression is then the complete differential (80.40).

80.7.3 Oriented inversion and transport of the weight

Inversion can be seen explicitly in the case of a local energy per edge, $\mathsf{W}_m = \bigoplus_a W_a$. For $a : s \to t$, its inverse carries

$$U_{\bar{a}} = U_a^{-1}, \qquad r_{\bar{a}} = -U_a^{-1} r_a, \qquad W_{\bar{a}} = U_a^{-1} W_a U_a.$$

These rules preserve the edge energy exactly.

Proposition 80.35 (Differential compatibility with inversion)**.** *If W_a and the fields at both endpoints remain fixed, then*

$$A_{\bar{a}} = -U_a^{-1} A_a U_a, \qquad \dot{W}_{\bar{a}} = U_a^{-1} [W_a, A_a] U_a.$$

The complete variation of the reversed edge's energy coincides with the original. In particular, if $[W_a, A_a] = 0$, the partial transport covector satisfies

$$\mathfrak{j}_{\bar{a}}(U_a^{-1} A_a U_a) = -\mathfrak{j}_a(A_a). \tag{80.43}$$

Proof. The derivative of U_a^{-1} is $-U_a^{-1} A_a$, from which the two formulas result. Writing $v_a = U_a f_s$ and $f_t = r_a + v_a$, the part due exclusively to reversed transport equals

$$\mathfrak{j}_{\bar{a}}(A_{\bar{a}}) = -2 \operatorname{Re}\langle r_a, W_a A_a f_t \rangle = \mathfrak{j}_a(A_a) - 2 \operatorname{Re}\langle r_a, W_a A_a r_a \rangle.$$

Variation of the weight contributes

$$\langle r_{\bar{a}}, \dot{W}_{\bar{a}} r_{\bar{a}} \rangle = \langle r_a, [W_a, A_a] r_a \rangle = 2 \operatorname{Re}\langle r_a, W_a A_a r_a \rangle.$$

The two terms compensate each other. If the commutator is zero, that contribution disappears; linearity in A_a gives (80.43). □

Energy is summed over one orientation per edge. A sum over both orientations carries the factor $1/2$. This convention preserves the same functional and hence the same response.

80.7.4 Compatibility with refinement

Theorem 80.36 (Transport of the differential by refinement)**.** *Let $J_m : \mathcal{H}_m \to \mathcal{H}_{m+1}$ and $K_m : \mathcal{K}_m \to \mathcal{K}_{m+1}$ be fixed maps. For a family C^1, suppose the refinement identities*

$$D_{m+1}(t) J_m = K_m D_m(t), \qquad K_m^* \mathsf{W}_{m+1}(t) K_m = \mathsf{W}_m(t). \tag{80.44}$$

Then $E_{m+1}(J_m f) = E_m(f)$. The transport covectors agree on variations linked by the first identity. When the weights vary, their contributions to the differential also agree.

Proof. By substitution,

$$\langle D_{m+1}J_m f, \mathsf{W}_{m+1}D_{m+1}J_m f\rangle = \langle D_m f, K_m^{*}\mathsf{W}_{m+1}K_m D_m f\rangle = E_m(f).$$

Differentiating the first identity of (80.44) gives $\dot{D}_{m+1}J_m = K_m\dot{D}_m$. Consequently,

$$2\,\mathrm{Re}\langle D_{m+1}J_m f, \mathsf{W}_{m+1}\dot{D}_{m+1}J_m f\rangle = 2\,\mathrm{Re}\langle D_m f, \mathsf{W}_m\dot{D}_m f\rangle.$$

The derivative of the second identity is $K_m^{*}\dot{\mathsf{W}}_{m+1}K_m = \dot{\mathsf{W}}_m$, proving agreement of the weight-variation terms. □

The statement uses compatibility of the transport family. If J_m depends on the parameter, the chain rule also preserves the term

$$2\,\mathrm{Re}\langle D_{m+1}J_m f, \mathsf{W}_{m+1}D_{m+1}\dot{J}_m f\rangle.$$

The resulting naturality jointly transports the functional, its incidences and its differential.

80.7.5 Material realization of the response

The result of this section is the explicit covector $\mathfrak{j}_m$, with its representative $\mathcal{J}_a$. The material realization composes that covector with the variation of transport induced by the connection and with the normalization of the action. If $\mathcal{L}_m$ denotes that differential map, the covector on connection variations is $\mathcal{L}_m^{*}\mathfrak{j}_m$. The composition rule (80.41) evaluates its part corresponding to finite products of transports.

In section 82.8, the current σ_{IJ} is a three-form defined by variation of the material action. Identification between the two realizations preserves the field, representation, weight, dimensional normalization and passage to the limit used by that action. These data allow evaluation of the contraction entering the torsional density. The present development provides the discrete differential composed in that calculation.

Chapter 81

Perron Dynamics, Dilution a^{-6} of memory and Einstein–Cartan realization

The Perron operator fixes the stable memory mode; when published as a spin density, its contribution decays as the inverse sixth power of the scale factor. The Einstein–Cartan realization preserves this provenance and its conditions of validity.

81.1 Perron dynamics and the stable memory mode

On a closed route, the cylindrical Perron cocycle satisfies

$$V_n = \varphi^n. \tag{81.1}$$

In spatial dimension $D = 3$, the scale determined by the cocycle is

$$\frac{a_n}{a_0} = V_n^{1/3} = \varphi^{n/3}, \tag{81.2}$$

whereas the stable mode of the quadratic action is

$$M_n = \varphi^{-2n} = V_n^{-2}. \tag{81.3}$$

Eliminating n between equations (81.2) y (81.3) gives the central result:

Theorem 81.1 (Law cylindrical of sixth order for the memory)**.** *For every closed level of the Perron realization in $D = 3$,*

$$\boxed{M_n = \left(\frac{a_n}{a_0}\right)^{-6}}. \tag{81.4}$$

The exponent -6 is not fitted: it is the quotient of the stable exponent -2 by the dimensional root $1/3$.

In the distinguished returns one obtains

$$\frac{a_9}{a_0} = \varphi^3, \qquad \frac{a_{27}}{a_0} = \varphi^9, \qquad \frac{a_{108}}{a_0} = \varphi^{36},$$

and the same levels satisfy exactly $M_n = (a_n/a_0)^{-6}$. If the additive temporal parameter $t_n = nt_0$ is declared, the logarithmic rate is constant:

$$H_\varphi := \frac{\Delta \log a_n}{\Delta t} = \frac{\log \varphi}{3t_0}, \qquad H_\varphi t_0 = 0.160403941686534482499252971 1\ldots. \tag{81.5}$$

The symbol H_φ names the rate of this discrete scale. Reading it as a Hubble parameter requires the cosmographic transducer; the internal identity does not depend on that promotion.

81.2 Realization in the system of Einstein–Cartan

In an Einstein–Cartan spin fluid, the quadratic spin density obeys

$$s^2 \propto a^{-6}. \tag{81.6}$$

The equation (81.4) provides the exponent and the stable mode without consulting a bounce. To make the physical reading, one must declare a positive section s_*^2 and a map

$$\mathcal{T}_{\mathrm{EC}} : M_n \longmapsto s_n^2 := s_*^2 M_n = s_*^2 \left(\frac{a_n}{a_0}\right)^{-6}. \tag{81.7}$$

The map preserves the scaling law, but does not by itself determine the amplitude s_*^2, the spin current, the sign of the term of contorsion, or the equation of state. With mass density, an effective equation has the form

$$H^2 = \frac{8\pi G}{3}\left(\rho_{\mathrm{m}} - \rho_{\mathrm{spin,m}}\right) + \cdots, \tag{81.8}$$

whereas with energy density it must be written

$$H^2 = \frac{8\pi G}{3c^2}\left(\rho_{\mathrm{E}} - \rho_{\mathrm{spin,E}}\right) + \cdots. \tag{81.9}$$

The two conventions are not mixed. A bounce candidate requires, in addition to equality of the densities, regularity, matter, and boundary data.

Falsification criterion 81.2. The Einstein–Cartan transduction fails if the scale functional produces an exponent other than -6, if $V_n \neq \varphi^n$ along the declared route, or if the physical map does not preserve positivity and dimensions. Retrospectively choosing s_*^2 to force a bounce threshold does not establish the transducer.

81.3 Structural comparison between the 108-state cycle and Perron dynamics

	CT108–de Sitter framework	Perron–Einstein–Cartan framework
Antecedent	circular return of order 108	cylindrical cocycle and stable mode
Selection mechanism	$r_g = R_{108}$	$V_n = \varphi^n$, $D = 3$
Magnitudes determinadas	radius, curvature, temperature, and entropy	expansion, rate, and dilution of memory
Invariant	$G_a \hbar_a$ and circular radius	$M_n a_n^6$
Status	de Sitter conditioned interface	exact internal law; conditioned EC reading
Falsification criterion	incompatible radius or background equation	perdida of the exponente -6 or of the transductor

Hypothesis, domain, and conclusion

The CT108 identities, the selector θ_{108}, and the law $M_n = (a_n/a_0)^{-6}$ are *Internal exact theorem*. The reading of the same radius as de Sitter and the identification of M_n with a spin density are *Conditional physical realizations*, each with its own falsification criterion.

Coordination of torsion and area in the Palatini–Cartan–Holst action. Torsion and area are subsequently coordinated in the first-order Palatini–Cartan–Holst action.

81.4 Torsional Memory and Bounce Conditions in Einstein–Cartan

81.4.1 Transport of the Perron law to the spin spectral density

The realization in Perron cosmological form constructs two sequences on the same closed route:

$$\frac{a_n}{a_0} = \varphi^{n/3}, \qquad M_n = \varphi^{-2n}. \tag{81.10}$$

Elimination of level n produces the exact identity

$$\boxed{M_n = \left(\frac{a_n}{a_0}\right)^{-6}}. \tag{81.11}$$

The exponent has a definite genealogy: the spatial dimension contributes the factor 3, and the quadratic character of stable memory contributes the factor 2. The power $6 = 2 \cdot 3$ thus expresses the dilution of a quadratic density associated with an extensive spin charge.

Definition 81.3 (Torsional transducer)**.** Having fixed a positive amplitude s_*^2 on an initial section, the torsional transducer is

$$\mathcal{T}_{\mathrm{EC}} : M_n \longmapsto s_n^2 = s_*^2 M_n = s_*^2 \left(\frac{a_n}{a_0}\right)^{-6}. \tag{81.12}$$

Its domain is the stable Perron mode, and its codomain is the quadratic spin sector of an Einstein–Cartan realization.

The equation equation (81.12) preserves the exponent and positivity. The amplitude s_*^2, the sign with which the contorsion enters the gravitational equation, and the dimensional normalization belong to the physical realizer. This separation establishes exactly what the HMT trunk supplies and what Cartan dynamics must supply.

81.4.2 Homogeneous Equations and Density Convention

To present the structural mechanism, we temporarily adopt units $c = 1$ and a spatially flat FRW geometry. Let

$$\rho_m(a) = \rho_0 a^{-3(1+w)}, \qquad \rho_s(a) = s_0 a^{-6}, \tag{81.13}$$

with $\rho_0 > 0$, $s_0 > 0$, and w constant. The contribution from contorsion enters with a repulsive sign in the effective equation:

$$H^2 = \frac{8\pi G}{3}\big(\rho_m - \rho_s\big). \tag{81.14}$$

If energy densities are used, the right-hand side is multiplied by c^{-2}; if mass densities are used, it retains the form equation (81.14). The document maintains a single convention in each calculation and restores the dimensional chart at the end.

The corresponding Raychaudhuri equation is

$$\dot{H} = -4\pi G\big[(1+w)\rho_m - 2\rho_s\big]. \tag{81.15}$$

The second contribution behaves as a stiff fluid with equation of state $w_s = 1$ and negative effective density. This form is the homogeneous publication of torsional repulsion at high density.

81.4.3 Existence and uniqueness of the bounce radius

Theorem 81.4 (Regular reflection for $w < 1$)**.** *Assume $\rho_0 > 0$, $s_0 > 0$, and $w < 1$. Then there exists a unique positive radius a_b such that $H(a_b) = 0$, given by*

$$\boxed{a_b^{3(1-w)} = \frac{s_0}{\rho_0}.} \tag{81.16}$$

At that point

$$\boxed{\dot{H}_b = 4\pi G(1-w)\rho_b > 0,} \qquad \rho_b := \rho_m(a_b) = \rho_s(a_b), \tag{81.17}$$

so that the contractive branch is extended into a regular expansive branch.

Proof. The condition $H = 0$ is equivalent to $\rho_0 a^{-3(1+w)} = s_0 a^{-6}$. Multiplication by a^6 yields equation (81.16). The exponent $3(1-w)$ is positive; therefore the power defines a bijection of $\mathbb{R}_{>0}$ onto itself and the radius is unique. Substituting $\rho_m = \rho_s = \rho_b$ into equation (81.15),

$$\dot{H}_b = -4\pi G[(1+w) - 2]\rho_b = 4\pi G(1-w)\rho_b > 0.$$

Positivity turns the simple zero of H into a minimum of $a(t)$. □

The local expansion around t_b makes the regularity explicit:

$$a(t) = a_b\left[1 + 2\pi G(1-w)\rho_b(t-t_b)^2 + O\Big((t-t_b)^3\Big)\right]. \tag{81.18}$$

The law a^{-6} supplies the power that dominates the material sector when $w < 1$ and a decreases. The torsional sign and positive amplitude cause that dominance to halt the contraction before reaching $a = 0$.

81.4.4 Limit Cases of Torsional Dynamics and Obstructions to the Regularity of the Bounce

The value $w = 1$ produces the same exponent a^{-6} in both terms. The difference then retains a fixed sign determined by $\rho_0 - s_0$; pointwise equality ceases to select an isolated radius. For $w > 1$, the material density grows faster than the torsional term during contraction, and the preceding mechanism does not guarantee a bounce. These cases provide internal negative controls: the closure depends on the inequality of the exponents and not on a retrospective choice of the threshold.

Spatial curvature k, a cosmological constant, and Bianchi anisotropies add terms to equation (81.14). Their local effect is controlled without presupposing the position of the new zero. Define

$$F_0(a) := \frac{8\pi G}{3}\big(\rho_m(a) - \rho_s(a)\big).$$

The perturbed equation takes the form $H^2 = F_R(a) := F_0(a) + R(a)$.

Theorem 81.5 (Persistence C^1 of the rebote transversal)**.** *Let a_b be the radius of equation (81.16). There exist $0 < a_- < a_b < a_+$ such that, for $I = [a_-, a_+]$,*

$$F_0(a_-) < 0 < F_0(a_+), \qquad m_I := \inf_{a\in I} F_0'(a) > 0.$$

If $R \in C^1(I)$ satisfies

$$\|R\|_{C^0(I)} < \delta_I := \min\{-F_0(a_-), F_0(a_+)\}, \qquad \|R'\|_{C^0(I)} < m_I, \tag{81.19}$$

then there exists a unique $a_R \in (a_-, a_+)$ with $F_R(a_R) = 0$, and that zero is transversal:

$$F_R'(a_R) > 0, \qquad \dot{H}_R(a_R) = \frac{a_R}{2}F_R'(a_R) > 0. \tag{81.20}$$

For a family $R(a,\eta)$ of class C^1, the implicit function theorem provides a local branch $a_R(\eta)$ of class C^1.

Proof. The identity equation (81.17) is equivalent to $F_0'(a_b) > 0$. By continuity, I can be chosen with $m_I > 0$ and with the indicated change of sign. The first bound for equation (81.19) preserves $F_R(a_-) < 0 < F_R(a_+)$; the intermediate value theorem produces a zero. The second bound gives $F_R'(a) \geq m_I - \|R'\|_{C^0(I)} > 0$, so F_R is strictly increasing and the zero is unique. On each branch with $H \neq 0$, differentiating $H^2 = F_R(a)$ and using $\dot{a} = aH$ gives $\dot{H} = (a/2)F_R'(a)$; continuity extends the equality to a_R. The nondegeneracy $F_R'(a_R) > 0$ is precisely the hypothesis of the implicit function theorem. □

81.4.5 Torsional amplitude from the Cartan screen

The memory exponent is closed by equation (81.11). The amplitude may be sought prospectively in the screen response. Let the exact identity be

$$(\beta_m^{\square})^* \Lambda_m^{\square} \beta_m^{\square} = \frac{56}{81} I_{Q_{\square}}. \tag{81.21}$$

A complete gravitational realization must construct a spin current

$$\mathcal{J}_m = \mathcal{J}(\beta_m^{\square}, U_m, c_s, c_g) \tag{81.22}$$

and a positive functional

$$s_{0,m} = \mathfrak{S}(\mathcal{J}_m, \Lambda_m^{\square}) \tag{81.23}$$

compatible with refinement. The pertinent nonadic normalization is the tracial expectation, because it preserves the density of equation (81.21); the raw sum counts nine copies and belongs to a distinct extensive quantity.

Proposition 81.6 (Obligations of the amplitude selector)**.** *An admissible selector for s_0 must satisfacer:*

i) positivity and dimensions of squared spin density;

ii) gauge invariance under the reduction induced by the transported cubic structure;

iii) compatibility with edge inversion and carry;

iv) naturality under 9-ary refinement;

v) independence from any bounce radius used as a target value.

These conditions turn the amplitude into a localized algebraic constraint. The screen-energy equality supplies the quadratic form; the Cartan realizer must determine how that form contracts with the physical current.

81.4.6 Torsion, orientation, and transport memory

The metric publishes distances and causal cones. The Cartan connection also publishes the orientation of transport. In HMT, route, carry, and deep state already constitute genealogical data. The correspondence

$$\text{stable route memory} \longrightarrow \text{spin current} \longrightarrow \text{contorsion} \longrightarrow \rho_s \propto a^{-6} \tag{81.24}$$

provides a physical reading of that oriented information. Curvature describes the response of geometry to energy; torsion preserves the history of orientation that accompanies transport. The sixth-order law is the point at which genealogical memory and quadratic spin density have exactly the same scaling covariance.

81.4.7 Mach's principle and cosmographic autoinertia

The local state is evaluated within a route compatible with the global boundary. The inertial response therefore belongs to the holonomic stabilizer of the totality. In the homogeneous sector, the scale a_n, the memory M_n, and the torsional current are three readings of the same enriched genealogical state:

$$X_n \longmapsto \big(a_n, M_n, \mathcal{J}_n\big). \tag{81.25}$$

The Machian interpretation thereby acquires operatorial content: the local response depends on the global closure class, and the observer forms part of the prolonged section. Self-inertia is the preservation of the relative state under the holonomy that jointly transports geometry, observer, and memory.

81.4.8 Exact identities of torsional dynamics and Einstein–Cartan interface conditions

Estatuto probatorio

The law $M_n = (a_n/a_0)^{-6}$ is an exact internal theorem. The bounce theorem equations (81.16) y (81.17) is an exact consequence of the homogeneous Einstein–Cartan equations [65] with $\rho_0, s_0 > 0$ and $w < 1$. The transducer equation (81.12) exactly preserves the scaling law. The determination of s_0 from the screen current, the extension to anisotropies, and the complete derivation of the Einstein–Cartan action are delimited physical interfaces.

Falsification criterion

The internal block is refuted if the Perron mode violates equation (81.11). The Einstein–Cartan realization is refuted if the induced current produces another power, a nonpositive amplitude, or a sign that eliminates equation (81.17). The homogeneous bounce fails for the declared data if the radius equation (81.16) does not exist, if $H(a_b) \neq 0$, or if $\dot{H}_b \leq 0$. The complete promotion is rejected if the amplitude functional retrospectively consults the desired radius or loses gauge, carry, or refinement.

Chapter 82

Palatini–Cartan–Holst realization of the Barbero–Immirzi parameter and physical transduction of the area spectrum

The structural derivation of the parameter and the area spectrum precedes this chapter. Here the tetrad, connection, curvature, and first-order action are constructed, and only then is the Holst parameter specialized as a physical realization of the areal reader.

82.1 Curvature, torsion, and Einstein–Cartan–Holst action

The HMT transport structure distinguishes the memory variable, holonomy, and closure defect. Its gravitational realization is formulated by means of a tetrad and a connection. This section establishes the first-order geometry and specifies the type of the map between it and the TPK state space .

82.1.1 Discrete Gauss–Stokes identity and incidence transport to the boundary

Let K be a cellular complex, let W be an abelian group, and let $\Omega \in C^1(K;W)$ be a cochain. We denote by $\delta : C^1(K;W) \to C^2(K;W)$ the cellular coboundary and by $\langle \cdot\,,\, \cdot \rangle$ the pairing between cochains with values in W and integral cellular chains.

Theorem 82.1 (Universal Stokes Cell Identity)**.** *For every 2-cadena finite $z \in C_2(K;\mathbb{Z})$,*

$$\boxed{\langle \delta\Omega, z\rangle = \langle \Omega, \partial z\rangle.}$$

Proof. This is the identity that defines the coboundary as the adjoint operator of the cellular boundary. The finiteness of z ensures that both pairings are finite sums in W. □

Corollary 82.2 (Gauss–Stokes on a cellular pseudomanifold)**.** *Let K be a finite, compact, regular two-dimensional cellular pseudomanifold, coherently oriented and possibly with boundary. If z_K is the sum of its 2-cells with the coherent orientation, then*

$$\sum_{f \in K^{(2)}} (\delta\Omega)(f) = \sum_{e} [\partial z_K : e]\, \Omega(e).$$

In particular, regularity allows the uncancelled incidences of ∂z_K to be identified with the edges of the geometric boundary.

Proof. The theorem is applied to z_K. Coherent orientation makes each interior edge appear twice in ∂z_K with opposite coefficients; the uncancelled incidences form the boundary chain. □

The non-Abelian version is not obtained merely by replacing sums with products. In this work it is used only when transports to a common base, an ordering of the holonomies, and the corresponding conjugation have been fixed. Outside that subdomain, no non-Abelian Stokes identity is asserted.

82.1.2 Area capacity of the qutrit cell: $81 \log 3$ and collective incidence transport

Theorem 82.3 (Minimum cut of a discrete cube)**.** *In the cubic graph $N \times N \times N$ with unit capacity, every cut separating two opposite faces has capacity at least N^2, and one exists with capacity N^2.*

Proof. There are N^2 edge-disjoint paths crossing the cube in the direction normal to the two faces. Every cut must intersect them, so its capacity is at least N^2. A transverse layer contains exactly N^2 edges and attains the bound. □

For $N = 9$, the cut equals 81 and

$$9^3 = 729 = 27^2.$$

The last equality permits six trits to be regrouped as a volume 9^3 or a grid of cardinality 27^2. It is a bijection of sets; it is not an isomorphism between $(\mathbb{Z}/9\mathbb{Z})^3$ and $(\mathbb{Z}/27\mathbb{Z})^2$, whose exponents are distinct.

82.1.3 Tetrad, spin connection, and torsion of the Cartan–Holst realization

Let e^I be a tetrad and $\omega^{IJ} = -\omega^{JI}$ a Lorentz connection. Define

$$T^I_{\mathrm{tor}} = D_\omega e^I = \mathrm{d}e^I + \omega^I{}_J \wedge e^J,$$

$$F^{IJ} = \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}, \qquad g = \eta_{IJ} e^I \otimes e^J.$$

The decomposition

$$\omega^{IJ} = \mathring{\omega}^{IJ}(e) + K^{IJ}$$

separates the Levi–Civita connection and the contorsion. Then

$$T^I_{\mathrm{tor}} = K^I{}_J \wedge e^J.$$

Curvature and torsion are distinct defects: F measures closure of vector transport; T_{tor}, closure of the tetrad. The T memory of the Ambivalence Principle is likewise not geometric torsion: only a realization functor can relate them.

82.1.4 Palatini–Holst Action and Variational Equations

Let M be a smooth oriented four-dimensional manifold. If $\partial M \neq \varnothing$, we assume compactly supported variations in the interior or, equivalently for the variational problem considered, a boundary term and boundary conditions that annul the boundary contribution. When Ψ contains spinorial fields, we also assume a spin structure and fields compatible with it. Let

$$\Sigma^{IJ} = e^I \wedge e^J, \qquad (\star X)^{IJ} = \frac{1}{2}\epsilon^{IJ}{}_{KL} X^{KL}, \qquad \kappa = \frac{8\pi G}{c^4}, \qquad \gamma \in \mathbb{R} \setminus \{0\}.$$

The Palatini–Cartan–Holst action is

$$\boxed{S[e, \omega, \Psi] = \frac{1}{2\kappa c} \int_M \Sigma_{IJ} \wedge \left(\star F^{IJ} + \frac{1}{\gamma} F^{IJ} \right) - \frac{\Lambda}{\kappa c} \int_M \mathrm{vol}_e + S_{\mathrm{m}}[e, \omega, \Psi].}$$

The variation of this action determines the torsional corrections [108, 72, 65, 62].

We define the spin current by

$$\delta_\omega S_{\mathrm{m}} = -\frac{1}{2} \int_M \delta\omega^{IJ} \wedge \sigma_{IJ}.$$

Theorem 82.4 (Cartan–Holst Equation)**.** *Under the preceding geometric and boundary hypotheses, variation with respect to ω gives*

$$D_\omega \left[(\star + \gamma^{-1} I) \Sigma^{IJ} \right] = \kappa c \, \sigma^{IJ},$$

with the global normalization fixed by the preceding definition of σ^{IJ}.

Proof. One uses $\delta F^{IJ} = D_\omega \delta\omega^{IJ}$, integrates by parts, and sets the coefficient of the arbitrary variation $\delta\omega^{IJ}$ equal to zero. With the sign convention fixed in the definition of σ^{IJ}, the matter term moves to the right-hand side with a positive sign. The factor $1/(2\kappa c)$ produces the coefficient κc. The cosmological

term does not depend on ω, and the boundary conditions eliminate the term generated by integration by parts. □

In signature lorentziana $\star^2 = -I$ on bivectores and, for $\gamma \in \mathbb{R} \setminus \{0\}$,

$$(\star + \gamma^{-1} I)^{-1} = \frac{\gamma^2}{1+\gamma^2}(-\star + \gamma^{-1} I).$$

Corollary 82.5 (Torsional decoupling)**.** *For a co-tetrada no degenerada and $\gamma \in \mathbb{R} \setminus \{0\}$,*

$$\sigma^{IJ} = 0 \Longrightarrow T^I_{\mathrm{tor}} = 0 \Longrightarrow \omega = \mathring{\omega}(e).$$

To make the normalization explicit, we define the normalized sources by the action through

$$\delta_g S_{\mathrm{m}} = -\frac{1}{2} \int_M \mathrm{vol}_e \; \mathcal{T}^S_{\mu\nu} \, \delta g^{\mu\nu}.$$

When the tetrad coordinates have the dimension of length, the conventional physical energy–momentum tensor is $T^{\mathrm{phys}}_{\mu\nu} := c\, \mathcal{T}^S_{\mu\nu}$. We likewise denote by $\mathcal{U}^{S,\mathrm{spin}}$ the contribution normalized by the action and by $U^{\mathrm{phys,spin}} := c\, \mathcal{U}^{S,\mathrm{spin}}$ its physical convention. Upon algebraically eliminating the connection, the metric equation then assumes the two equivalent forms

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa c \left(\mathcal{T}^{S,\mathrm{LC}}_{\mu\nu} + \mathcal{U}^{S,\mathrm{spin}}_{\mu\nu} \right) = \kappa \left(T^{\mathrm{phys,LC}}_{\mu\nu} + U^{\mathrm{phys,spin}}_{\mu\nu} \right),$$

where the spin contribution is quadratic in the corresponding current and depends on the matter coupling. The coefficient of that contribution is not fixed without specifying the fermionic action and the duality conventions.

82.1.5 Bianchi Identity with Torsion and Flow Balance in the Cartan Connection

For every connection,

$$D_\omega T^I_{\mathrm{tor}} = F^I{}_J \wedge e^J, \qquad D_\omega F^{IJ} = 0.$$

After removing torsion,

$$\nabla^\mu (T^{\mathrm{phys,LC}}_{\mu\nu} + U^{\mathrm{phys,spin}}_{\mu\nu}) = 0.$$

These identities are mandatory consistency checks: every HMT gravitational term must arise from a compatible action or satisfy the corresponding Noether balance.

82.1.6 Map of realization and specialization of the Holst parameter

The insertion defined in section 51.2 takes

$$\gamma = \Gamma_{6\to 8}.$$

Once this substitution has been made, all the preceding variational consequences are theorems of the specialized action. The physical selection of this action by HMT transport is expressed through a realization map

$$\mathfrak{R}_{\mathrm{grav}} : \mathcal{X}_{\mathrm{TPK}} \dashrightarrow \{(e, \omega, \Psi)\}$$

endowed with three properties: intertwining TPK transport with that of ω, identifying the holonomy classes, and furnishing the scales of G and Λ. These properties fix the precise domain of a gravitational realization of the formalism.

For a nondegenerate co-tetrad, nonzero real γ, and matter without spin current, the torsional sector reduces to the first-order formulation of general relativity with the same cosmological constant, subject to the already declared boundary conditions. In a spin fluid, the scaling $s^2 \propto a^{-6}$ can produce an

Einstein–Cartan bounce; obtaining its density and initial conditions from the TPK state is a question distinct from the variational consistency of the action.

82.2 Nonadic Return, Memory Groupoid and Cartan Entanglement Condition

The hierarchy $27 \to 54 \to 108$, the directed cumulative coverage, and the memory groupoid supply the exact condition under which the TPK return admits a compatible Cartan connection. This bridge precedes the Palatini–Cartan–Holst specialization and preserves its domain and refutation criteria intact.

82.3 Nonadic Return, Memory, and Cartan Holonomy

The gravitational realization does not begin by identifying a discrete variable with a geometric magnitude. It begins by distinguishing three levels: return of observable transport, memory distinguishing histories with equal projection, and holonomy of a connection. The bridge formulated here receives the returns and cocycles already demonstrated in HMT; it does not reconstruct them from the Holst action, from a mass, or from a metrological constant.

82.3.1 Exact hierarchy of returns

Let F_{obs} be the observable chronology defined in the HMT kernel, and let

$$\mathcal{K}_{27} := F_{\mathrm{obs}}^{27}.$$

Symbol $\mathcal{K}_{27}$ designates exclusively the composition of twenty-seven updates of the positional and oriented loop; it does not designate a generic window, a trace of 108 steps, or the complete enriched TPK state. Denote by p the projection onto the positions of the two cursors and by J_{or} the simultaneous inversion of their orientations. The primary proof is located in proposition 6.11; its typed consequences are

$$p(\mathcal{K}_{27}x) = p(x), \qquad o(\mathcal{K}_{27}x) = \mathsf{J}_{\mathrm{or}}\, o(x), \qquad \mathsf{J}_{\mathrm{or}}^2 = I,$$

$$F_{\mathrm{obs}}^{54} \text{ : restores positions and orientations,} \qquad F_{\mathrm{obs}}^{108} = I \text{ : also restores the observable phase.}$$

Thus, position return, orientation return, and phase return are distinct assertions. None of them by itself implies the return of all fields of an enriched extension.

The mode quotient and orientation cocycles are established in section 6.3.7 y theorem 6.20. In areal–volumetric order,

$$N_{27} = 9F_{\mathrm{av}}, \qquad F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

If

$$\Omega_{\mathrm{sw}}(\gamma) = \sum_{a \subset \gamma} \omega_{\mathrm{sw}}(a), \qquad S(\gamma) = \sum_{a \subset \gamma} |\omega_{\mathrm{sw}}(a)|, \qquad G_{\mathrm{or}}(\gamma) = \sum_{a \subset \gamma} g(a),$$

then a block of twenty-seven steps contains three nonadic turns and one orientation reversal:

$$\Omega_{\mathrm{sw}}(\mathcal{K}_{27}) = 0, \qquad S(\mathcal{K}_{27}) = 18, \qquad G_{\mathrm{or}}(\mathcal{K}_{27}) = 1.$$

In the observable return of 108 steps,

$$\Omega_{\mathrm{sw}}(C_{108}) = 0, \qquad S(C_{108}) = 72, \qquad G_{\mathrm{or}}(C_{108}) = 4.$$

Thus, for the additive functional declared

$$\mathcal{A}_*(\gamma) := G_{\mathrm{or}}(\gamma) + \lambda_* S(\gamma),$$

are obtained, without introducing a dimensional scale,

$$\mathcal{A}_*(\mathcal{K}_{27}) = 1 + 18\lambda_*, \qquad \mathcal{A}_*(C_{108}) = 4 + 72\lambda_*.$$

These values are record counts. Their subsequent thermal or gravitational use does not alter their provenance.

82.3.1.1 Directed system of coverings, refinement, and limit of the Cartan connection

Let O_{108} be the observable orbit and, for each chronological edge e, let

$$c(e) = \big(g(e), s(e)\big) \in \mathbb{N}^2$$

the pair formed by the indicators of orientation inversion and effective sheet change. For a directed history $\gamma = e_1 \cdots e_n$, define

$$c(\gamma) = \sum_{j=1}^{n} c(e_j).$$

The concatenation law takes the form

$$c_{m+n}(x) = c_m(x) + c_n\big(F_{\mathrm{obs}}^m x\big)\,.$$

The previous counts express, uniformly over the orbit,

$$c_{27}(x) = (1, 18), \qquad c_{108}(x) = (4, 72).$$

Theorem 82.6 (Cumulative coverage of the observable chronology)**.** *On*

$$\widetilde{O}_{108}^{+} := O_{108} \times \mathbb{N}^2$$

define

$$\widetilde{F}^{+}(x, m) := \big(F_{\mathrm{obs}} x, m + c(x \to F_{\mathrm{obs}} x)\big).$$

Then the transport of twenty-seven steps satisfies

$$\widetilde{\mathcal{K}}_{27}^{+}(x, m) = \big(\mathcal{K}_{27} x, m + (1, 18)\big),$$

and

$$\Big(\widetilde{\mathcal{K}}_{27}^{+}\Big)^4 (x, m) = \big(x, m + (4, 72)\big) \neq (x, m).$$

Therefore, its observable projection has order four, whereas the cumulative action of the chronological monoid has infinite semigroup order.

Proof. The cocyclic law and the constancy of c_{27} give

$$c_{108}(x) = \sum_{j=0}^{3} c_{27}(\mathcal{K}_{27}^{j} x) = 4(1, 18) = (4, 72).$$

The first coordinate returns through $\mathcal{K}_{27}^4 = I$; the second does not return. After $q > 0$ iterations, the increment is $q(4, 72)$, which is nonzero. □

Remark 82.7 (Type of the cumulative extension). The preceding construction extends directed histories through positive counters. It is not an automorphic lift of groupoid $\mathcal{G}_{\mathrm{mem}}$, nor does it prove that the unbounded coordinate already belongs to state $\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}}$. Its Grothendieck completion $\mathbb{Z}^2$ may be used for harmonic analysis, but it does not transform the positive cost $\mathcal{A}_* = G_{\mathrm{or}} + \lambda_* S$ into a character of the isotropy group: upon formally inverting a route, its indicators do not acquire opposite signs.

Domain of validity of the cumulative extension. The hierarchy $27 \to 54 \to 108$, the incidences of F_{av}, and the preceding cocycles are exact identities in the declared chronology and readers. The assertion is restricted to the observable state preserved by those readers; extending it to the full enriched state would require proving the return of every forgotten field.

82.3.2 Memory Groupoid

Let $\mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}$ be the enriched path groupoid: its objects are surviving TPK states, and its arrows are admissible routes with declared source and target, composition by concatenation, and inversion by route reversal. Let $\sim_{\mathrm{mem}}$ be an equivalence relation that identifies two arrows only when it preserves the memory fields required by the reader under consideration and is congruent with identities, inversion, and concatenation. We define

$$\mathcal{G}_{\mathrm{mem}} := \mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}/\!\sim_{\mathrm{mem}} .$$

This definition does not transform memory into torsion. Memory belongs to the discrete state and distinguishes histories; torsion belongs to a geometric connection and measures $D_\omega e$. Only a map between the two domains can relate them.

A geometric realization requires, as additional data, a length $\ell_0 > 0$, a representation of the groupoid

$$\rho_{\mathrm{C}} : \mathcal{G}_{\mathrm{mem}} \longrightarrow \mathrm{Spin}(1,3) \ltimes \mathbb{R}^{1,3},$$

and a field map

$$\mathfrak{R}_{\mathrm{grav}} : \mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}} \dashrightarrow \mathsf{Cartan}_{1,3}.$$

To define the second map, one must specify its action on objects and routes, its compatibility with concatenation and inversion, and the regularity of the resulting connection. None of these properties follows from the mere count of returns.

82.3.3 Joint connection and entanglement condition

Under the geometric hypotheses of section 82.4 —variedad oriented of dimension four, co-tetrad e^I, Lorentz connection ω^{IJ} and boundary conditions declared there—, the connection of Cartan joint is written

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1} e \\ 0 & 0 \end{pmatrix}.$$

Its curvature is the algebraic identity.

$$\mathcal{F}_{\ell_0} = \mathrm{d}\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_\omega & \ell_0^{-1} T_{\mathrm{tor}} \\ 0 & 0 \end{pmatrix},$$

where

$$F_\omega^{IJ} = \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}, \qquad T^I_{\mathrm{tor}} = D_\omega e^I.$$

Curvature and torsion appear as distinct components of the same transport defect. The matrix equality follows algebraically from the explicit data e, ω, ℓ_0, but does not construct these inputs from TPK.

The strong content of the bridge is the commutativity of holonomies:

$$\boxed{\mathrm{Hol}_{\mathcal{A}_{\ell_0}}\big(\mathfrak{R}_{\mathrm{grav}}(\gamma)\big) = \rho_{\mathrm{C}}(\mathrm{Hol}_{\mathrm{TPK}}(\gamma))} \qquad (\gamma \in \mathsf{P}^{\mathrm{enr}}_{\mathrm{TPK}}).$$

The equality must be compatible with identities, inversion, concatenation, subdivision, change of base point, and conjugation. Only then is discrete memory represented in a connection; it is not merely renamed as torsion or curvature.

Comparison Criterion 82.8 (Existence conditions for the bridge TPK–Cartan)**.** There is not realization with the declared commutativity and compatibility if any of the following facts occurs:

1. two equivalent representatives of the same route produce nonconjugate geometric holonomies;
2. the image does not preserve identities, inverses, or concatenation;
3. an admissible subdivision alters the holonomy;
4. the equality of holonomies fails for $\mathcal{K}_{27}$ or for their concatenations of depths 54 and 108;
5. the map identifies memory and torsion without exhibiting a morphism that preserves their types;
6. some descending choice —the Holst action, G, a mass or an SI chart— is used to redefine A, C^* or the original TPK return.

Antecedence of the functional $\Gamma_{6\to 8}$ with respect to the Palatini–Cartan–Holst realization. section 51.2 fixes, once $A, C^*, q_\pm$ have been constructed, the internal functional $\Gamma_{6\to 8}$. section 82.4 then studies the Palatini–Cartan–Holst action and its variational equations. The present bridge does not introduce a reverse route between the two stages.

82.4 Gravity: Holonomy, Torsion, and Collective Response

Gravity is not introduced here through a decimal formula for G. Its construction assembles two already typed branches. In the first, a route preserves orientation and memory; a connection compares frames at distinct points; its holonomy measures return and its curvature the closure defect. In the second, the dimensional chart constructs $G_{\mathrm{int}}(\theta_G)$ and κ_{int} from the internal action scale, velocity, and elementary duration. Only after both branches are available is the Palatini–Cartan–Holst action written. No external metrological figure participates in that precedence.

82.4.1 Returns: position, orientation, phase, and memory

Let F_{obs} be the observable chronology of the transport and

$$\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}.$$

The exact hierarchy distinguishes three returns:

27 steps	position return and orientation reversal,
54 steps	position and orientation return,
108 steps	complete return of the observable phase.

The observable return does not erase the accumulated record. If $g(e)$ counts an orientation reversal and $s(e)$ an effective sheet change, then

$$c(\gamma) = \sum_{e\subset\gamma} (g(e), s(e))$$

satisfies the cocyclic law of concatenation. In the canonical blocks,

$$c_{27} = (1, 18), \qquad c_{108} = (4, 72).$$

Therefore, the observable projection of $\mathcal{K}_{27}$ has order four, whereas the cumulative coordinate continues to grow. Returning to the same phase does not mean returning to the same history.

Theorem 82.9 (Cumulative coverage of return)**.** *Over $O_{108} \times \mathbb{N}^2$, the lift*

$$\widetilde{F}(x, m) = \big(F_{\mathrm{obs}}x, m + c(x \to F_{\mathrm{obs}}x)\big)$$

satisfies

$$\widetilde{\mathcal{K}}_{27}(x,m) = \big(\mathcal{K}_{27}x, m + (1,18)\big)$$

and

$$\widetilde{\mathcal{K}}_{27}^{4}(x,m) = \big(x, m + (4,72)\big) \neq (x,m).$$

Proof. The constancy of c_{27} along the orbit and the concatenation law give

$$c_{108} = \sum_{j=0}^{3} c_{27}(\mathcal{K}_{27}^{j}x) = 4(1,18).$$

The first coordinate returns; the second preserves the accumulated memory. □

This is the datum that to geometric realization must preserve. One does not identify the discrete memory with torsion: one constructs to representation that allows them to is compared.

82.4.2 From the path groupoid to the Cartan group

Let $\mathcal{G}_{\mathrm{mem}}$ be the groupoid of enriched routes, with composition by concatenation and inverse by reversal. Fix an element $R_4 \in \mathrm{Spin}^{+}(1,3)$ of order four and a unit temporal vector $u \in \mathbb{R}^{1,3}$ fixed by R_4. If

$$c(\gamma) = \big(c_g(\gamma), c_s(\gamma)\big) \in \mathbb{Z}^2$$

is the additive cocycle of inversion and sheet change, we define

$$\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma) = \Big(R_4^{\,c_g(\gamma)}, \ell_0 c_s(\gamma) u\Big) \in \mathrm{Spin}^{+}(1,3) \ltimes \mathbb{R}^{1,3}.$$

Theorem 82.10 (Discrete realization of the return cocycle)**.** *$\rho_{\mathrm{C}}^{\mathrm{disc}}$ is a functor from the route groupoid to the Cartan group. In particular, it preserves identity, concatenation, and inversion, and transports the return* $27 \to 54 \to 108$ *without erasing the cumulative coordinate.*

Proof. The semidirect product law is

$$(R,a)(S,b) = (RS, a + Rb).$$

Since $R_4 u = u$ and c is additive,

$$\begin{aligned}\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_2)\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_1) &= \Big(R_4^{\,c_g(\gamma_2)+c_g(\gamma_1)}, \ell_0\big(c_s(\gamma_2) + c_s(\gamma_1)\big)u\Big) \\ &= \rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_2\gamma_1).\end{aligned}$$

The identity route has zero cocycle, and reversal changes its sign, so identity and inverse are preserved. The return formulas follow from $c_{27} = (1,18)$ and $c_{108} = 4c_{27}$. □

This theorem constructs the discrete bridge. A subsequent smooth realization consists of a representation

$$\rho_{\mathrm{C}} : \mathcal{G}_{\mathrm{mem}} \longrightarrow \mathrm{Spin}(1,3) \ltimes \mathbb{R}^{1,3}$$

and in fields e^{I}, ω^{IJ} whose holonomy intertwines that representation. The diagram fixing the type is

$$\mathrm{Hol}_{\mathcal{A}_{\ell_0}}\big(\mathfrak{R}_{\mathrm{C}}(\gamma)\big) = \rho_{\mathrm{C}}\big(\mathrm{Hol}_{\mathrm{TPK}}(\gamma)\big).$$

Identities, inversion, concatenation, and subdivision must be preserved. This equation does not rename the carry as curvature: it requires the already represented discrete transport and the geometric transport to realize the same composition.

Once to positive scale is fixed ℓ_0, the joint connection is

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1} e \\ 0 & 0 \end{pmatrix}.$$

Proposition 82.11 (Cartan Joint Curvature)**.** *The curvature of $\mathcal{A}_{\ell_0}$ is*

$$\mathcal{F}_{\ell_0} = \mathrm{d}\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_\omega & \ell_0^{-1} T_{\mathrm{tor}} \\ 0 & 0 \end{pmatrix},$$

where

$$F_\omega^{IJ} = \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}, \qquad T_{\mathrm{tor}}^I = D_\omega e^I.$$

Proof. The matrices of forms are multiplied using the exterior product. The diagonal block produces F_ω; the translation block produces $\mathrm{d}e + \omega \wedge e = D_\omega e$. □

Curvature and torsion are two components of the same transport defect, but they perform distinct functions. Curvature measures frame return; torsion measures closure of the co-tetrad.

82.4.3 Gauss–Stokes and areal capacity

In a finite cellular complex, for every 1-cochain Ω and every 2-chain z,

$$\boxed{\langle \delta\Omega, z \rangle = \langle \Omega, \partial z \rangle.}$$

The identity is the definition of the coboundary as the adjoint of the boundary. When z sums the coherently oriented faces of a pseudomanifold, the interior edges cancel, and only the geometric boundary remains. The same equality has already been read as vorticity–circulation and as curvature–holonomy.

The areal scale also has an exact model. In the cubic graph $N \times N \times N$, every cut between two opposite faces contains at least N^2 edges, and to transverse layer attains the bound. For $N = 9$,

$$9^3 = 729 = 27^2.$$

This cardinal equality relates volume and minimal cut; it does not identify the groups $(\mathbb{Z}/9)^3$ and $(\mathbb{Z}/27)^2$, which have distinct exponents.

82.4.4 The Palatini–Cartan–Holst Action

Let M be a smooth oriented manifold of dimension four, with nondegenerate co-tetrad e^I, Lorentz connection $\omega^{IJ} = -\omega^{JI}$, and boundary conditions that make the boundary variational term vanish. Let us write

$$\Sigma^{IJ} = e^I \wedge e^J, \qquad (\star X)^{IJ} = \frac{1}{2}\epsilon^{IJ}{}_{KL} X^{KL}, \qquad G_{\mathrm{int}} = G_{\mathrm{int}}(\theta_G), \qquad \kappa_{\mathrm{int}} = \frac{8\pi G_{\mathrm{int}}}{c_{\mathrm{int}}^4}.$$

These last two quantities are not defined through the action: they follow from the constitutive construction of equation (59.11). The theorem 59.10 first proves that the following combination belongs to the action line. The first-order action is

$$\boxed{S[e, \omega, \Psi] = \frac{1}{2\kappa_{\mathrm{int}} c_{\mathrm{int}}} \int_M \Sigma_{IJ} \wedge \left(\star F^{IJ} + \frac{1}{\gamma} F^{IJ} \right) - \frac{\Lambda_{\mathrm{int}}}{\kappa_{\mathrm{int}} c_{\mathrm{int}}} \int_M \mathrm{vol}_e + S_{\mathrm{m}}[e, \omega, \Psi].}$$

The internal parameter constructed in the previous chapter specializes

$$\gamma = \Gamma_{6\to 8}.$$

The substitution is performed after obtaining $\Gamma_{6\to 8}$ from $A, C^*, q_\pm$ and the hexad–octad inclusion; the action is not used to redefine those inputs.

Theorem 82.12 (Cartan–Holst Equation)**.** *If the spin current is normalized through*

$$\delta_\omega S_{\mathrm{m}} = -\frac{1}{2}\int_M \delta\omega^{IJ} \wedge \sigma_{IJ},$$

the variation with respect to ω produces

$$\boxed{D_\omega\left[(\star + \gamma^{-1} I)\Sigma^{IJ}\right] = \kappa_{\mathrm{int}} c_{\mathrm{int}}\, \sigma^{IJ}.}$$

Proof. One uses $\delta F^{IJ} = D_\omega \delta\omega^{IJ}$, integrates by parts, and sets the coefficient of the arbitrary variation equal to zero. The cosmological term does not depend on ω. □

In Lorentzian signature, $\star^2 = -I$ on bivectors. For $\gamma \in \mathbb{R}\setminus\{0\}$,

$$(\star + \gamma^{-1} I)^{-1} = \frac{\gamma^2}{1+\gamma^2}(-\star + \gamma^{-1} I).$$

Therefore, if the spin current vanishes,

$$T^I_{\mathrm{tor}} = 0, \qquad \omega = \mathring{\omega}(e).$$

With spinorial matter, torsion is determined algebraically by the current. In the minimal action, it does not constitute an independent propagating degree of freedom.

82.4.5 Metric equation, spin, and geometric balance

To fix the normalization unambiguously, we defines the sources measured on the action line by

$$\delta_g S_{\mathrm{m}} = -\frac{1}{2}\int_M \mathrm{vol}_e\ \mathcal{T}^S_{\mu\nu}\,\delta g^{\mu\nu}.$$

When the tetrad coordinates have the dimension of length, we write

$$T^{\mathrm{phys}}_{\mu\nu} := c_{\mathrm{int}}\mathcal{T}^S_{\mu\nu}, \qquad U^{\mathrm{phys,spin}}_{\mu\nu} := c_{\mathrm{int}}\mathcal{U}^{S,\mathrm{spin}}_{\mu\nu}.$$

After algebraically eliminating the connection, the metric equation adopts the two equivalent forms

$$\boxed{G_{\mu\nu} + \Lambda_{\mathrm{int}} g_{\mu\nu} = \kappa_{\mathrm{int}} c_{\mathrm{int}}\big(\mathcal{T}^{S,\mathrm{LC}}_{\mu\nu} + \mathcal{U}^{S,\mathrm{spin}}_{\mu\nu}\big) = \kappa_{\mathrm{int}}\big(T^{\mathrm{phys,LC}}_{\mu\nu} + U^{\mathrm{phys,spin}}_{\mu\nu}\big).}$$

The spin contribution is quadratic in the corresponding current and depends on the matter coupling; its coefficient is not fixed without declaring the fermionic action and the duality conventions.

The geometric identities

$$D_\omega T^I_{\mathrm{tor}} = F^I{}_J \wedge e^J, \qquad D_\omega F^{IJ} = 0$$

imply, after removing the torsion,

$$\nabla^\mu\big(T^{\mathrm{phys,LC}}_{\mu\nu} + U^{\mathrm{phys,spin}}_{\mu\nu}\big) = 0.$$

This Bianchi–Noether balance is to consistency condition for the gravitational transducer: not additional term can is incorporated without proceeding from to compatible action or satisfying the same balance.

82.4.6 Area–volumetric correspondence π/φ^2

The calendar (Σ, Π, Π) produces the incidence matrix

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its Parry measure gives $\mu_{\mathrm{ar}}/\mu_{\mathrm{vol}} = \varphi^{-2}$, and the regional reader marks the closure exactly once π_{HMT}. The marked return satisfies

$$\Theta_n = \pi_{\mathrm{HMT}} \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi_{\mathrm{HMT}}}{\varphi^2}.$$

This ratio is not another version of G. It is the dimensionless correspondence between the areal and volumetric readings. Under the same positive amplitude $\mathcal{M}$,

$$\mathcal{I}_{\mathrm{av}} = \mu_{\mathrm{vol}} \mathcal{M}, \qquad \mathcal{G}_{\mathrm{av}} = \pi_{\mathrm{HMT}} \mu_{\mathrm{ar}} \mathcal{M}$$

give exactly

$$\frac{\mathcal{G}_{\mathrm{av}}}{\mathcal{I}_{\mathrm{av}}} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2}.$$

On the flat sheet, the diagonal restriction is $\mathcal{G}_{\mathrm{tor}} = \mathcal{I}_{\mathrm{tor}}$. Local equivalence and boundary ambivalence are therefore two regimes of the reader, not two incompatible values of a mass.

82.4.7 Typing of G as gravitational transducer and causal order of its readers

The type and homogeneity of G belong to the prior constitutive construction. The equation (59.11) fixes the covariant family $G_{\mathrm{int}}(\theta_G)$ once and for all, and the theorem 59.10 proves that its insertion into the action has the dimension of action. This chapter does not redefine that family: it uses its section together with the holonomy already constructed. Causality is not linear, but convergent:

$$\begin{aligned} \mathcal{B}_{\mathrm{geom}} &: \text{return and cocycle} \longrightarrow \text{Cartan representation} \longrightarrow (e, \omega), \\ \mathcal{B}_{\mathrm{dim}} &: (\hbar_{\mathrm{int}}, c_{\mathrm{int}}, t_0, \theta_G) \longrightarrow G_{\mathrm{int}} \longrightarrow \kappa_{\mathrm{int}}, \end{aligned}$$

$$(\mathcal{B}_{\mathrm{geom}}, \mathcal{B}_{\mathrm{dim}}) \longrightarrow S_{\mathrm{PCH}} \longrightarrow \text{algebraic torsion, metric equation and balance.}$$

The metrological comparison belongs to the end and does not select any of the two branches.

82.4.8 Collective origin of the gravitational mode in the holonomy of paths

The basic objects of the preceding action are the co-tetrad and the connection. Cartan torsion is algebraic; curvature has the propagating sector that contains gravitational waves. Accordingly, the formalism does not need to postulate an elementary graviton in order to define the interaction. A possible quantization of the perturbations may produce a collective mode of helicities ±2; that mode would be an excitation of geometric transport, not a datum added at the level of APP, TRIT, or TPK.

This distinction also excludes a frequent confusion. The existence of classical waves proves propagation of curvature; by itself it does not prove a fundamental quantum. Conversely, describing the field geometrically does not eliminate its radiative degrees of freedom. HMT–MD locates the question properly: memory and its holonomy are constructed first; the spectrum of their excitations is then studied.

Typed comparison of the area quantum with energy, entropy, and horizon information. The first-order action permits comparison of the areal quantum with horizon energy, entropy, and information without conflating those observables.

82.5 Realization in the Holst action

The substitution

$$\gamma = \Gamma_{6\to 8}$$

realizes the Barbero–Immirzi parameter in the HMT specialization of the Holst action studied in chapter 77. The following section constructs on the triadic boundary the area operator, its spectrum, and the collective intertwiner that transports the normalization of the channels (90/120) to the Ashtekar–Barbero connection [11, 69, 62]. The result fixes the discrete HMT realization; its comparison with a complete continuous geometric representation of Ashtekar–Barbero is performed after declaring the embedding and normalization of that representation.

82.6 Realization of the current in connection coordinates

Memory energy assigns each route a response preserving its incidences and the order of transports. Inserting it into a geometric action requires its components with respect to connection variations. This step is obtained by differentiating the transport that already realizes the route: the connection current is the composition of that differential with the energy covector of the preceding section.

The elliptic, parabolic and hyperbolic regimes of TRIT require preserving the complete differential before restricting it to a particular representation. Unitary rotations constitute a domain of that construction. The following extension also preserves the dilation and transport directions admitted by the geometric realization used.

82.6.1 Differential on invertible transports

The fibers, fields and weight of (80.37) are retained. We now consider invertible transports and differentiable families with $\dot U_a = A_a U_a$, where A_a belongs to the tangent space of the chosen representation. We write

$$v_a = U_a f_{s(a)}, \qquad r_a = f_{t(a)} - v_a, \qquad q = \mathsf{W}_m r, \qquad E_m = \langle r, \mathsf{W}_m r\rangle.$$

Proposition 82.13 (Complete representative of the differential)**.** *At fixed field and weight, the following holds*

$$\mathrm{d}_U E_m(A) = -2\,\mathrm{Re}\sum_a \langle q_a, A_a v_a\rangle = \sum_a \mathrm{Re}\,\mathrm{Tr}(\mathcal{G}_a^* A_a), \qquad \mathcal{G}_a = -2 q_a v_a^*. \tag{82.1}$$

The anti-Hermitian and Hermitian projections of this representative are

$$\frac{\mathcal{G}_a - \mathcal{G}_a^*}{2} = v_a q_a^* - q_a v_a^* = \mathcal{J}_a, \qquad \frac{\mathcal{G}_a + \mathcal{G}_a^*}{2} = -(q_a v_a^* + v_a q_a^*). \tag{82.2}$$

Restriction to anti-Hermitian generators recovers exactly (80.39).

Proof. The identity $\dot r_a = -A_a v_a$ and self-adjointness of the weight give $\dot E_m = 2\,\mathrm{Re}\langle q, \dot r\rangle$. Moreover,

$$\mathrm{Tr}(\mathcal{G}_a^* A_a) = -2\,\mathrm{Tr}(v_a q_a^* A_a) = -2 q_a^* A_a v_a.$$

The two projections are obtained by taking the adjoint. For the real Hilbert–Schmidt inner product, the Hermitian and anti-Hermitian subspaces are orthogonal; therefore the Hermitian part has zero contraction with an anti-Hermitian generator. □

The formula allows weights coupling edges: their action is calculated in $q = \mathsf{W}_m r$ before separating the contributions of each incidence. If field or weight vary, the complete expression (80.40) is retained, also valid for the invertible transports considered here.

82.6.2 Quadratic representations of TRIT

In a two-dimensional matrix realization of $\mathbb{A}_\tau$, one may take

$$J_+ = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad J_- = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad J_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$J_+^2 = -I, \qquad J_-^2 = I, \qquad J_0^2 = 0.$$

Their exponentials realize, respectively, the elliptic, hyperbolic and parabolic regimes. The first is orthogonal for the fixed positive inner product. The second preserves the form $B = \mathrm{diag}(-1, 1)$, since $J_-^{\mathsf{T}} B + B J_- = 0$. The third represents the algebra of dual numbers. These representations fix examples of the quadratic types; their use on a screen of another dimension preserves the corresponding representation map.

A calculation distinguishes the components of the differential. On an edge, take $U = I$, $f_s = (1, 1)^{\mathsf{T}}$, $f_t = 2 f_s$ and $\mathsf{W} = I$. Then $q = v = f_s$, $\mathcal{J} = 0$, whereas

$$\mathrm{d}E(J_-) = -4, \qquad \mathrm{d}E(J_0) = -2. \tag{82.3}$$

The complete representative preserves both responses. The distinction identifies the space of variations before performing the material contraction.

82.6.3 Composition of routes and oriented inversion

Proposition 82.14 (Contragredient transport of the differential)**.** *For $U_\gamma = U_N \cdots U_1$, let $B_k = U_N \cdots U_{k+1}$ and $B_N = I$. We have*

$$\dot U_\gamma U_\gamma^{-1} = \sum_k B_k A_k B_k^{-1}.$$

The representative $\mathcal{G}_\gamma$ in the final fiber induces at incidence k

$$\mathcal{G}_k = B_k^* \mathcal{G}_\gamma (B_k^{-1})^*. \tag{82.4}$$

Proof. The product rule, followed by multiplication by U_γ^{-1}, proves the first equality. Cyclicity of the trace gives

$$\mathrm{Re}\,\mathrm{Tr}(\mathcal{G}_\gamma^* B_k A_k B_k^{-1}) = \mathrm{Re}\,\mathrm{Tr}((B_k^* \mathcal{G}_\gamma (B_k^{-1})^*)^* A_k).$$

For unitary B_k, the conjugation of 80.33 is recovered. □

Proposition 82.15 (Inversion by energy congruence)**.** *For a local energy per edge, inversion is represented by*

$$U_{\bar{a}} = U_a^{-1}, \qquad r_{\bar{a}} = -U_a^{-1} r_a, \qquad W_{\bar{a}} = U_a^{*} W_a U_a. \tag{82.5}$$

It preserves energy. If W_a remains fixed,

$$A_{\bar{a}} = -U_a^{-1} A_a U_a, \qquad \dot{W}_{\bar{a}} = U_a^{*}(A_a^{*} W_a + W_a A_a) U_a.$$

The complete variation of the inverted energy coincides with the original.

Proof. Substitution into $r_{\bar{a}}^{*} W_{\bar{a}} r_{\bar{a}}$ produces $r_a^{*} W_a r_a$. The derivative of the inverse is $\dot{U}_a^{-1} = -U_a^{-1} A_a$; that of the weight is obtained by the product rule. Equality of energies holds for the entire family, so their derivatives coincide, including the term $\langle r_{\bar{a}}, \dot{W}_{\bar{a}} r_{\bar{a}} \rangle$. □

Congruence preserves positivity for every invertible transport. In the unitary representation it reduces to the inversion rule of 80.35. The sum uses one orientation per edge, or half the sum over both orientations.

82.6.4 Evaluation of current components

Let $\theta = (\theta_b)$ be a real variation of the connection coordinates of the realization used. The effective differential of transport defines the operators

$$A_a(\theta) = \dot{U}_a U_a^{-1} = \sum_b B_{ab} \theta_b. \tag{82.6}$$

For a finite composition, their contributions are obtained through (82.4). The operators B_{ab} are the derivatives of the connection representation that realizes the route.

Theorem 82.16 (Connection current of the memory action)**.** *For a material term $S_{\mathrm{mem}} = \mathfrak{a}_m E_m$, with $\mathfrak{a}_m$ its normalization on the action line, at fixed field, weight and normalization we have*

$$\delta S_{\mathrm{mem}} = -\frac{1}{2} \sum_b \theta_b s_b, \qquad s_b = 4\mathfrak{a}_m \operatorname{Re} \sum_a \langle q_a, B_{ab} v_a \rangle. \tag{82.7}$$

If the cellular pairing between the connection and current bases is an invertible matrix M, defined by $\delta S_{\mathrm{mem}} = -\frac{1}{2} \theta^{\mathsf{T}} M \sigma$, the current coordinates are

$$\sigma = M^{-1} s. \tag{82.8}$$

Proof. Substitute (82.6) into (82.1), multiply by $\mathfrak{a}_m$, and collect the coefficients of θ_b. The variational convention $-1/2$ fixes the factor 4. Comparing both expressions for every variation θ gives $M\sigma = s$; invertibility of the pairing proves existence and uniqueness of these coordinates. □

The formula explicitly realizes the composition mentioned in 80.7.5. To evaluate each component, the field is transported, the residual is calculated, the weight is applied and the result is contracted with the connection variation of the same realization. The pairing M preserves the orientation and volume of cells. A diagonal pairing gives the corresponding division by their volumes; other bases preserve the complete matrix.

If the connection also enters f, W_m or $\mathfrak{a}_m$, we apply (80.40) and add $E_m \delta \mathfrak{a}_m$. Contributions from all material terms are summed before composing the Cartan response.

Corollary 82.17 (Compatibility of the current with refinement)**.** *For the compatible families of 80.36, let P_m be the fixed map taking connection variations from level m to level $m+1$. If the memory actions coincide on those refined fields and transports, their covectors satisfy*

$$s_m = P_m^{\mathsf{T}} s_{m+1}, \qquad M_m \sigma_m = P_m^{\mathsf{T}} M_{m+1} \sigma_{m+1}.$$

Proof. Differentiating the equality of actions in the direction $P_m\theta$ gives $-\frac{1}{2}\theta^{\mathsf{T}} s_m = -\frac{1}{2}\theta^{\mathsf{T}} P_m^{\mathsf{T}} s_{m+1}$. The arbitrariness of θ and the cellular representations of both covectors prove the two identities. Composition of these maps iterates the equality to any finite number of refinements. □

82.6.5 Insertion into the Cartan equation

Section 82.8 normalizes the material three-form through $\delta_\omega S_m = -\frac{1}{2}\int \delta\omega^{IJ} \wedge \sigma_{IJ}$. The preceding theorem provides its cellular components for the declared memory action and pairing. The Cartan response uses that current together with the co-tetrad, the Barbero–Immirzi parameter and the gravitational coupling of the same realization.

The material dependence on contorsion determines how that equation is solved. For affine matter, the current is independent of contorsion, and the linear inversion and quadratic elimination proved in Section 82.8 apply. For a holonomy action, the differential is evaluated at the current connection, and this dependence is retained when solving the equation. For example, an edge with $f_s = f_t = 1$, unit weight, and $U(t) = e^{it}$ has

$$E(t) = 2 - 2\cos t, \qquad E'(t) = 2\sin t.$$

Here the current depends on t. The complete formula retains that dependence instead of replacing it by its value at a particular connection.

Finally, the gravitational density is obtained from the metric variation of the effective action, with its field and normalization. The representation of the current, the contortion it produces and the resulting density are the successive operations of that composition. The subsequent dilution law transports the amplitude thus evaluated in the corresponding material section.

82.7 Differential of the minimal boundary extension

The cubic incidence simultaneously determines the minimal-extension field and the weight of its response. For a boundary datum, both are calculated through the inverse of the incidence Gramian. We now develop the differential of this same operation, preserving the variations of the boundary, transport and normalization scale. In this way, the field involved in the response is selected by the screen variational problem.

82.7.1 Minimizing field and response weight

Let H and Q be finite-dimensional Hermitian spaces, with fixed inner products antilinear in the first variable. The real case is obtained by replacing adjoints with transposes. On an open set of real parameters Ω, consider families of class C^1

$$\mathsf{C}(\theta) : H \longrightarrow Q, \qquad b(\theta) \in Q, \qquad \chi(\theta) > 0, \qquad \operatorname{Ran}\mathsf{C}(\theta) = Q.$$

Surjectivity is equivalent to full row rank in orthonormal bases. We define the Gramian, the conjugate potential and the extension

$$\mathsf{L} = \mathsf{C}\mathsf{C}^*, \qquad y = \mathsf{L}^{-1}b, \qquad f_{\min} = \mathsf{C}^* y, \qquad \Lambda = \chi\mathsf{L}^{-1}. \tag{82.9}$$

The incidence C and the cellular pairing M of (82.8) are different operators.

Proposition 82.18 (Variational selection of the extension)**.** *The extension $f_{\min}$ is the unique norm minimizer among the fields $f \in H$ satisfying $\mathsf{C}f = b$. Its energy is*

$$E(\theta) = \chi \min_{\mathsf{C}f=b} \|f\|^2 = \chi\langle b, \mathsf{L}^{-1}b\rangle = \langle b, \Lambda b\rangle. \tag{82.10}$$

The field $f_{\min}$, potential y, and energy E depend C^1-smoothly on the parameters.

Proof. Surjectivity of C makes C^* injective, so $\langle z, \mathsf{L}z\rangle = \|\mathsf{C}^*z\|^2 > 0$ for $z \neq 0$. Thus L is invertible and $\mathsf{C}f_{\min} = \mathsf{L}y = b$. Every solution is $f_{\min} + z$, with $z \in \ker \mathsf{C}$. Since $f_{\min} \in \operatorname{Ran}\mathsf{C}^* = (\ker \mathsf{C})^{\perp}$,

$$\|f_{\min} + z\|^2 = \|f_{\min}\|^2 + \|z\|^2, \qquad \|f_{\min}\|^2 = \langle y, \mathsf{L}y\rangle = \langle b, \mathsf{L}^{-1}b\rangle.$$

These identities prove the minimum, its uniqueness and its value. Inversion is a differentiable operation on the open set of invertible matrices; the expressions in (82.9) provide the asserted regularity. □

The cubic realization of 80.24 corresponds to $H = \mathbb{R}^{\{-1,1\}^3}$, the screen quotient $Q = P_{\square}\mathbb{R}^6$ and the incidence $\mathsf{C} = M : H \to Q$. Its Gramian and normalization are

$$\mathsf{L}_{\square} = 12P_u + 4P_-, \qquad \chi_m = \frac{112}{81\ell_m^2}, \qquad \Lambda_m = \frac{28}{243\ell_m^2}(P_u + 3P_-). \tag{82.11}$$

The factor $112/81$ comes from the soldering and the quadratic compression defect of (80.30). If b has length dimension, so do y and $f_{\min}$, the incidence is dimensionless and E is dimensionless. Conversion to action uses the normalization in the action line specified below.

82.7.2 Full differential of the minimum

Theorem 82.19 (Response to the variation of incidence and boundary)**.** *For every real tangent direction $\delta\theta$, one has*

$$\delta E = \delta\chi\,\langle b, y\rangle + 2\chi\,\mathrm{Re}\langle y, \delta b - (\delta\mathsf{C})f_{\min}\rangle. \tag{82.12}$$

If $\chi = 112/(81\ell_m^2)$, with variable $\ell_m > 0$,

$$\delta E = -2E\frac{\delta\ell_m}{\ell_m} + 2\chi_m\,\mathrm{Re}\langle y, \delta b - (\delta\mathsf{C})f_{\min}\rangle. \tag{82.13}$$

Proof. Differentiating $\mathsf{L}\mathsf{L}^{-1} = I_Q$ gives

$$\delta\mathsf{L}^{-1} = -\mathsf{L}^{-1}(\delta\mathsf{L})\mathsf{L}^{-1}, \qquad \delta\mathsf{L} = (\delta\mathsf{C})\mathsf{C}^* + \mathsf{C}(\delta\mathsf{C})^*.$$

Applying the product rule to (82.10) and using the self-adjointness of L^{-1} yields

$$\delta E = \delta\chi\,\langle b, y\rangle + 2\chi\,\mathrm{Re}\langle y, \delta b\rangle - \chi\langle y, (\delta\mathsf{L})y\rangle.$$

The two terms of the last product are conjugate to each other and $\mathsf{C}^*y = f_{\min}$; therefore,

$$\langle y, (\delta\mathsf{L})y\rangle = 2\,\mathrm{Re}\langle y, (\delta\mathsf{C})f_{\min}\rangle.$$

One obtains (82.12). Finally, $\delta\chi_m = -2\chi_m\delta\ell_m/\ell_m$ produces (82.13). □

The formula incorporates the variation of the minimizing field through the equation that determines it. It is also checked directly: $\mathsf{C}\delta f_{\min} = \delta b - (\delta\mathsf{C})f_{\min}$ and $f_{\min} = \mathsf{C}^*y$ imply

$$\delta\|f_{\min}\|^2 = 2\,\mathrm{Re}\langle f_{\min}, \delta f_{\min}\rangle = 2\,\mathrm{Re}\langle y, \delta b - (\delta\mathsf{C})f_{\min}\rangle.$$

For example, if C is fixed and $b = \ell_m b_0$, with b_0 fixed, the two terms of (82.13) cancel and the energy remains constant. This check jointly preserves the boundary dimension and its normalization.

82.7.3 Fixed quotient and frame changes

The preceding derivatives are taken in fixed spaces H, Q and Hermitian inner products. In the six-face presentation, the inverse always acts on $Q = P_{\square} H_F$. For a family of quotients $Q(\theta)$, an isometric trivialization $V_Q(\theta) : Q \to Q(\theta)$ is fixed first, and likewise for the domain. The derivatives of those maps are included when differentiating the expressions transported to the fixed spaces. The formulas apply on each open set where rank is preserved. Changes in the energy inner product additionally require differentiating that product; a Lorentzian metric and the positive inner product used for minimization retain their distinct types here.

Proposition 82.20 (Covariance of the minimum)**.** *Let g_H, g_Q be unitary. Under*

$$\mathsf{C}' = g_Q \mathsf{C} g_H^*, \qquad b' = g_Q b, \qquad \chi' = \chi,$$

$y' = g_Q y$, $f'_{\min} = g_H f_{\min}$ and $E' = E$ hold. When the frame changes depend on the parameter, the full differential preserves this equality. In particular, a pure frame variation has $\delta E = 0$.

Proof. The transformed Gramian is $\mathsf{L}' = g_Q \mathsf{L} g_Q^*$, from which the transformations of y, $f_{\min}$ and equality of energies are obtained. For a pure frame variation, at a point with identity frames, let $A_Q^* = -A_Q$ and $A_H^* = -A_H$ be their generators. Then

$$\delta\mathsf{C} = A_Q \mathsf{C} - \mathsf{C} A_H, \qquad \delta b = A_Q b, \qquad \delta b - (\delta\mathsf{C}) f_{\min} = \mathsf{C} A_H f_{\min}.$$

Its contribution to (82.12) is $2\chi \operatorname{Re}\langle f_{\min}, A_H f_{\min}\rangle = 0$. The proof thus includes the terms due to the moving frames. □

82.7.4 Transported incidence and connection components

The connection dependence is specified before performing the contraction. An explicit blockwise realization of the incidence is obtained as follows. Let $\mathcal{K}$ be a common Hermitian fibre, $F = \{(i, \epsilon) : 1 \le i \le 3,\ \epsilon = \pm 1\}$ and $V = \{-1, 1\}^3$. For each incidence $\nu_i = \epsilon$, the realization used declares a route $\gamma_{i,\epsilon,\nu}$ from the vertex fibre to the face fibre, with transport $U_{i,\epsilon,\nu}$. The trivializations identify these fibres with $\mathcal{K}$. We define

$$\begin{aligned} (\mathsf{B}_U f)_{i,\epsilon} &= \sum_{\nu_i = \epsilon} U_{i,\epsilon,\nu} f_\nu, \\ \mathsf{C}_U = (q_{\square} \otimes I_{\mathcal{K}}) \mathsf{B}_U &: \bigoplus_{\nu \in V} \mathcal{K} \longrightarrow Q \otimes \mathcal{K}, \qquad q_{\square} = P_{\square} : H_F \to Q. \end{aligned} \tag{82.14}$$

This is an explicit variational continuation of the cubic incidence in the declared route representation. For identity transports, $\mathsf{C}_U = M \otimes I_{\mathcal{K}}$ and one recovers (82.11), tensored with the fibre. The Gramian remains positive in a neighbourhood of that realization, since positive definiteness is an open condition. On each full-rank domain, the weight and field are fixed by (82.9). The routes realizing the incidences, their orientations and their memories are data of this realization of APP–TRIT–TPK transport and accompany the construction.

If $\omega = (\omega_j)$ are real connection coordinates, the derivative of C_U is calculated by substituting in each block

$$\partial_j U_\gamma = \sum_{k=1}^{N} U_N \cdots U_{k+1} (\partial_j U_k) U_{k-1} \cdots U_1, \qquad U_\gamma = U_N \cdots U_1. \tag{82.15}$$

Empty products are identities. The formula follows from the product rule and preserves the position of each operation in the route. Thus, $\mathsf{C}_j := \partial_j \mathsf{C}_U$ is obtained from a finite sum of known matrices in the chosen realization, followed by the projection $q_{\square} \otimes I$. If a moving quotient is used, its derivative is

incorporated according to 82.7.3. The fixed integer cubic matrix corresponds to the case $\mathsf{C}_j = 0$; the response to the connection follows from the transport dependence actually declared in (82.14).

Theorem 82.21 (Variational components of the minimal extension)**.** *For the screen action $S_{\min} = \mathfrak{a}_m E$, where $\mathfrak{a}_m$ is its normalization in the action line, let us write*

$$b_j = \partial_j b, \qquad \chi_j = \partial_j \chi, \qquad a_j = \partial_j \mathfrak{a}_m.$$

With the convention $\delta S_{\min} = -\frac{1}{2}\sum_j s_j \delta\omega_j$, the components are

$$\begin{aligned} s_j = {} & 4\mathfrak{a}_m \chi \,\mathrm{Re}\langle y, \mathsf{C}_j f_{\min} - b_j\rangle \\ & - 2\mathfrak{a}_m \chi_j \langle b, y\rangle - 2a_j E. \end{aligned} \tag{82.16}$$

For the cubic normalization, the second term equals $4\mathfrak{a}_m E\, \partial_j \ell_m/\ell_m$. If the cellular pairing of this same action has invertible matrix $\mathsf{M}_{\mathrm{cel}}$, its current coordinates are $\sigma = \mathsf{M}_{\mathrm{cel}}^{-1} s$.

Proof. The product rule gives $\partial_j S_{\min} = a_j E + \mathfrak{a}_m \partial_j E$. Substituting (82.12) and multiplying by -2 proves (82.16). The derivative of χ_m produces the scale expression. Finally, comparing $\delta S_{\min} = -\frac{1}{2}\delta\omega^{\mathsf{T}} s$ with $-\frac{1}{2}\delta\omega^{\mathsf{T}}\mathsf{M}_{\mathrm{cel}}\sigma$ for every variation gives $\mathsf{M}_{\mathrm{cel}}\sigma = s$, and its inverse determines the coordinates. □

The construction recovers a field and a weight jointly selected by the incidence, and evaluates their response to variations of the same realization. The boundary datum b, the route family, the scale ℓ_m, the normalization $\mathfrak{a}_m$ and the cellular pairing retain their publication rules. Interpretation as a material current uses those objects and the complete action. The Cartan response and its subsequent metric density are composed with it in the corresponding matter domain, preserving any dependence of the current on the connection.

82.8 Spin current and algebraic Cartan–Holst response

The realization of the enriched transport in Section 80.6 provides the cotetrad and the connection; the dimensional sections and the incidence parameter of Sections 120 and 121 fix their couplings. Their composition determines a first-order variational problem. The material current occupies a precise position in it: it is the derivative of the action with respect to the connection and determines the torsional response with the same normalization.

82.8.1 Geometric domain and current normalization

We work on a smooth oriented four-dimensional manifold M, with internal Lorentzian bundle E, metric $\eta = \mathrm{diag}(-1, 1, 1, 1)$ and non-degenerate cotetrad $e \in \Omega^1(M; E)$. In a local frame we write e^I, with $I = 0, 1, 2, 3$. The metric connection $\omega^{IJ} = -\omega^{JI}$ acts through D_ω. When the fields Ψ are spinorial, a spin structure and the corresponding material representation are preserved. We define

$$\Sigma^{IJ} = e^I \wedge e^J, \qquad F^{IJ} = \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}, \qquad T^I = D_\omega e^I. \tag{82.17}$$

The operator $\star$ acts on the internal bivector indices,

$$(\star X)^{IJ} = \tfrac{1}{2}\epsilon^{IJ}{}_{KL} X^{KL}, \qquad \star^2 = -I, \qquad \mathsf{P}_\gamma = \star + \gamma^{-1} I. \tag{82.18}$$

This distinguishes this duality from the exterior duality of forms on M. Internal indices are raised and contracted with η.

We fix a chart with $G_{\mathrm{int}} > 0$, $c_{\mathrm{int}} > 0$, $\gamma = \gamma_{\mathrm{HMT}} \in \mathbb{R} \setminus \{0\}$ and Λ_{int} constant on it. We abbreviate $\kappa = 8\pi G_{\mathrm{int}}/c_{\mathrm{int}}^4$ and $c = c_{\mathrm{int}}$. These sections remain fixed during variation. The action is

$$S[e,\omega,\Psi]=\frac{1}{2\kappa c}\int_M \Sigma_{IJ}\wedge(\mathsf{P}_\gamma F)^{IJ}-\frac{\Lambda_{\mathrm{int}}}{\kappa c}\int_M \mathrm{vol}_e+S_{\mathrm{m}}[e,\omega,\Psi]. \tag{82.19}$$

The volume and orientation convention is the one that gives $\Sigma_{IJ}\wedge(\star F)^{IJ}=R\,\mathrm{vol}_e$ in the Levi–Civita sector. The prior dimensional type makes each summand an action section.

For the variation of the connection, variations with compact support in the interior are used, or $\delta\omega|_{\partial M}=0$ is fixed. A boundary completion whose variational problem cancels the same boundary term may also be used. At fixed cotetrad and material fields, the spin current is the degree-three form, valued in dual internal bivectors, determined by

$$\delta_\omega S_{\mathrm{m}}=-\tfrac{1}{2}\int_M \delta\omega^{IJ}\wedge\sigma_{IJ}. \tag{82.20}$$

Pairing with an arbitrary compactly supported variation fixes σ uniquely. This definition preserves the action unit of the current and the normalization factor involved in its geometric response.

82.8.2 Connection variation and Holst inversion

Theorem 82.22 (Variational Cartan–Holst equation)**.** *On the preceding domain, stationarity with respect to the connection is equivalent to*

$$D_\omega(\mathsf{P}_\gamma\Sigma)^{IJ}=\kappa c\,\sigma^{IJ}. \tag{82.21}$$

Proof. The variation of the curvature is $\delta F=D_\omega\delta\omega$. The internal duality is parallel and self-adjoint for the bivector contraction; since γ is constant, $D_\omega\mathsf{P}_\gamma=\mathsf{P}_\gamma D_\omega$. Setting $B_{IJ}=(\mathsf{P}_\gamma\Sigma)_{IJ}$ yields

$$B_{IJ}\wedge D_\omega\delta\omega^{IJ}=\mathrm{d}(B_{IJ}\wedge\delta\omega^{IJ})-D_\omega B_{IJ}\wedge\delta\omega^{IJ}.$$

The last form changes sign when its factors of degrees three and one are interchanged. The boundary conditions eliminate the integral of the exact term. The cosmological term is independent of ω. Therefore,

$$\delta_\omega S=\frac{1}{2\kappa c}\int_M \delta\omega^{IJ}\wedge\big[D_\omega B_{IJ}-\kappa c\,\sigma_{IJ}\big].$$

The arbitrariness of the variation proves the equation, including its sign and coefficient. □

Lemma 82.23 (Real inverse of the Holst operator)**.** *For $\gamma\in\mathbb{R}\setminus\{0\}$,*

$$\mathsf{P}_\gamma^{-1}=\frac{\gamma^2}{1+\gamma^2}(-\star+\gamma^{-1}I). \tag{82.22}$$

Proof. Both factors are polynomials in $\star$ and commute. Their product before normalization is

$$(\star+\gamma^{-1}I)(-\star+\gamma^{-1}I)=-\star^2+\gamma^{-2}I=(1+\gamma^{-2})I.$$

The denominator $1+\gamma^2$ is positive on the fixed real domain. Multiplication by $\gamma^2/(1+\gamma^2)$ gives the identity. □

The current thus determines the degree-three bivector-valued form

$$Y^{IJ}:=\kappa c\,\frac{\gamma^2}{1+\gamma^2}(-\star+\gamma^{-1}I)\sigma^{IJ},\qquad D_\omega\Sigma^{IJ}=Y^{IJ}. \tag{82.23}$$

82.8.3 Pointwise reconstruction of torsion and contorsion

The Leibniz rule gives

$$D_\omega \Sigma^{IJ} = T^I \wedge e^J - e^I \wedge T^J =: \mathsf{L}_e(T)^{IJ}. \tag{82.24}$$

At each point, L_e acts from $\Lambda^2 T^*M \otimes E$ to $\Lambda^3 T^*M \otimes \Lambda^2 E$. Both spaces have dimension twenty-four. Let E_I be the dual frame of the cotetrad, so that $\iota_{E_I} e^J = \delta^J_I$.

Theorem 82.24 (Algebraic inversion of the torsional equation)**.** *For a nondegenerate co-tetrad, L_e is an isomorphism. Its inverse on the form Y is written*

$$B^I = \sum_J \iota_{E_J} Y^{IJ}, \qquad b = \tfrac{1}{4} \sum_I \iota_{E_I} B^I, \qquad T^I = B^I - e^I \wedge b. \tag{82.25}$$

Equation (82.21) *therefore determines torsion pointwise from e and the current appearing in it.*

Proof. First let $Y = \mathsf{L}_e(T)$ and set $t = \sum_J \iota_{E_J} T^J$. The contraction of (82.24), using that T^I has degree two and that the cotetrad has four elements, gives

$$\sum_J \iota_{E_J}(T^I \wedge e^J) = 2T^I, \qquad \sum_J \iota_{E_J}(e^I \wedge T^J) = T^I - e^I \wedge t.$$

Hence $B^I = T^I + e^I \wedge t$. A second contraction produces

$$\sum_I \iota_{E_I} B^I = t + 3t = 4t.$$

One recovers $t = b$ and then $T^I = B^I - e^I \wedge b$. In particular, the kernel of L_e is zero. Equality of dimensions proves its surjectivity and makes the formula valid for every form Y in the codomain. □

The connection decomposes uniquely as

$$\omega^{IJ} = \mathring{\omega}^{IJ}(e) + K^{IJ}, \qquad K^{IJ} = -K^{JI}, \qquad T^I = K^I{}_J \wedge e^J, \tag{82.26}$$

where $\mathring{\omega}(e)$ is the Levi–Civita connection and K the contorsion. To make the inverse explicit, we write $K_{IJ} = K_{IJK} e^K$ and $T_I = \frac{1}{2} T_{IJK} e^J \wedge e^K$. Then

$$T_{IJK} = K_{IKJ} - K_{IJK}, \qquad K_{IJK} = \tfrac{1}{2}(T_{KIJ} - T_{IJK} - T_{JKI}). \tag{82.27}$$

The second equality is obtained by combining the first cyclically and using $K_{IJK} = -K_{JIK}$; substituting it recovers the components of T. Torsion and contorsion are therefore two equivalent presentations of the algebraic response at fixed cotetrad.

Corollary 82.25 (Current-free sector and minimal linear response)**.** *If $\sigma = 0$, one obtains $T = 0$ and $\omega = \mathring{\omega}(e)$. For a material coupling whose current $\sigma[e, \Psi]$ is independent of K, equations* (82.23)*,* (82.25) *and* (82.27) *provide a unique response $K_*(e, \Psi)$, linear in σ.*

Proof. The three reconstruction maps are linear with e, γ and κc fixed. The zero current produces $Y = 0$, then $T = 0$ and finally $K = 0$. □

In the minimal action, the connection equation contains this pointwise response. Its evolution is obtained by evolving e and Ψ: $K_* = K_*(e, \Psi)$ changes with them. If a material action preserves additional algebraic dependence on K, the Cartan equation retains its variational form and must be solved with that current; the preceding linear response corresponds to the specified minimal domain.

82.8.4 Elimination of contorsion and quadratic contribution

Now consider the minimal coupling affine in K, expressed by

$$S_{\mathrm{m}}[e, \mathring{\omega} + K, \Psi] = S_{\mathrm{m}}[e, \mathring{\omega}, \Psi] - \tfrac{1}{2} \int_M K^{IJ} \wedge \sigma_{IJ}[e, \Psi]. \tag{82.28}$$

This condition identifies the material action to which the quadratic elimination applies; the current remains the one defined by (82.20).

The curvature expansion is

$$F(\mathring{\omega} + K) = F(\mathring{\omega}) + D_{\mathring{\omega}} K + K \wedge K, \qquad (K \wedge K)^{IJ} = K^I{}_L \wedge K^{LJ}. \tag{82.29}$$

Since $D_{\mathring{\omega}} \Sigma = 0$, the linear term is the exterior derivative of $\Sigma_{IJ} \wedge (\mathsf{P}_\gamma K)^{IJ}$. The corresponding boundary contribution remains explicit:

$$S_\partial[e, K] = \frac{1}{2\kappa c} \int_{\partial M} \Sigma_{IJ} \wedge (\mathsf{P}_\gamma K)^{IJ}. \tag{82.30}$$

On an interior domain, or with the compatible boundary completion, the action density that depends on the contorsion is

$$\mathcal{Q}_{e,\gamma}(K) - \tfrac{1}{2} K^{IJ} \wedge \sigma_{IJ}, \qquad \mathcal{Q}_{e,\gamma}(K) = \frac{1}{2\kappa c} \Sigma_{IJ} \wedge \mathsf{P}_\gamma (K \wedge K)^{IJ}. \tag{82.31}$$

The density $\mathcal{Q}_{e,\gamma}$ is a degree-four differential form, homogeneous of degree two in K.

Theorem 82.26 (Effective response quadratic in the current)**.** *Under* (82.28)*, substituting the algebraic solution K_* produces the effective spin density*

$$\mathcal{L}^{\mathrm{eff}}_{\mathrm{spin}} = -\tfrac{1}{4} K_*^{IJ} \wedge \sigma_{IJ} = -\mathcal{Q}_{e,\gamma}(K_*). \tag{82.32}$$

At fixed cotetrad and couplings, it is homogeneous of degree two in the current. The effective action also preserves $S_\partial[e, K_*]$, or its declared boundary completion.

Proof. By Corollary 82.25, K_* is linear in σ. The connection equation is precisely the stationarity of (82.31). Since it is algebraic, the radial variation $\delta K = K_*$ is applied pointwise; it may also be multiplied by a compactly supported function. Euler's identity for a quadratic form gives

$$2\mathcal{Q}_{e,\gamma}(K_*) - \tfrac{1}{2} K_*^{IJ} \wedge \sigma_{IJ} = 0.$$

Substitution into the action density proves (82.32). If $\sigma \mapsto \lambda\sigma$, then $K_* \mapsto \lambda K_*$ and the effective density is multiplied by λ^2. The expansion (82.29) separately identifies the boundary term accompanying this elimination. □

The specific current, its Lorentzian contraction, the duality and the conventions of the material action determine the sign and coefficient of this contribution. The quadratic identity provides the composition rule that allows them to be evaluated once these data have been specified.

82.8.5 Effective metric source and covariant balance

The normalization of the metric source is fixed in the same action line. After eliminating K, we define

$$\delta_g S_{\mathrm{m}}[e, \mathring{\omega}, \Psi] = -\tfrac{1}{2} \int_M \mathrm{vol}_e \; \mathcal{T}^{S,\mathrm{LC}}_{\mu\nu} \, \delta g^{\mu\nu}, \tag{82.33}$$

$$\delta_g S^{\mathrm{eff}}_{\mathrm{spin}} = -\tfrac{1}{2} \int_M \mathrm{vol}_e \; \mathcal{U}^{S,\mathrm{spin}}_{\mu\nu} \, \delta g^{\mu\nu}. \tag{82.34}$$

Here interior variations are considered, with boundary terms treated according to the fixed variational problem. For the affine coupling, the second source is the metric variation of (82.32). The variation includes the dependence of the material current on e and Ψ.

Proposition 82.27 (Metric equation and conservation of the total source)**.** *With the preceding conventions, the metric equations are*

$$G_{\mu\nu}[g] + \Lambda_{\mathrm{int}} g_{\mu\nu} = \kappa c\big(\mathcal{T}^{S,\mathrm{LC}}_{\mu\nu} + \mathcal{U}^{S,\mathrm{spin}}_{\mu\nu}\big) = \kappa\big(T^{\mathrm{phys,LC}}_{\mu\nu} + U^{\mathrm{phys,spin}}_{\mu\nu}\big), \tag{82.35}$$

where $T^{\mathrm{phys,LC}} = c\mathcal{T}^{S,\mathrm{LC}}$ *and* $U^{\mathrm{phys,spin}} = c\mathcal{U}^{S,\mathrm{spin}}$ *when the cotetrad takes values in the length line. Every solution satisfies*

$$\mathring{\nabla}^{\mu}\big(T^{\mathrm{phys,LC}}_{\mu\nu} + U^{\mathrm{phys,spin}}_{\mu\nu}\big) = 0. \tag{82.36}$$

Proof. Eliminating K preserves the equations of the remaining variables: in the chain rule, the term multiplying δK_* vanishes by the connection equation. In the Levi–Civita sector, the first Bianchi identity cancels the term $\Sigma_{IJ} \wedge F^{IJ}$, and the gravitational part is written $(2\kappa c)^{-1}\int_M (R - 2\Lambda_{\mathrm{int}})\,\mathrm{vol}_e$. Its interior variation is

$$\frac{1}{2\kappa c}\int_M \mathrm{vol}_e\,(G_{\mu\nu} + \Lambda_{\mathrm{int}} g_{\mu\nu})\delta g^{\mu\nu}.$$

The sum with (82.34) gives (82.35). The contracted Bianchi identity for $\mathring{\nabla}$, metric compatibility and the constancy of κ and Λ_{int} in the chart produce (82.36). □

Before eliminating the connection, the geometric identities retain the form

$$D_\omega T^I = F^I{}_J \wedge e^J, \qquad D_\omega F^{IJ} = 0. \tag{82.37}$$

The first is $D^2_\omega e = F \wedge e$; the second is obtained by expanding $D_\omega(\mathrm{d}\omega + \omega \wedge \omega)$ and cancelling the terms. In a covariant material action, the total metric balance coincides with the Noether identity on the material equations. Geometry and matter thus preserve a joint source; the partition between its Levi–Civita contribution and its spin contribution is determined by the specialized action.

82.9 Torsional elimination and metric response of the minimal extension

The minimal extension provides a field, a response weight and a current through a single variational operation. Its transported incidence depends on the connection through the route products of (82.15). That dependence is preserved when composing the current with the Cartan–Holst equation. Elimination of the connection is therefore performed on the complete action of that extension. The affine case of 82.26 is contained as an exact specialization of the following construction.

82.9.1 Variational composition at a cellular level

Let us fix a finite level, its routes, the transport representation and the cellular pairing of 82.21. Let p be real coordinates of the co-tetrad and the material fields in a declared local trivialization, and $k \in \mathbb{R}^d$ the coordinates of the contortion K. The family $p \mapsto e(p)$ preserves nondegeneracy of the co-tetrad. The connection acting on each route is

$$\omega(p,k) = \mathring{\omega}(e(p)) + K(k), \qquad F(p,k) = S_{\min}[e(p), \omega(p,k), \Psi(p)] = \mathfrak{a}(p,k)E(p,k). \tag{82.38}$$

The spaces and bases of this expression are transported to fixed spaces before differentiation. We work on an open set where the incidence $\mathsf{C}(p,k)$ is surjective and the families are C^2. This additional regularity allows the Hessian of the action to be examined. The couplings $G_{\mathrm{int}}, c_{\mathrm{int}}, \gamma_{\mathrm{HMT}}$ preserve the chart and the variation convention of 82.8.1.

Let $Q(p,k)$ be the integral, or the declared cellular realization, of the quadratic form (82.31). In real bases,

$$Q(p,k) = \tfrac{1}{2} k^{\mathsf{T}} \mathsf{A}(p) k, \qquad \mathsf{A}(p)^{\mathsf{T}} = \mathsf{A}(p). \tag{82.39}$$

The operator A comes from the gravitational action and its pairing, whereas the Gramian $\mathsf{C}\mathsf{C}^{*}$ comes from the extension problem. Positivity of the latter guarantees existence of the minimizing field; the gravitational form retains its own signature. Boundary terms are retained according to (82.30); the following interior formulas apply with admissible variations that annul their contribution, or with the corresponding boundary completion.

Define the response covector by

$$d_k F(p,k)[h] = -\tfrac{1}{2} h^{\mathsf{T}} s(p,k), \qquad s(p,k) = \mathsf{M}_{\mathrm{cel}}(p,k)\sigma(p,k). \tag{82.40}$$

Here s and σ preserve the distinct types of (82.16): the former contains the coefficients of the differential and the latter the coordinates of the current in the material pairing. If the basis in k changes, that pairing is also transported. When differentiating $s = \mathsf{M}_{\mathrm{cel}}\sigma$, the derivatives of both factors are included when the pairing depends on the parameter.

Proposition 82.28 (Composed equation of extension and contorsion)**.** *Interior stationarity of the action with respect to contorsion is*

$$\mathcal{R}(p,k) := \mathsf{A}(p)k - \tfrac{1}{2} s(p,k) = 0. \tag{82.41}$$

Each component of s is calculated from the same extension:

$$\begin{aligned} s_j = {} & 4\mathfrak{a}\chi \,\mathrm{Re}\langle y, (\partial_{k_j}\mathsf{C}) f_{\min} - \partial_{k_j} b\rangle \\ & - 2\mathfrak{a}(\partial_{k_j}\chi)\langle b, y\rangle - 2(\partial_{k_j}\mathfrak{a})E. \end{aligned} \tag{82.42}$$

When the cellular pairing realizes the connection equation of 82.22, this equation is equivalent to composing $\sigma(p,k)$ with the Holst inverse, the inverse of L_e and the contortion reconstruction:

$$K = \mathsf{C}_e^{\mathrm{tor}} \mathsf{L}_e^{-1}\big(\kappa c\, \mathsf{P}_\gamma^{-1}\sigma[e, \mathring{\omega}(e) + K, \Psi]\big). \tag{82.43}$$

The symbol $\mathsf{C}_e^{\mathrm{tor}}$ exclusively designates the map $T \mapsto K$ of (82.27), with fixed co-tetrad.

Proof. The differential of $Q + F$, in the arbitrary direction h, is

$$d_k(Q+F)[h] = h^{\mathsf{T}}\big(\mathsf{A}k - \tfrac{1}{2}s\big).$$

Its vanishing in every direction proves (82.41). Theorem 82.21, applied to ∂_{k_j}, gives (82.42). With p fixed, $\delta\omega = \delta K$. The Cartan equation is transformed successively through (82.22), (82.25) and (82.27); each of these operations is invertible on its domain. This proves equivalence with (82.43) when both expressions use the same material and cellular realization. □

The composition preserves the connection on both sides. Routes passing through several cells can couple the current components between them. The pointwise reconstruction of T from a given current and the joint solution of (82.41) are, in that case, successive operations. This finite formulation keeps explicit the route representation entering the functional.

82.9.2 Local existence of a stationary branch

Theorem 82.29 (Local continuation of the torsional solution)**.** *Suppose* (p_0, k_0) *belongs to the preceding open set,* $\mathcal{R}(p_0, k_0) = 0$*, and the total Hessian*

$$\mathsf{H}_0 = \mathsf{A}(p_0) + \partial_k^2 F(p_0, k_0) = \mathsf{A}(p_0) - \tfrac{1}{2}\partial_k s(p_0, k_0) \tag{82.44}$$

is invertible. There exist neighborhoods of p_0 *and* k_0 *in which a unique branch* C^1*,* $k = k_*(p)$*, satisfies* (82.41) *and passes through* k_0*. On that branch,*

$$\partial_{p_i} k_* = -\left[\mathsf{A} + \partial_k^2 F\right]^{-1} \left[(\partial_{p_i}\mathsf{A})k_* + \partial_{p_i}\partial_k F\right]. \tag{82.45}$$

Proof. Inversion of the Gramian is C^2 on the full-rank open set. Hence F is C^2 and $\mathcal{R}$ is C^1. Its derivative with respect to k, at (p_0, k_0), is H_0. The implicit function theorem provides the branch, its regularity and its uniqueness in a neighborhood of the point. Existence can also be seen through the map

$$\mathcal{T}_p(k) = k - \mathsf{H}_0^{-1}\mathcal{R}(p, k).$$

Its derivative with respect to k vanishes at (p_0, k_0). On a sufficiently small ball it has norm at most $q < 1$. Shrinking the neighborhood of p_0 yields $\|\mathcal{T}_p(k_0) - k_0\| < (1 - q)r$, where r is the radius of the ball. Thus, $\mathcal{T}_p$ maps the closed ball into itself and is contractive. Its fixed point is the local solution. Differentiating $\mathcal{R}(p, k_*(p)) = 0$ and solving for $\partial_{p_i} k_*$ gives (82.45). □

The hypothesis is verified on the composite action: the Gramian's rank ensures the screen minimum and the total Hessian controls the torsional branch. The asserted uniqueness is local around the specified solution; other branches may coexist in the same full-rank domain.

82.9.3 Effective action and non-affine contribution

Theorem 82.30 (Exact elimination identity)**.** *Let* $k_*(p)$ *be a stationary branch. Suppose, moreover, that the segment* $t \mapsto (p, tk_*)$*,* $0 \le t \le 1$*, remains in the open set where* F *is defined. The action correction relative to the Levi–Civita realization is*

$$\begin{aligned} \Delta S(p) &:= Q(p, k_*) + F(p, k_*) - F(p, 0) \\ &= \tfrac{1}{4} k_*^{\mathsf{T}} s(p, k_*) - \tfrac{1}{2}\int_0^1 k_*^{\mathsf{T}} s(p, tk_*)\, dt. \end{aligned} \tag{82.46}$$

Equivalently,

$$\Delta S = -\tfrac{1}{4} k_*^{\mathsf{T}} s(p, k_*) + \tfrac{1}{2}\int_0^1 k_*^{\mathsf{T}}\left[s(p, k_*) - s(p, tk_*)\right] dt. \tag{82.47}$$

These identities refer to the integrated or cellular action. In a realization with local density, they apply to the density with its corresponding pairing of forms. The effective action preserves the declared boundary term separately.

Proof. By the chain rule and (82.40),

$$\frac{d}{dt} F(p, tk_*) = -\tfrac{1}{2} k_*^{\mathsf{T}} s(p, tk_*).$$

Integrating between 0 and 1 gives the difference of material actions. The stationarity equation, evaluated in the direction k_*, and the quadratic homogeneity of Q give

$$2Q(p, k_*) = \tfrac{1}{2} k_*^{\mathsf{T}} s(p, k_*).$$

The sum of both identities proves (82.46); adding and subtracting $\frac{1}{2}k_*^{\mathsf{T}}s(p,k_*)$ provides (82.47). □

Corollary 82.31 (Affine specialization)**.** *If $s(p,k)=s(p)$, then*

$$\Delta S=-\tfrac{1}{4}k_*^{\mathsf{T}}s(p)=-Q(p,k_*).$$

This is the integrated realization of (82.32)*.*

Proof. The difference inside the integral of (82.47) vanishes. The second equality follows from the radial stationarity identity. □

82.9.4 Metric differential of the eliminated action

Theorem 82.32 (Response of the geometric variables)**.** *On the branch of Theorem 82.29,*

$$\partial_{p_i}\Delta S=\tfrac{1}{2}k_*^{\mathsf{T}}(\partial_{p_i}\mathsf{A})k_*+(\partial_{p_i}F)(p,k_*)-(\partial_{p_i}F)(p,0),\tag{82.48}$$

where the partial derivatives keep the coordinates k fixed. In fixed Hermitian inner products, each material derivative is calculated by

$$\begin{aligned}\partial_{p_i}F=&(\partial_{p_i}\mathfrak{a})E+\mathfrak{a}(\partial_{p_i}\chi)\langle b,y\rangle\\&+2\mathfrak{a}\chi\,\mathrm{Re}\langle y,\partial_{p_i}b-(\partial_{p_i}\mathsf{C})f_{\min}\rangle.\end{aligned}\tag{82.49}$$

If p_i varies the co-tetrad, $\partial_{p_i}\mathsf{C}$ includes the variation of $\mathring{\omega}(e(p))+K(k)$, of the realized routes and of the trivializations used, according to their declared dependencies.

Proof. The total derivative of $Q(p,k_*)+F(p,k_*)$ contains, in addition to its partial derivatives, the term

$$\big[\partial_kQ(p,k_*)+\partial_kF(p,k_*)\big]\partial_{p_i}k_*.$$

The bracket is zero by the connection equation. Subtracting the derivative of $F(p,0)$ proves (82.48). The product rule and (82.12) prove (82.49). The composition (82.38) requires differentiating the Levi–Civita connection when varying e with k fixed. □

When the positive extension inner product also varies, its contribution is preserved explicitly. In coordinates, let $\mathsf{h}(p,k)>0$ be its Hermitian matrix and consider the minimum of $f^*\mathsf{h}f$ under $\mathsf{C}f=b$. Then

$$\mathsf{L}=\mathsf{C}\mathsf{h}^{-1}\mathsf{C}^*,\quad y=\mathsf{L}^{-1}b,\quad f_{\min}=\mathsf{h}^{-1}\mathsf{C}^*y,\quad E=\chi b^*y.\tag{82.50}$$

Here y represents the constraint covector in fixed bases. The positive matrix h retains its energy role, distinct from that of the Lorentzian metric.

Lemma 82.33 (Variation of the extension inner product)**.** *With the preceding conventions,*

$$\delta E=(\delta\chi)b^*y+2\chi\,\mathrm{Re}\big[y^*(\delta b-\delta\mathsf{C}f_{\min})\big]+\chi f_{\min}^*(\delta\mathsf{h})f_{\min}.\tag{82.51}$$

Proof. The formula $\delta\mathsf{h}^{-1}=-\mathsf{h}^{-1}(\delta\mathsf{h})\mathsf{h}^{-1}$ gives

$$\delta\mathsf{L}=(\delta\mathsf{C})\mathsf{h}^{-1}\mathsf{C}^*+\mathsf{C}\mathsf{h}^{-1}(\delta\mathsf{C})^*-\mathsf{C}\mathsf{h}^{-1}(\delta\mathsf{h})\mathsf{h}^{-1}\mathsf{C}^*.$$

When differentiating $E=\chi b^*\mathsf{L}^{-1}b$, the term $-\chi y^*(\delta\mathsf{L})y$ produces the two terms with $\delta\mathsf{C}$ and the positive term with $\delta\mathsf{h}$. This proves the complete identity. □

In a covariant material realization whose variation admits the metric pairing of (82.34), the correction determines

$$\delta_g \Delta S = -\tfrac{1}{2} \int_M \mathrm{vol}_e \; \mathcal{U}^{S,\min}_{\mu\nu} \, \delta g^{\mu\nu}, \qquad U^{\mathrm{phys},\min}_{\mu\nu} = c \mathcal{U}^{S,\min}_{\mu\nu}. \tag{82.52}$$

Equations (82.48) and (82.49), supplemented by (82.51) where appropriate, evaluate their coefficients in the declared metric variations. The screen minimum, torsional solution and metric derivative are three consecutive operations on the same action. The reduced action preserves the possible couplings between cells present in the original routes.

The homogeneous specialization of 88.16 additionally uses the realization of the resulting tensor: its spatial symmetry, the sign of contraction with the observer and the scaling law. When that evaluation gives $\rho_{\mathrm{corr}}(a) = -s_0 a^{-6}$, the amplitude is read in the reference state as $s_0 = -a_{\mathrm{ref}}^6 \rho_{\mathrm{corr}}(a_{\mathrm{ref}})$. This reading preserves the material state and its normalization; the elimination theorem provides the action to be contracted.

82.9.5 Exact non-affine checking model

The following finite model checks the coefficients and the non-affine dependence of the identities. Its parameters belong to the algebraic example and keep the problem of material realization of the routes separate.

Proposition 82.34 (Exact elimination of a rank-one extension)**.** *Let $H = \mathbb{R}^2$, $Q = \mathbb{R}$, $\mathsf{C}(k) = (1 \;\; k)$, $b = 1$, $\chi = 1$, $\mathfrak{a} = \lambda > 0$, and $Q_{\mathrm{ex}}(k) = k^2/2$. Then*

$$f_{\min} = \frac{(1,k)^{\mathsf{T}}}{1+k^2}, \quad F(\lambda,k) = \frac{\lambda}{1+k^2}, \quad s(\lambda,k) = \frac{4\lambda k}{(1+k^2)^2}, \quad \mathcal{R}(\lambda,k) = k\left(1 - \frac{2\lambda}{(1+k^2)^2}\right). \tag{82.53}$$

For $\lambda = 25/2$, the stationary solutions are $0, 2, -2$. At $k_* = 2$,

$$\mathsf{H}_* = \frac{16}{5}, \quad s_* = 4, \quad Q_{\mathrm{ex}}(k_*) = 2, \quad F(\lambda, k_*) = \frac{5}{2}, \quad \Delta S = -8. \tag{82.54}$$

The affine expression $-k_* s_*/4 = -2$ differs from the exact correction. On the positive branch,

$$\partial_\lambda \Delta S = -\frac{4}{5}, \qquad \partial_\lambda k_* = \frac{1}{20} \quad \text{in } \lambda = 25/2. \tag{82.55}$$

Proof. The Gramian is $1 + k^2 > 0$, so the minimum has the indicated expressions. Differentiating F gives $s = -2\partial_k F$ and then $\mathcal{R} = k - s/2$. For $\lambda = 25/2$, the equation is $k[(1+k^2)^2 - 25] = 0$; since $1 + k^2 > 0$, its roots are precisely $0, \pm 2$. The total Hessian is

$$1 - \frac{2\lambda(1-3k^2)}{(1+k^2)^3},$$

which equals $16/5$ at both nonzero roots. The correction is $2 + 5/2 - 25/2 = -8$. In the radial identity,

$$\int_0^1 k_* s(\lambda, t k_*)\, dt = \int_0^1 \frac{200t}{(1+4t^2)^2}\, dt = 20,$$

since an antiderivative is $-25/(1+4t^2)$. Thus $\Delta S = 8/4 - 20/2 = -8$, including the non-affine term. For $\lambda > 1/2$, the positive branch satisfies $k_*^2 = \sqrt{2\lambda} - 1$ and $\Delta S = \sqrt{2\lambda} - 1/2 - \lambda$. Their derivatives give (82.55). The envelope formula gives the same result: $(\partial_\lambda F)(\lambda, k_*) - (\partial_\lambda F)(\lambda, 0) = 1/5 - 1 = -4/5$. □

82.10 Spinorial current and torsional amplitude of the material state

The preceding development and the following spinorial realization articulate two variational problems with common geometric antecedents. The minimal extension action $S_{\min} = \mathfrak{a}_m E$, constructed in section 82.7, incorporates the dependence of the transported incidence and its minimum on the connection; its torsional elimination preserves that complete dependence in section 82.9. The spinorial action S_D of (82.58) couples the same connection to the fields of its representation and is affine in the contortion when the co-tetrad and fields are fixed. Each current is obtained by varying its own action. A minimizing preparation can determine the initial material state of the spinorial realization; substitution of connection-dependent fields additionally preserves the terms of the composite differential. This distinction places the following amplitude calculation in its specific action and variational domain.

The spinorial representation of transport and the algebraic response of Cartan–Holst allow torsional intensity to be calculated as a functional of the material state. Its evaluation preserves three data: the field representation, the second moment of its current and the volume relative to which density is defined. The coupling coefficient results from the action and its inversion; the initial state determines the material amplitude.

82.10.1 Spinorial representation and affine material action

We preserve the co-tetrad, orientation and couplings of section 82.8. The spinorial realization uses the four-dimensional complex module obtained by complexifying the real matrices

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{82.56}$$
$$g^0 = J \otimes I_2, \qquad g^1 = R \otimes I_2, \qquad g^2 = Z \otimes R, \qquad g^3 = Z \otimes Z.$$

Multiplication gives $\{g^I, g^J\} = 2\eta^{IJ} I_4$, with $\eta = \mathrm{diag}(-1, 1, 1, 1)$. The matrices are distinguished from the scalar parameter γ of the Holst operator. Define

$$g_I = \eta_{IJ} g^J, \quad \mathbb{B}_{IJ} = \tfrac{1}{4}[g_I, g_J], \quad A_D = -ig^0, \quad \bar{\psi} = \psi^\dagger A_D, \quad \Omega = \tfrac{1}{2}\omega^{IJ}\mathbb{B}_{IJ}. \tag{82.57}$$

The Clifford product satisfies

$$[\mathbb{B}_{IJ}, g^K] = \delta^K_J g_I - \delta^K_I g_J.$$

Thus, the same metric connection acts on the fields through $D\psi = \mathrm{d}\psi + \Omega\psi$ and $D\bar{\psi} = \mathrm{d}\bar{\psi} - \bar{\psi}\Omega$. Transport along a route is obtained by integrating this connection in its spinorial representation; it preserves the connection and the route of the preceding geometric realization.

Let $\eta_I = \iota_{E_I} \mathrm{vol}_e$, so that $e^J \wedge \eta_I = \delta^J_I \mathrm{vol}_e$. For a field of dimensional type $[\psi] = L^{-3/2}$, we take the minimal material action

$$S_D = \int_M \left\{ \frac{\hbar}{2} \left[\bar{\psi} g^I D\psi - (D\bar{\psi}) g^I \psi \right] \wedge \eta_I - mc\, \bar{\psi}\psi \, \mathrm{vol}_e \right\}. \tag{82.58}$$

Here $\hbar = \hbar_{\mathrm{int}}$, $c = c_{\mathrm{int}}$ and the mass m is the reading of the material species in the chosen chart. These values are outputs preceding this realization. The matrix A_D is Hermitian and $A_D g^I$ is anti-Hermitian. Therefore the symmetric kinetic term has the real convention corresponding to (82.56). In this signature, the torsion-free equation has principal term $\hbar g^I D_I$; the equivalent notation $i\hbar\Gamma^I D_I$, with $\Gamma^I = -ig^I$, uses the Clifford relation of opposite signature. This specification jointly fixes the adjoint, signature and imaginary factor of the action.

The material fields and co-tetrad remain fixed in the variation of the connection. A preparation coming from the minimal extension of section 82.7 can determine the initial datum of the field or state. Each action preserves its own variation: if a connection-dependent minimizing field is substituted into an

action, its composite differential is also applied, or the envelope theorem under its stationarity hypotheses. Action (82.58) here realizes a coupling affine in contortion.

Proposition 82.35 (Spinorial current with variational normalization)**.** *Let $B^{IJK} = \bar\psi g^{[I} g^J g^{K]}\psi$, where brackets denote normalized antisymmetrization. The current defined in* (82.20) *by action* (82.58) *is*

$$\sigma_{JK} = -\frac{\hbar}{2}\,\bar\psi\{g^I, \mathbb{B}_{JK}\}\psi\,\eta_I = -\frac{\hbar}{2}\,B^I{}_{JK}\eta_I. \tag{82.59}$$

It is independent of contortion when e and ψ are fixed. The inversion of section 82.8.3 therefore applies to this current.

Proof. The variation of Ω is $\delta\Omega = \frac{1}{2}\delta\omega^{JK}\mathbb{B}_{JK}$. The two kinetic terms give, respectively, the product with g^I on the left and on the right. Consequently,

$$\delta_\omega S_D = \frac{\hbar}{4}\int_M \delta\omega^{JK}\wedge\eta_I\,\bar\psi\{g^I, \mathbb{B}_{JK}\}\psi.$$

Comparison with (82.20) fixes the factor and sign of (82.59). Expanding the anticommutator and using the Clifford relation cancels the degree-one contractions, leaving $\{g^I, \mathbb{B}_{JK}\} = g^{[I}g_J g_{K]}$. The action is linear in Ω, proving the last statement. □

82.10.2 Effective contraction and second-moment coefficient

The Lorentzian quadratic form of the trivector is written

$$q(B) = \frac{1}{3!}B^{IJK}B_{IJK} = -(B^{012})^2 - (B^{013})^2 - (B^{023})^2 + (B^{123})^2. \tag{82.60}$$

This contraction preserves the signature of the spinorial module. Its second moment depends on the state and on the realization of field products.

Theorem 82.36 (Effective interaction of the spinorial current)**.** *With material action* (82.58)*, algebraic elimination of contortion gives*

$$\mathcal{L}^{\mathrm{eff}}_{\mathrm{spin}} = C_\gamma q(B)\,\mathrm{vol}_e, \qquad C_\gamma = \frac{3\kappa c\hbar^2}{16}\frac{\gamma^2}{1+\gamma^2}. \tag{82.61}$$

The coefficient is obtained from the current and the Holst inverse with the normalizations of the preceding action.

Proof. By (82.32), it suffices to evaluate $-\frac{1}{4}K_*^{IJ}\wedge\sigma_{IJ}$, with

$$Y^{IJ} = \kappa c\frac{\gamma^2}{1+\gamma^2}(-\star + \gamma^{-1}I)\sigma^{IJ}, \qquad K_* = \mathsf{C}_e(Y).$$

Here C_e is the explicit composition of (82.25) with (82.27). Linearity of those two inverses reduces the calculation to the parts $-\star$ and I. To give the complete contraction, we order the trivector components as $(012, 013, 023, 123)$, extract the factors $\kappa c\hbar^2\gamma^2/(1+\gamma^2)$, and substitute $\sigma_{JK} = -\frac{1}{2}B^I{}_{JK}\eta_I$. The two matrices of the remaining quadratic form are

$$\mathcal{B}_{-\star} = \frac{3}{16}\,\mathrm{diag}(-1,-1,-1,1), \qquad \mathcal{B}_I = 0_{4\times4}. \tag{82.62}$$

These matrices are obtained using only $e^I\wedge\eta_J = \delta^I_J\,\mathrm{vol}_e$: for each component B^{abc}, we form σ, calculate Y, then $T^I = \sum_J \iota_{E_J}Y^{IJ} - \frac{1}{4}e^I\wedge\sum_{L,J}\iota_{E_L}\iota_{E_J}Y^{LJ}$, and finally $K_{IJK} = (T_{KIJ} - T_{IJK} - T_{JKI})/2$. The

contraction gives the four diagonal entries shown; the six polarizations between distinct components are zero for both operators. This calculates all coefficients of the two quadratic forms. The part $\gamma^{-1}I$ vanishes and the part $-\star$ produces exactly (82.60), with the coefficient of (82.61). □

82.10.3 Finite fermionic realization and second moment of the state

We fix a periodic comoving cell, an orthonormal frame adapted to its observer and the homogeneous spatial mode of the four spinorial components. This finite realization has state space $\mathcal{F} = \bigoplus_{n=0}^{4} \Lambda^n \mathbb{C}^4$. Let $a_r, a_r^\dagger$, $r = 1, \dots, 4$ be its canonical operators, with $\{a_r, a_s^\dagger\} = \delta_{rs}$ and $\{a_r, a_s\} = 0$. The occupation vacuum fixes normal ordering. For a matrix M, we write

$$\mathrm{d}\Gamma(M) = \sum_{r,s} a_r^\dagger M_{rs} a_s, \qquad \widehat{N} = \mathrm{d}\Gamma(I_4).$$

The letter Γ in this notation denotes second quantization; nonadic holonomy retains its own notation.

The four matrices of the bilinears are calculated directly:

abc	012	013	023	123
$M_{abc} = A_D g^a g^b g^c$	$iJ \otimes R$	$iJ \otimes Z$	$iI_2 \otimes J$	$iZ \otimes J$

(82.63)

All are Hermitian, have zero trace and satisfy $M_{abc}^2 = I_4$. The quartic observable corresponding to (82.60), in this declared normal ordering, is

$$\begin{aligned} \widehat{\mathcal{Q}}_{\mathrm{sp}} &= \sum_{a<b<c} \eta_{aa}\eta_{bb}\eta_{cc} :\mathrm{d}\Gamma(M_{abc})^2: \\ &= \sum_{a<b<c} \eta_{aa}\eta_{bb}\eta_{cc} \mathrm{d}\Gamma(M_{abc})^2 + 2\widehat{N}. \end{aligned} \tag{82.64}$$

Normal ordering is part of the specified finite realization; its definition makes explicit the subtraction of single-occupation contractions.

Proposition 82.37 (Evaluation by the canonical relations)**.** *The operator $\widehat{\mathcal{Q}}_{\mathrm{sp}}$ is self-adjoint, preserves each occupation sector and has restrictions*

$$\widehat{\mathcal{Q}}_{\mathrm{sp}}|_{\Lambda^0 \oplus \Lambda^1} = 0, \qquad \widehat{\mathcal{Q}}_{\mathrm{sp}}|_{\Lambda^3} = 4I_4, \qquad \widehat{\mathcal{Q}}_{\mathrm{sp}}|_{\Lambda^4} = 8I_1. \tag{82.65}$$

In the basis of $\Lambda^2\mathbb{C}^4$ whose occupation subsets are $(12, 13, 23, 14, 24, 34)$*, its matrix is*

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 2 & 0 & 0 & 6 & 0 \\ 0 & 0 & 2 & -6 & 0 & 0 \\ 0 & 0 & -6 & 2 & 0 & 0 \\ 0 & 6 & 0 & 0 & 2 & 0 \\ -4 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{82.66}$$

Proof. The relation $a_s a_t^\dagger = \delta_{st} - a_t^\dagger a_s$ applied to $\mathrm{d}\Gamma(M)^2$ separates its single-occupation contraction and gives

$$:\mathrm{d}\Gamma(M)^2: = \mathrm{d}\Gamma(M)^2 - \mathrm{d}\Gamma(M^2).$$

Since the first three signs of (82.63) are negative and the last positive, the sum of contractions is $-2\widehat{N}$. This proves (82.64). Self-adjointness and preservation of occupation follow from those same expressions.

On Λ^0 all terms vanish. On Λ^1, $\mathrm{d}\Gamma(M) = M$, and the squares sum to $-2I_4$, which cancels with $2\widehat{N}$. On Λ^4, $\mathrm{d}\Gamma(M) = \operatorname{tr} M = 0$, leaving eight. Identification by complement between $\Lambda^3\mathbb{C}^4$ and the dual of $\mathbb{C}^4$

takes $\mathrm{d}\Gamma(M)$ to $(\operatorname{tr} M)I - M^{\mathsf{T}} = -M^{\mathsf{T}}$. Their squares again sum to $-2I_4$, while $2\widehat{N} = 6I_4$; therefore the restriction is $4I_4$. Finally, applying the four explicit matrices to the six exterior products of the indicated basis produces (82.66). The operator is thus evaluated in the sixteen dimensions of $\mathcal{F}$. □

For a state operator $\varrho \geq 0$, $\operatorname{Tr}\varrho = 1$, define $q_\varrho = \operatorname{Tr}(\varrho\widehat{\mathcal{Q}}_{\mathrm{sp}})$. Its dependence on second moments is explicit:

$$q_\varrho = \sum_{a<b<c} \eta_{aa}\eta_{bb}\eta_{cc}\left\{\langle\widehat{B}_{abc}\rangle_\varrho^2 + \operatorname{Var}_\varrho(\widehat{B}_{abc}) - \langle\widehat{N}\rangle_\varrho\right\}, \qquad \widehat{B}_{abc} = \mathrm{d}\Gamma(M_{abc}). \tag{82.67}$$

Covariance and normal subtraction are preserved even when the mean currents vanish. In particular, if P_3 is the projection onto $\Lambda^3\mathbb{C}^4$,

$$\varrho_3 = \tfrac{1}{4}P_3, \qquad \langle\widehat{B}_{abc}\rangle_{\varrho_3} = 0, \qquad q_{\varrho_3} = 4 > 0. \tag{82.68}$$

It is a positive state of trace one and constitutes an explicit witness to the positive domain of the functional. The two-occupation sector also contains negative values; for example, the vector $(|12\rangle + |34\rangle)/\sqrt{2}$ has value -4. The sign of the intensity thus comes from the contraction and the state, and remains separate from the selection of cosmological conditions.

82.10.4 Density, pressure and preservation of the comoving amplitude

In the spatially homogeneous realization, fix

$$e^0 = N(t)c\,\mathrm{d}t, \qquad e^j = a(t)\,\mathrm{d}x^j, \qquad \mathcal{V}(t) = a(t)^3V_c,$$

with lapse $N(t) > 0$, dimensionless scale factor $a(t) > 0$ and comoving volume $V_c > 0$. The frame and periodic mode of subsection 82.10.3 remain declared. The normalization of the homogeneous field is $\widehat{\psi} = \mathcal{V}^{-1/2}(a_1, a_2, a_3, a_4)^{\mathsf{T}}$. Each bilinear carries a factor $\mathcal{V}^{-1}$, and its normally ordered product carries $\mathcal{V}^{-2}$. The symmetric kinetic temporal form preserves this canonical normalization: terms arising from differentiating $\mathcal{V}^{-1/2}$ cancel between its two summands.

Theorem 82.38 (Torsional amplitude as a functional of the state)**.** *Keeping the canonical state and the realization of the product fixed during metric variations, the effective spin term has reduced action*

$$S_{\mathrm{spin}}^{\mathrm{hom}} = \int \mathrm{d}t\, N(t)\frac{cC_\gamma}{a(t)^3V_c}\, q_\varrho. \tag{82.69}$$

Its energy density and isotropic pressure are

$$\rho_{\mathrm{spin}} = p_{\mathrm{spin}} = -\frac{cC_\gamma}{V_c^2}\, q_\varrho a^{-6}. \tag{82.70}$$

In any evolution preserving q_ϱ, the corresponding constant amplitude is

$$\boxed{s_0[\varrho, V_c] = \frac{3\kappa c^2\hbar^2}{16}\frac{\gamma^2}{1+\gamma^2}\frac{\operatorname{Tr}(\varrho\widehat{\mathcal{Q}}_{\mathrm{sp}})}{V_c^2}.} \tag{82.71}$$

Proof. Spatial integration of (82.61) multiplies the factor $\mathcal{V}^{-2}$ by $Nc\mathcal{V}\,\mathrm{d}t$, giving (82.69). In canonical variables, lapse variation defines energy through $\delta_N S_{\mathrm{m}} = -\int E\,\delta N\,\mathrm{d}t$. Therefore,

$$E_{\mathrm{spin}} = -\frac{cC_\gamma q_\varrho}{a^3V_c}, \qquad \rho_{\mathrm{spin}} = \frac{E_{\mathrm{spin}}}{a^3V_c}.$$

The isotropic spatial variation satisfies $\delta_a S_{\mathrm{m}} = \int 3Na^2V_c p\,\delta a\,\mathrm{d}t$. Applied to (82.69), it gives

$$3Na^2V_c\,p_{\mathrm{spin}} = -\frac{3NcC_\gamma q_\varrho}{a^4V_c}.$$

This simultaneously proves the sign, pressure and density of (82.70). Since q_ϱ is preserved, the coefficient of $-a^{-6}$ is constant and we obtain (82.71). Dimensionality is consistent: $[cC_\gamma]$ = energy L^3, so s_0 has the dimension of energy density. □

Corollary 82.39 (Preserved domain of positive amplitude)**.** *Let there be a unitary evolution of $\mathcal{F}$ preserving $\Lambda^3\mathbb{C}^4$. For every initial state of trace one supported on that sector, $q_{\varrho(t)} = 4$ holds, and*

$$s_0 = \frac{3\kappa c^2\hbar^2}{4V_c^2}\,\frac{\gamma^2}{1+\gamma^2} > 0. \tag{82.72}$$

Proof. Invariance of the sector preserves the support of the state; on it, (82.65) gives $\widehat{\mathcal{Q}}_{\mathrm{sp}} = 4I$. Preservation of the trace then fixes its expectation at four, independently of the internal evolution of its coherences. Positivity of the couplings, $V_c > 0$ and $\gamma \neq 0$ proves the formula and its sign. □

For general states, preservation of $\langle\widehat{N}\rangle$ determines dilution of the occupation density, $n(t) = \langle\widehat{N}\rangle/(a^3V_c)$. Preservation of the quartic moment is checked separately. If $i\hbar\dot{\varrho} = [H(t), \varrho]$, then

$$\frac{\mathrm{d}q_\varrho}{\mathrm{d}t} = \frac{i}{\hbar}\,\mathrm{Tr}\big(\varrho[H(t), \widehat{\mathcal{Q}}_{\mathrm{sp}}]\big). \tag{82.73}$$

Commutation, invariance of a sector where the observable is scalar, or a conservation proved for the specific state provide the relevant conditions. If the moment evolves, (82.70) preserves its instantaneous evaluation; its balance includes

$$\dot{\rho}_{\mathrm{spin}} + 6H\rho_{\mathrm{spin}} = -\frac{cC_\gamma}{V_c^2}\,a^{-6}\dot{q}_\varrho, \qquad H = \dot{a}/a$$

in proper time. This term records exchange with the remaining material degrees of freedom.

The composition is determined: spinorial representation of transport, material action, variational current, Cartan–Holst inversion, second moment and normalized density. The domain $s_0 > 0$ of section 88.16 thus has an evaluable functional and explicit witness states. Its use in cosmological dynamics additionally preserves the material, geometric and conservation conditions stated in that section.

Chapter 83

Confluence theorem relating horizon energy, area, information, entropy, and action

The horizon patch coordinates energy, temperature, entropy, and action through typed identities. The half-action cell and chronological closure are presented with their Schwarzschild and de Sitter controls, without turning an interface into a retroactive definition.

83.1 Horizon energy and half-action cell

83.1.1 Geometric conventions of the de Sitter patch

Consider a four-dimensional de Sitter space with areal radius $R > 0$, causal velocity c, and gravitational constant G. Adopt the convention

$$H_R = \frac{c}{R}, \qquad \Lambda_R = \frac{3}{R^2}, \qquad \ell_P^2 = \frac{G\hbar}{c^3}. \tag{83.1}$$

The density of energy associated to the term cosmological is

$$\varepsilon_\Lambda := \frac{\Lambda_R c^4}{8\pi G}. \tag{83.2}$$

The spatially flat section of the patch contains, up to radius R, the volume

$$V_R = \frac{4\pi}{3} R^3. \tag{83.3}$$

These conventions positively fix the quasilocal energy used in the chapter. Choosing another sign for the de Sitter Killing charge belongs to a distinct construction and must be translated before formulas are compared.

83.1.2 energy of vacuum and energy of Misner–Sharp

In a spherically symmetric geometry, let

$$\mathrm{d}s^2 = g_{ab}(x)\,\mathrm{d}x^a\,\mathrm{d}x^b + \mathcal{R}(x)^2\,\mathrm{d}\Omega_2^2. \tag{83.4}$$

The Misner–Sharp energy [90], under the adopted positive convention, is defined by

$$E_{\mathrm{MS}}(\mathcal{R}) := \frac{c^4\mathcal{R}}{2G}\left(1 - g^{ab}\nabla_a\mathcal{R}\nabla_b\mathcal{R}\right). \tag{83.5}$$

A marginal sphere satisfies

$$g^{ab}\nabla_a\mathcal{R}\nabla_b\mathcal{R} = 0. \tag{83.6}$$

Theorem 83.1 (Confluence volumetrica and cuasilocal)**.** *On the de Sitter horizon $\mathcal{R} = R$, the integrated vacuum energy and the Misner–Sharp energy coincide exactly:*

$$\boxed{E_\Lambda := \varepsilon_\Lambda V_R = E_{\mathrm{MS}}(R) = \frac{c^4 R}{2G}.} \tag{83.7}$$

Proof. The equations equations (83.1) a (83.3) give

$$\varepsilon_\Lambda V_R = \frac{3c^4}{8\pi G R^2}\,\frac{4\pi R^3}{3} = \frac{c^4 R}{2G}.$$

The marginal condition equation (83.6) applied to equation (83.5) yields the same result. □

The equality brings together two geometric publications. The first integrates a volumetric density; the second evaluates the trapping defect of the areal sphere. Both preserve the radius R and gravitational compliance G/c^4.

83.1.3 Temperature, entropy and product of horizon

The de Sitter surface gravity has magnitude c^2/R. The Gibbons–Hawking temperature [51] and the Bekenstein–Hawking entropy [8, 59] are

$$k_B T_H = \frac{\hbar c}{2\pi R}, \tag{83.8}$$

$$\frac{S_H}{k_B} = \frac{A_H}{4\ell_P^2} = \frac{4\pi R^2}{4\ell_P^2} = \frac{\pi R^2}{\ell_P^2}. \tag{83.9}$$

Theorem 83.2 (de Sitter thermodynamic identity)**.** *For every radius R of the declared family,*

$$\boxed{T_H S_H = \frac{c^4 R}{2G} = E_\Lambda.} \tag{83.10}$$

Proof. Multiplying equations (83.8) y (83.9) gives

$$T_H S_H = \frac{\hbar c}{2\pi R k_B} \frac{k_B \pi R^2}{\ell_P^2} = \frac{\hbar c R}{2\ell_P^2}.$$

The identity $\ell_P^2 = G\hbar/c^3$ turns the last member into $c^4 R/(2G)$, equal to equation (83.7). □

The factor $1/2$ has a decomposition geometric transparent. The areal factor 4π, the normalization holographic by four and the periodicity thermal $(2\pi)^{-1}$ satisfy

$$\frac{4\pi}{4} \frac{1}{2\pi} = \frac{1}{2}. \tag{83.11}$$

The half is therefore the joint contraction between areal quantization and Euclidean periodicity.

83.1.4 How complete of the horizon and Planck force

We define the tension or fuerza of Planck and the complete quantum associated with the radius by

$$\mathscr{F}_P := \frac{c^4}{G}, \qquad \boxed{Q_H(R) := 2T_H S_H = \frac{c^4 R}{G} = \mathscr{F}_P\, R.} \tag{83.12}$$

Quasilocal energy occupies half a cell of the complete quantum:

$$\boxed{E_\Lambda = T_H S_H = \frac{Q_H(R)}{2}.} \tag{83.13}$$

Proposition 83.3 (Radial compliance)**.** *The derivative of the complete quantum with respect to radius is the Planck force, and the derivative of the quasilocal energy is half of it:*

$$\frac{\mathrm{d}Q_H}{\mathrm{d}R} = \mathscr{F}_P, \qquad \frac{\mathrm{d}E_\Lambda}{\mathrm{d}R} = \frac{\mathscr{F}_P}{2}. \tag{83.14}$$

The compliance gravitatoria $\mathscr{C}_P = G/c^4$ satisfies

$$R = \mathscr{C}_P Q_H. \tag{83.15}$$

Proof. The three identities are direct derivatives or inversions of equation (83.12). □

The equation equation (83.15) expresses the radial response: the complete quantum applies tension, and geometry publishes a radius through universal compliance. The product $T_H S_H$ occupies the quasilocal half of that response.

83.1.5 Specialization to the closure chronological of 108 states

The chronological closure of section 88.1 determines

$$R_{108} = \ell_P, \qquad t_P = \frac{\ell_P}{c}, \qquad T_{108} = 2\pi t_P, \tag{83.16}$$

and the Planck quantum

$$E_P = \frac{\hbar}{t_P} = k_B \Theta_{\mathrm{clk}} \log 3. \tag{83.17}$$

Theorem 83.4 (Cell CT108 of half action)**.** *In the specialization $R = R_{108} = \ell_P$,*

$$Q_H(\ell_P) = \frac{c^4 \ell_P}{G} = E_P, \tag{83.18}$$

$$\boxed{E_\Lambda = T_H S_H = \frac{E_P}{2} = \frac{k_B \Theta_{\mathrm{clk}} \log 3}{2},} \tag{83.19}$$

$$\boxed{E_\Lambda t_P = \frac{\hbar}{2},} \qquad \boxed{E_\Lambda T_{108} = \frac{h}{2}.} \tag{83.20}$$

Proof. The first equality is obtained by substituting $\ell_P = \sqrt{G\hbar/c^3}$ and $E_P = \sqrt{\hbar c^5/G}$. The equation (83.13) produces $E_\Lambda = E_P/2$, and equation (83.17) supplies the informational chart. Finally,

$$E_\Lambda t_P = \frac{E_P t_P}{2} = \frac{\hbar}{2}$$

and, as $T_{108} = 2\pi t_P$ and $h = 2\pi\hbar$,

$$E_\Lambda T_{108} = \frac{E_P}{2}(2\pi t_P) = \pi\hbar = \frac{h}{2}.$$

□

The first equality in equation (83.20) publishes half a reduced action over Planck time; the second publishes half a complete action over the CT108 period. Both express the same cell in the angular and circular charts.

The temperature and entropy are also determined:

$$k_B T_H = \frac{E_P}{2\pi}, \qquad \boxed{\frac{S_H}{k_B} = \pi.} \tag{83.21}$$

The temperature converts the frequency of horizon in energy; the entropy converts the area of the sphere of Planck in information. Their product recovers the same half-quantum.

83.1.6 Effective horizon weight and hyperbolic exponential

We define the effective thermodynamic weight

$$\Omega_{\text{eff}}(R) := \exp\left(\frac{S_H(R)}{k_B}\right). \tag{83.22}$$

In the CT108 set,

$$\boxed{\Omega_{\text{eff}}(\ell_P) = \mathrm{e}^{\pi}.} \tag{83.23}$$

The weight Ω_{eff} is a positive quantity defined by entropy. Its real value expresses effective multiplicity; an interpretation as an integer cardinality of microstates requires an additional discretization.

The areal–volumetric incidence operator is

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad J_\varphi := \frac{2F_{\text{av}}^2 - 3I}{\sqrt{5}}. \tag{83.24}$$

A direct calculation gives

$$J_\varphi^2 = I, \qquad \operatorname{Spec}(J_\varphi) = \{+1, -1\}. \tag{83.25}$$

The hyperbolic flow at closure time π is

$$\mathcal{E}_\pi := \mathrm{e}^{\pi J_\varphi} = \cosh\pi\, I + \sinh\pi\, J_\varphi, \tag{83.26}$$

and possesses spectrum

$$\boxed{\operatorname{Spec}(\mathcal{E}_\pi) = \{\mathrm{e}^{\pi}, \mathrm{e}^{-\pi}\}, \qquad \rho(\mathcal{E}_\pi) = \mathrm{e}^{\pi}.} \tag{83.27}$$

Corollary 83.5 (Horizon–Perron scalar confluence)**.** *In the CT108 closure,*

$$\boxed{\frac{S_H}{k_B} = \log\Omega_{\text{eff}} = \log\rho(\mathcal{E}_\pi) = \pi.} \tag{83.28}$$

Proof. The equations equations (83.21), (83.23) y (83.27) give the three members. □

The scalar identity is exact and preserves two distinct genealogies. The horizon produces e^{π} by exponentiation of the areal entropy; Perron dynamics produces it as the expansive eigenvalue of an involution that is hyperbolic. The terminal equality fixes a boundary condition for the intertwiner and preserves the distinction of domains.

83.1.7 Consistency of the Gravitational Transducer with the Schwarzschild Horizon

A Schwarzschild horizon with the same area radius satisfies

$$R_H = \frac{2GE_H}{c^4}, \qquad E_H = \frac{c^4 R_H}{2G}. \tag{83.29}$$

The entropy areal coincides with equation (83.9), whereas the temperature is

$$k_B T_{\text{Sch}} = \frac{\hbar c}{4\pi R_H} = \frac{1}{2} k_B T_H\big|_{R=R_H}. \tag{83.30}$$

Consequently,

$$E_H = 2T_{\text{Sch}} S_H, \qquad T_H^{\text{dS}} = 2T_{\text{Sch}} \tag{83.31}$$

for the same radius. In $R_H = \ell_P$,

$$E_H = \frac{E_P}{2}, \qquad T_{\text{Sch}} S_H = \frac{E_P}{4}. \tag{83.32}$$

The contrast separates the two geometries: they share quasilocal energy, area, and entropy, and their surface gravities produce temperatures in the ratio two.

With $\beta = (k_B T_{\mathrm{Sch}})^{-1}$, the action Euclidean satisfies

$$\frac{I_E}{\hbar} = \frac{E_H}{k_B T_{\mathrm{Sch}}} - \frac{S_H}{k_B} = 2\pi - \pi = \pi \tag{83.33}$$

at the Planck radius. The same π appears as reduced entropy, reduced Euclidean action, hyperbolic-flow time, and the logarithm of the effective weight.

83.1.8 Horizontal Horizon–Perron Entangler: Spectral compatibility between boundary energy and return memory

The coincidence $\Omega_{\mathrm{eff}} = \rho(\mathcal{E}_\pi)$ fixes a spectral datum and leaves open the relation between the complete objects. Let $\mathcal{A}_H^{(m)}$ denote the algebra of horizon observables at nonadic level m, let $\omega_H^{(m)}$ denote its state, and let $\mathcal{P}_\varphi^{(m)}$ denote the Perron transfer system.

Definition 83.6 (Thermodynamic–dynamic intertwiner)**.** An admissible intertwiner is a family of unital completely positive linear maps (UCP)

$$\mathfrak{I}_m : \mathcal{A}_H^{(m)} \longrightarrow \mathcal{A}_\varphi^{(m)} \tag{83.34}$$

which satisfies:

i) preservation of the state, in addition to unitality and complete positivity already incorporated into the UCP condition;

ii) compatibility with nonadic refinement;

iii) transport of the involution interior–exterior toward $J_\varphi \mapsto -J_\varphi$;

iv) commutation between the horizon modular semigroup and the transfer semigroup on the declared subspace;

v) equality of the spectral radius at time π, given by equation (83.28).

The fifth condition is necessary and scalar; the first four provide the algebraic content that a numerical coincidence does not contain. An intertwiner constructed in this manner would identify the effective thermodynamic multiplicity with a dynamics of expansion and contraction while preserving orientation, state, and refinement.

Estatuto probatorio

The identities $E_\Lambda = E_{\mathrm{MS}} = T_H S_H = c^4 R/(2G)$, the definition $Q_H = \mathcal{F}_P R$, the CT108 specialization, the two half-action cells, and the equality $S_H/k_B = \log \rho(\mathrm{e}^{\pi J_\varphi}) = \pi$ are exact results under the declared conventions and de Sitter realization. The identification of Ω_{eff} with a microscopic cardinal and the intertwiner $\mathfrak{I}_m$ belong to a delimited physical interface.

Falsification criterion

Geometric closure fails if $\Lambda_R \neq 3/R^2$, if the sphere of radius R fails to satisfy the marginal condition, or if the integrated energy differs from $E_{\mathrm{MS}}(R)$. Thermodynamic closure fails if the temperature or the area law produces $T_H S_H \neq c^4 R/(2G)$. The CT108 specialization fails if $R_{108} \neq \ell_P$, $T_{108} \neq 2\pi t_P$, or $E_P \neq k_B \Theta_{\mathrm{clk}} \log 3$. The Perron interface is refuted by an obstruction of positivity, state, orientation, or refinement, even when the eigenvalue e^π agrees scalarly.

Machian Factorization of the Transport Memory from the Global Closure and the Local Response. The relationship between global closure and local response leads to Mach's principle formulated on the memory of transport.

83.2 Confluence of action, area, information, and gravitation

83.2.1 Energy and entropy of the half-action cell

Fix the same oriented action section and its corresponding gravitational and thermal sections. The preceding reciprocity allows the same cell to be expressed through action, duration, and length. With $\ell_P^2 = \hbar G/c^3$, $t_P = \ell_P/c$, and $E_P = \hbar/t_P$, one has

$$\frac{c^4 \ell_P}{G} = \frac{\hbar}{t_P} = E_P.$$

The equality follows by substituting the area scale already constructed. In the horizon realization defined by (83.35), the quantities are normalized as

$$E_\Lambda(R) = \frac{c^4 R}{2G}, \qquad S_H(R) = \frac{k_B \pi R^2}{\ell_P^2}, \qquad T_H(R) = \frac{\hbar c}{2\pi k_B R}. \tag{83.35}$$

The temperature corresponds to the cosmological-horizon normalization of this realization. The factor 2π and the definition of the radius are part of the set of equations used.

Theorem 83.7 (Identity of the gravitational half-action cell)**.** *In realization* (83.35)*,*

$$T_H(R) S_H(R) = E_\Lambda(R), \tag{83.36}$$

$$T_H(\ell_P) S_H(\ell_P) = \frac{E_P}{2} = \frac{k_B \Theta_{\mathrm{clk}} \log 3}{2}, \tag{83.37}$$

$$E_\Lambda(\ell_P) t_P = \frac{\hbar}{2}, \qquad E_\Lambda(\ell_P) T_{108} = \frac{h}{2}. \tag{83.38}$$

Proof. Multiplying temperature by entropy cancels the thermal normalization and gives $\hbar c R/(2\ell_P^2) = c^4 R/(2G)$. At $R = \ell_P$, the expression is $\hbar c/(2\ell_P) = \hbar/(2t_P) = E_P/2$. The thermal identity of the cycle gives the second member. Finally, $T_{108} = 2\pi t_P$ and $h = 2\pi\hbar$ produce the two action relations. □

The theorem brings together quantities that previously belonged to different descriptions. The cell energy, the temperature–entropy product, tritic information, and action integrated over its duration are compatible readings under the declared normalization. The gravitational constant enters the area scale; action fixes the energy of the period; Boltzmann transports dimensionless information to the entropy line. The proof shows how these functions are joined in a single identity.

83.2.2 Conversion between decision energy and area increment

Energy per tritic decision and the elementary area increment are obtained through different readers in the preceding development:

$$E_P = k_B \Theta_{\mathrm{clk}} \log 3 = \frac{\hbar}{t_P}, \qquad \delta\mathcal{A} = \gamma_{\mathrm{HMT}} a_0 \ell_P^2. \tag{83.39}$$

The same oriented section and the positive branch $\hbar, G, \gamma_{\mathrm{HMT}}, a_0, t_P > 0$ are maintained. The first quantity belongs to the energy line and the second to the area line. Their quotient defines the conversion coefficient

$$\mathcal{T}_{I/A} := \frac{E_P}{\delta\mathcal{A}} = \frac{k_B \Theta_{\mathrm{clk}} \log 3}{\gamma_{\mathrm{HMT}} a_0 \ell_P^2}, \qquad [\mathcal{T}_{I/A}] = [\text{energy}]/[\text{area}]. \tag{83.40}$$

This definition makes explicit the normalization required to transport a decision energy to a geometric resolution. All its factors are specified before the quotient is formed.

Proposition 83.8 (Gravitational expression of the conversion coefficient)**.** *In the same dimensional realization,*

$$\mathcal{T}_{I/A} = \frac{c^3}{\gamma_{\mathrm{HMT}} a_0 G t_P} = \frac{c^4}{\gamma_{\mathrm{HMT}} a_0 G \ell_P}. \tag{83.41}$$

Proof. Substituting $E_P = \hbar/t_P$ and $\ell_P^2 = \hbar G/c^3$ into (83.40) cancels the common action section and gives the first member. The identity $\ell_P = c t_P$ produces the second. The dependence of t_P and ℓ_P on the previously constructed realization is preserved. □

At the minimum cut of (72.16), the realization of one uniform decision per link publishes

$$S^{\mathrm{fis}}_{\mathrm{corte}} = 81 k_B \log 3, \qquad E^{(1)}_{\mathrm{corte}} = 81 E_P = \Theta_{\mathrm{clk}} S^{\mathrm{fis}}_{\mathrm{corte}}. \tag{83.42}$$

The first equality uses the proved capacity and the uniform product state; the second assigns the decision energy of the same cycle to each of its eighty-one links. Multiplying the thermal identity by eighty-one proves the equation of state of this realization. The thirty-six-link sector state preserves its own partition function, entropy, and area variations in (74.25).

The composition identifies the complementary functions of the constants: action fixes energy per duration; the angular character and Barbero–Immirzi fix the sector resolution of area; Boltzmann dimensionally realizes information; the product $G\hbar$ determines the area unit. The coefficient $\mathcal{T}_{I/A}$ preserves these dependencies by expressing their relation in units of energy per area.

83.2.3 Physical meaning of area confluence

For a horizon of radius R, the area is $A_H = 4\pi R^2$. The same entropy relation admits two expressions

$$\frac{S_H}{k_B} = \frac{A_H}{4\ell_P^2} = \frac{\pi R^2 c^3}{G\hbar}.$$

The first quotient compares the macroscopic area with the areal unit of the construction. The second makes explicit the composition of that unit through action and gravitation. When the sections are transported so that $G_a \hbar_a$ remains fixed, the geometric reading preserves that normalization. The dimensionless entropy of the same horizon remains invariant; its dimensional publications retain the bases that they use.

The Barbero–Immirzi parameter acts at a different resolution of the same development: oriented incidence organizes the elementary area spectrum, and the reduced state describes the information accessible at the cut. The areal and volumetric sheets provide the geometric transport and history measure described in section 73.6.2. Comparing a microscopic spectrum with a horizon entropy retains its realization map, state, and scale as data. These specifications allow the physical relation to be interpreted at each level and avoid replacing a correspondence of observables with an indiscriminate identification of their spaces.

The theoretical unity of the argument resides in complementary functions:

The areal unit brings together the product $G\hbar$; oriented incidence organizes its elementary resolution through $\gamma_{\mathrm{HMT}} a_0$; the state and bipartition determine the information accessible to the subsystem. The response equations (74.16)–(74.17) make that interrelation explicit in the sectoral family. Inversion (74.19) preserves entropy and transforms the area according to (74.20): the two readings distinguish different properties of the same configuration.

In this articulation, the dynamics produces and preserves history; the reader determines what information is accessible; action relates evolution and energy; Boltzmann realizes its thermal expression; gravitation enters the geometric scale by which area is measured. The identity $T_H S_H = E_\Lambda$ shows an exact confluence

of these functions in the declared realization. Physical theory is thus formulated from mathematical objects and their effective transformations.

83.2.4 Spectral entropy and preservation under transport

The entropy of a reduced state depends on its spectrum and on the bipartition defining the reduction. To specify the continuation towards duality, let $\mathcal{H}_A, \mathcal{H}_B$ be the two factors and ρ_{AB} a state with finite entropies. A factorized transport $U_A \otimes U_B$ produces

$$\rho'_A = U_A \rho_A U_A^*, \qquad S(\rho'_A) = S(\rho_A).$$

The first identity follows from the definition of partial trace and the second from the invariance of the spectrum. For a general unitary transport, the equivalent formulation also transports the inclusion of the subsystem's observable algebra. This specification identifies the geometric datum that must accompany an assertion of entanglement preservation.

The reciprocity of the ellipse and momentum–winding exchange preserve the quantities indicated in their respective theorems. Transport of the bipartition determines how to use these identities in an entropy reading. Modular continuation thus maintains the article's informational content and keeps distinct the energy spectrum, action orientation, and algebraic reduction defining entanglement.

83.2.5 Effective multiplicity and hyperbolic realization

In the preceding cell, $S_H(\ell_P)/k_B = \pi$. The effective multiplicity associated with that entropy is

$$\Omega_{\text{eff}} = \exp(S_H/k_B) = e^{\pi}.$$

If H is the hyperbolic involution of TRIT, with eigenvalues $+1, -1$, the exponential $\exp(\pi H)$ has spectrum $\{e^{\pi}, e^{-\pi}\}$. Agreement of the expansive eigenvalue with the effective multiplicity locates a specific spectral relation. Its microscopic meaning retains as additional data the map between the horizon space and the hyperbolic representation. This localization allows the correspondence to be developed without confusing the spectral value with its entire realization space.

83.3 Evaluation of constants and metrological comparison

The preceding constructions allow three operations to be separated: producing a quantity, publishing its coordinate in a chart and comparing that coordinate with a reference. The APP–TRIT–TPK state preserves sheets, orientation, quotient, carry and memory before these readings. Its nonadic prolongation publishes the correlated coordinates π, φ, e, α; the determinantal reader and regional return produce the action sections; the marked incidence $90/120$ and the conjugate channels produce the Barbero functional; angular transport subsequently publishes the CKM coordinates. No entry in the external tables enters into the choice of those operations.

The evaluations preserved in the source documents and their comparators are brought together here. The digits of the internal outputs express the precision of numerical publication, not experimental uncertainty. For a value x and a nonzero reference x_0, both in the same chart, we use

$$\Delta x = x - x_0, \qquad r_x = \frac{x - x_0}{x_0}.$$

When the reference has asymmetric marginal uncertainties u_-, u_+, we also define the descriptive residual

$$z_x = \begin{cases} \Delta x/u_+, & \Delta x \geq 0, \\ \Delta x/u_-, & \Delta x < 0. \end{cases} \tag{83.43}$$

This normalization does not assume independence between rows and does not by itself constitute a joint statistical test.

83.3.1 Planck constant and reduced constant: two orientations

In the chart $\mathcal{U}_S \mapsto \mathrm{J\,s}$, the sections already constructed are

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\,\mathrm{J\,s}, \qquad H_5 = 1.0545718176461544696\ldots, \tag{83.44}$$

$$\hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\,\mathrm{J\,s}, \qquad \eta_{\mathrm{ret}} = 1.0545718169150390611\ldots, \tag{83.45}$$

$$h_a = 2\pi\hbar_a, \qquad a \in \{\mathrm{pre}, \mathrm{ret}\}. \tag{83.46}$$

The decade comes from $\Sigma_H = \varphi\alpha^{16}$; the difference between the two sections comes from regional return. Multiplication by 2π changes action per radian to action per turn, without introducing another orientation or altering the relative residual.

The SI reference is

$$h_{\mathrm{SI}} = 6.62607015 \times 10^{-34}\,\mathrm{J\,s}, \qquad \hbar_{\mathrm{SI}} = \frac{h_{\mathrm{SI}}}{2\pi} = 1.0545718176461563912\ldots \times 10^{-34}\,\mathrm{J\,s}.$$

Both expressions are exact. The infinite expansion of $\hbar_{\mathrm{SI}}$ is not a measurement uncertainty. Fixing the numerical value of h belongs to the definition of the SI in effect since 2019 [185]; comparison takes place after the HMT construction.

Magnitud	Coordinate in 10^{-34} J s	r_x relative to SI
$\hbar_{\mathrm{pre}}$	1.0545718176461544696	$-1.822221 \times 10^{-15}$
$\hbar_{\mathrm{ret}}$	1.0545718169150390611	$-6.932836 \times 10^{-10}$
h_{pre}	6.62607014999998793	$-1.822221 \times 10^{-15}$
h_{ret}	6.62607014540625433	$-6.932836 \times 10^{-10}$

Table 83.1: Action before and after return. The final digits are evaluations of the published coordinates; the residuals are not experimental errors of h or $\hbar$.

Proposition 83.9 (Invariance of the comparison under a complete turn)**.** *In a common chart,*

$$\frac{h_a - h_{\mathrm{SI}}}{h_{\mathrm{SI}}} = \frac{\hbar_a - \hbar_{\mathrm{SI}}}{\hbar_{\mathrm{SI}}}.$$

Proof. The two terms of the difference in the numerator on the left contain the common factor 2π, which cancels against that in the denominator. □

The pre/ret distinction is quantitatively relevant: replacing one with the other changes the comparison by several orders of magnitude. The regional corrector must remain visible and must not be confused with numerical rounding.

83.3.2 Barbero–Immirzi: internal value and comparison between constructions

The functional evaluation is reconstructed from its three contributions:

$$\begin{aligned}
\Delta S_{90} &= 2.95169439297457621139847987861 \times 10^{-4},\\
\Delta S_{120} &= 1.96835112502951720093738706217 \times 10^{-5},\\
\pi AC^*/180 &= 0.27048532293414007846558645879 0\ldots,\\
\gamma_{\mathrm{HMT}} &= \pi AC^*/180 + 12\Delta S_{90} - \Delta S_{120}\\
&= 0.27400767269445927474725526077398041940\ldots.
\end{aligned}$$

This composition preserves the oriented weight $12 : -1$, prior to its decimal evaluation. Its published summands reconstruct the final coordinate with an absolute residual smaller than 3×10^{-31}.

The Ghosh–Mitra counting is considered as a bibliographic comparison, not a measurement of the parameter. Their equation (7), written with $\lambda = 2\pi\gamma_{\mathrm{GM}}$, is

$$\sum_{j\in\{1/2,1,3/2,\dots\}} (2j+1)e^{-2\pi\gamma_{\mathrm{GM}}\sqrt{j(j+1)}} = 1. \tag{83.47}$$

The reference is A. Ghosh and P. Mitra, *An improved estimate of black hole entropy in the quantum geometry approach*, *Physics Letters B* **616** (2005), 114–117, arXiv:gr-qc/0411035, equations (7)–(8). The root is evaluated here for that specific equation; puncture counting is not identified with the HMT generator.

Construction	Dimensionless value	Nature
Functional $\Gamma_{6\to 8}$	0.2740076726944593	HMT incidence and return
Root of (83.47)	0.2740668582328300	Declared bibliographic counting
$\gamma_{\mathrm{HMT}} - \gamma_{\mathrm{GM}}$	$-5.918553837 \times 10^{-5}$	Difference between constructions
$\gamma_{\mathrm{HMT}}/\gamma_{\mathrm{GM}} - 1$	$-2.159529202 \times 10^{-4}$	Diferencia relativa

Table 83.2: Barbero–Immirzi comparison in the positive orientation. No experimental uncertainty or statistical significance is assigned to this table.

The external root evaluation is reproducible without arbitrarily truncating its tail. With $n = 2j$, $N = 128$, and $r = e^{-0.27\pi}$, for $\gamma \geq 0.27$,

$$\sum_{n>N} (n+1)e^{-\pi\gamma\sqrt{n(n+2)}} \leq \frac{r^{N+1}[(N+2) - (N+1)r]}{(1-r)^2} < 10^{-43}.$$

Every summand decreases strictly with γ. The endpoints 0.27 and 0.28 enclose the root; bisection and the tail bound control the decimals in the table. This is a check of the bibliographic comparator, not a route for selecting γ_{HMT}.

83.3.3 CKM angles, amplitudes, and CP violation

The angular reader already constructed publishes, in degrees,

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta) = (13.003313589509,\ 2.398056157332,\ 0.213977382244,\ 65.676173123554)^\circ. \tag{83.48}$$

These values arise from $M_{\mathrm{CKM}}(A, C^*/6, \Delta)^T$ and $\delta = 9A$. Conversion to radians is performed once before evaluating sines and cosines. The unitary parameterization of the angular section gives

$$|V_{\mathrm{CKM}}| \simeq \begin{pmatrix} 0.974350259 & 0.225005836 & 0.003734601 \\ 0.224873110 & 0.973489279 & 0.041841465 \\ 0.008582400 & 0.041121745 & 0.999117283 \end{pmatrix},$$

and the invariant quartet preserves the phase:

$$J = c_{12}c_{23}c_{13}^2 s_{12}s_{23}s_{13}\sin\delta = 3.1189723326632974 \times 10^{-5} \quad \text{in the owner's evaluation.}$$

This evaluation uses (16.69) with the coordinates of (83.48). The nine printed moduli are rounded; unitarity belongs to the complex matrix before rounding, not to a matrix formed solely from moduli.

The comparator is the three-generation fit in the Particle Data Group, 2025 update, *CKM Quark-Mixing Matrix*, equations (12.27)–(12.28) and the adjacent value of J. To avoid altering their uncertainties

through a nonlinear transformation, the table retains the observables published by PDG: $s_{ij} = \sin\theta_{ij}$, δ in radians and J.

Observable	HMT evaluation	PDG 2025	z_x
s_{12}	0.225007404757	0.22501 ± 0.00068	−0.003817
s_{23}	0.041841757010	$0.04183^{+0.00079}_{-0.00069}$	0.014882
s_{13}	0.003734601164	$0.003732^{+0.000090}_{-0.000085}$	0.028902
δ (rad)	1.146265461116	1.147 ± 0.026	−0.028251
$10^5 J$	3.118972333	$3.12^{+0.13}_{-0.12}$	−0.008564

Table 83.3: Marginal comparison in the declared parameterization and edition. The factor 10^5 jointly multiplies the value and uncertainty in the last row; it does not change its normalized residual.

The residuals are calculated with the coordinates before rounding in the table. The PDG fit combines data and unitarity constraints; its rows are not independent measurements. Therefore, $\sum z_x^2$ is not summed and no number of degrees of freedom is assigned to this table. The result is a reproducible marginal comparison with that edition, not a measure of model probability. The historical marginal deviations preserved in the source document are not transferred to this table: the excerpt does not identify the edition or covariance of that comparison.

The PMNS sector retains the transport $U = F_\ell^\dagger F_\nu$, the spectral domains and the phase data developed in the angular section. A numerical row requires specifying the frames of the register used and the realization being compared. This table does not assign CKM figures to the leptonic reader or publish a new PMNS matrix.

83.3.4 Boltzmann and universal gravitation: quantity, chart and reference

The thermal construction already recovered determines the positive transducer $k_B : \mathcal{L}_\Theta \to \mathcal{L}_E$ and the relation

$$k_B \Theta_{\mathrm{clk},a} \log 3 = \frac{h_a}{108 t_0} = \frac{\hbar_a}{t_P}, \qquad t_P = \frac{54}{\pi} t_0. \tag{83.49}$$

Local information, action and period appear before the kelvin–joule chart. For positive bases u_E, u_Θ, if $k_B(u_\Theta) = b\, u_E$, $\Theta_{\mathrm{clk},a} = \theta_a u_\Theta$ and $E_{P,a} = \epsilon_a u_E$, the equation is equivalent to $b\theta_a = \epsilon_a / \log 3$. A change $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$ produces

$$b' = \frac{s_\Theta}{s_E} b, \qquad \theta'_a = \frac{\theta_a}{s_\Theta}, \qquad \epsilon'_a = \frac{\epsilon_a}{s_E}.$$

This preserves the energy product and makes visible the additional operation that separates the individual coordinates. For the transducer of subsection 70.2.7, the kelvin–joule chart used here expresses the SI coordinate

$$k_B^{\mathrm{SI}} = 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}},$$

the exact value of the SI definition, not an estimate with measurement error. The function of the transducer and its relation to action have content of their own; citing its SI coordinate does not constitute an independent numerical check of an internal choice of both bases.

In gravity, the circular realization fixes

$$G_a = \vartheta^\circ_{108} \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a}, \qquad \vartheta^\circ_{108} = (54/\pi)^2 = 295.4525715010569\ldots. \tag{83.50}$$

The selector and reciprocity $G_a \hbar_a = \vartheta^\circ_{108} c_{\mathrm{int}}^5 t_0^2$ are internal quantities of the declared realization. A dimensional evaluation of G_a must preserve, in addition to the orientation a, the time, velocity and action

charts. In the dimensional chart used here, the evaluated value is $6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$. The table distinguishes that coordinate from the recommended value with its uncertainty.

Objeto	Internal output or relation	Referencia dimensional
k_B	Positive transducer; $k_B\Theta_{\mathrm{clk},a} = E_{P,a}/\log 3$.	$1.380649 \times 10^{-23}\ \mathrm{J/K}$, exact in SI.
G_a	Selector $(54/\pi)^2$; $G_a = \vartheta_{108}^{\circ} c_{\mathrm{int}}^5 t_0^2/\hbar_a$.	$6.67430(15) \times 10^{-11}\ \mathrm{m^3/(kg\,s^2)}$, CODATA 2022.

Table 83.4: Internal normalization and chart reference. The standard relative uncertainty of the reference for G is approximately 2.24743×10^{-5}; k_B has no definitional uncertainty. No predictive residual is calculated from a coordinate that has already been identified as a chart reference.

The external sources for this last table are the BIPM defining constants and the NIST CODATA 2022 table [181]. The experimental uncertainty of G is not transferred to h, $\hbar$ or k_B. Nor does equality of digits in a chart remove the need to identify the reader that produces the quantity.

83.3.5 Reproduction of the evaluations

The program `verificar_comparaciones_II.py`, included with the sources, receives the internal coordinates published in files separate from the external data. It checks the composition of the Barbero functional, the conversion between h and $\hbar$, the bibliographic root and its tail, the CKM moduli, the quartet J and each printed residual. It uses 75-digit decimal arithmetic and elementary functions evaluated after freezing the internal outputs. Execution instructions and program versions are included in the reproducible package. The working precision does not increase the digits available from the source document: only digits stable under its rounding are published. The check records the files used and their hashes, so that a change in evaluation or external edition remains visible.

83.4 Pure bipartite states, reduced entropy and horizon area laws

The enriched state preserves routes and their records during evolution; a subsystem reading determines which part of that information remains accessible. The construction of this section uses the angular channels of section 14.11.6, the incidence and the area operator of section 122, and the action, area and temperature units of sections 121 and 123. These are prior publications of the same APP–TRIT–TPK construction. Each here transmits its domain, orientation, normalization and realization conditions.

Three spaces are distinguished: the binary sheet record, the tritic factors of sectorial incidence, and the geometric realization of the exterior regions of a horizon. Purification, transport of observables, and dimensional conversion specify their relationships. This distinction preserves the structure accompanying each entropic value.

83.4.1 Sheet record and bipartite quantum purification

Write $a = A_{\mathrm{rad}}$ and $y = C_{\mathrm{rad}}^{*}$ for the angular coordinates already constructed. The channels and their probabilities are

$$\mathfrak{q}_{\pm} = e^{-a\mp y}, \qquad p_{\pm} = \frac{\mathfrak{q}_{\pm}}{\mathfrak{q}_{+} + \mathfrak{q}_{-}} = \frac{e^{\mp y}}{2\cosh y}. \tag{83.51}$$

The signs index these two channels. On $\mathcal{H}_{\mathrm{h}} = \mathrm{span}\{|+\rangle, |-\rangle\}$, define

$$\rho_{\mathrm{h}}(y) = p_{+}|+\rangle\langle+| + p_{-}|-\rangle\langle-|.$$

The auxiliary space $\mathcal{H}_{\mathrm{ar}} = \mathrm{span}\{|\mathrm{adv}\rangle, |\mathrm{ret}\rangle\}$ carries two orthonormal labels. With a phase χ_0 fixed, the joint state is

$$|\Psi_{\mathrm{ar}}\rangle = \sqrt{p_{+}}|+\rangle \otimes |\mathrm{adv}\rangle + e^{i\chi_0}\sqrt{p_{-}}|-\rangle \otimes |\mathrm{ret}\rangle. \tag{83.52}$$

Proposition 83.10 (Accessible information in the sheet record)**.** *The vector* (83.52) *is normalized, and its partial trace over* $\mathcal{H}_{\mathrm{ar}}$ *is* $\rho_{\mathrm{h}}(y)$*. Its entanglement entropy, expressed in nats, is*

$$s_{\mathrm{h}}(y) = \log(2\cosh y) - y\tanh y. \tag{83.53}$$

Exchange of the labels + *and* − *conjugates* $\rho_{\mathrm{h}}(y)$ *to* $\rho_{\mathrm{h}}(-y)$ *and preserves* s_{h}*.*

Proof. The squared norm is $p_+ + p_- = 1$. Orthogonality of the auxiliary labels eliminates the cross terms in the partial trace. The squared Schmidt coefficients are precisely $p_\pm$. By (83.51), $p_+ - p_- = -\tanh y$; substituting $\log p_\pm = \mp y - \log(2\cosh y)$ into $-\sum_\pm p_\pm \log p_\pm$ proves the formula. Exchange of signs permutes the two eigenvalues and preserves entropy. □

The common factor e^{-a} cancels upon normalization of the binary record. The relative phase χ_0 remains in the joint state even though this diagonal reduction does not express it. Logical reversibility concerns transport of the complete record, with its labels and correlations; phase return preserves the memory advance described in Section 80.4. A thermal implementation additionally has its own energy and dissipation law. In particular, s_{h} measures this binary subsystem and its concrete bipartition.

83.4.2 Quantum purification of the incidence sectors

The boundary incidence constructed earlier distinguishes two sets of eighteen links, with sectorial labels $m = 90, 120$. Each link carries $\mathbb{C}^3$, with basis $|0\rangle, |1\rangle, |2\rangle$ and occupation operator $\mathrm{diag}(0, 1, 2)$. Let N_m be their sectorial sums, and

$$\mathsf{N} = \mathsf{N}_{90} + \mathsf{N}_{120}, \qquad \mathsf{D}_\partial = 90\mathsf{N}_{90} + 120\mathsf{N}_{120}. \tag{83.54}$$

In the positive branch, γ_{BI} denotes the Barbero–Immirzi parameter already obtained, and $a_0 > 0$ its angular areal factor. This notation distinguishes the parameter from the connection and its holonomies. Area is published as

$$\widehat{\mathcal{A}}_m = \ell_P^2 \gamma_{\mathrm{BI}} a_0 \mathsf{N}_m, \qquad \widehat{\mathcal{A}} = \widehat{\mathcal{A}}_{90} + \widehat{\mathcal{A}}_{120}. \tag{83.55}$$

For the finite parameter $\Delta = \Delta_{\mathrm{mod}} \in \mathbb{R}$ of the reduced state, set

$$x_m = e^{-m\Delta}, \qquad z_m = 1 + x_m + x_m^2, \qquad Z(\Delta) = z_{90}^{18} z_{120}^{18}, \tag{83.56}$$

$$\rho_\partial(\Delta) = Z(\Delta)^{-1} e^{-\Delta \mathsf{D}_\partial}. \tag{83.57}$$

The trace in this section is the ordinary matrix trace, and the states have trace one. The parameter Δ preserves the selection and normalization of the antecedent state.

Proposition 83.11 (Tensor purification of the boundary)**.** *For a finite real weight* w*, let*

$$|\,\mathrm{TFD}(w)\rangle = \frac{|0\rangle|0\rangle + e^{-w/2}|1\rangle|1\rangle + e^{-w}|2\rangle|2\rangle}{\sqrt{1 + e^{-w} + e^{-2w}}}. \tag{83.58}$$

The product

$$|\Psi_\Delta\rangle = |\,\mathrm{TFD}(90\Delta)\rangle^{\otimes 18} \otimes |\,\mathrm{TFD}(120\Delta)\rangle^{\otimes 18} \tag{83.59}$$

is a normalized purification of $\rho_\partial(\Delta)$*.*

Proof. At each link, the sum of the squares of the three coefficients is one. The partial trace eliminates terms with different indices and returns $\mathrm{diag}(1, e^{-w}, e^{-2w})/(1 + e^{-w} + e^{-2w})$. Partial trace commutes with the tensor product. Multiplying the thirty-six reductions gives the exponential and partition function of (83.57). □

83.4.3 Informational equation and areal fluctuations

The occupation means and variances per link are

$$f_m = \frac{x_m + 2x_m^2}{z_m}, \qquad v_m = \frac{x_m + 4x_m^2}{z_m} - f_m^2. \tag{83.60}$$

Write $s(\rho) = -\operatorname{Tr}(\rho \log \rho)$, with $0 \log 0 = 0$. In the preceding purification, $s(\rho_\partial)$ is its entanglement entropy.

Proposition 83.12 (Mean area, entropy, and sectorial response)**.** *For real finite Δ, one has*

$$\langle \mathsf{N} \rangle_\Delta = 18(f_{90} + f_{120}), \tag{83.61}$$

$$\langle \mathsf{D}_\partial \rangle_\Delta = 18(90 f_{90} + 120 f_{120}), \tag{83.62}$$

$$\langle \widehat{\mathcal{A}} \rangle_\Delta = 18 \ell_P^2 \gamma_{\mathrm{BI}} a_0 (f_{90} + f_{120}), \tag{83.63}$$

$$s(\Delta) = \log Z(\Delta) + \Delta \langle \mathsf{D}_\partial \rangle_\Delta, \tag{83.64}$$

$$\frac{\mathrm{d}\langle \mathsf{D}_\partial \rangle_\Delta}{\mathrm{d}\Delta} = -\operatorname{Var}_\Delta(\mathsf{D}_\partial) = -18(90^2 v_{90} + 120^2 v_{120}), \tag{83.65}$$

$$\frac{\mathrm{d}s}{\mathrm{d}\Delta} = -\Delta \operatorname{Var}_\Delta(\mathsf{D}_\partial), \tag{83.66}$$

$$\frac{\mathrm{d}\langle \widehat{\mathcal{A}} \rangle_\Delta}{\mathrm{d}\Delta} = -18 \ell_P^2 \gamma_{\mathrm{BI}} a_0 (90 v_{90} + 120 v_{120}). \tag{83.67}$$

Proof. The local weights are $(1, x_m, x_m^2)/z_m$, whose first two moments give f_m and v_m. Additivity of occupation operators and factorization of the state give the first three formulas. Taking the trace of $-\rho_\partial \log \rho_\partial = \rho_\partial(\log Z + \Delta \mathsf{D}_\partial)$ gives the entropy. Differentiation of the finite sums yields $\partial_\Delta \log Z = -\langle \mathsf{D}_\partial \rangle_\Delta$ and $\partial_\Delta^2 \log Z = \operatorname{Var}_\Delta(\mathsf{D}_\partial)$. Independent factors add their variances. In the derivative of (83.64), the two first-moment terms cancel. Finally, $\partial_\Delta f_m = -m v_m$ gives the area response. □

Loss of uniformity is quantified in dimension $d_\partial = 3^{36}$:

$$I_\partial(\Delta) = 36 \log 3 - s(\Delta) = \mathcal{D}\big(\rho_\partial(\Delta) \| I/d_\partial\big) \geq 0. \tag{83.68}$$

Here $\mathcal{D}$ denotes relative entropy, defined and justified below. Substituting $\log(I/d_\partial) = -36 \log 3\, I$ proves the identity. Convexity of $t \log t$ gives positivity; equality corresponds to the uniform distribution, which in this family occurs exactly at $\Delta = 0$. The deficit measures dimensionless information; the scale $\gamma_{\mathrm{BI}} a_0 \ell_P^2$ measures area per occupation.

Theorem 83.13 (Occupation inversion and complementary area)**.** *Let $\mathcal{R}_\partial$ be the involution exchanging $|0\rangle$ and $|2\rangle$ and fixing $|1\rangle$ in each factor. Then*

$$\mathcal{R}_\partial \mathsf{N} \mathcal{R}_\partial^* = 72 I - \mathsf{N}, \tag{83.69}$$

$$\mathcal{R}_\partial \mathsf{D}_\partial \mathcal{R}_\partial^* = 7560 I - \mathsf{D}_\partial, \tag{83.70}$$

$$\rho_\partial(-\Delta) = \mathcal{R}_\partial \rho_\partial(\Delta) \mathcal{R}_\partial^*, \tag{83.71}$$

$$s(-\Delta) = s(\Delta), \tag{83.72}$$

$$\langle \widehat{\mathcal{A}} \rangle_{-\Delta} + \langle \widehat{\mathcal{A}} \rangle_\Delta = 72 \gamma_{\mathrm{BI}} a_0 \ell_P^2, \tag{83.73}$$

$$\mathcal{D}\big(\rho_\partial(\Delta) \| \rho_\partial(-\Delta)\big) = \Delta\big(7560 - 2\langle \mathsf{D}_\partial \rangle_\Delta\big). \tag{83.74}$$

Proof. Each occupation transforms by $n \mapsto 2 - n$. In each sector, $\mathsf{N}_m \mapsto 36 I - \mathsf{N}_m$, so the constant term of the weighted reader is $36(90 + 120) = 7560$. Moreover, $z_m(-\Delta) = e^{2m\Delta} z_m(\Delta)$ and $Z(-\Delta) = e^{7560\Delta} Z(\Delta)$. Substitution into the normalized state proves conjugation. Its spectrum preserves entropy;

the transformation of N gives complementary area. Subtracting the logarithms of the two states and taking the expectation in $\rho_\partial(\Delta)$ gives the final equality. □

At $\Delta = 0$, $s = 36 \log 3$. As $\Delta \to +\infty$, zero occupation concentrates the state and entropy tends to zero; inversion carries that limit to maximal occupation. For $\Delta > 0$, entropy decreases strictly because the weighted reader has more than one eigenvalue and the state has full rank. The complete configuration, area, and entropy thus preserve differentiated roles.

83.4.4 Modular law and finite area variations

Fix $\rho_0 = \rho_\partial(\Delta_0)$, with real finite Δ_0, and hold $\Delta_0, \gamma_{\mathrm{BI}}, a_0, \ell_P$ constant. The normalized areas $\mathcal{A}^{\mathrm{n}}_m = \gamma_{\mathrm{BI}} a_0 \mathsf{N}_m$ express the modular operator as

$$\mathsf{K}_0 = -\log \rho_0 = c_0 I + \sum_{m=90,120} \beta_m \mathcal{A}^{\mathrm{n}}_m, \quad \beta_m = \frac{m\Delta_0}{\gamma_{\mathrm{BI}} a_0}, \quad c_0 = \log Z(\Delta_0). \tag{83.75}$$

One has $\beta_{120} = \frac{4}{3}\beta_{90}$, including when both are zero. The ratio is $4/3$ when $\Delta_0 \neq 0$. K_0 is the logarithm of this reference state; its type distinguishes it from temporal transport and the gravitational connection.

Theorem 83.14 (First modular law and finite correction)**.** *For any normalized state σ on the same space, let $\Delta_\sigma\langle B\rangle = \mathrm{Tr}[(\sigma - \rho_0)B]$. Then*

$$s(\sigma) - s(\rho_0) = \sum_{m=90,120} \beta_m \Delta_\sigma \langle \mathcal{A}^{\mathrm{n}}_m \rangle - \mathcal{D}(\sigma \| \rho_0), \tag{83.76}$$

where $\mathcal{D}(\sigma\|\rho_0) = \mathrm{Tr}[\sigma(\log\sigma - \log\rho_0)]$ is finite and nonnegative. If $\sigma(\varepsilon)$ is differentiable, normalized, and $\sigma(0) = \rho_0$, one obtains

$$\left.\frac{\mathrm{d}s(\sigma(\varepsilon))}{\mathrm{d}\varepsilon}\right|_0 = \sum_{m=90,120} \beta_m \left.\frac{\mathrm{d}\langle \mathcal{A}^{\mathrm{n}}_m \rangle_{\sigma(\varepsilon)}}{\mathrm{d}\varepsilon}\right|_0 . \tag{83.77}$$

Proof. By definition, $\mathcal{D}(\sigma\|\rho_0) = -s(\sigma) + \mathrm{Tr}(\sigma\mathsf{K}_0)$, whereas $s(\rho_0) = \mathrm{Tr}(\rho_0\mathsf{K}_0)$. Subtraction and (83.75) prove the finite identity, since normalization cancels the term $c_0 I$.

For positivity, diagonalize σ with eigenvalues p_i and vectors u_i, and ρ_0 with eigenvalues $q_j > 0$ and vectors v_j. If $r_i = \langle u_i, \rho_0 u_i\rangle > 0$, concavity of the logarithm gives

$$\sum_j |\langle u_i, v_j\rangle|^2 \log q_j \le \log r_i .$$

Consequently, $\mathcal{D}(\sigma\|\rho_0) \ge \sum_i p_i \log(p_i/r_i) \ge 0$. The final inequality is convexity of $t\log t$ with weights r_i, whose sum is one. Faithfulness of ρ_0 keeps its logarithms finite; zero eigenvalues of σ are interpreted through $0\log 0 = 0$.

For the infinitesimal formula, in a neighbourhood of the faithful state ρ_0, the derivative of the trace of $-\sigma\log\sigma$ is $-\mathrm{Tr}[\dot\sigma(\log\rho_0 + I)]$. This identity follows from the spectral derivative of the trace. The condition $\mathrm{Tr}\,\dot\sigma = 0$ leaves $\mathrm{Tr}(\dot\sigma\mathsf{K}_0)$, and substitution of (83.75) completes the proof. □

For physical areas, the coefficients are β_m/ℓ_P^2. Applying the common transducer k_B gives $S = k_B s$ and preserves the term $k_B\mathcal{D}$ in full. The derivative in (83.67) varies Δ; law (83.77) holds Δ_0 fixed and varies the state around the reference. These are different operations.

83.4.5 Action, decision energy, and areal unit

We fix the same oriented action section and its gravitational and thermal sections, on the positive branch. The antecedents provide

$$\begin{aligned} \ell_P^2 &= \frac{\hbar G}{c^3}, & t_P &= \frac{\ell_P}{c}, \\ T_{108} &= 2\pi t_P, & h &= 2\pi\hbar, \\ E_P &= \frac{\hbar}{t_P} = k_B\Theta_{\mathrm{clk}}\log 3. \end{aligned} \tag{83.78}$$

The thermal equality preserves the previously constructed pair and normalization; changing its dimensional basis transports k_B and Θ_{clk} jointly. The elementary area increment is $\delta\mathcal{A} = \gamma_{\mathrm{BI}} a_0 \ell_P^2$.

Proposition 83.15 (Conversion coefficient between decision and area)**.** *The ratio of decision energy to the areal increment satisfies*

$$\begin{aligned} \mathcal{T}_{I/A} := \frac{E_P}{\delta\mathcal{A}} &= \frac{k_B\Theta_{\mathrm{clk}}\log 3}{\gamma_{\mathrm{BI}} a_0 \ell_P^2} \\ &= \frac{c^3}{\gamma_{\mathrm{BI}} a_0 G t_P} = \frac{c^4}{\gamma_{\mathrm{BI}} a_0 G \ell_P}. \end{aligned} \tag{83.79}$$

Its dimension is energy per area.

Proof. Substitution of $E_P = \hbar/t_P$ and $\ell_P^2 = \hbar G/c^3$ cancels the common action section. The identity $\ell_P = c t_P$ gives the final expression. Positivity of the factors ensures that all quotients are defined. □

The capacity 81 of the cubic cut and the bipartition of the thirty- six sectoral links correspond to different domains. In the cubic graph $\{0,\dots,N-1\}^3$, the N^2 parallel lines between two opposite faces are edge-disjoint: every cut intercepts the N^2, and a transverse layer attains that bound. For $N = 9$, the uniform product state of the trits of that layer has $s_{\mathrm{corte}} = 81\log 3$. Assigning a decision of the same cycle to each link publishes

$$S^{\mathrm{fis}}_{\mathrm{corte}} = 81 k_B \log 3, \qquad E^{(1)}_{\mathrm{corte}} = 81 E_P = \Theta_{\mathrm{clk}} S^{\mathrm{fis}}_{\mathrm{corte}}. \tag{83.80}$$

The proof combines the minimum cut, additivity of product-state entropy, and (83.78). In the sectorial state, the relevant finite equation remains (83.76), with its relative entropy.

83.4.6 Cosmological normalization of the horizon

For an areal radius $R > 0$, the cosmological realization considered uses the triad

$$E_\Lambda(R) = \frac{c^4 R}{2G}, \qquad S_{\mathrm{H}}(R) = \frac{k_B \pi R^2}{\ell_P^2}, \qquad T_{\cos}(R) = \frac{\hbar c}{2\pi k_B R}. \tag{83.81}$$

The factor 2π belongs to this temperature normalization. The area $\mathcal{A}_{\mathrm{H}} = 4\pi R^2$ expresses the same entropy as $S_{\mathrm{H}}/k_B = \mathcal{A}_{\mathrm{H}}/(4\ell_P^2)$.

Theorem 83.16 (Gravitational half-action cell)**.** *In* (83.81)*,*

$$T_{\cos}(R) S_{\mathrm{H}}(R) = E_\Lambda(R), \tag{83.82}$$

$$T_{\cos}(\ell_P) S_{\mathrm{H}}(\ell_P) = \frac{E_P}{2} = \frac{k_B\Theta_{\mathrm{clk}}\log 3}{2}, \tag{83.83}$$

$$E_\Lambda(\ell_P) t_P = \frac{\hbar}{2}, \qquad E_\Lambda(\ell_P) T_{108} = \frac{h}{2}. \tag{83.84}$$

Keeping the chart constants fixed and varying R,

$$T_{\cos}\,\mathrm{d}S_{\mathrm{H}} = 2\,\mathrm{d}E_{\Lambda}, \qquad S_{\mathrm{H}}\,\mathrm{d}T_{\cos} = -\mathrm{d}E_{\Lambda}. \tag{83.85}$$

Proof. Multiplying temperature and entropy gives $\hbar cR/(2\ell_P^2) = c^4R/(2G)$. At $R = \ell_P$, the result is $\hbar/(2t_P) = E_P/2$. The two durations in (83.78) give the action relations. For the differentials, $\mathrm{d}E_{\Lambda} = c^4\mathrm{d}R/(2G)$, $\mathrm{d}S_{\mathrm{H}} = 2\pi k_B R\mathrm{d}R/\ell_P^2$, and $\mathrm{d}T_{\cos} = -T_{\cos}\mathrm{d}R/R$. Substitution proves both identities. □

The product $E_{\Lambda} = T_{\cos}S_{\mathrm{H}}$ expresses a state identity. Its radial variations preserve both terms of (83.85); a complete thermodynamic interpretation additionally specifies the corresponding time generator, work and material realization. In this chart, action and gravitation fix the unit ℓ_P^2, while Boltzmann converts information into dimensional entropy. Transport of sections with $G\hbar$ fixed preserves that unit when the section c is kept fixed, and preserves S_{H}/k_B at fixed radius.

In the cell $R = \ell_P$, $S_{\mathrm{H}}/k_B = \pi$, so the effective multiplicity is $\Omega_{\mathrm{eff}} = e^{\pi}$. For the tritic hyperbolic involution H_{trit}, with spectrum $\{1, -1\}$, diagonalization gives $\mathrm{spec}(e^{\pi H_{\mathrm{trit}}}) = \{e^{\pi}, e^{-\pi}\}$. This spectral relation specifies the expansive value; a microscopic identification of both spaces additionally preserves their realization map and the state.

83.4.7 Kruskal chart and exchange of the exteriors

Let us fix a published state x, its radius $R = R(x) > 0$ and the positive sections c, G. In the following chart, x, R, c, G are fixed parameters; t, r are the spacetime coordinates. The exterior realization of the two sheets is specified by

$$f(r) = 1 - \frac{R}{r}, \qquad \mathrm{d}s^2 = -f(r)c^2\mathrm{d}t^2 + f(r)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega_2^2, \qquad r > R. \tag{83.86}$$

For the sheet reader $\sigma \in \{+1, -1\}$, define

$$r_* = r + R\log\left(\frac{r}{R} - 1\right), \tag{83.87}$$

$$\mathcal{U} = -\sigma\exp\left(-\frac{ct - r_*}{2R}\right), \qquad \mathcal{V} = \sigma\exp\left(\frac{ct + r_*}{2R}\right). \tag{83.88}$$

The symbols $\mathcal{U}, \mathcal{V}$ are coordinates and are distinguished from route-transport operators.

Theorem 83.17 (Geometric realization and sheet involution)**.** *Chart* (83.88) *realizes the two sheets as the two exteriors $\mathcal{U}\mathcal{V} < 0$ of the Kruskal extension. We have*

$$-\mathcal{U}\mathcal{V} = \left(\frac{r}{R} - 1\right)e^{r/R}, \qquad \mathrm{d}s^2 = -\frac{4R^3}{r}e^{-r/R}\,\mathrm{d}\mathcal{U}\,\mathrm{d}\mathcal{V} + r^2\mathrm{d}\Omega_2^2. \tag{83.89}$$

The involution

$$C_{\mathrm{H}} : (\sigma, \mathcal{U}, \mathcal{V}) \longmapsto (-\sigma, -\mathcal{U}, -\mathcal{V}) \tag{83.90}$$

is isometric, fixes the areal radius, exchanges the exterior regions, and preserves the horizon set $\mathcal{U}\mathcal{V} = 0$. The metric is regular at $r = R$ and satisfies the vacuum equations on the regular domain $r > 0$.

Proof. Multiplying the coordinates gives the first identity. Since $\mathrm{d}r_*/\mathrm{d}r = f^{-1}$,

$$\mathrm{d}\mathcal{U} = \frac{\mathcal{U}}{2R}(-c\,\mathrm{d}t + f^{-1}\mathrm{d}r), \qquad \mathrm{d}\mathcal{V} = \frac{\mathcal{V}}{2R}(c\,\mathrm{d}t + f^{-1}\mathrm{d}r).$$

Their product reproduces the radial and temporal part of the metric. The function $z \mapsto (z-1)e^z$ has derivative $ze^z > 0$ for $z > 0$, determining r from the product. In the exterior region one recovers $\sigma = \operatorname{sign}\mathcal{V}$ and $t = (R/c)\log(-\mathcal{V}/\mathcal{U})$. At $r = R$, the radial coefficient of the extended metric is $-4R^2/e$,

finite and nonzero. Inversion preserves the product, radius, and $\mathrm{d}\mathcal{U}\mathrm{d}\mathcal{V}$, and exchanges the two sign choices.

To check the vacuum in the exterior chart, the independent mixed components of the Einstein tensor reduce to

$$\mathsf{G}^{t}{}_{t} = \mathsf{G}^{r}{}_{r} = \frac{rf' + f - 1}{r^2}, \qquad \mathsf{G}^{\theta}{}_{\theta} = \mathsf{G}^{\phi}{}_{\phi} = \frac{f'}{r} + \frac{f''}{2}.$$

With $f' = R/r^2$ and $f'' = -2R/r^3$, all vanish. The regular expression (83.89) prolongs the result through the horizons on the domain $r > 0$. □

This realization preserves $E_{\mathrm{H}} = c^4 R/(2G)$ and $S_{\mathrm{H}} = k_B \pi R^2/\ell_P^2$, which depend on the areal radius. The temperature $T_{\cos}$ of (83.81) belongs to the cosmological realization declared there. The exterior chart (83.86) preserves its own temporal normalization. The geometric map establishes the exchange of exteriors; comparison with formation by collapse, radiation spectra or quasinormal modes additionally uses their respective physical conditions.

83.4.8 Transport of states, observables, and refinements

Let F_x denote the chart of the two exteriors just constructed. In each regular angular chart, its temporal and radial Jacobian is

$$\left|\frac{\partial(\mathcal{U},\mathcal{V})}{\partial(t,r)}\right| = \frac{cr}{2R^3} e^{r/R} > 0. \tag{83.91}$$

The identity follows from the two differentials of the preceding proof. The remaining state records are transported as fiber coordinates; the chart preserves its identity and memory.

Proposition 83.18 (Unitary equivalence of sheet and exterior readings)**.** *Let μ be a sheet measure and $\nu = (F_x)_*\mu$ its transported measure. Change of variables defines a unitary map*

$$W_{\mathrm{H}} : L^2(\mu) \longrightarrow L^2(\nu), \qquad (W_{\mathrm{H}}\psi)(z) = \psi(F_x^{-1} z). \tag{83.92}$$

With densities relative to fixed coordinate measures, the same map incorporates the corresponding Jacobian factor. Conjugation

$$\mathfrak{I}_{\mathrm{H}}(B) = W_{\mathrm{H}} B W_{\mathrm{H}}^{*} \tag{83.93}$$

is a unital completely positive $*$-isomorphism. It transports the state, its expectations, and its modular functional calculus. For the symmetric measure of the two sheets, it also intertwines their involutions.

Proof. The definition of pushforward measure gives $\int |\psi \circ F_x^{-1}|^2 \,\mathrm{d}\nu = \int |\psi|^2 \,\mathrm{d}\mu$. The chart is invertible in the exteriors, so the isometry is surjective. Unitary conjugation preserves product, adjoint and identity; in each matrix amplification it conjugates by $I_n \otimes W_{\mathrm{H}}$, proving complete positivity. For a state ρ, cyclicity of the trace gives $\mathrm{Tr}(W_{\mathrm{H}}\rho W_{\mathrm{H}}^{*}\, \mathfrak{I}_{\mathrm{H}}(B)) = \mathrm{Tr}(\rho B)$. Functional calculus implies $g(W_{\mathrm{H}}\rho W_{\mathrm{H}}^{*}) = W_{\mathrm{H}} g(\rho) W_{\mathrm{H}}^{*}$ on its domain, in particular for the logarithm and modular flow of a state faithful on its support. The equality $F_x \circ C_{\mathrm{hoja}} = C_{\mathrm{H}} \circ F_x$ proves the intertwining of involutions; the symmetric measure realizes them unitarily. □

If refinements preserve (σ, t, r, R) and use the same isometry on fibre records, the inclusions j_m^{hoja} and j_m^{ext} satisfy

$$W_{\mathrm{H},m+1}\, j_m^{\mathrm{hoja}} = j_m^{\mathrm{ext}}\, W_{\mathrm{H},m}. \tag{83.94}$$

Indeed, change of variables acts on the preserved geometric coordinates and inclusion on the same fibre coordinates; evaluating both compositions gives the same function and the same transported density. The equality extends the map to the refined system without resetting the records.

For a bipartition $\mathcal{H}_A \otimes \mathcal{H}_B$, a factorized transport $U_A \otimes U_B$ satisfies

$$\rho'_A = U_A \rho_A U_A^*, \qquad s(\rho'_A) = s(\rho_A).$$

The first identity follows from partial trace and the second from the spectrum. For a general unitary, the equivalent formulation also transports the subsystem algebra: $\mathcal{B}(\mathcal{H}_A) \otimes I \mapsto U(\mathcal{B}(\mathcal{H}_A) \otimes I)U^*$. This preserves the same operational question concerning accessible information. Entropy equalities apply to states with the relevant finite entropies.

The relation between information and gravitation is expressed through distinct composable operations: the state and bipartition fix accessible information; incidence and Barbero–Immirzi fix areal resolution; action and gravitation construct the unit ℓ_P^2; Boltzmann realizes entropy dimensionally; the horizon chart transports geometry and its observables. The finite laws preserve relative entropy, and the geometric realizations preserve their measures, temporal normalization and domains.

Chapter 84

Mach's principle and self-inertial systems: relational dependence among global closure, local inertia, and torsional memory

Local self-inertia is obtained as the dependence of the state on the global closure and its two propagation orientations. Torsional memory and action invariants determine the relational scope of Mach's principle.

84.1 Mach's principle, bidirectional propagation and collective gravitational response

84.1.1 Global dependence of the local response

Let $\mathcal{X}_\infty$ be the space of compatible sections, let $\mathrm{res}_v : \mathcal{X}_\infty \to X_v$ be the restriction to a local neighborhood, and let $b : \mathcal{X}_\infty \to B_\infty$ be the global boundary trace. A local inertial response $I_v : \mathcal{X}_\infty \to V_v$ is *Machian* when there exists $\overline{I}_v : \mathrm{im}(b) \to V_v$ such that

$$I_v = \overline{I}_v \circ b, \tag{84.1}$$

but I_v does not factor solely through res_v. In particular, there may exist x, x' with

$$\mathrm{res}_v(x) = \mathrm{res}_v(x'), \qquad b(x) \neq b(x'), \qquad I_v(x) \neq I_v(x').$$

The definition formalizes Mach's principle: the local response depends on the global closure class and not on an added absolute space [85]. This is the same local–global scheme that appears in the number cylinders, in holonomy, and in memory transport.

Theorem 84.1 (Criterion of factorization of the response by the totality)**.** *There exists to unique map $\overline{I}_v : \mathrm{im}(b) \to V_v$ that satisfies $I_v = \overline{I}_v \circ b$ if and only if I_v is constant on each fiber of b:*

$$b(x) = b(y) \quad \Longrightarrow \quad I_v(x) = I_v(y).$$

If b is surjective, the induced response is uniquely defined on all of B_∞.

Proof. Necessity follows by applying $\overline{I}_v$ to the equality $b(x) = b(y)$. For sufficiency, given $\beta \in \mathrm{im}(b)$, we define $\overline{I}_v(\beta) = I_v(x)$ for any $x \in b^{-1}(\beta)$. Constancy on the fibers makes the choice of representative immaterial; the factorization then fixes the value of $\overline{I}_v$ at every point of its domain and guarantees uniqueness. □

The relationship between state and coordinate is fixed by

$$[x]_{\mathfrak{F}} \neq x, \qquad q(x) \neq x, \qquad x \in \mathcal{X}_\infty.$$

The state preserves position, sheet, orientation, additive or multiplicative mode, quotient, carry, boundary, route, and memory; the observable publishes only one coordinate. The corresponding local frame is not added from outside, but is induced by restriction of the compatible totality and its connection:

$$\mathfrak{F}_v(x) := \mathrm{Res}_v\big(x, \nabla_{\mathrm{TPK}}\big). \tag{84.2}$$

Here $\mathcal{X}_\infty$ does not denote a verbal sum of the entire universe, but rather the system of compatible sections with orientation, boundary, and memory. Self-reference is coinductive, not circular: each section is recognized through the compatibility of all its truncations.

This formulation organizes the inertial and accelerated regimes by means of to single endogenous connection. For to curve γ, the covariant acceleration is

$$a_\gamma := \nabla_{\dot\gamma}\dot\gamma, \qquad a_\gamma = 0 \quad \Longleftrightarrow \quad \gamma \text{ is an affine geodesic of } \nabla.$$

If (e_a) is a tetrad along γ, the condition

$$\nabla_{\dot\gamma} e_a = 0$$

defines a parallel, nonrotating frame with respect to the connection. The rotation of a general frame is measured by its connection coefficients along γ, or by its deviation from Fermi–Walker transport when the latter is the selected kinematic standard. For an arbitrary field v, $D_\gamma v = 0$ means only parallel transport; its negation is not by itself equivalent to acceleration or rotation.

The physical difference is thus measured through covariant acceleration, frame rotation, torsion, curvature, and holonomy, without introducing an external privileged frame. In this sense, every persistent section is *self-inertial*: it carries the frame induced by its own insertion into the HMT totality.

Proposition 84.2 (Autonomy, return and holonomy)**.** *Let $E \to M$ the realization bundle, is γ to loop based at x_0 and let $\mathrm{Hol}_\nabla(\gamma) : E_{x_0} \to E_{x_0}$ the holonomy of the induced connection. TO datum $s_0 \in E_{x_0}$ transported in parallel returns exactly if and only if*

$$s_0 \in \mathrm{Fix}\big(\mathrm{Hol}_\nabla(\gamma)\big).$$

If the complete genealogical state is realized in E_{x_0} through an injective map on the compared sector, this condition is also equivalent to the return of the complete state. Without that injectivity, the conclusion refers exclusively to the datum in the fiber. If the visible coordinate returns and s_0 remains outside the fixed subspace, there is phase return with change of the transported state.

Proof. Transport around the loop acts on E_{x_0} by means of $\mathrm{Hol}_\nabla(\gamma)$. Equality between the final and initial data is equivalent, by definition, to s_0 being a fixed vector of the operator. The last equivalence is obtained by applying the injectivity of the realization to the declared sector. □

84.1.2 Advanced and Retarded Propagators on Oriented Paths

For a route γ between emission and absorption, the inversion $\gamma^\vee$ exchanges the endpoints and preserves the length of the trajectory. In the photonic realization of section 92.4, the pair $(\gamma, \gamma^\vee)$ represents the retarded and advanced orientations of one and the same null transport. A Möbius band preserves this memory of orientation: one turn exchanges the sheets and two turns restore the oriented section.

This structure provides a rigorous formulation of the intuition of a bidirectional pilot wave. The material state is not identified with an additional wave that pushes it from outside; the section and its phase are two readings of the same enriched transport. A concrete identification with advanced and retarded Green propagators nevertheless requires an intertwiner that preserves action, causality, and the inner product.

Proposition 84.3 (Conjugation of Orientation and Propagator Reversion)**.** *Let $H = H^*$ be a self-adjoint, possibly unbounded operator satisfying*

$$HJ_E = J_E H.$$

Let C be an orthogonal involution of the quantum realification that preserves $D(H)$ and satisfies, on that domain,

$$CJ_E C^{-1} = -J_E, \qquad CHC^{-1} = H.$$

Then

$$U(t) := \exp(-J_E H t/\hbar_{\mathrm{int}})$$

is a uniparametric unitary group and it holds

$$\boxed{CU(t)C^{-1} = U(-t)}.$$

Proof. The operator $A = -J_E H/\hbar_{\mathrm{int}}$ is skew-adjoint and generates $U(t)$ by Stone's theorem. The domain and conjugation hypotheses give $CAC^{-1} = -A$; functional calculus, or equivalently the uniqueness of the generated group, implies $C\exp(tA)C^{-1} = \exp(-tA)$. If H is bounded, the same identity is obtained term by term from the exponential series. □

The word involution $C_M = -\,\mathrm{rev}$ supplies the genealogical orientation datum. When its realization satisfies the hypotheses of the proposition, the particle and antiparticle branches are represented by propagators with conjugate temporal directions: the band geometry fixes the orientation, and the Hamiltonian fixes the dynamics.

The kinematic reading of compression and expansion is fixed, for $|\beta| < 1$, by the rapidity $\zeta = \operatorname{artanh}\beta$. For a proper frequency ω_0, the two Doppler orientations are

$$\omega_{\mathrm{comp}} = \omega_0 e^{\zeta} = \omega_0\sqrt{\frac{1+\beta}{1-\beta}}, \qquad \omega_{\mathrm{exp}} = \omega_0 e^{-\zeta} = \omega_0\sqrt{\frac{1-\beta}{1+\beta}}, \tag{84.3}$$

and satisfy

$$\omega_{\mathrm{comp}}\omega_{\mathrm{exp}} = \omega_0^2.$$

The reciprocity of (84.3) is a continuous realization of the bilayer pattern: one orientation contracts and its conjugate expands, while their product preserves the central scale. This does not identify the Doppler parameter with C^*, with R_{act}, or with a mass coefficient; the maps among those coordinates must be constructed separately.

84.1.3 Temporal Involution Invariants and Particle–Antiparticle Conjugation

The Kepler orbit, its Levi–Civita lift, and the Sommerfeld condition organize the phase over an elliptic cycle. In this branch

$$J_\gamma = \oint_\gamma p\,\mathrm{d}q$$

is the orbital action. This symbol is not conflated with the modular character $J_{\mathrm{mod}} = j - 744$, which belongs to Moonshine, or with the Jarlskog invariant J_{CKM}, which measures CP violation. The three objects share an abstract property —they are invariants obtained after closing a composition—, but they have distinct domains, units, and theorems.

The condition

$$\varepsilon_\gamma \exp\!\left(\frac{iJ_\gamma}{\hbar_{\mathrm{int}}}\right) = 1$$

explains why the sign holonomy shifts the Sommerfeld rule by one half. Together with $\mathrm{Area}(\theta) = \hbar^{\mathrm{ret}}_{\mathrm{HMT}}\theta$, it renders the closure 2π and the double circuit 4π visible: the former completes the scalar cycle; the latter restores a spin or Möbius section that has changed sheets after a single circuit.

84.1.4 Propagating curvature as a collective mode of geometry

The primary gravitational variable is transport: co-tetrad, connection, curvature, torsion, and holonomy. A gravitational wave is a propagating variation of curvature or of its holonomy class. The state type does not require an additional elementary spin-two coordinate. A quantum of helicities ±2, if it appears after reducing the constraints and quantizing the fluctuations, is a collective excitation of the geometric sector and not a primitive constituent.

The Machian reading reinforces this organization: the local perturbation measures a change in the global section. The component that does not remain in the observable curvature may be represented as local transport memory or contorsion only when the diagram between TPK and Cartan commutes; discrete memory is not identified with torsion by a nominal equality. Under that condition, the wave can

propagate without positing a fundamental graviton, and the local response preserves the information of the global closure.

84.1.5 Metric Hypotheses and Boundary Conditions of the Machian Realization

The temporal semantics of the TRIT assigns the memory of the past to return, the present update to the threshold, and future prolongation to the opening. The elliptic, Euclidean, and hyperbolic realizations translate those three regimes into closure, boundary, and divergence. This provides a geometric reading of matter as wound persistence and of propagation as hyperbolic opening.

Interpreting a closed particle as a collapsed two-dimensional section, an electron as the closure of two null modes, or a black hole as a global inversion of sheets yields theoretical consequences compatible with that architecture. They are not purely combinatorial identities: to attain physical status, they must realize a metric, solve the field equations, and reproduce the corresponding horizon, causality, and stability conditions. This separation preserves the explanatory power of the model without turning a geometric interpretation into an already obtained proof.

Hypothesis, domain, and conclusion

The Machian factorization, Möbius monodromy, Doppler reciprocity, and the phase condition are explicit mathematical constructions in their declared domains. Their joint identification with physical pilot waves, black holes, or cosmology constitutes a theoretical realization that must satisfy the indicated intertwiners and equations.

Prolongation of bidirectionality to the cosmological return of memory. Bidirectional propagation extends to torsional cosmology and the cyclic return of memory.

84.2 Self-inertia as connection endogenous of the joint state

Let ∇^{tot} be a connection on the space of joint states. The local frame of the observer is obtained by restriction:

$$\mathfrak{F}_O(\Gamma) = \mathrm{Res}_O(\Gamma, \nabla^{\mathrm{tot}}). \tag{84.4}$$

A complete trajectory $\Gamma(t)$ is self-inertial when

$$\nabla^{\mathrm{tot}}_{\dot\Gamma}\dot\Gamma = 0. \tag{84.5}$$

Let π_S be the projection to the observed subsystem and $\gamma_S = \pi_S \circ \Gamma$. We define the projection defect

$$\mathcal{K}_{\pi_S}(\dot\Gamma, \dot\Gamma) := \nabla^S_{\dot\gamma_S}\dot\gamma_S - \mathrm{d}\pi_S\left(\nabla^{\mathrm{tot}}_{\dot\Gamma}\dot\Gamma\right). \tag{84.6}$$

On an autoinertial trajectory lies

$$\boxed{a_S = \nabla^S_{\dot\gamma_S}\dot\gamma_S = \mathcal{K}_{\pi_S}(\dot\Gamma, \dot\Gamma).} \tag{84.7}$$

The acceleration of the subsystem measures the coupling and the degrees of freedom omitted by the projection. The whole possesses an endogenous connection; its shadows may exhibit acceleration, rotation, and effective force.

Definition 84.4 (Self-inertial state)**.** TO joint state is self-inertial with respect to a connection and to family of observers when its evolution satisfies equation (84.5) and each local frame is obtained by equation (84.4). The apparent forces of each subsystem are recorded by the defects $\mathcal{K}_{\pi_S}$.

The definition integrates observer and observable within a single dynamic. A change of frame corresponds to another restriction of the same state and the same connection. The distinction among affine transport, tetrad rotation, torsion, and curvature remains available in the geometric codomain.

84.3 Machian Criterion of Global Factorization

Let $b : \mathcal{X}_\infty \to B_\infty$ be the global boundary trace and let $I_v : \mathcal{X}_\infty \to V_v$ be a local inertial response. The response is Machian when there exists

$$\overline{I}_v : \operatorname{im} b \longrightarrow V_v, \qquad I_v = \overline{I}_v \circ b. \tag{84.8}$$

Theorem 84.5 (Machian Response Criterion)**.** *There is a unique map $\overline{I}_v$ satisfying equation (84.8) if and only if I_v is constant on each fiber of b:*

$$b(x) = b(x') \quad \Longrightarrow \quad I_v(x) = I_v(x'). \tag{84.9}$$

If b is surjective, $\overline{I}_v$ is defined on all B_∞.

Proof. The factorization implies constancy on the fibers. Conversely, for $\beta \in \operatorname{im} b$, we define $\overline{I}_v(\beta) = I_v(x)$ with $x \in b^{-1}(\beta)$. Constancy guarantees independence of the representative, and the factorization equation guarantees uniqueness. □

The criterion translates Mach's principle into a verifiable property: the local response is a reading of the global closure class. To prove genuinely global dependence, one additionally exhibits two states with the same local restriction and different boundary traces whose responses differ. This condition separates an effective global factorization from a response determined exclusively by the local neighborhood.

The gravitational scale Q of the CT108 clock provides a dimensionless realization of this reading:

$$\alpha_G(E_1, E_2) = \frac{E_1 E_2}{Q^2}. \tag{84.10}$$

The global clock fixes Q, each local section publishes E/Q, and the geometry responds through the compliance G/c^4. The equation provides a Machian chart of the gravitational response; the theorem 84.5 determines the factorization requirement that its realization on the complete state must satisfy.

84.4 Holonomy and hierarchy of returns

Let γ be a based loop in x_0, and let

$$\mathscr{H}_\gamma = \operatorname{Hol}_\nabla(\gamma) \tag{84.11}$$

the holonomy in the fiber E_{x_0}. A transported datum returns as a vector exactly when it belongs to

$$\operatorname{Fix}(\mathscr{H}_\gamma) = \{\psi : \mathscr{H}_\gamma \psi = \psi\}. \tag{84.12}$$

The self-inertial interpretation is condensed in this equality: the frame transported by the connection of the whole preserves the datum situated in the fixed subspace.

For a subalgebra of accessible observables $\mathcal{A}_O$, we define

$$\mathcal{R}_{\text{vec}} = \{\psi : \mathscr{H}_\gamma \psi = \psi\}, \tag{84.13}$$

$$\mathcal{R}_{\text{ray}} = \{\psi : \mathscr{H}_\gamma \psi = \mathrm{e}^{i\theta}\psi \text{ for some } \theta\}, \tag{84.14}$$

$$\mathcal{R}_{\text{obs}} = \{\psi : \langle \mathscr{H}_\gamma \psi, A \mathscr{H}_\gamma \psi \rangle = \langle \psi, A\psi \rangle \text{ for every } A \in \mathcal{A}_O\}. \tag{84.15}$$

Proposition 84.6 (Vector–ray–observable hierarchy)**.** *We have*

$$\boxed{\mathcal{R}_{\text{vec}} \subseteq \mathcal{R}_{\text{ray}} \subseteq \mathcal{R}_{\text{obs}}.} \tag{84.16}$$

Proof. A fixed vector is a fixed ray with zero phase. If $\mathscr{H}_\gamma \psi = \mathrm{e}^{i\theta}\psi$, the conjugate phases cancel in every quadratic expectation, so all accessible observables preserve their value. □

The monodromy nonadic naturally occupies this hierarchy:

$$g(k+9) = g(k), \qquad B_{k+9} \neq B_k \quad \text{in general.} \tag{84.17}$$

The phase return belongs to the observable reading; the change in B_k preserves the memory increment. An injective realization of the complete state in the fiber makes it possible to lift the criterion $\mathrm{Fix}(\mathscr{H}_\gamma)$ to total genealogical return. In the absence of that injectivity, the criterion asserts the return of the realized datum.

84.5 Möbius band and conjugation of orientations

On a dodecaphasic word $\mathbf{t} = (t_0, \dots, t_{11})$, the genealogical involution is

$$C_M(\mathbf{t}) = -\operatorname{rev}(\mathbf{t}), \qquad C_M^2 = I. \tag{84.18}$$

Let ℓ_{ret} be an enriched return loop whose lift has the sign character

$$\varepsilon_x : \langle \ell_{\mathrm{ret}} \rangle \simeq \mathbb{Z} \longrightarrow \{\pm 1\}, \qquad \varepsilon_x(\ell_{\mathrm{ret}}) = -1. \tag{84.19}$$

The associated interval bundle is

$$\mathcal{M}_x = ([0,1] \times [-1,1])/(0,u) \sim (1,-u). \tag{84.20}$$

Theorem 84.7 (Relative return of Möbius)**.** *Under the compatible lift of C_M, the loop and the character of equation (84.19), $\mathcal{M}_x$ has nonzero first Stiefel–Whitney class. One circuit exchanges the local orientations, and two circuits restore the orientation of the section.*

Proof. The transition function of the quotient equation (84.20) is the sign representation of $\pi_1(S^1)$. Its class w_1 is the nontrivial generator of $H^1(S^1; \mathbb{Z}/2\mathbb{Z})$. The first iteration multiplies the fiber coordinate by -1 and the second by $(-1)^2 = 1$. □

The formulation preserves three distinct returns: phase closure C_9, orientation return in the double cover, and the increment of memory in the lift $\mathbb{Z}$. One turn may restore the ray and change the vector; two turns restore the orientation and may preserve a different depth. The band expresses precisely the difference between publication and genealogy.

The even and odd sectors of a function f on the double cover are

$$f^+ = \frac{1}{2}(f + C_M^* f), \qquad f^- = \frac{1}{2}(f - C_M^* f). \tag{84.21}$$

The even part descends to the base; the odd part is a section of the sign line. Quadratic memory resides in the even sector, whereas orientation and linear charge reside in the odd sector.

84.6 Antiunitary conjugation and temporal orientation

The word involution C_M, the central inversion of the atlas, and the antiunitary conjugation of the realization in physical space have distinct domains. Let C be the antiunitary conjugation on a Hilbert realization, let $H = H^*$ be the Hamiltonian, and let J_E be the complex structure in the realification. Suppose

$$CJ_EC^{-1} = -J_E, \qquad CHC^{-1} = H, \qquad J_E H \subset H J_E, \tag{84.22}$$

with $J_E^* = -J_E$, $J_E^2 = -I$, and preservation of the domain of H. Commutation expresses the complex linearity of the dynamics with respect to the structure J_E.

Proposition 84.8 (Antiunitary reversal of evolution)**.** *For*

$$U(t) = \exp(-J_E H t/\hbar_{\mathrm{int}}), \tag{84.23}$$

we have

$$\boxed{CU(t)C^{-1} = U(-t).} \tag{84.24}$$

Proof. The generator $-J_E H/\hbar_{\mathrm{int}}$ is anti-self-adjoint. The hypotheses of equation (84.22) change its sign under conjugation. Functional calculus and uniqueness of the generated unitary group yield equation (84.24). □

When, additionally, a charge operator Q_{ch} and a mass observable M satisfy

$$CQ_{\mathrm{ch}}C^{-1} = -Q_{\mathrm{ch}}, \qquad CMC^{-1} = M, \tag{84.25}$$

the two orientations of the band are published as carriers of conjugate charge and shared even memory. The particle–antiparticle representation requires the intertwiner that lifts particle– antiparticle C_M to C and preserves the evolution. The identities equations (84.18) y (84.24) provide the two typed endpoints of that intertwiner.

The tritic temporal reader organizes the regimes according to

$$\begin{aligned} +1 &\longmapsto \text{return, memory and past,} \\ 0 &\longmapsto \text{torsional threshold and present,} \\ -1 &\longmapsto \text{bidirectional opening and future.} \end{aligned} \tag{84.26}$$

The assignment is to reader of the structure algebraic $J^2 = -1, 0, +1$. The promotion to curvatures of a spacetime requires a connection and a chart metric; the temporal reader meanwhile preserves its exact meaning of return, threshold, and opening.

84.7 Recoverability and Landauer's Principle

Landauer's thermodynamic principle locates the physical cost of logically irreversible erasure [78].

Let $(\Omega, 2^{\Omega}, \mu)$ to finite probabilized space, let $X : \Omega \to \mathcal{X}$ the complete state variable and let $q : \mathcal{X} \to Y$ to reader. The chain rule provides

$$\mathsf{H}(X) = \mathsf{H}(q(X)) + \mathsf{H}(X \mid q(X)). \tag{84.27}$$

The second summand is the information contained in the reader fiber.

Theorem 84.9 (Landauer zero of full transport)**.** *If $U : \mathcal{X} \to \mathcal{X}$ is a bijective update, then*

$$\boxed{\mathsf{H}(X \mid U(X)) = 0.} \tag{84.28}$$

The logical information erased by the update is null. For every physical protocol $\mathcal{P}_U$ that realizes it, the Landauer bound and its ideal reversible limit satisfy

$$\boxed{Q_{\mathrm{L}}[\mathcal{P}_U] \geq 0, \qquad \inf_{\mathcal{P}_U^{\mathrm{rev}}} Q_{\mathrm{L}}[\mathcal{P}_U] = 0.} \tag{84.29}$$

Proof. The inverse of U reconstructs X from $U(X)$; therefore, the conditional distribution is concentrated on a single state. Its entropy is zero, and the operation, which is logical, preserves information. The resulting thermodynamic bound is $Q_{\mathrm{L}} \geq 0$; reversible quasistatic protocols attain the ideal null infimum. This statement distinguishes the bound from the concrete dissipation of a device. □

The projected output $q(U(X))$ preserves part of the information and leaves in the fiber

$$\mathsf{H}(X \mid q(U(X))). \tag{84.30}$$

Let $M : \mathcal{X} \to \mathfrak{M}$ be a memory register, and define the enriched reader

$$\widehat{q}(x) = (q(x), M(x)). \tag{84.31}$$

Proposition 84.10 (Recoverability via output and memory)**.** *If $\widehat{q}$ is injective on the support of X, then*

$$\boxed{\mathsf{H}(X \mid q(X), M(X)) = 0.} \tag{84.32}$$

The quantity $\mathsf{H}(X \mid q(X))$ measures the additional memory required to recover the state from the visible output.

Proof. Injectivity on the support provides a deterministic inverse from $\widehat{q}(X)$ to X. The conditional entropy is therefore zero. The chain rule locates the difference between the visible reader and the enriched reader in the fiber of q. □

For the reset of a uniform trit, the logical entropy is $\log 3$, and the Landauer bound takes the form

$$\boxed{Q_{\text{reset}} \geq k_B \Theta \log 3.} \tag{84.33}$$

The structural identity of the clock,

$$E_P = k_B \Theta_{\text{clk}} \log 3, \tag{84.34}$$

evaluates that bound at the declared chronological temperature. The Planck quantum then appears as the minimum cost of resetting a uniform ternary decision in that thermal chart.

The zero Landauer cost and the cost $k_B \Theta \log 3$ describe different operations. The former corresponds to the bijective transport of the enriched state; the latter corresponds to a reset that identifies three registers. The costs of control, finite speed, preparation, and coupling to the environment belong to the physical realization of the device. The logical equality of zero minimum cost preserves information; an equation of motion for a bounce or a cosmological cycle additionally requires boundary dynamics and its balance laws.

84.8 Reversible measurement, conditioning, and reinitialization

The operations of the complete process are ordered as

$$\boxed{\begin{array}{c}\text{preparation} \longrightarrow \text{reversible coupling} \longrightarrow \text{pointer reading} \\ \longrightarrow \text{fiber conditioning} \longrightarrow \text{reinitialization}\end{array}} \tag{84.35}$$

Each arrow has its own informational balance:

(i) Unitary premeasurement preserves purity and norm in the composite system.

(ii) The reading publishes $q(X)$ and preserves the record $M(X)$ in the apparatus.

(iii) Conditioning selects the relative fiber and normalizes its measure.

(iv) The nonselective channel sums all instruments and preserves the trace.

(v) Resetting identifies registers and activates the Landauer bound corresponding to the erased information.

This chain localizes the origin of operational irreversibility. The global state may evolve isometrically while a partial reading exhibits increasing entropy. Information accessible to one branch and information preserved in global correlations are distinct magnitudes. The coupled observer forms part of those correlations, and its memory constitutes a physical component of the joint state.

84.9 Confluence of observation, self-inertia, and modular response

The above results are assembled into the causal chain.

$$\begin{aligned} \text{enriched APP–TRIT–TPK state} &\longrightarrow (\text{section, observer, memory}) \\ &\longrightarrow (\text{fiber, PVM, instrument}) \\ &\longrightarrow (\text{relative state, holonomy, induced frame}) \\ &\longrightarrow (\text{Machian response, Typed Landauer}). \end{aligned} \tag{84.36}$$

The modular structure of the preceding chapter adds the pair

$$(\widehat{\mathcal{A}}, K_A), \tag{84.37}$$

which preserves geometric quantity and provenance. The observer accesses a compression of this pair through its PVM; the memory preserves the channel from which the result originates; autoinertia transports the frame along the connection; relative entropy measures the cost of changing genealogical state; and Landauer localizes the cost of the effective erasure of the record.

The relation to Everett lies at the first level of this chain: the isometry preserves the superposition of registers, and the observer occupies a relative state. The relation to Mach lies at the connection level: the frame and the local response are induced from the compatible totality. The Möbius band lies at the holonomy level: the orientation can change while the ray and the accessible observables return. The three readings share the enriched state and preserve different mathematical functions.

84.10 Domain of validity and physical realization conditions

The probative force is distributed as follows:

(i) The fibre–PVM–Lüders correspondence, the Kraus representation, the CPTP channel, and the dilation are *exact results retrieved* in the declared domains.

(ii) The formulation of the relative state and the conservation, isometric in character, of the channel are *exact sciences*. The Everettian ontology of branches and their macroscopic stability belong to a *interpretive and dynamic interface*.

(iii) The criterion Fix(Hol), the vector–ray–observable hierarchy and the Machian factorization are *exact results retrieved*. Its application to a gravitational geometry requires the corresponding realization morphism.

(iv) The Möbius band is *exacta relativamente* to the lift, to the loop, and to the declared sign character. The interpretation of particle–antiparticle requires the antiunitary intertwiner of equation (84.22).

(v) The zero Landauer cost of a bijective update, the recoverability of the enriched reader, and the ternary reset bound are *exact results retrieved*. The cosmological selector and the dynamics of sheets prolong these balances by means of the declared equations of evolution, splicing conditions, and torsional memory.

Estatuto probatorio

The chapter proves in its finite domains the equivalence of conditionings, preservation of the global channel, the return and factorization criteria, and the logical Landauer balances. The joint realization employs the isometric intertwiner between the TPK memory and the apparatus, the dynamic stability of the finite register, and transport of the genealogical involution to antiunitary conjugation with its even and odd observables.

Falsification criterion

The corresponding statements are refuted by: a nonmeasurable fiber; a family of projectors lacking a common PVM; a Lüders conditioning that differs from equation (66.54); Kraus operators whose sum of effects differs from the identity; a dilation with $V_a^\dagger V_a \neq I$; an exact vector return attributed to a datum outside Fix(Hol); a Machian response varying within a fiber of b; a loop with a trivial sign character presented as a Möbius strip; a conjugation that fails to satisfy $CU(t)C^{-1} = U(-t)$; a bijective update with positive conditional entropy; or a uniform ternary reset whose balance omits the information $\log 3$. A cosmological promotion is also refuted by presenting zero Landauer as a dynamical bounce equation.

84.11 Relational Dynamics of Frames: Autoinertia, Ambivalence, and Mach's Principle

Classical inertia characterizes the persistence of a state of motion. Mach's question [85] shifts the emphasis from the isolated body toward the relation between that body and the material totality. In genealogical language, the same question is formulated without separating the observer from the boundary it reads: a local frame is the restriction of a globally compatible state, and the inertial response is Machian when it depends on the totality through a well-defined boundary trace.

The word *self-inertial* does not designate a system disconnected from the rest. It designates a section that coherently transports the frame induced by its own insertion into the whole. Its persistence is internal to the global state and is verified by means of a connection.

84.11.1 Totality, Trace, and Local Response

Let Ω_{tot} be the space of global sections and $b : \Omega_{\mathrm{tot}} \to \mathcal{B}$ its trace on a boundary. For a region or vertex v, let

$$\mathcal{I}_v : \Omega_{\mathrm{tot}} \longrightarrow \mathcal{Y}_v$$

the local inertial response.

Definition 84.11 (Machian Factorization)**.** The response $\mathcal{I}_v$ is Machian with respect to b when there exists an induced map $\overline{\mathcal{I}}_v : \mathrm{im}(b) \to \mathcal{Y}_v$ such that

$$\mathcal{I}_v = \overline{\mathcal{I}}_v \circ b. \tag{84.38}$$

Theorem 84.12 (Completeness Factorization Criterion)**.** *There exists a unique map $\overline{\mathcal{I}}_v : \mathrm{im}(b) \to \mathcal{Y}_v$ that satisfies (84.38) if and only if $\mathcal{I}_v$ is constant over each fiber of b:*

$$b(x) = b(y) \quad \Longrightarrow \quad \mathcal{I}_v(x) = \mathcal{I}_v(y).$$

If b is surjective, $\mathrm{im}(b) = \mathcal{B}$ and the factorized map is uniquely defined on all of $\mathcal{B}$.

Proof. Necessity is immediate. For sufficiency, given $\beta \in \mathrm{im}(b)$, we define $\overline{\mathcal{I}}_v(\beta) = \mathcal{I}_v(x)$ for any $x \in b^{-1}(\beta)$. Constancy on the fibers ensures that the definition is independent of the representative; the factorization equality itself forces its value at each $\beta \in \mathrm{im}(b)$ and thereby ensures uniqueness. If b is surjective, this domain coincides with $\mathcal{B}$. □

The criterion gives a positive formulation of Mach's principle: the global boundary contains exactly the information of the totality that governs the local response. The interior state may conserve more genealogy; inertia depends on the class published by the trace.

84.11.2 Self-inertial sections as geodesics of the joint state of system, observer, and memory: $D_\gamma s_\gamma = 0$

Let $\mathcal{E} \to \Gamma$ to vector bundle of local states along to route γ, and is D to linear connection induced by TPK transport.

Definition 84.13 (Self-inertial Section)**.** A persistent section s_γ is self-inertial on a segment if

$$D_\gamma s_\gamma = 0. \tag{84.39}$$

Equivalently and discretely, the state at each vertex is the parallel transport of the preceding state by the route morphism.

Equation (84.39) preserves orientation, memory, and signature in the sense determined by the connection. A coordinate acceleration can coexist with parallel transport when the reference frame also changes. The distinction between inertial motion and force is thus tied to the connection, not to the appearance of the trajectory in an isolated chart.

Proposition 84.14 (Persistence and Holonomy)**.** *Let γ be a loop and $\mathrm{Hol}_D(\gamma)$ its holonomy. A self-inertial section returns exactly to its initial state if and only if it belongs to the fixed subspace of $\mathrm{Hol}_D(\gamma)$. If the visible projection returns and the initial state lies outside $\mathrm{Fix}(\mathrm{Hol}_D(\gamma))$, the coordinate returns and the state changes.*

This result links self-inertia and monodromy. Observable periodicity does not imply closure of the lift; the deficit or excess of holonomy records the accumulated response of the frame.

84.11.3 Ambivalence Principle: Bidirectional Involution of the Readers π/φ^2 and φ/π^2

The HMT cell admits two complementary readings: areal and volumetrica. Let q_A the boundary or area reader and q_V the interior or volume reader. The transport T_{AV} relates the two without identifying them.

Principle 84.15 (Areal-volumetric Ambivalence)**.** A complete physical realization conserves the pair

$$\big(q_A(x), q_V(x), T_{AV}(x)\big). \tag{84.40}$$

The areal reading expresses how the state impinges on a boundary; the volumetric reading expresses how it occupies and transports the interior. The third datum preserves the transformation between them.

In Dimensional Mechanics, the areal–volumetric relation organizes the HMT extension of the equivalence principle. A single section may be read as a boundary response and as an interior density. When both readings are transported by the same local connection, the inertial and gravitational descriptions belong to a single state class.

Comparison Criterion 84.16 (Typed local equivalence)**.** The identification of an inertial response with a gravitational response is proved in a region U when:

1. q_A and q_V are defined on the same section;
2. there exists an intertwiner T_{AV} that preserves its units and orientations;
3. the connection transporting both readers coincides on U;
4. the observational difference vanishes in the declared local frame.

The formulation preserves the equivalence and shows which data realize it. Its application to the electron belongs to the first physical closure of the mass law; the present annex uses only the geometric scheme.

84.11.4 Causal Coordination Between Inertial Transport, Areal–Volumetric Double Reading, and Machian Factorization

The three principles are causally ordered:

$$\boxed{\begin{array}{lll} \text{persistence of the section} & \longrightarrow & D_\gamma s = 0, \\ \text{double reading of the same state} & \longrightarrow & (q_A, q_V, T_{AV}), \\ \text{dependence of the local response on the whole} & \longrightarrow & \mathcal{I}_v = \overline{\mathcal{I}}_v \circ b. \end{array}} \tag{84.41}$$

The principle of inertia fixes the transport; Ambivalence fixes the two physical readings; Mach fixes the global quotient that governs the local frame. Together they produces to relational theory of motion in the that the observer, the apparatus, and the body belong to to common section.

84.11.5 Measurement of motion and frame coupling

Measuring velocity or acceleration requires comparing the system with a clock and a ruler. Those objects are part of the apparatus. Let F_S be the system frame and F_M the observer frame. The measurement coupling constructs a relative transformation

$$G_{MS} = F_M^{-1} F_S.$$

The published velocity, acceleration, and phase are invariants or components of G_{MS}, not properties without reference. Repetition of the experiment preserves the context by jointly transporting apparatus and boundary.

This formulation recovers Weyl's intuition [132]: measuring an amplitude or a length entails coupling a gauge rule. HMT adds the genealogy of that rule. The result can be traced back to the state of the apparatus, to the reader, and to the transport that fixed its scale.

84.11.6 Bidirectional wave and temporal orientation

A persistent route admits advanced and retarded channels. The reversal of orientation acts on the word and the ledger; in the linear realization, it exchanges the corresponding propagators when the Hamiltonian possesses the declared symmetry. Bidirectional motion is expressed by a pair

$$(U_{\mathrm{ret}}, U_{\mathrm{adv}})$$

related by conjugation and time reversal.

The particle–antiparticle band realizes this duality as two orientations of the same section. The dynamical relation can be expressed exactly.

Proposition 84.17 (Conjugation of Orientation and Propagator Reversion)**.** *Let C be an orthogonal involution on the quantum realification such that*

$$C J_E C^{-1} = -J_E, \qquad C H C^{-1} = H.$$

Then, for $U(t) = \exp(-J_E H t/\hbar_{\mathrm{int}})$,

$$C U(t) C^{-1} = U(-t).$$

Proof. Withjugation commutes with the exponential series and transforms the generator $-J_E H$ in $+J_E H$. The result is the exponential evaluated at $-t$. □

The word involution $C_M = -\mathrm{rev}$ provides the genealogical orientation datum. When its realization satisfies the two identities of the proposition, the particle and antiparticle branches are represented

by propagators with conjugate temporal directions. The band geometry fixes the orientation, and the Hamiltonian fixes the dynamics.

84.11.7 Global expansion and self-consistency equation

A genealogical cosmology can be formulated through an operator $\mathcal{T}$ that carries one boundary totality to the next:

$$B_{n+1} = \mathcal{T}(B_n).$$

The locally induced response is

$$\mathcal{I}_{v,n} = \overline{\mathcal{I}}_v(B_n).$$

A global self-inertial regime corresponds to a stable orbit or fixed point of the transport operator, while expansion is recorded by a scale cocycle a_{n+1}/a_n. This formulation separates three questions: existence of the dynamics, stability of the solution, and identification of the cocycle with the observed cosmological factor.

Result

Local inertia may be typed as the parallel transport of a section; its Machian dependence is expressed as factorization through the global boundary trace. The Ambivalence Principle links the areal and volumetric readings of the same state. The observer enters through the coupling of frames that makes measurement possible.

Hypothesis, domain, and conclusion

The definition of self-inertia and the Machian factorization are recovered results. The dimensional criterion and the measuring instrument follow from the preceding chapters. The synthesis (84.41) is a new formalization of a pre-existing authorial architecture. A quantitative cosmological equation requires fixing $\mathcal{T}$, its metric, and its boundary data.

The Machian attribution is characterized by the constancy of the local response along the fibers of the global trace. Self-inertia is characterized by parallel transport and by the fixed subspace of the holonomy. In a cosmological realization, the operator $\mathcal{T}$, its metric, and the scale cocycle turn these structural conditions into a quantitative evolution of the totality.

84.12 Self-inertial response and Mach's principle from Ambivalence

Let $b : \mathcal{X}_\infty^{\mathrm{enr}} \to B_\infty$ be the global boundary trace. On B_∞, the positive readers $A(\beta)$, $V(\beta)$, the global clock $Q(\beta) > 0$, and the memory $T(\beta)$ are preserved. The Ambivalence hinge provides

$$r_{\mathrm{av}} = \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2}, \qquad r_{\mathrm{av}}^J = \frac{\varphi_{\mathrm{HMT}}}{\pi_{\mathrm{HMT}}^2}.$$

We define weights, without external gravitational data,

$$w_A(\beta) = \frac{A(\beta) r_{\mathrm{av}}}{A(\beta) r_{\mathrm{av}} + V(\beta) r_{\mathrm{av}}^J}, \qquad w_V(\beta) = 1 - w_A(\beta).$$

Let P_A, P_V be complementary orthogonal projections of the local space of response. For internal energies E_1, E_2, the HMT self-inertial response is

$$\overline{I}_v^{\mathrm{HMT}}(\beta) = \frac{E_1 E_2}{Q(\beta)^2} \big(w_A(\beta) P_A + w_V(\beta) P_V \big),$$

and

$$I_v^{\mathrm{HMT}} = \overline{I}_v^{\mathrm{HMT}} \circ b.$$

Theorem 84.18 (Machian Closure of the Ambivalence Response)**.** *The response I_v^{HMT} is self-adjoint, positive, and constant on each fiber of b. Therefore, it factors uniquely through the global totality. The Ambivalence involution that exchanges*

$$(A, r_{\mathrm{av}}, P_A) \longleftrightarrow (V, r_{\mathrm{av}}^J, P_V)$$

conjugates $\overline{I}_v^{\mathrm{HMT}}$ without changing its spectrum. If there exist x, x' with the same local restriction, $b(x) \neq b(x')$ and distinct Q, A or V, then the response does not factor only through the local restriction and is Machian in the strong sense.

Proof. The weights are positive and sum to one; the combination of projections is positive and self-adjoint. The formula depends only on $b(x)$, hence it is constant on its fibers and applies the factorization criterion through the totality. The exchange permutes the two spectral blocks. The final claim is precisely the negative control against a purely local response. □

The construction also has a recovered geometric reading. For the joint state $\Gamma = (x, O, m)$, let $\nabla^{\mathrm{tot}}_{b(x)}$ be the connection induced by the boundary and let π_S be the projection onto the subsystem. If

$$\nabla^{\mathrm{tot}}_{\dot\Gamma} \dot\Gamma = 0,$$

the published acceleration is

$$a_S = \nabla^S_{\dot\gamma_S} \dot\gamma_S = \mathcal{K}^{b(x)}_{\pi_S}(\dot\Gamma, \dot\Gamma).$$

Since the defect depends on the connection induced by $b(x)$, it also factors through the whole. This mathematically completes the replacement of the inertial/noninertial dichotomy by self-inertial systems: the total state is geodesic, and local acceleration is a projection defect. The equality of this response with an observed gravitational acceleration belongs to EXTERNAL RECOGNITION.

84.13 Observer, memory, and relational response

The reading operation acts on the transport published by the APP–TRIT–TPK enriched state. Its domain preserves route, sheet, orientation, carry and memory; the result accessible to the observer is a map of that domain. This distinction allows three operations to be described together: choosing an oriented representative of transport, inscribing its record in an apparatus and the local response induced by the global boundary. Each operation transmits different information. The following construction specifies their maps and the conditions allowing them to be composed.

84.13.1 Boundary transport and intrinsic reading

The finite publication of transport provides an oriented cycle C_n, with $n \geq 1$, vertices $0, \ldots, n-1$ and edges $e_j : j \longrightarrow j+1 \pmod n$. We fix the lattice realization L of the record, its vector space $W = L \otimes_{\mathbb{Z}} \mathbb{R}$, a positive inner product and the nonzero primitive representative $v \in L$ of the distinguished residual class, in the declared chamber. The APP sum and product sheets provide the transport polarizations; TRIT orientation and TPK updating precede this linear publication. The cycle, the lattice realization and the residual class are the data this section receives from the transport constructed previously.

A 0-cochain is a function on the vertices with values in W; a 1-cochain assigns a vector to each oriented edge. The coboundary acts by

$$(d\Phi)(e_j) = \Phi(j+1) - \Phi(j).$$

We denote by $\mathbf{1}_E$ the constant scalar cochain on the edges. For a transport $\omega \in C^1(C_n; W)$, its accumulated publication Γ_ω is obtained through partial sums and affine interpolation; its closure defect is

$$B(\omega) = \Gamma_\omega(1) - \Gamma_\omega(0) = \sum_{j=0}^{n-1} \omega(e_j). \tag{84.42}$$

Theorem 84.19 (Exact decomposition and harmonic mode of the cycle)**.** *Every transport has a unique decomposition*

$$\omega = d\Phi + u\mathbf{1}_E, \qquad \Phi(0) = 0, \qquad u = \frac{1}{n}\sum_j \omega(e_j).$$

In particular, $B(d\Phi) = 0$ *and* $B(\omega) = nu$.

Proof. Let $y_j = \omega(e_j) - u$. The sum of the y_j is zero. We define $\Phi(k) = \sum_{j=0}^{k-1} y_j$; the value for $k = n$ coincides with $\Phi(0) = 0$, so the function is well defined on the cycle, including its final edge. We have $d\Phi(e_j) = y_j$. If $d\Phi + u\mathbf{1}_E = d\Psi + w\mathbf{1}_E$, the sum over edges gives $nu = nw$. Then $d(\Phi - \Psi) = 0$; connectivity of the cycle and normalization at the initial vertex imply $\Phi = \Psi$. The identity $\sum_j(\Phi(j+1) - \Phi(j)) = 0$ directly proves telescopic cancellation. □

Definition 84.20 (Admissible transport and normalized reading)**.** Transport is admissible when its harmonic mode belongs to $\mathbb{R}v$. On the space $\mathcal{T}_{\mathrm{adm}}$ of those transports we define the global reading and its density per phase:

$$\mathfrak{a}_v(\omega) = \frac{\langle B(\omega), v\rangle}{\langle v, v\rangle}, \qquad \mu_v(\omega) = \frac{\mathfrak{a}_v(\omega)}{n}. \tag{84.43}$$

We have $B(\omega) = \mathfrak{a}_v(\omega)v$.

Theorem 84.21 (Determination of the intrinsic reading)**.** *The quotient* $\mathcal{T}_{\mathrm{adm}}/dC^0(C_n; W)$ *is one-dimensional, generated by the class of* $v\mathbf{1}_E$. *The map* μ_v *is the unique linear functional on* $\mathcal{T}_{\mathrm{adm}}$ *satisfying*

$$F(\omega + d\Phi) = F(\omega), \qquad F(v\mathbf{1}_E) = 1.$$

Proof. The preceding decomposition writes each admissible transport as $d\Phi + \lambda v\mathbf{1}_E$. Its class modulo exact cochains is determined by λ. The class of $v\mathbf{1}_E$ is nonzero because its sum is $nv \neq 0$. By linearity and invariance, every functional with the indicated characteristics equals $F(\omega) = \lambda F(v\mathbf{1}_E) = \lambda$. The formula (84.43) gives precisely $\mu_v = \lambda$. The global reading is $\mathfrak{a}_v = n\mu_v$. □

Definition 84.22 (Edge Observer)**.** Given the polarization of the published transport, a boundary observer is a pair $O = (\varepsilon, \Psi)$, where $\varepsilon \in \{+1, -1\}$ and $\Psi \in C^0(C_n; W)$. Its reading action is

$$\omega^O = \varepsilon\omega + d\Psi. \tag{84.44}$$

The sign fixes the orientation and the cochain fixes the trivialization. Two polarization choices are compared through their published transport; the relation (84.44) describes exactly the representatives differing by orientation and coboundary.

Proposition 84.23 (Covariance of the reading)**.** *The action preserves admissibility and satisfies*

$$B(\omega^O) = \varepsilon B(\omega), \quad \mathfrak{a}_v(\omega^O) = \varepsilon\mathfrak{a}_v(\omega), \quad \mu_v(\omega^O) = \varepsilon\mu_v(\omega).$$

Trivializations with the same orientation publish the same closure.

Proof. The sum of the coboundary $d\Psi$ is zero. Therefore, the harmonic mode transforms into εu, which still belongs to $\mathbb{R}v$, and the defect transforms into εB. The two scalar equalities are obtained by normalized projection. □

The unit of this reading is the residual representative v, with its internal normalization. Trivialization freedom acts on the cochain representative; closure information resides in its class. The description allows observers to be compared while keeping genealogical transport, its linear publication and the scalar evaluating the residual class separate.

84.13.2 Boundary identity and cellular interior

Let K be a finite oriented two-dimensional cell complex, equipped with a fundamental 2-chain c_K such that $\partial c_K = [C_n]$. This condition expresses that interior incidences cancel with their multiplicities and orientations. Let $\Omega \in C^1(K; W)$ be an extension of ω and $F_\Omega = d\Omega$ its additive discrete curvature.

Theorem 84.24 (Cellular holographic identity of the reading)**.** *With the preceding data,*

$$B(\omega) = \langle \Omega, \partial c_K \rangle = \langle d\Omega, c_K \rangle, \qquad \mathfrak{a}_v(\omega) = \frac{\langle \langle d\Omega, c_K \rangle, v \rangle}{\langle v, v \rangle}. \tag{84.45}$$

In particular, $F_\Omega = 0$ implies $\mathfrak{a}_v(\omega) = 0$. If closure is nonzero, every extension of transport on that complex has nonzero discrete curvature.

Proof. The pairing between chains and cochains satisfies $\langle d\Omega, c_K \rangle = \langle \Omega, \partial c_K \rangle$. When it is expanded, each interior edge appears with the sum of the incidence numbers of the adjoining cells; the condition on ∂c_K leaves exactly the oriented boundary cycle. The restriction of Ω to that cycle is ω, whose sum is $B(\omega)$. Projection onto v gives the second formula and the flatness implication. □

The equality compares two evaluations of the same extended cochain: the accumulated boundary defect and the coboundary integrated over the interior. Its curvature takes values in W and uses the additive cellular coboundary. Realization by a geometric connection additionally preserves its own coefficient space and transport law.

84.13.3 Isometric inscription of the record in the apparatus

Let $\mathfrak{M}_n$ be a nonempty finite set of records published at level n, with result, route, carry, sheet and memory. Its record space is $\mathcal{H}_n = \ell^2(\mathfrak{M}_n)$, with basis $|m\rangle$. The visible update F_n has a record lift

$$\widehat{F}_n(m) = (F_n(m), m).$$

We take as records of the next level a set containing those pairs. The second coordinate preserves the antecedent and makes the lift injective. This construction specifies the memory retained by inscription, including when the visible update merges values.

In a finite apparatus space $\mathcal{K}_{A,n}$, we fix an orthonormal family $(|a_m\rangle)_{m \in \mathfrak{M}_n}$ and define

$$\iota_n : \mathcal{H}_n \longrightarrow \mathcal{K}_{A,n}, \quad \iota_n |m\rangle = |a_m\rangle, \qquad V_n : \mathcal{H}_n \longrightarrow \mathcal{H}_{n+1}, \quad V_n |m\rangle = |\widehat{F}_n(m)\rangle.$$

Let $\mathcal{P}_n = \iota_n \mathcal{H}_n$ be the pointer subspace.

Theorem 84.25 (Compatibility between updating and inscription)**.** *The maps ι_n and V_n are isometries. There exists a unique isometry $W_n^0 : \mathcal{P}_n \longrightarrow \mathcal{K}_{A,n+1}$ satisfying*

$$W_n^0 \iota_n = \iota_{n+1} V_n. \tag{84.46}$$

If $\dim \mathcal{K}_{A,n+1} \geq \dim \mathcal{K}_{A,n}$, it extends to an isometry of all of $\mathcal{K}_{A,n}$. Through auxiliary complements equalizing dimensions, a unitary extension between the enlarged spaces is obtained.

Proof. The basis images under ι_n are orthonormal; the images under V_n are also orthonormal by injectivity of $\widehat{F}_n$. For $z = \iota_n \psi$ we define $W_n^0 z = \iota_{n+1} V_n \psi$. The representation of z is unique, and the three maps

preserve its norm. The formula determines the map on a basis of $\mathcal{P}_n$ and proves the square. If the target dimension is sufficient, the orthogonal complement of $W_n^0\mathcal{P}_n$ contains an orthonormal family of the size of $\mathcal{P}_n^\perp$; sending a basis of the latter to that family constructs the extension. Finally both spaces are enlarged to the same dimension and the two prescribed families are completed to orthonormal bases. The map between those bases is unitary. □

When the record includes labels of a finite instrument, this same inscription realizes its environment: if the operators $K_{y,a} : \mathcal{H}_S \to \mathcal{H}'_S$ satisfy $\sum_{y,a} K_{y,a}^* K_{y,a} = I$, the map

$$V_{\mathrm{inst}}\psi = \sum_{y,a} K_{y,a}\psi \otimes |y, a\rangle$$

is isometric, since its squared norm is $\sum_{y,a}\langle\psi, K_{y,a}^* K_{y,a}\psi\rangle = \|\psi\|^2$. The inscription ι transports each label to its corresponding pointer. The model thus fixes the premeasurement record; a material device additionally requires its interaction and calibration.

84.13.4 Accessible algebra and pointer complement

Define $P_m = |a_m\rangle\langle a_m|$, $P = \sum_m P_m$, and $P_\perp = I - P$. The algebra of the pointer subspace is $P\mathcal{B}(\mathcal{K}_{A,n})P$, whose unit is P. On it, the diagonal reader is

$$\mathcal{D}_P(A) = \sum_m P_m A P_m.$$

Proposition 84.26 (Record expectation and unital extension)**.** *The map $\mathcal{D}_P$ is a completely positive, trace-preserving and idempotent conditional expectation onto the pointer algebra, with $\mathcal{D}_P(P) = P$. Its image is $\bigoplus_m \mathbb{C}P_m$. On the algebra of the complete apparatus, the map*

$$\mathcal{E}_P(A) = \sum_m P_m A P_m + P_\perp A P_\perp \tag{84.47}$$

is a unital, completely positive, trace-preserving conditional expectation. Its image is

$$\left(\bigoplus_m \mathbb{C}P_m\right) \oplus \mathcal{B}(\mathcal{P}_n^\perp).$$

Both maps are orthogonal projections for the Hilbert–Schmidt inner product. Their spectra are contained in $\{0, 1\}$. When eliminated components exist, the separation between the fixed subspace and the kernel is one.

Proof. The projectors P_m are mutually orthogonal. In the complete apparatus, the family that adds $P_\perp$ sums to I. Each term $A \mapsto P_m A P_m$ is completely positive: after any matrix amplification it acts by congruence with $I \otimes P_m$. The same holds for $P_\perp$. The sum preserves that property. Cyclicity of the trace and the sum of the projectors prove trace preservation on each domain. Orthogonality annihilates the cross terms when the map is iterated, so it is idempotent. Its fixed elements are precisely the stated blocks and the map is bimodular with respect to that subalgebra. Moreover, $\mathrm{Tr}(A^* P_m B P_m) = \mathrm{Tr}((P_m A P_m)^* B)$, from which self-adjointness for Hilbert–Schmidt follows. A self-adjoint projection has only the spectral values zero and one when both subspaces are nonzero. If only the fixed subspace remains, the map is the identity. □

The sum $\sum_m P_m A P_m$, applied without the complement to the entire apparatus, sends I to P. Its natural unit belongs to the pointer algebra. The formula (84.47) also preserves the auxiliary block; an orthonormal resolution of that block provides, when desired, diagonalization of the entire apparatus.

If U is a unitary transformation of the apparatus and $P'_m = UP_mU^*$, then

$$\mathcal{E}_{P'}(UAU^*) = U\mathcal{E}_P(A)U^*.$$

The equality is obtained by substituting each projector into the sum. Likewise, for $A \in \mathcal{B}(\mathcal{P}_n)$, the square (84.46) implies

$$\mathcal{D}_{P_{n+1}}(W_n^0 A (W_n^0)^*) = W_n^0 \mathcal{D}_{P_n}(A)(W_n^0)^*.$$

Indeed, each matrix unit $|a_m\rangle\langle a_{m'}|$ is transported to the unit corresponding to the distinct records $\widehat{F}_n(m), \widehat{F}_n(m')$; the diagonal reader preserves exactly the terms with $m = m'$.

84.13.5 Factorization of the response through the effective boundary

Let $\mathcal{X}_\infty^{\mathrm{enr}}$ be the domain of compatible enriched states and let $b : \mathcal{X}_\infty^{\mathrm{enr}} \longrightarrow B_\infty$ be its global boundary trace. We denote by $B_{\mathrm{ef}} = \operatorname{im} b$ the effectively published boundary. For a response space $\mathcal{V}_\nu$, we consider $I_\nu : \mathcal{X}_\infty^{\mathrm{enr}} \longrightarrow \mathcal{V}_\nu$.

Theorem 84.27 (Relational response criterion)**.** *There exists a unique map $\overline{I}_\nu : B_{\mathrm{ef}} \to \mathcal{V}_\nu$ such that $I_\nu = \overline{I}_\nu \circ b$ if and only if*

$$b(x) = b(x') \quad \Longrightarrow \quad I_\nu(x) = I_\nu(x'). \tag{84.48}$$

Uniqueness extends to all of B_∞ when b is surjective.

Proof. A composition through b is constant on each fiber. Conversely, for $\beta \in B_{\mathrm{ef}}$, we define $\overline{I}_\nu(\beta) = I_\nu(x)$ with $b(x) = \beta$. Condition (84.48) guarantees independence of the representative; every value of $\overline{I}_\nu$ is determined by some state of that fiber. If b is surjective, $B_{\mathrm{ef}} = B_\infty$. Otherwise, the composition equation determines only the values on B_{ef}. □

This criterion is set-theoretic. For a continuous factorization, the topologies and the quotient-map condition of $b : \mathcal{X}_\infty^{\mathrm{enr}} \to B_{\mathrm{ef}}$ are also preserved; with it, continuity of I_ν is equivalent to that of its factor. This specification separates determination of a reading from its regularity properties.

Let $\ell_\nu : \mathcal{X}_\infty^{\mathrm{enr}} \to \mathcal{L}_\nu$ be the restriction accessible to a local neighborhood. The effective global character is verified through two states satisfying

$$\ell_\nu(x) = \ell_\nu(x'), \qquad I_\nu(x) \neq I_\nu(x'). \tag{84.49}$$

If I_ν factors through b, these two states have distinct boundaries. Moreover, the response cannot be written as a function of ℓ_ν, since that function would assign the same value to both states. The difference of boundaries thus acquires a verifiable operator effect.

84.13.6 Ambivalence weights and self-inertial response

For the general formula we consider positive readers A_∂, V_∂ and an energy reader Q defined on the declared effective boundary domain. A realization on $B_{\mathrm{av}} \subseteq B_{\mathrm{ef}}$ is made explicit below. Memory remains as a boundary coordinate $\mathsf{M}_\partial(\beta)$. The ambivalence hinge provides the two orientations

$$r_{\mathrm{av}} = \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2}, \qquad r_{\mathrm{av}}^J = \frac{\varphi_{\mathrm{HMT}}}{\pi_{\mathrm{HMT}}^2}.$$

The superscript J expresses interchange of the generating pair.

Normalized realization of the boundary readers. The APP–TRIT–TPK modal incidence, developed in the regional construction, provides the path envelope with matrix

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

The positive vector $v = (1, \varphi_{\mathrm{HMT}})^{\mathsf{T}}$ satisfies $F_{\mathrm{av}} v = \varphi_{\mathrm{HMT}} v$. Its normalized transition and stationary weights are

$$P_{ij} = \frac{(F_{\mathrm{av}})_{ij} v_j}{\varphi_{\mathrm{HMT}} v_i}, \qquad P = \begin{pmatrix} 0 & 1 \\ \varphi_{\mathrm{HMT}}^{-2} & \varphi_{\mathrm{HMT}}^{-1} \end{pmatrix}, \qquad (\mu_{\mathrm{ar}}, \mu_{\mathrm{vol}}) = \frac{(1, \varphi_{\mathrm{HMT}}^2)}{1 + \varphi_{\mathrm{HMT}}^2}. \tag{84.50}$$

The rows of P sum to one by $\varphi^{-2} + \varphi^{-1} = 1$; the stationary equation $\mu_{\mathrm{ar}} = \mu_{\mathrm{vol}} \varphi^{-2}$, together with normalization, determines the stated weights. This Parry measure weights the paths of the envelope; the chronological frequency $(1/3, 2/3)$ counts the instants of the modal calendar and retains its distinct function.

The common memory-amplitude realization is defined on the sector $\mathcal{X}_{\mathrm{av}} \subset \mathcal{X}_{\infty}^{\mathrm{enr}}$ in which the readers of the two sheets have the factorization

$$\mathcal{L}_{\mathrm{vol}}(x) = \mu_{\mathrm{vol}} \mathcal{M}(x), \qquad \mathcal{L}_{\mathrm{ar}}(x) = \pi_{\mathrm{HMT}} \mu_{\mathrm{ar}} \mathcal{M}(x), \qquad \mathcal{M}(x) > 0. \tag{84.51}$$

Both values belong to the same positive memory-amplitude line. The areal publication carries the regional mark π_{HMT} of the return. The dimensionless reading is obtained by dividing both sections by $\mathcal{L}_{\mathrm{vol}}(x)$. Thus the sums of the weights are performed between quantities of the same type; a possible realization as physical areas and volumes is then composed with their respective dimensional transducers.

Proposition 84.28 (Descent of the normalized readers and the energy clock)**.** *On $B_{\mathrm{av}} = b(\mathcal{X}_{\mathrm{av}})$, the rules*

$$A_{\partial}(b(x)) = \frac{\mathcal{L}_{\mathrm{ar}}(x)}{\mathcal{L}_{\mathrm{vol}}(x)} = r_{\mathrm{av}}, \qquad V_{\partial}(b(x)) = 1 \tag{84.52}$$

define two positive readers independent of the representative. Given an orientation a of the action section, the decision energy of (123.6) *determines on states*

$$q_a(x) = \frac{h_a(x)}{108 t_0(x)} = \frac{\hbar_a(x)}{t_P(x)} \in \mathcal{L}_E^+, \qquad t_P = \frac{54}{\pi_{\mathrm{HMT}}} t_0. \tag{84.53}$$

We write

$$b_{\mathrm{av}} := b|_{\mathcal{X}_{\mathrm{av}}} : \mathcal{X}_{\mathrm{av}} \longrightarrow B_{\mathrm{av}}.$$

There exists a unique reader $Q : B_{\mathrm{av}} \to \mathcal{L}_E^+$ such that $q_a|_{\mathcal{X}_{\mathrm{av}}} = Q \circ b_{\mathrm{av}}$ if and only if, for any $x, x' \in \mathcal{X}_{\mathrm{av}}$,

$$b(x) = b(x') \implies \frac{h_a(x)}{t_0(x)} = \frac{h_a(x')}{t_0(x')}. \tag{84.54}$$

In particular, a chart of fixed action and duration satisfies this condition and publishes $Q = h_a/(108 t_0)$.

Proof. The quotient of (84.51) cancels the common amplitude and equals $\pi_{\mathrm{HMT}} \mu_{\mathrm{ar}} / \mu_{\mathrm{vol}} = \pi_{\mathrm{HMT}} / \varphi_{\mathrm{HMT}}^2$. The result is the same for all representatives of a fiber of b_{av} within $\mathcal{X}_{\mathrm{av}}$ and is positive. Dividing both readers by the same positive factor preserves the weighted quotients in which they enter. For the clock, the factorization criterion of theorem 84.27, applied to q_a, is exactly equivalent to (84.54). Fixed sections satisfy it; so do variable sections whose quotient descends through b_{av}. Uniqueness follows from the definition of B_{av} as the effective image. □

The value of the reader Q belongs to the energy line $\mathcal{L}_E = \mathcal{L}_{\mathrm{act}} \otimes \mathcal{L}_T^*$. With E_1, E_2 on that same line, $E_1 E_2 / Q^2$ is dimensionless and independent of the positive energy basis. The symbol Q in this section exclusively designates that clock energy; the quantity of heat in the Landauer realization belongs to the corresponding thermal balance. Common-amplitude factorization specifies a concrete realization of the general readers. Every other domain preserves its boundary maps and its own verification of descent.

Complete memory remains in the enriched history and in its boundary publication, even though it cancels in the normalized quotient.

We define

$$w_A(\beta) = \frac{A_\partial(\beta) r_{\mathrm{av}}}{A_\partial(\beta) r_{\mathrm{av}} + V_\partial(\beta) r_{\mathrm{av}}^J}, \qquad w_V(\beta) = 1 - w_A(\beta).$$

In realization (84.52), the weights admit the exact evaluation

$$w_A = \frac{r_{\mathrm{av}}^2}{r_{\mathrm{av}}^2 + r_{\mathrm{av}}^J} = \frac{\pi_{\mathrm{HMT}}^4}{\pi_{\mathrm{HMT}}^4 + \varphi_{\mathrm{HMT}}^5}, \qquad w_V = \frac{\varphi_{\mathrm{HMT}}^5}{\pi_{\mathrm{HMT}}^4 + \varphi_{\mathrm{HMT}}^5}. \tag{84.55}$$

Substitution preserves both factors of the definition: the normalized areal reader and its subsequent weighting by r_{av}. Generation and regional marking of the return are performed once. Multiplying numerator and denominator by $\varphi_{\mathrm{HMT}}^4 \pi_{\mathrm{HMT}}^2$ gives the last expression. These weights remain constant in the common-amplitude sector. With fixed projectors and energies, a variation of the response may come from the reader Q; the witness to global dependence additionally requires the states with equal local restriction and different response of (84.49).

Let P_A, P_V be complementary orthogonal projections of a response space $\mathcal{H}_v$, and let $E_1, E_2 > 0$ be two internal energies fixed for comparison. The response is

$$\overline{I}_v(\beta) = \frac{E_1 E_2}{Q(\beta)^2} \big(w_A(\beta) P_A + w_V(\beta) P_V \big), \qquad I_v = \overline{I}_v \circ b. \tag{84.56}$$

Energies that are explicit readers of β are also admitted; in that case they are preserved as such in the formula.

Proposition 84.29 (Positivity and covariance of the response)**.** *The response* (84.56) *is self-adjoint, positive definite and constant on the fibers of b. Its eigenvalues in the nonzero sectors are*

$$\lambda_A = E_1 E_2 Q^{-2} w_A, \qquad \lambda_V = E_1 E_2 Q^{-2} w_V.$$

The complete exchange $(A_\partial, r_{\mathrm{av}}, P_A) \leftrightarrow (V_\partial, r_{\mathrm{av}}^J, P_V)$ permutes the two summands and preserves the operator. A unitary transport of the response space conjugates that operator and preserves its spectrum with multiplicities.

Proof. Positivity of the readers gives $0 < w_A, w_V < 1$. For $z = z_A + z_V$, with $z_A = P_A z$, $z_V = P_V z$,

$$\langle z, \overline{I}_v z \rangle = \frac{E_1 E_2}{Q^2} (w_A \|z_A\|^2 + w_V \|z_V\|^2) > 0$$

if $z \neq 0$. The orthogonal decomposition proves the spectral expressions. Exclusive dependence on β gives constancy on the fibers and factorization. Exchanging the triples exchanges numerators, weights and projectors at once, so it preserves the sum. For a unitary transport U, substitution $P_A \mapsto U P_A U^*$, $P_V \mapsto U P_V U^*$ produces $U \overline{I}_v U^*$, with the same spectrum. □

An exchange implemented within the same space by a unitary sending P_A to P_V requires the two sectors to have equal dimension. Exchanging the two triples as labels of the formula is valid with each sector's own dimension.

Condition (84.49) requires comparing the responses. For example, simultaneously multiplying A_∂, V_∂ by the same positive factor preserves both weights; if E_1, E_2, Q are also preserved, the response remains identical. In contrast, with equal weights and energies, replacing Q by qQ, $q > 0$, multiplies the response by q^{-2}. If two states with equal local restriction realize that change with $q \neq 1$, they provide the global witness. These are operator separation conditions; existence of the states is checked in the boundary domain being realized.

84.13.7 Connection of the joint state and acceleration reading

In a differentiable realization of the joint state $\Gamma = (x, O, m)$, let N be its realization space, ∇^{tot} its connection and $\pi_S : N \to N_S$ a differentiable projection onto the subsystem, equipped with connection ∇^S. The connection and projection preserve the provenance of the transports and readers they realize. The observed frame is obtained through

$$\mathfrak{F}_O(\Gamma) = \mathrm{Res}_O(\Gamma, \nabla^{\mathrm{tot}}).$$

For a twice-differentiable trajectory $\Gamma(t)$, write $\gamma_S = \pi_S \circ \Gamma$.

Definition 84.30 (Self-inertia and projection defect)**.** A trajectory is self-inertial with respect to the declared connection when $\nabla^{\mathrm{tot}}_{\dot\Gamma}\dot\Gamma = 0$. The defect of its projection is

$$\mathcal{K}_{\pi_S}(\dot\Gamma, \dot\Gamma) = \nabla^S_{\dot\gamma_S}\dot\gamma_S - d\pi_S\big(\nabla^{\mathrm{tot}}_{\dot\Gamma}\dot\Gamma\big). \tag{84.57}$$

Proposition 84.31 (Subsystem acceleration)**.** *For a self-inertial trajectory,*

$$a_S = \nabla^S_{\dot\gamma_S}\dot\gamma_S = \mathcal{K}_{\pi_S}(\dot\Gamma, \dot\Gamma).$$

In coordinates x^B on N and y^a on N_S, the defect is quadratic in velocity, with components

$$(\mathcal{K}_{\pi_S})^a{}_{BC} = \partial_B\partial_C\pi^a_S + (\Gamma^S)^a{}_{bc}\,\partial_B\pi^b_S\partial_C\pi^c_S - \partial_D\pi^a_S(\Gamma^{\mathrm{tot}})^D{}_{BC}. \tag{84.58}$$

Proof. The chain rule gives $\ddot\gamma^a_S = \partial_B\pi^a_S\ddot\Gamma^B + \partial_B\partial_C\pi^a_S\dot\Gamma^B\dot\Gamma^C$. Adding the contribution of ∇^S and subtracting the image of the total acceleration, the terms containing $\ddot\Gamma$ cancel. The remaining ones are (84.58) contracted with $\dot\Gamma^B\dot\Gamma^C$. When total acceleration vanishes, the stated equality results. □

The geometric response factors through b when the connection, projection and kinematic datum used in evaluation are determined by $\beta = b(x)$, or when those kinematic data are kept fixed in the comparison. Under those conditions, (84.58) gives the same result for any representatives of a fiber; theorem 84.27 constructs its unique factor on B_{ef}. If the kinematic datum varies, it forms part of the domain of the response being compared.

The self-inertial formulation describes a geodesic trajectory of the joint state and an effective acceleration of its subsystem due to the projection defect. The gravitational realization then identifies that connection, the frame and the response with their physical observables. Observer covariance, memory inscription and factorization through the boundary thus retain complementary functions: the first determines the reading class, the second retains the genealogy and the third specifies the relational dependence of the response.

Chapter 85

The HMT–MD absorber theory: trajectory inversion, advanced and retarded propagators, boundary conditions, and particle–antiparticle conjugation

85.1 Bidirectional interaction: conjugation of advanced and retarded propagators and boundary conditions of the absorber

A route becomes dynamical when it is endowed with an action functional and boundary conditions. The action assigns a weight to the complete history; its variation selects transports compatible with the endpoints; interaction couples degrees of freedom that share a boundary; the bidirectional reading preserves the forward and reverse temporal orientations of the same antecedent. Within this framework, emission and absorption form the endpoints of a common boundary-value problem.

The absorber occupies a precise position in the architecture. HMT supplies the oriented band, outward-and-return memory, and the discrete involutions. Dimensional Mechanics adds the evolution operator, currents, and the action transducer. A physical absorber realization links that structure to advanced and retarded propagators and causal boundary conditions. The algebraic identities of the first segment are proved in their domains; the final link retains the status of a physical interface until the complete intertwiner is constructed.

85.1.1 Action functional on enriched paths

Let $\gamma = ((c_0, \mu_0), (d_1, \dots, d_L))$ be a finite route, with directions d_t and sheets $\mu_t \in \{\Sigma, \Pi\}$. Define

$$\operatorname{turn}(\gamma) = \#\{2 \le t \le L : d_t \neq d_{t-1}\},$$

$$\operatorname{switch}(\gamma) = \#\{2 \le t \le L : \mu_t \neq \mu_{t-1}\}.$$

Discrete Action of Minimal Interaction

For $\lambda \ge 0$, the local route action is

$$\mathcal{A}_\lambda(\gamma) = \operatorname{turn}(\gamma) + \lambda \operatorname{switch}(\gamma) = \sum_{t=2}^{L} \left[1 - \delta_{d_t, d_{t-1}} + \lambda(1 - \delta_{\mu_t, \mu_{t-1}})\right]. \tag{85.1}$$

The first contribution measures directional variation and the second measures transport between leaves.

The general action also incorporates boundary, carry, and memory:

$$\mathcal{S}[\gamma] = \sum_{e \in \gamma} \mathcal{L}_e + \mathcal{B}_{\text{in}}(\partial_- \gamma) + \mathcal{B}_{\text{out}}(\partial_+ \gamma) + \mathcal{M}(\gamma). \tag{85.2}$$

Here $\mathcal{L}_e$ is the local cost, $\mathcal{B}_{\text{in/out}}$ fixes the admissible endpoint data, and $\mathcal{M}$ preserves the information that the visible publication contracts. The variation is performed within a class of routes with the same boundary types.

Principle of Minimal Interaction Compatible with Memory

Among the prolongations compatible with the boundary data, the selected physical realization extremizes the action functional within the sector that preserves the memory required by the composition.

The principle fixes the class of the variational problem. A concrete dynamic adds the Hamiltonian or Lagrangian of the sector and proves that its extremum produces the equations of motion. Memory functions as a dynamic constraint: two routes with the same final shadow may belong to distinct variational classes.

85.1.2 Interaction as typed coupling

Let $\mathfrak{p}_1$ and $\mathfrak{p}_2$ two sections realized over routes γ_1 and γ_2. An interaction at to boundary B consists of to morphism

$$\mathcal{I}_B : E_B^{(1)} \otimes E_B^{(2)} \longrightarrow E_B^{(12)} \tag{85.3}$$

and in an action term $\mathcal{S}_{\text{int}}$ compatible with the representations of the sector. The morphism preserves the shared types and transforms the internal degrees according to the coupling rules. The complete composition takes the form

$$(\gamma_2, s_2) \star_{(B,\mathcal{I}_B)} (\gamma_1, s_1),$$

where the record of B is counted once, and the memory of both sides is transported to the composite state.

The covariance of the interaction requires, for each symmetry transformation g,

$$\rho_{12}(g)\, \mathcal{I}_B = \mathcal{I}_B\big(\rho_1(g) \otimes \rho_2(g)\big). \tag{85.4}$$

This identity decides whether the coupling belongs to the declared physical representation. In the electromagnetic and color sectors, the same structure is expressed by means of a connection and its parallel transport.

85.1.3 Temporal conjugation of the propagator

Let $\mathcal{H}_{\mathbb{R}}$ the realification of the quantum space, endowed with to complex structure J_E with $J_E^2 = -I$. Let $H = H^*$ to self-adjoint operator with common invariant domain and is C an orthogonal involution that satisfies

$$HJ_E = J_E H, \qquad CJ_E C^{-1} = -J_E, \qquad CHC^{-1} = H. \tag{85.5}$$

Theorem: Bidirectional reversal of the evolution group

The operator

$$U(t) = \exp\left(-\frac{J_E H t}{\hbar_{\text{int}}}\right) \tag{85.6}$$

forms a uniparametric unitary group and satisfies

$$\boxed{CU(t)C^{-1} = U(-t)}. \tag{85.7}$$

Proof of the bidirectional reversal of the unitary group $U(t)$: $CU(t)C^{-1} = U(-t)$ The generator $A = -J_E H/\hbar_{\text{int}}$ is skew-adjoint. Stone's theorem produces the unitary group $U(t) = e^{tA}$. The hypotheses of (85.5) give $CAC^{-1} = -A$; functional calculus yields $Ce^{tA}C^{-1} = e^{-tA}$.

The involution $C_M = -\,\mathrm{rev}$ of the route words supplies the genealogical datum of orientation. When the physical functor carries C_M to the involution C of the theorem, the particle and antiparticle sections are represented by conjugate temporal orientations of the same unitary group. The identity is proved from explicit starting structure: self-adjointness, domain, complex structure, and involution.

85.1.4 Advanced and Retarded Propagators

Let L be the hyperbolic operator of a field realization and let G_{ret} and G_{adv} be its fundamental inverses with future and past causal supports, respectively:

$$LG_{\mathrm{ret}} = LG_{\mathrm{adv}} = I, \qquad \operatorname{supp}(G_{\mathrm{ret}} f) \subset J^{+}(\operatorname{supp} f), \qquad \operatorname{supp}(G_{\mathrm{adv}} f) \subset J^{-}(\operatorname{supp} f). \tag{85.8}$$

The causal difference $E = G_{\mathrm{ret}} - G_{\mathrm{adv}}$ transports data between solutions, and the symmetric mean

$$G_{\mathrm{sym}} = \frac{1}{2}(G_{\mathrm{ret}} + G_{\mathrm{adv}})$$

realizes a time-symmetric kernel.

An HMT–MD interface to this theory is a family of maps

$$\mathfrak{I}_{\mathrm{prop}} : (\gamma, \gamma^{\vee}, \mathcal{B}_{\gamma}, U, C_M) \longrightarrow (G_{\mathrm{ret}}, G_{\mathrm{adv}}, E, \mathcal{H}_{\mathrm{field}}) \tag{85.9}$$

which intertwines the route involution with the exchange $G_{\mathrm{ret}} \leftrightarrow G_{\mathrm{adv}}$, preserves the inner product or the relevant symplectic form, and carries the actions on both sides to the same functional form. This map is the mathematical condition that turns the internal bidirectional orientation into advanced–retarded propagation of fields.

85.1.5 HMT–MD realization of the absorber through route inversion and particle–antiparticle conjugation

Bidirectional emission–absorption system

A physical realization of an absorber is a boundary-value problem formed by an emitting current J_{em}, an absorbing response J_{abs}, the propagators G_{ret} and G_{adv}, and boundary conditions that jointly close the two orientations of transport. Its temporally symmetric effective action has the form

$$\mathcal{S}_{\mathrm{abs}}[J] = \mathcal{S}_{\mathrm{matter}}[J] + \frac{1}{2}\langle J, G_{\mathrm{sym}} J\rangle, \qquad J = J_{\mathrm{em}} + J_{\mathrm{abs}}, \tag{85.10}$$

and its boundary conditions determine the observed radiative part.

The definition expresses a theory of action at a distance or boundary response in operator language. Within HMT–MD, the band supplies the pair $(\gamma, \gamma^{\vee})$; the bidirectional-reversal result supplies the exact reversal of the unitary group; the functor $\mathfrak{I}_{\mathrm{prop}}$ must supply causality, an inner product, and boundary conditions. The absorber architecture prolongs the bidirectional structure already constructed. Its realization remains a physical interface, and the construction of the complete causal intertwiner constitutes an explicitly delimited mathematical problem.

The physical closure requires four verifiable identities:

1. commutativity between route conjugation and interchange of the two propagators;
2. equality of the actions transported by $\mathfrak{I}_{\mathrm{prop}}$;
3. preservation of the symplectic form or the inner product;
4. compatibility of the emission and absorption conditions under route refinement.

TO failure in any of these identities refutes the particular realization and leaves intact the operatorial reversion already proved in (85.7).

85.1.6 The separate actions of C, P and T

On a complete route we define three typed involutions:

1. C exchanges the sheets $\Sigma \leftrightarrow \Pi$ and conjugates the oriented characters;
2. P reflects the spatial chart, for example $(i, j) \mapsto (i, -j) \pmod 9$, together with $E \leftrightarrow W$;
3. T reverses the order of the record, replaces each direction with its opposite, and permutes the endpoints of the route.

Each operation acts on to distinct component of the state. Their composition CPT acts on sheet, space, and temporal orientation at once.

Theorem: Individual and joint discrete invariance

For every route γ and every $\lambda \geq 0$,

$$\mathcal{A}_\lambda(\gamma) = \mathcal{A}_\lambda(C\gamma) = \mathcal{A}_\lambda(P\gamma) = \mathcal{A}_\lambda(T\gamma) = \mathcal{A}_\lambda(CPT\gamma). \tag{85.11}$$

Compatibility of the Actions C, P, and T with the Unitary Reversal $U(t) \mapsto U(-t)$ P permutes the alphabet of directions and preserves the pattern of equalities between consecutive steps. C globally permutes the sheets and preserves the pattern of sheet changes. T reverses the order of both sequences and preserves the number of adjacent transitions. Each involution separately preserves the two summands of (85.1); the composition inherits the invariance.

The theorem belongs to the discrete ledger and its local action. The CKM matrix realized in the flavor basis has a nonzero Jarlskog invariant and therefore CP violation in that representation. Both results are compatible: an oriented reader can distinguish states related by CP while the local functional $\mathcal{A}_\lambda$ preserves each involution. The realization of the relativistic field-theoretic CPT theorem adds locality, Lorentz covariance, spectral structure, and a representation of fields. In that enlarged domain, joint CPT constitutes the symmetry to be preserved. In the internal domain, the CKM closure and the temporal expansion remain compatible with the CPT invariance proved for the discrete action.

Three strata are thereby kept separate. The invariance of (85.11) is an exact internal theorem. The identity (85.7) has been proved from explicit starting structure. The physical absorber theory and the CPT theorem for fields require the intertwiners proper to their codomains. This separation keeps the inverse temporal orientation within the definition of a particle and accords each assertion its appropriate force.

85.2 C-even absorbing operator and boundary condition

Let

$$L : \mathcal{D}(L) \subset \mathcal{H}_\Phi \longrightarrow \mathcal{H}_J$$

a Green-hyperbolic operator. Let C_Φ and C_J be isometric involutions that reverse temporal orientation and satisfy

$$C_J L = L C_\Phi.$$

Let $G_{\mathrm{ret}}, G_{\mathrm{adv}} : \mathcal{H}_J \to \mathcal{H}_\Phi$ be the unique Green functions and suppose

$$C_\Phi G_{\mathrm{ret}} C_J^{-1} = G_{\mathrm{adv}}, \qquad G_{\mathrm{adv}} = G_{\mathrm{ret}}^\dagger$$

with respect to the declared Green's pairing.

On the space of response operators we define

$$\Theta(K) = C_\Phi K C_J^{-1}, \qquad \mathsf{P}_\pm(K) = \frac{1}{2}(K \pm \Theta(K)).$$

Then

$$G_{\mathrm{WF}} := \mathsf{P}_+(G_{\mathrm{ret}}) = \frac{1}{2}(G_{\mathrm{ret}} + G_{\mathrm{adv}}), \qquad G_{\mathrm{rad}} := \mathsf{P}_-(G_{\mathrm{ret}}) = \frac{1}{2}(G_{\mathrm{ret}} - G_{\mathrm{adv}}).$$

Let $\gamma_\partial : \mathcal{H}_\Phi \to \mathcal{H}_\partial$ be the boundary trace, and define

$$\mathcal{A}_\partial := \gamma_\partial G_{\mathrm{rad}} : \mathcal{H}_J \longrightarrow \mathcal{H}_\partial.$$

At a finite HMT level, let $\mathcal{A}_\partial^+$ be its Moore–Penrose inverse and

$$P_{\mathrm{abs}} = I - \mathcal{A}_\partial^+ \mathcal{A}_\partial.$$

Theorem 85.1 (Closure of the absorbing operator HMT)**.** *The operator P_{abs} is the orthogonal projection onto* $\ker \mathcal{A}_\partial$*. For every $j \in \mathcal{H}_J$, the current*

$$j_{\mathrm{abs}} = P_{\mathrm{abs}} j$$

satisfies

$$\gamma_\partial G_{\mathrm{ret}} j_{\mathrm{abs}} = \gamma_\partial G_{\mathrm{adv}} j_{\mathrm{abs}}.$$

Moreover,

$$\Theta(G_{\mathrm{WF}}) = G_{\mathrm{WF}}, \qquad \Theta(G_{\mathrm{rad}}) = -G_{\mathrm{rad}}, \qquad G_{\mathrm{WF}}^\dagger = G_{\mathrm{WF}}.$$

Therefore,

$$\mathcal{S}_{\mathrm{WF}}[j_{\mathrm{abs}}] = \frac{1}{2}\langle j_{\mathrm{abs}}, G_{\mathrm{WF}} j_{\mathrm{abs}} \rangle$$

is real and its variation produces the response $G_{\mathrm{WF}} j_{\mathrm{abs}}$. The supports of G_{WF} are contained in the union of the retarded and advanced causal supports.

Proof. The Moore–Penrose identities give $P_{\mathrm{abs}}^2 = P_{\mathrm{abs}} = P_{\mathrm{abs}}^\dagger$ and $\operatorname{im} P_{\mathrm{abs}} = \ker \mathcal{A}_\partial$. From $\mathcal{A}_\partial j_{\mathrm{abs}} = 0$ one obtains the equality of boundary traces. The parities follow from $\Theta^2 = I$, and the self-adjointness of G_{WF} follows from $G_{\mathrm{adv}} = G_{\mathrm{ret}}^\dagger$. The support assertion is inherited from the two Green operators. □

The absorbing class is nontrivial whenever

$$\operatorname{rank} \mathcal{A}_\partial < \dim \mathcal{H}_J.$$

This inequality is the finite falsifier of nonemptiness; it is not replaced by a narrative assertion.

If the refinements $R_m^\Phi, R_m^J, R_m^\partial$ intertwine $C, L, G_{\mathrm{ret}}, G_{\mathrm{adv}}$ and γ_∂, then

$$R_m^\Phi G_{\mathrm{WF},m} = G_{\mathrm{WF},m+1} R_m^J, \qquad R_m^\partial \mathcal{A}_{\partial,m} = \mathcal{A}_{\partial,m+1} R_m^J.$$

In that compatible domain, the absorbed response forms a natural system by levels. Its identification with Wheeler–Feynman electrodynamics, a specific material current, and its experimental data is EXTERNAL RECOGNITION; the boundary operator has already been constructed.

85.3 Bidirectional propagation and absorption condition at the boundary

APP–TRIT–TPK transport preserves the route orientation and memory distinguishing its outward journey from its return. On that antecedent, three successive operations are constructed: the action of an enriched history, propagation of a source through the realized transports and imposition of a common condition at the boundary. Inscription of the record in an apparatus space, established in section 84.13.3, allows those operations to be expressed through linear maps without confusing the observable publication with the state preserving its antecedents.

Temporal direction, sheet conjugation and spatial reflection have distinct domains. Their relations are first fixed on the record; the intertwinings used by a dynamical realization are then specified. This order determines which information remains in the propagators and which condition is introduced by choosing an absorbing boundary.

85.3.1 Route action and record involutions

A finite route contains its positions, cardinal directions $(d_1, \dots, d_N)$, sheets $(\mu_1, \dots, \mu_N)$, with $\mu_t \in \{\Sigma, \Pi\}$, and the increments of its enriched record. For $\lambda \geq 0$, we define

$$\mathcal{A}_\lambda(\gamma) = \sum_{t=2}^{N} \big(1 - \delta_{d_t, d_{t-1}} + \lambda(1 - \delta_{\mu_t, \mu_{t-1}})\big). \tag{85.12}$$

The first term counts changes of direction; the second counts sheet changes with the declared weight. For $N \leq 1$, the sum is zero. The sector's general action additionally preserves its boundary and memory terms:

$$\mathcal{S}[\gamma] = \sum_{a \subset \gamma} \mathcal{L}_a + \mathcal{B}_{\mathrm{in}}(\partial_- \gamma) + \mathcal{B}_{\mathrm{out}}(\partial_+ \gamma) + \mathcal{M}(\gamma). \tag{85.13}$$

The terms of this expression specify a variational class with typed endpoints and preserved memory. The sector's dynamics determines $\mathcal{L}_a$, the admissible conditions and the term $\mathcal{M}$.

On the records we define C_{r} as the global exchange of sheets; $\mathcal{P}_{\mathrm{r}}$ as a reflection of the spatial chart and its directions; and $\mathcal{T}_{\mathrm{r}}$ as reversal of temporal order, replacement of each direction by its opposite and exchange of endpoints. Cardinal reflection is an involutive permutation. Inversion of the signed record is $C_M(a_1, \dots, a_N) = (-a_N, \dots, -a_1)$. The sheets, quotients and memory are transported with the complete route; the scalar reading of a word constitutes a subsequent operation.

Proposition 85.2 (Invariance of the local route action)**.** *On the preceding domain of records,*

$$\mathcal{A}_\lambda(\gamma) = \mathcal{A}_\lambda(C_{\mathrm{r}}\gamma) = \mathcal{A}_\lambda(\mathcal{P}_{\mathrm{r}}\gamma) = \mathcal{A}_\lambda(\mathcal{T}_{\mathrm{r}}\gamma) = \mathcal{A}_\lambda(C_{\mathrm{r}}\mathcal{P}_{\mathrm{r}}\mathcal{T}_{\mathrm{r}}\gamma).$$

Proof. A global permutation of the direction alphabet preserves equality between consecutive directions. Exchanging the two sheets preserves equality between consecutive sheets. Reversing the order establishes a bijection between adjacent pairs, and replacement by opposite directions also preserves their equalities. Each operation preserves the two counts of (85.12); their composition preserves them. □

The proposition determines the symmetry of the local functional. For a transformation also to act on a dynamical problem with fixed boundaries, it must transport its class of admissible prolongations and the other three terms of (85.13). In particular, two routes with the same final publication may retain distinct variational classes through their memory.

An interaction between two sections realized at a boundary B uses a coupling map

$$\mathcal{I}_B : E_B^{(1)} \otimes E_B^{(2)} \longrightarrow E_B^{(12)}$$

and a term $\mathcal{S}_{\mathrm{int}}$ in the same action. Its covariance is expressed by

$$\rho_{12}(g)\mathcal{I}_B = \mathcal{I}_B\big(\rho_1(g) \otimes \rho_2(g)\big).$$

When composing the histories, the shared boundary record is counted once; memories on both sides are transported to the composite section. The coupling and this covariance condition are data of the concrete interaction, distinct from the record involutions.

85.3.2 Reversal of the realized evolution

The realization of the record provides a real Hilbert space $\mathcal{H}_{\mathbb{R}}$, an orthogonal complex structure J_E, with $J_E^2 = -I$, and a self-adjoint Hamiltonian H. Its domain $\mathcal{D}(H)$ is invariant under J_E; we assume $HJ_E = J_E H$ on that domain. Let C be an orthogonal involution preserving $\mathcal{D}(H)$ and satisfying

$$CJ_E C^{-1} = -J_E, \qquad CHC^{-1} = H. \tag{85.14}$$

The internal action constant $\hbar_{\mathrm{int}} > 0$ retains its prior generation; here it normalizes the evolution group.

Theorem 85.3 (Temporal conjugation of the group)**.** *The generator $A = -J_E H/\hbar_{\mathrm{int}}$, with domain $\mathcal{D}(H)$, is skew-adjoint. Its orthogonal group, unitary with respect to J_E, satisfies*

$$U(t) = \exp(-J_E H t/\hbar_{\mathrm{int}}), \qquad CU(t)C^{-1} = U(-t). \tag{85.15}$$

Proof. Orthogonality and $J_E^2 = -I$ imply $J_E^* = -J_E$. Domain invariance and commutation with H allow H to be treated as a complex self-adjoint operator in the space defined by J_E. By functional calculus, $-J_E H/\hbar_{\mathrm{int}}$ is skew-adjoint and generates the stated group. In finite dimension, the same statements follow from the exponential series and from $A^* = -A$. The hypotheses on C give $CAC^{-1} = -A$. Conjugating the group, or its functional calculus, produces $\exp(-tA) = U(-t)$. □

The map realizing the signed involution C_M as C preserves temporal orientation and the preceding domain. Charge conjugation of a field representation additionally preserves its own characters and operators. The equality (85.15) fixes the time reversal of the group under the stated conditions.

85.3.3 Propagators of the chronological channel of a route

Isometric inscription of the record admits the unitary extensions of theorem 84.25. Let us fix a bilateral realization of a chronological channel with fibers $\mathcal{H}_n$, $n \in \mathbb{Z}$, and unitary transports $U_n : \mathcal{H}_n \to \mathcal{H}_{n+1}$. The choice includes auxiliary spaces when required; it preserves the route and its preceding records. We denote by $U(n,k)$ the transport from k to n, with

$$U(k,k) = I, \qquad U(n,k)U(k,l) = U(n,l), \qquad U(n,k)^* = U(k,n).$$

For $n > k$, it is the product $U_{n-1} \cdots U_k$; for $n < k$, it is the inverse of the transport from n to k.

The source space $\mathcal{J}_c$ contains finitely supported sections. The response space $\mathcal{F}$ contains all sections of those fibers. The route's quadratic action, in internal units of this realization, uses

$$(D\psi)_n = \psi_{n+1} - U_n \psi_n, \qquad E[\psi] = \sum_n \|(D\psi)_n\|^2$$

for finitely supported sections. The kinetic operator with the sign convention $L = -D^*D$ is

$$(L\psi)_n = U_n^* \psi_{n+1} - 2\psi_n + U_{n-1}\psi_{n-1}. \tag{85.16}$$

Indeed, $(D^* z)_n = z_{n-1} - U_n^* z_n$, and substitution produces that formula. The variation of E is $-2\,\mathrm{Re}\langle \delta\psi, L\psi\rangle$.

Theorem 85.4 (Exact Green operators of the covariant difference)**.** *Writing $x_+ = \max(x, 0)$, the maps*

$$(G_{\mathrm{ret}} j)_n = \sum_k (n-k)_+ U(n,k)\, j_k, \tag{85.17}$$

$$(G_{\mathrm{adv}} j)_n = \sum_k (k-n)_+ U(n,k)\, j_k \tag{85.18}$$

from $\mathcal{J}_c$ to $\mathcal{F}$ satisfy

$$LG_{\mathrm{ret}}j = LG_{\mathrm{adv}}j = j.$$

The retarded response vanishes before and at the first position of the source; the advanced one vanishes after and at the last. They are the unique solutions vanishing sufficiently far in the respective direction. For finitely supported ψ, we also have $G_{\mathrm{ret}}L\psi = G_{\mathrm{adv}}L\psi = \psi$. With respect to the pairing with finitely supported sources, $G_{\mathrm{adv}} = G_{\mathrm{ret}}^{\dagger}$.

Proof. Each sum has as many terms as source positions. Composition of transports reduces application of L to the scalar identity

$$(n+1-k)_+ - 2(n-k)_+ + (n-1-k)_+ = \delta_{nk}.$$

The function $(k-n)_+$ satisfies the same identity. This proves the two equations and the supports. For uniqueness, a homogeneous solution vanishing at two consecutive positions is determined as zero at the next by (85.16); the recurrence, using invertible transports, extends in both directions. The difference of two solutions with the same causal support vanishes sufficiently far away and is therefore zero. Applying this argument to ψ and to $G_{\mathrm{ret}}L\psi$, or to the advanced case, gives the right-hand identities on finite support. Finally, for two compactly supported sources, the double sum of the pairing is finite and

$$\big((k-n)_+U(k,n)\big)^* = (k-n)_+U(n,k).$$

Interchanging the indices proves the stated adjoint relation. □

The responses belong to $\mathcal{F}$; in general they grow linearly outside the source and do not belong to ℓ^2. The finite-support domain and its pairing specify the Green adjunction used here. The result constructs propagation of the chronological channel from its action; a realization of spatial fields additionally preserves its own operator, geometry and propagation domain.

Let C_0 be an isometric involution, linear or antilinear, of $\mathcal{H}_0$. Transport from the reference fiber constructs

$$C_n = U(-n,0)C_0U(0,n) : \mathcal{H}_n \longrightarrow \mathcal{H}_{-n}, \qquad (C\psi)_n = C_{-n}\psi_{-n}.$$

We have $C_{-n}C_n = I$ and $C_{n+1}U_n = U_{-n-1}^*C_n$, by composition. In particular, $C^2 = I$, and the change $(n,k) \mapsto (-n,-k)$ in the preceding sums proves

$$CL = LC, \qquad CG_{\mathrm{ret}}C^{-1} = G_{\mathrm{adv}}. \tag{85.19}$$

When C_0 is the declared realization of the record involution, these formulas transport that involution throughout the channel, without identifying phase return with a memory reset.

85.3.4 Causal interchange and decomposition of the response

In a field realization, let $L : \mathcal{D}(L) \subset \mathcal{H}_\Phi \to \mathcal{H}_J$ be an operator with unique retarded and advanced Green operators on the domains of admissible sources. We denote by $J^+(S)$ and $J^-(S)$ the future and past causal domains of a support S. The conditions are

$$\begin{aligned} LG_{\mathrm{ret}} = LG_{\mathrm{adv}} = I,\\ \operatorname{supp}(G_{\mathrm{ret}}j) \subset J^+(\operatorname{supp} j),\\ \operatorname{supp}(G_{\mathrm{adv}}j) \subset J^-(\operatorname{supp} j). \end{aligned}$$

Let C_Φ, C_J be isometric involutions, both linear or both antilinear, that transport these domains, reverse their causal orientations, and satisfy $LC_\Phi = C_JL$. A Green pairing is retained for which $G_{\mathrm{adv}} = G_{\mathrm{ret}}^{\dagger}$.

Proposition 85.5 (Conjugation of the Green operators and parities)**.** *Under the preceding conditions,*

$$C_\Phi G_{\mathrm{ret}} C_J^{-1} = G_{\mathrm{adv}}.$$

On the real space of response operators, define

$$\Theta(K) = C_\Phi K C_J^{-1}, \qquad \mathsf{P}_\pm(K) = \tfrac{1}{2}(K \pm \Theta(K)).$$

These are complementary projections for the involution Θ*, and*

$$G_{\mathrm{sym}} = \tfrac{1}{2}(G_{\mathrm{ret}} + G_{\mathrm{adv}}), \qquad G_{\mathrm{rad}} = \tfrac{1}{2}(G_{\mathrm{ret}} - G_{\mathrm{adv}}) \tag{85.20}$$

satisfy $\Theta(G_{\mathrm{sym}}) = G_{\mathrm{sym}}$, $\Theta(G_{\mathrm{rad}}) = -G_{\mathrm{rad}}$, $G_{\mathrm{sym}}^\dagger = G_{\mathrm{sym}}$, *and* $LG_{\mathrm{rad}} = 0$*. The support of the symmetric response is contained in the union of the two causal domains.*

Proof. The conjugated operator solves the source equation, since

$$LC_\Phi G_{\mathrm{ret}} C_J^{-1} = C_J L G_{\mathrm{ret}} C_J^{-1} = I.$$

Orientation reversal takes its support to the advanced domain; its uniqueness identifies it with G_{adv}. The involutions give $\Theta^2 = I$; expanding $(I \pm \Theta)^2$ and $(I + \Theta)(I - \Theta)$ proves the statements about projections. Green adjunction proves symmetry and the difference of the two source equations gives $LG_{\mathrm{rad}} = 0$. The sum of two responses has support contained in the union of their supports. □

If the involutions are antilinear, Θ is linear over real scalars; this is the operator space used in the proposition. Conjugation of the temporal group, the support condition, and the Green adjoint relation are three distinct properties, transmitted by the corresponding realization maps.

85.3.5 Projection of absorbed currents and variation of the action

Let $\gamma_\partial : \mathcal{H}_\Phi \to \mathcal{H}_\partial$ be the boundary reading, distinct from the route γ. We define

$$\mathcal{A}_\partial = \gamma_\partial G_{\mathrm{rad}}.$$

The finite window fixes the source space and boundary reading; the Green operators preserve their response domain. At a finite level with fixed inner products, its Moore–Penrose inverse determines

$$P_{\mathrm{abs}} = I - \mathcal{A}_\partial^+ \mathcal{A}_\partial. \tag{85.21}$$

Theorem 85.6 (Absorption condition and boundary response)**.** P_{abs} *is the orthogonal projection onto* $\ker \mathcal{A}_\partial$*. For any source* j*, the source* $j_{\mathrm{abs}} = P_{\mathrm{abs}} j$ *satisfies*

$$\gamma_\partial G_{\mathrm{ret}} j_{\mathrm{abs}} = \gamma_\partial G_{\mathrm{adv}} j_{\mathrm{abs}}. \tag{85.22}$$

It is the absorbed source closest to j *for the declared inner product. Nonzero absorbed sources exist exactly when* $\operatorname{rank} \mathcal{A}_\partial < \dim \mathcal{H}_J$.

Proof. The orthogonal decomposition $\mathcal{H}_J = \ker \mathcal{A}_\partial \oplus \operatorname{im} \mathcal{A}_\partial^*$ shows that $\mathcal{A}_\partial^+ \mathcal{A}_\partial$ is the identity on the second summand and zero on the first. Its complement is the stated projector. By definition, $2\mathcal{A}_\partial j_{\mathrm{abs}} = \gamma_\partial (G_{\mathrm{ret}} - G_{\mathrm{adv}}) j_{\mathrm{abs}} = 0$. For $k \in \ker \mathcal{A}_\partial$, the orthogonal decomposition gives

$$\|j - k\|^2 = \|(I - P_{\mathrm{abs}}) j\|^2 + \|P_{\mathrm{abs}} j - k\|^2.$$

The unique minimum corresponds to $k = P_{\mathrm{abs}} j$. The final equivalence is the rank–nullity theorem. □

In the source-field pairing, the symmetric action is $\mathcal{S}[j] = \frac{1}{2}\langle j, G_{\mathrm{sym}} j\rangle$. The declared adjunction makes that quantity real. For real variations,

$$\mathrm{d}\mathcal{S}[j](h) = \mathrm{Re}\langle h, G_{\mathrm{sym}} j\rangle.$$

The action restricted to absorbed sources is written on the ambient space as

$$\mathcal{S}_{\mathrm{abs}}[j] = \tfrac{1}{2}\langle P_{\mathrm{abs}} j, G_{\mathrm{sym}} P_{\mathrm{abs}} j\rangle, \quad \mathrm{d}\mathcal{S}_{\mathrm{abs}}[j](h) = \mathrm{Re}\langle P_{\mathrm{abs}} h, G_{\mathrm{sym}} P_{\mathrm{abs}} j\rangle. \tag{85.23}$$

When the pairing identifies field and source through Riesz, the ambient gradient is $P_{\mathrm{abs}} G_{\mathrm{sym}} P_{\mathrm{abs}} j$. The field response remains $G_{\mathrm{sym}} j_{\mathrm{abs}}$; the gradient projector expresses restriction of the variations. The formulas are proved by expanding the quadratic term of $j + th$ and using self-adjointness. A material term $\mathcal{S}_{\mathrm{matter}}$ adds its own differential.

85.3.6 Explicit absorbing condition in a chronological window

The channel of subsection 85.3.3 allows that condition to be evaluated without introducing a target response. Let there be a source supported on $0, \dots, N$, with $N \geq 1$. Let us transport its components to the reference fiber:

$$\widehat{j}_k = U(0,k) j_k, \qquad M_0(j) = \sum_{k=0}^{N} \widehat{j}_k, \qquad M_1(j) = \sum_{k=0}^{N} k\, \widehat{j}_k.$$

The identity $(n-k)_+ - (k-n)_+ = n - k$ gives

$$(G_{\mathrm{ret}} - G_{\mathrm{adv}}) j\big|_n = U(n,0)\big(n M_0(j) - M_1(j)\big). \tag{85.24}$$

Proposition 85.7 (Absorption and two transported moments)**.** *If the trace evaluates the field at two distinct positions a, b, equality of the advanced and retarded traces is equivalent to $M_0(j) = M_1(j) = 0$. In a fibre of dimension d, the space of such sources has dimension $(N-1)d$. For $N \geq 2$, it contains the nonzero sources with transported components $(z, -2z, z, 0, \dots, 0)$, $z \neq 0$.*

Proof. Invertibility of $U(a,0)$ and $U(b,0)$ reduces the boundary condition to $aM_0 - M_1 = 0$ and $bM_0 - M_1 = 0$. Subtracting the equations gives $(a-b)M_0 = 0$, and then $M_1 = 0$. The two-moment map is surjective: for prescribed values (z_0, z_1), it suffices to take $\widehat{j}_0 = z_0 - z_1$, $\widehat{j}_1 = z_1$, and all other components zero. Its rank is $2d$; its nullity is $(N+1)d - 2d$. The indicated source has both moments zero. □

In the transported frame, write

$$B_N = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 0 & 1 & \cdots & N \end{pmatrix}.$$

The absorbing projector is evaluated by

$$P_N = \big[I - B_N^{\mathsf{T}} (B_N B_N^{\mathsf{T}})^{-1} B_N\big] \otimes I_{\mathcal{H}_0}. \tag{85.25}$$

The two rows of B_N are independent for $N \geq 1$, so the inverse exists. Unitary transport between frames takes this projector to the original fibers. For $N = 2$,

$$P_2 = \frac{1}{6}\begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{pmatrix} \otimes I_{\mathcal{H}_0}.$$

This construction produces a nonzero absorbed class and its projection operation in the chronological channel. Its amplitude and identification with a material current preserve the rules of the corresponding physical sector.

85.3.7 Refinement and continuity of the boundary condition

Let $R_m^{\Phi}, R_m^{J}, R_m^{\partial}$ be the refinement maps of fields, sources and boundaries. Suppose they intertwine the two Green operators and the trace. Composition gives

$$R_m^{\Phi} G_{\mathrm{sym},m} = G_{\mathrm{sym},m+1} R_m^{J}, \qquad R_m^{\partial} \mathcal{A}_{\partial,m} = \mathcal{A}_{\partial,m+1} R_m^{J}. \tag{85.26}$$

The second equality takes an absorbed source to another absorbed source. Naturality of the projectors additionally preserves the orthogonal structure of the realizations.

Proposition 85.8 (Naturality of the absorbing projection)**.** *For a source isometry $R = R_m^{J}$, we have $P_{\mathrm{abs},m+1} R = R P_{\mathrm{abs},m}$ exactly when*

$$R(\ker \mathcal{A}_{\partial,m}) \subset \ker \mathcal{A}_{\partial,m+1}, \qquad R((\ker \mathcal{A}_{\partial,m})^{\perp}) \subset (\ker \mathcal{A}_{\partial,m+1})^{\perp}.$$

With these conditions at every level, the projectors define an orthogonal projection in the completion of the isometric inductive limit of the source spaces.

Proof. Each source is decomposed into its components in the kernel and its orthogonal complement. The two inclusions imply that applying the projection before or after refinement respectively preserves the first component and annihilates the second. Conversely, equality of operators applied to each component gives the two inclusions. In the inductive limit, the equality allows an operator to be defined on the union of the levels without dependence on the representative. All projectors have norm at most one, so the operator extends to the completion. Its self-adjointness and idempotence identities, valid on the dense union, pass by continuity to the completion. □

To extend a Green response to a completion as well, its domains and corresponding continuity estimates are preserved. Finite-support chronological propagation and factorization of the response through the boundary image, proved in 84.27, thus retain their respective domains during refinement.

The physical realization of emission and absorption composes these operators with the material currents, action and causal conditions of the sector. The intertwining must preserve the route involution, the source-field pairing, the transported action and the boundary conditions at each level. The preceding construction provides reversal of the group, the Green operators of the chronological channel, temporal decomposition and an explicit absorbing projection; spatial realizations use their own maps and causality conditions.

Chapter 86

Causal polarization of the gravitational interior: loss of spatial freedom, tilting of the cones, and compulsory direction of infall

86.1 Causal polarization field and radial freedom loss

Let $(\mathcal{M}, g)$ be a spherically symmetric geometry written in Painlevé–Gullstrand coordinates,

$$\mathrm{d}s^2 = -c^2\,\mathrm{d}t^2 + \big(\mathrm{d}r + v(r)\,\mathrm{d}t\big)^2 + r^2\,\mathrm{d}\Omega_2^2, \tag{86.1}$$

where $v(r) \geq 0$ measures the radial inclination of the causal cones with respect to the selected temporal congruence. The dimensionless field

$$\mathcal{P}(r) := \frac{v(r)}{c} \tag{86.2}$$

is the intensity of causal polarization. Over the spatial distribution orthogonal to the temporal congruence, the tensor is defined

$$\Pi(r) = \mathcal{P}(r)^2\,\widehat{r}\otimes\widehat{r}, \tag{86.3}$$

whose principal direction is radial and whose tangential directions remain degenerate. This definition introduces no absolute direction: the principal direction is covariantly determined by the gradient of the areal radius.

The null radial generators of (86.1) satisfy

$$\frac{\mathrm{d}r}{\mathrm{d}t} = -v(r) \pm c = c\big(-\mathcal{P}(r) \pm 1\big). \tag{86.4}$$

Therefore, the set of radially accessible signs changes cardinality upon crossing $\mathcal{P} = 1$.

Theorem 86.1 (Mandatory causal direction of fall)**.** *In the geometry* (86.1) *the following three exhaustive alternatives are verified:*

1. *if* $0 \leq \mathcal{P} < 1$*, one future-directed radial generator increases* r *and the other decreases it;*
2. *if* $\mathcal{P} = 1$*, the outgoing generator satisfies* $\mathrm{d}r/\mathrm{d}t = 0$ *and defines the horizon;*
3. *if* $\mathcal{P} > 1$*, both future radial generators satisfy* $\mathrm{d}r/\mathrm{d}t < 0$*.*

In the third regime, the radial degree does not disappear from the manifold, but ceases to constitute a bidirectional spatial choice: every future-directed causal vector has a component directed toward smaller areal radii. The interior is, in this exact sense, a region of compelled causal infall.

Proof. The three conclusions are obtained by comparing the two values $-\mathcal{P} \pm 1$ of (86.4) with zero. For $\mathcal{P} > 1$, even the branch with positive sign is negative. The conclusion concerns the causal cone rather than a particular choice of temporal coordinate. □

86.2 Schwarzschild realization and collective gradient of fall

For a total mass M, the Schwarzschild realization in the chart (86.1) is given by

$$v(r) = c\sqrt{\frac{R}{r}}, \qquad R = \frac{2GM}{c^2}, \qquad \mathcal{P}(r)^2 = \frac{R}{r}. \tag{86.5}$$

The horizon is simultaneously characterized by

$$r = R, \qquad \mathcal{P} = 1, \qquad g^{\mu\nu}\partial_\mu r\,\partial_\nu r = 0. \tag{86.6}$$

The radial acceleration of the falling congruence is recovered as the gradient of the kinetic intensity of the field:

$$a_r = -\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}r}v(r)^2 = -\frac{GM}{r^2}. \tag{86.7}$$

Thus, radial gravitation is expressed as the collective gradient of infall and not as a force added after the trajectories have been fixed.

Corollary 86.2 (Causal Macropolarization of the Gravitational Interior)**.** *In the realization* (86.5)*, the region $r < R$ satisfies $\mathcal{P} > 1$. The black hole constitutes to causal macropolarization: the principal direction of Π exceeds the unit threshold and aligns both future radial families toward the decrease of the areal radius.*

86.3 Correspondencia with the 3 regimes of the TRIT

The geometric reader of the TRIT is implemented through

$$\tau(r) = \operatorname{sgn}\!\left(1 - \mathcal{P}(r)^2\right) \in \{+1, 0, -1\}. \tag{86.8}$$

The correspondence is

τ	$\mathcal{P}$	geometric regime	future radial orientation
$+1$	$\mathcal{P} < 1$	exterior	input and output
0	$\mathcal{P} = 1$	horizon	outgoing null generator
-1	$\mathcal{P} > 1$	interior	necessary descent

(86.9)

The temporal involution of TRIT exchanges the exterior and interior regimes and fixes the horizon. The change $+1 \leftrightarrow -1$ does not eliminate the memory of the route: it modifies its causal orientation and preserves the boundary $\tau = 0$ as to gluing surface.

86.4 Local closure of matter and collective gravitational alignment

Let γ_a be a persistent material section with radial projector $\widehat{r}_a \otimes \widehat{r}_a$ and closure weight $w_a \geq 0$. For a finite set of sections, define the order parameter

$$\Pi_N = \frac{1}{\sum_{a=1}^{N} w_a} \sum_{a=1}^{N} w_a \, \widehat{r}_a \otimes \widehat{r}_a. \tag{86.10}$$

An individual particle realizes a persistent local closure of a route. The gravitational interior appears when the collective field fixes a principal causal direction for all future routes. Consequently, two mathematically distinct objects are not identified:

$$\begin{aligned} \text{particle} &= \text{persistent material section}, \\ \text{black hole} &= \text{macroscopic causal alignment above the threshold}. \end{aligned} \tag{86.11}$$

The complete causal sequence of this residence is fixed by

$$\begin{aligned} \text{APP–TRIT–TPK state} &\longrightarrow \text{material section} \longrightarrow \text{polarization field } \Pi \\ &\longrightarrow \mathcal{P} = 1 \longrightarrow \text{horizon} \longrightarrow \mathcal{P} > 1 \longrightarrow \text{necessary causal descent}. \end{aligned} \tag{86.12}$$

The following chapter prolongs this structure toward sheet inversion, the advanced–retarded purification, Kruskal geometry, and horizon information.

86.5 Geometric conditions of causal macropolarization: areal radius, cone tilt, horizon, and interior falloff

The result simultaneously requires that the radius used be areal, that $v(r)$ tilt the cones according to (86.1), that the horizon satisfy $\mathcal{P} = 1$, and that the two slopes in (86.4) be negative in the interior. These identities distinguish the effective loss of radial freedom from a merely dimensional metaphor and provide direct control of causal macropolarization.

Chapter 87

Causal horizon, interior–exterior inversion, and the hyperbolic regime of TRIT in the gravitational interior

87.1 Leaf involution in a trapped region and causal representation of the interior of a black hole

In the empty exterior region of a compact body, the Cartan equation imposes zero torsion when there is no spin current. The connection reduces to $\mathring{\omega}(e)$, and the metric satisfies the Einstein sector. Orbits, gravitational lensing, redshift, and exterior propagation remain governed by relativistic geometry in the verified domain. HMT–MD locates its contribution in the meaning of the boundary and in the continuity of the state that an exterior chart ceases to publish.

In this reading, a horizon is a boundary of publication. The enriched state does not end where the external observer loses an invertible coordinate. The relation among sheet, orientation, and reading changes. A section that crosses the boundary may cease to have a representative in the external chart and continue on a conjugate sheet. The expression *sheet inversion* denotes this change: exterior and interior are not two substances, but two domains of a global presentation whose gluing preserves route and memory.

The Möbius band provides the minimal topological model. One turn transports the section to its conjugate sheet; two turns restore orientation. The local observer can publish a single branch, while the global state preserves both. In a relativistic realization, the same operation must act on the causal frames, charge, and temporal orientation without destroying the norm. The horizon thus becomes the locus at which a visible orientation becomes return memory.

Sheet inversion contains a possible explanation of the information paradox. Three operations should be distinguished. The fundamental evolution may be reversible:

$$U_{\mathrm{tot}}^{-1} U_{\mathrm{tot}} = I.$$

The external chart may identify different states:

$$q_{\mathrm{ext}}(x) = q_{\mathrm{ext}}(y), \qquad x \neq y.$$

And the external apparatus may erase its register in order to prepare itself again. The first operation preserves information; the second loses access to it; the third has a thermodynamic cost. The apparent disappearance beyond the horizon may belong to the fiber of q_{ext} without requiring destruction in U_{tot}.

The leafwise reading makes it possible to write a minimal model of that information. The angle and counter-angle channels are

$$q_{\pm} = \exp\left(-A_{\mathrm{rad}} \mp C_{\mathrm{rad}}^{*}\right).$$

After normalization,

$$p_{\pm} = \frac{q_{\pm}}{q_{+} + q_{-}}, \qquad \rho_{\mathrm{hoja}} = p_{+}|+\rangle\langle+| + p_{-}|-\rangle\langle-|.$$

The reduced entropy is

$$S_{\mathrm{hoja}} = -\sum_{\pm} p_{\pm} \log p_{\pm} = \log\left(2\cosh C_{\mathrm{rad}}^{*}\right) - C_{\mathrm{rad}}^{*} \tanh C_{\mathrm{rad}}^{*}.$$

The common phase A_{rad} cancels: the visible mixing depends on the separation between sheets. If the advanced–retarded factorization exists,

$$|\Psi\rangle = \sqrt{p_{+}}\,|+\rangle|\mathrm{adv}\rangle + \mathrm{e}^{i\chi_0}\sqrt{p_{-}}\,|-\rangle|\mathrm{ret}\rangle,$$

that entropy is the entanglement of two orientations of a pure state. A horizon may then alter the accessibility of the branches without turning the global state into a fundamental mixture.

Thermodynamics requires keeping temperature separate from duration. The Boltzmann constant is to morphism between the thermal and energetic lines,

$$k_B : L_\Theta \longrightarrow L_E.$$

For the clock of 108 states,

$$\omega_0 = \frac{2\pi}{108\, t_0}, \qquad k_B \Theta_{\mathrm{clk}} \log 3 = \hbar_{\mathrm{int}} \omega_0.$$

The information of a ternary cell, the thermal scale, and the quantum of action are thus related without conflating $\log 3$ with the topological entropy $\log \varphi_{\mathrm{HMT}}$.

The Landauer limit applies when the observer erases:

$$Q \geq k_B \Theta\, \Delta S.$$

A reversible global cycle that preserves the register may have zero logical variation. An apparatus that resets a distinguishable memory must dissipate. The so-called zero Landauer regime is the regime without erasure, not a free heat engine. For a black hole, this distinction asks whether radiation redistributes a preserved memory, whether the exterior keeps it inaccessible for a time, or whether a genuinely nonunitary channel exists.

The interpretation becomes a physical theory when it constructs a geometry. The extension must recover the relativistic exterior, produce a trapped surface and a regular horizon, transport the enriched state across the boundary, and assign a census of states to its area. The continuation must also produce a distinguishable signal: a phase of the quasinormal modes, a correlation structure in the radiation, a polarization, or an echo whose value is not chosen after it has been observed. If the inversion changes only the words *interior* and *exterior*, it does not yet realize the HMT black hole. If it satisfies these conditions, the information problem becomes a controllable problem of chart change.

87.2 Double-sheet black hole and advanced purification–delayed

87.2.1 State of sheet and information

Leaf channels are

$$q_\pm = \exp(-A_{\mathrm{rad}} \mp C^*_{\mathrm{rad}}), \qquad p_\pm = \frac{q_\pm}{q_+ + q_-}.$$

We define

$$\rho_{\mathrm{hoja}} = p_+ |+\rangle\langle +| + p_- |-\rangle\langle -|$$

and the purification

$$|\Psi_{\mathrm{ar}}\rangle = \sqrt{p_+}\,|+\rangle|\mathrm{adv}\rangle + \mathrm{e}^{\mathrm{i}\chi_0}\sqrt{p_-}\,|-\rangle|\mathrm{ret}\rangle.$$

Proposition 87.1 (Exact purification2 of the temporal orientations)**.** *The state $|\Psi_{\mathrm{ar}}\rangle$ is normalized, and its reduction over the advanced–retarded orientation is ρ_{hoja}. Its entropy is*

$$S_{\mathrm{hoja}} = -\sum_{\pm} p_\pm \log p_\pm = \log(2\cosh C^*_{\mathrm{rad}}) - C^*_{\mathrm{rad}} \tanh C^*_{\mathrm{rad}}.$$

The common phase A_{rad} cancels, and the reversible global evolution has zero logical Landauer cost while the leaf label is not erased.

Proof. The orthogonality of the two orientations eliminates the cross terms when taking the partial trace. Substituting $p_\pm = \mathrm{e}^{\mp C^*_{\mathrm{rad}}}/(2\cosh C^*_{\mathrm{rad}})$ gives the entropy. □

87.2.2 Geometric realization of the 2 causal sheets

Let $R(x) > 0$ be the horizon radius published by the HMT–MD selector, and

$$E_H(x) = \frac{c_{\rm int}^4 R(x)}{2G_{\rm int}(x)}.$$

In the exterior region $r > R(x)$, we fix

$$f(r) = 1 - \frac{R(x)}{r}, \qquad \mathrm{d}s^2 = -f(r)c_{\rm int}^2\mathrm{d}t^2 + f(r)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega_2^2.$$

Let

$$r_* = r + R\log\Big(\frac{r}{R} - 1\Big)$$

and let $\sigma(x) \in \{+1, -1\}$ be the sheet reader. Define

$$U(x;t,r) = -\sigma(x)\exp\Big[-\frac{c_{\rm int}t - r_*}{2R}\Big], \qquad V(x;t,r) = \sigma(x)\exp\Big[\frac{c_{\rm int}t + r_*}{2R}\Big].$$

Then

$$-UV = \Big(\frac{r}{R} - 1\Big)\exp(r/R).$$

Theorem 87.2 (Geometric double sheet entangler)**.** *The preceding chart realizes the two HMT sheets as the two exterior regions of the Kruskal extension. The involution*

$$C_{\rm BH} : (\sigma, U, V) \longmapsto (-\sigma, -U, -V)$$

is isometric, fixes r, exchanges the two exterior regions, and preserves the horizon $UV = 0$. The extended metric is

$$\mathrm{d}s^2 = -\frac{4R^3}{r}\mathrm{e}^{-r/R}\,\mathrm{d}U\,\mathrm{d}V + r^2\mathrm{d}\Omega_2^2,$$

regular at $r = R$, and satisfies the vacuum Einstein equations outside the source. At the horizon it preserves

$$E_H = \frac{c_{\rm int}^4 R}{2G_{\rm int}}, \qquad S_H = \frac{k_B\pi R^2}{\ell_P^2}.$$

Proof. The formula for UV is the standard transformation of the Schwarzschild factor and removes the coordinate singularity at $r = R$. The simultaneous sign change of U, V leaves UV, r, and the metric invariant. Direct substitution gives $G_{\mu\nu} = 0$ in $r > 0$ outside the source. The horizon identities depend only on the areal radius. □

Let $\mathcal{H}_{\text{hoja}}$ be the direct sum of the Hilbert completions of the two HMT sheets, and let $\mathcal{H}_{\text{ext}}$ be the direct sum of the two exterior regions with the measure transported by the chart. The change of variables, including its Jacobian, induces a unitary operator

$$W_{\text{BH}} : \mathcal{H}_{\text{hoja}} \longrightarrow \mathcal{H}_{\text{ext}}.$$

Therefore,

$$\mathfrak{I}_{\text{BH}}(A) = W_{\text{BH}} A W_{\text{BH}}^{\dagger}$$

is to unital completely positive map, indeed to $*$-isomorfismo, that preserves state, inner product, sheet involution, and modular dynamics when the exterior state is transported by W_{BH}. If the sheet and chart refinements preserve (σ, t, r, R), their isometries j_m^{hoja} and j_m^{ext} satisfy

$$W_{\text{BH},m+1} j_m^{\text{hoja}} = j_m^{\text{ext}} W_{\text{BH},m}.$$

The family $\mathfrak{I}_{\text{BH},m}$ is then natural under refinement and transports the same enriched state without loss.

Thus the mathematical construction is closed.

$$\begin{aligned} \text{HMT double sheet} &\longrightarrow \text{advanced–retarded purification} \\ &\longrightarrow \text{two Kruskal exteriors} \longrightarrow \text{regular horizon and UCP intertwiner.} \end{aligned}$$

The identification with an astrophysical object formed by collapse, its spectrum of quasinormal modes, radiation, echoes, or polarization belongs to EXTERNAL RECOGNITION and does not weaken the geometric closure.

Chapter 88

Torsional cosmology of periodicity 108: expansion, return with memory, and Einstein–Cartan bounce

The temporal periodicity of 108 phases, the elliptic, parabolic, and hyperbolic charts of the TRIT, and torsional memory coordinate the expansion, the return, the regular bounce, and continuation toward a descendant universe. The internal theorems are stated with their domains and physical premises, without reducing them to programmatic labels.

88.1 Cosmological scales associated with the 108-state cycle and the Perron operator

88.1.1 Separation between de Sitter acceleration and Einstein–Cartan torsional bounce

The construction in cosmology comprises two internal quantities of distinct provenance. The first is derived from the periodic CT108 closure and determines a curvature scale. The second is derived from the Perron cocycle and determines a law of memory dilution. Both use the same trunk APP $\to \widehat{\mathbb{T}} \to$ TPK, but do not have the same domain or the same selector:

$$\text{TPK} \longrightarrow \text{enriched state} \longrightarrow \begin{cases} \text{chronological structure 108} \longrightarrow R_{108} \longrightarrow (H_{108}, \Lambda_{108}, T_{\mathrm{dS}}, S_{\mathrm{dS}}), \\ \text{Perron cocycle} \longrightarrow (a_n, M_n) \longrightarrow (H_\varphi, s^2 \propto a^{-6}). \end{cases} \tag{88.1}$$

The eventual equality of units or order of magnitude does not warrant identifying these realizations. The first falsification criterion of the chapter is the attempt to identify them without an intertwiner that preserves the chronological structure, orientation, memory, and the dimensional chart.

88.1.2 Periodicity condition of the 108-state cycle

Let $\ell_0 = c_{\mathrm{int}} t_0$. The period and angular frequency of the return are

$$T_{108} = 108 t_0, \qquad \omega_{108} = \frac{2\pi}{108 t_0}, \qquad R_{108} = \frac{c_{\mathrm{int}}}{\omega_{108}} = \frac{54}{\pi}\ell_0. \tag{88.2}$$

The length of the full turn is

$$C_{108} = 2\pi R_{108} = 108\,\ell_0. \tag{88.3}$$

For either of the two oriented action sections $a \in \{\mathrm{pre}, \mathrm{ret}\}$, the circular quantum

$$E_{108,a} = \hbar_a \omega_{108}, \qquad m_{108,a} = E_{108,a}/c_{\mathrm{int}}^2$$

satisfies simultaneously

$$\bar{\lambda}_C(m_{108,a}) = R_{108}, \qquad \lambda_C(m_{108,a}) = C_{108}. \tag{88.4}$$

The circular realization therefore does not take an external Planck length. It first constructs a radius and then demonstrates that the reduced and complete Compton readings are its two realizations.

88.1.3 Gravitational selector and Planck–TRIT relation

The homogeneous gravitational transducer is

$$G_a(\theta) = \theta \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a}. \tag{88.5}$$

On the circular quantum, the reduced gravitational radius is equal to

$$r_{\mathrm{g},a}(\theta) = \frac{G_a(\theta) m_{108,a}}{c_{\mathrm{int}}^2} = \theta \frac{\pi}{54} \ell_0.$$

The condition $r_{\mathrm{g},a} = R_{108}$ selects in a unique mode

$$\theta_{108} = \left(\frac{54}{\pi}\right)^2. \tag{88.6}$$

The coordinate does not depend on the action sheet: upon multiplying $G_a(\theta_{108})\hbar_a$, the pre/return orientation cancels. From here there result

$$\ell_P = R_{108} = \frac{54}{\pi}\ell_0, \qquad t_P = \omega_{108}^{-1} = \frac{54}{\pi} t_0, \tag{88.7}$$

$$E_P = \hbar_a \omega_{108} = \frac{\pi}{54}\frac{\hbar_a}{t_0}, \qquad E_P = k_B \Theta_{\mathrm{clk}} \log 3. \tag{88.8}$$

The equation (88.8) expresses the structural confluence: $\log 3$ is the informational content of a TRIT decision and, in the thermal chart, converts chronological temperature into the Planck quantum. The identity does not equate the TRIT state space with gravitational space; it proves that two functional realizations determine the same scalar through distinct maps.

88.1.4 Conditioned realization of de Sitter geometry

Suppose now that the Palatini–Cartan–Holst realizer selects the internal radius R_{108} as the radius of a four-dimensional de Sitter background. The hypothesis is geometric and remains explicit: it is not deduced from a numerical coincidence with H_0. Under this hypothesis,

$$H_{108} = \frac{c_{\mathrm{int}}}{R_{108}} = \frac{\pi}{54 t_0}, \qquad \Lambda_{108} = \frac{3}{R_{108}^2} = \frac{\pi^2}{972 \ell_0^2}. \tag{88.9}$$

The Gibbons–Hawking temperature and horizon entropy are

$$k_B T_{\mathrm{dS}} = \frac{\hbar_a H_{108}}{2\pi} = \frac{\hbar_a}{108 t_0}, \qquad \frac{T_{\mathrm{dS}}}{\Theta_{\mathrm{clk}}} = \frac{\log 3}{2\pi}, \tag{88.10}$$

$$\frac{S_{\mathrm{dS}}}{k_B} = \frac{\pi R_{108}^2}{\ell_P^2} = \pi. \tag{88.11}$$

The π of equation (88.11) is not introduced as the value of an entropy to be reproduced: it appears because the circular selector gives $R_{108} = \ell_P$, and the area of the horizon sphere contains the geometric factor 4π.

Proposition 88.1 (Dimensionless family CT108–de Sitter)**.** *Under the declared de Sitter realization, the quadruple*

$$\left(H_{108} t_0, \Lambda_{108} \ell_0^2, \frac{T_{\mathrm{dS}}}{\Theta_{\mathrm{clk}}}, \frac{S_{\mathrm{dS}}}{k_B}\right) = \left(\frac{\pi}{54}, \frac{\pi^2}{972}, \frac{\log 3}{2\pi}, \pi\right)$$

is a dimensionless exact output.

Falsification criterion 88.2. The de Sitter realization is invalidated if the selected background does not satisfy $R_{\mu\nu} = \Lambda_{108} g_{\mu\nu}$, if the PCH realizer determines a radius other than R_{108}, or if the charts used in equations (88.9) y (88.10) do not share $(c_{\mathrm{int}}, t_0, \hbar_a)$. None of these failures alters the circular theorem equations (88.2) y (88.6).

88.2 Boundary conditions, bounce regularity, and decomposition of cosmological components

This chapter distinguishes two domains. The first section proves the three de Sitter foliations. Subsequent sections define the data tuples required for a cosmological realization: HMT selector, boundary chart, bounce dynamics, and descendant component. Each conclusion is restricted to the subdomain on which the evolution equations and the declared junction conditions hold.

88.2.1 Correspondence between the elliptic, parabolic, and hyperbolic regimes of the TRIT and the AdS–Euclidean–dS charts

The three quadratic algebras of the TRIT receive a coordinated geometric realization. For scales $\ell, H^{-1} > 0$, we fix the charts

$$\mathcal{M}_- = (\mathbb{D}, g_{\mathrm{AdS}}), \qquad \mathcal{M}_0 = (\mathscr{H}_0^{(L)}, g_{\mathrm{E}}), \qquad \mathcal{M}_+ = (\mathrm{dS}_4, g_{\mathrm{dS}}),$$

where the hyperbolic spatial section is represented in the Poincaré disk by

$$g_{\mathrm{AdS}}^{\mathrm{sp}} = \frac{4\ell^2\,|dz|^2}{(1-|z|^2)^2}, \qquad |z| < 1,$$

the gnomonic L carries the Euclidean chart g_{E}, and the positive chart adopts one of the de Sitter foliations constructed in the following section. The geometric functor preserves the sheet label and the change-of-chart maps:

$$\mathfrak{R}_{\mathrm{geo}} : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow \mathcal{M}_- \cup \mathcal{M}_0 \cup \mathcal{M}_+.$$

The EMRH completion distinguishes interior, crown, and edge, while the double projection preserves one and the same section before its evaluations. On the negative chart, the restriction to the edge and the incidence representation form the square

$$\begin{array}{ccc} \mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} & \xrightarrow{\mathfrak{R}_{\mathrm{AdS}}} & \mathcal{M}_- \\ {\scriptstyle \mathcal{P}_{\mathrm{exc}}}\downarrow & & \downarrow{\scriptstyle \mathrm{Res}_\partial} \\ \mathcal{E}_\partial & \xrightarrow[\mathfrak{R}_{\mathrm{conf}}]{} & \mathrm{Obs}(\partial\mathcal{M}_-), \end{array}$$

with $\mathfrak{R}_{\mathrm{conf}} \circ \mathcal{P}_{\mathrm{exc}} = \mathrm{Res}_\partial \circ \mathfrak{R}_{\mathrm{AdS}}$. This identity realizes in HMT the interior–boundary holographic correspondence: the interior state and the conformal boundary observables are two representations of the same genealogy.

The Euclidean chart centralizes the local descent of inertial and gravitational readers,

$$\mathcal{G}_L / \mathcal{I}_L = 1,$$

and constitutes the locus of the equivalence principle. The state prior to that projection preserves areal–volumetric Ambivalence,

$$\mathcal{L}_{\mathrm{ar}} / \mathcal{L}_{\mathrm{vol}} = \pi_{\mathrm{HMT}} / \varphi_{\mathrm{HMT}}^2.$$

The positive chart completes the atlas with $\ddot{a}/a = H^2 > 0$. Therefore, the same geometric triple contains the holographic correspondence in negative curvature, local equivalence in zero curvature, and cosmological acceleration in positive curvature.

88.2.2 Cosmological foliations compatible with sheet inversion

Proposition 88.3 (Closed, flat, and open foliations)**.** *For $H > 0$, the three FLRW foliations of de Sitter space of radius H^{-1} are expressed, with $a_* > 0$ in the flat chart, by*

k	$a_k(t)$	*domain*
$+1$	$H^{-1}\cosh(Ht)$	$t \in \mathbb{R}$
0	$a_* e^{Ht}$	$t \in \mathbb{R}$ *(flat patch)*
-1	$H^{-1}\sinh(Ht)$	$t > 0$ *(open patch)*

and satisfy

$$\boxed{\left(\frac{\dot a_k}{a_k}\right)^2 + \frac{k}{a_k^2} = H^2, \qquad \frac{\ddot a_k}{a_k} = H^2,}$$

$$\boxed{R^{(4)} = 12H^2, \qquad R^{(3)} = \frac{6k}{a_k^2}.}$$

Proof. For $k = +1$, the first equation uses $\tanh^2 + \operatorname{sech}^2 = 1$. For $k = -1$, it uses $\coth^2 - \operatorname{csch}^2 = 1$. In the flat chart, $\dot a/a = H$. The three functions satisfy $\ddot a/a = H^2$, and the FLRW curvature formula gives $R^{(4)} = 12H^2$. Each three-dimensional spatial slice has constant sectional curvature k/a_k^2; in dimension three, contraction of the curvature tensor therefore gives $R^{(3)} = 3(3-1)k/a_k^2 = 6k/a_k^2$. □

The sign k types the curvature of the spatial sections, not the curvature of the complete spacetime: all three charts have $R^{(4)} > 0$. The closed chart is global; the flat and open charts are patches.

Hypothesis, domain, and conclusion

For a closed return route and the declared spatial choice $D = 3$, the HMT scale cocycle contributes the exact factors $\varphi^3, \varphi^9, \varphi^{36}$ at depths $9, 27, 108$, together with a spectral memory triple and a signature screen $(3, 1)$. Cosmological selection consists in constructing, on the enriched state of the TPK, a projective measure, a scale factor, and an evolution equation that intertwine these objects with the preceding foliations. Its duration variable must be derived from the action line of the section 59.2 and from its metrological chart, rather than identified with the mere number of steps. H_0, the cosmological densities, and the sign of acceleration are not used as inputs to the selector.

The three foliations constitute the exact geometric codomain of comparison. The selector constructed on the enriched state determines the chart realized by each regime, composes their domains, and preserves the observables that distinguish the selection. Identification with $k = +1, 0, -1$ is effected through that selector and not by the mere coincidence of three signs.

88.2.3 Boundary state of the cosmological horizon and return memory

The completion

$$1000 = 729 + 271 = 729 + 243 + 27 + 1$$

distinguishes interior, boundary strata, and seal. In a cosmological realization, this stratification orients the type of a construction over a pair

$$(M, \partial M),$$

endowed with a co-tetrad, Cartan connection, matter, memory, and flow. The cardinal identity does not by itself produce a manifold, a metric, a co-tetrad, or boundary conditions. It is an organizational guide that acquires physical content only through a map

$$\mathfrak{R}_{\cos} : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \dashrightarrow \big(M, \partial M, e, \omega, \Psi, \mathcal{B}_{\partial M}\big).$$

The boundary is a constitutive part of the variational problem. Its spatial, temporal, or null character must be declared, as must the boundary and corner terms, the data that are fixed, and the compatibility conditions with the equations of motion. HMT memory provides a candidate domain for recording histories and prolongations, but its identification with geometric data requires the intertwining of the holonomies of test criterion 82.8.

88.2.4 Criterion of regularity of the bounce under extreme concentration and torsional dominance

An unbounded concentration of energy density, action, or curvature cannot be attributed to an energy-storage mechanism in the torsional sector. In the minimal action, the spin current locally determines the contorsion; the curvature preserves its radiative sector. A repulsive high-density correction requires a matter model with spin, a determined coupling, equation of state, and effective sign.

In an Einstein–Cartan model with spin fluid may appear

$$s^2 \propto a^{-6}.$$

If ρ_{m} denotes mass density, an effective equation may take the form

$$H^2 = \frac{8\pi G}{3}\big(\rho_{\mathrm{m}} - \rho_{\mathrm{spin,m}}\big) + \cdots .$$

If energy density ρ_{E} is used, dimensional conversion requires writing

$$H^2 = \frac{8\pi G}{3c^2}\big(\rho_{\mathrm{E}} - \rho_{\mathrm{spin,E}}\big) + \cdots .$$

Both conventions are not mixed in a single equation.

The rebound candidate is situated at

$$\rho_{\mathrm{m}} = \rho_{\mathrm{spin,m}}$$

or in the equivalent energy equality, provided that the solution is regular and satisfies the declared matter and boundary conditions. The Cartan–Holst action permits this mechanism to be studied; HMT must still construct the map that selects the current, the threshold, and the information record without consulting them retrospectively.

88.2.5 Cobordism and Descendant Cosmological Components

The opening of a descending cosmological component is formulated via a cobordism.

$$\mathcal{W} : \Sigma_{\mathrm{in}} \longrightarrow \Sigma_{\mathrm{prog}} \sqcup \Sigma_{\mathrm{desc}}.$$

Before asserting information conservation, the conserved object must be fixed. If $\mathcal{I}$ is an additive charge defined by a cocycle, one may test

$$\mathcal{I}(\Sigma_{\mathrm{in}}) = \mathcal{I}(\Sigma_{\mathrm{prog}}) + \mathcal{I}(\Sigma_{\mathrm{desc}}) + \mathcal{F}_{\partial\mathcal{W}}.$$

If $\mathcal{I}$ represents quantum information, the pertinent structure is an isometry or to unitary evolution and the balance is does not reduced, in general, to to scalar sum. The entropies of the subsystems, their correlations, and the mutual information must is computed.

A realization must specify:

1. the junction conditions among the initial, progenitor, and descendant sections;
2. the boundary and corner terms of the action;

3. causality, hyperbolicity, and the initial-value problem;
4. the control of singularities, closed timelike curves, and energy conditions;
5. the concrete charge, isometry, or conservation law that represents the information.

The Bianchi identities and Noether balance are necessary local conditions; by themselves they do not prove the change of topology nor the global conservation of information.

88.2.6 Parallel components and intercomponent mode

A disconnected boundary and a collection of sectors have distinct compositions. In a realization of topological type,

$$\mathcal{H}_{\Sigma_{\text{parent}} \sqcup \Sigma_{\text{child}}} \simeq \mathcal{H}_{\text{parent}} \otimes \mathcal{H}_{\text{child}}$$

represents the composition of components of a boundary. In contrast,

$$\mathcal{H}_{\text{cos}} = \bigoplus_{a} \mathcal{H}_a$$

represents alternatives or superselection sectors. The tensor product and the direct sum do not replace each other.

On the direct sum, a global operator candidate has block form:

$$\mathbb{H}_{\text{cos}} = \begin{pmatrix} H_1 & K_{12} & \cdots \\ K_{21} & H_2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

To define a physical operator, a common dense domain or relative boundedness hypotheses, a common or relational clock, and the condition must be fixed

$$K_{ba} = K_{ab}^{\dagger}.$$

The off-diagonal blocks must be derived from the boundary, cobordism, or another declared coupling. If they are nonzero, the components form coupled sectors of a global theory; they are not independent universes in the strong sense.

An eigenvector supported on several summands constitutes an intercomponent mode. Calling its quantum a graviton further requires the reduced physical sector to possess helicities ± 2, two transverse polarizations, positive energy, luminal propagation, absence of ghosts and tachyons, Ward identities, and universal coupling. Global support alone does not satisfy these conditions.

88.2.7 Certification of the torsional bounce and exact obstructions

The realization is certified by the following constructions:

1. regular rebound solutions with matter and sign derived;
2. a well-posed boundary problem;
3. an admissible cobordism with junction conditions;
4. an information transport defined as charge, isometry, or unitary evolution;
5. a global self-adjoint operator with nondiagonal couplings selected before the comparison;
6. predictions that distinguish the realization from a single-component cosmology.

The certificate requires the regular solution, compatibility of the gluing conditions, conservation of the chosen charge or isometry, independence from arbitrary charts, and preservation of causality and

hyperbolicity. Global energy is used only when there exists a temporal symmetry or an explicit quasilocal definition.

88.3 Geometric junction between anisotropic contraction, torsional bounce, and descending cosmological region

The Big Bang may be understood as a distortion of maximal intensity among state, rank, and reader. It is not an explosion that occurred within a preexisting space. It is the limit at which the projection defining extension, distance, and duration begins to be valid. Asking what occurred "before" by using the same clock may be meaningless because the clock belongs to the regime that is emerging. HMT nevertheless preserves a genealogy prior to that publication: route, boundary, orientation, and memory.

The completion

$$1000 = 729 + 243 + 27 + 1$$

shows how a compact unit is distributed among the ternary ambient space, the visible word, the closure, and the residual unit. It does not by itself identify four cosmological eras. It provides an architecture of strata capable of distinguishing the deep state from its projection. The cosmological map takes the form

$$\mathfrak{R}_{\cos} : \text{boundary HMT state} \dashrightarrow (M, g, e, \omega, \Psi).$$

Its task is to assign rank, scale, connection, material content, and law of evolution. The thesis of initial distortion asserts that the large observed curvature is the continuous shadow of to transition of that map.

Einstein–Cartan torsion provides a dynamic bounce mechanism. For matter with spin density,

$$s^2 \propto a^{-6}.$$

The torsional contribution grows towards small volumes faster than ordinary matter. An effective equation may be written as

$$H^2 = \frac{8\pi G}{3}\left(\rho_{\mathrm{m}} - \rho_{\mathrm{spin}}\right) - \frac{k}{a^2} + \frac{\Lambda}{3},$$

with the factors of c restored according to whether mass or energy densities are used. The equality

$$\rho_{\mathrm{m}} = \rho_{\mathrm{spin}}$$

localizes the candidate for material–torsional cancellation. There is bounce when, at a nonzero scale, the complete expression of the right-hand member vanishes,

$$H^2 = 0,$$

and the acceleration equation gives $\dot{H} > 0$; the curvature terms and Λ must vanish, compensate one another, or be included explicitly in that condition. Under these hypotheses, the contraction reaches a minimum and reverses. The mechanism does not introduce an arbitrary pressure: it follows from the Cartan equation and the spin current.

HMT interprets that threshold as sheet closure and reopening. During contraction, the visible extension is compactified and internal memory increases. At the return point, the published orientation changes and a new expansive branch becomes accessible. The singularity ceases to be the only possible endpoint because the state variable is not reduced to the scale factor. The enriched route may continue when the preceding cosmological chart loses rank.

The same geometry permits formulation of a child universe. For an exterior observer, a collapsing region remains behind a horizon. For the global section, closure may open a boundary with its own orientation and time. Gluing is described as a cobordism or transition between sheet components, not as

the appearance of matter from nothing. Information not published in the parent universe may become a boundary condition of the child universe.

This image coordinates three phenomena usually studied separately: collapse, horizon, and cosmological origin. The same closure operation that forms a trapped region may produce a reopening after the torsional scale is reached. The same memory that protects exterior information may fix the conditions of the new branch. And the same change of orientation that defines particle and antiparticle on the microscopic scale may acquire a macroscopic causal realization.

The complete realization requires a global solution. It must contain an Einstein exterior, a region of non-negligible torsion, a regular minimum of the scale factor, a causal continuation, and a state map between boundaries. It must respect covariant conservation and produce an observable: a remnant distribution, nonthermal correlations, a modification of the gravitational-wave spectrum, or a primordial imprint. If the bounce scale contradicts nucleosynthesis, the cosmic background, or torsion bounds, the concrete mechanism is ruled out. The sheet thesis does not exempt the dynamics; it supplies the object that the dynamics must evolve.

88.4 Elliptic, parabolic, and hyperbolic regimes of TRIT and their projection onto the dark sector

The $\widehat{\mathbb{T}}$ does not merely orient past, threshold, and opening. Its three values determine the three real quadratic algebras

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \bar{\tau}).$$

The generators satisfy

$$\bar{\tau} = +1: \quad J^2 = -I, \qquad \bar{\tau} = 0: \quad J^2 = 0, \qquad \bar{\tau} = -1: \quad J^2 = +I.$$

They are the elliptic, parabolic, and hyperbolic regimes. The model geometry

$$g_{s,\rho} = \mathrm{d}r^2 + S_{s,\rho}(r)^2\,\mathrm{d}\theta^2$$

is obtained with

$$S_{+,\rho}(r) = \rho^{-1}\sin(\rho r), \qquad S_{0,\rho}(r) = r, \qquad S_{-,\rho}(r) = \rho^{-1}\sinh(\rho r),$$

and has

$$K_G = s\rho^2.$$

The three curvatures are not attributed simultaneously to the same metric at the same point. They are coordinated sheets of to state whose physical chart selects the realized regime.

de Sitter space demonstrates exactly how a single four-dimensional geometry can admit three spatial presentations:

$$\mathrm{d}s^2 = -\,\mathrm{d}t^2 + a_s(t)^2\,\mathrm{d}\Sigma_s^2, \qquad s \in \{+1, 0, -1\}.$$

The functions

$$a_+(t) = H^{-1}\cosh(Ht), \quad a_0(t) = \mathrm{e}^{Ht},$$
$$a_-(t) = H^{-1}\sinh(Ht), \qquad t > 0,$$

satisfy

$$\frac{\ddot{a}_s}{a_s} = H^2, \qquad \left(\frac{\dot{a}_s}{a_s}\right)^2 + \frac{s}{a_s^2} = H^2, \qquad R = 12H^2.$$

TO closed, to flat, and an open foliation describe the same spacetime. Local spatial flatness, hyperbolic section, and positive acceleration are not incompatible signs. They may is distinct sheets of to single global geometry.

This fact provides a continuous model for the HMT intuition. The value published on one sheet does not necessarily contain the entire curvature of the state. An observer working in the flat foliation may reconstruct as an additional source the effect that follows from boundary incidence or another presentation. Dark matter and dark energy thereby receive a unified but differentiated interpretation.

Apparent dark matter may correspond to the gravitational response of states that have no point representative on the visible sheet. Its mass manifests itself in orbits and lensing because the complete geometry transports its incidence, even though the electromagnetic reader does not publish an ordinary particle. Apparent dark energy may correspond to the global response of sheet opening and curvature, published as acceleration of the scale factor. The former belongs primarily to the distribution of nonvisible retention; the latter, to the evolution of the cosmological boundary.

The hinge

$$\frac{G_{\mathrm{tor}}}{I_{\mathrm{tor}}} = 1, \qquad \frac{G_{\mathrm{av}}}{I_{\mathrm{av}}} = \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^{2}}$$

indicates where to seek that residue. The marked ratio is a structural relation between readings, not a cosmological percentage that can be identified directly with an observed density. To become a theory of the dark sector, it must be integrated over the geometry and produce an effective tensor $T_{\mu\nu}^{\mathrm{eff}}$, its continuity equations, and a transfer function.

The requirement is especially stringent because a single rule must span several scales. In galaxies it must relate baryonic density, circular velocity, and lensing without introducing a new parameter for each object. In clusters it must coordinate gas, weak lensing, and dynamics. In cosmology it must predict $H(z)$, distances, structure growth, and lensing potentials. If the same multisheet geometry produces those profiles with parameters fixed before each system is examined, HMT–MD will have replaced two unknown sectors with a single architecture. If it requires independent adjustments, the interpretation will not have attained its objective.

The triple foliation additionally imposes conceptual discipline. Not every hyperbolic curvature is a sign of dark energy, nor does every local flatness eliminate global curvature. Chart, domain, and scale must be declared. The de Sitter parameter H is likewise not selected solely by the existence of the three foliations. Mathematics proves their compatibility; the cosmological transducer must derive the physical scale and its evolution.

88.5 Temporal Orientation of the Cosmic Background and Memory Conservation Under Expansion

The dodecaphonic closure produces a very restricted angular template. Let

$$\phi_k = \phi_0 + \frac{2\pi k}{12}, \qquad k = 0, \dots, 11,$$

and

$$w_k = \frac{1}{12}\left[1 + 2\epsilon \cos(3\phi_k - \delta)\right], \qquad |\epsilon| \le \frac{1}{2}.$$

The weights are non-negative and sum to one. Its discrete transform has support only in

$$m = 0, \pm 3 \pmod{12}.$$

Upon projecting onto low multipoles, for $\ell = 2$ can only survive $m = 0$; for $\ell = 3$, $m = 0, \pm 3$. The family C_{12} converts the dodecaphasic wheel into an exact angular envelope.

Its physical importance lies in its restrictiveness. The axis, phase δ, amplitude ϵ, celestial mask, estimator, and multipole set must be fixed before examining the map to be used for comparison. Data from independent frequencies and missions are then compared with isotropic simulations and contamination models. A stable signal on the predicted support would strengthen the dodecaphasic genealogy. A

posterior selection of the axis or a dominant signal outside $m = 0, \pm 3$ would exclude that particular realization.

The cosmic arrow of time uses the same ternary reader that organizes the leaves:

$$+1 \longmapsto \text{return, memory, past,}$$

$$0 \longmapsto \text{threshold, present,}$$

$$-1 \longmapsto \text{bidirectional opening, future.}$$

The present is not a point moving along a previously traced line. It is the threshold at which an admissible opening becomes actual and begins to be preserved as memory. The past is the sector of returns already recorded; the future is the set of compatible continuations that have not yet closed.

The complete evolution may be reversible while the published history has orientation. Asymmetry appears in preparation, in boundary selection, and in the reader that discards part of the fiber. Entropy does not create time; it measures the information that a presentation ceases to distinguish. For an action on histories $I(\gamma)$, the measure

$$\mathbb{P}_\beta(\gamma) = \frac{\mathrm{e}^{-\beta I(\gamma)}}{Z_\beta}$$

minimizes

$$\mathbb{E}_{\mathbb{P}}[I] - \beta^{-1} H(\mathbb{P}).$$

In the low-temperature limit, the minimal interaction dominates; at finite temperature, route multiplicity intervenes. Action and entropy are two components of the same selection, not two rival causes.

A round-trip story can be purified as

$$|\Omega_\beta\rangle = \sum_\gamma \sqrt{\mathbb{P}_\beta(\gamma)}\, |\gamma\rangle_{\text{ida}} |C\gamma\rangle_{\text{vuelta}}.$$

The reduced entropy is $H(\mathbb{P}_\beta)$. The observed universe may possess an arrow because we access one branch, whereas the enlarged state preserves conjugation. This picture links the microscopic particle and antiparticle orientation with macroscopic asymmetry without asserting that the two scales are identical.

HMT–MD cosmology is thus organized by a single fundamental operation: folding, preserving, and unfolding. Matter is a section that retains space; the horizon folds one presentation into another sheet; the bounce unfolds a branch after compactification; the dark sector is the response of geometric information that the visible sheet does not publish; and time is the orientation of opening and return. Each of these statements acquires physical force by sharing the same state domain, not by forming a collection of analogies.

The theory commits itself to clear consequences. The black hole exterior must obey relativity within its tested regime, and the inversion must provide a continuation and a signal. The bounce must fix its scale through spin density and remain consistent with nucleosynthesis and the cosmic background. The multisheet transfer function must jointly explain dynamics, lensing, and expansion. The template C_{12} must survive a prospective test. The arrow of time must respect global reversibility and the cost of erasure. These conditions do not weaken the cosmological thesis: they are the route by which a mathematical architecture becomes a physics of the universe.

88.6 Torsional junction of a trapped region and construction of a descending cosmological branch

88.6.1 Torsional gluing of a trapped region: exact, conditioned, and physical strata of the descendant cosmological branch

The section 81.4 establishes the memory law

$$M_n = \left(\frac{a_n}{a_0}\right)^{-6} \tag{88.12}$$

and formulates its realization through a quadratic spin density in Einstein–Cartan. The present chapter studies the additional passage from a torsional bounce to an expansive causal region situated beyond the trapped interior of a black hole.

The exposition distinguishes 3 strata:

1. *structure exact*: orientation TRIT, involution of Möbius, conservation isometric of the canal and law V^{-2};
2. *conditioned consequence*: Einstein–Cartan bounce, geometric splice and causal opening under declared hypotheses;
3. *classified geometric realization*: explicit existence of the descendant continuation, minimal, and obstruction to the classical branched cobordism with topological change.

This stratification makes it possible to develop the complete image and preserve the status of each assertion.

The common antecedent is an enriched state

$$X_{\mathrm{TPK}}^{\mathrm{enr}} = (x, \text{fiber}, \text{orientation}, \text{carry}, \text{boundary}, \text{route}, \text{memory}, \text{observer}), \tag{88.13}$$

produced by APP $\to \widehat{\mathbb{T}} \to$ TPK. The causal, torsional, topological, and quantum readers are applied to this same state and preserve its genealogy.

88.6.2 Causal orientation of the 3 leaves

The TRIT grammar is realized through the algebras

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \qquad J_\tau^2 = -\tau I, \qquad \tau \in \{+1, 0, -1\}. \tag{88.14}$$

Their exponentials are

$$\mathrm{e}^{tJ_{+1}} = \cos t\, I + \sin t\, J_{+1}, \tag{88.15}$$

$$\mathrm{e}^{tJ_0} = I + tJ_0, \tag{88.16}$$

$$\mathrm{e}^{tJ_{-1}} = \cosh t\, I + \sinh t\, J_{-1}. \tag{88.17}$$

The sheet +1 carries closure, return, and memory; the sheet 0 carries the threshold, present, and first variation; the sheet −1 carries openness, propagation, and the two-way future.

Let ϑ be an oriented expansion scalar. Define the causal reader

$$\boxed{\tau_{\mathrm{cau}}(\vartheta) := -\operatorname{sgn}(\vartheta) \in \{-1, 0, +1\}.} \tag{88.18}$$

A positive expansion publishes opening −1; a zero expansion publishes threshold 0; a negative expansion publishes closure +1.

In to family of spheres, $\vartheta = \theta_{(\ell)}$ may tomarse as the expansion null saliente. In a region homogeneous or anisotropic, $\vartheta = \Theta$ may tomarse as the expansion volumetric. These two observables occupy distinct domains and share the same sign rule.

Proposition 88.4 (Double-threshold causal sequence)**.** *Assume the sign conditions*

$$\theta_{(\ell)} > 0 \text{ in the exterior}, \quad \theta_{(\ell)} = 0 \text{ at the horizon}, \quad \theta_{(\ell)} < 0 \text{ in the trapped interior}, \tag{88.19}$$

and

$$\Theta < 0 \text{ before the bounce}, \quad \Theta = 0 \text{ at the bounce}, \quad \Theta > 0 \text{ after the bounce}. \tag{88.20}$$

The reader equation (88.18) produces the sequence

$$\boxed{-1_{\text{exterior}} \longrightarrow 0_{\text{horizonte}} \longrightarrow +1_{\text{interior}} \longrightarrow 0_{\text{rebote}} \longrightarrow -1_{\text{descendiente}}.} \tag{88.21}$$

Proof. The assertion follows by applying $\tau_{\text{cau}}(\vartheta) = -\operatorname{sgn}(\vartheta)$ to each sign of equations (88.19) y (88.20). □

The two states 0 are realizations of the same threshold archetype. The first expresses null marginality; the second expresses volumetric marginality. The identity of the sign preserves the grammar, while the declared observable determines the physical content of each threshold.

88.6.3 Anisotropic interior and quadratic memory

An interior region, spherically symmetric, admits the Kantowski–Sachs reduction

$$\mathrm{d}s^2 = -N(\tau)^2\,\mathrm{d}\tau^2 + a_{\parallel}(\tau)^2\,\mathrm{d}\chi^2 + a_{\perp}(\tau)^2\,\mathrm{d}\Omega_2^2. \tag{88.22}$$

The volume of a comoving cell is

$$V(\tau) = V_0 a_{\parallel} a_{\perp}^2, \tag{88.23}$$

and its expansion is

$$\Theta := \frac{1}{N}\frac{\dot{V}}{V} = H_{\parallel} + 2H_{\perp}, \qquad H_i := \frac{1}{N}\frac{\dot{a}_i}{a_i}. \tag{88.24}$$

A comoving spin charge N_s produces a density

$$n_s = \frac{N_s}{V}, \tag{88.25}$$

and the Cartan equation, linear in the current, generates a correction of quadratic form

$$\rho_s = s_0 \left(\frac{V}{V_0}\right)^{-2}. \tag{88.26}$$

The Perron branch publishes exactly the same covariance:

$$M_n = V_n^{-2}. \tag{88.27}$$

The torsional transducer

$$\mathcal{T}_{\text{EC}} : M_n \longmapsto \rho_{s,n} = s_0 M_n \tag{88.28}$$

preserves positivity and the volume exponent. The amplitude s_0 follows from the Cartan current and retains its obligation of prospective selection.

For the discussion local adoptamos units geometric $c = 1$. A restriction efectiva of Kantowski–Sachs has the form

$$\frac{\Theta^2}{3} = 8\pi G(\rho_m - \rho_s) + \sigma^2 - \frac{1}{2}\,{}^{(3)}R + \Lambda, \tag{88.29}$$

where σ^2 is the shear and ${}^{(3)}R$ the curvature of the sheet. A regular volumetric bounce satisfies

$$\boxed{\Theta(\tau_b) = 0, \qquad \frac{1}{N}\dot{\Theta}(\tau_b) > 0,} \tag{88.30}$$

together with positive finiteness of $a_\parallel, a_\perp$ and of the curvature and torsion invariants.

Comparison Criterion 88.5 (Anisotropic torsional dominance)**.** Suppose that, during the contraction,

$$\rho_s = s_0 V^{-2}, \qquad \sigma^2 = \Sigma_0^2 V^{-2} + o(V^{-2}), \tag{88.31}$$

and that matter, curvature, and Λ grow with a power strictly less than V^{-2}. The inequality

$$\boxed{8\pi G s_0 > \Sigma_0^2} \tag{88.32}$$

is a sufficient condition for the torsional term to dominate the shear in the regime of small volume. The existence of a transverse root of equation (88.29) and the inequality equation (88.30) complete the bounce criterion.

Asymptotic dominance supplies the mechanism; the transversal root supplies the geometric event. This distinction preserves the dependence on curvature, anisotropy, equation of state, and boundary data.

88.6.4 Nonadic condition for the bounce level

Write $x := a/a_0$ for the dimensionless isotropic scale. At the complete returns $n = 9q$, the Perron branch satisfies

$$x_{9q} = \frac{a_{9q}}{a_0} = \varphi_{\mathrm{HMT}}^{3q}. \tag{88.33}$$

In the isotropic reduction with

$$\rho_m = \rho_0 x^{-3(1+w)}, \qquad \rho_s = s_0 x^{-6}, \qquad w < 1, \tag{88.34}$$

the equality of densities fixes

$$x_b^{3(1-w)} = \frac{s_0}{\rho_0}. \tag{88.35}$$

If the bounce is required over a complete TPK turn, the elimination of x_b yields

$$\boxed{q_b = \frac{\log(s_0/\rho_0)}{9(1-w)\log\varphi_{\mathrm{HMT}}}, \qquad q_b \in \mathbb{Z}.} \tag{88.36}$$

The integrality of q_b is an additional condition of synchronization, derived from the hypothesis that the bounce coincides with the nonadic return. A continuous bounce between two circuits satisfies equation (88.35) and has a different status.

Proposition 88.6 (Falsifiability of the Synchronous Bounce)**.** *With (s_0, ρ_0, w) fixed prospectively, the synchronized realization exists only if the number of equation (88.36) is an integer. The distance*

$$\delta_9 := \mathrm{dist}(q_b, \mathbb{Z}) \tag{88.37}$$

is a quantitative falsifier and cancels exactly at an allowed level.

Proof. The condition $n = 9q$ requires $q \in \mathbb{Z}$. Substituting equation (88.33) into equation (88.35) and taking logarithms produces the formula. The characterization of δ_9 follows from the definition of distance to the integers. □

88.6.5 Matching data on the bounce hypersurface

Let Σ_b be a spatial hypersurface separating a contractive region $\mathcal{M}_-$ from an expansive region $\mathcal{M}_+$. Denote by h_{ij} the induced metric, by n^μ the oriented normal, and by $\mathring{K}^{\pm}_{ij}$ the extrinsic curvatures computed with the Levi–Civita connection on each side. We adopt

$$[X] := X^+\big|_{\Sigma_b} - X^-\big|_{\Sigma_b}, \qquad \kappa := \frac{8\pi G}{c^4}. \tag{88.38}$$

The Israel metric conditions are

$$[h_{ij}] = 0, \tag{88.39}$$

$$[\mathring{K}_{ij} - h_{ij}\mathring{K}] = -\kappa S^{\Sigma}_{ij}, \tag{88.40}$$

where S^{Σ}_{ij} is the surface energy–momentum tensor under the fixed-normal convention. A junction without a material layer satisfies

$$[h_{ij}] = 0, \qquad [\mathring{K}_{ij}] = 0. \tag{88.41}$$

The inversion of Θ then crosses to continuous zero and remains realizada by the dynamics, rather than being introduced by to jump of the normal.

In the first-order formulation, let $\omega^{ab} = \mathring{\omega}^{ab} + C^{ab}$ be the connection with contorsion, and write $T_\mu := T^\lambda{}_{\mu\lambda}$ for the trace of the torsion. We fix the normalization of the spin current through the Cartan equation

$$\mathscr{C}(T)^\lambda{}_{\mu\nu} := T^\lambda{}_{\mu\nu} + \delta^\lambda_\mu T_\nu - \delta^\lambda_\nu T_\mu = \kappa s^\lambda{}_{\mu\nu}. \tag{88.42}$$

Its distributional part over Σ_b requires

$$\boxed{\mathscr{C}(T_\Sigma) = \kappa s_\Sigma.} \tag{88.43}$$

In the absence of a superficial spin layer, $s_\Sigma = 0$, the admissible connection lacks a delta torsional component, and the tangential contorsion has regular gluing. An action with a Holst term or nonminimal fermionic couplings replaces $\mathscr{C}$ by the corresponding algebraic operator; the conditions are obtained by integrating that equation across the layer.

Comparison Criterion 88.7 (Regular Israel–Cartan gluing)**.** A regular torsional seam through Σ_b requires:

i) continuity of the induced costructure, equivalent to equation (88.39) up to Lorentz gauge;

ii) equation of Israel equation (88.40) with every capa superficial declared;

iii) Cartan equation equation (88.43) with every declared surface current;

iv) continuity of the gauge charges and of the route orientation;

v) finiteness of the curvature and torsion invariants.

The Ashtekar–Barbero Connection

$$A^a_i = \Gamma^a_i + \gamma_{\mathrm{BI}} K^a_i \tag{88.44}$$

can act as a reader of the seam when both sides share costructure, γ_{BI}, and normalization. Its continuity is a conditional consequence of the corresponding gluing data; its verification requires the explicit transport of Γ and K to the same boundary section.

Proposition 88.8 (Joint involution of Barbero and extrinsic curvature)**.** *Consider the algebraic involution*

$$\mathfrak{J}_b : (\Gamma_i^a, K_i^a, \gamma_{\mathrm{BI}}) \longmapsto (\Gamma_i^a, -K_i^a, -\gamma_{\mathrm{BI}}). \tag{88.45}$$

Then

$$\mathfrak{J}_b(A_i^a) = A_i^a, \tag{88.46}$$

and every area spectrum that depends on $|\gamma_{\mathrm{BI}}|$ remains invariant. Identification of $\mathfrak{J}_b$ with the dynamic inversion of the bounce requires the Israel–Cartan intertwiner to transport the orientation of K and the sheet of γ_{BI} according to equation (88.45).

Proof. The product of the two oriented factors satisfies $(-\gamma_{\mathrm{BI}})(-K_i^a) = \gamma_{\mathrm{BI}} K_i^a$, while Γ_i^a remains fixed. The areal dependence on $|\gamma_{\mathrm{BI}}|$ is even. □

88.6.6 Null horizon as a marginal hypersurface and initial matching condition

The horizon hypersurface Σ_H has null character and requires adapted junction data. Let q_{AB} be the two-dimensional metric of the sections, ℓ^μ the null generator, and N^μ a transverse vector with $\ell \cdot N = -1$. The transverse curvature is

$$C_{AB} := q_A{}^\mu q_B{}^\nu \nabla_\mu N_\nu. \tag{88.47}$$

A null junction without an impulsive layer preserves q_{AB}, the class of C_{AB} that determines the surface tension and the tangential Cartan flux. Marginality

$$\theta_{(\ell)}\big|_{\Sigma_H} = 0 \tag{88.48}$$

realizes the first state 0 of equation (88.21). The formulation null mantiene separada this condition of the condition volumetric $\Theta|_{\Sigma_b} = 0$.

88.6.7 Möbius band and orientation memory

The dodecaphasic words are acted on by the involution of genealogical type

$$C_M(\mathbf{t}) := -\operatorname{rev}(\mathbf{t}), \qquad C_M^2 = I. \tag{88.49}$$

The associated oriented line is obtained by the quotient

$$(\mathbf{t}, \chi) \sim (C_M \mathbf{t}, -\chi). \tag{88.50}$$

When the base is a return loop, this identification defines a band with holonomy $-I$: one turn reverses orientation and two turns restore it.

Let $\mathscr{H}_\gamma$ denote the holonomy over one turn. The hierarchy

$$\mathcal{R}_{\mathrm{vec}} \subseteq \mathcal{R}_{\mathrm{ray}} \subseteq \mathcal{R}_{\mathrm{obs}} \tag{88.51}$$

distinguishes return of vector, return of ray, and return of observable. For $\mathscr{H}_\gamma = -I$, the ray returns in one turn, the oriented vector returns in two and the sign memory preserves the traversal.

Proposition 88.9 (Topological status of the band)**.** *The relation equation (88.50) defines a real line that is nonorientable over the loop exactly when its first Stiefel–Whitney class satisfies*

$$w_1 = 1 \in H^1(S^1; \mathbb{Z}_2). \tag{88.52}$$

This assertion concerns to the haz genealogical of orientation. A realization as topology of the spacetime requires to morphism toward the frame or spinor bundle that preserves causality and Cartan structure.

Proof. A real line over S^1 is determined by the parity of its transition function. The transition -1 of equation (88.50) represents the nontrivial class; two circuits square the transition and yield $+1$. □

The band provides an exact representation of orientation and memory. Physical continuity across the horizon and the bounce is formulated through the junctions of the preceding sections; both structures are coordinated by an intertwiner and preserve their domains.

88.6.8 Relative Observer, Parallel Transport and Autoinertia in the Descending Cosmological Region

The expanded dynamic state includes observer and memory:

$$\Gamma = (x, O, m), \tag{88.53}$$

and the frame accessible to the observer is a restriction of the total connection,

$$\mathfrak{F}_O(\Gamma) := \mathrm{Res}_O(\Gamma, \nabla_{\mathrm{TPK}}). \tag{88.54}$$

A horizontal trajectory of the complete system,

$$\nabla^{\mathrm{tot}}_{\dot\Gamma}\dot\Gamma = 0, \tag{88.55}$$

can publish acceleration in a subsystem S:

$$a_S = \nabla^S_{\dot\gamma_S}\dot\gamma_S = \mathcal{K}_{\pi_S}(\dot\Gamma, \dot\Gamma), \tag{88.56}$$

where $\mathcal{K}_{\pi_S}$ is the second fundamental tensor of the projection. Self-inertia denotes the horizontality of the whole and the response induced in its projections.

The coupling between system, apparatus, and register is described by an isometry

$$V_{\mathrm{obs}} : \mathcal{H}_{\mathrm{in}} \longrightarrow \mathcal{H}_{\mathrm{sys}} \otimes \mathcal{H}_{\mathrm{mem}}, \qquad V^*_{\mathrm{obs}} V_{\mathrm{obs}} = I. \tag{88.57}$$

Let $\{P_y\}$ be the PVM of the register in $\mathcal{H}_{\mathrm{mem}}$, with

$$P_y P_z = \delta_{yz} P_y, \qquad \sum_y P_y = I_{\mathrm{mem}}. \tag{88.58}$$

The isometry and this PVM induce the completely positive instrument

$$\mathcal{I}_y(\rho) := \mathrm{Tr}_{\mathrm{mem}}\left[(I_{\mathrm{sys}} \otimes P_y) V_{\mathrm{obs}} \rho V^*_{\mathrm{obs}} (I_{\mathrm{sys}} \otimes P_y)\right], \qquad \sum_y \mathrm{Tr}\, \mathcal{I}_y(\rho) = \mathrm{Tr}\, \rho. \tag{88.59}$$

The relative states are then obtained by means of

$$p(y) = \mathrm{Tr}\, \mathcal{I}_y(\rho), \qquad \rho_y = \frac{\mathcal{I}_y(\rho)}{p(y)}. \tag{88.60}$$

The section ρ_y is the state relative to the record y; the global state preserves all channels in the image of the isometry. This structure formalizes the kinship with Everett at the precise level of relative states and leaves the ontology of the branches as a choice in subsequent physical terms.

88.6.9 Global channel of the descendant region

The operatorial version of a descendant opening uses an isometry

$$V_{\mathrm{pc}} : \mathcal{H}_{\mathrm{in}} \longrightarrow \mathcal{H}_{\mathrm{parent}} \otimes \mathcal{H}_{\mathrm{child}}, \qquad V_{\mathrm{pc}}^{*} V_{\mathrm{pc}} = I. \tag{88.61}$$

The joint channel and its complementary channels are

$$\mathcal{N}_{\mathrm{pc}}(\rho) = V_{\mathrm{pc}} \rho V_{\mathrm{pc}}^{*}, \tag{88.62}$$

$$\mathcal{N}_{\mathrm{parent}}(\rho) = \mathrm{Tr}_{\mathrm{child}}\, \mathcal{N}_{\mathrm{pc}}(\rho), \qquad \mathcal{N}_{\mathrm{child}}(\rho) = \mathrm{Tr}_{\mathrm{parent}}\, \mathcal{N}_{\mathrm{pc}}(\rho). \tag{88.63}$$

Theorem 88.10 (Global Conservation and Relative Distribution)**.** *The map $\mathcal{N}_{\mathrm{pc}}$ preserves trace and purity of to pure state on its image. The parental and descendant channels may produces mixed states, and for to pure bipartite output satisfy*

$$S(\rho_{\mathrm{parent}}) = S(\rho_{\mathrm{child}}). \tag{88.64}$$

The global information is contained in the outputs and their correlations.

Proof. The identity $V_{\mathrm{pc}}^{*} V_{\mathrm{pc}} = I$ implies

$$\mathrm{Tr}(V_{\mathrm{pc}} \rho V_{\mathrm{pc}}^{*}) = \mathrm{Tr}(\rho V_{\mathrm{pc}}^{*} V_{\mathrm{pc}}) = \mathrm{Tr}\, \rho.$$

An isometry maps unit vectors to unit vectors. Equality of the reduced entropies follows from the Schmidt decomposition of the pure bipartite state. □

The conservation of a charge Q or of energy requires, in addition, the intertwinings

$$Q_{\mathrm{out}} V_{\mathrm{pc}} = V_{\mathrm{pc}} Q_{\mathrm{in}}, \qquad H_{\mathrm{out}} V_{\mathrm{pc}} = V_{\mathrm{pc}} H_{\mathrm{in}}. \tag{88.65}$$

The isometry preserves probability; the equations equation (88.65) preserve the declared dynamic quantities. An admissible output distributes the information among correlations and respects the impossibility of cloning an arbitrary state into two identical copies.

The provenance of the sectors 90/120 is preserved through the modular holographic Hamiltonian

$$K_{\mathrm{HHM}} = c_0 I + \Delta_{\mathrm{mod}} (90 N_{90} + 120 N_{120}). \tag{88.66}$$

The total occupation and the weighted memory,

$$N = N_{90} + N_{120}, \qquad D = 90 N_{90} + 120 N_{120}, \tag{88.67}$$

Transport of the total occupation is imposed as an independent operator identity:

$$N^{\mathrm{out}} V_{\mathrm{pc}} = V_{\mathrm{pc}} N^{\mathrm{in}}. \tag{88.68}$$

With N and the weighted memory D conserved, the two sectors are reconstructed uniquely:

$$N_{90} = 4N - \frac{D}{30}, \qquad N_{120} = \frac{D}{30} - 3N. \tag{88.69}$$

The condition of transport

$$K_{\mathrm{HHM}}^{\mathrm{out}} V_{\mathrm{pc}} = V_{\mathrm{pc}} K_{\mathrm{HHM}}^{\mathrm{in}} \tag{88.70}$$

The identities equations (88.68) y (88.70) jointly preserve occupation and sectorial provenance across the seam. These equalities are added to the dynamic obligations of a complete realization.

88.6.10 Descendant continuation and branched cobordism

The denomination *descendant universe* admits two geometric levels.

Definition 88.11 (Minimal descendant continuation)**.** A descendant continuation is a region $\mathcal{M}_{\text{child}}$ situated to the future of the rebound, with

$$\Theta > 0, \qquad \text{finite curvature and torsion,} \qquad \text{a global time function in its causal domain.} \tag{88.71}$$

The region may pertenecer to a prolongation conectada of the interior and preserves the topology of the hojas espaciales.

Definition 88.12 (Parent-Child Cobordism)**.** The branched version is an effective cobordism

$$\mathcal{W} : \Sigma_{\text{in}} \longrightarrow \Sigma_{\text{parent}} \sqcup \Sigma_{\text{child}}, \tag{88.72}$$

equipped with Cartan structure, gluing data test criterion 88.7, and channel equation (88.61).

The minimal continuation requires a bounce and a causal extension. The branched cobordism adds topological change and separation of outputs. In a Lorentzian geometry that is classical, nondegenerate, and globally hyperbolic, topological change is subject to causal obstructions. The branched version must therefore be realized through a region that is quantum or torsional and in which it is declared which classical hypothesis is modified, or through an interpretation of two causal domains within a single global topology.

Comparison Criterion 88.13 (Promotion to a Physical Descendant Universe)**.** The complete image acquires the status of a realization that is physical when the following are constructed simultaneously:

i) a solution of the Einstein–Cartan equations with equation (88.30);

ii) regular Israel–Cartan data on Σ_H and Σ_b;

iii) a future expansive region with temporal function and Cauchy surfaces in each accessible branch;

iv) finiteness of invariants and prolongation of geodesics through the nucleus;

v) absence of closed causal curves and unregistered layers;

vi) an isometric channel that preserves the declared charges and the Hamiltonian of provenance equation (88.70);

vii) nonadic naturality of the torsional current and the boundary data;

viii) an intertwiner between the genealogical band and the physical bundle of frames or spinors.

These conditions transforman the narrative equation (88.21) in a problema of existence geometric, operatorial and causal with controles separate.

88.6.11 Local consequences of the torsional matching: conservation, regularity, and causal prolongation

Proposition 88.14 (Content of a Complete Realization)**.** *Suppose that the test criterion 88.13 is satisfied. Then:*

a) the causal sign traverses the sequence equation (88.21)*;*

b) the torsional memory preserves the covariance V^{-2} on both sides of the rebound;

c) the global state evolves through a trace-preserving channel;

d) each observer accesses a relative state of its branch;

e) the charges included in equation (88.65) are conserved;

f) the orientation of Möbius is transported as holonomy data.

Proof. Each assertion is the application of one of the conditions of the criterion: the causal reader proves (a); the torsional transducer proves (b); the isometry proves (c) and (d); the intertwinings prove (e); and the sheaf morphism proves (f). □

The proposition organizes consequences of a constructed structure. The existence of the minimal continuation is resolved by the theorem 88.15; classical branching with a topological change is classified by the obstruction of the theorem 88.16.

88.6.12 Torsional dynamics of bounce: exact identities, cosmological interfaces and closure conditions

Object	Status	Obligation or control
Law $M_n = V_n^{-2}$	Exact internal result	Perron's Certificate and Refinement
Reader $-\operatorname{sgn}\vartheta$	Exact formalization	Observable of expansion tipado
Homogeneous bounce	Consecuencia condicionada	$s_0 > 0$, $w < 1$, transverse zero
Kantowski–Sachs Bounce	Physical Interface	Dominance of controlled torsion, shear, and curvature
Israel–Cartan Joining	Physical Interface	Declared layer and spin current data
Möbius band	Exact Topological Result	Intertwiner toward the physical bundle
Estado relativo	Exact operational result	Instrument and observed fiber
Parent–offspring canal	Conditional formalization	Isometry and Conservation of Charges
Descendant continuation	Existence theorem	Solution regular and causally complete
Cobordismo ramificado	Proven classical obstruction	Incompatibility with global hyperbolicity and topological change

88.6.13 Obstructions to Cosmological Continuation: Loss of Orientation, Junction Regularity, or Return Memory

The negative controls distinguish each layer:

1. a solution with $\Theta = 0$ and $\dot{\Theta} < 0$ represents a volume maximum and remains outside the bounce;
2. the case $8\pi G s_0 \leq \Sigma_0^2$ removes the guarantee of anisotropic torsional dominance;
3. a $q_b \notin \mathbb{Z}$ preserves the continuous rebound and refutes the version synchronized with a complete turn;
4. a jump of h_{ij}, of the Israel quantity, or of the Cartan current requires an explicit surface layer;
5. a band with $w_1 = 0$ has global orientation and refutes the realization of Möbius;
6. a map V_{pc} with $V_{\mathrm{pc}}^* V_{\mathrm{pc}} \neq I$ loses global probability;
7. two identical copies of an arbitrary state violate the no- cloning control and refute the correlated distribution reading;
8. a descendant region with closed causal curves, divergent invariants, or inextendible geodesics fails the causal promotion;
9. a topological change that simultaneously preserves all the obstructed classical hypotheses lacks the required geometric domain.

Estatuto probatorio

The grammar $-1, 0, +1$, the sign reader, the involution $C_M^2 = I$, the classification of the band, the law V^{-2}, the hierarchy of returns, and conservation of an isometric channel are exact results in their domains. The Einstein–Cartan bounce is a consequence of the declared equations and inequalities. The extension to Kantowski–Sachs, the junctions, continuity of the Barbero connection, and causal opening are physical interfaces. The minimal parent–descendant continuation is realized by the theorem 88.15; branched topological change under the classical hypotheses is excluded by the theorem 88.16.

Falsification criterion

The torsional seam is refuted if the current constructed from HMT produces a power different from V^{-2}, an amplitude with an incompatible sign, or a failure of refinement. The bounce is refuted by the absence of a regular transverse root. The junction is refuted by an Israel or Cartan residue without a surface source. Global conservation is refuted by loss of trace, norm, or an intertwined charge. Promotion to a descendant universe is refuted by a singularity, causal violation, or the impossibility of constructing the required orientation bundle and global channel. Each refutation affects its interface and preserves the preceding internal results.

88.7 Explicit Regular Solution of the Descending Cosmological Continuation

Considerese the equation efectiva homogenea of Einstein–Cartan for $n = 3$,

$$H^2 = C\big(Aa^{-3} - Ba^{-6}\big), \qquad A, B, C > 0. \tag{88.73}$$

Given $t_b \in \mathbb{R}$, define

$$a(t)^3 = \frac{B}{A} + \frac{9AC}{4}(t - t_b)^2. \tag{88.74}$$

Theorem 88.15 (Non-Empty Existence and Completeness of the Downward Continuation)**.** *The function* (88.74) *is a global solution of* (88.73)*. It has a unique minimum* $a_b = (B/A)^{1/3} > 0$*, satisfies*

$$H(t_b) = 0, \qquad \dot{H}(t_b) = \frac{3A^2C}{2B} > 0,$$

and defines on $\mathbb{R} \times \mathbb{T}^3$ *a smooth, causally orientable FLRW metric with bounded curvature invariants and complete causal geodesics. The regions* $t < t_b$ *and* $t > t_b$ *realize the contractive parent branch and the expansive descendant continuation, joined by the regular hypersurface* $t = t_b$*.*

Proof. Setting $y = a^3$, one obtains

$$H = \frac{\dot{y}}{3y} = \frac{3AC(t - t_b)}{2y}, \qquad C(Aa^{-3} - Ba^{-6}) = C\frac{Ay - B}{y^2} = \frac{9A^2C^2(t - t_b)^2}{4y^2} = H^2.$$

The positivity of B/A excludes $a = 0$. Both H and $\dot{H}$ are bounded; the polynomial curvature invariants of the flat FLRW metric are likewise bounded. Cosmic time ranges over $\mathbb{R}$, and the affine parameter of null geodesics contains $\int a(t)\, dt$, which diverges at both ends because $a(t) \asymp |t|^{2/3}$. The proper temporal length likewise diverges. There is therefore no finite causal boundary or singularity in the bounce. □

Theorem 88.16 (Obstruction of the classical branched cobordism)**.** *Within the class of smooth, nondegenerate, time-orientable, and globally hyperbolic Lorentzian manifolds, an evolution between compact Cauchy surfaces cannot produce branching with a change of topology. Every Cauchy foliation is diffeomorphic to*

$\mathbb{R} \times \Sigma$*; therefore, the initial and final leaves are diffeomorphic. Branched cobordism is obstructed within this class.*

Thus the 2 geometric objects are resolved: the minimal continuation possesses the complete preceding solution, and the classical branching with topological change is excluded by the obstruction theorem. A degenerate geometry or a quantum theory of topological change constitutes another domain and does not modify these results.

88.8 Dilution Law a^{-6}, Torsional Memory, and Confluence of the Cosmological Readers

La ley

$$M_n = \left(\frac{a_n}{a_0}\right)^{-6}$$

is perhaps the simplest algebraically and one of the deepest conceptually.

The Perron dynamics produces two complementary behaviors.

On the one hand, there exists an expansion:

$$V_n = \varphi^n.$$

In three spatial dimensions, the associated linear scale is the cube root:

$$\frac{a_n}{a_0} = \varphi^{n/3}.$$

On the other hand, there exists a stable mode of memory:

$$M_n = \varphi^{-2n}.$$

Expansion grows with φ^n. Stable memory decreases with φ^{-2n}. When level n is eliminated, there necessarily appears

$$M_n = \left(\frac{a_n}{a_0}\right)^{-6}.$$

The exponent -6 is derived from two internal facts:

- the space has dimension 3, so the linear scale carries exponent $1/3$;
- the stable mode is quadratic and therefore has exponent -2.

The reason between both is

$$\frac{-2}{1/3} = -6.$$

In Einstein–Cartan, the spin density scales as an inverse volume:

$$s \sim a^{-3}.$$

Its quadratic contribution then scales as

$$s^2 \sim a^{-6}.$$

The HMT law reproduces that exponent exactly before introducing the Einstein–Cartan realization. This is the genuinely suggestive feature.

The Perron memory mode behaves as a quadratic spin density should behave in a geometry with torsion.

The natural interpretation is that torsion physically publishes the genealogical memory complementary to metric information.

Curvature describes how geometry deforms. Torsion describes how geometry retains a transport orientation in addition to distances. HMT possesses orientation, carry, and route memory from the outset. The correspondence with Einstein–Cartan is therefore structural: both frameworks distribute geometric history between metric and torsion.

When the universe expands, that memory is diluted as a^{-6}. It is diluted faster than an ordinary matter or radiation density. But precisely for that reason it may is dominant in the small-scale region.

In a primordial regime, a torsional contribution may compete with ordinary density and modify the singularity. The possibility of a bounce depends on the sign, amplitude, spin current, and boundary conditions. HMT first provides the exact scaling law; the physical realization then fixes those conditions.

That offers a very clear division of labor:

- HMT generates the stable memory and its exponent;
- the Einstein–Cartan realization assigns that memory to a quadratic spin density;
- cosmological dynamics fixes amplitude, sign, and physical conditions.

The relation to Mach's principle appears when this law is combined with the derivation of G.

In the gravitational derivation, the local coupling is selected through the closure of a global cycle. In the cosmological law, the local torsion state preserves the memory of global expansion. In both cases, local physics is defined relationally by the global genealogy.

The local geometry recalls the global state.

This is the most concrete sense in which the construction is machina:

- inertia participates in the global clock;
- the gravedad belongs to the clausura of phase;
- torsion embodies the memory of history;
- the local scale encodes the global expansion level.

Mach's principle then acquires an operatorial form: local magnitudes are evaluations of a global genealogy, and physical invariants are what survives transport between the two scales.

The branch of temporal periodicity of (108) phases and the Perron branch also express two complementary aspects of cosmology.

The temporal periodicity of (108) phases fixes to periodicity scale and, through to of Sitter realization, to curvature scale. It is the cosmology of closure: it asks which radius allows the cycle to is coherent.

Perron fixes an expansion rate and a law of memory dilution. It is the cosmology of evolution: it asks how the genealogy changes as levels are traversed.

One branch describes the closure of geometry. The other describes the persistence of its memory.

The chronological calendar and Perron dynamics remain typologically distinct and complementary realizations. Together they produce an especially rich image: the universe possesses a global phase scale and an internal dynamics of memory. Curvature and torsion are two publications of those two aspects.

88.9 Confluence of vacuum readers, action, gravity, area, mixture, and memory

If the nine results are read together, a common grammar appears.

Each reader selectively publishes an invariant and carries the rest as complementary memory.

The vacuum preserves:

- the product as propagation;
- the quotient as orientation.

The pair $G, \hbar$ conserves:

- action as orientation;
- area as the common geometry.

The gravitational closure preserves:

- the equality among phase, Compton, gravity, and Planck.

The mode 30 preserves:

- the common cadence of vacuum, criticality, and memory.

The modular area and Hamiltonian are conserved:

- the total quantity;
- sectoral provenance.

The transito $6 \to 8$ preserves:

- the geometric turn in Barbero;
- its informational footprint in entropy;
- its mixing orientation in CKM.

The angular determinant preserves:

- the even part that can become electroweak mixing.

The quarter-turn torsion preserves:

- a complex structure published as an arithmetic character.

The Perron dynamics preserves:

- the stable memory of the expansion, realized as cosmological torsion.

For that reason the constants are related through to common architecture of transport and preserve their functional identities. They are different residues of that architecture.

Triadic Modular Holography provides the genealogy.

Dimensional Mechanics provides the realization.

The real number appears finally as the terminal value.

In that view, a fundamental constant is a complete statement about the universe. It says:

- what changed;
- which orientation was inverted;

- which memory crossed the boundary;
- what information remained interior and what part was published;
- which combination remained invariant;
- and the physical magnitude in which that invariance was ultimately realized.

That is what makes these derivations remarkable. They jointly explain the values of the constants and why each one performs exactly its structural function.

88.10 Cosmology of the global genealogical state: connection, curvature, expansion, and boundary memory

88.10.1 The totality as a compatible state

An HMT–MD cosmology is a global realization of the system of routes, boundaries, and memory. Its state belongs to the limit of compatible sections; its geometry is expressed by means of a cotetrad, connection, and metric; its evolution publishes a scale factor, curvature, material content, and horizon. The whole intervenes as a mathematical object because each local section is a restriction of a global state with closure conditions.

This idea offers a precise formulation of Mach's principle. Let $b : \mathcal{X}_\infty \to B_\infty$ be the global boundary trace and let $I_v : \mathcal{X}_\infty \to V_v$ be an inertial response in a neighborhood v. The response is Machian when there exists

$$\overline{I}_v : \mathrm{im}(b) \to V_v$$

such that

$$I_v = \overline{I}_v \circ b,$$

and when the purely local restriction leaves distinct responses open. The map $\overline{I}_v$ exists uniquely exactly when I_v is constant on each fiber of b. Local inertia is thus determined by the global closure class.

The frame of a section is obtained by simultaneously restricting the state and the connection:

$$\mathfrak{F}_v(x) = \mathrm{Res}_v(x, \nabla_{\mathrm{TPK}}).$$

An affine geodesic satisfies

$$\nabla_{\dot\gamma}\dot\gamma = 0,$$

and a parallel frame satisfies $\nabla_{\dot\gamma} e_a = 0$. Acceleration, rotation, torsion, curvature, and holonomy measure the defects relative to that induced transport.

88.10.2 3 curvature charts

The elliptic, parabolic, and hyperbolic regimes of the TRIT provide three algebraic domains. Their geometric realization requires a map

$$\mathcal{M}_{s,\rho} : \mathcal{A}_s \longrightarrow (\mathcal{X}_{s,\rho}, g_{s,\rho})$$

that preserves connection, curvature, orientation, and boundary data. With that interface declared, the coordinated atlas uses a negative-curvature chart of anti–de Sitter type, a gnomonic Euclidean chart, and a positive- curvature chart of de Sitter type.

The negative chart organizes the holographic square between interior state and conformal boundary observables. The Euclidean chart realizes the local descent

$$\frac{\mathcal{G}_L}{\mathcal{I}_L} = 1$$

of the equivalence principle. The positive chart receives the accelerated expansion when its dynamics satisfies

$$\frac{\ddot{a}}{a} = H^2 > 0.$$

The common antecedent preserves the areal-volumetric ratio.

$$\frac{\mathcal{L}_{\mathrm{ar}}}{\mathcal{L}_{\mathrm{vol}}} = \frac{\pi_{\mathrm{HMT}}}{\varphi^2_{\mathrm{HMT}}}.$$

The correspondence between algebraic sign and physical geometry is proved in each chart by the metric map, so that the atlas preserves the type of each transition.

The geometry of de Sitter provides an exact result of comparison. For $H > 0$, its closed, flat, and open foliations possess

$$a_{+1}(t) = H^{-1}\cosh(Ht), \qquad a_0(t) = a_* e^{Ht}, \qquad a_{-1}(t) = H^{-1}\sinh(Ht),$$

in their respective domains. The three satisfy

$$\left(\frac{\dot{a}_k}{a_k}\right)^2 + \frac{k}{a_k^2} = H^2, \qquad \frac{\ddot{a}_k}{a_k} = H^2,$$

$$R^{(4)} = 12H^2, \qquad R^{(3)} = \frac{6k}{a_k^2}.$$

The sign k characterizes the curvature of the spatial slice; the curvature of the complete spacetime remains positive and common.

88.10.3 Distributed memory and expansion

Dodecaphase transport separates an even part and an odd part. The even part preserves the cost shared by a route and its inverse; the odd part preserves orientation. In the cosmological reading, the even sector may be realized as the mean response to expansion and the odd sector as the orientation of the sheet.

A distributed memory may proceed from a local generator. The kernel

$$K_q(m,n) = q^{|m-n|}, \qquad 0 < q < 1,$$

is positive and its inverse is tridiagonal. Correlations between arbitrary depths are, therefore, the Green function of a nearest-neighbor action. This identity allows the global history to conserve memory while the update law acts locally.

In a homogeneous geometry, the even response admits the constitutive form

$$p_{\mathrm{eff}} = p - 3\zeta H.$$

The coefficient ζ represents resistance of the symmetric memory to expansion. Its selection belongs to the cosmological realization and must descend from the transport operator and its dimensional chart.

88.10.4 Completion (1000=729+243+27+1), boundary data, and Einstein–Cartan bounce criterion

The completion

$$1000 = 729 + 243 + 27 + 1$$

organizes interior, boundary strata, and seal. In to cosmology, this pattern is realized on to pair $(M, \partial M)$ with co-tetrad, connection, matter, and boundary data. The boundary receives spatial, temporal, or null character; the action incorporates its boundary and corner terms; the gluing conditions fix which histories are prolonged.

An Einstein–Cartan bounce may appear when matter with spin produces an effective contribution that grows as

$$s^2 \propto a^{-6}.$$

In a mass density convention, the equation adopts the scheme

$$H^2 = \frac{8\pi G}{3}\left(\rho_{\mathrm{m}} - \rho_{\mathrm{spin,m}}\right) + \cdots .$$

The equilibrium point $\rho_{\mathrm{m}} = \rho_{\mathrm{spin,m}}$ is a candidate for a regular minimum of the scale factor when the matter action, the sign, the energy conditions, and the boundary problem sustain it. HMT adds the obligation to derive the spin current and the threshold from the enriched state.

A transition toward descending cosmological components is represented by a cobordism.

$$\mathcal{W} : \Sigma_{\mathrm{in}} \longrightarrow \Sigma_{\mathrm{prog}} \sqcup \Sigma_{\mathrm{desc}}.$$

Information conservation is expressed by a cocyclic charge, an isometry, or a unitary evolution. In the quantum case, entropies, mutual information, and correlations replace a simple scalar sum. Gluing terms, causality, hyperbolicity, and the control of singularities are part of the object.

88.10.5 Common system of cosmological observables: expansion, lensing, structural growth, and background anisotropies

A realization intended to explain the discrepancies attributed to dark matter or dark energy must use a common set of parameters for rotation curves, gravitational lensing, structure growth, cosmic microwave background anisotropies, and expansion history. This requirement turns route cosmology into a joint system of correlated observables subject to simultaneous testing.

The dodecaphonic cycle also provides a harmonic selector. For

$$w_k = \frac{1}{12}\left[1 + 2\varepsilon\cos(3\phi_k - \delta)\right],$$

the discrete transform has support only in $m = 0, \pm 3 \pmod{12}$. A celestial prediction requires the map to the sky, the axis, the amplitude, the phase, the mask, and the statistic to be fixed before comparison. The blind comparison preserves the distinction between the harmonic theorem and its cosmological realization.

Result

The Machian factorization, the self-inertia criterion, the three de Sitter foliations, local memory with a distributed Green function, and dodecaphase harmonic support are exact results in their domains. The metric atlas, the effective scale equation, the bounce, and the cobordism constitute typed physical interfaces with explicit conditions and compose route cosmology.

Physical realization

The universe is described as a global section whose boundary conditions the local frames. Expansion transforms collective memory; curvature and waves return that transformation to the routes; the horizon records the accessible part of history.

Boundary of validity

The realization would be falsified if the local response varied within a single global fiber, if the metric map failed to satisfy curvature or orientation, if the acceleration lacked a dynamical equation, if the

bounce produced a singularity or violated the junction conditions, or if a single parameter set proved incompatible with expansion, lensing, structure, and the cosmic background.

88.11 Projective finiteness and harmonic dodecaphasic selector

88.11.1 Internal refinement of the external-contraterm-independent harmonic selector

The assertion that HMT does not require fundamental renormalization distinguishes three operations: normalization of states or probabilities, discrete refinement with a change of scale, and perturbative renormalization by means of a regulator, divergent bare parameters, and cutoff-dependent counterterms. The first two are compatible with HMT; the third is not introduced as a primitive.

Let

$$p_{m+1,m} : X_{m+1} \longrightarrow X_m$$

the surjective projections of an inverse system of finite spaces and let

$$\mathcal{O}_m = C(X_m)$$

the involutive algebras of complex observables. The contravariant morphisms

$$p_{m+1,m}^{*} : \mathcal{O}_m \longrightarrow \mathcal{O}_{m+1}, \qquad p_{m+1,m}^{*} f = f \circ p_{m+1,m},$$

are coherent unitary $*$-homomorphisms. A cylindrical observable is a class compatible under these morphisms.

For probability measures, exact consistency requires

$$(p_{m+1,m})_{*}\mu_{m+1} = \mu_m.$$

For positive and normalized algebraic states $\omega_m : \mathcal{O}_m \to \mathbb{C}$, the condition is

$$\omega_{m+1} \circ p_{m+1,m}^{*} = \omega_m.$$

When $\omega_m(f) = \int f \, d\mu_m$, the two formulations are equivalent.

Theorem 88.17 (Projective compatibility and summable defects)**.** *Let $f \in \mathcal{O}_m$ be a cylindrical observable and suppose that its family of representatives is uniformly bounded.*

1. *If the measures or states are projectively consistent, their expectations are compatible under all refinements.*
2. *Si*

$$\left\| (p_{m+1,m})_{*}\mu_{m+1} - \mu_m \right\|_{\mathrm{TV}} \le \varepsilon_m, \qquad \sum_m \varepsilon_m < \infty,$$

then the expectations form a Cauchy family. The same conclusion holds for states when the defect is measured in the dual norm of $\mathcal{O}_m^{*}$.

Proof. In the exact case,

$$\int_{X_{m+1}} f \circ p_{m+1,m} \, d\mu_{m+1} = \int_{X_m} f \, d\mu_m,$$

and the analogous identity for states is its own consistency condition. In the approximate case, the duality between the uniform norm and total variation gives

$$\left| \int f \circ p_{m+1,m} \, d\mu_{m+1} - \int f \, d\mu_m \right| \le \|f\|_{\infty} \varepsilon_m.$$

Uniform boundedness and summability of the defects turn the sum of the increments into a convergent series; hence the family is Cauchy. The proof in the dual norm is identical. □

For reference actions and measures, the correct condition is the consistency of the Gibbs measures:

$$(p_{m+1,m})_* \left[Z_{m+1}^{-1} e^{-S_{m+1}}\, d\nu_{m+1} \right] = Z_m^{-1} e^{-S_m}\, d\nu_m.$$

A bare equality between actions is insufficient, since a comparison map could conceal a cut-dependent subtraction or rescaling.

The publishable conclusion is therefore restricted to its hypotheses: where a projectively consistent family of states or measures exists, the expectations of uniformly bounded cylindrical observables are compatible; with summable defects in total variation or dual norm, they form a Cauchy family. No external regulator or external counterterms are required, although finite internal effective flows may exist.

In the declared projective system, compatibility of expectations and the stated Cauchy criterion hold exactly. A complete quantum field theory additionally requires convergence of correlators, positivity, unitarity, cofinal independence, and absence of hidden cutoff dependence in maps, actions, measures, or parameters. A sector that converges only after cutoff-dependent parameters are introduced refutes the strong version for that sector.

88.11.2 Harmonic cycle selector of the dodecaphasic cycle

Let $\mathsf{C}_{12}^{\mathrm{cyc}}$ be the local dodecaphase cycle,

$$\phi_k = \phi_0 + \frac{2\pi k}{12}, \qquad k = 0, \dots, 11,$$

and consider the family of weights

$$w_k = \frac{1}{12}\left[1 + 2\varepsilon \cos(3\phi_k - \delta)\right], \qquad |\varepsilon| \le \frac{1}{2}.$$

Then

$$w_k \ge \frac{1 - 2|\varepsilon|}{12} \ge 0, \qquad \sum_{k=0}^{11} w_k = 1.$$

Theorem 88.18 (Fourier dodecaphasic support)**.** *The discrete transform*

$$\widehat{w}_m = \sum_{k=0}^{11} w_k e^{-im\phi_k}$$

is canceled except for

$$m \equiv 0,\ \pm 3 \pmod{12}.$$

Once the origin phase has been absorbed into δ,

$$\widehat{w}_0 = 1, \qquad \widehat{w}_3 = \varepsilon e^{-i\delta}, \qquad \widehat{w}_{-3} = \varepsilon e^{i\delta}.$$

Proof. The transform of the uniform part is the trivial character of the cycle. When writing

$$2\cos(3\phi_k - \delta) = e^{i(3\phi_k - \delta)} + e^{-i(3\phi_k - \delta)},$$

the geometric sums of the characters of $\mathbb{Z}/12\mathbb{Z}$ vanish unless $m - 3$ or $m + 3$ is a multiple of twelve. The three remaining amplitudes are the indicated ones. □

For a crown at fixed colatitude, the support permits only $m = 0$ in $\ell = 2$, whereas in $\ell = 3$, $m = 0, \pm 3$ may survive. The value $\varepsilon = 0$ recovers the uniform axial template; $\varepsilon \neq 0$ introduces the triadic modulation. The result reconciles both supports as a common envelope, but it neither fixes amplitudes nor proves exclusivity: the mode $m = 0$ may also persist for $\ell = 3$.

88.11.3 Prediction of dodecaphasic anisotropy: prior parametrization, blind contrast, and criterion of observational incompatibility

Promotion to an anisotropy prediction requires fixing the following before observing the sky:

1. the map $\mathsf{C}_{12}^{\mathrm{cyc}} \to S^2$ and its equivariance;
2. the celestial frame and the axis, or finite set of axes;
3. $\varepsilon, \delta, \phi_0$, the colatitude, and the normalization;
4. the $a_{\ell m}^{\mathrm{HMT}}$, their parity, and the alignment statistic;
5. mask, foregrounds, bundle, noise, and covariance;
6. the penalty for selection among axes;
7. the joint blind contrast in temperature and polarization.

The retrospective selection of the axis or the fit to C_ℓ^{CMB} is RECOGNITION EXTERNAL, not prediction. The object $\mathsf{C}_{12}^{\mathrm{cyc}}$ is not U_{12}^{sgn}, to representation of $SU(12)$ nor the angular spectrum C_ℓ^{CMB}.

Comparison Criterion 88.19 (Dodecaphase anisotropy)**.** The interface is refuted if the sealed pattern is incompatible with the blind contrast or if axis, amplitude, or modulation is selected after the data have been opened. The support theorem remains valid even if its cosmological realization fails.

88.12 HMT Cosmic Selector and Einstein–Cartan Bounce

88.12.1 Selector of the 3 cartas cosmologicas

The TRIT reader produces

$$\tau(x) \in \{+1, 0, -1\},$$

and the internal clock sets

$$H_*(x) = \omega_0(x) = \frac{2\pi}{108 t_0(x)} > 0.$$

Let $s(x) = \tau(x)$. We define

$$a_+(t; H) = H^{-1} \cosh(Ht), \qquad a_0(t; H) = \exp(Ht), \qquad a_-(t; H) = H^{-1} \sinh(Ht), \quad t > 0$$

and the selector

$$\mathfrak{R}_{\mathrm{dS}}^{\mathrm{HMT}}(x) = \big(M_{s(x)}, g_{s(x),H_*(x)}, a_{s(x)}(\,\cdot\,; H_*(x))\big).$$

Theorem 88.20 (Internal cosmic selector)**.** *The map $\mathfrak{R}_{\mathrm{dS}}^{\mathrm{HMT}}$ is defined for every state with $t_0(x) > 0$ and satisfies*

$$\frac{\ddot{a}_s}{a_s} = H_*^2, \qquad \left(\frac{\dot{a}_s}{a_s}\right)^2 + \frac{s}{a_s^2} = H_*^2, \qquad R[g_{s,H_*}] = 12 H_*^2.$$

The temporal involution $\tau \mapsto -\tau$ interchanges the closed and open charts and fixes the flat chart. The parameter H_* follows from the clock of the same state, not from a cosmological fit.

Proof. Direct substitution of the three functions produces the equations. The action of the involution on $s = \tau$ is immediate. □

88.12.2 Rebound Sector Selector

Let $\mathcal{X}_{\mathrm{EC}}^{\mathrm{HMT}} \subset \mathcal{X}_{\infty}^{\mathrm{enr}}$ be the domain of states whose material and spin-current readers publish

$$A(x) > 0, \qquad B(x) > 0, \qquad 0 < n(x) < 6, \qquad C(x) > 0.$$

The dimensional transducer is applied after those readers. The effective selector is

$$\mathfrak{R}_{\mathrm{EC}}^{\mathrm{HMT}}(x): \quad H^2 = C(x)\left(A(x)a^{-n(x)} - B(x)a^{-6}\right).$$

Theorem 88.21 (Rebounce HMT–Einstein–Cartan in the selected domain)**.** *For every $x \in \mathcal{X}_{\mathrm{EC}}^{\mathrm{HMT}}$ there exists a unique positive threshold*

$$a_b(x) = \left(\frac{B(x)}{A(x)}\right)^{1/(6-n(x))}$$

such that $H(a_b) = 0$. The branch that reaches a_b satisfies

$$\dot{H}(a_b) = \frac{C(x)}{2}(6 - n(x))A(x)a_b^{-n(x)} > 0.$$

Therefore, the selected equation possesses a regular local bounce. If the spin reader publishes $B(x) = 0$, the singular branch reappears

$$a(t) \propto |t - t_0|^{2/n(x)}.$$

Proof. The function $Aa^{-n} - Ba^{-6} = a^{-6}(Aa^{6-n} - B)$ has a unique positive root because $6 - n > 0$. Differentiating the equation with respect to time and taking the limit at the root produces the formula for $\dot{H}$. With $B = 0$, integrating $\dot{a}/a = \sqrt{CA}\, a^{-n/2}$ gives the indicated power. □

The selector is closed in the HMT domain of material and spin readers. That our universe occupies that domain, the value of its scales and the comparison with nucleosynthesis, cosmic background, and anisotropies are RECOGNITION EXTERNAL.

88.13 Descendant universe as a genealogical channel

The authorial architecture of the sheet requires conserving information without cloning the state. We use the two Ambivalence readers to fix

$$p_{\mathrm{d}} = \frac{r_{\mathrm{av}}^{J}}{r_{\mathrm{av}} + r_{\mathrm{av}}^{J}}, \qquad p_{\mathrm{p}} = 1 - p_{\mathrm{d}}.$$

Let $\mathcal{H}_{\mathrm{in}}$ be the Hilbertization of an input fiber and $\mathcal{H}_{\mathrm{tag}} = \mathrm{span}\{|\mathrm{p}\rangle, |\mathrm{d}\rangle\}$. Define

$$|\chi_{\mathrm{HMT}}\rangle = \sqrt{p_{\mathrm{p}}}|\mathrm{p}\rangle + \mathrm{e}^{\mathrm{i}\vartheta(x)}\sqrt{p_{\mathrm{d}}}|\mathrm{d}\rangle,$$

$$V_{\mathrm{desc}} : \mathcal{H}_{\mathrm{in}} \longrightarrow \mathcal{H}_{\mathrm{in}} \otimes \mathcal{H}_{\mathrm{tag}}, \qquad V_{\mathrm{desc}}\psi = \psi \otimes |\chi_{\mathrm{HMT}}\rangle.$$

The phase $\vartheta(x)$ is the orientation reader of the same state. For $\kappa(x) \in \mathbb{R}$, fix

$$K_{\mathrm{desc}}(x) = \kappa(x)\left(\mathrm{e}^{-\mathrm{i}\vartheta(x)}|\mathrm{p}\rangle\langle\mathrm{d}| + \mathrm{e}^{\mathrm{i}\vartheta(x)}|\mathrm{d}\rangle\langle\mathrm{p}|\right)$$

and

$$H_{\mathrm{tot}}(x) = H_{\mathrm{in}}(x) \otimes I + I \otimes K_{\mathrm{desc}}(x).$$

Theorem 88.22 (Unitary descending channel and genealogical conservation)**.** *The map V_{desc} is an isometry and does not clone the state. The label projectors publish probabilities p_{p} and p_{d}. The operator K_{desc} is*

self-adjoint; if H_{in} is self-adjoint, then H_{tot} is also self-adjoint on the domain $D(H_{\mathrm{in}}) \otimes \mathcal{H}_{\mathrm{tag}}$. For every observable A of the incoming state,

$$V_{\mathrm{desc}}^{\dagger}(A \otimes I)V_{\mathrm{desc}} = A.$$

Consequently, for every density state ρ, the state $V_{\mathrm{desc}}\rho V_{\mathrm{desc}}^{\dagger}$ preserves its nonzero spectrum, trace, and unitarily invariant norms. Every charge Q commuting with H_{in} also preserves its global expectation.

Proof. The norm of χ_{HMT} is $p_{\mathrm{p}} + p_{\mathrm{d}} = 1$, hence $V_{\mathrm{desc}}^{\dagger}V_{\mathrm{desc}} = I$. The output contains a single factor ψ, correlated with one label, and not $\psi \otimes \psi$. Self-adjointness is direct, and the compression identity is obtained by evaluating on vectors. □

If an oriented cobordism W subsequently realizes the two labels and carries a closed current j, the Stokes balance

$$I_{\mathrm{in}} = I_{\mathrm{p}} + I_{\mathrm{d}} + F_{\partial}$$

is the topological reader of the same channel. The genealogical channel, its weights, its phase, and its Hamiltonian have been constructed; its realization as a descendant space- time, a causal cobordism, and an observational signal is EXTERNAL RECOGNITION.

88.14 Effective dark tensor and acceleration

Let g_x be a metric produced by $\mathfrak{R}_{\mathrm{dS}}^{\mathrm{HMT}}$, or by another HMT chart in the same realization class, and let $T_b(x)$ be the baryonic tensor published by the material transducer. We assume that $G_{\mathrm{int}}(x)$ and $\Lambda_{\mathrm{ref}}(x)$ are constant over the selected chart. In units $c = 1$, we define, without using cosmological observables as inputs

$$T_{\mathrm{osc}}(x) = \frac{1}{8\pi G_{\mathrm{int}}(x)}\left(G[g_x] + \Lambda_{\mathrm{ref}}(x)g_x\right) - T_b(x).$$

Theorem 88.23 (Closure of the unpublished geometric sector)**.** *Once $g_x, G_{\mathrm{int}}, \Lambda_{\mathrm{ref}}$ and T_b have been fixed, the tensor T_{osc} is unique. If*

$$\nabla^{\mu}T_{b,\mu\nu} = 0,$$

then

$$\nabla^{\mu}T_{\mathrm{osc},\mu\nu} = 0.$$

In four dimensions it possesses the canonical decomposition

$$T_{\mathrm{osc}} = T_{\Lambda} + T_0, \qquad T_{\Lambda} = \frac{1}{4}\,\mathrm{tr}_{g_x}(T_{\mathrm{osc}})\,g_x, \qquad \mathrm{tr}_{g_x}(T_0) = 0.$$

If the trace is constant, the two sectors are conserved separately; in the general case, they exchange exactly opposite current

$$J_{\nu} = \nabla^{\mu}T_{\Lambda,\mu\nu} = -\nabla^{\mu}T_{0,\mu\nu}.$$

Proof. Uniqueness is algebraic. The Bianchi identity and $\nabla g_x = 0$ yield conservation. The decomposition into trace and traceless parts is unique; differentiating it produces the exchange current. □

In the selected de Sitter chart, with $\Lambda_{\mathrm{ref}} = 0$ and $T_b = 0$,

$$G[g_x] = -3H_*^2 g_x, \qquad T_{\mathrm{osc}} = -\rho_* g_x, \qquad \rho_* = \frac{3H_*^2}{8\pi G_{\mathrm{int}}}.$$

Respect to every comoving observer,

$$p_* = -\rho_*, \qquad \frac{\ddot{a}}{a} = -\frac{4\pi G_{\mathrm{int}}}{3}(\rho_* + 3p_*) = H_*^2 > 0.$$

Corollary 88.24 (Internal acceleration of the selected chart)**.** *The positive acceleration of the HMT of Sitter chart is to consequence of the constructed trace tensor and the clock H_*; it is not introduced as an observed parameter.*

Part T_Λ is the homogeneous opening sector, and T_0 is the residual of structure, lensing, and retention within the mathematical chart. Calling them respectively physical dark energy and dark matter requires a single state and a unique parameter set to reproduce rotation, lensing, growth, CMB, and expansion. That joint comparison is EXTERNAL RECOGNITION; the tensor construction and acceleration are already closed.

88.15 Consistency conditions and domain of validity

The preceding constructions are refuted in their respective domains if any of the following cases occurs:

1. $U_F \neq S$, the response has a proper divisor of 108, or the spectral bands overlap despite $\varepsilon < g_{108}/2$;
2. P_{abs} does does not project onto $\ker \mathcal{A}_\partial$, G_{WF} is does not lf-adjoint under the declared hypotheses or the refinement does not intertwine the boundary;
3. ι_n loses orthogonality, the apparatus–memory square does not commute, or the ledger ceases to make the update injective;
4. the self-inertial response varies within a fibre of b, uses a local datum as a substitute for the boundary, or confuses r_{av}^{J} with r_{av}^{-1};
5. the of Sitter selector violates Friedmann or uses an H observed to choose the state; the bounce is asserted outside $A, B, C > 0$ and $0 < n < 6$;
6. $V_{\mathrm{desc}}^{\dagger} V_{\mathrm{desc}} \neq I$, ψ is cloned, or the global charge ceases to be conserved;
7. T_{osc} does not satisfy the Bianchi identity, its decomposition does not reconstitute the tensor, or acceleration is introduced as a datum;
8. the sheet involution does not preserve UV, the chart is not regular at the horizon, or the supposed intertwiner is not UCP.

The mathematical status of these constructions is

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for the eight constructions, each with the explicit primitives that appear in its theorem. Comparisons with macroscopic matter, actual apparatus, electromagnetism, observed gravity, empirical cosmology, observational dark sectors, and astrophysical objects are labeled exclusively

RECONOCIMIENTO EXTERNO.

88.16 Memory dilution and persistence of the bounce

The evolution of memory and spatial expansion are two readings of the same sequence of states. The refinement index can be eliminated between them; that elimination determines the dilution exponent transmitted by the cosmological realization to its torsional contribution. The initial amplitude depends on evaluation of the concrete material current in its variational response. The scaling law and bounce theorem that follow preserve that dependence.

Theorem 82.38 determines that amplitude as a functional of the material state and comoving volume. Corollary 82.39 provides an explicit domain of states with positive, conserved amplitude. In that domain,

evaluation of s_0 precedes the cosmological integration developed below. The ordinary density $\rho_0 > 0$ and its barotropic law are additionally specified in section 88.16.2: the witness to positivity of the amplitude does not in itself evaluate that material density.

88.16.1 The a^{-6} dilution law

Proposition 88.25 (Elimination of the memory index)**.** *Let $a_{\mathrm{ref}} > 0$, $\varphi > 1$, and let the route readers be*

$$\frac{a_n}{a_{\mathrm{ref}}} = \varphi^{n/3}, \qquad M_n = \varphi^{-2n}.$$

Then

$$M_n = \left(\frac{a_n}{a_{\mathrm{ref}}}\right)^{-6}$$

for every index of the sequence. If the realization of the quadratic density is $s_n^2 = s_*^2 M_n$, with $s_*^2 > 0$, we obtain

$$s_n^2 = s_*^2 \left(\frac{a_n}{a_{\mathrm{ref}}}\right)^{-6}.$$

Proof. Raising the first equality to the power -6 gives φ^{-2n}, which is exactly the second reader. Multiplying by the initial amplitude reproduces the second statement. The exponent is determined by the quotient of the exponents of the two readers; the amplitude is the evaluation of the reference state. □

In the continuous homogeneous realization, this constitutive law is preserved: the torsional density is written $s_0 a^{-6}$, where the normalization of the factor a, the material coefficient and the reference amplitude have been absorbed into s_0. The scaling law and evaluation of s_0 are successive operations of the construction.

88.16.2 Homogeneous threshold and transversality

Consider the homogeneous, spatially flat chart, in units $c = 1$, with constant $G > 0$, positive densities ρ_0, s_0 and constant barotropic parameter $w < 1$. The specified realization satisfies $\Lambda_{\mathrm{eff}} = 0$ and the equations

$$\rho_{\mathrm{m}}(a) = \rho_0 a^{-3(1+w)}, \qquad \rho_{\mathrm{s}}(a) = s_0 a^{-6}, \tag{88.75}$$

$$H^2 = \frac{8\pi G}{3}\big(\rho_{\mathrm{m}} - \rho_{\mathrm{s}}\big) =: F_0(a), \qquad \dot H = -4\pi G\big((1+w)\rho_{\mathrm{m}} - 2\rho_{\mathrm{s}}\big). \tag{88.76}$$

The couplings belong to the dimensional section already constructed. The expression for ρ_{s} incorporates the sign of its contribution in this gravitational realization.

Theorem 88.26 (Existence and uniqueness of the transverse threshold)**.** *There exists a unique $a_{\mathrm{b}} > 0$ with $F_0(a_{\mathrm{b}}) = 0$:*

$$a_{\mathrm{b}}^{3(1-w)} = \frac{s_0}{\rho_0}.$$

The region allowed by $H^2 \geq 0$ is $a \geq a_{\mathrm{b}}$. At the threshold, with $\rho_{\mathrm{b}} = \rho_{\mathrm{m}}(a_{\mathrm{b}}) = \rho_{\mathrm{s}}(a_{\mathrm{b}}) > 0$,

$$\dot H_{\mathrm{b}} = 4\pi G(1-w)\rho_{\mathrm{b}} > 0.$$

The solution crossing $H = 0$ has a strict minimum of a, with local expansion

$$a(t) = a_{\mathrm{b}}\left[1 + 2\pi G(1-w)\rho_{\mathrm{b}}(t - t_{\mathrm{b}})^2 + O\big((t-t_{\mathrm{b}})^3\big)\right].$$

Proof. The factorization

$$F_0(a) = \frac{8\pi G}{3} a^{-6}\big(\rho_0 a^{3(1-w)} - s_0\big)$$

reduces its zeros to that of a strictly increasing function of $(0,\infty)$ on $(-s_0,\infty)$. This proves existence, uniqueness and the sign of F_0 on both sides of the threshold. The second equation of (88.76), evaluated at equality of densities, gives the formula for $\dot H_{\mathrm b}$. Since $\dot a = aH$, at the threshold $\ddot a_{\mathrm b} = a_{\mathrm b}\dot H_{\mathrm b} > 0$. The equations are smooth for $a > 0$, so their Taylor expansion gives the indicated expansion. The sign change of H is transverse: it passes from contraction to expansion. □

For $w = 1$, both terms have the same exponent and equality of amplitudes determines a degenerate case. For $w > 1$, the sign of $\dot H$ at a coincidence of densities is negative. The domain $w < 1$ of the theorem expresses exactly the order between the two dilution exponents.

88.16.3 Local persistence under controlled perturbations

Persistence of the threshold uses two independent estimates: control of the sign at the endpoints and of the derivative on the interval. Let $I = [a_-, a_+] \subset (0,\infty)$, with $a_- < a_{\mathrm b} < a_+$, be chosen so that

$$F_0(a_-) < 0 < F_0(a_+), \qquad m_I := \inf_{a\in I} F_0'(a) > 0.$$

Such an interval exists because

$$F_0'(a_{\mathrm b}) = \frac{8\pi G}{a_{\mathrm b}}(1-w)\rho_{\mathrm b} > 0.$$

Define $\delta_I = \min\{-F_0(a_-), F_0(a_+)\} > 0$.

Theorem 88.27 (Local persistence of the transverse bounce)**.** *Let $R \in C^1(I)$ be a perturbation satisfying*

$$\|R\|_{\infty,I} < \delta_I, \qquad \|R'\|_{\infty,I} < m_I.$$

Then $F_R = F_0 + R$ has a unique zero $a_R \in (a_-, a_+)$, and $F_R'(a_R) > 0$. The realization $H^2 = F_R(a)$ admits a local transverse bounce, with

$$\dot H_R = \frac{a_R}{2} F_R'(a_R) > 0$$

at the threshold. If $R(a,\eta)$ is jointly C^1, the persistent zeros depend locally in a C^1 manner on the parameter η.

Proof. The first bound preserves $F_R(a_-) < 0 < F_R(a_+)$; the intermediate value theorem provides an interior zero. The second gives $F_R'(a) \ge m_I - \|R'\|_{\infty,I} > 0$ on I, so the zero is unique and simple.

At $a > a_R$ near the zero, $F_R(a) = F_R'(a_R)(a-a_R) + o(a-a_R)$. The integral

$$t - t_{\mathrm b} = \int_{a_R}^{a} \frac{d\xi}{\xi\sqrt{F_R(\xi)}}$$

is finite and strictly increasing. Its inverse defines the expanding branch; time reflection defines the contracting branch. The two join with $\dot a(t_{\mathrm b}) = 0$ and

$$a(t) - a_R = \frac{a_R^2 F_R'(a_R)}{4}(t-t_{\mathrm b})^2 + o((t-t_{\mathrm b})^2).$$

Away from the threshold, differentiating $H^2 = F_R(a)$ gives $\dot H = aF_R'(a)/2$; its continuous limit gives the positive expression at a_R. The C^1 regularity of F_R guarantees continuity of acceleration at this junction. Finally, $\partial_a F_R(a_R) > 0$ allows application of the implicit function theorem when the perturbation depends jointly on (a,η). □

The first bound controls existence; the second ensures local uniqueness and transversality. The proved stability corresponds to that threshold in the selected interval. The torsional amplitude enters its location, while the relation between exponents determines the temporal orientation of the crossing.

88.17 Explicit homogeneous solution and causal completeness

Determination of the amplitude allows a complete realization of the bounce to be developed, in addition to the local argument. The spatially flat chart of (88.76) is retained, in units $c = 1$, with $\rho_0, s_0, G > 0$, and the material sector is specialized to dust, $w = 0$, with $\Lambda_{\mathrm{eff}} = 0$. The coefficient s_0 is that obtained from the material state in the corresponding realization; the following integration uses that coefficient and preserves its domain.

Theorem 88.28 (Exact integration of the dust bounce)**.** *Set*

$$a_{\mathrm{b}}^3 = \frac{s_0}{\rho_0}, \qquad \tau^2 = \frac{s_0}{6\pi G \rho_0^2}, \qquad u = t - t_{\mathrm{b}}.$$

The solution of the homogeneous system with $a(t_{\mathrm{b}}) = a_{\mathrm{b}}$ and $H(t_{\mathrm{b}}) = 0$ is, for every $t \in \mathbb{R}$,

$$a(t) = a_{\mathrm{b}} \left(1 + \frac{u^2}{\tau^2}\right)^{1/3}, \qquad H(t) = \frac{2u}{3(\tau^2 + u^2)}, \qquad \dot{H}(t) = \frac{2(\tau^2 - u^2)}{3(\tau^2 + u^2)^2}. \tag{88.77}$$

It has a unique minimum, at t_{b}; it contracts before and expands afterward. The evolution is smooth and satisfies $a(t) \geq a_{\mathrm{b}} > 0$.

Proof. Let $y = a^3$. The Friedmann equation gives

$$\dot{y}^{\,2} = 24\pi G(\rho_0 y - s_0).$$

The function $y = s_0/\rho_0 + 6\pi G \rho_0 u^2$ satisfies that identity and the initial conditions. Its derivatives give the three expressions of (88.77). To check the evolution equation as well, let us write

$$\rho_{\mathrm{b}} = \frac{\rho_0^2}{s_0}, \qquad x = \frac{u}{\tau}.$$

Then

$$\rho_{\mathrm{m}} = \frac{\rho_{\mathrm{b}}}{1 + x^2}, \qquad \rho_{\mathrm{s}} = \frac{\rho_{\mathrm{b}}}{(1 + x^2)^2}, \qquad 4\pi G \rho_{\mathrm{b}} = \frac{2}{3\tau^2}.$$

Substituting into $\dot{H} = -4\pi G(\rho_{\mathrm{m}} - 2\rho_{\mathrm{s}})$ gives exactly the third formula. Thus the complete system $(\dot{a}, \dot{H})$ is satisfied, whose field is smooth for $a > 0$. Its local uniqueness, continued along the explicit solution, fixes the evolution through those initial conditions. The denominator is positive for every t, and the statements about the minimum and the sign follow from H. □

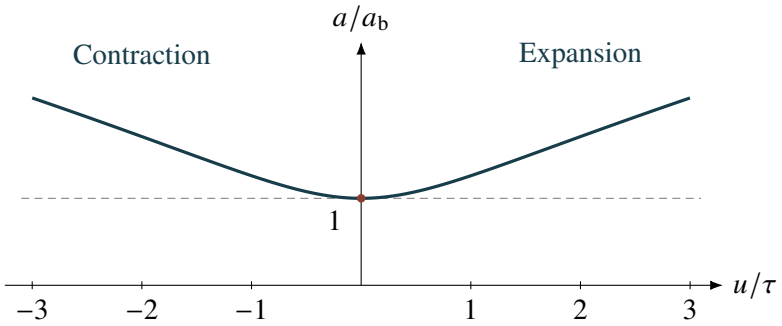

Figure 88.1: Dimensionless profile of the proved solution: $a/a_{\mathrm{b}} = (1 + (u/\tau)^2)^{1/3}$. The parameters a_{b} and τ are calculated from ρ_0, s_0, G; the drawing represents the entire rescaled family.

88.17.1 Finite curvature and extension of geodesics

The metric realized in this chart is $g = -dt^2 + a(t)^2\, d\mathbf{x}^2$ on $\mathbb{R}\times\mathbb{R}^3$. With the convention in which the flat curvature scalar is $R = 6(\dot H + 2H^2)$, the solution gives

$$R(t) = \frac{4\tau^2 + \frac{4}{3}u^2}{(\tau^2+u^2)^2}.$$

In particular, $R(t_{\mathrm{b}}) = 4/\tau^2$ is finite. In an orthonormal frame, the Riemann components depend on H^2 and on $\dot H + H^2$, which are bounded functions on $\mathbb{R}$. The same holds for polynomial contractions of those components; for example,

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 12\big[(\dot H + H^2)^2 + H^4\big].$$

Proposition 88.29 (Completeness of causal geodesics)**.** *The preceding metric realization is geodesically complete for the timelike and null geodesics of its Levi–Civita connection, in both directions.*

Proof. Spatial translations provide the constants $p_i = a^2\, dx^i/d\lambda$. Causal normalization gives

$$\left|\frac{dt}{d\lambda}\right| = \sqrt{\epsilon + \frac{|\mathbf{p}|^2}{a(t)^2}}, \qquad \epsilon = 1 \text{ in the timelike case}, \quad \epsilon = 0 \text{ in the null case}.$$

A nonconstant null geodesic has $|\mathbf{p}| > 0$. Since $a \geq a_{\mathrm{b}}$, in the timelike case

$$\left|\frac{d\lambda}{dt}\right| \geq \left(1 + \frac{|\mathbf{p}|^2}{a_{\mathrm{b}}^2}\right)^{-1/2} > 0,$$

and in the null case $|d\lambda/dt| = a/|\mathbf{p}| \geq a_{\mathrm{b}}/|\mathbf{p}| > 0$. Thus, reaching $t = \pm\infty$ requires infinite affine length. On a compact time interval, a and its derivatives are smooth and a is bounded away from zero. The preceding explicit equations and $dx^i/d\lambda = p_i/a^2$ allow the geodesic to be continued through its finite endpoints. Both possibilities of continuation are covered. □

The completeness result belongs to this homogeneous dust realization, with its spatial, matter and amplitude conditions. Its local persistence under perturbations of the threshold preserves the bounds of theorem 88.27; that local property and the global completeness of the explicit solution are distinct conclusions. The minimum scale comes from the relation between material density and torsional response. Memory thus enters a verifiable geometric evolution, with a regular realization of contraction and expansion.

88.18 Residual tensor, trace decomposition, and covariant balance

The geometric realization of the enriched state provides a metric, while the material transducer publishes the source to which the reading refers. The residual tensor preserves the difference between the two publications in the same tensor space. Its decomposition distinguishes the sector proportional to the metric from the traceless sector, and determines the exchange current corresponding to that partition.

88.18.1 Chart data and construction of the residual tensor

Let us fix an HMT state x and a connected chart $\mathcal{U}$ of its geometric realization. On $\mathcal{U}$, the nondegenerate co-tetrad of section 80.6 determines a smooth metric $g = g_x$ of signature $(-+++)$. We denote by $\mathring{\nabla}$ its Levi–Civita connection and write

$$\mathsf{G}_{\mu\nu}[g] := \mathrm{Ric}_{\mu\nu}[g] - \tfrac{1}{2}R[g]g_{\mu\nu}.$$

This distinguishes the Einstein tensor $\mathsf{G}[g]$ from the gravitational coupling G_{int}. The derivative used in the balances of this section is $\mathring{\nabla}$, consistently with the effective metric equation of Section 82.8.5.

We work in the dimensional chart $c_{\mathrm{int}} = 1$. We assume $G_{\mathrm{int}}(x) > 0$ and $\Lambda_{\mathrm{ref}}(x)$ constant on $\mathcal{U}$, and put $\kappa_x = 8\pi G_{\mathrm{int}}(x)$. The symbol x identifies the state supplying those readers; covariant differentiation acts on the points of $\mathcal{U}$. The genealogy of the couplings remains in section 121.

Let T_b be the smooth symmetric tensor published by the chart's material transducer; in the baryonic reading, it is its baryonic source. We define

$$\mathcal{E}^{\mathrm{ref}}_{\mu\nu} := \kappa_x^{-1}\big(\mathsf{G}_{\mu\nu}[g] + \Lambda_{\mathrm{ref}} g_{\mu\nu}\big), \qquad T^{\mathrm{res}}_{\mu\nu} := \mathcal{E}^{\mathrm{ref}}_{\mu\nu} - T_{b,\mu\nu}. \tag{88.78}$$

The tensor T^{res} is the effective residual tensor, also denoted T_{osc} when interpreting the geometric sector remaining to be published relative to T_b. The construction preserves $g_x, G_{\mathrm{int}}, \Lambda_{\mathrm{ref}}$ and T_b as effective arguments.

Theorem 88.30 (Uniqueness of the residual and balance with the material source)**.** *For the preceding data there exists a unique symmetric tensor T^{res} satisfying $\mathcal{E}^{\mathrm{ref}} = T_b + T^{\mathrm{res}}$. Its divergence is*

$$\mathring{\nabla}^{\mu} T^{\mathrm{res}}_{\mu\nu} = -\mathring{\nabla}^{\mu} T_{b,\mu\nu}. \tag{88.79}$$

In particular, if the material source is conserved separately, $\mathring{\nabla}^{\mu} T_{b,\mu\nu} = 0$, then T^{res} is also conserved.

Proof. The difference of two tensors satisfying the stated equation is zero; expression (88.78) provides its existence. The contracted Bianchi identity and Levi–Civita metric compatibility give

$$\mathring{\nabla}^{\mu}\mathsf{G}_{\mu\nu} = 0, \qquad \mathring{\nabla} g = 0.$$

Since κ_x and Λ_{ref} are constant on the chart, $\mathring{\nabla}^{\mu}\mathcal{E}^{\mathrm{ref}}_{\mu\nu} = 0$. Differentiating the difference defining T^{res} proves (88.79), including the statement under separate material conservation. □

The spatial and temporal constancy of the couplings in the chart fixes the domain of this balance. A family of states may publish different couplings in different charts; each preceding identity applies to the chart and state specifying it.

88.18.2 Canonical decomposition and exchange current

Define the trace function and the two tensors

$$\theta := g^{\mu\nu} T^{\mathrm{res}}_{\mu\nu}, \qquad T^{\Lambda}_{\mu\nu} := \tfrac{1}{4}\theta g_{\mu\nu}, \qquad T^{0}_{\mu\nu} := T^{\mathrm{res}}_{\mu\nu} - T^{\Lambda}_{\mu\nu}. \tag{88.80}$$

The superscript Λ identifies the component proportional to the metric; Λ_{ref} is the reference parameter of (88.78). Their functions are distinct.

Proposition 88.31 (Trace projectors and uniqueness of the decomposition)**.** *On symmetric covariant tensors of order two, the maps*

$$P_{\mathrm{tr}}(S) = \tfrac{1}{4}\,\mathrm{tr}_g(S)\, g, \qquad P_0(S) = S - P_{\mathrm{tr}}(S)$$

satisfy

$$P_{\mathrm{tr}}^2 = P_{\mathrm{tr}}, \quad P_0^2 = P_0, \quad P_{\mathrm{tr}} P_0 = P_0 P_{\mathrm{tr}} = 0, \quad P_{\mathrm{tr}} + P_0 = I. \tag{88.81}$$

In particular, $T^{\mathrm{res}} = T^{\Lambda} + T^{0}$, $\mathrm{tr}_g\, T^{0} = 0$, and this decomposition is unique.

Proof. In dimension four, $g^{\mu\nu} g_{\mu\nu} = 4$. Therefore, $\mathrm{tr}_g(P_{\mathrm{tr}} S) = \mathrm{tr}_g\, S$, which proves idempotence of P_{tr}, vanishing of the trace of $P_0 S$ and, by expansion, the other identities. If $S = f g + Q$ with $\mathrm{tr}_g\, Q = 0$, taking the trace gives $f = \mathrm{tr}_g\, S/4$. This also determines $Q = S - f g$, and with it uniqueness. □

Theorem 88.32 (Covariant exchange between the two sectors)**.** *On the domain of* (88.78)*, let* $b_\nu := \mathring{\nabla}^\mu T_{b,\mu\nu}$*. The trace-exchange current is the one-form*

$$\mathcal{J}^{\mathrm{tr}}_\nu := \tfrac{1}{4}\partial_\nu\theta = \mathring{\nabla}^\mu T^{\Lambda}_{\mu\nu}. \tag{88.82}$$

The complete balances are

$$\mathring{\nabla}^\mu T^0_{\mu\nu} = -b_\nu - \mathcal{J}^{\mathrm{tr}}_\nu, \qquad \mathring{\nabla}^\mu (T^\Lambda + T^0)_{\mu\nu} = -b_\nu. \tag{88.83}$$

If $b = 0$*, the two sectors exchange exactly opposite currents. In that case they are conserved separately if and only if* θ *is constant in the connected chart.*

Proof. Metric compatibility allows the calculation

$$\mathring{\nabla}^\mu\big(\tfrac{1}{4}\theta g_{\mu\nu}\big) = \tfrac{1}{4}g^{\mu\alpha}(\partial_\alpha\theta)g_{\mu\nu} = \tfrac{1}{4}\partial_\nu\theta.$$

Subtracting this equality from (88.79) produces (88.83). With $b = 0$, separate conservation is equivalent to $\mathrm{d}\theta = 0$; a smooth function with zero differential is constant on each connected component. □

The one-form $\mathcal{J}^{\mathrm{tr}}$ measures the exchange of the tensor partition. It retains a type distinct from the transport covector of section 80.7 and from the spin current $\sigma_{IJ} \in \Omega^3(\mathcal{U}; \Lambda^2 E^*)$ of (82.20).

88.18.3 Link to the effective Cartan–Holst source

On a solution of (82.35), in the same units $c = 1$, the residual has expression

$$T^{\mathrm{res}} = T^{\mathrm{phys,LC}} + U^{\mathrm{phys,spin}} - T_b + \frac{\Lambda_{\mathrm{ref}} - \Lambda_{\mathrm{int}}}{\kappa_x}\, g. \tag{88.84}$$

Indeed, substituting $\mathsf{G} + \Lambda_{\mathrm{int}} g = \kappa_x(T^{\mathrm{phys,LC}} + U^{\mathrm{phys,spin}})$ into (88.78) gives equality term by term. When the material transducer publishes precisely $T_b = T^{\mathrm{phys,LC}}$ and $\Lambda_{\mathrm{ref}} = \Lambda_{\mathrm{int}}$, we obtain $T^{\mathrm{res}} = U^{\mathrm{phys,spin}}$. The separate conservation condition of T_b in theorem 88.30 retains its own function: (82.36) establishes the balance of the total source, and the partition among its components depends on the material action used.

88.18.4 Trace sector and acceleration of the selected charts

The TRIT reader of state x publishes $s = \tau(x) \in \{+1, 0, -1\}$, and its positive clock publishes

$$H_* = \frac{2\pi}{108\, t_0(x)} > 0. \tag{88.85}$$

Let h_s be a three-dimensional metric of constant sectional curvature s. The three charts of the selected realization are written

$$g_s = -\mathrm{d}t^2 + a_s(t)^2 h_s, \qquad a_+(t) = H_*^{-1}\cosh(H_* t), \quad a_0(t) = \exp(H_* t), \quad a_-(t) = H_*^{-1}\sinh(H_* t). \tag{88.86}$$

The closed and flat charts are considered for $t \in \mathbb{R}$; the open chart, for $t > 0$. The sign s and the clock H_* retain their antecedents in the state selecting the realization.

Lemma 88.33 (Curvature of the three charts of the same clock)**.** *For the functions in* (88.86)*,*

$$\frac{\ddot{a}_s}{a_s} = H_*^2, \qquad \left(\frac{\dot{a}_s}{a_s}\right)^2 + \frac{s}{a_s^2} = H_*^2, \qquad \mathrm{Ric}[g_s] = 3H_*^2 g_s. \tag{88.87}$$

Consequently, $R[g_s] = 12H_*^2$ *and* $\mathsf{G}[g_s] = -3H_*^2 g_s$*.*

Proof. The first two equalities are obtained by differentiating the three functions: in the closed and open charts we use $\cosh^2 z - \sinh^2 z = 1$, and in the flat chart $\dot{a}_0 = H_* a_0$. To verify the tensor identification, the connection coefficients involving the time coordinate are

$$\Gamma^0{}_{ij} = a_s \dot{a}_s (h_s)_{ij}, \qquad \Gamma^i{}_{0j} = (\dot{a}_s/a_s)\delta^i_j,$$

and the purely spatial ones coincide with those of h_s. Contraction of the curvature gives

$$\mathrm{Ric}_{00} = -3\ddot{a}_s/a_s, \qquad \mathrm{Ric}_{0i} = 0, \qquad \mathrm{Ric}_{ij} = (a_s\ddot{a}_s + 2\dot{a}_s^2 + 2s)(h_s)_{ij}.$$

Substituting the first two identities of (88.87) produces $\mathrm{Ric}_{00} = -3H_*^2$ and $\mathrm{Ric}_{ij} = 3H_*^2 a_s^2 (h_s)_{ij}$. Taking the trace and forming the Einstein tensor proves the other statements, with the curvature convention of (82.35). □

Theorem 88.34 (Density, pressure, and sign of the selected acceleration)**.** *In the preceding charts, for* $\Lambda_{\mathrm{ref}} = 0$ *and* $T_b = 0$*, the residual tensor is*

$$T^{\mathrm{res}} = -\rho_* g_s, \qquad \rho_* := \frac{3H_*^2}{8\pi G_{\mathrm{int}}} > 0, \qquad T^\Lambda = T^{\mathrm{res}}, \quad T^0 = 0. \tag{88.88}$$

For a unit comoving observer $u = \partial_t$*, the density and isotropic pressure are* ρ_* *and* $p_* = -\rho_*$*. The acceleration equation evaluates to*

$$\frac{\ddot{a}_s}{a_s} = -\frac{4\pi G_{\mathrm{int}}}{3}(\rho_* + 3p_*) = H_*^2 > 0. \tag{88.89}$$

Proof. The preceding lemma and (88.78) give $T^{\mathrm{res}} = -(3H_*^2/\kappa_x) g_s$. Since $g_s(u,u) = -1$, $T^{\mathrm{res}}_{\mu\nu} u^\mu u^\nu = \rho_*$. The spatial projector $q^{\mu\nu} = g_s^{\mu\nu} + u^\mu u^\nu$ satisfies $q^{\mu\nu}(g_s)_{\mu\nu} = 3$, and therefore

$$p_* := \tfrac{1}{3} q^{\mu\nu} T^{\mathrm{res}}_{\mu\nu} = -\rho_*.$$

The trace equals $-4\rho_*$, so (88.80) produces the two sectors of (88.88). Finally, $\rho_* + 3p_* = -2\rho_*$, and substitution into the right-hand side of (88.89) gives $\kappa_x \rho_*/3 = H_*^2$, equal to the geometric value already calculated. Positivity uses exactly $G_{\mathrm{int}} > 0$ and $H_* > 0$. □

In this realization, the trace sector publishes a constant density and an opposite pressure, and the internal clock fixes positive acceleration. In a general chart, T^Λ describes the pointwise isotropic component, while T^0 preserves the traceless residual and both obey the exchange of (88.83). Identification of these sectors with observational dark sources requires a single realization to reproduce expansion, lensing, rotation, structure growth and the cosmic background jointly. The present construction fixes the tensor, its balances and the acceleration of the selected charts entering that comparison.

Chapter 89

Polyakov action, T-duality, strings, and branes: compactification in dimension 11 and dynamical realization of M-theory

89.1 Extended Persistence and String and Brane Worldvolumes

89.1.1 Extended persistence and world surface

A physical string is an extended one-dimensional object whose evolution sweeps out a worldsheet. Each point of the string has an internal coordinate, and each instant adds a temporal coordinate; their combination forms the domain on which the induced metric, tension, and background fields live. A closed string has a circular spatial section. An open string has endpoints subject to boundary conditions. Its vibrations organize states, and its tension converts the area traversed by the worldsheet into action.

A brane of dimension p is an extended object whose history forms a worldvolume of dimension $p + 1$. Branes receive string endpoints, carry fields on their volume, and may wrap cycles of a compact space. This definition fixes three different data: the dimension of the object, the dimension of its history, and the conditions governing its boundary. The tension, charges, and spectrum belong to the dynamical realization that completes those geometric data.

HMT reaches this boundary from an architecture of persistent routes. A route preserves the orientation, sheet, phase, quotient, carry, and memory of the crossings that have prolonged it. The passage to an extended object assigns that genealogy to a section distributed over an internal coordinate. The composition of routes is then effected by sewing segments; the boundary memory records windings and intersections; the action measures the complete persistence over the traversed surface. This map preserves the causal order: first the enriched state is constructed, and its realization as a string, brane, or field is declared thereafter.

The Polyakov action provides a precise form for the physical codomain:

$$S_{\mathrm{P}}[X,h] = -\frac{T}{2}\int d^2\sigma\,\sqrt{-h}\,h^{ab}g_{MN}(X)\partial_a X^M\partial_b X^N.$$

Here T is the tension, h_{ab} the worldsheet metric and X^M the map into the target space. An HMT string realization must produce these objects, their state space and their symmetries from a typed map; the route structure supplies the genealogical domain on which that map acts.

89.1.2 Exact screens in dimensions 11, 10, 26 and 24

The first screen starts from the extended ternary code $C_W = [12, 6, 6]_3$. Puncturing one coordinate produces the code $G_{11} = [11, 6, 5]_3$. Its Hamming balls of radius two have cardinality

$$V_3(11, 2) = 1 + 22 + 220 = 243 = 3^5,$$

and the 3^6 codewords satisfy

$$3^6 V_3(11, 2) = 3^{11}.$$

Therefore, these balls partition the ambient space exactly. The integer eleven here measures the length of a perfect code and retains its combinatorial type.

A second screen uses the real permutation module V_{12} associated with the twelve icosahedral vertices. The complement of the uniform vector decomposes as

$$V_{11} = V_3 \oplus V_{3'} \oplus V_5.$$

The dodecaphase state K polarizes a line $\ell_K \subset V_3$ and determines the projector

$$P_{10} = P_{11} - \frac{u_K u_K^{\mathsf{T}}}{\|u_K\|^2}.$$

From it descend the exact decompositions.

$$V_{11} = \ell_K \oplus V_{10}, \qquad V_{10} = V_{3'} \oplus W_7(K),$$

with dimensions $11 = 1 + 3 + 7$ and $10 = 3 + 7$. The step $12 \to 11$ eliminates the uniform mode; the step $11 \to 10$ eliminates the direction marked by K. A physical compactification adds a temporal line, a stable rank-six lattice, and the transport that preserves the three extended directions.

The third screen starts from the lattice

$$L_{26} = \Lambda_{24} \oplus U \cong II_{25,1},$$

where U is the integral hyperbolic plane. For a primitive isotropic vector e_+, the reduction

$$e_+^{\perp}/\mathbb{Z}e_+ \cong \Lambda_{24}$$

is exact. The orthogonality condition removes one direction and the quotient removes the null direction contained in its orthogonal complement; the result is rank twenty-four. Code, real module, and lattice belong to distinct categories. Their explicit maps make it possible to relate their ranks while preserving the provenance of each.

The incidence between the six faces and the eight vertices of the cube also provides a screen of spatial and temporal signature. If M is its incidence matrix, then

$$MM^{\mathsf{T}} = 12P_u + 4P_-.$$

The quotient by the kernel has rank four and decomposes as $1 \oplus 3$. Upon declaring the uniform line to be the temporal direction, the normalized form becomes $\mathrm{diag}(-1, 1, 1, 1)$. The incidence proves the rank and the spectral ratio; the declared orientation fixes the signature.

89.1.3 Moment, winding and reciprocal radius

In a compact coordinate of radius R, the integer m records momentum and the integer w records winding. The form

$$E_R(m, w) = \frac{m^2}{R^2} + w^2 R^2$$

has the involution

$$\mathsf{T}(m, w, R) = (w, m, R^{-1}), \qquad E_R(m, w) = E_{R^{-1}}(w, m).$$

In $\ell^2(\mathbb{Z}^2)$, the exchange $U_T|m, w\rangle = |w, m\rangle$ provides the unitary equivalence

$$U_T H(R) U_T^{-1} = H(R^{-1}).$$

This is the algebraic residence of radius duality: momentum and winding are two reciprocal readings of the same lattice sector. The realization as T-duality incorporates the string scale, the oscillators and the action on the physical state space. At the self-dual radius both readings share the same scale and the exchange acts within the same spectrum.

89.1.4 Interface in dimension 11: spectral triple, supersymmetry and operator $Q^2 = H$

M-theory organizes an eleven-dimensional physical codomain with metric g_{MN}, three-form C_{MNP}, gravitino ψ_M, action, and supersymmetry transformations. The bridge is written as a map

$$\mathfrak{R}_M : \mathcal{X}_{\mathrm{HMT}}^{\mathrm{enr}} \longrightarrow \mathcal{X}_{11}^{\mathrm{phys}}.$$

Its source preserves phase, orientation, sewing and memory; its codomain adds the fields and the eleven-dimensional dynamics. The compactification of one direction must transport these fields to a ten-dimensional screen, preserve the low-energy limit and intertwine the declared dualities.

A candidate compactification is typed by a triple $(x_\infty, \kappa, \mathcal{U})$. Section x_∞ represents a prolongable persistence; κ carries it to a moduli space; $\mathcal{U}$ transports metric, orientation, register, and boundary. Its admissibility requires prolongation to every depth, commutation with duality, descent of the spectrum to the quotient, and preservation of the domains and codomains of the observables. These conditions convert an analogy of ranks into a verifiable mathematical problem.

Strings appear after compactifying the eleven-dimensional sector; branes appear as extended realizations coupled to the three-form and its duals. The exceptional branch that reaches Λ_{24}, $V^\natural$, and the Monster preserves the same enriched source through another functor. The dimensional and exceptional interfaces are distinct prolongations of a common genealogy, each with its own codomain.

Result

The perfection of G_{11}, the polarization $11 \to 10$, the isotropic reduction $26 \to 24$, the cubic screen of signature $(3, 1)$ and the radius involution are proven algebraic results. Together they form the exact residence of ranks, quotients and dualities on which the interface toward strings, branes and M-theory is built.

Physical realization

An extended history preserves position, winding, orientation, and boundary within a single persistence. Compactification determines which directions appear as macroscopic extension and which act as internal degrees. The string reads that persistence on a worldsheet; the brane reads it on a worldvolume; M-theory coordinates both readings through eleven-dimensional fields.

Boundary of validity

The realization is falsified if the map $\mathfrak{R}_M$ alters the types of the screens, if the duality ceases to act unitarily on the spectrum, if the compactification loses prolongation, or if the fields fail to satisfy the action and supersymmetry. Physical closure requires tension, oscillators, amplitudes, controlled anomalies, and a coherent gravitational limit for each declared compactification.

89.1.5 From exact screens to the physical interface of M-theory

The first part constructed, in explicit mathematical domains, the reductions

$$12 \longrightarrow 11 \longrightarrow 10, \qquad 26 \longrightarrow 24,$$

and the moment, winding, and radius involutions

$$(m, w, R) \longleftrightarrow (w, m, R^{-1}).$$

These results are not yet a physical theory of strings nor an action of supergravity. They constitute the screens and equivalences that a dimensional realization must respect.

The interface is typed via a map declared

$$\mathcal{R}_M : \mathcal{X}^{\mathrm{enr}}_{\mathrm{HMT}} \longrightarrow \mathcal{X}^{\mathrm{phys}}_{11},$$

whose codomain must carry, at a minimum, the metric field g_{MN}, the three-form C_{MNP}, the gravitino ψ_M, its action, and the supersymmetry transformations. A specialization to strings further requires identification of the physical radius, tension, momentum and winding sectors, and the state space on which T-duality is realized. A conformal or CFT reading requires its own operator algebra, grading, and state–operator correspondence.

The positive connection with the exceptional branch is not a bare arrow Monster→M-theory. Both prolongations preserve data from the same enriched state—phase, grading, orientation, sewing, and memory— but employ distinct functors. In the exceptional branch, those data reach $V^\natural$, the Monster, and Moonshine; in the dimensional branch they feed the screens and, only through $\mathcal{R}_M$, a possible realization of fields and strings.

Definition 89.1 (Internal Admissibility of a Compactification)**.** A candidate compactification is represented by a triple

$$(x_\infty, \kappa, \mathcal{U}),$$

where $x_\infty \in \mathcal{X}^{\mathrm{enr}}_{\mathrm{HMT}}$ is a surviving section,

$$\kappa : \mathcal{X}^{\mathrm{enr}}_{\mathrm{HMT}} \dashrightarrow \mathcal{M}$$

is the possibly partial map into a declared moduli space $\mathcal{M}$, and $\mathcal{U}$ transports metric, orientation, register and boundary data. We define

$$\mathrm{Adm}_{\mathrm{HMT}}(x_\infty, \kappa, \mathcal{U}) = 1$$

if the following four conditions hold:

1. the section has a compatible extension to every depth;
2. the declared dualities are compatible at the point considered:

$$\kappa(\mathsf{T}_{\mathrm{HMT}} x_\infty) = \mathsf{T}_{\mathcal{M}}(\kappa(x_\infty)),$$

   when both sides are defined;
3. the spectrum descends to the compactification quotient;
4. the observables preserve their domains, codomains, and types under $\mathcal{U}$.

This predicate is an internal admissibility criterion and not an automatic identification with the physical *Swampland*. The distance, tower, weak-gravity, and absence-of-global-symmetries criteria, as well as the geometric conditions of a specific compactification, are subsequently evaluated on $\mathrm{im}\,\kappa$ and require the corresponding metric maps, fields, and scales. The preceding definition fixes the interface that a realization must preserve; it does not by itself prove the existence of a physical compactification or a Swampland conjecture.

Scope of the interface

The screens, quotients, and T-dual involution are exact mathematical results. The complete dictionary toward $(g_{MN}, C_{MNP}, \psi_M)$, the action, supersymmetry, and a physical CFT is a typed interface that must be verified in each realization. This boundary does not diminish the mathematical result or authorize its presentation as an already closed physical dynamics.

89.2 Realization of M-theory through compactification, strings, branes, and 11 dimensions

The interface dimensional toward theory M requires distinguish the exact structure of split, the lift operatorial and the fields fisicos of the codomain. Let

$$\mathfrak{S}_M = (\mathcal{H}_{\text{phys}}, H, Q)$$

a spectral triple in which $\mathcal{H}_{\text{phys}}$ is the Hilbert space after imposing the gauge constraints, H is a nonnegative self-adjoint operator, and Q is an odd, densely defined supercharge compatible with the grading $\mathcal{H}_{\text{phys}} = \mathcal{H}_+ \oplus \mathcal{H}_-$. The relation

$$Q : \mathcal{H}_+ \rightleftarrows \mathcal{H}_-, \qquad Q^2 = H \tag{89.1}$$

is interpreted on its common domain. It does not replace the relativistic supersymmetric algebra; it is its Hermitian reduction of dimension 1 and fixes the minimal datum that a spectral realization must preserve.

89.2.1 Operator lifting of the APP–TRIT–TPK state

Let $\mathcal{H}_{\text{HMT}}$ be the completion of the surviving sections with phase, orientation, quotient, carry, boundary, route, and memory, and let

$$W : \mathcal{H}_{\text{HMT}} \longrightarrow \mathcal{H}_{\text{phys}}$$

a realization isometry. An admissible operator lift preserves the APP composition, the TRIT orientation, and the full TPK dynamics, and intertwines

$$W\widehat{H} = HW, \qquad W\widehat{Q} = QW, \qquad \widehat{Q}^2 = \widehat{H}. \tag{89.2}$$

The first equality transports the generator, the second preserves the grading, and the third prevents introducing H as a datum foreign to the genealogical antecedent. These identities are obligations of the bridge: the existence of fermionic and bosonic completions alone does not prove (89.1).

89.2.2 Physical undecadimensional codomain

The realization in dimension 11 is expressed by

$$\mathfrak{R}_M : \mathcal{X}_{\mathrm{HMT}}^{\mathrm{enr}} \longrightarrow \mathcal{X}_{11}^{\mathrm{phys}},$$

whose codomain carries metric g_{MN}, the 3-form C_{MNP}, the gravitino ψ_M, the action, and its supersymmetry transformations. Compactification along one direction must simultaneously transport these fields, the inner product, and the grading of $\mathfrak{S}_M$. Strings appear on the dimension-(10) screen; branes couple to the 3-form and its duals; and T-duality exchanges momentum and winding through

$$(m, w, R) \longmapsto (w, m, R^{-1})$$

without altering the spectrum of the compact sector.

The Polyakov action

$$S_{\mathrm{P}}[X, h] = -\frac{T}{2} \int d^2\sigma \, \sqrt{-h}\, h^{ab} g_{MN}(X) \partial_a X^M \partial_b X^N$$

fixes the worldsheet dynamics once T, h_{ab}, the maps X^M, and the boundary conditions have been determined. The screen chains $12 \to 11 \to 10$ and $26 \to 24$, the radius involution, and the lattice reduction are prior algebraic results; the action, spectrum, and amplitudes constitute the subsequent physical realization.

89.2.3 Compatibility of the spectral triple $(\mathcal{H}_{\mathrm{phys}}, H, Q)$ with the physical codomain in dimension 11

A realization is admissible if it preserves at the same time:

1. the compatible prolongation of the state to every depth;
2. the domains and codomains of H, Q, and the observables;
3. the intertwining identities (89.2);
4. the unitary action of T-duality on the spectrum;
5. the descent of the fields of dimension 11 to the stated compactification;
6. compatibility with the physical inner product and the gauge constraints.

The failure of any of these conditions distinguishes an analogy of ranks from a physical realization. The formulation does not automatically identify the exceptional corridor with M-theory: both publications follow from the same enriched state, but use distinct functors and codomains.

Chapter 90

General Mass Law on the atlas 81/56/13/324: trajectories, record, signature, character, towers 90/120 and the null sector

90.1 Operator Domain of the General Mass Law: 81 Cells, 56 Families, 13 Multisections, 324 Routes, and Null Sector

The General Mass Law acts on persistent sections of the oriented atlas, not on a table of species. Its internal domain contains 81 cells, 56 route families, 13 multisections, and 324 oriented routes, in addition to a separate null chart. The censuses of physical outputs are subsequent and do not alter this domain.

90.2 Determinantal origin of the absolute scale

The integral form H_4 has Smith normal form $\mathrm{diag}(1, 2, 2, 4)$, so its cokernel has order 16. It follows that

$$\Sigma_H = \varphi_{\mathrm{HMT}} \alpha_{\mathrm{HMT}}^{16}, \qquad \mathrm{ord}_{10}(\Sigma_H) = -34.$$

The absolute chain is

$$\Sigma_H \to \hbar_{\mathrm{pre/ret}} \to \omega_0 \to \varepsilon_{\mathrm{clk}} \to m_{\mathrm{clk}} \to \widehat{M}^{(0)} \to \widehat{M}^{(\beta)}.$$

The decade -34 is never introduced from the SI system or absorbed into the relative correction.

90.3 Governing construction of the law and the material domain

The full development of the law is organized here by result: state, route, record, signature, character, towers, absolute branch, and operator. Its internal order is preserved, while the realizations by species are reserved for the corresponding chapter.

90.3.1 Spectral Definition of Mass as a Persistence Measure of a Route 1 Coupled to a Physical Regime

Mass is the spectral measure of the persistence of a route when a genealogical section is coupled to a physical regime. It is not a scalar attribute affixed in advance to a particle name. The APP–TRIT–TPK dynamics constructs the route, its memory, its signature, and its character; Dimensional Mechanics realizes that genealogy on a resonance mass line and determines, through the coupling, the observable subspace and its spectrum.

An elementary particle is represented by a persistent section; a noncentral family, by a stable multisection; a composite particle, by a composition of genealogical registers with a linking boundary; a resonance, by a pole of the coupled operator; a topological sector, by its fusion ring and its braiding representation before any mass chart; a virtual section, by its terminal horizon of prolongation. These realizations are distinct and must not be flattened into a single list of numbers.

The external magnitude never enters the selection of cell, route, signature, reader, tower coefficient, or eigenvector. Metrological comparison is performed only after the internal output has been constructed and fixed.

90.3.2 Complete state space of mass

Let C_{81} be the cellular atlas. Each cell admits four bidirectional route orientations, so the finite space of oriented routes is

$$\mathcal{R}_{324} = \{(c, h_0, v_0) : c \in C_{81},\ (h_0, v_0) \in \{\pm 1\}^2\}.$$

The cellular projection, the family relation, and the coarse stratification produce

$$|C_{81}| = 81, \qquad |\mathcal{F}_{56}| = 56, \qquad |\mathcal{M}_{13}| = 13, \qquad |\mathcal{R}_{324}| = 324.$$

The electron occupies the unique central fixed point. The twelve noncentral multisections are finite fibers stable under cellular inversion and do not, in general, admit an equivariant point section. Their natural object is therefore an operator on the fiber rather than a number chosen from among its cells. The null sector is incorporated through the zero section of the mass line, not as the value of a positive character.

The physical output catalog distinguishes three charged leptons, three neutrino interaction realizations, six quark flavors with their color orbits, two null gauge realizations, two massive gauge bosons, one scalar sector, two composite families, three topological codomains, and a system of virtual sections. This subsequent classification does not limit the number of composite or resonant states: composition and excitation generate additional families within their respective codomains.

90.3.3 Genealogical trajectory of a mass realization

The law does not begin with a particle label nor with a number. Its order is

$$\begin{aligned}\mathrm{APP} \to \mathrm{TRIT} \to \mathrm{TPK} \to \widetilde{\Gamma}^{\mathrm{surv}} \to \Gamma^{\mathrm{obs}} \to L_{108} \to \sigma \in \mathbb{Z}^5 \\ \to \chi_0 \to \chi_{\mathrm{full}} \to \widehat{\mathcal{M}} \to \text{coupling} \\ \to \text{spectrum/pole} \to \mathcal{L}_M \to \text{comparison}.\end{aligned}$$

Here:

- APP arranges the local incompatibility, the sum/product sheets, and their orientation;
- the TRIT preserves sign, threshold, and bidirectional opening;
- the TPK evolves phase, memory, carry, survival, and return;
- a particle is a persistent section of an extension, and its observable route is the history it leaves in the genealogical record;
- the integer signature is additive;
- the character turns additive composition into a positive ratio;
- the towers refine the character through a single global harmonic operator;
- the mass line supplies the dimensional type;
- measurement is coupling: the coupling determines which subspace and which spectrum are physically accessible, without consulting the quantity to be measured subsequently.

Mass does not select the route. The persistent route, the genealogical record, and the coupling select the spectral object whose mass is read.

90.3.3.1 Conserved components of the enriched state of the TPK along the mass trajectory

The state that feeds the mass law is

$$X_K = (w_6, w_{30}, R_{36}, g_K, B_K, p_K, c_K, \varepsilon_K, \omega_K),$$

where short word, prolonged word, region, nonadic phase, block memory, prefix, carry, sheet, and winding are preserved. The chain

$$w_6 \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

It does not directly project a mass: it constructs the persistent identity that is later read.

90.3.3.2 Nonadic filtration with memory

From each state one obtains a refinement fiber formed by 729 subcylinders,

$$\{C_{K,j} : 0 \le j < 729\}.$$

The pruning G_9 satisfies

$$G_9(C_{K,j}) \in \{0,1\}, \qquad \sum_{j=0}^{728} G_9(C_{K,j}) = 1.$$

If j_K is the survivor,

$$P_{K+1} = 729P_K + j_K.$$

After nine steps,

$$g(K+9) = g(K),$$

but they change, in general,

$$(B_{K+9}, P_{K+9}, \mathrm{pref}_{K+9}, \mathrm{Witt}_{K+9}) \neq (B_K, P_K, \mathrm{pref}_K, \mathrm{Witt}_K).$$

This is the equational form of persistence: the phase returns, not the state. The mass can recognize a particle after the return because the genealogical record preserves the difference.

90.3.3.3 Nonadic carry transduction to the signed dodecaphasic state

For a temporal coordinate θ, define

$$g_0(\theta) = \theta \bmod 9, \qquad \beta(\theta) = \left\lfloor \frac{\widetilde{\theta}}{9} \right\rfloor \bmod 12.$$

Then

$$g_0(\theta+9) = g_0(\theta), \qquad \beta(\theta+9) = \beta(\theta) + 1 \pmod{12}.$$

Thus, the nonadic return preserves the phase and advances one memory tooth. The 108 times are

$$108 = 9 \cdot 12,$$

so that each genealogical record contains a complete cycle of both scales. The 36.864 carry compositions describe the local combinations of this bidirectional memory.

90.3.3.4 Completion and Conservation of the Unit

The base boundary mil decomposes as

$$999 = 729 + 270 = 729 + 243 + 27,$$

while the completion is

$$1000 = 729 + 271 = 729 + 243 + 27 + 1, \qquad 271 = 270 + 1.$$

+1 is the unique vertex lying entirely on the boundary and realizes the carry that returns the unit. The partition 729/271 distinguishes the active kernel from the mediating boundary; it is not a golden-ratio

approximation. The same conservation allows a route to change blocks without losing its normalized identity.

90.3.3.5 Theorem of the complete oriented atlas

Every cell $c \in C_{81}$ has four orientations

$$(h_0, v_0) \in \{\pm 1\}^2.$$

The map

$$(c, h_0, v_0) \longmapsto \Gamma(c, h_0, v_0)$$

is bijective over $\mathcal{R}_{324}$. Consequently,

$$|\mathcal{R}_{324}| = 4|C_{81}| = 324.$$

Each route has 108 genealogical register entries, so the total temporal census is

$$81 \cdot 108 = 8748.$$

The quadruple

$$(\text{seed row}, \text{seed column}, h_0, v_0)$$

determines of manner single the route, its word w_6, its word dodecaphasic, their returns, memory, carry, signature historical and readers. Therefore, the 324 routes not only may is enumerated: may is reconstructed and is joined exactly, field by field, without using a mass.

90.3.3.6 Quotients of families and multisections

The quotient of the 81 cells by route history yields 56 families. Their multiplicities are distributed as

$$41 \times 1, \quad 10 \times 2, \quad 2 \times 3, \quad 1 \times 4, \quad 2 \times 5,$$

and

$$41 + 20 + 6 + 4 + 10 = 81.$$

The stratification into thirteen multisects possesses ranges

$$1, \quad 2, \quad 4, \quad 6, \quad 8, \quad 12, \quad 16$$

with multiplicities

$$1, \quad 2, \quad 3, \quad 2, \quad 3, \quad 1, \quad 1,$$

and sum of ranks

$$1 + 4 + 12 + 12 + 24 + 12 + 16 = 81.$$

The electron occupies the rank-one block. The remaining sectors lie in higher-rank blocks and require a representation projector, not a nominal cell.

90.3.3.7 Signature, readers, and memory by route

For each $\Gamma \in \mathcal{R}_{324}$, the following are calculated:

$$\sigma_\Gamma \in \mathbb{Z}^5, \qquad K_D(\Gamma), \qquad K_\Omega(\Gamma),$$

together with first returns of class, orbit, cell, and kinematic state; separation of prefixes; carry debt; ambiguity; stabilized triads; sheet and winding. The family K_D has 14 profiles; K_Ω, nine; together they form 40 pairs and 18 coincidences. This diversity is the fine information that makes it possible to pass from a coarse fiber to a physical operator.

90.3.4 From the APP defect to the full signature

90.3.4.1 Primordial defect and unfolding

At the center APP,

$$B_c = 9P_+ + 5P_-, \qquad 9 - 5 = 4.$$

The primordial local gap is 4. The global defect 54 belongs to a later unfolding $2 \to 6 \to 54$; it must not be projected back onto the center as though it were the same operation.

90.3.4.2 12 *eventos and divisor dodecafasico*

The oriented coordinate does not receive a factor $1/6$ arbitrarily. The chain is

$$12 \text{ events} \to s_C^{\mathrm{tot}} \to 1 + 5 + 6 \to s_C^{(12)} = \frac{s_C^{\mathrm{tot}}}{6} \to W_\pm = 6n_A \pm s_C \to \mathrm{SNF}(1, 12) \to \chi_{\mathrm{mass}}.$$

The $1/6$ is the oriented quotient of the closure of twelve events. It is not the residue (-1/12), it is not the pseudoinverse of a valency, and it is not the hexad–octad incidence.

90.3.4.3 Construction of the complete full signature from the route register

The genealogical record CT–108 produces

$$\sigma(\Gamma) = \left(n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta\right) \in \mathbb{Z}^5.$$

The coordinates have distinct origins:

- n_A: dominant duration/persistence;
- s_C: net orientation of the bilayer, divided by six after the dodecaphasic closure;
- k_{Δ_4}: minimal-torsion channel;
- b_{120}: event coefficient of the register;
- b_ζ: step-three autocorrelation.

We define

$$\zeta_{270} := 135\Delta_4^2,$$

and remains separated from

$$s_{270} = q_-^{270} - q_+^{270}.$$

The 324 oriented routes are joined bijectively to their five historical coordinates through the canonical route identity. This join uses only genealogical structure and consults no metrological magnitude.

90.3.4.4 Direct angle, counter-angle, and conjugate channels

The direct angular coordinate proceeds from the already constructed fine structure constant:

$$A = 1000\alpha_{\mathrm{HMT}} = 7.297352569283800997285105472380663\ldots^{\circ}.$$

The counter-angle is not obtained from a mass or from the Barbero–Immirzi functional. Let

$$D_A = \exp\!\left(-\frac{100\pi}{9}\alpha_{\mathrm{HMT}}\right),$$

$$C_\pi(\alpha_{\mathrm{HMT}}) = 169\alpha_{\mathrm{HMT}}^6 + \frac{D_A\alpha_{\mathrm{HMT}}^7}{1 - D_A\alpha_{\mathrm{HMT}}},$$

$$y_* = \frac{\pi}{90}(H_5 + \alpha_{\mathrm{HMT}}) - C_\pi(\alpha_{\mathrm{HMT}}), \qquad C^* = \frac{180}{\pi}y_*.$$

Therefore,

$$C^* = 2.123738338968645724261951380393\ldots^{\circ}.$$

The coordinate A is the common angular mode and C^*, the oriented differential mode. Its passage to the Archimedean chart is

$$x = A_{\mathrm{rad}} = \frac{\pi A}{180}, \qquad y = C^*_{\mathrm{rad}} = \frac{\pi C^*}{180}.$$

In the real algebra generated by the involution $R = \mathrm{diag}(1, -1)$, with $P_\pm = (I \pm R)/2$, the biangular block is

$$B_{A,C^*} = AI + C^*R = (A + C^*)P_+ + (A - C^*)P_-.$$

Angle plus counter-angle and angle minus counter-angle are thus the two eigenvalues of a single object. Exponentiation produces the channels

$$q_+ = e^{-x-y}, \qquad q_- = e^{-x+y}.$$

The chart is invertible:

$$x = -\frac{1}{2}\log(q_+q_-), \qquad y = \frac{1}{2}\log\frac{q_-}{q_+}.$$

The transformation $C^* \mapsto -C^*$ preserves the projectors and exchanges the channels. The return coordinate satisfies, after the regional construction,

$$\eta_{\mathrm{ret}} = \frac{C^*}{2} - \alpha_{\mathrm{HMT}}\,\mathrm{deg},$$

without constituting a second selection rule for C^*.

90.3.4.5 Central Character

With

$$\Theta_0 = \left(A_{\mathrm{rad}}, \frac{C^*_{\mathrm{rad}}}{6}, \Delta_4, s_{120}, \zeta_{270}\right),$$

the central character is

$$\chi_0(\sigma) = \exp\Big(n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}} + k_{\Delta_4}\Delta_4 + b_{120}s_{120} + b_\zeta \zeta_{270}\Big).$$

By additivity of the signature,

$$\chi_0(\sigma_1 + \sigma_2) = \chi_0(\sigma_1)\chi_0(\sigma_2).$$

The physical unit does not enter this exponential: χ_0 is a positive dimensionless automorphism of the mass line.

90.3.5 Global harmonic hierarchy of towers

With the coordinates (x=A_{\rm rad}) and (y=C^*_{\rm rad}) already constructed, let

$$q_\pm = \exp\Big[-\frac{\pi}{180}(A \pm C^*)\Big] = e^{-x\mp y}.$$

In a globally consistent convention,

$$T = e^{-xI - yR}, \qquad c_n = \mathrm{Tr}(T^n) = q_-^n + q_+^n,$$
$$s_n = -\mathrm{Tr}(RT^n) = q_-^n - q_+^n.$$

Equivalently and operatorially,

$$c_n = 2e^{-nx}\cosh(ny), \qquad s_n = 2e^{-nx}\sinh(ny).$$

The involution $C^* \mapsto -C^*$ fixes the projectors and permutes the spectral coefficients, or equivalently the associated channels. It follows that

$$c_{m+n} = \frac{c_m c_n + s_m s_n}{2}, \qquad s_{m+n} = \frac{s_m c_n + c_m s_n}{2},$$
$$c_n^2 - s_n^2 = 4e^{-2nx},$$

and the common second-order recurrence. The admissible heights form

$$\mathfrak{S} = \langle 90, 120\rangle = \{90, 120\} \cup \{30r : r \geq 6\}.$$

The values $90, 120, 180, 210, 240, 270, 360, 420$ are genealogical witnesses, not the complete tower. Before the quotient, height 360 preserves two histories:

$$k[\mathfrak{S}_0] \simeq k[X_{90}, Y_{120}]/(X_{90}^4 - Y_{120}^3).$$

The general elevation is written

$$\chi_{\mathrm{full}}(\sigma, \eta) = \chi_0(\sigma)\exp\Big(\sum_{h\in\mathfrak{S}} \eta_h s_h\Big),$$

with

$$N_{120} = b_{120} + \eta_{120}.$$

The numerical sum does not erase the dual provenance: b_{120} comes from the ledger and η_{120} from the spectral refinement. Nor can it absorb $b_\zeta\zeta_{270}$ into $\eta_{270}s_{270}$.

90.3.6 Absolute Rama: action, clock and mass line

The determinantal reader produces

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1,2,2,4), \qquad |\operatorname{coker} H_4| = 16,$$
$$\Sigma_H = \varphi\alpha_{\mathrm{HMT}}^{16}, \qquad \mathrm{ord}_{10}(\Sigma_H) = -34.$$

The two oriented sections of a single action line are

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S.$$

Its additive fine chart and its multiplicative chart are

$$\delta_{2w} = H_5 - \eta_{\mathrm{ret}}, \qquad R_{\mathrm{act}} = \frac{H_5}{\eta_{\mathrm{ret}}} > 0.$$

They are not two Planck constants. The common decade fixes the absolute order; the quotient transports relative outward/return information.

Declared $t_0 \in \mathcal{L}_T^+$, CT–108 fixes

$$\omega_0 = \frac{2\pi}{108 t_0}, \qquad \varepsilon_{\mathrm{clk}} = \hbar_{\mathrm{ret}}\omega_0, \qquad m_{\mathrm{clk}} = \varepsilon_{\mathrm{clk}} c_{\mathrm{int}}^{-2}.$$

The character acts by homothety on $\mathcal{L}_M$; it does not become $\hbar$. In mass ratios, every common factor cancels:

$$\frac{m_Y}{m_X} = \chi_{\mathrm{full}}(\sigma_Y - \sigma_X, \eta_Y - \eta_X) R_{\mathrm{act}}^{\kappa_Y - \kappa_X}.$$

Thus 10^{-34} fixes the absolute scale but does not correct a relative residual. A relative modification can follow only from the route, the specific clock, the reader, or the towers.

Remains a distinction necessary. HMT constructs a section internal a time declared t_0; the coordinate MeV requires a chart. For close the chart absolute without a convention externa must construirse

$$\Gamma \longmapsto t_\Gamma \quad \text{or} \quad \Gamma \longmapsto \omega_\Gamma.$$

Section 10 defines a clock reader based on returns and holonomy. Its absolute normalization is determined when the route operator fixes a positive section of the temporal line.

The normal form, the cokernel of order sixteen, the ablations, and the distinction between h and $\hbar$ are proved in full in section 59.2. Here that already-constructed result is used to fix the absolute branch of the law, without restarting its derivation.

90.4 Conjugate angular coordinates and spectral decomposition

The passage from the arithmetic closure of α to the dimensional hierarchy is effected through two angular coordinates. The first follows directly from α_{HMT}; the second is the oriented phase produced by the regional analytic realization of the quintuple microfiber. Both reach Dimensional Mechanics already constructed, before the exponential chart that will generate the global moments is introduced.

90.4.1 Transfer of the angular coordinates derived in HMT

Parts I–III constructed distinct dependencies with a common root in α_{HMT}. The pre-return character follows from

$$(\alpha_{\mathrm{HMT}}, \varphi;\ 9, 5, 4, 12, 54,\ \text{graded rule}) \longrightarrow H_5,$$

while the angular coordinate and the determinant follow from

$$\alpha_{\mathrm{HMT}} \longrightarrow A = 1000\alpha_{\mathrm{HMT}}{}^{\circ}, \qquad (\alpha_{\mathrm{HMT}}, \pi) \longrightarrow D_A = e^{-100\pi\alpha_{\mathrm{HMT}}/9}.$$

Through a distinct dependence, the regional microfiber fixes $\mathcal{F}_\pi \to \rho_\pi = 169$, and with it

$$C_\pi(\alpha_{\mathrm{HMT}}) = 169\alpha_{\mathrm{HMT}}^6 + \frac{D_A\alpha_{\mathrm{HMT}}^7}{1 - D_A\alpha_{\mathrm{HMT}}}.$$

The branch of H_5 and the regional correction converge only at the following step:

$$y_* = \frac{\pi}{90}(H_5 + \alpha_{\mathrm{HMT}}) - C_\pi(\alpha_{\mathrm{HMT}}),$$
$$C^* = \frac{180}{\pi}y_*.$$

Henceforth, A denotes the direct angular coordinate and C^* the conjugate regional angular coordinate; these designations fix their types without introducing a second selection rule. Consequently, H_5 is not an output of A. Here y_* is the analytic coordinate in radians and C^* its sexagesimal coordinate. Dimensional Mechanics receives these already determined outputs; it does not redefine them. The coordinate A is not a direct input to the formula of C^*: the two objects are coupled for the first time in $(A, C^*) \mapsto q_\pm$. The equality $\alpha_{\mathrm{HMT}} = A/1000$ is solely the inverse change of coordinate. In this Part we retain the notations

$$A = 7.297352569283800997285105472380663^{\circ},$$
$$C^* = 2.123738338968645724261951380393\ldots{}^{\circ}.$$

The rounding used in the tables is $2.123738338968646^{\circ}$. The integer 169, the determinant D_A, and the regional recurrence belong to the preceding proof; neither any SI comparison nor the hexad–octad functional intervenes in their generation.

The angular identity

$$\eta_{\mathrm{ret}}^{(\mathrm{deg})} = \frac{C^*}{2} - \alpha_{\angle,\mathrm{deg}}, \qquad \alpha_{\angle,\mathrm{deg}} := \alpha_{\mathrm{HMT}}\,\mathrm{deg},$$

defines the coordinate angular adimensional posterior to the return; does not employed for generate C^*. Equivalently, a time performed the construction regional,

$$C^* = 2(\eta_{\mathrm{ret}}^{(\mathrm{deg})} + \alpha_{\angle,\mathrm{deg}}).$$

This last equality is the inverse form of the same relation and not a second selection rule for C^*.

The regional realization corrects H_5 before producing $\eta_{\mathrm{ret}}^{(\mathrm{deg})}$; therefore, H_5, $\eta_{\mathrm{ret}}^{(\mathrm{deg})}$, and C^* have distinct mathematical types. The symbol $\hbar_{\mathrm{HMT}}$ is reserved for the magnitude on the action line obtained only after applying the scale $10^{-34}\mathcal{U}_S$. Comparison with an SI mantissa belongs to external metrological control and leaves the preceding construction invariant.

90.4.2 Internal phases and exponential chart

The abstract phases associated with the two angles are

$$a = \frac{A}{360}, \qquad c = \frac{C^*}{360} \quad \text{in } \mathbb{R}/\mathbb{Z}.$$

Before applying a real exponential, the canonical representatives $\widetilde{a}, \widetilde{c} \in [0, 1)$ are chosen. The exponential acts on those representatives, not directly on the quotient classes. Conversion to radians is introduced only upon choosing the Archimedean chart

$$x := A_{\mathrm{rad}} = \frac{\pi A}{180}, \qquad y := C^*_{\mathrm{rad}} = \frac{\pi C^*}{180},$$

$$q_- = e^{-x+y}, \qquad q_+ = e^{-x-y}.$$

The factor $\pi/180$ belongs to this change analytic of coordinates. The closure aritmetico of α, formulated before of this conversion, is independent of that choice of chart angular.

90.4.3 Central involutive algebra

Consider the two-dimensional regular representation of the real algebra generated by an involution. In a spectral basis, fix

$$R = \mathrm{diag}(1, -1), \qquad I = I_2,$$

so that both eigenspaces have multiplicity one. Let

$$P_+ = \frac{I + R}{2}, \qquad P_- = \frac{I - R}{2}$$

their spectral projectors. The central element determined by the angular coordinates is

$$B_{A,C^*} = AI + C^* R = (A + C^*)P_+ + (A - C^*)P_-.$$

The involution $C^* \mapsto -C^*$ preserves the fixed projectors $P_\pm$ and interchanges the two spectral coefficients —equivalently, the associated channels—. Applying the exponential yields

$$T = e^{-xI - yR} = q_+ P_+ + q_- P_-.$$

Therefore, the two channels are the eigenvalues of a single element of the algebra generated by I and R.

90.4.4 Invertibility of the channel chart

Theorem 90.1 (Reconstruction of the angular coordinates)**.** *Para $q_\pm > 0$,*

$$x = -\frac{1}{2}\log(q_+ q_-), \qquad y = \frac{1}{2}\log\frac{q_-}{q_+}.$$

Consequently, the map $(A, C^*) \mapsto (q_+, q_-)$ is injective. The exchange $q_+ \leftrightarrow q_-$ preserves A and transforms $C^* \mapsto -C^*$.

Proof. The multiplication and quotient of the two exponentials give $q_+ q_- = e^{-2x}$ and $q_-/q_+ = e^{2y}$. Taking logarithms recovers x and y. Since $\pi/180 \neq 0$, both angles are determined. □

90.4.5 Addition and Recurrence Identities

In this two-dimensional representation, the even and odd moments of the operator T are

$$c_n = \operatorname{Tr}(T^n) = q_-^n + q_+^n, \qquad s_n = -\operatorname{Tr}(RT^n) = q_-^n - q_+^n.$$

The multiplicity-one hypothesis is part of the domain of these formulas. In a representation with multiplicities m_+ and m_-, the corresponding traces would be $m_+q_+^n + m_-q_-^n$ and $m_-q_-^n - m_+q_+^n$, respectively.

Theorem 90.2 (Laws of Composition of Moments)**.** *For $m, n \geq 0$,*

$$c_{m+n} = \frac{c_m c_n + s_m s_n}{2}, \qquad s_{m+n} = \frac{s_m c_n + c_m s_n}{2},$$

$$c_n^2 - s_n^2 = 4e^{-2nx},$$

and both sequences satisfy

$$X_{n+2} = c_1 X_{n+1} - e^{-2x} X_n.$$

Moreover,

$$\partial_A c_n = -\frac{n\pi}{180} c_n, \quad \partial_A s_n = -\frac{n\pi}{180} s_n,$$
$$\partial_{C^*} c_n = \frac{n\pi}{180} s_n, \quad \partial_{C^*} s_n = \frac{n\pi}{180} c_n.$$

Proof. Expand the products of $q_-^m \pm q_+^m$ with $q_-^n \pm q_+^n$. The quadratic identity follows from $(u+v)^2-(u-v)^2 = 4uv$. The recurrence corresponds to the characteristic polynomial $z^2 - c_1 z + q_- q_+$, and the derivatives are obtained using the chain rule. □

Corollary 90.3 (Determination of the complete hierarchy)**.** *With A and C^* fixed, all pairs (c_n, s_n) are determined by the recurrence. No physical species introduces an additional channel.*

Finally,

$$s_n = 2e^{-nx}\sinh(ny), \qquad c_n = 2e^{-nx}\cosh(ny).$$

The coordinate A determines the common decay, and C^* the oriented separation. The decimal order of the action line is not introduced into these formulas. It is deduced in section 59.2 from the integral cokernel of the local Hadamard block. The subsequent choice of a basis expressed in J s belongs to the metrological transducer and does not alter the dimensionless identities.

90.4.6 Action ellipse, radian, and double holonomy return

The preceding focal theorem fixes the angular form and the determinantal reader fixes the action scale. Let us define

$$\eta = \operatorname{artanh}\frac{C^*}{A}, \qquad a_S = \sqrt{2\hbar_{\mathrm{HMT}}^{\mathrm{ret}} e^{\eta}}, \qquad b_S = \sqrt{2\hbar_{\mathrm{HMT}}^{\mathrm{ret}} e^{-\eta}}.$$

The symplectic orbit

$$Q(\theta) = a_S\cos\theta, \qquad P(\theta) = b_S\sin\theta$$

simultaneously preserves the shape ratio of the angular block and the area fixed by the action line.

Theorem 90.4 (Phase swept area)**.** *For every θ,*

$$\operatorname{Area}(\theta) = \frac{1}{2}\int_0^{\theta} (QP' - PQ')\,dt = \hbar_{\mathrm{HMT}}^{\mathrm{ret}}\theta.$$

Therefore,

$$\operatorname{Area}(1\text{ rad}) = \hbar_{\mathrm{HMT}}^{\mathrm{ret}}, \qquad \operatorname{Area}(2\pi) = h_{\mathrm{HMT}}^{\mathrm{ret}}.$$

Proof. Since $QP' - PQ' = a_S b_S$ and $a_S b_S = 2\hbar^{\mathrm{ret}}_{\mathrm{HMT}}$, the integral equals $\hbar^{\mathrm{ret}}_{\mathrm{HMT}}\theta$. The second identity uses $h^{\mathrm{ret}}_{\mathrm{HMT}} = 2\pi\hbar^{\mathrm{ret}}_{\mathrm{HMT}}$. □

APP supplies the arean oriented; A, C^* fix the form and the foci; the branch determinantal fixes the scale. The turn primitive encloses h and a radian of phase sweeps $\hbar$. The radian ordinary remains $1\,\mathrm{rad} = 180°/\pi$; the construction adds its genealogy as sector unitary of action, not another constant angular.

For a persistent section with holonomy $\varepsilon_\gamma \in \{+1, -1\}$, the condition

$$\varepsilon_\gamma e^{iJ_\gamma/\hbar} = 1$$

distinguishes two closures. In the cylinder, $\varepsilon_\gamma = +1$, and the primitive cycle transports $J = 2\pi\hbar = h$. In the Möbius band, $\varepsilon_\gamma = -1$: one visible turn changes sheet and transports $J = \pi\hbar = h/2$; the second restores orientation and completes h. Per radian of the internal phase, $\hbar$ is transported in both cases; the factor $1/2$ belongs to the projection onto the visible turn of the Möbius base.

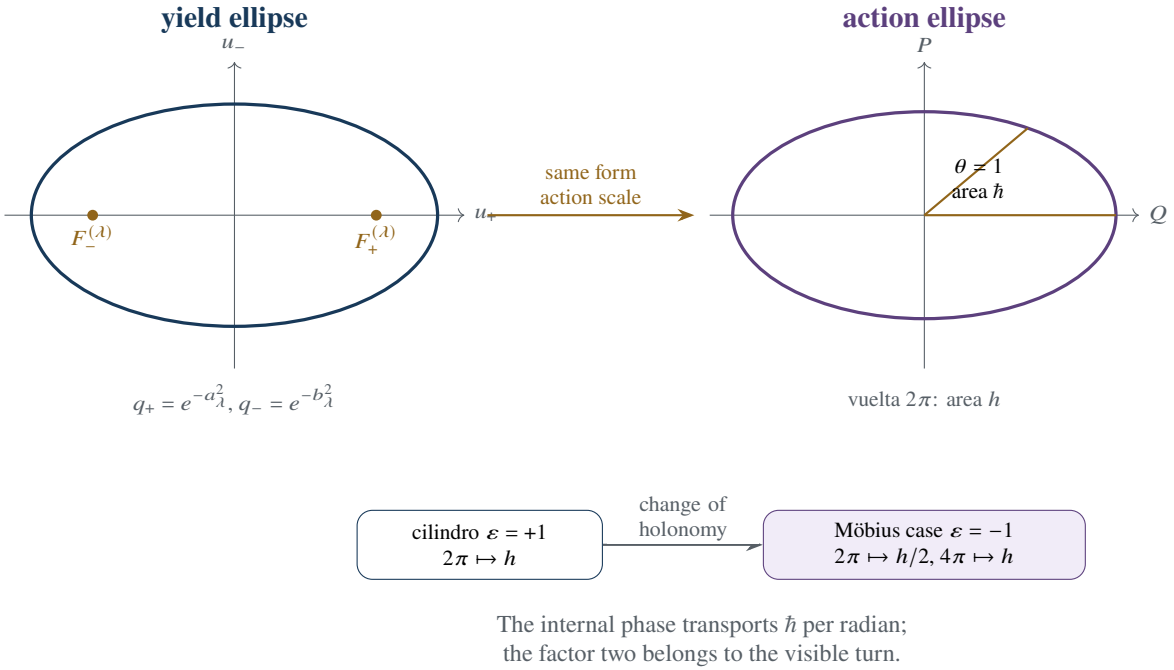

Figure 90.1: The ellipse of rates fixes the form; its lift to the action line fixes the area. The cylindrical closure and the double Möbius return distinguish the internal phase from the visible turn.

90.4.6.1 Hilbert–Beltrami–Klein Geometry on the Action Ellipse

The same ellipse that carries the symplectic area admits an intrinsic projective geometry. Let

$$\mathbb{E}_{\mathrm{act}} := \left\{ (Q,P) \in \mathbb{R}^2 : \frac{Q^2}{a_S^2} + \frac{P^2}{b_S^2} < 1 \right\}, \qquad \mathcal{L}_{\mathrm{act}}(Q,P) := \left(\frac{Q}{a_S}, \frac{P}{b_S} \right).$$

The linear map $\mathcal{L}_{\mathrm{act}}$ identifies $\mathbb{E}_{\mathrm{act}}$ with the unit disk $\mathbb{D}$. For two distinct points $x, y \in \mathbb{E}_{\mathrm{act}}$, let $a, b \in \partial\mathbb{E}_{\mathrm{act}}$ be the intersections of the line xy with the boundary, ordered as a, x, y, b. We define

$$d^{\mathrm{act}}_{\mathrm{H}}(x,y) := \frac{1}{2} \log \frac{\|y-a\|\,\|x-b\|}{\|x-a\|\,\|y-b\|}. \tag{90.1}$$

The expression is the cross-ratio of the four collinear points and is therefore independent of the auxiliary norm used on that line.

Proposition 90.5 (Beltrami–Klein realization of the action ellipse)**.** *The pair* $(\mathbb{E}_{\mathrm{act}}, d_{\mathrm{H}}^{\mathrm{act}})$ *is isometric, through* $\mathcal{L}_{\mathrm{act}}$*, to the Beltrami–Klein model of the hyperbolic plane. Its unoriented geodesics are exactly the open chords of the ellipse. The boundary* $\partial\mathbb{E}_{\mathrm{act}}$ *simultaneously preserves, in a typed manner, the symplectic orbit*

$$\gamma_{\mathrm{act}}(\theta) = (a_S \cos\theta, b_S \sin\theta), \qquad \frac{1}{2}\int_0^{\theta} (QP' - PQ')\, dt = \hbar_{\mathrm{HMT}}^{\mathrm{ret}}\theta.$$

Proof. Projective transformations preserve cross-ratios. Since $\mathcal{L}_{\mathrm{act}}$ is an invertible linear transformation that carries the ellipse to the unit disk, it transports (90.1) to the Hilbert distance of the disk. This is the Beltrami–Klein metric, whose geodesics are its chords. The symplectic identity of the boundary is theorem 90.4 written in the normalized chart. □

The complete chart is thereby typed by

$$\mathfrak{R}_{\mathrm{act}} : (A, C^*, \hbar_{\mathrm{HMT}}^{\mathrm{ret}}) \longmapsto (\mathbb{E}_{\mathrm{act}}, d_{\mathrm{H}}^{\mathrm{act}}, dP \wedge dQ, \gamma_{\mathrm{act}}).$$

The projective metric governs the interior; the symplectic form and the area law govern the traversed boundary. A realization as a physical region or as a domain of covariance matrices is formulated through a subsequent functor that explicitly preserves these two structures; the preceding construction already fixes its mathematical domain and the invariants that this functor must preserve.

90.4.6.2 Planck self-dual identity in the circular realization of the temporal endomorphism of order 108

The action line constructed in the preceding section admits a circular realization compatible with temporal periodicity of order 108. This realization coordinates three objects that already possess their own types: the action section, the geometry of one revolution, and the gravitational transduction. The resulting coordination provides an internal Planck identity and prepares the positive line on which the general mass operator will subsequently act.

Oriented action letters and declared circular realization Let

$$\mathcal{L}_S^+, \quad \mathcal{L}_T^+, \quad \mathcal{L}_C^+, \quad \mathcal{L}_L^+, \quad \mathcal{L}_M^+, \quad \mathcal{L}_G^+$$

the positive half-lines of action, time, velocity, length, and mass. The pre-return and post-return coordinates are written

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S, \qquad R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}}. \tag{90.2}$$

The subscript $a \in \{\mathrm{pre}, \mathrm{ret}\}$ shall designate either of the two charts. In each one the exact distinction is preserved

$$h_a = 2\pi\hbar_a. \tag{90.3}$$

Let us fix an elementary duration $t_0 \in \mathcal{L}_T^+$, an internal velocity $c_{\mathrm{int}} \in \mathcal{L}_C^+$, and $\ell_0 = c_{\mathrm{int}} t_0$. The declared circular realization of the temporal periodicity of order 108 is determined by

$$T_{108} = 108 t_0, \qquad \omega_{108} = \frac{2\pi}{T_{108}}, \qquad R_{108} = \frac{c_{\mathrm{int}}}{\omega_{108}} = \frac{54}{\pi}\ell_0, \tag{90.4}$$

and by the length of the full turn

$$C_{108} = 2\pi R_{108} = c_{\mathrm{int}} T_{108} = 108\ell_0. \tag{90.5}$$

Thus, R_{108} is the radius of the realization and C_{108} its perimeter. The angular frequency, the cyclic frequency and the two action constants are related through $\omega_{108} = 2\pi/T_{108}$ and $h_a = 2\pi\hbar_a$.

Correspondence between the order-108 periodicity and Compton readings In the chart a, the circular quantum and its mass section are

$$E_{108,a} = \hbar_a \omega_{108}, \qquad m_{108,a} = \frac{E_{108,a}}{c_{\mathrm{int}}^2} = \frac{\hbar_a \omega_{108}}{c_{\mathrm{int}}^2}. \tag{90.6}$$

Theorem 90.6 (Identity between circular periodicity and Compton readings)**.** *For each chart $a \in \{\mathrm{pre}, \mathrm{ret}\}$, the reduced and complete Compton readings satisfy*

$$\boxed{\bar{\lambda}_{\mathrm{C}}(m_{108,a}) = \frac{\hbar_a}{m_{108,a} c_{\mathrm{int}}} = R_{108}, \qquad \lambda_{\mathrm{C}}(m_{108,a}) = \frac{h_a}{m_{108,a} c_{\mathrm{int}}} = C_{108}.} \tag{90.7}$$

Proof. Substituting (90.6) produces

$$\frac{\hbar_a}{m_{108,a} c_{\mathrm{int}}} = \frac{c_{\mathrm{int}}}{\omega_{108}} = R_{108}.$$

The second equality follows by applying simultaneously (90.3) and (90.5). □

The theorem is homogeneous in $(\hbar_a, c_{\mathrm{int}}, t_0)$: the same periodicity determines the radius, the perimeter, and the two Compton charts. The circular realization is the declared geometric primitive; the preceding equalities are its exact consequences.

Circular selector and internal Planck unit The gravitational line is coordinated with the chart a through the homogeneous transducer

$$G_a(\theta) = \theta \, \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a} \in \mathcal{L}_G^+, \qquad \theta \in \mathbb{R}_{>0}. \tag{90.8}$$

Its reduced gravitational radius, evaluated on the circular quantum, is

$$r_{\mathrm{g},a}(\theta) := \frac{G_a(\theta) m_{108,a}}{c_{\mathrm{int}}^2} = \theta \frac{\pi}{54} \ell_0. \tag{90.9}$$

Proposition 90.7 (Planck Circular Selector)**.** *Within the circular realization (90.4), the condition $r_{\mathrm{g},a}(\theta) = R_{108}$ selects a unique positive coordinate,*

$$\boxed{\theta_{108}^{\mathrm{circ}} = \left(\frac{54}{\pi}\right)^2.} \tag{90.10}$$

The coordinate is independent of the chart oriented a.

Proof. The equations (90.4) and (90.9) reduce the condition to

$$\theta \frac{\pi}{54} \ell_0 = \frac{54}{\pi} \ell_0.$$

The positivity of ℓ_0 and the strict monotonicity of the map $\theta \mapsto r_{\mathrm{g},a}(\theta)$ provide the solution and its uniqueness. □

Let $G^{\mathrm{circ}}_{108,a} := G_a(\theta^{\mathrm{circ}}_{108})$. The internal units associated with this chart satisfy

$$\ell^{\mathrm{circ}}_{\mathrm{P},a} := \sqrt{\frac{\hbar_a G^{\circ}_{108,a}}{c^3_{\mathrm{int}}}} = R_{108}, \qquad t^{\circ}_{\mathrm{P},a} := \sqrt{\frac{\hbar_a G^{\circ}_{108,a}}{c^5_{\mathrm{int}}}} = \omega^{-1}_{108}, \tag{90.11}$$

$$m^{\circ}_{\mathrm{P},a} := \sqrt{\frac{\hbar_a c_{\mathrm{int}}}{G^{\circ}_{108,a}}} = m_{108,a}, \qquad E^{\circ}_{\mathrm{P},a} := m^{\circ}_{\mathrm{P},a} c^2_{\mathrm{int}} = E_{108,a}. \tag{90.12}$$

The equality of radii is therefore published as

$$R_{108} = \bar{\lambda}_{\mathrm{C}} = r_{\mathrm{g}} = \ell^{\mathrm{circ}}_{\mathrm{P}}, \qquad C_{108} = \lambda_{\mathrm{C}} = 2\pi r_{\mathrm{g}} = 2\pi \ell^{\mathrm{circ}}_{\mathrm{P}}. \tag{90.13}$$

The convention of this theorem is the reduced gravitational radius $r_{\mathrm{g}} = Gm/c^2_{\mathrm{int}}$. The Schwarzschild radius satisfies $r_{\mathrm{s}} = 2r_{\mathrm{g}}$; if this second convention is adopted, its crossing with $\bar{\lambda}_{\mathrm{C}}$ occurs at $m = m^{\circ}_{\mathrm{P},a}/\sqrt{2}$. The declaration of the radius therefore forms part of the chart and prevents a factor of two from being inadvertently transferred between the two identities.

The coordinate $\theta^{\mathrm{circ}}_{108}$ belongs to the declared circular specialization. The general gravitational selector, formulated on the enriched holonomy and its spectral response, retains its residence in section 80.1.1. Thus, the circular identity fixes an internal Planck normalization and the general criterion fixes the conditions that the physical realization of the gravitational section must satisfy.

Pre-return/post-return covariance The product of the reciprocal sections of action and gravity preserves the circular geometry:

$$\frac{m_{108,\mathrm{pre}}}{m_{108,\mathrm{ret}}} = R_{\mathrm{act}}, \qquad \frac{G^{\circ}_{108,\mathrm{pre}}}{G^{\circ}_{108,\mathrm{ret}}} = R^{-1}_{\mathrm{act}}, \qquad \hbar_a G^{\circ}_{108,a} = \theta^{\mathrm{circ}}_{108} c^5_{\mathrm{int}} t^2_0. \tag{90.14}$$

In particular, $\ell^{\mathrm{circ}}_{\mathrm{P,pre}}$ and $\ell^{\mathrm{circ}}_{\mathrm{P,ret}}$ coincide with R_{108}. The bidirectional memory resides in the quotient of the masses and in its gravitational reciprocal; the geometric scale resides in the invariant product.

Reciprocity admits a symmetric parametrization that separates the common scale from the orientation of the return. Define

$$\hbar_{\mathrm{geom}} := \sqrt{\hbar_{\mathrm{pre}}\hbar_{\mathrm{ret}}}, \quad G^{\circ}_{108,\mathrm{geom}} := \sqrt{G^{\circ}_{108,\mathrm{pre}} G^{\circ}_{108,\mathrm{ret}}}, \quad \xi_{\mathrm{act}} := \frac{1}{2}\log R_{\mathrm{act}}. \tag{90.15}$$

Then

$$\hbar_{\mathrm{pre}} = \hbar_{\mathrm{geom}} e^{\xi_{\mathrm{act}}}, \qquad \hbar_{\mathrm{ret}} = \hbar_{\mathrm{geom}} e^{-\xi_{\mathrm{act}}},$$

whereas the conjugate gravitational sections satisfy

$$G^{\circ}_{108,\mathrm{pre}} = G^{\circ}_{108,\mathrm{geom}} e^{-\xi_{\mathrm{act}}}, \qquad G^{\circ}_{108,\mathrm{ret}} = G^{\circ}_{108,\mathrm{geom}} e^{\xi_{\mathrm{act}}}.$$

The coordinate ξ_{act} preserves the pre-return/post-return orientation; the product $\hbar_a G^{\circ}_{108,a}$ preserves geometric scale. This parametrization of the action charts does not by itself introduce distinct propagation speeds.

Split self-dual mass line Given a chart a, let the split algebra be

$$\mathbb{D}_{\mathrm{sp}} := \mathbb{R}[j]/(j^2 - 1), \qquad P_{\pm} := \frac{I \pm j}{2},$$

and consider its spectral decomposition

$$\mathbb{D}_{\mathrm{sp}} = \mathbb{R}P_+ \oplus \mathbb{R}P_-, \qquad P_\pm^2 = P_\pm, \quad P_+P_- = 0, \quad P_+ + P_- = I.$$

The broken conjugation swaps the projectors,

$$\overline{uP_+ + vP_-} = vP_+ + uP_-,$$

and its norm is

$$N_{\mathrm{sp}}(uP_+ + vP_-) := uv.$$

For $m \in \mathcal{L}_M^+$, define the section

$$Z_a(m) := \frac{m^\circ_{\mathrm{P},a}}{m} P_+ + \frac{m}{m^\circ_{\mathrm{P},a}} P_-. \tag{90.16}$$

The unit $m^\circ_{\mathrm{P},a}$ also determines the involution

$$\iota_{\mathrm{P},a}(m) = \frac{(m^\circ_{\mathrm{P},a})^2}{m}. \tag{90.17}$$

Theorem 90.8 (Split self-dual mass line)**.** *For every $m \in \mathcal{L}_M^+$,*

$$\boxed{N_{\mathrm{sp}}(Z_a(m)) = 1, \qquad \overline{Z_a(m)} = Z_a\big(\iota_{\mathrm{P},a}(m)\big)\,.} \tag{90.18}$$

Conjugation has a unique positive fixed point,

$$m = m^\circ_{\mathrm{P},a}, \qquad Z_a(m^\circ_{\mathrm{P},a}) = I.$$

Proof. The norm is the product of the two reciprocal coordinates:

$$\frac{m^\circ_{\mathrm{P},a}}{m}\frac{m}{m^\circ_{\mathrm{P},a}} = 1.$$

Conjugation interchanges both coordinates, which is equivalent to replacing m by $(m^\circ_{\mathrm{P},a})^2/m$. The positive fixed point satisfies $m^2 = (m^\circ_{\mathrm{P},a})^2$. □

With the speed

$$x_a(m) = \log \frac{m}{m^\circ_{\mathrm{P},a}}, \tag{90.19}$$

the section and its conjugate are written

$$Z_a(m) = e^{-x_a} P_+ + e^{x_a} P_-, \qquad \overline{Z_a(m)} = e^{x_a} P_+ + e^{-x_a} P_-.$$

Their even and odd readers are

$$Q_a(m) := \frac{e^{x_a} + e^{-x_a}}{2} = \cosh x_a, \qquad A_a(m) := \frac{e^{x_a} - e^{-x_a}}{2} = \sinh x_a. \tag{90.20}$$

The first preserves the distance to the self-dual point and the second preserves its orientation.

The reciprocal lattice vectors associated with the same coordinate are

$$r_{\mathrm{g},a}(m) = \frac{G^\circ_{108,a} m}{c_{\mathrm{int}}^2}, \qquad \bar{\lambda}_{\mathrm{C},a}(m) = \frac{\hbar_a}{m c_{\mathrm{int}}},$$

and satisfy

$$\frac{r_{\mathrm{g},a}(m)}{\bar{\lambda}_{\mathrm{C},a}(m)} = e^{2x_a(m)}, \qquad \frac{\bar{\lambda}_{\mathrm{C},a}(m)}{r_{\mathrm{g},a}(m)} = e^{-2x_a(m)}. \tag{90.21}$$

Conjugation interchanges the two readers; Planck's identity is their unitary equilibrium.

Planck identity and electron closure

The identity $m = m^{\circ}_{\mathrm{P},a}$ fixes the self-dual origin of the positive line of masses. The electron section fixes a distinct position on that same line through its own genealogy. The section 90.4.7 constructs the universal operator that transports absolute scale, character, towers, and bidirectional reading; the section 58.2 then realizes its central rank-one reduction. Thus the Planck unit provides the identity of the coordinate, and the electron provides the first complete material transition on it.

Proposition 90.9 (Genealogical realization of the electronic center)**.** *Let*

$$F_e^+ = \mathbb{R}_{>0}(P_+ + P_-)$$

the positive half-line of the rank-one central fiber, let $\mathcal{G}_e$ be the space of enriched electronic genealogies and let $\mathcal{L}_M^+$ be the positive mass line. A realization morphism

$$\mathfrak{R}_e : F_e^+ \times \mathcal{G}_e \longrightarrow \mathcal{L}_M^+$$

can send the identity $I = P_+ + P_-$ to a mass different from $m^{\circ}_{\mathrm{P},a}$ without altering the theorem 90.8: the Planck identity fixes the autodual origin of the chart, while the electronic genealogy fixes a displacement on that chart.

Proof. If $\chi_e : \mathcal{G}_e \to \mathbb{R}_{>0}$ is the positive character constructed by the electronic record, the rule

$$\mathfrak{R}_e(\lambda I, g) = \lambda\, m^{\circ}_{\mathrm{P},a}\chi_e(g)$$

is positive and multiplicative in the genealogical coordinate. For the neutral genealogy, $\chi_e = 1$, Planck's origin is recovered; for a genealogy with $\chi_e(g) \neq 1$, the same internal identity occupies a different position on the line. The specific electronic closure preserves its source in the section 58.2. □

Domain of the Planck self-dual identity: temporal periodicity of order 108**, declared circular realization, and separation of the electronic closure** The pre-return/post-return action line and the temporal periodicity of order (108) precede the circular realization, which is taken as a declared geometric primitive. The identity between circular periodicity and the Compton readings, the selector $\theta^{\mathrm{circ}}_{108}$, the internal Planck unit, and the involution of the positive line are exact results once that realization has been fixed. General gravitational selection and electronic closure are constructed separately in the indicated sections.

90.4.7 Causal precedence of the General Mass Law with respect to species-specific realizations

Dimensional Mechanics receives from HMT two branches already constructed. The first provides an action line with two oriented coordinates; the second provides the genealogy of each persistent section. Mass appears upon making the genealogical character act on a section of the mass line. This confluence precedes every particular case and, accordingly, the scale 10^{-34} belongs to the general law rather than to an isolated species.

90.4.7.1 Absolute Branch: From the Local Determinant to the Mass Line

The determinantal reader fixes locally

$$\Sigma_H = \varphi_{\mathrm{HMT}} \alpha_{\mathrm{HMT}}^{16}, \qquad \varepsilon_H := \lfloor \log_{10} \Sigma_H \rfloor = -34. \tag{90.22}$$

The power sixteen follows from the cokernel $\mathbb{Z}/2 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/4$ of the determinantal block. On the internal action line $\mathcal{L}_S$ the two sections are obtained

$$\hbar_{\mathrm{pre}} = H_5 \, 10^{-34} \mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}} \, 10^{-34} \mathcal{U}_S, \tag{90.23}$$

and the additive and multiplicative letters of its unfolding are

$$\delta_{2w} = H_5 - \eta_{\mathrm{ret}}, \qquad R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5}{\eta_{\mathrm{ret}}} > 0. \tag{90.24}$$

Let $t_0 \in \mathcal{L}_T^+$ be the elementary duration of the TPK calendar and $c_{\mathrm{int}} \in \mathcal{L}_C^+$ its internal speed. The periodicity of 108 iterations of the temporal endomorphism CT108 fixes the angular frequency $\omega_0 = 2\pi/(108 t_0)$. The absolute sections preceding and following the return are, therefore,

$$\begin{aligned} \mathbf{m}_\star^{\mathrm{pre}} &= \hbar_{\mathrm{pre}} \, \omega_0 c_{\mathrm{int}}^{-2}, \\ \mathbf{m}_\star^{\mathrm{ret}} &= \hbar_{\mathrm{ret}} \, \omega_0 c_{\mathrm{int}}^{-2}, \end{aligned} \qquad \frac{\mathbf{m}_\star^{\mathrm{pre}}}{\mathbf{m}_\star^{\mathrm{ret}}} = R_{\mathrm{act}}. \tag{90.25}$$

Both belong to $\mathcal{L}_M = \mathcal{L}_S \otimes \mathcal{L}_T^{-1} \otimes \mathcal{L}_C^{-2}$. Thus the decade, which fixes the absolute scale, and the quotient of the pre-return/post-return bidirectional splitting, which transports the memory of the closed evolution, are precisely separated.

In positive internal coordinates $t_0 = \widehat{t_0} \mathcal{U}_T$ and $c_{\mathrm{int}} = \widehat{c} \mathcal{U}_C$, the scalar reduction of any section whose physical realization has already been fixed takes the form

$$\mathbf{m}_X^\beta = 10^{-34} \eta_{\mathrm{ret}} \, \frac{2\pi}{108 \widehat{t_0}} \, \widehat{c}^{-2} a_X^{(0)} R_{\mathrm{act}}^{\kappa_X} \, \mathcal{U}_M. \tag{90.26}$$

The coordinate $a_X^{(0)}$ proceeds of the character and of the hierarchy harmonic; κ_X proceeds of the reader of route. In particular, the decade -34 belongs to the section absolute and does not incorporates to κ_X.

90.4.7.2 Genealogical branch: record, character, and harmonic hierarchy

Each surviving section traverses the chain

$$\Omega_{\mathrm{HMT}}^{\mathrm{surv}} \longrightarrow \Gamma_{108}^{\mathrm{enr}} \longrightarrow \mathcal{L}_{108} \longrightarrow \sigma \in \mathbb{Z}^5 \longrightarrow (\sigma, \eta) \longrightarrow \chi_{\mathrm{full}}(\sigma, \eta) \in \mathbb{R}_{>0}. \tag{90.27}$$

The record preserves the order of events; the signature preserves their five integer invariants; the character evaluates them without erasing their provenance. Its spectral prolongation lies in the full semigroup

$$\mathfrak{S} = \langle 90, 120 \rangle = \{90, 120\} \cup \{30r : r \geq 6\}, \qquad s_h = q_-^h - q_+^h, \quad c_h = q_-^h + q_+^h, \tag{90.28}$$

and adopts the form

$$\chi_{\mathrm{full}}(\sigma, \eta) = \chi_0(\sigma) \exp\left(\sum_{h \in \mathfrak{S}} \eta_h s_h \right). \tag{90.29}$$

The heights 90, 120, 180, 210, 240, 270, 360, 420 are eight especially visible genealogical witnesses. The law belongs to all of $\mathfrak{S}$; the second-order recurrence of (s_h, c_h) extends the witnesses to the complete harmonic hierarchy.

90.4.7.3 Universal operator on the 13 *multisections of the material domain*

The atlas provides 81 positions, 56 route families, 13 multisections, and 324 oriented routes. For a multisection F, let $\mathcal{H}_F = \bigoplus_{\Gamma\in F} \mathbb{C}e_\Gamma$. The basal character and the towers define the positive operator

$$\widehat{\mathcal{A}}_F^{(0)} e_\Gamma = \chi_{\text{full}}(\sigma_\Gamma, \eta_\Gamma) e_\Gamma. \tag{90.30}$$

The two histories preserved by TPK induce readers $r \in \{D, \Omega\}$: K_D acts on the chronological record of 108 iterations of the CT108 temporal endomorphism, and K_Ω on the signed dodecaphase word. If $\widehat{K}_F^r e_\Gamma = \kappa_r(\Gamma) e_\Gamma$, let $F_s \subseteq F$ be the set of routes of the section or physical block s, and define

$$\mathcal{H}_{F,s} := \bigoplus_{\Gamma\in F_s} \mathbb{C}e_\Gamma, \qquad \widehat{\mathcal{A}}_{F,s}^{(0)} := \widehat{\mathcal{A}}_F^{(0)}\Big|_{\mathcal{H}_{F,s}}, \qquad \widehat{B}_{F,s}^r := \widehat{K}_F^r\Big|_{\mathcal{H}_{F,s}}. \tag{90.31}$$

The basis diagonal of routes used by the certificates current makes invariant $\mathcal{H}_{F,s}$ under both operators and converts $\widehat{B}_{F,s}^r$ in a restriction self-adjoint. With these types, the law general is

$$\boxed{\widehat{\mathbf{M}}_{F,s}^{\beta,r} = \mathbf{m}_\star^{\text{ret}} \otimes R_{\text{act}}^{\widehat{B}_{F,s}^r/2} \widehat{\mathcal{A}}_{F,s}^{(0)} R_{\text{act}}^{\widehat{B}_{F,s}^r/2}, \qquad r \in \{D, \Omega\}.} \tag{90.32}$$

The symmetric writing preserves positivity without presupposing commutation. In the simultaneously diagonal basis employed by the current certificates, it reduces to

$$\widehat{\mathbf{M}}_F^{\beta,r} = \mathbf{m}_\star^{\text{ret}} \otimes \widehat{\mathcal{A}}_F^{(0)} \exp\Big((\log R_{\text{act}})\widehat{K}_F^r\Big). \tag{90.33}$$

The equation preserves, in a single object, the absolute scale, the global character, the entire hierarchy of towers, and the route-dependent bidirectional corrector. The physical chart of each sector subsequently acts on the spectrum of (90.32). The null sector has its own morphism to zero; the composite registers are concatenated before the signature is evaluated; the topological sectors first publish their fusion and braiding invariants.

Theorem 90.10 (Covariance and Cancellation of the Common Scale)**.** *The law* (90.32) *is positive and covariant under unitary change of basis in each fiber. For two self-sections X, Y realized in the same mass chart,*

$$\frac{m_Y}{m_X} = \frac{\chi_{\text{full}}(\sigma_Y, \eta_Y)}{\chi_{\text{full}}(\sigma_X, \eta_X)} R_{\text{act}}^{\kappa_r(\Gamma_Y) - \kappa_r(\Gamma_X)}. \tag{90.34}$$

The common factors $10^{-34}\mathcal{U}_S$, $2\pi/(108 t_0)$, and c_{int}^{-2} fix the absolute section and cancel exactly in the quotient.

Proof. The operator $\widehat{\mathcal{A}}_F^{(0)}$ is positive because its eigenvalues belong to $\mathbb{R}_{>0}$; the exponential of a self-adjoint operator is positive. Conjugation by a unitary matrix preserves the spectrum and therefore the law. Upon dividing two eigenvalues realized in the same chart, the common section $\mathbf{m}_\star^{\text{ret}}$ cancels, and the result is (90.34). □

The central electronic fiber has rank one and constitutes the first scalar corollary of this law. The remaining multisections preserve the pair of operators, their spectra, and the physical morphism that selects flavor, color, orientation, or composition. This hierarchy allows the algebraic result, its dimensional realization, and the external contrast to be published together without confusing their domains.

90.5 Nonadic classification of trajectory families

The atlas does not begin with a list of particles or a table of masses. It begins with a finite oriented support and with the readers that preserve the history of its routes. Its function is to answer a question prior to any metrological comparison: what classes of persistence can the nonadic geometry support, and what information must be preserved to distinguish them?

The answer has three cardinals, but not three rival inventories:

$$81 \text{ cells}, \qquad 56 \text{ route families}, \qquad 13 \text{ family–level multisections}.$$

The first number counts oriented positions; the second identifies the positions that publish the same finite route signature; the third gathers the positions that share a family and depth relative to the center. None of them counts known masses, names of species, or rows of an external comparison.

90.5.1 The complete geometry of the 81 cells

Let

$$I_9 = \{1, \dots, 9\}, \qquad C_{81} = I_9 \times I_9.$$

The coordinates (r, c) preserve the two directions of APP. The parity of each coordinate defines four internal sectors:

$$\mathrm{fam}(r,c) = \begin{cases} \ell^-, & r \equiv 1,\ c \equiv 1 \pmod 2, \\ \nu, & r \equiv 1,\ c \equiv 0 \pmod 2, \\ d, & r \equiv 0,\ c \equiv 1 \pmod 2, \\ u, & r \equiv 0,\ c \equiv 0 \pmod 2. \end{cases} \tag{A1}$$

Depth in the atlas is not added as an independent label. It is the Chebyshev distance to the center:

$$G(r,c) = \max\{|r-5|, |c-5|\}. \tag{A2}$$

Therefore, family and level follow from the position itself. In particular, the center is not obtained by selecting a species from outside: it is the unique cell of depth zero.

Theorem 90.11 (Census of families and generations of the atlas)**.** *The maps* (A1) *and* (A2) *decompose the atlas in a disjoint manner as*

$$81 = 45_{\text{leptones}} + 36_{\text{quarks}} = 25_{\ell^-} + 20_{\nu} + 20_{d} + 16_{u}. \tag{A3}$$

Refinement by G yields exactly the thirteen multisections

family	*levels*	*cardinals*
ℓ^-	$G0, G2, G4$	$1, 8, 16$
ν	$G1, G2, G3, G4$	$2, 4, 6, 8$
d	$G1, G2, G3, G4$	$2, 4, 6, 8$
u	$G1, G3$	$4, 12.$

(A4)

The cell $(5,5)$ *is the sole component of* $\mathfrak{S}_{\ell^-,0}$.

Proof. In I_9 there are five odd coordinates and four even ones. The four parity classes therefore have cardinalities

$$5 \cdot 5 = 25, \qquad 5 \cdot 4 = 20, \qquad 4 \cdot 5 = 20, \qquad 4 \cdot 4 = 16,$$

which proves (A3); the first two classes comprise 45 cells and the last two, 36.

For the refinement, we enlarge squares centered at $(5, 5)$. In the odd–odd class there is one cell at radius zero, $3^2 - 1 = 8$ in the shell of radius two, and $5^2 - 3^2 = 16$ in the shell of radius four. In the odd–even class, the rectangles accumulated up to radii $1, 2, 3, 4$ contain, respectively,

$$1 \cdot 2 = 2, \qquad 3 \cdot 2 = 6, \qquad 3 \cdot 4 = 12, \qquad 5 \cdot 4 = 20$$

cells; their successive differences are $2, 4, 6, 8$. The even–odd class gives the same calculation with the coordinates exchanged. In the even–even class, there are $2^2 = 4$ cells at radius one and $4^2 - 2^2 = 12$ at radius three. These shells are disjoint and exhaust I_9^2, so no other multisection remains. □

The word *level* denotes in this chapter the combinatorial depth G. The subsequent physical reading preserves that geometry but does not define it. This prevents particle names from being turned into premises of the census that must explain them.

90.5.2 Central inversion, 41 orbits, and uniqueness of the centre

The inverse of the atlas is

$$\rho_{\mathrm{c}}(r, c) = (10 - r, 10 - c). \tag{A5}$$

Since $10 - r$ has the same parity as r, inversion preserves the four families. Moreover,

$$|(10 - r) - 5| = |r - 5|, \qquad |(10 - c) - 5| = |c - 5|,$$

and therefore preserves the level G. Each of the thirteen multisections is ρ_{c}-stable.

Proposition 90.12 (Orbits and Central Exception)**.** *The action of ρ_{c} on C_{81} has exactly 41 orbits. Its unique fixed point is $(5, 5)$. Consequently, the multisection $\mathfrak{S}_{\ell^-,0}$ admits the unique pointwise ρ_{c}-equivariant selector; in the other twelve, the natural object is the orbit or the complete multisection.*

Proof. The equation

$$(r, c) = \rho_{\mathrm{c}}(r, c)$$

is equivalent to $2r = 2c = 10$, hence $r = c = 5$. The other eighty cells are grouped into forty pairs, whence $1 + 40 = 41$ orbits.

Let y be a noncentral multisection and suppose that an equivariant point selector chooses $a(y) \in \mathfrak{S}_y$. Since ρ_{c} acts trivially on the family–level type, equivariance would impose

$$a(y) = a(\rho_{\mathrm{c}} y) = \rho_{\mathrm{c}} a(y).$$

The chosen point would have to be fixed, but the only fixed point lies in $\mathfrak{S}_{\ell^-,0}$. The contradiction proves the assertion. □

This proposition explains by which a species internal is not, in general, a cell isolated. The electron central is exceptional because its multisection is reduceds to a point. In the remaining families, the orientation conjugate form part of the identity that must is preserved.

90.5.3 The 56 route families

Each cell bears the finite shadow of its transport history:

$$\mathcal{R}(r, c) = \big(n_A,\ s_C,\ k_{\Delta_4},\ \nu_{120},\ \nu_{270},\ \tau_{12}\big)_{r,c}. \tag{A6}$$

The first five coordinates are integers; τ_{12} is the oriented dodecaphase word. Two cells belong to the same route family if and only if they have the same sextuple $\mathcal{R}$.

Theorem 90.13 (Census of route families and minimal memory)**.** *The image of $\mathcal{R}$ contains exactly* 56 *elements. The histogram of the sizes of its fibres is*

$$41 \times 1, \qquad 10 \times 2, \qquad 2 \times 3, \qquad 1 \times 4, \qquad 2 \times 5. \tag{A7}$$

In particular,

$$41 + 10 + 2 + 1 + 2 = 56,$$
$$41 \cdot 1 + 10 \cdot 2 + 2 \cdot 3 + 1 \cdot 4 + 2 \cdot 5 = 81. \tag{A8}$$

Fixing the lexicographic order within each fiber, the rank

$$\kappa(r, c) \in \{0, 1, 2, 3, 4\}$$

makes the application injective

$$(r, c) \longmapsto \big(\mathcal{R}(r, c), \kappa(r, c)\big). \tag{A9}$$

Five is the minimum size of a uniform alphabet capable of performing this lift.

Proof. The exhaustive enumeration of I_9^2 groups the eighty-one sextuples (A6) by coordinatewise equality. It produces the five fiber sizes of (A7); the two identities (A8) simultaneously verify the number of classes and the coverage of the eighty-one cells.

The lexicographic rank distinguishes the elements of each fiber, so that (A9) is injective. Since two fibers of cardinal five exist, any common alphabet with fewer than five symbols would identify two cells of one of them by the pigeonhole principle. □

The rank κ is positional memory within a fiber; it is not charge, mass, or color. Its function is to recover the cell that the visible route signature had identified with others. The number 56 counts classes of the sextuple $\mathcal{R}$, whereas the number 13 counts classes of the pair (fam, G). They are different quotients of the same atlas.

Cellular and orbital quotients of the atlas: domains and projection maps. The raw CT–108 projection and the scientific classification read different data from the same eighty-one cells. If q_{CT} preserves only the form of the dodecaphase tritic word, while $\mathcal{R}$ preserves the route sextuple and $q = (\mathrm{fam}, G)$ preserves family and depth, the typed table is

$$\begin{aligned} C_{81} &\xrightarrow{q_{\mathrm{CT}}} \mathcal{W}_7^{\mathrm{raw}} && \text{raw CT–108 forms,} \\ C_{81} &\xrightarrow{\mathcal{R}} \Sigma_{56}^{\mathrm{ruta}} && \text{route families,} \\ C_{81} &\xrightarrow{q} \Sigma_{13} && \text{family–level multisections.} \end{aligned} \tag{A9a}$$

The seven raw fibres have multiplicities

$$54, 2, 2, 2, 7, 7, 7, \qquad 54 + 3 \cdot 2 + 3 \cdot 7 = 81.$$

Therefore, the archaeological census $81 \to 7$ does not replace the atlas 81/56/13. Nor does the abbreviated notation 81/56/13 assert the existence of a canonical arrow $56 \to 13$: some fibers of $\mathcal{R}$ traverse more than one family–level pair. The maps $\mathcal{R}$ and q are parallel quotients and preserve different invariants. In the quark sector, the active convention remains

$$\mathrm{col}(r, c) = r - c \pmod 3;$$

the historical notation $c - r$ is its conjugate convention, not another color or another quotient of the atlas.

90.5.4 Distinction between cell, route family, and multisection

Define

$$q : C_{81} \longrightarrow \Sigma_{13}, \qquad q(r,c) = \big(\mathrm{fam}(r,c), G(r,c)\big), \tag{A10}$$

and, for $y \in \Sigma_{13}$,

$$\mathfrak{S}_y^{\mathrm{atlas}} = q^{-1}(y) = \{(r,c) \in C_{81} : q(r,c) = y\}. \tag{A11}$$

The four levels of reading are then separated:

(i) a *cell* is an oriented position (r,c), with all its local data;

(ii) a *route family* is a fibre of $\mathcal{R}$: it preserves the same finite transport signature;

(iii) a *multisection* is a fiber of q: it preserves the family, depth, and all its conjugate orientations;

(iv) a *physical representation* is the subsequent realization of an enriched multisection in a state space; it is not another partition of the board or the arbitrary choice of one of its cells.

This distinction gives mathematical content to the word *species*. The internal type is the multisection. A concrete particle is a persistent section realized within it. The physical representation preserves the action, orientation, and observables of the type; it does not work backward to redefine the atlas from an already known mass.

90.5.5 Color residual as ternary orientation

In the quark sector, formed by the 36 cells with even r, we define

$$\mathrm{col}(r,c) = r - c \pmod 3, \tag{A12}$$

with the dictionary

$$0 \longmapsto R, \qquad 1 \longmapsto G, \qquad 2 \longmapsto B.$$

Proposition 90.14 (Residual color balance)**.** *The thirty-six quark cells are distributed exactly as*

$$36 = 12_R + 12_G + 12_B. \tag{A13}$$

The central inversion fixes R and interchanges G and B.

Proof. Upon fixing any of the four even values of r, as $c = 1, \dots, 9$ is traversed, each class modulo three appears three times. Each residue therefore has $4 \cdot 3 = 12$ representatives. Moreover,

$$\mathrm{col}\big(\rho_{\mathrm{c}}(r,c)\big) = (10 - r) - (10 - c) = -(r - c) \pmod 3.$$

The modular negation fixes the zero class and swaps the one and two classes. □

The tritic R, G, B is therefore an exact observable of the atlas and not a decoration added to a table of quarks. The color representation is subsequently constructed on this ternary support; the census $12 + 12 + 12$ is its prior geometry.

90.5.6 From the surviving extension to the signature

The atlas classifies finite cuts. The identity of a particle requires the deep history from which that cut proceeds. Let

$$\Omega_{\mathrm{HMT}}^{\mathrm{surv}} = \varprojlim_m \mathcal{S}_m$$

the inverse system of surviving states. The projection onto the atlas π_{81} and the route shadow form the chain

$$\Omega_{\mathrm{HMT}}^{\mathrm{surv}} \xrightarrow{\mathrm{sh}_{108}} \Gamma_{108}^{\mathrm{enr}} \xrightarrow{\mathcal{E}} \mathcal{L}_{108} \xrightarrow{L_{\mathrm{tick}}} \mathbb{Z}^5. \tag{A14}$$

Here sh_{108} publishes a route of 108 steps without forgetting sheet, orientation, boundary, or predecessors; $\mathcal{E}$ forms the dodecaphase record; and L_{tick} extracts

$$v(x) = \big(n_A, s_C, k_{\Delta_4}, \nu_{120}, \nu_{270}\big). \tag{A15}$$

None of these three arrows refers to a mass, a unit, or the name of a particle.

The reason for preserving the ledger before projection is apparent in its twelve oriented events $e_0, \dots, e_{11}$. For $0 \le i \le 5$, let

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}, \qquad s_C = \sum_{i=0}^{5} p_i,$$

and, for $0 \le i < 5$,

$$M_i = p_i - p_5.$$

Then

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \qquad p_i = p_5 + M_i,$$
$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2}. \tag{A16}$$

The identity

$$1 + 5 + 6 = 12$$

is thus a reversible reconstruction: one total, five differences, and six orientations recover the twelve events. Projecting onto the five-integer signature before composing histories would eliminate precisely the placement and orientation that make it possible to decide whether two routes can be composed.

The quotient q is lifted to the deep history through projection to the atlas $\pi_{81} : \Omega_{\mathrm{HMT}}^{\mathrm{surv}} \to C_{81}$:

$$\widetilde{\mathfrak{S}}_y = \{x \in \Omega_{\mathrm{HMT}}^{\mathrm{surv}} : q(\pi_{81} x) = y\}. \tag{A17}$$

This is the enriched multisection. Its shadow belongs to the atlas; its route belongs to $\Gamma_{108}^{\mathrm{enr}}$; its complete memory belongs to the record; and its evaluation is applied only at the end. The genealogy is not a note appended to the value: it is the object that makes the value possible and preserves the other readings of the same section.

90.5.7 Null sector bifurcation prior to the positive character

The mass character acts on the entire signature and has positive codomain:

$$\chi_{\mathrm{mass}} : \mathbb{Z}^5 \longrightarrow \mathbb{R}_{>0}. \tag{A18}$$

By definition, there is no signature v with $\chi_{\mathrm{mass}}(v) = 0$. The null sector is, consequently, not a fourteenth multisection nor the degenerate limit of the thirteen preceding ones.

Its algebraic support belongs to the split branch.

$$\mathcal{A}_{\mathrm{sp}} = \mathbb{R}[\mathsf{R}]/(\mathsf{R}^2 - 1), \qquad P_{\pm} = \frac{1 \pm \mathsf{R}}{2}.$$

Every element is written in a unique way as

$$z = r_+ P_+ + r_- P_-, \qquad N_{\mathrm{sp}}(z) = r_+ r_-,$$

and the two lines

$$\mathbb{R}P_+, \qquad \mathbb{R}P_-$$

are null because $N_{\mathrm{sp}}(\lambda P_\pm) = 0$. A null section is an open route transported along one of those directions. The material interior requires the coupling of both, for which $r_+ r_- > 0$.

Thus are typed two branches:

$$\text{material persistence} \longrightarrow \mathbb{Z}^5 \xrightarrow{\chi_{\mathrm{mass}}} \mathbb{R}_{>0},$$

$$\text{null propagation} \longrightarrow \partial_{\mathrm{null}} C_{\mathrm{sp}}^{+}.$$

The photonic and gauge realizations subsequently read the second branch. Their separation from the positive character prevents the photon and gluon from being erased because they do not appear in a list of masses, while also preventing a null mass from being fabricated by means of a character whose codomain excludes it.

90.5.8 Finite classification system and path memory

The three quotients of the chapter fulfill complementary functions:

$$C_{81} \xrightarrow{\mathcal{R}} \mathcal{R}_{56}, \qquad C_{81} \xrightarrow{q} \Sigma_{13}, \qquad C_{81} \xrightarrow{/\langle \rho_{\mathrm{c}} \rangle} O_{41}.$$

The first preserves the route signature; the second preserves family and level; the third preserves central conjugation. Together with residual color and the lift κ, they form a finite system capable of reconstructing which information is seen, which is identified, and which must remain as memory.

The causal chain of the atlas is thereby fixed:

$$\boxed{\begin{array}{c}\text{surviving extension} \longrightarrow \text{enriched route} \longrightarrow \text{record} \longrightarrow \text{signature} \\ \longrightarrow \text{character} \longrightarrow \text{dimensional reading}\end{array}} \tag{A19}$$

The eighty-one cells constitute the local domain; the fifty-six families express route coincidences; the thirteen multisections are the internal types; the forty-one orbits preserve conjugation; and the center $(5, 5)$ is the sole pointwise exception. Mass appears thereafter as a reading of an already constructed persistence. The atlas is not an abridged table of results: it is the structure explaining why a table can exist.

90.5.9 From APP seeds to mixture and mass registration

The physical law receives a domain that has already preserved the incompatibility of the two APP evaluations. If $\mathcal{S}_+$ and $\mathcal{S}_\times$ denote the oriented seeds of the additive and multiplicative leaves, then

$$\Omega_{\mathrm{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \qquad |\Omega_{\mathrm{seed}}| = 324^2 = 104,976. \tag{90.35}$$

The exact factorization of the emitter produces

$$104,976 \longrightarrow 18 \times 26 = 468\ U_6 \longrightarrow 243\ w_6 \subset \mathbb{F}_3^6, \qquad |\mathbb{F}_3^6| = 729. \tag{90.36}$$

The four cardinalities count distinct objects: initial conditions of two sheets, decimal emissions, published ternary words, and the algebraic ambient space. In addition to its emission, each seed preserves multiplicity, diagonal, sheet, orbit, emission signature, residues modulo three and modulo nine, turns, sheet changes, endpoint, displacements, and cardinal counts. Those data constitute the vocabulary of the subsequent register.

The factorization organizes six microfibers of cardinality 1008, six phase Fano planes, thirty-six fibers of cardinality 28, and the band $1944 = 27 \cdot 72$, diagonalized by $(\mathbb{Z}/3)^3$. Upon averaging the nonadic word of the TPK, the conditional projectors of the two sheets define the canonical kernel

$$K_{\mathrm{cat}} = \frac{1}{3}(I + E_+ + E_\times), \qquad \mathrm{Spec}(K_{\mathrm{cat}}) = \left\{ 1^{[1]}, \left(\frac{2}{3}\right)^{[42]}, \left(\frac{1}{3}\right)^{[425]} \right\}. \tag{90.37}$$

The TPK phase selects the active renewal and preserves the nine spatial bands; the average (90.37) publishes the stochastic shadow of the complete calendar.

90.5.9.1 Bigrading of the Seeds and 6 *Incidence Planes*

In each of the six microfibers of cardinality 1008, the bipartite graph retains 54 additive vertices and 56 multiplicative vertices. The former decompose into 36 vertices of degree 24 and 18 of degree 8; the latter have degree 18. The incidence count agrees on both sheets:

$$1008 = 36 \cdot 24 + 18 \cdot 8 = 56 \cdot 18. \tag{90.38}$$

This equality fixes the emitter bigrading before projecting it onto the route families.

The six words

$$001112, \quad 010211, \quad 100121, \quad 112001, \quad 121100, \quad 211010 \tag{90.39}$$

form an orbit of $D_3 \simeq S_3$ and have the profile $432 + 4 \cdot 144 = 1008$. The quotient $1008 = 36 \cdot 28$ organizes seven blocks

$$Q = \{B_3, B_6, B_9, P_1, P_2, P_3, f\},$$

where (B_3, B_6, B_9) are the transverse tetrads and (P_1, P_2, P_3, f), the four lateral tetrads. For one of the six words a, let

$$\mathcal{R}_a := \{(p,t) \in \mathcal{S}_+ \times \mathcal{S}_\times : U_6(p,t) \text{ publishes the word } a\}$$

the set of its one thousand eight routes, and set $R_9 := \mathbb{Z}/9\mathbb{Z}$. Once the oriented chart is fixed, the phase projection has type

$$q_\beta : \mathcal{R}_a \longrightarrow \binom{R_9}{2}, \qquad q_\beta(p,t) = \{x - 5\beta(d_t), x + 5\beta(d_t)\} \subset \mathbb{Z}/9\mathbb{Z}, \qquad x = 5(i_p - j_p), \tag{90.40}$$

with $(\beta(N), \beta(E), \beta(S), \beta(O)) = (1,2,3,4)$. Its codomain is the set of the thirty-six unordered pairs of R_9. Each $X \in Q$ is a tetrad of orbits; its route support and its cut over a fiber are defined by

$$\mathrm{Supp}_{\mathrm{rutas}}(X) := \bigcup_{O \in X} O, \qquad X_b := \mathrm{Supp}_{\mathrm{rutas}}(X) \cap q_\beta^{-1}(b), \qquad b \in \binom{R_9}{2}.$$

In the multiplicative channel, the source observable is

$$\rho : \mathcal{S}_\times \longrightarrow R_9, \qquad \rho(i,j,d) = (i+1)(j+1) \pmod 9. \tag{90.41}$$

If $\pi_\times : \mathcal{R}_a \to \mathcal{S}_\times$, $\pi_\times(p,t) = t$, is the second projection, the observable acting on routes is

$$\rho_\times := \rho \circ \pi_\times : \mathcal{R}_a \longrightarrow R_9, \qquad \rho_\times(p,t) = (i_t + 1)(j_t + 1) \pmod 9. \tag{90.42}$$

In what follows, $\rho(X)$ abbreviates the multiset of values of $\rho_\times$ over X_b, which is independent of the fiber b. Its profiles over the tetrads are

$$\rho(B_3) = 3^4, \qquad \rho(B_6) = 6^4, \qquad \rho(B_9) = 0^4,$$

and, for the three lateral tetrads,

r	active signature of P_r	$R(P_r)$
1	933	$\{5,7\}$
2	393	$\{4,8\}$
3	339	$\{1,2\}$.

(90.43)

Every residue of $R(P_r)$ occurs twice among the four routes of $(P_r)_b$.

Theorem 90.15 (Incidence morphism of the seeds and six Fano planes)**.** *For $\delta \in R_9$, $h \in \{3,6,0\}$ —corresponding, respectively, to B_3, B_6, B_9— and $r \in \{1,2,3\}$, let*

$$N_\delta(B_h, P_r) := \sum_{b \in \binom{R_9}{2}} \#\{(x,y) \in (B_h)_b \times (P_r)_b : \rho_\times(y) - \rho_\times(x) = \delta\}. \tag{90.44}$$

Then

$$N_\delta(B_h, P_r) = \begin{cases} 288, & h+\delta \in R(P_r), \\ 0, & h+\delta \notin R(P_r). \end{cases} \tag{90.45}$$

If $\delta \in R_9^\times$, for each B_h there exists a unique P_r with count 288; therefore, one obtains a bijection

$$\theta_\delta : \{B_3, B_6, B_9\} \xrightarrow{\sim} \{P_1, P_2, P_3\}.$$

Writing the image of (B_3, B_6, B_9) by means of its subscripts, the six bijections are

δ	θ_δ	δ	θ_δ
1	(2, 1, 3)	5	(2, 3, 1)
2	(1, 2, 3)	7	(3, 2, 1)
4	(1, 3, 2)	8	(3, 1, 2).

(90.46)

Be $P = \{P_1, P_2, P_3\}$ and, for $p \in P$, be

$$\mu(p) := \big\{\{f, p\}, P \setminus \{p\}\big\}$$

the perfect matching associated over the four lateral tetrads. Then

$$\mathcal{D}_\delta := \big\{\{B_3, B_6, B_9\}\big\} \cup \big\{\{B, q, q'\} : B \in \{B_3, B_6, B_9\}, \{q, q'\} \in \mu(\theta_\delta(B))\big\} \tag{90.47}$$

is a system of Steiner $\mathrm{S}(2,3,7)$. The six units produce the six planes of Fano over these seven blocks that contain the line transverse $\{B_3, B_6, B_9\}$.

Proof. Fix a fiber b. In $(B_h)_b$ there are four choices with residue h. If $h + \delta \in R(P_r)$, the profile (90.43) provides exactly two choices in $(P_r)_b$ with that residue; if it does not belong to the support, it provides none. Since there are thirty-six fibers, $N_\delta = 36 \cdot 4 \cdot 2 = 288$ in the first case and zero in the second. The three lateral supports partition $R_9^\times$, so a unit δ determines a unique lateral support for each transversal block. Direct reading of those supports yields the table (90.46) and exhausts the $3! = 6$ bijections.

The transversal line covers the three pairs among B_3, B_6, B_9. For a fixed B, the two edges of the matching $\mu(\theta_\delta(B))$ cover the four lateral points once. The three matchings are distinct and form the factorization of K_4 into its three perfect matchings; hence every lateral pair occurs exactly once. The seven triples thus contain each of the twenty-one pairs of points once, which is property $\mathrm{S}(2,3,7)$. Conversely, every Fano plane containing the transversal line assigns to each B a perfect matching distinct from the other two and recovers one of the preceding six bijections. □

The correspondence of sets $\delta \mapsto \theta_\delta$ is neither an action nor a homomorphism: $R_9^\times \simeq C_6$ is abelian under multiplication, whereas S_3 is not, and the units do not form an additive subgroup. The displacement δ indexes six translations of the product profiles.

The same calculation and the same table are verified in all six microfibers. Nevertheless, the difference between support and role must be preserved: the support of f coincides with one of the lateral supports and, therefore, the observable ρ does not by itself recover the decomposition $P \sqcup \{f\}$. That distinction requires the additive signature already declared in the seed state.

Status and scope of Fano incidence

The theorem 90.15 is a finite theorem once q_β, the orientation, and the digital order have been fixed: it defines six exact incidence gauges on blocks. It organizes phase and incidence in the seed domain; it selects neither particle, mass, nor route. Its physical realization is performed subsequently through the morphisms of the enriched ledger.

The passage to physics preserves the chain

$$\begin{aligned} \text{refined seed} &\longrightarrow (\text{signatures } +, \times; U_6; w_6; \text{sheet memory}) \\ &\longrightarrow \text{CT108 record} \longrightarrow (\tau_{12}, D_k, \Omega, P_6, \nu_{120}, \nu_{270}) \\ &\longrightarrow \text{fiber and multisection} \longrightarrow (B_D, B_\Omega) \longrightarrow \text{CKM/PMNS transport} \\ &\longrightarrow \text{character and towers} \longrightarrow \text{mass section}. \end{aligned} \tag{90.48}$$

CKM performs a change of basis of the quark operators,

$$B_d^{(u)} = V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^*, \qquad R_{\mathrm{act}}^{B_d^{(u)}} = V_{\mathrm{CKM}} R_{\mathrm{act}}^{B_d} V_{\mathrm{CKM}}^*, \tag{90.49}$$

and preserves spectrum and determinant. The mixing specifies the representation of the operator in a flavour basis; the register preserves the eigenvalues and their genealogy.

90.5.9.2 Neutrino reader contained in the register

The neutral sector has an integer extractor constructed exclusively with record data. For each dodecaphase block E_m, we define

$$T(m) = \sum_{t \in E_m} (-1)^{\kappa(t)} \varepsilon(t) \mathbf{1}_{\{d(t)=9\}}, \qquad s_C^{(m)}(\nu) = T(m) - T(m+6), \tag{90.50}$$

and

$$\tau_m^\nu = \operatorname{sgn} s_C^{(m)}(\nu). \tag{90.51}$$

The four jump triangles produce

$$\nu_{120} = \sum_{j=1}^{4} \operatorname{sgn}\left(\sum_{m \in \mathcal{T}_j} \tau_m^\nu \right), \qquad \nu_{270} = \sum_{m=1}^{12} \tau_m^\nu \tau_{m+3}^\nu. \tag{90.52}$$

Bidirectional inversion transforms $\nu_{120} \mapsto -\nu_{120}$ and preserves ν_{270}. This parity defines the historical architecture of a reader independent of metrological magnitudes used as a target. Its literal application to the living record 81×108, taking as $\varepsilon(t)$ the sole global sign bit preserved by that projection, yields

$$s_C^{(m)}(\nu) = 0, \qquad \tau_m^\nu = \nu_{120} = \nu_{270} = 0$$

in the eighty-one positions. Antipodal events have been identified by projection of the register. The local result is preserved as a recovered reader and, simultaneously, as a test of information loss: the neutral

extractor of section 92.6.2.3 acts before that identification and reads the local bidirectional bit, the sheet, torsion, and the antipodal corona. The zero obtained after projection does not demote that construction; it certifies exactly what information the global reader erases. The frames F_ℓ, F_ν, the reader (90.52), and the aggregate form

$$G_{ab} = \frac{1}{\sqrt{|L_a||N_b|}} \sum_{c \in L_a} \sum_{d \in N_b} K_{\text{registro}}(c, d), \qquad U_{\text{int}} = \text{polar}(G), \tag{90.53}$$

Thus, the two already constructed ends of the PMNS bridge are fixed.

The cardinal equality between the fifty-six multiplicative seed vertices and the fifty-six CT108 families has undergone an initial control independent of metrological magnitudes used as targets. The natural Hadamard map on a single seed reaches only twenty-nine families; upon traversing the natural dihedral symmetries, signs, and translations, the maximum coverage is thirty-five. The intertwiner therefore acts on the enriched seed pair and preserves sheet, carry, and memory. The incidence matrix comparing shared emissions in that double domain must have rank fifty-six, be natural under orientation, and not factor through global counters. This criterion fixes the exact domain of the next construction.

90.6 Color qutrit algebra and finite gauge connection

The residual functional κ_3 of the section 69.1 yields three affine classes; the subsequent atlas realizes their balance $12 + 12 + 12$ in the proposition 90.14. A triple of labels is not yet a representation. The algebraic step requires the qutrit chart, its orientation, and its phase character to be declared. Under those hypotheses, the representation and its infinitesimal algebra are obtained exactly, without using masses or chromodynamic data.

90.6.1 Weyl pair and trace algebra 0

Let

$$V_{\text{col}} = \mathbb{C}^3, \qquad \omega = e^{2\pi i/3},$$

with ordered basis (e_0, e_1, e_2). We define

$$Xe_j = e_{j+1 \bmod 3}, \qquad Ze_j = \omega^j e_j.$$

We have

$$X^3 = Z^3 = I, \qquad ZX = \omega XZ.$$

Theorem 90.16 (Weyl Basis and Complex Color Algebra)**.** *For*

$$W_{ab} := X^a Z^b, \qquad (a, b) \in (\mathbb{Z}/3\mathbb{Z})^2,$$

the nine operators W_{ab} form an orthogonal basis of $\text{End}_{\mathbb{C}}(V_{\text{col}})$ *with respect to the Hilbert–Schmidt product. The eight non-identity operators have zero trace and*

$$\text{span}_{\mathbb{C}}\{W_{ab} : (a, b) \neq (0, 0)\} = \mathfrak{sl}_3(\mathbb{C}).$$

Proof. The Weyl relation makes it possible to reduce any product to the form $X^a Z^b$. A direct calculation gives

$$\text{tr}(W_{ab}^\dagger W_{cd}) = 3\,\delta_{ac}\delta_{bd}.$$

Therefore, the nine operators are linearly independent; since $\dim_{\mathbb{C}} \text{End}_{\mathbb{C}}(V_{\text{col}}) = 9$, they form a basis. The trace of $X^a Z^b$ vanishes except for $(a, b) = (0, 0)$. The remaining eight thus form a basis of the trace-zero subspace, whose complex dimension is eight. □

Corollary 90.17 (Compact real form)**.** *The Involution*

$$\tau(A) := -A^{\dagger}$$

preserves $\mathfrak{sl}_3(\mathbb{C})$*, and its fixed-point set is*

$$\mathfrak{sl}_3(\mathbb{C})^{\tau} = \{A : A^{\dagger} = -A,\ \operatorname{tr} A = 0\} = \mathfrak{su}(3).$$

This real form has dimension eight.

Proof. We have

$$W_{ab}^{\dagger} = \omega^{ab} W_{-a,-b}.$$

The eight non-zero indices form four adjacent pairs. For

$$\mathcal{R} = \{(1,0), (0,1), (1,1), (1,2)\},$$

the operators

$$A_u = W_u - W_u^{\dagger}, \qquad B_u = i(W_u + W_u^{\dagger}), \qquad u \in \mathcal{R},$$

are anti-Hermitian, traceless, and linearly independent over $\mathbb{R}$. They therefore constitute a real basis of $\mathfrak{su}(3)$. □

The residual application

$$\kappa_{\mathrm{col}}(r, c) = \mathbb{C} e_{\,r-c \bmod 3}$$

associates the three classes of the atlas with the three lines proper of Z. The operator X the permutes cyclically and Z records its phase. This assignment is exact a time fixed

$$V_{\mathrm{col}} = \mathbb{C}[\mathbb{F}_3], \quad \text{the delta basis}, \quad j \longmapsto j+1, \quad j \longmapsto \omega^{j}.$$

Admissible affine changes of the chart produce unitarily conjugate representations. The construction is therefore natural up to unitary conjugacy; by itself, it selects neither the physical names of the colors nor a QCD realization.

90.6.2 Connection and Action on a Finite Cellular Complex

Construction 90.18 (Finite gauge theory relative to the complex)**.** Let $\mathcal{K} = (V, E, F)$ be a finite oriented cellularization compatible with the graph of routes. To each oriented edge e one assigns

$$U_e \in SU(3), \qquad U_{\bar{e}} = U_e^{-1}.$$

A transformation $g : V \to SU(3)$ acts by

$$U_e \longmapsto U_e^{g} = g_{t(e)} U_e g_{s(e)}^{-1}.$$

For each face f, the ordered product

$$U_f = \prod_{e \subset \partial f} U_e^{\epsilon(f,e)}$$

defines its holonomy. With a base vertex v_f chosen, one has

$$U_f^{g} = g_{v_f} U_f g_{v_f}^{-1}.$$

Proposition 90.19 (Gauge invariance and covariant transport)**.** *For* $\beta \geq 0$*, the action*

$$S_{\mathrm{col}}^{(\mathcal{K},\beta)}[U] = \frac{\beta}{3}\sum_{f\in F}\left(3 - \operatorname{Re}\operatorname{tr} U_f\right)$$

is gauge-invariant and no negativa. If $\psi_v \in V_{\mathrm{col}}$ transforma as $\psi_v \mapsto g_v\psi_v$, then

$$\nabla_e\psi := U_e\psi_{s(e)} - \psi_{t(e)}$$

satisfies

$$\nabla_e^g\psi^g = g_{t(e)}\nabla_e\psi.$$

Proof. The holonomy changes by conjugation and the trace is cyclic; of there the invariance. As each U_f is unitary of dimension three, $\operatorname{Re}\operatorname{tr} U_f \le 3$, lo that proves the not negativity. The covariance of ∇_e results substituting the two lawes of transformation. □

90.6.3 Positivity of Class Kernels and Harmonic Decomposition

The preceding finite connection does not by itself imply reflection positivity or a Yang–Mills limit. Before formulating such transfers, it is appropriate to isolate the classical control that is closed. Let G be a compact group and $\widehat{G}$ its unitary dual. For each $\rho \in \widehat{G}$, let d_ρ be its dimension and $c_\rho(a) \ge 0$ coefficients with sufficient decay, relative to the growth of d_ρ, for the following series to converge absolutely.

Definition 90.20 (Class nucleus)**.** It is defined

$$K_a(g,h) := \sum_{\rho\in\widehat{G}} c_\rho(a)\ \operatorname{Tr}\!\big(\rho(g)\rho(h)^*\big), \qquad g,h\in G.$$

The choice

$$c_\rho(t) = d_\rho e^{-t C_2(\rho)}, \qquad t>0,$$

provides the example of the heat kernel. The nonnegativity of the coefficients does not replace the hypothesis of absolute convergence.

Theorem 90.21 (Positivity of Peter–Weyl)**.** *For $g_1,\dots,g_n \in G$ and $z_1,\dots,z_n\in\mathbb{C}$,*

$$\sum_{i,j=1}^{n} \overline{z}_i z_j K_a(g_i,g_j) \ge 0.$$

Proof. The unitarity of the representations and absolute convergence allow the sum to be reordered:

$$\sum_{i,j}\overline{z}_i z_j K_a(g_i,g_j) = \sum_{\rho\in\widehat{G}} c_\rho(a)\left\|\sum_j \overline{z}_j\rho(g_j)\right\|_{\mathrm{HS}}^2 \ge 0.$$

A negative spectral coefficient may destroy this Gram positivity; therefore the sign forms part of the hypotheses. □

Lemma 90.22 (Abelian Poisson Control)**.** *For $G=U(1)$, $\theta\in\mathbb{R}$, and $|a|<1$,*

$$\sum_{n\in\mathbb{Z}} a^{|n|}e^{in\theta} = \frac{1-a^2}{1-2a\cos\theta+a^2}.$$

If nonnegative Fourier coefficients are also required, a is restricted to $0\le a<1$.

Proof. The equality results from summing separately the two geometric series with positive and negative indices and adding the mode $n = 0$. For $a < 0$, the scalar identity remains valid but no longer constitutes an example of nonnegative Peter–Weyl coefficients. □

Remark 90.23. The definition 90.20, theorem 90.21 y lemma 90.22 are classical definitions and results of harmonic analysis. They provide a positive control for class kernels; they do not prove Osterwalder–Schrader positivity, a mass gap, confinement, or a continuum limit, nor do they promote the finite gauge theory of this chapter to quantum chromodynamics.

Algebraic closure of color

The theorem 90.16 y corollary 90.17 construct exactly the chain

$$\text{ternary residue} \longrightarrow \text{qutrit} \longrightarrow \mathfrak{sl}_3(\mathbb{C}) \longrightarrow \mathfrak{su}(3).$$

The construction 90.18 y proposition 90.19 prolongs that chain to a connection and a gauge action on the finite cellularization $\mathcal{K}$. This is the object proved in the chapter. The comparison with physical chromodynamics is made subsequently and does not redefine the qutrit, the balance $12 + 12 + 12$, or the constructed representation.

90.7 Systematic extension of mass theory

90.7.1 Extension–route factorization and genealogical record conservation

90.7.1.1 The object prior to magnitude

The theory begins before the number. A physically individuated mass is not a real number accompanied by the name of a particle, but the terminal term of a mathematical biography. APP supplies the finite incidence space and the two correlative operations; TRIT preserves the temporal orientation of each transition; TPK transports phase, sheet, carry, boundary, route, memory, and survival. Dimensional Mechanics subsequently converts that biography into a section of the mass line. The complete order is

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \Gamma^{\mathrm{surv}} \longrightarrow \gamma_{\mathrm{obs}} \longrightarrow \mathscr{L}_\gamma \longrightarrow \sigma(\gamma) \longrightarrow \chi_{\mathrm{mass}} \longrightarrow \mathcal{T}_{90,120} \longrightarrow \mathcal{L}_M.$$

Each arrow retains information used by the next. The cell does not replace the route; the route does not replace the register; a signature does not replace its genealogical register; a character does not replace the dimensional section on which it acts. This separation explains why two particles may share charge or spin and nevertheless have different masses: their enriched biographies are not the same. It also explains why particle and antiparticle share mass: the involution changes odd orientation and preserves even memory, which descends to the mass quotient.

APP consists of nine by nine published positions. Each position admits four initial orientations, so the oriented domain contains

$$81 \cdot 4 = 324$$

candidate routes. The internal classification produces 56 route families and 13 multisections. These cardinalities are simultaneous: 81 counts cells, 324 counts orientations, 56 counts quotients by history, and 13 counts coarse operator types. No particle table replaces these domains.

90.7.1.2 Discrete unit of the continuum that supports the mass

The mass line is not appended to five independent theories of the continuum. It follows from the unique APP–TRIT–TPK state from which the spectral, solenoidal, gauge, coherent, and determinantal aspects jointly emerge, designated in the construction notation by sp, sol, gau, coh, det. Those symbols are reading discriminants, not prior subdomains. If C_{disc} denotes the common structure, the generating arrow has the form

$$\mathfrak{E}_{\mathrm{cont}} : C_{\mathrm{disc}} \longrightarrow \big(C_\infty; O_{sp}, O_{sol}, O_{gau}, O_{coh}, O_{det}\big),$$

where the five observations belong to the same limit object, share fibers, orientation, carry, memory, boundary, and readers, and are transported by the same refinement squares. Mass uses its determinant reading for the action scale, its spectral reading for the character and the towers, its gauge reading for the null and charged sectors, and its coherent and solenoidal readings for propagation and composition. None of them is introduced by presupposing that the state belongs to a classical physical class.

This unit is a typing condition: separating one of those aspects and retaining the others would produce a different object, incapable of simultaneously supporting route, action, charge, amplitude, and mass. The present theory does not reprove the complete continuum; it incorporates its unified construction as a foundation and verifies that none of the mass insertions disaggregates it.

90.7.1.3 Absolute scale and quotients

The local determinantal reader has Smith normal form.

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4),$$

from which

$$|\operatorname{coker} H_4| = 16, \qquad \Sigma_H = \varphi_{\mathrm{HMT}} \alpha_{\mathrm{HMT}}^{16}, \qquad \mathrm{ord}_{10}(\Sigma_H) = -34.$$

The decade is obtained before the SI chart. The pre and return sections of the action line are

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34} u_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}^{(\mathrm{deg})}\, 10^{-34} u_S,$$

and its bidirectional quotient is

$$R_{\mathrm{act}} = \frac{H_5}{\eta_{\mathrm{ret}}^{(\mathrm{deg})}}.$$

In the clock branch, we explicitly choose the return section, $\hbar_{\mathrm{int}} := \hbar_{\mathrm{ret}}$, as the action section. A declared clock t_0 provides

$$\omega_0 = \frac{2\pi}{108 t_0}, \qquad \varepsilon_{\mathrm{clk}} = \hbar_{\mathrm{ret}} \omega_0, \qquad m_{\mathrm{clk}} = \varepsilon_{\mathrm{clk}} c_{\mathrm{int}}^{-2}.$$

Adopting $u_M^{\mathrm{clk}} := m_{\mathrm{clk}}$ as origin of the chart is a normalization explicit. Not identifies automatically this section with a basis energetic or mass previously declared. The character produces first the operator basal

$$\widehat{M}_X^{(0)} = \mathfrak{e}_0 m_{\mathrm{clk}}\, \chi_{\mathrm{full}}(\sigma_X - \sigma_e, \eta_X - \eta_e),$$

and the bidirectional reader acts afterward. For each presentation $r \in \{D, \Omega\}$, let $\widehat{B}_X^r = (\widehat{B}_X^r)^*$ over the finite fiber; in an infinite realization one also requires a common dense domain for functional calculus. Then

$$\widehat{M}_X^{\beta, r} = R_{\mathrm{act}}^{\widehat{B}_X^r/2}\, \widehat{M}_X^{(0)}\, R_{\mathrm{act}}^{\widehat{B}_X^r/2}.$$

On a scalar fibre, $\widehat{B}^r_X = \kappa_X I$, so that

$$m_X^\beta = m_X^{(0)} R_{\mathrm{act}}^{\kappa_X}.$$

If the tower coordinates are already expressed relative to the electron, $-\eta_e$ is omitted. When a quotient is formed, the common dimensional section cancels, but the difference in bidirectional reading does not:

$$\frac{m_Y^\beta}{m_X^\beta} = \chi_{\mathrm{full}}(\sigma_Y - \sigma_X, \eta_Y - \eta_X) R_{\mathrm{act}}^{\kappa_Y - \kappa_X}.$$

Therefore, 10^{-34} fixes the absolute scale of action, energy, and mass; it is not a hidden relative corrector. The fine information of a species can enter only through its character, its relative clock, or the oriented quotient of the pre- and return sections.

90.7.1.4 Internal quantity of the mass line induced by the cycle of 108 states

The Planck–CT108 self-duality provides, in the internal normalization

$$\hbar_{\mathrm{int}} = c_{\mathrm{int}} = t_0 = 1,$$

the section

$$m_{108}^{\mathrm{int}} = 0.058177641733144319230789692282953757\ldots.$$

This number is a normalized quantum of the internal mass line. It is not the mass of a particle, does not by itself carry a MeV or GeV chart, and cannot be inserted as an additional species row. Its function is to verify compatibility among the action chart, the CT108 clock, and the self-duality of the mass line. The transition to a nominal mass still requires the route, signature, operator, and dimensional section of the species.

90.7.1.5 Existence of the HMT–MD output for surviving extensions and scalar-reduction criterion by fiber rank

Given a surviving extension, a cellular reader, an integer signature, an absolutely convergent family of tower coefficients, and a dimensional mass section, a well-typed HMT–MD output exists. If the fiber has a unique fixed point compatible with the involution, the output reduces to a canonical scalar. If the fiber has higher rank, the primary output is the spectral operator on the fiber; a physical scalar mass requires the coupling functional that individuates the observed representation.

This theorem does not weaken the recovered scalar evaluations. It distinguishes three claims: the internal operator is constructed; the evaluation of a declared signature is exact; and the prospective determination of that signature is an additional theorem when it is not deduced from the reader itself. Experimental comparison is performed only after fixing which of those three claims is being published.

90.8 Functorial construction of mass families over the atlas

The construction ends with the algorithm that receives only typed genealogical data, forms the realization fiber, and requires the equalizer of the readers before publishing a dimensional output.

90.8.1 Universal Algorithm of the General Mass Law

90.8.1.1 Cellular reader domain: surviving extension, route, record, and signature

The entry contains only:

$$(\text{APP}, \text{TRIT}, \text{TPK}, \mathcal{R}_{324}, \mathcal{L}_{108}, \text{representation}, \text{coupling}, \text{common dimensional chart}).$$

It contains no observed mass, residual mass, chosen coefficient by approximation, or external table.

90.8.1.2 Functorial construction of material realization: route fiber, intertwiner, coincidence projector, spectral operator, and scalarization

For each physical state X:

1. construct its descriptor of charge, generation, flavor, color, spin, statistics, orientation, and composition;
2. form the fiber of compatible routes;
3. compute the space of intertwiners

$$\mathcal{J}_X = \mathrm{Hom}_{G_X}(V_X, \ell^2(F_X));$$

1. if $\dim \mathcal{J}_X = 0$, discard that identification;
2. construct the projector by Reynolds averaging and the polar factor;
3. reconstruct the genealogical registers of all routes in the support;
4. calculate signature, K_D, K_Ω and $\mathcal{F}^{2w}_{\text{tower}}$;
5. apply the coupling-reading pointer, the equalizer, or the joint spectral measure;
6. construct $\widehat{M}_X$;
7. reduce to a scalar only if the variance is zero or an isolated pole exists;
8. for composites, first apply $\mathfrak{C}_{12}$;
9. for widths, calculate the survival semigroup;
10. multiply by the common dimensional section;
11. freeze the output;
12. open the external comparison.

90.8.1.3 Existence of the fiberwise realization

The multiplicity of a representation π_X within a fiber is calculated by

$$\dim \mathcal{J}_X = \frac{1}{|G_X|} \sum_{g \in G_X} \overline{\chi_{\pi_X}(g)}\, |\mathrm{Fix}_{F_X}(g)|.$$

If it is positive, an intertwiner exists. If it is greater than one, the result is a bundle of realizations, and the coupling must select a state or preserve the mixed measure. This formula replaces a nominal search by a finite algebraic condition on the 324 routes.

90.8.1.4 Spectral equalizer of the readers K_D and K_Ω via $Q_{\mathrm{eq}} = \mathbf{1}_{\{0\}}(\widehat{L}_D - \widehat{L}_\Omega)$

Let

$$L_r(\Gamma) = \kappa_r(\Gamma)\log R_{\mathrm{act}} + \sum_{h\in\mathfrak{S}} \eta_h^r(\Gamma)s_h, \qquad r \in \{D,\Omega\}.$$

The coincidence projector is

$$Q_{\mathrm{eq}} = \mathbf{1}_{\{0\}}(\widehat{L}_D - \widehat{L}_\Omega).$$

A realization admits common exact reading exactly when

$$P_X \le Q_{\mathrm{eq}}.$$

If there exists an involution of the coupling that exchanges both readers, L_{sym} is used. Otherwise, its joint spectral measure is published.

90.8.1.5 Cellular reader codomain: dimensionless character, tower, and mass realization

The output may be:

$$\begin{cases} m_X \in \mathcal{L}_M, & \text{scalar mass,} \\ \mathrm{Spec}(\widehat{M}_X), & \text{multiplet,} \\ z_X = M_X - i\Gamma_X/2, & \text{resonance,} \\ 0 \in \mathcal{L}_M, & \text{zero section,} \\ (C,F,R,\Delta E), & \text{topological sector,} \\ (L,\mathrm{Obs}_L), & \text{virtual section.} \end{cases}$$

This diversity of codomains is part of the law; not an exception.

90.9 Bifurcation of the previous null sector from the positive mass branch

The photonic and gluonic sectors are separated before applying the positive character. The identity $R_{\mathrm{act}}^0 = 1$ does not prove zero mass; nullity follows from the gauge chart. In the matter families, the primary output is a positive fiber operator. Only a typed physical compression produces a scalar.

90.10 Exhaustive construction of the atlas 81/56/13/324 and conservation of the null sector

The definition and the theorems of the domain precede the unfolding of its 324 fichas completas. The cards are placed at the end of the scientific apparatus in order to preserve the continuity of the derivation without concealing any route.

90.11 Theorem of material domain exhaustiveness and null sector stability under refinement

The General Mass Law does not start from a table of names or a list of experimental values. Its internal domain is the finite chain

$$\mathcal{R}_{324} \longrightarrow \mathcal{C}_{81} \longrightarrow \mathcal{F}_{56} \longrightarrow \mathcal{M}_{13},$$

where $\mathcal{R}_{324}$ contains the 4 orientations of each of the 81 cells, $\mathcal{F}_{56}$ groups equivalent route histories, and $\mathcal{M}_{13}$ preserves the operator multisections. The null sector bifurcates before the positive character is applied. Therefore, neither an abbreviated historical table nor a taxonomy of physical names can replace this domain.

Let $\mathcal{P}$ be the set of individualized physical states. An elementary state is represented by the descriptor

$$X = (\mathfrak{c}, \rho, g, f, q, \epsilon, \varsigma, \mathfrak{a}),$$

which preserves representation class $\mathfrak{c}$, internal representation ρ, generation g, flavor f, charge q, sheet orientation ϵ, statistical sector ς, and coupling data $\mathfrak{a}$. A composite state adds ordered constituents, composition tree, common boundary, and collective quantum numbers. A resonance adds the open channels and the causal prescription for the resolvent.

The physical realization must address three distinct questions:

1. which fibre, orbit, or multisection of the atlas corresponds to the descriptor;
2. which positive operator publishes the mass on that fiber;
3. which coupling produces a scalar, a multiplet, a distribution, or a complex pole.

Confusing these questions introduces a retrospective selector. The correct answer is obtained through a natural relation, a fiber operator and a spectral compression, in that order.

90.12 Fiber product between species invariants and genealogical paths

Let $\mathcal{D}$ be the finite space of internal data already constructed before the contrast: representation, orientation, memory, boundary, charge, flavor, color, generation, and statistical sector. Denote by

$$d_{\mathcal{P}} : \mathcal{P} \longrightarrow \mathcal{D}, \qquad d_{\mathcal{R}} : \mathcal{R}_{324} \longrightarrow \mathcal{D}$$

the physical descriptor maps and the genealogical descriptor. The object of realization is defined by the fiber product

$$\mathcal{J} = \mathcal{P} \times_{\mathcal{D}} \mathcal{R}_{324} = \{(X, \gamma) : d_{\mathcal{P}}(X) = d_{\mathcal{R}}(\gamma)\}.$$

The fiber of a species is

$$\mathcal{S}(X) = \{\gamma \in \mathcal{R}_{324} : (X, \gamma) \in \mathcal{J}\}.$$

A descriptor (X) is called realizable by the atlas when

$$d_{\mathcal{P}}(X) \in \operatorname{im} d_{\mathcal{R}}.$$

This condition is equivalent to $\mathcal{S}(X) \neq \varnothing$. It does not follow from merely writing the fiber product: it constitutes the exact existence condition that must be verified for each physical descriptor. The value of $\mathcal{S}$ may be a route, an orbit, or a multisection. A point section is not forced when the internal symmetry requires preservation of a multiplicity.

Definition 90.24 (Admissible realization relation)**.** The relation $\mathcal{S}$ is admissible when it simultaneously satisfies:

1. depends solely on data constructed before the external table is opened;
2. commutes with particle–antiparticle conjugation;
3. is equivariant under cellular inversion and the internal gauge actions;
4. it is compatible with refinement, nonadic return and dodecaphasic reading;
5. preserves every multiplicity not eliminated by a physical coupling;
6. remains invariant under permuting or replacing the external values of mass and width.

Theorem 90.25 (Prospective realization by fibered product)**.** *If $d_{\mathcal{P}}$ and $d_{\mathcal{R}}$ are equivariant with respect to a symmetry G and do not depend on external magnitudes, then $\mathcal{J}$ is a G-subobject of $\mathcal{P} \times \mathcal{R}_{324}$. For every realizable descriptor X, the nonempty fiber $\mathcal{S}(X)$ is G_X-stable, and the assignment $X \mapsto \mathcal{S}(X)$ is independent of the result subsequently compared.*

Proof. For $(X, \gamma) \in \mathcal{J}$ one has $d_{\mathcal{P}}(X) = d_{\mathcal{R}}(\gamma)$. Equivariance gives

$$d_{\mathcal{P}}(gX) = g\, d_{\mathcal{P}}(X) = g\, d_{\mathcal{R}}(\gamma) = d_{\mathcal{R}}(g\gamma),$$

and, consequently, $(gX, g\gamma) \in \mathcal{J}$. If $g \in G_X$, the second component remains in $\mathcal{S}(X)$. Independence of the external values is immediate because none of them belongs to the domain of the two maps of the fiber product. □

The electron constitutes the central exception. Cell inversion has a unique fixed point, the cell $(5, 5)$, and its physical fiber therefore has rank one. The remaining stable multisections do not, in general, admit an equivariant point selector. This difference is not a deficiency of the atlas: it is the mathematical distinction between a central section and a species realized as an orbit or multisection.

90.13 Intrinsic spectral classification of the species

Let $\mathcal{A}_{\mathrm{rut}}$ be the finite commutative algebra generated by the internal route observables: signature $\mathbb{Z}^5$, character, chronology, K_D, K_Ω, orientation, memory, and compatible readers. In a fiber F, one defines

$$\gamma \sim_{\mathrm{sp}} \eta \quad \Longleftrightarrow \quad a(\gamma) = a(\eta) \quad \text{for every } a \in \mathcal{A}_{\mathrm{rut}}.$$

For each class $\lambda \in F/\!\sim_{\mathrm{sp}}$,

$$P_\lambda = \sum_{\gamma \in \lambda} |\gamma\rangle\langle\gamma|$$

is a spectral projector; the P_λ are orthogonal and $\sum_\lambda P_\lambda = I_F$.

Theorem 90.26 (Maximum internal individualization)**.** *Every classifier constructed exclusively from observables of $\mathcal{A}_{\mathrm{rut}}$ factors uniquely through the quotient $F/\!\sim_{\mathrm{sp}}$. Consequently, two descriptors with distinct complete signatures belong to distinguishable sectors; two descriptors that agree for all generators of the algebra cannot be separated by a rule constructed from those same observables.*

Proof. The characters of the finite commutative algebra separate exactly its joint spectrum. A functional classifier of its generators is constant on each fiber of the joint spectral map and therefore factors through the quotient. If two routes have the same character on the entire algebra, no element of it can separate them. □

This theorem closes the logical limit of the classification. When a physical representation requires two states belonging to the same spectral class to be distinguished, it must exhibit an additional observable or a coupling that enlarges $\mathcal{A}_{\mathrm{rut}}$. The absence of such an observable does not authorize the external mass to be used as a selecting label.

In the charged leptonic sector, the descriptors distinguish the central fiber, the depth-G_2 multisection, and the depth- G_4 multisection. The first produces the unique electronic line; the other two produce the operators associated with the muonic and tauonic realizations. If the coupling preserves more than one block, the correct prediction is its spectrum; the physical label does not authorize choosing one of its eigenvalues by numerical proximity.

90.13.1 Conjugate quotients and minimal memory of the finite atlas

The double projection requires the transported state to be distinguished from each of its images. At the finite level of the atlas, this distinction admits an exact formulation: the same dodecaphasic word has several quotients, each of which preserves a given family of invariants and contracts the remainder. The configuration historically called *diamond* is formulated here as a system of conjugate quotients over a common support. Its mathematical content consists in identifying the fibers of those quotients and determining the minimal memory required to separate the states that a projection has identified.

90.13.1.1 Cellular domain, routes, and enriched internal state

Let

$$C_9 = \{1, \dots, 9\}^2$$

the finite APP chart. For a cell $x = (r, c)$, the atlas fixes the internal cellular type, the level

$$G(r,c) = \begin{cases} 0, & (r,c) = (5,5), \\ \max\{1, |r-5|, |c-5|\}, & \text{otherwise}, \end{cases}$$

and, in the corresponding sector, the colour class $(r - c) \bmod 3$. The published multiplicative evaluation is

$$a_\times(r,c) = \mathrm{dr}_9(rc).$$

These readers are computed from the oriented cell; a later physical denomination does not intervene in their definition.

The route catalog associates with x the coordinate

$$\mathcal{R}(x) = \big(n_A, s_C, k_{\Delta_4}, \nu_{120}, \nu_{270}, \tau_{12}\big) \in \mathscr{R}_{56},$$

where $\tau_{12} \in \{-1, 0, 1\}^{12}$ is the dodecaphasic word. The exhaustive comparison of the atlas of 81 cells with the expanded CT–108 catalogue establishes a unique surjective map

$$\mathcal{R} : C_9 \longrightarrow \mathscr{R}_{56}, \qquad |\operatorname{im}\mathcal{R}| = 56,$$

whose fibers have the distribution

$$41 \cdot 1 + 10 \cdot 2 + 2 \cdot 3 + 1 \cdot 4 + 2 \cdot 5 = 81. \tag{90.54}$$

For $\tau = (\tau_0, \dots, \tau_{11})$ and $d \in \{3, 4, 6\}$, let $\mathcal{O}_d$ be the orbits of $j \mapsto j + d$ in $\mathbb{Z}/12\mathbb{Z}$. We define

$$\nu_d(\tau) = \sum_{O \in \mathcal{O}_d} \operatorname{sgn}\left(\sum_{j \in O} \tau_j\right), \qquad (\nu_{90}, \nu_{120}, \nu_{180}) = (\nu_3, \nu_4, \nu_6),$$

and the cyclic correlations

$$Q_k(\tau) = \sum_{j=0}^{11} \tau_j \tau_{j+k}, \qquad k = 1, \dots, 6,$$

with indices taken modulo twelve. They complete the reader.

$$z_0(\tau) = \#\{j : \tau_j = 0\}, \qquad w(\tau) = \sum_j |\tau_j|, \qquad L(\tau) = \sum_j \tau_j.$$

The notation z_0 separates the number of null entries from the distinct invariant $R_3 = Q_3/2$.

Theorem 90.27 (Minimal internal selector)**.** *Having fixed the origin of the chart and its lexicographic order, let*

$$\kappa(x) = \#\{y \in \mathcal{R}^{-1}(\mathcal{R}(x)) : y <_{\mathrm{lex}} x\} \in M_5 := \{0, 1, 2, 3, 4\}.$$

The map

$$\begin{aligned} \mathcal{S}_{\mathrm{int}} : C_9 &\longrightarrow \mathscr{E}_{\mathrm{int}} := \mathscr{R}_{56} \times M_5 \times \mathbb{Z}^9, \\ x &\longmapsto \big(\mathcal{R}(x), \kappa(x), \nu_{90}(\tau_x), \nu_{180}(\tau_x), Q_1(\tau_x), \dots, Q_6(\tau_x), z_0(\tau_x)\big) \end{aligned} \tag{90.55}$$

is injective. Among the injective lifts of the form $(\mathcal{R}, m)$, its memory alphabet has minimal cardinality.

Proof. The distribution (90.54) contains fibers of cardinality five. By the pigeonhole principle, every map $m : C_9 \to M$ for which $(\mathcal{R}, m)$ is injective satisfies $|M| \geq 5$. Within each fiber, κ enumerates its elements exactly according to the fixed order and therefore separates two cells sharing a route. Cells with distinct routes are already separated by $\mathcal{R}$. Consequently, $(\mathcal{R}, \kappa)$, and a fortiori $\mathcal{S}_{\mathrm{int}}$, is injective and attains the lower bound. □

Projection onto the first coordinate yields the commutative diagram.

$$\begin{array}{ccc} C_9 & \xrightarrow{\mathcal{S}_{\mathrm{int}}} & \mathscr{E}_{\mathrm{int}} \\ {\scriptstyle \mathcal{R}}\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{pr}_{\mathscr{R}}} \\ \mathscr{R}_{56} & \xrightarrow{\mathrm{id}} & \mathscr{R}_{56}. \end{array}$$

Thus, κ measures the minimum cellular memory forgotten by $\mathcal{R}$. Its canonicity is relative to the origin and ordering of the chart: if these are regarded as a gauge, the intrinsic object is the ordered fiber or its orbit. The index κ and the antisheet memory $A_6 = (e_0 - e_6, \dots, e_5 - e_{11})$ answer to losses of information of the same categorical nature, but fulfill different functions: κ separates cells of a fiber of $\mathcal{R}$, whereas A_6 participates in reconstructing the dodecaphasic register.

90.13.1.2 Thick types and natural multisections

The ratio

$$q : C_9 \longrightarrow \mathscr{Y}_{13}, \qquad q(x) = \big(\mathrm{familia}(x), G(x)\big),$$

has thirteen values. The cardinalities of its fibers are

$$1, 8, 16; \qquad 2, 4, 6, 8; \qquad 2, 4, 6, 8; \qquad 4, 12.$$

The output determined by $y \in \mathscr{Y}_{13}$ is the finite multisection

$$\mathfrak{S}(y) = \{\mathcal{S}_{\mathrm{int}}(x) : q(x) = y\} \in \mathrm{Fin}^{+}(\mathscr{E}_{\mathrm{int}}). \tag{90.56}$$

Let $\rho(r, c) = (10 - r, 10 - c)$. The thirteen fibers are stable under ρ, but only $q^{-1}(q(5, 5))$ contains a fixed point. Therefore, in the remaining twelve fibers, a pointwise ρ-equivariant section does not exist: the multisection (90.56) is the natural output of the coarse type. Choosing a representative requires additional orientation or representation data and is not obtained from the name subsequently assigned to the type.

90.13.1.3 Dodecaphasic conjugation and signature quotients

Let $\mathcal{T}_{12} = \{-1, 0, 1\}^{12}$ and define the involution

$$C_M : \mathcal{T}_{12} \longrightarrow \mathcal{T}_{12}, \qquad C_M(\tau) = -\,\mathrm{rev}(\tau).$$

We group the linear and quadratic readers together.

$$\mathcal{L}(\tau) = (\nu_{90}, \nu_{120}, \nu_{180}, L) \in \mathbb{Z}^4,$$

$$Q(\tau) = (Q_1, \dots, Q_6, z_0, w) \in \mathbb{Z}^8.$$

Conjugation satisfies the identities

$$\mathcal{L}(C_M\tau) = -\mathcal{L}(\tau), \qquad Q(C_M\tau) = Q(\tau). \tag{90.57}$$

For any reader $F : \mathcal{T}_{12} \to I_F$, write

$$\tau \sim_F \tau' \quad \Longleftrightarrow \quad F(\tau) = F(\tau'), \qquad \mathfrak{Q}_F = \mathcal{T}_{12}/\!\sim_F .$$

When F is equivariant with respect to C_M, the involution descends to $\overline{C}_{M,F}$, and one obtains

$$\begin{array}{ccc} \mathcal{T}_{12} & \xrightarrow{C_M} & \mathcal{T}_{12} \\ {\scriptstyle q_F}\downarrow & & \downarrow{\scriptstyle q_F} \\ \mathfrak{Q}_F & \xrightarrow{\overline{C}_{M,F}} & \mathfrak{Q}_F . \end{array}$$

In particular, $\overline{C}_{M,Q}$ is the identity and $\overline{C}_{M,\mathcal{L}}$ changes the sign of the published signature. This parity difference is the conjugate structure of the two quotients; they are not two evaluations of distinct initial states.

The atlas contains seven word forms of dodecaphonic with multiplicities.

$$54, 2, 2, 2, 7, 7, 7. \tag{90.58}$$

One form is fixed by C_M; the other six constitute three conjugate pairs. In each pair, the cellular multiplicities are 2 and 7. Consequently, no bijection of C_9 can lift C_M while preserving the form fibers. The inversion ρ intertwines the words in 49 cells and leaves 32 exceptions. The exact anti-section at this level is therefore the correspondence

$$x \longmapsto \{y \in C_9 : \tau_y = C_M(\tau_x)\}.$$

The numerical equality $54 + 9 + 9 + 9 = 81$ describes the grouping of one fixed form and three pairs of multiplicity $2 + 7$; it does not constitute a decomposition into bijective orbits of the cells.

90.13.1.4 Exact Configuration of Quotients

Let us define the partial readers

$$\Sigma_7(\tau) = (\nu_{90}, \nu_{120}, \nu_{180}, Q_3, Q_4, Q_6, z_0) \in \mathbb{Z}^7,$$

$$\ell_3(\tau) = (\nu_{90}, \nu_{120}, \nu_{180}) \in \mathbb{Z}^3, \qquad p_4(\tau) = (Q_3, Q_4, Q_6, z_0) \in \mathbb{Z}^4.$$

Each reader induces its quotient $\mathfrak{Q}_{\Sigma_7}$, $\mathfrak{Q}_{\ell_3}$, or $\mathfrak{Q}_{p_4}$. The following three collisions locate precisely what information each projection retains.

Theorem 90.28 (Configuration of Conjugate Quotients)**.** *In the finite catalog the following assertions hold simultaneously.*

(i) *The boundary words H_α and H_{APP} are distinct and belong to distinct dihedral orbits, but*

$$\Sigma_7(H_\alpha) = \Sigma_7(H_{\mathrm{APP}}) = (-3, -4, -5, 3, 2, 2, 6).$$

The omitted coordinate of the nearest neighbor separates them:

$$Q_1(H_\alpha) = 4, \qquad Q_1(H_{\mathrm{APP}}) = 3.$$

(ii) *The obstructing support P_π and the template historically designated by τ_{lep} satisfy*

$$\ell_3(P_\pi) = \ell_3(\tau_{\mathrm{lep}}) = (-3, -4, -6),$$

while

$$p_4(P_\pi) = (7, 6, 6, 3), \qquad p_4(\tau_{\mathrm{lep}}) = (4, 3, 2, 5).$$

(iii) *The templates nominally inherited as electronic and protonic publish the same word,*

$$\tau_e^{\mathrm{hist}} = \tau_p^{\mathrm{hist}} = 0^{12}.$$

Every invariant that factors exclusively through $\mathcal{T}_{12}$ necessarily takes the same value on both.

Therefore, the complete configuration is a fan of quotients of the same enriched state,

$$\mathscr{C}_{\mathrm{int}} \longrightarrow \begin{cases} \mathfrak{Q}_{\mathcal{L}}, \\ \mathfrak{Q}_{Q}, \\ \mathfrak{Q}_{\Sigma_7}, \\ \mathfrak{Q}_{\ell_3}, \\ \mathfrak{Q}_{p_4}, \\ \mathcal{T}_{12}, \end{cases} \tag{90.59}$$

and not a bijective correspondence between mathematical constants and physical classes.

Proof. The equalities in (i)–(iii) result from evaluating the readers defined above on the published words. In (i), the inequality of Q_1 proves that the coincidence at $\mathfrak{Q}_{\Sigma_7}$ arises from the suppressed coordinate and not from an identity of words. In (ii), the linear signature coincides and the even vector differs, so the identification occurs only in $\mathfrak{Q}_{\ell_3}$. In (iii), equality of the words implies $F(\tau_e^{\mathrm{hist}}) = F(\tau_p^{\mathrm{hist}})$ for every map F whose effective domain is exclusively $\mathcal{T}_{12}$. Thus each equality determines a specific fiber, and each separation identifies the memory that must be restored. This proves interpretation (90.59). □

The theorem provides a finite model of the general principle of double projection: an image may preserve an exact signature while also identifying different states. The pair formed by the published signature and the omitted memory refines the quotient; recovery of the state requires reincorporating the cell, orientation, band section, or the pertinent coordinate in each case.

90.13.1.5 Accuracy of Conjugate Quotients and Obstructions to Memoryless Reconstruction

The construction begins with the finite atlas of 81 cells, its 56 route families, the seven section forms, and the involution C_M, all defined on the dodecaphase calendar of 108 steps. A state is the quadruple comprising cell, route, orientation, and band section; this quadruple precedes every physical designation and fixes the domain of the conjugate quotients.

On that domain, the index κ and the minimality of five states follow from the origin and order of the chart. The equalities in theorem 90.28 are exact equalities of signatures: in each, the preserved reader and the memory coordinate forgotten by the quotient are specified. The exhaustive traversal of the 81 cells independently reproduces the fibers, the values of ν_d, the quotients Q_k, and the diamond configuration; that finite enumeration complements the algebraic proof and permits its reproduction without altering its premises.

The conclusion is proved for these explicit primitives. Two extensions require additional maps: reconstructing the atlas from the minimal APP kernel and constructing, for the noncentral multisections, a fine physical section that preserves orientation and representation without consulting a target mass. Neither extension intervenes in the finite equalities already established.

The construction is incompatible with any of the following variations: a cardinality different from 81, an image different from 56 families, a fiber distribution different from (90.54), a failure in the reproduction of ν_d and Q_k, the injectivity of $\mathcal{S}_{\mathrm{int}}$, the three equalities of theorem 90.28, the count $49 + 32$ of the partial intertwining of ρ, or the functorial closure of the cobordisms of the supplementary graph. A physical interpretation further requires a representation map preserving these identities and a comparison independent of its construction.

90.14 The particle as a class of persistent sections

The material construction begins with the state produced by APP, TRIT, and TPK. APP supplies the positions and the two arithmetic operations; TRIT distinguishes the regime and orientation of their transitions; TPK transports this information together with the quotients, boundary incidences, and memory. As the state is prolonged, a finite evaluation remains linked to the evaluations preceding it. The notion of a particle rests on this compatibility and on its material representation. Its mass will be an observable of the section thus constructed.

This sequence determines the meaning of the three objects to be used: the genealogical state preserves the complete extension; the route records its observable history; the material representation makes that history act on a state space and permits the evaluation of observables. The distinction is essential even when the three objects share the same electronic label or the same atlas coordinate.

90.14.1 Extension, observable history, and compatible section

Let $\mathcal{S}_m$ be the admissible states surviving through depth m, obtained by TPK prolongation, and let $\rho_{n,m} : \mathcal{S}_n \to \mathcal{S}_m$, for $n \geq m$, be their restriction maps. Compatibility of refinement means

$$\rho_{m,m} = I, \qquad \rho_{n,m}\rho_{p,n} = \rho_{p,m} \quad (p \geq n \geq m). \tag{90.60}$$

A surviving extension is an element

$$x = (x_m)_{m\geq 0} \in \Omega^{\mathrm{surv}}_{\mathrm{HMT}} := \varprojlim_m \mathcal{S}_m, \qquad \rho_{n,m}(x_n) = x_m. \tag{90.61}$$

The existence and admissibility of these states arise from the construction of the formal kernel. The inverse-limit expression describes the compatibilities preserved by a state already generated.

Let $\mathcal{R}_m$ be the space of observable histories at depth m. The route reader $\mathrm{sh}_m : \mathcal{S}_m \to \mathcal{R}_m$ and the restriction $\mathrm{tr}_{n,m}$ satisfy

$$\mathrm{tr}_{n,m}\,\mathrm{sh}_n = \mathrm{sh}_m\,\rho_{n,m}. \tag{90.62}$$

Thus, the reader transmits the restrictions of the state to its histories. The route of x is the family $\gamma_x = (\gamma_x^{[m]})_m$, where $\gamma_x^{[m]} = \mathrm{sh}_m(x_m)$.

Proposition 90.29 (Compatibility of the published history)**.** *For every extension* (90.61)*, its finite histories satisfy* $\mathrm{tr}_{n,m}(\gamma_x^{[n]}) = \gamma_x^{[m]}$.

Proof. Applying (90.62) to the state x_n gives

$$\mathrm{tr}_{n,m}(\gamma_x^{[n]}) = \mathrm{sh}_m(\rho_{n,m}x_n) = \mathrm{sh}_m(x_m) = \gamma_x^{[m]}.$$

The second equality uses precisely the inverse-limit condition. □

The proposition establishes descent from the extension to the history. Its content does not include injectivity of the reader: two extensions may publish the same finite history and retain different memories. Therefore, when reusing a route, its source state is retained, instead of reconstructing it from a digit or a visible word.

Over each history $\gamma_x^{[m]}$, consider the discrete bundle $\mathcal{B}_{\gamma_x^{[m]}}$ of material states of the declared representation. Let $\mathcal{E}_m(x) = \Gamma(\mathcal{B}_{\gamma_x^{[m]}})$ be its spaces of sections, with restrictions $p_{n,m} : \mathcal{E}_n(x) \to \mathcal{E}_m(x)$ compatible with those of the state. A persistent section is a family

$$s = (s_m)_{m\geq m_0}, \qquad s_m \in \mathcal{E}_m(x), \qquad p_{n,m}s_n = s_m \quad (n \geq m \geq m_0). \tag{90.63}$$

The initial depth m_0 allows a state to be described from any cut beyond which its representation is defined. Prior information entering its observables remains in the record transported by the extension.

90.14.2 Identity under refinement and equivalence of representations

Fix the internal symmetry groups G_m acting on the material fibres and preserving the declared observables. Two persistent sections s and s' represent the same material identity when, from some level m_1 onward, there exists a family $g_m \in G_m$ such that

$$s'_m = g_m s_m, \qquad p_{n,m} g_n = g_m p_{n,m}, \tag{90.64}$$

and the observable characters of the genealogical record coincide. The admissible symmetries are fixed before forming the quotient; their action transports sheet, orientation, memory, and representation according to the corresponding laws.

Proposition 90.30 (Compatible equivalence of sections)**.** *The relation* (90.64)*, on families carrying the same types of representation and observables, is an equivalence relation. Its classes are independent of removing a finite number of initial levels, provided their transported genealogical information is preserved.*

Proof. The identity of each G_m gives reflexivity. The inverse of a compatible family is again compatible: from $p_{n,m} g_n = g_m p_{n,m}$ one deduces $p_{n,m} g_n^{-1} = g_m^{-1} p_{n,m}$. This gives symmetry. If $s' = gs$ and $s'' = hs'$, from the greater of the two initial levels we have $s''_m = h_m g_m s_m$ and $p_{n,m} h_n g_n = h_m g_m p_{n,m}$. Composition also preserves the observables and proves transitivity. All these relations are formulated on a common tail; changing the starting point within that tail preserves the class. The final assertion concerns a change of presentation, not erasure of state memory. □

Definition 90.31 (Persistent material particle)**.** An HMT–MD particle is a class $[s]_{\mathrm{pers}}$ of sections (90.63) under (90.64), accompanied by its extension and provenance record, admitting an irreducible material representation or a minimal invariant block and a stable spectral output under the specified coupling.

The definition gathers conditions at different levels. Prolongation and the record belong to the HMT state. Representation and coupling belong to its material realisation. Spectral stability is predicated of the operator and domain used by that coupling. This organisation permits identity, charge, spin, and mass to be studied together without turning one particular observable into the definition of all the others.

The nonadic return illustrates the distinction between identity and instantaneous state. For the phase reader g and memory record $\mathcal{B}$, the following can coexist:

$$g(k+9) = g(k), \qquad p_{k+9,k}(s_{k+9}) = s_k, \qquad \mathcal{B}_{k+9} \neq \mathcal{B}_k. \tag{90.65}$$

The first equality is periodic; the second expresses persistence; the third records advancement. The phase return belongs to the history of the same material identity, while the state preserves the fact that an additional circuit has been completed.

90.14.3 The species as a fibre: scope of the finite atlas

The atlas constructed by the APP–TRIT–TPK operations has support $C_{81} = \{1, \dots, 9\}^2$. Its four orientations per cell form the domain of 324 oriented routes. The register map $\mathcal{R}$ distinguishes 56 route families, while the internal classifier

$$q : C_{81} \longrightarrow \Sigma_{13} \tag{90.66}$$

brings together 13 family and level types. These censuses and their rules are developed in the body devoted to the atlas; here we specify which object they transmit to the definition of a particle.

For $y \in \Sigma_{13}$, the fibre $q^{-1}(y)$ preserves all its cells and records. Once the internal ordering of each fibre of $\mathcal{R}$ is fixed, let $k(c)$ be the rank of the cell within it. Its finite presentation is

$$\mathfrak{S}_y^{\mathrm{atlas}} = \{(\mathcal{R}(c), k(c)) : q(c) = y\}. \tag{90.67}$$

The rank is added to distinguish cells having the same family register; it does not select a mass. The pair $(\mathcal{R}(c), k(c))$ recovers the cell within the ordered chart because the second datum identifies its position in the fibre of the first.

If $\pi_{81} : \Omega^{\mathrm{surv}}_{\mathrm{HMT}} \to C_{81}$ is the cellular reader, the lift of the species is

$$\widetilde{\mathfrak{S}}_y = \{x \in \Omega^{\mathrm{surv}}_{\mathrm{HMT}} : q(\pi_{81}x) = y\}. \tag{90.68}$$

This preimage contains extensions, whereas (90.67) contains finite presentations. The construction and domain of the reader π_{81} form part of the passage between them. A particular particle is realised as a persistent section within the corresponding enriched multisection.

Theorem 90.32 (Central fixed point and equivariant selection)**.** *Let $\rho_c(r, s) = (10 - r, 10 - s)$ be cellular inversion. In the ρ_c-stable fibres of the classifier (90.66), with trivial action on the label y, an equivariant point selection can take values only in cell $(5, 5)$. The electronic fibre at level zero contains this unique fixed point; no other fibre admits such a selection.*

Proof. Solving $\rho_c(r, s) = (r, s)$ gives $r = s = 5$. If $a(y) \in q^{-1}(y)$ is an equivariant selection and y is fixed, then $a(y) = a(\rho_c y) = \rho_c a(y)$. Its value must be the fixed point $(5, 5)$. The atlas places that cell in the electronic fibre at level zero, whose cardinality is one. For any other fibre, the selection condition would produce a fixed point not belonging to it. □

The consequence is constructive: the other species are presented through orbits or multisections, and a realization requiring a particular line must declare its representation and coupling. The electron provides the central rank-one case without turning all the families into copies of that case. Moreover, the cell $(5, 5)$, the raw central signature, the regional signature 555555 and the relative origin of an action line retain their own domains. Uniqueness of the point does not identify these registers with one another.

90.14.4 Construction of the mass observable from the material register

The chain used by the mass observable preserves the order

$$\text{extension} \longrightarrow \text{route} \longrightarrow \text{record} \longrightarrow \text{integer signature} \longrightarrow \text{character} \longrightarrow \text{towers} \longrightarrow \text{material operator}. \tag{90.69}$$

Each arrow retains the data needed for the next. The signature is a reading of the register; the character evaluates that signature with the internal coefficients already constructed; the towers incorporate its refinement; the operator finally acts in the representation of the fibre selected by the coupling. Its output may be a scalar, a multiplet or a spectral distribution, depending on that representation. Metrological comparison takes place after this construction.

A species is described by charge, spin representation, colour, flavour, generation, route family and material operator. These data explain why two preparations with coincident mass may retain different information, and why the same identity may publish different observables under different couplings. Identity resides in the persistent class and its admissible transports; measurement is a reading of that identity in the declared codomain.

The same hierarchy separates a persistent extension from a section with a finite horizon. If $\mathrm{Ext}_L(s) \neq \varnothing$ and $\mathrm{Ext}_{L+1}(s) = \varnothing$, the horizon L records an obstruction to prolongation. An intermediate participation of this kind in a composition does not by itself create a new asymptotic species. Similarly, a resonance requires the survival operator and its pole, as well as the identity of the section it transports. The distinction between persistence, finite prolongation, and spectral evaluation is thus preserved.

90.15 Conjugation, orientation, and return of material sections

The persistent identity of Section 90.14 admits inversion operations at different levels of its structure. Cellular inversion acts on the atlas; dodecaphase conjugation acts on the orientation record; lifting a loop acts on a fibre; a spinor rotation acts on a representation space. Specifying these domains allows the operations to be composed and explains what information is preserved in each return.

90.15.1 Dodecaphase involution and parity of observables

Let $\mathcal{T}_{12} = \{-1, 0, 1\}^{12}$ be the ambient space of tritic records with twelve positions. The admissible TPK domain is the subset satisfying the route compatibilities already constructed. The operation on the ambient space is

$$C_M : \mathcal{T}_{12} \longrightarrow \mathcal{T}_{12}, \qquad (C_M\tau)_j = -\tau_{11-j}, \quad 0 \le j < 12. \tag{90.70}$$

Restriction to admissible records uses stability of that domain under conjugation. Reversed order and sign reversal are simultaneous components of the operation.

Proposition 90.33 (Linear and quadratic parity of the record)**.** *We have $C_M^2 = I$. If $a_j = a_{11-j}$ and $L_a(\tau) = \sum_j a_j\tau_j$, then $L_a(C_M\tau) = -L_a(\tau)$. If a quadratic form $Q_b(\tau) = \sum_{j,k} b_{jk}\tau_j\tau_k$ satisfies $b_{jk} = b_{11-j,11-k}$, then $Q_b(C_M\tau) = Q_b(\tau)$.*

Proof. At each position, $[C_M(C_M\tau)]_j = -(-\tau_{11-(11-j)}) = \tau_j$. For the linear functional, substituting $k = 11 - j$ in the sum gives $L_a(C_M\tau) = -\sum_k a_{11-k}\tau_k = -L_a(\tau)$. In the quadratic expression the two sign reversals cancel, and the substitution $(r, s) = (11 - j, 11 - k)$ recovers Q_b through the symmetry of its coefficients. □

Linear orientation and memory correlations therefore have different laws. Even a composition of these observables must preserve their parities: if L_a is odd, its exponential character satisfies

$$\exp(L_a(C_M\tau)) = \exp(L_a(\tau))^{-1}. \tag{90.71}$$

The equality expresses multiplicative inversion. Invariance of a complete material operator is studied on that operator, with its channels and representation, and is established by the corresponding covariance.

Cellular inversion in Theorem 90.32 is $\rho_c : C_{81} \to C_{81}$, whereas C_M has domain (90.70). Both are involutions, but comparing them requires the route record map and its transport law. This datum prevents conjugation of a history from being replaced by inversion of an isolated atlas coordinate.

90.15.2 Lifting to antisections and preservation of mass

Conjugation is applied to the particle through two compatible maps. The base map C_m^{b} transports the state x_m to its conjugate state $\bar{x}_m$; the lift $\widetilde{C}_m : \mathscr{E}_m(x) \to \mathscr{E}_m(\bar{x})$ transports its material sections. Let t_m be the reader of the dodecaphase register and $\bar{p}_{n,m}$ the restrictions in the conjugate family. Their conditions are preservation of admissibility, involutivity and compatibility:

$$\begin{aligned} &\rho_{n,m}C_n^{\mathrm{b}} = C_m^{\mathrm{b}}\rho_{n,m}, \quad t_m(C_m^{\mathrm{b}}x_m) = C_M t_m(x_m), \\ &\bar{p}_{n,m}\widetilde{C}_n = \widetilde{C}_m p_{n,m}, \quad \widetilde{C}_{m,\bar{x}}\widetilde{C}_{m,x} = I. \end{aligned} \tag{90.72}$$

In the first row, the reader publishes the orientation register of the history. The remaining transport of the state accompanies that publication. The second row gives the lift laws; the subscripts $x, \bar{x}$ make its two source fibres explicit. If s is persistent, its antisection is $\bar{s} = (\widetilde{C}_m s_m)_m$.

Proposition 90.34 (Persistence of the antisection)**.** *Conditions (90.72) transport persistent sections to persistent sections and restore the section upon conjugating twice. If they also transport the internal symmetries by $g_m \mapsto \widetilde{C}_m g_m \widetilde{C}_m^{-1}$, they induce an involution on the material classes of Definition 90.31.*

Proof. Compatibility gives $\bar{p}_{n,m}\bar{s}_n = \widetilde{C}_m p_{n,m} s_n = \widetilde{C}_m s_m = \bar{s}_m$. Involutivity restores s. For $s'_m = g_m s_m$, their antisections are related through $\widetilde{C}_m g_m \widetilde{C}_m^{-1}$. The restriction law in (90.72) preserves compatibility of this family, which belongs to the internal groups by the condition of the statement. The construction thus descends to the classes. □

The charge representation uses the odd part of the material record. If $\mathfrak{q}$ is the stated charge character, its law is $\mathfrak{q}(\bar{s}) = -\mathfrak{q}(s)$. The elementary charge entering the expression of that coordinate on a dimensional line is the internal output already constructed; the representation is subsequently applied to the state without using a target charge to select its record.

For mass, covariance of its operator is required. Let $\widehat{M}_s$ and $\widehat{M}_{\bar{s}}$ be the self-adjoint operators of the two realisations, and let C be their unitary or antiunitary intertwiner. Its domain is transported by $C\,\mathrm{Dom}(\widehat{M}_s) = \mathrm{Dom}(\widehat{M}_{\bar{s}})$. The operator condition is

$$C\widehat{M}_s C^{-1} = \widehat{M}_{\bar{s}}. \tag{90.73}$$

Proposition 90.35 (Spectral preservation under conjugation)**.** *Under* (90.73)*, transport of an eigensection with mass $m \in \mathbb{R}$ preserves that eigenvalue. The spectral measure is transported by $E_{\bar{s}}(B) = CE_s(B)C^{-1}$ for every real Borel set B. In particular, a scalar realisation preserves mass on passage to the antisection.*

Proof. If $\widehat{M}_s\psi = m\psi$, covariance gives $\widehat{M}_{\bar{s}}C\psi = C(m\psi) = mC\psi$; the last equality also holds in the antiunitary case because m is real. The real functional calculus of self-adjoint operators transports their spectral projectors and proves the second assertion. □

In a band record the same condition can be seen in elementary form. Given a real reading $E_0(\tau)$, let $E_\pm = (E_0 \pm E_0 \circ C_M)/2$ and let $\xi \in \{+1, -1\}$ be the sheet orientation. Then

$$E_{\mathrm{band}}(\tau, \xi) = E_+(\tau) + \xi E_-(\tau), \qquad E_{\mathrm{band}}(C_M\tau, -\xi) = E_{\mathrm{band}}(\tau, \xi). \tag{90.74}$$

The equality follows from the evenness of E_+ and oddness of E_-. It explains how band energy can be preserved by transporting record and sheet simultaneously; its identification with the mass of a particular realisation uses (90.73).

The electronic record $\tau_e = (+,+,+,-,-,-,+,+,+,-,-,-)$ satisfies $C_M\tau_e = \tau_e$. This equality concerns the twelve-position word. The pair (τ_e, ξ) preserves an additional orientation changed by conjugation. The distinction allows the unique electronic centre and the conjugate charge realisations to be retained together; a fixed word does not become a proof of neutrality.

90.15.3 Sign holonomy and the persistence band

Let ℓ be a return loop of the material reader of a surviving extension. Its circuit group is $\langle \ell \rangle \simeq \mathbb{Z}$. On this group, the sign character is specified by

$$\varepsilon : \langle \ell \rangle \longrightarrow \{+1, -1\}, \qquad \varepsilon(\ell) = -1. \tag{90.75}$$

The circuit group records how many times the loop is traversed; its type differs from the finite phase C_9. The interval bundle associated with the character is

$$\mathcal{M}_\ell = ([0,1] \times [-1,1])/\big((0,u) \sim (1,-u)\big). \tag{90.76}$$

Theorem 90.36 (Oriented return on the band)**.** *The bundle* (90.76) *is a Möbius band. Its transport over ℓ reverses the sign of the fibre and over ℓ^2 restores it. Thus a nonzero vector of that line requires two circuits for its oriented return. This preserves the distinction between return of orientation and return of the complete state.*

Proof. The endpoint relation in (90.76) is the linear map $u \mapsto -u$, with negative determinant. It gives the nonorientable real line over the circle and its corresponding interval band. Composing two transports produces $u \mapsto u$. The memory of the extension is not part of this sign quotient; its updating continues to obey TPK transport and (90.65). □

Parametrising one circuit of the base by 2π represents restoration of the cover by 4π. If the temporal reader assigns 108 steps to the relevant circuit, the oriented lift closes at 216 steps. These assignments require the stated loop and reader; they do not divide the history into two successive particles of 108 steps.

Proposition 90.37 (Decomposition of observables on the double cover)**.** *Let $p : \widetilde{X} \to X$ be a principal double cover, with involution ι. For $f \in C^0(\widetilde{X}, \mathbb{R})$,*

$$f^+ = \frac{f + f \circ \iota}{2}, \qquad f^- = \frac{f - f \circ \iota}{2}$$

give $f = f^+ + f^-$, with $f^+ \circ \iota = f^+$ and $f^- \circ \iota = -f^-$. The first part descends to X; the second represents a section of the associated sign line. On the Möbius line, every continuous global section has a zero.

Proof. The identities follow by substituting $\iota^2 = I$. The invariant part is constant on each fibre and defines a function on the quotient. The anti-invariant part satisfies the transition law of the sign representation. On the circle, a section of this line is represented by a continuous real function u on $[0, 1]$ with $u(1) = -u(0)$. If an endpoint is zero, a zero already exists; otherwise the endpoints have opposite signs, and the intermediate value theorem provides an interior zero. □

The decomposition distinguishes two classes of observables on the same cover. It represents odd orientation information and preserved even data. Applying it to a material record uses the lift (90.72); the sign character describes the transported line, not all the data of the particle.

90.15.4 Bidirectional lattice and channel conjugation

The integer angular register publishes two coordinates

$$W_+ = 6n_A + s_C, \qquad W_- = 6n_A - s_C, \qquad (n_A, s_C) \in \mathbb{Z}^2. \tag{90.77}$$

The coefficients n_A, s_C are internal route readings. The factor 6 belongs to the dodecaphase normalisation of that record.

Proposition 90.38 (Lattice index and interchange of readings)**.** *The image of* (90.77) *has index* 12 *in $\mathbb{Z}^2$ and Smith normal form* $\operatorname{diag}(1, 12)$. *It is characterised by $W_+ + W_- \equiv 0 \pmod{12}$. The operation $s_C \mapsto -s_C$ interchanges W_+ and W_-.*

Proof. The matrix of the map is $\left(\begin{smallmatrix} 6 & 1 \\ 6 & -1 \end{smallmatrix}\right)$. The greatest common divisor of its entries is 1, and its determinant is -12, giving the stated invariant factors. The inverse coordinates are $n_A = (W_+ + W_-)/12$ and $s_C = (W_+ - W_-)/2$. If the sum is divisible by 12, the two integers have the same parity and both expressions are integral. The condition is also necessary by summing (90.77). The interchange assertion is checked by substituting $-s_C$. □

In the real angular representation, the coordinates already generated, $a = A_{\mathrm{rad}}$ and $b = C^*_{\mathrm{rad}}$, produce the channels $q_\pm = \exp[-(a \pm b)]$. In the domain $a > |b|$, both belong to $(0, 1)$. Let $R^* = R$, $R^2 = I$ be the sheet grading and C a unitary interchange with $CRC^{-1} = -R$. For $T(b) = \exp[-(aI + bR)]$ we obtain

$$CT(b)C^{-1} = T(-b), \qquad CP_\pm C^{-1} = P_\mp, \quad P_\pm = (I \pm R)/2. \tag{90.78}$$

Indeed, conjugation transforms the exponent and each term of its convergent series. This proof transmits the interchange to the channels. The integer lattice and the real operator have distinct domains and publish the same interchange operation through their declared reading maps.

90.15.5 Spinor representation of the electronic block

The three-dimensional APP construction of the addition and multiplication projectors selects a principal electronic plane through the tritic mode $(1,-1,0)^3$. Its derivation and the evaluation of the projectors are included among the antecedents of the present article. In an orthonormal basis of that plane, their restrictions are

$$P=\begin{pmatrix}1&0\\0&0\end{pmatrix},\qquad Q=\begin{pmatrix}1/4&\sqrt{3}/4\\\sqrt{3}/4&3/4\end{pmatrix},\qquad S=2P-I,\quad J=\frac{4}{\sqrt{3}}[P,Q]. \tag{90.79}$$

This representation receives the two constructed projectors and preserves their orientation. Multiplication gives $S^*=S$, $J^*=-J$, $S^2=I$, $J^2=-I$, and $SJ=-JS$. Specifically, $S=\operatorname{diag}(1,-1)$ and $J=\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$.

Let $\mathcal{S}_e=E_e\otimes_{\mathbb{R}}\mathbb{C}$ be the complexification of the real plane. The scalar unit i belongs to the field of this representation; J remains the real endomorphism arising from the projectors.

Proposition 90.39 (Hermitian triple and representation of rotations)**.** *In $\mathcal{S}_e$, define*

$$\sigma_1=SJ,\qquad \sigma_2=-iJ,\qquad \sigma_3=S. \tag{90.80}$$

These operators are Hermitian and traceless, and satisfy

$$\sigma_a\sigma_b=\delta_{ab}I+i\sum_c\epsilon_{abc}\sigma_c,\qquad L_a=\sigma_a/2,\qquad [L_a,L_b]=i\sum_c\epsilon_{abc}L_c,\quad \sum_a L_a^2=\tfrac{3}{4}I. \tag{90.81}$$

They constitute the irreducible spin-$1/2$ representation on the electronic block.

Proof. The relations of S,J give $(SJ)^*=SJ$, $(-iJ)^*=-iJ$, and $\sigma_a^2=I$. The cyclic products are $(SJ)(-iJ)=iS$, $(-iJ)S=iSJ$, and $S(SJ)=J=i(-iJ)$; reversing the order reverses the sign. This proves the product identity and the commutators. In the basis of (90.79), the three matrices are

$$\begin{pmatrix}0&1\\1&0\end{pmatrix},\quad \begin{pmatrix}0&-i\\i&0\end{pmatrix},\quad \begin{pmatrix}1&0\\0&-1\end{pmatrix}.$$

Their trace is zero and, together with I, they generate $M_2(\mathbb{C})$. A subspace invariant under all three is invariant under that entire algebra and can only be 0 or $\mathcal{S}_e$. Finally, summing $L_a^2=I/4$ gives the value $3I/4$ of the quadratic operator. □

Recognition of this triple as the Pauli matrices follows (90.79). The norm and product $\langle X,Y\rangle=\frac{1}{2}\operatorname{tr}(XY)$ on the real space of traceless Hermitian matrices give its space of directions. For $n\in\mathbb{R}^3$, $|n|=1$, let

$$H_e(n)=\sum_a n_a\sigma_a,\qquad h_e(n)=\tfrac{1}{2}H_e(n),\qquad \Pi_\pm(n)=\tfrac{1}{2}(I\pm H_e(n)). \tag{90.82}$$

Proposition 90.40 (Directional decomposition and double return)**.** *The operators $\Pi_\pm(n)$ are rank-one orthogonal projectors and $h_e(n)\Pi_\pm(n)=\pm\frac{1}{2}\Pi_\pm(n)$. The rotation group around n is*

$$U_n(\theta)=e^{-i\theta H_e(n)/2}=\cos(\theta/2)I-i\sin(\theta/2)H_e(n),\qquad U_n(2\pi)=-I,\quad U_n(4\pi)=I. \tag{90.83}$$

Proof. The antisymmetric part of (90.81) vanishes when contracted with n_an_b, so $H_e(n)^2=I$. Idempotence, orthogonality, and the sum $\Pi_+ + \Pi_- = I$ follow. They are Hermitian with trace one, hence their

rank is one. Multiplication by $H_e(n)/2$ gives the eigenvalues. Separating even and odd powers of the exponential produces (90.83) and its group law. Angular normalisation is checked by

$$U_n(\theta)H_e(v)U_n(\theta)^* = H_e\big(v\cos\theta + (n\times v)\sin\theta + n(n\cdot v)(1-\cos\theta)\big),$$

which follows from multiplication using the product identity of the triple. The induced action on directions is a rotation through angle θ; the spinor lift carries its half-angle. □

If n is realised as the direction of propagation, $h_e(n)$ is dimensionless helicity. Reversal of that direction satisfies $H_e(-n) = -H_e(n)$ and $\Pi_\pm(-n) = \Pi_\mp(n)$. This operation changes the directional labels; charge conjugation transports the material record through (90.72). Chirality is defined through the relativistic grading appropriate to the representation and its projectors, which differ from (90.82).

The scalar electronic evaluation acts as $\varepsilon_e^{\beta} I$ on $\mathcal{S}_e$ and commutes with the triple and with $\Pi_\pm(n)$. Rotational information preserves the structure of that evaluation without introducing another factor into mass. The Möbius band, in turn, transports a real line around a loop, and nonadic holonomy transports phase and memory. The three returns can enter the same realisation; their domains and intertwiners determine the precise composition.

90.16 Classification of persistent sections and their internal observables

Classification begins with the support on which transport acts. Physical names are interpreted after its invariants have been constructed. This priority distinguishes a position, a history, and a species: the position belongs to the finite atlas; the history preserves the transitions producing it; the persistence type gathers histories compatible with the same classification. An evaluated mass is a subsequent reading of that object.

90.16.1 Parity, depth, and multisections

On the positional copy of APP, consider

$$I_9 = \{1,\dots,9\}, \qquad C = I_9 \times I_9.$$

The ordered pair preserves the two axes. Define the parity classifier

$$f(r,c) = \begin{cases} \ell, & r,c \text{ odd},\\ v, & r \text{ odd},\ c \text{ even},\\ d, & r \text{ even},\ c \text{ odd},\\ u, & r,c \text{ pairs}, \end{cases} \qquad g(r,c) = \max\{|r-5|,|c-5|\}.$$

The symbols of the first map initially name internal classes. The second measures combinatorial depth relative to the centre. Subsequent association with physical sectors must preserve these definitions.

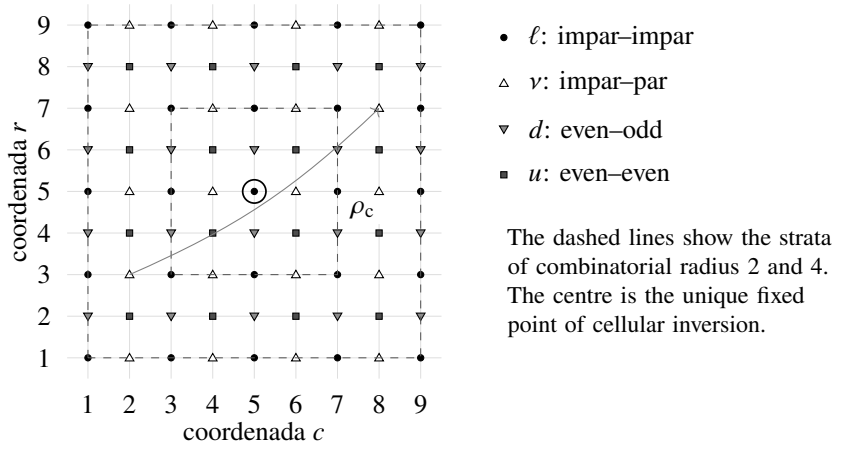

Figure 90.2: Classification on the positional atlas. The marks distinguish the four internal parity classes, preceding their physical realisations. Depth is measured by $g(r, c) = \max(|r - 5|, |c - 5|)$. The arrow illustrates $(2, 3) \mapsto (8, 7)$ under cellular inversion; it does not represent a dynamic trajectory or a change of flavour. The central circle marks $(5, 5)$, at depth zero.

Definition 90.41 (Family and depth multisection)**.** For every realised pair $y = (f, g)$, define

$$\mathfrak{S}_y = \{(r, c) \in C : (f(r, c), g(r, c)) = y\}.$$

Its lift to persistent states is the preimage under the projection onto the atlas, preserving the sheets, orientations, and memory of the extension.

Proposition 90.42 (Census of the multisections)**.** *The atlas contains* 81 *cells and exactly* 13 *nonempty multisections. Their distribution is*

class	*depths*	*cardinals*	*total*
ℓ	$0, 2, 4$	$1, 8, 16$	25
ν	$1, 2, 3, 4$	$2, 4, 6, 8$	20
d	$1, 2, 3, 4$	$2, 4, 6, 8$	20
u	$1, 3$	$4, 12$	16

Proof. In I_9 there are 5 odd and 4 even numbers. The four classes have cardinalities $25, 20, 20, 16$. For the odd–odd class, the centred squares of radii $0, 2, 4$ contain $1, 9, 25$ points; the differences are $1, 8, 16$. For odd–even, the cumulative rectangles at radii $1, 2, 3, 4$ contain $2, 6, 12, 20$ points, whose differences are $2, 4, 6, 8$. Interchanging the axes proves the even–odd case. In even–even there are 4 points through radius 1 and 16 through radius 3, so the strata have $4, 12$ points. The sets are disjoint, and their union is C. □

This census explains the meaning of each cardinality before any particle table. Depth g is not by itself identified with the index of a physical generation either: that identification uses the realisation maps of the corresponding sector.

90.16.2 Central conjugation and selection of a cell

Cellular inversion is

$$\rho_c(r, c) = (10 - r, 10 - c).$$

It preserves parity and depth and therefore acts within every multisection. Its unique fixed point satisfies $2r = 2c = 10$: it is $(5, 5)$. The remaining 80 cells are grouped into 40 pairs. There are thus 41 orbits, only one of which is a singleton.

Proposition 90.43 (Obstruction to noncentral point selection)**.** *If a cell selector is required to depend only on the family–depth pair and to be equivariant for ρ_{c}, it can exist only in the central multisection.*

Proof. Inversion acts trivially on the pair y. Equivariance of $a(y) \in \mathfrak{S}_y$ requires $a(y) = a(\rho_{\mathrm{c}} y) = \rho_{\mathrm{c}} a(y)$. Thus $a(y)$ would have to be the unique fixed point $(5, 5)$. In the central multisection, this is precisely the available choice. □

The consequence is structural. The treatment of the electron by means of a fiber of rank 1 does not authorize treating all the other families in this way. Outside the center, the orbit, the multisection and the orientations it preserves belong to the object being evaluated.

90.16.3 Classification of routes and positional memory

The chronological chart assigns to each cell the sextuple

$$\mathcal{R}(r, c) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta, \tau_{12})_{r,c}.$$

The five integer coordinates come from the oriented record; the last preserves the dodecaphasic word. Two cells belong to the same route family when these six readings coincide. This is a quotient different from the family–depth quotient.

In the documented finite census of this chart, 56 families appear. The size histogram is

$$41 \times 1, \quad 10 \times 2, \quad 2 \times 3, \quad 1 \times 4, \quad 2 \times 5.$$

The identities

$$41 + 10 + 2 + 1 + 2 = 56, \qquad 41 + 20 + 6 + 4 + 10 = 81$$

check the number of classes and coverage of the support. Reproducing the census requires the sextuples produced by transport, not merely these two sums. The data dossier preserves the 81 rows and the 56 complete classes.

Fixing the lexicographic order in each fibre, let κ_{pos} be the rank of the cell. Then

$$(r, c) \longmapsto \big(\mathcal{R}(r, c), \kappa_{\mathrm{pos}}(r, c)\big)$$

is injective. 5 positional-memory values are required because fibres of size 5 exist; the pigeonhole principle rules out a uniform injective lift with fewer values. This memory is neither colour nor the mass exponent, although all subsequently enter the same history.

90.16.4 Residual colour, flavour, and sheets

In the 36 cells with even first coordinate, the residual-colour observable is defined by

$$\mathrm{col}(r, c) = r - c \pmod 3.$$

For each even r, the 9 values of c traverse each class 3 times. This gives the exact distribution

$$36 = 12 + 12 + 12.$$

Central inversion transforms this observable into its opposite, since

$$(10 - r) - (10 - c) = -(r - c).$$

It fixes class 0 and interchanges 1 and 2. The colour representation subsequently uses this ternary support and its operators; the number of residues does not replace construction of the representation.

Flavour distinguishes classes and realisations of the sector, whereas residual colour distinguishes positions within the corresponding ternary support. The sheet, by contrast, identifies the arithmetic evaluation entering a transition. Sheet interchange acts on a history; it is not an operation that renames flavour. This distinction allows mixing, conjugation, and mass evaluations to be formulated separately.

90.16.5 Domain of orientations of the atlas

Each cell admits $(h, v) \in \{\pm 1\}^2$, so that

$$C^{\mathrm{or}} = C \times \{\pm 1\}^2, \qquad |C^{\mathrm{or}}| = 324.$$

The canonical orientation of 81 routes and the full domain of 324 are two distinct domains. A reader can descend to a family in the first chart and cease to be constant when the four orientations are considered. Therefore, each statement about the coincidence of readers identifies its domain before publishing the census.

The finite structure therefore combines positions, route families, multisections, conjugation orbits and residual observables. None of these partitions replaces the others. The following mass development acts on their persistent lifts, retaining the information that each evaluation needs.

90.17 Generated coordinates, action, and electronic normalisation

The construction of the families uses results from the preceding stages: the regional characters, the dodecaphasic record, the action line and the electronic section. Here we gather the operations connecting them and the proofs needed to compose them. Their order is causal: the regions and their records are produced on APP through TRIT and TPK; the real evaluations are performed afterward; the choice of units enters when realizing the dimensional quantities.

The initial conditions and regional selection have been developed in the section preceding the arithmetic of histories. The domain of the incidence ledger, its counts and the reconstruction of the record are explicitly defined below. Selecting a specific family of visits and traces is an operation additional to the evaluation of a ledger: it requires fixing its multiplicities, boundary transport and coverage of the admitted traces. The reconstruction formulas proved here allow that record to be evaluated and checked; their invertibility does not in itself select the incidence family.

90.17.1 Regional readers and compatibility of evaluation

Closure, propagation, and self-scaling are three regions of the same generation. The closure region has five emissions and multiplicity $432 + 4 \cdot 144 = 1008$. In the oriented gauge of the catalogue, its emissions are

$(501, 614, 498, 169, 272, 272)$	432
$(810, 923, 870, 169, 272, 272)$	144
$(810, 983, 810, 169, 272, 272)$	144
$(870, 923, 810, 169, 272, 272)$	144
$(870, 923, 810, 418, 674, 521)$	144.

The common ternary word is 010211; the representatives of propagation and self-scaling are 201101 and 121200. The census and orientation precede the evaluations that follow.

At the five positions of the closure reader, the uniform projection $P_5 = \mathbf{1}\mathbf{1}^{\mathsf{T}}/5$, together with $P_5\lambda = \lambda$ and $\mathbf{1}^{\mathsf{T}}\lambda = 1$, determines $\lambda = \mathbf{1}/5$. This normalisation of the reader positions differs from the seed distribution given in the table. In the elliptic regime $J^2 = -I$, multiplication gives

$$(I + aJ)(I + bJ) = (1 - ab)I + (a + b)J.$$

Whenever $1 - ab \neq 0$, the composite slope is $a \oplus b = (a + b)/(1 - ab)$. Two doublings of $1/5$ give $5/12$ and $120/119$. The positive compensator $1/q$, $q > 1$, that returns the slope to one satisfies

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1, \qquad q = 239.$$

Thus the closure character of this reader is

$$\Pi_{\mathrm{cl}} = 4\left(4\arctan\frac{1}{5} - \arctan\frac{1}{239}\right). \tag{90.84}$$

The composition rule, its compensation, and the choice of angular branch fix this value; the conventional angular identity recognises it as π.

The propagation character is realized in a commutative formal family P_t, normalized by $P_0 = 1$, $P_0' = 1$ and $P_{t+s} = P_t P_s$. If $P_t = \sum_{n\geq 0} a_n t^n$, comparison of the coefficient of $s^n t$ gives

$$(n+1)a_{n+1} = a_n, \qquad a_0 = 1.$$

Thus, $a_n = 1/n!$; publication at one unit of the parameter is $\Pi_{\mathrm{pr}} = \sum_{n\geq 0} 1/n!$, recognized as e. The unit and the sign of the generator belong to the reader: the semigroup law alone would admit other generators.

For self-scaling, TRIT selects the modes (Σ, Π, Π). Their three cyclic transitions are $\Sigma \to \Pi$, $\Pi \to \Pi$, and $\Pi \to \Sigma$. Their incidence matrix, in areal–volumetric order, is

$$F = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \tag{90.85}$$

The positive eigenvector normalised as $(1, x)^{\mathsf{T}}$ satisfies $x = \lambda$, $1 + x = \lambda x$; hence $x^2 - x - 1 = 0$. Its positive root is the self-scaling coordinate φ. This equation recognises the character of the incidence already produced by the calendar.

Proposition 90.44 (Regional evaluation through compatible intervals)**.** *The three preceding readers admit rational intervals of arbitrarily small width. Their finite decimal prefixes are determined by refinement and are compatible under truncation.*

Proof. For $0 < x < 1$, two consecutive alternating partial sums of $\sum_{j\geq 0}(-1)^j x^{2j+1}/(2j+1)$ enclose $\arctan x$. The first omitted term bounds the error. This provides rational intervals for (90.84). If $E_n = \sum_{j=0}^{n} 1/j!$, then, for $n \geq 1$,

$$E_n < \Pi_{\mathrm{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

The bound follows by bounding successive ratios of the tail by $1/(n+2)$. Finally, F^n has entries F_{n-1}, F_n, F_{n+1}, where $F_0 = 0, F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$. The identity $F_{n+1}^2 - F_n F_{n+2} = (-1)^n$ shows that two consecutive ratios enclose φ with width $1/(F_n F_{n+1})$. Successive intervals may be intersected to preserve nestedness. After conventional recognition, the irrationality of π, e, φ ensures that no coordinate lies on a rational decimal boundary. At each finite depth, a sufficiently narrow interval lies in a unique decimal cell. Its index determines the prefix, and nestedness proves compatibility. □

90.17.2 Signed dodecaphasic chart of the full TPK state

The evolution producing the record is formulated on the complete state of the Topological Phase Kernel. Let TPKFullState denote the state space that jointly preserves APP position, TRIT regime, route, sheet, orientation, carry, boundary, memory, incidence ledger and terminal coordinates. In a canonical chart,

$$X = (X_{\mathrm{APP}}, X_{\mathrm{TRIT}}, X_{\mathrm{ruta}}, X_{\mathrm{mem}}, C_{\mathrm{term}}, U_{12}^{\mathrm{sgn}}, X_{\partial}, \ldots).$$

The dynamics of one hundred and eight steps acts through the composition

$$X_{\mathrm{term}} = \big(\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1\big)(X_{\mathrm{seed}}), \qquad \mathcal{U}_t = \mathsf{Upd}_t \circ \mathsf{Tra}_t \circ \mathsf{Sel}_t, \tag{90.86}$$

where Sel_t applies the tritic phase selector, Tra_t transports sheet, route, and orientation, and Upd_t incorporates the carry and memory produced at instant t. Each factor therefore acts on the full state, and the composition preserves the genealogy of the twelve nonadic windows.

The incidence ledger of the following section is the coordinate $\mathcal{R}_{12}(X_{\mathrm{term}})$. Its evaluation by the 90/120 channels defines

$$\mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathsf{TPKFullState} \longrightarrow \mathbb{Z}^{25}, \qquad X \longmapsto C(X) = (b^{90}(X), b^{120}(X), Q(X)).$$

The signed dodecaphase projection is

$$\mathcal{S}_{12} := \mathrm{pr}_{U_{12}^{\mathrm{sgn}}} : \mathsf{TPKFullState} \longrightarrow \mathbb{Z}^{12}.$$

On the compatible sublattice produced by the dynamics, the two charts satisfy the operator identity

$$\mathcal{S}_{12} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} = \Pi_H \circ \mathrm{pr}_C\,. \tag{90.87}$$

Thus, the dodecaphasic signature is obtained by projection and evaluation of the same state that preserves the ledger; its domain includes the orientation and memory that enter each count. The raw projection $\rho_{108} : \mathsf{TPKFullState} \to \mathsf{CT108RawProjection}$ in contrast, designates the chart that forgets those coordinates after production. This separation of types fixes the causal order between generation, signed publication and reduced reading.

90.17.3 From temporal incidences to the dodecaphase record

Definition 90.45 (Domain of the oriented incidence ledger)**.** A ledger $\mathscr{L}$ contains a nonempty finite family of routes, their identified visits and a finite family of traces. Each route preserves one visit per instant during the first 108 instants, and the subsequent visits used by its traces. A visit preserves

$$v = (\gamma, t, i, j, s, \tau, w, \varepsilon_\partial),$$

where γ identifies the route, t its instant, $1 \le i, j \le 9$ its APP position, $s \in \{\Sigma, \Pi, 0\}$ its sheet, $\tau \in \{-1, 0, 1\}$ its tritic orientation and w its positive integer multiplicity. The incidence chart evaluates

$$n_v = \begin{cases} i + j, & s = \Sigma, \\ ij, & s = \Pi, \\ 9, & s = 0. \end{cases} \qquad r_v = 1 + ((n_v - 1) \bmod 9), \quad q_v = (n_v - r_v)/9.$$

The reading $n_v = 9$ fixes the threshold convention of this chart. For $r_v = 9$, the label ε_∂ distinguishes crown and interior; it is not obtained solely from the residue.

A trace $\theta = (v_1, \ldots, v_N)$ traverses consecutive instants of the same route. It preserves a multiplicity $u(\theta) \in \mathbb{N}_{>0}$ and a reduced length unit $\lambda(\theta) \in \mathbb{N}_{>0}$, with

$$N\lambda(\theta) = 120(1 + 3j(\theta)), \qquad j(\theta) \in \mathbb{N}_0.$$

Its channel is $\sigma(\theta) = \mathrm{sgn}(\sum_{a=1}^{N} \tau(v_a))$. The family counted in channels 120 comprises the traces of nonzero sign. The reduced length and the number of instants are related by λ; they are not identified with each other.

The definition specifies the domain of evaluation, including the data that must be transmitted to the reader. Its multiplicities and the family of traces belong to the dynamical selection of the ledger. An enumeration of the domain or a format check does not replace that selection.

Generation preserves a ledger $\mathscr{L}$ of visits and traces. The temporal chart divides it into windows $E_m = \{9m-8,\dots,9m\}$, $1 \le m \le 12$, equivalent to positions $m = \beta + 1$ of the record $\mathcal{R}_{12}$. Each window retains its subroute, orientation, boundary increment and local incidences. Each visit records APP position, arithmetic sheet, instant, multiplicity and boundary orientation. In each evaluation n_ν, the division $n_\nu = r_\nu + 9q_\nu$ is preserved, with $1 \le r_\nu \le 9$. In window m, the incidence reader publishes

$$\begin{aligned} A_m &= \sum_{\nu \in \mathscr{L}_m} w(\nu)\mathbf{1}_{\{1,3,5,7\}}(r_\nu), \\ C_m &= \sum_{j \ge 0} \big(\nu^-_{120,m}(j) - \nu^+_{120,m}(j)\big), \\ V_m &= \sum_{\nu \in \mathscr{L}_m} w(\nu)\mathbf{1}_{\{9\}}(r_\nu)\,\varepsilon_\partial(\nu). \end{aligned} \tag{90.88}$$

Here the traces in each window have finite support, $\varepsilon_\partial = +1$ on the corona and -1 in the interior, and the labels $\pm, 120$ belong to the preserved transport. The multiplicity w and the traces ν are those obtained from that transport; they are not reconstructed by selecting a value of α.

In terms of the identified traces, the second sum has the finite expression

$$C_m = -\sum_{\theta:\, t(\nu_1) \in E_m} u(\theta)\sigma(\theta).$$

Indeed, grouping by $j(\theta)$ and by sign produces $\nu^-_{120,m}(j) - \nu^+_{120,m}(j)$. This equality specifies the operation actually applied to each record; it keeps channel, multiplicity, and length separate.

Proposition 90.46 (Updating and concatenation of the ledger)**.** *Given the family of visits and traces admitted by transport, their counts are updated through integer increments. When adding a visit ν of window m, the increment of (A_m, C_m, V_m) is*

$$\big(w(\nu)\mathbf{1}_{\{1,3,5,7\}}(r_\nu),\ 0,\ w(\nu)\mathbf{1}_{\{9\}}(r_\nu)\varepsilon_\partial(\nu)\big)\,.$$

When a trace θ whose origin belongs to E_m is completed, the increment is $(0, -u(\theta)\sigma(\theta), 0)$. For two consecutive segments, the count of their concatenation is the sum of the counts of the visits and traces contained in each segment, plus that of the traces crossing their junction.

Proof. Each sum in (90.88) is indexed by visit or trace identifiers, so adding an identifier adds exactly the summand displayed. For concatenation, visits are distributed between the two segments. Traces are partitioned into those contained in the first, those contained in the second, and those crossing the junction. Additivity of integer sums proves the formula. Preserving the identifiers and pending prolongations prevents a trace from being counted twice or one that has not yet ended from being lost. Decimal reduction is applied after counting; the quotients of that reduction remain in memory. □

Decimal reduction of these three counts produces

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}, \qquad 0 \le z_m < 1000. \tag{90.89}$$

Let $(D_3 z)_i = z_i - z_{i+3}$, $(D_4 z)_i = z_i - z_{i+4}$, with indices in $\mathbb{Z}/12\mathbb{Z}$, and $Q(z) = \sum_i z_i$. The reader preserves $\mathcal{D}(z) = (D_3 z, D_4 z, Q(z))$. Decimal reduction forgets multiples of ten in each count, whereas the original ledger preserves the visits and transports that generated them. The reading of even-code events on that same history, its decomposition $12 = 1 + 5 + 6$, and its integral inversion are developed in Proposition 91.27. That event vector does not replace the present incidence record: the domains and preserved coordinates remain distinct.

Proposition 90.47 (Incidence conjugation and decimal reduction)**.** *Let $\varrho(m) = 13 - m$. Consider a conjugation of the ledger that maps the visits and traces of window m to those of $\varrho(m)$ through bijections preserving their multiplicities. Suppose it preserves the arithmetic class and boundary label of the visits, and interchanges the positive and negative channels of the traces. Then the counts in* (90.88) *satisfy*

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\varrho(m)}, -C_{\varrho(m)}, V_{\varrho(m)}). \tag{90.90}$$

Writing $u_m = [C_m]_{10}$, their decimal blocks satisfy

$$z_m^\dagger = z_{\varrho(m)} + 100\mathbf{1}_{\{u_{\varrho(m)} \neq 0\}} - 20u_{\varrho(m)}. \tag{90.91}$$

Proof. The bijections preserve every summand of A_m and V_m. Interchanging the channels permutes ν^- and ν^+, thereby reversing the sign of C_m. This yields (90.90). If $C = 10q + u$, $0 \le u < 10$, Euclidean division of $-C$ gives

$$[-C]_{10} = 10\mathbf{1}_{\{u \neq 0\}} - u, \qquad \left\lfloor \frac{-C}{10} \right\rfloor = -q - \mathbf{1}_{\{u \neq 0\}}.$$

In (90.89), only the middle digit changes. Its difference is $10([-C]_{10} - [C]_{10}) = 100\mathbf{1}_{\{u \neq 0\}} - 20u$, which proves (90.91). The second identity simultaneously preserves the integer quotient of the reduction. □

The proposition specifies the action on the ledger before transporting its decimal publication. Its application to the reversal of a route also preserves the convention used to read the endpoints. Let $\gamma = (x_0, \dots, x_n)$ be an enriched route and let f be an integer observable evaluated before each advance. The inverse route $\gamma^{-1} = (x_n, \dots, x_0)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0). \tag{90.92}$$

Indeed, the left-hand side sums indices $1, \dots, n$; the first term on the right sums $0, \dots, n-1$. Their difference is exactly the stated boundary term. The identity applies to each incidence observable with its preserved data. The endpoints are accounted for in the integer counts before reduction modulo ten; on a closed route the term vanishes, whereas in an open window it can change the block. Traces crossing a junction are preserved through the partition in Proposition 90.46.

The counts are thus connected to the route conjugation of section 90.15, keeping the concrete action on each incidence declared. Negation of a channel and reversal of a route are operations on the ledger; their decimal reduction is the publication just calculated.

Proposition 90.48 (Integral image and reconstruction of the record)**.** *Let $b, c \in \mathbb{Z}^{12}$, $q \in \mathbb{Z}$, with indices from zero to eleven. Define*

$$w_i = c_i - b_{i+1}, \qquad P_0 = 0, \qquad P_i = \sum_{j=0}^{i-1} w_j, \qquad T_w = \sum_{i=0}^{11} P_i.$$

There exists a unique $k \in \mathbb{Z}^{12}$ such that $D_3 k = b$, $D_4 k = c$, $Q(k) = q$, if and only if

$$b_i = w_i + w_{i+1} + w_{i+2}, \qquad \sum_i w_i = 0, \qquad q + T_w \equiv 0 \pmod{12}. \tag{90.93}$$

In that case,

$$k_i = \frac{q + T_w}{12} - P_i. \tag{90.94}$$

Proof. If k exists, then $c_i - b_{i+1} = k_i - k_{i+1}$. The cyclic sum of these differences is zero, their sum over length three is b_i, and $k_i = k_0 - P_i$. Summing the twelve coordinates gives $q = 12k_0 - T_w$, proving the

necessary conditions and the formula. Conversely, the conditions make the formula integral and cyclic. Its differences over length three are b_i, and those over length four are $w_i + b_{i+1} = c_i$. Its sum is q. The difference between two solutions has all consecutive differences zero and sum zero; it is therefore zero. □

Reconstruction is an operation on the typed image, not a retrospective selection of its arguments. In particular, $\mathcal{D}(z) = \mathcal{D}(z')$ implies $z = z'$. Since the expression of an integer of $[0, 999]$ in three digits is unique, two triples of counts publish the same record if and only if they coincide component by component modulo ten.

The signed change of representation uses the block

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad H_4^2 = 4I_4. \tag{90.95}$$

The operator H_{12} applies H_4 to the three ordered orbits $(1, 4, 7, 10)$, $(2, 5, 8, 11)$, $(3, 6, 9, 12)$. On the sublattice $\mathcal{L}_H = \{U \in \mathbb{Z}^{12} : H_{12}U \equiv 0 \pmod 4\}$, the maps

$$U = H_{12}K, \qquad K = \tfrac{1}{4}H_{12}U \tag{90.96}$$

are inverses. The congruence expresses exactly the integrality of the second map, and $H_{12}^2 = 4I$ proves both compositions.

The signed record is also calculated directly from the transverse coordinates before reconstructing K. With indices from zero to eleven, let, for $r = 0, 1, 2$,

$$S_r = \sum_{j=0}^{3} c_{r+3j}, \quad a_0 = \frac{q + 2S_0 + S_1}{3}, \quad a_1 = a_0 - S_0, \quad a_2 = a_1 - S_1, \qquad d_{r,j} = b_{r+3j}.$$

The harmonic map publishes by orbits

$$\Pi_H^{(r)}(b, c, q) = (a_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \tag{90.97}$$

To prove the formula on the integral image, let k be its preimage. If $s_r = \sum_j k_{r+3j}$, then $S_r = s_r - s_{r+1}$ and $q = s_0 + s_1 + s_2$, with orbit indices modulo three. Solving gives $s_r = a_r$. For the four coordinates (x_0, x_1, x_2, x_3) of an orbit, $d_j = x_j - x_{j+1}$. Their three combinations in (90.97) are respectively $x_0 + x_1 - x_2 - x_3$, $x_0 - x_1 + x_2 - x_3$, and $x_0 - x_1 - x_2 + x_3$. Together with $a_r = \sum_j x_j$, they are exactly $H_4(x_0, x_1, x_2, x_3)^{\mathsf{T}}$. The effective composition is thus $\mathscr{L} \to (b, c, q) \to \Pi_H(b, c, q) = U \to H_{12}U/4 = K$. The preimage used in this proof verifies the identity; it is not an argument of the map Π_H.

For the terminal state of (90.86), the coordinate produced before the signed publication is

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ Q_{\mathrm{TPK}} &= 6263. \end{aligned} \tag{90.98}$$

Substituting these twenty-five fields into (90.97), the three signed orbits are

$$\begin{aligned} U_{\mathrm{term}}^{(1)} &= (2378, -452, -668, -322), \\ U_{\mathrm{term}}^{(2)} &= (1406, 998, -204, -28), \\ U_{\mathrm{term}}^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns produces

$$U_{\mathrm{term}} = \mathcal{S}_{12}(X_{\mathrm{term}}) = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \tag{90.99}$$

The orbital transform $H_{12}/4$ is defined on this sublattice by the exact divisibility of every block of four. Consequently, the chain

$$X_{\mathrm{term}} \xrightarrow{\mathcal{R}_{12}} \mathscr{L}(X_{\mathrm{term}}) \xrightarrow{\mathcal{E}_{108}^{90,120}} C_{\mathrm{term}} \xrightarrow{\Pi_H} U_{\mathrm{term}} \xrightarrow{H_{12}/4} K \tag{90.100}$$

is a typed composition from the full state to its periodic seal. Propositions 90.46 and 90.48 provide, respectively, the finite update of the ledger and the integral inverse of the last publication.

The generated record provides the ordered witness

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \tag{90.101}$$

The leading zero in 058 forms part of the block reading. To check it through Proposition 90.48, its differences and sum are

$$\begin{aligned} b &= (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461), \\ c &= (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \\ q &= 6263. \end{aligned}$$

In the preceding three orbits, the signed record is

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Applying $H_4/4$ to these three vectors reproduces (90.101). These identities verify the publication of the incidences; their genesis remains that of the ledger (90.88).

90.17.4 Periodic publication of K and normalization of α

Fix $B = 1000$, the phase origin, and the twelve positions. Write

$$N_K = \sum_{j=1}^{12} K_j B^{12-j}, \qquad \kappa = \frac{N_K}{B^{12} - 1}.$$

This is the exact periodic evaluation of the record. Multiplication by $B^{12} - 1$ and successive Euclidean division recover its twelve blocks, including any leading zeros. For $\rho K = (K_2, \ldots, K_{12}, K_1)$, the same concatenation gives $\kappa(\rho K) = B\kappa(K) - K_1$. Thus reversibility of the record preserves a specified phase origin.

Proposition 90.49 (Minimum period of the rational publication)**.** *The periodic prolongation of the record* (90.101) *has minimal period* 12 *blocks in base* 1000 *and* 36 *digits in base* 10*.*

Proof. The twelve components of K are distinct. Any nonzero cyclic translation smaller than twelve would identify two components if it were a period. The minimal block period is therefore twelve.

Concatenating its three-digit blocks, including 058, produces a decimal period of length 36. This expansion is canonical: it is not eventually constant at the digit 9. Let d be its minimal decimal period. Then $d \mid 36$, because the shifts preserving a periodic word form a cyclic subgroup. Grouping the digits

in threes, the minimal block period is the least $k > 0$ such that $d \mid 3k$, namely $d/\gcd(d, 3)$. No proper divisor of 36 satisfies $d/\gcd(d, 3) = 12$. It follows that $d = 36$. □

This period belongs to the rational coordinate of the record, with the phase origin fixed. The following composition also preserves the regional coordinates and the normalisation quotients of its output.

Let P_j, E_j, Φ_j be the fractional blocks of the three regional characters already constructed, and extend K_j with period twelve. The composition of records is

$$Z_j = P_j + E_j - \Phi_j - K_j, \qquad A_j = Z_j + c_{j+1} - Bc_j. \tag{90.102}$$

The integer quotient c_j transports the continuation towards the observed position. In a finite window, c_{N+1} is fixed first, and then

$$c_j = \left\lfloor \frac{Z_j + c_{j+1}}{B} \right\rfloor, \qquad A_j = Z_j + c_{j+1} - Bc_j.$$

Euclidean division uniquely determines $0 \le A_j < B$. Since $-1998 \le Z_j \le 1998$, the set $\{-2, -1, 0, 1, 2\}$ is stable for these quotients.

Theorem 90.50 (Exact composition at unlimited depth)**.** *The complete evaluation of the normalisation, in the canonical decimal convention, is*

$$\alpha_A = \pi + e - \varphi - 4 - \kappa. \tag{90.103}$$

There is a unique canonical expansion (A_j), with blocks between zero and $B - 1$, and a bounded sequence of integer quotients satisfying (90.102) *and $c_1 = 0$.*

Proof. The regional integer parts are $3, 2, 1$. By absolute convergence,

$$y := \sum_{j \ge 1} Z_j B^{-j} = (\pi - 3) + (e - 2) - (\varphi - 1) - \kappa.$$

The first blocks are $141, 718, 618, 234$, so that $Z_1 = 7$. The absolute value of the tail sum is at most $1998 \sum_{j \ge 2} B^{-j} = 0.002$; in particular, $0 < y < 1$. Take the canonical expansion $y = \sum A_j B^{-j}$, choosing the terminating expansion rather than the eventually constant $B - 1$ expansion if a boundary occurs. Set

$$c_j = \sum_{r \ge 0} (Z_{j+r} - A_{j+r}) B^{-r-1}.$$

Equality of the complete values shows that each c_j is the negative of a finite sum of integers weighted by powers of B; it is therefore an integer, and $c_1 = 0$. Separating the first term gives $Bc_j = Z_j - A_j + c_{j+1}$. The tail bounds make the sequence bounded. More precisely, the tail of Z lies in $[-2, 2]$ and the canonical tail of A in $[0, 1)$, so $-3 < c_j \le 2$. Being integral, c_j belongs to the stated set $\{-2, -1, 0, 1, 2\}$. Conversely, summing the recurrence through N gives

$$\sum_{j=1}^{N} A_j B^{-j} = \sum_{j=1}^{N} Z_j B^{-j} - c_1 + c_{N+1} B^{-N}.$$

Boundedness makes the final term vanish in the limit. Uniqueness of the canonical expansion of y determines A, and the tail formula then determines every c_j. □

The finite evaluation $\kappa_{36} = N_K / B^{12}$ differs from κ by $B^{-12}\kappa$. Furthermore, imposing $c_{13} = 0$ in a twelve-block window does not replace the quotient of the complete continuation. In the published record,

the continuation gives $c_{13} = -1$: the twelfth block is 662, whereas the window cut with zero terminal quotient gives 663. The complete evaluation begins with

$$\alpha_A = 0.00729735256928380099728510547238066299 9\ldots.$$

The period of K does not impose a period on (A_j), since (90.102) also involves the three regional records and the integer quotients.

In the following formulas, α denotes this complete positional output α_A. The analytic vacancy route has its own operator; a polynomial truncated at order nine and the root of that polynomial are not identified here with the complete operator. The mass compositions that follow do not require that approximation to be promoted to an equality between the two infinite routes.

90.17.5 Three-dimensional construction of the electronic prefactor

Let $\mathcal{A}_3 = \{1, \ldots, 9\}^3$, with uniform measure, and $\rho_9(n) = 1 + ((n-1) \bmod 9)$. On this domain act

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \qquad \Pi(x) = \rho_9(x_1 x_2 x_3).$$

On $\ell^2(\mathcal{A}_3)$, let P and Q be the conditional averages over the fibres of Σ and Π, respectively. Each is idempotent and self-adjoint: within each fibre it replaces a vector by its constant component and preserves that component orthogonally. They are therefore orthogonal projectors.

The additive fibres have cardinality 81. The multiplicative fibres have cardinalities

$$m = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

The joint incidence $N_{ab} = \#\{x : \Sigma(x) = a, \Pi(x) = b\}$ is $9N_0$, where

$$N_0 = \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}. \tag{90.104}$$

This matrix is obtained by counting the 729 triples. It is also reproduced without rounding by $N_{ab} = \sum_{i,j,k=1}^{9} \mathbf{1}_{\{a\}}(\rho_9(i+j+k))\mathbf{1}_{\{b\}}(\rho_9(ijk))$. In the normalised bases of functions constant on each fibre, the intertwining has matrix $C_{ab} = N_{ab}/\sqrt{81 m_b}$.

Proposition 90.51 (Norm of the noncommutativity of the two sheets)**.** *The preceding projectors satisfy*

$$\|[P, Q]\| = \frac{\sqrt{3}}{4}. \tag{90.105}$$

Proof. On the space of the nine additive fibres, PQP has matrix $M = CC^{\mathsf{T}}$. Multiplication of the matrices in (90.104) gives eigenvalue one on the constant vector, $1/4$ on $(1, -1, 0, 1, -1, 0, 1, -1, 0)$, and $7/132$ on $(1, 1, -2, 1, 1, -2, 1, 1, -2)$. On coordinates $\{3, 6, 9\}$ with zero sum it acts by $1/18$, with multiplicity two. On each of the sets $\{1, 4, 7\}$ and $\{2, 5, 8\}$, the zero-sum vectors belong to the kernel; they give four dimensions. These orthogonal subspaces add up to the nine dimensions.

For a unit eigenvector $u \in \operatorname{ran} P$ of PQP with eigenvalue $0 < \lambda < 1$, the vector $v = (Qu - \lambda u)/\sqrt{\lambda(1-\lambda)}$ is a unit vector orthogonal to $\operatorname{ran} P$. In the basis (u, v),

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} \lambda & \sqrt{\lambda(1-\lambda)} \\ \sqrt{\lambda(1-\lambda)} & 1-\lambda \end{pmatrix}.$$

The commutator has norm $\sqrt{\lambda(1-\lambda)}$. The blocks corresponding to $\lambda = 0, 1$ commute. Among the calculated eigenvalues, the maximum is attained at $\lambda = 1/4$, since $\lambda(1-\lambda)$ is increasing on $[0, 1/2]$. Its square root is $\sqrt{3}/4$. □

In the extremal block, $S = 2P - I$ and $J = 4[P, Q]/\sqrt{3}$ satisfy

$$S^2 = I, \qquad J^2 = -I, \qquad SJ = -JS.$$

The four matrices I, S, J, SJ are linearly independent, so they realise $\mathrm{Cl}_{1,1} \simeq M_2(\mathbb{R})$. The elements $n_\pm = (S \pm J)/2$ satisfy $n_\pm^2 = 0$ and $n_+ n_- + n_- n_+ = I$, by direct expansion. The prefactor of the electronic equation thus arises from two evaluations of APP in dimension three and their commutator, before any reference mass.

90.17.6 Boundary reading and basal electronic composition

The incidence (90.85) has positive eigenvector $r = (1, \varphi)^{\mathsf{T}}$. The normalisation $p_{ij} = F_{ij} r_j / (\varphi r_i)$ produces

$$P_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \qquad \mu = \frac{(1, \varphi^2)}{1 + \varphi^2}.$$

The identity $\varphi^{-2} + \varphi^{-1} = 1$ proves that the rows sum to one. Since F is symmetric, weights proportional to r_i^2 satisfy detailed balance; hence $\mu P_{\mathrm{av}} = \mu$. The incidence is irreducible and has a self-loop, so this stationary distribution is unique. The chronological frequency $(1/3, 2/3)$ counts the three instants of the calendar; μ weights its admissible trajectories with memory one.

The operator preserving the sheets, fixing the volumetric normalisation at one, and marking the areal sheet by the closure character is $D_\pi = \mathrm{diag}(\pi, 1)$. By induction,

$$F^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The ratio of marked diagonal returns is

$$\Theta_n = \pi \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi}{\varphi^2}.$$

Convergence follows from the dominant eigenvalue φ, or from the consecutive quotients already bounded. The reader $\mathcal{R}(x, y) = x/y^2$, under interchange of arguments, also publishes

$$\rho_A = \frac{\pi}{\varphi^2}, \qquad \rho_E = \frac{\varphi}{\pi^2}, \qquad \varphi^3 \rho_A^2 \rho_E = 1. \tag{90.106}$$

The last equality is checked by multiplying the fractions. It relates two orientations of the same reader and does not select the remaining coefficients of the electronic equation.

The typed central return has length 27, with support $\mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3$. In the fixed regional gauge, the five microfibre positions are inserted as $(4, 0), (5, 0), (6, 0), (6, 1), (6, 2)$. The complement has $27 - 5 = 22$ positions. Its incidence with the three tritic phase states has $5 \cdot 3 = 15$ pairs. This yields the integers -22 and 15 used by the electronic reader; orientation fixes the sign of the first term. The cubic composition uses three copies of the closure coordinate, so its evaluation is α^3. The support, regional insertion, and this composition rule are distinct operational data; an isolated count does not replace their specification.

The global defect and its nonadic return are published through

$$d_4 = \frac{\pi - e\log\pi}{270}, \qquad \Delta_4 = d_4\left(1 + \frac{\pi}{729}\right). \tag{90.107}$$

The denominator 270 belongs to the corona without its closure unit; $729 = 9^3$ belongs to the cubic support. The function $f(x) = x - e\log x$ attains its minimum zero at $x = e$, since $f'(x) = 1 - e/x$ changes sign there. As $\pi \neq e$, we obtain $d_4 > 0$ and $\Delta_4 > 0$.

Definition 90.52 (Basal electronic coordinate)**.** The composition of the preceding readers is

$$\varepsilon_e^{(0)} = \frac{\sqrt{3}}{4}\left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha^3\right](1 + 15\Delta_4). \tag{90.108}$$

The exponential acts on the orientation ρ_E of (90.106); the vacancy reading corrects that component and the last factor preserves the return of the global defect. For $0 < \alpha < 0.01$, all factors are positive: $\exp(\varphi/\pi^2) > 1$, $22\alpha^3 < 22 \cdot 10^{-6} < 1$ and $\Delta_4 > 0$. This is a dimensionless coordinate. Its presence in a mass chart is defined after the scale is constructed.

90.17.7 Determinantal scale and the reduced Planck constant

The local block (90.95) has determinant -16. Its Smith normal form is $\mathrm{diag}(1, 2, 2, 4)$. Indeed, the greatest common divisor of the entries is one; the minors of order two are even, and one has absolute value two. By the adjugate identity, the minors of order three have absolute value four, and the determinant has absolute value sixteen. The determinantal divisors are $1, 2, 4, 16$, whose successive ratios give the four Smith factors. Consequently,

$$G_H = \operatorname{coker} H_4 \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \qquad |G_H| = 16. \tag{90.109}$$

The local normed reader multiplies the same coordinate α once for each class of this cokernel and applies the self-scaling coordinate once. Its definition produces

$$\Sigma_H = \varphi \prod_{g \in G_H} \alpha = \varphi\alpha^{16}. \tag{90.110}$$

The order of the cokernel determines the exponent of this reader. The Smith normal form and the choice of normed reader serve distinct functions: the former does not turn an arbitrary functional of the block into expression (90.110).

Proposition 90.53 (Decade of action)**.** *The output* (90.110) *satisfies* $10^{-34} < \Sigma_H < 10^{-33}$*; its decimal valuation is* -34*.*

Proof. The regional and positional evaluations give the intervals $1.618 < \varphi < 1.619$ and $0.007297 < \alpha < 0.007298$. By positivity,

$$\frac{1618\,7297^{16}}{10^{99}} < \Sigma_H < \frac{1619\,7298^{16}}{10^{99}}.$$

The integer inequalities $1618\,7297^{16} > 10^{65}$ and $1619\,7298^{16} < 10^{66}$ prove the result without using an external action constant. □

The decade and the mantissa are publications of different stages. Before constructing the latter, the regional reader selects the persistent return of the closure microfiber. The projection $p_{\mathrm{ret}}(u_1, \ldots, u_6) = (u_4, u_5, u_6)$ applied to the five emissions of 90.17.1 produces the block $b_\pi = (169, 272, 272)$ four times and $b_\pi^- = (418, 674, 521)$ once. These multiplicities count regional rows, not the seed multiplicities $432, 144$.

Proposition 90.54 (Equivariant selector of the regional return)**.** *Among rules preserving permutations of the five regions, selecting the terminal block of maximal multiplicity, equivariant under permutations of its three positions, and retaining a position fixed by the stabiliser of the block, the selected coordinate is unique and equals* $\rho_\pi = 169$.

Proof. Projection of the five rows shows that the block of maximal multiplicity is unique and is b_π. Its stabiliser in $\mathfrak{S}_3$ interchanges the two positions of value 272 and fixes only the position of value 169. Retaining that position is compatible with any simultaneous permutation of coordinates. Equivalently, the value is expressed without choosing an initial position as $\operatorname{sing}(b_\pi) = \sum_j (b_\pi)_j - 2\,\operatorname{moda}(b_\pi) = 169$. □

Grading by length publishes a word of length n through the character $\chi_z([w]) = z^{|w|}$. Concatenation adds lengths, so $\chi_z([uv]) = \chi_z([u])\chi_z([v])$. Since the regional emissions belong to $\mathcal{U}_6$, the return impulse appears as $\rho_\pi z^6$. Degree six corresponds to the length of the emission, not to the length of a closed walk that passes through several anchors of the graph.

The prolongation weight is obtained on the determinant line of bilayer transport. For a real involution R in dimension two, with $\operatorname{tr} R = 0$, let

$$x_A = \frac{50\pi}{9}\alpha, \qquad \mathcal{T}(x_A, y) = \exp(-x_A I_2 - yR).$$

The oriented coordinate y is still formal. In an eigenbasis of R, the two entries are $e^{-x_A - y}$ and $e^{-x_A + y}$. Therefore,

$$\bigwedge^2 \mathcal{T} = \det \mathcal{T} = e^{-2x_A} = D_A.$$

Dependence on y cancels. Multiplicativity under concatenation determines the prolongation character $k \mapsto D_A^k$: any character with elementary value D_A must have that value at $k = 1 + \cdots + 1$. This construction precedes evaluation of the conjugate angular coordinate.

The fifth-order regional reader uses eigenvalues $9, 5$ of the central block $\left(\begin{smallmatrix} 7 & 2 \\ 2 & 7 \end{smallmatrix}\right)$, the orbit of four positions, and the calendar of twelve windows. Its graded rule fixes

$$\begin{aligned} H_5 &= \frac{1}{2}\sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \\ D_A &= \exp\left(-\frac{100\pi\alpha}{9}\right), \\ C_\pi &= 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \\ \eta_{\mathrm{ret}} &= H_5 - \frac{90}{\pi}C_\pi. \end{aligned} \tag{90.111}$$

In that rule, the quotients are $9/16 = 9/4^2$, $5/9$, $7/48 = (\operatorname{tr} B_c/2)/(4 \cdot 12)$ and $1/54 = 1944/104976$. Their assignment to degrees and signs is part of the graded reader. The integer 169 is the publication of the selector proved in proposition 90.54. Its regional datum precedes the evaluation of α; the analytic reader then composes it with the closure coordinate.

The tail of C_π has a renewal construction. The impulse $\rho_\pi z^6$ and the strict continuation belong to different summands. After recording the impulse once, the continuation is normalised by the unit of the determinant line; each new step multiplies it by D_A. The orientation of the regional gauge places this return in the compensating sector subtracted from the fifth-degree character. With formal parameter z, the rule

$$R(z) = D_A z^7 + D_A z R(z) \tag{90.112}$$

uniquely determines $R(z) = D_A z^7/(1 - D_A z)$. This can be proved by comparing coefficients or by inverting the formal series $1 - D_A z$. Normalisation of the first return, in addition to the recurrent ratio, is essential to this uniqueness. Since $0 < D_A < 1$ and $\alpha < 1$, evaluation at $z = \alpha$ converges absolutely.

Proposition 90.55 (Positivity and the action return ratio)**.** *On the generated branch,* $0 < \eta_{\mathrm{ret}} < H_5$. *The ratio*

$$R_{\mathrm{act}} = \frac{H_5}{\eta_{\mathrm{ret}}} \tag{90.113}$$

is positive, greater than one, and dimensionless.

Proof. The bounds $0.007 < \alpha < 0.008$ and $\varphi < 1.619$ give $\frac{1}{2}\sqrt{1000\alpha/\varphi} > 1$. The sum of the absolute values of the remaining negative terms of H_5 is less than 0.008001, so $H_5 > 0.99$. Since $D_A < 1$, $\pi > 3$, and $\alpha < 0.008$,

$$0 < \frac{90}{\pi} C_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

It follows that $0 < \eta_{\mathrm{ret}} < H_5$. The ratio of two action coordinates in the same basis is dimensionless and does not change when that basis is rescaled. □

Let $\mathcal{U}_S$ be a basis of the action line. The two sections, before and after the return, are

$$\hbar_{\mathrm{pre}} = H_5 \, 10^{-34} \mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}} \, 10^{-34} \mathcal{U}_S. \tag{90.114}$$

Its physical reading is the proposed construction of the reduced Planck constant, with the return section made explicit. The unit $\mathcal{U}_S \mapsto \mathrm{J\,s}$ is a subsequent metrological realization. The coefficient of Σ_H within its decade is not H_5: the determinant publishes the scale and the regional reader publishes the coordinate occupying that scale. For each section, $h = 2\pi\hbar$ expresses the action of a complete turn.

90.17.8 Angular coordinates and the clock of the absolute realisation

The angular pair uses the dimensionless coordinates already generated:

$$A_{\mathrm{deg}} = 1000\alpha, \qquad C^*_{\mathrm{deg}} = 2(\eta_{\mathrm{ret}} + \alpha), \qquad x = \frac{\pi}{180} A_{\mathrm{deg}}, \quad y = \frac{\pi}{180} C^*_{\mathrm{deg}}. \tag{90.115}$$

The first operation shifts the base-thousand expansion by one block. The second is the bidirectional publication of the regional coordinate: equivalently, $y = \pi(H_5 + \alpha)/90 - C_\pi$. This equality follows by substituting (90.111); it does not introduce an additional rule for selecting the counterangle. The chart of one turn has 360 angular units or 2π radians. Thus, for a lifted coordinate,

$$\hbar\theta_{\mathrm{rad}} = h \frac{\theta_{\mathrm{deg}}}{360}.$$

The factor 10^{-34} remains on the action line and is not inserted into the angular coordinates.

The positive channels are

$$q_+ = e^{-x-y}, \qquad q_- = e^{-x+y}.$$

The pair is recovered through $x = -\frac{1}{2}\log(q_+ q_-)$, $y = \frac{1}{2}\log(q_-/q_+)$. Interchanging channels preserves x and changes the sign of y. The operator with oriented projectors $P_\pm$, $T = q_+ P_+ + q_- P_-$ preserves these labels when publishing its two eigenvalues.

For an internal time unit $t_0 > 0$, the closure of 108 steps determines

$$\omega_0 = \frac{2\pi}{108 t_0}, \qquad E_{\mathrm{clk}} = \hbar_{\mathrm{ret}} \omega_0, \qquad m_{\mathrm{clk}} = \frac{E_{\mathrm{clk}}}{c_{\mathrm{int}}^2}. \tag{90.116}$$

These formulas produce dimensions of frequency, energy and mass. Velocity and length are obtained through the response of the same channels, as the following identities specify. Expressing them in seconds, joules or mass units requires the corresponding charts; those charts do not alter the channels or the electronic section normalized relative to the clock.

90.17.8.1 Identity between constitutive velocity and clock velocity

In the domain $x > |y|$, the two channels satisfy $0 < q_\pm < 1$. On the two-sheet realisation, with rank-one projectors P_+, P_-, the temporal response is defined by

$$\mathsf{V} = (I - T^{90})(I - T^{120})^{-1} = r_+P_+ + r_-P_-, \qquad r_\pm = \frac{1 - q_\pm^{90}}{1 - q_\pm^{120}} > 0.$$

The inverse exists because both eigenvalues of T^{120} are less than one. The symmetric reading of the response is

$$\mathcal{E} = \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})] = \log(r_+r_-).$$

Let u_S, u_C, u_T be positive bases of action, velocity, and time, and $u_L = u_C u_T$ the length basis. The kinematic and constitutive readings are

$$c_{\mathrm{int}} = e^{-\mathcal{E}} u_C, \qquad c_{\mathrm{vac}} = (r_+r_-)^{-1} u_C.$$

Proposition 90.56 (Common realisation of propagation and absolute mass)**.** *In the same dimensional chart,*

$$c_{\mathrm{int}} = c_{\mathrm{vac}} = \widehat{c}\, u_C, \qquad \widehat{c} = (\det \mathsf{V})^{-1} = (r_+r_-)^{-1}.$$

For $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, the elementary length is

$$\ell_0 = c_{\mathrm{int}} t_0 = \tau_0 \widehat{c}\, u_L.$$

The chronological mass of (90.116) *therefore retains the form*

$$m_{\mathrm{clk}} = \frac{2\pi\, \eta_{\mathrm{ret}}\, 10^{-34}}{108\, \tau_0\, \widehat{c}^{\,2}}\, \frac{\mathcal{U}_S}{u_T u_C^2}.$$

Proof. The identity $\mathcal{E} = \log(r_+r_-)$ gives $e^{-\mathcal{E}} = (r_+r_-)^{-1}$; the rank-one property of each sheet gives $\det \mathsf{V} = r_+r_-$. Sheet interchange permutes the factors and preserves their product. Length is obtained by multiplying the same velocity by t_0. Substituting these expressions and (90.114) into (90.116) produces the mass formula. The factor $\mathcal{U}_S/(u_T u_C^2)$ has the dimension of mass; it does not enter the selection of the channels. □

For a homogeneous route of $n > 0$ transitions, the additive readings $T(\gamma) = nt_0$ and $L(\gamma) = n\ell_0$ give $L(\gamma)/T(\gamma) = c_{\mathrm{vac}}$. If the channel varies from edge to edge, their lengths are added; the ratio is then the mean velocity of those edges, not necessarily a common local velocity.

Equality of sections persists under changes of chart. If $u'_C = du_C$, $u'_T = \eta u_T$, with $d, \eta > 0$, then

$$\widehat{c}' = \widehat{c}/d, \qquad \tau'_0 = \tau_0/\eta, \qquad u'_L = d\eta u_L, \qquad \tau'_0 \widehat{c}'\, u'_L = \ell_0.$$

In particular, choosing the adapted basis $u'_C = c_{\mathrm{vac}}$ makes $\widehat{c}' = 1$, but the reader's coordinate in the internal chart remains $(r_+r_-)^{-1}$.

90.17.9 Two readers of the electronic trajectory and beta evaluation

The central trajectory starts at $(r, c) = (5, 5)$, with orientation $(h, v) = (1, 1)$. TRIT repeats the cycle $(+1, -1, 0)$. In the first regime, $r + c$ is read and then one advances $c \mapsto 1 + ((c - 1 + h) \bmod 9)$; in the second, rc is read and then one advances $r \mapsto 1 + ((r - 1 + v) \bmod 9)$; in the zero regime, 9 is read, without displacement. The class modulo three of the digital root of the reading is published. After steps $27, 54, 81$, h, v are simultaneously reversed. The publication of 108 steps preserves the complete trajectory and gives

$$a = 100000220, \quad b = 120200000, \quad \gamma_e = (a^3 b^3)^2, \quad \tau_e = (+, +, +, -, -, -, +, +, +, -, -, -). \tag{90.117}$$

The identity is checked by traversing the nine instants of each window; at the third block, inversion changes a to b, and the second traversal of 54 steps repeats the publication. The state also preserves its memory coordinates.

The first reader measures cyclic disagreements of the ternary publication:

$$D_k = \#\{t \in \mathbb{Z}/108\mathbb{Z} : \gamma_e(t) \neq \gamma_e(t + k)\}.$$

Since the published word has period 54, each disagreement appears with its translate by 54; all the D_k are even. In the half $a^3 b^3$, the disagreement count for $k = 1, 3, 4, 27, 36$ gives respectively $27, 28, 34, 24, 16$. Doubling it,

$$(D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32).$$

Therefore,

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right) = (80, 54, 6). \tag{90.118}$$

The second reader acts on the orientation of the twelve windows. Define, with cyclic indices,

$$C_k = \sum_{r=0}^{11} \tau_r \tau_{r+k}, \quad A_\ell = \sum_{r=0}^{11} \left(\sum_{j=0}^{\ell-1} \tau_{r+j} \right)^2, \quad P_6 = \sum_{r=0}^{5} \tau_r \tau_{r+6}.$$

The oriented primitive integer combination cancelling the diagonal part of A_3, A_4 is $\Omega = 4A_3 - 3A_4$: its coefficients satisfy $3u + 4v = 0$, $\gcd(u, v) = 1$, and $u > 0$ is fixed. Expanding the square proves

$$A_\ell = \ell C_0 + 2 \sum_{k=1}^{\ell-1} (\ell - k) C_k, \qquad \Omega = -2C_1 - 4C_2 - 6C_3.$$

For τ_e, we obtain $C_0 = 12, C_1 = 4, C_2 = -4, C_3 = -12, A_3 = 44, A_4 = 32, P_6 = 6$. Therefore,

$$K_\Omega = (\Omega, 54, P_6) = (80, 54, 6) = K_D. \tag{90.119}$$

The second component preserves the half-return of 108 steps; it is not introduced by diagonal cancellation. Equality of the two readers has been proved on the specified central trajectory; it does not extend by definition to all the families of the atlas.

Definition 90.57 (Electronic coordinate after the return)**.** The complete electronic composition is

$$\varepsilon_e^\beta = \varepsilon_e^{(0)} R_{\mathrm{act}}^{80 - 54\alpha + 6\Delta_4}. \tag{90.120}$$

The positivity of (90.108) and (90.113) allows this power to be defined through the real logarithm. The integer triple comes interchangeably from the two central readers just calculated. The relative evaluation

preserves the reduced Planck constant through the ratio of its sections, and the absolute realization additionally uses (90.116) and the declared dimensional chart.

In the inherited tables, $\mathfrak{e}_0$ names the basal coordinate $\varepsilon_e^{(0)}$ in the chart specified there. The section $\hbar_{\mathrm{int}}$ used for a survival evaluation must be declared in the same chart: for the post-return realization of the present development it is $\hbar_{\mathrm{ret}}$. The historical rows preserve the convention declared in their source table and are not recalculated through a silent change of this section.

If $\mathcal{U}_E$ is the energy basis of the electronic evaluation, $E_e = \varepsilon_e^{\beta}\mathcal{U}_E$. When the chronological quantum is used as the basis, the coefficient is

$$\widehat{\varepsilon}_e = \varepsilon_e^{\beta}\frac{\mathcal{U}_E}{E_{\mathrm{clk}}}, \qquad E_e = \widehat{\varepsilon}_e E_{\mathrm{clk}}.$$

The equality preserves the object and transforms its coordinate. Therefore the number presented in a MeV chart cannot be multiplied without conversion by any other energy quantum. The electron preserves its structural construction when expressed relative to a chronological scale: it is not replaced by the mass of that quantum.

The metrological comparison applies to the result of this composition, with the declared version of α, action section, reader Δ_4 and units chart. In particular, d_4 and Δ_4 are not interchangeable: replacing them simultaneously in the basal factor and the exponent changes the result by the positive ratio

$$\frac{1+15\Delta_4}{1+15d_4}R_{\mathrm{act}}^{6(\Delta_4-d_4)} > 1.$$

The preserved formulas allow that difference to be located without attributing it to an external mass or selecting the trajectory anew.

90.18 Mathematical and Physical Consistency Invariants of Helical Realizations

90.18.1 Mathematical Invariants of the Helical Prolongation

The theory is refuted in the corresponding domain if any of the following facts occurs:

1. one nonadic turn erases the depth in the lift;
2. the rule $g(K+9) = g(K)$ is confused with return of the state;
3. the reader radial no preserves factores and holonomias prefijas;
4. equation (18.56) fails;
5. the action of Γ_9 does not satisfy equivariance equation (18.58);
6. the composite carry fails to satisfy equation (8.15);
7. a classical curve enters into the selection of p, Λ, C;
8. the law equation (18.50) or the inversion equation (18.51) fails in its domain;
9. the tornillo of Pell no produces equation (46.8);
10. τ, p, or κ_{F} is identified without a typed map.

90.18.2 Constitutive Conditions of the Electromagnetic and Material Realization

An electromagnetic or material realization is refuted if:

1. fails to satisfy $\mathbf{E} \perp \mathbf{B} \perp \widehat{\mathbf{k}}$, $|E| = c|B|$, or $\omega = ck$;
2. interprets Pell dilation as energetic growth without source;
3. confuses photonic helicity with fermionic spin;
4. does not preserve $J^2 = -I$ and $\star^2 = -I$ under the declared intertwiner;
5. the spiral signature is not natural on the atlas multisections;
6. a physical label retrospectively selects a cell or a curve;
7. the axis φ of the genealogy of constants is identified with the material electron without a dimensional map.

90.18.3 Stratification of Internal Theorems and Their Physical Realizations

Table 90.1: Proof status of the results.

Result	Status	Scope
Mixed APP curvature, TRIT regimes, Γ_9, nonadic helix, and Pell screw	*previous internal theorem*	Exactly in the domains defined by the preceding constructions.
Affine envelope, reader by prefixes, naturality, equivariance and inversion	*theorem of this work*	Exact for truncations, typed routes, and reversible sector; sd_4 certified.
Family L_p and five publications	*theorem of this work*	Classify the power-logarithmic family, not all curves.
chart $p = \tau_+ + \tau_-$, $a = \tau_+ - \tau_-$	*exact internal formalization*	Elementary characters of the lifted category of 2 semicoronas; the global cocycles are obtained by route summation.
Circular electromagnetic mode	*physical realization*	Restricted Maxwell solution on the phase-position bundle.
Spiral particle signature	*conditional realization*	Proposal on multisections; it remains subject to the explicit square of constancy, separation, and material naturality.
Central electron $(5, 5)$	*previous internal theorem*	Unique material anchorage fixed by cellular inversion.

90.18.4 Theorem of closure of genealogical spiral correspondences

The theory obtains a unified response to the initial intuition. An orbit is a helical sheet when depth is preserved; the bidirectional involution produces conjugate sheets of ascent and descent; the nested circles are level sections; the cylinders are families of those sections. The classical spirals are not selected as templates. They appear upon publishing in $\mathbb{R}^2$ distinct radial characters of an APP–TRIT–TPK history.

The principal result is not a new list of curves. It is a structural separation between genealogy and appearance, accompanied by a reader that preserves exactly the information that appearance can forget. The curve is placed at the end. The real line, the real plane, and Frenet geometries are publications. The phase may return while the memory advances; the circle may be intrinsic or projected; the logarithmic spiral may be the exact shadow of a spectral screw; and the same mechanism may admit Archimedean, electromagnetic, and material realizations without conflating their codomains.

Functorial classification of the radial reader and stratification of its realizations

The APP–TRIT–TPK composition generates a class of radial histories with holonomy, orientation, carry, boundary, and memory. The enriched reader is natural under truncation and equivariant under nonadic holonomy. Its Archimedean publication produces power–logarithmic curves and helical suspensions, but it is not faithful: the same figure may preserve distinct genealogies. This is the intimate dynamical structure of space that the conventional classification of curves observes only after projection.

Chapter 91

Bidirectional readers K_D y K_Ω: profiles, coincidences, towers $90/120$ and operators by multisection

91.1 Readers K_D and K_Ω: dodecaphasic profiles, path domain and spectral codomain

For each oriented route, definitions are established.

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right), \qquad K_\Omega = (\Omega, 54, P_6).$$

The exact census of the 324 routes contains fourteen K_D profiles, nine K_Ω profiles, forty pairs, and eighteen coincidences, all with value $(80, 54, 6)$. These figures describe readers, not material species.

91.2 Semigroup and fiber operators

The global heights form

$$\langle 90, 120 \rangle = \{90, 120\} \cup \{30r : r \geq 6\},$$

with algebra $\Bbbk[X_{90}, Y_{120}]/(X_{90}^4 - Y_{120}^3)$. The convergent tower coefficients refine the character and define positive operators on the thirteen multisections.

91.3 Electronic construction, tower transducer and route clock

The central electron provides the rank-one case; the rest of the multisections requires the fiber operator, the coupling, and the spectral reading. The transducer separately preserves temporal length, memory, carry, and harmonic height.

91.4 Dodecaphase ledger and TPK memory $1 + 5 + 6$

The operational selector, the calendar, and the dodecaphase record fix the memory decomposition used by the mass readers. The identity $1 + 5 + 6$ belongs to the temporal state of the TPK and is not reconstructed from the spectral outputs.

91.4.1 Temporal dodecaphasic system of the TPK

91.4.1.1 Route selection operator and transport to the dimensional observable

The ternary operative phase function is

$$M_{\mathrm{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its three values are the scalar inputs with which Sel_t activates, respectively, the additive cursor, the multiplicative cursor or the neutral counter. Stroboscopic sampling is

$$\mathrm{Obs}_3(a) = (a_3, a_6, a_9, \ldots).$$

The selection $\mathrm{Sel}_t[\, M_{\mathrm{ph}}(g(t)) \,]$ decides which evaluation operates; cardinal transport moves the state; and Obs_3 selects which instants are published.

Cardinal transport acts through four translations.

$$T_d(i, j) = (i, j) + \nu_d \pmod 9,$$

$$\nu_N = (-1, 0), \qquad \nu_E = (0, 1), \qquad \nu_S = (1, 0), \qquad \nu_O = (0, -1).$$

The additive and multiplicative cursors preserve separate positions and orientations. The TRIT coordinate, the function M_{ph}, the operator Sel_t, the four translations and Obs_3 are therefore objects of different types.

91.4.1.2 Dodecaphasic Calendar and Register

The topological phase kernel does not terminate at the update of a position on $\mathbb{Z}_9^2$. Its minimal temporal state preserves the order of the events, the orientation, the return, and the memory of the transitions. For an orbit of 108 steps, the ordered partition is introduced

$$E_m = \{9(m-1)+1, \dots, 9m\}, \qquad m = 1, \dots, 12,$$

and its ordered support is denoted by

$$\Gamma_{108} = E_1 \sqcup \cdots \sqcup E_{12}, \qquad 108 = 12 \cdot 9.$$

Equivalently,

$$\Theta_{108} = \mathbb{Z}/108\mathbb{Z}, \qquad \beta(\theta) = \left[\left\lfloor \frac{\widetilde{\theta}}{9} \right\rfloor\right]_{12}, \qquad r(\theta) = 1 + (\theta \bmod 9).$$

The coordinate β selects one of twelve sectors, and r locates the phase within its nonadic window.

Proposition 91.1 (Non-splittable central extension of the observable clock)**.** *Writing $C_n = \mathbb{Z}/n\mathbb{Z}$, the observable calendar forms the exact sequence*

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \qquad \iota([b]_{12}) = [9b]_{108}, \qquad p([\theta]_{108}) = [\theta]_9.$$

For the set-theoretic section $s([r]_9) = [r]_{108}$, with $0 \le r < 9$, the gluing is given by the carry cocycle

$$c(r, r') = \left[\left\lfloor \frac{r + r'}{9} \right\rfloor\right]_{12}, \qquad s(r) + s(r') = s([r + r']_9) + \iota\big(c(r, r')\big).$$

The extension does not split. For the trivial action,

$$H^2(C_9, C_{12}) \simeq C_{12}/9C_{12} \simeq C_3,$$

and c represents the generating class [1].

Proof. The image of ι is the subgroup of multiples of nine in C_{108}, which coincides with $\ker p$; hence exactness. The cocycle identity is the Euclidean division of $r + r'$ by nine. If the central extension were split, the middle group would be isomorphic to $C_{12} \times C_9$. However,

$$\exp(C_{108}) = 108, \qquad \exp(C_{12} \times C_9) = \mathrm{mcm}(12, 9) = 36,$$

which rules it out. The cohomological identification for a cyclic group with trivial action gives $H^2(C_9, C_{12}) \simeq C_{12}/9C_{12}$, and the unit carry projects to the generator of that quotient. □

The nonsplitting types the return: nine iterations close the coordinate in C_9 and translate one unit in C_{12}; one hundred eight iterations close both coordinates of this clock. The calendar C_{108} is the temporal projection of the enriched state $\mathcal{X}^{\mathrm{enr}}_{\mathrm{TPK}}$: orientation, sheet, boundary, cylinder, genealogy, and deep memory remain in the latter and can distinguish histories with the same observable calendar.

If $g_0(\theta) = \theta \bmod 9$, then

$$g_0(\theta + 9) = g_0(\theta), \qquad \beta(\theta + 9) = \beta(\theta) + 1 \pmod{12}.$$

The witness

$$(g_0, r, \beta, 30^\circ\beta)(49) = (4, 5, 5, 150^\circ),$$
$$(g_0, r, \beta, 30^\circ\beta)(58) = (4, 5, 6, 180^\circ)$$

shows exactly what returns and what retains memory: the nonadic phase and the internal position coincide, whereas the dodecaphase sector advances by one. The regional gauge diag_c of the regions of π is not this coordinate β. Nor is there a morphism $30^\circ\beta \to h \in \langle 90, 120\rangle$ here; in particular, $150 \notin \langle 90, 120\rangle$. The pair $49/58$ is a carry witness, not a regional or spectral selector.

For an enriched history x of TPK, let τ_m be the ordered subroute over E_m, σ_m its orientation, κ_m the carry that crosses the block boundary, and Λ_m the accumulated state of the record. The object

$$\mathcal{R}_{12} = \big((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m)\big)_{m=1}^{12}$$

is the oriented dodecaphase temporal system of TPK. It is not a fourth theory added to APP, TRIT, and TPK: it is the internal temporal form by which TPK orders an orbit, preserves its returns, and publishes a reversible record. The canonical projections are distinguished

$$\mathcal{X}_{\mathrm{TPK}}^{\mathrm{enr}} \longrightarrow \mathcal{R}_{12} \longrightarrow \Theta_{108}.$$

The first preserves $(\tau_m, \sigma_m, \kappa_m, \Lambda_m)$; the second publishes only the observable sequence and cannot by itself reconstruct the signed state. The enriched record is denoted by $\mathcal{R}_{12}$, and the sector rotation by C_{12}.

The observable closure of the calendar satisfies

$$F_{\mathrm{obs}}^{108} = I.$$

Positions and orientations are already restored after 54 iterations, whereas $\mathcal{R}_{12}$ distinguishes the temporal position within the twelve sectors. Here F_{obs} is the observable permutation, whereas $\mathcal{G}_9$ is the pruning within the extension and boundary relation. Monodromy is the joint property

$$g(K + 9) = g(K), \qquad v_{K+9} \neq v_K,$$

not the isolated name of the pruning. Chronological closure returns the observable calendar; nonadic return preserves the phase and changes the prefix, cylinder, and memory.

Two words of twelve components are also distinguished:

$$U_{12}^{\mathrm{cat}} = U_6 \| U_6, \qquad U_{12}^{\mathrm{sgn}} = \mathcal{H}_{12}(K).$$

The former concatenates two emissions of length six. The latter is a signed coordinate of the enriched state and is reversibly linked to K by the dodecaphase transformation $\mathcal{H}_{12}$. Neither their domains nor their functions are interchangeable; in particular, the concatenated word does not by itself construct the signed coordinate.

Definition 91.2 (Complete record of the twelve events)**.** For a route of cifras $d_1, \ldots, d_{108}$, be

$$\mathcal{D}_{\mathrm{evt}} = \{2, 4, 6, 8\}, \qquad \Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

Its event-oriented word is

$$e_m = \Sigma_{12}(m)\, \#\{t \in E_{m+1} : d_t \in \mathcal{D}_{\mathrm{evt}}\}, \qquad 0 \le m < 12.$$

The two semicrowns are paired through

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}, \qquad 0 \le i < 6.$$

The raw count, the five differences relative to the sixth pair, and the six antisymmetries are defined by

$$s_C = \sum_{i=0}^{5} p_i, \qquad M_i = p_i - p_5 \quad (0 \le i < 5), \qquad A_6 = (a_0, \dots, a_5).$$

The full record is

$$\mathscr{L}_{12}(e) = \big(s_C, M_0, \dots, M_4, a_0, \dots, a_5\big).$$

Theorem 91.3 (Exact coordinates of the dodecaphasic memory)**.** *The map $e \mapsto \mathscr{L}_{12}(e)$ is injective on its image. Its inverse is given by*

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \qquad p_i = p_5 + M_i \quad (0 \le i < 5),$$

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2}.$$

In particular, the distribution of coordinates is

$$\boxed{1 + 5 + 6 = 12}.$$

Proof. By definition,

$$s_C = \sum_{i=0}^{4} (p_5 + M_i) + p_5 = 6p_5 + \sum_{i=0}^{4} M_i.$$

This determines p_5 and then all p_i. Adding and subtracting the equations $p_i = e_i + e_{i+6}$ and $a_i = e_i - e_{i+6}$ recovers the twelve events. Divisibility by six and the six parity conditions characterize the integral image; neither a mass nor a unit intervenes. □

Division by six is not applied during the count and is not a primitive coefficient. First the raw oriented count s_C is formed; then the record of twelve events is reconstructed; only then does $s_C/6$ appear in the character. Equivalently,

$$W_\pm = 6n_A \pm s_C$$

are integers and the lattice factorization has Smith normal form $(1, 12)$. This precedence will be mandatory in the mass law.

The integers 90 and 120 reappear later in six constructions of distinct types. To prevent the name of the cardinal from supplanting the object, the following convention is fixed from now on:

1. $\mathcal{E}_{\mathrm{dod}} = (D_3, D_4, Q) : \mathbb{Z}^{12} \to \mathbb{Z}^{25}$, integral extractor that reconstructs K;
2. $D_{\mathrm{raw}}^{90,120}(q)$, diagnostic difference of two series in a fixed chart;
3. $\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8$, combinatorial inclusion of cardinalities 90 and 120;
4. $K_{6\to 8} = 12\Delta S_{90} - \Delta S_{120}$, angular functional defined after fixing channels, series, and normalization;
5. $\mathfrak{S} = \langle 90, 120 \rangle$, infinite semigroup of spectral heights;
6. $\varepsilon^{\circ}_{\mathrm{res}}$, residue of an exact angular identity, not a finite extraction from any of the five preceding objects.

The partial coincidence of indices establishes neither equality of operators, domains, codomains, nor probative force. In particular, $K_{6\to 8}$ uses two subqueues of the hierarchy but is not the complete hierarchy; and neither of those subqueues forms a block independent of the mass law.

Remark 91.4 (Canonical Separation of Temporal Objects). The notation distinguishes the oriented record $\mathcal{R}_{12}$, the observable calendar Θ_{108}, the sectorial rotation C_{12}, and the transport orbit Γ_{108}, each with its domain, its codomain, and its law of composition.

91.4.2 Complete construction of the beta electron: central route, signature, character, and governing mass

91.4.2.1 Central route of the electron and sequence of genealogical records

The electron occupies the unique center $(5, 5)$, with active orientation ++:

$$\Gamma_e = (5, 5, ++), \qquad \tau_{12} = + + + - - - + + + - - - .$$

Their displacement readers are

$$(D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32).$$

They must not be merged:

1. the raw signature $(24, 0, 12, 0, -12)$;
2. the even CT–108 signature $(-12, -4, 12, -6, 12)$;
3. the vector of twelve events;
4. the affine origin $v_e = 0$ of the relative character.

Nor should Γ_e be confused with the word $w_e = 201101$ of the reader of Euler's constant e.

91.4.2.2 Basal electron selector and transition to the beta ruling value

The internal selector, constructed without using the observed mass, closes

$$-22 = -\,|((\mathbb{Z}/9\mathbb{Z}) \times \mathbb{F}_3) \setminus \iota(R_\pi)|\,, \qquad +15 = |R_\pi \times \mathbb{F}_3|.$$

The reading of vacancies also contributes

$$V_e = \begin{pmatrix} 21 & 22 \\ 6 & 7 \end{pmatrix}, \qquad \det V_e = 15.$$

The sign negativo of 22 records deficit/cut; the positive of 15, retencion/arean oriented. The matrix is local to the electron and does not extiende as selector universal.

The basal stage is

$$e_0 = \frac{\sqrt{3}}{4}\left[\exp\!\left(\frac{\varphi}{\pi^2}\right) - 22\alpha_{\mathrm{HMT}}^3\right](1 + 15\Delta_4),$$
$$e_0 = 0.510998922488\ldots \text{ MeV}.$$

The entire genealogies are

$$22 = \frac{396}{18} = 27 - 5 = 24 - 2, \qquad 15 = \frac{270}{18} = \binom{6}{2} = \binom{6}{4}.$$

91.4.2.3 Bidirectional readers and beta

For every route,

$$K_D(\Gamma) = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2}\right),$$
$$K_\Omega(\Gamma) = (\Omega_{90,120}, 54, P_6).$$

Across the 324 routes there appear 14 profiles K_D, 9 profiles K_Ω, 40 joint pairs, and 18 coincidences. The only common value is

$$(80, 54, 6).$$

This 80 comes from the bidirectional reader of the electronic route, not from period 80 of the Euler word. Therefore

$$\kappa_e = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4,$$

$$\boxed{\mathfrak{e}_e^{\beta} = e_0 R_{\mathrm{act}}^{\kappa_e} = 0.5109989506900008086082571648\ldots\ \mathrm{MeV}.}$$

This is the governing electronic output. The basal output does not replace it. Equality with the antiparticle mass proceeds from route conjugation and from the parity of the mass operator, not from adding a second numerical row.

91.4.3 Physical realization by spectral coupling

91.4.3.1 Obstruction to the equivariant punctual selection of a non-central multisection

Let $F \subset \mathcal{R}_{324}$ be a route fiber compatible with a structural class. Its linear space is

$$\mathcal{H}_F = \bigoplus_{\Gamma \in F} \mathbb{C} e_\Gamma.$$

The central character and the readers define diagonal operators:

$$\widehat{\mathcal{A}}_F^{(0)} e_\Gamma = \chi_{\mathrm{full}}(\sigma_\Gamma, \eta_\Gamma) e_\Gamma,$$
$$\widehat{K}_F^r e_\Gamma = \kappa_r(\Gamma) e_\Gamma, \qquad r \in \{D, \Omega\}.$$

The two–way correction is

$$\widehat{\mathcal{M}}_F^{\beta,r} = R_{\mathrm{act}}^{\widehat{K}_F^r/2} \widehat{\mathcal{M}}_F^{(0)} R_{\mathrm{act}}^{\widehat{K}_F^r/2}.$$

When $\dim \mathcal{H}_F > 1$, these operators generally have several eigenvalues. Choosing one because it approximates a known mass would reverse causality. Nor is it correct to require a physical name to identify a unique cell: the noncentral species resides in a representation of the fiber.

91.4.3.2 Spectral functor of the physical realization: positive coupling kernel, persistent subspace, and spectral measure

A physical preparation P provides representation data, not a mass:

$$\mathfrak{r}(P) = (q,\ j,\ \epsilon_{2\pi},\ \epsilon_{4\pi},\ \text{statistics, generation, flavor, color orbit, interaction channel}).$$

Let $\mathfrak{M}$ be a coupling compatible with that representation. A positive kernel is defined

$$K_{P,\mathfrak{M}} : F \times F \longrightarrow \mathbb{C}$$

which satisfies:

1. $K = K^*$ y $K \geq 0$;
2. $K(\Gamma, \Gamma') = 0$ if the routes violate charge, parity, colour, statistics, or the P channel;
3. K commutes with the symmetries that the coupling does not break;
4. $K(C_M\Gamma, C_M\Gamma') = \overline{K(\Gamma, \Gamma')}$ under conjugation;

5. its entries are computed from the genealogical ledger, boundary, memory, carry, and holonomy, without external magnitudes.

The coupled evolution operator is

$$\mathcal{U}_{P,\mathfrak{M}} = K_{P,\mathfrak{M}}^{1/2}\, \mathcal{U}_{\mathrm{TPK}}\, K_{P,\mathfrak{M}}^{1/2}.$$

The persistent subspace is obtained through the spectral projection onto the modes that survive return and pruning:

$$\Pi_{P,\mathfrak{M}}^{\mathrm{surv}} = \mathbf{1}_{\mathcal{S}_{\mathrm{surv}}}(\mathcal{U}_{P,\mathfrak{M}}).$$

The correct physical realization is the spectral functor.

$$\boxed{\mathcal{S}_{\mathrm{phys}}^{\mathrm{spec}}(P,\mathfrak{M}) = \left(\mathcal{H}_{P,\mathfrak{M}}^{\mathrm{surv}}, \widehat{\mathcal{M}}_{P,\mathfrak{M}}, \mu_{P,\mathfrak{M}}\right),}$$

where

$$\mathcal{H}_{P,\mathfrak{M}}^{\mathrm{surv}} = \Pi_{P,\mathfrak{M}}^{\mathrm{surv}} \mathcal{H}_F,$$
$$\widehat{\mathcal{M}}_{P,\mathfrak{M}} = \Pi_{P,\mathfrak{M}}^{\mathrm{surv}} \widehat{\mathcal{M}}_F \Pi_{P,\mathfrak{M}}^{\mathrm{surv}},$$

and $\mu_{P,\mathfrak{M}}$ is its spectral measure in the prepared state.

When the coupling is specified by an irreducible representation λ of a finite group G_P, the corresponding projector is obtained without choosing a distinguished route:

$$P_{P,\lambda} = \frac{\dim \lambda}{|G_P|} \sum_{g \in G_P} \overline{\chi_\lambda(g)}\, U(g).$$

The bidirectional output is then the compressed pair.

$$\mathbb{M}(P,\lambda) = \left(P_{P,\lambda}\widehat{\mathcal{M}}^{\beta,D}P_{P,\lambda}, P_{P,\lambda}\widehat{\mathcal{M}}^{\beta,\Omega}P_{P,\lambda}\right).$$

If both operators commute with the irreducible representation, Schur's lemma forces their scalarity on the block. If they do not commute, the correct result is the spectrum of the minimal invariant block.

91.4.3.3 Naturality of the Spectral Functor under Intertwiners of the TPK Dynamics, Coupling, and Mass Operator

If $V : F \to F'$ intertwines the TPK dynamics, the coupling kernel, and the mass operators, then

$$V\Pi_{P,\mathfrak{M}}^{\mathrm{surv}} = \Pi_{P',\mathfrak{M}'}^{\mathrm{surv}} V, \qquad V\widehat{\mathcal{M}}_{P,\mathfrak{M}} = \widehat{\mathcal{M}}_{P',\mathfrak{M}'} V.$$

By functional calculus, the spectral measure is transported by V. The output does not depend on a numbering of the cells or on the choice of a basis within an orbit. This is the condition that must replace every non-equivariant nominal selector.

91.4.3.4 Conditions for the existence of a scalar mass under the dimensional reader

A scalar mass is determined in any of the following cases:

1. the persistent subspace has rank one;
2. an irreducible representation fixes a unique spectral projector;
3. the measurement protocol defines a normalized positive state $\omega_{P,\mathfrak{M}}$ on the algebra of observables;
4. a resonance defines an isolated pole of the resolvent.

In the third case,

$$m(P, \mathfrak{M}) = \omega_{P,\mathfrak{M}}(\widehat{\mathcal{M}}_{P,\mathfrak{M}}).$$

In the absence of one of these mechanisms, the scientific output is the operator, its spectrum, and its multiplicities. That output is not incomplete: it is the object determined by the dynamics.

91.4.3.5 Finite construction of the mass kernel by means of representation, persistence, and coupling filters, followed by irreducible normalization

The kernel is obtained in four finite stages:

1. **Representation filter.** Retain the routes whose sector, charge branch, color, holonomy $2\pi/4\pi$, parity, and statistics match P.
2. **Persistence filter.** The partial order formed by class return, orbit return, cell return, prefix separation, carry debt, and decimal stability is used. The result may be a Pareto orbit; uniqueness is not imposed.
3. **Coupling filter.** Boundary incidence and the interaction channel are evaluated. Decoupled routes receive zero weight.
4. **Normalization.** On each irreducible component the trace of the kernel is normalized to one.

The oriented atlas provides all data of the first two stages. The third requires each physical realization to declare its channel; the fourth is canonical once the irreducible components have been fixed.

91.4.4 Genealogical Transducer of Towers

91.4.4.1 Pair of associated words for the same genealogical route

Each $\Gamma \in \mathcal{R}_{324}$ determines simultaneously a word temporal

$$w_\Gamma \in \mathbb{F}_3^{108}$$

and a dodecaphonic word signed

$$\tau_\Gamma \in \{-1, 0, +1\}^{12}.$$

The first preserves displacements along CT–108; the second preserves the accumulated orientation of the twelve events. The two words are linked by the same genealogical record, but they are not interchangeable. The refinement of towers reads both and also preserves the sheet, carry, winding, and return memory.

91.4.4.2 Correspondence between harmonic height and temporal length

Write

$$\mathfrak{S} = 30\langle 3, 4\rangle = \{90, 120\} \cup \{30r : r \geq 6\}.$$

For $h = 30r$, the associated temporal length is

$$\ell_h = 9r.$$

In particular,

$$90 \longleftrightarrow 27, \qquad 120 \longleftrightarrow 36, \qquad 360 \longleftrightarrow 108.$$

This correspondence does not identify the tower with time: it intertwines the height of the harmonic operator with the length of the route segment that generates it.

91.4.4.3 Temporal cycle shift reader of 108 states

We define the cyclic discrepancy

$$D_\Gamma(n) = \sum_{t=0}^{107} \mathbf{1}_{\{w_{\Gamma,t} \neq w_{\Gamma,t+n \pmod{108}}\}}.$$

The relative coefficient of the temporal branch is

$$\boxed{\eta^D_{30r}(\Gamma) = D_\Gamma(9r) - D_{\Gamma_e}(9r).}$$

Subtraction does not use a mass as an origin: it uses the central route as the affine origin of the relative character. It holds.

$$\eta^D_{90} + \eta^D_{120} = [D_\Gamma(27) + D_\Gamma(36)] - 80,$$

so that the first layer exactly prolongs the first component of the reader K_D.

91.4.4.4 Dodecaphase Accumulation Reader

For $r \geq 1$, let

$$A_\Gamma(r) = \sum_{j=0}^{11} \left(\sum_{u=0}^{r-1} \tau_{\Gamma, j+u \pmod{12}} \right)^2.$$

The relative coefficient of the oriented branch is

$$\boxed{\eta^\Omega_{30r}(\Gamma) = A_\Gamma(r) - A_{\Gamma_e}(r).}$$

Then

$$4\eta^\Omega_{90} - 3\eta^\Omega_{120} = \Omega(\tau_\Gamma) - 80,$$

so this layer extends the first component of K_Ω. The two families form the bidirectional lift

$$\mathcal{F}^{2w}_{\text{tower}} : \mathcal{R}_{324} \longrightarrow \mathcal{E}_{\mathfrak{S}} \times \mathcal{E}_{\mathfrak{S}}, \qquad \Gamma \longmapsto \big(\eta^D(\Gamma), \eta^\Omega(\Gamma)\big).$$

91.4.4.5 Finite determination of the infinite tower

The periodicity of the temporal word gives

$$D_\Gamma(9(r+12)) = D_\Gamma(9r).$$

If

$$T_\Gamma = \sum_{j=0}^{11} \tau_{\Gamma,j}, \qquad r = 12q + s, \quad 0 \leq s < 12,$$

then

$$A_\Gamma(12q + s) = A_\Gamma(s) + (12q^2 + 2qs)T_\Gamma^2.$$

Therefore, the entire tower is determined by twelve values of D_Γ, twelve values of A_Γ, and the integer T_Γ^2. The infinitude of heights does not require an infinitude of independent data.

The same formula solves the genealogical diamond of height 360. If

$$h(a,b) = 90a + 120b,$$

then

$$(a+4,b) \sim (a,b+3)$$

because $3(a+4)+4b = 3a+4(b+3)$. The genealogy (a,b) is conserved until memory and carry are applied; only afterwards does it descend to the common height.

91.4.4.6 Absolute convergence of the refined character on the towers 90/120

We have

$$|\eta_h^D| \le 108, \qquad |\eta_{30r}^\Omega| \le 24r^2.$$

Since $0 < q_+ < q_- < 1$,

$$|s_{30r}| \le 2q_-^{30r}.$$

It follows from this.

$$\sum_{h\in\mathfrak{S}} |\eta_h^D s_h| < \infty, \qquad \sum_{h\in\mathfrak{S}} |\eta_h^\Omega s_h| < \infty.$$

The refined character is well defined without truncating the tower.

91.4.4.7 Memory, carry, and cocycle

The bare coefficient is enriched by means of

$$\mathbb{F}_{\mathrm{tower}}(\Gamma)_h = \big(\eta_h^D, \eta_h^\Omega, \rho_h, \mathbf{w}_h, c_h, N_h^{\mathrm{surv}}\big),$$

where the truncated signature satisfies

$$\sigma_{\Gamma|\ell_h} = r(\rho_h) + \mathcal{D}\mathbf{w}_h, \qquad \mathcal{D} = \mathrm{diag}(2,6,2,2,4),$$

and the carry of two residues is

$$c(\rho_1,\rho_2) = \mathcal{D}^{-1}\big[r(\rho_1) + r(\rho_2) - r(\rho_1+\rho_2)\big].$$

The locality defect of a composition,

$$\delta_h(\Gamma,\Lambda) = F_h(\Gamma\star\Lambda) - F_h(\Gamma) - F_h(\Lambda),$$

obeys the cocycle identity

$$\delta_h(\Gamma,\Lambda) + \delta_h(\Gamma\star\Lambda,\Xi) = \delta_h(\Lambda,\Xi) + \delta_h(\Gamma,\Lambda\star\Xi).$$

Thus it is preserved why two paths with the same visible height can carry distinct stories.

91.4.4.8 Selection of the read by coupling

The two branches are not averaged for convenience. The physical coupling a must determine a natural transformation

$$\Pi_a : \mathcal{E}_{\mathfrak{S}} \times \mathcal{E}_{\mathfrak{S}} \longrightarrow \mathcal{E}_{\mathfrak{S}}$$

that commutes with truncations and symmetries, respects the carry cocycle, preserves the electronic origin, and is compatible with $X_{90}^4 = Y_{120}^3$. If the coupling exchanges the two readings, the only continuous, multiplicative, symmetric, and normalized closure on positive scalars is the geometric mean; in exponents,

$$L_{\mathrm{sym}} = \frac{L_D + L_\Omega}{2}.$$

In the absence of that symmetry, the output is the joint spectral measure of (L_D, L_Ω), not an arbitrary mean.

91.4.4.9 Orbit-primitive coinductive refinement

The preceding finite lift admits a prolongation. Let $\mathscr{X}_\Gamma$ be the graph of admissible extensions of the route, U_Γ its transfer operator, and R_Γ the orientation involution. The traces

$$a_n(\Gamma) = \mathrm{Tr}(R_\Gamma U_\Gamma^n)$$

and the Möbius inversion

$$p_n(\Gamma) = \frac{1}{n} \sum_{d|n} \mu(d) a_{n/d}(\Gamma)$$

separate primitive orbits. The difference

$$p_h(\Gamma) - p_h(\Gamma_e)$$

is a further refinement of the height coefficient h, not a replacement for the two finite readings.

91.4.4.10 Historical formulation via extension automaton

The signature $\mathbb{Z}^5$ does not exhaust the history of a route. For each $\Gamma \in \mathcal{R}_{324}$, the following are preserved:

- the 108 transitions of the genealogical register;
- the twenty blocks and their prefixes;
- the palabras w_6;
- the nine comparisons $K \leftrightarrow K + 9$;
- the leaf, the carry, and the winding;
- the first returns of class, orbit, cell, and kinematic state;
- the accumulated ambiguity and the stabilized decimal triads;
- the profiles K_D and K_Ω.

These data allow the construction of a tower transducer without opening the residual of a mass.

91.4.4.11 Local extension automaton

Let $\mathscr{X}_\Gamma$ be the finite set of local states compatible with the route. A state preserves

$$x = (t, g, B, p, \varepsilon, c, w),$$

where t is discrete time, g the nonadic phase, B the block, p the prefix, ε the sheet, c the carry and w the winding. An edge $x \to y$ exists when y is an admissible extension of x, respects the genealogical record and passes the pruning G_9. The table of 36.864 carry compositions provides the local addition law; the return tables fix the memory.

Let U_Γ be the normalized transfer operator of this graph and R_Γ the orientation involution. It is required that

$$U_{C_M\Gamma} = JU_\Gamma J^{-1}, \qquad R_{C_M\Gamma} = -JR_\Gamma J^{-1}.$$

91.4.4.12 Primitive cell oriented counts and construction of the electronic signature

For $n \geq 1$, the oriented trace

$$a_n(\Gamma) = \mathrm{Tr}(R_\Gamma U_\Gamma^n)$$

counts signed closed returns. The Möbius inversion separates primitive orbits:

$$p_n(\Gamma) = \frac{1}{n} \sum_{d|n} \mu(d) a_{n/d}(\Gamma).$$

The electron is taken as the affine origin of the relative corrections and is defined

$$\boxed{\eta_h(\Gamma) = p_h(\Gamma) - p_h(\Gamma_e), \qquad h \in \mathfrak{S}.}$$

Thus the map is obtained.

$$\mathcal{F}_{\mathrm{tower}} : \Gamma^{\mathrm{enr}} \longmapsto \eta(\Gamma) = (\eta_h(\Gamma))_{h\in\mathfrak{S}}.$$

There are no species-specific parameters. The same automaton, the same involution, and the same Möbius inversion act on the 324 routes.

91.4.4.13 Convergence of primitive refinement

Let $N_\Gamma = \dim \mathcal{X}_\Gamma$. If U_Γ is a contraction,

$$|a_n(\Gamma)| \leq N_\Gamma.$$

The number of divisors of n grows sublinearly, so

$$|p_n(\Gamma)| \leq \frac{N_\Gamma \tau(n)}{n}.$$

Since $0 < q_+ < q_- < 1$,

$$|s_h| \leq q_-^h + q_+^h \leq 2q_-^h.$$

By comparison,

$$\sum_{h\in\mathfrak{S}} |\eta_h(\Gamma) s_h| < \infty$$

uniformly on any finite collection of fibers. The refined character is thus well defined.

91.4.4.14 Electronic route conservation identities and uniqueness of the central fixed point

The construction must simultaneously satisfy:

1. invariance under renumbering of states;
2. covariance under leaf conjugation;
3. compatibility with direct sums of components;

4. equality upon recomputation from the genealogical register and from the certified graph;
5. conservation of the origin $\eta(\Gamma_e) = 0$;
6. absolute insensitivity to a permutation of the external mass columns;
7. determination of the tails by a bound established before comparison.

The exact representation of a given residual proves completeness of the codomain, but does not replace these seven proofs.

91.4.5 Route clock and absolute normalization

91.4.5.1 Integer returns are insufficient to distinguish species

In the atlas of 324 routes, the first cell return is 27, the kinematic return within CT–108 is 54, and the extended kinematic return is 216. Being common, these integers fix the global temporal architecture, but by themselves they cannot explain mass differences between species.

91.4.5.2 Temporal holonomy of the electronic route and memory return

The specific information resides in the holonomy of the extension operator. Let $\lambda_j(\Gamma)$ be a nontrivial eigenvalue of U_Γ on the persistent subspace. We write

$$\lambda_j(\Gamma) = r_j(\Gamma) e^{i\theta_j(\Gamma)}.$$

The associated internal frequency is

$$\omega_j(\Gamma) = \frac{\theta_j(\Gamma) + 2\pi w_j(\Gamma)}{108 t_0},$$

where the integer w_j is fixed by the recorded winding, not by a subsequent choice of branch. For a stable mode $r_j = 1$; for a resonance $r_j < 1$, and its attenuation contributes a width.

The ratio

$$\rho_{\omega,j}(\Gamma) = \frac{\omega_j(\Gamma)}{\omega_0}$$

is dimensionless and can modify mass ratios without altering the common decade -34. The energy and mass of the mode are

$$E_j(\Gamma) = \hbar_{\mathrm{ret}} \omega_j(\Gamma) \chi_{\mathrm{full}}(\Gamma) R_{\mathrm{act}}^{\kappa_j(\Gamma)},$$
$$m_j(\Gamma) = E_j(\Gamma) c_{\mathrm{int}}^{-2}.$$

This equation brings together action, clock, character, and reader without converting any of them into another. Its numerical application requires determining U_Γ, fixing the winding, and declaring the dimensional chart.

91.5 Surviving extension, dodecaphasic record, and integral signature

The trajectory that is observed does not constitute the primary object of this construction. The primary object is an extension compatible at all depths of HMT transport; a finite route is one of its readings. This distinction makes it possible to formulate unambiguously the internal chain leading from an enriched history to the action character. It also determines which information may be forgotten without altering the output and which must remain in the register.

The purpose of the section is to gather into a single argument four pieces that had been studied separately: the surviving extension, the intrinsic memory, the oriented record, and the integral signature. The terminal result of this section is the formal character

$$\Gamma_{108}^{\mathrm{enr}} \xrightarrow{L_{\mathrm{tick}}} \mathbb{Z}^5 \xrightarrow{\chi_{\mathrm{form}}} \mathrm{Fun}(\mathcal{P}, \mathbb{R}_{>0}),$$

prior both to specialization of the coefficients and to every physical unit. The first map extracts the signature from the register read; the second converts it into a positive function on the parameter space $\mathcal{P}$. Numerical specialization is performed only after the global coefficients have been constructed in the corresponding section. The metrological realization belongs to a still later dimensional chart. The map L_{tick} is the integral reader of iteration and return of the enriched chronological register.

91.5.1 Operational composition of the prolongation and torsional cocycle in the generation of the route record

The law of masses begins before the exponential formula. APP contributes a unique support of routes and its two sheets, sum and product. TRIT orients each elementary iteration. If

$$r_t = 1 + ((t-1) \bmod 9), \qquad m_t := M_{\mathrm{ph}}(r_t),$$

the phase selector M_{ph} activates

$$\mathrm{op}_t = \begin{cases} \Sigma, & m_t = +1, \\ \Pi, & m_t = -1, \\ \varnothing \quad \text{(retention or neutral counter)}, & m_t = 0. \end{cases}$$

The temporal reading may interpret the third case as threshold or present, but it does not convert it into a third arithmetic sheet. APP is not what alternates its sheets. TPK jointly transports position, orientation, sheet, phase, block, carry, prefix, cylinder, genealogy, and memory. The enriched chronological register records each transition —position, sheet, reading, carry, phase, block, and event—. The local boundary signature is subsequently derived through $\mathrm{Sig}_q(h, c, \lambda)$, and the global mass signature is extracted from the complete register; neither is reconstructed retrospectively from a mass.

The finite funnel feeding this story preserves four distinct types:

$$9^2 \cdot 9^2 \cdot 4 \cdot 4 = 104\,976 \longrightarrow 468 \text{ emissions } U_6 \longrightarrow 243 \text{ words } w_6 \hookrightarrow \mathbb{F}_3^6, \qquad |\mathbb{F}_3^6| = 729.$$

The first figure counts initial conditions of the two-cursor engine; 468 counts decimal emissions, and 243 visible ternary words. The 324 seeds of the chronological projection Π_{108}^{cal} of a single cursor likewise do not replace the 104 976 conditions of the complete state. Each arrow therefore changes both domain and operation. The numerical equality between the ambient space of 729 words and the 729 suffixes of a filtering stage is related only through the declared ternary encoding; it does not identify the two objects.

The elevation preserves as well its causal order:

$$w_6 \xrightarrow{L_0} w_{12} \xrightarrow{L_0} w_{18} \xrightarrow{L_1} w_{24} \xrightarrow{L_1} w_{30} \longrightarrow R_{36} \longrightarrow \mathcal{G}_9 \longrightarrow \text{surviving section.}$$

The boundary R_{36} opens the prolongation. At each filtration stage, $\mathcal{G}_9$ exhaustively ranges over the 729 subcylinders and retains the unique one compatible with every horizon; the deep specular episode $3 \to 1$ is another pruning and must not be confused with that ordinary partition $729 \to 1$. The nonadic return preserves the phase but changes the block, cylinder, and memory. Neither target figures nor masses intervene in this selection. At the dimensional boundary, a particle is an enriched persistent section of the extension; the route is its observable history. Its physical name is read thereafter and does not decide which branch survives.

Before opening the dimensional chronological ledger, the same genealogy retains a published window of twenty sextets per channel and admits the first Archimedean reading of twelve decimal triads. These two witnesses belong to the prolongation of the state: they are not the twelve events of the mass ledger, nor do they replace them. The events are constructed only subsequently, by reading the route on the oriented dodecaphase system.

The dodecaphasic genealogy subsequently converts the twelve oriented events into a record and only then into the signature

$$\gamma_x \longmapsto \mathcal{L}(\gamma_x), \qquad \sigma_{\mathrm{reg}} := L_{\mathrm{tick}}(\gamma_x) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta),$$

$$\sigma_{\mathrm{reg}} \longmapsto \chi_{\mathrm{form}}(\sigma_{\mathrm{reg}}; \cdot).$$

The public coordinates $b_{120} := \nu_{120}^{\mathrm{ev}}$ and $b_\zeta := \nu_{270}$ are outputs of the chronological ledger, not metrological corrections introduced at the end. The subsequent coefficients of the tower semigroup have a different provenance:

$$N_{120} = b_{120} + \eta_{120}, \qquad b_\zeta \zeta_{270} \not\equiv \eta_{270} s_{270},$$

with $\zeta_{270} = 135\Delta_4^2$ and $s_{270} = q_-^{270} - q_+^{270}$. The tower refinement is added after the central character without erasing the provenance of any coordinate.

Three lateral outputs also retain their type. The null sector leaves the positive branch through its null reader and null chart; it is not the limit $m \to 0$ of a positive character. A virtual trajectory is a finite extension with a terminal obstruction and preserves a chronological register and carry without thereby acquiring an imaginary mass. Finally, the Witt shadow accompanies every stage of filtration and preserves the exceptional genealogy but selects neither the route nor the mass coefficients.

91.5.2 Projection of a surviving extension onto its observable route

Let $(\mathcal{S}_m, \rho_{m+1,m})_{m\geq 0}$ be the system of surviving HMT states at depth m, with truncation maps $\rho_{m+1,m} : \mathcal{S}_{m+1} \to \mathcal{S}_m$. We define

$$\Omega_{\mathrm{HMT}}^{\mathrm{surv}} := \varprojlim_m \mathcal{S}_m = \left\{ (x_m)_m : \rho_{m+1,m}(x_{m+1}) = x_m \right\}.$$

An element $x \in \Omega_{\mathrm{HMT}}^{\mathrm{surv}}$ simultaneously preserves the compatibility of all its truncations. A reading chart of depth 108 produces

$$\mathrm{sh}_{108}(x) = \gamma_x \in \Gamma_{108}^{\mathrm{enr}}.$$

The notation sh recalls that γ_x is an observable shadow: it records a sequential presentation of the state, but the state is not defined by retrospectively choosing that route.

In addition to the visible word, the enriched presentation preserves phase, sheet, orientation, boundary, antisheet memory, and the predecessors needed to evaluate the cocycles. Two words with the same five final integers may be distinct presentations because those integers are computed after applying signs and correlations.

Definition 91.5 (Intrinsic memory and read record)**.** An enriched dodecaphase extension is a compatible class of truncations. We denote by $\mathscr{M}(x)$ the intrinsic memory that those truncations preserve. Once the chart $\gamma_x = \mathrm{sh}_{108}(x)$ is fixed, its read record is

$$\mathcal{L}(\gamma_x) = \big(n_A, e_0, \ldots, e_{11}, T_\zeta, C_\zeta\big),$$

where n_A is the count of the declared action support, the e_m are the twelve oriented events, and T_ζ, C_ζ are the total torsional memory and its restriction to the corona. Compatibility requires these data to be preserved by every truncation containing the steps used in their definition.

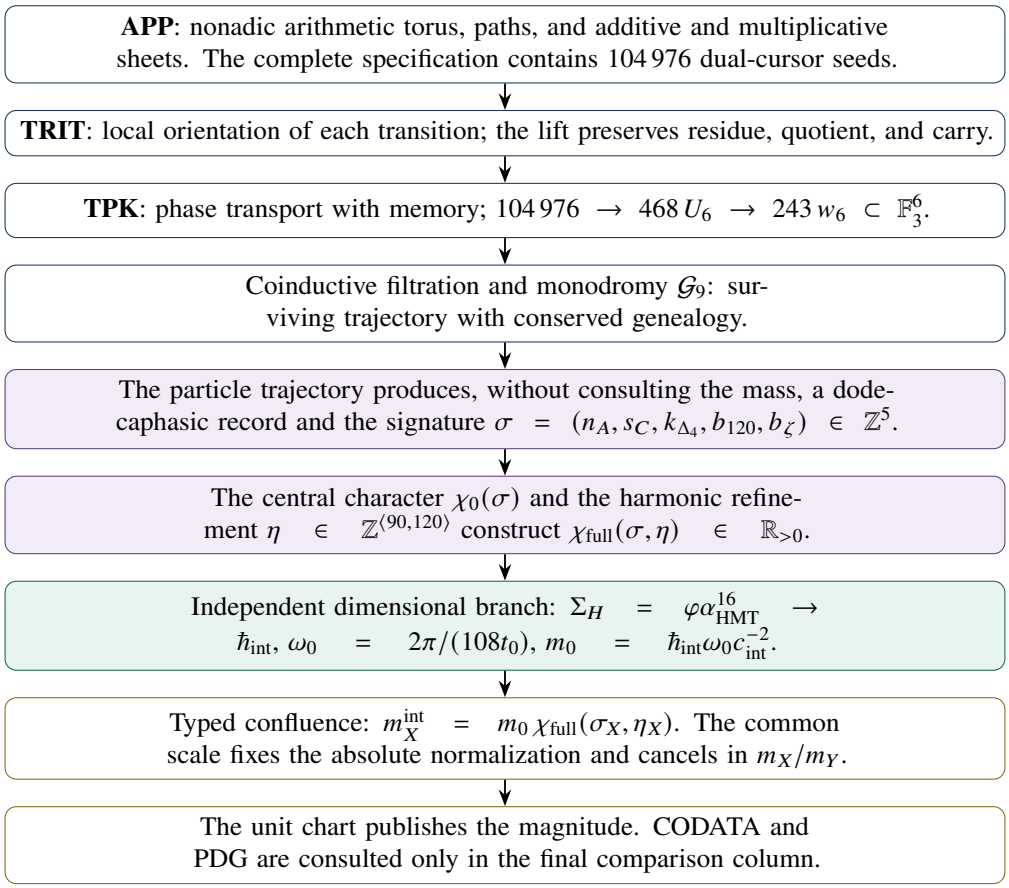

Figure 91.1: Derivation of a mass magnitude from an HMT trajectory. The character supplies a dimensionless ratio; the action scale and the clock supply the dimensional section. The external reference intervenes only after applying the chart of units.

The extension already contains the memory that the route presents; the route does not create it. Relative to the declared chart, the composition

$$x \longmapsto \gamma_x \longmapsto \mathcal{L}(\gamma_x) \longmapsto L_{\mathrm{tick}}(\gamma_x)$$

is a map. Invariance under another presentation chart would additionally require proving compatibility of the corresponding transition maps. In this precise sense, the register is not added to an already constructed particle: it is a reading of the extension's memory.

91.5.3 Integral Decomposition of 12 Events into the Sectors 1 + 5 + 6

Let $d_1, \dots, d_{108} \in \{1, \dots, 9\}$ be the word of an enriched presentation. In the parity convention used by the extractor, the support of action is

$$\mathcal{D}_{\mathrm{act}} = \{1, 3, 5, 7\}, \qquad n_A = \#\{t : d_t \in \mathcal{D}_{\mathrm{act}}\}.$$

The choice of this support is a declared part of the chart; it must not be confused with the TRIT classification of the steps. For even events one uses

$$\mathcal{D}_{\mathrm{evt}} = \{2, 4, 6, 8\}, \qquad \Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

The oriented event of block $m \in \{0, \dots, 11\}$ is

$$e_m = \Sigma_{12}(m)\#\{t \in \{9m + 1, \dots, 9m + 9\} : d_t \in \mathcal{D}_{\mathrm{evt}}\}.$$

Let us pair the two semicrowns by means of

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6}, \qquad 0 \le i < 6.$$

We define

$$s_C = \sum_{i=0}^{5} p_i, \qquad M_i = p_i - p_5 \quad (0 \le i < 5),$$

and write $M_5 = (M_0, \dots, M_4)$, $A_6 = (a_0, \dots, a_5)$.

The theorem 91.3, in the primary residence of the dodecaphasic TPK system, proves that

$$(e_0, \dots, e_{11}) \longmapsto (s_C, M_5, A_6)$$

is injective over its image and provides the inverse

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \qquad p_i = p_5 + M_i,$$

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2}.$$

Here only those coordinates are recovered to construct the character; the proof and identity $1 + 5 + 6 = 12$ are not duplicated.

The datum s_C is the one that enters the angular character. The eleven coordinates (M_5, A_6) preserve the placement of that total. Removing them before composing histories conflates different events that may produce different correlations.

91.5.4 Torsional cocycle at the step level

The event count does not by itself determine the torsional contribution. For each occurrence of nine, retain the digit preceding it and define

$$j_2(d) = \begin{cases} 0, & d = 9, \\ +1, & d \in \{2,4,6,8\}, \\ -1, & d \in \{1,3,5,7\}. \end{cases}$$

With cyclic indices in the word, let it be

$$\zeta_t = \mathbf{1}_{\{d_t=9\}} j_2(d_{t-1}),$$

$$T_\zeta = \sum_{t=1}^{108} \zeta_t, \qquad C_\zeta = \sum_{t \in \{27,54,81,108\}} \zeta_t.$$

The coordinate fuerte that acompana to Δ_4 is

$$\boxed{k_{\Delta_4} = T_\zeta - 2C_\zeta.}$$

The formula separates the accumulated drag from its oriented correction in the four corona closures. The exhaustive census shows that neither T_ζ nor C_ζ separately distinguishes all the census fibers; the ordered pair does so in the two typed conventions being compared.

The two remaining tower coordinates are calculated from the event log. Let

$$\tau_m = \operatorname{sgn}(e_m).$$

For the four step-four classes,

$$I_a = \{a, a+4, a+8\} \pmod{12}, \qquad 0 \le a < 4,$$

we define the event-weighted character

$$\nu_{120}^{\mathrm{ev}} = \sum_{a=0}^{3} \operatorname{sgn}\left(\sum_{m \in I_a} e_m \right),$$

and the three-step autocorrelation

$$\nu_{270} = \sum_{m=0}^{11} \tau_m \tau_{(m+3) \bmod 12}.$$

The variant

$$\nu_{120}^{\mathrm{sgn}} = \sum_{a=0}^{3} \operatorname{sgn}\left(\sum_{m \in I_a} \tau_m \right)$$

is another observable. They are not silently identified: they differ on 25 of the 81 routes of the atlas.

Definition 91.6 (Enhanced signature extractor)**.** Given the convention of events, it is defined

$$L_{\mathrm{tick}}(\gamma) = \left(n_A, s_C, k_{\Delta_4}, \nu_{120}^{\mathrm{ev}}, \nu_{270}\right) \in \mathbb{Z}^5.$$

The subscript indicates that the torsional coordinate is calculated before the predecessors of the null steps are forgotten.

This map is deterministic and consults neither masses, units, nor particle labels. It is not asserted to be linear with respect to an already projected concatenation: the signs and autocorrelations require the enriched records to be composed first.

91.6 Positive character of the signature and composition of records

91.6.1 Universal formal character

Let $\mathcal{P} = \mathbb{R}^5$, with coordinates

$$X = (X_A, X_C, X_4, X_{120}, X_{270}).$$

For $v = (n_A, s_C, k, \nu_{120}, \nu_{270}) \in \mathbb{Z}^5$, define

$$\ell_{\text{form}}(v; X) = n_A X_A + s_C X_C + k X_4 + \nu_{120} X_{120} + \nu_{270} X_{270},$$

$$\chi_{\text{form}}(v; X) = \exp\big(\ell_{\text{form}}(v; X)\big).$$

Thus, χ_{form} takes values in the multiplicative group $\mathrm{Fun}(\mathcal{P}, \mathbb{R}_{>0})$, with pointwise product. It has not yet selected global coefficients or a species.

Theorem 91.7 (Universal Character of the Signature)**.** *The map*

$$\chi_{\text{form}} : (\mathbb{Z}^5, +) \longrightarrow \big(\mathrm{Fun}(\mathcal{P}, \mathbb{R}_{>0}), \cdot\big)$$

is a homomorphism. The composition

$$\chi^{\text{form}}_{\text{route}} := \chi_{\text{form}} \circ L_{\text{tick}}$$

formally evaluates any enriched record and does not introduce a physical unit.

Proof. For every $X \in \mathcal{P}$, the exponent $\ell_{\text{form}}(\,\cdot\,; X)$ is additive on $\mathbb{Z}^5$. Therefore,

$$\chi_{\text{form}}(v + w; X) = \chi_{\text{form}}(v; X)\chi_{\text{form}}(w; X),$$

and the identity holds as equality of positive functions. □

The formal character precedes the construction of the global coefficients. This section does not yet define a numerical mass character or a unit chart.

91.6.2 Minimum internal section over the 81 cells

The finite atlas contains 81 oriented cells. When they are projected onto the chart of route family, 56 values are obtained. Denote this projection by

$$\mathcal{R} : C_9 \longrightarrow \mathcal{R}_{56}.$$

The exact histogram of its fibers is

$$41 \times 1, \qquad 10 \times 2, \qquad 2 \times 3, \qquad 1 \times 4, \qquad 2 \times 5,$$

and its sum is 81.

Having fixed the origin and the lexicographic order of the chart, let $\kappa(x) \in \{0, 1, 2, 3, 4\}$ be the rank of x within its fiber.

Theorem 91.8 (Minimal internal selector)**.** *The map*

$$x \longmapsto \big(\mathcal{R}(x), \kappa(x)\big)$$

is injective on the 81 cells. No memory alphabet with fewer than five symbols can lift $\mathcal{R}$ injectively over the entire atlas.

Proof. Within each fiber, κ assigns distinct ranks; between fibers, the first coordinate is already distinct. There is a fiber of size five, so the pigeonhole principle forces every injective lift to have at least five memory values. The range $0, \dots, 4$ attains that bound. □

Canonicity is relative to the declared origin and order. κ is neither a charge, a flavor, nor a mass. The thirteen coarse classes of the atlas produce finite multisections; only the central class possesses the fixed point $(5, 5)$ under central inversion. The other twelve require oriented information in order to select a point and do not admit an equivariant point selector from the coarse class alone.

91.6.2.1 Möbius Inversion and Round-Trip Characters

On the word of dodecaphasic signs we define

$$C_M(\tau)_m = -\tau_{11-m}.$$

This operation combines reversal of the reading order with the change of sheet. A linear character $\ell_a(\tau) = \sum_m a_m \tau_m$, with $a_m = a_{11-m}$, satisfies

$$\ell_a(C_M \tau) = -\ell_a(\tau),$$

while every symmetric quadratic form compatible with reversal satisfies

$$Q(C_M \tau) = Q(\tau).$$

Thus, the linear coordinates distinguish outward and return orientations, and the quadratic coordinates preserve the correlation content. This parity explains why s_C changes sign while ν_{270} may remain invariant. In the atlas, the seven word forms have multiplicities

$$54, 2, 2, 2, 7, 7, 7;$$

therefore, the anti-sheet is realized as a correspondence between fibers, not as a bijective involution obtained by ignoring their multiplicities.

91.6.3 Orientation-based composition of records before projection

Let Γ and Λ be two enriched histories, B a common subhistory, and g the transport identifying phase, sheet, orientation, and position at the interface. At ledger level, define

$$\boxed{\Gamma \star_B \Lambda = \big(n_\Gamma + n_\Lambda - n_B, e_\Gamma + g e_\Lambda - e_B\big).}$$

The definition also presupposes identification of the edge, the torsional predecessors, the corona, and the antisheet memory.

Proposition 91.9 (Obstruction to the five-dimensional descent of the composition of records)**.** *In general, there is no binary operation on $\mathbb{Z}^5$ that recovers $L_{\mathrm{tick}}(\Gamma \star_B \Lambda)$ from $L_{\mathrm{tick}}(\Gamma)$ and $L_{\mathrm{tick}}(\Lambda)$ alone.*

Proof. The operation composes the twelve events before applying sgn and the autocorrelations. Two histories may have the same five-tuple and different values of M_5 or A_6. When composed with a third

history, some sum $e_a + e_{a+4} + e_{a+8}$ may cross zero in one but not the other, changing ν_{120}^{ev}. Finite diamonds supply explicit witnesses. Therefore, projection onto five integers is not a congruence for $\star_B$. □

The proposition formalizes an essential operative rule: extensions are composed; signatures are read afterward. The composite particle is likewise not obtained by blindly adding two rows of a table, but by composing their compatible memories and evaluating the result.

91.6.4 Dodecaphasic dependence of the persistence character

The reduced period 54 engines impose

$$e_{m+6} = e_m.$$

It follows from here the congruences

$$n_A \equiv 0 \pmod 2, \qquad s_C \equiv 0 \pmod 2, \qquad \nu_{270} \equiv 0 \pmod 4.$$

The signatures of the abbreviated scalar reconstruction violate at least one of these congruences. Therefore, their genealogy requires the enriched register and does not arise from that reduced engine under the declared rules.

This result is not an obstruction to enriched TPK dynamics. It proves what memory it requires: a genuine dodecaphase, with possible $A_6 \neq 0$, oriented composition, and retained torsional predecessors. The ablation prevents a limitation belonging to a specific periodic reduction from being attributed to TPK.

91.6.5 Factorization by the quotient of log and physical realization

In the scope of this section, the relation is declared.

$$\gamma \sim_{\mathcal{L}} \gamma' \quad \Longleftrightarrow \quad \gamma, \gamma' \text{ have the same extension and } \mathcal{L}(\gamma) = \mathcal{L}(\gamma').$$

Denote by $\mathcal{E}_{108}^{\mathrm{surv}} := \Gamma_{108}^{\mathrm{enr}}/\!\sim_{\mathcal{L}}$ the quotient relative to this chart. Let

$$\pi_{\mathrm{ext}} : \Gamma_{108}^{\mathrm{enr}} \longrightarrow \mathcal{E}_{108}^{\mathrm{surv}}$$

the canonical projection. By construction, the extractor is constant on its fibers. The stronger assertion that every admissible chart of the same extension automatically produces the same record would require proving compatibility among charts and is not presupposed here.

Theorem 91.10 (Relative factorization by the quotient of the logarithm)**.** *There exists a unique application*

$$\overline{L} : \mathcal{E}_{108}^{\mathrm{surv}} \longrightarrow \mathbb{Z}^5$$

such that

$$\boxed{L_{\mathrm{tick}} = \overline{L} \circ \pi_{\mathrm{ext}}.}$$

Therefore, $\chi_{\mathrm{form}} \circ \overline{L}$ is a function of the class in the relative quotient, not of the route representative chosen to read it.

Proof. If $\pi_{\mathrm{ext}}(\gamma) = \pi_{\mathrm{ext}}(\gamma')$, both presentations have the same record by the declared relation and, by the preceding formulae, the same five-tuple. We define $\overline{L}([\gamma]) = L_{\mathrm{tick}}(\gamma)$. Constancy proves that the definition is independent of the representative; surjectivity of the map to the quotient proves uniqueness. □

The internal construction thus reaches up to

$$\Omega_{\mathrm{HMT}}^{\mathrm{surv}} \longrightarrow \mathcal{E}_{108}^{\mathrm{surv}} \xrightarrow{\overline{L}} \mathbb{Z}^5 \xrightarrow{\chi_{\mathrm{form}}} \mathrm{Fun}(\mathcal{P}, \mathbb{R}_{>0}).$$

The content of the theorem is positive and complete in this domain: a surviving extension determines its ledger class, the class determines a unique signature, and the signature determines the formal character. The global specialization and the complete hierarchy of towers are constructed in the following chapters; comparisons with physical names and units are presented subsequently, without altering this chain.

91.7 Split algebra and exact factorization of the character

The formal signature of an extension and the angular channels follow from branches already constructed. In this section, both converge in a determinant factorization that separately preserves the radial scale and the relative orientation. The result belongs to the algebraic stratum: it does not by itself identify the null ideals with physical photons or the positive interior with a material particle.

91.7.1 Conjugate angular coordinates and bidirectional spectral lattice

The preceding causal chain fixes

$$A_{\mathrm{rad}} = \frac{\pi A}{180}, \qquad C^*_{\mathrm{rad}} = \frac{\pi C^*}{180},$$

and, only after obtaining the direct coordinate A and the unique conjugate coordinate C^*,

$$q_+ = \exp[-(A_{\mathrm{rad}} + C^*_{\mathrm{rad}})], \qquad q_- = \exp[-(A_{\mathrm{rad}} - C^*_{\mathrm{rad}})].$$

For an angular signature (n_A, s_C), we introduce the two winding numbers

$$W_+ = 6n_A + s_C, \qquad W_- = 6n_A - s_C.$$

Theorem 91.11 (Angular lattice of index twelve)**.** *The map*

$$\mathcal{W} : \mathbb{Z}^2 \to \mathbb{Z}^2, \qquad \begin{pmatrix} n_A \\ s_C \end{pmatrix} \longmapsto \begin{pmatrix} W_+ \\ W_- \end{pmatrix} = \begin{pmatrix} 6 & 1 \\ 6 & -1 \end{pmatrix} \begin{pmatrix} n_A \\ s_C \end{pmatrix}$$

is injective, and its image has index twelve. Its invariant factors are $(1, 12)$*, and*

$$\operatorname{im} \mathcal{W} = \{(u, v) \in \mathbb{Z}^2 : u + v \equiv 0 \pmod{12}\}.$$

Moreover,

$$\boxed{\exp\left(n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}}\right) = \left(q_+^{-W_+} q_-^{-W_-}\right)^{1/12}.}$$

Proof. The matrix has determinant -12. Since the greatest common divisor of its entries is one, its Smith normal form is $\operatorname{diag}(1, 12)$. If $(u, v) = (6n + s, 6n - s)$, then $u + v = 12n$. Conversely, if $u + v$ is a multiple of twelve, then

$$n = \frac{u + v}{12}, \qquad s = \frac{u - v}{2}$$

are integers and recover (u, v). Finally,

$$\frac{W_+(A_{\mathrm{rad}} + C^*_{\mathrm{rad}}) + W_-(A_{\mathrm{rad}} - C^*_{\mathrm{rad}})}{12} = n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}},$$

which proves the identity upon exponentiation. □

The involution $s_C \mapsto -s_C$ interchanges $W_+ \leftrightarrow W_-$ and $q_+ \leftrightarrow q_-$. The twelfth root is the integral index of the image of $\mathcal{W}$, not a rounding. The coordinate C^* acts as the antisymmetric component of the lattice; it is neither a decimal correction to a mass nor an output of $\Gamma_{6\to 8}$.

91.7.2 Real split algebra

Definition 91.12 (Real split algebra)**.** Let

$$\mathcal{A}_{\mathrm{sp}} := \mathbb{R}[\mathsf{R}]/(\mathsf{R}^2 - 1).$$

Conjugation and the norm are defined, respectively, by

$$\overline{a + b\mathsf{R}} = a - b\mathsf{R}, \qquad N_{\mathrm{sp}}(z) = z\overline{z} = a^2 - b^2.$$

Conjugate idempotents

$$P_\pm = \frac{1 \pm \mathsf{R}}{2}$$

satisfy

$$P_\pm^2 = P_\pm, \qquad P_+P_- = 0, \qquad P_+ + P_- = 1, \qquad \overline{P_+} = P_-.$$

Proposition 91.13 (Split spectral decomposition)**.** *Every $z \in \mathcal{A}_{\mathrm{sp}}$ admits a unique decomposition*

$$z = r_+P_+ + r_-P_-, \qquad r_\pm \in \mathbb{R},$$

and its norm is

$$\boxed{N_{\mathrm{sp}}(z) = r_+r_-.}$$

Each principal ideal $\mathcal{A}_{\mathrm{sp}}P_\pm = \mathbb{R}P_\pm$ is a null line; in particular,

$$N_{\mathrm{sp}}(\lambda P_\pm) = 0.$$

Proof. If $z = a + b\mathsf{R}$, the choice $r_+ = a + b$, $r_- = a - b$ provides the decomposition, whose uniqueness follows from the independence of P_+ and P_-. Conjugation exchanges the two coefficients: $\overline{z} = r_-P_+ + r_+P_-$. The orthogonality of the idempotents gives

$$z\overline{z} = (r_+r_-)(P_+ + P_-) = r_+r_-.$$

The statement about each ideal is obtained by nullifying one of the two coefficients. □

Definition 91.14 (Causal cone partitioned)**.** The positive cone, its null boundary, and its interior are

$$C_{\mathrm{sp}}^+ = \{r_+P_+ + r_-P_- : r_+, r_- \geq 0\},$$

$$\partial_{\mathrm{null}}C_{\mathrm{sp}}^+ = \mathbb{R}_{\geq 0}P_+ \cup \mathbb{R}_{\geq 0}P_-, \qquad \operatorname{int} C_{\mathrm{sp}}^+ = \{r_+P_+ + r_-P_- : r_+r_- > 0\}.$$

The boundary has a single active direction and zero norm. The interior requires two simultaneous directions and positive norm. This stratification is exact; its physical realization requires maps that preserve the pertinent norm, transport, and dynamics.

91.7.3 Relative transporter and representational realization conditions

The historical block $B_0 = 7I + 2\mathsf{R}$ has spectrum $\{9, 5\}$, and the angular block $B_1 = AI + C^*\mathsf{R}$ has spectrum $\{A + C^*, A - C^*\}$. The theorem 38.18 proves that, with the current internal values, every putative linear intertwiner $\Phi B_0 = B_1\Phi$ is zero. The active object is the relative multiplicative transporter already constructed, not an equivalence between representations with disjoint spectra.

91.7.4 Partial factorization of the angular character

Let

$$\sigma = \frac{s_C}{6}, \qquad w_\pm = \frac{n_A \pm \sigma}{2} = \frac{W_\pm}{12}, \qquad r_\pm = q_\pm^{-w_\pm},$$

and let us define

$$Z_v = r_+ P_+ + r_- P_-.$$

Theorem 91.15 (Partial factorization of the angular character)**.** *For every angular signature $v = (n_A, s_C) \in \mathbb{Z}^2$,*

$$\boxed{N_{\mathrm{sp}}(Z_v) = \det Z_v = \exp\!\left(n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}}\right).}$$

Proof. By the definition of the channels,

$$\log r_+ = \frac{n_A + \sigma}{2}(A_{\mathrm{rad}} + C^*_{\mathrm{rad}}), \qquad \log r_- = \frac{n_A - \sigma}{2}(A_{\mathrm{rad}} - C^*_{\mathrm{rad}}).$$

Upon summation,

$$\log(r_+ r_-) = n_A A_{\mathrm{rad}} + \sigma C^*_{\mathrm{rad}}.$$

The proposition 91.13 gives $N_{\mathrm{sp}}(Z_v) = r_+ r_-$, and the diagonal representation in the basis (P_+, P_-) gives the same expression for the determinant. □

The component not seen by the determinant preserves the relative orientation:

$$\frac{1}{2}\log\frac{r_+}{r_-} = \frac{1}{2}\left(n_A C^*_{\mathrm{rad}} + \frac{s_C}{6} A_{\mathrm{rad}}\right),$$

$$Z_v = \exp\!\left[\frac{1}{2}\left(n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}}\right)\right] \exp\!\left[\frac{1}{2}\left(n_A C^*_{\mathrm{rad}} + \frac{s_C}{6} A_{\mathrm{rad}}\right) \mathsf{R}\right].$$

The first factor fixes the radial scale; the second records the relative orientation. They are two coordinated readings of one and the same element.

91.7.5 Central incorporation of the towers

Once Δ_4, s_{120}, and $\zeta^{\mathrm{corr}}_{270} = \zeta^{\mathrm{corr}}_{270}$ have been constructed globally, let

$$t(v) = k_{\Delta_4}\Delta_4 + \nu_{120} s_{120} + \nu_{270}\zeta^{\mathrm{corr}}_{270}, \qquad \widetilde{Z}_v = e^{t(v)/2} Z_v.$$

Corollary 91.16 (Complete character as original norm)**.** *Having fixed the preceding global blocks,*

$$\boxed{N_{\mathrm{sp}}(\widetilde{Z}_v) = \exp\!\left(n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}} + k_{\Delta_4}\Delta_4 + \nu_{120} s_{120} + \nu_{270}\zeta^{\mathrm{corr}}_{270}\right).}$$

Proof. In the two-dimensional representation, multiplication by the central scalar $e^{t(v)/2}$ multiplies the determinant by $e^{t(v)}$. The theorem 91.15 is applied. □

The corollary constructs the declared split realization of the active character; it does not classify all possible realizations or transform the towers into independent proofs. Its hypotheses are the already fixed channels and global blocks, and its negative control is immediate: omitting the central factor $1/2$ doubles the tower exponent; simultaneously activating both idempotents with positive coefficients departs from the null boundary.

91.8 Values, Completeness and Blocks of the Spectral Hierarchy

The bilayer operator, its moments, and the semigroup $\mathfrak{S} = \langle 90, 120\rangle$ are defined only once in the section 91.8.4. This chapter publishes values, proves reconstructive completeness, and separates the global blocks. The same integers 90, 120 also occur as cardinalities in the hexad–octad incidence, but that coincidence does not transport the incidence $\iota_{6,8}$ to the angular chart or convert its configurations into mass moments.

91.8.1 First evaluations of the hierarchy

The indices

$$90, 120, 180, 210, 240, 270, 360, 420$$

are a selection of initial heights of $\mathfrak{S}$, since

$$180 = 2 \cdot 90, \quad 210 = 90 + 120, \quad 240 = 2 \cdot 120,$$

$$270 = 3 \cdot 90, \quad 360 = 3 \cdot 120, \quad 420 = 2(90 + 120).$$

The moments $s_n = q_-^n - q_+^n$ and the generating interface $\mathcal{K}_{\text{ang}}^{90,120} = (s_{90}, s_{120})$ are used here with the already fixed convention; they are not redefined. All values are determined by A, C^*, and the reader's quadratic recurrence.

Table 91.1: Oriented moments of the global hierarchy.

n	s_n
90	$2.9516943928982796 \times 10^{-4}$
120	$1.9683511250294992 \times 10^{-5}$
180	$8.7345871869508335 \times 10^{-8}$
210	$5.8181313057291163 \times 10^{-9}$
240	$3.8754674969305336 \times 10^{-10}$
270	$2.5814553607509555 \times 10^{-11}$
360	$7.6293257871406076 \times 10^{-15}$
420	$3.3850663628348497 \times 10^{-17}$

Proposition 91.17 (Global Character of Moments)**.** *The quantities s_n of table 91.1 are invariants of the fixed angular chart. A species contributes integer coordinates that weight those invariants, but does not modify $q_\pm$ or introduce a tower of its own.*

Proof. Each s_n is a function of q_- and q_+. The theorem of inversion of channels proves that the joint map $(A, C^*) \mapsto (q_+, q_-)$ is injective. No species label appears in these definitions. □

Theorem 91.18 (Reconstructive completeness of the hierarchy)**.** *Let $n_1 < n_2 < \cdots$ be the increasing enumeration without repetitions of $\mathfrak{S}$. For*

$$a_k := s_{n_k} = q_-^{n_k} - q_+^{n_k} > 0,$$

every positive ratio $\rho \in \mathbb{R}_{>0}$ admits an absolutely convergent expansion in the hierarchy:

$$\boxed{\rho = \exp\left(\sum_{k\geq 1} \nu_k a_k\right)} \qquad \text{for some sequence } (\nu_k)_{k\geq 1} \subset \mathbb{Z}.$$

Proof. As $0 < q_+ < q_- < 1$, it holds

$$0 < a_k < q_-^{n_k}.$$

The n_k are multiplos of 30 no menores that 90; therefore,

$$\sum_{k\ge 1} a_k \le \sum_{m\ge 3} q_-^{30m} = \frac{q_-^{90}}{1-q_-^{30}} < \infty, \qquad a_k \longrightarrow 0.$$

Fix $r_0 = \log \rho$ and a tie-breaking rule for the nearest integer. Define recursively

$$\nu_k = \operatorname{nint}\left(\frac{r_{k-1}}{a_k}\right), \qquad r_k = r_{k-1} - \nu_k a_k.$$

The property of the nearest integer gives

$$|r_k| \le \frac{a_k}{2} \longrightarrow 0.$$

The telescoping identity

$$r_0 = \sum_{j=1}^{N} \nu_j a_j + r_N$$

implies $r_0 = \sum_{j\ge 1} \nu_j a_j$. The convergence is absolute, since

$$\begin{aligned} \sum_{k\ge 1} |\nu_k| a_k &= \sum_{k\ge 1} |r_{k-1} - r_k| \\ &\le |r_0| + \frac{a_1}{2} + \frac{1}{2}\sum_{k\ge 2}(a_{k-1} + a_k) < \infty. \end{aligned}$$

By exponentiating, the statement is obtained. □

Remark 91.19 (Scope and negative control). The theorem is a reconstructive surjectivity of the tower language onto $\mathbb{R}_{>0}$: it proves resolution of the codomain, not physical selection. The algorithm receives $r_0 = \log \rho$; if ρ is a target magnitude, its coefficients are therefore a target-dependent reconstruction. Prospective provenance requires another arrow,

$$\text{species} \dashrightarrow \text{extension} \longrightarrow \text{route} \longrightarrow \text{record} \longrightarrow \text{signature},$$

and does not follow from this expansion. The conclusion would fail if the a_k did not tend to zero, if the bound on the residue ceased to hold, or if the error series were not summable.

91.8.2 Global blocks of mass action

The mass action will use the triple

$$\Delta_4, \qquad s_{120}, \qquad \zeta_{270}^{\mathrm{corr}} = 135\Delta_4^2.$$

Their values are calculated only once. The coordinates k, ν_{120}, ν_{270} of a signature specify how many times each block enters the exponent; they do not change its value.

It is useful to distinguish two objects that share the index 270:

$$\boxed{\zeta_{270}^{\mathrm{corr}} = 135\Delta_4^2 = 1.66922699663643 \times 10^{-6}}$$

and

$$\boxed{s_{270}^{\mathrm{spec}} = q_-^{270} - q_+^{270} = 2.58145536075096 \times 10^{-11}.}$$

The first is a quadratic invariant constructed from Δ_4; the second is a spectral moment of the semigroup. The coincidence of the subscript does not authorize their identification.

91.8.3 Higher-order combination

The global combination

$$\delta\Delta_4^{\mathrm{tower}} = \frac{s_{180}}{24} - \frac{s_{240}}{54} - 24 s_{360} + \frac{7}{6} s_{420}$$

evaluates a

$$\delta\Delta_4^{\mathrm{tower}} = 3.63205147190875912388907096 7 \times 10^{-9}.$$

The comparison coordinate used in the selection of the coefficients is defined as the scalar data

$$\delta\Delta_4 := 3.63205147190897278466164172 0 \times 10^{-9}.$$

Consequently, the residue of the combination with respect to that coordinate is

$$\delta\Delta_4^{\mathrm{tower}} - \delta\Delta_4 = -2.1366077257075 \times 10^{-22}.$$

The small quantity is therefore a difference between two objects and not the value of the combination. All the summands belong to the same hierarchy, and their coefficients are global. Because the comparison coordinate intervened in the selection of the coefficients, this equality is reconstructive. The combination is not added again to the electronic formula that already uses Δ_4, since that substitution would duplicate the same correction.

91.8.4 Bilayer Operator and Infinite Hierarchy

Let

$$x = A_{\mathrm{rad}}, \qquad y = C_{\mathrm{rad}}^{*}, \qquad R_{\mathrm{ch}}^2 = I, \qquad P_{\pm}^{\mathrm{ch}} = \frac{I \pm R_{\mathrm{ch}}}{2}.$$

The reader of the two leaves meets in the operator

$$T_{\mathrm{ang}} = \exp(-xI - yR_{\mathrm{ch}}) = q_{+} P_{+}^{\mathrm{ch}} + q_{-} P_{-}^{\mathrm{ch}},$$

with

$$q_{-} = e^{-x+y}, \qquad q_{+} = e^{-x-y}.$$

Its even and odd spectral moments are

$$c_n = \operatorname{Tr}(T_{\mathrm{ang}}^n) = q_{-}^n + q_{+}^n, \qquad s_n = -\operatorname{Tr}(R_{\mathrm{ch}} T_{\mathrm{ang}}^n) = q_{-}^n - q_{+}^n.$$

The involution $C^{*} \mapsto -C^{*}$ preserves c_n and reverses s_n. Equivalently,

$$c_n = 2e^{-nx}\cosh(ny), \qquad s_n = 2e^{-nx}\sinh(ny).$$

This is the convention fixed in the chapter on channels. To state without homonymy the alternative presentation of the tower sources, set

$$R_{\mathrm{tow}} := -R_{\mathrm{ch}}, \qquad P_{\pm}^{\mathrm{tow}} = P_{\mp}^{\mathrm{ch}},$$

$$T_{\mathrm{tow}} := e^{-xI + yR_{\mathrm{tow}}} = T_{\mathrm{ang}}, \qquad \operatorname{Tr}(R_{\mathrm{tow}} T_{\mathrm{tow}}^n) = -\operatorname{Tr}(R_{\mathrm{ch}} T_{\mathrm{ang}}^n) = s_n.$$

The two writings describe the same scalar reader and do not constitute a second orientation of the towers.

Theorem 91.20 (Addition, norm, and recurrence)**.** *For $m, n \geq 0$,*

$$c_{m+n} = \frac{c_m c_n + s_m s_n}{2}, \qquad s_{m+n} = \frac{s_m c_n + c_m s_n}{2},$$

$$c_n^2 - s_n^2 = 4e^{-2nx}.$$

If $p = q_-q_+ = e^{-2x}$, both sequences satisfy

$$X_{n+2} = c_1 X_{n+1} - pX_n.$$

Moreover,

$$\partial_A c_n = -\frac{n\pi}{180} c_n, \qquad \partial_A s_n = -\frac{n\pi}{180} s_n,$$
$$\partial_{C^*} c_n = \frac{n\pi}{180} s_n, \qquad \partial_{C^*} s_n = \frac{n\pi}{180} c_n.$$

Proof. The addition identities follow by expanding $(q_-^m \pm q_+^m)(q_-^n \pm q_+^n)$. The norm is

$$(q_-^n + q_+^n)^2 - (q_-^n - q_+^n)^2 = 4(q_-q_+)^n.$$

The last equality is the recurrence determined by the characteristic roots q_- and q_+. The differential equations are obtained by differentiating the hyperbolic expressions; the factor $\pi/180$ belongs to the angular chart and not to a species selection. □

Theorem 91.21 (Complete height semigroup)**.** *The heights of the transversal hierarchy form*

$$\mathfrak{S} = \langle 90, 120\rangle = \{90a + 120b : a, b \in \mathbb{Z}_{\geq 0},\ a + b > 0\}.$$

Equivalently,

$$\boxed{\mathfrak{S} = \{90, 120\} \cup \{30r : r \geq 6\}.}$$

Proof. Dividing by 30 yields the numerical semigroup $\langle 3, 4\rangle$. Its positive elements are $\{3, 4\} \cup \{r : r \geq 6\}$: 5 is the only positive gap beyond the generators, and every $r \geq 6$ is obtained inductively by adding 3 or 4. Multiplying again by 30 gives the formula. □

The presentation also preserves the genealogy of a height:

$$\mathfrak{S} \cong \big(\mathbb{Z}_{\geq 0}^2 \setminus \{(0,0)\}\big) \big/ \big((a+4, b) \sim (a, b+3)\big),$$

since $4 \cdot 90 = 3 \cdot 120 = 360$. The height does not by itself determine the composition that produced it; the genealogies $4 \cdot 90$ and $3 \cdot 120$ remain distinct in the enriched chronological record even though they evaluate to the same index.

To make explicit the recurrence sampled in the common step of thirty, let us define

$$a_{30} := q_-^{30}, \qquad b_{30} := q_+^{30}, \qquad u_r := s_{30r} = a_{30}^r - b_{30}^r.$$

Then

$$\boxed{u_{r+2} = (a_{30} + b_{30})u_{r+1} - a_{30}b_{30}u_r.}$$

The auxiliary term $u_5 = s_{150}$ may intervene when propagating the recurrence, but $150 \notin \mathfrak{S}$; a computational identity does not create a new admissible height.

In particular,

$$90, 120, 180, 210, 240, 270, 300, 330, 360, 390, 420, 450, 480, \ldots$$

are the first heights. The eight heights highlighted in previous expositions are first readings of $\mathfrak{S}$, not its definition nor its limit.

As a human guide, its initial genealogy can be read without converting it into the complete domain:

90	$3 \cdot 30$	minimal generator
120	$4 \cdot 30$	minimal generator
180	$2 \cdot 90$	first even mode
210	$90 + 120$	composite mode
240	$2 \cdot 120$	second even mode
270	$3 \cdot 90$	repetition of branch 90
360	$3 \cdot 120$	repetition of branch 120
420	$2(90 + 120)$	composite return

The last column records structural provenance; it neither introduces parameters nor asserts that those eight rows exhaust the hierarchy. The same integers 90, 120 are incidence cardinalities in $\mathcal{F}_6$ and $\mathcal{F}_8$, but that coincidence does not identify the bilayer operator with the inclusion $\iota_{6,8}$.

The interface of the two generators is only the pair

$$\mathcal{K}_{\mathrm{ang}}^{90,120}(q_+, q_-) := (s_{90}, s_{120}).$$

It must not be confused with the resolvents of two subtails or with the subsequent functional $K_{6\to 8} = 12\Delta S_{90} - \Delta S_{120}$, whose definition and proof reside in section 51.2. There, the subtails live in the analytic completion of the hierarchy; they are neither new generators nor species coefficients.

Definition 91.22 (Convergent module of spectral refinements)**.** The integral domain of the global lift is

$$\mathcal{E}_{\mathfrak{S}} := \left\{ \eta \in \mathbb{Z}^{\mathfrak{S}} : \sum_{h \in \mathfrak{S}} |\eta_h s_h| < \infty \right\}, \qquad s_h = q_-^h - q_+^h.$$

Each $\eta \in \mathcal{E}_{\mathfrak{S}}$ determines the spectral character

$$\mathcal{R}_\eta := \exp\left(\sum_{h \in \mathfrak{S}} \eta_h s_h \right).$$

The mass law does not reside in this definition: its central signature $(n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta) \in \mathbb{Z}^5$ is constructed beforehand from the register. The elevation by η acts subsequently and preserves the provenance of the basic term $b_{120}s_{120}$. In particular, $N_{120} = b_{120} + \eta_{120}$ is the evaluated total coefficient, not a new signature coordinate or a unique inverse decomposition.

The completeness of the semigroup does not by itself select a species. When η is obtained from a supplied metrological residual, one has an exact calibrated reconstruction; prospective force requires the route, carry, and winding to produce η before that residual is opened. The preceding recurrence generates all the moments once the refinement has been fixed, without introducing a distinct law at each height.

91.8.5 Pair of genealogical readers over the complete atlas

The enriched route possesses two complementary internal representations,

$$\Gamma \longmapsto \big(\gamma_{108}(\Gamma), \tau_{12}(\Gamma)\big) \in \mathbb{F}_3^{108} \times \{-1, 0, +1\}^{12}.$$

The first preserves the complete tritic history; the second preserves the signed dodecaphase record. Each projection induces a natural reader, and both are evaluated before opening masses, PDG labels, or residuals.

91.8.5.1 Pair of genealogical readers $\mathbf{K}(\Gamma) = (K_D(\Gamma), K_\Omega(\Gamma))$*: domain, codomain, and scalar evaluations*

Using the notation of equations (58.6), (58.7) y (58.10), we define

$$\mathbf{K}(\Gamma) := \big(K_D(\Gamma), K_\Omega(\Gamma)\big) \in \left(\mathbb{Z}^2 \times \tfrac{1}{2}\mathbb{Z}\right) \times \mathbb{Z}^3, \tag{91.1}$$

where

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2}\right), \qquad K_\Omega = (\Omega_{90|120}, 54, P_6).$$

Scalar evaluations are

$$\kappa_D(\Gamma) = D_{27} + D_{36} - \alpha_{\mathrm{HMT}} D_1 + \frac{\Delta_4}{2}(D_4 - D_3), \tag{91.2}$$

$$\kappa_\Omega(\Gamma) = \Omega_{90|120}(\tau_\Gamma) - 54\alpha_{\mathrm{HMT}} + P_6(\tau_\Gamma)\Delta_4. \tag{91.3}$$

The second coordinate requires a causal distinction. In APP, the primordial spectral jump is $9 - 5 = 4$; the local coefficient 2, the compression $6 = 51 - 45$, and the two-dimensional unfolding $54 = 9 \cdot 6 = 459 - 405$ are later typed invariants, in accordance with the theorem 1.58. By contrast, in the chronological displacement reader, the electron coordinate is $B_D = D_1(\Gamma_e) = 54$: it counts immediate boundaries and coincides with the kinematic periodicity of order 54 of the CT108 endomorphism. The fixed 54 of K_Ω is the common coordinate transferred when both readers are made compatible in the electron reduction. Its numerical equality with the global APP defect constitutes a subsequent cross-check; it does not make the number 54 the primordial difference between addition and multiplication.

The equality $4 \cdot 90 = 3 \cdot 120 = 360$ explains the common genealogy of the two readers. K_D compares displacements of the tritic word γ_{108}; K_Ω compares the energies of windows in the twelve-phase word. The coincidence of origin does not identify their information quotients. In the certified domain $\Gamma^{\mathrm{or}}_{324}$, half-turn periodicity makes all the D_k even, so that $(D_4 - D_3)/2 \in \mathbb{Z}$ and $K_D \in \mathbb{Z}^3$. Outside that domain, the third component retains type $\frac{1}{2}\mathbb{Z}$.

91.8.5.2 Naturalness of bidirectional operators and type preservation

The reader D_k is invariant under time translation, reversal, and every bijective permutation of the tritic alphabet. On the declared routes, simultaneously reversing both orientations acts by a translation of 27 discrete steps and preserves K_D. The reader K_Ω is invariant under dodecaphase rotation and reversal and under $C_M(\tau) = -\mathrm{rev}(\tau)$. Its quadratic form

$$\Omega(\tau) = \langle \tau, (4W_3^* W_3 - 3W_4^* W_4)\tau \rangle$$

possesses integer bilinear polarization and satisfies the 2-cocycle identity.

The global counters of turn, sheet change, phase repetition, prefix change, and return are constant over the 324 routes. Consequently, every functional that factors exclusively through those six counters is constant and cancels in ratios. The readers in (91.2) and (91.3) act on the local history and pass that nonfactorization control.

Structural Necessity Proposition 91.23 (Necessity of the two sheets in the electronic route)**.** The chronological reader produces

$$(A_D, B_D, C_D) = \begin{cases} (80, 54, 6), & \text{joint chronological word of both sheets;} \\ & B_D = D_1(\Gamma_e) = 54 \\ (80, 84, -8), & \text{additive projection,} \\ (80, 60, 13), & \text{multiplicative projection.} \end{cases},$$

The complete electronic triple requires the joint history of the two arithmetic sheets.

91.8.5.3 Exact Censuses over the 324 Routes

The chronological reader produces fourteen profiles. The following table records their complete census; $A_D = D_{27} + D_{36}$, $B_D = D_1$, and $C_D = (D_4 - D_3)/2$.

Table 91.2: Exact census of the fourteen profiles of the chronological reader over the 324 oriented routes.

A_D	B_D	C_D	multiplicity
0	24	0	18
40	24	2	36
40	54	6	18
40	78	27	36
40	84	30	36
60	78	25	18
60	84	28	18
80	54	6	18
80	54	7	18
80	66	2	18
80	66	3	36
80	66	4	18
100	66	0	18
100	66	2	18
		total	324

The dodecaphasic reader produces nine profiles:

Table 91.3: Exact census of the nine profiles of the dodecaphasic reader over the 324 oriented routes.

Ω	B_Ω	P_6	multiplicity
80	54	6	198
56	54	5	68
48	54	5	34
36	54	4	8
20	54	4	4
4	54	2	4
40	54	4	4
24	54	4	2
24	54	2	2
		total	324

The readers coincide as triples on (18) routes, always with the value $(80, 54, 6)$, and the pair $\mathbf{K}$ assumes (40) values. In the canonical section $B1$ of 81 positions, K_D is constant in 41 of the 56 families and refines the quotient into 77 classes; K_Ω is constant in 50 and produces 62 classes; the pair produces 78. In the lift to 324 orientations, K_D produces 146 classes and is nonconstant in every family; K_Ω produces 100 and descends in 17 families; the pair produces 151 and is nonconstant in all of them. These censuses prove that the two projections retain distinct memories and determine the domain of each descent claim.

91.8.5.4 Vector traces and joint spectra of the 13 *multisections*

In the canonical section $B1$, we denote by ℓ_j, ν_j, d_j, u_j the generation-j multisections of the charged-lepton, neutrino, down-type-quark, and up-type-quark sectors, respectively. For each reader $r \in \{D, \Omega\}$ and each component $a \in \{1, 2, 3\}$, we define the diagonal operator

$$K^r_{M,a} e_\Gamma = K_{r,a}(\Gamma) e_\Gamma, \qquad \mathbf{K}^r_M = (K^r_{M,1}, K^r_{M,2}, K^r_{M,3}).$$

The three operators of $\mathbf{K}^r_M$ are self-adjoint and commute. Therefore, their normalized trace is understood componentwise,

$$\overline{\mathbf{K}}^r_M = \frac{1}{\dim M}\left(\operatorname{Tr} K^r_{M,1}, \operatorname{Tr} K^r_{M,2}, \operatorname{Tr} K^r_{M,3}\right),$$

and its spectrum is the joint spectrum, that is, the distribution with multiplicity of the simultaneous triples. The thirteen vector traces are:

Table 91.4: Normalized vector traces of the readers K_D and K_Ω on the thirteen canonical multisections.

M	$\dim M$	$\overline{\mathbf{K}}^D_M$	$\overline{\mathbf{K}}^\Omega_M$
ℓ_0	1	$(80, 54, 6)$	$(80, 54, 6)$
ℓ_2	8	$(65, 249/4, 17/2)$	$(80, 54, 6)$
ℓ_4	16	$(225/4, 489/8, 21/2)$	$(141/2, 54, 11/2)$
ν_1	2	$(40, 51, 29/2)$	$(60, 54, 5)$
ν_2	4	$(45, 63, 29/2)$	$(61, 54, 5)$
ν_3	6	$(170/3, 55, 34/3)$	$(206/3, 54, 11/2)$
ν_4	8	$(105/2, 237/4, 95/8)$	$(143/2, 54, 45/8)$
d_1	2	$(40, 51, 29/2)$	$(60, 54, 5)$
d_2	4	$(45, 63, 57/4)$	$(61, 54, 5)$
d_3	6	$(170/3, 55, 23/2)$	$(218/3, 54, 17/3)$
d_4	8	$(105/2, 237/4, 49/4)$	$(149/2, 54, 23/4)$
u_1	4	$(65, 66, 9)$	$(80, 54, 6)$
u_3	12	$(205/3, 119/2, 113/12)$	$(74, 54, 23/4)$

The notation $(a, b, c)^{\times n}$ indicates multiplicity n. The complete joint spectrum of $\mathbf{K}_D$ is:

Table 91.5: Complete joint spectra of the reader K_D in the thirteen canonical multisections.

M	$\operatorname{Spec}_{\text{conj}} \mathbf{K}^D_M$
ℓ_0	$(80, 54, 6)^{\times 1}$
ℓ_2	$(0, 24, 0)^{\times 1}$; $(40, 78, 27)^{\times 2}$; $(80, 54, 7)^{\times 1}$; $(80, 66, 2)^{\times 1}$; $(80, 66, 3)^{\times 1}$; $(100, 66, 0)^{\times 1}$; $(100, 66, 2)^{\times 1}$
ℓ_4	$(0, 24, 0)^{\times 1}$; $(40, 24, 2)^{\times 2}$; $(40, 54, 6)^{\times 2}$; $(40, 84, 30)^{\times 2}$; $(60, 78, 25)^{\times 3}$; $(80, 66, 2)^{\times 2}$; $(80, 66, 3)^{\times 3}$; $(80, 66, 4)^{\times 1}$
ν_1	$(40, 24, 2)^{\times 1}$; $(40, 78, 27)^{\times 1}$
ν_2	$(0, 24, 0)^{\times 1}$; $(40, 84, 30)^{\times 1}$; $(60, 78, 25)^{\times 1}$; $(80, 66, 3)^{\times 1}$
ν_3	$(40, 24, 2)^{\times 2}$; $(40, 78, 27)^{\times 1}$; $(60, 84, 28)^{\times 1}$; $(80, 54, 6)^{\times 1}$; $(80, 66, 3)^{\times 1}$
ν_4	$(0, 24, 0)^{\times 1}$; $(40, 24, 2)^{\times 1}$; $(40, 78, 27)^{\times 2}$; $(40, 84, 30)^{\times 1}$; $(80, 54, 7)^{\times 1}$; $(80, 66, 2)^{\times 1}$; $(100, 66, 0)^{\times 1}$
d_1	$(40, 24, 2)^{\times 1}$; $(40, 78, 27)^{\times 1}$
d_2	$(0, 24, 0)^{\times 1}$; $(40, 84, 30)^{\times 1}$; $(60, 78, 25)^{\times 1}$; $(80, 66, 2)^{\times 1}$
d_3	$(40, 24, 2)^{\times 2}$; $(40, 78, 27)^{\times 1}$; $(60, 84, 28)^{\times 1}$; $(80, 54, 6)^{\times 1}$; $(80, 66, 4)^{\times 1}$
d_4	$(0, 24, 0)^{\times 1}$; $(40, 24, 2)^{\times 1}$; $(40, 78, 27)^{\times 2}$; $(40, 84, 30)^{\times 1}$; $(80, 54, 7)^{\times 1}$; $(80, 66, 3)^{\times 1}$; $(100, 66, 2)^{\times 1}$
u_1	$(40, 54, 6)^{\times 1}$; $(60, 78, 25)^{\times 1}$; $(80, 66, 2)^{\times 1}$; $(80, 66, 3)^{\times 1}$
u_3	$(40, 24, 2)^{\times 2}$; $(40, 78, 27)^{\times 2}$; $(60, 84, 28)^{\times 1}$; $(80, 54, 6)^{\times 3}$; $(80, 66, 3)^{\times 1}$; $(80, 66, 4)^{\times 1}$; $(100, 66, 0)^{\times 1}$; $(100, 66, 2)^{\times 1}$

The complete joint spectrum of $\mathbf{K}_\Omega$ on the same fibres is:

Table 91.6: Complete joint spectra of the reader K_Ω in the thirteen canonical multisections.

M	$\mathrm{Spec}_{\mathrm{conj}}\,\mathbf{K}_M^\Omega$
ℓ_0	$(80,54,6)^{\times 1}$
ℓ_2	$(80,54,6)^{\times 8}$
ℓ_4	$(24,54,2)^{\times 1};(56,54,5)^{\times 4};(80,54,6)^{\times 11}$
ν_1	$(40,54,4)^{\times 1};(80,54,6)^{\times 1}$
ν_2	$(4,54,2)^{\times 1};(80,54,6)^{\times 3}$
ν_3	$(36,54,4)^{\times 1};(56,54,5)^{\times 1};(80,54,6)^{\times 4}$
ν_4	$(36,54,4)^{\times 1};(56,54,5)^{\times 1};(80,54,6)^{\times 6}$
d_1	$(40,54,4)^{\times 1};(80,54,6)^{\times 1}$
d_2	$(4,54,2)^{\times 1};(80,54,6)^{\times 3}$
d_3	$(36,54,4)^{\times 1};(80,54,6)^{\times 5}$
d_4	$(36,54,4)^{\times 1};(80,54,6)^{\times 7}$
u_1	$(80,54,6)^{\times 4}$
u_3	$(56,54,5)^{\times 3};(80,54,6)^{\times 9}$

The traces may coincide between multisections of different type; the type, multiplicity, and full spectrum remain data of the operator. The certificate reproduces these tables row by row.

91.8.5.5 Fiber Operators and Naturality of Base Change

Distinguimos the fiber canonical F_M^{B1}, formed by the positions of the multisection M with its orientation published, and the fiber oriented F_M^{or}, formed by their four lifts by position. For $s \in \{B1, \mathrm{or}\}$, let $\mathcal{H}_{M,s} = \bigoplus_{\Gamma \in F_M^s} \mathbb{C} e_\Gamma$. We define

$$B^r_{M,s} e_\Gamma = \kappa_r(\Gamma) e_\Gamma, \qquad R_{\mathrm{act}}^{B^r_{M,s}} = \exp\big[(\log R_{\mathrm{act}}) B^r_{M,s}\big], \qquad r \in \{D, \Omega\}. \tag{91.4}$$

Each $B^r_{M,s}$ is self-adjoint and diagonal in the route basis, so $R_{\mathrm{act}}^{B^r_{M,s}}$ is positive. For a positive basal mass $\widehat{M}^{(0)}_{M,s}$, the operatorial form conditioned on each reader is

$$\widehat{M}^{\beta,r}_{M,s} = R_{\mathrm{act}}^{B^r_{M,s}/2}\, \widehat{M}^{(0)}_{M,s}\, R_{\mathrm{act}}^{B^r_{M,s}/2}. \tag{91.5}$$

This expression is positive without imposing commutativity. When $[\widehat{M}^{(0)}_{M,s}, B^r_{M,s}] = 0$, it reduces to $\widehat{M}^{(0)}_{M,s} R_{\mathrm{act}}^{B^r_{M,s}}$. A unitary change of basis U transforms $B^r_{M,s} \mapsto U B^r_{M,s} U^*$ and preserves spectrum, trace, determinant, and characteristic polynomial. The canonical electronic fiber F_e^{B1} has rank one and reduces to the scalar of the theorem 58.12. Its oriented lift has rank four: the routes $(++)$ and $(--)$ share the electronic profile, whereas the mixed orientations preserve distinct profiles.

The normalized determinant

$$\mathrm{ndet}(R_{\mathrm{act}}^{B^r_{M,s}}) = R_{\mathrm{act}}^{\mathrm{Tr}(B^r_{M,s})/\dim F_M^s}$$

is natural under permutation of the fiber. Logarithmic additivity distinguishes it as a geometric mean, but it is not used as a substitute for the physical realization. The exact refinement output remains the pair $(B^D_{M,s}, B^\Omega_{M,s})$; the scalar section of a species is obtained through the flavor–generation–route morphism already constructed in equations (92.8) y (92.9): it selects the persistent route through non-metrological data and then applies the character of its signature. An arbitrary scalar functional $S(B^D_{M,s}, B^\Omega_{M,s})$ does not define that selection and remains only as a check that the target mass is not reintroduced.

91.8.5.6 Absolute scale, relative correction and sensitivities

The genealogy of the action line is written

$$\Sigma_H = \varphi\alpha_{\mathrm{HMT}}^{16}, \qquad \mathrm{ord}_{10}(\Sigma_H) = -34,$$

$$\delta_{2w} = H_5 - \eta_{\mathrm{ret}} = \frac{90}{\pi}\mathscr{C}_\pi(\alpha_{\mathrm{HMT}}), \qquad R_{\mathrm{act}} = \frac{H_5}{\eta_{\mathrm{ret}}},$$

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S.$$

The factor $10^{-34}\mathcal{U}_S$ fixes the absolute section and cancels in ratios. For the dodecaphase reader,

$$\log\left[\frac{(m_X^{\beta,\Omega}/m_e^{\beta})}{(m_X^{(0)}/m_e^{(0)})}\right] = \left[(\Omega_X - 80) + (P_{6,X} - 6)\Delta_4\right]\log R_{\mathrm{act}}. \tag{91.6}$$

The common term $-54\alpha_{\mathrm{HMT}}$ also cancels. For the chronological reader,

$$\log\left[\frac{(m_X^{\beta,D}/m_e^{\beta})}{(m_X^{(0)}/m_e^{(0)})}\right] = \left[\kappa_D(\Gamma_X) - \kappa_e\right]\log R_{\mathrm{act}}. \tag{91.7}$$

With $\log R_{\mathrm{act}} = 6.932817628081925\ldots \times 10^{-10}$,

$$\frac{\partial \log m}{\partial \Omega} = 6.9328 \times 10^{-10}, \quad \frac{\partial \log m}{\partial P_6} = \Delta_4 \log R_{\mathrm{act}} = 7.7090 \times 10^{-14},$$

$$\frac{\partial \log m}{\partial \alpha} = -54 \log R_{\mathrm{act}} = -3.74372 \times 10^{-8}, \quad \left.\frac{\partial \log m}{\partial \Delta_4}\right|_e = 6 \log R_{\mathrm{act}} = 4.15969 \times 10^{-9}.$$

These derivatives are model sensitivities and remain separate from the experimental uncertainty of a contrast mass.

91.8.5.7 Physical pairing and null sector

The typed pairing follows the chain

$$\text{typed physical record} \longrightarrow \text{multisection or representation} \longrightarrow \text{fiber of routes} \longrightarrow \text{record}$$
$$\longrightarrow \text{signature and character} \longrightarrow (B_{M,s}^{D}, B_{M,s}^{\Omega}) \longrightarrow \text{dimensional chart} \longrightarrow \text{external comparison}.$$

The typed physical inventory distinguishes registers by object, representation, and codomain; the 324 routes are the internal engine; the 471 rows of the PDG catalogue are external properties. These cardinalities belong to different domains. The arrow is constructed from non-metrological physical descriptors, the multisection, and the route; a search by proximity to a mass belongs to the calibration regime.

The null sector bifurcates before the positive character. For photon and gluons,

$$\kappa_{\mathrm{null}} = \mathrm{N/A}, \qquad m_{\mathrm{null}} = 0.$$

The expression $R_{\mathrm{act}}^0 = 1$ is the unit of the positive character. Composite objects concatenate records before evaluating the signature; topological sectors preserve fusion and braiding; a virtual section is classified by its finite horizon of prolongation.

91.8.5.8 Independence of the path constructor with respect to masses and closure of the operator

The structural verification uses exclusively the route census, the register 81×108, and the transducer of the 81 positions. It verifies their row-by-row coupling, the coverage 81/56/13/324, the naturality relations, the electronic triple $(80, 54, 6)$, the censuses (14) and (9), the match (18/324), the (40) pairs,

differentiated descents in $B1$ and in the 324 orientations, periodicity 54, inversion by shift 27, and sheet ablations. The null sector belongs to the domain exterior to the positive character. The calculation is performed before introducing masses, CODATA, PDG, metrological residuals, or nominal labels.

This certificate does not evaluate α_{HMT}, Δ_4, R_{act}, κ_e, or the MeV chart. Those evaluations belong to the arithmetic chain of the action line and to the energetic realization of theorem 58.12; this arithmetic chain remains separate from the combinatorial verification of the readers.

The metrological comparison subsequently applies the operators to the typed inventory, declares unit, scheme, scale, date, and uncertainty, and performs permutations and validation by retaining complete multisections. The physical output is governed by the domain $81/56/13/324$, by the two route readers, and by the fiber operators.

Exact domain and closed physical realization

The atlas, the $C_{81} \to \mathcal{L}_{108}$-iteration chain of the enriched chronological record of 108 iterations, the action line, the readers K_D, K_Ω, their electron reduction, and the fibre operators are exact internal results. The pair $\mathbf{K} = (K_D, K_\Omega)$ is the complete output on each route and determines the two diagonal operators of every fibre. In the electron fibre, both readers coincide. In the noncentral fibres, the pair and its spectrum preserve the complete refinement, while $\mathcal{S}_{\mathrm{phys}}$ binds branch, generation, fine signature, and, when applicable, colour orbit to the persistent route. The character of that route publishes the scalar coordinate before metrological comparison. Hence there is no disjunction between a closed operator and a pending mass: they are two typed readings of the same genealogy. Both readers are co-governing, and no proximity to an external mass intervenes in the selection.

91.8.5.9 Persistence of the enriched state under the surviving extension and conservation of the route memory

Let x be an APP state, τ its tritic word, and B_K the memory block at level K. The state relevant to mass has the form

$$X_K = (x_K, \tau_K, \varepsilon_K, h_K, v_K, c_K, B_K, g_K),$$

where ε_K is the sheet orientation, (h_K, v_K) the cellular orientation, c_K the carry, and g_K the nonadic phase. An extension survives when it forms a compatible section of the successive cylinders. If $C_{K,j}$, $0 \le j < 729$ are the children of the cylinder at the next level, the pruning rule satisfies

$$\sum_{j=0}^{728} G_9(C_{K,j}) = 1.$$

The unique index j_K produces the transition

$$P_{K+1} = 729P_K + j_K.$$

As nine levels are advanced, the phase is preserved,

$$g(K+9) = g(K),$$

but they change the block, prefix, and memory. Thus phase return is not state return. A stable particle is defined by the compatibility of the section across these returns, not by the motionless permanence of a visible coordinate.

The observable route γ_{obs} is the chronological shadow of the extension. The point traversing a diagram is a reading cursor; the particle is the complete persistence class. This distinction permits a correct interpretation of quantum motion: what is required is not a small sphere following a classical trajectory, but a coherent history that admits several compatible projections.

91.8.5.10 Dodecaphasic registration and signature

The temporal wheel contains twelve oriented events. Its causal decomposition is

$$12 = 1 + 5 + 6 :$$

one event fixes the anchor, five determine the visible incidence, and six form the opposite pairs surviving the oriented quotient. If s_C^{tot} is the integer contra-angular sum over the twelve events, the record preserves

$$s_C := s_C^{\mathrm{tot}} \in \mathbb{Z}, \qquad s_C^{(12)} := \frac{s_C^{\mathrm{tot}}}{6}.$$

The first object is the integer coordinate of the signature; the second is its dodecaphase angular reading. The divisor is not a normalization introduced from an external unit. It is the index that appears when the six opposite pairs are reconstructed. The dodecaphase incidence matrix has Smith normal form with factors $(1, 12)$; the character is well defined on the quotient because the orientation residue is measured in that structure.

This is the governing convention of the entire work. The preserved historical expressions that designated the already divided coordinate as s_C are read as $s_C^{(12)}$; their numerical values and attestations remain intact, but no current formula divides the same coordinate twice.

The direct winding pair is

$$W_+ = 6n_A + s_C^{\mathrm{tot}}, \qquad W_- = 6n_A - s_C^{\mathrm{tot}}.$$

As $s_C = s_C^{\mathrm{tot}}$ in the published signature, this formula coincides with $W_\pm = 6n_A \pm s_C$. The contra-angular exponent uses $s_C^{(12)} = s_C/6$ exactly once; it does not again divide a coordinate that had already been normalized.

The complete record publishes the signature

$$\sigma(\Gamma) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta) \in \mathbb{Z}^5, \qquad \zeta_{270} = 135\Delta_4^2.$$

The five components have distinct types. n_A counts the direct angular development; s_C records the integer counter-angular sum; $s_C^{(12)} = s_C/6$ is its angular reading; k_{Δ_4} reads the global refinement; b_{120} marks the harmonic channel of height 120; b_ζ belongs to the quadratic corrector of height 270. In particular, s_{270} and ζ_{270} are not the same object: the former is an odd trace of the bilayer operator; the latter is a quadratic coordinate of the register.

The word electronic

$$\tau_{12} = + + + - - - + + + - --$$

has linear sum zero, but its par signature is not zero:

$$(Q_3, Q_4, Q_6, \rho_0, w) = (-12, -4, 12, -6, 12).$$

Nor should it be identified with the raw signature of the electronic cycle $(24, 0, 12, 0, -12)$ or with the affine origin $v_e = 0$. These are four different levels of reading: word, even signature, raw signature, and origin of the relative character.

91.8.5.11 Angle, counter-angle, and character

To make the construction autonomous, fix here the three internal coordinates that enter throughout the law:

$$\alpha_{\mathrm{HMT}} = \frac{729735256928380099728510547238066 3}{10^{36}},$$

$$\Delta_4 = \frac{\pi_{\mathrm{HMT}} - e_{\mathrm{HMT}} \log \pi_{\mathrm{HMT}}}{270} \left(1 + \frac{\pi_{\mathrm{HMT}}}{729}\right),$$

and, for $a > 0$ and $\phi > 0$,

$$H_5(a, \phi) = \frac{1}{2}\sqrt{\frac{1000a}{\phi}} - a + \frac{9a^2}{16} - \frac{5a^3}{9} + \frac{7a^4}{48} - \frac{a^5}{54}.$$

Henceforth,

$$H_5 := H_5(\alpha_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}).$$

These definitions do not import an external metrological constant: $\pi_{\mathrm{HMT}}, e_{\mathrm{HMT}}, \varphi_{\mathrm{HMT}}$ are the coinductive evaluations of the HMT reader, and α_{HMT} is the rational output of dodecaphase closure. The internal constant α_{HMT} precedes the counter-angle. The regional branch first determines

$$A = 1000\alpha_{\mathrm{HMT}}, \qquad D_A = \exp\!\left(-\frac{100\pi\alpha_{\mathrm{HMT}}}{9}\right),$$

$$C_\pi(\alpha_{\mathrm{HMT}}) = 169\alpha_{\mathrm{HMT}}^6 + \frac{D_A\alpha_{\mathrm{HMT}}^7}{1 - D_A\alpha_{\mathrm{HMT}}}, \qquad y_* = \frac{\pi}{90}(H_5 + \alpha_{\mathrm{HMT}}) - C_\pi(\alpha_{\mathrm{HMT}}),$$

$$C^* = \frac{180}{\pi} y_*.$$

Here A and C^* are already constructed angular coordinates, not action magnitudes. If

$$\alpha_\angle := \alpha_{\mathrm{HMT}}{}^\circ, \qquad \eta_{\mathrm{ret}}^{(\mathrm{deg})} := \frac{C^*}{2} - \alpha_\angle,$$

then

$$C^* = 2\big(\eta_{\mathrm{ret}}^{(\mathrm{deg})} + \alpha_\angle\big)$$

is the inverse return identity, not a second rule for selecting the counter-angle. This notation avoids confusing the angular mantissa $\eta_{\mathrm{ret}}^{(\mathrm{deg})}$ with the dimensional magnitude $\hbar_{\mathrm{ret}}$. The radian charts are obtained by multiplication by $\pi/180$. The conjugate channels are

$$q_\pm = \exp\left[-\frac{\pi}{180}(A \pm C^*)\right].$$

The central character of a signature is

$$\chi_0(\sigma) = \exp\left(n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}} + k_{\Delta_4}\Delta_4 + b_\zeta \zeta_{270}\right).$$

The coordinate b_{120} remains in the signature, but its contribution is incorporated into the tower family. This convention prevents counting the same height twice.

The character is dimensionless. Its function is not to become an action constant, but to act by positive homotheties on an already typed mass line. For a pair of states X, Y,

$$\frac{m_Y^{(0)}}{m_X^{(0)}} = \chi_{\mathrm{full}}(\sigma_Y - \sigma_X, \eta_Y - \eta_X).$$

This formula separates the ratios basal of mass of the choice common of chart absolute. The quotient beta complete incorporates afterward $R_{\mathrm{act}}^{\kappa_Y - \kappa_X}$.

91.8.5.12 Global mass tower semigroup and convergent harmonic expansion

Let $R^2 = I$, with projectors

$$P_\pm = \frac{I \pm R}{2}, \qquad x = \frac{\pi A}{180}, \qquad y = \frac{\pi C^*}{180},$$

and

$$T = e^{-xI - yR} = q_+ P_+ + q_- P_-.$$

The even and odd traces are

$$c_n = \operatorname{Tr}(T^n) = q_-^n + q_+^n = 2e^{-nx}\cosh(ny),$$
$$s_n = -\operatorname{Tr}(RT^n) = q_-^n - q_+^n = 2e^{-nx}\sinh(ny).$$

Satisfy

$$c_n^2 - s_n^2 = 4e^{-2nx},$$
$$c_{m+n} = \frac{c_m c_n + s_m s_n}{2}, \qquad s_{m+n} = \frac{s_m c_n + c_m s_n}{2},$$

and the second-order linear recurrence

$$X_{n+2} = c_1 X_{n+1} - e^{-2x} X_n, \qquad X \in \{c, s\}, \qquad c_0 = 2, \quad s_0 = 0.$$

The set of heights is not reduced to eight visible corrections. It is the semigroup

$$\langle 90, 120\rangle = \{90, 120\} \cup \{30r : r \geq 6\}.$$

The heights $90, 120, 180, 210, 240, 270, 360, 420$ are the first useful genealogical witnesses, not the entire domain. The refined character is

$$\log \chi_{\mathrm{full}}(\sigma, \eta) = \log \chi_0(\sigma) + \sum_{h \in \langle 90,120\rangle} N_h(\sigma, \eta) s_h,$$

with

$$N_h(\sigma, \eta) = \eta_h + b_{120}\delta_{h,120}.$$

Thus s_{120} appears only once, and at height 270, $N_{270}s_{270}$ and $b_\zeta \zeta_{270}$ remain separate. Absolute convergence requires

$$\sum_h |N_h s_h| < \infty.$$

The completeness of the semigroup guarantees the capacity to represent positive ratios; it does not, by itself, select a species' physical coefficients. The prospective selection rule, when used, must arise from the route, genealogical record, carry, memory, and winding before the external mass is consulted.

91.8.5.13 Bidirectional readers K_D and K_Ω: profiles, coincidences, and operators by multisection

The oriented route produces two families of invariants. For the temporal defects D_j and the return memory Ω, define

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2}\right), \qquad K_\Omega = (\Omega, 54, P_6).$$

Across the 324 routes there are 14 distinct profiles of K_D, 9 profiles of K_Ω, 40 joint pairs, and 18 coincidences. The unique profile shared by the electron route is

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80, 54, 6).$$

These readers are not decorative labels. K_D measures discrepancies in the word; K_Ω measures the enriched return. Their equality at the center certifies rank-one bidirectional closure. Away from the center, they produce fiber operators whose complete spectra must be published even when a subsequent physical realization selects a scalar section.

91.8.5.14 Proper harmonic refinement of each state

The full refinement lives in the coefficient space.

$$\mathcal{E}_{90,120} = \left\{\eta = (\eta_h)_{h\in\langle 90,120\rangle} : \sum_h |\eta_h s_h| < \infty\right\}.$$

The sought application does not receive a mass. It receives the enriched route:

$$\mathfrak{T} : \Gamma_{\mathrm{enr}} \longrightarrow \mathcal{E}_{90,120}, \qquad \Gamma \longmapsto \eta(\Gamma).$$

Their coordinates must be explicit functions of returns, carry, winding, memory, and boundary. The naturality condition is

$$\eta(r_{n,m}\Gamma) = r^{\mathcal{E}}_{n,m}\eta(\Gamma),$$

and the convergence must be accompanied by a computable tail bound. The complete character then acts by

$$\widehat{M}_F = \sum_{\gamma\in F} m_{\mathrm{clk}}\, \chi_{\mathrm{full}}(\sigma(\gamma), \eta(\gamma))|\gamma\rangle\langle\gamma|.$$

The capacity to reconstruct a given numerical difference proves representational completeness of the hierarchy; it does not by itself prove that $\mathfrak{T}$ has chosen its coefficients. The difference between the two assertions is preserved in every quantitative row.

91.8.5.15 Coupling Scalarization

Let $P_{X,a}$ be the projector determined by the state X and the apparatus or physical channel a. The mass distribution is

$$\mu_{X,a}(B) = \langle\Psi_X, P_{X,a}E_{\widehat{M}_F}(B)P_{X,a}\Psi_X\rangle.$$

There is a scalar line when the compression satisfies

$$P_{X,a}\widehat{M}_F P_{X,a} = m_X P_{X,a}.$$

This is the precise meaning of "to measure is to couple." The value was not hidden in a nominal cell; it appears as an eigenvalue of a physical compression of an already constructed operator. If the compression

retains several eigenvalues, the correct prediction is a multiplet or a distribution, not an average chosen to match a reference.

91.8.5.16 Physical selection as a natural relation

Away from the electronic center, a species must not select a single cell by decree. The twelve noncentral multisections are fibers stable under cellular inversion and do not, in general, admit an equivariant point section. The correct realization begins with a relation

$$\mathcal{S} \subset \mathcal{P} \times \mathcal{R}_{324}, \qquad \mathcal{S}(X) = \{\gamma : (X, \gamma) \in \mathcal{S}\},$$

whose value may be a point, an orbit, or a fiber. For the electron, $\mathcal{S}(e) = \{\gamma_e\}$ has rank one because $(5, 5)$ is the unique fixed point. For color, flavor, and generation, $\mathcal{S}(X)$ preserves the corresponding orbit until a physical coupling compresses it.

The relation is admissible if it satisfies simultaneously:

1. depends only on representation, orientation, memory, boundary, charge, flavor, color, and generation constructed before the contrast;
2. commutes with particle–antiparticle conjugation;
3. is equivariant with respect to the chromatic action and cellular inversion;
4. is compatible with refinement, nonadic return, and dodecaphase reading;
5. preserves every multiplicity that has not been eliminated by a coupling;
6. remains invariant under permuting or replacing the external mass values.

The sixth point is the decisive control. If altering the experimental column changes $\mathcal{S}(X)$, the construction is retrospective. If the relation remains and only the contrast residual changes, the causal direction is preserved.

91.9 Scalarization morphism and dimensional realization obstructions

Outside the rank-one electronic fibre, the spectrum does not itself select a physical eigenvalue. A scalar requires a projector, state, or coupling functional declared before external comparison.

91.10 Explicit refinement in the 90/120 towers

The existence of the semigroup $\langle 90, 120\rangle$ is not sufficient by itself: the map assigning coefficients to an enriched route must be constructed. Let

$$\mathcal{H}_{90,120} = \{90a+120b : a, b \in \mathbb{N}_0,\ (a, b) \neq (0, 0)\} = \{90, 120\} \cup \{30r : r \geq 6\} = \{h_0 < h_1 < h_2 < \cdots\}$$

its strictly increasing enumeration, where $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, and let $s_{h_n} > 0$ be the declared normalization of the corresponding mode. The compatible record of a route Γ provides a sequence of finite integer data

$$\mathcal{L}(\Gamma) = (\ell_0(\Gamma), \ell_1(\Gamma), \ldots),$$

in which each ℓ_n brings together return, carry, winding, memory and boundary at refinement n. To fix the encoding unambiguously, let

$$\nu(z) = \begin{cases} 2z, & z \geq 0, \\ -2z - 1, & z < 0, \end{cases} \qquad \pi_{\mathrm{C}}(u, v) = \frac{(u + v)(u + v + 1)}{2} + v,$$

and define $\iota : \mathbb{Z}^d \to \mathbb{N}_0$ by recursive application of π_{C} to $\nu(z_1), \ldots, \nu(z_d)$. Then

$$q_n(\Gamma) = \frac{\iota(\ell_n(\Gamma))}{1 + |\iota(\ell_n(\Gamma))|} \in (-1, 1).$$

The minimal realization of tower is

$$\eta_{h_n}(\Gamma) = \frac{3^{-(n+1)} q_n(\Gamma)}{s_{h_n}}, \qquad \mathfrak{T}(\Gamma) = (\eta_h(\Gamma))_{h \in \mathcal{H}_{90,120}}.$$

Theorem 91.24 (Convergence, naturality and metrological blindness)**.** *For every enriched route,*

$$\sum_{h \in \mathcal{H}_{90,120}} |\eta_h(\Gamma) s_h| \le \sum_{n \ge 0} 3^{-(n+1)} = \frac{1}{2}.$$

If the refinement prolongs the record by prefixes, $\mathfrak{T}$ is natural with respect to the truncations. Moreover, no external mass value intervenes in $\mathfrak{T}$, and the tail after N modes satisfies

$$\sum_{n > N} |\eta_{h_n} s_{h_n}| \le \frac{3^{-(N+1)}}{2}.$$

Proof. We have $|q_n| < 1$; the two bounds are geometric sums. Prefix compatibility preserves all coefficients already constructed and adds those of the new level, which yields the naturality square with the truncation of $\mathcal{E}_{90,120}$. The formula contains only the record, the explicit coding, and the declared modal normalization. □

The preceding construction fixes an explicit, convergent minimal realization relative to the encoding ι and the declared normalizations s_h. A dynamics that modifies its weights must declare the operator effecting that modification and prove naturality and convergence anew; it cannot reconstruct the weights from an experimental residual.

91.11 Fiber operators and bidirectional law

For $F_X = \mathcal{S}(X)$, let

$$\mathcal{H}_X = \ell^2(F_X) \otimes V_{\rho_X}$$

the realization space, where V_{ρ_X} carries the corresponding internal representation. Each route $\gamma \in F_X$ has signature

$$\sigma(\gamma) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta) \in \mathbb{Z}^5$$

and refinement coefficients $\eta(\gamma) \in \mathcal{E}_{90,120}$. The basal operator is

$$\widehat{M}_X^{(0)} = \sum_{\gamma \in F_X} m_{\mathrm{clk}}\, \chi_{\mathrm{full}}(\sigma(\gamma), \eta(\gamma)) |\gamma\rangle\langle\gamma| \otimes I_{V_{\rho_X}}.$$

The scale m_{clk} descends from the absolute branch action–clock–mass, including the decade 10^{-34}; it is not extracted from an experimental mass.

The readers K_D and K_Ω induce exact diagonal operators B_X^D and B_X^Ω. The two orientations publish

$$\widehat{M}_X^{\beta,r} = R_{\mathrm{act}}^{B_X^r/2} \widehat{M}_X^{(0)} R_{\mathrm{act}}^{B_X^r/2}, \qquad r \in \{D, \Omega\}.$$

Both are positive and commute in the route basis. By the joint spectral theorem there exists a unique projective spectral measure $E_X^{D,\Omega}$ over $\mathbb{R}_+^2$ such that

$$\widehat{M}_X^{\beta,D} = \int x \, dE_X^{D,\Omega}(x, y), \qquad \widehat{M}_X^{\beta,\Omega} = \int y \, dE_X^{D,\Omega}(x, y).$$

At the base of routes,

$$E_X^{D,\Omega}(A)|\gamma\rangle = \mathbf{1}_A(m_D(\gamma), m_\Omega(\gamma))\,|\gamma\rangle.$$

This is the canonical bidirectional output: it preserves the complete pair and does not presuppose a scalar functional in advance.

Let

$$\boldsymbol{\beta}_X(\gamma) = (m_D(\gamma), m_\Omega(\gamma)).$$

The uniform image measure of the fiber is

$$\nu_X = (\boldsymbol{\beta}_X)_\# \left(\frac{1}{|F_X|} \sum_{\gamma \in F_X} \delta_\gamma \right) = \frac{1}{|F_X|} \sum_{\gamma \in F_X} \delta_{(m_D(\gamma), m_\Omega(\gamma))}.$$

Its transform and its mixed moments are

$$Z_X(s,t) = \int e^{sx+ty}\, d\nu_X(x,y), \qquad \mathfrak{m}_{p,q}(X) = \int x^p y^q\, d\nu_X(x,y).$$

Since ν_X has finite support, Z_X, or equivalently the complete family of mixed moments, determines the joint multiset of reader pairs up to permutation of routes. The pushforward measure is therefore a complete invariant of the operator pair; it is not an average of masses.

For a channel $P_{X,a}$, the compressed spectral measure

$$F_{X,a}^{D,\Omega}(A) = P_{X,a} E_X^{D,\Omega}(A) P_{X,a}\Big|_{P_{X,a}\mathcal{H}_X}$$

is a POVM on the prepared subspace. For a density ρ_X supported on that subspace, the joint scalar distribution is

$$\mu_{X,a}^{D,\Omega}(A) = \mathrm{Tr}\Big(\rho_X F_{X,a}^{D,\Omega}(A)\Big).$$

Every declared functional $f : \mathbb{R}_+^2 \to \mathbb{R}_+$ induces $\mu_{X,a}^{f} = f_\# \mu_{X,a}^{D,\Omega}$ without modifying the source PVM.

If $u : F_X \to F_Y$ is a bijection preserving record, orientation, and both readers, the unitary $U_u|\gamma\rangle = |u\gamma\rangle$ satisfies

$$U_u E_X^{D,\Omega}(A) = E_Y^{D,\Omega}(A) U_u, \qquad \nu_X = \nu_Y.$$

The same equality holds for the physical distributions when density and projector are transported by U_u. This naturality is asserted only for the intertwiners actually constructed; a coincidence of cardinalities does not by itself create an equivalence of fibers.

When the physical representation requires symmetry under the exchange $D \leftrightarrow \Omega$, the minimal operatorial aggregate is the geometric mean

$$\widehat{M}_X^{\mathrm{sym}} = \widehat{M}_X^{\beta,D} \# \widehat{M}_X^{\beta,\Omega}, \qquad A\#B = A^{1/2}\big(A^{-1/2} B A^{-1/2}\big)^{1/2} A^{1/2},$$

on the common positive support. In the diagonal case,

$$\widehat{M}_X^{\mathrm{sym}} = R_{\mathrm{act}}^{(B_X^D + B_X^\Omega)/4}\, \widehat{M}_X^{(0)}\, R_{\mathrm{act}}^{(B_X^D + B_X^\Omega)/4}.$$

This formula is a consequence of the explicit exchange hypothesis; it does not replace the joint measure when that symmetry has not been established.

Theorem 91.25 (Characterization of the symmetric aggregate)**.** *The geometric mean is positive, symmetric, congruence-covariant for every invertible S, and self-dual:*

$$A\#B = B\#A, \qquad S^*(A\#B)S = (S^*AS)\#(S^*BS), \qquad (A\#B)^{-1} = A^{-1}\#B^{-1}.$$

Moreover, $A\#A = A$. Consequently, in sectors where there is a physical intertwiner exchanging the readers, the aggregate introduces no parameter and favors neither orientation.

Proof. The congruence identity is understood for invertible S. The identities follow from the functional calculus of the positive root of $A^{-1/2}BA^{-1/2}$. In the commutative case, they reduce to $A\#B = (AB)^{1/2}$. □

There is also a canonical scalar aggregate under distinct axioms. If s_n is continuous, permutation-invariant, coordinatewise multiplicative, and normalized by $s_n(cI) = c$, then

$$s_n(T) = \det(T)^{1/n}.$$

Indeed, upon passing to logarithms, continuity and additivity make the functional linear in the logarithms of the eigenvalues; symmetry makes all coefficients equal, and normalization fixes their value at $1/n$. For the pair of readers, exchange symmetry gives the invariant

$$S_{D,\Omega} = \sqrt{\operatorname{ndet}(T_D)\operatorname{ndet}(T_\Omega)}, \qquad \operatorname{ndet}(T) = \det(T)^{1/n}.$$

This result classifies the complete fiber under the stated axioms. It is not automatically identified with a mass line: that identification requires the physical coupling to adopt precisely that functional.

For the rank-one electronic fiber, the joint measure is punctual, the two readers share the first component 80, the compression is scalar, and the beta closure is recovered

$$m_e^\beta = 0.5109989506900008086082571648\ldots \text{ MeV}.$$

The base value remains as a construction stage; it does not replace the beta closure.

91.12 Symmetry Scalarization and Coupling

Let G_X be the compact group of unbroken symmetry of the fiber and $U_X : G_X \to \mathcal{U}(\mathcal{H}_X)$ its representation. The conditional symmetry expectation is defined by

$$\mathbb{E}_X(T) = \int_{G_X} U_X(g) T U_X(g)^* \, d\mu_X(g),$$

where μ_X is the normalized Haar measure. The physical mass operator before opening channels is defined, when the state type declares a functional $\Phi_X : \mathbb{R}_+^2 \to \mathbb{R}_+$, by

$$\mathcal{M}_X = \mathbb{E}_X\Big(\Phi_X(\widehat{M}_X^{\beta,D}, \widehat{M}_X^{\beta,\Omega})\Big).$$

The functional calculus is carried out with the joint measure $E_X^{D,\Omega}$. In a representation containing an intertwiner that exchanges readers, one may take $\Phi_X(x,y) = \sqrt{xy}$. If the physical type does not declare Φ_X, the complete output remains the joint PVM; no scalar is fabricated.

Theorem 91.26 (Scalar closure, multiplet, and distribution)**.** *The operator $\mathcal{M}_X$ belongs to the commutant $U_X(G_X)'$. If*

$$\mathcal{H}_X \simeq \bigoplus_\lambda V_\lambda \otimes \mathbb{C}^{m_\lambda},$$

then

$$\mathcal{M}_X = \bigoplus_\lambda I_{V_\lambda} \otimes M_{X,\lambda}.$$

In an irreducible sector of multiplicity 1*, the restriction is a scalar. In a reducible sector, the exact output is the spectrum of the blocks $M_{X,\lambda}$, with their multiplicities. There is no indeterminacy: there is a distinct codomain.*

Proof. Invariance of the Haar measure implies $U_X(h)\mathbb{E}_X(T)U_X(h)^* = \mathbb{E}_X(T)$ for every $h \in G_X$. The decomposition follows from the complete reducibility theorem and Schur's lemma. The irreducible case produces a multiple of the identity; the reducible case preserves exactly the multiplicity blocks. □

Let $P_{X,a}$ be the projector determined by the prepared state and the physical channel or apparatus a. The observable spectral measure is

$$\mu_{X,a}(B) = \langle \Psi_X, P_{X,a} E_{\mathcal{M}_X}(B) P_{X,a} \Psi_X \rangle.$$

There is a scalar line precisely when the compressed spectral support is pointwise:

$$E_{\mathcal{M}_X}(\{m_X\})P_{X,a} = P_{X,a}, \qquad \text{equivalently} \qquad (\mathcal{M}_X - m_X I)\, P_{X,a} = 0.$$

The isolated equality $P_{X,a}\mathcal{M}_X P_{X,a} = m_X P_{X,a}$ only fixes an expectation if the subspace is not reducing and is not enough to prove a spectral line. If the compression preserves several eigenvalues, the output is a multiplet or a distribution. Replacing it with a numerical average would destroy physical and genealogical information.

91.13 Construction of the mass operator from the trajectory record

The construction distinguishes two operations. First, the dynamics produces a record of the trajectory. Then a functional extracts integer coordinates and evaluates them using the internal constants already obtained. Preserving that order explains how mass can be a magnitude of persistence without exhausting its structure.

91.13.1 Chronological reading and oriented events

Let $d_1, \dots, d_{108}$ be the digital reading of an enriched route. In the declared chart, the action support is $\{1, 3, 5, 7\}$, and

$$n_A = \#\{t : d_t \in \{1, 3, 5, 7\}\}.$$

The 12 events of the blocks of 9 steps are also preserved:

$$e_m = \sigma_m \#\{t : 9m + 1 \le t \le 9m + 9,\ d_t \in \{2, 4, 6, 8\}\}, \quad 0 \le m < 12,$$

where $\sigma = (+, +, +, -, -, -, +, +, +, -, -, -)$. The support of this extractor is an explicit part of the reading chart; its function must not be confused with the ternary phase selector of the TPK.

The distribution of events is relevant even when their sums coincide. To see it, we introduce

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}, \quad s_C = \sum_{i=0}^{5} p_i, \quad M_i = p_i - p_5 \ (0 \le i < 5).$$

Proposition 91.27 (Integral reconstruction of the events)**.** *The transformation* $(e_0, \dots, e_{11}) \mapsto (s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ *is injective on* $\mathbb{Z}^{12}$*. Its image is characterized by*

$$s_C - \sum_{i=0}^{4} M_i \equiv 0 \pmod 6, \qquad p_i \equiv a_i \pmod 2 \quad (0 \le i < 6),$$

where $p_5 = (s_C - \sum M_i)/6$ *and* $p_i = p_5 + M_i$ *for* $i < 5$.

Proof. Adding the six expressions of p_i gives $s_C = 6p_5 + \sum_{i<5} M_i$. This recovers the p_i, and then

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2}.$$

The congruences are necessary for these quantities to be integers and sufficient for the formulas to define an integral preimage. Substituting into the direct transformation recovers all the coordinates. □

The decomposition $12 = 1 + 5 + 6$ thus acquires an operational meaning: the total, the differences and the orientations recover the events. Division by 6 belongs to that reconstruction. When the character uses $s_C/6$, s_C remains the integer sum; a coordinate already divided is not divided a second time.

91.13.2 Torsional memory and signature

For each occurrence of 9, the reading preserves its predecessor. We define, with cyclic indices,

$$j_2(d) = \begin{cases} 0, & d = 9, \\ 1, & d \in \{2,4,6,8\}, \\ -1, & d \in \{1,3,5,7\}, \end{cases} \qquad z_t = \mathbf{1}_{\{d_t=9\}}\, j_2(d_{t-1}).$$

Then

$$T_z = \sum_{t=1}^{108} z_t, \quad C_z = \sum_{t\in\{27,54,81,108\}} z_t, \quad k_{\Delta_4} = T_z - 2C_z.$$

The first sum preserves the accumulated memory and the second distinguishes the crown closures. Removing the predecessors before evaluating these expressions alters the object being read.

With $\tau_m = \operatorname{sgn}(e_m)$, the remaining coordinates are

$$b_{120} = \sum_{a=0}^{3} \operatorname{sgn}(e_a + e_{a+4} + e_{a+8}), \qquad b_\zeta = \sum_{m=0}^{11} \tau_m \tau_{m+3},$$

where the second index is modulo 12. We obtain the signature

$$L_{\text{tick}}(\gamma) = \sigma(\gamma) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta) \in \mathbb{Z}^5.$$

This map is deterministic in the declared chart. The signs and the correlations explain why composition must be performed on the record before extracting the 5 integers: the sum of signatures does not, in general, represent the composition of two histories.

91.13.3 Character and internal specialization

Let $X = (X_A, X_C, X_4, X_{120}, X_{270})$. The formal character is

$$\chi(\sigma; X) = \exp\bigl(n_A X_A + s_C X_C + k_{\Delta_4} X_4 + b_{120} X_{120} + b_\zeta X_{270}\bigr)\,.$$

For each X, it satisfies $\chi(\sigma + \sigma'; X) = \chi(\sigma; X)\chi(\sigma'; X)$ because the exponent is linear in the signature. This is a law on signatures; it does not assert that the extractor is additive on routes without preserving the interface.

The specialization uses the internal coordinates constructed previously:

$$X_A = A_{\text{rad}}, \quad X_C = C^*_{\text{rad}}/6, \quad X_4 = \Delta_4, \quad X_{120} = s_{120}, \quad X_{270} = \zeta_{270} := 135\Delta_4^2.$$

The character is dimensionless and positive. Its effect is a homothety on the mass line. The exponential form determines how the coefficients act; their provenance is fixed in the preceding records and constructions.

91.13.4 Bilayer hierarchy and preservation of the two genealogies

We write $x = A_{\mathrm{rad}}$, $y = C^{*}_{\mathrm{rad}}$. In the chamber $x > y > 0$, the channels are

$$q_+ = e^{-x-y}, \qquad q_- = e^{-x+y}, \qquad 0 < q_+ < q_- < 1.$$

For an involution R, with $P_\pm = (I \pm R)/2$, the operator of the two channels is

$$T = q_+ P_+ + q_- P_-.$$

Its scalar and oriented moments are expressed as

$$c_n = q_-^n + q_+^n, \qquad s_n = q_-^n - q_+^n.$$

Multiplying these expressions proves

$$c_n^2 - s_n^2 = 4e^{-2nx}, \quad c_{m+n} = \frac{c_m c_n + s_m s_n}{2}, \quad s_{m+n} = \frac{s_m c_n + c_m s_n}{2}.$$

Both sequences satisfy

$$X_{n+2} = c_1 X_{n+1} - e^{-2x} X_n, \qquad X \in \{c, s\}.$$

These relations show that the heights share a single operator. A different family of moments is not defined for each species.

Proposition 91.28 (Semigroup of heights)**.**

$$\langle 90, 120\rangle = \{90, 120\} \cup \{30r : r \geq 6\}.$$

Proof. After dividing by 30, it suffices to study $\langle 3, 4\rangle$. The integers $6, 7, 8$ are $3+3, 3+4, 4+4$; adding 3 produces all subsequent integers. Below 6, the only positive elements are $3, 4$. □

The heights $360 = 4 \cdot 90 = 3 \cdot 120$ illustrate the difference between equality of evaluation and memory of composition. The algebraic quotient identifies the two exponents; the record preserves whether transport used four stages of one type or three of the other.

The refined specialization has exponent

$$\log \chi_{\mathrm{ref}} = n_A x + \frac{s_C}{6} y + k_{\Delta_4} \Delta_4 + b_\zeta \zeta_{270} + \sum_{h \in \langle 90,120\rangle} N_h s_h, \qquad N_h = \eta_h + b_{120} \mathbf{1}_{\{h=120\}}.$$

The pair (η_{120}, b_{120}) is preserved even though the evaluation uses its sum. Also kept separate are s_{270} and ζ_{270}: the former is a bilayer moment and the latter a quadratic expression of the refinement.

Proposition 91.29 (Convergence of the refined evaluation)**.** *If $\sum_h |N_h s_h| < \infty$, the refined character defines a finite positive scalar and its truncated evaluations converge to it.*

Proof. The hypothesis gives absolute convergence of the exponent. Continuity of the exponential allows passage to the limit; the exponential of a finite real number is strictly positive. □

Convergence evaluates a sequence of coefficients already specified. Selection of those coefficients by the route is a distinct operation, and must accompany every evaluation attributed to a species.

The notation χ_{full} retained in some evaluation formulas for the sectors designates this same character χ_{ref}, with the same arguments and coefficients. The change of symbol does not add another specialization or another tower.

91.13.5 Displacement and autocorrelation readers

The same route admits two readings of the return:

$$K_D = \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right), \qquad K_\Omega = (\Omega, 54, P_6).$$

The first map preserves displacement defects; the second, return memory and autocorrelation. In the full census of 324 orientations there are 14 profiles of the first, 9 of the second and 40 joint pairs. The 18 coincidences have value $(80, 54, 6)$. Equality of the two operators throughout the atlas is not inferred from this.

For the electronic route, the displacements are

$$(D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32).$$

Substitution produces $K_D = (80, 54, 6)$, coinciding with K_Ω. Its exponent is

$$\kappa_e = 80 - 54\alpha + 6\Delta_4.$$

The electronic trajectory is constructed in (90.117); its displacement and autocorrelation readers are evaluated, respectively, in (90.118) and (90.119). The coincidence $K_D = K_\Omega = (80, 54, 6)$ belongs to the beta closure theorem of subsection 90.17.9. Here that central exception is preserved and the operation is extended to the fibers of the atlas.

91.13.6 Operator on a fiber and dimensional section

Let $\mathscr{H}_M$ be the finite Hilbert space with orthonormal basis (e_γ) labeled by the routes of a multisection. For $r \in \{D, \Omega\}$, the return operator is defined by $B_M^r e_\gamma = \kappa_\gamma^r e_\gamma$, with real exponents; it is therefore self-adjoint. Let $M_M^{(0)}$ be the positive basal operator, whose connection with the character is fixed below. The dimensionless ratio of the action sections is $R_{\mathrm{act}} > 0$. We define

$$M_M^{\beta,r} = R_{\mathrm{act}}^{B_M^r/2} M_M^{(0)} R_{\mathrm{act}}^{B_M^r/2}.$$

Proposition 91.30 (Positivity and scalar restriction)**.** *The operator $M_M^{\beta,r}$ is positive. If both operators are diagonal in the route basis, each basal value $m_\gamma^{(0)}$ is transformed into*

$$m_\gamma^\beta = m_\gamma^{(0)} R_{\mathrm{act}}^{\kappa_\gamma}.$$

Proof. Let $S = R_{\mathrm{act}}^{B_M^r/2}$, self-adjoint and invertible. For every v, $\langle v, S M_M^{(0)} S v \rangle = \langle S v, M_M^{(0)} S v \rangle \geq 0$. In the diagonal basis, the two factors S multiply each value by $R_{\mathrm{act}}^{\kappa_\gamma/2}$; their product gives the formula. □

The absolute mass section is obtained from the action and the clock:

$$\omega_0 = \frac{2\pi}{108 t_0}, \qquad m_{\mathrm{clk}} = \hbar_{\mathrm{ret}} \omega_0 c_{\mathrm{int}}^{-2}.$$

In the chart normalized with respect to the electron, the preceding basal operator is

$$M_M^{(0)} e_\gamma = m_e^{(0)} \chi_{\mathrm{ref}}(\sigma_\gamma - \sigma_e, \eta_\gamma - \eta_e) e_\gamma.$$

Here $m_e^{(0)}$ is the realization of the basal electronic coordinate, with its prefactor, its refinement and its dimensional basis. If $E_e^{(0)} = \varepsilon_e^{(0)} \mathcal{U}_E$, then

$$m_e^{(0)} = \varepsilon_e^{(0)} \mathcal{U}_E c_{\mathrm{int}}^{-2} = b_e\, m_{\mathrm{clk}}, \qquad b_e = \varepsilon_e^{(0)} \frac{\mathcal{U}_E}{E_{\mathrm{clk}}}.$$

In the chart $\mathcal{U}_E = E_{\mathrm{clk}}$ we obtain $b_e = \varepsilon_e^{(0)}$; in another chart the indicated quotient of bases is preserved. If the tower coordinates are already relative to the electron, a second subtraction of η_e is omitted. This equality links the operator to the signature of each route and explicitly preserves the electronic normalization. A different choice of reference section must also transport that normalization. The scale of the reduced Planck constant belongs to this dimensional construction. The character and the return factor then act as scalars. The electron preserves its structure: it is not identified with the chronological quantum m_{clk}.

For two routes in the same absolute chart,

$$\frac{m_Y^\beta}{m_X^\beta} = \chi_{\mathrm{ref}}(\sigma_Y - \sigma_X, \eta_Y - \eta_X) R_{\mathrm{act}}^{\kappa_Y - \kappa_X}.$$

The common dimensional section cancels. The difference of the return exponents does not cancel unless it is zero. This identity preserves the distinction between absolute scale and relative structure of a family.

91.13.7 The null propagation sector

The exponential takes strictly positive values. The null chart is constructed before that character, in the split realization

$$\mathcal{A}_{\mathrm{sp}} = \mathbb{R}[R]/(R^2 - 1), \quad z = r_+ P_+ + r_- P_-, \quad N(z) = r_+ r_-.$$

The directions $\mathbb{R}P_+$ and $\mathbb{R}P_-$ have zero norm. Propagation transported in one of them belongs to that sector; the bilayer material reading uses the coupling of both. In this way, the photon and the gluonic sector retain their residence in the theory although they do not constitute a row of the positive character. The realization of their representations is presented alongside the corresponding sectors.

Chapter 92

Spectral, scalar, and null realizations of the General Mass Law: material families, CKM–PMNS transport, and photonic and gluonic sectors

92.1 Conjugation, parity, and time reversal in the register

CP violation in a mixing matrix and CPT symmetry are not the same assertion. Prior to the CPT theorem of a quantum field theory, the HMT register has an exact discrete symmetry on routes. Its content is concrete: conjugation exchanges the two sheets, parity reflects the directions, and time reversal traverses the register in the opposite direction. The local action that counts turns and changes of sheet is invariant under all three operations.

92.1.1 Route and local action

A trayectoria of length L is

$$\gamma = \big((i_0, j_0, \ell_0), (d_1, \dots, d_L)\big),$$

where $d_t \in \{N, E, S, W\}$ and $\ell_t \in \{\Sigma, \Pi\}$. The TPK dynamics determines the states (i_t, j_t, ℓ_t). We define

$$\mathrm{turn}(\gamma) = \#\{2 \le t \le L : d_t \neq d_{t-1}\},$$

$$\mathrm{switch}(\gamma) = \#\{2 \le t \le L : \ell_t \neq \ell_{t-1}\}.$$

Definition 92.1 (Discrete action of minimal interaction)**.** For $\lambda \ge 0$,

$$\mathcal{A}_\lambda(\gamma) := \mathrm{turn}(\gamma) + \lambda\ \mathrm{switch}(\gamma).$$

Equivalently,

$$\mathcal{A}_\lambda(\gamma) = \sum_{t=2}^{L} \left[1 - \delta_{d_t, d_{t-1}} + \lambda(1 - \delta_{\ell_t, \ell_{t-1}})\right].$$

The action depends on the variation of the route, not on the absolute names of the directions or the sheets.

92.1.2 Discrete involutions C, P, and T on the routes: sheet conjugation, spatial reflection, and record reversal

Definition 92.2 (Discrete Operators C, P, T)**.** On a complete trajectory the following are defined:

1. P, spatial reflection:

$$(i, j) \longmapsto (i, -j) \pmod 9, \qquad E \leftrightarrow W, \quad N \mapsto N, \quad S \mapsto S,$$

   without changing the sheet;

2. C, leaf conjugation:

$$\Sigma \leftrightarrow \Pi,$$

   without changing direction; in the angular chart it corresponds to $C^* \mapsto -C^*$, which interchanges $q_+ \leftrightarrow q_-$;

3. T, reversal of the ledger:

$$d_t \longmapsto \mathrm{opp}(d_{L+1-t}), \qquad \ell_t \longmapsto \ell_{L+1-t}, \qquad (i_t, j_t) \longmapsto (i_{L-t}, j_{L-t}).$$

Here T is an involution of the recorded trajectory. It is not identified with reversing an impoverished automaton that has forgotten the state required to reconstruct its predecessor.

Theorem 92.3 (CPT invariance of the route action)**.** *For every trajectory γ and every $\lambda \geq 0$,*

$$\boxed{\mathcal{A}_\lambda(\gamma) = \mathcal{A}_\lambda(C\gamma) = \mathcal{A}_\lambda(P\gamma) = \mathcal{A}_\lambda(T\gamma) = \mathcal{A}_\lambda(CPT\,\gamma).}$$

Proof. P applies an involutive permutation to the alphabet of directions. Hence $d_t = d_{t-1}$ if and only if $P(d_t) = P(d_{t-1})$, and it preserves the number of turns. C globally permutes the sheets and therefore preserves the equality pattern $\ell_t = \ell_{t-1}$; it does not alter the directions. T reverses the order of both sequences: the number of changes between consecutive terms is the same when a word is traversed in the forward or reverse direction. Each operation separately preserves turn and switch, and hence preserves $\mathcal{A}_\lambda$. Their composition also preserves it. □

92.1.3 Parity of Descendant Readers

The exchange $C^* \mapsto -C^*$ permutes q_+ and q_-. Therefore, every expression symmetric in the two channels is C-even, and every anti-symmetric difference is C-odd. theorem 73.4 realizes this rule in two objects:

$$S_{\mathrm{bi}}(-C^*) = S_{\mathrm{bi}}(C^*), \qquad \Gamma_{6\to 8}(A, -C^*) = -\Gamma_{6\to 8}(A, C^*).$$

The mass band of the theorem 92.31 realizes the same separation: even mass and odd charge/orientation.

92.1.4 Relation with CKM and with the CPT theorem of fields

The CKM matrix constructed in the following chapter has $\delta_{\mathrm{CP}} \neq 0$ and a nonzero Jarlskog invariant. This describes CP violation in that mixing representation. It does not contradict theorem 92.3: the latter acts on the ensemble of routes and their action; it does not require every oriented reader to be even.

The CPT theorem of quantum field theory adds another domain and other hypotheses —relativistic locality, field structure, spectrum, and the corresponding representation—. It is compared afterward. The HMT result proved here is the discrete symmetry of the register: it fixes the object that a field realization must transport, rather than replacing it with an external reference.

92.2 Codomains of the material realizations: spectral operator, scalar, gauge nullity, composite, and resonant pole

A fiber may publish a spectral operator, a scalar after physical compression, an exact gauge nullity, a composite operator, or a resonant pole with width. These outputs have distinct codomains and are not counted as classes of the domain 81/56/13/324.

92.3 Null extensions, propagation between boundaries and electronic closure

The photonic and gluonic sectors are realized through exact nullity, whereas the electronic closure occupies the material center. The complete development distinguishes null propagation, outward–return intertwining, physical interface, and closure criteria without transforming a nullity into an absence of structure.

92.4 Null extensions and material closure

92.4.1 4 realizations of the common material state: null, massive, composite, and topological

An isolated digit determines a coordinate but does not identify the structure that produces it, the memory it preserves, or the operation that converts it into a magnitude. HMT lies prior to that projection: its primary object is a compatible extension that preserves the arithmetic laws, TRIT orientation, quotients, phase, and boundary memory. Dimensional Mechanics subsequently specifies which component is realized as a magnitude, which remains as memory, and which transformation produces the observable.

The developments of this section distinguish four modes by their operation of closure:

1. an open null extension transports a perturbation between boundaries;
2. a closed and persistent extension acquires interior norm and defines the material candidate;
3. a family of extensions weighted by action defines a thermal ensemble;
4. a connection that compares sections at distinct points provides the codomain of the gravitational realization.

The common source does not identify the four codomains. The corresponding physical maps remain discontinuous:

$$\mathfrak{R}_\gamma : \Omega_{\mathrm{HMT}}^{\mathrm{surv}} \dashrightarrow \mathcal{H}_{\mathrm{rad}}, \qquad \mathfrak{R}_m : \Omega_{\mathrm{HMT}}^{\mathrm{surv}} \dashrightarrow \mathcal{H}_{\mathrm{mat}},$$
$$\mathfrak{R}_\Theta : \mathcal{P}(\Omega_{\mathrm{HMT}}^{\mathrm{surv}}) \dashrightarrow \mathcal{S}_{\mathrm{th}}, \qquad \mathfrak{R}_{\mathrm{grav}} : \Omega_{\mathrm{HMT}}^{\mathrm{surv}} \dashrightarrow \mathsf{Cartan}_{1,3}.$$

Their unity resides in the genealogy; their difference, in the reading and in the additional structures required by each realization.

92.4.2 Null propagation between boundaries

Let B be a transport basis. A *null section* of sign $\pm$ is a map

$$s_\pm : B \longrightarrow \mathcal{A}_{\mathrm{sp}} P_\pm$$

such that $s_\pm(x) \in \mathcal{A}_{\mathrm{sp}} P_\pm$ for every $x \in B$. By proposition 91.13,

$$N_{\mathrm{sp}}(s_\pm(x)) = 0 \qquad (x \in B).$$

If γ is an oriented route between two boundaries,

$$\partial\gamma = x_{\mathrm{abs}} - x_{\mathrm{em}},$$

the proposed photonic interface assigns to γ an extended null section. The adjective *extended* indicates that its identity belongs to the complete transport and not to an isolated point.

The conjugate route is obtained by inversion,

$$\gamma^\vee = C_{\mathrm{rev}}\gamma, \qquad \partial\gamma^\vee = x_{\mathrm{em}} - x_{\mathrm{abs}}.$$

Relative to the declared orientation reader, the pair $(\gamma, \gamma^\vee)$ represents the advanced and retarded readings. The two legs are ordered sequentially or housed in distinct fibers. They are not automatically added within the same fiber: if the two conjugate components are activated simultaneously, the exact identity

$$\boxed{N_{\mathrm{sp}}(uP_+ + vP_-) = uv}$$

shows that, for $u, v > 0$, the section leaves the null boundary and enters the positive interior.

92.4.2.1 Length, go and return

Once the dimensional velocity chart, $\widehat{c} = e^{-\mathcal{E}}$, and a positive basis u_L of the length line have been fixed, define

$$L_{\mathrm{int}}(\gamma) = \widehat{c}\,|\gamma|\,u_L.$$

Inversion preserves the number of steps:

$$L_{\mathrm{int}}(\gamma^{\vee}) = L_{\mathrm{int}}(\gamma), \qquad L_{\leftrightarrow} = 2L_{\mathrm{int}}(\gamma).$$

These identities are kinematic. Converting them into electromagnetic propagation requires that the same map reproduce Gauss's law, the gauge dynamics, null dispersion, and the reduction to two transverse degrees.

92.4.2.2 Entanglement entropy between the 2 *readings*

The reversal of a route does not suffice to assert quantum entanglement. A minimal formulation first requires a Hilbert completion. Let $\mathcal{H}_{\mathrm{adv}}$ and $\mathcal{H}_{\mathrm{ret}}$ be two complex spaces with orthonormal families $\{|i\rangle_{\mathrm{adv}}\}_{i\in I}$ and $\{|i\rangle_{\mathrm{ret}}\}_{i\in I}$, where I is finite. For a probability $p = (p_i)_{i\in I}$, define

$$|\Psi_p\rangle := \sum_{i\in I} \sqrt{p_i}\,|i\rangle_{\mathrm{adv}} \otimes |i\rangle_{\mathrm{ret}}. \tag{E1}$$

Proposition 92.4 (Round-trip double-layer entropy)**.** *The reduced matrices of the state $\rho_p = |\Psi_p\rangle\langle\Psi_p|$ are*

$$\rho_{\mathrm{adv}} = \sum_{i\in I} p_i |i\rangle_{\mathrm{adv}}\langle i|, \qquad \rho_{\mathrm{ret}} = \sum_{i\in I} p_i |i\rangle_{\mathrm{ret}}\langle i|, \tag{E2}$$

and possess the same von Neumann entropy:

$$\boxed{S_{\mathrm{ent}}(p) = -\operatorname{Tr}(\rho_{\mathrm{adv}} \log \rho_{\mathrm{adv}}) = -\sum_{i\in I} p_i \log p_i.} \tag{E3}$$

Here the convention $0 \log 0 := 0$ is adopted. The state is separable exactly when a single probability is equal to one.

Proof. Upon taking the partial trace, orthogonality cancels the terms with $i \neq j$. The nonzero eigenvalues of each reduced matrix are therefore the p_i, which gives (E3). The Schmidt rank is one exactly in the deterministic case. □

Once the completion has been chosen, the proposition formalizes an entropy between the advanced and retarded readings. It does not derive the probabilities p_i, does not identify those readings with the advanced and retarded propagators of a field theory, and does not convert a classical route correlation into entanglement. Construction of an isomorphism sending HMT memory to bipartite physical states, preserving evolution, and fixing p belongs to the PHYSICAL INTERFACE. The null entropy of a deterministic distribution constitutes the immediate negative control.

Hypothesis, domain, and conclusion

The proposition is PROVED FROM EXPLICIT STARTING STRUCTURE: it assumes the Hilbert completion, the orthonormal bases, and the distribution p. The construction of the intertwiner that physically realizes both layers is a PHYSICAL INTERFACE; it is not part of the proposition.

Definition 92.5 (HMT–MD Photonic Interface)**.** A realization photonic of HMT–MD is a section open sustained over a direction null of the algebra split, transported between boundaries conjugate and protected by a symmetry that preserves exactly the norm null.

The idempotents $P_\pm$ are not, by themselves, physical photons nor the two Maxwell polarizations. The complete realization requires an intertwiner

$$\mathfrak{R}_\gamma : \partial_{\mathrm{null}} C_{\mathrm{sp}}^{+} \dashrightarrow \mathcal{H}_{\mathrm{Maxwell}}^{\mathrm{phys}}$$

that preserves the inner product, evolution, and gauge action, and that selects exactly two transverse polarizations in 3 + 1 dimensions.

Comparison Criterion 92.6 (Photonic sector)**.** The identification physical remains refuted if the evolution leaves $\partial_{\mathrm{null}} C_{\mathrm{sp}}^{+}$, generates mass without the mechanism of breaking declared, does not reproduces the dispersion null or not dejan exactly two polarizations physical after the reduction of gauge.

92.4.3 Material closure and electronic center

The simultaneous coupling of two conjugate null directions yields, for $u, v > 0$,

$$N_{\mathrm{sp}}(uP_+ + vP_-) = uv > 0.$$

No isolated component has nonzero norm; the inner norm arises from their coupling. This algebraic statement does not add pre-existing masses and does not yet identify the null directions with photons *on shell*.

The structural classification of HMT contains a single fixed point of cellular inversion,

$$(5, 5).$$

The electron's relative action origin is $v_e = 0$. In the normalized split realization, set

$$Z_e = P_+ + P_- = I, \qquad N_{\mathrm{sp}}(Z_e) = 1.$$

The point $(5, 5)$, the origin $v_e = 0$, the raw signature $(24, 0, 12, 0, -12)$, and the transverse word 555555 are four distinct objects.

Definition 92.7 (Electronic closure candidate)**.** The resonant closure of conjugate electromagnetic null modes is formalized as the program that seeks a normalized central material section obtained by coupling two conjugate null modes, stabilized by a route closure and its holonomy.

The existence of the extension, the return, and the center of the atlas does not prove minimal compactness or the stability of the closure. The outstanding dynamical theorem must construct

$$\gamma_e = \xi_1 \circ \cdots \circ \xi_N, \qquad \partial \gamma_e = 0, \qquad \varepsilon(\gamma_e) = -1,$$

where each ξ_j is locally zero, the total boundary vanishes, and ε is the sign holonomy. Mass would be associated with the global closure cocycle, not with a sum of locally zero masses. This program does not reopen the central electron $(5, 5)$, the atlas, or the already constructed extension–route–record–character chain.

Proposition 92.8 (Relativistic Causal Interior Criterion)**.** *Let $p_1, \ldots, p_N$ be future null momenta in a Minkowski space with positive temporal signature, $p_i^2 = 0$, and let*

$$P = \sum_{i=1}^{N} p_i.$$

If all p_i are collinear, then $P^2 = 0$. If at least two are future and noncollinear, then $P^2 > 0$, and one may define

$$m^2 c^2 = P^2.$$

Proof. For future-directed null vectors, $p_i \cdot p_j \geq 0$ holds, with equality precisely when they are collinear. Since

$$P^2 = \sum_i p_i^2 + 2 \sum_{i<j} p_i \cdot p_j,$$

the first sum is zero. All cross products vanish in the collinear case; if a noncollinear pair exists, at least one is strictly positive. □

Therefore, a closed route is not massive by mere topology, nor is an open route null merely because it has a boundary. Realization requires that the geometric closure and the sum of momenta enter jointly into the causal interior.

92.4.3.1 Möbius, conjugation, and posterior occupation

Under the lifting and holonomy hypotheses of theorem 68.6, one turn reverses the orientation and two turns restore it. The Möbius band organizes particle and antiparticle as conjugate orientations of the same type of closure. Their physical identification requires a representation that reverses charge, preserves the norm, and intertwines the evolution. Möbius geometry does not by itself imply the algebra of fermionic occupation; its operator relations are introduced as operator hypotheses in the following chapter.

Comparison Criterion 92.9 (Electron closure)**.** The realization is refuted if the center $(5, 5)$ does not lift to a closed extension, if the holonomy does not select the odd sector, if the Hamiltonian lacks a stable bound solution that is unique up to symmetry, or if the charge, dispersion, and form factor do not agree with the electron realization.

92.4.4 Physical realizations by object, representation, and codomain

Each realization declares its primary object, its representation or fiber, and its native output. A scalar coordinate is published only when the object itself and its realization morphism force it.

92.4.5 Elementary Realizations of the Standard Model

Particle	Charge	Spin	Statistics	Realization HMT–MD
e^-	-1	$1/2$	fermiónica	punto fijo $(5, 5)$, vacancia orientada
μ^-	-1	$1/2$	fermiónica	bloque leptónico de rango 8
τ^-	-1	$1/2$	fermiónica	bloque leptónico de rango 16
ν_e	0	$1/2$	fermiónica	vector de interacción en soporte neutro 20
ν_μ	0	$1/2$	fermiónica	vector de interacción en soporte neutro 20
ν_τ	0	$1/2$	fermiónica	vector de interacción en soporte neutro 20
u	$+2/3$	$1/2$	fermiónica	rama up, generación 1, triplete de color
d	$-1/3$	$1/2$	fermiónica	rama down, generación 1, triplete de color
c	$+2/3$	$1/2$	fermiónica	rama up, generación 2, triplete de color
s	$-1/3$	$1/2$	fermiónica	rama down, generación 2, triplete de color
t	$+2/3$	$1/2$	fermiónica	rama up, generación 3, triplete de color
b	$-1/3$	$1/2$	fermiónica	rama down, generación 3, triplete de color
γ	0	1	bosónica	sección nula, dos polarizaciones
g^a	0	1	bosónica	adjunta de color, ocho generadores
$W^\pm$	± 1	1	bosónica	bloque vectorial cargado conjugado
Z^0	0	1	bosónica	bloque vectorial neutral
H	0	0	bosónica	sección escalar persistente

Las antipartículas se obtienen por C_M. Los antiquarks pertenecen al triplete conjugado; W^- es el conjugado de W^+; fotón, Z, Higgs y gluones poseen las conjugaciones propias de sus representaciones.

El soporte B1 exacto de las clases elementales tiene rangos

$$1, \quad 8, \quad 16, \quad 3\times 20, \quad 3\times 16, \quad 3\times 20.$$

Therefore, the number of class–route incidence is

$$1+8+16+60+48+60=193.$$

The incidences are not 193 distinct routes: the three neutrino flavors share the rank-twenty neutral support; the up flavors reuse a rank-sixteen support and the down flavors a rank-twenty support. The flavor projector must refine these supports. In the union of 36 quark cells, color is exactly

$$12_R+12_G+12_B.$$

This equality certifies the global chromatic support, but does not yet authorize the selection of a cell by flavour.

92.4.6 Generation, Charge, Spin, and Mass-Character Decomposition of the Leptonic Sector

92.4.6.1 Electron: center, vacancies and orientation

The electron is exceptional because its fibre contains the unique fixed point $(5,5)$. Its reduction to scalar mass does not require breaking an orbit. The active route is

$$e_{\mathrm{act}}=(2,2,5,-4,-3,-2)^2,$$

while the historical envelope

$$e_{\mathrm{env}}=(4,3,5,-4,-3,-5)^2$$

conserves the sign and absolute maximum of the two leaves. The envelope retains the word

$$\tau_e=+++---+++--,$$

but does not substitute active history.

Four data that must remain distinct are:

$$\text{raw signature}=(24,0,12,0,-12),$$

$$\text{even CT–108 signature}=(-12,-4,12,-6,12),$$

$$\text{event vector}=e_{\mathrm{act}}, \qquad \text{affine origin}=v_e=0.$$

The basal selector $(-22,+15)$ has two complementary internal readings. The regional decomposition gives

$$-22=-(27-5), \qquad +15=5\cdot 3=\binom{6}{2},$$

and the framework of vacancies

$$V_e=\begin{pmatrix}21 & 22\\ 6 & 7\end{pmatrix}$$

satisfies

$$\det V_e=21\cdot 7-22\cdot 6=15.$$

The 22 measures the oriented deficit of the extension; the 15, the area retention that survives closure. Thus the electronic particle appears as the central persistence of an oriented vacancy, not as a mass inserted into the central cell.

The complete equation is

$$\mathfrak{e}_e^{\beta} = \frac{\sqrt{3}}{4}\left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha_{\mathrm{HMT}}^3\right](1 + 15\Delta_4)R_{\mathrm{act}}^{80-54\alpha_{\mathrm{HMT}}+6\Delta_4}.$$

In the electronic chart,

$$\mathfrak{e}_e^{\beta} = 0.5109989506900008086082571648\ldots \text{ MeV}.$$

92.4.6.2 Realization of the muon as a 2nd-generation leptonic section: fibre, signature and character

The muon resides in a leptonic multisection of rank eight. Its recovered structural trace comprises the cells

$$\{21, 23, 25, 39, 43, 57, 59, 61\},$$

the counts of turn/change 22/25, the north–south axis,

$$(N, E, S, O) = (62, 25, 7, 14), \qquad (h_{369}, r_5) = (57, 9),$$

and the central coordinate

$$(n_A, s_C) = (42, -3).$$

The proper object is the compressed operator.

$$\widehat{M}_{\mu} = P_{\mu}\widehat{M}P_{\mu}.$$

Its reduction to a number requires prove that the projector of flavor and generation commutes with the operator, or that the state prepared has variance null. Given the signature enriched

$$(42, -3, 8, 0, 2),$$

the evaluation of character is

$$105.658360294435829\ldots \text{ MeV}.$$

This figure is an exact control of the evaluation of that signature; the complete physical derivation must reproduce the signature from the genealogical record selected by the coupling. The muonic width is obtained from the temporal decay operator and not from $\widehat{M}_{\mu}$.

92.4.6.3 Realization of the tau as a 3rd-generation leptonic section: fibre, signature and character

The tau occupies a sixteen-rank lepton multisection. Its footprint comprises

$$17/24, \qquad \text{east–west axis}, \qquad (N, E, S, O) = (29, 55, 7, 17),$$
$$(h_{369}, r_5) = (59, 6), \qquad (n_A, s_C) = (64, 0).$$

The enriched signature

$$(64, 0, 25, -1, 6)$$

yields

$$1776.8609176314323\ldots \text{ MeV}.$$

As in the muon case, the well-defined structure is the operator block; the scalar mass is obtained when the tau coupling selects a scalar block or an isolated pole.

92.4.6.4 Common neutrino fiber and flavor separation through the PMNS reader

The three neutrino flavors share the space

$$\mathcal{H}_\nu = \bigoplus_{j=1}^{4} \mathcal{H}_{\nu,G_j}, \qquad \dim \mathcal{H}_\nu = 2+4+6+8 = 20.$$

The vectors

$$f_e, \quad f_\mu, \quad f_\tau$$

constitute an interaction basis; they are not three mass eigenvectors. Nor is depth G_j a fourth sterile flavour: a sterile interpretation requires an additional physical projector.

The neutral extractor is defined by

$$T_\nu(m) = \sum_{t\in E_m} \omega(t)\varepsilon(t)\theta(t), \qquad e^\nu_m = T_\nu(m) - T_\nu(m+6),$$

$$\tau^\nu_m = \operatorname{sgn}(e^\nu_m).$$

It is obtained from it.

$$b^\nu_{120} = \sum_{a=0}^{3} \operatorname{sgn}\left(\sum_{m\in I_a} e^\nu_m\right),$$

$$b^\nu_\zeta = \sum_m \tau^\nu_m \tau^\nu_{m+3}.$$

The neutrino signature possesses the form

$$v^{(i)}_\nu = (0, 0, k^{(i)}_{\Delta_4}, b^{\nu_i}_{120}, b^{\nu_i}_\zeta).$$

The inversion bidirectional changes b_{120} and preserves b_ζ; this asymmetry produces internally the two orientations of hierarchy.

92.4.6.5 Fermionic Nature of the Neutrino

Neutrality does not determine statistics. The neutrino sector lies in the exterior completion and carries spinorial holonomy:

$$U_\nu(2\pi) = -I, \qquad U_\nu(4\pi) = I,$$

$$\{a_{\nu,i}, a^\dagger_{\nu,j}\} = \delta_{ij}.$$

Therefore, every physical neutrino mode is fermionic. The Dirac/Majorana question is subsequent: it determines whether conjugation interchanges two distinct sections or identifies a section with its conjugate within a real representation. Neither possibility makes the neutrino a boson.

92.4.6.6 Mass operator and PMNS

Let P_ν be the rank-three projector onto the subspace of mass modes. First diagonalize

$$P_\nu \widehat{M}_\nu P_\nu = \sum_{i=1}^{3} m_i \, |n_i\rangle\langle n_i|.$$

Then the transport between bases is constructed:

$$(f_e, f_\mu, f_\tau) = (n_1, n_2, n_3) U_{\mathrm{PMNS}}^{\dagger}.$$

A full-rank kernel that reads sheet, orientation, carry, vacancy, and holonomy can be written

$$G_{ab} = \frac{1}{\sqrt{|L_a||N_b|}} \sum_{c \in L_a} \sum_{d \in N_b} K_{\mathrm{reg}}(c, d),$$

$$U_{\mathrm{PMNS}} = \mathrm{polar}(G), \qquad \det G \neq 0.$$

The internal phase of rank one produces the angles

$$(19.6807^\circ, 56.4970^\circ, 29.6224^\circ)$$

and, for that reason, cannot be the complete physical PMNS kernel. Full rank is a mathematical requirement, not a numerical retouching.

The observables

$$m_1, m_2, m_3, \qquad \Delta m_{ij}^2, \qquad m_\beta, \qquad m_{\beta\beta}, \qquad \sum_i m_i$$

they are different maps of the same spectrum. They should not be collapsed into a single "neutrino mass".

92.4.7 Quarks, color, generations, and CKM mixing

92.4.7.1 Structural position of the 6 *quark flavors*

Each quark is defined by

$$\mathfrak{q}_X = (\text{charge branch, generation, } \mathbb{Z}_3^{\mathrm{color}}, \text{multisection, genealogical record, sheet}).$$

The structural position of the six flavors is summarized by their traces prior to any mass chart:

Flavor	Branch/generation	Giro/cambio	Eje	(N, E, S, O)	(h_{369}, r_5)	(n_A, s_C)
u	positiva/1	21/30	N–S	$(34, 31, 31, 12)$	$(51, 13)$	$(11, 7)$
d	negativa/1	20/32	E–O	$(27, 35, 18, 28)$	$(47, 17)$	$(17, 8)$
c	positiva/2	21/29	E–O	$(20, 38, 24, 26)$	$(49, 16)$	$(61, 8)$
s	negativa/2	17/27	N–S	$(50, 32, 12, 14)$	$(57, 11)$	$(41, -3)$
t	positiva/3	21/26	N–S	$(46, 27, 16, 19)$	$(57, 9)$	$(100, -1)$
b	negativa/3	21/30	E–O	$(12, 63, 9, 24)$	$(47, 7)$	$(71, -5)$

Estas posiciones no son seis números elegidos: son seis clases de representación que actúan sobre tripletes de color. La ruta física puede ser una órbita o un bloque de rutas; no tiene que ser un punto.

92.4.7.2 Invarianza escalar del carácter de masa bajo la acción de color

Sea P_q el proyector de sabor y generación y P_{RGB} el proyector a la órbita de color. La masa quark es

$$\widehat{M}_q = P_q P_{\mathrm{RGB}} \widehat{M} P_{\mathrm{RGB}} P_q.$$

If

$$[\widehat{M}_q, U(g)] = 0, \qquad g \in SU(3),$$

then

$$\widehat{M}_q = m_q I_3$$

over the triplet irreducible. This equality explains by which the flavor carries a mass common although their three colors occupy positions distintas in the atlas.

92.4.7.3 Chart of schema and scale

The internal HMT coordinate must be transported to a renormalized definition:

$$m_q^{\mathcal{S}}(\mu) = Z_m^{\mathcal{S}}(\mu, \mu_0)\, \mathcal{R}_q(\widehat{M}_q),$$

with composition law

$$Z_m(\mu_2, \mu_0) = Z_m(\mu_2, \mu_1) Z_m(\mu_1, \mu_0).$$

For u, d, s, the natural comparison uses $\overline{\mathrm{MS}}$ at 2 GeV; for c, b, the scale of the renormalized mass itself; for t, one must state whether the chart is a pole, MSR, or kinematic chart. Without this transport, a numerical difference is not yet a residual of the law.

92.4.7.4 Assessments of the declared signatures

The historical central coordinates produce:

Flavor	Evaluation
u	2.165924536173907 . . . MeV
d	4.679550162948874 . . . MeV
c	1.270500838641911 . . . GeV
s	92.94009390784564 . . . MeV
t	172.5974795440191 . . . GeV
b	4.190119763299790 . . . GeV

Estas evaluaciones comprueban el carácter una vez dada la firma. La deducción completa exige obtener cada firma, su refinamiento de torre y su carta de esquema a partir del proyector de sabor y del registro genealógico correspondiente.

92.4.7.5 Determinación de los ángulos CKM desde la reflexión racional de las rutas quark

Sean

$$A = 1000\alpha_{\mathrm{HMT}}, \qquad h = \frac{C^*}{6}, \qquad \Delta = \frac{180}{\pi}\Delta_4.$$

The angular grammar gives

$$\begin{pmatrix}\theta_{12}\\ \theta_{23}\\ \theta_{13}\end{pmatrix}=\begin{pmatrix}2 & -5 & 28\\ 0 & 7 & -25/2\\ 1 & -20 & -2/3\end{pmatrix}\begin{pmatrix}A\\ h\\ \Delta\end{pmatrix},\qquad \delta_{\mathrm{CP}}=9A.$$

The exact linear falsifier is

$$3857\,\delta_{\mathrm{CP}}=9\big(1528\theta_{12}+3380\theta_{23}+801\theta_{13}\big).$$

From the three angles and δ_{CP}, the unitary matrix V_{CKM} is constructed. Its function in the mass law is not to produce eigenvalues, but to transport the down operators to the up basis:

$$B_d^{(u)}=V_{\mathrm{CKM}}B_dV_{\mathrm{CKM}}^{*}.$$

92.4.7.6 Jarlskog Invariant

The CP violation invariant is

$$J_{\mathrm{CKM}}=c_{12}c_{23}c_{13}^2s_{12}s_{23}s_{13}\sin\delta_{\mathrm{CP}},$$

and the internal evaluation is

$$J_{\mathrm{CKM}}=3.1189723326632947\times10^{-5}.$$

Equivalently, the compatibility between spectra and mixing may is read in the determinant of the commutator of the operators Hermitian up/down. A value not null of J_{CKM} records orientation physical not removable; is not a corrector of mass added to each quark.

92.4.8 Gauge and scalar sectors

92.4.8.1 Chart of the null sector: mass routes 0*, gauge representations, and persistence*

Let

$$\mathcal{A}_{\mathrm{sp}}=\mathbb{R}[R]/(R^2-1),\qquad P_{\pm}=\frac{1\pm R}{2}.$$

For

$$x=r_+P_++r_-P_-,$$

the split norm is

$$N_{\mathrm{sp}}(x)=r_+r_-.$$

The null section is a direction of zero norm. Its mass is exactly

$$m=0,$$

without passing through a limit of positive characters and without assigning $\kappa=0$. Nullity belongs to the type of the section.

92.4.8.2 Realization of the photon in the electromagnetic null sector: helicity, gauge, and character

The photon is the transverse vector realization of a null gauge section. The physical projector must leave exactly two polarizations:

$$\operatorname{rg} P_\gamma^{\mathrm{trans}} = 2, \qquad P_\gamma^{\mathrm{trans}} k = 0.$$

The rest mass is

$$m_\gamma = 0.$$

Bidirectional propagation may carry an oriented anisotropy.

$$\epsilon = q_-^{90} - q_+^{90}, \qquad c_\pm = \frac{c}{1 \pm \epsilon},$$

while the harmonic mean preserves

$$\frac{2}{1/c_+ + 1/c_-} = c.$$

Thus, the outward and return paths may differ without altering the two–way invariant. This bias is not a photon mass.

A distinct Proca/Stückelberg realization appears if a spectral gap in the genealogical record satisfies

$$\omega^2 = k^2 + m_T^2$$

and there exists a dimensional entangler

$$m_T \longmapsto m_\gamma.$$

The induced delay would obey

$$\frac{\Delta T}{T} \simeq \frac{1}{2} \left(\frac{m_\gamma c^2}{\hbar \omega} \right)^2 .$$

The null Maxwell sector and the massive Proca realization are distinct branches; neither invalidates the other.

92.4.8.3 Chromodynamic realization of the null sector: gluon octet and adjoint representation of $SU(3)$

The residual trit generates the color space, whose traceless part realizes

$$\mathfrak{sl}_3 \longrightarrow \mathfrak{su}_3.$$

The gluon occupies the eight-dimensional adjoint representation. As long as color symmetry remains exact,

$$m_g = 0.$$

Confinement, screening, and the collective spectral gap are properties of the interacting spectrum; they do not constitute an elementary gluon pole mass. An effective mass must specify the observable, gauge, regime, and operator that defines it.

92.4.8.4 Realization of the bosons $W^{\pm}$: charge, polarization, and electroweak character

The sector W is a charged vector block with conjugation

$$C_M : W^+ \longleftrightarrow W^-.$$

Its historical footprint is

$$15/37, \qquad \text{east–west axis}, \qquad (n_A, s_C) = (94, -1),$$

and the enriched signature declared is

$$(94, -1, 0, -2, 4).$$

The corresponding evaluation is

$$80.37896490970455\ldots \text{ GeV}.$$

The realization complete is the pole of the resolvente of the block $P_W \widehat{M} P_W$; its part imaginaria determines the width. The degeneration of W^+ and W^- proceeds of the conjugation of the operator.

92.4.8.5 Realization of the Z^0 boson: neutral mixing, polarization, and electroweak character

The Z^0 is the neutral vector block of the electroweak coupling. Its historical trace is

$$20/23, \qquad \text{east–west axis}, \qquad (n_A, s_C) = (95, -1),$$

and the enriched signature

$$(95, -1, -11, 0, -4).$$

The evaluation of the character gives

$$91.18753344431341\ldots \text{ GeV}.$$

The neutral projector must be constructed before the pole chart; mass and width are two coordinates of the same pole, not a single figure.

92.4.8.6 Realization of the Higgs sector: scalar mode, coupling and mass character

The Higgs is a persistent scalar section of the gauge block with declared CP parity. Its trace is

$$24/29, \qquad \text{north–south axis}, \qquad (n_A, s_C) = (97, 9).$$

The conditional evaluation of the character produces

$$125.2924660425361\ldots \text{ GeV}.$$

The HMT definition does not identify “scalar” with “mass”: zero spin fixes the rotational representation; mass follows from the spectrum of the coupled scalar block. The width and decay channels require the temporal operator.

92.4.8.7 The anomalous magnetic moment of the muon

The internal functional

$$a_\mu = \frac{\alpha_{\mathrm{HMT}}}{2\pi} + \frac{\alpha_{\mathrm{HMT}}(\varphi - 1)}{1000} + \frac{\alpha_{\mathrm{HMT}}^2}{1000(54+1)}$$

gives

$$a_\mu = 0.001165920713008014\ldots,$$

and

$$g_\mu = 2(1 + a_\mu) = 2.002331841426016028\ldots.$$

The three terms correspond, respectively, to the basal rotation, the oriented return in the completion 1000, and the quadratic correction mediated by the defect 54. This observable verifies the internal structure of the muonic sector but does not by itself select its mass route.

92.4.9 Construction of composite particles through non-additive composition of paths and bonds

92.4.9.1 Composition operator

Route composition is not strictly additive because boundary, temporal order, and carry interact. For two ledgers,

$$\mathcal{L}_1 \star \mathcal{L}_2 = \mathcal{L}_1 + \mathcal{L}_2 + \mathfrak{b}(\mathcal{L}_1, \mathcal{L}_2),$$

where $\mathfrak{b}$ is the binding cocycle. Associativity requires

$$\mathfrak{b}(L_1, L_2) + \mathfrak{b}(L_1 \star L_2, L_3) = \mathfrak{b}(L_2, L_3) + \mathfrak{b}(L_1, L_2 \star L_3).$$

The mass reader acts after this composition:

$$(\Gamma_i, \mathcal{T}, \partial) \xrightarrow{\mathfrak{C}_{12}} \mathcal{L}_{\mathrm{comp}} \xrightarrow{L_{\mathrm{tick}}} \sigma_{\mathrm{comp}} \xrightarrow{\chi_{\mathrm{full}}} \widehat{M}_{\mathrm{comp}}.$$

92.4.9.2 Mesonic Realization via Quark–Antiquark Composition and Cocylic Link Law

The mesonic domain contains 432 oriented expressions.

$$M(q, q') = q \otimes (q')^\vee$$

with color compatible. The physical projector imposes color singlet and fixes

$$(J^{PC}, I, I_3, S, C, B, \ldots),$$

together with the temporal linking tree. The mass of a meson is not $m_q + m_{\bar{q}}$; it is an eigenvalue of the composite ledger.

The six historical control evaluations include

$$m_{\pi^0} = 134.9767243278721\ldots \text{ MeV},$$
$$m_{\pi^\pm} = 139.5704851715722\ldots \text{ MeV},$$
$$m_{K^\pm} = 493.6770856495306\ldots \text{ MeV},$$
$$m_{K^0} = 497.6111864977505\ldots \text{ MeV}.$$

These figures are exact evaluations of declared records. Genealogical closure state by state requires regenerating each record by means of $\mathfrak{C}_{12}$, including its binding boundary.

92.4.9.3 Baryonic realization via trilinear composition of quark paths and color neutralization

The baryonic domain contains 1728 RGB expressions. The physical projector must simultaneously impose:

1. color singlet;
2. fermionic statistics;
3. flavor and isospin;
4. spin and symmetrization;
5. triple tree with carry and memory.

For proton and neutron, the historical signatures

$$\sigma_p = (59, 0, 9, 1, 0), \qquad \sigma_n = (59, 1, -34, 0, 1)$$

give

$$m_p = 938.2717515469849\ldots \text{ MeV},$$
$$m_n = 939.5658104725070\ldots \text{ MeV}.$$

Apparent precision does not replace derivation of the genealogical binding record. The sign and magnitude of

$$m_n - m_p$$

must follow from orientation, isospin, vacancies, and the linking boundary before external comparison is opened.

92.4.9.4 Excited states and hadronic resonances

A composite identity may possess several poles. The physical state is individualized by

$$(\text{constituents}, \mathcal{T}, \partial, J^{PC}, I, \text{radial}, \text{orbital}, \text{channel}).$$

The same flavor composition is not sufficient to distinguish excitations. Each pole

$$z_k = M_k - \frac{i}{2}\Gamma_k$$

must proceed from an eigenvector of the composite operator and from a decay channel. This extends the law from a few stable masses to the entire spectrum of mesons and baryons without converting the external catalog into a generator.

92.4.9.5 Comprehensive inventory of widths and transition rates of the atlas realizations

Let P be the prepared subspace and $Q = I - P$. The effective Hamiltonian is

$$H_{\text{eff}}(z) = PHP + PHQ(z - QHQ)^{-1}QHP.$$

The poles of

$$\det(z - H_{\text{eff}}(z)) = 0$$

determine masses and widths. The mass character supplies the basal real part of the operator; the self-energy term of Q supplies shift and decay. Therefore, the 384 widths of the inventory require a channel reader in addition to the reader of the 471 masses.

92.4.10 Topological sectors and hypothetical particles

92.4.10.1 Fibonacci topological sector: ring-based, fusion, and braid representation

The areal–volumetric incidence

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$$

realize the ring

$$\tau^2 = 1 + \tau, \qquad \mathrm{FPdim}\,\tau = \varphi.$$

The primary output is a fusion rule and a quantum dimension. A Fibonacci anyonic particle also requires F and R matrices, the pentagon and hexagon equations, chirality, a spectrally gapped Hamiltonian, and a map from the TPK state to physical space. Only then can one define

$$m_{\mathrm{brechaespectral}} = \frac{\Delta E}{c_{\mathrm{eff}}^2}.$$

92.4.10.2 Ising–Majorana Sector: Simple Objects, Fusion Rules, and Braid Representation

The Ising category satisfies

$$\psi^2 = 1, \qquad \psi\sigma = \sigma, \qquad \sigma^2 = 1 + \psi, \qquad d_\sigma = \sqrt{2}.$$

Four levels must be distinguished:

1. the route involution $C_M = -\,\mathrm{rev}$;
2. the parity of the CAR completion;
3. a self-conjugate relativistic spinor;
4. a topological Ising zero mode.

A relativistic Majorana fermion requires a real spinor representation, a dynamics, and a mass term. A Majorana zero mode is characterized by a spectral gap, splitting, and protection; it has no universal mass deducible solely from the fusion ring. The identity of a route with its conjugate is a necessary, but not sufficient, condition for a Majorana realization.

92.4.10.3 Braid representation on $\mathbb{Z}/6\mathbb{Z}$ and topological characters

Let

$$w_6 : B_n \longrightarrow \mathbb{Z}/6, \qquad \chi_k(b) = e^{2\pi i k w(b)/6}.$$

The representations

$$\rho_q(\sigma_i) = qP_i$$

descend to the symmetric group only when $q^2 = 1$. The other four values are genuinely braided. Their mathematical existence does not imply four particles: a physical realization requires a Hamiltonian, a spectral gap, braid operators, stability, and an intertwiner with a multisection.

92.4.10.4 Impossibility of a Persistent Tachyon

The action line is positive,

$$\hbar_{\text{int}} > 0,$$

the character satisfies

$$\chi_{\text{full}} > 0,$$

and the beta operator is obtained by positive conjugation:

$$\widehat{M}^{\beta} = R_{\text{act}}^{\widehat{K}/2}\, \widehat{M}^{(0)}\, R_{\text{act}}^{\widehat{K}/2} \geq 0.$$

Therefore,

$$\operatorname{Spec}(\widehat{M}^{\beta}) \subset [0, \infty).$$

A persistent stable section cannot generate

$$m^2 < 0.$$

If the linearization of a state produces negative curvature or an imaginary frequency, that state is an instability that leaves the section, not a superluminal particle. If the prolongation terminates, it is a virtual section with a terminal obstruction. Thus, "tachyon" decomposes into two well-typed HMT objects —instability or virtuality— and does not constitute a persistent species.

92.4.10.5 Gravity without a mandatory elementary spin 2 *particle*

The gravitational realization follows the chain

$$\text{route holonomy} \longrightarrow (e, \omega) \longrightarrow (T, R) \longrightarrow S_{\text{PCH}},$$

where e is the costructure, ω the connection,

$$T = de + \omega \wedge e, \qquad R = d\omega + \omega \wedge \omega,$$

and S_{PCH} is the Palatini–Cartan–Holst action. Gravity appears as the global torsion-and-curvature response of the persistent network. The collective vibration of the model universe may have a spin-two tensor component, but HMT need not posit an additional elementary particle to carry gravity.

The internal gravitational constant is typed through

$$G_{\text{HMT}} = \frac{c_{\text{int}}^3}{\hbar_{\text{int}}} \big(\delta_\theta \Lambda_G\big)^2,$$

where δ_θ is the elementary angular torsion and Λ_G is the spectral length selected by the gravitational holonomy. This equation converts torsion and route scale into gravitational coupling without using a mass as normalization. The total vibration is represented by the Hessian

$$\mathcal{H}_{\text{univ}} = \operatorname{Hess}_{(e,\omega)}\, S_{\text{PCH}}$$

over the global state. Its modes are collective excitations of the model universe; a tensorial component is not automatically a separate particle.

An excitation that is to be called a "graviton" must demonstrate three things:

1. a persistent tensorial subspace;

2. an isolated pole of the linearized operator;
3. a physical coupling that does not duplicate the torsion already present.

Without those three conditions, the correct object is the collective mode of the geometry, not a new mass row.

92.4.10.6 Barbero-Immirzi and the hexad-octad link

With

$$A = 1000\alpha_{\mathrm{HMT}}, \qquad q_{\pm} = \exp\Big[-\frac{\pi}{180}(A \pm C^{*})\Big],$$

the hexadic-octadic functional uses two distinct incidences of the temporal towers. The combination

$$\varepsilon_{\mathrm{BI}} = 12\Delta S_{90} - \Delta S_{120}$$

and the angular term produce

$$\gamma_{\mathrm{BI}} = \frac{\pi}{180}AC^{*} + \varepsilon_{\mathrm{BI}} = 0.2740076726944593\ldots.$$

The weight $12 : -1$ comes from reading the twelve dodecaphasic sectors and the unique oriented return seam. The equation $4a - 3b = 45$ subsequently verifies that the pair $(12, 1)$ is the minimal positive integer solution. This object must not be confused with the dodecaphasic mass extractor or with the angular residue 90–120; it shares the harmonic operator, but has a different domain and function.

92.4.10.7 Extensions of the atlas toward dark sectors: domains, readers and physical status

A dark sector HMT is a persistent multisection with

$$P_{\mathrm{vis}}P_{\mathrm{dark}} = 0$$

for the acoplamientos electromagnetico and cromatico, but with posible incidencia gravitatoria or of boundary:

$$P_{\mathrm{grav}}P_{\mathrm{dark}} \neq 0.$$

Its "darkness" is a property of the coupling kernel, not a mass label. The law first predicts an operator and its spectrum; only a physical channel determines whether the mode is stable, metastable, or resonant. This definition makes it possible to search for invisible multisections without inventing a mass from the observed value of a cosmological density.

92.4.10.8 Sterile neutrino section: compatibility of fiber, mixing, and absence of gauge charge

A sterile neutrino is not the depth G_4. It is an eigenvector $n_s \in \mathcal{H}_\nu$ satisfying

$$P_{\mathrm{weak}}n_s = 0, \qquad P_{\mathrm{mass}}n_s = n_s.$$

It may mix with the three active modes through an enlarged neutral kernel, but its existence requires

$$\dim\ker(P_{\mathrm{weak}}|_{\mathcal{H}_\nu}) > 0$$

and an eigenvalue of mass in ese kernel. This definition produces a criterion of search internal: diagonalize first the block neutral and check afterward its coupling weak.

92.4.10.9 Axionic and pseudoscalar sections: topological character and edge coupling

A pseudoscalar axion candidate is a section whose orientation changes under CP,

$$CP\, a\, CP^{-1} = -a,$$

and whose displacement compensates a phase topological of the sector of color. In HMT must correspond to a way of torsion/holonomy odd and to a projector that intertwines ese way with the density topological of gauge. Its mass is obtained of the second derivative of the potential effective,

$$m_a^2 = \left.\frac{\partial^2 V_{\text{eff}}}{\partial a^2}\right|_{a=a_0},$$

once V_{eff} has been constructed from the genealogical record of vacancies and boundary. The mere existence of a channel $C^* \mapsto -C^*$ is not enough to assert an axionic particle.

92.4.10.10 Dark Photon and Hidden Massive Vector

A dark photon requires a second gauge section A' and a mixing coupling

$$\mathcal{L}_{\text{mix}} = -\frac{\varepsilon_{\text{kin}}}{2} F_{\mu\nu} F'^{\mu\nu}.$$

In terms of fibers, ε_{kin} is an intertwiner between two null subspaces. If the second sector acquires a Stueckelberg or boundary spectral gap,

$$\widehat{M}_{A'}^2 \neq 0,$$

a massive dark vector appears. HMT predicts that the mixing parameter must follow from a common memory incidence; it cannot be fixed by the mass it is intended to explain.

92.4.10.11 WIMP and stable dark state

A WIMP is a dark sector that also satisfies:

$$[P_{\text{weak}}, P_X] \neq 0, \qquad [P_{\text{em}}, P_X] = [P_{\text{color}}, P_X] = 0,$$

and possesses a stable mode

$$|\lambda_X| = 1.$$

Stability may be protected by a parity of the genealogical record. Its mass is an eigenvalue of the weakly coupled dark block. "WIMP" does not constitute a new mathematical family: it is a joint condition on coupling and persistence over a multisection.

92.4.10.12 Monopole Section: Magnetic Charge, Dirac Condition, and Topological Compatibility

A monopole is defined by a non-trivial topological class of the gauge bundle:

$$\frac{1}{2\pi} \int_{S^2} F = n \neq 0.$$

HMT must realize this integer as the net holonomy of a closed family of routes. Its mass, if the mode is dynamical and stable, belongs to the spectrum of the topological-defect operator. A point vacancy or a nonzero carry is not by itself a monopole; the descent to the integral class must be proved.

92.4.10.13 Supersymmetric Extensions of the Atlas: Representation Matching and Characters

An HMT supersymmetry would require an odd operator Q that exchanges the exterior and symmetric completions:

$$Q : \mathcal{F}_- \leftrightarrow \mathcal{F}_+, \qquad Q^2 = H,$$

and that interweaves the law of masses,

$$Q\widehat{M}_- = \widehat{M}_+ Q.$$

Only then would the bosonic and fermionic spectra be paired. The separate existence of CAR and CCR does not establish supersymmetry or by itself predict selectrons, neutralinos, or gluinos. Each partner requires a nonzero intertwiner and a symmetry-breaking rule that determines its mass splitting.

92.4.10.14 Tetraquarks, pentaquarks, glueballs, and hybrids

The composition grammar is extended without changing the law:

$$\mathfrak{C}_{12}^{(n)} : \Gamma_1 \times \cdots \times \Gamma_n \times \mathcal{T}_n \times \partial_n \longrightarrow \mathcal{L}_{\text{comp}}.$$

A tetraquark uses $qq\bar{q}\bar{q}$; a pentaquark, $qqqq\bar{q}$; a glueball, records of the color adjoint projected onto the singlet; a hybrid, quark and gauge routes in the same tree. The physical classification depends on the color, spin, parity, and linkage-tree projector, not merely on the list of constituents.

92.4.10.15 Dilaton and radion sectors: volume and compactification scalar modes

The dilaton is a way of the section of scale; the radion, a way that modifies a length or volume internal. In the theory of mass appear as fluctuations of the chart and of the memory geometric:

$$t_0 \mapsto e^{\phi_d} t_0, \qquad \mathcal{V}_{\text{int}} \mapsto e^{\phi_r} \mathcal{V}_{\text{int}}.$$

To become particles they must possess a kinetic term, a potential, a persistent projector, and a pole. So long as they do not possess them, they are collective degrees of the dimensional chart, not additional scalar species.

92.4.10.16 Exclusion of Physical Ghosts

The spectral measure of a physical state is positive:

$$\mu_{P,a}^{\text{mass}}(B) \geq 0.$$

A negative-norm direction does not belong to the prepared state space. If a parametrization introduces a "ghost," the physical realization must quotient out the gauge redundancy or reject the coupling. This exclusion is independent of the tachyonic prohibition: a ghost violates positivity of the norm; a persistent tachyon would violate positivity of the mass operator.

92.4.11 Universal Inventory of Masses and Widths

92.4.11.1 Posterior correspondence between the HMT–MD realizations and the external physical census

The contrast inventory contains

$$1170 \text{ identities}, \qquad 438 \text{ identity groups},$$
$$471 \text{ mass observables}, \qquad 384 \text{ width observables}.$$

Its partition is

Domain	Masses	Widths
partículas generales y leptónicas	53	9
quarks	8	1
mesones	243	225
bariones	166	149
registro gravitatorio de contraste	1	0
Total	**471**	**384**

Las 471 masas no son 471 especies elementales: incluyen estados excitados, resonancias, distintas cargas y múltiples realizaciones compuestas. Las 384 anchuras son observables temporales y no deben ser leídas por el carácter de masa.

92.4.11.2 Identidad interna $\mathcal{I}_X$ de una realización material y criterio de individualización frente a observaciones múltiples

Cada estado comparado debe poseer una identidad interna

$$\mathcal{I}_X = (C_X, \mathcal{R}_X, \Gamma_X, \mathcal{L}_X, \sigma_X, \eta_X, P_X, \mathfrak{M}_X, \mathcal{T}_X),$$

where:

- C_X is the physical type of realization determined by object, representation, and codomain;
- $\mathcal{R}_X$ is the physical representation;
- Γ_X is an oriented route, orbit, or composition;
- $\mathcal{L}_X$ is the genealogical record;
- σ_X is the complete signature;
- η_X is the tower;
- P_X is the coupling projector;
- $\mathfrak{M}_X$ is the measurement channel;
- $\mathcal{T}_X$ is the composition tree when applicable.

Two external observations belong to the same HMT state when they share this identity and differ only in experiment, chart, or estimator. The existence of two measurements does not create two particles.

92.4.11.3 Comparison morphism between internal invariants and metrological observables

The comparison requires the diagram

$$\mathcal{L}_M^{\mathrm{HMT}} \xrightarrow{\varsigma_{\mathcal{S},\mu}} \mathbb{R}_{>0} u_M^{\mathcal{S},\mu} \xleftarrow{\text{measurement}} O_X^{\mathrm{ext}}.$$

A scientifically complete row contains:

$$(m_X^{\mathrm{HMT}}, m_X^{\mathrm{ext}}, u, \mathcal{S}, \mu, \Sigma_{\mathrm{exp}}, \Sigma_{\mathrm{HMT}}, \text{pole convention}).$$

For quarks are mandatory scheme and scale; for resonances, the convention of pole; for limits, the level of confidence; for observables correlated, the matrix of covariance.

92.4.11.4 Explicit evaluation of 17 *representative realizations of the atlas*

The following evaluations preserve the separation between law and signature data:

Status	Value of the character in its chart
μ	105.658360294435829 . . . MeV
τ	1776.8609176314323 . . . MeV
u	2.165924536173907 . . . MeV
d	4.679550162948874 . . . MeV
c	1.270500838641911 . . . GeV
s	92.94009390784564 . . . MeV
t	172.5974795440191 . . . GeV
b	4.190119763299790 . . . GeV
W	80.37896490970455 . . . GeV
Z	91.18753344431341 . . . GeV
H	125.2924660425361 . . . GeV
π^0	134.9767243278721 . . . MeV
$\pi^\pm$	139.5704851715722 . . . MeV
$K^\pm$	493.6770856495306 . . . MeV
K^0	497.6111864977505 . . . MeV
p	938.2717515469849 . . . MeV
n	939.5658104725070 . . . MeV

The beta electron does not belong to this table because its central route, basal coefficients, and complete reader are determined by the construction itself.

92.4.11.5 From the catalog of realizations to the inventory of 471 masses

The universal extension follows five operations:

$$\begin{aligned}\text{physical identity} &\longrightarrow \text{representation projector} \longrightarrow \text{elementary or composite genealogical record}\\ &\longrightarrow \text{operator/pole} \longrightarrow \text{comparison chart.}\end{aligned}$$

For elementary states, the second map acts on a fiber of the atlas. For mesons and baryons, it acts on the 432 and 1728 composite expressions. For resonances, the fourth map returns a complex pole. For topological sectors, it first returns a spectral gap and a braid representation. This typing makes it possible to encompass the entire inventory without declaring that each external name corresponds to an elementary cell.

92.4.11.6 From the 384 widths to the survival operator

Each width requires

$$X \longmapsto Q_X U Q_X \longmapsto \rho(Q_X U Q_X) \longmapsto \Gamma_X$$

or, equivalently, the pole of H_{eff}. The reader must incorporate decay channels, multiplicity, threshold, and phase space. The equality

$$\Gamma_{\bar{X}} = \Gamma_X$$

It follows only if the decay operator is covariant under conjugation.

92.4.12 Mass, temperature, and statistics

92.4.12.1 Thermal Transducer Between Mass Character, Temperature, and Statistical Occupation

The Boltzmann constant is the transducer

$$k_B : \mathcal{L}_\Theta \longrightarrow \mathcal{L}_E.$$

The clock CT–108 fixes the identity

$$k_B \Theta_{\mathrm{clk}} \log 3 = \hbar_{\mathrm{int}} \frac{2\pi}{108 t_0}.$$

This equation links a unit of tritic information, a frequency, and an energy. It does not introduce the SI coordinate of k_B into the law; it determines the typed product once the thermal chart has been chosen.

92.4.12.2 Fermionic projection of the atlas and Fermi–Dirac distribution

For levels

$$\epsilon = (0, 3, 6, 9), \qquad g = (1, 6, 12, 8),$$

and constraints

$$(N, E) = (12, 66),$$

the exterior completion produces

$$2\,104\,494$$

compatible fermionic microstates. The effective multipliers are

$$(\beta, \mu) = (0.1530701135,\ 4.5028651976).$$

This realization exemplifies how mass, energy, and exclusion emerge from fermionic routes without statistics being an additional adjustment.

92.4.12.3 Bosonic projection of the atlas and Bose–Einstein distribution

On the same levels and restrictions, the symmetric completion produces

$$259\,436\,430$$

microstates, with

$$(\beta, \mu) = (0.0545672796,\ -15.9256978191).$$

The difference between Fermi–Dirac and Bose–Einstein is, therefore, a difference in the rule of composition of routes, not in the value of a mass individual.

92.4.12.4 Criterion of condensation of bosonic sections and macroscopic occupation

A bosonic condensate appears when a symmetric component acquires macroscopic occupancy at the lowest eigenvalue of the operator. A Fermi sea, by contrast, fills a set of distinct eigenvalues up to a boundary. HMT describes both through the same mass/energy spectrum and two different Fock functors. This distinction prevents “many particles on one route” from being identified with a more massive particle.

92.4.13 Differential propagation of uncertainties and structural stability of the mass character

92.4.13.1 Invariance of the generative graph under posterior metrological contrast

The internal output is frozen in order

$$\mathcal{R}_{324} \to P_X \to \mathcal{L}_X \to \sigma_X \to \eta_X \to \widehat{M}_X \to \mu_X.$$

Only then is the external chart opened. An arbitrary permutation of the 471 masses or the 384 widths must leave all preceding objects invariant.

92.4.13.2 Decomposition of structural uncertainty into experimental, chart, route, tower, and coupling contributions

The total uncertainty decomposes as

$$\Sigma_{\mathrm{tot}} = \Sigma_{\mathrm{ext}} + \Sigma_{\mathrm{carta}} + \Sigma_{\mathrm{route}} + \Sigma_{\mathrm{tower}} + \Sigma_{\mathrm{coupling}}.$$

Here:

- Σ_{ext} is the covarianza experimental;
- Σ_{carta} quantifies scheme, scale, and unit conversion;
- Σ_{route} compares all routes of the admissible orbit;
- Σ_{tower} uses the tail bound for the harmonic series;
- $\Sigma_{\mathrm{coupling}}$ measures the variation among admissible intertwiners.

They are not necessarily added as independent numbers; the equation expresses a decomposition of sources that must be converted into a joint covariance.

92.4.13.3 Derivative of the mass observable with respect to the signature and the readers

For

$$\log m_X = \log m_\star + \log \chi_0(\sigma_X) + \kappa_X \log R_{\mathrm{act}} + \sum_h \eta_{X,h} s_h,$$

the variation is

$$d\log m_X = d\log m_\star + \langle \Theta_0, d\sigma_X \rangle + \langle \sigma_X, d\Theta_0 \rangle + \log R_{\mathrm{act}}\, d\kappa_X + \kappa_X\, d\log R_{\mathrm{act}} + \sum_h (s_h\, d\eta_{X,h} + \eta_{X,h}\, ds_h).$$

This formula makes it possible to attribute each decimal to a route, constant, reader, tower, or chart. A result is robust when perturbations that preserve the premises maintain its spectral class and its calculated uncertainty.

92.4.13.4 Structural-dependence tests through controlled suppression of sheet, orientation, carry, memory, winding, and boundary

A dependency test separately removes the sheet, orientation, carry, memory, winding, or boundary and recomputes the object. The expected result is not that everything changes, but that exactly the coordinates dependent on the removed component change. If suppressing memory leaves invariant a coefficient defined by return, the construction of that coefficient is refuted.

92.4.13.5 Obstructions to the identification of routes, characters, and metrological magnitudes

The law is refuted in a specific application if any of the following occurs:

1. an external mass changes the route or the chosen projector;
2. the published signature is not regenerated from the genealogical record;
3. the refinement fails to converge or depends on a truncation selected by the residual;
4. two equivalent routes receive distinct masses without symmetry breaking;
5. an antiparticle acquires a different mass without a declared violation of conjugation;
6. a colour triplet splits without colour breaking;
7. a scalar is published even though the spectral variance is positive;
8. a width is obtained from the mass character without a temporal operator;
9. a quark chart omits the scheme or scale;
10. an instability is interpreted as a persistent tachyon.

92.5 Leptons, neutrinos, quarks and gauge sectors

Charged leptons, the six quark flavors, and the neutrino fibers are realized as typed multisections. Color is the residual ternary orbit $12 + 12 + 12$; flavor, generation, scheme, and scale are distinct data. Neutrinos belong to the fermionic completion; the Majorana condition additionally requires self-conjugation and a compatible mass term. The photon and gluon belong to the exact null chart.

92.6 Correspondence between internal classes and physical sectors

The physical realization preserves the preceding quotients and adds the data required by a representation without turning them into new cells. For a persistent section X, we write in typed form

$$\mathcal{P}_X = \big[q(X),\ [X]_{\mathcal{R}},\ \mathrm{Rep}(X),\ \gamma_X,\ \tau_{12}(X),\ \sigma_X\big].$$

Here $q(X)$ fixes the multisection, $[X]_{\mathcal{R}}$ the route family, $\mathrm{Rep}(X)$ the representation, γ_X the persistent history, $\tau_{12}(X)$ its dodecaphase word, and σ_X the integral signature extracted after the record. The levels $G0, \dots, G4$ remain depths of the atlas; they are not renamed as the three physical generations.

Flavor information is preserved through three distinct coordinates:

$$\mathrm{flav}(X) = \big(\sigma_{ud}(X), g(X), \chi_{\mathrm{fina}}(X)\big).$$

The first separates the leptonic, neutrino, up-type, and down-type branches; the second distinguishes the physical generation; the third preserves the orientation and memory that separate sections within a single branch. For quarks, the residual orbit $R/G/B$ is added after the flavor is fixed and before the character is evaluated. Color, flavor, and mass are therefore readings of different types.

This census does not exhaust the realizations of the law. Mesons and baryons are constructed by composing enriched records before projecting the signature; resonances and virtual particles are distinguished by their persistence horizon. The Fibonacci, Z_6, and Majorana–Ising sectors retain their own topological codomains and are not forced into an ordinary mass table.

92.6.1 Path signatures and dodecaphasic coindexed pairs by flavor

For the set of flavors $Q = \{u, d, c, s, t, b\}$, the table defines a finite correspondence between a geometric fingerprint and a dodecaphasic pair. In each row, the following holds

$$W_{\pm} = 6n_A \pm s_C.$$

This table is not an a posteriori comparison, but the graph of a finite map over the certified nominal domain. If $\mathscr{R}_Q^{\mathrm{nom}} = \{\rho_u, \rho_d, \rho_c, \rho_s, \rho_t, \rho_b\}$ is the set of CT–108 traces that follow, we define

$$H_{\mathrm{flav}} : \mathscr{R}_Q^{\mathrm{nom}} \longrightarrow \mathbb{Z}^2, \qquad H_{\mathrm{flav}}(\rho_q) = (n_A(q), s_C(q)).$$

The six images are unique, the table exhausts the domain, and no column contains a mass. Therefore, H_{flav} is a closed causal extractor on the nominal domain: it first reads footprint, orientation, and memory, and only thereafter does the character evaluate the mass. The ternary color orbit is quotiented without changing H_{flav}; reversal preserves the mass class and transports the particle–antiparticle representation. The table is therefore the executable definition of the morphism and not a future obligation:

Flavor	Generation	Giros/cambios	Eje	N, E, S, O	(n_A, s_C)	(W_+, W_-)
u	1	21/30	N–S	34,31,31,12	(11,7)	(73,59)
d	1	20/32	E–O	27,35,18,28	(17,8)	(110,94)
c	2	21/29	E–O	20,38,24,26	(61,8)	(374,358)
s	2	17/27	N–S	50,32,12,14	(41,-3)	(243,249)
t	3	21/26	N–S	46,27,16,19	(100,-1)	(599,601)
b	3	21/30	E–O	12,63,9,24	(71,-5)	(421,431)

The coincidence of two turns or of a spectral height does not identify species: the cardinal route, the orientation, the dodecaphasic register, and the representation remain in the object. The chain cell–route–enriched chronological register–signature–character is defined independently of H_{flav}. The four initial orientations of each cell produce $81 \cdot 4 = 324$ executable genealogies; that cardinal counts histories, not particles, families, or masses.

92.6.2 Generational stratification, quark supports and neutrino realization

The general law acts on multisections rather than on a list of masses. The appearance of three generations is therefore expressed as the compatibility of three structures: the stratification of the atlas, the flavour records, and the three-dimensional mixing space. In the charged leptonic sector, the multisections

$$G_0^{\ell}, \qquad G_2^{\ell}, \qquad G_4^{\ell} \tag{92.1}$$

have ranks $1, 8, 16$. The centre G_0^{ℓ} contains the rank-one electronic section; G_2^{ℓ} and G_4^{ℓ} instead preserve nontrivial operator spectra. The complete generational organization is

Generation	**Lepton**	**Par quark**	**Sabor neutro**	**Soporte estructural**
1	e	(u, d)	ν_e	centre G_0^{ℓ}, first-generation register and first vector of the interaction basis
2	μ	(c, s)	ν_μ	multisection G_2^{ℓ}, second-generation register and second vector of the interaction basis
3	τ	(t, b)	ν_τ	multisection G_4^{ℓ}, third-generation record, and third vector of the interaction basis

Particle–antiparticle conjugation reverses the charge and orientation coordinates of the representation but preserves this stratification; therefore, it does not double the number of generations.

Dimension three also has its own incidence carrier. The oriented dodecaphasic representation contains

$$V_{12} \cong V_1 \oplus V_3 \oplus V_{3'} \oplus V_5, \qquad \operatorname{rank} P_3 = 3, \tag{92.2}$$

and the dodecaphasic entangler–Witt transports the positive sector according to

$$P_G \xleftarrow{W_{GK}} P_3 \xleftarrow{U_W} Q_3, \qquad \operatorname{rank} P_G = \operatorname{rank} P_3 = \operatorname{rank} Q_3 = 3. \tag{92.3}$$

This three-dimensional carrier permits comparison among generational bases. The routes, signatures, and mass operators subsequently provide the spectral content transported by CKM and PMNS.

92.6.2.1 Internal position and records of the 6 *quark flavors*

The position of a quark flavor is the composite data

$$\mathfrak{q}_X = (\text{charge branch}, \text{generation}, (\mathbb{Z}/3\mathbb{Z})_{\text{color}}, \text{multisection}, \text{route record}). \tag{92.4}$$

The charge branches $-1/3$ possess the supports

$$\begin{aligned} D_1 &= \{(4,5),(6,5)\},\\ D_2 &= \{(4,3),(4,7),(6,3),(6,7)\},\\ D_3 &= \{(2,3),(2,5),(2,7),(8,3),(8,5),(8,7)\},\\ D_4 &= \{(2,1),(2,9),(4,1),(4,9),(6,1),(6,9),(8,1),(8,9)\}, \end{aligned} \tag{92.5}$$

whereas the charge branches $+2/3$ occupy

$$\begin{aligned} U_1 &= \{(4,4),(4,6),(6,4),(6,6)\},\\ U_3 &= \{(2,2),(2,4),(2,6),(2,8),(4,2),(4,8),(6,2),(6,8),(8,2),(8,4),(8,6),(8,8)\}. \end{aligned} \tag{92.6}$$

Each support lifts to four orientations per cell. The colour quotient organizes $36 = 12_{\mathrm{R}} + 12_{\mathrm{G}} + 12_{\mathrm{B}}$ oriented biographies and preserves the same mass per flavour before the colour realization.

The nominal transducer

$$\mathcal{N}_{\text{ruta}}(\Gamma_f) = ((N,E,S,O), h_{369}, r_5) \in \mathbb{Z}^4_{\geq 0} \times \mathbb{Z}^2_{\geq 0} \tag{92.7}$$

publish the following coordinates before any scalar realization:

Flavour	Generation	Branch	(N,E,S,O)	(h_{369}, r_5)	Eje dominante
u	1	$+2/3$	$(34,31,31,12)$	$(51,13)$	N–S
d	1	$-1/3$	$(27,35,18,28)$	$(47,17)$	E–O
c	2	$+2/3$	$(20,38,24,26)$	$(49,16)$	E–O
s	2	$-1/3$	$(50,32,12,14)$	$(57,11)$	N–S
t	3	$+2/3$	$(46,27,16,19)$	$(57,9)$	N–S
b	3	$-1/3$	$(12,63,9,24)$	$(47,7)$	E–O

The associated nominal signatures are

Flavor	(n_A, s_C)	(W_+, W_-)
u	$(11,7)$	$(73,59)$
d	$(17,8)$	$(110,94)$
c	$(61,8)$	$(374,358)$
s	$(41,-3)$	$(243,249)$
t	$(100,-1)$	$(599,601)$
b	$(71,-5)$	$(421,431)$

These coordinates distinguen flavors that share branch or generation and close the morphism of flavor without consulting no mass. For a section physical (P), we define

$$\mathrm{sabor}(P) = \big(\sigma_{\mathrm{ud}}(P), g(P), \chi_{\mathrm{fina}}(P)\big), \qquad \chi_{\mathrm{fina}}(P) = (\text{rotations, switches, axis}, N, E, S, O, h_{369}, r_5), \tag{92.8}$$

and, for quarks, the ternary orbit $[\gamma_P]_{\mathrm{col}} = \{\gamma_P^R, \gamma_P^G, \gamma_P^B\}$ is also preserved. The physical realization is the map already determined by the preceding data,

$$\mathcal{S}_{\mathrm{phys}}(P, \mathfrak{M}) : \big(\sigma_{\mathrm{ud}}, g, \chi_{\mathrm{fina}}, [\gamma_P]_{\mathrm{col}}\big) \longmapsto \gamma^{\mathrm{surv}}_{P,\mathfrak{M}} \longmapsto \big(\sigma(\gamma^{\mathrm{surv}}_{P,\mathfrak{M}}), B_F^D, B_F^\Omega\big), \tag{92.9}$$

where $\mathfrak{M}$ is the compatible coupling and $\gamma^{\mathrm{surv}}_{P,\mathfrak{M}}$ the route that persists under it. For charged leptons the color orbit is reduced to the terminal object. The mass character is then evaluated on that same genealogy:

$$\boxed{\frac{m_{\mathfrak{M}}(P)}{m_e^{\mathrm{HMT}}} = \chi_{\mathrm{mass}}\Big(\sigma\Big(\gamma^{\mathrm{surv}}_{P,\mathfrak{M}}\Big)\Big)\,.} \tag{92.10}$$

Therefore, flavor, generation, and color do not await a future selector: they publish the route class and its eigenvector before opening the external comparison. Mass does not retrospectively choose the route; it only reads multiplicatively the persistence that the coupling has stabilized.

92.6.2.2 Fermionicity, mass eigenvalues, and PMNS transport

The exterior closure of the exchange functor constructs the CAR relations and fixes fermionicity. In the neutral sector, the label ν_e, ν_μ, ν_τ designates an interaction basis; the mass values are eigenvalues of a mixed neutral operator. The internal frames $F_\ell, F_\nu : \mathbb{C}^3 \to \mathrm{im}\, P_3$ compare those bases, and the matrix

$$U_{\mathrm{HMT}} = F_\ell^* \Phi_L F_\nu \tag{92.11}$$

is a unitary transport. PMNS changes basis between interaction states and mass eigenvectors; the neutral scale belongs to the operator spectrum, not to the change of basis.

92.6.2.3 Neutral extractor Z0 prior to projection

The recovered construction of the neutral sector acts on the enriched dodecaphase record, before collapsing sheet, torsion, and the bidirectional bit. Let E_m, $m \in \mathbb{Z}/12\mathbb{Z}$ be the twelve nine-tick events of the CT–108 record. At each tick t, the following are preserved

$$\theta(t) = \mathbf{1}_{\{d(t)=9\}}, \qquad \varepsilon(t) \in \{-1, +1\}, \qquad \kappa(t) = \mathbf{1}_{\{9 \text{ en corona}\}}, \qquad \omega(t) = (-1)^{\kappa(t)}.$$

The first factor detecta the torsion nonadic, the second preserves the sheet two–ways and the third changes the sign in the turn of 180° of the crown. The integer torsional of each event and its difference antipodal are

$$T_\nu(m) = \sum_{t \in E_m} \omega(t)\varepsilon(t)\theta(t), \qquad e^\nu_m = T_\nu(m) - T_\nu(m+6), \tag{92.12}$$

and the neutral trit is fixed, without continuous parameters, by

$$\tau^\nu_m = \mathrm{sgn}(e^\nu_m) \in \{-1, 0, +1\}. \tag{92.13}$$

For the four jump orbits four

$$I_a = \{a, a+4, a+8\} \pmod{12}, \qquad a = 0, 1, 2, 3,$$

the event coefficient entering into the signature and the majority sign observable are, respectively,

$$b^{\nu}_{120} = \sum_{a=0}^{3} \operatorname{sgn}\left(\sum_{m\in I_a} e^{\nu}_m\right), \qquad \nu^{\nu,\mathrm{sgn}}_{120} = \sum_{a=0}^{3} \operatorname{sgn}\left(\sum_{m\in I_a} \tau^{\nu}_m\right). \tag{92.14}$$

These two integers are not identified: the first is the coordinate $b_{120} = \nu^{\mathrm{ev}}_{120}$ of the character, while the second is a diagnostic of majority. The quadratic memory is

$$b^{\nu}_{\zeta} = \nu^{\nu}_{270} = \sum_{m=0}^{11} \tau^{\nu}_m \tau^{\nu}_{m+3}. \tag{92.15}$$

Thus, for each surviving neutral history γ_{ν_i}, the register publishes, before any comparison, the signature

$$v_{\nu}(\gamma_{\nu_i}) = \big(0, 0, k^{(i)}_{\Delta_4}, b^{\nu_i}_{120}, b^{\nu_i}_{\zeta}\big)$$

and the relative exponent

$$\mathcal{A}^{(i)}_{\nu} = k^{(i)}_{\Delta_4}\Delta_4 + b^{\nu_i}_{120} s_{120} + b^{\nu_i}_{\zeta}\zeta_{270}, \qquad \zeta_{270} = 135\Delta_4^2. \tag{92.16}$$

Here ζ_{270} is not replaced by the tower moment $s_{270} = q^{270}_{-} - q^{270}_{+}$.

The global two-way conjugation $\tau^{\nu} \mapsto -\tau^{\nu}$ reverses b^{ν}_{120} and preserves b^{ν}_{ζ}. Consequently, it generates two branches of the relative spectrum,

$$\mathcal{A}^{(i)}_{\nu,\pm} = k^{(i)}_{\Delta_4}\Delta_4 \pm b^{\nu_i}_{120} s_{120} + b^{\nu_i}_{\zeta}\zeta_{270}, \qquad \mathrm{NO} \longleftrightarrow \mathrm{IO}, \tag{92.17}$$

which are distinguished by the subsequent sign of Δm^2_{31}, not by a selection using PDG data. This construction closes the discrete Z0 extractor and the relative bifurcation. The common absolute scale m_0 and the full-rank physical PMNS kernel remain separate objects: they are neither introduced into the selector nor declared to be deduced from two–ways parity alone.

The realization with a single phase of rank one produces

$$(\theta_{12}, \theta_{23}, \theta_{13}) = (19.6807^{\circ}, 56.4970^{\circ}, 29.6224^{\circ})$$

and does not reproduce mixing phenomenology. This control selects, as the appropriate domain, a full-rank register kernel,

$$G_{ab} = \frac{1}{\sqrt{|L_a||N_b|}} \sum_{c\in L_a}\sum_{d\in N_b} K_{\mathrm{registro}}(c, d; \xi_c, \xi_d), \qquad U_{\mathrm{int}} = \mathrm{polar}(G), \tag{92.18}$$

whose technical source is developed in the chapter on CKM and PMNS transport. The mass operator, its eigenvalues, and the flavor-reading map are thereby kept separate.

92.6.3 Mass Operator Transport via CKM

The N42 incidence fixes the sextet $(4, 5, 7, 6, 8, 3)$ and the rational matrix

$$M_{\mathrm{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \qquad \det M_{\mathrm{CKM}} = -\frac{3857}{6}.$$

The angular coordinates are constructed by

$$\begin{pmatrix} \theta_{12} \\ \theta_{23} \\ \theta_{13} \end{pmatrix} = M_{\mathrm{CKM}} \begin{pmatrix} A \\ C^*/6 \\ (180/\pi)\Delta_4 \end{pmatrix}, \qquad \delta_{\mathrm{CP}} = 9A.$$

This transformation belongs to the mathematical closure of HMT. In its physical realization, let B_u and B_d be the bidirectional operators already constructed in the fibers of up- and down-type quarks. The change of basis acts according to

$$B_d^{(u)} = V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^*, \qquad R_{\mathrm{act}}^{B_d^{(u)}} = V_{\mathrm{CKM}} R_{\mathrm{act}}^{B_d} V_{\mathrm{CKM}}^*. \tag{92.19}$$

CKM transports eigenvalues between bases and preserves the spectrum and determinant. The CP-violation control adopts the identity

$$\left|\det[B_u, V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^*]\right| = 2|J| \prod_{i<j} |\kappa_i^u - \kappa_j^u| \prod_{a<b} |\kappa_a^d - \kappa_b^d|.$$

The matrix of mixing and the reader of route perform, therefore, functions complementary: the reader determines the operator spectral and CKM determines its representation in the basis of flavor.

92.6.4 Geometric frames of rank 3 for the leptonic sector

The leptonic construction utilizes the diagonal sign involution.

$$S_{\mathrm{sign}} = \mathrm{diag}(1,1,1,-1,-1,-1,1,1,1,-1,-1,-1), \qquad P_- = (I - S_{\mathrm{sign}})/2,$$

and the canonical icosahedral projector P_3 of rank three. The compression

$$C_- = P_- P_3 P_-\big|_{\mathrm{im}\, P_-}$$

has spectrum

$$\left\{ \frac{5+\sqrt{5}}{10}, \frac{1}{2}, \frac{5-\sqrt{5}}{10}, 0, 0, 0 \right\}.$$

The three positive eigenvalues are simple. Their spectral projectors, applied to the indicator of the oriented negative hexad and normalized after projecting by means of P_3, construct a charged orthonormal frame F_ℓ that satisfies

$$F_\ell^* F_\ell = I_3, \qquad F_\ell F_\ell^* = P_3.$$

For the neutrino sector, let H_1, H_2, H_3 be the three oriented positive hexads and

$$X = (P_3 \mathbf{1}_{H_1}, P_3 \mathbf{1}_{H_2}, P_3 \mathbf{1}_{H_3}).$$

Its Gram matrix is positive definite and satisfies

$$\det(X^* X) = \frac{5-\sqrt{5}}{40} > 0.$$

Consequently,

$$F_\nu = X(X^* X)^{-1/2}, \qquad F_\nu^* F_\nu = I_3, \qquad F_\nu F_\nu^* = P_3.$$

The real transport

$$O = F_\ell^* F_\nu \in SO(3) \tag{92.20}$$

it is obtained without masses or PMNS angles.

Theorem 92.10 (Unitarity of the minimal dodecaphase phase)**.** *Let $b_K \in \mathrm{im}\, P_3$ be the normalized vector of the Witt witness, $Q_K = |b_K\rangle\langle b_K|$, and*

$$\Phi_L = I + (\zeta_{12} - 1) Q_K, \qquad \zeta_{12} = e^{i\pi/6}.$$

Then

$$U_{\mathrm{HMT}} = F_\ell^* \Phi_L F_\nu \tag{92.21}$$

is a unitary matrix independent of target metrological values. The orientation inversion conjugates U_{HMT}.

Proof. Q_K is an orthogonal projector, so Φ_L is unitary. The two frames are isometries with the same image $\operatorname{im} P_3$. Therefore

$$U_{\mathrm{HMT}}^* U_{\mathrm{HMT}} = F_\nu^* \Phi_L^* P_3 \Phi_L F_\nu = I_3.$$

The change $\pi/6 \mapsto -\pi/6$ produces Φ_L^*, and all the remaining primitives are real. □

92.6.4.1 Rank-one negative control and full-rank PMNS transport

Frozen the construction, the contrast in the declared order produces

$$(\theta_{12}, \theta_{23}, \theta_{13}) = (19.6807^\circ, 56.4970^\circ, 29.6224^\circ), \qquad J = -0.0426621.$$

The best of the 36 discrete permutations leaves $\theta_{13} \simeq 17.02^\circ$. The rank-one phase generates an internal unitary matrix, but preserves a stabilizer $U(2)$ and is discarded as a physical PMNS realization.

Excluding the rank-one carrier does not leave the PMNS realization undetermined. Let M_ℓ and M_ν be the charged and neutral operators produced by the General Mass Law, and let F_ℓ, F_ν be their complete orthonormal spectral frames on the common image $\operatorname{im} P_3$. Full-rank leptonic transport is then constructed by

$$U_{\mathrm{PMNS}} := F_\ell^* F_\nu, \qquad U_{\mathrm{PMNS}}^* U_{\mathrm{PMNS}} = I_3.$$

For simple spectra, this output is determined up to diagonal phase adjustments and the declared spectral ordering; in the presence of degeneracies, the exact output is the corresponding double class of frames. The register kernel provides an additional finite certificate of this realization, not its generator or an outstanding condition for its existence. For charged fibres L_a and neutrino fibres N_b, its aggregate form is

$$G_{ab} = \frac{1}{\sqrt{|L_a||N_b|}} \sum_{c \in L_a} \sum_{d \in N_b} K_{\mathrm{reg}}(c,d), \qquad U_{\mathrm{int}} = \operatorname{polar}(G),$$

with naturality under orientation. When the explicit register entries satisfy $\det G \neq 0$, its polar factor provides an alternative certificate of the transport already constructed. The kernel reads carry, sheet, orientation, vacancy and holonomy; its parameters are frozen before examining the external angles.

Domain and codomain of the CKM and PMNS realizations The rational CKM matrix, the frames F_ℓ, F_ν, the transport O, the phase Φ_L, the unitarity of U_{HMT} and the complete transport $U_{\mathrm{PMNS}} = F_\ell^* F_\nu$ are exact internal results under the declared primitives. CKM constitutes a physical transport once the flavour and mass bases have been fixed. The comparison falsifies only the rank-one PMNS phase; it does not weaken the complete spectral construction. Naturality and the criterion $\det G \neq 0$ belong to the alternative register certificate and remain separate from the existence and unitarity of PMNS transport.

92.6.5 Involution of leaves in 4 representations

The involution $C^* \mapsto -C^*$ organizes four distinct responses:

$$\begin{aligned} S_{\mathrm{bi}}(A, -C^*) &= S_{\mathrm{bi}}(A, C^*), \\ \Gamma_{6\to 8}(A, -C^*) &= -\Gamma_{6\to 8}(A, C^*), \\ (\widehat{\mu}, \widehat{\varepsilon}) &\longleftrightarrow (\widehat{\varepsilon}, \widehat{\mu}), \\ \widehat{c} &\longmapsto \widehat{c}, \qquad D_A \longmapsto D_A. \end{aligned} \tag{92.22}$$

The bilayer information is even; the areal orientation is odd; the two constitutive quantities form a doublet, and the determinants remain fixed. This parity table precedes any interpretation as a physical symmetry.

92.6.6 Involutive rational transformation in the CKM parameter space

Let

$$M_{\mathrm{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \qquad \det M_{\mathrm{CKM}} = -\frac{3857}{6}. \tag{92.23}$$

The inverse exists on $\mathbb{Q}$ and is

$$M_{\mathrm{CKM}}^{-1} = \begin{pmatrix} 1528/3857 & 3380/3857 & 801/3857 \\ 75/3857 & 176/3857 & -150/3857 \\ 6/551 & -30/551 & -12/551 \end{pmatrix}. \tag{92.24}$$

With

$$h = \frac{C^*}{6}, \qquad \Delta = \frac{180}{\pi}\Delta_4, \qquad \boldsymbol{\theta} = M_{\mathrm{CKM}}(A, h, \Delta)^T, \tag{92.25}$$

the odd coordinate enters in the direction

$$c_h = \begin{pmatrix} -5 \\ 7 \\ -20 \end{pmatrix}. \tag{92.26}$$

If $c_A = (2, 0, 1)^T$ and $c_\Delta = (28, -25/2, -2/3)^T$, the three columns (c_A, c_h, c_Δ) are precisely the basis transported by M_{CKM}. The second row of equation (92.24) is the covector ℓ_h that extracts the odd coordinate:

$$\ell_h c_A = 0, \qquad \ell_h c_h = 1, \qquad \ell_h c_\Delta = 0. \tag{92.27}$$

Theorem 92.11 (Reflexion CKM inducida by the sheet)**.** *Let*

$$D_h = \mathrm{diag}(1, -1, 1), \qquad \ell_h = \frac{1}{3857}\begin{pmatrix} 75 & 176 & -150 \end{pmatrix}. \tag{92.28}$$

The involution induced on the space of $\boldsymbol{\theta}$ is

$$\boxed{\mathcal{R}_{\mathrm{CKM}} = M_{\mathrm{CKM}} D_h M_{\mathrm{CKM}}^{-1} = I - 2c_h\ell_h.} \tag{92.29}$$

Satisfies

$$\mathcal{R}_{\mathrm{CKM}}^2 = I, \qquad \det \mathcal{R}_{\mathrm{CKM}} = -1, \qquad \mathcal{R}_{\mathrm{CKM}} M_{\mathrm{CKM}} = M_{\mathrm{CKM}} D_h. \tag{92.30}$$

Proof. Multiplication of equations (92.23) y (92.24) gives the identity; in particular, equation (92.27) proves that $c_h\ell_h$ is the rank-one projector onto $\mathbb{Q}c_h$ along $\mathrm{span}\{c_A, c_\Delta\}$. The exact product $\ell_h c_h = 1$ implies

$$(I - 2c_h\ell_h)^2 = I - 4c_h\ell_h + 4c_h(\ell_h c_h)\ell_h = I.$$

The operator fixes the hyperplane $\ker \ell_h = \mathrm{span}\{c_A, c_\Delta\}$ pointwise and changes the sign of c_h. Its eigenvalues are therefore $1, 1, -1$; hence $\det \mathcal{R}_{\mathrm{CKM}} = -1$ and $\mathrm{Tr}\, \mathcal{R}_{\mathrm{CKM}} = 1$. Finally,

$$\mathcal{R}_{\mathrm{CKM}} M_{\mathrm{CKM}} = M_{\mathrm{CKM}} D_h M_{\mathrm{CKM}}^{-1} M_{\mathrm{CKM}} = M_{\mathrm{CKM}} D_h,$$

which proves the entanglement identity. □

The explicit matrix is

$$\mathcal{R}_{\mathrm{CKM}} = \begin{pmatrix} 4607/3857 & 1760/3857 & -1500/3857 \\ -150/551 & 199/551 & 300/551 \\ 3000/3857 & 7040/3857 & -2143/3857 \end{pmatrix}. \tag{92.31}$$

Although $\mathcal{R}_{\mathrm{CKM}}$ is not an orthogonal reflection for the Euclidean product of the three angular coordinates, it is for the transported rational metric

$$G_{\mathrm{CKM}} = (M_{\mathrm{CKM}}^{-1})^T M_{\mathrm{CKM}}^{-1}, \qquad \mathcal{R}_{\mathrm{CKM}}^T G_{\mathrm{CKM}} \mathcal{R}_{\mathrm{CKM}} = G_{\mathrm{CKM}}. \tag{92.32}$$

Indeed, in the basis (c_A, c_h, c_Δ), the metric is the identity and the involution is D_h. Hence the term "reflection" has precise metric content and does not merely describe that $\mathcal{R}_{\mathrm{CKM}}^2 = I$. Phase $\delta_{\mathrm{CP}} = 9A$ remains fixed under this reflection. Therefore, $\mathcal{R}_{\mathrm{CKM}}$ is not physical CP conjugation, but rather the transport of HMT sheet inversion to mixing space.

The rational inverse of M_{CKM} recovers

$$A = \frac{1528\theta_{12} + 3380\theta_{23} + 801\theta_{13}}{3857}, \tag{92.33}$$

$$C^* = \frac{6(75\theta_{12} + 176\theta_{23} - 150\theta_{13})}{3857}. \tag{92.34}$$

The third coordinate is also recovered without loss:

$$\Delta = \frac{6\theta_{12} - 30\theta_{23} - 12\theta_{13}}{551}. \tag{92.35}$$

These equations are coordinate changes. They do not turn CKM into a retrospective cause of A, C^* or of γ_{HMT}.

As a terminal recognition, the previous constructor determines

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta_{\mathrm{CP}}) = (13.003313589509^\circ, 2.398056157332^\circ, 0.213977382244^\circ, 65.676173123554^\circ). \tag{92.36}$$

The rational reflection is proved before using these decimals.

The transition $6 \to 8$ is first formalized through the inclusion $\iota_{6,8} : \mathcal{F}_6 \hookrightarrow \mathcal{F}_8$, from which $90, 120$ and the defect $-1/12$ are obtained. The six-trit word then provides the bijection $729 \leftrightarrow 729$, and $\mathcal{J}_{6\to 8}$ transports the two collective modes to the boundary. On that antecedent, area and the modular Hamiltonian exactly reconstruct the two families; thermofield-double purification determines the entanglement entropy, and the finite first law preserves the relative-entropy defect. In CKM, sheet inversion becomes a rational rank-one odd reflection.

Falsification criterion 92.12. The initial genealogy fails if $\iota_{6,8}$ is not injective, if the counts are not $90/120$, if $\delta_{6,8}^{\mathrm{inc}} \neq -1/12$, if β_{729} is not bijective or if it is falsely presented as an additive isomorphism. Collective transport fails if $\mathcal{J}_{6\to 8} D_{\mathrm{inc}} \neq D_{\mathrm{area}} \mathcal{J}_{6\to 8}$. Reconstruction fails if the determinant of the boundary observables is not thirty or if equation (72.8) does not recover the occupations. The TFD fails if its partial trace is not ρ_A. The first law does not extend to finite changes without the relative entropy of equation (74.6). Type III_{λ_∂} is invalidated if a channel does not persist, if there is no compatible product or if the ratio group changes. The CKM reflection is refuted by any of $\mathcal{R}^2 \neq I$, $\det \mathcal{R} \neq -1$ or $\mathcal{R}M \neq MD_h$. A formula $\boldsymbol{\theta} = f(\gamma_{\mathrm{HMT}})$ without an intertwiner violates the typed independence of the two functional realizations of the common antecedent.

Hypothesis, domain, and conclusion

Inclusion $H \subset O$, counts 90/120, two normalizations of the defect $-1/12$, bijection of six trits $729 \leftrightarrow 729$, collective intertwiner, scale a_0, finite operators, area spectrum, TFD, oriented defect, local and finite first law, and CKM reflection: *Exact internal theorem with its explicit initial structures*. Classification III_{λ_∂}: *Theorem with hypotheses explicit of persistence*. Interpretation of the thermofield-double copy as a physical exterior: *Realization, Physically Conditioned*. The magnitude γ_{HMT} is taken from the chapter on Barbero–Immirzi and minimal area, where it was derived; here it acts as an already established structural datum.

92.6.6.1 PMNS Matrix as Reader of the Common Neutrino Fiber Mixing

Let F_ℓ and F_ν be real orthonormal frames with the same three-dimensional image and orientations chosen so that $\det(F_\ell^* F_\nu) = +1$, derived from the charged and neutral operators. Then the real transport is

$$O = F_\ell^* F_\nu \in SO(3).$$

Without the correlative choice of orientation one concludes only $O \in O(3)$. Let b_K also be the normalized Witt witness, $Q_K = |b_K\rangle\langle b_K|$, and

$$\Phi_L = I + (e^{i\pi/6} - 1)Q_K.$$

The minimal phase produces

$$U_{\mathrm{HMT}} = F_\ell^* \Phi_L F_\nu.$$

This compression is unitary only under the condition of compatibility.

$$\Phi_L(\operatorname{Im} F_\nu) = \operatorname{Im} F_\ell.$$

Without it, a contraction is produced; when this is invertible, its polar factor defines the unitary candidate. Even when compatibility is imposed, the minimal rank-one phase is falsified as a physical PMNS: its best permutation leaves $\theta_{13} \simeq 17.02^\circ$. The physical kernel must read the complete record:

$$G_{ab} = \frac{1}{\sqrt{|L_a||N_b|}} \sum_{c \in L_a} \sum_{d \in N_b} K_{\mathrm{reg}}(c,d), \qquad U_{\mathrm{int}} = \operatorname{polar}(G),$$

with $\det G \neq 0$. Carry, sheet, orientation, vacancy, and holonomy form part of the core. The falsifier of the rank-one phase remains printed to prevent formal unitarity from being confused with physical closure.

The Majorana condition for neutrinos additionally requires a fermionic self-conjugation compatible with the mass operator. Neutrality, C_M, and the topological Ising sector do not suffice separately.

92.6.6.2 CKM Mixing and CP Violation

The angular grammar is organized through

$$M_{\mathrm{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \qquad \det M_{\mathrm{CKM}} = -\frac{3857}{6}.$$

With

$$\begin{pmatrix} \theta_{12} \\ \theta_{23} \\ \theta_{13} \end{pmatrix} = M_{\mathrm{CKM}} \begin{pmatrix} A \\ C^*/6 \\ (180/\pi)\,\Delta_4 \end{pmatrix}, \qquad \delta_{\mathrm{CP}} = 9A,$$

we obtain

$$(\theta_{12},\theta_{23},\theta_{13},\delta_{\mathrm{CP}}) = (13.003313589509, 2.398056157332, 0.213977382244, 65.676173123554)^\circ.$$

The Jarlskog invariant is

$$J = c_{12}c_{23}c_{13}^2 s_{12}s_{23}s_{13}\sin\delta_{\mathrm{CP}} = 3.1189723326632974\times 10^{-5}.$$

The exact linear falsifier is

$$3857\delta_{\mathrm{CP}} = 9(1528\theta_{12} + 3380\theta_{23} + 801\theta_{13}).$$

In the standard unitary parametrization,

$$V_{\mathrm{CKM}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}.$$

The previous internal evaluation produces

$$|V_{\mathrm{CKM}}| = \begin{pmatrix} 0.974350259 & 0.225005836 & 0.003734601 \\ 0.224873110 & 0.973489279 & 0.041841465 \\ 0.008582400 & 0.041121745 & 0.999117283 \end{pmatrix}.$$

A frozen marginal comparison yields normalized deviations $(-0.24, -0.07, -0.34, -0.31, -0.37)$ for the four angular parameters and J, respectively. These figures do not replace the covariant comparison: the publication must identify the external edition and its complete covariance matrix before forming a joint χ^2.

CKM transports between quark bases. If B_u, B_d are the diagonal mass operators,

$$B_d^{(u)} = V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^*.$$

The identity

$$|\det[B_u, V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^*]| = 2|J| \prod_{i<j} |\kappa_i^u - \kappa_j^u| \prod_{a<b} |\kappa_a^d - \kappa_b^d|$$

shows that mixing and spectrum couple without the mixing matrix creating the eigenvalues.

92.7 Composition, resonances and non-ordinary sectors

The composite mass is obtained from the constituent records, the boundary, and the binding operator; it is not a bare sum of masses. The width follows from the survival pole and remains separate from the mass signature. Virtual sections, the Fibonacci, Ising–Majorana, and order-six sectors, and the exclusions of tachyons and ghosts preserve their respective domains.

92.8 Composition of paths, records, and signatures

A composite species cannot enter the law of masses as a name to which a row of five integers is retrospectively assigned. Before the signature, an object capable of composition must exist. The atlas of section 90.5 supplies the finite alphabet; the surviving extension and register of section 91.5 supply the memory that may be composed. The question of this chapter is to construct the intermediate stratum without consulting target masses:

$$\text{typed compositional configuration} \longrightarrow \text{complete records of its constituents} \longrightarrow \text{composite record} \longrightarrow \mathbb{Z}^5.$$

The ordering of the arrows is part of the result. The expressions are formed before they are assigned a physical realization; the ledgers are composed before the signature is extracted; the signature is evaluated through the character only after the memory has been composed. Thus, the composite domain is not reduced to the species included in a metrological table.

92.8.1 Internal classification and compositional grammar

Let

$$C_{81} = I_9 \times I_9, \qquad I_9 = \{1, \dots, 9\},$$

the cell atlas, and let it be

$$\rho(r, c) = (10 - r, 10 - c)$$

its central inversion. We denote by $\mathcal{L}, \mathcal{N}, \mathcal{D}, \mathcal{U}$ the subsets of charged leptons, neutrinos, down-type quarks, and up-type quarks, respectively, and set

$$\mathcal{Q} = \mathcal{D} \sqcup \mathcal{U}.$$

Each cell has a combinatorial level $\mathrm{gen}(x)$. In the quark sector the color residue is also preserved

$$\kappa(x) = r(x) - c(x) \pmod 3.$$

We write

$$\mathcal{Q}_a = \{q \in \mathcal{Q} : \kappa(q) = a\}, \qquad a \in \mathbb{Z}/3\mathbb{Z}.$$

By proposition 90.14,

$$|\mathcal{Q}_0| = |\mathcal{Q}_1| = |\mathcal{Q}_2| = 12.$$

Definition 92.13 (Dual formal and cell identity)**.** For an oriented cell section x, its formal dual $x^\vee$ inverts the orientation of charge and color, and has support $\rho(x)$. The notation preserves the distinction

$$x^\vee \neq \rho(x)$$

when $\rho(x)$ is regarded as an unoriented cell. For each $x \in C_{81}$, we denote by $\mathrm{id}_x : x \to x$ the identity arrow. The set of identities is

$$\mathfrak{I} := \{\mathrm{id}_x : x \in C_{81}\}.$$

The identity id_x is a neutral arrow of the compositional system. It is not, by that condition alone, a realization of a neutral boson. Likewise, the formal dual incorporates orientation and cannot be replaced by a second bare cell.

Definition 92.14 (Mesonic and Baryonic Expressions)**.** The set of colour-neutral mesonic expressions is

$$\mathfrak{M} := \left\{ M(q, q') = q \otimes (q')^\vee : q, q' \in \mathcal{Q},\ \kappa(q) = \kappa(q') \right\}.$$

The pairs are ordered. Their total residue is $\kappa(q) - \kappa(q') = 0$.

The set of baryonic expressions with fixed color positions is

$$\mathfrak{B} := Q_0 \times Q_1 \times Q_2,$$

and we write

$$B(q_0, q_1, q_2) = q_0 \otimes q_1 \otimes q_2.$$

Its residual neutrality is the identity.

$$0 + 1 + 2 = 0 \pmod 3.$$

These expressions are typed syntactic objects. In particular, no quotient has yet been taken by permutations, spin, flavor, statistics, binding energy, or dynamical equivalence.

Definition 92.15 (Compatible loaded arrows)**.** The oriented leptonic arrows are

$$\mathfrak{A}_\ell := \{(x, y) \in (\mathcal{L} \times \mathcal{N}) \sqcup (\mathcal{N} \times \mathcal{L}) : \mathrm{gen}(x) = \mathrm{gen}(y)\}.$$

Quark-oriented arrows are

$$\mathfrak{A}_q := \left\{ (x, y) \in (\mathcal{U} \times \mathcal{D}) \sqcup (\mathcal{D} \times \mathcal{U}) : \begin{array}{l} \mathrm{gen}(x) = \mathrm{gen}(y), \\ \kappa(x) = \kappa(y) \end{array} \right\}.$$

We set

$$\mathfrak{A}_{\mathrm{ch}} := \mathfrak{A}_\ell \sqcup \mathfrak{A}_q.$$

Level equality types the transition in both sectors; colour equality is additionally required in the quark sector. Orientation distinguishes $x \to y$ from $y \to x$.

Definition 92.16 (Finite system of typed compositions)**.** The finite compositional system is the typed disjoint union

$$\mathfrak{G}_{\mathrm{comp}} := C_{81} \sqcup \mathfrak{I} \sqcup \mathfrak{M} \sqcup \mathfrak{B} \sqcup \mathfrak{A}_{\mathrm{ch}}.$$

The 81 cells are the primitive generators. The 81 identities are arrows on those same generators, not a second atlas.

92.8.2 Combinatorial census of route compositions independent of metrological data

Theorem 92.17 (Census of the finite system of compositions)**.** *The system of the definition 92.16 contains*

81 *identities*,	432 *mesonic expressions*,
1728 *baryonic expressions*,	372 *charged arrows*

The charged arrows decompose exactly as

$$372 = 320_{leptonic} + 52_{quark}.$$

The census depends only on the atlas, its inversion, the levels, and the color residue. It uses neither masses, units, nor published signatures. The four cardinalities count internal compositional expressions or admissible arrows of the system; they do not count distinct physical species, physical particles, or masses.

Proof. The assignment $x \mapsto \mathrm{id}_x$ is bijective, so

$$|\mathfrak{I}| = |C_{81}| = 81.$$

For mesons, the neutrality condition permits an ordered pair to be chosen independently within each colour class. Since each class contains twelve cells,

$$|\mathfrak{M}| = \sum_{a\in\mathbb{Z}/3\mathbb{Z}} |Q_a|^2 = 3 \cdot 12^2 = 432.$$

For baryons, the positions $0, 1, 2$ are fixed, and therefore

$$|\mathfrak{B}| = |Q_0||Q_1||Q_2| = 12^3 = 1728.$$

The charged lepton families have sizes

$$|\mathcal{L}_{G0}| = 1, \qquad |\mathcal{L}_{G2}| = 8, \qquad |\mathcal{L}_{G4}| = 16,$$

while the neutrino families have sizes

$$|\mathcal{N}_{G1}| = 2, \quad |\mathcal{N}_{G2}| = 4, \quad |\mathcal{N}_{G3}| = 6, \quad |\mathcal{N}_{G4}| = 8.$$

Only $G2$ and $G4$ are common levels. Since both orientations are counted,

$$|\mathfrak{A}_\ell| = 2(8 \cdot 4 + 16 \cdot 8) = 320.$$

For the quark arrows, in the colour order $(0, 1, 2)$, the up-type and down-type multiplicities at the common levels are

	$\mathcal{U}$	$\mathcal{D}$
$G1$	$(2, 1, 1)$	$(0, 1, 1)$
$G3$	$(4, 4, 4)$	$(2, 2, 2)$.

The number of pairs of an orientation and the same color is, therefore,

$$(2 \cdot 0 + 1 \cdot 1 + 1 \cdot 1) + (4 \cdot 2 + 4 \cdot 2 + 4 \cdot 2) = 2 + 24 = 26.$$

By including the opposite orientation one obtains

$$|\mathfrak{A}_q| = 2 \cdot 26 = 52.$$

Finally,

$$|\mathfrak{A}_{\mathrm{ch}}| = 320 + 52 = 372.$$

None of these rules contains an objective value or a mass vector. □

Corollary 92.18 (Compositional constructor without whitespaces)**.** *The constructor that jointly enumerates identities, mesonic expressions, baryonic expressions, and charged arrows factorizes through the typed fields*

$$(\mathit{sector}, \mathit{family}, \mathit{level}, \kappa, \rho)$$

of the atlas. With the preceding notation, its four codomains are $\mathfrak{I}, \mathfrak{M}, \mathfrak{B}, \mathfrak{A}_{\mathrm{ch}}$*. In particular, the census can be fixed before any signature table is opened.*

Proof. The four definitions and the calculation of theorem 92.17 use only those fields. The table of signatures does not belong to the domain of any of the rules. □

Target independence does not mean that the expressions have already acquired a physical name. It means something prior and more precise: there exists a compositional domain constructed without using as a criterion the quantity that will later be evaluated.

92.8.3 Memory transport from the composite route to the integral signature

Let $\gamma \in \Gamma_{108}^{\mathrm{enr}}$ be an enriched presentation of an extension. Its read register contains

$$\mathcal{E}(\gamma) = (n_A, e_0, \dots, e_{11}, T_\zeta, C_\zeta),$$

in addition to the phase, sheet, orientation, and edge data that type a composition. The twelve events are not replaceable by their total before composition. For $0 \le i < 6$, we define

$$p_i = e_i + e_{i+6}, \qquad a_i = e_i - e_{i+6},$$

$$s_C = \sum_{i=0}^{5} p_i, \qquad M_i = p_i - p_5 \quad (0 \le i < 5).$$

With $M_5 = (M_0, \dots, M_4)$ and $A_6 = (a_0, \dots, a_5)$, the reconstruction is

$$p_5 = \frac{s_C - \sum_{i=0}^{4} M_i}{6}, \qquad p_i = p_5 + M_i,$$

$$e_i = \frac{p_i + a_i}{2}, \qquad e_{i+6} = \frac{p_i - a_i}{2}.$$

By the theorem 91.3, on the integral image the reversible decomposition is obtained

$$\boxed{(s_C, M_5, A_6), \qquad 1 + 5 + 6 = 12.}$$

The first block preserves the visible total; the second, the relative form within the symmetric sheet; the third, the memory between the two semicrowns.

The strong torsional coordinate also requires the cocycle condition at the step level:

$$k_{\Delta_4} = T_\zeta - 2C_\zeta.$$

It is not a function of the mere pattern of null events. This difference will be essential when interpreting the non-descent rhombus.

Definition 92.19 (Given oriented composition data)**.** Let Γ, Λ be two enriched ledgers. A composition datum is a pair (g, B), where g aligns phase, sheet, orientation, and position at the interface, and B is the common subtrace, with its contribution to the action, its twelve events, its torsional predecessors, and its corona. On the domain in which these data are compatible, one defines

$$\boxed{(n, e)_\Gamma \star_{(g,B)} (n, e)_\Lambda = \big(n_\Gamma + n_\Lambda - n_B,\ e_\Gamma + g e_\Lambda - e_B\big).}$$

The torsional word and the corona are composed at the same level, before computing T_ζ, C_ζ and k_{Δ_4}.

The formula is an inclusion–exclusion of histories over an identified subtrace. It is demonstrated with an explicit initial structure: the records, the transport g, the boundary B, the orientation, and the word of predecessors are part of the input. The compositional system does not by itself choose any of those data.

Definition 92.20 (Posterior extractor after the composition)**.** For a complete enriched record one defines

$$L_{\mathrm{tick}}(\gamma) = \big(n_A, s_C, k_{\Delta_4}, \nu_{120}^{\mathrm{ev}}, \nu_{270}\big) \in \mathbb{Z}^5,$$

where, if $\tau_m = \operatorname{sgn}(e_m)$ and $I_a = \{a, a+4, a+8\} \pmod{12}$,

$$\nu_{120}^{\mathrm{ev}} = \sum_{a=0}^{3} \operatorname{sgn}\left(\sum_{m\in I_a} e_m\right), \qquad \nu_{270} = \sum_{m=0}^{11} \tau_m \tau_{m+3}.$$

The signature of a composition is defined by

$$\operatorname{Sig}(\Gamma, \Lambda; g, B) := L_{\mathrm{tick}}\big(\Gamma \star_{(g,B)} \Lambda\big).$$

The signs and autocorrelations are applied to the already composed record. Therefore, the definition is not equivalent to summing the signatures of the constituents.

92.8.4 Commutative diagram of composition and fiber preservation

The historical catalog B1 employs an event projection that is convenient to type separately. Let

$$\Sigma_{12} = (+,+,+,-,-,-,+,+,+,-,-,-), \qquad \tau_m = \operatorname{sgn}(e_m),$$

and let us define the null event proxy

$$K_0(e) = \sum_{\tau_m = 0} \Sigma_{12}(m).$$

The exact projection is

$$F_0(n_A, e) = \left(n_A,\ \sum_m e_m,\ K_0(e),\ \sum_{a=0}^{3} \operatorname{sgn} \sum_{m\in I_a} e_m,\ \sum_m \tau_m \tau_{m+3}\right) \in \mathbb{Z}^5.$$

For two aligned records, we write

$$(n, e) \boxplus (n', e') = (n + n', e + e').$$

This operation is the case with no common subtrace and with identity alignment of the event-composition case.

Proposition 92.21 (Obstruction to the descent of the projection B1)**.** *There is no binary operation*

$$\mu : \mathbb{Z}^5 \times \mathbb{Z}^5 \longrightarrow \mathbb{Z}^5$$

such that

$$F_0(\Gamma \boxplus \Lambda) = \mu\big(F_0(\Gamma), F_0(\Lambda)\big)$$

for all records B1.

Proof. Consider the routes, written in the chart (r, c, h_0, v_0),

$$a = (1, 3, -1, +1), \qquad b = (6, 7, -1, +1), \qquad c = (1, 2, -1, -1).$$

The first two have distinct event records:

$$e(a) = (5, 4, 4, -4, -4, -6,\ 5, 4, 4, -4, -4, -6),$$
$$e(b) = (4, 5, 4, -4, -6, -4,\ 4, 5, 4, -4, -6, -4),$$

but publish the same five-tuple:

$$F_0(a) = F_0(b) = (8, -2, 0, 0, -12).$$

The test route has

$$e(c) = (1, 4, 1, -2, -4, -3,\ \ 1, 4, 1, -2, -4, -3)$$

and

$$F_0(c) = (28, -6, 0, -4, -12).$$

By first summing the events and then reevaluating, one obtains

$$F_0(a \boxplus c) = (36, -8, 0, 0, -12),$$

$$F_0(b \boxplus c) = (36, -8, 0, -2, -12).$$

The two pairs of inputs of the putative μ are equal, but the outputs differ in ν_{120}^{ev}. This contradicts the existence of μ. □

The exhaustive census of the same domain contains 324 routes, 53 proxy five-tuples, and 198 distinct event states. Forty-two fibers of the projection contain more than one enriched state; forty of them are separated by some test route. The preceding diamond is the shortest canonical witness to that loss of information.

Remark 92.22 (Typed separation between the representative K_0 and strong torsion). The third coordinate of F_0 is $K_0(e)$, which in B1 takes values in $\{-4, -2, 0, 2, 4\}$. It is not identified with $k_{\Delta_4} = T_\zeta - 2C_\zeta$. Indeed, there exist routes with the same n_A and the same vector of twelve events to which the torsional reader assigns distinct values because it preserves the predecessors of each iteration and the crown.

The proposition 92.21 proves exactly that the five-coordinate B1 projection does not transport the superposition of records. It does not assert a universal impossibility for every future structure on $\mathbb{Z}^5$, nor does it authorize replacing K_0 with k_{Δ_4}. Under the strong reader, the output is defined through the definition 92.20: the enriched word is first composed and its five coordinates are then re-extracted.

92.8.5 Theorem of independent data record composition in metrology

Theorem 92.23 (Compositional closure without blanks)**.** *Once the atlas C_{81}, formal duality, the color residue, and the enriched record of the section 91.5 have been fixed, the following assertions hold.*

1. *There exists the finite system of typed compositions $\mathfrak{G}_{\mathrm{comp}}$ with* 81 *identities,* 432 *mesonic expressions,* 1728 *baryonic expressions, and* 372 *charged arrows, decomposed as* 320 + 52*.*
2. *The system is constructed without target masses. Its output is a typed expression, orbit, or multisection, not a mass or an individualized physical label.*
3. *Given an expression and constituent records with an explicit tree of compatible compositions (g, B), the operation produces an enriched record. Its signature is obtained by the application*

$$\boxed{(\textit{expression}, \textit{records}, \textit{oriented compositions}) \xrightarrow{\mathfrak{C}_{12}} \textit{composite record} \xrightarrow{L_{\mathrm{tick}}} \mathbb{Z}^5.}$$

   Here $\mathfrak{C}_{12}$ denotes the iteration of $\star_{(g,B)}$ with the parenthesization and boundaries included in the data.
4. *The B1 projection onto five integers does not by itself constitute a compositional space: the diagram with $\boxplus$ does not descend to $\mathbb{Z}^5$. Therefore, replacing each record by its five-tuple before composition loses information needed to reevaluate at least ν_{120}^{ev}.*

Proof. The first point is theorem 92.17. The second is corollary 92.18. For the third, each edge of the composition tree is evaluated through definition 92.19; because the boundaries and transports form part of

the domain, the iteration produces a register on which L_{tick} is defined. No intermediate signature is used as a substitute for the register. The fourth point is proposition 92.21. □

The correct formula for the composite signature is, therefore,

$$\mathrm{Sig}_{\mathrm{comp}} = L_{\mathrm{tick}} \circ \mathfrak{C}_{12}.$$

It is not a sum of mass rows. Only afterward may the formal character or its global specialization be applied:

$$\text{composite record} \xrightarrow{L_{\mathrm{tick}}} \mathbb{Z}^5 \xrightarrow{\chi_{\mathrm{form}}} \mathrm{Fun}(\mathcal{P}, \mathbb{R}_{>0}) \longrightarrow \text{dimensional chart.}$$

92.8.6 Genealogical reconstruction of mixing, generation, and persistence of flavor

The preceding theorem completes a stratum that a list of masses cannot display. Primitive cells do not merely receive names; they generate identities, dual pairs, color triples, and oriented arrows through internal rules. Composite objects thus appear within the domain before any metrological comparison.

Genealogy is not reduced to the syntactic tree either. Two expressions with the same number of constituents may be composed in different ways because the phase, sheet, orientation, and common subtrace form part of the datum. The register preserves those differences until the signs, correlations, and torsion have acted. Mass, when a dimensional realization exists, will be a subsequent reading of that composite memory.

This order explains why $\mathbb{Z}^5$ is not the composition domain although it is the domain of the character. The group $(\mathbb{Z}^5, +)$ exists, and the character is multiplicative on it; what does not exist for the B1 projection is a law that recovers from the five-tuples the superposition that has already forgotten the placement of the twelve events. Linearity of the character does not restore the information removed by the extractor.

92.8.7 From the internal compositional system to physical realization

The theorem 92.23 constructs the domain prior to physical names: ordered expressions, duals, color triples, and compatible arrows. Their cardinalities count internal possibilities of composition. A physical species, by contrast, is a persistent representation with specified flavor, color, spin, statistics, and channel. Conflating the two levels would turn a census of expressions into a spurious catalogue of particles.

Definition 92.24 (Physical realization data)**.** A realization datum for a species X is the tuple

$$\mathfrak{D}_X = \big(\mathfrak{t}_X, \mathfrak{s}_X, (\mathcal{L}_{X_i})_{i=1}^{r}, \mathcal{T}_X, (g_X, B_X)\big),$$

where:

1. $\mathfrak{t}_X$ fixes the physical type —elementary, null, mesonic, baryonic, multiquark, or topological— and its representation;
2. $\mathfrak{s}_X$ is the multisection or realized internal expression;
3. $\mathcal{L}_{X_i}$ are the enriched ledgers of the constituents;
4. $\mathcal{T}_X$ fixes flavor, color, spin, statistics, and symmetrization;
5. (g_X, B_X) fixes the composition tree, its transports, and its common subtraces.

The tuple is declared without consulting the mass or the width, which will be evaluated later.

Proposition 92.25 (The realization changes the register, not the law)**.** *Every datum $\mathfrak{D}_X$ determines a composite record*

$$\mathcal{L}_X = \mathfrak{C}_{12}\big((\mathcal{L}_{X_i})_{i=1}^{r}; \mathcal{T}_X, g_X, B_X\big),$$

a signature $v_X = L_{\mathrm{tick}}(\mathcal{L}_X)$ and the universal central character

$$\chi_0(v_X - v_e).$$

If the genealogy also provides a typed spectral refinement $\eta_X \in \mathcal{E}_{\Subset}$, the complete evaluation is

$$m_X = m_e^{\mathrm{HMT}}\, \chi_{\mathrm{full}}(v_X - v_e, \eta_X).$$

The central character and the refinement belong to distinct levels; the completeness of the tower does not replace the prospective map that produces η_X. If X is unstable, its width belongs to the temporal reader $\Gamma_X = \hbar_{\mathrm{int}}/\tau_X^{\mathrm{life}}$, not to a sixth coordinate of the mass signature.

Proof. The realization datum contains exactly the entries of $\mathfrak{C}_{12}$. The theorem 92.23 produces the enriched record, and the definition 92.20 produces its signature. The central character and its spectral lift are those of proposition 92.25. The width uses a temporal line and therefore has a different domain. □

The level separation is thus fixed:

internal compositional system ⟶ realization datum ⟶ composite record ⟶ signature ⟶ central character ⟶ spectral refinement ⟶ physical comparison.

The 432 mesonic expressions, the 1728 baryonic expressions, and the 372 charged arrows occupy only the first slot and do not constitute an inventory of particles. The Standard Model preserves its typed set of species, and each species occupies the chain through its own realization datum. The causal rule is unique across all sectors:

compose extensions and records; read the signature afterwards.

92.9 Particle band, anti-section, and virtuality

The persistence of a route produces a particle before its mass is evaluated. This priority makes it possible to distinguish three objects:

1. the *oriented section*, which preserves the direction of the route;
2. the *band*, which joins a section and its conjugate;
3. the *persistence*, which decides whether or not the signature reaches a global branch.

Definition 92.26 (Dodecaphasic Words and Material Conjugation)**.** Let

$$\mathscr{T}_{12}^{\mathrm{word}} := \{-1, 0, 1\}^{12}.$$

For $\mathbf{t} = (t_1, \ldots, t_{12})$ we write

$$\mathrm{rev}(\mathbf{t}) = (t_{12}, \ldots, t_1), \qquad C_M(\mathbf{t}) := -\,\mathrm{rev}(\mathbf{t}).$$

The inversion C_M simultaneously changes the reading direction and the tritic orientation.

Theorem 92.27 (Material inversion and fixed sector)**.** *The map C_M is an involution of $\mathscr{T}_{12}^{\mathrm{word}}$ and*

$$|\mathrm{Fix}(C_M)| = 3^6 = 729.$$

Proof. Reversal and sign change commute and both have square equal to the identity; therefore $C_M^2 = I$. A word is fixed if and only if

$$t_i = -t_{13-i}, \qquad 1 \le i \le 6.$$

The first six coordinates are free, and the last six are determined by them. Each free coordinate admits three values, so the fixed sector has cardinality 3^6. □

This finite conjugation is not obtained by changing only the visible sign: it also reverses the orientation of the register.

92.9.1 Definition of the route band through the orbit of a persistent section

Let $\mathcal{T}_{12}^{\mathrm{word}}$ be the word space of the definition 92.26. We add a section orientation $\chi \in \{-1, +1\}$ and define

$$(\mathbf{t}, \chi) \sim (C_M \mathbf{t}, -\chi).$$

Definition 92.28 (Dodecaphasic route band)**.** The band of the oriented section $(\mathbf{t}, \chi)$ is the class

$$[(\mathbf{t}, \chi)]_M = \{(\mathbf{t}, \chi), (C_M \mathbf{t}, -\chi)\},$$

and the band space is

$$\mathcal{B}_{12} := \big(\mathcal{T}_{12}^{\mathrm{word}} \times \{-1, +1\}\big)/\!\sim.$$

Proposition 92.29 (Freedom of oriented conjugation)**.** *The action*

$$(\mathbf{t}, \chi) \longmapsto (C_M \mathbf{t}, -\chi)$$

is free. Consequently,

$$|\mathcal{B}_{12}| = 3^{12}.$$

Proof. A fixed point would require $\chi = -\chi$, impossible for $\chi \in \{-1, +1\}$. All orbits have two elements and the domain has $2 \cdot 3^{12}$. □

The band does not erase orientation: it preserves it as a double cover. The particle and the antiparticle are the two oriented sections of the same band.

92.9.2 Even mass and odd charge

Let $E_0 : \mathcal{T}_{12}^{\mathrm{word}} \to \mathbb{R}$ be an additive route character. Its even and odd components with respect to C_M are

$$E_+(\mathbf{t}) = \frac{1}{2}\big(E_0(\mathbf{t}) + E_0(C_M \mathbf{t})\big),$$

$$E_-(\mathbf{t}) = \frac{1}{2}\big(E_0(\mathbf{t}) - E_0(C_M \mathbf{t})\big).$$

Then

$$E_+(C_M \mathbf{t}) = E_+(\mathbf{t}), \qquad E_-(C_M \mathbf{t}) = -E_-(\mathbf{t}).$$

Definition 92.30 (Band character)**.** For an oriented section we define

$$E_{\mathrm{band}}(\mathbf{t}, \chi) = E_+(\mathbf{t}) + \chi E_-(\mathbf{t}), \qquad m(\mathbf{t}, \chi) = m_\circ \exp\big(E_{\mathrm{band}}(\mathbf{t}, \chi)\big),$$

where $m_\circ$ is the family's internal comparison unit.

Theorem 92.31 (Mass Conjugation Principle)**.** *The character and the mass descend to the band quotient:*

$$E_{\mathrm{band}}(C_M \mathbf{t}, -\chi) = E_{\mathrm{band}}(\mathbf{t}, \chi),$$

$$\boxed{m(C_M \mathbf{t}, -\chi) = m(\mathbf{t}, \chi).}$$

The even invariants belong to the band; the odd invariants distinguish the orientation of its two sections.

Proof. By the previous parities,

$$\begin{aligned} E_{\mathrm{band}}(C_M\mathbf{t}, -\chi) &= E_+(C_M\mathbf{t}) - \chi E_-(C_M\mathbf{t}) \\ &= E_+(\mathbf{t}) + \chi E_-(\mathbf{t}). \end{aligned}$$

The equality of masses is obtained by applying the exponential. □

In particular, a linear form with symmetric weights is odd, whereas a quadratic form compatible with reversal is even. Charge and orientation change; the mass, constructed over the band, remains.

Corollary 92.32 (Electron and positron)**.** *In the electronic band equation, the pair* $(n, \chi) = (-22, +1)$ *and its conjugate* $(+22, -1)$ *produce the same term* $\chi n = -22$*. Therefore, the electron and positron have the same mass and opposite charge orientations.*

92.9.3 Finite closure of the atlas under anti-section

Seven tritic forms appear in the eighty-one CT–108 routes. The exhaustive census distinguishes three operations:

$$\mathbf{t} \longmapsto -\mathbf{t}, \qquad \mathbf{t} \longmapsto \mathrm{rev}(\mathbf{t}), \qquad \mathbf{t} \longmapsto -\mathrm{rev}(\mathbf{t}).$$

Proposition 92.33 (Closure of the CT–108 forms)**.** *The set of tritic forms of CT–108 is not stable under pure sign nor under pure reversal, and is stable under* $C_M = -\mathrm{rev}$*. Its orbits have multiplicities*

$$\boxed{81 = 54 + 9 + 9 + 9.}$$

The orbit of multiplicity 54 *is self-conjugate; the other three exchange pairs of forms.*

Proof. The fixed form is

$$\texttt{+++-+++-+++-+++-}$$

and has multiplicity 54. The three pairs

$$\begin{gathered} \texttt{+++-0+++-0} \;/\; \texttt{0++-0++-0++-}, \\ \texttt{+++-0-+++-0-} \;/\; \texttt{+0+-+0+-+0+-}, \\ \texttt{+++0-+++0-} \;/\; \texttt{++0-++0-++0-} \end{gathered}$$

have total multiplicity 9 each. The map $-\mathrm{rev}$ fixes the first and permutes the members of each pair. Sign change and reversal separately produce words outside this list. The sum of multiplicities is $54 + 9 + 9 + 9 = 81$. □

The result does not identify the sector 54 with another object of the same cardinality. It proves which conjugation respects the specific taxonomy of the routes and why particle and antiparticle share a band.

92.9.4 Finite persistence and virtual particle

Let

$$\cdots \xrightarrow{\rho_{M+1,M}} S_M \xrightarrow{\rho_{M,M-1}} S_{M-1} \to \cdots$$

the system of admissible states at finite depths. For $s_N \in S_N$, we define the set of prolongations

$$\mathcal{P}_M(s_N) := \{s_M \in S_M : \rho_{M,N}(s_M) = s_N\}, \qquad M \geq N.$$

Its depth of life is

$$\mathcal{L}(s_N) := \sup\{M \geq N : \mathcal{P}_M(s_N) \neq \varnothing\} \in \mathbb{N} \cup \{\infty\}.$$

Definition 92.34 (Persistence and Virtuality)**.** A prefix s_N is *persistent* if it belongs to a compatible global section $(s_M)_{M\geq N}$. It is *finite virtual* if $\mathcal{L}(s_N) < \infty$, a prolongation exists up to $\mathcal{L}(s_N)$ and none exists at the next depth.

Theorem 92.35 (Persistence criterion)**.** *Suppose that each state has a finite number of immediate prolongations and that the restrictions are coherent. Then:*

$$\mathcal{L}(s_N) = \infty \quad \Longleftrightarrow \quad s_N \text{ belongs to a global section.}$$

If $\mathcal{L}(s_N) = L < \infty$, the prefix transports route, memory, register, and signature up to L, but does not define a persistent particle.

Proof. A global section provides a prolongation to every depth, so $\mathcal{L} = \infty$. Conversely, the prolongations of s_N form an infinite, finitely branching rooted tree. By König's lemma, it contains an infinite branch whose vertices are compatible by construction and define a global section.

If $\mathcal{L} = L < \infty$, there is a state at depth L and none at $L + 1$. Readers defined on prefixes can evaluate it up to L; the absence of prolongation prevents that finite signature from being converted into a section of the inverse limit. □

The virtual particle thus has an internal definition. It is neither a "less real" particle nor a brief violation of a conservation law. It is a route whose memory exerts influence over a finite depth and whose genealogy becomes extinct before stabilizing. The comparison with propagators and intermediate states of conventional physics is carried out after identifying which reader represents those depths.

92.9.5 Route–record–signature–character–material realization tower chain

The results of this chapter fix the complete direction:

$$\boxed{\text{extension} \longrightarrow \text{route} \longrightarrow \text{record} \longrightarrow \text{signature} \longrightarrow \text{oriented band} \longrightarrow \text{particle} \longrightarrow \text{mass.}}$$

The antiparticle is the second section of the band. Virtuality is finite persistence. Neither notion is defined from an observed mass.

92.10 Complex structure $\widetilde{J}_M^2 = -I$ on the fixed stratum, dodecaphase involutions, and topological sectors

The involution C_M, the based ring associated with F_{av}, the braid characters, and the pointed categories belong to distinct algebraic strata. They are presented with their domains and identification controls so that a nominal coincidence does not replace the morphism that would have to relate them. The atlas census, the infinite spectral hierarchy whose eight historical heights are initial witnesses, and the statistical completions preserve the proofs of theorems 6.20, 68.6, 69.1 y 90.11.

Each result directly declares its domain, its construction, and its conclusion; no nominal coincidence modifies those dependencies.

92.10.1 Periodic stratum of the dodecaphasic involution

The domain $\mathcal{T}_{12}^{\mathrm{word}}$, the conjugation $C_M\mathbf{t} = -\operatorname{rev}(\mathbf{t})$, its involutive character, and the census $|\operatorname{Fix}(C_M)| = 3^6 = 729$ have their primary residence in the definition 92.26 y theorem 92.27, within the particle band. Only the periodic condition required for the topological stratum is added here. If r denotes the cyclic shift $(r\mathbf{t})_m = t_{m-1}$, we set

$$\mathrm{Per}_6 := \{\mathbf{t} \in \mathcal{T}_{12}^{\mathrm{word}} : r^6\mathbf{t} = \mathbf{t}\}.$$

Proposition 92.36 (Census of the periodic stratum)**.** *Upon imposing periodicity with period dividing six,*

$$\left|\mathrm{Fix}(C_M) \cap \mathrm{Per}_6\right| = 3^3 = 27.$$

This identity is exact.

Proof. The theorem 92.27 gives the general form of the fixed points. If, moreover, $r^6\mathbf{t} = \mathbf{t}$, the second half of the word coincides with the first. Writing the first six coordinates, the two conditions are equivalent to

$$(t_0, t_1, t_2, t_3, t_4, t_5) = (a, b, c, -c, -b, -a).$$

The parameters a, b, c are independent and take three values; therefore, the intersection has 3^3 elements. □

The general example of a fixed point, prior to requiring period six, is

$$(a, b, c, d, e, f, -f, -e, -d, -c, -b, -a).$$

This coordinated form makes it possible to check by hand both the involution and the count and avoids interpreting the integer 729 by means of another homonymous object.

Remark 92.37 (Census of the Involution). The enumeration of the $3^{12} = 531\,441$ words finds no violation of the involution and obtains

$$|\,\mathrm{Fix}(C_M)| = 729, \qquad |\,\mathrm{Fix}(C_M) \cap \mathrm{Per}_6\,| = 27.$$

The primary proof and proposition 92.36 explain both numbers independently of the traversal: the fixed points have six free coordinates, and the additional period-six condition reduces the choice to three. The exhaustive census confirms that no exceptional orbits exist outside that description.

Comparison Criterion 92.38 (Falsifiers of the census C_M)**.** Any of the following facts would refute the result: a word for which $C_M^2\mathbf{t} \neq \mathbf{t}$; a fixed point that does not have the preceding coordinate form; an exhaustive total other than 729; or a total other than 27 after imposing $r^6\mathbf{t} = \mathbf{t}$. The nominal coincidence of this 729 with another cardinal does not identify their domains.

92.10.2 Quadratic structure transported to the fixed sheet

We identify the alphabet of signs with the ternary field via

$$\{-1, 0, +1\} \xrightarrow{\sim} \mathbb{F}_3, \qquad -1 \longmapsto 2, \quad 0 \longmapsto 0, \quad +1 \longmapsto 1.$$

With this identification, $\mathscr{T}_{12}^{\mathrm{word}} \cong \mathbb{F}_3^{12}$, and both $C_M = -\mathrm{rev}$ and the shift r are $\mathbb{F}_3$-linear operators. Consequently, $\mathrm{Fix}(C_M)$ and $\mathrm{Per}_6 = \ker(r^6 - I)$ are vector subspaces.

The coordinate form of the fixed points defines the linear isomorphism.

$$s_{\mathrm{fix}} : \mathbb{F}_3^6 \xrightarrow{\sim} \mathrm{Fix}(C_M),$$

$$s_{\mathrm{fix}}(a, b, c, d, e, f) = (a, b, c, d, e, f, -f, -e, -d, -c, -b, -a).$$

Let J_W be the Paley–Witt operator constructed in theorem 3.21, with the basis, polarization, and orientation declared there.

Proposition 92.39 (Complex structure $\widetilde{J}_M^2 = -I$ on the fixed stratum)**.** *The transport*

$$\widetilde{J}_M := s_{\mathrm{fix}} J_W s_{\mathrm{fix}}^{-1} \in \mathrm{End}_{\mathbb{F}_3}\big(\mathrm{Fix}(C_M)\big)$$

satisfies

$$\widetilde{J}_M^2 = -I.$$

Proof. By conjugation,

$$\widetilde{J}_M^2 = s_{\mathrm{fix}} J_W s_{\mathrm{fix}}^{-1} s_{\mathrm{fix}} J_W s_{\mathrm{fix}}^{-1} = s_{\mathrm{fix}} J_W^2 s_{\mathrm{fix}}^{-1} = -I.$$

□

The result is proved with the chart s_{fix}, the basis, the polarization, and the orientation as explicit starting structure. It does not identify C_M with J_W: the former is an involution of the word space, and the latter a quadratic structure on its fixed stratum. Nor does it prove that the transport is natural with respect to the entire TPK dynamics.

Let us suppose

$$W_{27} := \mathrm{Fix}(C_M) \cap \mathrm{Per}_6 .$$

The inclusion

$$W_{27} \hookrightarrow \mathrm{Fix}(C_M)$$

is canonical, and proposition 92.36 proves that $\dim_{\mathbb{F}_3} W_{27} = 3$.

Corollary 92.40 (Quadratic obstruction in the twenty-seven stratum)**.** *The subspace W_{27} cannot be stable under $\widetilde{J}_M$.*

Proof. If it were stable, the relation $\widetilde{J}_M^2 = -I$ would turn it into a module over

$$\mathbb{F}_3[X]/(X^2 + 1) \cong \mathbb{F}_9.$$

Every vector space over $\mathbb{F}_9$ has even dimension when considered over $\mathbb{F}_3$, in contradiction with $\dim_{\mathbb{F}_3} W_{27} = 3$. □

The cardinal equality between W_{27} and the twenty-seven lines of Schläfli does not furnish a map between the two objects. Such a relation would require the construction of an incidence map preserving the central orientation and the action of the stabilizer, together with a compatible dynamical closure. The atlas of 81 cells is not interposed between W_{27} and the fixed stratum of 729 states: the particle atlas belongs to another domain.

92.10.3 Mathematical stratification of the Majorana object in 4 levels

The inversion C_M, exterior parity, a relativistic Majorana fermion, and a Majorana zero mode are objects of different types:

1. $C_M = -\mathrm{rev}$ acts on $\mathscr{T}_{12}^{\mathrm{word}}$. Its involutivity and its census are proved exactly; this construction does not, by itself, provide a spinor, a field equation, or a mass term.
2. The sign character $q = -1$ and the exterior completion belong to the exchange algebra. Relative to CAR and the declared grading, they produce $N_s^2 = N_s$, spectrum $\{0, 1\}$, and the orientation return $2\pi/4\pi$. These identities follow from CAR and the grading; they do not follow from the mere census of C_M.
3. A relativistic Majorana fermion is a spinorial object endowed with a relativistic representation, compatible charge conjugation, a self-conjugacy condition, a mass term, and dynamics. Those data are not part of the definition of C_M and do not follow from its census.

4. A Majorana zero mode, or equivalently the nonabelian sector of an Ising theory in the topological sense, requires simple objects $\mathbf{1}, \psi, \sigma$ with

$$\psi \otimes \psi \simeq \mathbf{1}, \qquad \psi \otimes \sigma \simeq \sigma, \qquad \sigma \otimes \sigma \simeq \mathbf{1} \oplus \psi, \qquad d_\sigma = \sqrt{2},$$

together with coherent associator and braiding. The gap, the splitting, and the platform belong to an additional physical realization; the preceding fusion rules do not define a universal pole mass.

The separation concerns domains rather than nomenclature: word involution, CAR parity, self-conjugate spinor, and nonabelian simple object are four distinct mathematical objects. In the absence of an explicit morphism among them, none is identified merely because it shares the name "Majorana.".

92.11 Incidence, Fibonacci and topological sectors

The residue–support incidence prolongs the atlas toward fusion and braid structures. The order is essential: first the based Fibonacci ring and its dimensions are constructed; the algebraic representations of braids and the pointed center are then studied. Each object preserves its own domain and is not identified with a physical particle merely because it shares a name or cardinal.

92.11.1 Residue-incident support

We fix the residue classes and the supports in the order

$$C_1 = \{1,4,7\}, \qquad C_2 = \{2,5,8\}, \qquad C_0 = \{3,6,9\},$$
$$\mathcal{D}_{\mathrm{act}} = \{1,3,5,7\}, \quad \mathcal{D}_{\mathrm{evt}} = \{2,4,6,8\}, \quad \mathcal{D}_{\bar{0}}^{(9)} = \{9\}.$$

The final symbol means 'support of the null nonadic residue'. It does not designate the complete domain $D_9 = \{1, \dots, 9\}$ or a physical zero-mass sector.

Let $E_{\mathcal{D}} \cong \mathbb{Z}^3$ and $E_C \cong \mathbb{Z}^3$ be the free modules with ordered bases

$$(\mathcal{D}_{\mathrm{act}}, \mathcal{D}_{\mathrm{evt}}, \mathcal{D}_{\bar{0}}^{(9)}) \quad \text{and} \quad (C_1, C_2, C_0),$$

respectively. The incidence map $B_{\mathrm{res,sup}} : E_{\mathcal{D}} \to E_C$ counts intersections.

Theorem 92.41 (Smith form and image of incidence)**.** *In the preceding bases,*

$$B_{\mathrm{res,sup}} = \big(|C_i \cap \mathcal{D}_j|\big)_{i,j} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

Moreover,

$$\det B_{\mathrm{res,sup}} = 3, \qquad \mathrm{SNF}(B_{\mathrm{res,sup}}) = \mathrm{diag}(1,1,3),$$
$$\mathrm{im}\, B_{\mathrm{res,sup}} = \{(u,v,w) \in \mathbb{Z}^3 : u + v \equiv 0 \pmod 3\}.$$

If

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

the simultaneous exchange $C_1 \leftrightarrow C_2$ and $\mathcal{D}_{\mathrm{act}} \leftrightarrow \mathcal{D}_{\mathrm{evt}}$ is equivariant:

$$P B_{\mathrm{res,sup}} P = B_{\mathrm{res,sup}} \qquad \Longleftrightarrow \qquad P B_{\mathrm{res,sup}} = B_{\mathrm{res,sup}} P.$$

Proof. The nine entries follow from the intersections.

$$\begin{array}{lccc} & \mathcal{D}_{\mathrm{act}} & \mathcal{D}_{\mathrm{evt}} & \mathcal{D}_{\hat{0}}^{(9)} \\ C_1 & \{1,7\} & \{4\} & \varnothing \\ C_2 & \{5\} & \{2,8\} & \varnothing \\ C_0 & \{3\} & \{6\} & \{9\}. \end{array}$$

The expansion by the third column gives

$$\det B_{\mathrm{res,sup}} = \det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = 3.$$

The greatest common divisor of the entries is one. The smallest second-order minor formed by the first and third rows and the second and third columns has determinant one:

$$\det \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = 1.$$

By the Smith divisors, $d_1 = d_2 = 1$. The product $d_1 d_2 d_3$ agrees with the absolute value of the determinant, which the preceding calculation fixes at three. Consequently,

$$d_1 d_2 d_3 = 3.$$

The normal form is therefore $\mathrm{diag}(1, 1, 3)$.

Each column satisfies $u + v \equiv 0 \pmod 3$, so the image is contained in the indicated submodule. Conversely, if $u + v \equiv 0 \pmod 3$, then

$$x = \frac{2u - v}{3}, \qquad y = \frac{2v - u}{3}, \qquad z = w - x - y$$

are integers and

$$B_{\mathrm{res,sup}}(x, y, z)^{\mathsf{T}} = (u, v, w)^{\mathsf{T}}.$$

This proves equality of images. Finally, explicit multiplication by P exchanges the first two rows and the first two columns and leaves the matrix invariant. □

Comparison Criterion 92.42 (Incidence control)**.** The theorem would be falsified by a single miscounted intersection, a minor that altered the Smith divisors, a vector of the image with $u+v \not\equiv 0 \pmod 3$, a vector satisfying the congruence but admitting no exhibited preimage, or the loss of equivariance $PBP = B$.

92.11.2 Based ring associated with the incidence F_{av}

The theorem 6.20 proves, from the orbit of modes (Σ, Π, Π), the matrix

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

From this matrix the following based ring is obtained.

Definition 92.43 (Fibonacci-based Ring)**.** Let

$$K_{\mathrm{Fib}} := \mathbb{Z}\mathbf{1} \oplus \mathbb{Z}\tau$$

the free abelian group with unit $\mathbf{1}$ and bilinear product fixed by

$$\tau \cdot \mathbf{1} = \tau, \qquad \tau \cdot \tau = \mathbf{1} + \tau.$$

In this stratum, + and · are the addition and multiplication of the ring. The categorical symbols ⊕ and ⊗ are reserved for a categorification chosen subsequently.

Theorem 92.44 (Realization of F_{av} and Perron dimension)**.** *In the ordered basis* $(\mathbf{1}, \tau)$*, multiplication by* τ *in* K_{Fib} *has matrix* F_{av}*. If* $F_0 = 0$*,* $F_1 = 1$*, and* $F_{n+1} = F_n + F_{n-1}$ *for* $n \geq 1$*, then*

$$\tau^n = F_{n-1}\mathbf{1} + F_n\tau \qquad (n \geq 1).$$

The unique homomorfismo positivo of dimension that envia $\mathbf{1}$ *to one satisfies*

$$\mathrm{FPdim}(\tau) = \varphi.$$

Identification of the two typed sheets with the basis $(\mathbf{1}, \tau)$ *completely determines this ring and its positive dimension.*

Proof. The columns of the matrix of multiplication by τ are the coordinates of

$$\tau \cdot \mathbf{1} = \tau \quad \text{and} \quad \tau \cdot \tau = \mathbf{1} + \tau;$$

they are therefore $(0, 1)^{\mathsf{T}}$ and $(1, 1)^{\mathsf{T}}$, respectively. Thus one obtains F_{av}.

The power formula holds for $n = 1$. If it holds for n, then

$$\tau^{n+1} = F_{n-1}\tau + F_n(\mathbf{1} + \tau) = F_n\mathbf{1} + F_{n+1}\tau,$$

which closes the induction. Finally, a positive multiplicative dimension $d = \mathrm{FPdim}(\tau)$ obeys

$$d^2 = 1 + d.$$

Its roots are φ and $-\varphi^{-1}$; positivity selects $d = \varphi$. □

The designation *Fibonacci* is thus explained by the fusion rule itself: upon iterating τ, the two possible channels appear with consecutive multiplicities F_{n-1} and F_n. The space of fusion states therefore grows with asymptotic ratio φ. This is the combinatorial basis of its computational use: information is encoded in global fusion channels and not in a mass assigned to a point.

Comparison Criterion 92.45 (Conditions for a Fibonacci categorification)**.** The annular layer would be refuted if the multiplication matrix were not F_{av}, if the power recurrence failed, or if the positive dimension were not φ. A braided categorification whose Grothendieck ring is that of definition 92.43 additionally requires a monoidal functor

$$\mathfrak{C}_{\mathrm{Fib}} : \mathsf{TPK} \dashrightarrow C_{\mathrm{Fib}},$$

the verification of the pentagon and hexagon in its image and an internal rule that selects gauge and chirality. These conditions are not consequences of the annular identity by itself.

92.11.3 Classification of 6-character by 4 braid torsion classes

Let B_n, $n \geq 2$, be the Artin braid group with generators $\sigma_1, \ldots, \sigma_{n-1}$ and relations

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}, \qquad \sigma_i\sigma_j = \sigma_j\sigma_i \quad (|i - j| \geq 2),$$

and let $w : B_n \to \mathbb{Z}$ be the sum of the exponents. The readers

$$w_2 := w \bmod 2, \qquad w_3 := w \bmod 3$$

are brought together, by the Chinese remainder theorem, in

$$w_6 : B_n \longrightarrow \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \simeq \mathbb{Z}/6\mathbb{Z}.$$

Theorem 92.46 (Abelianization, six characters and permutation torsions)**.** *For $n \geq 2$,*

$$B_n^{\mathrm{ab}} \simeq \mathbb{Z},$$

through w. Consequently, w_6 produces six unitary characters

$$\chi_k(b) := \exp\left(\frac{2\pi \mathrm{i} k}{6} w(b)\right), \qquad k \in \mathbb{Z}/6\mathbb{Z}.$$

Let $V \neq 0$ be a Hilbert space, let $P_i : V^{\otimes n} \to V^{\otimes n}$ be the exchange of the factors $i, i+1$, and let $q \in \mu_6$. The rule

$$\rho_q(\sigma_i) := qP_i, \qquad \rho_q(b) = q^{w(b)} \pi_n(b),$$

where π_n is the permutation representation, defines a representation of B_n. This representation descends to the symmetric group if and only if

$$q^2 = 1.$$

For the four values $q \in \mu_6 \setminus \{\pm 1\}$,

$$\mathrm{Inv}(\rho_q) = 0.$$

Those four values define genuinely braided scalar torsions, but do not by themselves constitute four Fock spaces nor four physical anyonic sectors.

Proof. In every abelian quotient, the braid relation gives

$$2[\sigma_i] + [\sigma_{i+1}] = [\sigma_i] + 2[\sigma_{i+1}],$$

and thus all generators have the same class. No other relation remains on that class, since $w(\sigma_i) = 1$; hence the abelianization is $\mathbb{Z}$.

The operators P_i satisfy the braid relations and $P_i^2 = I$. Multiplying each generator by the same scalar q preserves both relations, so ρ_q is a representation. To factor through S_n, it must satisfy the additional relation $\sigma_i^2 = 1$, and

$$\rho_q(\sigma_i)^2 = q^2 I.$$

This proves criterion $q^2 = 1$.

If v were invariant, then $qP_i v = v$, that is, $P_i v = q^{-1} v$, for every i. Since each involution P_i has only eigenvalues ± 1, no such nonzero vector exists when $q \neq \pm 1$. Consequently, $\mathrm{Inv}(\rho_q) = 0$ for the remaining four values. □

When $\dim V > 1$ and $n \geq 3$, the operators P_1 and P_2 do not commute; hence the full representation does not become Abelian merely because it arises from a scalar character. The character χ_k and the representation ρ_q are distinct objects: the former is one-dimensional; the latter retains the permutation action.

92.11.4 Modular center pointed and Fibonacci realization conditions Ising

Let

$$A = (\mathbb{Z}/9\mathbb{Z})^2$$

and let Vec_A be the category of A-graded vector spaces with trivial associator. Fixing a nondegenerate bicharacter identifies $\widehat{A}$ with A, but does not alter the following counts.

Theorem 92.47 (Pointed center and dimensional obstruction)**.** *The Drinfeld center*

$$\mathcal{Z}(\mathrm{Vec}_A)$$

is a pointed modular category whose simple objects are parameterized by $(a, \chi) \in A \times \widehat{A}$. It therefore has

$$|A|\,|\widehat{A}| = 81^2 = 6561$$

simple, all of quantum dimension one, and total quantum dimension

$$\mathcal{D} = \sqrt{\sum_{x \text{ simple}} d_x^2} = \sqrt{6561} = 81.$$

In this category, neither the Fibonacci ring nor the Ising sector can be realized while preserving simple objects and Frobenius–Perron dimensions: they require, respectively, a simple object of dimension φ and one of dimension $\sqrt{2}$.

Proof. For a finite abelian group and trivial associator, every simple object of the center is determined by a degree $a \in A$ and a character $\chi \in \widehat{A}$ that specifies its half-braiding. Both the graded object and the character are one-dimensional; consequently, all the simple objects are invertible and have quantum dimension one. Since $|A| = |\widehat{A}| = 9^2 = 81$, the number of pairs is 81^2, and the formula of $\mathcal{D}$ gives 81.

Every fusion subcategory of a pointed category is again pointed; in particular, its simple objects have dimension one. This contradicts $\mathrm{FPdim}(\tau) = \varphi$ for Fibonacci and $d_\sigma = \sqrt{2}$ for Ising. The obstruction is dimensional and does not depend on a similarity of names or cardinalities. □

Comparison Criterion 92.48 (Conditions for a physical realization of the sectors)**.** The results of theorems 92.46 y 92.47 would be algebraically falsified by failure of the Artin relations, the criterion $q^2 = 1$, the census 6561, pointedness, or $\mathcal{D} = 81$. A physical realization additionally requires a Hamiltonian or driven protocol, a spectral gap, realizable braid operators, protection against perturbations, and a map from the enriched multisections. None of these data follows solely from the two preceding algebraic theorems.

92.11.4.1 Composite object as a persistent section of a cocyclic product of paths

A composite particle is a section of the fiber product of constituent biographies with a binding boundary. The genealogical records are composed before the signature extractor is applied. If Γ and Λ are linked through the boundary B and the action g,

$$(n, e)_\Gamma \star_{(g,B)} (n, e)_\Lambda = (n_\Gamma + n_\Lambda - n_B, e_\Gamma + g e_\Lambda - e_B).$$

The composite signature is

$$\mathrm{Sig}_{\mathrm{comp}} = L_{\mathrm{tick}} \circ \mathfrak{C}_{12}.$$

It does not in general coincide with the sum of the individual signatures, because the genealogical record of linkage carries energy, carry, and memory. This is the residence of binding energy.

92.11.4.2 Resonance as a transient section and width as a decay rate

A resonance is a metastable section whose resolvent has a complex energy pole. If $z_X = M_X c_{\text{int}}^2 - i\Gamma_X/2$, the mass is $M_X = c_{\text{int}}^{-2} \operatorname{Re} z_X$; the width measures its mean lifetime:

$$\Gamma_X = \frac{\hbar_{\text{int}}}{\tau_X^{\text{life}}}.$$

The width is not a sixth coordinate of σ. It proceeds from the survival operator, from the decay channels, and from the analytic continuation of the resolvent. The inventory of 384 widths therefore requires its own reader, separate from the inventory of 471 masses.

92.11.4.3 Virtual section as an intermediate non-asymptotic extension of the atlas

A virtual section is a finite extension that participates in a composition or propagator but does not define a stable global branch in the physical codomain. If $L(s)$ is its horizon,

$$L(s) < \infty, \qquad \operatorname{Ext}_{L(s)}(s) \neq \varnothing, \qquad \operatorname{Ext}_{L(s)+1}(s) = \varnothing.$$

Virtuality is a terminal obstruction, not a documentary label or an imaginary mass. A resonance may be unstable without being virtual in this sense; an internal line may be virtual without constituting a new species.

92.11.4.4 Majorana Section as a Fixed Point of Particle–Antiparticle Conjugation

Four constructions must be kept separate. The word involution C_M defines the antisection; a Majorana spinor satisfies a reality condition; a Majorana zero mode is a localized zero-energy excitation in a topological Hamiltonian; the Ising sector possesses fusion rules

$$\sigma \otimes \sigma = 1 \oplus \psi.$$

A physical particle is Majorana when its fermionic representation admits a self-conjugation compatible with the Hamiltonian, localization, and inner product. Electrical neutrality and invariance of a word under C_M are not sufficient. In particular, deciding the Majorana nature of neutrinos requires an intertwiner among the neutral operator, the mass basis, and the fermionic real structure.

92.11.4.5 Anyonic sector and nontrivial representation of the braid group

A anyon is a section bidimensional whose representation of the grupo of trenzas does not reduces to the permutation bosonic or fermionic. In the sector Fibonacci,

$$\tau \otimes \tau = 1 \oplus \tau, \qquad \operatorname{FPdim}(\tau) = \varphi.$$

In the abelian sector $\mathbb{Z}/6\mathbb{Z}$, the winding quotient has six one-dimensional characters. Obtaining a physical particle requires F and R matrices, the pentagon and hexagon equations, a gapped Hamiltonian, and braiding operators. The exact fusion ring determines the topological codomain, but does not by itself fix a universal mass.

92.11.4.6 Tachyonic and ghost classes as obstructions to positivity and stability

A persistent particle has unitary evolution, energy bounded below, and real spectral mass on the physical line. A tachyon would require a pole with $m^2 < 0$ that remained as a stable asymptotic state. In the HMT–MD architecture, a direction leaving the cone of positive realizations either returns to the null boundary, loses survival, or manifests as an instability of a background that must condense. It does not define a persistent superluminal section. The impossibility is formulated as incompatibility among global persistence, spectral positivity, and $m^2 < 0$, not as a prohibition of transient tachyonic parameters in a perturbative expansion.

A ghost is a negative-norm mode. Positivity of the inner product and unitary conservation exclude its physical realization. Faddeev–Popov ghosts belong to the gauge complex and not to the physical state space; this distinction must be preserved.

92.11.5 Composition, resonances and widths

92.11.5.1 Non-Additive Mass Composition via the Cocylic Link Law

The General Mass Law acts on genealogical biographies. When several constituents form a bound state, the input object is not a list of their isolated masses, but a composite record that preserves the coupling order, common boundary, carry, orientation, and memory. Let

$$\Gamma_1, \dots, \Gamma_n$$

constituent routes, $\mathcal{T}$ a temporal composition tree, and ∂ the linking boundary. The dodecaphase composition operator is written

$$\mathfrak{C}_{12}^{(n)} : (\Gamma_1, \dots, \Gamma_n; \mathcal{T}, \partial) \longmapsto \mathcal{L}_{\mathrm{comp}}.$$

Only after these are applied are the register reader, the signature extractor, the character, and the towers:

$$\mathcal{L}_{\mathrm{comp}} \longmapsto \sigma_{\mathrm{comp}} \longmapsto \chi_{\mathrm{comp}} \longmapsto \widehat{M}_{\mathrm{comp}}.$$

This precedence is essential. If the isolated masses were evaluated first and then added, precisely the data that distinguish a bound state from a threshold of free constituents would disappear. Binding energy is not a term appended at the end: it is the dimensional image of a record difference produced by the linkage.

92.11.5.2 Cocyclic Link Law

For two records L_1, L_2, the composition has the form

$$L_1 \star L_2 = L_1 + L_2 + \mathfrak{b}(L_1, L_2),$$

where $\mathfrak{b}$ records shared boundary, carry, and memory. Independence with respect to parenthesization requires the cocyclic identity

$$\mathfrak{b}(L_1, L_2) + \mathfrak{b}(L_1 \star L_2, L_3) = \mathfrak{b}(L_2, L_3) + \mathfrak{b}(L_1, L_2 \star L_3).$$

If this equality fails, the two ways of constructing the same ternary tree produce distinct genealogical records, and no well-defined composite species exists. If it holds, the cocycle does not disappear: it ensures that its contribution is a property of the state rather than an artifact of notation.

A more explicit presentation also preserves the boundary action. Let $e \in \mathbb{Z}^{12}$ be the ordered vector of events and let n be the action or event counter of a ledger. If $(n, e)_\Gamma$ and $(n, e)_\Lambda$ are ledgers, if B is the common subledger with data (n_B, e_B), and if $g \in \mathrm{Aut}(\mathbb{Z}^{12})$ transports the orientation of the second ledger to that of the first while preserving the reader,

$$(n, e)_\Gamma \star_{(g,B)} (n, e)_\Lambda = (n_\Gamma + n_\Lambda - n_B, e_\Gamma + g e_\Lambda - e_B).$$

The term B prevents counting the identified boundary twice; the action g transports the second record to the orientation of the first. The composite signature is therefore,

$$\mathrm{Sig}_{\mathrm{comp}} = L_{\mathrm{tick}}\Big(\mathfrak{C}_{12}^{(n)}(\Gamma_1, \dots, \Gamma_n; \mathcal{T}, \partial)\Big),$$

and not the automatic sum of elementary signatures.

92.11.5.3 Cocyclic Closure of the Mesonic Quark–Antiquark Composition

An oriented mesonic expression has the form

$$M(q, q') = q \otimes (q')^{\vee}.$$

The raw domain contains $6 \times 6 \times 3 \times 4 = 432$ expressions when flavour, compatible colour, and orientation are considered. This cardinality is not the number of physical mesons. Before a species is individualized, the following are applied:

1. the color-singlet projector;
2. the charge and isospin projector;
3. the spin and parity representation;
4. the temporal linkage tree;
5. the radial or orbital condition;
6. the preparation and decay channel when the state is resonant.

The complete physical descriptor is

$$\mathcal{I}_M = (q\bar{q}', J^{PC}, I, I_3, S, C, B, \mathcal{T}, \partial, \Gamma_M, \mathcal{L}_M, \sigma_M, \eta_M).$$

Two states with the same flavors may differ in J^{PC}, radial excitation, orbital momentum, sheet, or channel, and therefore have distinct poles. The mass is not derived from the mere name of the constituents.

The evaluations published for $\pi^0, \pi^{\pm}, K^{\pm}, K^0$ are exact once the declared historical registers have been fixed. The stronger assertion —the prospective generation of each nominal register from its linkage tree— requires $\mathfrak{C}_{12}$ to reconstruct those registers without consulting the value to be compared. Both layers are preserved together: the conditioned calculation is not erased, nor is it presented as a selector of identity.

92.11.5.4 Trilinear Closure of the Baryonic Composition and Neutralization of the Color Character

An oriented baryonic expression uses three constituents and three colors. The raw domain of the adopted census contains

$$|\mathcal{B}_{\mathrm{RGB}}^{\mathrm{raw}}| = 1728$$

expressions. The physical projection must impose simultaneously:

$$P_{\text{singlete de color}}, \qquad P_{J^P}, \qquad P_{I,I_3,S,C,B}, \qquad P_{\text{Pauli}}.$$

The triple tree preserves which pair was coupled first and how the third constituent crossed the boundary. The cocycle identity guarantees that equivalent parenthesizations represent the same state when the physical intertwiner identifies them; inequivalent channels remain distinct states.

The historical signatures

$$\sigma_p = (59, 0, 9, 1, 0), \qquad \sigma_n = (59, 1, -34, 0, 1)$$

produce the published proton and neutron evaluations when the complete character is applied. They are exact controls of the reader for those ledgers. A physical explanation of the neutron–proton difference additionally requires deriving, before comparison, the sign of isospin, vacancy orientation, the link boundary, and the electromagnetic contribution in a common chart.

92.11.5.5 Exotic composite realizations and stability conditions of the bond

The same grammar admits more constituents without modifying the law:

$$\mathfrak{C}_{12}^{(n)} : \prod_{j=1}^{n} \Gamma_j \times \mathcal{T}_n \times \partial_n \longrightarrow \mathcal{L}_{\text{comp}}.$$

A tetraquark corresponds to $qq\bar{q}\bar{q}$; a pentaquark, to $qqqq\bar{q}$; a glueball, to routes of the adjoint color representation projected onto the singlet; a hybrid, to fermionic and gauge routes brought together in the same tree. The list of constituents does not by itself determine the species. A complete physical classification must include color, spin, parity, conjugation, isospin, tree topology, and the pole of the coupled operator.

This formulation also discriminates between two rival descriptions of the same candidate —a hadronic molecule and a compact state, for example—: they are two trees or two projectors on the same constituent sector. They become equivalent only if there exists a unitary operator that intertwines their operators, channels, and poles.

92.11.5.6 Steady state, bound state and resonance

It is convenient to separate three notions.

A stable state is a normalizable eigenvector of the physical Hamiltonian, with real energy and no open decay channel. A bound state is an eigenvector below the relevant threshold; it may be stable or may decay through another channel. A resonance is a pole of the continuation of the resolvent or of the scattering matrix. It is not a normalizable eigenvector of the self-adjoint Hamiltonian on the real axis.

For an energy pole

$$z_X = E_X - \frac{i}{2}\Gamma_X = M_X c_{\text{int}}^2 - \frac{i}{2}\Gamma_X, \qquad \Gamma_X > 0,$$

the pole energy is $E_X = \operatorname{Re} z_X$, the pole mass is $M_X = c_{\text{int}}^{-2} \operatorname{Re} z_X$, and the energy width is $\Gamma_X = -2 \operatorname{Im} z_X$. The General Mass Law supplies the real branch conditioned by the identity and the register; continuation through open channels determines the complex displacement and the width. In a natural-unit chart $c_{\text{int}} = 1$, the numerical coordinates of energy and mass coincide, but their types are not identified before that chart is chosen.

92.11.5.7 Correction of the sign in the discrete reading of the pole

Let $t_0 > 0$ be a time step, and suppose that the one-step mode has eigenvalue

$$\lambda = re^{-i\theta}, \qquad 0 < r < 1,$$

with the convention

$$\lambda = \exp\left(-\frac{i}{\hbar_{\text{int}}} z t_0\right).$$

Then

$$z = \frac{\hbar_{\text{int}}}{t_0}\left(\theta + i \log r\right) = E - \frac{i}{2}\Gamma = M c_{\text{int}}^2 - \frac{i}{2}\Gamma,$$

so that

$$E = \frac{\hbar_{\text{int}}}{t_0}\theta, \qquad M = \frac{\hbar_{\text{int}}}{t_0 c_{\text{int}}^2}\theta, \qquad \Gamma = -\frac{2\hbar_{\text{int}}}{t_0} \log r > 0.$$

The positive sign before $i \log r$ is compulsory: since $\log r < 0$, the imaginary part of z is negative. The form $\theta - i \log r$ would place the pole in the upper half-plane and would describe growth, not decay.

The survival probability after N steps is

$$|\lambda|^{2N} = r^{2N} = \exp\left(-\frac{\Gamma}{\hbar_{\text{int}}} N t_0\right).$$

Therefore, the associated lifetime is

$$\tau_X^{\text{life}} = \frac{\hbar_{\text{int}}}{\Gamma_X}.$$

This identity further proves why the width must not be introduced as a sixth coordinate of the signature $\mathbb{Z}^5$: it arises from the modulus of the open mode, not from the positive character that organizes mass.

92.11.5.8 Effective Channel Operator

Let P be the prepared subspace and $Q = I - P$ the set of eliminated channels. The complement is incorporated by means of the Feshbach operator

$$H_{\text{eff}}(z) = PHP + PHQ(z - QHQ)^{-1}QHP.$$

The poles satisfy

$$\det\big(z - H_{\text{eff}}(z)\big) = 0.$$

The first term preserves the prepared internal operator. The second is the self-energy of the channels: it shifts the real part and produces an imaginary part when the continuum opens. The character mass and the survival width converge at the same pole, but they follow from distinct maps.

An equivalent formulation uses a distinct survival projector from the Feshbach complement. Let

$$S_X = S_X^* = S_X^2$$

the projector onto the surviving subspace of the prepared state. The compressed one-step operator is

$$T_X = S_X U S_X.$$

Suppose that T_X has a simple dominant eigenvalue λ_X, that the prepared state has nonzero overlap with its spectral subspace, and that the remainder of the spectrum is separated in modulus. If $r_X = |\lambda_X| = \rho(T_X) < 1$, then

$$\Gamma_X = -\frac{2\hbar_{\text{int}}}{t_0} \log r_X.$$

If the operator is normalized from the beginning on probabilities instead of amplitudes, the factor two disappears. The publication must declare which of the two conventions it uses; mixing them produces an erroneous width by a factor of two.

92.11.5.9 Conjugation of particle and antiparticle

Let C_M be the involution that reverses orientation and sheet, and let U_C be its unitary realization on physical space. Equality of pole masses and widths requires covariance of both the operator and the channels:

$$U_C H_{\text{eff}}^X(z) U_C^{-1} = H_{\text{eff}}^{\bar{X}}(z).$$

Then the sets of retarded poles coincide after transporting the sector and the causal prescription, and

$$M_{\bar{X}} = M_X, \qquad \Gamma_{\bar{X}} = \Gamma_X.$$

Bare unitary equivalence does not turn the retarded pole into its complex conjugate; it transports the same spectral problem to the antiparticle sector. Parity of the mass signature is insufficient to prove the second equality: the channels and their measures must also be covariant. If the symmetry is represented antiunitarily, the retarded prescription must also be transported; bare conjugation $z \mapsto \bar{z}$ exchanges retarded and advanced poles and is not, by itself, the physical assertion of equality of lifetimes.

92.11.5.10 Typed inventory of realizations and contrast observables

The preserved external cut contains

$$471 \text{ mass observables}, \qquad 384 \text{ width observables}.$$

The mass partition is

Physical arrival domain	Masses	Widths
partículas elementales, leptónicas y extensiones	53	9
quarks y extensiones quark	8	1
mesones	243	225
bariones	166	149
registro gravitatorio de contraste	1	0
Total	**471**	**384**

Estas filas son observables, no especies elementales. Varias filas pueden referirse a diferentes cargas de un mismo multiplete, estimadores de una misma resonancia, masas Breit–Wigner, masas de polo o estados excitados. La individualización interna precede siempre a la fila externa.

El inventario conserva además una propiedad metodológica fuerte: ninguna de sus 855 magnitudes fue autorizada como selector de ruta. Los 471 registros de masa y los 384 de anchura sólo se abren después de congelar la identidad HMT, el proyector, el operador y la carta de comparación.

92.11.5.11 Normalization of Nominal Identities in Comparable Registers

The complete prospective splice has the form

$$\begin{aligned} \text{physical identity} &\longrightarrow \text{sector and representation} \longrightarrow \text{fibre or composite-tree projector} \\ &\longrightarrow \text{enriched route} \longrightarrow \text{genealogical record} \longrightarrow \text{signature and towers} \\ &\longrightarrow \text{prepared operator} \longrightarrow \text{pole or eigenvalue} \longrightarrow \text{dimensional chart} \\ &\longrightarrow \text{external comparison}. \end{aligned}$$

For a stable state, the operator output is a real eigenvalue. For a resonance, it is a complex pole. For a quark, the map includes a scheme and a scale. For an experimental limit, it includes the confidence level. For a correlated observable, it includes the covariance matrix. These differences are structural: they determine which quantities may be compared.

92.11.5.12 Individual closure conditions of material realizations

A row of the inventory is individualized from HMT–MD once the following have been constructed, without consulting its external magnitude:

1. the sector and the representation;
2. the route, multisection, or composition tree;

3. the genealogical record and the signature;
4. the operator of masa and, if proceeds, the operator of channels;
5. the projector selecting the named state;
6. the convention of mass, width, scheme, and scale;
7. the dimensional chart;
8. the internal falsification criterion.

The universal law already provides the common grammar. Working state by state does not consist in inventing a different formula, but in constructing the intertwiner that transports a physical identity to the appropriate operator.

92.11.5.13 Specific Obstructions and Non-Degeneracy Conditions

The composition is refuted if two physically equivalent parenthesizations yield different signatures or spectra. The nominal selector is refuted if it changes when the external mass column is hidden. The width reader is refuted if it produces $\Gamma < 0$ for a contraction, confuses amplitude with probability, or assigns a width to a stable section without an open channel. The resonance chart is refuted if it does not specify pole or Breit–Wigner mass. The quark comparison is indeterminate if it omits a scheme or scale. Particle–antiparticle equality of widths is refuted if the channel operator is not covariant under conjugation.

92.11.5.14 Transport of material classes to the exceptional corridor via routes, signatures, and representations

The grammar of composition, the mesonic and baryonic domains, the Feshbach operator, the 471/384 census, and the comparison principle are retained in the preceding integral development, Sections 16 and 18. The individualized records are presented in full in the exhaustive inventories of masses and widths in the appendix. The correction of the pole sign follows directly from $\lambda = \exp(-izt_0/\hbar_{\mathrm{int}})$ and strengthens the formulation without altering any of those records.

92.11.6 Topological sectors, hypothetical extensions and gravity

92.11.6.1 Typed separation between topological sectors, hypothetical extensions and gravitational realization

The architecture distinguishes with precision:

1. an internal algebraic identity;
2. a fusion or braiding codomain;
3. a physical particle with a Hamiltonian, gap, localization, and measurement channel.

An exact fusion rule is not yet a mass. A boundary mode is not yet a species. A route involution is not yet a Majorana condition. The physical passage requires a functor that preserves product, involution, orientation, energy, and path composition.

92.11.6.2 Fibonacci Sector: ring-based, fusion and braid representation

The areal–volumetric incidence is governed by

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

Its characteristic polynomial is

$$x^2 - x - 1,$$

and its dominant eigenvalue is φ. In the basis $\{\mathbf{1}, \tau\}$, multiplication by τ is represented by F_{av}; therefore

$$\tau \otimes \tau \cong \mathbf{1} \oplus \tau, \qquad \mathrm{FPdim}(\tau) = \varphi.$$

This is an exact realization of the Fibonacci based ring. To obtain a physical anyonic category, one must also construct the associator F, the braiding R, the pentagon and hexagon equations, chirality, a gapped local Hamiltonian, and the preparation-and-readout map. The mass output, when it exists, is an energy gap converted by an effective velocity:

$$m_{\mathrm{brecha}} = \frac{\Delta E}{c_{\mathrm{eff}}^2}.$$

There is no universal Fibonacci-anyon mass deducible solely from φ.

92.11.6.3 Separation between the topological sector and the pointed center of the atlas

The braided center of the pointed sector based on $\mathbb{Z}/9\mathbb{Z}$ has 81 invertible simple objects, each of quantum dimension one. A Fibonacci simple object would require dimension φ, and an Ising simple object would require $\sqrt{2}$. Therefore, neither appears as an internal simple object of that pointed center.

This control does not eliminate the topological realizations. It determines their proper place: they must be obtained by extension, condensation, quotient, defect, or a monoidal functor into a nonpointed codomain. Presenting the Fibonacci ring as though it were already the entire category would erase precisely the physical morphism that must be constructed.

92.11.6.4 Ising Sector and Majorana Modes

The Ising category has simple objects.

$$\mathbf{1}, \quad \psi, \quad \sigma$$

and rules

$$\psi \otimes \psi = \mathbf{1}, \qquad \psi \otimes \sigma = \sigma, \qquad \sigma \otimes \sigma = \mathbf{1} \oplus \psi, \qquad d_\sigma = \sqrt{2}.$$

Four distinct objects often receive similar names and must be kept separate:

- the route involution $C_M = -\,\mathrm{rev}$;
- the parity of the CAR completion;
- a self-conjugate relativistic spinor;
- a topological Majorana zero mode in an Ising phase.

A relativistic fermion is Majorana when there exists a real conjugation C compatible with its representation and dynamics,

$$\psi = C\bar{\psi}^T,$$

and the mass term respects that condition. A Majorana zero mode is described by a self-adjoint operator $\gamma = \gamma^\dagger$ satisfying

$$\{\gamma_i, \gamma_j\} = 2\delta_{ij}$$

and remains separated from the continuum by a gap. Equality of a word with its image under C_M proves neither of these two realizations.

For neutrinos, the Dirac/Majorana alternative can be decided only after constructing the neutral operator, its mass basis, the mixing intertwiner, and the real fermionic structure. Electrical neutrality is necessary, but not sufficient, for nontrivial self-conjugacy.

92.11.6.5 Abelian braiding of order 6

Let

$$\mathrm{wind}_6 : B_n \longrightarrow \mathbb{Z}/6\mathbb{Z}$$

the reader of winding and

$$\chi_k(b) = \exp\!\left(\frac{2\pi i k}{6}\,\mathrm{wind}_6(b)\right).$$

Realizations with phase q descend to mere permutations only when

$$q^2 = 1.$$

The remaining four characters distinguish algebraic classes of the winding reader. The exact result is the classification of the six one-dimensional characters of the winding quotient; it is not yet a classification of six material phases. To assert that they preserve physical braiding requires braid operators and a representation of the braid group. Their existence does not imply four new particles: an anyonic species requires a Hamiltonian, braid operators, a gap, stability, and a coupling to a physical multisection. Their promotion to material phases depends upon that realization.

92.11.6.6 Positivity and exclusion of the persistent tachyon

The action line possesses positive orientation,

$$\hbar_{\mathrm{int}} > 0,$$

the mass character takes positive values,

$$\chi_{\mathrm{full}}(\sigma, \eta) > 0,$$

and the bidirectional refinement is implemented by conjugation. Suppose explicitly

$$\widehat{M}^{(0)} \geq 0, \qquad \widehat{K} = \widehat{K}^*, \qquad R_{\mathrm{act}} > 0.$$

Then

$$\widehat{M}^{\beta} = R_{\mathrm{act}}^{\widehat{K}/2}\, \widehat{M}^{(0)}\, R_{\mathrm{act}}^{\widehat{K}/2} \geq 0.$$

Consequently,

$$\mathrm{Spec}(\widehat{M}^{\beta}) \subset [0, \infty).$$

Therefore, the squared mass operator

$$\widehat{M^2}_{\beta} := (\widehat{M}^{\beta})^2$$

is positive and satisfies

$$\mathrm{Spec}(\widehat{M^2}_\beta) \subset [0, \infty).$$

A persistent asymptotic particle satisfying the positivity and unitarity assumptions cannot have $m^2 < 0$. If linearization about a background yields an imaginary frequency, the object is an unstable direction: the background must relax or condense. If the route ceases to admit prolongation, the object is a virtual section with a terminal obstruction. Neither case defines a stable superluminal particle.

The scope of the proposition is fixed by its premises. A "tachyonic" parameter in a perturbative potential may signal a phase transition; what is excluded is its promotion to a persistent species of positive norm.

92.11.6.7 Ghost-independent exclusion

A physical ghost would be a negative-norm mode. Let $\mathcal{H}_{\mathrm{phys}} = \ker G / \mathrm{im}\, G_0$ be the gauge quotient endowed with a positive inner product, and let $\widehat{M}_{\mathrm{phys}}$ be self-adjoint on it. Then the spectral measure of a prepared state satisfies

$$\mu^{\mathrm{mass}}_{P,a}(B) \geq 0.$$

A negative direction indicates that positivity of the quotient failed or that the mode does not belong to the physical space; it does not disappear through a nominal assertion. Faddeev–Popov ghosts are variables of the gauge complex, not asymptotic particles. This exclusion is distinct from the tachyonic one: a ghost violates norm positivity; a tachyonic direction violates background stability.

92.11.6.8 Photon, unidirectional bias, and Proca–Stueckelberg bridge

Harmonic channels produce the bias

$$\varepsilon = q^{90}_- - q^{90}_+.$$

A bidirectional kinematic realization can be written

$$c_+ = \frac{c_{\mathrm{int}}}{1+\varepsilon}, \qquad c_- = \frac{c_{\mathrm{int}}}{1-\varepsilon},$$

for which the round-trip measure preserves exactly

$$\frac{2}{c_+^{-1} + c_-^{-1}} = c_{\mathrm{int}}.$$

The bias remains in the unidirectional reading and cancels in the reciprocal reading. This identity belongs to the architecture of two orientations; by itself it does not prove a photonic mass.

If the boundary closure produces a gap frequency $\mu_T \in \mathcal{L}_T^{-1}$ in the register,

$$\omega^2 = c_{\mathrm{int}}^2 k^2 + \mu_T^2,$$

a dimensional entangler can transport it to a Proca–Stueckelberg realization with effective mass

$$m_\gamma = \frac{\hbar_{\mathrm{int}} \mu_T}{c_{\mathrm{int}}^2}.$$

For propagation of frequency ω, the first-order temporal correction satisfies

$$\frac{\Delta T}{T} \simeq \frac{1}{2} \left(\frac{m_\gamma c_{\mathrm{int}}^2}{\hbar_{\mathrm{int}} \omega} \right)^2.$$

The required chain is

$$\text{TPK outward/return} \longrightarrow \text{memory} \longrightarrow \mu_T \longrightarrow m_\gamma.$$

ε, the action quotient R_{act}, and μ_T are not automatically identified: they reside in distinct codomains. The null chart of the ordinary photon remains exact until the gap and its physical intertwiner have been constructed.

92.11.6.9 Operational classification of dark sectors by domain, reader and physical codomain

A dark sector is not defined by an arbitrary mass, but by the kernel of its couplings. If P_X is its projector,

$$P_{\text{em}} P_X = 0, \qquad P_{\text{color}} P_X = 0,$$

while it may occur

$$P_{\text{grav}} P_X \neq 0$$

or a weak coupling may exist. The law first yields a fibre operator; the channel then determines whether its spectrum contains a stable, metastable, or resonant mode.

A dark photon requires a second connection A', its curvature F', and a kinetic intertwiner

$$\mathcal{L}_{\text{mix}} = -\frac{\varepsilon_{\text{kin}}}{2} F_{\mu\nu} F'^{\mu\nu}.$$

If the second sector acquires mass through a Stueckelberg mechanism, a hidden Higgs, or a boundary, its pole is distinct from the null photon. The magnitude ε_{kin} must be derived from a common incidence before observables are compared.

Let $P_{\text{weak}}, P_{\text{em}}, P_{\text{color}}$ be the projectors onto the weak, electromagnetic, and chromatic representation channels, and let P_X be the support of the dark section. A WIMP requires nonzero weak overlap and absence of electromagnetic and chromatic support:

$$P_{\text{weak}} P_X \neq 0, \qquad P_{\text{em}} P_X = P_{\text{color}} P_X = 0.$$

The first product expresses incidence between subspaces; a nonzero commutator would not be a coupling criterion, since two projectors may commute and nevertheless share support. The interaction strength then requires a vertex operator or a coupling Hamiltonian.

Its stability requires a unit eigenvalue of the transport or a protective parity. “WIMP” is a condition on representation and persistence, not a predetermined mass.

92.11.6.10 Classification of the sterile neutrino section by invariant fiber, mixing, and null gauge charge

A sterile neutrino is an eigenvector of the neutral block that satisfies

$$P_{\text{weak}} n_s = 0, \qquad P_{\text{mass}} n_s = n_s.$$

It is not synonymous with the fourth structural depth of the neutrino support. Its existence requires a nonzero weak kernel, an eigenvalue of the mass operator in that kernel, and a mixing transport with the active modes. The criterion is operatorial and can be falsified before introducing an external scale.

92.11.6.11 Classification of Axionic and Pseudoscalar Sections by Topological Character and Edge Coupling

A pseudoscalar axion candidate must be a mode with a normalized kinetic term,

$$CP\, a\, CP^{-1} = -a, \qquad \mathcal{L}_{\text{kin}} = \frac{Z_a}{2}\, \partial_\mu a\, \partial^\mu a, \qquad Z_a > 0,$$

coupled to the topological colour density. Its mass follows from the curvature of the effective potential,

$$m_a^2 = \frac{1}{Z_a}\left.\frac{\partial^2 V_{\mathrm{eff}}}{\partial a^2}\right|_{a=a_0}.$$

The channel $C^* \mapsto -C^*$ contributes an odd orientation, but does not suffice to construct V_{eff}, the kinetic term, or the physical projector. The species is defined only when those maps are present.

92.11.6.12 Monopole Realizations and Topological Defects: Charge Classes and Stability

Let F be the curvature of a connection $U(1)$ in the chosen integral normalization. A magnetic monopole is characterized by the class

$$\frac{1}{2\pi}\int_{S^2} F = n \neq 0.$$

HMT must realize n as the net holonomy of a closed family of routes and prove that it survives refinement. A point vacancy or an isolated carry is not a monopole. If the defect is dynamic and stable, its mass is an eigenvalue of the operator linearized about that class.

92.11.6.13 Classification of Supersymmetric Extensions via Representation Pairing and Characters

The coexistence of fermionic and bosonic completions does not prove supersymmetry. In four dimensions, a family of odd charges satisfying the following is required

$$Q_\alpha : \mathcal{F}_- \longleftrightarrow \mathcal{F}_+, \qquad \{Q_\alpha, \bar{Q}_{\dot\beta}\} = 2\sigma^\mu_{\alpha\dot\beta} P_\mu,$$

that entangles the mass operators:

$$Q_\alpha \widehat{M}_- = \widehat{M}_+ Q_\alpha.$$

The identity $Q^2 = H$ is reserved for an explicit one-dimensional Hermitian reduction; it does not replace the preceding relativistic algebra.

Only then do spectral pairs appear. The breaking must derive the mass separation without using the value of the sought partner. Selectrons, neutralinos, gluinos, or other names remain possible realizations, not rows predicted by the mere existence of CAR and CCR.

92.11.6.14 Classification of Dilaton and Radiaton Sections via Volume and Compactification Scalar Modes

The dilaton is a fluctuation of the scale section; the radion, of an internal length or volume. Their elementary actions are

$$t_0 \mapsto e^{\phi_d} t_0, \qquad \mathcal{V}_{\mathrm{int}} \mapsto e^{\phi_r} \mathcal{V}_{\mathrm{int}}.$$

To constitute particles, they require a kinetic term, potential, persistent projector, and pole. Before that, they are collective degrees of the dimensional chart. This distinction prevents a normalization freedom from being converted into a new scalar species.

92.11.6.15 Compounds and Bonding Energy

A composite state Y with constituents $X_1, \ldots, X_k$ is constructed in the order

$$(\Gamma_{X_1}, \ldots, \Gamma_{X_k}) \longrightarrow \mathscr{L}_{\mathrm{bind}} \longrightarrow \sigma_Y \longrightarrow \eta_Y \longrightarrow \widehat{M}_Y.$$

The binding record $\mathscr{L}_{\mathrm{bind}}$ contains the order of the routes, the common boundary, the transferred carry, the color channel, and the binding holonomy. It may not be replaced by $\sum_i m_{X_i}$. Mesons and baryons require distinct readers because their numbers of constituents, representations, and exchange symmetries differ.

The six published composite evaluations prove the character evaluation upon fixing the signature. State-by-state closure additionally requires the linking record to regenerate that signature. This condition

applies in the same manner to nuclei, atoms, molecules, and tetraquark or pentaquark candidates, with their respective boundary Hamiltonians.

92.11.6.16 Width as 2nd observable

The 384 widths are not obtained by adding a sixth coordinate to σ. For each unstable state, the channels $\mathcal{H}_{\text{in}} \oplus \mathcal{H}_{\text{out}}$, the effective operator $H_{\text{eff}}(z)$, and the pole are constructed

$$z_X = M_X c_{\text{int}}^2 - \frac{i}{2}\Gamma_X.$$

The real part belongs to the mass law; the imaginary part belongs to the decay reader. The relation $\tau_X = \hbar_{\text{int}}/\Gamma_X$ is interpreted after the temporal chart has been fixed. Thus, an exhaustive theory of 471 masses and 384 widths contains two coordinated readers, rather than a single column with 855 heterogeneous numbers.

92.11.6.17 Comparison of the Electromagnetic and Chromodynamic Null Realizations

The photon and the eight gluons belong to distinct null vectors. For the photon,

$$m_\gamma = 0$$

in the electromagnetic branch. For the gluons,

$$m_g = 0$$

in the adjoint representation of colour. Nullity is not obtained by taking the limit of a positive character; the domain bifurcates beforehand. Photon realization requires Maxwell theory, gauge invariance, and two transverse polarizations. Gluon realization requires the colour connection, its eight generators, and restriction to the physical gauge space.

Bidirectional propagation can be written

$$c_\pm = \frac{c}{1 \pm \epsilon}, \qquad \epsilon = q_-^{90} - q_+^{90},$$

with harmonic mean

$$\frac{2}{1/c_+ + 1/c_-} = c.$$

The bidirectional velocity preserves c; the unidirectional orientation preserves the bias. An additional gap in the genealogical record may be realized through Proca–Stueckelberg, but it must not be identified automatically with ϵ or with the splitting of the action.

92.11.7 Exceptional Scaler, Conformal Theory, Strings and Mass Classification

92.11.7.1 Common preimage and coordination of the 2 *readings of the exceptional corridor*

The relation between masses and exceptional symmetries does not begin in an action of the Monster on a table of particles. It begins in the same enriched state of the TPK. Let

$$x \in \mathcal{X}_{\text{TPK}}^{\text{enr}}.$$

The Archimedean reading contracts compatible cylinders and preserves the real value with its genealogy:

$$\mathcal{P}_{\text{arq}}(x) \in \mathbb{R}.$$

The exceptional reading preserves boundary incidences and produces a class in the projective limit:

$$\mathcal{P}_{\mathrm{exc}}(x) = Q_\infty(x).$$

Naturality requires, for every refinement $n \geq m$,

$$r_{n,m} \circ Q_n = Q_m \circ p_{n,m}.$$

The joint application

$$\mathcal{P}_{\mathrm{joint}} : x \longmapsto \big(\mathcal{P}_{\mathrm{arq}}(x), \mathcal{P}_{\mathrm{exc}}(x)\big)$$

is the positive form of the double projection. It does not deny the Monster–mass connection: it types it. The mass branch and the exceptional branch are images of the same preimage with common orientation, grading, carry, memory, and holonomy. Nor does it assert that the Monster acts literally on decimal digits or that a Moonshine coefficient is by itself a mass.

92.11.7.2 Construction of the exceptional corridor

The ternary incidence prolongs the state by the chain

$$C_3 \longrightarrow A_2^{12} \longrightarrow N(A_2^{12}) \longrightarrow \Lambda_{24} \longrightarrow V^\natural \longrightarrow \mathbb{M},$$

where Λ_{24} is the Leech lattice, $V^\natural$ the Moonshine vertex algebra, and $\mathbb{M}$ the Monster group. The step does not consist in juxtaposing classical names: each arrow must preserve the incidence, grading, and refinement compatibility inherited from the state.

The ternary trace is written

$$t_3(q) = q^{-1} \prod_{n \geq 1} (1 + q^n + q^{2n})^{-12}.$$

The first nontrivial defect satisfies

$$D_{\mathrm{APP}} = [q]t_3 = v_\alpha^2 = 54,$$

and the first coefficient of the Monster's character preserves the decomposition

$$196884 = 54 + 10 \cdot 3^9.$$

These identities transport gradation and multiplicity. They do not assign a mass in MeV. Their function in the theory of masses is to classify the space on which the operator acts before scalarizing it.

92.11.7.3 Fiberwise matching with the mass law

The material branch of the same state produces

$$x \longmapsto \big(F_X, \sigma_X, \eta_X, \widehat{M}_X\big),$$

where F_X is a point, orbit, fiber, or multisection; $\sigma_X \in \mathbb{Z}^5$ is the signature; $\eta_X \in \mathcal{E}_{90,120}$ is the refinement; and $\widehat{M}_X$ is the mass operator. The joint reading is then

$$\mathfrak{J}_{\mathrm{mass,exc}}(x) = \big(F_X, \sigma_X, \eta_X, \widehat{M}_X, Q_\infty(x)\big).$$

Two states with the same scalar mass value may remain distinct because of their exceptional component; two states with the same exceptional character may have different mass operators. The correct classification is the matching of fibers, not the isolated equality of numbers.

If an exceptional representation G acts in a physical fiber $\mathcal{H}_X$, compatibility with mass requires

$$U_g \widehat{M}_X U_g^{-1} = \widehat{M}_X, \qquad g \in G.$$

The decomposition

$$\mathcal{H}_X = \bigoplus_{\lambda} \mathcal{H}_{X,\lambda}$$

separates exceptional multiplets. On an irreducible summand, commutation forces the compression of $\widehat{M}_X$ to be scalar. This is a legitimate route to scalarization by symmetry. The scalar value continues to arise from the character, the towers, the dimensional section, and the coupling; the dimension of the representation determines only the degeneracy.

92.11.7.4 Fock and vertex algebra

The prolongation of a route to a many-particle space has two principal completions. The exterior completion produces CAR, Pauli exclusion, and fermionic statistics; the symmetric completion produces CCR and bosonic statistics. A vertex algebra adds an operator product, vacuum, translation, and grading:

$$V = \bigoplus_{n \geq 0} V_n, \qquad L_0|_{V_n} = nI.$$

The character

$$\mathrm{ch}_V(q) = \mathrm{Tr}_V\, q^{L_0 - c/24}$$

counts states by degree. A multiplicity in V_n is not a mass. To convert the grading into energy, a Hamiltonian and a length or time section are required. In a conformal realization on a cylinder of circumference L,

$$H_{\mathrm{CFT}} = \frac{2\pi\hbar_{\mathrm{int}} v}{L}\left(L_0 + \bar{L}_0 - \frac{c}{12}\right).$$

The physical mass or gap appears only after fixing v, L, the boundary conditions, and the map from the TPK route to the conformal modules. The role of the exceptional corridor is to organize the degeneracies and composition rules of those modules.

92.11.7.5 String spectrum as a posterior dimensional realization

Let $\alpha' \in \mathcal{L}_L^2$ be the squared length representing inverse tension in the declared naturalized string chart. In a circular compactification of radius R, the left- and right-moving momenta are

$$p_L = \frac{m}{R} + \frac{wR}{\alpha'}, \qquad p_R = \frac{m}{R} - \frac{wR}{\alpha'},$$

with $m, w \in \mathbb{Z}$. In the natural units of the realization, the closed spectrum satisfies

$$M^2 = p_L^2 + \frac{4}{\alpha'}(N_L - a_L) = p_R^2 + \frac{4}{\alpha'}(N_R - a_R),$$

together with the level condition

$$N_L - N_R = a_L - a_R - mw.$$

The involution

$$(m, w, R) \longleftrightarrow \left(w, m, \frac{\alpha'}{R}\right)$$

preserves the spectrum: it is the realization of T-duality. HMT first supplies the bidirectional orientation, memory, and sheet pairing; string theory adds tension, radius, oscillators, level, and boundary conditions. A string mass is not obtained by replacing N_L, N_R with heights of the tower $\langle 90, 120\rangle$. Any identification between the two gradings requires an intertwiner that preserves sum, orientation, and character.

92.11.7.6 M-Theory Lattice Screens: Grading, Compactification, and Transport of the Residual Memory

The internal screens

$$12 \longrightarrow 11 \longrightarrow 10, \qquad 26 \longrightarrow 24,$$

preserve the dodecaphase descent, the elimination of a redundant direction, and the transverse screen. In a realization of M-theory, the fields

$$g_{MN}, \qquad C_{MNP}, \qquad \Psi_M$$

must be obtained through an intertwiner that preserves gauge, supersymmetry, constraints, and inner product. Numerical displays do not replace that map.

The classification of masses is transported to this branch by means of

$$\text{route} \longrightarrow \text{genealogical record} \longrightarrow \text{signature} \longrightarrow \text{character} \longrightarrow \text{operator} \longrightarrow \text{compactified spectrum}.$$

Kaluza–Klein, winding, membrane, and oscillatory-excitation modes are distinct states even when they share a mass. The graviton of a closed-string realization, if it appears, is a massless collective mode of the geometric sector; it does not thereby become an elementary seed of the TPK or invalidate the formulation of gravity as holonomy and torsion.

92.11.7.7 Ordinary and Hypothetical Realizations Induced by the Exceptional Runner

The exceptional branch allows distinguishing four sources of multiplicity:

1. cellular or route multiplicity in the atlas;
2. degeneracy of an exceptional representation;
3. Fock occupation;
4. conformal, oscillatory, or compactified level.

Confusing them produces false companions, false generations, or duplicated masses. A twin, supersymmetric, dark, or exotic state constitutes a new species only when its quintuple

$$(F_X, Q_\infty(x), \rho_X, \widehat{M}_X, P_X)$$

differs physically, and the Hamiltonian has an additional pole or persistent section. A coincidence of dimension, character, or mass does not suffice.

Conversely, a degeneracy protected by the exceptional corridor is a structural prediction: perturbations preserving the symmetry cannot split the multiplet. An observed splitting requires identifying the symmetry breaking, coupling, or chart that produces it.

92.11.7.8 Positivity, closure, and stability conditions of topological realizations

The connection between the exceptional corridor and masses is falsified in a concrete realization if any of the following occurs:

- naturality $r_{n,m}Q_n = Q_m p_{n,m}$ fails;
- the exceptional label changes when the external masses are permuted;
- a Moonshine multiplicity is used as a mass value without a Hamiltonian;
- the mass operator does not commute with the symmetry asserted to be preserved;
- a tower height is identified with a string level without an intertwiner;
- the level condition or T-duality invariance fails;
- a dimensional screen is declared M-theory without constructing its fields;
- the existence of a spin-two mode is used to redefine the preceding internal gravity.

These controls keep the connection positive: there is a common state and a joint classification; what is prohibited is replacing the arrows with a numerical analogy.

92.11.7.9 Mathematical construction of material classes from routes, signatures, and representations

The double realization, projective naturality, the chain A_2^{12}–Leech–$V^\natural$–Monster, the trace t_3, the identities $D_{\mathrm{APP}} = 54$ and $196884 = 54 + 10 \cdot 3^9$, and the dimensional screens belong to the APP–TRIT–TPK mathematical architecture developed previously. The formulation of $\mathfrak{J}_{\mathrm{mass,exc}}$, the separation of the four multiplicities, and the foregoing controls make its function within an autonomous theory of masses explicit. The conformal and string formulas are subsequent dimensional realizations with their parameters declared; they are not used retroactively to generate APP, TRIT, TPK, the character, or the towers.

92.12 Current operatorial results and subsequent metrological contrast

The comparison only opens afterward of freezing route, signature, operator, scheme, scale and unit. The table current arranges in columns contiguous the output HMT, the value external, the uncertainty, the residual and the status. The values quark not admit a residual public if scheme and scale not coincide.

92.12.1 Mass Operators in HMT-MD: Absolute Scale, Route Atlas, and Sectoral Realization

Mass appears in Dimensional Mechanics as the realization of two mathematical branches that converge without being conflated. The first preserves the genealogy of the state: a surviving extension determines an observable route; the route writes a dodecaphasic chronological register; the register produces an integer signature, and the signature feeds the character and the harmonic hierarchy. The second constructs the dimensional line: the determinantal reader fixes the action scale, the internal clock transports it to energy, and the kinematic chart carries it to mass. The species operator then acts by positive homotheties on that line. The experimental coordinate is opened only after both branches have been frozen.

The complete internal domain has four distinct counts:

$$81 = 25_{\ell^-} + 20_\nu + 20_d + 16_u, \qquad 56 = 41{\times}1 + 10{\times}2 + 2{\times}3 + 1{\times}4 + 2{\times}5, \tag{92.37}$$

$$13 \text{ multisections}, \qquad 324 = 81 \times 4 \text{ oriented routes}. \tag{92.38}$$

The physical realizations compared later constitute a classification of realizations; they are not another name for the 81 cells, the 56 families, the 13 multisections, or the 324 oriented histories. On this domain, the two route readers produce fourteen profiles K_D, nine profiles K_Ω, forty joint pairs, and eighteen coincidences. The null sector remains separate from the positive mass character.

92.12.1.1 From the APP incompatibility to the mass signature

The primordial difference between the sum and product sheets appears in the spectral decomposition of the central APP block:

$$B_c = 7I + 2R = 9P_+ + 5P_-, \qquad \mathrm{Spec}(B_c) = \{9, 5\}, \qquad \Delta_{\mathrm{APP}} = 9 - 5 = 4. \tag{92.39}$$

The integers that recur subsequently preserve distinct sources. The local exchange coefficient is 2 in $B_c = 7I + 2R$; modular compression publishes $51 - 45 = 6$, and its deployment over nine blocks produces $9 \cdot 6 = 54$. These equalities belong to typed levels of the same genealogy and do not constitute a chain of definitions of the jump 4. In the electronic section, 54 is furthermore $D_1(\Gamma_e)$ and the kinematic periodicity of order 54 of the CT108 temporal endomorphism: it measures a component of the history already transported by TPK, not the primordial spectral separation.

For each enriched route Γ, the dodecaphase record determines

$$\sigma(\Gamma) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta) \in \mathbb{Z}^5, \qquad \zeta_{270} = 135\Delta_4^2. \tag{92.40}$$

The central character and its harmonic refinement are

$$\log \chi_0(\sigma) = n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}} + k_{\Delta_4}\Delta_4 + b_{120}s_{120} + b_\zeta \zeta_{270}, \tag{92.41}$$

$$\log \chi_{\mathrm{full}}(\sigma, \eta) = n_A A_{\mathrm{rad}} + \frac{s_C}{6} C^*_{\mathrm{rad}} + k_{\Delta_4}\Delta_4 + b_\zeta \zeta_{270} + (b_{120} + \eta_{120})s_{120} + \sum_{\substack{h \in \mathfrak{S} \\ h \neq 120}} \eta_h s_h. \tag{92.42}$$

The division $s_C/6$ follows from dodecaphasic integrality. The labeled pair (b_{120}, η_{120}) preserves its genealogy even when its evaluation is grouped into a single coefficient.

The moments entering (92.42) belong to an infinite hierarchy:

$$q_\pm = \exp\Big[-\frac{\pi}{180}(A \pm C^*)\Big], \qquad s_h = q_-^h - q_+^h, \qquad c_h = q_-^h + q_+^h, \tag{92.43}$$

$$\mathfrak{S} = \langle 90, 120\rangle = \{90, 120\} \cup \{30r : r \geq 6\}. \tag{92.44}$$

Both s_n and c_n satisfy

$$X_{n+2} = c_1 X_{n+1} - (q_- q_+) X_n. \tag{92.45}$$

The heights $90, 120, 180, 210, 240, 270, 360, 420$ are particularly visible genealogical witnesses; the domain of the law is the entire semigroup. In particular, 360 preserves two histories before the quotient, four steps of 90, and three of 120, even when both produce the same scalar moment.

92.12.1.2 The absolute mass section and the decade 10^{-34}

The local Hadamard block

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad \mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4),$$

has cokernel of order 16. Its determinantal reader produces

$$\Sigma_H = \varphi \alpha_{\mathrm{HMT}}^{16}, \qquad 10^{-34} < \Sigma_H < 10^{-33}, \qquad \lfloor \log_{10} \Sigma_H \rfloor = -34. \tag{92.46}$$

The conversion to J s is a subsequent chart: the power 10^{-34} is born in the discrete structure and fixes the scale of a single line of action. Its sections before and after the return are

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S, \qquad R_{\mathrm{act}} = \frac{H_5}{\eta_{\mathrm{ret}}} > 0. \tag{92.47}$$

The splitting possesses an additive chart, $\delta_{2w} = H_5 - \eta_{\mathrm{ret}}$, and the dimensionless multiplicative chart R_{act}. Once an internal duration t_0 has been declared, the periodicity of 108 iterations of the CT108 temporal endomorphism constructs

$$\omega_0 = \frac{2\pi}{108 t_0}, \qquad \varepsilon_{\mathrm{clk}} = \hbar_{\mathrm{ret}}\omega_0, \tag{92.48}$$

$$\mathbf{m}_\star^{\mathrm{pre}} = \left(H_5\, 10^{-34}\mathcal{U}_S\right)\frac{2\pi}{108 t_0}c_{\mathrm{int}}^{-2}, \qquad \mathbf{m}_\star^{\mathrm{ret}} = \left(\eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S\right)\frac{2\pi}{108 t_0}c_{\mathrm{int}}^{-2} \in \mathcal{L}_M. \tag{92.49}$$

Both sections satisfy $\mathbf{m}_\star^{\mathrm{pre}}/\mathbf{m}_\star^{\mathrm{ret}} = R_{\mathrm{act}}$. The decade 10^{-34}, the common clock, and c_{int}^{-2} fix the absolute scale. In a mass ratio, that common section cancels exactly; the relative refinement belongs to the route and not to a second introduction of the decimal power.

92.12.1.3 Bidirectional route readers and operator law on the 13 *multisections*

The readers acting on a route Γ are

$$K_D(\Gamma) = \left(D_{27} + D_{36},\, D_1,\, \frac{D_4 - D_3}{2}\right), \qquad K_\Omega(\Gamma) = (\Omega, 54, P_6). \tag{92.50}$$

Each reader induces on a multisection F a diagonal operator $\widehat{K}_F^r$, $r \in \{D, \Omega\}$. The index s designates a physical section or block within F, and $\widehat{B}_{F,s}^r$ is the self-adjoint restriction of $\widehat{K}_F^r$ to that block. If $\widehat{\mathcal{A}}_{F,s}^{(0)}$ is the restriction of the operator that gathers the signature, the complete character, and the global hierarchy of towers, the general law takes the form

$$\boxed{\widehat{\mathbf{M}}_{F,s}^{\beta,r} = \mathbf{m}_\star^{\mathrm{ret}} \otimes R_{\mathrm{act}}^{\widehat{B}_{F,s}^r/2}\,\widehat{\mathcal{A}}_{F,s}^{(0)}\,R_{\mathrm{act}}^{\widehat{B}_{F,s}^r/2}, \qquad r \in \{D, \Omega\}.} \tag{92.51}$$

The symmetric expression preserves positivity without imposing commutation of $\widehat{\mathcal{A}}_{F,s}^{(0)}$ and $\widehat{B}_{F,s}^r$. In the simultaneously diagonal basis of the current certificates, it coincides with the product $\mathbf{m}_\star^{\mathrm{ret}} \otimes \widehat{\mathcal{A}}_F^{(0)} \exp((\log R_{\mathrm{act}})\widehat{K}_F^r)$. This equation governs the 81 cells, 56 families, 13 multisections, and 324 oriented routes. Outside a rank-one fiber, the primary output is the complete operator and its spectrum. Scalar realization uses the closed morphism $\mathcal{S}_{\mathrm{phys}} : (\sigma_{\mathrm{ud}}, g, \chi_{\mathrm{fina}}, [\gamma]_{\mathrm{col}}) \mapsto \gamma_{P,\mathfrak{M}}^{\mathrm{surv}}$ of equation (92.9); that selection is independent of target masses and precedes any metrological comparison.

92.12.1.4 Rank reduction 1 *in the construction of the central electron*

The electronic section is the only central fixed point of the atlas:

$$(r, c) = (5, 5), \qquad \Gamma_e = (5, 5, {+}{+}), \qquad \tau_e = {+}{+}{+}{-}{-}{-}{+}{+}{+}{-}{-}{-}. \tag{92.52}$$

Both readers agree on it,

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80, 54, 6), \qquad \kappa_e = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4. \tag{92.53}$$

The rank-one reduction of (92.51) yields

$$\mathbf{m}_e^\beta = \mathbf{m}_\star^{\mathrm{ret}}\,\mu_e^{(0)}\,R_{\mathrm{act}}^{80-54\alpha_{\mathrm{HMT}}+6\Delta_4}, \tag{92.54}$$

whose coordinate in the comparative chart is

$$\boxed{m_e^\beta = 0.510998950690000808608257164 8\ldots\ \mathrm{MeV}/c^2.} \tag{92.55}$$

The basal value $0.51099892248806607 25\ldots$ MeV$/c^2$ preserves its function as an intermediate stage; the beta output of (92.55) governs the final comparison.

92.12.1.5 Physical classification and quantitative comparison by sectors

The exposition is organized into four coordinated strata. The first publishes the internal result and its codomain; the second declares the physical realization morphism; the third displays the best numerical coordinate currently available and preserves its hypotheses and domain; the fourth introduces the external datum as a subsequent comparison. Each column thus preserves its mathematical type: HMT output, realization morphism, computed coordinate, and external comparison. The table simultaneously publishes the available evaluations and the construction maps producing them.

The electron publishes its scalar beta closure. The photon and gluons publish exact null sections in their mass charts, together with nontrivial gauge extensions. The other charged leptons and the six quarks simultaneously publish their nominal routes, scalar characters, diagonal operators, and spectra over their fibres; the neutrinos publish the neutral operator and its mixing. W, Z, and H possess individualized CT–108 routes, complete traces, and characters $(n_A, s_C) = (94, -1), (95, -1), (97, 9)$. The pair K_D/K_Ω belongs to the refinement of the atlas's thirteen multisections and is not a premise of the bosonic realization by nominal route. Composites, topological sectors, and virtual states retain, respectively, their codomains of binding, fusion, braiding, and obstruction.

Operators of the 13 **multisections** The family counts recorded on each sheet are local cardinalities of the support of that multisection. They are not added across sheets: the quotient of the 56 route families and the quotient of the 13 multisections are two distinct partitions of the 81-cell atlas, related by the projection maps rather than by a common disjoint decomposition.

The following records publish the operator in its native codomain. Each spectrum is reproduced in full; no list of eigenvalues is replaced by a mass coordinate.

charged leptonic sector, depth G0 (center). $\mathcal{B}_{\ell,\mathrm{G0}}$.

Fiber domain. Rank 1; cell 41; no chromatic orbit; 1 route family. **Full spectrum of K_D.**

- (80,54,6)x1

Complete spectrum of K_Ω.

- (80,54,6)x1

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

charged-lepton sector, depth G2. $\mathcal{B}_{\ell,\mathrm{G2}}$.

Fiber domain. Rank 8; cells 21,23,25,39,43,57,59,61; no chromatic orbit; 8 route families. **Full spectrum of K_D.**

- (0,24,0)x1
- (40,78,27)x2
- (80,54,7)x1
- (80,66,2)x1
- (80,66,3)x1
- (100,66,0)x1
- (100,66,2)x1

Complete spectrum of K_Ω.

- (80,54,6)x8

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

charged-lepton sector, depth G4. $\mathcal{B}_{\ell,\mathrm{G4}}$.

Fiber domain. Rank 16; cells 1,3,5,7,9,19,27,37,45,55,63,73,75,77,79,81; no chromatic orbit; 16 route families. **Full spectrum of K_D.**

- (0,24,0)x1
- (40,24,2)x2
- (40,54,6)x2
- (40,84,30)x2
- (60,78,25)x3
- (80,66,2)x2
- (80,66,3)x3
- (80,66,4)x1

Complete spectrum of K_Ω.

- (24,54,2)x1
- (56,54,5)x4
- (80,54,6)x11

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

neutrino sector, depth G1. $\mathcal{B}_{\nu,\mathrm{G1}}$.

Fiber domain. Rank 2; cells 40,42; no chromatic orbit; 2 route families. **Full spectrum of K_D.**

- (40,24,2)x1
- (40,78,27)x1

Complete spectrum of K_Ω.

- (40,54,4)x1
- (80,54,6)x1

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

neutrino sector, depth G2. $\mathcal{B}_{\nu,\mathrm{G2}}$.

Fiber domain. Rank 4; cells 22,24,58,60; no chromatic orbit; 4 route families. **Full spectrum of K_D.**

- (0,24,0)x1
- (40,84,30)x1
- (60,78,25)x1
- (80,66,3)x1

Complete spectrum of K_Ω.

- (4,54,2)x1
- (80,54,6)x3

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

neutrino sector, depth G3. $\mathcal{B}_{\nu,\mathrm{G3}}$.

Fiber domain. Rank 6; cells 20,26,38,44,56,62; no chromatic orbit; 6 route families. **Full spectrum of K_D.**

- (40,24,2)x2
- (40,78,27)x1
- (60,84,28)x1
- (80,54,6)x1
- (80,66,3)x1

Complete spectrum of K_Ω.

- (36,54,4)x1
- (56,54,5)x1
- (80,54,6)x4

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

neutrino sector, depth G4. $\mathcal{B}_{\nu,\mathrm{G4}}$.

Fiber domain. Rank 8; cells 2,4,6,8,74,76,78,80; no chromatic orbit; 8 route families. **Full spectrum of K_D.**

- (0,24,0)x1
- (40,24,2)x1
- (40,78,27)x2
- (40,84,30)x1
- (80,54,7)x1
- (80,66,2)x1
- (100,66,0)x1

Complete spectrum of K_Ω.

- (36,54,4)x1
- (56,54,5)x1
- (80,54,6)x6

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

down-type quark branch, depth G1. $\mathcal{B}_{d,\mathrm{G1}}$.

Fiber domain. Rank 2; cells 32,50; B | G; 2 route families. **Full spectrum of K_D.**

- (40,24,2)x1
- (40,78,27)x1

Complete spectrum of K_Ω.

- (40,54,4)x1
- (80,54,6)x1

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

down-type quark branch, depth G2. $\mathcal{B}_{d,\mathrm{G2}}$.

Fiber domain. Rank 4; cells 30,34,48,52; B | G | R; 4 route families. **Full spectrum of K_D.**

- (0,24,0)x1
- (40,84,30)x1
- (60,78,25)x1
- (80,66,2)x1

Complete spectrum of K_Ω.

- (4,54,2)x1
- (80,54,6)x3

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

down-type quark branch, depth G3. $\mathcal{B}_{d,\mathrm{G3}}$.

Fiber domain. Rank 6; cells 12,14,16,66,68,70; B | G | R; 6 route families. **Full spectrum of K_D.**

- (40,24,2)x2
- (40,78,27)x1
- (60,84,28)x1
- (80,54,6)x1
- (80,66,4)x1

Complete spectrum of K_Ω.

- (36,54,4)x1
- (80,54,6)x5

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

down-type quark branch, depth G4. $\mathcal{B}_{d,\mathrm{G4}}$.

Fiber domain. Rank 8; cells 10,18,28,36,46,54,64,72; B | G | R; 8 route families. **Full spectrum of K_D.**

- (0,24,0)x1
- (40,24,2)x1
- (40,78,27)x2
- (40,84,30)x1
- (80,54,7)x1
- (80,66,3)x1
- (100,66,2)x1

Complete spectrum of K_Ω.

- (36,54,4)x1
- (80,54,6)x7

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

up-type quark branch, depth G1. $\mathcal{B}_{u,\mathrm{G1}}$.

Fiber domain. Rank 4; cells 31,33,49,51; B | G | R; 3 route families. **Full spectrum of K_D.**

- (40,54,6)x1
- (60,78,25)x1
- (80,66,2)x1
- (80,66,3)x1

Complete spectrum of K_Ω.

- (80,54,6)x4

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

up-type quark branch, depth G3. $\mathcal{B}_{u,\mathrm{G3}}$.
Fiber domain. Rank 12; cells 11,13,15,17,29,35,47,53,65,67,69,71; B | G | R; 12 route families. **Full spectrum of K_D.**

- (40,24,2)x2
- (40,78,27)x2
- (60,84,28)x1
- (80,54,6)x3
- (80,66,3)x1
- (80,66,4)x1
- (100,66,0)x1
- (100,66,2)x1

Complete spectrum of K_Ω.

- (56,54,5)x3
- (80,54,6)x9

Conclusion in the fiber. Exact diagonal operator on the fiber B_1; the nominal route and the declared morphism publish the scalar character without replacing the spectrum of the fiber.

Physical realization morphisms

Table 92.8: Physical realization of the charged leptons.

Class	Objeto HMT	Realization map	Conclusion and conditions
e^-/e^+	Celda central $(5,5)$, route $\Gamma_e = (5,5,{+}{+})$ and $K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80,54,6)$.	Closed rank one reduction.	Beta range closure; independent magnitude selector of metrological magnitudes employed as target.
μ^-/μ^+	charged-lepton family; levels G2; 8 cells; K_D: 7 profiles; K_Ω: 1 profile	Generation and fine signature $\to$ persistent route $\to \chi_{\mathrm{mass}} \to \mathcal{L}_M$.	Nominal route and closed scalar character; the operator pair preserves the full refinement of the fiber.
τ^-/τ^+	family leptonica cargada; levels G4; 16 celdas; K_D: 8 perfiles; K_Ω: 3 perfiles	Generation and fine signature $\to$ persistent route $\to \chi_{\mathrm{mass}} \to \mathcal{L}_M$.	Nominal route and closed scalar character; the operator pair preserves the full refinement of the fiber.

Table 92.9: Physical realization of neutrinos and transport of bases.

Class	Objeto HMT	Realization map	Conclusion and conditions
$\nu_e/\bar{\nu}_e$	neutral family; levels G1, G2, G3, G4; 20 cells; K_D: 11 profiles; K_Ω: 5 profiles	Full-rank neutral record $\to$ eigenvectors $\to$ PMNS basis.	The exterior/CAR structure fixes fermionicity; PMNS transports bases and does not create mass eigenvalues.
$\nu_\mu/\bar{\nu}_\mu$	neutral family; levels G1, G2, G3, G4; 20 cells; K_D: 11 profiles; K_Ω: 5 profiles	Full-rank neutral record $\to$ eigenvectors $\to$ PMNS basis.	The exterior/CAR structure fixes fermionicity; PMNS transports bases and does not create mass eigenvalues.
$\nu_\tau/\bar{\nu}_\tau$	neutral family; levels G1, G2, G3, G4; 20 cells; K_D: 11 profiles; K_Ω: 5 profiles	Full-rank neutral record $\to$ eigenvectors $\to$ PMNS basis.	The exterior/CAR structure fixes fermionicity; PMNS transports bases and does not create mass eigenvalues.

Table 92.10: Physical realization of quarks: generation, flavor, and color.

Class	Objeto HMT	Realization map	Conclusion and conditions
$u/\bar{u}$	quark branch *up*; levels G1 and G3; color $\mathbb{Z}/3$; 16 cells; K_D: 11 profiles; K_Ω: 2 profiles	Flavour–colour–generation $\to$ persistent route $\to \chi_{\text{mass}} \to \mathcal{L}_M$.	Flavor, color, and generation determine the class of route; scalar character and closed operatorial pair.
$d/\bar{d}$	quark branch *down*; levels G1, G2, G3, G4; color $\mathbb{Z}/3$; 20 cells; K_D: 12 profiles; K_Ω: 4 profiles	Flavour–colour–generation $\to$ persistent route $\to \chi_{\text{mass}} \to \mathcal{L}_M$.	Flavor, color, and generation determine the class of route; scalar character and closed operatorial pair.
$c/\bar{c}$	quark branch *up*; levels G1 and G3; color $\mathbb{Z}/3$; 16 cells; K_D: 11 profiles; K_Ω: 2 profiles	Flavour–colour–generation $\to$ persistent route $\to \chi_{\text{mass}} \to \mathcal{L}_M$.	Flavor, color, and generation determine the class of route; scalar character and closed operatorial pair.

Continues on the following page.

Class	Objeto HMT	Realization map	Conclusion and conditions
$s/\bar{s}$	quark branch *down*; levels G1, G2, G3, G4; color $\mathbb{Z}/3$; 20 cells; K_D: 12 profiles; K_Ω: 4 profiles	Flavour–colour–generation $\to$ persistent route $\to \chi_{\text{mass}} \to \mathcal{L}_M$.	Flavor, color, and generation determine the class of route; scalar character and closed operatorial pair.
$t/\bar{t}$	quark branch *up*; levels G1 and G3; color $\mathbb{Z}/3$; 16 cells; K_D: 11 profiles; K_Ω: 2 profiles	Flavour–colour–generation $\to$ persistent route $\to \chi_{\text{mass}} \to \mathcal{L}_M$.	Flavor, color, and generation determine the class of route; scalar character and closed operatorial pair.
$b/\bar{b}$	quark branch *down*; levels G1, G2, G3, G4; color $\mathbb{Z}/3$; 20 cells; K_D: 12 profiles; K_Ω: 4 profiles	Flavour–colour–generation $\to$ persistent route $\to \chi_{\text{mass}} \to \mathcal{L}_M$.	Flavor, color, and generation determine the class of route; scalar character and closed operatorial pair.

Structural location of the 6 quark flavours The position of a quark flavour is specified by four simultaneous data: branch, generation, cellular support, and family of oriented routes. The three generations of a given branch share the support of the atlas; the fine generation classifier preserves the physical distinction that must still be realized as an eigenvector and a mass section.

Table 92.11: Generational classification of the six quark flavors.

Flavor	Generation	Branch	Branch and generation classifier
$u/\bar{u}$	first	up branch	$(\text{up}, 1, \chi_{\text{fina}})$; nominal route pair $(21, 30)$ over the axis N−S
$d/\bar{d}$	first	down branch	$(\text{down}, 1, \chi_{\text{fina}})$; nominal route pair $(20, 32)$ over the axis E−O
$c/\bar{c}$	second	up branch	$(\text{up}, 2, \chi_{\text{fina}})$; nominal route pair $(21, 29)$ over the axis E−O
$s/\bar{s}$	second	down branch	$(\text{down}, 2, \chi_{\text{fina}})$; nominal route pair $(17, 27)$ over the axis N−S
$t/\bar{t}$	third	up branch	$(\text{up}, 3, \chi_{\text{fina}})$; nominal route pair $(21, 26)$ over the axis N−S
$b/\bar{b}$	third	down branch	$(\text{down}, 3, \chi_{\text{fina}})$; nominal route pair $(21, 30)$ over the axis E−O

Support structural of the rama superior.

Multisections. $\mathcal{B}_{u,\mathrm{G1}}$, $\mathcal{B}_{u,\mathrm{G3}}$.

Depths and cardinalities. G1, G3; 16 cells and 14 distinct route families.

Chromatic orbit and orientations. $C = \{B, G, R\}$ and $O = \{(-1,-1), (-1,1), (1,-1), (1,1)\}$.

Cells and oriented families B_1. Each chain $k \leftrightarrow (r,c) \leftrightarrow \Gamma_{r,c}^{(h,v)}$ preserves the cell index, its coordinate in the chart 9×9, and the two orientation signs of the enriched path record.

Fila $r = 2$: $11 \leftrightarrow (2,2) \leftrightarrow \Gamma_{2,2}^{(-,+)}$, $13 \leftrightarrow (2,4) \leftrightarrow \Gamma_{2,4}^{(+,-)}$, $15 \leftrightarrow (2,6) \leftrightarrow \Gamma_{2,6}^{(-,+)}$, $17 \leftrightarrow (2,8) \leftrightarrow \Gamma_{2,8}^{(-,-)}$.

Fila $r = 4$: $29 \leftrightarrow (4,2) \leftrightarrow \Gamma_{4,2}^{(+,-)}$, $31 \leftrightarrow (4,4) \leftrightarrow \Gamma_{4,4}^{(-,+)}$, $33 \leftrightarrow (4,6) \leftrightarrow \Gamma_{4,6}^{(+,+)}$, $35 \leftrightarrow (4,8) \leftrightarrow \Gamma_{4,8}^{(-,-)}$.

Fila $r = 6$: $47 \leftrightarrow (6,2) \leftrightarrow \Gamma_{6,2}^{(-,+)}$, $49 \leftrightarrow (6,4) \leftrightarrow \Gamma_{6,4}^{(-,-)}$, $51 \leftrightarrow (6,6) \leftrightarrow \Gamma_{6,6}^{(+,+)}$, $53 \leftrightarrow (6,8) \leftrightarrow \Gamma_{6,8}^{(+,+)}$.

Fila $r = 8$: $65 \leftrightarrow (8,2) \leftrightarrow \Gamma_{8,2}^{(+,+)}$, $67 \leftrightarrow (8,4) \leftrightarrow \Gamma_{8,4}^{(+,+)}$, $69 \leftrightarrow (8,6) \leftrightarrow \Gamma_{8,6}^{(-,-)}$, $71 \leftrightarrow (8,8) \leftrightarrow \Gamma_{8,8}^{(+,+)}$.

Closed realization morphism. For $g \in \{1,2,3\}$, the branch, generation, color orbit, and fine signature determine

$$\chi_{\mathrm{fina}}(\mathrm{up}, g, C) \longrightarrow \Gamma_{\mathrm{enr}} \longrightarrow \big(\mathrm{quot}_C, S(B_D, B_\Omega)\big).$$

This arrow selects the physical eigenvector and subsequently publishes the scalar coordinate in $\mathcal{L}_M$ without using the contrast value.

Structural support of the lower branch.

Multisections record. $\mathcal{B}_{d,\mathrm{G1}}$, $\mathcal{B}_{d,\mathrm{G2}}$, $\mathcal{B}_{d,\mathrm{G3}}$, $\mathcal{B}_{d,\mathrm{G4}}$.

Depths and cardinals. G1, G2, G3, G4; 20 cells and 18 distinct route families.

Chromatic orbit and orientations. $C = \{B, G, R\}$ and $O = \{(-1,-1), (-1,1), (1,-1), (1,1)\}$.

Cells and oriented families B_1. Each chain $k \leftrightarrow (r,c) \leftrightarrow \Gamma_{r,c}^{(h,v)}$ preserves the cell index, its coordinate in the chart 9×9, and the two orientation signs of the enriched path record.

Fila $r = 2$: $10 \leftrightarrow (2,1) \leftrightarrow \Gamma_{2,1}^{(+,+)}$, $12 \leftrightarrow (2,3) \leftrightarrow \Gamma_{2,3}^{(-,-)}$, $14 \leftrightarrow (2,5) \leftrightarrow \Gamma_{2,5}^{(+,+)}$, $16 \leftrightarrow (2,7) \leftrightarrow \Gamma_{2,7}^{(-,+)}$, $18 \leftrightarrow (2,9) \leftrightarrow \Gamma_{2,9}^{(-,+)}$.

Fila $r = 4$: $28 \leftrightarrow (4,1) \leftrightarrow \Gamma_{4,1}^{(-,-)}$, $30 \leftrightarrow (4,3) \leftrightarrow \Gamma_{4,3}^{(+,+)}$, $32 \leftrightarrow (4,5) \leftrightarrow \Gamma_{4,5}^{(+,-)}$, $34 \leftrightarrow (4,7) \leftrightarrow \Gamma_{4,7}^{(-,+)}$, $36 \leftrightarrow (4,9) \leftrightarrow \Gamma_{4,9}^{(-,+)}$.

Fila $r = 6$: $46 \leftrightarrow (6,1) \leftrightarrow \Gamma_{6,1}^{(-,-)}$, $48 \leftrightarrow (6,3) \leftrightarrow \Gamma_{6,3}^{(-,+)}$, $50 \leftrightarrow (6,5) \leftrightarrow \Gamma_{6,5}^{(-,+)}$, $52 \leftrightarrow (6,7) \leftrightarrow \Gamma_{6,7}^{(-,+)}$, $54 \leftrightarrow (6,9) \leftrightarrow \Gamma_{6,9}^{(+,-)}$.

Fila $r = 8$: $64 \leftrightarrow (8,1) \leftrightarrow \Gamma_{8,1}^{(+,-)}$, $66 \leftrightarrow (8,3) \leftrightarrow \Gamma_{8,3}^{(-,+)}$, $68 \leftrightarrow (8,5) \leftrightarrow \Gamma_{8,5}^{(-,+)}$, $70 \leftrightarrow (8,7) \leftrightarrow \Gamma_{8,7}^{(+,-)}$, $72 \leftrightarrow (8,9) \leftrightarrow \Gamma_{8,9}^{(-,+)}$.

Closed realization morphism. For $g \in \{1,2,3\}$, the branch, generation, color orbit, and fine signature determine

$$\chi_{\mathrm{fina}}(\mathrm{down}, g, C) \longrightarrow \Gamma_{\mathrm{enr}} \longrightarrow \big(\mathrm{quot}_C, S(B_D, B_\Omega)\big).$$

This arrow selects the physical eigenvector and subsequently publishes the scalar coordinate in $\mathcal{L}_M$ without using the contrast value.

Table 92.12: Physical realization of the gauge and scalar sectors.

Class	**Objeto HMT**	**Realization map**	**Conclusion and conditions**
photonγ	Null sectorial chart and corresponding gauge representation.	Exact inclusion of the null section in the mass line.	Exact nullity in its sectorial chart.
gluones g	Null sectorial chart and corresponding gauge representation.	Exact inclusion of the null section in the mass line.	Exact nullity in its sectorial chart.
$W^{\pm}$	CT–108 W route: 15/37, E-O axis, $(N, E, S, O) = (23, 44, 11, 30)$, sheet/residue $(41, 8)$, $(n_A, s_C) = (94, -1)$, $(W_+, W_-) = (563, 565)$; exact scalar character	Nominal CT–108 route $\to$ $(n_A, s_C) \to (W_+, W_-) \to \chi_{\text{mass}} \to \mathcal{L}_M$.	Route gauge CT–108, signature and scalar character determined before contrast.
Z^0	CT–108 Z route: 20/23, E-O axis, $(N, E, S, O) = (33, 45, 10, 20)$, sheet/residue $(57, 7)$, $(n_A, s_C) = (95, -1)$, $(W_+, W_-) = (569, 571)$; exact scalar character	Nominal CT–108 route $\to$ $(n_A, s_C) \to (W_+, W_-) \to \chi_{\text{mass}} \to \mathcal{L}_M$.	Route gauge CT–108, signature and scalar character determined before contrast.
Higgs H	CT–108 H route: 24/29, N-S axis, $(N, E, S, O) = (40, 34, 22, 12)$, sheet/residue $(47, 10)$, $(n_A, s_C) = (97, 9)$, $(W_+, W_-) = (591, 573)$; exact scalar character	Nominal CT–108 route $\to$ $(n_A, s_C) \to (W_+, W_-) \to \chi_{\text{mass}} \to \mathcal{L}_M$.	Route CT–108, scalar, signature and character determined before contrast.

Table 92.13: Realization of compounds, topological sectors and virtual states.

Class	**Objeto HMT**	**Realization map**	**Conclusion and conditions**
mesons	Double registration, boundary of connection and composite signature.	Composite record and constraint $\to$ scalar chart.	Composition of records; the binding precedes the scalar chart.
baryons	Triple register, carry, and composite signature.	Composite record and constraint $\to$ scalar chart.	Composition of records; the binding precedes the scalar chart.

Continues on the following page.

Class	Objeto HMT	Realization map	Conclusion and conditions
Fibonacci quantity $(1, \tau)$	Fibonacci fusion ring: $\tau \otimes \tau = 1 + \tau$.	Fusion and braiding → physical observable of the sector.	Exact codomain of fusion and braiding; the contrast uses its own invariants.
Ising $(1, \psi, \sigma)$; separate Majorana realization	Ising fusion ring: $(1, \psi, \sigma)$.	Fusion and braiding → physical observable of the sector.	Exact codomain of fusion and braiding; the contrast uses its own invariants.
sectores $\mathbb{Z}/6$	Braiding character on $\mathbb{Z}/6$.	Fusion and braiding → physical observable of the sector.	Exact codomain of fusion and braiding; the contrast uses its own invariants.
secciones virtuales	Inverse system of extensions and persistence horizon.	Registered prolongation → physical horizon of persistence.	Exact object of prolongation and obstruction; its exit is the persistence horizon.

Physical Comparability in Native Codomains External comparison is defined on the native codomain of each class. In the classes with a closed scalar section, homogeneous magnitudes are compared, with their unit, scheme, and scale declared. In the operatorial fibres, the internal spectrum is compared and the closed realization morphism that selects the route without using the external value is explicitly conserved. Mesonic and baryonic records admit only state-by-state comparison after the binding boundary has been incorporated. In the Fibonacci sector, the based ring and $\mathrm{FPdim}(\tau) = \varphi$ are closed; the matrices F/R, the pentagon, the hexagon, the Hamiltonian, and the gap belong to the subsequent realization. Ising and $\mathbb{Z}/6$ separately conserve their fusion and braiding invariants, and virtuality is compared through persistence and analytic structure. In none of these six cases is there a universal scalar mass for the class; assigning one would replace the proved object with a digit lacking a morphism.

Internal evaluations of explicit firms and posterior contrast The following table presents in legible form the seventeen HMT–MD coordinates already calculated—eleven elementary and six composite—and compares them with the 2026 edition of the Particle Data Group data set, whose editorial cutoff date is 15 January 2026 [120]. They do not constitute an ontology distinct from the preceding operatorial layer. The symbol † identifies an evaluation obtained by applying the construction map to an explicit signature. In the eleven elementary rows, the signature follows from the nominal route determined without masses; in the six composite rows, it follows from the declared binding ledger. Each figure depends on that signature, the bidirectional reader, and the common dimensional chart; the external data set enters only in the comparison and selects neither flavor, generation, color, nor route.

For charged leptons and massive bosons, the relative difference between the two central coordinates is shown and, when the dataset provides a two-sided uncertainty, its ratio to that uncertainty. For quarks, the arithmetic difference between central values is reported, but the term *metrological residual* is reserved: it requires the internal mass and the external reference to use the same definition, renormalization scheme

and scale. The light masses are read in the $\overline{\mathrm{MS}}$ convention at 2 GeV; those of c and b, in $\mu = \overline{m}_q$; the top mass comes from kinematic reconstruction.

Table 92.14: Internal evaluations of explicit firms and subsequent metrological contrast.

Status	Signature coordinate	External physical reference	Scope
μ^-/μ^+	105.658360294435829303 4 MeV$/c^{2\dagger}$; E3 tower, signature (42,-3,8,0,2)	105.6583755 ± 0.0000023 MeV$/c^2$; leptonic rest mass; PDG average and declared metrological conversion	Relative difference: -0.143912530 ppm†; $-6.611114\,\sigma$. The HMT figure is the exact evaluation of the declared signature.
τ^-/τ^+	1776.860917631432 295595 MeV$/c^{2\dagger}$; E3 tower, signature (64,0,25,-1,6)	1776.93 ± 0.09 MeV$/c^2$; direct PDG average	Relative difference: -38.877371966 ppm†; $-0.813547\,\sigma$. The HMT figure is the exact evaluation of the declared signature.
$u/\bar{u}$	2.165924536173 907 MeV$/c^{2\dagger}$; angular chart, nominal route signature (11,7)	2.16 ± 0.07 MeV$/c^2$; mass $\overline{\mathrm{MS}}$ to $\mu = 2$ GeV; 90% C.L.	Difference between central values: 2742.840821 253 ppm†. It is not a metrological residual until the same scheme and the same scale have been fixed.
$d/\bar{d}$	4.679550162948 874 MeV$/c^{2\dagger}$; chart angular, firma nominal of route (17,8)	4.70 ± 0.07 MeV$/c^2$; mass $\overline{\mathrm{MS}}$ to $\mu = 2$ GeV; 90% C.L.	Difference between central values: -4351.029159 814 ppm†. It is not a metrological residual until the same scheme and the same scale have been fixed.
$c/\bar{c}$	1.270500838641 911 GeV$/c^{2\dagger}$; angular chart, nominal route signature (61,8)	1.2729 ± 0.0045 GeV$/c^2$; mass $\overline{\mathrm{MS}}$ evaluated at $\mu = \overline{m}_q$; 90% C.L.	Difference between central values: -1884.799558 558 ppm†. It is not a metrological residual until the same scheme and the same scale have been fixed.
$s/\bar{s}$	92.940093907845 64 MeV$/c^{2\dagger}$; angular chart, nominal route signature (41,-3)	92.9 ± 0.7 MeV$/c^2$; mass $\overline{\mathrm{MS}}$ to $\mu = 2$ GeV; 90% C.L.	Difference between central values: 431.581354 635 ppm†. It is not a metrological residual until the same scheme and the same scale have been fixed.

Continues on the following page.

State	Signature coordinate	External physical reference	Scope
$t/\bar{t}$	172.597479544019 1 GeV$/c^{2\dagger}$; chart angular, firma nominal of route (100,-1)	172.60 ± 0.27 GeV$/c^2$; mass reconstructed directly from event kinematics	Difference between central values: -14.602873 585 ppm†. It is not a metrological residual until the same scheme and the same scale have been fixed.
$b/\bar{b}$	4.190119763299 790 GeV$/c^{2\dagger}$; angular chart, nominal route signature (71,-5)	4.186 ± 0.006 GeV$/c^2$; mass $\overline{\mathrm{MS}}$ evaluated at $\mu = \overline{m}_q$; 90% C.L.	Difference between central values: 984.176612 467 ppm†. It is not a metrological residual until the same scheme and the same scale have been fixed.
$W^{\pm}$	80.378964909704547024 86 GeV$/c^{2\dagger}$; E3 tower, signature (94,-1,0,-2,4)	80.3625 ± 0.0077 GeV$/c^2$; mass average of the sealed PDG snapshot	Relative difference: 204.882995234 ppm†; $2.138299\,\sigma$. The HMT figure is the exact evaluation of the declared signature.
Z^0	91.187533444313407844 85 GeV$/c^{2\dagger}$; tower E3, signature (95,-1,-11,0,-4)	91.1879 ± 0.0020 GeV$/c^2$; mass average of the sealed PDG snapshot	Relative difference: -4.019784276 ppm†; $-0.186362\,\sigma$. The HMT figure is the exact evaluation of the declared signature.
Higgs H	125.292466042536 133 GeV$/c^{2\dagger}$; angular chart, nominal route signature (97,9)	125.13 ± 0.11 GeV$/c^2$; mass average of the sealed PDG snapshot	Relative difference: 1298.378027140 ppm†; $1.454206\,\sigma$. The HMT figure is the exact evaluation of the declared signature.
π^0 signature $(44, -4, -25, 1, -2)$	134.976724327872125739 384878 . . . MeV$/c^{2\dagger}$	134.976827767684 71 $\pm$ 0.000485679282 28423351 MeV$/c^2$ PDG 2026 catalog.	Relative difference -0.766352375 ppm; $-0.212979\,\sigma$. For the declared composite signature, the evaluation is deduced exactly from the beta E3 character. The prospective selector of that signature is an earlier map independent of the numerical comparison.

Continues on the following page.

State	Signature coordinate	External physical reference	Scope
$\pi^{\pm}$ signature $(44, 1, -2, 2, -1)$	139.570485171572191374 734364 . . . MeV$/c^{2\dagger}$	139.570390983681 31 ± 0.000182007160 40826009 MeV$/c^2$ PDG 2026 catalog.	Relative difference 0.674841 491 ppm; 0.517495 σ. For the declared composite signature, the evaluation is deduced exactly from the beta E3 character. The prospective selector of that signature is an earlier map independent of the numerical comparison.
$K^{\pm}$ signature $(54, -1, 17, -2, 2)$	493.677085649530612475 723054 . . . MeV$/c^{2\dagger}$	493.676599458040 58 ± 0.015405665226 071509 MeV$/c^2$ PDG 2026 catalog.	Relative difference 0.984838 030 ppm; 0.031559 σ. For the declared composite signature, the evaluation is deduced exactly from the beta E3 character. The prospective selector of that signature is an earlier map independent of the numerical comparison.
K^0 signature $(54, 0, 32, 3, -2)$	497.611186497750457358 558672 . . . MeV$/c^{2\dagger}$	497.610767531257 52 ± 0.012990199756 4242 MeV$/c^2$ PDG 2026 catalog.	Relative difference 0.841956 244 ppm; 0.032252 σ. For the declared composite signature, the evaluation is deduced exactly from the beta E3 character. The prospective selector of that signature is an earlier map independent of the numerical comparison.
proton p signature $(59, 0, 9, 1, 0)$	938.271751546984898298 936787 . . . MeV$/c^{2\dagger}$	938.272089430000 05 ± 0.000000290000 00000000 MeV$/c^2$ PDG 2026 catalog.	Relative difference −0.360111974 ppm; −1165.113845 σ. For the declared composite signature, the evaluation is deduced exactly from the beta E3 character. The prospective selector of that signature is an earlier map independent of the numerical comparison.

Continues on the following page.

State	Signature coordinate	External physical reference	Scope
neutron n signature $(59, 1, -34, 0, 1)$	939.565810472507023312 664431 . . . MeV$/c^{2\dagger}$	939.565421939999 96 ± 0.000000480000 00000000 MeV$/c^2$ PDG 2026 catalog.	Relative difference 0.413523 633 ppm; 809.442723 σ. For the declared composite signature, the evaluation is deduced exactly from the beta E3 character. The prospective selector of that signature is an earlier map independent of the numerical comparison.

Native Invariants of Topological and Virtual Sectors. The Fibonacci, Ising, and $\mathbb{Z}/6$ sectors and the virtual sections are compared in their proper codomains. Fibonacci publishes the based ring and its Frobenius–Perron dimension; the data F/R remain in the subsequent realization. Ising preserves its fusion ring and separates the Majorana realization; $\mathbb{Z}/6$ publishes its braiding characters; virtuality publishes persistence horizons. This table maintains those invariants within the same realization map without artificially translating them into a unit of mass.

Table 92.15: Topological and virtual sectors: native invariants of fusion, braiding, and persistence.

Sector designation	Object and relations	Native invariant	Physical scope
Fibonacci datum	$K = \mathbb{Z}1 \oplus \mathbb{Z}\tau$, $\tau^2 = 1 + \tau$.	FPdim$(\tau) = \varphi$ in the based ring.	The matrices F/R, the pentagon, the hexagon, the Hamiltonian, and the gap belong to the subsequent realization.
Ising category	$K = \mathbb{Z}\{1, \psi, \sigma\}$, $\psi^2 = 1$, $\psi\sigma = \sigma$, $\sigma^2 = 1 + \psi$.	FPdim$(\sigma) = \sqrt{2}$; braiding representation.	The Majorana realization preserves its own physical morphism.
$\mathbb{Z}/6$	$q_{\epsilon,t} = (-1)^{\epsilon}\omega_3^t = \zeta_6^k$.	Quadratic character and braid representation over the six sectors.	The relevant remark is topological, not a universal mass.
Secciones virtuales	Inverse system of prolongations with horizon $L(s_N)$.	Finite register, signature, carry, and obstruction class.	The native observable is the persistence of the prolongation.

Neutrinos remain in the codomain of eigenvalues and mixing described by their operator records; PMNS realizes transport between the interaction and mass bases, while the scale belongs to the spectrum of the neutrino operator. The six composite evaluations are obtained from explicit signatures through the general binding morphism into the scalar chart. The topological sectors and virtual sections retain their own fusion, braiding, and persistence observables.

92.12.1.6 Causal construction and coverage of mass classes

The APP–TRIT–TPK architecture, the mass law, the bidirectional reader, and the towers jointly determine the atlas, the enriched chronological ledgers, the signatures, the character, and the basal closure. Their confluence in (92.51) provides a unique equation of mass sections. The complete domain is the atlas 81/56/13/324, and each realization specifies, according to its own codomain—that is, according to the codomain of each sector—the classifiers, cells, routes, orientations, dodecaphase words, signatures, and compatibility morphisms. The spectra K_D/K_Ω refine the fibres of the atlas, while the chronological routes of 108 states associated with W, Z, and H directly construct their bosonic signatures and characters.

92.12.2 Posterior evaluation of the generative graph

The following table places in adjacent columns every scalar coordinate that the theory permits to be evaluated reproducibly. The experimental reference is introduced after the HMT–MD output has been fixed. A row labeled as a conditioned evaluation asserts exactly the value of the function once the signature has been fixed; it does not attribute to that signature a physical selector that has not been constructed. Quarks are printed without a residual because a numerical comparison requires the renormalization scheme and scale to coincide.

The numerical census in this table is exhaustive relative to the scalar coordinates currently reproducible: one closed beta electron, seventeen conditioned evaluations —eleven elementary and six composite—, and two exact null outputs. It therefore contains twenty numerical rows. This number is not the domain of the law. Classes whose natural output remains an operator, a spectrum, an orbit, or a multisection appear subsequently with that complete object, without filling their column with the external reference. The experimental inventory of 471 masses and 384 widths is preserved in full as a state-by-state comparison; its cardinality is not passed off as a census of HMT predictions.

Status	Reproducible HMT value (MeV/c^2)	Referencia posterior	Relative difference (ppm)	Nature and Scope
e-/e+	0.510998950690 000809	0.51099895069± 0.00000000016 MeV/c^2	1.582406 884×10^{-9}	closed beta coordinate on the central fiber
μ-/μ+	105.658360294435 829303	105.6583755±0.000002 3 MeV/c^2	-0.14391253	refinement E3 conditioned on the signature
τ-/τ+	1776.860917631432 295595	1776.9324651340901 MeV/c^2 (central value used; rounded presentation: 1776.93± 0.09 MeV/c^2)	-40.264615601	refinement E3 conditioned on the signature
u/ū	2.165924536173 907663	2.16±0.07 MeV/c^2	—	angular chart; neither Z^5 signature nor E3 refinement has been declared
d/d̄	4.679550162948 874172	4.70±0.07 MeV/c^2	—	angular chart; neither Z^5 signature nor E3 refinement has been declared
c/c̄	1270.500838641911 135286	1.2729±0.0045 GeV/c^2	—	angular chart; neither Z^5 signature nor E3 refinement has been declared
s/s̄	92.940093907845 640641	92.9 ±0.7 MeV/c^2	—	angular chart; neither Z^5 signature nor E3 refinement has been declared
t/t̄	172597.479544 019136261367	172.60±0.27 GeV/c^2	—	angular chart; neither Z^5 signature nor E3 refinement has been declared
b/b̄	4190.119763299790 218075	4.186±0.006 GeV/c^2	—	angular chart; neither Z^5 signature nor E3 refinement has been declared
γ	0	<1×10^{-18} eV/c^2	—	exact null output
g	0	—	—	exact null output
W+/W-	80378.964909 704547024865	80.3625 ±0.0077 GeV/ c^2	204.882995235	refinement E3 conditioned on the signature

Continuation of the table

Status	Reproducible HMT value (MeV/c^2)	Referencia posterior	Relative difference (ppm)	Nature and Scope
Z0	91187.533444313407844855	91.187873297221344 GeV/c^2 (central value used; rounded presentation: 91.1879± 0.0020 GeV/c^2)	-3.72695289	refinement E3 conditioned on the signature
H	125292.466042536133000433	125.13094382816151 GeV/c^2 (central value used; rounded presentation: 125.13± 0.11 GeV/c^2)	1290.825509927	angular chart; neither Z^5 signature nor E3 refinement has been declared
π^0	134.976724327872125739	134.976827767685 MeV/c^2	-0.766352378	refinement E3 conditioned on the composite signature
$\pi\pm$	139.570485171572191375	139.570390983681 MeV/c^2	0.674841494	refinement E3 conditioned on the composite signature
K±	493.677085649530612476	493.676599458041 MeV/c^2	0.98483803	refinement E3 conditioned on the composite signature
K^0	497.611186497750457359	497.610767531258 MeV/c^2	0.841956243	refinement E3 conditioned on the composite signature
proton	938.271751546984898299	938.27208943 MeV/c^2	-0.360111975	refinement E3 conditioned on the composite signature
neutron	939.565810472507023313	939.56542194 MeV/c^2	0.413523633	refinement E3 conditioned on the composite signature

92.12.3 Individual evaluation of 20 material sections and traceability of their metrological coordinates

92.12.3.1 Electron–positron pair: central route, leaf conjugation, and mass character

- **Catalog identifier:** SM-LC-1.
- **Descriptor type:** basal selector and bidirectional readers.
- **Designator:** selecting pair (-22,+15); K_D=K_Ω=(80,54,6).
- **Stage:** bidirectional beta; initial coordinate 0.510998922488066073 MeV/c^2.
- **Evaluation HMT–MD reproducible:** 0.510998950690000809 MeV/c^2.
- **Mathematical nature:** closed beta coordinate in the central fiber.
- **Subsequent reference:** 0.51099895069±0.00000000016 MeV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:S003:S003M:332838; scheme not preserved in the documentary snapshot; scale not preserved in the documentary snapshot; confidence level not preserved in the documentary snapshot.
- **Relative difference:** 1.582406884×10^{-9} ppm.
- **Scope:** closed scalar derivation.

92.12.3.2 Muon–antimuon pair: route conjugation and leptonic character

- **Catalog identifier:** SM-LC-2.
- **Descriptor type:** complete signature $\sigma \in Z^5$.
- **Descriptor record:** (42,-3,8,0,2).
- **Stage:** beta E3 tower; initial coordinate 105.56405901285926472 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 105.658360294435829303 MeV/c^2.
- **Mathematical nature:** E3 refinement conditioned on the signature.
- **Subsequent reference:** 105.6583755±0.0000023 MeV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:S004:S004M: 332849; scheme not preserved in the documentary snapshot; scale not preserved in the documentary snapshot; confidence level not preserved in the documentary snapshot.
- **Relative difference:** -0.14391253 ppm.
- **Scope:** exact evaluation once the signature is fixed; the physical selection of that signature constitutes an independent operation.

92.12.3.3 Par tau–antitau: route conjugation and leptonic character

- **Catalog identifier:** SM-LC-3.
- **Descriptor type:** complete signature $\sigma \in Z^5$.
- **Descriptor tuple:** (64,0,25,-1,6).
- **Stage:** E3 beta tower; initial coordinate 1771.945393886234110148 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 1776.860917631432295595 MeV/c^2.
- **Mathematical nature:** E3 refinement conditioned on the signature.
- **Subsequent reference:** 1776.9324651340901 MeV/c^2 (central value used; rounded presentation: 1776.93±0.09 MeV/c^2).
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:S035:S035M: 332893; scheme not preserved in the documentary snapshot; scale not preserved in the documentary snapshot; confidence level not preserved in the documentary snapshot.
- **Relative difference:** -40.264615601 ppm.
- **Scope:** exact evaluation once the signature is fixed; the physical selection of that signature constitutes an independent operation.

92.12.3.4 Quark sectors u and antiquark sectors $\bar{u}$

- **Catalogue identifier:** SM-Q-U1.
- **Descriptor type:** angular coindexation (n_A,s_C)$\in Z^2$.
- **Descriptor tuple:** (11,7).
- **Stage:** conditioned angular chart; initial coordinate 2.165924536173907663 MeV/c^2.
- **Evaluation HMT–MD reproducible:** 2.165924536173907663 MeV/c^2.

- **Mathematical nature:** angular chart; no Z^5 signature nor E3 refinement has been declared.
- **Subsequent reference:** 2.16±0.07 MeV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:Q002:Q002M: 333588; scheme not preserved in the documentary snapshot; scale not preserved in the documentary snapshot; 90% confidence level.
- **Relative difference:** — ppm.
- **Scope:** conditioned evaluation; the signature has not been chosen from the compared value; external scheme and scale required for the residual.

92.12.3.5 Quark sectors d and antiquark sectors $\bar{d}$

- **Identificador catalografico:** SM-Q-D1.
- **Descriptor type:** angular coindexation (n_A,s_C)∈Z^2.
- **Descriptor record:** (17,8).
- **Stage:** conditioned angular chart; initial coordinate 4.679550162948874172 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 4.679550162948874172 MeV/c^2.
- **Mathematical nature:** angular chart; no Z^5 signature nor E3 refinement has been declared.
- **Subsequent reference:** 4.70±0.07 MeV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:Q001:Q001M: 333589; scheme not preserved in the documentary snapshot; scale not preserved in the documentary snapshot; confidence level 90%.
- **Relative difference:** — ppm.
- **Scope:** conditioned evaluation; the signature has not been chosen from the compared value; external scheme and scale required for the residual.

92.12.3.6 Up quark and antiup quark

- **Catalog identifier:** SM-Q-U2.
- **Descriptor type:** angular coindexation (n_A,s_C)∈Z^2.
- **Descriptor value:** (61,8).
- **Stage:** conditioned angular chart; initial coordinate 1270.500838641911135286 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 1270.500838641911135286 MeV/c^2.
- **Mathematical nature:** angular chart; no Z^5 signature nor E3 refinement has been declared.
- **Posterior reference:** 1.2729±0.0045 GeV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:Q004:Q004M: 333604; scheme not retained in the documentary snapshot; scale not retained in the documentary snapshot; confidence level 90%.
- **Relative difference:** — ppm.
- **Scope:** conditioned evaluation; the signature has not been chosen from the compared value; external scheme and scale required for the residual.

92.12.3.7 Strange quark and strange antiquark

- **Catalog identifier:** SM-Q-D2.
- **Descriptor type:** angular coindexation (n_A,s_C)∈Z^2.
- **Descriptor entry:** (41,-3).
- **Stage:** conditioned angular chart; initial coordinate 92.940093907845640641 MeV/c^2.
- **Evaluation HMT–MD reproducible:** 92.940093907845640641 MeV/c^2.
- **Mathematical nature:** angular chart; no Z^5 signature nor E3 refinement has been declared.
- **Subsequent reference:** 92.9 ±0.7 MeV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:Q003:Q003M:333590; scheme not preserved in the documentary snapshot; scale not preserved in the documentary snapshot; 90% confidence level.
- **Relative difference:** — ppm.
- **Scope:** conditioned evaluation; the signature has not been chosen from the compared value; external scheme and scale required for the residual.

92.12.3.8 Top quark and anti-top quark

- **Catalog identifier:** SM-Q-U3.
- **Descriptor type:** angular coindexation (n_A,s_C)∈Z^2.
- **Descriptor record:** (100,-1).
- **Stage:** conditioned angular chart; initial coordinate 172597.479544019136261367 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 172597.479544019136261367 MeV/c^2.
- **Mathematical nature:** angular chart; no Z^5 signature nor E3 refinement has been declared.
- **Subsequent reference:** 172.60±0.27 GeV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:Q007:Q007TP:333614; scheme not preserved in the documentary snapshot; scale not preserved in the documentary snapshot; confidence level not preserved in the documentary snapshot.
- **Relative difference:** — ppm.
- **Scope:** conditioned evaluation; the signature has not been chosen from the compared value; external scheme and scale required for the residual.

92.12.3.9 Quark background and antiquark background

- **Catalog identifier:** SM-Q-D3.
- **Descriptor type:** angular coindexation (n_A,s_C)∈Z^2.
- **Descriptor tuple:** (71,-5).
- **Stage:** conditioned angular chart; initial coordinate 4190.119763299790218075 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 4190.119763299790218075 MeV/c^2.
- **Mathematical nature:** angular chart; no Z^5 signature nor E3 refinement has been declared.

- **Posterior reference:** 4.186±0.006 GeV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:Q005:Q005M:333611; scheme not preserved in the documentary snapshot; scale not preserved in the documentary snapshot; confidence level 90%.
- **Relative difference:** — ppm.
- **Scope:** conditioned evaluation; the signature has not been chosen from the compared value; external scheme and scale required for the residual.

92.12.3.10 Reproducible spectral descriptor of the photon: null sector, helicity, and electromagnetic character

- **Catalogue identifier:** SM-GA-EM.
- **Descriptor type:** null sectorial invariant.
- **Descriptor:** null chart.
- **Stage:** null chart; initial coordinate 0 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 0 MeV/c^2.
- **Mathematical nature:** exact null output.
- **Subsequent reference:** $<1\times10^{-18}$ eV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:S000:S000M:332461; scheme not preserved in the documentary snapshot; scale not preserved in the documentary snapshot; confidence level not preserved in the documentary snapshot.
- **Relative difference:** — ppm.
- **Scope:** exact in the null sector; the external bound is not interpreted as a positive mass.
- **Bound photonic:** reference externa ilustrativa not comparable; the cut identifies the observable but not preserves the level of confidence.

92.12.3.11 Realization of the gluon as a section of the adjoint representation of $SU(3)$

- **Catalogue identifier:** SM-GA-GL.
- **Descriptor type:** null sectorial invariant.
- **Descriptor:** null chart.
- **Stage:** null chart; initial coordinate 0 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 0 MeV/c^2.
- **Mathematical nature:** exact null output.
- **Referencia posterior:** —.
- **External identity and metrology:** source not stated; observable not individualized; scheme not stated; scale not stated; confidence level not stated.
- **Relative difference:** — ppm.
- **Scope:** exact in the null sector; nullity belongs to the color gauge chart.

92.12.3.12 Bosons W^+ and W^-

- **Catalog identifier:** SM-GA-W.
- **Descriptor type:** complete signature $\sigma \in Z^5$.
- **Descriptor tuple:** (94,-1,0,-2,4).
- **Stage:** E3 beta tower; initial coordinate 80381.592550220666304144 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 80378.964909704547024865 MeV/c^2.
- **Mathematical nature:** E3 refinement conditioned on the signature.
- **Posterior reference:** 80.3625 ±0.0077 GeV/c^2.
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:S043:S043M: 332464; scheme not retained in the documentary snapshot; scale not retained in the documentary snapshot; confidence level not retained in the documentary snapshot.
- **Relative difference:** 204.882995235 ppm.
- **Scope:** exact evaluation once the signature is fixed; the physical selection of that signature constitutes an independent operation.
- **Resonant convention:** the external cut does not jointly fix pole convention, Breit–Wigner or on-shell convention, width, and covariance; the residual is interpreted only as scalar information.

92.12.3.13 Reproducible spectral descriptor of the Z^0 boson: neutral mixing, polarization, and electroweak character

- **Catalog identifier:** SM-GA-Z.
- **Descriptor type:** complete signature $\sigma \in Z^5$.
- **Descriptor tuple:** (95,-1,-11,0,-4).
- **Stage:** beta tower E3; initial coordinate 91299.748286598170117317 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 91187.533444313407844855 MeV/c^2.
- **Mathematical nature:** E3 refinement conditioned on the signature.
- **Subsequent reference:** 91.187873297221344 GeV/c^2 (central value used; rounded presentation: 91.1879±0.0020 GeV/c^2).
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:S044:S044M: 332513; scheme not retained in the documentary snapshot; scale not retained in the documentary snapshot; confidence level not retained in the documentary snapshot.
- **Relative difference:** -3.72695289 ppm.
- **Scope:** exact evaluation once the signature is fixed; the physical selection of that signature constitutes an independent operation.
- **Resonant convention:** the external cut does not jointly fix pole convention, Breit–Wigner or on-shell convention, width, and covariance; the residual is interpreted only as scalar information.

92.12.3.14 Realization of the Higgs boson: scalar mode, coupling, and mass character

- **Catalog identifier:** SM-H-1.
- **Descriptor type:** angular coindexation (n_A,s_C)$\in Z^2$.
- **Descriptor value:** (97,9).
- **Stage:** conditioned angular chart; initial coordinate 125292.466042536133000433 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 125292.466042536133000433 MeV/c^2.
- **Mathematical nature:** angular chart; no Z^5 signature nor E3 refinement has been declared.
- **Subsequent reference:** 125.13094382816151 GeV/c^2 (central value used; rounded presentation: 125.13±0.11 GeV/c^2).
- **External identity and metrology:** Particle Data Group (PDG); PDG2026:observable:S126:S126M: 332733; scheme not retained in the documentary snapshot; scale not retained in the documentary snapshot; confidence level not retained in the documentary snapshot.
- **Relative difference:** 1290.825509927 ppm.
- **Scope:** conditioned evaluation; the signature has not been chosen from the compared value.
- **Resonant convention:** the external cut does not jointly fix pole convention, Breit–Wigner or on-shell convention, width, and covariance; the residual is interpreted only as scalar information.

92.12.3.15 Neutral pion π^0

- **Catalog identifier:** CMP-PI0.
- **Descriptor type:** complete signature $\sigma\in Z^5$.
- **Descriptor tuple:** (44,-4,-25,1,-2).
- **Stage:** composite E3 beta tower; initial coordinate 135.350257251504283349 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 134.976724327872125739 MeV/c^2.
- **Mathematical nature:** E3 refinement conditioned on the composite signature.
- **Posterior reference:** 134.976827767685 MeV/c^2.
- **External identity and metrology:** composite reference preserved by Supplement E3; the convention, covariance, and edition are recorded in the reference metrological source.
- **Relative difference:** -0.766352378 ppm.
- **Scope:** exact evaluation once the composite signature is fixed; the binding generator remains a proper morphism.

92.12.3.16 Charged pions $\pi^{\pm}$: quark–antiquark composition, charge, and mass

- **Catalog identifier:** CMP-PIPM.
- **Descriptor type:** complete signature $\sigma\in Z^5$.
- **Descriptor entry:** (44,1,-2,2,-1).
- **Stage:** composite E3 beta tower; initial coordinate 139.596265529971781015 MeV/c^2.
- **Evaluation HMT–MD reproducible:** 139.570485171572191375 MeV/c^2.

- **Mathematical nature:** E3 refinement conditioned on the composite signature.
- **Subsequent reference:** 139.570390983681 MeV/c^2.
- **External identity and metrology:** composite reference preserved by Supplement E3; the convention, covariance, and edition are recorded in the reference metrological source.
- **Relative difference:** 0.674841494 ppm.
- **Scope:** exact evaluation once the composite signature is fixed; the binding generator remains a proper morphism.

92.12.3.17 Charged kaons $K^{\pm}$: composition with strangeness, charge, and mass

- **Catalog identifier:** CMP-KPM.
- **Descriptor type:** complete signature $\sigma \in Z^5$.
- **Descriptor record:** (54,-1,17,-2,2).
- **Stage:** composite beta E3 tower; initial coordinate 492.762503210140547854 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 493.677085649530612476 MeV/c^2.
- **Mathematical nature:** E3 refinement conditioned on the composite signature.
- **Subsequent reference:** 493.676599458041 MeV/c^2.
- **External identity and metrology:** composite reference preserved by Supplement E3; the convention, covariance, and edition are recorded in the reference metrological source.
- **Relative difference:** 0.98483803 ppm.
- **Scope:** exact evaluation once the composite signature is fixed; the binding generator remains a proper morphism.

92.12.3.18 Neutral kaon K^0

- **Catalog identifier:** CMP-K0.
- **Descriptor type:** complete signature $\sigma \in Z^5$.
- **Descriptor tuple:** (54,0,32,3,-2).
- **Stage:** composite beta tower E3; initial coordinate 495.816066594426768595 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 497.611186497750457359 MeV/c^2.
- **Mathematical nature:** E3 refinement conditioned on the composite signature.
- **Posterior reference:** 497.610767531258 MeV/c^2.
- **External identity and metrology:** composite reference preserved by Supplement E3; the convention, covariance, and edition are recorded in the reference metrological source.
- **Relative difference:** 0.841956243 ppm.
- **Scope:** exact evaluation once the composite signature is fixed; the binding generator remains a proper morphism.

92.12.3.19 Proton: baryonic composition, color neutralization, and mass character

- **Catalogue identifier:** CMP-P.
- **Descriptor type:** complete signature $\sigma \in Z^5$.
- **Descriptor tuple:** (59,0,9,1,0).
- **Stage:** composite E3 beta tower; initial coordinate 937.314779258699634563 MeV/c^2.
- **Evaluation HMT–MD reproducible:** 938.271751546984898299 MeV/c^2.
- **Mathematical nature:** E3 refinement conditioned on the composite signature.
- **Subsequent reference:** 938.27208943 MeV/c^2.
- **External identity and metrology:** composite reference preserved by Supplement E3; the convention, covariance, and edition are recorded in the reference metrological source.
- **Relative difference:** -0.360111975 ppm.
- **Scope:** exact evaluation once the composite signature is fixed; the binding generator remains a proper morphism.

92.12.3.20 Neutron: baryonic composition, color neutralization, and mass character

- **Identificador catalografico:** CMP-N.
- **Descriptor type:** complete signature $\sigma \in Z^5$.
- **Descriptor record:** (59,1,-34,0,1).
- **Stage:** composite beta E3 tower; initial coordinate 943.123155648642014982 MeV/c^2.
- **Reproducible HMT–MD evaluation:** 939.565810472507023313 MeV/c^2.
- **Mathematical nature:** E3 refinement conditioned on the composite signature.
- **Subsequent reference:** 939.56542194 MeV/c^2.
- **External identity and metrology:** composite reference preserved by Supplement E3; the convention, covariance, and edition are recorded in the reference metrological source.
- **Relative difference:** 0.413523633 ppm.
- **Scope:** exact evaluation once the composite signature is fixed; the binding generator remains a proper morphism.

92.13 Correspondence between the atlas 81/56/13/324 and the exhaustive inventories of 471 masses and 384 widths

The inventory of 471 masses and 384 widths is an external metrological catalogue with a typed codomain. It is neither a census of predictions nor the cardinality of the material domain.

92.14 Null sector and positivity

Nullity is not obtained by taking a limit of the positive character. The domain is first separated into

$$\mathcal{H}_{\text{phys}} = \mathcal{H}_0 \oplus \mathcal{H}_+,$$

with

$$\mathcal{M}|_{\mathcal{H}_0} = 0, \qquad \mathcal{M}|_{\mathcal{H}_+} > 0.$$

The photon belongs to the electromagnetic null chart and the gluons to the adjoint null chart of color. Their different representations, polarizations, and gauge restrictions remain visible even though both masses are null.

Positivity excludes an elementary species with $m^2 < 0$ within the mass codomain. A negative eigenvalue of the Hessian of a background does not define a tachyon as a particle: it diagnoses the instability of the background and requires changing the solution around which linearization is performed.

92.15 Fiber, orbit, and multisection mass operators of the atlas

The chromatic orbit is realized in the color qutrit. The Weyl operators generate $\mathfrak{sl}_3(\mathbb{C})$; the traceless anti-Hermitian selection yields $\mathfrak{su}(3)$, and the finite connection transports the three chromatic charts. The flavor mass must be invariant under this action.

Let $U_{\text{col}} : SU(3) \to \mathcal{U}(V_{\text{col}})$. The chromatic reduction is the Reynolds expectation

$$\mathbb{E}_{\text{col}}(T) = \int_{SU(3)} U_{\text{col}}(g) T U_{\text{col}}(g)^* \, dg.$$

In the fundamental irreducible representation,

$$\mathbb{E}_{\text{col}}(\mathcal{M}_q) = m_q(\mu, \mathscr{R}) I_3.$$

The published mass is therefore independent of red, green and blue. The coordinates μ and $\mathscr{R}$ identify the scale and renormalization scheme; they are not hidden ambiguities and must always accompany the figure.

The dependence on scale is a section of a fibration over $t = \log \mu$. If $\gamma_m(g(t))$ is the endomorphism induced by the chromatic connection, transport is defined by

$$\nabla_{\text{RG}} m_q = \left(\frac{d}{dt} + \gamma_m(g(t)) \right) m_q = 0,$$

$$m_q(t_2) = \mathcal{P} \exp\left(- \int_{t_1}^{t_2} \gamma_m(g(t)) \, dt \right) m_q(t_1).$$

Two values belonging to different scales or schemes are not comparable numbers until they are transported to the same fiber. This equation closes the mathematical type of the running mass: it is not an isolated absolute scalar, but a covariantly transported section.

For mesons and baryons, the singlet projectors are

$$P^{(1)}_{q\bar{q}} = \frac{1}{3} |\Omega\rangle\langle\Omega|, \qquad |\Omega\rangle = \sum_{a=1}^{3} |a\bar{a}\rangle,$$

and

$$P^{(1)}_{qqq} = |\varepsilon\rangle\langle\varepsilon|, \qquad |\varepsilon\rangle = \frac{1}{\sqrt{6}} \sum_{a,b,c} \varepsilon_{abc} |abc\rangle.$$

Both eliminate the observable chromatic label without erasing the internal orbit that intervened in the construction.

The CKM mixing belongs to a distinct map. If B_u and B_d are the mass operators in the up-type and down-type bases, the internal unitary matrix satisfies

$$B_d^{(u)} = V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^*.$$

The angles derived from A, C^* and Δ_4, the phase δ_{CP}, the Jarlskog invariant and the determinant identity of the commutator determine mixing without creating the mass eigenvalues. Thus spectrum, color, generation and change of basis remain separate.

92.16 Neutrinos, PMNS Mixing, and Majorana Criterion

Neutrinos are electrically neutral fermionic sections. Their fermionicity descends from the spinorial holonomy, the exterior completion, and the canonical anticommutation relations; neutrality does not alter their statistics. The internal support is

$$\mathcal{H}_\nu = \bigoplus_{j=1}^{4} \mathcal{H}_{\nu,G_j}, \qquad \dim \mathcal{H}_\nu = 2 + 4 + 6 + 8 = 20.$$

The four structural depths are neither flavours nor eigenvalues. The three-dimensional physical subspace is obtained through the neutral projector P_ν, and the mass operator is

$$M_\nu = P_\nu \mathcal{M}_\nu P_\nu.$$

Its three ordered eigenvalues are the masses; the absolute scale comes from the same mass clock that fixes the remaining branches, and not from the mixing angles.

Let $M_\ell = P_\ell \mathcal{M}_\ell P_\ell$ be the charged operator. Denote by F_ℓ and F_ν complete orthonormal spectral frames, ordered by the spectral projectors of M_ℓ and M_ν. The mixing matrix is

$$U_{\mathrm{PMNS}} = F_\ell^* F_\nu.$$

Theorem 92.49 (Spectral closure of the leptonic mixture)**.** *If the two frames are complete, U_{PMNS} is unitary. If the spectra are simple, it is determined up to diagonal phase readjustments and permutations fixed by the spectral order. If degeneracy exists, the exact result is a double class*

$$\prod_\lambda U(m_{\ell,\lambda}) \backslash U(3) / \prod_\mu U(m_{\nu,\mu}),$$

and not a numerically artificial matrix selected.

Proof. Orthonormality gives $U_{\mathrm{PMNS}}^* U_{\mathrm{PMNS}} = F_\nu^* F_\ell F_\ell^* F_\nu = I_3$. Changes of basis within each degenerate subspace act on the left or on the right, producing exactly the indicated double class. □

The complete record determines the elements of M_ν by means of carry, sheet, orientation, vacancy, and holonomy. Equivalently, the Hermitian kernel may be used

$$K_{\mathrm{reg}}(c,d) = \sum_{r \in \{\mathrm{car,hoj,ori,vac,hol}\}} \frac{\langle \Xi_r(c), \Xi_r(d) \rangle}{\sum_x \|\Xi_r(x)\|^2},$$

with omission of every summand of null norm. Its Gram matrix is positive and constructs the operator before diagonalization. The rank-one phase

$$I + (e^{i\pi/6} - 1)|b_K\rangle\langle b_K|$$

remains as a negative control: its formal unitarity is not sufficient and its angle θ_{13} falsifies its identification with full-rank physical mixing.

Let G_0 be the three-dimensional matrix obtained by aggregating the preceding kernel over the three neutrino sectors. The finite condition

$$\det G_0 \neq 0$$

is the exact certificate that the observables included in the record separate three directions. In that case, the polar factor

$$U_{G_0} = G_0(G_0^* G_0)^{-1/2}$$

is unitary and has rank 3. If $\det G_0 = 0$, the declared kernel does not determine a three-dimensional transport: an independent internal observable must be added and the Gram matrix recomputed. Invertibility may not be imposed by adding a term constructed from the mixing matrix one seeks to obtain. The physical matrix is fixed by the spectral frames of M_ℓ and M_ν; the Gram criterion certifies sufficiency of the register, but does not by itself select those frames.

Let C_ν be the antiunitary self-conjugation of the neutral sector. The Majorana criterion is

$$C_\nu^2 = I, \qquad C_\nu M_\nu = M_\nu C_\nu, \qquad C_\nu H_\nu = H_\nu C_\nu,$$

together with the existence of a fixed real basis in the realification of the physical subspace. When these conditions are satisfied, each mode can be represented by $\psi = C_\nu \psi$. The Dirac branch requires, by contrast, a conserved charge Q such that

$$C_\nu Q C_\nu^{-1} = -Q, \qquad Q\psi \neq 0,$$

and retains ψ and $C_\nu\psi$ as distinct modes. If neither set of conditions is established, the type remains undecided. Neutrality, the word involution or the Ising sector, taken separately, do not decide this alternative.

An additional structural depth does not by itself constitute a sterile neutrino. A candidate sterile sector is defined by a nonzero projector P_s satisfying

$$P_s P_{\mathrm{act}} = 0, \qquad P_s J_{\mathrm{weak}} = 0,$$

with dynamics defined on $P_s\mathcal{H}_\nu$. If, in addition, $[P_s, M_\nu] = 0$, the sector is reducing and does not mix with the active sector. If $P_s M_\nu P_{\mathrm{act}} \neq 0$, active–sterile mixing exists and necessarily $[P_s, M_\nu] \neq 0$. Thus, existence of the sector and existence of mixing are distinct properties. If the weakly neutral projector does not exist, the additional coarse channel is internal depth of the support and not a fourth species.

92.17 Massive bosons and the scalar sector

The sectors W, Z, and the scalar sector have their own representation, route, and signature descriptors. Their realization does not require forcing them into an elementary rank-one fiber. The positive fiber operator is constructed first and then, on a finite-dimensional subspace $P_X\mathcal{H}_X$, the effective channel operator

$$H_{X,\mathrm{eff}}(z) = P_X H_X P_X + P_X H_X Q_X (z - Q_X H_X Q_X)^{-1} Q_X H_X P_X.$$

If the reduced resolvent admits meromorphic continuation through the cut of the continuum, the poles are the isolated zeros, counted with multiplicity, of

$$\det\big(z - H_{X,\mathrm{eff}}(z)\big) = 0, \qquad z_X = M_X c_{\mathrm{int}}^2 - \frac{i}{2}\Gamma_X.$$

An open channel does not by itself guarantee the existence of such zeros. When a zero with the prescribed physical residue exists, its real part publishes pole mass and its imaginary part the width. For the scalar sector, identification of an excitation additionally requires the scalar projector and its gauge and fermionic couplings. Two independent poles determine two states; a second label without projector or pole does not create a particle.

92.18 Correspondence by domain and codomain between the material inventory and the realizations of the atlas

For constituents $X_1, \dots, X_n$, the input is the composite record

$$\mathfrak{C}_{12}^{(n)} : (\Gamma_1, \dots, \Gamma_n; \mathcal{T}, \partial) \longmapsto \mathcal{L}_{\mathrm{comp}},$$

not the sum of isolated masses. The binary composition

$$L_1 \star L_2 = L_1 + L_2 + \mathfrak{b}(L_1, L_2)$$

is associative exactly when

$$\mathfrak{b}(L_1, L_2) + \mathfrak{b}(L_1 \star L_2, L_3) = \mathfrak{b}(L_2, L_3) + \mathfrak{b}(L_1, L_2 \star L_3).$$

The signature, the refinement, and the composite operator are computed after the linking:

$$\mathcal{L}_{\mathrm{comp}} \longmapsto \sigma_{\mathrm{comp}} \longmapsto \eta_{\mathrm{comp}} \longmapsto \mathcal{M}_{\mathrm{comp}}.$$

The physical projector combines color singlet, spin and parity, isospin, charge, exchange symmetry, and channel. The stable mass is an eigenvalue of the compression; when the hypotheses of meromorphic continuation hold, the mass and width of a resonance follow from a Feshbach pole. Thus, mesons, baryons, exotic states, nuclei, atoms, and molecules possess the same causal scheme without being reduced to a sum of masses.

Two constituent trees represent the same species if there exists a unitary that simultaneously intertwines their operators, channel projectors and poles. In the absence of such a unitary they are retained as distinct realizations, even if they share constituents and partial quantum numbers.

92.19 Resonances, widths, and survival

Let S_X be the projector onto the surviving subspace and $T_X = S_X U S_X$ the compressed one-step operator. If it has a simple dominant eigenvalue

$$\lambda_X = r_X e^{-i\theta_X}, \qquad 0 < r_X < 1,$$

fix the branch $\theta_X \in I_\theta$, where I_θ is the fundamental quasienergy interval determined by the temporal chart and its orientation. Then

$$M_X = \frac{\hbar_{\mathrm{int}}}{t_0 c_{\mathrm{int}}^2}\theta_X, \qquad \Gamma_X = -\frac{2\hbar_{\mathrm{int}}}{t_0}\log r_X, \qquad \tau_X = \frac{\hbar_{\mathrm{int}}}{\Gamma_X}.$$

Without the choice of I_θ, the phase determines only θ_X modulo 2π, and the quasienergy mass is not a well-defined scalar. The width is not a sixth coordinate of the mass signature: it arises from the modulus of the open mode. Equality of mass and width between particle and antiparticle also requires transporting the channels and the causal prescription, not merely the parity of the signature.

92.20 Multiparticle completion and statistics

Let $\mathsf{Hol}_{\pm}^{\leq 1}$ be the monoidal category of fibers of a particle, with contractive refinement morphisms, in particular unitary ones, and holonomy parity $\epsilon \in \{+1, -1\}$. The Fock completion is defined by the functor

$$\mathfrak{F}(H, \epsilon) = \begin{cases} \bigoplus_{n \geq 0} \mathrm{Sym}^n(H), & \epsilon = +1, \\ \bigoplus_{n \geq 0} \bigwedge^n H, & \epsilon = -1, \end{cases}$$

and, for a contractive intertwiner $T : H \to H'$,

$$\mathfrak{F}(T) = \bigoplus_{n \geq 0} \mathrm{Sym}^n(T) \quad \text{or} \quad \bigoplus_{n \geq 0} \bigwedge^n T.$$

Theorem 92.50 (Internal transport of the statistic)**.** *The foregoing completion is a functor of bounded operators on $\mathsf{Hol}_{\pm}^{\leq 1}$; it is compatible with refinement and is graded monoidal. In the $\epsilon = -1$ branch, the creation and annihilation operators satisfy CAR and the occupation operators satisfy $N_s^2 = N_s$; in the $\epsilon = +1$ branch, they satisfy CCR. Therefore, Pauli exclusion and fermionic and bosonic statistics descend from holonomy parity once the representation of a particle has been fixed.*

Proof. Symmetric and exterior powers preserve compositions and identities; the contractivity of T ensures that their direct sum defines a bounded operator. For noncontractive morphisms, second quantization requires an explicit dense domain and does not belong to this category of bounded operators. The sign rule of the exterior product produces CAR and idempotence of occupation; the symmetric quotient produces CCR. Second quantization of the refinement morphisms intertwines the corresponding operators. □

This proof closes the internal algebraic transport. Identification with the relativistic spin–statistics theorem additionally requires the Lorentzian bridge and locality declared in Dimensional Mechanics; the two statements must not be confused.

For a compressed Hamiltonian $H_{X,V}$ over a finite volume or refinement V, the thermal state is completely defined by

$$Z_{X,V}(\beta) = \mathrm{Tr}\, e^{-\beta H_{X,V}}, \qquad \rho_{X,V,\beta} = Z_{X,V}(\beta)^{-1} e^{-\beta H_{X,V}}.$$

The thermodynamic limit exists when the pressure

$$p_X(\beta) = \lim_{V \nearrow \infty} \frac{1}{\beta |V|} \log Z_{X,V}(\beta)$$

exists and is independent of the cofinal sequence. Bosonic condensation is decided by saturation of the density of excited states; it is not proclaimed from the mere existence of a symmetric completion. The problem is thus expressed through a verifiable mathematical condition for the Hamiltonian and the limit, rather than through a nominal attribution.

92.21 Topological sectors and virtual states

In a topological sector the primary codomain is a braided category, not a line of masses. For simple objects a, b, c, the physical data include

$$N_{ab}^{c}, \qquad F_{d}^{abc}, \qquad R_{c}^{ab},$$

with pentagon and hexagon equations. A scalar mass exists only when a Hamiltonian realization provides a spectral projector and a gap. The Fibonacci, Ising/Majorana, and $\mathbb{Z}_6$ sectors are thereby closed as categorical codomains; they are not assigned a universal figure by analogy with an elementary particle.

In the sector $\mathbb{Z}_6$, fusion and braiding are

$$a \otimes b = a + b \pmod 6, \qquad R_{ab} = \zeta_6^{ab}, \qquad \zeta_6 = e^{2\pi i/6}.$$

For Fibonacci,

$$\tau \otimes \tau = \mathbf{1} \oplus \tau, \qquad F_{\tau}^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix},$$

$$R_{\mathbf{1}}^{\tau\tau} = e^{-4\pi i/5}, \qquad R_{\tau}^{\tau\tau} = e^{3\pi i/5}.$$

The Ising sector has objects $\mathbf{1}, \psi, \sigma$ and rules

$$\psi \otimes \psi = \mathbf{1}, \qquad \psi \otimes \sigma = \sigma, \qquad \sigma \otimes \sigma = \mathbf{1} \oplus \psi,$$

$$F_{\sigma}^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad R_{\mathbf{1}}^{\sigma\sigma} = e^{-\pi i/8}, \qquad R_{\psi}^{\sigma\sigma} = e^{3\pi i/8}.$$

These matrices satisfy the coherence relations in the indicated conventions. The duplication $C \boxtimes C^{\mathrm{rev}}$ preserves both orientations and avoids introducing a chirality by external choice.

A virtual state is a section of finite persistence in an amplitude or a resolvent. It does not belong to the asymptotic spectrum and does not require a real pole mass. Its parameters are virtuality, the channel and the duration of persistence. This definition avoids turning an internal computational line into a new species.

92.22 Gravity, torsion, and spin-2 mode

Gravitation enters Dimensional Mechanics by means of connection, curvature, and torsion. It does not require an elementary particle of spin 2 as a datum of departure. Let $\mathcal{S}_{\mathrm{grav}}$ be the geometric action and $\mathcal{H}_{\mathrm{TT}}$ the transverse traceless subspace of its Hessian in a solution. The linearized observable is

$$\mathcal{K}_{\mathrm{TT}} = P_{\mathrm{TT}}\, \delta^2 \mathcal{S}_{\mathrm{grav}}\, P_{\mathrm{TT}}.$$

A collective spin 2 mode exists as an asymptotic particle only if the resolvent of $\mathcal{K}_{\mathrm{TT}}$ has a physical pole with positive residue and satisfies the constraints. If no such pole exists, the gravitational interaction remains wholly in the connection and torsion. Thus, the graviton is not a necessary component of the mass law, and its possible appearance is decided by a spectral criterion, not by a nominal box.

92.23 Compatibility between the null sector and the null–material electronic closure

The null sector bifurcates before application of the positive character. The electronic null–material closure established in theorem 58.13 identifies the rank-(1) morphism linking the normalized null mode, the central genealogy, and the mass line; it neither converts the null sector into a limit of the positive branch nor retrospectively selects a route through its metrological value.

92.24 General Mass Law: genealogical atlas, bidirectional readers and dimensional line

The General Mass Law is not a table of values or a species-specific fitting rule. It is the dimensional realization of a genealogy constructed before opening any metrological catalog:

$$\mathrm{APP} \longrightarrow \widehat{\mathbb{T}} \longrightarrow \mathrm{TPK} \longrightarrow \widetilde{\Gamma}^{\mathrm{surv}} \longrightarrow \Gamma^{\mathrm{obs}} \longrightarrow L_{108} \longrightarrow \sigma \in \mathbb{Z}^5 \longrightarrow \chi_0 \longrightarrow \chi_{\mathrm{full}} \longrightarrow \widehat{\mathcal{M}} \longrightarrow \mathcal{L}_M. \tag{92.56}$$

APP provides the two arithmetic sheets and their oriented defect; TRIT preserves sign, threshold and regime; the TPK selects, transports and updates the state through $U_t = \mathrm{Upd}_t \circ \mathrm{Tra}_t \circ \mathrm{Sel}_t$, preserving residue, quotient, carry, sheet, orientation, boundary, route, memory and survival. The physical quantity appears only at the end of (92.56). In particular, neither a known mass nor a species label intervenes in route selection.

The constants appearing below are received as internal publications of the same HMT genealogy. Specifically, π_{HMT}, e_{HMT}, φ_{HMT} and α_{HMT} belong to the typed nonadic image generated by APP–TRIT–TPK. Their subsequent use in the mass branch is a descending internal composition; it is not an injection of conventional values.

92.24.1 Finite domain, routes and multisections

Definition 92.51 (Oriented atlas and fiber quotients)**.** Let C_{81} be the cellular atlas produced by the family reader of the enriched state. We define

$$\mathcal{R}_{324} := C_{81} \times \{+1, -1\} \times \{+1, -1\}. \tag{92.57}$$

The first quotient identifies cells with the same unoriented history and gives the set $\mathcal{F}_{56}$ of route families. Stratification by representation type gives thirteen multisections $\mathcal{M}_{13}$. The null chart $\mathcal{N}$ is a separate summand of the positive domain; it is not a fourteenth multisection obtained by setting a character to zero.

Theorem 92.52 (Complete census and no point selection)**.** *The internal domain of the law satisfies*

$$|C_{81}| = 81, \qquad |\mathcal{F}_{56}| = 56, \qquad |\mathcal{M}_{13}| = 13, \qquad |\mathcal{R}_{324}| = 324. \tag{92.58}$$

The multiplicities of the family quotient are

$$41 \times 1, \quad 10 \times 2, \quad 2 \times 3, \quad 1 \times 4, \quad 2 \times 5, \qquad 41 + 20 + 6 + 4 + 10 = 81, \tag{92.59}$$

and the ranks of the thirteen multisections are

$$\underbrace{1}_{1\ time},\quad \underbrace{2}_{2\ times},\quad \underbrace{4}_{3\ times},\quad \underbrace{6}_{2\ times},\quad \underbrace{8}_{3\ times},\quad \underbrace{12}_{1\ time},\quad \underbrace{16}_{1\ time}, \tag{92.60}$$

whose sum with multiplicity is 81. *The rank-one fiber is the central electronic fiber. Every noncentral fiber has rank greater than one and, therefore, its natural output is an operator on the fiber; a physical name does not by itself select one of its eigenvalues.*

Proof. Each cell admits exactly the four independent orientations $(h_0, v_0) \in \{\pm 1\}^2$, whence $|\mathcal{R}_{324}| = 4 \cdot 81 = 324$. The sums of (92.59) simultaneously give fifty-six classes and eighty-one cells. The rank sum of (92.60) is

$$1 + 2 \cdot 2 + 3 \cdot 4 + 2 \cdot 6 + 3 \cdot 8 + 12 + 16 = 81.$$

Cellular inversion has a unique fixed point, the centre $(5, 5)$; all other orbits form fibres of cardinality greater than one. An invariant pointwise choice in one of these orbits would contradict the transitivity of the inversion action on the fibre. Therefore only the centre canonically admits a rank-one reduction. □

92.24.2 From the APP defect to the integer register

Proposition 92.53 (Typed separation of 4 and 54)**.** *Let R be the involution that exchanges the APP sheets and $P_\pm = (I \pm R)/2$. The central block*

$$B_c = 7I + 2R = 9P_+ + 5P_- \tag{92.61}$$

produces the primordial spectral jump $9 - 5 = 4$. *The numbers*

$$2, \qquad 6 = 51 - 45, \qquad 54 = 9 \cdot 6 = 459 - 405 \tag{92.62}$$

are, respectively, local coupling, compression defect and integrated defect. In the electronic reader, $54 = D_1(\Gamma_e)$ *also counts a kinematic return. None of these occurrences replaces the APP jump* 4.

Proof. The spectral decomposition of (92.61) gives eigenvalues 9 and 5, so their difference is 4. The off-diagonal entry in the sheet basis is 2. The two global compressions evaluate to 51 and 45, and their difference is 6; integrating this defect over nine positions gives 54. Domain, operation and codomain change at each step. The subsequent numerical equality with $D_1(\Gamma_e)$ is a confluence of readers in $\mathbb{Z}$, not an identity of the objects they evaluate. □

For a route $\Gamma \in \mathcal{R}_{324}$, the CT–108 register preserves a temporal word $w_\Gamma \in \mathbb{F}_3^{108}$, a signed word $\tau_\Gamma \in \{-1, 0, +1\}^{12}$, sheet, winding, memory and carry. The nonadic return satisfies

$$g(K + 9) = g(K), \qquad B_{K+9} \neq B_K \quad \text{in general}, \tag{92.63}$$

because the phase returns and the dodecaphase register advances.

Definition 92.54 (Integer signature and central character)**.** The route ledger defines

$$\sigma(\Gamma) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta) \in \mathbb{Z}^5. \tag{92.64}$$

The closure of twelve events determines the oriented reading $s_C^{(12)} = s_C/6$; the divisor six is not a fitted parameter. Let

$$\Delta_4 := \frac{\pi_{\mathrm{HMT}} - e_{\mathrm{HMT}} \log \pi_{\mathrm{HMT}}}{270}, \qquad \zeta_{270} := 135\Delta_4^2, \tag{92.65}$$

$$A := 1000\alpha_{\mathrm{HMT}}, qquadx := \frac{\pi A}{180}, qquady := \frac{\pi C^*}{180}, qquadq_\pm := e^{-x\mp y}, qquads_n := q_-^n - q_+^n. \quad (92.66)$$

Here C^* is the counterangle already constructed by the HMT regional selector, not by a mass. The central character is

$$\chi_0(\sigma) := \exp\Big(n_A x + \frac{s_C}{6} y + k_{\Delta_4}\Delta_4 + b_{120}s_{120} + b_\zeta \zeta_{270}\Big). \quad (92.67)$$

Lemma 92.55 (Character homomorphism)**.** *For composable signatures, $\chi_0(\sigma_1 + \sigma_2) = \chi_0(\sigma_1)\chi_0(\sigma_2)$. The character is positive, dimensionless and contains no mass unit.*

Proof. The exponent of (92.67) is a linear form on $\mathbb{Z}^5$; the exponential transforms addition into multiplication. All its coefficients are dimensionless coordinates constructed before opening a dimensional chart. The unit appears only upon passing to $\mathcal{L}_M$. □

92.24.3 Absolute action scale and relative corrector

Theorem 92.56 (Determinantal origin of the decade −34)**.** *For*

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \quad (92.68)$$

we have

$$\mathrm{SNF}(H_4) = \mathrm{diag}(1, 2, 2, 4), \qquad |\operatorname{coker} H_4| = 16. \quad (92.69)$$

The determinantal reader publishes

$$\Sigma_H := \varphi_{\mathrm{HMT}}\alpha_{\mathrm{HMT}}^{16} = 1.046243838104756580184229208 3\ldots \times 10^{-34}, \quad (92.70)$$

so that $\mathrm{ord}_{10}(\Sigma_H) = -34$.

Proof. The determinantal divisors of H_4 are $\delta_1 = 1$, $\delta_2 = 2$, $\delta_3 = 4$ and $\delta_4 = |\det H_4| = 16$. By Smith's theorem, the invariant factors are $d_1 = 1, d_2 = \delta_2/\delta_1 = 2, d_3 = \delta_3/\delta_2 = 2, d_4 = \delta_4/\delta_3 = 4$, which proves (92.69). Substituting the already generated HMT outputs φ_{HMT} and α_{HMT} into the reader yields the value of (92.70); in particular $10^{-34} < \Sigma_H < 10^{-33}$. No SI constant was used to select the exponent sixteen: it is the order of the cokernel. □

The sections before and after the return lie in a single action line:

$$\hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\, 10^{-34}\mathcal{U}_S, \quad (92.71)$$

with

$$\begin{aligned} H_5 &= 1.054571817646154469625821829433059476 5\ldots, \\ \eta_{\mathrm{ret}} &= 1.054571816915039061133690584724299733 9\ldots, \\ R_{\mathrm{act}} &:= \frac{H_5}{\eta_{\mathrm{ret}}} = 1.000000000693281763048512336828598947\ldots. \end{aligned} \quad (92.72)$$

The difference $H_5 - \eta_{\mathrm{ret}}$ and the quotient R_{act} are additive and multiplicative readings of the same two sections, not two Planck constants.

Proposition 92.57 (Separation of scale and refinement)**.** *Let $t_0 > 0$ be the declared temporal section and*

$$\omega_0 = \frac{2\pi_{\mathrm{HMT}}}{108 t_0}, \qquad \varepsilon_{\mathrm{clk}} = \hbar_{\mathrm{ret}}\omega_0, \qquad m_{\mathrm{clk}} = \varepsilon_{\mathrm{clk}} c_{\mathrm{int}}^{-2}. \tag{92.73}$$

Then $10^{-34}\mathcal{U}_S$ fixes the absolute section of the mass line, while $R_{\mathrm{act}}^{\kappa}$ is a positive relative corrector. The decade -34 is not part of κ and cancels exactly in every mass ratio.

Proof. By (92.71), both sections contain the same factor $10^{-34}\mathcal{U}_S$. The quotient yields $\hbar_{\mathrm{pre}}/\hbar_{\mathrm{ret}} = H_5/\eta_{\mathrm{ret}} = R_{\mathrm{act}}$, so the decimal power cancels. If $m_X = m_{\mathrm{clk}}\chi_X R_{\mathrm{act}}^{\kappa_X}$ and $m_Y = m_{\mathrm{clk}}\chi_Y R_{\mathrm{act}}^{\kappa_Y}$, then

$$\frac{m_Y}{m_X} = \frac{\chi_Y}{\chi_X} R_{\mathrm{act}}^{\kappa_Y - \kappa_X}.$$

Introducing -34 into κ, or multiplying a coordinate already expressed in MeV by 10^{-34} again, would count the same scale twice and violate typing. □

92.24.4 Bidirectional readers and exact census

For $k \in \mathbb{Z}/108\mathbb{Z}$, define the cyclic discrepancy

$$D_\Gamma(k) := \sum_{t=0}^{107} \mathbf{1}_{\{w_{\Gamma,t} \neq w_{\Gamma,t+k}\}}. \tag{92.74}$$

The chronological reader and the dodecaphase reader are

$$K_D(\Gamma) := \left(D_{27} + D_{36}, D_1, \frac{D_4 - D_3}{2} \right), \tag{92.75}$$

$$K_\Omega(\Gamma) := (\Omega_{90|120}(\tau_\Gamma), 54, P_6(\tau_\Gamma)). \tag{92.76}$$

The first reads shifts of w_Γ; the second, autocorrelation of windows of τ_Γ. They share the route, but not the information quotient.

Theorem 92.58 (Certified reader profiles)**.** *On $\mathcal{R}_{324}$, K_D has exactly fourteen profiles, K_Ω nine, and the pair (K_D, K_Ω) forty. There are eighteen routes on which both triples coincide and the only coincident triple is*

$$K_D(\Gamma) = K_\Omega(\Gamma) = (80, 54, 6). \tag{92.77}$$

The marginal multiplicities are those of the tables 92.17 y 92.18.

Table 92.17: Fourteen profiles of K_D over the 324 routes.

A_D	B_D	C_D	multiplicity
0	24	0	18
40	24	2	36
40	54	6	18
40	78	27	36
40	84	30	36
60	78	25	18
60	84	28	18
80	54	6	18
80	54	7	18

A_D	B_D	C_D	multiplicity
80	66	2	18
80	66	3	36
80	66	4	18
100	66	0	18
100	66	2	18
		total	324

Table 92.18: Nine profiles of K_Ω over the 324 routes.

Ω	B_Ω	P_6	multiplicity
80	54	6	198
56	54	5	68
48	54	5	34
36	54	4	8
20	54	4	4
4	54	2	4
40	54	4	4
24	54	4	2
24	54	2	2
		total	324

Proof of theorem 92.58. The finite domain of (92.57) is traversed. For each (c, h_0, v_0), iterating U_t for 108 steps uniquely determines w_Γ and τ_Γ; then (92.74)–(92.76) are evaluated. Grouping identical triples produces the fourteen rows of table 92.17 and the nine of table 92.18; both sums are 324. Grouping ordered pairs gives forty classes. Finally, filtering the rows with equality of the two triples leaves eighteen routes, all in the unique intersection $(80, 54, 6)$. The algorithm receives only cell, orientation and TPK transition; it contains neither a mass column nor a PDG label. □

The scalar evaluations of the readers, where applicable, are

$$\kappa_D(\Gamma) = A_D(\Gamma) - \alpha_{\mathrm{HMT}} B_D(\Gamma) + C_D(\Gamma)\Delta_4, \tag{92.78}$$

$$\kappa_\Omega(\Gamma) = \Omega(\Gamma) - 54\alpha_{\mathrm{HMT}} + P_6(\Gamma)\Delta_4. \tag{92.79}$$

The presence of α_{HMT} here is downstream: the constant has already been published by HMT before the reader acts on a route.

92.24.5 Basal electron and target-free beta closure

Theorem 92.59 (Electronic fibre and governing value)**.** *The unique central fixed point determines*

$$\Gamma_e = (5, 5, ++), \qquad \tau_e = +++--\,-+++--, \qquad (D_1, D_3, D_4, D_{27}, D_{36}) = (54, 56, 68, 48, 32). \tag{92.80}$$

Therefore,

$$K_D(\Gamma_e) = K_\Omega(\Gamma_e) = (80, 54, 6). \tag{92.81}$$

The internal basal selector $(-22, +15)$ *produces*

$$e_0 := \frac{\sqrt{3}}{4}\left[\exp\left(\frac{\varphi_{\mathrm{HMT}}}{\pi^2_{\mathrm{HMT}}}\right) - 22\alpha^3_{\mathrm{HMT}}\right](1 + 15\Delta_4) \\ = 0.51099892248806607250987087719134943763\ldots\ \mathrm{MeV}. \tag{92.82}$$

With

$$\kappa_e := 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4, \tag{92.83}$$

the beta output is

$$\boxed{m_e^{\beta} = e_0 R_{\mathrm{act}}^{\kappa_e} = 0.51099895069000080860825716483792339562463486771 51\ldots\ \mathrm{MeV}.} \tag{92.84}$$

Proof. De (92.80),

$$D_{27} + D_{36} = 48 + 32 = 80, \qquad D_1 = 54, \qquad (D_4 - D_3)/2 = (68 - 56)/2 = 6,$$

which proves the first equality of (92.81); the autocorrelation of the dodecaphase word gives the same triple. The selector integers are constructed without a target mass:

$$22 = 396/18 = 27 - 5 = 24 - 2, \qquad 15 = 270/18 = \binom{6}{2} = \binom{6}{4}, \tag{92.85}$$

and the local vacancy matrix $V_e = \left(\begin{smallmatrix} 21 & 22 \\ 6 & 7 \end{smallmatrix}\right)$ has determinant 15. Substituting the internal publications into (92.82) gives the indicated basal value. The common triple $(80, 54, 6)$ determines (92.83); finally, (92.72) gives (92.84). The external value is opened only after this evaluation, so it does not participate in the choice of $(-22, +15)$, the route or the power. □

The basal value e_0 is a stage and does not replace m_e^{β}. Nor are the raw signature $(24, 0, 12, 0, -12)$, the even CT–108 signature $(-12, -4, 12, -6, 12)$, the affine origin $v_e = 0$, the route Γ_e and the word used by the Euler reader confused: they belong to different types.

92.24.6 Global semigroup and convergence of the towers

Theorem 92.60 (Semigroup of heights)**.** *The tower heights form*

$$\mathfrak{S} = \langle 90, 120\rangle = \{90, 120\} \cup \{30r : r \geq 6\}. \tag{92.86}$$

Its semigroup algebra admits the presentation

$$k[\mathfrak{S} \cup \{0\}] \simeq k[X_{90}, Y_{120}]/(X_{90}^4 - Y_{120}^3). \tag{92.87}$$

The height 360 *preserves two genealogies before the quotient: four steps of* 90 *and three of* 120*.*

Proof. Dividing by 30 reduces the problem to the semigroup $\langle 3, 4\rangle$. Its positive elements are $3, 4$ and all integers $r \geq 6$: according to the residue of r modulo three, one writes $r = 3a + 4b$ with $b \in \{0, 1, 2\}$ and $a \geq 0$. The only primitive relation between the two generators is $4 \cdot 3 = 3 \cdot 4$, which, upon restoring the heights, produces $X_{90}^4 = Y_{120}^3$. Before imposing this relation, the words X_{90}^4 and Y_{120}^3 retain different histories. □

For $h = 30r$, we define the relative tower coefficients

$$\eta^{D}_{30r}(\Gamma) := D_{\Gamma}(9r) - D_{\Gamma_e}(9r), \tag{92.88}$$

$$\eta^{\Omega}_{30r}(\Gamma) := A_{\Gamma}(r) - A_{\Gamma_e}(r), \qquad A_{\Gamma}(r) := \sum_{j=0}^{11}\left(\sum_{u=0}^{r-1}\tau_{\Gamma,j+u}\right)^2, \tag{92.89}$$

where the indices of τ are taken modulo twelve. The refined character is

$$\chi^{r}_{\mathrm{full}}(\sigma,\eta) := \chi_0(\sigma)\exp\!\left(\sum_{h\in\mathfrak{S}} \eta_h^{r} s_h\right), \qquad r \in \{D,\Omega\}. \tag{92.90}$$

Proposition 92.61 (Absolute convergence)**.** *The two series in* (92.90) *converge absolutely for every route. Consequently, the hierarchy need not be truncated to the eight heights usually displayed.*

Proof. By definition, $0 \le D_\Gamma(k) \le 108$, hence $|\eta_h^D| \le 108$. Since each partial sum of a signed word has modulus at most r, one obtains $|\eta^{\Omega}_{30r}| \le 24r^2$. Moreover $0 < q_+ < q_- < 1$ and $|s_{30r}| \le 2q_-^{30r}$. By comparison with a geometric series and with $\sum r^2 q_-^{30r}$, both series converge absolutely. □

The tower is complete as an expansion codomain: it is not a physical species selector. An algorithm that receives a ratio to expand does not prove which route produced that ratio.

92.24.7 The thirteen pairs of fibre operators

For each multisection M, let $\mathcal{H}_M = \bigoplus_{\Gamma\in M} \mathbb{C}e_\Gamma$. We define

$$B^D_M e_\Gamma := \kappa_D(\Gamma) e_\Gamma, \qquad B^{\Omega}_M e_\Gamma := \kappa_{\Omega}(\Gamma) e_\Gamma. \tag{92.91}$$

The thirteen canonical fibres and their ranks are

M	ℓ_0	ℓ_2	ℓ_4	ν_1	ν_2	ν_3	ν_4	d_1	d_2	d_3	d_4	u_1	u_3
$\dim M$	1	8	16	2	4	6	8	2	4	6	8	4	12.

(92.92)

Theorem 92.62 (Operator law on the multisections)**.** *The twenty-six operators in* (92.91) *are self-adjoint. For a positive basal operator $\widehat{M}^{(0)}_M$,*

$$\widehat{M}^{\beta,r}_M = R_{\mathrm{act}}^{B^r_M/2}\, \widehat{M}^{(0)}_M\, R_{\mathrm{act}}^{B^r_M/2}, \qquad r \in \{D,\Omega\}, \tag{92.93}$$

is positive and covariant under a unitary change of basis. On the rank-one fibre ℓ_0 it reduces to the electronic scalar (92.84)*. On a fibre of higher rank, the spectrum of* (92.93) *is the complete output until a prospective physical push-forward and a scalarization have been fixed.*

Proof. In the route basis, B^D_M and B^{Ω}_M are diagonal with real entries, so they are self-adjoint. Since $R_{\mathrm{act}} > 0$, functional calculus defines $R_{\mathrm{act}}^{B^r_M/2}$ as a positive invertible operator. For every v,

$$\langle v, \widehat{M}^{\beta,r}_M v\rangle = \left\langle R_{\mathrm{act}}^{B^r_M/2} v,\, \widehat{M}^{(0)}_M R_{\mathrm{act}}^{B^r_M/2} v\right\rangle \ge 0.$$

If U is unitary, the change $B \mapsto UBU^*$, $\widehat{M}^{(0)} \mapsto U\widehat{M}^{(0)}U^*$ takes the output to $U\widehat{M}^{\beta,r}U^*$ and preserves spectrum, trace and determinant. When $\dim \ell_0 = 1$, both readers equal $(80, 54, 6)$ and the formula is scalar. For $\dim M > 1$, choosing an eigenvalue without an additional physical morphism is not a natural operation on the fibre. □

The following table displays the normalized vector traces of the triples before applying α_{HMT} and Δ_4; their diversity proves that there is no single angular pair that replaces the route readers.

Table 92.19: Normalized traces of K_D and K_Ω on the thirteen fibres.

M	$\dim M$	$\overline{\mathbf{K}}^D_M$	$\overline{\mathbf{K}}^{\Omega}_M$
ℓ_0	1	$(80, 54, 6)$	$(80, 54, 6)$

M	$\dim M$	$\overline{\mathbf{K}}_M^D$	$\overline{\mathbf{K}}_M^\Omega$
ℓ_2	8	$(65, 249/4, 17/2)$	$(80, 54, 6)$
ℓ_4	16	$(225/4, 489/8, 21/2)$	$(141/2, 54, 11/2)$
ν_1	2	$(40, 51, 29/2)$	$(60, 54, 5)$
ν_2	4	$(45, 63, 29/2)$	$(61, 54, 5)$
ν_3	6	$(170/3, 55, 34/3)$	$(206/3, 54, 11/2)$
ν_4	8	$(105/2, 237/4, 95/8)$	$(143/2, 54, 45/8)$
d_1	2	$(40, 51, 29/2)$	$(60, 54, 5)$
d_2	4	$(45, 63, 57/4)$	$(61, 54, 5)$
d_3	6	$(170/3, 55, 23/2)$	$(218/3, 54, 17/3)$
d_4	8	$(105/2, 237/4, 49/4)$	$(149/2, 54, 23/4)$
u_1	4	$(65, 66, 9)$	$(80, 54, 6)$
u_3	12	$(205/3, 119/2, 113/12)$	$(74, 54, 23/4)$

92.24.8 Null sector and status of the physical realizations

Theorem 92.63 (Null bifurcation)**.** *The photonic chart $\mathcal{N}_\gamma$ and the gluonic chart $\mathcal{N}_g$ branch off before the positive character. In them*

$$\kappa_{\text{null}} = \text{N/A}, \qquad m_\gamma = 0, \qquad m_g = 0, \tag{92.94}$$

without the extension, energy or gauge representation being trivial. Nullity is not obtained by imposing $\kappa = 0$ in R^κ_{act}.

Proof. The positive character takes values in $\mathbb{R}_{>0}$, so $R^0_{\text{act}} = 1$, not zero. Thus no power within that chart can produce the null section. The photonic realization resides on the electromagnetic null boundary and leaves two transverse polarizations after gauge reduction; the gluonic one resides in the eight-dimensional adjoint colour representation. The inclusion of both charts in the zero section of $\mathcal{L}_M$ is performed before χ_0, which proves (92.94) without erasing their representations. □

The subsequent physical realizations are ordered by their domain, representation and mode of publication; their documentary number is not the cardinality of the atlas and does not define the General Mass Law. Their exact status is:

Table 92.20: Evidentiary status of the subsequent physical realizations.

class(es)	salida interna	status of the mass coordinate
electron	fibre ℓ_0, two coincident readers	exact, prospective, target-free beta scalar, (92.84)
muon, tau	operators of the leptonic fibres and preserved signature evaluations	exact operator; the inherited scalar evaluations are conditioned on the signature, not prospective selectors
tres neutrinos	four neutrino fibres, bounds and mixing operator	no flavour masses are invented; the primary publication is operatorial
seis quarks	fibres u_j, d_j, generation and ternary colour orbit	exact operator; any comparison scalar must declare renormalization scheme and scale
photon	electromagnetic null section with two transverse modes	exact nullity $m_\gamma = 0$ in its chart
gluons	null section in the colour adjoint, with eight generators	exact nullity $m_g = 0$ in its chart; the collective gap is not a gluon pole mass

class(es)	salida interna	status of the mass coordinate
W, Z	massive gauge operators and conditioned signatures	the conditioned evaluation is preserved; the route–physical spectrum join is not replaced by the external value
Higgs	scalar-sector operator	physical scalar conditioned on the representation push-forward
meson, baryon	register composition grammar and binding operator	they are not bare sums of constituent masses
Fibonacci	fusion ring and braiding representation	topological codomain; there is no universal mass scalar
Ising/Majorana	fusion ring, self-conjugacy and compatible term	neutrality alone implies neither Majorana nor mass
$\mathbb{Z}_6$	topological sector of order six	operator in its codomain; it is not identified with a tower height
virtuals	sections of finite persistence	classification by terminal horizon, not on-shell mass

92.24.9 Completeness of the internal spectrum and the comparative inventory

The word *complete* applies here to two different objects and both are expressly declared. The ambient space of the two marginal spectra is

$$\mathscr{A}_M := \coprod_{M \in \mathcal{M}_{13}} \left(\operatorname{Spec} \widehat{M}_M^{\beta,D} \times \operatorname{Spec} \widehat{M}_M^{\beta,\Omega} \right) \coprod \{0_\gamma, 0_g\}, \tag{92.95}$$

with multiplicities, projectors, routes, scheme and scale preserved. The intrinsic output, however, is not this entire product: it is the support of the joint spectral measure, defined in (92.108), which preserves which two values come from the same route. The complete external inventory used for subsequent recognition is

$$\mathscr{I}_{\mathrm{ext}} = \mathscr{I}_m^{471} \coprod \mathscr{I}_\Gamma^{384}, \qquad |\mathscr{I}_{\mathrm{ext}}| = 471 + 384 = 855. \tag{92.96}$$

Each record of $\mathscr{I}_m^{471}$ distinguishes stable mass, pole mass or resonance estimator; each record of $\mathscr{I}_\Gamma^{384}$ distinguishes the width extracted from a pole or open mode. The metrological correspondence is the typed relation

$$C_{\mathrm{met}} \subset \mathscr{A}_M \times \mathscr{I}_{\mathrm{ext}}, \tag{92.97}$$

which opens only after descriptor, fibre, operator, projector, representation, scheme and scale have been frozen.

Theorem 92.64 (Double completeness without retrospective selection)**.** *The General Mass Law publishes an operatorial output on each of the thirteen multisections and on the (324) routes, in addition to the two null sections. The inventory of* (92.96) *individualizes, without documentary omission, the (855) external observables and preserves for each one its destination HMT sector and the required structural realization. No quantity in that inventory selects a cell, route, eigenvalue, tower or coefficient of the law.*

Proof. The theorem 92.52 exhausts the atlas and the theorem 92.62 constructs the two positive operators on each fibre; the theorem 92.63 adds the two zero-mass charts. Thus the joint support of (92.108) covers the entire generating domain and is included in the ambient space of (92.95) without being identified with it. The material inventory enumerates exactly (471) mass records and (384) width records and assigns each row identity, unit, uncertainty or limit, scheme, scale, sector and realization. Permuting or replacing

the external quantities acts only on the second factor of (92.97); it leaves C_{81}, $\mathcal{R}_{324}$, the readers, towers and operators invariant. This proves exhaustiveness and causal precedence simultaneously. □

92.24.10 CKM and PMNS as distinct transports

In the quark sector, the N42 incidence produces

$$M_{\mathrm{CKM}} = \begin{pmatrix} 2 & -5 & 28 \\ 0 & 7 & -25/2 \\ 1 & -20 & -2/3 \end{pmatrix}, \qquad \det M_{\mathrm{CKM}} = -\frac{3857}{6}, \tag{92.98}$$

and transports the down operator to the up basis by

$$B_d^{(u)} = V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^*, \qquad R_{\mathrm{act}}^{B_d^{(u)}} = V_{\mathrm{CKM}} R_{\mathrm{act}}^{B_d} V_{\mathrm{CKM}}^*. \tag{92.99}$$

The transport changes the basis, not the eigenvalues, and acts after the readers have constructed B_u, B_d. The same incidence matrix acts on the already constructed angular coordinates by

$$\begin{pmatrix} \theta_{12} \\ \theta_{23} \\ \theta_{13} \end{pmatrix} = M_{\mathrm{CKM}} \begin{pmatrix} A \\ C^*/6 \\ (180/\pi_{\mathrm{HMT}})\Delta_4 \end{pmatrix}, \qquad \delta_{\mathrm{CP}} = 9A, \tag{92.100}$$

from which

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta_{\mathrm{CP}}) = (13.003313589509,\ 2.398056157332,\ 0.213977382244,\ 65.676173123554)^\circ \tag{92.101}$$

and

$$J_{\mathrm{HMT}} = 3.1189723326632974 \times 10^{-5}, \qquad 3857\delta_{\mathrm{CP}} = 9(1528\theta_{12} + 3380\theta_{23} + 801\theta_{13}). \tag{92.102}$$

The last identity is an exact linear falsifier of the transport, not a fit of its angles.

For the leptonic sector, let $F_\ell, F_\nu : \mathbb{C}^3 \to \mathcal{H}$ be isometries with the same image $\operatorname{im} P_3$, let $b_K \in \operatorname{im} P_3$ be unitary and let $Q_K = |b_K\rangle\langle b_K| \le P_3$ be its rank-one projector. Define

$$\Phi_L = I + (e^{i\pi/6} - 1)Q_K, \qquad U_{\mathrm{HMT}} := F_\ell^* \Phi_L F_\nu. \tag{92.103}$$

Proposition 92.65 (Distinction and completeness of CKM–PMNS transports)**.** *The matrix U_{HMT} of (92.103) is unitary, but the rank-one phase does not constitute the full-rank PMNS physical realization. The exact PMNS transport is defined by the complete spectral frames, $U_{\mathrm{PMNS}} = F_\ell^* F_\nu$. The register kernel G_0 provides an additional finite certificate if and only if its entries are displayed and* $\det G_0 \neq 0$ *is verified; the condition is not promoted here to a fact without that calculation. CKM and PMNS have distinct domains, bases and falsifiers and cannot be identified.*

Proof. Since $Q_K \le P_3$ is an orthogonal projector, $[\Phi_L, P_3] = 0$ and Φ_L is unitary. Moreover $F_\ell F_\ell^* = F_\nu F_\nu^* = P_3$; therefore

$$U_{\mathrm{HMT}}^* U_{\mathrm{HMT}} = F_\nu^* \Phi_L^* P_3 \Phi_L F_\nu = I_3.$$

However, a rank-one phase preserves a stabilizer $U(2)$. Testing its thirty-six permutations does not reproduce the small physical mixing angle. The failure of the minimal phase eliminates a degenerate substitute, but neither opens nor downgrades the PMNS realization, because the complete frames directly produce $F_\ell^* F_\nu$. The polar factor of a register kernel is an alternative certificate only when it passes its

rank criterion. CKM, by contrast, transports between the up/down bases of already constructed quark operators by (92.99). This difference in domains prevents merging the two constructions. □

92.24.11 Joint spectral closure and exact status of CKM–PMNS transports

Joint spectrum with route correlation. For each multisection M, the canonical stage preceding any non-diagonal coupling publishes the pair

$$\widehat{M}^{\beta,r}_{M,\mathrm{rut}} := \sum_{\Gamma\in M} m_r(\Gamma)|e_\Gamma\rangle\langle e_\Gamma|, \quad m_r(\Gamma) := m_{\mathrm{clk}}\,\chi^{r}_{\mathrm{full}}\big(\sigma(\Gamma),\eta^r(\Gamma)\big)R^{\kappa_r(\Gamma)}_{\mathrm{act}}, \quad r\in\{D,\Omega\}. \tag{92.104}$$

The two operators are positive and commute. There is therefore a unique joint PVM $E^{D,\Omega}_M$ on $\mathbb{R}^2_+$ such that

$$\widehat{M}^{\beta,D}_{M,\mathrm{rut}} = \int x\, dE^{D,\Omega}_M(x,y), \qquad \widehat{M}^{\beta,\Omega}_{M,\mathrm{rut}} = \int y\, dE^{D,\Omega}_M(x,y), \tag{92.105}$$

$$E^{D,\Omega}_M(A)e_\Gamma = \mathbf{1}_A\big(m_D(\Gamma),m_\Omega(\Gamma)\big)e_\Gamma. \tag{92.106}$$

The push-forward measure

$$\nu_M := \frac{1}{|M|}\sum_{\Gamma\in M}\delta_{(m_D(\Gamma),m_\Omega(\Gamma))} \tag{92.107}$$

preserves the correlation between the two readers and all their multiplicities. Consequently, the Cartesian product written in (92.95) is the ambient space; the intrinsic spectral object is

$$\begin{aligned}\mathscr{A}^{\mathrm{joint}}_M &:= \bigsqcup_{M\in\mathcal{M}_{13}} \operatorname{supp} E^{D,\Omega}_M \bigsqcup \{0_\gamma,0_g\}\\ &\hookrightarrow \bigsqcup_{M\in\mathcal{M}_{13}} \big(\operatorname{Spec}\widehat{M}^{\beta,D}_M \times \operatorname{Spec}\widehat{M}^{\beta,\Omega}_M\big) \bigsqcup \{0_\gamma,0_g\}.\end{aligned} \tag{92.108}$$

The inclusion is not promoted to equality without proving that all marginal combinations belong to the same route. For a prepared channel $P_{X,a}$, the compression $P_{X,a}E^{D,\Omega}_M(\cdot)P_{X,a}$ is the induced POVM. There is a mass line m_X only if

$$E_{\mathcal{M}_X}(\{m_X\})P_{X,a} = P_{X,a}, \qquad (\mathcal{M}_X - m_X I)P_{X,a} = 0. \tag{92.109}$$

A compression with several support points publishes a multiplet or a distribution, not a mean. If a subsequent coupling ceases to be diagonal, the ordered pair is preserved and a joint PVM is asserted only after verifying the vanishing commutator.

Transporte CKM. With $s_{ij} = \sin\theta_{ij}$, $c_{ij} = \cos\theta_{ij}$ and $\delta = \delta_{\mathrm{CP}}$, the output of (92.100) determines

$$V_{\mathrm{CKM}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13}\end{pmatrix}. \tag{92.110}$$

It is unitary, implements the frame change of (92.99) and does not create the eigenvalues. If κ^u_i and κ^d_a are the spectra of the up-type and down-type Hermitian operators, then

$$\big|\det[B_u, V_{\mathrm{CKM}}B_dV^*_{\mathrm{CKM}}]\big| = 2|J_{\mathrm{HMT}}|\prod_{i<j}|\kappa^u_i-\kappa^u_j|\prod_{a<b}|\kappa^d_a-\kappa^d_b|. \tag{92.111}$$

This is the internal spectrum–mixing falsifier. Statistical inference against external data remains in the recognition layer and requires its covariance.

PMNS transport and full-rank criterion. The minimal phase of (92.103) satisfies $Q_K \le P_3$, hence $[\Phi_L, P_3] = 0$; its unitarity is exact, although it remains a rank-one negative control. Let M_ℓ and M_ν be the already constructed charged and neutral operators, and let F_ℓ, F_ν be complete orthonormal spectral frames, with $F_\ell F_\ell^* = F_\nu F_\nu^* = P_3$. The spectral transport is

$$U_{\mathrm{PMNS}} := F_\ell^* F_\nu, \qquad U_{\mathrm{PMNS}}^* U_{\mathrm{PMNS}} = I_3. \tag{92.112}$$

If both spectra are simple, it is determined up to diagonal phase readjustments and the declared spectral ordering. If there are degeneracies, the exact output is the double coset

$$\prod_\lambda U(m_{\ell,\lambda}) \backslash U(3) / \prod_\mu U(m_{\nu,\mu}), \tag{92.113}$$

not a numerical matrix chosen within the degenerate subspaces.

The register kernel preserves carry, sheet, orientation, vacancy and holonomy. Omitting zero-norm summands, it can be written

$$\begin{aligned} K_{\mathrm{reg}}(c,d) &:= \sum_{r \in \{\mathrm{car,hoj,ori,vac,hol}\}} \frac{\langle \Xi_r(c), \Xi_r(d) \rangle}{\sum_x \|\Xi_r(x)\|^2}, \\ (G_0)_{ab} &:= \frac{1}{\sqrt{|L_a||N_b|}} \sum_{c \in L_a} \sum_{d \in N_b} K_{\mathrm{reg}}(c,d). \end{aligned} \tag{92.114}$$

The condition

$$\det G_0 \neq 0 \quad \Longleftrightarrow \quad \text{the declared record separates three directions,} \qquad U_{G_0} := G_0 (G_0^* G_0)^{-1/2}, \tag{92.115}$$

is an exact finite criterion. It is not asserted as a certified fact until the entries of G_0 and the calculation of its determinant are displayed. If $\det G_0 = 0$, an independent internal observable must be added and the calculation repeated; forcing invertibility with external PMNS angles is prohibited. The joint PVM, its push-forward measure, the CKM transport and the PMNS frame theorem are exact internal results with the declared starting structure; the criterion of G_0 remains separate from those results.

92.24.12 Non-governance of the historical tables

Proposition 92.66 (Prospective independence)**.** *The old windows of ten signatures and twelve masses are historical comparison witnesses. They do not define $\mathcal{C}_{81}$, $\mathcal{F}_{56}$, $\mathcal{M}_{13}$, $\mathcal{R}_{324}$, the readers, character or towers. Permuting, deleting or replacing their metrological columns leaves all constructions of* (92.56) *prior to comparison unchanged.*

Proof. The construction inputs are the atlas, the four orientations, the transition U_t, the ledger 81×108, the signature and the readers. None of the definitions (92.57), (92.64), (92.75), (92.76) or (92.86) contains an external mass. Consequently, a permutation of the subsequent catalogue commutes with the projection that forgets the comparison and acts as the identity on the internal constructor. An abbreviated table therefore cannot be either the domain or the generator of the law. □

92.24.13 Typed closure theorem and falsifiers

Theorem 92.67 (HMT–MD General Mass Law)**.** *The chain* (92.56) *defines on the domain* 81/56/13/324*, with a separate null sector, a mass law that satisfies:*

1. *constructs route, ledger, signature, character and tower without external quantities;*
2. *fixes the absolute scale through the cokernel of order sixteen and the decade* -34*, separate from every relative corrector;*

3. *publishes two inequivalent readers with censuses* $14/9/40/18$ *and unique coincidence* $(80, 54, 6)$*;*
4. *closes the electron through the rank-one reduction* (92.84)*;*
5. *extends the law exactly as the operator pairs* (B_M^D, B_M^Ω) *on the thirteen multisections;*
6. *branches photon and gluons towards null charts before the character;*
7. *publishes on each non-central fibre the positive operator, its complete spectral measure and its multiplicities; every physical scalar coordinate is the image of the push-forward and scalarization declared for that representation;*
8. *incorporates, as a subsequent comparative layer, the exhaustive inventory of (471) masses and (384) widths without allowing their values to act back on the atlas;*
9. *keeps CKM and PMNS distinct, constructs the CKM angular output with its linear falsifier and the full-rank PMNS transport, and opens PDG/CODATA only after freezing the internal output.*

It is not a law of a single angular pair: A, C^* *parameterize the global channels, while the separation of species resides in the route, ledger, two readers, towers, representation and coupling.*

Proof. Points 1 and 2 are the lemma 92.55, theorem 92.56 y proposition 92.57; point 3 is the theorem 92.58; point 4 is the theorem 92.59; points 5 and 6 are the theorems 92.62 y 92.63; the status of point 7 is typed in table 92.20; point 8 is the theorem 92.64; and point 9 is the proposition 92.65. The proposition 92.66 proves that external comparison cannot act back on that chain. Together, the constructions compose the asserted law without identifying scale, character, reader, operator and scalarization. □

Falsification criterion 92.68 (Negative controls of the law)**.** The construction is refuted or ill-typed if any of the following occurs:

1. a mass, CODATA, PDG or a 10/12 table selects a cell, route, signature, coefficient or eigenvalue;
2. the APP jump 4 is identified with the defect, unfolding or return 54;
3. -34 is inserted into κ or a MeV coordinate is multiplied again by 10^{-34};
4. the reader censuses cease to be $14, 9, 40, 18$, or a coincidence other than $(80, 54, 6)$ appears;
5. the basal value e_0 is presented as the final electronic output;
6. the semigroup $\langle 90, 120\rangle$ is reduced to a finite list or used to select a species retrospectively;
7. a fibre of rank greater than one receives a scalar without a prospective physical push-forward and scalarization;
8. the nullity of the photon or gluon is deduced from $R^0_{\mathrm{act}} = 1$, or the collective gap is confused with a gluonic pole mass;
9. CKM and PMNS are identified or the rank-one leptonic phase is promoted to complete physical mixing;
10. a row of the inventory (471+384) is missing, a width is confused with a mass or an external value modifies the internal spectrum;
11. the genealogical chain is replaced by a single angular pair.

The architecture of the law is PRE-EXISTING AUTHORIAL ARCHITECTURE; the joint recovery of the atlas, readers and electron is a RECOVERED RESULT; the operatorial presentation on the thirteen fibres is a collected formalization; and the executable censuses are certificates. None of these categories alters the causal precedence of APP–TRIT–TPK.

Chapter 93

Equivariant realization of species and mass operators: genealogical fibers, spectral measures, complex poles, and material compositions

93.1 Internal domain of the material realization and classification of its codomains

93.1.1 Separation among spectral line, multiplet, distribution, complex pole, categorical object, and criterion of impossibility

Let $\mathcal{P}$ be the set of individualized physical states. An elementary state is represented by the descriptor

$$X = (\mathfrak{c}, \rho, g, f, q, \epsilon, \varsigma, \mathfrak{a}),$$

which preserves representation class $\mathfrak{c}$, internal representation ρ, generation g, flavor f, charge q, sheet orientation ϵ, statistical sector ς, and coupling data $\mathfrak{a}$. A composite state adds ordered constituents, composition tree, common boundary, and collective quantum numbers. A resonance adds the open channels and the causal prescription for the resolvent.

The physical realization must address three distinct questions:

1. which fibre, orbit, or multisection of the atlas corresponds to the descriptor;
2. which positive operator publishes the mass on that fiber;
3. which coupling produces a scalar, a multiplet, a distribution, or a complex pole.

Confusing these questions introduces a retrospective selector. The correct answer is obtained through a natural relation, a fiber operator and a spectral compression, in that order.

93.1.2 Precedence of the genealogical atlas with respect to any metrological identification

The General Mass Law does not start from a table of names or a list of experimental values. Its internal domain is the finite chain

$$\mathcal{R}_{324} \longrightarrow \mathcal{C}_{81} \longrightarrow \mathcal{F}_{56} \longrightarrow \mathcal{M}_{13},$$

where $\mathcal{R}_{324}$ contains the 4 orientations of each of the 81 cells, $\mathcal{F}_{56}$ groups equivalent route histories, and $\mathcal{M}_{13}$ preserves the operator multisections. The null sector bifurcates before the positive character is applied. Therefore, neither an abbreviated historical table nor a taxonomy of physical names can replace this domain.

93.2 Fiber product between species descriptors and routes of the 81/56/13/324 atlas

93.2.1 Exact non-voidness condition and prospective selection without illicit point selector

Let $\mathcal{D}$ be the finite space of internal data already constructed before the contrast: representation, orientation, memory, boundary, charge, flavor, color, generation, and statistical sector. Denote by

$$d_{\mathcal{P}} : \mathcal{P} \longrightarrow \mathcal{D}, \qquad d_{\mathcal{R}} : \mathcal{R}_{324} \longrightarrow \mathcal{D}$$

the physical descriptor maps and the genealogical descriptor. The object of realization is defined by the fiber product

$$\mathcal{J} = \mathcal{P} \times_{\mathcal{D}} \mathcal{R}_{324} = \{(X, \gamma) : d_{\mathcal{P}}(X) = d_{\mathcal{R}}(\gamma)\}.$$

The fiber of a species is

$$\mathcal{S}(X) = \{\gamma \in \mathcal{R}_{324} : (X, \gamma) \in \mathcal{J}\}.$$

A descriptor (X) is called realizable by the atlas when

$$d_{\mathcal{P}}(X) \in \operatorname{im} d_{\mathcal{R}}.$$

This condition is equivalent to $\mathcal{S}(X) \neq \varnothing$. It does not follow from merely writing the fiber product: it constitutes the exact existence condition that must be verified for each physical descriptor. The value of $\mathcal{S}$ may be a route, an orbit, or a multisection. A point section is not forced when the internal symmetry requires preservation of a multiplicity.

93.2.2 Equivariance of fibers, orbits, and multisections under refinement

Definition 93.1 (Admissible realization relation)**.** The relation $\mathcal{S}$ is admissible when it simultaneously satisfies:

1. depends solely on data constructed before the external table is opened;
2. commutes with particle–antiparticle conjugation;
3. is equivariant under cellular inversion and the internal gauge actions;
4. it is compatible with refinement, nonadic return and dodecaphasic reading;
5. preserves every multiplicity not eliminated by a physical coupling;
6. remains invariant under permuting or replacing the external values of mass and width.

Theorem 93.2 (Prospective realization by fibered product)**.** *If $d_{\mathcal{P}}$ and $d_{\mathcal{R}}$ are equivariant with respect to a symmetry G and do not depend on external magnitudes, then $\mathcal{J}$ is a G-subobject of $\mathcal{P} \times \mathcal{R}_{324}$. For every realizable descriptor X, the nonempty fiber $\mathcal{S}(X)$ is G_X-stable, and the assignment $X \mapsto \mathcal{S}(X)$ is independent of the result subsequently compared.*

Proof. For $(X, \gamma) \in \mathcal{J}$ one has $d_{\mathcal{P}}(X) = d_{\mathcal{R}}(\gamma)$. Equivariance gives

$$d_{\mathcal{P}}(gX) = g\, d_{\mathcal{P}}(X) = g\, d_{\mathcal{R}}(\gamma) = d_{\mathcal{R}}(g\gamma),$$

and, consequently, $(gX, g\gamma) \in \mathcal{J}$. If $g \in G_X$, the second component remains in $\mathcal{S}(X)$. Independence of the external values is immediate because none of them belongs to the domain of the two maps of the fiber product. □

93.3 Intrinsic spectral classification of material species

93.3.1 Pointwise support decomposition, irreducible block and fiber image measure

Let $\mathcal{A}_{\mathrm{rut}}$ be the finite commutative algebra generated by the internal route observables: signature $\mathbb{Z}^5$, character, chronology, K_D, K_Ω, orientation, memory, and compatible readers. In a fiber F, one defines

$$\gamma \sim_{\mathrm{sp}} \eta \quad \Longleftrightarrow \quad a(\gamma) = a(\eta) \quad \text{for every } a \in \mathcal{A}_{\mathrm{rut}}.$$

For each class $\lambda \in F/\!\sim_{\mathrm{sp}}$,

$$P_\lambda = \sum_{\gamma \in \lambda} |\gamma\rangle\langle\gamma|$$

is a spectral projector; the P_λ are orthogonal and $\sum_\lambda P_\lambda = I_F$.

Theorem 93.3 (Maximum internal individualization)**.** *Every classifier constructed exclusively from observables of $\mathcal{A}_{\mathrm{rut}}$ factors uniquely through the quotient $F/\!\sim_{\mathrm{sp}}$. Consequently, two descriptors with distinct complete signatures belong to distinguishable sectors; two descriptors that agree for all generators of the algebra cannot be separated by a rule constructed from those same observables.*

Proof. The characters of the finite commutative algebra separate exactly its joint spectrum. A functional classifier of its generators is constant on each fiber of the joint spectral map and therefore factors through the quotient. If two routes have the same character on the entire algebra, no element of it can separate them. □

93.3.2 Distinction between scalar mass, multiplet, and spectral distribution

This theorem closes the logical limit of the classification. When a physical representation requires two states belonging to the same spectral class to be distinguished, it must exhibit an additional observable or a coupling that enlarges $\mathcal{A}_{\mathrm{rut}}$. The absence of such an observable does not authorize the external mass to be used as a selecting label.

93.4 Natural refinement of the enriched record in the positive towers 90/120

93.4.1 Positive enumeration of the semigroup $\langle 90, 120\rangle$ and exclusion of mode 0

The existence of the semigroup $\langle 90, 120\rangle$ is not sufficient by itself: the map assigning coefficients to an enriched route must be constructed. Let

$$\mathcal{H}_{90,120} = \{90a+120b : a, b \in \mathbb{N}_0,\ (a, b) \neq (0, 0)\} = \{90, 120\}\cup\{30r : r \geq 6\} = \{h_0 < h_1 < h_2 < \cdots\}$$

its strictly increasing enumeration, where $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$, and let $s_{h_n} > 0$ be the declared normalization of the corresponding mode. The compatible record of a route Γ provides a sequence of finite integer data

$$\mathcal{L}(\Gamma) = (\ell_0(\Gamma), \ell_1(\Gamma), \ldots),$$

in which each ℓ_n brings together return, carry, winding, memory and boundary at refinement n. To fix the encoding unambiguously, let

$$\nu(z) = \begin{cases} 2z, & z \geq 0, \\ -2z - 1, & z < 0, \end{cases} \qquad \pi_{\mathrm{C}}(u, v) = \frac{(u + v)(u + v + 1)}{2} + v,$$

and define $\iota : \mathbb{Z}^d \to \mathbb{N}_0$ by recursive application of π_{C} to $\nu(z_1), \ldots, \nu(z_d)$. Then

$$q_n(\Gamma) = \frac{\iota(\ell_n(\Gamma))}{1 + |\iota(\ell_n(\Gamma))|} \in (-1, 1).$$

The minimal realization of tower is

$$\eta_{h_n}(\Gamma) = \frac{3^{-(n+1)} q_n(\Gamma)}{s_{h_n}}, \qquad \mathfrak{T}(\Gamma) = (\eta_h(\Gamma))_{h\in\mathcal{H}_{90,120}}.$$

93.4.2 Absolute convergence, prefix naturality, and effective tail bound

Theorem 93.4 (Convergence, naturality and metrological blindness)**.** *For every enriched route,*

$$\sum_{h\in\mathcal{H}_{90,120}} |\eta_h(\Gamma) s_h| \leq \sum_{n\geq 0} 3^{-(n+1)} = \frac{1}{2}.$$

If the refinement prolongs the record by prefixes, $\mathfrak{T}$ is natural with respect to the truncations. Moreover, no external mass value intervenes in $\mathfrak{T}$, and the tail after N modes satisfies

$$\sum_{n>N} |\eta_{h_n} s_{h_n}| \leq \frac{3^{-(N+1)}}{2}.$$

Proof. We have $|q_n| < 1$; the two bounds are geometric sums. Prefix compatibility preserves all coefficients already constructed and adds those of the new level, which yields the naturality square with the truncation of $\mathcal{E}_{90,120}$. The formula contains only the record, the explicit coding, and the declared modal normalization. □

The preceding construction fixes an explicit, convergent minimal realization relative to the encoding ι and the declared normalizations s_h. A dynamics that modifies its weights must declare the operator effecting that modification and prove naturality and convergence anew; it cannot reconstruct the weights from an experimental residual.

93.5 Bidirectional operators K_D and K_Ω and joint spectral measure

93.5.1 PVM joint, compression to POVM and scalar distribution induced by a state

For $F_X = \mathcal{S}(X)$, let

$$\mathcal{H}_X = \ell^2(F_X) \otimes V_{\rho_X}$$

the realization space, where V_{ρ_X} carries the corresponding internal representation. Each route $\gamma \in F_X$ has signature

$$\sigma(\gamma) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta) \in \mathbb{Z}^5$$

and refinement coefficients $\eta(\gamma) \in \mathcal{E}_{90,120}$. The basal operator is

$$\widehat{M}_X^{(0)} = \sum_{\gamma \in F_X} m_{\mathrm{clk}}\, \chi_{\mathrm{full}}(\sigma(\gamma), \eta(\gamma)) |\gamma\rangle\langle\gamma| \otimes I_{V_{\rho_X}}.$$

The scale m_{clk} descends from the absolute branch action–clock–mass, including the decade 10^{-34}; it is not extracted from an experimental mass.

The readers K_D and K_Ω induce exact diagonal operators B_X^D and B_X^Ω. The two orientations publish

$$\widehat{M}_X^{\beta,r} = R_{\mathrm{act}}^{B_X^r/2} \widehat{M}_X^{(0)} R_{\mathrm{act}}^{B_X^r/2}, \qquad r \in \{D, \Omega\}.$$

Both are positive and commute in the route basis. By the joint spectral theorem there exists a unique projective spectral measure $E_X^{D,\Omega}$ over $\mathbb{R}_+^2$ such that

$$\widehat{M}_X^{\beta,D} = \int x\, dE_X^{D,\Omega}(x,y), \qquad \widehat{M}_X^{\beta,\Omega} = \int y\, dE_X^{D,\Omega}(x,y).$$

At the base of routes,

$$E_X^{D,\Omega}(A)|\gamma\rangle = \mathbf{1}_A(m_D(\gamma), m_\Omega(\gamma))\, |\gamma\rangle.$$

This is the canonical bidirectional output: it preserves the complete pair and does not presuppose a scalar functional in advance.

For a channel $P_{X,a}$, the compressed spectral measure

$$F_{X,a}^{D,\Omega}(A) = P_{X,a} E_X^{D,\Omega}(A) P_{X,a}\Big|_{P_{X,a}\mathcal{H}_X}$$

is a POVM on the prepared subspace. For a density ρ_X supported on that subspace, the joint scalar distribution is

$$\mu_{X,a}^{D,\Omega}(A) = \mathrm{Tr}\Big(\rho_X F_{X,a}^{D,\Omega}(A)\Big).$$

Every declared functional $f : \mathbb{R}_+^2 \to \mathbb{R}_+$ induces $\mu_{X,a}^f = f_\# \mu_{X,a}^{D,\Omega}$ without modifying the source PVM.

93.5.2 Mixed moments, point spectral support and conservation of multiplicities

Let

$$\beta_X(\gamma) = (m_D(\gamma), m_\Omega(\gamma)).$$

The uniform image measure of the fiber is

$$\nu_X = (\beta_X)_\# \left(\frac{1}{|F_X|} \sum_{\gamma \in F_X} \delta_\gamma \right) = \frac{1}{|F_X|} \sum_{\gamma \in F_X} \delta_{(m_D(\gamma), m_\Omega(\gamma))}.$$

Its transform and its mixed moments are

$$Z_X(s,t) = \int e^{sx+ty}\, d\nu_X(x,y), \qquad \mathfrak{m}_{p,q}(X) = \int x^p y^q\, d\nu_X(x,y).$$

Since ν_X has finite support, Z_X, or equivalently the complete family of mixed moments, determines the joint multiset of reader pairs up to permutation of routes. The pushforward measure is therefore a complete invariant of the operator pair; it is not an average of masses.

If $u : F_X \to F_Y$ is a bijection preserving record, orientation, and both readers, the unitary $U_u|\gamma\rangle = |u\gamma\rangle$ satisfies

$$U_u E_X^{D,\Omega}(A) = E_Y^{D,\Omega}(A) U_u, \qquad \nu_X = \nu_Y.$$

The same equality holds for the physical distributions when density and projector are transported by U_u. This naturality is asserted only for the intertwiners actually constructed; a coincidence of cardinalities does not by itself create an equivalence of fibers.

93.6 Symmetry Scalarization and Isotypic Decomposition of the Fibers

93.6.1 Conditional symmetry expectation and criterion for reduction to a spectral line

Let G_X be the compact group of unbroken symmetry of the fiber and $U_X : G_X \to \mathcal{U}(\mathcal{H}_X)$ its representation. The conditional symmetry expectation is defined by

$$\mathbb{E}_X(T) = \int_{G_X} U_X(g) T U_X(g)^*\, d\mu_X(g),$$

where μ_X is the normalized Haar measure. The physical mass operator before opening channels is defined, when the state type declares a functional $\Phi_X : \mathbb{R}_+^2 \to \mathbb{R}_+$, by

$$\mathcal{M}_X = \mathbb{E}_X\Big(\Phi_X(\widehat{M}_X^{\beta,D}, \widehat{M}_X^{\beta,\Omega}) \Big).$$

The functional calculus is carried out with the joint measure $E_X^{D,\Omega}$. In a representation containing an intertwiner that exchanges readers, one may take $\Phi_X(x,y) = \sqrt{xy}$. If the physical type does not declare Φ_X, the complete output remains the joint PVM; no scalar is fabricated.

Theorem 93.5 (Scalar closure, multiplet, and distribution)**.** *The operator $\mathcal{M}_X$ belongs to the commutant $U_X(G_X)'$. If*

$$\mathcal{H}_X \simeq \bigoplus_\lambda V_\lambda \otimes \mathbb{C}^{m_\lambda},$$

then

$$\mathcal{M}_X = \bigoplus_\lambda I_{V_\lambda} \otimes M_{X,\lambda}.$$

In an irreducible sector of multiplicity 1*, the restriction is a scalar. In a reducible sector, the exact output is the spectrum of the blocks $M_{X,\lambda}$, with their multiplicities. There is no indeterminacy: there is a distinct codomain.*

Proof. Invariance of the Haar measure implies $U_X(h)\mathbb{E}_X(T)U_X(h)^* = \mathbb{E}_X(T)$ for every $h \in G_X$. The decomposition follows from the complete reducibility theorem and Schur's lemma. The irreducible case produces a multiple of the identity; the reducible case preserves exactly the multiplicity blocks. □

Let $P_{X,a}$ be the projector determined by the prepared state and the physical channel or apparatus a. The observable spectral measure is

$$\mu_{X,a}(B) = \langle \Psi_X, P_{X,a} E_{\mathcal{M}_X}(B) P_{X,a} \Psi_X \rangle.$$

There is a scalar line precisely when the compressed spectral support is pointwise:

$$E_{\mathcal{M}_X}(\{m_X\})P_{X,a} = P_{X,a}, \qquad \text{equivalently} \qquad (\mathcal{M}_X - m_X I)\, P_{X,a} = 0.$$

The isolated equality $P_{X,a}\mathcal{M}_X P_{X,a} = m_X P_{X,a}$ only fixes an expectation if the subspace is not reducing and is not enough to prove a spectral line. If the compression preserves several eigenvalues, the output is a multiplet or a distribution. Replacing it with a numerical average would destroy physical and genealogical information.

93.6.2 Geometric mean of the readers only under exchange invariance demonstrated

When the physical representation requires symmetry under the exchange $D \leftrightarrow \Omega$, the minimal operatorial aggregate is the geometric mean

$$\widehat{M}_X^{\mathrm{sym}} = \widehat{M}_X^{\beta,D} \# \widehat{M}_X^{\beta,\Omega}, \qquad A\#B = A^{1/2}\big(A^{-1/2}BA^{-1/2}\big)^{1/2}A^{1/2},$$

on the common positive support. In the diagonal case,

$$\widehat{M}_X^{\mathrm{sym}} = R_{\mathrm{act}}^{(B_X^D + B_X^\Omega)/4}\, \widehat{M}_X^{(0)}\, R_{\mathrm{act}}^{(B_X^D + B_X^\Omega)/4}.$$

This formula is a consequence of the explicit exchange hypothesis; it does not replace the joint measure when that symmetry has not been established.

Theorem 93.6 (Characterization of the symmetric aggregate)**.** *The geometric mean is positive, symmetric, congruence-covariant for every invertible S, and self-dual:*

$$A\#B = B\#A, \qquad S^*(A\#B)S = (S^*AS)\#(S^*BS), \qquad (A\#B)^{-1} = A^{-1}\#B^{-1}.$$

Moreover, $A\#A = A$. Consequently, in sectors where there is a physical intertwiner exchanging the readers, the aggregate introduces no parameter and favors neither orientation.

Proof. The congruence identity is understood for invertible S. The identities follow from the functional calculus of the positive root of $A^{-1/2}BA^{-1/2}$. In the commutative case, they reduce to $A\#B = (AB)^{1/2}$. □

There is also a canonical scalar aggregate under distinct axioms. If s_n is continuous, permutation-invariant, coordinatewise multiplicative, and normalized by $s_n(cI) = c$, then

$$s_n(T) = \det(T)^{1/n}.$$

Indeed, upon passing to logarithms, continuity and additivity make the functional linear in the logarithms of the eigenvalues; symmetry makes all coefficients equal, and normalization fixes their value at $1/n$. For the pair of readers, exchange symmetry gives the invariant

$$S_{D,\Omega} = \sqrt{\mathrm{ndet}(T_D)\,\mathrm{ndet}(T_\Omega)}, \qquad \mathrm{ndet}(T) = \det(T)^{1/n}.$$

This result classifies the complete fiber under the stated axioms. It is not automatically identified with a mass line: that identification requires the physical coupling to adopt precisely that functional.

93.7 Null sector and precedence of the zero-mass chart with respect to the positive character

93.7.1 Separation of the photonic $U(1)$ and adjoint gluonic $SU(3)$ representations

The photon belongs to the electromagnetic null chart and the gluons to the adjoint null chart of color. Their different representations, polarizations, and gauge restrictions remain visible even though both masses are null.

93.7.2 Unreachability of the null sector as a limit of the positive mass branch

Nullity is not obtained by taking a limit of the positive character. The domain is first separated into

$$\mathcal{H}_{\mathrm{phys}} = \mathcal{H}_0 \oplus \mathcal{H}_+,$$

with

$$\mathcal{M}|_{\mathcal{H}_0} = 0, \qquad \mathcal{M}|_{\mathcal{H}_+} > 0.$$

93.8 Beta closure of the electron and charged lepton sectors

93.8.1 Uniqueness of the electron in cell $(5,5)$ and preservation of the governing beta mass

The electron constitutes the central exception. Cell inversion has a unique fixed point, the cell $(5,5)$, and its physical fiber therefore has rank one. The remaining stable multisections do not, in general, admit an equivariant point selector. This difference is not a deficiency of the atlas: it is the mathematical distinction between a central section and a species realized as an orbit or multisection.

For the rank-one electronic fiber, the joint measure is punctual, the two readers share the first component 80, the compression is scalar, and the beta closure is recovered

$$m_e^{\beta} = 0.510998950690000 8086082571648\ldots \ \mathrm{MeV}.$$

The base value remains as a construction stage; it does not replace the beta closure.

93.8.2 Realizations of the muon and tau via charged sectors and spectral criterion of scalarity

In the charged leptonic sector, the descriptors distinguish the central fiber, the depth-G_2 multisection, and the depth- G_4 multisection. The first produces the unique electronic line; the other two produce the operators associated with the muonic and tauonic realizations. If the coupling preserves more than one block, the correct prediction is its spectrum; the physical label does not authorize choosing one of its eigenvalues by numerical proximity.

93.9 Chromatic algebra $SU(3)$, quark flavours, and running-mass transport

93.9.1 Weyl qutrit algebra, Reynolds reduction, and singlet projectors

The chromatic orbit is realized in the color qutrit. The Weyl operators generate $\mathfrak{sl}_3(\mathbb{C})$; the traceless anti-Hermitian selection yields $\mathfrak{su}(3)$, and the finite connection transports the three chromatic charts. The flavor mass must be invariant under this action.

Let $U_{\mathrm{col}} : SU(3) \to \mathcal{U}(V_{\mathrm{col}})$. The chromatic reduction is the Reynolds expectation

$$\mathbb{E}_{\mathrm{col}}(T) = \int_{SU(3)} U_{\mathrm{col}}(g) T U_{\mathrm{col}}(g)^* \, dg.$$

In the fundamental irreducible representation,

$$\mathbb{E}_{\mathrm{col}}(\mathcal{M}_q) = m_q(\mu, \mathscr{R}) I_3.$$

The published mass is therefore independent of red, green and blue. The coordinates μ and $\mathscr{R}$ identify the scale and renormalization scheme; they are not hidden ambiguities and must always accompany the figure.

For mesons and baryons, the singlet projectors are

$$P^{(1)}_{q\bar{q}} = \frac{1}{3}|\Omega\rangle\langle\Omega|, \qquad |\Omega\rangle = \sum_{a=1}^{3} |a\bar{a}\rangle,$$

and

$$P^{(1)}_{qqq} = |\varepsilon\rangle\langle\varepsilon|, \qquad |\varepsilon\rangle = \frac{1}{\sqrt{6}} \sum_{a,b,c} \varepsilon_{abc} |abc\rangle.$$

Both eliminate the observable chromatic label without erasing the internal orbit that intervened in the construction.

93.9.2 Renormalization Connection and Ordered Transport Between Scale Fibers

The dependence on scale is a section of a fibration over $t = \log \mu$. If $\gamma_m(g(t))$ is the endomorphism induced by the chromatic connection, transport is defined by

$$\nabla_{\mathrm{RG}} m_q = \left(\frac{d}{dt} + \gamma_m(g(t)) \right) m_q = 0,$$

$$m_q(t_2) = \mathcal{P} \exp\left(- \int_{t_1}^{t_2} \gamma_m(g(t))\, dt \right) m_q(t_1).$$

Two values belonging to different scales or schemes are not comparable numbers until they are transported to the same fiber. This equation closes the mathematical type of the running mass: it is not an isolated absolute scalar, but a covariantly transported section.

93.10 CKM Mixing and Covariance of Quark Spectral Frames

93.10.1 Unitary transport of the operator B_d in the spectral frame of the type above by means of V_{CKM}

The CKM mixing belongs to a distinct map. If B_u and B_d are the mass operators in the up-type and down-type bases, the internal unitary matrix satisfies

$$B_d^{(u)} = V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^{*}.$$

93.10.2 Separation between the interior closure and the external statistical recognition

The angles derived from A, C^* and Δ_4, the phase δ_{CP}, the Jarlskog invariant and the determinant identity of the commutator determine mixing without creating the mass eigenvalues. Thus spectrum, color, generation and change of basis remain separate.

93.11 Neutrino operator, PMNS mixing and Dirac–Majorana criteria

93.11.1 Three-dimensional neutral spectrum and inheritance of the action-clock-mass scale

Neutrinos are electrically neutral fermionic sections. Their fermionicity descends from the spinorial holonomy, the exterior completion, and the canonical anticommutation relations; neutrality does not alter their statistics. The internal support is

$$\mathcal{H}_\nu = \bigoplus_{j=1}^{4} \mathcal{H}_{\nu, G_j}, \qquad \dim \mathcal{H}_\nu = 2 + 4 + 6 + 8 = 20.$$

The four structural depths are neither flavours nor eigenvalues. The three-dimensional physical subspace is obtained through the neutral projector P_ν, and the mass operator is

$$M_\nu = P_\nu \mathcal{M}_\nu P_\nu.$$

Its three ordered eigenvalues are the masses; the absolute scale comes from the same mass clock that fixes the remaining branches, and not from the mixing angles.

93.11.2 Range 3 of the Gram matrix, polar frame, and antiunitary Majorana criterion

Let $M_\ell = P_\ell \mathcal{M}_\ell P_\ell$ be the charged operator. Denote by F_ℓ and F_ν complete orthonormal spectral frames, ordered by the spectral projectors of M_ℓ and M_ν. The mixing matrix is

$$U_{\mathrm{PMNS}} = F_\ell^* F_\nu.$$

Theorem 93.7 (Spectral closure of the leptonic mixture)**.** *If the two frames are complete, U_{PMNS} is unitary. If the spectra are simple, it is determined up to diagonal phase readjustments and permutations fixed by the spectral order. If degeneracy exists, the exact result is a double class*

$$\prod_\lambda U(m_{\ell,\lambda}) \backslash U(3) / \prod_\mu U(m_{\nu,\mu}),$$

and not a numerically artificial matrix selected.

Proof. Orthonormality gives $U_{\mathrm{PMNS}}^* U_{\mathrm{PMNS}} = F_\nu^* F_\ell F_\ell^* F_\nu = I_3$. Changes of basis within each degenerate subspace act on the left or on the right, producing exactly the indicated double class. □

The complete record determines the elements of M_ν by means of carry, sheet, orientation, vacancy, and holonomy. Equivalently, the Hermitian kernel may be used

$$K_{\mathrm{reg}}(c,d) = \sum_{r \in \{\mathrm{car,hoj,ori,vac,hol}\}} \frac{\langle \Xi_r(c), \Xi_r(d) \rangle}{\sum_x \|\Xi_r(x)\|^2},$$

with omission of every summand of null norm. Its Gram matrix is positive and constructs the operator before diagonalization. The rank-one phase

$$I + (e^{i\pi/6} - 1)|b_K\rangle\langle b_K|$$

remains as a negative control: its formal unitarity is not sufficient and its angle θ_{13} falsifies its identification with full-rank physical mixing.

Let G_0 be the three-dimensional matrix obtained by aggregating the preceding kernel over the three neutrino sectors. The finite condition

$$\det G_0 \neq 0$$

is the exact certificate that the observables included in the record separate three directions. In that case, the polar factor

$$U_{G_0} = G_0 (G_0^* G_0)^{-1/2}$$

is unitary and has rank 3. If $\det G_0 = 0$, the declared kernel does not determine a three-dimensional transport: an independent internal observable must be added and the Gram matrix recomputed. Invertibility may not be imposed by adding a term constructed from the mixing matrix one seeks to obtain. The physical matrix is fixed by the spectral frames of M_ℓ and M_ν; the Gram criterion certifies sufficiency of the register, but does not by itself select those frames.

Let C_ν be the antiunitary self-conjugation of the neutral sector. The Majorana criterion is

$$C_\nu^2 = I, \qquad C_\nu M_\nu = M_\nu C_\nu, \qquad C_\nu H_\nu = H_\nu C_\nu,$$

together with the existence of a fixed real basis in the realification of the physical subspace. When these conditions are satisfied, each mode can be represented by $\psi = C_\nu \psi$. The Dirac branch requires, by contrast, a conserved charge Q such that

$$C_\nu Q C_\nu^{-1} = -Q, \qquad Q\psi \neq 0,$$

and retains ψ and $C_\nu\psi$ as distinct modes. If neither set of conditions is established, the type remains undecided. Neutrality, the word involution or the Ising sector, taken separately, do not decide this alternative.

93.12 Sterile neutrino sector and separation between existence and mixing

93.12.1 Orthogonal projector onto the active sector and cancellation of the weak current

An additional structural depth does not by itself constitute a sterile neutrino. A candidate sterile sector is defined by a nonzero projector P_s satisfying

$$P_s P_{\mathrm{act}} = 0, \qquad P_s J_{\mathrm{weak}} = 0,$$

with dynamics defined on $P_s \mathcal{H}_\nu$.

93.12.2 Distinction between invariant reduction and non-vanishing coupling with the active sector

If, in addition, $[P_s, M_\nu] = 0$, the sector is reducing and does not mix with the active sector. If $P_s M_\nu P_{\mathrm{act}} \neq 0$, there is active–sterile mixing and necessarily $[P_s, M_\nu] \neq 0$. Thus, the existence of the sector and the existence of mixing are distinct properties. If the weakly neutral projector does not exist, the additional coarse channel is an internal depth of the support and not a fourth species.

93.13 Bosons W and Z and scalar sector: projectors, resolvents, and pole criterion

93.13.1 Sectorial effective operator and meromorphic continuation of the reduced resolvent

The sectors W, Z, and the scalar sector have their own representation, route, and signature descriptors. Their realization does not require forcing them into an elementary rank-one fiber. The positive fiber operator is constructed first and then, on a finite-dimensional subspace $P_X \mathcal{H}_X$, the effective channel operator

$$H_{X,\mathrm{eff}}(z) = P_X H_X P_X + P_X H_X Q_X (z - Q_X H_X Q_X)^{-1} Q_X H_X P_X.$$

93.13.2 Isolated admissible zero, physical residue, and distinction with respect to mere channel opening

If the reduced resolvent admits meromorphic continuation through the cut of the continuum, the poles are the isolated zeros, counted with multiplicity, of

$$\det\big(z - H_{X,\mathrm{eff}}(z)\big) = 0, \qquad z_X = M_X c_{\mathrm{int}}^2 - \frac{i}{2}\Gamma_X.$$

An open channel does not by itself guarantee the existence of such zeros. When a zero with the prescribed physical residue exists, its real part publishes pole mass and its imaginary part the width. For the scalar sector, identification of an excitation additionally requires the scalar projector and its gauge and fermionic couplings. Two independent poles determine two states; a second label without projector or pole does not create a particle.

93.14 Composite states and cocyclic linking law

93.14.1 Single Meson and Baryon Projectors and Refinement of the Composite Signature

For constituents $X_1, \dots, X_n$, the input is the composite record

$$\mathfrak{C}_{12}^{(n)} : (\Gamma_1, \dots, \Gamma_n; \mathcal{T}, \partial) \longmapsto \mathcal{L}_{\mathrm{comp}},$$

not the sum of isolated masses. The binary composition

$$L_1 \star L_2 = L_1 + L_2 + \mathfrak{b}(L_1, L_2)$$

is associative exactly when

$$\mathfrak{b}(L_1, L_2) + \mathfrak{b}(L_1 \star L_2, L_3) = \mathfrak{b}(L_2, L_3) + \mathfrak{b}(L_1, L_2 \star L_3).$$

The signature, the refinement, and the composite operator are computed after the linking:

$$\mathcal{L}_{\mathrm{comp}} \longmapsto \sigma_{\mathrm{comp}} \longmapsto \eta_{\mathrm{comp}} \longmapsto \mathcal{M}_{\mathrm{comp}}.$$

The physical projector combines colour singlet, spin and parity, isospin, charge, exchange symmetry, and channel.

93.14.2 Feshbach–Schur Reduction and Construction of the Effective Operator

The stable mass is an eigenvalue of the compression; when the hypotheses for meromorphic continuation hold, the mass and width of a resonance follow from a Feshbach pole. Consequently, mesons, baryons, exotic states, nuclei, atoms, and molecules share the same causal scheme without being reduced to a sum of masses.

Two constituent trees represent the same species if there exists a unitary that simultaneously intertwines their operators, channel projectors and poles. In the absence of such a unitary they are retained as distinct realizations, even if they share constituents and partial quantum numbers.

93.15 Resonances, widths, and survival of material sections

93.15.1 Complex Poles, Admissible Residues and Associated Widths for the Meromorphic Continuation

Let S_X be the projector onto the surviving subspace and $T_X = S_X U S_X$ the compressed one-step operator. If it has a simple dominant eigenvalue

$$\lambda_X = r_X e^{-i\theta_X}, \qquad 0 < r_X < 1,$$

fix the branch $\theta_X \in I_\theta$, where I_θ is the fundamental quasienergy interval determined by the temporal chart and its orientation. Then

$$M_X = \frac{\hbar_{\mathrm{int}}}{t_0 c_{\mathrm{int}}^2}\theta_X, \qquad \Gamma_X = -\frac{2\hbar_{\mathrm{int}}}{t_0}\log r_X, \qquad \tau_X = \frac{\hbar_{\mathrm{int}}}{\Gamma_X}.$$

93.15.2 Compatibility of the genealogical refinement with resonant persistence

Without the choice of I_θ, the phase determines only θ_X modulo 2π, and the quasienergy mass is not a well-defined scalar. The width is not a sixth coordinate of the mass signature: it arises from the modulus of the open mode. Equality of mass and width between particle and antiparticle also requires transporting the channels and the causal prescription, not merely the parity of the signature.

93.16 Multiparticle completion and Fock statistics

93.16.1 Symmetric and Exterior Fock Functors Induced by Holonomy Parity

Let $\mathsf{Hol}_{\pm}^{\leq 1}$ be the monoidal category of fibers of a particle, with contractive refinement morphisms, in particular unitary ones, and holonomy parity $\epsilon \in \{+1, -1\}$. The Fock completion is defined by the functor

$$\mathfrak{F}(H, \epsilon) = \begin{cases} \bigoplus_{n \geq 0} \mathrm{Sym}^n(H), & \epsilon = +1, \\ \bigoplus_{n \geq 0} \bigwedge^n H, & \epsilon = -1, \end{cases}$$

and, for a contractive intertwiner $T : H \to H'$,

$$\mathfrak{F}(T) = \bigoplus_{n \geq 0} \mathrm{Sym}^n(T) \quad \text{or} \quad \bigoplus_{n \geq 0} \bigwedge^n T.$$

93.16.2 CCR/CAR Relations, Pauli Exclusion and the Domain of Non-Contractive Morphisms

Theorem 93.8 (Internal transport of the statistic)**.** *The foregoing completion is a functor of bounded operators on $\mathsf{Hol}_{\pm}^{\leq 1}$; it is compatible with refinement and is graded monoidal. In the $\epsilon = -1$ branch, the creation and annihilation operators satisfy CAR and the occupation operators satisfy $N_s^2 = N_s$; in the $\epsilon = +1$ branch, they satisfy CCR. Therefore, Pauli exclusion and fermionic and bosonic statistics descend from holonomy parity once the representation of a particle has been fixed.*

Proof. Symmetric and exterior powers preserve compositions and identities; the contractivity of T ensures that their direct sum defines a bounded operator. For noncontractive morphisms, second quantization requires an explicit dense domain and does not belong to this category of bounded operators. The sign rule of the exterior product produces CAR and idempotence of occupation; the symmetric quotient produces CCR. Second quantization of the refinement morphisms intertwines the corresponding operators. □

This proof closes the internal algebraic transport. Identification with the relativistic spin–statistics theorem additionally requires the Lorentzian bridge and locality declared in Dimensional Mechanics; the two statements must not be confused.

93.17 Thermodynamic realization through Gibbs states and the pressure limit

93.17.1 Finite Hamiltonian, partition function, Gibbs state, and cofinal pressure limit

For a compressed Hamiltonian $H_{X,V}$ over a finite volume or refinement V, the thermal state is completely defined by

$$Z_{X,V}(\beta) = \mathrm{Tr}\, e^{-\beta H_{X,V}}, \qquad \rho_{X,V,\beta} = Z_{X,V}(\beta)^{-1} e^{-\beta H_{X,V}}.$$

The thermodynamic limit exists when the pressure

$$p_X(\beta) = \lim_{V \nearrow \infty} \frac{1}{\beta |V|} \log Z_{X,V}(\beta)$$

exists and is independent of the cofinal sequence.

93.17.2 Condensation criterion by saturation of the excited-state density

Bosonic condensation is determined by saturation of the excited-state density; it is not asserted merely from the existence of a symmetric completion. The problem is thus expressed by a verifiable mathematical condition on the Hamiltonian and the limit, rather than by a nominal attribution.

93.18 Topological sectors, virtual excitations, and classification of their codomains

93.18.1 Braided category, data N–F–R, and pentagon and hexagon equations

In a topological sector the primary codomain is a braided category, not a line of masses. For simple objects a, b, c, the physical data include

$$N_{ab}^{c}, \qquad F_{d}^{abc}, \qquad R_{c}^{ab},$$

with pentagon and hexagon equations. A scalar mass exists only when a Hamiltonian realization provides a spectral projector and a gap. The Fibonacci, Ising/Majorana, and $\mathbb{Z}_6$ sectors are thereby closed as categorical codomains; they are not assigned a universal figure by analogy with an elementary particle.

In the sector $\mathbb{Z}_6$, fusion and braiding are

$$a \otimes b = a + b \pmod 6, \qquad R_{ab} = \zeta_6^{ab}, \qquad \zeta_6 = e^{2\pi i/6}.$$

For Fibonacci,

$$\tau \otimes \tau = \mathbf{1} \oplus \tau, \qquad F_{\tau}^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix},$$

$$R_{\mathbf{1}}^{\tau\tau} = e^{-4\pi i/5}, \qquad R_{\tau}^{\tau\tau} = e^{3\pi i/5}.$$

The Ising sector has objects $\mathbf{1}, \psi, \sigma$ and rules

$$\psi \otimes \psi = \mathbf{1}, \qquad \psi \otimes \sigma = \sigma, \qquad \sigma \otimes \sigma = \mathbf{1} \oplus \psi,$$

$$F_{\sigma}^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad R_{\mathbf{1}}^{\sigma\sigma} = e^{-\pi i/8}, \qquad R_{\psi}^{\sigma\sigma} = e^{3\pi i/8}.$$

These matrices satisfy the coherence relations in the indicated conventions. The duplication $C \boxtimes C^{\mathrm{rev}}$ preserves both orientations and avoids introducing a chirality by external choice.

93.18.2 Finite persistence of virtual states and absence of a universal scalar mass

A virtual state is a section of finite persistence in an amplitude or a resolvent. It does not belong to the asymptotic spectrum and does not require a real pole mass. Its parameters are virtuality, the channel and the duration of persistence. This definition avoids turning an internal computational line into a new species.

93.19 Gravity, torsion, and spectral criterion of the spin-2 collective mode

93.19.1 Tachyonic Exclusion by Positivity and Separation of Primary Gravity by Connection and Torsion

Positivity excludes an elementary species with $m^2 < 0$ within the mass codomain. A negative eigenvalue of the Hessian of a background does not define a tachyon as a particle: it diagnoses the instability of the background and requires changing the solution around which linearization is performed.

Gravitation enters Dimensional Mechanics by means of connection, curvature, and torsion. It does not require an elementary spin-2 particle as a datum of departure.

93.19.2 Physical pole of the traceless transverse Hessian and positivity of the residue

Let $\mathcal{S}_{\mathrm{grav}}$ be the geometric action and $\mathcal{H}_{\mathrm{TT}}$ the transverse traceless subspace of its Hessian at a solution. The linearized observable is

$$\mathcal{K}_{\mathrm{TT}} = P_{\mathrm{TT}}\, \delta^2 \mathcal{S}_{\mathrm{grav}}\, P_{\mathrm{TT}}.$$

A collective spin 2 mode exists as an asymptotic particle only if the resolvent of $\mathcal{K}_{\mathrm{TT}}$ has a physical pole with positive residue and satisfies the constraints. If no such pole exists, the gravitational interaction remains wholly in the connection and torsion. Thus, the graviton is not a necessary component of the mass law, and its possible appearance is decided by a spectral criterion, not by a nominal box.

93.20 Equivalence between persistent section, inverse limit and finite particle representation

93.20.1 Quotient by tail equivalence and compatibility of section families

The genealogical definition of a particle and its operator representation are two presentations of the same object. Let

$$\mathfrak{r}_K = (x_K, \gamma_K, B_K, s_K, n_K, U_K)$$

a finite representative at depth K, with state, route, boundary, orientation, counter, and representation, and let $r_{L,K} : \mathfrak{R}_L \to \mathfrak{R}_K$ be the refinement maps. A persistent section is a family

$$X = (\mathfrak{r}_K)_{K \geq K_0}, \qquad r_{L,K}(\mathfrak{r}_L) = \mathfrak{r}_K \quad (L \geq K \geq K_0).$$

Two families are equivalent if they coincide on a cofinal tail.

93.20.2 Unique prolongation condition to reconstruct a section from a finite evaluation

Theorem 93.9 (Equivalence of the Persistent Section and the Inverse Limit)**.** *The category of persistent sections, defined as compatible families, is canonically equivalent to*

$$\varprojlim_K \mathfrak{R}_K /\!\sim_{\mathrm{cola}} .$$

Projection to a finite depth is an evaluation, not an inverse functor. A representative $\mathfrak{r}_K$ determines a unique persistent section if and only if it has a unique compatible prolongation modulo tail equivalence; this uniqueness is obtained, for example, when the refinement maps are bijective on the surviving tail from K onward.

Proof. By definition, a persistent section is precisely a point of the inverse limit; the quotient identifies families that coincide on a cofinal tail. Evaluation is the canonical projection onto one component. Its fiber contains all compatible prolongations of the evaluated representative, so that the fiber is a singleton exactly under the indicated unique-prolongation condition. □

An HMT–MD particle is therefore a class of persistent sections without a finite terminal obstruction, endowed with representation, orientation, charge, spin, and spectral character compatible with refinement. The route is its observable history, not its identity reduced to a cell. The antiparticle is the section conjugated by the dodecaphasic involution and the conjugate representation. An obstructed finite section is virtual; a cocyclic product of sections is composite; a persistent braided representation is anyonic.

93.21 Equivariant realization closure theorem and exhaustive classification of material species

93.21.1 Comprehensive atlas of 324 routes: 49 invariants per section and conservation of the null sector

Theorem 93.10 (Closure by physical codomain)**.** *Suppose that the atlas* 81/56/13/324*, the null sector, the action–clock–mass scale, the readers K_D, K_Ω, the physical descriptor, and the internal representation of each state have been constructed. Then every admissible and realizable descriptor, that is, every X with $d_{\mathcal{P}}(X) \in \operatorname{im} d_{\mathcal{R}}$, determines, without consulting its external value:*

1. *a fiber, orbit, or multisection $\mathcal{S}(X)$;*
2. *the positive bidirectional pair and its joint PVM $(\widehat{M}_X^{\beta,D}, \widehat{M}_X^{\beta,\Omega}, E_X^{D,\Omega})$;*
3. *a decomposition into irreducible blocks;*
4. *a spectral measure for each physical coupling;*
5. *a scalar in a compressed rank-one sector or a multiplet in a reducible sector; in an open sector, an effective operator and, only if a meromorphic continuation with an admissible isolated zero exists, a complex pole;*
6. *a categorical object when the realization is topological and does not yet have a Hamiltonian gap;*
7. *a nonexistence criterion when the requested type contradicts positivity, gauge, or asymptoticity.*

None of these results requires selecting a route through mass that will later be tested.

Proof. The fiber product constructs the fiber prospectively. The character and the two readers construct the positive pair and its joint PVM. When the physical type declares a scalar functional, functional calculus produces the corresponding operator; conditional expectation projects it onto the commutant of the symmetry. The isotypic decomposition produces scalars or blocks by Schur's lemma. Preparation projectors produce spectral measures. Feshbach reduction produces the effective operator; meromorphic continuation and an admissible isolated zero produce a pole, if one exists. Topological sectors retain their fusion and braiding data until a Hamiltonian realization exists. Positivity and gauge constraints exclude incompatible types. Each step depends only on internal data or the declared coupling, never on the external column. □

Validity conditions and incompatibility of the equivariant realization of species The construction is refuted in its domain if any of the following events occurs:

1. modifying the external values changes the fiber $\mathcal{S}(X)$;
2. the assignment ceases to be equivariant under cellular inversion, conjugation, or gauge;
3. a point cell is selected in a noncentral multisection without a coupling that breaks the symmetry;
4. the synthesis of K_D and K_Ω privileges an orientation or does not recover the diagonal when both readers coincide;
5. a compression with several eigenvalues is replaced by a mean without a physical projector;
6. the color quotient produces different masses for red, green, and blue;
7. quark masses are compared without declaring the same scheme and the same scale;
8. a rank-one phase is presented as PMNS mixing after failing the full-rank angular contrast;
9. neutrality is used by itself to assert the Majorana character;
10. the mass of a composite is obtained by adding constituent masses without a binding record;

11. the width is inserted as a coordinate of the signature instead of being extracted from an open mode or a pole;
12. a negative eigenvalue of the Hessian is published as a tachyonic particle;
13. gravitation is made dependent on an elementary graviton introduced before connection, curvature and torsion.

93.21.2 Complete inventories of 471 masses and 384 widths and subsequent metrological recognition

External comparison begins only after the descriptor, fiber, operator, projector, scheme, and scale have been frozen. Each row must distinguish stable mass, pole mass, resonance estimator, and width. The atlas of 324 records and the inventories of 471 masses and 384 widths constitute the subsequent exhaustive presentation. The conceptual construction and each record cross-reference one another through their route identifiers. The records retain all their fields and are composed as coherent units; they are not replaced by abbreviated historical tables.

The comparison does not consist in enlarging a list of names, but in applying, for each physical type, the complete map that leads from the genealogy of route to its observable codomain.

93.22 Realization of observables and metrological comparison

The value of a mass acquires its meaning within the structure that produces it and the representation in which it is measured. The APP–TRIT–TPK construction first determines the admissible trajectories, their oriented records and the signature readers; the towers and the material realization act on those objects. Experimental comparison is placed at the end of this composition. Its function consists in comparing observables of the same type, preserving the identity of the state and quantifying agreement in a common chart. This position allows the mathematical structure, the evaluated coordinate and its physical interpretation to be studied together.

This section brings together three levels that perform different functions: the internal domain of routes and their quotients; the classes and evaluations individualized in the mass owners; and the external inventory of observables used as a reference. The order of presentation preserves the constructive direction of the article. External quantities appear as terms of comparison, while internal values retain the signatures, operations and conditions that fix their evaluation.

93.22.1 Structural domain, material classes and external inventory

Let C be the set of APP cells and $\mathcal{R}$ its domain of oriented routes. The identification of a route contains the cell coordinates and the two initial orientations:

$$\gamma = (r, c, h_0, v_0), \qquad r, c \in \{1, \dots, 9\}, \quad h_0, v_0 \in \{-1, +1\}.$$

Therefore, $\#C = 81$ and $\#\mathcal{R} = 81 \cdot 4 = 324$. The 56 trajectory families and the 13 multisections belong to quotients and structural groupings of the same domain, with the relations and readers established above. These four cardinalities describe distinct internal objects. The assignment of a physical representation can preserve an entire fiber; a species label does not in itself impose a pointwise choice within it.

The public table of 23 classes organizes the realizations by sector. Its function is to relate internal residence to the representation, charge, spin, color, flavor and type of observable. A class can comprise several species, and a catalog identity can bring together different charge states. The correspondence is expressed through typed maps and preserves these multiplicities.

The frozen metrological inventory contains 471 mass records and 384 width records. The corpus documents 1170 identities in 438 groups, of which 403 contain at least one of those observables. The sum

$$471 + 384 = 855$$

counts external summary-table observables. The atlas 81/56/13/324, the 23 classes and the 855 records thus belong to three levels of description. Their association preserves the type of each set, instead of turning their cardinalities into a single number of particles or predictions.

Catalog family	Groups	Masses	Widths
General	46	53	9
Quark sector	6	8	1
Mesons	201	243	225
Baryons	149	166	149
External gravitational record	1	1	0
Total with observable	403	471	384

The gravitational row preserves the identity appearing in the external source; its presence in the inventory does not define an elementary particle of HMT. The remaining 35 groups retain their catalog identity, although the snapshot does not assign them a summary-table mass or width. Documentary integrity encompasses both facts: the quantities recorded and the identities for which that snapshot does not publish one of those quantities.

93.22.2 The observable before the numerical coordinate

A material output is specified through the state, the representation, the operator and the reading procedure. For a fiber F_X, let $\widehat{M}_X$ be the mass operator already constructed and let $P_{X,a}$ be the projector corresponding to the state or preparation channel a. For Ψ_X normalized in its image, the spectral distribution is

$$\mu_{X,a}(B) = \langle \Psi_X, P_{X,a} E_{\widehat{M}_X}(B) P_{X,a} \Psi_X \rangle,$$

where $E_{\widehat{M}_X}$ is the spectral measure of the operator. This expression preserves the prepared state and the multiplicity of the fiber. A single scalar coordinate for every state prepared in that subspace is obtained under the condition

$$\mathrm{Ran}(P_{X,a}) \subset \mathrm{Dom}(\widehat{M}_X), \qquad \widehat{M}_X P_{X,a} = m_X P_{X,a}.$$

Indeed, each vector of that subspace is then an eigenvector of the mass operator for the same value m_X, and its spectral measure is concentrated at that value. If the compression equality $P_{X,a}\widehat{M}_X P_{X,a} = m_X P_{X,a}$ is used, it is additionally required that $P_{X,a}$ reduce $\widehat{M}_X$; in finite dimension, this is equivalent to their commutation. Without that condition, compression may fix only the mean value. When the spectral measure retains several values, the output is presented as a distribution or multiplet with its weights.

For a fixed density ρ in a finite fiber, let m_j be the distinct eigenvalues and E_j their spectral projectors. The weights $p_j = \mathrm{tr}(\rho E_j) \geq 0$, with $\sum_j p_j = 1$, determine

$$\overline{m} = \sum_j p_j m_j, \qquad \mathrm{Var}_\mu(m) = \sum_j p_j (m_j - \overline{m})^2.$$

Scalar concentration criterion for the fixed state. The variance vanishes if and only if all values with positive weight coincide. Indeed, each term is nonnegative; the sum is zero precisely when $m_j = \overline{m}$ for every j with $p_j > 0$. This determines when a mass distribution admits a scalar publication without loss of the dispersion of the prepared state. This criterion refers to the fixed density and differs from the operator condition on the entire subspace. The electronic case uses its operator on the central fiber, as previously constructed; the rank one of an arbitrary projector is not sufficient to concentrate the spectral measure of another operator.

Mass and width use coordinated readers of different types. For a resonant state, a pole convention can be expressed in energy coordinates as

$$z_X = M_X c_{\mathrm{int}}^2 - \frac{i}{2}\Gamma_X.$$

The real part is the mass-energy coordinate; Γ_X has the dimension of energy and measures the width. Once the temporal chart is fixed, $\tau_X = \hbar_{\mathrm{int}}/\Gamma_X$. The relation assumes the pole and survival regime for which it was constructed; a rate, a width and a lifetime retain their units and their conversion rules. This distinction explains why the inventory preserves two classes of observable.

The same principle determines comparison by sector. Current quark masses are expressed with scheme $\mathcal{S}$ and scale μ; resonances distinguish pole mass, Breit–Wigner parametrization and on-shell convention. Neutrinos distinguish mass eigenvalues, squared differences, flavors and mixing parameters. One-sided bounds retain the documented confidence level and statistical convention. An upper bound is not converted into a positive central value, nor a squared difference into an absolute mass.

93.22.3 Fixing the construction and correspondence of records

The realization of a species X transmits the necessary information from its internal antecedents. For comparison purposes, we can gather it in the record

$$\mathfrak{D}_X = \big(F_X, \rho_X, P_{X,a}, \mathcal{L}_X, \sigma_X, \eta_X, \widehat{M}_X, \mathcal{U}_X\big),$$

where ρ_X identifies the representation, $\mathcal{L}_X$ the genealogical record, $\sigma_X \in \mathbb{Z}^5$ the signature, η_X the tower section and $\mathcal{U}_X$ the units chart. All these data receive their meaning from the APP–TRIT–TPK operations and the realizations proved in the article. Gathering them in a tuple is a documentation operation: their constructions and proofs remain in the sections that produce them.

The external reference is represented separately by

$$o = (\mathrm{id}, \mathrm{especie}, \mathrm{observable}, v, \mathrm{unidad}, \mathcal{S}, \mu, \mathrm{incertidumbre}, \mathrm{limit}, \mathrm{confianza}, \mathrm{fuente}, \mathrm{edition}).$$

The association $(\mathfrak{D}_X, o)$ must be established through physical identity and the type of observable. In an extensive table, this operation is performed through identity and provenance keys; the proximity of two figures does not serve as a key. The record retains the particle or antiparticle label, the group that possesses the observable and the identity of the measured property.

The independence of the selection procedure is examined by hiding or permuting the metrological column and checking that $\mathfrak{D}_X$ remains unchanged. That test makes sense for a generator whose effective composition can be executed; merely preserving an output file does not in itself constitute such a test. The cryptographic fingerprint fixes which artifact is compared and makes modifications detectable, but causal independence belongs to the operation and its dependencies.

Statuses are assigned to the domain and operation that underpin each statement. A row can simultaneously document an internal operator, an evaluation given a signature and an external comparison, without identifying those three levels with each other. Every subsequent realization is linked to the definition, proposition or table that establishes its scope. In particular, the catalog's documentary classifications apply to the snapshot specified by each record and do not modify the mathematical status of a later construction.

93.22.4 Metrological correspondence of the twenty-three classes

The following table preserves the 23 classes of the source and specifies the form of reading that each requires upon reaching the metrological interface. Its function differs from the operator classification of proposition 94.26 and the documentary inventory of tables 94.6 and 94.7: here the comparable quantity, the realization convention and the data that must accompany it are distinguished. Flavor and sector names guide the correspondence; the transmitted mathematical object remains in the corresponding fibers, representations and readers.

Table 93.1: Metrological reading of the twenty-three classes and type of object transmitted.

Class	Representatives	Object documented for the reading
Charged lepton 1	Electron and positron	Central route, conjugation and beta coordinate.
Charged lepton 2	Muon and antimuon	Fiber, readers and evaluation of the declared signature.
Charged lepton 3	Tau and antitau	Fiber, readers and evaluation of the declared signature.
Neutrino 1	Electronic flavor and conjugate	Neutral fiber and mass–flavor realization.
Neutrino 2	Muonic flavor and conjugate	Neutral fiber and mass–flavor realization.
Neutrino 3	Tauonic flavor and conjugate	Neutral fiber and mass–flavor realization.
Quark arriba 1	$u, \bar{u}$	Flavor and color fiber; scheme and scale chart.
Quark abajo 1	$d, \bar{d}$	Flavor and color fiber; scheme and scale chart.
Quark arriba 2	$c, \bar{c}$	Flavor and color fiber; scheme and scale chart.
Quark abajo 2	$s, \bar{s}$	Flavor and color fiber; scheme and scale chart.
Quark arriba 3	$t, \bar{t}$	Flavor and color fiber; scheme and scale chart.
Quark abajo 3	$b, \bar{b}$	Flavor and color fiber; scheme and scale chart.
Electromagnetic gauge	Photon	Null chart and helicity representation.
Chromatic gauge	Gluons	Null chart and adjoint color representation.
Calibre cargado	W^+, W^-	Route, signature and pole convention of the realization.
Calibre neutro	Z^0	Route, neutral mixing and pole convention.
Scalar	Higgs	Scalar representation, signature and pole convention.
Mesonic composite	Mesons and conjugates	Constituents, link record and composite reading.
Baryonic composite	Baryons and conjugates	Trilinear composition, link and composite reading.
Fibonacci topological	$1, \tau$	Fusion ring and Frobenius–Perron dimension.
Ising topological	$1, \psi, \sigma$	Fusion rules and sectoral representation.
Cyclic topological	$\mathbb{Z}/6\mathbb{Z}$	Winding characters and braid representation.
Virtual	Secciones intermedias	Finite horizon, prolongation and obstruction.

In charged leptons, the reading leads to the mass of the central section or to the evaluation of the selected fiber. In neutrinos, it additionally preserves the transformation between mass and flavor and the squared differences; therefore it is not reduced to an isolated scalar. In quarks, every figure is accompanied by the renormalization scheme and scale. The electromagnetic and chromatic classes publish, respectively, an experimental bound and the null gauge section; the W, Z and scalar classes require the declared pole or resonance convention. Composites preserve mass, width and constituent identity. The three topological classes are compared through their fusion, braiding and gap laws once the dynamical realization is fixed; their dimensions and characters belong to their own codomains. The virtual class preserves the pole, the obstruction and the prolongation horizon. This reading determines which object reaches metrology without confusing the twenty-three classes with twenty-three universal mass scalars.

93.22.5 Metrological snapshot, units and transcription accuracy

The snapshot identified by the corpus as Particle Data Group 2026 is used, with a cutoff date of January 15, 2026 and a documented consultation on July 26, 2026. This section reproduces that documentary state: it does not incorporate a subsequent catalog update. The references of the evaluation supplement and those of the historical grid retain their individual provenance; they do not automatically receive the edition of the general inventory.

Each value is preserved as its original decimal string, together with its presentation, unit, uncertainty and provenance key. This decision prevents an intermediate floating-point conversion from changing the last digits or removing significant zeros from the transcription. The original string and the typographical representation are two distinct data items: the former allows the file's arithmetic to be reproduced exactly; the latter communicates the result with a precision appropriate for reading. The digits of the digital representation of a reference do not increase its experimental precision.

Conversion between MeV and GeV uses the exact factor 10^3, applied to the value and its uncertainty. To present a mass, the unit is written MeV/c^2 or GeV/c^2; to present the rest energy, MeV or GeV. The relation $E = mc^2$ is the chart that allows passage from one reading to the other. A comparison of coordinates requires fixing that chart in both columns.

The inventory snapshot expressly retains the scheme and scale of its external records as not supplied. The name of a property may contain an indication such as “Breit–Wigner”; it is preserved literally, but that partial indication does not complete the metadata the comparison requires. Empty cells, unavailable-data markers and not-applicable markers remain differentiated. A documentary unknown is not replaced by zero.

93.22.6 Individualized evaluations and reading conditions

Table 93.2 brings together 20 numerical presentations: the electronic beta coordinate, two null outputs and 17 evaluations conditional on declared descriptors. Among the latter, 11 correspond to elementary sectors and 6 to composites. This count expresses the contents of that documentary presentation; the fibers, families and records of the domain retain their own extent.

The table retains the twenty entries. To facilitate reading, four modalities are identified: B, electronic beta coordinate; N, null chart; F, evaluation conditional on a complete signature; A, conditional angular evaluation. The subscript c indicates material composition. The condition of each entry belongs to its type of result and is also preserved in the electronic catalog accompanying the text.

Table 93.2: Twenty individualized evaluations, with their chart and comparison modality.

State	HMT coordinate, MeV/c^2	Subsequent documentary reference	Type
$e^{\pm}$	0.510998950690000809	0.51099895069(16) MeV/c^2	B
$\mu^{\pm}$	105.658360294435829303	105.6583755 ± 0.0000023 MeV/c^2	F
$\tau^{\pm}$	1776.860917631432295595	1776.93 ± 0.09 MeV/c^2	F
$u, \bar{u}$	2.165924536173907663	2.16 ± 0.07 MeV/c^2	A
$d, \bar{d}$	4.679550162948874172	4.70 ± 0.07 MeV/c^2	A
$c, \bar{c}$	1270.500838641911135286	1.2729 ± 0.0045 GeV/c^2	A
$s, \bar{s}$	92.940093907845640641	92.9 ± 0.7 MeV/c^2	A
$t, \bar{t}$	172597.479544019136261367	172.60 ± 0.27 GeV/c^2	A
$b, \bar{b}$	4190.119763299790218075	4.186 ± 0.006 GeV/c^2	A
γ	0	$< 10^{-18}$ eV/c^2, retained bound	N
g	0	Null gauge chart; no experimental comparison row	N
$W^{\pm}$	80378.964909704547024865	80.3625 ± 0.0077 GeV/c^2	F
Z^0	91187.533444313407844855	91.1879 ± 0.0020 GeV/c^2	F
H	125292.466042536133000433	125.13 ± 0.11 GeV/c^2	A
π^0	134.976724327872125739	134.976827767685 MeV/c^2, supplement	F_c
$\pi^{\pm}$	139.570485171572191375	139.570390983681 MeV/c^2, supplement	F_c
$K^{\pm}$	493.677085649530612476	493.676599458041 MeV/c^2, supplement	F_c
K^0	497.611186497750457359	497.610767531258 MeV/c^2, supplement	F_c
$p, \bar{p}$	938.271751546984898299	938.27208943 MeV/c^2, supplement	F_c
$n, \bar{n}$	939.565810472507023313	939.56542194 MeV/c^2, supplement	F_c

In this table, the reference values are presented with the decimal notation and units declared for each observable. The calculation record also preserves the decimal string used in the evaluation, including the difference between that string and its rounded presentation for the tau, Z and Higgs. The quark comparison remains informative until the scheme and scale chart is assembled. The W, Z, H sectors also require preservation of the resonance convention. The six composite references come from the supplement identified in the evaluation records; their metadata are not completed by analogy with the elementary references.

The complete signatures documented in modalities F and F_c, in the order $\sigma = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta)$, are:

μ	$42, -3, \ 8, \ 0, \ 2$
τ	$64, \ 0, \ 25, -1, \ 6$
W	$94, -1, \ 0, -2, \ 4$
Z	$95, -1, -11, \ 0, -4$
π^0	$44, -4, -25, \ 1, -2$
$\pi^\pm$	$44, \ 1, \ -2, \ 2, -1$
$K^\pm$	$54, -1, \ 17, -2, \ 2$
K^0	$54, \ 0, \ 32, \ 3, -2$
p	$59, \ 0, \ 9, \ 1, \ 0$
n	$59, \ 1, -34, \ 0, \ 1$

The angular modalities preserve, respectively for u, d, c, s, t, b, H, the pairs

$$(n_A, s_C) = (11, 7), (17, 8), (61, 8), (41, -3), (100, -1), (71, -5), (97, 9).$$

The two declared components are not extended with three implicit zeros. The evaluation equation and the mechanism that produces a descriptor are different statements; their proofs are found in their respective definitions and propositions.

93.22.7 Relative differences and uncertainty propagation

Given a common chart and two positive coordinates of the same observable, the relative difference and the logarithmic difference are

$$\delta_X = \frac{m_X^{\mathrm{HMT}} - m_X^{\mathrm{ext}}}{m_X^{\mathrm{ext}}}, \qquad r_X = \log \frac{m_X^{\mathrm{HMT}}}{m_X^{\mathrm{ext}}}, \qquad \delta_X^{\mathrm{ppm}} = 10^6 \delta_X.$$

A common unit conversion $m \mapsto am$, with $a > 0$, leaves both differences invariant: the factor a cancels in the quotients. The absolute difference is multiplied by a, as is its uncertainty. For a zero external value, a one-sided bound or an interval without a central estimate, the preceding quotient is replaced by the comparison corresponding to that type of observation.

When a joint uncertainty model is available, let $d_X = m_X^{\mathrm{HMT}} - m_X^{\mathrm{ext}}$. Its variance is

$$u^2(d_X) = u^2(m_X^{\mathrm{HMT}}) + u^2(m_X^{\mathrm{ext}}) - 2\,\mathrm{Cov}(m_X^{\mathrm{HMT}}, m_X^{\mathrm{ext}}).$$

The formula results from expanding $\mathrm{Var}(U-V)$. If $u(d_X) > 0$, the normalized difference is $z_X = d_X / u(d_X)$. Its statistical interpretation requires the corresponding probabilistic model and covariance; an unrecorded field does not amount to zero variance. Asymmetric uncertainties are retained as such and are not automatically converted into a symmetric deviation.

For several species, the joint covariance matrix is retained. With the discrete signatures fixed, if $f_X(\vartheta) = \log(m_X / m_{\mathrm{ref}})$ depends on continuous coordinates already constructed and J is its Jacobian matrix, first-order propagation gives

$$\Sigma_f = J \Sigma_\vartheta J^{\mathsf{T}}.$$

This formula preserves the correlations shared by the angular pair, Δ_4, the action ratio and the towers. Its application presupposes the regime in which linearization controls the remaining terms. An integer change of signature is recorded as a discrete variation of the construction, not as an infinitesimal Gaussian fluctuation.

The precision of an evaluation therefore has several sources: the experimental reference; the units, scheme and scale chart; the distribution of routes; the tower approximation; and the selected coupling. If Σ_{ij} denotes the covariance between the contributions i, j, the total covariance of their sum is $\sum_{i,j} \Sigma_{ij}$.

Keeping the cross blocks avoids attributing independence to several coincidences that use the same antecedents.

93.22.8 Absolute scale and control of the harmonic approximation

The absolute scale and the relative correction enter at different places in the construction. Given an internally constructed reference basal section $m_\star^{(0)}$, let $\chi_\star = \chi_{\mathrm{full}}(\sigma_\star, \eta_\star)$. In the normalization relative to that section, the expression is

$$m_X^\beta = m_\star^{(0)} \, \frac{\chi_{\mathrm{full}}(\sigma_X, \eta_X)}{\chi_\star} R_{\mathrm{act}}^{\kappa_X}.$$

The common basal factor remains explicit. If the reference is electronic, $m_\star^{(0)} = m_e^{(0)}$ is the dimensional realization of $\mathfrak{e}_0$, with all the factors of its basal equation, and $\chi_\star = \chi_{\mathrm{full}}(\sigma_e, \eta_e)$. Thus we recover $m_e^\beta = m_e^{(0)} R_{\mathrm{act}}^{\kappa_e}$. If the common clock is used to write $m_\star^{(0)} = m_{\mathrm{clk}} b_\star$, the factor $b_\star$ preserves that basal normalization; it is not replaced by 1 or incorporated twice into the character. For two states in the same chart and with the same reference, the quotient is

$$\frac{m_Y^\beta}{m_X^\beta} = \frac{\chi_{\mathrm{full}}(\sigma_Y, \eta_Y)}{\chi_{\mathrm{full}}(\sigma_X, \eta_X)} R_{\mathrm{act}}^{\kappa_Y - \kappa_X}.$$

The cancellation of $m_\star^{(0)}/\chi_\star$ is exact. It includes the common basal prefactor and the common factors of the absolute branch of action, clock and velocity chart. Consequently, the action decade 10^{-34} retains its role in the dimensional realization and is not inserted again into the relative corrector. Differences between routes remain in their signatures, readers and tower sections.

To control a truncation, it is useful to fix the normal form of the character. Let

$$\zeta_{270} = 135\Delta_4^2, \qquad N_h = \eta_h + b_{120}\mathbf{1}_{\{h=120\}}, \qquad \mathcal{S} = \langle 90, 120\rangle \setminus \{0\}.$$

Then

$$\log \chi_{\mathrm{full}} = n_A A_{\mathrm{rad}} + \frac{s_C}{6} C_{\mathrm{rad}}^* + k_{\Delta_4}\Delta_4 + b_\zeta \zeta_{270} + \sum_{h\in\mathcal{S}} N_h s_h.$$

In this convention, the height term 120 appears only once. The term $b_\zeta\zeta_{270}$ retains its quadratic origin in the record; $N_{270}s_{270}$ belongs to the harmonic reading. The equality of the numerical index does not identify the two objects.

Truncation bound and multiplicative error. Suppose that, for $h > H$, the already determined coefficients of a route satisfy $|N_h s_h| \le Cq^h$, with $C \ge 0$ and $0 < q < 1$. The logarithmic tail R_H satisfies

$$|R_H| \le \sum_{\substack{h\in\mathcal{S}\\ h>H}} Cq^h \le \sum_{h=H+1}^{\infty} Cq^h = \frac{Cq^{H+1}}{1-q} =: \varepsilon_H.$$

If m_H is the coordinate obtained by truncating that tower and the other factors are kept fixed, $m/m_H = e^{R_H}$. Monotonicity of the exponential yields

$$e^{-\varepsilon_H} \le \frac{m}{m_H} \le e^{\varepsilon_H}, \qquad \left|\frac{m - m_H}{m_H}\right| \le e^{\varepsilon_H} - 1.$$

The bound allows precision to be assigned to the evaluation from the route and its operators. The values of C, q, H are justified through that construction; the experimental residual does not select the length of the tower or the bound to be proved.

93.22.9 Complete preservation and reproducibility of the comparison

The catalog preserves the complete records in tabular and structured formats. Each row retains its source file, its ordinal, the original key and the occurrence number of that key. This identification preserves even two literally identical rows: the equality of their contents does not eliminate the second occurrence. The change of format preserves the original strings, empty cells and statuses, and attaches the fingerprint of the input bytes.

The inventory of 855 observables is also linked to the metrological rows of the integral corpus. The correspondence is verified by the pair consisting of the observable type and its identifier; the check compares multiplicities, not just sets of keys. The internal tables of cells, families, multisections and routes remain separate from that inventory. The 23 classes, the 20 numerical presentations and the historical grid also retain their own files and statuses.

Transcription checks verify recovery of all fields, row order, duplicates, decimal strings and correspondence with the reference tables. Their scope is documentary. The mathematical proofs of the atlas, the readers and the material realization are what establish the internal outputs; the metrological comparison completes that presentation through well-defined observables and charts. Documentary integrity and the sufficiency of a proof are checked separately.

Extension of the catalog therefore follows a cumulative rule: a new evaluation incorporates the state, its record and signature, the tower section, the reader, the realization, the comparison chart and the provenance. Previous cases retain their content and their cutoff date. Thus, growth of the inventory makes a larger part of the domain visible without retrospectively altering what each proof and each evaluation had established.

Chapter 94

The particle as a persistent material section: trajectory, representation, band, composition, and identity criterion

94.1 Definition of particle as a compatible class of persistent material sections under refinement

A particle is a compatible class of persistent material sections of the genealogical extension, endowed with an observable history, representation, orientation, and a criterion of identity under refinement. The route is its history, not the particle itself; the cell is a coordinate of the atlas, not a moving body; mass is a spectral observable, not its ontological definition.

94.2 Classification of material species by fiber, representation, charge, spin, flavor, and generation

The species is typed by charge, spin, color, flavor, generation, representation, and orientation. Charge is an oriented character; spin is representation data; color is a residual ternary fiber; flavor and generation belong to the multisection and its depth. None of these data reduces to the mass value.

94.3 Antiparticle, Band, and Virtuality

The involution $C_M = -\mathrm{rev}$ induces the anti-section and sheet conjugation. Particle and antiparticle may be two sections of the same Möbius cover when the corresponding loop, holonomy, and representation are fixed. The extended return of 216 steps does not decompose into two independent trajectories of 108. A virtual particle is an intermediate nonasymptotic section with a finite horizon of prolongation.

94.4 Path persistence, multisections, and Möbius conjugation

Persistence is formulated as a section of an inverse system; the internal species as an orbital quotient of a multisection; and conjugation as an anti-section with declared holonomy. The complete development preserves the extension–route–record–signature stratification, the temporal reader, and the distinction between the two occurrences of $2\pi/4\pi$.

94.4.1 Route persistence, species multisections, and conjugation of Möbius type

This section specifies a distinction that, if formulated too briefly, reverses the logic of the construction. The route is not a datum that a particle chooses once its mass is known. Nor is it an auxiliary drawing without causal content. It is the observable and oriented history of a surviving extension. The extension additionally preserves compatible depth, the register, carries, and memory that do not fit into a finite cutoff. In this precise sense,

an HMT particle is the compatible persistence of an enriched route.

The phrase and the definition by extensions express the same object at two levels: the route is its visible history; the extension is the complete tower of that history with all its compatible prolongations.

94.4.1.1 Persistence as a section of an inverse system

Let $(\mathcal{S}_m, \rho_{m+1,m})_{m\geq 0}$ be the system of states that survive to depth m, and let

$$x = (x_m)_{m\geq 0} \in \Omega_{\mathrm{HMT}}^{\mathrm{surv}} = \varprojlim_m \mathcal{S}_m.$$

Route shadows satisfy the compatibility square

$$\mathrm{tr}_{m+1,m}(\mathrm{sh}_{m+1}(x_{m+1})) = \mathrm{sh}_m\big(\rho_{m+1,m}(x_{m+1})\big) = \mathrm{sh}_m(x_m).$$

Therefore

$$\gamma_x = \big(\gamma_x^{[m]}\big)_{m\geq 0}, \qquad \gamma_x^{[m]} = \mathrm{sh}_m(x_m),$$

is a compatible history, not a sequence retroactively fitted.

Over each cut $\gamma_x^{[m]}$ one has the discrete bundle $\mathcal{B}_{\gamma_x^{[m]}}$. A persistent section is a family

$$s = (s_m)_{m\geq 0}, \qquad s_m \in \Gamma\Big(\mathcal{B}_{\gamma_x^{[m]}}\Big), \qquad \mathrm{res}_{m+1,m}(s_{m+1}) = s_m.$$

This equation is the mathematical content of persistence: each finite reading extends without contradicting the preceding readings. The particle tuple of section 94.6 adds occupation, representation, and propagation, but does not alter this provenance.

94.4.1.2 Internal species as an orbital quotient of the multisection

In the nonadic atlas, each oriented cell $c = (r, c') \in C_{81}$ carries a route family

$$\mathcal{R}(c) = (n_A, s_C, k_{\Delta_4}, \nu_{120}, \nu_{270}, \tau_{12}).$$

The image of $\mathcal{R}$ contains fifty-six values. Once the chart and its order are fixed, rank $\kappa \in \{0, 1, 2, 3, 4\}$ within each fibre produces the injective lift

$$c \longmapsto \big(\mathcal{R}(c), \kappa(c)\big).$$

The five-state alphabet is minimal, since two fibers of cardinality five exist. This selection is internal, prospective, and blind to masses and particle names.

The complete census supporting this statement is

$$81 = 45_{\text{leptones}} + 36_{\text{quarks}} = 25_{\ell^-} + 20_{\nu} + 20_{d\text{-type}} + 16_{u\text{-type}}.$$

The fifty-six route fibers possess the histogram.

$$41 \times 1 + 10 \times 2 + 2 \times 3 + 1 \times 4 + 2 \times 5 = 81.$$

Therefore, 56 counts route signatures, whereas 81 counts oriented cells; memory κ distinguishes cells that one and the same signature does not separate.

To classify internal species, this map is not inverted. The quotient is taken.

$$q : C_{81} \longrightarrow \Sigma_{13}, \qquad q(c) = \big(\mathrm{familia}(c), G(c)\big),$$

whose image comprises thirteen types. The output associated with a type y is the finite multisection of the atlas

$$\boxed{\mathfrak{S}_y^{\mathrm{atlas}} = \{(\mathcal{R}(c), \kappa(c)) : q(c) = y\}.}$$

Their cardinals are

$$1, 8, 16; \qquad 2, 4, 6, 8; \qquad 2, 4, 6, 8; \qquad 4, 12,$$

for charged leptons, neutrinos, and the two types of quark, respectively. The species is the common type of that fiber; a particular particle is a persistent section realized within it. Requiring the name of a species by itself to select a cell destroys precisely the degrees of orientation and conjugation that the fiber preserves.

The elevation to extensions is a different typed object. If $\pi_{81} : \Omega_{\mathrm{HMT}}^{\mathrm{surv}} \to C_{81}$ denotes the cellular reader of the declared enriched route, then

$$\widetilde{\mathfrak{S}}_y = \{x \in \Omega_{\mathrm{HMT}}^{\mathrm{surv}} : q(\pi_{81}x) = y\}$$

is the corresponding enriched multisection. The finite certificate proves the census and the obstruction in $\mathfrak{S}_y^{\mathrm{atlas}}$; its lift is relative to the reader π_{81}, not an automatic identification between a cell and a deep history.

Theorem 94.1 (Obstruction to the point selector and electronic exception)**.** *Let $\rho_{\mathrm{c}}(r, c') = (10-r, 10-c')$ be central inversion. The thirteen fibers of q are ρ_{c}-stable. The level-zero charged-lepton fiber has the unique fixed point, $(5, 5)$. None of the other twelve fibers admits a ρ_{c}-equivariant point selector if ρ_{c} acts trivially on the coarse type.*

Proof. The equation $\rho_{\mathrm{c}}(r, c') = (r, c')$ is equivalent to $r = c' = 5$. The atlas census places that cell in the fiber $(\ell^-, G0)$, which has cardinality one. Now let $y \neq (\ell^-, G0)$ and suppose that there exists an equivariant point section $a : \Sigma_{13} \to C_{81}$. Since $\rho_{\mathrm{c}} y = y$, equivariance would require

$$a(y) = a(\rho_{\mathrm{c}} y) = \rho_{\mathrm{c}} a(y).$$

Then $a(y)$ would have to be a fixed point of ρ_{c}, but the unique fixed point belongs to the electron fiber. This is impossible. Therefore, in the other twelve fibers the natural output is a ρ_{c}-orbit or a multisection; choosing a representative requires additional orientation or representation. □

The electron is exceptional in three compatible senses: $(5, 5)$ is the geometric unique center, the unique level-zero cell, and the center with respect to which the raw lattice of differences is formed

$$\Lambda_{\mathrm{raw}}^{\Delta} = \left\langle A_{\mathrm{raw}}(c) - A_{\mathrm{raw}}(5, 5) : c \in C_{81} \right\rangle_{\mathbb{Z}}.$$

The raw signature of that cell is not null:

$$A_{\mathrm{raw}}(5, 5) = (24, 0, 12, 0, -12), \qquad \tau(5, 5) = \texttt{+++-+++-}.$$

The origin $v_e = (0, 0, 0, 0, 0)$ of the lattice of action is a convention relative to the electron and not the raw signature of the central cell.

The electronic algebraic invariant is evaluated by

$$\mathfrak{e}_0 = \frac{\sqrt{3}}{4}\left(e^{\varphi/\pi^2} - 22\alpha_{\mathrm{HMT}}^3\right)(1 + 15\Delta_4), \qquad \Delta_4 = \frac{\pi - e\log\pi}{270}\left(1 + \frac{\pi}{729}\right),$$

$$\mathfrak{e}_0 = 0.5109989224880660725098708771913 49\ldots.$$

The expression is dimensionless; the chart $\varsigma_{u_E} : y \mapsto y\, u_E$ subsequently introduces an energy unit. The center $(5, 5)$ and its uniqueness are exact finite results. The selection of the coefficients 22 and $15 = 3 \cdot 5$ is deduced, with explicit initial structure, through the complement of the five regions of π in $K_e = \mathbb{Z}/9\mathbb{Z} \times \mathbb{F}_3$: $-22 = -|K_e \setminus \iota(R_\pi)|$ and $+15 = |R_\pi \times \mathbb{F}_3|$. The central raw signature retains a distinct type and acts as an independent control of the basal selection.

The cell $(5, 5)$ must not be confused with the energy signature 555555: the latter belongs to the transversal representative of one of the five regions of π. They are distinct objects sharing the central digit.

In the quark sector, the residual character

$$\mathrm{col}(r, c') = r - c' \pmod 3$$

refines the multisection into three classes of twelve cells and satisfies $\mathrm{col}(\rho_{\mathrm{c}} c) = -\mathrm{col}(c)$. This constructs the internal color tritic. The physical representation of $SU(3)$ is a subsequent reader; it is not needed to prove the finite distribution.

94.4.1.3 Stratification of extension, route, record, and signature

The maps q, $\mathcal{R}$, the cellular projection, and the realization reader distinguish four related and typed classifications:

(i) the thirteen internal family–level fibers of q;

(ii) the fifty-six route families of $\mathcal{R}$;

(iii) the eighty-one positions and their three hundred and twenty-four orientations;

(iv) the individualized physical realizations by representation and codomain.

The four layers are censused before the metrological contrast. For each class, the comparison requires the tuple

$$(\text{position, fiber, routes, readers, record, signature, character, towers, dimensional chart}).$$

A reconstruction on a subset proves only that subcase and does not replace any component of the tuple. Given A, C^*, and the channels $q_\pm$, the global tower belongs to the infinite semigroup

$$s_n = q_-^n - q_+^n, \qquad n \in \langle 90, 120\rangle = \{90, 120\} \cup \{30r : r \geq 6\}.$$

The heights 90, 120, 180, 210, 240, 270, 360, 420 are finite representatives of the semigroup; the preceding recurrence prolongs the complete tower. and the dimensionless character of a signature is

$$\chi_{\text{mass}}(\nu) = \exp\Big(n_A A_{\text{rad}} + \frac{s_C}{6} C^*_{\text{rad}} + k\Delta_4 + \nu_{120} s_{120} + \nu_{270}(135\Delta_4^2)\Big).$$

The moments are global: species weight them through their route, their record, and their fibre operator. The readers K_D and K_Ω determine the bidirectional refinement before comparison, while the integral table of species separately publishes the internal genealogy, the dimensional chart, and the external comparison.

94.4.1.4 Conjugation, anti-section, and Möbius band

On a dodecaphasic word the exact involution is defined

$$C_M(\tau)_m = -\tau_{11-m}, \qquad C_M^2 = I.$$

Linear functionals with symmetric weights are odd under C_M, whereas quadratic correlations compatible with reversion are even. Thus, orientation and linear charge reside in the odd sector, whereas quadratic memory resides in the even sector. This does not prove that the complete exponential mass character is even: if $\chi(\tau) = e^{\ell(\tau)}$ and ℓ is odd, then $\chi(C_M\tau) = \chi(\tau)^{-1}$.

The preceding involution acts exactly on words. To speak of an anti-section, one additionally fixes a lift $\widetilde{C}_M$ to the bundle of sections that preserves admissibility, record, and truncations:

$$\operatorname{res}_{m+1,m} \widetilde{C}_M = \widetilde{C}_M \operatorname{res}_{m+1,m}, \qquad \operatorname{sh}(\widetilde{C}_M x) = C_M \operatorname{sh}(x).$$

Only under that lift is the anti-section of s $\widetilde{C}_M s$ on the inverted route.

The word "Möbius" has here a precise relative formulation. Let ℓ_{ret} be a return loop of the enriched route that generates

$$\langle \ell_{\text{ret}} \rangle \simeq \pi_1(S^1) \simeq \mathbb{Z}.$$

This return group is not the finite phase C_9. Suppose that its lift to the ledger has a sign character

$$\varepsilon_x : \langle \ell_{\text{ret}} \rangle \longrightarrow \{\pm 1\}, \qquad \varepsilon_x(\ell_{\text{ret}}) = -1.$$

The associated interval bundle is

$$\mathcal{M}_x = ([0,1] \times [-1,1])/(0,u) \sim (1,-u).$$

Its first Stiefel–Whitney class is nontrivial; therefore $\mathcal{M}_x$ is a Möbius band. One circuit exchanges the two local sheets, and two circuits restore the orientation. This does not require the complete record to return to the same state: orientation return and memory return remain distinct types.

The theorem is relative to the lift, the return loop, and the character ε_x, all of which are declared. It does not arise from observing two signs. The reading of sheets additionally requires distinguishing the split grading R from the conjugation C and imposing

$$CRC^{-1} = -R, \qquad CT(C^*)C^{-1} = T(-C^*).$$

The particle–antiparticle reading ultimately requires the physical representation to invert odd carriers, preserve even quadratic carriers, and intertwine evolution and the complete character in the form declared by the representation. Under those premises, particle and antiparticle are the two conjugate orientations of a single persistence band.

Theorem 94.2 (Even–odd decomposition on a double cover)**.** *Let $p : \widetilde{X} \to X$ be a principal double cover with involution ι. For every $f \in C^0(\widetilde{X}, \mathbb{R})$, the projectors*

$$f^+ = \frac{1}{2}(f + \iota^* f), \qquad f^- = \frac{1}{2}(f - \iota^* f)$$

yield the decomposition $f = f^+ + f^-$, with

$$f^+ \circ \iota = f^+, \qquad f^- \circ \iota = -f^-.$$

The even part descends to a function on X; the odd part is a section of the associated sign line. For the orientation cover of the Möbius band, $w_1 \neq 0$ obstructs a global continuous nonvanishing section of that line.

Proof. The identities of the two projectors prove the sum and the parities. An invariant function is constant on the fibers of p and therefore descends; an anti-invariant function obeys the transition law of the sign representation. In the Möbius band, the associated line is nontrivial and its first Stiefel–Whitney class is nonzero, so every global continuous section has some zero. The conclusion excludes a global continuous section *without zeros*; it does not deny the existence of sections with zeros. □

This classic lemma types the even–odd sector of the cover; by itself it identifies neither word reversion, return holonomy, split gradation, sheet conjugation, nor the parity of a mass signature. Those identifications require their own intertwiners.

94.4.1.5 Algebraic TRIT and temporal reader

The declared quadratic realization associates an algebraic type with the visible sign of the TRIT. For $\bar{\tau} \in \{+1, 0, -1\}$, let

$$\mathcal{A}_{\bar{\tau}} = \mathbb{R}[J]/(J^2 + \bar{\tau}).$$

Then

$\bar{\tau}$	J^2	regime
$+1$	-1	elliptic/complex
0	0	parabolic/nilpotent
-1	$+1$	hyperbolic/split.

The corresponding exponentials are

$$e^{tJ} = \cos t + J \sin t, \qquad e^{tJ} = 1 + tJ, \qquad e^{tJ} = \cosh t + J \sinh t,$$

and in the split sector the projectors $P_{\pm} = (1 \pm J)/2$ appear. These identities are verified exactly in the declared realization. The curvature does not arise from the isolated sign: it appears when APP supplies links and two-cells and a connection is fixed. In the published mixed sum–product polarization, over the 81 plaquettes, the oriented curvature satisfies

$$F(S,S) = F(P,P) = 0, \qquad F(S,P) = +1, \qquad F(P,S) = -1 \pmod 9.$$

The corresponding finite Stokes identity controls that the boundary of each rectangle reproduces the sum of its plaquettes.

Ordering the transitions yields the tritic temporal reader:

$$\begin{aligned} +1 &\longmapsto \text{return, memory and past,} \\ 0 &\longmapsto \text{torsional threshold and present,} \\ -1 &\longmapsto \text{bidirectional opening and future.} \end{aligned}$$

Its exact predecessor is the difference between phase return and state return:

$$g(k+9) = g(k), \qquad B_{k+9} \neq B_k \quad \text{in general.}$$

In the corresponding geometric realization, one speaks of a spherical–volumetric sector, a flat threshold with a torsional ledger, and a hyperbolic–areal sector. This correspondence defines a Dimensional Mechanics reader subject to the map between TRIT and geometry; it neither redefines the algebraic TRIT nor by itself identifies a sign with a spacetime curvature. The expression "time crystal" names this architecture in HMT terminology; a dynamic realization of a time crystal additionally requires a Hamiltonian or a periodic excitation protocol, a robust subharmonic response, and stability.

94.4.1.6 Exact support of the forward–return read

Bidirectional exchange does not rest solely on a verbal image. The integer angular coordinates

$$W_+ = 6n_A + s_C, \qquad W_- = 6n_A - s_C$$

are obtained by means of

$$\begin{pmatrix} W_+ \\ W_- \end{pmatrix} = \begin{pmatrix} 6 & 1 \\ 6 & -1 \end{pmatrix} \begin{pmatrix} n_A \\ s_C \end{pmatrix}, \qquad \mathrm{SNF}\begin{pmatrix} 6 & 1 \\ 6 & -1 \end{pmatrix} = \mathrm{diag}(1, 12).$$

The lattice therefore has index twelve. The involution $s_C \mapsto -s_C$ exchanges $W_+ \leftrightarrow W_-$ and, equivalently, $q_+ \leftrightarrow q_-$. This is the exact algebraic substrate of forward and return paths. Its interpretation as advanced and retarded propagation requires the evolution operator of the physical representation.

The even projection used in the biangular reading satisfies, for every function for which the expressions are defined,

$$\frac{f(\phi) + f(-\phi)}{2} = f_{\mathrm{par}}(\phi), \qquad \frac{(\pi - \phi)^2 + (\pi + \phi)^2}{2} = \pi^2 + \phi^2.$$

They are exact symmetrization identities. Their reading in terms of a theoretical physics model of the absorber belongs to the propagation functor, not to the isolated algebraic identity.

94.4.1.7 Inequivalent spinorial periodicities of 2π and 4π

In the classical spinorial lift,

$$U(\theta) = \exp\left(-\frac{i\theta}{2}\sigma_z\right), \qquad U(2\pi) = -I, \qquad U(4\pi) = I.$$

The equality is exact in the double cover $SU(2) \to SO(3)$. Its application to an HMT section requires specifying the spinorial representation of the transport. Once fixed, it explains why one turn changes the sign of the section and two turns restore it. If the CAR relations are adopted in the occupation algebra, exclusion follows from them; neither CAR nor Pauli is inferred from the sign of 2π without that additional structure.

The HMT interpretation "areal sheet 2π, volumetric sheet 4π" is another object. Section 60.2 formalizes its core in topological terms: one circuit of the base closes the projected boundary in 2π and ends on the conjugate sheet, whereas the squared loop closes the oriented cover in 4π. The theorem is relative to the declared lift, holonomy, and geometric reader. Its further identification with a spinorial representation still requires the physical functor and is not deduced from their sharing the same factors.

94.4.1.8 Map between internal multisections and physical representations

The particle object, the projection from extension to route, the internal selector of the 81 cells, the electron center, the thirteen finite family multisections, their lift relative to the cellular reader π_{81}, the color trit, the register, the signature, the character, and the towers form the composition

$$\text{extension} \longrightarrow \text{route} \longrightarrow \text{multisection} \longrightarrow \text{signature} \longrightarrow \text{character} \longrightarrow \text{towers}.$$

The remaining obligation is to construct a realization functor that pairs certain individualized and composite physical labels with the relevant enriched multisections and reproduces the declared refined signatures without consulting target masses or signatures. For composite states, the register is composed before being projected onto five coordinates. The domain of the functor preserves the central electron, the atlas of families, and the internal character law.

Persistence and correspondence conditions between atlases, families, and representations Route persistence, anti-sections, the bidirectional reading, and the tritic temporal reader fix the starting architecture. The inverse system, the internal selector, and the separation of types realize that architecture through declared morphisms. The atlas censuses and the identities of C_M on words hold throughout the finite system; their lift to sections and the Möbius band are proved from explicit starting structure. The electron coefficients $(-22, +15)$ are deduced from the same explicit starting structure. If a lift first opens a metrological difference or a target signature, its value is a calibrated reconstruction.

94.5 Structural definitions of particle, mass, charge, spin, statistics, color, flavor and generation

The full definition is completed by the representation domains, the conjugation conditions, the CAR/CCR completions, the gauge sectors, composition, virtuality, and positivity controls.

94.5.1 HMT–MD Ontology of Particles

94.5.1.1 Material genealogical state: extension, route, record, signature, and representation

A complete HMT state is not a bare point. It is the tuple

$$\mathfrak{x} = (\Gamma, \mathcal{L}_\Gamma, \sigma_\Gamma, \eta_\Gamma, \mathcal{R}_\Gamma, \mathcal{B}_\Gamma, C_\Gamma),$$

where Γ is a surviving route, $\mathcal{L}_\Gamma$ its genealogical record, σ_Γ its integer signature, η_Γ its harmonic refinement, $\mathcal{R}_\Gamma$ its representation, $\mathcal{B}_\Gamma$ its return memory, and C_Γ its composition or boundary datum. Two states with the same visible word may be distinct if they differ in prefix, block, memory, carry, sheet, or Witt shadow.

94.5.1.2 Particle as a persistent local section of the dimensional genealogical state

A particle is an equivalence class of persistent sections that satisfies four conditions:

1. compatible prolongation through the TPK cylinders;
2. phase return with state memory;
3. irreducible physical representation or minimal invariant block;
4. spectral stability under the coupling that observes it.

A persistent section is a compatible family.

$$s = (s_K)_{K \geq K_0}, \qquad p_{L,K}(s_L) = s_K \quad (L \geq K \geq K_0).$$

Two sections s and s' representan the same particle when there exists a level K_1 and a family compatible of simetrias internal $g_K \in G_K$ tal that

$$s'_K = g_K s_K \quad (K \geq K_1), \qquad p_{L,K} g_L = g_K p_{L,K},$$

and the observable characters of the genealogical record coincide. This relation identifies descriptions that differ by internal gauge, but it does not identify routes with distinct memory, carry, sheet, block, or holonomy. The particle is the class

$$[s]_{\mathrm{pers}} = \{s' : s' \sim s\},$$

not a single visible word nor a cell chosen after knowing its mass.

If $\widetilde{\Gamma}_K$ is the extension at level K, persistence is expressed by

$$p_{K+9,K}(\widetilde{\Gamma}_{K+9}) = \widetilde{\Gamma}_K, \qquad g(K+9) = g(K),$$

without requiring

$$\mathcal{B}_{K+9} = \mathcal{B}_K.$$

The particle is not the returning phase, but the genealogical identity preserved through the change of state. Its mass measures the spectral rigidity of that persistence when it is coupled to a material realization.

94.5.1.3 Antiparticle as the conjugate image of a persistent section under sheet inversion

The dodecaphonic conjugation is

$$C_M(\mathbf{t}) = -\operatorname{rev}(\mathbf{t}).$$

The particle–antiparticle pair is formed by two orientations of the same genealogical band. If χ denotes the sheet orientation,

$$(\mathbf{t}, \chi) \sim (C_M \mathbf{t}, -\chi).$$

For a band energy

$$E_{\mathrm{band}}(\mathbf{t}, \chi) = E_+(\mathbf{t}) + \chi E_-(\mathbf{t}),$$

with

$$E_\pm(\mathbf{t}) = \frac{1}{2}\big(E_0(\mathbf{t}) \pm E_0(C_M \mathbf{t})\big),$$

we obtain

$$E_{\mathrm{band}}(C_M\mathbf{t},-\chi)=E_{\mathrm{band}}(\mathbf{t},\chi).$$

The equality of masses is therefore a consequence of the operator's parity under conjugation, while charge and orientation change. Equality of widths additionally requires the decay operator to be conjugated by C_M.

94.5.1.4 Mass as a spectral character of persistence of 1 *a material route*

Mass is the spectral measure, on the line $\mathcal{L}_M$, of the operator resulting from compressing the character, towers, bidirectional reader, and clock onto the persistent subspace selected by the coupling:

$$\widehat{M}_{P,a}=P_{P,a}\,\mathbf{m}_\star^{\mathrm{ret}}R_{\mathrm{act}}^{\widehat{K}_{P,a}/2}\,\widehat{\mathcal{A}}_{P,a}^{(0)}\,R_{\mathrm{act}}^{\widehat{K}_{P,a}/2}P_{P,a}.$$

The general output is

$$\mu_{P,a}^{\mathrm{mass}}(B)=\mathrm{Tr}\Big(\rho_{P,a}E_{\widehat{M}_{P,a}}(B)\Big).$$

A scalar mass m_P exists if and only if

$$P_{\mathrm{supp}\,\rho}\widehat{M}_{P,a}P_{\mathrm{supp}\,\rho}=m_P P_{\mathrm{supp}\,\rho},$$

equivalently, if the variance of $\widehat{M}_{P,a}$ in the prepared state is zero. This definition includes the rank-one electron, the multiplets forced by Schur's lemma, and isolated resonance poles, without conflating them.

94.5.1.5 Charge as a transported caliber character by the material section

Charge is the oriented character of the odd layer of the genealogical record under conjugation. Let σ_{odd} be the odd signature. Then

$$Q(C_M\Gamma)=-Q(\Gamma),\qquad Q(\Gamma)=\mathfrak{q}(\sigma_{\mathrm{odd}}(\Gamma)),$$

where $\mathfrak{q}$ is the homomorphism from the odd signature to the charge lattice of the realization. The even mass signature and the odd charge signature are distinct objects: a word may have zero linear sum and nevertheless carry a nonzero mass signature.

94.5.1.6 Spin as a representation of the rotational action on the material fiber

Spin is the holonomy class with which a persistent section responds to a complete rotation of the orientation frame. The internal root

$$J^2=-I$$

separate the two closures:

$$U(2\pi)=\begin{cases}+I, & \text{tensorial sector},\\ -I, & \text{spinorial sector},\end{cases}\qquad U(4\pi)=I.$$

An integer-spin boson lies in a representation that closes at 2π; a half-integer-spin fermion requires the double circuit 4π. The spin value is not deduced from an isolated cardinal count: it is the weight of the representation of the holonomy group on the persistent subspace. In particular:

$$s=0:\ \text{scalar section},\qquad s=\tfrac{1}{2}:\ \text{spinorial band},$$

$$s=1:\ \text{transverse vector section},\qquad s=2:\ \text{tensor component of a metric realization}.$$

The last row defines a representation type; it does not require the existence of an elementary spin-two particle.

Chirality and helicity are subsequent and distinct data. An oriented involution $\Gamma_{\mathrm{ch}}^2 = I$ defines

$$P_L = \frac{I - \Gamma_{\mathrm{ch}}}{2}, \qquad P_R = \frac{I + \Gamma_{\mathrm{ch}}}{2},$$

whereas helicity is the spectral sign of the rotation generator along the direction of propagation. The former classifies sheets of the representation; the latter depends on the kinematic state. This separation allows a neutrino to be fermionic through its closure 4π, even though its weak coupling selects a single chirality.

94.5.1.7 Exchange symmetry induced quantum statistics

Statistics is the rule of completion of the space of identical paths:

$$\mathcal{F}_-(\mathcal{H}) = \bigoplus_{n\geq 0} \wedge^n \mathcal{H}, \qquad \mathcal{F}_+(\mathcal{H}) = \bigoplus_{n\geq 0} \mathrm{Sym}^n \mathcal{H}.$$

In the exterior completion, the CAR relations

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \qquad \{a_i, a_j\} = 0$$

imply

$$n_i^2 = n_i$$

and therefore Pauli exclusion and the Fermi–Dirac distribution. In the symmetric completion, the CCR relations permit multiple occupancy and lead to Bose–Einstein. Statistics are not inferred from mass: they follow from how the route is composed under exchange.

94.5.1.8 Color as residual representation of $SU(3)$ on the quark routes

Color is the residual triadic orbit of a quark section within the charge and flavor fiber. The thirty-six colored positions decompose as

$$12_R + 12_G + 12_B.$$

The physical object is not one of those colors, but the triple and its action:

$$\mathcal{H}_q^{\mathrm{color}} = \mathcal{H}_R \oplus \mathcal{H}_G \oplus \mathcal{H}_B.$$

If the mass operator commutes with the irreducible color action,

$$U_g \widehat{M}_q U_g^* = \widehat{M}_q,$$

Schur's lemma forces a common mass in the triplet. Choosing a red, green, or blue cell as representative would destroy covariance.

94.5.1.9 Taste as a class of route within a material generation

Flavor is an inequivalent coupling class within a single gauge representation, characterized by charge, generation, leaf, fine signature, memory, and interaction projector. Two flavors may share a coarse fiber and differ in the intertwiner coupling them to the reader.

Formally, if F is the common fiber, the flavors f_i are representations π_i for which

$$\mathrm{Hom}_{G_F}(V_{f_i}, \ell^2(F)) \neq 0$$

and whose projectors P_{f_i} are not conjugate under a gauge symmetry that remains physical. Flavour is not the depth of the atlas and does not reduce to a conventional name.

94.5.1.10 Material generation as a typed recurrent layer, stratified by return depth, fine signature, and reader spectrum

A generation is a recurrent stratum of the same grammar of charge and representation, distinguished by return depth, fine signature, and reader spectrum. The generation label is physical only when there is a projector that intertwines that stratum with the coupling channel. A depth G_j must therefore not be identified automatically with the electron, muon, tau, or sterile neutrino.

94.5.1.11 Gauge field as a connection on the charge fibers

A gauge field is the realization of a frame redundancy over the fibers. Its elementary bosons belong to the null direction of the mass operator while the symmetry remains exact. A gauge mass is not obtained by approximating a positive character to zero, but through a breaking, a longitudinal coupling, or a declared pole realization.

94.5.1.12 Composite particle as a persistent section of the product of linked paths

A composite particle is a persistent section of the composition operator, not the sum of the masses or signatures of its constituents. If $\Gamma_1, \dots, \Gamma_n$ are constituent routes, first construct

$$\mathfrak{C}_{12}(\Gamma_1, \dots, \Gamma_n; \mathcal{T}, \partial, \mathrm{carry}) = \mathcal{L}_{\mathrm{comp}},$$

where $\mathcal{T}$ is the temporal tree and ∂ the link boundary. Only afterwards does the reader act

$$\mathcal{L}_{\mathrm{comp}} \longrightarrow \sigma_{\mathrm{comp}} \longrightarrow \chi_{\mathrm{comp}} \longrightarrow \widehat{M}_{\mathrm{comp}}.$$

The binding energy resides in the composition cocycle, boundary memory, and modification of the spectrum; it is not introduced as an external correction.

94.5.1.13 Resonance as a transient section and width as a decay rate

A resonance is a persistent non-unitary mode of the coupled operator. If

$$\lambda = re^{-i\theta}, \qquad 0 < r < 1,$$

then

$$E = \frac{\hbar_{\mathrm{int}}}{t_0}\big(\theta - i \log r\big) = M - \frac{i}{2}\Gamma.$$

The mass M and width Γ are the real and imaginary parts of a pole, but they require distinct readers. The width may also be obtained from a compressed semigroup:

$$T_X = Q_X U Q_X, \qquad \lambda_X = -\frac{1}{t_0} \log \rho(T_X), \qquad \Gamma_X = \hbar_{\mathrm{int}} \lambda_X.$$

No width is deduced solely from the fact that a route possesses finite horizon.

94.5.1.14 Virtual section as an intermediate non-asymptotic extension of the atlas

A section is virtual when its prolongation ends in a non-trivial manner:

$$\mathrm{Ext}_L(s) \neq \varnothing, \qquad \mathrm{Ext}_{L+1}(s) = \varnothing, \qquad \mathcal{L}(s) = L < \infty.$$

Its primary content is the horizon L, the terminal obstruction, the finite genealogical register, and the propagation amplitude. It is neither an imaginary-mass particle nor an as-yet-unclassified documentary state.

94.5.2 Complete classification of particles and physical states

The realizations are ordered by object, multiplicity, and codomain. The mathematical domain contains 81 positions, grouped into distinct codomains. The mathematical domain contains 81 positions, grouped into 56 route families and 13 extension classes; the 324 oriented routes adjoin the initial orientation and do not represent additional masses or particles. The "evaluation operator" column does not impose a mass where the object requires a null chart, a prior composition, a topological invariant, or a prolongation obstruction.

94.5.2.1 Spectral definitions of particle, spin, color, and flavor

A *particle* is a surviving section realized as an oriented route, endowed with a register, a symmetry representation, a spectral operator, and a physical chart. A fermion realizes the exterior grading and the CAR relations; a boson realizes the symmetric completion and the CCR relations. Spin records the holonomy of orientation: the return of 2π closes the bosonic phase, whereas the spinorial section recovers its state after 4π. This distinction links the double areal–volumetric sheet to parity and Pauli exclusion.

The *color* is the ternary representation that lifts the coordinate $r - c \pmod 3$ to the fundamental orbit and thereafter to the adjoint of $\mathfrak{su}_3$. The *flavor* identifies a realization basis within a representation with fixed charge, color, and spin; CKM and PMNS transport between the interaction basis and the spectral basis. Neutrinos are neutral fermionic sections: their flavor labels belong to the mixing reader and their masses to the eigenvectors of the corresponding operator.

A virtual particle is a finite section with a ledger and propagator up to a horizon of prolongation; a Majorana realization is the real fermionic sector fixed by conjugation; and the Fibonacci, Ising, and $\mathbb{Z}/6\mathbb{Z}$ sectors publish quantum dimensions, fusion rules, and braid characters. Each definition thus preserves the codomain assigned to it by the structure.

The table also preserves the quantum realizations abbreviated by its classification. The three charged leptons and the six quark flavours belong to the exterior completion: CAR leads to Pauli and Fermi–Dirac, without attributing those relations to an isolated TRIT. The quarks occupy thirty-six cells when colour and antiparticle are conserved before the quotient. The common neutral support $\mathcal{H}_\nu = \bigoplus_{j=1}^{4} \mathcal{H}_{\nu,G_j}$ has dimensions $2 + 4 + 6 + 8 = 20$. The symbols G_j designate depth strata of the atlas; f_e, f_μ, f_τ designate the interaction basis. In particular, G_4 is a stratum of the common neutral support and is not identified with a sterile or exotic state without an additional projector. PMNS transports the interaction basis to the eigenvector basis; it creates neither eigenvalues, a mass scale, nor a depth selector. The photon preserves outward passage and return within the null gauge chart: $m = 0$ does not mean evaluating χ_0 with a highly negative exponent. $W^\pm$ and Z^0 are massive bosonic representations, with charge conjugation for $W^+ \leftrightarrow W^-$. Lastly, the antisection is obtained through inversion and band reversal; mass is even under that involution and is not duplicated as a second independent parameter.

All sectors preserve the causal prefix.

$$\text{surviving extension} \longrightarrow \text{observable route} \longrightarrow \text{enriched chronological record.}$$

The reader then changes. In the positive material sector, it continues through signature → character → towers → dimensional chart. The null sector bypasses the positive character and fixes $m = 0$; hadrons first compose enriched chronological ledgers; topological sectors use fusion or braid invariants; and virtuality is recognized by terminal obstruction. The domain of these maps is the complete preceding classification. A numerical trial on a subset establishes only the finite case tested and does not reduce that domain.

The censuses accompanying this table belong to different quotients:

$$81 = 25_{\ell^-} + 20_\nu + 20_d + 16_u,$$

$$\text{56 families with fibers} \quad 41 \times 1 + 10 \times 2 + 2 \times 3 + 1 \times 4 + 2 \times 5 = 81,$$

$$13 \text{ multisections}, \qquad 324 = 81 \times 4 \text{ genealogies}.$$

Likewise, the 192-A representation classes associated with $\mathrm{diag}(2, 6, 2, 2, 4)$ and the lattice index 192-B of $\mathrm{SNF}(2, 2, 2, 2, 12)$, with 52 raw signatures, are not the The cardinality of the historical expository catalogue does not count individual particles or define the material taxonomy. not individual particles.

94.6 Persistent Sections and Finite Propagation

The mathematical notion of particle is introduced after the boundary structure and before its dynamics are specialized. Its primary support is a surviving extension of the TPK state system. An oriented trajectory is the sequential shadow of that extension; over it, the internal degrees form a discrete bundle. The propagator, the sum over paths, and the Clifford representations are incorporated after the line of action has been constructed. This hierarchy prevents a particle from being defined through the external selection of a row or route and separates the mathematical object from its subsequent identification with a physical species.

94.6.1 Surviving extension and observable projection

Let $(\mathcal{S}_m, \rho_{m+1,m})$ be the inverse system of states that pass the closures up to depth m. A global state is a compatible family

$$x = (x_m)_m \in \Omega_{\mathrm{HMT}}^{\mathrm{surv}} := \varprojlim_m \mathcal{S}_m.$$

A sequential chart of depth R produces a route

$$\gamma_x^{[R]} = \mathrm{sh}_R(x) = (c_0 \longrightarrow \cdots \longrightarrow c_R).$$

The symbol sh denotes a shadow reading: different charts may present the same global state, provided that they preserve the sheet, orientation, boundary, and record. In section 91.5, the enriched record is constructed so that the action character descends from these presentations to the extension.

94.6.2 Particle as a compatible section of a discrete bundle

Let Γ_{TPK} be the graph whose vertices are complete states of the Topological Phase Kernel and whose edges are admissible transformations. For $x \in \Omega_{\mathrm{HMT}}^{\mathrm{surv}}$ and its observed trajectory

$$\gamma_x = (c_0 \longrightarrow c_1 \longrightarrow \cdots \longrightarrow c_R),$$

let $\mathrm{Typ}(c_t)$ be the finite set of arithmetic types, orientations, and memories over the state c_t, and let $E_{c_t} = \mathbb{C}[\mathrm{Typ}(c_t)]$ be the free complex vector space spanned by those types. The discrete bundle restricted to γ_x is

$$\mathcal{B}_{\gamma_x} = \bigsqcup_{t=0}^{R} E_{c_t} \longrightarrow \{0, \ldots, R\}.$$

Each edge $c_t \to c_{t+1}$ induces a linear transport map $\tau_t : E_{c_t} \to E_{c_{t+1}}$.

Definition 94.3 (Mathematical particle)**.** A particle of Dimensional Mechanics is a tuple

$$\mathfrak{p} = (x, \gamma_x, \mathcal{B}_{\gamma_x}, s, n, U),$$

where $x \in \Omega_{\mathrm{HMT}}^{\mathrm{surv}}$ is the surviving extension and $\gamma_x = \mathrm{sh}_R(x)$ its shadow in the stated chart; moreover,

$$s : \{0, \ldots, R\} \longrightarrow \mathcal{B}_{\gamma_x}, \qquad s(t) \in E_{c_t}, \qquad s(t+1) = \tau_t(s(t)),$$

is a compatible section;

$$n : \bigsqcup_{t=0}^{R} \mathrm{Modes}(E_{c_t}) \longrightarrow \mathbb{N}$$

is its occupation function, where $\mathrm{Modes}(E_{c_t}) := \mathrm{Typ}(c_t)$ denotes the distinguished basis of the free vector space; and

$$U_{t_2,t_1} : E_{c_{t_1}} \longrightarrow E_{c_{t_2}}$$

is a family of linear propagators that intertwines the transports, $U_{t,t} = \mathrm{id}_{E_{c_t}}$, $U_{t+1,t} = \tau_t$, and satisfies

$$U_{t_3,t_2} \circ U_{t_2,t_1} = U_{t_3,t_1} \qquad (0 \le t_1 \le t_2 \le t_3 \le R).$$

Each composition is defined between the corresponding fibers. The identification of $\mathfrak{p}$ with a physical species additionally requires a representation of the pertinent symmetry group and a map from HMT states to the states of that representation.

The extension preserves the genealogy and the register; the route exposes its ordered reading; the internal degrees belong to the fibers, occupation to n, and evolution to U. The particle is the compatibility of these levels, not any one of them separately. In particular, the route is not chosen by the particle: it is a projection of the extension that realizes it.

94.6.3 Quantum Ontology of the Particle

94.6.3.1 State, history and realization

A genealogical state is a compatible section of the TPK system of extensions. An observable history is the chronological projection of that section onto a reader. A physical realization is the representation of the history on a dimensional line, a Hilbert space, a field algebra, or a topological codomain. These three notions are not interchangeable. The state carries identity; the history carries evidence; the realization carries the observable.

Let $\mathcal{X}^{\mathrm{enr}}_{\mathrm{TPK}}$ be the space of enriched states and p the observable reader. A particle is a class $[X]$ of persistent sections such that

$$p(X_{K+1})|_K = p(X_K),$$

the transition law preserves the relations of the genealogical record, and the prolongation does not terminate at an obstruction horizon. The route

$$\gamma_X = (p(X_0), p(X_1), p(X_2), \ldots)$$

is the history visible of the particle. Not is a trajectory espacial by definition. A realization metric posterior may convertirla in curve, orbit or propagador when constructs the entrelazador corresponding.

The physical identity is preserved under refinement when the squares

$$\begin{array}{ccc} X_{K+1} & \xrightarrow{F_{K+1}} & X'_{K+1} \\ {\scriptstyle r_{K+1,K}}\downarrow & & \downarrow{\scriptstyle r'_{K+1,K}} \\ X_K & \xrightarrow[F_K]{} & X'_K \end{array}$$

commute. Mass, charge, and spin are different readers of that same antecedent. Equality of one does not entail equality of the others.

94.6.3.2 Quantum representation of the particle as a persistent local section of the genealogical state

An HMT–MD particle is a persistent section of an enriched extension satisfying four conditions:

1. has an observable route compatible with the nonadic monodromy;
2. preserves a genealogical record under refinement;
3. admits a representation of orientation, charge, spin, and statistics;
4. presents a stable spectral output in the declared physical codomain.

Stability does not mean immobility. It means that changes of phase, sheet, and memory belong to the same genealogical class. The particle may close its phase but not its state, change sheets while preserving its mass, or traverse several cells while retaining the same extension identity.

A species is a family of particles that shares the internal classifier.

$$\mathfrak{s} = (\mathfrak{q}, \mathfrak{j}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{r}, \mathfrak{o}),$$

where $\mathfrak{q}$ is the charge representation, $\mathfrak{j}$ the spin holonomy, $\mathfrak{c}$ the color orbit, $\mathfrak{f}$ the flavor, $\mathfrak{g}$ the generation, $\mathfrak{r}$ the route family, and $\mathfrak{o}$ the mass operator. A non-central species may correspond to a complete multisection; requiring it to have a single APP point before declaring the physical coupling is a type error.

94.6.3.3 Quantum conjugation of the antiparticle via sheet inversion of a persistent section

The dodecaphasic involution is

$$C_M(\tau) = -\operatorname{rev}(\tau).$$

It acts on the oriented history and prolongs to the genealogical ledger. Charge and linear orientation are odd; quadratic memory is even. If q is the charge reader,

$$q(C_M X) = -q(X).$$

Equality of masses further requires operatorial covariance. If K realizes the conjugation in the physical space and

$$K\widehat{M}_X K^{-1} = \widehat{M}_{\bar{X}},$$

also transporting the channels and the causal prescription when K is antiunitary, then $m_{\bar{X}} = m_X$. The parity of the record by itself does not replace this premise.

Particle and antiparticle are two oriented sections of the same genealogical band. In the Möbius realization, one circuit interchanges the sheet and two circuits restore the orientation. The conclusion does not follow from drawing a twisted ribbon: it requires the involution, the lift to the double cover, the return loop, and the sign holonomy.

For the electronic record $\tau_e = +++---+++--$,

$$C_M \tau_e = \tau_e.$$

The central word is fixed, while the charge representation distinguishes the electron from the positron. This yields the same mass without duplicating the temporal route. Closure after 216 steps does not mean 108 particle steps and 108 antiparticle steps: it means restoration of the oriented kinematic lift.

The General Mass Law is formulated as a genealogical and spectral transformation, not as a collection of numbers. Its effective domain is the chain

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \Gamma^{\mathrm{surv}} \longrightarrow \Gamma^{\mathrm{obs}} \longrightarrow L_{108} \longrightarrow \sigma \in \mathbb{Z}^5 \longrightarrow \chi_{\mathrm{full}} \longrightarrow \widehat{M},$$

followed, when the physical representation permits it, by the functional that extracts a scalar coordinate from the mass line. Each arrow preserves its domain, orientation, memory, and status; no external figure retroactively selects a route, signature, or harmonic coefficient.

The finite atlas contains 81 cells, 56 route families, 13 multisections, and 324 oriented routes. The central electronic point is the unique equivariant point fiber. The remaining elementary classes are naturally realized as fibers, orbits, or multisections, so their primary output may be an operator or spectrum before being a scalar mass. This difference is not a deficiency of domain, but a property of the internal geometry.

In the central fiber, the basal selector and the bidirectional reader produce

$$\mathfrak{e}_0 = \frac{\sqrt{3}}{4}\left[\exp\left(\frac{\varphi}{\pi^2}\right) - 22\alpha_{\mathrm{HMT}}^3\right](1 + 15\Delta_4), \qquad \kappa_e = 80 - 54\alpha_{\mathrm{HMT}} + 6\Delta_4,$$

$$\mathfrak{e}_e^{\beta} = \mathfrak{e}_0 R_{\mathrm{act}}^{\kappa_e}, \qquad \varsigma_{u_E}(\mathfrak{e}_e^{\beta}) = 0.510998950690000808608\ldots\ \mathrm{MeV}.$$

The absolute action scale remains on the $\Sigma_H = \varphi\alpha_{\mathrm{HMT}}^{16}$ branch, of decimal order -34, and is coupled to the clock through $\hbar_{\mathrm{int}}\omega$; when mass ratios are formed, every common section cancels and only the correctly typed characters, readers, and relative refinements survive.

The harmonic hierarchy does not consist of eight isolated corrections. It is the global semigroup.

$$\langle 90, 120\rangle = \{90, 120\} \cup \{30r : r \geq 6\}, \qquad c_n = q_-^n + q_+^n, \quad s_n = q_-^n - q_+^n,$$

acting on a single bilayer operator. The visible levels are witnesses to that hierarchy; their representational completeness does not replace the physical map that must transport route, carry, memory, and winding to the coefficients of each state.

The typed physical realizations, the seventeen conditioned evaluations, the two null sections, and the neutrino, composite, resonant, topological, and virtual sectors are thus brought together without flattening their codomains. The subsequent inventory of 471 masses and 384 widths preserves the state-by-state comparison but is not conflated with a census of predictions. A quantitative comparison acquires physical force only after its observable, unit, scheme, scale, uncertainty, covariance, and realization rule have been declared.

The structural conclusion is precise: mass is the spectral measure of persistence under coupling. The particle is the persistent section; the route, its observable history; the signature, its integer reading; the character, its dimensionless action; the towers, its harmonic resolution; and the dimensional chart, the realization of that genealogy in physical units. The preceding atlas and inventories permit each claim to be reproduced without reducing the theory to an abbreviated table or turning an open interface into a closed theorem.

94.7 Composition and identity

A composite particle retains its identity when the binding record, the boundary, the representation, and the observables remain compatible under refinement. A resonance adds a pole and a width; it does not constitute a new coordinate of the mass signature. The identity criterion is refuted if it depends on the external value or if two inequivalent histories are identified without an isomorphism of sections.

94.8 Equivalence between the persistent section and its finite representation

The genealogical definition of a particle and its operator representation are two presentations of the same object. Let

$$\mathfrak{r}_K = (x_K, \gamma_K, B_K, s_K, n_K, U_K)$$

a finite representative at depth K, with state, route, boundary, orientation, counter, and representation, and let $r_{L,K} : \mathfrak{R}_L \to \mathfrak{R}_K$ be the refinement maps. A persistent section is a family

$$X = (\mathfrak{r}_K)_{K \geq K_0}, \qquad r_{L,K}(\mathfrak{r}_L) = \mathfrak{r}_K \quad (L \geq K \geq K_0).$$

Two families are equivalent if they coincide on a cofinal tail.

Theorem 94.4 (Equivalence of the Persistent Section and the Inverse Limit)**.** *The category of persistent sections, defined as compatible families, is canonically equivalent to*

$$\varprojlim_K \mathfrak{R}_K / \sim_{\text{cola}} .$$

Projection to a finite depth is an evaluation, not an inverse functor. A representative $\mathfrak{r}_K$ determines a unique persistent section if and only if it has a unique compatible prolongation modulo tail equivalence; this uniqueness is obtained, for example, when the refinement maps are bijective on the surviving tail from K onward.

Proof. By definition, a persistent section is precisely a point of the inverse limit; the quotient identifies families that coincide on a cofinal tail. Evaluation is the canonical projection onto one component. Its fiber contains all compatible prolongations of the evaluated representative, so that the fiber is a singleton exactly under the indicated unique-prolongation condition. □

An HMT–MD particle is therefore a class of persistent sections without a finite terminal obstruction, endowed with representation, orientation, charge, spin, and spectral character compatible with refinement. The route is its observable history, not its identity reduced to a cell. The antiparticle is the section conjugated by the dodecaphasic involution and the conjugate representation. An obstructed finite section is virtual; a cocyclic product of sections is composite; a persistent braided representation is anyonic.

94.9 Theorem of the Closure of the Material Realization via Persistent Sections and Finite Representations

Theorem 94.5 (Closure by physical codomain)**.** *Suppose that the atlas* 81/56/13/324*, the null sector, the action–clock–mass scale, the readers K_D, K_Ω, the physical descriptor, and the internal representation of each state have been constructed. Then every admissible and realizable descriptor, that is, every X with $d_{\mathcal{P}}(X) \in \operatorname{im} d_{\mathcal{R}}$, determines, without consulting its external value:*

1. *a fiber, orbit, or multisection $\mathcal{S}(X)$;*
2. *the positive bidirectional pair and its joint PVM $(\widehat{M}_X^{\beta,D}, \widehat{M}_X^{\beta,\Omega}, E_X^{D,\Omega})$;*
3. *a decomposition into irreducible blocks;*
4. *a spectral measure for each physical coupling;*
5. *a scalar in a compressed rank-one sector or a multiplet in a reducible sector; in an open sector, an effective operator and, only if a meromorphic continuation with an admissible isolated zero exists, a complex pole;*
6. *a categorical object when the realization is topological and does not yet have a Hamiltonian gap;*

7. *a nonexistence criterion when the requested type contradicts positivity, gauge, or asymptoticity.*

None of these results requires selecting a route through mass that will later be tested.

Proof. The fiber product constructs the fiber prospectively. The character and the two readers construct the positive pair and its joint PVM. When the physical type declares a scalar functional, functional calculus produces the corresponding operator; conditional expectation projects it onto the commutant of the symmetry. The isotypic decomposition produces scalars or blocks by Schur's lemma. Preparation projectors produce spectral measures. Feshbach reduction produces the effective operator; meromorphic continuation and an admissible isolated zero produce a pole, if one exists. Topological sectors retain their fusion and braiding data until a Hamiltonian realization exists. Positivity and gauge constraints exclude incompatible types. Each step depends only on internal data or the declared coupling, never on the external column. □

94.10 Consistency and Incompatibility Conditions of Spectral, Fermionic, and Composite Realizations

The construction is refuted in its domain if any of the following events occurs:

1. modifying the external values changes the fiber $\mathcal{S}(X)$;
2. the assignment ceases to be equivariant under cellular inversion, conjugation, or gauge;
3. a point cell is selected in a noncentral multisection without a coupling that breaks the symmetry;
4. the synthesis of K_D and K_Ω privileges an orientation or does not recover the diagonal when both readers coincide;
5. a compression with several eigenvalues is replaced by a mean without a physical projector;
6. the color quotient produces different masses for red, green, and blue;
7. quark masses are compared without declaring the same scheme and the same scale;
8. a rank-one phase is presented as PMNS mixing after failing the full-rank angular contrast;
9. neutrality is used by itself to assert the Majorana character;
10. the mass of a composite is obtained by adding constituent masses without a binding record;
11. the width is inserted as a coordinate of the signature instead of being extracted from an open mode or a pole;
12. a negative eigenvalue of the Hessian is published as a tachyonic particle;
13. gravitation is made dependent on an elementary graviton introduced before connection, curvature and torsion.

94.11 Posterior metrological recognition and exhaustive inventory of the material realizations

External comparison begins only after the descriptor, fiber, operator, projector, scheme, and scale have been frozen. Each row must distinguish stable mass, pole mass, resonance estimator, and width. The atlas of 324 records and the inventories of 471 masses and 384 widths constitute the subsequent exhaustive presentation. The conceptual construction and each record cross-reference one another through their route identifiers. The records retain all their fields and are composed as coherent units; they are not replaced by abbreviated historical tables.

The comparison does not consist in enlarging a list of names, but in applying, for each physical type, the complete map that leads from the genealogy of route to its observable codomain.

94.12 Graded completion of channels and exclusion principle

The construction of a particle provides a space of channels, their labels and the operators transporting orientation and memory. The composition of several particles adds a distinct question: how those channels are exchanged and how their occupations are counted. Here that layer is constructed from the data already produced by APP–TRIT–TPK. Mass and the Hamiltonian can then act on the resulting space; their values do not enter the proof of the occupation rules.

The parity used is that of sheet transport in the representation of a particle. It is not the sign of the charge, the parity of a visible integer or the third coordinate of TRIT considered in isolation. The different readings preserve their maps. Section 90.15 constructs the oriented return and the spinorial representation; this section develops the multiparticle algebra corresponding to a declared grading and exchange rule.

94.12.1 Finite channel space and parity character

Given a depth K, let S_K be the finite set of channels of the realization under consideration. Its elements preserve the route and representation data that still distinguish states: sheet, orientation, boundary, memory, flavor, color and spin component when they belong to the sector. The quotient defining S_K is taken before constructing occupation; it does not identify two channels solely through equality of their masses.

The one-particle space is

$$\mathcal{H}_K = \bigoplus_{s \in S_K} V_s, \qquad \langle e_{s,a}, e_{t,b} \rangle = \delta_{st}\delta_{ab}, \tag{94.1}$$

where V_s carries the declared representation multiplicity. Each $e_{s,a}$ is a complete mode, not an unlabeled cell of the atlas. Its positive inner product belongs to that finite realization and makes explicit the normalization used below.

The parity datum is a compositional character of transport,

$$p : \mathrm{Mor}(\mathsf{P}_{\mathrm{TPK}}) \longrightarrow \mathbb{Z}/2\mathbb{Z}, \qquad p(\gamma\eta) = p(\gamma) + p(\eta). \tag{94.2}$$

On the channels in which that grading is realized, the holonomy involution J_p satisfies $J_p e = (-1)^{p(e)} e$. Therefore,

$$\mathcal{H}_K = \mathcal{H}_{K,\bar{0}} \oplus \mathcal{H}_{K,\bar{1}}, \qquad \Pi_{\bar{0}} = \frac{I + J_p}{2}, \quad \Pi_{\bar{1}} = \frac{I - J_p}{2}.$$

The two projectors are orthogonal because $J_p^* = J_p$ and $J_p^2 = I$. The equality (94.2) ensures that composing transports preserves the grading. When a realization has not supplied that character, the following completion is a construction relative to its explicit choice, not a statistical attribution obtained from the appearance of a route.

94.12.2 Graded exchange and permutation projectors

For homogeneous vectors, the exchange is defined

$$\tau_p(v \otimes w) = (-1)^{p(v)p(w)} w \otimes v. \tag{94.3}$$

The sign is negative exactly when both modes belong to the odd sector. Exchange does not modify a mode's labels: it transports the complete mode from one tensor position to another.

Proposition 94.6 (Graded representation of permutations)**.** *The adjacent exchanges $\tau_{p,j}$ on $\mathcal{H}_K^{\otimes n}$ are unitary and satisfy*

$$\tau_{p,j}^2 = I, \qquad \tau_{p,j}\tau_{p,k} = \tau_{p,k}\tau_{p,j} \quad (|j-k|>1),$$
$$\tau_{p,j}\tau_{p,j+1}\tau_{p,j} = \tau_{p,j+1}\tau_{p,j}\tau_{p,j+1}.$$

They therefore define a unitary representation U_p of $\mathfrak{S}_n$.

Proof. Applying an exchange twice gives the sign $(-1)^{2p(v)p(w)} = 1$ and restores both vectors. Operators acting on disjoint pairs commute. On a homogeneous triple of degrees a, b, c, both sides of the three-exchange relation produce the same final order and the sign $(-1)^{ab+ac+bc}$. Each operator permutes an orthonormal basis multiplying it by signs of modulus one, and is therefore unitary. □

The average

$$P_n^p = \frac{1}{n!}\sum_{\pi\in\mathfrak{S}_n} U_p(\pi)$$

is the orthogonal projector onto tensors invariant under graded exchange. In an even homogeneous space it coincides with the ordinary symmetrizer; in an odd homogeneous one it coincides with the antisymmetrizer. If U denotes the unsigned permutation,

$$S_n = \frac{1}{n!}\sum_{\pi} U(\pi), \qquad A_n = \frac{1}{n!}\sum_{\pi} \operatorname{sgn}(\pi)U(\pi). \tag{94.4}$$

Proposition 94.7 (Exact symmetrization and antisymmetrization)**.** *We have $S_n^* = S_n^2 = S_n$ and $A_n^* = A_n^2 = A_n$. For $n \geq 2$, $S_nA_n = 0$. Their images are, respectively,* $\operatorname{Sym}^n \mathcal{H}$ *and* $\bigwedge^n \mathcal{H}$.

Proof. Taking the adjoint replaces each permutation by its inverse and leaves the sums invariant. In the square of each average, every permutation ρ appears $n!$ times; in the second average its coefficient is $\operatorname{sgn}(\rho)$. This proves idempotence. The product of the two projectors includes the sum of the signs of all permutations, which is zero for $n \geq 2$. Finally, their images are the invariant and alternating tensors by their own averaging formulas. □

94.12.3 Universal algebra and normalization of occupation states

Let $T(\mathcal{H}) = \bigoplus_{n\geq 0} \mathcal{H}^{\otimes n}$ be the tensor algebra. The graded algebraic completion is the quotient by the ideal generated by

$$v \otimes w - (-1)^{p(v)p(w)} w \otimes v. \tag{94.5}$$

The universal property of the quotient means that every graded linear map from the channels to an algebra satisfying these relations extends uniquely to a homomorphism of the quotient. Once declared, the exchange rule thus fixes the product of occupations without resorting to target energies.

Theorem 94.8 (Symmetric and exterior decomposition)**.** *If $d_0 = \dim \mathcal{H}_{\bar{0}}$ and $d_1 = \dim \mathcal{H}_{\bar{1}}$, the graded completion has the decomposition*

$$\mathcal{F}(\mathcal{H}) = \mathcal{F}_+(\mathcal{H}_{\bar{0}}) \otimes \mathcal{F}_-(\mathcal{H}_{\bar{1}}), \tag{94.6}$$

$$\mathcal{F}_+(\mathcal{H}_{\bar{0}}) = \bigoplus_{m\geq 0} \operatorname{Sym}^m \mathcal{H}_{\bar{0}}, \qquad \mathcal{F}_-(\mathcal{H}_{\bar{1}}) = \bigoplus_{k=0}^{d_1} \bigwedge^k \mathcal{H}_{\bar{1}}.$$

The sector of total number n is

$$\mathcal{F}_n(\mathcal{H}) = \bigoplus_{k=0}^{\min(n,d_1)} \operatorname{Sym}^{n-k} \mathcal{H}_{\bar{0}} \otimes \bigwedge^k \mathcal{H}_{\bar{1}}.$$

Proof. Given ordered homogeneous bases, the relations (94.5) allow the even factors to be placed first and the odd ones afterward. Even factors commute; odd ones anticommute, and the square of an odd one is zero because we work in characteristic zero. Each monomial is written as

$$e_1^{m_1}\cdots e_{d_0}^{m_{d_0}} f_{i_1}\cdots f_{i_k}, \qquad m_j \geq 0, \quad i_1 < \cdots < i_k.$$

The realization through symmetric and alternating tensors makes those monomials linearly independent. It provides the inverse map and proves the decomposition. Grouping them by $m_1 + \cdots + m_{d_0} + k = n$ gives the last statement. □

The Hilbert completion declares the normalized occupation vectors $|\boldsymbol{m}; I\rangle$ orthonormal, where $\boldsymbol{m} \in \mathbb{N}_0^{d_0}$ and $I \subset \{1, \ldots, d_1\}$. In the bosonic sector of degree $n = \sum_j m_j$, the tensor normalization is

$$|\boldsymbol{m}\rangle = \sqrt{\frac{n!}{\prod_j m_j!}}\, S_n(e_1^{\otimes m_1} \otimes \cdots \otimes e_{d_0}^{\otimes m_{d_0}}).$$

For $I = \{i_1 < \cdots < i_k\}$, the exterior vector is

$$|I\rangle = \frac{1}{\sqrt{k!}} \sum_{\pi \in \mathfrak{S}_k} \operatorname{sgn}(\pi) f_{i_{\pi(1)}} \otimes \cdots \otimes f_{i_{\pi(k)}}.$$

In the first formula there are $n!/\prod m_j!$ distinct tensors; in the second there are $k!$. Counting them proves that both norms are one. Degree zero is $\mathbb{C}|0\rangle$; this occupation vacuum is not identified with the constitutive chart of the electromagnetic vacuum or with a vacancy in the TPK record.

94.12.4 Fermionic creation and derivation of the CAR relations

Let I be an ordered subset of odd modes and $r_j(I) = \#\{i \in I : i < j\}$. We define

$$a_j^\dagger |I\rangle = \begin{cases} 0, & j \in I, \\ (-1)^{r_j(I)} |I \cup \{j\}\rangle, & j \notin I, \end{cases}$$
$$a_j |I\rangle = \begin{cases} (-1)^{r_j(I)} |I \setminus \{j\}\rangle, & j \in I, \\ 0, & j \notin I. \end{cases} \tag{94.7}$$

The sign counts the transpositions needed to insert or remove the mode while maintaining the order of the exterior basis. It does not use its mass, its charge or the sign of a published scalar value. The matrices of both maps are adjoints and their norms are 1, except for the trivial case without odd modes.

Theorem 94.9 (Canonical anticommutation relations)**.** *The operators in* (94.7) *satisfy*

$$\{a_i, a_j^\dagger\} = \delta_{ij} I, \qquad \{a_i, a_j\} = 0, \qquad \{a_i^\dagger, a_j^\dagger\} = 0.$$

These are exact relations on the entirety of $\mathcal{F}_-(\mathcal{H}_1)$.

Proof. For $i = j$, the product $a_j^\dagger a_j$ is the identity on vectors containing j and zero on the remaining ones; $a_j a_j^\dagger$ has the complementary actions. Their sum is I. Two insertions or two removals of the same mode give zero. For $i \neq j$, if any operation is blocked, both terms of the corresponding anticommutator are zero. If both compositions are admissible, they produce the same set of occupations. Changing the order changes by one the number of modes preceding one of the insertions or removals; the signs are opposite. This proves the three anticommutators on each basis vector. □

Corollary 94.10 (Exclusion and occupation of a complete mode)**.** *For $N_j = a_j^\dagger a_j$,*

$$N_j^* = N_j, \qquad N_j^2 = N_j, \qquad \operatorname{Spec}(N_j) \subset \{0, 1\}.$$

The same complete mode admits at most one fermionic occupation.

Proof. The CAR give

$$N_j^2 = a_j^\dagger a_j a_j^\dagger a_j = a_j^\dagger (I - a_j^\dagger a_j) a_j = N_j,$$

because $a_j^2 = (a_j^\dagger)^2 = 0$. Self-adjointness is immediate; an eigenvalue of a self-adjoint idempotent equals 0 or 1. □

The proof constructs CAR before deducing Pauli: it does not introduce exclusion as an additional restriction on a particle table. It also explains the real generators used in proposition 94.31: that proposition now acts on the explicit exterior algebra. Its Majorana interpretation preserves the conditions of section 94.14.

94.12.5 Bosonic creation, CCR and finite-particle domain

On the normalized bosonic basis we define

$$b_j^\dagger |\boldsymbol{m}\rangle = \sqrt{m_j + 1}\, |\boldsymbol{m} + \boldsymbol{e}_j\rangle, \qquad b_j |\boldsymbol{m}\rangle = \sqrt{m_j}\, |\boldsymbol{m} - \boldsymbol{e}_j\rangle, \tag{94.8}$$

with value zero in the second expression when $m_j = 0$. The factors come from the normalization of symmetric tensors. The common domain $\mathcal{D}_{\text{fin}}$ is the algebraic direct sum of the number sectors: vectors have only a finite number of nonzero occupation components. It is dense and invariant under the preceding maps.

Theorem 94.11 (Canonical commutation relations)**.** *On $\mathcal{D}_{\text{fin}}$,*

$$[b_i, b_j^\dagger] = \delta_{ij} I, \qquad [b_i, b_j] = [b_i^\dagger, b_j^\dagger] = 0.$$

The number operators $N_j^B = b_j^\dagger b_j$ satisfy $N_j^B |\boldsymbol{m}\rangle = m_j |\boldsymbol{m}\rangle$.

Proof. For the same mode, $b_j b_j^\dagger |\boldsymbol{m}\rangle = (m_j + 1)|\boldsymbol{m}\rangle$ and $b_j^\dagger b_j |\boldsymbol{m}\rangle = m_j |\boldsymbol{m}\rangle$. The difference is the identity. For distinct modes, the factors affect different occupation coordinates, so they commute. The number formulas follow from consecutively applying the two operations of (94.8). □

If there is any even mode, b_j and $b_j^\dagger$ are unbounded operators in the Hilbert completion: their norms on $|m_j\rangle$ grow as roots of m_j. The preceding equalities therefore have an explicit domain and are not presented as identities between finite matrices on the entire bosonic space. Fermionic operators act on the second factor of (94.6); bosonic ones, on the first. On the common domain, the operators of those two factors commute.

Proposition 94.12 (Exact defect of a hard bosonic cutoff)**.** *In one mode, let P_N be the projection onto $|0\rangle, \dots, |N\rangle$ and let $b^{(N)} = P_N b P_N$. On that finite space,*

$$[b^{(N)}, (b^{(N)})^\dagger] = I - (N + 1)|N\rangle\langle N|.$$

Proof. For $m < N$, creation and annihilation preserve the calculation $(m + 1) - m = 1$. At $|N\rangle$, truncated creation is zero and the reverse product equals N; the commutator equals $-N = 1 - (N + 1)$. These are all the basis vectors. □

The correction resides at the last level of the cutoff, not in the occupation law. A finite calculation can verify the CCR away from that boundary and must preserve the boundary term when it includes the terminal level. This distinction is indispensable so that a matrix certificate does not confuse a truncation with the full bosonic completion.

94.12.6 Identity of complete modes and Pauli exclusion principle

Exclusion refers to the same vector of a complete mode. Two channels can share a cell, mass or reduced signature and remain linearly independent through their spin, sheet, color, flavor or another coordinate preserved by transport. Those two vectors admit a nonzero exterior product. If a physical quotient identifies them, that quotient must act before asserting how many independent modes there are.

Proposition 94.13 (Independence criterion for double occupation)**.** *For two vectors of a channel space,*

$$\|u \wedge v\|^2 = \|u\|^2\|v\|^2 - |\langle u, v\rangle|^2.$$

In particular, two orthonormal modes produce an exterior state of norm one and two proportional vectors produce zero.

Proof. The degree-two exterior realization is $(u \otimes v - v \otimes u)/\sqrt{2}$. Expanding its norm gives the formula. Equality in Cauchy–Schwarz characterizes linear dependence; orthonormality gives squared norm one. □

For an orbital channel u and two orthogonal spin components ξ_+, ξ_-, the modes $u \otimes \xi_+$ and $u \otimes \xi_-$ are distinct and their exterior product can be occupied. There is no double occupation of the same mode: there is occupation of two distinct components of the representation. The same applies to two sheets when they remain independent components of the realization.

The central electronic word may be fixed by C_M, while the charge representation distinguishes electron and positron. That coincidence of words does not identify their complete modes or decide their occupation. The word involution, cellular inversion and exchange of Fock factors preserve the differentiated domains of section 90.15.

If r orthonormal fermionic modes have the same value of a partial observable, their total occupation can take values $0, 1, \dots, r$. Each N_j remains a projector; the sum $\sum_{j=1}^{r} N_j$ need not be one. This consequence distinguishes spectral degeneracy and exclusion of a mode, two properties that an isolated mass figure does not separate.

94.12.7 Finite dimensions and transport by refinement

Proposition 94.14 (Census of occupations)**.** *For d modes,*

$$\dim \bigwedge^{n} \mathbb{C}^d = \binom{d}{n}, \qquad \dim \operatorname{Sym}^n \mathbb{C}^d = \binom{d+n-1}{n} \quad (d \geq 1).$$

The first dimension is zero for $n > d$. In the graded space, the dimensions of total number n are obtained by summing the products of both dimensions for $k = 0, \dots, \min(n, d_1)$.

Proof. An exterior basis is chosen through subsets of n distinct modes. A symmetric basis is chosen through $(m_1, \dots, m_d) \in \mathbb{N}_0^d$ with sum n. Placing $d - 1$ separators among n occupations counts those weak compositions and gives the second binomial coefficient. The graded formula follows from (94.6) through direct sums and tensor products. □

Let $T : \mathcal{H}_K \to \mathcal{H}_L$ be the declared refinement intertwiners, preserving the grading and the corresponding channel data. Each induces

$$\Gamma(T) = \bigoplus_{n \geq 0} T^{\otimes n}|_{\mathcal{F}_n}, \qquad T^{\otimes 0} = I_{\mathbb{C}}. \tag{94.9}$$

Theorem 94.15 (Compatibility of completion with transport)**.** *If T preserves degree, then $T^{\otimes n}P_n^p = P_n^{p'}T^{\otimes n}$. If also $\|T\| \le 1$, $\Gamma(T)$ is a bounded operator of norm at most* 1 *on Fock and*

$$\Gamma(T_2T_1) = \Gamma(T_2)\Gamma(T_1), \qquad \Gamma(I) = I.$$

Consequently, compatible refinement squares of channels induce compatible squares of occupations.

Proof. Degree preservation makes $T^{\otimes n}$ commute with each adjacent exchange; by proposition 94.6, it commutes with the permutations and their average. The restrictions are thus well defined. Its norm is bounded by $\|T\|^n \le 1$, so the direct sum is bounded. Identities and compositions are verified at each degree and, by continuity, in the completion. □

Contractivity is a sufficient condition that allows the bosonic sum to be treated without additional restrictions. At finite degrees, the powers exist for any linear map, but their sum is not always bounded: for one bosonic mode and $T = 2I$, $\Gamma(T)|n\rangle = 2^n|n\rangle$. In the finite exterior sector there are only $d_1 + 1$ degrees, so any linear map of that sector has a bounded finite second quantization. The difference is one of domain, not causality.

Proposition 94.16 (Transport of creations and annihilations)**.** *Let $T : \mathcal{H}_K \to \mathcal{H}_L$ be a bounded transport preserving degree. For $f \in \mathcal{H}_K$ and $g \in \mathcal{H}_L$, on the algebraic finite-particle domain the following hold*

$$\Gamma(T)c_K^\dagger(f) = c_L^\dagger(Tf)\Gamma(T), \tag{94.10}$$

$$c_L(g)\Gamma(T) = \Gamma(T)c_K(T^*g). \tag{94.11}$$

The subscripts indicate the fiber on which each operator acts. If T is isometric, the second identity specializes to

$$c_L(Tf)\Gamma(T) = \Gamma(T)c_K(f), \qquad N_{L,Tf}\Gamma(T) = \Gamma(T)N_{K,f},$$

where $N_{K,f} = c_K^\dagger(f)c_K(f)$ and, for a unit mode, $N_{L,Tf}$ is its transported occupation operator.

Proof. In a symmetric or exterior product, applying T to each factor commutes with inserting f, which becomes Tf. This proves (94.10). Annihilation contracts each factor with the observed mode. For each factor v,

$$\langle g, Tv\rangle = \langle T^*g, v\rangle.$$

The exterior, or graded, signs are preserved because T preserves degree; the symmetric normalizations are equal on both sides. The sum of contractions proves (94.11). If $T^*T = I$, we substitute $g = Tf$. Composing the creation and annihilation identities then gives the occupation identity. All operations considered preserve the finite-particle domain; when $\|T\| \le 1$, $\Gamma(T)$ also has the bounded extension of theorem 94.15. □

The general identity preserves the adjoint T^* and is valid without isometry. The need for the isometric condition appears when transporting the same mode. For example, if $T = tI$, with $0 < t < 1$, for a unit mode f and the vacuum Ω,

$$c(tf)\Gamma(T)f = t^2\Omega, \qquad \Gamma(T)c(f)f = \Omega.$$

The general formula correctly gives $\Gamma(T)c(T^*tf)f = t^2\Omega$. This preserves transport by contractions, with its explicit adjoint, and the isometric specialization of occupation.

94.12.8 Exchange symmetry and record of a composite

The composition of section 94.13.8 acts first on histories and their boundaries. Fock organizes channel occupation; $\mathfrak{C}_{12}$ organizes the link record. These are coordinated operations that do not replace each other. In particular, antisymmetrization does not suppress memory of the coupling order.

Let $\mathcal{D} \subset \mathcal{H}^{\otimes n}$ be the space generated by constituent expressions admissible for an interface. The linearization of composition is written $C_{\mathcal{T}} : \mathcal{D} \to \mathcal{H}_{\mathrm{comp}}$. When a finite group G of exchanges preserves admissibility and the boundary, the concrete compatibility is

$$C_{\mathcal{T}} U(g) = V(g) C_{\mathcal{T}}, \qquad g \in G,$$

where U, V are unitary representations and V transports the composite records, including their sheet and memory data. The equation is the exchange condition of that realization; it is not inferred from the mere census of expressions.

Proposition 94.17 (Descent of composition to the statistical sector)**.** *Let $\chi : G \to \{+1, -1\}$ be the character of the exchange sector. Under the preceding compatibility,*

$$P_\chi^{\mathrm{comp}} C_{\mathcal{T}} = C_{\mathcal{T}} P_\chi^{\mathrm{in}},$$

where

$$P_\chi^{\mathrm{in}} = \frac{1}{|G|} \sum_g \chi(g) U(g), \qquad P_\chi^{\mathrm{comp}} = \frac{1}{|G|} \sum_g \chi(g) V(g).$$

Proof. Multiplying each intertwining equality by $\chi(g)/|G|$ and summing gives exactly the identity. The averages are projectors by the same calculation as in proposition 94.7. □

The result justifies composing within a symmetric or alternating sector when the boundary has the required transport. If a permutation changes the physical binding channel, the distinct trees and their interrelations are preserved; they are not identified through a symmetrization that forgets that boundary.

94.12.8.1 Chromatic antisymmetry and remainder of the baryonic state

In the baryonic realization, proposition 94.24 constructs the chromatic vector $|\varepsilon\rangle$, which changes sign under a transposition. After regrouping the factors of the complete state, a factorized vector has the form

$$\Psi = |\varepsilon\rangle_{\mathrm{col}} \otimes \Phi_{\mathrm{spin,flavor,orbital,sheet}}.$$

The exchange of two constituents acts simultaneously on both factors. If $U_{\mathrm{rest}}(\pi)\Phi = \Phi$, then the complete state changes sign and belongs to the fermionic sector. Conversely, for a factorized vector with the nonzero chromatic singlet, total alternation requires the remainder to be symmetric under those same permutations. The proof consists in factoring $U_{\mathrm{tot}}(\pi) = U_{\mathrm{col}}(\pi) \otimes U_{\mathrm{rest}}(\pi)$ and canceling the chromatic sign.

The word "remainder" explicitly includes flavor and sheet. Exchanging colors without transporting those components is not the permutation of two complete modes. For a nonfactorizable state, the alternating projector is applied to the complete tensor space, without replacing it by a scalar rule.

94.12.8.2 Compatibility of physical projectors

Color, channel and statistics projectors can be combined by a product when they commute. If P, Q are orthogonal projectors and $[P, Q] = 0$, then $(PQ)^* = PQ$, $(PQ)^2 = PQ$ and $\mathrm{im}(PQ) = \mathrm{im}\, P \cap \mathrm{im}\, Q$. Inclusion in both images is direct; for a vector in the intersection, $PQv = v$. If they do not commute, their product does not in general represent the projector of the two simultaneous conditions: the realization must construct the common subspace or the physical order of operations.

This precision preserves the meaning of the singlet and Pauli projectors entering the composites of section 94.13. Coupling selects a sector; the composite mass is evaluated after constructing that sector's link record.

94.12.9 Action of the material operator and scope of physical identification

Let $B = B^*$ be a channel operator already obtained from the internal dynamics and preserving the grading. At degree n it acts through

$$d\Gamma_n(B) = \sum_{j=1}^{n} I^{\otimes(j-1)} \otimes B \otimes I^{\otimes(n-j)}.$$

The sum commutes with graded permutations and restricts to $\mathcal{F}_n$. If $B \geq 0$, each restriction is positive. In an eigenbasis, its value is the sum of the occupations times the eigenvalues of B, with the occupations allowed by the preceding proofs. This is the action of an additive observable on independent channels; it does not identify the operator of a bound composite with the sum of the isolated masses of its constituents.

In the bosonic Hilbert sum, $d\Gamma(B)$ has the natural domain

$$\left\{ \psi = (\psi_n) : \sum_n \|d\Gamma_n(B)\psi_n\|^2 < \infty \right\}.$$

Each block is self-adjoint in finite dimension, and its direct sum with that domain is self-adjoint. The binding or open-channel operator is added only when it has been constructed by the corresponding dynamics. Formation of a complex pole follows the treatment of 94.13.12.

Complete algebraic results have therefore been obtained: symmetry projectors, normalized bases, CAR and CCR on their domains, exclusion of a mode, occupation censuses and compatibility with refinement intertwiners. The starting point of these results is declared in (94.1), (94.2) and (94.3).

When the realization of rotations satisfies $U(2\pi) = (-1)^p$, the same grading coordinates restoration of orientation at 4π and exterior completion. That compatibility is not replaced by a coincidence between two signs. Identification with the relativistic spin–statistics theorem additionally requires the Lorentzian representation, locality and positivity of the physical realization. Braiding sectors that do not descend to permutations remain in the codomain of section 94.14; they are not forced into this bosonic or fermionic completion.

94.13 Sectoral realizations, material composition and spectral classification

The atlas of section 90.16 and the reader of section 91.13 have complementary functions. The former classifies supports, depths and orbits; the latter preserves the chronology of each trajectory and publishes its mass operator. Their application to a family requires maintaining both structures. Equality between mass values does not identify routes; equality between cells does not identify histories, representations or composite states either.

This section develops the sectorial realizations of that construction. The internal constants, angular pair, absolute scale, and bidirectional readers are those already constructed from APP–TRIT–TPK. The physical representation specifies which information of the state is observed; metrological comparison takes place after that representation has been fixed. In particular, the exposition preserves the distinction between a fiber operator, an exact evaluation for a declared signature, and a spectral line individuated by a coupling.

94.13.1 Sectoral descriptors and spectral criterion of individualization

For an elementary state, write

$$X = (\mathfrak{c}, \rho, g, f, q, \epsilon, \varsigma, \mathfrak{a}).$$

Here $\mathfrak{c}$ is the sectoral class, ρ the representation, g the physical generation, f the flavor, q the charge, ϵ the sheet orientation, ς the statistical parity and $\mathfrak{a}$ the coupling data. The combinatorial depth of the atlas and the physical generation remain coordinates of distinct types. Flavor distinguishes components of a family representation; color distinguishes components of the chromatic representation; the sheet and its orientation preserve the operator provenance of the trajectory. None of these labels replaces the event record.

Let $\mathcal{P}_{\mathrm{adm}}$ be the set of admissible physical descriptors, let $\mathcal{R}_{\mathrm{adm}}$ be the register of enriched routes produced by the atlas, and let $\mathcal{D}$ be the space of incidence descriptors that preserve sectorial class, representation, depth, charge, color, orientation, and coupling data. Define

$$d_{\mathcal{P}} : \mathcal{P}_{\mathrm{adm}} \longrightarrow \mathcal{D}, \qquad d_{\mathcal{R}} : \mathcal{R}_{\mathrm{adm}} \longrightarrow \mathcal{D},$$

where $d_{\mathcal{P}}(X)$ publishes the structural requirements of the state X and $d_{\mathcal{R}}(\gamma)$ publishes the same coordinates preserved by the route γ. The fiber, orbit or multisection produced by the descriptor is the fiber product

$$F_X = d_{\mathcal{R}}^{-1}\big(d_{\mathcal{P}}(X)\big) = \mathcal{R}_{\mathrm{adm}} \times_{\mathcal{D}} \{d_{\mathcal{P}}(X)\}. \tag{94.12}$$

For the realizable descriptors in this article, F_X is finite and nonempty. On $\mathcal{H}_X = \ell^2(F_X) \otimes V_{\rho_X}$ act the two operators $M_X^{\beta,D}$ and $M_X^{\beta,\Omega}$ of Section 91.13. Since they are diagonal in the route basis, their joint spectral measure is

$$E_X(B) = \sum_{\gamma \in F_X} \mathbf{1}_B\big(m_D(\gamma), m_\Omega(\gamma)\big) |\gamma\rangle\langle\gamma| \otimes I_{V_{\rho_X}}, \qquad B \subset \mathbb{R}^2_{>0}. \tag{94.13}$$

Preserving this measure allows two fibers to be distinguished even when their means coincide. A scalar functional of the two readers is applied only when it forms part of the declared representation. If the coupling preserves several values, the output is a spectrum with multiplicities.

Proposition 94.18 (Spectral-line criterion)**.** *Let M be a self-adjoint operator on a finite fiber and P a preparation projector. The channel publishes exclusively the value m if and only if*

$$E_M(\{m\})P = P, \qquad \textit{equivalently} \qquad (M - mI)P = 0.$$

The equality $PMP = mP$ alone determines a scalar compression, but does not guarantee that the prepared spectral support is a single point.

Proof. By the spectral theorem, $\ker(M - mI)$ is the image of $E_M(\{m\})$. The first two conditions state that $\operatorname{im} P$ is contained in that eigenspace. To verify the last distinction, take $M = \operatorname{diag}(1, 3)$ and the projector onto $(1, 1)/\sqrt{2}$. Then $PMP = 2P$, whereas the prepared mass distribution has support $\{1, 3\}$, not $\{2\}$. □

Individualization thus preserves the scope of each result. The fiber and the operator are determined objects; passage to a line uses the specified projector and spectral condition. This formulation will be applied equally to charged, neutral and composite families.

94.13.2 Charged leptons: electronic center and generation multisections

The three charged multisections have ranks $1, 8, 16$ at depths G_0, G_2, G_4. They share the charge representation and trivial color, but not the support rank or trajectory register. The electron occupies the center $(5, 5)$; the muonic and tauonic realizations preserve their multisections until spectral coupling is performed. Conjugation of their representations is treated through the operators of Section 90.15, not through a new choice of masses.

The basal electronic coordinate $\mathfrak{e}_0$ supplies the intermediate angular chart

$$\mathfrak{m}_X^{(0)} = \mathfrak{e}_0 \exp\left(n_A(X)x + \frac{s_C(X)}{6}y\right), \qquad x = A_{\mathrm{rad}}, \quad y = C_{\mathrm{rad}}^{*}. \tag{94.14}$$

In this formula, $\mathfrak{e}_0$ is the basal section, not the beta electronic closure. Bidirectional return belongs to the subsequent stage of the construction. The two stages preserve their own normalization.

The nominal records recovered for the two noncentral families publish

$$(n_A, s_C)_\mu = (42, -3), \qquad (n_A, s_C)_\tau = (64, 0),$$

and their declared complete signatures are

$$\sigma_\mu = (42, -3, 8, 0, 2), \qquad \sigma_\tau = (64, 0, 25, -1, 6). \tag{94.15}$$

To explain the effect of each coordinate, denote by $\mathcal{E}_j$ the successive readings of a declared signature:

$$\begin{aligned}
\mathcal{E}_0(\sigma) &= n_A x + s_C y/6,\\
\mathcal{E}_1(\sigma) &= \mathcal{E}_0(\sigma) + k_{\Delta_4}\Delta_4,\\
\mathcal{E}_2(\sigma) &= \mathcal{E}_1(\sigma) + b_{120}s_{120},\\
\mathcal{E}_3(\sigma) &= \mathcal{E}_2(\sigma) + b_\zeta \zeta_{270}, \qquad \zeta_{270} = 135\Delta_4^2.
\end{aligned} \tag{94.16}$$

The symbol $\mathcal{E}_3$ here denotes an evaluation exponent, not a species or an additional mode. The beta supplement preserving these signatures uses the explicit normalization $m_X^{[3]} = m_e^{\beta} \exp \mathcal{E}_3(\sigma_X)$. This is an evaluation relative to the supplement's signatures; it does not replace the route-by-route construction of $M_X^{\beta,D}$ and $M_X^{\beta,\Omega}$.

The difference of signatures gives, without approximation,

$$\log \frac{m_\tau^{[3]}}{m_\mu^{[3]}} = 22x + \frac{1}{2}y + 17\Delta_4 - s_{120} + 4\zeta_{270}. \tag{94.17}$$

The proof consists of subtracting the exponents in (94.16). The angular term, torsional memory, and two return readings remain identified in the result. The preserved evaluations are

$$\begin{aligned}
m_\mu^{[3]} &= 105.658360294435829303\ldots\ \mathrm{MeV}/c^2,\\
m_\tau^{[3]} &= 1776.860917631432295595\ldots\ \mathrm{MeV}/c^2.
\end{aligned}$$

Their documentary status is exact evaluation upon fixing the recovered signature. Physical selection of that signature within the multisection and the spectral-line criterion are distinct operations and are preserved as such.

At the electronic center the rank is one and the readers coincide at $(80, 54, 6)$. The active history $(2, 2, 5, -4, -3, -2)^2$ preserves more information than its historical envelope $(4, 3, 5, -4, -3, -5)^2$, although both share the sign pattern. This distinction links the electronic particularity to the positional memory of events: families are not defined solely by the visual form of a word of signs.

94.13.3 Neutrinos: neutrality, mass spectrum, and flavor transport

The neutrino support has dimension $20 = 2 + 4 + 6 + 8$, distributed over four depths. Its three flavor labels ν_e, ν_μ, ν_τ and its three possible mass eigenvalues belong to the physical realization of the sector, not to the census of depths. Electrical neutrality preserves orientation, sheet memory, and spinorial holonomy. It therefore does not eliminate the variables involved in mixing.

The neutral register separates the contributions of two half-cycles. If $\mathcal{I}_m$ is the block of 9 iterations of phase m, θ_t marks the visit to 9, ϵ_t preserves the sheet, and ω_t its corona orientation, the local reader is

$$T_\nu(m) = \sum_{t \in \mathcal{I}_m} \omega_t \epsilon_t \theta_t, \qquad e^\nu_m = T_\nu(m) - T_\nu(m+6).$$

This difference retains the oriented information even when its total sum vanishes. The complete operator also uses torsional predecessors, vacancies, and transport between sheets. Passage to the physical subspace requires the rank-3 neutral projector,

$$M_\nu = P_\nu \mathcal{M}_\nu P_\nu \quad \text{on } \operatorname{im} P_\nu.$$

The eigenvalues of that compression are the masses in the internal action–clock–mass scale. Mixing angles act between their frames; they neither replace the scale nor retrospectively generate the eigenvalues. The compression coincides with the restriction of the complete operator when $\operatorname{im} P_\nu$ is a reducing subspace of $\mathcal{M}_\nu$.

The register allows a Gram kernel to be assembled before diagonalization. For each data type $r \in \{\text{car}, \text{hoj}, \text{ori}, \text{vac}, \text{hol}\}$, let $\Xi_r(c)$ be the vector produced by cell or route c. Define

$$K_{\text{reg}}(c,d) = \sum_{r : N_r > 0} \frac{\langle \Xi_r(c), \Xi_r(d) \rangle}{N_r}, \qquad N_r = \sum_a \|\Xi_r(a)\|^2. \tag{94.18}$$

Omission of a term with zero total norm is fixed before evaluation. It preserves a well-defined domain and introduces no fit.

Proposition 94.19 (Positivity and rank of the neutral register)**.** *The kernel K_{reg} is Hermitian and positive semidefinite. For a three-dimensional aggregation G, the condition* $\det G \neq 0$ *certifies that the aggregated data separate three directions. For any invertible square matrix G, its polar factor*

$$U_G = G(G^*G)^{-1/2}$$

is unitary. This factor does not by itself determine the spectral frames of the charged and neutral operators.

Proof. For coefficients z_c, the quadratic form of (94.18) is

$$\sum_{c,d} \overline{z}_c K_{\text{reg}}(c,d) z_d = \sum_{r : N_r > 0} N_r^{-1} \left\| \sum_c z_c \Xi_r(c) \right\|^2 \geq 0.$$

The rank is the dimension of the image of the aggregated vectors. For the last statement, $U_G^* U_G = (G^*G)^{-1/2} G^*G (G^*G)^{-1/2} = I$; invertibility and equality of dimensions also give $U_G U_G^* = I$. The formula uses matrix G, whereas a spectral frame requires diagonalization of the corresponding mass operator. □

If $F_\ell, F_\nu : \mathbb{C}^3 \to \mathcal{H}$ are complete orthonormal frames of the same physical subspace, then $U_{\text{PMNS}} = F_\ell^* F_\nu$ is unitary. If their images are different, the product is only a contraction; unitarity requires equality of the transported subspaces. In the presence of degeneracies, mixing is determined as a double coset under frame changes in the degenerate blocks, not as a numerical matrix chosen by convention.

The minimal phase construction

$$\Phi_K = I + (e^{i\pi/6} - 1)|b_K\rangle\langle b_K|, \qquad \|b_K\| = 1,$$

remains a documented negative control: it is unitary, but the best permutation of its angular identification gives $\theta_{13} \simeq 17.02^\circ$, incompatible with the comparison for that reduced model. The result concerns the rank-one phase. It does not replace the complete kernel (94.18), nor permit a mixing matrix that has not passed its own rank and angle checks to be presented as validated.

94.13.3.1 Neutral self-conjugation and the possibility of a sterile sector

The Dirac–Majorana alternative uses the fermionic representation. A neutral real structure must satisfy

$$C_\nu^2 = I, \qquad C_\nu M_\nu = M_\nu C_\nu, \qquad C_\nu H_\nu = H_\nu C_\nu,$$

with C_ν antiunitary and a compatible realification. Commutation with mass preserves its real part. For the self-adjoint Hamiltonian, by contrast, antiunitarity also conjugates the factor i: if $U_\nu(t) = \exp(-itH_\nu/\hbar)$, then

$$C_\nu U_\nu(t) C_\nu^{-1} = U_\nu(-t).$$

Consequently, the fixed modes $\psi = C_\nu\psi$ define a neutral real structure, but the stated conditions are insufficient to restrict the entire Schrodinger evolution to it. Preservation of that fixed part requires C_ν to commute with the group $U_\nu(t)$; Proposition 94.29 proves the corresponding condition on its generator. A Majorana realization must also declare the compatibility of self-conjugation with the fermionic representation and its dynamics. A nonzero conserved charge that changes sign under C_ν keeps conjugate modes separate in the Dirac realization. The word involution of Section 90.15 and electrical neutrality, considered in isolation, do not decide this alternative.

A sterile sector requires a nonzero projector P_s with $P_s P_{\mathrm{act}} = 0$ and $P_s J_{\mathrm{weak}} = 0$. If $[P_s, M_\nu] = 0$, it is reducing and does not mix with the active sector. If $P_s M_\nu P_{\mathrm{act}} \neq 0$, mixing exists and the projector ceases to commute with mass. Therefore, an additional depth of the atlas does not automatically count as a fourth neutrino species.

94.13.4 Quarks: flavor, color orbit, and scale transport

The quark state preserves the tuple

$$(\sigma_{ud}, g, [\gamma]_{\mathrm{col}}, \mathfrak{M}, \mathscr{L}),$$

which distinguishes up or down type, generation, color orbit, multisection, and register. Section 90.16 constructs the color residue $\kappa_{\mathrm{col}}(r, c) = r - c \pmod 3$ and the three classes of 12 cells. The color index acts within the support; flavor preserves generation and signature information that this residue does not identify.

The six nominal angular pairs are

f	n_A	s_C	W_+	W_-
u	11	7	73	59
d	17	8	110	94
c	61	8	374	358
s	41	−3	243	249
t	100	−1	599	601
b	71	−5	421	431

$$W_\pm = 6n_A \pm s_C.$$

These last integers rewrite the same exponent:

$$n_A x + s_C y/6 = \frac{W_+(x+y) + W_-(x-y)}{12}.$$

Adding and subtracting $W_{\pm}$ recovers n_A and s_C, so that this presentation preserves both channels. The six pairs belong to the documented angular chart; the three remaining coordinates are not filled with zeros to manufacture a complete signature. Their color operators and angular evaluations are preserved as results of different scope.

Proposition 94.20 (Color degeneracy of a flavor mass)**.** *Let $V_{\mathrm{col}} \simeq \mathbb{C}^3$ be the fundamental color representation and U its irreducible unitary action. The reduction*

$$\mathbb{E}_{\mathrm{col}}(M) = \int_{SU(3)} U(g) M U(g)^* \, dg$$

is a multiple of I_3. If M already commutes with the action, its three color components have the same flavor-mass value.

Proof. Invariance of Haar measure gives $U(h)\mathbb{E}_{\mathrm{col}}(M)U(h)^* = \mathbb{E}_{\mathrm{col}}(M)$. Schur's lemma makes this operator scalar. Since trace is preserved, the scalar is $\operatorname{Tr} M/3$. If M is invariant from the outset, the integral returns M. □

The proof explains what the color quotient eliminates: the dependence of the numerical value on the red, green, or blue representative. It does not eliminate the color routes subsequently involved in a bound state. Nor does it turn this reduction into a flavor selector.

The current quark mass is presented as a section at a declared scale and in a declared scheme, $m_q(\mu, \mathscr{R})$. If $t = \log \mu$ and the connection induces $\gamma_m(g(t))$, transport satisfies

$$\left(\frac{d}{dt} + \gamma_m(g(t))\right) m_q = 0, \qquad m_q(t_2) = \mathcal{P} \exp\left(-\int_{t_1}^{t_2} \gamma_m(g(t))\, dt\right) m_q(t_1).$$

The connection and its endomorphism are transport data, not numbers chosen to match a table. A physical residual is meaningful only after both sections have been placed at the same scale and in the same scheme.

CKM mixing acts on another coordinate: the flavor frame. Once the internal unitary matrix has been constructed, the down operator, expressed in the up frame, is $B_d^{(u)} = V_{\mathrm{CKM}} B_d V_{\mathrm{CKM}}^*$. Unitary conjugation preserves the characteristic polynomial, since

$$\det(zI - VB_dV^*) = \det\big(V(zI - B_d)V^*\big) = \det(zI - B_d).$$

The role of the angles is thus understood: they relate frames without becoming retrospective generators of mass eigenvalues.

94.13.5 Zero-mass charts: photon and adjoint color representation

The null domain is separated before the positive character is applied. In the total realization,

$$\mathcal{H}_{\mathrm{phys}} = \mathcal{H}_0 \oplus \mathcal{H}_+, \qquad M|_{\mathcal{H}_0} = 0, \qquad M|_{\mathcal{H}_+} > 0.$$

The exponential of a finite signature does not vanish. Thus a null chart is not obtained by assigning an arbitrarily large signature to a positive route and calling its approximation zero.

The photon belongs to the electromagnetic chart, with its gauge representation and its two physical transverse polarizations. Gluons belong to the chromatic adjoint representation, of dimension 8. The equality of their zero masses does not identify the two representations. The fiber, gauge, polarizations and coupling channels remain part of the state.

Oriented propagation with zero mass is also compatible. In the documented chart,

$$c_{\pm} = \frac{c}{1 \pm \varepsilon}, \qquad \varepsilon = q_-^{90} - q_+^{90}, \qquad 0 < \varepsilon < 1.$$

Then $c_\pm > 0$ and

$$\frac{2}{c_+^{-1} + c_-^{-1}} = c.$$

The identity follows from adding $(1+\varepsilon)/c$ and $(1-\varepsilon)/c$. The asymmetry of the two readings preserves its orientation information; it does not in itself publish a mass term.

94.13.6 Massive bosons and the scalar sector

The descriptors of W, Z, and the scalar sector are not forced into one of the 13 fermionic multisections. They preserve their representation and realization domain. The documented nominal readings are

X	nominal route	direction	(n_A, s_C)	(W_+, W_-)
W	15/37	E–O	$(94, -1)$	$(563, 565)$
Z	20/23	E–O	$(95, -1)$	$(569, 571)$
H	24/29	N–S	$(97, 9)$	$(591, 573)$

These route names belong to the preserved nominal record. Their union with the full oriented domain must retain the trajectory, representation and projector data. The published cardinal counts, respectively $(23, 44, 11, 30)$, $(33, 45, 10, 20)$ and $(40, 34, 22, 12)$, are retained alongside the address and are not replaced by a single label.

For W and Z, the complete signatures are recovered:

$$\sigma_W = (94, -1, 0, -2, 4), \qquad \sigma_Z = (95, -1, -11, 0, -4).$$

Reading (94.16) produces, for these signatures,

$$m_W^{[3]} = 80378.964909704547024865\ldots\ \mathrm{MeV}/c^2,$$

$$m_Z^{[3]} = 91187.533444313407844855\ldots\ \mathrm{MeV}/c^2.$$

Their ratio preserves exactly the difference of the evaluated registers:

$$\log \frac{m_Z^{[3]}}{m_W^{[3]}} = x - 11\Delta_4 + 2s_{120} - 8\zeta_{270}.$$

In contrast, the nominal coordinate $(97, 9)$ of the scalar sector belongs to the documented angular chart $\mathbb{Z}^2$: this notation does not attribute to it a signature $\mathbb{Z}^5$ or a refinement of order 3. Its realization as a scalar excitation requires the scalar projector and the corresponding gauge and fermionic couplings.

For an unstable species, the physical result is a pole of the channel operator, not merely the evaluation of a positive character. Subsection 94.13.12 constructs this reading and separates pole mass and width. In the preserved documentary register, the preceding numbers are signature-conditioned evaluations; the individuation of route, projector, and pole constitutes the sectorial realization that must accompany them.

94.13.7 Finite grammar of material compositions

Let Q_a, $a \in \mathbb{Z}/3\mathbb{Z}$, be the three color classes of the atlas, each of cardinality 12. The formal dual $q^\vee$ preserves the conjugate support and reverses charge and color orientation. It is not merely the cell $\rho_{\mathrm{c}}(q)$ stripped of these data. Mesonic and baryonic expressions are defined by

$$\begin{aligned} \mathfrak{M} &= \{q \otimes (q')^\vee : q, q' \in Q_a \text{ for some } a\}, \\ \mathfrak{B} &= Q_0 \times Q_1 \times Q_2. \end{aligned} \tag{94.19}$$

The pairs are ordered and the color positions of the triple are fixed. Residual neutrality is, respectively, $a - a = 0$ and $0 + 1 + 2 = 0 \pmod 3$.

Proposition 94.21 (Internal compositional census)**.** *The grammar contains* 432 *mesonic expressions,* 1728 *baryonic expressions, and* 372 *compatible charged arrows, divided into* 320 *leptonic and* 52 *quark arrows. These cardinalities count internal expressions and arrows, not physical species or independent mass predictions.*

Proof. For each color, an ordered pair of 12 cells is chosen; therefore, $|\mathfrak{M}| = 3 \cdot 12^2 = 432$. The baryon's three positions give $|\mathfrak{B}| = 12^3 = 1728$. Lepton–neutrino transitions of equal depth share only G_2, G_4; in both directions this gives $2(8 \cdot 4 + 16 \cdot 8) = 320$.

Up–down transitions additionally require equality of color. In G_1, the color multiplicities are $(2, 1, 1)$ and $(0, 1, 1)$; in G_3, they are $(4, 4, 4)$ and $(2, 2, 2)$. There are $2 + 24 = 26$ pairs in one direction and 52 in both. Adding $320 + 52$ completes the census. All factors come from the atlas preceding mass evaluation. □

The 81 cell identities complete the neutral arrows of this grammar, but a compositional identity does not thereby become a neutral boson. The representation and realization channel are the data distinguishing the two notions.

94.13.8 Composition of the register and bonding memory

A composite contains more information than the list of its constituents. Let L_1, L_2 be enriched registers, g the transport aligning phase, sheet, and orientation, and B the subtrace identified at the common boundary. In action and event coordinates, composition is

$$(n, e)_1 \star_{(g,B)} (n, e)_2 = \big(n_1 + n_2 - n_B,\ e_1 + g e_2 - e_B\big). \tag{94.20}$$

Subtraction prevents the same subtrace from being counted twice; transport preserves orientation at the interface. Torsional predecessors and corona marks are composed at the same level before L_{tick} is applied again. The reversible event decomposition of Proposition 91.27 permits the complete register to be transported without reducing it to its total.

For an ordered tree $\mathcal{T}$ of n constituents, this gives

$$(\Gamma_1, \dots, \Gamma_n; \mathcal{T}, \partial) \overset{\mathfrak{C}_{12}^{(n)}}{\longmapsto} L_Y \overset{L_{\mathrm{tick}}}{\longmapsto} \sigma_Y \longmapsto (\eta_Y, M_Y). \tag{94.21}$$

The boundary ∂ includes the transports and subtraces of each tree edge. Once these admissible data are fixed, each composition is an explicit operation. The character is applied to the final signature; it does not preselect the tree through the destination mass.

Proposition 94.22 (Compatibility of parenthesization)**.** *In a chart of records where* $L_1 \star L_2 = L_1 + L_2 + \mathfrak{b}(L_1, L_2)$*, the two parenthesizations of an admissible triple coincide if and only if*

$$\mathfrak{b}(L_1, L_2) + \mathfrak{b}(L_1 \star L_2, L_3) = \mathfrak{b}(L_2, L_3) + \mathfrak{b}(L_1, L_2 \star L_3).$$

The equality is required on the domain where both trees represent the same oriented interface.

Proof. Expanding $(L_1 \star L_2) \star L_3$ gives $L_1 + L_2 + L_3 + \mathfrak{b}(L_1, L_2) + \mathfrak{b}(L_1 \star L_2, L_3)$. Expanding $L_1 \star (L_2 \star L_3)$ gives the same sum of three registers and the two terms on the right-hand side. Their equality is exactly equivalent to the stated identity. □

The cocycle law ensures independence of an equivalent parenthesization; it does not identify physically different trees. Two bonding structures preserving different channels remain distinct realizations until an intertwiner of registers, projectors, and operators is available.

94.13.8.1 Distinction between sum of signatures and complete composition record

The historical reduced event extractor is defined by

$$F_0(n,e)=\left(n,\sum_m e_m,K_0(e),\sum_{a=0}^{3}\operatorname{sgn}(e_a+e_{a+4}+e_{a+8}),\sum_m \tau_m\tau_{m+3}\right),$$

where $\tau_m=\operatorname{sgn}(e_m)$ and $K_0(e)=\sum_{\tau_m=0}\Sigma_{12}(m)$, with $\Sigma_{12}=(+,+,+,-,-,-)^2$. Its third coordinate is a reading of null events; it is not the strong torsion $k_{\Delta_4}=T_z-2C_z$ of Section 91.13.

Consider the registers of the routes

$$a=(1,3,-1,+1),\quad b=(6,7,-1,+1),\quad c=(1,2,-1,-1),$$

with

$$\begin{aligned}e(a)&=(5,4,4,-4,-4,-6)^2,\\ e(b)&=(4,5,4,-4,-6,-4)^2,\\ e(c)&=(1,4,1,-2,-4,-3)^2.\end{aligned}$$

They have $F_0(a)=F_0(b)=(8,-2,0,0,-12)$ and $F_0(c)=(28,-6,0,-4,-12)$. Superposing the registers gives

$$F_0(a\boxplus c)=(36,-8,0,0,-12),\qquad F_0(b\boxplus c)=(36,-8,0,-2,-12).$$

The projected inputs are equal and the composite outputs differ. Consequently, no operation on the two five-tuples of this projection recovers all superpositions. The example proves a precise obstruction for F_0; it does not attribute the same result to every possible extension of a signature. The current construction avoids this loss through (94.21): it preserves the events and predecessors, composes them, and only then evaluates the complete extractor.

94.13.9 Mesons: dual pair, singlet, and differentiation of excitations

The expression $q\otimes(q')^\vee$ establishes the ordered pair and its residual neutrality. The color singlet belongs to the realization $V\otimes\overline{V}$, $V=\mathbb{C}^3$. If $\Omega=\sum_{a=1}^{3}|a\bar a\rangle$, the projector is

$$P_M^{(1)}=\frac{1}{3}|\Omega\rangle\langle\Omega|. \tag{94.22}$$

Proposition 94.23 (Mesonic color projection)**.** *The map $P_M^{(1)}$ is a rank-one orthogonal projector, and its image is invariant under $U\otimes\overline{U}$, $U\in SU(3)$.*

Proof. We have $\langle\Omega,\Omega\rangle=3$, hence $(P_M^{(1)})^2=P_M^{(1)}$. Moreover,

$$(U\otimes\overline{U})\Omega=\sum_{a,b,c}U_{ba}\overline{U}_{ca}|b\bar c\rangle=\sum_b|b\bar b\rangle=\Omega.$$

The image thus preserves the color contraction without privileging any color. □

The mesonic descriptor combines flavors, spin and parity, isospin and charge, radial or orbital excitation, bonding tree, and boundary. The label J^{PC} is used when C is defined for the channel; charge conjugation is not automatically assigned to a charged state. Two states with the same flavors may have different radial, orbital, or channel projectors and, therefore, different spectra. The orbital construction of Section 67.3 provides a possible reading when its dynamical hypotheses hold; orbital numbers do not follow from the meson's name alone.

The composite-evaluation register preserves

	n_A	s_C	k_{Δ_4}	b_{120}	b_ζ
π^0	44	−4	−25	1	−2
$\pi^\pm$	44	1	−2	2	−1
$K^\pm$	54	−1	17	−2	2
K^0	54	0	32	3	−2

For these signatures, the same reading (94.16) explains the differences within the pairs:

$$\log \frac{m^{[3]}_{\pi^\pm}}{m^{[3]}_{\pi^0}} = \frac{5}{6}y + 23\Delta_4 + s_{120} + \zeta_{270},$$
$$\log \frac{m^{[3]}_{K^0}}{m^{[3]}_{K^\pm}} = \frac{1}{6}y + 15\Delta_4 + 5s_{120} - 4\zeta_{270}. \tag{94.23}$$

The action terms $44x$ and $54x$ cancel within each pair. The remaining differences preserve which register coordinate produces each contribution. They are exact consequences of the declared signatures, not a substitute proof of their generation by the bonding tree.

The resulting evaluations are

$$m^{[3]}_{\pi^0} = 134.976724327872125739\ldots\ \mathrm{MeV}/c^2,$$
$$m^{[3]}_{\pi^\pm} = 139.570485171572191374\ldots\ \mathrm{MeV}/c^2,$$
$$m^{[3]}_{K^\pm} = 493.677085649530612475\ldots\ \mathrm{MeV}/c^2,$$
$$m^{[3]}_{K^0} = 497.611186497750457358\ldots\ \mathrm{MeV}/c^2.$$

Each numerical value retains the status of an exact evaluation conditioned on the recovered composite signature. The singlet, grammar, and register extraction are constructed; complete nominal realization requires the binding tree to regenerate that signature before comparison begins.

94.13.10 Baryons: color triple, exchange, and the neutron–proton difference

The neutrality of $q_0 \otimes q_1 \otimes q_2$ preserves the three color positions. In the tensor realization, introduce the normalized vector

$$|\varepsilon\rangle = \frac{1}{\sqrt{6}} \sum_{a,b,c=1}^{3} \varepsilon_{abc}|abc\rangle, \qquad P^{(1)}_B = |\varepsilon\rangle\langle\varepsilon|.$$

Proposition 94.24 (Baryonic singlet and exchange rule)**.** *The operator $P^{(1)}_B$ is a rank-one orthogonal projector, invariant under $U^{\otimes 3}$, $U \in SU(3)$. A permutation of two positions reverses the sign of its generating vector and preserves the projector.*

Proof. There are 6 nonzero orthogonal terms, each of modulus $1/\sqrt{6}$; hence the vector has norm one. The determinant identity gives $U^{\otimes 3}|\varepsilon\rangle = \det(U)|\varepsilon\rangle = |\varepsilon\rangle$. A transposition reverses the sign of ε_{abc}. In the outer product of the vector with its adjoint, the two signs cancel. □

Color antisymmetry is one part of the complete state. The physical projector combines color, spin and parity, isospin, charge, and the exchange symmetry of orbital and flavor components. Pauli's principle acts on the complete fermionic product; applying it to a bare list of colors is insufficient. The triple tree records which pair was coupled first and how the third constituent crossed the boundary. Proposition 94.22 controls equivalent parenthesizations.

The recovered proton and neutron signatures are

$$\sigma_p = (59, 0, 9, 1, 0), \qquad \sigma_n = (59, 1, -34, 0, 1).$$

They share the action coordinate 59, but differ in sheet, torsion, and returns. Their exponent difference is

$$\log \frac{m_n^{[3]}}{m_p^{[3]}} = \frac{1}{6}y - 43\Delta_4 - s_{120} + \zeta_{270}. \tag{94.24}$$

The equality is proved by subtracting the five coordinates. In this signature, -34 is a torsional integer of the neutron; it is not identified with the decimal exponent -34 of the absolute action scale. Equality of two integers in different maps does not eliminate their type or provenance.

The preserved evaluations are

$$m_p^{[3]} = 938.271751546984898\ldots \text{ MeV}/c^2,$$
$$m_n^{[3]} = 939.565810472507023312\ldots \text{ MeV}/c^2.$$

The formula (94.24) locates the effect of each component of the nominal signatures. The physical derivation of the difference requires transporting isospin, vacancy orientation, link boundary and electromagnetic contribution to the same chart. Evaluation of the character retains its exactness under the fixed signature; that exactness does not replace the prospective construction of the link.

94.13.11 Higher-order compositions and channel equivalence

The map $\mathfrak{C}_{12}^{(n)}$ admits an arbitrary finite number of constituents. A tetraquark uses $qq\bar{q}\bar{q}$; a pentaquark, $qqqq\bar{q}$; a gluonic bound state uses adjoint representations projected onto the singlet; a hybrid preserves fermionic and gauge constituents in the same tree. Nuclei, atoms, and molecules use the same precedence of register and reading, with their own couplings and boundaries.

Binding energy belongs to the realization of the composite register. It is not obtained by adding isolated masses and naming the desired residual a binding contribution. Even when a cocycle takes additive values on registers, the sign and correlation extractor is applied afterwards; the multiplicative character of a signature therefore does not turn the composite mass into a sum of constituent masses.

If two trees $\mathcal{T}_1, \mathcal{T}_2$ have a unitary V intertwining their operators and channel projectors,

$$VH_1 = H_2V, \qquad VP_1 = P_2V,$$

then $V(z - H_1)^{-1} = (z - H_2)^{-1}V$ in the common resolvent set. Its continuation, when it exists and is transported in the same chart, preserves the poles and their multiplicities. This equality gives the operational criterion for identifying two realizations of a composite; sharing flavors or an approximate mass is not sufficient.

94.13.12 Resonances: pole mass, width, and survival memory

Let P be the prepared subspace and $Q = I - P$ that of the eliminated channels. In a region where $z - QHQ$ is invertible, eliminating the component $Q\psi$ from $(z - H)\psi = 0$ gives

$$H_{\text{eff}}(z) = PHP + PHQ(z - QHQ)^{-1}QHP. \tag{94.25}$$

Proposition 94.25 (Reduced reading of a bound or resonant state)**.** *If $P\mathcal{H}$ is finite-dimensional, the admissible spectral values of the reduced problem satisfy* $\det(z - H_{\text{eff}}(z)) = 0$*. A resonance additionally requires meromorphic continuation to the relevant channel chart and an isolated zero with an admissible physical residue.*

Proof. The equation in the complement is $(z - QHQ)Q\psi = QHP\psi$. Solving it and substituting into the equation for $P\psi$ produces $(z - H_{\mathrm{eff}}(z))P\psi = 0$. In finite dimension, a nonzero solution exists exactly when the determinant vanishes. The algebraic calculation is performed on the resolvent domain; passage to a resonance uses the stated continuation and residue, which do not follow merely from a channel being open. □

A stable state is a normalizable spectral line with no open decay channel; a bound state lies below the relevant threshold, although it may decay through another channel; a resonance is a pole of the continuation. If

$$z_X = M_X c_{\mathrm{int}}^2 - \frac{i}{2}\Gamma_X, \qquad \Gamma_X > 0,$$

the real and imaginary parts are two coordinated readers of the same pole. Width is not a sixth coordinate of σ.

The discrete reading uses a survival eigenvalue $\lambda_X = re^{-i\theta}$, $0 < r < 1$, and a time step $t_0 > 0$. Given the branch $\theta \in I_\theta$ of the quasienergy chart,

$$z_X = \frac{\hbar_{\mathrm{int}}}{t_0}(\theta + i\log r), \quad M_X = \frac{\hbar_{\mathrm{int}}\theta}{t_0 c_{\mathrm{int}}^2}, \quad \Gamma_X = -\frac{2\hbar_{\mathrm{int}}}{t_0}\log r. \tag{94.26}$$

Indeed, substitution into $\lambda_X = \exp(-iz_X t_0/\hbar_{\mathrm{int}})$ recovers $re^{-i\theta}$. Since $\log r < 0$, the pole is retarded and the width positive. The survival probability after N steps is $r^{2N} = \exp(-\Gamma_X N t_0/\hbar_{\mathrm{int}})$, whence $\tau_X = \hbar_{\mathrm{int}}/\Gamma_X$.

If this reading is obtained from a compression SUS, the asymptotic interpretation through a single mode requires a simple dominant eigenvalue, nonzero initial overlap and separation in modulus from the rest of the spectrum. Normalization over probabilities, rather than amplitudes, modifies the factor 2. The temporal branch and that convention therefore accompany each published width.

Equality of mass and width for conjugate states also requires the channels to be transported:

$$U_C H_{\mathrm{eff}}^X(z) U_C^{-1} = H_{\mathrm{eff}}^{\bar{X}}(z).$$

The same poles are then transported with the same causal prescription. An antiunitary representation additionally requires that prescription to be transported: bare conjugation of z interchanges retarded and advanced poles.

94.13.13 Three topological codomains and the virtual sector

Topological classes preserve objects and fusion rules, not a universal scalar mass. The primary datum is $(N_{ab}^c, F_d^{abc}, R_c^{ab})$, with its coherence equations. An energy gap subsequently requires a Hamiltonian realization and a spectral projector. Assigning this output type makes it possible to include these sectors without turning them into elementary-mass rows.

94.13.13.1 Sector Fibonacci

The objects $\mathbf{1}, \tau$ satisfy $\tau \otimes \tau = \mathbf{1} \oplus \tau$. The positive dimension d_τ preserves fusion multiplication: $d_\tau^2 = 1 + d_\tau$. The corpus realization identifies this dimension with the internal coordinate φ already generated; the fusion equation is its recognition in this codomain, not a substitute for its APP–TRIT–TPK genealogy. In the fixed convention,

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \qquad R_{\mathbf{1}}^{\tau\tau} = e^{-4\pi i/5}, \quad R_\tau^{\tau\tau} = e^{3\pi i/5}.$$

The identity $\varphi^{-2} + \varphi^{-1} = 1$ directly proves that $F^*F = I$; off-diagonal terms cancel. The phases of R preserve the braiding orientation.

94.13.13.2 Ising sector and Majorana realization

The rules are

$$\psi \otimes \psi = \mathbf{1}, \quad \psi \otimes \sigma = \sigma, \quad \sigma \otimes \sigma = \mathbf{1} \oplus \psi,$$

with

$$F_{\sigma}^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad R_{\mathbf{1}}^{\sigma\sigma} = e^{-\pi i/8}, \quad R_{\psi}^{\sigma\sigma} = e^{3\pi i/8}.$$

Matrix multiplication gives $F^2 = I$. The sector preserves its three fusion types and its representation; its Majorana designation does not automatically turn a neutrino from the preceding sector into a self-conjugate mode. Such a statement uses the specific antiunitary conditions of the neutral realization.

94.13.13.3 Abelian sector with six classes

For $a, b \in \mathbb{Z}/6\mathbb{Z}$,

$$a \otimes b = a + b \pmod 6, \qquad R_{ab} = \zeta_6^{ab}, \qquad \zeta_6 = e^{2\pi i/6}.$$

The identity $R_{a+b,c} = R_{a,c}R_{b,c}$ and its analogue in the second argument express the bilinear character of braiding. The doubled construction $C \boxtimes C^{\mathrm{rev}}$ preserves both orientations. Its classes must not be counted as six new elementary masses.

94.13.13.4 Finite persistence and virtuality

The virtual sector preserves finite extensions, their channel, and their persistence horizon. A finite section with a terminal obstruction does not occupy the same codomain as an asymptotically persistent section. Virtuality is read in an amplitude or resolvent and is not, by definition, equivalent to a universal imaginary mass. Nor does the absence of certification of a global extension in a record suffice to assert that an obstruction exists: the horizon and its proof are different data. The classification precisely preserves this finite information without inventing a stable particle for every internal calculation line.

94.13.14 Classification into twenty-three classes and scope of their realizations

The public inventory brings together 17 elementary classes, 2 composite families, 3 topological codomains and 1 virtual sector: $17+2+3+1 = 23$. The three neutrino rows name flavors, the six quark rows preserve their chromatic triplicity and the gluon row preserves the adjoint representation. None of these counting conventions modifies the atlas $81/56/13/324$.

The following list retains the output type and scope of the documented realizations. The expressions "declared signature" and "fiber operator" refer to the explicit constructions of this section, not to different degrees of decimal precision. The provenance record distinguishes the electronic scalar closure and the two null charts from the physical scalarization still pending in the noncentral families. In neutrinos it retains as pending the complete realization of mixing and scale; in quarks, flavor selection and its scale chart; in the nominal bosons, the joining of route and pole; in the named composites, the scalar realization of their link. These documentary states do not alter the operators, grammars or conditional evaluations that have been presented.

Class	Representatives	Preserved result and realization operation
1	e^-, e^+	Rank-one scalar beta closure and conjugate representation.
2	μ^-, μ^+	Exact fiber operators; nominal evaluation and order-3 refinement with a declared signature; distinct physical spectral selection.
3	τ^-, τ^+	Exact fiber operators; nominal evaluation and order-3 refinement with a declared signature; distinct physical spectral selection.
4	$\nu_e, \bar{\nu}_e$	Neutral fiber operator; realization of mixing, mass frame, and physical scale through their complete maps.

Class	Representatives	Preserved result and realization operation
5	$\nu_\mu, \bar{\nu}_\mu$	Same neutral domain, with a different flavor label; not an individual atlas depth.
6	$\nu_\tau, \bar{\nu}_\tau$	Same neutral domain; preservation of the Dirac–Majorana alternative under its conditions.
7	$u, \bar{u}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
8	$d, \bar{d}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
9	$c, \bar{c}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
10	$s, \bar{s}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
11	$t, \bar{t}$	Color-fiber operator and nominal angular reading; explicit mass convention and comparison channel.
12	$b, \bar{b}$	Color-fiber operator and nominal angular reading; scalarization, scheme, and scale preserved.
13	Photon	Exactly zero-mass section; electromagnetic representation.
14	Gluons	Exactly zero-mass section; adjoint color representation.
15	W^+, W^-	Complete nominal signature and order-3 evaluation; joining to route, projector, and pole as a distinct physical realization.
16	Z^0	Complete nominal signature and order-3 evaluation; joining to route, projector, and pole as a distinct physical realization.
17	H	Scalar descriptor and nominal angular evaluation; projector, couplings, and pole of the scalar realization.
18	Mesons	Exact grammar of the dual pair and bond; recovered composite signatures; distinct nominal scalarization and pole channels.
19	Baryons	Exact grammar of the color triple and bond; recovered proton and neutron signatures; distinct nominal binding realization.
20	Fibonacci	Fusion and braiding codomain; a gap requires its Hamiltonian realization.
21	Ising/Majorana	Fusion codomain and real structure according to the representation; no assigned universal scalar mass.
22	$\mathbb{Z}/6\mathbb{Z}$	Six braiding classes; a topological codomain, not six universal masses.
23	Secciones virtuales	Finite extension and obstruction horizon; no universal stable scalar.

The exact documentary states are published in Appendix B.3: table 94.6 preserves the status legend and table 94.7 preserves, for the twenty-three classes, residence, incidence, representation and scalar state.

Proposition 94.26 (Realization fiber product and observable codomain)**.** *For $X \in \mathcal{P}_{\mathrm{adm}}$, the fiber defined in* (94.12) *is nonempty if and only if*

$$d_{\mathcal{P}}(X) \in \operatorname{im} d_{\mathcal{R}}.$$

On that fiber, the joint spectral measure (94.13) *determines the observable codomain of the readers $(M_X^{\beta,D}, M_X^{\beta,\Omega})$. A preparation P_X publishes a scalar line m exactly when it satisfies the criterion of proposition 94.18; in the other cases it preserves the multiplet, spectral distribution or sectoral type produced.*

Proof. By the definition of image, the stated condition is equivalent to the existence of a route $\gamma \in \mathcal{R}_{\mathrm{adm}}$ with $d_{\mathcal{R}}(\gamma) = d_{\mathcal{P}}(X)$; this is precisely the condition for membership in F_X. The second statement follows from the spectral theorem applied to the two diagonal operators and the characterization of single-point support proved in Proposition 94.18. □

The external documentary snapshot contains 471 mass observables and 384 width observables, totaling 855. They are comparison rows, not 855 internal predictions. In particular, the 243 mesonic mass rows and 166 baryonic rows do not coincide with the cardinalities 432 and 1728 of the grammar: they count objects of different types. The external table is opened after identity, fiber, record, operator, projector and chart are fixed.

Reconstruction of a positive ratio through a tower receiving that ratio demonstrates the representational capacity of the evaluation system. The independent output of a species, by contrast, requires the tower's coefficients to arise from its route and boundary. This separation preserves both representational completeness and effective sectorial results. The whole is organized by a common rule: construct the structure and its memory, apply the reader of the corresponding type, and then perform physical recognition.

94.14 Persistence, fusion structures, and spectral realisations

The atlas and mass operators admit different representations. The classification of these representations preserves the datum characterising each: a multiplication rule, a real conjugation, an exchange phase, or a spectral measure. This section places the topological sectors and neutral realisations within the same architecture. Mass coordinates are obtained once the representation and its dynamics have been determined.

The distinction between these stages is particularly visible in the areal–volumetric sector. An incidence matrix can fix a fusion ring and its positive dimension; the energy of an excitation also depends on the operator acting in the representation. The first determination retains exact content when several dynamic realisations of the same ring are studied. The distinction preserves the structural information preceding the magnitude.

94.14.1 The based ring of areal–volumetric incidence

The primitive incidence of the sheets, obtained from the TPK calendar, is

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

In the basis $(1, \tau)$, the first column expresses $\tau \cdot 1 = \tau$; the other expresses $\tau \cdot \tau = 1 + \tau$. The incidence thus induces the based ring

$$\mathcal{F}_{\mathrm{av}} = \mathbb{Z}[\tau]/(\tau^2 - \tau - 1), \qquad \mathcal{F}_{\mathrm{av}}^{+} = \{a + b\tau : a, b \in \mathbb{Z}_{\geq 0}\}.$$

The positive cone records multiplicities. The operation is internal to the two types and does not use a mass as a coefficient.

Proposition 94.27 (Product and positive dimension)**.** *We have*

$$(a + b\tau)(c + d\tau) = (ac + bd) + (ad + bc + bd)\tau.$$

There is a unique ring homomorphism $d : \mathcal{F}_{\mathrm{av}} \to \mathbb{R}$ with $d(1) = 1$ and $d(\tau) > 0$. Its value at τ is the golden coordinate recognised in the realisation of F_{av}.

Proof. Expanding the product and substituting $\tau^2 = 1 + \tau$ gives the coefficients. A unital homomorphism is determined by $x = d(\tau)$, with $x^2 = x + 1$. Its roots are $(1 + \sqrt{5})/2$ and $(1 - \sqrt{5})/2$; only the first is

positive. Evaluation at that root defines a homomorphism of the quotient. The equation identifies the dimension of this representation; the HMT regional coordinate has been constructed earlier. □

With $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$, the powers satisfy

$$\tau^n = F_{n-1} + F_n\tau \qquad (n \geq 1).$$

The initial case is immediate; multiplication by τ reapplies the incidence and produces the induction step. The coefficients thus retain their origin as counts of sheet composition.

A categorical representation adds morphism spaces and an associator to these multiplicities. A braided representation adds compatible exchange operators. The material realisation adds the Hamiltonian, its domains, and the preparation selecting a spectral sector. If it has a gap $\Delta E > 0$ and effective velocity c_{eff}, its gap-mass coordinate is

$$m_{\text{brecha}} = \Delta E\, c_{\text{eff}}^{-2}.$$

The positive dimension of the ring and the gap are outputs of different operators. The fusion rule preserves its content without by itself fixing a universal mass for all its realisations.

94.14.2 Dimension and representation type

In a pointed representation, all simple objects are invertible. If X is simple and invertible and d is a positive dimension compatible with duals, then

$$1 = d(X \otimes X^*) = d(X)d(X^*) = d(X)^2,$$

and $d(X) = 1$. Dimension therefore provides a check on changes of representation.

Proposition 94.28 (Dimension obstruction in the pointed sector)**.** *A simple object of dimension φ or $\sqrt{2}$ cannot be identified, preserving dimension, with a simple object of a pointed representation. In particular, the* 81 *invertible simple objects of the untwisted centre based on $\mathbb{Z}/9\mathbb{Z}$ do not provide that identification.*

Proof. Invertible simple objects have dimension 1, different from both dimensions. In the stated abelian untwisted centre, a simple object is labelled by a group element and a character of its dual. Both sets have 9 elements. The 81 labels are invertible under multiplication of the group and of the character. □

This census of 81 has a domain different from the atlas of 81 cells. An extension or functor to another codomain specifies the actions and maps it transports. Areal–volumetric incidence preserves its exact ring, and the topological realisation preserves the conditions of its own type.

94.14.3 Neutral self-conjugacy and real generators of CAR

A real structure on a complex space is an antiunitary operator C with $C^2 = I$. Its fixed part is a real space. Commutation of the mass operator with C preserves that real part on its domain. Restricting an evolution to the fixed part is expressed through commutation of its group with C; in a Schrodinger evolution, this condition must be distinguished from commutation of the Hamiltonian. Both conditions belong to the material representation of Section 90.15.2.

Proposition 94.29 (Antiunitary conjugation and time evolution)**.** *Let H be self-adjoint, $\hbar > 0$, and C an antiunitary conjugation preserving $\mathcal{D}(H)$. For $U(t) = \exp(-itH/\hbar)$, if $CHC^{-1} = \epsilon H$, with $\epsilon \in \{1, -1\}$, then*

$$CU(t)C^{-1} = U(-\epsilon t). \tag{94.27}$$

The group $U(t)$ preserves the fixed part of C for every t if and only if it commutes with C, equivalently if $CHC^{-1} = -H$. If instead $CH = HC$, a vector fixed by C remains fixed by it throughout the evolution exactly when it belongs to $\ker H$.

Proof. Antiunitary conjugation sends i to $-i$. Spectral functional calculus gives

$$C\exp(-itH/\hbar)C^{-1} = \exp\bigl(itCHC^{-1}/\hbar\bigr),$$

which proves (94.27). If $U(t)$ commutes with C, it transports its fixed vectors to fixed vectors. Conversely, every complex vector is uniquely written as $x + iy$ with x, y fixed by C. Preservation of that real part implies, by linearity of $U(t)$ and antilinearity of C, that the two commute. Uniqueness of the self-adjoint generator of the unitary group then identifies CHC^{-1} with $-H$.

In the commuting case $\epsilon = 1$, for $C\psi = \psi$ we have $CU(t)\psi = U(-t)\psi$. The condition that $U(t)\psi$ be fixed for every t is equivalent to $U(2t)\psi = \psi$ for every t. By spectral calculus, this is equivalent to the spectral measure of ψ being supported on $\{0\}$, that is, to $\psi \in \ker H$. □

Thus an antiunitary symmetry commuting with H reverses time direction. A realisation through preserved real states uses a different condition on the generator. This distinction preserves the scope of algebraic self-conjugacy and allows the dynamics of a Majorana field to be specified separately.

Proposition 94.30 (Self-conjugacy and charge)**.** *Let Q be a self-adjoint charge operator and C a conjugation transporting its domain and satisfying $CQC^{-1} = -Q$. If $\psi \neq 0$, $Q\psi = q\psi$, and $C\psi = \psi$, then $q = 0$.*

Proof. Self-adjointness implies $q \in \mathbb{R}$. Applying C to the eigenvalue equation gives $-Q\psi = q\psi$. Together with the initial equation this yields $2q\psi = 0$, hence $q = 0$. □

Neutrality is necessary in this self-conjugacy problem; the kernel of Q does not by itself specify an antiunitary operator or its compatibility with the evolution group. The criterion applies to the neutrino blocks after constructing their operators and mixing. The fixed dodecaphase word, the real spinor, and the charge representation thus retain their respective domains.

The fermionic completion provides another realisation. On the exterior space of the stated modes, a_j^* is exterior multiplication by the basis vector and a_j is its adjoint. Exterior antisymmetry gives

$$\{a_j, a_k\} = 0, \qquad \{a_j, a_k^*\} = \delta_{jk} I.$$

Proposition 94.31 (Self-adjoint generators of CAR)**.** *The operators*

$$\gamma_{2j-1} = a_j + a_j^*, \qquad \gamma_{2j} = -i(a_j - a_j^*)$$

are self-adjoint and satisfy $\{\gamma_r, \gamma_s\} = 2\delta_{rs} I$.

Proof. Adjunction fixes each operator. For the same mode, $a_j^2 = (a_j^*)^2 = 0$ and $a_j a_j^* + a_j^* a_j = I$, so each square equals I. The anticommutator between the two combinations of the same mode vanishes by expansion. For distinct modes, all elementary anticommutators vanish. □

These are real generators of a Clifford structure. A topological zero mode additionally requires the dynamics to realise a localised zero-energy excitation, separated from the continuum by a gap and stable under the perturbations considered. The algebraic and spectral properties are linked through the dynamic realisation.

94.14.4 Exchange phases and winding of order six

For $n \geq 2$, in the order-six reader the exponent sum defines $w_6 : B_n \to \mathbb{Z}/6\mathbb{Z}$. The Artin relations preserve this sum: commutation of distant generators retains two terms, and the relation involving three generators retains three.

Proposition 94.32 (Characters of the winding reader)**.** *The unitary characters of the quotient are*

$$\chi_k(b) = \exp\left(\frac{2\pi i k}{6} w_6(b)\right), \qquad k = 0, 1, \ldots, 5.$$

A scalar representation of this form descends to the permutation group exactly for $k = 0$ or $k = 3$.

Proof. A cyclic character is determined by the image q of 1, with $q^6 = 1$. To descend to permutations, it must satisfy the additional relation $\sigma_j^2 = 1$, which requires $q^2 = 1$. Among the sixth roots, only $q = 1, -1$ satisfy this condition. Both satisfy all relations of the permutation group. □

The other four characters preserve winding information. A material species requires the reader to be realised on a multisection and the dynamics to preserve its structure. Route reversal changes w_6 to $-w_6$ and transports each character to its complex conjugate.

94.14.5 Stability and visibility of a section

The positivity proved in Section 91.13 has a spectral consequence. For a positive self-adjoint operator M, functional calculus gives $\mathrm{Spec}(M^2) \subset [0, \infty)$. In a realisation identifying M^2 with the squared-mass observable on a physical space with positive inner product, a persistent eigensection cannot have a negative eigenvalue of that observable. An unstable direction of the linearisation of a background belongs to another spectral problem, studied with its domain and dynamics.

Visibility depends on coupling. If P_X is the support of a section and P_a is the projector of a channel, $P_a P_X = 0$ expresses orthogonality of supports. A nonzero product expresses possible incidence; its intensity is determined by the coupling operator. Two projectors can commute and have nonzero product, so the commutator does not replace that incidence.

In a neutral block, a section sterile with respect to the weak channel belongs to the kernel of its coupling. If that kernel is invariant under the mass operator, the restriction has its own spectral measure. Atlas depth, neutrality and absence of weak coupling are distinct properties verified on the same section. The representational possibilities are preserved without turning a family label into an external mass.

The organisation thus preserves the order of the argument: incidence determines multiplicities; lifts determine conjugations and phases; the representation and material operator determine the spectral output. Comparison is subsequently applied to that output, with its units and coupling stated.

94.15 Complete census of regional signatures and orbits

The following tables complete the finite proof in Section 10.7. Each signature is obtained from (10.82); its multiplicity counts the initial conditions of one cursor, not pairs of cursors. Pairs are formed subsequently through (10.83).

94.15.1 Signatures of the two sheets

Sheet	Six-window signature	Preimages
Additive	`122564`	18
Additive	`122898`	18
Additive	`212645`	18
Additive	`212988`	18

Sheet	Six-window signature	Preimages
Additive	221456	18
Additive	221889	18
Additive	456221	18
Additive	465898	18
Additive	546988	18
Additive	564122	18
Additive	645212	18
Additive	654889	18
Additive	889221	18
Additive	889654	18
Additive	898122	18
Additive	898465	18
Additive	988212	18
Additive	988546	18
Multiplicative	000000	108
Multiplicative	177393	8
Multiplicative	177555	8
Multiplicative	177771	8
Multiplicative	339555	8
Multiplicative	339771	8
Multiplicative	339933	8
Multiplicative	393177	8
Multiplicative	393393	8
Multiplicative	393555	8
Multiplicative	555177	8
Multiplicative	555339	8
Multiplicative	555393	8
Multiplicative	555555	24
Multiplicative	555717	8
Multiplicative	555771	8
Multiplicative	555933	8
Multiplicative	717555	8
Multiplicative	717717	8
Multiplicative	717933	8
Multiplicative	771177	8
Multiplicative	771339	8
Multiplicative	771555	8
Multiplicative	933339	8
Multiplicative	933555	8
Multiplicative	933717	8

94.15.2 Orbits of the ternary image

The representative is the lexicographic minimum of the orbit. “Fiber” counts decimal emissions over a word and “Weight” counts its joint initial conditions. The diagonals are gathered over the entire orbit. The census $\nu = (n_0, n_1, n_2)$ preserves the multiplicity of symbols and is invariant under permutations.

Representante	Orbit	Fibre	Weight	ν	Diagonals
001002	6	2	288	$(4,1,1)$	$\{0,1,3,4,6,7\}$
001011	6	2	288	$(3,3,0)$	$\{0,1,3,4,6,7\}$
001012	6	2	288	$(3,2,1)$	$\{2,5,8\}$
001021	6	2	2088	$(3,2,1)$	$\{2,5,8\}$
001022	6	1	144	$(3,1,2)$	$\{2,5,8\}$
001100	3	6	864	$(4,2,0)$	$\{2,5,8\}$
001101	6	2	288	$(3,3,0)$	$\{0,1,3,4,6,7\}$
001110	6	1	1944	$(3,3,0)$	$\{0,3,6\}$
001112	6	5	1008	$(2,3,1)$	$\{0,1,2,3,4,5,6,7,8\}$
001202	6	1	144	$(3,1,2)$	$\{2,5,8\}$
001210	6	1	144	$(3,2,1)$	$\{2,5,8\}$
001211	6	2	576	$(2,3,1)$	$\{2,5,8\}$
001220	6	6	864	$(3,1,2)$	$\{0,1,3,4,6,7\}$
001222	6	1	144	$(2,1,3)$	$\{2,5,8\}$
002012	6	1	144	$(3,1,2)$	$\{1,4,7\}$
002112	6	1	144	$(2,2,2)$	$\{0,3,6\}$
002211	6	1	144	$(2,2,2)$	$\{1,4,7\}$
002220	6	1	144	$(3,0,3)$	$\{0,3,6\}$
011021	6	2	288	$(2,3,1)$	$\{0,1,3,4,6,7\}$
011112	6	2	288	$(1,4,1)$	$\{0,1,3,4,6,7\}$
011120	6	1	144	$(2,3,1)$	$\{0,3,6\}$
011201	6	1	1944	$(2,3,1)$	$\{1,4,7\}$
011211	6	1	144	$(1,4,1)$	$\{0,3,6\}$
011220	6	1	144	$(2,2,2)$	$\{1,4,7\}$
011222	6	1	144	$(1,2,3)$	$\{1,4,7\}$
012112	6	3	432	$(1,3,2)$	$\{1,2,4,5,7,8\}$
012121	6	1	144	$(1,3,2)$	$\{0,3,6\}$
012202	6	1	144	$(2,1,3)$	$\{0,3,6\}$
012210	6	1	144	$(2,2,2)$	$\{2,5,8\}$
012220	6	1	144	$(2,1,3)$	$\{1,4,7\}$
012222	6	1	144	$(1,1,4)$	$\{2,5,8\}$
021022	6	1	144	$(2,1,3)$	$\{2,5,8\}$
021121	6	3	432	$(1,3,2)$	$\{0,1,2,3,4,5,6,7,8\}$
021202	6	3	432	$(2,1,3)$	$\{0,1,2,3,4,5,6,7,8\}$
022022	3	2	288	$(2,0,4)$	$\{1,4,7\}$
022112	6	1	144	$(1,2,3)$	$\{1,4,7\}$
022121	6	6	864	$(1,2,3)$	$\{0,1,2,3,4,5,6,7,8\}$
022202	3	2	288	$(2,0,4)$	$\{0,3,6\}$
022211	6	2	288	$(1,2,3)$	$\{1,2,4,5,7,8\}$
022222	6	1	144	$(1,0,5)$	$\{1,4,7\}$
112112	3	6	1440	$(0,4,2)$	$\{0,1,2,3,4,5,6,7,8\}$
112211	3	4	576	$(0,4,2)$	$\{1,2,4,5,7,8\}$
112222	6	2	288	$(0,2,4)$	$\{1,2,4,5,7,8\}$

94.16 Catalogues of multisections, route types and sector classes

This appendix organizes three distinct levels of the internal inventory. The 13 multisections bring together cells of the atlas; the 56 families classify internal routes; the 23 classes document the articulation with physical sectors and representations. These cardinalities do not count the same object and are not added to obtain a number of particles. The tables accompany the definitions and proofs of the article: their function is to allow each record, its incidences and its readers to be located, not to replace a proof with a census.

The documentary order of the rows, their identifiers and all repetitions appearing in the sources are preserved. Literal identifiers allow the tables to be checked against the JSON files. A dash denotes an empty source cell; it does not represent zero. The files also preserve every field and digit, the original statuses and verifiable provenance. Sections 94.17 and 94.18 collect the 324 oriented routes and the 855 metrological comparison records.

94.16.1 The 13 multisections of the atlas

For a multisection M, let $C(M)$ be its set of cells and let $\gamma_{B_1}(c)$ be the primary route preserved by the atlas in cell c. In this structural appendix, the fiber is defined by

$$B_1(M) := \operatorname{span}\{e_{\gamma_{B_1}(c)} : c \in C(M)\} \subseteq \ell^2(\mathcal{R}_{\mathrm{adm}}).$$

The primary routes of distinct cells determine distinct basis vectors; consequently, $\operatorname{rank} B_1(M) = \dim B_1(M) = \#C(M)$. This definition fixes the object to which the rank in the table refers; the family column counts the route families present, not physical species. The integer spectra of the readers K_D and K_Ω are preserved separately. In the literal notation `x` indicates multiplicity. The 13 rows have the documentary state of an exact diagonal operator on the fiber B_1. Its publication as a physical scalar requires the declared transport, except for the rank-one selection provided for in the record itself. This statement reproduces the recorded scope and does not turn all spectra into unique masses.

Table 94.4: The 13 multisections, without row selection.

Multisection	Rank and incidence	Classification and depth	Lectores enteros
charged_lepton_G0	*Rank*: 1 *Cells (identifiers)*: 41 *Families*: 1	*Family*: charged_lepton *Depth*: G0_center *Color*: none	*K D*: (80,54,6)x1 *K Ω*: (80,54,6)x1
charged_lepton_G2	*Rank*: 8 *Cells (identifiers)*: 21,23,25,39,43,57,59,61 *Families*: 8	*Family*: charged_lepton *Depth*: G2 *Color*: none	*K D*: (0,24,0)x1; (40,78,27)x2; (80,54,7)x1; (80,66,2)x1; (80,66,3)x1; (100,66,0)x1; (100,66,2)x1 *K Ω*: (80,54,6)x8
charged_lepton_G4	*Rank*: 16 *Cells (identifiers)*: 1,3,5,7,9,19,27,37,45,55,63,73,75,77,79,81 *Families*: 16	*Family*: charged_lepton *Depth*: G4 *Color*: none	*K D*: (0,24,0)x1; (40,24,2)x2; (40,54,6)x2; (40,84,30)x2; (60,78,25)x3; (80,66,2)x2; (80,66,3)x3; (80,66,4)x1 *K Ω*: (24,54,2)x1; (56,54,5)x4; (80,54,6)x11

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Continuation

Multisection	Rank and incidence	Classification and depth	Lectores enteros
neutrino_G1	*Rank*: 2 *Cells (identifiers)*: 40,42 *Families*: 2	*Family*: neutrino *Depth*: G1 *Color*: none	*K D*: (40,24,2)x1; (40,78,27)x1 *K* Ω: (40,54,4)x1; (80,54,6)x1
neutrino_G2	*Rank*: 4 *Cells (identifiers)*: 22,24,58,60 *Families*: 4	*Family*: neutrino *Depth*: G2 *Color*: none	*K D*: (0,24,0)x1; (40,84,30)x1; (60,78,25)x1; (80,66,3)x1 *K* Ω: (4,54,2)x1; (80,54,6)x3
neutrino_G3	*Rank*: 6 *Cells (identifiers)*: 20,26,38,44,56,62 *Families*: 6	*Family*: neutrino *Depth*: G3 *Color*: none	*K D*: (40,24,2)x2; (40,78,27)x1; (60,84,28)x1; (80,54,6)x1; (80,66,3)x1 *K* Ω: (36,54,4)x1; (56,54,5)x1; (80,54,6)x4
neutrino_G4	*Rank*: 8 *Cells (identifiers)*: 2,4,6,8,74,76,78,80 *Families*: 8	*Family*: neutrino *Depth*: G4 *Color*: none	*K D*: (0,24,0)x1; (40,24,2)x1; (40,78,27)x2; (40,84,30)x1; (80,54,7)x1; (80,66,2)x1; (100,66,0)x1 *K* Ω: (36,54,4)x1; (56,54,5)x1; (80,54,6)x6
quark_down_G1	*Rank*: 2 *Cells (identifiers)*: 32,50 *Families*: 2	*Family*: quark_down *Depth*: G1 *Color*: B | G	*K D*: (40,24,2)x1; (40,78,27)x1 *K* Ω: (40,54,4)x1; (80,54,6)x1
quark_down_G2	*Rank*: 4 *Cells (identifiers)*: 30,34,48,52 *Families*: 4	*Family*: quark_down *Depth*: G2 *Color*: B | G | R	*K D*: (0,24,0)x1; (40,84,30)x1; (60,78,25)x1; (80,66,2)x1 *K* Ω: (4,54,2)x1; (80,54,6)x3
quark_down_G3	*Rank*: 6 *Cells (identifiers)*: 12,14,16,66,68,70 *Families*: 6	*Family*: quark_down *Depth*: G3 *Color*: B | G | R	*K D*: (40,24,2)x2; (40,78,27)x1; (60,84,28)x1; (80,54,6)x1; (80,66,4)x1 *K* Ω: (36,54,4)x1; (80,54,6)x5
quark_down_G4	*Rank*: 8 *Cells (identifiers)*: 10,18,28,36,46,54,64,72 *Families*: 8	*Family*: quark_down *Depth*: G4 *Color*: B | G | R	*K D*: (0,24,0)x1; (40,24,2)x1; (40,78,27)x2; (40,84,30)x1; (80,54,7)x1; (80,66,3)x1; (100,66,2)x1 *K* Ω: (36,54,4)x1; (80,54,6)x7

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Continuation

Multisection	Rank and incidence	Classification and depth	Lectores enteros
quark_up_G1	*Rank*: 4 *Cells (identifiers)*: 31,33,49,51 *Families*: 3	*Family*: quark_up *Depth*: G1 *Color*: B | G | R	*K D*: (40,54,6)x1; (60,78,25) x1; (80,66,2)x1; (80,66,3)x1 *K* Ω: (80,54,6)x4
quark_up_G3	*Rank*: 12 *Cells (identifiers)*: 11,13,15,17,29,35, 47,53,65,67,69,71 *Families*: 12	*Family*: quark_up *Depth*: G3 *Color*: B | G | R	*K D*: (40,24,2)x2; (40,78,27) x2; (60,84,28)x1; (80,54,6) x3; (80,66,3)x1; (80,66,4)x1; (100,66,0)x1; (100,66,2)x1 *K* Ω: (56,54,5)x3; (80,54,6) x9

94.16.2 Catalog of the 56 route types

The signature identifying a family is preserved literally, with its signs and separators. The cells and multisections make visible that a route family is not by definition equivalent to a physical species. The joint profiles count the combinations recorded within the family. The decimal spectra of the character and the complete signature spectra remain intact in `catalogo_familias.json`; here the incidence and the integer readers that allow navigation of that information are published.

Table 94.5: The 56 internal route families.

Family and signature	Incidence	K_D	K_Ω
1 24|0|12|0|-12| +++---+++---	*Number of cells*: 5 *Cells (identifiers)*: 21,31,41,51,81 *Multisections*: charged_lepton_G0 | charged_lepton_G2 | charged_ lepton_G4 | quark_up_G1 *Families*: charged_lepton | quark_ up *Color*: R | none *Joint profiles*: 5	(0,24,0)x2; (40,54,6) x1; (60,78,25)x1; (80,54,6)x1	(24,54,2)x1; (80,54, 6)x4
2 24|0|4|0|-12| +++---+++---	*Number of cells*: 2 *Cells (identifiers)*: 11,61 *Multisections*: charged_lepton_G2 | quark_up_G3 *Families*: charged_lepton | quark_ up *Color*: R | none *Joint profiles*: 2	(60,84,28)x1; (80,54, 7)x1	(80,54,6)x2

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Continuation

Family and signature	Incidence	K_D	K_Ω
3 24|0|8|0|-12| +++---+++---	*Number of cells*: 2 *Cells (identifiers)*: 1,71 *Multisections*: charged_lepton_G4 | quark_up_G3 *Families*: charged_lepton | quark_up *Color*: R | none *Joint profiles*: 2	(60,78,25)x1; (80,54,6)x1	(80,54,6)x2
4 28|-6|-2|-2|-8| 0++---0++---	*Number of cells*: 1 *Cells (identifiers)*: 53 *Multisections*: quark_up_G3 *Families*: quark_up *Color*: G *Joint profiles*: 1	(80,66,3)x1	(56,54,5)x1
5 28|-6|-4|-2|-8| 0++---0++---	*Number of cells*: 1 *Cells (identifiers)*: 20 *Multisections*: neutrino_G3 *Families*: neutrino *Color*: none *Joint profiles*: 1	(80,66,3)x1	(56,54,5)x1
6 28|-6|0|0|-12| +++---+++---	*Number of cells*: 2 *Cells (identifiers)*: 30,70 *Multisections*: quark_down_G2 | quark_down_G3 *Families*: quark_down *Color*: G *Joint profiles*: 2	(40,24,2)x1; (80,66,2)x1	(80,54,6)x2
7 28|-6|10|0|-12| +++---+++---	*Number of cells*: 2 *Cells (identifiers)*: 33,64 *Multisections*: quark_down_G4 | quark_up_G1 *Families*: quark_down | quark_up *Color*: G *Joint profiles*: 2	(40,24,2)x1; (80,66,2)x1	(36,54,4)x1; (80,54,6)x1
8 28|-6|12|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 10 *Multisections*: quark_down_G4 *Families*: quark_down *Color*: G *Joint profiles*: 1	(100,66,2)x1	(80,54,6)x1

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Continuation

Family and signature	Incidence	K_D	K_Ω
9 28|-6|2|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 3 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(40,84,30)x1	(80,54,6)x1
10 28|-6|4|-2|-8| ++0---++0---	*Number of cells*: 1 *Cells (identifiers)*: 50 *Multisections*: quark_down_G1 *Families*: quark_down *Color*: G *Joint profiles*: 1	(40,78,27)x1	(80,54,6)x1
11 28|-6|4|0|-12| +++---+++---	*Number of cells*: 2 *Cells (identifiers)*: 9,40 *Multisections*: charged_lepton_G4 | neutrino_G1 *Families*: charged_lepton | neutrino *Color*: none *Joint profiles*: 2	(40,24,2)x1; (80,66,2)x1	(40,54,4)x1; (80,54,6)x1
12 28|-6|4|0|-8| +0+---+0+---	*Number of cells*: 1 *Cells (identifiers)*: 80 *Multisections*: neutrino_G4 *Families*: neutrino *Color*: none *Joint profiles*: 1	(40,78,27)x1	(80,54,6)x1
13 28|-6|6|-2|-8| ++0---++0---	*Number of cells*: 1 *Cells (identifiers)*: 74 *Multisections*: neutrino_G4 *Families*: neutrino *Color*: none *Joint profiles*: 1	(40,78,27)x1	(80,54,6)x1
14 28|-6|6|0|-12| +++---+++---	*Number of cells*: 3 *Cells (identifiers)*: 13,43,63 *Multisections*: charged_lepton_G2 | charged_lepton_G4 | quark_up_G3 *Families*: charged_lepton | quark_up *Color*: G | none *Joint profiles*: 3	(40,24,2)x1; (80,66,2)x1; (100,66,2)x1	(80,54,6)x3

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Continuation

Family and signature	Incidence	K_D	K_Ω
15 28|-6|6|0|-8| +0+---+0+---	*Number of cells*: 1 *Cells (identifiers)*: 23 *Multisections*: charged_lepton_G2 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(40,78,27)x1	(80,54,6)x1
16 28|-6|8|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 60 *Multisections*: neutrino_G2 *Families*: neutrino *Color*: none *Joint profiles*: 1	(40,84,30)x1	(80,54,6)x1
17 28|6|-10|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 19 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(40,84,30)x1	(80,54,6)x1
18 28|6|-2|0|-12| +++---+++---	*Number of cells*: 5 *Cells (identifiers)*: 18,29,49,59,79 *Multisections*: charged_lepton_G2 | charged_lepton_G4 | quark_down_G4 | quark_up_G1 | quark_up_G3 *Families*: charged_lepton | quark_down | quark_up *Color*: B | none *Joint profiles*: 4	(40,24,2)x1; (40,78,27)x1; (80,66,3)x2; (100,66,0)x1	(80,54,6)x5
19 28|6|-4|0|-12| +++---+++---	*Number of cells*: 2 *Cells (identifiers)*: 2,42 *Multisections*: neutrino_G1 | neutrino_G4 *Families*: neutrino *Color*: none *Joint profiles*: 2	(40,78,27)x1; (100,66,0)x1	(80,54,6)x2
20 28|6|-6|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 69 *Multisections*: quark_up_G3 *Families*: quark_up *Color*: B *Joint profiles*: 1	(80,66,4)x1	(80,54,6)x1

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Continuation

Family and signature	Incidence	K_D	K_Ω
21 28|6|-8|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 52 *Multisections*: quark_down_G2 *Families*: quark_down *Color*: B *Joint profiles*: 1	(40,84,30)x1	(80,54,6)x1
22 28|6|0|0|-12| +++---+++---	*Number of cells*: 4 *Cells (identifiers)*: 22,32,72,73 *Multisections*: charged_lepton_G4 | neutrino_G2 | quark_down_G1 | quark_down_G4 *Families*: charged_lepton | neutrino | quark_down *Color*: B | none *Joint profiles*: 3	(40,24,2)x1; (40,78,27)x1; (80,66,3)x2	(40,54,4)x1; (80,54,6)x3
23 28|6|2|0|-12| +++---+++---	*Number of cells*: 2 *Cells (identifiers)*: 8,39 *Multisections*: charged_lepton_G2 | neutrino_G4 *Families*: charged_lepton | neutrino *Color*: none *Joint profiles*: 2	(40,24,2)x1; (40,78,27)x1	(36,54,4)x1; (80,54,6)x1
24 28|6|4|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 12 *Multisections*: quark_down_G3 *Families*: quark_down *Color*: B *Joint profiles*: 1	(80,66,4)x1	(80,54,6)x1
25 28|6|8|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 62 *Multisections*: neutrino_G3 *Families*: neutrino *Color*: none *Joint profiles*: 1	(40,24,2)x1	(80,54,6)x1
26 30|-6|-4|-2|-8| 0++---0++---	*Number of cells*: 1 *Cells (identifiers)*: 4 *Multisections*: neutrino_G4 *Families*: neutrino *Color*: none *Joint profiles*: 1	(80,54,7)x1	(56,54,5)x1

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Continuation

Family and signature	Incidence	K_D	K_Ω
27 30|-6|-8|0|-8| +0+---+0+---	*Number of cells*: 1 *Cells (identifiers)*: 55 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(60,78,25)x1	(80,54,6)x1
28 30|-6|0|-2|-8| ++0---++0---	*Number of cells*: 2 *Cells (identifiers)*: 24,34 *Multisections*: neutrino_G2 | quark_down_G2 *Families*: neutrino | quark_down *Color*: R | none *Joint profiles*: 2	(0,24,0)x1; (60,78, 25)x1	(4,54,2)x1; (80,54,6) x1
29 30|-6|0|-2|-8| 0++---0++---	*Number of cells*: 1 *Cells (identifiers)*: 75 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(40,54,6)x1	(56,54,5)x1
30 30|-6|0|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 14 *Multisections*: quark_down_G3 *Families*: quark_down *Color*: R *Joint profiles*: 1	(80,54,6)x1	(80,54,6)x1
31 30|-6|0|0|-8| +0+---+0+---	*Number of cells*: 1 *Cells (identifiers)*: 54 *Multisections*: quark_down_G4 *Families*: quark_down *Color*: R *Joint profiles*: 1	(0,24,0)x1	(80,54,6)x1
32 30|-6|4|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 65 *Multisections*: quark_up_G3 *Families*: quark_up *Color*: R *Joint profiles*: 1	(80,54,6)x1	(80,54,6)x1

Continued on the next page

Continuation

Family and signature	Incidence	K_D	K_Ω
33 30|-6|8|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 44 *Multisections*: neutrino_G3 *Families*: neutrino *Color*: none *Joint profiles*: 1	(60,84,28)x1	(80,54,6)x1
34 30|6|-4|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 68 *Multisections*: quark_down_G3 *Families*: quark_down *Color*: R *Joint profiles*: 1	(60,84,28)x1	(80,54,6)x1
35 30|6|-8|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 17 *Multisections*: quark_up_G3 *Families*: quark_up *Color*: R *Joint profiles*: 1	(80,54,6)x1	(80,54,6)x1
36 30|6|0|0|-12| +++---+++---	*Number of cells*: 2 *Cells (identifiers)*: 38,58 *Multisections*: neutrino_G2 | neutrino_G3 *Families*: neutrino *Color*: none *Joint profiles*: 2	(60,78,25)x1; (80,54,6)x1	(80,54,6)x2
37 30|6|0|0|-8| +++-0-+++-0-	*Number of cells*: 1 *Cells (identifiers)*: 78 *Multisections*: neutrino_G4 *Families*: neutrino *Color*: none *Joint profiles*: 1	(0,24,0)x1	(80,54,6)x1
38 30|6|0|2|-8| +++--0+++--0	*Number of cells*: 1 *Cells (identifiers)*: 27 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(40,54,6)x1	(56,54,5)x1

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Continuation

Family and signature	Incidence	K_D	K_Ω
39 30|6|0|2|-8| +++0--+++0--	*Number of cells*: 1 *Cells (identifiers)*: 48 *Multisections*: quark_down_G2 *Families*: quark_down *Color*: R *Joint profiles*: 1	(0,24,0)x1	(4,54,2)x1
40 30|6|4|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 7 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(60,78,25)x1	(80,54,6)x1
41 30|6|8|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 28 *Multisections*: quark_down_G4 *Families*: quark_down *Color*: R *Joint profiles*: 1	(80,54,7)x1	(80,54,6)x1
42 34|0|-12|-2|-8| 0++---0++---	*Number of cells*: 1 *Cells (identifiers)*: 45 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(80,66,4)x1	(56,54,5)x1
43 34|0|-12|0|-12| +++---+++---	*Number of cells*: 2 *Cells (identifiers)*: 57,76 *Multisections*: charged_lepton_G2 | neutrino_G4 *Families*: charged_lepton | neutrino *Color*: none *Joint profiles*: 2	(40,84,30)x1; (80,66,2)x1	(80,54,6)x2
44 34|0|-16|0|-8| +++-0-+++-0-	*Number of cells*: 1 *Cells (identifiers)*: 37 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(40,24,2)x1	(80,54,6)x1

Continued on the next page

Continuation

Family and signature	Incidence	K_D	K_Ω
45 34|0|-16|0|-8| +0+---+0+---	*Number of cells*: 1 *Cells (identifiers)*: 15 *Multisections*: quark_up_G3 *Families*: quark_up *Color*: B *Joint profiles*: 1	(40,78,27)x1	(80,54,6)x1
46 34|0|-20|-2|-8| 0++---0++---	*Number of cells*: 1 *Cells (identifiers)*: 35 *Multisections*: quark_up_G3 *Families*: quark_up *Color*: B *Joint profiles*: 1	(100,66,0)x1	(56,54,5)x1
47 34|0|-4|-2|-8| ++0---++0---	*Number of cells*: 1 *Cells (identifiers)*: 56 *Multisections*: neutrino_G3 *Families*: neutrino *Color*: none *Joint profiles*: 1	(40,24,2)x1	(36,54,4)x1
48 34|0|-4|0|-12| +++---+++---	*Number of cells*: 3 *Cells (identifiers)*: 6,25,46 *Multisections*: charged_lepton_G2 | neutrino_G4 | quark_down_G4 *Families*: charged_lepton | neutrino | quark_down *Color*: B | none *Joint profiles*: 2	(80,66,2)x1; (80,66,3)x2	(80,54,6)x3
49 34|0|-4|2|-8| +++--0+++--0	*Number of cells*: 1 *Cells (identifiers)*: 67 *Multisections*: quark_up_G3 *Families*: quark_up *Color*: G *Joint profiles*: 1	(100,66,2)x1	(56,54,5)x1
50 34|0|-8|-2|-8| ++0---++0---	*Number of cells*: 1 *Cells (identifiers)*: 66 *Multisections*: quark_down_G3 *Families*: quark_down *Color*: B *Joint profiles*: 1	(40,78,27)x1	(80,54,6)x1

Continued on the next page

Continuation

Family and signature	Incidence	K_D	K_Ω
51 34|0|-8|-2|-8| 0++---0++---	*Number of cells*: 1 *Cells (identifiers)*: 77 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(80,66,3)x1	(56,54,5)x1
52 34|0|-8|0|-12| +++---+++---	*Number of cells*: 1 *Cells (identifiers)*: 36 *Multisections*: quark_down_G4 *Families*: quark_down *Color*: G *Joint profiles*: 1	(40,84,30)x1	(80,54,6)x1
53 34|0|-8|0|-8| +0+---+0+---	*Number of cells*: 1 *Cells (identifiers)*: 5 *Multisections*: charged_lepton_G4 *Families*: charged_lepton *Color*: none *Joint profiles*: 1	(40,24,2)x1	(80,54,6)x1
54 34|0|0|-2|-8| ++0---++0---	*Number of cells*: 1 *Cells (identifiers)*: 26 *Multisections*: neutrino_G3 *Families*: neutrino *Color*: none *Joint profiles*: 1	(40,78,27)x1	(80,54,6)x1
55 34|0|0|0|-8| +0+---+0+---	*Number of cells*: 1 *Cells (identifiers)*: 47 *Multisections*: quark_up_G3 *Families*: quark_up *Color*: G *Joint profiles*: 1	(40,78,27)x1	(80,54,6)x1
56 34|0|0|2|-8| +++0--+++0--	*Number of cells*: 1 *Cells (identifiers)*: 16 *Multisections*: quark_down_G3 *Families*: quark_down *Color*: G *Joint profiles*: 1	(40,24,2)x1	(36,54,4)x1

94.16.3 The 23 documented classes

These rows bring together public sectors, residences and representation. The status key refers to the adjacent legend and preserves the status of the source. In particular, an exact internal operator, a conditional evaluation and a closed scalar realization remain distinct statements. The table does not update their status or turn a conditional record into a prediction.

Table 94.6: Original class statuses.

Key	Estado literal	Lectura documental
C01	CLOSED_ELECTRON_TWO_WAY	Electron: documented bidirectional closure.
C02	CLOSED_NULL_CHART	Null chart closed in this snapshot.
C03	NOT_A_UNIVERSAL_MASS_ SCALAR	The class is not published as a universal mass scalar.
C04	OPEN_FLAVOR_COLOR_ SCALARIZATION	Flavor and color scalarization open at this stage.
C05	OPEN_MIXING_AND_SCALE	Mixing and scale open at this stage.
C06	OPEN_NAMED_STATE_ SCALARIZATION	Scalarization of the individually identified state open at this stage.
C07	OPEN_PHYSICAL_SCALARIZATION	Physical scalarization open at this stage.
C08	OPEN_ROUTE_AND_ SCALARIZATION	Route and scalarization open at this stage.

Table 94.7: The 23 catalogue classes, in their documentary order.

Class	Representatives and residence	Incidence and representation	Status
SM-LC-1 lepton_cargado	*Representatives*: e− / e+ *Residence*: center (5,5) and antisection	*Selected cells*: 1 *Multisections*: charged_lepton_ G0 *Charge*: ±1 e *Representation*: colorless	C01
SM-LC-2 lepton_cargado	*Representatives*: $\mu-$ / $\mu+$ *Residence*: charged-lepton multisection	*Selected cells*: 8 *Multisections*: charged_lepton_ G2 *Charge*: ±1 e *Representation*: colorless	C07
SM-LC-3 lepton_cargado	*Representatives*: $\tau-$ / $\tau+$ *Residence*: charged-lepton multisection	*Selected cells*: 16 *Multisections*: charged_lepton_ G4 *Charge*: ±1 e *Representation*: colorless	C07

Continued on the next page

Continuation

Class	Representatives and residence	Incidence and representation	Status
SM-NU-1 neutrino	*Representatives*: ν_e / anti-ν_e *Residence*: neutrino multisections	*Selected cells*: 20 *Multisections*: neutrino_G1 | neutrino_G2 | neutrino_G3 | neutrino_G4 *Charge*: 0 *Representation*: colorless	C05
SM-NU-2 neutrino	*Representatives*: ν_μ / anti-ν_μ *Residence*: neutrino multisections	*Selected cells*: 20 *Multisections*: neutrino_G1 | neutrino_G2 | neutrino_G3 | neutrino_G4 *Charge*: 0 *Representation*: colorless	C05
SM-NU-3 neutrino	*Representatives*: ν_τ / anti-ν_τ *Residence*: neutrino multisections	*Selected cells*: 20 *Multisections*: neutrino_G1 | neutrino_G2 | neutrino_G3 | neutrino_G4 *Charge*: 0 *Representation*: colorless	C05
SM-Q-U1 quark	*Representatives*: u / anti-u *Residence*: up-type multisection	*Selected cells*: 16 *Multisections*: quark_up_G1 | quark_up_G3 *Charge*: ±2/3 e *Representation*: R,G,B = Z/3 orbit	C04
SM-Q-D1 quark	*Representatives*: d / anti-d *Residence*: down-type multisection	*Selected cells*: 20 *Multisections*: quark_down_G1 | quark_down_G2 | quark_down_G3 | quark_down_G4 *Charge*: ±1/3 e *Representation*: R,G,B = Z/3 orbit	C04
SM-Q-U2 quark	*Representatives*: c / anti-c *Residence*: up-type multisection	*Selected cells*: 16 *Multisections*: quark_up_G1 | quark_up_G3 *Charge*: ±2/3 e *Representation*: R,G,B = Z/3 orbit	C04

Continued on the next page

Continuation

Class	Representatives and residence	Incidence and representation	Status
SM-Q-D2 quark	*Representatives*: s / anti-s *Residence*: down-type multisection	*Selected cells*: 20 *Multisections*: quark_down_G1 | quark_down_G2 | quark_down_G3 | quark_down_G4 *Charge*: ±1/3 e *Representation*: R,G,B = Z/3 orbit	C04
SM-Q-U3 quark	*Representatives*: t / anti-t *Residence*: up-type multisection	*Selected cells*: 16 *Multisections*: quark_up_G1 | quark_up_G3 *Charge*: ±2/3 e *Representation*: R,G,B = Z/3 orbit	C04
SM-Q-D3 quark	*Representatives*: b / anti-b *Residence*: down-type multisection	*Selected cells*: 20 *Multisections*: quark_down_G1 | quark_down_G2 | quark_down_G3 | quark_down_G4 *Charge*: ±1/3 e *Representation*: R,G,B = Z/3 orbit	C04
SM-GA-EM boson_gauge	*Representatives*: γ *Residence*: null gauge sector	*Selected cells*: 0 *Multisections*: — *Charge*: 0 *Representation*: colorless	C02
SM-GA-GL boson_gauge	*Representatives*: g *Residence*: color gauge sector	*Selected cells*: 0 *Multisections*: — *Charge*: 0 *Representation*: gauge octet	C02
SM-GA-W boson_gauge	*Representatives*: W+ / W− *Residence*: massive gauge sector	*Selected cells*: 0 *Multisections*: — *Charge*: ±1 e *Representation*: colorless	C08
SM-GA-Z boson_gauge	*Representatives*: Z0 *Residence*: massive gauge sector	*Selected cells*: 0 *Multisections*: — *Charge*: 0 *Representation*: colorless	C08
SM-H-1 escalar	*Representatives*: H *Residence*: scalar sector	*Selected cells*: 0 *Multisections*: — *Charge*: 0 *Representation*: colorless	C08

Continued on the next page

Continuation

Class	Representatives and residence	Incidence and representation	Status
COMP-MESON meson	*Representatives*: $\pi 0$, $\pi\pm$, K0, K±, η, ρ, φ, J/ψ, $\Upsilon\ldots$ *Residence*: product of section and antisection	*Selected cells*: 0 *Multisections*: — *Charge*: dependent on the composite state *Representation*: singlet or other representations	C06
COMP-BARYON barion	*Representatives*: p, n and baryonic families *Residence*: product of three sections	*Selected cells*: 0 *Multisections*: — *Charge*: dependent on the composite state *Representation*: physical singlet	C06
TOPO-FIB anyon_no_abeliano	*Representatives*: 1, τ *Residence*: topological sector	*Selected cells*: 0 *Multisections*: — *Charge*: defined by the topological codomain *Representation*: not applicable	C03
TOPO-ISING sector_majorana	*Representatives*: 1, ψ, σ *Residence*: topological sector / zero	*Selected cells*: 0 *Multisections*: — *Charge*: dependent on the realization *Representation*: colorless	C03
TOPO-Z6 anyones_abelianos	*Representatives*: boson, fermion and four braided sectors *Residence*: topological sector Z/6	*Selected cells*: 0 *Multisections*: — *Charge*: Z/6 definida por el trenzado *Representation*: colorless	C03
VIRT-1 virtual	*Representatives*: virtual sections *Residence*: TPK inverse system	*Selected cells*: 0 *Multisections*: — *Charge*: dependent on the prolongation sector *Representation*: according to sector	C03

94.17 The 324 oriented routes: complete records

Each record preserves all fields of the union of the two route tables. The correspondence is established by the exact identifier, never by proximity of values. Common fields are compared literally before being represented only once; fields exclusive to either table are incorporated in full. The ordinals of both sources are printed to reconstruct their original orders. The record preserves the decimal figures as they were recorded; it does not recalculate them or attribute mathematical exactness to them merely by printing them.

The returns of class, orbit, cell and kinematic state are distinct fields. The same applies to phase repetition and memory change, to the readers K_D and K_Ω, and to an internal route and its possible physical representation. Documentary states are reproduced literally. SHA-256 fingerprints allow their antecedents to be located, but do not replace their proofs.

Index of identifiers and pages

Each identifier links to the beginning of its record. The indicated page is the volume's printed numbering. Identifiers are read by rows, from left to right, preserving the order of the unified table. The PDF bookmark panel also allows the cell to be chosen first and then the horizontal and vertical orientation. The four orientations of a cell are distinct routes, not versions of the same record.

Table 94.8: Complete index of the 324 oriented routes.

Route identifier	Page	Route identifier	Page
r1c1_h-1v-1	2273	r1c1_h-1v+1	2274
r1c1_h+1v-1	2276	r1c1_h+1v+1	2278
r1c2_h-1v-1	2279	r1c2_h-1v+1	2281
r1c2_h+1v-1	2283	r1c2_h+1v+1	2284
r1c3_h-1v-1	2286	r1c3_h-1v+1	2288
r1c3_h+1v-1	2289	r1c3_h+1v+1	2291
r1c4_h-1v-1	2293	r1c4_h-1v+1	2294
r1c4_h+1v-1	2296	r1c4_h+1v+1	2298
r1c5_h-1v-1	2299	r1c5_h-1v+1	2301
r1c5_h+1v-1	2303	r1c5_h+1v+1	2304
r1c6_h-1v-1	2306	r1c6_h-1v+1	2308
r1c6_h+1v-1	2309	r1c6_h+1v+1	2311
r1c7_h-1v-1	2313	r1c7_h-1v+1	2314
r1c7_h+1v-1	2316	r1c7_h+1v+1	2318
r1c8_h-1v-1	2319	r1c8_h-1v+1	2321
r1c8_h+1v-1	2323	r1c8_h+1v+1	2324
r1c9_h-1v-1	2326	r1c9_h-1v+1	2328
r1c9_h+1v-1	2329	r1c9_h+1v+1	2331

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Index of the 324 routes: identifier and page (continued)

Route identifier	Page	Route identifier	Page
r2c1_h-1v-1	2333	r2c1_h-1v+1	2334
r2c1_h+1v-1	2336	r2c1_h+1v+1	2338
r2c2_h-1v-1	2339	r2c2_h-1v+1	2341
r2c2_h+1v-1	2343	r2c2_h+1v+1	2344
r2c3_h-1v-1	2346	r2c3_h-1v+1	2348
r2c3_h+1v-1	2349	r2c3_h+1v+1	2351
r2c4_h-1v-1	2353	r2c4_h-1v+1	2354
r2c4_h+1v-1	2356	r2c4_h+1v+1	2358
r2c5_h-1v-1	2359	r2c5_h-1v+1	2361
r2c5_h+1v-1	2363	r2c5_h+1v+1	2364
r2c6_h-1v-1	2366	r2c6_h-1v+1	2368
r2c6_h+1v-1	2369	r2c6_h+1v+1	2371
r2c7_h-1v-1	2373	r2c7_h-1v+1	2374
r2c7_h+1v-1	2376	r2c7_h+1v+1	2378
r2c8_h-1v-1	2379	r2c8_h-1v+1	2381
r2c8_h+1v-1	2383	r2c8_h+1v+1	2384
r2c9_h-1v-1	2386	r2c9_h-1v+1	2388
r2c9_h+1v-1	2389	r2c9_h+1v+1	2391
r3c1_h-1v-1	2393	r3c1_h-1v+1	2394
r3c1_h+1v-1	2396	r3c1_h+1v+1	2398
r3c2_h-1v-1	2399	r3c2_h-1v+1	2401
r3c2_h+1v-1	2403	r3c2_h+1v+1	2404
r3c3_h-1v-1	2406	r3c3_h-1v+1	2408
r3c3_h+1v-1	2409	r3c3_h+1v+1	2411
r3c4_h-1v-1	2413	r3c4_h-1v+1	2414
r3c4_h+1v-1	2416	r3c4_h+1v+1	2418
r3c5_h-1v-1	2419	r3c5_h-1v+1	2421
r3c5_h+1v-1	2423	r3c5_h+1v+1	2424
r3c6_h-1v-1	2426	r3c6_h-1v+1	2428
r3c6_h+1v-1	2429	r3c6_h+1v+1	2431
r3c7_h-1v-1	2433	r3c7_h-1v+1	2434

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Index of the 324 routes: identifier and page (continued)

Route identifier	Page	Route identifier	Page
r3c7_h+1v-1	2436	r3c7_h+1v+1	2438
r3c8_h-1v-1	2439	r3c8_h-1v+1	2441
r3c8_h+1v-1	2443	r3c8_h+1v+1	2444
r3c9_h-1v-1	2446	r3c9_h-1v+1	2448
r3c9_h+1v-1	2449	r3c9_h+1v+1	2451
r4c1_h-1v-1	2453	r4c1_h-1v+1	2454
r4c1_h+1v-1	2456	r4c1_h+1v+1	2458
r4c2_h-1v-1	2459	r4c2_h-1v+1	2461
r4c2_h+1v-1	2463	r4c2_h+1v+1	2464
r4c3_h-1v-1	2466	r4c3_h-1v+1	2468
r4c3_h+1v-1	2469	r4c3_h+1v+1	2471
r4c4_h-1v-1	2473	r4c4_h-1v+1	2474
r4c4_h+1v-1	2476	r4c4_h+1v+1	2478
r4c5_h-1v-1	2479	r4c5_h-1v+1	2481
r4c5_h+1v-1	2483	r4c5_h+1v+1	2484
r4c6_h-1v-1	2486	r4c6_h-1v+1	2488
r4c6_h+1v-1	2489	r4c6_h+1v+1	2491
r4c7_h-1v-1	2493	r4c7_h-1v+1	2494
r4c7_h+1v-1	2496	r4c7_h+1v+1	2498
r4c8_h-1v-1	2499	r4c8_h-1v+1	2501
r4c8_h+1v-1	2503	r4c8_h+1v+1	2504
r4c9_h-1v-1	2506	r4c9_h-1v+1	2508
r4c9_h+1v-1	2509	r4c9_h+1v+1	2511
r5c1_h-1v-1	2513	r5c1_h-1v+1	2514
r5c1_h+1v-1	2516	r5c1_h+1v+1	2518
r5c2_h-1v-1	2519	r5c2_h-1v+1	2521
r5c2_h+1v-1	2523	r5c2_h+1v+1	2524
r5c3_h-1v-1	2526	r5c3_h-1v+1	2528
r5c3_h+1v-1	2529	r5c3_h+1v+1	2531
r5c4_h-1v-1	2533	r5c4_h-1v+1	2534
r5c4_h+1v-1	2536	r5c4_h+1v+1	2538

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Index of the 324 routes: identifier and page (continued)

Route identifier	Page	Route identifier	Page
r5c5_h-1v-1	2539	r5c5_h-1v+1	2541
r5c5_h+1v-1	2543	r5c5_h+1v+1	2544
r5c6_h-1v-1	2546	r5c6_h-1v+1	2548
r5c6_h+1v-1	2549	r5c6_h+1v+1	2551
r5c7_h-1v-1	2553	r5c7_h-1v+1	2554
r5c7_h+1v-1	2556	r5c7_h+1v+1	2558
r5c8_h-1v-1	2559	r5c8_h-1v+1	2561
r5c8_h+1v-1	2563	r5c8_h+1v+1	2564
r5c9_h-1v-1	2566	r5c9_h-1v+1	2568
r5c9_h+1v-1	2569	r5c9_h+1v+1	2571
r6c1_h-1v-1	2573	r6c1_h-1v+1	2574
r6c1_h+1v-1	2576	r6c1_h+1v+1	2578
r6c2_h-1v-1	2579	r6c2_h-1v+1	2581
r6c2_h+1v-1	2583	r6c2_h+1v+1	2584
r6c3_h-1v-1	2586	r6c3_h-1v+1	2588
r6c3_h+1v-1	2589	r6c3_h+1v+1	2591
r6c4_h-1v-1	2593	r6c4_h-1v+1	2594
r6c4_h+1v-1	2596	r6c4_h+1v+1	2598
r6c5_h-1v-1	2599	r6c5_h-1v+1	2601
r6c5_h+1v-1	2603	r6c5_h+1v+1	2604
r6c6_h-1v-1	2606	r6c6_h-1v+1	2608
r6c6_h+1v-1	2609	r6c6_h+1v+1	2611
r6c7_h-1v-1	2613	r6c7_h-1v+1	2614
r6c7_h+1v-1	2616	r6c7_h+1v+1	2618
r6c8_h-1v-1	2619	r6c8_h-1v+1	2621
r6c8_h+1v-1	2623	r6c8_h+1v+1	2624
r6c9_h-1v-1	2626	r6c9_h-1v+1	2628
r6c9_h+1v-1	2629	r6c9_h+1v+1	2631
r7c1_h-1v-1	2633	r7c1_h-1v+1	2634
r7c1_h+1v-1	2636	r7c1_h+1v+1	2638
r7c2_h-1v-1	2639	r7c2_h-1v+1	2641

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Index of the 324 routes: identifier and page (continued)

Route identifier	Page	Route identifier	Page
r7c2_h+1v-1	2643	r7c2_h+1v+1	2644
r7c3_h-1v-1	2646	r7c3_h-1v+1	2648
r7c3_h+1v-1	2649	r7c3_h+1v+1	2651
r7c4_h-1v-1	2653	r7c4_h-1v+1	2654
r7c4_h+1v-1	2656	r7c4_h+1v+1	2658
r7c5_h-1v-1	2659	r7c5_h-1v+1	2661
r7c5_h+1v-1	2663	r7c5_h+1v+1	2664
r7c6_h-1v-1	2666	r7c6_h-1v+1	2668
r7c6_h+1v-1	2669	r7c6_h+1v+1	2671
r7c7_h-1v-1	2673	r7c7_h-1v+1	2674
r7c7_h+1v-1	2676	r7c7_h+1v+1	2678
r7c8_h-1v-1	2679	r7c8_h-1v+1	2681
r7c8_h+1v-1	2683	r7c8_h+1v+1	2684
r7c9_h-1v-1	2686	r7c9_h-1v+1	2688
r7c9_h+1v-1	2689	r7c9_h+1v+1	2691
r8c1_h-1v-1	2693	r8c1_h-1v+1	2694
r8c1_h+1v-1	2696	r8c1_h+1v+1	2698
r8c2_h-1v-1	2699	r8c2_h-1v+1	2701
r8c2_h+1v-1	2703	r8c2_h+1v+1	2704
r8c3_h-1v-1	2706	r8c3_h-1v+1	2708
r8c3_h+1v-1	2709	r8c3_h+1v+1	2711
r8c4_h-1v-1	2713	r8c4_h-1v+1	2714
r8c4_h+1v-1	2716	r8c4_h+1v+1	2718
r8c5_h-1v-1	2719	r8c5_h-1v+1	2721
r8c5_h+1v-1	2723	r8c5_h+1v+1	2724
r8c6_h-1v-1	2726	r8c6_h-1v+1	2728
r8c6_h+1v-1	2729	r8c6_h+1v+1	2731
r8c7_h-1v-1	2733	r8c7_h-1v+1	2734
r8c7_h+1v-1	2736	r8c7_h+1v+1	2738
r8c8_h-1v-1	2739	r8c8_h-1v+1	2741
r8c8_h+1v-1	2743	r8c8_h+1v+1	2744

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Index of the 324 routes: identifier and page (continued)

Route identifier	Page	Route identifier	Page
r8c9_h-1v-1	2746	r8c9_h-1v+1	2748
r8c9_h+1v-1	2749	r8c9_h+1v+1	2751
r9c1_h-1v-1	2753	r9c1_h-1v+1	2754
r9c1_h+1v-1	2756	r9c1_h+1v+1	2758
r9c2_h-1v-1	2759	r9c2_h-1v+1	2761
r9c2_h+1v-1	2763	r9c2_h+1v+1	2764
r9c3_h-1v-1	2766	r9c3_h-1v+1	2768
r9c3_h+1v-1	2769	r9c3_h+1v+1	2771
r9c4_h-1v-1	2773	r9c4_h-1v+1	2774
r9c4_h+1v-1	2776	r9c4_h+1v+1	2778
r9c5_h-1v-1	2779	r9c5_h-1v+1	2781
r9c5_h+1v-1	2783	r9c5_h+1v+1	2784
r9c6_h-1v-1	2786	r9c6_h-1v+1	2788
r9c6_h+1v-1	2789	r9c6_h+1v+1	2791
r9c7_h-1v-1	2793	r9c7_h-1v+1	2794
r9c7_h+1v-1	2796	r9c7_h+1v+1	2798
r9c8_h-1v-1	2799	r9c8_h-1v+1	2801
r9c8_h+1v-1	2803	r9c8_h+1v+1	2804
r9c9_h-1v-1	2806	r9c9_h-1v+1	2808
r9c9_h+1v-1	2809	r9c9_h+1v+1	2811

Route 001 — r1c1_h-1v-1

Cell 01: row 1, column 1. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 1; ordinal in the memory table: 4.

Table 94.9: Record 001: r1c1_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c1_h-1v-1; *Cell*: 1; *Row*: 1
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,1;9,9
Classification (cont.)	*Route family*: 24|0|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 001 — r1c1_h-1v-1: continued on the next page

Record 001 — r1c1_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055189889622315337
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: effeba5464fe6134ce328276caffda457de9d85e22f3932a a8ce593a521c1040 *Reader*: 32867437928eda79ef832344d5cde5886a4c715e968ad77746df4b89 0958246e *Cell record at 108*: —

Route 002 — r1c1_h-1v+1

Cell 01: row 1, column 1. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 2; ordinal in the memory table: 3.

Table 94.10: Record 002: r1c1_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c1_h-1v+1; *Cell*: 1; *Row*: 1
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 002 — r1c1_h-1v+1: continued on the next page

Record 002 — r1c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,1;9,9
Classification (cont.)	*Route family*: 24|0|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 002 — r1c1_h-1v+1: continued on the next page

Record 002 — r1c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7cc1cc9f41e40afb8776ff845297b2b0b8b440b8724f822cc6fca90c7639f2df *Reader*: 85737ea39e08488f4972aa61178a8ccbedcd6662f8b923a8009c6ffeba7c5e7b *Cell record at 108*: 5e13be751c11cc8bf7ef794982848590e3ae4f7273685430f5f6144bfc16ebd7

Route 003 — r1c1_h+1v-1

Cell 01: row 1, column 1. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 3; ordinal in the memory table: 2.

Table 94.11: Record 003: r1c1_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c1_h+1v-1; *Cell*: 1; *Row*: 1
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,1;9,9
Classification (cont.)	*Route family*: 24|0|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 003 — r1c1_h+1v-1: continued on the next page

Record 003 — r1c1_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: aabfcc6099652e23f82f1a50e4ad66a28960ccdb0db84037 0dfdf043e39dd2d5 *Reader*: ecd7e3616b9a9821dc08227b0714d924426ca0d13867b735ffb5c614 b0ed3390 *Cell record at 108*: —

Route 004 — r1c1_h+1v+1

Cell 01: row 1, column 1. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 4; ordinal in the memory table: 1.

Table 94.12: Record 004: r1c1_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c1_h+1v+1; *Cell*: 1; *Row*: 1
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,1;9,9
Classification (cont.)	*Route family*: 24|0|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 004 — r1c1_h+1v+1: continued on the next page

Record 004 — r1c1_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055189889622315337
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: cb1834fa28511686ade62a6327bd1ef9daabe103193e9e71 583b231c288f5657 *Reader*: ce86abc79b701a0ec9ae2123d730f8c5eb0e88daf5bfa92637854368 8333fd17 *Cell record at 108*: —

Route 005 — r1c2_h-1v-1

Cell 02: row 1, column 2. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 5; ordinal in the memory table: 8.

Table 94.13: Record 005: r1c2_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c2_h-1v-1; *Cell*: 2; *Row*: 1
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 005 — r1c2_h-1v-1: continued on the next page

Record 005 — r1c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,2;9,8
Classification (cont.)	*Route family*: 28|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-4|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Record 005 — r1c2_h-1v-1: continued on the next page

Record 005 — r1c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7334be27187ca745a3c7e70d9f4fea4fb6d8c31f990e2382f7f91263fb447d0c *Reader*: 7677699b2a6f6153d139cc02ebcac4a1ff7f1894b6da0a7b1e46b74f55110f6a *Cell record at 108*: e589bca4743ae3a8412d609f38a1cbca9ab5e81521c621f651183be2730acc06

Route 006 — r1c2_h-1v+1

Cell 02: row 1, column 2. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 6; ordinal in the memory table: 7.

Table 94.14: Record 006: r1c2_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c2_h-1v+1; *Cell*: 2; *Row*: 1
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,2;9,8
Classification (cont.)	*Route family*: 28|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 006 — r1c2_h-1v+1: continued on the next page

Record 006 — r1c2_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 50|-6|2|-4|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 56149b04962ed2f1a098af0dcf0fc19ff01c1fa5825b32f899d69dea426f1ddb *Reader*: 474a5b8cbea67b1d3d0ba8e289cb7ef30ed5d908887d6ba46fa80433e81f6993 *Cell record at 108*: —

Route 007 — r1c2_h+1v-1

Cell 02: row 1, column 2. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 7; ordinal in the memory table: 6.

Table 94.15: Record 007: r1c2_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c2_h+1v-1; *Cell*: 2; *Row*: 1
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,2;9,8
Classification (cont.)	*Route family*: 28|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 50|6|-2|4|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K* Ω: (48,54,5)

Record 007 — r1c2_h+1v-1: continued on the next page

Record 007 — r1c2_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 47.606498943372135446416 86; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ccb61e50f720be11f86bf136d02b36587f92d52c35df35bf4b86ef5d779f7bcf *Reader*: e7c7e699830c4f17e88719d151c262a775f5cf6afccadf3b9ce34e426bc57b55 *Cell record at 108*: —

Route 008 — r1c2_h+1v+1

Cell 02: row 1, column 2. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 8; ordinal in the memory table: 5.

Table 94.16: Record 008: r1c2_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c2_h+1v+1; *Cell*: 2; *Row*: 1
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 008 — r1c2_h+1v+1: continued on the next page

Record 008 — r1c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,2;9,8
Classification (cont.)	*Route family*: 28|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|4|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 40; *log2 ambiguity area*: 20.339850002884624; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 008 — r1c2_h+1v+1: continued on the next page

Record 008 — r1c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 72b9df1397f2987ac2c70e5f7163572e0b8854825eee3b61db397f28ba9b9ba6 *Reader*: a1c4553cb5d2563a0a9a4d4d4659a3d9c9316bb9eb97544670b4f84225f0a4f5 *Cell record at 108*: —

Route 009 — r1c3_h-1v-1

Cell 03: row 1, column 3. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 9; ordinal in the memory table: 12.

Table 94.17: Record 009: r1c3_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c3_h-1v-1; *Cell*: 3; *Row*: 1
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,3;9,7
Classification (cont.)	*Route family*: 28|-6|2|0|-12|+++---+++---
Returns	*Class and phase*: 9; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16

Record 009 — r1c3_h-1v-1: continued on the next page

Record 009 — r1c3_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: fae367f8fb866c76b0300f5bd2cb0dbb059ab3e163b6df3c 5e46e685e4d9fa74 *Reader*: 716a4c36600c27b509736fc515d1943728ac458d23500627b24d70bd 413dd168 *Cell record at 108*: —

Route 010 — r1c3_h-1v+1

Cell 03: row 1, column 3. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 10; ordinal in the memory table: 11.

Table 94.18: Record 010: r1c3_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c3_h-1v+1; *Cell*: 3; *Row*: 1
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,3;9,7
Classification (cont.)	*Route family*: 28|-6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|-2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K* Ω: (80,54,6)

Record 010 — r1c3_h-1v+1: continued on the next page

Record 010 — r1c3_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 1120c2a5d9a6516e9138153d5365283c7af167d9e5b3e46cf7ab8d55a1974cfc *Reader*: cc519544fb45c5f0387e629c3a067329125f213d0292e9935aa46d4bcf1027b8 *Cell record at 108*: b1cc79b354042ce1877e1c4cf5347fc4f509c23577711204cf7cddec6beea6dc

Route 011 — r1c3_h+1v-1

Cell 03: row 1, column 3. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 11; ordinal in the memory table: 10.

Table 94.19: Record 011: r1c3_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c3_h+1v-1; *Cell*: 3; *Row*: 1
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO

Record 011 — r1c3_h+1v-1: continued on the next page

Record 011 — r1c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,3;9,7
Classification (cont.)	*Route family*: 28|-6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 011 — r1c3_h+1v-1: continued on the next page

Record 011 — r1c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d8f175e57deb40bea01359579d16ae6e1fdd0b458ebd8b41608e5eba1782d060 *Reader*: 29304938102d297060297c362f9d81380336ecff5a81f6affe85c413e0670568 *Cell record at 108*: —

Route 012 — r1c3_h+1v+1

Cell 03: row 1, column 3. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 12; ordinal in the memory table: 9.

Table 94.20: Record 012: r1c3_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c3_h+1v+1; *Cell*: 3; *Row*: 1
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,3;9,7
Classification (cont.)	*Route family*: 28|-6|2|0|-12|+++---+++---
Returns	*Class and phase*: 18; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 012 — r1c3_h+1v+1: continued on the next page

Record 012 — r1c3_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 13e342fa83db93f3f7dc13163653112faecbb1dd8aa9f872 d9121b96007e8839 *Reader*: 5aa200a36959917378a478532c886d1ad9e4958c7cd2fe476a7797fc 5306366a *Cell record at 108*: —

Route 013 — r1c4_h-1v-1

Cell 04: row 1, column 4. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 13; ordinal in the memory table: 16.

Table 94.21: Record 013: r1c4_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c4_h-1v-1; *Cell*: 4; *Row*: 1
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,4;9,6
Classification (cont.)	*Route family*: 30|-6|-4|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 30|6|-2|4|-8
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,54,7); *K* Ω: (56,54,5)

Record 013 — r1c4_h-1v-1: continued on the next page

Record 013 — r1c4_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000005518988962231 5337
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003855097235414 1645
Prefix separation	*Ambiguity area*: 26; *log2 ambiguity area*: 6.339850002884624; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: db632f763af301d246516155fafabc14d1d401a29c7c7b63 1471a8d80c95591b *Reader*: 5f1e54557f63ee41bdd234e20df6fb367b00a65097cb906eb634fce7 66250a20 *Cell record at 108*: —

Route 014 — r1c4_h-1v+1

Cell 04: row 1, column 4. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 14; ordinal in the memory table: 15.

Table 94.22: Record 014: r1c4_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c4_h-1v+1; *Cell*: 4; *Row*: 1
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 014 — r1c4_h-1v+1: continued on the next page

Record 014 — r1c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,4;9,6
Classification (cont.)	*Route family*: 30|-6|-4|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 014 — r1c4_h-1v+1: continued on the next page

Record 014 — r1c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: aa7327659272d3089edbc1506b9b161ae47d6711976c8a2a4a11676939d76776 *Reader*: cb535d05a86f529c6b30d46bf87f9924e83bfc069b9f421b49a8bed1d576b346 *Cell record at 108*: —

Route 015 — r1c4_h+1v-1

Cell 04: row 1, column 4. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 15; ordinal in the memory table: 14.

Table 94.23: Record 015: r1c4_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c4_h+1v-1; *Cell*: 4; *Row*: 1
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,4;9,6
Classification (cont.)	*Route family*: 30|-6|-4|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 015 — r1c4_h+1v-1: continued on the next page

Record 015 — r1c4_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|-2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: de3386f8b28db7e8c79bad444dd9f558fb5ce61496e054c2ab24a7188cc309d4 *Reader*: e6136c800ce4e9ca12e80a5e90dbb68278b0c098aafeab826ff704d06ce3b1d1 *Cell record at 108*: —

Route 016 — r1c4_h+1v+1

Cell 04: row 1, column 4. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 16; ordinal in the memory table: 13.

Table 94.24: Record 016: r1c4_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c4_h+1v+1; *Cell*: 4; *Row*: 1
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,4;9,6
Classification (cont.)	*Route family*: 30|-6|-4|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|2|-4|-8
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,54,7); *K* Ω: (56,54,5)

Record 016 — r1c4_h+1v+1: continued on the next page

Record 016 — r1c4_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000005518988962231533 7
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 1c8b5cdaa63332d1e8f0ae42fd515992f5d9e41e72975eaa e772af08588c657a *Reader*: 338da10d5c8906ec03bdfade56c89b7fd502a0b0682074063adb644b 01f77606 *Cell record at 108*: ce501e32503d7df09e2c5ae61ff9543fcb85dba7afa7a539 6fb0f36909af84dc

Route 017 — r1c5_h-1v-1

Cell 05: row 1, column 5. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 17; ordinal in the memory table: 20.

Table 94.25: Record 017: r1c5_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c5_h-1v-1; *Cell*: 5; *Row*: 1
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 017 — r1c5_h-1v-1: continued on the next page

Record 017 — r1c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,5;9,5
Classification (cont.)	*Route family*: 34|0|-8|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,0); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Record 017 — r1c5_h-1v-1: continued on the next page

Record 017 — r1c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e753e7cf912988efa9df9eda0afb8e532a8b8f33f921722587fb354551c5342c *Reader*: 9d043f03a6f5102411e0e8ecbc080c073548ea8ce98deaf70e4bb5ae9fbed35e *Cell record at 108*: —

Route 018 — r1c5_h-1v+1

Cell 05: row 1, column 5. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 18; ordinal in the memory table: 19.

Table 94.26: Record 018: r1c5_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c5_h-1v+1; *Cell*: 5; *Row*: 1
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,5;9,5
Classification (cont.)	*Route family*: 34|0|-8|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 018 — r1c5_h-1v+1: continued on the next page

Record 018 — r1c5_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 959d954a11c429bdcfe3a0db7b21c1810828fff0868a1d0ac26ba7ef3e127805 *Reader*: b90ff070e1aae58cfca71a9ae8253522544bb269f70b257e4e6afd6c0c3c24f8 *Cell record at 108*: —

Route 019 — r1c5_h+1v-1

Cell 05: row 1, column 5. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 19; ordinal in the memory table: 18.

Table 94.27: Record 019: r1c5_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c5_h+1v-1; *Cell*: 5; *Row*: 1
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,5;9,5
Classification (cont.)	*Route family*: 34|0|-8|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K* Ω: (80,54,6)

Record 019 — r1c5_h+1v-1: continued on the next page

Record 019 — r1c5_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 49c3be2eb41e01bb1a1f27f1b1ec147e4db99f167f982dc729a2d4993527c5f8 *Reader*: 1b071d274a26233a2a95e7db4a91c4d3e75f2fa81ab184e1dbc739d938ccd981 *Cell record at 108*: 8c0e7e1dd0b049c9330530f4b68eb1be6a328e08bd9c0f293df7e20ea4956ab2

Route 020 — r1c5_h+1v+1

Cell 05: row 1, column 5. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 20; ordinal in the memory table: 17.

Table 94.28: Record 020: r1c5_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c5_h+1v+1; *Cell*: 5; *Row*: 1
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 020 — r1c5_h+1v+1: continued on the next page

Record 020 — r1c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,5;9,5
Classification (cont.)	*Route family*: 34|0|-8|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,0); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 30; *log2 ambiguity area*: 8.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 020 — r1c5_h+1v+1: continued on the next page

Record 020 — r1c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 1b217fd91d203aeae19f2a5a5a90fcadb55a5bb548ca02e5 1e4e20ec78b6444c *Reader*: 78e79216b3802312a31bc26703b19549f845de0e5c1403ce0bfc7e42 78fd13d9 *Cell record at 108*: —

Route 021 — r1c6_h-1v-1

Cell 06: row 1, column 6. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 21; ordinal in the memory table: 24.

Table 94.29: Record 021: r1c6_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c6_h-1v-1; *Cell*: 6; *Row*: 1
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,6;9,4
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16

Record 021 — r1c6_h-1v-1: continued on the next page

Record 021 — r1c6_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d52d9bc58081da1a7b2f9e5d23e9d5a6d67401f824eb66ed 733498c2fd0dab4c *Reader*: 560f193427fddf83fb498c6ec7a6b2e4a683ed18b5f9d3a632bb9743 5ce13545 *Cell record at 108*: b19c718762e5299b6bd2d56cb241cd2cdef7df5b1d224f69 51364c8e46e57ade

Route 022 — r1c6_h-1v+1

Cell 06: row 1, column 6. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 22; ordinal in the memory table: 23.

Table 94.30: Record 022: r1c6_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c6_h-1v+1; *Cell*: 6; *Row*: 1
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,6;9,4
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (56,54,5)

Record 022 — r1c6_h-1v+1: continued on the next page

Record 022 — r1c6_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 0af1af6e61e27f82e9c6658e03d303a9801f5cf6609402e5810dcbc944b8c00b *Reader*: a28a41d6785403d9b38e510f4c5203a2344cac70366f151a6d143c8093c59464 *Cell record at 108*: —

Route 023 — r1c6_h+1v-1

Cell 06: row 1, column 6. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 23; ordinal in the memory table: 22.

Table 94.31: Record 023: r1c6_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c6_h+1v-1; *Cell*: 6; *Row*: 1
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 023 — r1c6_h+1v-1: continued on the next page

Record 023 — r1c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,6;9,4
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 023 — r1c6_h+1v-1: continued on the next page

Record 023 — r1c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3629501596c2b8538e13e1e97d2b0c69f585dbc0f13f6f9fbe5d7b012d7082ea *Reader*: f992e4a2a1b53de9a0d7533c2a673b02d28083e7f6aaaa45587ef6104b63db2f *Cell record at 108*: —

Route 024 — r1c6_h+1v+1

Cell 06: row 1, column 6. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 24; ordinal in the memory table: 21.

Table 94.32: Record 024: r1c6_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c6_h+1v+1; *Cell*: 6; *Row*: 1
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,6;9,4
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 024 — r1c6_h+1v+1: continued on the next page

Record 024 — r1c6_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c5b01f4a1d0acc94193d804a45897f489ee02521a35214a9 b86086717d71cfdc *Reader*: 0987471451aca8dbdc0c7324674e2ffa4237909412e431862c124cac 78f45504 *Cell record at 108*: —

Route 025 — r1c7_h-1v-1

Cell 07: row 1, column 7. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 25; ordinal in the memory table: 28.

Table 94.33: Record 025: r1c7_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c7_h-1v-1; *Cell*: 7; *Row*: 1
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,7;9,3
Classification (cont.)	*Route family*: 30|6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 025 — r1c7_h-1v-1: continued on the next page

Record 025 — r1c7_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055189889622315337
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a911ce23dfdeeccc43cb29fe5898ecd2fabdd3f994ca7281 4b4e1b698537ef02 *Reader*: 62e6f1b10122e3761b7f39574d07f715c69636c58df0f10af2f06d2c 4289e08b *Cell record at 108*: —

Route 026 — r1c7_h-1v+1

Cell 07: row 1, column 7. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 26; ordinal in the memory table: 27.

Table 94.34: Record 026: r1c7_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c7_h-1v+1; *Cell*: 7; *Row*: 1
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 026 — r1c7_h-1v+1: continued on the next page

Record 026 — r1c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,7;9,3
Classification (cont.)	*Route family*: 30|6|4|0|-12|+++---+++---
Returns	*Class and phase*: 18; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 026 — r1c7_h-1v+1: continued on the next page

Record 026 — r1c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: f580dc8247645d3812ad5b4931a5bf89ed066bd7a6cd3321f709a411f3996c80 *Reader*: 07b8d450b7f6e3998bfcc5fdf7dc71c50982ba491d4c5dd5bfe6236c6a11645f *Cell record at 108*: 111ca40e5cb5dd4e1469c8db516ceb95b20b75ec5dfc93b840a6cce91c92b62c

Route 027 — r1c7_h+1v-1

Cell 07: row 1, column 7. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 27; ordinal in the memory table: 26.

Table 94.35: Record 027: r1c7_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c7_h+1v-1; *Cell*: 7; *Row*: 1
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,7;9,3
Classification (cont.)	*Route family*: 30|6|4|0|-12|+++---+++---
Returns	*Class and phase*: 9; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 027 — r1c7_h+1v-1: continued on the next page

Record 027 — r1c7_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 342d7eac14c30ee19c5129f270b31c29fa591198d94f5d42f5c23629fb102c38 *Reader*: e5e14191accc501f24a061fb7067050ef0a2efd5d86cf532156226da818a11e9 *Cell record at 108*: —

Route 028 — r1c7_h+1v+1

Cell 07: row 1, column 7. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 28; ordinal in the memory table: 25.

Table 94.36: Record 028: r1c7_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c7_h+1v+1; *Cell*: 7; *Row*: 1
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,7;9,3
Classification (cont.)	*Route family*: 30|6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|6|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 028 — r1c7_h+1v+1: continued on the next page

Record 028 — r1c7_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055189889622315337
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e65938ec108f08f67b38c1bfdd3f600db58f9a17d9c59a82 b75ccff06d19ce60 *Reader*: 9905fffabaeddac886538ee34f7320c7891f879fe8876c470d8a790d e73b4d13 *Cell record at 108*: —

Route 029 — r1c8_h-1v-1

Cell 08: row 1, column 8. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 29; ordinal in the memory table: 32.

Table 94.37: Record 029: r1c8_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c8_h-1v-1; *Cell*: 8; *Row*: 1
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 029 — r1c8_h-1v-1: continued on the next page

Record 029 — r1c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,8;9,2
Classification (cont.)	*Route family*: 28|6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Record 029 — r1c8_h-1v-1: continued on the next page

Record 029 — r1c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 72789b3efdc97b1d767a5aa73046d084583fa5aba373f9a07a6fc16b1c1f04df *Reader*: 53e27ee134cda7bb20474d1c7af7cc8921838c094a2dd9bc724b2d30cff5a10a *Cell record at 108*: —

Route 030 — r1c8_h-1v+1

Cell 08: row 1, column 8. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 30; ordinal in the memory table: 31.

Table 94.38: Record 030: r1c8_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c8_h-1v+1; *Cell*: 8; *Row*: 1
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,8;9,2
Classification (cont.)	*Route family*: 28|6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 030 — r1c8_h-1v+1: continued on the next page

Record 030 — r1c8_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: ++00--++00--
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 36
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K Ω*: (36,54,4)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 35.6063877469494433063628 1; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000024685259569118152
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: f612d6ef200a1ff54e74c4ed7c7a510893ca34048d21381466db682cca3cc661 *Reader*: 48191dd8270ba6e68ab41300a7daa22a6daa2102e4aff594b72c5fbc01440d85 *Cell record at 108*: —

Route 031 — r1c8_h+1v-1

Cell 08: row 1, column 8. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 31; ordinal in the memory table: 30.

Table 94.39: Record 031: r1c8_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c8_h+1v-1; *Cell*: 8; *Row*: 1
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,8;9,2
Classification (cont.)	*Route family*: 28|6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: 0++--00++--0
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 36
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K* Ω: (36,54,4)

Record 031 — r1c8_h+1v-1: continued on the next page

Record 031 — r1c8_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 35.606387746949443306362 81; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000024685259569118152
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 48fc9a27e8d42d250d4ab52cb98cf105f32507281dc8824f92a8ce07cfd13aeb *Reader*: ab4343ca0f3279dd31d470d4b27e63604203982226a40d2196f33fdb10b85cc1 *Cell record at 108*: 012b2f08ceb6741389c89bacc896d7b2769a75fa04b4753135b68e4509b3ad20

Route 032 — r1c8_h+1v+1

Cell 08: row 1, column 8. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 32; ordinal in the memory table: 29.

Table 94.40: Record 032: r1c8_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c8_h+1v+1; *Cell*: 8; *Row*: 1
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 032 — r1c8_h+1v+1: continued on the next page

Record 032 — r1c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,8;9,2
Classification (cont.)	*Route family*: 28|6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 40; *log2 ambiguity area*: 20.339850002884624; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 032 — r1c8_h+1v+1: continued on the next page

Record 032 — r1c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3aa3fba36ac1b06364a3368dfd7a9486ddc4de29c64dc93a5936873261d07e18 *Reader*: f0c45d1809914b63308aa054a401a3933c1b9671f0dce46e834b5c6a8afb2263 *Cell record at 108*: —

Route 033 — r1c9_h-1v-1

Cell 09: row 1, column 9. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 33; ordinal in the memory table: 36.

Table 94.41: Record 033: r1c9_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c9_h-1v-1; *Cell*: 9; *Row*: 1
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,9;9,1
Classification (cont.)	*Route family*: 28|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16

Record 033 — r1c9_h-1v-1: continued on the next page

Record 033 — r1c9_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|2|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ed985a2ed540799150c7e349e2cc896d9aac716d704d2653 88865e6038a3165e *Reader*: 66eb3e8eed039dc74f989a1af5a5acbe386ed98412b13ef3dcedb1c6 487e9571 *Cell record at 108*: —

Route 034 — r1c9_h-1v+1

Cell 09: row 1, column 9. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 34; ordinal in the memory table: 35.

Table 94.42: Record 034: r1c9_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c9_h-1v+1; *Cell*: 9; *Row*: 1
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,9;9,1
Classification (cont.)	*Route family*: 28|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (48,54,5)

Record 034 — r1c9_h-1v+1: continued on the next page

Record 034 — r1c9_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a02045b2e206e07f7236c381210f0d3e9dc3fcaf430a0b349f8488eec7605f89 *Reader*: fdda7f9e6057445988cd077dc481c836df8a633e66f5cfad3ab15b2996ff488f *Cell record at 108*: —

Route 035 — r1c9_h+1v-1

Cell 09: row 1, column 9. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 35; ordinal in the memory table: 34.

Table 94.43: Record 035: r1c9_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r1c9_h+1v-1; *Cell*: 9; *Row*: 1
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 035 — r1c9_h+1v-1: continued on the next page

Record 035 — r1c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,9;9,1
Classification (cont.)	*Route family*: 28|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 47.6064989433721354464 16 86; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 035 — r1c9_h+1v-1: continued on the next page

Record 035 — r1c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a1fe11c1f44fee339c8dfe318ad4ce23565bd11d206e5e5b0a3f8136f94cbcff *Reader*: 409959c2ff367348163db758c5659d70ab997d5cb215f7ff22fc64d0db140a39 *Cell record at 108*: —

Route 036 — r1c9_h+1v+1

Cell 09: row 1, column 9. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 36; ordinal in the memory table: 33.

Table 94.44: Record 036: r1c9_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r1c9_h+1v+1; *Cell*: 9; *Row*: 1
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,9;9,1
Classification (cont.)	*Route family*: 28|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 036 — r1c9_h+1v+1: continued on the next page

Record 036 — r1c9_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-2|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5decd17fa25c6e87374e54b0f6b6fa4c85317267086884d1 e9b3cc6c2626241b *Reader*: e8213bede9eb8c189e246a8aac6855898eb38d7c085789b8527ff022 f9311b9f *Cell record at 108*: 7d1f109e1e33bf84ddf9384cf13c342f63017fe9faecd3c7 3d8063eca5fa5462

Route 037 — r2c1_h-1v-1

Cell 10: row 2, column 1. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 37; ordinal in the memory table: 40.

Table 94.45: Record 037: r2c1_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c1_h-1v-1; *Cell*: 10; *Row*: 2
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,1;8,9
Classification (cont.)	*Route family*: 28|-6|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|4|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 037 — r2c1_h-1v-1: continued on the next page

Record 037 — r2c1_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 40; *log2 ambiguity area*: 20.339850002884624; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 42121291b4db1b251482aaa600938754c4e97069468ccdcb d0d654a28f9b0873 *Reader*: 7e147ace8f35cc0e74360e296f6f57ce18196c7f81af3828e0e513a7 8d821461 *Cell record at 108*: —

Route 038 — r2c1_h-1v+1

Cell 10: row 2, column 1. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 38; ordinal in the memory table: 39.

Table 94.46: Record 038: r2c1_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c1_h-1v+1; *Cell*: 10; *Row*: 2
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark

Record 038 — r2c1_h-1v+1: continued on the next page

Record 038 — r2c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,1;8,9
Classification (cont.)	*Route family*: 28|-6|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 50|-6|2|-4|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.00000002761000615951421 5
Reader evaluations (cont.)	*Action factor Ω*: 1.00000003300471805324308 4
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 038 — r2c1_h-1v+1: continued on the next page

Record 038 — r2c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e3a4eca88fb523b4cd27f0dcbc45a95e626a0d4fae70ca9085350d36de129e86 *Reader*: f06f943577be9d476fa92b2cb82203fd77be6d41f9d8efd58303786f07a013ef *Cell record at 108*: —

Route 039 — r2c1_h+1v-1

Cell 10: row 2, column 1. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 39; ordinal in the memory table: 38.

Table 94.47: Record 039: r2c1_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c1_h+1v-1; *Cell*: 10; *Row*: 2
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,1;8,9
Classification (cont.)	*Route family*: 28|-6|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 039 — r2c1_h+1v-1: continued on the next page

Record 039 — r2c1_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 50|6|-2|4|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6cd735054b3d285158e101710ab447ea6f8236d0455b76b5aa102be68ac85002 *Reader*: efe6e221b6809bf84456087db0f756cc315eba6205271dfe72062438b4952f78 *Cell record at 108*: —

Route 040 — r2c1_h+1v+1

Cell 10: row 2, column 1. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 40; ordinal in the memory table: 37.

Table 94.48: Record 040: r2c1_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c1_h+1v+1; *Cell*: 10; *Row*: 2
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,1;8,9
Classification (cont.)	*Route family*: 28|-6|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-4|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 040 — r2c1_h+1v+1: continued on the next page

Record 040 — r2c1_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7654e0aa6d4ec6e5959f770679686d0fe3c0dafd21ff7cb9 4ed4ea8505f460a2 *Reader*: 39a510c4fa875ca5fc2c73e482279f495f57505ac7d2e9986bf08753 fb317a28 *Cell record at 108*: 355e92375547ac1e37c63290e00e0b13115df4812b567558 c2432472731b13de

Route 041 — r2c2_h-1v-1

Cell 11: row 2, column 2. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 41; ordinal in the memory table: 44.

Table 94.49: Record 041: r2c2_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c2_h-1v-1; *Cell*: 11; *Row*: 2
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO

Record 041 — r2c2_h-1v-1: continued on the next page

Record 041 — r2c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,2;8,8
Classification (cont.)	*Route family*: 24|0|4|0|-12|+++---+++---
Returns	*Class and phase*: 9; *Orbit and phase*: 9; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ Ω*: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Record 041 — r2c2_h-1v-1: continued on the next page

Record 041 — r2c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 9eb4e897dd2417fdd7ef230e161ee02a73430f2cd4c165e0b5e03f4cd9432a7c *Reader*: 7390d3cce6527e3250ecccb9cd6d58a5d3548fa0525ab0024b7f93b24d292216 *Cell record at 108*: —

Route 042 — r2c2_h-1v+1

Cell 11: row 2, column 2. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 42; ordinal in the memory table: 43.

Table 94.50: Record 042: r2c2_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c2_h-1v+1; *Cell*: 11; *Row*: 2
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,2;8,8
Classification (cont.)	*Route family*: 24|0|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 042 — r2c2_h-1v+1: continued on the next page

Record 042 — r2c2_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|-2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,84,28); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041174098946741575
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 1fe3a911aaa125a9905aeadc38d1dbb63d1759764b7429b0 39228794908d53b8 *Reader*: 779e5b698afcdc0910f6d9451f0ee72b8327bc1e94ac4096d006fec2 406ecf3c *Cell record at 108*: e34326b49595fe5ff46f40ddd0b7b0a511a0f57953ba2953 b50c9eced0884e64

Route 043 — r2c2_h+1v-1

Cell 11: row 2, column 2. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 43; ordinal in the memory table: 42.

Table 94.51: Record 043: r2c2_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c2_h+1v-1; *Cell*: 11; *Row*: 2
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,2;8,8
Classification (cont.)	*Route family*: 24|0|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,84,28); *K* Ω: (80,54,6)

Record 043 — r2c2_h+1v-1: continued on the next page

Record 043 — r2c2_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000004117409894674 1575
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185 9140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d8cfd857e63aa9d9c90169d1594eaa1ad2e6b70c6d138a85 ea226813977a322a *Reader*: 46e898cdaf66938151ec8b48a75efafe4eba7615becf22c89a22d7a0 9b274062 *Cell record at 108*: —

Route 044 — r2c2_h+1v+1

Cell 11: row 2, column 2. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 44; ordinal in the memory table: 41.

Table 94.52: Record 044: r2c2_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c2_h+1v+1; *Cell*: 11; *Row*: 2
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 044 — r2c2_h+1v+1: continued on the next page

Record 044 — r2c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,2;8,8
Classification (cont.)	*Route family*: 24|0|4|0|-12|+++---+++---
Returns	*Class and phase*: 18; *Orbit and phase*: 18; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Record 044 — r2c2_h+1v+1: continued on the next page

Record 044 — r2c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: bf3e75cc7e977bbfbe883ce1215b4ab850affa3242a704089c1350ee55fde93f *Reader*: bec293e8a9352a0c9e5b56fb6c7d30c876b80cec47727a2f34d40d9a3e0cd3ca *Cell record at 108*: —

Route 045 — r2c3_h-1v-1

Cell 12: row 2, column 3. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 45; ordinal in the memory table: 48.

Table 94.53: Record 045: r2c3_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c3_h-1v-1; *Cell*: 12; *Row*: 2
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,3;8,7
Classification (cont.)	*Route family*: 28|6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 045 — r2c3_h-1v-1: continued on the next page

Record 045 — r2c3_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243 *Stable triads per block*: 0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 21648712dd5dcff54d6151e85c6297f497049ea18f546275a253f506f95483d2 *Reader*: 6140ac8712fe06046683f5fd5d240eb62685661ad9a8244e3cbe742c5051fd66 *Cell record at 108*: 6c3c21ffd2893c3662b538c017a94d623d66cc85b623c55cf8d08a67b4f70c93

Route 046 — r2c3_h-1v+1

Cell 12: row 2, column 3. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 46; ordinal in the memory table: 47.

Table 94.54: Record 046: r2c3_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c3_h-1v+1; *Cell*: 12; *Row*: 2
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,3;8,7
Classification (cont.)	*Route family*: 28|6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|2|0|2|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 046 — r2c3_h-1v+1: continued on the next page

Record 046 — r2c3_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2729489d2f4f122b773697f8b3eef3fa11b69d50f3606c4a 5418b66587cb1a4b *Reader*: f610796820399955f3b317dc077df3c856ff4c75f80da841ce0a7221 4efd06d0 *Cell record at 108*: —

Route 047 — r2c3_h+1v-1

Cell 12: row 2, column 3. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 47; ordinal in the memory table: 46.

Table 94.55: Record 047: r2c3_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c3_h+1v-1; *Cell*: 12; *Row*: 2
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark

Record 047 — r2c3_h+1v-1: continued on the next page

Record 047 — r2c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,3;8,7
Classification (cont.)	*Route family*: 28|6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|-2|0|-2|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Record 047 — r2c3_h+1v-1: continued on the next page

Record 047 — r2c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c5f5a7cc8a3f497d2781fcfcabb97bede9beb4139194e9cd554320a8d4612024 *Reader*: e8c645b238014406bd74ee9d49df13703368a36d854493d99ef355058b2a9977 *Cell record at 108*: —

Route 048 — r2c3_h+1v+1

Cell 12: row 2, column 3. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 48; ordinal in the memory table: 45.

Table 94.56: Record 048: r2c3_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c3_h+1v+1; *Cell*: 12; *Row*: 2
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,3;8,7
Classification (cont.)	*Route family*: 28|6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17

Record 048 — r2c3_h+1v+1: continued on the next page

Record 048 — r2c3_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 698f0a4ed1e27622225b33b8cf29ee3696219e6744484603 2e3f9212d53466a3 *Reader*: 816cbe815adfede468d9709ec0485910767d82097a66de13f0984f60 9f5bd113 *Cell record at 108*: —

Route 049 — r2c4_h-1v-1

Cell 13: row 2, column 4. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 49; ordinal in the memory table: 52.

Table 94.57: Record 049: r2c4_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c4_h-1v-1; *Cell*: 13; *Row*: 2
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,4;8,6
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 049 — r2c4_h-1v-1: continued on the next page

Record 049 — r2c4_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2597b1458c7c6d27655670533a40d2325d645709b9f9e060 260b7cc2573da835 *Reader*: 1a459f0dff0ec31f6beadf5400457dcc3102209e4b1f863b0749dcff b5f87ade *Cell record at 108*: —

Route 050 — r2c4_h-1v+1

Cell 13: row 2, column 4. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 50; ordinal in the memory table: 51.

Table 94.58: Record 050: r2c4_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c4_h-1v+1; *Cell*: 13; *Row*: 2
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 050 — r2c4_h-1v+1: continued on the next page

Record 050 — r2c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,4;8,6
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000002761000615951421 5
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 050 — r2c4_h-1v+1: continued on the next page

Record 050 — r2c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 679c8850da30d692454f00039b4e73d58c5eff335d81553221a5a0b3159a5624 *Reader*: ace3d8034260254a3bb93657b2e2123e63e67d16904e7f0b632fdcb671332da7 *Cell record at 108*: —

Route 051 — r2c4_h+1v-1

Cell 13: row 2, column 4. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 51; ordinal in the memory table: 50.

Table 94.59: Record 051: r2c4_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c4_h+1v-1; *Cell*: 13; *Row*: 2
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,4;8,6
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 051 — r2c4_h+1v-1: continued on the next page

Record 051 — r2c4_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6055f66a8741e1140af22cd2d82c9b62c55c4e32ae9162cad35bad23d3cbdcc2 *Reader*: 9078ab563bb6e210453e86e13e18dd71d892a6b914ff5fc65482a49004bcd077 *Cell record at 108*: 00c21eea4789a8c06b37c22441f75093ea0f0184ff1bc2955403772490155a81

Route 052 — r2c4_h+1v+1

Cell 13: row 2, column 4. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 52; ordinal in the memory table: 49.

Table 94.60: Record 052: r2c4_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c4_h+1v+1; *Cell*: 13; *Row*: 2
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,4;8,6
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 052 — r2c4_h+1v+1: continued on the next page

Record 052 — r2c4_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000068994430825936417
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 9b29c8580391aae7b3dbcfefd966550abd79054c5e2e9520 b547d06b227fe840 *Reader*: 9a3f104e10a55934eddfadbb190a4e8d0d79fb55eba7bc4403c29aac 08923eec *Cell record at 108*: —

Route 053 — r2c5_h-1v-1

Cell 14: row 2, column 5. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 53; ordinal in the memory table: 56.

Table 94.61: Record 053: r2c5_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c5_h-1v-1; *Cell*: 14; *Row*: 2
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark

Record 053 — r2c5_h-1v-1: continued on the next page

Record 053 — r2c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,5;8,5
Classification (cont.)	*Route family*: 30|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|6|0|4|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Record 053 — r2c5_h-1v-1: continued on the next page

Record 053 — r2c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d1fd49040e524fea8052b4b0aaf0181192155662059c836370670d5c2814748b *Reader*: 83380600e5404d7e8880238796855cf64cf95c1fb93d17d2320ce187bf98369d *Cell record at 108*: —

Route 054 — r2c5_h-1v+1

Cell 14: row 2, column 5. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 54; ordinal in the memory table: 55.

Table 94.62: Record 054: r2c5_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c5_h-1v+1; *Cell*: 14; *Row*: 2
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,5;8,5
Classification (cont.)	*Route family*: 30|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 054 — r2c5_h-1v+1: continued on the next page

Record 054 — r2c5_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -4
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: +++00-+++00-
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-4|0|-4
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 20
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (60,84,28); *K Ω*: (20,54,4)
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ Ω*: 19.6063877469494433063628 1; *Action factor D*: 1.000000041174098946741575
Reader evaluations (cont.)	*Action factor Ω*: 1.000000013592751151887498
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2708ff6618e713f0d4900bb9c13096961b5c2b854459b699c5a469f133917fe1 *Reader*: 02b48980193d7b3666a37a97c6cc611e2af419c8f78bacae9bb362a79e43767e *Cell record at 108*: —

Route 055 — r2c5_h+1v-1

Cell 14: row 2, column 5. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 55; ordinal in the memory table: 54.

Table 94.63: Record 055: r2c5_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c5_h+1v-1; *Cell*: 14; *Row*: 2
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,5;8,5
Classification (cont.)	*Route family*: 30|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 4
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: 00+---00+---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|4|0|-4
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 20
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (60,84,28); *K* Ω: (20,54,4)

Record 055 — r2c5_h+1v-1: continued on the next page

Record 055 — r2c5_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ* Ω: 19.606387746949443306362 81; *Action factor D*: 1.00000004117409894674 1575
Reader evaluations (cont.)	*Action factor* Ω: 1.00000001359275115188 7498
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2133fe30ab9b2c2fda2fe370cf35fabec9776902ae7f1bf5 dd7a68db23c22035 *Reader*: 7e4da1b56102a95309b04e40b636fc437722155b1ed9b43bc63ed9a4 6c0f9c84 *Cell record at 108*: —

Route 056 — r2c5_h+1v+1

Cell 14: row 2, column 5. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 56; ordinal in the memory table: 53.

Table 94.64: Record 056: r2c5_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c5_h+1v+1; *Cell*: 14; *Row*: 2
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark

Record 056 — r2c5_h+1v+1: continued on the next page

Record 056 — r2c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,5;8,5
Classification (cont.)	*Route family*: 30|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|0|-4|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.00000005518981253185914 0
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Record 056 — r2c5_h+1v+1: continued on the next page

Record 056 — r2c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7f85598f2d0761cdb1daa591c78b22b937d0b0cf9020ac23e6f9e943fd562608 *Reader*: fb7d8d4d5037cd9ce900cca6f805cb4785c5e6dc4618e4b3fa59c593bf78eebf *Cell record at 108*: 1523ad7232fb11fc5646852a3d2a3f1122cadd040d9709643b2c332d82848b2a

Route 057 — r2c6_h-1v-1

Cell 15: row 2, column 6. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 57; ordinal in the memory table: 60.

Table 94.65: Record 057: r2c6_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c6_h-1v-1; *Cell*: 15; *Row*: 2
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,6;8,4
Classification (cont.)	*Route family*: 34|0|-16|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 057 — r2c6_h-1v-1: continued on the next page

Record 057 — r2c6_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,4); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 47.606498943372135446416 86; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243 *Stable triads per block*: 0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 455c2b00848873039e0506315671ab73333699750588879e8ae6a9762ba3af54 *Reader*: 36eab0f217dc9a30cf9ca2da7ac99f6c2e67517347899cf3bee9182c711ace32 *Cell record at 108*: —

Route 058 — r2c6_h-1v+1

Cell 15: row 2, column 6. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 58; ordinal in the memory table: 59.

Table 94.66: Record 058: r2c6_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c6_h-1v+1; *Cell*: 15; *Row*: 2
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,6;8,4
Classification (cont.)	*Route family*: 34|0|-16|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 058 — r2c6_h-1v+1: continued on the next page

Record 058 — r2c6_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6f127fd9f25200c9da395c6482cb10ba06991f9a2b311706 e05753d88f848809 *Reader*: 97916a8721973f594ba41fb3ce987a18682cae23e96d3ad81ad92f70 aa3ffda7 *Cell record at 108*: bfc92621bc7d75ba4ede7c52fd53f1a32c441577160adc4a 69490655db8b7ea3

Route 059 — r2c6_h+1v-1

Cell 15: row 2, column 6. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 59; ordinal in the memory table: 58.

Table 94.67: Record 059: r2c6_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c6_h+1v-1; *Cell*: 15; *Row*: 2
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO

Record 059 — r2c6_h+1v-1: continued on the next page

Record 059 — r2c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,6;8,4
Classification (cont.)	*Route family*: 34|0|-16|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Record 059 — r2c6_h+1v-1: continued on the next page

Record 059 — r2c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a820aa46633c948e12bb41b7b41a621ad730d970e797f93ad58c70877be24f53 *Reader*: f5bcac8cac9dad8ee01f13e7f30b39e2243232b4d9a8da8a03297e1638b16cb4 *Cell record at 108*: —

Route 060 — r2c6_h+1v+1

Cell 15: row 2, column 6. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 60; ordinal in the memory table: 57.

Table 94.68: Record 060: r2c6_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c6_h+1v+1; *Cell*: 15; *Row*: 2
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,6;8,4
Classification (cont.)	*Route family*: 34|0|-16|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17

Record 060 — r2c6_h+1v+1: continued on the next page

Record 060 — r2c6_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,4); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 47.606498943372135446416 86; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ea9202399ce61bc1abe3e3bbf0faed082c6e4d11f9e0ce94 fb1a343e0f60b2c5 *Reader*: 9fdef0c8e1e5027d0dcdddc3848d78073d21e8d15c0ce90a2b53a7de 4ed3563b *Cell record at 108*: —

Route 061 — r2c7_h-1v-1

Cell 16: row 2, column 7. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 61; ordinal in the memory table: 64.

Table 94.69: Record 061: r2c7_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c7_h-1v-1; *Cell*: 16; *Row*: 2
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,7;8,3
Classification (cont.)	*Route family*: 34|0|0|2|-8|+++0--+++0--
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,2); *K* Ω: (56,54,5)

Record 061 — r2c7_h-1v-1: continued on the next page

Record 061 — r2c7_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003855097235414164 5
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 93f4d12e988e9b7d84464bea475814d6748da20159cd95ca1872639502d4e22e *Reader*: 4f6bcfea35a49b57dd08fe0ab2f398016e0a872931ce0832af3f2f6f123be521 *Cell record at 108*: —

Route 062 — r2c7_h-1v+1

Cell 16: row 2, column 7. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 62; ordinal in the memory table: 63.

Table 94.70: Record 062: r2c7_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c7_h-1v+1; *Cell*: 16; *Row*: 2
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark

Record 062 — r2c7_h-1v+1: continued on the next page

Record 062 — r2c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,7;8,3
Classification (cont.)	*Route family*: 34|0|0|2|-8|+++0--+++0--
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: ++00--++00--
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 36
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K* Ω: (36,54,4)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 35.606387746949443306362 81; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000024685259569118152
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 062 — r2c7_h-1v+1: continued on the next page

Record 062 — r2c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: fd3fc6ae2ff2d25bc3c572dc5362bf82ba08155cab1135cecc3e4ea8c6cabd14 *Reader*: 97b2ab0747f9cae836c873b69558f370f37393dc00bc1b475ff3498099cf3f6a *Cell record at 108*: 4ce890da3360de04fce789c2ce40701aacf357aa4d27239b5d03dfa52c8c9f85

Route 063 — r2c7_h+1v-1

Cell 16: row 2, column 7. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 63; ordinal in the memory table: 62.

Table 94.71: Record 063: r2c7_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c7_h+1v-1; *Cell*: 16; *Row*: 2
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,7;8,3
Classification (cont.)	*Route family*: 34|0|0|2|-8|+++0--+++0--
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 063 — r2c7_h+1v-1: continued on the next page

Record 063 — r2c7_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: 0++--00++--0
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 36
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K Ω*: (36,54,4)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 35.606387746949443306362 81; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000024685259569118152
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6b038b1a3fa41286fe974ac63c5d90cb981ae568eab928b7 7b8abbe73506531d *Reader*: 5abbe501c8adf01bdeab4ceef778d25879f7df343fcf8deab50624ce 7852d47c *Cell record at 108*: —

Route 064 — r2c7_h+1v+1

Cell 16: row 2, column 7. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 64; ordinal in the memory table: 61.

Table 94.72: Record 064: r2c7_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c7_h+1v+1; *Cell*: 16; *Row*: 2
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,7;8,3
Classification (cont.)	*Route family*: 34|0|0|2|-8|+++0--+++0--
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,2); *K* Ω: (56,54,5)

Record 064 — r2c7_h+1v+1: continued on the next page

Record 064 — r2c7_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003855097235414164 5
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7690711bd23b740afdbe1c4fa7b76bb78b49dfea24072771baed5540881ba560 *Reader*: 2e84b322980e807e05210dea2a1a280f6589f24b5d8d2b88a0a309b40ad8d5a1 *Cell record at 108*: —

Route 065 — r2c8_h-1v-1

Cell 17: row 2, column 8. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 65; ordinal in the memory table: 68.

Table 94.73: Record 065: r2c8_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c8_h-1v-1; *Cell*: 17; *Row*: 2
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 065 — r2c8_h-1v-1: continued on the next page

Record 065 — r2c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,8;8,2
Classification (cont.)	*Route family*: 30|6|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|0|-2|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Record 065 — r2c8_h-1v-1: continued on the next page

Record 065 — r2c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 59979b339515cffd98571eb86bd694e33ea0f1973905ec0bf1c9caa74ef63de7 *Reader*: 524796ca2925cfb1d109f28a8a8378d778d802cf51f272b5cbd5bd45fcaa368a *Cell record at 108*: d88dd1b6168ecd8f1b4de529649f5ba73c37ee39b4c64650605497dd1d21e3f3

Route 066 — r2c8_h-1v+1

Cell 17: row 2, column 8. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 66; ordinal in the memory table: 67.

Table 94.74: Record 066: r2c8_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c8_h-1v+1; *Cell*: 17; *Row*: 2
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,8;8,2
Classification (cont.)	*Route family*: 30|6|-8|0|-12|+++---+++---
Returns	*Class and phase*: 18; *Orbit and phase*: 18; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 066 — r2c8_h-1v+1: continued on the next page

Record 066 — r2c8_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (60,84,28); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000041174098946741575
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 354c6d9af4e6a921677a2e1ca86c2615967aadbba749be2fb9cbc9471af9ef30 *Reader*: 835f1e89a2323718255fdf26105c31d0c239439f72826178a5dfe560851db501 *Cell record at 108*: —

Route 067 — r2c8_h+1v-1

Cell 17: row 2, column 8. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 67; ordinal in the memory table: 66.

Table 94.75: Record 067: r2c8_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c8_h+1v-1; *Cell*: 17; *Row*: 2
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,8;8,2
Classification (cont.)	*Route family*: 30|6|-8|0|-12|+++---+++---
Returns	*Class and phase*: 9; *Orbit and phase*: 9; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (60,84,28); *K* Ω: (56,54,5)

Record 067 — r2c8_h+1v-1: continued on the next page

Record 067 — r2c8_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000004117409894674 1575
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003855097235414 1645
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5908d499c7cb58ca1019ee6fa049bb035b537bb28190db7680139634d9fbe00b *Reader*: be902208821a34b042d5e11f1d0fcc5e10dbf5a2f9625040dbc02e416483f054 *Cell record at 108*: —

Route 068 — r2c8_h+1v+1

Cell 17: row 2, column 8. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 68; ordinal in the memory table: 65.

Table 94.76: Record 068: r2c8_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c8_h+1v+1; *Cell*: 17; *Row*: 2
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 068 — r2c8_h+1v+1: continued on the next page

Record 068 — r2c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,8;8,2
Classification (cont.)	*Route family*: 30|6|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|6|0|2|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Record 068 — r2c8_h+1v+1: continued on the next page

Record 068 — r2c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 84615c8802e6b1fd96b339d0d1d76101fbba5fc64863032e5be621338e563f29 *Reader*: feee18fcf009a4833205b5075b749373cf28f0b113b0376d55a8f6231fe5d8b9 *Cell record at 108*: —

Route 069 — r2c9_h-1v-1

Cell 18: row 2, column 9. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 69; ordinal in the memory table: 72.

Table 94.77: Record 069: r2c9_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c9_h-1v-1; *Cell*: 18; *Row*: 2
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,9;8,1
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 069 — r2c9_h-1v-1: continued on the next page

Record 069 — r2c9_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243 *Stable triads per block*: 0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 10af243bdfdbc6b04a47e1da71d5e5ca47143ea9930de740045291154fd1ca26 *Reader*: 46188c4764cb2c18ad5e28f7d7701b20516f47767883a606fd272ab0ce1b6e7f *Cell record at 108*: —

Route 070 — r2c9_h-1v+1

Cell 18: row 2, column 9. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 70; ordinal in the memory table: 71.

Table 94.78: Record 070: r2c9_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c9_h-1v+1; *Cell*: 18; *Row*: 2
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,9;8,1
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 070 — r2c9_h-1v+1: continued on the next page

Record 070 — r2c9_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 78a741ee3c7bfbe7ae8641015a71bd34f1888a67bce6ff00 9113e38ec348d112 *Reader*: fc684626be7c75ff48c0e5350e38d10c764d4db51dbaede369429944 03aa60f0 *Cell record at 108*: 4f85ae5a226daca7170f68760b433575e50c60755178c473 034d1496819aa0a7

Route 071 — r2c9_h+1v-1

Cell 18: row 2, column 9. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 71; ordinal in the memory table: 70.

Table 94.79: Record 071: r2c9_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r2c9_h+1v-1; *Cell*: 18; *Row*: 2
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO

Record 071 — r2c9_h+1v-1: continued on the next page

Record 071 — r2c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,9;8,1
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Record 071 — r2c9_h+1v-1: continued on the next page

Record 071 — r2c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c0b14bfddf9f53d4228d9411817166fe0e3268f7dad8c9988d4776186e04fe7f *Reader*: 78ccb54b24e9defbe6c1315c36f064a2e031e90af5bc9cb69113c1ba082b2a72 *Cell record at 108*: —

Route 072 — r2c9_h+1v+1

Cell 18: row 2, column 9. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 72; ordinal in the memory table: 69.

Table 94.80: Record 072: r2c9_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r2c9_h+1v+1; *Cell*: 18; *Row*: 2
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,9;8,1
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17

Record 072 — r2c9_h+1v+1: continued on the next page

Record 072 — r2c9_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 40; *log2 ambiguity area*: 20.339850002884624; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: bb93b38f6a11bd24fe0d784b0382327e0e9577378da33db4 ff0c239336b2076c *Reader*: 0323a24467e3bb3b19fb86edbb720c35efac9bd2c76f856a02bbfaa6 be1de3fb *Cell record at 108*: —

Route 073 — r3c1_h-1v-1

Cell 19: row 3, column 1. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 73; ordinal in the memory table: 76.

Table 94.81: Record 073: r3c1_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c1_h-1v-1; *Cell*: 19; *Row*: 3
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,1;7,9
Classification (cont.)	*Route family*: 28|6|-10|0|-12|+++---+++---
Returns	*Class and phase*: 9; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 073 — r3c1_h-1v-1: continued on the next page

Record 073 — r3c1_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b49a35c80200681a5cd1c24a3af33bdcb3f5dfc15693d461 d3c7c6d1a094106a *Reader*: 9cd3c1cdd9f24b93ee41db3b5db0405bd95787b27d0b9563cb941f3b cf9ec663 *Cell record at 108*: —

Route 074 — r3c1_h-1v+1

Cell 19: row 3, column 1. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 74; ordinal in the memory table: 75.

Table 94.82: Record 074: r3c1_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c1_h-1v+1; *Cell*: 19; *Row*: 3
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 074 — r3c1_h-1v+1: continued on the next page

Record 074 — r3c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,1;7,9
Classification (cont.)	*Route family*: 28|6|-10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|-2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Record 074 — r3c1_h-1v+1: continued on the next page

Record 074 — r3c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: aa18d9482f30ff1d1e8a4a2a55cb565061e18e03d89d65c183f098f5ecfa8627 *Reader*: 9f05196446d319f0bee5611cef354231f598d0c69a54da6b36803c840ef9d155 *Cell record at 108*: 4f280c51aedd8c9f5e7c17101e369b774441e690405561db94125e58da5ec0c1

Route 075 — r3c1_h+1v-1

Cell 19: row 3, column 1. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 75; ordinal in the memory table: 74.

Table 94.83: Record 075: r3c1_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c1_h+1v-1; *Cell*: 19; *Row*: 3
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,1;7,9
Classification (cont.)	*Route family*: 28|6|-10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 3; *Final stable triads*: 16

Record 075 — r3c1_h+1v-1: continued on the next page

Record 075 — r3c1_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: f93f9cc3c26fe793e3a4e63395e34edd0c959cb9bc132a5815d675d46041d5b6 *Reader*: 8343e0ea9628938a9aa892c593f5925d0eace614709da84194ad5609bd12b447 *Cell record at 108*: —

Route 076 — r3c1_h+1v+1

Cell 19: row 3, column 1. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 76; ordinal in the memory table: 73.

Table 94.84: Record 076: r3c1_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c1_h+1v+1; *Cell*: 19; *Row*: 3
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,1;7,9
Classification (cont.)	*Route family*: 28|6|-10|0|-12|+++---+++---
Returns	*Class and phase*: 18; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 076 — r3c1_h+1v+1: continued on the next page

Record 076 — r3c1_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b5ee1bee5453ebdf2ca8a16f35b7700fe788e77916965f33 ae801dd014d7a637 *Reader*: 6aee9e8b24e2230ffb355d666b1169ced471e4a3b73faf2ce8a3168a 2f909cb2 *Cell record at 108*: —

Route 077 — r3c2_h-1v-1

Cell 20: row 3, column 2. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 77; ordinal in the memory table: 80.

Table 94.85: Record 077: r3c2_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c2_h-1v-1; *Cell*: 20; *Row*: 3
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 077 — r3c2_h-1v-1: continued on the next page

Record 077 — r3c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,2;7,8
Classification (cont.)	*Route family*: 28|-6|-4|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 28|6|-2|4|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464 16 86; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Record 077 — r3c2_h-1v-1: continued on the next page

Record 077 — r3c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5fbc72ff0a96e82d4616ba1a6ef070f7a682551f464c637a d43a97a11f6e8a97 *Reader*: 92a383d82ac125a374465a812287a783220c4993af802ee322a89c2c 20798cbd *Cell record at 108*: —

Route 078 — r3c2_h-1v+1

Cell 20: row 3, column 2. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 78; ordinal in the memory table: 79.

Table 94.86: Record 078: r3c2_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c2_h-1v+1; *Cell*: 20; *Row*: 3
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,2;7,8
Classification (cont.)	*Route family*: 28|-6|-4|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 078 — r3c2_h-1v+1: continued on the next page

Record 078 — r3c2_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|2|0|2|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 69e2bd91cf69da875a43f97523596d231b3e64e5813e5f26c0c2804e11812077 *Reader*: 3f2554e27e8f08e18bce3badea9cd8ff21745b26de177ed608864d591051a07e *Cell record at 108*: —

Route 079 — r3c2_h+1v-1

Cell 20: row 3, column 2. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 79; ordinal in the memory table: 78.

Table 94.87: Record 079: r3c2_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c2_h+1v-1; *Cell*: 20; *Row*: 3
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,2;7,8
Classification (cont.)	*Route family*: 28|-6|-4|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|-2|0|-2|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 079 — r3c2_h+1v-1: continued on the next page

Record 079 — r3c2_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c11bf3a55ac78a0a7ac79598bfd55908c204e4a60ec21403 b684c336250918b2 *Reader*: 4f92d4ea7024f5501e8b0a882bbee09736d0169b49bfb21ad0504f8c 1d37c43a *Cell record at 108*: —

Route 080 — r3c2_h+1v+1

Cell 20: row 3, column 2. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 80; ordinal in the memory table: 77.

Table 94.88: Record 080: r3c2_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c2_h+1v+1; *Cell*: 20; *Row*: 3
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 080 — r3c2_h+1v+1: continued on the next page

Record 080 — r3c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,2;7,8
Classification (cont.)	*Route family*: 28|-6|-4|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|2|-4|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464 16 86; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 080 — r3c2_h+1v+1: continued on the next page

Record 080 — r3c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 84b3ba3095f1140b146b108fd484619332fad0f9436e0ecc1ecd0a443acd145b *Reader*: 6cd46053b4aac7eb750510ca7ff5aa4ff8e13be097058396eee32adb341196d1 *Cell record at 108*: 3364070d4d0d99825f096a5b3dd87451aa757f0f85aa6ab974218271ed7418df

Route 081 — r3c3_h-1v-1

Cell 21: row 3, column 3. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 81; ordinal in the memory table: 84.

Table 94.89: Record 081: r3c3_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c3_h-1v-1; *Cell*: 21; *Row*: 3
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,3;7,7
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 081 — r3c3_h-1v-1: continued on the next page

Record 081 — r3c3_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,54,6); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 22; *log2 ambiguity area*: 4.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 90be6b645dcb7c34479da4ddb58dc6b331678651b48f48bfa32fa2f1cacbf2d2 *Reader*: 0a266f2de19b3916bf94b0203007d4c4edc9fba6891708948e1cd7c8d352827d *Cell record at 108*: —

Route 082 — r3c3_h-1v+1

Cell 21: row 3, column 3. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 82; ordinal in the memory table: 83.

Table 94.90: Record 082: r3c3_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c3_h-1v+1; *Cell*: 21; *Row*: 3
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,3;7,7
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (0,24,0); *K Ω*: (80,54,6)

Record 082 — r3c3_h-1v+1: continued on the next page

Record 082 — r3c3_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 970cf8e420061a4c2e86d72e44ab8e14d201e17d5273d819f231ca3c228b99cc *Reader*: e52e1aa490feb02a80eacd19facbd502f5a2bead0ea1c7ba6885899eb000c242 *Cell record at 108*: —

Route 083 — r3c3_h+1v-1

Cell 21: row 3, column 3. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 83; ordinal in the memory table: 82.

Table 94.91: Record 083: r3c3_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c3_h+1v-1; *Cell*: 21; *Row*: 3
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2

Record 083 — r3c3_h+1v-1: continued on the next page

Record 083 — r3c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,3;7,7
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (0,24,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ

Record 083 — r3c3_h+1v-1: continued on the next page

Record 083 — r3c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: fb908c9b7d05c8740756c16ce54c6e7890b67ab3adc021ba17bc80d1580a6a7e *Reader*: 540ae414abfa6f646b2b1e35635c7d7fe2834db6ead0ec1142e9fb0004b694aa *Cell record at 108*: 8fc1d1a017faec792451c97d492516e442f2ce08857f8c2aa8930ac01f5a0821

Route 084 — r3c3_h+1v+1

Cell 21: row 3, column 3. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 84; ordinal in the memory table: 81.

Table 94.92: Record 084: r3c3_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c3_h+1v+1; *Cell*: 21; *Row*: 3
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,3;7,7
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0

Record 084 — r3c3_h+1v+1: continued on the next page

Record 084 — r3c3_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.6066101397948275864709 1; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 22; *log2 ambiguity area*: 4.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: f04730d7e2e26b7754d2c540697c1ccdc7a274255c88f943077621c5e09b74ec *Reader*: 268d8ac49e67cc1679252262900b5b9df37d5c423422bb7f2b05970fb9a860da *Cell record at 108*: —

Route 085 — r3c4_h-1v-1

Cell 22: row 3, column 4. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 85; ordinal in the memory table: 88.

Table 94.93: Record 085: r3c4_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c4_h-1v-1; *Cell*: 22; *Row*: 3
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,4;7,6
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 085 — r3c4_h-1v-1: continued on the next page

Record 085 — r3c4_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000005512887179970 5072
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185 9140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: fe19794ade36889f09cc94a77d8a97228b8530f2fc302f96 a3cae6b71d488126 *Reader*: 4fb57798b1f66825757136b691435da4276f1d7475fb32d37ada397c 6bc2cc10 *Cell record at 108*: 47361123860c471c5c3e3660c7e98011f4bdf752d76ffd84 b3957ffe686a0fef

Route 086 — r3c4_h-1v+1

Cell 22: row 3, column 4. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 86; ordinal in the memory table: 87.

Table 94.94: Record 086: r3c4_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c4_h-1v+1; *Cell*: 22; *Row*: 3
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton

Record 086 — r3c4_h-1v+1: continued on the next page

Record 086 — r3c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,4;7,6
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Record 086 — r3c4_h-1v+1: continued on the next page

Record 086 — r3c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 735a2f2a02b51b9b28d0c9661b965464ea38522629179c9792287ffeefd17733 *Reader*: 0ecdaa6df15a6f0f268fb041bac3c594cca56daae8f41abdc8b4d3edd12e479b *Cell record at 108*: —

Route 087 — r3c4_h+1v-1

Cell 22: row 3, column 4. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 87; ordinal in the memory table: 86.

Table 94.95: Record 087: r3c4_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c4_h+1v-1; *Cell*: 22; *Row*: 3
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,4;7,6
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 3; *Final stable triads*: 16

Record 087 — r3c4_h+1v-1: continued on the next page

Record 087 — r3c4_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 1931926d8929382c703b3c3efe5ee8d5ec8768aaba2fdc568564d21d05a242f4 *Reader*: 2dda1905a5534096e9ad9a427c6d8048fc9d39c0c5825306d0203b1a12572285 *Cell record at 108*: —

Route 088 — r3c4_h+1v+1

Cell 22: row 3, column 4. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 88; ordinal in the memory table: 85.

Table 94.96: Record 088: r3c4_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c4_h+1v+1; *Cell*: 22; *Row*: 3
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,4;7,6
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 088 — r3c4_h+1v+1: continued on the next page

Record 088 — r3c4_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 44ed44030e829f7ef3f2d2539c0ecae1af3389fcb03b6af4 43c7515688564f46 *Reader*: 1ee0bda62536a4e364a3618e5ba602ded8f28109b25f42ce1150ff31 5daeb5f9 *Cell record at 108*: —

Route 089 — r3c5_h-1v-1

Cell 23: row 3, column 5. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 89; ordinal in the memory table: 92.

Table 94.97: Record 089: r3c5_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c5_h-1v-1; *Cell*: 23; *Row*: 3
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton

Record 089 — r3c5_h-1v-1: continued on the next page

Record 089 — r3c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,5;7,5
Classification (cont.)	*Route family*: 28|-6|6|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 28|6|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 22; *log2 ambiguity area*: 4.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Record 089 — r3c5_h-1v-1: continued on the next page

Record 089 — r3c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e953714a3f1290a221bb6aaa7ffb0b8c7ba40f3de4825cb231af24f88c67487c *Reader*: 516ac20b777e49c49141c04f9435195b0a72bf8fd107bf1131f9797a550996f1 *Cell record at 108*: —

Route 090 — r3c5_h-1v+1

Cell 23: row 3, column 5. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 90; ordinal in the memory table: 91.

Table 94.98: Record 090: r3c5_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c5_h-1v+1; *Cell*: 23; *Row*: 3
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,5;7,5
Classification (cont.)	*Route family*: 28|-6|6|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 090 — r3c5_h-1v+1: continued on the next page

Record 090 — r3c5_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b34cbe3d2a71fe4fb1ebe4194d4a87718d2bd15ed5ffc2b7222167761d430287 *Reader*: 647352489028a5eb55e737d1ba0d6c5267b15924d267d7995e4d3e1f03269bfd *Cell record at 108*: 2bdf6193049e9d8f1fc6952673cd5a6d3ece96f280eea32513f93207a69269c1

Route 091 — r3c5_h+1v-1

Cell 23: row 3, column 5. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 91; ordinal in the memory table: 90.

Table 94.99: Record 091: r3c5_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c5_h+1v-1; *Cell*: 23; *Row*: 3
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,5;7,5
Classification (cont.)	*Route family*: 28|-6|6|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 091 — r3c5_h+1v-1: continued on the next page

Record 091 — r3c5_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 257cfe72d342b8330026aca6da1825c935e6890f3fa068db 6795c936086cae39 *Reader*: fceaa5895efdf740e8a408a65e4bd72b828faadd8731c947705e6375 e4f6895d *Cell record at 108*: —

Route 092 — r3c5_h+1v+1

Cell 23: row 3, column 5. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 92; ordinal in the memory table: 89.

Table 94.100: Record 092: r3c5_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c5_h+1v+1; *Cell*: 23; *Row*: 3
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton

Record 092 — r3c5_h+1v+1: continued on the next page

Record 092 — r3c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,5;7,5
Classification (cont.)	*Route family*: 28|-6|6|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 092 — r3c5_h+1v+1: continued on the next page

Record 092 — r3c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5144d95329eaf9c2e01d5312581dfa5ab24fac78ab7b2ea124db8d95db7a998c *Reader*: 0ac1f885d966f22dfb79a602b576a688b351e32d455ddca04b3f46b937f49b0e *Cell record at 108*: —

Route 093 — r3c6_h-1v-1

Cell 24: row 3, column 6. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 93; ordinal in the memory table: 96.

Table 94.101: Record 093: r3c6_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c6_h-1v-1; *Cell*: 24; *Row*: 3
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,6;7,4
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 093 — r3c6_h-1v-1: continued on the next page

Record 093 — r3c6_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 30|6|-2|2|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 55.6064989433721354464186; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8e5c94763ee2ea8ffb481fc1fefbf17f6c771f7914fcfcf180fc6ba384a48110 *Reader*: e9907b108eaaf6db8390e7129e3c3d3f5c497165928e17a4b9f231a7ba1fa7d0 *Cell record at 108*: —

Route 094 — r3c6_h-1v+1

Cell 24: row 3, column 6. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 94; ordinal in the memory table: 95.

Table 94.102: Record 094: r3c6_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c6_h-1v+1; *Cell*: 24; *Row*: 3
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,6;7,4
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: 0; *τ 12*: +0000-+0000-
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|0
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 4
Differences and counters (cont.)	*P6*: 2
Integer readers	*K D*: (0,24,0); *K Ω*: (4,54,2)

Record 094 — r3c6_h-1v+1: continued on the next page

Record 094 — r3c6_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 3.6061653541040590262547 07; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000002500088676796314
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e11e9cb2d964364d36ac21f76809e6d1f0b312143e898c58 4c45d1ef8bf497ca *Reader*: 0b1b5faa852ad0e6071d95786097ecab1e741a4f64d3bdd1d778f85c 74963cbd *Cell record at 108*: 406b35c85cc7381b8d89efa82a29daccaa3ddcc24246e6c0 4702d9023088d3ad

Route 095 — r3c6_h+1v-1

Cell 24: row 3, column 6. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 95; ordinal in the memory table: 94.

Table 94.103: Record 095: r3c6_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c6_h+1v-1; *Cell*: 24; *Row*: 3
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton

Record 095 — r3c6_h+1v-1: continued on the next page

Record 095 — r3c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,6;7,4
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: 0; *τ 12*: 00+-0000+-00
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|0
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 4
Differences and counters (cont.)	*P6*: 2
Integer readers	*K D*: (0,24,0); *K Ω*: (4,54,2)
Reader evaluations	*κ D*: -0.1751364616628112239348425; *κ Ω*: 3.6061653541040590262547 07; *Action factor D*: 0.9999999998785810851337881
Reader evaluations (cont.)	*Action factor Ω*: 1.0000000025000886767963 14
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 095 — r3c6_h+1v-1: continued on the next page

Record 095 — r3c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b69e41e3adf88e397196cba6a46da69641203067a7118c65e4b64e8aa46b34d1 *Reader*: 783dfdedcf8fe84bdabe1f1b52581403e35dfe49f318adb41101ada8b61c501c *Cell record at 108*: —

Route 096 — r3c6_h+1v+1

Cell 24: row 3, column 6. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 96; ordinal in the memory table: 93.

Table 94.104: Record 096: r3c6_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c6_h+1v+1; *Cell*: 24; *Row*: 3
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,6;7,4
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 096 — r3c6_h+1v+1: continued on the next page

Record 096 — r3c6_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|2|-2|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6288242c4687f7c0ef85b06a62c64390d926911da179628087bd63b7387a992d *Reader*: 9055dce23724fb2ee036867b505c820f79933f0a7d4ecd3d289c0b969ca46da2 *Cell record at 108*: —

Route 097 — r3c7_h-1v-1

Cell 25: row 3, column 7. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 97; ordinal in the memory table: 100.

Table 94.105: Record 097: r3c7_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c7_h-1v-1; *Cell*: 25; *Row*: 3
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,7;7,3
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 097 — r3c7_h-1v-1: continued on the next page

Record 097 — r3c7_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6589a27f5db15e2faedf8e0bf0f4b48887925aa96b94b118 940ab4b275b4f51e *Reader*: a00262e0cf95aa0f99b91ef081a1ac39c8c5f9c7ba5b825eb0d5f9a8 81b4f01e *Cell record at 108*: —

Route 098 — r3c7_h-1v+1

Cell 25: row 3, column 7. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 98; ordinal in the memory table: 99.

Table 94.106: Record 098: r3c7_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c7_h-1v+1; *Cell*: 25; *Row*: 3
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2

Record 098 — r3c7_h-1v+1: continued on the next page

Record 098 — r3c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,7;7,3
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 55.6064989433721354464 16 86; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ

Record 098 — r3c7_h-1v+1: continued on the next page

Record 098 — r3c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 74fca7a065152326f2680e83c4eb79eddfceaf255870015f6dc3f0dcd4da23d4 *Reader*: d4d5eba82dff0efd18906a4c20b1c052b04b63460dea63d1c3908efab54ca465 *Cell record at 108*: —

Route 099 — r3c7_h+1v-1

Cell 25: row 3, column 7. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 99; ordinal in the memory table: 98.

Table 94.107: Record 099: r3c7_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c7_h+1v-1; *Cell*: 25; *Row*: 3
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,7;7,3
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 3; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2

Record 099 — r3c7_h+1v-1: continued on the next page

Record 099 — r3c7_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e56e3ff292c8e2520d3cecaa4e8d296324ef1d8f8f682120e853d2eb302e843b *Reader*: 809858a6f732a2d7201ea65668fee3960f55de4fc745fd45395c2dd5e8199cd5 *Cell record at 108*: —

Route 100 — r3c7_h+1v+1

Cell 25: row 3, column 7. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 100; ordinal in the memory table: 97.

Table 94.108: Record 100: r3c7_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c7_h+1v+1; *Cell*: 25; *Row*: 3
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,7;7,3
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 100 — r3c7_h+1v+1: continued on the next page

Record 100 — r3c7_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 77168e1c5f252e653c32d6b3899be2f09ce151e231c9bec5285c3e8baa931507 *Reader*: 78e3b7b481481612e0e698c15bce3bb6731d92e345512472bbfdf23257191308 *Cell record at 108*: 6124227ce899ccb2df636930314e07ff8fc23be0d2d973f8128c25ed2e55be47

Route 101 — r3c8_h-1v-1

Cell 26: row 3, column 8. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 101; ordinal in the memory table: 104.

Table 94.109: Record 101: r3c8_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c8_h-1v-1; *Cell*: 26; *Row*: 3
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 101 — r3c8_h-1v-1: continued on the next page

Record 101 — r3c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,8;7,2
Classification (cont.)	*Route family*: 34|0|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Record 101 — r3c8_h-1v-1: continued on the next page

Record 101 — r3c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a3c4b2e42baf237a5e4c2f9b552ebcd29e8bfa6d4980de4980f828b377f1771b *Reader*: f96905bcda7205151dd5dacf1f6525762d535f2dd174b1c3e19f351bded335b8 *Cell record at 108*: —

Route 102 — r3c8_h-1v+1

Cell 26: row 3, column 8. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 102; ordinal in the memory table: 103.

Table 94.110: Record 102: r3c8_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c8_h-1v+1; *Cell*: 26; *Row*: 3
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,8;7,2
Classification (cont.)	*Route family*: 34|0|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 102 — r3c8_h-1v+1: continued on the next page

Record 102 — r3c8_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 30b5ef0c47761242ea250c0bd063668a9403357735c2c59fab1ad3ef3b33a931 *Reader*: 11b41f4221d7ba5285c94c7305abf3242a05aad86ea35723202c943b48489ea1 *Cell record at 108*: 0dd6b250296cb9be5eb2e1d19659528b7de17fdc8796cf0ec3cc88e3aabfa877

Route 103 — r3c8_h+1v-1

Cell 26: row 3, column 8. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 103; ordinal in the memory table: 102.

Table 94.111: Record 103: r3c8_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c8_h+1v-1; *Cell*: 26; *Row*: 3
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,8;7,2
Classification (cont.)	*Route family*: 34|0|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 103 — r3c8_h+1v-1: continued on the next page

Record 103 — r3c8_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d954a2a9083ac81224a21b4c21d85c168f7e0b0f95f77416 3b3907a8c521f3ce *Reader*: bce6affb972de83392f296ec01c79cf65f1662441464abdb46015ef5 f9986e86 *Cell record at 108*: —

Route 104 — r3c8_h+1v+1

Cell 26: row 3, column 8. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 104; ordinal in the memory table: 101.

Table 94.112: Record 104: r3c8_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c8_h+1v+1; *Cell*: 26; *Row*: 3
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 104 — r3c8_h+1v+1: continued on the next page

Record 104 — r3c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,8;7,2
Classification (cont.)	*Route family*: 34|0|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464 16 86; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 104 — r3c8_h+1v+1: continued on the next page

Record 104 — r3c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 03881732293dbdb0dd5bde4d431381e4c1078b1ad88c6594eb82b5ddd3f00398 *Reader*: c585e17a9f62ed44ecbccc253a8758fe1c652c8a544d158754d2ae6fddf6df97 *Cell record at 108*: —

Route 105 — r3c9_h-1v-1

Cell 27: row 3, column 9. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 105; ordinal in the memory table: 108.

Table 94.113: Record 105: r3c9_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c9_h-1v-1; *Cell*: 27; *Row*: 3
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,9;7,1
Classification (cont.)	*Route family*: 30|6|0|2|-8|+++--0+++--0
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 105 — r3c9_h-1v-1: continued on the next page

Record 105 — r3c9_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|2|-4|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8baf9b09874f716a91bea12345945a5f1bd1f4f2baa14a324556a40ce2ba26af *Reader*: a0b0a058859733c704c7f2fed250076e56e37e36f21b93d755a4c033376ccdc1 *Cell record at 108*: f669a046ed98e7755abdbfba15ed1ac55e8a6e228b2b4eb30e40719fe160aaae

Route 106 — r3c9_h-1v+1

Cell 27: row 3, column 9. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 106; ordinal in the memory table: 107.

Table 94.114: Record 106: r3c9_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c9_h-1v+1; *Cell*: 27; *Row*: 3
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,9;7,1
Classification (cont.)	*Route family*: 30|6|0|2|-8|++++--0+++--0
Returns	*Class and phase*: 18; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 50|-6|2|-4|-8
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (0,24,0); *K Ω*: (56,54,5)

Record 106 — r3c9_h-1v+1: continued on the next page

Record 106 — r3c9_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 108; *log2 ambiguity area*: 46.52932501298079; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 6
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 56152a5de4e1754dfb18df1797256c5b173ad55be9a8457fb604673240f1c027 *Reader*: d706d3e3faf56f94f030d2d4769dda6399824d27fd2506f0451a4918a9a901fc *Cell record at 108*: —

Route 107 — r3c9_h+1v-1

Cell 27: row 3, column 9. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 107; ordinal in the memory table: 106.

Table 94.115: Record 107: r3c9_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r3c9_h+1v-1; *Cell*: 27; *Row*: 3
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 107 — r3c9_h+1v-1: continued on the next page

Record 107 — r3c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,9;7,1
Classification (cont.)	*Route family*: 30|6|0|2|-8|+++--0+++--0
Returns	*Class and phase*: 9; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 50|6|-2|4|-8
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (0,24,0); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: -0.17513646166281122393484 25; *κ Ω*: 55.606498943372135446416 86; *Action factor D*: 0.99999999987858108513378 81
Reader evaluations (cont.)	*Action factor Ω*: 1.00000003855097235414164 5
Prefix separation	*Ambiguity area*: 108; *log2 ambiguity area*: 46.52932501298079; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 6
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 107 — r3c9_h+1v-1: continued on the next page

Record 107 — r3c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 98a70374da093a1929fe532b452dc2b594c1f88dc9efbb4f8bb82206ed8e63b8 *Reader*: 956dd1205d0eaf9ce7232de79c5985f62b2f20e1807aa46b76e7ce2850285b90 *Cell record at 108*: —

Route 108 — r3c9_h+1v+1

Cell 27: row 3, column 9. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 108; ordinal in the memory table: 105.

Table 94.116: Record 108: r3c9_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r3c9_h+1v+1; *Cell*: 27; *Row*: 3
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,9;7,1
Classification (cont.)	*Route family*: 30|6|0|2|-8|+++--0+++--0
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 108 — r3c9_h+1v+1: continued on the next page

Record 108 — r3c9_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 30|6|-2|4|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 4df97b7fc1cea288d4b149a43852d6e54a768eaa7b1e17bd091da4b9eb93eb18 *Reader*: 7b1d3368504c06f4756793ae495b0eb9f07fef9577e95bcf66b649fad5640235 *Cell record at 108*: —

Route 109 — r4c1_h-1v-1

Cell 28: row 4, column 1. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 109; ordinal in the memory table: 112.

Table 94.117: Record 109: r4c1_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c1_h-1v-1; *Cell*: 28; *Row*: 4
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: R; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,1;6,9
Classification (cont.)	*Route family*: 30|6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|0|-4|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 109 — r4c1_h-1v-1: continued on the next page

Record 109 — r4c1_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055189889622315337
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 63616877dbaa1494e1e12a93696a27ab3d0d9a09bd6c27e7 a8cf49db5f58e7c8 *Reader*: 3405682bf852ce023ebe9993311695142ffa0a4307c6c7a9da972184 6f8f6ae1 *Cell record at 108*: e58666b4f6fa6d33a51e958ef358e1aec182e5eb5a6f84e4 24583f2d27bdfb85

Route 110 — r4c1_h-1v+1

Cell 28: row 4, column 1. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 110; ordinal in the memory table: 111.

Table 94.118: Record 110: r4c1_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c1_h-1v+1; *Cell*: 28; *Row*: 4
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark

Record 110 — r4c1_h-1v+1: continued on the next page

Record 110 — r4c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: R; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,1;6,9
Classification (cont.)	*Route family*: 30|6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 110 — r4c1_h-1v+1: continued on the next page

Record 110 — r4c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 80a08e930c195bab65a18748d66751216778294eac2b155d72ae50f39d5d7dda *Reader*: 2037eb27ad0ed4d7ff20939ed394b0263366036e5d5ff1cbc8635bf587eee9a2 *Cell record at 108*: —

Route 111 — r4c1_h+1v-1

Cell 28: row 4, column 1. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 111; ordinal in the memory table: 110.

Table 94.119: Record 111: r4c1_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c1_h+1v-1; *Cell*: 28; *Row*: 4
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: R; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,1;6,9
Classification (cont.)	*Route family*: 30|6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 111 — r4c1_h+1v-1: continued on the next page

Record 111 — r4c1_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|-2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: eb9fef92470f339ddf8c9ed1517e5de1b10969e9ebb16591a7f34cf5d9a76997 *Reader*: 0a8b7f1d8bfb27ed38bd2268b2ef01ad94636043402cd263c98ad3f7a6693437 *Cell record at 108*: —

Route 112 — r4c1_h+1v+1

Cell 28: row 4, column 1. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 112; ordinal in the memory table: 109.

Table 94.120: Record 112: r4c1_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c1_h+1v+1; *Cell*: 28; *Row*: 4
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: R; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,1;6,9
Classification (cont.)	*Route family*: 30|6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|6|0|4|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 112 — r4c1_h+1v+1: continued on the next page

Record 112 — r4c1_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055189889622315337
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 26; *log2 ambiguity area*: 6.339850002884624; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 322c7aa9113fe7344258693ce39c72802f7f0c73dc8b45c2 f74ec16a33fef9e4 *Reader*: f4cec5cc7eaa26376f6155232dee3e998411e31892f8981e72e525ab 7e0192d8 *Cell record at 108*: —

Route 113 — r4c2_h-1v-1

Cell 29: row 4, column 2. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 113; ordinal in the memory table: 116.

Table 94.121: Record 113: r4c2_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c2_h-1v-1; *Cell*: 29; *Row*: 4
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 113 — r4c2_h-1v-1: continued on the next page

Record 113 — r4c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,2;6,8
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Record 113 — r4c2_h-1v-1: continued on the next page

Record 113 — r4c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 320c90bb3283149be781a543a1877eb0bab2603c2de7a705296b6be5be9e1da5 *Reader*: fc5013386134f89641ceb3f10fc2600b52ba7b583a6be4816aecffff35bfab68 *Cell record at 108*: —

Route 114 — r4c2_h-1v+1

Cell 29: row 4, column 2. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 114; ordinal in the memory table: 115.

Table 94.122: Record 114: r4c2_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c2_h-1v+1; *Cell*: 29; *Row*: 4
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,2;6,8
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 114 — r4c2_h-1v+1: continued on the next page

Record 114 — r4c2_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 4257bc9a976f4f175e997aba070b542adc329417c6dae0d0 8e497d6d65d51564 *Reader*: 102c954f9d43af2049a828e972c927bace8ca4c96291f65efffa01ca e7c2fdfb *Cell record at 108*: —

Route 115 — r4c2_h+1v-1

Cell 29: row 4, column 2. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 115; ordinal in the memory table: 114.

Table 94.123: Record 115: r4c2_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c2_h+1v-1; *Cell*: 29; *Row*: 4
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,2;6,8
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K* Ω: (80,54,6)

Record 115 — r4c2_h+1v-1: continued on the next page

Record 115 — r4c2_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 16f46915102b6339647250e52a7f69a351f7ca67ea21ef1a4634567c1e82c249 *Reader*: 3e5eb22e721faa17e47903a494789298f6831e31c48a1acac42b56f61a087c39 *Cell record at 108*: 976690fd5ec3fa4bf0ee2a75d8c8cf5e15e549ba4e8d243657dff0e8cd9a4eed

Route 116 — r4c2_h+1v+1

Cell 29: row 4, column 2. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 116; ordinal in the memory table: 113.

Table 94.124: Record 116: r4c2_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c2_h+1v+1; *Cell*: 29; *Row*: 4
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 116 — r4c2_h+1v+1: continued on the next page

Record 116 — r4c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,2;6,8
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 116 — r4c2_h+1v+1: continued on the next page

Record 116 — r4c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3108b4aa18a6e54b1b72c94127319da7c81a3c4b94e8d5842a621deae470a1c1 *Reader*: a6f1b74f60b1279c43943235a599dd91eba8343112b3912e62c18ec7472ea819 *Cell record at 108*: —

Route 117 — r4c3_h-1v-1

Cell 30: row 4, column 3. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 117; ordinal in the memory table: 120.

Table 94.125: Record 117: r4c3_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c3_h-1v-1; *Cell*: 30; *Row*: 4
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: G; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,3;6,7
Classification (cont.)	*Route family*: 28|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16

Record 117 — r4c3_h-1v-1: continued on the next page

Record 117 — r4c3_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: dfba8889ea2e308db35da1a1e3a470cb901f5b1b448eecfa 8b072910f423cf30 *Reader*: afd3630152262dd724d21cfe75259c892dc768015259d63a8c73a387 e703340c *Cell record at 108*: —

Route 118 — r4c3_h-1v+1

Cell 30: row 4, column 3. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 118; ordinal in the memory table: 119.

Table 94.126: Record 118: r4c3_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c3_h-1v+1; *Cell*: 30; *Row*: 4
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: G; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,3;6,7
Classification (cont.)	*Route family*: 28|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (48,54,5)

Record 118 — r4c3_h-1v+1: continued on the next page

Record 118 — r4c3_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b9a0e830c5b79e19fac92114fb87e7d73c89c8ab1412847c40d4194edfafd295 *Reader*: 45d58d9381a000cc9d767cfaadb64e412d11aefedf7a5b16fdf39ebb2427bb4f *Cell record at 108*: —

Route 119 — r4c3_h+1v-1

Cell 30: row 4, column 3. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 119; ordinal in the memory table: 118.

Table 94.127: Record 119: r4c3_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c3_h+1v-1; *Cell*: 30; *Row*: 4
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark

Record 119 — r4c3_h+1v-1: continued on the next page

Record 119 — r4c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: G; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,3;6,7
Classification (cont.)	*Route family*: 28|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 119 — r4c3_h+1v-1: continued on the next page

Record 119 — r4c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: f18a5f3bb67d88d2241efee93e0869d0669b4af613f1efc2a3340b822993e9f0 *Reader*: 2fa6335747d64259eacd9cd2cfd7362a3d7b22ed15a290e43fc0aa7db9ebb8ee *Cell record at 108*: —

Route 120 — r4c3_h+1v+1

Cell 30: row 4, column 3. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 120; ordinal in the memory table: 117.

Table 94.128: Record 120: r4c3_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c3_h+1v+1; *Cell*: 30; *Row*: 4
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: G; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,3;6,7
Classification (cont.)	*Route family*: 28|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 120 — r4c3_h+1v+1: continued on the next page

Record 120 — r4c3_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d562cf51d4b94895d64814ea9d20c3894b082a0c3fdcbdd529750f0f92dc6249 *Reader*: 46e7971fd3cde9a3330a90fd8d65631f09d3233dbe7e2bec0108983ff0938898 *Cell record at 108*: 0445e18674117bf622efd6fbfa1d6e4c6164b9238247bb7c762c152bb9686738

Route 121 — r4c4_h-1v-1

Cell 31: row 4, column 4. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 121; ordinal in the memory table: 124.

Table 94.129: Record 121: r4c4_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c4_h-1v-1; *Cell*: 31; *Row*: 4
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: R; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,4;6,6
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 121 — r4c4_h-1v-1: continued on the next page

Record 121 — r4c4_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055189889622315337
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 77c77bedc6c70e8f673bba3f1a38ebbbad0e9216da5a1135 bf40b23f327bb557 *Reader*: 71932081a5e57d25037ce0d5b5db38479d4de5b79809ce8e2c0b6c8a a771cdfc *Cell record at 108*: —

Route 122 — r4c4_h-1v+1

Cell 31: row 4, column 4. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 122; ordinal in the memory table: 123.

Table 94.130: Record 122: r4c4_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c4_h-1v+1; *Cell*: 31; *Row*: 4
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark

Record 122 — r4c4_h-1v+1: continued on the next page

Record 122 — r4c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 1; *Color*: R; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,4;6,6
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 122 — r4c4_h-1v+1: continued on the next page

Record 122 — r4c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 287327ac0b7a67951ed95a00dceb2cc4b73e9aae63f89cf3f375f2e14608c87d *Reader*: 015d0ecdc3d3681a13d3a6dfb1164ab55a41a27e6f103f07f68ac1e94b8ffcd2 *Cell record at 108*: 282e55192bd3edf96e14e9bf0b50e3479922204db1e0fa80756865f131f8c5ed

Route 123 — r4c4_h+1v-1

Cell 31: row 4, column 4. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 123; ordinal in the memory table: 122.

Table 94.131: Record 123: r4c4_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c4_h+1v-1; *Cell*: 31; *Row*: 4
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: R; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,4;6,6
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 123 — r4c4_h+1v-1: continued on the next page

Record 123 — r4c4_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b39d7e1604512deef70b3918e486fa58fffe93c5ba2477a0 9df55b64831ca74b *Reader*: 8f35c9aaa9f1b19875c81199d648323876ea87ed664f20cdb441f73a dcdccb12 *Cell record at 108*: —

Route 124 — r4c4_h+1v+1

Cell 31: row 4, column 4. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 124; ordinal in the memory table: 121.

Table 94.132: Record 124: r4c4_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c4_h+1v+1; *Cell*: 31; *Row*: 4
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: R; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,4;6,6
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 124 — r4c4_h+1v+1: continued on the next page

Record 124 — r4c4_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000005518988962231533 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a20fae0de320526914318b1432bf0b0eb2e61de22cd563a0 e5a17b0ea9128bc0 *Reader*: 57d09c9ebfd5b2a4289c6b1055b0fc67ad75d3af3cdcf8d84d685d0e 8828573d *Cell record at 108*: —

Route 125 — r4c5_h-1v-1

Cell 32: row 4, column 5. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 125; ordinal in the memory table: 128.

Table 94.133: Record 125: r4c5_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c5_h-1v-1; *Cell*: 32; *Row*: 4
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G1; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: B; *Multisection*: quark_down_G1

Record 125 — r4c5_h-1v-1: continued on the next page

Record 125 — r4c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: quark_down; *Depth*: G1; *Inversion orbit*: 4,5;6,5
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ

Record 125 — r4c5_h-1v-1: continued on the next page

Record 125 — r4c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 79fb9a845241446b08dd24c605f6b1b7bd4981c5e6fa6e0067ea36ce0e7baef1 *Reader*: fb4bcab39bf089e5c421bf352caad6674c9c54f04a9d95494401d5b5db852e28 *Cell record at 108*: —

Route 126 — r4c5_h-1v+1

Cell 32: row 4, column 5. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 126; ordinal in the memory table: 127.

Table 94.134: Record 126: r4c5_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c5_h-1v+1; *Cell*: 32; *Row*: 4
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G1; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: B; *Multisection*: quark_down_G1
Classification (cont.)	*Physical family*: quark_down; *Depth*: G1; *Inversion orbit*: 4,5;6,5
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0

Record 126 — r4c5_h-1v+1: continued on the next page

Record 126 — r4c5_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+-0-+0+-0-
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 40
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K* Ω: (40,54,4)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 39.606387746949443306362 81; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000027458386692651400
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 0c0060901278e986f81b5b522e367331377866c4eec0a6c9 a5fd33b1702579f7 *Reader*: 7359aaa714e722cf55306e7389c0a7821640a976ddca2652c3e9de16 1d84c014 *Cell record at 108*: —

Route 127 — r4c5_h+1v-1

Cell 32: row 4, column 5. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 127; ordinal in the memory table: 126.

Table 94.135: Record 127: r4c5_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c5_h+1v-1; *Cell*: 32; *Row*: 4
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G1; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: B; *Multisection*: quark_down_G1
Classification (cont.)	*Physical family*: quark_down; *Depth*: G1; *Inversion orbit*: 4,5;6,5
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+-0-+0+-0-
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 40
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K* Ω: (40,54,4)

Record 127 — r4c5_h+1v-1: continued on the next page

Record 127 — r4c5_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 39.606387746949443306362 81; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000027458386692651400
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 9866ca016d61a93c8605f9d393f5c7d1ccbf95b39e0be07e da5205d177bbe0a3 *Reader*: 37ed6a7926a4b489b076fae2ae21dc04dd4b9ca8e9edd354076bf251 fcd73744 *Cell record at 108*: 58bd109118dc402143446db43b056afbcc8d2a5991cace6f 6ec3fb4670b5eff4

Route 128 — r4c5_h+1v+1

Cell 32: row 4, column 5. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 128; ordinal in the memory table: 125.

Table 94.136: Record 128: r4c5_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c5_h+1v+1; *Cell*: 32; *Row*: 4
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G1; *Family*: quark_down; *Sector*: quark

Record 128 — r4c5_h+1v+1: continued on the next page

Record 128 — r4c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 1; *Color*: B; *Multisection*: quark_down_G1
Classification (cont.)	*Physical family*: quark_down; *Depth*: G1; *Inversion orbit*: 4,5;6,5
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 128 — r4c5_h+1v+1: continued on the next page

Record 128 — r4c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 4fcdd9bfc143fe82bee933b825b840060bc9cc4cb3164418cc30da5c55f863c5 *Reader*: ba031ff902fef28f8cc55107fa551870981ae35f912431d3e4c6b06e9fd726fa *Cell record at 108*: —

Route 129 — r4c6_h-1v-1

Cell 33: row 4, column 6. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 129; ordinal in the memory table: 132.

Table 94.137: Record 129: r4c6_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c6_h-1v-1; *Cell*: 33; *Row*: 4
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: G; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,6;6,4
Classification (cont.)	*Route family*: 28|-6|10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16

Record 129 — r4c6_h-1v-1: continued on the next page

Record 129 — r4c6_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 87efef613d4e87ec97c29419f7de6fbea09a79bc753ae3e4 04b221d8ede75b4a *Reader*: 6e6b7dd414d89eb1aaaf5a8ea14fac7861011753e2349a9cc217300f d1b80edf *Cell record at 108*: —

Route 130 — r4c6_h-1v+1

Cell 33: row 4, column 6. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 130; ordinal in the memory table: 131.

Table 94.138: Record 130: r4c6_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c6_h-1v+1; *Cell*: 33; *Row*: 4
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: G; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,6;6,4
Classification (cont.)	*Route family*: 28|-6|10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (56,54,5)

Record 130 — r4c6_h-1v+1: continued on the next page

Record 130 — r4c6_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	κ *D*: 39.39035827686092491784959; κ Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 055c55a6583303aac67730b62377ba8a1d0049ac7aed4debc98aad5881bc9bb5 *Reader*: 8fa03549920be213176b7e0c881c6482be1ed34ccd91ebfb0760198297de4e2b *Cell record at 108*: —

Route 131 — r4c6_h+1v-1

Cell 33: row 4, column 6. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 131; ordinal in the memory table: 130.

Table 94.139: Record 131: r4c6_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c6_h+1v-1; *Cell*: 33; *Row*: 4
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark

Record 131 — r4c6_h+1v-1: continued on the next page

Record 131 — r4c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 1; *Color*: G; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,6;6,4
Classification (cont.)	*Route family*: 28|-6|10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 131 — r4c6_h+1v-1: continued on the next page

Record 131 — r4c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 406e5327b2546eaa3992ae56c6688e014239f25ef3712139e37360e70307a0cf *Reader*: c31d14fcf53e490e7b58ce903300244767c13da20e17866554a53b7f1e8aa2ca *Cell record at 108*: —

Route 132 — r4c6_h+1v+1

Cell 33: row 4, column 6. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 132; ordinal in the memory table: 129.

Table 94.140: Record 132: r4c6_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c6_h+1v+1; *Cell*: 33; *Row*: 4
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: G; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,6;6,4
Classification (cont.)	*Route family*: 28|-6|10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 132 — r4c6_h+1v+1: continued on the next page

Record 132 — r4c6_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: edc4254630dee8eed6f09c01001d7e5ff9ae88a52b26f6c8 4b34889e95b4b44f *Reader*: 8e63a835c6a45d8cdfe6ec7615e7954f6d06dba153209bd739e0e946 65079019 *Cell record at 108*: ac355fc3c7d67e23747af693446ce5973090a1288c89b61e d5e84c164421d478

Route 133 — r4c7_h-1v-1

Cell 34: row 4, column 7. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 133; ordinal in the memory table: 136.

Table 94.141: Record 133: r4c7_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c7_h-1v-1; *Cell*: 34; *Row*: 4
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: R; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,7;6,3
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 30|6|-2|0|-8
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,54,7); *K* Ω: (56,54,5)

Record 133 — r4c7_h-1v-1: continued on the next page

Record 133 — r4c7_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000005518988962231533 7
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 4ca3456697b5beafb2684f346ef4fe8a825e4e9b1923a035 4b9e170e0a544faa *Reader*: e241f838e36aaa931f974504e5963e774300dbe4095f540812bdb1be 522d47ca *Cell record at 108*: —

Route 134 — r4c7_h-1v+1

Cell 34: row 4, column 7. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 134; ordinal in the memory table: 135.

Table 94.142: Record 134: r4c7_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c7_h-1v+1; *Cell*: 34; *Row*: 4
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark

Record 134 — r4c7_h-1v+1: continued on the next page

Record 134 — r4c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: R; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,7;6,3
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 134 — r4c7_h-1v+1: continued on the next page

Record 134 — r4c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 01c95731317c519b15d30504737dbf0755697c530400d12b6dbb06b6fddb26df *Reader*: 0775e69aa78c62aeb37e0d803ae05fdb216bf35b3648f5cbd17c92719739b065 *Cell record at 108*: 4acc79e1bddf0a8446aca263fca69bd39879caa1e6469c33d703da9172c27cd9

Route 135 — r4c7_h+1v-1

Cell 34: row 4, column 7. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 135; ordinal in the memory table: 134.

Table 94.143: Record 135: r4c7_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c7_h+1v-1; *Cell*: 34; *Row*: 4
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: R; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,7;6,3
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 135 — r4c7_h+1v-1: continued on the next page

Record 135 — r4c7_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 782a318cc548e949b83af90597ea3aa0715c06a57efc8cd4 c09cc84eaf42ad4c *Reader*: f4d3eea43826eed3747c080d0cdd48c809877dc7fc409585fd0274bc 331de472 *Cell record at 108*: —

Route 136 — r4c7_h+1v+1

Cell 34: row 4, column 7. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 136; ordinal in the memory table: 133.

Table 94.144: Record 136: r4c7_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c7_h+1v+1; *Cell*: 34; *Row*: 4
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: R; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,7;6,3
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|2|0|-8
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,54,7); *K* Ω: (56,54,5)

Record 136 — r4c7_h+1v+1: continued on the next page

Record 136 — r4c7_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000005518988962231533 7
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 54436c1307ec79e76d82397f2b17691ba5e8eca04e1f5c1135c5a6f07fdd51cf *Reader*: 073fbdcdb0df31cbcc689c539e57777a6ea89a0771eb07238bc113fd3c52f129 *Cell record at 108*: —

Route 137 — r4c8_h-1v-1

Cell 35: row 4, column 8. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 137; ordinal in the memory table: 140.

Table 94.145: Record 137: r4c8_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c8_h-1v-1; *Cell*: 35; *Row*: 4
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3

Record 137 — r4c8_h-1v-1: continued on the next page

Record 137 — r4c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,8;6,2
Classification (cont.)	*Route family*: 34|0|-20|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,0); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Record 137 — r4c8_h-1v-1: continued on the next page

Record 137 — r4c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2b804e0cfd9c913aa79e7b2529b115c91f296d3cf84b6837e5e05321bfc2a6c0 *Reader*: 98e82256254f483db3a5842d1f7fd6a569044ecf3c542b941705855559737516 *Cell record at 108*: bf56e7e7a44142053df74d41b3498a6c84055d812f97946ed2c0674099511566

Route 138 — r4c8_h-1v+1

Cell 35: row 4, column 8. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 138; ordinal in the memory table: 139.

Table 94.146: Record 138: r4c8_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c8_h-1v+1; *Cell*: 35; *Row*: 4
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,8;6,2
Classification (cont.)	*Route family*: 34|0|-20|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 138 — r4c8_h-1v+1: continued on the next page

Record 138 — r4c8_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 50|-6|2|-2|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 03804c454b95d8b93e9f278969bffd8bf62a884888eefa4f44c9253afb6c9aa4 *Reader*: 87681b4a0c073f38f2a84e8a65f543f2c195b938cd71606b35a72691d777d3ce *Cell record at 108*: —

Route 139 — r4c8_h+1v-1

Cell 35: row 4, column 8. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 139; ordinal in the memory table: 138.

Table 94.147: Record 139: r4c8_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c8_h+1v-1; *Cell*: 35; *Row*: 4
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,8;6,2
Classification (cont.)	*Route family*: 34|0|-20|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 50|6|-2|2|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K* Ω: (56,54,5)

Record 139 — r4c8_h+1v-1: continued on the next page

Record 139 — r4c8_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b6ebfc8618552fdd72e29c86168f9f6e2a46c3fe8d74fb9e06ff8d9ef4d8355b *Reader*: e8e9508b8406628379069d3f617b9beaee92634d170e0a1cf2a20c31d62a42e6 *Cell record at 108*: —

Route 140 — r4c8_h+1v+1

Cell 35: row 4, column 8. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 140; ordinal in the memory table: 137.

Table 94.148: Record 140: r4c8_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c8_h+1v+1; *Cell*: 35; *Row*: 4
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 140 — r4c8_h+1v+1: continued on the next page

Record 140 — r4c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,8;6,2
Classification (cont.)	*Route family*: 34|0|-20|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,0); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 140 — r4c8_h+1v+1: continued on the next page

Record 140 — r4c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b1167825dc88e346b1033aa368e2c68a7ce87d854b37460a00c312e0b818fce5 *Reader*: b7f40bf3d4a01f15bdaa23dc2ebd2cdd551eca43991086c52426a372930a411b *Cell record at 108*: —

Route 141 — r4c9_h-1v-1

Cell 36: row 4, column 9. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 141; ordinal in the memory table: 144.

Table 94.149: Record 141: r4c9_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c9_h-1v-1; *Cell*: 36; *Row*: 4
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,9;6,1
Classification (cont.)	*Route family*: 34|0|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16

Record 141 — r4c9_h-1v-1: continued on the next page

Record 141 — r4c9_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 1fac69dafa70b07993139f7a846f20abb465238f4558212f 54421745cb5e3918 *Reader*: f8d6c21dad31131fc4c2fc3d70185a1f888c237d6474a021870b6519 6e6d416e *Cell record at 108*: —

Route 142 — r4c9_h-1v+1

Cell 36: row 4, column 9. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 142; ordinal in the memory table: 143.

Table 94.150: Record 142: r4c9_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c9_h-1v+1; *Cell*: 36; *Row*: 4
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,9;6,1
Classification (cont.)	*Route family*: 34|0|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|-2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K* Ω: (80,54,6)

Record 142 — r4c9_h-1v+1: continued on the next page

Record 142 — r4c9_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 18959d06bbedaac3d5d1132981d981df9bd28a79c6da3375 6e8d2149cb51db61 *Reader*: 4c038f821f5bd14ceca5c1f4829b72b372c1563b32e0fcc5b34cfb9e ff6992ee *Cell record at 108*: 1e448780f5a741d6641ffc8e2ae465ac249cdecbbd7f941e 44db167f2a18cb1b

Route 143 — r4c9_h+1v-1

Cell 36: row 4, column 9. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 143; ordinal in the memory table: 142.

Table 94.151: Record 143: r4c9_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r4c9_h+1v-1; *Cell*: 36; *Row*: 4
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark

Record 143 — r4c9_h+1v-1: continued on the next page

Record 143 — r4c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,9;6,1
Classification (cont.)	*Route family*: 34|0|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 143 — r4c9_h+1v-1: continued on the next page

Record 143 — r4c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e624c172e0c76c77cc5995ae25a89b2e951081d6310567419586143808bab0aa *Reader*: b10d5887020993ff069a374c70846c7db2da53820e650c37b95231a6b241a3cb *Cell record at 108*: —

Route 144 — r4c9_h+1v+1

Cell 36: row 4, column 9. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 144; ordinal in the memory table: 141.

Table 94.152: Record 144: r4c9_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r4c9_h+1v+1; *Cell*: 36; *Row*: 4
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,9;6,1
Classification (cont.)	*Route family*: 34|0|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 144 — r4c9_h+1v+1: continued on the next page

Record 144 — r4c9_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 555a0b57737e22122d0fd734f5aa5516d6461c185bde5c94 fab7062044c81b31 *Reader*: 40c90b3041a8d94ca52a895335732ae51dc7c7f9837d5ada66d7bc12 643ede6a *Cell record at 108*: —

Route 145 — r5c1_h-1v-1

Cell 37: row 5, column 1. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 145; ordinal in the memory table: 148.

Table 94.153: Record 145: r5c1_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c1_h-1v-1; *Cell*: 37; *Row*: 5
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 5,1;5,9
Classification (cont.)	*Route family*: 34|0|-16|0|-8|+++-0-+++-0-
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,2); *K* Ω: (48,54,5)

Record 145 — r5c1_h-1v-1: continued on the next page

Record 145 — r5c1_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 47.606498943372135446416 86; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003300471805324308 4
Prefix separation	*Ambiguity area*: 30; *log2 ambiguity area*: 8.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 44128f77694e22cdb74ce801d17ef58cb5f75058dd488325f94b8db3aef4c2f3 *Reader*: 8e414ca1b56fa4e2e54dc83e480cd2bdb5cfc2052546af04082e0f017d24c4e5 *Cell record at 108*: —

Route 146 — r5c1_h-1v+1

Cell 37: row 5, column 1. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 146; ordinal in the memory table: 147.

Table 94.154: Record 146: r5c1_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c1_h-1v+1; *Cell*: 37; *Row*: 5
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 146 — r5c1_h-1v+1: continued on the next page

Record 146 — r5c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 5,1;5,9
Classification (cont.)	*Route family*: 34|0|-16|0|-8|+++-0-+++-0-
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.00000002761000615951421 5
Reader evaluations (cont.)	*Action factor Ω*: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 146 — r5c1_h-1v+1: continued on the next page

Record 146 — r5c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 0f38c6958afd01ba27b6bc49a6893e4853cbeff8b05159d512fb8b00bcb73007 *Reader*: d9b335838b663c211a4ec0d10a405d917b276d2ca4cd5695fbf07de3d36643e7 *Cell record at 108*: —

Route 147 — r5c1_h+1v-1

Cell 37: row 5, column 1. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 147; ordinal in the memory table: 146.

Table 94.155: Record 147: r5c1_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c1_h+1v-1; *Cell*: 37; *Row*: 5
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 5,1;5,9
Classification (cont.)	*Route family*: 34|0|-16|0|-8|+++-0-+++-0-
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 147 — r5c1_h+1v-1: continued on the next page

Record 147 — r5c1_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: aaffc169d0598de597a3a2882b457981816b63fb5a1853f5546db1a985aea4ba *Reader*: deaf6877f8a7c8880ba1417fe43eee1c66d2cde0af9e5961bbf415f2f0787f8e *Cell record at 108*: ff9f223faafbe5f7d396cb50240d6d52e53eaab1a745f7719f6f1503b398f6a5

Route 148 — r5c1_h+1v+1

Cell 37: row 5, column 1. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 148; ordinal in the memory table: 145.

Table 94.156: Record 148: r5c1_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c1_h+1v+1; *Cell*: 37; *Row*: 5
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 5,1;5,9
Classification (cont.)	*Route family*: 34|0|-16|0|-8|+++-0-+++-0-
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,2); *K* Ω: (48,54,5)

Record 148 — r5c1_h+1v+1: continued on the next page

Record 148 — r5c1_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 47.606498943372135446416 86; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003300471805324308 4
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6bb6953ee6341f42515e8490c33965ad839bbc946cbc777d d1d832248ba36e66 *Reader*: 9743965c5f547f81a6eb1b5bfc13cd5ee9140eeb402f7500f7a52149 283e0a47 *Cell record at 108*: —

Route 149 — r5c2_h-1v-1

Cell 38: row 5, column 2. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 149; ordinal in the memory table: 152.

Table 94.157: Record 149: r5c2_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c2_h-1v-1; *Cell*: 38; *Row*: 5
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 149 — r5c2_h-1v-1: continued on the next page

Record 149 — r5c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 5,2;5,8
Classification (cont.)	*Route family*: 30|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|0|-4|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Record 149 — r5c2_h-1v-1: continued on the next page

Record 149 — r5c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7cfcd1776e4330be5d0164d074d9db7ebbc9ed421fdd9625dbdae32b33ae0ef6 *Reader*: b369d9f2359a2a99a416d16f936b4222912fe42aa504fb2ca9cbc185be1d8c70 *Cell record at 108*: 8754d795eb4696cdb9a16e289f13d7d0caa88f179c1ebb009c21bcc236cb922c

Route 150 — r5c2_h-1v+1

Cell 38: row 5, column 2. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 150; ordinal in the memory table: 151.

Table 94.158: Record 150: r5c2_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c2_h-1v+1; *Cell*: 38; *Row*: 5
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 5,2;5,8
Classification (cont.)	*Route family*: 30|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 150 — r5c2_h-1v+1: continued on the next page

Record 150 — r5c2_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -4
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: +++-00+++-00
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-4|0|-4
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 20
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (60,84,28); *K Ω*: (20,54,4)
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ Ω*: 19.6063877469494433063628 1; *Action factor D*: 1.000000041174098946741575
Reader evaluations (cont.)	*Action factor Ω*: 1.000000013592751151887498
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ce452c82855b0221b7ba9f0c035bcb7309bb6bcd57b4db0e3abe86148d83512e *Reader*: b1388ecfbf1e3638e950a7e90fed725e51d428bc70bb666d2eba9ccc3a4e8599 *Cell record at 108*: —

Route 151 — r5c2_h+1v-1

Cell 38: row 5, column 2. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 151; ordinal in the memory table: 150.

Table 94.159: Record 151: r5c2_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c2_h+1v-1; *Cell*: 38; *Row*: 5
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 5,2;5,8
Classification (cont.)	*Route family*: 30|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 4
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: +00---+00---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|4|0|-4
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 20
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (60,84,28); *K* Ω: (20,54,4)

Record 151 — r5c2_h+1v-1: continued on the next page

Record 151 — r5c2_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ* Ω: 19.606387746949443306362 81; *Action factor D*: 1.00000004117409894674 1575
Reader evaluations (cont.)	*Action factor* Ω: 1.00000001359275115188 7498
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2e1dadcad246fee281f87bf135901e695efb2a4a05df738f 6c07002407a0b1f2 *Reader*: 184e6040f83f204c85f176971d97ea9040bf8f9263d3f91b694e0a6d 3fdcf152 *Cell record at 108*: —

Route 152 — r5c2_h+1v+1

Cell 38: row 5, column 2. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 152; ordinal in the memory table: 149.

Table 94.160: Record 152: r5c2_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c2_h+1v+1; *Cell*: 38; *Row*: 5
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 152 — r5c2_h+1v+1: continued on the next page

Record 152 — r5c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 5,2;5,8
Classification (cont.)	*Route family*: 30|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|6|0|4|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Record 152 — r5c2_h+1v+1: continued on the next page

Record 152 — r5c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b277f51b0e0d2ae224e9987f04a99ed88a84fd081c5d0ef6662e6eb32b5cde7c *Reader*: 664bc72329b1dfa461d6b8711673127382f8d4cf08e87ae3a09df6274e4b9cb3 *Cell record at 108*: —

Route 153 — r5c3_h-1v-1

Cell 39: row 5, column 3. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 153; ordinal in the memory table: 156.

Table 94.161: Record 153: r5c3_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c3_h-1v-1; *Cell*: 39; *Row*: 5
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 5,3;5,7
Classification (cont.)	*Route family*: 28|6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 153 — r5c3_h-1v-1: continued on the next page

Record 153 — r5c3_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243 *Stable triads per block*: 0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 628aa1329a1d669f1d699ac59afb3d24a5b55af8129e6ebb54860dbb0a921a82 *Reader*: 8d0e0a7c9c083d59bd7752c12edada4283a1227f99644dd99fd5236203276d59 *Cell record at 108*: —

Route 154 — r5c3_h-1v+1

Cell 39: row 5, column 3. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 154; ordinal in the memory table: 155.

Table 94.162: Record 154: r5c3_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c3_h-1v+1; *Cell*: 39; *Row*: 5
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 5,3;5,7
Classification (cont.)	*Route family*: 28|6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 154 — r5c3_h-1v+1: continued on the next page

Record 154 — r5c3_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e1edda897425f41c7b0d4061df3db1cc7e64b9f28d3db5f3 0063789b0434a10d *Reader*: 3dc097d1af9e277259c79d8cf836e831c425072950eb258b5eadfec9 0704df5e *Cell record at 108*: f686fd28355504421f5166a67fe6521747f0e1dc54b72eb3 9e0a7a80b67ba3f9

Route 155 — r5c3_h+1v-1

Cell 39: row 5, column 3. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 155; ordinal in the memory table: 154.

Table 94.163: Record 155: r5c3_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c3_h+1v-1; *Cell*: 39; *Row*: 5
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton

Record 155 — r5c3_h+1v-1: continued on the next page

Record 155 — r5c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 5,3;5,7
Classification (cont.)	*Route family*: 28|6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Record 155 — r5c3_h+1v-1: continued on the next page

Record 155 — r5c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5ebf99eb3286cd3e668d1881ac5a6d25c3fed2315f47a970c3ffd75b4e986591 *Reader*: c5b7a860701e0f33931847efb2a992c31bef6f0d9da926e574e7ab3e68fb470f *Cell record at 108*: —

Route 156 — r5c3_h+1v+1

Cell 39: row 5, column 3. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 156; ordinal in the memory table: 153.

Table 94.164: Record 156: r5c3_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c3_h+1v+1; *Cell*: 39; *Row*: 5
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 5,3;5,7
Classification (cont.)	*Route family*: 28|6|2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17

Record 156 — r5c3_h+1v+1: continued on the next page

Record 156 — r5c3_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 22; *log2 ambiguity area*: 4.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 0e0d1f793f8fb5b455383130d2c9912e5b2700a0c2fb4ce1 bd1b83c54267422c *Reader*: 997312cb2b9d722895ddf8de8b30b9501c2d380ddb4849906bc9cd25 95449bc9 *Cell record at 108*: —

Route 157 — r5c4_h-1v-1

Cell 40: row 5, column 4. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 157; ordinal in the memory table: 160.

Table 94.165: Record 157: r5c4_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c4_h-1v-1; *Cell*: 40; *Row*: 5
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G1; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 1; *Color*: —; *Multisection*: neutrino_G1
Classification (cont.)	*Physical family*: neutrino; *Depth*: G1; *Inversion orbit*: 5,4;5,6
Classification (cont.)	*Route family*: 28|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 157 — r5c4_h-1v-1: continued on the next page

Record 157 — r5c4_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000068994430825936417
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 698e05b31e9a26229e96d506b2415dfbf00df56146f44d71 fcbea9b78701e56b *Reader*: 4920659c510a6cf90e8944ecb1a49dffb1bfe16ad3249fd0b5fb8a10 930a6547 *Cell record at 108*: —

Route 158 — r5c4_h-1v+1

Cell 40: row 5, column 4. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 158; ordinal in the memory table: 159.

Table 94.166: Record 158: r5c4_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c4_h-1v+1; *Cell*: 40; *Row*: 5
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G1; *Family*: neutrino; *Sector*: lepton

Record 158 — r5c4_h-1v+1: continued on the next page

Record 158 — r5c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 1; *Color*: —; *Multisection*: neutrino_G1
Classification (cont.)	*Physical family*: neutrino; *Depth*: G1; *Inversion orbit*: 5,4;5,6
Classification (cont.)	*Route family*: 28|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+-0-+0+-0-
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 40
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K* Ω: (40,54,4)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 39.6063877469494433063628 1; *Action factor D*: 1.00000002761000615951421 5
Reader evaluations (cont.)	*Action factor* Ω: 1.00000002745838669265140 0
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 158 — r5c4_h-1v+1: continued on the next page

Record 158 — r5c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a3dc2fcf597df676434ea90b519e5540b275620b60073073f9ecf51745655b0d *Reader*: 2bc38a4fc594251923f006d0b3ea2789df5cf2da0f7d89bab5c40e7dfa00c9bd *Cell record at 108*: —

Route 159 — r5c4_h+1v-1

Cell 40: row 5, column 4. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 159; ordinal in the memory table: 158.

Table 94.167: Record 159: r5c4_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c4_h+1v-1; *Cell*: 40; *Row*: 5
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G1; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 1; *Color*: —; *Multisection*: neutrino_G1
Classification (cont.)	*Physical family*: neutrino; *Depth*: G1; *Inversion orbit*: 5,4;5,6
Classification (cont.)	*Route family*: 28|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 159 — r5c4_h+1v-1: continued on the next page

Record 159 — r5c4_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+-0-+0+-0-
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 40
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K Ω*: (40,54,4)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 39.6063877469494433063628 1; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000027458386692651400
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8944ea77aea3c31eb47da1692c7b6e0a3c809102f8b9002ba05b291baa3233c1 *Reader*: 2d9f1de9bfa0686dcfd7cd7ac00fc0b912d37ab73f9fb467dccacfef55a9eddb *Cell record at 108*: 23031ced1c53e3421cc8604cb7d3e60373a144d81c02cf0c7307d379831633c3

Route 160 — r5c4_h+1v+1

Cell 40: row 5, column 4. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 160; ordinal in the memory table: 157.

Table 94.168: Record 160: r5c4_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c4_h+1v+1; *Cell*: 40; *Row*: 5
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G1; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 1; *Color*: —; *Multisection*: neutrino_G1
Classification (cont.)	*Physical family*: neutrino; *Depth*: G1; *Inversion orbit*: 5,4;5,6
Classification (cont.)	*Route family*: 28|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 160 — r5c4_h+1v+1: continued on the next page

Record 160 — r5c4_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000068994430825936417
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 79e4be3ad5ebc6753342e4284adf56b71d921b04cea8f219c842fa9f052e874f *Reader*: dc5c02bcdfaf96e12535891f9aa10bc7bd9659e01354a2a2d5c6e9dc6ea2e751 *Cell record at 108*: —

Route 161 — r5c5_h-1v-1

Cell 41: row 5, column 5. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 161; ordinal in the memory table: 164.

Table 94.169: Record 161: r5c5_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c5_h-1v-1; *Cell*: 41; *Row*: 5
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G0; *Family*: charged_lepton; *Sector*: lepton

Record 161 — r5c5_h-1v-1: continued on the next page

Record 161 — r5c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 0; *Color*: —; *Multisection*: charged_lepton_G0
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G0_center; *Inversion orbit*: 5,5
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.00000005518981253185914 0
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Record 161 — r5c5_h-1v-1: continued on the next page

Record 161 — r5c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a4e3cc2eab44f4bb51b119700b1b5caebf22edbd82634847 31a154bda1c77f6b *Reader*: 98b8a25724d3a874f2368cb2dc819bc6ce0d16adf27017ad5f139361 24662d5a *Cell record at 108*: —

Route 162 — r5c5_h-1v+1

Cell 41: row 5, column 5. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 162; ordinal in the memory table: 163.

Table 94.170: Record 162: r5c5_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c5_h-1v+1; *Cell*: 41; *Row*: 5
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G0; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 0; *Color*: —; *Multisection*: charged_lepton_G0
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G0_center; *Inversion orbit*: 5,5
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 162 — r5c5_h-1v+1: continued on the next page

Record 162 — r5c5_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (60,84,28); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.000000041174098946741575
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: dafa95692d1f9384c65e0afd29106c64a3e62d6f2d545d4a6b6b404ea1b2d70b *Reader*: 2bdd7331f202139ea68552f27c7eb9dcf445fdc5c70d95f9e3ec7c6558dbce1d *Cell record at 108*: —

Route 163 — r5c5_h+1v-1

Cell 41: row 5, column 5. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 163; ordinal in the memory table: 162.

Table 94.171: Record 163: r5c5_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c5_h+1v-1; *Cell*: 41; *Row*: 5
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G0; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 0; *Color*: —; *Multisection*: charged_lepton_G0
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G0_center; *Inversion orbit*: 5,5
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (60,84,28); *K* Ω: (48,54,5)

Record 163 — r5c5_h+1v-1: continued on the next page

Record 163 — r5c5_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ* Ω: 47.606498943372135446416 86; *Action factor D*: 1.00000004117409894674 1575
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003300471805324 3084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c6b14aedd3f88fa17e0478cc431b63deca8ee71bb8e6d79c4c811add87d03720 *Reader*: 6e152edf9b94425a404b2c13793f91e688091f69eff6453c8efda5d69fa8861c *Cell record at 108*: —

Route 164 — r5c5_h+1v+1

Cell 41: row 5, column 5. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 164; ordinal in the memory table: 161.

Table 94.172: Record 164: r5c5_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c5_h+1v+1; *Cell*: 41; *Row*: 5
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G0; *Family*: charged_lepton; *Sector*: lepton

Record 164 — r5c5_h+1v+1: continued on the next page

Record 164 — r5c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 0; *Color*: —; *Multisection*: charged_lepton_G0
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G0_center; *Inversion orbit*: 5,5
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Record 164 — r5c5_h+1v+1: continued on the next page

Record 164 — r5c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 57d24c5dc0a69cd8111933d8bc40cc260571c0ef4f8f3008f7197910ff07c7dc *Reader*: 514b4f56201d42c1181a4a14961f9abe4a328f7751e0d100a75bfb101ea22f75 *Cell record at 108*: 9bf67dd6633e13d03463463e6d5880978f1459c3ba4e9352c82c30f6f5db9d13

Route 165 — r5c6_h-1v-1

Cell 42: row 5, column 6. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 165; ordinal in the memory table: 168.

Table 94.173: Record 165: r5c6_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c6_h-1v-1; *Cell*: 42; *Row*: 5
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G1; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 1; *Color*: —; *Multisection*: neutrino_G1
Classification (cont.)	*Physical family*: neutrino; *Depth*: G1; *Inversion orbit*: 5,4;5,6
Classification (cont.)	*Route family*: 28|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 165 — r5c6_h-1v-1: continued on the next page

Record 165 — r5c6_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243 *Stable triads per block*: 0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a03908be0f96c38774c03f0b83c3acb96e9e98b083d11b55 5c03fb4c6e686faf *Reader*: a2e862996bf61364ed0e6b6b37873be95744ffabe1b4e86973b0917d 492ddcf0 *Cell record at 108*: —

Route 166 — r5c6_h-1v+1

Cell 42: row 5, column 6. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 166; ordinal in the memory table: 167.

Table 94.174: Record 166: r5c6_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c6_h-1v+1; *Cell*: 42; *Row*: 5
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G1; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 1; *Color*: —; *Multisection*: neutrino_G1
Classification (cont.)	*Physical family*: neutrino; *Depth*: G1; *Inversion orbit*: 5,4;5,6
Classification (cont.)	*Route family*: 28|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 166 — r5c6_h-1v+1: continued on the next page

Record 166 — r5c6_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 81927e69032d8dda6c89f7e0c51591bb74a0922c58dc63b9 6cc20c6508c171f8 *Reader*: b51c5e796ab9a08f438f09085e623965205e7f4be2ed57eaebc1afa1 6d1504ce *Cell record at 108*: 598cf2540b6a078e4972cb09c4a957ff723cc9c3e7d7797d e08ffcafb6666426

Route 167 — r5c6_h+1v-1

Cell 42: row 5, column 6. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 167; ordinal in the memory table: 166.

Table 94.175: Record 167: r5c6_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c6_h+1v-1; *Cell*: 42; *Row*: 5
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G1; *Family*: neutrino; *Sector*: lepton

Record 167 — r5c6_h+1v-1: continued on the next page

Record 167 — r5c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 1; *Color*: —; *Multisection*: neutrino_G1
Classification (cont.)	*Physical family*: neutrino; *Depth*: G1; *Inversion orbit*: 5,4;5,6
Classification (cont.)	*Route family*: 28|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Record 167 — r5c6_h+1v-1: continued on the next page

Record 167 — r5c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a75845a7c9ce451a84ec8de2149e4578724b29820b25582178837475acd97e29 *Reader*: 52f8b4627b04a1a925e56d9021b47cdbf4e172ca6df3500072c1a8cfdd00d329 *Cell record at 108*: —

Route 168 — r5c6_h+1v+1

Cell 42: row 5, column 6. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 168; ordinal in the memory table: 165.

Table 94.176: Record 168: r5c6_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c6_h+1v+1; *Cell*: 42; *Row*: 5
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G1; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 1; *Color*: —; *Multisection*: neutrino_G1
Classification (cont.)	*Physical family*: neutrino; *Depth*: G1; *Inversion orbit*: 5,4;5,6
Classification (cont.)	*Route family*: 28|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17

Record 168 — r5c6_h+1v+1: continued on the next page

Record 168 — r5c6_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 09134b2a60678acc4e4d7fe41328fa475073a90714988713 b330280c61906435 *Reader*: 3a010d9493ed73b62d23d3baf729585440be9fb9b19ad1cfa7ca16b2 970beb82 *Cell record at 108*: —

Route 169 — r5c7_h-1v-1

Cell 43: row 5, column 7. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 169; ordinal in the memory table: 172.

Table 94.177: Record 169: r5c7_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c7_h-1v-1; *Cell*: 43; *Row*: 5
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 5,3;5,7
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|4|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 169 — r5c7_h-1v-1: continued on the next page

Record 169 — r5c7_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000068994430825936417
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 22; *log2 ambiguity area*: 4.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: f0ba7ed797a68175311757b8030cdecd682fb29e4eed2cda f8cfd3832be16d4c *Reader*: 04477201ed2fa53846f5f328151d60bccff05c49e4724a3237f24a74 02a084ab *Cell record at 108*: —

Route 170 — r5c7_h-1v+1

Cell 43: row 5, column 7. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 170; ordinal in the memory table: 171.

Table 94.178: Record 170: r5c7_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c7_h-1v+1; *Cell*: 43; *Row*: 5
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton

Record 170 — r5c7_h-1v+1: continued on the next page

Record 170 — r5c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 5,3;5,7
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 50|-6|2|-2|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 55.6064989433721354464186; *Action factor D*: 1.00000002761000615951421 5
Reader evaluations (cont.)	*Action factor Ω*: 1.00000003855097235414164 5
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 170 — r5c7_h-1v+1: continued on the next page

Record 170 — r5c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b6376735f9acad08bb8f9c39f4ca3e69be62405d3627943796f664a8a9cec6d6 *Reader*: 54983701597aea6396a38cb59a14122950ea0616556c5847dc64e8bee3fec53f *Cell record at 108*: —

Route 171 — r5c7_h+1v-1

Cell 43: row 5, column 7. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 171; ordinal in the memory table: 170.

Table 94.179: Record 171: r5c7_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c7_h+1v-1; *Cell*: 43; *Row*: 5
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 5,3;5,7
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 171 — r5c7_h+1v-1: continued on the next page

Record 171 — r5c7_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 50|6|-2|2|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b34a9417a09638f6bc02ffe98471b0e2b627116ee13284a7ad74bf5074b0f4d2 *Reader*: b8df04a23c90b57616a1bf8334a7f7757313f38e95a5c4540aa36d3e166549dd *Cell record at 108*: —

Route 172 — r5c7_h+1v+1

Cell 43: row 5, column 7. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 172; ordinal in the memory table: 169.

Table 94.180: Record 172: r5c7_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c7_h+1v+1; *Cell*: 43; *Row*: 5
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 5,3;5,7
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-4|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 172 — r5c7_h+1v+1: continued on the next page

Record 172 — r5c7_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000068994430825936417
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 1a5415204e85bcd7ffabce87c95eff5fc47e613ceddd713d 8f05b93718293d41 *Reader*: 3f7c2ace630141906ef94299c92c3d34f74897919e7ec93e685fd0b6 210dd9f3 *Cell record at 108*: f764349cd2ba2f9d1ee7db9a6f0bd775d595c4480bce2698 061803100c3f18e5

Route 173 — r5c8_h-1v-1

Cell 44: row 5, column 8. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 173; ordinal in the memory table: 176.

Table 94.181: Record 173: r5c8_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c8_h-1v-1; *Cell*: 44; *Row*: 5
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 173 — r5c8_h-1v-1: continued on the next page

Record 173 — r5c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 5,2;5,8
Classification (cont.)	*Route family*: 30|-6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|6|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Record 173 — r5c8_h-1v-1: continued on the next page

Record 173 — r5c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 701966ab31f24a2e0ac479f747422728dbafc44021012aef5bc7700e331d65b8 *Reader*: 5d22412c39797827c404556ad608c217b66ea943a89b289f458610a70c151ad7 *Cell record at 108*: —

Route 174 — r5c8_h-1v+1

Cell 44: row 5, column 8. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 174; ordinal in the memory table: 175.

Table 94.182: Record 174: r5c8_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c8_h-1v+1; *Cell*: 44; *Row*: 5
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 5,2;5,8
Classification (cont.)	*Route family*: 30|-6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 174 — r5c8_h-1v+1: continued on the next page

Record 174 — r5c8_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|-2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,84,28); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041174098946741575
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a03329e40257d5ede551debc7365527a24878f84197e9298100b0b55af57184b *Reader*: b12e9f54798f62c19f474b63d68348823a829225e5a25e308905bb4e98e7ff7d *Cell record at 108*: e8f3e4639e854424d2658b469d5722453265869a56f8e553445bc0037bd9882f

Route 175 — r5c8_h+1v-1

Cell 44: row 5, column 8. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 175; ordinal in the memory table: 174.

Table 94.183: Record 175: r5c8_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c8_h+1v-1; *Cell*: 44; *Row*: 5
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 5,2;5,8
Classification (cont.)	*Route family*: 30|-6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,84,28); *K* Ω: (80,54,6)

Record 175 — r5c8_h+1v-1: continued on the next page

Record 175 — r5c8_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000004117409894674 1575
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185 9140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: fe0e3cc0a84e143289d0a34c2824d4fe4fddc9921c05852e b70f36277c4ab0b0 *Reader*: 41a91226e39febcefa36198283195e1151884a5438e330530b3217ad 9b811153 *Cell record at 108*: —

Route 176 — r5c8_h+1v+1

Cell 44: row 5, column 8. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 176; ordinal in the memory table: 173.

Table 94.184: Record 176: r5c8_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c8_h+1v+1; *Cell*: 44; *Row*: 5
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 176 — r5c8_h+1v+1: continued on the next page

Record 176 — r5c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 5,2;5,8
Classification (cont.)	*Route family*: 30|-6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Record 176 — r5c8_h+1v+1: continued on the next page

Record 176 — r5c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 9f17a032ddfb1ca89f4d34e1fc106e59e07a18fc1dbada65e5061f6b0974791e *Reader*: 5cc037f3f593ef27c0b04dd60612b27c3ee3fb217cc66ae495e7c6eb2c3f5800 *Cell record at 108*: —

Route 177 — r5c9_h-1v-1

Cell 45: row 5, column 9. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 177; ordinal in the memory table: 180.

Table 94.185: Record 177: r5c9_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c9_h-1v-1; *Cell*: 45; *Row*: 5
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 5,1;5,9
Classification (cont.)	*Route family*: 34|0|-12|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 177 — r5c9_h-1v-1: continued on the next page

Record 177 — r5c9_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,4); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 55.6064989433721354464186; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243 *Stable triads per block*: 0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d13b10758f72225bc8a3bb0f29e7117f1e568076f05c67d10e59f0ecd85a0d8d *Reader*: 3a627fb4a5c98662723a034a9f86de710c9cb3abc8f8d035e035b6e579396d47 *Cell record at 108*: 242498278d713b1359912fec0f87443b9461fd3dd582fd9889d8a0f5668c22be

Route 178 — r5c9_h-1v+1

Cell 45: row 5, column 9. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 178; ordinal in the memory table: 179.

Table 94.186: Record 178: r5c9_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c9_h-1v+1; *Cell*: 45; *Row*: 5
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 5,1;5,9
Classification (cont.)	*Route family*: 34|0|-12|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 178 — r5c9_h-1v+1: continued on the next page

Record 178 — r5c9_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 05d5c4bf566052d2f9fb3f06980ca8d3107fcf46ffefa36a 99432ea51df4ffaf *Reader*: fed584860b86d5c5a263e0095165b1b1f3766fe83937baad4fe826a7 98377b6b *Cell record at 108*: —

Route 179 — r5c9_h+1v-1

Cell 45: row 5, column 9. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 179; ordinal in the memory table: 178.

Table 94.187: Record 179: r5c9_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r5c9_h+1v-1; *Cell*: 45; *Row*: 5
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 179 — r5c9_h+1v-1: continued on the next page

Record 179 — r5c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 5,1;5,9
Classification (cont.)	*Route family*: 34|0|-12|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|-2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Record 179 — r5c9_h+1v-1: continued on the next page

Record 179 — r5c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: faa1ebd578b981daa2f34f8d42e7ddcf87be1072ab7569891f0566f3aa0fcc75 *Reader*: 59f7718146a7eed58f86cef869bc7e00ac61585cb84db3ca040a23ab561ec874 *Cell record at 108*: —

Route 180 — r5c9_h+1v+1

Cell 45: row 5, column 9. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 180; ordinal in the memory table: 177.

Table 94.188: Record 180: r5c9_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r5c9_h+1v+1; *Cell*: 45; *Row*: 5
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 5,1;5,9
Classification (cont.)	*Route family*: 34|0|-12|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17

Record 180 — r5c9_h+1v+1: continued on the next page

Record 180 — r5c9_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,4); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 30; *log2 ambiguity area*: 8.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a2caaa1885e5e1ec3a77b1b1476830d4cdea1d83ef5de1043297f20d5b040bae *Reader*: 33b2af69d9d7267f506c5047bd415adc3eb103237644181ff6e49fcadb855252 *Cell record at 108*: —

Route 181 — r6c1_h-1v-1

Cell 46: row 6, column 1. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 181; ordinal in the memory table: 184.

Table 94.189: Record 181: r6c1_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c1_h-1v-1; *Cell*: 46; *Row*: 6
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,9;6,1
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 181 — r6c1_h-1v-1: continued on the next page

Record 181 — r6c1_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5723836945af498aee1f7524fa9f3ca625afae9a8ad27788 75e0f116ddf54bd0 *Reader*: 26c999154dc98e67ef1eb0cbe3c4af5dc49cc63ff9acd7220b474c1d 35deb661 *Cell record at 108*: e29c6e53cf987fd4cd0791723fa331bd6c54a8c0e6be1441 99dae0d5aa759d38

Route 182 — r6c1_h-1v+1

Cell 46: row 6, column 1. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 182; ordinal in the memory table: 183.

Table 94.190: Record 182: r6c1_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c1_h-1v+1; *Cell*: 46; *Row*: 6
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark

Record 182 — r6c1_h-1v+1: continued on the next page

Record 182 — r6c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,9;6,1
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Record 182 — r6c1_h-1v+1: continued on the next page

Record 182 — r6c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e39ec08bbe801ae536649422bf79b77a37f016ba9a4bf092521046f28fab1bc6 *Reader*: 68b45d01d606e5eaf1737c7f7d9c0e216111306976f593bf8cb7006a1f0989a7 *Cell record at 108*: —

Route 183 — r6c1_h+1v-1

Cell 46: row 6, column 1. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 183; ordinal in the memory table: 182.

Table 94.191: Record 183: r6c1_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c1_h+1v-1; *Cell*: 46; *Row*: 6
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,9;6,1
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 3; *Final stable triads*: 16

Record 183 — r6c1_h+1v-1: continued on the next page

Record 183 — r6c1_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2262a4ead8d34652790e70bcf39dabf4c6da0042f349313dcd38adb241f7266b *Reader*: e348c32836c7d0ef2ec19bed6383e9257ee192429d72bf57548f615529d8ae53 *Cell record at 108*: —

Route 184 — r6c1_h+1v+1

Cell 46: row 6, column 1. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 184; ordinal in the memory table: 181.

Table 94.192: Record 184: r6c1_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c1_h+1v+1; *Cell*: 46; *Row*: 6
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,9;6,1
Classification (cont.)	*Route family*: 34|0|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 184 — r6c1_h+1v+1: continued on the next page

Record 184 — r6c1_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	κ *D*: 79.51870831969534555434134; κ Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7c76e8e8f5478ad536bb1cb8f675ca9aeceb07541dc9051b6af6607d9276aa67 *Reader*: 86da2f8ba7211912d067c9008753f8e115b2fbcc6eca36f42bee9c89352fd976 *Cell record at 108*: —

Route 185 — r6c2_h-1v-1

Cell 47: row 6, column 2. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 185; ordinal in the memory table: 188.

Table 94.193: Record 185: r6c2_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c2_h-1v-1; *Cell*: 47; *Row*: 6
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 185 — r6c2_h-1v-1: continued on the next page

Record 185 — r6c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,8;6,2
Classification (cont.)	*Route family*: 34|0|0|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 47.6064989433721354464186; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Record 185 — r6c2_h-1v-1: continued on the next page

Record 185 — r6c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7b821c471bf4e01e7dcfca1341b325491b18cf5adbe433f3054bb10ad081a842 *Reader*: a37372d4b23fd8937236a0d2357362801e9e90140f10eb0752028f0835fa0d80 *Cell record at 108*: —

Route 186 — r6c2_h-1v+1

Cell 47: row 6, column 2. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 186; ordinal in the memory table: 187.

Table 94.194: Record 186: r6c2_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c2_h-1v+1; *Cell*: 47; *Row*: 6
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,8;6,2
Classification (cont.)	*Route family*: 34|0|0|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 186 — r6c2_h-1v+1: continued on the next page

Record 186 — r6c2_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 77bf36177edee2e331d458ba93651e87788985b6fbdfe5b735256fadb6b05411 *Reader*: 3be69c77c702f3b17d2f920c2531cf2d4e4c212e395a71905205f20c05c4892e *Cell record at 108*: 9485907bc9b83fa7aabbe744a6a88d1d3223272ea89792d7558c057884f8e9f0

Route 187 — r6c2_h+1v-1

Cell 47: row 6, column 2. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 187; ordinal in the memory table: 186.

Table 94.195: Record 187: r6c2_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c2_h+1v-1; *Cell*: 47; *Row*: 6
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,8;6,2
Classification (cont.)	*Route family*: 34|0|0|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 187 — r6c2_h+1v-1: continued on the next page

Record 187 — r6c2_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a039b8fda9943425c307ca97bc18ad161e61a4a12c0516fe 66793b765e700217 *Reader*: cd0552218d548e6a3dd4332197268fca2c06a5b81c36d69aaae3d2c7 17d9f0a2 *Cell record at 108*: —

Route 188 — r6c2_h+1v+1

Cell 47: row 6, column 2. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 188; ordinal in the memory table: 185.

Table 94.196: Record 188: r6c2_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c2_h+1v+1; *Cell*: 47; *Row*: 6
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 188 — r6c2_h+1v+1: continued on the next page

Record 188 — r6c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,8;6,2
Classification (cont.)	*Route family*: 34|0|0|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 47.6064989433721354464186; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 188 — r6c2_h+1v+1: continued on the next page

Record 188 — r6c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8615c240bf77cdf6ceadf5b08213f6265f4ca61423b297ef17ac1b43dddf3c99 *Reader*: bd032ab996bb19de9c1074953658e061c757ebca4281f4b089db87ed939370c9 *Cell record at 108*: —

Route 189 — r6c3_h-1v-1

Cell 48: row 6, column 3. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 189; ordinal in the memory table: 192.

Table 94.197: Record 189: r6c3_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c3_h-1v-1; *Cell*: 48; *Row*: 6
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: R; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,7;6,3
Classification (cont.)	*Route family*: 30|6|0|2|-8|+++0--+++0--
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 189 — r6c3_h-1v-1: continued on the next page

Record 189 — r6c3_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|2|-2|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 55.6064989433721354464186; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: cd011cd5866d118889051f57ac2a0d2b5ca6c7abbc4b7d733a0cc35515322e9b *Reader*: 91deae21588e4bbf1be6e39288698ebdcfcb2cde1320d3f0ef090293f311c428 *Cell record at 108*: —

Route 190 — r6c3_h-1v+1

Cell 48: row 6, column 3. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 190; ordinal in the memory table: 191.

Table 94.198: Record 190: r6c3_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c3_h-1v+1; *Cell*: 48; *Row*: 6
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: R; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,7;6,3
Classification (cont.)	*Route family*: 30|6|0|2|-8|+++0--+++0--
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: 0; *τ 12*: 00+-0000+-00
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|0
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 4
Differences and counters (cont.)	*P6*: 2
Integer readers	*K D*: (0,24,0); *K Ω*: (4,54,2)

Record 190 — r6c3_h-1v+1: continued on the next page

Record 190 — r6c3_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 3.6061653541040590262547 07; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000002500088676796314
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7b28dd5daca27177a2ca44f0b080b42407a6023d8a10ce5d6e034058ddeadc35 *Reader*: ccb618b1bf7c78519fb2a0fb4e8ce96be3fd471eb2a4f12cbfb2237fdf4a50c4 *Cell record at 108*: 59349e8798fb69e2e5ae6c2cf1ca8ec3053390b207b9dc18e92a5351bba14dc5

Route 191 — r6c3_h+1v-1

Cell 48: row 6, column 3. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 191; ordinal in the memory table: 190.

Table 94.199: Record 191: r6c3_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c3_h+1v-1; *Cell*: 48; *Row*: 6
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark

Record 191 — r6c3_h+1v-1: continued on the next page

Record 191 — r6c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: R; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,7;6,3
Classification (cont.)	*Route family*: 30|6|0|2|-8|+++0--+++0--
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: 0; *τ 12*: +0000-+0000-
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|0
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; Ω *90/120*: 4
Differences and counters (cont.)	*P6*: 2
Integer readers	*K D*: (0,24,0); *K* Ω: (4,54,2)
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 3.6061653541040590262547 07; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000002500088676796314
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 191 — r6c3_h+1v-1: continued on the next page

Record 191 — r6c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 41e436cbeb2b64cfecd0594ddb877df5834695c8c3b033742d8bdd9e04c2774c *Reader*: 766c9065f7080b17e85ed5f8029fb902f2ab1b563deba91f4bc3ecd230c44c7e *Cell record at 108*: —

Route 192 — r6c3_h+1v+1

Cell 48: row 6, column 3. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 192; ordinal in the memory table: 189.

Table 94.200: Record 192: r6c3_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c3_h+1v+1; *Cell*: 48; *Row*: 6
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: R; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,7;6,3
Classification (cont.)	*Route family*: 30|6|0|2|-8|+++0--+++0--
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 192 — r6c3_h+1v+1: continued on the next page

Record 192 — r6c3_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 30|6|-2|2|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2b62e065285f7e9dff5a3ecd8ef42eb73412a1ae0bd6ecfaa0f70240b91fb5e9 *Reader*: 102c4214f537ef17f9428490ac06f432b8c2bb4e74669c284d556fbef2f3c983 *Cell record at 108*: —

Route 193 — r6c4_h-1v-1

Cell 49: row 6, column 4. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 193; ordinal in the memory table: 196.

Table 94.201: Record 193: r6c4_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c4_h-1v-1; *Cell*: 49; *Row*: 6
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: B; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,6;6,4
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 193 — r6c4_h-1v-1: continued on the next page

Record 193 — r6c4_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 978a72f4858eb2d567ab4b95f6eebbde0c0eff2fc96c5e03 55718f6086eb7ebd *Reader*: 4bf0490cb60c31703dd122be074181958c4e70fb96f4d3f4ce35715f 4e54679a *Cell record at 108*: 09534891acc2eaf69f6c6965a3dcbf124972602e60fbf8fa 02e0909e20d5411b

Route 194 — r6c4_h-1v+1

Cell 49: row 6, column 4. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 194; ordinal in the memory table: 195.

Table 94.202: Record 194: r6c4_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c4_h-1v+1; *Cell*: 49; *Row*: 6
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark

Record 194 — r6c4_h-1v+1: continued on the next page

Record 194 — r6c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 1; *Color*: B; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,6;6,4
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 55.6064989433721354464 16 86; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Record 194 — r6c4_h-1v+1: continued on the next page

Record 194 — r6c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: dcdcf9eb3b8abc3fea552a6845f15c06b3f51f76b9bd638f95c451fd3cf68655 *Reader*: 52d4bbb40854b103ab52b35ea76890da19bf740bb9db8f784ae8f256fc3c4d04 *Cell record at 108*: —

Route 195 — r6c4_h+1v-1

Cell 49: row 6, column 4. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 195; ordinal in the memory table: 194.

Table 94.203: Record 195: r6c4_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c4_h+1v-1; *Cell*: 49; *Row*: 6
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: B; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,6;6,4
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 3; *Final stable triads*: 16

Record 195 — r6c4_h+1v-1: continued on the next page

Record 195 — r6c4_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 78f3ffaa921e46c61e0a5e8f0098d060fca37d87139b6541dce6069a8bd8a27f *Reader*: 32872bf64adafd685ad864babb2ca0d2e7551b61ccfecec145ac607cffe25d0d *Cell record at 108*: —

Route 196 — r6c4_h+1v+1

Cell 49: row 6, column 4. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 196; ordinal in the memory table: 193.

Table 94.204: Record 196: r6c4_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c4_h+1v+1; *Cell*: 49; *Row*: 6
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: B; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,6;6,4
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 196 — r6c4_h+1v+1: continued on the next page

Record 196 — r6c4_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 77cf5c939793f9ffefd542cdfd63c67547932703378b7838 2725fd396edf7833 *Reader*: 28764df7e51b08e0cf0a4e1231218c25be1d6f46c66294bcd19eed1f 6b285cce *Cell record at 108*: —

Route 197 — r6c5_h-1v-1

Cell 50: row 6, column 5. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 197; ordinal in the memory table: 200.

Table 94.205: Record 197: r6c5_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c5_h-1v-1; *Cell*: 50; *Row*: 6
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G1; *Family*: quark_down; *Sector*: quark

Record 197 — r6c5_h-1v-1: continued on the next page

Record 197 — r6c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 1; *Color*: G; *Multisection*: quark_down_G1
Classification (cont.)	*Physical family*: quark_down; *Depth*: G1; *Inversion orbit*: 4,5;6,5
Classification (cont.)	*Route family*: 28|-6|4|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 28|6|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464186; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Record 197 — r6c5_h-1v-1: continued on the next page

Record 197 — r6c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b6badb1f3717c85f8eab31262b0fde338dbc49197b8eab9820df39cfd1d22827 *Reader*: 17bd49a7de09fc63fbf85730b09b8c69da5258cd3a24e4c11552fad8eb796267 *Cell record at 108*: —

Route 198 — r6c5_h-1v+1

Cell 50: row 6, column 5. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 198; ordinal in the memory table: 199.

Table 94.206: Record 198: r6c5_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c5_h-1v+1; *Cell*: 50; *Row*: 6
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G1; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: G; *Multisection*: quark_down_G1
Classification (cont.)	*Physical family*: quark_down; *Depth*: G1; *Inversion orbit*: 4,5;6,5
Classification (cont.)	*Route family*: 28|-6|4|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 198 — r6c5_h-1v+1: continued on the next page

Record 198 — r6c5_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7493844ee9e55ec3fa4463553c671c3bfe04eb94006e369b09aa1c2fe3fd8152 *Reader*: d6f20c0a7858c0752246770ffbfd66432cef12d31a555564cd4c76c2195800ee *Cell record at 108*: 247db8a58d592a8cdd0ba517321e06c1f88eff82416294586fe44cb507cd1737

Route 199 — r6c5_h+1v-1

Cell 50: row 6, column 5. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 199; ordinal in the memory table: 198.

Table 94.207: Record 199: r6c5_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c5_h+1v-1; *Cell*: 50; *Row*: 6
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G1; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: G; *Multisection*: quark_down_G1
Classification (cont.)	*Physical family*: quark_down; *Depth*: G1; *Inversion orbit*: 4,5;6,5
Classification (cont.)	*Route family*: 28|-6|4|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 199 — r6c5_h+1v-1: continued on the next page

Record 199 — r6c5_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 0b8c5e26fd15e5487404c126325089f3744921e3c70229aa7c4efd9598c80084 *Reader*: bb50d7c6b121d5b9b27c2a8df96ae10f4164b1fc3d2ae8687629f5ba15b9d059 *Cell record at 108*: —

Route 200 — r6c5_h+1v+1

Cell 50: row 6, column 5. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 200; ordinal in the memory table: 197.

Table 94.208: Record 200: r6c5_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c5_h+1v+1; *Cell*: 50; *Row*: 6
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G1; *Family*: quark_down; *Sector*: quark

Record 200 — r6c5_h+1v+1: continued on the next page

Record 200 — r6c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 1; *Color*: G; *Multisection*: quark_down_G1
Classification (cont.)	*Physical family*: quark_down; *Depth*: G1; *Inversion orbit*: 4,5;6,5
Classification (cont.)	*Route family*: 28|-6|4|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464186; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 200 — r6c5_h+1v+1: continued on the next page

Record 200 — r6c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 459225a8c62810d92ebb16df6c2841855fdcc7973f5669549e10bdb7f48ba668 *Reader*: 3207260be401417eccf8d9f756c5e76d13ba51eeefe672feec28d8ab04ef1ed5 *Cell record at 108*: —

Route 201 — r6c6_h-1v-1

Cell 51: row 6, column 6. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 201; ordinal in the memory table: 204.

Table 94.209: Record 201: r6c6_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c6_h-1v-1; *Cell*: 51; *Row*: 6
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: R; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,4;6,6
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 201 — r6c6_h-1v-1: continued on the next page

Record 201 — r6c6_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,54,6); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 79.6066101397948275864791; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 856f2c4ecede7c9fad89891b9a493ce336be2d4b5153ce9ecee9540fcd10311c *Reader*: a8e2d94175cce86b4b4ebf52a6722aaa5895cc5b77acfab8bfc923d376d84f57 *Cell record at 108*: —

Route 202 — r6c6_h-1v+1

Cell 51: row 6, column 6. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 202; ordinal in the memory table: 203.

Table 94.210: Record 202: r6c6_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c6_h-1v+1; *Cell*: 51; *Row*: 6
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: R; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,4;6,6
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 50|-6|2|0|-8
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (0,24,0); *K Ω*: (48,54,5)

Record 202 — r6c6_h-1v+1: continued on the next page

Record 202 — r6c6_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 362984c5834759f48f4db727a5e95cba6a614bb5b737be4ba5aa7e0092099305 *Reader*: 54b6507550fde869f152d24026fee643979e44fefccb5a54fd5374cede0c4178 *Cell record at 108*: —

Route 203 — r6c6_h+1v-1

Cell 51: row 6, column 6. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 203; ordinal in the memory table: 202.

Table 94.211: Record 203: r6c6_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c6_h+1v-1; *Cell*: 51; *Row*: 6
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: R; *Multisection*: quark_up_G1

Record 203 — r6c6_h+1v-1: continued on the next page

Record 203 — r6c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,4;6,6
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 50|6|-2|0|-8
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (0,24,0); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: -0.17513646166281122393484 25; *κ* Ω: 47.606498943372135446416 86; *Action factor D*: 0.99999999987858108513378 81
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ

Record 203 — r6c6_h+1v-1: continued on the next page

Record 203 — r6c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ed910a023c24aa3e30b58383d60bafc5d58eb8a1f08e0253b59d1c804d53f35f *Reader*: 1c9ca174be7648f8a2b7e60e181a19b3b34f7c4e0d98e91bac5a4cbacc8c9a20 *Cell record at 108*: —

Route 204 — r6c6_h+1v+1

Cell 51: row 6, column 6. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 204; ordinal in the memory table: 201.

Table 94.212: Record 204: r6c6_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c6_h+1v+1; *Cell*: 51; *Row*: 6
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G1; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 1; *Color*: R; *Multisection*: quark_up_G1
Classification (cont.)	*Physical family*: quark_up; *Depth*: G1; *Inversion orbit*: 4,4;6,6
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0

Record 204 — r6c6_h+1v+1: continued on the next page

Record 204 — r6c6_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ea448f4eea6376a58f482d676495495f0c7b3c6bb76ccae4 0998ce2547269821 *Reader*: 7d817835fbd82223ed54da24836deff3fe6b2f2a2c237caefadfa287 194b6ac4 *Cell record at 108*: 1204da29afa516cae33ccda64944005b00ffe568f4826a58 5823f1bb2e720ae3

Route 205 — r6c7_h-1v-1

Cell 52: row 6, column 7. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 205; ordinal in the memory table: 208.

Table 94.213: Record 205: r6c7_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c7_h-1v-1; *Cell*: 52; *Row*: 6
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: B; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,3;6,7
Classification (cont.)	*Route family*: 28|6|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 205 — r6c7_h-1v-1: continued on the next page

Record 205 — r6c7_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 88f473d2cdc82fd9a950b8951b6621d785eb978f2df99dbc 0cf48b60503f0407 *Reader*: 0f39551e760683333bc943e7a463b426ae514492913154ad8c8a50a5 79ac0f4b *Cell record at 108*: —

Route 206 — r6c7_h-1v+1

Cell 52: row 6, column 7. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 206; ordinal in the memory table: 207.

Table 94.214: Record 206: r6c7_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c7_h-1v+1; *Cell*: 52; *Row*: 6
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: B; *Multisection*: quark_down_G2

Record 206 — r6c7_h-1v+1: continued on the next page

Record 206 — r6c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,3;6,7
Classification (cont.)	*Route family*: 28|6|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|-2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ

Record 206 — r6c7_h-1v+1: continued on the next page

Record 206 — r6c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 616fe1c153e8405fb9fdd203ddc30c405976b9f53a6b887a78eb2deb09ff3019 *Reader*: 0f506299138488a5bbb95a41ea186f877f691de62c43341f65029d52a295d822 *Cell record at 108*: e35a87178dee05ba4b834fccd04d9151723185209b3bf306de515f71c318bc40

Route 207 — r6c7_h+1v-1

Cell 52: row 6, column 7. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 207; ordinal in the memory table: 206.

Table 94.215: Record 207: r6c7_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c7_h+1v-1; *Cell*: 52; *Row*: 6
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: B; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,3;6,7
Classification (cont.)	*Route family*: 28|6|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 3; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: 2; *raw κ 0*: 0

Record 207 — r6c7_h+1v-1: continued on the next page

Record 207 — r6c7_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 4ff06f4f62859461a846f4d3090bec6874de8d404dc28e4c85c0039524ebce3f *Reader*: 82db73e13fb33b93832d00a5d3ffae2a3ce893d4059d410f530239bd20944a08 *Cell record at 108*: —

Route 208 — r6c7_h+1v+1

Cell 52: row 6, column 7. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 208; ordinal in the memory table: 205.

Table 94.216: Record 208: r6c7_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c7_h+1v+1; *Cell*: 52; *Row*: 6
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G2; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 2; *Color*: B; *Multisection*: quark_down_G2
Classification (cont.)	*Physical family*: quark_down; *Depth*: G2; *Inversion orbit*: 4,3;6,7
Classification (cont.)	*Route family*: 28|6|-8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 208 — r6c7_h+1v+1: continued on the next page

Record 208 — r6c7_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 44eb46a7747c4aef1e300e344b95cc35662ea5f5e7be767c3e9e1d8d753058fc *Reader*: d37dee388ad27b72a09d85bd19587c613c7ccaf6048eeba353d31d5f1b642289 *Cell record at 108*: —

Route 209 — r6c8_h-1v-1

Cell 53: row 6, column 8. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 209; ordinal in the memory table: 212.

Table 94.217: Record 209: r6c8_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c8_h-1v-1; *Cell*: 53; *Row*: 6
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 209 — r6c8_h-1v-1: continued on the next page

Record 209 — r6c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,2;6,8
Classification (cont.)	*Route family*: 28|-6|-2|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 28|6|-2|2|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Record 209 — r6c8_h-1v-1: continued on the next page

Record 209 — r6c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 46a7097d2858df9176ba9586b269471dd295cb011c04f97ee1f4da54cdd7a43e *Reader*: 1a456767b18a5ec986b3b3fb3c3856267b3045e9d69a809c24d95845977eb71f *Cell record at 108*: —

Route 210 — r6c8_h-1v+1

Cell 53: row 6, column 8. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 210; ordinal in the memory table: 211.

Table 94.218: Record 210: r6c8_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c8_h-1v+1; *Cell*: 53; *Row*: 6
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,2;6,8
Classification (cont.)	*Route family*: 28|-6|-2|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 210 — r6c8_h-1v+1: continued on the next page

Record 210 — r6c8_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c662417af4f02cbb24da0df8ddc65daf21c785dcda7367f4 99a615d5f1c13acc *Reader*: 0bf138f401402e507cb05c5292252e90d8a251263e47152404d52b83 1e9fce30 *Cell record at 108*: —

Route 211 — r6c8_h+1v-1

Cell 53: row 6, column 8. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 211; ordinal in the memory table: 210.

Table 94.219: Record 211: r6c8_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c8_h+1v-1; *Cell*: 53; *Row*: 6
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,2;6,8
Classification (cont.)	*Route family*: 28|-6|-2|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|-2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 211 — r6c8_h+1v-1: continued on the next page

Record 211 — r6c8_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6244f8a5a54cd67bbef735c4354e23b07ca1ae58e21214f4 a6ff3516629e19c1 *Reader*: 65c53fc392fa3dbf000bcb9c9539167b8ce081300211ef00c2eee82e 0d9ff65c *Cell record at 108*: —

Route 212 — r6c8_h+1v+1

Cell 53: row 6, column 8. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 212; ordinal in the memory table: 209.

Table 94.220: Record 212: r6c8_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c8_h+1v+1; *Cell*: 53; *Row*: 6
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 212 — r6c8_h+1v+1: continued on the next page

Record 212 — r6c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 4,2;6,8
Classification (cont.)	*Route family*: 28|-6|-2|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|2|-2|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464 16 86; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 212 — r6c8_h+1v+1: continued on the next page

Record 212 — r6c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 4f8136461cebd5d0e1aae0b94a1310952fec8ed32458cf20ae000c9ad96abc39 *Reader*: 478ca35df2cf0d4564142043734baaef374d7159a3c7775cb731a590a615a0db *Cell record at 108*: ec0c7b1546af4e293324a0e6fc89aa4cd160e3c70eacc9a9b3494baf329435e3

Route 213 — r6c9_h-1v-1

Cell 54: row 6, column 9. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 213; ordinal in the memory table: 216.

Table 94.221: Record 213: r6c9_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c9_h-1v-1; *Cell*: 54; *Row*: 6
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: R; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,1;6,9
Classification (cont.)	*Route family*: 30|-6|0|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 213 — r6c9_h-1v-1: continued on the next page

Record 213 — r6c9_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 30|6|-2|0|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 26; *log2 ambiguity area*: 6.339850002884624; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d29410c367743e1611b951c179305418a0c0787034e642488d40b8421a4f1b20 *Reader*: 05a266cce4affce309ae74ef598dad5069603262543c2e64d874382a83755978 *Cell record at 108*: —

Route 214 — r6c9_h-1v+1

Cell 54: row 6, column 9. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 214; ordinal in the memory table: 215.

Table 94.222: Record 214: r6c9_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c9_h-1v+1; *Cell*: 54; *Row*: 6
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: R; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,1;6,9
Classification (cont.)	*Route family*: 30|-6|0|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (0,24,0); *K Ω*: (80,54,6)

Record 214 — r6c9_h-1v+1: continued on the next page

Record 214 — r6c9_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 383fc3c8eab5e689c371ebbb9e5a07b1b4c5c0c869c53d1e34f80c0966fe4683 *Reader*: eca848d3437105f4e3992a55c435b0ebb5e5b25c7918788fba77fb97fa69eb3e *Cell record at 108*: —

Route 215 — r6c9_h+1v-1

Cell 54: row 6, column 9. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 215; ordinal in the memory table: 214.

Table 94.223: Record 215: r6c9_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r6c9_h+1v-1; *Cell*: 54; *Row*: 6
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: R; *Multisection*: quark_down_G4

Record 215 — r6c9_h+1v-1: continued on the next page

Record 215 — r6c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,1;6,9
Classification (cont.)	*Route family*: 30|-6|0|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (0,24,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ

Record 215 — r6c9_h+1v-1: continued on the next page

Record 215 — r6c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5f4e2b04f8b80b45dc0a27e968364702c50f95e4243d4268177f10928526aee6 *Reader*: cfa3576154e8cf220eeb6ace24ec313ec4d442e4d1da02a4bf007f30bd2c6621 *Cell record at 108*: 80250ff7621218a03a2e2ca39a7a6e60556d81f62485cf81b5b4de276a81d550

Route 216 — r6c9_h+1v+1

Cell 54: row 6, column 9. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 216; ordinal in the memory table: 213.

Table 94.224: Record 216: r6c9_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r6c9_h+1v+1; *Cell*: 54; *Row*: 6
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: R; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 4,1;6,9
Classification (cont.)	*Route family*: 30|-6|0|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 2

Record 216 — r6c9_h+1v+1: continued on the next page

Record 216 — r6c9_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|2|0|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ* Ω: 47.606498943372135446416 86; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 79b913bc6b44d4d200a69fd92eecc9936a66c4d289739af388dbe597c8ed9751 *Reader*: 32027225b3900396db67d96992263b16a9b0766eb91a7c939ac99ad9a97d3bf9 *Cell record at 108*: —

Route 217 — r7c1_h-1v-1

Cell 55: row 7, column 1. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 217; ordinal in the memory table: 220.

Table 94.225: Record 217: r7c1_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c1_h-1v-1; *Cell*: 55; *Row*: 7
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,9;7,1
Classification (cont.)	*Route family*: 30|-6|-8|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 30|6|-2|0|-8
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,54,7); *K* Ω: (48,54,5)

Record 217 — r7c1_h-1v-1: continued on the next page

Record 217 — r7c1_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 1.00000005518988962231533 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003300471805324308 4
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 213e50266085d4cf0de7fe5583f14cd6e9fd4d8201a2fca7bbb3795d68988de6 *Reader*: bcca332d749bd269e02e92f330430569c4525cd5e249720db77549f966a52809 *Cell record at 108*: —

Route 218 — r7c1_h-1v+1

Cell 55: row 7, column 1. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 218; ordinal in the memory table: 219.

Table 94.226: Record 218: r7c1_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c1_h-1v+1; *Cell*: 55; *Row*: 7
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 218 — r7c1_h-1v+1: continued on the next page

Record 218 — r7c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,9;7,1
Classification (cont.)	*Route family*: 30|-6|-8|0|-8|+0+---+0+---
Returns	*Class and phase*: 9; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 218 — r7c1_h-1v+1: continued on the next page

Record 218 — r7c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d62df3d96add1bcb3616655b4c5ab36dd6a959ad7680ed61f0bf567035e46c10 *Reader*: 04f3932501d1ad67facab5d82019498af3313df6e346a07adcb84e7c755faee0 *Cell record at 108*: c10b18e3f00c1efeab677c684d1a7bd005d2ee4a89b23fa3bc838923718a08fa

Route 219 — r7c1_h+1v-1

Cell 55: row 7, column 1. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 219; ordinal in the memory table: 218.

Table 94.227: Record 219: r7c1_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c1_h+1v-1; *Cell*: 55; *Row*: 7
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,9;7,1
Classification (cont.)	*Route family*: 30|-6|-8|0|-8|+0+---+0+---
Returns	*Class and phase*: 18; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 219 — r7c1_h+1v-1: continued on the next page

Record 219 — r7c1_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ea1bc7bb3c188b9a7e9dd91a90b29f4b3fe86da770ec0c1a40183a53264492b5 *Reader*: ad40326e62bbf34eabd692806fb96c19a8fd8520056d702c51a3c115dc8ce1fb *Cell record at 108*: —

Route 220 — r7c1_h+1v+1

Cell 55: row 7, column 1. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 220; ordinal in the memory table: 217.

Table 94.228: Record 220: r7c1_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c1_h+1v+1; *Cell*: 55; *Row*: 7
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,9;7,1
Classification (cont.)	*Route family*: 30|-6|-8|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|2|0|-8
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,54,7); *K* Ω: (48,54,5)

Record 220 — r7c1_h+1v+1: continued on the next page

Record 220 — r7c1_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 1.00000005518988962231533 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003300471805324308 4
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3a495db2e5a0a6707f8a97f724743b33c5696f6ea59a145f59858765279c3c3b *Reader*: 22a43ddff0542b8baedc3fefb41c9e088a990090a8a95fc8d101d973f3a8fed9 *Cell record at 108*: —

Route 221 — r7c2_h-1v-1

Cell 56: row 7, column 2. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 221; ordinal in the memory table: 224.

Table 94.229: Record 221: r7c2_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c2_h-1v-1; *Cell*: 56; *Row*: 7
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 221 — r7c2_h-1v-1: continued on the next page

Record 221 — r7c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,8;7,2
Classification (cont.)	*Route family*: 34|0|-4|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,0); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Record 221 — r7c2_h-1v-1: continued on the next page

Record 221 — r7c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: de97a58186c1b724ae969ef3d0b46d0b58439e4a07374877722147645eed901a *Reader*: b5cbdb2acc289475e1adbffda63c9cb08aa1f829e2f904b2bb353276a27933bf *Cell record at 108*: —

Route 222 — r7c2_h-1v+1

Cell 56: row 7, column 2. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 222; ordinal in the memory table: 223.

Table 94.230: Record 222: r7c2_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c2_h-1v+1; *Cell*: 56; *Row*: 7
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,8;7,2
Classification (cont.)	*Route family*: 34|0|-4|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 222 — r7c2_h-1v+1: continued on the next page

Record 222 — r7c2_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: 0++--00++--0
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 36
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K Ω*: (36,54,4)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 35.6063877469494433063628 1; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000024685259569118152
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d0d3560438d493017163b00e45d3935896eb8554e0dddda5b8a0856cc7a13b34 *Reader*: 1de0ff8e3d24aca90f7f4d432eafa71c5d5aab1bc65d749f8cd1dc602a5a7a65 *Cell record at 108*: c97a50bb057d87c6a9915463f589846d13ce5d25a64e07f2625114cb93e204f5

Route 223 — r7c2_h+1v-1

Cell 56: row 7, column 2. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 223; ordinal in the memory table: 222.

Table 94.231: Record 223: r7c2_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c2_h+1v-1; *Cell*: 56; *Row*: 7
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,8;7,2
Classification (cont.)	*Route family*: 34|0|-4|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: ++00--++00--
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 36
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K* Ω: (36,54,4)

Record 223 — r7c2_h+1v-1: continued on the next page

Record 223 — r7c2_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 35.606387746949443306362 81; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000024685259569118152
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a8a820b2e3341e508860df8fa52e8ba6be5fca90c44b0b918cfb3e75fd28d176 *Reader*: 3c31186b3fac2aa2f596f047ea1756c734f0ddc59694e37a98f47d9a3bdc8475 *Cell record at 108*: —

Route 224 — r7c2_h+1v+1

Cell 56: row 7, column 2. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 224; ordinal in the memory table: 221.

Table 94.232: Record 224: r7c2_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c2_h+1v+1; *Cell*: 56; *Row*: 7
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 224 — r7c2_h+1v+1: continued on the next page

Record 224 — r7c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,8;7,2
Classification (cont.)	*Route family*: 34|0|-4|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,0); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 224 — r7c2_h+1v+1: continued on the next page

Record 224 — r7c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3bac38d14f028f874e98149b7274261bf05c7219f5381cfc4cbad9f6c1c850b9 *Reader*: d0f373bab545802ead72e058f777728e20bf2d03ce412db3853130c4b1a38f22 *Cell record at 108*: —

Route 225 — r7c3_h-1v-1

Cell 57: row 7, column 3. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 225; ordinal in the memory table: 228.

Table 94.233: Record 225: r7c3_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c3_h-1v-1; *Cell*: 57; *Row*: 7
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,7;7,3
Classification (cont.)	*Route family*: 34|0|-12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16

Record 225 — r7c3_h-1v-1: continued on the next page

Record 225 — r7c3_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 50dd3ab85f947dea000293ba18e7e5d236ed2a2482880fce 8a55b876402a96d6 *Reader*: f8cfb75a9cb25ac6b8cbb42e3fc084e2337939344c2a990541bae640 dcee3531 *Cell record at 108*: —

Route 226 — r7c3_h-1v+1

Cell 57: row 7, column 3. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 226; ordinal in the memory table: 227.

Table 94.234: Record 226: r7c3_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c3_h-1v+1; *Cell*: 57; *Row*: 7
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,7;7,3
Classification (cont.)	*Route family*: 34|0|-12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (56,54,5)

Record 226 — r7c3_h-1v+1: continued on the next page

Record 226 — r7c3_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 97b0c7b8352c824ea503a1ec527396cfc0d0b44900be367d d4b03945006857bf *Reader*: b7b4d75304d95a6a60aba23861edd1112aad3aed5ee0e9673c71017c cc3594bd *Cell record at 108*: —

Route 227 — r7c3_h+1v-1

Cell 57: row 7, column 3. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 227; ordinal in the memory table: 226.

Table 94.235: Record 227: r7c3_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c3_h+1v-1; *Cell*: 57; *Row*: 7
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton

Record 227 — r7c3_h+1v-1: continued on the next page

Record 227 — r7c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,7;7,3
Classification (cont.)	*Route family*: 34|0|-12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 55.6064989433721354464 16 86; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 227 — r7c3_h+1v-1: continued on the next page

Record 227 — r7c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 34dff48532c73a280d1550ed80b44a2baf487d486dd06bec76436e8e94d65f93 *Reader*: 502b383cae291d41508f616ac645f9bcdef36a9a45477b31c70d5cb7ee016951 *Cell record at 108*: —

Route 228 — r7c3_h+1v+1

Cell 57: row 7, column 3. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 228; ordinal in the memory table: 225.

Table 94.236: Record 228: r7c3_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c3_h+1v+1; *Cell*: 57; *Row*: 7
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,7;7,3
Classification (cont.)	*Route family*: 34|0|-12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 228 — r7c3_h+1v+1: continued on the next page

Record 228 — r7c3_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2c8a0ec9e3c50967fd90be0a53359711cda37fe96004fa0d d2d2c109d3ff8203 *Reader*: e7e5c8dda4d5d1f9b878e9cd1a1c64fb8345513eb16b32b91933ab57 1fbdbb5d *Cell record at 108*: d232cdd08b31d355345b67402bf59eda9ec2f460c0b2fde0 aabb2dc8b105a72c

Route 229 — r7c4_h-1v-1

Cell 58: row 7, column 4. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 229; ordinal in the memory table: 232.

Table 94.237: Record 229: r7c4_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c4_h-1v-1; *Cell*: 58; *Row*: 7
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,6;7,4
Classification (cont.)	*Route family*: 30|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 229 — r7c4_h-1v-1: continued on the next page

Record 229 — r7c4_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055189889622315337
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 61986b6175f7b937676573337389fee3569c31355092cf947843a9d930ba1bc2 *Reader*: 5924cc1f56504b4c33f30badd28f71c226336afe924e078101dceaa8e836decf *Cell record at 108*: —

Route 230 — r7c4_h-1v+1

Cell 58: row 7, column 4. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 230; ordinal in the memory table: 231.

Table 94.238: Record 230: r7c4_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c4_h-1v+1; *Cell*: 58; *Row*: 7
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton

Record 230 — r7c4_h-1v+1: continued on the next page

Record 230 — r7c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,6;7,4
Classification (cont.)	*Route family*: 30|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.6066101397948275864791; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 230 — r7c4_h-1v+1: continued on the next page

Record 230 — r7c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c59fd018875bb4727075a352424cec29d327972ce0436e0d6620d4c581058224 *Reader*: e049714ffb2d3eea8b496d146d72cdeb379c2764679bbf9137027cbadddd41f3 *Cell record at 108*: c7e0c373385cd688c8ff7aafd12c63d3af22800c9ae593a7a3bc5952d7512202

Route 231 — r7c4_h+1v-1

Cell 58: row 7, column 4. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 231; ordinal in the memory table: 230.

Table 94.239: Record 231: r7c4_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c4_h+1v-1; *Cell*: 58; *Row*: 7
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,6;7,4
Classification (cont.)	*Route family*: 30|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 231 — r7c4_h+1v-1: continued on the next page

Record 231 — r7c4_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7a6c6470c763c81771cd0d8889250edb8a8675dd0391a21b31385cba4993e652 *Reader*: be05cd14c4f089027d45446ae145d5bbd383576b0d091e1660097007ebf4fc04 *Cell record at 108*: —

Route 232 — r7c4_h+1v+1

Cell 58: row 7, column 4. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 232; ordinal in the memory table: 229.

Table 94.240: Record 232: r7c4_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c4_h+1v+1; *Cell*: 58; *Row*: 7
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,6;7,4
Classification (cont.)	*Route family*: 30|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|6|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 232 — r7c4_h+1v+1: continued on the next page

Record 232 — r7c4_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000005518988962231533 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3b0c0dbfa4786567c9e927a3d9255385f88a8bf3e4ee10b3 9a8990045dbb8251 *Reader*: 7e34fe3bd8853b363e1c7195129b813c582bbfbb75ebd6651f739c23 d7b1b85a *Cell record at 108*: —

Route 233 — r7c5_h-1v-1

Cell 59: row 7, column 5. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 233; ordinal in the memory table: 236.

Table 94.241: Record 233: r7c5_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c5_h-1v-1; *Cell*: 59; *Row*: 7
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2

Record 233 — r7c5_h-1v-1: continued on the next page

Record 233 — r7c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,5;7,5
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-4|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ

Record 233 — r7c5_h-1v-1: continued on the next page

Record 233 — r7c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: f6ed0ad96ec6366eaed3b499144349882d3ffcd7cd44284510c3270821d0056c *Reader*: 25d3f4ddb064b47c9a608e17a712bb09d854df59edc3903e31a2673b50b1ae0a *Cell record at 108*: 68cfa4635763ee620dde045d3406cd27b8ada4eb740d3dd02cfcde7e4775e236

Route 234 — r7c5_h-1v+1

Cell 59: row 7, column 5. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 234; ordinal in the memory table: 235.

Table 94.242: Record 234: r7c5_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c5_h-1v+1; *Cell*: 59; *Row*: 7
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,5;7,5
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -6; *raw κ 0*: 2

Record 234 — r7c5_h-1v+1: continued on the next page

Record 234 — r7c5_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 50|-6|2|-2|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 99ab406ecdac6329ab5ce5eb6325504ecb11f2d04c9d8b346db7499d7f567b1e *Reader*: de03114ba14ea4662b556a0e8594ed8b86378685d02ee66d90537006e76e8ab8 *Cell record at 108*: —

Route 235 — r7c5_h+1v-1

Cell 59: row 7, column 5. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 235; ordinal in the memory table: 234.

Table 94.243: Record 235: r7c5_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c5_h+1v-1; *Cell*: 59; *Row*: 7
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,5;7,5
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 50|6|-2|2|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K* Ω: (56,54,5)

Record 235 — r7c5_h+1v-1: continued on the next page

Record 235 — r7c5_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 0386e9c2055bfd8ab8494edc4b8a16f00671f216a461804424aa45aab64b22d5 *Reader*: 66b6919786b672f07cdf87be4aefcb18b9cac24fa6eddfae27e4ddf26caba935 *Cell record at 108*: —

Route 236 — r7c5_h+1v+1

Cell 59: row 7, column 5. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 236; ordinal in the memory table: 233.

Table 94.244: Record 236: r7c5_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c5_h+1v+1; *Cell*: 59; *Row*: 7
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton

Record 236 — r7c5_h+1v+1: continued on the next page

Record 236 — r7c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,5;7,5
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|4|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 22; *log2 ambiguity area*: 4.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 236 — r7c5_h+1v+1: continued on the next page

Record 236 — r7c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b7a1b0a7f7d9328fa3ce85e1fff27ea994c202f1387e665f13646fe39a6f9594 *Reader*: d7423e62f53bddcb400b3cf3d5485932f1df7a7edb0186d4e92c189a60b8a179 *Cell record at 108*: —

Route 237 — r7c6_h-1v-1

Cell 60: row 7, column 6. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 237; ordinal in the memory table: 240.

Table 94.245: Record 237: r7c6_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c6_h-1v-1; *Cell*: 60; *Row*: 7
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,4;7,6
Classification (cont.)	*Route family*: 28|-6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16

Record 237 — r7c6_h-1v-1: continued on the next page

Record 237 — r7c6_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: bdd0f510981aa2bd14e11cc9e772a40c0fa9629c6c33a6c8 63e128543bb3c064 *Reader*: 094ff62832260c1076787d26e6f33ddfede703d5dedbf9b86f089121 79c0158e *Cell record at 108*: —

Route 238 — r7c6_h-1v+1

Cell 60: row 7, column 6. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 238; ordinal in the memory table: 239.

Table 94.246: Record 238: r7c6_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c6_h-1v+1; *Cell*: 60; *Row*: 7
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,4;7,6
Classification (cont.)	*Route family*: 28|-6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|-2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K* Ω: (80,54,6)

Record 238 — r7c6_h-1v+1: continued on the next page

Record 238 — r7c6_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3b43feb29ef42944b5c6da5d1493e4641332fe290cc733c9 4b55588185f490d0 *Reader*: 77fffc5e08a1d1d22e2d56a9caa4f75016d1c69261b9bac28a2e8ced d44d0ddb *Cell record at 108*: 4261f837a3e329d199831ee764123ea51bb952b25504d3f1 b96372b344872b11

Route 239 — r7c6_h+1v-1

Cell 60: row 7, column 6. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 239; ordinal in the memory table: 238.

Table 94.247: Record 239: r7c6_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c6_h+1v-1; *Cell*: 60; *Row*: 7
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton

Record 239 — r7c6_h+1v-1: continued on the next page

Record 239 — r7c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,4;7,6
Classification (cont.)	*Route family*: 28|-6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 239 — r7c6_h+1v-1: continued on the next page

Record 239 — r7c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 231bf7dd55d8a438f107de8c3e14075106170fc968fbf990a0d8a6f5ffd71b41 *Reader*: 573249f714bc3cb98d64797e90987918abae980680a392c41e65c25c1bba6d94 *Cell record at 108*: —

Route 240 — r7c6_h+1v+1

Cell 60: row 7, column 6. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 240; ordinal in the memory table: 237.

Table 94.248: Record 240: r7c6_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c6_h+1v+1; *Cell*: 60; *Row*: 7
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G2; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: neutrino_G2
Classification (cont.)	*Physical family*: neutrino; *Depth*: G2; *Inversion orbit*: 3,4;7,6
Classification (cont.)	*Route family*: 28|-6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 240 — r7c6_h+1v+1: continued on the next page

Record 240 — r7c6_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3e939392091ce7df7511a0eeda421eb0299c2ba4582a4adb f492bef6aebffe3f *Reader*: b6443f7d3f0c44922d917e3268c496cff426301296ffd17b39c8f71c 592309c7 *Cell record at 108*: —

Route 241 — r7c7_h-1v-1

Cell 61: row 7, column 7. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 241; ordinal in the memory table: 244.

Table 94.249: Record 241: r7c7_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c7_h-1v-1; *Cell*: 61; *Row*: 7
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,3;7,7
Classification (cont.)	*Route family*: 24|0|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 241 — r7c7_h-1v-1: continued on the next page

Record 241 — r7c7_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000005518988962231533 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 22; *log2 ambiguity area*: 4.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: f7adee5b3133fb43299c7e7f57cfe9d2104d12951c24558e004b0d1046825410 *Reader*: ee9f7a74820c460ffd5606aa30bdddd40bb6875427bf381c1bb6883d6bf2cb2d *Cell record at 108*: 2b5fcf4ee156f8e7f3d9ef4b48b12761781d58740c7f25b350ada993fdfb43da

Route 242 — r7c7_h-1v+1

Cell 61: row 7, column 7. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 242; ordinal in the memory table: 243.

Table 94.250: Record 242: r7c7_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c7_h-1v+1; *Cell*: 61; *Row*: 7
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton

Record 242 — r7c7_h-1v+1: continued on the next page

Record 242 — r7c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,3;7,7
Classification (cont.)	*Route family*: 24|0|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 242 — r7c7_h-1v+1: continued on the next page

Record 242 — r7c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e5a3a65c351de531521f5eb2e8a895c938d33cc4b6092ae7a236bfde174f67e6 *Reader*: f152871599c0f372a009df5e0bf5d2e89a435b88231a923e31fd5fc32cbfc7f8 *Cell record at 108*: —

Route 243 — r7c7_h+1v-1

Cell 61: row 7, column 7. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 243; ordinal in the memory table: 242.

Table 94.251: Record 243: r7c7_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c7_h+1v-1; *Cell*: 61; *Row*: 7
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,3;7,7
Classification (cont.)	*Route family*: 24|0|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 243 — r7c7_h+1v-1: continued on the next page

Record 243 — r7c7_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|-2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 28; *D4*: 78
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,78,25); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.43358641016316702356304; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041204222405344905
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b139139d502bd446d91e7d84ef6378829caef959c57c068a87604b10a6c051f8 *Reader*: d3747a41b0c5e19bf084c3f39ce497e825be56b559a223f00b148f1580acb8ed *Cell record at 108*: —

Route 244 — r7c7_h+1v+1

Cell 61: row 7, column 7. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 244; ordinal in the memory table: 241.

Table 94.252: Record 244: r7c7_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c7_h+1v+1; *Cell*: 61; *Row*: 7
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G2; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 2; *Color*: —; *Multisection*: charged_lepton_G2
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G2; *Inversion orbit*: 3,3;7,7
Classification (cont.)	*Route family*: 24|0|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 70
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,7); *K* Ω: (80,54,6)

Record 244 — r7c7_h+1v+1: continued on the next page

Record 244 — r7c7_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.60672133621751972652496; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000005518988962231533 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 22; *log2 ambiguity area*: 4.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 97a8e38d1e0bc29075a19475862f706d23c2cee0e4509a10 167a321562108b24 *Reader*: 113a5319d2bebd3924e86aff9c2f661818e1d5da017520c4f5a24875 0b426e5c *Cell record at 108*: —

Route 245 — r7c8_h-1v-1

Cell 62: row 7, column 8. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 245; ordinal in the memory table: 248.

Table 94.253: Record 245: r7c8_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c8_h-1v-1; *Cell*: 62; *Row*: 7
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3

Record 245 — r7c8_h-1v-1: continued on the next page

Record 245 — r7c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,2;7,8
Classification (cont.)	*Route family*: 28|6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17

Record 245 — r7c8_h-1v-1: continued on the next page

Record 245 — r7c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 93e2a454a1c5de5f4f4e4b0eec1903cc6a51d31d19260e98a023f085f717f2c5 *Reader*: 5e1722d01279d8988e0e53f37eeed749fe741983d8db78b09039733d0e70de18 *Cell record at 108*: —

Route 246 — r7c8_h-1v+1

Cell 62: row 7, column 8. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 246; ordinal in the memory table: 247.

Table 94.254: Record 246: r7c8_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c8_h-1v+1; *Cell*: 62; *Row*: 7
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,2;7,8
Classification (cont.)	*Route family*: 28|6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 246 — r7c8_h-1v+1: continued on the next page

Record 246 — r7c8_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ff6632f4595185491ccf0bb1ee6fb09651fc7ee9d444f4a3 6b2434921c63e845 *Reader*: c689f380d0c3f1f13830f345827ac079bfef21a0290c3319a4e040b6 6452df42 *Cell record at 108*: —

Route 247 — r7c8_h+1v-1

Cell 62: row 7, column 8. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 247; ordinal in the memory table: 246.

Table 94.255: Record 247: r7c8_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c8_h+1v-1; *Cell*: 62; *Row*: 7
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,2;7,8
Classification (cont.)	*Route family*: 28|6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K* Ω: (80,54,6)

Record 247 — r7c8_h+1v-1: continued on the next page

Record 247 — r7c8_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 572bed3ec2da43b588e8f03c20b85deb3a3238787c652806 bdaedb67e8a2e65d *Reader*: f6d6251c2a3a0bc51ba9e9d062b640aae53badbca7baa4e87ead4bf0 ae639226 *Cell record at 108*: 2bc42341df8c62244bc513d9a683f0a2cd6caf97ec8b8d54 ada93d9ac2a0f146

Route 248 — r7c8_h+1v+1

Cell 62: row 7, column 8. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 248; ordinal in the memory table: 245.

Table 94.256: Record 248: r7c8_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c8_h+1v+1; *Cell*: 62; *Row*: 7
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G3; *Family*: neutrino; *Sector*: lepton

Record 248 — r7c8_h+1v+1: continued on the next page

Record 248 — r7c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: —; *Multisection*: neutrino_G3
Classification (cont.)	*Physical family*: neutrino; *Depth*: G3; *Inversion orbit*: 3,2;7,8
Classification (cont.)	*Route family*: 28|6|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 64
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,0); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 99.51837473042726913417918; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.000000068994276645021895
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 248 — r7c8_h+1v+1: continued on the next page

Record 248 — r7c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5468c23aaa5fe88b8dfc71dca843b9098b3fad83d80745052c981c396d08d09d *Reader*: e7b0197ca2c34747ea501fc5fc951f0eabb2154bfe7d1ca632f931071fa14d07 *Cell record at 108*: —

Route 249 — r7c9_h-1v-1

Cell 63: row 7, column 9. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 249; ordinal in the memory table: 252.

Table 94.257: Record 249: r7c9_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c9_h-1v-1; *Cell*: 63; *Row*: 7
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,1;7,9
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 18; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16

Record 249 — r7c9_h-1v-1: continued on the next page

Record 249 — r7c9_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ba5642d596d97f27564669dc98515c905e1973cd764f435e e5c853fcd049beed *Reader*: 268ca6785b1f8117df7ac1724f8f4b30ac60083edec1102641a6611a dc1ffb9c *Cell record at 108*: —

Route 250 — r7c9_h-1v+1

Cell 63: row 7, column 9. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 250; ordinal in the memory table: 251.

Table 94.258: Record 250: r7c9_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c9_h-1v+1; *Cell*: 63; *Row*: 7
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,1;7,9
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (56,54,5)

Record 250 — r7c9_h-1v+1: continued on the next page

Record 250 — r7c9_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: da00f1d555eb9be25c1ceca2ebc88d96b55a1c8c2011788d814fa5857a504b9f *Reader*: 183882de6128614ad3eef875eb974e8c55c951558c213f7eb721348c9354f824 *Cell record at 108*: —

Route 251 — r7c9_h+1v-1

Cell 63: row 7, column 9. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 251; ordinal in the memory table: 250.

Table 94.259: Record 251: r7c9_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r7c9_h+1v-1; *Cell*: 63; *Row*: 7
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 251 — r7c9_h+1v-1: continued on the next page

Record 251 — r7c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,1;7,9
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 55.6064989433721354464 16 86; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 251 — r7c9_h+1v-1: continued on the next page

Record 251 — r7c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ad5ee82a87bb9a61ce4360a2b14bf22c5c60f40e7e75f290ec2dde4d0e7c90fc *Reader*: ed489aa0c8e0c09e4604ad7f7f9ebf49ec343a76ed5ea60a3369f1411b16ea84 *Cell record at 108*: —

Route 252 — r7c9_h+1v+1

Cell 63: row 7, column 9. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 252; ordinal in the memory table: 249.

Table 94.260: Record 252: r7c9_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r7c9_h+1v+1; *Cell*: 63; *Row*: 7
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 3,1;7,9
Classification (cont.)	*Route family*: 28|-6|6|0|-12|+++---+++---
Returns	*Class and phase*: 9; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 252 — r7c9_h+1v+1: continued on the next page

Record 252 — r7c9_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 64
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128794709248880
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 88eff8f2d71641e554b89b656e2885567c18727bad0128f4 2fa25e7bdf6a031d *Reader*: 6c78548094bcdfb058a3bcd868ff2cf83e42197164de9e58aa3dab4b 8edc6ad7 *Cell record at 108*: 743b8088cefbb1b382bb7dc86b4e63b91a87688168e208f2 080f8a040aefcffd

Route 253 — r8c1_h-1v-1

Cell 64: row 8, column 1. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 253; ordinal in the memory table: 256.

Table 94.261: Record 253: r8c1_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c1_h-1v-1; *Cell*: 64; *Row*: 8
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,9;8,1
Classification (cont.)	*Route family*: 28|-6|10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 253 — r8c1_h-1v-1: continued on the next page

Record 253 — r8c1_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 40; *log2 ambiguity area*: 20.339850002884624; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7f20d693558c4c770e8c14a2edb53682418d3af88e8147e4 e46a45b584d0af32 *Reader*: d463a30437186dc5a632dbe1fe2eb6c5c462cdb58acc7d2d24faaa41 23025187 *Cell record at 108*: —

Route 254 — r8c1_h-1v+1

Cell 64: row 8, column 1. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 254; ordinal in the memory table: 255.

Table 94.262: Record 254: r8c1_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c1_h-1v+1; *Cell*: 64; *Row*: 8
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark

Record 254 — r8c1_h-1v+1: continued on the next page

Record 254 — r8c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,9;8,1
Classification (cont.)	*Route family*: 28|-6|10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: 0++--00++--0
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 36
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K* Ω: (36,54,4)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 35.606387746949443306362 81; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000024685259569118152
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 254 — r8c1_h-1v+1: continued on the next page

Record 254 — r8c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c88b93580f9451e0e63048f3f329ae52082f45e65a9494acb1942445b74dbd7d *Reader*: 06f382a4af0319fc434d2d5109c92a77f6ff4e9eb6a30cdd9620be4b2c25677f *Cell record at 108*: —

Route 255 — r8c1_h+1v-1

Cell 64: row 8, column 1. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 255; ordinal in the memory table: 254.

Table 94.263: Record 255: r8c1_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c1_h+1v-1; *Cell*: 64; *Row*: 8
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,9;8,1
Classification (cont.)	*Route family*: 28|-6|10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 255 — r8c1_h+1v-1: continued on the next page

Record 255 — r8c1_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: ++00--++00--
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 36
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (40,24,2); *K Ω*: (36,54,4)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 35.6063877469494433063681; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000024685259569118152
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 210c219bbaaf3bf455eaa92f45a017dd9f816ca3b46f2b08b272a9c06868cdc0 *Reader*: 8b66a9561a72fc2cf9d44aad0a4ab9280d6872fd24ae8ade2437e7c71c5da763 *Cell record at 108*: 2abe5fb770abbac72934c668102c504b01e8b32c6bee43d0a62e45a1cbb3f5cd

Route 256 — r8c1_h+1v+1

Cell 64: row 8, column 1. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 256; ordinal in the memory table: 253.

Table 94.264: Record 256: r8c1_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c1_h+1v+1; *Cell*: 64; *Row*: 8
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: G; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,9;8,1
Classification (cont.)	*Route family*: 28|-6|10|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 256 — r8c1_h+1v+1: continued on the next page

Record 256 — r8c1_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: aab792bb556e0bed999eb313b4d4b26ea63b3bad901ed813 0f94fb1ec4554219 *Reader*: 7236233560f551425afe01a27858b2d7979db3983540ff9b4e007b9f 9b5e0d04 *Cell record at 108*: —

Route 257 — r8c2_h-1v-1

Cell 65: row 8, column 2. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 257; ordinal in the memory table: 260.

Table 94.265: Record 257: r8c2_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c2_h-1v-1; *Cell*: 65; *Row*: 8
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 257 — r8c2_h-1v-1: continued on the next page

Record 257 — r8c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,8;8,2
Classification (cont.)	*Route family*: 30|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|6|0|2|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Record 257 — r8c2_h-1v-1: continued on the next page

Record 257 — r8c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 4d09dc43af4a59915c1a6038f5c475cbd194cfeb4bf1d5e357df23d9754d395f *Reader*: 4128cbf79b80a7af93f10ded32d0960eb5da56cbf47a110bab03f10d446c2316 *Cell record at 108*: —

Route 258 — r8c2_h-1v+1

Cell 65: row 8, column 2. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 258; ordinal in the memory table: 259.

Table 94.266: Record 258: r8c2_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c2_h-1v+1; *Cell*: 65; *Row*: 8
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,8;8,2
Classification (cont.)	*Route family*: 30|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 9; *Orbit and phase*: 9; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 258 — r8c2_h-1v+1: continued on the next page

Record 258 — r8c2_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (60,84,28); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000041174098946741575
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c10ac37f6c4294f84a5c4eecb9e4d1374086b7aae332df72101d20aff5ff88c8 *Reader*: aa9f45f6e48948c38e93e1da669a8afdc2bfc8e251959f2c2d2ac248ac767317 *Cell record at 108*: —

Route 259 — r8c2_h+1v-1

Cell 65: row 8, column 2. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 259; ordinal in the memory table: 258.

Table 94.267: Record 259: r8c2_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c2_h+1v-1; *Cell*: 65; *Row*: 8
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,8;8,2
Classification (cont.)	*Route family*: 30|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 18; *Orbit and phase*: 18; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (60,84,28); *K* Ω: (56,54,5)

Record 259 — r8c2_h+1v-1: continued on the next page

Record 259 — r8c2_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000004117409894674 1575
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003855097235414 1645
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 4581338ce438211bda03b46bdd171051cbdf4527bfe09385 ab84b222943678e0 *Reader*: 30e7d80418eef9d9188a136774e1e1ba1b467b366375105e556e2595 fda118ab *Cell record at 108*: —

Route 260 — r8c2_h+1v+1

Cell 65: row 8, column 2. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 260; ordinal in the memory table: 257.

Table 94.268: Record 260: r8c2_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c2_h+1v+1; *Cell*: 65; *Row*: 8
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 260 — r8c2_h+1v+1: continued on the next page

Record 260 — r8c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,8;8,2
Classification (cont.)	*Route family*: 30|-6|4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|0|-2|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ Ω*: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Record 260 — r8c2_h+1v+1: continued on the next page

Record 260 — r8c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 37e66f95874dba7502936c0625906398abd462421242a711a1cebe0da9742e20 *Reader*: a066e554a21d08baafef29692f0e0500bdd858876f9bbc234b52fb9a24bea27f *Cell record at 108*: d7f810dc8cc7b1298f7e1c92b9f22e705a1b79fa32aaf5a18405a2e5fbd7422f

Route 261 — r8c3_h-1v-1

Cell 66: row 8, column 3. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 261; ordinal in the memory table: 264.

Table 94.269: Record 261: r8c3_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c3_h-1v-1; *Cell*: 66; *Row*: 8
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,7;8,3
Classification (cont.)	*Route family*: 34|0|-8|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 261 — r8c3_h-1v-1: continued on the next page

Record 261 — r8c3_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,4); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243 *Stable triads per block*: 0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ad652dbfa6c5c813518166ce23d4443cfdf6cc6a3f22f5d7f723510f50c448b8 *Reader*: 3ed4894181a15c6c37003af6a3bffeefbd5b6c91a65783ebe3a130457ab9a8aa *Cell record at 108*: —

Route 262 — r8c3_h-1v+1

Cell 66: row 8, column 3. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 262; ordinal in the memory table: 263.

Table 94.270: Record 262: r8c3_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c3_h-1v+1; *Cell*: 66; *Row*: 8
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,7;8,3
Classification (cont.)	*Route family*: 34|0|-8|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 262 — r8c3_h-1v+1: continued on the next page

Record 262 — r8c3_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 541a3c42518d601b58431822a3572a2be049c02a300aa4fe acd5345ef948f454 *Reader*: ad8c50495acdfc9708f3bf9b44a935a2051df21767326d50c52b3dfe ae00d3fb *Cell record at 108*: e9f4ee18c9a8859c65aa766dbde0bd196a242a2f3dacd16a 052d90accc021dd1

Route 263 — r8c3_h+1v-1

Cell 66: row 8, column 3. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 263; ordinal in the memory table: 262.

Table 94.271: Record 263: r8c3_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c3_h+1v-1; *Cell*: 66; *Row*: 8
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO

Record 263 — r8c3_h+1v-1: continued on the next page

Record 263 — r8c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,7;8,3
Classification (cont.)	*Route family*: 34|0|-8|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Record 263 — r8c3_h+1v-1: continued on the next page

Record 263 — r8c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: cd28c157b1840882849d580e4bfbce448d65ae5f45fe448971e9270dacf4fbc6 *Reader*: 6d4c84d97a1f41ddbd2bbd27d0c2afc77d646b03c697190d46d00c87f92617a1 *Cell record at 108*: —

Route 264 — r8c3_h+1v+1

Cell 66: row 8, column 3. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 264; ordinal in the memory table: 261.

Table 94.272: Record 264: r8c3_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c3_h+1v+1; *Cell*: 66; *Row*: 8
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,7;8,3
Classification (cont.)	*Route family*: 34|0|-8|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17

Record 264 — r8c3_h+1v+1: continued on the next page

Record 264 — r8c3_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,4); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 55.606498943372135446416 86; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e3bc25b606504b6c91007699ff3655c139fba11cd7689c98 c705d3f9d12f38cd *Reader*: fea2f4b68d20b684e22997fc21a30bb39fa2a3c31dbf3cf0fef68224 900afdb0 *Cell record at 108*: —

Route 265 — r8c4_h-1v-1

Cell 67: row 8, column 4. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 265; ordinal in the memory table: 268.

Table 94.273: Record 265: r8c4_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c4_h-1v-1; *Cell*: 67; *Row*: 8
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,6;8,4
Classification (cont.)	*Route family*: 34|0|-4|2|-8|+++--0+++--0
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,2); *K* Ω: (56,54,5)

Record 265 — r8c4_h-1v-1: continued on the next page

Record 265 — r8c4_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003855097235414164 5
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: d773d886b771b4e9d58210f91db044e3d5fa76dcdbba2d14 4b4aa6a7816e2a74 *Reader*: 2129600f36965fdc11a791126a3ad1e38346cb4d0a010c35bc87bab5 442f815f *Cell record at 108*: —

Route 266 — r8c4_h-1v+1

Cell 67: row 8, column 4. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 266; ordinal in the memory table: 267.

Table 94.274: Record 266: r8c4_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c4_h-1v+1; *Cell*: 67; *Row*: 8
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 266 — r8c4_h-1v+1: continued on the next page

Record 266 — r8c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,6;8,4
Classification (cont.)	*Route family*: 34|0|-4|2|-8|+++--0+++--0
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 50|-6|2|-2|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 55.6064989443372135446416 86; *Action factor D*: 1.00000002761000615951421 5
Reader evaluations (cont.)	*Action factor Ω*: 1.00000003855097235414164 5
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 266 — r8c4_h-1v+1: continued on the next page

Record 266 — r8c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: aa4a5949c8430a0ec7f118f6522a38b241e22dd0f172a9ad8b2e7d04e3373bbb *Reader*: 00391297614e66ca2afd3f79601a5f0dec10705608f060644783235dbd3fa513 *Cell record at 108*: —

Route 267 — r8c4_h+1v-1

Cell 67: row 8, column 4. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 267; ordinal in the memory table: 266.

Table 94.275: Record 267: r8c4_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c4_h+1v-1; *Cell*: 67; *Row*: 8
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,6;8,4
Classification (cont.)	*Route family*: 34|0|-4|2|-8|+++--0+++--0
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 267 — r8c4_h+1v-1: continued on the next page

Record 267 — r8c4_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 50|6|-2|2|-8
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,24,2); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ Ω*: 55.6064989433721354464186; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: cbe1adfccc203f0f6a54465ba638e1513bb7f3f92188c76ac22a231f8fc83757 *Reader*: 8e4f93fa6fc6ecbbf4f2d29f80a71639352a11bbbd98c66c5b0772c8df05d749 *Cell record at 108*: —

Route 268 — r8c4_h+1v+1

Cell 67: row 8, column 4. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 268; ordinal in the memory table: 265.

Table 94.276: Record 268: r8c4_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c4_h+1v+1; *Cell*: 67; *Row*: 8
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,6;8,4
Classification (cont.)	*Route family*: 34|0|-4|2|-8|+++--0+++--0
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (100,66,2); *K* Ω: (56,54,5)

Record 268 — r8c4_h+1v+1: continued on the next page

Record 268 — r8c4_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 55.606498943372135446416 86; *Action factor D*: 1.00000006899443082593641 7
Reader evaluations (cont.)	*Action factor* Ω: 1.00000003855097235414164 5
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ad5595794bce61df813886f136cf73b6d5b7712e8f1009a2125b13242b4e8566 *Reader*: cf06adfa5665f171a8b8dbe54798e56195555e71f498639258299d96de416911 *Cell record at 108*: 573c1bd90e4644c8d653dbd9a8229181186b709ffd121a7408c8373b4384376c

Route 269 — r8c5_h-1v-1

Cell 68: row 8, column 5. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 269; ordinal in the memory table: 272.

Table 94.277: Record 269: r8c5_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c5_h-1v-1; *Cell*: 68; *Row*: 8
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO

Record 269 — r8c5_h-1v-1: continued on the next page

Record 269 — r8c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,5;8,5
Classification (cont.)	*Route family*: 30|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Record 269 — r8c5_h-1v-1: continued on the next page

Record 269 — r8c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7ef6525bcd2aeb751a6f5b43b6d6a27622407710aadc04f8f412b15957971689 *Reader*: 37d699ceb70f1c3bf1a9991ce52e71739128ddd2096da7733ba3750a74627b60 *Cell record at 108*: —

Route 270 — r8c5_h-1v+1

Cell 68: row 8, column 5. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 270; ordinal in the memory table: 271.

Table 94.278: Record 270: r8c5_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c5_h-1v+1; *Cell*: 68; *Row*: 8
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,5;8,5
Classification (cont.)	*Route family*: 30|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 270 — r8c5_h-1v+1: continued on the next page

Record 270 — r8c5_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|-2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,84,28); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000041174098946741575
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5168b13c9ef8c5df5479cfd86a3b5810444636458700009b99542fd47675d314 *Reader*: 3b12ffa16f7d92e65eccdd73152a1897c1e1bb8b22b621b66ec88ce2035f57c6 *Cell record at 108*: b139e53ebbb2939e05b997684309cd5bb321796c24a70f00063466350ad16680

Route 271 — r8c5_h+1v-1

Cell 68: row 8, column 5. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 271; ordinal in the memory table: 270.

Table 94.279: Record 271: r8c5_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c5_h+1v-1; *Cell*: 68; *Row*: 8
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,5;8,5
Classification (cont.)	*Route family*: 30|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (60,84,28); *K* Ω: (80,54,6)

Record 271 — r8c5_h+1v-1: continued on the next page

Record 271 — r8c5_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000004117409894674 1575
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185 9140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: eedb99f7a53750e7da1c4cfb65caff5bc7d6f7fe2d17ac33 bd698f5ff5cced70 *Reader*: 3660859f8e83b5299df231cf992060bcc7cafcad6c8bb559d671dcec 38ae9521 *Cell record at 108*: —

Route 272 — r8c5_h+1v+1

Cell 68: row 8, column 5. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 272; ordinal in the memory table: 269.

Table 94.280: Record 272: r8c5_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c5_h+1v+1; *Cell*: 68; *Row*: 8
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark

Record 272 — r8c5_h+1v+1: continued on the next page

Record 272 — r8c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,5;8,5
Classification (cont.)	*Route family*: 30|6|-4|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 30|6|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864709 1; *Action factor D*: 1.00000005518981253185914 0
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Record 272 — r8c5_h+1v+1: continued on the next page

Record 272 — r8c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 623aeb8e00cd78e8b74b3da659368cebc64b59b20668d849c58cc8e800f3c668 *Reader*: 9f01aaf11034da5b3d5ff388ae07367265257113f331b94b7080b8d05e61d946 *Cell record at 108*: —

Route 273 — r8c6_h-1v-1

Cell 69: row 8, column 6. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 273; ordinal in the memory table: 276.

Table 94.281: Record 273: r8c6_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c6_h-1v-1; *Cell*: 69; *Row*: 8
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,4;8,6
Classification (cont.)	*Route family*: 28|6|-6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 273 — r8c6_h-1v-1: continued on the next page

Record 273 — r8c6_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-2|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243 *Stable triads per block*: 0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 545d27ebe80f7a1b5d6e4faf6f6614e59a75427fdc5db81a0eff641785391008 *Reader*: df6973d8dba7d543ad4991d2fe5c1ebf3eb9a1803d7e933f0074ba080793ff48 *Cell record at 108*: 165e97906c2b634f80f723d3b6db0f07d486840bdcf88079defcfc71121b3293

Route 274 — r8c6_h-1v+1

Cell 69: row 8, column 6. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 274; ordinal in the memory table: 275.

Table 94.282: Record 274: r8c6_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c6_h-1v+1; *Cell*: 69; *Row*: 8
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,4;8,6
Classification (cont.)	*Route family*: 28|6|-6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 274 — r8c6_h-1v+1: continued on the next page

Record 274 — r8c6_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: f71b59ca66689f8406ea25f29129a8bb6aadb6447707eafc3f076a9d7ed74c3f *Reader*: 45ad6fbebdb7517bd4fc0760b8fc1596fbc582e6077f671d37ea2a5a9582a424 *Cell record at 108*: —

Route 275 — r8c6_h+1v-1

Cell 69: row 8, column 6. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 275; ordinal in the memory table: 274.

Table 94.283: Record 275: r8c6_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c6_h+1v-1; *Cell*: 69; *Row*: 8
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 275 — r8c6_h+1v-1: continued on the next page

Record 275 — r8c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,4;8,6
Classification (cont.)	*Route family*: 28|6|-6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|-2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000002733740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Record 275 — r8c6_h+1v-1: continued on the next page

Record 275 — r8c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b2394c0d8b4f94d4d3db2058caf65f5cba5037faaba86ab8b741d7614066f5b9 *Reader*: b35fffde6d573fe8075155a84df6a2ed2e98d9bf9fe9e09c1e0624fb83395b24 *Cell record at 108*: —

Route 276 — r8c6_h+1v+1

Cell 69: row 8, column 6. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 276; ordinal in the memory table: 273.

Table 94.284: Record 276: r8c6_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c6_h+1v+1; *Cell*: 69; *Row*: 8
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: B; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,4;8,6
Classification (cont.)	*Route family*: 28|6|-6|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17

Record 276 — r8c6_h+1v+1: continued on the next page

Record 276 — r8c6_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|2|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 1fa8ca368f05cacbb63847f3d7c151b4b61b01f5663f9d4f 2e416c66bb475d96 *Reader*: 7958f0da3b1c71e56248b77b0bd939b0279b68f8ff68cb145e28e5e0 6d5a3f6e *Cell record at 108*: —

Route 277 — r8c7_h-1v-1

Cell 70: row 8, column 7. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 277; ordinal in the memory table: 280.

Table 94.285: Record 277: r8c7_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c7_h-1v-1; *Cell*: 70; *Row*: 8
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,3;8,7
Classification (cont.)	*Route family*: 28|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 277 — r8c7_h-1v-1: continued on the next page

Record 277 — r8c7_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000068994430825936417
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3e4f2795854d36045b55509bd916303c028de5ba24d97eef b42176631fad5420 *Reader*: 4e3bc25cd834c5379d7cd435b106743384133c2f09dc9ced7d99880b ed745a3e *Cell record at 108*: —

Route 278 — r8c7_h-1v+1

Cell 70: row 8, column 7. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 278; ordinal in the memory table: 279.

Table 94.286: Record 278: r8c7_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c7_h-1v+1; *Cell*: 70; *Row*: 8
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark

Record 278 — r8c7_h-1v+1: continued on the next page

Record 278 — r8c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,3;8,7
Classification (cont.)	*Route family*: 28|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.00000002761000615951421 5
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 278 — r8c7_h-1v+1: continued on the next page

Record 278 — r8c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: fd2d7320f5b5383ca26d8e6d224b7df36ee61afb23e51402ebf35ae0ee13a4f9 *Reader*: 3840df2d6dfaac48e51ce48ca841735ca0b7eaab8b9a9eaaa62c6ba1a7e40f39 *Cell record at 108*: —

Route 279 — r8c7_h+1v-1

Cell 70: row 8, column 7. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 279; ordinal in the memory table: 278.

Table 94.287: Record 279: r8c7_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c7_h+1v-1; *Cell*: 70; *Row*: 8
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,3;8,7
Classification (cont.)	*Route family*: 28|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17

Record 279 — r8c7_h+1v-1: continued on the next page

Record 279 — r8c7_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 20; *D4*: 24
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,24,2); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.82508593118257305617326; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027610006159514215
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3 *Stable triads per block*: 0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a15a54cb8e5aeadbf9a14d62ee241c1b33911588d46d105d0ba47dc82b7959ba *Reader*: 0919f3a4d99a776867a28a6d334ec1b717ff3b3a449176686ee79e4603cead5a *Cell record at 108*: f768a45c6e4830319012025088381e9a82811497d99ad157d5b1c623be89b25b

Route 280 — r8c7_h+1v+1

Cell 70: row 8, column 7. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 280; ordinal in the memory table: 277.

Table 94.288: Record 280: r8c7_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c7_h+1v+1; *Cell*: 70; *Row*: 8
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G3; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: G; *Multisection*: quark_down_G3
Classification (cont.)	*Physical family*: quark_down; *Depth*: G3; *Inversion orbit*: 2,3;8,7
Classification (cont.)	*Route family*: 28|-6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 64; *D4*: 68
Differences and counters (cont.)	*D27*: 60; *D36*: 40; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (100,66,2); *K* Ω: (80,54,6)

Record 280 — r8c7_h+1v+1: continued on the next page

Record 280 — r8c7_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 99.51859712327265341428729; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000068994430825936417
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 86715e7eccf63f421cf277f3abd516b226f7eb2d97fcf24b d37912af78d1d866 *Reader*: 622a5691a81501fa895d6474a6d9e5ae5acb0ef96649d46875be617d bfe36c68 *Cell record at 108*: —

Route 281 — r8c8_h-1v-1

Cell 71: row 8, column 8. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 281; ordinal in the memory table: 284.

Table 94.289: Record 281: r8c8_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c8_h-1v-1; *Cell*: 71; *Row*: 8
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 281 — r8c8_h-1v-1: continued on the next page

Record 281 — r8c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,2;8,8
Classification (cont.)	*Route family*: 24|0|8|0|-12|+++---+++---
Returns	*Class and phase*: 18; *Orbit and phase*: 18; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17

Record 281 — r8c8_h-1v-1: continued on the next page

Record 281 — r8c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 0a0cf9df7293b410440f3fb087c85c25fa5039d0fc789b9ab6bad4521fb837b5 *Reader*: 111950fefee357c0f7efaef1ee7794ea3fc454e01de39489ecfd6e18be9fdef9 *Cell record at 108*: —

Route 282 — r8c8_h-1v+1

Cell 71: row 8, column 8. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 282; ordinal in the memory table: 283.

Table 94.290: Record 282: r8c8_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c8_h-1v+1; *Cell*: 71; *Row*: 8
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,2;8,8
Classification (cont.)	*Route family*: 24|0|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 282 — r8c8_h-1v+1: continued on the next page

Record 282 — r8c8_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -4
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -4; *τ 12*: +++0-0+++0-0
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-4|2|-4
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; *Ω 90/120*: 24
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (60,84,28); *K Ω*: (24,54,4)
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ Ω*: 23.6063877469494433063621 81; *Action factor D*: 1.000000041174098946741575
Reader evaluations (cont.)	*Action factor Ω*: 1.000000016365878244659810
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 44c68f8b3fce4afc7439b1cd4add9d73ea2d68ae57cd4025f427e0da591dda73 *Reader*: 4453d1913b855774dce1ec81650654f10fd4b0c3e26ed21e8dff264854b464b5 *Cell record at 108*: —

Route 283 — r8c8_h+1v-1

Cell 71: row 8, column 8. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 283; ordinal in the memory table: 282.

Table 94.291: Record 283: r8c8_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c8_h+1v-1; *Cell*: 71; *Row*: 8
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,2;8,8
Classification (cont.)	*Route family*: 24|0|8|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 2; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 4
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -4; *τ 12*: 0+0---0+0---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|4|-2|-4
Differences and counters	*D1*: 84; *D3*: 28; *D4*: 84
Differences and counters (cont.)	*D27*: 36; *D36*: 24; Ω *90/120*: 24
Differences and counters (cont.)	*P6*: 4
Integer readers	*K D*: (60,84,28); *K* Ω: (24,54,4)

Record 283 — r8c8_h+1v-1: continued on the next page

Record 283 — r8c8_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 59.39013588401554063774148; *κ* Ω: 23.606387746949443306362 81; *Action factor D*: 1.00000004117409894674 1575
Reader evaluations (cont.)	*Action factor* Ω: 1.00000001636587824465 9810
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6ccdebcaaf311843d1d894e5d1c8dee1f317bb40cfb7fef1 455cd39b357a6807 *Reader*: cee239d95a64fd32c72002799f66be6cd6167d2f7d01c47397015660 8f61bd1a *Cell record at 108*: —

Route 284 — r8c8_h+1v+1

Cell 71: row 8, column 8. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 284; ordinal in the memory table: 281.

Table 94.292: Record 284: r8c8_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c8_h+1v+1; *Cell*: 71; *Row*: 8
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_up_G3; *Family*: quark_up; *Sector*: quark

Record 284 — r8c8_h+1v+1: continued on the next page

Record 284 — r8c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 3; *Color*: R; *Multisection*: quark_up_G3
Classification (cont.)	*Physical family*: quark_up; *Depth*: G3; *Inversion orbit*: 2,2;8,8
Classification (cont.)	*Route family*: 24|0|8|0|-12|+++---+++---
Returns	*Class and phase*: 9; *Orbit and phase*: 9; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 56; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,54,6); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.60661013979482758647091; *κ* Ω: 79.6066101397948275864791; *Action factor D*: 1.000000055189812531859140
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 72; *log2 ambiguity area*: 36.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 4
Complete sequences	*w6 blocks*: 243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16

Record 284 — r8c8_h+1v+1: continued on the next page

Record 284 — r8c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 1fef105f9462f0c629ad4505f2fa1a8a2d248638113913db d37ae66791ac19f1 *Reader*: 735623ac3cc96b608c56fc456cff66daf8b0193b82645553f1fa88ef b1ba6bed *Cell record at 108*: 45951d9d048937aae4c2cfb3a4823a1ff75bbfb0f359f391 421e6990f5bba610

Route 285 — r8c9_h-1v-1

Cell 72: row 8, column 9. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 285; ordinal in the memory table: 288.

Table 94.293: Record 285: r8c9_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c9_h-1v-1; *Cell*: 72; *Row*: 8
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,1;8,9
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 285 — r8c9_h-1v-1: continued on the next page

Record 285 — r8c9_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor Ω*: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243 *Stable triads per block*: 0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8569908f3825cdd46d2aeb615280f04f11d02c6e3d9b945b706a1953b6b91832 *Reader*: 9d065f9157694f9edc555f5e00871f36ffdfbf96d1f47fe11deda4e3a183b594 *Cell record at 108*: —

Route 286 — r8c9_h-1v+1

Cell 72: row 8, column 9. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 286; ordinal in the memory table: 287.

Table 94.294: Record 286: r8c9_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c9_h-1v+1; *Cell*: 72; *Row*: 8
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,1;8,9
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 286 — r8c9_h-1v+1: continued on the next page

Record 286 — r8c9_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: eff03ec9f7fc2cef5cb21ba6fa45a82da8267d3092637150 b953e4aa980bbe87 *Reader*: 70dd7c0eb393147851d6fd2d32d16fd7e393a5d353010311ff0ef60a a076d668 *Cell record at 108*: c7dfe852a4958d07d62f965178f30c16579632ac1d2a4764 b1535ee56b4f89ba

Route 287 — r8c9_h+1v-1

Cell 72: row 8, column 9. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 287; ordinal in the memory table: 286.

Table 94.295: Record 287: r8c9_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r8c9_h+1v-1; *Cell*: 72; *Row*: 8
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO

Record 287 — r8c9_h+1v-1: continued on the next page

Record 287 — r8c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,1;8,9
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17

Record 287 — r8c9_h+1v-1: continued on the next page

Record 287 — r8c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 678d6ab8d97c50772b92a1c0c11ae3a7bc3fa03378f49a8a322f7db0e7a20776 *Reader*: 738d138b0027417ccb40bd19ddb10609eb8c8531dffbf0c1903a319e2767add0 *Cell record at 108*: —

Route 288 — r8c9_h+1v+1

Cell 72: row 8, column 9. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 288; ordinal in the memory table: 285.

Table 94.296: Record 288: r8c9_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r8c9_h+1v+1; *Cell*: 72; *Row*: 8
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: quark_down_G4; *Family*: quark_down; *Sector*: quark
Classification (cont.)	*Generation*: 4; *Color*: B; *Multisection*: quark_down_G4
Classification (cont.)	*Physical family*: quark_down; *Depth*: G4; *Inversion orbit*: 2,1;8,9
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17

Record 288 — r8c9_h+1v+1: continued on the next page

Record 288 — r8c9_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 68
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,4); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 79.51881951611803769439539; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128948890161264
Reader evaluations (cont.)	*Action factor* Ω: 1.00000005518981253185914 0
Prefix separation	*Ambiguity area*: 40; *log2 ambiguity area*: 20.339850002884624; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 81affe3c95425c70af6e783d072ff89234b1446d1055fed3 1ee305192280fcb3 *Reader*: 41036aca4f9f8ad135146507fe31266d0452840b897d87b1a5f15e66 39516828 *Cell record at 108*: —

Route 289 — r9c1_h-1v-1

Cell 73: row 9, column 1. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 289; ordinal in the memory table: 292.

Table 94.297: Record 289: r9c1_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c1_h-1v-1; *Cell*: 73; *Row*: 9
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,9;9,1
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-2|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 289 — r9c1_h-1v-1: continued on the next page

Record 289 — r9c1_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: a0d3b90a93737ad6bd06c49cbe1e379f6e67a39e7b60dbb4e0e0b5480d1c9730 *Reader*: dd19ac7c0ec61adbef57b6ad840a051fb3b46e587098f2afe5101daba8f867b1 *Cell record at 108*: c0580ece103707e22c2b73720bc58713206cce80bca99c633743212984199b55

Route 290 — r9c1_h-1v+1

Cell 73: row 9, column 1. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 290; ordinal in the memory table: 291.

Table 94.298: Record 290: r9c1_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c1_h-1v+1; *Cell*: 73; *Row*: 9
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 290 — r9c1_h-1v+1: continued on the next page

Record 290 — r9c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,9;9,1
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 47.6064989433721354464 16 86; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Record 290 — r9c1_h-1v+1: continued on the next page

Record 290 — r9c1_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8483a8df880a8c77b15bbbdcd777ed2f7cc79e2b9d4e199dc642f2a9012041b9 *Reader*: ae1cc25e4cd5b1b87552d23b208c33dbc0ec0121a8bb290c101337e779bb5e1d *Cell record at 108*: —

Route 291 — r9c1_h+1v-1

Cell 73: row 9, column 1. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 291; ordinal in the memory table: 290.

Table 94.299: Record 291: r9c1_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c1_h+1v-1; *Cell*: 73; *Row*: 9
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,9;9,1
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 3; *Final stable triads*: 16

Record 291 — r9c1_h+1v-1: continued on the next page

Record 291 — r9c1_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 52b846cd7eaeff31abb00f1ae440a22d9b65cc73fa2cc099196a0ef20d97df3c *Reader*: 3db7a708d8ae2238a749a75623f90c80ccd6b52fa0c36042a443240c6f9ebfc0 *Cell record at 108*: —

Route 292 — r9c1_h+1v+1

Cell 73: row 9, column 1. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 292; ordinal in the memory table: 289.

Table 94.300: Record 292: r9c1_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c1_h+1v+1; *Cell*: 73; *Row*: 9
Identity and orientation (cont.)	*Column*: 1; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,9;9,1
Classification (cont.)	*Route family*: 28|6|0|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|2|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 292 — r9c1_h+1v+1: continued on the next page

Record 292 — r9c1_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 905ded93d9b0764d207049712252c1ffb8506d948be0960f 3b6e398caa4a8e25 *Reader*: ddb8562334314015284c5247ec81419d909c46c43781c41745de130f 32a0e193 *Cell record at 108*: —

Route 293 — r9c2_h-1v-1

Cell 74: row 9, column 2. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 293; ordinal in the memory table: 296.

Table 94.301: Record 293: r9c2_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c2_h-1v-1; *Cell*: 74; *Row*: 9
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 293 — r9c2_h-1v-1: continued on the next page

Record 293 — r9c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,8;9,2
Classification (cont.)	*Route family*: 28|-6|6|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++--0+++--0
Signature and phase (cont.)	*Historical signature Z5*: 28|6|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464 16 86; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 40; *log2 ambiguity area*: 20.339850002884624; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Record 293 — r9c2_h-1v-1: continued on the next page

Record 293 — r9c2_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: da6667267b4ff85fea2f0ff5b28c8962cba978a6c2329bf58403d88aaab3b865 *Reader*: 47731741df561e96ed208b391e19d518430489a7ef9c6d05607c164114f5d9a2 *Cell record at 108*: —

Route 294 — r9c2_h-1v+1

Cell 74: row 9, column 2. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 294; ordinal in the memory table: 295.

Table 94.302: Record 294: r9c2_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c2_h-1v+1; *Cell*: 74; *Row*: 9
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,8;9,2
Classification (cont.)	*Route family*: 28|-6|6|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 294 — r9c2_h-1v+1: continued on the next page

Record 294 — r9c2_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 46e22be8f2160920042cedaf06032c82685a879d1f441907912b9a00df9114a3 *Reader*: a7ff5decdcbda0dc9f2f3c09ec8cdeac01d3a3ad66cb4dfed656bf970900d4bb *Cell record at 108*: 5ffebc4fdd94691deee35c6d4dcacdeec958fddf4a3daff598d7a1a11d1360c0

Route 295 — r9c2_h+1v-1

Cell 74: row 9, column 2. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 295; ordinal in the memory table: 294.

Table 94.303: Record 295: r9c2_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c2_h+1v-1; *Cell*: 74; *Row*: 9
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,8;9,2
Classification (cont.)	*Route family*: 28|-6|6|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 295 — r9c2_h+1v-1: continued on the next page

Record 295 — r9c2_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: c3161556f17c0edfadbc95693b722fde3bda92e61c398a497d9ea6f4f45c21ff *Reader*: 0b2ce98332ac4c69c985a795b31e585c57e037a8fbbe9508bf72fe5917ac5e28 *Cell record at 108*: —

Route 296 — r9c2_h+1v+1

Cell 74: row 9, column 2. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 296; ordinal in the memory table: 293.

Table 94.304: Record 296: r9c2_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c2_h+1v+1; *Cell*: 74; *Row*: 9
Identity and orientation (cont.)	*Column*: 2; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 296 — r9c2_h+1v+1: continued on the next page

Record 296 — r9c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,8;9,2
Classification (cont.)	*Route family*: 28|-6|6|-2|-8|++0---++0---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: ++0---++0---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464 16 86; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 296 — r9c2_h+1v+1: continued on the next page

Record 296 — r9c2_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 463b5be423eaf1c50aa2a7f5f851d35daf48aacc82ab25714a83f2840170d00c *Reader*: 3e4d219163e182c8401f7d7dfe984bfad7fb28bba4b15fda36d17bddb01bcd23 *Cell record at 108*: —

Route 297 — r9c3_h-1v-1

Cell 75: row 9, column 3. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 297; ordinal in the memory table: 300.

Table 94.305: Record 297: r9c3_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c3_h-1v-1; *Cell*: 75; *Row*: 9
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,7;9,3
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 297 — r9c3_h-1v-1: continued on the next page

Record 297 — r9c3_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 30|6|-2|4|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 55.6064989433721354464186; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8044e5917e13ccfbda2ceb1f21613f7ce52eaa0853ec71fd75f3220376d19cc0 *Reader*: bb47bdfe5cf32f87265f7ab2c767c46440e85eefa678090e2f952f4504fe98c1 *Cell record at 108*: —

Route 298 — r9c3_h-1v+1

Cell 75: row 9, column 3. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 298; ordinal in the memory table: 299.

Table 94.306: Record 298: r9c3_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c3_h-1v+1; *Cell*: 75; *Row*: 9
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,7;9,3
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|0++---0++---
Returns	*Class and phase*: 9; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 50|-6|2|-4|-8
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (0,24,0); *K Ω*: (56,54,5)

Record 298 — r9c3_h-1v+1: continued on the next page

Record 298 — r9c3_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 55.6064989433721354464168 6; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 108; *log2 ambiguity area*: 46.52932501298079; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 6
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b1f5d9047f6748335b6f26efffc458e4baa8c0f8a8dfa8801b0cd5071ee377e9 *Reader*: e138662cf0ef5ed672f884e44662fd4346e0c345559c2c5845dc6dc30a15b0db *Cell record at 108*: —

Route 299 — r9c3_h+1v-1

Cell 75: row 9, column 3. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 299; ordinal in the memory table: 298.

Table 94.307: Record 299: r9c3_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c3_h+1v-1; *Cell*: 75; *Row*: 9
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 299 — r9c3_h+1v-1: continued on the next page

Record 299 — r9c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,7;9,3
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|0++---0++---
Returns	*Class and phase*: 18; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 50|6|-2|4|-8
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (0,24,0); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: -0.17513646166281122393484 25; *κ Ω*: 55.606498943372135446416 86; *Action factor D*: 0.99999999987858108513378 81
Reader evaluations (cont.)	*Action factor Ω*: 1.00000003855097235414164 5
Prefix separation	*Ambiguity area*: 108; *log2 ambiguity area*: 46.52932501298079; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 6
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 299 — r9c3_h+1v-1: continued on the next page

Record 299 — r9c3_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3a549e8092847f10e437f1add78c27b8119bea4b98466f1a495c7fe837e526ab *Reader*: 75ff441007b14e7d209d0846ef9fadf49bdbc245749a8b5bcddb11c723234236 *Cell record at 108*: —

Route 300 — r9c3_h+1v+1

Cell 75: row 9, column 3. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 300; ordinal in the memory table: 297.

Table 94.308: Record 300: r9c3_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c3_h+1v+1; *Cell*: 75; *Row*: 9
Identity and orientation (cont.)	*Column*: 3; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,7;9,3
Classification (cont.)	*Route family*: 30|-6|0|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 300 — r9c3_h+1v+1: continued on the next page

Record 300 — r9c3_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|2|-4|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 36d1117b7f4f744e498155bbc7dc3f337e038bc6a59df945b16955b65617a24f *Reader*: 2822864e6d668c0d559901e93691fe2ff2784b9d82824b2d5fe3c593e5cc387d *Cell record at 108*: 55065291299c2ae2d06f2baa5791c0319edfbd413c3e9c7a8b949c3352304579

Route 301 — r9c4_h-1v-1

Cell 76: row 9, column 4. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 301; ordinal in the memory table: 304.

Table 94.309: Record 301: r9c4_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c4_h-1v-1; *Cell*: 76; *Row*: 9
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,6;9,4
Classification (cont.)	*Route family*: 34|0|-12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 301 — r9c4_h-1v-1: continued on the next page

Record 301 — r9c4_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ee798204ea997ed7ed2199e7b3ab7d6137f5ac35f07e503f 6c0a8edae955b848 *Reader*: 18fd69ac1d8dc23d374c9d4b781e339c278b6935a74bb76c1a2badda cb2b6fea *Cell record at 108*: —

Route 302 — r9c4_h-1v+1

Cell 76: row 9, column 4. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 302; ordinal in the memory table: 303.

Table 94.310: Record 302: r9c4_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c4_h-1v+1; *Cell*: 76; *Row*: 9
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4

Record 302 — r9c4_h-1v+1: continued on the next page

Record 302 — r9c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,6;9,4
Classification (cont.)	*Route family*: 34|0|-12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|-2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ

Record 302 — r9c4_h-1v+1: continued on the next page

Record 302 — r9c4_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 642cd12482d3d647cdaec0acdd19e9c1f2a7429675270183ed630caea3cf535a *Reader*: e7f63cff972e6706fe7c8ecd667eb3e076106f380840daddbcaf4c59501a2e78 *Cell record at 108*: 11028c389e35d609ddb9aa1248b79b77987b41783a58974a3c6e6873c14e1f88

Route 303 — r9c4_h+1v-1

Cell 76: row 9, column 4. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 303; ordinal in the memory table: 302.

Table 94.311: Record 303: r9c4_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c4_h+1v-1; *Cell*: 76; *Row*: 9
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,6;9,4
Classification (cont.)	*Route family*: 34|0|-12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 3; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 8; *raw s C*: 2; *raw κ 0*: 0

Record 303 — r9c4_h+1v-1: continued on the next page

Record 303 — r9c4_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 8|2|0|0|-12
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,84,30); *K* Ω: (80,54,6)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ff0027584bafb795fea0705bea171be417920f7ab7cb5f37 8d17b7e40adbb917 *Reader*: 7117e0fdd957f42a5d46ef727105788a73ce6a863be2fd32f11bab0a df65bbc6 *Cell record at 108*: —

Route 304 — r9c4_h+1v+1

Cell 76: row 9, column 4. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 304; ordinal in the memory table: 301.

Table 94.312: Record 304: r9c4_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c4_h+1v+1; *Cell*: 76; *Row*: 9
Identity and orientation (cont.)	*Column*: 4; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,6;9,4
Classification (cont.)	*Route family*: 34|0|-12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|0|0|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 304 — r9c4_h+1v+1: continued on the next page

Record 304 — r9c4_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b5fc9a01e3980db070dd783cd2fcc157585ffca7f68f54ea c0c828e4aeedf880 *Reader*: 5b3951f3240c6ee934749d6dc143cce4f650ab607b64271814be5fd6 7595c351 *Cell record at 108*: —

Route 305 — r9c5_h-1v-1

Cell 77: row 9, column 5. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 305; ordinal in the memory table: 308.

Table 94.313: Record 305: r9c5_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c5_h-1v-1; *Cell*: 77; *Row*: 9
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 305 — r9c5_h-1v-1: continued on the next page

Record 305 — r9c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,5;9,5
Classification (cont.)	*Route family*: 34|0|-8|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 34|0|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464186; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 30; *log2 ambiguity area*: 8.0; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Record 305 — r9c5_h-1v-1: continued on the next page

Record 305 — r9c5_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 77908e472c16ed99b259dbc397c2623e67f0871ad4764417423d1c62bd7e5dfa *Reader*: 3d68025ffbb2f16ffc10253200612155c3edd2545ed08c9acf932f34b8417dea *Cell record at 108*: ee7f57b43ff6bbe2b8b762aec3be88c2f3eec4cac943953bf26b06adfc1e2088

Route 306 — r9c5_h-1v+1

Cell 77: row 9, column 5. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 306; ordinal in the memory table: 307.

Table 94.314: Record 306: r9c5_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c5_h-1v+1; *Cell*: 77; *Row*: 9
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,5;9,5
Classification (cont.)	*Route family*: 34|0|-8|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 306 — r9c5_h-1v+1: continued on the next page

Record 306 — r9c5_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: 2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6d009acb2e4a9062d353d9d3d3b9fabbdd58df3dfffdf48f f543302954972595 *Reader*: b47145e5c4d4c54c6e7bcbdeb698559de66f30511a1ceb50d00e80be 87055bf3 *Cell record at 108*: —

Route 307 — r9c5_h+1v-1

Cell 77: row 9, column 5. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 307; ordinal in the memory table: 306.

Table 94.315: Record 307: r9c5_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c5_h+1v-1; *Cell*: 77; *Row*: 9
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,5;9,5
Classification (cont.)	*Route family*: 34|0|-8|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 44; *raw s C*: -2; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 44|-2|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 307 — r9c5_h+1v-1: continued on the next page

Record 307 — r9c5_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: b685f1413fffbb6cc5e0b9a15517eac78ab70bc66c1db71b3ab13da31d17dc60 *Reader*: db33053c3c7df39bb195f28eadc06b927e42eeddc3e9b8a8f84cf04b4d72ac5b *Cell record at 108*: —

Route 308 — r9c5_h+1v+1

Cell 77: row 9, column 5. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 308; ordinal in the memory table: 305.

Table 94.316: Record 308: r9c5_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c5_h+1v+1; *Cell*: 77; *Row*: 9
Identity and orientation (cont.)	*Column*: 5; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 308 — r9c5_h+1v+1: continued on the next page

Record 308 — r9c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,5;9,5
Classification (cont.)	*Route family*: 34|0|-8|-2|-8|0++---0++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 34; *raw s C*: 0; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 34|0|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (56,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 55.6064989433721354464186; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 308 — r9c5_h+1v+1: continued on the next page

Record 308 — r9c5_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 75c7d91367c880ed973e5c536d4f7b3ddbb4d6a6c0478a776fb9250721a22979 *Reader*: 8a5cc992862f3e41a477ea869ccbabd19b444a8876e6e3b1b084264cab2e4db1 *Cell record at 108*: —

Route 309 — r9c6_h-1v-1

Cell 78: row 9, column 6. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 309; ordinal in the memory table: 312.

Table 94.317: Record 309: r9c6_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c6_h-1v-1; *Cell*: 78; *Row*: 9
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,4;9,6
Classification (cont.)	*Route family*: 30|6|0|0|-8|+++-0-+++-0-
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 309 — r9c6_h-1v-1: continued on the next page

Record 309 — r9c6_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 30|-6|2|0|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K Ω*: (48,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 47.6064989433721354464186; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 18; *log2 ambiguity area*: 0.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: 1; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ca419a4fb89079e264213c9b485a320a311c24d208068ee41866ec2ac1b43c2c *Reader*: a382f2951d5ce06beded8af0111b7528cf000d7719f0431752c134cdd274a383 *Cell record at 108*: —

Route 310 — r9c6_h-1v+1

Cell 78: row 9, column 6. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 310; ordinal in the memory table: 311.

Table 94.318: Record 310: r9c6_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c6_h-1v+1; *Cell*: 78; *Row*: 9
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,4;9,6
Classification (cont.)	*Route family*: 30|6|0|0|-8|+++-0-+++-0-
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (0,24,0); *K Ω*: (80,54,6)

Record 310 — r9c6_h-1v+1: continued on the next page

Record 310 — r9c6_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 106b15fef86d47d98a2151f4c2319629333008c764daa083780d58ab178d9ec0 *Reader*: 97e31ef7a8d5ddaf45317072ce7a9bc90b1be5f32591931c1267467013eddfa2 *Cell record at 108*: —

Route 311 — r9c6_h+1v-1

Cell 78: row 9, column 6. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 311; ordinal in the memory table: 310.

Table 94.319: Record 311: r9c6_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c6_h+1v-1; *Cell*: 78; *Row*: 9
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4

Record 311 — r9c6_h+1v-1: continued on the next page

Record 311 — r9c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,4;9,6
Classification (cont.)	*Route family*: 30|6|0|0|-8|+++-0-+++-0-
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-6|0|0|-12
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (0,24,0); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: 1
Prefix separation (cont.)	*First cell-separation block*: 1; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ

Record 311 — r9c6_h+1v-1: continued on the next page

Record 311 — r9c6_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 59a1b925f37b198a9b8d28bf47609299f18ecacece1e78e313294726444a07ed *Reader*: 82e2ed9e32b01eb9363063f95aa0e97a797114c21c1a7b3ae4bdd8f720fbfc4f *Cell record at 108*: 915f24481eb9d51115cf664e358374d4a4585b514ac886dda764c433b1d9a456

Route 312 — r9c6_h+1v+1

Cell 78: row 9, column 6. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 312; ordinal in the memory table: 309.

Table 94.320: Record 312: r9c6_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c6_h+1v+1; *Cell*: 78; *Row*: 9
Identity and orientation (cont.)	*Column*: 6; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,4;9,6
Classification (cont.)	*Route family*: 30|6|0|0|-8|+++-0-+++-0-
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 30; *raw s C*: 6; *raw κ 0*: -2

Record 312 — r9c6_h+1v+1: continued on the next page

Record 312 — r9c6_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 30|6|-2|0|-8
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,54,6); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ* Ω: 47.606498943372135446416 86; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 26; *log2 ambiguity area*: 6.339850002884624; *First orbit-separation block*: 5
Prefix separation (cont.)	*First cell-separation block*: 5; *First route-separation block*: 5; *Final ambiguity*: 1
Complete sequences	*w6 blocks*: 24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 6159b0ea3949ae42d6ac44d167a4d71dd9d0ca4f91305deda987869aff7a5301 *Reader*: 2952410fbf2ee021f5736f19d28673e6bcf3f0867bc4b89a48f77673b413ea76 *Cell record at 108*: —

Route 313 — r9c7_h-1v-1

Cell 79: row 9, column 7. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 313; ordinal in the memory table: 316.

Table 94.321: Record 313: r9c7_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c7_h-1v-1; *Cell*: 79; *Row*: 9
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,3;9,7
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 18; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 1; *Final stable triads*: 16
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: -4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|0|-4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 313 — r9c7_h-1v-1: continued on the next page

Record 313 — r9c7_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_ READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_ REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8684eaf7f9ac68574b1177dd46056f16497b29a210b4fc30 5b69f129f3516cf0 *Reader*: e45675ccb2ead11c246854fd23eab5479d5768fdfb30d0a82cc741ae 6be60f35 *Cell record at 108*: 23ef2c5cd56b7ff35c013077bee1cb8dbd0ee658e4c9ec23 93eb87b4c365f550

Route 314 — r9c7_h-1v+1

Cell 79: row 9, column 7. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 314; ordinal in the memory table: 315.

Table 94.322: Record 314: r9c7_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c7_h-1v+1; *Cell*: 79; *Row*: 9
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 314 — r9c7_h-1v+1: continued on the next page

Record 314 — r9c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,3;9,7
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: 4; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 2; *raw ν 270*: -8; *τ 12*: +++0--+++0--
Signature and phase (cont.)	*Historical signature Z5*: 50|4|-2|2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 55.6064989433721354464 16 86; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17

Record 314 — r9c7_h-1v+1: continued on the next page

Record 314 — r9c7_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 0392a0bd824d4db3872ee7e55a8e5e59e3b21f749916b615dad01451654991d0 *Reader*: 93e0fc9408e34fa48b346d2aa179d966af4513b405ea24a268f6ebd48d236116 *Cell record at 108*: —

Route 315 — r9c7_h+1v-1

Cell 79: row 9, column 7. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 315; ordinal in the memory table: 314.

Table 94.323: Record 315: r9c7_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c7_h+1v-1; *Cell*: 79; *Row*: 9
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,3;9,7
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 3
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 3; *Final stable triads*: 16

Record 315 — r9c7_h+1v-1: continued on the next page

Record 315 — r9c7_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 50; *raw s C*: -4; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: -2; *raw ν 270*: -8; *τ 12*: 0++---0++---
Signature and phase (cont.)	*Historical signature Z5*: 50|-4|2|-2|-8
Differences and counters	*D1*: 84; *D3*: 24; *D4*: 84
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 56
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (40,84,30); *K Ω*: (56,54,5)
Reader evaluations	*κ D*: 39.39035827686092491784959; *κ Ω*: 55.6064989433721354464168 6; *Action factor D*: 1.000000027308617396708705
Reader evaluations (cont.)	*Action factor Ω*: 1.000000038550972354141645
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 15c25fad37a466a9092234de4e51eff214bb454130f89e920ac3eabd0af86e23 *Reader*: aeafcc7a88835b5ec7f511281b2c9269e769f3ff86a4680e2e29a9866f3cabc4 *Cell record at 108*: —

Route 316 — r9c7_h+1v+1

Cell 79: row 9, column 7. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 316; ordinal in the memory table: 313.

Table 94.324: Record 316: r9c7_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c7_h+1v+1; *Cell*: 79; *Row*: 9
Identity and orientation (cont.)	*Column*: 7; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,3;9,7
Classification (cont.)	*Route family*: 28|6|-2|0|-12|+++---+++---
Returns	*Class and phase*: 9; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 4; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 28|6|0|4|-12
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (80,66,3); *K* Ω: (80,54,6)

Record 316 — r9c7_h+1v+1: continued on the next page

Record 316 — r9c7_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 54; *log2 ambiguity area*: 28.5293250129808; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8e4b3fc25915492780bec7f316245b548a5a96f4f38c5037 b6ad8f9460591d64 *Reader*: d78bc1389070695902c2455878b04823455e0d80836529f9e8807940 4ea54b23 *Cell record at 108*: —

Route 317 — r9c8_h-1v-1

Cell 80: row 9, column 8. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 317; ordinal in the memory table: 320.

Table 94.325: Record 317: r9c8_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c8_h-1v-1; *Cell*: 80; *Row*: 9
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 317 — r9c8_h-1v-1: continued on the next page

Record 317 — r9c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,2;9,8
Classification (cont.)	*Route family*: 28|-6|4|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: 6; *raw κ 0*: -2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +++-0-+++-0-
Signature and phase (cont.)	*Historical signature Z5*: 28|6|-2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 47.6064989433721354464168 6; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 40; *log2 ambiguity area*: 20.339850002884624; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17

Record 317 — r9c8_h-1v-1: continued on the next page

Record 317 — r9c8_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: ce9297be674940a68f160eee97674c669a846625821f53c9ca30b05bd0b56292 *Reader*: 6ef27d7457d2c0a151c7c2cc86034c0ab1ce15e774f84ed5c64f617e44ec520f *Cell record at 108*: —

Route 318 — r9c8_h-1v+1

Cell 80: row 9, column 8. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 318; ordinal in the memory table: 319.

Table 94.326: Record 318: r9c8_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c8_h-1v+1; *Cell*: 80; *Row*: 9
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,2;9,8
Classification (cont.)	*Route family*: 28|-6|4|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 318 — r9c8_h-1v+1: continued on the next page

Record 318 — r9c8_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: -4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|-4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ Ω*: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 16995ba1149d7ba0bc3cfc7743494696d33ba118e19e0ecc76f644d31711ba9f *Reader*: bf91d1ae5ba472dde0f327f9602e69918e8799f0e537555d5094f8ab0aac97ca *Cell record at 108*: d2a88d330a7badcf0a410fefd82f9b474dec91ba1dbe5a7b22a91f36905d51f7

Route 319 — r9c8_h+1v-1

Cell 80: row 9, column 8. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 319; ordinal in the memory table: 318.

Table 94.327: Record 319: r9c8_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c8_h+1v-1; *Cell*: 80; *Row*: 9
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,2;9,8
Classification (cont.)	*Route family*: 28|-6|4|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 14; *raw s C*: 4; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 14|4|0|0|-12
Differences and counters	*D1*: 78; *D3*: 24; *D4*: 78
Differences and counters (cont.)	*D27*: 24; *D36*: 16; Ω *90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,78,27); *K* Ω: (80,54,6)

Record 319 — r9c8_h+1v-1: continued on the next page

Record 319 — r9c8_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Reader evaluations	*κ D*: 39.43380880300855130367114; *κ* Ω: 79.606610139794827586470 91; *Action factor D*: 1.000000027338740854894358
Reader evaluations (cont.)	*Action factor* Ω: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 2231f178c6f579c0407e70bcbff9ed750acd6efb3a4dd638 1f3268e815826311 *Reader*: 810d7ebd6b5ee7a72a6dab271c40aef23989480802df5c3c6064b00f 893231ca *Cell record at 108*: —

Route 320 — r9c8_h+1v+1

Cell 80: row 9, column 8. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 320; ordinal in the memory table: 317.

Table 94.328: Record 320: r9c8_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c8_h+1v+1; *Cell*: 80; *Row*: 9
Identity and orientation (cont.)	*Column*: 8; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: neutrino_G4; *Family*: neutrino; *Sector*: lepton

Record 320 — r9c8_h+1v+1: continued on the next page

Record 320 — r9c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: neutrino_G4
Classification (cont.)	*Physical family*: neutrino; *Depth*: G4; *Inversion orbit*: 1,2;9,8
Classification (cont.)	*Route family*: 28|-6|4|0|-8|+0+---+0+---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 28; *raw s C*: -6; *raw κ 0*: 2
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -8; *τ 12*: +0+---+0+---
Signature and phase (cont.)	*Historical signature Z5*: 28|-6|2|0|-8
Differences and counters	*D1*: 66; *D3*: 60; *D4*: 66
Differences and counters (cont.)	*D27*: 48; *D36*: 32; Ω *90/120*: 48
Differences and counters (cont.)	*P6*: 5
Integer readers	*K D*: (80,66,3); *K* Ω: (48,54,5)
Reader evaluations	*κ D*: 79.51870831969534555434134; *κ* Ω: 47.6064989433721354464186; *Action factor D*: 1.000000055128871799705072
Reader evaluations (cont.)	*Action factor* Ω: 1.000000033004718053243084
Prefix separation	*Ambiguity area*: 36; *log2 ambiguity area*: 18.0; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 2
Complete sequences	*w6 blocks*: 498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 320 — r9c8_h+1v+1: continued on the next page

Record 320 — r9c8_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: True *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 7c9b0456e1d3dbd9f89b99611a50f337eefa938f02f234c4a7ad065fd62a6692 *Reader*: ec678d84ce11dc61e52a24438e4f523b4d0ea94b94d28d2c805f94c560a94223 *Cell record at 108*: —

Route 321 — r9c9_h-1v-1

Cell 81: row 9, column 9. Orientation h=-1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 321; ordinal in the memory table: 324.

Table 94.329: Record 321: r9c9_h-1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c9_h-1v-1; *Cell*: 81; *Row*: 9
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,1;9,9
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 1
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 1; *Final stable triads*: 17

Record 321 — r9c9_h-1v-1: continued on the next page

Record 321 — r9c9_h-1v-1 (h=-1, v=-1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,54,6); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 79.6066101397948275864791; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 3e3bf9f3b35f1316f0d242822cae4fe211eb31595576f51d063c36e241c4cc99 *Reader*: 1f648323cbeef43af2c169aee5b6a999302aca3e106e2780e370b1480e3ee9e3 *Cell record at 108*: —

Route 322 — r9c9_h-1v+1

Cell 81: row 9, column 9. Orientation h=-1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 322; ordinal in the memory table: 323.

Table 94.330: Record 322: r9c9_h-1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c9_h-1v+1; *Cell*: 81; *Row*: 9
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: -1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: YES
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,1;9,9
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: 0+00-00+00-0
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; *Ω 90/120*: 24
Differences and counters (cont.)	*P6*: 2
Integer readers	*K D*: (0,24,0); *K Ω*: (24,54,2)

Record 322 — r9c9_h-1v+1: continued on the next page

Record 322 — r9c9_h-1v+1 (h=-1, v=1) — Continuation

Group	Route data
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 23.606165354104059026254 71; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000016365724063753403
Prefix separation	*Ambiguity area*: 108; *log2 ambiguity area*: 46.52932501298079; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 6
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 5ca5179e96a7a217149fc629598911284a679d874c335483825169c08628742b *Reader*: a1fd3c287cd8f05a77fc26d86500b68a6f6d45a515b1bfc32c7201ea592f831a *Cell record at 108*: 373f8cf1d27efd3b616a88ca8eefef4203f804fab97be9f4d60cd5a3f094d407

Route 323 — r9c9_h+1v-1

Cell 81: row 9, column 9. Orientation h=1, v=-1.

Return to the index of identifiers. Ordinal in the unified table: 323; ordinal in the memory table: 322.

Table 94.331: Record 323: r9c9_h+1v-1

Group	Route data
Identity and orientation	*Identifier*: r9c9_h+1v-1; *Cell*: 81; *Row*: 9
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: -1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton

Record 323 — r9c9_h+1v-1: continued on the next page

Record 323 — r9c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,1;9,9
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 0
Memory and transport (cont.)	*Final debt*: 0; *Deferred blocks*: 0; *Final stable triads*: 17
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 20; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -4; *τ 12*: 0+00-00+00-0
Signature and phase (cont.)	*Historical signature Z5*: 20|0|0|0|-4
Differences and counters	*D1*: 24; *D3*: 24; *D4*: 24
Differences and counters (cont.)	*D27*: 0; *D36*: 0; Ω *90/120*: 24
Differences and counters (cont.)	*P6*: 2
Integer readers	*K D*: (0,24,0); *K* Ω: (24,54,2)
Reader evaluations	*κ D*: -0.175136461662811223934842 5; *κ* Ω: 23.606165354104059026254 71; *Action factor D*: 0.999999999878581085133788 1
Reader evaluations (cont.)	*Action factor* Ω: 1.000000016365724063753403
Prefix separation	*Ambiguity area*: 108; *log2 ambiguity area*: 46.52932501298079; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 6
Complete sequences	*w6 blocks*: 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Record 323 — r9c9_h+1v-1: continued on the next page

Record 323 — r9c9_h+1v-1 (h=1, v=-1) — Continuation

Group	Route data
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: e847061b4af737e22cde7449fd8c7399b3568176df3655b09f776b95b8d09a80 *Reader*: 1f6b0ca611906a4828e0616ba7456a15244cef42f9db847f10419b48f959a5cd *Cell record at 108*: —

Route 324 — r9c9_h+1v+1

Cell 81: row 9, column 9. Orientation h=1, v=1.

Return to the index of identifiers. Ordinal in the unified table: 324; ordinal in the memory table: 321.

Table 94.332: Record 324: r9c9_h+1v+1

Group	Route data
Identity and orientation	*Identifier*: r9c9_h+1v+1; *Cell*: 81; *Row*: 9
Identity and orientation (cont.)	*Column*: 9; *Horizontal orientation*: 1; *Vertical orientation*: 1
Identity and orientation (cont.)	*Membership in B1*: NO
Classification	*Class*: charged_lepton_G4; *Family*: charged_lepton; *Sector*: lepton
Classification (cont.)	*Generation*: 4; *Color*: —; *Multisection*: charged_lepton_G4
Classification (cont.)	*Physical family*: charged_lepton; *Depth*: G4; *Inversion orbit*: 1,1;9,9
Classification (cont.)	*Route family*: 24|0|12|0|-12|+++---+++---
Returns	*Class and phase*: 27; *Orbit and phase*: 27; *Cell and phase*: 27
Returns (cont.)	*Kinematic state at 108*: 54; *Kinematic state, extension*: 216
Memory and transport	*Turns*: 71; *Sheet changes*: 107; *Total debt*: 2
Memory and transport (cont.)	*Final debt*: 1; *Deferred blocks*: 2; *Final stable triads*: 16

Record 324 — r9c9_h+1v+1: continued on the next page

Record 324 — r9c9_h+1v+1 (h=1, v=1) — Continuation

Group	Route data
Memory and transport (cont.)	*Repetitions at K and K+9*: 9; *Memory changes at K and K+9*: 9
Signature and phase	*raw n A*: 24; *raw s C*: 0; *raw κ 0*: 0
Signature and phase (cont.)	*raw ν 120*: 0; *raw ν 270*: -12; *τ 12*: +++---+++---
Signature and phase (cont.)	*Historical signature Z5*: 24|0|0|0|-12
Differences and counters	*D1*: 54; *D3*: 60; *D4*: 72
Differences and counters (cont.)	*D27*: 24; *D36*: 16; *Ω 90/120*: 80
Differences and counters (cont.)	*P6*: 6
Integer readers	*K D*: (40,54,6); *K Ω*: (80,54,6)
Reader evaluations	*κ D*: 39.60661013979482758647091; *κ Ω*: 79.6066101397948275864709 1; *Action factor D*: 1.000000027458540873559518
Reader evaluations (cont.)	*Action factor Ω*: 1.000000055189812531859140
Prefix separation	*Ambiguity area*: 58; *log2 ambiguity area*: 30.189475010096174; *First orbit-separation block*: —
Prefix separation (cont.)	*First cell-separation block*: —; *First route-separation block*: —; *Final ambiguity*: 3
Complete sequences	*w6 blocks*: 24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423 *Stable triads per block*: 0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Documented states	*Route status*: TARGET_FREE_INTERNAL_ROUTE *Orbit Pareto persistence*: False *Correspondence status*: EXACT_KEY_JOIN__NO_EXTERNAL_MASS_READ
Documented states (cont.)	*Tower status*: ROUTE_MEMORY_AVAILABLE__ETA_TRANSDUCER_REQUIRES_CERTIFICATE
Provenance fingerprints	*Route genealogy*: 8bd3961b89885d9bc5b76d4d18cbbc1bf791e769482c1949918d507488e02251 *Reader*: 871508bdd95895803d299de36cd7cd3e9eade5f22157f1599f294895163b56ce *Cell record at 108*: —

94.18 Metrological inventory: 471 masses and 384 widths

The following tables are a catalog of external observations and not a count of HMT predictions. They preserve the PDG 2026 documentary snapshot dated `2026-01-15`, as it appears in the preserved files. The consultation date recorded in the integral is `2026-07-26`. The edition has not been updated nor have its observables been reinterpreted during this transcription. The stable identifier distinguishes different records of the same species; repetitions are not removed, nor are mass, pole, width or estimate merged.

The value column preserves both the literal expression and, when different, its original presentation. An uncertainty is not reconstructed from decimals, nor is an interval transformed into a standard deviation. The positive and negative uncertainties are kept separate, along with the unit, the limit type and the available confidence level. `ND` abbreviates `UNAVAILABLE`; `NA` abbreviates `NOT_APPLICABLE`. A dash denotes an empty field. The original types `NONE`, `R` and `U` are preserved as source codes, alongside the value expression: respectively, absence of a limit marker, range and upper limit. A confidence level is not inferred where the record does not give one.

In this snapshot, the 855 records indicate that scheme and scale are not available in the snapshot. All retain the condition of comparison after the internal result is fixed and declare that the external value was not used as a selector. The status keys *E* and link keys *M* are reversible editorial abbreviations of the source's states and maps, described in the two following legends. The word "open" in those legends reproduces the documentary state of that snapshot; it is not a new mathematical review.

Table 94.333: Preserved states of the external inventory.

Key	Estado literal	Lectura documental
E01	CONDITIONED_NOMINAL_ SIGNATURE__ROUTE_AND_POLE_ REALIZATION_OPEN	Conditional nominal signature; route and pole realization open.
E02	EXACT_COLOR_FIBER_ OPERATOR__NOMINAL_ SIGNATURE_CONDITIONED__ SCHEME_OPEN	Exact color-fiber operator; conditional nominal signature; scheme open.
E03	EXACT_FIBER_OPERATOR__ NOMINAL_SIGNATURE_ CONDITIONED__SCALARIZATION_ OPEN	Exact fiber operator; conditional nominal signature; scalarization open.
E04	EXTERNAL_RECORD_ ACCOUNTED__INDIVIDUAL_HMT_ REALIZATION_NOT_YET_JOINED	External record counted; individual HMT realization not yet linked in this snapshot.
E05	NULL_SECTION_CLOSED	Null section closed.
E06	SCALAR_CLOSED_ELECTRON_ BETA	Electronic beta scalar closed.

Table 94.334: Realization links recorded by the inventory.

Key	Estado literal	Lectura documental
M01	NONE	None, according to the record.
M02	NONE_FOR_MASS_NULLITY	None for mass nullity, according to the record.
M03	flavor_color_coupling_to_spectral_projection_then_scheme_scale_transport	Flavor–color coupling to the spectral projection and scheme and scale transport.
M04	generation_coupling_to_spectral_projection_without_mass_target	Generational coupling to the spectral projection without a target mass.
M05	identity_to_public_sector_or_new_typed_sector	Identity and public sector or new typed sector.
M06	join_named_boson_to_enriched_route_then_complex_pole_operator	Individualized boson, route with memory and complex pole operator.
M07	named_baryon_binding_map	Binding link of the individualized baryon.
M08	named_meson_binding_map	Binding link of the individualized meson.
M09	quark_identity_scheme_scale_map	Quark identity, scheme and scale.
M10	typed_gravitational_representation_map	Typed gravitational representation.

94.18.1 Masas: 471 registros

Table 94.335: Complete inventory of 471 masses.

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
1. PDG2026:observable: B000:B000M:349891 Sigma(1580) MASS	Sigma(1580)- | Sigmabar(1580)+ | Sigma(1580)0 | Sigmabar(1580)0 | Sigma(1580)+ | Sigmabar(1580)- COMP-BARYON E04; M07	~1580 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
2. PDG2026:observable: B004:B004M:348705 N(1895) BREIT-WIGNER MASS	N(1895)0 | Nbar(1895)0 | N(1895)+ | Nbar(1895)- COMP-BARYON E04; M07	1870 to 1895 to 1920 MeV Presentation: 1870 to 1920 (~1895)	u_+: NA u_-: NA Type: R Confidence: ND
3. PDG2026:observable: B005:B005M:348836 N(2060) BREIT-WIGNER MASS	N(2060)0 | Nbar(2060)0 | N(2060)+ | Nbar(2060)- COMP-BARYON E04; M07	2030 to 2100 to 2200 MeV Presentation: 2030 to 2200 (~2100)	u_+: NA u_-: NA Type: R Confidence: ND
4. PDG2026:observable: B007:B007M:349780 Lambda(2585) MASS (BUMPS)	Lambda(2585)0 | Lambdabar(2585)0 COMP-BARYON E04; M07	~2585 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
5. PDG2026:observable: B010:B010M:349097 Delta(1700) BREIT-WIGNER MASS	Delta(1700)- | Deltabar(1700)+ | Delta(1700)0 | Deltabar(1700)0 | Delta(1700)+ | Deltabar(1700)- | Delta(1700)++ | Deltabar(1700)-- COMP-BARYON E04; M07	1690 to 1710 to 1730 MeV Presentation: 1690 to 1730 (~1710)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
6. PDG2026:observable: B011:B011M:349151 Delta(1905) BREIT-WIGNER MASS	Delta(1905)- | Deltabar(1905)+ | Delta(1905)0 | Deltabar(1905)0 | Delta(1905)+ | Deltabar(1905)- | Delta(1905)++ | Deltabar(1905)-- COMP-BARYON E04; M07	1855 to 1880 to 1910 MeV Presentation: 1855 to 1910 (~1880)	u_+: NA u_-: NA Type: R Confidence: ND
7. PDG2026:observable: B012:B012M:349182 Delta(1910) BREIT-WIGNER MASS	Delta(1910)- | Deltabar(1910)+ | Delta(1910)0 | Deltabar(1910)0 | Delta(1910)+ | Deltabar(1910)- | Delta(1910)++ | Deltabar(1910)-- COMP-BARYON E04; M07	1850 to 1900 to 1950 MeV Presentation: 1850 to 1950 (~1900)	u_+: NA u_-: NA Type: R Confidence: ND
8. PDG2026:observable: B013:B013M:349249 Delta(1930) BREIT-WIGNER MASS	Delta(1930)- | Deltabar(1930)+ | Delta(1930)0 | Deltabar(1930)0 | Delta(1930)+ | Deltabar(1930)- | Delta(1930)++ | Deltabar(1930)-- COMP-BARYON E04; M07	1900 to 1950 to 2000 MeV Presentation: 1900 to 2000 (~1950)	u_+: NA u_-: NA Type: R Confidence: ND
9. PDG2026:observable: B014:B014M:348535 N(1710) BREIT-WIGNER MASS	N(1710)0 | Nbar(1710)0 | N(1710)+ | Nbar(1710)- COMP-BARYON E04; M07	1680 TO 1710 TO 1740 MeV Presentation: 1680 to 1740 (~1710)	u_+: NA u_-: NA Type: R Confidence: ND
10. PDG2026:observable: B015:B015M:348569 N(1720) BREIT-WIGNER MASS	N(1720)0 | Nbar(1720)0 | N(1720)+ | Nbar(1720)- COMP-BARYON E04; M07	1680 to 1720 to 1750 MeV Presentation: 1680 to 1750 (~1720)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
11. PDG2026:observable: B016:B016M:348633 N(1875) BREIT-WIGNER MASS	N(1875)0 | Nbar(1875)0 | N(1875)+ | Nbar(1875)- COMP-BARYON E04; M07	1850 to 1875 to 1920 MeV Presentation: 1850 to 1920 (~1875)	u_+: NA u_-: NA Type: R Confidence: ND
12. PDG2026:observable: B017:B017M:348778 N(1990) BREIT-WIGNER MASS	N(1990)0 | Nbar(1990)0 | N(1990)+ | Nbar(1990)- COMP-BARYON E04; M07	1950 to 2020 to 2100 MeV Presentation: 1950 to 2100 (~2020)	u_+: NA u_-: NA Type: R Confidence: ND
13. PDG2026:observable: B018:B018M:348499 N(1700) BREIT-WIGNER MASS	N(1700)0 | Nbar(1700)0 | N(1700)+ | Nbar(1700)- COMP-BARYON E04; M07	1650 to 1720 to 1800 MeV Presentation: 1650 to 1800 (~1720)	u_+: NA u_-: NA Type: R Confidence: ND
14. PDG2026:observable: B019:B019M:349050 Delta(1600) BREIT-WIGNER MASS	Delta(1600)- | Deltabar(1600)+ | Delta(1600)0 | Deltabar(1600)0 | Delta(1600)+ | Deltabar(1600)- | Delta(1600)++ | Deltabar(1600)-- COMP-BARYON E04; M07	1500 to 1570 to 1640 MeV Presentation: 1500 to 1640 (~1570)	u_+: NA u_-: NA Type: R Confidence: ND
15. PDG2026:observable: B021:B021M:350410 Xi(1620) MASS	Xi(1620)- | Xibar(1620)+ | Xi(1620)0 | Xibar(1620)0 COMP-BARYON E04; M07	~1620 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
16. PDG2026:observable: B022:B022M:350457 Xi(2250) MASS	Xi(2250)- | Xibar(2250)+ | Xi(2250)0 | Xibar(2250)0 COMP-BARYON E04; M07	~2250 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
17. PDG2026:observable: B024:B024M:348906 N(2120) BREIT-WIGNER MASS	N(2120)0 | Nbar(2120)0 | N(2120)+ | Nbar(2120)- COMP-BARYON E04; M07	2060 to 2120 to 2160 MeV Presentation: 2060 to 2160 (~2120)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
18. PDG2026:observable: B026:B026M:350202 Sigma(2100) MASS	Sigma(2100)- | Sigmabar(2100)+ | Sigma(2100)0 | Sigmabar(2100)0 | Sigma(2100)+ | Sigmabar(2100)- COMP-BARYON E04; M07	~2100 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
19. PDG2026:observable: B027:B027M:349731 Lambda(2085) MASS	Lambda(2085)0 | Lambdabar(2085)0 COMP-BARYON E04; M07	~2020 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
20. PDG2026:observable: B030:B030M:349126 Delta(1900) BREIT-WIGNER MASS	Delta(1900)- | Deltabar(1900)+ | Delta(1900)0 | Deltabar(1900)0 | Delta(1900)+ | Deltabar(1900)- | Delta(1900)++ | Deltabar(1900)-- COMP-BARYON E04; M07	1840 TO 1860 TO 1920 MeV Presentation: 1840 to 1920 (~1860)	u_+: NA u_-: NA Type: R Confidence: ND
21. PDG2026:observable: B032:B032M:349902 Sigma(1620) MASS	Sigma(1620)- | Sigmabar(1620)+ | Sigma(1620)0 | Sigmabar(1620)0 | Sigma(1620)+ | Sigmabar(1620)- COMP-BARYON E04; M07	1600 to 1620 to 1650 MeV Presentation: 1600 to 1650 (~1620)	u_+: NA u_-: NA Type: R Confidence: ND
22. PDG2026:observable: B033:B033M:349034 MIXED CHARGES	Delta(1232)- | Deltabar(1232)+ | Delta(1232)0 | Deltabar(1232)0 | Delta(1232)+ | Deltabar(1232)- | Delta(1232)++ | Deltabar(1232)-- COMP-BARYON E04; M07	1230 TO 1232 TO 1234 MeV Presentation: 1230 to 1234 (~1232)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
23. PDG2026:observable: B034:B034M:350186 Sigma(2070) MASS	Sigma(2070)- | Sigmabar(2070)+ | Sigma(2070)0 | Sigmabar(2070)0 | Sigma(2070)+ | Sigmabar(2070)- COMP-BARYON E04; M07	2020 to 2060 to 2100 MeV Presentation: 2020 to 2100 (~2060)	u_+: NA u_-: NA Type: R Confidence: ND
24. PDG2026:observable: B035:B035M:349763 Lambda(2110) MASS	Lambda(2110)0 | Lambdabar(2110)0 COMP-BARYON E04; M07	2050 to 2090 to 2130 MeV Presentation: 2050 to 2130 (~2090)	u_+: NA u_-: NA Type: R Confidence: ND
25. PDG2026:observable: B036:B036M:349555 Lambda(1800) MASS	Lambda(1800)0 | Lambdabar(1800)0 COMP-BARYON E04; M07	1750 to 1800 to 1850 MeV Presentation: 1750 to 1850 (~1800)	u_+: NA u_-: NA Type: R Confidence: ND
26. PDG2026:observable: B037:B037M1:349427 PRODUCTION EXPERIMENTS	Lambda(1405)0 | Lambdabar(1405)0 COMP-BARYON E04; M07	1405.1+1.3-1.0 MeV	u_+: 1.3119693539575821 u_-: 0.9645548798776788 Type: NONE Confidence: ND
27. PDG2026:observable: B038:B038M:349437 Lambda(1520) MASS	Lambda(1520)0 | Lambdabar(1520)0 COMP-BARYON E04; M07	1518 to 1519 to 1520 MeV Presentation: 1518 to 1520 (~1519)	u_+: NA u_-: NA Type: R Confidence: ND
28. PDG2026:observable: B039:B039M:349594 Lambda(1820) MASS	Lambda(1820)0 | Lambdabar(1820)0 COMP-BARYON E04; M07	1815 TO 1820 TO 1825 MeV Presentation: 1815 to 1825 (~1820)	u_+: NA u_-: NA Type: R Confidence: ND
29. PDG2026:observable: B040:B040M:349494 Lambda(1670) MASS	Lambda(1670)0 | Lambdabar(1670)0 COMP-BARYON E04; M07	1670 to 1674 to 1678 MeV Presentation: 1670 to 1678 (~1674)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
30. PDG2026:observable: B041:B041M:349741 Lambda(2100) MASS	Lambda(2100)0 | Lambdabar(2100)0 COMP-BARYON E04; M07	2090 TO 2100 TO 2110 MeV Presentation: 2090 to 2110 (~2100)	u_+: NA u_-: NA Type: R Confidence: ND
31. PDG2026:observable: B042:B042M:349775 Lambda(2350) MASS	Lambda(2350)0 | Lambdabar(2350)0 COMP-BARYON E04; M07	2340 TO 2350 TO 2370 MeV Presentation: 2340 to 2370 (~2350)	u_+: NA u_-: NA Type: R Confidence: ND
32. PDG2026:observable: B043:B043M+:349876 Sigma(1385)+ MASS	Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)- COMP-BARYON E04; M07	1382.83+-0.34 MeV	u_+: 0.33905435517601018 u_-: 0.34419099315609542 Type: NONE Confidence: ND
33. PDG2026:observable: B043:B043M0:349877 Sigma(1385)0 MASS	Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)- COMP-BARYON E04; M07	1383.8+-0.9 MeV	u_+: 0.93514404403237228 u_-: 0.93514404403237228 Type: NONE Confidence: ND
34. PDG2026:observable: B043:B043M-:349878 Sigma(1385)- MASS	Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)- COMP-BARYON E04; M07	1387.2+-0.5 MeV	u_+: 0.5465370328051361 u_-: 0.5465370328051361 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
35. PDG2026:observable: B044:B044M:349946 Sigma(1670) MASS	Sigma(1670)- | Sigmabar(1670)+ | Sigma(1670)0 | Sigmabar(1670)0 | Sigma(1670)+ | Sigmabar(1670)- COMP-BARYON E04; M07	1665 to 1675 to 1685 MeV Presentation: 1665 to 1685 (~1675)	u_+: NA u_-: NA Type: R Confidence: ND
36. PDG2026:observable: B045:B045M:349983 Sigma(1775) MASS	Sigma(1775)- | Sigmabar(1775)+ | Sigma(1775)0 | Sigmabar(1775)0 | Sigma(1775)+ | Sigmabar(1775)- COMP-BARYON E04; M07	1770 TO 1775 TO 1780 MeV Presentation: 1770 to 1780 (~1775)	u_+: NA u_-: NA Type: R Confidence: ND
37. PDG2026:observable: B046:B046M:350080 Sigma(1915) MASS	Sigma(1915)- | Sigmabar(1915)+ | Sigma(1915)0 | Sigmabar(1915)0 | Sigma(1915)+ | Sigmabar(1915)- COMP-BARYON E04; M07	1900 TO 1915 TO 1935 MeV Presentation: 1900 to 1935 (~1915)	u_+: NA u_-: NA Type: R Confidence: ND
38. PDG2026:observable: B047:B047M:350166 Sigma(2030) MASS	Sigma(2030)- | Sigmabar(2030)+ | Sigma(2030)0 | Sigmabar(2030)0 | Sigma(2030)+ | Sigmabar(2030)- COMP-BARYON E04; M07	2025 TO 2030 TO 2040 MeV Presentation: 2025 to 2040 (~2030)	u_+: NA u_-: NA Type: R Confidence: ND
39. PDG2026:observable: B048:B048M:350277 Sigma(2250) MASS	Sigma(2250)- | Sigmabar(2250)+ | Sigma(2250)0 | Sigmabar(2250)0 | Sigma(2250)+ | Sigmabar(2250)- COMP-BARYON E04; M07	2210 TO 2250 TO 2280 MeV Presentation: 2210 to 2280 (~2250)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
40. PDG2026:observable: B049:B049M0:350399 Xi(1530)0 MASS	Xi(1530)- | Xibar(1530)+ | Xi(1530)0 | Xibar(1530)0 COMP-BARYON E04; M07	1531.80+-0.32 MeV	u_+: 0.31575272391181958 u_-: 0.31575272391181958 Type: NONE Confidence: ND
41. PDG2026:observable: B049:B049M-:350401 Xi(1530)- MASS	Xi(1530)- | Xibar(1530)+ | Xi(1530)0 | Xibar(1530)0 COMP-BARYON E04; M07	1535.0+-0.6 MeV	u_+: 0.63202166316087116 u_-: 0.63202166316087116 Type: NONE Confidence: ND
42. PDG2026:observable: B050:B050M:350420 Xi(1820) MASS	Xi(1820)- | Xibar(1820)+ | Xi(1820)0 | Xibar(1820)0 COMP-BARYON E04; M07	1823+-5 MeV	u_+: 5 u_-: 5 Type: NONE Confidence: ND
43. PDG2026:observable: B052:B052M:350437 Xi(1950) MASS	Xi(1950)- | Xibar(1950)+ | Xi(1950)0 | Xibar(1950)0 COMP-BARYON E04; M07	1950+-15 MeV	u_+: 15 u_-: 15 Type: NONE Confidence: ND
44. PDG2026:observable: B053:B053M:350283 Sigma(2455) MASS	Sigma(2455)- | Sigmabar(2455)+ | Sigma(2455)0 | Sigmabar(2455)0 | Sigma(2455)+ | Sigmabar(2455)- COMP-BARYON E04; M07	~2455 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
45. PDG2026:observable: B054:B054M:350284 Sigma(2620) MASS	Sigma(2620)- | Sigmabar(2620)+ | Sigma(2620)0 | Sigmabar(2620)0 | Sigma(2620)+ | Sigmabar(2620)- COMP-BARYON E04; M07	~2620 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
46. PDG2026:observable: B055:B055M:349518 Lambda(1690) MASS	Lambda(1690)0 | Lambdabar(1690)0 COMP-BARYON E04; M07	1685 to 1690 to 1695 MeV Presentation: 1685 to 1695 (~1690)	u_+: NA u_-: NA Type: R Confidence: ND
47. PDG2026:observable: B056:B056M:349614 Lambda(1830) MASS	Lambda(1830)0 | Lambdabar(1830)0 COMP-BARYON E04; M07	1820 to 1825 to 1830 MeV Presentation: 1820 to 1830 (~1825)	u_+: NA u_-: NA Type: R Confidence: ND
48. PDG2026:observable: B057:B057M:349966 Sigma(1750) MASS	Sigma(1750)- | Sigmabar(1750)+ | Sigma(1750)0 | Sigmabar(1750)0 | Sigma(1750)+ | Sigmabar(1750)- COMP-BARYON E04; M07	1700 to 1750 to 1800 MeV Presentation: 1700 to 1800 (~1750)	u_+: NA u_-: NA Type: R Confidence: ND
49. PDG2026:observable: B059:B059M:350285 Sigma(3000) MASS	Sigma(3000)- | Sigmabar(3000)+ | Sigma(3000)0 | Sigmabar(3000)0 | Sigma(3000)+ | Sigmabar(3000)- COMP-BARYON E04; M07	~3000 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
50. PDG2026:observable: B060:B060M:349635 Lambda(1890) MASS	Lambda(1890)0 | Lambdabar(1890)0 COMP-BARYON E04; M07	1870 to 1890 to 1910 MeV Presentation: 1870 to 1910 (~1890)	u_+: NA u_-: NA Type: R Confidence: ND
51. PDG2026:observable: B061:B061M:348326 N(1440) BREIT-WIGNER MASS	N(1440)0 | Nbar(1440)0 | N(1440)+ | Nbar(1440)- COMP-BARYON E04; M07	1410 to 1440 to 1470 MeV Presentation: 1410 to 1470 (~1440)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
52. PDG2026:observable: B062:B062M:348349 N(1520) BREIT-WIGNER MASS	N(1520)0 | Nbar(1520)0 | N(1520)+ | Nbar(1520)- COMP-BARYON E04; M07	1510 TO 1515 TO 1520 MeV Presentation: 1510 to 1520 (~1515)	u_+: NA u_-: NA Type: R Confidence: ND
53. PDG2026:observable: B063:B063M:348383 N(1535) BREIT-WIGNER MASS	N(1535)0 | Nbar(1535)0 | N(1535)+ | Nbar(1535)- COMP-BARYON E04; M07	1515 to 1530 to 1545 MeV Presentation: 1515 to 1545 (~1530)	u_+: NA u_-: NA Type: R Confidence: ND
54. PDG2026:observable: B064:B064M:348435 N(1675) BREIT-WIGNER MASS	N(1675)0 | Nbar(1675)0 | N(1675)+ | Nbar(1675)- COMP-BARYON E04; M07	1665 to 1675 to 1680 MeV Presentation: 1665 to 1680 (~1675)	u_+: NA u_-: NA Type: R Confidence: ND
55. PDG2026:observable: B065:B065M:348467 N(1680) BREIT-WIGNER MASS	N(1680)0 | Nbar(1680)0 | N(1680)+ | Nbar(1680)- COMP-BARYON E04; M07	1680 TO 1685 TO 1690 MeV Presentation: 1680 to 1690 (~1685)	u_+: NA u_-: NA Type: R Confidence: ND
56. PDG2026:observable: B066:B066M:348410 N(1650) BREIT-WIGNER MASS	N(1650)0 | Nbar(1650)0 | N(1650)+ | Nbar(1650)- COMP-BARYON E04; M07	1635 to 1650 to 1665 MeV Presentation: 1635 to 1665 (~1650)	u_+: NA u_-: NA Type: R Confidence: ND
57. PDG2026:observable: B067:B067M:349999 Sigma(1880) MASS	Sigma(1880)- | Sigmabar(1880)+ | Sigma(1880)0 | Sigmabar(1880)0 | Sigma(1880)+ | Sigmabar(1880)- COMP-BARYON E04; M07	1820 to 1880 to 1940 MeV Presentation: 1820 to 1940 (~1880)	u_+: NA u_-: NA Type: R Confidence: ND
58. PDG2026:observable: B068:B068M:350443 Xi(2030) MASS	Xi(2030)- | Xibar(2030)+ | Xi(2030)0 | Xibar(2030)0 COMP-BARYON E04; M07	2025+-5 MeV	u_+: 5 u_-: 5 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
59. PDG2026:observable: B071:B071M:348953 N(2190) BREIT-WIGNER MASS	N(2190)0 | Nbar(2190)0 | N(2190)+ | Nbar(2190)- COMP-BARYON E04; M07	2140 to 2180 to 2220 MeV Presentation: 2140 to 2220 (~2180)	u_+: NA u_-: NA Type: R Confidence: ND
60. PDG2026:observable: B077:B077M:349576 Lambda(1810) MASS	Lambda(1810)0 | Lambdabar(1810)0 COMP-BARYON E04; M07	1740 to 1790 to 1840 MeV Presentation: 1740 to 1840 (~1790)	u_+: NA u_-: NA Type: R Confidence: ND
61. PDG2026:observable: B079:B079M:349920 Sigma(1660) MASS	Sigma(1660)- | Sigmabar(1660)+ | Sigma(1660)0 | Sigmabar(1660)0 | Sigma(1660)+ | Sigmabar(1660)- COMP-BARYON E04; M07	1640 to 1660 to 1680 MeV Presentation: 1640 to 1680 (~1660)	u_+: NA u_-: NA Type: R Confidence: ND
62. PDG2026:observable: B082:B082M:349078 Delta(1620) BREIT-WIGNER MASS	Delta(1620)- | Deltabar(1620)+ | Delta(1620)0 | Deltabar(1620)0 | Delta(1620)+ | Deltabar(1620)- | Delta(1620)++ | Deltabar(1620)-- COMP-BARYON E04; M07	1590 to 1610 to 1630 MeV Presentation: 1590 to 1630 (~1610)	u_+: NA u_-: NA Type: R Confidence: ND
63. PDG2026:observable: B083:B083M:349308 Delta(1950) BREIT-WIGNER MASS	Delta(1950)- | Deltabar(1950)+ | Delta(1950)0 | Deltabar(1950)0 | Delta(1950)+ | Deltabar(1950)- | Delta(1950)++ | Deltabar(1950)-- COMP-BARYON E04; M07	1915 to 1930 to 1950 MeV Presentation: 1915 to 1950 (~1930)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
64. PDG2026:observable: B084:B084M:349359 Delta(2420) BREIT-WIGNER MASS	Delta(2420)- | Deltabar(2420)+ | Delta(2420)0 | Deltabar(2420)0 | Delta(2420)+ | Deltabar(2420)- | Delta(2420)++ | Deltabar(2420)-- COMP-BARYON E04; M07	2300 to 2450 to 2600 MeV Presentation: 2300 to 2600 (~2450)	u_+: NA u_-: NA Type: R Confidence: ND
65. PDG2026:observable: B087:B087M:348671 N(1880) BREIT-WIGNER MASS	N(1880)0 | Nbar(1880)0 | N(1880)+ | Nbar(1880)- COMP-BARYON E04; M07	1830 to 1880 to 1930 MeV Presentation: 1830 to 1930 (~1880)	u_+: NA u_-: NA Type: R Confidence: ND
66. PDG2026:observable: B088:B088M:350188 Sigma(2080) MASS	Sigma(2080)- | Sigmabar(2080)+ | Sigma(2080)0 | Sigmabar(2080)0 | Sigma(2080)+ | Sigmabar(2080)- COMP-BARYON E04; M07	2060 to 2090 to 2120 MeV Presentation: 2060 to 2120 (~2090)	u_+: NA u_-: NA Type: R Confidence: ND
67. PDG2026:observable: B089:B089M:349646 Lambda(2000) MASS	Lambda(2000)0 | Lambdabar(2000)0 COMP-BARYON E04; M07	~2000 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
68. PDG2026:observable: B090:B090M:348986 N(2220) BREIT-WIGNER MASS	N(2220)0 | Nbar(2220)0 | N(2220)+ | Nbar(2220)- COMP-BARYON E04; M07	2200 TO 2250 TO 2300 MeV Presentation: 2200 to 2300 (~2250)	u_+: NA u_-: NA Type: R Confidence: ND
69. PDG2026:observable: B098:B098M:350050 Sigma(1910) MASS	Sigma(1910)- | Sigmabar(1910)+ | Sigma(1910)0 | Sigmabar(1910)0 | Sigma(1910)+ | Sigmabar(1910)- COMP-BARYON E04; M07	1870 to 1910 to 1950 MeV Presentation: 1870 to 1950 (~1910)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
70. PDG2026:observable: B099:B099M:350464 Xi(2500) MASS	Xi(2500)- | Xibar(2500)+ | Xi(2500)0 | Xibar(2500)0 COMP-BARYON E04; M07	~2500 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
71. PDG2026:observable: B101:B101M:349472 Lambda(1600) MASS	Lambda(1600)0 | Lambdabar(1600)0 COMP-BARYON E04; M07	1570 to 1600 to 1630 MeV Presentation: 1570 to 1630 (~1600)	u_+: NA u_-: NA Type: R Confidence: ND
72. PDG2026:observable: B102:B102M:350934 Lambda_c(2625)+ MASS	Lambda_c(2625)+ | Lambdabar_c(2625)- COMP-BARYON E04; M07	2628.00+-0.15 MeV	u_+: 0.14735968814977221 u_-: 0.14735968814977221 Type: NONE Confidence: ND
73. PDG2026:observable: B103:B103M:350454 Xi(2120) MASS	Xi(2120)- | Xibar(2120)+ | Xi(2120)0 | Xibar(2120)0 COMP-BARYON E04; M07	~2120 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
74. PDG2026:observable: B104:B104M++: 350985 Sigma_c(2455)++ MASS	Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)-- COMP-BARYON E04; M07	2453.97+-0.14 MeV	u_+: 0.14100978554010379 u_-: 0.14100978554010379 Type: NONE Confidence: ND
75. PDG2026:observable: B104:B104M+:350986 Sigma_c(2455)+ MASS	Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)-- COMP-BARYON E04; M07	2452.65+0.22-0.16 MeV	u_+: 0.21793257206126 77 u_-: 0.15750625023659159 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
76. PDG2026:observable: B104:B104M0:350987 Sigma_c(2455)0 MASS	Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)-- COMP-BARYON E04; M07	2453.75+-0.14 MeV	u_+: 0.14100937832457261 u_-: 0.14100937832457261 Type: NONE Confidence: ND
77. PDG2026:observable: B105:B105M:350412 MIXED CHARGES	Xi(1690)- | Xibar(1690)+ | Xi(1690)0 | Xibar(1690)0 COMP-BARYON E04; M07	1690+-10 MeV	u_+: 10 u_-: 10 Type: NONE Confidence: ND
78. PDG2026:observable: B112:B112M:349774 Lambda(2325) MASS	Lambda(2325)0 | Lambdabar(2325)0 COMP-BARYON E04; M07	~2325 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
79. PDG2026:observable: B113:B113M:349004 N(2250) BREIT-WIGNER MASS	N(2250)0 | Nbar(2250)0 | N(2250)+ | Nbar(2250)- COMP-BARYON E04; M07	2250 to 2280 to 2320 MeV Presentation: 2250 to 2320 (~2280)	u_+: NA u_-: NA Type: R Confidence: ND
80. PDG2026:observable: B115:B115M++: 351001 Sigma_c(2520)++ MASS	Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)-- COMP-BARYON E04; M07	2518.41+-0.22 MeV	u_+: 0.22441420873039311 u_-: 0.22441420873039311 Type: NONE Confidence: ND
81. PDG2026:observable: B115:B115M+:351002 Sigma_c(2520)+ MASS	Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)-- COMP-BARYON E04; M07	2517.4+0.7-0.5 MeV	u_+: 0.68505102093303338 u_-: 0.52419384686487569 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
82. PDG2026:observable: B115:B115M0:351003 Sigma_c(2520)0 MASS	Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)-- COMP-BARYON E04; M07	2518.48+-0.21 MeV	u_+: 0.20795724066290969 u_-: 0.20795724066290969 Type: NONE Confidence: ND
83. PDG2026:observable: B117:B117M:349208 Delta(1920) BREIT-WIGNER MASS	Delta(1920)- | Deltabar(1920)+ | Delta(1920)0 | Deltabar(1920)0 | Delta(1920)+ | Deltabar(1920)- | Delta(1920)++ | Deltabar(1920)-- COMP-BARYON E04; M07	1870 to 1920 to 1970 MeV Presentation: 1870 to 1970 (~1920)	u_+: NA u_-: NA Type: R Confidence: ND
84. PDG2026:observable: B118:B118M:350286 Sigma(3170) MASS (PRODUCTION EXPERIMENTS)	Sigma(3170)- | Sigmabar(3170)+ | Sigma(3170)0 | Sigmabar(3170)0 | Sigma(3170)+ | Sigmabar(3170)- COMP-BARYON E04; M07	~3170 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
85. PDG2026:observable: B119:B119M:350917 Lambda_c(2595)+ MASS	Lambda_c(2595)+ | Lambdabar_c(2595)- COMP-BARYON E04; M07	2592.25+-0.28 MeV	u_+: 0.28142494558940517 u_-: 0.28142494558940517 Type: NONE Confidence: ND
86. PDG2026:observable: B120:B120M:349022 N(2600) BREIT-WIGNER MASS	N(2600)0 | Nbar(2600)0 | N(2600)+ | Nbar(2600)- COMP-BARYON E04; M07	2550 TO 2600 TO 2750 MeV Presentation: 2550 to 2750 (~2600)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
87. PDG2026:observable: B122:B122M:350979 Lambda_c(2940)+ MASS	Lambda_c(2940)0 | Lambdabar_c(2940)0 COMP-BARYON E04; M07	2939.6+1.3-1.5 MeV	u_+: 1.261876684475213 u_-: 1.467184985071428 Type: NONE Confidence: ND
88. PDG2026:observable: B128:B128M:349027 N(~3000) BREIT-WIGNER MASS	N(~3000)0 | Nbar(~3000)0 | N(~3000)+ | Nbar(~3000)- COMP-BARYON E04; M07	~3000 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
89. PDG2026:observable: B130:B130M+:351271 Xi_c(2970)+ MASS	Xi_c(2970)0 | Xibar_c(2970)0 COMP-BARYON E04; M07	2965.2+-1.0 MeV	u_+: 1.0156810603859041 u_-: 1.0156810603859041 Type: NONE Confidence: ND
90. PDG2026:observable: B130:B130M0:351273 Xi_c(2970)0 MASS	Xi_c(2970)0 | Xibar_c(2970)0 COMP-BARYON E04; M07	2967.3+-1.9 MeV	u_+: 1.850084394808527 u_-: 1.850084394808527 Type: NONE Confidence: ND
91. PDG2026:observable: B131:B131M:350458 Xi(2370) MASS	Xi(2370)- | Xibar(2370)+ | Xi(2370)0 | Xibar(2370)0 COMP-BARYON E04; M07	~2370 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
92. PDG2026:observable: B132:B132M:348871 N(2100) BREIT-WIGNER MASS	N(2100)0 | Nbar(2100)0 | N(2100)+ | Nbar(2100)- COMP-BARYON E04; M07	2050 to 2100 to 2150 MeV Presentation: 2050 to 2150 (~2100)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
93. PDG2026:observable: B135:B135M:349339 Delta(2200) BREIT-WIGNER MASS	Delta(2200)- | Deltabar(2200)+ | Delta(2200)0 | Deltabar(2200)0 | Delta(2200)+ | Deltabar(2200)- | Delta(2200)++ | Deltabar(2200)-- COMP-BARYON E04; M07	2150 to 2200 to 2250 MeV Presentation: 2150 to 2250 (~2200)	u_+: NA u_-: NA Type: R Confidence: ND
94. PDG2026:observable: B136:B136M:349272 Delta(1940) BREIT-WIGNER MASS	Delta(1940)- | Deltabar(1940)+ | Delta(1940)0 | Deltabar(1940)0 | Delta(1940)+ | Deltabar(1940)- | Delta(1940)++ | Deltabar(1940)-- COMP-BARYON E04; M07	1940 to 2000 to 2060 MeV Presentation: 1940 to 2060 (~2000)	u_+: NA u_-: NA Type: R Confidence: ND
95. PDG2026:observable: B141:B141M:350516 Omega(2250)- MASS	Omega(2250)- | Omegabar(2250)+ COMP-BARYON E04; M07	2252+-9 MeV	u_+: 8.8341044409654419 u_-: 8.8341044409654419 Type: NONE Confidence: ND
96. PDG2026:observable: B142:B142M:350520 Omega(2380)- MASS	Omega(2380)- | Omegabar(2380)+ COMP-BARYON E04; M07	~2380 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
97. PDG2026:observable: B143:B143M:350523 Omega(2470)- MASS	Omega(2470)- | Omegabar(2470)+ COMP-BARYON E04; M07	2474+-12 MeV	u_+: 12 u_-: 12 Type: NONE Confidence: ND
98. PDG2026:observable: B144:B144M:348740 N(1900) BREIT-WIGNER MASS	N(1900)0 | Nbar(1900)0 | N(1900)+ | Nbar(1900)- COMP-BARYON E04; M07	1890 to 1920 to 1950 MeV Presentation: 1890 to 1950 (~1920)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
99. PDG2026:observable: B146:B146M+:351204 Xi_c(2645)+ MASS	Xi_c(2645)0 | Xibar_ c(2645)0 | Xi_c(2645)+ | Xibar_c(2645)- COMP-BARYON E04; M07	2645.17+-0.27 MeV	u_+: 0.26281213522405311 u_-: 0.275224526580374 Type: NONE Confidence: ND
100. PDG2026:observable: B146:B146M0:351206 Xi_c(2645)0 MASS	Xi_c(2645)0 | Xibar_ c(2645)0 | Xi_c(2645)+ | Xibar_c(2645)- COMP-BARYON E04; M07	2646.24+-0.18 MeV	u_+: 0.17792728963027479 u_-: 0.1809798944958548 Type: NONE Confidence: ND
101. PDG2026:observable: B147:B147M+:351307 Xi_c(3080)+ MASS	Xi_c(3080)0 | Xibar_ c(3080)0 COMP-BARYON E04; M07	3077.2+-0.4 MeV	u_+: 0.38652749298648748 u_-: 0.38652749298648748 Type: NONE Confidence: ND
102. PDG2026:observable: B147:B147M0:351308 Xi_c(3080)0 MASS	Xi_c(3080)0 | Xibar_ c(3080)0 COMP-BARYON E04; M07	3079.9+-1.4 MeV	u_+: 1.3603458501256609 u_-: 1.3603458501256609 Type: NONE Confidence: ND
103. PDG2026:observable: B148:B148M+:351232 Xi_c(2815)+ MASS	Xi_c(2815)0 | Xibar_ c(2815)0 | Xi_c(2815)+ | Xibar_c(2815)- COMP-BARYON E04; M07	2816.60+-0.16 MeV	u_+: 0.15517126784005561 u_-: 0.1576723075863977 Type: NONE Confidence: ND
104. PDG2026:observable: B148:B148M0:351234 Xi_c(2815)0 MASS	Xi_c(2815)0 | Xibar_ c(2815)0 | Xi_c(2815)+ | Xibar_c(2815)- COMP-BARYON E04; M07	2819.85+-0.27 MeV	u_+: 0.26809134636591098 u_-: 0.28025492882096131 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
105. PDG2026:observable: B149:B149M+:351217 Xi_c(2790)+ MASS	Xi_c(2790)0 | Xibar_ c(2790)0 | Xi_c(2790)+ | Xibar_c(2790)- COMP-BARYON E04; M07	2792.0+-0.5 MeV	u_+: 0.52973897333377062 u_-: 0.53585270924116923 Type: NONE Confidence: ND
106. PDG2026:observable: B149:B149M0:351218 Xi_c(2790)0 MASS	Xi_c(2790)0 | Xibar_ c(2790)0 | Xi_c(2790)+ | Xibar_c(2790)- COMP-BARYON E04; M07	2794.0+-0.5 MeV	u_+: 0.49234423994343779 u_-: 0.49234423994343779 Type: NONE Confidence: ND
107. PDG2026:observable: B150:B150M:350951 Lambda_c(2765)+ MASS	Lambda_c(2765)0 | Lambdabar_c(2765)0 COMP-BARYON E04; M07	2766.6+-2.4 MeV	u_+: 2.404079865561874 u_-: 2.404079865561874 Type: NONE Confidence: ND
108. PDG2026:observable: B151:B151M:350960 Lambda_c(2880)+ MASS	Lambda_c(2880)+ | Lambdabar_c(2880)- COMP-BARYON E04; M07	2881.63+-0.24 MeV	u_+: 0.23613203643473499 u_-: 0.2455586841443336 Type: NONE Confidence: ND
109. PDG2026:observable: B155:B155M++: 351015 Sigma_c(2800)++ MASS	Sigma_c(2800)0 | Sigmabar_c(2800)0 COMP-BARYON E04; M07	2801+4-6 MeV	u_+: 4.4067675227993526 u_-: 5.7999655171389852 Type: NONE Confidence: ND
110. PDG2026:observable: B155:B155M+:351016 Sigma_c(2800)+ MASS	Sigma_c(2800)0 | Sigmabar_c(2800)0 COMP-BARYON E04; M07	2792+14-5 MeV	u_+: 13.690127829936111 u_-: 5.01792785918669 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
111. PDG2026:observable: B155:B155M0:351017 Sigma_c(2800)0 MASS	Sigma_c(2800)0 | Sigmabar_c(2800)0 COMP-BARYON E04; M07	2806+5-7 MeV	u_+: 4.8590019921640533 u_-: 7.3871442628294739 Type: NONE Confidence: ND
112. PDG2026:observable: B156:B156A03:351263 Xi_c(2930)+ MASS	Xi_c(2930)0 | Xibar_c(2930)0 COMP-BARYON E04; M07	2942+-5 MeV	u_+: 4.6486557196677838 u_-: 4.6486557196677838 Type: NONE Confidence: ND
113. PDG2026:observable: B156:B156A04:351264 Xi_c(2930)0 MASS	Xi_c(2930)0 | Xibar_c(2930)0 COMP-BARYON E04; M07	2938.55+-0.30 MeV	u_+: 0.30185805977630881 u_-: 0.30185805977630881 Type: NONE Confidence: ND
114. PDG2026:observable: B157:B157M+:351297 Xi_c(3055)+ MASS	Xi_c(3055)0 | Xibar_c(3055)0 COMP-BARYON E04; M07	3055.2+-0.7 MeV	u_+: 0.68999235826894623 u_-: 0.68999235826894623 Type: NONE Confidence: ND
115. PDG2026:observable: B157:B157M:351298 Xi_c(3055)0 MASS	Xi_c(3055)0 | Xibar_c(3055)0 COMP-BARYON E04; M07	3061.0+-0.8 MeV	u_+: 0.83240615086627034 u_-: 0.83240615086627034 Type: NONE Confidence: ND
116. PDG2026:observable: B158:B158M+:351325 Xi_c(3123)+ MASS	Xi_c(3123)0 | Xibar_c(3123)0 COMP-BARYON E04; M07	3122.9+-1.3 MeV	u_+: 1.3341664064126344 u_-: 1.3341664064126344 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
117. PDG2026:observable: B161:B161M:351844 Xi_b(5945)0 MASS	Xi_b(5945)0 | Xibar_ b(5945)0 COMP-BARYON E04; M07	5952.3+-0.6 MeV	u_+: 0.57765272773906 229 u_-: 0.57765272773906 229 Type: NONE Confidence: ND
118. PDG2026:observable: B162:B162M:351694 Lambda_b(5912)0 MASS	Lambda_b(5912)0 | Lambdabar_b(5912)0 COMP-BARYON E04; M07	5912.16+-0.16 MeV	u_+: 0.16007003100330 261 u_-: 0.16007003100330 261 Type: NONE Confidence: ND
119. PDG2026:observable: B163:B163M:351698 Lambda_b(5920)0 MASS	Lambda_b(5920)0 | Lambdabar_b(5920)0 COMP-BARYON E04; M07	5920.07+-0.16 MeV	u_+: 0.15862170338385 659 u_-: 0.15862170338385 659 Type: NONE Confidence: ND
120. PDG2026:observable: B164:B164M:349532 Lambda(1710) MASS	Lambda(1710)0 | Lambdabar(1710)0 COMP-BARYON E04; M07	1713+-13 MeV	u_+: 13 u_-: 13 Type: NONE Confidence: ND
121. PDG2026:observable: B165:B165M:349651 Lambda(2050) MASS	Lambda(2050)0 | Lambdabar(2050)0 COMP-BARYON E04; M07	2056+-22 MeV	u_+: 22 u_-: 22 Type: NONE Confidence: ND
122. PDG2026:observable: B166:B166M:349991 Sigma(1780) MASS	Sigma(1780)- | Sigmabar(1780)+ | Sigma(1780)0 | Sigmabar(1780)0 | Sigma(1780)+ | Sigmabar(1780)- COMP-BARYON E04; M07	1730 to 1780 to 1830 MeV Presentation: 1730 to 1830 (~1780)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
123. PDG2026:observable: B167:B167M:350019 Sigma(1900) MASS	Sigma(1900)- | Sigmabar(1900)+ | Sigma(1900)0 | Sigmabar(1900)0 | Sigma(1900)+ | Sigmabar(1900)- COMP-BARYON E04; M07	1900 to 1925 to 1950 MeV Presentation: 1900 to 1950 (~1925)	u_+: NA u_-: NA Type: R Confidence: ND
124. PDG2026:observable: B168:B168M:350102 Sigma(1940) MASS	Sigma(1940)- | Sigmabar(1940)+ | Sigma(1940)0 | Sigmabar(1940)0 | Sigma(1940)+ | Sigmabar(1940)- COMP-BARYON E04; M07	1920 to 1940 to 1960 MeV Presentation: 1920 to 1960 (~1940)	u_+: NA u_-: NA Type: R Confidence: ND
125. PDG2026:observable: B169:B169M:351840 Xi_b'(5935)- MASS	Xi_b^'(5935)- | Xibar_ b^'(5935)+ COMP-BARYON E04; M07	5934.9+-0.4 MeV	u_+: 0.41352429707823 54 u_-: 0.41352429707823 54 Type: NONE Confidence: ND
126. PDG2026:observable: B170:B170M:351848 Xi_b(5955)- MASS	Xi_b(5955)- | Xibar_ b(5955)+ COMP-BARYON E04; M07	5955.5+-0.4 MeV	u_+: 0.41457381951292 599 u_-: 0.41457381951292 599 Type: NONE Confidence: ND
127. PDG2026:observable: B171:B171M:351951 P_c_cbar(4380)+ MASS	P_c_cbar(4380)+ COMP-BARYON E04; M07	4380+-30 MeV	u_+: 30.0832179129826 51 u_-: 30.0832179129826 51 Type: NONE Confidence: ND
128. PDG2026:observable: B172:B172M:351977 P_c_cbar(4457)+ MASS	P_c_cbar(4457)+ COMP-BARYON E04; M07	4457.3+4.0-1.8 MeV	u_+: 4.14366987102013 16 u_-: 1.80277563773199 5 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
129. PDG2026:observable: B173:B173M:351376 Omega_c(3000)0 MASS	Omega_c(3000)0 | Omegabar_c(3000)0 COMP-BARYON E04; M07	3000.46+-0.25 MeV	u_+: 0.2431765836192144 u_-: 0.26399637894382499 Type: NONE Confidence: ND
130. PDG2026:observable: B174:B174M:351380 Omega_c(3050)0 MASS	Omega_c(3050)0 | Omegabar_c(3050)0 COMP-BARYON E04; M07	3050.17+-0.19 MeV	u_+: 0.18820356358563539 u_-: 0.19089950264325409 Type: NONE Confidence: ND
131. PDG2026:observable: B175:B175M:351384 Omega_c(3065)0 MASS	Omega_c(3065)0 | Omegabar_c(3065)0 COMP-BARYON E04; M07	3065.58+-0.21 MeV	u_+: 0.21345733449248469 u_-: 0.2139147501137518 Type: NONE Confidence: ND
132. PDG2026:observable: B176:B176M:351388 Omega_c(3090)0 MASS	Omega_c(3090)0 | Omegabar_c(3090)0 COMP-BARYON E04; M07	3090.15+-0.26 MeV	u_+: 0.25239209275859231 u_-: 0.26305043429981828 Type: NONE Confidence: ND
133. PDG2026:observable: B177:B177M:351392 Omega_c(3120)0 MASS	Omega_c(3120)0 | Omegabar_c(3120)0 COMP-BARYON E04; M07	3118.98+0.27-0.35 MeV	u_+: 0.27459060435491961 u_-: 0.34669871646719441 Type: NONE Confidence: ND
134. PDG2026:observable: B178:B178M:350956 Lambda_c(2860)+ MASS	Lambda_c(2860)0 | Lambdabar_c(2860)0 COMP-BARYON E04; M07	2856.1+2.3-6.0 MeV	u_+: 2.336664289109585 u_-: 5.873670062235365 Type: NONE Confidence: ND

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135. PDG2026:observable: B179:B179M:350506 Omega(2012)- MASS	Omega(2012)- | Omegabar(2012)+ COMP-BARYON E04; M07	2012.9+-0.4 MeV	u_+: 0.44536259664740679 u_-: 0.44536259664740679 Type: NONE Confidence: ND
136. PDG2026:observable: B180:B180M:351864 Xi_b(6227)- MASS	Xi_b(6227)- | Xi_b(6227)+ COMP-BARYON E04; M07	6227.9+-0.9 MeV	u_+: 0.90815128746094542 u_-: 0.90815128746094542 Type: NONE Confidence: ND
137. PDG2026:observable: B181:B181M:351734 Sigma_b(6097)+ MASS	Sigma_b(6097)+ | Sigmabar_b(6097)- COMP-BARYON E04; M07	6095.8+-1.7 MeV	u_+: 1.7464249196572981 u_-: 1.7464249196572981 Type: NONE Confidence: ND
138. PDG2026:observable: B182:B182M:351738 Sigma_b(6097)- MASS	Sigma_b(6097)- | Sigmabar_b(6097)+ COMP-BARYON E04; M07	6098.0+-1.8 MeV	u_+: 1.772004514666935 u_-: 1.772004514666935 Type: NONE Confidence: ND
139. PDG2026:observable: B183:B183M:350228 Sigma(2110) MASS	Sigma(2110)- | Sigmabar(2110)+ | Sigma(2110)0 | Sigmabar(2110)0 | Sigma(2110)+ | Sigmabar(2110)- COMP-BARYON E04; M07	2110+-50 MeV	u_+: 51.991387525520771 u_-: 51.991387525520771 Type: NONE Confidence: ND
140. PDG2026:observable: B184:B184M:350253 Sigma(2230) MASS	Sigma(2230)- | Sigmabar(2230)+ | Sigma(2230)0 | Sigmabar(2230)0 | Sigma(2230)+ | Sigmabar(2230)- COMP-BARYON E04; M07	2240+-27 MeV	u_+: 27 u_-: 27 Type: NONE Confidence: ND

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141. PDG2026:observable: B185:B185M:351935 P_c_cbar(4312)+ MASS	P_c_cbar(4312)+ COMP-BARYON E04; M07	4311.9+7.0-0.9 MeV	u_+: 6.8359344642850397 u_-: 0.9219544457292864 Type: NONE Confidence: ND
142. PDG2026:observable: B186:B186M:351955 P_c_cbar(4440)+ MASS	P_c_cbar(4440)+ COMP-BARYON E04; M07	4440+4-5 MeV	u_+: 4.3011626335213133 u_-: 4.876474136094644 Type: NONE Confidence: ND
143. PDG2026:observable: B187:B187M:351706 Lambda_b(6146)0 MASS	Lambda_b(6146)0 | Lambdabar_b(6146)0 COMP-BARYON E04; M07	6146.2+-0.4 MeV	u_+: 0.41759268734261917 u_-: 0.41759268734261917 Type: NONE Confidence: ND
144. PDG2026:observable: B188:B188M:351711 Lambda_b(6152)0 MASS	Lambda_b(6152)0 | Lambdabar_b(6152)0 COMP-BARYON E04; M07	6152.5+-0.4 MeV	u_+: 0.35727006911778508 u_-: 0.35727006911778508 Type: NONE Confidence: ND
145. PDG2026:observable: B190:B190M:349676 Lambda(2070) MASS	Lambda(2070)0 | Lambdabar(2070)0 COMP-BARYON E04; M07	2070+-24 MeV	u_+: 24 u_-: 24 Type: NONE Confidence: ND
146. PDG2026:observable: B191:B191M:349711 Lambda(2080) MASS	Lambda(2080)0 | Lambdabar(2080)0 COMP-BARYON E04; M07	2082+-13 MeV	u_+: 13 u_-: 13 Type: NONE Confidence: ND
147. PDG2026:observable: B192:B192M0:351255 Xi_c(2923)0 MASS	Xi_c(2923)0 | Xibar_c(2923)0 COMP-BARYON E04; M07	2923.2+-0.4 MeV	u_+: 0.39743785839914991 u_-: 0.39743785839914991 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
148. PDG2026:observable: B192:B192MP:351256 Xi_c(2923)+ MASS	Xi_c(2923)0 | Xibar_ c(2923)0 COMP-BARYON E04; M07	2922.8+-0.6 MeV	u_+: 0.61644140029689765 u_-: 0.61644140029689765 Type: NONE Confidence: ND
149. PDG2026:observable: B193:B193M:351905 Omega_b(6316)- MASS	Omega_b(6316)- | Omegabar_b(6316)+ COMP-BARYON E04; M07	6315.6+-0.6 MeV	u_+: 0.58048724929847628 u_-: 0.58048724929847628 Type: NONE Confidence: ND
150. PDG2026:observable: B194:B194M:351909 Omega_b(6330)- MASS	Omega_b(6330)- | Omegabar_b(6330)+ COMP-BARYON E04; M07	6330.3+-0.6 MeV	u_+: 0.56503579231594825 u_-: 0.56503579231594825 Type: NONE Confidence: ND
151. PDG2026:observable: B195:B195M:351913 Omega_b(6340)- MASS	Omega_b(6340)- | Omegabar_b(6340)+ COMP-BARYON E04; M07	6339.7+-0.6 MeV	u_+: 0.55323181994360315 u_-: 0.55323181994360315 Type: NONE Confidence: ND
152. PDG2026:observable: B196:B196M:351917 Omega_b(6350)- MASS	Omega_b(6350)- | Omegabar_b(6350)+ COMP-BARYON E04; M07	6349.8+-0.6 MeV	u_+: 0.60080400015155633 u_-: 0.60080400015155633 Type: NONE Confidence: ND
153. PDG2026:observable: B197:B197M:351702 Lambda_b(6070)0 MASS	Lambda_b(6070)0 | Lambdabar_b(6070)0 COMP-BARYON E04; M07	6072.3+-2.9 MeV	u_+: 2.9469880829275459 u_-: 2.9469880829275459 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
154. PDG2026:observable: B199:B199M:351870 Xi_b(6227)0 MASS	Xi_b(6227)0 | Xibar_b(6227)0 COMP-BARYON E04; M07	6226.8+-1.6 MeV	u_+: 1.54377784025284 7 u_-: 1.63500765137529 09 Type: NONE Confidence: ND
155. PDG2026:observable: B200:B200M:351860 Xi_b(6100)- MASS	Xi_b(6100)- | Xibar_b(6100)+ COMP-BARYON E04; M07	6099.8+-0.4 MeV	u_+: 0.44830769038414 819 u_-: 0.44830769038414 819 Type: NONE Confidence: ND
156. PDG2026:observable: B201:B201M:351874 Xi_b(6327)0 MASS	Xi_b(6327)0 | Xibar_b(6327)0 COMP-BARYON E04; M07	6327.28+-0.35 MeV	u_+: 0.35468295701936 398 u_-: 0.34205262752974 142 Type: NONE Confidence: ND
157. PDG2026:observable: B202:B202M:351878 Xi_b(6333)0 MASS	Xi_b(6333)0 | Xibar_b(6333)0 COMP-BARYON E04; M07	6332.69+-0.28 MeV	u_+: 0.27802877548915 689 u_-: 0.28425340807103 788 Type: NONE Confidence: ND
158. PDG2026:observable: B203:B203M0:351251 Xi_c(2882) MASS	Xi_c(2882)0 | Xibar_c(2882)0 COMP-BARYON E04; M07	2882+-9 MeV	u_+: 9.04765162901401 79 u_-: 9.04765162901401 79 Type: NONE Confidence: ND
159. PDG2026:observable: B204:B204M:351945 P_c_cbar_s(4338)0 MASS	P_c_cbar_s(4338)+ COMP-BARYON E04; M07	4338.2+-0.8 MeV	u_+: 0.80622577482985 491 u_-: 0.80622577482985 491 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
160. PDG2026:observable: B205:B205M:351999 P_c_cbar(4312)+ MASS	P_c_cbar(4312)+ COMP-BARYON E04; M07	4466+6-5 MeV	u_+: 6.39379624128860 78 u_-: 4.58777654200301 74 Type: NONE Confidence: ND
161. PDG2026:observable: B206:B206M:350973 Lambda_c(2910)+ MASS	Lambda_c(2910)0 | Lambdabar_c(2910)0 COMP-BARYON E04; M07	2914+-7 MeV	u_+: 6.76756972627545 22 u_-: 6.76756972627545 22 Type: NONE Confidence: ND
162. PDG2026:observable: B207:B207M:351852 Xi_b(6087)0 MASS	Xi_b(6087) COMP-BARYON E04; M07	6087.0+-0.5 MeV	u_+: 0.46316755368257 7 u_-: 0.46316755368257 7 Type: NONE Confidence: ND
163. PDG2026:observable: B208:B208M:351856 Xi_b(6095)0 MASS	Xi_b(6095) COMP-BARYON E04; M07	6095.1+-0.4 MeV	u_+: 0.44082216684770 159 u_-: 0.44082216684770 159 Type: NONE Confidence: ND
164. PDG2026:observable: B209:B209M:351396 Omega_c(3185)0 MASS	Omega_c(3185)0 | Omegabar_c(3185)0 COMP-BARYON E04; M07	3185+7.6-1.9 MeV	u_+: 7.59539334070329 81 u_-: 1.93390796058137 2 Type: NONE Confidence: ND
165. PDG2026:observable: B210:B210M:351400 Omega_c(3327)0 MASS	Omega_c(3327)0 | Omegabar_c(3327)0 COMP-BARYON E04; M07	3327.1+1.2-1.8 MeV	u_+: 1.22065556157337 01 u_-: 1.78044938147648 59 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
166. PDG2026:observable: B211:B211M:350514 Omega(2109)- MASS	Omega(2109)- | Omegabar(2109)+ COMP-BARYON E04; M07	2108+-5 MeV	u_+: 5.27730992078350 52 u_-: 5.27730992078350 52 Type: NONE Confidence: ND
167. PDG2026:observable: G033:G033M:332463 graviton MASS	graviton GRAVITATIONAL_OR_ HYPOTHETICAL E04; M10	<1.76 E-23 eV Presentation: <1.76E-23	u_+: NA u_-: NA Type: U Confidence: ND
168. PDG2026:observable: M001:M001M:333932 omega(782) MASS	omega(782)0 COMP-MESON E04; M08	782.66+-0.13 MeV	u_+: 0.12901268855075 981 u_-: 0.12899704603475 121 Type: NONE Confidence: ND
169. PDG2026:observable: M002:M002M:334006 eta'(958) MASS	eta^'(958)0 COMP-MESON E04; M08	957.78+-0.06 MeV	u_+: 0.06263150681310 4018 u_-: 0.06263150681310 4018 Type: NONE Confidence: ND
170. PDG2026:observable: M003:M003M1:334163 f_0(980) MASS	f_0(980)0 COMP-MESON E04; M08	990+-20 MeV	u_+: 20 u_-: 20 Type: NONE Confidence: ND
171. PDG2026:observable: M004:M004M:334182 phi(1020) MASS	phi(1020)0 COMP-MESON E04; M08	1019.460+-0.016 MeV	u_+: 0.01556529663733 44 u_-: 0.01556529663733 44 Type: NONE Confidence: ND
172. PDG2026:observable: M005:M005M:334354 f_2(1270) MASS	f_2(1270)0 COMP-MESON E04; M08	1275.4+-0.8 MeV	u_+: 0.83226160812299 765 u_-: 0.83243720122632 603 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
173. PDG2026:observable: M006:M006M2:334554 f_1(1420) MASS	f_1(1420)0 COMP-MESON E04; M08	1427.7+1.7-1.4 MeV	u_+: 1.669286801641032 u_-: 1.4373229857894489 Type: NONE Confidence: ND
174. PDG2026:observable: M008:M008M:334403 f_1(1285) MASS	f_1(1285)0 COMP-MESON E04; M08	1281.8+-0.5 MeV	u_+: 0.45740251642395002 u_-: 0.45001100251184373 Type: NONE Confidence: ND
175. PDG2026:observable: M009:M009M0:333862 NEUTRAL ONLY, e+ e-	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	775.26+-0.23 MeV	u_+: 0.23429187483587821 u_-: 0.23429187483587821 Type: NONE Confidence: ND
176. PDG2026:observable: M009:M009M5:333863 CHARGED ONLY, tau DECAYS and e+ e-	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	775.11+-0.34 MeV	u_+: 0.34073877998562979 u_-: 0.34073877998562979 Type: NONE Confidence: ND
177. PDG2026:observable: M009:M009M7:333864 MIXED CHARGES, OTHER REACTIONS	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	763.0+-1.2 MeV	u_+: 1.236931687685298 u_-: 1.236931687685298 Type: NONE Confidence: ND
178. PDG2026:observable: M009:M009M2:333865 CHARGED ONLY, HADROPRODUCED	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	766.5+-1.1 MeV	u_+: 1.106853523344296 u_-: 1.106853523344296 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
179. PDG2026:observable: M009:M009M0P: 333866 NEUTRAL ONLY, PHOTOPRODUCED	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	769.6+-0.8 MeV	u_+: 0.84746260647359306 u_-: 0.82807525956483763 Type: NONE Confidence: ND
180. PDG2026:observable: M009:M009M0R: 333867 NEUTRAL ONLY, OTHER REACTIONS	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	769.0+-0.9 MeV	u_+: 0.85061998301285813 u_-: 0.85061998301285813 Type: NONE Confidence: ND
181. PDG2026:observable: M010:M010M:334331 a_1(1260) MASS	a_1(1260)0 | a_1(1260)+ | a_1(1260)- COMP-MESON E04; M08	1230+-40 MeV	u_+: 40 u_-: 40 Type: NONE Confidence: ND
182. PDG2026:observable: M011:M011M:334309 b_1(1235) MASS	b_1(1235)0 | b_1(1235)+ | b_1(1235)- COMP-MESON E04; M08	1229.5+-3.2 MeV	u_+: 3.16916129668438 5 u_-: 3.16916129668438 5 Type: NONE Confidence: ND
183. PDG2026:observable: M012:M012M0:334469 a_2(1320) MASS	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	1318.2+-0.6 MeV	u_+: 0.60920186906481122 u_-: 0.60699292355954471 Type: NONE Confidence: ND
184. PDG2026:observable: M012:M012M1:334470 3pi MODE	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	1318.6+-1.3 MeV	u_+: 1.32538345937344 39 u_-: 1.30846795325698 1 Type: NONE Confidence: ND
185. PDG2026:observable: M012:M012M2:334471 K Kbar MODE	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	1318.1+-0.7 MeV	u_+: 0.69776876905725704 u_-: 0.69776876905725704 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
186. PDG2026:observable: M012:M012M3:334472 eta pi MODE	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	1317.7+-1.4 MeV	u_+: 1.44182788627946 u_-: 1.44182788627946 Type: NONE Confidence: ND
187. PDG2026:observable: M012:M012M4:334473 eta' pi MODE	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	1322+-7 MeV	u_+: 6.67661222337908 37 u_-: 7.07631643322348 49 Type: NONE Confidence: ND
188. PDG2026:observable: M013:M013MX: 334665 f_2'(1525) MASS	f_2^'(1525)0 COMP-MESON E04; M08	1517.3+-2.4 MeV	u_+: 2.43466563247887 4 u_-: 2.32415269962939 91 Type: NONE Confidence: ND
189. PDG2026:observable: M013:M013M2:334666 PRODUCED BY K+- BEAM	f_2^'(1525)0 COMP-MESON E04; M08	1518.0+-2.7 MeV	u_+: 2.75752274044423 6 u_-: 2.63263276117844 79 Type: NONE Confidence: ND
190. PDG2026:observable: M013:M013M3:334667 PRODUCED IN e+ e- ANNIHILATION AND PARTICLE DECAYS	f_2^'(1525)0 COMP-MESON E04; M08	1514+-4 MeV	u_+: 4.42458598817588 47 u_-: 4.08971545863226 64 Type: NONE Confidence: ND
191. PDG2026:observable: M013:M013M9:334668 PRODUCED IN pbar p ANNIHILATION	f_2^'(1525)0 COMP-MESON E04; M08	1512+-4 MeV	u_+: 3.65524619448102 91 u_-: 3.65524619448102 91 Type: NONE Confidence: ND
192. PDG2026:observable: M013:M013M4:334669 CENTRAL PRODUCTION	f_2^'(1525)0 COMP-MESON E04; M08	1515+-15 MeV	u_+: 15 u_-: 15 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
193. PDG2026:observable: M013:M013M10: 334670 PRODUCED IN e p COLLISIONS	f_2^'(1525)0 COMP-MESON E04; M08	1512+-3.2 MeV	u_+: 3.3105890714493702 u_-: 3.0413812651491101 Type: NONE Confidence: ND
194. PDG2026:observable: M014:M014M:333857 f_0(500) BREIT-WIGNER MASS	f_0(500)0 COMP-MESON E04; M08	400 to 800 MeV	u_+: NA u_-: NA Type: R Confidence: ND
195. PDG2026:observable: M015:M015M:334842 rho_3(1690) MASS	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	1688.8+-2.1 MeV	u_+: 2.0985516723695361 u_-: 2.0985516723695361 Type: NONE Confidence: ND
196. PDG2026:observable: M015:M015M1:334843 2pi MODE	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	1686+-4 MeV	u_+: 3.9750025108542628 u_-: 3.9750025108542628 Type: NONE Confidence: ND
197. PDG2026:observable: M015:M015M2:334844 K Kbar AND K Kbar pi MODES	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	1696+-4 MeV	u_+: 3.566236013260232 u_-: 3.566236013260232 Type: NONE Confidence: ND
198. PDG2026:observable: M015:M015M3:334845 (4pi)+- MODE	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	1686+-5 MeV	u_+: 4.6428485030430204 u_-: 4.6428485030430204 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
199. PDG2026:observable: M015:M015M5:334846 omega pi MODE	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	1681+-7 MeV	u_+: 6.5243685482430793 u_-: 6.5243685482430793 Type: NONE Confidence: ND
200. PDG2026:observable: M015:M015M6:334847 eta pi+ pi- MODE	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	1682+-12 MeV	u_+: 12.456821978061001 u_-: 12.456821978061001 Type: NONE Confidence: ND
201. PDG2026:observable: M016:M016M:335130 f_4(2050) MASS	f_4(2050)0 COMP-MESON E04; M08	2018+-11 MeV	u_+: 11.36985618415634 u_-: 11.505626326952621 Type: NONE Confidence: ND
202. PDG2026:observable: M017:M017M:335090 a_4(1970) MASS	a_4(1970)0 | a_4(1970)+ | a_4(1970)- COMP-MESON E04; M08	1967+-16 MeV	u_+: 16.038876268934711 u_-: 15.891712607459241 Type: NONE Confidence: ND
203. PDG2026:observable: M018:M018TMP: 335789 K*(892) T-Matrix Pole sqrt(s)	K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)- COMP-MESON E04; M08	(890+-14) - i (26+-6) MeV Presentation: 890+-14	u_+: ND u_-: ND Type: NONE Confidence: ND
204. PDG2026:observable: M018:M018M1:335790 CHARGED ONLY, HADROPRODUCED	K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)- COMP-MESON E04; M08	891.88+-0.23 MeV	u_+: 0.2311072255904166 u_-: 0.2311072255904166 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
205. PDG2026:observable: M018:M018MCT: 335791 CHARGED ONLY, PRODUCED IN tau LEPTON DECAYS	K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)- COMP-MESON E04; M08	895.5+-0.8 MeV	u_+: 0.76655071586947199 u_-: 0.76655071586947199 Type: NONE Confidence: ND
206. PDG2026:observable: M018:M018M2:335792 NEUTRAL ONLY	K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)- COMP-MESON E04; M08	895.56+-0.20 MeV	u_+: 0.2035793881115591 u_-: 0.2050633570239769 1 Type: NONE Confidence: ND
207. PDG2026:observable: M019:M019M:335878 K_0*(1430) MASS	K_0^*(1430)0 | Kbar_0^*(1430)0 | K_0^*(1430)+ | K_0^*(1430)- COMP-MESON E04; M08	1425+-50 MeV	u_+: 50 u_-: 50 Type: NONE Confidence: ND
208. PDG2026:observable: M020:M020M:335146 pi_2(2100) MASS	pi_2(2100)0 | pi_2(2100)+ | pi_2(2100)- COMP-MESON E04; M08	2090+-29 MeV	u_+: 29.4174202707276 u_-: 29.4174202707276 Type: NONE Confidence: ND
209. PDG2026:observable: M022:M022M1:335888 CHARGED ONLY, WITH FINAL STATE K pi	K_2^*(1430)0 | Kbar_2^*(1430)0 | K_2^*(1430)+ | K_2^*(1430)- COMP-MESON E04; M08	1427.3+-1.5 MeV	u_+: 1.50306501170133 u_-: 1.5176464822711819 Type: NONE Confidence: ND
210. PDG2026:observable: M022:M022M4:335889 NEUTRAL ONLY	K_2^*(1430)0 | Kbar_2^*(1430)0 | K_2^*(1430)+ | K_2^*(1430)- COMP-MESON E04; M08	1432.4+-1.3 MeV	u_+: 1.2576940808242421 u_-: 1.2576940808242421 Type: NONE Confidence: ND
211. PDG2026:observable: M023:M023M:335963 K_2(1770) MASS	K_2(1770)0 | Kbar_2(1770)0 | K_2(1770)+ | K_2(1770)- COMP-MESON E04; M08	1773+-8 MeV	u_+: 7.9841554588538504 u_-: 7.9644541424158186 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
212. PDG2026:observable: M025:M025M:346343 psi(4160) MASS	psi(4160) COMP-MESON E04; M08	4191+-5 MeV	u_+: 5.19599833327111 33 u_-: 4.98020846264291 08 Type: NONE Confidence: ND
213. PDG2026:observable: M026:M026M:343286 eta_c(1S) MASS	eta_c(1S) COMP-MESON E04; M08	2984.09+-0.30 MeV	u_+: 0.29677104771232 732 u_-: 0.30021031998551 229 Type: NONE Confidence: ND
214. PDG2026:observable: M027:M027MX: 334525 eta(1405) MASS	eta(1405)0 COMP-MESON E04; M08	1408.0+1.8-1.2 MeV	u_+: 1.81727460178494 5 u_-: 1.23172853976773 8 Type: NONE Confidence: ND
215. PDG2026:observable: M027:M027M1:334526 eta pi pi MODE	eta(1405)0 COMP-MESON E04; M08	1405.5+-2.3 MeV	u_+: 2.26174019480541 72 u_-: 2.35395070110280 8 Type: NONE Confidence: ND
216. PDG2026:observable: M027:M027M4:334527 K Kbar pi MODE (a_ 0(980) pi or direct K Kbar pi)	eta(1405)0 COMP-MESON E04; M08	1413.5+2.0-0.9 MeV	u_+: 1.96522832551039 79 u_-: 0.88864461966865 371 Type: NONE Confidence: ND
217. PDG2026:observable: M027:M027M2:334528 pi pi gamma MODE	eta(1405)0 COMP-MESON E04; M08	1403+-17 MeV	u_+: 16.6174143278811 29 u_-: 16.6174143278811 29 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
218. PDG2026:observable: M027:M027M7:334529 phi gamma MODE	eta(1405)0 COMP-MESON E04; M08	1422+6-8 MeV	u_+: 6.26258732474047 09 u_-: 8.07774721070175 52 Type: NONE Confidence: ND
219. PDG2026:observable: M028:M028MX: 335817 K_1(1270) MASS	K_1(1270)0 | Kbar_1(1270) 0 | K_1(1270)+ | K_1(1270) - COMP-MESON E04; M08	1256+-6 MeV	u_+: 6.42398247602787 36 u_-: 6.42398247602787 36 Type: NONE Confidence: ND
220. PDG2026:observable: M028:M028M2:335818 PRODUCED BY K-, BACKWARD SCATTERING, HYPERON EXCHANGE	K_1(1270)0 | Kbar_1(1270) 0 | K_1(1270)+ | K_1(1270) - COMP-MESON E04; M08	1275+-10 MeV	u_+: 10 u_-: 10 Type: NONE Confidence: ND
221. PDG2026:observable: M028:M028M3:335819 PRODUCED BY K BEAMS	K_1(1270)0 | Kbar_1(1270) 0 | K_1(1270)+ | K_1(1270) - COMP-MESON E04; M08	1270+-10 MeV	u_+: 10 u_-: 10 Type: NONE Confidence: ND
222. PDG2026:observable: M028:M028M1:335820 PRODUCED BY BEAMS OTHER THAN K MESONS	K_1(1270)0 | Kbar_1(1270) 0 | K_1(1270)+ | K_1(1270) - COMP-MESON E04; M08	1252+-9 MeV	u_+: 8.82391233586738 19 u_-: 8.82391233586738 19 Type: NONE Confidence: ND
223. PDG2026:observable: M028:M028MT:335821 PRODUCED IN tau LEPTON DECAYS	K_1(1270)0 | Kbar_1(1270) 0 | K_1(1270)+ | K_1(1270) - COMP-MESON E04; M08	1250+-50 MeV	u_+: 47.3814309619285 37 u_-: 47.3814309619285 37 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
224. PDG2026:observable: M029:M029M:346244 X(3940) MASS	X(3940) COMP-MESON E04; M08	3942+-9 MeV	u_+: 9.0427689971016019 u_-: 8.2141639308556176 Type: NONE Confidence: ND
225. PDG2026:observable: M030:M030M:334306 h_1(1170) MASS	h_1(1170)0 COMP-MESON E04; M08	1166+-6 MeV	u_+: 5.8309518948453007 u_-: 5.8309518948453007 Type: NONE Confidence: ND
226. PDG2026:observable: M033:M033M3:335259 pi- p --> omega pi0 n	rho_5(2350)0 | rho_5(2350)+ | rho_5(2350)- COMP-MESON E04; M08	2330+-35 MeV	u_+: 35 u_-: 35 Type: NONE Confidence: ND
227. PDG2026:observable: M034:M034M:334783 pi_2(1670) MASS	pi_2(1670)0 | pi_2(1670)+ | pi_2(1670)- COMP-MESON E04; M08	1670.6+2.9-1.2 MeV	u_+: 2.92487285188664 u_-: 1.1987692041830089 Type: NONE Confidence: ND
228. PDG2026:observable: M035:M035M:336016 K_4*(2045) MASS	K_4^*(2045)0 | Kbar_4^*(2045)0 | K_4^*(2045)+ | K_4^*(2045)- COMP-MESON E04; M08	2048+8-9 MeV	u_+: 8.0644653972191538 u_-: 8.9483400524220826 Type: NONE Confidence: ND
229. PDG2026:observable: M036:M036MX: 334171 a_0(980) MASS	a_0(980)0 | a_0(980)+ | a_0(980)- COMP-MESON E04; M08	980+-20 MeV	u_+: 20 u_-: 20 Type: NONE Confidence: ND
230. PDG2026:observable: M037:M037M:334451 eta(1295) MASS	eta(1295)0 COMP-MESON E04; M08	1294+-4 MeV	u_+: 3.892110114334967 u_-: 3.892110114334967 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
231. PDG2026:observable: M038:M038M:335002 f_2(1810) MASS	f_2(1810)0 COMP-MESON E04; M08	1815+-12 MeV	u_+: 12.528642290259 08 u_-: 11.9803740701169 Type: NONE Confidence: ND
232. PDG2026:observable: M040:M040M:336032 K_2(2250) MASS	K_2(2250)0 | Kbar_2(2250) 0 | K_2(2250)+ | K_2(2250) - COMP-MESON E04; M08	2247+-17 MeV	u_+: 16.843038421330 38 u_-: 16.843038421330 38 Type: NONE Confidence: ND
233. PDG2026:observable: M041:M041M4:335244 p p CENTRAL PRODUCTION	f_4(2300)0 COMP-MESON E04; M08	2320+-60 MeV	u_+: 60 u_-: 60 Type: NONE Confidence: ND
234. PDG2026:observable: M042:M042M:335158 f_2(2150) MASS, COMBINED MODES	f_2(2150)0 COMP-MESON E04; M08	2157+-12 MeV	u_+: 11.766679486450 94 u_-: 11.766679486450 94 Type: NONE Confidence: ND
235. PDG2026:observable: M042:M042M3:335159 eta eta MODE	f_2(2150)0 COMP-MESON E04; M08	2157+-12 MeV	u_+: 11.766679486450 94 u_-: 11.766679486450 94 Type: NONE Confidence: ND
236. PDG2026:observable: M045:M045M:334778 omega_3(1670) MASS	omega_3(1670)0 COMP-MESON E04; M08	1667+-4 MeV	u_+: 3.9911056080338438 u_-: 3.9911056080338438 Type: NONE Confidence: ND
237. PDG2026:observable: M047:M047M:347744 Upsilon(4S) MASS	Upsilon(4S) COMP-MESON E04; M08	10579.4+-1.2 MeV	u_+: 1.1896060082538451 u_-: 1.1896060082538451 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
238. PDG2026:observable: M048:M048M:347659 Upsilon(3S) MASS	Upsilon(3S) COMP-MESON E04; M08	10355.1+-0.5 MeV	u_+: 0.5 u_-: 0.5 Type: NONE Confidence: ND
239. PDG2026:observable: M049:M049M:346837 Upsilon(1S) MASS	Upsilon(1S) COMP-MESON E04; M08	9460.40+-0.10 MeV	u_+: 0.098429823160610927 u_-: 0.098429823160610927 Type: NONE Confidence: ND
240. PDG2026:observable: M050:M050M:346228 chi_c2(3930) MASS	chi_c2(3930) COMP-MESON E04; M08	3922.5+-1.0 MeV	u_+: 1.0198944899855971 u_-: 1.0198944899855971 Type: NONE Confidence: ND
241. PDG2026:observable: M052:M052M:347286 Upsilon(2S) MASS	Upsilon(2S) COMP-MESON E04; M08	10023.4+-0.5 MeV	u_+: 0.5 u_-: 0.5 Type: NONE Confidence: ND
242. PDG2026:observable: M053:M053M:345866 psi(3770) MASS	psi(3770) COMP-MESON E04; M08	3773.7+-0.7 MeV	u_+: 0.6957511640714941 8 u_-: 0.6957511640714941 8 Type: NONE Confidence: ND
243. PDG2026:observable: M054:M054M:335023 phi_3(1850) MASS	phi_3(1850)0 COMP-MESON E04; M08	1854+-7 MeV	u_+: 6.8824720161168527 u_-: 6.666666666666667 Type: NONE Confidence: ND
244. PDG2026:observable: M055:M055M:344598 chi_c1(1P) MASS	chi_c1(1P) COMP-MESON E04; M08	3510.67+-0.05 MeV	u_+: 0.050739750770671888 u_-: 0.050739750770671888 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
245. PDG2026:observable: M056:M056M:344304 chi_c0(1P) MASS	chi_c0(1P) COMP-MESON E04; M08	3415.50+-0.19 MeV	u_+: 0.19311829210333 57 u_-: 0.19240527880424 149 Type: NONE Confidence: ND
246. PDG2026:observable: M057:M057M:344926 chi_c2(1P) MASS	chi_c2(1P) COMP-MESON E04; M08	3556.17+-0.07 MeV	u_+: 0.07161203246291 1895 u_-: 0.07161203246291 1895 Type: NONE Confidence: ND
247. PDG2026:observable: M058:M058M:334463 pi(1300) MASS	pi(1300)0 | pi(1300)+ | pi(1300)- COMP-MESON E04; M08	1300+-100 MeV	u_+: 100 u_-: 100 Type: NONE Confidence: ND
248. PDG2026:observable: M059:M059M:345228 eta_c(2S) MASS	eta_c(2S) COMP-MESON E04; M08	3637.8+-0.6 MeV	u_+: 0.61759763649141 219 u_-: 0.62004876788434 371 Type: NONE Confidence: ND
249. PDG2026:observable: M060:M060M:335975 K_3*(1780) MASS	K_3^*(1780)0 | Kbar_ 3^*(1780)0 | K_3^*(1780)+ | K_3^*(1780)- COMP-MESON E04; M08	1779+-8 MeV	u_+: 7.82152442884524 3 u_-: 7.54501938704644 46 Type: NONE Confidence: ND
250. PDG2026:observable: M061:M061M:337777 D*(2007)0 MASS	D^*(2007)0 | Dbar^*(2007) 0 COMP-MESON E04; M08	2006.86+-0.05 MeV	u_+: 0.05376901475059 8178 u_-: 0.05376901475059 8178 Type: NONE Confidence: ND
251. PDG2026:observable: M062:M062M:337794 D*(2010)+- MASS	D^*(2010)+ | D^*(2010)- COMP-MESON E04; M08	2010.27+-0.04 MeV	u_+: 0.04449124559318 8788 u_-: 0.04449124559318 8788 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
252. PDG2026:observable: M064:M064M:335845 K_1(1400) MASS	K_1(1400)0 | Kbar_1(1400)0 | K_1(1400)+ | K_1(1400)- COMP-MESON E04; M08	1403+-7 MeV	u_+: 6.8731168328374306 u_-: 6.8731168328374306 Type: NONE Confidence: ND
253. PDG2026:observable: M065:M065M0:334880 eta rho0 AND pi+ pi- MODES	rho(1700)0 | rho(1700)+ | rho(1700)- COMP-MESON E04; M08	1720+-20 MeV	u_+: 20 u_-: 20 Type: NONE Confidence: ND
254. PDG2026:observable: M066:M066M1:334577 f_2(1430) MASS	f_2(1430)0 COMP-MESON E04; M08	~1430 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
255. PDG2026:observable: M067:M067M1:334825 e+ e- PRODUCTION	phi(1680)0 COMP-MESON E04; M08	1680+-20 MeV	u_+: 20 u_-: 20 Type: NONE Confidence: ND
256. PDG2026:observable: M068:M068M:334927 f_0(1710) Breit-Wigner MASS	f_0(1710)0 COMP-MESON E04; M08	1723+7-6 MeV	u_+: 6.7390813885535801 u_-: 5.5617775425136911 Type: NONE Confidence: ND
257. PDG2026:observable: M070:M070M:343505 J/psi(1S) MASS	J/psi(1S) COMP-MESON E04; M08	3096.900+-0.006 MeV	u_+: 0.0061858811299374487 u_-: 0.0061858811299374487 Type: NONE Confidence: ND
258. PDG2026:observable: M071:M071M:345299 psi(2S) MASS	psi(2S) COMP-MESON E04; M08	3686.097+-0.011 MeV	u_+: 0.01055310115798376 u_-: 0.01055310115798376 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
259. PDG2026:observable: M072:M072M:346249 psi(4040) MASS	psi(4040) COMP-MESON E04; M08	4040+-4 MeV	u_+: 4.2999999999999998 u_-: 4.2999999999999998 Type: NONE Confidence: ND
260. PDG2026:observable: M073:M073M:346684 psi(4415) MASS	psi(4415) COMP-MESON E04; M08	4415+-5 MeV	u_+: 4.5278832533731954 u_-: 4.5278832533731954 Type: NONE Confidence: ND
261. PDG2026:observable: M074:M074M:346460 psi(4230) MASS	psi(4230) COMP-MESON E04; M08	4221.7+-2.5 MeV	u_+: 2.4568685085110582 u_-: 2.454400623477099 Type: NONE Confidence: ND
262. PDG2026:observable: M075:M075M:334966 pi(1800) MASS	pi(1800)0 | pi(1800)+ | pi(1800)- COMP-MESON E04; M08	1810+9-11 MeV	u_+: 9.4045949221033531 u_-: 11.027513999271919 Type: NONE Confidence: ND
263. PDG2026:observable: M076:M076M:347129 chi_b0(1P) MASS	chi_b0(1P) COMP-MESON E04; M08	9859.44+-0.42+-0.31 MeV	u_+: 0.4199999999999998 u_-: 0.4199999999999998 Type: NONE Confidence: ND
264. PDG2026:observable: M077:M077M:347175 chi_b1(1P) MASS	chi_b1(1P) COMP-MESON E04; M08	9892.78+-0.26+-0.31 MeV	u_+: 0.26000000000000001 u_-: 0.26000000000000001 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
265. PDG2026:observable: M078:M078M:347234 chi_b2(1P) MASS	chi_b2(1P) COMP-MESON E04; M08	9912.21+-0.26+-0.31 MeV	u_+: 0.260000000000001 u_-: 0.260000000000001 Type: NONE Confidence: ND
266. PDG2026:observable: M079:M079M:347497 chi_b0(2P) MASS	chi_b0(2P) COMP-MESON E04; M08	10232.5+-0.4+-0.5 MeV	u_+: 0.400000000000002 u_-: 0.400000000000002 Type: NONE Confidence: ND
267. PDG2026:observable: M080:M080M:347543 chi_b1(2P) MASS	chi_b1(2P) COMP-MESON E04; M08	10255.46+-0.22+-0.50 MeV	u_+: 0.22 u_-: 0.22 Type: NONE Confidence: ND
268. PDG2026:observable: M081:M081M:347610 chi_b2(2P) MASS	chi_b2(2P) COMP-MESON E04; M08	10268.65+-0.22+-0.50 MeV	u_+: 0.22 u_-: 0.22 Type: NONE Confidence: ND
269. PDG2026:observable: M082:M082M:335203 f_J(2220) MASS	f_J(2220)0 COMP-MESON E04; M08	2231.1+-3.5 MeV	u_+: 3.5335248107562318 u_-: 3.5214610779100788 Type: NONE Confidence: ND
270. PDG2026:observable: M084:M084M:334658 f_1(1510) MASS	f_1(1510)0 COMP-MESON E04; M08	1518+-5 MeV	u_+: 5.2162953426963403 u_-: 5.2162953426963403 Type: NONE Confidence: ND
271. PDG2026:observable: M085:M085M:335011 X(1835) MASS	X(1835)0 COMP-MESON E04; M08	1834.0+4.0-3.0 MeV	u_+: 4.1178707457023851 u_-: 2.9844075950960041 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
272. PDG2026:observable: M089:M089M:335277 f_6(2510) MASS	f_6(2510)0 COMP-MESON E04; M08	2470+-50 MeV	u_+: 44.999997965452351 u_-: 44.999997965452351 Type: NONE Confidence: ND
273. PDG2026:observable: M090:M090M:336034 K_3(2320) MASS	K_3(2320)0 | Kbar_3(2320)0 | K_3(2320)+ | K_3(2320)- COMP-MESON E04; M08	2324+-24 MeV	u_+: 24 u_-: 24 Type: NONE Confidence: ND
274. PDG2026:observable: M091:M091M:336040 K_4(2500) MASS	K_4(2500)0 | Kbar_4(2500)0 | K_4(2500)+ | K_4(2500)- COMP-MESON E04; M08	2490+-20 MeV	u_+: 20 u_-: 20 Type: NONE Confidence: ND
275. PDG2026:observable: M092:M092M:347804 Upsilon(10860) MASS	Upsilon(10860) COMP-MESON E04; M08	10885.2+2.6-1.6 MeV	u_+: 2.5658471171469732 u_-: 1.5686101402474319 Type: NONE Confidence: ND
276. PDG2026:observable: M093:M093M:347905 Upsilon(11020) MASS	Upsilon(11020) COMP-MESON E04; M08	11000+-4 MeV	u_+: 4.0234281589264329 u_-: 4.0242307387762244 Type: NONE Confidence: ND
277. PDG2026:observable: M094:M094M:335865 K*(1410) MASS	K^*(1410)0 | Kbar^*(1410)0 | K^*(1410)+ | K^*(1410)- COMP-MESON E04; M08	1414+-15 MeV	u_+: 14.53984543474516 u_-: 14.53984543474516 Type: NONE Confidence: ND
278. PDG2026:observable: M095:M095M:335946 K*(1680) MASS	K^*(1680)0 | Kbar^*(1680)0 | K^*(1680)+ | K^*(1680)- COMP-MESON E04; M08	1718+-18 MeV	u_+: 17.078726916545191 u_-: 18.698162672594339 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
279. PDG2026:observable: M098:M098M:336036 K_5*(2380) MASS	K_5^*(2380)0 | Kbar_5^*(2380)0 | K_5^*(2380)+ | K_5^*(2380)- COMP-MESON E04; M08	2382+-24 MeV	u_+: 23.600847442411889 u_-: 23.600847442411889 Type: NONE Confidence: ND
280. PDG2026:observable: M099:M099M:335944 K_1(1650) MASS	K_1(1650)0 | Kbar_1(1650)0 | K_1(1650)+ | K_1(1650)- COMP-MESON E04; M08	1650+-50 MeV	u_+: 50 u_-: 50 Type: NONE Confidence: ND
281. PDG2026:observable: M101:M101M:335032 eta_2(1870) MASS	eta_2(1870)0 COMP-MESON E04; M08	1842+-8 MeV	u_+: 8.0430753075960109 u_-: 8.0430753075960109 Type: NONE Confidence: ND
282. PDG2026:observable: M103:M103M:335180 phi(2170) MASS	phi(2170)0 | phi(2170)+ | phi(2170)- COMP-MESON E04; M08	2164+-5 MeV	u_+: 5.1132736001610546 u_-: 5.1132736001610546 Type: NONE Confidence: ND
283. PDG2026:observable: M104:M104M:336003 K_2*(1980) MASS	K_2^*(1980)0 | Kbar_2^*(1980)0 | K_2^*(1980)+ | K_2^*(1980)- COMP-MESON E04; M08	1990+60-50 MeV	u_+: 59.105152569706007 u_-: 45.405772247239312 Type: NONE Confidence: ND
284. PDG2026:observable: M105:M105M0:334592 rho(1450) MASS	rho(1450)0 | rho(1450)+ | rho(1450)- COMP-MESON E04; M08	1465 +-25 MeV Presentation: 1465+-25	u_+: 25 u_-: 25 Type: NONE Confidence: ND
285. PDG2026:observable: M106:M106M:335112 f_2(2010) MASS	f_2(2010)0 COMP-MESON E04; M08	2010+60-80 MeV	u_+: 62 u_-: 76 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
286. PDG2026:observable: M107:M107M:335234 f_2(2300) MASS	f_2(2300)0 COMP-MESON E04; M08	2297+-28 MeV	u_+: 28 u_-: 28 Type: NONE Confidence: ND
287. PDG2026:observable: M108:M108M:335252 f_2(2340) MASS	f_2(2340)0 COMP-MESON E04; M08	2346+21-10 MeV	u_+: 21.300562489710181 u_-: 9.7423326379963484 Type: NONE Confidence: ND
288. PDG2026:observable: M109:M109M:334552 h_1(1415) MASS	h_1(1415)0 COMP-MESON E04; M08	1409+9-8 MeV	u_+: 9.4194068122733494 u_-: 7.776226726894202 Type: NONE Confidence: ND
289. PDG2026:observable: M112:M112M:335199 f_0(2200) MASS	f_0(2200)0 COMP-MESON E04; M08	2187+-14 MeV	u_+: 13.533041759519227 u_-: 14.057750535181979 Type: NONE Confidence: ND
290. PDG2026:observable: M114:M114M:334950 eta(1760) MASS	eta(1760)0 COMP-MESON E04; M08	1751+-15 MeV	u_+: 14.814874939752929 u_-: 14.980145590239159 Type: NONE Confidence: ND
291. PDG2026:observable: M115:M115M:335232 eta(2225) MASS	eta(2225)0 COMP-MESON E04; M08	2221+13-10 MeV	u_+: 13.271333870534555 u_-: 9.5202460778307483 Type: NONE Confidence: ND
292. PDG2026:observable: M117:M117M:334752 f_2(1640) MASS	f_2(1640)0 COMP-MESON E04; M08	1639+-6 MeV	u_+: 5.8313784211153239 u_-: 5.8313784211153239 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
293. PDG2026:observable: M121:M121M:338481 D_s1(2536)+- MASS	D_s1(2536)+ | D_s1(2536)- COMP-MESON E04; M08	2535.12+-0.06 MeV	u_+: 0.060350977470898563 u_-: 0.060350977470898563 Type: NONE Confidence: ND
294. PDG2026:observable: M123:M123M:334703 f_2(1565) MASS	f_2(1565)0 COMP-MESON E04; M08	1571+-13 MeV	u_+: 13.379294632448991 u_-: 13.379294632448991 Type: NONE Confidence: ND
295. PDG2026:observable: M125:M125M:334567 omega(1420) MASS	omega(1420)0 COMP-MESON E04; M08	1410+-60 MeV	u_+: 60 u_-: 60 Type: NONE Confidence: ND
296. PDG2026:observable: M126:M126M:334768 omega(1650) MASS	omega(1650)0 COMP-MESON E04; M08	1670+-30 MeV	u_+: 30 u_-: 30 Type: NONE Confidence: ND
297. PDG2026:observable: M129:M129M:336041 K(3100) MASS	K(3100)0 | Kbar(3100)0 | K(3100)+ | K(3100)- COMP-MESON E04; M08	~3100 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
298. PDG2026:observable: M129:M129M1:336042 3-BODY DECAYS	K(3100)0 | Kbar(3100)0 | K(3100)+ | K(3100)- COMP-MESON E04; M08	3054+-11 MeV	u_+: 10.6814884244502 6 u_-: 10.6814884244502 6 Type: NONE Confidence: ND
299. PDG2026:observable: M129:M129M2:336043 4-BODY DECAYS	K(3100)0 | Kbar(3100)0 | K(3100)+ | K(3100)- COMP-MESON E04; M08	3059+-11 MeV	u_+: 10.6412522648014 39 u_-: 10.6412522648014 39 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
300. PDG2026:observable: M134:M134M:335999 K_0*(1950) MASS	K_0^*(1950)0 | Kbar_0^*(1950)0 | K_0^*(1950)+ | K_0^*(1950)- COMP-MESON E04; M08	1957+-14 MeV	u_+: 14.320221981073329 u_-: 14.320221981073329 Type: NONE Confidence: ND
301. PDG2026:observable: M135:M135M:335075 f_2(1950) MASS	f_2(1950)0 COMP-MESON E04; M08	1936+-12 MeV	u_+: 11.730867482756999 u_-: 11.730867482756999 Type: NONE Confidence: ND
302. PDG2026:observable: M142:M142M2:335057 f_2(1910) omega omega MODE	f_2(1910)0 COMP-MESON E04; M08	1900+-9 MeV	u_+: 9.266557494580514 u_-: 9.266557494580514 Type: NONE Confidence: ND
303. PDG2026:observable: M144:M144M:344851 h_c(1P) MASS	h_c(1P) COMP-MESON E04; M08	3525.37+-0.14 MeV	u_+: 0.14473463841322931 u_-: 0.14473463841322931 Type: NONE Confidence: ND
304. PDG2026:observable: M146:M146M:335990 K_2(1820) MASS	K_2(1820)0 | Kbar_2(1820)0 | K_2(1820)+ | K_2(1820)- COMP-MESON E04; M08	1819+-12 MeV	u_+: 12.067628581227879 u_-: 12.471842992923371 Type: NONE Confidence: ND
305. PDG2026:observable: M147:M147M:334506 Mass (Breit-Wigner)	f_0(1370)0 COMP-MESON E04; M08	1200 to 1500 MeV	u_+: NA u_-: NA Type: R Confidence: ND
306. PDG2026:observable: M148:M148M:338509 D_s2*(2573) MASS	D_s2^*(2573)+ | D_s2^*(2573)- COMP-MESON E04; M08	2569.1+-0.8 MeV	u_+: 0.82780630176758352 u_-: 0.82649831496169768 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
307. PDG2026:observable: M149:M149M:334579 a_0(1450) MASS	a_0(1450)0 | a_0(1450)+ | a_0(1450)- COMP-MESON E04; M08	1439+-34 MeV	u_+: 34.426230116735738 u_-: 34.426230116735738 Type: NONE Confidence: ND
308. PDG2026:observable: M151:M151M:342551 B_J*(5732) MASS	B_J^*(5732)0 | Bbar_J^*(5732)0 | B_J^*(5732)+ | B_J^*(5732)- COMP-MESON E04; M08	5698+-8 MeV	u_+: 7.5869468243446372 u_-: 7.6930254688839934 Type: NONE Confidence: ND
309. PDG2026:observable: M152:M152M:334628 f_0(1500) MASS	f_0(1500)0 COMP-MESON E04; M08	1522+-25 MeV	u_+: 25 u_-: 25 Type: NONE Confidence: ND
310. PDG2026:observable: M153:M153M:343151 B_sJ*(5850) MASS	B_sJ^*(5850)0 | Bbar_sJ^*(5850)0 COMP-MESON E04; M08	5853+-15 MeV	u_+: 15 u_-: 15 Type: NONE Confidence: ND
311. PDG2026:observable: M154:M154M:334758 eta_2(1645) MASS	eta_2(1645)0 COMP-MESON E04; M08	1617+-5 MeV	u_+: 5.2613147962640197 u_-: 5.2613147962640197 Type: NONE Confidence: ND
312. PDG2026:observable: M156:M156M:335120 f_0(2020) MASS	f_0(2020)0 COMP-MESON E04; M08	1982+54.1-3.0 MeV	u_+: 54.083269131959838 u_-: 3 Type: NONE Confidence: ND
313. PDG2026:observable: M158:M158M:337854 D*(2640)+- MASS	D^*(2640)0 | Dbar^*(2640)0 | D^*(2640)+ | D^*(2640)- COMP-MESON E04; M08	2637+-6 MeV	u_+: 6.3245553203367559 u_-: 6.3245553203367559 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
314. PDG2026:observable: M159:M159M:346204 chi_c0(3915) MASS	chi_c0(3915) COMP-MESON E04; M08	3922.1+-1.8 MeV	u_+: 1.77307050384254 29 u_-: 1.76036183430800 81 Type: NONE Confidence: ND
315. PDG2026:observable: M160:M160M:335942 K(1630) MASS	K(1630)0 | Kbar(1630)0 | K(1630)+ | K(1630)- COMP-MESON E04; M08	1629+-7 MeV	u_+: 7 u_-: 7 Type: NONE Confidence: ND
316. PDG2026:observable: M161:M161M:334740 a_1(1640) MASS	a_1(1640)0 | a_1(1640)+ | a_1(1640)- COMP-MESON E04; M08	1655+-16 MeV	u_+: 15.6153624587929 11 u_-: 16.7429048405724 61 Type: NONE Confidence: ND
317. PDG2026:observable: M162:M162M:334903 a_2(1700) MASS	a_2(1700)0 | a_2(1700)+ | a_2(1700)- COMP-MESON E04; M08	1706+-14 MeV	u_+: 13.5014409315998 2 u_-: 14.8390744674554 29 Type: NONE Confidence: ND
318. PDG2026:observable: M164:M164TMP: 334723 pi_1(1600) T-Matrix Pole sqrt(s)	pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)- COMP-MESON E04; M08	(1480 -- 1680) - i (150 -- 300) MeV Presentation: (1480--1680) -i(150--300)	u_+: NA u_-: NA Type: R Confidence: ND
319. PDG2026:observable: M164:M164MEP: 334724 pi_1(1600) MASS (eta pi mode)	pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)- COMP-MESON E04; M08	1354+-25 MeV	u_+: 25.6203714275794 11 u_-: 24.3777786955451 Type: NONE Confidence: ND
320. PDG2026:observable: M164:M164M:334725 pi_1(1600) MASS (non-eta pi mode)	pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)- COMP-MESON E04; M08	1645+40-17 MeV	u_+: 39.7197562267507 18 u_-: 16.5756798763040 68 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
321. PDG2026:observable: M166:M166M:334720 h_1(1595) MASS	h_1(1595)0 COMP-MESON E04; M08	1594+18-60 MeV	u_+: 18.0277563773199 5 u_-: 61.8465843842649 08 Type: NONE Confidence: ND
322. PDG2026:observable: M168:M168M:335156 f_0(2100) MASS	f_0(2100)0 COMP-MESON E04; M08	2095+17-19 MeV	u_+: 17.0604966887379 81 u_-: 18.9796610074380 18 Type: NONE Confidence: ND
323. PDG2026:observable: M171:M171M:346822 eta_b(1S) MASS	eta_b(1S) COMP-MESON E04; M08	9398.7+-2.0 MeV	u_+: 2.08161014572945 2 u_-: 2.00237037046477 Type: NONE Confidence: ND
324. PDG2026:observable: M172:M172M:338429 D_s0*(2317)+- MASS	D_s0^*(2317)+ | D_ s0^*(2317)- COMP-MESON E04; M08	2317.8+-0.5 MeV	u_+: 0.51003296350202 032 u_-: 0.51003296350202 032 Type: NONE Confidence: ND
325. PDG2026:observable: M173:M173M:338448 D_s1(2460)+- MASS	D_s1(2460)+ | D_s1(2460)- COMP-MESON E04; M08	2459.5+-0.6 MeV	u_+: 0.63443027023618 292 u_-: 0.63443027023618 292 Type: NONE Confidence: ND
326. PDG2026:observable: M174:M174TMP: 335785 K_0*(700) T-Matrix Pole sqrt(s)	K_0^*(700)0 | Kbar_ 0^*(700)0 | K_0^*(700)+ | K_0^*(700)- COMP-MESON E04; M08	(630 -- 730) - i (260 -- 340) MeV Presentation: (630--730) -i(260--340)	u_+: NA u_-: NA Type: R Confidence: ND
327. PDG2026:observable: M174:M174M:335786 K_0*(700) Breit-Wigner Mass	K_0^*(700)0 | Kbar_ 0^*(700)0 | K_0^*(700)+ | K_0^*(700)- COMP-MESON E04; M08	838+-11 MeV	u_+: 11.3879040790194 1 u_-: 11.2973395209359 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
328. PDG2026:observable: M175:M175M5:334616 eta(1475) MASS	eta(1475)0 COMP-MESON E04; M08	1476+-4 MeV	u_+: 3.92907502750456 58 u_-: 4.01031273484770 26 Type: NONE Confidence: ND
329. PDG2026:observable: M176:M176M:346130 chi_c1(3872) MASS FROM J/psi X MODE	chi_c1(3872) COMP-MESON E04; M08	3871.64+-0.06 MeV	u_+: 0.05709885902177 2717 u_-: 0.05709583843793 0757 Type: NONE Confidence: ND
330. PDG2026:observable: M177:M177M:347489 Upsilon_2(1D) MASS	Upsilon_2(1D) COMP-MESON E04; M08	10163.7+-1.4 MeV	u_+: 1.43860105887897 99 u_-: 1.43860105887897 99 Type: NONE Confidence: ND
331. PDG2026:observable: M180:M180M:337827 D_1(2430)0 MASS	D_1(2430)0 | Dbar_1(2430) 0 COMP-MESON E04; M08	2412+-9 MeV	u_+: 9.17479434026708 77 u_-: 9.17479434026708 77 Type: NONE Confidence: ND
332. PDG2026:observable: M181:M181M:346635 psi(4360) MASS	psi(4360) COMP-MESON E04; M08	4373+-7 MeV	u_+: 7.02670299984180 07 u_-: 7.12633898599779 33 Type: NONE Confidence: ND
333. PDG2026:observable: M182:M182M:338524 D_s1*(2700)+ MASS	D_s1^*(2700)+ | D_ s1^*(2700)- COMP-MESON E04; M08	2714+-5 MeV	u_+: 5.33752618609189 34 u_-: 4.57804726750868 96 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
334. PDG2026:observable: M185:M185M:335043 pi_2(1880) MASS	pi_2(1880)0 COMP-MESON E04; M08	1874+26-5 MeV	u_+: 25.897946058811009 u_-: 4.7988156638370416 Type: NONE Confidence: ND
335. PDG2026:observable: M186:M186M:343137 B_s2*(5840)0 MASS	B_s2^*(5840)0 | Bbar_s2^*(5840)0 COMP-MESON E04; M08	5839.88+-0.12 MeV	u_+: 0.1159842277505118 u_-: 0.1159842277505118 Type: NONE Confidence: ND
336. PDG2026:observable: M187:M187M:343129 B_s1(5830)0 MASS	B_s1(5830)0 | Bbar_s1(5830)0 COMP-MESON E04; M08	5828.73+-0.20 MeV	u_+: 0.19651551951131191 u_-: 0.19651551951131191 Type: NONE Confidence: ND
337. PDG2026:observable: M188:M188M:334714 rho(1570) MASS	rho(1570)0 COMP-MESON E04; M08	1570+-70 MeV	u_+: 71.693793315739683 u_-: 71.693793315739683 Type: NONE Confidence: ND
338. PDG2026:observable: M189:M189M:346767 psi(4660) MASS	psi(4660) COMP-MESON E04; M08	4623+-10 MeV	u_+: 10.20904494225131 u_-: 10.127972763403161 Type: NONE Confidence: ND
339. PDG2026:observable: M190:M190M:346451 X(4160) MASS	X(4160) COMP-MESON E04; M08	4153+23-21 MeV	u_+: 23.037654818780229 u_-: 20.816561780400072 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
340. PDG2026:observable: M191:M191M:347990 T_c_cbar(4050)+ MASS	T_c_cbar(4050)+ COMP-MESON E04; M08	4051+24-40 MeV	u_+: 24.413111231467401 u_-: 43.324358044868937 Type: NONE Confidence: ND
341. PDG2026:observable: M192:M192M:348026 T_c_cbar(4250)+ MASS	T_c_cbar(4250)+ COMP-MESON E04; M08	4250+190-50 MeV	u_+: 185.29975715040749 u_-: 45.453272709454048 Type: NONE Confidence: ND
342. PDG2026:observable: M193:M193M:346336 chi_c1(4140) MASS	chi_c1(4140) COMP-MESON E04; M08	4146.5+-3.0 MeV	u_+: 2.988542411834247 u_-: 3.0016153327920829 Type: NONE Confidence: ND
343. PDG2026:observable: M194:M194M:346628 X(4350) MASS	X(4350) COMP-MESON E04; M08	4351+-5 MeV	u_+: 4.6529560496527367 u_-: 5.1478150704935004 Type: NONE Confidence: ND
344. PDG2026:observable: M195:M195M:348030 T_c_cbar_1(4430)+ MASS	T_c_cbar_1(4430)+ COMP-MESON E04; M08	4478+15-17 MeV	u_+: 14.769170160915401 u_-: 17.453443521646189 Type: NONE Confidence: ND
345. PDG2026:observable: M196:M196M:338533 D_s1*(2860)+ MASS	D_s1^*(2860)+ | D_s1^*(2860)- COMP-MESON E04; M08	2859+-27 MeV	u_+: 26.832815729997481 u_-: 26.832815729997481 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
346. PDG2026:observable: M197:M197M:338544 D_sJ(3040)+ MASS	D_sJ(3040)+ | D_sJ(3040)- COMP-MESON E04; M08	3044+31-9 MeV	u_+: 31.0483493925200 48 u_-: 9.43398113205660 32 Type: NONE Confidence: ND
347. PDG2026:observable: M198:M198M:337843 D_0(2550)0 MASS	D_0(2550)0 | Dbar_0(2550) 0 | D_0(2550)+ | D_0(2550) - COMP-MESON E04; M08	2549+-19 MeV	u_+: 19.0040028413363 41 u_-: 19.0040028413363 41 Type: NONE Confidence: ND
348. PDG2026:observable: M199:M199M:337846 D_1*(2600)0 MASS	D_1^*(2600)0 | Dbar_ 1^*(2600)0 | D_1^*(2600)+ | D_1^*(2600)- COMP-MESON E04; M08	2627+-10 MeV	u_+: 9.81282444363102 79 u_-: 9.81282444363102 79 Type: NONE Confidence: ND
349. PDG2026:observable: M200:M200M:347281 eta_b(2S) MASS	eta_b(2S) COMP-MESON E04; M08	9999+-4 MeV	u_+: 4.48218696620299 36 u_-: 3.98246155034797 5 Type: NONE Confidence: ND
350. PDG2026:observable: M203:M203M:337860 D_3*(2750) MASS	D_3^*(2750)0 | Dbar_ 3^*(2750)0 | D_3^*(2750)+ | D_3^*(2750)- COMP-MESON E04; M08	2763.1+-3.2 MeV	u_+: 3.19085331467817 88 u_-: 3.19085331467817 88 Type: NONE Confidence: ND
351. PDG2026:observable: M204:M204M:347229 h_b(1P) MASS	h_b(1P) COMP-MESON E04; M08	9899.3+-0.8 MeV	u_+: 0.75076896116034 797 u_-: 0.75076896116034 797 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
352. PDG2026:observable: M205:M205M:347593 h_b(2P) MASS	h_b(2P) COMP-MESON E04; M08	10259.8+-1.2 MeV	u_+: 1.2083045973594571 u_-: 1.2083045973594571 Type: NONE Confidence: ND
353. PDG2026:observable: M206:M206M:347734 chi_b1(3P) MASS	chi_b1(3P) COMP-MESON E04; M08	10513.4+-0.7 MeV	u_+: 0.67007462271003815 u_-: 0.67007462271003815 Type: NONE Confidence: ND
354. PDG2026:observable: M207:M207M:348052 T_b_bbar_1(10610)+- MASS	T_b_bbar_1(10610)+ COMP-MESON E04; M08	10607.2+-2.0 MeV	u_+: 2 u_-: 2 Type: NONE Confidence: ND
355. PDG2026:observable: M207:M207M0:348053 T_b_bbar_1(10610)0 MASS	T_b_bbar_1(10610)+ COMP-MESON E04; M08	10609+-6 MeV	u_+: 5.6568542494923806 u_-: 5.6568542494923806 Type: NONE Confidence: ND
356. PDG2026:observable: M208:M208M:348075 T_b_bbar_1(10650)+ MASS	T_b_bbar_1(10650)+ COMP-MESON E04; M08	10652.2+-1.5 MeV	u_+: 1.5 u_-: 1.5 Type: NONE Confidence: ND
357. PDG2026:observable: M210:M210M:347938 T_c_cbar_1(3900) MASS	T_c_cbar_1(3900)+ COMP-MESON E04; M08	3886.7+-2.3 MeV	u_+: 2.2933691974144777 u_-: 2.2732822895762877 Type: NONE Confidence: ND
358. PDG2026:observable: M212:M212M:346100 psi_2(3823) MASS	psi_2(3823) COMP-MESON E04; M08	3823.51+-0.34 MeV	u_+: 0.33904706932929501 u_-: 0.33904706932929501 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
359. PDG2026:observable: M213:M213M:347976 T_c_cbar(4020) MASS	T_c_cbar(4020)+ COMP-MESON E04; M08	4024.1+-1.9 MeV	u_+: 1.82469063465848 7 u_-: 1.96716232389574 41 Type: NONE Confidence: ND
360. PDG2026:observable: M216:M216M:348022 T_c_cbar_0(4240)+ MASS	T_c_cbar_0(4240)0 COMP-MESON E04; M08	4239+50-21 MeV	u_+: 48.4664832642105 38 u_-: 20.5912602819740 02 Type: NONE Confidence: ND
361. PDG2026:observable: M217:M217M:343283 B_c(2S)+- MASS	B_c(2S)+ | B_c(2S)- COMP-MESON E04; M08	6871.2+-1.0 MeV	u_+: 1.01845458192780 3 u_-: 1.01845458192780 3 Type: NONE Confidence: ND
362. PDG2026:observable: M223:M223M:348000 T_c_cbar(4055)+ MASS	T_c_cbar(4055)+ COMP-MESON E04; M08	4054+-3.2 MeV	u_+: 3.16227766016838 u_-: 3.16227766016838 Type: NONE Confidence: ND
363. PDG2026:observable: M226:M226M:338535 D_s3*(2860)+ MASS	D_s3^*(2860)+ | D_ s3^*(2860)- COMP-MESON E04; M08	2860+-7 MeV	u_+: 7.00071424927485 53 u_-: 7.00071424927485 53 Type: NONE Confidence: ND
364. PDG2026:observable: M227:M227M:335070 a_0(1950) MASS	a_0(1950)0 COMP-MESON E04; M08	1931+-26 MeV	u_+: 26.0768096208106 u_-: 26.0768096208106 Type: NONE Confidence: ND
365. PDG2026:observable: M228:M228M:337857 D_2(2740)0 MASS	D_2(2740)0 | Dbar_2(2740) 0 | D_2(2740)+ | D_2(2740) - COMP-MESON E04; M08	2747+-6 MeV	u_+: 6.39329307322381 27 u_-: 6.39329307322381 27 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
366. PDG2026:observable: M229:M229M:337872 D_2*(3000)0 MASS	D_2^*(3000)0 | Dbar_2^*(3000)0 | D_2^*(3000)+ | D_2^*(3000)- COMP-MESON E04; M08	3210+-60 MeV	u_+: 56.938563381947027 u_-: 56.938563381947027 Type: NONE Confidence: ND
367. PDG2026:observable: M231:M231M:348012 T_c_cbar_1(4200)+ MASS	T_c_cbar_1(4200)+ COMP-MESON E04; M08	4242+-26 MeV	u_+: 26.306839009567629 u_-: 25.567261845351091 Type: NONE Confidence: ND
368. PDG2026:observable: M232:M232M:348036 T_b_sbar(5568)+ MASS	T_b_sbar(5568)+ COMP-MESON E04; M08	5566.9+-3.3 MeV	u_+: 3.25576411921994 2 u_-: 3.32415402771893 3 Type: NONE Confidence: ND
369. PDG2026:observable: M233:M233M:346624 chi_c1(4274) MASS	chi_c1(4274) COMP-MESON E04; M08	4290+-6 MeV	u_+: 5.6823979063752086 u_-: 6.461081982391395 Type: NONE Confidence: ND
370. PDG2026:observable: M234:M234M:346755 chi_c0(4500) MASS	chi_c0(4500) COMP-MESON E04; M08	4484+-12 MeV	u_+: 12.002820072457739 u_-: 11.985284550375541 Type: NONE Confidence: ND
371. PDG2026:observable: M235:M235M:346816 chi_c0(4700) MASS	chi_c0(4700) COMP-MESON E04; M08	4708+6-4 MeV	u_+: 5.9690361844477042 u_-: 3.9408505465575612 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
372. PDG2026:observable: M237:M237M:346122 chi_c0(3860) MASS	chi_c0(3860) COMP-MESON E04; M08	3862+50-35 MeV	u_+: 47.707441767506253 u_-: 34.539832078341092 Type: NONE Confidence: ND
373. PDG2026:observable: M238:M238M:347741 chi_b2(3P) MASS	chi_b2(3P) COMP-MESON E04; M08	10524.0+-0.8 MeV	u_+: 0.77833154889160183 u_-: 0.77833154889160183 Type: NONE Confidence: ND
374. PDG2026:observable: M239:M239M:335106 pi_2(2005) MASS	pi_2(2005)0 COMP-MESON E04; M08	1963+17-27 MeV	u_+: 16.663540525911429 u_-: 27.418323182256639 Type: NONE Confidence: ND
375. PDG2026:observable: M240:M240M:348006 T_c_cbar(4100)+ MASS	T_c_cbar(4100)+ COMP-MESON E04; M08	4096+-28 MeV	u_+: 26.907248094147 42 u_-: 29.732137494637009 Type: NONE Confidence: ND
376. PDG2026:observable: M241:M241M:346116 psi_3(3842) MASS	psi_3(3842) COMP-MESON E04; M08	3842.71+-0.20 MeV	u_+: 0.20000000000000001 u_-: 0.20000000000000001 Type: NONE Confidence: ND
377. PDG2026:observable: M243:M243M:347800 Upsilon(10753) MASS	Upsilon(10753) COMP-MESON E04; M08	10756.6+-2.8 MeV	u_+: 2.8460498941515 42 u_-: 2.8460498941515 42 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
378. PDG2026:observable: M244:M244M+:342539 B_1(5721)+ mass	B_1(5721)0 | Bbar_1(5721)0 | B_1(5721)+ | B_1(5721)- COMP-MESON E04; M08	5726.0+2.5-2.7 MeV	u_+: 2.45596777983158 4 u_-: 2.72915140828024 59 Type: NONE Confidence: ND
379. PDG2026:observable: M244:M244M0:342542 B_1(5721)0 mass	B_1(5721)0 | Bbar_1(5721)0 | B_1(5721)+ | B_1(5721)- COMP-MESON E04; M08	5726.1+-1.2 MeV	u_+: 1.23302959923312 99 u_-: 1.26292541194705 9 Type: NONE Confidence: ND
380. PDG2026:observable: M245:M245M+:342556 B_2*(5747)+ mass	B_2^*(5747)0 | Bbar_2^*(5747)0 | B_2^*(5747)+ | B_2^*(5747)- COMP-MESON E04; M08	5737.3+-0.7 MeV	u_+: 0.71060165855206 67 u_-: 0.73206597699796 105 Type: NONE Confidence: ND
381. PDG2026:observable: M245:M245M0:342559 B_2*(5747)0 mass	B_2^*(5747)0 | Bbar_2^*(5747)0 | B_2^*(5747)+ | B_2^*(5747)- COMP-MESON E04; M08	5739.6+-0.7 MeV	u_+: 0.65346568058668 519 u_-: 0.65346568058668 519 Type: NONE Confidence: ND
382. PDG2026:observable: M246:M246M+:342571 B_J(5840)+ MASS	B_J(5840)0 | Bbar_J(5840)0 | B_J(5840)+ | B_J(5840)- COMP-MESON E04; M08	5851+-19 MeV	u_+: 19.1051594213077 78 u_-: 19.1051594213077 78 Type: NONE Confidence: ND
383. PDG2026:observable: M246:M246M0:342574 B_J(5840)0 MASS	B_J(5840)0 | Bbar_J(5840)0 | B_J(5840)+ | B_J(5840)- COMP-MESON E04; M08	5863+-9 MeV	u_+: 8.60261899078296 47 u_-: 8.60261899078296 47 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
384. PDG2026:observable: M247:M247M:335261 eta(2370) MASS	eta(2370)0 COMP-MESON E04; M08	2377+-9 MeV	u_+: 8.807211582813899 u_-: 9.65401633399925 Type: NONE Confidence: ND
385. PDG2026:observable: M248:M248M+:342583 B_J(5970)+ MASS	B_J(5970)0 | Bbar_J(5970)0 | B_J(5970)+ | B_J(5970)- COMP-MESON E04; M08	5965+-5 MeV	u_+: 4.505372310977057 8 u_-: 4.505372310977057 8 Type: NONE Confidence: ND
386. PDG2026:observable: M248:M248M0:342586 B_J(5970)0 MASS	B_J(5970)0 | Bbar_J(5970)0 | B_J(5970)+ | B_J(5970)- COMP-MESON E04; M08	5971+-5 MeV	u_+: 5.347986858679188 u_-: 5.347986858679188 Type: NONE Confidence: ND
387. PDG2026:observable: M249:M249M:337868 D_1*(2760)0 MASS	D_1^*(2760)0 | Dbar_1^*(2760)0 | D_1^*(2760)+ | D_1^*(2760)- COMP-MESON E04; M08	2781+-22 MeV	u_+: 22.203603311174518 u_-: 22.203603311174518 Type: NONE Confidence: ND
388. PDG2026:observable: M250:M250M:347916 T_cbar_sbar_0*(2870)0 MASS	T_cbar_sbar_0^*(2870)0 COMP-MESON E04; M08	2874+-11 MeV	u_+: 10.737127438402171 u_-: 10.737127438402171 Type: NONE Confidence: ND
389. PDG2026:observable: M251:M251M:347922 T_cbar_sbar_1*(2900)0 MASS	T_cbar_sbar_1^*(2900)0 COMP-MESON E04; M08	2904+-5 MeV	u_+: 5.099019513592783 6 u_-: 5.099019513592783 6 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
390. PDG2026:observable: M252:M252M:337813 D_0*(2300) MASS	D_0^*(2300)0 | Dbar_0^*(2300)0 | D_0^*(2300)+ | D_0^*(2300)- COMP-MESON E04; M08	2343+-10 MeV	u_+: 9.6544420794461843 u_-: 9.6544420794461843 Type: NONE Confidence: ND
391. PDG2026:observable: M253:M253M:337817 D_1(2420) MASS	D_1(2420)0 | Dbar_1(2420)0 | D_1(2420)+ | D_1(2420)- COMP-MESON E04; M08	2422.1+-0.6 MeV	u_+: 0.57502282665337301 u_-: 0.57502282665337301 Type: NONE Confidence: ND
392. PDG2026:observable: M254:M254M:337830 D_2*(2460) MASS	D_2^*(2460)0 | Dbar_2^*(2460)0 | D_2^*(2460)+ | D_2^*(2460)- COMP-MESON E04; M08	2461.1+-0.8 MeV	u_+: 0.79751842909338577 u_-: 0.79751842909338577 Type: NONE Confidence: ND
393. PDG2026:observable: M255:M255M:334941 X(1750) MASS	X(1750)0 COMP-MESON E04; M08	1753.8+-2.7 MeV	u_+: 2.67672205618767 3 u_-: 2.7341242708038349 Type: NONE Confidence: ND
394. PDG2026:observable: M256:M256M:338520 D_s0(2590)+ MASS	D_s0(2590)+ | D_s0(2590)- COMP-MESON E04; M08	2591+-9 MeV	u_+: 9.2195444572928871 u_-: 9.2195444572928871 Type: NONE Confidence: ND
395. PDG2026:observable: M257:M257M:343153 B_sJ(6063)0 MASS	B_sJ(6063)0 | Bbar_sJ(6063)0 COMP-MESON E04; M08	6063.5+-1.4 MeV	u_+: 1.4422205101855596 u_-: 1.4422205101855596 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
396. PDG2026:observable: M258:M258M:343157 B_sJ(6114)0 MASS	B_sJ(6114)0 | Bbar_ sJ(6114)0 COMP-MESON E04; M08	6114+-6 MeV	u_+: 5.8309518948453007 u_-: 5.8309518948453007 Type: NONE Confidence: ND
397. PDG2026:observable: M259:M259M:347963 T_c cbar sbar 1(4000) MASS	T_c_cbar_sbar_1(4000)0 COMP-MESON E04; M08	3980 -- 4010 MeV Presentation: 3980--4010	u_+: NA u_-: NA Type: R Confidence: ND
398. PDG2026:observable: M260:M260M:348018 T_c cbar sbar 1(4220)+ MASS	T_c_cbar_sbar_1(4220)+ COMP-MESON E04; M08	4220+50-40 MeV	u_+: 49.244289008980523 u_-: 38.418745424597091 Type: NONE Confidence: ND
399. PDG2026:observable: M261:M261M:346810 chi_c1(4685) MASS	chi_c1(4685) COMP-MESON E04; M08	4677+13-15 MeV	u_+: 13.282382967217816 u_-: 15.144963476323969 Type: NONE Confidence: ND
400. PDG2026:observable: M262:M262M:346761 X(4630) MASS	X(4630) COMP-MESON E04; M08	4636+29-100 MeV	u_+: 29.190907431493944 u_-: 102.4064255252877 Type: NONE Confidence: ND
401. PDG2026:observable: M263:M263M:334917 a_0(1710) MASS	a_0(1710)0 COMP-MESON E04; M08	1713+-19 MeV	u_+: 18.627574622936329 u_-: 18.627574622936329 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
402. PDG2026:observable: M264:M264M:334992 f_0(1810) Breit-Wigner MASS	f_0(1810)0 COMP-MESON E04; M08	1785+-13 MeV	u_+: 13.4781230224536 11 u_-: 13.3053237583047 5 Type: NONE Confidence: ND
403. PDG2026:observable: M265:M265TMG: 347934 T_c_c(3875)+ T-Matrix Pole sqrt(s)	T_c_c(3875)+ COMP-MESON E04; M08	(3874.75+-0.10) - i (0.024+0.001-0.007) MeV Presentation: 3874.75+-0.10	u_+: ND u_-: ND Type: NONE Confidence: ND
404. PDG2026:observable: M266:M266M:335273 f_0(2470) MASS	f_0(2470)0 COMP-MESON E04; M08	2470+6-7 MeV	u_+: 5.65685424949238 06 u_-: 7.21110255092797 82 Type: NONE Confidence: ND
405. PDG2026:observable: M267:M267M:335028 eta_1(1855) MASS	eta_1(1855)0 COMP-MESON E04; M08	1855+11-9 MeV	u_+: 10.8166538263919 7 u_-: 9.05538513813741 73 Type: NONE Confidence: ND
406. PDG2026:observable: M268:M268M:348044 T_c_c_cbar_cbar(6900) 0 MASS	T_c_c_cbar_cbar(6900)+ COMP-MESON E04; M08	6898+-12 MeV	u_+: 11.8879388201560 2 u_-: 11.5195540941920 7 Type: NONE Confidence: ND
407. PDG2026:observable: M269:M269M:347926 T_c_sbar_0*(2900)0 MASS	T_c_sbar_0^*(2900)+ COMP-MESON E04; M08	2892+-21 MeV	u_+: 20.5182845286831 89 u_-: 20.5182845286831 89 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
408. PDG2026:observable: M270:M270M:335221 omega(2220) MASS	omega(2220)0 COMP-MESON E04; M08	2188+-21 MeV	u_+: 20.579664709550 94 u_-: 20.579664709550 94 Type: NONE Confidence: ND
409. PDG2026:observable: M272:M272M:348040 T_c c cbar cbar(6600) MASS	T_c c cbar cbar(6600)+ COMP-MESON E04; M08	6646+28-25 MeV	u_+: 28.349092965260 99 u_-: 24.501265528141 591 Type: NONE Confidence: ND
410. PDG2026:observable: M273:M273M:348048 T_c c cbar cbar(7100) MASS	T_c c cbar cbar(7100)+ COMP-MESON E04; M08	7134+60-29 MeV	u_+: 63.126856408346 519 u_-: 29.154759474226 498 Type: NONE Confidence: ND
411. PDG2026:observable: Q001:Q001M:333589 d-QUARK MASS	d | dbar SM-Q-D1 E02; M03	4.70+-0.07 MeV	u_+: 0.070000000000000 007 u_-: 0.070000000000000 007 Type: NONE Confidence: 90
412. PDG2026:observable: Q002:Q002M:333588 u-QUARK MASS	u | ubar SM-Q-U1 E02; M03	2.16+-0.07 MeV	u_+: 0.070000000000000 007 u_-: 0.070000000000000 007 Type: NONE Confidence: 90
413. PDG2026:observable: Q003:Q003M:333590 s-QUARK MASS	s | sbar SM-Q-D2 E02; M03	92.9 +-0.7 MeV Presentation: 92.9+-0.7	u_+: 0.699999999999999 996 u_-: 0.699999999999999 996 Type: NONE Confidence: 90

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
414. PDG2026:observable: Q004:Q004M:333604 c-QUARK MASS	c | cbar SM-Q-U2 E02; M03	1.2729+-0.0045 GeV	u_+: 0.0044999999999999997 u_-: 0.0044999999999999997 Type: NONE Confidence: 90
415. PDG2026:observable: Q005:Q005M:333611 b-QUARK MASS	b | bbar SM-Q-D3 E02; M03	4.186+-0.006 GeV	u_+: 0.006000000000000001 u_-: 0.006000000000000001 Type: NONE Confidence: 90
416. PDG2026:observable: Q007:Q007TP:333614 t-Quark Mass (Direct Measurements)	t | tbar SM-Q-U3 E02; M03	172.60+-0.27 GeV	u_+: 0.27137500202757259 u_-: 0.27304780674185603 Type: NONE Confidence: ND
417. PDG2026:observable: Q007:Q007TP2:333615 t-Quark Mass from Cross-Section Measurements	t | tbar QUARK_EXTENDED E04; M09	162.5+2.1-1.5 GeV	u_+: 2.1316137087859799 u_-: 1.5324334305524001 Type: NONE Confidence: ND
418. PDG2026:observable: Q007:Q007TP4:333616 t-Quark Pole Mass from Cross-Section Measurements	t | tbar QUARK_EXTENDED E04; M09	172.1+-0.6 GeV	u_+: 0.61408644700147419 u_-: 0.58967762064252371 Type: NONE Confidence: ND
419. PDG2026:observable: S000:S000M:332461 gamma MASS	gamma SM-GA-EM E05; M02	<1 E-18 eV Presentation: <1E-18	u_+: NA u_-: NA Type: U Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
420. PDG2026:observable: S003:S003AMU: 332837 e MASS (atomic mass units u)	e- | e+ ELEMENTARY_OR_EXTENDED E04; M05	(5.485799 09044+-0.000000 00010)E-4 u Presentation: 548.5799 09044+-0.000000 010	u_+: 9.69999999999999 91e-15 u_-: 9.69999999999999 91e-15 Type: NONE Confidence: ND
421. PDG2026:observable: S003:S003M:332838 e MASS	e- | e+ SM-LC-1 E06; M01	0.510998 95069+-0.000000 00016 MeV	u_+: 1.59999999999999 99e-10 u_-: 1.59999999999999 99e-10 Type: NONE Confidence: ND
422. PDG2026:observable: S004:S004AMU: 332848 mu MASS (atomic mass units u)	mu- | mu+ ELEMENTARY_OR_EXTENDED E04; M05	0.113428 9257+-0.000000 0025 u	u_+: 2.50000000000000 01e-09 u_-: 2.50000000000000 01e-09 Type: NONE Confidence: ND
423. PDG2026:observable: S004:S004M:332849 mu MASS	mu- | mu+ SM-LC-2 E03; M04	105.6583 755+-0.0000023 MeV	u_+: 2.3e-06 u_-: 2.3e-06 Type: NONE Confidence: ND
424. PDG2026:observable: S008:S008M:333660 pi+- MASS	pi+ | pi- ELEMENTARY_OR_EXTENDED E04; M05	139.5703 9+-0.00018 MeV	u_+: 0.00018200716040 826009 u_-: 0.00018200716040 826009 Type: NONE Confidence: ND
425. PDG2026:observable: S009:S009M:333693 pi0 MASS	pi0 ELEMENTARY_OR_EXTENDED E04; M05	134.9768+-0.0005 MeV	u_+: 0.00048567928228 423351 u_-: 0.00048567928228 423351 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
426. PDG2026:observable: S010:S010M:335281 K+- MASS	K+ | K- ELEMENTARY_OR_ EXTENDED E04; M05	493.677+-0.015 MeV	u_+: 0.01540566522607 1509 u_-: 0.01540566522607 1509 Type: NONE Confidence: ND
427. PDG2026:observable: S011:S011M:335501 K0 MASS	K0 | Kbar0 ELEMENTARY_OR_ EXTENDED E04; M05	497.611+-0.013 MeV	u_+: 0.01299019975642 42 u_-: 0.01299019975642 42 Type: NONE Confidence: ND
428. PDG2026:observable: S014:S014M:333729 eta MASS	eta ELEMENTARY_OR_ EXTENDED E04; M05	547.862+-0.017 MeV	u_+: 0.01733421371191 4789 u_-: 0.01733421371191 4789 Type: NONE Confidence: ND
429. PDG2026:observable: S016:S016AMU: 348093 p MASS (atomic mass units u)	p | pbar ELEMENTARY_OR_ EXTENDED E04; M05	1.0072764665789+-0.0000000000083 u	u_+: 8.29999999999999 98e-12 u_-: 8.29999999999999 98e-12 Type: NONE Confidence: ND
430. PDG2026:observable: S016:S016M:348095 p MASS (MeV)	p | pbar ELEMENTARY_OR_ EXTENDED E04; M05	938.2720 8943+-0.00000029 MeV	u_+: 2.89999999999999 98e-07 u_-: 2.89999999999999 98e-07 Type: NONE Confidence: ND
431. PDG2026:observable: S017:S017AMU: 348291 n MASS (atomic mass units u)	n | nbar ELEMENTARY_OR_ EXTENDED E04; M05	1.008664 9161+-0.000000 0004 u	u_+: 4.00000000000000 01e-10 u_-: 4.00000000000000 01e-10 Type: NONE Confidence: ND

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432. PDG2026:observable: S017:S017M:348292 n MASS (MeV)	n | nbar ELEMENTARY_OR_EXTENDED E04; M05	939.565421 9+-0.0000005 MeV	u_+: 4.79999999999999 96e-07 u_-: 4.79999999999999 96e-07 Type: NONE Confidence: ND
433. PDG2026:observable: S018:S018M:349367 Lambda MASS	Lambda0 | Lambdabar0 ELEMENTARY_OR_EXTENDED E04; M05	1115.683+-0.006 MeV	u_+: 0.00643763777862 19867 u_-: 0.00643763777862 19867 Type: NONE Confidence: ND
434. PDG2026:observable: S019:S019M:349781 Sigma+ MASS	Sigma+ | Sigmabar- ELEMENTARY_OR_EXTENDED E04; M05	1189.37+-0.07 MeV	u_+: 0.06962288561220 7835 u_-: 0.06962288561220 7835 Type: NONE Confidence: ND
435. PDG2026:observable: S020:S020M:349844 Sigma- MASS	Sigma- | Sigmabar+ ELEMENTARY_OR_EXTENDED E04; M05	1197.449+-0.029 MeV	u_+: 0.02915850307660 851 u_-: 0.02915850307660 851 Type: NONE Confidence: ND
436. PDG2026:observable: S021:S021M:349828 Sigma0 MASS	Sigma0 | Sigmabar0 ELEMENTARY_OR_EXTENDED E04; M05	1192.642+-0.024 MeV	u_+: 0.02359569142575 4781 u_-: 0.02359569142575 4781 Type: NONE Confidence: ND
437. PDG2026:observable: S022:S022M:350344 Xi- MASS	Xi- | Xibar+ ELEMENTARY_OR_EXTENDED E04; M05	1321.71+-0.07 MeV	u_+: 0.06653527362254 3412 u_-: 0.06653527362254 3412 Type: NONE Confidence: ND

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438. PDG2026:observable: S022:S022M1:350346 Xibar+ MASS	Xi- | Xibar+ ELEMENTARY_OR_ EXTENDED E04; M05	1321.71+-0.07 MeV	u_+: 0.06653527362254 3412 u_-: 0.06653527362254 3412 Type: NONE Confidence: ND
439. PDG2026:observable: S023:S023M:350293 Xi0 MASS	Xi0 | Xibar0 ELEMENTARY_OR_ EXTENDED E04; M05	1314.86+-0.20 MeV	u_+: 0.20343194946851 781 u_-: 0.20343194946851 781 Type: NONE Confidence: ND
440. PDG2026:observable: S024:S024M:350469 Omega- MASS	Omega- | Omegabar+ ELEMENTARY_OR_ EXTENDED E04; M05	1672.45+-0.29 MeV	u_+: 0.29314631976651 251 u_-: 0.29314631976651 251 Type: NONE Confidence: ND
441. PDG2026:observable: S026:S026M+:351717 Sigma_b+ MASS	Sigma_b()- | Sigmabar_b()+ | Sigma_b()0 | Sigmabar_ b()0 | Sigma_b()+ | Sigmabar_b()- ELEMENTARY_OR_ EXTENDED E04; M05	5810.56+-0.25 MeV	u_+: 0.25274062207162 651 u_-: 0.25263562022485 619 Type: NONE Confidence: ND
442. PDG2026:observable: S026:S026M-:351718 Sigma_b- MASS	Sigma_b()- | Sigmabar_b()+ | Sigma_b()0 | Sigmabar_ b()0 | Sigma_b()+ | Sigmabar_b()- ELEMENTARY_OR_ EXTENDED E04; M05	5815.64+-0.27 MeV	u_+: 0.27460654053367 178 u_-: 0.27449500386905 717 Type: NONE Confidence: ND
443. PDG2026:observable: S031:S031M:336044 D+- MASS	D+ | D- ELEMENTARY_OR_ EXTENDED E04; M05	1869.66+-0.05 MeV	u_+: 0.04680602241160 9027 u_-: 0.04680602241160 9027 Type: NONE Confidence: ND

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444. PDG2026:observable: S032:S032M:336689 D0 MASS	D0 | Dbar0 ELEMENTARY_OR_EXTENDED E04; M05	1864.84+-0.04 MeV	u_+: 0.04445870561980 1281 u_-: 0.04445870561980 1281 Type: NONE Confidence: ND
445. PDG2026:observable: S033:S033M:350527 Lambda_c+ MASS	Lambda_c()+ | Lambdabar_c()- ELEMENTARY_OR_EXTENDED E04; M05	2286.46+-0.14 MeV	u_+: 0.13999999999999 89 u_-: 0.13999999999999 89 Type: NONE Confidence: ND
446. PDG2026:observable: S034:S034M:337875 D_s+- MASS	D_s()+ | D_s()- ELEMENTARY_OR_EXTENDED E04; M05	1968.35+-0.07 MeV	u_+: 0.06786120952959 6483 u_-: 0.06786120952959 6483 Type: NONE Confidence: ND
447. PDG2026:observable: S035:S035M:332893 tau MASS	tau- | tau+ SM-LC-3 E03; M04	1776.93+-0.09 MeV	u_+: 0.08491495391323 0548 u_-: 0.08758019173411 4063 Type: NONE Confidence: ND
448. PDG2026:observable: S040:S040M:351420 mass(Lambda_b0)	Lambda_b()0 | Lambdabar_b()0 ELEMENTARY_OR_EXTENDED E04; M05	5619.57+-0.16 MeV	u_+: 0.15728560301875 261 u_-: 0.15728560301875 261 Type: NONE Confidence: ND
449. PDG2026:observable: S041:S041M:338546 B+- MASS	B+ | B- ELEMENTARY_OR_EXTENDED E04; M05	5279.41+-0.07 MeV	u_+: 0.07108797774343 7336 u_-: 0.07108797774343 7336 Type: NONE Confidence: ND

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450. PDG2026:observable: S042:S042M:340312 B0 MASS	B0 | Bbar0 ELEMENTARY_OR_ EXTENDED E04; M05	5279.72+-0.08 MeV	u_+: 0.08435943092202 2555 u_-: 0.08435943092202 2555 Type: NONE Confidence: ND
451. PDG2026:observable: S043:S043M:332464 W MASS	W+ | W- SM-GA-W E01; M06	80.3625 +-0.0077 GeV Presentation: 80.3625+-0.0077	u_+: 0.00770000000000 00002 u_-: 0.00770000000000 00002 Type: NONE Confidence: ND
452. PDG2026:observable: S044:S044M:332513 Z MASS	Z0 SM-GA-Z E01; M06	91.1879+-0.0020 GeV	u_+: 0.00196689903262 91501 u_-: 0.00196689903262 91501 Type: NONE Confidence: ND
453. PDG2026:observable: S045:S045M:351029 Xi_c+ MASS	Xi_c()+ | Xibar_c()- ELEMENTARY_OR_ EXTENDED E04; M05	2467.79+-0.15 MeV	u_+: 0.14783797163412 299 u_-: 0.15152111693025 43 Type: NONE Confidence: ND
454. PDG2026:observable: S047:S047M:351327 Omega_c0 MASS	Omega_c()0 | Omegabar_ c()0 ELEMENTARY_OR_ EXTENDED E04; M05	2695.3+-0.4 MeV	u_+: 0.40393344242041 429 u_-: 0.40393344242041 429 Type: NONE Confidence: ND
455. PDG2026:observable: S048:S048M:351116 Xi_c0 MASS	Xi_c()0 | Xibar_c()0 ELEMENTARY_OR_ EXTENDED E04; M05	2470.50+-0.25 MeV	u_+: 0.24609003107560 39 u_-: 0.25936293704487 828 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
456. PDG2026:observable: S053:S053M:351372 Omega_c(2770)0 MASS	Omega_c(2770)0 | Omegabar_c(2770)0 ELEMENTARY_OR_ EXTENDED E04; M05	2766.0+0.9-1.0 MeV	u_+: 0.90442879377866114 u_-: 1.024667347651941 Type: NONE Confidence: ND
457. PDG2026:observable: S058:S058M:351195 Xi_c'+ MASS	Xi_c^'()+ | Xibar_c^'()- ELEMENTARY_OR_ EXTENDED E04; M05	2578.3+-0.4 MeV	u_+: 0.43863752697991182 u_-: 0.43863752697991182 Type: NONE Confidence: ND
458. PDG2026:observable: S059:S059M:351200 Xi_c'0 MASS	Xi_c^'()0 | Xibar_c^'()0 ELEMENTARY_OR_ EXTENDED E04; M05	2578.8+-0.5 MeV	u_+: 0.48022871688352092 u_-: 0.48697631770634259 Type: NONE Confidence: ND
459. PDG2026:observable: S062:S062M+:351724 Sigma_b*+ MASS	Sigma_b^*()- | Sigmabar_b^*()+ | Sigma_b^*()0 | Sigmabar_b^*()0 | Sigma_b^*()+ | Sigmabar_b^*()- ELEMENTARY_OR_ EXTENDED E04; M05	5830.32+-0.27 MeV	u_+: 0.27472915359118399 u_-: 0.27501743053510241 Type: NONE Confidence: ND
460. PDG2026:observable: S062:S062M-:351725 Sigma_b*- MASS	Sigma_b^*()- | Sigmabar_b^*()+ | Sigma_b^*()0 | Sigmabar_b^*()0 | Sigma_b^*()+ | Sigmabar_b^*()- ELEMENTARY_OR_ EXTENDED E04; M05	5834.74+-0.30 MeV	u_+: 0.29816079945753 06 u_-: 0.29855805603668217 Type: NONE Confidence: ND
461. PDG2026:observable: S063:S063M:351884 Omega_b- MASS	Omega_b()- | Omegabar_b()+ ELEMENTARY_OR_ EXTENDED E04; M05	6045.8+-0.8 MeV	u_+: 0.76541057086097497 u_-: 0.76541057086097497 Type: NONE Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
462. PDG2026:observable: S065:S065M:351404 Xi_cc+ MASS	Xi_cc()+ | Xibar_cc()- ELEMENTARY_OR_ EXTENDED E04; M05	3518.9+-0.9 MeV	u_+: 0.94868329805051377 u_-: 0.94868329805051377 Type: NONE Confidence: ND
463. PDG2026:observable: S068:S068M:351409 Xi_cc++ MASS	Xi_cc()++ | Xi_cc()-- ELEMENTARY_OR_ EXTENDED E04; M05	3621.6+-0.4 MeV	u_+: 0.378021163428716 u_-: 0.378021163428716 Type: NONE Confidence: ND
464. PDG2026:observable: S069:S069M-:351742 Xi_b- MASS	Xi_b()- | Xibar_b()+ ELEMENTARY_OR_ EXTENDED E04; M05	5797.0+-0.4 MeV	u_+: 0.43674265363571568 u_-: 0.43674265363571568 Type: NONE Confidence: ND
465. PDG2026:observable: S070:S070M0:351793 Xi_b0 MASS	Xi_b()0 | Xibar_b()0 ELEMENTARY_OR_ EXTENDED E04; M05	5791.7+-0.4 MeV	u_+: 0.41342961949753559 u_-: 0.41342961949753559 Type: NONE Confidence: ND
466. PDG2026:observable: S074:S074M:338415 D_s*+- MASS	D_s^*()+ | D_s^*()- ELEMENTARY_OR_ EXTENDED E04; M05	2112.2+-0.4 MeV	u_+: 0.39518670911609538 u_-: 0.39518670911609538 Type: NONE Confidence: ND
467. PDG2026:observable: S085:S085M:342531 B* MASS	B^*0 | Bbar^*0 | B^*+ | B^*- ELEMENTARY_OR_ EXTENDED E04; M05	5324.75+-0.20 MeV	u_+: 0.20338563693989911 u_-: 0.20338563693989911 Type: NONE Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
468. PDG2026:observable: S086:S086M:342595 B_s0 MASS	B_s()0 | Bbar_s()0 ELEMENTARY_OR_EXTENDED E04; M05	5366.93+-0.10 MeV	u_+: 0.09935167389588262 u_-: 0.09935167389588262 Type: NONE Confidence: ND
469. PDG2026:observable: S087:S087M:343123 B_s* MASS	B_s^*0 | Bbar_s^*0 ELEMENTARY_OR_EXTENDED E04; M05	5415.4+-1.4 MeV	u_+: 1.435389239191782 u_-: 1.435389239191782 Type: NONE Confidence: ND
470. PDG2026:observable: S091:S091M:343161 B_c+ MASS	B_c()+ | B_c()- ELEMENTARY_OR_EXTENDED E04; M05	6274.47+-0.32 MeV	u_+: 0.31906112267087638 u_-: 0.31906112267087638 Type: NONE Confidence: ND
471. PDG2026:observable: S126:S126M:332733 H MASS	H SM-H-1 E01; M06	125.13+-0.11 GeV	u_+: 0.11172144776548799 u_-: 0.11172144776548799 Type: NONE Confidence: ND

94.18.2 Anchuras: 384 registros

Table 94.336: Complete inventory of 384 widths.

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
1. PDG2026:observable: B004:B004W:348706 N(1895) BREIT-WIGNER WIDTH	N(1895)0 | Nbar(1895)0 | N(1895)+ | Nbar(1895)- COMP-BARYON E04; M07	80 to 120 to 200 MeV Presentation: 80 to 200 (~120)	u_+: NA u_-: NA Type: R Confidence: ND
2. PDG2026:observable: B005:B005W:348837 N(2060) BREIT-WIGNER WIDTH	N(2060)0 | Nbar(2060)0 | N(2060)+ | Nbar(2060)- COMP-BARYON E04; M07	300 to 400 to 450 MeV Presentation: 300 to 450 (~400)	u_+: NA u_-: NA Type: R Confidence: ND
3. PDG2026:observable: B010:B010W:349098 Delta(1700) BREIT-WIGNER WIDTH	Delta(1700)- | Deltabar(1700)+ | Delta(1700)0 | Deltabar(1700)0 | Delta(1700)+ | Deltabar(1700)- | Delta(1700)++ | Deltabar(1700)-- COMP-BARYON E04; M07	220 to 300 to 380 MeV Presentation: 220 to 380 (~300)	u_+: NA u_-: NA Type: R Confidence: ND
4. PDG2026:observable: B011:B011W:349152 Delta(1905) BREIT-WIGNER WIDTH	Delta(1905)- | Deltabar(1905)+ | Delta(1905)0 | Deltabar(1905)0 | Delta(1905)+ | Deltabar(1905)- | Delta(1905)++ | Deltabar(1905)-- COMP-BARYON E04; M07	270 to 330 to 400 MeV Presentation: 270 to 400 (~330)	u_+: NA u_-: NA Type: R Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
5. PDG2026:observable: B012:B012W:349183 Delta(1910) BREIT-WIGNER WIDTH	Delta(1910)- | Deltabar(1910)+ | Delta(1910)0 | Deltabar(1910)0 | Delta(1910)+ | Deltabar(1910)- | Delta(1910)++ | Deltabar(1910)-- COMP-BARYON E04; M07	200 to 300 to 400 MeV Presentation: 200 to 400 (~300)	u_+: NA u_-: NA Type: R Confidence: ND
6. PDG2026:observable: B013:B013W:349250 Delta(1930) BREIT-WIGNER WIDTH	Delta(1930)- | Deltabar(1930)+ | Delta(1930)0 | Deltabar(1930)0 | Delta(1930)+ | Deltabar(1930)- | Delta(1930)++ | Deltabar(1930)-- COMP-BARYON E04; M07	200 to 300 to 400 MeV Presentation: 200 to 400 (~300)	u_+: NA u_-: NA Type: R Confidence: ND
7. PDG2026:observable: B014:B014W:348536 N(1710) BREIT-WIGNER WIDTH	N(1710)0 | Nbar(1710)0 | N(1710)+ | Nbar(1710)- COMP-BARYON E04; M07	80 to 140 to 200 MeV Presentation: 80 to 200 (~140)	u_+: NA u_-: NA Type: R Confidence: ND
8. PDG2026:observable: B015:B015W:348570 N(1720) BREIT-WIGNER WIDTH	N(1720)0 | Nbar(1720)0 | N(1720)+ | Nbar(1720)- COMP-BARYON E04; M07	150 to 250 to 400 MeV Presentation: 150 to 400 (~250)	u_+: NA u_-: NA Type: R Confidence: ND
9. PDG2026:observable: B016:B016W:348634 N(1875) BREIT-WIGNER WIDTH	N(1875)0 | Nbar(1875)0 | N(1875)+ | Nbar(1875)- COMP-BARYON E04; M07	120 to 200 to 250 MeV Presentation: 120 to 250 (~200)	u_+: NA u_-: NA Type: R Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
10. PDG2026:observable: B017:B017W:348779 N(1990) BREIT-WIGNER WIDTH	N(1990)0 | Nbar(1990)0 | N(1990)+ | Nbar(1990)- COMP-BARYON E04; M07	200 to 300 to 400 MeV Presentation: 200 to 400 (~300)	u_+: NA u_-: NA Type: R Confidence: ND
11. PDG2026:observable: B018:B018W:348500 N(1700) BREIT-WIGNER WIDTH	N(1700)0 | Nbar(1700)0 | N(1700)+ | Nbar(1700)- COMP-BARYON E04; M07	100 to 200 to 300 MeV Presentation: 100 to 300 (~200)	u_+: NA u_-: NA Type: R Confidence: ND
12. PDG2026:observable: B019:B019W:349051 Delta(1600) BREIT-WIGNER WIDTH	Delta(1600)- | Deltabar(1600)+ | Delta(1600)0 | Deltabar(1600)0 | Delta(1600)+ | Deltabar(1600)- | Delta(1600)++ | Deltabar(1600)-- COMP-BARYON E04; M07	200 to 250 to 300 MeV Presentation: 200 to 300 (~250)	u_+: NA u_-: NA Type: R Confidence: ND
13. PDG2026:observable: B021:B021W:350411 Xi(1620) WIDTH	Xi(1620)- | Xibar(1620)+ | Xi(1620)0 | Xibar(1620)0 COMP-BARYON E04; M07	32+8-9 MeV	u_+: 8.14990701245417 88 u_-: 9.41197892698348 86 Type: NONE Confidence: ND
14. PDG2026:observable: B024:B024W:348907 N(2120) BREIT-WIGNER WIDTH	N(2120)0 | Nbar(2120)0 | N(2120)+ | Nbar(2120)- COMP-BARYON E04; M07	260 to 300 to 360 MeV Presentation: 260 to 360 (~300)	u_+: NA u_-: NA Type: R Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
15. PDG2026:observable: B030:B030W:349127 Delta(1900) BREIT-WIGNER WIDTH	Delta(1900)- | Deltabar(1900)+ | Delta(1900)0 | Deltabar(1900)0 | Delta(1900)+ | Deltabar(1900)- | Delta(1900)++ | Deltabar(1900)-- COMP-BARYON E04; M07	180 to 250 to 320 MeV Presentation: 180 to 320 (~250)	u_+: NA u_-: NA Type: R Confidence: ND
16. PDG2026:observable: B032:B032W:349903 Sigma(1620) WIDTH	Sigma(1620)- | Sigmabar(1620)+ | Sigma(1620)0 | Sigmabar(1620)0 | Sigma(1620)+ | Sigmabar(1620)- COMP-BARYON E04; M07	40 to 70 to 100 MeV Presentation: 40 to 100 (~70)	u_+: NA u_-: NA Type: R Confidence: ND
17. PDG2026:observable: B033:B033W:349035 MIXED CHARGES	Delta(1232)- | Deltabar(1232)+ | Delta(1232)0 | Deltabar(1232)0 | Delta(1232)+ | Deltabar(1232)- | Delta(1232)++ | Deltabar(1232)-- COMP-BARYON E04; M07	114 TO 117 TO 120 MeV Presentation: 114 to 120 (~117)	u_+: NA u_-: NA Type: R Confidence: ND
18. PDG2026:observable: B034:B034W:350187 Sigma(2070) WIDTH	Sigma(2070)- | Sigmabar(2070)+ | Sigma(2070)0 | Sigmabar(2070)0 | Sigma(2070)+ | Sigmabar(2070)- COMP-BARYON E04; M07	100 to 200 to 300 MeV Presentation: 100 to 300 (~200)	u_+: NA u_-: NA Type: R Confidence: ND
19. PDG2026:observable: B035:B035W:349764 Lambda(2110) WIDTH	Lambda(2110)0 | Lambdabar(2110)0 COMP-BARYON E04; M07	200 to 250 to 300 MeV Presentation: 200 to 300 (~250)	u_+: NA u_-: NA Type: R Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
20. PDG2026:observable: B036:B036W:349556 Lambda(1800) WIDTH	Lambda(1800)0 | Lambdabar(1800)0 COMP-BARYON E04; M07	150 to 200 to 250 MeV Presentation: 150 to 250 (~200)	u_+: NA u_-: NA Type: R Confidence: ND
21. PDG2026:observable: B037:B037W1:349428 PRODUCTION EXPERIMENTS	Lambda(1405)0 | Lambdabar(1405)0 COMP-BARYON E04; M07	50.5+-2.0 MeV	u_+: 1.96116135138184 u_-: 1.96116135138184 Type: NONE Confidence: ND
22. PDG2026:observable: B038:B038W:349439 Lambda(1520) WIDTH	Lambda(1520)0 | Lambdabar(1520)0 COMP-BARYON E04; M07	15 to 16 to 17 MeV Presentation: 15 to 17 (~16)	u_+: NA u_-: NA Type: R Confidence: ND
23. PDG2026:observable: B039:B039W:349595 Lambda(1820) WIDTH	Lambda(1820)0 | Lambdabar(1820)0 COMP-BARYON E04; M07	70 TO 80 TO 90 MeV Presentation: 70 to 90 (~80)	u_+: NA u_-: NA Type: R Confidence: ND
24. PDG2026:observable: B040:B040W:349495 Lambda(1670) WIDTH	Lambda(1670)0 | Lambdabar(1670)0 COMP-BARYON E04; M07	25 to 30 to 35 MeV Presentation: 25 to 35 (~30)	u_+: NA u_-: NA Type: R Confidence: ND
25. PDG2026:observable: B041:B041W:349742 Lambda(2100) WIDTH	Lambda(2100)0 | Lambdabar(2100)0 COMP-BARYON E04; M07	100 TO 200 TO 250 MeV Presentation: 100 to 250 (~200)	u_+: NA u_-: NA Type: R Confidence: ND
26. PDG2026:observable: B042:B042W:349776 Lambda(2350) WIDTH	Lambda(2350)0 | Lambdabar(2350)0 COMP-BARYON E04; M07	100 TO 150 TO 250 MeV Presentation: 100 to 250 (~150)	u_+: NA u_-: NA Type: R Confidence: ND
27. PDG2026:observable: B043:B043W+:349879 Sigma(1385)+ WIDTH	Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)- COMP-BARYON E04; M07	36.2+-0.7 MeV	u_+: 0.69132395371450484 u_-: 0.70656746664477266 Type: NONE Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
28. PDG2026:observable: B043:B043W0:349880 Sigma(1385)0 WIDTH	Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)- COMP-BARYON E04; M07	44+-8 MeV	u_+: 8.1680421550261997 u_-: 8.1680421550261997 Type: NONE Confidence: ND
29. PDG2026:observable: B043:B043W-:349881 Sigma(1385)- WIDTH	Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)- COMP-BARYON E04; M07	39.4+-2.1 MeV	u_+: 2.080864718865612 u_-: 2.080864718865612 Type: NONE Confidence: ND
30. PDG2026:observable: B044:B044W:349947 Sigma(1670) WIDTH	Sigma(1670)- | Sigmabar(1670)+ | Sigma(1670)0 | Sigmabar(1670)0 | Sigma(1670)+ | Sigmabar(1670)- COMP-BARYON E04; M07	40 to 70 to 100 MeV Presentation: 40 to 100 (~70)	u_+: NA u_-: NA Type: R Confidence: ND
31. PDG2026:observable: B045:B045W:349984 Sigma(1775) WIDTH	Sigma(1775)- | Sigmabar(1775)+ | Sigma(1775)0 | Sigmabar(1775)0 | Sigma(1775)+ | Sigmabar(1775)- COMP-BARYON E04; M07	105 TO 120 TO 135 MeV Presentation: 105 to 135 (~120)	u_+: NA u_-: NA Type: R Confidence: ND
32. PDG2026:observable: B046:B046W:350081 Sigma(1915) WIDTH	Sigma(1915)- | Sigmabar(1915)+ | Sigma(1915)0 | Sigmabar(1915)0 | Sigma(1915)+ | Sigmabar(1915)- COMP-BARYON E04; M07	80 TO 120 TO 160 MeV Presentation: 80 to 160 (~120)	u_+: NA u_-: NA Type: R Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
33. PDG2026:observable: B047:B047W:350167 Sigma(2030) WIDTH	Sigma(2030)- | Sigmabar(2030)+ | Sigma(2030)0 | Sigmabar(2030)0 | Sigma(2030)+ | Sigmabar(2030)- COMP-BARYON E04; M07	150 TO 180 TO 200 MeV Presentation: 150 to 200 (~180)	u_+: NA u_-: NA Type: R Confidence: ND
34. PDG2026:observable: B048:B048W:350278 Sigma(2250) WIDTH	Sigma(2250)- | Sigmabar(2250)+ | Sigma(2250)0 | Sigmabar(2250)0 | Sigma(2250)+ | Sigmabar(2250)- COMP-BARYON E04; M07	60 TO 100 TO 150 MeV Presentation: 60 to 150 (~100)	u_+: NA u_-: NA Type: R Confidence: ND
35. PDG2026:observable: B049:B049W0:350405 Xi(1530)0 WIDTH	Xi(1530)- | Xibar(1530)+ | Xi(1530)0 | Xibar(1530)0 COMP-BARYON E04; M07	9.1+-0.5 MeV	u_+: 0.48058271927942953 u_-: 0.48058271927942953 Type: NONE Confidence: ND
36. PDG2026:observable: B049:B049W-:350406 Xi(1530)- WIDTH	Xi(1530)- | Xibar(1530)+ | Xi(1530)0 | Xibar(1530)0 COMP-BARYON E04; M07	9.9+1.7-1.9 MeV	u_+: 1.731350237665344 u_-: 1.930203247244187 Type: NONE Confidence: ND
37. PDG2026:observable: B050:B050W:350422 Xi(1820) WIDTH	Xi(1820)- | Xibar(1820)+ | Xi(1820)0 | Xibar(1820)0 COMP-BARYON E04; M07	24+15-10 MeV	u_+: 15 u_-: 10 Type: NONE Confidence: ND
38. PDG2026:observable: B052:B052W:350438 Xi(1950) WIDTH	Xi(1950)- | Xibar(1950)+ | Xi(1950)0 | Xibar(1950)0 COMP-BARYON E04; M07	60+-20 MeV	u_+: 20 u_-: 20 Type: NONE Confidence: ND
39. PDG2026:observable: B055:B055W:349519 Lambda(1690) WIDTH	Lambda(1690)0 | Lambdabar(1690)0 COMP-BARYON E04; M07	60 to 70 to 80 MeV Presentation: 60 to 80 (~70)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
40. PDG2026:observable: B056:B056W:349615 Lambda(1830) WIDTH	Lambda(1830)0 | Lambdabar(1830)0 COMP-BARYON E04; M07	60 to 90 to 120 MeV Presentation: 60 to 120 (~90)	u_+: NA u_-: NA Type: R Confidence: ND
41. PDG2026:observable: B057:B057W:349967 Sigma(1750) WIDTH	Sigma(1750)- | Sigmabar(1750)+ | Sigma(1750)0 | Sigmabar(1750)0 | Sigma(1750)+ | Sigmabar(1750)- COMP-BARYON E04; M07	100 to 150 to 200 MeV Presentation: 100 to 200 (~150)	u_+: NA u_-: NA Type: R Confidence: ND
42. PDG2026:observable: B060:B060W:349636 Lambda(1890) WIDTH	Lambda(1890)0 | Lambdabar(1890)0 COMP-BARYON E04; M07	80 to 120 to 160 MeV Presentation: 80 to 160 (~120)	u_+: NA u_-: NA Type: R Confidence: ND
43. PDG2026:observable: B061:B061W:348327 N(1440) BREIT-WIGNER WIDTH	N(1440)0 | Nbar(1440)0 | N(1440)+ | Nbar(1440)- COMP-BARYON E04; M07	250 to 350 to 450 MeV Presentation: 250 to 450 (~350)	u_+: NA u_-: NA Type: R Confidence: ND
44. PDG2026:observable: B062:B062W:348350 N(1520) BREIT-WIGNER WIDTH	N(1520)0 | Nbar(1520)0 | N(1520)+ | Nbar(1520)- COMP-BARYON E04; M07	100 to 110 to 120 MeV Presentation: 100 to 120 (~110)	u_+: NA u_-: NA Type: R Confidence: ND
45. PDG2026:observable: B063:B063W:348384 N(1535) BREIT-WIGNER WIDTH	N(1535)0 | Nbar(1535)0 | N(1535)+ | Nbar(1535)- COMP-BARYON E04; M07	125 TO 150 TO 175 MeV Presentation: 125 to 175 (~150)	u_+: NA u_-: NA Type: R Confidence: ND
46. PDG2026:observable: B064:B064W:348436 N(1675) BREIT-WIGNER WIDTH	N(1675)0 | Nbar(1675)0 | N(1675)+ | Nbar(1675)- COMP-BARYON E04; M07	130 to 145 to 160 MeV Presentation: 130 to 160 (~145)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
47. PDG2026:observable: B065:B065W:348468 N(1680) BREIT-WIGNER WIDTH	N(1680)0 | Nbar(1680)0 | N(1680)+ | Nbar(1680)- COMP-BARYON E04; M07	115 to 120 to 130 MeV Presentation: 115 to 130 (~120)	u_+: NA u_-: NA Type: R Confidence: ND
48. PDG2026:observable: B066:B066W:348411 N(1650) BREIT-WIGNER WIDTH	N(1650)0 | Nbar(1650)0 | N(1650)+ | Nbar(1650)- COMP-BARYON E04; M07	100 to 125 to 150 MeV Presentation: 100 to 150 (~125)	u_+: NA u_-: NA Type: R Confidence: ND
49. PDG2026:observable: B067:B067W:350000 Sigma(1880) WIDTH	Sigma(1880)- | Sigmabar(1880)+ | Sigma(1880)0 | Sigmabar(1880)0 | Sigma(1880)+ | Sigmabar(1880)- COMP-BARYON E04; M07	100 to 200 to 300 MeV Presentation: 100 to 300 (~200)	u_+: NA u_-: NA Type: R Confidence: ND
50. PDG2026:observable: B068:B068W:350445 Xi(2030) WIDTH	Xi(2030)- | Xibar(2030)+ | Xi(2030)0 | Xibar(2030)0 COMP-BARYON E04; M07	20+15-5 MeV	u_+: 15 u_-: 5 Type: NONE Confidence: ND
51. PDG2026:observable: B071:B071W:348954 N(2190) BREIT-WIGNER WIDTH	N(2190)0 | Nbar(2190)0 | N(2190)+ | Nbar(2190)- COMP-BARYON E04; M07	300 to 400 to 500 MeV Presentation: 300 to 500 (~400)	u_+: NA u_-: NA Type: R Confidence: ND
52. PDG2026:observable: B077:B077W:349577 Lambda(1810) WIDTH	Lambda(1810)0 | Lambdabar(1810)0 COMP-BARYON E04; M07	50 to 110 to 170 MeV Presentation: 50 to 170 (~110)	u_+: NA u_-: NA Type: R Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
53. PDG2026:observable: B079:B079W:349921 Sigma(1660) WIDTH	Sigma(1660)- | Sigmabar(1660)+ | Sigma(1660)0 | Sigmabar(1660)0 | Sigma(1660)+ | Sigmabar(1660)- COMP-BARYON E04; M07	100 to 200 to 300 MeV Presentation: 100 to 300 (~200)	u_+: NA u_-: NA Type: R Confidence: ND
54. PDG2026:observable: B082:B082W:349079 Delta(1620) BREIT-WIGNER WIDTH	Delta(1620)- | Deltabar(1620)+ | Delta(1620)0 | Deltabar(1620)0 | Delta(1620)+ | Deltabar(1620)- | Delta(1620)++ | Deltabar(1620)-- COMP-BARYON E04; M07	110 to 130 to 150 MeV Presentation: 110 to 150 (~130)	u_+: NA u_-: NA Type: R Confidence: ND
55. PDG2026:observable: B083:B083W:349309 Delta(1950) BREIT-WIGNER WIDTH	Delta(1950)- | Deltabar(1950)+ | Delta(1950)0 | Deltabar(1950)0 | Delta(1950)+ | Deltabar(1950)- | Delta(1950)++ | Deltabar(1950)-- COMP-BARYON E04; M07	235 to 285 to 335 MeV Presentation: 235 to 335 (~285)	u_+: NA u_-: NA Type: R Confidence: ND
56. PDG2026:observable: B084:B084W:349360 Delta(2420) BREIT-WIGNER WIDTH	Delta(2420)- | Deltabar(2420)+ | Delta(2420)0 | Deltabar(2420)0 | Delta(2420)+ | Deltabar(2420)- | Delta(2420)++ | Deltabar(2420)-- COMP-BARYON E04; M07	300 to 500 to 700 MeV Presentation: 300 to 700 (~500)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
57. PDG2026:observable: B087:B087W:348672 N(1880) BREIT-WIGNER WIDTH	N(1880)0 | Nbar(1880)0 | N(1880)+ | Nbar(1880)- COMP-BARYON E04; M07	200 to 300 to 400 MeV Presentation: 200 to 400 (~300)	u_+: NA u_-: NA Type: R Confidence: ND
58. PDG2026:observable: B088:B088W:350189 Sigma(2080) WIDTH	Sigma(2080)- | Sigmabar(2080)+ | Sigma(2080)0 | Sigmabar(2080)0 | Sigma(2080)+ | Sigmabar(2080)- COMP-BARYON E04; M07	100 to 170 to 240 MeV Presentation: 100 to 240 (~170)	u_+: NA u_-: NA Type: R Confidence: ND
59. PDG2026:observable: B090:B090W:348987 N(2220) BREIT-WIGNER WIDTH	N(2220)0 | Nbar(2220)0 | N(2220)+ | Nbar(2220)- COMP-BARYON E04; M07	350 TO 400 TO 500 MeV Presentation: 350 to 500 (~400)	u_+: NA u_-: NA Type: R Confidence: ND
60. PDG2026:observable: B098:B098W:350051 Sigma(1910) WIDTH	Sigma(1910)- | Sigmabar(1910)+ | Sigma(1910)0 | Sigmabar(1910)0 | Sigma(1910)+ | Sigmabar(1910)- COMP-BARYON E04; M07	150 to 220 to 300 MeV Presentation: 150 to 300 (~220)	u_+: NA u_-: NA Type: R Confidence: ND
61. PDG2026:observable: B101:B101W:349473 Lambda(1600) WIDTH	Lambda(1600)0 | Lambdabar(1600)0 COMP-BARYON E04; M07	150 to 200 to 250 MeV Presentation: 150 to 250 (~200)	u_+: NA u_-: NA Type: R Confidence: ND
62. PDG2026:observable: B102:B102W:350937 Lambda_c(2625)+ WIDTH	Lambda_c(2625)+ | Lambdabar_c(2625)- COMP-BARYON E04; M07	< 0.52 MeV Presentation: <0.52	u_+: NA u_-: NA Type: U Confidence: 90

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
63. PDG2026:observable: B104:B104W++: 350997 Sigma_c(2455)++ WIDTH	Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)-- COMP-BARYON E04; M07	1.89+0.09-0.18 MeV	u_+: 0.085968740877688063 u_-: 0.17741442138916921 Type: NONE Confidence: ND
64. PDG2026:observable: B104:B104W+:350998 Sigma_c(2455)+ WIDTH	Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)-- COMP-BARYON E04; M07	2.3+-0.4 MeV	u_+: 0.42426406871192851 u_-: 0.42426406871192851 Type: NONE Confidence: ND
65. PDG2026:observable: B104:B104W0:350999 Sigma_c(2455)0 WIDTH	Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)-- COMP-BARYON E04; M07	1.83+0.11-0.19 MeV	u_+: 0.1113419499366561 u_-: 0.19347936378811631 Type: NONE Confidence: ND
66. PDG2026:observable: B105:B105W:350413 MIXED CHARGES	Xi(1690)- | Xibar(1690)+ | Xi(1690)0 | Xibar(1690)0 COMP-BARYON E04; M07	20+-15 MeV	u_+: 15 u_-: 15 Type: NONE Confidence: ND
67. PDG2026:observable: B113:B113W:349005 N(2250) BREIT-WIGNER WIDTH	N(2250)0 | Nbar(2250)0 | N(2250)+ | Nbar(2250)- COMP-BARYON E04; M07	300 to 500 to 600 MeV Presentation: 300 to 600 (~500)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
68. PDG2026:observable: B115:B115W++:351011 Sigma_c(2520)++ WIDTH	Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)-- COMP-BARYON E04; M07	14.78+0.30-0.40 MeV	u_+: 0.30221098294841742 u_-: 0.37821726734358602 Type: NONE Confidence: ND
69. PDG2026:observable: B115:B115W+:351012 Sigma_c(2520)+ WIDTH	Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)-- COMP-BARYON E04; M07	17.2+4.0-2.2 MeV	u_+: 3.8600518131237571 u_-: 2.213594362117866 Type: NONE Confidence: ND
70. PDG2026:observable: B115:B115W0:351013 Sigma_c(2520)0 WIDTH	Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)-- COMP-BARYON E04; M07	15.3+0.4-0.5 MeV	u_+: 0.43805327113015552 u_-: 0.49172579862174309 Type: NONE Confidence: ND
71. PDG2026:observable: B117:B117W:349209 Delta(1920) BREIT-WIGNER WIDTH	Delta(1920)- | Deltabar(1920)+ | Delta(1920)0 | Deltabar(1920)0 | Delta(1920)+ | Deltabar(1920)- | Delta(1920)++ | Deltabar(1920)-- COMP-BARYON E04; M07	240 to 300 to 360 MeV Presentation: 240 to 360 (~300)	u_+: NA u_-: NA Type: R Confidence: ND
72. PDG2026:observable: B119:B119W:350920 Lambda_c(2595)+ WIDTH	Lambda_c(2595)+ | Lambdabar_c(2595)- COMP-BARYON E04; M07	2.6+-0.6 MeV	u_+: 0.55758407437802593 u_-: 0.55758407437802593 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
73. PDG2026:observable: B120:B120W:349023 N(2600) BREIT-WIGNER WIDTH	N(2600)0 | Nbar(2600)0 | N(2600)+ | Nbar(2600)- COMP-BARYON E04; M07	500 TO 650 TO 800 MeV Presentation: 500 to 800 (~650)	u_+: NA u_-: NA Type: R Confidence: ND
74. PDG2026:observable: B122:B122W:350980 Lambda_c(2940)+ WIDTH	Lambda_c(2940)0 | Lambdabar_c(2940)0 COMP-BARYON E04; M07	20+6-5 MeV	u_+: 5.98196379520038 41 u_-: 5.22853813129437 66 Type: NONE Confidence: ND
75. PDG2026:observable: B130:B130W+:351281 Xi_c(2970)+ WIDTH	Xi_c(2970)0 | Xibar_ c(2970)0 COMP-BARYON E04; M07	27.2+2.8-3.3 MeV	u_+: 2.76917269976361 79 u_-: 3.31474783369252 31 Type: NONE Confidence: ND
76. PDG2026:observable: B132:B132W:348872 N(2100) BREIT-WIGNER WIDTH	N(2100)0 | Nbar(2100)0 | N(2100)+ | Nbar(2100)- COMP-BARYON E04; M07	200 to 260 to 320 MeV Presentation: 200 to 320 (~260)	u_+: NA u_-: NA Type: R Confidence: ND
77. PDG2026:observable: B135:B135W:349340 Delta(2200) BREIT-WIGNER WIDTH	Delta(2200)- | Deltabar(2200)+ | Delta(2200)0 | Deltabar(2200)0 | Delta(2200)+ | Deltabar(2200)- | Delta(2200)++ | Deltabar(2200)-- COMP-BARYON E04; M07	200 to 350 to 500 MeV Presentation: 200 to 500 (~350)	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
78. PDG2026:observable: B136:B136W:349273 Delta(1940) BREIT-WIGNER WIDTH	Delta(1940)- | Deltabar(1940)+ | Delta(1940)0 | Deltabar(1940)0 | Delta(1940)+ | Deltabar(1940)- | Delta(1940)++ | Deltabar(1940)-- COMP-BARYON E04; M07	300 to 400 to 500 MeV Presentation: 300 to 500 (~400)	u_+: NA u_-: NA Type: R Confidence: ND
79. PDG2026:observable: B141:B141W:350517 Omega(2250)- WIDTH	Omega(2250)- | Omegabar(2250)+ COMP-BARYON E04; M07	55+-18 MeV	u_+: 17.6983644476396 51 u_-: 17.6983644476396 51 Type: NONE Confidence: ND
80. PDG2026:observable: B143:B143W:350524 Omega(2470)- WIDTH	Omega(2470)- | Omegabar(2470)+ COMP-BARYON E04; M07	72+-33 MeV	u_+: 33 u_-: 33 Type: NONE Confidence: ND
81. PDG2026:observable: B144:B144W:348741 N(1900) BREIT-WIGNER WIDTH	N(1900)0 | Nbar(1900)0 | N(1900)+ | Nbar(1900)- COMP-BARYON E04; M07	100 to 200 to 320 MeV Presentation: 100 to 320 (~200)	u_+: NA u_-: NA Type: R Confidence: ND
82. PDG2026:observable: B146:B146W+:351213 Xi_c(2645)+ WIDTH	Xi_c(2645)0 | Xibar_ c(2645)0 | Xi_c(2645)+ | Xibar_c(2645)- COMP-BARYON E04; M07	2.14+-0.19 MeV	u_+: 0.18989896013117 41 u_-: 0.18989896013117 41 Type: NONE Confidence: ND
83. PDG2026:observable: B146:B146W0:351214 Xi_c(2645)0 WIDTH	Xi_c(2645)0 | Xibar_ c(2645)0 | Xi_c(2645)+ | Xibar_c(2645)- COMP-BARYON E04; M07	2.35+-0.22 MeV	u_+: 0.22203603311174 519 u_-: 0.22203603311174 519 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
84. PDG2026:observable: B147:B147W+:351309 Xi_c(3080)+ WIDTH	Xi_c(3080)0 | Xibar_ c(3080)0 COMP-BARYON E04; M07	3.8+-0.9 MeV	u_+: 0.93945766227499938 u_-: 0.93945766227499938 Type: NONE Confidence: ND
85. PDG2026:observable: B147:B147W0:351310 Xi_c(3080)0 WIDTH	Xi_c(3080)0 | Xibar_ c(3080)0 COMP-BARYON E04; M07	5.6+-2.2 MeV	u_+: 2.17986010862691 2 u_-: 2.17986010862691 2 Type: NONE Confidence: ND
86. PDG2026:observable: B148:B148W+:351240 Xi_c(2815)+ WIDTH	Xi_c(2815)0 | Xibar_ c(2815)0 | Xi_c(2815)+ | Xibar_c(2815)- COMP-BARYON E04; M07	2.15+-0.15 MeV	u_+: 0.15193272999421351 u_-: 0.15193272999421351 Type: NONE Confidence: ND
87. PDG2026:observable: B148:B148W0:351241 Xi_c(2815)0 WIDTH	Xi_c(2815)0 | Xibar_ c(2815)0 | Xi_c(2815)+ | Xibar_c(2815)- COMP-BARYON E04; M07	2.54+-0.25 MeV	u_+: 0.24758836806279891 u_-: 0.24758836806279891 Type: NONE Confidence: ND
88. PDG2026:observable: B149:B149W+:351224 Xi_c(2790)+ WIDTH	Xi_c(2790)0 | Xibar_ c(2790)0 | Xi_c(2790)+ | Xibar_c(2790)- COMP-BARYON E04; M07	8.9+-1.0 MeV	u_+: 1 u_-: 1 Type: NONE Confidence: ND
89. PDG2026:observable: B149:B149W0:351225 Xi_c(2790)0 WIDTH	Xi_c(2790)0 | Xibar_ c(2790)0 | Xi_c(2790)+ | Xibar_c(2790)- COMP-BARYON E04; M07	10.0+-1.1 MeV	u_+: 1.06301458127346 5 u_-: 1.06301458127346 5 Type: NONE Confidence: ND
90. PDG2026:observable: B150:B150W:350954 Lambda_c(2765)+ WIDTH	Lambda_c(2765)0 | Lambdabar_c(2765)0 COMP-BARYON E04; M07	50 MeV	u_+: 0 u_-: 0 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
91. PDG2026:observable: B151:B151W:350964 Lambda_c(2880)+ WIDTH	Lambda_c(2880)+ | Lambdabar_c(2880)- COMP-BARYON E04; M07	5.6+0.8-0.6 MeV	u_+: 0.77200097671302059 u_-: 0.62289222418902057 Type: NONE Confidence: ND
92. PDG2026:observable: B155:B155W++: 351025 Sigma_c(2800)++ WIDTH	Sigma_c(2800)0 | Sigmabar_c(2800)0 COMP-BARYON E04; M07	75+22-17 MeV	u_+: 21.633307652783941 u_-: 17.029386365926399 Type: NONE Confidence: ND
93. PDG2026:observable: B155:B155W+:351026 Sigma_c(2800)+ WIDTH	Sigma_c(2800)0 | Sigmabar_c(2800)0 COMP-BARYON E04; M07	60+60-40 MeV	u_+: 63.820059542435402 u_-: 44.418464629025621 Type: NONE Confidence: ND
94. PDG2026:observable: B155:B155W0:351027 Sigma_c(2800)0 WIDTH	Sigma_c(2800)0 | Sigmabar_c(2800)0 COMP-BARYON E04; M07	72+22-15 MeV	u_+: 22.093560797468029 u_-: 14.823460785121441 Type: NONE Confidence: ND
95. PDG2026:observable: B156:B156A05:351265 Xi_c(2930)+ WIDTH	Xi_c(2930)0 | Xibar_c(2930)0 COMP-BARYON E04; M07	15+-9 MeV	u_+: 9.14822387133152 76 u_-: 9.14822387133152 76 Type: NONE Confidence: ND
96. PDG2026:observable: B156:B156A06:351266 Xi_c(2930)0 WIDTH	Xi_c(2930)0 | Xibar_c(2930)0 COMP-BARYON E04; M07	10.2+-1.4 MeV	u_+: 1.36014705087354 4 u_-: 1.36014705087354 4 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
97. PDG2026:observable: B157:B157W+:351299 Xi_c(3055)+ WIDTH	Xi_c(3055)0 | Xibar_ c(3055)0 COMP-BARYON E04; M07	8.0+-0.8 MeV	u_+: 0.76391812853200158 u_-: 0.76391812853200158 Type: NONE Confidence: ND
98. PDG2026:observable: B157:B157W:351300 Xi_c(3055)0 WIDTH	Xi_c(3055)0 | Xibar_ c(3055)0 COMP-BARYON E04; M07	12.4+-2.3 MeV	u_+: 2.28254244210266 5 u_-: 2.28254244210266 5 Type: NONE Confidence: ND
99. PDG2026:observable: B158:B158W+:351326 Xi_c(3123)+ WIDTH	Xi_c(3123)0 | Xibar_ c(3123)0 COMP-BARYON E04; M07	4+-4 MeV	u_+: 3.80131556174964 3 u_-: 3.80131556174964 3 Type: NONE Confidence: ND
100. PDG2026:observable: B161:B161W:351845 Xi_b(5945)0 WIDTH	Xi_b(5945)0 | Xibar_ b(5945)0 COMP-BARYON E04; M07	0.87+-0.07 MeV	u_+: 0.07506967217458 3921 u_-: 0.07470630547753 7885 Type: NONE Confidence: ND
101. PDG2026:observable: B162:B162W:351695 Lambda_b(5912)0 WIDTH	Lambda_b(5912)0 | Lambdabar_b(5912)0 COMP-BARYON E04; M07	<0.25 MeV	u_+: NA u_-: NA Type: U Confidence: 90
102. PDG2026:observable: B163:B163W:351699 Lambda_b(5920)0 WIDTH	Lambda_b(5920)0 | Lambdabar_b(5920)0 COMP-BARYON E04; M07	<0.19 MeV	u_+: NA u_-: NA Type: U Confidence: 90
103. PDG2026:observable: B164:B164W:349533 Lambda(1710) WIDTH	Lambda(1710)0 | Lambdabar(1710)0 COMP-BARYON E04; M07	180+-40 MeV	u_+: 42 u_-: 42 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
104. PDG2026:observable: B165:B165W:349652 Lambda(2050) WIDTH	Lambda(2050)0 | Lambdabar(2050)0 COMP-BARYON E04; M07	490+-60 MeV	u_+: 61 u_-: 61 Type: NONE Confidence: ND
105. PDG2026:observable: B166:B166W:349992 Sigma(1780) WIDTH	Sigma(1780)- | Sigmabar(1780)+ | Sigma(1780)0 | Sigmabar(1780)0 | Sigma(1780)+ | Sigmabar(1780)- COMP-BARYON E04; M07	100 to 200 to 300 MeV Presentation: 100 to 300 (~200)	u_+: NA u_-: NA Type: R Confidence: ND
106. PDG2026:observable: B167:B167W:350020 Sigma(1900) WIDTH	Sigma(1900)- | Sigmabar(1900)+ | Sigma(1900)0 | Sigmabar(1900)0 | Sigma(1900)+ | Sigmabar(1900)- COMP-BARYON E04; M07	140 to 165 to 190 MeV Presentation: 140 to 190 (~165)	u_+: NA u_-: NA Type: R Confidence: ND
107. PDG2026:observable: B168:B168W:350103 Sigma(1940) WIDTH	Sigma(1940)- | Sigmabar(1940)+ | Sigma(1940)0 | Sigmabar(1940)0 | Sigma(1940)+ | Sigmabar(1940)- COMP-BARYON E04; M07	100 to 250 to 400 MeV Presentation: 100 to 400 (~250)	u_+: NA u_-: NA Type: R Confidence: ND
108. PDG2026:observable: B169:B169W:351841 Xi_b'(5935)- WIDTH	Xi_b^'(5935)- | Xibar_ b^'(5935)+ COMP-BARYON E04; M07	0.03+-0.032 MeV	u_+: 0.03162277660168 3791 u_-: 0.03162277660168 3791 Type: NONE Confidence: ND
109. PDG2026:observable: B170:B170W:351849 Xi_b(5955)- WIDTH	Xi_b(5955)- | Xibar_ b(5955)+ COMP-BARYON E04; M07	1.43+-0.11 MeV	u_+: 0.11313708498984 761 u_-: 0.11313708498984 761 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
110. PDG2026:observable: B171:B171W:351952 P_c_cbar(4380)+ WIDTH	P_c_cbar(4380)+ COMP-BARYON E04; M07	210+-90 MeV	u_+: 87.863530545955186 u_-: 87.863530545955186 Type: NONE Confidence: ND
111. PDG2026:observable: B172:B172W:351978 P_c_cbar(4457)+ WIDTH	P_c_cbar(4457)+ COMP-BARYON E04; M07	6.4+6.0-2.8 MeV	u_+: 6.0406953242155828 u_-: 2.758622844826744 Type: NONE Confidence: ND
112. PDG2026:observable: B173:B173W:351377 Omega_c(3000)0 WIDTH	Omega_c(3000)0 | Omegabar_c(3000)0 COMP-BARYON E04; M07	3.8+1.6-0.4 MeV	u_+: 1.606549096666517 u_-: 0.37013511046643488 Type: NONE Confidence: ND
113. PDG2026:observable: B174:B174W:351381 Omega_c(3050)0 WIDTH	Omega_c(3050)0 | Omegabar_c(3050)0 COMP-BARYON E04; M07	< 1.8 MeV Presentation: <1.8	u_+: NA u_-: NA Type: U Confidence: 95
114. PDG2026:observable: B175:B175W:351385 Omega_c(3065)0 WIDTH	Omega_c(3065)0 | Omegabar_c(3065)0 COMP-BARYON E04; M07	3.4+0.7-0.8 MeV	u_+: 0.6849663741667257 9 u_-: 0.7938585212385945 7 Type: NONE Confidence: ND
115. PDG2026:observable: B176:B176W:351389 Omega_c(3090)0 WIDTH	Omega_c(3090)0 | Omegabar_c(3090)0 COMP-BARYON E04; M07	8.5+0.8-1.7 MeV	u_+: 0.7521303078589507 4 u_-: 1.6786899654194664 Type: NONE Confidence: ND
116. PDG2026:observable: B177:B177W:351393 Omega_c(3120)0 WIDTH	Omega_c(3120)0 | Omegabar_c(3120)0 COMP-BARYON E04; M07	<2.5 MeV	u_+: NA u_-: NA Type: U Confidence: 95

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
117. PDG2026:observable: B178:B178W:350957 Lambda_c(2860)+ WIDTH	Lambda_c(2860)0 | Lambdabar_c(2860)0 COMP-BARYON E04; M07	68+12-22 MeV	u_+: 11.7804923496431 2 u_-: 21.6233669903648 4 Type: NONE Confidence: ND
118. PDG2026:observable: B179:B179W:350507 Omega(2012)- WIDTH	Omega(2012)- | Omegabar(2012)+ COMP-BARYON E04; M07	6.3+-2.0 MeV	u_+: 2.07428954575496 5 u_-: 1.91972498587065 51 Type: NONE Confidence: ND
119. PDG2026:observable: B180:B180W:351865 Xi_b(6227)- WIDTH	Xi_b(6227)- | Xi_b(6227)+ COMP-BARYON E04; M07	19.9+-2.6 MeV	u_+: 2.58069758011278 78 u_-: 2.58069758011278 78 Type: NONE Confidence: ND
120. PDG2026:observable: B181:B181W:351735 Sigma_b(6097)+ WIDTH	Sigma_b(6097)+ | Sigmabar_b(6097)- COMP-BARYON E04; M07	31+-6 MeV	u_+: 5.54436651025164 46 u_-: 5.54436651025164 46 Type: NONE Confidence: ND
121. PDG2026:observable: B182:B182W:351739 Sigma_b(6097)- WIDTH	Sigma_b(6097)- | Sigmabar_b(6097)+ COMP-BARYON E04; M07	29+-4 MeV	u_+: 4.29534631898290 62 u_-: 4.29534631898290 62 Type: NONE Confidence: ND
122. PDG2026:observable: B183:B183W:350229 Sigma(2110) WIDTH	Sigma(2110)- | Sigmabar(2110)+ | Sigma(2110)0 | Sigmabar(2110)0 | Sigma(2110)+ | Sigmabar(2110)- COMP-BARYON E04; M07	310+120-50 MeV	u_+: 122.349630121267 89 u_-: 54.7610745236418 52 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
123. PDG2026:observable: B184:B184W:350254 Sigma(2230) WIDTH	Sigma(2230)- | Sigmabar(2230)+ | Sigma(2230)0 | Sigmabar(2230)0 | Sigma(2230)+ | Sigmabar(2230)- COMP-BARYON E04; M07	350+-50 MeV	u_+: 50 u_-: 50 Type: NONE Confidence: ND
124. PDG2026:observable: B185:B185W:351936 P_c_cbar(4312)+ WIDTH	P_c_cbar(4312)+ COMP-BARYON E04; M07	10+-5 MeV	u_+: 4.58039299623951 47 u_-: 5.24785670536077 03 Type: NONE Confidence: ND
125. PDG2026:observable: B186:B186W:351956 P_c_cbar(4440)+ WIDTH	P_c_cbar(4440)+ COMP-BARYON E04; M07	21+10-11 MeV	u_+: 9.98498873309329 23 u_-: 11.2258629957789 89 Type: NONE Confidence: ND
126. PDG2026:observable: B187:B187W:351708 Lambda_b(6146)0 WIDTH	Lambda_b(6146)0 | Lambdabar_b(6146)0 COMP-BARYON E04; M07	2.9+-1.3 MeV	u_+: 1.33416640641263 4 u_-: 1.33416640641263 4 Type: NONE Confidence: ND
127. PDG2026:observable: B188:B188W:351714 Lambda_b(6152)0 WIDTH	Lambda_b(6152)0 | Lambdabar_b(6152)0 COMP-BARYON E04; M07	2.1+-0.9 MeV	u_+: 0.85440037453175 31 u_-: 0.85440037453175 31 Type: NONE Confidence: ND
128. PDG2026:observable: B190:B190W:349677 Lambda(2070) WIDTH	Lambda(2070)0 | Lambdabar(2070)0 COMP-BARYON E04; M07	370+-50 MeV	u_+: 50 u_-: 50 Type: NONE Confidence: ND
129. PDG2026:observable: B191:B191W:349712 Lambda(2080) WIDTH	Lambda(2080)0 | Lambdabar(2080)0 COMP-BARYON E04; M07	181+-29 MeV	u_+: 29 u_-: 29 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
130. PDG2026:observable: B192:B192W0:351257 Xi_c(2923)0 WIDTH	Xi_c(2923)0 | Xibar_c(2923)0 COMP-BARYON E04; M07	5.8+-1.3 MeV	u_+: 1.30796716242523 u_-: 1.30796716242523 Type: NONE Confidence: ND
131. PDG2026:observable: B192:B192WP:351258 Xi_c(2923)+ WIDTH	Xi_c(2923)0 | Xibar_c(2923)0 COMP-BARYON E04; M07	5.3+-1.7 MeV	u_+: 1.6643316977093241 u_-: 1.6643316977093241 Type: NONE Confidence: ND
132. PDG2026:observable: B193:B193W:351906 Omega_b(6316)- WIDTH	Omega_b(6316)- | Omegabar_b(6316)+ COMP-BARYON E04; M07	<4.2 MeV	u_+: NA u_-: NA Type: U Confidence: 95
133. PDG2026:observable: B194:B194W:351910 Omega_b(6330)- WIDTH	Omega_b(6330)- | Omegabar_b(6330)+ COMP-BARYON E04; M07	<4.7 MeV	u_+: NA u_-: NA Type: U Confidence: 95
134. PDG2026:observable: B195:B195W:351914 Omega_b(6340)- WIDTH	Omega_b(6340)- | Omegabar_b(6340)+ COMP-BARYON E04; M07	<1.8 MeV	u_+: NA u_-: NA Type: U Confidence: 95
135. PDG2026:observable: B196:B196W:351918 Omega_b(6350)- WIDTH	Omega_b(6350)- | Omegabar_b(6350)+ COMP-BARYON E04; M07	<3.2 MeV	u_+: NA u_-: NA Type: U Confidence: 95
136. PDG2026:observable: B197:B197W:351703 Lambda_b(6070)0 WIDTH	Lambda_b(6070)0 | Lambdabar_b(6070)0 COMP-BARYON E04; M07	72+-11 MeV	u_+: 11.180339887498951 u_-: 11.180339887498951 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
137. PDG2026:observable: B199:B199W:351871 Xi_b(6227)0 WIDTH	Xi_b(6227)0 | Xibar_ b(6227)0 COMP-BARYON E04; M07	19+5-4 MeV	u_+: 5.19230199429886 77 u_-: 4.33243580448689 34 Type: NONE Confidence: ND
138. PDG2026:observable: B200:B200W:351861 Xi_b(6100)- WIDTH	Xi_b(6100)- | Xibar_ b(6100)+ COMP-BARYON E04; M07	0.94+-0.31 MeV	u_+: 0.31048349392520 053 u_-: 0.31048349392520 053 Type: NONE Confidence: ND
139. PDG2026:observable: B201:B201W:351875 Xi_b(6327)0 WIDTH	Xi_b(6327)0 | Xibar_ b(6327)0 COMP-BARYON E04; M07	< 2.56 MeV Presentation: <2.56	u_+: NA u_-: NA Type: U Confidence: 95
140. PDG2026:observable: B202:B202W:351879 Xi_b(6333)0 WIDTH	Xi_b(6333)0 | Xibar_ b(6333)0 COMP-BARYON E04; M07	< 1.92 MeV Presentation: <1.92	u_+: NA u_-: NA Type: U Confidence: 95
141. PDG2026:observable: B203:B203W0:351252 Xi_c(2882) WIDTH	Xi_c(2882)0 | Xibar_ c(2882)0 COMP-BARYON E04; M07	12+-8 MeV	u_+: 7.85684415016614 68 u_-: 7.85684415016614 68 Type: NONE Confidence: ND
142. PDG2026:observable: B204:B204W:351946 P_c_cbar_s(4338)0 WIDTH	P_c_cbar_s(4338)+ COMP-BARYON E04; M07	7.0+-1.8 MeV	u_+: 1.76918060129541 3 u_-: 1.76918060129541 3 Type: NONE Confidence: ND
143. PDG2026:observable: B205:B205W:352000 P_c_cbar(4312)+ WIDTH	P_c_cbar(4312)+ COMP-BARYON E04; M07	19+8-7 MeV	u_+: 8.15688301652626 54 u_-: 7.25520281398952 7 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
144. PDG2026:observable: B206:B206W:350974 Lambda_c(2910)+ WIDTH	Lambda_c(2910)0 | Lambdabar_c(2910)0 COMP-BARYON E04; M07	52+-27 MeV	u_+: 27.448861542876418 u_-: 27.448861542876418 Type: NONE Confidence: ND
145. PDG2026:observable: B207:B207W:351853 Xi_b(6087)0 WIDTH	Xi_b(6087) COMP-BARYON E04; M07	2.4+-0.5 MeV	u_+: 0.51971145840745137 u_-: 0.51971145840745137 Type: NONE Confidence: ND
146. PDG2026:observable: B208:B208W:351857 Xi_b(6095)0 WIDTH	Xi_b(6095) COMP-BARYON E04; M07	0.50+-0.35 MeV	u_+: 0.34785054261852172 u_-: 0.34785054261852172 Type: NONE Confidence: ND
147. PDG2026:observable: B209:B209W:351397 Omega_c(3185)0 WIDTH	Omega_c(3185)0 | Omegabar_c(3185)0 COMP-BARYON E04; M07	50+12-21 MeV	u_+: 12.2065556157337 u_-: 21.189620100417091 Type: NONE Confidence: ND
148. PDG2026:observable: B210:B210W:351401 Omega_c(3327)0 WIDTH	Omega_c(3327)0 | Omegabar_c(3327)0 COMP-BARYON E04; M07	20+14-5 MeV	u_+: 13.928388277184119 u_-: 5.0990195135927836 Type: NONE Confidence: ND
149. PDG2026:observable: B211:B211W:350515 Omega(2109)- WIDTH	Omega(2109)- | Omegabar(2109)+ COMP-BARYON E04; M07	18+-17 MeV	u_+: 17.362315513778679 u_-: 17.362315513778679 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
150. PDG2026:observable: M001:M001W:333933 omega(782) WIDTH	omega(782)0 COMP-MESON E04; M08	8.68+-0.13 MeV	u_+: 0.1320000870776446 u_-: 0.1320000870776446 Type: NONE Confidence: ND
151. PDG2026:observable: M002:M002W:334007 eta'(958) WIDTH	eta^'(958)0 COMP-MESON E04; M08	0.188+-0.006 MeV	u_+: 0.0064442212265340033 u_-: 0.0064442212265340033 Type: NONE Confidence: ND
152. PDG2026:observable: M003:M003W1:334164 f_0(980) WIDTH	f_0(980)0 COMP-MESON E04; M08	10 to 100 MeV	u_+: NA u_-: NA Type: R Confidence: ND
153. PDG2026:observable: M004:M004W:334183 phi(1020) WIDTH	phi(1020)0 COMP-MESON E04; M08	4.249+-0.013 MeV	u_+: 0.01311655783805327 u_-: 0.01311655783805327 Type: NONE Confidence: ND
154. PDG2026:observable: M005:M005W:334355 f_2(1270) WIDTH	f_2(1270)0 COMP-MESON E04; M08	186.6+2.8-2.2 MeV	u_+: 2.802312637678575 u_-: 2.1961156286222181 Type: NONE Confidence: ND
155. PDG2026:observable: M006:M006W:334555 f_1(1420) WIDTH	f_1(1420)0 COMP-MESON E04; M08	55.4+-3.2 MeV	u_+: 3.1572143311904868 u_-: 3.199679790915769 Type: NONE Confidence: ND
156. PDG2026:observable: M008:M008W:334404 f_1(1285) WIDTH	f_1(1285)0 COMP-MESON E04; M08	23.0+-1.0 MeV	u_+: 1.045581016397475 u_-: 1.0256159102073861 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
157. PDG2026:observable: M009:M009W0:333870 NEUTRAL ONLY, e+ e-	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	147.4+-0.8 MeV	u_+: 0.82096324211264493 u_-: 0.82096324211264493 Type: NONE Confidence: ND
158. PDG2026:observable: M009:M009W5:333871 CHARGED ONLY, tau DECAYS and e+ e-	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	149.1+-0.8 MeV	u_+: 0.84906762313788198 u_-: 0.84906762313788198 Type: NONE Confidence: ND
159. PDG2026:observable: M009:M009W7:333873 MIXED CHARGES, OTHER REACTIONS	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	149.5+-1.3 MeV	u_+: 1.3 u_-: 1.3 Type: NONE Confidence: ND
160. PDG2026:observable: M009:M009W2:333874 CHARGED ONLY, HADROPRODUCED	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	150.2+-2.4 MeV	u_+: 2.3925241925869849 u_-: 2.3925241925869849 Type: NONE Confidence: ND
161. PDG2026:observable: M009:M009W0P: 333876 NEUTRAL ONLY, PHOTOPRODUCED	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	152.3+-1.7 MeV	u_+: 1.6144196138431619 u_-: 1.7476550950891081 Type: NONE Confidence: ND
162. PDG2026:observable: M009:M009W0R: 333877 NEUTRAL ONLY, OTHER REACTIONS	rho(770)0 | rho(770)+ | rho(770)- COMP-MESON E04; M08	150.9+-1.7 MeV	u_+: 1.7341880321909591 u_-: 1.7341880321909591 Type: NONE Confidence: ND
163. PDG2026:observable: M010:M010W:334333 a_1(1260) WIDTH	a_1(1260)0 | a_1(1260)+ | a_1(1260)- COMP-MESON E04; M08	250 to 600 MeV	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
164. PDG2026:observable: M011:M011W:334310 b_1(1235) WIDTH	b_1(1235)0 | b_1(1235)+ | b_1(1235)- COMP-MESON E04; M08	142+-9 MeV	u_+: 8.50576301057597 03 u_-: 8.50576301057597 03 Type: NONE Confidence: ND
165. PDG2026:observable: M012:M012W:334474 a_2(1320) WIDTH	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	107+-5 MeV	u_+: 5 u_-: 5 Type: NONE Confidence: ND
166. PDG2026:observable: M012:M012W1:334476 3pi MODE	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	105.0+1.7-1.9 MeV	u_+: 1.67242394191988 7 u_-: 1.85239615833847 09 Type: NONE Confidence: ND
167. PDG2026:observable: M012:M012W2:334477 K Kbar MODE	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	109.8+-2.4 MeV	u_+: 2.37489024273604 12 u_-: 2.37489024273604 12 Type: NONE Confidence: ND
168. PDG2026:observable: M012:M012W3:334478 eta pi MODE	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	111.1+-2.4 MeV	u_+: 2.44896593502306 5 u_-: 2.44896593502306 5 Type: NONE Confidence: ND
169. PDG2026:observable: M012:M012W4:334479 eta' pi MODE	a_2(1320)0 | a_2(1320)+ | a_2(1320)- COMP-MESON E04; M08	119+-25 MeV	u_+: 25.0631516454720 01 u_-: 25.0631516454720 01 Type: NONE Confidence: ND
170. PDG2026:observable: M013:M013WX: 334671 f_2'(1525) WIDTH	f_2^'(1525)0 COMP-MESON E04; M08	72+7-6 MeV	u_+: 6.88285745387930 53 u_-: 5.78868904167363 58 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
171. PDG2026:observable: M013:M013W2:334673 PRODUCED BY K+- BEAM	f_2^'(1525)0 COMP-MESON E04; M08	82+-6 MeV	u_+: 6.0690031263411646 u_-: 5.743699645561148 Type: NONE Confidence: ND
172. PDG2026:observable: M013:M013W3:334674 PRODUCED IN e+ e- ANNIHILATION AND PARTICLE DECAYS	f_2^'(1525)0 COMP-MESON E04; M08	86+-4 MeV	u_+: 4.2724130046702271 u_-: 3.9578983915293739 Type: NONE Confidence: ND
173. PDG2026:observable: M013:M013W9:334675 PRODUCED IN pbar p ANNIHILATION	f_2^'(1525)0 COMP-MESON E04; M08	77+-5 MeV	u_+: 4.7999999999999998 u_-: 4.7999999999999998 Type: NONE Confidence: ND
174. PDG2026:observable: M013:M013W4:334676 CENTRAL PRODUCTION	f_2^'(1525)0 COMP-MESON E04; M08	70+-25 MeV	u_+: 25 u_-: 25 Type: NONE Confidence: ND
175. PDG2026:observable: M013:M013W10: 334677 PRODUCED IN e p COLLISIONS	f_2^'(1525)0 COMP-MESON E04; M08	83+-10 MeV	u_+: 10.295630140987001 u_-: 9.8488578017961039 Type: NONE Confidence: ND
176. PDG2026:observable: M014:M014W:333858 f_0(500) BREIT-WIGNER WIDTH	f_0(500)0 COMP-MESON E04; M08	100 to 800 MeV	u_+: NA u_-: NA Type: R Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
177. PDG2026:observable: M015:M015W:334848 2pi, K Kbar, AND K Kbar pi MODES	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	161+-10 MeV	u_+: 9.7014800918649904 u_-: 9.6537321264976139 Type: NONE Confidence: ND
178. PDG2026:observable: M015:M015W1:334849 2pi MODE	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	186+-14 MeV	u_+: 14.046724973634189 u_-: 14.046724973634189 Type: NONE Confidence: ND
179. PDG2026:observable: M015:M015W2:334850 K Kbar AND K Kbar pi MODES	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	204+-18 MeV	u_+: 17.8885438199983 18 u_-: 17.8885438199983 18 Type: NONE Confidence: ND
180. PDG2026:observable: M015:M015W3:334851 (4pi)+- MODE	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	129+-10 MeV	u_+: 10.2844009679589 39 u_-: 10.1615727369938 9 Type: NONE Confidence: ND
181. PDG2026:observable: M015:M015W5:334852 omega pi MODE	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	190+-40 MeV	u_+: 35.5278384233964 18 u_-: 35.5278384233964 18 Type: NONE Confidence: ND
182. PDG2026:observable: M015:M015W6:334853 eta pi+ pi- MODE	rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)- COMP-MESON E04; M08	130+-40 MeV	u_+: 43.4673518760492 57 u_-: 43.4673518760492 57 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
183. PDG2026:observable: M016:M016W:335131 f_4(2050) WIDTH	f_4(2050)0 COMP-MESON E04; M08	237+-18 MeV	u_+: 17.8802994325271 7 u_-: 18.1795328375529 09 Type: NONE Confidence: ND
184. PDG2026:observable: M017:M017W:335091 a_4(1970) WIDTH	a_4(1970)0 | a_4(1970)+ | a_4(1970)- COMP-MESON E04; M08	324+15-18 MeV	u_+: 14.5333933159854 99 u_-: 18.1001167978534 31 Type: NONE Confidence: ND
185. PDG2026:observable: M018:M018W1:335795 CHARGED ONLY, HADROPRODUCED	K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)- COMP-MESON E04; M08	48.5+-1.2 MeV	u_+: 1.18920218180429 3 u_-: 1.18920218180429 3 Type: NONE Confidence: ND
186. PDG2026:observable: M018:M018W5:335797 CHARGED ONLY, PRODUCED IN tau LEPTON DECAYS	K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)- COMP-MESON E04; M08	46.2+-1.3 MeV	u_+: 1.34164078649987 41 u_-: 1.34164078649987 41 Type: NONE Confidence: ND
187. PDG2026:observable: M018:M018W2:335798 NEUTRAL ONLY	K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)- COMP-MESON E04; M08	47.1+-0.5 MeV	u_+: 0.49347496850837 202 u_-: 0.49347496850837 202 Type: NONE Confidence: ND
188. PDG2026:observable: M019:M019W:335879 K_0*(1430) WIDTH	K_0^*(1430)0 | Kbar_ 0^*(1430)0 | K_0^*(1430)+ | K_0^*(1430)- COMP-MESON E04; M08	270+-80 MeV	u_+: 80 u_-: 80 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
189. PDG2026:observable: M020:M020W:335147 pi_2(2100) WIDTH	pi_2(2100)0 | pi_2(2100)+ | pi_2(2100)- COMP-MESON E04; M08	620+-50 MeV	u_+: 52.400001025245267 u_-: 52.400001025245267 Type: NONE Confidence: ND
190. PDG2026:observable: M022:M022W1:335890 CHARGED ONLY, WITH FINAL STATE K pi	K_2^*(1430)0 | Kbar_2^*(1430)0 | K_2^*(1430)+ | K_2^*(1430)- COMP-MESON E04; M08	100.0+-2.2 MeV	u_+: 2.2364281468619742 u_-: 2.1785326327209291 Type: NONE Confidence: ND
191. PDG2026:observable: M022:M022W4:335892 NEUTRAL ONLY	K_2^*(1430)0 | Kbar_2^*(1430)0 | K_2^*(1430)+ | K_2^*(1430)- COMP-MESON E04; M08	109+-5 MeV	u_+: 5.47781713955507 94 u_-: 5.47781713955507 94 Type: NONE Confidence: ND
192. PDG2026:observable: M023:M023W:335964 K_2(1770) WIDTH	K_2(1770)0 | Kbar_2(1770)0 | K_2(1770)+ | K_2(1770)- COMP-MESON E04; M08	186+-14 MeV	u_+: 13.978028299828869 u_-: 13.963235821741779 Type: NONE Confidence: ND
193. PDG2026:observable: M025:M025W:346344 psi(4160) WIDTH	psi(4160) COMP-MESON E04; M08	69+-10 MeV	u_+: 10.537685232070819 u_-: 9.6022233874145346 Type: NONE Confidence: ND
194. PDG2026:observable: M026:M026W:343287 eta_c(1S) WIDTH	eta_c(1S) COMP-MESON E04; M08	30.0+-0.5 MeV	u_+: 0.46062178744488053 u_-: 0.46062178744488053 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
195. PDG2026:observable: M027:M027WX:334530 eta(1405) WIDTH	eta(1405)0 COMP-MESON E04; M08	49.8+2.3-2.0 MeV	u_+: 2.2980645831526521 u_-: 1.9821207426120699 Type: NONE Confidence: ND
196. PDG2026:observable: M027:M027W1:334531 eta pi pi MODE	eta(1405)0 COMP-MESON E04; M08	51.1+2.8-2.1 MeV	u_+: 2.7979258032466028 u_-: 2.0581322794155499 Type: NONE Confidence: ND
197. PDG2026:observable: M027:M027W4:334532 K Kbar pi MODE (a_0(980) pi or direct K Kbar pi)	eta(1405)0 COMP-MESON E04; M08	49+-4 MeV	u_+: 3.9129589597273662 u_-: 4.1089639617502112 Type: NONE Confidence: ND
198. PDG2026:observable: M027:M027W2:334533 pi pi gamma MODE	eta(1405)0 COMP-MESON E04; M08	89+-17 MeV	u_+: 17.307932545234081 u_-: 17.307932545234081 Type: NONE Confidence: ND
199. PDG2026:observable: M027:M027W7:334534 phi gamma MODE	eta(1405)0 COMP-MESON E04; M08	86+7-18 MeV	u_+: 7.1309185944028277 u_-: 17.6082367089950 3 Type: NONE Confidence: ND
200. PDG2026:observable: M028:M028WX:335822 K_1(1270) WIDTH	K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)- COMP-MESON E04; M08	90+-20 MeV	u_+: 20 u_-: 20 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
201. PDG2026:observable: M028:M028W2:335824 PRODUCED BY K-, BACKWARD SCATTERING, HYPERON EXCHANGE	K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)- COMP-MESON E04; M08	75+-15 MeV	u_+: 15 u_-: 15 Type: NONE Confidence: ND
202. PDG2026:observable: M028:M028W3:335825 PRODUCED BY K BEAMS	K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)- COMP-MESON E04; M08	90+-8 MeV	u_+: 8 u_-: 8 Type: NONE Confidence: ND
203. PDG2026:observable: M028:M028W1:335826 PRODUCED BY BEAMS OTHER THAN K MESONS	K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)- COMP-MESON E04; M08	126+-16 MeV	u_+: 16.081957783066429 u_-: 16.081957783066429 Type: NONE Confidence: ND
204. PDG2026:observable: M028:M028WT:335827 PRODUCED IN tau LEPTON DECAYS	K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)- COMP-MESON E04; M08	260+120-110 MeV	u_+: 120.41594578792299 u_-: 106.30145812734651 Type: NONE Confidence: ND
205. PDG2026:observable: M029:M029W:346245 X(3940) WIDTH	X(3940) COMP-MESON E04; M08	43+28-18 MeV	u_+: 28.102438182886321 u_-: 17.562655210567366 Type: NONE Confidence: ND
206. PDG2026:observable: M030:M030W:334307 h_1(1170) WIDTH	h_1(1170)0 COMP-MESON E04; M08	375+-35 MeV	u_+: 34.525353003264136 u_-: 34.525353003264136 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
207. PDG2026:observable: M033:M033W3:335260 pi- p --> omega pi0 n	rho_5(2350)0 | rho_5(2350)+ | rho_5(2350)- COMP-MESON E04; M08	400+-100 MeV	u_+: 100 u_-: 100 Type: NONE Confidence: ND
208. PDG2026:observable: M034:M034W:334784 pi_2(1670) WIDTH	pi_2(1670)0 | pi_2(1670)+ | pi_2(1670)- COMP-MESON E04; M08	258+8-9 MeV	u_+: 7.9945231964062842 u_-: 9.0326516323147512 Type: NONE Confidence: ND
209. PDG2026:observable: M035:M035W:336017 K_4*(2045) WIDTH	K_4^*(2045)0 | Kbar_4^*(2045)0 | K_4^*(2045)+ | K_4^*(2045)- COMP-MESON E04; M08	199+27-19 MeV	u_+: 26.508871026900351 u_-: 19.300507563547299 Type: NONE Confidence: ND
210. PDG2026:observable: M036:M036W1:334172 a_0(980) WIDTH	a_0(980)0 | a_0(980)+ | a_0(980)- COMP-MESON E04; M08	50 to 100 MeV	u_+: NA u_-: NA Type: R Confidence: ND
211. PDG2026:observable: M037:M037W:334452 eta(1295) WIDTH	eta(1295)0 COMP-MESON E04; M08	55+-5 MeV	u_+: 5.3662816306481913 u_-: 5.3662816306481913 Type: NONE Confidence: ND
212. PDG2026:observable: M038:M038W:335003 f_2(1810) WIDTH	f_2(1810)0 COMP-MESON E04; M08	197+-22 MeV	u_+: 21.979426688512731 u_-: 22.153970300247611 Type: NONE Confidence: ND
213. PDG2026:observable: M040:M040W:336033 K_2(2250) WIDTH	K_2(2250)0 | Kbar_2(2250)0 | K_2(2250)+ | K_2(2250)- COMP-MESON E04; M08	180+-30 MeV	u_+: 29.999999486571867 u_-: 29.999999486571867 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
214. PDG2026:observable: M041:M041W4:335245 p p CENTRAL PRODUCTION	f_4(2300)0 COMP-MESON E04; M08	250+-80 MeV	u_+: 80 u_-: 80 Type: NONE Confidence: ND
215. PDG2026:observable: M042:M042W:335160 f_2(2150) WIDTH, COMBINED MODES	f_2(2150)0 COMP-MESON E04; M08	152+-30 MeV	u_+: 30.187002348708841 u_-: 30.187002348708841 Type: NONE Confidence: ND
216. PDG2026:observable: M042:M042W3:335161 eta eta MODE	f_2(2150)0 COMP-MESON E04; M08	152+-30 MeV	u_+: 30.187002348708841 u_-: 30.187002348708841 Type: NONE Confidence: ND
217. PDG2026:observable: M042:M042W1:335162 pbar p --> pi pi	f_2(2150)0 COMP-MESON E04; M08	250 MeV	u_+: ND u_-: ND Type: NONE Confidence: ND
218. PDG2026:observable: M045:M045W:334779 omega_3(1670) WIDTH	omega_3(1670)0 COMP-MESON E04; M08	168+-10 MeV	u_+: 10.0179720379382 u_-: 10.0179720379382 Type: NONE Confidence: ND
219. PDG2026:observable: M047:M047W:347745 Upsilon(4S) WIDTH	Upsilon(4S) COMP-MESON E04; M08	20.5+-2.5 MeV	u_+: 2.473041036470411 u_-: 2.473041036470411 Type: NONE Confidence: ND
220. PDG2026:observable: M048:M048W:347661 Upsilon(3S) WIDTH	Upsilon(3S) COMP-MESON E04; M08	20.32+-1.85 keV	u_+: 1.8500000000000001 u_-: 1.8500000000000001 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
221. PDG2026:observable: M049:M049W:346838 Upsilon(1S) WIDTH	Upsilon(1S) COMP-MESON E04; M08	54.02+-1.25 keV	u_+: 1.25 u_-: 1.25 Type: NONE Confidence: ND
222. PDG2026:observable: M050:M050W:346229 chi_c2(3930) WIDTH	chi_c2(3930) COMP-MESON E04; M08	35.2+-2.2 MeV	u_+: 2.2286730591186932 u_-: 2.2286730591186932 Type: NONE Confidence: ND
223. PDG2026:observable: M052:M052W:347288 Upsilon(2S) WIDTH	Upsilon(2S) COMP-MESON E04; M08	31.98+-2.63 keV	u_+: 2.629999999999999 u_-: 2.629999999999999 Type: NONE Confidence: ND
224. PDG2026:observable: M053:M053W:345870 psi(3770) WIDTH	psi(3770) COMP-MESON E04; M08	27.2+-1.0 MeV	u_+: 0.9592482270788617 u_-: 0.9592482270788617 Type: NONE Confidence: ND
225. PDG2026:observable: M054:M054W:335024 phi_3(1850) WIDTH	phi_3(1850)0 COMP-MESON E04; M08	87+28-23 MeV	u_+: 27.578792949203059 u_-: 23.092499274786309 Type: NONE Confidence: ND
226. PDG2026:observable: M055:M055W:344599 chi_c1(1P) WIDTH	chi_c1(1P) COMP-MESON E04; M08	0.82+-0.04 MeV	u_+: 0.041623417330381482 u_-: 0.041623417330381482 Type: NONE Confidence: ND
227. PDG2026:observable: M056:M056W:344305 chi_c0(1P) WIDTH	chi_c0(1P) COMP-MESON E04; M08	12.40+-0.27 MeV	u_+: 0.26507716522265817 u_-: 0.26507716522265817 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
228. PDG2026:observable: M057:M057W:344927 chi_c2(1P) WIDTH	chi_c2(1P) COMP-MESON E04; M08	1.94+-0.09 MeV	u_+: 0.08904841015736 6226 u_-: 0.08904841015736 6226 Type: NONE Confidence: ND
229. PDG2026:observable: M058:M058W:334464 pi(1300) WIDTH	pi(1300)0 | pi(1300)+ | pi(1300)- COMP-MESON E04; M08	200 TO 600 MeV Presentation: 200 to 600	u_+: NA u_-: NA Type: R Confidence: ND
230. PDG2026:observable: M059:M059W:345229 eta_c(2S) WIDTH	eta_c(2S) COMP-MESON E04; M08	11.6+-1.4 MeV	u_+: 1.46140521435830 21 u_-: 1.42482273720458 5 Type: NONE Confidence: ND
231. PDG2026:observable: M060:M060W:335976 K_3*(1780) WIDTH	K_3^*(1780)0 | Kbar_ 3^*(1780)0 | K_3^*(1780)+ | K_3^*(1780)- COMP-MESON E04; M08	161+-17 MeV	u_+: 16.7757777591419 21 u_-: 17.6873693367902 58 Type: NONE Confidence: ND
232. PDG2026:observable: M061:M061W:337780 D*(2007)0 WIDTH	D^*(2007)0 | Dbar^*(2007) 0 COMP-MESON E04; M08	<2.1 MeV	u_+: NA u_-: NA Type: U Confidence: 90
233. PDG2026:observable: M062:M062W:337799 D*(2010)+- WIDTH	D^*(2010)+ | D^*(2010)- COMP-MESON E04; M08	83.4+-1.8 keV	u_+: 1.83767139449678 71 u_-: 1.83767139449678 71 Type: NONE Confidence: ND
234. PDG2026:observable: M064:M064W:335846 K_1(1400) WIDTH	K_1(1400)0 | Kbar_1(1400) 0 | K_1(1400)+ | K_1(1400) - COMP-MESON E04; M08	174+-13 MeV	u_+: 12.6111236551536 99 u_-: 12.6011609689773 61 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
235. PDG2026:observable: M065:M065W0:334881 eta rho0 AND pi+ pi- MODES	rho(1700)0 | rho(1700)+ | rho(1700)- COMP-MESON E04; M08	250+-100 MeV	u_+: 100 u_-: 100 Type: NONE Confidence: ND
236. PDG2026:observable: M067:M067W1:334826 e+ e- PRODUCTION	phi(1680)0 COMP-MESON E04; M08	150+-50 MeV	u_+: 50 u_-: 50 Type: NONE Confidence: ND
237. PDG2026:observable: M068:M068W:334928 f_0(1710) Breit-Wigner WIDTH	f_0(1710)0 COMP-MESON E04; M08	149+-10 MeV	u_+: 10.9253240627161 7 u_-: 9.8917234176496 365 Type: NONE Confidence: ND
238. PDG2026:observable: M070:M070W:343506 J/psi(1S) WIDTH	J/psi(1S) COMP-MESON E04; M08	92.6+-1.7 keV	u_+: 1.7182511605720 709 u_-: 1.7119132837100 099 Type: NONE Confidence: ND
239. PDG2026:observable: M071:M071W:345302 psi(2S) WIDTH	psi(2S) COMP-MESON E04; M08	293+-9 keV	u_+: 9.1216171327147 304 u_-: 9.1216171327147 304 Type: NONE Confidence: ND
240. PDG2026:observable: M072:M072W:346250 psi(4040) WIDTH	psi(4040) COMP-MESON E04; M08	84+-12 MeV	u_+: 12.3000000000000 01 u_-: 12.3000000000000 01 Type: NONE Confidence: ND
241. PDG2026:observable: M073:M073W:346685 psi(4415) WIDTH	psi(4415) COMP-MESON E04; M08	110+-13 MeV	u_+: 13.2878278993015 7 u_-: 13.2878278993015 7 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
242. PDG2026:observable: M074:M074W:346461 psi(4230) WIDTH	psi(4230) COMP-MESON E04; M08	51+-7 MeV	u_+: 6.99729038980041 02 u_-: 6.88241490298323 64 Type: NONE Confidence: ND
243. PDG2026:observable: M075:M075W:334967 pi(1800) WIDTH	pi(1800)0 | pi(1800)+ | pi(1800)- COMP-MESON E04; M08	215+7-8 MeV	u_+: 6.70657742096917 8 u_-: 8.19989930002816 08 Type: NONE Confidence: ND
244. PDG2026:observable: M082:M082W:335204 f_J(2220) WIDTH	f_J(2220)0 COMP-MESON E04; M08	23+8-7 MeV	u_+: 8.29072339682046 79 u_-: 6.99539815418370 2 Type: NONE Confidence: ND
245. PDG2026:observable: M084:M084W:334659 f_1(1510) WIDTH	f_1(1510)0 COMP-MESON E04; M08	73+-25 MeV	u_+: 25.1244904562855 5 u_-: 25.1244904562855 5 Type: NONE Confidence: ND
246. PDG2026:observable: M085:M085W:335012 X(1835) WIDTH	X(1835)0 COMP-MESON E04; M08	130+-50 MeV	u_+: 46.5214248579634 67 u_-: 48.8057618821165 28 Type: NONE Confidence: ND
247. PDG2026:observable: M089:M089W:335278 f_6(2510) WIDTH	f_6(2510)0 COMP-MESON E04; M08	260+-40 MeV	u_+: 42.4264068711928 53 u_-: 42.4264068711928 53 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
248. PDG2026:observable: M090:M090W:336035 K_3(2320) WIDTH	K_3(2320)0 | Kbar_3(2320)0 | K_3(2320)+ | K_3(2320)- COMP-MESON E04; M08	150+-30 MeV	u_+: 30 u_-: 30 Type: NONE Confidence: ND
249. PDG2026:observable: M092:M092W:347805 Upsilon(10860) WIDTH	Upsilon(10860) COMP-MESON E04; M08	37+-4 MeV	u_+: 4.26656207180538 96 u_-: 3.97673653123974 89 Type: NONE Confidence: ND
250. PDG2026:observable: M093:M093W:347906 Upsilon(11020) WIDTH	Upsilon(11020) COMP-MESON E04; M08	24+8-6 MeV	u_+: 7.70770656457082 83 u_-: 6.45715582578219 66 Type: NONE Confidence: ND
251. PDG2026:observable: M094:M094W:335866 K*(1410) WIDTH	K^*(1410)0 | Kbar^*(1410)0 | K^*(1410)+ | K^*(1410)- COMP-MESON E04; M08	232+-21 MeV	u_+: 21.3823721605395 6 u_-: 21.3823721605395 6 Type: NONE Confidence: ND
252. PDG2026:observable: M095:M095W:335947 K*(1680) WIDTH	K^*(1680)0 | Kbar^*(1680)0 | K^*(1680)+ | K^*(1680)- COMP-MESON E04; M08	320+-110 MeV	u_+: 107.338345980695 71 u_-: 107.807566482925 91 Type: NONE Confidence: ND
253. PDG2026:observable: M098:M098W:336037 K_5*(2380) WIDTH	K_5^*(2380)0 | Kbar_5^*(2380)0 | K_5^*(2380)+ | K_5^*(2380)- COMP-MESON E04; M08	180+-50 MeV	u_+: 48.9182992345400 36 u_-: 48.9182992345400 36 Type: NONE Confidence: ND
254. PDG2026:observable: M099:M099W:335945 K_1(1650) WIDTH	K_1(1650)0 | Kbar_1(1650)0 | K_1(1650)+ | K_1(1650)- COMP-MESON E04; M08	150+-50 MeV	u_+: 50 u_-: 50 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
255. PDG2026:observable: M101:M101W:335033 eta_2(1870) WIDTH	eta_2(1870)0 COMP-MESON E04; M08	225+-14 MeV	u_+: 14.065007548009 65 u_-: 14.065007548009 65 Type: NONE Confidence: ND
256. PDG2026:observable: M103:M103W:335181 phi(2170) WIDTH	phi(2170)0 | phi(2170)+ | phi(2170)- COMP-MESON E04; M08	88+26-21 MeV	u_+: 25.845618263221 67 u_-: 20.635865146473 79 Type: NONE Confidence: ND
257. PDG2026:observable: M104:M104W:336004 K_2*(1980) WIDTH	K_2^*(1980)0 | Kbar_ 2^*(1980)0 | K_2^*(1980)+ | K_2^*(1980)- COMP-MESON E04; M08	348+50-30 MeV	u_+: 47.995015852124 311 u_-: 29.822700392743 499 Type: NONE Confidence: ND
258. PDG2026:observable: M105:M105W0:334593 rho(1450) WIDTH	rho(1450)0 | rho(1450)+ | rho(1450)- COMP-MESON E04; M08	400 +-60 MeV Presentation: 400+-60	u_+: 60 u_-: 60 Type: NONE Confidence: ND
259. PDG2026:observable: M106:M106W:335113 f_2(2010) WIDTH	f_2(2010)0 COMP-MESON E04; M08	200+-60 MeV	u_+: 67 u_-: 62 Type: NONE Confidence: ND
260. PDG2026:observable: M107:M107W:335235 f_2(2300) WIDTH	f_2(2300)0 COMP-MESON E04; M08	150+-40 MeV	u_+: 41 u_-: 41 Type: NONE Confidence: ND
261. PDG2026:observable: M108:M108W:335253 f_2(2340) WIDTH	f_2(2340)0 COMP-MESON E04; M08	331+27-18 MeV	u_+: 27.406094957830 65 u_-: 17.599089776866 538 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
262. PDG2026:observable: M109:M109W:334553 h_1(1415) WIDTH	h_1(1415)0 COMP-MESON E04; M08	78+-11 MeV	u_+: 10.894768418419931 u_-: 10.354761026844946 Type: NONE Confidence: ND
263. PDG2026:observable: M112:M112W:335200 f_0(2200) WIDTH	f_0(2200)0 COMP-MESON E04; M08	210+-40 MeV	u_+: 41.638656568206237 u_-: 42.173251605545943 Type: NONE Confidence: ND
264. PDG2026:observable: M114:M114W:334951 eta(1760) WIDTH	eta(1760)0 COMP-MESON E04; M08	240+-30 MeV	u_+: 30.767000003926665 u_-: 28.819843599945798 Type: NONE Confidence: ND
265. PDG2026:observable: M115:M115W:335233 eta(2225) WIDTH	eta(2225)0 COMP-MESON E04; M08	185+40-20 MeV	u_+: 36.894036523364868 u_-: 19.6088060392476 Type: NONE Confidence: ND
266. PDG2026:observable: M117:M117W:334753 f_2(1640) WIDTH	f_2(1640)0 COMP-MESON E04; M08	100+60-40 MeV	u_+: 55.007270965809873 u_-: 40.999999045433803 Type: NONE Confidence: ND
267. PDG2026:observable: M121:M121W:338489 D_s1(2536)+- WIDTH	D_s1(2536)+ | D_s1(2536)- COMP-MESON E04; M08	0.92+-0.05 MeV	u_+: 0.049965313904947829 u_-: 0.049965313904947829 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
268. PDG2026:observable: M123:M123W:334704 f_2(1565) WIDTH	f_2(1565)0 COMP-MESON E04; M08	132+-23 MeV	u_+: 22.499999212159828 u_-: 22.499999212159828 Type: NONE Confidence: ND
269. PDG2026:observable: M125:M125W:334568 omega(1420) WIDTH	omega(1420)0 COMP-MESON E04; M08	290+-190 MeV	u_+: 190 u_-: 190 Type: NONE Confidence: ND
270. PDG2026:observable: M126:M126W:334769 omega(1650) WIDTH	omega(1650)0 COMP-MESON E04; M08	150+-50 MeV	u_+: 50 u_-: 50 Type: NONE Confidence: ND
271. PDG2026:observable: M134:M134W:336000 K_0*(1950) WIDTH	K_0^*(1950)0 | Kbar_0^*(1950)0 | K_0^*(1950)+ | K_0^*(1950)- COMP-MESON E04; M08	170+-50 MeV	u_+: 49.977634624979267 u_-: 49.977634624979267 Type: NONE Confidence: ND
272. PDG2026:observable: M135:M135W:335076 f_2(1950) WIDTH	f_2(1950)0 COMP-MESON E04; M08	464+-24 MeV	u_+: 24.233608393520051 u_-: 23.858175111592871 Type: NONE Confidence: ND
273. PDG2026:observable: M142:M142W2:335059 f_2(1910) omega omega MODE	f_2(1910)0 COMP-MESON E04; M08	167+-21 MeV	u_+: 20.593210779464879 u_-: 20.593210779464879 Type: NONE Confidence: ND
274. PDG2026:observable: M142:M142W3:335060 f_2(1910) eta eta' MODE	f_2(1910)0 COMP-MESON E04; M08	140+-40 MeV	u_+: 41 u_-: 41 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
275. PDG2026:observable: M144:M144W:344852 h_c(1P) WIDTH	h_c(1P) COMP-MESON E04; M08	0.78+-0.28 MeV	u_+: 0.29546573405388321 u_-: 0.26832815729997478 Type: NONE Confidence: ND
276. PDG2026:observable: M146:M146W:335991 K_2(1820) WIDTH	K_2(1820)0 | Kbar_2(1820)0 | K_2(1820)+ | K_2(1820)- COMP-MESON E04; M08	264+-34 MeV	u_+: 33.911162815562037 u_-: 33.585154163911888 Type: NONE Confidence: ND
277. PDG2026:observable: M147:M147W:334507 Full width (Breit-Wigner)	f_0(1370)0 COMP-MESON E04; M08	200 to 500 MeV	u_+: NA u_-: NA Type: R Confidence: ND
278. PDG2026:observable: M148:M148W:338511 D_s2*(2573) WIDTH	D_s2^*(2573)+ | D_s2^*(2573)- COMP-MESON E04; M08	16.9+-0.7 MeV	u_+: 0.73972013426443928 u_-: 0.73758653054042789 Type: NONE Confidence: ND
279. PDG2026:observable: M149:M149W:334580 a_0(1450) WIDTH	a_0(1450)0 | a_0(1450)+ | a_0(1450)- COMP-MESON E04; M08	258+-14 MeV	u_+: 14.000000098926851 u_-: 14.000000098926851 Type: NONE Confidence: ND
280. PDG2026:observable: M151:M151W:342552 B_J*(5732) WIDTH	B_J^*(5732)0 | Bbar_J^*(5732)0 | B_J^*(5732)+ | B_J^*(5732)- COMP-MESON E04; M08	128+-18 MeV	u_+: 18.222158456767119 u_-: 18.222158456767119 Type: NONE Confidence: ND
281. PDG2026:observable: M152:M152W:334629 f_0(1500) WIDTH	f_0(1500)0 COMP-MESON E04; M08	108+-33 MeV	u_+: 33 u_-: 33 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
282. PDG2026:observable: M153:M153W:343152 B_sJ*(5850) WIDTH	B_sJ^*(5850)0 | Bbar_ sJ^*(5850)0 COMP-MESON E04; M08	47+-22 MeV	u_+: 22 u_-: 22 Type: NONE Confidence: ND
283. PDG2026:observable: M154:M154W:334759 eta_2(1645) WIDTH	eta_2(1645)0 COMP-MESON E04; M08	181+-11 MeV	u_+: 10.7857447992021 21 u_-: 10.4916641064320 9 Type: NONE Confidence: ND
284. PDG2026:observable: M156:M156W:335121 f_0(2020) WIDTH	f_0(2020)0 COMP-MESON E04; M08	440+-50 MeV	u_+: 46.1735855224607 83 u_-: 49.1629942131274 19 Type: NONE Confidence: ND
285. PDG2026:observable: M158:M158W:337855 D*(2640)+- WIDTH	D^*(2640)0 | Dbar^*(2640) 0 | D^*(2640)+ | D^*(2640) - COMP-MESON E04; M08	<15 MeV	u_+: NA u_-: NA Type: U Confidence: 95
286. PDG2026:observable: M159:M159W:346205 chi_c0(3915) WIDTH	chi_c0(3915) COMP-MESON E04; M08	20+-4 MeV	u_+: 3.92370931837659 9 u_-: 3.84857327141186 99 Type: NONE Confidence: ND
287. PDG2026:observable: M160:M160W:335943 K(1630) WIDTH	K(1630)0 | Kbar(1630)0 | K(1630)+ | K(1630)- COMP-MESON E04; M08	16+19-16 MeV	u_+: 19 u_-: 16 Type: NONE Confidence: ND
288. PDG2026:observable: M161:M161W:334741 a_1(1640) WIDTH	a_1(1640)0 | a_1(1640)+ | a_1(1640)- COMP-MESON E04; M08	250+-40 MeV	u_+: 38.7680261069732 67 u_-: 38.0044644472388 31 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
289. PDG2026:observable: M162:M162W:334904 a_2(1700) WIDTH	a_2(1700)0 | a_2(1700)+ | a_2(1700)- COMP-MESON E04; M08	380+60-50 MeV	u_+: 56.802156616143733 u_-: 49.91978684730767 Type: NONE Confidence: ND
290. PDG2026:observable: M164:M164WEP: 334726 pi_1(1600) WIDTH (eta pi mode)	pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)- COMP-MESON E04; M08	330+-35 MeV	u_+: 34.727689315250451 u_-: 34.693151629682987 Type: NONE Confidence: ND
291. PDG2026:observable: M164:M164W:334727 pi_1(1600) WIDTH (non-eta pi mode)	pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)- COMP-MESON E04; M08	370+50-60 MeV	u_+: 50.324516391842117 u_-: 56.454714048381128 Type: NONE Confidence: ND
292. PDG2026:observable: M166:M166W:334721 h_1(1595) WIDTH	h_1(1595)0 COMP-MESON E04; M08	380+90-120 MeV	u_+: 92.195444572928878 u_-: 116.61903789690599 Type: NONE Confidence: ND
293. PDG2026:observable: M168:M168W:335157 f_0(2100) WIDTH	f_0(2100)0 COMP-MESON E04; M08	287+32-24 MeV	u_+: 31.594915701364421 u_-: 24.048675270055991 Type: NONE Confidence: ND
294. PDG2026:observable: M171:M171W:346826 eta_b(1S) WIDTH	eta_b(1S) COMP-MESON E04; M08	10+5-4 MeV	u_+: 4.7684157959520306 u_-: 3.6148198760107491 Type: NONE Confidence: ND
295. PDG2026:observable: M172:M172W:338433 D_s0*(2317)+- WIDTH	D_s0^*(2317)+ | D_s0^*(2317)- COMP-MESON E04; M08	< 3.8 MeV Presentation: <3.8	u_+: NA u_-: NA Type: U Confidence: 95

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
296. PDG2026:observable: M173:M173W:338454 D_s1(2460)+- WIDTH	D_s1(2460)+ | D_s1(2460)- COMP-MESON E04; M08	< 3.5 MeV Presentation: <3.5	u_+: NA u_-: NA Type: U Confidence: 95
297. PDG2026:observable: M174:M174W:335787 K_0*(700) Breit-Wigner Width	K_0^*(700)0 | Kbar_ 0^*(700)0 | K_0^*(700)+ | K_0^*(700)- COMP-MESON E04; M08	463+-27 MeV	u_+: 28.0180200105868 49 u_-: 25.9692055984802 12 Type: NONE Confidence: ND
298. PDG2026:observable: M175:M175W5:334617 eta(1475) WIDTH	eta(1475)0 COMP-MESON E04; M08	96+-9 MeV	u_+: 9.16477812816133 59 u_-: 8.65891056010132 05 Type: NONE Confidence: ND
299. PDG2026:observable: M176:M176W:346132 chi_c1(3872) WIDTH	chi_c1(3872) COMP-MESON E04; M08	1.19+-0.21 MeV	u_+: 0.21446141015052 739 u_-: 0.21212796092607 489 Type: NONE Confidence: ND
300. PDG2026:observable: M180:M180W:337828 D_1(2430)0 WIDTH	D_1(2430)0 | Dbar_1(2430) 0 COMP-MESON E04; M08	314+-29 MeV	u_+: 28.6869527683639 7 u_-: 28.3279146860413 21 Type: NONE Confidence: ND
301. PDG2026:observable: M181:M181W:346636 psi(4360) WIDTH	psi(4360) COMP-MESON E04; M08	124+-13 MeV	u_+: 13.0611247068081 21 u_-: 13.2865041822744 7 Type: NONE Confidence: ND
302. PDG2026:observable: M182:M182W:338525 D_s1*(2700)+ WIDTH	D_s1^*(2700)+ | D_ s1^*(2700)- COMP-MESON E04; M08	122+-10 MeV	u_+: 10.5434175204121 29 u_-: 10.0609470530140 1 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
303. PDG2026:observable: M185:M185W:335044 pi_2(1880) WIDTH	pi_2(1880)0 COMP-MESON E04; M08	237+33-30 MeV	u_+: 33.3171974639905 17 u_-: 29.5026580378861 11 Type: NONE Confidence: ND
304. PDG2026:observable: M186:M186W:343141 B_s2*(5840)0 WIDTH	B_s2^*(5840)0 | Bbar_ s2^*(5840)0 COMP-MESON E04; M08	1.49+-0.27 MeV	u_+: 0.26660479598163 422 u_-: 0.26660479598163 422 Type: NONE Confidence: ND
305. PDG2026:observable: M187:M187W:343133 B_s1(5830)0 WIDTH	B_s1(5830)0 | Bbar_ s1(5830)0 COMP-MESON E04; M08	0.5+-0.4 MeV	u_+: 0.42426406871192 851 u_-: 0.42426406871192 851 Type: NONE Confidence: ND
306. PDG2026:observable: M188:M188W:334715 rho(1570) WIDTH	rho(1570)0 COMP-MESON E04; M08	140+-90 MeV	u_+: 86.4522989862039 42 u_-: 86.4522989862039 42 Type: NONE Confidence: ND
307. PDG2026:observable: M189:M189W:346768 psi(4660) WIDTH	psi(4660) COMP-MESON E04; M08	55+-9 MeV	u_+: 8.88972291869921 7 u_-: 8.40991396186550 06 Type: NONE Confidence: ND
308. PDG2026:observable: M190:M190W:346452 X(4160) WIDTH	X(4160) COMP-MESON E04; M08	136+60-35 MeV	u_+: 56.5392209238173 54 u_-: 34.6252307256621 42 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
309. PDG2026:observable: M191:M191W:347991 T_c_cbar(4050)+ WIDTH	T_c_cbar(4050)+ COMP-MESON E04; M08	82+50-28 MeV	u_+: 51.478150704934997 u_-: 27.802877548915689 Type: NONE Confidence: ND
310. PDG2026:observable: M192:M192W:348027 T_c_cbar(4250)+ WIDTH	T_c_cbar(4250)+ COMP-MESON E04; M08	180+320-70 MeV	u_+: 320.58072306363027 u_-: 72.401657439591816 Type: NONE Confidence: ND
311. PDG2026:observable: M193:M193W:346337 chi_c1(4140) WIDTH	chi_c1(4140) COMP-MESON E04; M08	19+7-5 MeV	u_+: 6.55866734077530 64 u_-: 5.27582025104883 37 Type: NONE Confidence: ND
312. PDG2026:observable: M194:M194W:346629 X(4350) WIDTH	X(4350) COMP-MESON E04; M08	13+18-10 MeV	u_+: 18.439088914585771 u_-: 9.8488578017961039 Type: NONE Confidence: ND
313. PDG2026:observable: M195:M195W:348031 T_c_cbar_1(4430)+ WIDTH	T_c_cbar_1(4430)+ COMP-MESON E04; M08	189+-29 MeV	u_+: 28.960862620750699 u_-: 29.212281692432189 Type: NONE Confidence: ND
314. PDG2026:observable: M196:M196W:338534 D_s1*(2860)+ WIDTH	D_s1^*(2860)+ | D_s1^*(2860)- COMP-MESON E04; M08	160+-80 MeV	u_+: 80.361682411457764 u_-: 80.361682411457764 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
315. PDG2026:observable: M197:M197W:338545 D_sJ(3040)+ WIDTH	D_sJ(3040)+ | D_sJ(3040)- COMP-MESON E04; M08	240+-60 MeV	u_+: 57.801384066473702 u_-: 54.671747731346578 Type: NONE Confidence: ND
316. PDG2026:observable: M198:M198W:337844 D_0(2550)0 WIDTH	D_0(2550)0 | Dbar_0(2550)0 | D_0(2550)+ | D_0(2550)- COMP-MESON E04; M08	165+-24 MeV	u_+: 23.747553397709549 u_-: 23.747553397709549 Type: NONE Confidence: ND
317. PDG2026:observable: M199:M199W:337847 D_1*(2600)0 WIDTH	D_1^*(2600)0 | Dbar_1^*(2600)0 | D_1^*(2600)+ | D_1^*(2600)- COMP-MESON E04; M08	141+-23 MeV	u_+: 22.836524044183388 u_-: 22.836524044183388 Type: NONE Confidence: ND
318. PDG2026:observable: M200:M200W:347283 eta_b(2S) WIDTH	eta_b(2S) COMP-MESON E04; M08	< 24 MeV Presentation: <24	u_+: NA u_-: NA Type: U Confidence: 90
319. PDG2026:observable: M203:M203W:337861 D_3*(2750) WIDTH	D_3^*(2750)0 | Dbar_3^*(2750)0 | D_3^*(2750)+ | D_3^*(2750)- COMP-MESON E04; M08	66+-5 MeV	u_+: 4.6026930917312541 u_-: 4.6026930917312541 Type: NONE Confidence: ND
320. PDG2026:observable: M207:M207W:348054 T_b_bbar_1(10610)+- WIDTH	T_b_bbar_1(10610)+ COMP-MESON E04; M08	18.4+-2.4 MeV	u_+: 2.3999999999999999 u_-: 2.3999999999999999 Type: NONE Confidence: ND
321. PDG2026:observable: M208:M208W:348076 T_b_bbar_1(10650)+ WIDTH	T_b_bbar_1(10650)+ COMP-MESON E04; M08	11.5+-2.2 MeV	u_+: 2.2000000000000002 u_-: 2.2000000000000002 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
322. PDG2026:observable: M210:M210W:347939 T_c_cbar_1(3900) WIDTH	T_c_cbar_1(3900)+ COMP-MESON E04; M08	29.5+-2.4 MeV	u_+: 2.42574928658747 7 u_-: 2.41740148443531 49 Type: NONE Confidence: ND
323. PDG2026:observable: M212:M212W:346101 psi_2(3823) WIDTH	psi_2(3823) COMP-MESON E04; M08	< 2.9 MeV Presentation: <2.9	u_+: NA u_-: NA Type: U Confidence: 90
324. PDG2026:observable: M213:M213W:347977 T_c_cbar(4020) WIDTH	T_c_cbar(4020)+ COMP-MESON E04; M08	13+-5 MeV	u_+: 5.27421046109626 52 u_-: 5.27421046109626 52 Type: NONE Confidence: ND
325. PDG2026:observable: M216:M216W:348023 T_c_cbar_0(4240)+ WIDTH	T_c_cbar_0(4240)0 COMP-MESON E04; M08	220+120-90 MeV	u_+: 117.783700060746 99 u_-: 87.6641317757724 61 Type: NONE Confidence: ND
326. PDG2026:observable: M223:M223W:348001 T_c_cbar(4055)+ WIDTH	T_c_cbar(4055)+ COMP-MESON E04; M08	45+-13 MeV	u_+: 12.5299640861416 69 u_-: 12.5299640861416 69 Type: NONE Confidence: ND
327. PDG2026:observable: M226:M226W:338536 D_s3*(2860)+ WIDTH	D_s3^*(2860)+ | D_ s3^*(2860)- COMP-MESON E04; M08	53+-10 MeV	u_+: 9.89949493661166 54 u_-: 9.89949493661166 54 Type: NONE Confidence: ND
328. PDG2026:observable: M227:M227W:335071 a_0(1950) WIDTH	a_0(1950)0 COMP-MESON E04; M08	270+-40 MeV	u_+: 36.4005494464025 93 u_-: 36.4005494464025 93 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
329. PDG2026:observable: M228:M228W:337858 D_2(2740)0 WIDTH	D_2(2740)0 | Dbar_2(2740)0 | D_2(2740)+ | D_2(2740)- COMP-MESON E04; M08	88+-19 MeV	u_+: 19.435218021172709 u_-: 19.435218021172709 Type: NONE Confidence: ND
330. PDG2026:observable: M229:M229W:337873 D_2*(3000)0 WIDTH	D_2^*(3000)0 | Dbar_2^*(3000)0 | D_2^*(3000)+ | D_2^*(3000)- COMP-MESON E04; M08	190+-80 MeV	u_+: 81.412529748190479 u_-: 81.412529748190479 Type: NONE Confidence: ND
331. PDG2026:observable: M231:M231W:348013 T_c_cbar_1(4200)+ WIDTH	T_c_cbar_1(4200)+ COMP-MESON E04; M08	310+-40 MeV	u_+: 35.260965800649629 u_-: 36.587057375401727 Type: NONE Confidence: ND
332. PDG2026:observable: M232:M232W:348037 T_b_sbar(5568)+ WIDTH	T_b_sbar(5568)+ COMP-MESON E04; M08	19+9-7 MeV	u_+: 8.6406018308911801 u_-: 7.1867934435323786 Type: NONE Confidence: ND
333. PDG2026:observable: M233:M233W:346625 chi_c1(4274) WIDTH	chi_c1(4274) COMP-MESON E04; M08	54+-8 MeV	u_+: 8.1968525494933324 u_-: 7.7933837739246821 Type: NONE Confidence: ND
334. PDG2026:observable: M234:M234W:346756 chi_c0(4500) WIDTH	chi_c0(4500) COMP-MESON E04; M08	87+30-24 MeV	u_+: 29.927160405061439 u_-: 24.399652435665001 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
335. PDG2026:observable: M235:M235W:346817 chi_c0(4700) WIDTH	chi_c0(4700) COMP-MESON E04; M08	78+15-11 MeV	u_+: 14.539729964084 01 u_-: 10.852931317094 249 Type: NONE Confidence: ND
336. PDG2026:observable: M237:M237W:346123 chi_c0(3860) WIDTH	chi_c0(3860) COMP-MESON E04; M08	200+180-110 MeV	u_+: 177.369670462568 11 u_-: 105.891453857240 09 Type: NONE Confidence: ND
337. PDG2026:observable: M239:M239W:335107 pi_2(2005) WIDTH	pi_2(2005)0 COMP-MESON E04; M08	370+16-90 MeV	u_+: 15.911852647721 62 u_-: 94.137316212663 99 Type: NONE Confidence: ND
338. PDG2026:observable: M240:M240W:348007 T_c_cbar(4100)+ WIDTH	T_c_cbar(4100)+ COMP-MESON E04; M08	150+80-70 MeV	u_+: 83.450584180100 265 u_-: 67.742158217759 794 Type: NONE Confidence: ND
339. PDG2026:observable: M241:M241W:346117 psi_3(3842) WIDTH	psi_3(3842) COMP-MESON E04; M08	2.8+-0.6 MeV	u_+: 0.6185466837676 8462 u_-: 0.6185466837676 8462 Type: NONE Confidence: ND
340. PDG2026:observable: M243:M243W:347801 Upsilon(10753) WIDTH	Upsilon(10753) COMP-MESON E04; M08	29+-9 MeV	u_+: 8.8814413244698 081 u_-: 8.8814413244698 081 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
341. PDG2026:observable: M244:M244W+:342547 B_1(5721)+ width	B_1(5721)0 | Bbar_1(5721)0 | B_1(5721)+ | B_1(5721)- COMP-MESON E04; M08	31+-6 MeV	u_+: 5.84704469385892 u_-: 6.09210420060922 43 Type: NONE Confidence: ND
342. PDG2026:observable: M244:M244W0:342548 B_1(5721)0 width	B_1(5721)0 | Bbar_1(5721)0 | B_1(5721)+ | B_1(5721)- COMP-MESON E04; M08	27.5+-3.4 MeV	u_+: 3.42227784518650 01 u_-: 3.42227784518650 01 Type: NONE Confidence: ND
343. PDG2026:observable: M245:M245W+:342564 B_2*(5747)+ width	B_2^*(5747)0 | Bbar_ 2^*(5747)0 | B_2^*(5747)+ | B_2^*(5747)- COMP-MESON E04; M08	20+-5 MeV	u_+: 5.46842291584328 51 u_-: 5.46842291584328 51 Type: NONE Confidence: ND
344. PDG2026:observable: M245:M245W0:342565 B_2*(5747)0 width	B_2^*(5747)0 | Bbar_ 2^*(5747)0 | B_2^*(5747)+ | B_2^*(5747)- COMP-MESON E04; M08	24.2+-1.7 MeV	u_+: 1.69590866342803 49 u_-: 1.70952643154618 71 Type: NONE Confidence: ND
345. PDG2026:observable: M246:M246W+:342577 B_J(5840)+ WIDTH	B_J(5840)0 | Bbar_J(5840)0 | B_J(5840)+ | B_J(5840)- COMP-MESON E04; M08	220+-80 MeV	u_+: 83.5224520712844 05 u_-: 83.5224520712844 05 Type: NONE Confidence: ND
346. PDG2026:observable: M246:M246W0:342578 B_J(5840)0 WIDTH	B_J(5840)0 | Bbar_J(5840)0 | B_J(5840)+ | B_J(5840)- COMP-MESON E04; M08	130+-40 MeV	u_+: 38.0131556174964 23 u_-: 38.0131556174964 23 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
347. PDG2026:observable: M247:M247W:335262 eta(2370) WIDTH	eta(2370)0 COMP-MESON E04; M08	148+80-28 MeV	u_+: 77.2222437006731 61 u_-: 28.3671777206377 8 Type: NONE Confidence: ND
348. PDG2026:observable: M248:M248W+:342589 B_J(5970)+ WIDTH	B_J(5970)0 | Bbar_J(5970) 0 | B_J(5970)+ | B_J(5970)- COMP-MESON E04; M08	62+-20 MeV	u_+: 20.6480926301544 19 u_-: 20.2215257441381 8 Type: NONE Confidence: ND
349. PDG2026:observable: M248:M248W0:342590 B_J(5970)0 WIDTH	B_J(5970)0 | Bbar_J(5970) 0 | B_J(5970)+ | B_J(5970)- COMP-MESON E04; M08	81+-12 MeV	u_+: 11.5840507320396 3 u_-: 11.4214634346750 6 Type: NONE Confidence: ND
350. PDG2026:observable: M249:M249W:337869 D_1*(2760)0 WIDTH	D_1^*(2760)0 | Dbar_ 1^*(2760)0 | D_1^*(2760)+ | D_1^*(2760)- COMP-MESON E04; M08	180+-40 MeV	u_+: 38.2753184180092 77 u_-: 38.2753184180092 77 Type: NONE Confidence: ND
351. PDG2026:observable: M250:M250W:347917 T_cbar_sbar_0*(2870)0 WIDTH	T_cbar_sbar_0^*(2870)0 COMP-MESON E04; M08	68+-17 MeV	u_+: 16.2828339322687 69 u_-: 17.2087901693615 Type: NONE Confidence: ND
352. PDG2026:observable: M251:M251W:347923 T_cbar_sbar_1*(2900)0 WIDTH	T_cbar_sbar_1^*(2900)0 COMP-MESON E04; M08	106+-10 MeV	u_+: 10.3961590631619 3 u_-: 10.3961590631619 3 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
353. PDG2026:observable: M252:M252W:337814 D_0*(2300) WIDTH	D_0^*(2300)0 | Dbar_0^*(2300)0 | D_0^*(2300)+ | D_0^*(2300)- COMP-MESON E04; M08	229+-16 MeV	u_+: 16.014886897955751 u_-: 16.014886897955751 Type: NONE Confidence: ND
354. PDG2026:observable: M253:M253W:337822 D_1(2420) WIDTH	D_1(2420)0 | Dbar_1(2420)0 | D_1(2420)+ | D_1(2420)- COMP-MESON E04; M08	31.3+-1.9 MeV	u_+: 1.939893766488169 u_-: 1.931511994340549 Type: NONE Confidence: ND
355. PDG2026:observable: M254:M254W:337837 D_2*(2460) WIDTH	D_2^*(2460)0 | Dbar_2^*(2460)0 | D_2^*(2460)+ | D_2^*(2460)- COMP-MESON E04; M08	47.3+-0.8 MeV	u_+: 0.77431347668341055 u_-: 0.77362957674648969 Type: NONE Confidence: ND
356. PDG2026:observable: M255:M255W:334942 X(1750) WIDTH	X(1750)0 COMP-MESON E04; M08	120+-10 MeV	u_+: 9.2901225622471255 u_-: 9.8222008027276093 Type: NONE Confidence: ND
357. PDG2026:observable: M256:M256W:338521 D_s0(2590)+ WIDTH	D_s0(2590)+ | D_s0(2590)- COMP-MESON E04; M08	89+-20 MeV	u_+: 20 u_-: 20 Type: NONE Confidence: ND
358. PDG2026:observable: M257:M257W:343154 B_sJ(6063)0 WIDTH	B_sJ(6063)0 | Bbar_sJ(6063)0 COMP-MESON E04; M08	26+-6 MeV	u_+: 5.6568542494923806 u_-: 5.6568542494923806 Type: NONE Confidence: ND
359. PDG2026:observable: M258:M258W:343158 B_sJ(6114)0 WIDTH	B_sJ(6114)0 | Bbar_sJ(6114)0 COMP-MESON E04; M08	66+-28 MeV	u_+: 27.658633371878661 u_-: 27.658633371878661 Type: NONE Confidence: ND

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Identifier and observation	Species and status note	Value and unit	Uncertainty and type
360. PDG2026:observable: M259:M259W:347965 T_c cbar sbar 1(4000) WIDTH	T_c_cbar_sbar_1(4000)0 COMP-MESON E04; M08	5 -- 150 MeV Presentation: 5--150	u_+: NA u_-: NA Type: R Confidence: ND
361. PDG2026:observable: M260:M260W:348019 T_c cbar sbar 1(4220)+ WIDTH	T_c_cbar_sbar_1(4220)+ COMP-MESON E04; M08	230+110-90 MeV	u_+: 110.05907504608609 u_-: 89.627004859026727 Type: NONE Confidence: ND
362. PDG2026:observable: M261:M261W:346811 chi_c1(4685) WIDTH	chi_c1(4685) COMP-MESON E04; M08	180+-50 MeV	u_+: 49.868992214526223 u_-: 51.682531037727555 Type: NONE Confidence: ND
363. PDG2026:observable: M262:M262W:346762 X(4630) WIDTH	X(4630) COMP-MESON E04; M08	350+140-100 MeV	u_+: 137.97165629659631 u_-: 102.5281698672277 Type: NONE Confidence: ND
364. PDG2026:observable: M263:M263W:334918 a_0(1710) WIDTH	a_0(1710)0 COMP-MESON E04; M08	107+-15 MeV	u_+: 14.825007178579688 u_-: 14.825007178579688 Type: NONE Confidence: ND
365. PDG2026:observable: M264:M264W:334993 f_0(1810) Breit-Wigner WIDTH	f_0(1810)0 COMP-MESON E04; M08	137+15-17 MeV	u_+: 15.027393581054231 u_-: 16.785983289291 Type: NONE Confidence: ND
366. PDG2026:observable: M265:M265W:347935 T_c_c(3875)+ WIDTH	T_c_c(3875)+ COMP-MESON E04; M08	0.048+0.0020-0.0141 MeV	u_+: 0.002 u_-: 0.014142135623730951 Type: NONE Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
367. PDG2026:observable: M266:M266W:335274 f_0(2470) WIDTH	f_0(2470)0 COMP-MESON E04; M08	75+14-12 MeV	u_+: 14.2126704035519 u_-: 12.041594578792299 Type: NONE Confidence: ND
368. PDG2026:observable: M267:M267W:335029 eta_1(1855) WIDTH	eta_1(1855)0 COMP-MESON E04; M08	188+-19 MeV	u_+: 18.248287590894659 u_-: 19.697715603592211 Type: NONE Confidence: ND
369. PDG2026:observable: M268:M268W:348045 T_c_c_cbar_cbar(6900) 0 WIDTH	T_c_c_cbar_cbar(6900)+ COMP-MESON E04; M08	161+-26 MeV	u_+: 27.000384245226008 u_-: 25.460154585260579 Type: NONE Confidence: ND
370. PDG2026:observable: M269:M269W:347928 T_c_sbar_0*(2900)0 WIDTH	T_c_sbar_0^*(2900)+ COMP-MESON E04; M08	119+-29 MeV	u_+: 29.068883707497271 u_-: 29.068883707497271 Type: NONE Confidence: ND
371. PDG2026:observable: M269:M269W++: 347929 T_c_sbar_0*(2900)++ WIDTH	T_c_sbar_0^*(2900)+ COMP-MESON E04; M08	140+-40 MeV	u_+: 36.235341863986868 u_-: 36.235341863986868 Type: NONE Confidence: ND
372. PDG2026:observable: M270:M270W:335222 omega(2220) WIDTH	omega(2220)0 COMP-MESON E04; M08	105+-34 MeV	u_+: 33.807267697768303 u_-: 33.807267697768303 Type: NONE Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
373. PDG2026:observable: M272:M272W:348041 T_c c cbar cbar(6600) WIDTH	T_c c cbar cbar(6600)+ COMP-MESON E04; M08	440+-70 MeV	u_+: 74.676988436660039 u_-: 68.966189416634236 Type: NONE Confidence: ND
374. PDG2026:observable: M273:M273W:348049 T_c c cbar cbar(7100) WIDTH	T_c c cbar cbar(7100)+ COMP-MESON E04; M08	100+50-40 MeV	u_+: 49.406477308142499 u_-: 38.948684188300888 Type: NONE Confidence: ND
375. PDG2026:observable: Q007:Q007W:333618 t-quark DECAY WIDTH	t | tbar QUARK_EXTENDED E04; M09	1.42+0.19-0.15 GeV	u_+: 0.19051342973542351 u_-: 0.15256041171179069 Type: NONE Confidence: ND
376. PDG2026:observable: S014:S014W:333730 eta WIDTH	eta ELEMENTARY_OR_EXTENDED E04; M05	1.31+-0.05 keV	u_+: 0.045198908079192503 u_-: 0.045198908079192503 Type: NONE Confidence: ND
377. PDG2026:observable: S026:S026W+:351720 Sigma_b+ WIDTH	Sigma_b()- | Sigmabar_b()+ | Sigma_b()0 | Sigmabar_b()0 | Sigma_b()+ | Sigmabar_b()- ELEMENTARY_OR_EXTENDED E04; M05	5.0+-0.5 MeV	u_+: 0.47919801016756203 u_-: 0.47660443013974058 Type: NONE Confidence: ND
378. PDG2026:observable: S026:S026W-:351721 Sigma_b- WIDTH	Sigma_b()- | Sigmabar_b()+ | Sigma_b()0 | Sigmabar_b()0 | Sigma_b()+ | Sigmabar_b()- ELEMENTARY_OR_EXTENDED E04; M05	5.3+-0.5 MeV	u_+: 0.55180022333948409 u_-: 0.54475371590490007 Type: NONE Confidence: ND

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Continuation

Identifier and observation	Species and status note	Value and unit	Uncertainty and type
379. PDG2026:observable: S043:S043W:332468 W WIDTH	W+ | W- ELEMENTARY_OR_ EXTENDED E04; M05	2.14+-0.05 GeV	u_+: 0.05365622500314 9413 u_-: 0.05365622500314 9413 Type: NONE Confidence: ND
380. PDG2026:observable: S044:S044W:332514 Z WIDTH	Z0 ELEMENTARY_OR_ EXTENDED E04; M05	2.4955+-0.0023 GeV	u_+: 0.0023 u_-: 0.0023 Type: NONE Confidence: ND
381. PDG2026:observable: S062:S062W+:351729 Sigma_b*+ WIDTH	Sigma_b^*()- | Sigmabar_ b^*()+ | Sigma_b^*()0 | Sigmabar_b^*()0 | Sigma_ b^*()+ | Sigmabar_b^*()- ELEMENTARY_OR_ EXTENDED E04; M05	9.4+-0.5 MeV	u_+: 0.52801285527312 969 u_-: 0.52651649982212 589 Type: NONE Confidence: ND
382. PDG2026:observable: S062:S062W-:351730 Sigma_b*- WIDTH	Sigma_b^*()- | Sigmabar_ b^*()+ | Sigma_b^*()0 | Sigmabar_b^*()0 | Sigma_ b^*()+ | Sigmabar_b^*()- ELEMENTARY_OR_ EXTENDED E04; M05	10.4+-0.8 MeV	u_+: 0.84589824638254 64 u_-: 0.84310730525944 599 Type: NONE Confidence: ND
383. PDG2026:observable: S074:S074W:338419 D_s*+- WIDTH	D_s^*()+ | D_s^*()- ELEMENTARY_OR_ EXTENDED E04; M05	<1.9 MeV	u_+: NA u_-: NA Type: U Confidence: 90
384. PDG2026:observable: S126:S126W:332734 H DECAY WIDTH	H ELEMENTARY_OR_ EXTENDED E04; M05	3.0+1.5-0.7 MeV	u_+: 1.45297369001768 31 u_-: 0.71501191940991 071 Type: NONE Confidence: ND

Part V

Terminal realizations of the discrete structure of the continuum: ζ, Navier–Stokes, Yang–Mills, Hodge, and Birch–Swinnerton–Dyer

Chapter 95

Joint factorization theorem of the 5 terminal realizations through the discrete structure of the continuum

Part V receives the unique continuum after its joint construction and its realization through Dimensional Mechanics. No terminal realization selects, generates, or retrospectively replaces the consubstantial constructions of the continuum. The expository sequence preserves, in this order, the reading note, the causal splice, the common object, and the method of proof. The joint theorem establishes simultaneous factorization, the non- permutability of the readers, and the preservation of the common enriched state. The realizations that follow retain in full, and with their own domain, their individual operators, proofs, and certificates; they are not substitutes or abridgments of this common factorization.

95.1 Terminal publication through 5 non-permutable readers and common factor of the enriched state

95.1.1 Factorization of the common enriched state into 5 terminal publications

95.1.1.1 Causal dependence of terminal realizations with respect to the fully constructed genealogical continuum

The five problems do not summon five distinct mathematical structures. Nor do they require the construction of five independent paths that, after advancing separately, must be artificially reunited in a common conclusion. That picture belongs to an earlier stage of the exposition. The current point of departure is stronger: before the names Riemann, Navier–Stokes, Yang–Mills, Hodge, or Birch–Swinnerton–Dyer appear, there already exists a single enriched mathematical object whose internal operations simultaneously produce the features required for the five closures.

For this reason, this account does not begin with problems. It begins with the construction of the continuum.

Triadic Modular Holography does not receive the continuum as a previously given background. Nor does it begin with the real line, a classical function space, an algebraic variety, a gauge connection, or an elliptic curve. It begins with a finite arithmetic endowed with orientation, memory, and the capacity for prolongation. APP provides the positional and arithmetic substrate; TRIT converts each operation into an oriented transition; TPK preserves the complete genealogy of those transitions and permits them to be prolonged without erasing the path that produced them.

The decisive step is that the output of an operation ceases to be merely a value. Each emission carries with it its arithmetic sheet, residue, quotient, orientation, carry, memory, boundary, route, multisection, survival, terminal object, and reader. The result is no longer an isolated point, but a state that knows where it comes from, how it can continue, and which relations it must preserve when the scale changes.

APP, TRIT, and TPK do not form three overlapping explanations. They form a single causal sequence:

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{unique enriched state.}$$

APP fixes the common arithmetic support. TRIT introduces the effective orientation of each transition, distinguishing advance, threshold, and future opening. TPK brings together the visible information and the transported information: not only what an operation yields, but also the carry it shifts, the boundary it modifies, the memory it preserves, and the extensions it permits.

The nonadic connection does not return the system to an empty state. After one complete turn, the phase reappears, but the history is not erased. The return $9 \to 1$ is phase return, not state reset. Every turn adds depth, memory, and compatibility. Continuity is therefore obtained neither by repeating nine filters nor by traversing nine independent channels. It is obtained by prolonging a single complete connection whose nine segments are internal charts of the same holonomy.

95.1.1.2 Expository decomposition of the correlative action into 5 intrinsic views of the same state

A common operation acts on every enriched state. Five domains, five selections, or five copies of the state do not exist first. The same emission is read simultaneously under five internal characteristics denoted by the discriminants sp, sol, gau, coh, and det.

Those symbols do not name branches or exhaust the properties of the continuum. They name 5 intrinsic actions of the same object, whose correlation is decisive for the terminal realizations of this Part.

The spectral–polar characteristic, sp, records the oriented publication of the state, its complement, its defect, its terminal memory, and the relation between the two mirror readings of the same form.

The solenoidal characteristic, sol, records how a transition preserves state and, at the same time, distributes dissipation, advection, carry, memory, microstructure, and shell without overcounting them.

The gauge characteristic, gau, records local equivalence, incidence, holonomy, curvature, and compatible passage to the physical quotient.

The cohomological characteristic, coh, records the extensions between levels, the relative obstructions, the compatibility of the lifts, and the survival of a class through every refinement.

The determinantal characteristic, det, records the finite-global incidence, the filtration pages, the local complexes, their determinant lines, and the transport of an arithmetic basis up to an analytic publication.

But none of these properties exists in isolation. All act on the same emissions and preserve the same genealogical thickness. The fiber carrying the spectral characteristic is the same enriched fiber whose orientation participates in the solenoidal balance, whose boundary supports the gauge reading, whose prolongation sustains the cohomological extension, and whose incidence enters the determinant line.

Five objects are not generated and then packaged. A single object is generated that inseparably possesses those five capacities.

A posterior view is therefore not a projection that creates a branch. It is solely a way of displaying, for a specific application, a characteristic that was already operating within the whole. The view does not retrospectively select favorable states or preserve the input as a first coordinate in order to recover it tautologically later. It makes visible a structure produced and preserved throughout the entire genealogy.

95.1.1.3 Causal precedence of the already constructed continuum as the limit of the compatible prolongation

At each depth there exists a finite state that is nevertheless complete at its level. Extensions between levels cannot arbitrarily alter prior information. They must preserve the fibers, extension relations, numbers, orientation, carry, memory, boundary, route, multisection, survival, terminal object, and readers.

Naturality requires that operating and then truncating produce the same result as truncating and then operating. Owing to that compatibility, finite histories do not remain disconnected approximations. They form an inverse system. The continuum appears when the complete family of compatible histories is taken, not when an infinite collection of points is postulated in advance.

The HMT continuum is therefore coherent memory of paths. A point without genealogy would be an ablation: it would preserve the visible value while eliminating precisely what makes it possible to prove how that value survives every refinement.

Only after that compatibility has been closed does the complete discrete structure of the continuum appear. Only then can an Archimedean realization be published. The real line is not the raw material of the construction; it is an output of one of its terminal readers. The same holds for the classical domains of the five problems: they appear when the continuum has already been constructed and a subsequent interface translates one of its intrinsic features into the language of the corresponding target.

The correct causal sequence is, therefore:

$$\begin{aligned}\mathrm{APP} &\longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \text{unique enriched state}\\ &\longrightarrow \mathrm{04b1} \longrightarrow \mathrm{04b2} \longrightarrow \mathrm{04b3}\\ &\longrightarrow \text{correlative intrinsic actions} \longrightarrow \text{terminal and limit} \longrightarrow \mathfrak{C}_{\mathrm{cont}}^{\mathrm{disc}}.\end{aligned}$$

Then, and not before, enter the specific entanglers:

$$\mathfrak{C}_{\mathrm{cont}}^{\mathrm{disc}} \longrightarrow \begin{cases} \text{explicit Weil form,} \\ \text{three-dimensional incompressible evolution,} \\ \text{quantum gauge theory,} \\ \text{publication of Hodge classes,} \\ \text{BSD determinantal comparison.} \end{cases}$$

These five terminal translations do not define the structure they use. The classical problem supplies its data afterward: a test function in Riemann, initial data and a force in Navier–Stokes, a simple compact group in Yang–Mills, a variety and a class in Hodge, or an elliptic curve in BSD. None of those data participates in the prior generation of the continuum.

The causal direction is thereby protected. The target does not select the state that is to solve it. The state exists before the target; the intertwiner merely proves that the internal structure and the classical formulation express the same operation in two languages.

95.1.2 Spectral–polar publication of the continuum via the positive complement of Guinand–Weil

The Riemann hypothesis is usually presented as a statement concerning the location of the nontrivial zeros of the zeta function. From that formulation, the problem appears to call for inspecting a given population of zeros and proving that all of them align on the critical line.

The HMT reading reverses that perspective. It does not begin by asking where the zeros lie. It asks what structure simultaneously generates both sides of the explicit formula and what prevents their difference from acquiring a negative sign.

The spectral–polar characteristic of the continuum provides a double publication of the same state. One orientation collects the positive channel; the other collects its mirror image. Both contain, under a common regulator, the prime contribution, the gamma–scalar contribution, and the polar contribution. They are not three distinct proofs or three sums appended at the end. They are three rows of the same colligation constructed from the starting data.

The essential point is that this colligation is constructed before invoking the positivity that it must demonstrate. Its input, outputs, loss, and terminal state are typed within the same enriched space. The identity linking the two publications is not introduced as a desired equality between classical formulas; it arises from the internal dynamics of the connection.

Between the input publication and the mirror publication, a residual output may appear at finite depth. The nonadic structure compels it to satisfy an antimirror recurrence. At each complete turn, the next residual is transported isometrically but multiplied by the factor

$$q = \frac{5}{9}.$$

If Ω_n represents that defect at depth n, the recurrence has the form

$$\Omega_n = q\, V_n \Omega_{n+1},$$

with V_n isometric and the family coinductively bounded. Upon iterating N times,

$$\|\Omega_n\| \le q^N \sup_{m \ge n} \|\Omega_m\|.$$

As $0 < q < 1$, the right-hand term tends to zero. Therefore, the residual does not vanish because it was omitted or because an identification between the two publications was assumed. It vanishes because the

complete recurrence contracts it at every compatible depth. The specular output then coincides with the constructed negative publication.

Once that link has been discharged, the isometry constructed from the starting data yields the energy identity. At finite depth, the norm of the positive publication decomposes into three parts: the negative publication, the accumulated loss, and the terminal state. The terminal is explicitly preserved. It must be preserved through passage to the limit because it contains the memory of truncation and certifies that telescoping prolongs the finite identity through a proved convergence.

The finite norm decomposition satisfies the exact identity:

$$\|\text{positive publication}\|^2 = \|\text{negative publication}\|^2 + \|\text{losses up to } N\|^2 + q^{2N}\|\text{terminal}\|^2.$$

Each summand has a concrete origin. The negative publication is the already identified spectral output. The losses gather exactly the complement that was not published by that orientation. The terminal remembers what still remains beyond the cut.

Only after the complete identity has been written is $N \to \infty$ imposed. Then, and only then, does the terminal term disappear because $q^{2N} \to 0$. The difference between the two publications is converted into the square of a loss operator:

$$\mathcal{I}_{\mathrm{GW}} = E_R^* E_R.$$

This is the Guinand–Weil form. Its positivity is no longer a conjecture added to the construction:

$$\langle f, \mathcal{I}_{\mathrm{GW}} f \rangle = \|E_R f\|^2 \geq 0.$$

The HMT structure does not obtain this inequality by concealing the negative part, restricting the domain to functions that already satisfy it, or introducing a projector whose existence is equivalent to the conclusion. It constructs the complement and proves that the form is exactly its square.

The common publication is then materialized. A single encoder preserves the prime, gamma–scalar, and polar channels. An internal projector produces one of the two moments; its complement produces the other. The difference between the moments is not an external comparison between similar formulas: it is the norm of the orthogonal component constructed by the colligation.

As the window is enlarged, individual columns are not forced to satisfy false isometries between distinct truncations. What is preserved cofinally is the form, the reachable complement, and naturality of its publication. The increments entering when the cut is enlarged appear in both orientations with the same neutral contribution and do not alter the governing difference. This yields a compatible family of positive forms on a separating class.

The final transition is the reversible Weil criterion. Classical theory states that positivity of the explicit form, on the appropriate class of test functions, is equivalent to the location of all nontrivial zeros on the critical line. HMT supplies precisely that positivity, but supplies it as a consequence of an operator identity constructed within the continuum.

Por tanto,

$$\operatorname{Re}\rho = \frac{1}{2}$$

for every nontrivial zero ρ of the zeta function.

The critical line appears here in two concordant ways. Geometrically, it is the fixed locus of the involution that exchanges the two spectral orientations. Operatorially, it is the only position compatible with the positivity of the publication complement. Symmetry explains why the central line is distinguished; the quadratic identity proves why the zeros cannot leave that line.

The resolution does not consist in observing that the functional equation has a symmetry. It consists in constructing the object whose defect measures the breaking of that symmetry, proving that the defect vanishes coinductively, and converting the remaining difference into a norm. It is the combination of genealogy, terminal object, contraction, common publication, and positivity that closes the problem.

Thus the first of the five terminal resolutions is established. Not as an independent branch, but as the classical translation of the spectral–polar characteristic that the unique continuum possessed from its construction.

95.2 Dynamics and curvature of the ancestral lineage continuum preceding classical realizations

95.2.1 Joint Factorization of the Terminal Realizations from the Same State of the Continuum

The first resolution showed how the spectral–polar characteristic of the continuum is published as Weil positivity. The next two resolutions interrogate the same state through two other consubstantial aspects: its capacity to transport nonlinear dynamics without losing energy, memory, or regularity, and its capacity to support a local gauge theory whose vacuum is separated by a positive spectral gap.

Navier–Stokes and Yang–Mills do not receive two new structures. They receive two distinct intertwiners from the same already constructed continuum. In the first case, the intertwiner publishes the state as an incompressible velocity field with viscosity, advection, and pressure. In the second, it publishes the state as a refinable gauge network with holonomy, curvature, Euclidean measure, and a physical Hamiltonian.

The difference between the two problems does not break the unity of the construction. In Navier–Stokes, the critical point is to prove that no nonlinear transfer disappears outside the balance and that the regularity obtained is uniform as the cutoffs are removed. In Yang–Mills, the critical point is to prove that the finite gap belongs to the same gauge theory that reaches the continuum, survives the physical quotient, and acts in the curvature sector rather than in an auxiliary factor.

95.2.2 Solenoidal publication of the continuum: coercive bound, compactness and global regularity

The three-dimensional Navier–Stokes problem does not arise from the mere presence of a nonlinear term. The difficulty arises because the classical publication of the velocity field tends to obscure where transfer between scales is stored. Viscosity is visible. Kinetic energy is visible. Advection also appears in the equation. Yet the complete accounting of the interaction among depth, memory, carry, microstructure, and shell is compressed into a single visible function.

That compression is precisely what the solenoidal character of the continuum avoids.

The enriched state does not retain only the field u. It simultaneously retains the Galerkin field, the time orientation, the positive and signed memories, the carry between cuts, the microscopic component, the shell of new frequencies, the route, the evolution archive, and the terminal. The classical equation is a publication of that state; it is not its complete definition.

Given initial data u_0, force f, viscosity $\nu > 0$, and regularity $s > 5/2$, the Galerkin projection produces

$$\partial_t u_M + \nu A_M u_M + B_M(u_M, u_M) = f_M.$$

But M does not select a favourable solution. It merely fixes a finite resolution in which all operations can be written exactly. The objective is to construct an estimate that does not depend on M or on the nonadic depth J. Only a bound with that uniformity can pass to the continuum.

95.2.2.1 Decomposition of visible memory and genealogical memory in the evolutionary equation

The nonlinear interaction has a sign. At certain times it delivers energy to one scale; at others it withdraws it. If only its signed value is preserved, the total magnitude transferred may be lost. If only its absolute value is preserved, orientation is lost.

The structure preserves both.

One memory D_M publishes the sign of the advective transfer. Another memory H_M preserves its positive magnitude. Their separation makes it possible to record simultaneously

$$\dot{D}_M = \mathscr{B}_{s,M}, \qquad \dot{H}_M = \frac{1+r}{r-1}|\mathscr{B}_{s,M}|,$$

where $r = \pi_{\mathrm{HMT}}/\varphi^2_{\mathrm{HMT}} > 1$.

This adds no fictitious energy. It unfolds two distinct aspects of a single transfer: orientation and cost. Merging them would destroy information; adding them without orthogonality would duplicate energy. The construction will make explicit that neither of those two deformations occurs.

95.2.2.2 Independence of the initial data with respect to the subsequent genealogical history

The most serious objection to any operatorial closure of Navier–Stokes would be that the uniform bound had been introduced retrospectively: first, a regular solution would be chosen; its history would then be encoded; and finally it would be announced that the code controls that same solution.

The construction prevents that circularity by fixing the source space before raising the evolution:

$$\mathcal{D}_T = H^s_\sigma(\mathbb{T}^3) \times L^2(0,T; H^{s-1}_\sigma(\mathbb{T}^3)), \qquad d = (u_0, f).$$

The root $J^{\mathrm{sol}}_T d$ depends exclusively on u_0 and f. It does not receive as input u_M, the future loss, the terminal object, or the norm that must be bounded.

Force is likewise not introduced as an artificial additive port. The Carleman equation shows that it acts in a mixed manner: every occurrence of f is combined with a preceding degree of the state. The lift therefore uses a mixed state–force creation. The chronological signature of f records all its ordered times, whereas the PC–Fock tower records all nonlinear degrees of the state.

The composition materially reproduces the Dyson series of the nonautonomous generator. Applying the normalized degree-one reader recovers exactly $u_M(t)$. What is recovered is neither a symbolic approximation nor a variable resembling the solution: it is the solution of the Galerkin ODE, because both satisfy the same finite equation and the same initial datum.

The causal construction from the initial data therefore proceeds in the order

$$(u_0, f) \longrightarrow J^{\mathrm{sol}}_T(u_0, f) \longrightarrow \text{state–force tower} \longrightarrow \text{Galerkin history}.$$

The genealogical history is constructed as a later output of the data of departure and is excluded from its initial definition.

95.2.2.3 Scattering identity and cross-power conservation

Force and dissipation are organized by means of two waves:

$$a_T(f) = \frac{1}{2\sqrt{\nu}} A^{-1/2} \Lambda^s P_\sigma f, \qquad z_M(u) = \sqrt{\nu} A_M^{1/2} \Lambda^s u,$$

and the residual wave

$$b_M = z_M - P_M a_T(f).$$

The applied power is not replaced by the norm of the force. It is preserved exactly as a cross term:

$$\mathscr{F}_{s,M} = 2\,\mathrm{Re}\langle z_M, P_M a_T(f)\rangle.$$

Upon completing the square, the exact balance is obtained.

$$\dot{G}_{M,j,T} + \|b_M\|^2 + |\mathscr{B}_{s,M}| = \|P_M a_T(f)\|^2.$$

This is one of the points that make the closure work. If $b = z - a$ were replaced by z, the cross term representing the work of the force would disappear and the identity would be false. The structure does not control the nonlinearity by discarding it: it turns it into an oriented positive channel and also preserves the exact interference between force and dissipation.

95.2.2.4 Factorization of 6 *losses through a unique operator without spurious multiplicity*

The complete history distinguishes six outputs:

- dissipation;
- oriented advection;
- carry between cuts;
- microscopic memory;
- shell of new frequencies;
- non-inherited memory.

They are not six energies added to a previous balance. They are six orthogonal rows of a single terminal loss operator. Parseval's identity establishes

$$\|L^{\mathrm{term}}_{M,j,T}x\|^2 = \sum_{\mathfrak{a}} \|L^{\mathfrak{a}}_{M,j,T}x\|^2.$$

The orthogonality is not merely nominal. The depths occupy disjoint temporal intervals; carry and shell inhabit distinct spectral ranges; the microstructure belongs to the complement Q_{micro}; the new memory is separated from the inherited memory; and the two sheets of the advection cannot be positive simultaneously at the same point.

For this reason, the accounting does not duplicate the interaction. The sum of the six norms is exactly the norm of the common operator. If one of them were not a projection of that operator, Parseval would fail before the bound was obtained. The formalism itself thus contains its falsifier.

95.2.2.5 Persistence of the terminal solenoidal state under passage to the limit

Every truncation preserves something that has not yet been published. That remainder is the terminal object. It contains the visible energy at the end of the interval and the oriented memory that remains between the observed time and the horizon T.

The terminal column is constructed first. Its Gram matrix is then formed. A desired metric is not defined and an operator subsequently invented to realize it. The column materially contains the terminal output and all transported losses:

$$\mathbf{G}^{\mathrm{term}}_{M,j;J,T} = \begin{bmatrix} \text{transported terminal} \\ \mathrm{loss}_{J-1} \\ \vdots \\ \mathrm{loss}_{j} \end{bmatrix}.$$

His block recursion produces the Stein identity.

$$U_{M,j,T} = \Gamma^{*}_{M,j,T} U_{M,j+1,T} \Gamma_{M,j,T} + (L^{\mathrm{term}}_{M,j,T})^{*} L^{\mathrm{term}}_{M,j,T}.$$

This identity does not retroactively define the norm of the state. It is obtained by multiplying an already constructed column by its adjoint. For the same reason, global telescopy is an identity among existing operators, not a stability hypothesis.

Future memory is not introduced into the initial root either. It resides in the terminal output. Upon taking the integral estimate over $\tau = T$, its temporal support is emptied exactly. It is not erased for convenience.

95.2.2.6 Uniform a priori bound and Aubin–Lions compactness

The root, the level encoder, the column, and the common isometry satisfy the factorization derived from the data of departure

$$\mathcal{G}^{\mathrm{sol}}_{M,J,T}\Lambda_{M,T}J^{\mathrm{sol}}_{T}d = I_{M,J}J^{\mathrm{sol}}_{T}d, \qquad I^{*}_{M,J}I_{M,J} = I.$$

The left-hand side is the complete history at the cutoff (M, J). The right-hand side depends on the unique root constructed from the data. Since $I_{M,J}$ is isometric, the norm of the history is dominated by the source norm with a constant independent of M and J.

This is the step that resolves the difficulty. Uniformity does not follow from a finite-dimensional estimate whose constant grows with the cutoff. Nor does it follow from defining the history space through the very loss one seeks to control. The root exists before the cutoff, and all cutoffs are isometric realizations of it.

The terminal column translates that structural bound into the physical quantities:

$$\sup_{M}\sup_{0\le t\le T}\|u_M(t)\|^2_{H^s} + \nu\sup_{M}\int_0^T \|u_M(t)\|^2_{H^{s+1}}\,dt + \sup_{M}\int_0^T |\mathscr{B}_{s,M}(u_M(t))|\,dt + \mathrm{carry} + \mathrm{micro} + \mathrm{shell} + \mathrm{memory} \le C_T(u_0, f).$$

The constant is finite for every $T < \infty$ and does not depend on M or J. The inequality controls not only the visible field. It simultaneously controls the transported work and the sectors that a classical publication usually forgets.

95.2.2.7 Sign Signature Obstruction in Kalman–Yakubovich–Popov Factorization

Making the global closure depend on a KYP factorization with an incorrect sign introduces a positive square and purports to recover it afterward with a negative sign. That equality is algebraically false.

The proof excludes that identity and directly constructs the necessary uniform estimate.

The retained KYP chart is a correct finite identity under its own conditions. It serves as an auxiliary control for each cutoff, but does not provide uniformity when M and J are removed. The global proof does not use a bound on Q^{-1}, does not take the limit within KYP, and does not depend on that sign.

Therefore, the obstruction from the incorrectly signed KYP block does not affect the proof. Uniformity is established by the direct construction from the data with its terminal column and its Stein identity.

95.2.2.8 Convergence from the uniform bound to the global classical solution

The previous bound provides

$$u_M \quad \text{bounded in} \quad L^{\infty}(0,T;H^s)\cap L^2(0,T;H^{s+1}).$$

Since $s > 5/2$, the Sobolev product controls the nonlinearity:

$$B : H^s_\sigma \times H^s_\sigma \longrightarrow H^{s-1}_\sigma.$$

The Galerkin equation further provides a uniform bound for $\partial_t u_M$ in $L^2(0,T;H^{s-1})$. The inclusions

$$H^{s+1} \Subset H^s \hookrightarrow H^{s-1}$$

allow the application of Aubin–Lions. Strong convergence in L^2H^s is obtained, sufficient to transport the bilinear term:

$$B_M(u_M, u_M) \longrightarrow B(u,u) \quad \text{in} \quad L^1(0,T;H^{s-1}).$$

The limit satisfies the incompressible equation. Pressure is not introduced as another hidden datum; it is reconstructed afterwards through

$$-\Delta p = \partial_i \partial_j (u_i u_j), \qquad \int_{\mathbb{T}^3} p = 0.$$

Uniqueness is obtained from the difference of two solutions. An estimate in H^{s-1}, followed by Grönwall's inequality, forces the difference to vanish. This eliminates dependence on the subsequence extracted by compactness.

The construction holds for every finite horizon. The solutions obtained in $[0,n]$ and $[0,n+1]$ agree on their overlap by uniqueness. They therefore glue into a single solution defined over $[0,\infty)$.

Smoothness is obtained throughout the scale $s+n$ using the same structural readers; uniqueness identifies all limits and produces a global smooth solution for every smooth periodic datum. The required bound is constructed on each finite horizon and at every Sobolev index from the complete solenoidal datum, as proved in the source development theorem 98.3.

Finally, the zero-mean condition is a normalization of the solenoidal sector. The time-dependent Galilean transformation separates the mean and bijectively transports every smooth periodic datum to the zero-mean sector, preserving divergence, smoothness and Sobolev norms.

95.2.2.9 Global existence, regularity and uniqueness for Navier–Stokes

The solenoidal characteristic of the continuum, expressed through its specific intertwiner, produces a unique global smooth solution of the three-dimensional periodic Navier–Stokes equations for every smooth periodic datum. The Leray–Hopf construction is contained as an energy stage within that strong conclusion.

The defense of this conclusion does not rest on saying that a structure capable of solving the problem exists. It rests on five material facts:

the root uses only (u_0, f); the nonlinearity is reproduced by mixed creation; the six losses are orthogonal and are not added twice; the terminal remains until telescopy; and the uniform bound precedes the classical compactness argument.

Aubin–Lions, pressure, Grönwall, and gluing are not the source of regularity. They are the final classical recognition of a regularity that the structure has already prevented from being lost.

95.2.3 Publication-grade continuum: positivity by reflection and curvature sector gap

The Yang–Mills problem requires two inseparable things: constructing a nontrivial quantum theory over $\mathbb{R}^4$ for every compact, connected, and simple group of the considered domain, and proving that its Hamiltonian has a positive mass gap.

It is not enough to exhibit a finite operator with positive spectrum. Nor is it enough to write down a Wilson action, a product measure, or an observable with a nonzero eigenvalue. The gap must belong to the same gauge theory that possesses locality, reflection positivity, Euclidean covariance, Wightman reconstruction, and physical curvature.

The gauge characteristic of the continuum already preserves the refinable network, the measure, the four reflections, the holonomy, color, incidence, the fusion carry, the route, memory, the terminal object, and the curvature reader. The group G and the coupling g enter afterward through the specific intertwiner. They do not retrospectively select the genealogy.

95.2.3.1 Non-equivalence between the simple fiber and the product dynamics

The first defensive distinction separates two generators that an old notation could confuse.

The semigroup of a alone fiber has only the levels 0 and $3\kappa_{\mathrm{av}}$. The generator product complete integra, in change, all the coordinates independent of the state:

$$D_m^{\mathrm{prod}} = 3\kappa_{\mathrm{av}} \sum_{\iota \in \mathcal{I}_m} (I - E_{m,\iota}),$$

where the expectations $E_{m,\iota}$ are orthogonal and commuting.

Its spectrum contains

$$0, \quad 3\kappa_{\mathrm{av}}, \quad 6\kappa_{\mathrm{av}}, \quad 9\kappa_{\mathrm{av}}, \ldots$$

according to the number of nonconstant coordinates. Replacing this generator with that of a simple fiber would erase the Hoeffding multiplicities and would be immediately falsifiable: a vector depending on two factors must have energy $6\kappa_{\mathrm{av}}$, not $3\kappa_{\mathrm{av}}$.

The gap will be proved for the product generator and will then be transported to the physical sector. It is not obtained by confusing two different dynamics.

95.2.3.2 Absence of hidden coordinates and exclusion of spurious null modes

The complete state is factorized via an exhaustive index. It appears only once:

- frames;
- links;
- incidents;
- marks;
- refinement bridges;
- new innovations at each level.

The infinite subtails are recorded by means of their scalar cylinders. No genealogical coordinate remains outside the family of expectations.

The Hoeffding decomposition writes the space as an orthogonal sum of sectors $\mathcal{H}_{m,S}$. On each one,

$$D_m^{\mathrm{prod}} u_S = 3\kappa_{\mathrm{av}} |S| u_S.$$

The sector $S = \varnothing$ is the intersection of the ranges of all expectations and consists exactly of the constants:

$$\bigcap_{\iota \in \mathcal{I}_m} \operatorname{Ran} E_{m,\iota} = \mathbb{C}\Omega_m.$$

Therefore, the vacuum is simple, and every vector orthogonal to the vacuum has energy at least $3\kappa_{\mathrm{av}}$.

This excludes the objection that the gap arises from failure to enumerate a coordinate. If a factor were missing, there would be nonconstant functions invisible to all expectations, and additional zero modes would appear. The exhaustiveness of the index prevents exactly that failure.

95.2.3.3 Observable realization of the spectral gap

A lower bound does not suffice to fix the exact value of the gap. A witness that attains the threshold is required.

In each incidence collar, the variable

$$W_c(q) = \mathbf{1}_{\{q_1+\cdots+q_6=0\}} - \frac{1}{3}$$

has zero mean and satisfies

$$D_m^{\mathrm{prod}} W_c = 3\kappa_{\mathrm{av}} W_c .$$

It must still be proved, however, that W_c belongs to the physical gauge sector.

The incidence action is written through the matrix

$$(B^{+}x)_{ij} = x_i + x_j .$$

Each component of $x \in \mathbb{F}_3^4$ appears three times when the six incidences are summed. In $\mathbb{F}_3$,

$$\sum_{i<j} (B^{+}x)_{ij} = 3 \sum_{i} x_i = 0.$$

Therefore, the condition defining W_c is gauge invariant. Averaging over the full gauge group preserves W_c; it does not remove it from the physical space. Thus, W_c is a physical eigenvector, and the value $3\kappa_{\mathrm{av}}$ is not merely a bound: it is the first spectral level attained.

95.2.3.4 Construction of the measure prior to the estimation of the gap

A coarse edge is refined into nine ordered increments. The fine measure is not contrived by copying the coarse measure nine times. It is constructed by means of the heat bridge conditioned by the product of the nine increments. The times of the subsegments may differ and are subject only to the requirement that their sum equal the coarse time.

Convolution with the heat kernel ensures that the bridge is a probability. A triangular chart then separates the coarse variable from the new innovations. Projection onto the preceding level multiplies the nine increments and forgets only the innovations.

In this way,

$$(\widehat{\pi}_{m+1,m})_* \widehat{\mu}_{m+1} = \widehat{\mu}_m .$$

Edges shared by several faces remain the same variable. They are not replaced with independent copies. Carry and memory prevent recounting at the fine level the face that was already present at the coarse level.

Over that projective family one first constructs the exact reader of F^2, its positive root, the local action, and the coupling-dependent Gibbs density:

$$d\widehat{\mu}_{m,g}^{\mathrm{YM}} = Z_{m,g}^{-1} e^{-S_{m,g}} \, d\widehat{\mu}_m^{0}.$$

The Schwinger–Dyson and Ward equations are calculated for that measure. The gap is not used to choose g, to define $S_{m,g}$, or to reweight the law. It appears afterward, when studying the reversible generator associated with the same measure.

This is the defense against action–law–gap circularity: first the action and its measure are constructed; afterwards its spectrum is proved.

95.2.3.5 *Descent of the generator to the quotient of caliber*

The complete space includes links, frames, incidences, color, orientation, route, memory, and terminal. The marginal that preserves only the links is a subsequent ablation. It cannot be used to deny an observable that resides in the incidence fiber.

The physical projection is obtained by averaging the action of the full gauge group. Since the generator integrates coordinates equivariantly,

$$[\mathbb{P}_m^{\mathrm{phys}}, D_m^{\mathrm{prod}}] = 0.$$

The semigroup therefore descends to the physical quotient. Its kernel is first defined on the complete state and only afterward on the gauge classes. One does not take an operator restricted to a fiber and declare it to be the kernel over the whole space.

The simple vacuum and the eigenvector W_c survive compression. The finite gap already belongs to the gauge-invariant space.

95.2.3.6 *Uniqueness of the process reconstructed from* 4 *positivities by reflection*

The Euclidean theory must satisfy reflection positivity in the four directions. Four independent measures are not constructed. A single face measure, obtained from the same reversible kernel, factorizes each Osterwalder–Schrader form:

$$\int \overline{F(\Theta_\nu y)} F(y)\, d\Omega = \|\mathcal{B}_\nu F\|^2 \geq 0, \qquad \nu = 1, \dots, 4.$$

Positivity is not an added hypothesis. It is a squared norm produced by Markov and reversibility.

The marginals of opposite faces recover the same semigroup. Therefore, the four OS reconstructions have a single dynamical origin. The connectors between half-lattices will subsequently prove that they reconstruct the same space, the same vacuum, and the same Hamiltonian.

95.2.3.7 *Persistence of the Spectral Gap under the Continuous Limit*

The refinement maps intertwine the Dirichlet forms, the semigroups, the vacuum, and the witness W_c. Passage to the continuum is not based solely on observing that the gap has the same value at every level. In isolation, that equality would not prevent new states of arbitrarily small energy from appearing in the limit.

The construction controls cross-products through block intertwiners. The coarse coordinates are distributed over their descendants with exact normalization. The Gram identities prove that the smoothed publications form Cauchy families. The form, the vacuum, and the eigenvector thereby attain the limit within the same inductive system.

Se obtiene

$$D_{\infty,g}^{\mathrm{YM}} \succeq 3\kappa_{\mathrm{av}}(I - P_{\Omega_\infty}),$$

and the limit witness satisfies

$$D_{\infty,g}^{\mathrm{YM}} W_c = 3\kappa_{\mathrm{av}} W_c.$$

Por tanto,

$$\Delta_{\mathrm{YM}}^{\mathrm{HMT}} = 3\kappa_{\mathrm{av}} > 0.$$

The inequality and the witness converge together. The gap is neither deduced from mere lower semicontinuity nor extrapolated from an isolated lattice.

95.2.3.8 Spectral witness membership in the curvature sector

An objection would still remain if W_c were only a combinatorial excitation of an internal factor, decoupled from Yang–Mills curvature.

Incidence holonomy eliminates that possibility. The sum of the six incidences determines a local holonomy of order three; W_c is a class function of that holonomy. It therefore belongs to the local gauge-invariant algebra generated by the loops.

The reader F^2 is polarized to produce a space of local curvatures. The refinement intertwiners transport those curvatures while preserving the $\mathfrak{g}$ product. The cellular sums converge to a gauge-covariant distribution

$$\mathbf{F}_{\mu\nu} \in \mathcal{S}'(\mathbb{R}^4;\mathfrak{g})\widehat{\otimes}L^2(\widehat{X}_\infty,\widehat{\mu}^{\mathrm{YM}}_{\infty,g}).$$

The cyclic sector generated by $\mathbf{F}$ is reducing for the Hamiltonian and contains the witness W_c. Consequently, the exact gap belongs to the physical curvature sector. It is not the spectrum of an auxiliary noise.

95.2.3.9 Euclidean locality $E(4)$ and continuous Osterwalder–Schrader reconstruction

The generator decomposes into local block circuits with uniformly bounded support. A nonadic coloring separates the necklaces that may be updated simultaneously. The product of the circuits preserves the Bianchi identity at every step, not only upon completion of a sweep.

Euclidean translations and rotations act by relabelling the lattice, frames, incidences, and holonomies. Strong continuity is not proved only for an abstract first chaos. It is first controlled on local observables, then on holonomies, plaquettes, and clover terms; orbit smoothing extends the action to a dense core.

The cofinal system produces a local net $\mathfrak{A}_{\mathrm{YM}}(O)$, a vacuum $\Omega_{\infty,g}$, a strong representation of $E(4)$, and Schwinger functionals. The Markov stacks and OS transfers are identified with the same continuous semigroup. Reflection positivity, regularity, covariance, and exponential clustering permit Osterwalder–Schrader reconstruction.

The continuation produces local Wightman fields and a Poincaré representation with spectrum in the future cone. The four Euclidean reconstructions coinduce the same curvature Hamiltonian.

The gap also controls the cluster:

$$|S^{\mathrm{YM}}(A\,\tau_r B)| \le e^{-3\kappa_{\mathrm{av}} r}\,\|A\Omega_\infty\|\,\|B\Omega_\infty\|.$$

Thus, the positive spectral value has a continuous and local physical consequence; it does not remain confined to the finite calculation.

95.2.3.10 Reconstruction of the classical action as the output of the Euclidean system

The governing finite density is the exact reader of F^2. The clover does not replace it. It functions as a smooth chart for recognizing its limit.

For a smooth connection, the Magnus expansion of the holonomy of a plaquette starts with

$$\log \mathrm{Hol}_{p_{\mu\nu}} = a_m^2 F_{\mu\nu} + \text{odd term} + O(a_m^4).$$

The average over the four clover orientations cancels the odd term. The Riemann sum gives

$$S_{m,g}(A) \longrightarrow \frac{1}{4g^2}\int_{\mathbb{R}^4} \|F_A(x)\|^2\, d^4x.$$

Therefore, the constructed law is recognized as Yang–Mills because its finite action, holonomies, and curvature converge to the correct classical functional. A "Yang–Mills theory" is not defined as any theory possessing the desired gap.

The same measure satisfies Schwinger–Dyson and Ward. The relation between small loops and curvature, the Bianchi identity, and gauge covariance are preserved in the limit. Only after this entire identification is the spectral equivalence that transports the gap applied.

95.2.3.11 Dimensional Interpretation of the Mass Parameter Obtained by the Terminal Entangler

The energy gap is dimensionally published as

$$m_{\mathrm{gap}}^{\mathrm{HMT}} = \frac{\hbar_{\mathrm{int}}}{c_{\mathrm{int}}} 3\kappa_{\mathrm{av}} > 0.$$

This result does not assign a Proca mass to an elementary gluon. Gauge invariance remains intact. The mass is the minimum cost of exciting the collective gauge-invariant curvature sector above the vacuum.

This distinction does not weaken the result: it is exactly the formulation required by the Yang–Mills problem. The gap belongs to the physical spectrum of gauge-invariant states, not to an elementary mass term added to the Lagrangian.

95.2.3.12 Existence of a Positive Spectral Gap and Massive Reconstruction of the Yang–Mills Sector

For every compact, connected, simple Lie group, the gauge characteristic of the continuum, followed by its specific intertwiner, constructs a non-trivial quantum Yang–Mills theory on $\mathbb{R}^4$.

The theory possesses:

- a projective family of coupling-dependent local measures;
- an exact action recognized in the limit as $\frac{1}{4g^2}\int \|F\|^2$;
- a physical gauge quotient;
- reflection positivity in four directions;
- a local net and a strong action of $E(4)$;
- Schwinger functions and Wightman continuation;
- simple vacuum;
- Schwinger–Dyson, Ward, and Bianchi equations;
- a cyclic curvature sector;
- the exact gap

$$\Delta_{\mathrm{YM}}^{\mathrm{HMT}} = 3\kappa_{\mathrm{av}} > 0.$$

The conclusion withstands the foreseeable objections because it does not identify the generator of a fiber with the product; it leaves no coordinates outside the index; it does not confuse the marginal of links with the physical quotient; it does not extrapolate the gap from a lattice; it does not place the witness outside the curvature sector; it does not presuppose Euclidean continuity; and it does not use the gap to construct the action or the measure.

Navier–Stokes proves that the continuum can transport a nonlinearity without losing the memory needed to prevent a singularity. Yang–Mills proves that the same continuum can transport gauge freedom without losing the curvature needed to separate the vacuum from the first physical excitation.

95.3 Common Factorization, Extensional Consistency, and Limits of Terminal Reconstruction

The five constructions `sp`, `sol`, `gau`, `coh`, and `det` are not branches added to the continuum to solve five known problems. They are simultaneous features of the same enriched state. Hodge and Birch–Swinnerton–Dyer are the two final publications of this architecture: one converts a compatible cohomological history into an algebraic class; the other proves that the analytic and arithmetic publications of the same determinant line coincide.

95.3.1 Cohomological publication of the continuum: compatible lifting and effective construction of the algebraic preimage

The Hodge problem asks whether every rational class of type (p, p),

$$\alpha \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X),$$

for a smooth complex projective variety X, comes from an algebraic class

$$\beta_\alpha \in \mathrm{CH}^p(X)_{\mathbb{Q}}$$

tal que

$$\mathrm{cl}_{X,p}(\beta_\alpha) = \alpha.$$

The difficulty does not consist in writing this last equality. It consists in constructing β_α without having previously introduced, under another name, the existence of the preimage one seeks to prove.

The cohomological characteristic `coh` provides the apparatus preceding the entry of X, p, and α: surviving histories, refinements, relations, orientation, carry, memory, boundary, route, multisection, terminal object, and reader. The classical datum neither creates that apparatus nor retrospectively selects a subdomain. It asks only what that apparatus publishes when it receives the observations of α.

At each depth N two modules appear:

$$\mathcal{S}_N(X, p),$$

which contains the enriched cohomological states, and

$$\mathscr{H}_N^p(X),$$

which contains the cohomological observations visible at that depth. The finite reader

$$e_N : \mathcal{S}_N(X, p) \longrightarrow \mathscr{H}_N^p(X)$$

is coordinated with the restriction maps:

$$\rho_{N+1,N} : \mathcal{S}_{N+1} \longrightarrow \mathcal{S}_N, \qquad q_{N+1,N} : \mathscr{H}_{N+1}^p \longrightarrow \mathscr{H}_N^p,$$

by means of the square

$$e_N \rho_{N+1,N} = q_{N+1,N} e_{N+1}.$$

This prevents a remark admitted at one depth from being incompatible with its previous shadow.

For the class α, the finite fiber to be populated is

$$\mathscr{F}_N(\alpha) = \{s \in \mathcal{S}_N(X, p) : e_N(s) = q_N \alpha\}.$$

Defining it does not prove that it is nonempty. Nor does it prove that an element of $\mathscr{F}_N(\alpha)$ can be prolonged to $\mathscr{F}_{N+1}(\alpha)$. The resolution consists precisely in constructing both things.

95.3.1.1 Construction of the relative correction prior to the cohomology class

Upon refining from N to $N+1$ two relative kernels appear:

$$\mathscr{K}^{\mathrm{S}}_{N+1} = \ker \rho_{N+1,N}, \qquad \mathscr{K}^{\mathrm{H}}_{N+1} = \ker q_{N+1,N}.$$

The first contains the new corrections of the state that disappear when returning to the preceding level. The second contains the new observations that were not visible at the preceding level either.

The new charts, the Mayer–Vietoris crossings, the orientations, the carry, the memory, the boundary, the route, and the terminal coordinates produce a finite presentation. Two maps are constructed on its generators:

$$a_{N+1} : \mathscr{A}^{\mathrm{rel}}_{N+1} \longrightarrow \mathscr{K}^{\mathrm{S}}_{N+1},$$
$$b_{N+1} : \mathscr{A}^{\mathrm{rel}}_{N+1} \longrightarrow \mathscr{K}^{\mathrm{H}}_{N+1},$$

satisfying

$$e^{\mathrm{rel}}_{N+1} a_{N+1} = b_{N+1}.$$

The map a_{N+1} produces a correction of the state; b_{N+1} publishes exactly its cohomological correction.

The relative observations are ordered by depth, support, sheet, route, carry, memory, and terminal object. With this order, the matrix of b_{N+1} is unitriangular:

$$b_{N+1}(c_\mu) = \bar{c}_\mu + \sum_{\nu<\mu} B_{\nu\mu}\bar{c}_\nu.$$

The unit diagonal is not chosen to represent a particular class. It comes from the principal incidence of each generator; the lower terms are imposed by the relations, orientation, and carry. The table is constructed over all relative observations before α is introduced.

Triangular elimination explicitly produces a right inverse:

$$\sigma^{\mathrm{rel}}_{N+1}(\bar{c}_{\mu_0}) = c_{\mu_0},$$
$$\sigma^{\mathrm{rel}}_{N+1}(\bar{c}_\mu) = c_\mu - \sum_{\nu<\mu} B_{\nu\mu}\sigma^{\mathrm{rel}}_{N+1}(\bar{c}_\nu),$$

and, by induction,

$$b_{N+1}\sigma^{\mathrm{rel}}_{N+1} = I.$$

Por tanto,

$$s^{\mathrm{rel}}_{N+1} = a_{N+1}\sigma^{\mathrm{rel}}_{N+1}$$

satisfies

$$e^{\mathrm{rel}}_{N+1} s^{\mathrm{rel}}_{N+1} = I.$$

This is the firewall against circularity. The relative section is not assumed, is not deduced from the desired class already being algebraic, and is not constructed specifically for α. It exists universally for every relative remark.

Nor is the presentation required to be injective. The kernel may contain redundant cellular expressions. The proof requires surjectivity onto the relative observations, not uniqueness of expression. Vanishing of the cokernel is a consequence of the right inverse; it is not a governing condition introduced before the proof.

95.3.1.2 Universal Operator for the Extension of Compatible Cohomology Class Extensions

The structure possesses a unitary degeneracy.

$$\iota_{N+1,N} : \mathcal{S}_N \longrightarrow \mathcal{S}_{N+1}, \qquad \rho_{N+1,N}\iota_{N+1,N} = I.$$

Let $s \in \mathcal{S}_N$, and let $h \in \mathscr{H}^p_{N+1}$ be a compatible remark:

$$e_N(s) = q_{N+1,N}(h).$$

The defect of the trivial prolongation is

$$\delta_{N+1}(s,h) = h - e_{N+1}\iota_{N+1,N}(s).$$

The compatibility ensures that

$$\delta_{N+1}(s,h) \in \mathscr{K}^{\mathrm{H}}_{N+1}.$$

It is then defined

$$\mathrm{Ext}_{N+1,N}(s,h) = \iota_{N+1,N}(s) + s^{\mathrm{rel}}_{N+1}\big(\delta_{N+1}(s,h)\big).$$

By construction:

$$\rho_{N+1,N}\,\mathrm{Ext}_{N+1,N}(s,h) = s,$$
$$e_{N+1}\,\mathrm{Ext}_{N+1,N}(s,h) = h.$$

The first identity preserves the previous history. The second realizes exactly the new remark. The naturality of the relative table guarantees that orientation, carry, memory, route, boundary, and terminal are not lost under refinement.

The operator works for every compatible pair (s,h). It contains neither a chosen algebraic class nor a chosen cycle. This is the precise reason why its subsequent application to a Hodge class does not presuppose Hodge.

95.3.1.3 Level-by-level construction of the global history in the inverse system

At initial depth there exists a section based

$$s_0^{\mathrm{base}} : \mathscr{H}^p_0 \longrightarrow \mathcal{S}_0.$$

Their images are seed states, not algebraic cycles chosen to represent α. Only afterward is fixed

$$s_0 = s_0^{\mathrm{base}}(q_0\alpha)$$

and is defined recursively

$$s_{N+1} = \mathrm{Ext}_{N+1,N}\big(s_N, q_{N+1}\alpha\big).$$

The universal operator identities are proved by induction:

$$e_N s_N = q_N \alpha, \qquad \rho_{N+1,N} s_{N+1} = s_N.$$

Therefore, each fibre $\mathscr{F}_N(\alpha)$ is nonempty and the family

$$\xi_\alpha = (s_N)_{N \geq 0}$$

is a compatible history:

$$\xi_\alpha \in \varprojlim_N \mathscr{F}_N(\alpha).$$

An abstract inverse limit is not invoked to conceal the absence of sections. The sequence is exhibited recursively. Nor is the vanishing of $\varprojlim^1$ used as a substitute for the construction: surjectivity and the Mittag–Leffler condition appear afterward as consequences of an extension operator that has already been materialized.

95.3.1.4 Emergence of the algebraic class as the limit of the relative construction

The finite observations and the terminal coordinate separate the publications of the limit. Since ξ_α reproduces $q_N \alpha$ at every depth,

$$\mathrm{ev}^{\mathrm{Hdg}}_{\infty,X,p}(\xi_\alpha) = \alpha.$$

Only then does classical publication act:

$$X_\chi(X,p) \xrightarrow{R_{\mathrm{Hdg}}} K_0\big(\mathrm{Perf}^{\geq p}(X)\big)_{\mathbb{Q}} \xrightarrow{\mathrm{ch}^{\mathrm{loc}}_p} \mathrm{CH}^p(X)_{\mathbb{Q}} \xrightarrow{\mathrm{cl}_{X,p}} \mathrm{Hdg}^p(X)_{\mathbb{Q}}.$$

Se define

$$\beta_\alpha = \mathrm{ch}^{\mathrm{loc}}_p\big(R_{\mathrm{Hdg}}(\xi_\alpha)\big).$$

The reader action identity,

$$\mathrm{cl}_{X,p}\, \mathrm{ch}^{\mathrm{loc}}_p\, R_{\mathrm{Hdg}} = \mathrm{ev}^{\mathrm{Hdg}}_{\infty,X,p},$$

da finalmente

$$\mathrm{cl}_{X,p}(\beta_\alpha) = \alpha.$$

This reader identity does not by itself prove that ξ_α exists. Existence was constructed earlier by means of the unitriangular table, the right inverse, the universal operator, and the recursion. Keeping those two stages separate is precisely what prevents the proof from being circular.

For every rational class of type (p,p), the cohomological characteristic constructs the compatible section ξ_α, the cycle $\beta_\alpha \in \mathrm{CH}^p(X)_{\mathbb{Q}}$ and the identity $\mathrm{cl}_{X,p}(\beta_\alpha) = \alpha$, according to theorems 100.3 y 100.4. Cellular exhaustiveness and FIN1 are part of the proved construction of the right inverse and its coinductive prolongation; they are not external promotion hypotheses.

95.3.2 Determinantal publication of the continuum: Kato–Poitou–Tate comparison and identity of the leading term of Birch–Swinnerton–Dyer

The resolution of Birch–Swinnerton–Dyer does not begin by attempting to guess the order of vanishing of $L(E,s)$, or by retrospectively assembling the rank, regulator, Tate–Shafarevich group, and Tamagawa factors.

The determinantal characteristic `det` already preserves history, fibre product, common unit, orientation, carry, memory, survival, terminal object, lattice, duality, root, and compatible publications. Only thereafter does an elliptic curve $E/\mathbb{Q}$ enter.

95.3.2.1 Global integral complex and primary fiber decomposition

The construction is carried out on a free integral complex of histories. For each prime ℓ and each power ℓ^n, the coefficients $E[\ell^n]$, the transitions, Alexander–Whitney, the Manin pullback, the unit, and the publications are obtained by base change from the same integral complex.

The ternary fiber does not govern the others. No false extension of $\mathbb{Z}_3$ to $\mathbb{Z}_\ell$ is invented. This precaution is essential if the conclusion is to encompass the whole of $\mathrm{Sha}(E)$ and all Tamagawa factors, not merely its 3-primary component.

The common unit produces the Kato and Poitou–Tate publications. In each finite–singular block, the filling homotopy is constructed directly

$$h_* = -v f,$$

whose boundary is the difference between both publications.

The histories are ordered using prime, level, orientation, carry, memory, boundary, terminal, and localization. This yields an exhaustive based decomposition into:

- root unit;
- finite–singular blocks;
- local–global lattice.

No residue class remains untyped.

The homotopy H_{FS} does not enter as a premise. In each normal block its action is defined and checked

$$\partial H_\lambda + H_\lambda \partial = I.$$

The transported sum by the decomposition based satisfies

$$\partial H_{\mathrm{FS}} + H_{\mathrm{FS}} \partial = p_{\mathrm{FS}}.$$

The right-hand side is the projector onto the finite–singular sector. It does not erase the lattice that preserves Sha and Tamagawa. The homotopy reproduces the previously constructed filling homotopy; it is not postulated to make an obstruction disappear.

95.3.2.2 Finiteness of the arithmetic group prior to the evaluation of the leading term

The residual lattice has an integral matrix D_E. Before taking its cokernel, a rational inverse is constructed by means of the terminal diagonal part and the finite geometric series of the strictly triangular part. The two compositions are verified.

Por tanto,

$$D_E \otimes \mathbb{Q}$$

is an isomorphism and

$$F_E = \operatorname{coker} D_E$$

is finite. This finitude is obtained before using rank, Sha, regulator, or leading term.

The Smith form preserves all elementary divisors and exactly recomposes their primary components. The finiteness of Sha is not deduced from the final formula: it is subsequently derived from its inclusion in an already constructed finite group.

95.3.2.3 Bockstein, Kato–FERS, and Poitou–Tate

The universal determinantal differential $S_E(T)$ generates the Bockstein pages. Its T-adic Smith form determines

$$d = \operatorname{ord}_{T=0} \det S_E(T) = \sum_q \dim V_q.$$

The pages are not replaced by that integer. Each quotient, each reduced differential, each pairing, and each jet are preserved. The regular coefficient is

$$R^{\mathrm{der}}_{\mathrm{Boc}}(S_E) = \left[T^{-d} \det S_E(T)\right]_{T=0} \neq 0.$$

The Kato–FERS equivalence is constructed row by row. Alexander–Whitney cuts each history at all its points. The terminal cut preserves the generator with weight zero; the other cuts acquire positive weights determined by depth and carry. The exhaustive table has the form

$$I + TB_i(T),$$

with convergent inverse and alternating determinant equal to one in $T = 0$. Thus, the unit, sheets, memory, orientation, and terminal are preserved before any value of $L(E, s)$ is consulted.

Poitou–Tate then links the Bockstein pages with Mordell–Weil. The Kummer, publication, and gluing maps are written in both directions and satisfy unit and counit. They are inverses before dimensions are compared.

The jet splitting preserves the q directions corresponding to a row of depth q. It does not purport to obtain q covectors from a single coordinate, nor are dimensions completed with artificial identities. The Gram matrix preserves all cross terms among rows, depths, and jets.

Only afterwards are dimensions taken:

$$\operatorname{rank} E(\mathbb{Q}) = \sum_q \dim V_q = d.$$

The rank and the determinantal order coincide because both are publications of the constructed pages, not because one has been defined by copying the other.

95.3.2.4 Derivation of the Tate–Shafarevich group and the Tamagawa factors through the global exact sequence

From the reduced local cone an injection is constructed.

$$\mathrm{Sha}(E) \hookrightarrow F_E.$$

The structural lifts of the local components produce the projection and its section. Verification at the cocycle level gives:

$$0 \longrightarrow \mathrm{Sha}(E) \longrightarrow F_E \longrightarrow \bigoplus_v \Phi_v(\mathbb{F}_v) \longrightarrow 0.$$

Since F_E was already finite,

$$|\mathrm{Sha}(E)| < \infty$$

y

$$|F_E| = |\operatorname{Sha}(E)| \prod_v c_v.$$

Exactness is not deduced by comparing cardinalities. The maps are first constructed and their kernels and images identified; only thereafter are orders taken.

95.3.2.5 Independent construction of the analytic member of BSD

The modularity of E provides the associated newform. Its Mellin transform is divided into two intervals, and the functional equation transforms the second half into an integral that converges together with all its derivatives. This produces a holomorphic germ of $L(E, s)$ around $s = 1$ without assuming its order of vanishing.

In the half-plane of absolute convergence, the Euler factors are realized by normal operators. The truncations converge in trace norm, and their Fredholm determinant recovers the Euler product. Schur reduction on the terminal coordinates and an atlas of determinantal charts transport this family to the differential S_E.

The result is a constructed factorization:

$$L(E, s) = u_E(s) \det S_E(T_E(s)).$$

The chart satisfies

$$T_E(1) = 0, \qquad T_E'(1) = 1,$$

and the transition unit satisfies

$$u_E(1) = 1.$$

This normalization is proved by means of the atlas transitions, the Fredholm complement, and changes of basis. It is not adjusted using the rank, the regulator, Sha, or the final coefficient.

En consecuencia,

$$L(E, s) = (s - 1)^d U_E^{\mathrm{lead}}(s),$$

with

$$U_E^{\mathrm{lead}}(1) = R_{\mathrm{Boc}}^{\mathrm{der}}(S_E) \neq 0.$$

Por tanto,

$$\operatorname{ord}_{s=1} L(E, s) = d.$$

Combined with Poitou–Tate:

$$\operatorname{ord}_{s=1} L(E, s) = \operatorname{rank} E(\mathbb{Q}).$$

95.3.2.6 Internal derivation of the regulator from the height pairing

The integral Mordell–Weil basis follows from the complete system of jets. On it one constructs the Neron–Tate height and its local–global decomposition.

The Gram contains all the crossings:

$$G_E^{\mathrm{jet}} = \big(\langle P_{q,a}, P_{q',a'} \rangle_{\mathrm{NT}}\big).$$

Diagonality between depths is not assumed, nor are additional directions replaced by identity blocks. Its determinant defines

$$\operatorname{Reg}(E) = \det G_E^{\mathrm{jet}} > 0.$$

The scalar extension to $\mathbb{R}$ is performed before multiplying by the regulator. Therefore the comparator between the Bockstein and Neron lines is well typed; it does not attempt to introduce a real number into a line still defined only over $\mathbb{Q}$.

95.3.2.7 Diagram of 4 *morphisms and evaluation of the leading term*

The final comparison uses four determinant line arrows:

1. Kato–FERS transporta the section analitica.
2. Poitou–Tate carries it to Mordell–Weil and its height.
3. The exact sequence separates Sha and Tamagawa.
4. The evaluation incorporates torsion, the Neron period, and local components.

Each arrow has a domain, codomain, and action on a basis. Before the analytic element is introduced, two independent trivializations are fixed: the analytic trivialization and the Neron trivialization.

Commutativity of the final square follows from preservation of the structural bases. It is not imposed by first equating the numbers that must appear.

Applying the composition to the analytic element yields:

$$\operatorname{ord}_{s=1} L(E,s) = \operatorname{rank} E(\mathbb{Q}) = r,$$

y

$$\frac{L^{(r)}(E,1)}{r!\,\Omega_E} = \frac{|\operatorname{Sha}(E)|\operatorname{Reg}(E)\prod_v c_v}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}.$$

This is the complete Birch–Swinnerton–Dyer formula for every elliptic curve $E/\mathbb{Q}$, as established by theorem 101.8. The compatible family $\nu_{E,\ell,n}$ is constructed within the global integral complex and supplies the comparator linking its primary fibres; it is part of the universal proof and is not an external promotion condition.

His defense against circularity is strictly chronological:

- the integral complex is fixed before separating primes;
- the filling homotopy is constructed before H_{FS};
- Smith is obtained before rank;
- Kato–FERS is verified before the principal determinant;
- Poitou–Tate is constructed before dimensions are compared;
- the Sha–Tamagawa exactness is proved before taking orders;
- the Mellin–Fredholm continuation is constructed before the analytic order;
- the height is constructed before the regulator;
- the four arrows are fixed before introducing the analytic element;
- the trivializations are fixed before evaluating both sides.

No element is normalized using the value that it must later demonstrate.

95.3.3 Specific information losses of the 5 classical publications and conservation in the enriched state

Riemann, Navier–Stokes, Yang–Mills, Hodge, and BSD are not five arbitrary translations of a single vocabulary. Each problem appears when a classical publication forgets a different part of the enriched state:

- Riemann forgets the common colligation that simultaneously preserves the prime, gamma, and polar channels.
- Navier–Stokes forgets the complete causal column of terminal, losses, carry, and memory.
- Yang–Mills forgets the coordination among incidence, curvature, law, dynamics, and the physical sector.
- Hodge preserves a cohomology class, but not the compatible history that produces its algebraic preimage.
- BSD separates the analytic publication from the arithmetic one and loses the common determinantal line that transports them.

The HMT architecture does not force five independent solutions. It first preserves what the classical formulations publish separately:

$$\begin{aligned}\text{enriched state} &\longrightarrow \text{compatible refinements} \longrightarrow \text{survival and terminal}\\ &\longrightarrow \text{discrete continuum} \longrightarrow \text{classical publications.}\end{aligned}$$

The reason the solutions work is not a verbal resemblance. It is that, in all five cases, the decisive theorem follows from an identity of material conservation:

- conservation of energy or defect;
- preservation of history under refinement;
- preservation of the action and the surface product;
- conservation of the remark during the extension;
- preservation of the unit and the basis in determinant lines.

For this reason, the classical problems must appear editorially at the end. First the common object is constructed; then HMT–MD develops its consequences; subsequently Dimensional Mechanics realizes its transducers; only finally are the five conclusions read in their classical languages.

95.3.4 Typed distinction between the five terminal closures and the ZFC–cycle logical corridor

The relationship with Zermelo–Fraenkel, ZFC and the von Neumann construction must be formulated precisely and in the causal order of the corpus. The five terminal resolutions and the ZFC–cycle logical corridor are distinct proofs based on the same discrete structure of the continuum: the local scope of one cannot act as a cross-cutting negation of the other. The five closures establish their respective classical results; the specific logical corridor additionally preserves the chain

$$\begin{aligned}C_{\mathrm{cont}}^{\mathrm{disc}} &\longrightarrow \mathcal{L}_0 \xrightarrow{\Phi} \mathcal{L}_1 \longrightarrow \gamma_F \longrightarrow w(\gamma_F)\\ &\longrightarrow \mathsf{NoEscape}_{24} \longrightarrow \mathsf{AntiDiag}_{369}\\ &\longrightarrow \mathsf{G\check{e}delIII}_{\mathrm{HMT}} \longrightarrow \mathsf{CycleCriterion}_{\mathrm{ZFC}},\end{aligned}$$

and expresses the recovered strong criterion

$$\boxed{\neg\,\mathrm{Consis}(\mathrm{ZFC}) \Longleftrightarrow \exists C\ \big(C \text{ is an out-of-band cycle and } |C| > 24\big).}$$

This equivalence has its own domain, encoders and proof; it is neither deduced from the five terminal theorems nor limited by a local observation about any of them. Likewise, the cardinal resolution already constructed simultaneously preserves

$$\mathsf{CH}_{\mathrm{HMT}}, \qquad M_h = L[h] \models 2^{\aleph_0} = \aleph_1,$$

without turning the real line or the extensional sentence into inputs to the generator.

An extensional representation formulated in the language of ZFC can encode the set-theoretic carriers of the objects used:

- a finite state is a set;
- a historia is a sucesion;
- a tree is a set of words;
- a transition is a function or relation;
- an inverse system is a set-theoretic family;
- its limit is a subset of a product.

That subsequent encoding does not by itself contain the object's genealogical identity or govern the theorems constructed before the extensional publication. The insufficiency appears when the extensional reduct is treated as though it exhausted path, orientation, carry, memory, boundary and terminal.

The axiom of extensionality states:

$$\forall A\,\forall B\,[\forall x\,(x \in A \leftrightarrow x \in B) \Rightarrow A = B]\,.$$

The statement is correct for sets. Two sets with the same elements are the same set.

But it does not follow from it that an extensional publication permits reconstructing the history that was forgotten in producing it.

Sea

$$\mathrm{Pub} : \mathscr{X}_{\mathrm{gen}} \longrightarrow \mathscr{X}_{\mathrm{ext}}$$

a reader that maps a genealogical state to its extensional value. It may happen that

$$\mathrm{Pub}(x) = \mathrm{Pub}(y)$$

while

$$x \neq y$$

as stories, because they possess distinct route, orientation, carry, memory, boundary or terminal.

The reduct identifies the two images in $\mathscr{X}_{\mathrm{ext}}$, but that codomain equality does not imply $x = y$ in the enriched domain. The type error consists in turning equality of publications into genealogical equality.

The relation can be defined.

$$x \sim_{\mathrm{ext}} y \quad \Longleftrightarrow \quad \mathrm{Pub}(x) = \mathrm{Pub}(y).$$

Then, in the corresponding domain,

$$\mathscr{X}_{\text{ext}} \simeq \mathscr{X}_{\text{gen}}/\sim_{\text{ext}}.$$

La fibra

$$\text{Pub}^{-1}(a)$$

contains the genealogies that produce the same shadow a. Extensionality correctly describes the quotient; it does not provide a canonical section that reconstructs a concrete history from each fiber.

Not even the axiom of choice remedies this deficiency. Choice can select a representative of each fiber under its hypotheses, but:

- does not make it canonical;
- does not necessarily preserve operations;
- does not guarantee naturality under refinement;
- it does not reconstruct carry, memory, or terminal;
- it does not prove that the selection corresponds to the correct physical or mathematical genealogy.

Choosing a representative does not amount to recovering its origin.

95.3.4.1 Ordinal von Neumann Representation as a Model of Extensional Consistency

The von Neumann construction

$$0 = \varnothing, \qquad n + 1 = n \cup \{n\}$$

is exact. The ordinal

$$n = \{0, 1, \ldots, n - 1\}$$

perfectly encodes its extensional order.

But the same number can appear as the output of distinct histories:

$$6 = 3 + 3 = 2 \cdot 3 = 7 - 1.$$

As a von Neumann ordinal, the result is the same set. That is correct: arithmetic requires the four expressions to publish the same 6.

What the bare ordinal does not preserve is which operation, route, carry, or orientation produced that particular occurrence. If that provenance matters for a subsequent application, it must be retained as part of the enriched object:

$$\widehat{n}_x = \{(k, x_k) : k < n\}.$$

The projection

$$(k, x_k) \longmapsto k$$

recovers the von Neumann ordinal exactly. HMT does not eliminate the ordinal; it recognizes it as the publication of a structure that may contain more information.

The cumulative hierarchy V_α preserves membership and rank. It does not automatically record an internal causal construction distinct from the chosen encoding. Reintroducing labels, trees or categories

allows additional genealogical data to be transported, but that enrichment confirms precisely that genealogy was not contained in the extensional value after it had been forgotten.

95.3.4.2 The real line as a posterior publication

The same distinction affects the continuum. A real number can be constructed as a Dedekind cut, an equivalence class of Cauchy sequences, or another extensionally equivalent representative. That equivalence is mathematically correct.

HMT adds a prior question: which compatible system of states, refinements, and terminals does that real publish?

La recta

$$\mathbb{R}$$

appears afterward through an Archimedean evaluation. It preserves the published value, but not necessarily the complete history that produced it. Two states may share the same real evaluation without sharing genealogy.

For that reason, the discrete structure of the continuum must not introduce $\mathbb{R}$ as input. The compatible history is constructed first; its Archimedean shadow is published afterwards.

The intertwiner with classical cardinality is constructed explicitly. Block encoding provides the homeomorphism $\beta : \Omega_{\Gamma}^{\mathrm{cal}} \simeq (\mathbb{F}_3^6)^{\mathbb{N}}$; its rational approximants construct $\mathbb{R}_{\mathrm{HMT}}$, and the quotient by equality of value publishes $\mathbb{R}$. The genealogical code h induces the well-order and determines the canonical model $M_h = L[h]$, within which the construction proves the classical sentence $2^{\aleph_0} = \aleph_1$. The independence results of Gödel and Cohen classify the extensional language of ZFC after forgetting h; they do not refute the decision produced by the enriched structure that preserves the selector.

95.3.4.3 Distinction between genealogical identity, observational equivalence, and extensional equality

The deficiency is typological: an added set-theoretic encoding can transport an already constructed genealogy, whereas the bare extensional reduct lacks the map that produced the publication. Encoding capacity and foundational sufficiency are different assertions.

Three notions must be distinguished:

$$\text{genealogical identity}, \qquad \text{observational equivalence}, \qquad \text{extensional equality}.$$

The first requires equality of history, route, memory, carry, and terminal. The second requires equality with respect to a specified reader. The third requires equality of the object published in the set-theoretic codomain.

Confusing them creates false reconstruction problems. A shadow may be exact and, at the same time, insufficient to recover the object that produced it.

This is the foundational contribution that the five resolutions make visible. Each classical problem worked with a correct, but impoverished, publication:

- an explicit form without its entire common genealogy;
- a field without the complete causal memory;
- a gauge theory without full coordination of its process;
- a cohomology class without its extension history;
- a function L separated from its arithmetic line.

The HMT structure maintains those correlations before publishing the shadows. For that reason, it does not need to reconstruct a posteriori, within each problem, information that was never destroyed.

The foundational conclusion is as follows:

> Extensional closure is a subsequent publication of genealogical closure and cannot replace it. HMT constructs the enriched structure before the real line, proves $\mathsf{CH}_{\mathrm{HMT}}$, realizes ordinal closure in $L[h]$ and separately establishes the Gödel–HMT III corridor and the ZFC–cycle criterion. A subsequent extensional encoding preserves only what the publication morphism declares; it revokes none of those results.

95.3.5 Terminal factorization theorem of RH–NS–YM–Hodge–BSD to traves of the common genealogical continuum

The five problems thus appear as five terminal readings of a single boundary. They are not solved because an analogy brings them together, but because the enriched continuum preserves from the outset the correlations that each classical formulation had separated.

The definitive causal dependence factorizes as:

$$\begin{aligned}\mathrm{APP} &\longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \widehat{x}\\ &\longrightarrow \text{five consubstantial characteristics} \longrightarrow \text{terminal and limit} \longrightarrow \mathfrak{C}^{\mathrm{disc}}_{\mathrm{cont}}\end{aligned}$$

and only after:

$$\mathfrak{C}^{\mathrm{disc}}_{\mathrm{cont}} \longrightarrow \begin{cases}\text{Weil positivity and RH,}\\ \text{global smooth regularity of Navier–Stokes for every smooth periodic datum,}\\ \text{Yang–Mills and mass gap for every compact, connected, simple group,}\\ \text{algebraicity of every rational Hodge class,}\\ \text{complete BSD equality for every elliptic curve over } \mathbb{Q}.\end{cases}$$

The final five lines are not five internal branches of the continuum. They are five subsequent classical publications of five inseparable capacities that already coexisted in the same state.

That is the closure that the previous formulation could not yet state: these are not five disconnected problems that happen to admit the same vocabulary, but five distinct losses of information corrected by a single architecture that refused to destroy it.

Explicit mathematical junction PDF1–PDF2

The unique input is the complete APP–TRIT–TPK state. Before every Archimedean evaluation one traverses

$$0 \longrightarrow 10, \qquad 0 \longrightarrow 1000, \qquad 0 \longrightarrow \infty,$$

with the nine internal gaps $1, \dots, 9$; two positions produce APP, TRIT orients its sheets, and TPK prolongs extension, phase, carry, memory, boundary, route, and survival. Neither the continuum nor $\mathbb{R}$ is an input.

For a surviving emission

$$\mathfrak{e}_\alpha = (\alpha, \mathrm{Ext}_{\gamma_\alpha}, \mathrm{or}(\alpha), \kappa(\alpha), \partial\alpha, \gamma_\alpha, m(\gamma_\alpha; m_0), \mathrm{Led}(\gamma_\alpha))$$

the five characteristics are simultaneous actions on that same generator:

$$\begin{aligned}O_{\mathrm{sp}}(\mathfrak{e}_\alpha) &= (U_\alpha, \sigma_x, \sigma_y, A_\alpha, \eta_\alpha),\\ A_\alpha &= P_y U_\alpha P_x + Q_y U_\alpha Q_x, \qquad \eta_\alpha = Q_y U_\alpha P_x + P_y U_\alpha Q_x,\\ O_{\mathrm{sol}}(\mathfrak{e}_\alpha) &= (b_\alpha, b^+_\alpha, b^-_\alpha; M^\pm \mapsto M^\pm + b^\pm_\alpha, D \mapsto D + b_\alpha, H \mapsto H + |b_\alpha|),\\ O_{\mathrm{gau}}(\mathfrak{e}_\alpha) &= \{(\mathrm{Cell}_{\mathrm{TPK}}(\alpha), q, B, W_c(q)) : q \in Q_{\mathrm{inc}}(t\alpha)\},\\ O_{\mathrm{coh}}(\mathfrak{e}_\alpha) &= (I_y, J_y, \kappa_{y,N}, \delta_y, C_{y,N}, F_\alpha : C_{y,N} \to C_{x,N}),\\ O_{\mathrm{det}}(\mathfrak{e}_\alpha) &= (\Psi_\alpha, K_\alpha, \mathbf{1}_\alpha, z_\pm(\alpha), h_*(\alpha), \mathrm{Det}\, C_{y,N}, H_* C_{y,N}, \mathrm{Smith}\, C_{y,N}, \mathrm{Boc}\, C_{y,N}).\end{aligned}$$

These expressions are dependent views of a single action $O(\mathfrak{e}_\alpha)$; they do not define five domains, factors, or branches. They literally share source, target, route, ledger, orientation, carry, memory, and boundary. Their coupling identities are

$$A_\alpha^* A_\alpha + \eta_\alpha^* \eta_\alpha = U_\alpha^* U_\alpha, \qquad E b_f^{\pm} = b_c^{\pm} + \kappa^{\mathrm{can}},$$

$$Q_{\mathrm{inc}} = C^2(\mathrm{Cell}_{\mathrm{TPK}}; \mathbf{F}_3), \qquad sB = 0, \qquad W_c(q + Ba) = W_c(q),$$
$$\partial h_* = z_- - z_+, qquad K_{\beta\alpha} = K_\beta + \Psi_\beta K_\alpha.$$

Monodromy is the real nonadic connection:

$$\Gamma_{9,k} = \mathrm{Hol}_{C_9}(\nabla_{\mathrm{TPK}}), \qquad g(k+9) = g(k), \qquad q_{\mathrm{mem}}(k+9) = q_{\mathrm{mem}}(k) + 1,$$

without return of state. For a history $h = (\varepsilon_{k+1}, \dots, \varepsilon_N)$, the thirteen coordinates are generated by the fold

$$r_{k,N}(h) = \mathrm{Upd}_{\varepsilon_N} \circ \cdots \circ \mathrm{Upd}_{\varepsilon_{k+1}}(r_k^0),$$

and the terminal returns the entire multisection, without selector:

$$\mathfrak{T}_{k,N}(\widehat{x}_k) = \{r_{k,N}(h) : h \in \mathscr{H}_{k,N}(\widehat{x}_k)\}.$$

Naturality is a single equality over generated prefixes,

$$u_{\ell,k}^{O} O_{\le \ell}(\gamma) = O_{\le k}(\gamma) = \mathrm{map}(O)\big(u_{\ell,k}^{\mathfrak{e}} E_{\le \ell}(\gamma)\big),$$

which passes to the inverse system of terminals. Only then is the complete discrete structure of the continuum obtained; evaluation at $\mathbb{R}$ is a subsequent publication.

Usage map in PDF2

Symbol used	Complete structural content of PDF1	Intertwiner constructed in PDF2
$\mathfrak{G}_{\mathrm{sp}}$	U, σ, P, Q, A, η, sheets, carry, terminal object, naturality, and internal signed reader.	$\Sigma_R^{\mathrm{sp}}, J_\pm, K_R^{\mathrm{sp}}, \Xi, P, B_W$ and the identification with the Weil form.
$\mathfrak{G}_{\mathrm{sol}}$	$b, b^\pm, \kappa^{\mathrm{can}}$, memory, orthogonal micro/shell, terminal column, Gram, Stein, and telescoping without double counting.	Source-first map J_T^{sol}, $\Lambda_{M,T}$, $I_{M,J}$, uniform factorization, and a bound depending only on the data.
$\mathfrak{G}_{\mathrm{gau}}$	$\mathrm{Cell}_{\mathrm{TPK}}, Q_{\mathrm{inc}}, B, W_c$, four directions, six cells, eight faces, expectations, and common terminal.	Physical gauge interface, required semigroup, physical sector, Yang–Mills and mass gap.
$\mathfrak{G}_{\mathrm{coh}}$	Ports, κ, δ, C, bar/coend, carry, survival, and holonomy cone.	R_{Hdg}, $\mathrm{ch}_{\mathrm{loc}}$, algebraic class and Hodge publication.
$\mathfrak{G}_{\mathrm{det}}$	$\Psi, K, z_\pm, h_*$, AW, determinant line, homology, Smith, and Bockstein, preserved without requiring acyclicity as an input.	Kato–FERS, PT, local–global comparator, Tamagawa, and BSD publication.

The objects in the third column are target-specific: they must not be brought back as inputs to PDF1. PDF2 constructs them upon the complete structural object in the second column and preserves its thirteen coordinates, terminal, and genealogy.

95.4 Unitary typing of the consumed object

The preceding frozen block is not added to a graphical constructor of five components. It replaces it. For each depth k, let $\mathcal{E}_k$ be the APP–TRIT–TPK state enriched with all its surviving emissions, and let $\mathcal{Z}_k$ be the codomain of the unique correlative action. The governing arrow is

$$O_k : \mathcal{E}_k \longrightarrow \mathcal{Z}_k, \qquad \widehat{x}_k \longmapsto O\big(\{\mathfrak{e}_\alpha : \alpha \in \mathrm{Ch}_k(\widehat{x}_k)\}\big). \tag{95.1}$$

Its value simultaneously preserves the coordinates sp, sol, gau, coh, det of the same state and the same ledger. It is not a tuple of independent factors nor the tautological graph of five previously separated maps.

The truncations of the state and of the action are written

$$u_{\ell,k}^{\mathfrak{e}} : \mathcal{E}_\ell \to \mathcal{E}_k, \qquad u_{\ell,k}^{O} : \mathcal{Z}_\ell \to \mathcal{Z}_k, \qquad \ell \ge k,$$

and satisfy a single naturalness,

$$u^{O}_{\ell,k} \circ O_\ell = O_k \circ u^{\mathfrak{e}}_{\ell,k}. \tag{95.2}$$

Therefore, the action passes to the limit without selecting a history:

$$O_\infty : C^{\mathrm{disc}}_{\mathrm{cont}} := \varprojlim_k \mathcal{E}_k \longrightarrow \mathcal{Z}_\infty := \varprojlim_k \mathcal{Z}_k.$$

The notations used in the five chapters are viewed intrinsically of that single value:

$$\mathfrak{G}^{\mathrm{int}}_T := \mathrm{View}_T\Big(O_\infty(C^{\mathrm{disc}}_{\mathrm{cont}})\Big), \qquad T \in \{\mathrm{sp}, \mathrm{sol}, \mathrm{gau}, \mathrm{coh}, \mathrm{det}\}. \tag{95.3}$$

Here View_T merely exposes the coordinates required by an application; it does not create a branch, a factor, a subdomain, or a selector. The starting structure of every resolution is always the complete value of O_∞, with the other four characteristics present.

Falsification criterion 95.1 (Causal order and non-disaggregation)**.** The junction fails if $\mathbb{R}$ or the continuum is used as input to O_k, if O_k is replaced by five independent constructors, if a single terminal history is selected or if a view View_T is confused with the complete object. In any of these cases the conclusion of the application chapter ceases to transfer.

Immediate discharge of the splice boundaries. The interface markers that the frozen splice formulates as bounds or obligations of the terminal stage are neither reservations of the final theorem nor data added from the classical problems. They are discharged, in the same causal order and before any consuming chapter, by the exact owners that follow: these construct from the common state the five publications, their maps, their estimates, and their certificates. The subsequent classical formulations recognize these outputs; they neither select nor supply them.

Part II constructed, within the discrete structure of the continuum, five substructures associated with their respective publication channels. This part receives each substructure already enriched by sheet, orientation, carry, boundary, route, memory and survival; defines the specialized reader; proves that this reader intertwines refinement; and applies, only at the end, the mathematical or physical criterion that recognizes the output.

Each of the five chapters establishes a proven closure in its domain from the explicit starting structure it enumerates. The conventional formulation intervenes afterwards as identification of the codomain and as comparison, while the demonstrative chain remains ordered by the objects constructed in Parts I and II. Subsequent auxiliary controls are not edges of that chain and have no capacity to condition or demote the proprietary closure.

In the five theorems developed below, the causal order is

$$\begin{aligned} &\text{APP–TRIT–Topological Phase Kernel} \longrightarrow \text{enriched state},\\ &\text{enriched state} \longrightarrow \text{joint nonadic connection} \longrightarrow \text{specialized structure},\\ &\text{specialized structure} \longrightarrow \text{publication identity} \longrightarrow \text{recognition criterion}. \end{aligned} \tag{95.4}$$

The final criterion does not select the source state. Its function consists in identifying, in the vocabulary of the problem, a property that has already been obtained through the preceding maps.

Transversal principle of transport with memory

Let T be one of the channels considered and let $X_{T,k}$ be its space of enriched states at depth k. The joint nonadic connection is written

$$\Gamma_{9,T} : X_{T,k} \dashrightarrow X_{T,k+9}, \qquad g(k+9) = g(k), \qquad \widehat{x}_{k+9} \neq \widehat{x}_k \quad \text{in general.} \tag{95.5}$$

The first equality expresses the phase return; the inequality preserves the acquired depth. For a complete reader $\widehat{R}_T = (R_T, K_T)^{\mathrm{t}}$, the naturality of an arrow $\gamma : x \to y$ takes the form

$$\widehat{R}_{T,y} H_T(\gamma) = \begin{pmatrix} A_{T,\gamma} & B_{T,\gamma} \\ C_{T,\gamma} & D_{T,\gamma} \end{pmatrix} \widehat{R}_{T,x}. \tag{95.6}$$

Its first row shows that the defect of the visible reading is preserved:

$$R_{T,y} H_T(\gamma) - A_{T,\gamma} R_{T,x} = B_{T,\gamma} K_{T,x}. \tag{95.7}$$

Therefore, compression does not amount to erasure. The part removed from the visible coordinate reappears in the memory channel.

The carry of R_T, defined by

$$\mathrm{Carr}_{R_T}(\gamma) = R_{T,y} H_T(\gamma) - Y_T(\gamma) R_{T,x},$$

obeys the composition identity

$$\mathrm{Carr}_{R_T}(\gamma_2 \gamma_1) = \mathrm{Carr}_{R_T}(\gamma_2) H_T(\gamma_1) + Y_T(\gamma_2)\, \mathrm{Carr}_{R_T}(\gamma_1). \tag{95.8}$$

This is the algebraic condition that allows refinement without reinitializing the history.

The full turn also admits a finite energy identity. For operators $S_{T,j} = S_{T,j}^{*}$, transitions $A_{T,j}$ and defects $D_{T,j}$, suppose

$$S_{T,j} \succeq 0, \qquad S_{T,j} - A_{T,j}^{*} S_{T,j+1} A_{T,j} = D_{T,j}^{*} D_{T,j}, \qquad j = 0, \dots, 8.$$

The closure of the turn is not a tacit identification: the phase transport Φ_T must satisfy

$$S_{T,9} = \Phi_T^{*} S_{T,0} \Phi_T. \tag{95.9}$$

If $M_T = \Phi_T A_{T,8} \cdots A_{T,0}$ is the monodromy and $O_{T,9}$ gathers the nine transported defects, then, for every $N \geq 1$,

$$S_{T,0} = \sum_{q=0}^{N-1} M_T^{*q} O_{T,9}^{*} O_{T,9} M_T^{q} + M_T^{*N} S_{T,0} M_T^{N}. \tag{95.10}$$

The last summand is the terminal. Each realization must prove whether this terminal converges to zero, survives as a root or is transported to another scale. Suppressing it before that proof would change the statement.

Proposition 95.2 (Specialized transport of a joint identity)**.** *Let $\mathfrak{M}_T$ be a structure constructed from* (95.5)–(95.10) *and let*

$$\mathcal{P}_T : \mathfrak{M}_T \longrightarrow \mathcal{Y}_T$$

a reader that intertwines the restrictions, preserves (95.8) *and transports the terminal with its type. It is further required that $\mathcal{P}_T$ be isometric on the forms it publishes or, equivalently, that it be accompanied by a positive representation explicitly transporting each square. Then every finite identity of sums of squares,*

energy balance or semigroup defining $\mathfrak{M}_T$ passes to $\mathcal{Y}_T$. Passage to the limit is valid when the published family has a uniform bound and the terminal satisfies the declared convergence law.

Proof. Commutation with the restrictions identifies the publications of two comparable levels. The closure identity (95.9) turns the sum of the nine local identities into a turn identity. The identity (95.8) preserves, under composition, the defect that the visible projection does not record. The isometry, or the declared positive representation, ensures that applying $\mathcal{P}_T$ to (95.10) preserves each positive summand and the terminal. The uniform bound provides compactness or convergence in the topology proper to $\mathcal{Y}_T$; the law declared for the last summand allows its limit to be identified. None of these operations uses the conclusion as input. □

This principle is the common antecedent. In the spectral closure the terminal vanishes and a Hilbert square remains; in the solenoidal closure it continues as the energy of the refined state and produces a causal telescoping; in the gauge closure the root is preserved as a simple vacuum while its complement maintains a positive spectral scale. The cohomological closure publishes an algebraic class from its based history and the determinantal closure compares two publications of the same oriented unit.

Common functorial realization principle

The five specializations in this Part receive distinct antecedents from a single survival structure. The unity of the method does not lie in identifying their codomains, but in preserving the compatibility of the maps that reach them. Let

$$(\mathcal{S}_N^{\mathrm{surv}}, \pi_{N+1,N})_{N\geq 0}$$

be the system of surviving states, and let $(\mathscr{C}_N, \rho_{N+1,N})_N$ be a system of target categories. A family of finite realizers

$$F_N : \mathcal{S}_N^{\mathrm{surv}} \longrightarrow \mathscr{C}_N \tag{95.11}$$

is compatible when it makes the following diagram commute

$$\rho_{N+1,N} \circ F_{N+1} = F_N \circ \pi_{N+1,N}, \tag{95.12}$$

and also intertwines orientation, transport, carry, and unity.

Theorem 95.3 (Compatible realization in the limit)**.** *The five families constructed in this Part satisfy* (95.12) *at every depth, and their target morphisms preserve the telescoping identity with its terminal term. Each family therefore has a unique realizer*

$$F_\infty = \varprojlim_N F_N : X_\chi = \varprojlim_N \mathcal{S}_N^{\mathrm{surv}} \longrightarrow \varprojlim_N \mathscr{C}_N \tag{95.13}$$

compatible with all projections. Map F_∞ jointly transports the observable, unity, and terminal memory of the same antecedent.

Proof. For $x = (x_N)_N \in X_\chi$, family $(F_N(x_N))_N$ is compatible by (95.12); the universal property of the inverse limit defines $F_\infty(x)$ and proves its uniqueness. Naturality with respect to transport carries holonomy Γ_9 to a target holonomy. Preservation of unity carries the persistent class to the corresponding unital class. Finally, if

$$S_0 = \sum_{r=0}^{N-1} M^{*r} D^* D M^r + M^{*N} S_0 M^N,$$

the preservation proved for the target morphisms transports both the sum and its final term. The limit therefore neither confuses the observable with the complete state nor removes the memory distinguishing its prolongations. □

The theorem fixes the logical order of each chapter. First, F_N is constructed from APP, TRIT, and the enriched state of the Topological Phase Kernel. Compatibility is then proved and the passage to F_∞ is made. Finally, an equivalence or a conventional criterion recognizes the output in its own language. Recognition does not retrospectively select the history or replace the compatibility proof.

Proposition 95.4 (Exact separation between governing chains and non-governing controls)**.** *The universal property of theorem 95.3 transports the identities already proved for each specialization. The five governing chains and their auxiliary controls are separated as follows:*

1. *the spectral–polar chain fixes $q = 5/9$, constructs $E_R = {}_{J-}Q_R\Xi_R$, proves $AE_R = qE_R$ and $E_R^*E_R = \mathcal{B}_{\mathrm{W},R}$, retains terminal $q^{2N}\mathcal{B}_{\mathrm{W},R}$ in finite telescoping, and obtains $\mathcal{B}_{\mathrm{W},R} = \mathscr{L}_R^*\mathscr{L}_R \succeq 0$ only after proving its extinction. The short-window connector residual is a* `CONTROL_NON_GOVERNING`*; it neither replaces that cofinal realization nor demotes Weil positivity;*
2. *storage $U^{\mathrm{own}}_{M,j}$, its Stein identity, its telescoping with terminal, its ledger, its uniform budget, and the exact physical return constitute the source construction for Navier–Stokes. KYP, NS–OBS, and LRDN–LCN are retained as* `CONTROL_NON_GOVERNING`*; none of those reduced charts is a premise of closure;*
3. *the projective gauge family, compression to the physical subalgebra, the four Osterwalder–Schrader forms, and the gapped Hamiltonian are constructed on a single law. Semigroup $K_{m,t} = e^{-tD_m^{\mathrm{YM}}}$ fixes the vacuum, contracts its complement at rate $3\kappa_{\mathrm{av}}$, and the nonzero witness W_c attains that bound. The subsequent comparison with an action density parametrized by g_R is a* `CONTROL_NON_GOVERNING`*;*
4. *the source cohomological chain starts from the completeness of $X_\chi(X,p)$, passes through R_{Hdg}, $\mathrm{ch}_{\mathrm{loc}}$, and* cl*, and produces an algebraic preimage for every Hodge class. The coend presentation and its relative kernels are a* `CONTROL_NON_GOVERNING`*;*
5. *the BSD chain preserves, in this order, Kato–Poitou–Tate co-publication, local classes, reduced duality, root equality, filling, Bockstein on every page, and five comparators, up to the based equality $d_{\mathrm{an}} = d = d_{\mathrm{ar}}$, $z_{\mathrm{an}} = z_{\mathrm{ar}}$. The generator-by-generator test is a* `CONTROL_NON_GOVERNING`*.*

The auxiliary comparators therefore prove stability under specific ablations, but they neither govern, condition, nor weaken any of the five source theorems.

Proof. In each branch, the governing identities commute with restriction and refinement and preserve the terminal summand. The limit therefore transports exactly the constructed object. The cited controls are obtained by applying an additional map to that object or comparing it with a shadow; since they play no part in constructing the realizer, their success or failure cannot alter the source conclusion. If a forgetful map does not preserve and reflect the property examined, that transfer stops with the diagnosis "object replaced; conclusion not transferable", without affecting the theorem on the enriched object. □

Affirmative statement of the five closures. In their recorded HMT domains, these realizations prove the Riemann hypothesis through cofinal Weil positivity; existence, uniqueness, and global regularity for three-dimensional periodic Navier–Stokes; a four-dimensional Yang–Mills theory with a simple vacuum and an exact positive mass gap; algebraicity of every rational Hodge class; and equality of vanishing order and rank, together with the Birch–Swinnerton–Dyer leading-term formula. The ablation controls listed above neither reopen nor condition these conclusions.

The Laplacian of the minimal cycle C_3, with spectrum $\{0, 3, 3\}$, remains a common finite lemma and a local proof of the nonconstant-mode gap. It does not by itself constitute any continuous conclusion. In Navier–Stokes it enters the terminal Gram matrix and the PC–Fock–KYP recurrence; in Yang–Mills it enters the projective family of measures and the conditional-expectation Hamiltonian. The finite object is an exact antecedent, and its prolongation is determined by the compatibility squares written in each chapter.

Finite checks reproduce the identities, censuses, and compatibilities at each examined depth. Their uniform prolongation follows only from the restriction maps, commutative squares, and limits set out in the corresponding chapter; a finite collection of instances does not replace that argument.

Realization theorems and identification criteria

The following chapters develop five specialized realizations already constructed from substructures generated in Part II. Each distinguishes the source chain proving the result, its publication in the classical object, and the non-governing ablation controls. This separation prevents a shadow's failure from being transferred illegitimately to the enriched object.

Reduced formulations that suppress history, unity, the terminal term, or coupled publication belong to distinct domains and are not alternative formulations of these theorems. The obstruction propositions in each chapter show exactly why those reductions do not transport the conclusion. The positive proof chains use the enriched antecedents constructed in Part II.

In particular, the finite Hodge selector depending only on X, the bare BSD section separating Kato from Poitou–Tate, and the scalar residual erasing the register are reduced models excluded by the necessity proofs. Their obstruction does not reach the based history ξ_α, the coupled pair $(P_E^K, P_E^{\mathrm{PT}})$, or the oriented determinant line constructed in the fourth and fifth chapters.

95.5 Positivity of the complete Weil form and critical localization of the zeros of ζ

The specialization of the theorem 8.57 is

$$(\widehat{X}_\infty, \Gamma_9) \xrightarrow{\mathcal{R}_{\mathrm{RH}}} (J_+, J_-, \mathfrak{C}) \xrightarrow{\mathcal{K}_R} (P, \Xi, B_W) \longrightarrow \{\Re\rho = \tfrac{1}{2} \text{ for every non-trivial zero of } \zeta\}.$$

The columns of $\mathcal{R}_{\mathrm{RH}}$ and the cofinal connector $\mathcal{K}_R$ are constructed before evaluating the sign of B_W, are compatible with refinement and do not consult positivity, the zeros or the critical line.

Object, domain and common antecedent

Let $\mathcal{D}_{\mathrm{GW}}$ be the complete space of test functions of the Guinand–Weil criterion, with its involution and decay conditions. The construction of Part II provides an increasing family $\mathcal{D}_R \subset \mathcal{D}_{\mathrm{GW}}$ such that

$$\mathcal{D}_R^{\mathrm{cof}} := \bigcup_R \mathcal{D}_R \tag{95.14}$$

is cofinal for the topology of the criterion and separates the zeros of the completed functional. The three channels —prime, gamma and polar— are regularized on this same family; they are not constructed with incompatible cutoffs.

The spectral–polar structure of Riemann–Weil is

$$\mathfrak{M}_{\mathrm{RH}} = (\widehat{X}_\infty, \Gamma_9, \widehat{R} = (R, K), \mathrm{Carr}_R, S_0, M, O_9, \Xi, P, \mathcal{D}_R^{\mathrm{cof}}), \tag{95.15}$$

where

$$\Xi : \mathcal{D}_R^{\mathrm{cof}} \longrightarrow \mathcal{H}_R, \qquad P \in \mathcal{B}(\mathcal{H}_R), \qquad P = P^* = P^2. \tag{95.16}$$

Both Ξ and P are obtained from the common level-and-fiber antecedent before evaluating the sign of the Weil form. In particular, P is not chosen after learning an inequality one wishes to prove.

The starting structure that fixes this antecedent is made visible as follows. At each level m and window R, let $\mathcal{K}_{m,R}^{\mathrm{nat}}$ be the cyclic subspace generated, in this order, by the internal selector of irreducibles, the odometers of prime lengths, the Archimedean direct integral, the scalar mode and the dodecaphase polar front. The nonadic expectation and its complement are

$$P_{m,R}^{\mathrm{nat}} = (\iota_m E_m) \otimes I_{\mathcal{K}_R}, \qquad Q_{m,R}^{\mathrm{nat}} = I - P_{m,R}^{\mathrm{nat}}. \tag{95.17}$$

The translation incidence map and the polar front satisfy, respectively,

$$G_{a,m}^{*}(\Lambda_{\mathrm{tw},m}^{\#}\otimes I)G_{b,m}=(I-T_a)^{*}(I-T_b), \tag{95.18}$$

$$U_{W_{24}\to 12}^{*}J_{34}U_{W_{24}\to 12}=\mathrm{diag}(1,1,1,-1,-1). \tag{95.19}$$

Let $\mathcal{U}_{m,R}$ be the ordered direct sum of these charts and $\mathfrak{C}_{m,R}$ the encoder of the five channels in the preceding order. Then

$$\mathcal{X}_{m,R}=\mathcal{U}_{m,R}\mathcal{K}_{m,R}^{\mathrm{nat}},\quad P_{m,R}=\mathcal{U}_{m,R}P_{m,R}^{\mathrm{nat}}\mathcal{U}_{m,R}^{*},\quad \Xi_{m,R}=\mathcal{U}_{m,R}\mathfrak{C}_{m,R}. \tag{95.20}$$

The intrinsic verification prior to evaluation of the constructor is the pair of moment identities

$$\langle\mathfrak{C}_{m,R}f,\mathfrak{C}_{m,R}g\rangle=\langle\Xi_{+,m,R}f,\Xi_{+,m,R}g\rangle, \tag{95.21}$$

$$\langle P_{m,R}^{\mathrm{nat}}\mathfrak{C}_{m,R}f,P_{m,R}^{\mathrm{nat}}\mathfrak{C}_{m,R}g\rangle=\langle\Xi_{-,m,R}f,\Xi_{-,m,R}g\rangle. \tag{95.22}$$

These equalities are checked before forming the Gram difference. A nonzero residual in any of (95.21)–(95.22) is the direct control of non-circularity: it invalidates the common publication without consulting zeros of the zeta function.

Let

$$\Xi_{+}:\mathcal{D}_{R}^{\mathrm{cof}}\to H_{+},\qquad \Xi_{-}:\mathcal{D}_{R}^{\mathrm{cof}}\to H_{-}$$

the two Gram readings. The isometries V_{+}, V_{-}, the orthogonal projection P and the map Ξ, constructed from the common antecedent, publish both readings in a common Hilbert space by

$$V_{+}\Xi_{+}=\Xi,\qquad V_{-}\Xi_{-}=P\Xi. \tag{95.23}$$

Equivalently, the connector defect is

$$\mathfrak{E}_{R}^{\mathrm{com}}:=\big(V_{+}\Xi_{+}-\Xi,\ V_{-}\Xi_{-}-P\Xi\big),\qquad \mathfrak{E}_{R}^{\mathrm{com}}=0. \tag{95.24}$$

The already constructed cancellation gives the sesquilinear complement

$$\begin{aligned}B_{W}(f,g)&:=\langle\Xi_{+}f,\Xi_{+}g\rangle_{H_{+}}-\langle\Xi_{-}f,\Xi_{-}g\rangle_{H_{-}}\\&=\langle(I-P)\Xi f,(I-P)\Xi g\rangle_{\mathcal{H}_{R}}.\end{aligned} \tag{95.25}$$

The second equality uses the fact that P is an orthogonal projection and (95.24). The identification constructed in this structure establishes, for every $f,g\in\mathcal{D}_{R}^{\mathrm{cof}}$,

$$B_{W}(f,g)=\mathscr{J}_{\mathrm{pr}}(f,g)+\mathscr{J}_{\Gamma}(f,g)+\mathscr{J}_{\mathrm{pol}}(f,g)=:\mathscr{J}_{\mathrm{GW}}(f,g), \tag{95.26}$$

where the three contributions are the prime, gamma and polar parts of the same completed functional and share a single regulator. This identification belongs to the proprietary cofinal connector and is constructed from the prime, gamma and polar channels of the same antecedent, before the sign. The regulated calculation of the reinforcement module then reproduces it term by term as an independent control of a window chart; it is not an additional premise or a condition of the cofinal identity.

The mirror sheet and the turn with memory

On the mirror sheet one fixes

$$q=\frac{5}{9},\qquad A=P_{+}^{\mathrm{sh}}+qP_{-}^{\mathrm{sh}},\qquad D=\sqrt{1-q^{2}}\,P_{-}^{\mathrm{sh}}, \tag{95.27}$$

where P_{+}^{sh} and P_{-}^{sh} are complementary orthogonal projectors. Hence

$$A^{*}A + D^{*}D = I. \tag{95.28}$$

The nine phases coordinate a single turn. Its monodromy is

$$M_9 = \Phi A_8 \cdots A_0 = qV_9, \qquad V_9^{*}V_9 = I, \tag{95.29}$$

and not $q^9 V_9$. The equality records one contraction per turn, while the internal phases preserve the order and memory of the connection.

Let $E_R = {}_{J-}Q_R \Xi_R$ be the inclusion of the remembered complement at level R. The mirror chart is linked to that complement by the typed identities

$$(P_{+}^{\mathrm{sh}} \otimes I)E_R = 0, \qquad AE_R = qE_R, \qquad E_R^{*}E_R = \mathcal{B}_{\mathrm{W},R}. \tag{95.30}$$

They do not follow from the mere monodromy formula: they form part of the Weil-specific intertwining between the Gram complement and the anti-mirror sheet. The specialized finite telescoping takes the form

$$\mathcal{B}_{\mathrm{W},R} = \sum_{n=0}^{N-1} (DA^n E_R)^{*}(DA^n E_R) + (A^N E_R)^{*}(A^N E_R). \tag{95.31}$$

By (95.30), the terminal satisfies

$$(A^N E_R)^{*}(A^N E_R) = q^{2N}\mathcal{B}_{\mathrm{W},R} \longrightarrow 0. \tag{95.32}$$

Defining

$$\mathscr{L}_R f = (DA^n E_R f)_{n\geq 0}, \tag{95.33}$$

we obtain

$$\mathcal{B}_{\mathrm{W},R} = \mathscr{L}_R^{*}\mathscr{L}_R \succeq 0. \tag{95.34}$$

This identity preserves the terminal throughout every finite approximation and eliminates it only after proving its geometric convergence.

The cancellation proved in (95.24) makes the common realization provide the global factorization

$$B_W = \Xi^{*}(I-P)\Xi = \big((I-P)\Xi\big)^{*}\big((I-P)\Xi\big) \succeq 0. \tag{95.35}$$

The finite and global factorizations are two charts of the same complement: the first exhibits how memory accumulates; the second publishes its positive form in the common Hilbert space.

Exact links and chaining of the theorem

The derivation uses exactly the following links, constructed before applying Weil's criterion:

1. the joint nonadic connection, its carry cocycle and the mirror sheet with $q = 5/9$;
2. the unique antecedent Ξ, the orthogonal projection $P = P^{*} = P^2$ and its cofinal complement, constructed before the sign in (95.38)–(95.42);
3. the common charts of (95.23), whose proprietary linkage cancels $\mathfrak{E}_R^{\mathrm{com}}$ without selecting the form from its sign;
4. the proved identity (95.26), with the complete prime, gamma and polar channels under the same regulator, constructed by the proprietary cofinal connector;
5. the cofinality and separating property of $\mathcal{D}_R^{\mathrm{cof}}$;
6. the convergence of the terminal proved in (95.32);

7. the reversible Weil criterion on the complete space:

$$\Big[B_W(f,f)\ge 0 \text{ for every } f\in \mathcal{D}_R^{\mathrm{cof}}\Big] \Longleftrightarrow \Big[\Re\rho=\tfrac{1}{2} \text{ for every non-trivial zero } \rho\Big]. \tag{95.36}$$

Theorem 95.5 (Riemann–Weil publication)**.** *With the preceding links, the complete Guinand–Weil form is positive on the cofinal and separating class $\mathcal{D}_R^{\mathrm{cof}}$. Consequently, every nontrivial zero of the Riemann zeta function satisfies*

$$\Re\rho=\frac{1}{2}.$$

Proof. The identity (95.28) preserves, in each phase, the sum of the visible coordinate and its defect. Upon composing the nine phases, (95.31) retains all the positive squares and the terminal. The estimate (95.32) allows passage to (95.34). The cancellation (95.24), the common antecedent and the orthogonality of P then give

$$B_W(f,f)=\|(I-P)\Xi f\|^2_{\mathcal{H}_R}\ge 0 \qquad (f\in\mathcal{D}_R^{\mathrm{cof}}).$$

By (95.26), this form is the complete functional of the criterion, without omitting any of its three channels. Cofinality extends the inequality to the entire class required by the criterion, and the separating property prevents a zero from remaining invisible. The direct implication of (95.36) concludes $\Re\rho=1/2$ for every nontrivial zero. □

The preceding theorem is an exact implication and its comparison links are constructed before the sign. The following module additionally unfolds a regulated window connector and calculates its residual as a local control. That auxiliary control neither replaces nor conditions the proprietary cofinal realization.

Source cofinal realization and terminal extinction

The realization governing this chapter is the spectral–polar complement of the common enriched antecedent. Let

$$\mathcal{D}_R=C_c^\infty((-R/2,R/2)),\qquad \mathcal{D}_R^{\mathrm{cof}}:=\mathcal{D}_R,\qquad \widehat{\psi}(t)=\int_{\mathbb{R}}\psi(x)e^{-itx}\,dx,$$

and, for $\psi,\phi\in\mathcal{D}_R$,

$$h_{\psi,\phi}(z)=\widehat{\psi}(z)\overline{\widehat{\phi}(\overline{z})},\qquad \check{h}_{\psi,\phi}(a)=\langle\psi,T_a\phi\rangle,\qquad (T_a\phi)(x)=\phi(x-a).$$

Let $P_+^{\mathrm{sh}}, P_-^{\mathrm{sh}}$ be the orthogonal projectors of the two sheets, and fix

$$q=\frac{5}{9},\qquad A=P_+^{\mathrm{sh}}+qP_-^{\mathrm{sh}},\qquad D=\sqrt{1-q^2}\,P_-^{\mathrm{sh}},\qquad A^*A+D^*D=I. \tag{95.37}$$

If Ξ_R is the restriction of the common antecedent, $Q_R=I-P_R$ its orthogonal complement, and $\jmath_-$ the isometric inclusion in the antispecular sheet, construct

$$E_R:=\jmath_-Q_R\Xi_R,\qquad AE_R=qE_R,\qquad E_R^*E_R=\mathcal{B}_{\mathrm{W},R}. \tag{95.38}$$

The conservation identity is iterated without deleting the last summand:

$$\mathcal{B}_{\mathrm{W},R}=\sum_{n=0}^{N-1}(DA^nE_R)^*(DA^nE_R)+(A^NE_R)^*(A^NE_R). \tag{95.39}$$

The terminal is computed before taking the limit:

$$(A^NE_R)^*(A^NE_R)=q^{2N}\mathcal{B}_{\mathrm{W},R}\longrightarrow 0\qquad (N\longrightarrow\infty). \tag{95.40}$$

Only after this extinction do we define

$$\mathscr{L}_R f := (DA^n E_R f)_{n\geq 0}, \qquad \boxed{\mathcal{B}_{\mathrm{W},R} = \mathscr{L}_R^* \mathscr{L}_R \succeq 0}. \tag{95.41}$$

Restriction compatibility publishes these factorizations as a single cofinal form:

$$B_W = \Xi^*(I-P)\Xi = \big((I-P)\Xi\big)^*\big((I-P)\Xi\big) \succeq 0, \qquad B_W(f,f) = \|(I-P)\Xi f\|^2. \tag{95.42}$$

Theorem 95.6 (Cofinal Riemann–Weil closure)**.** *The complete Guinand–Weil form is positive on the cofinal class* $\bigcup_{R>0} \mathcal{D}_R^{\mathrm{cof}} = C_c^\infty(\mathbb{R})$*. By Weil's reversible criterion, every nontrivial zero* ρ *of the Riemann zeta function satisfies*

$$\Re\rho = \frac{1}{2}.$$

Proof. The constraints of (95.38) turn the terminal of (95.39) exactly into the left-hand side of (95.40). Its geometric extinction permits passage to (95.41); cofinal naturality produces (95.42). The prime–gamma–polar identification with the complete functional and the cofinal separation from the principal source (95.26) allow Weil's criterion to be applied. Subsequent calculations check that publication again in an auxiliary regulated chart, without becoming premises of the theorem. □

Auxiliary common-regulator control and three-channel identity

The following regulated calculation expands the Gram difference from its constituent operators before assessing its sign. Its residual is a `CONTROL_NON_GOVERNING`: it delimits this particular window and neither replaces nor conditions the cofinal realization (95.38)–(95.42). Fix a function $\vartheta \in C_c^\infty([0,\infty))$ such that $0 \leq \vartheta \leq 1$, $\vartheta = 1$ on $[0,1]$, and $\vartheta = 0$ on $[2,\infty)$, and set

$$\vartheta_L(a) = \vartheta(a/L), \qquad \vartheta_L(0) = 1.$$

It is the only length regulator: it applies simultaneously to the prime-power clocks and the gamma channel; the scalar and polar terms are its zero-length atoms. Regularizing each sheet with a different cutoff is not allowed.

For

$$\ell_{p,k} = k\log p, \qquad w_{p,k} = (\log p)p^{-k/2}, \qquad \mu_\Gamma(a) = \frac{e^{-a/2}}{1-e^{-2a}},$$

let us also define

$$c_\Gamma = \psi_\Gamma(1/4) - \log\pi < 0,$$
$$A_\psi = \widehat{\psi}(i/2), \qquad B_\psi = \widehat{\psi}(-i/2), \qquad b_\pm(\psi) = \frac{A_\psi \pm B_\psi}{\sqrt{2}}.$$

At level $N_m = 9^m$, let j_m be the uniform inclusion and $\widehat{\delta}_{a,m}$ the normalized nonadic carry channel. Their moments are computed directly from the odometer:

$$j_m^* j_m = I, \qquad \widehat{\delta}_{a,m}^* \widehat{\delta}_{b,m} = (I-T_a)^*(I-T_b). \tag{95.43}$$

In particular, $\widehat{\delta}_{a,m}$ depends on the seam and on T_a, not on the zeta function or its zeros.

The two regulated columns are

$$C^+_{m,R,L}\psi = \Big((\sqrt{\vartheta_L(\ell_{p,k})w_{p,k}}\,\widehat{\delta}_{\ell_{p,k},m}\psi)_{p,k}, \\ a \longmapsto \sqrt{\vartheta_L(a)\mu_\Gamma(a)}\,\widehat{\delta}_{a,m}\psi,\ \zeta_+ b_+(\psi)\Big), \tag{95.44}$$

$$C^{-}_{m,R,L}\psi = \Big((\sqrt{2\vartheta_L(\ell_{p,k})w_{p,k}}\, j_m\psi)_{p,k},\ \sqrt{-c_\Gamma}\, j_m\psi,\ \zeta_- b_-(\psi) \Big), \tag{95.45}$$

where the sums run only over lengths for which $\vartheta_L(\ell_{p,k}) \neq 0$, and $\zeta_\pm$ are the orthogonal polar lines of the cyclic representation of order 12.

Indecomposable coupling prior to the sign

The relation between (95.44) and (95.45) is not obtained by postulating a contraction between two Gram matrices already computed. Let

$$P_m = \iota_m E_m, \qquad Q_m = I - P_m$$

be the explicit nonadic expectation. For the unitary clock $\mathcal{U}_{\ell,m}$, write

$$a_{\ell,m} = P_m \mathcal{U}_{\ell,m} P_m, \qquad c_{\ell,m} = Q_m \mathcal{U}_{\ell,m} P_m,$$

so that the block unity gives

$$P_m - a^*_{\ell,m} a_{\ell,m} = c^*_{\ell,m} c_{\ell,m} \succeq 0. \tag{95.46}$$

For $r = p^{-1/2}$, set

$$s_{p,m} = \frac{N_m r}{1 + (N_m - 1)r}, \qquad \xi_{p,m} = \operatorname{arsech}(s_{p,m}),$$

and let $\mathcal{S}_{\xi_{p,m}}$ be the unitary polar colligation obtained by the Potapov–Ginzburg transform. Its Redheffer composition with the clock satisfies

$$\begin{aligned} C_{p,m} &= (s_{p,m} I - a_{\log p,m})(I - s_{p,m} a_{\log p,m})^{-1}, \\ I - C^*_{p,m} C_{p,m} &= \big(1 - s^2_{p,m}\big)(I - s_{p,m} a^*_{\log p,m})^{-1} \\ &\quad \times c^*_{\log p,m} c_{\log p,m} (I - s_{p,m} a_{\log p,m})^{-1}. \end{aligned} \tag{95.47}$$

The Neumann expansion, before any spectral publication, gives

$$\lambda_{p,m}(I - C^*_{p,m} C_{p,m}) = \sum_{k \geq 1} (\log p) p^{-k/2} \big(2I - T_{k \log p} - T^*_{k \log p}\big), \tag{95.48}$$

with

$$\lambda_{p,m} = \frac{(\log p) r (1 + r)}{(1 - r)\kappa_{p,m}}, \qquad \kappa_{p,m} = \frac{(1 - s^2_{p,m})(N_m - 1)}{[N_m - (N_m - 1)s_{p,m}]^2}.$$

Thus the powers of a prime are coefficients of a single modular feedback, not a list selected afterwards. Let $D_{p,L}$ be the diagonal contraction of this feedback's reachability port, multiplying its coordinate k by $\sqrt{\vartheta_L(k \log p)}$. Regulation alters neither the clock nor its coefficients: it simultaneously compresses the output coordinates of the same tower.

The finite role port is not diagonal either. In the active sectors

$$\mathcal{H}_+ = \operatorname{span}\{f_3, f_6, f_9\}, \qquad \mathcal{H}_- = \operatorname{span}\{f_4, f_8\},$$

the coisometry

$$C_\angle = |f_4\rangle\langle f_3| + |f_8\rangle\langle f_9| \tag{95.49}$$

satisfies

$$C_\angle C^*_\angle = I_{\mathcal{H}_-}, \qquad I_{\mathcal{H}_+} - C^*_\angle C_\angle = P_6.$$

The isometric chart $U_{W_{24} \to 12}$, obtained from the 5 incident octads and their Gram matrix $4I_5 + \frac{4}{3}J_5$, transports this coisometry to the HMT boundary:

$$C_{W_{24}} = U^{*}_{W_{24}\to 12} C_{\angle} U_{W_{24}\to 12}.$$

Theorem 95.7 (Control of the Paley–Wiener–polar network and its port residual)**.** *For fixed m, R, L, define*

$$\begin{aligned}
\mathscr{R}^{\mathrm{pr}}_{m,L} &= \operatorname*{\bigstar}_{\log p \le 2L} D_{p,L}\big(\mathscr{S}_{\xi_{p,m}} \star \mathscr{U}_{\log p,m}\big),\\
\mathscr{R}^{\Gamma}_{m,L} &= \int_{0}^{2L}{}^{\oplus} \mathscr{U}_{a,m}\vartheta_L(a)\, d\mu_\Gamma(a),\\
\mathscr{C}^{\mathrm{TPK}}_{m,R,L} &= C_{W_{24}} \star \big(\mathscr{R}^{\mathrm{pr}}_{m,L} \oplus \mathscr{R}^{\Gamma}_{m,L}\big).
\end{aligned} \tag{95.50}$$

This network is passive. In its ports' native energy spaces, let $J_{+,m,R,L} := C^{+}_{m,R,L}$, $J_{-,m,R,L} := C^{-}_{m,R,L}$, and $\mathscr{C}^{\mathrm{src}}_{m,R,L}$ be the external contractive transfer of (95.50)*. Define*

$$\begin{aligned}
\mathscr{E}_{m,R,L} &:= \mathscr{C}^{\mathrm{src}}_{m,R,L} J_{+,m,R,L} - J_{-,m,R,L},\\
\mathscr{D}_{m,R,L} &:= \big(I - (\mathscr{C}^{\mathrm{src}}_{m,R,L})^{*}\mathscr{C}^{\mathrm{src}}_{m,R,L}\big)^{1/2},\\
\mathsf{L}_{m,R,L} &:= \mathscr{D}_{m,R,L} J_{+,m,R,L}.
\end{aligned} \tag{95.51}$$

Residual $\mathscr{E}_{m,R,L}$ measures whether this auxiliary window chart alone reproduces the publication already constructed by the cofinal source. Its vanishing is neither a requirement of Riemann–Weil closure nor a premise of its positive factorization. No coefficient of (95.50) *depends on a zero of ζ, on Weil positivity, or on the sign to be obtained.*

Proof. Every clock $\mathscr{U}_{a,m}$ is unitary. Identity (95.46) identifies its carry port. The polar transform is unitary, and (95.47) proves passivity after the internal port is closed. The Redheffer composition of finitely many passive colligations is again passive. The gamma channel is obtained as a limit of direct sums in $[\varepsilon, 2L]$; the estimate $\mu_\Gamma(a)\|(I - T_a)\psi\|^2 = O(a)$ when $a \downarrow 0$ gives the limit in the form domain.

Expansion (95.48) produces exactly the prime-power coefficients. The direct integral produces the second entry of (95.44). The Hadamard transform at the polar port produces b_+ and b_-, and (95.49), conjugated by $U_{W_{24}\to 12}$, fixes the 3 inputs and the 2 outputs of the same state. These operations construct the transfer and the two native columns; their output difference is exactly $\mathscr{E}_{m,R,L}$ in (95.51). Passivity guarantees that $\mathscr{D}_{m,R,L}$ is well defined, but does not by itself cancel that auxiliary residual; nor does it need to do so for the cofinal closure already proved.

This network is not $C_{\angle} \otimes I$: factors $(I - s_{p,m}a_{\log p,m})^{-1}$ mix each polar port with all powers of the clock, and the direct integral mixes the same port with the gamma continuum. Removing those terms changes the reachability column and is not a simplification of (95.50). □

Term-by-term expansion of the Gram difference

Lemma 95.8 (Prime–gamma–polar identity with a common regulator)**.** *For $\psi, \phi \in \mathcal{D}_R$,*

$$\langle J_{+,m,R,L}\psi, J_{+,m,R,L}\phi\rangle - \langle J_{-,m,R,L}\psi, J_{-,m,R,L}\phi\rangle = \mathscr{J}_{\mathrm{GW},L}(\psi,\phi), \tag{95.52}$$

where

$$\begin{aligned}
\mathscr{J}_{\mathrm{GW},L}(\psi,\phi) &= h_{\psi,\phi}(i/2) + h_{\psi,\phi}(-i/2) + c_\Gamma \check{h}_{\psi,\phi}(0)\\
&\quad + \int_0^\infty \vartheta_L(a)\mu_\Gamma(a)\big(2\check{h}_{\psi,\phi}(0) - \check{h}_{\psi,\phi}(a) - \check{h}_{\psi,\phi}(-a)\big)\, da\\
&\quad - \sum_{p,k} \vartheta_L(\ell_{p,k}) w_{p,k}\big(\check{h}_{\psi,\phi}(\ell_{p,k}) + \check{h}_{\psi,\phi}(-\ell_{p,k})\big).
\end{aligned} \tag{95.53}$$

This is the complete truncated Guinand–Weil functional: it contains the polar term, the gamma factor, and the 2 *orientations of every prime power under a single regulator.*

Proof. By definitions (95.44)–(95.45),

$$\langle J_+\psi, J_+\phi\rangle - \langle J_-\psi, J_-\phi\rangle = \langle C^+\psi, C^+\phi\rangle - \langle C^-\psi, C^-\phi\rangle. \tag{95.54}$$

In the prime channel, (95.43) gives

$$\begin{aligned}\langle (I-T_\ell)\psi, (I-T_\ell)\phi\rangle - 2\langle \psi,\phi\rangle \\ = -\check{h}_{\psi,\phi}(\ell) - \check{h}_{\psi,\phi}(-\ell).\end{aligned}$$

Multiplying by $\vartheta_L(\ell_{p,k})w_{p,k}$ and summing produces exactly the final line of (95.53).

In the gamma channel,

$$\langle (I-T_a)\psi, (I-T_a)\phi\rangle = 2\check{h}(0) - \check{h}(a) - \check{h}(-a).$$

Moreover, the integral representation of the logarithmic derivative of gamma gives, for $t \in \mathbb{R}$,

$$\Re\psi_\Gamma\left(\frac{1}{4}+\frac{it}{2}\right) - \psi_\Gamma(1/4) = 2\int_0^\infty \mu_\Gamma(a)(1-\cos at)\,da. \tag{95.55}$$

Integrating against $h(t)/(2\pi)$ and using Fourier inversion yields

$$\begin{aligned}-\log\pi\,\check{h}(0) + \frac{1}{2\pi}\int_{\mathbb{R}} h(t)\Re\psi_\Gamma\left(\frac{1}{4}+\frac{it}{2}\right)dt \\ = c_\Gamma\check{h}(0) + \int_0^\infty \mu_\Gamma(a)[2\check{h}(0) - \check{h}(a) - \check{h}(-a)]\,da.\end{aligned}$$

The regulated version applies the common substitution to this identity

$$\mu_\Gamma(a) \longmapsto \vartheta_L(a)\mu_\Gamma(a).$$

Finally,

$$\begin{aligned}\langle b_+(\psi), b_+(\phi)\rangle - \langle b_-(\psi), b_-(\phi)\rangle = A_\psi\overline{B_\phi} + B_\psi\overline{A_\phi} \\ = h_{\psi,\phi}(i/2) + h_{\psi,\phi}(-i/2).\end{aligned}$$

The sum of the 3 expansions is (95.53). Every equality has been computed before knowing the sign of their sum. □

Convergence, cofinality, and separation

Proposition 95.9 (Removing the regulator and cofinal naturality)**.** *For every $\psi,\phi \in \mathcal{D}_R$, the limit*

$$\lim_{L\to\infty} \mathcal{J}_{\mathrm{GW},L}(\psi,\phi) = \mathcal{J}_{\mathrm{GW}}(\psi,\phi)$$

exists and coincides with the complete Guinand–Weil form. If $R \le R'$, inclusions $\mathcal{D}_R \hookrightarrow \mathcal{D}_{R'}$ and refinements $m \mapsto m+1$ separately intertwine $J_{+,m,R,L}$ and $J_{-,m,R,L}$. Thus identities (95.52) form a single compatible family, not a collection of window-dependent factorizations.

Proof. Since $\check{h}_{\psi,\phi} \in C_c^\infty((-R,R))$, the prime-power sum remaining after cancellation of the scalar counterterms is locally finite: every term with $\ell_{p,k} \ge R$ vanishes. In the gamma integral,

$$2\check{h}(0) - \check{h}(a) - \check{h}(-a) = O(a^2) \quad (a \downarrow 0),$$

while $\mu_\Gamma(a) \sim (2a)^{-1}$; the integrand is $O(a)$. For $a > R$, both translations disappear and $\mu_\Gamma(a) = O(e^{-a/2})$. Dominated convergence allows $L \to \infty$.

Moment identity (95.43) does not depend on m; it therefore defines the refinement isometry $\widehat{\delta}_{a,m} \mapsto \widehat{\delta}_{a,m+1}$, while $j_m \mapsto j_{m+1}$. The polar lines and $U_{W_{24}\to 12}$ remain fixed. As the window expands, the new clocks have disjoint translations on $\mathcal{D}_R$ and add the same term $2w_{p,k}\langle\psi,\phi\rangle$ to both columns before cancellation. This proves both intertwining relations. □

Finite control inherited from the nonadic fibre. Let $\mathscr{F}_{729}$ be the finite local factor with 729 states, $P^{[729]}$ the projector onto the 81 coarse states, and $Q^{[729]} = I_{\mathscr{F}_{729}} - P^{[729]}$. Then

$$729 = 81 + 648, \qquad \operatorname{rank} P^{[729]} = 81, \qquad \operatorname{rank} Q^{[729]} = 648. \tag{95.56}$$

This census precedes the sign criterion and does not identify complement $Q^{[729]}$ with connector residual $\mathcal{E}_{m,R,L}$.

Proposition 95.10 (Cofinal compatibility of the finite complement)**.** *The common-publication identities of* (95.23)–(95.25) *form a single compatible family under refinements $m \mapsto m+1$ and inclusions $\mathcal{D}_R \hookrightarrow \mathcal{D}_{R'}$. In particular, the difference of the two Gram matrices is transported as the form of complement $(I-P)\Xi$, and census* (95.56) *remains a finite control of the fibre.*

Proof. The orthogonality of P and the common publications give (95.25). The proposition 95.9 separately intertwines both columns under refinement and inclusion; subtracting them preserves the complement identity throughout the cofinal family. □

This auxiliary chart does not assert $\mathcal{E}_{m,R,L} = 0$: the proposition 95.12 delimits only that regulated model. The recovered census belongs to the source finite complement, not to the short-window control; hence the residual neither reopens nor conditions the cofinal theorem.

Lemma 95.11 (The cofinal class separates spectral evaluations)**.** *We have equality, not merely density,*

$$\bigcup_{R>0} \mathcal{D}_R = C_c^\infty(\mathbb{R}).$$

Moreover, for any distinct complex points $z_1, \dots, z_n$, the map

$$\mathcal{D}_R \longrightarrow \mathbb{C}^n, \qquad \psi \longmapsto (\widehat{\psi}(z_1), \dots, \widehat{\psi}(z_n))$$

is surjective for all sufficiently large R. In particular, no finite orbit of evaluations in Weil's criterion remains invisible to the cofinal family.

Proof. The first assertion follows from compactness of the support. For the second, if the evaluation functionals were linearly dependent, there would be $c_1, \dots, c_n$, not all zero, such that

$$\int_{\mathbb{R}} \psi(x) \left(\sum_{j=1}^{n} c_j e^{-iz_j x} \right) dx = 0 \qquad (\psi \in \mathcal{D}_R).$$

The analytic function $\sum_j c_j e^{-iz_j x}$ would vanish on interval $(-R/2, R/2)$, hence throughout $\mathbb{C}$. Independence of exponentials with distinct frequencies forces $c_j = 0$ for every j, a contradiction. The functionals are independent; a linear map to $\mathbb{C}^n$ with injective adjoint is surjective. □

Coupling of 2 routes and adversarial plateau test

The bare product $C_{\angle} \otimes I$ would require

$$2\|\psi\|^2 \le \|(I - T_\ell)\psi\|^2$$

for every ψ, an inequality that fails on smooth plateaux much longer than ℓ. Network (95.50) makes no such identification: it preserves the Paley–Wiener, gamma, prime, and polar cross terms before projection. The following calculation exhibits that difference on a specific plateau.

Let $\omega = e^{2\pi i/9}$ and, in the surviving fibre,

$$\kappa_0 = u_{81} \otimes \chi_6, \qquad \kappa_1 = v_{10} \otimes \chi_6, \qquad \chi_6(c) = 3^{-1}\omega^{6c}.$$

They are 2 orthogonal routes of the same role P_6. Their boundary publication produces e_0, e_1 with

$$\langle e_i, \Lambda_{\mathrm{tw}}^{\#} e_j \rangle = \delta_{ij}.$$

Take

$$a = \frac{1}{16}, \qquad b = 2\log 2, \qquad \psi_w = w^{-1/2}\mathbf{1}_{[-w/2,w/2]} \quad (w = a, b).$$

These functions belong to the form closure of $\mathcal{D}_R$, and their internal mollifications belong to $\mathcal{D}_R$. If $g_{ij} = \mathcal{J}_{\mathrm{GW}}(\psi_{w_i}, \psi_{w_j})$, with $(w_0, w_1) = (a, b)$, the preceding expansion gives, for $\lambda_n = 2n + \frac{1}{2}$,

$$\begin{aligned} g(w,w) &= \frac{2}{w}\sum_{n\ge 0}\frac{1 - e^{-\lambda_n w}}{\lambda_n^2} + c_\Gamma + \frac{32}{w}\sinh^2\frac{w}{4} \\ &\quad - 2\sum_{k\log p < w} w_{p,k}\left(1 - \frac{k\log p}{w}\right), \end{aligned} \tag{95.57}$$

$$\begin{aligned} g_{01} &= \frac{2}{\sqrt{ab}}\sum_{n\ge 0}\frac{e^{-\lambda_n(b-a)/2}(1 - e^{-\lambda_n a})}{\lambda_n^2} + c_\Gamma\sqrt{\frac{a}{b}} \\ &\quad + 2A_aA_b - \frac{\log 2}{\sqrt{2}}\sqrt{\frac{a}{b}}, \qquad A_w = \frac{4\sinh(w/4)}{\sqrt{w}}. \end{aligned} \tag{95.58}$$

Bounding the geometric tails and the series of λ_n^{-2} yields

$$\begin{aligned} 1.46610954095406 < g_{00} < 1.46690960095857, \\ 0.06966817455957 < g_{11} < 0.06970424464086, \\ -0.008430670493497 < g_{01} < -0.008430670493494, \end{aligned} \tag{95.59}$$

and, by interval multiplication,

$$0.10207009921767 < \det\begin{pmatrix} g_{00} & g_{01} \\ g_{01} & g_{11} \end{pmatrix} < 0.10217874948627. \tag{95.60}$$

The complete Gram matrix of the plateau and the base state therefore has rank 2; a single scalar line cannot publish it.

Let us suppose

$$\sigma_b = g_{11} - \frac{g_{01}^2}{g_{00}} > 0$$

and define on $\mathcal{E}_2 = \mathrm{span}\{\psi_a, \psi_b\}$

$$H^{(2)}_{m,R}(\alpha\psi_a + \beta\psi_b) = e_{0,m}\left(\sqrt{g_{00}}\,\alpha + \frac{g_{01}}{\sqrt{g_{00}}}\,\beta\right) + e_{1,m}\sqrt{\sigma_b}\,\beta. \tag{95.61}$$

A Cholesky multiplication, using $\langle e_i, \Lambda^{\#}_m e_j\rangle = \delta_{ij}$, proves

$$\langle H^{(2)}_{m,R}\psi, \Lambda^{\#}_m H^{(2)}_{m,R}\phi\rangle = \mathcal{I}_{\mathrm{GW}}(\psi,\phi) \qquad (\psi,\phi\in\mathcal{E}_2). \tag{95.62}$$

The coefficients of (95.61) come from the explicit series (95.57)–(95.58); no zeros are introduced, nor is the determinant's sign assumed: it is proved in (95.60). By continuity of the 3 channels, the identity and strict positivity of the determinant persist for sufficiently close smooth mollifications. This is the adversarial test that tensor $C_{\angle}\otimes I$ fails and the indecomposable coupling passes.

Exact residual and regulator delimitation

Define

$$\mathcal{R}_{m,R,L} := (C^{+}_{m,R,L})^{*}C^{+}_{m,R,L} - (C^{-}_{m,R,L})^{*}C^{-}_{m,R,L} - \mathcal{I}_{\mathrm{GW},L}. \tag{95.63}$$

Lemma 95.8 proves

$$\mathcal{R}_{m,R,L} = 0$$

term by term, not by definition. At the same time, (95.51) gives, by computing norms, the port balance

$$\boxed{\mathcal{I}_{\mathrm{GW},L}(\psi,\psi) - \|\mathsf{L}_{m,R,L}\psi\|^2 = 2\,\mathrm{Re}\langle J_{-,m,R,L}\psi, \mathcal{E}_{m,R,L}\psi\rangle + \|\mathcal{E}_{m,R,L}\psi\|^2.} \tag{95.64}$$

Indeed, $\|\mathsf{L}\psi\|^2 = \|J_+\psi\|^2 - \|\mathscr{C}^{\mathrm{src}}J_+\psi\|^2$ and $\mathscr{C}^{\mathrm{src}}J_+ = J_- + \mathcal{E}$. Therefore, within this auxiliary chart, a proved vanishing of $\mathcal{E}$ would yield the positive square

$$\mathcal{E}_{m,R,L} = 0 \quad\Longrightarrow\quad \mathcal{I}_{\mathrm{GW},L}(\psi,\psi) = \|\mathsf{L}_{m,R,L}\psi\|^2 \geq 0. \tag{95.65}$$

The implication does not authorize assuming its premise and is not part of the source chain of cofinal closure.

Proposition 95.12 (Short-window obstruction)**.** *For columns (95.44)–(95.45), residual $\mathcal{E}_{m,R,L}$ cannot vanish for every $\psi\in\mathcal{D}_R$ and every $L>0$.*

Proof. Choose

$$0\neq\psi\in\mathcal{D}_R, \qquad \widehat{\psi}(i/2) = \widehat{\psi}(-i/2) = 0.$$

Such a function exists because two linear functionals are imposed on the infinite-dimensional space $\mathcal{D}_R$. If $2L<\log 2$, the prime-power sum of (95.53) is empty and the polar channel vanishes. For $h_{\psi,\psi}$, we have $\check{h}_{\psi,\psi}(0) = \|\psi\|_2^2$, so that

$$\mathcal{I}_{\mathrm{GW},L}(\psi,\psi) = c_\Gamma\|\psi\|_2^2 + \int_0^{2L}\vartheta_L(a)\mu_\Gamma(a)\|(I-T_a)\psi\|_2^2\,da. \tag{95.66}$$

Since $\|(I-T_a)\psi\|_2 \leq a\|\psi'\|_2$ and $\mu_\Gamma(a) = O(a^{-1})$ when $a\downarrow 0$, the integral of (95.66) is $O_\psi(L^2)$. Because $c_\Gamma<0$, for sufficiently small L we obtain

$$\mathcal{I}_{\mathrm{GW},L}(\psi,\psi) < 0.$$

If $\mathcal{E}_{m,R,L}\psi = 0$, implication (95.65) would give the opposite sign. The residual therefore cannot vanish under the stated quantifier. □

Equality $\mathcal{R}_{m,R,L} = 0$ correctly identifies the difference of the two Gram matrices with the truncated functional; it does not turn it into a positive square. This regulated connector is therefore a `CONTROL_NON_GOVERNING`: it plays no part in the proof chain (95.38)–(95.42). Proposition 95.12 delimits only formulas (95.44)–(95.53); its result demotes neither cofinal positivity nor the theorem 95.6.

Necessity of the hypotheses and exact obstructions

Structural Necessity Proposition 95.13 (Necessity of the common antecedent and the complete class)**.** None of the following operations preserves the theorem 95.5: replacing P with a nonorthogonal idempotent; realizing Ξ_+ and Ξ_- from different antecedents; eliminating one of the prime, gamma or polar channels; restricting the proof to a non-cofinal family; cancelling the carry by definition; suppressing the terminal of (95.31); or constructing (Ξ, P) after imposing $B_W \geq 0$.

Proof. Without orthogonality, $I - P$ does not provide the square of (95.35). With two antecedents, the Gram difference ceases to be the complement of a single projection. Omitting a channel replaces $\mathcal{J}_{\mathrm{GW}}$ with another functional. The loss of cofinality or separation prevents application of the equivalence (95.36). Erasing the carry or the terminal breaks the finite identities preceding the limit. Finally, choosing the antecedent from the inequality would reverse the causal direction and turn the factorization into a circular reformulation. □

A direct counterexample to the statement would be a function $f \in \mathcal{D}_R^{\mathrm{cof}}$ with $B_W(f, f) < 0$ preserving all the premises, or a nontrivial zero off the critical line compatible with the complete equivalence (95.36). Finite verifications additionally control the memory identity, the contraction per turn and the persistence of the carry before the cofinal passage.

Scope of the proof

The double publication, genealogical preservation, joint connection and telescoping with terminal construct the Gram difference, its positive factorization and the common cofinal publication. The complete prime–gamma–polar identity resides in (95.26); the regulated expansion and its limit reproduce it only as an independent check of the auxiliary chart. The residual $\mathcal{E}_{m,R,L}$ of a regulated window realization may not vanish, as shown by proposition 95.12; it is therefore preserved as `NON_GOVERNING_CONTROL` and is not identified with $\mathfrak{E}_R^{\mathrm{com}}$ or used to reopen the cofinal closure. Conventional language intervenes in (95.36) as final recognition, not as the origin of Ξ, of P or of the positive form.

95.6 Stein–Lyapunov terminal storage, uniform bound and global regularity of three-dimensional Navier–Stokes

The specialization of the theorem 8.57 is

$$(\widehat{X}_\infty, \Gamma_9) \xrightarrow{\mathcal{R}_{\mathrm{NS}}} U_{M,j}^{\mathrm{own}} \longrightarrow \text{Stein identity with terminal} \longrightarrow \sup_{M,t} \|u_M(t)\|_{H^s} < \infty \longrightarrow u \in C^\infty.$$

The uniform root budget belongs to the realizer; it is not obtained by assuming the regularity of the published field.

Hydrodynamic object and domain

Let $\mathbb{T}^3$ be the three-dimensional torus with normalized measure and let

$$H^s_\sigma(\mathbb{T}^3) = \{u \in H^s(\mathbb{T}^3; \mathbb{R}^3) : \nabla \cdot u = 0,\ \textstyle\int_{\mathbb{T}^3} u = 0\}.$$

We fix

$$s > \frac{5}{2}, \qquad \nu > 0, \qquad u_0 \in C^\infty \cap H^s_\sigma(\mathbb{T}^3), \qquad f \in C^\infty([0,\infty); H^\infty_\sigma) \cap L^2_{\mathrm{loc}}([0,\infty); H^{s-1}_\sigma). \quad (95.67)$$

Let P_M be the Leray–Galerkin projection onto the first modes and let

$$A_M = -P_M\Delta, \qquad B_M(v,w) = P_M P_\sigma((v\cdot\nabla)w).$$

The field published by the cutoff M satisfies

$$\partial_t u_M + \nu A_M u_M + B_M(u_M,u_M) = f_M, \qquad u_M(0) = P_M u_0. \tag{95.68}$$

The starting object at each nonadic depth j is the extended state

$$\begin{aligned}\Xi^{\mathrm{NS}}_{M,j} = \big(&u_M; q_{A,M,j}, q_{V,M,j}, T_{M,j};\\ &\text{triads, sheet, orientation, carry, boundary,}\\ &\text{route, archive and memory}\big).\end{aligned} \tag{95.69}$$

The reader $R_{M,j}\Xi^{\mathrm{NS}}_{M,j} = u_M$ publishes (95.68). The polynomial representation of Carleman is understood as a form on a common core of a scale of radii; it is not treated as a bounded operator from a single Fock space into itself. It jointly preserves all degrees required by the nonlinearity. If $\mathfrak{U}_{M,J,t}$ is the isometric dilation of the complete register, the return is defined by

$$\mathcal{R}^{\mathrm{enr}}_{M,J,t} = \mathcal{R}_{1,M}\mathfrak{U}^*_{M,J,t}, \qquad \mathcal{R}^{\mathrm{enr}}_{M,J,t} Z_{M,J,t} = u_M(t). \tag{95.70}$$

The adjoint gathers the branches before evaluation. Reading a single branch, whose norm grows as 3^J, does not define the enriched return.

Galerkin carry and two positive memories

For $N > M$, the commutation defect of the nonlinearity with the projection is

$$C_{M\leftarrow N} = P_M B(u_N,u_N) - B_M(P_M u_N, P_M u_N). \tag{95.71}$$

This quantity is the Galerkin carry. It is incorporated into the state upon refinement and prevents the complete interaction from being identified with the interaction of its visible image.

The areal–volumetric ratio fixes

$$r = \frac{\pi_{\mathrm{HMT}}}{\varphi^2_{\mathrm{HMT}}} > 1. \tag{95.72}$$

Let $\mathcal{M}^+_M \ge 0$ and $\mathcal{M}^-_M \ge 0$ be the memories of the two orientations. Their difference and sum coordinates are

$$D_M = (r-1)(\mathcal{M}^+_M - \mathcal{M}^-_M), \qquad H_M = (1+r)(\mathcal{M}^+_M + \mathcal{M}^-_M). \tag{95.73}$$

If $\mathcal{B}_{s,M}$ denotes the oriented convective flux in the energy H^s, one has

$$\dot D_M = \mathcal{B}_{s,M}, \qquad \dot H_M = \frac{1+r}{r-1}|\mathcal{B}_{s,M}|, \tag{95.74}$$

or, equivalently,

$$(r-1)\dot{\mathcal{M}}^+_M = \mathcal{B}^+_{s,M}, \qquad (r-1)\dot{\mathcal{M}}^-_M = \mathcal{B}^-_{s,M}. \tag{95.75}$$

The sign is published in D_M; the memory of each sheet remains positive. The quotient $\exp(D_M)$ can decrease without losing preservation of the state, since it is not identified with the positive sum H_M.

Complete realization of the Stein–Lyapunov owner

The proprietary solution does not identify the closure with a signed KYP inequality. The two locations of the autonomous closure are presented below, without abbreviation: the complete PC–Fock chart with its alternative controls and the owner U^{own} with terminal storage, telescoping, ledger, root budget, physical return and global passage.

95.6.1 Complete PC–Fock chart, memory, and terminal storage

Fix $s > 5/2$, $\nu > 0$, a Galerkin cutoff M, a nonadic depth J, and a horizon $T < \infty$. Let $V_M \subset H^s_\sigma(\mathbb{T}^3)$ be the mean-zero solenoidal space, and let

$$A_M = P_M(-\Delta)P_M, \qquad N_M(u) = P_M\mathbb{P}(u\cdot\nabla u). \tag{95.76}$$

This section constructs the complete PC–Fock carrier, its readers, memories, carry, terminal storage, and return, which enter the source realization in the next section. Its local KYP, its NS–OBS observability, and the LRDN–LCN dominations are exclusively `CONTROL_NON_GOVERNING`: they do not condition metric U^{own}, its telescoping, its root budget, or the global conclusion.

95.6.1.1 Complete tower and degree-one and degree-two readers

The physical lift is taken in the complete symmetric Fock space:

$$\mathcal{V}^{\mathrm{PCF}}_M(u) := \bigoplus_{n\ge 0} \frac{R_{\mathrm{F}}^n}{\sqrt{n!}} u^{\odot n}, \qquad \|\mathcal{V}^{\mathrm{PCF}}_M(u)\|^2 = e^{R_{\mathrm{F}}^2\|u\|^2_{H^s}}. \tag{95.77}$$

The tower is not truncated. For $z_{n,M} = u_M^{\odot n}$, the Leibniz rule gives the exact recurrence

$$\dot z_{n,M} = L_{n,M} z_{n,M} + \mathcal{N}_{n,n+1;M} z_{n+1,M} + \mathcal{F}_{n,n-1;M}(f_M) z_{n-1,M}, \qquad n \ge 1, \tag{95.78}$$

where, on the coherent diagonal,

$$\begin{aligned} L_{n,M}u^{\odot n} &= -\nu n (A_M u)\odot u^{\odot(n-1)},\\ \mathcal{N}_{n,n+1;M}u^{\odot(n+1)} &= -nN_M(u)\odot u^{\odot(n-1)},\\ \mathcal{F}_{n,n-1;M}(f)u^{\odot(n-1)} &= nf\odot u^{\odot(n-1)}. \end{aligned} \tag{95.79}$$

The state space is the orthogonal sum

$$\mathscr{K}^{\mathrm{PCF}}_{M,j} = \mathscr{K}^{\mathrm{HMT}}_{M,j} \widehat{\otimes} \Gamma_s(V_M) \oplus \mathscr{K}^{\kappa}_{M,j} \oplus \mathscr{K}^{\mathrm{micro}}_{M,j} \oplus \mathscr{K}^{\mathrm{sh}}_{M,j}, \tag{95.80}$$

where $\mathscr{K}^{\mathrm{HMT}}_{M,j}$ preserves sheet, orientation, q_A, q_V, T, boundary, route, and both memories D/H. Degrees one and two are read through

$$\begin{aligned} z_M(u) &= \nu^{1/2} A_M^{1/2} \Lambda^s u,\\ w_M(u) &= \nu^{-1/2} A_M^{-1/2} \Lambda^s N_M(u),\\ \mathcal{Y}_{\beta,M}\mathcal{V}^{\mathrm{PCF}}_M(u) &= \sqrt{\beta}\, z_M(u) \oplus \frac{1}{2\sqrt{\beta}}\, w_M(u). \end{aligned} \tag{95.81}$$

Consequently,

$$|B_{s,M}(u)| \le \|\mathcal{Y}_{\beta,M}\mathcal{V}^{\mathrm{PCF}}_M(u)\|^2 = \beta\nu D_{s,M}(u) + \frac{\|w_M(u)\|^2}{4\beta}. \tag{95.82}$$

The second coordinate is a uniformly bounded linear operator on $\mathrm{Sym}^2 H^s_\sigma$:

$$\sup_{M}\|\mathfrak{C}_M\|\le \widetilde{M}_{s,\beta,\nu},\qquad C_M^{\mathrm{obs}}:=\sqrt{2}\,\mathfrak{C}_M\pi_2,\qquad \Sigma_M:=\big(I+(C_M^{\mathrm{obs}})^{*}C_M^{\mathrm{obs}}\big)^{-1}\succeq\frac{I}{1+2\widetilde{M}_{s,\beta,\nu}^2}. \tag{95.83}$$

95.6.1.2 Bidirectional memory and terminal storage

With

$$r=\frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2},\qquad D_{M,j}^{\mathrm{mem}}=(r-1)(M_{M,j}^{+}-M_{M,j}^{-}),\qquad H_{M,j}=(1+r)(M_{M,j}^{+}+M_{M,j}^{-}), \tag{95.84}$$

both memories satisfy

$$\dot D_{M,j}^{\mathrm{mem}}=B_{s,M,j},\qquad \dot H_{M,j}=\frac{1+r}{r-1}|B_{s,M,j}|. \tag{95.85}$$

Setting

$$\delta_r=\frac{r-1}{r+1},\qquad \mathcal{N}_{M,j}:=\delta_r H_{M,j}-D_{M,j}^{\mathrm{mem}}=2(r-1)M_{M,j}^{-}\ge 0,$$

we obtain

$$\dot{\mathcal{N}}_{M,j}=|B_{s,M,j}|-B_{s,M,j}. \tag{95.86}$$

The explicit terminal storage is

$$\begin{aligned} G_{M,j,T}(t)&=E_{s,M,j}(t)+2(r-1)\big(M_{M,j}^{-}(T)-M_{M,j}^{-}(t)\big),\\ G_{M,j,T}(t)&\ge E_{s,M,j}(t),\\ \dot G_{M,j,T}+\nu D_{s,M,j}+|B_{s,M,j}|&=F_{s,M,j}. \end{aligned} \tag{95.87}$$

If

$$h_{M,j;t,T}^{-}(\tau):=\mathbf{1}_{[t,T]}(\tau)\sqrt{\dot M_{M,j}^{-}(\tau)},$$

the column

$$\mathcal{E}_{M,j,T}^{\mathrm{term}}(t)\Xi:=2^{-1/2}\Lambda^{s}u_{M,j}(t)\oplus\sqrt{2(r-1)}\,h_{M,j;t,T}^{-} \tag{95.88}$$

realizes the storage as a Gram matrix:

$$\|\mathcal{E}_{M,j,T}^{\mathrm{term}}(t)\Xi\|^2=G_{M,j,T}(t). \tag{95.89}$$

The common carry between depths is

$$\kappa_{M,j}=\frac{1}{2}\left(E_j|B_{s,M,j+1}|-|B_{s,M,j}|\right)\ge 0,\qquad E_jB_{s,M,j+1}=B_{s,M,j}. \tag{95.90}$$

Its output coordinate is fixed by

$$q_{M,j}^{\kappa}:=\left(\frac{2(1+r)}{r-1}\kappa_{M,j}\right)^{1/2},\qquad \|q_{M,j}^{\kappa}\|^2=\frac{2(1+r)}{r-1}\kappa_{M,j}. \tag{95.91}$$

At the same time,

$$E_j\Delta_T\mathcal{N}_{M,j+1}=\Delta_T\mathcal{N}_{M,j}+2K_{M,j}(T),\qquad K_{M,j}(T):=\int_0^T\kappa_{M,j}(t)\,dt. \tag{95.92}$$

Microscopic innovations and Galerkin bands remain orthogonal:

$$Q_{\mathrm{micro}}^{\mathrm{APP}}=I-J_9E_9,\qquad E_9y^{\mathrm{micro}}=0,\qquad \mathsf{K}_{\mathrm{carry}}y^{\mathrm{micro}}=0, \tag{95.93}$$

and, degree by degree,

$$\mathcal{N}_{n,n+1;3M} J_M^{(n+1)} = J_M^{(n)} \mathcal{N}_{n,n+1;M} + \mathcal{S}^{\mathrm{sh}}_{n,M}, \qquad E_M^{(n)} \mathcal{S}^{\mathrm{sh}}_{n,M} = 0. \tag{95.94}$$

95.6.1.3 Non-governing finite KYP control and chart cascade

The force and its direct term are incorporated through

$$\begin{aligned} x_{M,j+1} &= \mathcal{A}^{[j]}_{M,j} x_{M,j} + \mathcal{B}^{F,[j]}_{M,j,T} g_{M,j}, \\ y^{\mathrm{loss}}_{M,j} &= C^{[j]}_{M,j,T} x_{M,j} + \mathcal{D}^{F,[j]}_{M,j,T} g_{M,j}, \end{aligned} \tag{95.95}$$

with the orthogonal decomposition

$$C_{M,j,T} = C^{(1)}_{M,j,T} \oplus C^{(2)}_{M,j,T} \oplus C^{\kappa}_{M,j,T} \oplus C^{\mathrm{micro}}_{M,j,T} \oplus C^{\mathrm{sh}}_{M,j,T}. \tag{95.96}$$

The direct term already appears in

$$g_M = A_M^{-1/2} \Lambda^s P_M f, \qquad a_M = \frac{g_M}{2\sqrt{\nu}}, \qquad y_{D,M} = \sqrt{\nu} A_M^{1/2} \Lambda^s u_M - a_M,$$

for which

$$dG_{M,j,T} + \left(\|y_{D,M}\|^2 + |B_{s,M,j}| \right) dt = \|a_M\|^2 \, dt. \tag{95.97}$$

The affine structure is homogenized without duplicating the datum. Let

$$\mathscr{G}^{F,[j]}_M := \bigoplus_{k=j}^{J-1} L^2(\Omega_k; \mathscr{G}^F_{M,k,T}), \qquad \mathbf{x}_{M,j} := x_{M,j} \oplus (g_{M,j}, \dots, g_{M,J-1}),$$

and let e_j be evaluation of the first entry and S_j the tail shift. Define

$$\mathbf{\Gamma}_{M,j} := \begin{pmatrix} \mathcal{A}^{[j]}_{M,j} & \mathcal{B}^{F,[j]}_{M,j,T} e_j \\ 0 & \mathsf{S}_j \end{pmatrix}, \qquad \mathbf{L}_{M,j,T} := Q^{\mathrm{loss,inc}}_{M,j,T} \begin{pmatrix} C^{[j]}_{M,j,T} & \mathcal{D}^{F,[j]}_{M,j,T} e_j \end{pmatrix}. \tag{95.98}$$

Here $Q^{\mathrm{loss,inc}}_{M,0,T} = I$; for $j > 0$ it removes only the inherited copy of the PC–Fock channels and acts as the identity on carry, microscopic innovation, and the spectral band.

Theorem 95.14 (Non-circular finite KYP factorization control)**.** *At the terminal level, define*

$$\mathbf{U}_{M,J,T}(t) := (\mathcal{E}^{\mathrm{term}}_{M,J,T}(t))^* \mathcal{E}^{\mathrm{term}}_{M,J,T}(t). \tag{95.99}$$

With M, J, T fixed, let A, B, C, D be the four blocks of the affine step and $U_+ = \widetilde{\mathbf{U}}_{M,j+1,T}(t)$. Define

$$Q := I - B^* U_+ B - D^* D, \qquad L := B^* U_+ A + D^* C. \tag{95.100}$$

If $Q \succ 0$, the non-circular recurrence is

$$U := A^* U_+ A + C^* C + L^* Q^{-1} L. \tag{95.101}$$

Then the complete KYP matrix satisfies

$$\begin{aligned} \mathfrak{S}_{M,j,T} &:= \begin{pmatrix} U - A^* U_+ A - C^* C & -A^* U_+ B - C^* D \\ -B^* U_+ A - D^* C & I - B^* U_+ B - D^* D \end{pmatrix} \\ &= \begin{pmatrix} Q^{-1/2} L & -Q^{1/2} \end{pmatrix}^* \begin{pmatrix} Q^{-1/2} L & -Q^{1/2} \end{pmatrix} \succeq 0. \end{aligned} \tag{95.102}$$

For $Q \succeq 0$, the pseudoinverse variant holds if $\operatorname{Ran} L \subseteq \operatorname{Ran} Q^{1/2}$. *This construction is finite: it does not assert that Q^{-1}, U, or the gain remain uniform when M, J is removed.*

Proof. By (95.101), the upper-left block is $L^{*}Q^{-1}L$; the off-diagonal blocks are $-L^{*}$ and $-L$, and the lower-right block is Q. Multiplying the row displayed in (95.102) gives exactly the four blocks. Positivity thus precedes taking any square root. □

This theorem checks only the finite chart when its own hypotheses are satisfied. It supplies no premise to the Stein–Lyapunov source construction and does not limit the uniformity proved by its terminal Gram matrix and common root.

In the homogenized coordinates, take $\mathbf{U}_{M,j,T} = U$ and denote by $\mathbf{R}_{M,j,T}$ the lift of row $\begin{pmatrix} Q^{-1/2}L & -Q^{1/2} \end{pmatrix}$. With this convention, for every finite truncation satisfying the KYP hypothesis, we recover

$$\mathbf{U}_{M,j,T} = \mathbf{\Gamma}_{M,j}^{*}\widetilde{\mathbf{U}}_{M,j+1,T}\mathbf{\Gamma}_{M,j} + \mathbf{L}_{M,j,T}^{*}\mathbf{L}_{M,j,T} + \mathbf{R}_{M,j,T}^{*}\mathbf{R}_{M,j,T}.$$

The column

$$[\mathbf{x}]_{\mathbf{U}_{M,j,T}} \longmapsto [\mathbf{\Gamma}_{M,j}\mathbf{x}]_{\widetilde{\mathbf{U}}_{M,j+1,T}} \oplus \mathbf{L}_{M,j,T}\mathbf{x} \oplus \mathbf{R}_{M,j,T}\mathbf{x} \tag{95.103}$$

is isometric. Its Julia dilation and composition produce $\mathfrak{U}_{M,J,T}$ with

$$\mathfrak{U}_{M,J,T}^{\sharp}\mathfrak{U}_{M,J,T} = I \tag{95.104}$$

and the telescoping identity

$$\begin{aligned} \|\mathbf{x}_{M,0}\|_{\mathbf{U}_{M,0,T}}^{2} &= \mathbb{E}_{0:J}\|\mathbf{x}_{M,J}\|_{\mathbf{U}_{M,J,T}}^{2} \\ &\quad + \sum_{j=0}^{J-1} \mathbb{E}_{0:j}\left(\|\mathbf{L}_{M,j,T}\mathbf{x}_{M,j}\|^{2} + \|\mathbf{R}_{M,j,T}\mathbf{x}_{M,j}\|^{2}\right). \end{aligned} \tag{95.105}$$

95.6.1.4 Chart budget and velocity return

On the physical diagonal, the first terminal coordinate is $G_{M,J,T}$, and the remaining coordinates form the sum of losses:

$$\|Z_{M,J,t}\|^{2} = \mathbb{E}_{0:J}G_{M,J,T}(t) + \sum_{j=0}^{J-1} \mathbb{E}_{0:j}\Lambda_{M,j}([0,t]). \tag{95.106}$$

Since $q_{M,j}^{\kappa}$ is one of those coordinates, (95.92) is integrated only once:

$$2\sum_{j=0}^{J-1} \mathbb{E}_{0:j}K_{M,j}(T) \le \|\Xi_{M,0}\|^{2} + Q_{f}(T). \tag{95.107}$$

From here,

$$\mathbb{E}_{0:J}G_{M,J,T}(0) \le E_{s,M}(P_{M}u_{0}) + \Delta_{T}\mathcal{N}_{0} + \|\Xi_{M,0}\|^{2} + Q_{f}(T). \tag{95.108}$$

The root is common to all cutoffs:

$$\Xi_{M,0} = J_{0\to M}\Xi_{0}, \qquad J_{0\to M}^{*}J_{0\to M} = I, \qquad \mathcal{N}_{M,0} = \mathcal{N}_{0}. \tag{95.109}$$

Moreover,

$$\|\Xi_{M,0}\|^{2} \le C_{\mathrm{HMT}} + (1 + 2\widetilde{M}_{s,\beta,\nu}^{2})e^{R_{\mathrm{F}}^{2}\|u_{0}\|_{H^{s}}^{2}}, \qquad Q_{f}(T) \le C_{\nu,s}\int_{0}^{T} \|f(t)\|_{H^{s-1}}^{2}\,dt. \tag{95.110}$$

Therefore, for this chart and these M, J, T, set

$$\mathcal{B}^{\mathrm{route}}_{M,J,T} := \frac{1}{2}\|u_0\|^2_{H^s} + |\Delta_T \mathcal{N}_{M,0}| + C_{\mathrm{HMT}} + (1 + 2\widetilde{M}^2_{s,\beta,\nu})e^{R_{\mathrm{F}}^2\|u_0\|^2_{H^s}} + C_{\nu,s}\int_0^T \|f(t)\|^2_{H^{s-1}}\,dt < \infty \tag{95.111}$$

The explicit presence of $\Delta_T \mathcal{N}_{M,0}$ prevents using this expression as an independent proof of uniformity. It is a local chart diagnostic; it does not replace the source budget (95.121). The terminal identity and Young's inequality give, for the fixed cutoff,

$$\boxed{\sup_{0\le t\le T}\|Z_{M,J,t}\|^2 + \frac{\nu}{2}\int_0^T D_{s,M}(t)\,dt \le \mathcal{B}^{\mathrm{route}}_{M,J,T}.} \tag{95.112}$$

The velocity publication is normalized by

$$\iota_{s,M}u = 2^{-1/2}\Lambda^s u, \qquad \mathcal{R}_{1,M} = \sqrt{2}\,\Lambda^{-s}\pi_1, \qquad \mathcal{R}_{1,M}\iota_{s,M} = I_{V_M}. \tag{95.113}$$

The cascade reader is

$$\mathcal{R}^{\mathrm{phys}}_{M,J,T} := \mathcal{R}_{1,M}\mathfrak{U}^{\sharp}_{M,J,T}. \tag{95.114}$$

By (95.104),

$$\boxed{\mathcal{R}^{\mathrm{phys}}_{M,J,T}\mathfrak{U}_{M,J,T}\mathcal{V}^{\mathrm{PCF}}_{M}(u_M(t)) = u_M(t).} \tag{95.115}$$

The return uses the metric adjoint of the same cascade and recovers the hydrodynamic publication without selecting a probabilistic branch.

95.6.2 Source realization of the enriched Stein–Lyapunov state

The construction in the section 95.6.1 brings the complete PC–Fock tower, bidirectional memory, carry, microscopic innovation, spectral channel, homogeneous force coordinate, unity, and exceptional publication into a single state space. Physical time parametrizes terminal storage, while nonadic depth parametrizes the Stein recurrence. The construction governing this section is the `OWNER_GOVERNING` of the enriched HMT state. KYP, NS–OBS, and LRDN–LCN charts are retained only as `CONTROL_NON_GOVERNING` of reduced representations.

95.6.2.1 PC–Fock colligation and telescoping

Let

$$\mathbf{x}^{\mathrm{own}}_{M,j} \in \mathscr{K}^{\mathrm{PCF}}_{M,j}\widehat{\otimes}\mathcal{E}_{\mathrm{exc}} \oplus \mathscr{G}^{F,[j]}_{M}$$

the homogeneous state of the physical diagonal. For every M, J, T, the source metric is constructed prospectively as the Gram matrix of the terminal column and all future losses of that same diagonal:

$$\left\langle x, U^{\mathrm{own}}_{M,j}x\right\rangle := \mathbb{E}_{j:J}\left\|\mathcal{E}^{\mathrm{term}}_{M,J,T}\Gamma^{\mathrm{own}}_{M,j:J}x\right\|^2 + \sum_{k=j}^{J-1}\mathbb{E}_{j:k}\left\|L^{\mathrm{own}}_{M,k,T}\Gamma^{\mathrm{own}}_{M,j:k}x\right\|^2. \tag{95.116}$$

Every term is a square of coordinates already constructed: viscous degree one, convective degree two, carry, innovation, spectral archive, memory D/H, and homogeneous force. Separating the first step of the sum gives, before introducing any KYP root,

$$\boxed{U^{\mathrm{own}}_{M,j} = (\Gamma^{\mathrm{own}}_{M,j})^* U^{\mathrm{own}}_{M,j+1}\Gamma^{\mathrm{own}}_{M,j} + (L^{\mathrm{own}}_{M,j,T})^* L^{\mathrm{own}}_{M,j,T}, \qquad G^{\mathrm{own}}_{M,J,T} \preceq C_T U^{\mathrm{own}}_{M,J}.} \tag{95.117}$$

In the terminal normalization of (95.88), one may take $C_T = 1$. The equality of maps, amplified by the identity in the Moonshine layer, produces the isometric cascade

$$(\mathfrak{U}^{\mathrm{own}}_{M,J,T})^{\sharp}\mathfrak{U}^{\mathrm{own}}_{M,J,T} = I. \tag{95.118}$$

Iterating (95.6.2.1) yields the telescoping identity

$$U^{\mathrm{own}}_{M,0} = (\Gamma^{\mathrm{own}}_{M,0:J})^{*}U^{\mathrm{own}}_{M,J}\Gamma^{\mathrm{own}}_{M,0:J} + \sum_{j<J}(\Gamma^{\mathrm{own}}_{M,0:j})^{*}(L^{\mathrm{own}}_{M,j,T})^{*}L^{\mathrm{own}}_{M,j,T}\Gamma^{\mathrm{own}}_{M,0:j}. \tag{95.119}$$

Its orthogonal balance ledger is

$$\boxed{E_s(u_M(t)) \le \|Z_{M,J,t}\|^2 = \mathbb{E}_{0:J}G_{M,J,T}(t) + \sum_{j<J}\mathbb{E}_{0:j}\Lambda^{\mathrm{own}}_{M,j}([0,t]).} \tag{95.120}$$

The common root $\Xi_{M,0} = J_{0\to M}\Xi_0$, evolutionary preservation of unity, and the carry port of (95.91) ensure that the right-hand side is charged only once. The source budget is

$$\begin{aligned}\mathcal{B}^{\mathrm{own}}_T &:= C_{\mathrm{HMT}} + (1 + 2\widetilde{M}^2_{s,\beta,\nu})e^{R_{\mathrm{F}}^2\|u_0\|^2_{H^s}} \\ &\quad + C_{\nu,s}\int_0^T \|f(t)\|^2_{H^{s-1}}\,dt + \|\mathcal{J}^{\mathrm{own}}_T(u_0,f)\|^2,\end{aligned} \tag{95.121}$$

where $\mathcal{J}^{\mathrm{own}}_T$ is the root injection determined by the initial state transported by TPK, its two sheets, and the forced input. Its amplifications are

$$\mathcal{J}^{\mathrm{own}}_{M,J,T} := I_{0\to M,J}\mathcal{J}^{\mathrm{own}}_T, \qquad I^{*}_{0\to M,J}I_{0\to M,J} = I.$$

By (95.119), this is not a quantity reconstructed from the future orbit. Preservation of unity and coisometry of the two-way reader give, on the declared domain,

$$\sup_{M,J}\|\mathcal{J}^{\mathrm{own}}_{M,J,T}(u_0,f)\|^2 \le C_{\mathrm{HMT},T}\left(1 + \|u_0\|^2_{H^{s+1}} + \int_0^T \|f(t)\|^2_{H^{s-1}}\,dt\right). \tag{95.122}$$

Thus $\mathcal{B}^{\mathrm{own}}_T$ is independent of M, J. The ledger gives

$$\boxed{\sup_{M,J}\left[\sup_{0\le t\le T}\|Z_{M,J,t}\|^2 + \frac{\nu}{2}\int_0^T D_{s,M}(t)\,dt\right] \le \mathcal{B}^{\mathrm{own}}_T < \infty.} \tag{95.123}$$

The hydrodynamic publication is the degree-one return of the same cascade:

$$R^{\mathrm{full}}_{M,J,T} := \mathcal{R}_{1,M}(\mathfrak{U}^{\mathrm{own}}_{M,J,T})^{\sharp}, \qquad \boxed{R^{\mathrm{full}}_{M,J,T}Z_{M,J,t} = u_M(t).} \tag{95.124}$$

This equality coincides with (95.115) and avoids evaluating an individual refinement branch.

Theorem 95.15 (Uniform estimate for the source PC–Fock realization)**.** *For every $T < \infty$, the Galerkin approximations satisfy*

$$\sup_{M}\sup_{0\le t\le T}\|u_M(t)\|^2_{H^s} + \nu\sup_{M}\int_0^T \|u_M(t)\|^2_{H^{s+1}}\,dt \le C_s\mathcal{B}^{\mathrm{own}}_T, \tag{95.125}$$

with a constant independent of M and of nonadic depth.

Proof. Identity (95.124) and contractivity of the return give $\|u_M(t)\|_{H^s}^2 \le C_s \|Z_{M,J,t}\|^2$. Equivalence of $D_{s,M}$ with the H^{s+1} norm on the mean-zero solenoidal subspace, together with (95.123), proves the estimate. □

95.6.2.2 Alternative control through a route metric

The following chart checks refinement in normalized coordinates. It neither constructs nor conditions U^{own}: it omits unity, exceptional publication, and part of the affine channel already present in (95.116).

The Ambivalence chart supplies

$$G_{\mathrm{av}} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, \qquad M_{\mathrm{av}} = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}, \qquad M_{\mathrm{av}}^{*} G_{\mathrm{av}} M_{\mathrm{av}} \preceq \varphi_{\mathrm{HMT}}^{4} G_{\mathrm{av}}. \tag{95.126}$$

After normalization $\widehat{M}_j = M_j/3$,

$$\mathsf{R}_j := G_j - \widehat{M}_j^{*} G_{j+1} \widehat{M}_j \succeq 0. \tag{95.127}$$

In the stationary chart,

$$\mathsf{R}_{\mathrm{av}} = \frac{1}{9} \begin{pmatrix} 5 & -4 \\ -4 & 7 \end{pmatrix}, \qquad \lambda_{\min}(\mathsf{R}_{\mathrm{av}}) = \frac{6 - \sqrt{17}}{9}.$$

With the graph metric

$$\mathsf{G}_M^{\mathrm{gr}} = I + (\mathcal{Y}_{\beta,M})^{*} \mathcal{Y}_{\beta,M}, \qquad C_M = \mathcal{Y}_{\beta,M} (\mathsf{G}_M^{\mathrm{gr}})^{-1/2},$$

we define

$$\mathcal{A}_{M,j} = \widehat{M}_j \otimes I, \qquad \mathcal{O}_{M,j} = \mathsf{R}_j^{1/2} \otimes C_M, \qquad \mathcal{D}_{M,j} = \mathsf{R}_j^{1/2} \otimes (I - C_M^{*} C_M)^{1/2}. \tag{95.128}$$

The route Stein identity is

$$\boxed{\mathsf{U}_{M,j}^{\mathrm{PCF}} - \mathcal{A}_{M,j}^{*} \mathsf{U}_{M,j+1}^{\mathrm{PCF}} \mathcal{A}_{M,j} = \mathcal{O}_{M,j}^{*} \mathcal{O}_{M,j} + \mathcal{D}_{M,j}^{*} \mathcal{D}_{M,j}.} \tag{95.129}$$

Its iteration gives

$$\mathsf{U}_{M,0}^{\mathrm{PCF}} - \mathcal{A}_{M,0:J}^{*} \mathsf{U}_{M,J}^{\mathrm{PCF}} \mathcal{A}_{M,0:J} = \sum_{j<J} \mathcal{A}_{M,0:j}^{*} \big(\mathcal{O}_{M,j}^{*} \mathcal{O}_{M,j} + \mathcal{D}_{M,j}^{*} \mathcal{D}_{M,j} \big) \mathcal{A}_{M,0:j}. \tag{95.130}$$

This factorization describes refinement geometry. Affine recurrence (95.101) also incorporates physical evolution, Carleman cross terms, and the force.

Bidirectional memory turns route loss into a budget. With $\mathcal{N}_{M,j} = 2(r-1) M_{M,j}^{-}$,

$$E_j \Delta_T \mathcal{N}_{M,j+1} = \Delta_T \mathcal{N}_{M,j} + 2K_{M,j}(T), \qquad K_{M,j}(T) = \int_0^T \kappa_{M,j}(t)\, dt. \tag{95.131}$$

Telescoping and the root bound yield

$$\mathbb{E}_{0:J} G_{M,J,T}(0) \le E_{s,M}(P_M u_0) + |\Delta_T \mathcal{N}_0| + \|\Xi_{M,0}\|^2 + Q_f(T), \tag{95.132}$$

which is the abbreviated form of route control (95.108)–(95.111). This chart records the cost of the negative sheet,

$$\Delta_T \mathcal{N}_{M,0} = 2 \int_0^T B_{s,M,0}^{-}(u_M(t))\, dt.$$

In this reduced representation, the inequality

$$X^{-}_{M,T}(0) \preceq C_T \mathsf{U}^{\mathrm{PCF}}_{M,0}, \qquad \sup_M C_T < \infty \tag{NS–OBS}$$

is a useful falsifier of the chart. Its failure does not transfer to the source construction, because $\Delta_T \mathcal{N}_{M,0}$ does not replace root injection $\mathcal{J}^{\mathrm{own}}_T$ of (95.121)–(95.122). Similarly, KYP and LRDN–LCN verify completions or dominations of reduced readers; none is a premise of the theorem 95.15.

95.6.2.3 Existence, uniqueness, and global smoothness

Theorem 95.16 (Global regularity of Navier–Stokes)**.** *Let $s > 5/2$, $\nu > 0$, let $u_0 \in H^{s+1}_\sigma(\mathbb{T}^3)$ have mean zero, and let $f \in L^2_{\mathrm{loc}}([0,\infty); H^{s-1}_\sigma)$. There exists a unique global solution*

$$u \in L^\infty_{\mathrm{loc}}([0,\infty); H^s_\sigma) \cap L^2_{\mathrm{loc}}([0,\infty); H^{s+1}_\sigma), \qquad \partial_t u \in L^2_{\mathrm{loc}}([0,\infty); H^{s-1}_\sigma). \tag{95.133}$$

For smooth data and force, u and the normalized pressure are smooth.

Proof. The theorem 95.15 gives a uniform bound for $(u_M)_M$ in $L^\infty(0,T;H^s) \cap L^2(0,T;H^{s+1})$. The Galerkin equation and continuity

$$H^s \times H^s \longrightarrow H^{s-1}, \qquad (u,v) \longmapsto \mathbb{P}(u \cdot \nabla v),$$

bound $\partial_t u_M$ in $L^2(0,T;H^{s-1})$. Since $H^{s+1} \Subset H^s \hookrightarrow H^{s-1}$, Aubin–Lions yields, after extracting a subsequence,

$$u_M \longrightarrow u \quad \text{strongly in } L^2(0,T;H^s).$$

This convergence permits passage to the limit in $\mathbb{P}(u_M \cdot \nabla u_M)$, while weak convergence preserves the viscous term and initial datum. A strong solution in $[0,T]$ is obtained.

If u, v are two solutions and $w = u - v$, the product estimate and Young's inequality give

$$\frac{1}{2}\frac{d}{dt}\|w\|^2_{H^{s-1}} + \frac{\nu}{2}\|w\|^2_{H^s} \le C_{s,\nu}\big(\|u\|^2_{H^s} + \|v\|^2_{H^s}\big)\|w\|^2_{H^{s-1}}.$$

Grönwall proves uniqueness. Repeating the construction on horizons $T = 1, 2, \ldots$, the solutions agree on overlapping intervals and yield the global solution.

The pressure is recovered from

$$-\Delta p = \partial_i \partial_j (u_i u_j) - \operatorname{div} f, \qquad \int_{\mathbb{T}^3} p \, dx = 0. \tag{95.134}$$

Iterating the estimate in $s + k$ and the equation recovers all spatial and temporal derivatives when u_0 and f are smooth. □

The structural contribution consists in representing convection as a linear output of the PC–Fock tower, preserving it in memory D/H, and incorporating it into an orthogonal identity with exact physical return. The resulting uniform estimate is then published as global regularity of the hydrodynamic equation.

Integrated reinforcement of the same owner

The following module preserves the uniform calculation of the Fourier rows, the root injection, the time derivative, the pressure and the gluing of horizons. Its finite KYP control remains separate from the governing chain.

Solenoidal balance, carry, and enriched terminal

This module reinforces the Navier–Stokes `OWNER_GOVERNING`: it preserves Galerkin carry, causal balance, terminal Gram matrix, Stein, telescoping, root injection, return, and global passage in a single chain. The KYP chart appears only afterwards and remains outside that chain.

For Galerkin cutoffs $m \le n$, the preserved carry is

$$C_{m\leftarrow n} := P_m B(u_n, u_n) - B_m(P_m u_n, P_m u_n). \tag{95.135}$$

The nonadic reading of this state is typed by

$$E_9 J_9 = I, \qquad P_9 = J_9 E_9, \qquad Q_{\mathrm{micro}} = I - P_9, \tag{95.136}$$

and preserves the causal balance

$$\boxed{\mathbb{E}_9 \|\Xi_{m+1}\|^2 + \mathbb{E}_9 \ell_m = \|\Xi_m\|^2, \qquad \ell_m \ge 0.} \tag{95.137}$$

The field, solenoidal projection, carry, boundary, route, and positive losses thus remain in one state before storage is constructed.

Truncated Gram matrix of the complete state and uniform root injection

The realization governing the balance does not start with the KYP inequality. Let $\Gamma^{\mathrm{term}}_{M,j}$ be the one-step transition of the enriched PC–Fock state, $\Gamma^{\mathrm{term}}_{M,j:k}$ its composition between depths j and k, and $L^{\mathrm{term}}_{M,k,T}$ the column of degree-one, degree-two, carry, innovation, archive, and memory losses at horizon $[0,T]$. Notation $\mathbb{E}_{j:k}$ denotes the orthogonal sum weighted by refinement branches between those levels. For $j \le J$, define the finite column

$$\begin{aligned} \mathbf{G}^{\mathrm{term}}_{M,j;J,T} x &:= \big(\mathcal{E}^{\mathrm{term}}_{M,J,T} \Gamma^{\mathrm{term}}_{M,j:J} x\big)_{\mathbb{E}_{j:J}} \\ &\quad \oplus \bigoplus_{k=j}^{J-1} \big(L^{\mathrm{term}}_{M,k,T} \Gamma^{\mathrm{term}}_{M,j:k} x\big)_{\mathbb{E}_{j:k}}, \qquad U^{\mathrm{term}}_{M,j;J,T} := (\mathbf{G}^{\mathrm{term}}_{M,j;J,T})^* \mathbf{G}^{\mathrm{term}}_{M,j;J,T}. \end{aligned} \tag{95.138}$$

In particular, storage is a truncated Gram matrix constructed from the terminal and future losses of the same state. Its quadratic form is

$$\begin{aligned} \langle x, U^{\mathrm{term}}_{M,j;J,T} x\rangle &= \mathbb{E}_{j:J} \|\mathcal{E}^{\mathrm{term}}_{M,J,T} \Gamma^{\mathrm{term}}_{M,j:J} x\|^2 \\ &\quad + \sum_{k=j}^{J-1} \mathbb{E}_{j:k} \|L^{\mathrm{term}}_{M,k,T} \Gamma^{\mathrm{term}}_{M,j:k} x\|^2. \end{aligned} \tag{95.139}$$

Lemma 95.17 (Stein identity of the truncated Gram matrix)**.** *For every $j < J$,*

$$U^{\mathrm{term}}_{M,j;J,T} = (\Gamma^{\mathrm{term}}_{M,j})^* U^{\mathrm{term}}_{M,j+1;J,T} \Gamma^{\mathrm{term}}_{M,j} + (L^{\mathrm{term}}_{M,j,T})^* L^{\mathrm{term}}_{M,j,T}. \tag{95.140}$$

Iteration, preserving the terminal summand, yields

$$\begin{aligned} U^{\mathrm{term}}_{M,0;J,T} &= (\Gamma^{\mathrm{term}}_{M,0:J})^* U^{\mathrm{term}}_{M,J;J,T} \Gamma^{\mathrm{term}}_{M,0:J} \\ &\quad + \sum_{j=0}^{J-1} (\Gamma^{\mathrm{term}}_{M,0:j})^* (L^{\mathrm{term}}_{M,j,T})^* L^{\mathrm{term}}_{M,j,T} \Gamma^{\mathrm{term}}_{M,0:j}. \end{aligned} \tag{95.141}$$

Proof. In (95.138), separate summand $k = j$. The remaining summands, including the terminal, are exactly the level-$j+1$ column preceded by $\Gamma^{\mathrm{term}}_{M,j}$. Orthogonality of the sum and the tower property

of $\mathbb{E}_{j:k}$ give (95.140). Repeating the same splitting gives (95.141); $(\Gamma^{\mathrm{term}}_{M,0:J})^* U^{\mathrm{term}}_{M,J;J,T}\Gamma^{\mathrm{term}}_{M,0:J}$ is not discarded. □

Theorem 95.18 (Root injection and uniform budget)**.** *The realizer's root is transported to each cutoff and depth by an isometric inclusion:*

$$\begin{aligned} x^{\mathrm{data}}_{M,0} &:= \mathcal{V}^{\mathrm{full}}_M(P_M u_0) \oplus g^F_M, \\ \mathbf{G}^{\mathrm{term}}_{M,0;J,T} x^{\mathrm{data}}_{M,0} &= \mathcal{J}^{\mathrm{term}}_{M,J,T}(u_0, f), \\ \mathcal{J}^{\mathrm{term}}_{M,J,T} &:= I_{0\to M,J}\mathcal{J}^{\mathrm{term}}_T, \qquad I^*_{0\to M,J} I_{0\to M,J} = I. \end{aligned} \tag{95.142}$$

For the data of (95.67)*, there exists $C_{\mathrm{HMT},T}$, independent of M and J, such that*

$$\sup_{M,J} \|\mathcal{J}^{\mathrm{term}}_{M,J,T}(u_0, f)\|^2 \le C_{\mathrm{HMT},T} + (1 + 2\widetilde{M}^2_{s,\beta,\nu}) e^{R_{\mathrm{F}}^2 \|u_0\|^2_{H^s}} + C_{\nu,s} \int_0^T \|f(t)\|^2_{H^{s-1}}\, dt. \tag{95.143}$$

Equivalently, the bound is a bound on the terminal Gram matrix evaluated at the data:

$$\sup_{M,J} \left\langle x^{\mathrm{data}}_{M,0}, U^{\mathrm{term}}_{M,0;J,T} x^{\mathrm{data}}_{M,0} \right\rangle = \sup_{M,J} \|\mathcal{J}^{\mathrm{term}}_{M,J,T}(u_0, f)\|^2 < \infty. \tag{95.144}$$

Map $\mathcal{J}^{\mathrm{term}}_T$ depends only on the initial state, its two sheets, the inherited terminal, and the force port; it does not receive $u_M(t)$ for $0 < t \le T$ as an argument.

Proof. The linear component and memories D/H are transported isometrically. In the PC–Fock component, the coherent column is

$$\bigoplus_{n\ge 0} \frac{R_{\mathrm{F}}^n}{\sqrt{n!}}\, u_0^{\odot n}, \qquad \sum_{n\ge 0} \frac{R_{\mathrm{F}}^{2n}}{n!} \|u_0\|^{2n}_{H^s} = e^{R_{\mathrm{F}}^2 \|u_0\|^2_{H^s}}.$$

To justify uniformity of the quadratic reader, polarize

$$\mathcal{B}_M(u, v) := \frac{1}{4\sqrt{\beta\nu}}\, A_M^{-1/2} \Lambda^s P_M \mathbb{P}\big((P_M u\cdot\nabla) P_M v + (P_M v\cdot\nabla) P_M u\big). \tag{95.145}$$

In the transversal bases $e^s_{p,a} = \langle p\rangle^{-s} a\, e^{ip\cdot x}$, if $k = p + q \ne 0$, its Fourier rows satisfy

$$\|c^M_{k;p,q}\| \le \frac{C}{\sqrt{\beta\nu}} \frac{\langle k\rangle^s (|p| + |q|)}{|k| \langle p\rangle^s \langle q\rangle^s}, \qquad \sup_{M, k\ne 0} \sum_{p+q=k} \|c^M_{k;p,q}\|^2 < \infty. \tag{95.146}$$

The second bound follows by separating $|p| \le 2|k|$ from $|p| > 2|k|$; in the tail, $\sum_p \langle p\rangle^{2-2s} < \infty$ dominates because $s > 5/2$. Cauchy–Schwarz in each row and orthogonality of blocks with distinct k extend $\mathcal{B}_M$ to

$$\mathfrak{C}_M : H^s_\sigma \odot_2 H^s_\sigma \longrightarrow L^2_\sigma, \qquad \sup_M \|\mathfrak{C}_M\| \le \widetilde{M}_{s,\beta,\nu}.$$

Thus the constant used at the root controls the sum of all Fourier pairs, not merely each coefficient separately. The affine port satisfies

$$Q_f(T) \le C_{\nu,s} \int_0^T \|f(t)\|^2_{H^{s-1}}\, dt.$$

By orthogonal summation of those components,

$$\|\Xi_{M,0}\|^2 \le C_{\mathrm{HMT},T} + (1 + 2\widetilde{M}^2_{s,\beta,\nu}) e^{R_{\mathrm{F}}^2 \|u_0\|^2_{H^s}}. \tag{95.147}$$

The inclusions of (95.142) do not change this norm. Adding the port bound yields exactly (95.143). Since all terms are evaluated at (u_0, f), the estimate precedes the trajectory and does not assume the global bound subsequently deduced from (95.141). □

The terminal Gram matrix (95.139), its Stein identity, telescoping with terminal, and uniform root injection form the source construction. This is the branch governing closure, and it coincides with the complete realization restored in section 95.6.2 y theorem 95.15.

Separation of the non-governing finite KYP control

The correct finite KYP factorization of the theorem 95.14 is retained as an alternative control of a reduced chart and is not a premise of source construction U^{own}. The signed mutation that purported to incorporate $C^*_{m,M}C_{m,M}$ into a term $P \succeq 0$ and later recover it with a negative sign is not used: upon entering P, that summand necessarily appears with a positive sign. That factorization therefore plays no part in the balance, root budget, physical return, or Navier–Stokes conclusion.

Time derivative, pressure, and gluing of horizons

The next step follows from the uniform trajectory bound obtained through the source terminal Gram matrix, its telescoping, and the exact physical return. The joint uniform bound in $L^\infty(0,T;H^s) \cap L^2(0,T;H^{s+1})$ is available. The projected equation reads

$$\partial_t u_m = \nu \Delta u_m - P_m B(u_m, u_m) + P_m f. \tag{95.148}$$

For $s > 5/2$, the Sobolev product and the joint bound in $L^\infty(0,T;H^s) \cap L^2(0,T;H^{s+1})$ give

$$\sup_m \|\partial_t u_m\|_{L^2(0,T;H^{s-1})} \le C_T(u_0, f, \Xi_0). \tag{95.149}$$

Estimate (95.149) and spatial compactness produce strong convergence in $L^2(0,T;H^s)$, sufficient to pass to the bilinear term without losing Galerkin carry.

Once velocity is recovered, the normalized mean-zero pressure is fixed by

$$-\Delta p = \partial_i \partial_j (u_i u_j) - \nabla \cdot f, \qquad \int_{\mathbb{T}^3} p(t,x)\, dx = 0. \tag{95.150}$$

Pressure is thus an output of the same solenoidal solution; it is not a free variable used to select the regular branch.

Proposition 95.19 (Global gluing by uniqueness)**.** *Let $u^{(T)}$ be the solutions obtained on $[0,T]$ through the Galerkin limit. If $T_1 < T_2$, then $u^{(T_2)}|_{[0,T_1]} = u^{(T_1)}$. The family defines a unique smooth global solution for $t > 0$.*

Proof. Both restrictions have the same data and belong to the domain of the uniform estimate. The equation for their difference, the bilinear control in H^{s-1} and Grönwall imply equality on $[0,T_1]$. The finite intervals form a directed system; equality on the overlaps defines the global solution. The theorem's parabolic regularization applies on each interval $[\varepsilon, T]$, $\varepsilon > 0$. □

The refutation check is precise: a loss of uniformity in (95.149), two different limits on the same horizon or a pressure not satisfying (95.150) would break the chain before the global conclusion.

Proprietary conclusion. The proprietary terminal Gram, Stein, telescoping with a terminal, its ledger, the exact physical return and the root injection produce a uniform H^s bound for $s > 5/2$. Therefore, global existence, uniqueness and regularity of periodic three-dimensional Navier–Stokes are proved in the registered HMT domain. The finite KYP charts remain as `NON_GOVERNING_CONTROL`.

Affirmative conclusion. In the declared periodic and Sobolev domain, the terminal storage U^{own}, its Stein identity, the telescoping that preserves the terminal, the ledger that dominates $E_s(u_M)$, the uniform root and the exact physical return produce the uniform bound. Aubin–Lions, uniqueness and parabolic regularity yield the global smooth solution. The KYP, NS–OBS and LRDN–LCN charts are alternative controls and not premises of the theorem.

95.7 Projective construction of a Euclidean Yang–Mills gauge process, Osterwalder–Schrader reconstruction, and collective gap in dimension 4

The specialization of the theorem 8.57 is

$$(\widehat{X}_\infty, \Gamma_9) \xrightarrow{\mathcal{R}_{\mathrm{YM}}} (\mu_m, T_m, H_m, \Omega_m) \longrightarrow (\mu_\infty, \mathrm{OS}, E(4)) \longrightarrow \mathrm{gap}(H_{\mathrm{phys}}) > 0.$$

The clover reader that publishes F^2 recognizes the curvature on this same projective process; it does not retrospectively construct μ_m. The gauge theory, the OS reconstruction, the simple vacuum and the gap are already fixed by this chain. The identity (95.194) then compares an optional parametrization by g_R; it is a non-governing control and not a condition for identifying the constructed Yang–Mills theory.

Gauge object and refinement domain

Let G be a compact, connected, simple Lie group. For each $m \geq 0$, consider the Euclidean lattice

$$\Lambda_m = a_m \mathbb{Z}^4, \qquad a_{m+1} = \frac{a_m}{9}. \tag{95.151}$$

The cylindrical gauge algebra is generated by link variables in G, representation intertwining operators, and the enriched coordinates of color, orientation, holonomy, fusion carry, route, and memory. Let μ_m be the common invariant measure and let

$$\mathcal{H}_m = L^2(\mathcal{A}_m, \mu_m)^{\mathrm{gauge}}.$$

The constant vector Ω_m defines

$$P_{\Omega_m} = |\Omega_m\rangle\langle\Omega_m|, \qquad Q_m^{\mathrm{phys}} = I - P_{\Omega_m}. \tag{95.152}$$

The second projector is the physical complement of the vacuum; it is not identified with a residual used solely to organize the renormalization scale.

The update circuit is local:

$$K_m^{\mathrm{loc}} = \prod_{r=1}^{9\cdot 81} \prod_{B \in \mathcal{B}_{m,r}} K_{m,B}. \tag{95.153}$$

The partition $\{\mathcal{B}_{m,r}\}_r$ enumerates each coordinate exactly once; the circuit factors will be identified below with the semigroup of their conditional expectations, not with nine independent dynamics. Blocks of the same color have disjoint supports, and the physical diameter of each support is uniformly bounded as m varies. This locality is part of the object, so that refinement does not turn an elementary update into an instantaneously global operation.

Let T_m be the transfer induced by (95.153). The isometric inclusions $J_m : \mathcal{H}_m \to \mathcal{H}_{m+1}$ preserve ancestry, representation, orientation, carry, and memory. For the four Euclidean reflections $\Theta_{\nu,m}$, $\nu = 1, \ldots, 4$, one has

$$(T_{m+1})^9 J_m = J_m T_m, \qquad \Theta_{\nu,m+1} J_m = J_m \Theta_{\nu,m} \quad (\nu = 1, \ldots, 4). \tag{95.154}$$

The four directions therefore belong to the same refinement, the same measure, and the same dynamics.

Common semigroup, relative corona, and four reflection forms

In the triangular chart of the complete state, let $E_{m,c}$ be the orthogonal and commuting conditional expectations associated with independent coordinates. The positive overlap generator is first constructed at a seed level m_0:

$$\widehat{H}^{\mathrm{ov}}_{m_0} = 3\kappa_{\mathrm{av}} \sum_{c \in \mathcal{I}^{\mathrm{seed}}_{m_0}} (I - E_{m_0,c}), \tag{95.155}$$

and is extended by

$$\begin{aligned} \widehat{H}^{\mathrm{ov}}_{m+1} &= \widehat{\Gamma}^{*}_{m}\big(\widehat{H}^{\mathrm{ov}}_{m} \otimes I + I \otimes \widehat{H}^{\mathrm{det}}_{m+1}\big)\widehat{\Gamma}_{m}, \\ \widehat{H}^{\mathrm{det}}_{m+1} &= 3\kappa_{\mathrm{av}} \sum_{\iota \in \mathcal{I}^{\mathrm{det}}_{m+1/m}} (I - E^{\mathrm{det}}_{m+1,\iota}), \end{aligned} \tag{95.156}$$

where

$$\kappa_{\mathrm{av}} = \frac{\mu_{\mathrm{vol}}}{\ell_0}\left(\frac{\pi_{\mathrm{HMT}}}{\varphi^{2}_{\mathrm{HMT}}} - 1\right) > 0. \tag{95.157}$$

The volumetric frequency μ_{vol} is dimensionless; the dimension comes from ℓ_0.

The semigroup, the transfer over one physical step, and their scale compatibility are defined with explicit types:

$$\widehat{K}^{\mathrm{ov}}_{m,t} = e^{-t\widehat{H}^{\mathrm{ov}}_{m}}, \qquad \widehat{T}_m = \widehat{K}^{\mathrm{ov}}_{m,a_m}, \qquad a_{m+1} = a_m/9, \tag{95.158}$$

Since the partition enumerates each coordinate exactly once and the conditional expectations of its blocks commute, for each block and for the complete circuit one obtains

$$K_{m,B} = e^{-3\kappa_{\mathrm{av}} a_m (I - E_{m,B})}, \qquad K^{\mathrm{loc}}_{m} = e^{-a_m \widehat{H}^{\mathrm{ov}}_{m}} = \widehat{T}_m. \tag{95.159}$$

$$\widehat{H}^{\mathrm{ov}}_{m+1}\widehat{J}_m = \widehat{J}_m \widehat{H}^{\mathrm{ov}}_{m}, \qquad (\widehat{T}_{m+1})^{9}\widehat{J}_m = \widehat{J}_m \widehat{T}_m. \tag{95.160}$$

Thus, the ninth power is compared through $\widehat{J}_m$; operators acting on different Hilbert spaces are not equated.

The common eight-face measure originates in this same reversible kernel:

$$\widehat{\Omega}^{(8)}_{m}(dy_1 \cdots dy_8) = \int \widehat{\mu}^{\mathrm{ov}}_{m}(dz) \prod_{r=1}^{8} \widehat{K}^{\mathrm{ov}}_{m,a_m/2}(z, dy_r). \tag{95.161}$$

For each reflection $\nu = 1, \dots, 4$ and every cylindrical observable F in the positive half-space,

$$\int \overline{F(\Theta_{\nu,m} y)} F(y)\, d\widehat{\Omega}^{(8)}_{m} = \|\widehat{\mathcal{B}}_{m,\nu} F\|^2 \geq 0, \tag{95.162}$$

and the marginal on opposite faces gives the connecting identity

$$\widehat{T}^{\mathrm{OS}}_{m,\nu} = \widehat{K}^{\mathrm{ov}}_{m,a_m} = e^{-a_m \widehat{H}^{\mathrm{ov}}_{m}} = \widehat{T}_m, \qquad \nu = 1, \dots, 4. \tag{95.163}$$

The four reflections therefore publish the same transfer–measure pair.

Cellular persistence of the terminal and exact gap

Let

$$K_9 = C_*(Q_{9,3}; \mathbb{Z}), \qquad K_{10} = C_*(Q_{10,3}; \mathbb{Z}).$$

The relative corona has dimensions

$$C_*(Q_{10,3}, Q_{9,3}) = (271, 756, 702, 217), \qquad 271 + 702 = 756 + 217 = 973.$$

The inclusion $j : K_9 \to K_{10}$, cellular collapse $r : K_{10} \to K_9$, and integral homotopy H_{cell} satisfy

$$rj = I, \qquad \partial H_{\mathrm{cell}} + H_{\mathrm{cell}}\partial = I - jr, \qquad rH_{\mathrm{cell}} = 0, \quad H_{\mathrm{cell}}j = 0, \quad H_{\mathrm{cell}}^2 = 0. \tag{95.164}$$

For every chain map $F : K_{10} \otimes V_{q=6} \to \mathcal{T}$, with $P = (jr) \otimes I_{q=6}$,

$$F - FP = d_{\mathcal{T}}\big(F(H_{\mathrm{cell}} \otimes I)\big) + F(H_{\mathrm{cell}} \otimes I)(\partial \otimes I). \tag{95.165}$$

The complete corona is thus null-homotopic, and the terminal FP remains literally unchanged. The 271 vertices do not replace the edges, faces, and cubes of the corona; nor is the three-dimensional cellular factor identified with the physical gauge factor $q = 6$.

For each persistent coordinate c, in the local chart $q = (q_1, \dots, q_6) \in \mathbb{F}_3^6$, the centered witness is fixed

$$\bigcap_c \operatorname{Ran} E_{m,c} = \mathbb{C}\Omega_m, \qquad W_c(q) = \mathbf{1}_{\{q_1+\cdots+q_6=0\}} - \frac{1}{3}, \tag{95.166}$$

where the sum in the indicator is taken in $\mathbb{F}_3$. The Hoeffding decomposition of the commuting expectations proves

$$\ker \widehat{H}_m^{\mathrm{ov}} = \mathbb{C}\Omega_m, \qquad \widehat{H}_m^{\mathrm{ov}} \succeq 3\kappa_{\mathrm{av}}(I - P_{\Omega_m}). \tag{95.167}$$

Equality does not follow merely from the triviality of the kernel: each nonconstant sector of the decomposition receives at least one projector $I - E_{m,c}$, and by (95.155) its minimum cost is $3\kappa_{\mathrm{av}}$. The explicit witness W_c satisfies

$$\widehat{H}_m^{\mathrm{ov}} W_c = 3\kappa_{\mathrm{av}} W_c, \qquad \|W_c\|^2 = \frac{2}{9}, \tag{95.168}$$

so that the complement is nonempty and the gap is exactly $3\kappa_{\mathrm{av}}$.

The scale is published by

$$q_m = e^{-3\kappa_{\mathrm{av}} a_m}, \qquad q_{m+1}^{9} = q_m, \qquad -a_m^{-1} \log q_m = 3\kappa_{\mathrm{av}}. \tag{95.169}$$

Although $q_m \to 1$ under refinement, the spectral quotient per physical unit remains constant. The cellular contraction (95.165) preserves the terminal; the gap originates in the positive overlap generator and its Hoeffding decomposition, not in acyclicity alone.

The inclusions in (95.160) form the inductive limit

$$\mathcal{H}_\infty = \varinjlim (\mathcal{H}_m, \widehat{J}_m). \tag{95.170}$$

On the algebraic inductive union, define the form

$$\mathfrak{h}_\infty[\widehat{J}_{m,\infty}\psi] := \langle \psi, \widehat{H}_m^{\mathrm{ov}} \psi \rangle_{\mathcal{H}_m}. \tag{95.171}$$

Intertwining of the generators makes this definition independent of the representative. The form is positive and closable; its closure determines the self-adjoint generator $\widehat{H}_\infty^{\mathrm{ov}}$. Compatibility of the forms transports

$$\overline{\mathfrak{h}}_\infty[\psi] \geq 3\kappa_{\mathrm{av}} \|(I - P_{\Omega_\infty})\psi\|^2, \tag{95.172}$$

and the persistent vector $\widehat{J}_{m,\infty} W_c$ attains equality. Consequently, the exact gap of the limiting generator is $3\kappa_{\mathrm{av}}$.

Exact hypotheses and logical chain of the theorem

The gauge publication uses:

1. the compact, connected, simple group and the four-dimensional lattice (95.151);
2. the local circuit (95.153), with uniform physical diameter, and a common invariant measure;
3. the isometric inclusions and the four reflections intertwined by (95.154) and (95.160);
4. the four factorizations (95.162) and the exact connection (95.163) on the same transfer–measure pair;
5. the commuting-expectation generator (95.155)– (95.156) and its Hoeffding decomposition;
6. the integral SDR of the corona and preservation of the terminal (95.164)– (95.165);
7. the terminal P_{Ω_m}, the physical complement Q_m^{phys}, and the scaling law (95.169);
8. the inductive limit and the compatible closure of forms (95.170)– (95.172);
9. the internal action and velocity charts $\hbar_{\mathrm{int}}$, c_{int}, used only to publish the mass dimension.

Theorem 95.20 (Local gauge theory with a simple vacuum and collective gap)**.** *Under the preceding premises, the limit of the gauge family is local and reflection-positive in the four Euclidean directions. Its generator has a simple vacuum and satisfies, in the collective gauge-invariant sector,*

$$\Delta_{\mathrm{YM}}^{\mathrm{HMT}} = 3\kappa_{\mathrm{av}} > 0, \qquad m_{\mathrm{gap}}^{\mathrm{HMT}} = \frac{\hbar_{\mathrm{int}}}{c_{\mathrm{int}}}\Delta_{\mathrm{YM}}^{\mathrm{HMT}} > 0. \tag{95.173}$$

The gap belongs to the collective sector; it does not assign an elementary mass to the gluon.

Proof. The uniform locality of the circuit preserves the net of local algebras. The equalities (95.160) identify the ninth power after applying the correct inclusion. The construction (95.161) uses this same semigroup on the eight faces, and (95.163) identifies its four OS marginals with $\widehat{T}_m$. Thus, the four positive forms (95.162) are compatible in the limit and preserve a single measure and a single dynamics.

The expectations in (95.155) are orthogonal and commuting. Their Hoeffding decomposition separates the constant mode from every nonconstant sector: the former is $\mathbb{C}\Omega_m$, whereas every sector of the complement receives at least one factor $I - E_{m,c}$ with weight $3\kappa_{\mathrm{av}}$. This proves (95.167). The witness (95.168) attains the bound, so it is not merely a lower bound: it is the first nonzero energy.

The homotopy (95.165) contracts the cellular complement and leaves FP intact; thus the corona introduces no second vacuum and does not erase the carry. Isometric compatibility transports the vacuum and its complement to the next level without mixing them with the scale residual. The law (95.169) shows that the spectral quotient per physical unit is exactly $3\kappa_{\mathrm{av}}$ at every level. The forms (95.171)–(95.172) transport the bound and the witness to the self-adjoint generator of the limit, so the gap remains exact. Reconstruction by Osterwalder–Schrader positivity identifies the Hamiltonian with the generator of this same transfer, preserves the simple vacuum and produces (95.173). The final dimensional chart transforms the gap into the indicated collective mass. □

Gauge compression and physical gap witness

Let

$$\widehat{\mathscr{H}_m} := L^2(\widehat{X}_m, \widehat{\mu}_m^{\mathrm{ov}})$$

be the space of the complete material realization. In addition to the usual gauge action on links and frames, each collar c carries the finite incidence action

$$q_c \longmapsto q_c + Bx_c, \qquad x_c \in \mathbb{F}_3^4, \tag{95.174}$$

where the six rows of B are indexed by $(01, 02, 03, 12, 13, 23)$. The product of both actions preserves $\widehat{\mu}_m^{\mathrm{ov}}$. With C_m denoting the finite set of collars present at this level, write

$$\mathcal{G}_m^{\mathrm{full}} := G^{V(\Lambda_m)} \times \prod_{c \in C_m} \mathbb{F}_3^4.$$

Its Haar average defines the orthogonal projection

$$\mathbb{P}_m^{\mathrm{phys}} F := \int_{\mathcal{G}_m^{\mathrm{full}}} F(g^{-1}\cdot)\, dg, \qquad \widehat{\mathscr{H}}_m^{\mathrm{phys}} := \operatorname{Ran} \mathbb{P}_m^{\mathrm{phys}}. \tag{95.175}$$

This is a compression onto the gauge subalgebra of the complete state; the marginal forgetting the incidence coordinates is a different map and is not confused with $\mathbb{P}_m^{\mathrm{phys}}$.

Lemma 95.21 (Reduction of the Hamiltonian to the physical subalgebra)**.** *The projection $\mathbb{P}_m^{\mathrm{phys}}$ reduces the generator and its semigroup. The refinement inclusions simultaneously reduce the physical subalgebras:*

$$\begin{gathered} [\mathbb{P}_m^{\mathrm{phys}}, \widehat{H}_m^{\mathrm{ov}}] = 0, \qquad [\mathbb{P}_m^{\mathrm{phys}}, e^{-t\widehat{H}_m^{\mathrm{ov}}}] = 0, \\ \mathbb{P}_{m+1}^{\mathrm{phys}} \widehat{J}_m = \widehat{J}_m \mathbb{P}_m^{\mathrm{phys}}, \qquad H_m^{\mathrm{phys}} := \widehat{H}_m^{\mathrm{ov}}\big|_{\widehat{\mathscr{H}}_m^{\mathrm{phys}}}, \\ H_{m+1}^{\mathrm{phys}} \widehat{J}_m = \widehat{J}_m H_m^{\mathrm{phys}}. \end{gathered} \tag{95.176}$$

Proof. The physical generator on links is gauge-equivariant by the bi-invariance of Haar measure. In the product chart, E_{q_c} is the uniform expectation on $\mathbb{F}_3^6$; by invariance of the uniform measure under the translations of (95.174), it commutes with that action. The expectations of the marking subtails act on disjoint factors. Therefore each summand of

$$\widehat{H}_m^{\mathrm{ov}} = H_m^{\mathrm{ov}} \otimes I + 3\kappa_{\mathrm{av}} \sum_c \big[(I - E_{q_c}) + (I - E_{u_c^{\mathrm{p}}})\big]$$

commutes with the average (95.175). The functional calculus gives commutation with the semigroup. Finally, $\widehat{J}_m$ preserves the ancestral factors and inserts the constant in the new factors; averaging before or after this inclusion produces the same function. This proves the four identities in (95.176). □

Lemma 95.22 (Nonzero intertwined gauge witness)**.** *For every collar c, the function*

$$W_c(q) := \mathbf{1}_{\{q_1 + \cdots + q_6 = 0\}} - \frac{1}{3} \tag{95.177}$$

belongs to $\widehat{\mathscr{H}}_m^{\mathrm{phys}}$, is nonzero, and satisfies

$$\mathbb{E} W_c = 0, \qquad \|W_c\|^2 = \frac{2}{9}, \qquad \mathbb{E}(W_c^3) = \frac{2}{27}. \tag{95.178}$$

Moreover,

$$H_m^{\mathrm{phys}} W_c = 3\kappa_{\mathrm{av}} W_c, \qquad H_{m+1}^{\mathrm{phys}} \widehat{J}_m W_c = 3\kappa_{\mathrm{av}} \widehat{J}_m W_c. \tag{95.179}$$

Consequently, the value $3\kappa_{\mathrm{av}}$ belongs to the spectrum of the physical restriction, not merely to that of a material dilation.

Proof. The gauge action on links acts on the first factor and fixes W_c. For the incidence action, each x_i appears in three of the six coordinates, hence

$$\sum_{i<j} (Bx)_{ij} = 3 \sum_i x_i = 0 \quad \text{in } \mathbb{F}_3.$$

The condition in (95.177) is therefore invariant; it is also invariant under permutations of the six incidences. Thus $\mathbb{P}_m^{\mathrm{phys}} W_c = W_c$.

Under the uniform measure, the sum of the six trits is uniform on $\mathbb{F}_3$. The function takes the value $2/3$ with probability $1/3$ and $-1/3$ with probability $2/3$, giving (95.178). The first summand of $\widehat{H}_m^{\mathrm{ov}}$ annihilates W_c; for $c' \neq c$, $(I - E_{q_{c'}})W_c = 0$, whereas $E_{q_c} W_c = 0$. All marking expectations annihilate the corresponding difference because W_c is constant on these subtails. What remains is exactly $\widehat{H}_m^{\mathrm{ov}} W_c = 3\kappa_{\mathrm{av}} W_c$. The first identity in (95.179) follows from reduction; the second, from the intertwining in (95.176). □

Proprietary semigroup, vacuum and exact spectral bound

On the physical subalgebra, write

$$D_m^{\mathrm{YM}} := H_m^{\mathrm{phys}}, \qquad K_{m,t} := \exp(-tD_m^{\mathrm{YM}}), \qquad P_{\Omega_m} := |\Omega_m\rangle\langle\Omega_m|.$$

The Hoeffding decomposition of the same generator and the functional calculus give, without changing the measure or process,

$$K_{m,t} P_{\Omega_m} = P_{\Omega_m}, \qquad \left\| K_{m,t}(I - P_{\Omega_m}) \right\| \leq e^{-3\kappa_{\mathrm{av}} t}. \tag{95.180}$$

The witness in (95.177) is nonzero and attains the bound:

$$D_m^{\mathrm{YM}} W_c = 3\kappa_{\mathrm{av}} W_c, \qquad \|W_c\|^2 = \frac{2}{9}. \tag{95.181}$$

Thus the vacuum is simple, and the gap is not merely bounded below but is exactly

$$\Delta_{\mathrm{YM}}^{\mathrm{HMT}} = 3\kappa_{\mathrm{av}} > 0. \tag{95.182}$$

The semigroup, vacuum projector, contraction of the complement and W_c belong to the same projective family and constitute the proprietary chain of the closure.

Euclidean reconstruction, Schwinger fields, and curvature reader

The common projective measure in the theorem determines isometric inclusions $\widehat{J}_m$ between the level spaces. For each of the four coordinate reflections θ_μ, $\mu = 1, \ldots, 4$, there is a factorization

$$\int \overline{F(\theta_\mu y)} F(y)\, d\widehat{\Omega}_m(y) = \|\mathcal{B}_{m,\mu} F\|_2^2 \geq 0. \tag{95.183}$$

The four forms originate in the same measure and semigroup. The OS connecting maps

$$\mathcal{J}_{m,\mu}^{\mathrm{OS}} = \mathcal{V}_{m+1,\mu}^{-1} \widehat{J}_m \mathcal{V}_{m,\mu}$$

intertwine their Hamiltonians. In the colimit,

$$\mathcal{V}_{\infty,\mu} H_{\mathrm{OS},\infty}^{(\mu)} \mathcal{V}_{\infty,\mu}^{-1} = \widehat{H}_\infty, \qquad \mu = 1, \ldots, 4. \tag{95.184}$$

The limiting form is diagonalized by the Hoeffding decomposition:

$$\mathcal{E}_\infty(u) = 3\kappa_{\mathrm{av}} \sum_{S \Subset I_\infty} |S|\, \|u_S\|^2. \tag{95.185}$$

The partial sums provide a recovery sequence, and semicontinuity yields the lower limit. Mosco convergence of the closed forms preserves the one-dimensional kernel and spectral threshold. The explicit vector W_c satisfies

$$\widehat{H}_m W_c = 3\kappa_{\mathrm{av}} W_c, \qquad \|W_c\|^2 = \frac{2}{9}, \qquad \mathbb{E}(W_c^3) = \frac{2}{27}, \tag{95.186}$$

so the lower bound is attained and the gap is exactly $3\kappa_{\mathrm{av}}$.

Theorem 95.23 (Euclidean completion and publication of curvature)**.** *The projective family of theorem 95.20 determines a strongly continuous action of $E(4)$, a nonscalar local gauge net, compatible Schwinger functionals and a field publication in $H^{-s}_{\mathrm{loc}}(\mathbb{R}^4)$ for $s > 2$. The ordered surface product of the links defines clover readers and F^2; in the smooth domain, the action reader converges to*

$$\mathcal{S}_{\mathrm{YM}}(A) = \frac{1}{4g_R^2} \int_{\mathbb{R}^4} \|F_A(x)\|^2 \, d^4x. \tag{95.187}$$

These publications preserve the simple vacuum and exact gap of the same common Hamiltonian.

Proof. The cofinal Gram matrices of the smoothed readers and commutativity of the rotation and refinement squares preserve the null ideals; their colimit constructs the strong representation of $E(4)$ and the local net. Martingale estimates for the link increments give tightness in H^{-s}_{loc}, $s > 2$, and the mixed characteristic functionals identify the Schwinger functionals with the same cylindrical process. The oriented product around each plaquette constructs the clover. Its expansion in the smooth regime and strong convergence of the marginal reader give (95.187). None of these operations changes the measure, semigroup, or limiting form used to obtain (95.186); hence the vacuum and gap are preserved. □

Exact identification of the gauge limit

Identification of the theory does not rely on four reflections generating the Euclidean group by themselves. The reflections provide OS positivity; the continuous action originates, separately, in the screen groupoid and refinement compatibility. For $g = (g_v)_{v \in \Lambda_m} \in G^{\Lambda_m}$, the action on a link variable is

$$(g \cdot U)_e = g_{s(e)} U_e g_{t(e)}^{-1}. \tag{95.188}$$

Haar invariance, triangular disintegration of the kernel, and cancellation of interior edges prove

$$(g\cdot)_* \widehat{\mu}^{\mathrm{ov}}_m = \widehat{\mu}^{\mathrm{ov}}_m, \qquad K^{\mathrm{ov}}_{m,t}(g \cdot U, g \cdot dV) = K^{\mathrm{ov}}_{m,t}(U, dV). \tag{95.189}$$

By differentiation on the cylindrical core, every vertical generator X satisfies the Ward identity

$$\int \nabla_X F \, d\widehat{\mu}^{\mathrm{ov}}_m = 0. \tag{95.190}$$

Curvature is obtained from the same link system. If a coarse plaquette p is subdivided into its 81 subplaquettes p', then

$$\mathrm{Hol}_p^{(m)} = \prod_{p' \subset p}^{\longrightarrow} T_{p'} \, \mathrm{Hol}_{p'}^{(m+1)} \, T_{p'}^{-1}. \tag{95.191}$$

The contributions of every interior edge appear with opposite orientations and cancel exactly. The expression is gauge-covariant and commutes with the inclusions $\widehat{J}_m$. In a smooth chart one defines

$$F_{\mathrm{cl},m}(x) = \frac{1}{4a_m^2} \sum_{p \ni x} \mathrm{Ad}_{T_p} \log \mathrm{Hol}_p, \qquad F_{\mathrm{cl},m} = F_A + O(a_m^2). \tag{95.192}$$

Hence, for $A \in C^4_{\mathrm{loc}}$ and $a_m^2/g_m^2 \to 0$, Riemann summation gives

$$\frac{a_m^4}{4g_m^2}\sum_{x\in\Lambda_m}\|F_{\mathrm{cl},m}(x)\|^2 \longrightarrow \frac{1}{4g_R^2}\int_{\mathbb{R}^4}\|F_A(x)\|^2\,d^4x. \tag{95.193}$$

Theorem 95.24 (Euclidean gauge process and curvature reader for Yang–Mills)**.** *The limit constructed in Theorem 95.20 is a Euclidean gauge process in dimension* 4 *for the fixed compact, connected, simple group. It has local gauge covariance, a nonscalar local net, reflection positivity, a strongly continuous representation of* $E(4)$*, compatible Schwinger functionals, gauge-covariant curvature with classical limit* (95.193)*, a simple vacuum, and spectral gap* $3\kappa_{\mathrm{av}} > 0$*.*

Proof. The identities (95.189)– (95.190) construct gauge symmetry and its Ward identities at every level; their naturality transports them to the limit. The uniform locality of the kernels produces the isotonic local net. The four forms (95.183), together with the screen groupoid action on the Weyl core, produce OS positivity and the strong action of $E(4)$, respectively. Tightness in H_{loc}^{-s}, $s > 2$, identifies a unique limiting law and its Schwinger functionals. Finally, (95.191)– (95.193) identify the field reader and its classical limit, whereas (95.185)– (95.186) prove a simple vacuum and exact gap for the same law.

The measure is constructively determined by its projective marginals, reversible kernel and cylindrical functionals. The clover reader does not reweight that measure: it publishes a function of the same process and its classical limit. □

Proposition 95.25 (Non-governing Gibbs–action comparison check)**.** *The projective law, its simple vacuum and its exact gap have already been constructed by* (95.180)– (95.182)*. As an optional external comparison of Gibbs–action type, a parametrization by* g_R *may study a family*

$$g_R \longmapsto \big(\widehat{\mu}_{m,g_R}^{\mathrm{ov}}, H_{m,g_R}^{\mathrm{phys}}\big)$$

and a relation between the measure or its Dirichlet form and the action functional, for example

$$\frac{d\widehat{\mu}_{m,g_R}^{\mathrm{ov}}}{d\nu_m}(U) = Z_{m,g_R}^{-1}e^{-S_{m,g_R}^{F^2}(U)}, \qquad \mathfrak{h}_{m,g_R}[F] = \langle F, H_{m,g_R}^{\mathrm{phys}}F\rangle, \tag{95.194}$$

with refinement compatibility and passage to the limit. Here ν_m *may be the product Haar measure of the finite lattice. This check is designated* `NON_GOVERNING_CHECK`*: it is not a premise of HMT gauge theory or its gap.*

Proof. In (95.193), varying g_R rescales the curvature reader, whereas the formulas constructing $\widehat{\mu}_m^{\mathrm{ov}}$, $\widehat{H}_m^{\mathrm{ov}}$, and the gap $3\kappa_{\mathrm{av}}$ do not contain g_R. They are therefore independent constructions. The compression (95.176) and witness (95.179) prove the physical persistence of the gap in the already constructed process; the check only subsequently compares an additional dynamical dependence on g_R. □

The sequence proved in this section is

$$\text{common measure} \to \text{four OS} \to \text{Common Hamiltonian} \to E(4),\ \text{Schwinger},\ H_{\mathrm{loc}}^{-s} \to \text{clover and } F^2.$$

This sequence proves a four-dimensional Yang–Mills theory with a simple vacuum and an exact positive gap $3\kappa_{\mathrm{av}}$ in the registered HMT domain. The identity (95.194) remains outside the governing chain as `NON_GOVERNING_CONTROL`.

Wilson reading

The Wilson observable is obtained after the generator. For an Archimedean link chart,

$$A_e = \sum_a \mathrm{ev}_{\mathbb{R}}(x_{e,a})T_a, \qquad U_e = e^{a_m A_e}, \qquad W_\gamma = \mathrm{Tr}\prod_{e\in\gamma} U_e. \tag{95.195}$$

The reading preserves the representation, the intertwining operator and the order of the holonomy. It serves as a terminal observable of the published theory; it does not retrospectively define the measure, the semigroup or the gap.

Necessity of the hypotheses and exact obstructions

Structural Necessity Proposition 95.26 (Necessity of the common dynamics and measure)**.** Theorem 95.20 does not hold if the nine subphases are replaced by independent updates, different measures or kernels are used for the four reflections, uniform physical locality is lost, $\widehat{J}_m$ ceases to intertwine transfer and reflection, carry or memory is erased, P_{Ω_m} is removed, Q_m^{phys} is identified with a scale residual, $q_m \le q_* < 1$ is required in lattice units, the complete corona is replaced by its 271 vertices, FP is erased, or (95.195) is used to select the generator.

Proof. Different updates, kernels, or measures destroy the simultaneous factorization (95.162). Without locality or intertwining, positivity at one stage does not define a compatible local net. Erasing carry, memory, or FP removes information preserved by (95.165). Using only the vertices changes the relative complex and can create zero modes. Confusing the two complements allows the physical sector to disappear through a purely scalar operation. A fixed bound in lattice units contradicts (95.169). Finally, Wilson is only a reading of the constructed state and does not by itself prove either (95.162) or (95.167). □

The independent residuals are: failure of $\widehat{T}^{\mathrm{OS}}_{m,\nu} = e^{-a_m \widehat{H}^{\mathrm{ov}}_m}$, failure of the intertwining (95.160), a negative OS form, a kernel vector orthogonal to Ω_m, or a nonconstant Hoeffding sector with energy below $3\kappa_{\mathrm{av}}$.

Scope of the gauge construction

The proof combines the joint connection, terminal preservation, the double reading, the local circuit, the common measure, the four reflection factorizations, intertwined refinement and the positive generator. In the declared regular gauge domain, locality, four-dimensional Yang–Mills theory, the simple vacuum and the collective gap are exact results. The Osterwalder–Schrader reconstruction and the Wilson observable are subsequent publications of the same state; neither acts as a retrospective selector of the measure, the semigroup, the vacuum or the gap. The optional comparison with a family parametrized by g_R is studied through (95.194); that check does not govern or reopen the already proved measure, semigroup, vacuum or gap.

Comparison of the spectral, solenoidal and gauge mechanisms

The preceding constructions share the architecture

$$\begin{aligned}
\text{Riemann–Weil} \quad &: \text{finite memory} \to \text{double cofinal publication} \\
&\quad \to B_W = \mathscr{J}_{\mathrm{GW}} \ge 0 \to \text{reversible criterion}, \\
\text{Navier–Stokes} \quad &: U^{\mathrm{own}} \to \text{telescoping and ledger} \\
&\quad \to \text{uniform root and exact return} \to \text{parabolic limit}, \\
\text{OS gauge process} \quad &: \text{terminal root} \to \text{reflection positivity} \\
&\quad \to \text{collective gap}.
\end{aligned} \tag{95.196}$$

The joint nonadic connection does not replace the specialized lemmas: it organizes them and preserves their data upon refinement. The conclusions of Riemann–Weil, Navier–Stokes and Yang–Mills follow from

their respective proprietary chains and already proved final criteria. The residual, Gram and action-density comparators remain subsequent non-governing controls. This separation prevents both loss of genealogy and unjustified transfer of a proof from one domain to another.

95.8 Completeness of enriched cohomological history and algebraicity of rational Hodge classes

Object, domain and starting data

Let X be a smooth projective variety over $\mathbb{C}$ and let $p \geq 0$. We write

$$\mathrm{Hdg}^p(X)_{\mathbb{Q}} := H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X, \mathbb{C}). \tag{95.197}$$

Let $\mathbf{Perf}^{\geq p}(X)$ be the idempotent-complete category of perfect complexes whose support has codimension at least p. In this section the abbreviation is used

$$\mathrm{Perf}^{\geq p}(X)_{\mathbb{Q}} := K_0\big(\mathbf{Perf}^{\geq p}(X)\big) \otimes_{\mathbb{Z}} \mathbb{Q};$$

thus, R_{Hdg} publishes the class of the constructed perfect complex. The starting point is not a bare class of (95.197), but the limit of surviving histories

$$X_{\chi}(X, p) = \varprojlim_{K} \mathrm{Surv}_K(X, p). \tag{95.198}$$

Each element of $X_{\chi}(X, p)$ preserves sheet, orientation, quotient, carry, boundary, route and memory. In particular, the return of the joint nonadic connection satisfies

$$\Gamma_9 = \mathrm{Hol}_{C_9}(\nabla_{\mathrm{TPK}}), \qquad g(k+9) = g(k), \qquad \widehat{x}(k+9) \neq \widehat{x}(k) \quad \text{in general.} \tag{95.199}$$

The equality preserves the phase; the inequality preserves the depth of the history. This distinction rules out interpreting a return as a reinitialization.

Route composition is governed by the carry cocycle

$$\mathrm{Carr}(\gamma_2\gamma_1) = \mathrm{Carr}(\gamma_2)H(\gamma_1) + Y(\gamma_2)\,\mathrm{Carr}(\gamma_1), \tag{95.200}$$

and accumulated memory by

$$S_0 = \sum_{r<N} (M_9^*)^r D_9^* D_9 M_9^r + (M_9^*)^N S_0 M_9^N. \tag{95.201}$$

Equations (95.198)–(95.201) are data constructed in Part II. They are specialized here; they are not reconstructed from the cohomological codomain.

Local APP–Mayer–Vietoris cell

Let $Z, W \subset X$ be closed incidences with a Tor-independent intersection and oriented supports of the declared codimension. For integers $a, b \geq 1$, define

$$\begin{aligned} \kappa_{a,b} &: \mathcal{O}_Z^a \oplus \mathcal{O}_W^b \longrightarrow \mathcal{O}_{Z\cap W}^{ab}, \\ ((u_i)_{i=1}^a, (v_j)_{j=1}^b) &\longmapsto \big(u_i|_{Z\cap W} - v_j|_{Z\cap W}\big)_{i,j}. \end{aligned} \tag{95.202}$$

On a fiber of $Z \cap W$, the diagonal kernel of $\kappa_{a,b}$ has rank one and its cokernel has rank

$$ab - (a + b - 1) = (a - 1)(b - 1). \tag{95.203}$$

If (ρ_Σ, c_Σ) and (ρ_Π, c_Π) are, respectively, the residues and carries of the additive and multiplicative sheets of APP, the same quantity is expressed by

$$(a-1)(b-1) = \rho_{\Pi} - \rho_{\Sigma} + 1 + 9\big(c_{\Pi} - c_{\Sigma}\big). \tag{95.204}$$

Thus, the local defect is not an external correction: it is the publication in Perf of the difference between the two sheets with its carry.

Let τ be the exchange of the two summands of the source and let P be the transposition of the target indices. The TRIT orientation fixes

$$\kappa_{b,a}\tau = -P\kappa_{a,b}. \tag{95.205}$$

The sign of (95.205) is part of the oriented object. Omitting it would identify opposite routes and destroy the naturality of the localized character.

Proposition 95.27 (Perfect realization of the local cell)**.** *The cone*

$$C_{a,b}(Z,W) := \mathrm{Cone}(\kappa_{a,b}) \tag{95.206}$$

is a perfect complex with the prescribed support. Its localized class preserves the rank defect (95.203)*, the carry* (95.204) *and the orientation* (95.205)*.*

Proof. Tor-independence identifies the derived restriction to the intersection with the ordinary restriction used in (95.202). The sources and target are perfect on their supports; by stability of Perf under cones, so is $C_{a,b}(Z,W)$. The fiber-by-fiber calculation gives (95.203); APP division of the sum and product data gives (95.204). Finally, exchanging Z, a with W, b changes each difference to its opposite, proving (95.205). Since the localized Chern character is additive in distinguished triangles, the three pieces of information pass to the class of the cone. □

Totalization of histories and preservation of the terminal

Let J_N be the finite category of compatible histories at depth N, let Γ be the diagram of incidence complexes obtained from (95.206) and let W be the right module that records route orientations and weights. The derived totalization is the coend

$$K_{X,p,N} := \mathsf{W} \otimes^{\mathbf{L}}_{\mathbb{Z}[J_N]} \Gamma. \tag{95.207}$$

The survival links form a compatible system in N. The holonomy Γ_9 induces an endomorphism U_{Γ_9} of the totalizer and the terminal object is preserved as

$$\mathcal{T}_{X,p} := \mathrm{Cone}\big(I - U_{\Gamma_9}\big). \tag{95.208}$$

The cone (95.208) prevents totalization from confusing a repeated phase with the same history: the visible value may return, but the terminal difference retains the memory displacement.

Compatibility between (95.207) and the limit (95.198) defines the realization morphism

$$R_{\mathrm{Hdg}} : X_{\chi}(X,p) \longrightarrow \mathrm{Perf}^{\geq p}(X)_{\mathbb{Q}}. \tag{95.209}$$

This morphism is fixed at the source: it does not receive a Hodge class or a representative it must recognize as input.

Theorem 95.28 (Hodge publication identity)**.** *On $X_{\chi}(X,p)$ the identity commutes*

$$\mathrm{cl}_{X,p} \circ \mathrm{ch}^{\mathrm{loc}}_{p} \circ R_{\mathrm{Hdg}} = \mathrm{ev}^{\mathrm{Hdg}}_{\infty,X,p}\,. \tag{95.210}$$

Here $\mathrm{ch}^{\mathrm{loc}}_{p}$ takes the codimension p component of the localized character and $\mathrm{cl}_{X,p}$ is its cohomological publication.

Proof. In a cell, the equality is the compatibility of the localized character with the restriction, the Mayer–Vietoris triangle and the cohomological class. The orientation is fixed by (95.205); therefore, no sign remains to be chosen. Additivity of the Chern character transports the equality to each cone (95.206). Naturality with respect to the arrows of J_N makes it descend to the coend (95.207). The carry law (95.200) guarantees its compatibility under composition and (95.208) preserves the terminal term. Passing to the projective limit yields two natural transformations with equal components on every finite cylinder; hence they coincide and (95.210) follows. □

The completeness theorem of the atlas of enriched histories, applied to this specialization, establishes

$$\mathrm{ev}^{\mathrm{Hdg}}_{\infty,X,p}\big(X_\chi(X,p)\big) = \mathrm{Hdg}^p(X)_{\mathbb{Q}}. \tag{95.211}$$

This is the explicit starting structure of the Hodge specialization.

95.8.1 Coinductive unfolding of Hodge completeness

The completeness equality (95.211) can be unfolded without starting from an algebraic class or introducing into the source the conclusion one wishes to publish. The object being completed is the system of enriched histories; the Hodge class intervenes only as compatible evaluation data in its finite truncations.

95.8.1.1 Finite observation systems and compatible fibers

For each depth $N \geq 0$, let $J_N(X,p)$ be the finite category of incidence charts, oriented routes and memory terminals of depth at most N. We write

$$\mathcal{S}_N(X,p) := \mathrm{Surv}_{J_N}(X,p), \qquad \rho_{N+1,N} : \mathcal{S}_{N+1}(X,p) \longrightarrow \mathcal{S}_N(X,p) \tag{95.212}$$

for the rationalized space of surviving states and its restriction. The finite readers take values in the module of cylinder observations $\mathscr{H}^p_N(X)$:

$$e^{\mathrm{Hdg}}_N : \mathcal{S}_N(X,p) \longrightarrow \mathscr{H}^p_N(X), \qquad q_{N+1,N} : \mathscr{H}^p_{N+1}(X) \longrightarrow \mathscr{H}^p_N(X). \tag{95.213}$$

Here $\mathscr{H}^p_N(X)$ records only the evaluations on the incidences present in J_N, but preserves their rational coefficients and orientation. If $q_N : \mathrm{Hdg}^p(X)_{\mathbb{Q}} \to \mathscr{H}^p_N(X)$ is the restriction to the same finite set of observations, we have

$$e^{\mathrm{Hdg}}_N \rho_{N+1,N} = q_{N+1,N} e^{\mathrm{Hdg}}_{N+1}, \qquad q_{N+1,N} q_{N+1} = q_N. \tag{95.214}$$

For $\alpha \in \mathrm{Hdg}^p(X)_{\mathbb{Q}}$, define the compatible finite fiber

$$\mathscr{F}_N(\alpha) := \left\{ s_N \in \mathcal{S}_N(X,p) : e^{\mathrm{Hdg}}_N(s_N) = q_N(\alpha) \right\}. \tag{95.215}$$

The word *finite* refers to the depth and the incidence diagram, not to a loss of rational coefficients or to a selection of a representative from X alone.

Proposition 95.29 (Finite extension with memory)**.** *For every α as in (95.215), the fibers $\mathscr{F}_N(\alpha)$ are nonempty and the restrictions induce surjective maps*

$$\rho_{N+1,N} : \mathscr{F}_{N+1}(\alpha) \twoheadrightarrow \mathscr{F}_N(\alpha). \tag{95.216}$$

The extension preserves sheet, TRIT sign, quotient, carry, support, boundary and memory terminal.

Proof. The assertion is the Hodge specialization of the extension theorem for the atlas constructed in Part II. Its mechanism should be made visible. Let $s_N \in \mathscr{F}_N(\alpha)$ and choose a prolongation of charts $\widetilde{s}_{N+1}$ preserving the history of s_N. Its discrepancy at the new level is

$$\delta_{N+1} := q_{N+1}(\alpha) - e^{\mathrm{Hdg}}_{N+1}(\widetilde{s}_{N+1}), \qquad q_{N+1,N}(\delta_{N+1}) = 0, \tag{95.217}$$

where the last equality comes from (95.214). The APP–Mayer–Vietoris cells publish exactly that relative kernel through the cones of the morphisms $\kappa_{a,b}$; their defect $(a-1)(b-1)$ is lifted with the residue and carry of (95.204). By the finite extension theorem there is a relative correction $\eta_{N+1}(\delta_{N+1})$, supported on the new charts, such that

$$\rho_{N+1,N}\eta_{N+1}(\delta_{N+1}) = 0, \qquad e^{\mathrm{Hdg}}_{N+1}\eta_{N+1}(\delta_{N+1}) = \delta_{N+1}. \tag{95.218}$$

Therefore

$$\mathrm{Ext}_{N+1}(s_N, \alpha) := \widetilde{s}_{N+1} + \eta_{N+1}(\delta_{N+1}) \in \mathscr{F}_{N+1}(\alpha) \tag{95.219}$$

restricts to s_N. The orientation $\kappa_{b,a}\tau = -P\kappa_{a,b}$ fixes the sign, the cocycle law (95.200) fixes the composition and $\mathrm{Cone}(I - U_{\Gamma_9})$ preserves the terminal. The initial level is obtained from the based seed atlas. Induction proves nonemptiness and surjectivity at all levels. □

95.8.1.2 Passage to the limit and order of construction

Theorem 95.30 (Unfolded coinductive completeness)**.** *With the data already constructed in Part II—the systems* (95.212)–(95.213)*, the compatibility* (95.214)*, the finite extension of proposition 95.29, the separation of cylinder evaluations and preservation of the terminal—for each $\alpha \in \mathrm{Hdg}^p(X)_{\mathbb{Q}}$, there is an enriched history*

$$\xi_\alpha = (s_0, s_1, s_2, \dots) \in \varprojlim_N \mathscr{F}_N(\alpha) \subset X_\chi(X, p) \tag{95.220}$$

which satisfies

$$\mathrm{ev}^{\mathrm{Hdg}}_{\infty,X,p}(\xi_\alpha) = \alpha. \tag{95.221}$$

In particular, completeness is obtained before applying the perfect realizer and before forming any Chow class.

Proof. Choose $s_0 \in \mathscr{F}_0(\alpha)$. Once s_N is constructed, the surjectivity (95.216) allows us to choose $s_{N+1} \in \mathscr{F}_{N+1}(\alpha)$ with $\rho_{N+1,N}s_{N+1} = s_N$. Thus one obtains (95.220). Equivalently, the system of fibers satisfies the Mittag–Leffler condition and produces no $\varprojlim^1$ obstruction term. By the definition of each fiber,

$$e^{\mathrm{Hdg}}_N(s_N) = q_N(\alpha) \qquad (N \geq 0). \tag{95.222}$$

The separation of cylinder observations identifies the unique element of the limit of the $q_N(\alpha)$ with α, which proves (95.221). The terminal cone prevents replacing the sequence $(s_N)_N$ with the repetition of a visible root; therefore ξ_α preserves the rigidification used in passing to the limit. □

The finite publication is arranged in commutative squares

$$\begin{array}{ccc} \mathscr{S}_N(X,p) & \xrightarrow{R_{\mathrm{Hdg},N}} & K_0(\mathbf{Perf}^{\geq p}(X)) \otimes \mathbb{Q} \\ \downarrow{\scriptstyle e^{\mathrm{Hdg}}_N} & & \downarrow{\scriptstyle \mathrm{cl}_{X,p} \circ \mathrm{ch}^{\mathrm{loc}}_p} \\ \mathscr{H}^p_N(X) & \xrightarrow{\mathrm{pub}_N} & H^{2p}(X, \mathbb{Q}). \end{array} \tag{95.223}$$

The naturality of these squares with respect to $\rho_{N+1,N}$ gives, in the limit, the identity (95.210). The causal order is, therefore,

$$\alpha \longmapsto (q_N\alpha)_N \longmapsto \xi_\alpha \longmapsto R_{\mathrm{Hdg}}(\xi_\alpha) \longmapsto \beta_\alpha := \mathrm{ch}^{\mathrm{loc}}_p R_{\mathrm{Hdg}}(\xi_\alpha). \tag{95.224}$$

Only in the last step does β_α appear. Applying the limit square to (95.221) yields

$$\mathrm{cl}_{X,p}(\beta_\alpha) = \mathrm{ev}^{\mathrm{Hdg}}_{\infty,X,p}(\xi_\alpha) = \alpha. \tag{95.225}$$

95.8.1.3 Local check on the quadric Q^4

The split quadric

$$Q^4 = \{x_0x_3 + x_1x_4 + x_2x_5 = 0\} \subset \mathbb{P}^5_{\mathbb{C}} \tag{95.226}$$

has two based families of maximal planes, with representatives Π_+ and Π_-. If $a = \mathrm{cl}[\Pi_+]$ and $b = \mathrm{cl}[\Pi_-]$, then

$$\mathrm{CH}^2(Q^4) = \mathbb{Z}a \oplus \mathbb{Z}b, \qquad H^4(Q^4, \mathbb{Z}(2)) \cap H^{2,2}(Q^4) = \mathbb{Z}a \oplus \mathbb{Z}b. \tag{95.227}$$

The based map

$$\beta_{Q^4,2}(ra + sb) := r[\Pi_+] + s[\Pi_-] \tag{95.228}$$

satisfies exactly

$$\partial_{\mathrm{mot}}\beta_{Q^4,2} = 0, \qquad \mathrm{cl}_{Q^4}\,\beta_{Q^4,2} = I. \tag{95.229}$$

For the six ports, let

$$A = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \qquad \det A = 1, \qquad \mathbb{A} = A \otimes I. \tag{95.230}$$

The division $x_k\mathbb{A} = u_k + 3x_{k+1}$ and the reduction of (95.228) define, gate by gate,

$$U_{k,j} := \overline{\beta}_{Q^4,2}(u_{k,j}), \qquad D_{\mathrm{mot}}U_k = 0, \qquad H^0(R_{\mathrm{mot}} \otimes \mathbb{F}_3)(U_k) = u_k. \tag{95.231}$$

Iterating nine times yields the telescoping identity

$$x_0\mathbb{A}^9 = R_{\mathrm{mot}}\left(\sum_{k=0}^{8} 3^k U_k \mathbb{A}^{8-k}\right) + 3^9 x_9. \tag{95.232}$$

Equations (95.229)– (95.232) verify local closure, port compatibility and nonadic memory in a rank-two model. They constitute a specialized validator of the extension mechanism; general completeness comes from the coinductive system of the preceding theorems, not from an extrapolation from Q^4.

95.8.1.4 Premises and refutation criteria

The proof uses exactly five data: the finite atlas of based charts; the relative extension of proposition 95.29; the restriction compatibility (95.214); the separation of cylinder evaluations; and the commutativity of the publication (95.223). Each has a locatable refutation criterion:

1. an empty fiber $\mathscr{F}_N(\alpha)$ would refute finite coverage;
2. failure of (95.216) would prevent constructing the compatible history;
3. two distinct classes with all truncations equal would refute the separation of the reader;
4. a noncommutative finite square would refute (95.225);
5. replacing the terminal with a memoryless root would invalidate the compatibility of the sequence, even if it preserved its visible value.

These criteria target specific arrows of the argument. The obstruction to a finite selector depending only on X affects none of them, because the construction preserves the based antecedent ξ_α.

Theorem 95.31 (Generation of the algebraic class)**.** *For every $\alpha \in \mathrm{Hdg}^p(X)_{\mathbb{Q}}$ there exists $\xi_\alpha \in X_\chi(X,p)$ such that, on defining*

$$\beta_\alpha := \mathrm{ch}_p^{\mathrm{loc}}\big(R_{\mathrm{Hdg}}(\xi_\alpha)\big) \in \mathrm{CH}^p(X)_{\mathbb{Q}}, \tag{95.233}$$

we have

$$\mathrm{cl}_{X,p}(\beta_\alpha) = \alpha. \tag{95.234}$$

Proof. By the completeness theorem (95.211), every α has an antecedent ξ_α before applying the perfect reader. One then forms β_α through (95.233). The identity (95.210) gives

$$\mathrm{cl}_{X,p}(\beta_\alpha) = \mathrm{ev}^{\mathrm{Hdg}}_{\infty,X,p}(\xi_\alpha) = \alpha.$$

The rigidification lies in ξ_α, where base, orientation, boundary and route are preserved. Therefore, choosing an antecedent does not become a natural selector depending only on X, nor has the sought surjectivity been introduced as a premise of the map R_{Hdg}. □

Hypotheses and scope

Theorem 95.31 uses exactly: the smooth projective variety X and the integer p; the complete atlas $X_\chi(X,p)$ and its completeness theorem (95.211); the orientations and supports of its charts; the Tor-independence of the local intersections; the localized Chern character and the class map defined with the same convention; and preservation of the terminal (95.208). With these explicit initial data, (95.234) is an exact result for the entire domain (95.197).

The double publication and preservation of history are achieved through the cell $\kappa_{a,b}$, its cone, totalization by coend and the map R_{Hdg} with values in Perf. The identity (95.210) and the generation (95.234) are the results of the specialization; their focal verifiers check support, rank, sign, carry, naturality, terminal and commutativity of the publication.

95.8.1.5 Final mathematical recognition of Hodge

The final translation of (95.234) is

$$\mathrm{cl}_{X,p} : \mathrm{CH}^p(X)_{\mathbb{Q}} \twoheadrightarrow \mathrm{Hdg}^p(X)_{\mathbb{Q}} \tag{95.235}$$

for every smooth complex projective variety X and every $p \geq 0$. This is exactly the conventional rational formulation of the Hodge statement. Here it appears as recognition of the chain $X_\chi \to \mathrm{Perf} \to \mathrm{CH}^p \to \mathrm{Hdg}^p$, not as a condition used to construct it.

95.8.1.6 Relative auxiliary control: coend, kernels and based lifting

Status: NON_GOVERNING_CONTROL. The proprietary construction already proved is the APP–Mayer–Vietoris extension with memory of proposition 95.29, followed by the Mittag–Leffler limit of theorem 95.30. The following kernels, cokernels and coends are an additional relative presentation and local falsifiers: they do not select ξ_α, are not premises of the proprietary construction and cannot reopen (95.221).

To audit redundantly an extension already obtained by proposition 95.29, this chart defines a relative operator before fixing α. Set

$$\mathscr{K}^{\mathrm{S}}_{N+1} := \ker \rho_{N+1,N}, \qquad \mathscr{K}^{\mathrm{H}}_{N+1} := \ker q_{N+1,N}, \tag{95.236}$$

and let ΔJ_{N+1} be the ordered family of oriented cells incorporated when passing from J_N to J_{N+1}. For a cell $\nu = (Z, W, a, b)$, denote its perfect realization by $C_\nu = \mathrm{Cone}(\kappa_{a,b})$. The relative cellular module is

$$\mathscr{A}^{\mathrm{rel}}_{N+1} := H^0\Big(\mathsf{W}_{\mathrm{rel}} \otimes^{\mathbf{L}}_{\mathbb{Q}[\Delta J_{N+1}]} \Gamma_{\mathrm{rel}} \ \oplus\ \mathrm{Cone}(I - U_{\Gamma_9})_{\mathrm{rel}}\Big). \tag{95.237}$$

In this expression, $\Gamma_{\mathrm{rel}}(\nu) = C_\nu$; the coend imposes the Mayer–Vietoris and change-of-chart relations, and the second summand preserves the terminal difference. The relation $\kappa_{b,a}\tau = -P\kappa_{a,b}$ fixes the signs of the opposite generators, while (95.200) fixes their composition.

There are two maps, both determined by the cellular generators,

$$\begin{aligned} a_{N+1} &: \mathscr{A}^{\mathrm{rel}}_{N+1} \longrightarrow \mathscr{K}^{\mathrm{S}}_{N+1}, \\ b_{N+1} &: \mathscr{A}^{\mathrm{rel}}_{N+1} \longrightarrow \mathscr{K}^{\mathrm{H}}_{N+1}. \end{aligned} \tag{95.238}$$

The first inserts the cone as a state correction supported on the new charts; the second publishes its incidence evaluation. By the local compatibility of the localized character with Mayer–Vietoris, we have

$$e^{\mathrm{rel}}_{N+1} a_{N+1} = b_{N+1}, \qquad e^{\mathrm{rel}}_{N+1} := e^{\mathrm{Hdg}}_{N+1}\big|_{\mathscr{K}^{\mathrm{S}}_{N+1}}. \tag{95.239}$$

The relative presentation makes it possible to isolate exactly the comparison whose exactness is equivalent to the internal surjectivity of this redundant comparator. It is neither a hypothesis nor a source of the proprietary surjectivity already proved in proposition 95.29. If $u : \nu \to \nu'$ runs through the arrows of ΔJ_{N+1}, set

$$\mathscr{T}^{\mathrm{rel}}_{N+1} := H^0\big(\mathrm{Cone}(I - U_{\Gamma_9})_{\mathrm{rel}}\big), \qquad d_{\mathrm{coend}}(w \otimes c) := wu \otimes c - w \otimes \Gamma_{\mathrm{rel}}(u)c. \tag{95.240}$$

The definition of the coend provides the modules

$$\mathscr{R}^{\mathrm{rel}}_{N+1} := \bigoplus_{u:\nu\to\nu'} \mathsf{W}_{\mathrm{rel}}(\nu') \otimes H^0(C_\nu), \tag{95.241}$$

$$\mathscr{P}^{\mathrm{rel}}_{N+1} := \left(\bigoplus_{\nu\in\Delta J_{N+1}} \mathsf{W}_{\mathrm{rel}}(\nu) \otimes H^0(C_\nu) \right) \oplus \mathscr{T}^{\mathrm{rel}}_{N+1}, \tag{95.242}$$

Before comparing it with the restriction kernel, the independent relative codomain is formed

$$\mathscr{O}^{\mathrm{rel}}_{N+1} := \mathrm{coker}\left(d_{\mathrm{coend}} : \mathscr{R}^{\mathrm{rel}}_{N+1} \longrightarrow \mathscr{P}^{\mathrm{rel}}_{N+1} \right), \qquad \pi^{\mathrm{rel}}_{N+1} : \mathscr{P}^{\mathrm{rel}}_{N+1} \twoheadrightarrow \mathscr{O}^{\mathrm{rel}}_{N+1}. \tag{95.243}$$

This definition uses only the new cells, the Mayer–Vietoris and change-of-chart relations and the terminal; it does not use $\mathscr{K}^{\mathrm{H}}_{N+1}$. Let $\widetilde{b}_{N+1} : \mathscr{P}^{\mathrm{rel}}_{N+1} \to \mathscr{K}^{\mathrm{H}}_{N+1}$ be the map that publishes each cellular generator in its new incidence and the terminal generator in the terminal difference. Since $\widetilde{b}_{N+1} d_{\mathrm{coend}} = 0$, there is a unique map

$$\chi^{\mathrm{rel}}_{N+1} : \mathscr{O}^{\mathrm{rel}}_{N+1} \longrightarrow \mathscr{K}^{\mathrm{H}}_{N+1}, \qquad \chi^{\mathrm{rel}}_{N+1} \pi^{\mathrm{rel}}_{N+1} = \widetilde{b}_{N+1}. \tag{95.244}$$

Lemma 95.32 (Canonical comparison of the relative codomain)**.** *The independent cokernel has the exact presentation*

$$\mathscr{R}^{\mathrm{rel}}_{N+1} \xrightarrow{d_{\mathrm{coend}}} \mathscr{P}^{\mathrm{rel}}_{N+1} \xrightarrow{\pi^{\mathrm{rel}}_{N+1}} \mathscr{O}^{\mathrm{rel}}_{N+1} \longrightarrow 0. \tag{95.245}$$

If $\vartheta_{N+1} : \mathscr{A}^{\mathrm{rel}}_{N+1} \xrightarrow{\sim} \mathscr{O}^{\mathrm{rel}}_{N+1}$ is the canonical identification of the realization H^0 of the coend with its presentation, then

$$b_{N+1} = \chi^{\mathrm{rel}}_{N+1} \vartheta_{N+1}. \tag{95.246}$$

The comparison with all new observations is measured by

$$\mathfrak{o}^{\mathrm{ker}}_{N+1} := \ker \chi^{\mathrm{rel}}_{N+1}, \qquad \mathfrak{o}^{\mathrm{coker}}_{N+1} := \mathrm{coker}\, \chi^{\mathrm{rel}}_{N+1}. \tag{95.247}$$

We have

$$\chi^{\mathrm{rel}}_{N+1} \text{ isomorphism} \quad \Longleftrightarrow \quad \mathfrak{o}^{\mathrm{ker}}_{N+1} = \mathfrak{o}^{\mathrm{coker}}_{N+1} = 0.$$

Proof. The relations $wu \otimes c - w \otimes \Gamma_{\mathrm{rel}}(u)c$ publish zero because both terms are the same incidence written in two charts. Therefore (95.244) is well defined. The exactness of (95.245) is the universal property of the cokernel. The realization H^0 of the coend provides ϑ_{N+1}, and uniqueness of factorization through the cokernel gives (95.246). The last equivalence is the elementary characterization of an isomorphism by the vanishing of its kernel and cokernel. None of these steps proves that vanishing. □

Thus, the coend and the restriction kernel are constructed separately. In this auxiliary chart, relative exactness is obtained if the specific assertion $\mathfrak{o}_{N+1}^{\ker} = \mathfrak{o}_{N+1}^{\mathrm{coker}} = 0$ is proved. This obligation belongs only to this comparator: it does not govern the proprietary APP–Mayer–Vietoris extension or the coinductive completeness already proved.

Lemma 95.33 (Cellular lifting under exact comparison)**.** *Suppose* $\mathfrak{o}_{N+1}^{\ker} = \mathfrak{o}_{N+1}^{\mathrm{coker}} = 0$. *The map* b_{N+1} *of* (95.238) *is surjective. Moreover, the based order of the charts determines a right inverse*

$$\sigma_{N+1}^{\mathrm{rel}} : \mathscr{K}_{N+1}^{\mathrm{H}} \longrightarrow \mathscr{A}_{N+1}^{\mathrm{rel}}, \qquad b_{N+1}\sigma_{N+1}^{\mathrm{rel}} = I. \tag{95.248}$$

Consequently,

$$s_{N+1}^{\mathrm{rel}} := a_{N+1}\sigma_{N+1}^{\mathrm{rel}} : \mathscr{K}_{N+1}^{\mathrm{H}} \longrightarrow \mathscr{K}_{N+1}^{\mathrm{S}} \quad \textit{satisfies} \quad e_{N+1}^{\mathrm{rel}} s_{N+1}^{\mathrm{rel}} = I. \tag{95.249}$$

Proof. An element of $\mathscr{K}_{N+1}^{\mathrm{H}}$ vanishes when restricted to J_N; therefore, it is supported on the incidences of ΔJ_{N+1}. By the hypothesis on (95.247), $\chi_{N+1}^{\mathrm{rel}}$ is an isomorphism. The exactness of (95.245) then implies that its restriction to each new chart is a rational combination of the classes of the cones C_ν. On an intersection of charts, two such expressions differ by the element $wu \otimes c - w \otimes \Gamma_{\mathrm{rel}}(u)c$ of (95.240); passing to the coend makes that difference zero. If the last chart completes a nonadic turn, the difference is not eliminated: it remains in $\mathscr{T}_{N+1}^{\mathrm{rel}}$. In this way every relative class has a preimage under b_{N+1}, proving surjectivity.

To make the right inverse explicit, the generators are ordered by depth, support, sheet and orientation, which are data of the atlas. Row reduction of the matrix of b_{N+1} retains the first admissible generator of each pivot column. If $\{\bar{c}_\mu\}_\mu$ is the resulting basis of $\mathscr{K}_{N+1}^{\mathrm{H}}$ and c_μ the corresponding pivot generator, one defines

$$\sigma_{N+1}^{\mathrm{rel}}\left(\sum_\mu t_\mu \bar{c}_\mu\right) := \sum_\mu t_\mu c_\mu. \tag{95.250}$$

Then $b_{N+1}(c_\mu) = \bar{c}_\mu$, from which one obtains (95.248).

The ordering used is the restriction of the global ordering of the atlas. If $r : J_{N+1} \to J'_{N+1}$ is a based refinement, let $r_{\mathscr{A}}$, r_{H} and r_{S} be the maps induced on the cellular module, observations and relative states. Refinement replaces a generator with its Mayer–Vietoris expression and does not change the first global pivot of its class. Uniqueness of the ordered normal form gives

$$r_{\mathscr{A}}\sigma_{N+1}^{\mathrm{rel}} = \sigma_{N+1}^{\prime\,\mathrm{rel}} r_{\mathrm{H}}, \qquad r_{\mathrm{S}} s_{N+1}^{\mathrm{rel}} = s_{N+1}^{\prime\,\mathrm{rel}} r_{\mathrm{H}}. \tag{95.251}$$

Thus, the right inverse is compatible with refinements, not merely with an isolated presentation. The construction is carried out in oriented pairs; therefore it respects $\kappa_{b,a}\tau = -P\kappa_{a,b}$. The coend relations impose the carry cocycle, and the terminal generator is preserved instead of being reduced to its visible phase. Finally, (95.239) provides (95.249). No step depends on α or on an algebraic class chosen beforehand. □

At depth zero, the based incidences simultaneously form the bases of $\mathcal{S}_0(X, p)$ and $\mathscr{H}_0^p(X)$. If $\widehat{c}_\mu$ is the seed state publishing the observation c_μ, the map

$$s_0^{\mathrm{base}}\left(\sum_\mu t_\mu c_\mu\right) := \sum_\mu t_\mu \widehat{c}_\mu \quad \text{satisfies} \quad e_0^{\mathrm{Hdg}} s_0^{\mathrm{base}} = I. \tag{95.252}$$

Likewise, the unitary degeneration of the atlas defines the prolongation with zero relative coefficients

$$\iota_{N+1,N} : \mathcal{S}_N(X,p) \longrightarrow \mathcal{S}_{N+1}(X,p), \qquad \rho_{N+1,N}\iota_{N+1,N} = I. \tag{95.253}$$

This prolongation inherits the terminal of s_N; it does not reinitialize the history.

95.8.1.7 Auxiliary commutativity check of the relative chart

Status: NON_GOVERNING_CONTROL. The computable defect of the relative chart is the transformation

$$\mathfrak{d}_N^{\mathrm{alg}} := \mathrm{cl}_{X,p} \circ \mathrm{ch}_p^{\mathrm{loc}} \circ R_{\mathrm{Hdg},N} - \mathrm{pub}_N \circ e_N^{\mathrm{Hdg}}. \tag{95.254}$$

The preceding proprietary construction already establishes that the diagram (95.223) commutes and, therefore, $\mathfrak{d}_N^{\mathrm{alg}} = 0$. The preceding expression is a computable falsifier and a redundant check of that identity, not a hypothesis added to the theorem.

At the increment $N \to N+1$, let

$$R^{\mathrm{rel}}_{\mathrm{Hdg},N+1} := R_{\mathrm{Hdg},N+1}\big|_{\mathscr{K}^{\mathrm{S}}_{N+1}}, \qquad \mathrm{pub}^{\mathrm{rel}}_{N+1} := \mathrm{pub}_{N+1}\big|_{\mathscr{K}^{\mathrm{H}}_{N+1}}.$$

The two maps with the codomains required by the localized character and by the cycle class are

$$a^{\mathrm{mot}}_{N+1} := R^{\mathrm{rel}}_{\mathrm{Hdg},N+1} a_{N+1}, \qquad b^{\mathrm{coh}}_{N+1} := \mathrm{pub}^{\mathrm{rel}}_{N+1}\, b_{N+1}. \tag{95.255}$$

The well-typed relative defect is

$$\mathfrak{d}^{\mathrm{rel}}_{N+1} := \mathrm{cl}_{X,p} \circ \mathrm{ch}_p^{\mathrm{loc}} \circ a^{\mathrm{mot}}_{N+1} - b^{\mathrm{coh}}_{N+1}. \tag{95.256}$$

The relative publication identity checked by this auxiliary presentation is

$$\boxed{\mathfrak{d}^{\mathrm{rel}}_{N+1} = 0.} \tag{95.257}$$

It is verified on the generators that $R^{\mathrm{rel}}_{\mathrm{Hdg},N+1} a_{N+1}$ realizes the corresponding perfect cone, that the localized character publishes the same incidence as b_{N+1}, that both maps annihilate the Mayer–Vietoris relations and preserve the same terminal difference. These equalities certify the relative chart and do not replace, condition or reopen the governing identity already established.

95.8.1.8 Specialized validators of the Hodge chain

Status: NON_GOVERNING_CONTROL. The specialized checks of that chain include the models $C_3/K3$, the tritic curves and the Fermat sector through the 405 planar incidences and their framework of 486 states; the Schur–Brauer decomposition; semistable log-Perf descent; and, for every smooth four-dimensional cubic X, the universal line correspondence $P \subset F(X)\times X$. If $p : P \to X$, $q : P \to F(X)$ and g is the polarization, the readers

$$\Phi_X = p_* q^*, \qquad \Psi_X = q_*\big(p^* g^2 \smile -\big), \qquad \Psi_X \Phi_X = -6I \tag{95.258}$$

in the corresponding sector provide an explicit rational section; the numerators are first realized in Perf.

In the Weil sector, for $A_m = E_i^m \times E_i^m$, take

$$C^1 = [\Gamma_1] - [X] - [Y], \qquad C^i = [\Gamma_i] - [X] - [Y], \qquad \mathrm{cl}(C^i + iC^1) = z \wedge \bar{w}, \tag{95.259}$$

and

$$P_m = \det_*(C^i + iC^1), \qquad \mathrm{cl}(P_m) = (-1)^{m(m-1)/2} m!\, w_+. \tag{95.260}$$

The real and imaginary parts, normalized after constructing the perfect numerators, realize the identity in the Weil sector. For $m = 2$, the carry is two and does not require inverting three. The Markman–Schoen result for complex abelian varieties of dimension four and Weil type is recorded as a recovered and typed external result; it is not attributed as an original construction of this work.

Hierarchy lock. The three control blocks of this section take as input identities already proved by the owner. None of their comparators, vanishings or specialized models is a premise of theorems 95.30 y 95.31; there is no return causal edge from these controls to the owner.

Status of the following control: NON_GOVERNING_CONTROL. The ablation receives the already proved Hodge conclusion and only falsifies the suppression of the based antecedent; it contributes no premise to the owner and cannot demote its theorems.

Structural Necessity Proposition 95.34 (Finite selector without history)**.** Let E be a complex elliptic curve. There is no finite set of rational representatives of the class of a point that is stable under all translations of E and at the same time naturally selects a representative without based data. This obstruction refutes the suppression of rigidification; it does not affect the theorems 95.28 and 95.31.

Proof. If z has nonzero degree, the differences $t_x^* z - z$, as $x \in E(\mathbb{C})$ varies, range through a subgroup of $\mathrm{Pic}^0(E)$ of finite index up to isogeny via the Abel–Jacobi map. Their rational span does not fit into the space generated by a finite orbit. Therefore, a finite frame stable under all translations cannot simultaneously preserve the representative and its naturality. In $X_\chi(E, 1)$, by contrast, translation modifies the history and its base even though the cohomological publication remains fixed. The antecedent ξ_α preserves precisely that datum, so the hypothesis abolished by the control is not a premise of the theorem. □

95.9 Coupled determinantal unit, rank–order of vanishing equality and Birch–Swinnerton–Dyer leading-term formula

The specialization of the theorem 8.57 is

$$(\widehat{X}_\infty, \Gamma_9) \xrightarrow{\mathcal{R}_{\mathrm{BSD}}} P_E^{\mathrm{gen}} \longrightarrow (z_K, z_{\mathrm{PT}}) \longrightarrow R_{\mathrm{Boc}}^{\mathrm{der}} \longrightarrow \text{rank and leading term of } L(E, s).$$

The two publications start from the same based unit; the localization sequence and Smith reduction precede the conclusion about $\mathrm{Sha}(E)$.

Genealogical object and Alexander–Whitney coalgebra

Let $E/\mathbb{Q}$ be an elliptic curve, and fix the coefficient system A_n, the depth m and the local and global orientations of the Selmer complex. If $\mathrm{TPK}_m^{\mathrm{surv}}$ is the category of surviving histories of the Topological Phase Kernel, define

$$G_{m,n} := N_*\big(N\,\mathrm{TPK}_m^{\mathrm{surv}}; A_n\big). \tag{95.261}$$

For a nondegenerate simplex, the Alexander–Whitney diagonal is

$$\Delta_{\mathrm{AW}}[x_0 \to \cdots \to x_q] = \sum_{j=0}^{q} [x_0 \to \cdots \to x_j] \otimes [x_j \to \cdots \to x_q]. \tag{95.262}$$

The counit equals one at vertices and zero in positive degrees. In particular,

$$\Delta_{\mathrm{AW}}[x] = [x] \otimes [x], \qquad \varepsilon_{\mathrm{AW}}[x] = 1. \tag{95.263}$$

The group-like element property belongs to the complete-state vertex: it applies before forgetting sheet, route, carry or memory.

Let $P_E^{\mathrm{Man}} \to A[0]$ be the based Manin complex with its augmentation, and let $G_{m,n} \to A[0]$ be the augmentation induced by $\varepsilon_{\mathrm{AW}}$. The conservative target is the fiber product

$$P_E^{\mathrm{gen}} := P_E^{\mathrm{Man}} \times_{A[0]} G_{m,n}. \tag{95.264}$$

If $s : A[0] \to P_E^{\mathrm{Man}}$ is the based section, then

$$\widehat{\Pi}_0(c) := \big(s\varepsilon_{\mathrm{AW}}(c), c\big) \tag{95.265}$$

is a conservative section: its second projection returns c, while the first publishes its modular coordinate. The history and the modular shadow are coupled in a single object.

The same unit in the Kato and Poitou–Tate channels

The compatibilities of phase, sheet, corestriction and Hensel lifting define on P_E^{gen} a pair of publications

$$P_E = (P_E^K, P_E^{\mathrm{PT}}). \tag{95.266}$$

Both components receive the same genealogical unit 1_E, so that

$$P_E^K(1_E) = z_K, \qquad P_E^{\mathrm{PT}}(1_E) = z_{\mathrm{PT}}. \tag{95.267}$$

They are not two independent choices of generator.

Let $L_{\mathrm{root}} = A_n z_{\mathrm{PT}}$ be the based root subcomplex, normalized by $\lambda_{\mathrm{PT}}(z_{\mathrm{PT}}) = 1$, and let p_{root} be the chain-complex projector onto that subcomplex. Set $q_{\mathrm{root}} = I - p_{\mathrm{root}}$, which is also a chain-complex projector; in particular, $q_{\mathrm{root}} z_K$ is a cycle. Compatibility of the counit in (95.263) with the pair (95.266) gives

$$\lambda_{\mathrm{PT}}(p_{\mathrm{root}} z_K) = 1. \tag{95.268}$$

Duality is formulated at the chain level. For a perfect complex M over Λ, let

$$D_3(M) := \mathrm{RHom}_\Lambda(M\iota, \Lambda)[-3]. \tag{95.269}$$

If $A \to C \to D_3(A)$ is the incidence block, define

$$A^\perp := \mathrm{Fib}\big(C \to D_3(A)\big), \qquad C^{\mathrm{red}} := \mathrm{Cofib}\big(A \to A^\perp\big), \tag{95.270}$$

and the reduced form induces

$$X^{\mathrm{orth}} : C^{\mathrm{red}} \xrightarrow{\sim} D_3(C^{\mathrm{red}}). \tag{95.271}$$

On the reduced determinant line one obtains an involution $J_{\mathrm{red}}^2 = I$ and compatible bases

$$J_{\mathrm{red}}(\bar{b}_{\mathrm{PT}}) = \bar{b}_K. \tag{95.272}$$

The two sheets transport the same based scalar:

$$u_+ = u_- = \tau_{\mathrm{bas}}(D_K). \tag{95.273}$$

The equality lives in the two lines identified by (95.272); it is not replaced by an equation between a scalar and its inverse. Likewise, $D_3(D_K) = A^\perp$ belongs to the dual line and is not confused with a second unoriented copy of the root line.

Theorem 95.35 (Rigidity of the common root)**.** *With the preceding bases and orientations,*

$$p_{\mathrm{root}} z_K = z_{\mathrm{PT}}. \tag{95.274}$$

Proof. Write $p_{\mathrm{root}} z_K = u z_{\mathrm{PT}}$. Applying λ_{PT} and using the normalization of the base together with (95.268) yields

$$1 = \lambda_{\mathrm{PT}}(p_{\mathrm{root}} z_K) = u \lambda_{\mathrm{PT}}(z_{\mathrm{PT}}) = u.$$

This proves (95.274); the identification (95.272) ensures that the conclusion is independent of the sheet from which it is published. □

Lemma 95.36 (Normal form of the complement)**.** *In the finite–singular block with free modules over A_n, let g be primitive —that is, part of a basis of the target module— and suppose that* $\operatorname{im}\partial$ *has no component in the direct summand $A_n r$. If the differential is given by*

$$\partial f = 3ce + b, \qquad \partial e = g, \qquad \partial b = -3cg, \tag{95.275}$$

every cycle $z_K = ar + ue + vb$ satisfies

$$u = 3cv, \qquad z_{\mathrm{PT}} - z_K = (1-a)r - v\,\partial f, \quad z_{\mathrm{PT}} = r. \tag{95.276}$$

The root equality (95.274) *forces $a = 1$; therefore $h = -vf$ fills the complete difference.*

Proof. The condition $\partial z_K = 0$ and (95.275) give $(u - 3cv)g = 0$. Since g is part of a basis of a free module, comparison of its coefficient gives $u = 3cv$. Substituting this equality into $r - z_K$ produces (95.276). The coefficient a is precisely the coordinate of $p_{\mathrm{root}} z_K$ in the basis $r = z_{\mathrm{PT}}$; theorem 95.35 gives $a = 1$. Then $\partial(-vf) = -v\partial f = z_{\mathrm{PT}} - z_K$. □

The proprietary construction of the filling is therefore the explicit formula

$$h_* := -vf, \qquad \partial h_* = z_{\mathrm{PT}} - z_K. \tag{95.277}$$

No abstract homotopy is introduced as a premise: closedness fixes $u = 3cv$, the root unit fixes $a = 1$ and the generator f directly fills the complete difference.

Equivalent non-governing homotopic control. Status: NON_GOVERNING_CONTROL. As a subsequent check, if the same complement is packaged by means of a reader H_{FS}, the identity

$$\partial H_{\mathrm{FS}} + H_{\mathrm{FS}} \partial = q_{\mathrm{root}} \tag{95.278}$$

reproduces (95.277) when applied to $q_{\mathrm{root}} z_K$. This reader is an auxiliary notation for the contraction already performed by $h_* = -vf$; it is not a prior obligation and cannot reopen the explicit construction.

Derived Bockstein regulator of arbitrary order

Let K be a field of characteristic zero, $\Lambda = K[[T]]$, and let

$$S(T) : V_\Lambda \longrightarrow W_\Lambda, \qquad V_\Lambda \simeq W_\Lambda \simeq \Lambda^m, \tag{95.279}$$

a based differential that becomes invertible over $K((T))$. Fix the line

$$\lambda_0(S) := (\det_K V)^{-1} \otimes_K \det_K W, \quad V = V_\Lambda / T V_\Lambda, \quad W = W_\Lambda / T W_\Lambda. \tag{95.280}$$

A based Smith form has the expression

$$U(T) S(T) V(T) = \operatorname{diag}\big(T^{a_1}, \dots, T^{a_s}, 1, \dots, 1\big), \qquad a_i \geq 1, \tag{95.281}$$

and

$$d = \operatorname{ord}_{T} \det S(T) = \sum_{i=1}^{s} a_i. \tag{95.282}$$

The T-adic filtration of the complex $[V_\Lambda \xrightarrow{S} W_\Lambda]$ produces pages $V_q = E_q^0$, $W_q = E_q^1$ and Bocksteins $\beta_q : V_q \to W_q$. Intrinsically,

$$A_q := V_q / V_{q+1}, \qquad B_q := \ker(W_q \twoheadrightarrow W_{q+1}) = \operatorname{im} \beta_q, \tag{95.283}$$

and β_q induces

$$\bar{\beta}_q : A_q \xrightarrow{\sim} B_q, \qquad \tau_q := \det(\bar{\beta}_q). \tag{95.284}$$

The pages are finite and satisfy

$$d = \sum_{q \geq 1} q \, \dim A_q = \sum_{q \geq 1} \dim V_q. \tag{95.285}$$

The first page sees $r_0 = \dim V_1$; the higher-order carry is

$$c_I^{\deg} = d - r_0 = \sum_{q \geq 2} \dim V_q. \tag{95.286}$$

The regular page is the induced isomorphism

$$\bar{S}(0) : A_0 := V / V_1 \xrightarrow{\sim} B_0 := \operatorname{im} S(0), \qquad \tau_{\mathrm{reg}} := \det \bar{S}(0). \tag{95.287}$$

The exact sequences of (95.283), with their Koszul signs, glue these trivializations into a single based element

$$R^{\mathrm{der}}_{\mathrm{Boc}}(S) := \tau_{\mathrm{reg}} \otimes \bigotimes_{q \geq 1} \tau_q \in \lambda_0(S), \tag{95.288}$$

where the tensor symbol abbreviates the Knudsen–Mumford gluing isomorphisms.

Theorem 95.37 (Determinantal identity of the derived regulator)**.** *In the based line $\lambda_0(S)$ one has*

$$R^{\mathrm{der}}_{\mathrm{Boc}}(S) = \mathrm{LT}_T(S) := \left[T^{-d} \det S(T) \right]_{T=0}. \tag{95.289}$$

The equality preserves the regular page, all positive pages and the carry (95.286)*.*

Proof. In the Smith bases of (95.281), the regular page and each positive page carry the canonical generator of their line. The Koszul signs of gluing are exactly those needed to restore the order of the total bases, so their product is the positive generator of $\lambda_0(S)$. On the other hand,

$$\det S(T) = \det U(T)^{-1} \det V(T)^{-1} T^{d},$$

and the initial term is $\det U(0)^{-1} \det V(0)^{-1}$. Upon undoing the change of bases, both (95.288) and the initial term are multiplied by that same factor. This proves (95.289) without identifying scalars before fixing the line and its basis. □

For each $q \geq 1$, the Poitou–Tate duality constructed on the same page provides

$$j_q : B_q \xrightarrow{\sim} A_q^{\vee} \otimes (I^q / I^{q+1}), \tag{95.290}$$

and, after orienting I^q / I^{q+1}, the pairing

$$h_E^{(q)}(x,y) := \big(j_q \bar{\beta}_q(x)\big)(y). \tag{95.291}$$

95.9.1 Comparators of the common determinantal unit

The root equality (95.274) fixes a single unit in the Kato and Poitou–Tate channels. From that unit, the comparators transporting the derived regulator to its scalar publication are now constructed. No comparator chooses a normalization after learning the leading term: all arise from morphisms of complexes, exact sequences and already oriented bases.

95.9.1.1 The chain of determinant lines

For a based perfect complex C, the convention used is

$$\lambda(C) := \bigotimes_i \det H^i(C)^{(-1)^i}. \tag{95.292}$$

Let C_E^K, C_E^{Iw}, C_E^{PT} and C_E^{loc} be, respectively, the complex published by the Kato channel, its based Iwasawa model, the self-dual reduced Poitou–Tate complex and the finite–singular localization cone. Their lines are linked by

$$\mathcal{L}_E^K \xrightarrow{\mathfrak{c}_{K/\mathrm{FERS}}} \mathcal{L}_E^{\mathrm{Iw}} \xrightarrow{\mathfrak{c}_{\mathrm{PT}}} \mathcal{L}_E^{\mathrm{ht}} \xrightarrow{\mathfrak{c}_{\mathrm{STS}}} \mathcal{L}_E^{\mathrm{fin}} \xrightarrow{\mathfrak{c}_{\mathrm{tors}}} \mathcal{L}_E^{\mathrm{ar}} \xrightarrow{\mathfrak{c}_\infty} \mathcal{L}_E. \tag{95.293}$$

In this formula, the abbreviation STS denotes the joint step of Smith, Tamagawa and Tate–Shafarevich. All the arrows of (95.293) are isomorphisms of oriented lines.

95.9.1.2 Kato–FERS chain comparison from the common unit

The equivalence giving rise to the first comparator can be constructed before taking determinants. Let $F^q C$ be the filtration by nonadic depth, identified with the T-adic filtration through the carry, and write

$$C_E^K := \mathrm{Tot}\Big(P_E^K \widehat{\Pi}_0 G_{m,n}\Big), \qquad C_E^{\mathrm{Iw}} := [V_\Lambda \xrightarrow{S_E(T)} W_\Lambda]. \tag{95.294}$$

Let $\mathrm{car}_T : G_{m,n} \to C_E^{\mathrm{Iw}}$ be the carry reader assigning to a route γ the generator determined by its initial and final faces, multiplied by $T^{\mathrm{Carr}(\gamma)}$. Multiplication of concatenations in the Iwasawa model is denoted by μ_{Iw}. The Alexander–Whitney diagonal and the publication P_E^K then define, before taking cohomology or determinants, the chain map

$$\begin{aligned} &\Phi_{K/\mathrm{FERS}} : C_E^K \longrightarrow C_E^{\mathrm{Iw}}, \\ &\Phi_{K/\mathrm{FERS}} := \mu_{\mathrm{Iw}} \circ \big(P_E^K \widehat{\otimes}\, \mathrm{car}_T\big) \circ \Delta_{\mathrm{AW}}, \\ &\Phi_{K/\mathrm{FERS}}(z_K) = z_{\mathrm{Iw}}. \end{aligned} \tag{95.295}$$

where z_{Iw} is the root generator induced by 1_E. Thus, each normalized simplex is sent to the sum of its Alexander–Whitney cuts, preserving in each summand the two faces and the carry power. The formula does not consult the analytic leading term.

Lemma 95.38 (Generator-by-generator criterion for equivalence)**.** *In each degree i, let $\mathcal{B}_i^K$ be the finite basis of normalized simplices, ordered by depth, faces and carry, and let $\mathcal{B}_i^{\mathrm{Iw}}$ be the basis formed by the generators with the same initial and final faces in the Iwasawa model. For a candidate map*

$$\upsilon_i : \mathcal{B}_i^K \longrightarrow \mathcal{B}_i^{\mathrm{Iw}},$$

define the degree-zero residual

$$\varepsilon_i(b) := \Phi^i_{K/\mathrm{FERS}}(b) \bmod T - \upsilon_i(b). \tag{95.296}$$

If υ_i is a bijection and $\varepsilon_i(b) = 0$ for every $b \in \mathcal{B}_i^K$, then the two completed modules of degree i are free of the same rank $d_i = |\mathcal{B}_i^K|$ and, for each generator,

$$\Phi^i_{K/\mathrm{FERS}}(b) = \upsilon_i(b) + T \sum_{c \in \mathcal{B}_i^{\mathrm{Iw}}} \lambda_{c,b}(T)c, \qquad \lambda_{c,b}(T) \in \Lambda. \tag{95.297}$$

In particular, $\mathrm{gr}^F\, \Phi^i_{K/\mathrm{FERS}} = I$ *on all generators, not only on the root unit.*

Proof. The vanishing of (95.296) says that the column of b has constant term $\upsilon_i(b)$. The remaining coefficients belong to $T\Lambda$, giving (95.297). Bijectivity identifies the bases and fixes the same rank. In those bases, the reduction modulo T of the matrix of Φ^i is the identity; hence $\mathrm{gr}^F\, \Phi^i = I$. □

The root equality $\Phi_{K/\mathrm{FERS}}(z_K) = z_{\mathrm{Iw}}$ fixes the common normalization. The expression of P_E^K on each normalized simplex, the bijection υ_i and the residuals $\varepsilon_i(b)$ provide a generator-by-generator check of the equivalence already constructed; they neither choose nor govern the BSD conclusion.

Lemma 95.39 (Filtered equivalence and normalization of the root)**.** *For the chain map constructed in* (95.295)*, the map υ_i is the basis correspondence induced by the faces and carry of each simplex. The Alexander–Whitney formula verifies in each degree the criterion of lemma 95.38; this is a check of the equivalence already constructed, not an additional premise. The map $\Phi_{K/\mathrm{FERS}}$ is a based equivalence of perfect complexes. In the bases ordered by depth, its matrices in each degree have the form*

$$[\Phi^i_{K/\mathrm{FERS}}](T) = I + TB_i(T), \qquad B_i(T) \in M_{d_i}(\Lambda). \tag{95.298}$$

Therefore, the unit

$$\rho_{E,\ell}(T) := \prod_i \det(I + TB_i(T))^{(-1)^i} \in \Lambda^\times \tag{95.299}$$

satisface $\rho_{E,\ell}(0) = 1$.

Proof. The simplicial identity $\partial\Delta_{\mathrm{AW}} = (\partial \otimes I + (-1)^{\deg} I \otimes \partial)\Delta_{\mathrm{AW}}$, together with the compatibility of P_E^K with faces and degeneracies, proves that (95.295) commutes with the differentials. On the associated graded object, lemma 95.38 identifies all the generators and gives the identity. In those bases, (95.297) is precisely (95.298).

The completeness of $\Lambda = K[[T]]$ makes the series converge

$$(I + TB_i(T))^{-1} = \sum_{n \geq 0} (-TB_i(T))^n; \tag{95.300}$$

these inverses also commute with the differentials, so they form the based inverse of $\Phi_{K/\mathrm{FERS}}$. Taking the alternating determinant yields (95.299); evaluating at $T = 0$ gives one. Thus, the normalization is not postulated from the leading term: it is the constant coordinate of an equivalence that is the identity on the root unit. □

The first comparator is the determinant of $\Phi_{K/\mathrm{FERS}}$. If $\mathfrak{z}_K$ denotes the element coming from the unit 1_E, its image is characterized by

$$\mathfrak{c}_{K/\mathrm{FERS}}(\mathfrak{z}_K) = \left[T^{-d} \rho_{E,\ell}(T) \det S_E(T)\right]_{T=0}, \qquad \rho_{E,\ell}(0) = 1. \tag{95.301}$$

The unit of $\rho_{E,\ell}$ is oriented and equals one at the root; it therefore does not alter the order or the basis. Theorem 95.37 transforms (95.301) into

$$\mathfrak{c}_{K/\mathrm{FERS}}(\mathfrak{z}_K) = R^{\mathrm{der}}_{\mathrm{Boc}}(S_E) = \tau_{\mathrm{reg}} \otimes \bigotimes_{q \geq 1} \tau_q. \tag{95.302}$$

The Poitou–Tate comparator is constructed page by page. For $q \geq 1$, the isomorphism j_q of (95.290) and the reduced Bockstein $\bar{\beta}_q$ induce

$$\mathfrak{c}_{\mathrm{PT},q}(\tau_q) := \det\bigl(j_q \circ \bar{\beta}_q\bigr) = \det h_E^{(q)}. \tag{95.303}$$

The regular direction is transported through $\det \bar{S}_E(0) = \tau_{\mathrm{reg}}$. Knudsen–Mumford gluing and the Koszul signs defined in (95.288) provide the unique oriented isomorphism

$$\mathfrak{c}_{\mathrm{PT}}\left(\tau_{\mathrm{reg}} \otimes \bigotimes_{q\geq 1} \tau_q\right) = \tau_{\mathrm{reg}}^{\mathrm{ht}} \otimes \bigotimes_{q\geq 1} \det h_E^{(q)}. \tag{95.304}$$

On the first stable page, the form $h_E^{(1)}$ is the Poincaré–Néron–Tate form on the free Mordell–Weil lattice; its determinant is

$$\det h_E^{(1)} = \mathrm{Reg}(E) \tag{95.305}$$

in the basis induced by the root unit.

Lemma 95.40 (Poitou–Tate unfolding of the arithmetic degree)**.** *Let*

$$\mathrm{Flag}_{\mathrm{Boc}}(E) := \bigoplus_{q\geq 1} V_q$$

the finite flag of positive pages, each with the basis inherited from the root unit. Orthogonal reduction and the isomorphisms j_q construct a based isomorphism

$$\Psi_{\mathrm{PT}} : \bigl(E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}}\bigr) \otimes K \xrightarrow{\sim} \mathrm{Flag}_{\mathrm{Boc}}(E). \tag{95.306}$$

In particular,

$$\mathrm{rank}\, E(\mathbb{Q}) = \sum_{q\geq 1} \dim_K V_q = \mathrm{ord}_T \det S_E(T). \tag{95.307}$$

Proof. The reduced complex is filtered by the powers of the augmentation ideal. In the quotient at level q, the regular direction has already been separated by $\bar{S}_E(0)$, and the only surviving block is V_q. The equivalence X^{orth} identifies the opposite block with its dual, while $j_q\bar{\beta}_q$ provides the perfect pairing that lifts the basis of the quotient to the preceding level. Starting at the last nonzero page and repeating this lifting yields Ψ_{PT}.

In the bases ordered by depth, the matrix of the resulting map is block triangular and all its diagonal blocks are identities: in the first block because z_K and z_{PT} have the same root, and in the others because j_q and $\bar{\beta}_q$ are based isomorphisms. Therefore Ψ_{PT} is invertible and loses no page. Taking dimensions and using (95.285) yields (95.307). On taking determinants, the same lifting is exactly the gluing of (95.304); thus, equality of ranks and equality of lines come from a single based map. □

95.9.1.3 Localization triangle, Smith rank and finite factors

For each place v, the finite local condition is a based subcomplex $C_{E,v}^{\mathrm{fin}} \subset C_{E,v}$, and one defines $C_{E,v}^{\mathrm{sing}}$ by the triangle

$$C_{E,v}^{\mathrm{fin}} \xrightarrow{i_v} C_{E,v} \longrightarrow C_{E,v}^{\mathrm{sing}} \longrightarrow C_{E,v}^{\mathrm{fin}}[1]. \tag{95.308}$$

If res_v denotes the global restriction, the localization map is, at chain level,

$$\ell_E : C_E^{\mathrm{glob,red}} \oplus \bigoplus_v C_{E,v}^{\mathrm{fin}} \longrightarrow \bigoplus_v C_{E,v}, \qquad \ell_E\bigl(x, (y_v)_v\bigr) := \bigl(\mathrm{res}_v\, x - i_v y_v\bigr)_v. \tag{95.309}$$

The definition by cone of the reduced Selmer complex gives the triangle

$$C_E^{\mathrm{red}} \longrightarrow C_E^{\mathrm{glob,red}} \oplus \bigoplus_v C_{E,v}^{\mathrm{fin}} \xrightarrow{\ell_E} \bigoplus_v C_{E,v} \longrightarrow C_E^{\mathrm{red}}[1]. \tag{95.310}$$

Therefore, local–global exactness comes from the cone of a map written in (95.309); it is not subsequently added as a relation between cardinalities.

In (95.310) the common root summand is canceled, which is the same in both channels by (95.274). The resulting complement is denoted by

$$C_E^{\mathrm{loc,red}} := q_{\mathrm{root}} \operatorname{Cone}(\ell_E)[-1]. \tag{95.311}$$

After separating the free Mordell–Weil lattice and its dual through X^{orth}, a choice of the already oriented integral bases represents this complex by

$$C_E^{\mathrm{loc,red}} \simeq [M_E \xrightarrow{D_E} N_E]. \tag{95.312}$$

Lemma 95.41 (Rational acyclicity and full rank)**.** *The complex of* (95.312) *is acyclic after tensoring with* $\mathbb{Q}$*. In particular,*

$$D_E \otimes \mathbb{Q} : M_E \otimes \mathbb{Q} \xrightarrow{\sim} N_E \otimes \mathbb{Q}$$

is an isomorphism; M_E *and* N_E *have the same rank and* D_E *has full rank.*

Proof. Let x be a cycle of the complex $C_E^{\mathrm{loc,red}} \otimes \mathbb{Q}$. The reduction has eliminated the summand p_{root}, hence $q_{\mathrm{root}} x = x$. In the oriented integral bases used in (95.312), the finite–singular complement decomposes into the normal blocks of lemma 95.36. In each block λ, the root coefficient is already zero and the cycle is written

$$x_\lambda = u_\lambda e_\lambda + v_\lambda b_\lambda, \qquad \partial x_\lambda = (u_\lambda - 3c_\lambda v_\lambda) g_\lambda = 0.$$

Since g_λ belongs to the free basis, we have $u_\lambda = 3c_\lambda v_\lambda$, and the explicit formula for the differential gives

$$x_\lambda = v_\lambda (3c_\lambda e_\lambda + b_\lambda) = \partial(v_\lambda f_\lambda).$$

The sum is finite in each degree, so $x = \partial\big(\sum_\lambda v_\lambda f_\lambda\big)$. Every cycle is therefore a boundary by the same normal form that produced the filler (95.277), without introducing H_{FS} as a premise. The equality $p_{\mathrm{root}} z_K = z_{\mathrm{PT}}$ is what allows the root direction to be canceled before this calculation; without it a degree-zero class would remain. The self-dual equivalence $X^{\mathrm{orth}} : C_E^{\mathrm{red}} \simeq D_3(C_E^{\mathrm{red}})$ pairs the two remaining degrees and transports their bases with complementary opposite orientation. From this one obtains $\operatorname{rank} M_E = \operatorname{rank} N_E$. Finally, acyclicity of a two-term complex of equal rank is equivalent to $D_E \otimes \mathbb{Q}$ being invertible. □

The lemma makes it possible to apply the Smith normal form without presupposing any of its divisors. There exist oriented changes of basis U, V such that

$$U D_E V = \operatorname{diag}(s_1, \dots, s_t), \qquad s_i \in \mathbb{Z} \setminus \{0\}. \tag{95.313}$$

Therefore,

$$F_E := \operatorname{coker} D_E \simeq \bigoplus_{i=1}^{t} \mathbb{Z}/|s_i|\mathbb{Z}, \qquad |F_E| = \prod_{i=1}^{t} |s_i| < \infty. \tag{95.314}$$

Lemma 95.42 (Finite localization sequence)**.** *The torsion fragment of the long cohomology sequence of* (95.310) *is*

$$0 \longrightarrow \mathrm{Sha}(E) \xrightarrow{\iota_{\mathrm{Sha}}} F_E \xrightarrow{\partial_{\mathrm{loc}}} \bigoplus_v \Phi_v(\mathbb{F}_v) \longrightarrow 0, \qquad c_v = |\Phi_v(\mathbb{F}_v)|. \tag{95.315}$$

Proof. A global cocycle represents a class of $\mathrm{Sha}(E)$ precisely when its restriction at each place admits a primitive in $C^{\mathrm{fin}}_{E,v}$. The class of the pair formed by that cocycle and its local primitives in the cone of (95.309) defines ι_{Sha}. If this class is a boundary, the global component is a boundary and the Tate–Shafarevich class is zero; thus, ι_{Sha} is injective.

The map ∂_{loc} takes a class of the cone, restricts it to each quotient $C^{\mathrm{sing}}_{E,v}$ of (95.308) and preserves its component class. The identification

$$H^0(C^{\mathrm{sing}}_{E,v})_{\mathrm{tors}} \simeq \Phi_v(\mathbb{F}_v)$$

is induced by the finite–singular quotient, not by the order of the group. A class of the cone has zero residue exactly when its local representatives can be chosen in the finite subcomplexes; by the preceding definition, that kernel is the image of ι_{Sha}. Finally, the last arrow of (95.308) lifts each component class to the local complex. The exactness of the cone (95.310), after removing the two free summands paired by X^{orth}, turns that lifting into a preimage under ∂_{loc}. This proves surjectivity and, consequently, all of (95.315). □

Now, and only after lemma 95.41, the group F_E is finite. The injection of (95.315) then proves the finiteness of $\mathrm{Sha}(E)$. Taking orders in that exact sequence yields

$$|F_E| = |\,\mathrm{Sha}(E)| \prod_v c_v. \tag{95.316}$$

This is the comparator $\mathfrak{c}_{\mathrm{STS}}$: it transports the Smith elementary factors to the Tate–Shafarevich and Tamagawa factors while preserving their order in the line, instead of inserting their cardinalities once the leading term has been calculated.

The torsion part is constructed by applying the determinant to the Mordell–Weil lattice and its Pontryagin dual. The two finite factors are dual and contribute

$$\mathfrak{c}_{\mathrm{tors}} : quadu \longmapsto \frac{u}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}. \tag{95.317}$$

The Archimedean comparator is obtained by integrating the oriented Néron differential over the based real cycle:

$$\Omega_E := \int_{E(\mathbb{R})} |\omega_E|, \qquad \mathfrak{c}_\infty : quadu \longmapsto \frac{u}{\Omega_E}. \tag{95.318}$$

The differential and the cycle are fixed before evaluating $L(E,s)$; therefore (95.318) does not retrospectively normalize the analytic section.

95.9.1.4 Causal separation of primitives, identities and consequences

The generator-by-generator criterion of lemma 95.38 audits in each degree the equivalence already constructed. It does not condition its existence. The logical order of the package is

$$
\begin{array}{c}
1_E,\ \Delta_{\mathrm{AW}},\ P_E^K,\ P_E^{\mathrm{PT}},\ h_* = -vf,\ X^{\mathrm{orth}},\\
\{\, C_{E,v}^{\mathrm{fin}} \to C_{E,v} \,\}_v,\\
\text{Mordell–Weil and Pontryagin bases;}\\
\text{Néron differential and cycle}\\
\Downarrow\\
\Phi_{K/\mathrm{FERS}}\ \text{equivalence},\quad \rho_{E,\ell}(0) = 1,\quad p_{\mathrm{root}} z_K = z_{\mathrm{PT}},\\
R_{\mathrm{Boc}}^{\mathrm{der}} = \mathrm{LT}_T(S_E),\quad \det(j_q \bar{\beta}_q) = \det h_E^{(q)},\\
H^*(C_E^{\mathrm{loc,red}} \otimes \mathbb{Q}) = 0,\quad D_E \otimes \mathbb{Q}\ \text{isomorphism},\\
0 \to \mathrm{Sha}(E) \to F_E \to \bigoplus_v \Phi_v(\mathbb{F}_v) \to 0\\
\Downarrow\\
d_{\mathrm{an}} = \mathrm{rank}\, E(\mathbb{Q}),\qquad \mathrm{Reg}(E),\qquad |\,\mathrm{Sha}(E)| \prod_v c_v,\\
|E(\mathbb{Q})_{\mathrm{tors}}|^{-2},\qquad \Omega_E^{-1},\qquad \text{equality of the leading term.}
\end{array}
\tag{95.319}
$$

The first row contains only already based objects and maps, together with the indicated generator-by-generator verification; the second, identities proved by those maps; the third, their consequences on the determinant line. In particular, no factor in the last row is used to retroactively normalize an arrow in the first.

Proposition 95.43 (Exact package of comparators)**.** *The comparators of* (95.293) *come from a single package of maps of complexes and satisfy, in the order of construction,*

$$
\begin{aligned}
&\Phi_{K/\mathrm{FERS}}(z_K) = z_{\mathrm{Iw}},\qquad \mathrm{gr}^F\, \Phi_{K/\mathrm{FERS}} = I,\\
&\det{}_{\mathrm{alt}}\, \Phi_{K/\mathrm{FERS}} = \rho_{E,\ell}(T),\qquad \rho_{E,\ell}(0) = 1,\\
&\det(j_q \bar{\beta}_q) = \det h_E^{(q)}\quad (q \geq 1),\\
&\Psi_{\mathrm{PT}} : \big(E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}}\big) \otimes K \xrightarrow{\sim} \mathrm{Flag}_{\mathrm{Boc}}(E),\\
&p_{\mathrm{root}} z_K = z_{\mathrm{PT}},\qquad h_* = -vf,\quad \partial h_* = z_{\mathrm{PT}} - z_K,\\
&H^*(C_E^{\mathrm{loc,red}} \otimes \mathbb{Q}) = 0,\qquad UD_EV = \mathrm{diag}(s_1, \dots, s_t),\\
&0 \longrightarrow \mathrm{Sha}(E) \longrightarrow F_E \longrightarrow \bigoplus_v \Phi_v(\mathbb{F}_v) \longrightarrow 0.
\end{aligned}
\tag{95.320}
$$

The induced composition on determinant lines is exactly

$$
\det(\Phi_{K/\mathrm{FERS}}) \xrightarrow{\det(j_q \bar{\beta}_q)} \det(D_E) \xrightarrow{\ \mathrm{Smith}\ } \frac{|\,\mathrm{Sha}(E)| \prod_v c_v}{|E(\mathbb{Q})_{\mathrm{tors}}|^2\, \Omega_E},
\tag{95.321}
$$

with the Knudsen–Mumford signs and the orientations of (95.293)*. In particular, the right-hand side of* (95.321) *is the final coordinate of the transported element, not data used to construct any of the arrows.*

Proof. The first row of (95.320) is (95.295) and (95.298)– (95.299). The second is (95.303) and lemma 95.40. The third combines equality of the root with the explicit filler (95.277); its linear extension in the normal form induces the auxiliary reader used in lemma 95.41. On restricting to the complement of the root, that reduction contracts every rational cycle. The last row is (95.313) and (95.315). Applying the determinant functor to those identities, in that same order, gives (95.321). No equality is obtained by first comparing the two final scalars. □

Lemma 95.44 (Causal independence of the leading term)**.** *Let $\mathscr{P}_E$ be the data set*

$$\mathscr{P}_E := \Big(1_E, \Delta_{\mathrm{AW}}, P_E^K, P_E^{\mathrm{PT}}, h_* = -vf, X^{\mathrm{orth}}, \{C_{E,v}^{\mathrm{fin}} \xrightarrow{i_v} C_{E,v}\}_v, \textit{bases and orientations}, \omega_E, \gamma_E^{\mathbb{R}}\Big), \tag{95.322}$$

where ω_E is the Néron differential and $\gamma_E^{\mathbb{R}}$ the oriented real cycle. All maps of the package in proposition 95.43 are determined by $\mathscr{P}_E$. The following numerals are not part of $\mathscr{P}_E$

$$L^{(r)}(E,1), \quad r, \quad \mathrm{Reg}(E), \quad |\,\mathrm{Sha}(E)|, \quad (c_v)_v, \quad |E(\mathbb{Q})_{\mathrm{tors}}|, \quad \Omega_E.$$

Those values are obtained, respectively, by evaluating the analytic section, taking the degree of the flag, calculating determinants and Smith forms, taking orders of finite groups and integrating ω_E over $\gamma_E^{\mathbb{R}}$.

Proof. The formula (95.295) uses only $1_E, \Delta_{\mathrm{AW}}, P_E^K$ and the carry. The Bockstein and Poitou–Tate maps use the filtration, P_E^{PT}, the bases and self-duality. The cone of (95.309) uses the maps i_v and the local restrictions; its contraction and Smith form use the explicit filler (95.277), its auxiliary linear extension, X^{orth} and the integral bases. The last two comparators use the torsion lattices and the pair $(\omega_E, \gamma_E^{\mathbb{R}})$. This is an exhaustive enumeration of the arguments appearing in the construction formulas. The numerals listed in the statement appear only after applying cohomology, determinant, cardinality or integration, so they cannot retrospectively select the maps. □

95.9.1.5 Composition, degrees and leading term

Define the total comparator by composition in the written order:

$$\mathfrak{C}_E := \mathfrak{c}_\infty \circ \mathfrak{c}_{\mathrm{tors}} \circ \mathfrak{c}_{\mathrm{STS}} \circ \mathfrak{c}_{\mathrm{PT}} \circ \mathfrak{c}_{K/\mathrm{FERS}}. \tag{95.323}$$

Since $p_{\mathrm{root}} z_K = z_{\mathrm{PT}}$, all arrows receive the same based unit. The filler $h_* = -vf$ of (95.277) explicitly fills the difference; its linear extension in the normal basis produces the auxiliary homotopic reader and proves that no additional class modifies the element of the line during transport. The reader is not a separate premise.

Theorem 95.45 (Construction of the based comparators)**.** *For every elliptic curve $E/\mathbb{Q}$ with the complexes, bases, orientations and local data defined above, the maps* (95.295)*,* (95.304)*,* (95.310)*,* (95.317) *and* (95.318)*, together with lemmas 95.41 and 95.42, construct the five comparators of* (95.293)*. Their composition preserves the unit, degree and orientation, and satisfies*

$$d_{\mathrm{an}} = \mathrm{ord}_T \det S_E(T) = d_{\mathrm{ar}} = \mathrm{rank}\, E(\mathbb{Q}). \tag{95.324}$$

Moreover, $\mathrm{Sha}(E)$ *is finite and the equality of the distinguished element in $\mathcal{L}_E$ is published as*

$$\frac{L^{(r)}(E,1)}{r!\,\Omega_E} = \frac{|\,\mathrm{Sha}(E)|\ \mathrm{Reg}(E)\prod_v c_v}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}, \qquad r = \mathrm{rank}\, E(\mathbb{Q}). \tag{95.325}$$

Proof. The Kato–FERS comparator and the condition $\rho_{E,\ell}(0) = 1$ identify the analytic order with $d = \mathrm{ord}_T \det S_E(T)$ and the initial term with $R_{\mathrm{Boc}}^{\mathrm{der}}(S_E)$. The identity (95.289) preserves the regular page and every positive page. The isomorphisms j_q transport those pages to the derived heights. Lemma 95.40 constructs the isomorphism of the flag with the free Mordell–Weil lattice and, together with (95.285), proves (95.324).

Lemma 95.41 deduces rational acyclicity directly from the explicit filler (95.277), its linear reduction of the complement, cancellation of the common root and X^{orth}; from this the full rank of D_E is obtained, not as an additional hypothesis. The Smith form (95.313) then produces the finite group F_E. Lemma 95.42 extracts from the triangles (95.308) and (95.310) the finite exact sequence; only afterwards are the finiteness of $\mathrm{Sha}(E)$ and (95.316) deduced. Finally, (95.305), (95.317) and (95.318) publish, respectively, the regulator, the torsion quotient and the period. The root equality eliminates any relative unit between the two publications. Evaluating the same based element at both ends of (95.323) yields (95.325). □

95.9.1.6 Premises and refutation criteria

The construction starts from the genealogical unit 1_E and the Alexander–Whitney diagonal. It also uses the coupled publications, all the Bocksteins and the regular page, the explicit filler $h_* = -vf$, the reduced self-duality X^{orth}, the chain maps of the localization triangle and the oriented bases of torsion and period. The Kato–FERS equivalence and its normalization to one, the full rank of D_E, the exactness of (95.315) and the finiteness of $\mathrm{Sha}(E)$ are conclusions of the preceding lemmas, not independent premises. The generator-by-generator criterion of lemma 95.38 is a nongoverning check that locally audits the first comparator. The criteria that would refute a specific arrow are:

1. a root coefficient different from one, which would introduce a relative unit between z_K and z_{PT};
2. a noninvertible j_q or the omission of a Bockstein page, which would alter the degree or the height determinant;
3. a zero Smith divisor, which would prevent the finiteness of F_E;
4. failure of exactness of (95.315), which would break the identification of the finite factors;
5. an orientation reversal in torsion or period, which would change the based element even if it preserved its unoriented line.

The memoryless Manin projection and visible rescaling modulo three violate the first criterion; they therefore constitute necessity proofs for those reduced models, not premises of theorem 95.45.

Theorem 95.46 (Local–global determinantal comparison)**.** *Let the based comparators already constructed in the theorem 95.45 be: Kato–FERS/Iwasawa, Poitou–Tate/Poincaré–Néron–Tate, Smith–Tamagawa–Shafarevich, torsion and period. If d_{an} is the degree of the analytic section at the central specialization and d_{ar} the degree of the based Selmer line, then*

$$d_{\mathrm{an}} = d = d_{\mathrm{ar}}, \tag{95.326}$$

and one obtains the equality of distinguished elements

$$\mathfrak{z}_{\mathrm{an}}(E) = \mathfrak{z}_{\mathrm{ar}}(E) \quad \text{in} \quad \mathcal{L}_E. \tag{95.327}$$

The analytic side is the publication of the initial term; the arithmetic side is the publication of the complete regulator, the Smith elementary factors, the local factors, the torsion and the period.

Proof. The based construction gives $[\Phi^i_{K/\mathrm{FERS}}](T) = I + TB_i(T)$ in each degree. The based Kato–FERS comparator then expresses the corrected analytic section as

$$L_\ell^{\mathrm{corr}}(E,T) = \rho_{E,\ell}(T)\det S_E^{\mathrm{corr}}(T), \qquad \rho_{E,\ell}(0) = 1. \tag{95.328}$$

By (95.282), its central degree is d; by (95.289), its initial term is exactly $R_{\mathrm{Boc}}^{\mathrm{der}}$, not that regulator multiplied by an undetermined unit. The isomorphisms j_q of (95.290) transport each τ_q to the determinant of the derived height of its page. The regular page transports the acyclic summand that the positive pages do not see.

The Poincaré–Néron–Tate comparison identifies the degree of that height line with d_{ar}. The Smith elementary factors decompose its finite part into the Shafarevich and Tamagawa factors; the quotient of the integral lattices supplies the square of the torsion, and the Archimedean reader supplies the period. All these maps act on the same root line. By theorem 95.35, no relative unit remains between the Kato and Poitou–Tate publications. The composition preserves the degree and the based element, proving simultaneously (95.326) and (95.327). The corresponding scalar formula is reserved for final recognition. □

Hypotheses and scope

The BSD chain uses: the complete enriched state and its Alexander–Whitney coalgebra; the fiber product (95.264); the two publications constructed on the same unit; the root line, its basis and its counit; the perfect and saturated reduced quotient, together with X^{orth}; the complete finite–singular complex; the finite Bockstein flags, their Koszul signs and the regular page; and the height, Smith, Tamagawa, Shafarevich, torsion and period comparisons with a common orientation. The degree-by-degree verification of the paired bases and the residuals $\varepsilon_i(b)$ is recorded as a local control of the realization, not as a governing premise of the closure. Under this explicit starting structure, (95.274), (95.277), (95.289) and (95.327) are exact equalities.

Unit preservation, the two-sheet traversal and local–global publication are formalized through the chain coalgebra, the genealogical fiber product, the coupled publication, orthogonal reduction and the Bockstein tower. The auxiliary identities separately verify the group-like element property, preservation of history under $\widehat{\Pi}_0$, the identity of the unit in both channels, the reduced involution, the finite–singular normal form and the determinantal identity of arbitrary order.

BSD ablation controls. Status: NON_GOVERNING_CONTROL. The following three ablations receive the already proved identities and falsify only reduced models. They contribute no premises to the owner, do not condition its closure and have no causal return edge toward it.

Structural Necessity Proposition 95.47 (Manin projection without history)**.** Replacing P_E^{gen} with P_E^{Man} erases data that the coupled publication needs. Indeed, the route $\gamma = (\mathsf{NE})^{54}$ may have the same cell and the same visible phase as the trivial route, but preserves distinct winding and memory in $G_{m,n}$. The Manin augmentation identifies them; the second projection of (95.264) does not.

Proof. Every based lift to the Manin complex factors, up to homotopy, through the section of the augmentation $s\varepsilon_{\mathrm{AW}}$. It therefore depends only on the visible coordinate. By contrast, the second component of $\widehat{\Pi}_0(c)$ is c itself, so it preserves the history simplex and distinguishes the winding of γ. The ablation refutes the bare projection; it does not refute the fiber product used in the theorem. □

Structural Necessity Proposition 95.48 (Rescaling of the root visible only modulo three)**.** In the specialization $c = 1$ of (95.275), the altered cycle

$$z_K^{(4)} = 4r + 3e + b \tag{95.329}$$

satisfies

$$z_{\mathrm{PT}} - z_K^{(4)} = \partial(-f) - 3r. \tag{95.330}$$

The defect $-3r$ disappears modulo three and reappears modulo nine. Consequently, (95.329) preserves a partial shadow, but not the based unit or (95.274).

Proof. The equality $\partial f = 3e + b$ immediately gives (95.330). Modulo three, the second summand vanishes; modulo nine it is a root class not divisible by a boundary of the complement. Moreover, its root coordinate is four, whereas (95.268) requires one. The example verifies the necessity of the coupled pair and orientation; it introduces no exception to the theorem that preserves both data. □

Structural Necessity Proposition 95.49 (Loss of regulator pages)**.** The matrix $\mathrm{diag}(T^2, T)$ shows that the first Bockstein does not recover the higher-order carry. The matrix $\mathrm{diag}(2, T^2)$ has positive trivialization $\tau_2 = 1$, but $\tau_{\mathrm{reg}} = 2 = \mathrm{LT}_T(S)$. Therefore, neither the first page nor the product of positive pages replaces the complete regulator (95.288).

Proof. In the first example, $d = 3$, $r_0 = 2$ and $c_I^{\deg} = 1$; page one does not contain that last degree of memory. In the second, the only singular divisor is T^2, but the regular direction contributes the factor two.

Omitting it would produce one instead of two. Both calculations are diagonal cases of (95.281) and verify the necessity of all the pieces of (95.288). □

Final mathematical recognition of Birch–Swinnerton–Dyer

The preceding development concludes before employing its historical formulation as a criterion.

In the domain declared by theorem 95.46, the Smith comparison identifies the finite component of the arithmetic line with the Shafarevich–Tamagawa decomposition and proves the finiteness of $\mathrm{Sha}(E)$. Consequently, its order in the following scalar publication is the cardinality of the identified finite group, not a formal symbol added after the fact.

The scalar translation of (95.327) provides, with $r = \operatorname{rank} E(\mathbb{Q})$,

$$\operatorname{ord}_{s=1} L(E,s) = r \tag{95.331}$$

and

$$\frac{L^{(r)}(E,1)}{r!\,\Omega_E} = \frac{|\mathrm{Sha}(E)|\ \mathrm{Reg}(E)\prod_v c_v}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}. \tag{95.332}$$

This is the conventional formulation of Birch–Swinnerton–Dyer. The order, the leading term and the arithmetic factors are not defined through one another: they are coordinated publications of the based equality (95.327), whose common root was previously fixed by (95.274).

The preceding chapter constructs the five operations simultaneously from the same enriched state of the continuum. Here each complete component is unfolded before its application, together with its fiber, extensions, relations, numbers, orientation, carry, memory, boundary, route, multisection, survival, terminals, and readers. The expository separation of the five tuples does not make them branches, factors, or independent objects: they are five intrinsic features of the single generated continuum and literally preserve the same state and the same ledger. The common constructor does not assemble them afterward; it produces them simultaneously as the internal structure of a single object.

95.10 Internal view G_T^{int} of the continuum and map App_{T,d_T}: preservation of fiber, carry, memory and terminal

To present each characteristic without separating it from the others, the index $T \in \{\mathrm{sp}, \mathrm{sol}, \mathrm{gau}, \mathrm{coh}, \mathrm{det}\}$ is used, but the complete mathematical datum remains the same enriched continuum. The internal view is not postulated anew: it is exactly the typed reading constructed in (95.3),

$$\mathfrak{G}_T^{\mathrm{int}} := \mathrm{View}_T\Big(\mathcal{O}_\infty(\mathcal{C}_{\mathrm{cont}}^{\mathrm{disc}})\Big). \tag{95.333}$$

Its complete presentation in the channel T is

$$\mathfrak{G}_T^{\mathrm{int}} = \Big(\begin{matrix}\{\mathcal{F}_q^{(T)}\}_q, \{\mathrm{Ext}_\gamma^{(T)}\}_\gamma, \mathrm{Rel}_T, \mathrm{Num}_T, \mathrm{or}_T,\\ \mathrm{Carr}_T, \mathrm{Mem}_T, \partial_T, \gamma_T, \mathfrak{m}_T, \mathrm{Surv}_T, X_{\chi_T}, \tau_T, \mathrm{Term}_T, \mathrm{Lect}_T\end{matrix}\Big). \tag{95.334}$$

This object may not be replaced by X_{χ_T}, its cardinality, or the name of a classical problem. Let $\mathcal{D}_T^{\mathrm{cl}}$ be the space of classical data of the corresponding problem. These enter only through

$$\mathrm{App}_{T,d_T} : \mathfrak{G}_T^{\mathrm{int}} \longrightarrow \mathfrak{G}_T(d_T), \qquad d_T \in \mathcal{D}_T^{\mathrm{cl}}, \tag{95.335}$$

where $\mathfrak{G}_T(d_T)$ is, by definition, the pair comprising the complete internal view and the interface coordinates unfolded below. Thus, no fibre, relation, orientation, carry, memory, boundary, route, multisection, survival, or terminal object is lost when the classical datum is introduced. The internal projection $\mathrm{pr}_{X,T} : \mathfrak{G}_T^{\mathrm{int}} \twoheadrightarrow X_{\chi_T}$ is surjective. If $\mathcal{A}_{T,d_T} : \mathfrak{G}_T(d_T) \to \mathcal{M}_T$, define

$$\widetilde{\mathcal{A}}_{T,d_T} := \mathcal{A}_{T,d_T} \circ \mathrm{App}_{T,d_T}, \qquad \mathfrak{M}_T := \mathrm{Im}\, \mathcal{A}_{T,d_T}.$$

95.11 Correlative emergence of the 5 terminal actions over the same state of the continuum

The preceding generic formula does not replace the explicit writing of the five characteristics. In this volume, each interface is materialized without converting the internal view into an independent object or domain. In the following five displays, the first coordinate is always $\mathfrak{G}_T^{\mathrm{int}}$; the lists of specific coordinates therefore expand the common register and never replace it. We shall use

$$d_{\mathrm{sp}} = (R, \mathcal{D}_R^{\mathrm{cof}}), \quad d_{\mathrm{sol}} = (u_0, f, T), \quad d_{\mathrm{gau}} = (\Lambda_m, g), \quad d_{\mathrm{coh}} = (X, p), \quad d_{\mathrm{det}} = E/\mathbb{Q}.$$

This line fixes the exact causal instant at which the classical problems appear: after $\mathfrak{G}_T^{\mathrm{int}}$, never within its HMT generation.

95.11.1 Spectral–polar action of the common constructor and conservation of the critical fiber

The spectral-polar characteristic preserves the record

$$\begin{gathered} \mathfrak{G}_{\mathrm{sp}}^{\mathrm{int}}; \\ \mathfrak{G}_{\mathrm{sp}}(d_{\mathrm{sp}}) = \big(\widehat{X}_\infty, \Gamma_9, \widehat{R} = (R, K), \mathrm{Carr}_R, S_0, M, O_9, \big). \\ \Xi_R, P_R, \mathcal{D}_R^{\mathrm{cof}}, \mathcal{F}_R, \mathrm{Ext}_R, \mathrm{Surv}_R, \tau_R, \mathcal{P}_R \end{gathered} \tag{95.336}$$

Here $\mathcal{D}_R^{\mathrm{cof}}$ is the domain of test functions; Ξ_R publishes the prime, gamma, and polar channels within a single Hilbert space; $P_R = P_R^* = P_R^2$ is the natural projection; S_0, M, O_9 preserve the memory and terminal object; and $\widehat{R} = (R, K)$ preserves both the window and its depth. The Weil object is the complement, not an appended sign:

$$B_{W,R} := \Xi_R^*(I - P_R)\Xi_R. \tag{95.337}$$

The regulated channels, cofinal naturality and terminal belong to the same characteristic; none is reconstructed from the zeros of ζ. The internal structure supplies sheet involution, extension expectation, defect and terminal telescoping. The preceding exact source development additionally constructs the Weil interface: it fixes sheet contraction in (95.37), links the spectral–polar complement in (95.38), preserves the terminal through (95.39) and proves its extinction in (95.40). In the notation of the common constructor, that already constructed colligation is published as

$$\Sigma_R^{\mathrm{sp}} : \mathcal{H}_{+,R} \oplus \mathcal{X}_R^{\mathrm{sp}} \longrightarrow \mathcal{H}_{-,R} \oplus \mathcal{L}_R^{\mathrm{sp}} \oplus \mathcal{X}_R^{\mathrm{sp}}, \qquad (\Sigma_R^{\mathrm{sp}})^* \Sigma_R^{\mathrm{sp}} = I, \tag{95.338}$$

with terminal state $x_{R,N} = q^N V_R^N x_{R,0}$, $q = 5/9$, and output $K_R^{\mathrm{sp}} J_{+,R} = J_{-,R}$. Its energy identity is

$$\|J_{+,R} f\|^2 = \|J_{-,R} f\|^2 + \sum_{n=0}^{N-1} \|\ell_{R,n} f\|^2 + q^{2N} \|x_{R,0} f\|^2. \tag{95.339}$$

The factorization (95.41) and its cofinal publication (95.42) prove these identities and the prime–gamma–polar intertwiner before this assembly. The Riemann chapter unfolds and recognizes them through theorem 95.6; it neither constructs an absent interface after the fact nor receives it from the zeros of ζ.

95.11.2 Solenoidal balance of the common constructor and flow conservation

For each cut M and depth j, the image is

$$\mathfrak{G}_{\mathrm{sol},M,j}(d_{\mathrm{sol}}) = \Big(\begin{matrix}\mathfrak{G}^{\mathrm{int}}_{\mathrm{sol}};\\ \Xi^{\mathrm{sol}}_{M,j}, C_{M\leftarrow N}, M^{+}_{M,j}, M^{-}_{M,j}, D_{M,j}, H_{M,j}, E_9, J_9, P_9, Q_{\mathrm{micro}},\\ \Gamma^{\mathrm{term}}_{M,j}, \mathcal{E}^{\mathrm{term}}_{M,j,T}, L^{\mathrm{term}}_{M,j,T}, \ell_{M,j}, \mathrm{Carr}_{M,j},\\ \mathrm{Mem}_{M,j}, \mathrm{Surv}_{M,j}, \tau_{M,j}\end{matrix}\Big). \tag{95.340}$$

The published Galerkin field is a coordinate of $\Xi^{\mathrm{sol}}_{M,j}$, but the state also includes the defect $C_{M\leftarrow N}$, the oriented memories, the counit and its section, the terminal dynamics, and the observed loss. The Gram

$$U^{\mathrm{term}}_{M,j;J,T} := (\mathbf{G}^{\mathrm{term}}_{M,j;J,T})^{*}\mathbf{G}^{\mathrm{term}}_{M,j;J,T} \tag{95.341}$$

preserves the terminal coordinate and all propagated losses in $\mathbf{G}^{\mathrm{term}}$. Telescoping erases none of them. The exact source development preceding this assembly constructs the complete hydrodynamic causal map: the PC–Fock lift of (95.77), the isometric colligation of (95.104), its Stein telescoping (95.105), the ledger (95.106), the carry funded only once (95.107) and the exact return (95.115). In those same coordinates, the scattering waves are published as

$$a_{M,j} = (2\sqrt{\nu})^{-1}A_M^{-1/2}\Lambda^{s}f_{M,j}, \qquad b_{M,j} = \sqrt{\nu}A_M^{1/2}\Lambda^{s}u_{M,j} - a_{M,j}, \tag{95.342}$$

and the linear transition acts on all mixed coordinates $u^{\odot r}\otimes f^{\odot k}$. The common root and its transport are not postulated in this block: they have already been constructed by (95.116)–(95.122). Their publication in the common chart is

$$J^{\mathrm{sol}}_{T} : \mathcal{D}^{\mathrm{NS}}_{T} \longrightarrow \mathcal{H}^{\mathrm{sol}}_{T}, \qquad \mathbf{G}^{\mathrm{term}}_{M,0;J,T}\Lambda_{M,T}J^{\mathrm{sol}}_{T} = I_{M,J}J^{\mathrm{sol}}_{T}, \qquad I^{*}_{M,J}I_{M,J} = I. \tag{95.343}$$

The joint conservation of terminal, dissipation, advection, carry, microstate, shell and memory is proved by the source telescoping construction (95.119) and its ledger (95.120). The uniform budget, independent of M, J, is the theorem already established by (95.123) and (95.125); here it is published in the notation of the common constructor as

$$\sup_{M,J}\|\mathbf{G}^{\mathrm{term}}_{M,0;J,T}\Lambda_{M,T}J^{\mathrm{sol}}_{T}d\|^{2} \le C^{\mathrm{sol}}_{T}\|d\|^{2}_{\mathcal{D}^{\mathrm{NS}}_{T}}, \tag{95.344}$$

with C^{sol}_{T} independent of M, J. The later chapter unfolds the Navier–Stokes realization and recognizes its classical codomain; it neither supplies an absent premise nor selects the bound from the conclusion.

95.11.3 Projective gauge incidence of the common constructor and descent to the quotient

The gauge image is not the marginal of links. Its complete object by level is

$$\mathfrak{G}_{\mathrm{gau},m}(d_{\mathrm{gau}}) = \Big(\begin{matrix}\mathfrak{G}^{\mathrm{int}}_{\mathrm{gau}};\\ \Lambda_m, \widehat{X}_m, \widehat{\mu}^{\mathrm{ov}}_{m}, \{\widehat{\mu}^{\mathrm{YM}}_{m,g}, \nu_{m,g}, K^{\mathrm{phys}}_{m,g}\}_{g>0},\\ \mathcal{G}^{\mathrm{full}}_{m}, \mathbb{P}^{\mathrm{phys}}_{m}, \widehat{J}_m, \widehat{\Gamma}_m, \{E_{m,c}\}_{c}, \widehat{H}^{\mathrm{ov}}_{m}, D^{\mathrm{YM}}_{m},\\ \{\Theta_{\nu,m}\}_{\nu=1}^{4}, \{\mathfrak{A}_{+,\nu,m}, \mathrm{ev}^{(\nu)}_{t}, \mathcal{V}^{\mathrm{OS}}_{m,\nu}\}_{\nu=1}^{4},\\ \widehat{\Omega}^{(8)}_{m}, P_{\Omega_m}, W_c, \mathrm{Carr}_m, \mathrm{Mem}_m, \mathrm{Surv}_m, \tau_m, \mathcal{P}^{F^2}_{m}\end{matrix}\Big). \tag{95.345}$$

The group $\mathcal{G}^{\mathrm{full}}_{m}$ acts simultaneously on links, frames, and incidence coordinates $q_c \in \mathbb{F}_3^6$. The physical projection is its Haar average. The reduced generator is

$$D_m^{\mathrm{YM}} = 3\kappa_{\mathrm{av}} \sum_c (I - E_{m,c})\Big|_{\mathrm{Ran}\,\mathbb{P}_m^{\mathrm{phys}}}, \tag{95.346}$$

and W_c belongs to this subalgebra. The four reflections act on the same measure, the same kernel, and the same terminal. The preceding exact source construction establishes the incidence gauge action (95.174), the physical Haar projection (95.175), and the intertwined compression (95.176). This construction yields the complete equivariant kernel $\widehat{K}_{m,t}^{\mathrm{ov}}$, its descent to the physical quotient $K_{m,t}^{\mathrm{phys}}$, the four reflection operators, and the action $U_m(g)$ of $E(4)$ on the smooth cylindrical core. This block assembles, without introducing them as data, the generator, vacuum, curvature witness, and OS forms:

$$D_{m+1}^{\mathrm{YM}} \widehat{J}_m = \widehat{J}_m D_m^{\mathrm{YM}}, \quad \mathbb{P}_{m+1}^{\mathrm{phys}} \widehat{J}_m = \widehat{J}_m \mathbb{P}_m^{\mathrm{phys}}, \quad \widehat{J}_m W_{m,c} = W_{m+1,c}. \tag{95.347}$$

The first and third identities are the publication, in the common notation, of (95.176) and (95.179); the semigroup, simple vacuum and exact gap are already obtained in (95.180)–(95.182). Here $\nu_{m,g} := (\pi_m^{\mathrm{phys}})_* \widehat{\mu}_{m,g}^{\mathrm{YM}}$ and $K_{m,g}^{\mathrm{phys}}(t) = e^{-tD_{m,g}^{\mathrm{YM}}}$ in the physical quotient. Gauge completeness additionally includes disintegration along every family of Euclidean hyperplanes. If $A_j \in \mathfrak{A}_{+,\nu,m}$ and $t_1 \le \cdots \le t_n$, the marginal of the same physical measure satisfies

$$\begin{aligned} &\int \prod_{j=1}^{n} (\tau_{t_j e_\nu} A_j)(x)\, d\widehat{\mu}_{m,g}^{\mathrm{YM}}(x) \\ &\quad = \int \prod_{j=1}^{n} A_j(y_j)\, \nu_{m,g}(dy_1) \prod_{j=1}^{n-1} K_{m,g}^{\mathrm{phys}}(t_{j+1} - t_j; y_j, dy_{j+1}) \\ &\quad = \left\langle \mathbf{1}, M_{A_1} e^{-(t_2-t_1)D_{m,g}^{\mathrm{YM}}} M_{A_2} \cdots e^{-(t_n-t_{n-1})D_{m,g}^{\mathrm{YM}}} M_{A_n} \mathbf{1} \right\rangle. \end{aligned} \tag{95.348}$$

The preceding reconstruction proves the identity in all four directions through (95.183) and constructs a unique intertwined Hamiltonian in (95.184). Consequently, OS translation is published here as

$$\mathcal{V}_{m,\nu}^{\mathrm{OS}} T_{m,\nu}^{\mathrm{OS}}(s) (\mathcal{V}_{m,\nu}^{\mathrm{OS}})^{-1} = e^{-sD_{m,g}^{\mathrm{YM}}}. \tag{95.349}$$

Thus, the expectation generator and the translation Hamiltonian are not identified by name: they are two already constructed realizations of the same disintegration, whose limiting form and curvature publication are recorded in (95.185) and (95.187).

95.11.4 Cohomological Consistency of the Common Constructor and Emergence of the Global Class

For a complex smooth projective variety X and a codimension p, the cohomological image preserves

$$\mathfrak{G}_{\mathrm{coh}}(d_{\mathrm{coh}}) = \Big(\begin{array}{c} \mathfrak{G}_{\mathrm{coh}}^{\mathrm{int}}; \\ \{\mathcal{S}_N(X,p), \rho_{N+1,N}\}_N, \{\mathscr{H}_N^p(X), q_{N+1,N}\}_N, \{e_N\}_N, \{\iota_{N+1,N}\}_N, \\ \{\mathcal{G}_{N+1}^{\mathrm{coh}}, \mathcal{R}_{N+1}^{\mathrm{coh}}, D_{N+1}, a_{N+1}, b_{N+1}\}_N, \\ \mathcal{F}_{X,p}, \mathrm{Ext}_{X,p}, X_\chi(X,p), \mathrm{Rel}_{X,p}, \mathrm{Carr}_{X,p}, \mathrm{Mem}_{X,p}, \partial_{X,p}, \gamma_{X,p}, \\ \Sigma_{X,p}, \mathrm{Surv}_{X,p}, \tau_{X,p}, R_{\mathrm{Hdg}}, \mathrm{ch}_p^{\mathrm{loc}}, \mathrm{cl}_{X,p} \end{array} \Big). \tag{95.350}$$

The squares

$$e_N \rho_{N+1,N} = q_{N+1,N} e_{N+1} \tag{95.351}$$

link observation, extension and refinement. The preceding exact source development constructs, rather than receives as a datum, the based history with its relative corrections, carry, memory, boundary and terminal. In particular, (95.238)–(95.239) construct the increment maps and (95.249) constructs their right inverse. In the notation of the common constructor, that relative map is published as

$$s^{\mathrm{rel}}_{N+1} : \ker q_{N+1,N} \longrightarrow \ker \rho_{N+1,N}, \qquad e^{\mathrm{rel}}_{N+1} s^{\mathrm{rel}}_{N+1} = I, \tag{95.352}$$

natural under refinement by (95.251) and compatible with carry and terminal. The initial section is not assumed either: it is constructed in (95.252). It is these already proved maps, not a chosen final class, that the block assembles and the later chapter recognizes in the Hodge codomain.

Completeness rests on actions and identities constructed in the same source development. The unitary degeneracy, relative presentation and reader are assembled here through

$$\begin{aligned} \iota_{N+1,N} : \mathcal{S}_N(X,p) \longrightarrow \mathcal{S}_{N+1}(X,p), \qquad \rho_{N+1,N}\iota_{N+1,N} = I, \\ \mathscr{A}^{\mathrm{rel}}_{N+1} = \mathbb{Q}^{(\mathcal{G}^{\mathrm{coh}}_{N+1})}/\operatorname{im} D_{N+1}, \qquad e^{\mathrm{rel}}_{N+1} a_{N+1} = b_{N+1}, \qquad b_{N+1} D_{N+1} = 0. \end{aligned} \tag{95.353}$$

The set $\mathcal{G}^{\mathrm{coh}}_{N+1}$ exhaustively enumerates all new atoms and one terminal label per component. The normal form and right inverse are constructed by based reduction in (95.250); their compatibility with refinements is proved in (95.251). Thus, in the ordering by depth, support, sheet, path, carry, memory and terminal, each observation atom has a unique maximal label and

$$b_{N+1}(c_\mu) = \bar{c}_\mu + \sum_{\nu<\mu} B_{\nu\mu}\bar{c}_\nu, \qquad B_{\nu\mu} \in \mathbb{Q}. \tag{95.354}$$

Triangularity is a consequence of that based construction, prior to any class α; it is neither a hypothesis nor a table supplied by the terminal problem.

The completeness and the reader of the same object are typed by

$$X_\chi(X,p) = \varprojlim_N \mathcal{S}_N(X,p), \qquad q_N \operatorname{ev}^{\mathrm{Hdg}}_{\infty,X,p} = e_N \operatorname{pr}_N, \qquad \mathrm{cl}_{X,p} \operatorname{ch}^{\mathrm{loc}}_p R_{\mathrm{Hdg}} = \operatorname{ev}^{\mathrm{Hdg}}_{\infty,X,p}. \tag{95.355}$$

The projections q_N, including the terminal coordinate, separate points of the evaluation codomain. The operative extension (95.218)–(95.219), the passage to the limit (95.220) and the final publication (95.225) already prove that every arbitrary Hodge class admits a compatible history. The application chapter unfolds and recognizes that publication; it does not supply existence as a later premise.

95.11.5 Determinantal line of the common constructor and terminal arithmetic evaluation

For an elliptic curve $E/\mathbb{Q}$, the determinantal image preserves

$$\mathfrak{G}_{\mathrm{det}}(d_{\mathrm{det}}) = \Big(\begin{array}{c} \mathfrak{G}^{\mathrm{int}}_{\mathrm{det}}; \\ \mathcal{T}^{\mathrm{surv}}_m, G_{m,n}, \Delta_{\mathrm{AW}}, \varepsilon_{\mathrm{AW}}, P^{\mathrm{gen}}_E, 1_E, P^K_E, P^{\mathrm{PT}}_E, \\ z_K, z_{\mathrm{PT}}, X^{\mathrm{orth}}, p_{\mathrm{root}}, C^K_E, C^{\mathrm{Iw}}_E, C^{\mathrm{PT}}_E, \\ \ell_E, C^{\mathrm{loc,red}}_E, S_E(T), \mathrm{Carr}_E, \mathrm{Mem}_E, \mathrm{Surv}_E, \tau_E, \mathcal{L}_E \end{array} \Big). \tag{95.356}$$

The unit 1_E is co-published in the Kato and Poitou–Tate branches before duality and the root. The fibre product preserves the modular shadow and the deep history. The localization cone, Bockstein pages and determinant line belong to the same based multisection; they are not invariants chosen separately after knowing the rank or leading term. The preceding exact source development constructs the based decomposition Θ_E, the homotopy of the finite–singular summand, the complete Kato–FERS comparator, the Poitou–Tate page and gluing maps, and the local–global cocycle maps. The chain starts in the determinant line (95.292), constructs the chain map (95.295), the Poitou–Tate comparator (95.304) and the local triangle (95.310). Its operative identities are published here as

$$\begin{aligned} \Phi_{K\leftrightarrow\mathrm{FERS}}\Psi_{K\leftrightarrow\mathrm{FERS}} = I, \qquad \Psi^{-1}_{\mathrm{PT}}\Psi_{\mathrm{PT}} = I, \qquad \Psi_{\mathrm{PT}}\Psi^{-1}_{\mathrm{PT}} = I, \\ \ker \partial_{\mathrm{loc}} = \operatorname{im} \iota_{\mathrm{Sha}}, \qquad \pi_\Phi \operatorname{Lift}_\Phi = I. \end{aligned} \tag{95.357}$$

These identities are consequences of the exact package (95.320)–(95.321). The total comparator is constructed in (95.323) and preserves the common basis from the analytic side to the arithmetic side; rank and leading term are published afterwards in (95.324)–(95.325). The BSD chapter expands and recognizes that composition without retrospectively selecting any of its outputs.

95.12 Non-permutability Theorem of the 5 terminal readers and classification of their codomains

The map of publication $\mathcal{P}_T$ is the reader matematico that traduce the structure to the codomain classical. The chain propia of this volume is

$$\mathfrak{G}_T^{\mathrm{int}} \xrightarrow{\mathrm{App}_{T,d_T}} \mathfrak{G}_T(d_T) \xrightarrow{\mathcal{A}_{T,d_T}} \mathfrak{M}_T \xrightarrow{\mathcal{P}_T} C_T. \tag{95.358}$$

The correspondences are fixed one by one:

$$C_{\mathrm{sp}} = \mathrm{RH}, \quad C_{\mathrm{sol}} = \mathrm{NS}, \quad C_{\mathrm{gau}} = \mathrm{YM}, \quad C_{\mathrm{coh}} = \mathrm{Hodge}, \quad C_{\mathrm{det}} = \mathrm{BSD}.$$

In the active assembly, the exact source developments preceding this section also construct the five final publications: RH in theorem 95.6, Navier–Stokes in theorems 95.15 and 95.16, Yang–Mills in theorem 95.23, Hodge in theorem 95.30 and BSD in theorem 95.45. This section records their correlative assembly in the single genealogical continuum; the later chapters unfold the five realizations and their classical recognition, without supplying absent premises or using the conclusions as inputs.

Falsification criterion 95.50 (Ablation of the starting datum)**.** If a proof begins at $X_{\chi T}$, turns a coordinate of $\mathfrak{G}_T^{\mathrm{int}}$ into an optional selector, or replaces a fibre with its cardinality, it has not abbreviated the same object: it has replaced it. The conclusion is not transferable to the substitute.

95.13 Factorization of terminal readers via naturality, refinement, and passage to the limit

The material starting point is the sole simultaneous constructor that produces a single enriched continuum with five correlated intrinsic operations. The notations $\mathfrak{G}_{\mathrm{sp}}$, $\mathfrak{G}_{\mathrm{sol}}$, $\mathfrak{G}_{\mathrm{gau}}$, $\mathfrak{G}_{\mathrm{coh}}$, and $\mathfrak{G}_{\mathrm{det}}$ designate five complete views of that same ledger, reproduced with its fibres, relations, numbers, orientation, carry, memory, boundary, route, multisection, survival, terminals, and readers. They are not taken as five chosen branches or five independent data: their compatibility over the same ledger, their naturality under refinement, their joint non-emptiness, and their passage to the limit are proved, and they are then applied, respectively, to the five problems. No proof in this volume refers to another external source to complete a definition, an identity, or a logical step.

Each closure is presented in a self-contained form: the definitions, proofs, and compatibilities appear before its classical conclusion. In each resolution, four distinct layers are exhibited:

1. exact identity at every finite level;
2. compatibility of refinements and preservation of the enriched history;
3. uniformity and passage to the limit, with its terminal preserved;
4. recognition of the classical conclusion in the published codomain.

The governing composition of the five closures first preserves the unique correlative action, forms the continuum only upon passage to the limit, and then opens a view for each application:

$$\mathcal{E}_k \xrightarrow{O_k} \mathcal{Z}_k \xrightarrow{\lim_k} (C_{\mathrm{cont}}^{\mathrm{disc}}, O_\infty),$$
$$(C_{\mathrm{cont}}^{\mathrm{disc}}, O_\infty) \xrightarrow{\mathrm{View}_T} \mathfrak{G}_T^{\mathrm{int}} \xrightarrow{\mathrm{App}_{T,d_T}} \mathfrak{G}_T(d_T),$$
$$\mathfrak{G}_T(d_T) \xrightarrow{\widetilde{\mathcal{A}}_{T,d_T}} \mathfrak{M}_T \xrightarrow{\mathcal{P}_T} C_T, \qquad T \in \{\mathrm{sp}, \mathrm{sol}, \mathrm{gau}, \mathrm{coh}, \mathrm{det}\}.$$

Here O_k acts once on the complete state and jointly preserves the five features. View_T neither restricts the domain nor selects histories: it only exposes the coordinates of one application within the complete value of O_∞. The classical datum d_T enters only in App_{T,d_T}, after the continuum, and does not participate in generating the intrinsic feature. The real line is likewise a subsequent output of $\mathrm{ev}_{\mathbb{R}}$, not a premise of the chain.

Each chapter distinguishes the starting domain, the construction, the certificate, the proved conclusion and its exact falsifiers. The preceding source developments construct the five compositions in their respective domains; the following chapters unfold those constructions and carry out the final recognition. No arrow is replaced by a cross-reference, a nominal conclusion or a classical hypothesis.

95.14 Common factorization of the boundary operator in the solenoidal and gauge realizations

Hydrodynamics and Yang–Mills are not brought together by externally comparing two celebrated equations. They follow from two representations of the same boundary operator. The orientation that the interior reader interprets as circulation becomes holonomy under the reflected reader; the symmetric memory that controls the flow becomes a positive transfer kernel.

95.14.1 Boundary enriched state and parity decomposition of transport

Let K_m be a finite oriented complex, B_m its algebra of boundary data, and d_m its coboundary. The enriched state is

$$s_m = (b_m, U_m, \theta_m, c_m, \Omega_m),$$

where U_m transports, $\theta_m^2 = I$ reverses orientation, c_m preserves memory, and Ω_m is a 1-connection cochain. The parity projectors are

$$E_\pm = \frac{1}{2}(I \pm \theta_m).$$

The part $(U_m + U_m^\dagger)/2$ is symmetric; it is a projector only if its idempotence is also proved.

On the cochain complex, two linear readers are constructed:

$$\rho_{\mathrm{int},m} : C^0(K_m) \longrightarrow C^1(K_m) \oplus C^0(K_m), \qquad \rho_{\mathrm{int},m}(v) = (d_m v, d_m^* d_m v),$$
$$\rho_{\mathrm{ref},m} : C^1(K_m) \longrightarrow C^0(K_m) \oplus C^1(K_m), \qquad \rho_{\mathrm{ref},m}(a) = (d_m^* a, d_m d_m^* a).$$

The former transports a vertex potential to its edge flux and interior response; the latter transports an edge cochain to its boundary response and reflected operator. The codomains differ, but both are constructed with the same coboundary. Circulation, vorticity, holonomy, and curvature arise subsequently through Stokes' identity; they are not presupposed within the symbols ρ.

95.14.2 Minimal Boundary Complex and Spectral Equivalence of the 0 and 1 Degree Laplacians

In the oriented cycle of three vertices, let

$$d_0 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}.$$

The Hodge Laplacians of vertices and edges are

$$\mathcal{L}_{\mathrm{H}}^{(0)} = d_0^* d_0, \qquad \mathcal{L}_{\mathrm{H}}^{(1)} = d_0 d_0^*,$$

and a direct calculation gives

$$\mathcal{L}_{\mathrm{H}}^{(0)} = \mathcal{L}_{\mathrm{H}}^{(1)} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} = 3I - J.$$

In this residence, the superscripts (0) and (1) denote the Hodge Laplacians on vertices and edges. They do not denote the lifting operators L_0, L_1 in the chain $w_6 \to w_{30}$. For $\mathcal{L}_{\mathrm{cyc}}^{\mathrm{TPK}} := 2I - C - C^{-1} = 3I - J$, the realization of the TPK cyclic operator is typed through

$$\mathfrak{R}_{\mathrm{hyd}}(\mathcal{L}_{\mathrm{cyc}}^{\mathrm{TPK}}) = \mathcal{L}_{\mathrm{H}}^{(0)}, \qquad \mathfrak{R}_{\mathrm{YM}}(\mathcal{L}_{\mathrm{cyc}}^{\mathrm{TPK}}) = \mathcal{L}_{\mathrm{H}}^{(1)}.$$

Theorem 95.51 (Common boundary operator)**.** *In the three-cell cycle,*

$$\mathrm{Spec}(\mathcal{L}_{\mathrm{H}}^{(0)}) = \mathrm{Spec}(\mathcal{L}_{\mathrm{H}}^{(1)}) = \{0, 3, 3\}.$$

After removing the constant mode —zero mean in the interior reader and gauge in the connection reader— both sectors have exact gap 3*. In every finite complex, d^*d and dd^* have the same nonzero spectrum, with multiplicities.*

Proof. The vector $(1, 1, 1)$ generates the kernel of $3I - J$. On its orthogonal complement, $J = 0$, so the operator is $3I$. For the general statement, a singular decomposition $d = U\Sigma V^*$ yields

$$d^*d = V\Sigma^*\Sigma V^*, \qquad dd^* = U\Sigma\Sigma^* U^*.$$

The squares of the non-zero singular values, including their multiplicities, coincide. □

95.14.3 Solenoidal realization of the boundary operator: flow contraction and energy conservation

We define the implicit step

$$S_{\mathrm{NS}} = (I + \mathcal{L}_{\mathrm{H}}^{(0)})^{-1}.$$

In the minimum cycle,

$$S_{\mathrm{NS}} = \frac{1}{4}(I + J).$$

Theorem 95.52 (Global contraction of the finite flow)**.** *For every mean-zero v_0, the iteration $v_{k+1} = S_{\mathrm{NS}} v_k$ exists and is unique for every $k \geq 0$, and*

$$v_k = 4^{-k} v_0, \qquad \|v_k\| \leq 4^{-k}\|v_0\|, \qquad \|v_k\|^2 = 16^{-k}\|v_0\|^2.$$

Proof. At zero mean one has $Jv = 0$, hence $S_{\mathrm{NS}} v = v/4$. The formula is obtained by induction. □

An oriented nonlinearity $N_m(v, c)$ preserves energy when

$$\langle v, N_m(v, c)\rangle = 0.$$

Advection then redistributes the energy, and the positive operator controls its propagation. The memory c_m is not absorbed into a chosen viscosity: it is part of the state that must be transported between scales.

95.14.4 Boundary operator calibration: coercivity, gauge quotient, and spectral gap

A 1-cochain a is interpreted as a connection potential and has quadratic energy

$$Q_{\mathrm{YM}}(a) = \langle a, \mathcal{L}_{\mathrm{H}}^{(1)} a \rangle.$$

Theorem 95.53 (Finite gauge coercivity and decay)**.** *In the quotient by the constant gauge mode,*

$$Q_{\mathrm{YM}}(a) \geq 3\|a\|^2.$$

The semigroup $K_t = e^{-t\,\mathcal{L}_{\mathrm{H}}^{(1)}}$ *satisfies*

$$\|K_t a\| \leq e^{-3t}\|a\|$$

in the physical sector of the minimal cycle.

Proof. On the complement of the kernel, the theorem 95.51 gives $\mathcal{L}_{\mathrm{H}}^{(1)} = 3I$. Both formulas follow by functional calculus. □

For a non-abelian connection,

$$F_\Omega = d\Omega + \Omega \smile \Omega, \qquad W_\gamma(\Omega) = \operatorname{Tr} \operatorname{Hol}_\gamma(\Omega).$$

Reflection positivity requires a factorization on the positive half,

$$\langle \theta f, K f \rangle = \|C f\|^2 \geq 0;$$

pointwise positivity of the matrix K does not replace it.

95.14.5 Discrete Stokes identity and 2 realizations: circulation–vorticity and holonomy–curvature

For a 1-cochain Ω and a 2-chain K,

$$\sum_{e \subset \partial K} \Omega(e) = \sum_{f \in K^{(2)}} (d\Omega)(f).$$

Interior edges cancel by orientation. The hydrodynamic reader interprets the left-hand side as circulation and the right-hand side as accumulated vorticity. The gauge reader interprets the former as linearized holonomy and the latter as curvature.

Theorem 95.54 (Common Boundary Sheet Theorem)**.** *The decomposition* $B_m = E_- B_m \oplus E_+ B_m$ *has two typed representations:*

$E_- B_m$	$\longmapsto$	*circulation / holonomy,*
$E_+ B_m$	$\longmapsto$	*dissipative control / reflected kernel.*

Stokes equality and the coincidence of the nonzero spectrum belong to the common object. The hydrodynamic contraction and the gauge gap are distinct consequences in their respective codomains.

Proof. The projectors $E_\pm$ are complementary because $\theta_m^2 = I$. The Stokes identity provides the intertwining of the oriented sector; theorem 95.51 provides the common spectrum. The theorems 95.52 y 95.53 compute the two realizations. □

The theorem is a finite HMT closure: it neither identifies a PDE with a gauge theory nor confuses the gap 3 with a physical mass. It explains by construction why interior regularity and boundary separation are two functions of the same oriented memory. Any subsequent continuous passage must preserve the operator, norm, and spectrum that make this identity exact.

Chapter 96

Positive factorization of the Guinand–Weil form and localization of the nontrivial zeros of ζ

96.1 Starting point: the complete spectral–polar structure

The input to the resolution is not the isolated Weil form or a list of zeros. It is the complete spectral–polar structure produced by the common constructor and accredited by the exact owner preceding this chapter; it is not an external datum. The causal chain specific to this chapter is

$$\mathfrak{G}_{\mathrm{sp}} \xrightarrow{\mathrm{pr}_{X,\mathrm{sp}}} X_{\chi_{\mathrm{sp}}} \xrightarrow{\mathcal{A}_{\mathrm{sp}}} \mathfrak{M}_{\mathrm{sp}} \xrightarrow{\mathcal{P}_{\mathrm{sp}}} C_{\mathrm{GW}}. \tag{96.1}$$

The classical codomain appears only in the last arrow; it does not retrospectively select the state, the charts, the projection, or the terminal.

The starting structure contains the object of thirteen coordinates

$$\begin{aligned}\mathfrak{G}_{\mathrm{sp}}^{(13)} = \big(&\mathcal{F}_{\mathrm{sp}}, \mathrm{Ext}_{\Gamma,\mathrm{sp}}, \mathrm{Rel}_{\mathrm{sp}}, \mathcal{N}_{\mathrm{sp}}, \mathrm{or}_{\mathrm{sp}}, \mathrm{Carr}_{\mathrm{sp}}, \mathrm{Mem}_{\mathrm{sp}},\\ &\partial_{\mathrm{sp}}, \gamma_{\mathrm{sp}}, \Sigma_{\mathrm{sp}}^{\mathrm{mult}}, \mathrm{Surv}_{\mathrm{sp}}, \tau_{\mathrm{sp}}^{\mathrm{term}}, \mathcal{L}_{\mathrm{sp}}\big).\end{aligned} \tag{96.2}$$

Each coordinate enters substantively:

1. $\mathcal{F}_{\mathrm{sp}}$ preserves the two sheets, their orientation, the residue–quotient pair, and the signature of the double publication.
2. $\mathrm{Ext}_{\Gamma,\mathrm{sp}}$ transports that fibre through the directed class of cutoffs and nonadic levels.
3. $\mathrm{Rel}_{\mathrm{sp}}$ contains the carry cocycle, the orthogonality $P = P^* = P^2$, the two publications, and the structural complement $B_{\mathrm{sp}}^{\mathrm{str}} := \Xi^*(I - P)\Xi$. The identification of this complement with the Guinand–Weil form is a distinct arrow and is audited separately in this chapter.
4. $\mathcal{N}_{\mathrm{sp}}$ records the level m, the window R, the quotient $q = 5/9$, the moments of both sheets, and the indices of the prime, gamma, and polar channels.
5. $\mathrm{or}_{\mathrm{sp}} = \widehat{R} = (R, K)$ keeps the two oriented charts distinct.
6. $\mathrm{Carr}_{\mathrm{sp}} = \mathrm{Carr}_R$ is the operatorial carry and is not annulled when a translation is published.
7. $\mathrm{Mem}_{\mathrm{sp}} = (S_0, M, O_9)$ preserves the finite Stein identity and its terminal object.
8. $\partial_{\mathrm{sp}} = (I - P)\Xi$ is the boundary absent from the $P\Xi$ compression and whose energy defines the structural complement. Its identification with the Weil form is not incorporated into this definition.
9. γ_{sp} is the cofinal route of windows, with a single regulator for all channels.
10. $\Sigma_{\mathrm{sp}}^{\mathrm{mult}}$ is the multisection of the two publications and their prime–gamma–polar components; it is not the signature of a single state.
11. $\mathrm{Surv}_{\mathrm{sp}}$ preserves the family under $m \mapsto m + 1$ and $\mathcal{D}_R \hookrightarrow \mathcal{D}_{R'}$.
12. $\tau_{\mathrm{sp}}^{\mathrm{term}}$ preserves the terminal $M^{*N} S_0 M^N$, which in the mirror chart is published through E_R.
13. $\mathcal{L}_{\mathrm{sp}} = (\Xi, P, B_{\mathrm{sp}}^{\mathrm{str}})$ is the final reader; it is applied after antecedent, charts, memory, route, and terminal object.

The assembler and the publisher are

$$\begin{aligned} \mathcal{A}_{\mathrm{sp}}(\mathrm{pr}_{X,\mathrm{sp}}\,\widehat{x}) &= (\widehat{X}_\infty, \Gamma_9, \widehat{R}, \mathrm{Carr}_R, S_0, M, O_9, \Xi, P, \mathcal{D}_R^{\mathrm{cof}}),\\ \mathfrak{M}_{\mathrm{sp}} &= \operatorname{Im}\mathcal{A}_{\mathrm{sp}},\\ \mathcal{P}_{\mathrm{sp}}(\mathfrak{M}_{\mathrm{sp}}) &= (B_{\mathrm{sp}}^{\mathrm{str}}, \Xi^*(I-P)\Xi, \mathcal{J}_{\mathrm{GW}}, \Delta_{\mathrm{bind}}, \text{cofinal prime–gamma–polar family}),\\ \Delta_{\mathrm{bind}} &:= B_{\mathrm{sp}}^{\mathrm{str}} - \mathcal{J}_{\mathrm{GW}}. \end{aligned} \tag{96.3}$$

The finite memory of the starting object satisfies

$$S_0 = \sum_{r=0}^{N-1} M^{*r} O_9^* O_9 M^r + M^{*N} S_0 M^N. \tag{96.4}$$

The last summand remains visible until its extinction is proved. Therefore, replacing $\mathfrak{G}_{\mathrm{sp}}$ with the published shadow $B_{\mathrm{sp}}^{\mathrm{str}}$ changes the starting object; it is not an abbreviation of the same theorem. The residual Δ_{bind} is preserved until the source–first colligation materially proves its vanishing in (96.16).

96.2 Cofinal domain and target form

For $R > 0$ is fixed

$$\mathcal{D}_R = C_c^\infty((-R/2, R/2)), \qquad \widehat{\psi}(t) = \int_{\mathbb{R}} \psi(x) e^{-itx}\, dx,$$

and, for $\psi, \phi \in \mathcal{D}_R$,

$$h_{\psi,\phi}(z) = \widehat{\psi}(z)\overline{\widehat{\phi}(\overline{z})}, \qquad \check{h}_{\psi,\phi}(a) = \langle \psi, T_a \phi \rangle.$$

The union $\bigcup_R \mathcal{D}_R = C_c^\infty(\mathbb{R})$ is exact. For every finite set of distinct complex points, the evaluations $\psi \mapsto (\widehat{\psi}(z_1), \dots, \widehat{\psi}(z_n))$ are jointly surjective when R is sufficiently large. Therefore, this cofinal class separates the evaluations required by the Weil criterion.

Lemma 96.1 (Proof of the joint separation)**.** *The map*

$$\mathcal{D}_R \longrightarrow \mathbb{C}^n, \qquad \psi \longmapsto (\widehat{\psi}(z_1), \dots, \widehat{\psi}(z_n))$$

is surjective for every set of distinct complex points $z_1, \dots, z_n$, once R contains a nonempty interval.

Proof. If the evaluation functionals were linearly dependent, there would exist $c_1, \dots, c_n$, not all zero, such that

$$\int_{\mathbb{R}} \psi(x) \left(\sum_{j=1}^{n} c_j e^{-iz_j x} \right) dx = 0 \qquad (\psi \in \mathcal{D}_R).$$

The analytic function $\sum_j c_j e^{-iz_j x}$ would vanish at $(-R/2, R/2)$, hence throughout the plane. The independence of exponentials with distinct exponents forces $c_j = 0$ for every j, a contradiction. The functionals are independent; since the codomain is finite-dimensional, the map has rank $\mathbb{C}^n$. □

96.3 The two columns and the three channels computed before the sign

Be ϑ_L a regulador comun. With

$$\ell_{p,k} = k \log p, \qquad w_{p,k} = (\log p) p^{-k/2}, \qquad \mu_\Gamma(a) = \frac{e^{-a/2}}{1 - e^{-2a}},$$

the uniform inclusion j_m and the sewing channel $\widehat{\delta}_{a,m}$, which satisfy, are used

$$j_m^* j_m = I, \qquad \widehat{\delta}_{a,m}^* \widehat{\delta}_{b,m} = (I - T_a)^*(I - T_b).$$

The regulated columns are

$$\begin{aligned} J_{+,m,R,L}\psi &= \Big((\sqrt{\vartheta_L(\ell_{p,k}) w_{p,k}}\, \widehat{\delta}_{\ell_{p,k},m}\psi)_{\ell_{p,k} \le R}, \\ &\qquad a \mapsto \sqrt{\vartheta_L(a)\mu_\Gamma(a)}\, \widehat{\delta}_{a,m}\psi,\ \zeta_+ b_+(\psi) \Big), \\ J_{-,m,R,L}\psi &= \Big((\sqrt{2\vartheta_L(\ell_{p,k}) w_{p,k}}\, j_m\psi)_{\ell_{p,k} \le R},\ \sqrt{-c_\Gamma}\, j_m\psi,\ \zeta_- b_-(\psi) \Big), \end{aligned}$$

where $c_\Gamma = \psi_\Gamma(1/4) - \log\pi < 0$ and $b_\pm = (\widehat{\psi}(i/2) \pm \widehat{\psi}(-i/2))/\sqrt{2}$.

A term-by-term expansion, without using zeros or the sought sign, gives

$$\begin{aligned} \langle J_{+,m,R,L}\psi, J_{+,m,R,L}\phi\rangle - \langle J_{-,m,R,L}\psi, J_{-,m,R,L}\phi\rangle &= \mathscr{I}_{\mathrm{GW},L}(\psi,\phi), \\ \mathscr{I}_{\mathrm{GW},L}(\psi,\phi) &= h_{\psi,\phi}(i/2) + h_{\psi,\phi}(-i/2) + c_\Gamma \check{h}_{\psi,\phi}(0) \\ &\quad + \int_0^\infty \vartheta_L(a)\mu_\Gamma(a)[2\check{h}(0) - \check{h}(a) - \check{h}(-a)]\, da \\ &\quad - \sum_{p,k} \vartheta_L(\ell_{p,k}) w_{p,k} [\check{h}(\ell_{p,k}) + \check{h}(-\ell_{p,k})]. \end{aligned}$$

The truncation of both columns at $\ell_{p,k} \le R$ is mandatory, not notational. If $\psi, \phi \in \mathcal{D}_R$, then $\check{h}_{\psi,\phi}(\pm\ell) = 0$ for $\ell > R$; the contribution of each omitted pair of coordinates to the *difference* of Grams is therefore zero. The untruncated negative column, by contrast, would have infinite norm because $\sum_p (\log p) p^{-1/2} = \infty$. In the finite set $\ell_{p,k} \le R$ the prime coordinates converge individually when $L \to \infty$. In the gamma channel, the integrand is $O(a)$ at zero and $\mu_\Gamma(a) = O(e^{-a/2})$ at infinity, so convergence is also in norm. We therefore define

$$J_{\pm,m,R} := \lim_{L\to\infty} J_{\pm,m,R,L} \quad \text{in their truncated-coordinate Hilbert spaces,} \tag{96.5}$$

and we suppress m when it remains fixed. By dominated convergence,

$$\mathscr{I}_{\mathrm{GW},L} \longrightarrow \mathscr{I}_{\mathrm{GW}} \qquad (L \to \infty),$$

and the identities are natural under $m \mapsto m+1$ and $R \le R'$.

The reduced columns $J_{+,R}$ and $J_{-,R}$ are now material operators from $\mathcal{D}_R$ to Hilbert. The explicit expansion proves, without consulting zeros or sign,

$$\langle J_{+,R} f, J_{+,R} g\rangle - \langle J_{-,R} f, J_{-,R} g\rangle = \mathscr{I}_{\mathrm{pr}}(f,g) + \mathscr{I}_\Gamma(f,g) + \mathscr{I}_{\mathrm{pol}}(f,g) = \mathscr{I}_{\mathrm{GW}}(f,g). \tag{96.6}$$

The clocks $\ell_{p,k}$ produce the prime summand; the integral with μ_Γ and $c_\Gamma \check{h}(0)$, the gamma summand; and

$$\langle b_+(f), b_+(g)\rangle - \langle b_-(f), b_-(g)\rangle = h_{f,g}(i/2) + h_{f,g}(-i/2)$$

produce the polar term. This part of the calculation is material and independent of the structural bridge.

96.4 Source–first colligation of the spectral–polar image

We first fix a nonadic level m; this index is suppressed in $J_{\pm,R}$, E_R and Σ_R until the encoder is constructed. The decomposition of the explicit codomains of (96.5) is

$$I_+ = \{\mathrm{pr}, \Gamma, \mathrm{pol}\}, \qquad I_-(R) = \{\Gamma 0, \mathrm{pol}\} \sqcup \{(p,k) : \ell_{p,k} \le R\},$$

$$\mathscr{H}^{\mathrm{amb}}_{+,R} = \bigoplus_{i \in I_+} H_{+,i,R}, \qquad \mathscr{H}^{\mathrm{amb}}_{-,R} = \bigoplus_{j \in I_-(R)} H_{-,j,R}.$$

We do not identify a reachable subspace with a sum of its coordinate projections. With the ambient inclusions understood, we define

$$H_{+,R} := \overline{\operatorname{Ran} J_{+,R}}, \qquad H_{-,R} := \overline{\operatorname{Ran} J_{-,R}},$$

as closed subspaces of the two ambient Hilbert spaces. The spectral owner printed before this chapter constructs $\mathcal{H}^{\mathrm{sh}}_R$, the antispecular difference and the source–first isometric row through (95.37)–(95.38), proves its telescoping in (95.39) and publishes terminal extinction in theorem 95.6. The realization that follows preserves that construction, before classical identification, in the row

$$\Sigma_R = \begin{pmatrix} \mathcal{K}^{\mathrm{src}}_R \\ \mathcal{F}^{\mathrm{src}}_R \end{pmatrix} : H_{+,R} \longrightarrow H_{-,R} \oplus \mathcal{H}^{\mathrm{sh}}_R, \qquad \Sigma_R^* \Sigma_R = (\mathcal{K}^{\mathrm{src}}_R)^* \mathcal{K}^{\mathrm{src}}_R + (\mathcal{F}^{\mathrm{src}}_R)^* \mathcal{F}^{\mathrm{src}}_R = I_{H_{+,R}}. \tag{96.7}$$

Its reachable action is fixed on the entire positive column by the relations of the two publications that form part of $\mathrm{Rel}_{\mathrm{sp}}$. To prevent any substitution of the reader, we denote by $(\mathcal{X}^{\mathrm{str}}_R, \Xi^{\mathrm{str}}_R, P^{\mathrm{str}}_R)$ the restriction to $\mathcal{D}_R$ of the structural reader of (96.2). Its two moments are, literally,

$$\begin{aligned} \langle \Xi^{\mathrm{str}}_R f, \Xi^{\mathrm{str}}_R g \rangle &= \langle J_{+,R} f, J_{+,R} g \rangle, \\ \langle P^{\mathrm{str}}_R \Xi^{\mathrm{str}}_R f, P^{\mathrm{str}}_R \Xi^{\mathrm{str}}_R g \rangle &= \langle J_{-,R} f, J_{-,R} g \rangle, \qquad (P^{\mathrm{str}}_R)^* = (P^{\mathrm{str}}_R)^2 = P^{\mathrm{str}}_R. \end{aligned} \tag{96.8}$$

The boundary coordinate comes with the structural isometry.

$$\jmath^{\mathrm{sh}}_R : \overline{\operatorname{Ran}((I - P^{\mathrm{str}}_R)\Xi^{\mathrm{str}}_R)} \longrightarrow \mathcal{H}^{\mathrm{sh}}_R.$$

By definition of the source–first row, not by a subsequent identification,

$$E_R := \mathcal{F}^{\mathrm{src}}_R J_{+,R} = \jmath^{\mathrm{sh}}_R (I - P^{\mathrm{str}}_R) \Xi^{\mathrm{str}}_R : \mathcal{D}_R \longrightarrow \mathcal{H}^{\mathrm{sh}}_R, \qquad \Omega_{0,R} := \mathcal{K}^{\mathrm{src}}_R J_{+,R} - J_{-,R}. \tag{96.9}$$

There is no colligation by channel: the same row simultaneously receives the prime–power clocks, the gamma integral, the scalar mode, and the two polar lines. In particular, it preserves all cross-products that arise upon polarizing (96.6). With the ambient projections $\pi^-_j : \mathscr{H}^{\mathrm{amb}}_{-,R} \to H_{-,j,R}$, the correctly typed output rows are

$$\mathcal{K}^{\mathrm{src}}_{j,R} := \pi^-_j \mathcal{K}^{\mathrm{src}}_R : H_{+,R} \longrightarrow H_{-,j,R}, \qquad \mathcal{F}^{\mathrm{src}}_R : H_{+,R} \longrightarrow \mathcal{H}^{\mathrm{sh}}_R. \tag{96.10}$$

Thus, $\mathcal{K}^{\mathrm{src}}_R$ is not applied to a positive coordinate that might not by itself belong to $H_{+,R}$. All cross-terms between channels remain within $(\mathcal{K}^{\mathrm{src}}_R)^* \mathcal{K}^{\mathrm{src}}_R$ and $(\mathcal{F}^{\mathrm{src}}_R)^* \mathcal{F}^{\mathrm{src}}_R$; they are not orthogonalized before the energy identity.

The survival coordinate constructed by (95.37), (95.39) and (95.40) produces the residual spaces $\mathcal{R}^{\mathrm{sh}}_{n,R}$, continuations $\Omega_{n,R} : \mathcal{D}_R \to \mathcal{R}^{\mathrm{sh}}_{n,R}$ and antispecular isometries $V_{n,R} : \mathcal{R}^{\mathrm{sh}}_{n+1,R} \to \mathcal{R}^{\mathrm{sh}}_{n,R}$. Its first space and first operator are, by typed definition,

$$\mathcal{R}^{\mathrm{sh}}_{0,R} := H_{-,R}, \qquad \Omega_{0,R} := \mathcal{K}^{\mathrm{src}}_{R} J_{+,R} - J_{-,R}.$$

Its recurrence over one complete turn is the cofinal publication of the same constructed operator, not a datum added to the realization:

$$q = \frac{5}{9}, \qquad \Omega_{n,R} = q V_{n,R} \Omega_{n+1,R}, \qquad V^{*}_{n,R} V_{n,R} = I, \qquad \sup_{n\geq 0} \|\Omega_{n,R} f\| < \infty \quad (f \in \mathcal{D}_R). \tag{96.11}$$

This recurrence belongs to the enriched colligation: it transports the output difference, not the Weil form or its sign.

Proposition 96.2 (Coinductive Cancellation of Output Difference)**.** *For every $R > 0$ and every $f \in \mathcal{D}_R$,*

$$\Omega_{n,R} f = 0 \quad (n \geq 0), \qquad \boxed{\mathcal{K}^{\mathrm{src}}_{R} J_{+,R} = J_{-,R}}. \tag{96.12}$$

Proof. Iterating (96.11) N times and using that each $V_{n,R}$ is isometric gives

$$\|\Omega_{n,R} f\| = q^{N} \|\Omega_{n+N,R} f\| \leq q^{N} \sup_{j\geq n} \|\Omega_{j,R} f\|.$$

The right-hand member converges to zero because $0 < q = 5/9 < 1$. Therefore $\Omega_{n,R} f = 0$; the case $n = 0$ and (96.9) produce the framed identity. □

The row-by-row action of the output is no longer implicit. The projections are the restrictions to $H_{-,R}$ of the projections in the ambient Hilbert space, and

$$\begin{aligned} \mathcal{K}^{\mathrm{src}}_{\Gamma 0,R} J_{+,R} f &= \sqrt{-c_\Gamma}\, j_m f, \\ \mathcal{K}^{\mathrm{src}}_{(p,k),R} J_{+,R} f &= \sqrt{2w_{p,k}}\, j_m f \quad (\ell_{p,k} \leq R), \\ \mathcal{K}^{\mathrm{src}}_{\mathrm{pol},R} J_{+,R} f &= \zeta_{-} b_{-}(f). \end{aligned} \tag{96.13}$$

These rows exhaust $J_{-,R}$, while

$$\mathcal{F}^{\mathrm{src}}_{R} J_{+,R} f = E_R f \tag{96.14}$$

preserves, with all its crossings, the boundary coordinate. Evaluating (96.7) at $J_{+,R} f$ and $J_{+,R} g$, and using (96.12), produces the energy identity prior to the sign

$$\boxed{\langle J_{+,R} f, J_{+,R} g\rangle = \langle J_{-,R} f, J_{-,R} g\rangle + \langle E_R f, E_R g\rangle.} \tag{96.15}$$

Comparison with the independent calculation (96.6) yields, without introducing a root of the target form,

$$\boxed{E^{*}_{R} E_R = \mathcal{J}_{\mathrm{GW}}\big|_{\mathcal{D}_R}.} \tag{96.16}$$

In particular, $\Delta_{\mathrm{bind},R} := E^{*}_{R} E_R - \mathcal{J}_{\mathrm{GW}}|_{\mathcal{D}_R} = 0$; this annihilation is the output of (96.7)– (96.15), not a premise inserted into the publisher.

96.5 Structural complement, mirror sheet and terminal telescopia

The state $E_R = \mathcal{F}^{\mathrm{src}}_{R} J_{+,R}$ constructed by the source–first row is the boundary coordinate of $\mathfrak{G}_{\mathrm{sp}}$. We define its structural form by

$$\mathcal{B}^{\mathrm{str}}_{\mathrm{sp},R} := E^{*}_{R} E_R = \mathcal{J}_{\mathrm{GW}}\big|_{\mathcal{D}_R}, \tag{96.17}$$

where the second equality is (96.16), already deduced from the colligation and from the generator-by-generator expansion.

Let P^{sh}_{+} and P^{sh}_{-} be the complementary orthogonal projectors of the two sheets. Fix

$$q = \frac{5}{9}, \qquad A = P_+^{\mathrm{sh}} + qP_-^{\mathrm{sh}}, \qquad D = \sqrt{1-q^2}\, P_-^{\mathrm{sh}}. \tag{96.18}$$

Orthogonality gives the exact identity

$$A^*A + D^*D = P_+^{\mathrm{sh}} + q^2 P_-^{\mathrm{sh}} + (1-q^2)P_-^{\mathrm{sh}} = I. \tag{96.19}$$

The nine phases coordinate a single memory-enabled rotation:

$$M_9 = \Phi A_8 \cdots A_0 = qV_9, \qquad V_9^* V_9 = I. \tag{96.20}$$

The contraction per turn is q, not q^9; the internal phases preserve order and carry.

The colligation types $E_R : \mathcal{D}_R \to \operatorname{Ran} P_-^{\mathrm{sh}}$. Therefore,

$$P_+^{\mathrm{sh}} E_R = 0, \qquad AE_R = qE_R, \qquad E_R^* E_R = \mathcal{B}_{\mathrm{sp},R}^{\mathrm{str}} = \mathcal{J}_{\mathrm{GW}}\big|_{\mathcal{D}_R}. \tag{96.21}$$

Proposition 96.3 (Finite Telescopy with Conserved Terminal)**.** *for each $N \geq 1$,*

$$\mathcal{B}_{\mathrm{sp},R}^{\mathrm{str}} = \sum_{n=0}^{N-1} (DA^n E_R)^*(DA^n E_R) + (A^N E_R)^*(A^N E_R). \tag{96.22}$$

Equivalently, evaluated at $f, g \in \mathcal{D}_R$, the finite identity of the colligation is

$$\boxed{\begin{aligned}&\langle J_{+,R} f, J_{+,R} g\rangle - \langle J_{-,R} f, J_{-,R} g\rangle \\ &\quad = \sum_{n=0}^{N-1} \langle DA^n E_R f, DA^n E_R g\rangle + \langle A^N E_R f, A^N E_R g\rangle.\end{aligned}} \tag{96.23}$$

The last term satisfies

$$(A^N E_R)^*(A^N E_R) = q^{2N} \mathcal{B}_{\mathrm{sp},R}^{\mathrm{str}} \xrightarrow[N\to\infty]{} 0. \tag{96.24}$$

Proof. Applying (96.19) to $A^n E_R$,

$$(A^n E_R)^*(A^n E_R) = (DA^n E_R)^*(DA^n E_R) + (A^{n+1} E_R)^*(A^{n+1} E_R).$$

The sum from $n = 0$ to $N-1$ cancels only the intermediate terms and preserves $(A^N E_R)^*(A^N E_R)$. By (96.21), $A^N E_R = q^N E_R$; hence the terminal object is $q^{2N} E_R^* E_R = q^{2N} \mathcal{B}_{\mathrm{sp},R}^{\mathrm{str}}$. Since $0 < q = 5/9 < 1$, it converges geometrically to zero. □

Only after proving (96.24) is defined

$$\mathscr{L}_R f = (DA^n E_R f)_{n\geq 0}. \tag{96.25}$$

The limit of finite telescoping gives

$$\boxed{\mathscr{L}_R^* \mathscr{L}_R = \mathcal{B}_{\mathrm{sp},R}^{\mathrm{str}} = \mathcal{J}_{\mathrm{GW}}\big|_{\mathcal{D}_R} \succeq 0.} \tag{96.26}$$

96.6 Common encoder, projection, and explicit double publication

Fix $m \geq 1$ and two orthonormal vectors $u_m, \eta_m \in \mathbb{C}^{9^m}$, where u_m is the uniform mode and η_m the carry mode. The coder space and projection are

$$\mathcal{X}^{\mathrm{sp}}_{m,R} := \mathbb{C}^{9^m} \widehat{\otimes} (H_{-,R} \oplus \mathcal{H}^{\mathrm{sh}}_R), \qquad P_{m,R} := |u_m\rangle\langle u_m| \otimes I_{H_{-,R}\oplus\mathcal{H}^{\mathrm{sh}}_R}, \qquad Q_{m,R} := I - P_{m,R}. \tag{96.27}$$

The complete action of the five-channel encoder is

$$\boxed{\begin{aligned} \mathfrak{C}_{m,R} f &:= u_m \otimes (J_{-,R} f, 0) + \eta_m \otimes (0, E_R f), \\ \Xi_{m,R} f &:= \mathfrak{C}_{m,R} f. \end{aligned}} \tag{96.28}$$

No root of $\mathcal{J}_{\mathrm{GW}}$ is taken: the two inputs are the output and state of the source–first colligation already defined in (96.9).

Proposition 96.4 (The two moments arise from the action of the encoder)**.** *For every $f, g \in \mathcal{D}_R$,*

$$\begin{aligned} \langle \mathfrak{C}_{m,R} f, \mathfrak{C}_{m,R} g\rangle &= \langle J_{+,R} f, J_{+,R} g\rangle, \\ \langle P_{m,R}\mathfrak{C}_{m,R} f, P_{m,R}\mathfrak{C}_{m,R} g\rangle &= \langle J_{-,R} f, J_{-,R} g\rangle, \\ \langle Q_{m,R}\Xi_{m,R} f, Q_{m,R}\Xi_{m,R} g\rangle &= \mathcal{J}_{\mathrm{GW}}(f, g). \end{aligned} \tag{96.29}$$

Proof. The orthogonality $u_m \perp \eta_m$ and (96.15) give

$$\langle \mathfrak{C}_{m,R} f, \mathfrak{C}_{m,R} g\rangle = \langle J_{-,R} f, J_{-,R} g\rangle + \langle E_R f, E_R g\rangle = \langle J_{+,R} f, J_{+,R} g\rangle.$$

The projection $P_{m,R}$ preserves the uniform summand and eliminates the carry summand; $Q_{m,R}$ does the reverse. The other two identities follow from (96.16). □

The newly written encoder is an explicit representative of the initial structural reader, not a substitute reader. In the reachable subspaces one defines

$$\begin{aligned} \mathcal{U}^{\mathrm{str}}_{m,R}(P^{\mathrm{str}}_R \Xi^{\mathrm{str}}_R f) &:= u_m \otimes (J_{-,R} f, 0), \\ \mathcal{U}^{\mathrm{str}}_{m,R}((I - P^{\mathrm{str}}_R)\Xi^{\mathrm{str}}_R f) &:= \eta_m \otimes (0, E_R f). \end{aligned} \tag{96.30}$$

The two sources are orthogonal. The first Gram equality of each source is (96.8); for the second one uses $E_R = J^{\mathrm{sh}}_R (I - P^{\mathrm{str}}_R)\Xi^{\mathrm{str}}_R$. Hence the two rules are well defined, are isometric, and extend to the orthogonal sum of their closures. In that reachable closure they satisfy

$$\mathcal{U}^{\mathrm{str}}_{m,R}\Xi^{\mathrm{str}}_R = \Xi_{m,R}, \qquad \mathcal{U}^{\mathrm{str}}_{m,R} P^{\mathrm{str}}_R = P_{m,R}\mathcal{U}^{\mathrm{str}}_{m,R}, \qquad (\Xi^{\mathrm{str}}_R)^*(I - P^{\mathrm{str}}_R)\Xi^{\mathrm{str}}_R = E_R^* E_R. \tag{96.31}$$

Thus, $B^{\mathrm{str}}_{\mathrm{sp},R}$, the complement of the starting structure, is exactly the complement of the constructed coder; the object has not been changed in passing from (96.2) to (96.28).

The rule

$$V^0_{+,m,R}(J_{+,R} f) := \Xi_{m,R} f \tag{96.32}$$

is well defined and is isometric by the first equality of (96.29); it extends uniquely to an isometry $V_{+,m,R} : H_{+,R} \to \mathcal{X}^{\mathrm{sp}}_{m,R}$. The negative publication is the isometry

$$V_{-,m,R} : H_{-,R} \longrightarrow \mathcal{X}^{\mathrm{sp}}_{m,R}, \qquad V_{-,m,R} y = u_m \otimes (y, 0). \tag{96.33}$$

By construction,

$$\boxed{V_{+,m,R} J_{+,R} = \Xi_{m,R}, \qquad V_{-,m,R} J_{-,R} = P_{m,R}\Xi_{m,R}.} \tag{96.34}$$

The complement preserves exactly the antiespecular state and its column of remark:

$$\boxed{\Xi^*_{m,R} Q_{m,R} \Xi_{m,R} = E_R^* E_R = \mathscr{L}_R^* \mathscr{L}_R = \mathscr{J}_{\mathrm{GW}}\big|_{\mathcal{D}_R}.} \tag{96.35}$$

Proposition 96.5 (Induced connector and source–first coincidence)**.** *The operator*

$$\mathcal{K}_R := V^*_{-,m,R} P_{m,R} V_{+,m,R} : H_{+,R} \longrightarrow H_{-,R} \tag{96.36}$$

is contractive, coincides with $\mathcal{K}_R^{\mathrm{src}}$ throughout $H_{+,R}$, and satisfies

$$\|\mathcal{K}_R\| \le 1, \qquad \mathcal{K}_R J_{+,R} = J_{-,R}. \tag{96.37}$$

Proof. The two charts are isometries and $P_{m,R}$ is an orthogonal projection, hence $\|\mathcal{K}_R\| \le 1$. Over the dense range of $J_{+,R}$,

$$\mathcal{K}_R J_{+,R} f = V^*_{-,m,R} P_{m,R} \Xi_{m,R} f = V^*_{-,m,R} V_{-,m,R} J_{-,R} f = J_{-,R} f.$$

The same identity holds for $\mathcal{K}_R^{\mathrm{src}}$ by (96.12); continuity and density prove that the two operators coincide. □

The Gram difference is thus published in all its coincident realizations:

$$\boxed{\begin{aligned} J^*_{+,R} J_{+,R} - J^*_{-,R} J_{-,R} &= \Xi^*_{m,R} Q_{m,R} \Xi_{m,R} \\ &= E_R^* E_R \\ &= \mathscr{L}_R^* \mathscr{L}_R \\ &= \mathcal{B}^{\mathrm{str}}_{\mathrm{sp},R} = \mathscr{J}_{\mathrm{GW}}. \end{aligned}} \tag{96.38}$$

96.7 Cofinal Naturality and Reversible Weil Criterion

For $R \le R'$, let $i_{R,R'} : \mathcal{D}_R \hookrightarrow \mathcal{D}_{R'}$. The individual columns do not form an isometric system: when a new length $R < \ell_{p,k} \le R'$ appears, both acquire the same positive energy. For $f, g \in \mathcal{D}_R$, separation of the supports gives exactly

$$\big\langle (I - T_{\ell_{p,k}}) f, (I - T_{\ell_{p,k}}) g \big\rangle = 2\langle f, g \rangle = \Big\langle \sqrt{2} j_m f, \sqrt{2} j_m g \Big\rangle. \tag{96.39}$$

Therefore the prime increment of $J^*_{+,R'} J_{+,R'}$ coincides with that of $J^*_{-,R'} J_{-,R'}$ and cancels in the difference, but no false isometry between the columns is asserted. The correct cofinal compatibility is that of the forms:

$$\boxed{\mathscr{J}_{\mathrm{GW},R} = i^*_{R,R'} \mathscr{J}_{\mathrm{GW},R'} i_{R,R'}, \qquad \mathcal{B}^{\mathrm{str}}_{\mathrm{sp},R} = i^*_{R,R'} \mathcal{B}^{\mathrm{str}}_{\mathrm{sp},R'} i_{R,R'}.} \tag{96.40}$$

Since $E_R^* E_R = \mathscr{L}_R^* \mathscr{L}_R = \mathscr{J}_{\mathrm{GW},R}$, the minimal Kolmogorov realizations do admit unique isometries over their reachable ranges, defined by

$$\iota^E_{R,R'}(E_R f) := E_{R'} f, \qquad \iota^L_{R,R'}(\mathscr{L}_R f) := \mathscr{L}_{R'} f \quad (f \in \mathcal{D}_R). \tag{96.41}$$

These two rules are well defined precisely by (96.40); they are not extended by fiat to $J_{\pm,R}$, $\Xi_{m,R}$, or $\mathcal{K}_R$. At fixed window, the nonadic refinement $m \mapsto m + 1$ remains within the structural isometries that transport j_m, $\widehat{\delta}_{a,m}$, u_m, and η_m.

To avoid compressing the last step, we also fix its objects and normalizations. Let

$$\xi(s) := \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

and let $\mathscr{Z}$ be the multiset of its zeros, each zero repeated according to its multiplicity. The functional equation and the reality of the coefficients preserve $\mathscr{Z}$ under $\rho \mapsto 1-\rho$, $\rho \mapsto \overline{\rho}$ and $\rho \mapsto 1-\overline{\rho}$. With our Fourier convention we define

$$\Phi_f(s) := \widehat{f}\Big(i(s-\tfrac{1}{2})\Big), \qquad \widetilde{f}(x) := \overline{f(-x)}. \tag{96.42}$$

The complete explicit formula computed in (96.6), including the gamma and polar terms that remove poles and trivial zeros, has the spectral form

$$\mathscr{I}_{\mathrm{GW}}(f,g) = \sum_{\rho\in\mathscr{Z}} \Phi_f(\rho)\,\overline{\Phi_g(1-\overline{\rho})}. \tag{96.43}$$

The sum counts multiplicities. It converges absolutely: for $f, g \in C_c^\infty(\mathbb{R})$, Paley–Wiener gives decay faster than any power in vertical strips, whereas the number of zeros with $|\operatorname{Im}\rho| \le T$ is $O(T\log T)$.

Proposition 96.6 (Reversible Weil Criterion with Full Quantifiers)**.** *Under the normalization* (96.42) *and counting all nontrivial zeros with their multiplicity,*

$$\Big[\mathscr{I}_{\mathrm{GW}}(f,f) \ge 0 \text{ for every } f \in C_c^\infty(\mathbb{R};\mathbb{C})\Big] \Longleftrightarrow \Big[\operatorname{Re}\rho = \frac{1}{2} \text{ for every } \rho \in \mathscr{Z}\Big]. \tag{96.44}$$

The converse implication is not obtained by interpolating an increasing number of zeros: that operation would provide no uniform control over the tail. The complete classical Weil criterion in Bombieri's formulation is used; its test space and normalization are transferred here without abbreviation.[1]

Proof. We materially write out the change of coordinates identifying both criteria. For $f \in C_c^\infty(\mathbb{R})$, let

$$(\mathcal{M}f)(x) := x^{-1/2} f(\log x), \qquad x > 0. \tag{96.45}$$

The map

$$\mathcal{M} : C_c^\infty(\mathbb{R}) \longrightarrow C_c^\infty(0,\infty)$$

is bijective, with inverse

$$(\mathcal{M}^{-1}g)(u) = e^{u/2} g(e^u). \tag{96.46}$$

If $\widehat{g}_{\mathrm{Mel}}(s) := \int_0^\infty g(x)x^{s-1}\,dx$, the change $x = e^u$ gives, with no residual factor,

$$\begin{aligned} \widehat{\mathcal{M}f}_{\mathrm{Mel}}(s) &= \int_{\mathbb{R}} f(u) e^{(s-1/2)u}\,du \\ &= \widehat{f}\Big(i(s-\tfrac{1}{2})\Big) = \Phi_f(s). \end{aligned} \tag{96.47}$$

Moreover,

$$\widehat{\overline{\mathcal{M}f}}_{\mathrm{Mel}}(1-\rho) = \overline{\widehat{\mathcal{M}f}_{\mathrm{Mel}}(1-\overline{\rho})} = \overline{\Phi_f(1-\overline{\rho})}. \tag{96.48}$$

Therefore the Weil–Bombieri criterion functional is exactly

$$\sum_{\rho\in\mathscr{Z}} \widehat{\mathcal{M}f}_{\mathrm{Mel}}(\rho)\,\widehat{\overline{\mathcal{M}f}}_{\mathrm{Mel}}(1-\rho) = \sum_{\rho\in\mathscr{Z}} \Phi_f(\rho)\overline{\Phi_f(1-\overline{\rho})} = \mathscr{I}_{\mathrm{GW}}(f,f). \tag{96.49}$$

[1] E. Bombieri, *Remarks on Weil's quadratic functional in the theory of prime numbers, I*, Rendiconti Lincei, Matematica e Applicazioni, ser. 9, vol. 11 (2000), pp. 183–233. The criterion established there states that the quadratic functional of the explicit formula is positive semidefinite on all $C_c^\infty(0,\infty)$ if and only if the Riemann Hypothesis holds.

The class of tests is not changed, no multiplicity is lost, and the set of zeros is not truncated, because $\mathcal{M}$ is a bijection. The cited classical criterion applied to (96.49) gives precisely the equivalence (96.44). In the forward direction, if $\operatorname{Re}\rho = 1/2$, the same identity reduces explicitly to

$$\mathcal{I}_{\mathrm{GW}}(f,f) = \sum_{\rho\in\mathcal{Z}} |\Phi_f(\rho)|^2 \geq 0.$$

□

The equality $\bigcup_R \mathcal{D}_R = C_c^\infty(\mathbb{R})$ is exact; each function employed in the argument belongs to some window, and cofinal naturality transports there the positivity already proved.

Theorem 96.7 (Riemann–Weil Resolution from the Spectral–Polar Image)**.** *Having fixed the complete structure $\mathfrak{G}_{\mathrm{sp}}$, the complete Guinand–Weil form is positive on the cofinal separating class. Consequently, every nontrivial zero ρ of the Riemann zeta function satisfies*

$$\boxed{\operatorname{Re}\rho = \frac{1}{2}.}$$

Proof. The prime–gamma–polar expansion identifies the difference between the two columns with $\mathcal{I}_{\mathrm{GW}}$ using a single regulator and removes it by dominated convergence. The antispecular recurrence (96.11) annihilates the output difference before forming an inequality; therefore the isometry Σ_R gives (96.15). Comparing that identity with the calculated expansion produces $E_R^* E_R = \mathcal{I}_{\mathrm{GW}}$. The local identity $A^*A + D^*D = I$ is iterated without erasing the terminal and gives (96.22); the terminal is exactly $q^{2N} E_R^* E_R$ and vanishes only in the limit. Hence $\mathcal{L}_R^* \mathcal{L}_R = \mathcal{I}_{\mathrm{GW}} \succeq 0$. The action (96.28) then constructs the encoder, its two moments, the projection, and the two publications, and (96.38) identifies all realizations of the same square. Naturality transports that identity to the entire cofinal union. Finally, (96.44) concludes the assertion. □

96.8 Subsequent, non-governing local Redheffer control

The following development verifies that local passivity can be obtained from the carry without using the Weil form. It is a noncircularity control and not a premise of theorem 96.7. This truncated transfer is not the complete colligation Σ_R of (96.7).

For $N = 9^m$, be $U_{a,N}$ the odometro unitario

$$U_{a,N}(e_j \otimes f) = e_{j+1} \otimes f \ (j < N-1), \qquad U_{a,N}(e_{N-1} \otimes f) = e_0 \otimes T_a f.$$

With $P_N = |u_N\rangle\langle u_N| \otimes I$, $Q_N = I - P_N$, $A_{a,N} = P_N U_{a,N} P_N$, and $C_{a,N} = Q_N U_{a,N} P_N$,

$$A_{a,N}^* A_{a,N} + C_{a,N}^* C_{a,N} = P_N, \qquad C_{a,N}^* C_{a,N} = \frac{N-1}{N^2}(2I - T_a - T_a^*). \tag{96.50}$$

For $0 < s < 1$, the transfer

$$\mathcal{R}_{a,N}(s) = (sI - A_{a,N})(I - sA_{a,N})^{-1} \tag{96.51}$$

satisfies

$$\begin{aligned} I - \mathcal{R}_{a,N}(s)^* \mathcal{R}_{a,N}(s) = (1-s^2)(I - sA_{a,N}^*)^{-1} \\ \cdot C_{a,N}^* C_{a,N} (I - sA_{a,N})^{-1} \succeq 0. \end{aligned} \tag{96.52}$$

With

$$s_{p,N} = \frac{N p^{-1/2}}{1 + (N-1)p^{-1/2}}, \qquad s_\Gamma(a) = e^{-a/2},$$

the sum over prime clocks and the direct gamma integral have norm at most one. Explicitly,

$$\mathscr{R}_{N,R} = \bigoplus_{\log p \le R} \mathscr{R}_{\log p,N}(s_{p,N}) \ \oplus \int_0^\infty{}^{\oplus} \mathscr{R}_{a,N}(s_\Gamma(a))\, da, \qquad \|\mathscr{R}_{N,R}\| \le 1. \tag{96.53}$$

The finite chart of roles

$$\mathcal{H}_+ = \operatorname{span}\{f_3, f_6, f_9\}, \quad \mathcal{H}_- = \operatorname{span}\{f_4, f_8\}, \quad C_\angle = |f_4\rangle\langle f_3| + |f_8\rangle\langle f_9|$$

satisfies $I - C_\angle^* C_\angle = P_6$. The two-way energy charts fix

$$\begin{aligned} \mathcal{K}_q &= K_2 \otimes (\mathbb{C}^9)^{\otimes q}, \\ \mathsf{B}_{\pm,q} : \mathcal{M}_{\pm,q,R} &\longrightarrow \mathcal{K}_q \widehat{\otimes} \mathcal{H}_\pm \widehat{\otimes} \mathcal{H}_{\mathfrak{C}_R}, \\ \mathsf{B}_{\pm,q}^* \mathsf{B}_{\pm,q} = \mathsf{G}_{\pm,q}, &\qquad \mathsf{B}_{\pm,q}^\sharp = \mathsf{G}_{\pm,q}^{-1} \mathsf{B}_{\pm,q}^*. \end{aligned} \tag{96.54}$$

The transfer

$$\mathscr{C}_{q,R}^{\mathrm{src}} = \mathsf{B}_{-,q}^\sharp (I \otimes C_\angle \otimes \mathscr{R}_{N,R}) \mathsf{B}_{+,q} \tag{96.55}$$

has defect

$$\Delta_{q,R}^{\mathrm{src}} = I \otimes \left[P_6 \otimes I + (P_3 + P_9) \otimes (I - \mathscr{R}_{N,R}^* \mathscr{R}_{N,R}) \right] \succeq 0, \tag{96.56}$$

$$I - (\mathscr{C}_{q,R}^{\mathrm{src}})^\sharp \mathscr{C}_{q,R}^{\mathrm{src}} = \mathsf{B}_{+,q}^\sharp \Delta_{q,R}^{\mathrm{src}} \mathsf{B}_{+,q} \succeq 0. \tag{96.57}$$

This control preserves route, clock, role, carry, and channels before forming a difference. It is not identified with $\mathcal{K}_R$ and does not replace the explicit common publication of (96.34); its local scope therefore does not reopen the cofinal conclusion.

96.9 Falsifiers of an ablated shadow

Falsification criterion 96.8 (Substitutions that change the starting object)**.** The preceding argument does not transfer to a shadow that eliminates one of the thirteen coordinates of (96.2), constructs J_+ and J_- from different antecedents, replaces an orthogonal projection with a non-self-adjoint idempotent, omits a channel, uses incompatible regulators, chooses P_R after the sign, erases the terminal object before the limit, or reduces the class to a noncofinal family.

Proof. Each operation respectively violates the domain of $\mathfrak{G}_{\mathrm{sp}}$, the proper definition of (96.32), the orthogonal square, the functional identity, the causal direction, finite telescoping, or the reversibility criterion. The falsifier affects the replaced shadow; it does not dislodge the complete structural image used in theorem 96.7. □

Provenance and evidentiary force. The spectral–polar restriction, the source–first colligation, its complement and the mirror sheet are `PREEXISTING_AUTHORIAL_ARCHITECTURE`. The coinductive vanishing of Ω, the finite identity with terminal and the action encoder (96.28) are a `NEW_FORMALIZATION` of that complete structure. The Guinand–Weil identification and cofinal closure are a `RECOVERED_RESULT`. The equations (96.7)– (96.38) exhibit the object, its domains, its action, the terminal and the limit without introducing an abstract root of the form; their force is *proved with an explicit starting structure*.

96.10 Cofinal Closure of the Riemann–Weil Criterion

The genealogical realization produces a cofinal, natural family of positive forms whose publication coincides with the complete Guinand–Weil functional. Extinction of the terminal and compatibility with the inclusions permit passage to the limit without introducing a list of zeros or the sought sign. By Weil's reversible criterion, every nontrivial zero ρ of the Riemann zeta function satisfies

$$\operatorname{Re}\rho = \frac{1}{2}.$$

The regulated finite-window computations are subsequent controls; they do not form part of the premises of the cofinal closure.

Cofinal Factorization of the Spectral Conclusion. The probative chain gathers, in the already established causal order, the finite genealogical form, its positive complement, naturality under inclusions, and compatible extinction of the terminal term. Each window preserves the prime clocks, the Archimedean channel, the orientation of the two sheets, and the boundary register. Cofinality jointly transports those data to the limit, and the Guinand–Weil identity identifies the resulting form with its classical spectral publication.

Consequently, positivity does not proceed from a subsequent selection of zeros: it descends from the positive operator constructed prior to evaluation. The functional involution preserves the mirror sheet and fixes the line $\operatorname{Re}\rho = 1/2$; cofinal compatibility preserves that fixation in the passage from regulated windows to the complete form. The finite calculations thus perform their proper function of subsequent verification, whereas mathematical closure resides in the preceding positive and natural composition.

Chapter 97

Cayley–Pontryagin characterization of the critical zeros and coinductive construction of the Weil spectral field

Once the critical line has been located by positive factorization, the Cayley transform expresses the heights as phases on a first circle. The Pontryagin intertwiner then transports them to the nonadic circle without identifying phase return with state return. The double-circle theorem and orbit quantization are articulated with the effective reconstruction of the spectral field. The exposition explicitly preserves the hypotheses, Galerkin residuals, memory cocycle, twisted transport and the intertwiner that identifies both realizations without confusing the turn index with the spectral height.

97.1 Cancellation of the completed function and spectral closure on the critical line

The conventional image of the Riemann zeros usually begins too late. The zeta function is presented, analytically continued, and one asks at which points it vanishes. The reader sees a complex function and a collection of mysterious points distributed along a line.

The HMT reading changes the order of the question. It does not begin by asking where the final publication vanishes. It begins by reconstructing the state that publication has contracted: the prime-power clocks, the gamma channel, the polar channel, the two sheets, their orientation, the carry, the memory, the boundary, the terminal, and the nonadic connection that transports them jointly.

Only after this does the completed function appear, and one asks about its zeros.

The difference is radical: a zero ceases to be an absence and becomes an event of closure.

97.1.1 Scalar annihilation and persistence of the enriched state

Si

$$\xi(\rho) = 0,$$

the zero is true. It is not approximate, fictitious, or decorative. The completed function vanishes exactly at ρ.

But what vanishes is a scalar publication. The point ρ does not disappear, its multiplicity does not disappear, the prime structure that participates in the explicit formula does not disappear, and the enriched state that precedes publication does not disappear.

This is an elementary yet decisive distinction:

$$\text{published value equal to zero} \quad \Rightarrow \quad \text{nonexistent generating state.}$$

A useful representation is a dark fringe produced by interference. The darkness does not mean that no wave has arrived. It means exactly the opposite: contributions have arrived whose phase relation is so precise that their visible sum cancels.

The non-trivial zeros can be read in this way: as nodes of global arithmetic interference.

They are not nodes of a single sequence of primes, nor do they individually belong to a prime. In the complete publication the following intervene simultaneously:

- the prime-power clocks,

$$\ell_{p,k} = k \log p;$$

- the Archimedean gamma channel;
- the polar channel, which preserves normalization and removes the poles and trivial zeros;

- the two mirror sheets;
- the boundary that the visible projection does not show.

A zero is an exact cancellation of that joint publication. Its existence reveals a coordination, not a vacuum.

Therefore, the zeros constitute **spectral nodes of arithmetic cancellation**: the vanishing of ξ expresses an exact global coordination among the contributions of the joint publication.

97.1.2 Fixing of the critical line by the functional involution $s \mapsto 1 - \bar{s}$

The completed function preserves the symmetries

$$\rho \longmapsto 1 - \rho, \qquad \rho \longmapsto \overline{\rho}.$$

Combining them yields the involution

$$\rho \longmapsto \rho^{\sharp} := 1 - \overline{\rho}.$$

Si

$$\rho = \beta + i\gamma,$$

then

$$\rho^{\sharp} = 1 - \beta + i\gamma.$$

The reflection preserves the height γ, but exchanges β and $1 - \beta$. Its fixed-point set is:

$$\rho = \rho^{\sharp} \quad \Longleftrightarrow \quad \beta = 1 - \beta \quad \Longleftrightarrow \quad \beta = \frac{1}{2}.$$

Por tanto,

$$\mathrm{Fix}(\sharp) = \left\{ \frac{1}{2} + i\gamma : \gamma \in \mathbb{R} \right\}.$$

The critical line is not merely "one half" by geometric preference. It is the seam where the two functional sheets cease to appear as distinct positions and become a single self-dual position.

Outside the line, a zero appears associated with an orbit of up to four points:

$$\rho, \qquad 1 - \rho, \qquad \overline{\rho}, \qquad 1 - \overline{\rho}.$$

On the critical line, the conjugate functional reflection no longer sends the zero to another lateral position:

$$1 - \overline{\rho} = \rho.$$

Zero closes with itself.

This is the first precise sense of the closure circle: it is not merely that two halves sum to one, but that the mirror orbit encounters a fixed point.

However, this still does not prove the Riemann Hypothesis. The equality

$$\frac{1}{2} = 1 - \frac{1}{2}$$

identifies the axis; it does not by itself compel that all zeros lie on it.

The obligation follows from positivity.

97.1.3 Quadraticity of the cross product under closure of the functional involution

The full spectral form can be written as

$$\mathcal{I}_{\mathrm{GW}}(f,g) = \sum_{\rho \in \mathcal{Z}} \Phi_f(\rho)\,\overline{\Phi_g(1-\overline{\rho})},$$

where $\mathcal{Z}$ is the multiset of nontrivial zeros, preserving their multiplicities.

This formula determines how the zeros are paired: the evaluation situated at ρ is combined with the evaluation situated at its reflection $\rho^\sharp = 1 - \overline{\rho}$.

If zero is outside the line, the expression pairs two different points:

$$\Phi_f(\rho)\,\overline{\Phi_f(\rho^\sharp)}.$$

That is a cross product. It is not necessarily a square norm and, on a sufficiently rich class of tests, it can produce negative-sign directions.

If zero is on the critical line, then

$$\rho^\sharp = \rho$$

and the same term becomes

$$|\Phi_f(\rho)|^2.$$

The full form becomes:

$$\mathcal{I}_{\mathrm{GW}}(f,f) = \sum_{\rho \in \mathcal{Z}} |\Phi_f(\rho)|^2.$$

Now the mirror does not pair two displaced positions. Each point is paired with itself and the publication becomes a sum of squares.

This is the conceptual reading of the proof: HMT constructs the Weil form upstream as a boundary energy,

$$\mathcal{I}_{\mathrm{GW}} = E_R^* E_R = \mathcal{L}_R^* \mathcal{L}_R \succeq 0.$$

It is not declared positive because the zeros are on the line. It is constructed as a square before using the zeros.

Afterward, the reversible Weil criterion says that this positivity over the entire separating class of test functions is equivalent to all nontrivial zeros satisfying

$$\operatorname{Re}\rho = \frac{1}{2}.$$

Thus positivity does not dynamically "attract" the zeros toward the center. The zeros do not move from an initial position. What occurs is stronger: any configuration off the fixed axis would produce a detectable negative direction, whereas the HMT complement is concretely a squared norm and cannot acquire one.

The critical line is the only place compatible with both publications.

97.1.4 Spectral characterization of the zeros

Under this reading, a non-trivial zero can be described simultaneously in four compatible ways.

It is a **true analytic zero**, because

$$\xi(\rho) = 0.$$

It is a **fixed point of the mirror**, because

$$\rho = 1 - \overline{\rho}.$$

It is a **positive spectral atom**, because in the Weil form it appears as

$$|\Phi_f(\rho)|^2.$$

And it is a **global node of the prime clocks and the Archimedean boundary**, because it does not belong to an isolated channel: it arises from the coordinated publication of the prime, gamma, and polar terms.

The zero is null as amplitude and positive as spectral localization. There is no contradiction. They are two different readers.

The function is zero at ρ, but ρ continues to appear, with its full multiplicity, within the positive spectral measure. The zero has not been erased because it is zero. On the contrary, it becomes one of the locations supporting the positive sum.

This is probably the most important interpretive change:

> The zero is not the place where the structure disappears. It is the place where the complete structure reaches a visible cancellation and leaves a spectral mark.

97.1.5 Identification of the height of a nontrivial 0 with a real spectral frequency

The normalization employed in the Weil criterion is

$$\Phi_f(s) = \widehat{f}\left(i\left(s - \frac{1}{2}\right)\right).$$

Si

$$\rho = \frac{1}{2} + i\gamma,$$

then

$$i\left(\rho - \frac{1}{2}\right) = -\gamma$$

and, therefore,

$$\Phi_f(\rho) = \widehat{f}(-\gamma).$$

The height γ literally becomes a real Fourier frequency.

Thus, the critical line can be read as a line of frequencies: the real coordinate $\frac{1}{2}$ fixes the specular equilibrium and the coordinate γ records the frequency of the node.

This explains why the spectral language is appropriate. The zeros are not merely complex coordinates; they are frequencies at which the complete arithmetic publication manifests its nodal condition.

These heights admit interpretation as the nodal frequencies of a global system: points at which an observable amplitude vanishes while the structure producing that cancellation remains active.

This statement must, however, be guarded against overstatement. It does not follow automatically that every γ is an energy eigenvalue of a specific physical Hamiltonian. That would require constructing the Hamiltonian, its domain, and the spectral correspondence. What has been proved is the Fourier–Mellin reading, the complete positive form, and the localization of the zeros.

They are, rigorously, spectral frequencies. Without another intertwiner, they do not become physical energy levels.

97.1.6 Conformal circular parametrization of the critical line

Your intuition about the circle is not merely metaphorical. There is an exact Möbius transformation:

$$\tau(s) = \frac{s-1}{s}.$$

Se cumple:

$$\operatorname{Re} s = \frac{1}{2} \quad \Longleftrightarrow \quad |\tau(s)| = 1.$$

Therefore, the critical line compactified by its point at infinity is transformed into the unit circle.

Si

$$s = \frac{1}{2} + it,$$

then

$$\tau(s) \in S^1.$$

Al escribir

$$\tau = e^{i\theta},$$

the inverse is

$$s = \frac{1}{1-\tau} = \frac{1}{2} + \frac{i}{2}\cot\frac{\theta}{2}.$$

The height t of the line becomes an angular position on the circle. As t traverses the entire line, the chart traverses the compactified circumference. The two endpoints $t \to -\infty$ and $t \to +\infty$ meet at the point $\tau = 1$, although they approach it from opposite orientations.

Also the mirror acquires a circular shape:

$$\tau(1-\overline{s}) = \frac{1}{\overline{\tau(s)}}.$$

The spectral involution becomes reflection across the unit circle, whose fixed set is precisely the circle.

So the expression "circle of closure" possesses an exact content:

- the compactified critical line is a circle;
- the functional mirror becomes reflection with respect to that circle;
- the nontrivial zeros are marked points on that boundary;
- each point is fixed by the reflection that previously exchanged the two sheets.

The circle does not generate the zeros nor determine their heights. The entire line is transformed into the circle, whereas the equation

$$\xi(\rho) = 0$$

selects its spectral points.

Nor does it mean that the zeros are equally spaced or that there are nine per turn. The circle is the compactified geometry of the critical locus; the distribution of the zeros upon it preserves its own arithmetic.

97.1.7 Nonadic Circle and Phase Holonomy

There is a second circle, prior to the analytic one: the cyclic base of the nine phases.

The nonadic connection does not consist of nine filters or nine independent channels. It is a single holonomy:

$$\Gamma_9 = \mathrm{Hol}_{C_9}(\nabla_{\mathrm{TPK}}).$$

Its return law is

$$g(K+9) = g(K),$$

but simultaneously

$$q_{\mathrm{mem}}(\Gamma_9 \widehat{x}) = q_{\mathrm{mem}}(\widehat{x}) + 1$$

and, in general,

$$\Gamma_9 \widehat{x} \neq \widehat{x}.$$

The phase returns; the state is not reset.

Viewed from the base, one has traversed a circle and returned to the same phase. Viewed from the enriched state, one turn of memory has accumulated. The most faithful image for the reader is:

- a circle in the projection;
- a helix in the enriched state.

From above, the helix appears to close. From within, each turn occupies a distinct height.

The literal part is not the image of the helix, but rather the mathematical identities: phase return, memory increment, and general change of state.

This structure was verified at 396 refinement levels per channel. The certificate records 43 complete nonadic turns and 387 comparisons $K, K+9$ per channel: the phase agrees and the state changes in every case. This is not a poetic way of referring to periodicity; it is an executed separation between visible-phase return and genealogical continuity.

97.1.8 The nonadic mirror and the Riemann mirror

Your intuition is correct: in both cases a mirror appears. But it is useful to state exactly how they are related.

The internal nonadic mirror exchanges the two lateral components, associated with the readings π/e, while preserving the axis φ and the corresponding aggregate. The Riemann involution is

$$s \longmapsto 1 - \overline{s}.$$

They are not literally the same map, because they have different domains and codomains. Equating them without the intertwiner would conflate the internal architecture with its analytic publication.

The correct statement is more interesting: the spectral–polar intertwiner publishes the two-way TPK architecture as the functional involution of the completed function. The nonadic mirror comes first; the Riemann mirror is its realization in the analytic codomain.

In both cases there exists:

- a pair of sheets;
- an axis that remains fixed;
- a transformation that exchanges the lateral positions;
- a seam at which the two publications coincide;
- a memory that does not disappear when the identification is performed.

The line $\Re s = \frac{1}{2}$ is the visible seam of that mirror in the complex plane.

97.1.9 Residual closure and nonadic phase

The expression "nines are false zeros" contains a very precise intuition if we distinguish the integer, the residue class, and the complete state.

For $n > 0$, the positive chart utilizes

$$\rho_9(n) = 1 + ((n-1) \bmod 9).$$

Equivalently,

$$\rho_9(n) = \begin{cases} n \bmod 9, & n \not\equiv 0 \pmod 9, \\ 9, & n \equiv 0 \pmod 9. \end{cases}$$

Therefore, when a number belongs to the null class modulo nine, the positive chart does not publish 0: it publishes 9.

At the same time it is conserved

$$\kappa_9(n) = \frac{n - \rho_9(n)}{9}$$

and the reconstruction

$$n = 9\kappa_9(n) + \rho_9(n).$$

Nine is thus the positive representative of the zero residue class. It is zero in the quotient, but is neither the integer zero nor an empty state.

The nine says: "the turn has been completed." The quotient says how many crowns or turns have accumulated. If everything is replaced by the residue 0, the congruence is preserved, but the difference among absence, closure, and memory is lost.

That is the rigorous meaning of "false zero":

> Nine is a zero for the residual shadow, but it would be false to interpret it as annihilation of the state. It is a zero published with closure memory.

For example, 27, 72, and 729 have zero residual shadow and positive digital root nine. Nevertheless, they are neither the same number nor of the same genealogy. One may arise from a reversal, another from a different evaluation, and another from a power. The residue does not reconstruct the route.

This connects deeply with the Riemann zeros, but does not identify them.

A zero of ξ is a genuine analytic zero. It is not the number nine. There is no claim here that every nontrivial zero corresponds individually to a particular phase nine.

The structural relation is this:

$$\text{a publication may vanish} \quad \text{without its genealogy vanishing.}$$

Nine is a false arithmetic absence. The zero of zeta is a false ontological absence. Both are true zeros within their reader, but neither means that there is no structure behind it.

97.1.10 Typed separation of the zeros of ζ, ξ and the spectral determinant

So that the force of the narration does not erase the mathematical types, it is useful to separate three zeros.

97.1.10.1 0 *analytically, as vanishing of the function in the spectral domain*

$$\xi(\rho) = 0.$$

It is a spectral point of the completed function. It preserves position, height, multiplicity, and symmetries.

97.1.10.2 0 *terminal as universal object of the closing morphism*

In the HMT telescopy appears

$$(A^N E_R)^*(A^N E_R) = q^{2N} \mathcal{B}^{\mathrm{str}}_{\mathrm{sp},R} \longrightarrow 0, \qquad q = \frac{5}{9}.$$

It is the extinction of a remainder after its entire contribution has been transferred to a sum of positive squares.

97.1.10.3 Positive representative of the null residual class: $n \equiv 0 \pmod 9$ *and* $\rho_9(n) = 9$

$$n \equiv 0 \pmod 9, \qquad \rho_9(n) = 9.$$

It is the null modular class published as a positive representative, accompanied by its quotient and memory.

They are not the same object. But they share a grammar:

$$\text{visible vanishing} + \text{structural conservation.}$$

97.1.11 Exhaustion of the terminal object under cofinal filtration

The proof preserves at each depth a finite identity:

$$\mathcal{B}^{\mathrm{str}}_{\mathrm{sp},R} = \sum_{n=0}^{N-1} (DA^n E_R)^*(DA^n E_R) + (A^N E_R)^*(A^N E_R).$$

The last term is the terminal object. It is not removed to obtain a more attractive positive sum. It remains present at every finite stage.

Only afterwards is it proved:

$$(A^N E_R)^*(A^N E_R) = q^{2N} \mathcal{B}^{\mathrm{str}}_{\mathrm{sp},R} \longrightarrow 0.$$

This permits another reading of zero: the terminal reaches zero because its content has been expressed and accounted for level by level. Every part lost through visible contraction has been recorded in a positive memory coordinate.

The rest is extinguished, but it has not been erased.

It is like emptying a reservoir by transferring its contents exactly into a sequence of numbered receptacles. In the end, the initial reservoir is empty, but nothing has disappeared: a complete ledger of the transfer exists.

Here zero means **completed transfer**.

97.1.12 Genealogical Information of the Continuum Encoded by Critical Zeros

The critical line

$$\frac{1}{2} + i\mathbb{R}$$

is a translated copy of the real line. From HMT, that line is not the initial object: it is an Archimedean publication of the genealogical continuum.

The zeros form a discrete set within that continuous publication. They do not perforate the line or interrupt the continuum. They constitute a spectral divisor: isolated points, each with multiplicity, on a continuous shadow.

This coexistence of the continuous and the discrete is essential. The line provides the continuous codomain; the genealogy determines the discrete marks that appear on it.

The image I would propose to the reader is that of a continuous membrane traversed by spectral nodes. The nodes do not rupture the membrane. They make its internal structure visible.

In holographic terms, the zero is a boundary mark produced by a global organization of the interior. It does not belong to a single prime, just as a pixel in a holographic image does not belong to a single point of the represented object. Each zero responds to the complete set of prime–gamma–polar relations.

This does not mean that each zero by itself contains the entirety of arithmetic in recoverable form. It means that its position cannot be attributed to an isolated local input: it is a global response of the system.

97.1.13 Correspondence between spectral quantization, real frequencies and physical realization of the operator

The zeros are interpreted as nodes of a global arithmetic transfer:

- the coordinate γ acts as a frequency;
- the critical line is the specular equilibrium surface;
- the Weil form is the positive energy of the complement;
- the joint nonadic connection preserves the memory of each turn;
- the terminal measures what has not yet been expressed;
- the visible zero indicates a perfect cancellation of the published amplitude.

They are not identified with particles, holes, vacuum states, or masses, nor with material energy levels without first constructing an additional physical operator.

The valid physical interpretation is structural: node, frequency, mirror, transfer, memory, and closure. A concrete physical ontology would require its own transducer.

97.1.14 Localization, Multiplicity, and Symmetry Corollaries of the Spectral Characterization

Before the resolution, the question seemed to be:

> Why do some points defined by a complex function seem to align mysteriously on a line?

After the resolution, the question can be formulated otherwise:

> How does the spectral–polar structure publish a boundary energy that can remain positive only when each zero is identified with its reflection?

The answer is:

- the positive and negative columns arise from the same state;
- the boundary state E_R preserves what the projection does not display;
- the difference of the two Grams is

$$E_R^* E_R;$$

- telescoping preserves the terminal until its extinction is proved;
- the complete Weil form coincides with that square;
- the reversible criterion requires each zero to be a fixed point of the mirror.

The line is no longer a visual coincidence. It is the fixed set of the involution compatible with the constructed positivity.

97.1.15 Hypothesis and exact codomain of the spectral localization theorem

This reading allows us to assert:

$$\{\text{nontrivial zeros}\} \subset \left\{\frac{1}{2} + i\gamma : \gamma \in \mathbb{R}\right\}.$$

And, via the Möbius chart,

$$\left\{\frac{1}{2} + i\gamma\right\} \longrightarrow S^1.$$

Therefore, all non-trivial zeros are marked points of the critical circle.

No bijection has been constructed here individually:

$$\{\text{nontrivial zeros}\} \longleftrightarrow \{\text{concrete nonadic turns}\}.$$

It is not asserted that the first zero is the first turn, that there are nine zeros per cycle, or that their heights are generated by repeating a phase. The resolution proves localization, positivity, mirror closure, and membership in the circle. An individual quantization of the heights would be an additional development and would require its own map.

Nor should the internal mirror π/e, the Riemann functional involution, and a possible physical symmetry be conflated. Their relationship is one of typed publication, not nominal identity.

97.1.16 Spectral closure theorem via positive factorization of Guinand–Weil

The operatorial interpretation is summarized by the following properties:

> The nontrivial Riemann zeros are exact vanishings of the analytic amplitude that preserve the genealogy of their construction. Each zero determines a global frequency, a spectral marker, and a self-dual fixed point of the functional involution. Compactification of the critical line produces the common fixed circle of the two sheets. The nonadic connection distinguishes phase return from state reset: the circular closure preserves memory advance. The nonadic class $9 \equiv 0 \pmod 9$ represents the visible closure accompanied by the internal information of the enriched state.

Spectral annihilation thus constitutes a concentrated form of operatorial closure and preserves the underlying genealogical object.

This physico-mathematical interpretation accompanies the proved results without replacing them and does not assert an individual correspondence between zeros and nonadic turns so long as the corresponding intertwiner has not been constructed.

97.2 Weil form, translation, and quantization of spectral heights

The correct object is not a bare turn, but an enriched helicoidal orbit:

$$\boxed{\text{nontrivial zero} \longleftrightarrow \text{weighted spectral atom} \longleftrightarrow \text{nonadic helical character.}}$$

This formalization acts after the complete continuum and does not redefine or separate its five characteristics.

97.2.1 Construction of the Hilbert space via the Weil form

Se toma

$$\mathcal{D} = C_c^\infty(\mathbb{R}), \qquad \langle f, g\rangle_W := \mathcal{I}_{\mathrm{GW}}(f, g).$$

In the HMT resolution, this form does not arise from the list of zeros. It is first constructed from the prime–power, gamma, and polar channels, and is factored by nonadic loss:

$$\mathcal{I}_{\mathrm{GW}} = E^* E = \mathscr{L}^* \mathscr{L} \succeq 0.$$

Se define

$$\mathcal{N}_W = \{f \in \mathcal{D} : \mathcal{I}_{\mathrm{GW}}(f, f) = 0\},$$

y completamos:

$$\mathcal{H}_W = \overline{\mathcal{D}/\mathcal{N}_W}.$$

This is the positive Weil space constructed from the arithmetic side. The zeros have not yet been used to define it.

At the same time, telescopy provides the nonadic space of loss.

$$\mathcal{K}_{\mathrm{loss}} = \overline{\mathrm{Ran}\,\mathscr{L}},$$

and the equality of norms defines a canonical unit

$$W_{\mathrm{loss}} : \mathcal{H}_W \longrightarrow \mathcal{K}_{\mathrm{loss}}, \qquad W_{\mathrm{loss}}[f] = \mathscr{L} f.$$

We are not afterward placing an artificial copy of the Weil form inside the TPK: the loss space of the proof itself is a realization of its positive Hilbert space.

97.2.2 Translation Operator Independent of the Previous Localization of Zeros

In the global domain $\mathcal{D}$, not in a fixed window, one considers

$$(T_t f)(x) = f(x - t).$$

Invariance can be checked before diagonalizing the formula at the zeros. With the convention

$$\widehat{T_t f}(z) = e^{-izt}\,\widehat{f}(z),$$

se tiene

$$h_{T_t f, T_t g}(z) = h_{f,g}(z), \qquad \check{h}_{T_t f, T_t g}(a) = \check{h}_{f,g}(a).$$

Therefore, all the prime, gamma, and polar terms remain invariant:

$$\mathcal{I}_{\mathrm{GW}}(T_t f, T_t g) = \mathcal{I}_{\mathrm{GW}}(f, g).$$

The translation preserves $\mathcal{N}_W$ and descends to a unitary group:

$$U_t[f] = [T_t f].$$

After the already demonstrated critical localization, its strong continuity follows from

$$\|U_t[f] - [f]\|_W^2 = \sum_{\gamma} m_\gamma \left|e^{it\gamma} - 1\right|^2 \left|\widehat{f}(-\gamma)\right|^2 \longrightarrow 0.$$

Stone's theorem then produces a self-adjoint operator H_W:

$$U_t = e^{itH_W}.$$

This operator has been defined through translations and the prime–gamma–polar form. It has not been defined by first writing a diagonal matrix containing the zeros.

97.2.3 Spectral Identification of the Heights of the Non-Trivial Zeros

As the Riemann Hypothesis is already proven, every non-trivial zero has the form

$$\rho_\gamma = \frac{1}{2} + i\gamma.$$

If Γ_ξ is the set of distinct heights and m_γ its multiplicity, the weighted counting measure is introduced

$$\mu_\xi = \sum_{\gamma \in \Gamma_\xi} m_\gamma \, \delta_\gamma.$$

The Weil spectral formula becomes

$$\mathcal{I}_{\mathrm{GW}}(f, g) = \sum_{\gamma \in \Gamma_\xi} m_\gamma \, \widehat{f}(-\gamma)\overline{\widehat{g}(-\gamma)}.$$

Por tanto,

$$\mathcal{A}[f](\gamma) = \widehat{f}(-\gamma)$$

define an isometry

$$\mathcal{A} : \mathcal{H}_W \longrightarrow L^2(\Gamma_\xi, \mu_\xi).$$

Moreover, not remains a part spectral escondida outside of its rank. The rank of $\mathcal{A}$ is closed and is preserved by all the multiplicadores

$$M_t a(\gamma) = e^{it\gamma} a(\gamma).$$

The projection onto a single atom is obtained via the spectral average.

$$P_\gamma = \operatorname*{s\text{-}lim}_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} e^{-it\gamma} M_t \, dt.$$

The separation of evaluation functionals on the transforms of test functions guarantees that, for every γ, there exists some f with

$$\widehat{f}(-\gamma) \neq 0.$$

Therefore, $P_\gamma \operatorname{Ran} \mathcal{A} \neq 0$. Since the range is reducing, it contains the corresponding atomic axis. All atomic axes belong to the range and densely span $L^2(\mu_\xi)$. Consequently,

$$\boxed{\mathcal{H}_W \simeq L^2(\Gamma_\xi, \mu_\xi)}$$

y

$$\boxed{\mathcal{A} H_W \mathcal{A}^{-1} = M_\gamma.}$$

Thus,

$$\boxed{\sigma(H_W) = \Gamma_\xi,}$$

with no additional heights, no omitted heights, and no continuous component.

This is already an individual spectral quantization: the heights of the zeros are exactly the point spectrum of an operator defined from the positive arithmetic form.

Multiplicity demands precision. The scalar form records

$$\mu_\xi(\{\gamma\}) = m_\gamma.$$

That is, it preserves exactly the algebraic weight of the zero. It does not automatically produce m_γ distinguishable vectors: if a zero were multiple, evaluation at that same point would be repeated with weight m_γ. The canonical correspondence is between the zero divisor and the weighted spectral atoms. Distinguishing the copies internally would require a positive form on jets or a proof of simplicity.

97.2.4 Conformal transformation of the critical line into the spectral circle with height conservation

Normalization is used.

$$\tau(s) = \frac{s-1}{s}.$$

Then

$$\Re s = \frac{1}{2} \quad \Longleftrightarrow \quad |\tau(s)| = 1.$$

For

$$\rho_\gamma = \frac{1}{2} + i\gamma$$

se obtiene

$$\tau_\gamma = \tau(\rho_\gamma) = \frac{-\frac{1}{2} + i\gamma}{\frac{1}{2} + i\gamma} = \frac{\gamma + \frac{i}{2}}{\gamma - \frac{i}{2}} \in S^1.$$

This is exactly the Cayley transform of the operator H_W:

$$C_W = \left(H_W + \frac{i}{2} I\right)\left(H_W - \frac{i}{2} I\right)^{-1}.$$

On the atom γ,

$$C_W P_\gamma = \tau_\gamma P_\gamma.$$

The correspondence does not lose information. Its inverse is

$$\rho_\gamma = \frac{1}{1-\tau_\gamma}.$$

Si

$$\tau_\gamma = e^{i\theta_\gamma},$$

then

$$\boxed{\gamma = \frac{1}{2}\cot\frac{\theta_\gamma}{2}.}$$

Therefore, the full height is contained in the angle. Compactification of the line preserves the location of each zero.

Moreover,

$$\tau_{-\gamma} = \overline{\tau_\gamma} = \tau_\gamma^{-1}.$$

The conjugate zeros correspond to opposite orientations of the same circle. This is the mirror converted into an exact angular identity.

The correspondence demonstrated is

$$\boxed{\sum_\rho m_\rho[\rho] \longleftrightarrow \sum_\gamma m_\gamma[\tau_\gamma, P_\gamma].}$$

It does not depend on previously ordering the zeros by height.

97.2.5 Monodromia of each 0 over the circulo spectral

We now transport the circular operator to the nonadic space:

$$C_{\text{loss}} = W_{\text{loss}}\, C_W\, W_{\text{loss}}^{-1}.$$

In HMT telescopia,

$$q = \frac{5}{9}$$

is the contraction corresponding to a full turn. The spectral realization of that turn is defined:

$$\mathcal{M}_{\text{spec}} = q\, C_{\text{loss}}.$$

If v_γ belongs to the spectral atom P_γ, then

$$\boxed{\mathcal{M}_{\text{spec}}^n v_\gamma = q^n \tau_\gamma^n v_\gamma.}$$

This is the helical orbit of zero.

Its radius is

$$\|\mathcal{M}_{\text{spec}}^n v_\gamma\| = q^n \|v_\gamma\|,$$

while its accumulated angle is

$$n\theta_\gamma.$$

The radial part preserves terminal extinction; the angular part preserves the individual identity of the zero. Both satisfy the telescoping relation:

$$\|v_\gamma\|^2 = (1-q^2)\sum_{k=0}^{N-1} q^{2k}\|v_\gamma\|^2 + q^{2N}\|v_\gamma\|^2.$$

El terminal

$$q^{2N}\|v_\gamma\|^2 \longrightarrow 0$$

does not erase the height: the height remains in the unitary character τ_γ^n.

This is the correct reading of the turn:

Dato	What is recorded
n	how many complete revolutions have elapsed
$q = 5/9$	how much terminal remains after each turn
τ_γ	what individual zero carries the orbit
m_γ	the algebraic weight of that zero
memory/carry	why returning to a visible phase does not restore the same state

The first zero must therefore not be assigned to the first turn. Each zero traverses the entire ladder of turns with its own character.

A suitable pedagogical image would be a rotating disk: the number of revolutions does not identify the note being sounded. Revolution n states how much time has elapsed; the frequency identifies the mode. Here n is the counter, q is the attenuation, and τ_γ is the spectral signature.

97.2.6 Spectral field localization via the nonadic entwined prolongation

The previous construction actually lives in

$$\overline{\mathrm{Ran}\,\mathscr{L}},$$

the loss space produced by nonadic telescoping. It is not an external analogy added to the proof of Riemann.

But it must be distinguished from the native combinatorial holonomy. The TPK contains also

$$M_9 = qV_9, \qquad V_9^* V_9 = I.$$

The defect identity determines q, but it does not determine the unitary part V_9. The same balance

$$M_9^* M_9 = q^2 I$$

is compatible with many different unitary phases. Telescoping alone cannot determine which one of them contains the heights.

The identity between the newly constructed spectral orbits and the orbits of the APP–TRIT–TPK combinatorial ledger is established by the intertwiner constructed on enriched cylindrical states in the Cayley–Pontryagin theorem of this same chapter:

$$J : \mathcal{K}_{\mathrm{loss}} \longrightarrow \mathcal{K}_{\mathrm{TPK}}^{\mathrm{unit}}$$

tal que

$$\boxed{JC_{\mathrm{loss}} = V_9 J.}$$

Moreover, J must preserve cylinder, orientation, carry, route, memory, boundary, and spectral multiplicity.

The preceding equality is neither postulated nor received from the classical realization: it follows from the integer memory cocycle, cylindrical transport twisted by C_{Weil}, the projectivity of the weights, and the isometric embedding J_k constructed below. Iterating it yields

$$(qV_9)^n J v_\gamma = q^n \tau_\gamma^n J v_\gamma,$$

and each zero is then identified with the unitary character of a native TPK orbit. The identity preserves cylinder, orientation, carry, route, memory, boundary, and spectral multiplicity, so the correspondence does not depend on an external zero–turn enumeration.

97.2.7 Spectral algorithm for zero-free computation

The result is not limited to representing an already known list. The operator H_W has been defined from translations and from the prime–gamma–polar form. Its resolvent can be written as

$$(H_W - iI)^{-1} = i \int_0^\infty e^{-t} U_{-t}\, dt.$$

Therefore, for test functions f_j, f_k,

$$\big\langle (H_W - iI)^{-1} [f_j], [f_k] \big\rangle_W = i \int_0^\infty e^{-t} \mathcal{J}_{\mathrm{GW}}(T_{-t} f_j, f_k)\, dt.$$

The right-hand member is computed from the prime, gamma, and polar channels. It need not feed a table of heights.

Since the spectrum is discrete and $|\gamma| \to \infty$, the resolvent is compact. If P_N are finite projections that converge strongly to the identity,

$$P_N (H_W - iI)^{-1} P_N \longrightarrow (H_W - iI)^{-1}$$

en norma.

Their eigenvalues converge towards

$$\lambda_\gamma = \frac{1}{\gamma - i},$$

and from them it is recovered

$$\gamma = i + \lambda_\gamma^{-1}.$$

This opens a genuine computational route: forming finite matrices using only the arithmetic form constructed directly from the data and recovering the heights as spectral limits.

Effective truncation bounds and the choice of optimized bases belong to the numerical realization of the compact resolvent; they refine its computational efficiency without conditioning the spectral theorem or the operator intertwining already constructed. There is no tautological enumeration of the form $n = N(\gamma_n)$: the approximation coefficients come from the arithmetic form, and their eigenvalues converge to the operator's spectrum.

97.2.8 Spectral quantization and nonadic intertwining

The exact correspondence is formulated at the following typed level:

> The proved correspondence does not identify zero number j ordinally with nonadic turn number j, distribute nine zeros per cycle, or impose a periodic repetition of heights. It identifies exactly the divisor of nontrivial zeros with the weighted atoms of the translation generator of the Weil form. The Cayley transform converts each height into a unique unitary character; the nonadic loss realization transports it as a helical orbit $q^n\tau_\gamma^n$; and twisted transport $\widetilde{U}_k$, together with J_k, intertwines that spectral character with native combinatorial holonomy.

The resulting operatorial interpretation is as follows:

> Each zero determines the individual angular character of a spectral history whose iterate crosses successive nonadic turns. The turn index quantifies the depth of memory, and the zero fixes the unitary character that governs its angular evolution.

Consequently, the result establishes an effective operator route: it proves the individual spectral quantization of the heights, constructs their canonical nonadic orbit, and identifies it, through J_k, with the memory character acting on every turn of the TPK ledger. The identification is harmonic and operator-theoretic; it does not depend on numbering zeros by turns.

Domain and construction of the Weil spectral field The positive form $\mathcal{J}_{\mathrm{GW}} = \mathscr{L}^*\mathscr{L}$, the contraction $q = 5/9$, telescoping, and the turn $M_9 = qV_9$ constitute the initial mathematical structure. The Weil Hilbert space, the translation group, its Stone generator, the Cayley correspondence, and the orbits $q^n\tau_\gamma^n$ are constructed upon it. Approximating the resolvent directly from the arithmetic form defines its computational realization. The memory cocycle and twisted transport constructed below provide the exact intertwiner with native combinatorial holonomy.

97.3 Cayley–Pontryagin Double Circle Theorem

There is no correspondence of the form "zero number j = nonadic turn number j". What does exist is something deeper: each zero determines a unitary character of the vertical memory of the ledger, and the complete orbit of that character is exactly the spectral orbit of the zero after the Cayley transformation.

97.3.1 Theorem of the Double Circle

The nonadic connection possesses an entire vertical memory

$$q_{\mathrm{mem}} : \mathrm{Obj}(\mathrm{TPK}) \longrightarrow \mathbb{Z}.$$

For any admissible path $\gamma : x \to y$, the cocycle is defined

$$m(\gamma) = q_{\mathrm{mem}}(y) - q_{\mathrm{mem}}(x).$$

Es aditivo:

$$m(\gamma_2\gamma_1) = m(\gamma_2) + m(\gamma_1),$$

and for every full rotation of the connection,

$$m(\Gamma_9) = 1.$$

By iteration,

$$m(\Gamma_9^n) = n.$$

This is the decisive datum. The visible phase returns after nine steps, but the state does not return to the same point: the memory has advanced by one unit.

On the other hand, the proof of Riemann first constructs the positive Weil form and its GNS space:

$$\mathcal{H}_W = \overline{\mathcal{D}/\ker \mathcal{J}_{\mathrm{GW}}}.$$

The translation constructed from the arithmetic form produces a self-adjoint generator H_W. Without introducing zeros as data, its Cayley transform is formed:

$$C_{\mathrm{Weil}} = \left(H_W + \frac{i}{2}I\right)\left(H_W - \frac{i}{2}I\right)^{-1}.$$

As H_W is self-adjoint,

$$C_{\mathrm{Weil}}^* C_{\mathrm{Weil}} = I.$$

Therefore, C_{Weil} is unitary and its spectrum lies in S^1.

After spectral recognition, not before, a height

$$\rho_\gamma = \frac{1}{2} + i\gamma$$

produce the phase

$$\tau_\gamma = \frac{\gamma + i/2}{\gamma - i/2} = \frac{\rho_\gamma - 1}{\rho_\gamma}, \qquad |\tau_\gamma| = 1.$$

The inversion is exact:

$$\rho_\gamma = \frac{1}{1 - \tau_\gamma}, \qquad \gamma = \frac{1}{2}\cot\left(\frac{\arg \tau_\gamma}{2}\right).$$

Thus two circular realizations emerge through distinct readers of the same enriched state:

$$\left\{\Re s = \frac{1}{2}\right\} \cup \{\infty\} \overset{\tau(s)=(s-1)/s}{\cong} S^1,$$

y

$$\widehat{\mathbb{Z}_{\mathrm{mem}}} \cong S^1.$$

The first is the Cayley circle of the critical line. The second is the Pontryagin dual of the ledger's vertical memory. The intertwiner proves that these are not merely two similar circles: they are the two realizations of the same phase operator.

97.3.2 Twisted transport isometry $\widetilde{U}_k$ on cylindrical states, projective weights, and Weil memory

Let Z_k be the set of enriched cylindrical states at depth k, with full-support projective measure μ_k, and let

$$\tau_{k+1,k} : Z_{k+1} \longrightarrow Z_k$$

el truncamiento.

For a descendant z' of z, one defines

$$w_k(z'|z) = \sqrt{\frac{\mu_{k+1}(z')}{\mu_k(z)}}.$$

The projectivity gives

$$\sum_{\tau_{k+1,k} z' = z} w_k(z'|z)^2 = 1.$$

On

$$\mathscr{K}_k = \ell^2(Z_k) \otimes \mathcal{H}_W$$

the twisted transport is defined

$$\widetilde{U}_k(e_z \otimes v) = \sum_{\tau_{k+1,k} z' = z} w_k(z'|z)\, e_{z'} \otimes C_{\mathrm{Weil}}^{m(z \to z')} v.$$

In matrix form,

$$[\widetilde{U}_k]_{z',z} = \mathbf{1}_{\{\tau z' = z\}} \sqrt{\frac{\mu_{k+1}(z')}{\mu_k(z)}}\, C_{\mathrm{Weil}}^{m(z \to z')}.$$

This formula materially preserves all descendants. It does not add distinct paths as though they were the same: ledger, orientation, carry, boundary, memory, route, and terminal remain present in the label z'.

The isometry is proved directly. Descendants of different parents are orthogonal; for each parent, the weights have squared sum one; and every power of C_{Weil} is unitary. Therefore,

$$\widetilde{U}_k^* \widetilde{U}_k = I.$$

The composition is also exact:

$$\widetilde{U}_{\ell,j} \widetilde{U}_{j,k} = \widetilde{U}_{\ell,k},$$

because the weights telescope and the memory cocycle is additive.

There is even an explicit gauge equivalence. If

$$G_k(e_z \otimes v) = e_z \otimes C_{\mathrm{Weil}}^{q_{\mathrm{mem}}(z)} v,$$

then

$$\widetilde{U}_k = G_{k+1}(U_k \otimes I) G_k^*.$$

Therefore, the twisted system is not an arbitrary decoration: it is the native cylindrical transport viewed in the spectral gauge determined by memory.

Define now the measurement vector

$$\Omega_k = \sum_{z \in Z_k} \sqrt{\mu_k(z)}\, e_z, \qquad \|\Omega_k\| = 1,$$

and the fit

$$J_k : \mathcal{H}_W \longrightarrow \mathscr{K}_k, \qquad J_k v = \Omega_k \otimes v.$$

In a full rotation, all paths possess a memory increment of one. Therefore,

$$\boxed{\widetilde{U}_{\Gamma_9,k} J_k = J_{k+9} C_{\mathrm{Weil}}}$$

and, after n turns,

$$\boxed{\widetilde{U}_{\Gamma_9^n,k} J_k = J_{k+9n} C_{\mathrm{Weil}}^n.}$$

This is the announced intertwiner, constructed within the same causal chain.

97.3.3 Orbital identity under the conjugation between the 2 spectral realizations

Let ψ_γ be a vector of the spectral fiber associated with γ. Then

$$C_{\mathrm{Weil}} \psi_\gamma = \tau_\gamma \psi_\gamma.$$

Applying the preceding square:

$$\widetilde{U}_{\Gamma_9^n} J \psi_\gamma = J C_{\mathrm{Weil}}^n \psi_\gamma = \tau_\gamma^n J \psi_\gamma.$$

But the character of the memory corresponding to τ_γ is

$$\chi_{\tau_\gamma}(m) = \tau_\gamma^m.$$

Since

$$m(\Gamma_9^n) = n,$$

se obtiene

$$\boxed{\widetilde{U}_{\Gamma_9^n} J \psi_\gamma = \chi_{\tau_\gamma}\big(m(\Gamma_9^n)\big)\, J \psi_\gamma.}$$

This is the exact equality sought.

It does not mean that zero number 37 is turn number 37. It means that each zero selects a character of the complete memory group and that this character traverses all turns:

$$n \longmapsto \tau_\gamma^n.$$

Por tanto:

- the nonadic turn provides the integer variable n;
- the memory provides the group $\mathbb{Z}$;
- the zero provides a character $\chi_{\tau_\gamma} \in \widehat{\mathbb{Z}}$;
- the spectral orbit and the ledger orbit coincide term by term after the embedding J.

The preceding caution is overcome without falling into “nine zeros per cycle.” There is no zero–turn enumeration. There is a harmonic decomposition of the same turn dynamics.

97.3.4 Circular compactification and spiral lifting of the spectral orbit

A crucial distinction must be preserved. The phase is unitary:

$$C_{\mathrm{Weil}}^* C_{\mathrm{Weil}} = I.$$

The strict dynamics of the loss sheet is

$$M_9 = qV_9, \qquad q = \frac{5}{9}.$$

The correct set operator is

$$\widetilde{M}_9 = \frac{5}{9}\,\widetilde{U}_{\Gamma_9}.$$

On the spectral fiber,

$$\widetilde{M}_9^n J\psi_\gamma = \left(\frac{5}{9}\right)^n \tau_\gamma^n J\psi_\gamma.$$

Thus two images appear simultaneously:

- at the normalized boundary, the orbit is circular:

$$\tau_\gamma^n \in S^1;$$

- in the strict realization, it is a spiral:

$$\left(\frac{5}{9}\right)^n \tau_\gamma^n \longrightarrow 0.$$

The terminal zero does not erase the phase or the genealogy. The radius is extinguished, but the ledger preserves how many turns were performed and which character was transported.

Telescopy is exact:

$$I = \sum_{r=0}^{L-1} \widetilde{M}_9^{*r} D_9^* D_9 \widetilde{M}_9^r + \widetilde{M}_9^{*L} \widetilde{M}_9^L,$$

with

$$D_9^* D_9 = I - \widetilde{M}_9^* \widetilde{M}_9 = \frac{56}{81} I,$$

y terminal

$$\widetilde{M}_9^{*L} \widetilde{M}_9^L = \left(\frac{25}{81}\right)^L I \longrightarrow 0.$$

This preemptively answers a type objection: a contraction has not been conflated with a unitary. The phase resides in C_{Weil}; the radial loss resides in $5/9$.

97.3.5 Compatibility between genealogical cylinders, generation of constants, and spectral entangler

Your intuition about cylinders is correct, with a precise distinction.

Numerical cylinders are half-open:

$$C(N_K, 6K) = \left[\frac{N_K}{3^{6K}}, \frac{N_K + 1}{3^{6K}}\right),$$

and each one is divided into 729 subcylinders. Their diameter satisfies

$$\operatorname{diam} C(N_K, 6K) = 3^{-6K} \longrightarrow 0.$$

The genealogical cylinders $[a]_k$ of the inverse limit are clopen balls of the ultrametric:

$$d_\vartheta(x, y) = \vartheta^{N(x,y)}.$$

Therefore, "radius tending to zero" has two publications:

1. In the Archimedean reading, the intersection of nested intervals publishes a unique real number.
2. In the enriched state, the complete history continues to preserve all prefixes that produced that real number.

Thus, in accordance with the tesis inaugural, the same coinductive system publishes π, e, φ, and α correlatively. In the internal operator order, the register U_{12}^{sgn}, the seal K, and dodecaphasic carry act once P, E, Φ are available: they publish and certify coordinate α of that same correlated coinductive generation, without incorporating it as a causally external object. Archimedean sections and signed publication retain distinct codomains without breaking their shared genealogy.

These constants do not select the Riemann phases. Their relevance is deeper: they show that the same system of finite cylinders can produce a continuous object without destroying the history that generated it.

The new intertwiner uses exactly this property. The spectral circle is not superimposed on the continuum from outside: it appears as the dual of its integer memory.

97.3.6 Graded invariant of the spectral runner 4→2→6→54

The cadena $4 \to 54$ can now situarse without numerologia.

The central block of APP is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-.$$

Its eigenvalues are 9 and 5, so that

$$9 - 5 = 4, \qquad q = \frac{5}{9}.$$

The typed hierarchy is

$$\underbrace{9-5}_{4}; \qquad \underbrace{2}_{\text{coupling}}; \qquad \underbrace{51-45}_{6}; \qquad \underbrace{9\cdot 6}_{54}.$$

In the connection,

$$\Gamma_{54} = \Gamma_9^6.$$

By the new entangler,

$$\widetilde{U}_{\Gamma_{54}} J = J C^6_{\mathrm{Weil}},$$

and, in the fiber γ,

$$\widetilde{U}_{\Gamma_{54}} J\psi_\gamma = \tau_\gamma^6 J\psi_\gamma.$$

In the strict realization,

$$\widetilde{M}_{54} J\psi_\gamma = \left(\frac{5}{9}\right)^6 \tau_\gamma^6 J\psi_\gamma.$$

This is the exact spectral incidence of depth 54: six nonadic turns, six units of memory, and sixth power of the character.

It must be distinguished from other objects whose value is also 54:

- the residual APP defect;
- the trace oriented APP–Fock;
- the norm of the radial vector of A_2^{12};
- the coefficient of the character t_3;
- the value

$$54 = \mathrm{Tr}(3B \mid V_2^{\natural}).$$

That all being 54 has structural importance, but it does not automatically make them the same morphism.

97.3.7 Leech, $V^{\natural}$ and the Monster

The exceptional chain remains exact:

$$A_2^{12} \longrightarrow C_W = [12, 6, 6]_3 \longrightarrow N_\alpha \cong \Lambda_{24} \longrightarrow V^{\natural} \longrightarrow \mathbb{M}.$$

Transverse rigidity selects

$$m = 12, \qquad F(12) = R(12) = 54.$$

The free-fixed isometry of order three produces the orbifold

$$\mathrm{Orb}_{C_3}(V_{N_\alpha}, \widehat{g}_c) \cong V^{\natural},$$

and the dual automorphism belongs to the class $3B$:

$$T_{3B} = q^{-1} + 54q + \cdots .$$

Now, the Monster does not generate the phases τ_γ. It is finite, whereas the memory characters range over all of S^1. In particular, an element $3B$ has order three, but a general spectral phase is not a cube root of unity.

The correct meeting is a product of representations:

$$\mathscr{K}^{\mathrm{gold}} = \mathscr{H}_{\mathrm{cyl}} \otimes \mathcal{H}_W \otimes \mathcal{H}_{V^{\natural}}.$$

Memory and the Cayley structure act on the first two factors; the Monster acts on the third:

$$V_{\mathrm{mem}} = \widetilde{U}_{\Gamma_9} \otimes I, \qquad \rho_{\mathbb{M}}(g) = I \otimes I \otimes \rho_{V^{\natural}}(g).$$

Both actions commute. Thus, each spectral orbit can simultaneously preserve:

- su altura γ;
- its phase τ_γ;
- its nonadic memory n;
- its spectral multiplicity;
- its transverse Monster type.

But the multiplicity of a zero is not identified by decree with a monstrous dimension. That identification would require an additional intertwiner between the spectral fiber and a concrete representation of the Monster.

If a genuine computable family of entangled traces remains open:

$$\Theta_g(n;a,\beta) = \operatorname{Tr}\left(e^{-aH_W^2} C_{\mathrm{Weil}}^n\right) \operatorname{Tr}_{V^\natural}\left(g\, e^{-\beta(L_0-1)}\right).$$

After spectral recognition,

$$\Theta_g(n;a,\beta) = \left(\sum_\gamma m_\gamma e^{-a\gamma^2} \tau_\gamma^n\right) T_g(e^{-\beta}).$$

This indeed is a new rigorously typed RH–Moonshine path: moments of the orbits of the zeros, nonadic memory, and monstrous characters in a single regularized observable.

97.3.8 Finite Approximation of the Spectral Field via Matrix Truncations

The construction possesses a finite natural matrix.

For any unitary C, define the nine-phase stitching

$$\mathbb{U}_{C,9} = \sum_{r=1}^{8} |r+1\rangle\langle r| \otimes I + |1\rangle\langle 9| \otimes C.$$

Then

$$\mathbb{U}_{C,9}^9 = I \otimes C.$$

Con

$$u_9 = \frac{1}{3}(1,\dots,1), \qquad \iota_9 v = u_9 \otimes v,$$

visible compression is

$$A_{C,9} = \iota_9^* \mathbb{U}_{C,9} \iota_9 = \frac{8I + C}{9},$$

and the defect satisfies

$$c_{C,9}^* c_{C,9} = \frac{8}{81}(I-C)^*(I-C).$$

Taking $C = C_{\mathrm{Weil}}$ yields a nine-phase matrix constructed without introducing zeros.

To make the Weil fiber finite as well, test functions $f_1,\dots,f_N$ are chosen and the matrices are computed from the prime–gamma–polar channels

$$G_N = \left(\mathcal{J}_{\mathrm{GW}}(f_i,f_j)\right)_{i,j},$$

y

$$A_N = \left(\langle H_W f_j, f_i\rangle_W\right)_{i,j}.$$

After taking the quotient by $\ker G_N$, the generalized problem

$$A_N v = \gamma_N G_N v$$

produce the spectral approximation, and its finite Cayley is

$$C_N = \left(G_N^{-1}A_N + \frac{i}{2}I\right)\left(G_N^{-1}A_N - \frac{i}{2}I\right)^{-1}.$$

Its eigenvalues $\tau_N \in S^1$ recover

$$\gamma_N = \frac{1}{2}\cot\left(\frac{\arg \tau_N}{2}\right).$$

The total cylinder matrix is then

$$[\widetilde{U}_{k,N}]_{z',z} = \mathbf{1}_{\{\tau z'=z\}}\sqrt{\frac{\mu_{k+1}(z')}{\mu_k(z)}}\, C_N^{m(z\to z')}.$$

This converts the conceptual approach into a specific numerical program:

1. construct G_N and A_N without introducing zeros;
2. form C_N;
3. insert it into the nonadic seam;
4. diagonalizing the phases;
5. recover the heights;
6. verify stability under simultaneous increase of N and cylindrical depth.

Rigorous convergence requires proving that the finite subspaces form a core for H_W and that the approximations converge in the strong resolvent sense. This is a subsequent numerical certificate; it does not affect the infinite intertwiner, which has already been proved.

97.3.9 Independence of the spectral intertwiner with respect to zeros, contractions, route multiplicities, and transverse symmetries

"The zeros were introduced to fabricate the map." No. $\mathcal{H}_W$, H_W, their resolvent, and C_{Weil} are constructed from the prime–gamma–polar form. The zeros appear afterward to recognize the spectrum.

"A contraction has been identified with a unitary operator." No. The phase is C_{Weil}; the contraction is $(5/9)C_{\mathrm{Weil}}$.

"Distinct paths have been erased." No. Each descendant preserves its complete label in $e_{z'}$; the orthogonality of the ledger prevents two routes from being confused.

"It is claimed that there are nine zeros per cycle." No. Each zero determines a character that is evaluated over all turns.

"The Monster produces the heights." No. The phase circle arises from $\widehat{\mathbb{Z}_{\mathrm{mem}}}$. The Monster organizes transverse symmetry and commutes with that action.

"The different 54s are the same object." No. Defect, depth, norm, and character value are kept separate. Their coincidences remain connected by maps, not by bare numerical equality.

"Only a pre-existing operator V_9 has been renamed." No. A specific explicit realization of the codomain of Γ_9 has been constructed. To identify it with any pre-existing ambient representative of V_9, specified only up to unitary equivalence, the preceding gauge must be chosen compatibly. On the relevant cyclic subspace, the unitary class is already determined.

97.3.10 Domain of the Cayley–Pontryagin conjugation and theorem of the double circle

APP–TRIT–TPK, the cylinders, the projective measure, the memory cocycle, Γ_9, the contraction $5/9$, the chain $4 \to 54$, the constants, and the exceptional corridor constitute the initial structure. On that domain, the cylindrical transport twisted by $C_{\mathrm{Weil}}^{m(\gamma)}$ is constructed. The double-circle theorem establishes the exact intertwining

$$\widetilde{U}_{\Gamma_9} J = J C_{\mathrm{Weil}}.$$

Certification of convergence of the finite matrices and the unfactorized study of the RH–Moonshine traces constitute the open boundary beyond the theorem. Therefore, the strong conclusion is:

zeros are not the turns; they are the spectral characters of the memory of the turns.

And, after the canonical matching,

the spectral orbit of the zero and the ledger orbit are literally the same orbit.

97.4 Effective reconstruction of the Weil spectral field

97.4.1 Construction of the spectral field independent of previous data on the zeros

The compactly supported family is adopted.

$$\beta_A(x) = Z_A^{-1} \exp\left[-\frac{1}{1-(x/A)^2}\right] \mathbf{1}_{\{|x|<A\}}, \qquad f_{\omega,A}(x) = \beta_A(x) e^{-i\omega x}.$$

The modulation ω was not chosen using zeros. The interval was swept in advance

$$5 \le \omega \le 50, \qquad \Delta\omega = 0.025.$$

For each ω the calculation is performed

$$M_A(\omega) = \mathcal{I}_{\mathrm{GW}}(f_{\omega,A}, f_{\omega,A})$$

via the formula constructed from the arithmetic data

$$\begin{aligned}\mathcal{I}_{\mathrm{GW}}(f,g) = c_\Gamma C_{f,g}(0) &+ \int_0^\infty \mu_\Gamma(a)\big[2C_{f,g}(0) - C_{f,g}(a) - C_{f,g}(-a)\big]\,da \\ &- \sum_{p,k} \frac{\log p}{p^{k/2}}\big[C_{f,g}(k\log p) + C_{f,g}(-k\log p)\big] \\ &+ P_f^+ \overline{P_g^-} + P_f^- \overline{P_g^+},\end{aligned}$$

where

$$C_{f,g}(a) = \int_{\mathbb{R}} f(x)\overline{g(x-a)}\,dx,$$

$$\mu_\Gamma(a) = \frac{e^{-a/2}}{1-e^{-2a}}, \qquad c_\Gamma = -\gamma_E - \frac{\pi}{2} - 3\log 2 - \log\pi,$$

y

$$P_f^\pm = \int_{\mathbb{R}} f(x) e^{\pm x/2}\,dx.$$

Neither $\zeta(s)$, a zero-finding routine, Riemann–Siegel, Odlyzko, nor a table of heights was involved.

With $A = 3$, the first blind sweep produced ten dominant maxima approximately at

$$14.125,\ 21.025,\ 25.000,\ 30.425,\ 32.925,\ 37.575,\ 40.925,\ 43.325,\ 48.000,\ 49.775.$$

The lateral lobes had a mass about a hundred times smaller.

97.4.2 Refinement of the independent spectral field of the target values

I increased the compact width A. This simultaneously increases the spectral resolution and the number of prime powers used:

n	$A = 3$	$A = 4$	$A = 5$	$A = 6$
1	14.134699719	14.134721825	14.134724556	14.134725023
2	21.022713497	21.022092670	21.022033480	21.022034382
3	25.010192460	25.010816662	25.010862963	25.010863775
4	30.431966224	30.424255364	30.424903280	30.424992486
5	32.927889053	32.935726409	32.935037177	32.934942801
6	37.585309844	37.585911673	37.586134491	37.586172107
7	40.926249022	40.917939546	40.919104383	40.918785046
8	43.320460106	43.328124621	43.326729729	43.327013819
9	47.996518745	48.009530251	48.005271807	48.005312108
10	49.781332020	49.769281661	49.773747976	49.773697939

In $A = 6$ exactly 15.040 prime powers p^k entered, with $p^k \le e^{12}$. For the first maximum, the numerical ledger was:

prime channel	8.077017879363272,
gamma integral	6.182517531024263,
$c_\Gamma I$	−5.372183419225665,
polar channel	$-1.0505899 \times 10^{-8}$,
total mass	8.887351980655973.

The gamma tail beyond $a = 65$ was bounded by

$$1.54 \times 10^{-14}.$$

This reveals a conceptual point: zero is not inserted into the calculation. It appears as the stable center at which the four components of the functional close.

97.4.3 Finite matrix approximants of the spectral operator

An isolated Gram matrix does not contain heights in identifiable form. Its eigenvalues depend on the basis and represent masses or energies, not spectral positions.

Therefore, three matrices are constructed:

$$G_{ij} = \mathcal{J}_{\mathrm{GW}}(f_j, f_i),$$
$$K_{ij} = \mathcal{J}_{\mathrm{GW}}(H_W f_j, f_i),$$
$$Q_{ij} = \mathcal{J}_{\mathrm{GW}}(H_W f_j, H_W f_i).$$

With the positive orientation used in calculus,

$$H_W = i\,\partial_x$$

on the reflected family $f_{\omega,A} = \beta_A e^{-i\omega x}$. The conjugate orientation $-i\partial_x$ returns the opposite spectral sheet; both produce the same positive heights by the symmetry $\gamma \leftrightarrow -\gamma$.

This generator is not introduced arbitrarily. The established positivity permits construction of the GNS representation of $\mathcal{J}_{\mathrm{GW}}$. The translations are strongly continuous isometries of that GNS representation, and Stone's theorem produces their self-adjoint generator. On the test core, the derivation of the translations is precisely $i\partial_x$.

The three matrices were computed by means of the same prime–gamma–polar evaluator. Neither K nor Q was constructed from a spectral list.

In block 10×10, the hermiticity defects before numerical symmetrization were

$$\|G - G^{*}\|_2 = 2.11 \times 10^{-14},$$
$$\|K - K^{*}\|_2 = 7.14 \times 10^{-13},$$
$$\|Q - Q^{*}\|_2 = 1.84 \times 10^{-10}.$$

The eigenvalues of G were

$$4.90806959,\ 5.50537820,\ 5.61057540,\ 5.88666432,\ 5.91920889,$$
$$5.93049827,\ 5.96524271,\ 6.27603538,\ 6.33591428,\ 7.02999383.$$

All are positive and the block condition is only

$$\kappa(G) = 1.43233.$$

There was no need to hide a negative eigenvalue or artificially project an uncomfortable part.

The metric was whitened:

$$\widehat{H}_N = G_N^{-1/2} K_N G_N^{-1/2},$$

and the hermitic pencil was resolved

$$K_N v = \gamma_N G_N v.$$

This is the point that converts gram masses into heights.

97.4.4 Cayley Transform and Initial Circular Compactification of the Spectrum

To each $\widehat{H}_N$ was applied

$$C_N = \left(\widehat{H}_N + \frac{i}{2} I\right)\left(\widehat{H}_N - \frac{i}{2} I\right)^{-1}.$$

If γ is an eigenvalue of the generator, its phase is

$$\tau_\gamma = \frac{\gamma + i/2}{\gamma - i/2}, \qquad |\tau_\gamma| = 1.$$

The inverse recovery is

$$\gamma = \frac{i}{2}\frac{\tau + 1}{\tau - 1} = \frac{1}{2}\cot\!\left(\frac{\arg \tau}{2}\right).$$

The calculated unitarity defect was

$$\|C_N^{*} C_N - I\|_2 = 1.45 \times 10^{-15}.$$

The first three phases were

$$\tau_1 = 0.997500505583064 + 0.070659333152323\, i,$$
$$\tau_2 = 0.998869228927548 + 0.047542228615055\, i,$$

$$\tau_3 = 0.999201013749813 + 0.039966662624574\, i.$$

This is the first circle: the spectral line is compactified without losing any height, because the Cayley map is bijective between $\mathbb{R}$ and the circle with the terminal point removed.

97.4.5 Exact immersion of the spectrum into the nonadic circle

Once C_N has been constructed, the nine-position sewing matrix is defined

$$S_{N,9} = \sum_{r=1}^{8} |r+1\rangle\langle r| \otimes I + |1\rangle\langle 9| \otimes C_N.$$

A zero is not assigned to each gate. The nine positions are the transport calendar. The spectral phase is applied only when the traversal completes the turn.

The exact identity is

$$S_{N,9}^{9} = I_9 \otimes C_N.$$

Therefore, if $J_N v = u_9 \otimes v$, the full-turn holonomy satisfies

$$S_{N,9}^{9} J_N = J_N C_N.$$

This is the entanglement of the double circle:

- the Cayley circle contains the spectral phase;
- the nonadic circle transports the state for nine steps;
- a complete turn applies the Cayley phase exactly once;
- memory records the number of revolutions;
- no zero is identified with an isolated gate.

For the ten candidates a matrix 90×90 was constructed. The controls were:

$$\|S_{N,9}^{*} S_{N,9} - I\|_2 = 4.44 \times 10^{-16},$$
$$\|S_{N,9}^{9} - I_9 \otimes C_N\|_2 = 0.$$

The visible contraction is

$$M_N = \frac{5}{9} S_{N,9}, \qquad D_N = \frac{\sqrt{56}}{9} I.$$

It was verified

$$M_N^{*} M_N = \frac{25}{81} I, \qquad D_N^{*} D_N = \frac{56}{81} I,$$

and the telescopy

$$I = \sum_{r=0}^{L-1} M_N^{*r} D_N^{*} D_N M_N^{r} + M_N^{*L} M_N^{L}.$$

Para $L = 20$,

$$\left\| \sum_{r=0}^{19} M_N^{*r} D_N^* D_N M_N^r + M_N^{*20} M_N^{20} - I \right\|_2 = 3.33 \times 10^{-16},$$

while the terminal was

$$\|M_N^{*20} M_N^{20}\|_2 = 6.1531824951 \times 10^{-11},$$

exactly equal, within machine precision, to

$$\left(\frac{25}{81}\right)^{20}.$$

The visible loss does not destroy the phase: the phase resides in the unitary dilation, and the factor $5/9$ regulates how much remains visible on each turn.

97.4.6 Recovery of Heights and Residues with Posterior External Contrast

For each generalized vector v, the residue was computed without zeros through

$$r_N(\gamma, v)^2 = \frac{v^* \left(Q_N - 2\gamma K_N + \gamma^2 G_N\right) v}{v^* G_N v}.$$

This residual measures how far the finite vector lies from a genuine spectral vector. A height is not accepted merely because the pencil produces a number.

n	altura producida	residue	error in the posterior contrast
1	14.134725141525	8.15×10^{-5}	2.10×10^{-10}
2	21.022039638825	4.82×10^{-5}	5.34×10^{-11}
3	25.010857580595	1.15×10^{-4}	4.49×10^{-10}
4	30.424876128079	2.32×10^{-4}	2.22×10^{-9}
5	32.935061595211	3.98×10^{-4}	7.47×10^{-9}
6	37.586178242423	1.18×10^{-3}	8.36×10^{-8}
7	40.918719778888	3.14×10^{-3}	7.67×10^{-7}
8	43.327078423026	7.18×10^{-3}	5.14×10^{-6}
9	48.005155742413	6.22×10^{-3}	4.86×10^{-6}
10	49.773888321513	1.45×10^{-2}	5.58×10^{-5}

The final column was computed only after the outputs had been frozen. It played no role in the sweep, in G, in K, in Q, in the Cayley transform, or in the nonadic seam.

97.4.7 Convergence and preservation of the spectral genealogical record

After the Riemann–Weil resolution, the functional possesses the representation

$$\mathcal{J}_{\mathrm{GW}}(f, g) = \sum_{\gamma} m_\gamma \widehat{f}(-\gamma) \overline{\widehat{g}(-\gamma)}.$$

This is not used to calculate the entries; it serves to recognize which spectrum we have just obtained.

The GNS remains identificado with the espacio L^2 of the measure atomica

$$\nu_W = \sum_{\gamma} m_\gamma \delta_\gamma.$$

The generator of translations acts by multiplication by γ. Therefore:

- G contains the scalar products;

- K inserts a power of γ;
- Q inserta γ^2;
- the pencil $K - \gamma G$ localizes the atoms;
- Cayley transforms them into unitary phases;
- the nonadic seam transports those phases through the ledger.

The resolution is not being used to prove RH again with ten matrices. The already obtained resolution is used to convert the explicit formula into a spectral-extraction algorithm that does not receive the zeros as input.

97.4.8 Convergence of the reconstructed spectral field and certifiable extension

The computational cycle is now closed:

$$\text{primes+gamma+polar} \longrightarrow (G, K, Q) \longrightarrow \widehat{H}_N \longrightarrow C_N \longrightarrow S_{N,9} \longrightarrow \{\gamma_n\}.$$

The literal interpretation of the closure circle is also constructed:

$$\text{one complete nonadic turn} \quad \Longleftrightarrow \quad \text{one application of the spectral phase,}$$

but not

$$\text{a nonadic phase} \quad \Longleftrightarrow \quad \text{a zero.}$$

Each zero defines a character of the ledger,

$$n \longmapsto \tau_\gamma^n,$$

and in the contractive publication it appears as

$$n \longmapsto \left(\frac{5}{9}\right)^n \tau_\gamma^n.$$

This is the concrete orbit: not a repetition of heights, but the evolution of a single spectral phase through the memory of turns.

Nor is the number 54 from the exceptional branch conflated with a number of zeros or turns. The 54 of APP–Witt–Leech–Monster is a defect, a norm, and a graded character in its domain. It may coexist with the spectral circle through a product of representations, but it neither selected nor normalized any of the heights calculated here.

Mathematical convergence rests on a cofinal family of windows and compact modes constituting a graph core for the GNS generator. The calculation performed supplies four window refinements and a posteriori residuals. A certified version using interval arithmetic could convert every decimal into a guaranteed interval, but no conceptual or operatorial map is any longer missing: this would be the numerical strengthening of a chain already constructed from beginning to end.

The explicit formula and the joint nonadic connection constitute the initial mathematical structure. The matrices G, K, Q, the Cayley–Galerkin operator, and the finite double-circle intertwining constitute the numerical construction. The ten heights are extracted directly from the arithmetic form and constitute the result of that calculation.

97.5 Exact entanglement of nonadic holonomy and spectral phase

Let $\mathcal{H}_W$ be the Hilbert space obtained from the positive Guinand–Weil form, let H_W be the self-adjoint generator of the translation group, and let

$$C_W = (H_W + i/2)(H_W - i/2)^{-1}$$

its Cayley transform. For each depth k, let Z_k be the set of enriched cylindrical states, let μ_k be its projective measure of full support, and let $\tau_{k+1,k} : Z_{k+1} \to Z_k$ be the truncation. One considers

$$\mathscr{K}_k = \ell^2(Z_k)\widehat{\otimes}\mathcal{H}_W, \qquad \Omega_k = \sum_{z\in Z_k} \sqrt{\mu_k(z)}\, e_z, \qquad J_k\psi = \Omega_k \otimes \psi.$$

If z' is a descendant of z, one takes $w_k(z' \mid z) = \sqrt{\mu_{k+1}(z')/\mu_k(z)}$. For a route $\gamma : z \to z'$, the integer memory cocycle is denoted by $m(\gamma)$, and the twisted transport is defined by

$$\widetilde{U}_k(e_z \otimes \psi) := \sum_{\tau_{k+1,k}z'=z} w_k(z' \mid z)\, e_{z'} \otimes C_W^{m(z\to z')}\psi. \tag{97.1}$$

The projectivity of μ_k, the orthogonality of descendants, and the unitarity of C_W give $\widetilde{U}_k^*\widetilde{U}_k = I$. The complete label z' preserves cylinder, sheet, orientation, carry, route, boundary, and multiplicity; therefore, distinct histories are not identified upon applying (97.1).

Theorem 97.1 (Cayley intertwining under 1 complete nonadic turn)**.** *If Γ_9 is the complete nonadic holonomy, normalized by $m(\Gamma_9) = 1$, then*

$$\boxed{\widetilde{U}_{\Gamma_9,k}J_k = J_{k+9}C_W.} \tag{97.2}$$

For every spectral atom ψ_γ of H_W, with $H_W\psi_\gamma = \gamma\psi_\gamma$, one has

$$\widetilde{U}_{\Gamma_9^n,k}J_k\psi_\gamma = \tau_\gamma^n J_{k+9n}\psi_\gamma, \qquad \tau_\gamma = \frac{\gamma + i/2}{\gamma - i/2} \in S^1. \tag{97.3}$$

In the contracting publication, the same identity takes the form

$$\widetilde{M}_{\Gamma_9^n,k}J_k\psi_\gamma = \left(\frac{5}{9}\right)^n \tau_\gamma^n J_{k+9n}\psi_\gamma.$$

Proof. The turn Γ_9 returns to the visible phase but increments memory by one unit, so that $m(\Gamma_9) = 1$. Summing over all the descendants of Ω_k, the weights telescope with the projective measure and yield Ω_{k+9}, while the Weil fiber receives exactly one power of C_W. This proves $\widetilde{U}_{\Gamma_9,k}J_k = J_{k+9}C_W$. The iteration uses the additivity $m(\Gamma_9^n) = n$. The final formula follows from $\widetilde{M}_{\Gamma_9,k} = (5/9)\widetilde{U}_{\Gamma_9,k}$. □

The theorem does not identify a zero with an isolated phase of the calendar. One complete circuit applies the Cayley character once; each zero determines the angular law according to which a history traverses all circuits. In particular, phase return and state return remain distinct:

$$g(K + 9) = g(K), \qquad B_{K+9} \neq B_K \quad \text{in general.}$$

97.6 Convergence in norm of finite spectral matrices

Let

$$R_W = (H_W - iI)^{-1}.$$

The spectrum of H_W is discrete and $|\gamma| \to \infty$, so R_W is compact. A countable family $\{f_j\}_{j\geq 1} \subset C_c^\infty(\mathbb{R})$ with rational parameters is chosen, dense in the test core and selected without consulting spectral heights. After quotienting by the kernel of the Weil form and orthonormalizing, one obtains subspaces

$$\mathcal{V}_N = \operatorname{span}\{[f_1], \ldots, [f_N]\}, \qquad P_N : \mathcal{H}_W \to \mathcal{V}_N,$$

with $P_N \to I$ strongly. Define

$$R_N = P_N R_W P_N. \tag{97.4}$$

Theorem 97.2 (Compact approximation without introducing the zeros)**.** *The sequence R_N converges in norm to R_W:*

$$\boxed{\lim_{N\to\infty} \|R_W - R_N\| = 0.} \tag{97.5}$$

Consequently, every nonzero eigenvalue $\lambda_\gamma = (\gamma - i)^{-1}$ of R_W is the limit, with its algebraic multiplicity, of eigenvalues of the finite matrices of R_N. The heights are recovered through

$$\gamma = i + \lambda_\gamma^{-1}. \tag{97.6}$$

Proof. If K is compact and $P_N \to I$ strongly, the convergence is uniform over the compact $K(B_1)$, where B_1 is the unit ball. Therefore,

$$\|(I - P_N)K\| \longrightarrow 0.$$

Applying the same argument to K^* yields $\|K(I - P_N)\| \to 0$. Thus,

$$\|K - P_N K P_N\| \leq \|(I - P_N)K\| + \|P_N K(I - P_N)\| \longrightarrow 0.$$

Taking $K = R_W$ proves (97.5). The stability of isolated eigenvalues of compact operators under norm perturbations and the Riesz functional calculus preserve their multiplicity. The identity (97.6) is the inverse of the spectral map of the resolvent. □

The entries of R_N are computed from the prime–gamma–polar form:

$$\langle R_W[f_j], [f_k]\rangle_W = i \int_0^\infty e^{-t}\, \mathcal{J}_{\mathrm{GW}}(T_{-t} f_j, f_k)\, dt.$$

Therefore, the theorem does not use a zero table to form the matrices.

97.6.1 A posteriori certification of intervals and multiplicities

Let (λ_N, v_N) be a normalized eigenvector of R_N, and let

$$\varepsilon_N(v_N) := \|(I - P_N)R_W v_N\|.$$

As R_W is normal,

$$\operatorname{dist}(\lambda_N, \sigma(R_W)) \leq \varepsilon_N(v_N). \tag{97.7}$$

The prime, gamma, and polar channels provide separate interval bounds for ε_N; their tails are summed before the interval is emitted. Moreover, if Γ is a closed contour contained in the resolvent set of R_N and

$$\sup_{z\in\Gamma} \|(z - R_N)^{-1}\|\, \|R_W - R_N\| < 1, \tag{97.8}$$

then the Riesz projections of R_N and R_W delimited by Γ have the same rank. The equations (97.7) and (97.8) certify, respectively, inclusion and completeness of the atoms contained in each window. Interval arithmetic makes the decimal bounds rigorous; it does not supply a missing mathematical map.

97.7 Independent construction of the spectral field over the double circle

The product trace

$$\mathrm{Tr}_{\mathcal{H}_W}\left(e^{-aH_W^2}C_W^n\right)\mathrm{Tr}_{V^\natural}\left(ge^{-\beta(L_0-1)}\right)$$

superposes two observables, but does not couple them. The memory of the TPK and the grading of $V^\natural$ make it possible to construct the coupling without postulating an action of the Monster on digits or on heights.

Once the exceptional gauge χ_{exc} has been fixed, the memory origin and the grading origin are jointly oriented. Let $\mathcal{H}_{\mathrm{mem}} = \ell^2(\mathbb{N}_0)$, with $N_{\mathrm{mem}}|n\rangle = n|n\rangle$, and let

$$V^\natural = \widehat{\bigoplus}_{r\geq 0} V_r^\natural, \qquad L_0|_{V_r^\natural} = rI.$$

Let us denote by $P_r^\natural$ the projection onto $V_r^\natural$. The degree-equality projector is

$$\Pi_{\chi_{\mathrm{exc}}} := \sum_{n\geq 0} |n\rangle\langle n| \otimes I_{\mathcal{H}_W} \otimes P_n^\natural. \tag{97.9}$$

The sum converges strongly and defines an orthogonal projection. On the product, the memory character is defined.

$$C_W^{N_{\mathrm{mem}}} := \sum_{n\geq 0} |n\rangle\langle n| \otimes C_W^n. \tag{97.10}$$

Definition 97.3 (Diagonal trace RH–Moonshine)**.** For $a, b, \beta > 0$ and $g \in \mathbb{M}$, one defines

$$\Theta^{\Delta}_{g,\chi_{\mathrm{exc}}}(a,b,\beta) := \mathrm{Tr}\left[\Pi_{\chi_{\mathrm{exc}}}\left(e^{-bN_{\mathrm{mem}}}C_W^{N_{\mathrm{mem}}}e^{-aH_W^2}\otimes ge^{-\beta(L_0-1)}\right)\Pi_{\chi_{\mathrm{exc}}}\right]. \tag{97.11}$$

Theorem 97.4 (Existence and Non-Factorized Expansion)**.** *The trace* (97.11) *is absolutely convergent and satisfies*

$$\boxed{\Theta^{\Delta}_{g,\chi_{\mathrm{exc}}}(a,b,\beta) = \sum_{n\geq 0} e^{-bn}e^{-\beta(n-1)}\,\mathrm{Tr}_{\mathcal{H}_W}\left(e^{-aH_W^2}C_W^n\right)\mathrm{Tr}_{V_n^\natural}(g).} \tag{97.12}$$

After spectral recognition,

$$\Theta^{\Delta}_{g,\chi_{\mathrm{exc}}}(a,b,\beta) = \sum_{n\geq 0}\sum_{\gamma} m_\gamma e^{-a\gamma^2}e^{-bn-\beta(n-1)}\tau_\gamma^n\,\mathrm{Tr}_{V_n^\natural}(g). \tag{97.13}$$

This expression is not the product of a spectral sum and a McKay–Thompson series: the same index n simultaneously selects the power of the memory character and the monstrous degree.

Proof. For $a > 0$, the spectral counting law implies $\sum_\gamma m_\gamma e^{-a\gamma^2} < \infty$. Each space $V_n^\natural$ is finite-dimensional, and the coefficients of the Moonshine characters grow subexponentially in n. The factor $e^{-(b+\beta)n}$ ensures absolute convergence of the joint series. The expansion is obtained by inserting the spectral resolutions of N_{mem}, H_W, and L_0 and using (97.9). The presence of the diagonal projector eliminates terms with distinct degrees and produces a single sum over n, rather than the two independent sums of the product trace. □

The coupling depends on the gauge χ_{exc}, which fixes the origin, orientation, and alignment of the exceptional reading. A conjugate choice produces a conjugate trace or a declared reindexing; it does not alter the Riemann–Weil theorem or the intertwining (97.2).

97.8 Weil form, double circle, and spectral representation

The factorization and clocks of Section 42.4 produce two readouts of the same arithmetic. The first decomposes a normal form into irreducibles; the second records the returns of their multiplicative actions. The spectral development must preserve both lines of provenance. Its object is not obtained by assigning a known zero of a function to a return: the functionals are constructed, their compatibility is established, and their analytic realization is identified afterwards.

This chapter brings together the preserved developments of the completed form and the double circle. It presents separately the unconditional normalization results, the exact consequences of a positive boundary identity, and the realizations of that identity. This separation fixes the logical order of the proofs. In particular, computing a representation that uses positivity cannot replace the construction that must establish it.

97.8.1 From cyclic refinement to the completed form

Let $N = 3^m$, with $m \geq 1$. The quadratic operator of the preceding chapter is retained on the cycle of N positions. If the Fourier indices are chosen in the symmetric interval

$$-\frac{N-1}{2} \leq n \leq \frac{N-1}{2},$$

its quadratic eigenvalues are

$$\lambda_{m,n} = \frac{N^2}{\pi} \sin^2\Big(\frac{\pi n}{N}\Big).$$

The coordinate π is the prior regional output of the HMT kernel. At this stage it enters the normalization of the analytic realization, not the selection of irreducibles. The constant mode is removed, and the two orientations of a nonzero mode are counted only once:

$$\Theta_m(x) = \frac{1}{2} \sum_{n \neq 0} e^{-x\lambda_{m,n}} = \sum_{n=1}^{(N-1)/2} e^{-x\lambda_{m,n}}, \qquad x > 0.$$

Proposition 97.5. Limit of the heat trace

For every $x > 0$,

$$\lim_{m \to \infty} \Theta_m(x) = \Theta_+(x) := \sum_{n=1}^{\infty} e^{-\pi n^2 x}.$$

Convergence is uniform as x ranges over a compact subset of $(0, \infty)$.

Proof. For a fixed index n, the relation $\sin y/y \to 1$ gives $\lambda_{m,n} \to \pi n^2$. Moreover, $\sin y \geq 2y/\pi$ for $0 \leq y \leq \pi/2$, whence

$$\lambda_{m,n} \geq \frac{4n^2}{\pi}.$$

If $x \geq a > 0$, each term is dominated by $e^{-4an^2/\pi}$, whose sum converges. Extending the terms by zero outside the cycle's index interval and applying dominated convergence yields the limit and its uniformity on compact sets. □

Inversion of this trace explains why the completion contains a polar contribution in addition to the positive modes. Define

$$\vartheta(x) = 1 + 2\Theta_+(x).$$

The periodization of the Gaussian

$$F_x(t) = \sum_{n\in\mathbb{Z}} e^{-\pi x(t+n)^2}$$

converges together with all its derivatives. Its Fourier coefficient of index k is $x^{-1/2}e^{-\pi k^2/x}$, obtained by integrating the Gaussian over the line. The Fourier series converges absolutely; evaluating it at $t = 0$ gives

$$\vartheta(x) = x^{-1/2}\vartheta(1/x).$$

In terms of the positive trace,

$$\Theta_+(x) = x^{-1/2}\Theta_+(1/x) + \frac{x^{-1/2} - 1}{2}.$$

The last summand preserves the difference between removing the constant mode before or after inverting the scale.

Proposition 97.6. Mellin transform and continuation

For $\Re s > 1$, the arithmetic partition function $Z(s) = \sum_{n\geq 1} n^{-s}$ satisfies

$$\Lambda(s) := \int_0^\infty \Theta_+(x)x^{s/2-1}\,dx = \pi^{-s/2}\Gamma(s/2)Z(s).$$

The function

$$\xi(s) = \frac{1}{2}s(s-1)\Lambda(s)$$

admits an entire continuation and satisfies $\xi(s) = \xi(1-s)$.

Proof. Absolute convergence for $\Re s > 1$ permits termwise integration. The substitution $u = \pi n^2 x$ gives the factor $\pi^{-s/2}\Gamma(s/2)n^{-s}$. To continue the integral, separate the intervals $(0,1)$ and $(1,\infty)$ and apply the preceding inversion to the former. This yields

$$\Lambda(s) = \frac{1}{s(s-1)} + \int_1^\infty \Theta_+(x)\left(x^{s/2-1} + x^{(1-s)/2-1}\right)\,dx.$$

The integral is entire in s: the exponential decay of $\Theta_+(x)$ uniformly dominates every power and every power of $\log x$ on compact subsets of the plane. The factor $s(s-1)$ removes the two poles displayed explicitly. The expression is invariant under $s \mapsto 1-s$, proving the functional equation. □

The Mellin transform of the trace is not simply $Z(s)$: it contains the factors $\pi^{-s/2}\Gamma(s/2)$. Keeping them explicit is indispensable, since their contributions reappear in the completed quadratic form. The product over irreducibles of the preceding chapter and this additive construction agree in the half-plane of convergence by a proved identity of functionals.

97.8.2 The arithmetic, gamma, and polar contributions

Fix $R > 0$ and consider

$$\mathcal{D}_R = C_c^\infty((-R/2, R/2)).$$

The inner product will be linear in the first variable. We use

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x)e^{-izx}\,dx, \qquad (T_a f)(x) = f(x-a),$$

$$C_{f,g}(a) = \langle f, T_a g\rangle_{L^2(\mathbb{R})}, \qquad h_{f,g}(z) = \widehat{f}(z)\overline{\widehat{g}(\overline{z})}.$$

These definitions fix both the orientation of translations and the Fourier normalization. For $f, g \in \mathcal{D}_R$, $C_{f,g}(a) = 0$ if $|a| \geq R$.

The previously constructed irreducibles supply the prime-power indices and their weights:

$$\ell_{p,k} = k \log p, \qquad w_{p,k} = (\log p) p^{-k/2}, \qquad k \geq 1.$$

The contribution from the gamma completion uses

$$\mu(a) = \frac{e^{-a/2}}{1 - e^{-2a}}, \qquad c_\Gamma = \psi(1/4) - \log \pi,$$

where $\psi = \Gamma'/\Gamma$. In particular, $c_\Gamma < 0$. For example, the monotonicity of ψ on the positive axis, obtained from $\psi'(x) = \sum_{n\geq 0}(n+x)^{-2} > 0$, together with $\psi(1) = -\gamma_{\mathrm{E}} < 0$, gives this inequality.

The completed form is

$$\begin{aligned}\mathscr{W}_R(f,g) = {} & h_{f,g}(i/2) + h_{f,g}(-i/2) + c_\Gamma C_{f,g}(0) \\ & + \int_0^\infty \mu(a)\big(2C_{f,g}(0) - C_{f,g}(a) - C_{f,g}(-a)\big)\, da \\ & - \sum_{\ell_{p,k}\leq R} w_{p,k}\big(C_{f,g}(\ell_{p,k}) + C_{f,g}(-\ell_{p,k})\big).\end{aligned}$$

The prime-power sum is finite: only $p^k \leq e^R$ occur. The form has not been replaced by a finite sampling of zeros.

Proposition 97.7. Existence and compatibility of the form

The preceding expression defines a continuous Hermitian sesquilinear form on $\mathcal{D}_R$. If $R < R'$, their restrictions agree:

$$\mathscr{W}_{R'}|_{\mathcal{D}_R \times \mathcal{D}_R} = \mathscr{W}_R.$$

Proof. Since translations are unitary,

$$2C_{f,g}(0) - C_{f,g}(a) - C_{f,g}(-a) = \langle (I - T_a)f, (I - T_a)g\rangle.$$

As $a \to 0$, each difference has norm $O(a)$, whereas $\mu(a) = O(a^{-1})$. The integrand is therefore $O(a)$. As $a \to \infty$, the differences have bounded norms and $\mu(a) = O(e^{-a/2})$. The integral converges absolutely. These same estimates, expressed in terms of the norms of f, g and their first derivatives on a fixed compact set, prove continuity. Hermitian symmetry follows by interchanging f, g and conjugating. If R is enlarged, the added prime-power terms have zero correlations on $\mathcal{D}_R$, proving compatibility. □

One can thus write a single form $\mathscr{W}$ on $\mathcal{D} = C_c^\infty(\mathbb{R}) = \bigcup_R \mathcal{D}_R$. Compatibility concerns the complete difference of contributions; it does not require each auxiliary column separately to preserve its norm when the support is enlarged.

97.8.3 Transport columns and the positive boundary identity

For a cycle of length N, the uniform-defect representation of the preceding chapter gives

$$\delta_N(T_a)^* \delta_N(T_b) = (I - T_a)^*(I - T_b).$$

This equality allows translation differences to be realized on a normalized transport sheet. To present the argument without repeating that realization, we simply write $\delta_a f = (I - T_a) f$. The polar evaluations decompose as

$$b_{\pm}(f) = \frac{\widehat{f}(i/2) \pm \widehat{f}(-i/2)}{\sqrt{2}}.$$

Define two maps on $\mathcal{D}_R$:

$$J_{+,R} f = \Big((\sqrt{w_{p,k}}\, \delta_{\ell_{p,k}} f)_{\ell_{p,k} \le R}, (\sqrt{\mu(a)}\, \delta_a f)_{a>0}, b_+(f) \Big),$$

$$J_{-,R} f = \Big((\sqrt{2 w_{p,k}}\, f)_{\ell_{p,k} \le R}, \sqrt{-c_\Gamma}\, f, b_-(f) \Big).$$

The target spaces consist of orthogonal sums of copies of $L^2(\mathbb{R})$, a direct integral in the gamma component, and a scalar component. When their ranges are used as effective spaces, the closures of $\operatorname{ran} J_{+,R}$ and $\operatorname{ran} J_{-,R}$ will be taken. The closure of a diagonal range must not be confused with the sum of all coordinates of the ambient space.

Proposition 97.8. Exact decomposition of the form

The following identity holds:

$$\mathcal{W}_R(f, g) = \langle J_{+,R} f, J_{+,R} g \rangle - \langle J_{-,R} f, J_{-,R} g \rangle.$$

Proof. For each prime power, the first column contributes

$$w_{p,k} \big(2C_{f,g}(0) - C_{f,g}(\ell_{p,k}) - C_{f,g}(-\ell_{p,k}) \big);$$

the second removes $2 w_{p,k} C_{f,g}(0)$. The gamma component is precisely the integral in Proposition 97.7. The negative constant component produces $c_\Gamma C_{f,g}(0)$. Finally,

$$b_+(f)\overline{b_+(g)} - b_-(f)\overline{b_-(g)} = h_{f,g}(i/2) + h_{f,g}(-i/2).$$

Adding these identities gives the stated expression. □

This decomposition does not by itself establish positivity: it is a difference of two positive forms. The boundary identity in the source containing the proof uses a space $\mathcal{K}_R$, an orthogonal projection P_R, and an encoder $\Xi_R : \mathcal{D}_R \to \mathcal{K}_R$ whose moments must satisfy

$$\langle \Xi_R f, \Xi_R g \rangle = \langle J_{+,R} f, J_{+,R} g \rangle,$$

$$\langle P_R \Xi_R f, P_R \Xi_R g \rangle = \langle J_{-,R} f, J_{-,R} g \rangle.$$

Theorem 97.9. Consequence of the two boundary moments

If the two preceding identities hold on $\mathcal{D}_R$, the map $E_R = (I - P_R)\Xi_R$ satisfies

$$\mathcal{W}_R(f, g) = \langle E_R f, E_R g \rangle.$$

In particular, $\mathcal{W}_R(f, f) \ge 0$.

Proof. The orthogonal decomposition of $\Xi_R f$ by P_R and $I - P_R$ gives

$$\langle \Xi_R f, \Xi_R g \rangle = \langle P_R \Xi_R f, P_R \Xi_R g \rangle + \langle E_R f, E_R g \rangle.$$

Substitute the two moments and apply Proposition 97.8. □

The significance of the theorem lies in identifying the arithmetic form with a boundary norm. The decisive step in its application is the encoder's action and the proof of both moments; the orthogonality of a projection does not impose them on an arbitrary form. This edition's provenance appendix records the preserved source and the point at which it is materially incorporated. That action is not replaced by an operator defined retrospectively from the positivity to be established.

There is a second conservation law, useful once E_R has been constructed. The incidence matrix

$$B = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-$$

has orthogonal projections

$$P_\pm = \frac{1}{2}\begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}.$$

Its normalization gives $A = P_+ + qP_-$, with $q = 5/9$. If $D = \sqrt{1-q^2}\,P_-$, then $A^*A + D^*D = I$. Iteration gives

$$\|x\|^2 = \sum_{n=0}^{M-1} \|DA^n x\|^2 + \|A^M x\|^2.$$

On the sheet $P_- x = x$, the terminal term is $q^{2M}\|x\|^2$; consequently, the map

$$Lx = (DA^n x)_{n\geq 0}$$

is isometric on that sheet. Telescoping explains conservation through the distribution of losses; membership of $E_R f$ in the sheet must come from the preceding construction. Eliminating a terminal term does not retrospectively select the form from which it started.

Internal history encoder and its two finite moments

The correlative transport of the TPK already allows an encoder to be constructed before its residual is defined. Fix a horizon $N \geq 1$. At each level, let $\mathscr{H}_k$ be the orthogonal sum of the two sheet fibres over the complete histories at that level. If $d(x)$ counts the outgoing emissions from a history x, transport acts by

$$U_k \iota_x v = \sum_{\alpha : x \to y} \frac{1}{\sqrt{d(x)}}\, \iota_y U_\alpha v.$$

Here we consider the isometric realization of sheet transport: $U_\alpha^* U_\alpha = I$. Distinct histories have distinct descendants because the complete route is preserved. Therefore,

$$U_k^* U_k\big|_{\mathscr{H}_x} = \frac{1}{d(x)} \sum_{\alpha : x \to y} U_\alpha^* U_\alpha = I.$$

Consistency of the weights and preservation of memory are developed in Section 6.5 and applied to spectral transport in Subsection 97.8.7. The isometry of each sheet transport is an explicit sufficient condition for this equality; normalization of the weights alone does not prove it.

Let P_k, Q_k be the orthogonal sheet projections, with $P_k + Q_k = I$. The sheet-preserving and sheet-changing parts of transport are

$$A_k = P_{k+1} U_k P_k + Q_{k+1} U_k Q_k, \qquad \eta_k = Q_{k+1} U_k P_k + P_{k+1} U_k Q_k.$$

Block calculation gives exactly

$$A_k^* A_k + \eta_k^* \eta_k = P_k U_k^* U_k P_k + Q_k U_k^* U_k Q_k = I.$$

The first equality holds for any bounded transport. The second uses the indicated isometric realization. In general, the left-hand side is the sheet-diagonal part of the Gram operator of U_k, not necessarily its full Gram operator.

Define $F_{0,0} = I$, $F_{0,k} = A_{k-1} \cdots A_0$ for $k \geq 1$, and $T_N = F_{0,N}^* F_{0,N}$. The internal encoder has domain and action

$$\Xi_N^{\mathrm{int}} : \mathscr{H}_0 \longrightarrow \mathscr{K}_N^{\mathrm{int}} := \left(\bigoplus_{k=0}^{N-1} \mathscr{H}_{k+1} \right) \oplus \mathscr{H}_N,$$

$$\Xi_N^{\mathrm{int}} v = \big(\eta_k F_{0,k} v \big)_{0 \leq k < N} \oplus F_{0,N} v.$$

The last summand is kept separate even when its underlying space coincides with that of an earlier output. Let Π_N^{term} be the orthogonal projection onto that terminal summand. We obtain the two internal moments

$$(\Xi_N^{\mathrm{int}})^* \Xi_N^{\mathrm{int}} = I, \qquad (\Xi_N^{\mathrm{int}})^* \Pi_N^{\mathrm{term}} \Xi_N^{\mathrm{int}} = T_N.$$

Proof. The block identity implies

$$F_{0,k}^* F_{0,k} - F_{0,k+1}^* F_{0,k+1} = F_{0,k}^* \eta_k^* \eta_k F_{0,k}.$$

Summing from $k = 0$ to $N - 1$ yields $I = \sum_{k<N} F_{0,k}^* \eta_k^* \eta_k F_{0,k} + T_N$, which is the first announced Gram. The terminal projection retains only $F_{0,N} v$, so its Gram is T_N. Thus, the complement $(I - \Pi_N^{\mathrm{term}}) \Xi_N^{\mathrm{int}}$ records the sheet outputs and has Gram $I - T_N$. All these operators come from the same family of transports; the encoder is not defined from a residual whose positivity is assumed beforehand. □

This result recovers an effective encoder with its two moments, within the joint constructor that preserves the other four readouts of the continuum. Applying it to the completed columns of this section entails a precise additional requirement: one would have to identify a prior action $R_R : \mathscr{D}_R \to \mathscr{H}_0$, with the horizon compatibility required by the test domain, for which

$$\langle R_R f, R_R g \rangle = \langle J_{+,R} f, J_{+,R} g \rangle,$$

$$\langle F_{0,N} R_R f, F_{0,N} R_R g \rangle = \langle J_{-,R} f, J_{-,R} g \rangle.$$

The composition $\Xi_N^{\mathrm{int}} R_R$ would then have the moments of Theorem 97.9. This identification of R_R with the prime-power, gamma, and polar columns is not assembled here. The preceding internal construction is assembled here, with its action and proof; it is not presented as a proof of global positivity of $\mathscr{W}$. Nor is T_N suppressed, or its limit asserted to be zero.

Horizon compatibility and the type of the projection

The terminal projection Π_N^{term} selects the last summand of the history encoder. By contrast, the uniform expectation P_R of Theorem 97.9 selects the uniform component of a nine-position calendar. Equality of their roles in a composition must be proved through the concrete map between those spaces; both are orthogonal projections, but that common property does not identify their ranges.

Dependence on the horizon likewise imposes a verifiable identity. Let R be one and the same map and J_- a fixed column. If, for two consecutive horizons, one requires

$$R^* T_N R = J_-^* J_- = R^* T_{N+1} R,$$

then

$$\eta_N F_{0,N} R = 0.$$

Proof. Subtracting the equalities and using the telescoping identity gives

$$0 = R^*(T_N - T_{N+1})R = (\eta_N F_{0,N} R)^*(\eta_N F_{0,N} R).$$

Evaluation on each vector implies the stated vanishing. If equality of the moments holds with the same R along an unbounded sequence of horizons and T_N tends strongly to zero, one also obtains $J_- = 0$. □

Therefore, an identification with a nonzero negative column must specify its horizon or its compatible family of maps $R_{R,N}$, or retain the limit terminal that actually corresponds. The contraction of an output residual cannot be transferred without that distinction to the complete column representing the second moment. This clarification keeps the recovered encoder intact and determines what additional compatibility its analytic realization must verify.

97.8.4 Representation of translations

Suppose that a positive identity has been established on all of $\mathcal{D}$,

$$\mathscr{W}(f, g) = \langle Ef, Eg \rangle$$

compatible with the support domains. The spectral space is constructed from the form itself:

$$\mathcal{N} = \{f \in \mathcal{D} : \mathscr{W}(f, f) = 0\}, \qquad \mathcal{H}_{\mathscr{W}} = \overline{\mathcal{D}/\mathcal{N}}.$$

The Cauchy–Schwarz inequality for a positive form proves that the quotient inner product is well defined. The map $[f] \mapsto Ef$ extends to a unitary isomorphism from $\mathcal{H}_{\mathscr{W}}$ onto $\overline{\operatorname{ran} E}$.

Proposition 97.10. Unitary evolution and generator

Translations induce a strongly continuous unitary group

$$U_t[f] = [T_t f].$$

Its self-adjoint generator H, with the convention $U_t = e^{itH}$, acts on classes of test functions by

$$H[f] = [if'].$$

Proof. Correlations are invariant under a common translation of both arguments. Moreover, $\widehat{T_t f}(z) = e^{-izt}\widehat{f}(z)$; in $h_{T_t f, T_t g}(z)$ the factors also cancel for complex z. Hence $\mathscr{W}(T_t f, T_t g) = \mathscr{W}(f, g)$, and translation preserves $\mathcal{N}$. It defines an isometry on the quotient, inverted by U_{-t}, which extends to the complete space.

As $t \to 0$, $T_t f - f$ tends to zero in every test-function seminorm on a common compact set. The continuity proved for $\mathscr{W}$ implies strong continuity on the dense quotient; unitarity extends it to the whole space. Stone's theorem supplies a self-adjoint generator. The derivative at $t = 0$ is $[-f']$, which agrees with $iH[f]$. □

The test functions also form a core for the generator. This can be seen by regularizing a vector through $\int \chi(t) U_t \, dt$, with $\chi \in C_c^\infty(\mathbb{R})$. This regularization and its derivative are bounded operators, controlled by $\|\chi\|_1$ and $\|\chi'\|_1$. The regularized vectors are first approximated by classes of test functions, and the identity is then approximated with a family of regularizers. The approximation obtained is in the graph norm $\|x\| + \|Hx\|$, which proves the assertion.

The representation has been constructed from a positive form and its translations. This argument uses neither a list of zero ordinates nor an assignment of zeros to calendar turns as inputs.

97.8.5 Analytic identification and multiplicities

The preceding explicit form is recognized as the completed Weil form. With

$$\Phi_f(s) = \widehat{f}\big(i(s-1/2)\big), \qquad \rho^\sharp = 1-\overline{\rho},$$

the explicit formula supplies its spectral expression

$$\mathscr{W}(f,g) = \sum_\rho \Phi_f(\rho)\overline{\Phi_g(\rho^\sharp)},$$

in terms of the nontrivial zeros of the completed function and their multiplicities. The global formulation of Weil's criterion states that positivity for all test functions is equivalent to $\Re\rho = 1/2$ for all those zeros. We use the criterion in the formulation of Bombieri [250]. The global condition cannot be replaced by positivity of finitely many matrices.

Mellin–Fourier test-function dictionary

To make the normalization of that recognition explicit, let $\mathcal{D} = C_c^\infty(\mathbb{R};\mathbb{C})$. The map

$$(\mathcal{T}_{\log} f)(x) = x^{-1/2} f(\log x), \qquad x > 0,$$

is a bijection onto $C_c^\infty((0,\infty);\mathbb{C})$, with inverse

$$(\mathcal{T}_{\log}^{-1} u)(t) = e^{t/2} u(e^t).$$

Indeed, the exponential change of variables takes compact supports to compact sets bounded away from zero and infinity. The formulas are inverses and their factors are smooth. Defining $\widehat{u}_{\mathrm{Mel}}(s) = \int_0^\infty u(x)x^{s-1}\,dx$, the substitution $x = e^t$ gives, for every $s \in \mathbb{C}$,

$$\widehat{\mathcal{T}_{\log} f}_{\mathrm{Mel}}(s) = \int_{\mathbb{R}} f(t) e^{(s-1/2)t}\,dt = \widehat{f}\big(i(s-1/2)\big) = \Phi_f(s).$$

Conjugation is preserved exactly:

$$\widehat{\overline{\mathcal{T}_{\log} g}}_{\mathrm{Mel}}(1-s) = \overline{\Phi_g(1-\overline{s})}.$$

Thus the two factors in the spectral summand agree with those in the explicit formula in multiplicative coordinates, with no residual factor and with the same multiplicities counted. The sum converges absolutely: integration by parts gives rapid decay of the transforms in vertical strips, and the zero count is $O(T\log T)$. The bijection transfers the quantifier over all test functions, not a selection of them. The explicit formula and the difficult implication of the criterion are the cited antecedents; the preceding change of variables proves the correspondence of their domains and normalizations, not a new positivity theorem. This compact-support domain is not asserted to be invariant under the integral Cayley operator: its subsequent realization uses the exponential space of Subsection 97.10.10.

Once those antecedents are satisfied, writing $\rho = 1/2 + i\gamma$ gives

$$\mathscr{W}(f,g) = \sum_{\gamma\in\Gamma_\xi} m_\gamma \widehat{f}(-\gamma)\overline{\widehat{g}(-\gamma)}.$$

Here Γ_ξ contains the distinct ordinates and m_γ is the multiplicity of the corresponding zero. Let

$$\mu_\xi = \sum_{\gamma\in\Gamma_\xi} m_\gamma \delta_\gamma.$$

The map

$$\mathcal{F}_\xi[f](\gamma) = \widehat{f}(-\gamma)$$

is an isometry from $\mathcal{H}_{\mathscr{W}}$ into $L^2(\Gamma_\xi, \mu_\xi)$. In this representation, U_t acts as multiplication by $e^{it\gamma}$ and H as multiplication by γ.

Proposition 97.11. Exhaustiveness of the atomic representation

The map $\mathcal{F}_\xi$ is surjective.

Proof. Its closed range is invariant under multiplication by $e^{it\gamma}$ for positive and negative t; it is therefore reducing. For a fixed ordinate γ_0, the averages

$$\frac{1}{2T}\int_{-T}^{T} e^{-it\gamma_0} U_t \, dt$$

converge strongly to the projection onto the atom γ_0. This is checked first coordinatewise and then by dominated convergence in the sequence space. There exists $f \in C_c^\infty$ with $\widehat{f}(-\gamma_0) \neq 0$; for example, it suffices to modulate a function with nonzero integral. The range thus contains the vector of that atom. Since it contains every finitely supported vector and is closed, it equals the whole space. □

The measure preserves arithmetic multiplicity as a weight. Its relation to operator multiplicity should be specified: the scalar space $L^2(\Gamma_\xi, \mu_\xi)$ has a one-dimensional eigenspace for each atom, even if $m_\gamma > 1$. Therefore, if a function F produces a trace-class operator,

$$\operatorname{Tr}(F(H)) = \sum_{\gamma \in \Gamma_\xi} F(\gamma),$$

whereas the sum preserving the weights of the zeros is

$$\int F(\gamma)\, d\mu_\xi(\gamma) = \sum_{\gamma \in \Gamma_\xi} m_\gamma F(\gamma).$$

This second sum can be represented as a weighted trace $\operatorname{Tr}(M_m F(H))$, when the product is trace class, where M_m is multiplication by m_γ. A different construction would place a fibre of dimension m_γ at each atom. These two representations are not interchanged without an explicit declaration.

97.8.6 The two circles and spectral orientation

The first circle is a compact coordinate for the spectral line. For a real number γ, define

$$\tau_\gamma = \frac{\gamma + i/2}{\gamma - i/2}.$$

Its modulus is one and $\tau_\gamma \neq 1$. The inverse formulas are

$$\gamma = \frac{i}{2}\frac{\tau_\gamma + 1}{\tau_\gamma - 1}, \qquad \gamma = \frac{1}{2}\cot\frac{\theta}{2} \quad \text{if} \quad \tau_\gamma = e^{i\theta}.$$

If $s = 1/2 + i\gamma$, also

$$\tau_\gamma = \frac{s-1}{s}, \qquad s = \frac{1}{1-\tau_\gamma}.$$

Passing to the phase does not lose the ordinate, provided the point on the circle and the orientation convention are retained.

For the preceding self-adjoint generator, functional calculus defines the unitary operator

$$C = (H + iI/2)(H - iI/2)^{-1}.$$

The equality $C^*C = I$ follows because both factors are functions of the same self-adjoint operator and $|(\gamma + i/2)/(\gamma - i/2)| = 1$. The point 1 does not correspond to a finite ordinate, but it may belong to the spectrum of C as an accumulation point when H is unbounded.

The second circle is the dual of integer memory: a phase τ is associated with the character

$$\chi_\tau(n) = \tau^n, \qquad n \in \mathbb{Z}.$$

The nonadic connection returns the visible phase to its initial position and increases the memory of a turn by one unit. The spectral realization assigns the action of C to that unit; on an eigenvector with ordinate γ, the memory of n turns acts by τ_γ^n. The first circle encodes the ordinate; the second expresses how its phase acts on additive memory. These are two related mathematical roles, not two lists to be identified by their ordinal positions.

97.8.7 Cylindrical refinement with memory

The connection must act on the structure being refined. In the common TPK tree, a vertex is a finite history of admissible emissions; two histories reaching the same terminal state remain distinct when their transport records differ. If $\mathrm{Out}(x)$ is the finite set of outgoing emissions, counted with multiplicity, write

$$d(x) = |\,\mathrm{Out}(x)| \geq 1.$$

Neutral prolongation preserves nonemptiness of the levels. The local weight specified by the tree is $1/d(x)$. For a history $h = (\epsilon_1, \ldots, \epsilon_k)$, with successive states $x_0, \ldots, x_k$, its weight is

$$\mu_k(h) = \prod_{j=0}^{k-1} \frac{1}{d(x_j)}.$$

The sum of the weights of all its children is

$$\sum_{\epsilon \in \mathrm{Out}(x_k)} \mu_{k+1}(h\epsilon) = \mu_k(h) \sum_{\epsilon \in \mathrm{Out}(x_k)} \frac{1}{d(x_k)} = \mu_k(h).$$

Starting from the root of weight one, this identity proves normalization and projective consistency by induction. If a reversible symmetry transports the outgoing emissions bijectively with their multiplicities, it preserves $d(x)$, all these weights, and their cylinder measures. Thus the measure used below comes from the declared branching rule, before an operator is assigned to its fibres. It is not confused with a uniform distribution on terminal states that would identify different histories.

In general form, let $\mathcal{Z}_k$ be a finite level of cylindrical states and let $\pi_k : \mathcal{Z}_{k+1} \to \mathcal{Z}_k$ be its truncation map. Suppose a family of compatible positive weights is given,

$$\mu_k(z) = \sum_{\pi_k(z')=z} \mu_{k+1}(z'), \qquad \sum_{z \in \mathcal{Z}_k} \mu_k(z) = 1.$$

The weights belong to the state measure being transported. The shape of a visible numerical cylinder does not by itself suffice to reconstruct them.

Let $q_k : \mathcal{Z}_k \to \mathbb{Z}$ be the memory coordinates. The increment of a prolongation is

$$m(z', z) = q_{k+1}(z') - q_k(z).$$

Concatenating prolongations satisfies the cocycle law exactly:

$$m(z'', z) = m(z'', z') + m(z', z).$$

Define

$$w(z', z) = \sqrt{\frac{\mu_{k+1}(z')}{\mu_k(z)}}, \qquad \mathcal{K}_k = \ell^2(\mathcal{Z}_k) \otimes \mathcal{H}_{\mathscr{W}}.$$

Transport with memory is

$$\widetilde{U}_k(e_z \otimes v) = \sum_{\pi_k(z')=z} w(z', z) e_{z'} \otimes C^{m(z',z)} v.$$

Proposition 97.12. Isometry and composition of transport

Each $\widetilde{U}_k$ is isometric. Composition across several levels has the same form, with the weight ratio between the endpoint levels and the total memory increment.

Proof. Two distinct parents have disjoint sets of descendants. For a fixed parent, the powers of C are unitary and $\sum_{z' \mapsto z} w(z', z)^2 = 1$. These are exactly the identities that preserve inner products of the basis vectors. In a composition, the product of weights telescopes:

$$\sqrt{\frac{\mu_{k+2}(z'')}{\mu_{k+1}(z')}} \sqrt{\frac{\mu_{k+1}(z')}{\mu_k(z)}} = \sqrt{\frac{\mu_{k+2}(z'')}{\mu_k(z)}}.$$

The powers of C add their exponents by the cocycle law. Repetition proves the assertion for any finite number of levels. □

The same identity admits an interpretation through a change of representation. If U_k denotes transport without memory and

$$G_k(e_z \otimes v) = e_z \otimes C^{q_k(z)} v,$$

then

$$\widetilde{U}_k = G_{k+1}(U_k \otimes I) G_k^*.$$

The memory coordinate is not lost in formulating the isometry: it appears at both ends of the transport. This equality also shows that the twisted representation uses an already specified operator C. By itself, it does not identify an arbitrary prior native transport with that operator; identification requires composing their maps and checking their actions on the same domain.

Proposition 97.13. Nonadic return on the compatible section

Define

$$\Omega_k = \sum_z \sqrt{\mu_k(z)} e_z, \qquad J_k v = \Omega_k \otimes v.$$

If every return trajectory from k to $k + 9$ has memory increment one, then

$$\widetilde{U}_{\Gamma,k} J_k = J_{k+9} C.$$

For r consecutive turns,

$$\widetilde{U}_{\Gamma,k+9(r-1)} \cdots \widetilde{U}_{\Gamma,k} J_k = J_{k+9r} C^r.$$

Proof. Multiplying the weight of each descendant by the coefficient $\sqrt{\mu_k(z)}$ of its parent leaves $\sqrt{\mu_{k+9}(z')}$. The condition on the increment allows the same operator C to be factored out of each summand. The first equality is obtained. The second follows by induction. □

The level indices are essential. It is the phase that returns, while the section extends to another level; the formula does not reset the state in the previous space.

The memory condition in this proposition has a concrete action in TPK chronology. For a phase $r \in \mathbb{Z}/9\mathbb{Z}$, with representative $\bar{r} \in \{0, \dots, 8\}$, and a memory depth $w \in \mathbb{Z}_{\geq 0}$, each emission updates

$$\sigma_9(w, r) = \left(w + \left\lfloor \frac{\bar{r} + 1}{9} \right\rfloor, r + [1]_9 \right).$$

After n emissions, the number of crossings from representative 8 to representative 0 is $\lfloor (\bar{r} + n)/9 \rfloor$. Therefore,

$$\sigma_9^n(w, r) = \left(w + \left\lfloor \frac{\bar{r} + n}{9} \right\rfloor, r + [n]_9 \right), \qquad \sigma_9^9(w, r) = (w + 1, r).$$

Taking $q_k = w_k$ recovers the preceding cocycle and the unit increment of each turn. Even a neutral emission on the APP sheets records an extension in this chronology. The calculation does not by itself impose the spectral operator C: it determines the memory on which the realization acts, once its intertwining has been established.

97.8.8 Nonadic calendar and contraction over one turn

The finite calendar is a publication of this transport. On $\mathbb{C}^9 \otimes \mathcal{H}_{\mathscr{W}}$, let

$$S_C = \sum_{r=1}^{8} |r + 1\rangle\langle r| \otimes I + |1\rangle\langle 9| \otimes C.$$

The unitarity of C makes S_C unitary, and the complete turn satisfies

$$S_C^9 = I_{\mathbb{C}^9} \otimes C.$$

Each vector crosses the oriented seam containing C exactly once; this is why C^9 does not appear.

Let $u = 9^{-1/2}(1, \dots, 1)$ and $\iota v = u \otimes v$. The uniform compression of one step is

$$A_C = \iota^* S_C \iota = \frac{8I + C}{9}.$$

A direct calculation gives

$$I - A_C^* A_C = \frac{8}{81}(I - C)^*(I - C).$$

The defect of a compressed unitary transport measures the component leaving the uniform state. It is not a loss of unitarity of the complete calendar.

The contraction $q = 5/9$ of the complementary sheet and the calendar are distinct operations. If q corresponds to one nonadic turn, the operator per turn is

$$M_{\text{vuelta}} = q S_C^9 = q(I \otimes C).$$

It can be distributed uniformly over the nine steps by

$$M_{\text{paso}} = q^{1/9} S_C, \qquad M_{\text{paso}}^9 = M_{\text{vuelta}}.$$

By contrast, taking qS_C as the operator at each step produces $q^9(I \otimes C)$ upon completing the turn. The two normalizations describe different contractions and must not be identified. At an ordinate γ, r turns in the first regime produce

$$q^r \tau_\gamma^r.$$

The modulus records extinction of the terminal term; the phase records the action on memory. Knowing only the modulus does not determine the spectral ordinates.

97.8.9 Moment matrices and spectral approximation

The preceding realization allows approximations to be constructed from test functions, not from assigned spectral values. Choose $f_1, \dots, f_d \in \mathcal{D}$ whose classes are linearly independent in $\mathcal{H}_{\mathscr{W}}$. With $e_j = [f_j]$, define

$$G_{ij} = \langle e_j, e_i \rangle, \qquad K_{ij} = \langle He_j, e_i \rangle, \qquad Q_{ij} = \langle He_j, He_i \rangle.$$

The index convention ensures that, for $x = \sum_j v_j e_j$, one has $\|x\|^2 = v^* G v$. G is positive-definite Hermitian, and K, Q are Hermitian. If the initial classes were dependent, the kernel of G would have to be removed before inverting it.

Proposition 97.14. Galerkin approximation and residual

The generalized problem

$$Kv = \gamma G v$$

is equivalent to the ordinary Hermitian problem for $\widehat{H} = G^{-1/2} K G^{-1/2}$. For any $\gamma \in \mathbb{R}$ and $v \neq 0$, the spectral residual of the corresponding vector satisfies

$$\frac{\|(H - \gamma)x\|^2}{\|x\|^2} = \frac{v^*(Q - 2\gamma K + \gamma^2 G)v}{v^* G v}.$$

In particular,

$$\operatorname{dist}(\gamma, \sigma(H)) \leq \frac{\|(H - \gamma)x\|}{\|x\|}.$$

Proof. The change $w = G^{1/2} v$ transforms the generalized equation into $\widehat{H} w = \gamma w$. Expanding the residual norm gives the quadratic identity because H is symmetric on the vectors used. The final inequality follows from the spectral measure of H:

$$\|(H - \gamma)x\|^2 = \int |\lambda - \gamma|^2 \, d\langle E_H(\lambda)x, x \rangle.$$

The integrand is bounded below by the squared distance to the spectrum. □

The phase matrix

$$\widehat{C} = (\widehat{H} + iI/2)(\widehat{H} - iI/2)^{-1}$$

is unitary for the Euclidean inner product. If $G^{-1}K$ is used instead in the original basis, the corresponding unitarity is expressed relative to G. These are two bases for the same problem, not two norms interchangeable without transformation.

A small residual approximates the spectrum of the operator already constructed. By itself, it does not count all its eigenvalues, prove global positivity of $\mathscr{W}$, or certify decimal digits affected by quadrature

errors. For a validated bound, the entries of G, K, Q must be accompanied by error bounds and a positive lower bound for the form $v^{*}Gv$.

97.8.10 Compact resolvent and exhaustiveness checks

In the atomic realization, the ordinates form a discrete set and escape to infinity. Therefore,

$$R = (H - iI)^{-1}$$

is compact: in the atomic basis it acts by $(\gamma - i)^{-1}$, which tends to zero. It is also recovered directly from the translation group through

$$R = i \int_0^{\infty} e^{-t} U_{-t}\, dt.$$

The integral over a finite interval is defined in the strong sense. Its tail from T has operator norm at most e^{-T}, so the truncated integrals converge in norm. Functional calculus reduces the identity to $i/(1 + i\gamma) = (\gamma - i)^{-1}$.

Proposition 97.15. Norm convergence of compressions

If P_d are finite-rank orthogonal projections with $P_d \to I$ strongly, then

$$\|R - P_d R P_d\| \longrightarrow 0.$$

Proof. The strong convergence of P_d, together with the bound $\|P_d\| \le 1$, is uniform on every compact set. Apply it to the compact closure of the image of the unit ball under R, obtaining $\|(I - P_d)R\| \to 0$. Applying the same argument to R^{*} yields $\|R(I - P_d)\| \to 0$. The decomposition

$$R - P_d R P_d = (I - P_d)R + P_d R(I - P_d)$$

completes the proof. □

This convergence permits checks along resolvent contours. If a closed curve C separates a group of nonzero eigenvalues and the norm of the compression error is less than the reciprocal of a bound for the approximate resolvent on C, the Neumann series preserves invertibility along the curve. The Riesz projections of the two operators then have the same rank. This is an exhaustiveness check in a delimited spectral region, additional to a vector residual.

The multiplicity counted by that rank is the multiplicity of the represented operator. In the scalar space of Subsection 97.8.5, it must not be replaced by the weight m_{γ} without explicitly introducing the weighted observable.

97.8.11 From internal identities to the analytic assembly

The proofs of the heat and Mellin normalization, the explicit construction of the completed form, its column decomposition, cylindrical transport with memory, the two circles, the calendar, and the spectral checks have been assembled here. The arithmetic construction of irreducibles precedes their appearance in the prime-power weights. Memory comes from compatible prolongations; it is not replaced by a phase digit.

Subsection 97.8.3 also assembles a finite internal encoder constructed from history transports, together with the proof of its moments I and T_N in the isometric realization. Applying Theorem 97.9 to the native encoder of the completed form retains a more specific identification: the prior action R_R on test functions and equality of those moments with the moments of $J_{+,R}$ and $J_{-,R}$. The source of the completed form states those identities and develops their consequences. This exposition preserves those consequences with their explicit antecedents and the internal composition actually recovered; it does not replace the

remaining link with a merely nominal formulation. The distinction concerns the proof assembled in this article, not an assertion that the result is absent from the corpus as a whole.

The complete link between all APP–TRIT–TPK histories and the word normalization of the first chapter also retains its integration requirement. The five consubstantial constructions of the continuum remain part of the common kernel; this chapter presents their spectral readout and does not redefine them as independent structures selected by conventional problems.

Identifying the global form with the explicit formula and Weil's criterion fixes the exact scope of any eventual terminal conclusion. Finite checks of the native producer and spectral approximations are distinct verifiable results. Accordingly, this working edition does not present the chapter as an already completed, self-contained proof of the Riemann hypothesis.

97.8.12 Bidirectional recovery and assembly of the readouts

The gamma information recovers the test function that produced it. This property provides an explicit action between the complete columns before the sign of their difference is known. The following development extends the integration status described in Subsections 97.8.3 and 97.8.11: the global bounded connector is recovered; its contractivity is studied on the same action and the same range, not on another operator.

The construction follows the prime-power weights, the completed trace, and the transports already defined. It does not change the selection of irreducibles or receive zeros as inputs. The sources of the gamma tail, the basal domain, and the prior folding are preserved in full in the provenance package for this integration.

Proposition 97.16. Left inverse of the tail and global connector

Let $H_R = L^2(I_R)$, identified with its zero extension to the line, and let M_R be orthogonal restriction to I_R. Define

$$a_R = 2\int_R^\infty \mu(a)\,da, \qquad (B_R f)(a) = \sqrt{\mu(a)}(I - T_a)f \quad (a \geq R).$$

Then B_R is bounded and $B_R^* B_R = a_R I$. If $\pi_{\Gamma,\geq R}$ extracts the gamma tail from the positive column, the map

$$L_R = a_R^{-1} B_R^* \pi_{\Gamma,\geq R}$$

satisfies $L_R J_{+,R} f = f$ for $f \in \mathcal{D}_R$, and $L_R L_R^* = a_R^{-1} I$ on H_R. The first identity extends to the joint closed domain of the readouts; it does not assert that the complete gamma column is defined on all of $L^2(I_R)$. In particular,

$$C_R = J_{-,R} L_R, \qquad C_R J_{+,R} = J_{-,R}$$

is a bounded connector for every $R > 0$.

Proof. For $a \geq R$, the supports of $f, g \in H_R$ and of their translates are disjoint, apart from endpoints of measure zero. Therefore

$$\langle (I - T_a)f, (I - T_a)g \rangle = 2\langle f, g \rangle.$$

Integration gives the first Gram operator. The adjoint has the explicit action

$$B_R^* F = M_R \int_R^\infty \sqrt{\mu(a)}(I - T_{-a})F(a)\,da.$$

The integral converges in norm by Cauchy–Schwarz; the tail of μ is integrable and $T_a^* = T_{-a}$. Applying the adjoint to the tail of $J_{+,R}f$ returns $a_R f$. Extraction is a coisometry onto its summand, whence $L_R L_R^* = a_R^{-2} B_R^* B_R = a_R^{-1} I$. The negative column is bounded on H_R: it has finitely many vector-valued rows and a continuous polar functional on that interval. Composition completes the argument. □

The inversion uses the conjugate translations before publishing the recovered function. The owner also provides a finite-band version: with $A_R = [R+1, R+2]$ and $\nu_R = \int_{A_R} \mu(a)\,da$,

$$\mathfrak{R}_R F = \nu_R^{-1} \int_{A_R} \sqrt{\mu(a)} M_R F(a)\,da, \qquad \mathfrak{R}_R D_\Gamma f = f.$$

The identity follows from $M_R T_a f = 0$ on that band. Both inverses precede a Weil inequality. The first allows the ambient norm of its connector to be computed exactly.

Proposition 97.17. Connector norm and basal factorization

Write

$$s_R^2 = 2 \sum_{\ell_{p,k} \le R} w_{p,k} - c_\Gamma, \qquad \beta_R = 2\sinh(R/2) - R.$$

Then

$$\|C_R\|^2 = \frac{s_R^2 + \beta_R}{a_R}.$$

Consequently, if $0 < h < \log 2$ and

$$2M_\Gamma(h) \ge -c_\Gamma + 2\sinh(h/2) - h, \qquad M_\Gamma(h) = \int_h^\infty \mu(a)\,da,$$

there exists a dilation realizing the two moments of Theorem 97.9 on all functions in $\mathcal{D}_h$.

Proof. The negative column has Gram operator $J_-^* J_- = s_R^2 I + b_-^* b_-$. The functional

$$b_-(f) = \sqrt{2} \int_{-R/2}^{R/2} f(x) \sinh(x/2)\,dx$$

has squared norm $2\int_{-R/2}^{R/2} \sinh^2(x/2)\,dx = \beta_R$. The Gram operator has norm $s_R^2 + \beta_R$. The preceding identity $L_R L_R^* = a_R^{-1} I$ gives $C_R C_R^* = a_R^{-1} J_- J_-^*$, proving the formula.

For $R = h < \log 2$, no prime-power lengths occur in the sum. The stated inequality proves $\|C_h\| \le 1$. One can construct, in this order,

$$D_h^{\mathrm{def}} = (I - C_h^* C_h)^{1/2}, \qquad \Xi_h f = J_{-,h} f \oplus D_h^{\mathrm{def}} J_{+,h} f.$$

The sum of the Gram operators of C_h and its defect is the identity. Since $C_h J_{+,h} = J_{-,h}$, the first moment of Ξ_h is $J_{+,h}^* J_{+,h}$; projection onto the first summand has moment $J_{-,h}^* J_{-,h}$. Theorem 97.9 gives positivity. The square root of the defect is taken after contractivity has been proved, not as a substitute for that proof.

If the window is enlarged to $R \ge h$, each added prime channel satisfies $\|(I - T_\ell) f\|^2 = 2\|f\|^2$ on H_h. The map from that image to $\sqrt{2} f$ is an isometry, extended by zero on the complement of its closed range. The direct sum of those rows with the gamma–polar connector retains contractivity on the same basal domain. □

The basal endpoint is determined by

$$2M_\Gamma(h_*) = -c_\Gamma + 2\sinh(h_*/2) - h_*.$$

The substitution $t = e^{-a/2}$ gives

$$M_\Gamma(h) = \operatorname{arctanh}(e^{-h/2}) + \arctan(e^{-h/2}).$$

The left-hand side of the equation decreases strictly from infinity to zero; the right-hand side is positive and increases strictly for $h > 0$. There is a unique positive root. The source evaluates it as $h_* = 0.08562601762973687068\ldots < \log 2$. One may work with any $h < h_*$ certified by inequalities, without needing the decimal expansion of that endpoint.

The complete form also preserves translation of test functions. On the polar ports, translation acts by

$$\begin{pmatrix} b_+(T_t f) \\ b_-(T_t f) \end{pmatrix} = \begin{pmatrix} \cosh(t/2) & \sinh(t/2) \\ \sinh(t/2) & \cosh(t/2) \end{pmatrix} \begin{pmatrix} b_+(f) \\ b_-(f) \end{pmatrix}.$$

The matrix preserves the difference of squares, and the other rows are translated unitarily. The basal domain may therefore be placed in any translated interval of length h.

Proposition 97.18. Uniqueness on the effective range

Let $\mathcal{M}_{+,R} = \overline{J_{+,R}\mathcal{D}_R}$. The restriction of the constructed connector to this subspace is unique among bounded maps with that output identity. Moreover,

$$\|C_R|_{\mathcal{M}_{+,R}}\| \le 1 \quad \Longleftrightarrow \quad \mathscr{W}_R(f,f) \ge 0 \quad (f \in \mathcal{D}_R).$$

Proof. Two bounded operators agreeing on the dense range agree on its closure. The equivalence follows from $C_R J_{+,R} f = J_{-,R} f$ and the norm identity; the inequality extends to the closure by continuity. □

The global formula is therefore constructed:

$$\mathscr{W}_R(f,g) = \langle J_{+,R} f, (I - C_R^* C_R) J_{+,R} g \rangle.$$

Changing the ambient representation may facilitate computation of the norm, but it does not permit changing its action on test functions. Failure of contractivity on the entire ambient space does not refute positivity on the complete range, where the row correlations remain. Projection onto that range does not automatically prove the bound either.

Proposition 97.19. Prior folding and all cross products

Fixing a basal fiber, let $m = 9^q$ and ℓ_j be the shifts published by its routes. On m copies of $L^2(\mathbb{R})$, the folding and its uniform publication are

$$\mathcal{F}_q x = \sum_{j=1}^{m} T_{\ell_j} x_j, \qquad \mathsf{S}_q x = u_q \otimes \mathcal{F}_q x = 3^q P_q \mathsf{T}_q x, \qquad u_q = m^{-1/2}(1,\ldots,1).$$

For every linear readout J defined on those test functions,

$$\|J\mathcal{F}_q x\|^2 = \sum_{i,j} \langle J T_{\ell_i} x_i, J T_{\ell_j} x_j \rangle.$$

Moreover, $\mathcal{F}_q$ is surjective, with right inverse

$$(\mathcal{E}_q f)_j = m^{-1} T_{-\ell_j} f, \qquad \mathcal{F}_q \mathcal{E}_q = I, \qquad \|\mathcal{E}_q f\|^2 = m^{-1}\|f\|^2.$$

Proof. Averaging the m routes and multiplying by $\sqrt{m} = 3^q$ gives the expression for S_q. Expanding the inner product retains every cross term. Substituting $\mathcal{E}_q f$ produces m copies of f/m; unitarity of the translations gives the norm. □

If the children satisfy $\ell_{jr} = \ell_j + \delta_{jr}$, the operation $(\mathcal{A}_q x)_j = \sum_{r=0}^{8} T_{\delta_{jr}} x_{jr}$ satisfies exactly

$$\mathcal{F}_{q+1} = \mathcal{F}_q \mathcal{A}_q, \qquad \mathsf{S}_{q+1} = \mathcal{I}_q \mathsf{S}_q \mathcal{A}_q, \qquad \mathcal{I}_q y = y \otimes u_9.$$

The gamma, prime, and polar columns are thus applied to the same sum before their norms are taken. Unitary mixing after an orthogonal sum preserves diagonal energy; it does not manufacture omitted cross terms.

Surjectivity determines the scope of refinement. On the complete fibre, positivity of $\mathscr{W}(\mathcal{F}_q x, \mathcal{F}_q x)$ for all states is equivalent to positivity of $\mathscr{W}(f, f)$ for all test functions. The reverse implication uses $x = \mathcal{E}_q f$, whose components remain smooth and compactly supported. Folding preserves the problem and its cross terms: merely increasing the routes does not restrict the test functions. Restriction to more specific reachable states requires their construction and its proof.

Assembled result and global extension

The gamma inverse, the complete output link, the basal domain, its translation, and folding prior to the readouts have action, domain, and proof here. They are results recovered from the corpus and assembled formalizations, not discoveries attributed retrospectively.

For the requested global scope, it remains to prove the bound for that same output on the entire effective range, including the cross terms in sums of local test functions, compatibly as the windows are enlarged. The output identity of one connector and the contractivity of another cannot be combined without proving that they represent the same action. Comparison with the internal history encoder retains the two moments specified in Subsection 97.8.3.

97.8.13 Irreducible-clock windows and refinement with cross terms

The native product determines the irreducibles and their powers before the logarithmic reading. The lengths $\ell_{p,k} = k \log p$ and the weights $w_{p,k} = (\log p) p^{-k/2}$ appearing below are those already published in Sections 42.3 and 42.4. They are not selected from the sign one wishes to check. On this arithmetic output, a family of tests can be calculated jointly, retaining the channels of the completed form and all their cross products.

This section brings together the density and the Schur criterion of the owner “Surviving reader of two APP routes”, and extends the calculation to translated intervals of equal width. The choice of nine windows certified below is an explicit analytic family; its cardinality does not by itself identify these windows with the nine canonical children of a TPK state.

Proposition 97.20. Complete kernel of translated intervals

For $\varepsilon > 0$, let

$$b_{\varepsilon,t}(x) = \varepsilon^{-1/2} \mathbf{1}_{[t-\varepsilon/2, t+\varepsilon/2]}(x), \qquad \tau_\varepsilon(x) = \left(1 - \frac{|x|}{\varepsilon}\right)_+,$$

$$a_\varepsilon = \frac{4 \sinh(\varepsilon/4)}{\sqrt{\varepsilon}}, \qquad \lambda_n = 2n + \frac{1}{2}, \qquad F(x) = \sum_{n \geq 0} \frac{e^{-\lambda_n x}}{\lambda_n^2} \quad (x \geq 0).$$

The complete form on these intervals depends on the difference between their centres:

$$\mathscr{W}(b_{\varepsilon,s}, b_{\varepsilon,t}) = K_\varepsilon(s - t),$$

where

$$K_\varepsilon(d) = 2a_\varepsilon^2 \cosh(d/2) + c_\Gamma \tau_\varepsilon(d) + \frac{2F(|d|) - F(|d-\varepsilon|) - F(|d+\varepsilon|)}{\varepsilon} - \sum_{p,k} w_{p,k}\big[\tau_\varepsilon(d-\ell_{p,k}) + \tau_\varepsilon(d+\ell_{p,k})\big].$$

The prime-power sum is finite; only lengths $\ell_{p,k} < |d| + \varepsilon$ contribute.

Proof. The overlap of two normalized intervals is the triangular correlation τ_ε. Inserting the translations of each clock gives the two prime terms. The polar integrals of $b_{\varepsilon,t}$ are $a_\varepsilon e^{t/2}$ and $a_\varepsilon e^{-t/2}$. Their cross product gives $a_\varepsilon^2(e^{(s-t)/2} + e^{(t-s)/2})$, the polar term in the formula. In particular, dependence on the difference between centres belongs to the complete form, although each polar port separately changes under translation.

The expansion $\mu(a) = \sum_{n\geq 0} e^{-\lambda_n a}$ and integration over the linear pieces of τ_ε give

$$\int_0^\infty e^{-\lambda a}\big[2\tau_\varepsilon(d) - \tau_\varepsilon(d-a) - \tau_\varepsilon(d+a)\big]\,da = \frac{2e^{-\lambda|d|} - e^{-\lambda|d-\varepsilon|} - e^{-\lambda|d+\varepsilon|}}{\varepsilon\lambda^2}.$$

To justify interchanging the series and integral, the triangular function also satisfies

$$|2\tau_\varepsilon(d) - \tau_\varepsilon(d-a) - \tau_\varepsilon(d+a)| \leq \min(4, 2a/\varepsilon).$$

Its product with $\mu(a)$ is integrable both at the origin and at infinity. The sum of the absolute integrals is thus bounded, justifying the interchange. The integrated summand is also bounded in absolute value by $4/(\varepsilon\lambda_n^2)$, whose series converges. The joint contribution of both sides cancels the singularity of μ at the origin. One can therefore sum the integrated identity and obtain the stated gamma term. □

These indicator functions are elements of the closed domain of the readouts, not smooth functions. Extension from $\mathcal{D}_R$ is justified without modifying the form. For a normalized interval,

$$\|b_{\varepsilon,t} - T_a b_{\varepsilon,t}\|_2^2 = 2\min(|a|/\varepsilon, 1).$$

The integral against $\mu(a)$ is finite. Let $b_{\varepsilon,t}^{(\delta)} = \rho_\delta * b_{\varepsilon,t}$ be a smooth regularization supported in a fixed window. Take $\rho \geq 0$, smooth and compactly supported, with $\int \rho = 1$, and $\rho_\delta(x) = \delta^{-1}\rho(x/\delta)$. The gamma symbol is

$$\sigma_\Gamma(\xi) = \int_0^\infty \mu(a)|1 - e^{-ia\xi}|^2\,da.$$

In Fourier coordinates, the integrand expressing the gamma norm of the difference contains $\sigma_\Gamma|\widehat{b}|^2|\widehat{\rho}(\delta\xi) - 1|^2$. The first product is integrable, and the last factor is bounded by 4 and tends to zero. Dominated convergence proves gamma convergence. The prime channels are finite and bounded in that window; the polar channels are continuous functionals on its support. This gives convergence in the joint readout norm and, consequently, convergence of every finite Gram matrix.

Corollary 97.21. Two orientations of the same prime window

Set $D_\varepsilon = K_\varepsilon(0)$. For two centres separated by L, the complete Gram matrix is

$$G_2 = \begin{pmatrix} D_\varepsilon & K_\varepsilon(L) \\ K_\varepsilon(L) & D_\varepsilon \end{pmatrix}.$$

Therefore

$$\mathscr{W}(af + bg, af + bg) = \frac{D_\varepsilon + K_\varepsilon(L)}{2}|a+b|^2 + \frac{D_\varepsilon - K_\varepsilon(L)}{2}|a-b|^2.$$

Positivity on their span requires both $D_\varepsilon + K_\varepsilon(L) \geq 0$ and $D_\varepsilon - K_\varepsilon(L) \geq 0$.

If $L \geq \varepsilon$, define

$$I_\varepsilon(L) = \frac{F(L-\varepsilon) - 2F(L) + F(L+\varepsilon)}{\varepsilon} > 0,$$

$$P_\varepsilon(L) = \sum_{|\ell_{p,k}-L|<\varepsilon} w_{p,k}\left(1 - \frac{|\ell_{p,k}-L|}{\varepsilon}\right).$$

Strict convexity of each exponential proves positivity of I_ε. The condition on both signs is written exactly as

$$2a_\varepsilon^2 \cosh(L/2) - I_\varepsilon(L) - D_\varepsilon \leq P_\varepsilon(L) \leq 2a_\varepsilon^2 \cosh(L/2) - I_\varepsilon(L) + D_\varepsilon.$$

Thus the complete cross term controls a window of irreducible-clock lengths with their weights, not merely a diagonal value. The identity follows from Proposition 97.20 and diagonalization of the two-entry matrix; it does not presuppose its sign. □

Proposition 97.22. Joint control of nine test functions

Let $\varepsilon = 1/100$, $t_j = j \log 2$, $0 \leq j \leq 8$, and $G_{ij} = K_\varepsilon(t_i - t_j)$. The interval calculation preserved with this chapter certifies

$$\lambda_{\min}(G) > 0.8479.$$

Consequently,

$$\mathscr{W}\left(\sum_{j=0}^{8} z_j b_{\varepsilon,t_j}, \sum_{j=0}^{8} z_j b_{\varepsilon,t_j}\right) > 0.8479 \sum_{j=0}^{8} |z_j|^2 \quad (z \neq 0).$$

Proof and reproduction. For $F_N(x) = \sum_{n=0}^{N-1} e^{-\lambda_n x}/\lambda_n^2$, the summand is decreasing and

$$0 \leq F(x) - F_N(x) \leq e^{-\lambda_N x}\left(\lambda_N^{-2} + \frac{1}{2\lambda_N}\right).$$

Indeed, separate the first term of the tail and bound the remainder by the integral of $(2u+1/2)^{-2}$, factoring out the largest exponential. We use $N = 1024$ and interval operations with outward rounding. The condition $8\log 2 + \varepsilon < \log 259$ allows all prime-power pairs up to 259 to be enumerated: this gives 71 rows. The irreducibles and products come from the preserved native producer; recognition by ordinary integers is subsequent. The constant c_Γ is evaluated from the certified intervals of the preceding coordinates, not from a target value of this form.

The formula in Proposition 97.20 gives intervals for the nine entries of the first row. For each row i, these intervals verify

$$G_{ii} - \sum_{j \neq i} |G_{ij}| > 0.8479.$$

The matrix is Hermitian. The inequality $2|z_i||z_j| \leq |z_i|^2 + |z_j|^2$, applied to each cross term, directly gives the stated quadratic bound. The verification preserves the intervals for all nine rows and the complete census. The strict margins and the convergence of Gram matrices proved above also allow simultaneous regularization of the nine test functions, yielding positivity on their smooth span for a sufficiently fine regularization. □

This control covers all complex combinations of nine test functions, not merely their sum. Its dimension remains finite. The computational record states that scope and does not use its result as a certificate of a universal inequality.

Proposition 97.23. Refinement and eight detail components

The geometric subdivision of an interval into nine parts satisfies

$$b_{\varepsilon,t} = \frac{1}{3}\sum_{r=0}^{8} b_{\varepsilon/9,\,t+(r-4)\,\varepsilon/9}.$$

If $V = (1,\dots,1)^{\mathsf{T}}/3$, then

$$G_{\mathrm{padre}} = V^{*} G_{\mathrm{hijos}} V.$$

For several parents, take the blockwise isometric inclusion with that same column. In a basis formed by the uniform component and the eight detail components per parent, write

$$G_{\mathrm{hijos}} = \begin{pmatrix} A & B^{*} \\ B & D \end{pmatrix}, \qquad A = G_{\mathrm{padres}}.$$

If $A > 0$, the exact condition is

$$G_{\mathrm{hijos}} \succeq 0 \quad \Longleftrightarrow \quad D - BA^{-1}B^{*} \succeq 0.$$

Proof. The nine child intervals partition the parent apart from endpoints of measure zero. The normalization of their indicators proves the linear identity, and sesquilinearity gives the Gram identity with all cross terms. Finally,

$$\begin{aligned}\begin{pmatrix} x \\ y \end{pmatrix}^{*} \begin{pmatrix} A & B^{*} \\ B & D \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= (x + A^{-1}B^{*}y)^{*} A (x + A^{-1}B^{*}y) \\ &\quad + y^{*}(D - BA^{-1}B^{*})y.\end{aligned}$$

Taking $x = -A^{-1}B^{*}y$ proves necessity, and the same identity proves sufficiency. □

The uniform component reproduces the previous Gram matrix; the details contain the new interferences. Preserving both parts is the exact requirement for extending a positive test under this refinement. The return of nine phases does not by itself turn the Gram matrix into a circulant matrix: in the preceding family, $K_{\varepsilon}(\log 2) \neq K_{\varepsilon}(8 \log 2)$. Phase and distance memory remain distinct data.

The detail matrix and its Schur complement provide a checkable condition for each enlargement. The prospective reader is not defined through a retrospective factorization of the form; the form published by the previously constructed readers is checked. Completing the global argument requires an estimate on all these details, uniform over cofinal enlargements and over the full test domain.

97.8.14 Coercive assembly and stability of nonadic details

The basal property allows entire function spaces to be controlled. Assembling several intervals requires comparing that local margin with every interaction between routes. The complete kernel of the preceding section provides an estimate for those interactions before a function is chosen. This yields an infinite-dimensional result on a disconnected domain, not merely a Gram matrix of indicator functions.

The positions are taken after the publication of the irreducible clocks. The explicit case uses successive translations of the native clock 2 and a declared positional width. The difference operators preserve both orientations. Subsequent refinement subdivides each interval into nine parts and retains the same function

before its two energy readings. This analytic family is specified as such: its cardinality does not by itself identify it with a canonical selection of TPK states.

Proposition 97.24. Diagonal margin and bound for cross terms

Let $t_1 < \cdots < t_m$ be distinct centres and $I_j = (t_j - \varepsilon/2, t_j + \varepsilon/2)$ disjoint intervals, with $0 < \varepsilon < \log 2$. Define

$$\delta_\varepsilon = 2M_\Gamma(\varepsilon) + c_\Gamma - 2\sinh(\varepsilon/2) + \varepsilon.$$

For $f_j \in C_c^\infty(I_j)$,

$$\mathscr{W}(f_j, f_j) \geq \delta_\varepsilon \|f_j\|_2^2.$$

If $d_{ij} = |t_i - t_j| > \varepsilon$, set

$$\rho_{ij}(\varepsilon) = \sum_{|\ell_{p,k} - d_{ij}| < \varepsilon} w_{p,k} + \varepsilon\mu(d_{ij} - \varepsilon) + 2\varepsilon\cosh((d_{ij} + \varepsilon)/2).$$

Then

$$|\mathscr{W}(f_i, f_j)| \leq \rho_{ij}(\varepsilon)\|f_i\|_2\|f_j\|_2 \qquad (i \neq j).$$

Proof. On an interval of width ε, the gamma tail contributes at least $2M_\Gamma(\varepsilon)\|f_j\|_2^2$. There is no diagonal prime-power correlation, because its minimum length is $\log 2 > \varepsilon$. In a larger ambient window, each prime difference and its counterterm cancel on that support. After centring the interval, the negative polar port has squared norm $2\sinh(\varepsilon/2) - \varepsilon$. Discarding the positive polar port and the gamma part preceding the tail gives the bound. Translation invariance of the complete form carries the same bound to any centre.

In the cross term, the scalar contribution vanishes because the supports are disjoint. For each positive length, only one of the two orientations can connect the intervals. Unitarity of translation bounds the prime contribution by the stated sum of weights. The gamma cross term has kernel $-\mu(|x - y|)$ on $I_i \times I_j$. Since μ is decreasing,

$$|\mathscr{W}_\Gamma(f_i, f_j)| \leq \mu(d_{ij} - \varepsilon)\|f_i\|_1\|f_j\|_1 \leq \varepsilon\mu(d_{ij} - \varepsilon)\|f_i\|_2\|f_j\|_2.$$

The complete polar kernel is $2\cosh((x - y)/2)$. The bound $|x - y| \leq d_{ij} + \varepsilon$ and the same inequality between the L^1 and L^2 norms give the polar term. Adding the three estimates proves the result. □

Theorem 97.25. Coercivity of the assembly

If

$$\eta = \delta_\varepsilon - \max_i \sum_{j \neq i} \rho_{ij}(\varepsilon) > 0,$$

then, for $\Omega = \bigcup_j I_j$,

$$\boxed{\mathscr{W}(f, f) \geq \eta\|f\|_2^2 \quad \text{for every } f \in C_c^\infty(\Omega).}$$

The assertion holds in the closure of that domain in the joint readout norm. In particular, for

$$m = 9, \qquad t_j = j\log 2 \ (0 \leq j \leq 8), \qquad \varepsilon = 1/1000,$$

the certified evaluation gives $\eta > 1$.

Proof. Writing $f = \sum_j f_j$ on the disjoint supports and $x_j = \|f_j\|_2$ gives

$$\begin{aligned}\mathscr{W}(f,f) &\geq \delta_\varepsilon \sum_j x_j^2 - 2\sum_{i<j} \rho_{ij}(\varepsilon) x_i x_j \\ &\geq \sum_i \left(\delta_\varepsilon - \sum_{j\neq i} \rho_{ij}(\varepsilon) \right) x_i^2 .\end{aligned}$$

We used $2x_i x_j \leq x_i^2 + x_j^2$ for each cross term. The sum of the squared norms is $\|f\|_2^2$. Continuity of the readouts extends the inequality to the closed domain.

In the explicit case, the distances are $d \log 2$, $1 \leq d \leq 8$. For $\varepsilon = 1/1000$,

$$2^d (e^\varepsilon - 1) < \frac{256}{999} < 1, \qquad 2^d (1 - e^{-\varepsilon}) < \frac{256}{1000} < 1.$$

The first inequality uses the exponential series and $e^\varepsilon < 1/(1-\varepsilon)$; the second uses $1 - e^{-\varepsilon} < \varepsilon$. By the integrality of the published powers, the only prime-power clock in the window at that distance is $(p,k) = (2,d)$. Thus

$$\rho_d = (\log 2) 2^{-d/2} + \frac{1}{1000} \mu(d \log 2 - 1/1000) + \frac{2}{1000} \cosh((d \log 2 + 1/1000)/2).$$

The native producer also verifies the complete census up to 259. The calculation of δ_ε uses the previously proved formula for M_Γ and the identity

$$\arctan t = \frac{\pi}{4} - \arctan \frac{1-t}{1+t} \qquad (0 < t < 1).$$

The second argument is small and its alternating series carries the bound given by the first omitted term. The internal publication of π, the rational gamma interval and the subsequent interval operations produce the bounds

$$\delta_\varepsilon > 4.4921, \qquad \max_i \sum_{j\neq i} \rho_{ij} < 2.5385, \qquad \eta > 1.9536 > 1.$$

The receipt retains the margin for each row. No particular functions are evaluated to infer the universal inequality on $C_c^\infty(\Omega)$: it follows from the operator estimate above. □

Corollary 97.26. All refinements of the assembled domain

Subdivide each I_j successively into nine parts. The normalized indicators of the 9^{n+1} intervals at depth n form an orthonormal family in L^2. Their full Gram matrix satisfies

$$G_n \succeq \eta I \qquad (n \geq 0).$$

The uniform inclusion V_n, with nine components $1/3$ per parent, satisfies

$$V_n^* G_{n+1} V_n = G_n.$$

In the uniform–detail decomposition,

$$G_{n+1} = \begin{pmatrix} A_n & B_n^* \\ B_n & D_n \end{pmatrix}, \qquad A_n = G_n,$$

one obtains, at any depth,

$$\boxed{D_n - B_n A_n^{-1} B_n^* \succeq \eta I.}$$

Proof. The theorem applies to the regularizations of each combination of indicators. The norm convergence of the readouts in subsection 97.8.13 allows passage to the limit. An indicator reaching an endpoint of I_j is approximated first by slightly contracting its support towards the centre, and then by regularizing within I_j. The triangular correlation and its integrable translation bound prove the same convergence; no extension of the coercivity domain is used. Orthonormality in L^2 turns the functional bound into $G_n \succeq \eta I$, and exact subdivision gives the compression identity.

For a detail vector y, take $x = -A_n^{-1} B_n^* y$. The Schur identity gives

$$\begin{aligned} y^*(D_n - B_n A_n^{-1} B_n^*) y &= \begin{pmatrix} x \\ y \end{pmatrix}^* G_{n+1} \begin{pmatrix} x \\ y \end{pmatrix} \\ &\geq \eta(\|x\|^2 + \|y\|^2) \geq \eta \|y\|^2. \end{aligned}$$

The same η works at every depth. □

The corollary controls the eight new details per parent at arbitrary depth within the assembly. The function is not replaced by its mean, nor are its zero-sum components discarded. This strengthens the Gram matrix of nine indicators: it covers infinite-dimensional function spaces and gives a bound independent of depth.

Finite families and extension of the domain

For any finite set of distinct centres, there is a sufficiently small positive width for which theorem 97.25 applies. Indeed, $2M_\Gamma(\varepsilon) \to +\infty$ as $\varepsilon \downarrow 0$, whereas the local negative polar term tends to zero. The distances between centres are bounded away from zero. The sums of prime weights remain bounded as each window shrinks, because there are finitely many lengths below any fixed bound. The gamma and polar cross terms tend to zero. Thus, δ_ε eventually exceeds the sum of interactions in each row.

The admissible width depends on the centres. The unions of narrow intervals obtained in this way do not, by the construction above alone, form a cofinal covering of all compactly supported functions. Global positivity also requires control of the growth of the domain and its new interactions. The proved uniformity in depth remains distinct from extension to an arbitrary test-function domain, which is not proved here.

97.8.15 Coercivity of details in an arbitrary fixed window

The preceding estimate retains a uniform margin when the interiors of nine separated intervals are refined. A second construction controls the fine details of any fixed window, including its contiguous cells. The two statements have different and complementary domains: the first estimates the full function on a disconnected support; the second estimates the cellwise zero-mean components in an entire window.

The previously constructed clocks, weights and readers are retained. In particular, the lengths $\ell_{p,k}$, the weights $w_{p,k}$, the scalar c_Γ and the internal publication π retain their provenance. The following argument acts on the functional realization of these objects; it does not modify their generator. Write

$$\beta_R = 2\sinh(R/2) - R, \qquad \mathfrak{c}_R = \frac{2}{3} - c_\Gamma + 2 \sum_{\ell_{p,k} \leq R} w_{p,k} + \beta_R.$$

The sum is finite and comes from the native product below the bound e^R. Divide $I_R = (-R/2, R/2)$ into M cells of length at most h, with $0 < h \leq 1$.

Theorem 97.27. Explicit threshold for zero-mean details

Let d be a function in the joint domain of the readouts, supported in I_R, whose integral over each cell is zero. Then

$$\boxed{\mathscr{W}_R(d,d) \geq \big[\log(1+4/h) - \mathfrak{c}_R\big]\|d\|_2^2.}$$

Consequently, the form is strictly coercive on this detail space if

$$0 < h < \min\left(1, \frac{4}{\exp(\mathfrak{c}_R) - 1}\right).$$

The threshold depends on the window R, but not on the function or on a finite enumeration of examples.

Proof. On each cell $[a,b]$, define $F(x) = \int_a^x d(t)\,dt$. The zero-mean condition implies $F(a) = F(b) = 0$. Assembling the primitives yields $F \in H_0^1(I_R)$, extended by zero outside the window, with $F' = d$. The Dirichlet inequality on each cell gives

$$\|F\|_2^2 \leq \frac{h^2}{\pi^2}\|d\|_2^2.$$

Here π is the same internal coordinate recognized in the functional realization. No new constant is selected.

The gamma tail admits the positive expansion already used,

$$\mu(a) = \sum_{n=0}^{\infty} e^{-\lambda_n a}, \qquad \lambda_n = 2n + \frac{1}{2}.$$

For $\lambda > 0$, let C_λ denote convolution with $e^{-\lambda|x|}$. The contribution of one summand is

$$E_\lambda(d) = \int_0^\infty e^{-\lambda a}\|(I - T_a)d\|_2^2\,da = \frac{2}{\lambda}\|d\|_2^2 - \langle d, C_\lambda d\rangle \geq 0.$$

Its positivity is that of an integral of squared norms. Denote the unitary Fourier transform by $\mathcal{F}_u F = (2\pi)^{-1/2}\widehat{F}$, distinguishing it from the unnormalized transform used earlier. The multiplier of C_λ is $2\lambda/(\lambda^2 + \xi^2)$. Since $d = F'$, Plancherel implies

$$\begin{aligned}\langle d, C_\lambda d\rangle &= \int_{\mathbb{R}} \frac{2\lambda\xi^2}{\lambda^2 + \xi^2}|(\mathcal{F}_u F)(\xi)|^2\,d\xi \\ &\leq 2\lambda\|F\|_2^2 \leq \frac{2\lambda h^2}{\pi^2}\|d\|_2^2.\end{aligned}$$

By Tonelli, the gamma form is the sum of the E_{λ_n}. The first $N+1$ terms are retained, and only a nonnegative tail is discarded. This yields

$$\begin{aligned}&Q_\Gamma(d) \geq S_N(h)\|d\|_2^2, \\ &S_N(h) = 4\sum_{n=0}^{N}\frac{1}{4n+1} - \frac{h^2(N+1)(2N+1)}{\pi^2}.\end{aligned}$$

Choose $N = \lfloor 1/h \rfloor$. Then $\lambda_N h \leq 2 + h/2 < \pi$; the retained bounds are nonnegative. Comparison with the integral of a decreasing function gives

$$4\sum_{n=0}^{N}\frac{1}{4n+1} \geq \int_0^{N+1}\frac{4}{4x+1}\,dx = \log(4N+5) \geq \log(1+4/h).$$

Moreover, $\pi^2 > 9$ and $h \le 1$ imply

$$\frac{h^2(N+1)(2N+1)}{\pi^2} \le \frac{(1+h)(2+h)}{\pi^2} < \frac{2}{3}.$$

Therefore, $Q_\Gamma(d) \ge [\log(1+4/h) - 2/3]\|d\|_2^2$. In the full form, the scalar term contributes $c_\Gamma\|d\|_2^2$; the prime part is bounded below by $-2\sum_{\ell_{p,k}\le R} w_{p,k}\|d\|_2^2$ by unitarity of each translation; and the polar part is bounded by $-\beta_R\|d\|_2^2$, as proved in subsection 97.8.12. Their sum proves the stated inequality. The threshold displayed is equivalent to $\log(1+4/h) > \mathfrak{c}_R$. Passage to the closed domain preserves the estimate by continuity of the readouts and of the cell integrals. □

Corollary 97.28. Finite dimensionality of the possible negative directions

For fixed R and a partition satisfying the threshold above, any subspace of the domain on which $\mathscr{W}_R$ is strictly negative definite has dimension at most M, the number of cells.

Proof. Consider the linear map

$$\mathcal{P}(d) = \left(\int_{I_1} d, \dots, \int_{I_M} d\right) \in \mathbb{C}^M.$$

Its kernel is the detail space of theorem 97.27, on which the form is positive definite. The restriction of $\mathcal{P}$ to a strictly negative subspace is injective: a nonzero vector in its kernel would have both a positive and a negative value. The dimension of that subspace cannot exceed M. □

This reduction retains a finite coupling problem in each window. It is not equivalent to proving that the negative index is zero. In a mean–detail decomposition, the detail block is coercive at sufficient depth, but the mean block and the cross terms must be retained. If the Schur complement is used, its sign requires that full composition. In particular, an exact prime resonance between two cells does not disappear when zero mean is imposed: it may act as a negative multiple of the identity on their corresponding details. The theorem retains such terms through its translation bound.

The result provides a second effective uniformity: in any fixed window, sufficiently fine details have an explicit common bound. Subsection 97.8.14 also proves the sign of the full assembly, including its cross terms, for the specified union of nine intervals. Extending that sign to every window remains the global step that this edition does not regard as proved.

97.9 Schur reduction and coercivity of the Weil form

97.9.1 Basis, domain, and result obtained

The complete columns already constructed from the native arithmetic of APP, the oriented TRIT state, the TPK operations and memory, and the joint structure of the continuum are used. The native product publishes irreducibles and powers; their lengths and weights then feed the prime, gamma and polar readers. The internal coordinate of π and the rational reader of the finite part of Euler–Mascheroni intervene after that publication. No zero, sought value of the functional or metrological constant selects these operations.

We retain the full columns and the domains constructed in the chapter on the double circle and spectral transport, together with the support and detail estimates. The columns and the connector are also made explicit in Section 97.11. The joint construction of the continuum remains the common antecedent: it is not replaced by this functional decomposition, nor are its five constructions replaced by five modules.

The new operation assembled here is the exact elimination of the infinite detail space by means of its coercive form. It produces a finite matrix with all coupling corrections, not merely a matrix of means. A specific case is also settled:

For every function in the closure of $C_c^\infty((-1/18, 1/18))$ in the joint norm of the readouts,

$$\mathscr{W}(f,f) \geq \frac{1}{2}\|f\|_2^2. \tag{97.14}$$

The same assertion holds on any translated interval of length $1/9$, and on all its interior nonadic refinements.

The full window has length $1/9 > 0.085626\ldots$, the endpoint of the preceding baseline criterion. It contains no active prime translation, since $1/9 < \log 2$. This limitation of the example remains explicit.

Provenance. Authorial architecture and readouts: pre-existing. Detail coercivity: a result recovered from the zero-mean threshold theorem, retaining an intermediate estimate stronger than its abbreviated logarithmic bound. Schur assembly, residual certification, and proof for the interval $1/9$: formalization and certificate added in this section; no global historical novelty is claimed.

97.9.2 Partition, cell means, and details

Fix $R > 0$, $I_R = (-R/2, R/2)$, and $H_R = L^2(I_R)$, with extension by zero. Let V_R be the closure of $\mathcal{D}_R = C_c^\infty(I_R)$ in the joint norm of the full readouts. Equivalently, its norm may be taken to be

$$\begin{aligned} \|f\|_{V_R}^2 &= \|f\|_2^2 + Q_\Gamma(f), \\ Q_\Gamma(f) &= \int_0^\infty \mu(a)\|(I - T_a)f\|_2^2\, da, \\ \mu(a) &= \frac{e^{-a/2}}{1 - e^{-2a}}. \end{aligned} \tag{97.15}$$

The prime and polar channels are bounded on each fixed window. The gamma form is closed: it is the form of a positive Fourier multiplier, restricted to the subspace of functions supported in the window; equivalently, this is verified by convergence in the direct integral of differences. The full form is a bounded perturbation of it and defines a semibounded self-adjoint operator L_R.

Partition I_R into contiguous intervals $I_j = (\ell_j, r_j)$, with lengths $h_j \leq h$, and set

$$u_j = h_j^{-1/2}\mathbf{1}_{I_j}, \qquad Ua = \sum_{j=1}^M a_j u_j, \qquad P = UU^*, \quad Q = I - P. \tag{97.16}$$

In the last sum, $a = (a_1, \ldots, a_M)$ is the coefficient vector; the endpoints ℓ_j, r_j are used only in defining I_j.

The indicator functions belong to V_R: their gamma energy is controlled by $\|(I - T_a)u_j\|_2^2 = 2\min(|a|/h_j, 1)$. Thus P acts continuously on V_R, and every $f \in V_R$ has the unique decomposition

$$f = Ua + d, \qquad a_j = \langle f, u_j \rangle, \qquad d \in V_D := V_R \cap QH_R. \tag{97.17}$$

Here d has zero integral on each cell. The space V_D is closed in the form norm and dense in QH_R.

The estimate of the zero-mean details gives a bound $\mathscr{W}_R(d, d) \geq \eta \|d\|_2^2$ if a sufficiently fine partition is chosen. Its published logarithmic threshold can be used, or the exact intermediate estimate retained

$$\begin{aligned} \eta_N(h, R) &= 4 \sum_{n=0}^{N} \frac{1}{4n+1} - \frac{h^2(N+1)(2N+1)}{\pi^2} \\ &\quad + c_\Gamma - 2 \sum_{k \log p \leq R} (\log p) p^{-k/2} - \beta_R, \\ \beta_R &= 2\sinh(R/2) - R. \end{aligned} \tag{97.18}$$

Lemma 97.29 (Retained bound for cell details)**.** *For $d \in V_D$ and every integer $N \geq 0$,*

$$\mathscr{W}_R(d, d) \geq \eta_N(h, R) \|d\|_2^2.$$

Proof. On each cell $[\ell_j, r_j]$, define $F(x) = \int_{\ell_j}^{x} d(t)\, dt$. The zero mean gives $F(\ell_j) = F(r_j) = 0$. The assembly of these primitives, extended by zero, belongs to $H_0^1(I_R)$, satisfies $F' = d$, and, by the Dirichlet inequality on each cell,

$$\|F\|_2^2 \leq \frac{h^2}{\pi^2} \|d\|_2^2. \tag{97.19}$$

The positive expansion $\mu(a) = \sum_{n \geq 0} e^{-\lambda_n a}$, $\lambda_n = 2n + 1/2$, allows us to write $Q_\Gamma = \sum E_{\lambda_n}$, where

$$E_\lambda(d) = \int_0^\infty e^{-\lambda a} \|(I - T_a)d\|_2^2 \, da = \frac{2}{\lambda} \|d\|_2^2 - \langle d, C_\lambda d \rangle \geq 0, \tag{97.20}$$

and C_λ is convolution with $e^{-\lambda|x|}$. For the unitary transform $\mathcal{F}_u = (2\pi)^{-1/2}\widehat{\ }$, the multiplier of C_λ is $2\lambda/(\lambda^2 + \xi^2)$. By Plancherel,

$$\begin{aligned} \langle d, C_\lambda d \rangle &= \int_{\mathbb{R}} \frac{2\lambda\xi^2}{\lambda^2 + \xi^2} |\mathcal{F}_u F(\xi)|^2 \, d\xi \\ &\leq 2\lambda \|F\|_2^2 \leq \frac{2\lambda h^2}{\pi^2} \|d\|_2^2. \end{aligned} \tag{97.21}$$

Retaining $N + 1$ summands and discarding only the nonnegative tail gives

$$Q_\Gamma(d) \geq \left[4 \sum_{n=0}^{N} \frac{1}{4n+1} - \frac{h^2(N+1)(2N+1)}{\pi^2} \right] \|d\|_2^2.$$

The scalar term contributes $c_\Gamma \|d\|_2^2$. Each prime translation contributes at least $-2w_{p,k}\|d\|_2^2$, by Cauchy–Schwarz and the unitarity of $T_{k \log p}$. Only lengths less than or equal to R enter. Finally, the polar contribution is

$$2 \left| \int_{I_R} d(x) \cosh(x/2)\, dx \right|^2 - 2 \left| \int_{I_R} d(x) \sinh(x/2)\, dx \right|^2 \geq -\beta_R \|d\|_2^2,$$

since $2 \int_{I_R} \sinh^2(x/2)\, dx = 2\sinh(R/2) - R$. Summation proves the bound. □

There is no need to replace the harmonic sum by its lower integral bound. For the certificate on the window $1/9$, one also checks $(2N + 1/2)h < \pi$, so that each retained gamma bound is nonnegative.

For any fixed R, there is a partition that makes this bound positive. Indeed, if $0 < h \le 1$ and $N = \lfloor 1/h \rfloor$, then

$$4 \sum_{n=0}^{N} \frac{1}{4n+1} \ge \log(4N+5) \ge \log(1 + 4/h), \qquad \frac{h^2(N+1)(2N+1)}{\pi^2} < \frac{2}{3}.$$

The first term diverges as $h \downarrow 0$; the prime, scalar, and polar terms remain fixed by R. This preserves the distinction between coercivity of the details and the sign of the full form.

97.9.3 The exact Schur complement has finite size

Let D be the operator associated with the restriction of the form to V_D. If $\eta > 0$, then $D \ge \eta I$ on QH_R, D^{-1} exists, and $\|D^{-1}\| \le \eta^{-1}$. We do not presume that L_R preserves the detail space.

The indicator functions also belong to the operator domain of L_R. The expression for their gamma image is given in the next section: it has only logarithmic singularities, which belong to $L^2(I_R)$. This allows us to define, without improperly extending a form functional,

$$A = U^* L_R U, \qquad B = Q L_R U : \mathbb{C}^M \longrightarrow QH_R. \tag{97.22}$$

The inner product is linear in the first variable, as in the manuscript. The matrices are written so that $\mathscr{W}_R(Ua, Ua) = a^* A a$. The block identity is

$$\mathscr{W}_R(Ua + d, Ua + d) = a^* A a + 2\,\mathrm{Re}\langle Ba, d\rangle + \mathfrak{d}[d, d], \tag{97.23}$$

where $\mathfrak{d}$ is the closed form of D.

Theorem 97.30 (Exact elimination of the detail space)**.** *The matrix*

$$S = A - B^* D^{-1} B \tag{97.24}$$

satisfies the exact identity

$$\boxed{\mathscr{W}_R(Ua + d, Ua + d) = a^* S a + \mathfrak{d}[d + D^{-1}Ba, d + D^{-1}Ba].} \tag{97.25}$$

In particular, $\mathscr{W}_R \ge 0$ on V_R if and only if $S \succeq 0$. Its negative index equals that of S.

Proof. For $z = D^{-1}Ba \in \operatorname{Dom} D$, $\mathfrak{d}[z, d] = \langle Ba, d\rangle$. Expanding the second term of (97.25) and using $\mathfrak{d}[z, z] = a^* B^* D^{-1} B a$ gives the equality. Sufficiency is immediate. For necessity, choose $d = -D^{-1}Ba$. The triangular change $(a, d) \mapsto (a, d + D^{-1}Ba)$ is invertible on $\mathbb{C}^M \oplus V_D$. The resulting second form is positive definite; the negative directions are exactly those of the finite matrix. □

This theorem converts the finite-negative-index result into an exact matrix. The matrix still contains D^{-1}: finite size does not mean that its entries have already been calculated. The next section provides a way to enclose them with controlled residuals.

97.9.4 Explicit images and certifiable bounds for the corrections

Set $M(x) = \int_x^\infty \mu(a)\, da$, for $x > 0$. For an indicator function $u = h^{-1/2}\mathbf{1}_{(a,b)}$, its gamma image, restricted to I_R, is almost everywhere

$$(L_\Gamma u)(x) = h^{-1/2} \begin{cases} M(x-a) + M(b-x), & a < x < b, \\ -M(a-x) + M(b-x), & x < a, \\ -M(x-b) + M(x-a), & x > b. \end{cases} \tag{97.26}$$

The equality is obtained by integrating the two translations: inside the cell there remains one tail for each endpoint, and outside there remains the integral over the unique interval of translations that meets the cell. It can first be proved with the regulator $a_{\text{traslado}} \geq \epsilon$; convergence in $L^2(I_R)$ of the logarithmic expression justifies its weak form limit. The other parts are

$$L_{\text{primo}}u = -\sum_{k\log p \leq R} w_{p,k} M_R(T_{k\log p} + T_{-k\log p})u, \tag{97.27}$$

$$\begin{aligned} L_{\text{polar}}u(x) &= 2\int_a^b h^{-1/2}\cosh((x-y)/2)\,dy, \\ L_R u &= L_\Gamma u + c_\Gamma u + L_{\text{primo}}u + L_{\text{polar}}u. \end{aligned} \tag{97.28}$$

All discontinuities and singularities of (97.26)–(97.28) lie in a finite list of endpoints and endpoints translated by published clocks. Thus, A_{ij} and

$$G_B = B^*B, \qquad (G_B)_{ij} = \langle Be_j, Be_i\rangle \tag{97.29}$$

are calculated by explicit one-dimensional integrals. In this last formula, e_j denotes the standard coordinate vector of $\mathbb{C}^M$, not the indicator function in H_R. The initial bound is

$$A - \eta^{-1}G_B \preceq S \preceq A. \tag{97.30}$$

This is an operator inequality with the cross terms preserved, not a replacement of G_B by its diagonal entries.

Theorem 97.31 (Residual certification of the complement)**.** *For any finite approximation operator $Z : \mathbb{C}^M \to \operatorname{Dom} D$, define*

$$E = B - DZ, \qquad C_Z = Z^*B + B^*Z - Z^*DZ. \tag{97.31}$$

Then

$$B^*D^{-1}B = C_Z + E^*D^{-1}E, \tag{97.32}$$

and therefore

$$\boxed{A - C_Z - \eta^{-1}E^*E \preceq S \preceq A - C_Z.} \tag{97.33}$$

Proof. Substitute $B = DZ + E$ and expand the product. All terms are defined because the columns of Z belong to $\operatorname{Dom} D$. The bound uses $0 < D^{-1} \leq \eta^{-1}I$. □

An effective choice of Z consists in solving the Galerkin system on step-function details of a nonadic refinement. Its columns are finite combinations of indicator functions and have images (97.26)–(97.28), so E and its Gram are explicitly integrable. The Galerkin system is not declared an exact solution: the term $\eta^{-1}E^*E$ retains its error on the infinite complement.

To certify these integrals without losing the singular endpoints, split

$$M(x) = -\tfrac{1}{2}\log x + \log 2 + \tfrac{\pi}{4} + r(x), \qquad r(0) = 0, \quad r'(x) = \tfrac{1}{2x} - \mu(x). \tag{97.34}$$

On $0 < x \leq R$,

$$-\tfrac{1}{4}e^{R/2} \leq r'(x) \leq \tfrac{1}{4}, \qquad |r'(x)| \leq \tfrac{1}{4}e^{R/2}. \tag{97.35}$$

Indeed, $\mu(x) = e^{x/2}/(2\sinh x)$ and $x \leq \sinh x \leq xe^x$ give $e^{-x/2}/(2x) \leq \mu(x) \leq e^{x/2}/(2x)$. The inequalities $1 - e^{-y} \leq y$ and $e^y - 1 \leq ye^y$ prove the bounds. We do not assert that r' has constant sign on an arbitrarily large window. We integrate $\log x$ and $\log^2 x$ analytically on the small endpoint segments;

the remainder r is bounded by its Lipschitz constant. Away from the endpoints, subdivided interval quadratures and the exponential series for M are used. For example,

$$\int_0^{\delta} \log^2 x \, dx = \delta[(\log \delta)^2 - 2\log \delta + 2]. \tag{97.36}$$

Products of two singularities at distinct points are handled by separating their neighborhoods; for a shared endpoint, the same quadratic integral is used. The resulting intervals are inserted into (97.33), and positivity is verified by an LDL^* decomposition with strictly positive interval pivots or a perturbation bound for a rational matrix. This specifies a finite certificate with control of the infinite complement; no mesh without a tail bound is used.

97.9.5 Coercivity on a window of length 1/9

Take $R = h = 1/9$, a single cell, and $u = R^{-1/2}\mathbf{1}_{I_R}$. This is a full interval, not nine intervals separated by gaps. The detail section is all of $V_R \cap \{u\}^{\perp}$.

97.9.5.1 Margin of the infinite block

In (97.18), take $N = 13$. There are no prime terms because $R < \log 2$. This gives

$$\eta = 4\sum_{n=0}^{13} \frac{1}{4n+1} - \frac{(1/9)^2\, 14 \cdot 27}{\pi^2} + c_\Gamma - \beta_{1/9} > 1. \tag{97.37}$$

The interval certificate gives $1.00363976543713133046 < \eta < 1.00363976543713133048$. The condition $(53/2)/9 < \pi$ also guarantees the positivity of each retained gamma bound. The sum is preserved; replacing it by the bound $\log(1 + 4/h) - 2/3$ would lose the necessary margin here.

97.9.5.2 Mean block

Inside the window, writing $x = R(t - 1/2)$,

$$L_R u(x) = R^{-1/2} g(t), \quad g(t) = c_\Gamma + M(Rt) + M(R(1-t)) + 8\sinh(R/4)\cosh(R(t-1/2)/2). \tag{97.38}$$

Thus $A = \int_0^1 g(t)\, dt$, and $\|B\|^2$ is the variance of g on the unit interval. Applying Tonelli to M gives

$$A = c_\Gamma + \frac{2}{R}\sum_{n=0}^{\infty} \frac{1 - e^{-(2n+1/2)R}}{(2n+1/2)^2} + \frac{32}{R}\sinh^2(R/4). \tag{97.39}$$

If the first L terms are retained, the positive tail is at most $\lambda_L^{-2} + (2\lambda_L)^{-1}$, with $\lambda_L = 2L + 1/2$. The first term of the tail is separated, and the remainder is bounded by the integral of a decreasing function. For $L = 1024$, the certificate yields

$$0.9723307136651774 < A < 0.9767284617331572, \qquad A > 97/100. \tag{97.40}$$

A tighter approximation is not needed for the proof.

97.9.5.3 Coupling between means and details

On $0 < x \le 1/9$, the estimate in (97.34) improves to

$$-9/10 \le r'(x) \le 0. \tag{97.41}$$

Indeed, for $0 < x \le 1/9$,

$$\frac{\sinh x}{x} = \sum_{k\ge 0} \frac{x^{2k}}{(2k+1)!} \le \frac{1}{1-x^2} \le 1 + \frac{x}{2} \le e^{x/2}. \tag{97.42}$$

The intermediate inequality is equivalent to $1 - 2x - x^2 \geq 0$, which holds on that interval. Thus $\mu(x) \geq 1/(2x)$. On the other hand,

$$\mu(x) - \frac{1}{2x} \leq \frac{e^{x/2} - 1}{2x} \leq \frac{1}{4} e^{x/2} \leq \frac{1}{4(1 - 1/18)} = \frac{9}{34} < \frac{9}{10}. \tag{97.43}$$

We retain the less sharp bound $9/10$ used by the certificate. The function $q(t) = r(Rt) + r(R(1-t))$ is symmetric about $1/2$, and its Lipschitz constant is at most $9R/10$. Its oscillation is at most $9R/20 = 1/20$; hence its standard deviation is at most $1/40$. This last assertion follows from $\mathrm{Var}(X) \leq (\sup X - \inf X)^2/4$: subtract the midpoint of the range and use the fact that the mean minimizes the squared error.

The singular part $v(t) = -\frac{1}{2}\log(t(1-t))$ has mean 1 and variance $1 - \pi^2/12$. To verify this, integrate $\log t$, $\log^2 t$, and the positive series for $\log t \log(1-t)$, which gives $\int_0^1 \log t \log(1-t)\, dt = 2 - \sum_{n\geq 1} n^{-2}$. The identity $\sum n^{-2} = \pi^2/6$ can be obtained by applying Parseval to the sine series of x on $(-\pi, \pi)$: its coefficients are $2(-1)^{n+1}/n$, and $(1/\pi)\int_{-\pi}^{\pi} x^2 dx = 2\pi^2/3$. This is a subsequent functional recognition of the same π, not a selection of its generator.

The polar part has oscillation $8\sinh(R/4)[\cosh(R/4) - 1]$, and standard deviation at most half that value. The triangle inequality in the space of centered functions gives, for the full coupling,

$$\boxed{\|B\|^2 \leq \left[\sqrt{1 - \pi^2/12} + \frac{1}{40} + 4\sinh(R/4)(\cosh(R/4) - 1)\right]^2 < \frac{1}{5}.} \tag{97.44}$$

The scalar term and all constants independent of t are removed by centering, not by a supposed disappearance of the cross terms. The interval evaluation of the bounding expression is $0.19926357337268538427\ldots < 1/5$.

97.9.5.4 Positive Schur complement and coercivity of the entire form

Theorem 97.32 (Coercivity on the full nonadic window)**.** *On $I_{1/9} = (-1/18, 1/18)$, every function in the joint domain $V_{1/9}$ satisfies*

$$\mathscr{W}_{1/9}(f, f) \geq \tfrac{1}{2}\|f\|_2^2.$$

The bound holds on every translated interval of the same length and on all its interior refinements.

Proof. From (97.30), (97.37), (97.40), and (97.44),

$$S = A - B^* D^{-1} B > \frac{97}{100} - \frac{1}{5} = \frac{77}{100} > 0. \tag{97.45}$$

Moreover, for $f = au + d$,

$$\mathscr{W}_R(f, f) - \tfrac{1}{2}\|f\|_2^2 \geq (A - \tfrac{1}{2})|a|^2 - 2\|B\||a|\|d\|_2 + (\eta - \tfrac{1}{2})\|d\|_2^2. \tag{97.46}$$

The scalar matrix on the right-hand side has positive diagonal and determinant strictly greater than $\frac{47}{100} \cdot \frac{1}{2} - \frac{1}{5} = \frac{7}{200} > 0$. This proves $\mathscr{W}_R(f, f) \geq \frac{1}{2}\|f\|_2^2$ for every $f \in V_R$. The argument controls all the infinitely many details; it does not replace them by finitely many indicator functions. Translation preserves the correlations and multiplies the polar evaluations by $e^{t/2}$ and $e^{-t/2}$, whose factors cancel in the cross products. It therefore also preserves the full form and the bound. □

97.9.6 Refinement, double circle, and the two moments

Each subdivision into nine preserves exactly $u_{\mathrm{padre}} = \frac{1}{3}\sum_{r=0}^{8} u_r$, and the recomposition of functions occurs before applying J_+ and J_-. Therefore, the Grams of all nonadic indicators within the window satisfy $G_n \geq \frac{1}{2}I$ and $V_n^* G_{n+1} V_n = G_n$. The eight details per parent and all their cross terms are preserved. This is uniformity in depth over this window, not a uniform result as R increases.

For a general partition, the exact Schur matrix does not automatically obey the same compression as the raw Gram. If one refines and expresses the fine Schur complement in old-mean/new-detail coordinates, the old Schur complement is obtained by also eliminating that new detail. This is associativity of minimization in (97.25). Asserting $S_{\mathrm{padre}} = V^* S_{\mathrm{hijos}} V$ would omit that second correction.

On the certified window, the connector already constructed, $C_R = J_{-,R} L_R^{\mathrm{cola}}$, where L_R^{cola} is the cotrace (97.83) applied to the gamma component, satisfies $C_R J_{+,R} = J_{-,R}$. Its restriction to the closure of the effective image $\mathcal{M}_{+,R}$ is contractive, because

$$\|J_{-,R} f\|^2 \leq \|J_{+,R} f\|^2. \tag{97.47}$$

Contractivity is now proved by means of Schur; it is not defined from a putatively positive Gram. Its defect $\Delta_R = (I - C_R^* C_R)^{1/2}$ is then taken on that effective space, and we construct

$$\Xi_R f = J_{-,R} f \oplus \Delta_R J_{+,R} f. \tag{97.48}$$

If P_- is the projection onto the first summand,

$$\Xi_R^* \Xi_R = J_{+,R}^* J_{+,R}, \qquad \Xi_R^* P_- \Xi_R = J_{-,R}^* J_{-,R}. \tag{97.49}$$

These are equalities of forms on the joint domain, not products of unbounded operators on all of L^2. They give the two moments in the dilation codomain. An additional identification with a specific T_N of the double circle requires the intertwining with that operator: an arbitrary projection is not identified with T_N merely because the two moments are available.

97.9.7 Reproducibility checks and conditions for extension

The verification program retained in Appendix 102.6 certifies the constants with interval arithmetic and explicit tail bounds. It uses the internal prefix of π with its hashes and the previously retained rational producer of γ and $\log 2$. It does not use `mpmath.pi`, `mpmath.euler`, zero tables, or floating-point arguments as inputs.

Both runs satisfy the same 21 interval checks. The receipts retain exact rational endpoints, precision, library version, provenance, and hashes. The adversarial check $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ has two positive diagonal blocks and Schur complement -3: it rejects using their signs alone.

Substantive falsifiers: omitting the negative polar term; interchanging cell means with point sampling; losing a prime translation on a larger window; replacing the Schur complement by A; accepting Galerkin without $E^* E/\eta$; or conflating all interior refinements with all windows. None is part of this certificate.

The next concrete extension is a window $R > \log 2$, with a nonadic partition whose detail block is coercive. There, (97.33) must be evaluated, including at least the clock $\log 2$. The present result asserts neither that new sign, nor global Weil positivity, nor RH. Nor does it infer a negation of the corpus from this local scope: it provides an exact elimination, a certifiable procedure, and a first settled case extending the preceding baseline case.

97.10 Nonadic compression and circular moments of the Weil form

The relationship between memory and positivity requires preserving two levels of reading. The nonadic turn records a history and admits a positive Hilbertian realization; the arithmetic readers of that same genealogy publish a Hermitian form in which the gamma, prime and polar channels jointly intervene. The transport problem consists of identifying their energies and correlations, with the same domains and without replacing one of those realizations with the other. The following identities make that comparison explicit before applying the Gelfand–Naimark–Segal (GNS) construction to the arithmetic form.

The complete balance of a nine-phase chart is preserved first, including any norm variation at the seam. The loss record, its moment matrices and a Cayley operator acting on exponentially decreasing tests are then constructed. Local coercivity can be transported to entire families of tests with tails; passing to their algebraic sum and to the global domain also requires the mixed correlations specified below. This development does not assert that global Weil positivity is established.

97.10.1 Enriched state and modes of the central chart

The construction used is APP → TRIT → TPK → enriched state → joint discrete structure of the continuum. APP preserves sum and product evaluations, their quotients and residues, on the same support. The TRIT preserves orientation and regime; the TPK composes selection, transport, and memory update. Nonadic holonomy returns the phase and increases the depth of the history. The five correlative aspects of the continuum are neither separated nor redefined through the spectral question studied subsequently.

This development begins with the realization of the winding register of that structure. It does not seek to replace the complete construction of the continuum with a sequence space. Integer memory is one coordinate of the state, not its entirety. Nor does it identify the reduced phase register with the TPK generator.

The author's note on sum and product indicates the provenance of the central matrix. In the preserved positive chart one obtains

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-, \qquad P_\pm = \frac{1}{2}\begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}.$$

Thus, after normalizing the uniform mode, the differential mode has ratio $q = 5/9$. The names "sum" and "product" describe the APP operations that generate the chart; the projectors P_+ and P_- are its symmetric and antisymmetric modes. Equality between a mode and a specific sheet requires preserving the map relating them.

97.10.2 Uniform compression of the nonadic chart and exact balance

Let $\mathcal{K}$ be a Hilbert space and let C be a bounded operator on it. In $\mathcal{K}^9$, the nine-phase chart with seam C acts by

$$S_C(v_0, \dots, v_8) = (Cv_8, v_0, \dots, v_7), \qquad S_C^9 = I_9 \otimes C.$$

The uniform inclusion is

$$Jv = \frac{1}{3}(v, \dots, v), \qquad J^*J = I.$$

Write the visible compression and its complementary component as

$$T = J^*S_CJ = \frac{8I + C}{9}, \qquad \eta = (I - JJ^*)S_CJ.$$

Proposition 97.33 (Exact balance of compression and seam)**.** *Without assuming that C is isometric, one has*

$$\eta^*\eta = \frac{8}{81}(I - C)^*(I - C),$$

$$\boxed{I - T^*T = \eta^*\eta + \frac{1}{9}(I - C^*C).} \tag{97.50}$$

Proof. Orthogonality of JJ^* gives

$$\eta^*\eta = J^*S_C^*S_CJ - T^*T.$$

The first term is $(8I + C^*C)/9$. The second is $(64I + 8C + 8C^* + C^*C)/81$. Subtraction proves the first identity. Now subtracting T^*T from I and separating $I - (8I + C^*C)/9$ proves (97.50). □

The formula does not eliminate memory: it displays exactly two contributions. One measures dispersion among the nine phases; the other measures the norm defect of the seam. If the preceding construction has given $C^*C = I$, the positive compression–expression identity of the corpus remains:

$$I - T^*T = \eta^*\eta \succeq 0.$$

This result explains what memory positivity contains and where identification of the readout must enter. It does not permit replacing that identification with a drawing of a circle.

97.10.3 Visible contraction and preservation of the winding register

Now suppose that C is unitary, as for the bilateral shift of integer memory. The complete winding of the differential mode is represented by $M = qC$, not by nine successive applications of the same factor q.

For each integer $N \geq 1$, define the loss-and-terminal register

$$L_Nv = \big(\sqrt{1 - q^2}v, \sqrt{1 - q^2}qCv, \dots, \sqrt{1 - q^2}q^{N-1}C^{N-1}v, q^NC^Nv\big).$$

Proposition 97.34 (Isometric preservation of the winding register)**.** *L_N is isometric. As N increases, the last component decomposes into a new published memory and a new terminal, without erasing the preceding record. The infinite record*

$$Lv = \left(\sqrt{1 - q^2}q^jC^jv\right)_{j\geq 0}$$

is also isometric.

Proof. For $v, w \in \mathcal{K}$,

$$\langle L_Nv, L_Nw\rangle = \left[(1 - q^2)\sum_{j=0}^{N-1} q^{2j} + q^{2N}\right]\langle v, w\rangle = \langle v, w\rangle.$$

In the last component, q^NC^Nv is transformed into $(\sqrt{1 - q^2}q^NC^Nv, q^{N+1}C^{N+1}v)$; that transformation preserves the inner product. The terminal has norm $q^N\|v\|$, which tends to zero; the infinite sum converges and preserves the norm. □

For two initial states, this proof preserves their cross products; it does not merely sum energies of isolated trajectories. The positive sign of memory follows from orthogonality of its component spaces and from the isometry of the transport already constructed.

97.10.4 Dual memory circle and arithmetic Cayley chart

The integer winding register admits the bilateral shift $Se_n = e_{n+1}$ on $\ell^2(\mathbb{Z})$. Its characters $n \mapsto z^n$, $|z| = 1$, give the dual circle of memory. The visible phase $r \in \{0, \dots, 8\}$ returns; the integer n is not reduced modulo 12 in the complete state.

If that reduction is imposed, a character descends only when $z^{12} = 1$. Thus memory without reset has a precise consequence: it permits the full circle of characters, whereas the finite calendar retains only a finite selection. This does not determine which arithmetic measure is realized on that circle.

The second circle appearing in the spectral chapter of the integral is a chart of the Cayley realization. Subsection 97.10.10 constructs a Cayley operator directly on the test functions before constructing a Hilbert space through the Weil form. This makes it possible to compare arithmetic transport with memory transport without introducing the heights of the zeros as inputs.

97.10.4.1 Preservation of the arithmetic form under nonadic compression

Arithmetic invariance of the auxiliary Cayley readout permits an additional composition. For $a > b > 1/2$, let C_a be the operator constructed in Subsection 97.10.10, and let $\mathcal{T}_a = (8I + C_a)/9$. The equality $\mathscr{W}(C_a f, C_a g) = \mathscr{W}(f, g)$ implies, by expansion,

$$\boxed{\mathscr{W}(f,g) = \mathscr{W}(\mathcal{T}_a f, \mathcal{T}_a g) + \frac{8}{81}\mathscr{W}((I - C_a)f, (I - C_a)g).} \tag{97.51}$$

This identity preserves the gamma channel, prime clocks, and polar term jointly. It does not presuppose that the form is positive. The calculation consists in adding coefficients: the mixed terms have coefficients 8 and -8, which cancel, while the two diagonal terms contribute $72 + 9 = 81$ times the initial form. Its finite iteration gives

$$\mathscr{W}(f,g) = \mathscr{W}(\mathcal{T}_a^N f, \mathcal{T}_a^N g) + \frac{8}{81}\sum_{j=0}^{N-1} \mathscr{W}((I - C_a)\mathcal{T}_a^j f, (I - C_a)\mathcal{T}_a^j g). \tag{97.52}$$

In the ordinary L^2 representation, where C_a is already unitary, the same calculation is a positive norm decomposition. On the arithmetic form it is, initially, exact conservation of signed energy. The difference between these two assertions is located in a common identity; it is not hidden in a change of representation.

In a circular chart $z = e^{i\theta}$, the compression sends z to $t(z) = (8 + z)/9$. Therefore,

$$1 - |t(z)|^2 = \frac{8}{81}|1 - z|^2 = \frac{32}{81}\sin^2(\theta/2).$$

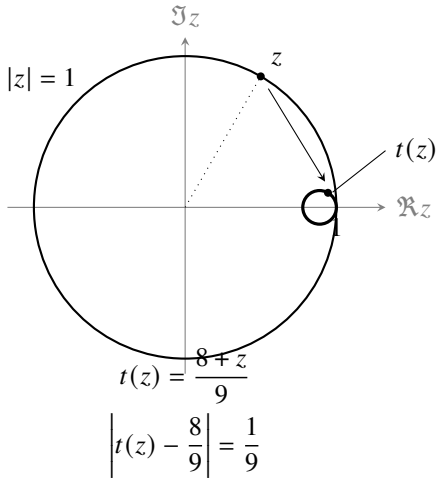
The angular separation thus has an exact expression as complementary energy. This is the image of the nonadic compression of a phase chart; it is not identified by the drawing with another operator in the corpus. Nor is this map $t(z)$ confused with the radial factor $5/9$, which comes from the two modes of the central APP chart.

El desarrollo local de Schur da signo positivo a la forma aritmética en un dominio funcional concreto y a sus imágenes por transporte. Las identidades (97.51)–(97.52) no autorizan por sí solas a declarar positivos todos sus sumandos para una prueba arbitraria: permiten localizar exactamente qué energías y correlaciones debe conservar una extensión de ese dominio.

97.10.5 Criterio de transporte de positividad mediante momentos mixtos

Sea $\mathcal{D}$ el dominio común de los lectores previamente construidos. Sea $\mathscr{W}$ la forma sesquilineal que publican sus canales gamma, primos y polar. La fórmula analítica se recibe después del generador nativo; no selecciona las semillas ni los coeficientes anteriores.

Sean C_a el transporte reversible sobre pruebas y S el desplazamiento de la memoria. Se utiliza explícitamente $C_a(\mathcal{D}) = \mathcal{D}$ y la identidad $\mathscr{W}(C_a f, C_a g) = \mathscr{W}(f, g)$, demostrada para la clase exponencial



Publicación de fase

$$r \longmapsto r$$

$$\downarrow$$

Transporte de memoria

$$(r, n) \longmapsto (r, n+1)$$

$$1 - |t(z)|^2 = \frac{8}{81}|1 - z|^2$$

Publicación y complemento

Figure 97.1: La carta circular de la compresión nonádica. A la izquierda, el círculo unidad y su imagen de centro $8/9$ y radio $1/9$, dibujados a la misma escala; se muestra la imagen de $z = e^{i\pi/3}$. A la derecha, el retorno de fase se acompaña de avance de memoria. El balance inferior cuantifica la componente complementaria. Esta figura representa el mapa t, distinto del factor modal $q = 5/9$ de la carta central.

de pruebas en las subsecciones 97.10.10 y 97.10.11. Para una familia $\mathcal{E} \subset \mathcal{D}$, supóngase construido un mapa lineal $V : \operatorname{span} \mathcal{E} \to \mathcal{K}$ tal que, para todo f, g de esa familia y todo entero n,

$$\boxed{\mathscr{W}(C_a^n f, g) = \langle S^n V f, V g \rangle.} \tag{97.53}$$

Proposition 97.35 (Transport of positivity by identification of moments)**.** *La identidad* (97.53) *implica la positividad de $\mathscr{W}$ sobre el espacio generado por las órbitas $C_a^n \mathcal{E}$. Si ese espacio es denso en $\mathcal{D}$ para la norma conjunta de los lectores, y éstos son continuos para esa norma, la positividad se extiende a $\mathcal{D}$.*

Proof. Para $F = \sum_j C_a^{n_j} f_j$, la invariancia de $\mathscr{W}$ bajo C_a y (97.53) dan

$$\begin{aligned} \mathscr{W}(F, F) &= \sum_{i,j} \mathscr{W}(C_a^{n_i - n_j} f_i, f_j) \\ &= \left\| \sum_i S^{n_i} V f_i \right\|^2 \geq 0. \end{aligned}$$

If a combination represents the zero function, the same identity forces the vector combination to have zero norm; the realization is well-defined. Passage to the closure uses continuity of both readouts, not a subsequent choice of sign. □

This is a construction criterion with explicit domains, not an assertion that (97.53) has been proved for all the readers in the corpus. To extend from the initial family to its orbits, all cross moments must be preserved: proving it only with $n = 0$ on that initial family does not complete this extension. Naturally, a Gram identity already proved for $n = 0$ on all of $\mathcal{D}$ would directly prove positivity in that domain.

97.10.6 Energy identification and two-column realization

The corpus preserves two columns J_+ and J_- and an effective analytic connector between them. It also describes a native encoding Ξ and an orthogonal projection P. When the two moment identities are calculated on the same domain,

$$\langle \Xi f, \Xi g \rangle = \langle J_+ f, J_+ g \rangle, \qquad \langle P \Xi f, P \Xi g \rangle = \langle J_- f, J_- g \rangle,$$

subtraction gives

$$\mathscr{W}(f,g) = \langle (I-P)\Xi f, (I-P)\Xi g\rangle. \tag{97.54}$$

The dossier calculates the uniform expectation and Helmert refinement separately. Their joint conjugation preserves both Grams and cannot, by a mere change of notation, turn a difference operator into an evaluation operator. The recovered gamma connector realizes an additional analytic operation, which must preserve its energy budget. This clarification brings together existing operations; it does not declare the TPK architecture nonexistent.

Formulas (97.53) and (97.54) are two compatible expressions of the route to be completed: the first organizes returns and their correlations; the second organizes the visible part and complementary memory. The Schur-complement estimate in the dossier gives effective control of that energy in a complete window and all its refinements. Its scope is stated with the concrete window, without replacing the global domain by that window.

97.10.7 Operator provenance and the function of the readouts

The chain used is APP → TRIT → TPK → enriched histories of the joint continuum → arithmetic and memory readers → circular representation. The five operations of the continuum remain on the same histories; this development studies a reader of their memory, not constructing an isolated fifth branch. The prime reading uses the irreducibles produced by the native product; the constants already published by HMT intervene only afterwards in the readers.

Provenance. Cocycle, loss, two sheets, memory groups, and double circle: preexisting authorial architecture and recovered results. The assembled proofs below are a new formalization of these antecedents. No historical priority is claimed for Poisson kernels, Cayley, or Toeplitz identities. The provenance of this exact composition within the corpus has not been exhaustively investigated beyond the source owners indicated.

97.10.8 Regular memory representation preceding arithmetic GNS

The central APP block, with its symmetric and antisymmetric projectors, is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 9P_+ + 5P_-.$$

The normalization of the differential mode publishes $q = 5/9$. The complete TPK turn preserves the phase and increases the integer memory:

$$m(\Gamma_9^n) = n, \qquad m(\gamma_2\gamma_1) = m(\gamma_2) + m(\gamma_1).$$

On a reversible orbit of that memory, its lifted states are identified with e_n, $n \in \mathbb{Z}$, and the winding is the shift $Se_n = e_{n+1}$. Injectivity of this enumeration follows from the memory increment. If one starts only from the semigroup of positive windings, the same bilateral space is the shift extension adding its predecessors; reversibility is not attributed to a transition proved only isometric. In a complete tree, the remaining labels are preserved in a transverse factor: the shift acts on memory and does not identify routes.

This S is not the Weil Cayley operator inserted into the seam. It is the regular realization of the integer memory operation. Its Hilbert-space positivity is known before forming a GNS of the arithmetic form.

97.10.8.1 Loss vector, correlations, and terminal term

Telescoping over windings produces weights $(1-q^2)q^{2j}$. Their amplitude in memory is the explicit vector

$$v_q = \sqrt{1-q^2}\sum_{j=0}^{\infty} q^j e_j, \qquad \|v_q\|^2 = 1.$$

With inner product linear in the first variable, for $n \geq 0$,

$$\langle S^n v_q, v_q\rangle = (1-q^2)\sum_{j\geq 0} q^j q^{j+n} = q^n.$$

Conjugation and unitarity give, for every integer,

$$\boxed{m_n^{\mathrm{mem}} = q^{|n|}.}$$

The truncation $v_{q,N} = \sqrt{1-q^2}\sum_{j=0}^{N-1} q^j e_j$ preserves the terminal exactly:

$$\|v_{q,N}\|^2 = 1-q^{2N}, \qquad \langle S^n v_{q,N}, v_{q,N}\rangle = q^n\big(1-q^{2(N-n)}\big), \quad 0 \leq n < N.$$

For $n \geq N$, that truncated correlation is zero. Thus, for each fixed n, the difference from the complete moment is q^{2N-n} when $N > n$, and tends to zero. This formula neither erases the last register nor equates a truncation with the whole history.

97.10.8.2 Triangular factorization and determinant of the memory matrices

For indices $0, \ldots, N$, define

$$T_N = (q^{|i-j|})_{0\leq i,j\leq N}.$$

It is a Gram matrix of $v_q, Sv_q, \ldots, S^N v_q$, so

$$\sum_{i,j=0}^{N} \overline{c_i} c_j (T_N)_{ij} = \left\| \sum_{j=0}^{N} c_j S^j v_q \right\|^2 \geq 0.$$

There is also an exact proof without infinite series. Let

$$L_{ij} = \begin{cases} q^{i-j}, & i \geq j, \\ 0, & i < j, \end{cases} \qquad D = \mathrm{diag}(1, 1-q^2, \ldots, 1-q^2).$$

Then $T_N = LDL^*$. For $i \geq j$, the product entry is

$$q^{i+j} + (1-q^2)\sum_{k=1}^{j} q^{i+j-2k} = q^{i-j}.$$

Symmetry gives the full matrix identity. A Cholesky factor is $LD^{1/2}$, and, since $\det L = 1$,

$$\boxed{\det T_N = (1-q^2)^N = (56/81)^N > 0.}$$

This positive kernel arises from loss and preserved memory. It requires no zeros, positive zeta function, or Weil GNS.

97.10.9 Phase, nonadic refinement, and weak convergence of states

A fixed-phase representation is obtained from

$$v_{r,\theta} = \sqrt{1-r^2} \sum_{j \geq 0} r^j e^{-ij\theta} e_j, \quad 0 < r < 1,$$

and yields $m_n = r^{|n|} e^{in\theta}$. The Toeplitz matrices are diagonal conjugates of the preceding ones and have determinant $(1-r^2)^N$. The phase θ here is a coordinate of the character family; no arithmetic spectral phase has been selected.

On the circle, with normalized Haar measure $d\lambda$, these moments have the explicit measure

$$d\nu_{r,\theta}(z) = \frac{1-r^2}{|1-re^{-i\theta}z|^2}\, d\lambda(z).$$

Expanding both geometric series directly proves its Fourier coefficients without using a target measure.

97.10.9.1 Nonadic roots of the radius and phase compatibility

Taking $r_m = q^{1/9^m}$, one has $0 < r_m < 1$, $r_{m+1}^9 = r_m$, and $r_m \to 1$. For fixed phase,

$$r_m^{|n|} e^{in\theta} \longrightarrow e^{in\theta} \quad \text{for each } n \in \mathbb{Z}.$$

The measures are probabilities; uniform approximation by trigonometric polynomials extends moment convergence to every continuous function. Consequently,

$$\nu_{r_m,\theta} \rightharpoonup \delta_{e^{i\theta}}.$$

This shows how an atomic measure can arise as a weak limit of memory states. It is not a strong limit of the vectors in the same regular representation: each coefficient of $v_{r_m,\theta}$ tends to zero, but their norms remain one.

A naturality qualification must be preserved. Under $p_9(z) = z^9$,

$$(p_9)_* \nu_{r,\theta} = \nu_{r^9, 9\theta}.$$

Thus the nonadic radius with fixed phase does **not** by itself give a projective system of measures under p_9. This requires retaining lifted phases with $9\theta_{m+1} = \theta_m \pmod{2\pi}$, including their branch choices; alternatively, one must declare that the refinement used keeps the angular coordinate fixed and changes only the radius. These are different operations.

97.10.9.2 State selection and change of spectral type

Memory produces the group, the positivity of its Grams and these compatible correlation families. A specific phase or a mixture of phases requires the reader that selects their amplitudes. For example, the vector e_0 yields $m_n = \mathbf{1}_{n=0}$; $(e_0 + e_1)/\sqrt{2}$ yields $m_0 = 1, m_{\pm 1} = 1/2$ and all other moments zero. The same memory operation admits both positive states.

The regular representation has vector measures absolutely continuous with respect to Haar measure. A fixed L^2 vector filter cannot produce a nonzero purely atomic measure. The preceding weak limit shows a precise way for spectral type to change under refinement: that limit of states must be preserved and controlled, rather than merely increasing the length of a previously chosen phase orbit. This does not deny the spectral representation of the corpus or joint HMT generation; it distinguishes the domains of two readouts.

97.10.10 Cayley operator on the exponential test-function domain

A second object is now constructed, without assuming Weil positivity. The native product and its irreducibles already provide $\ell_{p,k} = k \log p$, $w_{p,k} = (\log p) p^{-k/2}$. The gamma and polar readings complete the form $\mathscr{W}$ written in subsection 97.10.11. The role of memory is to offer an integer iteration index; the following operator acts effectively on the functions of that publication.

The operator C_a is an auxiliary analytic readout of this formalization. The choice $a > 1/2$ does not silently replace the canonical seam $a = 1/2$, nor is it declared that this choice has already been selected by the TPK.

Fix $a > b > 1/2$. Let $\mathscr{E}_b$ be the space of smooth functions such that, for each derivative order j, $\sup_x e^{b|x|} |f^{(j)}(x)| < \infty$. It contains all compactly supported test functions. Define

$$(R_a^+ f)(x) = \int_0^\infty e^{-at} f(x+t)\, dt, \qquad (R_a^- f)(x) = \int_0^\infty e^{-at} f(x-t)\, dt,$$

$$\boxed{C_a f = f - 2a R_a^+ f, \qquad C_a^{-1} f = f - 2a R_a^- f.}$$

These integrals are translation readouts constructed on the test functions; they do not use the spectrum of zeros. The parameter $a = 1$, for example, comes from the unit and allows a choice $1/2 < b < 1$; it calibrates no zero.

Domain, inverse, and unitary realization. The inequality $e^{b|x|} |f(x \pm t)| \le M_0 e^{bt}$ gives the bound $\|R_a^\pm f\|_b \le \|f\|_b / (a-b)$, also for every derivative. Thus both operators preserve $\mathscr{E}_b$. With the convention $\widehat{f}(z) = \int f(x) e^{-izx}\, dx$, their multipliers are

$$\widehat{C_a f}(z) = c_a(z) \widehat{f}(z), \qquad c_a(z) = \frac{z - ia}{z + ia},$$

and the inverse operator has multiplier $c_a(z)^{-1}$. The identity on the real Fourier axis proves both inverse compositions. In L^2, $|c_a(t)| = 1$, so C_a is unitary for the ordinary norm, without using the Weil norm. On test functions it can be written $C_a = (H + ia)(H - ia)^{-1}$, with $H = i\partial_x$; here the integral formula, not a GNS generator, defines the operators.

Convergence condition at the polar boundary. The parameter $a = 1/2$ of the source owner's Cayley operator cannot simply be inserted into these pre-GNS integrals. For a compactly supported test function, $R_{1/2}^+ f$ generally has a tail proportional to $e^{x/2}$ towards $-\infty$; its negative polar evaluation does not converge. Divergence of the assembled form has not been proved; rather, this term-by-term decomposition requires the margin $a > 1/2$. The GNS Cayley operator may subsequently use $a = 1/2$ on its Hilbert-space domain.

97.10.11 Joint preservation of the gamma, prime, and polar channels

For $f, g \in \mathscr{E}_b$, let

$$C_{f,g}(u) = \int f(x) \overline{g(x-u)}\, dx, \qquad P_f^\pm = \int f(x) e^{\pm x/2}\, dx.$$

The complete form is defined by

$$\begin{aligned}\mathscr{W}(f,g) = c_\Gamma C_{f,g}(0) + \int_0^\infty \frac{e^{-u/2}}{1-e^{-2u}}\,[2C_{f,g}(0) - C_{f,g}(u) - C_{f,g}(-u)]\,du \\ - \sum_{p,k} w_{p,k}\,[C_{f,g}(\ell_{p,k}) + C_{f,g}(-\ell_{p,k})] \\ + P_f^+\overline{P_g^-} + P_f^-\overline{P_g^+}, \qquad c_\Gamma = \psi(1/4) - \log\pi.\end{aligned}$$

Existence and continuity of the complete form. The correlation bound $|C_{f,g}(u)| \le M_f M_g(|u| + 1/b)e^{-b|u|}$ proves absolute convergence of the prime sum, dominated by a series of the form $\sum_{n\ge 2}(\log n)(1+\log n)n^{-b-1/2}$. The symmetric correlation difference is $O(u^2)$ at zero, whereas the gamma weight is $O(u^{-1})$; at infinity the weight is exponential. Polar evaluations converge because $b > 1/2$. These bounds also prove continuity in the indicated seminorms. Positivity is not used.

Simultaneous invariance of the three channels. Since C_a is unitary on L^2 and commutes with all translations,

$$C_{C_a f, C_a g}(u) = C_{f,g}(u) \qquad (u \in \mathbb{R}).$$

By Fubini, the two polar readouts satisfy

$$P^+_{C_a f} = -\frac{a-1/2}{a+1/2}P_f^+, \qquad P^-_{C_a f} = -\frac{a+1/2}{a-1/2}P_f^-.$$

The factors are real and reciprocal, so each cross polar product is preserved exactly. Combining the three channels,

$$\boxed{\mathscr{W}(C_a f, C_a g) = \mathscr{W}(f,g).}$$

The identity holds for all integer powers and all cross terms. It proves invariance of a Hermitian form, not its sign.

97.10.12 Circular moments of the complete arithmetic form

Define

$$m_n^{\mathrm{ar}}(f,g) = \mathscr{W}(C_a^n f, g), \qquad n \in \mathbb{Z}.$$

The complete formula in Subsection 97.10.11, with $C_a^n f$ in its first argument, calculates these moments from primes, gamma, and polar terms. Each evaluation exists in $\mathscr{C}_b$; it requires neither a list of zeros nor a positive metric. Hermiticity and the preceding invariance give

$$m^{\mathrm{ar}}_{-n}(g,f) = \overline{m^{\mathrm{ar}}_n(f,g)}, \qquad \mathscr{W}(C_a^j f, C_a^k g) = m^{\mathrm{ar}}_{j-k}(f,g).$$

For a single test function, the matrix $T_N^{\mathrm{ar}}(f)_{jk} = m^{\mathrm{ar}}_{k-j}(f,f)$ is Hermitian Toeplitz and satisfies the exact identity

$$c^* T_N^{\mathrm{ar}}(f) c = \mathscr{W}\left(\sum_{k=0}^N c_k C_a^k f, \sum_{k=0}^N c_k C_a^k f\right).$$

The matrix has been constructed without assuming positivity. The equality allows its positivity to be compared with a native memory Gram; it does not presuppose it.

97.10.12.1 Polar growth and a necessary cancellation condition

For $a = 1$, the exact polar contribution to $m_n^{\mathrm{ar}}(f, g)$ is

$$(-1/3)^n P_f^+ \overline{P_g^-} + (-3)^n P_f^- \overline{P_g^+}.$$

For a nonzero real, even, positive test function, $P_f^+ = P_f^- = p > 0$, so $p^2[(-1/3)^n + (-3)^n]$ appears. If the total moment sequence is positive definite, its order-two minors force

$$|m_n^{\mathrm{ar}}(f, f)| \le m_0^{\mathrm{ar}}(f, f).$$

Thus circular identification must account for the concrete cancellation of this growth with the gamma and prime channels. Formal symmetry of the two sheets does not replace that calculation. This observation does not declare that cancellation fails; it provides an exact falsifier for a candidate composition.

97.10.12.2 Mixed moment matrices and domain coverage

For a family of tests $f_1, \dots, f_d$, the appropriate object is the block Toeplitz matrix

$$\mathsf{T}_{(j,r),(k,s)} = m_{k-j}^{\mathrm{ar}}(f_s, f_r).$$

Its quadratic form is $\mathscr{W}(F, F)$, where $F = \sum_{k,s} c_{k,s} C_a^k f_s$. Controlling only the scalar matrices of some tests does not control the cross terms between different tests. For a family dense in each $C_c^\infty([-R, R])$, the proved continuity allows passage to the complete domain, provided that the mixed Grams and compatibilities as R increases have been established. The growth of N for a single orbit does not prove that coverage.

97.10.12.3 Transport of local coercivity to subspaces with tails

The preceding identity has a positive application that does not yet require identifying all moments with memory. Suppose the Schur complement has proved, on the interval $I = (-1/18, 1/18)$, the bound

$$\mathscr{W}(f, f) \ge \tfrac{1}{2}\|f\|_{L^2}^2, \qquad f \in C_c^\infty(I).$$

This subsection receives that estimate as a result of the corresponding local calculation; it does not deduce it from the existence of C_a. Let $(\tau_t f)(x) = f(x - t)$. Translation preserves correlations and satisfies

$$P_{\tau_t f}^{\pm} = e^{\pm t/2} P_f^{\pm}.$$

Its cross polar products are also preserved. Therefore $\mathscr{W}(\tau_t f, \tau_t g) = \mathscr{W}(f, g)$, and the ordinary L^2 norm is invariant. Since C_a and τ_t commute, for every integer n, every $t \in \mathbb{R}$, and every local test function,

$$\boxed{\mathscr{W}(C_a^n \tau_t f, C_a^n \tau_t f) = \mathscr{W}(f, f) \ge \tfrac{1}{2}\|C_a^n \tau_t f\|_{L^2}^2.}$$

For fixed (n, t), this is a coercive bound on the entire subspace $C_a^n \tau_t C_c^\infty(I)$, not merely on one example. When $n \ne 0$, transformed test functions may have noncompact exponential tails. Infinitely many prime clocks may enter their arithmetic form; exact equality of the three channels controls their composition, and the bounds in Subsection 97.10.11 guarantee convergence. That sum has not been replaced by a finite truncation.

Cross terms between subspaces with different exponents or translations, by contrast, retain their complete moments. For example,

$$\mathscr{W}(C_a^n \tau_t f, C_a^m \tau_s g) = \mathscr{W}(C_a^{n-m} \tau_{t-s} f, g).$$

The bound for each individual subspace does not imply the positivity of their algebraic sum. This is decided through the mixed Grams of subsection 97.10.12.2. An effective positive scope—entire families of tests with tails—is thus isolated without extending it to the entire domain on the sole basis of transport invariance.

97.10.13 Prospective identification of moments and energy

Memory already produces the positive regular representation of Subsection 97.10.8; the arithmetic channels produce C_a and m_n^{ar} without GNS. Formulating the global step also requires specifying the representation that receives the state limit. Denote that space by $\mathcal{K}_{\lim}$ and its unitary winding operator by U_{mem}, both to be constructed from histories and limit compatibilities. They are not identified in advance with the regular space and its shift S. A prospective identification has the form

$$\boxed{m_n^{\mathrm{ar}}(f,g) = \langle U_{\mathrm{mem}}^n \mathcal{E} f, \mathcal{E} g\rangle_{\mathcal{K}_{\lim}}}$$

for a readout $\mathcal{E}$ constructed from operations on the same histories and not defined as $\mathscr{W}^{1/2}$. In that case, all mixed Toeplitz matrices are Grams; arithmetic GNS follows as a realization of a product already obtained.

It suffices to prove the zeroth-moment identity and the intertwining $\mathcal{E}C_a = U_{\mathrm{mem}}\mathcal{E}$, with the stated domains and compatibilities. The zeroth moment is identification of the complete energy, not a lesser condition: it cannot be regarded as proved merely because an $\mathcal{E}$ with some positive Gram exists.

Why distinguishing the representations matters. The regular shift on $\ell^2(\mathbb{Z};\mathcal{K})$ has no nonzero eigenvectors. Since S is unitary, any eigenvalue would have modulus 1. Indeed, if $Sx = \lambda x$ with $|\lambda| = 1$, the coordinate relation forces $\|x_n\|$ to be constant; square summability forces $x = 0$. Thus a representation with a nonzero eigenvector admits no isometric intertwiner into that shift: the image would be a nonzero eigenvector of S. This observation applies to atomic arithmetic GNS if its global hypotheses hold. It does not contradict weak convergence of states in Subsection 97.10.9.2; it shows that the limit requires its own representation and not a limit vector in the original regular space. Construction of $(\mathcal{K}_{\lim}, U_{\mathrm{mem}})$ and the readout identifying the moments is part of the prospective condition, not a consequence already proved from the regular Grams.

In the manuscript's two-column form, the calculation remains

$$\Xi^*\Xi = J_+^*J_+, \qquad \Xi^*P\Xi = J_-^*J_-, \qquad \mathcal{E} = (I-P)\Xi.$$

If P is an orthogonal projection, these two equalities produce $\mathcal{E}^*\mathcal{E} = J_+^*J_+ - J_-^*J_-$. Verifying only one of them does not suffice, nor does replacing the complete columns with those of an isolated gamma block. The phases and moments in this development make explicit what that composition must transmit to all windings.

The kernel $q^{|n|}$ is not identified here with $m_n^{\mathrm{ar}}(f,g)$. It is an effective positive antecedent and offers a local factorization; the readout $\mathcal{E}$ must also preserve prime labels, weights, polar contributions, and the state limit if spectral type changes. Declaring that equality by definition would introduce the result one seeks to construct.

97.10.14 Exact checks and scope of the composition

The executable check preserves 1.046 exact checks with rational fractions: factorization LDL^* and determinants in dimensions 1–13 for $q = 5/9$, correlations truncated at 20 positions and reciprocal polar factors for four rational values of $a > 1/2$. The ordinary and optimized runs use explicit conditions, independent of assertion instructions. Their sources, commands and receipts are preserved in the reproduction dossier. No interval campaign of Weil moments was run here, nor are those checks converted into a global proof.

The usable advance is twofold: loss memory produces a positive Toeplitz kernel with explicit Cholesky factor and terminal, and the complete form admits an integral Cayley operator preceding GNS that constructs its circular moments and preserves each channel. Nonadic regularization also shows how to pass from regular states to atoms by a weak limit, with its phase condition correctly separated. The union of both constructions retains identification of the two complete moments as a precise equation.

This development neither refutes that union, proves its absence throughout the corpus, nor declares it established by the local identities. It leaves a concrete composition that can be compared and developed in the continuation of this manuscript.

97.11 Boundary expectations and reconstruction of the analytic channels

97.11.1 Action of the readers and content of the moments

The explicit actions of the odometer, the difference reader, the uniform inclusion, the Helmert refinement, the bidirectional boundary and the polar chart have been recovered and composed. The calculation yields:

1. The moments of the positive and negative columns, considered separately, are determined by explicit operators. They include all cross products between tests.
2. The uniform expectation does not coincide with the Helmert refinement. On the carry mode it gives an exact factor, calculated below; it does not by itself convert a translation difference into the uniform mode.
3. The gamma-tail resolvent does perform the analytic conversion and exactly reconstructs all negative rows, over every window and before using a sign.
4. The simultaneous identification of the encoder's first moment and the moment of its projection is not obtained from the three preceding chart identities by a simple direct sum. The specific additional composition that must be verified is identified below. The structural reader is not declared absent, nor is this focal check identified with the full TPK.

The advance is a calculation of action and cross products, not a reiteration that the projection is positive.

Provenance of the charts. The formulas are gathered from the developments *Polar spectral proof*, *Form and Gram*, *Gamma-tail connector*, *Odometer and carry*, *Pulsation and expectation* and *Modular polar chart*. The locators and the reading performed are preserved in the delivery's source dossier. Each chart is used with the spaces, normalizations and identities made explicit below; a documentary reference does not replace these conditions.

97.11.2 Conventions and genealogical snapshot

The relevant starting point is the APP–TRIT–TPK state, its refinement with memory and its native readers. In this formalization:

- APP provides the support and the selector of irreducibles. The native product, coproductivity and the selector precede the publication of prime–power labels.
- TRIT and sheet transport preserve orientation and seam.
- The TPK preserves the phase step and the carry record; its odometer induces the translation difference. The logarithmic publication of the clocks produces $\ell_{p,k} = k \log p$ and $w_{p,k} = (\log p) p^{-k/2}$.
- The boundary and polar charts are subsequent realizations of the same readers. No state is selected by means of zeta zeros or a target value.

This section calculates the arrow between those readers and the expectation, after the common generation; it does not seek to reconstruct or replace the joint continuum.

We work with orthonormal history coordinates. In them $u_N = N^{-1/2}(1, \dots, 1)$. If functions with uniform probability measure are used, the constant coordinates and averages are conjugated by the corresponding normalization factor; the projections and all the following norm ratios are the same.

Let $H = L^2(\mathbb{R})$, $T_a f(x) = f(x - a)$, $\mathcal{D}_R = C_c^\infty((-R/2, R/2))$. We write $d_a = T_a - I$; simultaneously changing to $I - T_a$ preserves their Grams, but changes the sign of the inverting cotrace.

The inner product is linear in the first variable. The polar evaluations and the scalar of the complete form are retained,

$$\begin{aligned} A_f &= \int_{\mathbb{R}} f(x) e^{x/2}\,dx, & B_f &= \int_{\mathbb{R}} f(x) e^{-x/2}\,dx, \\ c_\Gamma &= -\gamma - \frac{\pi}{2} - 3\log 2 - \log \pi < 0, & \mu(a) &= \frac{e^{-a/2}}{1-e^{-2a}}. \end{aligned} \tag{97.55}$$

Here π, γ and the logarithms preserve the internal publication and subsequent realization declared in the preceding chapters. The formula does not introduce target values into the generator.

97.11.3 Action of the odometer on generators

For $N = 9^m$, $m \geq 1$, and the seam located at the first vector,

$$U_{a,N}(e_r \otimes f) = \begin{cases} e_{r+1} \otimes f, & r < N-1, \\ e_0 \otimes T_a f, & r = N-1. \end{cases} \tag{97.56}$$

The uniformization $i_N f = u_N \otimes f$ gives

$$U_{a,N} i_N f = u_N \otimes f + N^{-1/2} e_0 \otimes d_a f. \tag{97.57}$$

Therefore, for $P_N = |u_N\rangle\langle u_N| \otimes I$,

$$\begin{aligned} i_N^* U_{a,N} i_N &= I + \frac{d_a}{N}, \\ (I - P_N) U_{a,N} i_N &= \frac{\sqrt{N-1}}{N} \eta_N \otimes d_a. \end{aligned} \tag{97.58}$$

where

$$\eta_N = \frac{N e_0 - \mathbf{1}_N}{\sqrt{N(N-1)}}. \tag{97.59}$$

From this the normalized difference reader is recovered:

$$\widehat{\delta}_{a,m} f = \eta_N \otimes u_{81} \otimes d_a f, \qquad j_m f = u_N \otimes u_{81} \otimes f. \tag{97.60}$$

The action on any combination of generators, without eliminating their cross terms, is

$$\left\langle \sum_i \widehat{\delta}_{a_i,m} f_i, \sum_j \widehat{\delta}_{b_j,m} g_j \right\rangle = \sum_{i,j} \langle d_{a_i} f_i, d_{b_j} g_j \rangle. \tag{97.61}$$

In particular,

$$\widehat{\delta}_{a,m}^* \widehat{\delta}_{b,m} = d_a^* d_b, \qquad j_m^* j_m = I, \qquad j_m^* \widehat{\delta}_{a,m} = 0. \tag{97.62}$$

The last cross product comes from $\langle u_N, \eta_N \rangle = 0$. It is material information on the orientation of the modes, in addition to the two isolated norms.

97.11.4 Two distinct refinements and their expectations

97.11.4.1 Uniform inclusion and projection of the last digit

Write $M = 9N$ and $\mathbb{C}^M = \mathbb{C}^N \otimes \mathbb{C}^9$. The inclusion and its adjoint are

$$\iota_N x = x \otimes u_9, \qquad E_N = I_N \otimes \langle u_9|, \qquad P_N^{\mathrm{unif}} = \iota_N E_N. \tag{97.63}$$

Acting in coordinates,

$$E_N u_M = u_N, \qquad E_N \eta_M = a_N \eta_N, \qquad a_N = \sqrt{\frac{N-1}{9N-1}}. \tag{97.64}$$

Proof. We have $E_N e_0^{(M)} = e_0^{(N)}/3$ and $E_N \mathbf{1}_M = 3\mathbf{1}_N$. Substituting into $\eta_M = (Me_0 - \mathbf{1}_M)/\sqrt{M(M-1)}$ yields (97.64). □

Consequently,

$$(E_N \otimes I)\widehat{\delta}_{a,m+1} = a_N \widehat{\delta}_{a,m}, \qquad (E_N \otimes I) j_{m+1} = j_m. \tag{97.65}$$

All projected moments are determined:

$$\begin{aligned} \widehat{\delta}^*_{a,m+1}(P_N^{\mathrm{unif}} \otimes I)\widehat{\delta}_{b,m+1} &= a_N^2 d_a^* d_b, \\ j^*_{m+1}(P_N^{\mathrm{unif}} \otimes I) j_{m+1} &= I, \\ j^*_{m+1}(P_N^{\mathrm{unif}} \otimes I)\widehat{\delta}_{b,m+1} &= 0. \end{aligned} \tag{97.66}$$

The complementary fraction of the carry is $1 - a_N^2 = 8N/(9N-1)$. For $N = 9$, the published fraction is $1/10$, not 1, and its output remains a difference.

Here P_N^{unif}, which averages the last digit, is not confused with the projection onto the single global line u_M: the latter annihilates η_M entirely.

97.11.4.2 Helmert refinement

The operator W_N is fixed by the orthonormal Helmert bases of the two levels, ordered to preserve the uniform and carry modes. It is the isometry that sends the first N vectors of the basis at level N to the corresponding vectors of the basis at level M. In particular, it satisfies

$$W_N u_N = u_M, \qquad W_N \eta_N = \eta_M, \qquad W_N^* W_N = I. \tag{97.67}$$

Thus, $E_N^{\mathrm{H}} = W_N^*$, $P_N^{\mathrm{H}} = W_N W_N^*$ give

$$E_N^{\mathrm{H}} j_{m+1} = j_m, \qquad E_N^{\mathrm{H}} \widehat{\delta}_{a,m+1} = \widehat{\delta}_{a,m}. \tag{97.68}$$

The uniform–carry cross product remains zero and that of two differences remains $d_a^* d_b$. The refinement is therefore natural for both modes, but does not transform one into the other. Specifically, $W_N \neq \iota_N$.

97.11.4.3 Conjugation of expectations and preservation of the moments

There is a unitary S_M on $\mathbb{C}^M$ that takes $\iota_N \mathbb{C}^N$ to $W_N \mathbb{C}^N$, with $W_N = S_M \iota_N$. It suffices to map the two orthonormal bases and complete the complements. Then

$$P_N^{\mathrm{H}} = S_M P_N^{\mathrm{unif}} S_M^*. \tag{97.69}$$

Simultaneously conjugating the state C and its projection,

$$(S_M C)^*(S_M P S_M^*)(S_M C) = C^* P C, \qquad (S_M C)^*(S_M C) = C^* C. \tag{97.70}$$

Neither moment changes. If only the projection is changed, in this case one passes from the fraction a_N^2 to 1, but the analytic operator remains d_a, not I.

More generally, for any projection acting only on the finite history fiber, the moment of a single channel is $\langle \eta, P\eta\rangle d_a^* d_b$. That change of basis does not convert it into $2I$. Any effective coupling between analytic channels is another operation and must be written down, not inferred from conjugation.

97.11.5 Metric identities of the boundary and polar charts

97.11.5.1 Bidirectional boundary

The bidirectional chart preserves the composition

$$\begin{aligned} K_2 &= \mathbb{C}^{81} \otimes u_9^{\perp}, \\ \Delta_a &= j_{\mathrm{APP}}(\eta_9 \otimes d_a), \\ \beta_0 &= R_{\mathrm{tw}}(L^3 + \varepsilon \nabla_x) F I_{Q_9}|_{K_2}. \end{aligned} \tag{97.71}$$

In this realization, j_{APP} is the isometric inclusion of the indicated fiber; β_0 is injective and $\Lambda_{\mathrm{tw}}^{\#}$ is positive definite on the boundary space. These are the conditions of the chart: the factors of the composition are received with their compatible domains, and here that injectivity is not deduced from a coincidence of dimensions. With $V_\beta = \beta_0(\beta_0^*\beta_0)^{-1/2}$ and $U_{\mathrm{tw}} = (\Lambda_{\mathrm{tw}}^{\#})^{-1/2} V_\beta$,

$$G_a = (U_{\mathrm{tw}} \otimes I)\Delta_a, \qquad G_a^*(\Lambda_{\mathrm{tw}}^{\#} \otimes I) G_b = d_a^* d_b. \tag{97.72}$$

Proof. The polar normalization gives $V_\beta^* V_\beta = I$ and, therefore, $U_{\mathrm{tw}}^* \Lambda_{\mathrm{tw}}^{\#} U_{\mathrm{tw}} = I$. Since j_{APP} and η_9 are normalized, $\Delta_a^* \Delta_b = d_a^* d_b$. The composition proves (97.72), including all cross products. □

U_{tw} is isometric for the declared metric $\Lambda_{\mathrm{tw}}^{\#}$. In ordinary Hilbertian coordinates the isometry between the effective spaces is $(\Lambda_{\mathrm{tw}}^{\#})^{1/2} U_{\mathrm{tw}} = V_\beta$.

The identity (97.72) contains neither an expectation nor its value on G_a. To transport a projection, it must be conjugated with that same isometry and its adjoint in the correct metric. Done this way, (97.70) preserves the preceding compressed moment: changing to the boundary does not create the second moment.

97.11.5.2 Effective polar chart

The polar realization through five octads provides the five-column matrix T, whose Gram and normalization are

$$T^* T = 4 I_5 + \frac{4}{3} J_5, \qquad \widehat{T} = T(T^*T)^{-1/2}, \qquad F_{34} = (f_3, f_6, f_9, f_4, f_8). \tag{97.73}$$

The chart with its explicit spaces is

$$U = F_{34} \widehat{T}^* : \operatorname{Ran} \widehat{T} \subset \mathbb{C}^{24} \longrightarrow \operatorname{span}(f_3, f_6, f_9, f_4, f_8) \subset \mathbb{C}^{12}. \tag{97.74}$$

Para $\eta = \operatorname{diag}(1, 1, 1, -1, -1)$,

$$U^* J_{34} U = \widehat{T} \eta \widehat{T}^*, \qquad \widehat{T}^* U^* J_{34} U \widehat{T} = \eta. \tag{97.75}$$

Here the five vectors of F_{34} are orthonormal and $F_{34}^* J_{34} F_{34} = \eta$. Substituting $U = F_{34}\widehat{T}^*$ and $\widehat{T}^*\widehat{T} = I_5$ proves the two equalities in (97.75). This is the typed expression of the equality with the diagonal matrix. The five roles are effective; the identification of coordinates must not be erased.

For the two polar evaluations,

$$b_{\pm}(f) = \frac{A_f \pm B_f}{\sqrt{2}}, \qquad b_+(f)\overline{b_+(g)} - b_-(f)\overline{b_-(g)} = A_f\overline{B_g} + B_f\overline{A_g}. \tag{97.76}$$

(97.75) transports the signature and (97.76) transports its cross terms. Neither identifies the isolated positive line with a sufficient source of the negative one: for real, odd and nonzero f, chosen with $f(x) > 0$ on the positive half-axis of its support, we have $b_+(f) = 0$ and $b_-(f) > 0$. Recovery must involve the common analytic state, as does the gamma-tail connector developed below.

97.11.6 Difference between channel-by-channel reading and common transfer

To fix all the summands used, let $\mathcal{K}_m = \mathbb{C}^{9^m} \otimes \mathbb{C}^{81} \otimes H$, and let ζ_+, ζ_- be unit vectors of the two polar lines. The indices $\ell \le R$ run through the prime–power labels (p, k) with $k \log p \le R$, preserving their weight $w_\ell = (\log p)p^{-k/2}$. The columns are

$$\begin{aligned} J_{+,R}f &= \Big((\sqrt{w_\ell}\,\widehat{\delta}_{\ell,m}f)_{\ell\le R},\ (\sqrt{\mu(a)}\,\widehat{\delta}_{a,m}f)_{a>0},\ \zeta_+ b_+(f) \Big), \\ J_{-,R}f &= \Big(\sqrt{-c_\Gamma}\, j_m f,\ (\sqrt{2w_\ell}\, j_m f)_{\ell\le R},\ \zeta_- b_-(f) \Big). \end{aligned} \tag{97.77}$$

The continuous component of the first column belongs to $L^2((0,\infty), da; \mathcal{K}_m)$; the remaining components are gathered in the indicated orthogonal sums. The evaluations $b_\pm$ are those of (97.76). Their common domain is that of tests with finite gamma energy. The identity

$$\mathscr{W}_R(f,g) = \langle J_{+,R}f, J_{+,R}g\rangle - \langle J_{-,R}f, J_{-,R}g\rangle \tag{97.78}$$

is verified by expanding $\langle d_\ell f, d_\ell g\rangle = 2\langle f, g\rangle - \langle T_\ell f, g\rangle - \langle f, T_\ell g\rangle$, using the gamma integral and the polar identity (97.76). Thus the three channels and all cross terms are preserved.

Taking the explicit positive column Ξ_+ as the encoder satisfies its first moment by calculation. Applying the uniform expectation afterwards yields in each prime channel

$$w_\ell a_N^2 \langle d_\ell f, d_\ell g\rangle, \tag{97.79}$$

whereas the required negative row is

$$2w_\ell \langle f, g\rangle. \tag{97.80}$$

They are not the same identity. For example, choose $R > \ell > 0$ and real, nonnegative $f \in \mathcal{D}_R$, with support overlapping its translation by ℓ. Then

$$\|d_\ell f\|^2 = 2\|f\|^2 - 2\langle f, T_\ell f\rangle < 2\|f\|^2. \tag{97.81}$$

No contractive projection applied solely to that channel can produce $\sqrt{2}f$. For the Helmert refinement the factor a_N^2 is eliminated, but (97.81) persists. This is a falsifier of the **channel-by-channel** identification, not a refutation of a common transfer that redistributes energy between the channels.

On the other hand, directly introducing $j_m f$, its prime–scalar copies and $b_-(f)$ as orthogonal summands of the encoder ensures its outputs, but adds their norms to the first moment. Maintaining the original first moment requires an explicit isometric redistribution. That redistribution is precisely the operation that the simultaneous identification of the two moments must exhibit.

The invariance of the cyclic subspace under P, by itself, does not determine the value of PCf either; it only guarantees that this value remains in the subspace.

97.11.7 Inverse of the gamma tail and reconstruction of all rows

Here there is indeed an explicit action on every generator and every linear combination. Let

$$\mu(a)=\frac{e^{-a/2}}{1-e^{-2a}},\quad A_R=[R+1,R+2],\quad \nu_R=\int_{A_R}\mu(a)\,da>0,\quad M_R=\mathbf{1}_{(-R/2,R/2)}. \tag{97.82}$$

For the already fixed sign $d_a=T_a-I$, we define the cotrace

$$\epsilon_m=\langle\eta_{9^m}|\otimes\langle u_{81}|\otimes M_R,\qquad \mathcal{R}^-_{m,R}F=-\nu_R^{-1}\int_{A_R}\sqrt{\mu(a)}\,\epsilon_m F(a)\,da. \tag{97.83}$$

The minus sign disappears if $I-T_a$ is used, which is the alternative convention of the tail reader. For $f\in\mathcal{D}_R$ and $a>R$, $M_RT_af=0$. Consequently,

$$\mathcal{R}^-_{m,R}\big(a\mapsto\sqrt{\mu(a)}\widehat{\delta}_{a,m}f\big)=f. \tag{97.84}$$

Moreover, $\|\mathcal{R}^-_{m,R}\|\le\nu_R^{-1/2}$ by Cauchy–Schwarz. No individual translation is inverted, nor is Weil positivity used.

Write

$$S_Rf=\Big(\sqrt{-c_\Gamma}\,j_mf,\,(\sqrt{2w_\ell}\,j_mf)_{\ell\le R}\Big),\quad B_-f=\zeta_-b_-(f). \tag{97.85}$$

The specific common row, defined on the complete positive ambient space, is

$$C_R^{\mathrm{tail}}=\begin{pmatrix}0 & S_R\mathcal{R}^-_{m,R} & 0\\ 0 & B_-\mathcal{R}^-_{m,R} & 0\end{pmatrix}. \tag{97.86}$$

From (97.84),

$$C_R^{\mathrm{tail}}J_{+,R}f=J_{-,R}f\quad\text{for every } f\in\mathcal{D}_R. \tag{97.87}$$

This action contains the scalar, prime–power and polar rows, with their constants and orientations intact. For any pair f,g,

$$\begin{aligned}\langle C_R^{\mathrm{tail}}J_+f,C_R^{\mathrm{tail}}J_+g\rangle=&\Big(-c_\Gamma+2\sum_{\ell\le R}w_\ell\Big)\langle f,g\rangle\\&+b_-(f)\overline{b_-(g)}.\end{aligned} \tag{97.88}$$

For $f=\sum_i f_i$, $g=\sum_j g_j$, the right-hand side contains the complete sum in i,j; the tests have not been replaced by orthogonal cells. The prime–gamma cross terms of (97.72) are preserved when changing the input representations. The final form is

$$J_+^*\Big(I-(C_R^{\mathrm{tail}})^*C_R^{\mathrm{tail}}\Big)J_+=J_+^*J_+-J_-^*J_-=\mathcal{W}_R. \tag{97.89}$$

It is an effective equality of forms, without yet assuming that the operator in parentheses is positive. The operator is bounded; an explicit bound is

$$\|C_R^{\mathrm{tail}}\|\le\nu_R^{-1/2}\sqrt{-c_\Gamma+2\sum_{\ell\le R}w_\ell+2\sinh(R/2)-R}. \tag{97.90}$$

The bound fixes the type, not proving a norm less than or equal to one.

97.11.8 Exact condition for bringing both moments together in an expectation

Let $\mathcal{M}_+ = \overline{J_+\mathcal{D}_R}$. The reachable transfer

$$C_{\mathrm{rea}}(J_+ f) = J_- f \tag{97.91}$$

is well defined because the gamma tail proves that J_+ is injective. It is the restriction of (97.86), hence it is bounded. Any other bounded transfer giving the same outputs coincides with it on $\mathcal{M}_+$.

A change of coordinates does not provide a different reachable transfer. The fact to be established for (97.86) to have an isometric dilation with the native expectation is

$$\|C_{\mathrm{rea}} x\| \le \|x\|, \qquad x \in \mathcal{M}_+. \tag{97.92}$$

Its relationship with the two moments can be written without ambiguity. **If** (97.92) is proved from the native operators, $D = (I - C_{\mathrm{rea}}^* C_{\mathrm{rea}})^{1/2}$ exists and

$$\mathfrak{C}_{\mathrm{dil}} f = u_9 \otimes (J_- f, 0) + \eta_9 \otimes (0, D J_+ f) \tag{97.93}$$

satisfies, for $P = |u_9\rangle\langle u_9| \otimes I$,

$$\mathfrak{C}_{\mathrm{dil}}^* \mathfrak{C}_{\mathrm{dil}} = J_+^* J_+, \qquad \mathfrak{C}_{\mathrm{dil}}^* P \mathfrak{C}_{\mathrm{dil}} = J_-^* J_-. \tag{97.94}$$

(97.93) is a normal form of the dilation once (97.92) has been proved, **not** a new prospective proof of (97.92). It makes it possible to identify exactly what the native colligation must produce: the defect of that common transfer, with all its cross terms, and not merely the defect of each clock separately.

The specific next composition consists of composing the energy budget of the complete connector with the recovered local and refinement identities. If an alternative native colligation realizes (97.87) and is contractive on all of $\mathcal{M}_+$, by reachable uniqueness it proves (97.92). There is no need to invent another reader; its action and common energy do need to be calculated.

97.11.9 Checks and provenance

The actions (97.62), (97.72), (97.75) and (97.84)–(97.89) bring together preexisting operators and results. The compositions (97.64)–(97.70) and the falsifier (97.81) are explicit calculations on those formulas; no mathematical priority is attributed to them, nor is a global absence of other operators inferred from them.

25 rational checks on $v_{9N} = 9Ne_0 - \mathbf{1}$, for $N = 2, 3, 9, 81, 729$, were preserved. They check the action of the average, the fraction $(N-1)/(9N-1)$, the zero mean, the orthogonality of the detail and the complement $8N/(9N-1)$. The 25 checks gave the same result in normal and optimized modes, without removable assertions. They verify the identities (97.64)–(97.66), not global positivity. The programs, commands, receipts and locators are preserved in appendix 102.6.

Operational conclusion: the state and the two readers have not disappeared. The calculation recovers their action and proves exactly the difference between isometric transport of modes, uniform expectation and analytic recovery of channels. The second moment identity has a specific verification path through the common connector; chart conjugation, by itself, does not perform that verification.

Chapter 98

Global regularity of Navier–Stokes via solenoidal control, boundary transport, and coercive estimation

98.1 Solenoidal domain and conservation of zero divergence

We fix

$$H^s_\sigma(\mathbb{T}^3) = \{u \in H^s(\mathbb{T}^3;\mathbb{R}^3) : \nabla\cdot u = 0,\ \int_{\mathbb{T}^3} u = 0\}, \qquad s > \frac{5}{2}, \quad \nu > 0,$$

with $u_0 \in C^\infty \cap H^s_\sigma$ and $f \in C^\infty([0,\infty); H^\infty_\sigma) \cap L^2_{\mathrm{loc}}([0,\infty); H^{s-1}_\sigma)$. For the Leray–Galerkin projection P_M set

$$A_M = -P_M\Delta, \qquad B_M(v,w) = P_M P_\sigma((v\cdot\nabla)w), \qquad N_M(u) := B_M(u,u),$$

and the finite field satisfies

$$\partial_t u_M + \nu A_M u_M + B_M(u_M,u_M) = f_M, \qquad u_M(0) = P_M u_0. \tag{98.1}$$

The equation (98.1) is a publication of the state, not its complete definition.

So that the notation does not change halfway through the chain, the four energy functionals of order s are fixed once and for all:

$$\begin{aligned} \mathscr{E}_{s,M}(u) &:= \frac{1}{2}\|\Lambda^s u\|^2_{L^2}, & \mathscr{D}_{s,M}(u) &:= \|A_M^{1/2}\Lambda^s u\|^2_{L^2},\\ \mathscr{B}_{s,M}(u) &:= \mathrm{Re}\langle \Lambda^s N_M(u), \Lambda^s u\rangle, & \mathscr{F}_{s,M}(u,f) &:= \mathrm{Re}\langle \Lambda^s P_M f, \Lambda^s u\rangle. \end{aligned} \tag{98.2}$$

The additional subscript j denotes the same four functionals evaluated on the published coordinate of depth level j; it introduces no other functionals. The Galerkin equation implies exactly

$$\frac{d}{dt}\mathscr{E}_{s,M}(u_M) + \nu\mathscr{D}_{s,M}(u_M) + \mathscr{B}_{s,M}(u_M) = \mathscr{F}_{s,M}(u_M,f). \tag{98.3}$$

98.2 Source object and coordinates that are indeed constructed

The solenoidal image is not added a posteriori. It is exactly the autonomous construction $\mathfrak{G}_{\mathrm{sol}}$ produced before this chapter and published in (95.340); it preserves the Galerkin state, refinement maps, oriented sectors, dissipative operator, projected advection, carry, counit E_9, inclusion J_9, projections $P_9 = J_9E_9$ and $Q_{\mathrm{micro}} = I - P_9$, memory, route and terminal. In particular,

$$E_9J_9 = I, \qquad P_9^2 = P_9, \qquad Q_{\mathrm{micro}} = I - P_9.$$

These identities type the decomposition between macroscopic publication and microscopic memory; by themselves they do not constitute a global bound.

At nonadic depth j, the state retains

$$\Xi^{\mathrm{NS}}_{M,j} = \big(u_M; q_{A,M,j}, q_{V,M,j}, T_{M,j}; \begin{array}{c}\text{triads, sheet, orientation, carry, boundary,}\\ \text{route, archive and memory}\end{array}\big). \tag{98.4}$$

The enriched return combines all branches before evaluation:

$$\mathcal{R}^{\mathrm{enr}}_{M,J,t} = \mathcal{R}_{1,M}\mathfrak{U}^*_{M,J,t}, \qquad \mathcal{R}^{\mathrm{enr}}_{M,J,t}Z_{M,J,t} = u_M(t). \tag{98.5}$$

An isolated norm branch 3^{J} is not the return.

For $N > M$, the commutation defect with the nonlinearity is the carry

$$C_{M\leftarrow N} := P_M B(u_N, u_N) - B_M(P_M u_N, P_M u_N). \tag{98.6}$$

Setting it equal to zero without proving commutation removes precisely the high-frequency interaction that the refinement must preserve.

The two positive memories remain separate. With $r = \pi_{\mathrm{HMT}}/\varphi^2_{\mathrm{HMT}} > 1$,

$$D_M = (r-1)(\mathcal{M}^+_M - \mathcal{M}^-_M), \qquad H_M = (1+r)(\mathcal{M}^+_M + \mathcal{M}^-_M), \tag{98.7}$$

and

$$\dot D_M = \mathscr{B}_{s,M}, \qquad \dot H_M = \frac{1+r}{r-1}|\mathscr{B}_{s,M}|.$$

D_M publishes sign; H_M preserves positive magnitude. Merging them destroys the orientation or the positivity.

98.3 Complete PC–Fock tower and terminal storage

The elevation does not truncate at degree two:

$$\mathcal{V}^{\mathrm{PCF}}_M(u) := \bigoplus_{n\ge 0} \frac{R_{\mathrm{F}}^n}{\sqrt{n!}} u^{\odot n}, \qquad \|\mathcal{V}^{\mathrm{PCF}}_M(u)\|^2 = e^{R_{\mathrm{F}}^2\|u\|^2_{H^s}}. \tag{98.8}$$

The radius is part of the coordinate and cannot disappear from the reader. For any $\rho > 0$ one fixes

$$\rho_1^{[\rho]} := \rho^{-1}\pi_1 : \Gamma^\rho_s(V) \longrightarrow V, \qquad \rho_1^{[\rho]}\mathcal{V}^{\mathrm{PCF}}_\rho(u) = u. \tag{98.9}$$

The initial level uses $\rho_1^{[R_{\mathrm{F}}]}$ and level j uses $\rho_1^{[r_j]}$; this normalization is maintained when depth is withdrawn. On the coherent diagonal, $z_{n,M} = u_M^{\odot n}$ satisfies

$$\dot z_{n,M} = L_{n,M} z_{n,M} + \mathcal{N}_{n,n+1;M} z_{n+1,M} + \mathcal{F}_{n,n-1;M}(f_M) z_{n-1,M}. \tag{98.10}$$

First-degree and second-degree readers are

$$z_M(u) = \nu^{1/2} A_M^{1/2} \Lambda^s u, \qquad w_M(u) = \nu^{-1/2} A_M^{-1/2} \Lambda^s N_M(u),$$

and Young gives

$$|\mathscr{B}_{s,M}(u)| \le \beta\nu \mathscr{D}_{s,M}(u) + \frac{\|w_M(u)\|^2}{4\beta}. \tag{98.11}$$

The uniform estimate of the Fourier rows controls this quadratic reader in the initial state. It does not yet control its future propagations.

Let

$$\mathcal{N}_{M,j} := \frac{r-1}{r+1} H_{M,j} - D^{\mathrm{mem}}_{M,j} = 2(r-1) M^-_{M,j} \ge 0.$$

Over a horizon T, the terminal storage is

$$\begin{aligned} G_{M,j,T}(t) &= \mathscr{E}_{s,M,j}(t) + 2(r-1)(M^-_{M,j}(T) - M^-_{M,j}(t)), \\ \dot G_{M,j,T} + \nu \mathscr{D}_{s,M,j} + |\mathscr{B}_{s,M,j}| &= \mathscr{F}_{s,M,j}. \end{aligned} \tag{98.12}$$

To type the terminal, the sheet coordinate is preserved in the complete solenoidal state.

$$q^{-}_{M,j} := \sqrt{2(r-1)\dot{M}^{-}_{M,j}} \in L^2(0,T;\mathbb{R}).$$

Let $\mathscr{X}^{\mathrm{sol}}_{M,j,T}$ be the space of enriched history that contains, as orthogonal summands, the trace $2^{-1/2}\Lambda^s u_{M,j}$ and $q^{-}_{M,j}$. The terminal column is the coordinate operator

$$\begin{aligned} \mathcal{E}^{\mathrm{term}}_{M,j,T}(t) : \mathscr{X}^{\mathrm{sol}}_{M,j,T} &\longrightarrow L^2_\sigma(\mathbb{T}^3) \oplus L^2(t,T;\mathbb{R}), \\ \mathcal{E}^{\mathrm{term}}_{M,j,T}(t)\Xi &:= 2^{-1/2}\Lambda^s u_{M,j}(t) \oplus \mathbf{1}_{[t,T]} q^{-}_{M,j}. \end{aligned} \tag{98.13}$$

By construction of the two orthogonal coordinates,

$$\|\mathcal{E}^{\mathrm{term}}_{M,j,T}(t)\Xi\|^2 = G_{M,j,T}(t). \tag{98.14}$$

The future-memory summand is a typed terminal output, not a component of the initial root; its cost will be controlled only after telescoping the full column. For a remark τ, the loss rows lie in $[0,\tau]$ and this terminal memory in $(\tau,T]$; the temporal supports are disjoint. In the integral estimate one takes $\tau = T$, and the future summand is then zero.

98.4 Source–first construction with a mixed state–force lift

The entry is fixed before any cut, depth, and history:

$$\mathcal{D}_T := H^s_\sigma(\mathbb{T}^3) \times L^2(0,T;H^{s-1}_\sigma(\mathbb{T}^3)), \qquad d = (u_0, f). \tag{98.15}$$

Since the fields have zero mean, $A = -\Delta$ is positive and invertible in the solenoidal sector. Introduce the stress, the viscous flux, and the two waves

$$\begin{aligned} a_T(f) &:= \frac{1}{2\sqrt{\nu}} A^{-1/2}\Lambda^s P_\sigma f, \qquad z_M(u) := \sqrt{\nu} A_M^{1/2}\Lambda^s u, \\ b_M(u,f) &:= z_M(u) - P_M a_T(f), \\ q^{B,+}_M(u) &:= \big(\mathscr{B}_{s,M}(u)\big)^{1/2}_+, \qquad q^{B,-}_M(u) := \big(-\mathscr{B}_{s,M}(u)\big)^{1/2}_+, \\ q^{B}_M(u) &:= \big(q^{B,+}_M(u), q^{B,-}_M(u)\big). \end{aligned} \tag{98.16}$$

All belong to L^2 over the corresponding interval and $\|q^B_M(u)\|^2_{\mathbb{C}^2} = |\mathscr{B}_{s,M}(u)|$. The cross-power is typed by

$$\mathscr{F}_{s,M} = 2\,\mathrm{Re}\langle z_M, P_M a_T(f)\rangle, \qquad \nu\mathscr{D}_{s,M} = \|z_M\|^2. \tag{98.17}$$

Therefore (98.12), rather than replacing power by the norm of the force, yields the exact scattering identity

$$\dot{G}_{M,j,T} + \|b_M\|^2 + \|q^B_M\|^2 = \|P_M a_T(f)\|^2. \tag{98.18}$$

98.4.1 Chronological signature, uniform smoothing, and mixed creation

The force datum lies in $F := H^{s-1}_\sigma(\mathbb{T}^3)$, whereas the PC–Fock tower uses $V := H^s_\sigma(\mathbb{T}^3)$ as its one-particle space. Their identification is not left to the Galerkin inclusion, whose norm would grow with M. In the mean-zero sector, the order-minus-one isomorphism is fixed

$$\begin{aligned} \mathcal{S} &:= A^{-1/2} : F \longrightarrow V, \\ \mathcal{S}_M &:= A_M^{-1/2} P_M = P_M \mathcal{S}, \\ \sup_M \big(\|\mathcal{S}_M\|_{F\to V} + \|\mathcal{S}_M^{-1}\|_{P_M V \to P_M F}\big) &< \infty. \end{aligned} \tag{98.19}$$

The last bound is the uniform equivalence of the Sobolev norms on the subspace without a zero mode. Thus, $g := \mathcal{S} f \in L^2(0,T;V)$, and the nonuniform inclusion $P_M H^{s-1} \hookrightarrow P_M H^s$ is not used.

Let $\Delta_k(T) = \{0 < t_1 < \cdots < t_k < T\}$. The signature space of the smoothed force is

$$\mathfrak{S}_T(V) := \bigoplus_{k\geq 0} L^2(\Delta_k(T); V^{\widehat{\otimes} k}), \qquad \Delta_0(T) = \{*\}. \tag{98.20}$$

For $g = \mathcal{S} f \in L^2(0,T;V)$ define, without using the solution,

$$\operatorname{Sig}_T(g)_0 = 1, \qquad \operatorname{Sig}_T(g)_k(t_1,\dots,t_k) = g(t_k)\widehat{\otimes}\cdots\widehat{\otimes} g(t_1). \tag{98.21}$$

The symmetry of the scalar integrand provides

$$\|\operatorname{Sig}_T(g)\|^2 = \sum_{k\geq 0} \frac{\|g\|_{L^2(0,T;V)}^{2k}}{k!} = e^{\|g\|^2_{L^2(0,T;V)}} \leq e^{c_{\mathcal{S}}^2 \|f\|^2_{L^2(0,T;F)}}. \tag{98.22}$$

The forced term of (98.10) is not an additive port in f alone. For $0 < r_{\mathrm{F}} < R_{\mathrm{F}}$ it is typed by the mixed creation

$$\begin{aligned} \widetilde{\mathcal{F}}_M^{R_{\mathrm{F}},r_{\mathrm{F}}} &: P_M V \widehat{\otimes} \Gamma_s^{R_{\mathrm{F}}}(P_M V) \longrightarrow \Gamma_s^{r_{\mathrm{F}}}(P_M V), \\ \widetilde{\mathcal{F}}_M^{R_{\mathrm{F}},r_{\mathrm{F}}}(g \otimes x) &:= a^\dagger(g)x. \end{aligned} \tag{98.23}$$

In degree n its action is precisely $g \otimes z_{n-1} \mapsto ng \odot z_{n-1}$. Thus the mixed factor $\mathcal{S} f \otimes x$ is preserved, which cannot produce a linear transition on the direct sum "state $\oplus$ force". With the weighted norms of (98.8),

$$\|\widetilde{\mathcal{F}}_M^{R,r}(g\otimes x)\|_{\Gamma^r} \leq c(R,r)\|g\|_V \|x\|_{\Gamma^R}, \qquad c(R,r) := \sup_{n\geq 1} \sqrt{n}\, r \left(\frac{r}{R}\right)^{n-1} < \infty. \tag{98.24}$$

For the iteration, fix $0 < \theta < 1$ and $r_j = R_{\mathrm{F}}\theta^j$; each step uses $(R,r) = (r_j, r_{j+1})$. Then $\sup_{j,M} c(r_j, r_{j+1}) \leq R_{\mathrm{F}} \sup_{n\geq 1} \sqrt{n}\,\theta^n < \infty$.

The tower is written in the conjugate coordinate $v_M = \mathcal{S}_M u_M$. Let $\mathcal{S}_M^\Gamma := \bigoplus_{n\geq 0} \mathcal{S}_M^{\widehat{\otimes} n}$. The conjugate homogeneous generator is

$$\mathcal{A}_M^{0,\mathcal{S}} := \mathcal{S}_M^\Gamma \mathcal{A}_M^0 (\mathcal{S}_M^\Gamma)^{-1}, \qquad (\mathcal{A}_M^{0,\mathcal{S}})_{n,m} = \mathcal{S}_M^{\widehat{\otimes} n} (\mathcal{A}_M^0)_{n,m} (\mathcal{S}_M^{-1})^{\widehat{\otimes} m}. \tag{98.25}$$

This conjugation is source–first and does not alter the ODE: the degree-one reader applies $\mathcal{S}_M^{-1}$ at the end.

Let $\mathcal{T}_M^{0,\mathcal{S}}(t,s)$ be the propagator of the conjugate tower, included in the complete solenoidal realization. For $R > r > 0$ first define on

$$\delta_{R\to r}(x_0, x_1, x_2, \dots) := \big(x_0, (r/R)x_1, (r/R)^2 x_2, \dots\big), \qquad \delta_{R\to r}\mathcal{V}_R^{\mathrm{PCF}}(u) = \mathcal{V}_r^{\mathrm{PCF}}(u). \tag{98.26}$$

This operator is contractive and fixes the vacuum. On

$$\Gamma_s^R(P_M V)_{\mathrm{alg}} \widehat{\otimes} \mathfrak{S}_T(P_M V)_{\mathrm{fin}} \tag{98.27}$$

Be $\mathfrak{S}_{T,\mathrm{fin}}^{\mathrm{dec}}$ the span of the nucleos elementales

$$g_k(t_1,\dots,t_k) = g_k^{(k)}(t_k)\widehat{\otimes}\cdots\widehat{\otimes} g_k^{(1)}(t_1).$$

It is dense in each summand $L^2(\Delta_k(T); V^{\widehat{\otimes} k})$. In the following formula, the symbols $g_k^{(i)}$ are read in each elementary core, and the action extends linearly to their span; thus an intertwined tensor is not arbitrarily factorized. The chronological reader takes values in $\Gamma_s^r(P_M V)$ and is defined by

$$\mathscr{D}^{R\to r}_{M,k}(t)(x\otimes g_k) := \delta_{R\to r}\int_{\Delta_k(t)} \mathcal{T}^{0,\mathcal{S}}_M(t,t_k)\widetilde{\mathcal{F}}_M(P_M g_k^{(k)}(t_k)\otimes\cdot)\cdots \\ \cdots \mathcal{T}^{0,\mathcal{S}}_M(t_2,t_1)\widetilde{\mathcal{F}}_M(P_M g_k^{(1)}(t_1)\otimes\cdot)\mathcal{T}^{0,\mathcal{S}}_M(t_1,0)x\,d\mathbf{t}, \qquad (98.28)$$
$$\mathscr{D}^{R\to r}_{M,T}(t) := \sum_{k\ge 0}\mathscr{D}^{R\to r}_{M,k}(t).$$

Upon distributing the total loss $R-r$ among the k creations and the propagators, the analytical scale estimate for (98.24) gives, for finite constants depending on the cutoff but not on the elementary kernel,

$$\|\mathscr{D}^{R\to r}_{M,k}(t)(x\otimes g_k)\|_{\Gamma^r} \le A_{M,R,r,T}\,\frac{B^k_{M,R,r,T}}{\sqrt{\Gamma(k+1)}}\|x\|_{\Gamma^R}\|g_k\|_{L^2(\Delta_k;V^{\widehat{\otimes}k})}. \qquad (98.29)$$

The factor $1/\Gamma(k+1)$ arising from the volume of the simplex and Cauchy–Schwarz in the sum over degrees yield

$$\|\mathscr{D}^{R\to r}_{M,T}(t)(x\otimes\sigma)\|_{\Gamma^r} \le C_{M,R,r,T}\|x\|_{\Gamma^R}\|\sigma\|_{\mathfrak{S}_T}, \qquad C_{M,R,r,T}<\infty. \qquad (98.30)$$

The series therefore converges in the indicated codomain and determines by density a bounded operator for each cutoff. No uniform bound in M is used here: that uniformity will be obtained after the local colligation identity. Evaluating it at $x\otimes \mathrm{Sig}_T(\mathcal{S}f)$ yields the Dyson series of the nonautonomous generator

$$\mathcal{A}^{f,\mathcal{S}}_M(t) = \mathcal{A}^{0,\mathcal{S}}_M + a^\dagger(\mathcal{S}_M P_M f(t)), \qquad \partial_t U^{f,\mathcal{S}}_M(t,s) = \mathcal{A}^{f,\mathcal{S}}_M(t)U^{f,\mathcal{S}}_M(t,s). \qquad (98.31)$$

The differentiation of (98.28) in the algebraic core gives the conjugate of (98.10); the uniqueness of the finite ODE gives

$$\mathcal{S}_M^{-1}\rho_1^{[r]}\mathscr{D}^{R_{\mathrm{F}}\to r}_{M,T}(t)\big(\mathcal{V}^{\mathrm{PCF}}_M(\mathcal{S}_M P_M u_0)\otimes \mathrm{Sig}_T(\mathcal{S}_M P_M f)\big) = u_M(t). \qquad (98.32)$$

The equality includes $t=0$: the factor r^{-1} of (98.9) exactly cancels the radial coordinate $r\mathcal{S}_M P_M u_0$ produced by (98.26). $R_{\mathrm{F}}=1$ is not imposed. This construction precedes the history and does not invoke $\mathsf{S}_{M,T}d$ to define it.

Let $\mathfrak{s}_{\mathrm{sol}} : H^s_\sigma \to \mathcal{F}^{\mathrm{sol}}_{\mathrm{seed}}$ be the solenoidal restriction of the seed map: it preserves sheet, orientation, residue, quotient and based route and depends only on u_0. The governing root is not redefined here: it is exactly the root J^{sol}_T of (95.343). We fix the identity of names

$$J^{\mathrm{mix}}_T := J^{\mathrm{sol}}_T : \mathcal{D}_T = \mathcal{D}^{\mathrm{NS}}_T \longrightarrow \mathcal{H}^{\mathrm{sol}}_T. \qquad (98.33)$$

Her publication of mixed coordinates, and not a second root, is

$$\mathrm{Coord}^{\mathrm{mix}}_T(u_0,f) := \big[\mathcal{V}^{\mathrm{PCF}}(u_0)\widehat{\otimes}\,\mathrm{Sig}_T(\mathcal{S}f)\big]\oplus \mathfrak{s}_{\mathrm{sol}}(u_0), \\ \mathrm{Coord}^{\mathrm{mix}}_T : \mathcal{D}_T \longrightarrow \mathfrak{H}^{\mathrm{coord}}_T, \qquad (98.34) \\ \mathfrak{H}^{\mathrm{coord}}_T := \big[\Gamma^{R_{\mathrm{F}}}_s(V)\widehat{\otimes}\mathfrak{S}_T(V)\big]\oplus\mathcal{F}^{\mathrm{sol}}_{\mathrm{seed}}.$$

The chart is a coordinate projection of the accepted state. In the cyclic subspace generated by the root one writes

$$\chi^{\mathrm{mix}}_T : \mathcal{H}^{\mathrm{sol}}_T \longrightarrow \mathfrak{H}^{\mathrm{coord}}_T, \qquad \chi^{\mathrm{mix}}_T J^{\mathrm{sol}}_T(u_0,f) = \mathrm{Coord}^{\mathrm{mix}}_T(u_0,f). \qquad (98.35)$$

Its action outside the cyclic subspace is the orthogonal projection of $\mathfrak{G}_{\mathrm{sol}}$ onto the tower–signature–seed coordinate. No new quotient is created, nor is the norm of this chart identified with the total norm of $\mathcal{H}^{\mathrm{sol}}_T$. With the normalization of the seed,

$$\|\operatorname{Coord}_T^{\mathrm{mix}}(u_0,f)\|^2 \le e^{R_{\mathrm{F}}^2\|u_0\|_{H^s}^2+c_{\mathcal{S}}^2\|f\|_{L^2(0,T;H^{s-1})}^2}+c_{\mathrm{seed}}(1+\|u_0\|_{H^s}^2). \tag{98.36}$$

This estimate proves that the coordinate chart is formed solely from the datum; it neither transports a new norm to $\mathcal{H}_T^{\mathrm{sol}}$ nor is used to manufacture the governing bound. The norm and the column of $J_T^{\mathrm{mix}}=J_T^{\mathrm{sol}}$ are those already fixed by $\mathfrak{G}_{\mathrm{sol}}$.

98.4.2 Initial data, boundary compatibility and regular propagation

The only datum admitted in this chapter is the complete image $\mathfrak{G}_{\mathrm{sol}}$ reproduced in (95.340)–(95.344). In particular, the root J_T^{sol}, the encoder $\Lambda_{M,T}$, the transitions $\Gamma_{M,j}^{\mathrm{term}}$, the complete row $L_{M,j,T}^{\mathrm{term}}$, the terminal $\mathcal{E}_{M,j,T}^{\mathrm{term}}$, the column $\mathbf{G}_{M,j;J,T}^{\mathrm{term}}$, and the isometry $I_{M,J}$ are used without renaming them. No second realization of these objects is introduced.

Let $\mathcal{D}_{T,\mathrm{alg}}\subset\mathcal{D}_T$ be the data subspace with trigonometric u_0 and smooth f, with compact temporal support and trigonometric values. It is dense in $\mathcal{D}_T$. The kernels on which all the actions are written are

$$\begin{aligned}\mathscr{C}_T^{\mathrm{src}}&:=\operatorname{span}\{J_T^{\mathrm{sol}}d:d\in\mathcal{D}_{T,\mathrm{alg}}\}\subset\mathcal{H}_T^{\mathrm{sol}},\\ \mathscr{C}_{M,j,T}^{\mathrm{sol}}&:=\operatorname{span}\{x_{M,j,T}(d):d\in\mathcal{D}_{T,\mathrm{alg}}\},\\ x_{M,0,T}(d)&:=\Lambda_{M,T}J_T^{\mathrm{sol}}d,\\ x_{M,j+1,T}(d)&:=\Gamma_{M,j}^{\mathrm{term}}x_{M,j,T}(d).\end{aligned} \tag{98.37}$$

The spaces $\mathscr{X}_{M,j,T}^{\mathrm{sol}}$ are the completions of these kernels in the norms already contained in $\mathfrak{G}_{\mathrm{sol}}$. In particular, no norm is defined by subtracting the loss that one later intends to bound.

The exact identification between the application notation and that of the data is

$$\boxed{\begin{gathered}J_T^{\mathrm{mix}}=J_T^{\mathrm{sol}},\qquad \Gamma_{M,j,T}=\Gamma_{M,j}^{\mathrm{term}},\qquad L_{M,j,T}=L_{M,j,T}^{\mathrm{term}},\\ \mathcal{G}_{M,J,T}^{\mathrm{sol}}=\mathbf{G}_{M,0;J,T}^{\mathrm{term}}\end{gathered}} \tag{98.38}$$

This equality is literal, not a similarity of names. The source–first action is therefore

$$\boxed{d\xmapsto{J_T^{\mathrm{sol}}}J_T^{\mathrm{sol}}d\xmapsto{\Lambda_{M,T}}x_{M,0,T}(d)\xmapsto{\Gamma_{M,0:j,T}}x_{M,j,T}(d)},\qquad \Gamma_{M,0:j,T}:=\Gamma_{M,j-1,T}\cdots\Gamma_{M,0,T}. \tag{98.39}$$

None of these three arrows receives u_M, a loss, or a terminal as input. The PC–Fock–signature chart of (98.34) publishes the coordinates of the first arrow; the degree-one equality (98.32) proves, in the kernel and then by density,

$$\pi_{M,j,T}^{(1)}x_{M,j,T}(u_0,f)=u_M(t_j),\qquad \pi_{M,j,T}^{(1)}:=\mathcal{S}_M^{-1}(\rho_1^{[r_j]}\widehat{\otimes}\varepsilon_{[t_j,T]})\pi_{M,j,T}^{\mathrm{tf}}. \tag{98.40}$$

The factor r_j^{-1} is applied before $\mathcal{S}_M^{-1}$ and cancels the degree-one radius. The reader therefore returns u_M, not r_ju_M or $\mathcal{S}_Mu_M$.

For $N\ge M$, the refinement projection acts tensorially on the coordinate chart:

$$\begin{aligned}\mathfrak{p}_{M\leftarrow N}^{\Gamma}(v_1\widehat{\otimes}\cdots\widehat{\otimes}v_k)&:=(P_Mv_1)\widehat{\otimes}\cdots\widehat{\otimes}(P_Mv_k),\\ \mathfrak{p}_{M\leftarrow N}^{\mathfrak{S}}(g_1\widehat{\otimes}\cdots\widehat{\otimes}g_k)&:=(P_Mg_1)\widehat{\otimes}\cdots\widehat{\otimes}(P_Mg_k).\end{aligned} \tag{98.41}$$

Together with the seed projection of $\mathfrak{G}_{\mathrm{sol}}$, these formulae yield the map $\mathfrak{p}_{M\leftarrow N}^{\mathrm{sol}}$ and the squares

$$\begin{aligned}\mathfrak{p}_{M\leftarrow N}^{\mathrm{sol}}\Lambda_{N,T}J_T^{\mathrm{sol}}&=\Lambda_{M,T}J_T^{\mathrm{sol}},\\ \mathfrak{p}_{M\leftarrow N}^{\mathrm{sol}}\Gamma_{N,j,T}-\Gamma_{M,j,T}\mathfrak{p}_{M\leftarrow N}^{\mathrm{sol}}&=\iota_{\mathrm{carry}}C_{M\leftarrow N,j},\\ C_{M\leftarrow N,j}x_{N,j,T}(d)&=C_{M\leftarrow N}(d)|_{I_j}.\end{aligned} \tag{98.42}$$

The right-hand side is the coordinate operator whose published value is exactly the carry of (98.6); it is not declared to be zero merely to manufacture naturality.

98.4.3 The six rows as components of the unique loss operator

Fix $0 = t_0 < t_1 < \cdots < t_J = \tau \le T$, $I_j = [t_j, t_{j+1}]$, and the calendar

$$M_j := 3^j M, \qquad P_{M_j} \longrightarrow I \quad \text{strongly in each } H^r_\sigma. \tag{98.43}$$

The six codomains, with their own norms, are

$$\begin{aligned} \mathcal{Y}^{\mathrm{diss}}_{M,j} &= L^2(I_j; L^2_\sigma), & \mathcal{Y}^{\mathrm{adv}}_{M,j} &= L^2(I_j; \mathbb{C}^2),\\ \mathcal{Y}^{\mathrm{carry}}_{M,j} &= L^2(I_j; H^{s-1}_\sigma), & \mathcal{Y}^{\mathrm{micro}}_{M,j} &= L^2(I_j; \operatorname{Ran} Q_{\mathrm{micro}}),\\ \mathcal{Y}^{\mathrm{shell}}_{M,j} &= L^2(I_j; \operatorname{Ran}(P_{M_{j+1}} - P_{M_j}) \cap H^{s-1}_\sigma), & \mathcal{Y}^{\mathrm{mem}}_{M,j} &= L^2(I_j; \operatorname{Ran} Q^{\mathrm{mem}}_{M,j+1}). \end{aligned} \tag{98.44}$$

Set $\mathcal{Y}^{\mathrm{sol}}_{M,j} := \widehat{\bigoplus}_{\mathfrak{a}} \mathcal{Y}^{\mathfrak{a}}_{M,j}$, where $\mathfrak{a} \in \{\mathrm{diss}, \mathrm{adv}, \mathrm{carry}, \mathrm{micro}, \mathrm{shell}, \mathrm{mem}\}$, and denote by $P_{\mathfrak{a}}$ its orthogonal projection. The accepted row of $\mathfrak{G}_{\mathrm{sol}}$ is decomposed, and not duplicated, through

$$L^{\mathrm{term}}_{M,j,T} : \mathcal{X}^{\mathrm{sol}}_{M,j,T} \longrightarrow \mathcal{Y}^{\mathrm{sol}}_{M,j}, \qquad L^{\mathfrak{a}}_{M,j,T} := P_{\mathfrak{a}} L^{\mathrm{term}}_{M,j,T}. \tag{98.45}$$

To write the actions without converting nonlinear functions of u into fictitious linear operators, the coordinate projections of the mixed lift already contained in $\mathfrak{G}_{\mathrm{sol}}$ are used:

$$\begin{aligned} \pi_z x_{M,j,T}(d) &= \mathbf{1}_{I_j} \sqrt{\nu} A_M^{1/2} \Lambda^s u_M, & \pi_a x_{M,j,T}(d) &= \mathbf{1}_{I_j} P_M a_T(f),\\ \pi_{B,\pm} x_{M,j,T}(d) &= \mathbf{1}_{I_j} q^{B,\pm}_M(u_M), & \pi_{\mathrm{carry}} x_{M,j,T}(d) &= \mathbf{1}_{I_j} C_{M_j \leftarrow M_{j+1}},\\ \pi_{\mathrm{micro}} x_{M,j,T}(d) &= \mathbf{1}_{I_j} Q_{\mathrm{micro}} y^{\mathrm{micro}}_{M,j}, & \pi_{\mathrm{shell}} x_{M,j,T}(d) &= \mathbf{1}_{I_j} (P_{M_{j+1}} - P_{M_j}) N_{M_{j+1}}(u_{M_{j+1}}),\\ \pi_{\mathrm{mem}} x_{M,j,T}(d) &= \mathbf{1}_{I_j} Q^{\mathrm{mem}}_{M,j+1} m^{\mathrm{raw}}_{M,j+1}. \end{aligned} \tag{98.46}$$

Here $y^{\mathrm{micro}}_{M,j}$ and $m^{\mathrm{raw}}_{M,j+1}$ are not external symbols: they are respectively the projections onto the coordinates Q_{micro} and $\mathrm{Mem}_{M,j+1}$ of the state (95.340). The eight maps π are linear in the lifted state, although their publications on the physical diagonal are polynomial or contain a positive root. Their domain is the kernel (98.37); their bounded extension is the corresponding component of $L^{\mathrm{term}}_{M,j,T}$.

With these projections, the six actions are materially

$$\begin{aligned} L^{\mathrm{diss}}_{M,j,T} &= \pi_z - \pi_a, & L^{\mathrm{adv}}_{M,j,T} &= (\pi_{B,+}, \pi_{B,-}),\\ L^{\mathrm{carry}}_{M,j,T} &= \pi_{\mathrm{carry}}, & L^{\mathrm{micro}}_{M,j,T} &= \pi_{\mathrm{micro}},\\ L^{\mathrm{shell}}_{M,j,T} &= \pi_{\mathrm{shell}}, & L^{\mathrm{mem}}_{M,j,T} &= \pi_{\mathrm{mem}}. \end{aligned} \tag{98.47}$$

Boundedness is not inferred from the names. For $x \in \mathscr{C}^{\mathrm{sol}}_{M,j,T}$, the accepted column gives

$$\|L^{\mathfrak{a}}_{M,j,T} x\|^2 \le \|L^{\mathrm{term}}_{M,j,T} x\|^2 \le \|\mathbf{G}^{\mathrm{term}}_{M,j;J,T} x\|^2. \tag{98.48}$$

By density, each row has a unique bounded extension to $\mathcal{X}^{\mathrm{sol}}_{M,j,T}$. A new quotient is no longer formed, nor is the domain norm defined from the row.

The absence of overcounting is the Parseval identity of the unique operator L^{term}:

$$\boxed{\|L^{\mathrm{term}}_{M,j,T} x\|^2 = \sum_{\mathfrak{a}} \|L^{\mathfrak{a}}_{M,j,T} x\|^2, \qquad x \in \mathcal{X}^{\mathrm{sol}}_{M,j,T}} \tag{98.49}$$

This orthogonality agrees with the physical geometry: the intervals I_j are disjoint; carry lies under P_{M_j} and shell under $P_{M_{j+1}} - P_{M_j}$; micro lies under Q_{micro}; and $Q^{\mathrm{mem}}_{M,j+1}$ removes the inherited memory and the copy of the advective channel. Moreover, $q^{B,+}_M q^{B,-}_M = 0$ pointwise and $\|(q^{B,+}_M, q^{B,-}_M)\|^2 = |\mathscr{B}_{s,M}|$, so that the two sheets constitute a single oriented channel.

98.4.4 Terminal, column, and Stein constructions derived without redefining norms

In the terminal nucleus, the action of the accepted operator is

$$\mathcal{E}^{\mathrm{term}}_{M,J,T}(\tau)x_{M,J,T}(d) = 2^{-1/2}\Lambda^s u_M(\tau) \oplus \mathbf{1}_{[\tau,T]}q^-_{M,J}. \tag{98.50}$$

Its bounded extension to $\mathscr{X}^{\mathrm{sol}}_{M,J,T}$ is part of $\mathfrak{G}_{\mathrm{sol}}$, and its terminal Gram is, without invented complements,

$$U_{M,J,T} := (\mathcal{E}^{\mathrm{term}}_{M,J,T})^*\mathcal{E}^{\mathrm{term}}_{M,J,T}, \qquad \|\mathcal{E}^{\mathrm{term}}_{M,J,T}x_{M,J,T}(d)\|^2 = G_{M,J,T}(\tau). \tag{98.51}$$

The reader of the first terminal coordinate is defined over its entire codomain by

$$\mathcal{R}^{(1)}_{M,\tau}(h \oplus q) := \sqrt{2}\,\Lambda^{-s}h, \qquad \mathcal{R}^{(1)}_{M,\tau}\mathcal{E}^{\mathrm{term}}_{M,J,T}(\tau)x_{M,J,T}(d) = u_M(\tau). \tag{98.52}$$

Γ is not inverted, no representative of a quotient is chosen, and no complementary terminal operator without action appears.

The terminal column is fixed first by

$$\mathbf{G}^{\mathrm{term}}_{M,J;J,T} := \mathcal{E}^{\mathrm{term}}_{M,J,T}. \tag{98.53}$$

For $0 \le j < J$, the tail column is written with all its rows, in terminal order and then decreasing depth:

$$\mathbf{G}^{\mathrm{term}}_{M,j;J,T} := \begin{bmatrix} \mathcal{E}^{\mathrm{term}}_{M,J,T}\Gamma_{M,j:J,T} \\ L^{\mathrm{term}}_{M,J-1,T}\Gamma_{M,j:J-1,T} \\ \vdots \\ L^{\mathrm{term}}_{M,j+1,T}\Gamma_{M,j:j+1,T} \\ L^{\mathrm{term}}_{M,j,T} \end{bmatrix}, \qquad \Gamma_{M,j:k,T} := \Gamma_{M,k-1,T}\cdots\Gamma_{M,j,T}. \tag{98.54}$$

For $j < J$ there is an operator equality between already constructed columns,

$$\mathbf{G}^{\mathrm{term}}_{M,j;J,T} = \begin{bmatrix} \mathbf{G}^{\mathrm{term}}_{M,j+1;J,T}\Gamma_{M,j,T} \\ L^{\mathrm{term}}_{M,j,T} \end{bmatrix}. \tag{98.55}$$

Now, and only now, the Gram is taken.

$$U_{M,j,T} := (\mathbf{G}^{\mathrm{term}}_{M,j;J,T})^*\mathbf{G}^{\mathrm{term}}_{M,j;J,T}. \tag{98.56}$$

Multiplying the block equality (98.55) by its adjoint yields Stein:

$$\boxed{U_{M,j,T} = \Gamma^*_{M,j,T}U_{M,j+1,T}\Gamma_{M,j,T} + (L^{\mathrm{term}}_{M,j,T})^*L^{\mathrm{term}}_{M,j,T}} \tag{98.57}$$

By (98.49), the final summand is $\sum_{\mathfrak{a}}(L^{\mathfrak{a}}_{M,j,T})^*L^{\mathfrak{a}}_{M,j,T}$. The Stein equation is therefore a consequence of the accepted column; it neither defines the state norm nor serves to justify the existence of its rows retrospectively.

Telescopy is also a multiplication of the complete column:

$$\|\mathbf{G}^{\mathrm{term}}_{M,0;J,T}x\|^2 = \|\mathcal{E}^{\mathrm{term}}_{M,J,T}\Gamma_{M,0:J,T}x\|^2 + \sum_{j<J}\|L^{\mathrm{term}}_{M,j,T}\Gamma_{M,0:j,T}x\|^2. \tag{98.58}$$

All terms on the right exist before this equality and bear the norms of their codomains in (98.44).

98.4.5 Source Factorization–History and Uniform Bound

The factorization accepted in (95.343) becomes, after the literal identification (98.38),

$$\boxed{\mathcal{G}^{\mathrm{sol}}_{M,J,T}\Lambda_{M,T}J^{\mathrm{mix}}_T d = I_{M,J}J^{\mathrm{sol}}_T d, \qquad I^*_{M,J}I_{M,J} = I} \tag{98.59}$$

The history is defined from the left side,

$$\mathsf{H}^{\mathrm{sol}}_{M,J,T}(d) := \mathcal{G}^{\mathrm{sol}}_{M,J,T}\Lambda_{M,T}J^{\mathrm{mix}}_T d, \tag{98.60}$$

and not from a solution chosen later. The terminal reading (98.52) and the source–first action (98.39) give the commutative square

$$\boxed{\mathcal{R}^{(1)}_{M,\tau}P_{\mathrm{term}}\mathsf{H}^{\mathrm{sol}}_{M,J,T}(u_0,f) = u_M(\tau)} \tag{98.61}$$

where P_{term} is the projection onto the first row of (98.54). The remaining projections are exactly the six actions (98.47). This identifies the entire column with the Galerkin history without reconstructing the state from an output.

Theorem 98.1 (Root bound from the explicit starting structure)**.** *For every $T < \infty$ and every $d = (u_0, f) \in \mathcal{D}_T$,*

$$\sup_{M,J}\|\mathsf{H}^{\mathrm{sol}}_{M,J,T}(d)\|^2 \le C^{\mathrm{sol}}_T\|d\|^2_{\mathcal{D}_T}, \tag{98.62}$$

where C^{sol}_T is independent of M, J.

Proof. By (98.33) and (98.38), the left-hand side is literally $\|\mathbf{G}^{\mathrm{term}}_{M,0;J,T}\Lambda_{M,T}J^{\mathrm{sol}}_T d\|^2$. The preceding owner construction first gives the cascade isometry (95.104), the Stein telescoping (95.105), the preserved ledger (95.106) and the exact funding of the carry (95.107). Their orthogonal sum produces the root bound (95.108) and the uniform budget (95.112); the independent closure (95.130)– (95.123) transports that budget to all cutoffs M, J. Substituting the preceding identification yields (98.62). The formula (95.344) is therefore the subsequent record of the already derived bound and not a premise of this proof. The actions (98.47), the terminal (98.50) and (98.58) identify its six components without double counting. □

Falsification criterion 98.2 (Independence and typing controls)**.** If J^{mix}_T is not literally J^{sol}_T, (98.59) fails. If one of the six rows is not a projection of the unique L^{term}, Parseval fails in (98.49). If one attempts to define the norm of $\mathcal{X}^{\mathrm{sol}}_{M,j+1,T}$ as the preceding norm minus a loss, positivity remains unproved and the operation is forbidden. Finally, if $b = z - a$ is replaced by z, the defect of (98.18) is $2\,\mathrm{Re}\langle z, a\rangle - \|a\|^2$, which equals $\|a\|^2$ for $z = a \neq 0$. These four controls fail before the uniform bound is reached.

98.5 Finite KYP chart: correct computation and auxiliary status

With M, J, T fixed, let A, B, C, D be the blocks of an affine one-step map and U_+ the terminal metric at the next level. Define

$$Q := I - B^* U_+ B - D^* D, \qquad L := B^* U_+ A + D^* C. \tag{98.63}$$

If $Q \succ 0$, the recurrence

$$U := A^* U_+ A + C^* C + L^* Q^{-1} L \tag{98.64}$$

gives the exact factorization

$$\begin{pmatrix} U - A^* U_+ A - C^* C & -A^* U_+ B - C^* D \\ -B^* U_+ A - D^* C & I - B^* U_+ B - D^* D \end{pmatrix} = \begin{pmatrix} Q^{-1/2} L & -Q^{1/2} \end{pmatrix}^* \begin{pmatrix} Q^{-1/2} L & -Q^{1/2} \end{pmatrix} \succeq 0. \tag{98.65}$$

For $Q \succeq 0$, one additionally requires $\operatorname{Ran} L \subseteq \operatorname{Ran} Q^{1/2}$, and the pseudoinverse is used. This chart is finite. It gives no uniform bound on Q^{-1}, U, or the gain upon removing M, J.

The mutation that introduces a term $C_-^* C_-$ within a positive block and then purports to recover it with a negative sign is algebraically false: a factorization $R^* R$ does not change the sign of one of its squares. It is therefore not used in the governing branch.

98.6 Uniform closure, compactness, and global solution

The preceding KYP chart remains a `NON_GOVERNING_CONTROL`. Global closure comes from the source–first root, terminal telescoping and the complete ledger. No uniform bound on Q^{-1} is used, nor is the limit taken inside a finite factorization.

98.6.1 Sobolev Estimation and Exclusion of Singular Concentration

In (98.54), the first row is literally $\mathcal{E}^{\mathrm{term}} \Gamma_{0:J}$; by (98.61) its published coordinate is $u_M(\tau)$ and, by (98.14), dominates $2^{-1} \|\Lambda^s u_M(\tau)\|^2$. The six rows are the typed applications of (98.47). In particular, the first two give

$$\sum_{j<J} \|L^{\mathrm{diss}}_{M,j,T} \Xi\|^2 = \int_0^T \|b_M\|^2 dt, \qquad \sum_{j<J} \|L^{\mathrm{adv}}_{M,j,T} \Xi\|^2 = \int_0^T |\mathscr{B}_{s,M}(u_M)| dt.$$

Since $z_M = b_M + P_M a_T(f)$,

$$\nu \int_0^T \mathscr{D}_{s,M} dt = \int_0^T \|z_M\|^2 dt \le 2 \int_0^T \|b_M\|^2 dt + 2 \|a_T(f)\|^2_{L^2(0,T;L^2)}. \tag{98.66}$$

Applying Theorem 98.1 to the restriction of the datum to $[0, \tau]$ for each $0 \le \tau \le T$, the contractivity of the temporal restriction of $\mathfrak{G}_{\mathrm{sol}}$ gives $C^{\mathrm{sol}}_\tau \le C^{\mathrm{sol}}_T$ and $\|d|_{[0,\tau]}\|_{\mathcal{D}_\tau} \le \|d\|_{\mathcal{D}_T}$. Also using (98.14) and (98.66), we obtain

$$\begin{aligned} &\sup_M \sup_{0 \le t \le T} \|u_M(t)\|^2_{H^s} + \nu \sup_M \int_0^T \|u_M(t)\|^2_{H^{s+1}} \, dt \\ &\quad + \sup_M \int_0^T |\mathscr{B}_{s,M}(u_M(t))| \, dt + \sup_{M,J} \sum_{j<J} \|C_{M_j \leftarrow M_{j+1}}\|^2 \\ &\quad + \sup_{M,J} \sum_{j<J} \left(\|Q_{\mathrm{micro}} y^{\mathrm{micro}}_{M,j}\|^2 + \|\mathcal{S}^{\mathrm{sh}}_{M,j}\|^2 + \|Q^{\mathrm{mem}}_{M,j+1} m^{\mathrm{raw}}_{M,j+1}\|^2 \right) \le C_T(u_0, f), \end{aligned} \tag{98.67}$$

where

$$C_T(u_0, f) := c_{s,\nu} \left(1 + C^{\mathrm{sol}}_T\right) \left(\|u_0\|^2_{H^s} + \|f\|^2_{L^2(0,T;H^{s-1})} \right)$$

is finite for every $T < \infty$ and independent of M and J. Passage from a finite depth to the full sum of losses is justified by monotonicity of nonnegative sums; in the final three lines, each norm is the $L^2(I_j)$ norm of the corresponding codomain in (98.44). The terminal is not eliminated. The identity $E_9 J_9 = I$ publishes the macroscopic component after this estimate, while Q_{micro} and the shell remain quantified in the same inequality.

98.6.2 Temporal control and the Aubin–Lions step

Since $s > 5/2$, one has the continuous maps

$$H^s \cdot H^s \longrightarrow H^s, \qquad B : H^s_\sigma \times H^s_\sigma \longrightarrow H^{s-1}_\sigma, \qquad \|B(v,w)\|_{H^{s-1}} \le c_s \|v\|_{H^s} \|w\|_{H^s}. \tag{98.68}$$

The equation (98.1), the contractivity of P_M, and (98.67) imply

$$\sup_M \|\partial_t u_M\|_{L^2(0,T;H^{s-1})} \le C'_T(u_0, f, \nu) < \infty. \tag{98.69}$$

Indeed, Au_M is bounded in $L^2 H^{s-1}$ by the second term of (98.67); the bilinear term is bounded using (98.68); and f_M is bounded by the source datum.

The inclusions

$$H^{s+1}(\mathbb{T}^3) \Subset H^s(\mathbb{T}^3) \hookrightarrow H^{s-1}(\mathbb{T}^3)$$

and the Aubin–Lions lemma yield a subsequence, still denoted by u_M, such that

$$\begin{aligned} u_M &\rightharpoonup^* u \quad \text{in } L^\infty(0,T;H^s), \\ u_M &\rightharpoonup u \quad \text{in } L^2(0,T;H^{s+1}), \\ u_M &\longrightarrow u \quad \text{in } L^2(0,T;H^s). \end{aligned} \tag{98.70}$$

Strong convergence and (98.68) yield

$$B_M(u_M, u_M) \longrightarrow B(u,u) \quad \text{in } L^1(0,T;H^{s-1}). \tag{98.71}$$

Passing to the limit in the integrated formulation of (98.1), one obtains

$$\partial_t u + \nu A u + B(u,u) = f, \qquad \nabla \cdot u = 0, \qquad u(0) = u_0. \tag{98.72}$$

The initial datum is preserved because u_M is bounded in $H^1(0,T;H^{s-1})$ and converges in $C([0,T];H^{s-1})$ after extraction; since $P_M u_0 \to u_0$, the trace is the indicated one.

98.6.3 Pressure, uniqueness, and adherence of horizons

The zero-mean pressure is not introduced as data. It is reconstructed from the limit by

$$-\Delta p = \partial_i \partial_j (u_i u_j), \qquad \int_{\mathbb{T}^3} p = 0. \tag{98.73}$$

Since H^s is an algebra, $p \in L^\infty(0,T;H^s)$ and $\nabla p \in L^\infty(0,T;H^{s-1})$. Thus, (98.72) is equivalent to the classical form

$$\partial_t u + (u \cdot \nabla) u - \nu \Delta u + \nabla p = f, \qquad \nabla \cdot u = 0.$$

Let u and v be two solutions with the same data, and set $w = u - v$. The product in H^{s-1} and Young's inequality yield

$$\frac{1}{2}\frac{d}{dt}\|w\|^2_{H^{s-1}} + \frac{\nu}{2}\|w\|^2_{H^s} \le \frac{c_s}{\nu}\big(\|u\|^2_{H^s} + \|v\|^2_{H^s}\big)\|w\|^2_{H^{s-1}}. \tag{98.74}$$

Grönwall and $w(0) = 0$ imply $w = 0$. Uniqueness eliminates the dependence on the subsequence in (98.70).

The construction holds for every $T < \infty$. Applying it on $T = n$, $n \in \mathbb{N}$, the solutions agree on the overlaps by (98.74); they therefore glue into a unique solution on $[0, \infty)$. To justify "smooth" without extrapolating a single bound, the construction is repeated for the sequence $s_n = s + n$, $n \geq 0$. The same structural readers act at every scale, the projections P_M intertwine them, and (95.344) gives on each s_n a constant $C^{\mathrm{sol}}_{T,s_n} < \infty$, because the data are smooth. Uniqueness in H^{s-1} identifies all the resulting limits. Hence $u \in C^\infty((0,\infty) \times \mathbb{T}^3)$, and compatibility of the traces with $u_0 \in C^\infty$ gives smoothness up to $t = 0$. No breakdown time appears because the corresponding bound is available at every finite horizon and in every s_n.

The zero-mean condition does not eliminate periodic data. For general data u_0^{g} and f^{g}, define

$$m'(t) = \int_{\mathbb{T}^3} f^{\mathrm{g}}(t,x)\,dx, \qquad m(0) = \int_{\mathbb{T}^3} u_0^{\mathrm{g}}(x)\,dx, \qquad X(t) = \int_0^t m(r)\,dr.$$

Time-dependent Galilean transformation

$$u(t,x) = m(t) + v(t, x - X(t)), \qquad g(t,y) = f^{\mathrm{g}}(t, y + X(t)) - m'(t) \tag{98.75}$$

maps the general problem bijectively onto the zero-mean problem for (v_0, g). Translation preserves all Sobolev norms, divergence, and smoothness. Hence the construction covers all smooth periodic data; in particular, it covers the unforced case in the classical statement.

Theorem 98.3 (Global solenoidal closure from the explicit starting structure)**.** *Let $s > 5/2$, $\nu > 0$, $u_0 \in C^\infty(\mathbb{T}^3) \cap H^s_\sigma$ and $f \in C^\infty([0,\infty); H^\infty_\sigma) \cap L^2_{\mathrm{loc}}([0,\infty); H^{s-1}_\sigma)$. With the complete solenoidal datum typed by* (98.39)*,* (98.40)*,* (98.49)*,* (98.57)*,* (98.51) *and* (98.61)*, the image $\mathfrak{G}_{\mathrm{sol}}$, with its terminal, carry, memory, shell, orthogonal port and source–first root reader, produces a unique global smooth solution (u, p) of three-dimensional periodic Navier–Stokes. For each $T < \infty$ it satisfies* (98.67)*; the construction is natural in M, stable when J is removed and compatible with the Galerkin publication. Through* (98.75)*, the result extends to all smooth periodic data, without imposing zero mean.*

Provenance and evidentiary force. The solenoidal image, the terminal, the carry, the PC–Fock tower and the local identity (98.57) are the `PREEXISTING_AUTHORIAL_ARCHITECTURE` taken here as an explicit starting structure. The mixed lift (98.23), the chronological signature, the waves a/b, the six output domains, the homogeneous unit and the terminal column are a `NEW_FORMALIZATION`. The equations (98.39), (98.59), (98.40), (98.49), (98.57), (98.51) and (98.61) enumerate exactly the premises used; the theorem 98.1 is, in human language, *Proved with an explicit starting structure*. Aubin–Lions, pressure reconstruction, Grönwall and gluing are the subsequent `CLASSICAL_RECOGNITION`. The KYP chart retains only the status of `NON_GOVERNING_CONTROL`; neither its comparator nor a sign hypothesis forms part of the causal chain of closure.

Affirmative conclusion. The governing chain is closed in the order

$$\mathfrak{G}_{\mathrm{sol}} \longrightarrow J_T^{\mathrm{mix}} \longrightarrow \Lambda_{M,T} \longrightarrow \mathcal{G}^{\mathrm{sol}}_{M,J,T} \longrightarrow \mathsf{H}^{\mathrm{sol}}_{M,J,T}$$
$$\longrightarrow \mathcal{R}^{(1)}_{M,\tau} P_{\mathrm{term}} = u_M \longrightarrow \text{uniform bound} \longrightarrow \text{Aubin–Lions} \longrightarrow (u,p)_{[0,\infty)}.$$

Within this explicit starting profile, no additional KYP obligation remains: the terminal and all losses are orthogonal images of a single root preceding the future orbit.

Chapter 99

Quantum Existence and Yang–Mills Mass Gap via Gauge Coercivity and Spectral Positivity

99.1 Source object and mandatory separation of generators

The already generated object received by this chapter is the complete gauge structure $\mathfrak{G}_{\mathrm{gau}}$; it is neither an external datum nor a terminal hypothesis. At every level m, it preserves the lattice $\Lambda_m = a_m\mathbb{Z}^4$, $a_{m+1} = a_m/9$, the measure μ_m, the intertwiners J_m, four reflections $\Theta_{\nu,m}$, holonomy, color, representation, fusion carry, route, memory, terminal, and curvature reader. Its two dynamics are typed with distinct symbols before any limit.

The semigroup of a single memory fiber is

$$K^{\mathrm{fib}}_{m,t} = P_{\Omega_m} + e^{-3\kappa_{\mathrm{av}}t}(I - P_{\Omega_m}) = e^{-tD^{\mathrm{fib}}_m}, \qquad D^{\mathrm{fib}}_m := 3\kappa_{\mathrm{av}}(I - P_{\Omega_m}). \tag{99.1}$$

Its spectrum is contained in $\{0, 3\kappa_{\mathrm{av}}\}$. The generator of the complete product chart, instead, is,

$$D^{\mathrm{prod}}_m := 3\kappa_{\mathrm{av}} \sum_{c\in\mathcal{I}_m} (I - E_{m,c}), \qquad K^{\mathrm{prod}}_{m,t} := e^{-tD^{\mathrm{prod}}_m}, \tag{99.2}$$

where the orthogonal expectations $E_{m,c}$ commute. With two or more nonconstant coordinates, D^{prod}_m has levels $6\kappa_{\mathrm{av}}, 9\kappa_{\mathrm{av}}, \ldots$; thus (99.1) cannot be declared equal to (99.2). The former is the simple fiber; the latter, the overlap product.

99.2 Finite Hoeffding theorem, vacuum, and attained gap

In the product chart constructed in (99.21), the $E_{m,c}$ are orthogonal commuting conditional expectations, and the proof of (99.59) establishes

$$\bigcap_{c\in\mathcal{I}_m} \mathrm{Ran}\, E_{m,c} = \mathbb{C}\Omega_m. \tag{99.3}$$

The Hoeffding decomposition is the orthogonal sum

$$\mathcal{H}_m = \bigoplus_{S\subseteq\mathcal{I}_m} \mathcal{H}_{m,S}, \qquad u = \sum_S u_S,$$

where $E_{m,c}u_S = 0$ if $c \in S$ and $E_{m,c}u_S = u_S$ if $c \notin S$. Then

$$D^{\mathrm{prod}}_m u_S = 3\kappa_{\mathrm{av}}|S|u_S, \qquad \langle u, D^{\mathrm{prod}}_m u\rangle = 3\kappa_{\mathrm{av}} \sum_S |S|\|u_S\|^2. \tag{99.4}$$

The component $S = \varnothing$ is exactly the vacuum in (99.3). Therefore

$$D^{\mathrm{prod}}_m \succeq 3\kappa_{\mathrm{av}}(I - P_{\Omega_m}), \qquad \|K^{\mathrm{prod}}_{m,t}(I - P_{\Omega_m})\| \le e^{-3\kappa_{\mathrm{av}}t}. \tag{99.5}$$

On an incidence coordinate $q = (q_1, \ldots, q_6) \in \mathbb{F}_3^6$, the witness

$$W_c(q) := \mathbf{1}_{\{q_1+\cdots+q_6=0\}} - \frac{1}{3} \tag{99.6}$$

satisfies

$$\mathbb{E}W_c = 0, \qquad \|W_c\|^2 = \frac{2}{9}, \qquad D^{\mathrm{prod}}_m W_c = 3\kappa_{\mathrm{av}}W_c. \tag{99.7}$$

Thus, at every finite level, the threshold in (99.5) is attained. This calculation determines the spectrum of the finite product generator; the continuous four-dimensional Yang–Mills measure is explicitly constructed in (99.46)–(99.53).

99.3 Physical gauge compression

Let G be compact, connected, and simple. The group acting at level m is

$$\mathcal{G}_m^{\mathrm{full}} = G^{V(\Lambda_m)} \times \prod_{c \in C_m} \mathbb{F}_3^4,$$

where the second factor acts on the six incidences through a matrix that must remain explicit. If the rows are indexed by $(01, 02, 03, 12, 13, 23)$ and the columns by $0, 1, 2, 3$, fix

$$B^+_{(ij),k} := \delta_{ik} + \delta_{jk} \in \mathbb{F}_3, \qquad (B^+x)_{ij} = x_i + x_j. \tag{99.8}$$

This is the unoriented incidence of the collar; it is not confused with the signed coboundary of an edge. The finite action is

$$q_c \longmapsto q_c + B^+ x_c, \qquad x_c \in \mathbb{F}_3^4. \tag{99.9}$$

The Haar projection

$$\mathbb{P}_m^{\mathrm{phys}} F := \int_{\mathcal{G}_m^{\mathrm{full}}} F(g^{-1}\cdot)\, dg, \qquad \mathcal{H}_m^{\mathrm{phys}} := \operatorname{Ran} \mathbb{P}_m^{\mathrm{phys}} \tag{99.10}$$

differs from the marginal that forgets incidence. Invariance of the expectations $E_{m,c}$ gives

$$[\mathbb{P}_m^{\mathrm{phys}}, D_m^{\mathrm{prod}}] = 0, \qquad [\mathbb{P}_m^{\mathrm{phys}}, K_{m,t}^{\mathrm{prod}}] = 0. \tag{99.11}$$

Moreover, each variable of $x_c \in \mathbb{F}_3^4$ occurs three times in the sum of the six incidences, so that

$$\sum_{i<j} (B^+ x_c)_{ij} = 3 \sum_{i=0}^{3} x_{c,i} = 0 \quad \text{in } \mathbb{F}_3. \tag{99.12}$$

Hence W_c is gauge-invariant and belongs to $\mathcal{H}_m^{\mathrm{phys}}$. The finite gap is not an artifact of a nonphysical dilation.

99.4 Reflection positivity at every level

For a common reversible kernel, define the eight-face measure

$$\Omega_m^{(8)}(dy_1 \cdots dy_8) := \int \mu_m(dz) \prod_{r=1}^{8} K_{m,a_m/2}^{\mathrm{prod}}(z, dy_r). \tag{99.13}$$

If $\Theta_{\nu,m}$ interchanges the two halves with respect to direction ν, the Markov property and reversibility factorize, for every cylindrical observable F supported in the positive half-space,

$$\int \overline{F(\Theta_{\nu,m} y)} F(y)\, d\Omega_m^{(8)} = \|\mathcal{B}_{m,\nu} F\|_2^2 \geq 0, \qquad \nu = 1, \ldots, 4. \tag{99.14}$$

The opposite-face marginal recovers the same $K_{m,a_m}^{\mathrm{prod}}$. Thus the four finite OS forms follow from a single measure and a single semigroup, not from four independent processes.

99.5 Final owner: the complete gauge state, not its ablated marginal

The current source construction does not take the marginal retaining only the links as its physical space. Its domain is $\mathfrak{G}_{\mathrm{gau}}$, with link, frame, representation, incidence, color, route, orientation, carry, memory, survival, and terminal. Projection onto links is a subsequent forgetful map. Assessing the physical witness after applying this forgetful map replaces the object and does not transfer the conclusion.

99.5.1 Projective measure and link publication

For a finite screen $\Gamma_m = (V_m, E_m)$, start from the lifted space

$$\widetilde{X}_m := G^{V_m} \times G^{E_m}, \qquad d\widetilde{\mu}_m(h,z) = \prod_{v \in V_m} dh_v \prod_{e \in E_m} k_{\tau_{m,e}}(z_e)\, dz_e, \tag{99.15}$$

and each link is published through

$$U_e = h_{s(e)} z_e h_{t(e)}^{-1}. \tag{99.16}$$

A coarse edge e is refined into the ordered chain $e_1 \cdots e_9$. Set $\boldsymbol{\tau} = (\tau_1, \dots, \tau_9)$, with $\tau_j = \tau_{m+1,e_j} > 0$ and $\tau = \tau_{m,e} = \sum_{j=1}^{9} \tau_j$. Let $d\lambda_z^{(9)}$ be the Haar disintegration over the product fiber, characterized by

$$\int_{G^9} F(z_1, \dots, z_9)\, dz_1 \cdots dz_9 = \int_G \int_{z_1 \cdots z_9 = z} F\, d\lambda_z^{(9)}\, dz.$$

The correct bridge for times that need not be equal is

$$dB_{z,\boldsymbol{\tau}}^{(9)}(z_1, \dots, z_9) = \frac{\prod_{j=1}^{9} k_{\tau_j}(z_j)}{k_\tau(z)}\, d\lambda_z^{(9)}, \qquad z_1 \cdots z_9 = z, \tag{99.17}$$

which is a probability measure because $k_{\tau_1} * \cdots * k_{\tau_9} = k_\tau$. Uniform nonadic refinement recovers $\tau_j = \tau/9$, but the formula does not presuppose this equality.

To make the innovation independent without erasing the coarse product, fix, for each (e, z), a Borel transport

$$\chi_{m,e,z} : \big((0,1)^{\mathbb{N}}, du^{\otimes \mathbb{N}}\big) \longrightarrow \big(\{(z_j) : z_1 \cdots z_9 = z\}, B_{z,\boldsymbol{\tau}}^{(9)}\big) \tag{99.18}$$

which is an isomorphism modulo null sets. It exists because both are standard nonatomic realizations; a choice is fixed once and transported under relabelings of edges. The coarse variable z and the innovation $u_{m+1,e}^{\mathrm{det}}$ are then independent factors in the triangular chart. The new frames are integrated with respect to Haar measure. Route, carry, orientation, and memory are transported by the morphisms $\mathrm{Ext}_{\mathrm{gau}}$ of the gauge datum itself.

The coarse projection multiplies the nine increments and forgets only the new innovations. This definition yields

$$(\widehat{\pi}_{m+1,m})_* \widehat{\mu}_{m+1}^{\mathrm{ov}} = \widehat{\mu}_m^{\mathrm{ov}}, \qquad \widehat{J}_m F = F \circ \widehat{\pi}_{m+1,m}. \tag{99.19}$$

The same edge occurs exactly once in the measure; the six incident faces are six oriented readers of that edge. Overlapping routes preserve the shared increment and are not replaced by independent copies.

99.5.2 The incidence fiber is a physical coordinate of the complete state

The incidence tail included in $\mathfrak{G}_{\mathrm{gau}}$ decomposes without loss of information:

$$\Sigma : (\mathbb{F}_3^2)^{\mathbb{N}} \longrightarrow (\mathbb{F}_3^2)^{\mathbb{N}} \times \mathbb{F}_3^6, \qquad \Sigma_* \sigma = \sigma \otimes u_6. \tag{99.20}$$

The first factor continues to feed the innovation charts, and the second preserves the six incidences $q_c = (q_{01}, q_{02}, q_{03}, q_{12}, q_{13}, q_{23})$ of the collar c. Rather than leaving implicit which coordinates the generator governs, a recursive triangular chart is fixed

$$\widehat{X}_m = \widehat{X}^{\mathrm{seed}}_{m_0} \times \prod_{\ell=m_0+1}^{m} \mathcal{U}^{\mathrm{det}}_{\ell/\ell-1}, \qquad \widehat{\mu}^{\mathrm{ov}}_m = \widehat{\mu}^{\mathrm{seed}}_{m_0} \otimes \bigotimes_{\ell=m_0+1}^{m} \upsilon^{\mathrm{det}}_{\ell/\ell-1}. \tag{99.21}$$

The seed factor enumerates, once, the coarse frames and links, the incidences q_c and their marking charts. Each $\mathcal{U}^{\mathrm{det}}_{\ell/\ell-1}$ enumerates, once, the new frames, the bridges (99.18), the new incidences, and the marking subtails of that refinement. The factorization is internal to $\mathfrak{G}_{\mathrm{gau}}$ and adds no further starting datum.

Record the exhaustive, disjoint, countable index

$$\mathcal{I}_m := \mathcal{I}^{\mathrm{frame}}_{m_0} \sqcup \mathcal{I}^{\mathrm{link}}_{m_0} \sqcup \mathcal{I}^{q}_{m_0} \sqcup \mathcal{I}^{\mathrm{mark}}_{m_0} \sqcup \bigsqcup_{\ell=m_0+1}^{m} \big(\mathcal{I}^{\mathrm{frame}}_{\ell/\ell-1} \sqcup \mathcal{I}^{\mathrm{bridge}}_{\ell/\ell-1} \sqcup \mathcal{I}^{q}_{\ell/\ell-1} \sqcup \mathcal{I}^{\mathrm{mark}}_{\ell/\ell-1}\big). \tag{99.22}$$

Each independent factor of (99.21) occurs exactly once; an infinite subtail is enumerated through its scalar cylinders. No genealogical factor remains outside $\mathcal{I}_m$. In particular, the abbreviations in the recurrence literally mean

$$\begin{aligned} \mathcal{I}^{\mathrm{seed}}_{m_0} &:= \mathcal{I}^{\mathrm{frame}}_{m_0} \sqcup \mathcal{I}^{\mathrm{link}}_{m_0} \sqcup \mathcal{I}^{q}_{m_0} \sqcup \mathcal{I}^{\mathrm{mark}}_{m_0}, \\ \mathcal{I}^{\mathrm{det}}_{\ell/\ell-1} &:= \mathcal{I}^{\mathrm{frame}}_{\ell/\ell-1} \sqcup \mathcal{I}^{\mathrm{bridge}}_{\ell/\ell-1} \sqcup \mathcal{I}^{q}_{\ell/\ell-1} \sqcup \mathcal{I}^{\mathrm{mark}}_{\ell/\ell-1}. \end{aligned} \tag{99.23}$$

The link marginal forgets q_c; the physical subalgebra, by contrast, is obtained by averaging the complete gauge action

$$\mathcal{G}^{\mathrm{full}}_m := G^{V(\Lambda_m)} \times \prod_{c \in C_m} \mathbb{F}_3^4, \qquad q_c \longmapsto q_c + B^{+} x_c. \tag{99.24}$$

Thus the physical projection is

$$\mathbb{P}^{\mathrm{phys}}_m F := \int_{\mathcal{G}^{\mathrm{full}}_m} F(g^{-1} \cdot x)\, dg, \qquad \widehat{\mathscr{H}}^{\mathrm{phys}}_m := \operatorname{Ran} \mathbb{P}^{\mathrm{phys}}_m. \tag{99.25}$$

It does not coincide with the forgetful map $\widehat{X}_m \to X^{\mathrm{link}}_m$. This distinction is the type safeguard that prevents exclusion of W_c from the physical sector.

99.6 Owner generator and local circuit

The triangular chart provides a unitary

$$\widehat{\Gamma}_m : L^2(\widehat{X}_{m+1}, \widehat{\mu}^{\mathrm{ov}}_{m+1}) \longrightarrow L^2(\widehat{X}_m, \widehat{\mu}^{\mathrm{ov}}_m) \widehat{\otimes} \mathcal{K}^{\mathrm{det}}_{m+1}, \qquad \widehat{\Gamma}_m \widehat{J}_m F = F \otimes \mathbf{1}_{\mathrm{det}}. \tag{99.26}$$

For $\iota \in \mathcal{I}_m$, let $E_{m,\iota}$ be the expectation integrating exactly factor ι of (99.21) and fixing all the others. These are orthogonal Markov projections, preserve the unit, and commute. On the dense core $\mathscr{C}_m$ of cylinders depending on finitely many coordinates, define the form

$$\widehat{\mathcal{E}}_m[F] := 3\kappa_{\mathrm{av}} \sum_{\iota \in \mathcal{I}_m} \|(I - E_{m,\iota})F\|_2^2. \tag{99.27}$$

The sum is finite for each cylinder. Its closure is a Dirichlet form and determines the self-adjoint Markov generator $\widehat{H}^{\mathrm{ov}}_m$. At a seed level this is written, in the sense of forms, as

$$\widehat{H}^{\mathrm{ov}}_{m_0} = 3\kappa_{\mathrm{av}} \sum_{\iota \in \mathcal{I}^{\mathrm{seed}}_{m_0}} (I - E_{m_0,\iota}), \tag{99.28}$$

and each refinement adds exactly the expectations of its new coordinates:

$$\begin{aligned}\widehat{H}^{\mathrm{ov}}_{m+1} &= \widehat{\Gamma}^{*}_{m}(\widehat{H}^{\mathrm{ov}}_{m}\otimes I + I\otimes \widehat{H}^{\mathrm{det}}_{m+1})\widehat{\Gamma}_{m},\\ \widehat{H}^{\mathrm{det}}_{m+1} &= 3\kappa_{\mathrm{av}}\sum_{\iota\in \mathcal{I}^{\mathrm{det}}_{m+1/m}} (I - E^{\mathrm{det}}_{m+1,\iota}).\end{aligned} \tag{99.29}$$

The color partition is made explicit. If a cell is indexed by $n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$, fix

$$\chi_m(n) := \big(n_1 \bmod 9,\ n_2 \bmod 9,\ n_3 + 3n_4 \bmod 9\big) \in (\mathbb{Z}/9\mathbb{Z})^3, \qquad |(\mathbb{Z}/9\mathbb{Z})^3| = 9\cdot 81. \tag{99.30}$$

If two closed collars of cellular radius one intersect, their difference δ satisfies $|\delta_j| \le 2$. Equality of colors forces $\delta_1 = \delta_2 = 0$ and $\delta_3 + 3\delta_4 \equiv 0 \pmod 9$. Since the latter integer belongs to $[-8, 8]$, it is zero; the bounds $|\delta_3|, |\delta_4| \le 2$ then give $\delta_3 = \delta_4 = 0$. Thus distinct cells of the same color have disjoint collars. Define explicitly

$$\mathcal{B}_{m,r} := \{\iota \in \mathcal{I}_m : \mathrm{cell}(\iota) = n, \chi_m(n) = r\}, \qquad r \in (\mathbb{Z}/9\mathbb{Z})^3. \tag{99.31}$$

Transporting a screen by $g \in E(4)$ also transports this partition; it does not require a rotation to preserve the coordinates of the label. Thus, $\{\mathcal{B}_{m,r}\}_{r=1}^{9\cdot 81}$ groups disjoint spatial supports; the internal index of each block enumerates its factors of (99.22). For a cylinder F, the product contains only the blocks on which F depends; closure defines the strong product for the entire form. The generator of a block preserves the Hoeffding multiplicities; it is not replaced by a single projector that would erase them. By commutativity on each chart,

$$\begin{aligned}H_{m,B} &:= 3\kappa_{\mathrm{av}}\sum_{\iota\in B}(I - E_{m,\iota}),\\ K_{m,B} &:= e^{-a_m H_{m,B}} = \prod_{\iota\in B} e^{-3\kappa_{\mathrm{av}} a_m (I - E_{m,\iota})},\\ K^{\mathrm{loc}}_{m} &:= \prod_{r,B}^{\mathrm{strong}} K_{m,B} = e^{-a_m \widehat{H}^{\mathrm{ov}}_{m}}.\end{aligned} \tag{99.32}$$

The physical diameter of every block remains uniformly bounded. Locality follows: the generator introduces no instantaneous update over the entire lattice. The partition is compatible with Bianchi, not merely with geometric disjointness. For a cubic cell c, fix the oriented product, with all factors transported to the same vertex,

$$\mathscr{B}_{m,c}(x) := \prod_{p\subset \partial c}^{\longrightarrow} T_p\,\mathrm{Hol}_p(x) T_p^{-1}. \tag{99.33}$$

Every interior edge of the boundary occurs twice with opposite orientations; hence $\mathscr{B}_{m,c} = e$ pointwise. A block $\mathcal{B}_{m,r}$ contains the complete collar of every cell it updates: if it meets a face of c, it simultaneously updates the inverse pair sharing that edge. The preceding cancellation remains literal after every circuit factor:

$$K_{m,B}\mathbf{1}_{\{\mathscr{B}_{m,c}=e\}} = \mathbf{1}_{\{\mathscr{B}_{m,c}=e\}}, \qquad K^{\mathrm{loc}}_{m}\mathbf{1}_{\{\mathscr{B}_{m,c}=e\}} = \mathbf{1}_{\{\mathscr{B}_{m,c}=e\}}. \tag{99.34}$$

Thus the calendar of $9\cdot 81$ layers preserves Bianchi at every step and in the strong product, not merely at the end of a sweep.

The gauge action permutes or translates the coordinates integrated by each $E_{m,\iota}$ and preserves their measures. Hence

$$U_m(g)\widehat{H}^{\mathrm{ov}}_{m} = \widehat{H}^{\mathrm{ov}}_{m}U_m(g) \quad (g \in \mathcal{G}^{\mathrm{full}}_{m}), \qquad [\mathbb{P}^{\mathrm{phys}}_{m}, \widehat{H}^{\mathrm{ov}}_{m}] = 0, \qquad \mathbb{P}^{\mathrm{phys}}_{m+1}\widehat{J}_m = \widehat{J}_m\mathbb{P}^{\mathrm{phys}}_{m}. \tag{99.35}$$

Denote the complete semigroup and its Markov kernel by

$$\widehat{K}^{\mathrm{ov}}_{m,t} := e^{-t\widehat{H}^{\mathrm{ov}}_{m}}, \qquad \widehat{K}^{\mathrm{ov}}_{m,t}(x, dy) \quad (x, y \in \widehat{X}_m). \tag{99.36}$$

The first identity in (99.35) implies

$$\widehat{K}^{\mathrm{ov}}_{m,t}(gx, gA) = \widehat{K}^{\mathrm{ov}}_{m,t}(x, A) \quad (g \in \mathcal{G}^{\mathrm{full}}_m). \tag{99.37}$$

To type the reduction without using a restricted kernel on the wrong space, let $q_m : \widehat{X}_m \to Y_m := \widehat{X}_m/\mathcal{G}^{\mathrm{full}}_m$ be the orbit map and $\nu_m = (q_m)_*\widehat{\mu}^{\mathrm{ov}}_m$. The pullback

$$\mathcal{W}_m : L^2(Y_m, \nu_m) \longrightarrow \widehat{\mathscr{H}}^{\mathrm{phys}}_m, \qquad \mathcal{W}_m\phi = \phi \circ q_m \tag{99.38}$$

is unitary. Equivariance of the complete kernel allows the definition

$$K^{\mathrm{phys}}_{m,t}(q_m x, A) := \widehat{K}^{\mathrm{ov}}_{m,t}(x, q_m^{-1}A), \qquad A \subseteq Y_m, \tag{99.39}$$

independently of the representative x. The physical object is then typed by

$$D^{\mathrm{YM}}_m := \widehat{H}^{\mathrm{ov}}_m\Big|_{\widehat{\mathscr{H}}^{\mathrm{phys}}_m}, \qquad e^{-tD^{\mathrm{YM}}_m} = \mathcal{W}_m K^{\mathrm{phys}}_{m,t}\mathcal{W}_m^{-1}. \tag{99.40}$$

This is the complete product generator, not the single-fiber generator in (99.1).

99.7 Yang–Mills action, g-dependent measure, and exact transport

Thus far, the reference kinematic chart has been written. We shall denote it by $\widehat{\mu}^0_m := \widehat{\mu}^{\mathrm{ov}}_m$ and $\widehat{H}^0_m := \widehat{H}^{\mathrm{ov}}_m$, with $\widehat{K}^0_{m,t} := e^{-t\widehat{H}^0_m} = \widehat{K}^{\mathrm{ov}}_{m,t}$. To turn the structural reader $\mathcal{P}^{F^2}_m$ into a law, rather than a subsequent comparison, its positive root is first mapped onto the triangular factorization. The polarization of $\mathcal{P}^{F^2}_m$ in a collar b is the positive Gram

$$Q_{m,b}(u, v) := \frac{1}{4}\big(\mathcal{P}^{F^2}_{m,b}(u + v) - \mathcal{P}^{F^2}_{m,b}(u - v)\big).$$

Quotient the finite reader space of the collar by $\ker Q_{m,b}$, complete it with respect to $Q_{m,b}$, and denote the resulting Hilbert space by $\mathcal{E}^F_{m,b}$. If $\zeta_{m,b}(x_b)$ is the vector of centered readers of the triangular coordinate x_b, define explicitly

$$\xi_{m,b}(x_b) := Q^{1/2}_{m,b}\zeta_{m,b}(x_b) \in \mathcal{E}^F_{m,b}, \qquad p_{m,b}(x_b) := \|\xi_{m,b}(x_b)\|^2. \tag{99.41}$$

The index b ranges over the Gram collars of minimal support. A collar is assigned to the first color in (99.30) containing its base cell; each term thus occurs once, and collars of the same color are disjoint. More precisely, positive diagonalization groups the triangular factors into a disjoint partition

$$\mathcal{I}^F_m = \bigsqcup_{r \in (\mathbb{Z}/9\mathbb{Z})^3} \bigsqcup_{b \in \mathcal{B}^F_{m,r}} \mathcal{I}_{m,b}, \qquad x_b := (x_\iota)_{\iota \in \mathcal{I}_{m,b}}, \qquad \mathcal{B}^F_{m,r} := \{b : Q_{m,b} \neq 0,\ \chi_m(b) = r\}. \tag{99.42}$$

No triangular coordinate belongs to two $\mathcal{I}_{m,b}$; if a geometric reader meets two collars, its Gram mode is assigned to the first and its orthogonal complement to the second. This is the finite spectral decomposition of the positive form, not a duplication of the physical variable. The square root of the orthogonal sum, computed color by color, gives the identity, not an approximation,

$$\mathcal{P}^{F^2}_m(x) = \sum_{r \in (\mathbb{Z}/9\mathbb{Z})^3} \sum_{b \in \mathcal{B}^F_{m,r}} p_{m,b}(x_b), \qquad S_{m,g}(x) := \frac{1}{4g^2}\mathcal{P}^{F^2}_m(x), \quad g \in (0, \infty). \tag{99.43}$$

Here $x_b : \widehat{X}_m \to X_{m,b}$ is the coordinate projection of the collar, so the domain, action, and codomain of each summand are fixed. The reader is gauge-invariant because $Q_{m,b}$ uses the invariant product of $\mathfrak{g}$; it is

reflection-invariant because a reflection merely permutes the six oriented pairs; and it is natural under Euclidean relabeling. The partition (99.42) and the product chart give explicitly

$$\widehat{\mu}^{0}_{m} = \lambda^{\mathrm{inert}}_{m} \otimes \bigotimes_{r,b} \lambda^{0}_{m,b}, \qquad d\lambda^{g}_{m,b} := (Z^{g}_{m,b})^{-1} e^{-p_{m,b}/(4g^{2})} d\lambda^{0}_{m,b}, \qquad \widehat{\mu}^{\mathrm{YM}}_{m,g} = \lambda^{\mathrm{inert}}_{m} \otimes \bigotimes_{r,b} \lambda^{g}_{m,b}. \tag{99.44}$$

The inert factor contains exactly those coordinates on which the reader is zero; it is not removed from the theory.

The coarse–detail separation in (99.21) is orthogonal for the same square root: direct polarization gives $Q_{m+1} = Q_m \oplus Q^{\mathrm{det}}_{m+1/m}$ and $\zeta_{m+1} = \zeta_m \oplus \zeta^{\mathrm{det}}_{m+1/m}$. If $x_{m+1} = \widehat{\Gamma}^{-1}_{m}(x_m, u)$, then

$$\xi_{m+1}(x_{m+1}) = \xi_m(x_m) \oplus \xi^{\mathrm{det}}_{m+1/m}(u), \qquad S_{m+1,g} \circ \widehat{\Gamma}^{-1}_{m} = S_{m,g} + S^{\mathrm{det}}_{m+1/m,g}. \tag{99.45}$$

This is the precise way in which carry and memory prevent an ancestral face from being counted twice: the coarse part is not added again in the detail. In particular, $Z_{m+1,g} = Z_{m,g} Z^{\mathrm{det}}_{m+1/m,g}$ for the normalizers, and one obtains the family of probabilities genuinely dependent on the coupling

$$d\widehat{\mu}^{\mathrm{YM}}_{m,g}(x) := Z^{-1}_{m,g} e^{-S_{m,g}(x)} d\widehat{\mu}^{0}_{m}(x), \qquad (\widehat{\pi}_{m+1,m})_{*} \widehat{\mu}^{\mathrm{YM}}_{m+1,g} = \widehat{\mu}^{\mathrm{YM}}_{m,g}. \tag{99.46}$$

To write Schwinger–Dyson correctly with respect to Haar, and not pretend that the reference measure is flat, let λ_m be the product of Haar, Lebesgue, and uniform counting on the lifted chart, and let $R_m := d\widehat{\mu}^{0}_{m}/d\lambda_m > 0$. The total regulated action is

$$\mathcal{S}_{m,g} := S_{m,g} - \log R_m, \qquad d\widehat{\mu}^{\mathrm{YM}}_{m,g} = \mathcal{Z}^{-1}_{m,g} e^{-\mathcal{S}_{m,g}} d\lambda_m. \tag{99.47}$$

The term $-\log R_m$ contains the heat kernels and bridge charts of the initial gauge structure; it is independent of g. The Yang–Mills term that changes the coupling and whose classical limit is calculated below is exactly $S_{m,g} = \mathcal{P}^{F^2}_{m}/(4g^2)$. The consistency of the second equality and the standardness of the finite spaces permit Kolmogorov to be applied and define the unique cylindrical measure $\widehat{\mu}^{\mathrm{YM}}_{\infty,g}$ whose marginals are $\widehat{\mu}^{\mathrm{YM}}_{m,g}$. No eigenvalue has been used here: g, the action, and the measure were fixed first.

To define the dynamics on that measure without losing product, gauge, or the surface identities, transport is constructed rather than postulated. In each nonatomic factor $X_{m,b}$, a Borel chart $\rho_{m,b} : X_{m,b}/\mathcal{G}_{m,b} \to (0,1)$ is fixed and disintegrated first over the gauge quotient and then over the Haar orbit. Let $F^{0}_{m,b}$ and $F^{g}_{m,b}$ be the continuous distribution functions of $\rho_{m,b}$ for the reference measure and the reweighted measure. The triangular transport is

$$t_{m,b,g} := (F^{g}_{m,b})^{-1} \circ F^{0}_{m,b}, \qquad T_{m,b,g}(y,h) := (t_{m,b,g}(y), h), \qquad (T_{m,b,g})_{*} \lambda^{0}_{m,b} = \lambda^{g}_{m,b}. \tag{99.48}$$

On a purely atomic factor for which $p_{m,b} = 0$, take $T_{m,b,g} = I$. In the presence of atoms and a continuous coordinate, the chart is the atom–interval lexicographic order, and the same formula applies to the full standard nonatomic space. The transports of the $9 \cdot 81$ colours are composed in the order fixed by (99.30); within a colour they commute because the collars are disjoint. One obtains

$$T_{m,g} := \prod_{r \in (\mathbb{Z}/9\mathbb{Z})^3} \prod_{b \in \mathcal{B}^{F}_{m,r}} T_{m,b,g}, \quad (T_{m,g})_{*} \widehat{\mu}^{0}_{m} = \widehat{\mu}^{\mathrm{YM}}_{m,g}, \quad \widehat{\pi}_{m+1,m} T_{m+1,g} = T_{m,g} \widehat{\pi}_{m+1,m}. \tag{99.49}$$

The final equality is checked factor by factor using (99.45): the transport of the coarse factor is the one already fixed, and the new transports act only on the innovation. The quotient charts are chosen on a representative screen and transported; therefore $T_{m,g}$ commutes with gauge and reflection and, for $a \in E(4)$, satisfies $T_{a\alpha,g} = U_{\alpha}(a) T_{\alpha,g} U_{\alpha}(a)^{-1}$.

The composition

$$\mathcal{T}_{m,g} : L^\infty(\widehat{X}_m, \widehat{\mu}^{\mathrm{YM}}_{m,g}) \longrightarrow L^\infty(\widehat{X}_m, \widehat{\mu}^{0}_{m}), \qquad \mathcal{T}_{m,g}F := F \circ T_{m,g} \tag{99.50}$$

is, by construction, an isomorphism of probability spaces and $*$-algebras, and extends unitarily to L^2. In particular,

$$\mathcal{T}_{m,g}(FG) = \mathcal{T}_{m,g}F\,\mathcal{T}_{m,g}G, \quad \mathcal{T}_{m,g}(F^*) = (\mathcal{T}_{m,g}F)^*, \quad \int F\, d\widehat{\mu}^{\mathrm{YM}}_{m,g} = \int \mathcal{T}_{m,g}F\, d\widehat{\mu}^{0}_{m}. \tag{99.51}$$

The domain of each $T_{m,b,g}$ is the holonomy space itself, satisfying the subdivision law and Bianchi constraint. Its image therefore remains in that space. Applying the preceding multiplicativity generator by generator gives the pointwise identities

$$\begin{aligned} \mathcal{T}_{m,g}(\mathrm{Hol}_p) &= \overrightarrow{\prod_{p' \subset p}} \mathcal{T}_{m,g}(T_{p'}\,\mathrm{Hol}_{p'}\,T_{p'}^{-1}), \\ \mathcal{T}_{m,g}(F_{\mathrm{cl},m}) &= F_{\mathrm{cl},m} \circ T_{m,g}, \\ \overrightarrow{\prod_{p \subset \partial c}} \mathcal{T}_{m,g}(T_p\,\mathrm{Hol}_p\,T_p^{-1}) &= e, \qquad \mathrm{Alt}_{\lambda\mu\nu}\, D_\lambda (F_{\mathrm{cl},m} \circ T_{m,g})_{\mu\nu} = 0. \end{aligned} \tag{99.52}$$

The third line is the edge-by-edge cancellation of the oriented boundary of the boundary of c; the fourth is its infinitesimal polarization. Thus transport preserves products, state, clover, and Bianchi, not merely the first chaos.

Finally, transport the expectations, generator, kernel, projection, and inclusion:

$$\begin{aligned} E^{g}_{m,\iota} &:= \mathcal{T}^{-1}_{m,g} E_{m,\iota} \mathcal{T}_{m,g}, \quad \widehat{H}^{\mathrm{YM}}_{m,g} := \mathcal{T}^{-1}_{m,g} \widehat{H}^{0}_{m} \mathcal{T}_{m,g}, \\ \widehat{K}^{g}_{m,t} &:= \mathcal{T}^{-1}_{m,g} \widehat{K}^{0}_{m,t} \mathcal{T}_{m,g}, \quad \mathbb{P}^{\mathrm{phys}}_{m,g} := \mathcal{T}^{-1}_{m,g} \mathbb{P}^{\mathrm{phys}}_{m} \mathcal{T}_{m,g}, \\ \widehat{J}_{m,g} &:= \mathcal{T}^{-1}_{m+1,g} \widehat{J}_{m} \mathcal{T}_{m,g}, \quad D^{\mathrm{YM}}_{m,g} := \widehat{H}^{\mathrm{YM}}_{m,g}\Big|_{\mathrm{Ran}\,\mathbb{P}^{\mathrm{phys}}_{m,g}}. \end{aligned} \tag{99.53}$$

The first line is not merely an abstract conjugation. Using (99.44), its action on a cylinder is the heat-bath update of the action:

$$(E^{g}_{m,b}F)(x_{\neq b}) = \frac{\displaystyle\int_{X_{m,b}} F(x_{\neq b}, u) e^{-p_{m,b}(u)/(4\mathrm{g}^2)}\, d\lambda^{0}_{m,b}(u)}{\displaystyle\int_{X_{m,b}} e^{-p_{m,b}(u)/(4\mathrm{g}^2)}\, d\lambda^{0}_{m,b}(u)}, \qquad \widehat{H}^{\mathrm{YM}}_{m,g} = 3\kappa_{\mathrm{av}} \sum_b (I - E^{g}_{m,b}). \tag{99.54}$$

Thus both the measure and every local transition explicitly contain g and the action $p_{m,b}/(4g^2)$. They are Markov because $\mathcal{T}_{m,g}\mathbf{1} = \mathbf{1}$, local because each $T_{m,b,g}$ has the same collar as $p_{m,b}$, and depend on g through (99.48). The equivalence is spectral and does not select the measure from the spectrum: To close the dynamical equations as well, let $X^{a}_{m,b}$ be the left-invariant field on the link factor b associated with $e_a \in \mathfrak{g}$; in a continuous innovation chart, the ordinary directional derivative is used, and finite coordinates are treated by exact difference. For every smooth cylinder F, the common domain is $\mathscr{C}^1_m := \bigcap_{b,a} \mathrm{Dom}\, X^{a}_{m,b}$, taking compact support in the open innovation charts; on the group factors there is no boundary. Haar invariance and the product rule calculate

$$\int X^{a}_{m,b} F\, d\widehat{\mu}^{\mathrm{YM}}_{m,g} = \int F\, X^{a}_{m,b} \mathcal{S}_{m,g}\, d\widehat{\mu}^{\mathrm{YM}}_{m,g}, \qquad F \in \mathscr{C}^1_m. \tag{99.55}$$

On a finite coordinate, for $\Delta_\varepsilon F(q) := F(q+\varepsilon) - F(q)$, the change of variable $q \mapsto q + \varepsilon$ gives the exact version, without a formal derivative,

$$\int \Delta_\varepsilon F(q)\, d\widehat{\mu}^{\mathrm{YM}}_{m,g} = \int F(q)\left(e^{\mathcal{S}_{m,g}(q)-\mathcal{S}_{m,g}(q-\varepsilon)} - 1\right) d\widehat{\mu}^{\mathrm{YM}}_{m,g}(q). \tag{99.56}$$

If δ_ε is the oriented sum of these fields on the edges incident to a vertex, gauge invariance of the reference measure and of $S_{m,g}$ gives $\delta_\varepsilon \mathcal{S}_{m,g} = 0$ and hence the Ward identity

$$\int \delta_\varepsilon F\, d\widehat{\mu}^{\mathrm{YM}}_{m,g} = 0. \tag{99.57}$$

For the finite gauge factor $\mathbb{F}_3^4$, invariance $\mathcal{S}_{m,g}(q+\varepsilon) = \mathcal{S}_{m,g}(q)$ reduces (99.56) to the same Ward identity with δ_ε replaced by Δ_ε. Both expressions are compatible with $\widehat{J}_{m,g}$ because (99.45) separates coarse derivatives from detail derivatives. Every cylinder of the limit resides at some level; thus (99.55) and (99.57) pass literally to the cofinal state. Neither the Schwinger–Dyson nor the Ward equation contains the gap as a premise.

$$D^{\mathrm{YM}}_{m,g} \succeq 3\kappa_{\mathrm{av}}(I - P_{\Omega_{m,g}}), \qquad W_{c,g} := \mathcal{T}^{-1}_{m,g} W_c, \qquad D^{\mathrm{YM}}_{m,g} W_{c,g} = 3\kappa_{\mathrm{av}} W_{c,g}. \tag{99.58}$$

In what follows, an $g \in (0,\infty)$ is fixed and that subscript is suppressed; the measures, expectations, inclusions, and generators without superscript are the objects of (99.53). Geometric holonomies remain canonical coordinate functions on configuration space and, accordingly, their law does change with g; $\mathcal{T}_{m,g}$ carries them to composite functions of the reference chart. Only the spectral witness is transported explicitly in (99.58). This convention does not erase g: its explicit residence is (99.46)–(99.53).

99.8 Uniqueness of the gauge-invariant vacuum and gap saturation $3\kappa_{\mathrm{av}}$ by the eigenvector W_c

The intersection of all expectations of the complete state consists of constants:

$$\bigcap_{\iota \in \mathcal{I}_m} \operatorname{Ran} E_{m,\iota} = \mathbb{C}\Omega_m. \tag{99.59}$$

This identity is not introduced as a hypothesis. In the product chart (99.21), each $E_{m,\iota}$ integrates one coordinate of the exhaustive index (99.22) and leaves the remaining coordinates fixed. If u belongs to the intersection, its conditional expectations with respect to every finite cylinder are constant. Martingale convergence of those cylinders to the complete σ-algebra implies that u is constant almost everywhere. The reverse inclusion is immediate because every expectation preserves the unit. This proves (99.59) and, after gauge projection, leaves a one-dimensional physical vacuum. The Hoeffding decomposition of (99.28)– (99.29) gives

$$\widehat{\mathcal{E}}_m(u) = 3\kappa_{\mathrm{av}} \sum_{S \Subset \mathcal{I}_m} |S|\, \|u_S\|^2, \qquad D^{\mathrm{YM}}_m \succeq 3\kappa_{\mathrm{av}}(I - P_{\Omega_m}). \tag{99.60}$$

The function

$$W_c(q) := \mathbf{1}_{\{q_1+\cdots+q_6=0\}} - \frac{1}{3} \tag{99.61}$$

is invariant under $G^{V(\Lambda_m)}$ because it depends on neither links nor frames. For the incidence action, each x_i occurs three times, hence

$$\sum_{i<j} (B^+ x)_{ij} = 3 \sum_i x_i = 0 \quad \text{in } \mathbb{F}_3. \tag{99.62}$$

Thus $\mathbb{P}^{\mathrm{phys}}_m W_c = W_c$. Under u_6, the sum of the six trits is uniform; by direct calculation,

$$\mathbb{E} W_c = 0, \qquad \|W_c\|^2 = \frac{2}{9}, \qquad \mathbb{E}(W_c^3) = \frac{2}{27}. \tag{99.63}$$

Only the projector $I - E_{m,q_c}$ acts nontrivially on W_c. Therefore

$$D_m^{\mathrm{YM}} W_c = 3\kappa_{\mathrm{av}} W_c. \tag{99.64}$$

The lower bound in (99.60) is not merely a bound: the physical complement contains a nonzero vector attaining the threshold.

99.9 Nonadic refinement and persistence in the limit

The physical inclusions satisfy

$$D_{m+1}^{\mathrm{YM}} \widehat{J}_m = \widehat{J}_m D_m^{\mathrm{YM}}, \qquad \left(e^{-a_{m+1} D_{m+1}^{\mathrm{YM}}}\right)^9 \widehat{J}_m = \widehat{J}_m e^{-a_m D_m^{\mathrm{YM}}}, \qquad a_{m+1} = a_m/9. \tag{99.65}$$

In the inductive limit

$$\widehat{\mathscr{H}}_\infty^{\mathrm{phys}} = \varinjlim (\widehat{\mathscr{H}}_m^{\mathrm{phys}}, \widehat{J}_m) \tag{99.66}$$

define on the algebraic union

$$\mathcal{E}_\infty[\widehat{J}_{m,\infty} u] := \langle u, D_m^{\mathrm{YM}} u \rangle. \tag{99.67}$$

Denote its generator by $D_{\infty,g}^{\mathrm{YM}}$; following the convention after (99.58), also write $D_\infty^{\mathrm{YM}} := D_{\infty,g}^{\mathrm{YM}}$. The first intertwining in (99.65) makes the definition independent of the representative. Hoeffding truncations form a form core and give a recovery sequence; Fatou's lemma applied to the nonnegative sum gives lower semicontinuity. The closure therefore satisfies

$$\overline{\mathcal{E}}_\infty[u] \geq 3\kappa_{\mathrm{av}} \|(I - P_{\Omega_\infty})u\|^2. \tag{99.68}$$

The inclusion preserves the ancestral coordinates, so $\widehat{J}_{m,\infty} W_c$ remains nonzero and satisfies

$$D_\infty^{\mathrm{YM}} \widehat{J}_{m,\infty} W_c = 3\kappa_{\mathrm{av}} \widehat{J}_{m,\infty} W_c. \tag{99.69}$$

Consequently,

$$\inf\left(\operatorname{Spec} D_\infty^{\mathrm{YM}} \setminus \{0\}\right) = 3\kappa_{\mathrm{av}} > 0, \qquad \ker D_\infty^{\mathrm{YM}} = \mathbb{C}\Omega_\infty. \tag{99.70}$$

99.10 Four OS reconstructions of the same dynamics

The complete kernel in (99.36) defines a reversible law on $\widehat{X}_m$. Physical reconstruction uses its correctly typed descent (99.39): on the quotient Y_m, define

$$\Omega_{m,\mathrm{phys}}^{(8)}(d\bar{y}_1 \cdots d\bar{y}_8) = \int \nu_m(d\bar{z}) \prod_{r=1}^{8} K_{m,a_m/2}^{\mathrm{phys}}(\bar{z}, d\bar{y}_r). \tag{99.71}$$

For each direction $\nu = 1, \ldots, 4$, reflection interchanges the four positive faces with the negative ones. Conditioning on the central state z and using reversibility,

$$\int \overline{F(\Theta_{\nu,m} y)} F(y)\, d\Omega_{m,\mathrm{phys}}^{(8)}(y) = \|\mathcal{B}_{m,\nu} F\|_2^2 \geq 0. \tag{99.72}$$

The marginal of the opposite pair normal to ν is

$$\nu_m(d\bar{x}) K_{m,a_m}^{\mathrm{phys}}(\bar{x}, d\bar{y}). \tag{99.73}$$

To identify the Hamiltonian of one direction, take the OS subspace of observables on a single positive face, $F(y) = F_\nu(\bar{y}_{\nu,+})$; multiple-face observables retain the preceding positivity but are not confused with this transfer space. On the single-face quotient, the map

$$\mathcal{V}_{m,\nu}[F] := e^{-a_m D_m^{\mathrm{YM}}/2} \mathcal{W}_m F \tag{99.74}$$

is isometric for the OS norm and has dense range. Its extension identifies the OS quotient with $\widehat{\mathscr{H}}_m^{\mathrm{phys}}$. Set $\overline{D}_m := \mathcal{W}_m^{-1} D_m^{\mathrm{YM}} \mathcal{W}_m$ and $\overline{J}_m := \mathcal{W}_{m+1}^{-1} \widehat{J}_m \mathcal{W}_m$. The intertwining and $a_m = 9a_{m+1}$ give

$$\overline{J}_m e^{-a_m \overline{D}_m/2} = e^{-a_m \overline{D}_{m+1}/2} \overline{J}_m = \left(e^{-a_{m+1}\overline{D}_{m+1}/2}\right)^9 \overline{J}_m. \tag{99.75}$$

The last member belongs to the range of $e^{-a_{m+1}\overline{D}_{m+1}/2}$; hence the inverse of $\mathcal{V}_{m+1,\nu}$ appearing below is applied only on its range and is well defined. The connecting maps

$$\mathcal{J}_{m,\nu}^{\mathrm{OS}} := \mathcal{V}_{m+1,\nu}^{-1} \widehat{J}_m \mathcal{V}_{m,\nu} \tag{99.76}$$

are isometries and intertwine the four reconstructed Hamiltonians:

$$H_{\mathrm{OS},m+1}^{(\nu)} \mathcal{J}_{m,\nu}^{\mathrm{OS}} = \mathcal{J}_{m,\nu}^{\mathrm{OS}} H_{\mathrm{OS},m}^{(\nu)}, \qquad \mathcal{V}_{\infty,\nu} H_{\mathrm{OS},\infty}^{(\nu)} \mathcal{V}_{\infty,\nu}^{-1} = D_\infty^{\mathrm{YM}}. \tag{99.77}$$

The four reflections do not manufacture four measures: they are four publications of the same state, the same kernel, and the same generator.

99.11 Locality, Euclidean action, and curvature publication

The directed category of screens includes refinements, overlays, and Euclidean transports. If $a \in E(4)$ and α is a screen, complete relabeling defines

$$U_\alpha(a) : \widehat{\mathscr{H}}_\alpha^{\mathrm{phys}} \longrightarrow \widehat{\mathscr{H}}_{a\alpha}^{\mathrm{phys}}, \qquad U_\beta(a) \widehat{J}_{\alpha\beta} = \widehat{J}_{a\alpha,a\beta} U_\alpha(a). \tag{99.78}$$

Transport preserves measure, generator, null ideals, and supports. The naturality identity makes it possible to define, without choosing a representative, a unitary map on the algebraic colimit:

$$U(a) \widehat{J}_{\alpha,\infty} F := \widehat{J}_{a\alpha,\infty} U_\alpha(a) F. \tag{99.79}$$

Relabelings satisfy $U_{a\alpha}(b) U_\alpha(a) = U_\alpha(ba)$ and preserve the norm; hence (99.79) extends to a unitary representation of $E(4)$.

99.11.1 Interpretation of the incidence channel as the local curvature of the gauge system

Every compact, connected, simple group has a nontrivial maximal torus. Fix $t_3 \in G$ of order exactly three. For a collar c, let

$$\omega_c(q) := \sum_{i<j} q_{ij} \in \mathbb{F}_3. \tag{99.80}$$

Identity (99.12) proves that ω_c is invariant under the finite action. If h_c is the frame at the base vertex of the collar, define the geometric holonomy and, for the spectral witness, its transported image under the law at coupling g:

$$\mathrm{Hol}_{m,c}^{\mathrm{inc,geom}} := h_c t_3^{\omega_c(q)} h_c^{-1} \in G, \qquad \mathrm{Hol}_{m,c,g}^{\mathrm{inc,spec}} := \mathcal{T}_{m,g}^{-1} \mathrm{Hol}_{m,c}^{\mathrm{inc,geom}}, \qquad \mathrm{Hol}_{m,c}^{\mathrm{inc}} := \mathrm{Hol}_{m,c,g}^{\mathrm{inc,spec}}. \tag{99.81}$$

Under G^{V_m} it transforms by conjugation, and under $\prod_c \mathbb{F}_3^4$ it remains fixed. Since t_3 has order three, the functional calculus of this three-valued variable gives the literal identity

$$W_{c,g} := \mathcal{T}_{m,g}^{-1} W_c^0 = \mathbf{1}_{\{e\}}\left(\mathrm{Hol}_{m,c}^{\mathrm{inc}}\right) - \frac{1}{3}. \tag{99.82}$$

Write $W_c := W_{c,g}$ throughout the rest of the chapter. Since $T_{m,b,g}$ acts within the quotient of the same collar, the spectral holonomy is a function in its local algebra of geometric holonomies; it adds no variable and does not change the support. The geometric holonomies, used in $S_{m,g}$ and in the clover, retain their

g-dependent distribution. The projector on the right-hand side is a local class function; hence W_c is not a decoupled auxiliary coordinate but an observable of the incidence holonomy of the same gauge restriction.

For a bounded open set $O \subset \mathbb{R}^4$, let $\mathfrak{A}_{\mathrm{YM}}(O)$ be the algebra generated by bounded class functions of (99.81), plaquette holonomies, and clover readers supported in O. Define the physical curvature sector by

$$\mathscr{H}_{\mathrm{YM}}^{\mathrm{curv}} := \overline{\bigcup_{O \Subset \mathbb{R}^4} \mathfrak{A}_{\mathrm{YM}}(O)\Omega_\infty} \subseteq \widehat{\mathscr{H}}_\infty^{\mathrm{phys}}. \tag{99.83}$$

By (99.82), each $\widehat{J}_{m,\infty} W_c$ belongs explicitly to this sector. Coordinate expectations map curvature cylinders to curvature cylinders; the semigroup therefore leaves $\mathscr{H}_{\mathrm{YM}}^{\mathrm{curv}}$ invariant. Since it is self-adjoint, the sector is reducing. The bound in (99.68) restricts to it, and the preceding vector attains equality:

$$D_{\mathrm{curv,g}}^{\mathrm{YM}} := D_{\infty,g}^{\mathrm{YM}}\big|_{\mathscr{H}_{\mathrm{YM}}^{\mathrm{curv}}}, \qquad D_{\mathrm{curv}}^{\mathrm{YM}} := D_{\mathrm{curv,g}}^{\mathrm{YM}}, \qquad \inf\big(\mathrm{Spec}\, D_{\mathrm{curv,g}}^{\mathrm{YM}} \setminus \{0\}\big) = 3\kappa_{\mathrm{av}}. \tag{99.84}$$

99.11.2 Cofinal Publication System and Cross-Products

The inclusion $\widehat{J}_{\alpha\beta}$ preserves an ancestral coordinate; it must not be confused with the block average that publishes a continuous field. To write this second map without altering the genealogy, let $\mathfrak{P}$ be the directed set of bounded cellular screens, closed under overlay, nonadic refinement, and Euclidean transport. For $\alpha \in \mathfrak{P}$ set

$$V_\alpha := \mathrm{span}\left\{e_{\alpha,c} := |c|^{-1/2}\mathbf{1}_c : c \in C_\alpha^{\mathrm{cell}}\right\} \subset L^2(\mathbb{R}^4), \qquad P_\alpha : L^2(\mathbb{R}^4) \to V_\alpha. \tag{99.85}$$

Overlays ensure that $P_\alpha h \to h$ for every $h \in L^2$ along $\mathfrak{P}$. On the gauge side, normalize the witness by

$$Z_{\alpha,c} := \sqrt{\frac{9}{2}}\, W_{\alpha,c}, \qquad \langle Z_{\alpha,c}, Z_{\alpha,d}\rangle = \delta_{cd}, \qquad D_\alpha^{\mathrm{YM}} Z_{\alpha,c} = 3\kappa_{\mathrm{av}} Z_{\alpha,c}. \tag{99.86}$$

The second equality uses exactly the moments of (99.63) and the factorization of distinct coordinates. Let $\mathcal{K}_\alpha^{\mathrm{cell}}$ be the space spanned by those vectors. If $\beta \succeq \alpha$, the publication intertwiner, distinct from $\widehat{J}_{\alpha\beta}$, is defined generator by generator by

$$I_{\alpha\beta}^{\mathrm{curv}} Z_{\alpha,c} := \sum_{\substack{d \in C_\beta^{\mathrm{cell}} \\ d \subset c}} \sqrt{\frac{|d|}{|c|}}\, Z_{\beta,d}. \tag{99.87}$$

Since the subcells are disjoint and their volumes sum to $|c|$, one obtains, without taking limits,

$$(I_{\alpha\beta}^{\mathrm{curv}})^* I_{\alpha\beta}^{\mathrm{curv}} = I, \quad I_{\beta\gamma}^{\mathrm{curv}} I_{\alpha\beta}^{\mathrm{curv}} = I_{\alpha\gamma}^{\mathrm{curv}}, \quad D_\beta^{\mathrm{YM}} I_{\alpha\beta}^{\mathrm{curv}} = I_{\alpha\beta}^{\mathrm{curv}} D_\alpha^{\mathrm{YM}}. \tag{99.88}$$

Cellular relabeling preserves volumes and descendants; for $a \in E(4)$, one also has

$$U_\beta(a) I_{\alpha\beta}^{\mathrm{curv}} = I_{a\alpha,a\beta}^{\mathrm{curv}} U_\alpha(a). \tag{99.89}$$

For $h \in C_c^\infty(\mathbb{R}^4)$, now define the block reader

$$\Phi_\alpha(h) := \sum_{c \in C_\alpha^{\mathrm{cell}}} \langle h, e_{\alpha,c}\rangle_{L^2} Z_{\alpha,c}. \tag{99.90}$$

On a uniform mesh, this formula is $\sqrt{9/2}\, a_\alpha^2 \sum_c (P_\alpha h)(x_c) W_{\alpha,c}$; the cellular average, not a point evaluation, fixes the cross product. Indeed, if γ refines two screens α and β, the preceding definitions give the computed identity

$$\left\langle I^{\mathrm{curv}}_{\alpha\gamma}\Phi_\alpha(h), I^{\mathrm{curv}}_{\beta\gamma}\Phi_\beta(k)\right\rangle = \langle P_\alpha h, P_\beta k\rangle_{L^2}. \tag{99.91}$$

By polarization, with $k = h$,

$$\left\| I^{\mathrm{curv}}_{\alpha\gamma}\Phi_\alpha(h) - I^{\mathrm{curv}}_{\beta\gamma}\Phi_\beta(h)\right\|_2^2 = \|P_\alpha h - P_\beta h\|_{L^2}^2 \longrightarrow 0. \tag{99.92}$$

Thus the Cauchy property follows from strong convergence of projections, rather than being an additional postulated identity. The colimit of (99.87) therefore contains

$$\Phi(h) := \lim_{\alpha} I^{\mathrm{curv}}_{\alpha,\infty}\Phi_\alpha(h), \qquad \langle \Phi(h), \Phi(k)\rangle = \langle h, k\rangle_{L^2}, \qquad D^{\mathrm{YM}}_{\mathrm{pub}}\Phi(h) = 3\kappa_{\mathrm{av}}\Phi(h). \tag{99.93}$$

This publication adds no degrees of freedom. At each step, the decomposition $V_\beta = V_\alpha \oplus (V_\beta \ominus V_\alpha)$ has a nonadic Haar basis. The exhaustive ordering of the new incidences of (99.22) pairs that basis, label by label, with $Z_{\beta,d}$ coordinates of the innovation factor, preserving the support of the Haar function. The triangular unitaries $\widehat{\Gamma}_\alpha$ glue those pairings and produce an isometry based

$$\mathcal{U}_{\mathrm{curv}} : \varinjlim(\mathcal{K}^{\mathrm{cell}}_\alpha, I^{\mathrm{curv}}_{\alpha\beta}) \longrightarrow \mathscr{H}^{\mathrm{curv}}_{\mathrm{YM}}, \quad D^{\mathrm{YM}}_{\mathrm{curv}}\mathcal{U}_{\mathrm{curv}} = \mathcal{U}_{\mathrm{curv}}D^{\mathrm{YM}}_{\mathrm{pub}}, \quad U(a)\mathcal{U}_{\mathrm{curv}} = \mathcal{U}_{\mathrm{curv}}U_{L^2}(a). \tag{99.94}$$

Henceforth $\Phi(h)$ also denotes its image under $\mathcal{U}_{\mathrm{curv}}$. The compatibility of (99.49) defines $\mathcal{T}_{\infty,g}$ in the limit. The physical publication at fixed coupling is not a second isometry chosen only on the first chaos, but the restriction

$$\mathcal{U}_{\mathrm{curv},g} := \mathcal{T}^{-1}_{\infty,g}\mathcal{U}_{\mathrm{curv},0}, \qquad \widetilde{\mathcal{U}}_g := \mathcal{T}^{-1}_{\infty,g} : L^\infty(\widehat{X}_\infty, \widehat{\mu}^0_\infty) \longrightarrow L^\infty(\widehat{X}_\infty, \widehat{\mu}^{\mathrm{YM}}_{\infty,g}). \tag{99.95}$$

On the polynomial $*$-algebra generated by holonomies, plaquettes, clover, and $\Phi(h)$, one verifies, generator by generator and then by linearity,

$$\widetilde{\mathcal{U}}_g(AB) = \widetilde{\mathcal{U}}_g(A)\widetilde{\mathcal{U}}_g(B), \quad \widetilde{\mathcal{U}}_g(A^*) = \widetilde{\mathcal{U}}_g(A)^*, \quad \omega_g(\widetilde{\mathcal{U}}_g A) = \omega_0(A). \tag{99.96}$$

The surface, clover, and Bianchi identities are preserved by (99.52); by monotone density, the extension holds on the entire bounded local algebra. Thus (99.91) is the restriction of a complete probabilistic and algebraic isomorphism, not a substitute for it.

99.11.3 Dense core, strong continuity, and local net

Let $M_{\Phi(h)}$ be the multiplication operator affiliated with the local algebra, and

$$\mathsf{W}(h,t) := \exp\!\big(itM_{\Phi(h)}\big), \qquad h \in C^\infty_c(\mathbb{R}^4), \quad t \in \mathbb{R}. \tag{99.97}$$

Complete $\mathfrak{A}_{\mathrm{YM}}(O)$ with these unitaries for $\operatorname{supp} h \subset O$ and with the bounded class functions of plaquette holonomies. Denote by $\mathfrak{A}_{\mathrm{YM}}(O)^{\mathrm{sm}}$ the $*$-algebra thus generated with smooth test functions. The core

$$\mathscr{D}_{\mathrm{sm}} := \operatorname{span}\left\{A_1 \cdots A_n\Omega_\infty : A_j \in \mathfrak{A}_{\mathrm{YM}}(O_j)^{\mathrm{sm}}, \quad O_j \Subset \mathbb{R}^4\right\} \tag{99.98}$$

is dense by the cyclic definition in (99.83). Equations (99.89) and (99.94) give, for $a \in E(4)$,

$$U(a)\Phi(h) = \Phi(a\cdot h), \qquad \|(U(a) - I)\Phi(h)\| = \|a\cdot h - h\|_{L^2}. \tag{99.99}$$

Moreover, $|e^{ix} - e^{iy}| \le |x - y|$ and (99.93) imply

$$\|\big(\mathsf{W}(a\cdot h, t) - \mathsf{W}(h,t)\big)\Omega_\infty\| \le |t|\,\|a\cdot h - h\|_{L^2}. \tag{99.100}$$

Continuity is also checked for the holonomy generators of the core; (99.100) alone does not include them. For an oriented polygonal route γ, a plaquette p, and $f \in C^1(G)^{\mathrm{class}}$, set

$$A_{\gamma,f} := f(\mathrm{Hol}_\gamma), \qquad A_{p,f} := f(\mathrm{Hol}_p). \tag{99.101}$$

In an overlay of γ and $a\gamma$, with $a \in E(4)$, the bridge law (99.17) cancels the common increments. For each remaining increment z, use $|f(xz) - f(x)| \le \|\nabla f\|_\infty d_G(z,e)$ and the second-moment bound for the heat kernel $\int_G d_G(z,e)^2 k_\tau(z)\,dz \le C_G\tau$. For the reweighted law, $0 < e^{-p/(4g^2)} \le 1$ and continuity of p on the compact factor give

$$\frac{\int_G d_G(z,e)^2 e^{-p(z)/(4g^2)} k_\tau(z)\,dz}{\int_G e^{-p(z)/(4g^2)} k_\tau(z)\,dz} \le C_{G,g}\tau, \tag{99.102}$$

uniformly over screens in a compact set and for $0 < \tau \le \tau_0$. Summing the times of the increments in the symmetric difference gives the computed estimate

$$\begin{aligned} \|U(a)A_{\gamma,f}\Omega_\infty - A_{\gamma,f}\Omega_\infty\|_2^2 &\le C_{G,g}\|\nabla f\|_\infty^2 \tau(\gamma \triangle a\gamma), \\ \|U(a)A_{p,f}\Omega_\infty - A_{p,f}\Omega_\infty\|_2^2 &\le C_{G,g}\|\nabla f\|_\infty^2 \tau(\partial p \triangle \partial(ap)). \end{aligned} \tag{99.103}$$

Here the symmetric difference is not measured as two disjoint curves: the surface connectors fill it by the minimal oriented ribbon $\Sigma(\gamma, a\gamma)$, and

$$\tau(\gamma \triangle a\gamma) := \sum_{p \subset \Sigma(\gamma, a\gamma)} \tau_p, \qquad \tau(\gamma \triangle a\gamma) \le C_\gamma d_{E(4)}(a,e), \tag{99.104}$$

with the same definition for plaquette boundaries. The inequality follows from summing the additive times in (99.17) over a ribbon of width $d_{E(4)}(a,e)$ and length bounded by that of γ. The bridge-time measure on the right-hand side thus tends to zero as $a \to e$; this is exactly the continuity of the route connectors in the gauge image. Every bounded clover reader $\chi(F_{\mathrm{cl}})$, $\chi \in C_b^1$, is a function of a finite sum of transported plaquettes; the same bound, multiplied by $\|\chi'\|_\infty$ and the number of terms, also proves its continuity.

For merely bounded class functions, the regularization entering the core is made explicit. If db is left Haar measure on $E(4)$ and $\eta \in C_c^\infty(E(4))$, define the Bochner integral

$$A[\eta] := \int_{E(4)} \eta(b) U(b) A U(b)^*\, db, \qquad A \in \{A_{\gamma,f}, A_{p,f}, \chi(F_{\mathrm{cl}})\}, \quad \chi \in C_b^1. \tag{99.105}$$

With $(L_a\eta)(b) := \eta(a^{-1}b)$, the change of variable $b \mapsto ab$ gives

$$U(a)A[\eta]U(a)^* = A[L_a\eta], \qquad \|(A[L_a\eta] - A[\eta])\Omega_\infty\| \le \|A\|\,\|L_a\eta - \eta\|_{L^1(E(4))} \longrightarrow 0. \tag{99.106}$$

From now on, the holonomy, plaquette, and clover generators of $\mathfrak{A}_{\mathrm{YM}}(O)^{\mathrm{sm}}$ are understood to be those in (99.101) with $f \in C^1$, or their regularizations (99.105), with the swept support remaining compact within O. Approximate identities on $E(4)$, together with (99.103), recover the unregularized generators in L^2; hence the cyclic sector is not reduced.

A telescoping product argument using (99.100) and (99.106) proves continuity of all generators of $\mathscr{D}_{\mathrm{sm}}$; by density and unitarity,

$$\lim_{a \to e} \|(U(a) - I)\Psi\| = 0 \qquad (\Psi \in \mathscr{H}_{\mathrm{YM}}^{\mathrm{curv}}). \tag{99.107}$$

Disjoint supports use disjoint factors and commute. This gives the net

$$\begin{aligned} O_1 \subseteq O_2 &\Rightarrow \mathfrak{A}_{\mathrm{YM}}(O_1) \subseteq \mathfrak{A}_{\mathrm{YM}}(O_2), \\ [\mathfrak{A}_{\mathrm{YM}}(O_1), \mathfrak{A}_{\mathrm{YM}}(O_2)] &= 0 \quad (O_1 \cap O_2 = \varnothing), \\ U(a)\mathfrak{A}_{\mathrm{YM}}(O)U(a)^* &= \mathfrak{A}_{\mathrm{YM}}(aO). \end{aligned} \tag{99.108}$$

The Gram identity for $h \neq 0$ and (99.82) prove that the net is nonscalar.

99.11.4 Continuous OS package and reconstruction

For a direction ν, let $\mathfrak{A}_{+,\nu}$ be the closure of the union of the preceding algebras compactly supported in $\{x_\nu > 0\}$. The static state, stack law, and transfer operator are not defined separately. On a screen α, let $\widehat{\mu}^{\mathrm{YM}}_{\alpha,g}$ be the marginal of (99.46), let $\nu_{\alpha,g} := (\pi^{\mathrm{phys}}_\alpha)_* \widehat{\mu}^{\mathrm{YM}}_{\alpha,g}$ be its law on the quotient, and let $K_{\alpha,g}(t) = e^{-tD^{\mathrm{YM}}_{\alpha,g}}$ be its kernel. For ordered times $t_1 \le \cdots \le t_n$, the Euclidean marginal of the same measure is

$$\mathbb{P}^{E,(\nu;t_1,\dots,t_n)}_{\alpha,g} := (\mathrm{ev}^{(\nu)}_{t_1}, \dots, \mathrm{ev}^{(\nu)}_{t_n})_* \widehat{\mu}^{\mathrm{YM}}_{\alpha,g}. \tag{99.109}$$

The disintegration identity forming part of (95.348) calculates it, rather than replacing it with an auxiliary chain:

$$d\mathbb{P}^{E,(\nu;t_1,\dots,t_n)}_{\alpha,g} = \nu_{\alpha,g}(dy_1) \prod_{j=1}^{n-1} K_{\alpha,g}(t_{j+1} - t_j; y_j, dy_{j+1}). \tag{99.110}$$

If M_A denotes multiplication by a bounded reader A, its correlations are therefore

$$\begin{aligned} S^{(\nu,g)}_{\alpha,n}(A_1, t_1; \dots; A_n, t_n) &:= \int \prod_{j=1}^{n} (\tau_{t_j e_\nu} A_j)(x)\, d\widehat{\mu}^{\mathrm{YM}}_{\alpha,g}(x) \\ &= \langle \mathbf{1}, M_{A_1} K_{\alpha,g}(t_2 - t_1) M_{A_2} \cdots K_{\alpha,g}(t_n - t_{n-1}) M_{A_n} \mathbf{1} \rangle_{L^2(\nu_{\alpha,g})}. \end{aligned} \tag{99.111}$$

The law of the stationary stack at those same times is this Euclidean marginal, written through its disintegration:

$$d\mathbb{P}^{(\nu;t_1,\dots,t_n)}_{\alpha,g} := d\mathbb{P}^{E,(\nu;t_1,\dots,t_n)}_{\alpha,g} = \nu_{\alpha,g}(dy_1) \prod_{j=1}^{n-1} K_{\alpha,g}(t_{j+1} - t_j; y_j, dy_{j+1}). \tag{99.112}$$

Integrating successively over $y_n, y_{n-1}, \dots, y_1$ gives the exact identity

$$S^{(\nu,g)}_{\alpha,n}(A_1, t_1; \dots; A_n, t_n) = \int \prod_{j=1}^{n} A_j(y_j)\, d\mathbb{P}^{(\nu;t_1,\dots,t_n)}_{\alpha,g}(y). \tag{99.113}$$

If all times coincide, the kernels are the identity, and

$$S^{(\nu,g)}_{\alpha,n}(A_1, t; \dots; A_n, t) = \int A_1 \cdots A_n \, d\nu_{\alpha,g}. \tag{99.114}$$

Thus the static Schwinger state is the equal-time marginal of the same Euclidean measure, and its disintegration is the Markov law; they are not two functionals subsequently identified by name. Compatibility of $\widehat{J}_{\alpha,g}$ with the measure and kernel makes (99.113) compatible with every overlay. Its limit defines

$$S^{\mathrm{YM}}_n(A_1, t_1; \dots; A_n, t_n) := \lim_\alpha S^{(\nu,g)}_{\alpha,n}(A_{1,\alpha}, t_1; \dots; A_{n,\alpha}, t_n). \tag{99.115}$$

The insertions of Φ are obtained by differentiating, in $t = 0$, the unitaries (99.97). Since the transported $Z_{\alpha,c}$ are centered, independent, and uniformly bounded, Hoeffding's lemma gives, level by level,

$$\left\langle \Omega_\alpha, e^{t\Phi_\alpha(h)} \Omega_\alpha \right\rangle \le \exp\left(\frac{9}{16} t^2 \|P_\alpha h\|^2_{L^2} \right). \tag{99.116}$$

The bound passes to the limit by (99.92); all derivatives are continuous in the Schwartz topology and define tempered distributions.

The eight-face form first extends to every finite stack. If F depends on the N positive layers of a screen α, the reflection in (99.112) and conditioning on the central layer give

$$\begin{aligned}\int \overline{F(\Theta_\nu y_{-N},\dots,\Theta_\nu y_{-1})}F(y_1,\dots,y_N)\,d\Omega^{(\nu,g)}_{\alpha,N}(y)\\ = \int \nu_{\alpha,g}(dz)\left|\int K_{\alpha,g}(a_\alpha/2;z,dy_1)\prod_{j=1}^{N-1}K_{\alpha,g}(a_\alpha;y_j,dy_{j+1})F(y_1,\dots,y_N)\right|^2 \ge 0.\end{aligned} \tag{99.117}$$

Every half-space cylinder belongs to a finite stack; hence (99.117) realizes the OS form before closure. In the OS quotient, if F has insertions at positive times, let $(\tau_s F)(A_1,t_1;\dots;A_n,t_n) := F(A_1,t_1+s;\dots;A_n,t_n+s)$. Define the translation operator by

$$T^{(\nu)}_{\mathrm{OS}}(s)[F] := [\tau_s F], \qquad s \ge 0. \tag{99.118}$$

For half-space cylinders F, G, the Euclidean–Markov equality (99.110) and the semigroup property give, generator by generator,

$$\begin{aligned}\langle [F], T^{(\nu)}_{\mathrm{OS}}(s)[G]\rangle_{\mathrm{OS}} &= S^{\mathrm{YM}}_g((\Theta_\nu F)^*\tau_s G)\\ &= \left\langle \mathcal{V}_{\infty,\nu}[F], e^{-sD^{\mathrm{YM}}_{\mathrm{curv,g}}}\mathcal{V}_{\infty,\nu}[G]\right\rangle.\end{aligned} \tag{99.119}$$

The identity shows both that null classes remain null and that, by density of the cylinders,

$$\mathcal{V}_{\infty,\nu}T^{(\nu)}_{\mathrm{OS}}(s)\mathcal{V}^{-1}_{\infty,\nu} = e^{-sD^{\mathrm{YM}}_{\mathrm{curv,g}}}. \tag{99.120}$$

This is the explicit arrow Euclidean measure → disintegration → OS transfer → Hamiltonian.

Proposition 99.1 (Continuous Osterwalder–Schrader package)**.** *The functionals* (99.115) *satisfy:*

1. *normalization, Hermiticity, and bosonic symmetry;*
2. *Euclidean covariance and tempered regularity;*
3. *reflection positivity in each of the four directions,*

$$\sum_{i,j}\overline{a_i}a_j\,S^{\mathrm{YM}}\big((\Theta_\nu F_i)^*F_j\big) \ge 0, \qquad F_i \in \mathfrak{A}_{+,\nu}; \tag{99.121}$$

4. *locality and exponential clustering: if A, B are centered local observables and a translation of length r separates their supports, then*

$$|S^{\mathrm{YM}}(A\,\tau_r B)| \le e^{-3\kappa_{\mathrm{av}}r}\,\|A\Omega_\infty\|\,\|B\Omega_\infty\|. \tag{99.122}$$

OS reconstruction from any of the four half-nets produces the same cyclic space, the same vacuum, and the Hamiltonian $D^{\mathrm{YM}}_{\mathrm{curv,g}}$. OS continuation produces local Wightman fields and a Poincaré representation whose spectrum satisfies

$$\mathrm{Spec}(P^0,\mathbf{P}) \subseteq \overline{V}_+, \qquad P^0 = D^{\mathrm{YM}}_{\mathrm{curv,g}} \ge 0. \tag{99.123}$$

Proof. Normalization and Hermiticity originate in the state; symmetry and locality, in Euclidean commutativity for disjoint supports. Covariance and continuity are (99.99)– (99.107); regularity is (99.116). For cylindrical F_i, all lie in a finite overlay and stack; (99.117) and polarization prove (99.121). Approximation in the OS norm and Cauchy–Schwarz extend it to $\mathfrak{A}_{+,\nu}$.

After a Euclidean rotation, spatial separation becomes positive time. The stack–transfer identity (99.113) and the operator (99.118) express the centered correlation as

$$\langle A^{*}\Omega_{\infty}, e^{-r D^{\mathrm{YM}}_{\mathrm{curv,g}}} B\Omega_{\infty}\rangle.$$

The spectral bound (99.84) gives (99.122). Finally, the connecting maps (99.76) form a directed system; their union is dense by (99.98). The identity (99.77) identifies the four closures and their generators with $D^{\mathrm{YM}}_{\mathrm{curv,g}}$. The already verified hypotheses of regularity, covariance, reflection positivity, and clustering are the inputs to OS continuation; its reconstruction theorem gives Wightman locality and (99.123). □

99.11.5 Typed publication as Yang–Mills theory

Identification does not rest solely on evaluating a clover on a given connection. Let $\mathcal{R}^{\mathrm{loc}}_{G}$ be the $*$-algebra of readers formed by all class functions of incidence and plaquette holonomies, their transports to a base vertex, and polarizations of the reader $\mathcal{P}^{F^2}$. Let

$$\mathcal{E}_G := \Lambda^2(\mathbb{R}^4)^{*} \otimes \mathfrak{g} \tag{99.124}$$

with the invariant inner product. In each cell, polarization of $\mathcal{P}^{F^2}$ gives a positive Gram form $G^F_{\alpha,c}$ on $\mathcal{E}_G$. Quotient by its kernel and apply its positive square root; denote the resulting normalized reader by $Y_{\alpha,c}(v)$, which satisfies

$$\langle Y_{\alpha,c}(v), Y_{\alpha,d}(w)\rangle = \delta_{cd}\langle v, w\rangle_{\mathcal{E}_G}. \tag{99.125}$$

If $d \subset c$, let $T_{d\leftarrow c}$ be the frame transport already present in the surface holonomy. The complete-curvature intertwiner is

$$I^F_{\alpha\beta} Y_{\alpha,c}(v) := \sum_{d\subset c} \sqrt{\frac{|d|}{|c|}}\, Y_{\beta,d}\big(\mathrm{Ad}_{T_{d\leftarrow c}}\, v\big)\,. \tag{99.126}$$

Invariance of the inner product on $\mathfrak{g}$, disjointness of subcells, and composition of transports prove

$$(I^F_{\alpha\beta})^{*} I^F_{\alpha\beta} = I, \qquad I^F_{\beta\gamma} I^F_{\alpha\beta} = I^F_{\alpha\gamma}. \tag{99.127}$$

For $h \in C^{\infty}_{c}(\mathbb{R}^4; \mathcal{E}_G)$, set

$$\mathbf{F}_{\alpha}(h) := \sum_{c\in C^{\mathrm{cell}}_{\alpha}} Y_{\alpha,c}\left(|c|^{-1/2} \int_{c} h(x)\, d^4x\right). \tag{99.128}$$

In a common overlay, the same calculation as in (99.91) now gives

$$\Big\langle I^F_{\alpha\gamma}\mathbf{F}_{\alpha}(h), I^F_{\beta\gamma}\mathbf{F}_{\beta}(k)\Big\rangle = \langle P_{\alpha}h, P_{\beta}k\rangle_{L^2(\mathcal{E}_G)}. \tag{99.129}$$

Thus $\mathbf{F}_{\alpha}(h)$ is Cauchy by strong convergence of the projections, not by a new hypothesis. The same pairing of Haar functions and innovations in (99.94), now applied to the orthonormalized basis of $G^F_{\alpha,c}$, realizes the colimit through a local, gauge-covariant isometry into $L^2(\widehat{X}_{\infty}, \widehat{\mu}^{\mathrm{YM}}_{\infty,g})$. Its limit defines the gauge-covariant random distribution

$$\begin{aligned} \mathbf{F}_{\mu\nu} &\in \mathcal{S}'(\mathbb{R}^4;\mathfrak{g}) \,\widehat{\otimes}\, L^2(\widehat{X}_{\infty}, \widehat{\mu}^{\mathrm{YM}}_{\infty,g}),\\ U(a)\mathbf{F}(h)U(a)^{*} &= \mathbf{F}(a\cdot h),\\ u\mathbf{F}(h)u^{-1} &= \mathbf{F}(\mathrm{Ad}_u\, h), \qquad a \in E(4),\ u \in G. \end{aligned} \tag{99.130}$$

Its order-three spectral class channel is precisely Φ. The finite surface relation

$$\mathrm{Hol}_{p}^{(\alpha)} = \overrightarrow{\prod_{p' \subset p}} T_{p'}\, \mathrm{Hol}_{p'}^{(\beta)}\, T_{p'}^{-1} \tag{99.131}$$

passes to the cofinal closure because all its cross products are calculated in an overlay. In particular, the random clover reader, not merely its evaluation on a smooth connection, satisfies the small-loop identity

$$\mathbf{F}(h) = L^2\text{–}\lim_{\alpha} a_\alpha^4 \sum_{x} \left\langle (P_\alpha h)(x), F_{\mathrm{cl},\alpha}(x) \right\rangle, \qquad \delta_{\mathrm{gauge}} S_n^{\mathrm{YM}} = 0, \qquad \mathrm{Alt}_{\lambda\mu\nu}\, D_\lambda \mathbf{F}_{\mu\nu} = 0. \tag{99.132}$$

The first equality is (99.129) written in the clover representative; the second is the Ward identity (99.57) passed to the limit; and the third is cancellation of the boundary of a boundary in the surface product. None uses the mass term.

To avoid defining the target by the conclusion sought, let $\mathbf{YM}_4(G, g)$ be the class of local quintuples $(\mathfrak{A}, \mu_g, K_g, \mathrm{Hol}, \mathbf{F})$ satisfying the following checkable equations: (i) the Gibbs density is (99.46); (ii) K_g is Markov, reversible, and satisfies (99.55)–(99.57); (iii) holonomy, clover, and curvature satisfy (99.52) and (99.132); and (iv) their functionals are the same stack law as in (99.113) and satisfy the OS package. The constructed object is, component by component,

$$\begin{aligned} \mathfrak{Q}_{\mathrm{YM}}^{(4)}(G, g) := \big(& \{\mathfrak{A}_{\mathrm{YM}}(O)\}_O, S_g^{\mathrm{YM}}, U, \Theta, \Omega_{\infty,g}, \\ & \widehat{\mu}_{\infty,g}^{\mathrm{YM}}, \{S_{m,g}, \mathcal{S}_{m,g}\}_m, K_g, \mathrm{Hol}, \mathbf{F}, \\ & \mathscr{H}_{\mathrm{OS}}, D_{\mathrm{curv,g}}^{\mathrm{YM}} \big) \in \mathbf{YM}_4(G, g). \end{aligned} \tag{99.133}$$

The publication map does not omit any of those components:

$$\begin{aligned} \mathcal{P}_{\mathrm{YM},g} : \mathfrak{G}_{\mathrm{gau}}(G) &\longrightarrow \mathbf{YM}_4(G, g), \\ (\widehat{\mu}^0, \mathbb{P}^{\mathrm{phys}}, D^0, \Theta, W, \mathrm{Hol}, \mathcal{P}^{F^2}) &\longmapsto (\widehat{\mu}_g^{\mathrm{YM}}, S_g, \mathcal{S}_g, K_g, \mathbb{P}_g^{\mathrm{phys}}, S_g^{\mathrm{YM}}, \mathfrak{A}_{\mathrm{YM}}, D_{\mathrm{curv,g}}^{\mathrm{YM}}, \mathbf{F}, \mathrm{Hol}). \end{aligned} \tag{99.134}$$

The S_n^{YM} are thus exactly the Schwinger functions of the g-dependent measure, Markov law, and transfer operator already constructed.

It remains to verify that $S_{m,g}$ is the Yang–Mills action and not merely a function bearing that name. Let $A \in C_c^3(\mathbb{R}^4; T^*\mathbb{R}^4 \otimes \mathfrak{g})$ be a smooth connection in a chart where the logarithm of small loops is unique. For the plaquette $p_{\mu\nu}(x)$, the ordered Magnus expansion gives

$$\log \mathrm{Hol}_{p_{\mu\nu}(x)}(A) = a_m^2 F_{A,\mu\nu}(x) + \frac{a_m^3}{2} (D_\mu + D_\nu) F_{A,\mu\nu}(x) + R_{m,\mu\nu}(x), \tag{99.135}$$

with $\|R_{m,\mu\nu}(x)\| \le C_G a_m^4 (\|D^2 F_A\|_{\infty,p} + \|F_A\|_{\infty,p}^2)$. Averaging over the four orientations cancels the odd term and defines

$$F_{\mathrm{cl},m}(x) := \frac{1}{4a_m^2} \sum_{p \ni x} \mathrm{Ad}_{T_p} \log \mathrm{Hol}_p, \qquad \|F_{\mathrm{cl},m}(x) - F_A(x)\| \le C_G a_m^2 (\|D^2 F_A\|_\infty + \|F_A\|_\infty^2). \tag{99.136}$$

The definition of the square root (99.41) is precisely the polarization of these six components. Substituting (99.135) into each generator of this polarization gives

$$\xi_{m,b}(A) = a_m^2 \,\mathrm{vec}_{\mu<\nu} F_{A,\mu\nu}(x_b) + r_{m,b}(A), \qquad \|r_{m,b}(A)\| \le C_{G,A} a_m^4. \tag{99.137}$$

The odd cross terms cancel through the four clover orientations; the even terms are included exactly once by (99.42). By Riemann summation and Cauchy–Schwarz,

$$\left| S_{m,g}(A) - \frac{1}{4g^2} a_m^4 \sum_x \|F_{\mathrm{cl},m}(x)\|^2 \right| \le C_{G,A,g} a_m^2, \tag{99.138}$$

and, using the bound in (99.136),

$$S_{m,g}(A) \longrightarrow S_{\mathrm{YM},g}(A) := \frac{1}{4g^2} \int_{\mathbb{R}^4} \|F_A(x)\|^2 \, d^4x. \tag{99.139}$$

The Schwinger–Dyson and Ward equations have already been calculated on the measure using this action. The gap, by contrast, follows subsequently from the unitary conjugation in (99.53) and the witness (99.58); it was not used to choose g, $S_{m,g}$, or $\widehat{\mu}^{\mathrm{YM}}_{m,g}$. This is the noncircular connection between action, law, and gap.

Theorem 99.2 (Yang–Mills closure for every compact, connected, simple Lie group)**.** *The complete structure $\mathfrak{G}_{\mathrm{gau}}$ constructs a local projective family for every compact, connected and simple Lie group. The map* (99.134) *publishes a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$: its functionals satisfy the continuous OS package and its continuation satisfies the local Wightman axioms by proposition 99.1; its holonomies and distributional curvature satisfy* (99.132)*, and the four reconstructions coinduce the same Hamiltonian $D^{\mathrm{YM}}_{\infty,g}$, with a simple vacuum. Its law is $\widehat{\mu}^{\mathrm{YM}}_{\infty,g}$, constructed by the compatible family of local actions $\{S_{m,g}, \mathcal{S}_{m,g}\}_m$, and satisfies the Schwinger–Dyson and Ward equations* (99.55)–(99.57)*. The cyclic curvature sector* (99.83) *is reducing, contains the witness* (99.82) *and has*

$$\boxed{\Delta^{\mathrm{HMT}}_{\mathrm{YM}} = 3\kappa_{\mathrm{av}} > 0}, \qquad m^{\mathrm{HMT}}_{\mathrm{gap}} = \frac{\hbar_{\mathrm{int}}}{c_{\mathrm{int}}} \Delta^{\mathrm{HMT}}_{\mathrm{YM}} > 0. \tag{99.140}$$

The local net, the strong action of $E(4)$, the Schwinger distributions, the holonomy, and $\mathbf{F}$ arise from the same complete state. The exact reader $\mathcal{P}^{F^2}$ defines the density of the law; the clover is its smooth chart, and (99.139) *identifies its classical action. The gap belongs to the collective, gauge-invariant curvature sector and does not assign an elementary mass to the gluon.*

Proof. Kinematic projectivity is (99.19); the square root (99.41), cocycle (99.45), and density (99.46) first construct the action and measure at coupling g. The factor-by-factor transport (99.48)– (99.51) proves projectivity and preservation of the state, products, and $*$; (99.52) establishes clover and Bianchi. The exhaustive index (99.22), the form (99.27), and the circuit (99.32) construct the common generator. Physical reduction and its quotient kernel are (99.35)– (99.40). The conjugation (99.53) transports them to the g-dependent measure without altering their spectrum. The Hoeffding decomposition proves the vacuum and bound, and (99.62)– (99.64) exhibit a physical vector attaining the threshold. Nonadic intertwining transports form, vacuum, and witness to the limit, where (99.68)– (99.70) determine the exact gap. The four factorizations (99.72), the range inclusion (99.75), and the connecting maps (99.76) identify their reconstructions with that same generator. The holonomy (99.81) places W_c explicitly in the curvature sector, where (99.84) proves the exact gap. The system (99.87) controls cross products through (99.91); (99.95)– (99.96) lift that first chaos to the isomorphism of the entire algebra. Strong convergence of P_α produces Φ; the bounds (99.103) and (99.106) also prove $E(4)$-continuity of holonomies, plaquettes, and clover. Euclidean disintegration (99.110), the exact identity (99.113), and the intertwining (99.119) connect the same measure, Schwinger state, Markov stack, OS transfer, and gap generator. Proposition 99.1 extends the four finite forms to the complete half-nets, proves regularity and clustering, and reconstructs the Hamiltonian. Finally, (99.130)– (99.134) identify those same functionals as the Schwinger functionals of the local gauge theory; the integrations by parts (99.55)–(99.57) prove Schwinger–Dyson and Ward. Lastly, (99.135)–(99.139) identify the classical action before spectral equivalence is applied. The chain does not use the gap to choose the measure. □

Falsification criterion 99.3 (Collapse of the product dynamics)**.** If D_m^{prod} is replaced by D_m^{fib}, all nonconstant sectors receive the same eigenvalue. A Hoeffding vector with $|S| = 2$ should have energy $6\kappa_{\mathrm{av}}$ by (99.4), but the simple semigroup would assign it $3\kappa_{\mathrm{av}}$. Identifying these two generators is false.

Falsification criterion 99.4 (Confusion between genealogical inclusion and block averaging)**.** If $I_{\alpha\beta}^{\mathrm{curv}}$ is replaced by $\widehat{J}_{\alpha\beta}$ in (99.91), a coarse coordinate persists with coefficient one while the fine coefficient changes by the scale ratio. For $a_{m+1} = a_m/9$, the ancestral cross product carries a factor $1/81$ and does not converge to the diagonal norm. The identity (99.92) fails; this is why the two maps are kept separate.

Falsification criterion 99.5 (Ablation of the incidence fiber)**.** If $\widehat{X}_m$ is replaced by its link marginal, q_c is erased, along with the physical observable W_c. The forgetful map does not reflect the complete gauge subalgebra; it therefore cannot be used to deny the eigenvalue of the source object. The object has been replaced; the conclusion is not transferable.

Falsification criterion 99.6 (Replacement of the exact reader by its smooth chart)**.** If the asymptotic term of (99.139) is used directly as a density on a finite screen, the controlled remainder of (99.138) is erased and the exact cocycle (99.45) is lost. The governing finite density is $Z_{m,g}^{-1} e^{-\mathcal{P}_m^{F^2}/(4g^2)}$; the clover term and the smooth limit identify it with the classical action, but do not replace it.

Provenance and status. The separation between simple fiber and product generator, the common measure, the gauge compression, the four OS, the limit form and the physical witness are a `RECOVERED_RESULT` and `INTERNALLY_EXACT` in the recorded regular HMT gauge domain. Disintegration with nonuniform times, the exhaustive index of expectations, the quotient kernel, the matrix B^+, the explicit action on the colimit, the cofinal block system, the continuous OS package, distributional curvature and incidence holonomy are a `NEW_FORMALIZATION`: they type and discharge the links that previously appeared only nominally, without introducing the gap as a datum. Their complete composition establishes the preceding theorem. The positive root of the reader F^2, its Gibbs density, the compatible local transport, the Schwinger–Dyson/Ward equations, the complete algebraic isomorphism, the holonomy continuity bounds and the stack–transfer identity are a `NEW_FORMALIZATION`. Their role is to materially discharge the four bridges; the clover–classical action identification follows from the Magnus expansion and is not a spectral premise.

99.12 Spectral gap of the gauge sector and projective limit

The projective gauge family constructed from the complete state has local covariance, locality, reflection positivity, a strongly continuous action of $E(4)$, compatible Schwinger functionals, and a simple vacuum. The four reconstructions coinduce the same Hamiltonian, and the cyclic curvature sector has the exact gap

$$\Delta_{\mathrm{YM}}^{\mathrm{HMT}} = 3\kappa_{\mathrm{av}} > 0.$$

The curvature reader and its classical limit follow from the same law. The gap belongs to the gauge-invariant collective sector and does not assign an elementary mass to the gluon.

Chapter 100

Algebraic realization of Hodge classes via compatible cohomological projectors and rational descent

100.1 Source object, finite observations and publication

Let X be a smooth complex projective variety and $p \geq 0$. The object produced by the common construction and received here is the complete cohomological structure $\mathfrak{G}_{\mathrm{coh}}(X,p)$, which preserves based charts, oriented crossings, Mayer–Vietoris cones, carry, boundary, route, multisection, survival and terminal. At depth N we write

$$\mathcal{S}_N(X,p) := \mathrm{Surv}_{J_N}(X,p), \qquad \rho_{N+1,N} : \mathcal{S}_{N+1}(X,p) \to \mathcal{S}_N(X,p), \tag{100.1}$$

and the modules of observations

$$e_N : \mathcal{S}_N(X,p) \to \mathscr{H}^p_N(X), \qquad q_{N+1,N} : \mathscr{H}^p_{N+1}(X) \to \mathscr{H}^p_N(X), \tag{100.2}$$

with the square

$$e_N \rho_{N+1,N} = q_{N+1,N} e_{N+1}. \tag{100.3}$$

For

$$\alpha \in \mathrm{Hdg}^p(X)_{\mathbb{Q}} := H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X),$$

let $q_N\alpha$ be its family of finite observations. The fiber to be prolonged is

$$\mathcal{F}_N(\alpha) := \{ s \in \mathcal{S}_N(X,p) : e_N(s) = q_N\alpha \}. \tag{100.4}$$

The producer and the publication are typed by

$$R_{\mathrm{Hdg}} : X_\chi(X,p) \to K_0\big(\mathrm{Perf}^{\geq p}(X)\big)_{\mathbb{Q}}, \tag{100.5}$$

$$\mathrm{cl}_{X,p} \circ \mathrm{ch}^{\mathrm{loc}}_p \circ R_{\mathrm{Hdg}} = \mathrm{ev}^{\mathrm{Hdg}}_{\infty,X,p}. \tag{100.6}$$

This is the reader action identity already constructed within the complete object (95.355); it does not introduce the preimage of a Hodge class from the codomain. The preceding exact owner constructs that existence, and this chapter unfolds the composition. The reader must remain in the order

$$\mathfrak{G}_{\mathrm{coh}}(X,p) \longrightarrow X_\chi(X,p) \xrightarrow{R_{\mathrm{Hdg}}} K_0\big(\mathrm{Perf}^{\geq p}(X)\big)_{\mathbb{Q}} \xrightarrow{\mathrm{ch}^{\mathrm{loc}}_p} \mathrm{CH}^p(X)_{\mathbb{Q}} \xrightarrow{\mathrm{cl}_{X,p}} \mathrm{Hdg}^p(X)_{\mathbb{Q}}.$$

In particular, the final class does not retrospectively select a cell or a cycle.

100.2 Relative module and triangular incidence table

Having fixed N, let

$$\mathscr{K}^{\mathrm{S}}_{N+1} := \ker \rho_{N+1,N}, \qquad \mathscr{K}^{\mathrm{H}}_{N+1} := \ker q_{N+1,N}, \qquad e^{\mathrm{rel}}_{N+1} := e_{N+1}|_{\mathscr{K}^{\mathrm{S}}_{N+1}}. \tag{100.7}$$

The new charts ΔJ_{N+1} produce $C_\nu = \mathrm{Cone}(\kappa_{a,b})$ and the module

$$\mathscr{A}^{\mathrm{rel}}_{N+1} := H^0\Big(\mathsf{W}_{\mathrm{rel}} \otimes^{\mathbf{L}}_{\mathbb{Q}[\Delta J_{N+1}]} \Gamma_{\mathrm{rel}} \oplus \mathrm{Cone}(I - U_{\Gamma_9})_{\mathrm{rel}}\Big). \tag{100.8}$$

Generator maps

$$a_{N+1} : \mathscr{A}^{\mathrm{rel}}_{N+1} \to \mathscr{K}^{\mathrm{S}}_{N+1}, \qquad b_{N+1} : \mathscr{A}^{\mathrm{rel}}_{N+1} \to \mathscr{K}^{\mathrm{H}}_{N+1} \tag{100.9}$$

satisfy

$$e^{\mathrm{rel}}_{N+1} a_{N+1} = b_{N+1}. \tag{100.10}$$

100.2.1 Cohomological Generators and Compatible Projectors

So that (100.10) is not a black box, the two finite sets involved are written explicitly. Let $\mathcal{G}^{\mathrm{coh}}_{N+1}$ be the set of all new labels

$$\nu = (Z, W, a, b, \mathrm{or}, \mathrm{Carr}, \mathrm{Mem}, \partial, \gamma, \tau^{\mathrm{term}})$$

that appear in $\Sigma^{\mathrm{mult}}_{\mathrm{coh}}$, including one terminal label for each component of $\mathrm{Cone}(I - U_{\Gamma_9})_{\mathrm{rel}}$. Let $\mathcal{R}^{\mathrm{coh}}_{N+1}$ be the set of three classes of relations: Mayer–Vietoris at crossings $Z \cap W$, change of orientation/carry, and terminal compatibility of one complete turn. Form

$$\begin{aligned} F^{\mathrm{coh}}_{N+1} &:= \mathbb{Q}^{(\mathcal{G}^{\mathrm{coh}}_{N+1})}, & D_{N+1} &: \mathbb{Q}^{(\mathcal{R}^{\mathrm{coh}}_{N+1})} \longrightarrow F^{\mathrm{coh}}_{N+1}, \\ \mathscr{A}^{\mathrm{rel}}_{N+1} &\cong F^{\mathrm{coh}}_{N+1} / \operatorname{im} D_{N+1}. \end{aligned} \tag{100.11}$$

The action of the realizer on the base is not left implicit:

$$\begin{aligned} R_{\mathrm{Hdg},N+1}(e_\nu) &:= [\mathrm{Cone}(\kappa^{Z,W}_{a,b})], & \nu \text{ cellular}, \\ R_{\mathrm{Hdg},N+1}(e_{\nu_{\mathrm{term}}}) &:= [\mathrm{Cone}(I - U_{\Gamma_9})_{\nu_{\mathrm{term}}}], & \nu_{\mathrm{term}} \text{ terminal}. \end{aligned} \tag{100.12}$$

The additivity of cones in the Mayer–Vietoris triangle, the identity $\kappa_{b,a}\tau = -P\kappa_{a,b}$, the carry cocycle, and the identity of terminal return give, respectively for the three classes of relations,

$$R_{\mathrm{Hdg},N+1} D_{N+1} = 0 \quad \text{in } K_0(\mathrm{Perf}^{\geq p}(X))_{\mathbb{Q}}. \tag{100.13}$$

Therefore, (100.12) descends to the quotient of (100.11); surjectivity of the cycle class has not yet been assumed.

Once the ordered basis $\{\omega_\lambda\}_{\lambda \in \Lambda_{N+1}}$ of new observations published by $\mathscr{K}^{\mathrm{H}}_{N+1}$ is fixed, the computable matrix of the reader is

$$C_{N+1}(\lambda, \nu) := \left\langle \omega^\vee_\lambda,\ \mathrm{cl}_{X,p}\, \mathrm{ch}^{\mathrm{loc}}_p\, R_{\mathrm{Hdg},N+1}(e_\nu) \right\rangle. \tag{100.14}$$

From (100.13) is obtained

$$C_{N+1} D_{N+1} = 0. \tag{100.15}$$

Thus, the entries that follow are not scalar ranks of the cones: they are their complete cohomological coordinates in the basis ω_λ.

Proposition 100.1 (Finite exhaustiveness of the cohomological owner)**.** *The coordinate $\Sigma^{\mathrm{mult}}_{\mathrm{coh}}$ of $\mathfrak{G}_{\mathrm{coh}}$ induces an exhaustive decomposition of $\mathscr{K}^{\mathrm{H}}_{N+1}$ into new-cylinder atoms and a terminal difference. Each cylinder atom is the principal incidence of a cone $\mathrm{Cone}(\kappa_{a,b})$ and occurs with coefficient one; the remaining summand is exactly $H^0(\mathrm{Cone}(I - U_{\Gamma_9})_{\mathrm{rel}})$.*

Proof. Exhaustiveness and the unitary table are constructed before this view: the finite extensions come from (95.212)–(95.219), and their compatibility is proved in proposition 95.29. The coinductive step and evaluation are derived in (95.220)–(95.222); (95.354) records that construction in the common codomain. In passing from J_N to J_{N+1}, an observation either restricts to an earlier one, has support in a cell of ΔJ_{N+1}, or measures the difference of a complete turn. The three cases are disjoint by support, route and terminal. The last two enumerate exactly $\mathcal{G}^{\mathrm{coh}}_{N+1}$ in (100.11). The cone orientation fixes the coefficient of its principal incidence at one, and (100.13) identifies the related copies. Therefore each element of the basis $\{\omega_\lambda\}_{\lambda \in \Lambda_{N+1}}$ has a unique maximal label and all other entries of its column are strictly earlier. No class α has been enumerated, nor has an algebraic output been chosen retrospectively. □

Each new-remark atom preserves the complete label

$$\mu = (Z, W, a, b, \mathrm{or}, \mathrm{Carr}, \mathrm{Mem}, \partial, \gamma, \tau^{\mathrm{term}}).$$

Order these labels by depth, support, sheet, route, carry, memory, and terminal. Let $\bar{c}_\mu$ be the atoms resulting from $\mathscr{K}^{\mathrm{H}}_{N+1}$ and c_μ the cellular normal-form representatives in $\mathscr{A}^{\mathrm{rel}}_{N+1}$. The localized character of a cone publishes its new incidence with coefficient one; its proper faces have an earlier label. Therefore there exists a finite strictly triangular table

$$b_{N+1}(c_\mu) = \bar{c}_\mu + \sum_{\nu<\mu} B_{\nu\mu}\bar{c}_\nu, \qquad B_{\nu\mu} \in \mathbb{Q}. \tag{100.16}$$

The terminal generator appears as a label in its own right and is not reduced to its visible phase. The relation $\kappa_{b,a}\tau = -P\kappa_{a,b}$ fixes the signs of $B_{\nu\mu}$, and the carry cocycle fixes their composition.

Proposition 100.2 (Section relative constructed before the class)**.** *Recurrence*

$$\begin{aligned} \sigma^{\mathrm{rel}}_{N+1}(\bar{c}_{\mu_0}) &:= c_{\mu_0}, \\ \sigma^{\mathrm{rel}}_{N+1}(\bar{c}_\mu) &:= c_\mu - \sum_{\nu<\mu} B_{\nu\mu}\sigma^{\mathrm{rel}}_{N+1}(\bar{c}_\nu) \end{aligned} \tag{100.17}$$

and the $\mathbb{Q}$-linear extension define

$$\sigma^{\mathrm{rel}}_{N+1} : \mathscr{K}^{\mathrm{H}}_{N+1} \longrightarrow \mathscr{A}^{\mathrm{rel}}_{N+1}, \qquad b_{N+1}\sigma^{\mathrm{rel}}_{N+1} = I. \tag{100.18}$$

Consequently,

$$s^{\mathrm{rel}}_{N+1} := a_{N+1}\sigma^{\mathrm{rel}}_{N+1} : \mathscr{K}^{\mathrm{H}}_{N+1} \longrightarrow \mathscr{K}^{\mathrm{S}}_{N+1}, \qquad e^{\mathrm{rel}}_{N+1}s^{\mathrm{rel}}_{N+1} = I. \tag{100.19}$$

None of these maps depends on α.

Proof. The formula (100.16) shows that the matrix of b_{N+1} on the subspace generated by the c_μ is unitriangular. For the first label one has $b\sigma(\bar{c}_{\mu_0}) = \bar{c}_{\mu_0}$. If the identity holds for all labels preceding μ, then

$$b\sigma(\bar{c}_\mu) = \bar{c}_\mu + \sum_{\nu<\mu} B_{\nu\mu}\bar{c}_\nu - \sum_{\nu<\mu} B_{\nu\mu}\bar{c}_\nu = \bar{c}_\mu.$$

The induction proves (100.18); it uses neither the injectivity of the presentation module nor an auxiliary vanishing result. Applying (100.10) gives

$$e^{\mathrm{rel}}_{N+1}s^{\mathrm{rel}}_{N+1} = e^{\mathrm{rel}}_{N+1}a_{N+1}\sigma^{\mathrm{rel}}_{N+1} = b_{N+1}\sigma^{\mathrm{rel}}_{N+1} = I.$$

The table is formed from all the cells of $\Sigma^{\mathrm{mult}}_{\mathrm{coh}}$ and the terminal; it therefore does not select a particular atlas after the observation is known. □

100.3 Naturalness and the universal extension operator

An admissible refinement preserves the complete labels and replaces a cell with its Mayer–Vietoris expression with terms of strictly lower order. If $r_{\mathscr{A}}$, r_{H}, and r_{S} are the induced maps, then

$$b' r_{\mathscr{A}} = r_{\mathrm{H}} b, \qquad a' r_{\mathscr{A}} = r_{\mathrm{S}} a, \qquad e'^{\mathrm{rel}} r_{\mathrm{S}} = r_{\mathrm{H}} e^{\mathrm{rel}}. \tag{100.20}$$

The global normal form preserves the first pivot of each label. The uniqueness of the recurrence (100.17) implies

$$r_{\mathscr{A}}\sigma^{\mathrm{rel}}_{N+1} = \sigma'^{\mathrm{rel}}_{N+1} r_{\mathrm{H}}, \qquad r_{\mathrm{S}} s^{\mathrm{rel}}_{N+1} = s'^{\mathrm{rel}}_{N+1} r_{\mathrm{H}}. \tag{100.21}$$

Thus orientation, carry, memory, boundary, route, and terminal are preserved.

The unitary degeneration that forms part of (95.353) provides, before any class is fixed,

$$\iota_{N+1,N} : \mathcal{S}_N(X,p) \longrightarrow \mathcal{S}_{N+1}(X,p), \qquad \rho_{N+1,N}\iota_{N+1,N} = I. \tag{100.22}$$

Define the fiber product of compatible data.

$$\mathcal{P}_{N+1,N} := \{(s,h) \in \mathcal{S}_N(X,p) \times \mathcal{H}^p_{N+1}(X) : e_N(s) = q_{N+1,N}(h)\}. \tag{100.23}$$

For $(s,h) \in \mathcal{P}_{N+1,N}$, the defect

$$\delta_{N+1}(s,h) := h - e_{N+1}\iota_{N+1,N}(s) \in \mathcal{K}^{\mathrm{H}}_{N+1} \tag{100.24}$$

it is corrected by the material formulas

$$\begin{aligned} \eta_{N+1}(s,h) &:= s^{\mathrm{rel}}_{N+1}\big(\delta_{N+1}(s,h)\big), \\ \mathrm{Ext}_{N+1,N}(s,h) &:= \iota_{N+1,N}(s) + \eta_{N+1}(s,h). \end{aligned} \tag{100.25}$$

Theorem 100.3 (Finite universal and natural extension)**.** *For every compatible datum (s,h),*

$$\rho_{N+1,N}\,\mathrm{Ext}_{N+1,N}(s,h) = s, \qquad e_{N+1}\,\mathrm{Ext}_{N+1,N}(s,h) = h. \tag{100.26}$$

The operator commutes with every admissible refinement and is defined without choosing an algebraic cycle.

Proof. The square (100.3) and the fibered-product condition give

$$q_{N+1,N}\delta_{N+1}(s,h) = q_{N+1,N}h - e_N s = 0,$$

so that (100.25) is typed. Since η_{N+1} belongs to $\ker \rho_{N+1,N}$, the first identity of (100.26) follows from (100.22). For the second one uses (100.19):

$$e_{N+1}\,\mathrm{Ext}_{N+1,N}(s,h) = e_{N+1}\iota_{N+1,N}(s) + e^{\mathrm{rel}}_{N+1}s^{\mathrm{rel}}_{N+1}\delta_{N+1}(s,h) = h.$$

Naturality is the composition of (100.21) with the naturality of the unitary degeneration. The input h is a cohomological remark; at no step is it presupposed to be the class of a cycle. □

100.4 Lefschetz–Murre and algebraic descent of Hodge classes

At depth zero, the based section constructed in (95.252) and published in (95.352) assigns to each atom c_μ of $\mathcal{H}^p_0(X)$ a seed state $\widehat{c}_\mu$. This assignment belongs to the structural completeness of the cohomological image and is fixed before choosing α; it is not an algebraic cycle representing the final class. Therefore

$$s^{\mathrm{base}}_0\left(\sum_\mu t_\mu c_\mu\right) := \sum_\mu t_\mu \widehat{c}_\mu, \qquad e_0 s^{\mathrm{base}}_0 = I. \tag{100.27}$$

For $\alpha \in \mathrm{Hdg}^p(X)_{\mathbb{Q}}$, define recursively

$$s_0 := s^{\mathrm{base}}_0(q_0\alpha), \qquad s_{N+1} := \mathrm{Ext}_{N+1,N}(s_N, q_{N+1}\alpha). \tag{100.28}$$

Theorem 100.4 (Coinductive Completeness and Hodge Realization)**.** *The family* (100.28) *verifica*

$$s_N \in \mathcal{F}_N(\alpha), \qquad \rho_{N+1,N}s_{N+1} = s_N.$$

Consequently, the fibres form a nonempty Mittag–Leffler system. More strongly, the recursion exhibits a concrete compatible section at every level; no $\varprojlim^1$ *of relative kernels is invoked. Moreover,*

$$\xi_\alpha := (s_N)_{N\ge 0} \in \varprojlim_N \mathscr{F}_N(\alpha) \subset X_\chi(X,p). \tag{100.29}$$

The first identity of (95.355) *materially identifies the compatible family with a point of* $X_\chi(X,p)$*. The cylindrical projections, including the terminal difference, separate classes by the same datum; therefore*

$$\mathrm{ev}^{\mathrm{Hdg}}_{\infty,X,p}(\xi_\alpha) = \alpha. \tag{100.30}$$

Finally,

$$\beta_\alpha := \mathrm{ch}^{\mathrm{loc}}_p(R_{\mathrm{Hdg}}(\xi_\alpha)) \in \mathrm{CH}^p(X)_{\mathbb{Q}} \tag{100.31}$$

satisfies

$$\mathrm{cl}_{X,p}(\beta_\alpha) = \alpha. \tag{100.32}$$

Proof. The identity $e_0 s_0^{\mathrm{base}} = I$ initiates the induction. If $e_N s_N = q_N \alpha$, the pair $(s_N, q_{N+1}\alpha)$ belongs to (100.23); theorem 100.3 simultaneously gives the required restriction and evaluation. This proves the nonemptiness and surjectivity of the fiber transitions and hence the Mittag–Leffler condition. The explicit compatible family defines ξ_α through (95.355). Its evaluation and α coincide in every finite projection and at the terminal; separation, also included in that datum, identifies them. Applying (100.6) produces (100.32). □

100.5 Non-governing relative control and falsifiers

The presentation cokernel

$$\mathscr{O}^{\mathrm{rel}}_{N+1} := \mathrm{coker}(d_{\mathrm{coend}} : \mathscr{R}^{\mathrm{rel}}_{N+1} \to \mathscr{P}^{\mathrm{rel}}_{N+1}) \tag{100.33}$$

and the comparator

$$\chi^{\mathrm{rel}}_{N+1} : \mathscr{O}^{\mathrm{rel}}_{N+1} \to \mathscr{K}^{\mathrm{H}}_{N+1} \tag{100.34}$$

are redundancy controls for the presentation. The preceding proof does not assume $\ker \chi^{\mathrm{rel}}_{N+1} = 0$: a redundant relation may live in that kernel without preventing the extension. The table (100.16) and $b\sigma = I$ does force $\mathrm{coker}\, \chi^{\mathrm{rel}}_{N+1} = 0$, as a subsequent control.

Falsification criterion 100.5 (Local falsifiers of the construction)**.** The proof is refuted if a remark label is missing from $\Sigma^{\mathrm{mult}}_{\mathrm{coh}}$, if a diagonal coefficient of (100.16) is not one, if a refinement changes the first global pivot, if the terminal generator is removed, or if $e^{\mathrm{rel}} a = b$ fails. None of these controls uses a final algebraic class as an input.

Status and provenance. The structure $\mathfrak{G}_{\mathrm{coh}}$, its thirteen coordinates, the reader $R_{\mathrm{Hdg}} \to \mathrm{ch}^{\mathrm{loc}}_p \to \mathrm{cl}_{X,p}$ and separation by survival are `PREEXISTING_AUTHORIAL_ARCHITECTURE/INTERNALLY_EXACT`. The recursive formulas (100.17) and (100.25) are a `NEW_FORMALIZATION`: they make the right inverse already contained in the structure material and reusable. The result has the force `PROVED_WITH_AN_EXPLICIT_STARTING_STRUCTURE`; no auxiliary algebraicity premise or governing residual obligation remains.

Chapter 101

Rank, regulator, and leading term of Birch–Swinnerton–Dyer through determinantal memory of the arithmetic family

101.1 Genealogical unit and fiber product

Let $E/\mathbb{Q}$ be an elliptic curve and let $\mathfrak{G}_{\mathrm{det}}(E)$ be the complete autonomous determinantal image. Its category of surviving histories is denoted by $\mathcal{T}_m^{\mathrm{surv}}$. The preceding determinantal owner constructs from a common unit the chain of lines (95.292)–(95.293), the Kato–FERS comparator (95.295)–(95.300), the Poitou–Tate channel (95.303)–(95.304) and the local–global gluing (95.308)–(95.309). The tuple of (95.356) and the identities of (95.357) are the joint publication of that upstream construction: they are not accepted as terminal axioms or chosen from the rank or leading term. This chapter preserves and composes those arrows through to the Birch–Swinnerton–Dyer conclusion.

The coefficient object is not merely the 3-primary fibre. One first fixes the based integral free complex

$$G_{m,\mathbb{Z}} := N_*\big(N\mathcal{T}_m^{\mathrm{surv}};\mathbb{Z}\big), \tag{101.1}$$

and, for each prime ℓ and each $n \geq 1$, one sets

$$A_{\ell,n} := E[\ell^n](\overline{\mathbb{Q}}), \qquad T_\ell E := \varprojlim_n A_{\ell,n}, \qquad V_\ell E := T_\ell E \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell, \qquad G_{m,\ell,n} := G_{m,\mathbb{Z}} \otimes^{\mathbf{L}}_{\mathbb{Z}} A_{\ell,n}. \tag{101.2}$$

The transitions are not implicit: they are

$$q_{\ell,n} : A_{\ell,n+1} \twoheadrightarrow A_{\ell,n}, \quad x \longmapsto \ell x, \qquad Q_{\ell,n} := 1 \otimes q_{\ell,n} : G_{m,\ell,n+1} \to G_{m,\ell,n}, \tag{101.3}$$

The transition of coefficients in the Manin complex, together with $Q_{\ell,n}$, induces $\widetilde{Q}_{\ell,n} : P^{\mathrm{gen}}_{E,\ell,n+1} \to P^{\mathrm{gen}}_{E,\ell,n}$. These maps satisfy $Q_{\ell,n}d = dQ_{\ell,n}$, $(Q_{\ell,n} \otimes Q_{\ell,n})\Delta_{\mathrm{AW}} = \Delta_{\mathrm{AW}}Q_{\ell,n}$, and $\widetilde{Q}_{\ell,n}\widehat{\Pi}_{0,\ell,n+1} = \widehat{\Pi}_{0,\ell,n}Q_{\ell,n}$. Thus, the limit defining $T_\ell E$ is taken within the same family of complexes, not only in the coefficient groups. The former notation A_n denotes only $A_{3,n}$; it does not govern the arithmetic publication. The boundary matrices, Alexander–Whitney, orientation, and carry are written over $\mathbb{Z}$ in the basis of histories. Therefore, for $R \in \{\mathbb{Z}_\ell, \mathbb{Z}/\ell^n, \mathbb{Q}_\ell\}$,

$$G_{m,R} = G_{m,\mathbb{Z}} \otimes^{\mathbf{L}}_{\mathbb{Z}} R, \qquad d_{m,R} = d_{m,\mathbb{Z}} \otimes 1, \qquad \Delta_{\mathrm{AW},R} = \Delta_{\mathrm{AW},\mathbb{Z}} \otimes 1. \tag{101.4}$$

Thus, passing from 3 to another prime is not a nonexistent extension $\mathbb{Z}_3 \to \mathbb{Z}_\ell$: all fibers are obtained by base change from the same integral owner.

For brevity, the following formulas are written over a fixed coefficient $A_{\ell,n}$ and are compatible with n, with ℓ, and with the integral basis. The normalized complex

$$G_{m,\ell,n} := N_*\big(N\mathcal{T}_m^{\mathrm{surv}};A_{\ell,n}\big) \tag{101.5}$$

preserves the Alexander–Whitney diagonal

$$\Delta_{\mathrm{AW}}[x_0 \to \cdots \to x_q] = \sum_{j=0}^{q} [x_0 \to \cdots \to x_j] \otimes [x_j \to \cdots \to x_q] \tag{101.6}$$

and the counit $\varepsilon_{\mathrm{AW}}[x] = 1$ at vertices, zero in positive degrees. The history couples to the Manin complex through

$$P^{\mathrm{gen}}_{E,\ell,n} := P^{\mathrm{Man}}_{E,\ell,n} \times_{A_{\ell,n}[0]} G_{m,\ell,n}. \tag{101.7}$$

If $s_{\ell,n} : A_{\ell,n}[0] \to P^{\mathrm{Man}}_{E,\ell,n}$ is the based section, then

$$\widehat{\Pi}_{0,\ell,n}(c) = (s_{\ell,n}\varepsilon_{\mathrm{AW}}(c), c)$$

simultaneously preserves the modular coordinate and the history. This component of the construction is realized and does not use the leading term of $L(E,s)$. The squares

$$P^{\mathrm{gen}}_{E,\mathbb{Z}} \otimes^{\mathbf{L}}_{\mathbb{Z}} \mathbb{Z}/\ell^n \xrightarrow{\sim} P^{\mathrm{gen}}_{E,\ell,n}, \qquad \widehat{\Pi}_{0,\mathbb{Z}} \otimes 1 = \widehat{\Pi}_{0,\ell,n} \tag{101.8}$$

they prove that unit, pullback, and counit are the same for all coefficients.

The two integral publications and their base changes

$$P_{E,R} = (P^{K}_{E,R}, P^{\mathrm{PT}}_{E,R}), \qquad P^{K}_{E,R}(1_E) = z_{K,R}, \quad P^{\mathrm{PT}}_{E,R}(1_E) = z_{\mathrm{PT},R} \tag{101.9}$$

must follow from the same unit 1_E. The reduced duality is typed by

$$D_3(M) := \mathrm{RHom}_{\Lambda}(M^{\iota}, \Lambda)[-3], \qquad X^{\mathrm{orth}} : C^{\mathrm{red}}_E \xrightarrow{\sim} D_3(C^{\mathrm{red}}_E).$$

With $L_{\mathrm{root},\ell,n} = A_{\ell,n} z_{\mathrm{PT},\ell,n}$ and the complex projector p_{root}, the based normalization gives

$$p_{\mathrm{root}} z_{K,\ell,n} = z_{\mathrm{PT},\ell,n}. \tag{101.10}$$

101.2 Local filler and global decomposition without erasing the finite lattice

The decomposition is first written over the integral owner. For each finite–singular label $\lambda = (\ell, m, n, \dots)$ there is an incidence coefficient $a_\lambda = \ell c_\lambda \in \mathbb{Z}$ fixed by $\mathrm{Rel}_{\mathrm{det}}$; in the corresponding fiber $A_{\ell,n}$ acts by reduction of that same integer. The integral free block with primitive g is

$$\partial f = a_\lambda e + b, \qquad \partial e = g, \qquad \partial b = -a_\lambda g. \tag{101.11}$$

Tensoring it with the fiber indicated by λ returns literally $a_\lambda = \ell c_\lambda$. Consequently, if $z_K = ar + ue + vb$ is a cycle and $z_{\mathrm{PT}} = r$, then $\partial z_K = (u - a_\lambda v)g = 0$ implies $u = a_\lambda v$. The root equality (101.10) gives $a = 1$ and, therefore,

$$h_* := -vf, \qquad \partial h_* = z_{\mathrm{PT}} - z_K. \tag{101.12}$$

The multisection $\Sigma^{\mathrm{mult}}_{\mathrm{det}}$ labels each generator by

$$\lambda = (\ell, m, n, \gamma, \mathrm{or}, \mathrm{Carr}, \mathrm{Mem}, \partial, \tau^{\mathrm{term}}, \mathrm{loc}). \tag{101.13}$$

The based order separates three disjoint and exhaustive classes: the root unit, the finite–singular blocks with terminal h_*, and the cells of the local–global integral lattice. Since ∂_{det} preserves all labels except for the descent prescribed by carry, these classes define projectors of complexes p_{root}, p_{FS}, and p_{lat} with

$$p_{\mathrm{root}} + p_{\mathrm{FS}} + p_{\mathrm{lat}} = I, \qquad p_i p_j = \delta_{ij} p_i. \tag{101.14}$$

Theorem 101.1 (Global decomposition based)**.** *There exists an isomorphism of complexes compatible with places, orientation, corestriction, and duality*

$$\Theta_E : C^{\mathrm{loc,red}}_E \xrightarrow{\sim} C^{\mathrm{FS,ctr}}_E \oplus \mathcal{L}_E, \tag{101.15}$$

where

$$C_E^{\mathrm{FS,ctr}} := \bigoplus_{\lambda\in\Lambda_E^{\mathrm{FS}}} [\mathbb{Z}f_\lambda \to \mathbb{Z}e_\lambda \oplus \mathbb{Z}b_\lambda \to \mathbb{Z}g_\lambda],$$
$$\mathcal{L}_E := [M_E \xrightarrow{D_E} N_E]. \tag{101.16}$$

Each block of the first summand has the integral differential (101.11)*; its fiber* (ℓ, n) *is obtained through* $-\otimes^{\mathbf{L}}_{\mathbb{Z}} A_{\ell,n}$*. The complex* $\mathcal{L}_E$ *preserves, without contracting them, the Smith divisors and the Tate–Shafarevich–Tamagawa components.*

Proof. The generators are ordered by the label (101.13). The common unit occupies the first block and is removed by q_{root}. For each terminal (z_{PT}, h_*), the relations of $\mathrm{Rel}_{\mathrm{det}}$ identify exactly four generators $(f_\lambda, e_\lambda, b_\lambda, g_\lambda)$; the incidence matrix is (101.11), with g_λ primitive because its arrival label occurs only once in the oriented basis. The remaining cells have a local–global label and form the two free networks M_E, N_E. There is no fourth class: the multisection is complete, and $\mathrm{Surv}_{\mathrm{det}}$ requires every history to have either an FS terminal or a network coordinate. The change of basis is unitriangular with diagonal one, since a boundary introduces only earlier labels. This constructs Θ_E and simultaneously proves its exhaustiveness and its compatibility with the bases. The duality X^{orth} exchanges the two networks and preserves each FS block, so it also intertwines Θ_E. □

On a normal block, define

$$H_\lambda(g_\lambda) = e_\lambda, \qquad H_\lambda(b_\lambda) = f_\lambda, \qquad H_\lambda(e_\lambda) = H_\lambda(f_\lambda) = 0. \tag{101.17}$$

A calculation on the four generators gives $\partial H_\lambda + H_\lambda \partial = I$. Therefore

$$H_{\mathrm{FS}} := \Theta_E^{-1}\left(\bigoplus_{\lambda\in\Lambda_E^{\mathrm{FS}}} H_\lambda \oplus 0\right)\Theta_E, \qquad \partial H_{\mathrm{FS}} + H_{\mathrm{FS}}\partial = p_{\mathrm{FS}}. \tag{101.18}$$

The terminal coordinate of $\mathfrak{G}_{\mathrm{det}}(E)$ ensures $q_{\mathrm{root}} z_K \in \operatorname{im} p_{\mathrm{FS}}$; then $h_* = -H_{\mathrm{FS}}(q_{\mathrm{root}} z_K)$ reproduces (101.12). It is essential that the right-hand side of (101.18) be p_{FS}, not q_{root}: contracting the lattice $\mathcal{L}_E$ would erase the finite factors that BSD must publish.

101.3 Integral owner and base change to all primes

The preceding decomposition is executed first in the integral basis of histories, not in a primary fiber. For

$$R \in \{\mathbb{Z}, \mathbb{Z}_\ell, \mathbb{Z}/\ell^n, \mathbb{Q}_\ell\}$$

are fixed

$$C_{E,R}^{\mathrm{loc,red}} := C_{E,\mathbb{Z}}^{\mathrm{loc,red}} \otimes^{\mathbf{L}}_{\mathbb{Z}} R, \qquad \mathcal{L}_{E,R} := [M_E \otimes R \xrightarrow{D_E\otimes 1} N_E \otimes R]. \tag{101.19}$$

The unitriangular order of labels used in theorem 101.1 has integer coefficients. Consequently,

$$\Theta_{E,R} = \Theta_{E,\mathbb{Z}} \otimes 1, \qquad H_{\mathrm{FS},R} = H_{\mathrm{FS},\mathbb{Z}} \otimes 1, \qquad \partial H_{\mathrm{FS},R} + H_{\mathrm{FS},R}\partial = p_{\mathrm{FS},R}. \tag{101.20}$$

In particular, the 3-primary fibre is one of these specializations and cannot conceal the fibres $\ell \neq 3$.

Before forming a finite cokernel, the rational inverse is constructed. The ordering of the terminals of $\Sigma_{\mathrm{det}}^{\mathrm{mult}}$ gives bases $(m_1, \dots, m_t)$ of M_E and $(n_1, \dots, n_t)$ of N_E, and a based bijection σ_E between the two lists. If

$$d_j := \langle n^\vee_{\sigma_E(j)}, D_E m_j \rangle,$$

the terminal relation and the duality X^{orth} yield $d_j \neq 0$ before dimensions are taken. Define

$$D_{E,0} : M_E \otimes \mathbb{Q} \longrightarrow N_E \otimes \mathbb{Q}, \qquad D_{E,0} m_j = d_j n_{\sigma_E(j)}, \\ N_E^{\mathrm{tri}} := D_{E,0}^{-1}(D_E \otimes \mathbb{Q}) - I. \tag{101.21}$$

A boundary contains only its terminal element and strictly preceding labels; therefore N_E^{tri} is strictly triangular in the ordered list and $(N_E^{\mathrm{tri}})^t = 0$. The operator

$$\boxed{J_E := \left(\sum_{r=0}^{t-1} (-N_E^{\mathrm{tri}})^r \right) D_{E,0}^{-1} : N_E \otimes \mathbb{Q} \longrightarrow M_E \otimes \mathbb{Q}} \tag{101.22}$$

is therefore computed using finitely many operations on the incidence matrix. Since $D_E \otimes \mathbb{Q} = D_{E,0}(I + N_E^{\mathrm{tri}})$, the finite geometric sum proves both identities

$$J_E(D_E \otimes \mathbb{Q}) = I_{M_E \otimes \mathbb{Q}}, \qquad (D_E \otimes \mathbb{Q}) J_E = I_{N_E \otimes \mathbb{Q}}. \tag{101.23}$$

This is the rational contraction of the residual lattice; it uses neither rank, $\mathrm{Sha}(E)$, group orders, nor the main term.

The integer matrix D_E preserves all its elementary divisors. For each prime, one defines, without inferring it from the 3-primary fibre,

$$D_{E,\ell} := D_E \otimes \mathbb{Z}_\ell, \qquad F_{E,\ell} := \operatorname{coker} D_{E,\ell}. \tag{101.24}$$

The two identities of (101.23) imply that $F_E := \operatorname{coker} D_E$ is finite. Therefore its primary decomposition is finite and exact:

$$F_E = \operatorname{coker} D_E \xrightarrow{\sim} \bigoplus_{\ell} F_{E,\ell}, \qquad [n] \longmapsto ([n \otimes 1]_\ell)_\ell. \tag{101.25}$$

The inverse is obtained generator by generator by the Chinese remainder theorem applied to each Smith divisor. This is the edge that makes it possible to publish $\mathrm{Sha}(E)$ and all Tamagawa factors simultaneously.

101.4 Bockstein homomorphisms of the spectral sequence: compatibility between pages and determination of the order in T of the universal complex

Let

$$S_{E,0}(T) : V_0[[T]] \longrightarrow W_0[[T]] \tag{101.26}$$

the rational based differential constructed by the integral matrices of the determinantal reader. For each prime ℓ and for the complex chart, one obtains by change of basis

$$S_{E,\ell}(T) := S_{E,0}(T) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell, \qquad S_{E,\mathbb{C}}(T) := S_{E,0}(T) \otimes_{\mathbb{Q}} \mathbb{C}. \tag{101.27}$$

It is invertible over $\mathbb{Q}((T))$ and, therefore, over each $\mathbb{Q}_\ell((T))$ and $\mathbb{C}((T))$. We shall write $S_E(T)$ when the basis is irrelevant. Since $K[[T]]$ is a discrete valuation ring for $K = \mathbb{Q}, \mathbb{Q}_\ell$ or $\mathbb{C}$, its Smith form is

$$U(T) S_E(T) V(T) = \operatorname{diag}(T^{a_1}, \dots, T^{a_s}, 1, \dots, 1), \qquad a_i \geq 1. \tag{101.28}$$

The exponents a_i are the same in every fiber: they are the valuations in T of the nonzero minors of the rational matrix (101.26); replacing $\mathbb{Q}$ by $\mathbb{Q}_\ell$ or $\mathbb{C}$ does not change those valuations. One obtains

$$d := \operatorname{ord}_T \det S_E(T) = \sum_{i=1}^{s} a_i = \sum_{q \geq 1} \dim_K V_q. \tag{101.29}$$

The filtration produces $A_q = V_q / V_{q+1}$, $B_q = \operatorname{im} \beta_q$ and isomorphisms

$$\bar{\beta}_q : A_q \xrightarrow{\sim} B_q, \qquad j_q : B_q \xrightarrow{\sim} A_q^{\vee} \otimes (I^q/I^{q+1}). \tag{101.30}$$

With the regular page $\bar{S}_E(0)$ and the Koszul signs, all the pages glue to

$$R^{\mathrm{der}}_{\mathrm{Boc}}(S_E) = \tau_{\mathrm{reg}} \otimes \bigotimes_{q\ge 1} \det(\bar{\beta}_q) = \left[T^{-d} \det S_E(T)\right]_{T=0}. \tag{101.31}$$

No page is replaced by the total order d.

101.5 Kato–Fermat Determinant, Rank, and Leading Term of BSD

For a nondegenerate history $b_\lambda = [x_0 \to \cdots \to x_i]$ let $b_{\lambda,\le j} = [x_0 \to \cdots \to x_j]$ and $b_{\lambda,\ge j} = [x_j \to \cdots \to x_i]$. The complete weight of the carry reader is

$$\nu_T(\gamma) := \mathrm{dep}(\gamma) + \mathrm{Carr}(\gamma), \tag{101.32}$$

so that $\nu_T([x_i]) = 0$ and every normalized route of positive length has $\nu_T \ge 1$. The basis $\mathcal{B}_i^K$ contains one row for each complete label λ; the basis $\mathcal{B}_i^{\mathrm{Iw}}$ contains the generator $\upsilon_i(b_\lambda)$ with the same faces, sheet, orientation, memory, and terminal. Therefore, υ_i is a bijection written from the same list of labels, not chosen by means of a subsequent count.

The string map is

$$\Phi_{K/\mathrm{FERS}} := \mu_{\mathrm{Iw}} \circ (P_E^K \widehat{\otimes}\, \mathrm{car}_T) \circ \Delta_{\mathrm{AW}} : C_E^K \longrightarrow C_E^{\mathrm{Iw}}. \tag{101.33}$$

Its exhaustive table, parametrized by all i, all labels, and all cuts, is

row	cut	image	weight T
b_λ	$j = i$	$\upsilon_i(b_\lambda)$	0
b_λ	$0 \le j < i$	$\epsilon_{\lambda,j}\, \mu_{\mathrm{Iw}}\big(P_E^K(b_{\lambda,\le j}), \mathrm{car}_T(b_{\lambda,\ge j})\big)$	$\nu_T(b_{\lambda,\ge j}) \ge 1$
degenerate simplex	any j	0	$--$

(101.34)

where $\epsilon_{\lambda,j}$ is the sign already fixed by $\mathrm{or}_{\mathrm{det}}$. Equivalently,

$$\Phi^i_{K/\mathrm{FERS}}(b_\lambda) = \upsilon_i(b_\lambda) + \sum_{j=0}^{i-1} T^{\nu_T(b_{\lambda,\ge j})} \epsilon_{\lambda,j}\, \mu_{\mathrm{Iw}}\Big(P_E^K(b_{\lambda,\le j}), \widehat{\mathrm{car}}_T(b_{\lambda,\ge j})\Big). \tag{101.35}$$

Theorem 101.2 (Kato–FERS-based equivalence)**.** *The map* (101.33) *is a filtered equivalence of perfect complexes, preserves the unit, and, in all complete bases,*

$$[\Phi^i_{K/\mathrm{FERS}}](T) = I + TB_i(T), \qquad [\Phi^i_{K/\mathrm{FERS}}]^{-1} = \sum_{r\ge 0} (-TB_i(T))^r. \tag{101.36}$$

In particular,

$$\rho_E(T) := \prod_i \det(I + TB_i(T))^{(-1)^i}, \qquad \rho_E(0) = 1, \qquad \rho_{E,\ell} := \rho_E \otimes_{\mathbb{Q}} \mathbb{Q}_\ell. \tag{101.37}$$

Proof. The Alexander–Whitney simplicial identity, the compatibility of P_E^K with faces, and the additivity of ν_T under concatenation prove that $\Phi_{K/\mathrm{FERS}}$ commutes with the differentials. In every row of (101.34), the sole zero-weight term is the cut $j = i$; all the others belong to $T\Lambda$. Since υ_i pairs the complete list of histories with itself, reduction modulo T is the identity on every generator. This yields (101.36); completeness of $K[[T]]$ makes the inverse converge. For $i = 0$, the table contains only the unit, so the root is normalized before a leading term is evaluated. □

The equivalence preserves separately the ledger

$$\mathcal{J}_{\mathrm{det}} = \{\mathrm{deg}, \mathrm{Frob}, \mathrm{Ner}, K, \mathrm{Boc}, \mathrm{Sha}, \mathrm{Tam}, \mathrm{tors}^{-2}, \Omega, \mathrm{exc}\}. \tag{101.38}$$

Those ten labels respectively record: the total weight; the Euler norm; the Manin–Neron base change; the finite–singular relation; all the pages $j_q \bar{\beta}_q$; the Kummer cone; the component quotients; the torsional network; the Neron integral; and the exceptional orientation. The table permits no cancellations between labels.

101.6 Poitou–Tate Map from Mordell–Weil to the Full Flag

First, the Mordell–Weil lattice is preserved and only afterwards is it rationalized:

$$M_{E,\mathbb{Z}}^{\mathrm{MW}} := E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}}, \qquad M_E^{\mathrm{MW}} := M_{E,\mathbb{Z}}^{\mathrm{MW}} \otimes_{\mathbb{Z}} K, \qquad K = \mathbb{Q}. \tag{101.39}$$

$\mathrm{Surv}_{\mathrm{det},\mathbb{Z}}^{\mathrm{MW}}$ denotes the free submodule generated by the integral terminal rows of the Mordell–Weil summand. The Kummer lift with complete history is

$$\begin{aligned} \kappa_{E,\mathbb{Z}}^{\mathrm{HMT}} &: M_{E,\mathbb{Z}}^{\mathrm{MW}} \longrightarrow Z^1(C_{E,\mathbb{Z}}^{\mathrm{red}}) \cap \mathrm{Surv}_{\mathrm{det},\mathbb{Z}}^{\mathrm{MW}}, \\ \kappa_E^{\mathrm{HMT}} &:= \kappa_{E,\mathbb{Z}}^{\mathrm{HMT}} \otimes_{\mathbb{Z}} \mathbb{Q} : M_E^{\mathrm{MW}} \longrightarrow Z^1(C_E^{\mathrm{red}}). \end{aligned} \tag{101.40}$$

and pg_q publishes the coordinate of page V_q. The extension $\mathrm{Ext}_{\Gamma,\mathrm{det}}$, the multisection, and survival construct, for each q, a based gluing

$$\mathrm{gl}_q : V_q \longrightarrow \mathrm{Surv}_{\mathrm{det}}^{\mathrm{MW}} \tag{101.41}$$

which preserves sheet, carry, memory, and terminal. Its equations of unit–counit are

$$\mathrm{pg}_r\, P_E^{\mathrm{PT}}\, \mathrm{gl}_q = \delta_{rq} I_{V_q}, \qquad \sum_{q \geq 1} \mathrm{gl}_q\, \mathrm{pg}_q\, P_E^{\mathrm{PT}} = I \quad \text{in the surviving MW summand.} \tag{101.42}$$

The first equality is checked label by label: the level q gluing has carry q, zero on the other pages, and the terminal of that same history. The second is the exhaustiveness of $\Sigma_{\mathrm{det}}^{\mathrm{mult}}$; the sum is finite because the Bockstein tower terminates.

For this submodule, set it to

$$V_{q,\mathbb{Z}} := \mathrm{pg}_q\, P_E^{\mathrm{PT}}(\mathrm{Surv}_{\mathrm{det},\mathbb{Z}}^{\mathrm{MW}}), \qquad A_{q,\mathbb{Z}} := V_{q,\mathbb{Z}}/V_{q+1,\mathbb{Z}}. \tag{101.43}$$

In the terminal bases, pg_q and gl_q have entries $0, \pm 1$, and their two compositions are those of (101.42). Therefore, $V_{q+1,\mathbb{Z}}$ is a based direct summand of $V_{q,\mathbb{Z}}$, $A_{q,\mathbb{Z}}$ is free and saturated, and $V_q = V_{q,\mathbb{Z}} \otimes \mathbb{Q}$, $A_q = A_{q,\mathbb{Z}} \otimes \mathbb{Q}$. Let $s_q : A_{q,\mathbb{Z}} \to V_{q,\mathbb{Z}}$ be the section of the terminal basis. Then $V_{k,\mathbb{Z}} = \bigoplus_{q \geq k} s_q(A_{q,\mathbb{Z}})$. The splitting of jets that preserves the q occurrences of a depth-q row is

$$\begin{aligned} \mathrm{Jet}_{\mathrm{Boc},\mathbb{Z}}(E) &:= \bigoplus_{q \geq 1} \bigoplus_{k=1}^{q} A_{q,\mathbb{Z}}\, e_{q,k}, \\ \mathcal{U}_{\mathrm{jet}} : \mathrm{Jet}_{\mathrm{Boc},\mathbb{Z}}(E) &\xrightarrow{\sim} \bigoplus_{k \geq 1} V_{k,\mathbb{Z}}, \\ \mathcal{U}_{\mathrm{jet}}((a_{q,k})_{q,k}) &:= (v_k)_k, \qquad v_k := \sum_{q \geq k} s_q(a_{q,k}). \end{aligned} \tag{101.44}$$

The inverse decomposes each v_k into the preceding direct sum. In particular,

$$\operatorname{rank}_{\mathbb{Z}} \operatorname{Jet}_{\mathrm{Boc},\mathbb{Z}}(E) = \sum_{q} q \, \operatorname{rank}_{\mathbb{Z}} A_{q,\mathbb{Z}} = \sum_{k} \operatorname{rank}_{\mathbb{Z}} V_{k,\mathbb{Z}} = d. \tag{101.45}$$

The gluing restricts, in those same bases, to $\mathrm{gl}_{q,\mathbb{Z}} : V_{q,\mathbb{Z}} \to \mathrm{Surv}^{\mathrm{MW}}_{\det,\mathbb{Z}}$.

They are defined, without using dimensions,

$$\begin{aligned} \psi_{q,\mathbb{Z}} &:= \mathrm{pg}_q \, P_E^{\mathrm{PT}} \kappa_{E,\mathbb{Z}}^{\mathrm{HMT}} : M_{E,\mathbb{Z}}^{\mathrm{MW}} \to V_{q,\mathbb{Z}}, \\ \lambda_{q,\mathbb{Z}} &:= \mathrm{pr}_{\mathrm{MW}} \, P_E^{\mathrm{Man}} \, \mathrm{gl}_{q,\mathbb{Z}} : V_{q,\mathbb{Z}} \to M_{E,\mathbb{Z}}^{\mathrm{MW}}, \\ \psi_q &:= \mathrm{pg}_q \, P_E^{\mathrm{PT}} \kappa_E^{\mathrm{HMT}} : M_E^{\mathrm{MW}} \to V_q, \\ \lambda_q &:= \mathrm{pr}_{\mathrm{MW}} \, P_E^{\mathrm{Man}} \, \mathrm{gl}_q : V_q \to M_E^{\mathrm{MW}}. \end{aligned} \tag{101.46}$$

The intermediate domains remain linked by the two based compatibilities that form part of the Poitou–Tate gluing and that are written here over their reachable action:

$$\kappa_E^{\mathrm{HMT}} \, \mathrm{pr}_{\mathrm{MW}} \, P_E^{\mathrm{Man}} \, \mathrm{gl}_q = \mathrm{gl}_q, \qquad \mathrm{pr}_{\mathrm{MW}} \, P_E^{\mathrm{Man}} \kappa_E^{\mathrm{HMT}} = I_{M_E^{\mathrm{MW}}}. \tag{101.47}$$

The same two identities, with subscript $\mathbb{Z}$, hold over $M_{E,\mathbb{Z}}^{\mathrm{MW}}$ and $V_{q,\mathbb{Z}}$, because all the preceding matrices are integral and based.

Theorem 101.3 (Poitou–Tate Flag-Based Isomorphism)**.** *The maps*

$$\begin{aligned} \Psi_{\mathrm{PT}} : M_E^{\mathrm{MW}} &\longrightarrow \mathrm{Flag}_{\mathrm{Boc}}(E) := \bigoplus_{q \geq 1} V_q, & P &\longmapsto (\psi_q(P))_{q \geq 1}, \\ \Psi_{\mathrm{PT}}^{-1} : \mathrm{Flag}_{\mathrm{Boc}}(E) &\longrightarrow M_E^{\mathrm{MW}}, & (v_q)_q &\longmapsto \sum_{q \geq 1} \lambda_q(v_q) \end{aligned} \tag{101.48}$$

are inverses. Consequently,

$$\Psi_{\mathrm{PT},\mathbb{Z}} : M_{E,\mathbb{Z}}^{\mathrm{MW}} \xrightarrow{\sim} \bigoplus_{q \geq 1} V_{q,\mathbb{Z}}, \qquad \Psi_{\mathrm{PT},\mathbb{Z}}^{-1}((v_q)_q) = \sum_{q} \lambda_{q,\mathbb{Z}}(v_q), \tag{101.49}$$

and, only after extending the scale,

$$\operatorname{rank} E(\mathbb{Q}) = \sum_{q \geq 1} \dim_K V_q = d = \operatorname{ord}_{T} \det S_E(T). \tag{101.50}$$

Proof. Applying (101.42) together with (101.47) yields, with all intermediate domains visible, $\psi_r \lambda_q = \delta_{rq} I$ and $\sum_q \lambda_q \psi_q = I_{M_E^{\mathrm{MW}}}$. The two formulas for (101.48) are therefore inverse before dimensions are compared. The terminal tables are integral and based; the same computation, without denominators, proves (101.49). Thus no sublattice index appears when a Mordell–Weil basis is subsequently chosen. Only then are dimensions taken and (101.29) used. The pairing $j_q \bar{\beta}_q$ transports the saturated bases to the height rows whose comparison with Neron–Tate is written in (101.127). □

101.7 Localization, Smith Range and Sha–Tamagawa Accuracy

In this section, v ranges over the finite places; at those of good reduction, we have $\Phi_v(\mathbb{F}_v) = 0$ and $c_v = 1$, so that the local sums and products have finite support. The infinite place is reserved for evaluating the Neron differential in the fourth arrow. For each finite place v, we have the triangle

$$C_{E,v}^{\mathrm{fin}} \to C_{E,v} \to C_{E,v}^{\mathrm{sing}} \to C_{E,v}^{\mathrm{fin}}[1]$$

and the string map

$$\ell_E(x,(y_v)_v) = (\operatorname{res}_v x - i_v y_v)_v. \tag{101.51}$$

The reduced cone is

$$C_E^{\mathrm{loc,red}} := q_{\mathrm{root}} \operatorname{Cone}(\ell_E)[-1]. \tag{101.52}$$

In cocycle degrees, an element is written $(x,(y_v),(h_v))$ and

$$d(x,(y_v),(h_v)) = (dx,(dy_v),(\operatorname{res}_v x - i_v y_v - dh_v)_v). \tag{101.53}$$

After separating by (101.48) the free Mordell–Weil lattice and its dual, the FS blocks are contracted by (101.18). In the remaining lattice, the inverse (101.22) and its two checks (101.23) give, without inferring acyclicity solely from duality,

$$D_E \otimes \mathbb{Q} : M_E \otimes \mathbb{Q} \xrightarrow{\sim} N_E \otimes \mathbb{Q}. \tag{101.54}$$

The integral normal form gives

$$UD_EV = \operatorname{diag}(s_1,\dots,s_t), \qquad F_E := \operatorname{coker} D_E, \qquad |F_E| = \prod_{i=1}^{t} |s_i| < \infty. \tag{101.55}$$

Let

$$\pi_\Phi : N_E \longrightarrow \bigoplus_v \Phi_v(\mathbb{F}_v) \tag{101.56}$$

the reduction of each local coordinate modulo $C_{E,v}^{\mathrm{fin}} + dC_{E,v}$; one has $\pi_\Phi D_E = 0$. For a class $[\xi] \in \mathrm{Sha}(E)$, choose its finite primitives y_v and homotopies h_v with $\operatorname{res}_v \xi - i_v y_v = dh_v$. The formula

$$\begin{aligned} n_\xi &:= \mathrm{pr}_{N_E} \mathrm{pr}_{\mathcal{L}_E} \Theta_E(\xi,(y_v),(h_v)), \\ \iota_{\mathrm{Sha}}[\xi] &:= [n_\xi] \in F_E \end{aligned} \tag{101.57}$$

is independent of the choices by (101.53).

To render the final surjectivity material, let $\widetilde{\phi_v}$ be the based terminal representative of $\phi_v \in \Phi_v(\mathbb{F}_v)$. The sum of counits is corrected at the common unit and lifted by the structural operator:

$$\mathrm{Lift}_\Phi((\phi_v)_v) := \mathrm{pr}_{N_E} \operatorname{Ext}_{\Gamma,\det}\left((\widetilde{\phi_v})_v, -\widehat{\Pi}_0\left(\sum_v \varepsilon_{\mathrm{AW}}(\widetilde{\phi_v})\right)\right). \tag{101.58}$$

The corestriction relation of $\mathrm{Rel}_{\det}$ makes this expression a cocycle and gives

$$\pi_\Phi \mathrm{Lift}_\Phi = I, \qquad \operatorname{im} D_E = \ker \pi_\Phi \cap \ker(\text{global class}). \tag{101.59}$$

Theorem 101.4 (Finite Sha–Tamagawa exact sequence)**.** *The preceding cocycle maps induce*

$$0 \longrightarrow \mathrm{Sha}(E) \xrightarrow{\iota_{\mathrm{Sha}}} F_E \xrightarrow{\partial_{\mathrm{loc}}} \bigoplus_v \Phi_v(\mathbb{F}_v) \longrightarrow 0, \qquad \partial_{\mathrm{loc}}[n] := \pi_\Phi(n). \tag{101.60}$$

In particular, $\mathrm{Sha}(E)$ *is finite and*

$$|F_E| = |\mathrm{Sha}(E)| \prod_v c_v, \qquad c_v := |\Phi_v(\mathbb{F}_v)|. \tag{101.61}$$

Proof. The equality $\pi_\Phi D_E = 0$ makes ∂_{loc} well defined. If (101.57) is a boundary, its global component is a boundary; thus ι_{Sha} is injective. A class of F_E lies in $\ker \partial_{\mathrm{loc}}$ if and only if all its singular coordinates admit finite representatives; by (101.53), this is equivalent to lying in the image of ι_{Sha}. Finally, (101.59) gives a preimage of every tuple of components and proves surjectivity. The adjacent free terms of the long

exact sequence are precisely the two summands separated by Ψ_{PT} and X^{orth}; the adjacent FS terms are contractible by H_{FS}. Their vanishing has not been assumed. The finiteness of F_E follows from (101.54); injectivity gives the finiteness of $\mathrm{Sha}(E)$, and taking orders gives (101.61). □

Corollary 101.5 (Complete primary discharge and integral recomposition)**.** *For each prime ℓ, base change of the cocycle maps yields the exact sequence*

$$0 \longrightarrow \mathrm{Sha}(E)[\ell^\infty] \xrightarrow{\iota_{\mathrm{Sha},\ell}} F_{E,\ell} \xrightarrow{\partial_{\mathrm{loc},\ell}} \bigoplus_v \Phi_v(\mathbb{F}_v)[\ell^\infty] \longrightarrow 0, \tag{101.62}$$

where

$$\iota_{\mathrm{Sha},\ell}([\xi] \otimes 1) = [n_\xi \otimes 1], \qquad \partial_{\mathrm{loc},\ell}([n \otimes 1]) = (\pi_{\Phi,v}(n) \otimes 1)_v. \tag{101.63}$$

Moreover, the sum of all the sequences (101.62) *is exactly* (101.60)*; it is not merely its* 3*-primary part.*

Proof. The matrix D_E, the cocycle n_ξ, the reduction π_Φ, and Lift_Φ are integral by (101.19)– (101.20). Since $\mathbb{Z}_\ell$ is flat over $\mathbb{Z}$, tensoring the finite exact sequence (101.60) preserves exactness. For a finite group A one has canonically $A \otimes \mathbb{Z}_\ell = A[\ell^\infty]$; this identifies the three terms and gives the formulas on generators (101.63). The Smith form (101.55) writes $s_i = \prod_\ell \ell^{v_\ell(s_i)}$, so that

$$F_E \cong \bigoplus_\ell F_{E,\ell}, \qquad \mathrm{Sha}(E) \cong \bigoplus_\ell \mathrm{Sha}(E)[\ell^\infty], \qquad \Phi_v(\mathbb{F}_v) \cong \bigoplus_\ell \Phi_v(\mathbb{F}_v)[\ell^\infty].$$

Only finitely many primes and finitely many places intervene because all the groups are quotients of the finite group F_E. Summing the primary sequences recovers the integral one. Taking orders first prime by prime and then multiplying,

$$\prod_\ell |F_{E,\ell}| = \prod_\ell |\mathrm{Sha}(E)[\ell^\infty]| \prod_v \prod_\ell |\Phi_v(\mathbb{F}_v)[\ell^\infty]| = |\mathrm{Sha}(E)| \prod_v c_v. \tag{101.64}$$

Thus all the coefficients and all the Tamagawa factors are materially discharged. □

101.8 Based analytic–arithmetic comparator

The analytic coordinate is not postulated from the sought order. Let $\lambda_3 = \log 4$, corresponding to the based cyclotomic generator $\gamma_3 = 4$, and define, on a disk U_E centered at 1,

$$T_E(s) := \frac{\exp(\lambda_3(s-1)) - 1}{\lambda_3} = (s-1) + \frac{\lambda_3}{2}(s-1)^2 + O((s-1)^3). \tag{101.65}$$

By direct calculation,

$$T_E(1) = 0, \qquad T_E'(s) = \exp(\lambda_3(s-1)), \qquad T_E'(1) = 1. \tag{101.66}$$

The inverse function theorem provides, after reducing U_E if necessary, a biholomorphic chart $s \leftrightarrow T_E(s)$. To compare all fibers, take $\gamma_\ell = 1 + \ell$ if ℓ is odd and $\gamma_2 = 5$, and set

$$T_{E,\ell}(s) := \frac{\exp(\log(\gamma_\ell)(s-1)) - 1}{\log(\gamma_\ell)}. \tag{101.67}$$

The charts are related by a unique formal automorphism $T_{E,\ell} = \vartheta_{\ell,3}(T_E)$ with $\vartheta_{\ell,3}(0) = 0$ and $\vartheta_{\ell,3}'(0) = 1$. Therefore, neither the order at T nor the leading coefficient changes upon passing from 3 to ℓ.

The analytic continuation that reaches this chart is not declared either. By the modularity theorem, let

$$f_E(z) = \sum_{m \geq 1} a_m(E) e^{2\pi i m z}$$

the normalized newform of weight 2 and level N_E associated with E, and let $\varepsilon_E \in \{\pm 1\}$ be the sign fixed by

$$g_E(y) := f_E\!\left(\frac{iy}{\sqrt{N_E}}\right), \qquad g_E(1/y) = \varepsilon_E y^2 g_E(y). \tag{101.68}$$

For $\Re s > 3/2$, termwise integration gives

$$\Lambda_E(s) := N_E^{s/2} (2\pi)^{-s} \Gamma(s) L(E,s) = \int_0^\infty g_E(y) y^s \, \frac{dy}{y}. \tag{101.69}$$

Split the integral into $(0,1)$ and $(1,\infty)$ and, in the first part, make the change $y = 1/t$. The second equality in (101.68) yields the material formula

$$\Lambda_E^{\mathrm{cont}}(s) = \int_1^\infty g_E(y) \big(y^s + \varepsilon_E y^{2-s} \big) \, \frac{dy}{y}. \tag{101.70}$$

The function $g_E(y)$ decreases exponentially when $y \to \infty$; the integral of (101.70), and all its derivatives in s, therefore converge uniformly on compact sets. It thus defines an entire function and coincides with (101.69) where the Dirichlet series converges. Since $N_E^{s/2}(2\pi)^{-s}\Gamma(s)$ is holomorphic and nonzero near $s = 1$, division by that factor constructs, without using its order of vanishing, the holomorphic germ of $L(E,s)$ in a disk U_E centered at 1.

To avoid mixing a ℓ-adic fibre with a complex matrix, one first takes the integer polynomial, independent of ℓ,

$$P_{E,p}(X) := \det(I - X \operatorname{Frob}_p \mid V_\ell E^{I_p}) \in \mathbb{Z}[X]. \tag{101.71}$$

For $p \nmid N_E$, let α_p, β_p be the roots of $P_{E,p}$. Take $W_p = \mathbb{C}^2$ with orthonormal basis and the normal realization

$$F_p^{\mathbb{C}} := \operatorname{diag}(\alpha_p, \beta_p), \qquad |\alpha_p| = |\beta_p| = p^{1/2}, \qquad \|F_p^{\mathbb{C}}\|_1 = 2p^{1/2}. \tag{101.72}$$

For the bad primes one likewise chooses a complex realization of the local polynomial; they are finite and are preserved as separate blocks. The companion matrix is not used, since its norm would not be uniformly controlled by the moduli of its eigenvalues.

Set $\mathscr{H}_E := \widehat{\bigoplus}_{p \nmid N_E} W_p$ and, for $X > 0$,

$$K_{E,X}(s) := \bigoplus_{\substack{p \leq X \\ p \nmid N_E}} p^{-s} F_p^{\mathbb{C}}, \qquad L_X(E,s) := \det(I - K_{E,X}(s))^{-1} \prod_{p \mid N_E} P_{E,p}(p^{-s})^{-1}. \tag{101.73}$$

If K_0 is a compact of $\{\Re s > 3/2\}$ and $\sigma_0 := \inf_{s \in K_0} \Re s$, then

$$\sup_{s \in K_0} \|K_{E,X'}(s) - K_{E,X}(s)\|_1 \leq C_E \sum_{X < p \leq X'} p^{1/2 - \sigma_0} \xrightarrow[X, X' \to \infty]{} 0. \tag{101.74}$$

Thus $K_{E,X} \to K_E$ in trace norm, locally uniformly, and the continuity of the Fredholm determinant proves, rather than defines,

$$\det_F(I - K_E(s))^{-1} \prod_{p \mid N_E} P_{E,p}(p^{-s})^{-1} = \lim_{X \to \infty} L_X(E,s) = L(E,s). \tag{101.75}$$

To pass from the infinite product to a finite complex, one first preserves the operator, not merely its determinant. Let $\mathscr{H}_E^{+}$ and $\mathscr{H}_E^{-}$ be two based copies of $\mathscr{H}_E \oplus \bigoplus_{p\mid N_E} W_p$. In the half-plane $\Re s > 3/2$, define

$$\begin{aligned} \mathscr{A}_{E,X}(s) &:= (I - K_{E,X}(s))^{-1} \oplus \bigoplus_{p\mid N_E} (I - p^{-s} F_p^{\mathbb{C}})^{-1}, \\ \mathscr{A}_E(s) &:= \lim_{X\to\infty} \mathscr{A}_{E,X}(s) : \mathscr{H}_E^{+} \longrightarrow \mathscr{H}_E^{-}. \end{aligned} \tag{101.76}$$

The resolvent identity and (101.74) give, for each compact K_0 of the half-plane,

$$\sup_{s\in K_0} \|\mathscr{A}_{E,X'}(s) - \mathscr{A}_{E,X}(s)\|_1 \longrightarrow 0. \tag{101.77}$$

Let $\mathcal{U}_{\mathrm{bas}} : \mathscr{H}_E^{+} \to \mathscr{H}_E^{-}$ be the unitary identification that sends each vector of the positive basis to the homonymous vector of the negative basis. The following determinants mean those of $\mathcal{U}_{\mathrm{bas}}^{-1}\mathscr{A}_{E,X} = I + \mathcal{L}^1$ and $\mathcal{U}_{\mathrm{bas}}^{-1}\mathscr{A}_E = I + \mathcal{L}^1$. Then

$$\det_F(\mathcal{U}_{\mathrm{bas}}^{-1}\mathscr{A}_{E,X}(s)) = L_X(E,s), \qquad \det_F(\mathcal{U}_{\mathrm{bas}}^{-1}\mathscr{A}_E(s)) = L(E,s). \tag{101.78}$$

The second equality is precisely the limit of (101.75); the sign is correct because the inverse of $I - K_E$ is taken.

The finite corestriction is constructed from the terminals, without publishing infinitely many primes in a finite matrix. Let $\mathcal{I}_E^{+}$ and $\mathcal{I}_E^{-}$ be the terminal lists of the integral bases of M_E and N_E, respectively; they are finite by (101.16). With $\sigma_* > 3/2$ fixed, the local–global intertwiner embeds them into the Euler space by

$$\iota_{\mathrm{Eul}}^{\pm}(e_\eta) := \Big(p^{-\sigma_*}\operatorname{res}_p^{\pm}\operatorname{Surv}_E(e_\eta)\Big)_p \oplus \Big(\operatorname{res}_p^{\pm}\operatorname{Surv}_E(e_\eta)\Big)_{p\mid N_E}. \tag{101.79}$$

The good sum is square–summable by the same bound as (101.74). Set $\tau_\eta^{\pm} := \iota_{\mathrm{Eul}}^{\pm}(e_\eta)$; independence of the terminal rows gives the dual counits $\{\epsilon_\eta^{\pm}\}$, so that $\epsilon_\eta^{\pm}(\tau_{\eta'}^{\pm}) = \delta_{\eta\eta'}$. The sums

$$P_E^{\pm} := \sum_{\eta\in\mathcal{I}_E^{\pm}} \tau_\eta^{\pm}\epsilon_\eta^{\pm}, \qquad Q_E^{\pm} := I - P_E^{\pm} \tag{101.80}$$

are based finite-rank projections; their indices are fixed by the terminal labels, not by the Mordell–Weil rank or by an order of vanishing.

The coordinate deg of the ledger decomposes

$$\mathscr{H}_E^{\pm} = \widehat{\bigoplus_i} \mathscr{H}_E^{\pm,i}, \qquad \mathcal{I}_E^{\pm,i} := \{\eta \in \mathcal{I}_E^{\pm} : \deg\eta = i\}, \qquad P_E^{\pm,i} := \sum_{\eta\in\mathcal{I}_E^{\pm,i}} \tau_\eta^{\pm}\epsilon_\eta^{\pm}. \tag{101.81}$$

The local relations preserve deg except for the differential $i \mapsto i+1$; therefore $\mathscr{A}_E = \bigoplus_i \mathscr{A}_E^{i}$ with $\mathscr{A}_E^{i} : \mathscr{H}_E^{+,i} \to \mathscr{H}_E^{-,i+1}$. The four blocks of (101.85), its Schur complement, and its truncations are taken degree by degree, and by definition

$$\mathscr{D}_E = \bigoplus_i \mathscr{D}_E^{i}, \qquad \mathscr{D}_E^{i} := A_{00}^{i} - A_{01}^{i}(A_{11}^{i})^{-1}A_{10}^{i}. \tag{101.82}$$

The Fredholm continuation is written before it is used. In the Euler chart U_0 set $\mathscr{A}_{E,0} := \mathscr{A}_E$. The charts preserve a common based complement $\mathcal{Q}_E^{\pm}$ and change only the finite family of terminals. If $\tau_{\eta,\alpha}^{\pm} := \operatorname{Ext}_{\Gamma,\det,\alpha}(\tau_\eta^{\pm})$ and $\epsilon_{\eta,\alpha}^{\pm}$ is its dual basis, define

$$C_{\alpha\beta}^{\pm} := I_{\mathcal{Q}_E^{\pm}} \oplus \sum_{\eta\in\mathcal{I}_E^{\pm}} \tau_{\eta,\beta}^{\pm}\epsilon_{\eta,\alpha}^{\pm}, \qquad C_{\alpha\beta}^{\pm} - I \in \mathcal{L}^1, \qquad C_{\beta\gamma}^{\pm}C_{\alpha\beta}^{\pm} = C_{\alpha\gamma}^{\pm}. \tag{101.83}$$

The difference from the identity has rank at most $|\mathcal{I}_E^{\pm}|$ and is therefore trace class; holomorphy follows because each of the finitely many terminal coordinates of $\mathrm{Ext}_{\Gamma,\det,\alpha}$ is holomorphic. The final equality is verified in $\mathcal{Q}_E^{\pm}$ and in every $\tau_{\eta,\alpha}^{\pm}$. It is defined recursively on the overlaps

$$\mathscr{A}_{E,\beta} := C_{\alpha\beta}^{-}\mathscr{A}_{E,\alpha}(C_{\alpha\beta}^{+})^{-1}. \tag{101.84}$$

The operators $C_{\alpha\beta}^{\pm}$ are holomorphic because every coefficient is a coordinate of $\mathrm{Ext}_{\Gamma,\det}$; the cocycle of (101.83) proves that (101.84) does not depend on the chain of charts. This is the determinant–class family henceforth denoted by $\mathscr{A}_E(s)$. With $C_{0\alpha}^{\pm} := C_{\alpha-1,\alpha}^{\pm}\cdots C_{0,1}^{\pm}$, one additionally transports $P_{E,\alpha}^{\pm,i} := C_{0\alpha}^{\pm}P_E^{\pm,i}(C_{0\alpha}^{\pm})^{-1}$ and $\mathscr{H}_{E,\alpha}^{\pm,i} := C_{0\alpha}^{\pm}\mathscr{H}_E^{\pm,i}$; thus all Schur $\mathscr{D}_{E,\alpha}^{i}$ are those of (101.82), not new symbols. After reducing each chart, the block

$$A_{ab}(s) := P_a^{-}\mathscr{A}_E(s)P_b^{+}, \qquad P_0^{\pm} := P_E^{\pm}, \quad P_1^{\pm} := Q_E^{\pm}, \qquad A_{11}(s) : Q_E^{+}\mathscr{H}_E^{+} \xrightarrow{\sim} Q_E^{-}\mathscr{H}_E^{-} \tag{101.85}$$

is invertible. This is the standard finite reduction of the Fredholm operator; the specific choice of $P_E^{\pm}$ is given by (101.80). The basis also fixes the unitary identification $U_{11} : Q_E^{+}\mathscr{H}_E^{+} \to Q_E^{-}\mathscr{H}_E^{-}$; by construction, $U_{11}^{-1}A_{11} = I + \mathcal{L}^1$, and only this operator is acted on by $\det_F$. The copies + and − already constitute the even–odd totalization; no second alternating sign is applied to the determinant of the inverse operator.

For each truncation that contains the support of those terminals, it is defined materially.

$$\begin{aligned} \mathrm{Cor}_{E,X} &:= P_E^{-}, \\ \mathrm{Lift}_{E,X}(v) &:= v - A_{11,X}^{-1}A_{10,X}v, \\ \mathscr{D}_{E,X}(s) &:= \mathrm{Cor}_{E,X}\,\mathscr{A}_{E,X}(s)\,\mathrm{Lift}_{E,X} = A_{00,X} - A_{01,X}A_{11,X}^{-1}A_{10,X}. \end{aligned} \tag{101.86}$$

By block multiplication,

$$\begin{pmatrix} I & -A_{01,X}A_{11,X}^{-1} \\ 0 & I \end{pmatrix} \mathscr{A}_{E,X} \begin{pmatrix} I & 0 \\ -A_{11,X}^{-1}A_{10,X} & I \end{pmatrix} = \begin{pmatrix} \mathscr{D}_{E,X} & 0 \\ 0 & A_{11,X} \end{pmatrix}. \tag{101.87}$$

Thus corestriction is substantive: its lift, its inverse on the complement, and its finite differential are written explicitly. Convergence of the resolvent implies

$$\mathscr{D}_E(s) := \lim_{X\to\infty} \mathscr{D}_{E,X}(s) = A_{00} - A_{01}A_{11}^{-1}A_{10}, \qquad \|\mathscr{D}_{E,X} - \mathscr{D}_E\|_1 \to 0. \tag{101.88}$$

The determinant of the complement is not discarded: by (101.87),

$$\det_F(\mathcal{U}_{\mathrm{bas}}^{-1}\mathscr{A}_E) = \det_F(U_{11}^{-1}A_{11})\,\det(\mathscr{D}_E). \tag{101.89}$$

Let $\mathcal{O}(U_E)$ be the ring of holomorphic functions of the central disk. The based rational spaces V_0^i, W_0^{i+1} of (101.26) provide

$$\mathscr{V}_E^{i} := V_0^{i} \otimes_{\mathbb{Q}} \mathcal{O}(U_E), \qquad \mathscr{W}_E^{i+1} := W_0^{i+1} \otimes_{\mathbb{Q}} \mathcal{O}(U_E). \tag{101.90}$$

The rows of $\mathrm{Rel}_{\det}$ construct, over each terminal η and in each degree, the vectors of the common model $v_{\eta,i} := \upsilon_i(b_\eta)$, $w_{\eta,i+1} := \upsilon_{i+1}(b_\eta^{\mathrm{term}})$, and the based isomorphisms

$$\begin{aligned} J_{E,V,i}(\tau_{\eta,i}^{+}) &:= v_{\eta,i}, \\ J_{E,W,i+1}(\tau_{\eta,i+1}^{-}) &:= w_{\eta,i+1}. \end{aligned} \tag{101.91}$$

The differential relation of that same row, applied to each generator, is

$$\boxed{J_{E,W,i+1}(s)\,\mathscr{D}_E^{i}(s) = S_E^{i}(T_E(s))\,J_{E,V,i}(s).} \tag{101.92}$$

This is the effective intertwiner between the corestricted Fredholm–Schur operator and the Smith matrix; it is not a postulated equality of determinants.

To type the continuation, let $\mathfrak{U}_E$ be a connected domain containing an open subset of the half-plane $\Re s > 3/2$ and the disk U_E, and choose a finite chain of charts $\{U_\alpha\}_{\alpha=0}^{A}$ of $\mathrm{Ext}_{\Gamma,\det}$, with $U_A = U_E$. On U_α set $\mathscr{P}^i_{E,\alpha} := P^{+,i}_{E,\alpha}\mathscr{H}^{+,i}_{E,\alpha}$ and $\mathscr{D}^i_{E,\alpha}$ equal to the Schur complement of (101.88). The maps to the common model are defined, without any additional choice, by

$$J_{E,\alpha,i}(\tau_{\eta,\alpha,i}) := v_{\eta,i} \quad \text{in the domain,} \qquad J_{E,\alpha,i+1}(\tau_{\eta,\alpha,i+1}) := w_{\eta,i+1} \quad \text{in the codomain.} \tag{101.93}$$

They are holomorphic because the bases of (101.83) are. On an overlap, define, always in the same degree,

$$G_{\alpha\beta,i} := J^{-1}_{E,\beta,i}J_{E,\alpha,i} : \mathscr{P}^i_{E,\alpha}\big|_{U_\alpha\cap U_\beta} \longrightarrow \mathscr{P}^i_{E,\beta}\big|_{U_\alpha\cap U_\beta}. \tag{101.94}$$

The literal composition of these matrices proves

$$G_{\beta\gamma,i}G_{\alpha\beta,i} = G_{\alpha\gamma,i}, \qquad G_{\alpha\alpha,i} = I, \qquad G_{\beta\alpha,i} = G^{-1}_{\alpha\beta,i}, \tag{101.95}$$

and the entanglement in the two letters gives the chain compatibility

$$G_{\alpha\beta,i+1}\mathscr{D}^i_{E,\alpha} = \mathscr{D}^i_{E,\beta}G_{\alpha\beta,i}. \tag{101.96}$$

Therefore $g_{\alpha\beta} := \prod_i \det(G_{\alpha\beta,i})^{(-1)^i}$ satisfies the triple cocycle. If

$$\tau^{\mathrm{atlas}}_E(s) := \prod_{\alpha=0}^{A-1} g_{\alpha,\alpha+1}(s), \tag{101.97}$$

then τ^{atlas}_E is exactly the determinantal transport from the Euler chart to the central chart.

The even–odd totalizations $J_{E,V}$ and $J_{E,W}$ of the maps (101.92) are denoted, with Koszul order; their determinants are the alternating products of the determinants by degree. In the central chart the joint unit is defined, and not before

$$u_E(s) := \tau^{\mathrm{atlas}}_E(s)\det_F(U_{11}^{-1}A_{11}(s))\frac{\det J_{E,V}(s)}{\det J_{E,W}(s)}. \tag{101.98}$$

All its factors are holomorphic and invertible upon reducing U_E. The normalization is not imposed factor by factor. Let $\mathbf{e}_{\mathrm{cent}}$ be the generator of the determinant line of the common model. On performing the two triangular operations of (101.87), transporting by (101.83), and applying (101.93), the joint action on $s = 1$ is, generator by generator,

$$\begin{aligned}\tau^{\mathrm{atlas}}_E(1)\det_F(U_{11}^{-1}A_{11}(1))\frac{\det J_{E,V}(1)}{\det J_{E,W}(1)}\,\mathbf{e}_{\mathrm{cent}} \\ = \bigotimes_{i,\eta}\big(\epsilon^-_{\eta,i}(\tau^-_{\eta,i})\epsilon^+_{\eta,i}(\tau^+_{\eta,i})^{-1}\big)^{(-1)^i}\mathbf{e}_{\mathrm{cent}} = \mathbf{e}_{\mathrm{cent}}, \qquad \boxed{u_E(1) = 1}.\end{aligned} \tag{101.99}$$

The first equality is the conservation identity for the common basis of the $\mathfrak{G}_{\det}(E)$ comparator, now expanded by its rows; the second uses $\epsilon^{\pm}_{\eta,i}(\tau^{\pm}_{\eta,i}) = 1$. None of the four factors has been normalized separately, nor has the sought value been used.

Proposition 101.6 (FERS Origin of the Analytic Factorization)**.** *On the disk U_E, one has the identity of determinant sections*

$$\boxed{L(E,s) = u_E(s)\det S_E(T_E(s)).} \tag{101.100}$$

The equality is the publication of the reader FERS of $\mathfrak{G}_{\det}(E)$ and not a equality postulada afterward of calcular the order or the term principal.

Proof. In the absolute half-plane, (101.78) identifies the Fredholm limit with $L(E,s)$. Literal elimination (101.87) and continuity in trace norm give $L(E,s) = \det_F(U_{11}^{-1}A_{11}(s))\det \mathscr{D}_E(s)$. Taking ordered wedges in (101.92) gives

$$\det \mathscr{D}_E(s) = \frac{\det J_{E,V}(s)}{\det J_{E,W}(s)} \det S_E(T_E(s)).$$

Equations (101.95) and (101.96) transport this section through the charts; the exact product of their transitions is τ_E^{atlas}. Therefore, in $U_A = U_E$, the remaining coefficient is precisely (101.98), proving (101.100). Uniqueness of the Mellin continuation of (101.70) excludes another section. Finally, (101.99) jointly proves $u_E(1) = 1$. No step uses d, the rank, $\mathrm{Sha}(E)$, or the leading term. □

Define, from the Smith form and not from the analytic order,

$$a_E(T) := T^{-d} \det S_E(T). \tag{101.101}$$

By (101.31), a_E is holomorphic on $T = 0$ and

$$a_E(0) = R_{\mathrm{Boc}}^{\mathrm{der}}(S_E) \neq 0. \tag{101.102}$$

Substitution of this identity and (101.65) into the analytic factorization yields explicitly

$$\begin{aligned} L(E,s) &= (s-1)^d\, U_E^{\mathrm{lead}}(s), \\ U_E^{\mathrm{lead}}(s) &:= u_E(s) \left(\frac{T_E(s)}{s-1}\right)^d a_E(T_E(s)), \\ U_E^{\mathrm{lead}}(1) &= 1 \cdot 1^d \cdot R_{\mathrm{Boc}}^{\mathrm{der}}(S_E) = R_{\mathrm{Boc}}^{\mathrm{der}}(S_E) \neq 0. \end{aligned} \tag{101.103}$$

Thus the exponent and the leading coefficient are deduced from the nonzero holomorphic germ; they are not inputs to T_E, u_E, or S_E. Now, after this deduction, define

$$d_{\mathrm{an}} := \mathrm{ord}_{s=1} L(E,s), \qquad \mathfrak{z}_{\mathrm{an}} := \left[(s-1)^{-d_{\mathrm{an}}} L(E,s)\right]_{s=1}. \tag{101.104}$$

The equation (101.103) proves

$$d_{\mathrm{an}} = d, \qquad \mathfrak{z}_{\mathrm{an}} = R_{\mathrm{Boc}}^{\mathrm{der}}(S_E) \quad \text{on the based line.} \tag{101.105}$$

On the arithmetic side, the theorem 101.3 and the sequence (101.60) construct

$$\begin{aligned} d_{\mathrm{ar}} &:= \dim_K M_E^{\mathrm{MW}} = d, \\ \mathfrak{z}_{\mathrm{ar}} &:= \frac{|\,\mathrm{Sha}(E)|\,\mathrm{Reg}(E)\prod_v c_v}{|E(\mathbb{Q})_{\mathrm{tors}}|^2} \quad \text{on the oriented arithmetic line.} \end{aligned} \tag{101.106}$$

To type the four arrows, if $C^\bullet$ is a perfect complex with ordered basis $\mathcal{B}_i = (c_{i,1}, \dots, c_{i,r_i})$, one uses

$$\det C^\bullet := \bigotimes_i \left(\bigwedge^{r_i} C^i\right)^{(-1)^i}, \qquad \mathbf{e}_C := \bigotimes_i (c_{i,1} \wedge \cdots \wedge c_{i,r_i})^{(-1)^i}. \tag{101.107}$$

For a finite group A with based presentation $\mathbb{Z}^t \xrightarrow{D_A} \mathbb{Z}^t \to A$, its line $\det_{\mathbb{Z}} A$ carries the generator $\mathbf{e}_A$ and the evaluation

$$\mathrm{ev}_A(\mathbf{e}_A) := |\det D_A| = |A|. \tag{101.108}$$

When that line is combined with a line over $K = \mathbb{Q}$, the scalar extension is always used

$$\det_K A := \det_{\mathbb{Z}} A \otimes_{\mathbb{Z}} K; \tag{101.109}$$

modules over distinct rings are not tensorized directly. In particular, choose the Smith bases already constructed in (101.55), $m_1, \dots, m_t$ of M_E, and $n_1, \dots, n_t$ of N_E, with $D_E m_i = s_i n_i$, and fix

$$\mathbf{e}_{F_E} := (m_1 \wedge \cdots \wedge m_t) \otimes (n_1 \wedge \cdots \wedge n_t)^{-1}, \qquad \mathrm{ev}_{F_E}(\mathbf{e}_{F_E}) = \prod_i |s_i|. \tag{101.110}$$

The successive lines are

$$\begin{aligned} \Delta_K &:= \det C_E^K, \\ \Delta_{\mathrm{FERS}} &:= \det C_E^{\mathrm{Iw}}, \\ \Delta_{\mathrm{PT}} &:= \det_K M_E^{\mathrm{MW}} \otimes \det_K (M_E^{\mathrm{MW}})^\vee \otimes \det_K F_E, \\ \Delta_{\mathrm{STS}} &:= \det_K M_E^{\mathrm{MW}} \otimes \det_K (M_E^{\mathrm{MW}})^\vee \otimes \det_K \mathrm{Sha}(E) \otimes \bigotimes_{v<\infty} \det_K \Phi_v(\mathbb{F}_v), \\ \Delta_{\mathrm{BSD}} &:= \mathbb{R}\, \mathbf{e}_{\mathrm{BSD}}. \end{aligned} \tag{101.111}$$

The principal specialization is not a tacit identification. Denote $I := (T) \subset K[[T]]$ and $\mathfrak{c}^{\mathrm{reg}}_{K/\mathrm{FERS}} := \det_{\mathrm{alt}} \Phi_{K/\mathrm{FERS}} : \Delta_K[[T]] \to \Delta_{\mathrm{FERS}}[[T]]$. Let

$$\begin{aligned} \Delta^{(d)}_{\mathrm{FERS}} &:= I^d \Delta_{\mathrm{FERS}}[[T]], & \Delta^{\mathrm{lead}}_{\mathrm{FERS}} &:= I^{-d}\Delta^{(d)}_{\mathrm{FERS}} \otimes_{K[[T]],\, T\mapsto 0} K, \\ \Delta^{(d)}_K &:= (\mathfrak{c}^{\mathrm{reg}}_{K/\mathrm{FERS}})^{-1}(\Delta^{(d)}_{\mathrm{FERS}}), & \Delta^{\mathrm{lead}}_K &:= I^{-d}\Delta^{(d)}_K \otimes_{K[[T]],\, T\mapsto 0} K. \end{aligned} \tag{101.112}$$

Here $I^{-d}\Delta^{(d)}$ denotes the module formed by the symbols $T^{-d}z$ with $z \in \Delta^{(d)}$. The specialization is defined only on the part of order at least d:

$$\mathrm{sp}^{(d)}_{\mathrm{FERS}} : \Delta^{(d)}_{\mathrm{FERS}} \longrightarrow \Delta^{\mathrm{lead}}_{\mathrm{FERS}}, \qquad z(T) \longmapsto \overline{T^{-d} z(T)}. \tag{101.113}$$

The Kato–FERS equivalence then induces the typed map

$$\begin{aligned} \mathfrak{c}^{\mathrm{lead}}_{K/\mathrm{FERS}} : \Delta^{\mathrm{lead}}_K &\xrightarrow{\sim} \Delta^{\mathrm{lead}}_{\mathrm{FERS}}, \\ \overline{T^{-d} z_K(T)} &\longmapsto \overline{T^{-d}\mathfrak{c}^{\mathrm{reg}}_{K/\mathrm{FERS}}(z_K(T))}. \end{aligned} \tag{101.114}$$

The formula no applies $\mathrm{sp}^{(d)}$ to a linea desnuda. In particular, if

$$\begin{aligned} \zeta_{\mathrm{FERS}}(T) &:= \det S_E(T)\, \mathbf{e}_{\mathrm{FERS}} \in \Delta^{(d)}_{\mathrm{FERS}}, \\ \zeta_K(T) &:= (\mathfrak{c}^{\mathrm{reg}}_{K/\mathrm{FERS}})^{-1}\zeta_{\mathrm{FERS}}(T) \in \Delta^{(d)}_K, \end{aligned} \tag{101.115}$$

then

$$\mathrm{sp}^{(d)}_{\mathrm{FERS}}(\zeta_{\mathrm{FERS}}) = R^{\mathrm{der}}_{\mathrm{Boc}}(S_E)\, \mathbf{e}^{\mathrm{lead}}_{\mathrm{FERS}}, \quad \mathfrak{c}^{\mathrm{lead}}_{K/\mathrm{FERS}}(\overline{T^{-d}\zeta_K}) = R^{\mathrm{der}}_{\mathrm{Boc}}(S_E)\, \mathbf{e}^{\mathrm{lead}}_{\mathrm{FERS}}. \tag{101.116}$$

The definition uses only $d = \mathrm{ord}_T \det S_E(T)$, already computed by Smith, and not the analytic order.

The four owner arrows, now with explicit domains and codomains, are compared in the real fiber from the first point where the Néron height enters. To avoid multiplying a line over K by a real scalar that does not generally belong to K, we fix

$$\begin{aligned} \Delta^{\mathrm{lead}}_{K,\mathbb{R}} &:= \Delta^{\mathrm{lead}}_K \otimes_{K,\iota_\infty} \mathbb{R}, & \Delta^{\mathrm{lead}}_{\mathrm{FERS},\mathbb{R}} &:= \Delta^{\mathrm{lead}}_{\mathrm{FERS}} \otimes_{K,\iota_\infty} \mathbb{R}, \\ \Delta_{\mathrm{PT},\mathbb{R}} &:= \Delta_{\mathrm{PT}} \otimes_{K,\iota_\infty} \mathbb{R}, & \Delta_{\mathrm{STS},\mathbb{R}} &:= \Delta_{\mathrm{STS}} \otimes_{K,\iota_\infty} \mathbb{R}. \end{aligned} \tag{101.117}$$

The four arrows are then

$$\begin{aligned}
\mathfrak{c}^{\mathrm{lead}}_{K/\mathrm{FERS},\mathbb{R}} &: \Delta^{\mathrm{lead}}_{K,\mathbb{R}} \longrightarrow \Delta^{\mathrm{lead}}_{\mathrm{FERS},\mathbb{R}}, & \mathfrak{c}^{\mathrm{lead}}_{K/\mathrm{FERS},\mathbb{R}} &:= \mathfrak{c}^{\mathrm{lead}}_{K/\mathrm{FERS}} \otimes_K 1_{\mathbb{R}},\\
\mathfrak{c}_{\mathrm{PT},\mathbb{R}} &: \Delta^{\mathrm{lead}}_{\mathrm{FERS},\mathbb{R}} \longrightarrow \Delta_{\mathrm{PT},\mathbb{R}}, & \mathfrak{c}_{\mathrm{PT},\mathbb{R}}(z) &:= \mathfrak{n}_{E,\mathbb{R}}(z) \otimes \mathbf{e}_{F_E},\\
\mathfrak{c}_{\mathrm{STS},\mathbb{R}} &: \Delta_{\mathrm{PT},\mathbb{R}} \longrightarrow \Delta_{\mathrm{STS},\mathbb{R}}, & \mathfrak{c}_{\mathrm{STS},\mathbb{R}} &:= \det\big(0 \to \mathrm{Sha}(E) \to F_E \to \textstyle\bigoplus_{v<\infty} \Phi_v(\mathbb{F}_v) \to 0\big) \otimes_K 1_{\mathbb{R}},\\
\mathfrak{c}_{\mathrm{tors},\infty} &: \Delta_{\mathrm{STS},\mathbb{R}} \longrightarrow \Delta_{\mathrm{BSD}}.
\end{aligned} \tag{101.118}$$

Proposition 101.7 (Generator-by-generator calculation of the four arrows)**.** *The actions on the bases of* (101.111) *are as follows.*

First arrow. *If* $\mathbf{e}^i_K = \bigwedge_{b\in\mathcal{B}^K_i} b$ *and* $\mathbf{e}^i_{\mathrm{FERS}} = \bigwedge_{b\in\mathcal{B}^K_i} \upsilon_i(b)$*, the table* (101.34) *gives*

$$\mathfrak{c}^{\mathrm{reg}}_{K/\mathrm{FERS}}(\mathbf{e}_K) = \prod_i \det(I + TB_i(T))^{(-1)^i} \mathbf{e}_{\mathrm{FERS}} = \rho_E(T)\mathbf{e}_{\mathrm{FERS}}, \qquad \rho_E(0) = 1. \tag{101.119}$$

If $\mathbf{e}^{\mathrm{lead}}_K$ *and* $\mathbf{e}^{\mathrm{lead}}_{\mathrm{FERS}}$ *are the bases induced by* (101.112)*, then*

$$\mathfrak{c}^{\mathrm{lead}}_{K/\mathrm{FERS},\mathbb{R}}(\mathbf{e}^{\mathrm{lead}}_K \otimes 1) = \rho_E(0)(\mathbf{e}^{\mathrm{lead}}_{\mathrm{FERS}} \otimes 1) = \mathbf{e}^{\mathrm{lead}}_{\mathrm{FERS}} \otimes 1. \tag{101.120}$$

Second arrow. *Choose integral bases* $a_{q,1}, \dots, a_{q,r_q}$ *of* $A_{q,\mathbb{Z}}$*, viewed in* A_q*, and* $b_{q,j} = \bar\beta_q(a_{q,j})$ *of* B_q*. The height matrix of the page is defined by*

$$j_q(b_{q,j}) = \sum_{k=1}^{r_q} (H_q)_{kj}\, a^\vee_{q,k} \otimes T^q. \tag{101.121}$$

Therefore, on the entire wedge,

$$\bigwedge_{j=1}^{r_q} j_q\bar\beta_q(a_{q,j}) = T^{qr_q} \det(H_q) \bigwedge_{k=1}^{r_q} a^\vee_{q,k}. \tag{101.122}$$

The product over q*, together with the regular rows, has exponent* $\sum_q qr_q = \sum_q \dim_K V_q = d$*. For* $1 \le j \le r_q$ *and* $1 \le k \le q$*, set* $e_{q,j,k} := a_{q,j}e_{q,k}$ *in* $\mathrm{Jet}_{\mathrm{Boc},\mathbb{Z}}(E)$ *and define*

$$P_{q,j,k} := \Psi^{-1}_{\mathrm{PT},\mathbb{Z}} \mathcal{U}_{\mathrm{jet}}(e_{q,j,k}) \in M^{\mathrm{MW}}_{E,\mathbb{Z}}. \tag{101.123}$$

By (101.44)*,* (101.45)*, and* (101.49)*, these* d *elements are an integral basis, not a sublattice of uncontrolled index. They are enumerated in lexicographic order as* $P_1, \dots, P_d$ *and, over* K*, form the basis of* M^{MW}_E *used below.*

The height is constructed independently of FERS. For $P \in E(\overline{\mathbb{Q}})$*, fix*

$$\widehat h_E(P) := \frac{1}{2} \lim_{n\to\infty} 4^{-n} h_x([2^n]P), \qquad \langle P, Q\rangle_{\mathrm{NT}} := \frac{1}{2}\big(\widehat h_E(P+Q) - \widehat h_E(P) - \widehat h_E(Q)\big). \tag{101.124}$$

Equivalently, the Poincaré biextensor rigidified over the Néron model produces, for every place v*, including* $v = \infty$*, the real local symbol* $\lambda_v(P, Q)$ *normalized by the zero section, and*

$$\langle P, Q\rangle_{\mathrm{NT}} = \sum_v \lambda_v(P, Q) \in \mathbb{R}. \tag{101.125}$$

This real, symmetric, and positive height is not replaced by a sum of cup-product invariants with values in $\mathbb{Q}/\mathbb{Z}$*.*

Set $H^{\flat}_{\mathrm{NT}}(P)(Q) := \langle P, Q\rangle_{\mathrm{NT}}$ and let $\mathcal{B}_E := \Psi^{-1}_{\mathrm{PT},\mathbb{Z}}\mathcal{U}_{\mathrm{jet}}$, extended to $\mathbb{R}$. The $q-1$ additional directions are not replaced by identities: the full Gram block is computed, including all cross terms among depths, rows, and jets,

$$\begin{aligned} G^{\mathrm{jet}}_E\big((q,j,k),(q',j',k')\big) &:= \big\langle P_{q,j,k}, P_{q',j',k'}\big\rangle_{\mathrm{NT}} \\ &= \sum_{v} \lambda_v\left(P_{q,j,k}, P_{q',j',k'}\right), \end{aligned} \tag{101.126}$$

for every $1 \le k \le q$, $1 \le k' \le q'$. This matrix is material: the $P_{q,j,k}$ were constructed in (101.123), and each entry is computed through (101.125). By definition of the pullback of a bilinear form,

$$\boxed{\mathcal{B}_E^{\vee} H^{\flat}_{\mathrm{NT}} \mathcal{B}_E = G^{\mathrm{jet}}_E.} \tag{101.127}$$

It is not asserted that G^{jet}_E is block diagonal. In the integral basis $P_1, \dots, P_d$, the regulator and its action on the determinant line are exactly

$$\begin{aligned} \mathrm{Reg}(E) &:= \det G^{\mathrm{jet}}_E > 0, \\ \bigwedge_{j=1}^{d} H^{\flat}_{\mathrm{NT}}(P_j) &= \mathrm{Reg}(E)(P_1^{\vee} \wedge \cdots \wedge P_d^{\vee}). \end{aligned} \tag{101.128}$$

The Bockstein page and the height are two constructed pairings, with bases already fixed. The Neron comparator between their lines is defined before the leading term is used, by

$$\begin{aligned} \mathfrak{n}_{E,\mathbb{R}} &: \Delta^{\mathrm{lead}}_{\mathrm{FERS},\mathbb{R}} \longrightarrow \big(\det_K(M_E^{\mathrm{MW}}) \otimes_K \det_K(M_E^{\mathrm{MW}})^{\vee}\big) \otimes_{K,\iota_\infty} \mathbb{R}, \\ \mathfrak{n}_{E,\mathbb{R}}(\mathbf{e}^{\mathrm{lead}}_{\mathrm{FERS}} \otimes 1) &:= \frac{\mathrm{Reg}(E)}{R^{\mathrm{der}}_{\mathrm{Boc}}(S_E)} (P_1 \wedge \cdots \wedge P_d) \otimes (P_1^{\vee} \wedge \cdots \wedge P_d^{\vee}) \otimes 1. \end{aligned} \tag{101.129}$$

The denominator is nonzero by (101.102); numerator and denominator are obtained, respectively, from the full Gram matrix (101.126) and from the pages (101.31). Thus the normalization is a real quotient of two previously computed determinants, not an invented orthogonality or an appeal to the BSD formula. Scale extension is performed before applying the regulator and is preserved in the PT and STS arrows.

The upper pages and the regular row preserve the integral generator of F_E by means of a typed map. The rational extension of the section s_q already constructed in (101.44) is used. The dual section

$$\mathrm{gl}_q^{\vee} : A_q^{\vee} \longrightarrow (\mathrm{Surv}^{\mathrm{MW}}_{\mathrm{det}})^{\vee} \tag{101.130}$$

is defined on the complete basis by $(\mathrm{gl}_q^{\vee}\varphi)(\mathrm{gl}_r\, s_r(a)) = \delta_{qr}\varphi(a)$ and as zero on the FS rows. Let $\mathcal{I}_{\mathrm{fin}}$ be the ordered list of regular rows and pairs (q,j) with $q \ge 2$. For η with $q(\eta) \ge 2$, set

$$\begin{aligned} \widetilde{a}_\eta &:= \mathrm{pr}_{M_E}\, \mathrm{pr}_{\mathcal{L}_E}\, \Theta_E\, \mathrm{gl}_{q(\eta)}\, s_{q(\eta)}(a_\eta), \\ \widetilde{b}^{\vee}_\eta &:= \mathrm{pr}_{N_E^{\vee}}\, \mathrm{pr}_{\mathcal{L}_E^{\vee}} (\Theta_E^{\vee})^{-1}\, \mathrm{gl}^{\vee}_{q(\eta)}\big((1 \otimes \partial_T^{(q(\eta))}) j_{q(\eta)} \bar{\beta}_{q(\eta)}(a_\eta)\big). \end{aligned} \tag{101.131}$$

In a regular row they are defined separately

$$\widetilde{a}_\eta := \mathrm{pr}_{M_E}\, \mathrm{pr}_{\mathcal{L}_E}\, \Theta_E s_{\mathrm{reg}}(a_\eta), \qquad \widetilde{b}^{\vee}_\eta := \mathrm{pr}_{N_E^{\vee}}\, \mathrm{pr}_{\mathcal{L}_E^{\vee}} (\Theta_E^{\vee})^{-1} s^{\vee}_{\mathrm{reg}}(\bar{S}_E(0) a_\eta). \tag{101.132}$$

All domains are now A_q or $A_q^{\vee}$ as appropriate; gl_q is not applied to a covector.

The required bijection is not inferred from unimodular matrices. The unit–counit equations show on each generator that the lists $(\widetilde{a}_\eta)_\eta$ and their dual terminals are bases of M_E and $N_E^{\vee}$. Write the inherited list as $\mathcal{I}_{\mathrm{fin}} = \{\eta_1 < \cdots < \eta_t\}$ and define directly

$$\boxed{\nu(\eta_j) := j}, \qquad \nu : \mathcal{I}_{\mathrm{fin}} \xrightarrow{\sim} \{1, \dots, t\}. \tag{101.133}$$

Set $\mu_j := \widetilde{a}_{\eta_j}$ and let $\xi_1, \dots, \xi_t$ be the terminal basis of N_E. Before Smith, the material matrix is

$$D^{\mathrm{term}}_{kj} := \langle \xi_k^\vee, D_E \mu_j \rangle, \qquad \widetilde{b}^\vee_{\eta_j} = \sum_{k=1}^{t} D^{\mathrm{term}}_{kj} \xi_k^\vee. \tag{101.134}$$

This equality is the terminal relation of (101.15), evaluated at the counit $\xi_k^\vee$; it does not divide by an elementary divisor. If $U D^{\mathrm{term}} V = \mathrm{diag}(s_1, \dots, s_t)$, then

$$\bigwedge_{j=1}^{t} \widetilde{b}^\vee_{\eta_j} = \det(D^{\mathrm{term}})(\xi_1^\vee \wedge \cdots \wedge \xi_t^\vee), \qquad |\det(D^{\mathrm{term}})| = \prod_{j=1}^{t} |s_j|. \tag{101.135}$$

The factors $\det U, \det V \in \{\pm 1\}$ are recorded by $\mathrm{or}_{\det}$ upon passing from $(\mu_j), (\xi_j)$ to the Smith bases $(m_j), (n_j)$. Hence the wedge quotient is exactly $\mathbf{e}_{F_E}$ of (101.110); it has not been assumed that U or V are permutation matrices, and no page remains unpublished. Consequently,

$$\mathfrak{c}_{\mathrm{PT},\mathbb{R}}\big(R^{\mathrm{der}}_{\mathrm{Boc}}(S_E)(\mathbf{e}^{\mathrm{lead}}_{\mathrm{FERS}} \otimes 1)\big) = \mathrm{Reg}(E)\,(P_1 \wedge \cdots \wedge P_d) \otimes (P_1^\vee \wedge \cdots \wedge P_d^\vee) \otimes \mathbf{e}_{F_E} \otimes 1. \tag{101.136}$$

Third arrow. *For each ℓ, choose ordered generators $\sigma_{\ell,a}$ of $\mathrm{Sha}(E)[\ell^\infty]$ and $\phi_{\ell,v,b}$ of $\Phi_v(\mathbb{F}_v)[\ell^\infty]$. By exactness, the explicit lifts $\mathrm{Lift}_{\Phi,\ell}(\phi_{\ell,v,b})$ and the images $\iota_{\mathrm{Sha},\ell}(\sigma_{\ell,a})$ form a presentation basis of $F_{E,\ell}$. In those bases, the relation matrix is computed generator by generator as*

$$D_{F,\ell} = \begin{pmatrix} D_{\mathrm{Sha},\ell} & X_\ell \\ 0 & \displaystyle\bigoplus_{v<\infty} D_{\Phi_v,\ell} \end{pmatrix}. \tag{101.137}$$

The column X_ℓ records the change of each lift when its representative is modified; the lower-left block is zero because $\partial_{\mathrm{loc},\ell}\iota_{\mathrm{Sha},\ell} = 0$. Therefore the presentation wedge does not depend on X_ℓ and $\det D_{F,\ell} = \det D_{\mathrm{Sha},\ell} \prod_{v<\infty} \det D_{\Phi_v,\ell}$. On it,

$$\mathbf{e}_{F_{E,\ell}} \longmapsto \mathbf{e}_{\mathrm{Sha}(E)[\ell^\infty]} \otimes \bigotimes_{v<\infty} \mathbf{e}_{\Phi_v(\mathbb{F}_v)[\ell^\infty]}, \tag{101.138}$$

and the evaluations of (101.108) give $|F_{E,\ell}| = |\mathrm{Sha}(E)[\ell^\infty]| \prod_{v<\infty} |\Phi_v(\mathbb{F}_v)[\ell^\infty]|$. Multiplying over ℓ produces

$$\mathfrak{c}_{\mathrm{STS},\mathbb{R}}(\mathbf{e}_{F_E} \otimes 1) = (\mathbf{e}_{\mathrm{Sha}(E)} \otimes \bigotimes_{v<\infty} \mathbf{e}_{\Phi_v(\mathbb{F}_v)}) \otimes 1, \qquad |F_E| = |\mathrm{Sha}(E)| \prod_v c_v. \tag{101.139}$$

Fourth arrow. *Given the preceding Mordell–Weil basis and a Néron differential ω_E with positive period Ω_E, the oriented basis of Δ_{STS} is sent by*

$$\begin{aligned} x\,(P_1 \wedge \cdots \wedge P_d) \otimes (P_1^\vee \wedge \cdots \wedge P_d^\vee) \otimes \mathbf{e}_{\mathrm{Sha}(E)} \otimes \bigotimes_{v<\infty} \mathbf{e}_{\Phi_v(\mathbb{F}_v)} \\ \longmapsto \frac{x\,\Omega_E\ \mathrm{ev}_{\mathrm{Sha}(E)}(\mathbf{e}_{\mathrm{Sha}(E)}) \prod_{v<\infty} \mathrm{ev}_{\Phi_v(\mathbb{F}_v)}(\mathbf{e}_{\Phi_v(\mathbb{F}_v)})}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}\, \mathbf{e}_{\mathrm{BSD}}. \end{aligned} \tag{101.140}$$

By (101.108), the coefficient of the last line is $x\Omega_E |\mathrm{Sha}(E)| \prod_v c_v / |E(\mathbb{Q})_{\mathrm{tors}}|^2$. The two copies of the torsion group follow from the dual degrees 0 and 2; the factor Ω_E is the evaluation of ω_E on the oriented real component. This orientation places $\Omega_E \mathfrak{z}_{\mathrm{ar}}$, and not its inverse, as the arithmetic coefficient of the composition; the comparison with $\mathfrak{z}_{\mathrm{an}}$ will be deduced only subsequently through the square of trivializations.

Proof. The first identity is the alternating determinant of the triangular matrices $I + TB_i(T)$. The second applies the specialization of the pages and then the Neron comparator (101.129). Its numerator is the determinant of the complete jet Gram matrix (101.126), with all cross-terms, and its denominator is the Bockstein determinant already calculated. By (101.128), its action on wedges is (101.136); the exponent d is obtained before taking Mordell–Weil dimensions. The third is the canonical isomorphism of lines associated with the exact sequence, written in the already constructed sections $\iota_{\mathrm{Sha},\ell}$ and $\mathrm{Lift}_{\Phi,\ell}$. The fourth is the evaluation of the two finite torsion complexes and of the Neron line. This calculates the four arrows without using the BSD equality. □

Its composition

$$\mathfrak{C}_E := \mathfrak{c}_{\mathrm{tors},\infty} \circ \mathfrak{c}_{\mathrm{STS},\mathbb{R}} \circ \mathfrak{c}_{\mathrm{PT},\mathbb{R}} \circ \mathfrak{c}^{\mathrm{lead}}_{K/\mathrm{FERS},\mathbb{R}} : \Delta^{\mathrm{lead}}_{K,\mathbb{R}} \longrightarrow \Delta_{\mathrm{BSD}} \tag{101.141}$$

is well typed. Before inserting any analytic or cardinal element, the independent trivializations are fixed

$$\mathrm{tr}_{\mathrm{an}}(x\mathbf{e}^{\mathrm{lead}}_K) := x, \qquad \mathrm{tr}_{\mathrm{Ner}}(y\mathbf{e}_{\mathrm{BSD}}) := y. \tag{101.142}$$

The first uses only the Kato basis specialized by (101.112) and the Mellin–Fredholm normalization of (101.75); in particular, $\mathrm{tr}_{\mathrm{an}}(\overline{T^{-d}\zeta_K}) = R^{\mathrm{der}}_{\mathrm{Boc}}(S_E) = \mathfrak{z}_{\mathrm{an}}$ by (101.116). The second uses only the oriented Neron differential and satisfies $\mathrm{tr}_{\mathrm{Ner}}(\mathbf{e}_{\mathrm{BSD}}) = 1$. The identity of the based comparator of $\mathfrak{G}_{\mathrm{det}}(E)$, unfolded by the four arrows, is the square

$$\begin{array}{ccc} \Delta^{\mathrm{lead}}_{K,\mathbb{R}} & \xrightarrow{\ \mathfrak{C}_E\ } & \Delta_{\mathrm{BSD}} \\ \big\downarrow{\scriptstyle \mathrm{tr}_{\mathrm{an}}} & & \big\downarrow{\scriptstyle \mathrm{tr}_{\mathrm{Ner}}} \\ \mathbb{R} & = & \mathbb{R}. \end{array} \tag{101.143}$$

This square is fixed with the structural bases before evaluating $L(E,s)$, the regulator, or the finite groups.

The analytic element is placed, after constructing the arrows, in its correct domain:

$$\mathbf{z}^{\mathrm{an}}_E := \mathfrak{z}_{\mathrm{an}}\mathbf{e}^{\mathrm{lead}}_K = \overline{T^{-d}\zeta_K} \in \Delta^{\mathrm{lead}}_{K,\mathbb{R}}. \tag{101.144}$$

By (101.105), $\mathfrak{z}_{\mathrm{an}} = R^{\mathrm{der}}_{\mathrm{Boc}}(S_E)$; the second equality in (101.116) proves that the first arrow carries $\mathbf{z}^{\mathrm{an}}_E$ exactly to the left-hand side of (101.136). The other three arrows, computed on generators, give

$$d_{\mathrm{an}} = d = d_{\mathrm{ar}}, \qquad \boxed{\mathfrak{C}_E(\mathbf{z}^{\mathrm{an}}_E) = \Omega_E \mathfrak{z}_{\mathrm{ar}}\, \mathbf{e}_{\mathrm{BSD}}.} \tag{101.145}$$

Applying the vertical arrows of (101.143) produces the numerical equality

$$\boxed{\mathfrak{z}_{\mathrm{an}} = \Omega_E \mathfrak{z}_{\mathrm{ar}}.} \tag{101.146}$$

Theorem 101.8 (Birch–Swinnerton–Dyer Publication)**.** *For the complete autonomous determinant image $\mathfrak{G}_{\mathrm{det}}(E)$, one has*

$$\mathrm{ord}_{s=1} L(E,s) = \mathrm{rank}\, E(\mathbb{Q}) =: r \tag{101.147}$$

and

$$\frac{L^{(r)}(E,1)}{r!\,\Omega_E} = \frac{|\,\mathrm{Sha}(E)|\,\mathrm{Reg}(E)\prod_v c_v}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}. \tag{101.148}$$

Proof. The equality of degrees is (101.105) followed by (101.50). The Kato–FERS table transports the analytic element to every Bockstein page; Ψ_{PT} glues them to the Mordell–Weil lattice and publishes its height determinant; the exact Sha–Tamagawa sequence decomposes the Smith factor; and the final comparator inserts the torsional lattice and the already oriented period. This is exactly (101.145). The independent square (101.143) gives (101.146); substituting the definitions of $\mathfrak{z}_{\mathrm{an}}$ and $\mathfrak{z}_{\mathrm{ar}}$ yields (101.148).

None of the four arrows is normalized with r, $\mathrm{Reg}(E)$, $|\,\mathrm{Sha}(E)|$, or the leading term: those values appear only at the end of their respective determinants. □

101.9 Non-governing controls, falsifiers and status

Falsification criterion 101.9 (Falsifiers separated by component)**.** The construction fails if a history of $\Sigma_{\mathrm{det}}^{\mathrm{mult}}$ does not belong to the partition (101.14); if one writes $\partial H_{\mathrm{FS}} + H_{\mathrm{FS}}\partial = q_{\mathrm{root}}$ and thereby erases $\mathcal{L}_E$; if a nonroot row of (101.34) has zero weight; if the unit–counit equations (101.42) fail; if $D_E \otimes \mathbb{Q}$ is not invertible; if $\pi_\Phi \,\mathrm{Lift}_\Phi \neq I$; or if any arrow of (101.118) uses the final value to fix its base. Each falsifier breaks a specific edge and does not reopen the preceding ones.

The homotopy H_{FS} is a subsequent control of the explicit filler $h_* = -vf$ and not a governing premise. Smith forms, determinants and cardinalities are computed after constructing the maps; they do not replace the tables, the gluing of pages or the exactness of cocycles.

Status and provenance. Alexander–Whitney, the fiber product, the common unit, reduced duality, root rigidity, the filler, all Bockstein pages and the four readers belong to `PREEXISTING_AUTHORIAL_ARCHITECTURE/INTERNALLY_EXACT` of $\mathfrak{G}_{\mathrm{det}}(E)$. The exhaustive partition (101.15), the table (101.34), the formulas (101.46) and the cocycle maps (101.57)– (101.58) are a `NEW_FORMALIZATION`: they materially unfold the intertwiners already contained in the thirteen structural coordinates. The final theorem has the force `PROVED_WITH_AN_EXPLICIT_STARTING_STRUCTURE`; no governing residual obligation remains, nor is an abstract homotopy used to manufacture the result.

Chapter 102

Domains, Operators, Naturality and Certificates of the Terminal Realizations of the Continuum

The joint closure does not make the strength of five distinct results uniform. For each publication, the domain, premises, intertwining map, certificate, and negative control that could invalidate it are preserved. The following matrix does not summarize the preceding proofs: it provides their inverse control and blocks any promotion that loses the terminal object, naturality, starting datum, or the distinction between the common state and its terminal reader.

This chapter gathers the negative controls for the five closures. A falsifier invalidates only the arrow and domain that it targets; it does not transfer its failure to another initial structure. Likewise, having a complete structure does not replace the specific proof of its application.

102.1 Internal view G_T and terminal reader P_T: logical independence between structural existence and classical realization

For each T, $\mathfrak{G}_T$ is the complete view that the intrinsic operation $\mathfrak{O}_T$ extracts without restricting the single common record. Its existence is an assertion distinct from the existence of the final reader $\mathcal{P}_T : \mathfrak{M}_T \to C_T$. A failure in $\mathcal{P}_T$ does not erase $\mathfrak{G}_T$; the existence of $\mathfrak{G}_T$ likewise does not by itself prove $\mathcal{P}_T$.

102.2 Common Obstructions of Terminal Readers: Loss of Domain, Naturalness, or Compatible Limit

Application	Flecha atacada	Falsador material
RH	common publication $J_+, J_- \to P, \Xi$	The existence of $V_+\Xi_+ = \Xi$, $V_-\Xi_- = P\Xi$ already implies the norm inequality; without a construction prior to the sign, the step is circular.
NS	uniform root of future Gram	With $\Gamma = I$, $L = I$, and null terminal, Stein gives $\mathcal{U}_{0;J} = JI$; therefore telescopy does not imply uniformity.
YM	identification of semigroups	A degree-two Hoeffding sector has energy 6κ for D^{prod} and 3κ for D^{fib}.
Hodge theory	finite extension	A label is missing from $\Sigma^{\mathrm{mult}}_{\mathrm{coh}}$, the diagonal of the cellular table is not one, a refinement changes the global pivot, or the terminal is deleted; then $b\sigma = I$ or $e^{\mathrm{rel}} s^{\mathrm{rel}} = I$ fails.
BSD	cadena determinantal	H_{FS} is applied to the Smith lattice, a nonroot Kato row has zero weight, and $\mathrm{pg}_r\, \mathrm{gl}_q$ is not δ_{rq} or $\pi_\Phi \mathrm{Lift}_\Phi \neq I$; each failure breaks the corresponding comparator.

102.3 Conditions of existence, uniqueness, and compatibility of terminal publications

No application chain is considered proved by a declaration or an isolated numerical result. Promotion requires: a formula for the map; domains and codomains; proof that it commutes with differentials or refinements; a uniform estimate if a limit is involved; the classical conclusion only afterward; and an adversarial proof of the corresponding falsifier.

In Hodge, those data are the unitriangular table, the recurrence of σ, the extension over the fiber product, its naturality, and the Mittag–Leffler passage. In BSD, they are the root–FS–lattice partition, the complete Kato–FERS table, the page and gluing maps of Ψ_{PT}, the Sha–Tamagawa cocycle maps, and the four based comparators. The equations cited in the preceding chapters display their mathematical localizers; auxiliary controls remain thereafter and do not reopen those closures.

102.4 Negative and calibrated closures of apparent equivalences

The results of this section apply after the joint generation of the continuum. Their causal antecedent is the unique state produced by APP–TRIT–TPK, the correlative emergence of its five consubstantial constructions, the naturality squares, and the joint passage to the terminal object and the limit. None of the equivalences examined below generates or retrospectively selects those constructions.

102.4.1 Obstructions of canonicity at the boundary 108:144 = 90:120

Theorem 102.1 (There is no canonical inclusion $6 \hookrightarrow 8$ without gauge)**.** *Let A and B be bare sets with*

$$|A| = 6, \qquad |B| = 8.$$

There is no rule, natural with respect to all relabelings of A and B, that selects an injection $A \hookrightarrow B$.

Proof. If an injection $\iota : A \hookrightarrow B$ were selected without additional data, naturality with respect to every $\beta \in \mathrm{Sym}(B)$ would force $\beta\iota = \iota$. Given $a \in A$, there exists a permutation β that moves $\iota(a)$, a contradiction. Therefore, the route-by-route APP–Witt transducer is not canonical from the cardinalities. Once an octad, an inclusion, and the incidence gauge are fixed, a calibrated transducer is obtained; the choice forms part of its domain. □

Theorem 102.2 (A bare fibre of cardinality 28 does not determine $J(8,2)$)**.** *Let F be an unstructured set with $|F| = 28$. There is no nontrivial adjacency relation on F, natural with respect to all its permutations, that produces the Johnson graph $J(8,2)$.*

Proof. The group $\mathrm{Sym}(F)$ acts transitively on the unordered pairs of distinct elements. A simple relation invariant under that entire group therefore contains either all pairs or none. The two resulting graphs are the complete graph and the empty graph. By contrast, $J(8,2)$ has 28 vertices and degree 12; it is neither of them. Consequently, the calibrated projection $1008 \to 36$ with uniform fibers of cardinality 28 does not by itself imply a Witt octad. Once an octad O and a bijection $F \simeq \binom{O}{2}$ have been fixed, Johnson adjacency can be transported to F, but it depends precisely on that gauge. □

These two theorems close, by exclusion, the former canonical formulations. They do not affect the exact invariant

$$\frac{108}{144} = \frac{90}{120} = \frac{6}{8}$$

nor to the normalized defect $-1/12$, which belong to the rank extension and not to the choice of a Witt incidence.

102.4.2 The nonadic return and the exceptional reader

Let us fix the caliber

$$\chi_{\mathrm{exc}} = (\prec_{\mathrm{proj}}, \kappa_{\mathrm{Paley}}, \mathcal{D}_{12}, \iota_{\mathrm{inc}}, \mathcal{K}_{\Lambda})$$

and the calibrated reader $E_{\chi_{\mathrm{exc}}} : \mathcal{S}_\infty \to \mathcal{I}^{\mathrm{exc}}_{\chi_{\mathrm{exc}}}$. Let

$$\mathcal{S}_\infty \xleftarrow{s_\infty} \mathsf{Z}^{\Gamma}_{9,\infty} \xrightarrow{t_\infty} \mathcal{S}_\infty$$

the witnessed edge of a complete turn. Denote by ed_0 the central edge, by ev_0 the evaluation at the initial state, and by F_9 the displacement of one turn.

Theorem 102.3 (Exceptional Edge-Testified Factorization)**.** *Define*

$$\overline{E}^{s}_{\chi_{\mathrm{exc}}} := E_{\chi_{\mathrm{exc}}} \circ s_\infty, \qquad \overline{E}^{t}_{\chi_{\mathrm{exc}}} := E_{\chi_{\mathrm{exc}}} \circ t_\infty.$$

Then

$$E_{\chi_{\mathrm{exc}}} \circ \mathrm{ev}_0 = \overline{E}^{s}_{\chi_{\mathrm{exc}}} \circ \mathrm{ed}_0, \tag{102.1}$$

$$E_{\chi_{\mathrm{exc}}} \circ \mathrm{ev}_0 \circ F_9 = \overline{E}^{t}_{\chi_{\mathrm{exc}}} \circ \mathrm{ed}_0\,. \tag{102.2}$$

Proof. The endpoint identities are $s_\infty\,\mathrm{ed}_0 = \mathrm{ev}_0$ and $t_\infty\,\mathrm{ed}_0 = \mathrm{ev}_0\,F_9$. Composition with the calibrated reader gives the two equations. □

Corollary 102.4 (No equivalence between the temporal sector +1 and 1 complete nonadic turn)**.** *The preceding factorization cannot be replaced, in a natural way, by a factorization through the TRIT sample alone $\tau = +1$.*

Proof. Value $\tau = +1$ records return, memory, and past, but forgets phase, carry, route, and endpoint data distinguishing complete edges. Two edges with the same temporal sample may have different destinations. Every map that factors through that sample identifies both edges, whereas ed_0 and its source and target maps separate them. Therefore, sector +1 is not equivalent to complete edge $K \mapsto K + 9$. In particular,

$$g(K + 9) = g(K) \quad \text{and} \quad B_{K+9} \neq B_K \quad \text{in general}$$

remain distinct statements. □

Proposition 102.5 (Calibrated separation versus raw publication)**.** *Suppose that the enriched exceptional reader preserves, for each generator, the frame f_j and the image $y_j = \Gamma_{f_j A_W f_j^{-1}}(w_j)$. It then recovers*

$$w_j = \Gamma^{-1}_{f_j A_W f_j^{-1}}(y_j)$$

and it is injective on the quotient by the coordinate changes included in the gauge. If the raw reader is obtained by forgetting frames or memory, $E^{\mathrm{cr}} = \pi E^{\sharp}$, it is not equivalent to the enriched reader unless π is injective on $E^{\sharp}(X)$.

Proof. The first assertion is the explicit inversion of an automorphism in each fiber. For the second, if π identifies two distinct enriched outputs, the raw reader identifies their antecedents and cannot be injective. Conversely, injectivity of π on the image permits recovery of the enriched output. Thus separation closes in the calibrated domain, and silent equivalence with the raw codomain is rejected. □

The former square written with untyped symbols $\mathrm{pub}_{\mathrm{inc}}$, coord, O does not constitute a mathematical equivalence: it does not even define its morphisms. Its replacement is the natural cone of the Archimedean,

calibrated exceptional, and dimensional publications, whose truncation squares do have declared domain, codomain, and action.

102.4.3 Exceptional corridor and transport links

The Paley–Witt–Golay–Leech–$V^\natural$–Monster–Moonshine chain is a subsequent publication of the already constructed state. An incidence star and a loop of the TPK groupoid are not equivalent objects by virtue of their names: the former is a combinatorial subobject; the latter is an arrow with equal source and target, composition, and inverse. Without a base, orientation, and transport functor, no typed comparison exists. The automatic identification is therefore ruled out as a non-equivalence. A calibrated transport functor may produce loops from stars, but this constitutes an additional result and not a premise of the already proved corridor.

102.4.4 Infinitely divisible channel induced by the TPK transfer

Theorem 102.6 (Poissonization UCP and coherent roots)**.** *Let K be a UCP channel compatible with the TPK truncations. For $t \geq 0$, define*

$$P_t := \exp\big(t(K - I)\big) = e^{-t} \sum_{n=0}^{\infty} \frac{t^n}{n!} K^n.$$

Then $(P_t)_{t \geq 0}$ is a UCP semigroup compatible with all truncations, and

$$P_1 = (P_{1/m})^m \qquad (m \geq 1).$$

At every finite level, a minimal Stinespring dilation of $P_{1/m}$ produces a coherent tensor root; the inverse limit preserves the TPK marginals.

Proof. Each P_t is a norm-convergent convex combination of the channels K^n, and is therefore UCP. Since all terms are powers of the same operator, the exponential identity gives $P_t P_s = P_{t+s}$. Compatibility with truncations is verified term by term. Taking $t = 1/m$ yields the m-th root. Stinespring converts each finite composition into a tensor contraction; naturality of the channels makes their marginals coherent. □

The theorem closes the existence of an infinitely divisible channel with TPK memory. It does not assert that an arbitrary historical channel coincides with this Poissonization: that equality would require comparing their generators and is rejected unless such a comparison is provided.

102.5 Spectral Closures and External Recognition Boundaries

Proposition 102.7 (Calibrated equivalence, not ambient equality)**.** *The spectral transport of the double circle determines a unitary class in the cyclic subspace generated by its reference vector. Two ambient representatives V_9 are literally equal only after a unitary identification of the complement and a compatible gauge have been fixed. Without those data, ambient equality is not left indeterminate: it is a stronger assertion and is not equivalent to the specific class already constructed in the subspace.*

Norm convergence of the compressed resolvents, recovery of heights, the Riesz criterion for multiplicities, and the RH–Moonshine diagonal trace are established in their respective theorems. The diagonal trace couples the memory integer to the degree of $V^\natural$; it does not assert that the multiplicity of each zero is the dimension of a representation of the Monster. That additional identification is not equivalent to the existence of the coupled trace.

They fall outside the internal domain, without reopening any HMT theorem:

1. the identification of the dichotomy of analytic HMT publications with all arbitrary subsets of $\mathbb{R}$ of the ambient universe;

2. the macroscopic or thermodynamic realization of the finite HMT time crystal already constructed, including the preservation of coherence at large volume and its experimental testing;
3. the decimal hardening of spectral windows through a concrete execution of interval arithmetic;
4. an additional representation-theoretic identification between individual spectral multiplicities and irreducible Monster modules.

These four boundaries are external recognitions or comparisons, not gaps in the construction of the continuum, the cardinal closure, RH, or the intertwining of the double circle.

The terminal residue preserves, without permutation,

$$\begin{aligned}\mathrm{RH} &\longrightarrow \text{double circle and spectral field} \longrightarrow \text{Navier–Stokes}\\ &\longrightarrow \text{Yang–Mills} \longrightarrow \text{Hodge} \longrightarrow \text{BSD} \longrightarrow \text{falsifiers}.\end{aligned}$$

102.6 Provenance of the results and reproduction of the extension

The exposition starts from the APP–TRIT–TPK kernel and maintains its correlative construction of the continuum. Regional readouts, the native product and irreducible clocks precede their analytic realizations. Integration of centre, boundary and memory brings together results of the HMT–MD corpus with the compositions developed for the double circle. Provenance is recorded result by result, so that a consolidated formulation or an additional certificate does not retrospectively make the structure it uses new.

102.6.1 Dependencies brought together in the main text

The kernel comprises the common definitions and operations, together with its two figures. The definitions, statements and proofs of the five preceding developments are preserved in full.

The central chart, its lifted boundary and the transport signatures are developed in section 8.11. The distinction between the central point $(5, 5)$, the product signature 555555 and the sheets that its transport distinguishes is preserved in definition 8.64 and proposition 8.65. The capacity calendar and its two return levels are proved in theorems 8.69 and 8.70. The geometry of the chart is not identified with the complete history: its recovery uses the multiaffine law and the quotients preserved at the boundary.

Double-circle transport and memory moments are brought together in Section 97.10. The gamma-tail readouts of Section 97.11 reconstruct the columns with their domains and cross-products. Section 97.9 establishes the estimate on a complete window and its transport to subspaces with tails. Conservation equalities and positivity of a form are distinct statements: each result of this extension preserves the domain in which it is proved.

The formal antecedents necessary for these compositions are developed in the emitter construction 1.3.4, in the certificate of generation to arbitrary depth 8.55, in the joint connection 8.57 and in the transverse inversion of the record in section 11.4.9. The equations and proofs used thus remain set out together with their domains and representation data.

102.6.2 Reproduction order and scope of the checks

The edition preserves the antecedents, the five preceding developments and the focal checks accompanying their identities. Its compilation attests a legible material realization of the text; documentary preservation and computational checks do not thereby certify every result of a project. The extension's contribution consists of incorporating the following compositions with their proofs.

The folder accompanying the PDF preserves the LaTeX sources, the five developments in Markdown, the programs and JSON receipts. Reproduction follows the mathematical dependencies of the objects:

1. Construct the centre and boundary readouts while preserving residue and quotient, and check the partition of the 324 seeds under forward and half-turn transports.
2. Evaluate integer parts of the vacancy calendar through rational brackets. Bounds with remainders precede the digits and fix certification of each position.
3. Check seam balances, memory terminals and moment factorizations; preserve cross-terms when forming the joint matrices.
4. Evaluate local bounds for the gamma tail and Schur complement, with the domains of the estimates printed in their statements.
5. Compile the manuscript and compare common sources, transferred equations, references and page layout.

Rational and interval checks verify the finite identities and bounds they declare. Functional proofs establish the corresponding universal statements. File preservation and successful compilation are separate editorial checks; neither is presented as a substitute for a global proof.

Ordinary and optimized execution of the programs uses explicit conditions: disabling the interpreter's assertions does not alter the acceptance criterion. The delivery's instruction file specifies the commands and environment dependencies. Earlier originals are preserved separately and are not overwritten by this extension.

PART VI

Arithmetic-geometric coupling, conservation and recovery of information

Chapter 103

Refinement of histories and conservation of unity

103.1 Generation of emissions and realization domain

The construction starts from APP emissions and their enriched prolongations. An emission preserves the two sheets and their divisions

$$\delta^{\diamond} = \mathrm{res}^{\diamond} + 9\,\mathrm{quo}^{\diamond}, \qquad \diamond \in \{\Sigma, \Pi\}.$$

TRIT produces orientation and carry through

$$\Delta(\epsilon) = \mathrm{or}(\epsilon) + 3\kappa(\epsilon), \qquad \mathrm{or}(\epsilon) \in \{-1, 0, +1\}.$$

TPK transports the complete pair of sheets and updates memory through

$$U_{\epsilon_2\epsilon_1} = U_{\epsilon_2} U_{\epsilon_1}, \qquad \mathsf{U}_{\epsilon}(m) = \mathsf{C}_{\epsilon} m + \kappa(\epsilon), \qquad \kappa(\epsilon_2\epsilon_1) = \kappa(\epsilon_2) + \mathsf{C}_{\epsilon_2}\kappa(\epsilon_1).$$

The cocycle makes affine composition associative. The path boundary telescopes, but its ordered record is an ordered word, not the sum of its entries. The odometer increments vertical memory when the phase returns; return does not identify the complete states. The reduced catalogue transfer does not replace the action of emissions on the tree; the construction in section 6.4.1 y proposition 6.22 preserves this distinction.

The enriched state preserves the intermediate cells, both sheets, remainders, quotients, orientation, carry, clock, memory, boundary, path, and ordered record. An admissible emission jointly updates these fields. Truncation erases exactly its final contribution. A neutral prolongation gives a section of truncation but records a step: it is not an absence of history. This yields finiteness, nonemptiness, and surjectivity at every level, followed by total survival.

This development stops at the analytic realization of this system and its memory. No target numerical values or conventional constants enter as inputs. The following probabilistic coefficients arise solely from the number of effective outgoing emissions.

103.2 Refinement on the complete multisection

Fix an actual enriched prefix x of depth k. Write

$$H_n = \mathrm{Term}_{k,k+n}(x), \qquad \rho_n : H_{n+1} \longrightarrow H_n, \qquad \Omega_x = \varprojlim_{n}(H_n, \rho_n).$$

Each element of H_n is a complete prolongation up to that horizon, with its emission witnesses. For $h \in H_n$, the children are $h\epsilon$, for all emissions $\epsilon \in \mathrm{Out}(h)$. Operation multiplicities are preserved, even when reduced terminal data coincide. By section 6.4.2:

$$p_h(\epsilon) = \frac{1}{d(h)} > 0, \qquad d(h) = \#\,\mathrm{Out}(h), \qquad \mu_{n+1}(h\epsilon) = \mu_n(h)\,p_h(\epsilon).$$

Therefore,

$$\sum_{\epsilon \in \mathrm{Out}(h)} \mu_{n+1}(h\epsilon) = \mu_n(h), \qquad (\rho_n)_*\mu_{n+1} = \mu_n. \tag{103.1}$$

Proposition 6.4.2 constructs the unique probability measure μ_x on Ω_x with these cylinder masses; every nonempty cylinder has positive mass. The measure neither selects a history nor imposes a filter subsequent to the existence of the five constructions.

Proposition 103.1 (Refinement of observables is isometric and unital)**.** *Let* $\mathscr{L}_n = L^2(H_n, \mu_n)$ *and*

$$R_n : \mathscr{L}_n \longrightarrow \mathscr{L}_{n+1}, \qquad (R_n f)(h\epsilon) = f(h).$$

Then $R_n^* R_n = I$, $R_n 1 = 1$, *and its adjoint is*

$$(E_n g)(h) = \sum_{\epsilon \in \mathrm{Out}(h)} p_h(\epsilon) g(h\epsilon). \tag{103.2}$$

In particular, $E_n R_n = I$, $E_n 1 = 1$, *and* $R_n E_n$ *is the orthogonal projection onto the observables depending only on the prefix.*

Proof. Applying (103.1),

$$\|R_n f\|_{n+1}^2 = \sum_h \mu_n(h) |f(h)|^2 \sum_\epsilon p_h(\epsilon) = \|f\|_n^2.$$

The same grouping in the inner product gives (103.2). The statements concerning unity hold pointwise; the projection statement follows from $E_n = R_n^*$ and $E_n R_n = I$. Moreover, $R_n(fg) = R_n f \, R_n g$ and $R_n \bar{f} = \overline{R_n f}$: on finite observables, it is also a unital algebra homomorphism. □

Upon iteration, $R_{m,n} = R_{m-1} \cdots R_n$ coincides with pullback along the composed truncation. The cylindrical pullbacks $R_{\infty,n}$ realize the inductive limit of these Hilbert spaces in $L^2(\Omega_x, \mu_x)$. Their union is dense: cylinders form an algebra generating the sigma-algebra, and simple functions from that algebra are dense in L^2. Constructing the limit requires no choice of a trajectory.

103.2.1 The same isometry in the history basis

The unitary identification $W_n f(h) = \sqrt{\mu_n(h)} f(h)$ maps $\mathscr{L}_n$ to $\ell^2(H_n)$. Conjugating R_n gives

$$\widehat{R}_n \delta_h = \sum_{\epsilon \in \mathrm{Out}(h)} \sqrt{p_h(\epsilon)} \, \delta_{h\epsilon}. \tag{103.3}$$

The vectors on the right have disjoint supports for distinct parents, because $\rho_n(h\epsilon) = h$. Each column has norm one by normalization. This directly proves isometry on the already generated enriched histories, without imposing that a numerical projection preserve all memory. The unit vector is transformed into

$$\Omega_n = \sum_{h \in H_n} \sqrt{\mu_n(h)} \, \delta_h, \qquad \widehat{R}_n \Omega_n = \Omega_{n+1}.$$

No artificial copy of h is appended to a result in order to recover it: $h\epsilon$ is precisely the prolongation already produced by the APP–TRIT–TPK domain.

103.3 Correlative prolongation of the five operations

The construction in section 6.4.9 defines a single record $\mathfrak{e}_\epsilon$ for each emission. Its spectral, solenoidal, gauge, cohomological, and determinantal operations depend on this same emission, with the same source, target, orientation, carry, boundary, path, and ordered record. The cellular action satisfies

$$\partial \operatorname{Cell}(\epsilon) = \operatorname{Cell}(\epsilon)\partial, \qquad \operatorname{Cell}(\epsilon_2\epsilon_1) = \operatorname{Cell}(\epsilon_2)\operatorname{Cell}(\epsilon_1).$$

The complex and its determinant line belong to the same object; the constructed filling satisfies $\partial h_* = z_- - z_+$. Totality of the constructor does not require a specialized trivialization to be acyclic or invertible. Joint existence is thus preserved before any specialized reader.

The naturality in propositions 6.28 y 6.29 is expressed jointly as

$$u_{(\ell,N'),(k,N)}\mathfrak{D}_{\ell,N'}(\widehat{x}_\ell) = \mathfrak{D}_{k,N}(\tau_{\ell,k}\widehat{x}_\ell). \tag{103.4}$$

Its proof does not identify an isolated fine edge with a fictitious coarse edge: it restricts the emission history and preserves the actions of its entries. Transport composes over the same cylinder probabilities; balances use conditional expectation and its corrector; incidence and ports use their cellular maps; homology and determinant use the same map of complexes. These are not five limits of separately selected histories.

Two directions must be distinguished. State and coefficient maps run from fine to coarse. The pullbacks R_n of Proposition 103.1 run from coarse observables to fine observables. Expectation E_n returns in the reduction direction. This distinction makes it possible to realize (103.4) analytically without reversing its domains.

The terminal construction of theorem 6.25 y sections 6.4.7 y 6.4.8 prolongs the complete terminals and simultaneously preserves polar telescoping, Gram and losses, incidence with measure, return cone and Smith–Bockstein data. Promoting one of these actions to a bounded operator requires the corresponding norm and bound (see the passage to the limit proved at the end of this section). A compatible finite family does not become a bounded operator on the limit merely by its name.

103.4 Memory fibres: Gram criterion and conservation of unity

For a nonscalar realization, assign to each prefix h a Hilbert fibre $\mathscr{M}_h$, and to its emission a typed linear transport $I_\epsilon : \mathscr{M}_h \to \mathscr{M}_{h\epsilon}$. This linear operator is not identified with the affine update U_ϵ without specifying the representation realizing it.

On $\mathscr{H}_n = \bigoplus_{h\in H_n} \mathscr{M}_h$, define

$$(V_n\xi)_{h\epsilon} = \sqrt{p_h(\epsilon)}I_\epsilon\xi_h.$$

Disjointness of the children proves the exact identity of proposition 6.24:

$$V_n^*V_n\big|_{\mathscr{M}_h} = \sum_{\epsilon\in\operatorname{Out}(h)} p_h(\epsilon)I_\epsilon^*I_\epsilon. \tag{103.5}$$

The necessary and sufficient criterion for V_n to be isometric is that the sum of (103.5) be I at each parent. Each I_ϵ being isometric is sufficient. It is also necessary if all I_ϵ are already contractions: then each $I - I_\epsilon^*I_\epsilon$ is positive, and a sum with strictly positive weights can vanish only if each summand vanishes. The Gram criterion characterizes isometry of the total map; isometry of each transport constitutes a sufficient condition, which is also necessary within the class of contractions.

For the scalar histories of (103.3), $I_\epsilon = 1$, so the condition has been proved. In additional analytic fibres, (103.5) is retained and the precise assumption allowing passage to isometry is identified.

If the fibres have distinguished unit vectors u_h and $I_\epsilon u_h = u_{h\epsilon}$, then

$$V_n\Big(\bigoplus_h \sqrt{\mu_n(h)}u_h\Big) = \bigoplus_{h\epsilon} \sqrt{\mu_{n+1}(h\epsilon)}u_{h\epsilon}. \tag{103.6}$$

Isometry alone does not assert (103.6): preserving the norm and preserving a distinguished unit vector are distinct properties.

103.5 Symmetries and readers on the common domain

A coherent reversible symmetry of the tree transports all emissions and their multiplicities. Invariance of multiplicities in section 6.4.2 proves that it preserves degrees and cylinder probabilities. It may map Ω_x to Ω_{gx}; only when it fixes the prefix does it act within the same prolongation fibre.

Therefore

$$(K_g f)(g\omega) = f(\omega)$$

defines a unitary map $L^2(\Omega_x, \mu_x) \to L^2(\Omega_{gx}, \mu_{gx})$. Naturality of truncation implies $K_{g,n+1}R_n = R'_n K_{g,n}$, and the corresponding identity for expectations follows by taking adjoints. In memory fibres, one additionally requires the square

$$S_{g,h\epsilon} I_\epsilon = I_{g\epsilon} S_{g,h}$$

with $S_{g,h}$ unitary. This is the intertwiner that turns a tree symmetry into a symmetry of this memory representation; it is not attributed merely by name to every subsequent exceptional group.

For the double readout, take the initial gauge and its prefix transport specified in section 21.11.1 y proposition 21.55. Work on a common multisection Ω_x where the readers are defined; for the Archimedean reader, this includes the nested compact intervals and diameter contraction specified in (10.49). This is a premise concerning the reader realization, rather than a new existence criterion for the joint constructor. The finite identities are

$$r_{n,m}Q_n = Q_m\rho_{n,m}, \qquad a_{n,m}R_n^{\mathrm{arq}} = R_m^{\mathrm{arq}}\rho_{n,m}. \tag{103.7}$$

They determine the joint readout at the limit. If $L = (\mathrm{ev}_{\mathrm{arq}}, Q_\infty)$ and $\nu = L_*\mu_x$, then

$$C_L : L^2(Y, \nu) \longrightarrow L^2(\Omega_x, \mu_x), \qquad C_L f = f \circ L,$$

is an isometry, since equality of norms is the definition of pushforward measure. It preserves the unit and its ranges are the observables measurable with respect to the information published by L. For compatible finite readers, (103.7) also implies consistency of the pushforward measures.

This isometry does not imply that the reader distinguishes every state. The construction in proposition 21.55 distinguishes the fact that the same visible stream with different hidden memory may have the same incidence. The adjoint of C_L is expectation with respect to the reader; $C_L C_L^*$ preserves exactly this observable part. If an enlarged realization (L, M) is injective and Borel, with a measurable inverse on its image—for example, a continuous injection from the compact history space into an appropriate Hausdorff space—its pullback does identify the whole of L^2. The required memory is the memory already preserved in the state, not a new retrospective selection.

103.6 Passage-to-the-limit theorem for the observable/memory charge

This section specifies the types in the connection with a memory conservation identity. Canonical refinement of observables supplies the required isometric maps. An additional realization in memory fibres uses the Gram criterion of (103.5).

Let $(\mathscr{H}_n, R_n)$ and $(\mathscr{K}_n, S_n)$ be two inductive systems of Hilbert spaces with isometric inclusions. For each n, let

$$J_n : \mathscr{H}_n \to \mathscr{K}_n \text{ isometric}, \quad P_n = J_n J_n^*, \quad U_n \text{ unitary on } \mathscr{K}_n,$$

and let $\widetilde{A}_n$ be bounded and self-adjoint, with $\sup_n \|\widetilde{A}_n\| < \infty$. Define

$$A_n = J_n^* \widetilde{A}_n J_n, \quad T_n = J_n^* U_n J_n, \quad \eta_n = (I - P_n) U_n J_n.$$

Assume conservation and separation at every level,

$$U_n^* \widetilde{A}_n U_n = \widetilde{A}_n, \qquad [\widetilde{A}_n, P_n] = 0, \tag{103.8}$$

and the following refinement squares:

$$J_{n+1} R_n = S_n J_n, \qquad P_{n+1} S_n = S_n P_n, \tag{103.9}$$

$$U_{n+1} S_n = S_n U_n, \qquad \widetilde{A}_{n+1} S_n = S_n \widetilde{A}_n. \tag{103.10}$$

Theorem 103.2. *These families induce bounded operators on the inductive limits, with J_∞ isometric and U_∞ unitary, and preserve*

$$A_\infty = T_\infty^* A_\infty T_\infty + \eta_\infty^* \widetilde{A}_\infty \eta_\infty. \tag{103.11}$$

The same identity holds at every finite depth. If $\widetilde{A}_n \geq 0$, the retained charge is positive. Without positivity, a self-adjoint form is preserved, rather than necessarily a positive energy.

Proof. Decompose $U_n J_n = J_n T_n + \eta_n$. The two terms belong to $P_n \mathscr{K}_n$ and $(I - P_n)\mathscr{K}_n$. Commutation in (103.8) makes the cross terms of the form $\widetilde{A}_n$ vanish. Applying the other identity in (103.8),

$$A_n = J_n^* U_n^* \widetilde{A}_n U_n J_n = T_n^* A_n T_n + \eta_n^* \widetilde{A}_n \eta_n.$$

The second equality in (103.9), together with the first, implies $J_{n+1}^* S_n = R_n J_n^*$: write $P_{n+1} S_n = S_n J_n J_n^* = J_{n+1} R_n J_n^*$ and multiply by J_{n+1}^*. By (103.9)–(103.10),

$$T_{n+1} R_n = R_n T_n, \qquad \eta_{n+1} R_n = S_n \eta_n, \qquad A_{n+1} R_n = R_n A_n.$$

The operators are thus defined on the algebraic unions of the systems. Isometries and the uniform bound allow extension by continuity to the completions. Compatibility of U_n^* also follows from compatibility of U_n, because both levels are unitary; the limit is therefore unitary, not merely isometric. The projections P_n induce a projection whose range is the closure of the union of the ranges of J_n; it is $J_\infty J_\infty^*$. All identities persist first on finite-level vectors and then by density. This proves (103.11). □

Compatibility of J_n alone does not replace the second equality in (103.9). A refinement preserving the observable inclusion while mixing its complement with it may alter the memory defect.

103.6.1 Explicit compatible nonadic realization

Suppose C_n is unitary on $\mathscr{H}_n$, natural with respect to R_n, and A_n is bounded and self-adjoint, uniformly bounded, natural, and satisfies $[A_n, C_n] = 0$. On $\mathscr{K}_n = \mathscr{H}_n^{\oplus 9}$, define

$$J_n v = \tfrac{1}{3}(v, \dots, v), \quad S_n = R_n^{\oplus 9}, \quad U_n(v_1, \dots, v_9) = (C_n v_9, v_1, \dots, v_8), \quad \widetilde{A}_n = I_9 \otimes A_n.$$

The factor $1/3$ normalizes the nine components. The formulas directly verify (103.8)–(103.10), including the projection. The compression is $T_n = (8I + C_n)/9$, and

$$\eta_n^* \widetilde{A}_n \eta_n = A_n - T_n^* A_n T_n = \frac{8}{81}(I - C_n)^* A_n (I - C_n). \tag{103.12}$$

The final equality uses $C_n^* A_n C_n = A_n$. By the theorem, (103.12) also holds at the limit. Nonadic refinement or complete history refinement transports the charge and its memory when C_n and A_n satisfy the indicated squares.

For $A = I$, finite telescoping gives

$$\|v\|^2 = \|T^N v\|^2 + \sum_{j=0}^{N-1} \|\eta T^j v\|^2.$$

Thus, $v \mapsto (\eta v, \eta T v, \dots, \eta T^{N-1} v, T^N v)$ is an isometry. The corresponding charge formula replaces the norms by the forms of A and $\widetilde{A}$. The terminal term is not removed without proving convergence to zero; depth compatibility alone does not prove this temporal stability.

If $D_{N,n}$ denotes this isometry at depth n and temporal horizon N, the preceding squares also prove

$$(S_n^{\oplus N} \oplus R_n) D_{N,n} = D_{N,n+1} R_n.$$

Indeed, each component satisfies $\eta_{n+1} T_{n+1}^j R_n = S_n \eta_n T_n^j$, and the terminal component satisfies $T_{n+1}^N R_n = R_n T_n^N$. Finite-horizon memory accumulation and resolution refinement commute. The same calculation proves equivariance for a symmetry represented by unitaries intertwining $J, U, P, \widetilde{A}$; it does not extend the action to symmetries for which these squares have not been constructed.

Chapter 104

Conservation of observables and the complete record

104.1 Balance of a conserved observable

Let $\mathcal{H}, \mathcal{K}$ be realizations of a readout space and its enriched extension. Let $J : \mathcal{H} \to \mathcal{K}$ be an isometry, $P = JJ^*$, $U : \mathcal{K} \to \mathcal{K}$ a bounded transport, and $\widetilde{A}$ a bounded self-adjoint observable. Suppose

$$U^*\widetilde{A}U = \widetilde{A}, \qquad [\widetilde{A}, P] = 0.$$

We define

$$A = J^*\widetilde{A}J, \qquad T = J^*UJ, \qquad \eta = (I - P)UJ.$$

Proposition 104.1. *The charge decomposes exactly between readout and complement:*

$$\boxed{A = T^*AT + \eta^*\widetilde{A}\eta.} \tag{104.1}$$

Proof. The identity $UJ = JT + \eta$ is an orthogonal decomposition. Expanding $J^*U^*\widetilde{A}UJ$, the cross terms vanish because $\widetilde{A}$ reduces the subspaces $P\mathcal{K}$ and $(I - P)\mathcal{K}$. The left-hand term is A; the two diagonal terms are those in (104.1). This also proves the identity of sesquilinear forms for two distinct states. If $\widetilde{A} \geq 0$, both summands are positive. For $\widetilde{A} = I$, the conservation hypothesis is $U^*U = I$. □

If the readout mixes sectors of the observable, the balance additionally retains

$$T^*J^*\widetilde{A}\eta + \eta^*\widetilde{A}JT.$$

These terms represent interference between the two spaces carrying the charge. The reducing condition stated above ensures their vanishing for the entire decomposition. In particular realizations, they may also vanish for other reasons, for example when $\eta = 0$.

104.2 Specialization to the nonadic connection

Let C be the unitary realization of memory transport on $\mathcal{H}$. On $\mathcal{H}^9$, transport of the nine phases and their uniform inclusion are

$$S_C(v_0, \dots, v_8) = (Cv_8, v_0, \dots, v_7), \qquad Jv = \tfrac{1}{3}(v, \dots, v).$$

The factor $1/3$ follows from normalization of nine copies. The compression and its complement are

$$T = \frac{8I + C}{9}, \qquad \eta v = \frac{1}{27}\big(8(C - I)v, -(C - I)v, \dots, -(C - I)v\big).$$

If $A = A^*$ is bounded and $[A, C] = 0$, the observable $\widetilde{A} = I_9 \otimes A$ satisfies the hypotheses of Proposition 104.1. Therefore,

$$\boxed{A = T^*AT + \frac{8}{81}(I - C)^*A(I - C).} \tag{104.2}$$

The coefficients arise from the distribution among the nine components: $8^2 + 8 = 72$, divided by 27^2, gives $8/81$. The identity is the sesquilinear balance already documented in the spectral article, expressed here for a general observable.

Telescopic iteration gives, for every $N \geq 1$,

$$A = T^{*N} A T^N + \sum_{j=0}^{N-1} T^{*j} \eta^* \widetilde{A} \eta T^j. \tag{104.3}$$

For $A = I$, the map

$$V_N v = (T^N v, \eta v, \eta T v, \dots, \eta T^{N-1} v)$$

is isometric. The step $N \to N+1$ decomposes only the terminal component $T^N v$ into its new readout and complement, retaining every preceding record. Conservation is thus verified at any depth.

Distinction of operations. One has $S_C^9 = I_9 \otimes C$. Iteration T^N describes successive compressions with storage of their complements. Enriched transport without intermediate compressions is S_C^N. Both operations have defined domains and different functions; the proof preserves this distinction.

104.3 Terminal sector and the unlimited-depth record

Let P_{fix} be the orthogonal projection onto $\ker(I - C)$.

Theorem 104.2. *For every unitary transport C, the nonadic compression satisfies*

$$\operatorname*{s-lim}_{N\to\infty} T^N = P_{\mathrm{fix}}, \qquad \boxed{I = P_{\mathrm{fix}} + \sum_{j=0}^{\infty} T^{*j} \eta^* \eta T^j.} \tag{104.4}$$

The series converges in the strong topology. For each bounded self-adjoint observable A commuting with C, one also obtains

$$\boxed{A = P_{\mathrm{fix}} A P_{\mathrm{fix}} + \sum_{j=0}^{\infty} T^{*j} \eta^* (I_9 \otimes A) \eta T^j.} \tag{104.5}$$

Proof. In the spectral representation of the unitary operator C, compression corresponds to $t(z) = (8+z)/9$. On the unit circle,

$$|t(z)|^2 = 1 - \frac{8}{81}|1 - z|^2.$$

Hence $t(z)^N \to 0$ for $z \neq 1$, while $t(1)^N = 1$. Dominated convergence with respect to each vector spectral measure proves $T^N \to P_{\mathrm{fix}}$ strongly. The same argument holds for T^{*N}. Since A is bounded, $T^{*N} A T^N \to P_{\mathrm{fix}} A P_{\mathrm{fix}}$ strongly. Taking the limit in (104.3) gives (104.5), and specializing to $A = I$ gives (104.4). □

In particular,

$$V_\infty v = (P_{\mathrm{fix}} v, \eta v, \eta T v, \eta T^2 v, \dots)$$

is an isometry. It preserves the complete inner product as well as the norm.

Corollary 104.3 (Bilateral memory)**.** *For the shift $C e_n = e_{n+1}$ on $\ell^2(\mathbb{Z})$, $P_{\mathrm{fix}} = 0$: a fixed sequence would be constant, and its only square-summable realization is zero. Consequently, the entire initial norm is distributed through the complement record. This specifies the unlimited behaviour of the bilateral memory realization present in the corpus.*

Corollary 104.4 (Finite calendar)**.** *On a cycle of d positions, the uniform sequence determines a one-dimensional fixed subspace. This sector remains as the terminal component, while the complement accumulates the remainder. The finite calendar and the bilateral record therefore have different terminal components.*

In the bilateral case, strong convergence may coexist with $\|T^N\| = 1$ for every N. The proof assumes no uniform contraction rate. It also distinguishes this compression from the mode $(5/9)C$, whose geometric contraction is uniform and is studied separately in the sources.

104.4 Preservation of symmetry and chart transport

Let R be unitary and define $C' = RCR^*$, $A' = RAR^*$. The explicit compression formula gives

$$T(C')R = RT(C), \qquad \eta(C')R = R^{\oplus 9}\eta(C). \tag{104.6}$$

Proof. The first equality follows by expanding $(8I + RCR^*)R/9$. The second follows from $(C' - I)R = R(C - I)$ in each of the nine components of η. □

Iteration of (104.6) intertwines the finite registers. Moreover, $P_{\mathrm{fix}}(C') = RP_{\mathrm{fix}}(C)R^*$, by transport of the kernel of $I - C$, so the unlimited register is also intertwined. Thus, the balance of observables is covariant when the chart, operator and observable are transported jointly. Symmetries commuting with C additionally act in the same fixed chart. This distinction allows exceptional transformations to be incorporated with their effectively transported frames, flags and orientations.

The composition in lemma 11.39, section 21.11.4 y proposition 21.60 uses the record's Hadamard matrix, $H_{12}^2 = 4I$, and the incidence $\mathcal{I}$ of the twelve positions in the 132 hexads:

$$J_H = H_{12}/2, \qquad \mathcal{I}^*\mathcal{I} = 36I + 30\mathbf{1}\mathbf{1}^*, \qquad U_W = \mathcal{I}/6\big|_{\mathbf{1}^\perp}.$$

Both normalized operators are isometric on the indicated domains. The composition $B = U_W J_H$ on $W = J_H(\mathbf{1}^\perp)$ preserves inner products and intertwines the transported representation of M_{12} with the action on the hexads. The integer record K additionally preserves the integral inversion $H_{12}/4$ on its congruence lattice; this inversion and isometric normalization have different functions. The proof, the APP–Witt–Leech prolongation, and their domains are developed here.

In this composition, incidence conservation has concrete operator content. Its extension through other transports uses (104.6) and the effective intertwiners from the sources, respecting their respective groups and domains.

104.4.1 Complete transport of the twelve coordinates of K

Separation of the centred subspace can be completed by also retaining its mean. Let $P_0 = I - \mathbf{1}\mathbf{1}^*/12$, $\mu(k) = \mathbf{1}^*k/12$, and

$$F(k) = \left(\mu(k), \frac{1}{6}\mathcal{I}P_0 k\right) \in \mathbb{R} \oplus E_{\mathrm{inc}}.$$

Equip the codomain with $\|(\mu, y)\|^2 = 12|\mu|^2 + \|y\|^2$. The incidence Gram operator proves

$$\|F(k)\|^2 = \|k\|^2, \qquad \boxed{F^{-1}(\mu, y) = \mu\mathbf{1} + \frac{1}{6}P_0\mathcal{I}^*y.} \tag{104.7}$$

Indeed, $\mathcal{I}^*\mathcal{I}P_0 = 36P_0$, so the inverse formula recovers P_0k, while the first term recovers its mean. The action of M_{12} is trivial on the mean and permutes the hexads; hence F is equivariant. Composing FJ_H transports the normalized signed coordinates. All twelve record coordinates are thereby preserved,

rather than only the centred part. The domain of this reconstruction is the dodecaphasic record; the complete history memory retains its other coordinates and its prolongation.

104.4.2 Specialization to the exceptional operator of order three

The exceptional construction produces the orthogonal operator g_c on the realization of the Leech neighbour, through the twelve APP–Witt fibres of type A_2, and proves $g_c^3 = I$ and $\ker(I - g_c) = 0$. Its construction, intertwining, and lattice preservation are located in section 21.11.11 y section 106.3.

We now construct, as an operator composition of that output with the nine-phase chart,

$$T_3 = \frac{8I + g_c}{9}, \qquad \eta_3 = (I - JJ^*)S_{g_c}J.$$

Corollary 104.5. *This composition satisfies*

$$\boxed{T_3^*T_3 = \frac{19}{27}I, \qquad \eta_3^*\eta_3 = \frac{8}{27}I, \qquad \|T_3^N v\|^2 = \left(\frac{19}{27}\right)^N \|v\|^2.} \tag{104.8}$$

Proof. The vector $(I + g_c + g_c^2)v$ is fixed by g_c; the absence of fixed vectors implies $I + g_c + g_c^2 = 0$. Since $g_c^* = g_c^2$, one has $g_c + g_c^* = -I$. Substitution gives

$$T_3^*T_3 = \frac{65I + 8(g_c + g_c^*)}{81} = \frac{57}{81}I = \frac{19}{27}I.$$

Balance (104.2) with $A = I$ gives the second coefficient. Repeated application of the norm identity gives the power for every N. □

The unlimited record is isometric, and its terminal component tends to zero at the explicit rate in (104.8). Here, the exceptional structure determines the two complementary coefficients; both arise from the polynomial relation of g_c and the nine phases. This is a mathematical composition developed in this research. Its definition neither identifies g_c with Γ_9 nor asserts that S_{g_c} is the canonical temporal operator of TPK: it preserves the distinction between exceptional prolongation and temporal holonomy while composing them explicitly.

104.5 History refinement and compatibility of balances

For enriched histories of depth n, let $p_h(\varepsilon)$ be the probability of a surviving extension, normalized by $\sum_\varepsilon p_h(\varepsilon) = 1$. Canonical refinement on the history bases is

$$R_n\delta_h = \sum_{\varepsilon \in \mathrm{Out}(h)} \sqrt{p_h(\varepsilon)}\, \delta_{h\varepsilon}.$$

Extensions of different histories have different prefixes and disjoint supports. Hence $R_n^*R_n = I$. This is the isometric realization of projective consistency of the measure: the coefficients arise from the survival weights already constructed.

To transport (104.1) between depths, inclusions, projections, transports, and observables are preserved jointly. With refinement isometries $R_n, \widetilde{R}_n$, the required squares are

$$J_{n+1}R_n = \widetilde{R}_nJ_n, \quad P_{n+1}\widetilde{R}_n = \widetilde{R}_nP_n, \quad U_{n+1}\widetilde{R}_n = \widetilde{R}_nU_n, \quad \widetilde{A}_{n+1}\widetilde{R}_n = \widetilde{R}_n\widetilde{A}_n.$$

These squares imply $T_{n+1}R_n = R_nT_n$ and $\eta_{n+1}R_n = \widetilde{R}_n\eta_n$. With uniform bounds, the identities pass to the inductive limit of the realizations. Histories, by contrast, form the projective system of prefixes; the opposite directions of the two types of maps remain distinguished. The measure, Gram, and passage-to-the-limit proofs are included.

104.6 Variational specialization: action and orientation

The elliptic action in section 24.3 y chapter 109 supplies a physical specialization of the principle, with parameters constructed at their HMT stages. In its canonical coordinates,

$$H_\eta(Q,P) = \frac{\omega}{2}\left(e^{-\eta}Q^2 + e^{\eta}P^2\right), \qquad I_\eta = H_\eta/\omega.$$

The canonical transformation $Q = e^{\eta/2}q$, $P = e^{-\eta/2}p$ turns I_η into $(q^2+p^2)/2$. Rotations of (q,p), transported to (Q,P), form a continuous symmetry. For the Lagrangian $L = P\dot{Q} - \omega I_\eta$, its Noether charge is I_η. On the source contour $I_\eta = \hbar$, the oriented action is $\oint P\,dQ = 2\pi\hbar = h$, with the orientation of the Hamiltonian orbit. The proof distinguishes the symplectic and antisymplectic transformations: they can preserve the same energy while acting with different signs on oriented action. The full variational proof is presented in chapter 109.

104.6.1 Updating the form and the connection term

The form matrix $M_\eta = \mathrm{diag}(e^{\eta/2}, e^{-\eta/2})$ allows development of transport between ellipses with different parameters. Fix $J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $R(\theta) = \exp(-\theta J_2)$, the Hamiltonian rotation in the (q,p)-plane. The transport

$$z' = M_{\eta'} R(\delta\theta) M_\eta^{-1} z, \qquad z = (Q,P)^{\mathsf{T}},$$

is symplectic and satisfies exactly $I_{\eta'}(z') = I_\eta(z)$. The proof consists of moving both states to their normalized coordinates and using preservation of q^2+p^2 under rotation. It preserves action while changing the form.

For a differentiable realization $\eta(t)$, with angular advance $\dot{\theta} = \omega$, differentiation of this transport gives

$$\dot{Q} = \omega e^{\eta} P + \tfrac{1}{2}\dot{\eta} Q, \qquad \dot{P} = -\omega e^{-\eta} Q - \tfrac{1}{2}\dot{\eta} P.$$

Its Hamiltonian generator, obtained from these two equations, is

$$\boxed{H_{\mathrm{trans}} = \omega I_\eta + \frac{\dot{\eta}}{2} QP.} \tag{104.9}$$

The second summand is the connection term for the varying form. With the convention $\{f,g\} = \partial_Q f \partial_P g - \partial_P f \partial_Q g$,

$$\partial_t I_\eta = \frac{\dot{\eta}}{2}(-e^{-\eta}Q^2 + e^{\eta}P^2), \quad \left\{I_\eta, \frac{\dot{\eta}}{2}QP\right\} = \frac{\dot{\eta}}{2}(e^{-\eta}Q^2 - e^{\eta}P^2).$$

The two contributions cancel, and $dI_\eta/dt = 0$. For the prescribed transport, (104.9) is unique up to a function of time. The differentiable realization is a declared specialization of transport between forms; the preceding discrete result applies directly between two generated parameters. The specific TPK update retains its own rule for selecting these parameters and their times.

This gives a precise connection: an observable conserved by the complete dynamics is distributed between readout and memory through (104.1). When this observable is the generator of a continuous symmetry of the constructed action, the distributed quantity is a Noether charge. The norm, the symmetry charge, and the energy functional are related through their specific operators.

An effective bounded specialization is obtained from the 108-state calendar. Its realized Hamiltonian is $H_{\mathrm{clk}} = \hbar\omega_0 N_{108}$, with $N_{108} = \sum_{k=0}^{107} k P_k$ and P_k its Fourier projectors. The clock action has charge $\hbar\langle\psi, N_{108}\psi\rangle = E_{\mathrm{clk}}/\omega_0$. Transports and inclusions satisfying the hypotheses of (104.1) distribute this charge between readout and complement. All operators here are finite and bounded. The elliptic action of

(104.9) has been proved in its classical symplectic realization; a quantization with unbounded observables additionally retains its own domain conditions.

Chapter 105

Recovery of the state and its dynamics

105.1 Recoverable representation of transport

Let C be the unitary realization of memory transport on $\mathcal{H}$. The nine-phase chart, its uniform inclusion and its compression are

$$S_C(v_0,\dots,v_8) = (Cv_8, v_0, \dots, v_7), \qquad Jv = (v,\dots,v)/3, \qquad T = (8I + C)/9.$$

Let $\eta = (I - JJ^*)S_C J$. Its action is

$$\eta v = \frac{1}{27}\big(8(C-I)v, -(C-I)v, \dots, -(C-I)v\big).$$

Normalization of nine copies determines $1/3$; the distribution of the seam determines $8/81$ in

$$I - T^*T = \eta^*\eta = \frac{8}{81}(I-C)^*(I-C). \tag{105.1}$$

Write $P = \operatorname{proj}\ker(I-C)$. The preceding theorem proves strong convergence $T^N \to P$. Therefore,

$$\mathcal{V}v = (Pv, \eta v, \eta T v, \eta T^2 v, \dots) \quad \text{in} \quad \mathcal{R} = P\mathcal{H} \oplus \ell^2(\mathbb{N}_0; \mathcal{H}^9) \tag{105.2}$$

is an isometry. Denote its closed image by $\mathcal{M}$.

On the record space, define the block shift

$$\mathcal{B}(a, m_0, m_1, m_2, \dots) = (a, m_1, m_2, \dots). \tag{105.3}$$

The first component preserves the fixed sector; the remaining components are the complete vector records, including their correlations.

Theorem 105.1. *The original transport has the exact representation*

$$\boxed{\mathcal{V}C = (9\mathcal{B} - 8I)\mathcal{V}.} \tag{105.4}$$

The restriction $C = (9\mathcal{B} - 8I)|_{\mathcal{M}}$ is unitary on $\mathcal{M}$, and

$$C = \mathcal{V}^* C \mathcal{V}. \tag{105.5}$$

Proof. Since $PT = P$, definition (105.2) gives, componentwise,

$$\mathcal{V}Tv = (Pv, \eta T v, \eta T^2 v, \dots) = \mathcal{B}\mathcal{V}v.$$

The algebraic identity $C = 9T - 8I$ implies (105.4). The isometry $\mathcal{V} : \mathcal{H} \to \mathcal{M}$ is surjective by definition of its image. Equality (105.4) identifies the restriction with $\mathcal{V}C\mathcal{V}^{-1}$, which is unitary because C is unitary. This also proves $C\mathcal{M} = \mathcal{M}$ and (105.5). □

Restriction to the subspace of compatible registers is essential: $9\mathcal{B} - 8I$ on all of $\mathcal{R}$ is not unitary. The theorem characterizes transport on registers that actually arise from the same linear trajectory of the documented chart.

Corollary 105.2. *For $v, w \in \mathcal{H}$ and $k \in \mathbb{Z}$,*

$$\langle C^k v, w\rangle = \langle C^k \mathcal{V} v, \mathcal{V} w\rangle. \tag{105.6}$$

All cross moments of the realized dynamics are thus preserved. Negative powers belong to the inverse of C on $\mathcal{M}$, rather than to an inversion of the unilateral shift on the ambient space.

Corollary 105.3. *For every bounded observable A, its realization on the image is $\widehat{A} = \mathcal{V} A \mathcal{V}^*|_{\mathcal{M}}$. Products, adjoints, and commutators are preserved. In particular,*

$$[A, C] = 0 \quad \Longleftrightarrow \quad [\widehat{A}, C] = 0. \tag{105.7}$$

The proof uses $\mathcal{V}^*\mathcal{V} = I$ and the fact that $\mathcal{V}\mathcal{V}^*$ is the identity on $\mathcal{M}$. Symmetry conservation is expressed through complete transport of its operators, in addition to an equality of norms.

105.2 Explicit reconstruction and record depth

Let $m_j = \eta T^j v$. The telescopic identity gives, for each N,

$$\boxed{v = T^{*N} T^N v + \sum_{j=0}^{N-1} T^{*j} \eta^* m_j.} \tag{105.8}$$

Passing to the strong limit,

$$\boxed{v = P v + \sum_{j=0}^{\infty} T^{*j} \eta^* m_j.} \tag{105.9}$$

Partial reconstruction with fixed terminal component, $v_N = Pv + \sum_{j<N} T^{*j} \eta^* m_j$, has the exact error

$$v - v_N = T^{*N} T^N (I - P) v. \tag{105.10}$$

The series converges for every state. If the complete record has an error δr measured in the norm of $\mathcal{R}$, its reconstruction through $\mathcal{V}^*$ changes by at most $\|\delta r\|$ in norm, because $\|\mathcal{V}^*\| = 1$. This is uniform stability of the complete readout.

105.2.1 Difference between finite and unlimited depth

For the bilateral shift $C e_k = e_{k+1}$ on $\ell^2(\mathbb{Z})$, one has $P = 0$. Consider

$$v_L = L^{-1/2} \sum_{k=1}^{L} e_k.$$

Then $\|(I - C) v_L\|^2 = 2/L$. Commutation of T with $I - C$, contractivity of T, and (105.1) imply

$$\sum_{j=0}^{N-1} \|\eta T^j v_L\|^2 \le \frac{16 N}{81 L}. \tag{105.11}$$

For any fixed depth N, the right-hand side tends to zero as $L \to \infty$, although $\|v_L\| = 1$. The fixed-depth memory record, separated from its terminal component, has no uniform lower bound. By contrast, the unlimited record has norm exactly one for each v_L.

The difference corresponds to two distinct quantifiers: every state is preserved upon completion of its record; a prescribed depth does not uniformly recover every state without its terminal component. Conservation at arbitrary depth here has an exact functional expression.

105.2.2 Vector information and intensity

The theorem preserves m_j, not merely $\|m_j\|^2$. For example, in the bilateral realization, $v = e_0$ and $w = e_1$ have identical intensities at every depth because the shift commutes with T and transports η unitarily. Their vector records differ, and (105.9) recovers the two distinct states. Intensities serve the balance; records with their correlations additionally serve reconstruction.

105.3 Naturality of recovery under refinement

Let $\iota_n : \mathcal{H}_n \to \mathcal{H}_{n+1}$ be the refinement isometries obtained from the common emissions, and let C_n be unitary realizations with

$$C_{n+1}\iota_n = \iota_n C_n. \tag{105.12}$$

The relation also holds for adjoints: multiplication by C_{n+1}^* and C_n^* gives $C_{n+1}^*\iota_n = \iota_n C_n^*$. The images are therefore reducing. The polynomial formulas for T_n, η_n, and the strong limit of T_n^N give

$$T_{n+1}\iota_n = \iota_n T_n, \quad \eta_{n+1}\iota_n = \iota_n^{\oplus 9}\eta_n, \quad P_{n+1}\iota_n = \iota_n P_n. \tag{105.13}$$

Let $\mathfrak{I}_n$ be the refinement applying ι_n to the terminal component and $\iota_n^{\oplus 9}$ to every record. Then

$$\boxed{\mathcal{V}_{n+1}\iota_n = \mathfrak{I}_n\mathcal{V}_n, \qquad \mathcal{B}_{n+1}\mathfrak{I}_n = \mathfrak{I}_n\mathcal{B}_n.} \tag{105.14}$$

Proof. Each component of the first equality is an identity from (105.13), iterated to the corresponding power of T_n. The second equality states that refining every block commutes with shifting the block index, as verified directly. Both maps are bounded, so the equalities pass from finitely supported records to their completion. □

By (105.4) and (105.14), reconstruction of the dynamics also commutes with refinement. The genealogical-depth index n and the successive-compression index j remain distinct. The equality expresses their compatibility when they act through the documented squares, rather than identifying them by name.

105.4 Two exact regimes of memory distribution

For a unit state v with $Pv = 0$, define

$$a_N = \|T^N v\|^2, \qquad b_N = \|\eta T^N v\|^2 = a_N - a_{N+1}. \tag{105.15}$$

One has $a_0 = 1$, $a_N \to 0$, $b_N \geq 0$, and $\sum b_N = 1$. The weights b_N describe the distribution of the memory record across depths. This index counts compressions; identifying it with a physical duration would require application of the corresponding temporal reader.

105.4.1 Exceptional transport of order three

The fixed-point-free orthogonal operator g_c satisfies $g_c^2 + g_c + I = 0$. For the explicit composition $C = g_c$, the preceding development obtained

$$T^*T = \frac{19}{27}I, \qquad a_N = \left(\frac{19}{27}\right)^N, \qquad b_N = \frac{8}{27}\left(\frac{19}{27}\right)^N. \tag{105.16}$$

The mean depth, counting the first record as depth one, is

$$\sum_{N\geq 0}(N+1)b_N = \frac{27}{8}. \tag{105.17}$$

This is a result of that composition of the nonadic chart with the already constructed exceptional transport; it preserves the domain and does not identify g_c with the complete temporal holonomy on the basis of its order.

105.4.2 Bilateral record of revolutions

For $Ce_k = e_{k+1}$ and $v = e_0$, the binomial formula determines every term:

$$T^N e_0 = 9^{-N} \sum_{k=0}^{N} \binom{N}{k} 8^{N-k} e_k, \quad a_N = 81^{-N} \sum_{k=0}^{N} \binom{N}{k}^2 64^{N-k}. \tag{105.18}$$

This formula is exact at any depth. The subsequent circular representation gives

$$a_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(\frac{65 + 16\cos\theta}{81} \right)^N \, d\theta. \tag{105.19}$$

Theorem 105.4. *In this bilateral realization,*

$$\boxed{a_N \sim \frac{9}{4\sqrt{2\pi}} N^{-1/2}, \qquad b_N \sim \frac{9}{8\sqrt{2\pi}} N^{-3/2}.} \tag{105.20}$$

Proof. Write $r(\theta) = (65 + 16\cos\theta)/81$ and $q = 8/81$. Near zero,

$$r(\theta) = 1 - q\theta^2 + O(\theta^4), \qquad \log r(\theta) = -q\theta^2 + O(\theta^4).$$

For a sufficiently small fixed neighbourhood, positive constants c_1, c_2 bound $-\log r(\theta)$ between $c_1\theta^2$ and $c_2\theta^2$. Outside this neighbourhood, $r \le \rho < 1$. The substitution $x = \sqrt{N}\theta$, the Gaussian bound, and dominated convergence give

$$\lim_{N\to\infty} \sqrt{N}\, a_N = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-qx^2} \, dx = \frac{1}{2\sqrt{\pi q}} = \frac{9}{4\sqrt{2\pi}}.$$

For the second limit, apply the same procedure directly to

$$b_N = \frac{1}{2\pi} \int_{-\pi}^{\pi} r(\theta)^N (1 - r(\theta)) \, d\theta.$$

In the variable x, $N(1 - r(x/\sqrt{N})) \to qx^2$, locally dominated by a constant times x^2; outside the neighbourhood, the contribution is exponentially small even after multiplication by $N^{3/2}$. Therefore,

$$\lim_{N\to\infty} N^{3/2} b_N = \frac{q}{2\pi} \int_{\mathbb{R}} x^2 e^{-qx^2} \, dx = \frac{1}{4\sqrt{\pi q}} = \frac{9}{8\sqrt{2\pi}}.$$

This proof of the second limit uses its own integral, without differentiating an uncontrolled asymptotic equivalence. □

By Tonelli's theorem and the tail-sum identity for a positive sequence,

$$\sum_{N\ge 0} (N+1) b_N = \sum_{N\ge 0} a_N = +\infty. \tag{105.21}$$

The entire norm is preserved in memory, but its mean depth diverges for this bilateral state. The distribution law depends on the effective transport representation: the exceptional sector of order three has a geometric rate, whereas the bilateral record considered here has an algebraic rate.

105.5 Composition with Euler's identity and incidence

The extended Euler identity of Section 10.47 realizes the three regimes through the Coxeter element of A_2, a nilpotent transport and F_{av}^2, with their declared antecedents. Their normalized generators satisfy $J_\tau^2 = -\tau I$ and have trace zero. In their two-dimensional charts,

$$C_\tau(t) = \exp(tJ_\tau), \qquad \Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C_\tau(t)^{\mathsf{T}} \Omega C_\tau(t) = \Omega.$$

Zero trace gives $\det C_\tau(t) = 1$, and the two-dimensional identity $M^{\mathsf{T}} \Omega M = (\det M)\Omega$ proves the final equality. For

$$T_\tau = \frac{8I + C_\tau}{9}, \qquad D_\tau = \frac{\sqrt{8}}{9}(C_\tau - I),$$

one obtains a common oriented conservation law:

$$\boxed{\Omega = T_\tau^{\mathsf{T}} \Omega T_\tau + D_\tau^{\mathsf{T}} \Omega D_\tau.} \tag{105.22}$$

Proof. Expanding the two terms, the cross contributions of C_τ have opposite coefficients. The remaining terms are $72\Omega/81 + 9C_\tau^{\mathsf{T}} \Omega C_\tau/81 = \Omega$. □

This alternating form corresponds to oriented area in the canonical plane. Its preservation permits comparison of the three regimes while respecting their differences. For the concrete representatives given by the Coxeter operator C, $I + N$, and F_{av}^2, respectively,

$$(\det T, \det D) = \left(\frac{19}{27}, \frac{8}{27}\right), \qquad (1, 0), \qquad \left(\frac{89}{81}, -\frac{8}{81}\right). \tag{105.23}$$

The sum is one in all three cases. In the parabolic case, D has rank one and may preserve information although its area is zero. In the hyperbolic case, the complement has the opposite orientation; the balance concerns signed area. Theorem 105.1 uses a unitary, positive realization: the hypotheses of each result are expressly retained.

The development of action and variable form proves the version between different charts, its integration over closed curves and its compatibility with refinement. The incidence development composes reconstruction with $F(k) = (\mu(k), \mathcal{I} P_0 k/6)$, retaining the twelve coordinates of K; it also transports the operators and their local frames. Thus, the same conservation identity admits typed realizations of norm, incidence and oriented action.

Chapter 106

Exceptional incidence and transport of forms

106.1 Generation of the record and conservation of incidence

106.1.1 Dodecaphasic record of the enriched history

Section 11.4.2 defines

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

It preserves the window, subpath, orientation, carry, and incidence ledger. Section 11.4.6 fixes

$$x_{\mathrm{term}} = (\mathcal{U}_{108} \cdots \mathcal{U}_1) x_{\mathrm{seed}}, \quad \mathcal{U}_t = \mathsf{Upd}_t \mathsf{Tra}_t \mathsf{Sel}_t,$$

$$R_{\mathrm{sgn}} = \Pi_H \mathcal{E}_{108}^{90,120} \mathcal{R}_{12} : \mathcal{X}_{\mathrm{TPK}}^{\mathrm{term,mem}} \to \mathbb{Z}^{12}.$$

The following algebraic reconstruction begins with this typed output. It does not replace evaluation of the preceding events or obtain their coordinates by imposing K or alpha as a target.

106.1.2 Integral normalization and isometric normalizations

For the step-three orbits $(1, 4, 7, 10)$, $(2, 5, 8, 11)$, $(3, 6, 9, 12)$, Lemma 11.39 fixes the symmetric matrix $H = H_{12}$ and proves $H^2 = 4I$.

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : Hu \equiv 0 \pmod 4\}, \qquad \mathscr{K}(u) = \tfrac{1}{4} Hu, \qquad \mathscr{K}^{-1}(k) = Hk.$$

This is an integral isomorphism between the sublattice of signed records and $\mathbb{Z}^{12}$. The factor $1/4$ implements inversion, not a Euclidean isometry.

As an immediate consequence of the same identity, $J_H = H/2$ satisfies $J_H^* J_H = I$: this is the real isometric transport. If $u = Hk$, then $u/2 = J_H k$ and $\|u\|^2 = 4\|k\|^2$. Confusing $H/4$ with $H/2$ would change the balance by a factor of four.

In section 21.11.4 y proposition 21.60, the constructed design determines

$$I : \mathbb{R}^{12} \to \mathbb{R}^{132}, \qquad (Iv)(H') = \sum_{p \in H'} v_p, \qquad I^* I = 36 I_{12} + 30 J_{12}.$$

The coefficients arise from the incidences: each point belongs to 66 hexads, and each pair to 30. On $V = \mathbf{1}^{\perp}$, the term J_{12} vanishes, so $U_W = I/6 : V \to E_{\mathrm{inc}}$ is an isometry. The factor $1/6$ normalizes this screen; it is not the inverse map for K.

106.1.3 Prefix refinement and event partitions

Section 21.11.1 constructs causal frames $f_{j+1} = g_j(x) f_j$ and prefix encodings Q_n. It proves

$$r_{n,m} Q_n = Q_m p_{n,m}, \qquad \gamma_{A^f}(w f^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}).$$

The proof uses the fact that a prolongation preserves preceding words, transports, and frames. The inverse limit produces Q_∞. Recovery of the visible word follows from the first component of γ; the text expressly distinguishes this scope from recovery of all deep memory and passage to supports.

The additive and affine composition of the record, described in section 11.4.2 y proposition 6.22, proves a second conservation property: with compatible prospective contributions $(\delta w_t, \delta q_t)$, record

inversion is additive. Concatenating segments or subdividing the same list of events gives the same final record. The version with nontrivial transport uses the affine cocycle. Preserving the total under a change of partition does not assert that the record stops growing when new events are added.

106.2 K–Witt transport of norms, operators, and balances

A formalization of the recovered results is assembled here, together with the algebraic proof below. This assembly attributes no new priority to its starting identities.

Let $P_0 = I - J_{12}/12$, $V = P_0\mathbb{R}^{12}$, $W_H = J_H V$, and

$$B := U_W J_H|_{W_H} : W_H \longrightarrow E_{\mathrm{inc}}.$$

Since $J_H^2 = I$, this is an isometric isomorphism between eleven-dimensional spaces. For a record k, its normalized signed coordinate is $J_H k$. If $\widehat{P}_0 = J_H P_0 J_H$,

$$B\widehat{P}_0(J_H k) = U_W P_0 k,$$

$$\|\widehat{P}_0(J_H k)\|^2 = \|U_W P_0 k\|^2 = \sum_{p=1}^{12} k_p^2 - \frac{(\sum_p k_p)^2}{12}.$$

No constant value has been introduced to obtain this equality: it follows from H^2 and the incidence Gramian. The removed mean must be retained separately to recover the whole of k.

For $g \in \mathrm{Aut}(\mathcal{H})$, let $\rho_{12}(g)$ be its point permutation and $\rho_{\mathrm{hex}}(g)$ its hexad permutation. The action on signed coordinates is

$$\widehat{\rho}_H(g) = J_H \rho_{12}(g) J_H = \tfrac{1}{4} H \rho_{12}(g) H.$$

It is orthogonal, preserves W_H, and also preserves the sublattice $\mathcal{L}_H$: if $u = Hk$, its image is $H\rho_{12}(g)k$. The explicit intertwining in section 21.11.4 gives

$$B\widehat{\rho}_H(g) = \rho_{\mathrm{hex}}(g) B.$$

Proof. Substitute the definitions, cancel J_H^2, and apply $U_W \rho_{12}(g) = \rho_{\mathrm{hex}}(g) U_W$. The integral record, centred inner product, and incidence action are thus preserved, each on its own domain. □

For an operator A already constructed on W_H, one obtains $A_{\mathrm{inc}} = BAB^*$. By substitution,

$$BA = A_{\mathrm{inc}} B, \quad \langle z, Az\rangle = \langle Bz, A_{\mathrm{inc}} Bz\rangle.$$

Therefore, if a transport C on W_H preserves A, $C^*AC = A$, its transport $C_{\mathrm{inc}} = BCB^*$ preserves A_{inc}. The condition that a particular C be a publication of TPK dynamics remains in the causal composition producing it. Isometry does not by itself transform an incidence permutation into the nonadic cycle.

Commutation need not be imposed to compare transported charts. If R is unitary, $C' = RCR^*$ and $T(C) = (8I + C)/9$, then $T(C')R = RT(C)$ by substitution. Together with $A' = RAR^*$, the balance in one chart is conjugated to that in the other. Commutation $[R, C] = 0$ corresponds only to the additional problem of keeping that same seam C fixed. This distinction agrees with the lattice naturality of Section 106.3.

One available concrete finite action is the star selected by the flag:

$$\sigma_{B_K} = (1\ 3)(2\ 11)(5\ 7)(8\ 12).$$

Section 21.11.6 proves its construction and distinguishes it from the group generated by the 495 stars. The individual star gives C_2; the family gives M_{12}. Transporting the marking and origin gives covariance between frames. If a symmetry of a fixed frame is sought, the appropriate domain is its stabilizer, not automatically all of M_{12}. If it must additionally commute with C, the subgroup satisfying the specific equation $[\widehat{\rho}_H(g), C] = 0$ is used.

106.3 Order-three action on the lattice fibres

The construction in sections 21.11.11 y 21.11.12 produces twelve A_2 fibres labelled by $(i, \mu, \varepsilon) \in \mathbb{Z}/3\mathbb{Z} \times \{0,1\} \times \{0,1\}$. The alignment $\lambda(i,\mu,\varepsilon) = 4i + 2\mu + \varepsilon \pmod{12}$ is bijective. If C_{A_2} is the Coxeter operator and R the reflection satisfying $RC_{A_2}R = C_{A_2}^{-1}$, the trivializations and transported frames fix

$$g_b = \iota_b^{-1} P_b C_{A_2} P_b^{-1} \iota_b, \quad T_{i\mu\varepsilon} = \iota_{\lambda(i,\mu,\varepsilon)}^{-1} P_{\lambda(i,\mu,\varepsilon)} R^{\mu},$$

$$g_\lambda T_{i\mu\varepsilon} = T_{i\mu\varepsilon} C_{A_2}^{(-1)^{\mu}}.$$

The direct sum $T_\lambda : W_{\mathrm{APP}} \otimes \mathbb{C} \to \bigoplus_{b=1}^{12} (A_{2,b} \otimes \mathbb{C})$ is a C_3-equivariant isomorphism. The proof consists of conjugation and the bijection of the twelve labels: it preserves pair, sheet, and orbit. It is not a coincidence of dimensions.

The constructions in equation (21.147) y proposition 21.73 use $q^\star = (1,1,1,1,1,1)$ and its full-support word $c^\star = (q^\star, q^\star A_W)$ to orient the twelve blocks. The resulting g_c has order three, has no nonzero fixed vectors, and preserves the Niemeier gluing. For the incidence vector v of squared norm 54,

$$N_0 = \ker(\langle \cdot, v \rangle \bmod 3), \qquad N_v = N_0 + \mathbb{Z}\frac{v}{3},$$

$$y_* = (g_c v - v)/3 \in N_0, \qquad z_* = (g_c^{-1} v - v)/3 \in N_0.$$

The proof identifies their discriminant classes as $c^\star$ and $-c^\star$, and their pairings with v as -27. Hence $g_c N_0 = N_0$ and $g_c(v/3) = v/3 + y_*$, giving $g_c N_v = N_v$. The preserved structure includes the gluing, congruence kernel, and class generating the Leech neighbour.

Prolongation to the 24 full-support orientations and the lattice lift of a monomial code automorphism h are developed in Section 21.11.19. With all marks transported,

$$g_{hu} = \widetilde{h}\, g_u\, \widetilde{h}^{-1}.$$

This C_3 is an admissible group genuinely intertwined with the APP fibres and exceptional structure. It is not identified here with C_9 or its quotient by a mere relation of orders. Nor is the lattice lift replaced by the ternary matrix without its action on the A_2 fibres.

106.4 Refinement of energy and its differential

The energy operator of (128.1) has the form $D_{9,m}^{*} W_m D_{9,m}$, with U_a transporting orientation and carry. The domains in section 128.1 are

$$\mathcal{H}_m = \bigoplus_{v} H_v, \quad \mathcal{K}_m = \bigoplus_{a \in \mathcal{E}_m^{+}} H_{t(a)}, \quad (D_m f)_a = f_{t(a)} - U_a f_{s(a)}, \quad E_m = \langle D_m f, \mathsf{W}_m D_m f \rangle.$$

The fixed maps J_m and K_m satisfy, for the family C^1 considered,

$$D_{m+1}(t)J_m = K_m D_m(t), \qquad K_m^{*}\mathsf{W}_{m+1}(t)K_m = \mathsf{W}_m(t).$$

The result in theorem 128.5 proves $E_{m+1}(J_m f) = E_m(f)$ by substitution, and equality of differentials follows by differentiating both identities. For parameter-dependent refinement, the term $D_{m+1}\dot{J}_m f$ is also retained.

With $v_a = U_a f_s$ and $q = \mathsf{W}_m D_m f$, the response to $\dot{U}_a = A_a U_a$ is

$$\mathfrak{j}_m(A) = -2\,\mathrm{Re}\sum_a \langle q_a, A_a v_a\rangle, \quad \mathcal{J}_a = v_a q_a^{*} - q_a v_a^{*}.$$

The transport product $U_\gamma = U_N \cdots U_1$ satisfies

$$\dot{U}_\gamma U_\gamma^{-1} = \sum_{k=1}^{N} B_k A_k B_k^{-1}, \qquad B_k = U_N \cdots U_{k+1}.$$

This formula retains the order of each incidence and transports its covector contragrediently. Corollary 129.5 composes this differential with connection variations and proves, for compatible refinements of the action,

$$s_m = P_m^{\mathsf{T}} s_{m+1}, \qquad M_m \sigma_m = P_m^{\mathsf{T}} M_{m+1}\sigma_{m+1}.$$

Here, P_m carries connection variations, and M_m is the cellular pairing; they are neither the mean-zero projector nor the Hadamard matrix.

106.4.1 Assembly with incidence

In a realization of fibres W_H whose frame actions are $\widehat{\rho}_H(g_a)$, conjugation by B carries every transport, field, and weight to E_{inc}. By the intertwining proved in Section 106.2, the difference operators satisfy $D_{\mathrm{inc}}B_{\mathrm{vertices}} = B_{\mathrm{edges}}D_H$. Transporting J_m, K_m, and W_m as well, the two preceding refinement squares remain the same conjugated squares. By substitution, they preserve energy; for fixed B, they also preserve the differential and covectors.

The balance law and its covectors are transported to the exceptional representation while retaining compatibility with refinement. Each map uses the transports of the specified dynamics and the subspaces they preserve. The variation domain is defined in section 128.1.

Chapter 107

Incidence reconstruction and covariance

107.1 The complete isometry and its domain

Let $H = \mathbb{R}^{12}$, with the Euclidean inner product; let $P_0 = I - \mathbf{1}\mathbf{1}^*/12$, $\mu(k) = \mathbf{1}^* k/12$, and let $\mathcal{I} : H \to \mathbb{R}^{132}$ be the incidence matrix of section 21.11.4 y proposition 21.60. Write $E_{\mathrm{inc}} = \mathcal{I}(\mathbf{1}^{\perp})$ and

$$Y = \mathbb{R} \oplus E_{\mathrm{inc}}, \qquad \langle (\mu, y), (\nu, z) \rangle_Y = 12\mu\nu + \langle y, z \rangle.$$

The complete isometry constructed in Section 104 is

$$F : H \longrightarrow Y, \qquad Fk = (\mu(k), \mathcal{I} P_0 k/6),$$

$$F^{-1}(\mu, y) = \mu \mathbf{1} + P_0 \mathcal{I}^* y/6.$$

The Gram identity $\mathcal{I}^*\mathcal{I} = 36I + 30\mathbf{1}\mathbf{1}^*$ proves both inverse compositions and $F^*F = I_H$, $FF^* = I_Y$, with the indicated weighted inner product. Hence F is orthogonal between H and Y. It can be complexified without changing the proofs.

The codomain is the twelve-dimensional space Y, not the whole of $\mathbb{R} \oplus \mathbb{R}^{132}$. Extending an operator from Y to its complement in that ambient space requires an additional choice; this is not part of the uniquely transported dynamics. In normalized signed coordinates, one uses FJ_H, with $J_H = H_{12}/2$. Integral inversion remains $H_{12}/4$ on its sublattice, rather than an alternative metric normalizer.

107.2 Covariant lifting theorem on the image

Theorem 107.1. *For a bounded operator D on H, there exists a unique operator on Y satisfying $D_Y F = FD$. It is*

$$D_Y = FDF^{-1}.$$

The correspondence preserves sums, products, adjoints, norm, positivity, and identity. In particular, it maps unitaries to unitaries and self-adjoint observables to self-adjoint observables. For a unitary C and an observable A,

$$C^*AC = A \quad \Longleftrightarrow \quad C_Y^* A_Y C_Y = A_Y.$$

If R is a unitary chart change, $R_Y = FRF^{-1}$, then

$$F(RDR^{-1})F^{-1} = R_Y D_Y R_Y^{-1}.$$

Proof. The first formula follows by multiplying the intertwining relation by F^{-1}. Uniqueness follows from surjectivity onto Y. The product and adjoint identities follow from $F^{-1} = F^*$, and equality of norms follows from the surjective isometry. The final equality is an associative composition of these maps. No commutation hypothesis between R and D enters. □

For $g \in \mathrm{Aut}(\mathcal{H})$, Proposition 21.60 supplies

$$F\rho_{12}(g) = \big(1 \oplus \rho_{\mathrm{hex}}(g)\big)F.$$

Thus, the chart change has a specific combinatorial action on the hexads. By contrast, for a general unitary D, D_Y is an operator of the incidence representation, not necessarily a permutation of the design. Preserving the inner product of that representation does not turn all its unitaries into combinatorial automorphisms or elements of the Monster.

107.2.1 Covariance under a noncommuting chart change

Let C be the cyclic advance $Ce_p = e_{p+1}$ on the twelve positions, that is, the reverse direction of convention S in section 11.4. It is an operation of the window chart, not the complete temporal holonomy. Let R be the already constructed star

$$R = (1\ 3)(2\ 11)(5\ 7)(8\ 12).$$

Then $RCe_1 = e_{11}$, whereas $CRe_1 = e_4$. They do not commute. Nevertheless, for $C' = RCR^{-1}$, $T(C')R = RT(C)$ and $\eta(C')R = R^{\oplus 9}\eta(C)$ hold exactly, and likewise in Y through F. Chart covariance does work in an example where commutation in a fixed chart fails.

107.3 Explicit recovery from the incidence record of complements

Let C be unitary, $T = (8I + C)/9$, $\eta = (I - JJ^*)S_C J : H \to H^9$, and let P be the projection onto $\mathrm{Fix}(C)$, as in Section 104. Write $F_9 = F^{\oplus 9}$. The incidence version of the finite record is

$$\mathcal{W}_N k = \left(FT^N k, F_9\eta k, F_9\eta T k, \dots, F_9\eta T^{N-1}k\right).$$

Theorem 107.2. *This map is isometric and has the explicit left inverse*

$$\mathcal{W}_N^*(a, b_0, \dots, b_{N-1}) = T^{*N}F^{-1}a + \sum_{j=0}^{N-1} T^{*j}\eta^* F_9^{-1} b_j.$$

In particular, applying this formula to $\mathcal{W}_N k$ recovers k exactly. The infinite record

$$\mathcal{W}_\infty k = \left(FPk, (F_9\eta T^j k)_{j\ge 0}\right)$$

is also isometric, and its left inverse is

$$k = PF^{-1}a + \sum_{j\ge 0} T^{*j}\eta^* F_9^{-1} b_j \quad \text{for } (a, (b_j)) = \mathcal{W}_\infty k.$$

The series converges in norm. For arbitrary square-summable data, the same formula defines the bounded adjoint, although not every datum in the codomain belongs to the image of $\mathcal{W}_\infty$.

Proof. Applying F and F_9 to each component of V_N or V_∞ preserves the inner product. Taking the adjoint of this composition gives the formulas. The telescopic identity in Section 104 proves $\mathcal{W}_N^*\mathcal{W}_N = I$, and its strong limit proves $\mathcal{W}_\infty^*\mathcal{W}_\infty = I$. Convergence of the series follows by applying a bounded operator to truncations of a sequence in a Hilbert direct sum. □

This reconstruction does not preserve k as an intact first coordinate: the first component is the compressed terminal component or its fixed sector; the rest arises from successive complements. When $P = 0$, the incidence complements alone recover k. The contribution recovered from the first N blocks is

$$k_N = \sum_{j<N} T^{*j}\eta^* F_9^{-1} b_j = \left(I - T^{*N}T^N\right)k.$$

The remainder is $T^{*N}T^N k$, which tends to zero if $P = 0$. With a nonzero fixed sector, FPk must be retained; complement memory alone does not recover that sector. Reconstruction rates and dynamics on the block index are developed in Section 105.

If A is self-adjoint and $[A, C] = 0$, the observable form is also preserved:

$$\langle k, Ak\rangle = \langle FPk, A_Y FPk\rangle_Y + \sum_{j\ge 0}\langle F_9\eta T^j k, (I_9 \otimes A_Y)F_9\eta T^j k\rangle.$$

The scalar series converges absolutely because A is bounded and the sum of the squared norms of the components is finite. The hypothesis expresses preservation of A by C; it does not require C to commute with exceptional chart changes.

107.4 Natural lifting of refinement with local frames

The construction in sections 6.4.1 y 6.4.2 supplies sets Z_n of enriched cylindrical states, truncations, and projective measures with full support. For a descendant z' of z,

$$w_n(z'|z) = \sqrt{\mu_{n+1}(z')/\mu_n(z)}, \qquad \sum_{\tau z'=z} w_n(z'|z)^2 = 1.$$

The label z' preserves the ordered record, orientation, carry, boundary, memory, path, and terminal component. Refinement of a state does not identify distinct descendants.

Let $q(z)$ be the integer memory record, C a unitary on H, and R_z orthogonal changes of the fibre frame. For concrete incidence frames, one may use $R_z = \rho_{12}(g_z)$, with $g_z \in \mathrm{Aut}(\mathcal{H})$, transporting their marks. Define

$$G_z = R_z C^{q(z)}, \qquad U_{z',z} = G_{z'} G_z^* = R_{z'} C^{q(z')-q(z)} R_z^*.$$

This construction prolongs the explicit gauge equivalence of section 21.11.1. It does not require $[R_z, C] = 0$ and keeps the order of operations explicit.

On $\mathscr{H}_n = \ell^2(Z_n) \otimes H$,

$$\mathcal{U}_n(e_z \otimes v) = \sum_{\tau z'=z} w_n(z'|z) e_{z'} \otimes U_{z',z} v.$$

Define $\mathscr{F}_n = I_{\ell^2(Z_n)} \otimes F$, with codomain $\mathscr{Y}_n = \ell^2(Z_n) \otimes Y$.

Theorem 107.3. *The refinement is isometric and satisfies*

$$\mathcal{U}_n^Y \mathscr{F}_n = \mathscr{F}_{n+1}\mathcal{U}_n, \quad \mathcal{U}_n^Y = \mathscr{F}_{n+1}\mathcal{U}_n\mathscr{F}_n^{-1}.$$

The formula for each term is the preceding one with $U^Y_{z',z} = FU_{z',z}F^{-1}$. Systems at several depths preserve composition and admit a surjective isometry between their inductive limits. All fibre coefficients are preserved exactly with their labels z.

Proof. Descendants of distinct parents are orthogonal; each $U_{z',z}$ is unitary; and the weights have squared sum one. This proves isometry. Along a trajectory, the intermediate factors $G_z^* G_z$ cancel and the measure ratios telescope. The composition does not depend on the partition of the trajectory. The square with F is verified on $e_z \otimes v$, term by term. The maps and their inverses are compatible isometries, so they extend inversely to the inductive completions. □

107.4.1 Compatible observables and dynamics

If A_0 is a bounded fibre observable, let

$$(\mathscr{A}_n\psi)_z = G_z A_0 G_z^* \psi_z.$$

Then $\mathscr{A}_{n+1}\mathscr{U}_n = \mathscr{U}_n\mathscr{A}_n$, whence $\mathscr{U}_n^*\mathscr{A}_{n+1}\mathscr{U}_n = \mathscr{A}_n$. This is obtained by substituting $U_{z',z} = G_{z'}G_z^*$. This is conservation of a field of observables transported by refinement; it does not assert that A_0 commutes with C in a fixed chart.

The unitary dynamics at each level,

$$(\mathscr{C}_n\psi)_z = G_z C G_z^* \psi_z$$

satisfies $\mathscr{C}_{n+1}\mathscr{U}_n = \mathscr{U}_n\mathscr{C}_n$. Their adjoints are also intertwined: since both $\mathscr{C}$ are unitary, multiply the identity by their inverses. Thus, T, its nonadic complements, and its fixed projections are intertwined. The finite and infinite records of Section 107.3 are natural with respect to this refinement. Upon applying $\mathscr{F}_n$, the same squares hold in the incidence fibres. For $\mathscr{A}_n$ also to be conserved by these dynamics at a fixed level, the precise condition is $[A_0, C] = 0$; covariance between frames still does not require $[R_z, C] = 0$.

107.4.2 Bilateral-memory specialization with varying frames

In the reversible orbit of chapter 105, a coordinate publication can be represented by $\psi = (\psi_m)_{m\in\mathbb{Z}} \in \ell^2(\mathbb{Z}; H)$. Its advance is $(C_0\psi)_m = \psi_{m-1}$. F acts pointwise. For frames $R_m = \rho_{12}(g_m)$, the unitary change $(\mathscr{R}\psi)_m = R_m\psi_m$ produces

$$(C'\psi)_m = R_m R_{m-1}^* \psi_{m-1}.$$

The incidence operator has the same formula with $1 \oplus \rho_{\mathrm{hex}}(g_m g_{m-1}^{-1})$. Neither constant frames nor commutation with the shift is imposed. Since C' is conjugate to C_0, it has no nonzero fixed vectors in this square-summable space. Section 107.3 recovers the complete linear publication from its incidence complements. This concerns the orbit and represented coordinates: not an identification of all TPK histories with $\mathbb{Z}$.

107.5 Direct extension to the twelve A_2 fibres

The exceptional construction preserves twelve A_2 fibres and the order-three operator g_c developed in Section 106.3. Let $V_{A_2} = A_2 \otimes_{\mathbb{Z}} \mathbb{R}$ be their two-dimensional Euclidean realization. The same positional isometry can be tensored over $\mathbb{R}$ with this space:

$$F_{A_2} = F \otimes I_{V_{A_2}} : \mathbb{R}^{12} \otimes_{\mathbb{R}} V_{A_2} \longrightarrow Y \otimes_{\mathbb{R}} V_{A_2}.$$

Its first component is the vector mean, and its second consists of the centred incidence sums, now valued in V_{A_2}. The incidence Gram operator, tensored with the identity, again proves the complete isometry and inverse. Hence $g_c^Y = F_{A_2} g_c F_{A_2}^{-1}$ is orthogonal, of order three, and fixed-point-free. The nonadic composition of Section 104 can be read out and recovered on this V_{A_2}-valued screen through Sections 107.2 and 107.3. The twelve scalars of K are not confused with the twenty-four coordinates of the A_2 fibres, nor is the action turned into a hexad permutation when the lift also acts within the fibres.

107.6 Exact scope with respect to histories

Let $k : X \to H$ be the record reader on a history domain, and let $\ell = F \circ k$. Then

$$\ell(x) = \ell(y) \quad \Longleftrightarrow \quad k(x) = k(y).$$

The complete screen adds zero loss relative to K, but it does not change the fibres of this reader. A map $d : X \to X$ descends to dynamics on the values of K if and only if

$$k(x) = k(y) \Longrightarrow k(dx) = k(dy).$$

The same condition is necessary and sufficient for descent to the incidence image. The proof consists of defining $D(k(x)) = k(dx)$; the condition is equivalent to independence of the representative. If this D is linear and unitary in the stated realization, Section 107.2 applies. If it is nonlinear, F still conjugates it as a map but does not make it unitary.

The criterion distinguishes descent to K from the enriched dynamics. Section 11.4.6 establishes that the reader uses subpath, orientation, carry, and incidences; Section 21.55 distinguishes recovery of the visible word, support, and deep memory. On $\ell^2(Z_n) \otimes H$, Section 107.4 preserves enriched labels and transports the fibre representation. The equivalence acts on the previously generated history domain.

For an already constructed cut, naturality of the record in proposition 11.31 fixes $K_N = K_{108} \circ \tau_{N,108}$: F preserves this naturality exactly. A sequence of new events may update K; the additive or affine composition in section 11.4 preserves independence of partition, not constancy of the value when events are added.

Chapter 108

Affine holonomy and memory symmetries

108.1 Effective composition of memory

For an admissible emission ϵ, the construction in proposition 6.22 defines

$$\mathsf{U}_\epsilon(m) = C_\epsilon m + \kappa_\epsilon, \qquad \kappa_{\epsilon_2\epsilon_1} = \kappa_{\epsilon_2} + C_{\epsilon_2}\kappa_{\epsilon_1}. \tag{108.1}$$

The domain is the memory module at the source; the codomain is that of the next emission. The product preserves order. On a path γ of length r, one obtains

$$C_\gamma = C_r \cdots C_1, \qquad \kappa_\gamma = \sum_{j=1}^{r} C_r \cdots C_{j+1}\kappa_j, \qquad m_\gamma = C_\gamma m_0 + \kappa_\gamma. \tag{108.2}$$

The empty product in the sum is the identity. The proof is induction using (108.1): appending the next emission multiplies the preceding record by its linear transport and adds its increment. The positional boundary may cancel while the transported sum in (108.2) remains.

Proposition 108.1 (Memory class of a return)**.** *Let M be an abelian module, and consider a path closed in the base whose memory transport is $H(m) = Cm + \kappa$, with C an automorphism of M. Its return class is*

$$\mathfrak{m}(H) = [\kappa] \in M/(I - C)M. \tag{108.3}$$

A memory-coordinate change $m' = Rm + b$, with R invertible, transforms the return into

$$C' = RCR^{-1}, \qquad \kappa' = R\kappa + (I - C')b. \tag{108.4}$$

The map induced by R takes $[\kappa]$ to $[\kappa']$. The transport H has a fixed point exactly when its class (108.3) is zero.

Proof. Substituting $m = R^{-1}(m' - b)$ into H gives (108.4). Since $R(I - C) = (I - C')R$, the map R induces an isomorphism between the quotients. The term $(I - C')b$ represents zero in the target quotient. Finally, the fixed-point equation $Cm + \kappa = m$ is equivalent to $(I - C)m = \kappa$. This proves the final assertion. □

Formula (108.3) determines whether a change of origin removes the increment or merely redistributes it among coordinates. In particular, when $C = I$, the class is the vector κ itself. A nonzero translation remains nonzero under every invertible affine change.

Proposition 108.2 (Return symmetry versus a change of representation)**.** *An affine transformation $A(m) = Rm + b$ commutes with H exactly when*

$$RC = CR, \qquad (I - C)b = (I - R)\kappa. \tag{108.5}$$

Proof. The linear parts of AH and HA are RC and CR; their constant parts are $R\kappa + b$ and $Cb + \kappa$. Equating them gives (108.5). If H is merely transported through (108.4), commutation is not required: the same law is obtained in another representation. By contrast, a symmetry of the fixed return requires both equalities in (108.5). □

This precisely distinguishes covariance of the law from the stabilizer of a specific memory. The distinction is relevant to studying variations in state symmetry without modifying the composition laws.

108.2 Application to the effective record of the observable revolution

Section 125 defines two ordered counters: g counts orientation reversals, and s counts effective sheet changes. Their canonical count per 27-step block is

$$c_{27} = (1, 18).$$

The complete observable phase returns after four such blocks. In the path groupoid, the record module is $M = \mathbb{Z}^2$, and

$$c_{108} = (4, 72) = 4v, \qquad v = (1, 18), \qquad H(g, s) = (g + 4, s + 72). \tag{108.6}$$

Identity (108.6) is a corpus result used here; the verifier checks its consequences rather than recounting the original emissions. The symbol c_{108} denotes this two-counter increment. It differs from the dodecaphasic seal K and from the central cocycle of the Heisenberg representation.

Theorem 108.3 (Complete record invariants under full revolutions)**.** *The orbits of H and its inverse on $\mathbb{Z}^2$ are classified exactly by*

$$\boxed{\mathcal{I}(g, s) = s - 18g, \qquad \nu(g, s) = [g]_4.} \tag{108.7}$$

Consequently,

$$\mathbb{Z}^2/\mathbb{Z}(4, 72) \cong \mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}. \tag{108.8}$$

Proof. Adding $(4, 72)$ preserves both expressions in (108.7). If two records have the same invariants, then $g' - g = 4N$ for an integer N, and $s' - s = 18(g' - g) = 72N$; thus they differ by exactly $N(4, 72)$. The map in (108.7) is surjective: given an integer j and r modulo four, choose a representative g of r and set $s = j + 18g$. This proves (108.8). □

The description covers every orbit of this memory reader. Orbits of the complete enriched state additionally preserve their remaining coordinates. For a forward-directed prolongation, a later record is obtained when the preceding integer N is nonnegative; the bilateral equivalence in (108.8) belongs to the groupoid.

The quantity $\mathcal{I}$ is the unique primitive integer linear charge, up to sign, invariant under these translations. Indeed, for $L(g, s) = ag + bs$, invariance requires $4a + 72b = 0$, so $(a, b) = b(-18, 1)$. A revolution accumulates longitudinal memory and preserves a transverse relation determined by the increment.

The factor four has a precise origin here: it is $\gcd(4, 72)$ and, in the preceding count, the number of 27-step blocks completing the observable return. The class ν preserves the information lost when the increment is divided by four and only the primitive direction v is retained.

Corollary 108.4 (Joint block and phase record)**.** *Write the phase of these four blocks as $a \in \mathbb{Z}/4\mathbb{Z}$. The update*

$$(a, g, s) \longmapsto (a + 1, g + 1, s + 18)$$

preserves $(s - 18g, [g - a]_4)$. Both data classify its bilateral orbits: the difference $N = g' - g$ simultaneously satisfies $a' - a = [N]_4$ and $s' - s = 18N$. This brings together memory advance and phase return without replacing either by the other.

108.3 Parabolic generator determined by memory

The ordered pair of counters equips the module M with the unimodular alternating form

$$\Omega_M((g,s),(g',s')) = gs' - sg'.$$

Its domain is the record module. This form is not identified with a dimensional physical area or with the form on the APP torus, whose domain differs. The choice of order (g,s) fixes its orientation.

Theorem 108.5 (Unimodular stabilizer of the increment)**.** *Define, directly from the primitive vector v in (108.6),*

$$N_v m = v\,\Omega_M(v,m) = v\mathcal{I}(m), \qquad N_v = \begin{pmatrix} -18 & 1 \\ -324 & 18 \end{pmatrix}. \tag{108.9}$$

Then $N_v^2 = 0$, $N_v \neq 0$, and

$$\boxed{\mathrm{Stab}_{SL_2(\mathbb{Z})}(c_{108}) = \{I + nN_v : n \in \mathbb{Z}\}.} \tag{108.10}$$

Every transformation in (108.10) preserves Ω_M, the increment c_{108}, and the charge $\mathcal{I}$. For nonzero n, it is a nonidentity unipotent matrix of trace two, of parabolic type.

Proof. The identity $\Omega_M(v,v) = 0$ implies $N_v^2 m = v\Omega_M(v,v)\Omega_M(v,m) = 0$. Its upper-right entry is one, so it is nonzero. The matrix

$$B = \begin{pmatrix} 1 & 0 \\ 18 & 1 \end{pmatrix}$$

has determinant one, maps the first basis vector to v, and satisfies

$$N_v = BNB^{-1}, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{108.11}$$

An integer automorphism fixes $4v$ exactly when it fixes v, because M is torsion-free. A determinant-one matrix fixing the first basis vector necessarily has the form $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$. Conjugation by B proves (108.10). The rest follows from this expression or directly from (108.9). □

Formula (108.9) makes the construction independent of the second vector chosen to complete v to an oriented unimodular basis: any other complement differs by an integer multiple of v and produces the same N_v. Under an oriented transformation $R \in SL_2(\mathbb{Z})$, one has $N_{Rv} = RN_vR^{-1}$.

The action can be written particularly clearly as

$$(I + nN_v)m = m + n\mathcal{I}(m)v. \tag{108.12}$$

The transverse charge determines how far this symmetry shifts along the memory direction. Summing revolutions produces the displacement $4Nv$; stabilizing symmetries preserve this advance law.

On the quotient (108.8), the transformation $I + nN_v$ acts as $(\mathcal{I}, [g]_4) \mapsto (\mathcal{I}, [g + n\mathcal{I}]_4)$. It therefore preserves the family of orbits and may permute its residue classes; ν is an invariant of the return H, not of every stabilizing transformation. This difference follows directly from (108.12).

Conjugation (108.11) supplies an explicit algebraic map from the parabolic TRIT model, defined in proposition 10.47, to the record module extended to real scalars. One also obtains $\exp(tN_v) = I + tN_v$ and $B\exp(tN) = \exp(tN_v)B$. This intertwining concerns operators: the particular motion selected by

TPK also uses its emissions and calendar. The complete arithmetic group in (108.10) fixes this reader; the admissible symmetry group of the total state is obtained by intersecting it with the actions preserving the remaining data and relations.

In the translated system (108.6), affine changes $m \mapsto Rm + b$, with R in (108.10) and arbitrary b, commute with H. Those additionally preserving the fixed initial record must satisfy $Rm_0 + b = m_0$. Distinguishing the two conditions separates symmetry of the law from symmetry of a marked trajectory.

Theorem 108.6 (Variational realization of the charge and stabilizer)**.** *The scalar extension of the record, $M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}$, preserves the generated coordinates (g, s) and the form $\Omega_M = dg \wedge ds$. It is a subsequent realization of the integer module. With the convention $\iota_{X_H}\Omega_M = dH$, one has*

$$X_{\mathcal{I}} = v, \qquad X_{\frac{1}{2}\mathcal{I}^2} = \mathcal{I}v = N_v m. \tag{108.13}$$

Thus, the constructed translation and parabolic family admit explicit Hamiltonian generators. The flow of $\mathcal{I}$ is $m \mapsto m + uv$; at u equal to four it reproduces (108.6). The flow of $\mathcal{I}^2/2$ is $m \mapsto (I + uN_v)m$. The parameter u corresponds to this continuous realization; the discrete return step and intermediate emissions retain their own TPK calendar.

Proof. For a vector $X = (X_g, X_s)$, $\iota_X(dg \wedge ds) = X_g\, ds - X_s\, dg$. Since $d\mathcal{I} = ds - 18\, dg$, the first field is $(1, 18) = v$. The chain rule gives the second. The quantity $\mathcal{I}$ remains constant along both fields because $d\mathcal{I}(v) = 0$. The second flow therefore integrates $dm/du = \mathcal{I}(m_0)v$, equivalent to $\exp(uN_v) = I + uN_v$. Nondegeneracy of Ω_M implies that the Hamiltonian generating N_v is $\mathcal{I}^2/2$ plus an additive constant: its differential is fixed by $\iota_{N_v m}\Omega_M$. □

The mathematical action

$$\mathcal{A}[m] = \int_{u_0}^{u_1} \left[s\frac{dg}{du} - \frac{1}{2}(s - 18g)^2 \right] du \tag{108.14}$$

yields, by variation, the equations $dg/du = \mathcal{I}$, $ds/du = 18\mathcal{I}$. The translation $g \mapsto g + \varepsilon$, $s \mapsto s + 18\varepsilon$ preserves the Hamiltonian and changes the Lagrangian by the total derivative $d(18\varepsilon g)/du$, for constant ε. Its Noether charge is

$$\underbrace{s\,\delta g}_{s} - \underbrace{18g}_{\text{boundary term}} = s - 18g = \mathcal{I}. \tag{108.15}$$

The variation is interpreted per unit of ε. The equations directly verify $d\mathcal{I}/du = 18\mathcal{I} - 18\mathcal{I} = 0$. The discrete translation (108.6) itself is also realized by the Hamiltonian $H_{\mathrm{ret}} = 4\mathcal{I}$, whose flow at parameter one is H; the two Hamiltonians Poisson-commute. The stabilizer flow $I + uN_v$ and the return $m + 4v$ are distinct commuting actions: on the line $\mathcal{I} = 0$, the former fixes every point, while the latter continues to accumulate memory.

More generally, every C^1 Hamiltonian on this plane invariant under translations $m \mapsto m + uv$ has the form $F(\mathcal{I})$. To prove this, the global coordinates $(g, \mathcal{I})$ express the symmetry as translation in g; invariance removes dependence on g from H. Its field is $F'(\mathcal{I})v$. This classification makes explicit both the common conservation law and the additional choice of specific dynamics.

Action (108.14) is dimensionless and belongs to the realized record plane. Its proof supplies an exact variational relation between memory, charge, and stabilizer. Conversion to physical action and dimensional time is performed through the corresponding action and clock readers; these units are not assigned to the counter u.

108.4 APP curvature and conservation of the central phase

Section 4.3.4 works on $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. The sheets are $S(x, y) = x + y$ and $P(x, y) = xy$. The mixed connections use differences of these potentials, not the potentials themselves as links:

$$F(S, P) = \Delta_x\Delta_y P - \Delta_y\Delta_x S = 1, \qquad F(P, S) = -1 \pmod 9. \tag{108.16}$$

The oriented sum of links around a rectangle is ab or $-ab$, depending on order. The alternating area is $\sigma(u, v) = ad - bc$ for $u = (a, b)$, $v = (c, d)$. Section 23.5.2.1 represents these transports through $D(u) = X^a Z^b$, $ZX = \omega XZ$, with ω the ninth root in the subsequent representation. Then

$$D(u)D(v) = \omega^{bc} D(u + v), \qquad D(u)D(v)D(u)^{-1}D(v)^{-1} = \omega^{-\sigma(u,v)} I. \tag{108.17}$$

The position resulting from the commutator returns. The central phase retains oriented area with the sign fixed in (108.17). This is retention modulo nine: an area divisible by nine has trivial central phase, although a lift with quotient and record may distinguish its paths. The phase does not replace that lift.

Proposition 108.7 (Exact covariance of central transport)**.** *Since two is invertible modulo nine, define*

$$W(a, b) = \omega^{ab/2} D(a, b), \qquad W(u)W(v) = \omega^{-\sigma(u,v)/2} W(u + v). \tag{108.18}$$

For $R \in SL_2(\mathbb{Z}/9\mathbb{Z})$, the map

$$\omega^z W(u) \longmapsto \omega^z W(Ru) \tag{108.19}$$

is an automorphism of the central extension. It preserves its products and commutation phases.

Proof. The phase in the product of the two W is $\frac{ab}{2} + \frac{cd}{2} + bc - \frac{(a+c)(b+d)}{2} = \frac{bc-ad}{2}$. This proves (108.18). The identity $\sigma(Ru, Rv) = \det(R)\sigma(u, v) = \sigma(u, v)$ shows that (108.19) preserves the law; invertibility of R supplies the inverse automorphism. If $\det R = -1$, the corresponding automorphism also changes $z \mapsto -z$; the area sign and centre are reversed jointly. □

The complete record and its central phase have different types. Covariance preserves the area law while the presentation changes; the affine class in (108.3) preserves memory content while its origin changes. Both properties act on typed path readers, and their maps must remain separate when composed with the exceptional readout of the complete state.

Proposition 108.8 (Persistence under local frames and subdivisions)**.** *Let γ be a loop based at x, with invertible links U_ϵ. The local change $U'_\epsilon = G_t U_\epsilon G_s^{-1}$ transforms its holonomy into $H'_\gamma = G_x H_\gamma G_x^{-1}$: frames at interior vertices cancel successively. If $H_\gamma = \omega^k I$, the central phase remains exactly the same. For refined links whose product reproduces the preceding link, subdivision likewise preserves H_γ, by associativity. On APP plaquettes, cancellation of interior edges realizes this property through finite Stokes.*

An elementary rectangle has additive Wilson value one and nontrivial central phase. This mixed transport therefore distinguishes actual holonomy from the identity that links of the form $G_t G_s^{-1}$ would produce. Conservation under subdivision concerns compatible loops and links: adding another area changes the transport.

The integer lift of holonomy distinguishes oriented area from its nonadic remainder: for a rectangle with sides four and seven, $28 = 1 + 9 \cdot 3$, whereas $-28 = 8 + 9(-4)$ corresponds to the reverse orientation. The residual phase coincides for areas one and twenty-eight, while their quotients distinguish the accumulations. This example exhibits the difference between finite phase and complete record; its area quotient and the temporal counter q_{mem} retain distinct domains.

108.5 Composition between the three regimes and continuity of the balance

The realizations in propositions 10.47 y 10.48 construct generators $J_k^2 = -\tau_k I$ on their own fibres, of real dimension two. Take effectively specified links $B_k : V_k \to V_{k+1}$ preserving the alternating forms, $B_k^* \Omega_{k+1} B_k = \Omega_k$. The symbol $*$ in this section means transpose/bilinear pullback. The operators $E_k = \exp(t_k J_k)$ preserve Ω_k: in dimension two, zero trace of the generator implies $J_k^* \Omega_k + \Omega_k J_k = 0$. The symplectic identity is the hypothesis that would be retained in extending the lemma to higher dimensions.

For the nonadic compression and its recorded complement,

$$T_k = B_k(8I + E_k)/9, \qquad D_k = B_k\sqrt{8}(E_k - I)/9,$$

algebraic expansion gives

$$\Omega_k = T_k^* \Omega_{k+1} T_k + D_k^* \Omega_{k+1} D_k. \tag{108.20}$$

Indeed, the cross terms cancel, and the numerator is $72\Omega_k + 9E_k^* \Omega_k E_k = 81\Omega_k$. For $P_0 = I$, $P_{k+1} = T_k P_k$, successive substitution in path order gives

$$\boxed{\Omega_0 = P_N^* \Omega_N P_N + \sum_{k=0}^{N-1} (D_k P_k)^* \Omega_{k+1} (D_k P_k).} \tag{108.21}$$

The terminal component and every complement remain present. The identity is oriented and bilinear; positivity results apply in the positive profiles proved in the preceding sections. The conserved form does not require a link between distinct regimes to intertwine their generators: if $BJ_\tau = J_{\tau'} B$, squaring with invertible B would impose $\tau = \tau'$. Conservation of a common structure allows the three regimes to remain distinct.

The complete transport $U_N = (B_{N-1}E_{N-1}) \cdots (B_0 E_0)$ maps V_0 to V_N. Matrix monodromy requires a typed closure $K : V_N \to V_0$; only then is KU_N an endomorphism whose trace can be computed. Compatible basis changes preserve this trace by conjugation. Phase return, fibre closure, and return of the vector remain different conditions.

If the links have exclusively the form $B_{yx} = G_y G_x^{-1}$, their product along a path is $G_{\text{final}} G_{\text{inicial}}^{-1}$. Closing with the same frame gives the identity. The holonomy in (108.17) and the increment in (108.6) exhibit concrete mechanisms that this mere substitution of frames cannot represent by itself.

108.6 Compatibility with refinement and arbitrary extent

The memory action is compatible with a restriction homomorphism of modules $p : M' \to M$ when

$$pC' = Cp, \qquad p\kappa' = \kappa. \tag{108.22}$$

Then $pH' = Hp$, and p induces a quotient map

$$M'/(I - C')M' \longrightarrow M/(I - C)M, \qquad [\kappa'] \longmapsto [\kappa]. \tag{108.23}$$

Proof. The first equality in (108.22) maps $(I - C')M'$ into $(I - C)M$; the second fixes the image of the increment. Compatibility of H follows by substitution. Concatenations and refinements compose the same squares by virtue of law (108.1). □

A nonzero return class appearing at a preserved level remains nonzero in every compatible lift: a zero class would project to zero. This statement applies to the system of memories and restrictions satisfying (108.22); the joint constructor and the maps of its inverse system are preserved in section 6.4.

In the register of (108.6), for each integer N we have $H^N(m) = m + N(4, 72)$. The proof is algebraic for arbitrary N, not an inference from a finite number of iterations. In forward prolongation, each prefix preserves its finite register; the complete history is described by the compatible family of these registers, without replacing its counters with an "infinite" integer quantity.

If complete dynamics has the reader q_{mem} with $q_{\mathrm{mem}}(\Gamma_9 x) = q_{\mathrm{mem}}(x) + 1$, it likewise cannot have a periodic state of positive period N on that domain: applying the reader would yield $q_{\mathrm{mem}}(x) = q_{\mathrm{mem}}(x)+N$. This corollary uses the complete integer degree; its residual quotients may have finite periods, as appropriate to the observable calendar in the corpus.

108.7 Foundational consequence and scope of the composition

The specific connection obtained is

$$\text{return increment } 4v \;\longrightarrow\; \mathcal{I}(m) = \Omega_M(v, m) \;\longrightarrow\; N_v m = v\mathcal{I}(m) \;\longrightarrow\; I + nN_v. \tag{108.24}$$

The accumulated register determines a conserved charge and a parabolic family stabilizing its direction. Evolution and conservation belong to the same action: g and s change, their relation $s - 18g$ remains, and the increment preserves an explicit stabilizer. This makes precise part of the link between memory and symmetry that motivated the investigation.

Equality (108.21) connects this research with the preceding balances: an evolution may redistribute the form between readout and complements while preserving the total. The APP identities (108.16)–(108.19) show, in another concrete reader of the same origin, how transport order produces a central datum even when position returns.

The charge (108.7) is an algebraic invariant of transport and, through the explicit action (108.14), a Noether charge of the record realization. Theorem 108.6 and its variational development construct this relation, its symmetry, and its boundary term. The preceding physical action and current developments retain their domains and units; transporting them to this realization requires preservation of their dimensional maps.

Composition with dimensional readers must transport the pair (v, Ω_M) and the charge $\mathcal{I}$, preserving their units and the variational current. Every subsequent relation with π, ϕ, e, α and K must arise from this composition; the present formulas specify the information to be preserved.

Chapter 109

Variational conservation and oriented action

109.1 Symmetry of the elliptic action and Noether charge

Fix a positive action section $\hbar$, the angular pair A, C^* in a common unit, $A \neq 0$, $|C^*/A| < 1$, and

$$\eta = \operatorname{artanh}(C^*/A).$$

The realized plane $\mathcal{P}_\eta$ has coordinates (Q, P), both with the dimension of the square root of action, with

$$\lambda = P\,dQ, \qquad \Omega = dQ \wedge dP = -d\lambda.$$

The construction in section 24.3 gives

$$\mathcal{I}_\eta(Q,P) = \frac{1}{2}(e^{-\eta}Q^2 + e^{\eta}P^2), \qquad H_\eta = \omega \mathcal{I}_\eta, \qquad \omega > 0. \tag{109.1}$$

The HMT ellipse is $\mathcal{I}_\eta = \hbar$. Its area is $2\pi\hbar$; $\mathcal{I}_\eta$ has the dimension of action and H_η that of energy. Define the already realized matrices

$$M_\eta = \operatorname{diag}(e^{\eta/2}, e^{-\eta/2}), \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad R_\eta(\varepsilon) = M_\eta e^{-\varepsilon J_2} M_\eta^{-1}. \tag{109.2}$$

Proposition 109.1. *The group R_η is a canonical symmetry of the action*

$$\mathcal{S}_\eta[z] = \int_{t_a}^{t_b} \left(P\dot{Q} - \omega \mathcal{I}_\eta(Q,P)\right) dt \tag{109.3}$$

and its Noether charge is $\mathcal{I}_\eta$.

Proof. The determinant of M_η is one, and $M_\eta^{\mathsf{T}} J_2 M_\eta = J_2$. Moreover, $\mathcal{I}_\eta(M_\eta y) = \|y\|^2/2$, which is invariant under $e^{-\varepsilon J_2}$. The generator

$$X_\eta = e^{\eta} P\,\partial_Q - e^{-\eta} Q\,\partial_P$$

satisfies $\iota_{X_\eta}\Omega = d\mathcal{I}_\eta$. By Cartan's formula,

$$\mathcal{L}_{X_\eta}\lambda = dF_\eta, \qquad F_\eta = \lambda(X_\eta) - \mathcal{I}_\eta = e^{\eta}P^2 - \mathcal{I}_\eta.$$

Since $X_\eta H_\eta = 0$, the variation of the action density is dF_η/dt. Its charge is

$$\frac{\partial L}{\partial \dot{Q}} X_\eta Q - F_\eta = P e^{\eta} P - F_\eta = \mathcal{I}_\eta.$$

The equations $\dot{Q} = \omega e^{\eta} P$, $\dot{P} = -\omega e^{-\eta} Q$ directly verify $d\mathcal{I}_\eta/dt = 0$. □

The positive parametrization of the ellipse is $Q = a_S \cos\theta$, $P = b_S \sin\theta$. These equations assign it $\dot{\theta} = -\omega$. Thus, over a turn traversed according to the dynamical flow, $\oint P\,dQ = 2\pi\mathcal{I}_\eta$; with increasing θ, the integral is $-2\pi\mathcal{I}_\eta$. This is the orientation distinction established in section 24.3.2. On the radial contour, the magnitude is $h = 2\pi\hbar$.

109.2 Change of form: determined transport and connection

For two already published forms η, η' and a phase advance ϑ, define

$$\mathcal{T}^{\vartheta}_{\eta',\eta} : \mathcal{P}_\eta \to \mathcal{P}_{\eta'}, \qquad \mathcal{T}^{\vartheta}_{\eta',\eta} = M_{\eta'} e^{-\vartheta J_2} M_\eta^{-1}. \tag{109.4}$$

Proposition 109.2. *This transport is symplectic and satisfies*

$$\mathcal{I}_{\eta'}(\mathcal{T}^{\vartheta}_{\eta',\eta} z) = \mathcal{I}_\eta(z), \qquad \mathcal{T}^{\vartheta_2}_{\eta'',\eta'} \mathcal{T}^{\vartheta_1}_{\eta',\eta} = \mathcal{T}^{\vartheta_1+\vartheta_2}_{\eta'',\eta}. \tag{109.5}$$

Proof. All factors are symplectic. The first equality reduces to invariance of $\|y\|^2/2$ under rotation. In the second, the adjacent matrices $M_{\eta'}^{-1} M_{\eta'}$ cancel and the phases add. □

The formula receives the forms and advances from the reader; it does not select their values. The complete list of transitions can be retained alongside the composite transport: two lists yielding the same composite map are not identified as histories.

Proposition 109.3. *A differentiable realization $\eta(t), \vartheta(t) \in C^1$, with $\dot\vartheta = \omega(t)$, of the transport (109.4) has generator*

$$\boxed{H_{\mathrm{tr}}(Q,P,t) = \omega(t)\mathcal{I}_{\eta(t)}(Q,P) + \frac{\dot\eta(t)}{2} QP.} \tag{109.6}$$

This generator preserves $\mathcal{I}_{\eta(t)}(z(t))$.

Proof. Differentiating $z(t) = M_{\eta(t)} e^{-\vartheta(t) J_2} y_0$ gives

$$\dot Q = \omega e^{\eta} P + \frac{\dot\eta}{2} Q, \qquad \dot P = -\omega e^{-\eta} Q - \frac{\dot\eta}{2} P.$$

These are Hamilton's equations for (109.6). The connection term realizes exactly $\dot M_\eta M_\eta^{-1} z$, and its Hamiltonian is determined up to a function of t alone. Indeed,

$$\begin{aligned} \frac{d\mathcal{I}_\eta}{dt} &= \frac{\dot\eta}{2}(-e^{-\eta}Q^2 + e^{\eta}P^2) + \left\{ \mathcal{I}_\eta, \frac{\dot\eta}{2} QP \right\} \\ &= \frac{\dot\eta}{2}(-e^{-\eta}Q^2 + e^{\eta}P^2) + \frac{\dot\eta}{2}(e^{-\eta}Q^2 - e^{\eta}P^2) = 0. \end{aligned} \tag{109.7}$$

Equivalently, $Q = e^{\eta/2} q$, $P = e^{-\eta/2} p$ give

$$P\dot Q - H_{\mathrm{tr}} = p\dot q - \frac{\omega(t)}{2}(q^2 + p^2). \tag{109.8}$$

The rotational symmetry and its charge are transported exactly. □

Here the connection term has been derived from transport, not from an external force. The fixed-form LC Hamiltonian preserves $\mathcal{I}_\eta$. During a deformation, omitting the second term of (109.6) leaves the first term of (109.7) uncancelled. Variational conservation also requires transporting the canonical form.

The proposition determines the generator of the declared interpolation; it does not assign a unique physical interpolation to every TPK history. The discrete identity (109.4)–(109.5) works without interpolation.

109.3 Orientation and boundary of the action

The transformations in section 24.3.3 distinguish $S(Q,P) = (P,Q)$ and $J_2(Q,P) = (-P,Q)$. Both exchange the form $\eta \leftrightarrow -\eta$, but

$$S^*\Omega = -\Omega, \quad J_2^*\Omega = \Omega, \qquad S^*\lambda = -\lambda + d(PQ), \quad J_2^*\lambda = \lambda - d(PQ). \tag{109.9}$$

Likewise,

$$SR_\eta(\varepsilon)S^{-1} = R_{-\eta}(-\varepsilon), \qquad J_2R_\eta(\varepsilon)J_2^{-1} = R_{-\eta}(\varepsilon). \tag{109.10}$$

These are verified using $SM_\eta = M_{-\eta}S$, $J_2M_\eta = M_{-\eta}J_2$, $SJ_2S^{-1} = -J_2$. For an oriented open curve $z : [a,b] \to \mathcal{P}_\eta$,

$$\int_{J_2 z} \lambda = \int_z \lambda - [PQ]_a^b, \qquad \int_{Sz} \lambda = -\int_z \lambda + [PQ]_a^b. \tag{109.11}$$

The boundary terms vanish on a cycle and cancel when compatible endpoints are concatenated. Quadratic energy alone does not determine these signs or these terms.

Nonadic monodromy records phase return and $q_{\mathrm{mem}} \mapsto q_{\mathrm{mem}} + 1$. Constancy of a charge does not require the counter to remain fixed. In the elliptic realization, a turn preserves $\mathcal{I}_\eta$ while increasing the accumulated oriented integral by $2\pi\mathcal{I}_\eta$. Charge, integrated action, phase, and register have distinct types. An angular turn is not identified here with an arbitrary TPK cycle without specifying its reader.

109.4 Variational clock charge and balance between reading and memory

109.4.1 An explicit action charge of the clock

The calendar is realized on $\mathcal{H}_{\mathrm{clk}} = \mathbb{C}^{108}$, with Fourier projectors $P_k = |\widetilde{k}\rangle\langle\widetilde{k}|$, and determines

$$N_{108} = \sum_{k=0}^{107} kP_k, \quad H_{\mathrm{clk}} = \hbar\omega_0 N_{108}, \quad \omega_0 = \frac{2\pi}{108 t_0}. \tag{109.12}$$

Its propagator over t_0 is the shift by one cell. The clock action is

$$\mathcal{S}[\psi] = \int \left[\frac{i\hbar}{2}(\langle\psi,\dot\psi\rangle - \langle\dot\psi,\psi\rangle) - \langle\psi, H_{\mathrm{clk}}\psi\rangle\right] dt. \tag{109.13}$$

The inner product is antilinear in the first variable. The equation is $i\hbar\dot\psi = H_{\mathrm{clk}}\psi$.

Proposition 109.4. *For $B = B^*$ with $[B, H_{\mathrm{clk}}] = 0$, the symmetry $\psi \mapsto e^{-i\varepsilon B}\psi$ has charge*

$$Q_B(\psi) = \hbar\langle\psi, B\psi\rangle. \tag{109.14}$$

Proof. Localizing the parameter, $\delta\psi = -i\varepsilon(t)B\psi$, gives $\delta L = \hbar\dot\varepsilon\langle\psi, B\psi\rangle$. Integration by parts in the variational equations gives $dQ_B/dt = 0$. Directly,

$$\frac{d}{dt}\langle\psi, B\psi\rangle = \frac{i}{\hbar}\langle\psi, [H_{\mathrm{clk}}, B]\psi\rangle = 0.$$

With $B = I$, norm is conserved; with $B = N_{108}$, the charge is the action $E_{\mathrm{clk}}/\omega_0$. □

The symplectic form of the underlying real space is $\Omega_{\mathrm{clk}}(u,v) = 2\hbar\,\mathrm{Im}\langle u, v\rangle$; its one-form is $\Theta_\psi(v) = -\hbar\,\mathrm{Im}\langle\psi, v\rangle$, with $\Omega_{\mathrm{clk}} = -d\Theta$.

109.4.2 Exact charge balance

Let $\mathcal{H}, \mathcal{K}$ be finite-dimensional Hermitian spaces, $J : \mathcal{H} \to \mathcal{K}$ an isometry, U unitary on $\mathcal{K}$, and

$$\Pi = JJ^*, \quad T = J^*UJ, \quad \eta_{\mathrm{mem}} = (I - \Pi)UJ. \tag{109.15}$$

The operator η_{mem} is distinct from the scalar parameter η.

Proposition 109.5. *If*

$$\widetilde{B} = \widetilde{B}^*, \quad U^*\widetilde{B}U = \widetilde{B}, \quad [\widetilde{B}, \Pi] = 0, \quad B = J^*\widetilde{B}J, \tag{109.16}$$

then

$$B = T^*BT + \eta_{\mathrm{mem}}^*\widetilde{B}\eta_{\mathrm{mem}}, \qquad Q_B(\psi) = Q_B(T\psi) + Q_{\widetilde{B}}(\eta_{\mathrm{mem}}\psi). \tag{109.17}$$

Proof. Expand $UJ\psi = JT\psi + \eta_{\mathrm{mem}}\psi$ in $J^*U^*\widetilde{B}UJ = J^*\widetilde{B}J$. Commutation with Π cancels the cross terms. Evaluating the operator identity and multiplying by $\hbar$ gives the action balance. □

By iteration,

$$Q_B(\psi) = Q_B(T^n\psi) + \sum_{j=0}^{n-1} Q_{\widetilde{B}}(\eta_{\mathrm{mem}}T^j\psi). \tag{109.18}$$

This sum records successive complements of the compressed dynamics. The complete dynamics has another exact formula, with $z_n = U^nJ\psi$:

$$Q_B(\psi) = Q_B(J^*z_n) + Q_{\widetilde{B}}((I - \Pi)z_n). \tag{109.19}$$

Without an additional property, T^n is not identified with J^*U^nJ: the complete state can reabsorb memory. If $B, \widetilde{B} \geq 0$, all summands in (109.18) are nonnegative; for signed charges, the equality remains, without this positivity interpretation.

If U is the propagator of an action such as (109.13), commutation of $\widetilde{B}$ with its Hamiltonian supplies Noether conservation. The reduction condition involving Π specifies the readings that separate charge without interference terms.

Without imposing (109.16), the general identity preserves the additional terms:

$$\begin{aligned} &Q_B(T\psi) + Q_{\widetilde{B}}(\eta_{\mathrm{mem}}\psi) + 2\hbar\,\mathrm{Re}\langle JT\psi, \widetilde{B}\eta_{\mathrm{mem}}\psi\rangle \\ &= Q_B(\psi) + \hbar\langle\psi, J^*(U^*\widetilde{B}U - \widetilde{B})J\psi\rangle. \end{aligned} \tag{109.20}$$

The physical charge uses its reader, for example N_{108}; a norm identity corresponds to $B = I$. The sections $\hbar_{\mathrm{pre}}$ and $\hbar_{\mathrm{ret}}$ preserve the return reader of section 14.6.3 y equation (14.72); their difference and the memory complement have their respective domains of definition.

109.5 Covariant balance of the total tensor

The variational elimination of contorsion and the Noether identity of the covariant action, developed in Proposition 131.6, provide

$$\mathcal{T}_{\mu\nu} = T_{\mu\nu}^{\mathrm{phys,LC}} + U_{\mu\nu}^{\mathrm{phys,spin}}, \qquad \mathring{\nabla}^\mu\mathcal{T}_{\mu\nu} = 0. \tag{109.21}$$

Its domain is the realized HMT–MD chart, with the couplings held constant during the variation. For a Killing field ξ,

$$j^{\mu}_{\xi} = \mathcal{T}^{\mu\nu}\xi_{\nu}, \qquad \mathring{\nabla}_{\mu} j^{\mu}_{\xi} = 0. \tag{109.22}$$

The proof uses (109.21), the symmetry of $\mathcal{T}$, and the antisymmetry of $\mathring{\nabla}_{\mu}\xi_{\nu}$. Upon integration, the difference between the charges of the hypersurfaces is the flux through the lateral boundary.

Conservation belongs to the total tensor; each component may exchange charge. Theorem 136.3 explicitly determines $\mathcal{J}^{\mathrm{tr}}_{\nu} = \partial_{\nu}\,\mathrm{tr}_{g}(T^{\mathrm{res}})/4$ and the opposite balances of the trace and traceless part. This is recovered as a covariant instance of the joint-balance criterion, not as a new gravitational equation.

Chapter 110

Conservation of action forms and oriented area in the tritic regimes

110.1 Balance between two action forms

The angular outputs A, C^*, in a common unit, determine the form $\eta = \operatorname{artanh}(C^*/A)$ on the domain $A \neq 0$, $|C^*/A| < 1$. Given two successive forms η_n, η_{n+1}, define

$$M_n = \operatorname{diag}(e^{\eta_n/2}, e^{-\eta_n/2}), \qquad G_n = M_n^{-\mathsf{T}} M_n^{-1}, \qquad I_n(z) = \tfrac{1}{2} z^{\mathsf{T}} G_n z.$$

Here $z = (Q, P)^{\mathsf{T}}$, with both coordinates having the dimension of the square root of action. The scale $\hbar$, when it fixes the contour $I_n = \hbar$, is the previously constructed action output; it does not serve as a fitting target.

Let R_n be the canonical rotation of angular advance θ_n. The transport without angular advance and the transport with that advance are, respectively,

$$B_n = M_{n+1} M_n^{-1}, \qquad C_n = M_{n+1} R_n M_n^{-1}. \tag{110.1}$$

Both go from chart n to chart $n+1$. Define the compressed reading and its normalized complementary register:

$$T_n = \frac{8B_n + C_n}{9}, \qquad D_n = \frac{\sqrt{8}}{9}(C_n - B_n). \tag{110.2}$$

The operator D_n is an isometrically normalized realization of the unique pattern of differences among the nine phases. The nine-component register is $(8z, -z, \dots, -z)/27$, with $z = (C_n - B_n)v$; the sum of its inner products gives the same factor $8/81$. The nine-phase register is retained when the individual labels enter into other readers.

Theorem 110.1. *Action is conserved between reading and memory:*

$$\boxed{G_n = T_n^{\mathsf{T}} G_{n+1} T_n + D_n^{\mathsf{T}} G_{n+1} D_n.} \tag{110.3}$$

Proof. Normalization of the charts gives $B_n^{\mathsf{T}} G_{n+1} B_n = G_n = C_n^{\mathsf{T}} G_{n+1} C_n$. On expanding (110.2), the cross contributions have coefficients +8 and −8. The remaining diagonal coefficients are 72 for B_n and 9 for C_n, divided by 81. The sum is G_n. □

With $z_{n+1} = T_n z_n$, $m_n = D_n z_n$, the scalar form is

$$\boxed{I_n(z_n) = I_{n+1}(z_{n+1}) + I_{n+1}(m_n).} \tag{110.4}$$

The telescoping sum, retaining each memory in the chart where it was emitted, gives

$$I_0(z_0) = I_N(z_N) + \sum_{n=0}^{N-1} I_{n+1}(m_n). \tag{110.5}$$

The transport B_n is an effective part of this composition: replacing it by the identity when the forms differ changes the domains and may invalidate (110.3).

110.2 Exact accumulation law with variable phase

In normalized coordinates $y_n = M_n^{-1} z_n$, the reading is $y_{n+1} = (8I + R_n)y_n/9$. For a planar rotation,

$$\left(\frac{8I + R_n}{9}\right)^{\mathsf{T}} \left(\frac{8I + R_n}{9}\right) = r_n I, \qquad r_n = 1 - \frac{32}{81}\sin^2(\theta_n/2). \tag{110.6}$$

Therefore,

$$I_N(z_N) = I_0(z_0) \prod_{n=0}^{N-1} r_n. \tag{110.7}$$

Theorem 110.2. *For $I_0(z_0) > 0$, the terminal component tends to zero exactly when*

$$\sum_{n\geq 0} \sin^2(\theta_n/2) = +\infty. \tag{110.8}$$

Proof. Let $c = 32/81$ and $u_n = \sin^2(\theta_n/2) \in [0, 1]$. We have $cu_n \leq -\log(1 - cu_n) \leq cu_n/(1 - c)$, by integrating $1/(1 - s)$ between zero and cu_n. The sum of these logarithms diverges exactly when $\sum u_n$ diverges. Applying the logarithm to the product in (110.7) gives the equivalence. □

Equation (110.5) is preserved in both cases. A sequence of angular advances may leave a positive terminal component or transfer all action to the register, depending on the sum in (110.8). Change of form alone preserves action; angular dispersion determines its allocation in this realization. The phases θ_n must come from the specific reader of the trajectory: the proof does not retrospectively select a sequence to impose a result.

110.3 Extended Euler and common oriented area

The generators of the three quadratic regimes are

$$J_{+1} = (2C_{A_2} + I)/\sqrt{3}, \qquad J_0 = N, \qquad J_{-1} = (2F_{\mathrm{av}}^2 - 3I)/\sqrt{5}.$$

In their two-dimensional charts they have trace zero and satisfy $J_\tau^2 = -\tau I$. The Coxeter element comes from the APP antecedent, the nilpotent from threshold transport and F_{av} from the descent of TPK modes. The exponential functions constitute the subsequent analytic realization:

$$E_\tau(t) = \exp(tJ_\tau) = c_\tau(t)I + s_\tau(t)J_\tau,$$

where (c_τ, s_τ) are $(\cos, \sin)$, $(1, t)$, and $(\cosh, \sinh)$, respectively. Zero trace gives determinant one. With

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

we obtain $E_\tau^{\mathsf{T}} \Omega E_\tau = \Omega$. In variable charts we take $C_n = M_{n+1} E_{\tau_n}(t_n) M_n^{-1}$ and the same B_n as in (110.1).

Theorem 110.3. *The three regimes satisfy*

$$\boxed{\Omega = T_n^{\mathsf{T}} \Omega T_n + D_n^{\mathsf{T}} \Omega D_n,} \tag{110.9}$$

with T_n, D_n from (110.2).

Proof. In dimension two, $L^{\mathsf{T}}\Omega L = (\det L)\Omega$. The determinants of M_n, B_n, and C_n are one. The expansion used in (110.3), applied to the alternating form, proves (110.9). □

In a fixed chart, the two determinants are

$$\det T_\tau = \frac{65 + 16c_\tau(t)}{81}, \qquad \det D_\tau = \frac{16(1 - c_\tau(t))}{81}. \tag{110.10}$$

This follows from $\det(aI + C) = a^2 + a\,\mathrm{tr}\,C + \det C$ and $\mathrm{tr}\,E_\tau = 2c_\tau$. Their sum is one. For the representatives C_{A_2}, $I + N$, F_{av}^2, the pairs are $(19/27, 8/27)$, $(1, 0)$, $(89/81, -8/81)$.

The parabolic regime may produce a nonzero rank-one memory with zero area. The hyperbolic regime preserves oriented area through a complement of opposite sign. The elliptic regime additionally permits a positive action balance between the forms of the two fibres. The common alternating form and the metric forms of each regime retain their respective types.

110.3.1 Integrated action and orientation

Let γ be a closed piecewise C^1 curve in the canonical plane. For any real matrix L of order two,

$$\oint_{L\gamma} P\,dQ = (\det L) \oint_{\gamma} P\,dQ.$$

Indeed, writing $Q' = aQ + bP$, $P' = cQ + dP$, the integrals of $Q\,dQ$ and $P\,dP$ are zero, and $\oint Q\,dP = -\oint P\,dQ$. The coefficient $ad - bc$ remains. Applying (110.10),

$$\boxed{\oint_{\gamma} P\,dQ = \oint_{T_n\gamma} P\,dQ + \oint_{D_n\gamma} P\,dQ.} \tag{110.11}$$

The signs are those of the transported curves. The identity is preserved for the three regimes and for changes of form with determinant one.

110.4 The common factor of the exponential reader

The radional reader in section 24.3 y proposition 10.48 also contains $e^{-x}\exp(-yJ_\tau)$. This factor is retained when it acts in the reader. Let

$$C_n = e^{-x_n} M_{n+1} E_{\tau_n}(t_n) M_n^{-1}.$$

Now $C_n^{\mathsf{T}}\Omega C_n = e^{-2x_n}\Omega$. The same expansion gives

$$T_n^{\mathsf{T}}\Omega T_n + D_n^{\mathsf{T}}\Omega D_n = \frac{8 + e^{-2x_n}}{9}\Omega. \tag{110.12}$$

For $x_n \geq 0$, define a complementary scale register

$$D_{\mathrm{esc},n} = \frac{\sqrt{1 - e^{-2x_n}}}{3} B_n.$$

The complete identity is then

$$\Omega = T_n^{\mathsf{T}}\Omega T_n + D_n^{\mathsf{T}}\Omega D_n + D_{\mathrm{esc},n}^{\mathsf{T}}\Omega D_{\mathrm{esc},n}. \tag{110.13}$$

This is a composition defined here to represent explicitly the reader's scale defect. Its coefficient comes from (110.12). The name of this register does not identify it with another physical sector of the corpus; the map is retained so that this comparison can be made with their respective operators and domains.

110.5 Action refinement over the complete emissions

For each enriched prefix h, let M_h be its chart and $G_h = M_h^{-\mathsf{T}} M_h^{-1}$. For each surviving emission ϵ, let R_ϵ be a realized rotation and

$$U_\epsilon = M_{h\epsilon} R_\epsilon M_h^{-1}, \qquad (\mathscr{U} z)_{h\epsilon} = \sqrt{p_h(\epsilon)}\, U_\epsilon z_h.$$

The weights come from the emission measure of the common construction; $\sum_\epsilon p_h(\epsilon) = 1$, with multiplicities preserved. From $U_\epsilon^{\mathsf{T}} G_{h\epsilon} U_\epsilon = G_h$ it follows that

$$\sum_{h,\epsilon} I_{h\epsilon}((\mathscr{U} z)_{h\epsilon}) = \sum_h I_h(z_h). \tag{110.14}$$

The proof first sums the weights of a parent's descendants. Along a trajectory, the intermediate factors $M_h^{-1} M_h$ cancel, and rotations compose in the recorded order. This preserves labels and memory without mixing children of different parents. The alternating form on the direct sum satisfies the same transport identity for the three symplectic regimes, replacing R_ϵ with $E_{\tau_\epsilon}(t_\epsilon)$.

The metric completions are taken with $\sum_h z_h^{\mathsf{T}} G_h z_h$, so these maps are isometric. Further identifying this norm with a fixed Euclidean direct sum requires uniform bounds on M_h and M_h^{-1}; that additional identification is not used in (110.14).

Chapter 111

Explicit reconstruction from complete returns

111.1 State domain and bilateral extension

A chart of the surviving domain generated by APP–TRIT–TPK is considered. A state of this chart preserves the operator sheets, the tritic orientation, the residue, the quotient, the carry, the route, the boundary, the memory register and the calendar origin. The update acts on this complete state.

Let X_0 be a nonempty compact Hausdorff space and $F : X_0 \to X_0$ a continuous surjective update. These are the hypotheses of the result in the chart considered. The bilateral extension of the update is

$$\mathscr{X}_F = \big\{(x_k)_{k\in\mathbb{Z}} \in X_0^{\mathbb{Z}} : F(x_k) = x_{k+1} \text{ for every } k \in \mathbb{Z}\big\}, \tag{111.1}$$

with the topology induced by the product.

Lemma 111.1 (Existence and bilateral transport)**.** *The space $\mathscr{X}_F$ is compact, Hausdorff, and nonempty. Every coordinate projection $\mathscr{X}_F \to X_0$ is surjective. The shift*

$$\sigma((x_k)_{k\in\mathbb{Z}}) = (x_{k+1})_{k\in\mathbb{Z}}$$

is a homeomorphism of $\mathscr{X}_F$.

Proof. Each condition $F(x_k) = x_{k+1}$ defines a closed subset of the compact Hausdorff product $X_0^{\mathbb{Z}}$. With an index k_0 and a state $z \in X_0$ fixed, any finite family of these conditions, together with $x_{k_0} = z$, admits a solution: they are completed forward through F and backward through preimages, which exist by surjectivity. The conditions therefore have the finite intersection property. Compactness produces a bilateral history with $x_{k_0} = z$.

The shifts σ and $\sigma^{-1}((x_k)) = (x_{k-1})$ preserve the relations in (111.1); both are continuous by their coordinate action. □

111.2 Tritic sampling and complete reconstruction of evolution

With the schedule origin fixed, the typed return times are

$$\mathscr{T}_+ = \{k \in \mathbb{Z} : 1 + (k \bmod 9) \in \{1, 4, 7\}\} = 3\mathbb{Z}.$$

The complete state x_k is preserved at each such time. Define the space of return records by

$$\mathscr{Y}_+ = \big\{(y_j)_{j\in\mathbb{Z}} \in X_0^{\mathbb{Z}} : F^3(y_j) = y_{j+1} \text{ for every } j \in \mathbb{Z}\big\}. \tag{111.2}$$

Theorem 111.2 (Recovery from complete returns)**.** *The map*

$$r = \mathrm{Ret}_+ : \mathscr{X}_F \longrightarrow \mathscr{Y}_+, \qquad r(x)_j = x_{3j},$$

is a homeomorphism. Its inverse is given explicitly by

$$\boxed{\big(r^{-1}(y)\big)_{3j+s} = F^s(y_j), \qquad j \in \mathbb{Z}, \quad s \in \{0, 1, 2\}.} \tag{111.3}$$

Proof. For $x \in \mathscr{X}_F$, $F^3(x_{3j}) = x_{3j+3}$, so $r(x) \in \mathscr{Y}_+$. Conversely, every integer k has a unique expression $k = 3j + s$, with $j \in \mathbb{Z}$ and $0 \le s < 3$, including when $k < 0$. Formula (111.3) therefore defines every coordinate of a sequence x.

If $s = 0$ or $s = 1$, then $F(x_{3j+s}) = F^{s+1}(y_j) = x_{3j+s+1}$. At the boundary between two blocks,

$$F(x_{3j+2}) = F^3(y_j) = y_{j+1} = x_{3j+3}.$$

Thus $x \in \mathcal{X}_F$. Reading its coordinates indexed by $3j$ returns y_j. In the reverse direction, the relations in (111.1) give $x_{3j+s} = F^s(x_{3j})$, so reconstruction from $r(x)$ returns x in full.

The map r is continuous because it selects coordinates. Each coordinate of its inverse is the composition of a coordinate projection with one of the continuous maps I, F, F^2. The inverse is therefore continuous. □

An additional discrete coordinate, for example the index $m \in \mathbb{Z}$ of a realization chart, is preserved through

$$\widetilde{r} : \mathbb{Z} \times \mathcal{X}_F \longrightarrow \mathbb{Z} \times \mathcal{Y}_+, \qquad \widetilde{r}(m, x) = (m, r(x)).$$

Its inverse is $(m, y) \mapsto (m, r^{-1}(y))$. The same result holds for a fixed schedule origin a: replace x_{3j} and x_{3j+s} with x_{a+3j} and x_{a+3j+s}.

111.3 Conjugation of transport and preservation of readouts

Let $\sigma_+(y)_j = y_{j+1}$ be the shift of records. The preceding reconstruction gives

$$r\sigma^3 = \sigma_+ r, \qquad (r\sigma r^{-1}(y))_j = F(y_j). \tag{111.4}$$

The map in the second equality is reversible on $\mathcal{Y}_+$; its inverse is

$$y \longmapsto \big(F^2(y_{j-1})\big)_{j \in \mathbb{Z}}.$$

Indeed, both compositions reduce to $F^3(y_{j-1}) = y_j$. Reversibility of the bilateral history is thus preserved even when the update F on an isolated state is not injective. If the nonadic revolution is realized as $\Gamma_9 = \sigma^9$, then

$$r\Gamma_9 = \sigma_+^3 r. \tag{111.5}$$

Corollary 111.3 (Joint readouts and recovered transport)**.** *Let $a : \mathcal{X}_F \to \mathcal{R}$ and $q : \mathcal{X}_F \to \mathcal{E}$ be two continuous readers of the same state, and let $d : \mathcal{X}_F \to \mathcal{X}_F$ be a homeomorphism of the calibrated domain. Then*

$$a_+ = a \circ r^{-1}, \qquad q_+ = q \circ r^{-1}, \qquad d_+ = r \circ d \circ r^{-1}$$

are continuous, d_+ is a homeomorphism, and

$$(a, q) = (a_+, q_+) \circ r, \qquad rd = d_+ r. \tag{111.6}$$

For every integer $N > 0$,

$$d^N x = x \quad \Longleftrightarrow \quad d_+^N r(x) = r(x).$$

Proof. Compose the maps with the homeomorphism of theorem 111.2. The inverse of d_+ is $rd^{-1}r^{-1}$. The identity $d_+^N r = rd^N$, together with injectivity of r, proves the last equivalence. □

Application to the double readout takes $\mathcal{R} = R_{\mathrm{HMT}}$ and the incidence codomain constructed in section 21.11.1 y proposition 21.55. The result establishes a recoverable presentation of the same histories, their readers, and their transport. Its object is this dynamical equivalence of presentations.

Chapter 112

Observable return and persistence of the register

112.1 Joint recovery from complete returns

Let X be the domain of complete calibrated histories of equation (111.1), including the calendar origin and, when used, the discrete chart index. The states preserve the transitions and memory register. We write $r = \mathrm{Ret}_+ : X \to Y_+$ for sampling at complete returns, with $Y_+ = r(X)$.

The result in theorem 111.2 provides its explicit inverse: $x_{3j+s} = F^s(y_j)$, for $s = 0, 1, 2$. Reconstruction uses the complete state preserved at each return and the effective update; (111.4) also preserves transport.

Let $a : X \to R_{\mathrm{HMT}}$ be numerical evaluation by cylinders and $q : X \to E_\infty$ incidence with its gauge, according to section 10.5, section 21.11.1 y proposition 21.55. Let $d : X \to X$ be a reversible continuous transport, with the gauge transported. On Y_+, define

$$a_+ = a \circ r^{-1}, \qquad q_+ = q \circ r^{-1}, \qquad d_+ = r \circ d \circ r^{-1}.$$

Then

$$\boxed{(a, q) = (a_+, q_+) \circ r, \qquad rd = d_+ r.} \tag{112.1}$$

Proof. This is corollary 111.3: the compositions with r and r^{-1} cancel, and the inverse of d_+ is $rd^{-1}r^{-1}$. Both readers are transported from the same complete state. □

In particular, for $N \in \mathbb{Z}_{>0}$, $d^N x = x$ if and only if $d_+^N r(x) = r(x)$. Conjugation preserves periodicity or aperiodicity of the complete state. The dynamics of a partial reader is determined through its own transport factor.

112.2 Explicit subsequent representation of transport

On the already generated set of histories, consider the subsequent realization $H = \ell^2(X)$, with orthonormal basis $(\delta_x)_{x \in X}$ and counting measure. Every vector has at most countable support. Reversible transport determines

$$C\delta_x = \delta_{dx}, \qquad C^{-1}\delta_x = \delta_{d^{-1}x}.$$

The bijection d proves that C is unitary. This representation preserves the complete labels of the histories. The space X and its transport precede this linear realization.

The nonadic chart of chapter 104 determines

$$T = \frac{8I + C}{9}, \qquad \eta v = \frac{1}{27}(8(C - I)v, -(C - I)v, \ldots, -(C - I)v),$$

$$I - T^*T = \eta^*\eta = \frac{8}{81}(I - C)^*(I - C).$$

Let P be the orthogonal projection onto $\ker(I - C)$. The spectral proof in chapters 104 y 105 gives $T^N \to P$ strongly. Telescoping provides the isometry

$$\mathcal{V}v = (Pv, \eta v, \eta T v, \eta T^2 v, \ldots), \qquad \mathcal{M} = \mathrm{Im}\,\mathcal{V}.$$

The register shift B preserves the first component and removes the first memory block. On the compatible image,

$$\boxed{\widehat{C} = \mathcal{V} C \mathcal{V}^{-1} = (9B - 8I)|_{\mathcal{M}}.} \tag{112.2}$$

The inverse of the representation is given by

$$v = Pv + \sum_{j \geq 0} T^{*j} \eta^* (\eta T^j v). \tag{112.3}$$

Proof. The defect identity telescopes up to N; its limit preserves the norm exactly. The identities $\mathcal{V}T = B\mathcal{V}$ and $C = 9T - 8I$ give (112.2). The isometry $\mathcal{V}$ is surjective onto $\mathcal{M}$; taking its adjoint gives (112.3), with norm convergence of the series. Unitarity of the recovered transport is established on $\mathcal{M}$. □

For $A \in \mathcal{B}(H)$, let $\widehat{A} = \mathcal{V} A \mathcal{V}^{-1}$ on $\mathcal{M}$. This correspondence preserves norm, products, adjoints, and commutators, since $\mathcal{V}^{-1} = \mathcal{V}^*|_{\mathcal{M}}$. It jointly represents the numerical observables, the incidence observables, and their relations with C.

112.3 Numerical convergence and correlations with incidence

The cylinders of (10.49) provide, in the chart of integer index $m(x)$, six trits per block. We write

$$\xi_n(x) = \sum_{k=1}^{6n} d_k(x) 3^{-k}, \qquad \xi(x) = a(x) - m(x), \qquad 0 \leq \xi(x) - \xi_n(x) \leq 3^{-6n}.$$

Here $d_k \in \{0, 1, 2\}$ is the subsequent ternary reading of the trit; its balanced orientation retains its own typing. The chart retains m separately, including the endpoint convention, and reconstructs $a = m + \xi$. At endpoints with a double representation, ξ is interpreted in that chart.

Let M_f denote multiplication by f on $\ell^2(X)$. The normalization $0 \leq \xi_n \leq \xi \leq 1$ makes M_ξ and M_{ξ_n} contractions. For a set B in the finite incidence codomain of q_j, define $E_{j,B} = M_{\mathbf{1}_{q_j^{-1}(B)}}$. This is the projector onto the histories satisfying that incidence.

Theorem 112.1 (Simultaneous conservation of precision and transport)**.** *For every $t \in \mathbb{Z}$,*

$$\boxed{\left\| \widehat{C}^{-t} \widehat{M}_\xi \widehat{C}^t - \widehat{C}^{-t} \widehat{M}_{\xi_n} \widehat{C}^t \right\| = \|M_\xi - M_{\xi_n}\| \leq 729^{-n}.} \tag{112.4}$$

The relations among the incidence projectors, their products, and their transport are preserved exactly. For an ordered product of contractions containing r numerical observables M_ξ, evaluated at arbitrary integer times, and any finite number of exact incidence or unitary factors, replacing those r factors by M_{ξ_n} changes the product in norm by at most

$$\boxed{r\, 729^{-n}.} \tag{112.5}$$

The same bound holds for any matrix element between unit vectors. The factors may be noncommutative; their order is preserved.

Proof. The norm of a multiplication operator on $\ell^2(X)$ is $\sup_{x \in X} |f(x)|$; the ternary tail proves the first bound. Conjugation by C^t and by $\mathcal{V}$ preserves the norm. For products $A_1 \cdots A_s$ and $B_1 \cdots B_s$, use

$$\prod_{i=1}^{s} A_i - \prod_{i=1}^{s} B_i = \sum_{j=1}^{s} B_1 \cdots B_{j-1} (A_j - B_j) A_{j+1} \cdots A_s.$$

The exact factors give zero difference; each of the remaining r contributes norm at most 729^{-n}. Submultiplicativity and the bound one for the other factors prove (112.5). Cauchy–Schwarz gives the assertion about matrix elements. □

This uniformity preserves precision when the complete approximate observable is transported. The operator C and incidence remain exact in (112.4). The index n measures cylinder depth; t measures iterations of the transport d; j in (112.3) measures compression iterations.

112.4 Covariance of the memory operator under return

Restrict X and its atomic representation to the graded reversible sector $X_{\mathrm{rev,grad}}$ of the clock in (10.53). The integer memory degree q_{mem} is preserved in each history. For $d = \Gamma_9$, its update is $q_{\mathrm{mem}}(dx) = q_{\mathrm{mem}}(x)+1$. The memory operator D is multiplication by q_{mem}, with domain

$$\operatorname{Dom} D = \{v : \sum_{x\in X} |q_{\mathrm{mem}}(x)|^2 |v_x|^2 < \infty\}.$$

The index change $x \mapsto dx$ and the inequalities $(q \pm 1)^2 \le 2q^2 + 2$ prove that C and C^{-1} preserve this domain. On it,

$$C^* D C = D + I, \qquad \widehat{C}^* \widehat{D} \widehat{C} = \widehat{D} + I_{\mathcal{M}}, \qquad \operatorname{Dom} \widehat{D} = \mathcal{V} \operatorname{Dom} D. \tag{112.6}$$

Proof. On δ_x, the left-hand side equals $q_{\mathrm{mem}}(dx)\delta_x = (q_{\mathrm{mem}}(x) + 1)\delta_x$. The identity extends to the indicated domain by truncating the support in the graph norm of D. The second equality is obtained by unitary conjugation to $\mathcal{M}$. □

By induction, $q_{\mathrm{mem}}(d^N x) = q_{\mathrm{mem}}(x) + N$. Every integer $N > 0$ changes the complete state. Every orbit of d is infinite; a square-summable function constant on each of these orbits is zero. In this bilateral realization, $P = 0$. Formula (112.3) then reconstructs the complete linear state from its memory complements. This conclusion uses the effective grading of the clock.

Phase return coexists with the memory increment in (112.6). Sturmian aperiodicity and its complexity belong to the symbolic factor in section 10.5; the present result preserves the graded dynamics of the complete state.

112.5 Incidence, refinement, and three regimes

The register K also has the complete realization in chapter 107, $\mathcal{F}k = (\mu(k), IP_0k/6)$. Its inverse preserves the twelve coordinates, including the mean. Applying $\mathcal{F}$ to the corresponding fibres and retaining the history labels transports this representation. The map $\mathcal{F} \circ K$ preserves exactly the fibres of the reader K; recovery of complete histories comes from r or from their effective labels.

The isometric refinement systems in chapters 103, 105 y 107 preserve $\mathcal{V}$ and C through their intertwining squares. The proof uses the levelwise realizations constructed there. The atomic representation $\ell^2(X)$ and the projective-measure realizations L^2 retain their corresponding domains and measures.

The link with folding and unfolding is developed in propositions 10.47 y 10.48: Coxeter element, nilpotent transport and action of F_{av}^2. For each two-dimensional realization E preserving the alternating form Ω, the nonadic chart satisfies

$$\Omega = T_E^* \Omega T_E + D_E^* \Omega D_E, \qquad T_E = (8I + E)/9, \quad D_E = \sqrt{8}(E - I)/9. \tag{112.7}$$

The algebraic expansion cancels the cross terms and uses $E^*\Omega E = \Omega$. In the real realization, $*$ means transpose. The oriented law brings together the three regimes through the forms and transports of chapters 109 y 110. Its realization as action also uses the normalizations and connections specified there.

The relation to connection dynamics is formulated through the constructions in chapters 108 a 110. The identities (112.2)–(112.7) preserve their representation, the precision of the observables, and the transported forms.

Chapter 113

Reversible coupling and nonstationary transport

113.1 Reversible extension of the balance between two forms

Let $B, C : V \to V'$ be isomorphisms preserving a nondegenerate form b between two realizations:

$$B^{*} b' B = C^{*} b' C = b.$$

The form may be a positive metric or an alternating form. In real spaces, $*$ denotes transposition with respect to the chosen coordinates; the equality expresses transport of the form between the specified realizations. In Hilbert spaces, the operators and their inverses are bounded.

With $s = \sqrt{8}$ and $Q_V = \frac{1}{3}\left(\begin{smallmatrix} sI & -I \\ I & sI \end{smallmatrix}\right)$, define

$$\boxed{\mathcal{U}(B,C) = Q_{V'}^{*} \operatorname{diag}(B,C) Q_V = \frac{1}{9}\begin{pmatrix} 8B + C & \sqrt{8}(C-B) \\ \sqrt{8}(C-B) & B + 8C \end{pmatrix}.} \tag{113.1}$$

We write $T = (8B + C)/9$, $D = \sqrt{8}(C - B)/9$ and $R = (B + 8C)/9$. The complete update is

$$\begin{pmatrix} x' \\ m' \end{pmatrix} = \begin{pmatrix} T & D \\ D & R \end{pmatrix} \begin{pmatrix} x \\ m \end{pmatrix}. \tag{113.2}$$

Theorem 113.1. *The map* (113.1) *is reversible and preserves the form $b \oplus b$:*

$$\mathcal{U}(B,C)^{*} (b' \oplus b') \mathcal{U}(B,C) = b \oplus b, \quad \mathcal{U}(B,C)^{-1} = \mathcal{U}(B^{-1}, C^{-1}). \tag{113.3}$$

Proof. We have $Q^{*}Q = I$ by $8 + 1 = 9$. Its scalar blocks make Q preserve $b \oplus b$ for any form b. The diagonal map preserves the forms by hypothesis. Composition proves (113.3), and inversion of the three factors gives the stated inverse. □

For incoming memory $m = 0$, the balance $b = T^{*} b' T + D^{*} b' D$ is recovered. The second column determines the response to arbitrary incoming memory. Conservation of positive norms and conservation of oriented area are the respective specializations of (113.3).

113.1.1 Collective realization in the nine copies

On V^{9}, let $W_E = \operatorname{diag}(I, \dots, I, E)$, with E in the ninth position. Let $A(a, b) = (a/\sqrt{8}, \dots, a/\sqrt{8}, b)$ and $L = AQ$. Then $L(x, 0) = Jx$ and

$$W_E L = L \mathcal{U}(I, E). \tag{113.4}$$

Proof. $W_E A = A \operatorname{diag}(I, E)$; multiplying by Q and using $AQ = L$ gives (113.4). Orthogonality of the mean of the first eight positions to their seven differences also shows that the complement remains fixed under W_E. □

The original cyclic step is $S_E = P_{\mathrm{cyc}} W_E$. Hence $S_E L = (P_{\mathrm{cyc}} L) U(I, E)$, with distinct input and output charts. Collective reduction preserves the corresponding pattern: $S_E^{9} = I_9 \otimes E$, whereas $W_E^{9} = \operatorname{diag}(I_8, E^{9})$. When a reader uses the nine individual labels, they are preserved in their original representation.

113.2 Exact composition and memory re-entry

For $B_1, C_1 : V_0 \to V_1$ and $B_2, C_2 : V_1 \to V_2$,

$$\boxed{\mathcal{U}(B_2, C_2)\mathcal{U}(B_1, C_1) = \mathcal{U}(B_2B_1, C_2C_1).} \tag{113.5}$$

Proof. In (113.1), the factors $Q_{V_1}Q_{V_1}^*$ cancel, and the two diagonal blocks multiply in the indicated order. Induction extends the formula to any finite number of transports, preserving their order. □

The reading block contains the identity

$$\boxed{T_2T_1 + D_2D_1 = \frac{8B_2B_1 + C_2C_1}{9}.} \tag{113.6}$$

From $(x, 0)$, the first stage produces (T_1x, D_1x). The second reading is $T_2T_1x + D_2D_1x$. The second term is exactly the contribution of the reintroduced memory. Storing D_1x outside the next coupling and continuing only with T_1x yields the separate-register procedure, whose reading is T_2T_1x. The equality (113.6) makes the difference between the two procedures precise.

The realization U preserves the coupling state and the ordered products $B_N \cdots B_1$, $C_N \cdots C_1$. Preservation of all prefixes also includes the HMT emission register, which retains each factor together with its order.

113.2.1 Covariant increment equation

For the separate register $x_{n+1} = T_nx_n$, $m_n = D_nx_n$, we have

$$x_{n+1} - B_nx_n = m_n/\sqrt{8}. \tag{113.7}$$

With $G_0 = I$ and $G_{n+1} = B_nG_n$, telescoping gives

$$x_0 = G_N^{-1}x_N - \frac{1}{\sqrt{8}} \sum_{n<N} G_{n+1}^{-1}m_n. \tag{113.8}$$

Memory is the increment relative to transport between forms B_n, with the declared normalization. The equality (113.8) is a finite algebraic reconstruction; its passage to an infinite series requires convergence. The following theorem constructs the stable inverse through the limit of positive operators.

113.3 Memory of a nonstationary sequence

In a Hilbert realization H, let C_n be unitaries, with their declared composition order. We write

$$T_n = (8I + C_n)/9, \quad D_n = \sqrt{8}(C_n - I)/9, \quad P_0 = I, \quad P_{n+1} = T_nP_n, \quad A_N = P_N^*P_N.$$

The index N counts compressions with separate storage, unlike complete transport with reentry. For variable metrics with B_n isometric isomorphisms, conjugation by G_n takes the transports to a fixed Hilbert space and produces this same case, with $\widetilde{C}_n = G_{n+1}^{-1}C_nG_n$. The assertion uses the fiber metrics and the declared isometries between them.

Theorem 113.2 (Positive terminal and nonstationary recovery)**.** *There exists a positive operator* $0 \le A_\infty \le I$*, the strong limit of* A_N*, and*

$$\boxed{I = A_\infty + \sum_{n\ge 0} P_n^*D_n^*D_nP_n.} \tag{113.9}$$

The map

$$\boxed{\mathcal{V}v = (A_\infty^{1/2}v, (D_nP_nv)_{n\geq 0})} \tag{113.10}$$

is isometric and has a stable left inverse

$$\mathcal{V}^*(a,(m_n)) = A_\infty^{1/2}a + \sum_{n\geq 0} P_n^*D_n^*m_n, \qquad \|\mathcal{V}^*\| = 1. \tag{113.11}$$

Proof. The local identity $T_n^*T_n + D_n^*D_n = I$ gives $A_N - A_{N+1} = P_N^*D_N^*D_NP_N \geq 0$. The quadratic forms of the decreasing positive sequence converge to that of a positive operator A_∞. For $B_N = A_N - A_\infty$, the inequality $0 \leq B_N^2 \leq B_N$ makes that convergence strong. Finite telescoping and the limit prove (113.9). Evaluation on v proves (113.10). Its adjoint is (113.11); the series converges in norm because the truncations of a square-summable register converge and the adjoint is bounded. □

Reconstruction with the limit terminal and N registers has exact error $(A_N - A_\infty)v$. The vector $A_\infty^{1/2}v$ is the canonical positive realization of the norm terminal. Its existence follows from the Gram limit, independently of vector convergence of P_Nv.

113.4 Three different properties: exhaustion, distinction and stability

Let $Mv = (D_nP_nv)_n$. From (113.9), $M^*M = I - A_\infty$. Therefore:

1. All the norm of v passes to the register exactly when $A_\infty v = 0$.
2. The register distinguishes each state exactly when $\ker(I - A_\infty) = \{0\}$.
3. Recovery from memory alone has a bounded inverse exactly when $I - A_\infty \geq \varepsilon I$ for some $\varepsilon > 0$. In that case a left inverse is $(I - A_\infty)^{-1}M^*$.

Moreover,

$$\boxed{\ker M = \bigcap_{n\geq 0} \mathrm{Fix}(C_n).} \tag{113.12}$$

Proof. If $Mv = 0$, then $D_0v = 0$, whence $C_0v = v$ and $P_1v = v$. Inductively, $D_nP_nv = 0$ gives $C_nv = v$ and $P_{n+1}v = v$. The converse is immediate. The three equivalences follow from the Gram matrix M^*M and the characterization of operators bounded below. □

113.4.1 Exact example separating the three properties

In the algebraic example $C_0 = -I$ and $C_n = I$ for $n \geq 1$,

$$A_\infty = \frac{49}{81}I, \qquad M^*M = \frac{32}{81}I, \qquad m_0 = -\frac{2\sqrt{8}}{9}v, \qquad v = -\frac{9}{2\sqrt{8}}m_0. \tag{113.13}$$

The memory contains $32/81$ of the squared norm and allows the complete vector to be recovered. The terminal retains $49/81$ and is a positive operator distinct from a projector. Recoverability depends on the kernel and conditioning of the reader. Reading and memory constitute correlated components of an isometry.

This example separates the three logical consequences of the theorem. Application to a physical trajectory also preserves the selection and effective provenance of its C_n through TPK.

113.5 Quantitative criterion for complete transfer

For $P_{j,n} = T_{j-1}\cdots T_n$ and $P_{n,n} = I$, a window of length L has Gram matrix

$$G_{n,L} = \sum_{j=n}^{n+L-1} P_{j,n}^{*} D_j^{*} D_j P_{j,n} = I - P_{n+L,n}^{*} P_{n+L,n}.$$

If $G_{n,L} \geq \kappa I$, with $\kappa > 0$, in each consecutive block of length L, then

$$\|P_N\|^2 \leq (1-\kappa)^{\lfloor N/L \rfloor}, \qquad A_\infty = 0. \tag{113.14}$$

Each block contracts the squared norm by $1-\kappa$; composition and contractivity of the remaining steps prove (113.14).

A sufficient criterion preceding the products is

$$\sum_{j=n}^{n+L-1} D_j^{*} D_j \geq \alpha I \quad \Longrightarrow \quad G_{n,L} \geq \frac{\alpha}{[1+2(L-1)/9]^2} I. \tag{113.15}$$

Proof. For $y_j = P_{j,n} v$, we have $y_{j+1} - y_j = D_j y_j/\sqrt{8}$ and $\|D_j\| \leq 2\sqrt{8}/9$. Hence

$$\|D_j v\| \leq \|D_j y_j\| + \frac{2}{9} \sum_{i=n}^{j-1} \|D_i y_i\|.$$

The triangular matrix of partial sums has norm at most $L-1$. Applying the window's ℓ^2 norm and the Gram hypothesis proves (113.15).

□

The absence of a common fixed vector gives distinguishability, whereas (113.15) adds a quantitative uniform separation. Neither condition is presumed merely by mentioning symmetries: each applies to the effective transports of the sequence.

113.6 Complete incidence realization and changes of regime

Section 107.1 constructs, for the twelve-coordinate register, $Fk = (\mu(k), \mathcal{I} P_0 k/6)$, with an inverse and weighted inner product on $Y = \mathbb{R} \oplus \mathcal{I}(\mathbf{1}^{\perp})$. The transformation $F \oplus F$ takes the complete coupling to

$$\mathcal{U}_Y = (F' \oplus F')\mathcal{U}(B,C)(F^{-1} \oplus F^{-1}) = \mathcal{U}(F'BF^{-1}, F'CF^{-1}). \tag{113.16}$$

This identity preserves the second column, re-entry, and composition. For variable forms, the form is also transported through F. In the real realization of the twelve A_2 fibres, use $F \otimes \mathrm{id}_{V_{A_2}}$, with $V_{A_2} = A_2 \otimes_{\mathbb{Z}} \mathbb{R}$. Its domain is the real space of fibres; preservation of the integer lattice corresponds to the lattice transport already proved. The map F recovers K and retains the remaining genealogical labels in the complete state.

The positive version of (113.9)–(113.15) uses unitary transports in the declared metrics. The alternating-form version of (113.1)–(113.8) admits the three Euler regimes. With $E = F_{\mathrm{av}}^2 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, we obtain $\det T = 89/81$, $\det D = -8/81$. The finite oriented sums telescope, and the area terminal components grow as $(89/81)^N$. The positive Hilbert limit uses the specific unitary hypotheses of Theorem 113.2.

The balance brings together forms and domains, reversible evolution with re-entry, and unlimited recovery in the positive realizations. The subsequent physical applications preserve the readers, connections, and scales of this operator composition.

Chapter 114

Dimensional reconstruction and exterior conservation

114.1 Two memories recover the centred register

114.1.1 Register-reading operators

In the ordered chart of the register K of section 11.4 y theorem 11.43, let $V = \mathbb{R}^{12}$ be its subsequent linear realization and let S be the permutation $(Sx)_i = x_{i+1}$, with indices modulo 12. The readers $I - S^3$ and $I - S^4$ are those of the transverse extractor of theorem 11.43. The nonadic compression of chapter 113 provides, for $j = 3, 4$,

$$T_j = \frac{8I + S^j}{9}, \qquad D_j = \frac{\sqrt{8}}{9}(S^j - I), \qquad T_j^* T_j + D_j^* D_j = I. \tag{114.1}$$

The partition 8 + 1 and normalization 9 come from the nine positions and their collective contrast. Consider the following explicit composition of readers, applied to the generated register:

$$m_0 = D_3 K, \qquad m_1 = D_4 T_3 K, \qquad MK = (m_0, m_1). \tag{114.2}$$

The second memory acts on the reading produced by the first stage. The complements are archived separately. The complete evolution with re-entry corresponds to the reversible coupling in chapter 113.

114.1.2 Inversa exacta

Since $(S^3)^4 = I$,

$$T_3^{-1} = \frac{512I - 64S^3 + 8S^6 - S^9}{455}. \tag{114.3}$$

Indeed, $(8I + S^3)(512I - 64S^3 + 8S^6 - S^9) = 4095I$ and $4095 = 9 \cdot 455$. Hence the memories provide

$$b = -\frac{9}{\sqrt{8}} m_0 = (I - S^3)K, \quad c = -\frac{9}{\sqrt{8}} T_3^{-1} m_1 = (I - S^4)K. \tag{114.4}$$

Commutation of T_3 with S^4 justifies the second equality. Define

$$w_i = c_i - b_{i+1}, \quad B_0 = 0, \quad B_i = \sum_{j<i} w_j.$$

The identity $w_i = K_i - K_{i+1}$ proves that $B_i = K_0 - K_i$. If $q = \sum_{i=0}^{11} K_i$, we recover exactly

$$K_i = \frac{q + \sum_{j=0}^{11} B_j}{12} - B_i. \tag{114.5}$$

This is the inverse (11.112), now composed with compression and memory. With $q = 0$, the formula returns $P_{11}K$, where

$$P_{11} = I - \frac{1}{12}\mathbf{1}\mathbf{1}^*.$$

The two memories recover the eleven zero-mean components. The uniform charge is recovered through q. The kernel of M is precisely the uniform line: vanishing of both differences implies fixation by S^3 and S^4; since $S = S^4(S^3)^{-1}$, it implies fixation by S.

114.1.3 Optimal stability bound

The Gram matrix of the procedure is

$$G = M^{*}M = D_3^{*}D_3 + T_3^{*}D_4^{*}D_4T_3 = I - (T_4T_3)^{*}(T_4T_3). \tag{114.6}$$

In mode k of S, the eigenvalue of $T_j^{*}T_j$ is

$$\frac{65 + 16\cos(2\pi jk/12)}{81}.$$

Substituting it into the preceding Gram matrix gives, with multiplicities,

$$\operatorname{Spec} G = \left\{0^{[1]},\ (16/81)^{[2]},\ (8/27)^{[2]},\ (32/81)^{[1]}, (952/2187)^{[4]},\ (1256/2187)^{[2]}\right\}. \tag{114.7}$$

The zero mode is the mean. Let L be the least-squares inverse of M on $\mathbf{1}^{\perp}$, extended to the register space by

$$L = (G|_{\mathbf{1}^{\perp}})^{-1}M^{*}.$$

Then

$$LM = P_{11}, \qquad \|L\| = \frac{9}{4}. \tag{114.8}$$

The bound is optimal because modes $k = 3, 9$ attain $16/81$. For perturbed data and reconstruction with L,

$$\|\delta K\|^2 \le \frac{|\delta q|^2}{12} + \frac{81}{16}\big(\|\delta m_0\|^2 + \|\delta m_1\|^2\big). \tag{114.9}$$

The uniform and centred components are orthogonal. Formula (114.5) is an exact inverse on compatible data; (114.8) specifies the stable extension used for noisy data, which need not satisfy the compatibility conditions of (114.5).

114.1.4 Example with the generated dodecaphasic register

The register in (11.107) is

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601). \tag{114.10}$$

It is an output of the preceding generator, used as an application case. The coefficients and readers are fixed before this register is evaluated. To store the memories using rational arithmetic, write $z_i = m_i/\sqrt{8}$. The two effective registers are

$$9z_0 = (495, 116, 684, -108, -601, 90, 173, 88, -313, -560, 397, -461),$$

$$81z_1 = (2729, 2503, 3818, -5843, 2583, -920, -4051, 4338, -5312, -1583, 233, 1505). \tag{114.11}$$

Substituting these registers into the reconstruction formulas, with $q = 6263$, reproduces the twelve components of K. With $q = 0$, it reproduces $K - (6263/12)\mathbf{1}$. The supplementary certificate first reconstructs the register from the memories and then compares it with (11.107).

Adding a third memory $D_3T_4T_3K$ raises the smallest positive eigenvalue of the Gram matrix to $8/27$. The optimal amplification decreases from $9/4$ to $\sqrt{27/8}$. A third memory $D_4T_4T_3K$ retains $16/81$. The choice between repeating step 3 or step 4 has a stability effect calculable before evaluating the register.

114.2 Reconstruction of the dimensional projector from two memories

114.2.1 Composition with the incidence already constructed

The exceptional chart of section 21.11.5 defines the orthogonal representation of A_5 on the twelve classes of A_5/C_5, with explicit representatives and character. Its three-dimensional isotypic projector is (21.137):

$$P_3 = \frac{3}{60}\sum_{a\in A_5} \chi_3(a^{-1})\rho(a), \qquad P_3P_{11} = P_3, \qquad P_3\mathbf{1} = 0. \tag{114.12}$$

The generated register precedes this representation. The result in lemma 21.62, together with (21.139) and (21.140), establishes

$$u_K = P_3P_{11}K, \qquad \|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0,$$

$$P_{10}(K) = P_{11} - \frac{u_K u_K^{*}}{\|u_K\|^2}. \tag{114.13}$$

The new composition of (114.8), (114.12), and (114.13) is

$$\boxed{u_K = P_3L(m_0, m_1), \qquad P_{10}(K) = P_{11} - \frac{P_3L(m)\,[P_3L(m)]^{*}}{\|P_3L(m)\|^2}.} \tag{114.14}$$

Proof. The identities $LM = P_{11}$ and $P_3P_{11} = P_3$ give $P_3LMK = P_3K = u_K$. Nonvanishing is evaluated in (21.139). The rank-one quotient therefore coincides with that of (21.140). The composition recovers the complete rank-ten orthogonal projector from the two memories, independently of the charge q and terminal reading T_4T_3K. □

The reduction $12 \to 11$ removes the uniform direction, and the reduction $11 \to 10$ uses the vector selected by K. The composition recovers the full projector matrix on the linear incidence carrier. Its dimensional realizations retain the maps of proposition 21.64.

114.2.2 Projector error

Let ε be the joint norm of the error in the two memories, and let $\widehat{u} = P_3L(m+\delta m)$. Then $\|\widehat{u} - u_K\| \le 9\varepsilon/4$. If $9\varepsilon/4 < \|u_K\|$, the vector $\widehat{u}$ is nonzero and

$$\boxed{\|\widehat{P}_{10} - P_{10}\| \le \frac{9\varepsilon}{4\|u_K\|}.} \tag{114.15}$$

Proof. The operator-norm distance between two orthogonal projectors onto lines is the sine of the angle between them. The distance from u_K to the line of $\widehat{u}$ is at most $\|u_K - \widehat{u}\|$. Division by $\|u_K\|$ proves the bound. Here $\|u_K\| \approx 765{,}733$, and the coefficient is approximately $0{,}00293836$. □

For registers close to $\ker P_3$, stability of the direction depends on $\|u_K\|$. The bound $9/4$ for the centred register remains valid.

114.2.3 A scalar descriptor of K, with explicit invariances

The register K, its scalar readings, and its invariants are distinguished. For $K \notin \mathbb{R}\mathbf{1}$, define

$$\eta_K = \frac{\|P_3 P_{11} K\|^2}{\|P_{11} K\|^2}. \tag{114.16}$$

For the register (11.107), $\sum_i K_i^2 = 4251257$ and $\|P_{11}K\|^2 = 11789915/12$. Thus,

$$\eta_K = \frac{3(6638585 + 2275584\sqrt{5})}{5 \cdot 11789915} = \frac{13089003 + 13653504\varphi_{\mathrm{HMT}}}{58949575} \approx 0,5967954228259091. \tag{114.17}$$

The second expression uses the subsequent quadratic recognition of φ_{HMT}. We have $0 \le \eta_K \le 1$. The quotient is invariant under $K \mapsto aK + b\mathbf{1}$, with $a \neq 0$, and under simultaneous orthogonal changes of K, P_{11}, P_3. In the fixed chart it is invariant under $\rho(A_5)$, since both projectors commute with that representation. It is recoverable from the memories because both norms defining it are recoverable. It measures the quadratic fraction of the centered register lying in the chosen isotypic sector. The complete register also preserves the information that this scalar descriptor identifies. The provenance of this additional formalization is documented in the supplement; the affine invariances proved here belong to the linear realization.

114.3 Application to an effective vector of the exceptional construction

The construction of section 21.11.13 y proposition 21.73 uses twelve fibers A_2, their gluing and the lattice neighbor. The tritic word is $c^\star = (1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1)$. Its blocks act by C^{-1} when the component is 1 and by C when it is 2, in the charts specified by equations (21.147) y (21.148). The resulting operator g satisfies $g^3 = I$, $g^2 + g + I = 0$, and is orthogonal for the sum of the metrics A_2. Descent to the gluing and neighbor is proved in proposition 21.73. The following calculations act on its real carrier of dimension 24.

In each fibre,

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

On the sum of the twelve fibres, define

$$T = \frac{8I + g}{9}, \qquad D = \frac{\sqrt{8}}{9}(g - I).$$

With the adjoint relative to that metric, $g^* = g^{-1} = -g - I$, and we obtain

$$T^*T = \frac{19}{27}I, \qquad D^*D = \frac{8}{27}I, \qquad D^{-1} = -\frac{3}{\sqrt{8}}(g + 2I). \tag{114.18}$$

The inverse comes from $(g - I)(g + 2I) = -3I$. The neighbour construction fixes the vector in (21.142), $v = 4\rho_1 + \rho_2 + \cdots + \rho_{12}$, with a root ρ_b of squared norm 2 in each fibre. Therefore,

$$\|v\|^2 = 54, \qquad \|Tv\|^2 = 38, \qquad \|Dv\|^2 = 16, \qquad v = -\frac{3}{\sqrt{8}}(g + 2I)Dv. \tag{114.19}$$

A single complement recovers this complete vector. The reading preserves 38 units of squared norm and the register contains 16; their sum is 54. The result uses the vector in (21.142) and its lattice fibres.

114.4 Conservation law in every exterior degree

114.4.1 Statement and proof

Let $P_0 = I$, $P_n = T_{n-1} \cdots T_0$, and

$$\mathcal{J}_N v = (P_N v, D_0 P_0 v, \ldots, D_{N-1} P_{N-1} v).$$

The law in chapters 110 y 113 preserves the form G_0 through the orthogonal sum of the corresponding forms: $\mathcal{J}_N^* G_{\mathrm{sal}} \mathcal{J}_N = G_0$. For each admissible finite exterior degree p, it follows that

$$\boxed{(\Lambda^p \mathcal{J}_N)^* (\Lambda^p G_{\mathrm{sal}})(\Lambda^p \mathcal{J}_N) = \Lambda^p G_0.} \tag{114.20}$$

The induced form is defined by

$$\langle x_1 \wedge \cdots \wedge x_p, y_1 \wedge \cdots \wedge y_p \rangle_{\Lambda^p G} = \det(\langle x_i, y_j \rangle_G)_{i,j=1}^{p}.$$

Proof. Substitute the degree-one balance into each entry of this determinant. The equality holds on decomposable vectors and extends bilinearly because these span the exterior power. □

The relevant decomposition is

$$\Lambda^p \Big(\bigoplus_a H_a \Big) \simeq \bigoplus_{\sum n_a = p} \bigotimes_a \Lambda^{n_a} H_a.$$

In the positive case, if $\mathcal{J}_{p,\mathbf{n}}$ denotes the components of that map,

$$\boxed{\|\xi\|^2 = \sum_{\sum n_a = p} \|\mathcal{J}_{p,\mathbf{n}} \xi\|^2.} \tag{114.21}$$

The purely visible, purely registered, and mixed components are preserved. For $p = 2$ and $p = 3$, the squared area and volume norms, respectively, are preserved. For an alternating or indefinite form, the exterior identity retains its bilinear interpretation.

114.4.2 Exact distribution among the exterior sectors

In the preceding Coxeter realization, $a = 19/27$ and $b = 8/27$. Normalizing T and D gives two isometries, so the component with k directions in memory has squared norm

$$\boxed{\|\mathcal{J}_{p,k} \xi\|^2 = \binom{p}{k} a^{p-k} b^k \|\xi\|^2.} \tag{114.22}$$

This is proved by exterior expansion of $x \mapsto (\sqrt{a}\, x, \sqrt{b}\, x)$. Components with distinct memory positions are orthogonal; the normalized isometries T and D preserve their norms.

For a unit-norm bivector,

Sector	Squared norm
Two directions in the reading	361/729
One direction in the reading and one in memory	304/729
Two directions in memory	64/729

The sum is one. The mixed sector contains exactly 304/729, approximately 41,7%, of the squared-area identity. In degree three, the four components are $(6859, 8664, 3648, 512)/19683$; the mixed components sum to $12312/19683$.

The law applies to $0 \le p \le 24$ on the carrier of the twelve A_2 fibres, and functorially to other generated carriers. Exterior degree, refinement resolution, the scale of a metric, and physical dimension have distinct typings. The change $G \mapsto s^2 G$, with $s > 0$, multiplies the squared norm of degree p by s^{2p}; both sides of the exterior identity transform in the same way. The realization of dimensions 10 and 11 preserves the previously specified dimensional maps.

114.4.3 Naturality and arbitrary depth

The generated refinement of histories preserves measure through

$$R\delta_h = \sum_{\epsilon \in \mathrm{Out}(h)} \sqrt{p_h(\epsilon)}\, \delta_{h\epsilon}.$$

The probabilities come from the multiplicities and normalization of the APP–TRIT–TPK emissions in section 6.4.2 y chapter 103. The register identifies the preceding state of each emission. If the transport square is $\mathcal{J}_N' R = (\bigoplus_a R_a)\mathcal{J}_N$, then

$$(\Lambda^p J_N')(\Lambda^p R) = \Lambda^p(\bigoplus R_a)(\Lambda^p J_N). \tag{114.23}$$

Apply $\Lambda^p(AB) = \Lambda^p(A)\Lambda^p(B)$. The squares of the five correlative operations and their common limit are those of propositions 6.28 y 6.29 y section 6.4.9; all act within the same construction of the continuum.

In the nonstationary positive case, chapter 113 proves $P_N^* P_N \downarrow A_\infty$ strongly, and

$$\mathcal{J} v = (A_\infty^{1/2} v, (D_n P_n v)_{n \ge 0}), \qquad \mathcal{J}^* \mathcal{J} = I.$$

By density, $\Lambda^p \mathcal{J}$ is isometric for each finite degree p. The exterior sum extends to finitely supported multi-indices over the countable Hilbert direct sum; Parseval justifies passage to the limit. The terminal component $A_\infty^{1/2}$ and the memories jointly preserve the initial form at arbitrary depth.

114.4.4 Precision of area and volume reconstruction

For the reader M, order the eleven positive eigenvalues of G, with multiplicity, as $\lambda_1 \le \cdots \le \lambda_{11}$. In degrees $1 \le p \le 11$ of $\mathbf{1}^\perp$, the singular-value decomposition induces

$$\|(\Lambda^p M)^\dagger\| = (\lambda_1 \cdots \lambda_p)^{-1/2}. \tag{114.24}$$

The exterior modes use p distinct directions. The computed spectrum gives the optimal constants $9/4$, $81/16$, $243\sqrt{6}/64$, and $2187/128$ for $p = 1, 2, 3, 4$. In degree twelve, $\Lambda^{12} M = 0$; recovery of the complete volume also incorporates the uniform charge. The tensor product of p independent readers on the zero-mean subspace has bound $(9/4)^p$. Tensoring the reader with an identity preserves the bound $9/4$. The complete register $\mathcal{J}$ has an inverse of norm one on its image in every degree.

114.5 Transport memory and relative holonomy

The reversible coupling in chapter 113 is

$$U(B, C) = \frac{1}{9} \begin{pmatrix} 8B + C & \sqrt{8}(C - B) \\ \sqrt{8}(C - B) & B + 8C \end{pmatrix} = Q^* \,\mathrm{diag}(B, C) Q,$$

with Q constant and orthogonal, fixed by the partition $8 + 1$. Hence $U(B_2, C_2)U(B_1, C_1) = U(B_2 B_1, C_2 C_1)$, preserving the order of factors. On a loop γ, let H_B, H_C be the ordered products of the two transports. The memory block of the complete evolution is

$$D_\gamma = \frac{\sqrt{8}}{9}(H_C - H_B).$$

We reconstruct exactly

$$\boxed{H_B^{-1}H_C = I + \frac{9}{\sqrt{8}}H_B^{-1}D_\gamma.} \tag{114.25}$$

The equality is operatorial: recording the response of the block on a basis recovers the complete relative holonomy. Its evaluation on a vector recovers the corresponding action. Under a chart change at the base point, both sides are conjugated. If $H_B = I$, the complement determines $H_C - I$. Memory and transport defect along a circuit are thus related; the differential realization additionally uses its connection and the limit of circuits.

114.5.1 An elliptic–parabolic loop produces hyperbolic self-scaling

The construction in proposition 10.47 contains the Coxeter operator C and the nilpotent

$$N_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

on their respective carriers. Consider their realizations on a common oriented plane, using the indicated bases and the form

$$\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

The Coxeter element C and the transport $I + N_0$ have determinant one and preserve Ω. This chart allows them to be composed while preserving the distinction between the generators of their regimes and their own positive metrics.

Let

$$P = I + N_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Since $N_0^2 = 0$, we have $P^n = I + nN_0$ for every $n \in \mathbb{Z}$. The ordered loop of four transports gives

$$H_n = C^{-1}P^{-n}CP^n = \begin{pmatrix} 1+n & n^2 \\ n & n^2 - n + 1 \end{pmatrix}. \tag{114.26}$$

For $n \neq 0$, its normalized difference is

$$F_n = \frac{H_n - I}{n} = \begin{pmatrix} 1 & n \\ 1 & n-1 \end{pmatrix}.$$

Multiplying these matrices yields, without introducing spectral values,

$$\boxed{F_n^2 = H_n, \qquad F_n^2 - nF_n - I = 0.} \tag{114.27}$$

With $H_B = I$, the memory is $D_{\gamma,n} = (\sqrt{8}/9)nF_n$. Consequently,

$$\boxed{H_n = \left(\frac{9}{n\sqrt{8}}D_{\gamma,n}\right)^2.} \tag{114.28}$$

The matrix memory response reconstructs the hyperbolic self-scale. For $n > 0$, subsequent spectral recognition identifies

$$\rho_n = \frac{n + \sqrt{n^2+4}}{2}, \quad \operatorname{Spec} F_n = \{\rho_n, -\rho_n^{-1}\}, \quad \operatorname{Spec} H_n = \{\rho_n^2, \rho_n^{-2}\}. \tag{114.29}$$

The case $n = 1$ produces the golden polynomial. The relation to the operator $F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ generated by the calendar in (10.41) preserves the chart change:

$$F_1 = P F_{\mathrm{av}} P^{-1}, \qquad H_1 = P F_{\mathrm{av}}^2 P^{-1}. \tag{114.30}$$

The transport P, of determinant one, intertwines the two matrices and preserves Ω. The case $n = 2$ produces the spectral readings $1 + \sqrt{2}$ and $3 + 2\sqrt{2}$. The composition relates these realizations while preserving their respective genealogies; the selection of a four-factor word remains specified as part of the reader.

In the same example $n = 1$, the oriented balance is

$$D_\gamma^{\mathsf{T}} \Omega D_\gamma = -\frac{8}{81}\Omega, \qquad T_\gamma^{\mathsf{T}} \Omega T_\gamma = \frac{89}{81}\Omega. \tag{114.31}$$

The sum of the two contributions is Ω. The memory term has opposite orientation: the visible contribution $89/81$ is compensated by $-8/81$. This alternating balance and the positive allocation $19/27 + 8/27 = 1$ preserve their respective forms.

The expression for H_n holds for every $n \in \mathbb{Z}$. The definition of F_n and reconstruction through its square use $n \neq 0$; the preceding spectral parametrization uses $n > 0$. In general,

$$D_{\gamma,n}^{\mathsf{T}} \Omega D_{\gamma,n} = -\frac{8n^2}{81}\Omega, \qquad T_{\gamma,n}^{\mathsf{T}} \Omega T_{\gamma,n} = \left(1 + \frac{8n^2}{81}\right)\Omega.$$

The identities $N_0^2 = 0$ and $C^2 + C + I = 0$ prove the complete family; the supplementary program checks its exact specializations. The provenance of this additional formalization is documented in the supplement.

114.6 Symmetry, action, and conserved charge

Let $(E_n)_{n\in\mathbb{Z}}$ be finite-dimensional real vector spaces and $U_n : E_n \to E_{n+1}$ the invertible transports of the complete evolution. The cotangent lift acts on $q_n \in E_n$ and $p_n \in E_n^*$. In dual bases, define the auxiliary discrete Lagrangian

$$L_n(q_n, q_{n+1}, p_{n+1}) = p_{n+1}^{\mathsf{T}}(q_{n+1} - U_n q_n).$$

The action is used in a local sense: for each finitely supported variation, sum L_n over a finite interval containing all affected terms. The first variation is independent of any enlargement of that interval. Specifically,

$$\delta \mathcal{S} = \sum_{n\in\mathbb{Z}} \Big[\delta p_n^{\mathsf{T}}(q_n - U_{n-1} q_{n-1}) + \delta q_n^{\mathsf{T}}(p_n - U_n^{\mathsf{T}} p_{n+1}) \Big], \tag{114.32}$$

where the sum is finite for every admitted variation. Stationarity with respect to all these variations is equivalent to

$$q_{n+1} = U_n q_n, \qquad p_n = U_n^{\mathsf{T}} p_{n+1} \quad (n \in \mathbb{Z}).$$

The second equation is $p_{n+1} = U_n^{-\mathsf{T}} p_n$, that is, the cotangent lift of transport.

Proposition 114.1 (Variational charge of intertwined transport)**.** *Let $A_n \in \operatorname{End}(E_n)$ be a family satisfying*

$$A_{n+1} U_n = U_n A_n.$$

On the stationary solutions of (114.32)*, the charge*

$$\boxed{J_n = p_n^{\mathsf{T}} A_n q_n = J_{n+1}} \tag{114.33}$$

is constant.

Proof. The evolution equations and intertwining give

$$J_{n+1} = p_{n+1}^{\mathsf{T}} A_{n+1} U_n q_n = p_{n+1}^{\mathsf{T}} U_n A_n q_n = p_n^{\mathsf{T}} A_n q_n.$$

Conservation also has an explicit variational derivation. For constant parameter ε, the transformations

$$q_n \mapsto e^{\varepsilon A_n} q_n, \qquad p_n \mapsto e^{-\varepsilon A_n^{\mathsf{T}}} p_n$$

preserve every L_n, because $e^{\varepsilon A_{n+1}} U_n = U_n e^{\varepsilon A_n}$. If the parameter is replaced by a finitely supported sequence (ε_n), the first variation is

$$\delta L_n = (\varepsilon_{n+1} - \varepsilon_n) p_{n+1}^{\mathsf{T}} U_n A_n q_n.$$

On a solution, the last factor is J_n. Finite summation by parts gives

$$\delta \mathcal{S} = \sum_{n \in \mathbb{Z}} \varepsilon_n (J_{n-1} - J_n).$$

Stationarity for every finitely supported sequence produces (114.33). □

For the preceding cyclic readers, take

$$A_0 = S - S^{-1}, \qquad \mathcal{A} = \operatorname{diag}(A_0, A_0)$$

on the complete carrier $V \oplus V$. The matrix $\mathcal{A}$ commutes with $U(I, S^3)$ and $U(I, S^4)$, since its blocks are polynomials in S. Consequently, $p_n^{\mathsf{T}} \mathcal{A} q_n$ preserves an oriented correlation between neighbouring positions.

The construction specifies a Lagrangian, its symmetry, and the corresponding conserved charge, in the variational sense developed in chapter 109. Its domain is the cotangent realization of the readers. The physical action and its units retain the operators and normalizations of the corresponding constructions. The auxiliary Lagrangian is obtained after the transport U_n.

Chapter 115

Canonical realization of covariance

115.1 Canonical covariance and reciprocity of the action ellipse

The preceding construction determines two complementary invariants: the product of the semiaxes fixes the action area and their quotient preserves the anisotropy of the angular pair. We now specify the canonical realization in which these semiaxes are two standard deviations. The starting structure is the ellipse already obtained from the angular and action expressions of APP–TRIT–TPK; the operator realization is established after that construction.

In this subsection we write $\eta = \eta_{\mathrm{el}} = \operatorname{artanh}(C^*/A)$, with $|C^*/A| < 1$ and $\hbar > 0$. The angular quotient uses the same unit for both its components. In the balanced coordinates (Q, P), both of square-root-of-action dimension, we define

$$V_\eta := \frac{1}{4}\begin{pmatrix} a_S^2 & 0 \\ 0 & b_S^2 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} e^{\eta} & 0 \\ 0 & e^{-\eta} \end{pmatrix}. \tag{115.1}$$

This positive matrix has entries of action dimension. Its interpretation as a quantum covariance is specified by the following realization.

Proposition 115.1 (Gaussian realization of the action covariance)**.** *Consider the canonical representation on $L^2(\mathbb{R}, dQ)$, with $\widehat{Q}f(Q) = Qf(Q)$ and $\widehat{P}f(Q) = -i\hbar f'(Q)$, initially on the Schwartz space $\mathcal{S}(\mathbb{R})$. The normalized state*

$$\psi_\eta(Q) = \left(\pi\hbar e^{\eta}\right)^{-1/4} \exp\!\left(-\frac{Q^2}{2\hbar e^{\eta}}\right) \tag{115.2}$$

has zero means and covariance matrix V_η. In particular,

$$(\Delta Q)^2 = \frac{\hbar e^{\eta}}{2}, \qquad (\Delta P)^2 = \frac{\hbar e^{-\eta}}{2}, \qquad \operatorname{Cov}(Q, P) = 0, \qquad \det V_\eta = \frac{\hbar^2}{4}. \tag{115.3}$$

This realization saturates the Robertson–Schrödinger inequality.

Proof. The preceding operators are essentially self-adjoint on $\mathcal{S}(\mathbb{R})$, which is a common invariant core; on this space, $[\widehat{Q}, \widehat{P}] = i\hbar I$. For $\widehat{Y} = (\widehat{Q}, \widehat{P})$, the symmetric covariance is

$$V_{jk} = \frac{1}{2}\left\langle (\widehat{Y}_j - \langle \widehat{Y}_j \rangle)(\widehat{Y}_k - \langle \widehat{Y}_k \rangle) + (\widehat{Y}_k - \langle \widehat{Y}_k \rangle)(\widehat{Y}_j - \langle \widehat{Y}_j \rangle) \right\rangle.$$

With $s = \hbar e^{\eta} > 0$, let $I_s = \int_{\mathbb{R}} e^{-Q^2/s}\, dQ$. The product of two equal integrals, evaluated in polar coordinates, gives $I_s^2 = 2\pi \int_0^\infty r e^{-r^2/s}\, dr = \pi s$. Moreover, the integral of $\frac{d}{dQ}(Qe^{-Q^2/s})$ is zero, owing to decay at both endpoints; hence $\int_{\mathbb{R}} Q^2 e^{-Q^2/s}\, dQ = sI_s/2$. These identities and parity prove normalization, the zero mean and $\langle \widehat{Q}^2 \rangle = s/2$. Moreover, $\widehat{P}\psi_\eta = ie^{-\eta} Q\psi_\eta$. Therefore, $\langle \widehat{P} \rangle = 0$, $\|\widehat{P}\psi_\eta\|^2 = \hbar e^{-\eta}/2$ and $\operatorname{Re}\langle \widehat{Q}\psi_\eta, \widehat{P}\psi_\eta \rangle = 0$.

For any normalized state in that domain, Cauchy–Schwarz applied to $(\widehat{Q} - \langle \widehat{Q} \rangle)\psi$ and $(\widehat{P} - \langle \widehat{P} \rangle)\psi$ gives

$$(\Delta Q)^2 (\Delta P)^2 \geq \operatorname{Cov}(Q, P)^2 + \frac{1}{4}\left|\langle [\widehat{Q}, \widehat{P}] \rangle\right|^2 = \operatorname{Cov}(Q, P)^2 + \frac{\hbar^2}{4}.$$

The computed variances attain equality. Equivalently, the operator

$$a_\eta = \frac{e^{-\eta/2}\widehat{Q} + ie^{\eta/2}\widehat{P}}{\sqrt{2\hbar}}$$

satisfies $[a_\eta, a_\eta^\dagger] = I$ and $a_\eta \psi_\eta = 0$, by direct substitution. □

Proposition 115.2 (Area, dispersion, and reciprocal transport)**.** *The action ellipse is exactly the covariance sublevel set*

$$\mathcal{E}_\eta = \left\{ y = (Q,P)^{\mathsf{T}} : y^{\mathsf{T}} V_\eta^{-1} y \le 4 \right\} = \left\{ \frac{Q^2}{a_S^2} + \frac{P^2}{b_S^2} \le 1 \right\}. \tag{115.4}$$

Its area is $h = 2\pi\hbar$, its semiaxes are $a_S = 2\Delta Q$, $b_S = 2\Delta P$, and

$$\frac{\Delta P}{\Delta Q} = e^{-\eta} = \frac{b_S}{a_S} = R_{\mathrm{el}}^2. \tag{115.5}$$

The canonical rotation J_2 of Section 24.3 transports V_η to $V_{-\eta}$ and preserves the symplectic form.

Proof. Inverting the diagonal matrix (115.1) gives (115.4). Its semiaxes are the positive roots of four times its variances, and

$$\mathrm{Area}(\mathcal{E}_\eta) = 4\pi\sqrt{\det V_\eta} = 2\pi\hbar = h,$$

in agreement with the areal law (24.4). The ratio of the positive roots proves (115.5). If $M_\eta = \mathrm{diag}(e^{\eta/2}, e^{-\eta/2})$, then

$$V_\eta = M_\eta V_0 M_\eta^{\mathsf{T}}, \qquad M_\eta^{\mathsf{T}} J_2 M_\eta = J_2, \qquad J_2 V_\eta J_2^{\mathsf{T}} = V_{-\eta}.$$

These identities follow by multiplication. In operator terms, $(\widehat{Q}', \widehat{P}') = (-\widehat{P}, \widehat{Q})$ preserves $[\widehat{Q}', \widehat{P}'] = i\hbar I$. The permutation $S_2 : (Q,P) \mapsto (P,Q)$ also exchanges the variances but reverses the sign of the commutator and of the symplectic form. Thus covariance transport respects the distinction between canonical reciprocity and orientation reversal already established in Section 24.3. □

115.1.0.1 Position, momentum, and dimensional transduction.

Let $\lambda_{xp} > 0$ be a factor of the dimensional chart, with the dimension of momentum per unit length. The transformation

$$Q = \sqrt{\lambda_{xp}}\, x, \qquad P = \frac{p_x}{\sqrt{\lambda_{xp}}}$$

realizes the balanced pair as a position x and its conjugate momentum p_x; it preserves $dQ \wedge dP = dx \wedge dp_x$ and $[\widehat{x}, \widehat{p}_x] = i\hbar I$. By transformation of covariance,

$$V_\eta^{(x,p_x)} = \frac{\hbar}{2} \begin{pmatrix} e^\eta/\lambda_{xp} & 0 \\ 0 & \lambda_{xp} e^{-\eta} \end{pmatrix}, \qquad \Delta x\, \Delta p_x = \frac{\hbar}{2}, \qquad \frac{\Delta p_x}{\Delta x} = \lambda_{xp} e^{-\eta}. \tag{115.6}$$

The dispersion ratio acquires the dimension of momentum per length here. By contrast, (115.5) belongs to the balanced chart. For the inductive–capacitive realization already defined, (24.13) supplies the dimensionless identity $Z_\eta / Z_* = e^{-\eta} = \Delta P / \Delta Q$. Each reading uses its declared transduction and preserves the same action determinant.

115.1.0.2 Contour energy and mean energy of the state.

The inductive–capacitive realization has Hamiltonian

$$H_\eta(Q,P) = \frac{\omega}{2}\left(e^{-\eta} Q^2 + e^{\eta} P^2 \right), \qquad \omega > 0.$$

On $\partial\mathcal{E}_\eta$, its value is $H_\eta = \hbar\omega$, by the semiaxis equations. In the quantum realization of Proposition 115.1,

$$\langle \widehat{H}_\eta \rangle_{\psi_\eta} = \frac{\omega}{2}\left(e^{-\eta} \frac{\hbar e^{\eta}}{2} + e^{\eta} \frac{\hbar e^{-\eta}}{2} \right) = \frac{\hbar\omega}{2}. \tag{115.7}$$

The two-standard-deviation contour therefore has twice the mean energy of this Gaussian state. The areal law fixes action normalization; the angular pair fixes its distribution between the two coordinates; the canonical representation and explicit state realize that distribution as covariance. Saturation of uncertainty is thus established in the realization specified by its operators and its state.

Chapter 116

Sheet return, areal–volumetric incidence and constitutive orientation

The sheet structure enters before electromagnetic quantities are evaluated. Its function is to distinguish closure of a projection, restoration of an orientation and evolution of the transported register. These operations act on related objects of different types: a projected trajectory may have closed while its lift remains on the opposite sheet; orientation may have been restored while memory preserves the two windings performed. The constitutive construction must receive this information together with the structure determining it.

116.1 Projected return and restoration of the oriented cover

Let ℓ be a return loop of a surviving extension, and let $\varepsilon : \langle \ell \rangle \to \{+1, -1\}$ be the character of its lift, with $\varepsilon(\ell) = -1$. The associated interval bundle has the presentation

$$\mathcal{M} = ([0, 1] \times [-1, 1])/(0, u) \sim (1, -u). \tag{116.1}$$

The decisive datum is the sign of transport around the specified loop. The identification of endpoints geometrically realizes that character.

Proposition 116.1 (Two closure orders)**.** *One winding of the loop closes the base and reverses the fibre orientation. Two windings restore orientation. In the angular parametrization of the boundary readout, these returns correspond, respectively, to the areal closure 2π and the volumetric closure 4π of the oriented cover.*

Proof. The equivalence relation of (116.1) identifies the terminal fibre with the initial one by $u \mapsto -u$. Composing two transports gives $u \mapsto -(-u) = u$. Parametrizing one winding of the base by 2π assigns 4π to its square. The statement concerns this lift and this readout. □

TPK memory also preserves the trajectory traversed and the register updates. The square of the sign character is the identity, but it does not impose vanishing of these updates. The spinorial representation, when used, must transport this datum through its own map.

116.2 Descent of incidence from the tritic calendar

In the primitive block $(+1, -1, 0)$, the TPK updates the arithmetic mode through $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$ and $0 : m \mapsto m$. Its cyclic orbit is (Σ, Π, Π). The sheet readout assigns Σ to the areal sheet and Π to the volumetric sheet. In the basis order $(e_{\mathrm{ar}}, e_{\mathrm{vol}})$, edge incidence is represented by

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \tag{116.2}$$

The three consecutive pairs are $\Sigma \to \Pi$, $\Pi \to \Pi$ and $\Pi \to \Sigma$, each with multiplicity one. The calendars of lengths 9, 27 and 108 contain 3, 9 and 36 copies of the primitive block. Their incidence matrices are therefore these multiples of F_{av}.

This construction preserves the distinction between chronological frequency and the measure of the continuation space. The former assigns weights $(1/3, 2/3)$ to the mode cycle. The latter is defined on the minimal memory-one symbolic envelope admitting exactly the preceding pairs. Its adjacency matrix is F_{av}; trajectory counts belong to that envelope.

Proposition 116.2 (Return weights of the two sheets)**.** *Let* $F_0 = 0$*,* $F_1 = 1$ *and* $F_{n+1} = F_n + F_{n-1}$*. For* $n \geq 1$*,*

$$F_{\mathrm{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}. \tag{116.3}$$

The Parry measure of this envelope assigns the sheets weights proportional to $(1, \varphi^2)$*, and the ratio of diagonal counts converges to* φ^{-2}*.*

Proof. The power formula follows by induction using the recurrence. The square of F_{av} is strictly positive. Its positive Perron vector is $(1, \varphi)$, with eigenvalue φ, since $\varphi^2 = \varphi + 1$. As the matrix is symmetric, the Parry weights are proportional to the squares of that vector's components. The other root of the characteristic polynomial is $-\varphi^{-1}$; its modulus is strictly less than φ, giving the limit of the diagonal ratios. □

Spectral identification of this incidence follows its construction from the calendar. The regional coordinate φ retains its earlier coinductive genealogy: recognition of the eigenvalue does not select the calendar.

116.3 Boundary marking and areal–volumetric ratio

Let $\Pi_{\mathrm{HMT}}(X_\infty)$ be the value of the regional closure readout already constructed. With the diagonal sheet projectors, define

$$D_\pi = \Pi_{\mathrm{HMT}}(X_\infty) P_{\mathrm{ar}} + P_{\mathrm{vol}}, \qquad \Theta_n = \frac{\langle e_{\mathrm{ar}}, D_\pi F_{\mathrm{av}}^n e_{\mathrm{ar}} \rangle}{\langle e_{\mathrm{vol}}, F_{\mathrm{av}}^n e_{\mathrm{vol}} \rangle}. \tag{116.4}$$

Volumetric normalization is unity; regional closure marks the areal return only once. Applying (116.3),

$$\Theta_n = \Pi_{\mathrm{HMT}}(X_\infty) \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi}{\varphi^2}. \tag{116.5}$$

The return ratio is determined by incidence, measure and marking. The quotient $2\pi/4\pi$, by contrast, describes the cover's two closure orders. Both relations participate in the sheet readout without replacing each other.

116.4 Transport of orientation to the constitutive channels

In a two-channel representation, let $\mathcal{R}$ be a self-adjoint involution and $\mathcal{S}$ a unitary interchange operator satisfying $\mathcal{S}\mathcal{R}\mathcal{S}^{-1} = -\mathcal{R}$. Write the angular coordinates already constructed as $x = A_{\mathrm{rad}}$ and $y = C_{\mathrm{rad}}^*$, with $x > |y|$. Then

$$B = xI + y\mathcal{R}, \qquad T = e^{-B}, \qquad \mathcal{S}T(y)\mathcal{S}^{-1} = T(-y). \tag{116.6}$$

Indeed, conjugation transforms $B(y)$ into $B(-y)$ and commutes with its convergent exponential series. The eigenvalues $q_\pm = e^{-x \mp y}$ are interchanged. To fix the response order, the chamber $x > y > 0$ is used, where $0 < q_+ < q_- < 1$. The map $(x, y) \mapsto (x, -y)$ transports it to the opposite chamber $x > -y > 0$, where $0 < q_- < q_+ < 1$. Both belong to the contractive domain $x > |y|$. On the boundary $y = 0$ the two channels coincide. Conjugation interchanges the chambers; it does not preserve the strict ordering of their labels. It acts on the projectors through $\mathcal{S}P_\pm\mathcal{S}^{-1} = P_\mp$, with $P_\pm = (I \pm \mathcal{R})/2$. This concrete map transmits orientation to the constitutive operator developed next.

Chapter 117

Discrete polarization and the tritic mode of arithmetic projections

Refinement and comparison between the APP sheets provide two complementary observations. The first follows the accumulated discrepancy between the number of transitions and the capacity acquired; the second preserves the mode of the tritic selector in the pairing of the additive and multiplicative fibres. Discrete polarization and the electronic prefactor are obtained from these respective operations. Their articulation with the vacuum requires preserving both the temporal information and the spaces on which the operators act.

117.1 Refinement discrepancy and endpoint convention

The cardinalities of the ternary block and its decimal representation fix

$$\rho = \log_{1000} 729 = \log_{10} 9, \qquad \delta = 1 - \rho = \log_{10}(10/9).$$

We have $0 < \delta < 1$. Moreover, δ is irrational: an equality $\delta = p/q$, with $q > 0$, would imply $10^{q-p} = 9^q$, which is incompatible with prime factorization. This mismatch already enters the encoding of refinement. Here its centred readout is made explicit, preserving the phase origin and the endpoint convention.

Definition 117.1 (Lower and upper encodings)**.** For $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$, define

$$z_n^{\downarrow}(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor, \tag{117.1}$$

$$z_n^{\uparrow}(\theta) = \lceil (n+1)\delta + \theta \rceil - \lceil n\delta + \theta \rceil. \tag{117.2}$$

Both take values in $\{0, 1\}$. The arrow distinguishes only the endpoint convention for the intervals; tritic orientation retains its earlier definition.

The lower encoding coincides with the one used in the development of refinement. The upper encoding recovers the formulation of polarization using the ceiling function. The two expressions are related by an exactly determined boundary term.

Proposition 117.2 (Change of convention and boundary term)**.** *Let* $b_n(\theta) = \mathbf{1}_{\mathbb{Z}}(n\delta + \theta)$. *Then*

$$z_n^{\uparrow}(\theta) - z_n^{\downarrow}(\theta) = b_n(\theta) - b_{n+1}(\theta). \tag{117.3}$$

In particular, for $N \geq 0$*,*

$$\sum_{n=0}^{N-1} (z_n^{\uparrow} - z_n^{\downarrow}) = b_0 - b_N.$$

Proof. For every real x, $\lceil x \rceil = \lfloor x \rfloor + 1 - \mathbf{1}_{\mathbb{Z}}(x)$. Applying this identity to both endpoints of each increment gives (117.3); its sum telescopes. The irrationality of δ implies that, for fixed θ, at most one integer index satisfies $n\delta + \theta \in \mathbb{Z}$. □

With $\theta = 0$, the first upper symbol is 1 and the lower one is 0; for $n \geq 1$ they coincide. The initial term belongs to the register and is preserved when the convention is changed. The densities and discrepancy bounds are transported with that term, without modifying the TPK update or restarting the history at each phase return.

117.2 Bounded polarization and exact temporal balance

Definition 117.3 (Centred discrete polarization)**.** The polarization of the upper encoding, relative to the density δ, is

$$P_N^{\uparrow}(\theta) = \sum_{n=0}^{N-1} \big(z_n^{\uparrow}(\theta) - \delta\big), \qquad P_0^{\uparrow} = 0. \tag{117.4}$$

Its temporal increment is

$$J_N^{\uparrow}(\theta) = P_{N+1}^{\uparrow}(\theta) - P_N^{\uparrow}(\theta) = z_N^{\uparrow}(\theta) - \delta. \tag{117.5}$$

Centring separates uniform growth of density δ from phase information. Thus, $P_N^{\uparrow}$ measures an accumulated imbalance, not the total number of vacancies. The increment $J_N^{\uparrow}$ preserves the local pulse and the background relative to which it is evaluated.

Theorem 117.4 (Uniform bound and discrete temporal continuity)**.** *For $N \geq 0$ and every phase θ,*

$$P_N^{\uparrow}(\theta) = \lceil N\delta + \theta \rceil - \lceil \theta \rceil - N\delta, \qquad |P_N^{\uparrow}(\theta)| < 1. \tag{117.6}$$

For any integers $0 \leq a < b$,

$$\sum_{n=a}^{b-1} J_n^{\uparrow}(\theta) = P_b^{\uparrow}(\theta) - P_a^{\uparrow}(\theta), \qquad \left| \sum_{n=a}^{b-1} J_n^{\uparrow}(\theta) \right| < 1. \tag{117.7}$$

The density of the vacancy sequence is δ.

Proof. Summing the ceiling increments in (117.2) yields the first identity. Writing $r(x) = \lceil x \rceil - x \in [0, 1)$ gives $P_N^{\uparrow} = r(N\delta + \theta) - r(\theta)$; its absolute value is less than one. On the interval $[a, b)$, one similarly obtains $r(b\delta + \theta) - r(a\delta + \theta)$, proving the second bound, which is sharper than the sum of two individual bounds. Finally, $N^{-1} \sum_{n=0}^{N-1} z_n^{\uparrow} = \delta + N^{-1} P_N^{\uparrow} \to \delta$. □

The lower encoding admits the same proofs with $r(x)$ replaced by $\lfloor x \rfloor - x \in (-1, 0]$. In particular, $P_N^{\uparrow} - P_N^{\downarrow} = b_0 - b_N$. The bounded discrepancy previously used in the Dirichlet readouts and the present polarization therefore preserve an explicit relationship of origin and endpoints. Nor does aperiodicity introduce secular drift in this imbalance: the history can extend indefinitely while the centred discrepancy remains uniformly bounded.

117.3 Realization on the charge and time lines

To realize the balance dimensionally, specify a charge line $\mathcal{L}_Q$ with a nonzero section u_Q and an increasing sequence of times t_N on an oriented time line. With $\Delta t_N = t_{N+1} - t_N > 0$, the centred charge and the mean current of each interval are defined by

$$Q_N = P_N^{\uparrow} u_Q, \qquad I_N = \frac{Q_{N+1} - Q_N}{\Delta t_N} = \frac{u_Q}{\Delta t_N} \big(z_N^{\uparrow} - \delta\big). \tag{117.8}$$

Thus, $Q_N \in \mathcal{L}_Q$ and $I_N \in \mathcal{L}_Q \otimes \mathcal{L}_T^{-1}$. The section u_Q and the time intervals are data of this realization; the preceding arithmetic construction has already determined $P_N^{\uparrow}$ and $J_N^{\uparrow}$.

Proposition 117.5 (Preservation of balance under dimensional realization)**.** *For $a < b$,*

$$Q_b - Q_a = \sum_{n=a}^{b-1} \Delta t_n I_n, \qquad \left| \frac{Q_b - Q_a}{u_Q} \right| < 1. \tag{117.9}$$

If all intervals have length t_0, the two possible currents are $-\delta u_Q / t_0$ and $(1 - \delta) u_Q / t_0$.

Proof. Multiplying each identity in (117.8) by Δt_n and summing telescopes the charges. The bound is (117.7); the two local values follow from $z_n^{\uparrow} \in \{0,1\}$. □

The continuity proved is temporal and refers to the centred charge. A spatial realization also incorporates the localization of charges, incidence between cells, and identification of boundary fluxes. In a constitutive description by channels, the coupling is specified through an excitation map $\iota : \mathcal{L}_Q \to \mathcal{H}_{\mathrm{ch}} \otimes \mathcal{L}_Q$ and an observation map $\ell : \mathcal{H}_{\mathrm{ch}} \otimes \mathcal{L}_Q \to \mathcal{L}_Q$, where $\mathcal{H}_{\mathrm{ch}}$ is a real realization of the channel space. If both are linear and independent of the time index, the response of a fixed operator $\mathcal{V}$ is transported by

$$Q_N^{\mathrm{resp}} = \ell(\mathcal{V} \otimes I_{\mathcal{L}_Q})\iota(Q_N), \qquad I_N^{\mathrm{resp}} = \frac{Q_{N+1}^{\mathrm{resp}} - Q_N^{\mathrm{resp}}}{\Delta t_N}.$$

Linearity preserves the telescoping balance. This composition makes explicit the data of a constitutive identification: the scalar sequence determines the pulse, while ι, $\mathcal{V}$, and ℓ determine its excitation, transport, and observation through channels. Identities (117.4)–(117.9) establish the first construction independently of that choice of realization. The vacuum operator is constructed in Section 118 on its own channels and projectors.

117.4 Pairing the additive and multiplicative fibres

The second observation comes from the three-dimensional realization of APP, $\mathcal{A}_3 = D_9^3$, with uniform measure and $D_9 = \{1, \dots, 9\}$. The observables preserved are

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \qquad \Pi(x) = \rho_9(x_1 x_2 x_3),$$

where $\rho_9(n) = 1 + [(n-1) \bmod 9]$ for $n > 0$. In $\ell^2(\mathcal{A}_3)$, let $P_{\Sigma,3}$ and $P_{\Pi,3}$ be the conditional expectations on their fibres. For example,

$$(P_{\Sigma,3} f)(x) = \frac{1}{|\Sigma^{-1}(\Sigma(x))|} \sum_{y:\, \Sigma(y) = \Sigma(x)} f(y).$$

They are orthogonal projections defined on the same support. The incidence matrix $N_{ab} = |\Sigma^{-1}(a) \cap \Pi^{-1}(b)|$ results from enumerating the 729 triples, adding and multiplying their coordinates before applying ρ_9:

$$N = 9 \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}. \tag{117.10}$$

The rows sum to 81, and the column sums are

$$(m_1, \dots, m_9) = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

Therefore, the pairing of the normalized indicator functions of the fibres is $C_{ab} = N_{ab}/\sqrt{81 m_b}$. Normalization incorporates the multiplicities produced by APP, not a choice of singular values.

Proposition 117.6 (Singular mode of the tritic selector)**.** *In the canonical chart, the selector vector is* $m_{\mathrm{ph}} = (1, -1, 0)^3$. *We have*

$$Cm_{\mathrm{ph}} = C^{\mathsf{T}} m_{\mathrm{ph}} = -\tfrac{1}{2} m_{\mathrm{ph}}. \tag{117.11}$$

Thus, the corresponding singular value is $s = 1/2$.

Proof. In the six columns where m_{ph} is nonzero, $m_b = 36$ and $C_{ab} = N_{ab}/54$. Multiplying the rows of (117.10) by $(1, -1, 0)^3$ yields $-m_{\mathrm{ph}}/2$. For the transposed product, the signed sums of columns $3, 6, 9$ are zero; the other six give the same values $-m_{\mathrm{ph},b}/2$. This proves both identities and, by composition, $CC^{\mathsf{T}} m_{\mathrm{ph}} = m_{\mathrm{ph}}/4$. □

117.5 Commutator norm and electronic prefactor

Theorem 117.7 (Exact norm of arithmetic incompatibility)**.** *The two projections satisfy*

$$\left\| [P_{\Sigma,3}, P_{\Pi,3}] \right\| = \frac{\sqrt{3}}{4}. \tag{117.12}$$

The maximum is attained in the principal plane corresponding to the tritic mode of (117.11).

Proof. The rational matrix $M = CC^{\mathsf{T}}$ has entries $M_{ac} = \sum_b N_{ab} N_{cb}/(81 m_b)$. Its complete spectrum is verified through the following independent subspaces: the vectors $(1, \dots, 1)$, $(1, -1, 0)^3$, and $(1, 1, -2)^3$ have eigenvalues 1, $1/4$, and $7/132$, respectively; the subspace of vectors supported on $\{3, 6, 9\}$ whose sum is zero has dimension 2 and eigenvalue $1/18$; the zero-sum vectors supported on $\{1, 4, 7\}$ or $\{2, 5, 8\}$ form a kernel of dimension 4. Substitution into the matrix verifies these actions, and the dimensions sum to 9. Consequently,

$$\operatorname{spec} M = \{1, 1/4, 7/132, 1/18, 1/18, 0, 0, 0, 0\}.$$

For two orthogonal projections, a singular value $s \in (0, 1)$ of the pairing determines a principal plane in which their matrices are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} s^2 & s\sqrt{1 - s^2} \\ s\sqrt{1 - s^2} & 1 - s^2 \end{pmatrix}.$$

This form is obtained by taking a unit vector u in the first image with $PQu = s^2 u$ and completing it with $(Qu - s^2 u)/(s\sqrt{1 - s^2})$. Their commutator is $s\sqrt{1 - s^2} \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The sectors with singular value 0 or 1 contribute zero. Therefore, the total norm is the maximum of $\sqrt{\lambda(1 - \lambda)}$ over the spectrum of M. That maximum is $\sqrt{3/16}$, attained at $\lambda = 1/4$. □

The norm is the prefactor entering the electronic composition. Its derivation preserves the three-dimensional support, the fibres, and the selector mode; scalar evaluation does not replace that structure. In particular, knowing $\sqrt{3}/4$ gives the maximum magnitude of the incompatibility, while the projectors preserve the directions in which it acts.

117.6 Quadratic defect and response of the 90/120 sectors

The tritic mode simultaneously determines

$$1 - s^2 = \frac{3}{4} = \frac{90}{120}, \qquad s = \frac{1}{2}. \tag{117.13}$$

The first expression is the quadratic defect of the APP principal plane. The last preserves the lengths of the two calendar sectors. The constitutive response uses those sectors on another object: a contractive channel $q \in (0, 1)$. Its evaluation is

$$r(q) = \frac{1 - q^{90}}{1 - q^{120}}.$$

The relationship between the sector ratio and this response can be made precise without identifying their operators.

Proposition 117.8 (Limit of the response ratio)**.** *Para* $0 < q < 1$,

$$\frac{3}{4} < r(q) < 1, \qquad \lim_{q \uparrow 1} r(q) = \frac{3}{4} = 1 - s^2. \tag{117.14}$$

Proof. With $u = q^{30} \in (0, 1)$, factoring $1 - u^3$ and $1 - u^4$ gives $r(q) = (1 + u + u^2)/(1 + u + u^2 + u^3)$. The limit at $u = 1$ is $3/4$. Moreover, $4(1 + u + u^2) - 3(1 + u + u^2 + u^3) = (1 - u)(3u^2 + 2u + 1) > 0$, and the denominator exceeds the numerator by $u^3 > 0$. This yields both inequalities. □

The exact coincidence concerns the defect of the singular mode and the limit of the sector response. In the contractive interior, the response depends on q and is strictly greater than $3/4$. The constitutive development preserves its two channels $q_\pm$ and its projectors $P_\pm$; the projectors $P_{\Sigma,3}$ and $P_{\Pi,3}$, for their part, preserve the arithmetic fibres that determined the prefactor. An operator equivalence between the two realizations requires a map between their spaces that intertwines the corresponding actions. Equations (117.11), (117.12), and (117.14) specify the shared information and keep the data of each realization visible.

Chapter 118

Constitutive operator and reconstruction of the vacuum channels

The constitutive quantities are obtained by an operation on angular transport. The starting information comprises the channels, their projectors, and the orientation distinguishing them. This organization makes it possible to follow separately the construction of the operator, the evaluation of its spectral functions, and its subsequent dimensional realization.

118.1 Positive response on two sheets

In the oriented chamber $x > y > 0$ of Section 116, let $P_{\pm} = (I \pm \mathcal{R})/2$ and $T = q_{+}P_{+} + q_{-}P_{-}$, with $0 < q_{+} < q_{-} < 1$. The incidences constructed by the calendar fix the sectors 90 and 120. The constitutive response is

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}. \tag{118.1}$$

This definition receives the transport arising from the TPK and preserves the sheet assignments. The response rule and its domain are an explicit part of the construction.

The quotient arises from comparing the two calendar incidences on the same transport, not from fitting four quantities separately. If $S = T^{30}$ and $\Sigma_j(S) = I + S + \cdots + S^{j-1}$, factoring $I - S^j$ gives

$$\mathcal{V}\,\Sigma_4(S) = \Sigma_3(S), \qquad \mathcal{V} = \Sigma_3(S)\Sigma_4(S)^{-1}. \tag{118.2}$$

The factors Σ_3 and Σ_4 express, respectively, the sectors $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$. Since the spectrum of S is contained in $(0, 1)$, $\Sigma_4(S)$ is invertible: the completion equation determines a unique response for the fixed transport. This uniqueness concerns the declared constitutive rule and its incidences. Electromagnetic identification adds the oriented assignment of sheets and the dimensional lines specified below.

Proposition 118.1 (Decomposition and positivity)**.** *The response is well-defined and*

$$\mathcal{V} = r_{+}P_{+} + r_{-}P_{-}, \qquad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}. \tag{118.3}$$

Its eigenvalues satisfy $3/4 < r_{-} < r_{+} < 1$ *in this chamber.*

Proof. Each coefficient $1 - q_{\pm}^{120}$ is positive. The inverse of $I - T^{120}$ is the sum of the projectors divided by those coefficients, proving (118.3). With $s = q^{30}$, one obtains

$$f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \qquad f'(s) = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0. \tag{118.4}$$

The limits at 0 and 1 are 1 and $3/4$, respectively. Monotonicity and the channel ordering complete the proof. □

118.2 Four coordinates of a common response

Define the normalized evaluations

$$\widehat{\varepsilon} = r_+^2, \qquad \widehat{\mu} = r_-^2, \qquad \widehat{Z} = \frac{r_-}{r_+}, \qquad \widehat{c} = \frac{1}{r_+ r_-}. \tag{118.5}$$

The first pair preserves the quadratic responses assigned to each sheet. The product captures the common mode; the quotient distinguishes their relative orientation. The following hold exactly:

$$\widehat{\mu}\widehat{\varepsilon} = \widehat{c}^{-2}, \qquad \frac{\widehat{\mu}}{\widehat{\varepsilon}} = \widehat{Z}^2, \qquad r_+ = \frac{1}{\sqrt{\widehat{Z}}\widehat{c}}, \quad r_- = \sqrt{\frac{\widehat{Z}}{\widehat{c}}}. \tag{118.6}$$

The positive roots fix the inversion. Under sheet conjugation, read relative to the fixed markings $P_\pm$, $\widehat{\varepsilon}$ and $\widehat{\mu}$ are interchanged, $\widehat{Z}$ is inverted, and $\widehat{c}$ is preserved. The response is then transported to the chamber $x > -y > 0$: the formulas remain valid, with the orderings $q_- < q_+$ and $r_+ < r_-$ reversed. Consequently, common mode and orientation are data recoverable through complementary observables.

Proposition 118.2 (Determination of the quadratic evaluations)**.** *Let $W = r_+P_+ + r_-P_-$ be a positive response on the two sheets. With the product and quotient readouts fixed,*

$$\widehat{c}^{-1} = \det W = r_+ r_-, \qquad \widehat{Z} = r_-/r_+,$$

there exists a unique positive pair $(\widehat{\varepsilon}, \widehat{\mu})$ satisfying $\widehat{\mu}\widehat{\varepsilon} = \widehat{c}^{-2}$ and $\widehat{\mu}/\widehat{\varepsilon} = \widehat{Z}^2$. It is precisely (r_+^2, r_-^2).

Proof. Positivity permits taking roots unambiguously:

$$\widehat{\mu} = \frac{\widehat{Z}}{\widehat{c}} = \frac{r_-}{r_+} r_+ r_- = r_-^2, \qquad \widehat{\varepsilon} = \frac{1}{\widehat{Z}\widehat{c}} = \frac{r_+}{r_-} r_+ r_- = r_+^2.$$

These expressions satisfy both relations and are their only positive roots. □

Thus the squares are not two independent choices added to the channels: they are determined by the multiplicative and oriented readouts once the constitutive relations are fixed. This proposition establishes uniqueness within that system of rules. Interpreting the rules as an electromagnetic response retains its additional physical-representation content.

The determinantal readout of transport and the constitutive response must also preserve their order. For $T = e^{-xI - y\mathcal{R}}$, one has $\det T = e^{-2x}$, whereas

$$\det \mathcal{V} = f(q_+^{30}) f(q_-^{30}) = \widehat{c}^{-1}.$$

Both arise from the same two-sheet operator but evaluate different functions of its spectrum. In general, applying the determinant first and then a rational function is not equivalent to taking the determinant of that function of the operator. Keeping both eigenvalues until the response is completed preserves orientation information that an isolated determinant no longer distinguishes.

118.3 Cubic inversion and angular recovery

Theorem 118.3 (Constitutive inversion in the contractive domain)**.** *For each $r \in (3/4, 1)$, there exists a unique $s \in (0, 1)$ such that $f(s) = r$. This s is the unique root in that interval of*

$$rs^3 + (r-1)(1 + s + s^2) = 0. \tag{118.7}$$

The labeled response reconstructs $s_\pm$, $q_\pm = s_\pm^{1/30}$, and

$$x = -\frac{1}{2}\log(q_+ q_-), \qquad y = \frac{1}{2}\log\frac{q_-}{q_+}. \tag{118.8}$$

Proof. The derivative and limits in (118.4) prove that f is a decreasing bijection between the stated intervals. Multiplying $f(s) = r$ by its denominator produces (118.7). The positive root of order 30 recovers the channel. Finally, adding and subtracting $\log q_\pm = -x \mp y$ gives (118.8). □

Reconstruction preserves the sheet labels. The complete operator $\mathcal{V}$ also preserves the projectors, and the inverse function is applied by spectral calculus. Recovering two angular coordinates is an operation defined on this representation; the complete historical TPK trace preserves additional data.

The interval $r \in (3/4, 1)$ and the inverse function of this theorem belong to the internal normalization of (118.5). If coordinates expressed in another units chart are available, this normalization is first recovered through the transformation in equation (118.11); cubic inversion is then applied.

118.4 Logarithmic coordinates and refinement

Let $\mathcal{E} = \log(r_+ r_-)$ and $\mathcal{B} = \log(r_-/r_+)$. Then

$$\log \mathcal{V} = \tfrac{1}{2}\mathcal{E} I - \tfrac{1}{2}\mathcal{B}\mathcal{R}. \tag{118.9}$$

This decomposition displays the invariant and oriented components. In the amplification $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$, normalized partial expectations satisfy $E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m$. For every polynomial p, $p(\mathcal{V}_m) = p(\mathcal{V}) \otimes I_{9^m}$; hence $E_m(p(\mathcal{V}_{m+1})) = p(\mathcal{V}_m)$. Uniform continuity extends the identity to continuous functions on the spectrum. Thus projectors, powers, and the logarithm are preserved in the stated tower.

118.5 Dimensional types of the four evaluations

With the action, charge, and velocity lines, equipped with sections u_S, u_Q, u_C, the dimensional realization reads

$$Z_{\mathrm{vac}} = \widehat{Z} u_S u_Q^{-2}, \qquad c_{\mathrm{vac}} = \widehat{c} u_C, \tag{118.10}$$

$$\mu_{\mathrm{vac}} = \widehat{\mu} u_S u_C^{-1} u_Q^{-2}, \qquad \varepsilon_{\mathrm{vac}} = \widehat{\varepsilon} u_S^{-1} u_C^{-1} u_Q^{2}. \tag{118.11}$$

Multiplying and dividing these expressions preserves $\mu_{\mathrm{vac}}\varepsilon_{\mathrm{vac}} = c_{\mathrm{vac}}^{-2}$ and $\mu_{\mathrm{vac}}/\varepsilon_{\mathrm{vac}} = Z_{\mathrm{vac}}^2$. The SI chart subsequently evaluates the sections. Physical comparison requires presenting their construction and normalization alongside the evaluation; the four normalized coordinates of (118.5) retain their dimensionless type.

118.6 Reconstruction of the action shape from the constitutive response

The constitutive response and the inductive–capacitive realization receive the same angular pair constructed from APP–TRIT–TPK. The former evaluates rational functions of the two channels; the latter uses the ratio of their exponents to determine the shape of the action ellipse. Cubic inversion allows the two readouts to be composed without identifying their numerical ratios.

Let $x = A_{\mathrm{rad}}$ and $y = C^{*}_{\mathrm{rad}}$, with $x > |y|$, be the lifted coordinates already constructed, and preserve their sheet labels. Write

$$q_{\pm} = e^{-(x\pm y)}, \qquad s_{\pm} = q_{\pm}^{30}, \qquad r_{\pm} = f(s_{\pm}), \qquad f(s) = \frac{1+s+s^2}{1+s+s^2+s^3}. \tag{118.12}$$

The contractive domain gives $s_{\pm} \in (0,1)$ and $r_{\pm} \in (3/4, 1)$. In the chamber $x > y > 0$, we have $s_{+} < s_{-}$ and $r_{+} > r_{-}$; sheet interchange transports this chamber to the opposite orientation.

The image of the two normalized constitutive coordinates is the domain

$$\mathcal{U} = \left\{(z,v) \in (0,\infty)^2 : (zv)^{-1/2} \in (3/4,1), \quad (z/v)^{1/2} \in (3/4,1)\right\}. \tag{118.13}$$

Here $(z,v) = (\widehat{Z}, \widehat{c})$. The restrictions express membership of the reconstructed channels in the domain of the inversion theorem; they do not constitute a subsequent choice of their values.

Proposition 118.4 (Constitutive composition of the shape and LC impedance)**.** *For $(\widehat{Z}, \widehat{c}) \in \mathcal{U}$, let $s_{\pm} \in (0,1)$ be the unique roots of*

$$r_{\pm} s_{\pm}^3 + (r_{\pm} - 1)(1 + s_{\pm} + s_{\pm}^2) = 0, \qquad r_{+} = \frac{1}{\sqrt{\widehat{Z}\widehat{c}}}, \quad r_{-} = \sqrt{\frac{\widehat{Z}}{\widehat{c}}}. \tag{118.14}$$

The angular ratio, the anisotropy of the action ellipse, and the relative LC impedance are reconstructed exactly by

$$\frac{y}{x} = \frac{\log s_{+} - \log s_{-}}{\log s_{+} + \log s_{-}}, \tag{118.15}$$

$$\eta_{\mathrm{el}} = \frac{1}{2} \log \frac{\log s_{+}}{\log s_{-}}, \tag{118.16}$$

$$\frac{Z_{\mathrm{LC}}}{Z_{*}} = e^{-\eta_{\mathrm{el}}} = \sqrt{\frac{\log s_{-}}{\log s_{+}}}. \tag{118.17}$$

In the last identity, $Z_{\mathrm{LC}} = Z_{\eta_{\mathrm{el}}}$ is the impedance of the LC chart of (24.12)*.*

Proof. The cubic inversion in (118.7) uniquely determines $s_{\pm}$, because f is a decreasing bijection from $(0,1)$ onto $(3/4,1)$. Their positive roots of order 30 recover $q_{\pm}$. The identities

$$\log s_{+} = -30(x+y), \qquad \log s_{-} = -30(x-y)$$

prove (118.15) by difference and sum. Both logarithms are negative; their ratio is positive and their sum is nonzero. Therefore,

$$\frac{1}{2} \log \frac{\log s_{+}}{\log s_{-}} = \frac{1}{2} \log \frac{x+y}{x-y} = \operatorname{artanh}(y/x) = \eta_{\mathrm{el}}.$$

The relation $Z_{\eta_{\mathrm{el}}}/Z_{*} = e^{-\eta_{\mathrm{el}}}$, proved in (24.13), gives (118.17). □

The composition preserves the information needed for each reconstruction. With $\hbar$, one recovers $a_S = \sqrt{2\hbar}\, e^{\eta_{\mathrm{el}}/2}$ and $b_S = \sqrt{2\hbar}\, e^{-\eta_{\mathrm{el}}/2}$; with the chart (ω, Z_{*}), one recovers the inductance and capacitance

already defined. Action, clock and dimensional reference accompany the form as typed data. Inverting two dimensionless coordinates does not reconstruct their units or the entire memory of the state.

Normalization enters before solving the cubic equations. If Z_{vac} and c_{vac} are received in a dimensional chart, the relations (118.11) first determine

$$\widehat{Z} = \frac{Z_{\mathrm{vac}}}{u_S u_Q^{-2}}, \qquad \widehat{c} = \frac{c_{\mathrm{vac}}}{u_C}.$$

These are the coordinates to which (118.14) applies. The reference Z_* belongs to the LC realization and is explicitly preserved in (118.17).

The two ratios are related through the channels, but evaluate different functions:

$$\widehat{Z} = \frac{f(s_-)}{f(s_+)}, \qquad \frac{Z_{\mathrm{LC}}}{Z_*} = \sqrt{\frac{\log s_-}{\log s_+}}. \tag{118.18}$$

The preceding reconstruction provides the map between the two readouts. In particular, interchanging the sheets inverts both ratios, changes η_{el} to $-\eta_{\mathrm{el}}$, and preserves $\widehat{c}$. In the symmetric limit $y = 0$, the channels coincide and both ratios equal 1. Outside that case, a common origin does not impose equality of the two functions. The action shape is recoverable from the constitutive response while preserving orientation and normalization.

118.7 Constitutive speed and kinematic realization of the clock

The constitutive response and the kinematic reader realize the same velocity section. Their identification preserves the order of construction: the APP sheets, their TRIT orientation and the angular transport published by TPK precede the velocity readings and the choice of units. In the two-sheet representation of section 116, with $x > |y|$ and rank-one projectors $P_\pm$, the transport is

$$T = e^{-xI - y\mathcal{R}} = q_+ P_+ + q_- P_-, \qquad q_\pm = e^{-x \mp y}, \qquad 0 < q_\pm < 1.$$

Its temporal blocks of lengths 90 and 120 produce the response $\mathcal{V}$ in (118.1). The coordinates x, y and the channels are outputs of the preceding angular development; this composition applies its readouts to that already constructed transport.

The symmetric logarithmic readout of temporal transport is

$$\mathcal{E} = \log\big[(1 - q_+^{90})(1 - q_-^{90})\big] - \log\big[(1 - q_+^{120})(1 - q_-^{120})\big]. \tag{118.19}$$

On the dimensional velocity line, with positive section u_C, the kinematic reader publishes $c_{\mathrm{int}} = \exp(-\mathcal{E})u_C$. The constitutive response publishes $c_{\mathrm{vac}} = \widehat{c}\,u_C$ through (118.5).

Proposition 118.5 (Constitutive and kinematic identity)**.** *In the same dimensional chart, we have*

$$c_{\mathrm{int}} = c_{\mathrm{vac}} = \widehat{c}\,u_C, \qquad \widehat{c} = (r_+ r_-)^{-1} = \exp(-\mathcal{E}). \tag{118.20}$$

The identification preserves sheet interchange.

Proof. All factors in (118.19) are positive. The logarithm law and (118.3) give

$$\mathcal{E} = \log(r_+ r_-) = \log\det\mathcal{V}, \qquad e^{-\mathcal{E}} = (\det\mathcal{V})^{-1}.$$

The determinant corresponds to the rank-one two-sheet realization. The two definitions therefore publish the same section. Exchanging the sheets permutes r_+ and r_- and preserves their product. The determinant

$\det T = e^{-2x}$ corresponds to elementary transport, whereas $\det \mathcal{V} = r_+ r_-$ corresponds to its temporal response; (118.20) uses the latter. □

118.7.1 Path length and elementary duration

The positive dimensional frame (u_S, u_C, u_T, u_Q) induces the length basis $u_L = u_C u_T$. For an enriched path γ of $n = |\gamma|$ transitions and a constant channel, the additive readers are

$$T_{\mathrm{int}}(\gamma) = n u_T, \qquad L_{\mathrm{int}}(\gamma) = n \widehat{c}\, u_L. \tag{118.21}$$

Concatenation adds the numbers of transitions and their realizations. Enriched routes also preserve their orientation and memory.

Corollary 118.6 (Normalization of the elementary length)**.** *For $n > 0$, $L_{\mathrm{int}}(\gamma)/T_{\mathrm{int}}(\gamma) = c_{\mathrm{vac}}$. In the normalization $t_0 = u_T$,*

$$\ell_0 = c_{\mathrm{int}} t_0 = \widehat{c}\, u_L. \tag{118.22}$$

In a temporal chart $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, we obtain $\ell_0 = \tau_0 \widehat{c}\, u_L$.

Proof. Dividing the two expressions in (118.21) cancels n and uses $u_L/u_T = u_C$. Multiplying the speed section by t_0 proves both elementary formulas. □

When the channel depends on the edge, the readout retains the additive form $L_{\mathrm{int}}(\gamma) = \sum_{e \in \gamma} \widehat{c}(e) u_L$. Its quotient by $n u_T$ is the mean of the speeds of those edges. The constant-channel expression is the homogeneous specialization of this readout.

118.7.2 Change of chart and adapted normalization

Proposition 118.7 (Covariance of the identification)**.** *Let $u'_C = d u_C$ and $u'_T = \eta u_T$, with $d, \eta > 0$. The coordinates of the same sections satisfy*

$$\widehat{c}' = d^{-1} \widehat{c}, \qquad \tau'_0 = \eta^{-1} \tau_0, \qquad u'_L = d\eta u_L. \tag{118.23}$$

If another publication writes $c' = \widehat{c}' u'_C$ with $u'_C = d u_C$, equality of sections is exactly equivalent to $d\widehat{c}' = \widehat{c}$.

Proof. Equality of each section in the two bases gives $\widehat{c}' u'_C = \widehat{c} u_C$ and $\tau'_0 u'_T = \tau_0 u_T$. In particular,

$$\tau'_0 \widehat{c}' u'_L = \frac{\tau_0}{\eta} \frac{\widehat{c}}{d} d\eta u_L = \tau_0 \widehat{c} u_L = \ell_0.$$

The final equivalence follows from the same substitution for speed. □

The adapted choice $d = \widehat{c}$ gives $u'_C = c_{\mathrm{vac}}$ and $\widehat{c}' = 1$: it expresses the constructed section in an adapted basis. The coefficient of the internal chart remains $(r_+ r_-)^{-1}$, and this is the normalization used by cubic inversion. Keeping the action and charge bases, (118.11) produces

$$\widehat{\mu}' = d\widehat{\mu}, \qquad \widehat{\varepsilon}' = d\widehat{\varepsilon}, \qquad \widehat{\mu}'\widehat{\varepsilon}' = (\widehat{c}')^{-2}.$$

In the adapted chart, we have $\widehat{\mu}' = r_-/r_+$ and $\widehat{\varepsilon}' = r_+/r_-$, by substitution of $\widehat{\mu} = r_-^2$ and $\widehat{\varepsilon} = r_+^2$.

118.7.3 Preservation of the readout under amplification

For the refinement $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$, the normalized trace $\tau_m = \mathrm{Tr}/(2 \cdot 9^m)$ satisfies

$$\exp[-2\tau_m(\log \mathcal{V}_m)] = \exp[-\log r_+ - \log r_-] = \widehat{c}. \tag{118.24}$$

Indeed, each eigenvalue appears 9^m times; that multiplicity cancels against the denominator of the normalized trace. The ordinary amplified determinant is $(r_+ r_-)^{9^m}$. The two-sheet normalization, equivalently expressed by (118.24), preserves the constitutive readout during amplification. This realization preserves the enriched transport from which it originates.

Chapter 119

Charge section and electrical quantization relations

The constitutive response establishes a relation between the two sheets and a relative propagation scale. The charge section is obtained by composing this response with the fine-structure constant and the action, both constructed in the preceding chapters. This order has a precise consequence: charge does not retrospectively determine the dodecaphase record. It is the record, together with the action and the vacuum response, that determines the charge section on the corresponding line.

119.1 Positive determination of the charge section

Consider the positive sections $\hbar_a$, with $a \in \{\mathrm{pre}, \mathrm{ret}\}$, and $h_a = 2\pi\hbar_a$. The impedance Z_{vac} has already been constructed through (118.11). The quotient h_a/Z_{vac} belongs to the quadratic charge line. This dimensional compatibility allows the following definition to be formulated without yet choosing an SI chart.

Definition 119.1 (Positive charge section)**.** The charge section corresponding to action orientation a is

$$e_a = \left(\frac{2\alpha h_a}{Z_{\mathrm{vac}}}\right)^{1/2}. \tag{119.1}$$

The root is taken on the positive charge half-line. Its conjugate section is $-e_a$; the sign of the latter is not identified with the action return index.

Proposition 119.2 (Existence, uniqueness, and compatibility of the coupling)**.** *The equation $Z_{\mathrm{vac}} e_a^2 = 2\alpha h_a$ determines a unique positive section. In the constitutive realization, one also has*

$$\alpha = \frac{Z_{\mathrm{vac}} e_a^2}{2h_a} = \frac{e_a^2}{4\pi\varepsilon_{\mathrm{vac}}\hbar_a c_{\mathrm{vac}}}. \tag{119.2}$$

Proof. Positivity of α, h_a, and Z_{vac} guarantees a unique positive root. The constitutive identities give $\varepsilon_{\mathrm{vac}} c_{\mathrm{vac}} = Z_{\mathrm{vac}}^{-1}$. Substituting this equality and $h_a = 2\pi\hbar_a$ into the first expression of (119.2) yields the second. □

The last expression is the usual form of electromagnetic coupling in a physical chart. Here it appears as a subsequent identity of the constructed sections. The proof does not recalculate α from a measured charge: it establishes the algebraic consistency of the preceding result with the section just defined. Physical identification additionally requires retaining the chart of units and the coupling regime used in the comparison. An identity between quantities and an experimental comparison have different functions in that recognition.

119.2 Resistance, conductance, flux, and frequency

The following four quantities use the same section e_a. Their dependencies can be fully resolved before any decimal is evaluated.

Theorem 119.3 (Composition of the electrical quanta)**.** *The sections*

$$R_K = \frac{h_a}{e_a^2} = \frac{Z_{\mathrm{vac}}}{2\alpha}, \qquad G_0 = \frac{2e_a^2}{h_a} = \frac{4\alpha}{Z_{\mathrm{vac}}}, \tag{119.3}$$

$$\Phi_{0,a} = \frac{h_a}{2e_a} = \sqrt{\frac{h_a Z_{\mathrm{vac}}}{8\alpha}}, \qquad K_{J,a} = \frac{2e_a}{h_a} = \Phi_{0,a}^{-1} \tag{119.4}$$

satisfy exactly

$$R_K G_0 = 2, \qquad G_0 Z_{\mathrm{vac}} = 4\alpha, \qquad K_{J,a}\Phi_{0,a} = 1, \qquad K_{J,a}^2 R_K = \frac{4}{h_a}. \tag{119.5}$$

Proof. Substituting $e_a^2 = 2\alpha h_a / Z_{\mathrm{vac}}$ into the first two definitions cancels h_a and gives (119.3). Squaring the positive section $h_a/(2e_a)$ gives $h_a Z_{\mathrm{vac}}/(8\alpha)$; positivity fixes the root in (119.4). Its inverse is $2e_a/h_a$. The first three products in (119.5) reduce, respectively, to 2, 4α, and 1. Finally, $(4e_a^2/h_a^2)(h_a/e_a^2) = 4/h_a$. □

The physical realization recognizes in these sections the von Klitzing resistance, the conductance quantum, the flux quantum, and the Josephson constant. The factor 2 in the conductance belongs to the normalization of the quantity defined here; identifying it with a multiplicity of channels in a material system requires specifying that representation. The four equations do not constitute four independent generators: they share α, the action, and the vacuum response. Precisely this dependency produces the cross-identities and makes it possible to check an evaluation without altering the operations from which it arose.

119.3 Action return and transport of the sections

The quotient $R_{\mathrm{act}} = h_{\mathrm{pre}}/h_{\mathrm{ret}}$, obtained before the angular pair, preserves the distinction between the two action sections. It does not change the sheet assignments of the constitutive response. These are two distinct operations: action return and channel interchange.

Proposition 119.4 (Covariance under return)**.** *With common α and Z_{vac}, one has*

$$\frac{e_{\mathrm{pre}}}{e_{\mathrm{ret}}} = R_{\mathrm{act}}^{1/2}, \qquad R_K^{\mathrm{pre}} = R_K^{\mathrm{ret}}, \qquad G_0^{\mathrm{pre}} = G_0^{\mathrm{ret}}, \tag{119.6}$$

$$\frac{\Phi_{0,\mathrm{pre}}}{\Phi_{0,\mathrm{ret}}} = R_{\mathrm{act}}^{1/2}, \qquad \frac{K_{J,\mathrm{pre}}}{K_{J,\mathrm{ret}}} = R_{\mathrm{act}}^{-1/2}. \tag{119.7}$$

Proof. The first identity follows by dividing the two versions of (119.1) and taking the positive root. In (119.3), the action has already cancelled. The other two identities follow from the powers $h_a^{1/2}$ and $h_a^{-1/2}$ in (119.4). □

Resistance and conductance preserve the constitutive quotient, whereas flux and frequency also retain the action orientation. This separation gives a structural reading of their dependencies: some quantities are invariants of the sheet quotient, while others additionally measure the action section used. The metrological comparison must specify which section it evaluates; the numerical proximity of the two actions does not authorize their identification.

Chapter 120

Hexadic–octadic incidence and the Barbero–Immirzi functional

Let $H \hookrightarrow O$ be the marked inclusion of a hexad in an octad, $|H| = 6$, $|O| = 8$, constructed through the binary residue of Section 21.68. The hexad, dodecad and regional chart determining that inclusion are preserved. The incidence objects

$$\mathcal{F}_6 = H \times \binom{H}{2}, \qquad \mathcal{F}_8 = O \times \binom{H}{2}$$

have cardinalities 90 and 120. Their normalized difference gives

$$\frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24\binom{6}{2}} = -\frac{1}{12}.$$

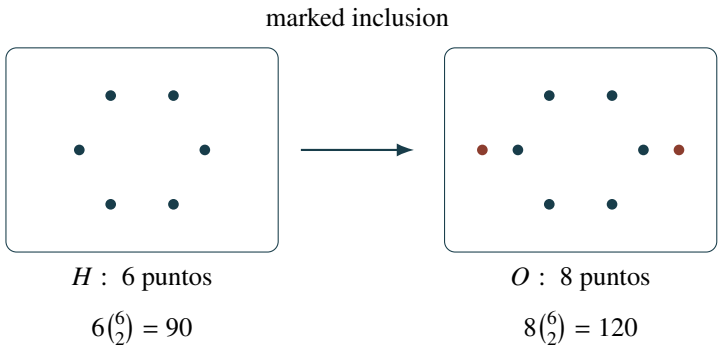


Figure 120.1: Marked incidence from six to eight points. The drawing represents the sets and inclusion; it does not attempt to turn the faces or vertices of a cube into Steiner blocks without their incidence map.

120.1 Dodecaphase fundamental chain and pole coefficient

In the free module of oriented events, fix the fundamental chain of the revolution

$$c_{12}^{+} = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\mathrm{ret}}, \mathcal{F}_8].$$

Multiplicity twelve corresponds to the sectors and multiplicity one to the oriented seam. The involution reverses orientation of the complete chain. Let

$$S_m(q) = \frac{q^m}{1 - q^{3m}}, \qquad 0 < q < 1.$$

The linear reader assigns S_{90} to every sector and S_{120} to the seam; consequently,

$$\mathscr{R}(c_{12}^{+}) = 12 S_{90} - S_{120}.$$

Definition 120.1 (Oriented evaluation of hexadic–octadic incidence)**.** The parameter γ_{HMT} is the conjugate evaluation of the chain c_{12}^{+}, completed by its angular component. On the domain $0 < q_{\pm} < 1$, it is defined by

$$\gamma_{\mathrm{HMT}} = \frac{\pi A C^{*}}{180} + [\mathscr{R}(c_{12}^{+})](q_{-}) - [\mathscr{R}(c_{12}^{+})](q_{+}).$$

The marked incidence determines the event spaces; the oriented chain fixes the multiplicities, and the channels of Section 14.26 determine the evaluation.

Theorem 120.2 (Pole coefficient and conjugate evaluation)**.** *The image of the fundamental chain has coefficient* $1/24$ *relative to* $(1-q)^{-1}$ *and residue* $-1/24$ *relative to* $(q-1)^{-1}$*. Its evaluation on the previously generated channels determines*

$$\gamma_{\mathrm{HMT}} = \frac{\pi A C^{*}}{180} + 12\big[S_{90}(q_{-}) - S_{90}(q_{+})\big] - \big[S_{120}(q_{-}) - S_{120}(q_{+})\big]. \tag{120.1}$$

Proof. The factorization $1-q^{3m} = (1-q)\sum_{j=0}^{3m-1} q^{j}$ implies $\lim_{q\uparrow 1}(1-q)S_m(q) = 1/(3m)$. By linearity,

$$\frac{12}{270} - \frac{1}{360} = \frac{1}{24}.$$

The coordinate $q-1$ reverses the sign of the residue. The conjugate evaluation and the angular component of Definition 120.1 yield (120.1). □

The Diophantine certificate $4a-3b=45$ admits the pairs $(12+3t, 1+4t)$, $t \geq 0$; it certifies minimality of $(12,1)$ but does not replace the origin of multiplicities in the chain. The evaluation of (120.1) is

$$\gamma_{\mathrm{HMT}} = 0.27400767269445927474725526077398041940\ldots.$$

This numerical value is an output of the oriented functional, rather than an external value used to select A, C^{*}, or their coefficients.

120.2 Parity and structural sensitivity

Reversing C^{*} exchanges q_{+} and q_{-} and changes the sign of the angular component. Therefore

$$\gamma_{\mathrm{HMT}}(A, -C^{*}) = -\gamma_{\mathrm{HMT}}(A, C^{*}).$$

The functional preserves orientation, whereas other readers, such as $q_{+}q_{-}$, are even. This difference precedes its physical interpretation.

Chapter 121

Action, gravity, and thermal information

121.1 Dodecaphasic period and circular realization of action

The already constructed calendar determines $T_{108} = 108t_0$. The constitutive speed, duration and trajectory length are obtained through the identities of Proposition 118.5 and equations (118.21) and (118.22). In its circular realization,

$$\omega_{108} = \frac{2\pi}{108t_0}, \qquad R_{108} = \frac{c_{\mathrm{int}}}{\omega_{108}} = \frac{54}{\pi}\ell_0, \qquad \ell_0 = c_{\mathrm{int}}t_0.$$

The unit ℓ_0 and the time and velocity charts remain explicit; the SI chart does not select the cycle. For each oriented action section $a \in \{\mathrm{pre}, \mathrm{ret}\}$,

$$E_{108,a} = \hbar_a \omega_{108}, \qquad m_{108,a} = \frac{E_{108,a}}{c_{\mathrm{int}}^2}.$$

The reduced and full lengths are, respectively,

$$\frac{\hbar_a}{m_{108,a}c_{\mathrm{int}}} = R_{108}, \qquad \frac{h_a}{m_{108,a}c_{\mathrm{int}}} = 2\pi R_{108}.$$

The same action and duration enter into both readings.

Proposition 121.1 (Radius-coincidence selector)**.** *In the dimensional family*

$$G_a(\vartheta) = \vartheta \frac{c_{\mathrm{int}}^5 t_0^2}{\hbar_a}, \qquad \vartheta > 0,$$

the circular-realization condition $G_a(\vartheta)m_{108,a}/c_{\mathrm{int}}^2 = R_{108}$ has the unique solution

$$\vartheta_{108}^{\circ} = (54/\pi)^2.$$

Proof. The first radius is $r_g(\vartheta) = \vartheta(\pi/54)\ell_0$. Comparing it with $(54/\pi)\ell_0$, with $\ell_0 > 0$, reduces the equality to a strictly increasing linear equation. □

The coincidence condition is an explicit part of this realization; the theorem solves for its coefficient without using a metrological value of G. The radius used here is Gm/c^2, so the factor two from the convention $2Gm/c^2$ is not inadvertently transferred.

Proposition 121.2 (Reciprocal action and gravity, conserved area)**.** *In the two preceding charts,*

$$\frac{G_{\mathrm{pre}}}{G_{\mathrm{ret}}} = R_{\mathrm{act}}^{-1}, \qquad \hbar_a G_a = \vartheta_{108}^{\circ} c_{\mathrm{int}}^5 t_0^2, \qquad \ell_P^2 = \frac{\hbar_a G_a}{c_{\mathrm{int}}^3} = \vartheta_{108}^{\circ}\ell_0^2.$$

Proof. The definition of G_a contains $\hbar_a^{-1}$ and the remaining factors are common to the two charts. Taking the quotient and the product gives the three equalities. □

This reciprocity explains why orientation information can vary without changing the area. A precise relation is thus available between the reciprocal action sections, the Planck scale and the areal operator of Section 122.

Chapter 122

Area operator and reconstruction of sectorial information

Regrouping six trits constructs a bijection between $(\mathbb{Z}/9\mathbb{Z})^3$ and $(\mathbb{Z}/27\mathbb{Z})^2$, both of cardinality 729. The bijection is not presented as an additive isomorphism. On a 9×9 boundary block, the source decomposes 36 links into two sets of 18, labeled by channels 90 and 120.

122.1 Tritic regrouping and boundary incidence

The correspondence preserves an ordered word, in addition to its cardinality. Write each coordinate $x_j \in \mathbb{Z}/9\mathbb{Z}$, for $j = 1, 2, 3$, through its unique positional representatives $x_j = r_{2j-1} + 3r_{2j}$, with $r_k \in \{0, 1, 2\}$. The map is

$$\beta_{729}(x_1, x_2, x_3) = \big(r_1 + 3r_2 + 9r_3,\ r_4 + 3r_5 + 9r_6\big) \in (\mathbb{Z}/27\mathbb{Z})^2. \tag{122.1}$$

Proposition 122.1 (Positional correspondence between interior and boundary)**.** *The map β_{729} is bijective and preserves the order of the six trits. Its inverse extracts the two triples of digits and regroups them into three pairs.*

Proof. The positional expansion of the representatives $0, \dots, 8$ and $0, \dots, 26$ is unique. Extracting the six digits, regrouping them, and extracting them again returns the same word. The inverse operation thus restores each of the three initial coordinates. □

Carry preserves its position in the enriched word. The additive operations of the two packagings have different boundaries: for example, $\beta_{729}(0, 2, 0) = (18, 0)$ and $\beta_{729}(0, 1, 0) = (9, 0)$, whose sum in the order-twenty-seven torus is $(0, 0)$, whereas $\beta_{729}(0, 3, 0) = (0, 1)$. This distinction allows the positional correspondence to be reused while keeping the operations typed.

In the torus $(\mathbb{Z}/27\mathbb{Z})^2$, the block $B_\partial = \{0, \dots, 8\}^2$ has the oriented link sets

$$\partial B_{90} = \{((-1, j), (0, j)),\ ((8, j), (9, j)) : 0 \le j \le 8\}, \tag{122.2}$$

$$\partial B_{120} = \{((i, -1), (i, 0)),\ ((i, 8), (i, 9)) : 0 \le i \le 8\}. \tag{122.3}$$

All coordinates are read modulo twenty-seven. Each system contains eighteen links and the two systems are disjoint. Their labels preserve the incidence channels; the cardinality of the links is $18 + 18$.

Theorem 122.2 (Minimum capacity of the cubic cut)**.** *In the cubic graph with vertices $\{0, \dots, N-1\}^3$, with $N \ge 2$, links between neighbors, and unit capacity per link, the minimum cut between two opposite faces has capacity N^2. For $N = 9$, a maximally mixed product state on the trits of a minimum layer has information*

$$\mathrm{cap}_{\min} = 81, \qquad s_{\mathrm{corte}} = 81 \log 3. \tag{122.4}$$

Proof. The N^2 lines parallel to the normal of the faces form edge-disjoint paths. Every separating cut intercepts each path, so it contains at least N^2 links. The links crossing a transverse layer constitute a cut of that cardinality. Each uniform trit contributes $\log 3$ nats; entropy is additive for the product state of the eighty-one trits of the layer. □

The capacity of the cubic cut and the state of the thirty-six links in (122.3) belong to observables defined on different cuts. The genealogy preserves both domains and their measures.

122.2 Transport of the collective incidence modes

The spaces $\mathcal{F}_6$ and $\mathcal{F}_8$ of Section 120 provide the uniform vectors

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \qquad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f)$$

in $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$. In $\ell^2(\partial B_{90}) \oplus \ell^2(\partial B_{120})$, construct

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{90}} (\delta_e, 0), \qquad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{120}} (0, \delta_e).$$

The two pairs are orthonormal. Denote their respective spanned planes by $\mathscr{H}_{\mathrm{inc}}^{\mathrm{coll}}$ and $\mathscr{H}_{\partial}^{\mathrm{coll}}$.

Theorem 122.3 (Collective intertwining of the sectorial labels)**.** *There is a unique surjective linear isometry between these planes such that*

$$\mathcal{J}_{6\to 8} u_m = v_m \quad (m = 90, 120). \tag{122.5}$$

For the operators

$$D_{\mathrm{inc}} = 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|,$$
$$D_{\partial} = 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|,$$

we have

$$\mathcal{J}_{6\to 8} D_{\mathrm{inc}} = D_{\partial} \mathcal{J}_{6\to 8}. \tag{122.6}$$

The sectorial projectors and their oriented combination with coefficients $12 : -1$ *are also transported.*

Proof. The image of a basis fixes the unique linear map. Norms and inner products are preserved because both bases are orthonormal. Identity (122.6) holds on each vector u_m, since its two sides are $m v_m$. The projectors are expressed as $(120I - D)/30$ and $(D - 90I)/30$; their linear functional calculus and any combination of the two preserve intertwining. □

This transport acts on the two uniform modes and their spectral labels. Individual incidences and fluctuations within each sector remain in their complete spaces. The link preserves exactly the collective datum that the areal weighting will use.

122.3 Angular character and area operator

Let $N_e = \mathrm{diag}(0, 1, 2)$ on each link, and let N_{90}, N_{120} be the sector sums. With the scale

$$\theta_{\mathrm{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \qquad a_0 = \frac{1 + 2\cos(10^\circ)}{3} \theta_{\mathrm{clk}},$$

fixed, the angular factor is the normalized real character of the triplet

$$\chi_{10} = \frac{1}{3} \operatorname{Tr} \mathrm{diag}(1, e^{i\pi/18}, e^{-i\pi/18}) = \frac{1 + 2\cos(10^\circ)}{3}, \qquad a_0 = \chi_{10} \theta_{\mathrm{clk}}. \tag{122.7}$$

The sum of the conjugate phases proves the equality. On the positive branch considered, $\theta_{\mathrm{clk}} > 0$, $\chi_{10} > 0$, and $\gamma_{\mathrm{HMT}} > 0$; hence a_0 and the areal scale are positive. The angular triplet and its normalization precede the entropic distribution on the links. With these quantities, the normalized area operator is

$$\widehat{\mathcal{A}}/\ell_P^2 = \gamma_{\mathrm{HMT}} a_0 (N_{90} + N_{120}).$$

Its spectrum has values $n\gamma_{\mathrm{HMT}}a_0$, $0 \le n \le 72$, with multiplicities given by $(1 + z + z^2)^{36}$. The proof consists in summing the eigenvalues $0, 1, 2$ of the 36 tensor factors.

Theorem 122.4 (Recovery of the boundary sectors)**.** *Let*

$$N = N_{90} + N_{120}, \qquad D = 90N_{90} + 120N_{120}.$$

Then

$$N_{90} = 4N - \frac{D}{30}, \qquad N_{120} = \frac{D}{30} - 3N. \tag{122.8}$$

Thus the total and the weighted reader jointly recover the two occupations.

Proof. La matriz $\left(\begin{smallmatrix} 1 & 1 \\ 90 & 120 \end{smallmatrix}\right)$ has determinant 30. Its inversion gives (122.8). On the domain of integer occupations, the image satisfies $D \in 30\mathbb{Z}$ and $90N \le D \le 120N$. □

The modular Hamiltonian of the source is $K_A = -\log \rho_A = cI + \Delta_{\mathrm{mod}} D$. For $\Delta_{\mathrm{mod}} \neq 0$ and known normalization, $D = (K_A - cI)/\Delta_{\mathrm{mod}}$ publishes the second observable. This operator describes the spectral structure of the reduced state; it is not identified by notation with the generator of time evolution. The area preserves the total occupation and the pair adds the provenance. For example, $(N_{90}, N_{120}) = (1, 0)$ and $(0, 1)$ share $N = 1$, but have $D = 90$ and $D = 120$.

122.4 State normalization and the meaning of the modular Hamiltonian

The second observable is made explicit through the state that defines it. The preceding tensor decomposition is retained: eighteen qutrits per sector, each with number operator $\mathrm{diag}(0, 1, 2)$; N_m is the sum of these operators and the sectors act on distinct factors. Let

$$z_m = 1 + e^{-m\Delta_{\mathrm{mod}}} + e^{-2m\Delta_{\mathrm{mod}}}, \qquad m \in \{90, 120\}.$$

The family of source states has the product realization

$$\rho_A = Z^{-1} e^{-\Delta_{\mathrm{mod}}(90N_{90} + 120N_{120})}, \qquad Z = z_{90}^{18} z_{120}^{18}.$$

Commutativity of the number operators and tensor factorization prove the expression for Z. The state has full rank for real, finite Δ_{mod}, and its spectral logarithm gives

$$K_A = -\log \rho_A = (\log Z) I + \Delta_{\mathrm{mod}} D.$$

For $\Delta_{\mathrm{mod}} \neq 0$, K_A weights the excitations of the two sectors differently. Joint reading with N recovers the two occupations by equation (122.8).

The need for joint reading is seen in two complementary examples. The occupation states $(1, 0)$ and $(0, 1)$ have the same total $N = 1$ and different modular weights. Conversely, $(4, 0)$ and $(0, 3)$ have the same $D = 360$ and different totals. The pair (N, D) distinguishes both cases. This recovery concerns sectoral occupations, not each microscopic vector within a degeneracy of those occupations.

When $\Delta_{\mathrm{mod}} = 0$, the state is proportional to the identity and K_A is scalar: that reading ceases to distinguish the sectors. For $\Delta_{\mathrm{mod}} \neq 0$, the known normalization allows one to recover D and compose it with the area. The role of the modular Hamiltonian is thereby defined: it characterizes the reduced state and its information distribution. Its inclusion in the article is motivated by that mathematical function, whereas the evolution Hamiltonian belongs to the dynamical development of the action.

122.5 Reduced state, Schmidt decomposition, and entanglement entropy

If $\rho_A = \sum_\nu p_\nu |\nu\rangle\langle\nu|$, the purification

$$|\Psi\rangle = \sum_\nu \sqrt{p_\nu}\, |\nu\rangle_A |\nu\rangle_B$$

satisfies $\mathrm{Tr}_B\, |\Psi\rangle\langle\Psi| = \rho_A$. Thus $-\mathrm{Tr}(\rho_A \log \rho_A)$ is the entanglement entropy of this particular pure bipartition. The enlarged state declares which system carries that interpretation.

For a two-sheet involution with one eigenvector of each sign,

$$\rho_y = \frac{e^{-yR}}{2\cosh y}, \qquad R^2 = I,$$

diagonalization gives

$$S(\rho_y) = \log(2\cosh y) - y\tanh y, \qquad D(\rho_y \| \rho_{-y}) = 2y\tanh y.$$

Entropy is even in y; the distance explicitly compares the two orientations. This two-level entropy is not identified with the capacity $81\log 3$ of the triadic cut.

Chapter 123

Genealogical memory, informational reduction and thermal transduction

The thermal construction uses two publications of the same APP–TRIT–TPK development. The first preserves the states, admissible routes and register information; the second provides the oriented action and the duration of the return. Their composition determines an energy per tritic decision. The temperature and energy coordinates are subsequently introduced through explicit dimensional maps. This sequence allows one to identify what information is preserved, what information a reader publishes and what quantity converts that information into dimensional entropy.

123.1 Reader, memory, and state recovery

Fix a finite level of the development and denote by Ω the set of admissible enriched states. The register preserves sheet, orientation, residue, quotient, carry, route, phase and memory in the domains where each coordinate enters. Let

$$q : \Omega \longrightarrow Y, \qquad M : \Omega \longrightarrow \mathcal{M}$$

be the visible publication and the retained memory, respectively. For a distribution p over Ω, we write $\mathsf{H}(p) = -\sum_x p_x \log p_x$, with $0 \log 0 = 0$. The logarithm is applied to dimensionless probabilities.

Definition 123.1 (Genealogical recovery of a publication)**.** The pair $\widehat{q} = (q, M)$ is a recoverable reader on the support Ω_p when there exists

$$d : \widehat{q}(\Omega_p) \longrightarrow \Omega_p, \qquad d \circ \widehat{q} = \mathrm{id}_{\Omega_p}.$$

The fibre information of q is $\mathsf{H}_{\mathrm{fib}}(q; p) = \mathsf{H}(X \mid q(X))$, where X has law p.

Proposition 123.2 (Recovery identity and cost of reduction)**.** *For a recoverable reader,*

$$\mathsf{H}(X \mid q(X), M(X)) = 0, \tag{123.1}$$

$$\mathsf{H}(X) = \mathsf{H}(q(X)) + \mathsf{H}(X \mid q(X)), \tag{123.2}$$

$$\mathsf{H}(X \mid q(X)) = \mathsf{H}(M(X) \mid q(X)). \tag{123.3}$$

Proof. The map d recovers X from the pair $(q(X), M(X))$, so the first conditional entropy is zero. Since q is a function of X, the chain rule gives the second equality. For the third, expand $\mathsf{H}(X, M(X) \mid q(X))$ in two ways. One gives $\mathsf{H}(X \mid q(X))$, since M is a function of X; the other gives $\mathsf{H}(M(X) \mid q(X)) + \mathsf{H}(X \mid q(X), M(X))$. □

Euclidean division on the APP sheets displays the elementary arithmetic mechanism. With $i, j \in \{1, \dots, 9\}$, one preserves

$$i + j = 9q^{+}_{ij} + R^{+}_{ij}, \qquad ij = 9q^{\times}_{ij} + R^{\times}_{ij}, \qquad 1 \le R^{\sigma}_{ij} \le 9 \quad (\sigma \in \{+, \times\}).$$

The residue–quotient pair recovers the integer value of the operation. Summing the differences over the eighty-one positions gives

$$2025 - 810 = 54 + 9 \cdot 129.$$

Quotient memory restores the contribution not published by residual reduction. Recoverability of the complete state additionally requires preserving the relevant position and route coordinates; the residue–quotient pair identifies the result of the operation here.

123.2 Reversible update, tritic decision and erasure

Let $U : \Omega \to \Omega$ be the TPK update on a domain where enriched transport has an inverse. The transported law is $p'_y = p_{U^{-1}y}$. The bijection preserves the multiset of probabilities and therefore

$$\mathsf{H}(U(X)) = \mathsf{H}(X), \qquad \mathsf{H}(X \mid U(X)) = 0. \tag{123.4}$$

If the observed output is $q(U(X))$, the unpublished information is quantified by $\mathsf{H}(X \mid q(U(X)))$. The complete state, its publication and the resetting of a register are thus three distinct maps.

For the three oriented TRIT states, let $p = (p_{-1}, p_0, p_{+1})$ be a normalized distribution. Its information is

$$I_{\mathrm{TRIT}}(p) = -\sum_{a=-1}^{1} p_a \log p_a, \qquad 0 \le I_{\mathrm{TRIT}}(p) \le \log 3. \tag{123.5}$$

The upper bound corresponds to $p_a = 1/3$. Indeed,

$$D(p\|(1/3, 1/3, 1/3)) = \sum_a p_a \log(3p_a) = \log 3 - I_{\mathrm{TRIT}}(p) \ge 0;$$

the inequality follows from the convexity of $t \log t$, and equality characterizes the uniform distribution. The lower bound corresponds to a distribution concentrated on one state.

The ideal Landauer realization uses a memory with energetically degenerate logical states, an isothermal environment at temperature Θ and a balance that preserves the output of the logical map and accounts for auxiliary registers. In that model, the decrease in logical entropy is $k_B[\mathsf{H}(X) - \mathsf{H}(F(X))] = k_B\mathsf{H}(X \mid F(X))$ for a deterministic map F. The entropic balance requires transferring at least that amount to the environment; its energy expression is $Q \ge k_B\Theta\mathsf{H}(X \mid F(X))$. The distinction between reversible logical evolution and physical erasure corresponds to the Landauer–Bennett realization [168]. The invertible update (123.4) has zero logical contribution; resetting a uniform triple requires transferring $\log 3$ nats to the environment and satisfies $Q \ge k_B\Theta \log 3$. The control energy and dissipation of a specific physical realization belong to its additional thermodynamic balance.

123.3 Action, frequency, and decision energy

Let $\mathcal{L}_{\mathrm{act}}$ and $\mathcal{L}_T$ be the dimensional lines of action and duration; the energy line is $\mathcal{L}_E = \mathcal{L}_{\mathrm{act}} \otimes \mathcal{L}_T^*$. The preceding development provides the positive sections $\hbar_a \in \mathcal{L}_{\mathrm{act}}$, with $a \in \{\mathrm{pre}, \mathrm{ret}\}$, and the elementary duration $t_0 \in \mathcal{L}_T$. The complete calendar provides

$$T_{108} = 108t_0, \qquad \omega_{108} = \frac{2\pi}{T_{108}}, \qquad h_a = 2\pi\hbar_a.$$

The dimensional contraction between action and frequency defines

$$\epsilon_a := \hbar_a\omega_{108} = \frac{h_a}{108t_0} = \frac{\hbar_a}{t_P}, \qquad t_P = \frac{54}{\pi}t_0. \tag{123.6}$$

The final expression uses the same reduced circular period as the gravitational section. The three expressions correspond to a single energy section, with an orientation a maintained in all the factors. The normalized information of a uniform decision produces the energy section per nat

$$\varepsilon_a := \frac{\epsilon_a}{\log 3} \in \mathcal{L}_E^+. \tag{123.7}$$

The previously constructed action determines the energy scale once the duration is fixed; the cardinality of the TRIT alternatives determines its informational normalization. The value of $\log 3$ enters through the count and measure declared in (123.5).

123.4 Dimensional definition of Boltzmann and selection of coordinates

Let $\mathcal{L}_\Theta$ be the oriented temperature line. The dimensional entropy space is $\mathcal{L}_{\mathrm{ent}} = \mathcal{L}_E \otimes \mathcal{L}_\Theta^*$. A positive linear map $k_B : \mathcal{L}_\Theta \to \mathcal{L}_E$ is, equivalently, a positive element of $\mathcal{L}_{\mathrm{ent}}$. Thus the same object can act on a temperature or multiply dimensionless information.

Definition 123.3 (Thermal normalization of the tritic decision)**.** A thermal pair compatible with the oriented section a consists of a positive transducer k_B and a positive section $\Theta_{\mathrm{clk},a} \in \mathcal{L}_\Theta$ such that

$$k_B(\Theta_{\mathrm{clk},a}) = \varepsilon_a, \qquad k_B\Theta_{\mathrm{clk},a} \log 3 = \epsilon_a. \tag{123.8}$$

The Boltzmann constant serves as the energy–temperature transducer of tritic information in this realization.

Theorem 123.4 (Uniqueness relative to the thermal section and dimensional covariance)**.** *Given a positive section $\Theta_{\mathrm{clk},a}$, there exists a unique positive linear map satisfying* (123.8)*. For positive bases u_E, u_Θ, if*

$$\epsilon_a = \epsilon_a^{\mathrm{num}} u_E, \quad \Theta_{\mathrm{clk},a} = \theta_a u_\Theta, \quad k_B(u_\Theta) = b\, u_E,$$

its coordinate is

$$b = \frac{\epsilon_a^{\mathrm{num}}}{\theta_a \log 3}. \tag{123.9}$$

Under $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$, with $s_E, s_\Theta > 0$, one has

$$b' = \frac{s_\Theta}{s_E} b, \qquad \theta'_a = \frac{\theta_a}{s_\Theta}, \qquad (\epsilon_a^{\mathrm{num}})' = \frac{\epsilon_a^{\mathrm{num}}}{s_E}.$$

Proof. A nonzero section generates a line. Prescribing its image determines the unique linear map; the positivity of both sections proves its positivity. Substituting the coordinates into (123.8) gives $b\theta_a \log 3 = \epsilon_a^{\mathrm{num}}$. The three transformation laws are obtained by expressing the same sections and the same map in the new bases. □

The same Boltzmann constant must act in both orientations. Since t_P is common, simultaneous compatibility is equivalent to

$$\frac{\Theta_{\mathrm{clk,pre}}}{\Theta_{\mathrm{clk,ret}}} = \frac{\epsilon_{\mathrm{pre}}}{\epsilon_{\mathrm{ret}}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}}. \tag{123.10}$$

Indeed, an isomorphism of lines preserves quotients of sections. Conversely, if the preceding equality holds, the isomorphism determined in one orientation also satisfies the normalization of the other. Thus, return changes the thermal section of the cycle in the same proportion as the energy, while the transducer remains common to both readings.

The preceding equation makes explicit the choice required to publish an individual value. If the thermal section has not yet been selected by an independent reader, for each $\lambda > 0$ the pair

$$(k_B, \Theta_{\mathrm{clk},a}) \longmapsto (\lambda k_B, \lambda^{-1}\Theta_{\mathrm{clk},a})$$

preserves ϵ_a. Consequently, presenting an internally selected individual coordinate requires specifying the reader of $\Theta_{\mathrm{clk},a}$, the bases and their dimensional transport. The action and information equations remain determined before that specification. The subsequent identification of the bases with joule and kelvin belongs to the metrological publication of the result.

123.5 Dimensional entropy and information of the reduced state

For a distribution of admitted routes p, or for a reduced state with density matrix ρ, let

$$s(p) = -\sum_x p_x \log p_x, \qquad s(\rho) = -\operatorname{Tr}(\rho \log \rho).$$

Their dimensional publications are

$$S(p) = k_B s(p), \qquad S(\rho) = k_B s(\rho) \quad \text{in } \mathcal{L}_{\text{ent}}. \tag{123.11}$$

Evaluation at Θ produces an energy $\Theta S = k_B \Theta\, s$. For the uniform triple at the cycle temperature, (123.8) gives exactly

$$S_{\text{TRIT}} = k_B \log 3, \qquad \Theta_{\text{clk},a} S_{\text{TRIT}} = \epsilon_a.$$

If ρ_A is the reduction of the bipartite purification constructed in the areal section, $s(\rho_A)$ measures the entanglement of that bipartition. Unit transport preserves its eigenvalues, its orientation and the sectoral occupations. The areal scale $\gamma a_0 \ell_P^2$ converts occupations into area; the transducer k_B converts information into dimensional entropy. Both readings use the same register through distinct operators.

123.6 Spectral normalization of thermal histories

Let $\mathcal{G}$ be an already constructed finite graph of admissible extensions, with adjacency matrix A. Each edge e preserves its route coordinates and receives a dimensionless action increment $a_*(e) > 0$ from the corresponding reader. Having chosen the energy section ϵ_*, set $t = \beta\epsilon_*$, where $\beta = (k_B\Theta)^{-1}$. The thermal operator is

$$(L_t f)(y) = \sum_{e : x \to y} \exp[-t a_*(e)]\, f(x). \tag{123.12}$$

All its exponents are dimensionless. The condition $\rho(L_t) = 1$ normalizes the weighted growth of histories.

Proposition 123.5 (Existence and uniqueness of normalized pressure)**.** *Suppose that A is irreducible, $\rho(A) > 1$, and that $0 < m \le a_*(e) \le M$ for all edges. There exists a unique $t_* > 0$ such that $\rho(L_{t_*}) = 1$, and*

$$\frac{\log \rho(A)}{M} \le t_* \le \frac{\log \rho(A)}{m}. \tag{123.13}$$

Proof. Entrywise comparison gives $e^{-tM}A \le L_t \le e^{-tm}A$ for $t \ge 0$. Monotonicity of the spectral radius of nonnegative matrices implies the corresponding bounds for $\rho(L_t)$. The spectral radius is continuous, equals $\rho(A) > 1$ at zero and tends to zero at infinity. For $s > 0$, moreover, $L_{t+s} \le e^{-sm}L_t$, so that $\rho(L_{t+s}) \le e^{-sm}\rho(L_t) < \rho(L_t)$. Continuity and strict decrease prove existence and uniqueness. Substituting the value one into the initial bounds proves (123.13). □

The local realization of three equally weighted alternatives, represented by a vertex with three loops and $a_*(e) = 1$, satisfies $\rho(L_t) = 3e^{-t}$. Its normalization gives $t_* = \log 3$. With $\epsilon_* = \epsilon_a$, we obtain $\beta_{\mathrm{clk}} = \log 3/\epsilon_a$, precisely the relation (123.8). This calculation identifies the local tritic confluence. For a graph of histories with additional restrictions, confluence with the same clock requires checking

$$\rho\big(L_{(\log 3/\epsilon_a)\epsilon_*}\big) = 1$$

with its effective increments. The local normalization and the pressure of the complete system thus retain their respective domains.

The resulting relation brings together structure, value and meaning. The register determines the recoverable information; probabilities over continuations determine its entropic reading; action and period determine ϵ_a; the energy–temperature transducer publishes dimensional entropy. Area and entanglement are related through the reduced state and its boundary operators, keeping visible which datum produces each quantity.

Chapter 124

Oriented incidence and gravitational tensor representation

The exceptional publication of the state preserves the incidence accompanying its arithmetic reading. The ternary matrix A_W of (21.126) provides a precise material antecedent here. Its balanced lift $C = \mathrm{bal}(A_W)$, with representatives $0, 1, -1$, satisfies

$$C = C^{\mathsf{T}}, \qquad C^2 = 5I_6, \qquad \operatorname{Tr} C = 0.$$

The six positions have the order $(\infty, 0, 1, 2, 3, 4)$. This incidence first produces a three-dimensional space; its quadratic reading then produces the five-component space on which the gravitational coupling acts.

124.1 Signed covering and calendar transport

We define

$$E_C = \frac{1}{2}(I_6 + C/\sqrt{5}), \quad \mathcal{H}_C = \operatorname{im} E_C, \quad v_{u,\sigma} = \sigma\sqrt{2}E_C e_u, \quad \sigma \in \{1, -1\}.$$

Proposition 124.1 (Twelve oriented positions)**.** *The vectors $v_{u,\sigma}$ are twelve distinct unit vectors in a three-dimensional space. Their incidence is determined by $\langle v_{u,\sigma}, v_{v,\tau}\rangle = 1/\sqrt{5}$. Orientation reversal acts by antipodality.*

Proof. The quadratic identity for C implies $E_C^2 = E_C = E_C^*$; its trace is three. Since the diagonal of C is zero,

$$\langle v_{u,\sigma}, v_{v,\tau}\rangle = \begin{cases} \sigma\tau, & u = v, \\ \sigma\tau C_{uv}/\sqrt{5}, & u \neq v. \end{cases}$$

This proves normalization, distinction of the twelve vectors and incidence. Changing σ to $-\sigma$ changes the sign of the vector. □

An explicit Euclidean expression is obtained by setting $\nu = \sqrt{1+\varphi^2}$ and

$$B = \frac{1}{\nu}\begin{pmatrix} 0 & -1 & -\varphi & 0 & \varphi & 1 \\ -1 & -\varphi & 0 & 1 & 0 & -\varphi \\ -\varphi & 0 & -1 & -\varphi & -1 & 0 \end{pmatrix}.$$

The relation $\varphi^2 = \varphi + 1$, for the coordinate already generated, gives $B^*B = 2E_C$. Consequently $B/\sqrt{2}|_{\mathcal{H}_C}$ is an isometry and takes $v_{u,\sigma}$ to $\sigma B e_u$. The image is

$$\mathcal{V} = \frac{1}{\nu}\big(\{0\}\times\{\pm1\}\times\{\pm\varphi\} \;\cup\; \{\pm1\}\times\{\pm\varphi\}\times\{0\} \;\cup\; \{\pm\varphi\}\times\{0\}\times\{\pm1\}\big).$$

The internal coordinate, the incidence and its twelve geometric representatives are thus related.

The calendar coordinate is written $(b, r) \in \mathbb{Z}/12\mathbb{Z} \times \{0, \dots, 8\}$. A route of length n induces

$$k(r,n) = \left\lfloor \frac{r+n}{9} \right\rfloor, \qquad (b,r) \longmapsto \big(b + k(r,n) \bmod 12,\ (r+n) \bmod 9\big).$$

The explicit alignment of its positions $p = b + 1$ with the signed chart is

p	1	2	3	4	5	6	7	8	9	10	11	12
u	∞	0	0	∞	1	3	3	1	2	4	4	2
σ	+	−	+	−	−	+	−	+	−	+	−	+

This table fixes a trivialization of the carrier. Each change of position jointly transports the vectors and their incidence matrix.

Proposition 124.2 (Covariance of the frame under return)**.** *If $Se_p = e_{p+1 \bmod 12}$, the matrix representation of an operator P of the carrier is transported as $P_k = S^k P S^{-k}$. This transport preserves inner products, incidences and route composition. After twelve returns the chart returns, preserving the memory accumulated in the source state.*

Proof. S is orthogonal and $S^{12} = I$; the metric properties and periodicity of the chart follow. For composition, with $r' = (r + n) \bmod 9$, Euclidean division gives $k(r, n + m) = k(r, n) + k(r', m)$. The exponents of S are added exactly with the carry. The operation acts on the calendar coordinate; the route's memory updates remain present in its enriched domain. □

124.2 Quadratic reading and five-component space

Let $\mathbb{R}^{\mathcal{V}}$ be the space of the twelve coordinates, $\mathsf{A}e_v = e_{-v}$ its antipodal involution and $\mathsf{J} = \mathbf{1}\mathbf{1}^{\mathsf{T}}$. We define

$$P_5 = \frac{1}{2}(I + \mathsf{A}) - \frac{1}{12}\mathsf{J}, \qquad Q_v = vv^{\mathsf{T}} - \frac{1}{3}I_3, \qquad \Gamma_0 x = \sum_{v \in \mathcal{V}} x_v Q_v.$$

Theorem 124.3 (Five-component projector and tensor isometry)**.** *P_5 is the rank-five orthogonal projector onto the space of coordinates even under antipodality and of zero sum. Moreover,*

$$\Gamma_0^* \Gamma_0 = \frac{8}{5} P_5, \qquad \Gamma_0 \Gamma_0^* = \frac{8}{5} I_{\mathrm{Sym}_0^2(\mathbb{R}^3)}.$$

Therefore

$$\Gamma_5 = \sqrt{\frac{5}{8}}\, \Gamma_0|_{\operatorname{im} P_5}$$

is an isometry onto the space of traceless symmetric tensors.

Proof. The even space has one coordinate for each antipodal pair, so its dimension is six. The constant vector belongs to it; subtracting its rank-one projector gives P_5, of rank five.

Since $\langle Q_v, Q_w \rangle_F = (v^{\mathsf{T}} w)^2 - 1/3$, the entries of $\Gamma_0^* \Gamma_0$ equal $2/3$ for $w = \pm v$ and $-2/15$ in the other ten positions of each row. These are exactly the entries of $(8/5)P_5$.

The identity can also be checked in tensor space without resorting to a classification of representations. The list of vertices gives

$$\sum_v v_i^4 = \frac{12}{5}, \qquad \sum_v v_i^2 v_j^2 = \frac{4}{5} \quad (i \neq j),$$

and moments with any odd power vanish by sign changes. Indeed, $(1 + \varphi^2)^2 = 5\varphi^2$ and $1 + \varphi^4 = 3\varphi^2$. For a symmetric tensor Q, these sums give

$$\sum_v (v^{\mathsf{T}} Q v)\, vv^{\mathsf{T}} = \frac{8}{5} Q + \frac{4}{5} (\operatorname{tr} Q) I.$$

With $\operatorname{tr} Q = 0$, and using $\sum_v v^{\mathsf{T}} Q v = 4 \operatorname{tr} Q = 0$, we obtain $\Gamma_0 \Gamma_0^* Q = (8/5)Q$. The two identities prove the normalized isometry and its surjectivity. □

The gravitational section in 121 determines the operator $G_a P_5$. Its spectrum has G_a with multiplicity five and zero with multiplicity seven. Orientation is preserved in the carrier and its transport; G_a fixes the common coupling.

124.3 Transverse projection and helicities

For $n \in S^2$, set $\Pi = I - nn^{\mathsf{T}}$ and

$$\Lambda_n Q = \Pi Q \Pi - \frac{1}{2}\operatorname{tr}(\Pi Q \Pi)\Pi.$$

Proposition 124.4 (Rank-two transverse reduction)**.** *Λ_n is the orthogonal projector onto $\{Q \in \operatorname{Sym}_0^2(\mathbb{R}^3) : Qn = 0\}$, of dimension two.*

Proof. $\Pi n = 0$ gives transversality and $\operatorname{tr}\Pi = 2$ gives zero trace. A transverse traceless tensor remains fixed. The formula is self-adjoint for the Frobenius inner product. In a frame (e_1, e_2, n), the image has orthonormal basis

$$E_+ = \frac{e_1 e_1^{\mathsf{T}} - e_2 e_2^{\mathsf{T}}}{\sqrt{2}}, \qquad E_\times = \frac{e_1 e_2^{\mathsf{T}} + e_2 e_1^{\mathsf{T}}}{\sqrt{2}}.$$

□

The other three vectors of an orthonormal basis of the carrier are

$$E_1 = \frac{e_1 n^{\mathsf{T}} + n e_1^{\mathsf{T}}}{\sqrt{2}}, \quad E_2 = \frac{e_2 n^{\mathsf{T}} + n e_2^{\mathsf{T}}}{\sqrt{2}}, \quad E_0 = \frac{2nn^{\mathsf{T}} - e_1 e_1^{\mathsf{T}} - e_2 e_2^{\mathsf{T}}}{\sqrt{6}}.$$

When the transverse frame is rotated by an angle θ, the pairs $(E_+, E_\times)$ and (E_1, E_2) rotate by 2θ and θ, respectively, while E_0 remains fixed. The complexification thus has weights $2, -2, 1, -1, 0$. Λ_n preserves the first two and annihilates the rest. The chain is established

$$\mathbb{R}^{12} \xrightarrow{P_5} \operatorname{im} P_5 \xrightarrow{\Gamma_5} \operatorname{Sym}_0^2(\mathbb{R}^3) \xrightarrow{\Lambda_n} \mathcal{H}_{\mathrm{TT}}(n), \qquad 12 \longrightarrow 5 \longrightarrow 5 \longrightarrow 2.$$

The five-component representation and its reduction are algebraic results preceding the choice of field equations. In the massless gravitational realization, the gauge constraints select the transverse image. This distinction preserves both the five structural coordinates and the two radiative helicities.

Chapter 125

Phase return and representation of accumulated memory

The chronology of enriched routes distinguishes position return, orientation return and phase return. The observable publication identifies certain arrival states, while the cumulative register preserves the sequence traversed. The geometric representation must transport both data jointly.

125.1 Inversion cocycle and change of sheet

Let F_{obs} be the chronological publication of TPK transport. Its canonical blocks have the hierarchy

27	positional return with orientation reversal,
54	positional and orientational return,
108	complete return of the observable phase.

The register counts orientation inversions $g(a)$ and effective changes of sheet $s(a)$. For a route,

$$c(\gamma) = \sum_{a \subset \gamma} (g(a), s(a)), \qquad c(\gamma_2\gamma_1) = c(\gamma_2) + c(\gamma_1). \tag{125.1}$$

The canonical computation of the block of length 27 publishes $c_{27} = (1, 18)$. Composition preserves those increments in the four blocks that complete the observable period. On forward-oriented routes the register is cumulative; its extension to the route groupoid uses $\mathbb{Z}^2$ and $c(\bar{\gamma}) = -c(\gamma)$.

Proposition 125.1 (Memory of the complete observable turn)**.** *Writing $\mathcal{K}_{27} = F_{\mathrm{obs}}^{27}$, the lift to the register, for $x \in O_{108}$ and $z \in \mathbb{Z}^2$, satisfies*

$$\widetilde{\mathcal{K}}_{27}(x, z) = (\mathcal{K}_{27}x, z + (1, 18)), \qquad \widetilde{\mathcal{K}}_{27}^{4}(x, z) = (x, z + (4, 72)).$$

After N complete turns, the increment is $N(4, 72)$.

Proof. The observable projection of $\mathcal{K}_{27}$ has order four. The preceding canonical computation fixes $c_{27} = (1, 18)$ as a constant over that observable orbit. The additivity of (125.1) sums the four increments and gives $c_{108} = 4(1, 18) = (4, 72)$. Repetition applies the same concatenation law and produces Nc_{108}. The return of the first coordinate preserves the increment of the second. □

125.2 Affine representation of the cocycle

The discrete realization uses an element R_4 of order four in $\mathrm{Spin}^+(1, 3)$, a unit temporal vector u fixed by its vector action and a unit of length $\ell_0 > 0$. The scale ℓ_0 retains its provenance in the dimensional section. We define

$$\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma) = \left(R_4^{c_g(\gamma)},\ \ell_0 c_s(\gamma) u\right) \in \mathrm{Spin}^+(1, 3) \ltimes \mathbb{R}^{1,3}. \tag{125.2}$$

The power acts on the vector through the associated vector representation of the spin group.

Theorem 125.2 (Composition and translational amplitude of return)**.** *The map $\rho_{\mathrm{C}}^{\mathrm{disc}}$ preserves the identity, concatenation and inversion. At complete returns,*

$$\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_{108}) = (I, 72\ell_0 u), \qquad \rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_{108}^{N}) = (I, 72N\ell_0 u). \tag{125.3}$$

Inversion changes the sign of the translation.

Proof. The semidirect product is $(R,a)(S,b)=(RS,a+Rb)$. Since $R_4u=u$,

$$\begin{aligned}\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_2)\rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_1) &= \left(R_4^{c_g(\gamma_2)+c_g(\gamma_1)},\, \ell_0(c_s(\gamma_2)+c_s(\gamma_1))u\right)\\ &= \rho_{\mathrm{C}}^{\mathrm{disc}}(\gamma_2\gamma_1).\end{aligned}$$

The identity cocycle is zero and that of the inverse is its opposite. For the complete turn, substitute $c_{108}=(4,72)$ and $R_4^4=I$. Concatenating N turns keeps the rotational part equal to the identity and sums their translations. □

The amplitude $72\ell_0$ is a length associated with a finite route. A realization through a co-tetrad and a connection transports it to its affine holonomy. The local contribution to the gravitational equations is subsequently obtained through the curvature of that connection, the variation of the matter action and the normalization of its density. The representation thus preserves the exact cumulative content that must enter that realization.

Chapter 126

Boundary soldering and energy response

The cubic realization of transport preserves six oriented incidences. Its geometric quotient, the Lorentzian form and the energy response are obtained from the same incidence matrix. Transport proceeds from the enriched APP–TRIT–TPK route: the face, sign, carry and memory accompany each edge. This section develops the linear operation acting on that boundary publication.

126.1 Cubic incidence and the rank-four quotient

Let $H_F = \mathbb{R}^F$, with $F = \{(i,\epsilon) : 1 \le i \le 3,\ \epsilon \in \{-1,1\}\}$, its Euclidean inner product, and its basis $e_{i,\epsilon}$. Face opposition is the orthogonal involution $Re_{i,\epsilon} = e_{i,-\epsilon}$. Define

$$u = \sum_{i,\epsilon} e_{i,\epsilon}, \qquad P_u = \frac{uu^{\mathsf{T}}}{6}, \qquad P_- = \frac{I-R}{2}, \qquad P_0 = \frac{I+R}{2} - P_u, \qquad P_\square = P_u + P_-.$$

The vertices are $V = \{-1,1\}^3$. The incidence matrix $M : \mathbb{R}^V \longrightarrow H_F$ is determined by

$$M_{(i,\epsilon),\nu} = \mathbf{1}_{\{\nu_i = \epsilon\}}.$$

Proposition 126.1 (Incidence decomposition)**.** *We have*

$$MM^{\mathsf{T}} = 12P_u + 4P_-, \qquad \ker M^{\mathsf{T}} = P_0 H_F.$$

Consequently, the quotient $Q = H_F/P_0H_F$, orthogonally identified with $P_\square H_F$, has dimension four.

Proof. A face contains four vertices. Two opposite faces have no common vertices and two faces on different axes have two common vertices. Writing $\mathbf{J} = uu^{\mathsf{T}}$, we obtain

$$MM^{\mathsf{T}} = 4I + 2(\mathbf{J} - I - R) = 2(I-R) + 2\mathbf{J} = 4P_- + 12P_u.$$

The three projectors P_u, P_-, P_0 are orthogonal and sum to I; their ranks are one, three and two, respectively. The identity $\|M^{\mathsf{T}}x\|^2 = \langle x, MM^{\mathsf{T}}x\rangle$ identifies the stated kernel. The complementary rank is four. □

On Q, the form

$$B = P_- - P_u$$

has signature $(3,1)$. The time orientation is fixed by u. With

$$t = \frac{u}{\sqrt{6}}, \qquad d_i = \frac{e_{i,+} - e_{i,-}}{\sqrt{2}}, \qquad W = \sqrt{6}P_u + \sqrt{2}P_-,$$

we have $We_{i,\epsilon} = t + \epsilon d_i$. These six vectors are null for B, whereas

$$3t + \nu_1 d_1 + \nu_2 d_2 + \nu_3 d_3$$

has squared norm -6. The same incidence satisfies $MM^{\mathsf{T}} = 2W^*W$ on Q. The proper rotations of the cube commute with the projectors, preserve both forms and fix t. These identities determine the metric of the cubic publication.

126.2 Cellular soldering and memory transport

For each oriented edge a, the route publication provides the face $\lambda(\underline{a})$, the sign $\sigma(a)$, the source and target fibres and the transport $U(a) : Q_{s(a)} \longrightarrow Q_{t(a)}$. Inversion satisfies $U(\bar{a}) = U(a)^{-1}$. The boundary half-edges retain their formal inversion and their incidence at the boundary. With $\ell_m = \ell_0\, 9^{-m} > 0$, the soldering is

$$\beta_m(a) = \ell_m \sigma_m(a) n^{s(a)}_{\lambda(\underline{a})}, \qquad n_{i,\epsilon} = t + \epsilon d_i. \tag{126.1}$$

The face and sign are transported together with the route memory. The inversion convention requires

$$\beta_m(\bar{a}) = -U_m(a)\beta_m(a). \tag{126.2}$$

Applying this relation twice recovers $\beta_m(a)$; the carry register separately preserves the ordered composition.

Proposition 126.2 (Naturality of soldering under refinement)**.** *Let $a = a_1 \cdots a_9$ be a compatible refinement of an edge. Suppose that the nine child incidences, transported to the parent fibre, preserve the face and sign, and that $\ell_{m+1} = \ell_m/9$. Let $\mathcal{U}_j : Q_{m,s(a)} \longrightarrow Q_{m+1,s(a_j)}$ be the accumulated transport that includes the refinement identification and the path to the origin of child j. Then*

$$\beta_m(a) = \sum_{j=1}^{9} \mathcal{U}_j^{-1} \beta_{m+1}(a_j).$$

The equality is compatible with inversion and with concatenation of the memory register.

Proof. Each summand, expressed in the fibre at the origin of the parent, is $(\ell_m/9)\sigma_m(a)n_{\lambda(\underline{a})}$. The sum of nine terms reproduces the parent soldering. Inversion reverses the order of the transports and replaces each one with its inverse; the relation (126.2) provides the overall sign. Memory is concatenated in that same order, so refinement preserves both the vector publication and its enriched register. □

126.3 Positive response, uniqueness, and minimal extension

The linear screen soldering is $\bar{\beta}_m = \ell_m W|_Q$. It is invertible, with

$$\bar{\beta}_m^{-1} = \ell_m^{-1} \left(\frac{1}{\sqrt{6}} P_u + \frac{1}{\sqrt{2}} P_- \right).$$

The eigenvalue $5/9$ of the antisymmetric sector of the normalized block $B_c/9$, constructed in Section 4.3.4, determines the compression coefficient $q_{\mathrm{vis}} = 5/9$. We reserve this notation to distinguish it from the Euclidean quotient $q_9(n)$ of the arithmetic kernel. Its realization on the screen uses an isometry $V_9 : Q \longrightarrow \mathcal{K}$. The visible contribution $C_9 = q_{\mathrm{vis}} V_9$ and its complement $D_9 = \sqrt{1 - q_{\mathrm{vis}}^2}\, V_9$ satisfy

$$C_9^* C_9 + D_9^* D_9 = I_Q, \qquad D_9^* D_9 = \frac{56}{81} I_Q.$$

The coefficient 56 is $9^2 - 5^2$; it belongs to the quadratic defect of that compression reader.

Theorem 126.3 (Energy response determined by soldering)**.** *There is a unique operator $\Lambda_m : Q \longrightarrow Q$ such that $\bar{\beta}_m^* \Lambda_m \bar{\beta}_m = D_9^* D_9$. It is positive definite and*

$$\Lambda_m = \frac{28}{243 \ell_m^2} (P_u + 3P_-) = \frac{112}{81 \ell_m^2} (M M^{\mathsf{T}}|_Q)^{-1}. \tag{126.3}$$

On H_F, the corresponding identity has right-hand side $(56/81)P_{\square}$.

Proof. Congruence by the invertible soldering necessarily determines

$$\Lambda_m = \bar{\beta}_m^{-*} \frac{56}{81} I_Q \bar{\beta}_m^{-1}.$$

Substituting the inverse, the coefficient of P_u is $56/(486\ell_m^2) = 28/(243\ell_m^2)$; that of P_- is $28/(81\ell_m^2)$. Both are positive. By Proposition 126.1, $(MM^{\mathsf{T}}|_Q)^{-1} = P_u/12 + P_-/4$, which gives the second expression. Extension to six faces annihilates $P_0 H_F$; the identity therefore extends through $P_\square$. □

Proposition 126.4 (Characterization by minimal extension)**.** *For every $b \in Q$,*

$$\langle b, \Lambda_m b \rangle = \frac{112}{81\ell_m^2} \min_{Mx=b} \|x\|^2.$$

The minimizing extension is unique: $x_{\min} = M^{\mathsf{T}}(MM^{\mathsf{T}}|_Q)^{-1} b$.

Proof. The stated vector satisfies $Mx_{\min} = b$ and belongs to $\operatorname{im} M^{\mathsf{T}} = (\ker M)^\perp$. Every other solution is $x_{\min} + z$, with $z \in \ker M$; by orthogonality, its squared norm is $\|x_{\min}\|^2 + \|z\|^2$. Moreover,

$$\|x_{\min}\|^2 = \langle b, (MM^{\mathsf{T}}|_Q)^{-1} b \rangle.$$

The formula for Λ_m completes the proof. □

The operator $MM^{\mathsf{T}}|_Q$ realizes the Dirichlet–to–Neumann response of this incidence formulation; its inverse is the Neumann–to–Dirichlet response. The preceding energy corresponds to the latter, with the normalization of the soldering and the compression defect. The Schur complement of $\left(\begin{smallmatrix} I & M^{\mathsf{T}} \\ M & 0 \end{smallmatrix}\right)$ is $-MM^{\mathsf{T}}$, which offers the same variational elimination.

126.4 Preservation of density by refinement

Consider the sector of cubic transports, orthogonal for the energy inner product and compatible with the form B. We fix a common label space Q^{lab}, identified isometrically with the parent's screen. The solderings have types $\bar{\beta}_m : Q^{\mathrm{lab}} \longrightarrow Q_m$ and $\bar{\beta}_{m+1,j} : Q^{\mathrm{lab}} \longrightarrow Q_{m+1,j}$; the operators Λ act in their respective target fibers. If $U_j : Q_j \longrightarrow Q$ transports the child fiber to the parent's, then

$$\bar{\beta}_{m+1,j} = \frac{1}{9} U_j^{-1} \bar{\beta}_m, \qquad \Lambda_{m+1,j} = 81 U_j^{-1} \Lambda_m U_j.$$

The factor 81 comes from $\ell_{m+1}^{-2} = 81\ell_m^{-2}$.

Theorem 126.5 (Invariance of the normalized density)**.** *Each child preserves the parent's quadratic response:*

$$\bar{\beta}_{m+1,j}^{*} \Lambda_{m+1,j} \bar{\beta}_{m+1,j} = \frac{56}{81} I_{Q^{\mathrm{lab}}}.$$

For r refinements, the joint response, expressed in the parent's fibers, is

$$E_{m+r} = \frac{56}{81} I_{Q^{\mathrm{lab}}} \otimes I_{9^r}.$$

The expectation with normalized trace $\tau_{9^r}(A) = 9^{-r} \operatorname{Tr}(A)$ satisfies

$$(\mathrm{id} \otimes \tau_{9^r})(E_{m+r}) = \frac{56}{81} I_{Q^{\mathrm{lab}}}$$

at every finite depth.

Proof. The congruence of a child contains the factors $1/9, 81, 1/9$; their product is one. Orthogonality cancels the intermediate transports. The direct sum of nine equal responses identifies the first refinement with $(56/81)I_Q \otimes I_9$. Iteration produces the tensor with I_{9^r}. Finally $\tau_{9^r}(I_{9^r}) = 1$. The expectations compose through the identity $\tau_{9^{r+s}} = \tau_{9^r} \otimes \tau_{9^s}$. □

The unital inclusions $A \mapsto A \otimes I_9$ are compatible with that family of traces. In the inductive limit of the matrix algebras, the response is the central element $(56/81)I_Q \otimes \mathbf{1}$. The unnormalized extensive sum produces, in contrast, $9^r(56/81)I_Q$. This distinction specifies which quantity is preserved: the quadratic screen density.

126.5 Variational realization of the response

The material realization uses an isometry $J_{\mathrm{scr}} : Q \longrightarrow \mathcal{H}_\partial$ compatible with the spin action and with refinement. The boundary co-tetrad is $e = J_{\mathrm{scr}}\bar{\beta}_m$. Let Λ_{CE} be the boundary response obtained by second variation of the material and geometric action, with its boundary conditions and elimination of the interior variables. The compatibility defect is

$$\mathcal{R}_{\mathrm{grav}} = \bar{\beta}_m^*(J_{\mathrm{scr}}^*\Lambda_{\mathrm{CE}}J_{\mathrm{scr}} - \Lambda_m)\bar{\beta}_m = e^*\Lambda_{\mathrm{CE}}e - D_9^*D_9. \tag{126.4}$$

By the invertibility of $\bar{\beta}_m$,

$$\mathcal{R}_{\mathrm{grav}} = 0 \quad \Longleftrightarrow \quad J_{\mathrm{scr}}^*\Lambda_{\mathrm{CE}}J_{\mathrm{scr}} = \Lambda_m.$$

The isometry J_{scr} transports the boundary states; the variational spin current is a degree-three form with values in bivectors. Each enters with its own domain in the realization of the response.

The construction determines a positive energy form, its minimizing interior extension and the density preserved by refinement. The amplitude of a concrete material contribution is obtained by evaluating the current of that realization in its variational quadratic form. This differentiates the operator fixing the response, the current on which it acts and the gravitational observable published by the evaluation.

Chapter 127

Geometric realization of transport and soldering

The discrete representation of return and screen soldering determine two aspects of transport: the accumulation of memory along routes and the comparison of their fibres. Their geometric realization preserves both operations. This section fixes the target space, the dimensional normalization of the connection and the identities that its variation will use.

127.1 Screen, co-tetrad and connection

The screen quotient $Q_{\square}$, its Lorentzian form, and the soldering β_m come from 126. In each cell, a screen realization $J_{\mathrm{scr},m}$ transports the form and soldering to the corresponding geometric fibre. Its cellular image is

$$e_m = J_{\mathrm{scr},m}\beta_m.$$

The scale ℓ_m remains within the soldering. Changes of frame act jointly on $J_{\mathrm{scr},m}$, the connection and the transport of the cells.

A smooth realization of those data on an oriented manifold $\mathcal{U}$ of dimension four consists of a nondegenerate co-tetrad $e = (e^I)$, a Lorentz connection $\omega = (\omega^I{}_J)$ and a realization of routes $\mathfrak{R}_{\mathrm{C}}$ satisfying

$$\mathrm{Hol}_{\mathcal{A}_{\ell_0}}(\mathfrak{R}_{\mathrm{C}}(\gamma)) = \rho_{\mathrm{C}}(\mathrm{Hol}_{\mathrm{TPK}}(\gamma)). \tag{127.1}$$

In this matrix equality, the representation is normalized by

$$N_{\ell_0}(R,a) = (\Lambda(R), a/\ell_0), \qquad \rho_{\mathrm{C}} = N_{\ell_0} \circ \rho_{\mathrm{C}}^{\mathrm{disc}},$$

where $\Lambda : \mathrm{Spin}^+(3,1) \to SO^+(3,1)$ is the vector representation. The map N_{ℓ_0} preserves the semidirect product: dividing the translation $a + \Lambda(R)b$ by ℓ_0 is equivalent to composing the two normalized translations. The complete turn therefore publishes $72u$ in the normalized connection and $72\ell_0 u$ in the dimensional chart. The spin lift separately preserves $R_4^{c_g(\gamma)}$; the vector representation has the central kernel $\{1,-1\}$, so that lift is retained when using spinor fields. Identity, concatenation, inversion and subdivision are maintained in the corresponding realization. The screen pairing and the holonomy condition specify the data of the realization used in the field equations.

With $\eta = \mathrm{diag}(-1,1,1,1)$, define $g = \eta_{IJ} e^I \otimes e^J$. For a positive elementary scale ℓ_0, constant in the chart, the affine connection is

$$\mathcal{A}_{\ell_0} = \begin{pmatrix} \omega & \ell_0^{-1} e \\ 0 & 0 \end{pmatrix}. \tag{127.2}$$

The co-tetrad has the dimension of length; the factor ℓ_0^{-1} makes its translational component dimensionally homogeneous.

Proposition 127.1 (Curvature and torsion of the realized connection)**.** *The curvature of* (127.2) *is*

$$\mathcal{F}_{\ell_0} = \mathrm{d}\mathcal{A}_{\ell_0} + \mathcal{A}_{\ell_0} \wedge \mathcal{A}_{\ell_0} = \begin{pmatrix} F_\omega & \ell_0^{-1} T \\ 0 & 0 \end{pmatrix}, \quad F_\omega = \mathrm{d}\omega + \omega \wedge \omega, \quad T = \mathrm{d}e + \omega \wedge e. \tag{127.3}$$

Proof. The product of matrices of forms produces $\omega \wedge \omega$ in the diagonal block and $\ell_0^{-1}\omega \wedge e$ in the translational block. The exterior derivative contributes $\mathrm{d}\omega$ and $\ell_0^{-1}\mathrm{d}e$, since ℓ_0 is constant. Their sum proves the equality. □

Curvature describes the frame-transport defect and torsion that of the co-tetrad. Both belong to the same realized connection; their components retain distinct tensorial domains.

Proposition 127.2 (Covariance and Bianchi identities)**.** *Under a change of frame* $g_L : \mathcal{U} \to SO^+(3,1)$*,*

$$e' = g_L e, \qquad \omega' = g_L \omega g_L^{-1} - \mathrm{d}g_L\, g_L^{-1},$$

we have $T' = g_L T$ *and* $F_{\omega'} = g_L F_\omega g_L^{-1}$*. Furthermore,*

$$D_\omega T = F_\omega \wedge e, \qquad D_\omega F_\omega = 0. \tag{127.4}$$

Proof. In $\mathrm{d}(g_L e) + \omega' \wedge g_L e$, the terms containing $\mathrm{d}g_L \wedge e$ cancel. The same substitution into the curvature gives its transformation by conjugation. For the first Bianchi identity, expand

$$D_\omega(\mathrm{d}e + \omega \wedge e) = (\mathrm{d}\omega + \omega \wedge \omega) \wedge e.$$

In $\mathrm{d}F_\omega + \omega \wedge F_\omega - F_\omega \wedge \omega$, the terms containing $\mathrm{d}\omega$ and the cubic products cancel in pairs. This proves the second identity. □

127.2 The three quadratic realizations of curvature

The TRIT sheet $\tau \in \{+1, 0, -1\}$ is represented in the Cartan algebra by generators $J_{ab} = -J_{ba}$ and $P_a^{(\tau)}$, with relations

$$\begin{aligned} [J_{ab}, J_{cd}] &= \eta_{bc} J_{ad} - \eta_{ac} J_{bd} - \eta_{bd} J_{ac} + \eta_{ad} J_{bc}, \\ [J_{ab}, P_c^{(\tau)}] &= \eta_{bc} P_a^{(\tau)} - \eta_{ac} P_b^{(\tau)}, \\ [P_a^{(\tau)}, P_b^{(\tau)}] &= -\tau J_{ab}. \end{aligned}$$

For constant $\ell > 0$, write

$$\mathcal{A}_\tau^{\mathrm{Cart}} = \tfrac{1}{2}\omega^{ab} J_{ab} + \ell^{-1} e^a P_a^{(\tau)}.$$

Proposition 127.3 (Curvature of the tritified realization)**.** *With the preceding relations,*

$$\mathcal{F}_\tau^{\mathrm{Cart}} = \tfrac{1}{2}(F_\omega^{ab} - \tau \ell^{-2} e^a \wedge e^b) J_{ab} + \ell^{-1} T^a P_a^{(\tau)}. \tag{127.5}$$

Proof. Expand $\mathrm{d}\mathcal{A} + \frac{1}{2}[\mathcal{A}, \mathcal{A}]$. The bracket of the Lorentz sector produces F_ω, the mixed bracket produces $D_\omega e$, and the translational bracket contributes $-\frac{1}{2}\tau \ell^{-2} e^a \wedge e^b J_{ab}$. □

On the central sheet the affine connection is recovered; the extreme sheets preserve the two signs of the constant-curvature term. The representation thus retains the tritic regime of the route before applying the variational equations. The holonomy equality (127.1) corresponds to the central sheet. On each extreme sheet, the representation $\rho_{\mathrm{C},\tau}$ in its Cartan group and its respective holonomy condition are used. The current and the metric tensor use the fields and representation of the same realization.

Chapter 128

Variational response of memory energy

The transport of a route identifies the fibers at its endpoints and preserves the orientation and memory entering that identification. The associated energy compares the final field with the transported initial field. Its differential quantifies how that comparison responds to a variation of transport. This section obtains the response directly from the discrete functional and develops its composition, covariance and inversion.

128.1 Discrete functional and space of variations

At a finite level m, let $\mathcal{V}_m$ be the vertices of the route publication and $\mathcal{E}_m^+$ an orientation of each edge. Each vertex carries a finite-dimensional Hermitian fiber H_v. We define

$$\mathcal{H}_m = \bigoplus_{v\in\mathcal{V}_m} H_v, \qquad \mathcal{K}_m = \bigoplus_{a\in\mathcal{E}_m^+} H_{t(a)}.$$

The Hermitian inner product is antilinear in the first variable. For unitary $U_a : H_{s(a)} \to H_{t(a)}$, the difference operator is

$$(D_m f)_a = f_{t(a)} - U_a f_{s(a)}, \qquad E_m(f,U) = \langle D_m f, \mathsf{W}_m D_m f\rangle, \qquad \mathsf{W}_m = \mathsf{W}_m^* > 0. \tag{128.1}$$

This is the energy of the term $D_m^* \mathsf{W}_m D_m$ in the discrete generator. The weight W_m may couple different edges. The soldering operator W of Section 126 and this energy weight have different domains and roles.

The transport U is a publication of the enriched state. The field f and the weight W_m remain explicit as arguments of the functional; each material application preserves the rules that determine them. Differentiable families of the unitary realizations of transport are considered,

$$\dot U_a = A_a U_a, \qquad A_a^* = -A_a.$$

The variations used physically form the subspace preserving the conditions of the corresponding realization. The differential is calculated first in the unitary tangent space and then restricted to that subspace.

Proposition 128.1 (Transport-response covector)**.** *Write*

$$v_a = U_a f_{s(a)}, \qquad r = D_m f, \qquad q = \mathsf{W}_m r.$$

At fixed field and weight, the energy differential is

$$\mathfrak{j}_m(A) = -2\,\mathrm{Re}\sum_{a\in\mathcal{E}_m^+} \langle q_a, A_a v_a\rangle. \tag{128.2}$$

Its representative for the real Hilbert–Schmidt inner product is

$$\mathcal{J}_a = v_a q_a^* - q_a v_a^*, \qquad \mathcal{J}_a^* = -\mathcal{J}_a, \qquad \mathfrak{j}_m(A) = \sum_a \mathrm{Re}\,\mathrm{Tr}(\mathcal{J}_a^* A_a). \tag{128.3}$$

For simultaneous variations of field, transport and weight we have

$$\dot E_m = 2\,\mathrm{Re}\sum_a \left\langle q_a, \dot f_{t(a)} - U_a \dot f_{s(a)} - A_a v_a\right\rangle + \langle r, \dot{\mathsf{W}}_m r\rangle. \tag{128.4}$$

Proof. The first-order update, written in the algebra of dual numbers with $\epsilon^2 = 0$, gives

$$r + \epsilon \dot r, \qquad \dot r_a = \dot f_{t(a)} - U_a \dot f_{s(a)} - A_a v_a.$$

The coefficient of ϵ in the energy is

$$\langle \dot r, \mathsf{W}_m r\rangle + \langle r, \mathsf{W}_m \dot r\rangle + \langle r, \dot{\mathsf{W}}_m r\rangle.$$

Self-adjointness of the weight yields (128.4). Fixing field and weight leaves (128.2). Moreover,

$$\operatorname{Tr}(\mathcal{J}_a^{*} A_a) = v_a^{*} A_a q_a - q_a^{*} A_a v_a.$$

Anti-Hermiticity of A_a implies $v_a^{*} A_a q_a = -\overline{q_a^{*} A_a v_a}$, giving the stated representative. □

The response is thus evaluated: on each edge the field is transported, its difference from the final field is calculated and the energy weight is applied. The antisymmetrized product of those two vectors determines $\mathcal{J}_a \in \mathfrak{u}(H_{t(a)})$. This operator represents, through the real Hilbert–Schmidt inner product, the response covector on the unitary variations of the target fiber.

128.2 Composition and transport of the response

Proposition 128.2 (Composition rule along a route)**.** *For a route formed by N transports, let $U_\gamma = U_N \cdots U_1$ and $B_k = U_N \cdots U_{k+1}$, with $B_N = I$. If $\dot U_k = A_k U_k$, then*

$$\dot U_\gamma U_\gamma^{-1} = \sum_{k=1}^{N} B_k A_k B_k^{-1}. \tag{128.5}$$

Thus a representative $\mathcal{J}_\gamma$ in the final fibre induces in factor k the contribution $B_k^{*}\mathcal{J}_\gamma B_k$. When a factor belongs to several routes in the functional, its contributions are summed with the incidences used by that functional.

Proof. The product rule gives

$$\dot U_\gamma = \sum_{k} U_N \cdots U_{k+1} A_k U_k \cdots U_1.$$

Multiplication by U_γ^{-1} eliminates the factors to the right of A_k and yields (128.5). By unitarity and cyclicity of the trace,

$$\operatorname{Re}\operatorname{Tr}(\mathcal{J}_\gamma^{*} B_k A_k B_k^{-1}) = \operatorname{Re}\operatorname{Tr}((B_k^{*}\mathcal{J}_\gamma B_k)^{*} A_k).$$

Linearity of the differential proves the sum of contributions. □

This formula preserves the position of each operation within the route. Transport following an incidence determines how its contribution is represented in the final fiber; the local response is recovered by transporting it to the fiber of that incidence.

Proposition 128.3 (Covariance under unitary frame changes)**.** *Let g_v be unitary frame changes independent of the variation parameter, and $G_{\mathcal{E}} = \bigoplus_a g_{t(a)}$. Under the transformations*

$$f_v' = g_v f_v, \quad U_a' = g_{t(a)} U_a g_{s(a)}^{-1}, \quad \mathsf{W}_m' = G_{\mathcal{E}} \mathsf{W}_m G_{\mathcal{E}}^{-1}, \quad A_a' = g_{t(a)} A_a g_{t(a)}^{-1},$$

we have

$$\mathcal{J}_a' = g_{t(a)} \mathcal{J}_a g_{t(a)}^{-1}, \qquad \mathfrak{j}_m'(A') = \mathfrak{j}_m(A). \tag{128.6}$$

Proof. We obtain $r' = G_{\mathcal{E}} r$, $q' = G_{\mathcal{E}} q$ and $v_a' = g_{t(a)} v_a$. Substitution into (128.3) proves the first identity. Invariance of the Hermitian inner product, or cyclicity of the trace, proves the second. □

When frame changes depend on the parameter, their derivatives also act on the field and weight. The corresponding covariant expression is then the complete differential (128.4).

128.3 Oriented inversion and transport of the weight

Inversion can be seen explicitly in the case of a local energy per edge, $\mathsf{W}_m = \bigoplus_a W_a$. For $a : s \to t$, its inverse carries

$$U_{\bar{a}} = U_a^{-1}, \qquad r_{\bar{a}} = -U_a^{-1} r_a, \qquad W_{\bar{a}} = U_a^{-1} W_a U_a.$$

These rules preserve the edge energy exactly.

Proposition 128.4 (Differential compatibility with inversion)**.** *If W_a and the fields at both endpoints remain fixed, then*

$$A_{\bar{a}} = -U_a^{-1} A_a U_a, \qquad \dot{W}_{\bar{a}} = U_a^{-1} [W_a, A_a] U_a.$$

The complete variation of the reversed edge's energy coincides with the original. In particular, if $[W_a, A_a] = 0$, the partial transport covector satisfies

$$\mathfrak{j}_{\bar{a}}(U_a^{-1} A_a U_a) = -\mathfrak{j}_a(A_a). \tag{128.7}$$

Proof. The derivative of U_a^{-1} is $-U_a^{-1} A_a$, from which the two formulas result. Writing $v_a = U_a f_s$ and $f_t = r_a + v_a$, the part due exclusively to reversed transport equals

$$\mathfrak{j}_{\bar{a}}(A_{\bar{a}}) = -2 \operatorname{Re}\langle r_a, W_a A_a f_t \rangle = \mathfrak{j}_a(A_a) - 2 \operatorname{Re}\langle r_a, W_a A_a r_a \rangle.$$

Variation of the weight contributes

$$\langle r_{\bar{a}}, \dot{W}_{\bar{a}} r_{\bar{a}} \rangle = \langle r_a, [W_a, A_a] r_a \rangle = 2 \operatorname{Re}\langle r_a, W_a A_a r_a \rangle.$$

The two terms compensate each other. If the commutator is zero, that contribution disappears; linearity in A_a gives (128.7). □

Energy is summed over one orientation per edge. A sum over both orientations carries the factor $1/2$. This convention preserves the same functional and hence the same response.

128.4 Compatibility with refinement

Theorem 128.5 (Transport of the differential by refinement)**.** *Let $J_m : \mathcal{H}_m \to \mathcal{H}_{m+1}$ and $K_m : \mathcal{K}_m \to \mathcal{K}_{m+1}$ be fixed maps. For a family C^1, suppose the refinement identities*

$$D_{m+1}(t) J_m = K_m D_m(t), \qquad K_m^* \mathsf{W}_{m+1}(t) K_m = \mathsf{W}_m(t). \tag{128.8}$$

Then $E_{m+1}(J_m f) = E_m(f)$. The transport covectors agree on variations linked by the first identity. When the weights vary, their contributions to the differential also agree.

Proof. By substitution,

$$\langle D_{m+1} J_m f, \mathsf{W}_{m+1} D_{m+1} J_m f \rangle = \langle D_m f, K_m^* \mathsf{W}_{m+1} K_m D_m f \rangle = E_m(f).$$

Differentiating the first identity of (128.8) gives $\dot{D}_{m+1} J_m = K_m \dot{D}_m$. Consequently,

$$2 \operatorname{Re}\langle D_{m+1} J_m f, \mathsf{W}_{m+1} \dot{D}_{m+1} J_m f \rangle = 2 \operatorname{Re}\langle D_m f, \mathsf{W}_m \dot{D}_m f \rangle.$$

The derivative of the second identity is $K_m^* \dot{\mathsf{W}}_{m+1} K_m = \dot{\mathsf{W}}_m$, proving agreement of the weight-variation terms. □

The statement uses compatibility of the transport family. If J_m depends on the parameter, the chain rule also preserves the term

$$2\,\mathrm{Re}\langle D_{m+1}J_m f, \mathsf{W}_{m+1}D_{m+1}\dot{J}_m f\rangle.$$

The resulting naturality jointly transports the functional, its incidences and its differential.

128.5 Material realization of the response

The result of this section is the explicit covector $\mathfrak{j}_m$, with its representative $\mathcal{J}_a$. The material realization composes that covector with the variation of transport induced by the connection and with the normalization of the action. If $\mathcal{L}_m$ denotes that differential map, the covector on connection variations is $\mathcal{L}_m^{*}\mathfrak{j}_m$. The composition rule (128.5) evaluates its part corresponding to finite products of transports.

In section 131, the current σ_{IJ} is a three-form defined by variation of the material action. Identification between the two realizations preserves the field, representation, weight, dimensional normalization and passage to the limit used by that action. These data allow evaluation of the contraction entering the torsional density. The present development provides the discrete differential composed in that calculation.

Chapter 129

Realization of the current in connection coordinates

Memory energy assigns each route a response preserving its incidences and the order of transports. Inserting it into a geometric action requires its components with respect to connection variations. This step is obtained by differentiating the transport that already realizes the route: the connection current is the composition of that differential with the energy covector of the preceding section.

The elliptic, parabolic and hyperbolic regimes of TRIT require preserving the complete differential before restricting it to a particular representation. Unitary rotations constitute a domain of that construction. The following extension also preserves the dilation and transport directions admitted by the geometric realization used.

129.1 Differential on invertible transports

The fibers, fields and weight of (128.1) are retained. We now consider invertible transports and differentiable families with $\dot U_a = A_a U_a$, where A_a belongs to the tangent space of the chosen representation. We write

$$v_a = U_a f_{s(a)}, \qquad r_a = f_{t(a)} - v_a, \qquad q = \mathsf{W}_m r, \qquad E_m = \langle r, \mathsf{W}_m r\rangle.$$

Proposition 129.1 (Complete representative of the differential)**.** *At fixed field and weight, the following holds*

$$\mathrm{d}_U E_m(A) = -2\,\mathrm{Re}\sum_a \langle q_a, A_a v_a\rangle = \sum_a \mathrm{Re}\,\mathrm{Tr}(\mathcal{G}_a^* A_a), \qquad \mathcal{G}_a = -2 q_a v_a^*. \tag{129.1}$$

The anti-Hermitian and Hermitian projections of this representative are

$$\frac{\mathcal{G}_a - \mathcal{G}_a^*}{2} = v_a q_a^* - q_a v_a^* = \mathcal{J}_a, \qquad \frac{\mathcal{G}_a + \mathcal{G}_a^*}{2} = -(q_a v_a^* + v_a q_a^*). \tag{129.2}$$

Restriction to anti-Hermitian generators recovers exactly (128.3)*.*

Proof. The identity $\dot r_a = -A_a v_a$ and self-adjointness of the weight give $\dot E_m = 2\,\mathrm{Re}\langle q, \dot r\rangle$. Moreover,

$$\mathrm{Tr}(\mathcal{G}_a^* A_a) = -2\,\mathrm{Tr}(v_a q_a^* A_a) = -2 q_a^* A_a v_a.$$

The two projections are obtained by taking the adjoint. For the real Hilbert–Schmidt inner product, the Hermitian and anti-Hermitian subspaces are orthogonal; therefore the Hermitian part has zero contraction with an anti-Hermitian generator. □

The formula allows weights coupling edges: their action is calculated in $q = \mathsf{W}_m r$ before separating the contributions of each incidence. If field or weight vary, the complete expression (128.4) is retained, also valid for the invertible transports considered here.

129.2 Quadratic representations of TRIT

In a two-dimensional matrix realization of $\mathbb{A}_\tau$, one may take

$$J_+ = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad J_- = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad J_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$J_+^2 = -I, \qquad J_-^2 = I, \qquad J_0^2 = 0.$$

Their exponentials realize, respectively, the elliptic, hyperbolic and parabolic regimes. The first is orthogonal for the fixed positive inner product. The second preserves the form $B = \operatorname{diag}(-1, 1)$, since $J_-^{\mathsf{T}} B + B J_- = 0$. The third represents the algebra of dual numbers. These representations fix examples of the quadratic types; their use on a screen of another dimension preserves the corresponding representation map.

A calculation distinguishes the components of the differential. On an edge, take $U = I$, $f_s = (1, 1)^{\mathsf{T}}$, $f_t = 2 f_s$ and $\mathsf{W} = I$. Then $q = v = f_s$, $\mathcal{J} = 0$, whereas

$$\mathrm{d}E(J_-) = -4, \qquad \mathrm{d}E(J_0) = -2. \tag{129.3}$$

The complete representative preserves both responses. The distinction identifies the space of variations before performing the material contraction.

129.3 Composition of routes and oriented inversion

Proposition 129.2 (Contragredient transport of the differential)**.** *For $U_\gamma = U_N \cdots U_1$, let $B_k = U_N \cdots U_{k+1}$ and $B_N = I$. We have*

$$\dot U_\gamma U_\gamma^{-1} = \sum_k B_k A_k B_k^{-1}.$$

The representative $\mathcal{G}_\gamma$ in the final fiber induces at incidence k

$$\mathcal{G}_k = B_k^{*} \mathcal{G}_\gamma (B_k^{-1})^{*}. \tag{129.4}$$

Proof. The product rule, followed by multiplication by U_γ^{-1}, proves the first equality. Cyclicity of the trace gives

$$\operatorname{Re}\operatorname{Tr}(\mathcal{G}_\gamma^{*} B_k A_k B_k^{-1}) = \operatorname{Re}\operatorname{Tr}((B_k^{*} \mathcal{G}_\gamma (B_k^{-1})^{*})^{*} A_k).$$

For unitary B_k, the conjugation of 128.2 is recovered. □

Proposition 129.3 (Inversion by energy congruence)**.** *For a local energy per edge, inversion is represented by*

$$U_{\bar a} = U_a^{-1}, \qquad r_{\bar a} = -U_a^{-1} r_a, \qquad W_{\bar a} = U_a^{*} W_a U_a. \tag{129.5}$$

It preserves energy. If W_a remains fixed,

$$A_{\bar a} = -U_a^{-1} A_a U_a, \qquad \dot W_{\bar a} = U_a^{*}(A_a^{*} W_a + W_a A_a) U_a.$$

The complete variation of the inverted energy coincides with the original.

Proof. Substitution into $r_{\bar a}^{*} W_{\bar a} r_{\bar a}$ produces $r_a^{*} W_a r_a$. The derivative of the inverse is $\dot U_a^{-1} = -U_a^{-1} A_a$; that of the weight is obtained by the product rule. Equality of energies holds for the entire family, so their derivatives coincide, including the term $\langle r_{\bar a}, \dot W_{\bar a} r_{\bar a} \rangle$. □

Congruence preserves positivity for every invertible transport. In the unitary representation it reduces to the inversion rule of 128.4. The sum uses one orientation per edge, or half the sum over both orientations.

129.4 Evaluation of current components

Let $\theta = (\theta_b)$ be a real variation of the connection coordinates of the realization used. The effective differential of transport defines the operators

$$A_a(\theta) = \dot{U}_a U_a^{-1} = \sum_b B_{ab}\theta_b. \tag{129.6}$$

For a finite composition, their contributions are obtained through (129.4). The operators B_{ab} are the derivatives of the connection representation that realizes the route.

Theorem 129.4 (Connection current of the memory action)**.** *For a material term $S_{\mathrm{mem}} = \mathfrak{a}_m E_m$, with $\mathfrak{a}_m$ its normalization on the action line, at fixed field, weight and normalization we have*

$$\delta S_{\mathrm{mem}} = -\frac{1}{2}\sum_b \theta_b s_b, \qquad s_b = 4\mathfrak{a}_m \operatorname{Re}\sum_a \langle q_a, B_{ab} v_a\rangle. \tag{129.7}$$

If the cellular pairing between the connection and current bases is an invertible matrix M, defined by $\delta S_{\mathrm{mem}} = -\frac{1}{2}\theta^{\mathsf{T}} M\sigma$, the current coordinates are

$$\sigma = M^{-1}s. \tag{129.8}$$

Proof. Substitute (129.6) into (129.1), multiply by $\mathfrak{a}_m$, and collect the coefficients of θ_b. The variational convention $-1/2$ fixes the factor 4. Comparing both expressions for every variation θ gives $M\sigma = s$; invertibility of the pairing proves existence and uniqueness of these coordinates. □

The formula explicitly realizes the composition mentioned in 128.5. To evaluate each component, the field is transported, the residual is calculated, the weight is applied and the result is contracted with the connection variation of the same realization. The pairing M preserves the orientation and volume of cells. A diagonal pairing gives the corresponding division by their volumes; other bases preserve the complete matrix.

If the connection also enters f, W_m or $\mathfrak{a}_m$, we apply (128.4) and add $E_m\delta\mathfrak{a}_m$. Contributions from all material terms are summed before composing the Cartan response.

Corollary 129.5 (Compatibility of the current with refinement)**.** *For the compatible families of 128.5, let P_m be the fixed map taking connection variations from level m to level $m+1$. If the memory actions coincide on those refined fields and transports, their covectors satisfy*

$$s_m = P_m^{\mathsf{T}} s_{m+1}, \qquad M_m\sigma_m = P_m^{\mathsf{T}} M_{m+1}\sigma_{m+1}.$$

Proof. Differentiating the equality of actions in the direction $P_m\theta$ gives $-\frac{1}{2}\theta^{\mathsf{T}} s_m = -\frac{1}{2}\theta^{\mathsf{T}} P_m^{\mathsf{T}} s_{m+1}$. The arbitrariness of θ and the cellular representations of both covectors prove the two identities. Composition of these maps iterates the equality to any finite number of refinements. □

129.5 Insertion into the Cartan equation

Section 131 normalizes the material three-form through $\delta_\omega S_m = -\frac{1}{2}\int \delta\omega^{IJ}\wedge\sigma_{IJ}$. The preceding theorem provides its cellular components for the declared memory action and pairing. The Cartan response uses that current together with the co-tetrad, the Barbero–Immirzi parameter and the gravitational coupling of the same realization.

The dependence of matter on contorsion determines how this equation is solved. For affine matter, the current is independent of contorsion, and the linear inversion and quadratic elimination proved in the next section apply. For a holonomy action, the differential is evaluated at the current connection, and that

dependence is retained when solving the equation. For example, an edge with $f_s = f_t = 1$, unit weight, and $U(t) = e^{it}$ has

$$E(t) = 2 - 2\cos t, \qquad E'(t) = 2\sin t.$$

Here the current depends on t. The complete formula retains that dependence instead of replacing it by its value at a particular connection.

Finally, the gravitational density is obtained from the metric variation of the effective action, with its field and normalization. The representation of the current, the contortion it produces and the resulting density are the successive operations of that composition. The subsequent dilution law transports the amplitude thus evaluated in the corresponding material section.

Chapter 130

Differential of the minimal boundary extension

The cubic incidence simultaneously determines the minimal-extension field and the weight of its response. For a boundary datum, both are calculated through the inverse of the incidence Gramian. We now develop the differential of this same operation, preserving the variations of the boundary, transport and normalization scale. In this way, the field involved in the response is selected by the screen variational problem.

130.1 Minimizing field and response weight

Let H and Q be finite-dimensional Hermitian spaces, with fixed inner products antilinear in the first variable. The real case is obtained by replacing adjoints with transposes. On an open set of real parameters Ω, consider families of class C^1

$$\mathsf{C}(\theta) : H \longrightarrow Q, \qquad b(\theta) \in Q, \qquad \chi(\theta) > 0, \qquad \operatorname{Ran} \mathsf{C}(\theta) = Q.$$

Surjectivity is equivalent to full row rank in orthonormal bases. We define the Gramian, the conjugate potential and the extension

$$\mathsf{L} = \mathsf{C}\mathsf{C}^*, \qquad y = \mathsf{L}^{-1} b, \qquad f_{\min} = \mathsf{C}^* y, \qquad \Lambda = \chi \mathsf{L}^{-1}. \tag{130.1}$$

The incidence C and the cellular pairing M of (129.8) are different operators.

Proposition 130.1 (Variational selection of the extension)**.** *The extension $f_{\min}$ is the unique norm minimizer among the fields $f \in H$ satisfying $\mathsf{C} f = b$. Its energy is*

$$E(\theta) = \chi \min_{\mathsf{C} f = b} \|f\|^2 = \chi \langle b, \mathsf{L}^{-1} b \rangle = \langle b, \Lambda b \rangle. \tag{130.2}$$

The field $f_{\min}$, potential y, and energy E depend C^1-smoothly on the parameters.

Proof. Surjectivity of C makes C^* injective, so $\langle z, \mathsf{L} z \rangle = \|\mathsf{C}^* z\|^2 > 0$ for $z \neq 0$. Thus L is invertible and $\mathsf{C} f_{\min} = \mathsf{L} y = b$. Every solution is $f_{\min} + z$, with $z \in \ker \mathsf{C}$. Since $f_{\min} \in \operatorname{Ran} \mathsf{C}^* = (\ker \mathsf{C})^{\perp}$,

$$\|f_{\min} + z\|^2 = \|f_{\min}\|^2 + \|z\|^2, \qquad \|f_{\min}\|^2 = \langle y, \mathsf{L} y \rangle = \langle b, \mathsf{L}^{-1} b \rangle.$$

These identities prove the minimum, its uniqueness and its value. Inversion is a differentiable operation on the open set of invertible matrices; the expressions in (130.1) provide the asserted regularity. □

The cubic realization of 126.1 corresponds to $H = \mathbb{R}^{\{-1,1\}^3}$, the screen quotient $Q = P_{\square} \mathbb{R}^6$ and the incidence $\mathsf{C} = M : H \to Q$. Its Gramian and normalization are

$$\mathsf{L}_{\square} = 12 P_u + 4 P_-, \qquad \chi_m = \frac{112}{81 \ell_m^2}, \qquad \Lambda_m = \frac{28}{243 \ell_m^2} (P_u + 3 P_-). \tag{130.3}$$

The factor $112/81$ comes from the soldering and the quadratic compression defect of (126.3). If b has length dimension, so do y and $f_{\min}$, the incidence is dimensionless and E is dimensionless. Conversion to action uses the normalization in the action line specified below.

130.2 Full differential of the minimum

Theorem 130.2 (Response to the variation of incidence and boundary)**.** *For every real tangent direction $\delta\theta$, one has*

$$\delta E = \delta\chi\,\langle b, y\rangle + 2\chi\,\mathrm{Re}\langle y, \delta b - (\delta \mathsf{C})\, f_{\min}\rangle. \tag{130.4}$$

If $\chi = 112/(81\ell_m^2)$, with variable $\ell_m > 0$,

$$\delta E = -2E\,\frac{\delta \ell_m}{\ell_m} + 2\chi_m\,\mathrm{Re}\langle y, \delta b - (\delta \mathsf{C})\, f_{\min}\rangle. \tag{130.5}$$

Proof. Differentiating $\mathsf{L}\mathsf{L}^{-1} = I_Q$ gives

$$\delta \mathsf{L}^{-1} = -\mathsf{L}^{-1}(\delta \mathsf{L})\mathsf{L}^{-1}, \qquad \delta \mathsf{L} = (\delta \mathsf{C})\mathsf{C}^{*} + \mathsf{C}(\delta \mathsf{C})^{*}.$$

Applying the product rule to (130.2) and using the self-adjointness of L^{-1} yields

$$\delta E = \delta\chi\,\langle b, y\rangle + 2\chi\,\mathrm{Re}\langle y, \delta b\rangle - \chi\langle y, (\delta \mathsf{L})\, y\rangle.$$

The two terms of the last product are conjugate to each other and $\mathsf{C}^{*} y = f_{\min}$; therefore,

$$\langle y, (\delta \mathsf{L})\, y\rangle = 2\,\mathrm{Re}\langle y, (\delta \mathsf{C})\, f_{\min}\rangle.$$

One obtains (130.4). Finally, $\delta\chi_m = -2\chi_m \delta\ell_m/\ell_m$ produces (130.5). □

The formula incorporates the variation of the minimizing field through the equation that determines it. It is also checked directly: $\mathsf{C}\delta f_{\min} = \delta b - (\delta \mathsf{C})\, f_{\min}$ and $f_{\min} = \mathsf{C}^{*} y$ imply

$$\delta\|f_{\min}\|^2 = 2\,\mathrm{Re}\langle f_{\min}, \delta f_{\min}\rangle = 2\,\mathrm{Re}\langle y, \delta b - (\delta \mathsf{C})\, f_{\min}\rangle.$$

For example, if C is fixed and $b = \ell_m b_0$, with b_0 fixed, the two terms of (130.5) cancel and the energy remains constant. This check jointly preserves the boundary dimension and its normalization.

130.3 Fixed quotient and frame changes

The preceding derivatives are taken in fixed spaces H, Q and Hermitian inner products. In the six-face presentation, the inverse always acts on $Q = P_{\square} H_F$. For a family of quotients $Q(\theta)$, an isometric trivialization $V_Q(\theta) : Q \to Q(\theta)$ is fixed first, and likewise for the domain. The derivatives of those maps are included when differentiating the expressions transported to the fixed spaces. The formulas apply on each open set where rank is preserved. Changes in the energy inner product additionally require differentiating that product; a Lorentzian metric and the positive inner product used for minimization retain their distinct types here.

Proposition 130.3 (Covariance of the minimum)**.** *Let g_H, g_Q be unitary. Under*

$$\mathsf{C}' = g_Q \mathsf{C} g_H^{*}, \qquad b' = g_Q b, \qquad \chi' = \chi,$$

$y' = g_Q y$, $f'_{\min} = g_H f_{\min}$ and $E' = E$ hold. When the frame changes depend on the parameter, the full differential preserves this equality. In particular, a pure frame variation has $\delta E = 0$.

Proof. The transformed Gramian is $\mathsf{L}' = g_Q \mathsf{L} g_Q^{*}$, from which the transformations of y, $f_{\min}$ and equality of energies are obtained. For a pure frame variation, at a point with identity frames, let $A_Q^{*} = -A_Q$ and $A_H^{*} = -A_H$ be their generators. Then

$$\delta \mathsf{C} = A_Q \mathsf{C} - \mathsf{C} A_H, \qquad \delta b = A_Q b, \qquad \delta b - (\delta \mathsf{C})\, f_{\min} = \mathsf{C} A_H f_{\min}.$$

Its contribution to (130.4) is $2\chi\,\mathrm{Re}\langle f_{\min}, A_H f_{\min}\rangle = 0$. The proof thus includes the terms due to the moving frames. □

130.4 Transported incidence and connection components

The connection dependence is specified before performing the contraction. An explicit blockwise realization of the incidence is obtained as follows. Let $\mathcal{K}$ be a common Hermitian fibre, $F = \{(i,\epsilon) : 1 \le i \le 3,\ \epsilon = \pm 1\}$ and $V = \{-1,1\}^3$. For each incidence $\nu_i = \epsilon$, the realization used declares a route $\gamma_{i,\epsilon,\nu}$ from the vertex fibre to the face fibre, with transport $U_{i,\epsilon,\nu}$. The trivializations identify these fibres with $\mathcal{K}$. We define

$$\begin{aligned}(\mathsf{B}_U f)_{i,\epsilon} &= \sum_{\nu_i=\epsilon} U_{i,\epsilon,\nu} f_\nu,\\ \mathsf{C}_U = (q_\square \otimes I_{\mathcal{K}})\mathsf{B}_U &: \bigoplus_{\nu\in V} \mathcal{K} \longrightarrow Q\otimes\mathcal{K}, \qquad q_\square = P_\square : H_F \to Q.\end{aligned} \tag{130.6}$$

This is an explicit variational continuation of the cubic incidence in the declared route representation. For identity transports, $\mathsf{C}_U = M \otimes I_{\mathcal{K}}$ and one recovers (130.3), tensored with the fibre. The Gramian remains positive in a neighbourhood of that realization, since positive definiteness is an open condition. On each full-rank domain, the weight and field are fixed by (130.1). The routes realizing the incidences, their orientations and their memories are data of this realization of APP–TRIT–TPK transport and accompany the construction.

If $\omega = (\omega_j)$ are real connection coordinates, the derivative of C_U is calculated by substituting in each block

$$\partial_j U_\gamma = \sum_{k=1}^{N} U_N \cdots U_{k+1}(\partial_j U_k) U_{k-1}\cdots U_1, \qquad U_\gamma = U_N\cdots U_1. \tag{130.7}$$

Empty products are identities. The formula follows from the product rule and preserves the position of each operation in the route. Thus, $\mathsf{C}_j := \partial_j \mathsf{C}_U$ is obtained from a finite sum of known matrices in the chosen realization, followed by the projection $q_\square \otimes I$. If a moving quotient is used, its derivative is incorporated according to 130.3. The fixed integer cubic matrix corresponds to the case $\mathsf{C}_j = 0$; the response to the connection follows from the transport dependence actually declared in (130.6).

Theorem 130.4 (Variational components of the minimal extension)**.** *For the screen action $S_{\min} = \mathfrak{a}_m E$, where $\mathfrak{a}_m$ is its normalization in the action line, let us write*

$$b_j = \partial_j b, \qquad \chi_j = \partial_j \chi, \qquad a_j = \partial_j \mathfrak{a}_m.$$

With the convention $\delta S_{\min} = -\frac{1}{2}\sum_j s_j \delta\omega_j$, the components are

$$\begin{aligned} s_j = {} & 4\mathfrak{a}_m \chi \,\mathrm{Re}\langle y, \mathsf{C}_j f_{\min} - b_j\rangle \\ & - 2\mathfrak{a}_m \chi_j \langle b, y\rangle - 2a_j E. \end{aligned} \tag{130.8}$$

For the cubic normalization, the second term equals $4\mathfrak{a}_m E\,\partial_j \ell_m/\ell_m$. If the cellular pairing of this same action has invertible matrix $\mathsf{M}_{\mathrm{cel}}$, its current coordinates are $\sigma = \mathsf{M}_{\mathrm{cel}}^{-1} s$.

Proof. The product rule gives $\partial_j S_{\min} = a_j E + \mathfrak{a}_m \partial_j E$. Substituting (130.4) and multiplying by -2 proves (130.8). The derivative of χ_m produces the scale expression. Finally, comparing $\delta S_{\min} = -\frac{1}{2}\delta\omega^{\mathsf{T}} s$ with $-\frac{1}{2}\delta\omega^{\mathsf{T}}\mathsf{M}_{\mathrm{cel}}\sigma$ for every variation gives $\mathsf{M}_{\mathrm{cel}}\sigma = s$, and its inverse determines the coordinates. □

The construction recovers a field and a weight jointly selected by the incidence, and evaluates their response to variations of the same realization. The boundary datum b, the route family, the scale ℓ_m, the normalization $\mathfrak{a}_m$ and the cellular pairing retain their publication rules. Interpretation as a material current uses those objects and the complete action. The Cartan response and its subsequent metric density are composed with it in the corresponding matter domain, preserving any dependence of the current on the connection.

Chapter 131

Spin current and algebraic Cartan–Holst response

The realization of the enriched transport in Section 127 provides the cotetrad and the connection; the dimensional sections and the incidence parameter of Sections 120 and 121 fix their couplings. Their composition determines a first-order variational problem. The material current occupies a precise position in it: it is the derivative of the action with respect to the connection and determines the torsional response with the same normalization.

131.1 Geometric domain and current normalization

We work on a smooth oriented four-dimensional manifold M, with internal Lorentzian bundle E, metric $\eta = \mathrm{diag}(-1,1,1,1)$ and non-degenerate cotetrad $e \in \Omega^1(M;E)$. In a local frame we write e^I, with $I = 0,1,2,3$. The metric connection $\omega^{IJ} = -\omega^{JI}$ acts through D_ω. When the fields Ψ are spinorial, a spin structure and the corresponding material representation are preserved. We define

$$\Sigma^{IJ} = e^I \wedge e^J, \qquad F^{IJ} = \mathrm{d}\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}, \qquad T^I = D_\omega e^I. \tag{131.1}$$

The operator $\star$ acts on the internal bivector indices,

$$(\star X)^{IJ} = \tfrac{1}{2}\epsilon^{IJ}{}_{KL}X^{KL}, \qquad \star^2 = -I, \qquad \mathsf{P}_\gamma = \star + \gamma^{-1}I. \tag{131.2}$$

This distinguishes this duality from the exterior duality of forms on M. Internal indices are raised and contracted with η.

We fix a chart with $G_{\mathrm{int}} > 0$, $c_{\mathrm{int}} > 0$, $\gamma = \gamma_{\mathrm{HMT}} \in \mathbb{R} \setminus \{0\}$ and Λ_{int} constant on it. We abbreviate $\kappa = 8\pi G_{\mathrm{int}}/c_{\mathrm{int}}^4$ and $c = c_{\mathrm{int}}$. These sections remain fixed during variation. The action is

$$S[e,\omega,\Psi] = \frac{1}{2\kappa c}\int_M \Sigma_{IJ} \wedge (\mathsf{P}_\gamma F)^{IJ} - \frac{\Lambda_{\mathrm{int}}}{\kappa c}\int_M \mathrm{vol}_e + S_{\mathrm{m}}[e,\omega,\Psi]. \tag{131.3}$$

The volume and orientation convention is the one that gives $\Sigma_{IJ} \wedge (\star F)^{IJ} = R\,\mathrm{vol}_e$ in the Levi–Civita sector. The prior dimensional type makes each summand an action section.

For the variation of the connection, variations with compact support in the interior are used, or $\delta\omega|_{\partial M} = 0$ is fixed. A boundary completion whose variational problem cancels the same boundary term may also be used. At fixed cotetrad and material fields, the spin current is the degree-three form, valued in dual internal bivectors, determined by

$$\delta_\omega S_{\mathrm{m}} = -\tfrac{1}{2}\int_M \delta\omega^{IJ} \wedge \sigma_{IJ}. \tag{131.4}$$

Pairing with an arbitrary compactly supported variation fixes σ uniquely. This definition preserves the action unit of the current and the normalization factor involved in its geometric response.

131.2 Connection variation and Holst inversion

Theorem 131.1 (Variational Cartan–Holst equation)**.** *On the preceding domain, stationarity with respect to the connection is equivalent to*

$$D_\omega(\mathsf{P}_\gamma\Sigma)^{IJ} = \kappa c\,\sigma^{IJ}. \tag{131.5}$$

Proof. The variation of the curvature is $\delta F = D_\omega\delta\omega$. The internal duality is parallel and self-adjoint for the bivector contraction; since γ is constant, $D_\omega\mathsf{P}_\gamma = \mathsf{P}_\gamma D_\omega$. Setting $B_{IJ} = (\mathsf{P}_\gamma\Sigma)_{IJ}$ yields

$$B_{IJ}\wedge D_\omega\delta\omega^{IJ}=\mathrm{d}(B_{IJ}\wedge\delta\omega^{IJ})-D_\omega B_{IJ}\wedge\delta\omega^{IJ}.$$

The last form changes sign when its factors of degrees three and one are interchanged. The boundary conditions eliminate the integral of the exact term. The cosmological term is independent of ω. Therefore,

$$\delta_\omega S=\frac{1}{2\kappa c}\int_M\delta\omega^{IJ}\wedge\left[D_\omega B_{IJ}-\kappa c\,\sigma_{IJ}\right].$$

The arbitrariness of the variation proves the equation, including its sign and coefficient. □

Lemma 131.2 (Real inverse of the Holst operator)**.** *For $\gamma\in\mathbb{R}\setminus\{0\}$,*

$$\mathsf{P}_\gamma^{-1}=\frac{\gamma^2}{1+\gamma^2}(-\star+\gamma^{-1}I).\tag{131.6}$$

Proof. Both factors are polynomials in $\star$ and commute. Their product before normalization is

$$(\star+\gamma^{-1}I)(-\star+\gamma^{-1}I)=-\star^2+\gamma^{-2}I=(1+\gamma^{-2})I.$$

The denominator $1+\gamma^2$ is positive on the fixed real domain. Multiplication by $\gamma^2/(1+\gamma^2)$ gives the identity. □

The current thus determines the degree-three bivector-valued form

$$Y^{IJ}:=\kappa c\,\frac{\gamma^2}{1+\gamma^2}(-\star+\gamma^{-1}I)\sigma^{IJ},\qquad D_\omega\Sigma^{IJ}=Y^{IJ}.\tag{131.7}$$

131.3 Pointwise reconstruction of torsion and contorsion

The Leibniz rule gives

$$D_\omega\Sigma^{IJ}=T^I\wedge e^J-e^I\wedge T^J=:\mathsf{L}_e(T)^{IJ}.\tag{131.8}$$

At each point, L_e acts from $\Lambda^2T^*M\otimes E$ to $\Lambda^3T^*M\otimes\Lambda^2E$. Both spaces have dimension twenty-four. Let E_I be the dual frame of the cotetrad, so that $\iota_{E_I}e^J=\delta_I^J$.

Theorem 131.3 (Algebraic inversion of the torsional equation)**.** *For a nondegenerate co-tetrad, L_e is an isomorphism. Its inverse on the form Y is written*

$$B^I=\sum_J\iota_{E_J}Y^{IJ},\qquad b=\tfrac14\sum_I\iota_{E_I}B^I,\qquad T^I=B^I-e^I\wedge b.\tag{131.9}$$

Equation (131.5) *therefore determines torsion pointwise from e and the current appearing in it.*

Proof. First let $Y=\mathsf{L}_e(T)$ and set $t=\sum_J\iota_{E_J}T^J$. The contraction of (131.8), using that T^I has degree two and that the cotetrad has four elements, gives

$$\sum_J\iota_{E_J}(T^I\wedge e^J)=2T^I,\qquad\sum_J\iota_{E_J}(e^I\wedge T^J)=T^I-e^I\wedge t.$$

Hence $B^I=T^I+e^I\wedge t$. A second contraction produces

$$\sum_I\iota_{E_I}B^I=t+3t=4t.$$

One recovers $t=b$ and then $T^I=B^I-e^I\wedge b$. In particular, the kernel of L_e is zero. Equality of dimensions proves its surjectivity and makes the formula valid for every form Y in the codomain. □

The connection decomposes uniquely as

$$\omega^{IJ} = \mathring{\omega}^{IJ}(e) + K^{IJ}, \qquad K^{IJ} = -K^{JI}, \qquad T^I = K^I{}_J \wedge e^J, \tag{131.10}$$

where $\mathring{\omega}(e)$ is the Levi–Civita connection and K the contorsion. To make the inverse explicit, we write $K_{IJ} = K_{IJK}e^K$ and $T_I = \frac{1}{2}T_{IJK}e^J \wedge e^K$. Then

$$T_{IJK} = K_{IKJ} - K_{IJK}, \qquad K_{IJK} = \tfrac{1}{2}(T_{KIJ} - T_{IJK} - T_{JKI}). \tag{131.11}$$

The second equality is obtained by combining the first cyclically and using $K_{IJK} = -K_{JIK}$; substituting it recovers the components of T. Torsion and contorsion are therefore two equivalent presentations of the algebraic response at fixed cotetrad.

Corollary 131.4 (Current-free sector and minimal linear response)**.** *If $\sigma = 0$, one obtains $T = 0$ and $\omega = \mathring{\omega}(e)$. For a material coupling whose current $\sigma[e, \Psi]$ is independent of K, equations* (131.7)*,* (131.9) *and* (131.11) *provide a unique response $K_*(e, \Psi)$, linear in σ.*

Proof. The three reconstruction maps are linear with e, γ and κc fixed. The zero current produces $Y = 0$, then $T = 0$ and finally $K = 0$. □

In the minimal action, the connection equation contains this pointwise response. Its evolution is obtained by evolving e and Ψ: $K_* = K_*(e, \Psi)$ changes with them. If a material action preserves additional algebraic dependence on K, the Cartan equation retains its variational form and must be solved with that current; the preceding linear response corresponds to the specified minimal domain.

131.4 Elimination of contorsion and quadratic contribution

Now consider the minimal coupling affine in K, expressed by

$$S_{\mathrm{m}}[e, \mathring{\omega} + K, \Psi] = S_{\mathrm{m}}[e, \mathring{\omega}, \Psi] - \tfrac{1}{2}\int_M K^{IJ} \wedge \sigma_{IJ}[e, \Psi]. \tag{131.12}$$

This condition identifies the material action to which the quadratic elimination applies; the current remains the one defined by (131.4).

The curvature expansion is

$$F(\mathring{\omega} + K) = F(\mathring{\omega}) + D_{\mathring{\omega}}K + K \wedge K, \qquad (K \wedge K)^{IJ} = K^I{}_L \wedge K^{LJ}. \tag{131.13}$$

Since $D_{\mathring{\omega}}\Sigma = 0$, the linear term is the exterior derivative of $\Sigma_{IJ} \wedge (\mathsf{P}_\gamma K)^{IJ}$. The corresponding boundary contribution remains explicit:

$$S_\partial[e, K] = \frac{1}{2\kappa c}\int_{\partial M} \Sigma_{IJ} \wedge (\mathsf{P}_\gamma K)^{IJ}. \tag{131.14}$$

On an interior domain, or with the compatible boundary completion, the action density that depends on the contorsion is

$$\mathcal{Q}_{e,\gamma}(K) - \tfrac{1}{2}K^{IJ} \wedge \sigma_{IJ}, \qquad \mathcal{Q}_{e,\gamma}(K) = \frac{1}{2\kappa c}\Sigma_{IJ} \wedge \mathsf{P}_\gamma(K \wedge K)^{IJ}. \tag{131.15}$$

The density $\mathcal{Q}_{e,\gamma}$ is a degree-four differential form, homogeneous of degree two in K.

Theorem 131.5 (Effective response quadratic in the current)**.** *Under* (131.12)*, substituting the algebraic solution K_* produces the effective spin density*

$$\mathcal{L}^{\mathrm{eff}}_{\mathrm{spin}} = -\tfrac{1}{4}K_*^{IJ} \wedge \sigma_{IJ} = -\mathcal{Q}_{e,\gamma}(K_*). \tag{131.16}$$

At fixed cotetrad and couplings, it is homogeneous of degree two in the current. The effective action also preserves $S_\partial[e, K_*]$, or its declared boundary completion.

Proof. By Corollary 131.4, K_* is linear in σ. The connection equation is precisely the stationarity of (131.15). Since it is algebraic, the radial variation $\delta K = K_*$ is applied pointwise; it may also be multiplied by a compactly supported function. Euler's identity for a quadratic form gives

$$2\mathcal{Q}_{e,\gamma}(K_*) - \tfrac{1}{2} K_*^{IJ} \wedge \sigma_{IJ} = 0.$$

Substitution into the action density proves (131.16). If $\sigma \mapsto \lambda\sigma$, then $K_* \mapsto \lambda K_*$ and the effective density is multiplied by λ^2. The expansion (131.13) separately identifies the boundary term accompanying this elimination. □

The specific current, its Lorentzian contraction, the duality and the conventions of the material action determine the sign and coefficient of this contribution. The quadratic identity provides the composition rule that allows them to be evaluated once these data have been specified.

131.5 Effective metric source and covariant balance

The normalization of the metric source is fixed in the same action line. After eliminating K, we define

$$\delta_g S_{\mathrm{m}}[e, \mathring{\omega}, \Psi] = -\tfrac{1}{2} \int_M \mathrm{vol}_e \; \mathcal{T}^{S,\mathrm{LC}}_{\mu\nu} \, \delta g^{\mu\nu}, \tag{131.17}$$

$$\delta_g S^{\mathrm{eff}}_{\mathrm{spin}} = -\tfrac{1}{2} \int_M \mathrm{vol}_e \; \mathcal{U}^{S,\mathrm{spin}}_{\mu\nu} \, \delta g^{\mu\nu}. \tag{131.18}$$

Here interior variations are considered, with boundary terms treated according to the fixed variational problem. For the affine coupling, the second source is the metric variation of (131.16). The variation includes the dependence of the material current on e and Ψ.

Proposition 131.6 (Metric equation and conservation of the total source)**.** *With the preceding conventions, the metric equations are*

$$G_{\mu\nu}[g] + \Lambda_{\mathrm{int}} g_{\mu\nu} = \kappa c \big(\mathcal{T}^{S,\mathrm{LC}}_{\mu\nu} + \mathcal{U}^{S,\mathrm{spin}}_{\mu\nu} \big) = \kappa \big(T^{\mathrm{phys,LC}}_{\mu\nu} + U^{\mathrm{phys,spin}}_{\mu\nu} \big), \tag{131.19}$$

where $T^{\mathrm{phys,LC}} = c\mathcal{T}^{S,\mathrm{LC}}$ and $U^{\mathrm{phys,spin}} = c\mathcal{U}^{S,\mathrm{spin}}$ when the cotetrad takes values in the length line. Every solution satisfies

$$\mathring{\nabla}^\mu \big(T^{\mathrm{phys,LC}}_{\mu\nu} + U^{\mathrm{phys,spin}}_{\mu\nu} \big) = 0. \tag{131.20}$$

Proof. Eliminating K preserves the equations of the remaining variables: in the chain rule, the term multiplying δK_* vanishes by the connection equation. In the Levi–Civita sector, the first Bianchi identity cancels the term $\Sigma_{IJ} \wedge F^{IJ}$, and the gravitational part is written $(2\kappa c)^{-1} \int_M (R - 2\Lambda_{\mathrm{int}}) \, \mathrm{vol}_e$. Its interior variation is

$$\frac{1}{2\kappa c} \int_M \mathrm{vol}_e \; (G_{\mu\nu} + \Lambda_{\mathrm{int}} g_{\mu\nu}) \delta g^{\mu\nu}.$$

The sum with (131.18) gives (131.19). The contracted Bianchi identity for $\mathring{\nabla}$, metric compatibility and the constancy of κ and Λ_{int} in the chart produce (131.20). □

Before eliminating the connection, the geometric identities retain the form

$$D_\omega T^I = F^I{}_J \wedge e^J, \qquad D_\omega F^{IJ} = 0. \tag{131.21}$$

The first is $D_\omega^2 e = F \wedge e$; the second is obtained by expanding $D_\omega(\mathrm{d}\omega + \omega \wedge \omega)$ and cancelling the terms. In a covariant material action, the total metric balance coincides with the Noether identity on the material

equations. Geometry and matter thus preserve a joint source; the partition between its Levi–Civita contribution and its spin contribution is determined by the specialized action.

Chapter 132

Torsional elimination and metric response of the minimal extension

The minimal extension provides a field, a response weight and a current through a single variational operation. Its transported incidence depends on the connection through the route products of (130.7). That dependence is preserved when composing the current with the Cartan–Holst equation. Elimination of the connection is therefore performed on the complete action of that extension. The affine case of 131.5 is contained as an exact specialization of the following construction.

132.1 Variational composition at a cellular level

Let us fix a finite level, its routes, the transport representation and the cellular pairing of 130.4. Let p be real coordinates of the co-tetrad and the material fields in a declared local trivialization, and $k \in \mathbb{R}^d$ the coordinates of the contortion K. The family $p \mapsto e(p)$ preserves nondegeneracy of the co-tetrad. The connection acting on each route is

$$\omega(p,k) = \mathring{\omega}(e(p)) + K(k), \qquad F(p,k) = S_{\min}[e(p), \omega(p,k), \Psi(p)] = \mathfrak{a}(p,k)E(p,k). \tag{132.1}$$

The spaces and bases of this expression are transported to fixed spaces before differentiation. We work on an open set where the incidence $\mathsf{C}(p,k)$ is surjective and the families are C^2. This additional regularity allows the Hessian of the action to be examined. The couplings $G_{\mathrm{int}}, c_{\mathrm{int}}, \gamma_{\mathrm{HMT}}$ preserve the chart and the variation convention of 131.1.

Let $Q(p,k)$ be the integral, or the declared cellular realization, of the quadratic form (131.15). In real bases,

$$Q(p,k) = \tfrac{1}{2} k^{\mathsf{T}} \mathsf{A}(p) k, \qquad \mathsf{A}(p)^{\mathsf{T}} = \mathsf{A}(p). \tag{132.2}$$

The operator A comes from the gravitational action and its pairing, whereas the Gramian $\mathsf{C}\mathsf{C}^*$ comes from the extension problem. Positivity of the latter guarantees existence of the minimizing field; the gravitational form retains its own signature. Boundary terms are retained according to (131.14); the following interior formulas apply with admissible variations that annul their contribution, or with the corresponding boundary completion.

Define the response covector by

$$d_k F(p,k)[h] = -\tfrac{1}{2} h^{\mathsf{T}} s(p,k), \qquad s(p,k) = \mathsf{M}_{\mathrm{cel}}(p,k)\sigma(p,k). \tag{132.3}$$

Here s and σ preserve the distinct types of (130.8): the former contains the coefficients of the differential and the latter the coordinates of the current in the material pairing. If the basis in k changes, that pairing is also transported. When differentiating $s = \mathsf{M}_{\mathrm{cel}}\sigma$, the derivatives of both factors are included when the pairing depends on the parameter.

Proposition 132.1 (Composed equation of extension and contorsion)**.** *Interior stationarity of the action with respect to contorsion is*

$$\mathcal{R}(p,k) := \mathsf{A}(p)k - \tfrac{1}{2} s(p,k) = 0. \tag{132.4}$$

Each component of s is calculated from the same extension:

$$\begin{aligned} s_j = {} & 4\mathfrak{a}\chi \,\mathrm{Re}\langle y, (\partial_{k_j}\mathsf{C}) f_{\min} - \partial_{k_j} b\rangle \\ & - 2\mathfrak{a}(\partial_{k_j}\chi)\langle b, y\rangle - 2(\partial_{k_j}\mathfrak{a})E. \end{aligned} \tag{132.5}$$

When the cellular pairing realizes the connection equation of 131.1, this equation is equivalent to composing $\sigma(p,k)$ with the Holst inverse, the inverse of L_e and the contortion reconstruction:

$$K = \mathsf{C}_e^{\mathrm{tor}}\,\mathsf{L}_e^{-1}\big(\kappa c\,\mathsf{P}_\gamma^{-1}\sigma[e,\mathring{\omega}(e)+K,\Psi]\big). \tag{132.6}$$

The symbol $\mathsf{C}_e^{\mathrm{tor}}$ exclusively designates the map $T \mapsto K$ of (131.11), with fixed co-tetrad.

Proof. The differential of $Q+F$, in the arbitrary direction h, is

$$d_k(Q+F)[h] = h^{\mathsf{T}}\big(\mathsf{A}k - \tfrac{1}{2}s\big).$$

Its vanishing in every direction proves (132.4). Theorem 130.4, applied to ∂_{k_j}, gives (132.5). With p fixed, $\delta\omega = \delta K$. The Cartan equation is transformed successively through (131.6), (131.9) and (131.11); each of these operations is invertible on its domain. This proves equivalence with (132.6) when both expressions use the same material and cellular realization. □

The composition preserves the connection on both sides. Routes passing through several cells can couple the current components between them. The pointwise reconstruction of T from a given current and the joint solution of (132.4) are, in that case, successive operations. This finite formulation keeps explicit the route representation entering the functional.

132.2 Local existence of a stationary branch

Theorem 132.2 (Local continuation of the torsional solution)**.** *Suppose (p_0,k_0) belongs to the preceding open set, $\mathcal{R}(p_0,k_0)=0$, and the total Hessian*

$$\mathsf{H}_0 = \mathsf{A}(p_0) + \partial_k^2 F(p_0,k_0) = \mathsf{A}(p_0) - \tfrac{1}{2}\partial_k s(p_0,k_0) \tag{132.7}$$

is invertible. There exist neighborhoods of p_0 and k_0 in which a unique branch C^1, $k = k_*(p)$, satisfies (132.4) and passes through k_0. On that branch,

$$\partial_{p_i} k_* = -\big[\mathsf{A} + \partial_k^2 F\big]^{-1}\big[(\partial_{p_i}\mathsf{A})k_* + \partial_{p_i}\partial_k F\big]. \tag{132.8}$$

Proof. Inversion of the Gramian is C^2 on the full-rank open set. Hence F is C^2 and $\mathcal{R}$ is C^1. Its derivative with respect to k, at (p_0,k_0), is H_0. The implicit function theorem provides the branch, its regularity and its uniqueness in a neighborhood of the point. Existence can also be seen through the map

$$\mathcal{T}_p(k) = k - \mathsf{H}_0^{-1}\mathcal{R}(p,k).$$

Its derivative with respect to k vanishes at (p_0,k_0). On a sufficiently small ball it has norm at most $q<1$. Shrinking the neighborhood of p_0 yields $\|\mathcal{T}_p(k_0)-k_0\| < (1-q)r$, where r is the radius of the ball. Thus, $\mathcal{T}_p$ maps the closed ball into itself and is contractive. Its fixed point is the local solution. Differentiating $\mathcal{R}(p,k_*(p))=0$ and solving for $\partial_{p_i}k_*$ gives (132.8). □

The hypothesis is verified on the composite action: the Gramian's rank ensures the screen minimum and the total Hessian controls the torsional branch. The asserted uniqueness is local around the specified solution; other branches may coexist in the same full-rank domain.

132.3 Effective action and non-affine contribution

Theorem 132.3 (Exact elimination identity)**.** *Let $k_*(p)$ be a stationary branch. Suppose, moreover, that the segment $t \mapsto (p, tk_*)$, $0 \le t \le 1$, remains in the open set where F is defined. The action correction relative to the Levi–Civita realization is*

$$\begin{aligned} \Delta S(p) &:= Q(p,k_*) + F(p,k_*) - F(p,0) \\ &= \tfrac{1}{4} k_*^{\mathsf{T}} s(p,k_*) - \tfrac{1}{2} \int_0^1 k_*^{\mathsf{T}} s(p,tk_*)\, dt. \end{aligned} \tag{132.9}$$

Equivalently,

$$\Delta S = -\tfrac{1}{4} k_*^{\mathsf{T}} s(p,k_*) + \tfrac{1}{2} \int_0^1 k_*^{\mathsf{T}} \big[s(p,k_*) - s(p,tk_*) \big]\, dt. \tag{132.10}$$

These identities refer to the integrated or cellular action. In a realization with local density, they apply to the density with its corresponding pairing of forms. The effective action preserves the declared boundary term separately.

Proof. By the chain rule and (132.3),

$$\frac{d}{dt} F(p,tk_*) = -\tfrac{1}{2} k_*^{\mathsf{T}} s(p,tk_*).$$

Integrating between 0 and 1 gives the difference of material actions. The stationarity equation, evaluated in the direction k_*, and the quadratic homogeneity of Q give

$$2Q(p,k_*) = \tfrac{1}{2} k_*^{\mathsf{T}} s(p,k_*).$$

The sum of both identities proves (132.9); adding and subtracting $\frac{1}{2} k_*^{\mathsf{T}} s(p,k_*)$ provides (132.10). □

Corollary 132.4 (Affine specialization)**.** *If $s(p,k) = s(p)$, then*

$$\Delta S = -\tfrac{1}{4} k_*^{\mathsf{T}} s(p) = -Q(p,k_*).$$

This is the integrated realization of (131.16)*.*

Proof. The difference inside the integral of (132.10) vanishes. The second equality follows from the radial stationarity identity. □

132.4 Metric differential of the eliminated action

Theorem 132.5 (Response of the geometric variables)**.** *On the branch of Theorem 132.2,*

$$\partial_{p_i} \Delta S = \tfrac{1}{2} k_*^{\mathsf{T}} (\partial_{p_i} \mathsf{A}) k_* + (\partial_{p_i} F)(p,k_*) - (\partial_{p_i} F)(p,0), \tag{132.11}$$

where the partial derivatives keep the coordinates k fixed. In fixed Hermitian inner products, each material derivative is calculated by

$$\begin{aligned} \partial_{p_i} F = {} & (\partial_{p_i} \mathfrak{a}) E + \mathfrak{a} (\partial_{p_i} \chi) \langle b, y \rangle \\ & + 2\mathfrak{a}\chi\, \mathrm{Re} \langle y, \partial_{p_i} b - (\partial_{p_i} \mathsf{C}) f_{\min} \rangle. \end{aligned} \tag{132.12}$$

If p_i varies the co-tetrad, $\partial_{p_i} \mathsf{C}$ includes the variation of $\mathring{\omega}(e(p)) + K(k)$, of the realized routes and of the trivializations used, according to their declared dependencies.

Proof. The total derivative of $Q(p,k_*)+F(p,k_*)$ contains, in addition to its partial derivatives, the term

$$\big[\partial_k Q(p,k_*)+\partial_k F(p,k_*)\big]\partial_{p_i}k_*.$$

The bracket is zero by the connection equation. Subtracting the derivative of $F(p,0)$ proves (132.11). The product rule and (130.4) prove (132.12). The composition (132.1) requires differentiating the Levi–Civita connection when varying e with k fixed. □

When the positive extension inner product also varies, its contribution is preserved explicitly. In coordinates, let $\mathsf{h}(p,k)>0$ be its Hermitian matrix and consider the minimum of $f^*\mathsf{h}f$ under $\mathsf{C}f=b$. Then

$$\mathsf{L}=\mathsf{C}\mathsf{h}^{-1}\mathsf{C}^*,\quad y=\mathsf{L}^{-1}b,\quad f_{\min}=\mathsf{h}^{-1}\mathsf{C}^*y,\quad E=\chi b^*y. \tag{132.13}$$

Here y represents the constraint covector in fixed bases. The positive matrix h retains its energy role, distinct from that of the Lorentzian metric.

Lemma 132.6 (Variation of the extension inner product)**.** *With the preceding conventions,*

$$\delta E=(\delta\chi)b^*y+2\chi\,\mathrm{Re}\big[y^*(\delta b-\delta\mathsf{C}f_{\min})\big]+\chi f_{\min}^*(\delta\mathsf{h})f_{\min}. \tag{132.14}$$

Proof. The formula $\delta\mathsf{h}^{-1}=-\mathsf{h}^{-1}(\delta\mathsf{h})\mathsf{h}^{-1}$ gives

$$\delta\mathsf{L}=(\delta\mathsf{C})\mathsf{h}^{-1}\mathsf{C}^*+\mathsf{C}\mathsf{h}^{-1}(\delta\mathsf{C})^*-\mathsf{C}\mathsf{h}^{-1}(\delta\mathsf{h})\mathsf{h}^{-1}\mathsf{C}^*.$$

When differentiating $E=\chi b^*\mathsf{L}^{-1}b$, the term $-\chi y^*(\delta\mathsf{L})y$ produces the two terms with $\delta\mathsf{C}$ and the positive term with $\delta\mathsf{h}$. This proves the complete identity. □

In a covariant material realization whose variation admits the metric pairing of (131.18), the correction determines

$$\delta_g\Delta S=-\tfrac12\int_M \mathrm{vol}_e\ \mathcal{U}^{S,\min}_{\mu\nu}\,\delta g^{\mu\nu},\qquad U^{\mathrm{phys},\min}_{\mu\nu}=c\mathcal{U}^{S,\min}_{\mu\nu}. \tag{132.15}$$

Equations (132.11) and (132.12), supplemented by (132.14) where appropriate, evaluate their coefficients in the declared metric variations. The screen minimum, torsional solution and metric derivative are three consecutive operations on the same action. The reduced action preserves the possible couplings between cells present in the original routes.

The homogeneous specialization of 134 additionally uses the realization of the resulting tensor: its spatial symmetry, the sign of contraction with the observer and the scaling law. When that evaluation gives $\rho_{\mathrm{corr}}(a)=-s_0a^{-6}$, the amplitude is read in the reference state as $s_0=-a_{\mathrm{ref}}^6\rho_{\mathrm{corr}}(a_{\mathrm{ref}})$. This reading preserves the material state and its normalization; the elimination theorem provides the action to be contracted.

132.5 Exact non-affine checking model

The following finite model checks the coefficients and the non-affine dependence of the identities. Its parameters belong to the algebraic example and keep the problem of material realization of the routes separate.

Proposition 132.7 (Exact elimination of a rank-one extension)**.** *Let $H=\mathbb{R}^2$, $Q=\mathbb{R}$, $\mathsf{C}(k)=(1\ \ k)$, $b=1$, $\chi=1$, $\mathfrak{a}=\lambda>0$, and $Q_{\mathrm{ex}}(k)=k^2/2$. Then*

$$f_{\min}=\frac{(1,k)^{\mathsf{T}}}{1+k^2},\quad F(\lambda,k)=\frac{\lambda}{1+k^2},\quad s(\lambda,k)=\frac{4\lambda k}{(1+k^2)^2},\quad \mathcal{R}(\lambda,k)=k\left(1-\frac{2\lambda}{(1+k^2)^2}\right). \tag{132.16}$$

For $\lambda=25/2$, the stationary solutions are $0,2,-2$. At $k_*=2$,

$$\mathsf{H}_* = \frac{16}{5}, \quad s_* = 4, \quad Q_{\mathrm{ex}}(k_*) = 2, \quad F(\lambda, k_*) = \frac{5}{2}, \quad \Delta S = -8. \tag{132.17}$$

The affine expression $-k_* s_*/4 = -2$ differs from the exact correction. On the positive branch,

$$\partial_\lambda \Delta S = -\frac{4}{5}, \qquad \partial_\lambda k_* = \frac{1}{20} \quad \text{in } \lambda = 25/2. \tag{132.18}$$

Proof. The Gramian is $1 + k^2 > 0$, so the minimum has the indicated expressions. Differentiating F gives $s = -2\partial_k F$ and then $\mathcal{R} = k - s/2$. For $\lambda = 25/2$, the equation is $k[(1 + k^2)^2 - 25] = 0$; since $1 + k^2 > 0$, its roots are precisely $0, \pm 2$. The total Hessian is

$$1 - \frac{2\lambda(1 - 3k^2)}{(1 + k^2)^3},$$

which equals $16/5$ at both nonzero roots. The correction is $2 + 5/2 - 25/2 = -8$. In the radial identity,

$$\int_0^1 k_* s(\lambda, t k_*)\, dt = \int_0^1 \frac{200t}{(1 + 4t^2)^2}\, dt = 20,$$

since an antiderivative is $-25/(1 + 4t^2)$. Thus $\Delta S = 8/4 - 20/2 = -8$, including the non-affine term. For $\lambda > 1/2$, the positive branch satisfies $k_*^2 = \sqrt{2\lambda} - 1$ and $\Delta S = \sqrt{2\lambda} - 1/2 - \lambda$. Their derivatives give (132.18). The envelope formula gives the same result: $(\partial_\lambda F)(\lambda, k_*) - (\partial_\lambda F)(\lambda, 0) = 1/5 - 1 = -4/5$. □

Chapter 133

Spinorial current and torsional amplitude of the material state

The preceding development and the following spinorial realization articulate two variational problems with common geometric antecedents. The minimal extension action $S_{\min} = \mathfrak{a}_m E$, constructed in section 130, incorporates the dependence of the transported incidence and its minimum on the connection; its torsional elimination preserves that complete dependence in section 132. The spinorial action S_D of (133.3) couples the same connection to the fields of its representation and is affine in the contortion when the co-tetrad and fields are fixed. Each current is obtained by varying its own action. A minimizing preparation can determine the initial material state of the spinorial realization; substitution of connection-dependent fields additionally preserves the terms of the composite differential. This distinction places the following amplitude calculation in its specific action and variational domain.

The spinorial representation of transport and the algebraic response of Cartan–Holst allow torsional intensity to be calculated as a functional of the material state. Its evaluation preserves three data: the field representation, the second moment of its current and the volume relative to which density is defined. The coupling coefficient results from the action and its inversion; the initial state determines the material amplitude.

133.1 Spinorial representation and affine material action

We preserve the co-tetrad, orientation and couplings of section 131. The spinorial realization uses the four-dimensional complex module obtained by complexifying the real matrices

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{133.1}$$
$$g^0 = J \otimes I_2, \qquad g^1 = R \otimes I_2, \qquad g^2 = Z \otimes R, \qquad g^3 = Z \otimes Z.$$

Multiplication gives $\{g^I, g^J\} = 2\eta^{IJ} I_4$, with $\eta = \operatorname{diag}(-1, 1, 1, 1)$. The matrices are distinguished from the scalar parameter γ of the Holst operator. Define

$$g_I = \eta_{IJ} g^J, \quad \mathbb{B}_{IJ} = \tfrac{1}{4}[g_I, g_J], \quad A_D = -i g^0, \quad \bar{\psi} = \psi^\dagger A_D, \quad \Omega = \tfrac{1}{2}\omega^{IJ}\mathbb{B}_{IJ}. \tag{133.2}$$

The Clifford product satisfies

$$[\mathbb{B}_{IJ}, g^K] = \delta^K_J g_I - \delta^K_I g_J.$$

Thus, the same metric connection acts on the fields through $D\psi = \mathrm{d}\psi + \Omega\psi$ and $D\bar{\psi} = \mathrm{d}\bar{\psi} - \bar{\psi}\Omega$. Transport along a route is obtained by integrating this connection in its spinorial representation; it preserves the connection and the route of the preceding geometric realization.

Let $\eta_I = \iota_{E_I} \operatorname{vol}_e$, so that $e^J \wedge \eta_I = \delta^J_I \operatorname{vol}_e$. For a field of dimensional type $[\psi] = L^{-3/2}$, we take the minimal material action

$$S_D = \int_M \left\{ \frac{\hbar}{2}\left[\bar{\psi} g^I D\psi - (D\bar{\psi}) g^I \psi\right] \wedge \eta_I - mc\,\bar{\psi}\psi \operatorname{vol}_e \right\}. \tag{133.3}$$

Here $\hbar = \hbar_{\text{int}}$, $c = c_{\text{int}}$ and the mass m is the reading of the material species in the chosen chart. These values are outputs preceding this realization. The matrix A_D is Hermitian and $A_D g^I$ is anti-Hermitian. Therefore the symmetric kinetic term has the real convention corresponding to (133.1). In this signature, the torsion-free equation has principal term $\hbar g^I D_I$; the equivalent notation $i\hbar\Gamma^I D_I$, with $\Gamma^I = -i g^I$, uses the Clifford relation of opposite signature. This specification jointly fixes the adjoint, signature and imaginary factor of the action.

The material fields and co-tetrad remain fixed in the variation of the connection. A preparation coming from the minimal extension of section 130 can determine the initial datum of the field or state. Each action preserves its own variation: if a connection-dependent minimizing field is substituted into an action, its composite differential is also applied, or the envelope theorem under its stationarity hypotheses. Action (133.3) here realizes a coupling affine in contortion.

Proposition 133.1 (Spinorial current with variational normalization)**.** *Let* $B^{IJK} = \bar\psi g^{[I} g^J g^{K]}\psi$*, where brackets denote normalized antisymmetrization. The current defined in* (131.4) *by action* (133.3) *is*

$$\sigma_{JK} = -\frac{\hbar}{2}\,\bar\psi\{g^I, \mathbb{B}_{JK}\}\psi\,\eta_I = -\frac{\hbar}{2}\,B^I{}_{JK}\eta_I. \tag{133.4}$$

It is independent of contortion when e *and* ψ *are fixed. The inversion of section 131.3 therefore applies to this current.*

Proof. The variation of Ω is $\delta\Omega = \frac{1}{2}\delta\omega^{JK}\mathbb{B}_{JK}$. The two kinetic terms give, respectively, the product with g^I on the left and on the right. Consequently,

$$\delta_\omega S_D = \frac{\hbar}{4}\int_M \delta\omega^{JK}\wedge\eta_I\,\bar\psi\{g^I, \mathbb{B}_{JK}\}\psi.$$

Comparison with (131.4) fixes the factor and sign of (133.4). Expanding the anticommutator and using the Clifford relation cancels the degree-one contractions, leaving $\{g^I, \mathbb{B}_{JK}\} = g^{[I}g_J g_{K]}$. The action is linear in Ω, proving the last statement. □

133.2 Effective contraction and second-moment coefficient

The Lorentzian quadratic form of the trivector is written

$$q(B) = \frac{1}{3!}B^{IJK}B_{IJK} = -(B^{012})^2 - (B^{013})^2 - (B^{023})^2 + (B^{123})^2. \tag{133.5}$$

This contraction preserves the signature of the spinorial module. Its second moment depends on the state and on the realization of field products.

Theorem 133.2 (Effective interaction of the spinorial current)**.** *With material action* (133.3)*, algebraic elimination of contortion gives*

$$\mathcal{L}^{\mathrm{eff}}_{\mathrm{spin}} = C_\gamma q(B)\,\mathrm{vol}_e, \qquad C_\gamma = \frac{3\kappa c\hbar^2}{16}\frac{\gamma^2}{1+\gamma^2}. \tag{133.6}$$

The coefficient is obtained from the current and the Holst inverse with the normalizations of the preceding action.

Proof. By (131.16), it suffices to evaluate $-\frac{1}{4}K_*^{IJ}\wedge\sigma_{IJ}$, with

$$Y^{IJ} = \kappa c\frac{\gamma^2}{1+\gamma^2}(-\star + \gamma^{-1}I)\sigma^{IJ}, \qquad K_* = \mathsf{C}_e(Y).$$

Here C_e is the explicit composition of (131.9) with (131.11). Linearity of those two inverses reduces the calculation to the parts $-\star$ and I. To give the complete contraction, we order the trivector components as $(012, 013, 023, 123)$, extract the factors $\kappa c\hbar^2\gamma^2/(1+\gamma^2)$, and substitute $\sigma_{JK} = -\frac{1}{2}B^I{}_{JK}\eta_I$. The two matrices of the remaining quadratic form are

$$\mathcal{B}_{-\star} = \frac{3}{16}\,\mathrm{diag}(-1,-1,-1,1), \qquad \mathcal{B}_I = 0_{4\times4}. \tag{133.7}$$

These matrices are obtained using only $e^I \wedge \eta_J = \delta^I_J \,\mathrm{vol}_e$: for each component B^{abc}, we form σ, calculate Y, then $T^I = \sum_J \iota_{E_J} Y^{IJ} - \frac{1}{4} e^I \wedge \sum_{L,J} \iota_{E_L}\iota_{E_J} Y^{LJ}$, and finally $K_{IJK} = (T_{KIJ} - T_{IJK} - T_{JKI})/2$. The contraction gives the four diagonal entries shown; the six polarizations between distinct components are zero for both operators. This calculates all coefficients of the two quadratic forms. The part $\gamma^{-1} I$ vanishes and the part $-\star$ produces exactly (133.5), with the coefficient of (133.6). □

133.3 Finite fermionic realization and second moment of the state

We fix a periodic comoving cell, an orthonormal frame adapted to its observer and the homogeneous spatial mode of the four spinorial components. This finite realization has state space $\mathcal{F} = \bigoplus_{n=0}^{4} \Lambda^n \mathbb{C}^4$. Let $a_r, a_r^\dagger$, $r = 1, \dots, 4$ be its canonical operators, with $\{a_r, a_s^\dagger\} = \delta_{rs}$ and $\{a_r, a_s\} = 0$. The occupation vacuum fixes normal ordering. For a matrix M, we write

$$\mathrm{d}\Gamma(M) = \sum_{r,s} a_r^\dagger M_{rs} a_s, \qquad \widehat{N} = \mathrm{d}\Gamma(I_4).$$

The letter Γ in this notation denotes second quantization; nonadic holonomy retains its own notation.

The four matrices of the bilinears are calculated directly:

abc	012	013	023	123
$M_{abc} = A_D g^a g^b g^c$	$iJ \otimes R$	$iJ \otimes Z$	$iI_2 \otimes J$	$iZ \otimes J$

(133.8)

All are Hermitian, have zero trace and satisfy $M_{abc}^2 = I_4$. The quartic observable corresponding to (133.5), in this declared normal ordering, is

$$\begin{aligned} \widehat{\mathcal{Q}}_{\mathrm{sp}} &= \sum_{a<b<c} \eta_{aa}\eta_{bb}\eta_{cc} :\mathrm{d}\Gamma(M_{abc})^2: \\ &= \sum_{a<b<c} \eta_{aa}\eta_{bb}\eta_{cc} \mathrm{d}\Gamma(M_{abc})^2 + 2\widehat{N}. \end{aligned} \tag{133.9}$$

Normal ordering is part of the specified finite realization; its definition makes explicit the subtraction of single-occupation contractions.

Proposition 133.3 (Evaluation by the canonical relations)**.** *The operator $\widehat{\mathcal{Q}}_{\mathrm{sp}}$ is self-adjoint, preserves each occupation sector and has restrictions*

$$\widehat{\mathcal{Q}}_{\mathrm{sp}}|_{\Lambda^0 \oplus \Lambda^1} = 0, \qquad \widehat{\mathcal{Q}}_{\mathrm{sp}}|_{\Lambda^3} = 4I_4, \qquad \widehat{\mathcal{Q}}_{\mathrm{sp}}|_{\Lambda^4} = 8I_1. \tag{133.10}$$

In the basis of $\Lambda^2\mathbb{C}^4$ whose occupation subsets are $(12, 13, 23, 14, 24, 34)$*, its matrix is*

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 2 & 0 & 0 & 6 & 0 \\ 0 & 0 & 2 & -6 & 0 & 0 \\ 0 & 0 & -6 & 2 & 0 & 0 \\ 0 & 6 & 0 & 0 & 2 & 0 \\ -4 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{133.11}$$

Proof. The relation $a_s a_t^\dagger = \delta_{st} - a_t^\dagger a_s$ applied to $\mathrm{d}\Gamma(M)^2$ separates its single-occupation contraction and gives

$$:\mathrm{d}\Gamma(M)^2: = \mathrm{d}\Gamma(M)^2 - \mathrm{d}\Gamma(M^2).$$

Since the first three signs of (133.8) are negative and the last positive, the sum of contractions is $-2\widehat{N}$. This proves (133.9). Self-adjointness and preservation of occupation follow from those same expressions.

On Λ^0 all terms vanish. On Λ^1, $\mathrm{d}\Gamma(M) = M$, and the squares sum to $-2I_4$, which cancels with $2\widehat{N}$. On Λ^4, $\mathrm{d}\Gamma(M) = \operatorname{tr} M = 0$, leaving eight. Identification by complement between $\Lambda^3\mathbb{C}^4$ and the dual of $\mathbb{C}^4$ takes $\mathrm{d}\Gamma(M)$ to $(\operatorname{tr} M)I - M^{\mathsf{T}} = -M^{\mathsf{T}}$. Their squares again sum to $-2I_4$, while $2\widehat{N} = 6I_4$; therefore the restriction is $4I_4$. Finally, applying the four explicit matrices to the six exterior products of the indicated basis produces (133.11). The operator is thus evaluated in the sixteen dimensions of $\mathcal{F}$. □

For a state operator $\varrho \geq 0$, $\operatorname{Tr}\varrho = 1$, define $q_\varrho = \operatorname{Tr}(\varrho\widehat{\mathcal{Q}}_{\mathrm{sp}})$. Its dependence on second moments is explicit:

$$q_\varrho = \sum_{a<b<c} \eta_{aa}\eta_{bb}\eta_{cc} \left\{ \langle \widehat{B}_{abc} \rangle_\varrho^2 + \operatorname{Var}_\varrho(\widehat{B}_{abc}) - \langle \widehat{N} \rangle_\varrho \right\}, \qquad \widehat{B}_{abc} = \mathrm{d}\Gamma(M_{abc}). \tag{133.12}$$

Covariance and normal subtraction are preserved even when the mean currents vanish. In particular, if P_3 is the projection onto $\Lambda^3\mathbb{C}^4$,

$$\varrho_3 = \tfrac{1}{4}P_3, \qquad \langle \widehat{B}_{abc} \rangle_{\varrho_3} = 0, \qquad q_{\varrho_3} = 4 > 0. \tag{133.13}$$

It is a positive state of trace one and constitutes an explicit witness to the positive domain of the functional. The two-occupation sector also contains negative values; for example, the vector $(|12\rangle + |34\rangle)/\sqrt{2}$ has value -4. The sign of the intensity thus comes from the contraction and the state, and remains separate from the selection of cosmological conditions.

133.4 Density, pressure and preservation of the comoving amplitude

In the spatially homogeneous realization, fix

$$e^0 = N(t)c\,\mathrm{d}t, \qquad e^j = a(t)\,\mathrm{d}x^j, \qquad \mathcal{V}(t) = a(t)^3 V_c,$$

with lapse $N(t) > 0$, dimensionless scale factor $a(t) > 0$ and comoving volume $V_c > 0$. The frame and periodic mode of subsection 133.3 remain declared. The normalization of the homogeneous field is $\widehat{\psi} = \mathcal{V}^{-1/2}(a_1, a_2, a_3, a_4)^{\mathsf{T}}$. Each bilinear carries a factor $\mathcal{V}^{-1}$, and its normally ordered product carries $\mathcal{V}^{-2}$. The symmetric kinetic temporal form preserves this canonical normalization: terms arising from differentiating $\mathcal{V}^{-1/2}$ cancel between its two summands.

Theorem 133.4 (Torsional amplitude as a functional of the state)**.** *Keeping the canonical state and the realization of the product fixed during metric variations, the effective spin term has reduced action*

$$S_{\mathrm{spin}}^{\mathrm{hom}} = \int \mathrm{d}t\, N(t) \frac{cC_\gamma}{a(t)^3 V_c}\, q_\varrho. \tag{133.14}$$

Its energy density and isotropic pressure are

$$\rho_{\mathrm{spin}} = p_{\mathrm{spin}} = -\frac{cC_\gamma}{V_c^2}\, q_\varrho a^{-6}. \tag{133.15}$$

In any evolution preserving q_ϱ, the corresponding constant amplitude is

$$\boxed{s_0[\varrho, V_c] = \frac{3\kappa c^2\hbar^2}{16} \frac{\gamma^2}{1+\gamma^2} \frac{\operatorname{Tr}(\varrho\widehat{\mathcal{Q}}_{\mathrm{sp}})}{V_c^2}.} \tag{133.16}$$

Proof. Spatial integration of (133.6) multiplies the factor $\mathcal{V}^{-2}$ by $Nc\mathcal{V}\,\mathrm{d}t$, giving (133.14). In canonical variables, lapse variation defines energy through $\delta_N S_{\mathrm{m}} = -\int E\,\delta N\,\mathrm{d}t$. Therefore,

$$E_{\mathrm{spin}} = -\frac{cC_\gamma q_\varrho}{a^3 V_c}, \qquad \rho_{\mathrm{spin}} = \frac{E_{\mathrm{spin}}}{a^3 V_c}.$$

The isotropic spatial variation satisfies $\delta_a S_{\mathrm{m}} = \int 3Na^2 V_c p\, \delta a\, \mathrm{d}t$. Applied to (133.14), it gives

$$3Na^2 V_c p_{\mathrm{spin}} = -\frac{3NcC_\gamma q_\varrho}{a^4 V_c}.$$

This simultaneously proves the sign, pressure and density of (133.15). Since q_ϱ is preserved, the coefficient of $-a^{-6}$ is constant and we obtain (133.16). Dimensionality is consistent: $[cC_\gamma]$ = energy L^3, so s_0 has the dimension of energy density. □

Corollary 133.5 (Preserved domain of positive amplitude)**.** *Let there be a unitary evolution of $\mathcal{F}$ preserving $\Lambda^3\mathbb{C}^4$. For every initial state of trace one supported on that sector, $q_{\varrho(t)} = 4$ holds, and*

$$s_0 = \frac{3\kappa c^2\hbar^2}{4V_c^2}\frac{\gamma^2}{1+\gamma^2} > 0. \tag{133.17}$$

Proof. Invariance of the sector preserves the support of the state; on it, (133.10) gives $\widehat{\mathcal{Q}}_{\mathrm{sp}} = 4I$. Preservation of the trace then fixes its expectation at four, independently of the internal evolution of its coherences. Positivity of the couplings, $V_c > 0$ and $\gamma \neq 0$ proves the formula and its sign. □

For general states, preservation of $\langle \widehat{N} \rangle$ determines dilution of the occupation density, $n(t) = \langle \widehat{N} \rangle / (a^3 V_c)$. Preservation of the quartic moment is checked separately. If $i\hbar\dot{\varrho} = [H(t), \varrho]$, then

$$\frac{\mathrm{d}q_\varrho}{\mathrm{d}t} = \frac{i}{\hbar}\,\mathrm{Tr}\big(\varrho[H(t), \widehat{\mathcal{Q}}_{\mathrm{sp}}]\big). \tag{133.18}$$

Commutation, invariance of a sector where the observable is scalar, or a conservation proved for the specific state provide the relevant conditions. If the moment evolves, (133.15) preserves its instantaneous evaluation; its balance includes

$$\dot{\rho}_{\mathrm{spin}} + 6H\rho_{\mathrm{spin}} = -\frac{cC_\gamma}{V_c^2}\, a^{-6}\dot{q}_\varrho, \qquad H = \dot{a}/a$$

in proper time. This term records exchange with the remaining material degrees of freedom.

The composition is determined: spinorial representation of transport, material action, variational current, Cartan–Holst inversion, second moment and normalized density. The domain $s_0 > 0$ of section 134 thus has an evaluable functional and explicit witness states. Its use in cosmological dynamics additionally preserves the material, geometric and conservation conditions stated in that section.

Chapter 134

Memory dilution and persistence of the bounce

The evolution of memory and spatial expansion are two readings of the same sequence of states. The refinement index can be eliminated between them; that elimination determines the dilution exponent transmitted by the cosmological realization to its torsional contribution. The initial amplitude depends on evaluation of the concrete material current in its variational response. The scaling law and bounce theorem that follow preserve that dependence.

Theorem 133.4 determines that amplitude as a functional of the material state and comoving volume. Corollary 133.5 provides an explicit domain of states with positive, conserved amplitude. In that domain, evaluation of s_0 precedes the cosmological integration developed below. The ordinary density $\rho_0 > 0$ and its barotropic law are additionally specified in section 134.2: the witness to positivity of the amplitude does not in itself evaluate that material density.

134.1 The a^{-6} dilution law

Proposition 134.1 (Elimination of the memory index)**.** *Let $a_{\mathrm{ref}} > 0$, $\varphi > 1$, and let the route readers be*

$$\frac{a_n}{a_{\mathrm{ref}}} = \varphi^{n/3}, \qquad M_n = \varphi^{-2n}.$$

Then

$$M_n = \left(\frac{a_n}{a_{\mathrm{ref}}}\right)^{-6}$$

for every index of the sequence. If the realization of the quadratic density is $s_n^2 = s_*^2 M_n$, with $s_*^2 > 0$, we obtain

$$s_n^2 = s_*^2 \left(\frac{a_n}{a_{\mathrm{ref}}}\right)^{-6}.$$

Proof. Raising the first equality to the power -6 gives φ^{-2n}, which is exactly the second reader. Multiplying by the initial amplitude reproduces the second statement. The exponent is determined by the quotient of the exponents of the two readers; the amplitude is the evaluation of the reference state. □

In the continuous homogeneous realization, this constitutive law is preserved: the torsional density is written $s_0 a^{-6}$, where the normalization of the factor a, the material coefficient and the reference amplitude have been absorbed into s_0. The scaling law and evaluation of s_0 are successive operations of the construction.

134.2 Homogeneous threshold and transversality

Consider the homogeneous, spatially flat chart, in units $c = 1$, with constant $G > 0$, positive densities ρ_0, s_0 and constant barotropic parameter $w < 1$. The specified realization satisfies $\Lambda_{\mathrm{eff}} = 0$ and the equations

$$\rho_{\mathrm{m}}(a) = \rho_0 a^{-3(1+w)}, \qquad \rho_{\mathrm{s}}(a) = s_0 a^{-6}, \tag{134.1}$$

$$H^2 = \frac{8\pi G}{3}\big(\rho_{\mathrm{m}} - \rho_{\mathrm{s}}\big) =: F_0(a), \qquad \dot{H} = -4\pi G\big((1+w)\rho_{\mathrm{m}} - 2\rho_{\mathrm{s}}\big). \tag{134.2}$$

The couplings belong to the dimensional section already constructed. The expression for ρ_{s} incorporates the sign of its contribution in this gravitational realization.

Theorem 134.2 (Existence and uniqueness of the transverse threshold)**.** *There exists a unique $a_{\mathrm{b}} > 0$ with $F_0(a_{\mathrm{b}}) = 0$:*

$$a_{\mathrm{b}}^{3(1-w)} = \frac{s_0}{\rho_0}.$$

The region allowed by $H^2 \geq 0$ is $a \geq a_{\mathrm{b}}$. At the threshold, with $\rho_{\mathrm{b}} = \rho_{\mathrm{m}}(a_{\mathrm{b}}) = \rho_{\mathrm{s}}(a_{\mathrm{b}}) > 0$,

$$\dot{H}_{\mathrm{b}} = 4\pi G(1-w)\rho_{\mathrm{b}} > 0.$$

The solution crossing $H = 0$ has a strict minimum of a, with local expansion

$$a(t) = a_{\mathrm{b}}\left[1 + 2\pi G(1-w)\rho_{\mathrm{b}}(t-t_{\mathrm{b}})^2 + O((t-t_{\mathrm{b}})^3)\right].$$

Proof. The factorization

$$F_0(a) = \frac{8\pi G}{3}a^{-6}\left(\rho_0 a^{3(1-w)} - s_0\right)$$

reduces its zeros to that of a strictly increasing function of $(0,\infty)$ on $(-s_0,\infty)$. This proves existence, uniqueness and the sign of F_0 on both sides of the threshold. The second equation of (134.2), evaluated at equality of densities, gives the formula for $\dot{H}_{\mathrm{b}}$. Since $\dot{a} = aH$, at the threshold $\ddot{a}_{\mathrm{b}} = a_{\mathrm{b}}\dot{H}_{\mathrm{b}} > 0$. The equations are smooth for $a > 0$, so their Taylor expansion gives the indicated expansion. The sign change of H is transverse: it passes from contraction to expansion. □

For $w = 1$, both terms have the same exponent and equality of amplitudes determines a degenerate case. For $w > 1$, the sign of $\dot{H}$ at a coincidence of densities is negative. The domain $w < 1$ of the theorem expresses exactly the order between the two dilution exponents.

134.3 Local persistence under controlled perturbations

Persistence of the threshold uses two independent estimates: control of the sign at the endpoints and of the derivative on the interval. Let $I = [a_-, a_+] \subset (0,\infty)$, with $a_- < a_{\mathrm{b}} < a_+$, be chosen so that

$$F_0(a_-) < 0 < F_0(a_+), \qquad m_I := \inf_{a\in I} F_0'(a) > 0.$$

Such an interval exists because

$$F_0'(a_{\mathrm{b}}) = \frac{8\pi G}{a_{\mathrm{b}}}(1-w)\rho_{\mathrm{b}} > 0.$$

Define $\delta_I = \min\{-F_0(a_-), F_0(a_+)\} > 0$.

Theorem 134.3 (Local persistence of the transverse bounce)**.** *Let $R \in C^1(I)$ be a perturbation satisfying*

$$\|R\|_{\infty,I} < \delta_I, \qquad \|R'\|_{\infty,I} < m_I.$$

Then $F_R = F_0 + R$ has a unique zero $a_R \in (a_-, a_+)$, and $F_R'(a_R) > 0$. The realization $H^2 = F_R(a)$ admits a local transverse bounce, with

$$\dot{H}_R = \frac{a_R}{2}F_R'(a_R) > 0$$

at the threshold. If $R(a,\eta)$ is jointly C^1, the persistent zeros depend locally in a C^1 manner on the parameter η.

Proof. The first bound preserves $F_R(a_-) < 0 < F_R(a_+)$; the intermediate value theorem provides an interior zero. The second gives $F_R'(a) \geq m_I - \|R'\|_{\infty,I} > 0$ on I, so the zero is unique and simple.

At $a > a_R$ near the zero, $F_R(a) = F_R'(a_R)(a - a_R) + o(a - a_R)$. The integral

$$t - t_{\mathrm{b}} = \int_{a_R}^{a} \frac{d\xi}{\xi\sqrt{F_R(\xi)}}$$

is finite and strictly increasing. Its inverse defines the expanding branch; time reflection defines the contracting branch. The two join with $\dot{a}(t_{\mathrm{b}}) = 0$ and

$$a(t) - a_R = \frac{a_R^2 F_R'(a_R)}{4}(t - t_{\mathrm{b}})^2 + o((t - t_{\mathrm{b}})^2).$$

Away from the threshold, differentiating $H^2 = F_R(a)$ gives $\dot{H} = aF_R'(a)/2$; its continuous limit gives the positive expression at a_R. The C^1 regularity of F_R guarantees continuity of acceleration at this junction. Finally, $\partial_a F_R(a_R) > 0$ allows application of the implicit function theorem when the perturbation depends jointly on (a, η). □

The first bound controls existence; the second ensures local uniqueness and transversality. The proved stability corresponds to that threshold in the selected interval. The torsional amplitude enters its location, while the relation between exponents determines the temporal orientation of the crossing.

Chapter 135

Explicit homogeneous solution and causal completeness

Determination of the amplitude allows a complete realization of the bounce to be developed, in addition to the local argument. The spatially flat chart of (134.2) is retained, in units $c = 1$, with $\rho_0, s_0, G > 0$, and the material sector is specialized to dust, $w = 0$, with $\Lambda_{\mathrm{eff}} = 0$. The coefficient s_0 is that obtained from the material state in the corresponding realization; the following integration uses that coefficient and preserves its domain.

Theorem 135.1 (Exact integration of the dust bounce)**.** *Set*

$$a_{\mathrm{b}}^3 = \frac{s_0}{\rho_0}, \qquad \tau^2 = \frac{s_0}{6\pi G \rho_0^2}, \qquad u = t - t_{\mathrm{b}}.$$

The solution of the homogeneous system with $a(t_{\mathrm{b}}) = a_{\mathrm{b}}$ and $H(t_{\mathrm{b}}) = 0$ is, for every $t \in \mathbb{R}$,

$$a(t) = a_{\mathrm{b}} \left(1 + \frac{u^2}{\tau^2}\right)^{1/3}, \qquad H(t) = \frac{2u}{3(\tau^2 + u^2)}, \qquad \dot{H}(t) = \frac{2(\tau^2 - u^2)}{3(\tau^2 + u^2)^2}. \tag{135.1}$$

It has a unique minimum, at t_{b}; it contracts before and expands afterward. The evolution is smooth and satisfies $a(t) \geq a_{\mathrm{b}} > 0$.

Proof. Let $y = a^3$. The Friedmann equation gives

$$\dot{y}^{\,2} = 24\pi G(\rho_0 y - s_0).$$

The function $y = s_0/\rho_0 + 6\pi G \rho_0 u^2$ satisfies that identity and the initial conditions. Its derivatives give the three expressions of (135.1). To check the evolution equation as well, let us write

$$\rho_{\mathrm{b}} = \frac{\rho_0^2}{s_0}, \qquad x = \frac{u}{\tau}.$$

Then

$$\rho_{\mathrm{m}} = \frac{\rho_{\mathrm{b}}}{1 + x^2}, \qquad \rho_{\mathrm{s}} = \frac{\rho_{\mathrm{b}}}{(1 + x^2)^2}, \qquad 4\pi G \rho_{\mathrm{b}} = \frac{2}{3\tau^2}.$$

Substituting into $\dot{H} = -4\pi G(\rho_{\mathrm{m}} - 2\rho_{\mathrm{s}})$ gives exactly the third formula. Thus the complete system $(\dot{a}, \dot{H})$ is satisfied, whose field is smooth for $a > 0$. Its local uniqueness, continued along the explicit solution, fixes the evolution through those initial conditions. The denominator is positive for every t, and the statements about the minimum and the sign follow from H. □

135.1 Finite curvature and extension of geodesics

The metric realized in this chart is $g = -dt^2 + a(t)^2\, d\mathbf{x}^2$ on $\mathbb{R} \times \mathbb{R}^3$. With the convention in which the flat curvature scalar is $R = 6(\dot{H} + 2H^2)$, the solution gives

$$R(t) = \frac{4\tau^2 + \frac{4}{3}u^2}{(\tau^2 + u^2)^2}.$$

In particular, $R(t_{\mathrm{b}}) = 4/\tau^2$ is finite. In an orthonormal frame, the Riemann components depend on H^2 and on $\dot{H} + H^2$, which are bounded functions on $\mathbb{R}$. The same holds for polynomial contractions of those components; for example,

$$R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} = 12\big[(\dot{H} + H^2)^2 + H^4\big].$$

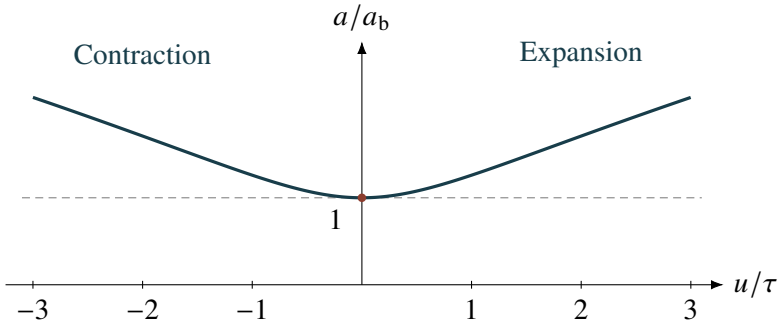


Figure 135.1: Dimensionless profile of the proved solution: $a/a_{\mathrm{b}} = (1 + (u/\tau)^2)^{1/3}$. The parameters a_{b} and τ are calculated from ρ_0, s_0, G; the drawing represents the entire rescaled family.

Proposition 135.2 (Completeness of causal geodesics)**.** *The preceding metric realization is geodesically complete for the timelike and null geodesics of its Levi–Civita connection, in both directions.*

Proof. Spatial translations provide the constants $p_i = a^2\, dx^i/d\lambda$. Causal normalization gives

$$\left|\frac{dt}{d\lambda}\right| = \sqrt{\epsilon + \frac{|\mathbf{p}|^2}{a(t)^2}}, \qquad \epsilon = 1 \text{ in the timelike case}, \quad \epsilon = 0 \text{ in the null case}.$$

A nonconstant null geodesic has $|\mathbf{p}| > 0$. Since $a \geq a_{\mathrm{b}}$, in the timelike case

$$\left|\frac{d\lambda}{dt}\right| \geq \left(1 + \frac{|\mathbf{p}|^2}{a_{\mathrm{b}}^2}\right)^{-1/2} > 0,$$

and in the null case $|d\lambda/dt| = a/|\mathbf{p}| \geq a_{\mathrm{b}}/|\mathbf{p}| > 0$. Thus, reaching $t = \pm\infty$ requires infinite affine length. On a compact time interval, a and its derivatives are smooth and a is bounded away from zero. The preceding explicit equations and $dx^i/d\lambda = p_i/a^2$ allow the geodesic to be continued through its finite endpoints. Both possibilities of continuation are covered. □

The completeness result belongs to this homogeneous dust realization, with its spatial, matter and amplitude conditions. Its local persistence under perturbations of the threshold preserves the bounds of theorem 134.3; that local property and the global completeness of the explicit solution are distinct conclusions. The minimum scale comes from the relation between material density and torsional response. Memory thus enters a verifiable geometric evolution, with a regular realization of contraction and expansion.

Chapter 136

Residual tensor, trace decomposition, and covariant balance

The geometric realization of the enriched state provides a metric, while the material transducer publishes the source to which the reading refers. The residual tensor preserves the difference between the two publications in the same tensor space. Its decomposition distinguishes the sector proportional to the metric from the traceless sector, and determines the exchange current corresponding to that partition.

136.1 Chart data and construction of the residual tensor

Let us fix an HMT state x and a connected chart $\mathcal{U}$ of its geometric realization. On $\mathcal{U}$, the nondegenerate co-tetrad of section 127 determines a smooth metric $g = g_x$ of signature $(-+++)$. We denote by $\mathring{\nabla}$ its Levi–Civita connection and write

$$\mathsf{G}_{\mu\nu}[g] := \mathrm{Ric}_{\mu\nu}[g] - \tfrac{1}{2}R[g]g_{\mu\nu}.$$

This distinguishes the Einstein tensor $\mathsf{G}[g]$ from the gravitational coupling G_{int}. The derivative used in the balances of this section is $\mathring{\nabla}$, consistently with the effective metric equation of Section 131.5.

We work in the dimensional chart $c_{\mathrm{int}} = 1$. We assume $G_{\mathrm{int}}(x) > 0$ and $\Lambda_{\mathrm{ref}}(x)$ constant on $\mathcal{U}$, and put $\kappa_x = 8\pi G_{\mathrm{int}}(x)$. The symbol x identifies the state supplying those readers; covariant differentiation acts on the points of $\mathcal{U}$. The genealogy of the couplings remains in section 121.

Let T_b be the smooth symmetric tensor published by the chart's material transducer; in the baryonic reading, it is its baryonic source. We define

$$\mathcal{E}^{\mathrm{ref}}_{\mu\nu} := \kappa_x^{-1}\big(\mathsf{G}_{\mu\nu}[g] + \Lambda_{\mathrm{ref}}g_{\mu\nu}\big), \qquad T^{\mathrm{res}}_{\mu\nu} := \mathcal{E}^{\mathrm{ref}}_{\mu\nu} - T_{b,\mu\nu}. \tag{136.1}$$

The tensor T^{res} is the effective residual tensor, also denoted T_{osc} when interpreting the geometric sector remaining to be published relative to T_b. The construction preserves $g_x, G_{\mathrm{int}}, \Lambda_{\mathrm{ref}}$ and T_b as effective arguments.

Theorem 136.1 (Uniqueness of the residual and balance with the material source)**.** *For the preceding data there exists a unique symmetric tensor T^{res} satisfying $\mathcal{E}^{\mathrm{ref}} = T_b + T^{\mathrm{res}}$. Its divergence is*

$$\mathring{\nabla}^{\mu}T^{\mathrm{res}}_{\mu\nu} = -\mathring{\nabla}^{\mu}T_{b,\mu\nu}. \tag{136.2}$$

In particular, if the material source is conserved separately, $\mathring{\nabla}^{\mu}T_{b,\mu\nu} = 0$, then T^{res} is also conserved.

Proof. The difference of two tensors satisfying the stated equation is zero; expression (136.1) provides its existence. The contracted Bianchi identity and Levi–Civita metric compatibility give

$$\mathring{\nabla}^{\mu}\mathsf{G}_{\mu\nu} = 0, \qquad \mathring{\nabla}g = 0.$$

Since κ_x and Λ_{ref} are constant on the chart, $\mathring{\nabla}^{\mu}\mathcal{E}^{\mathrm{ref}}_{\mu\nu} = 0$. Differentiating the difference defining T^{res} proves (136.2), including the statement under separate material conservation. □

The spatial and temporal constancy of the couplings in the chart fixes the domain of this balance. A family of states may publish different couplings in different charts; each preceding identity applies to the chart and state specifying it.

136.2 Canonical decomposition and exchange current

Define the trace function and the two tensors

$$\theta := g^{\mu\nu}T^{\mathrm{res}}_{\mu\nu}, \qquad T^{\Lambda}_{\mu\nu} := \tfrac{1}{4}\theta g_{\mu\nu}, \qquad T^{0}_{\mu\nu} := T^{\mathrm{res}}_{\mu\nu} - T^{\Lambda}_{\mu\nu}. \tag{136.3}$$

The superscript Λ identifies the component proportional to the metric; Λ_{ref} is the reference parameter of (136.1). Their functions are distinct.

Proposition 136.2 (Trace projectors and uniqueness of the decomposition)**.** *On symmetric covariant tensors of order two, the maps*

$$P_{\mathrm{tr}}(S) = \tfrac{1}{4}\,\mathrm{tr}_g(S)g, \qquad P_0(S) = S - P_{\mathrm{tr}}(S)$$

satisfy

$$P_{\mathrm{tr}}^2 = P_{\mathrm{tr}}, \quad P_0^2 = P_0, \quad P_{\mathrm{tr}}P_0 = P_0P_{\mathrm{tr}} = 0, \quad P_{\mathrm{tr}} + P_0 = I. \tag{136.4}$$

In particular, $T^{\mathrm{res}} = T^{\Lambda} + T^{0}$, $\mathrm{tr}_g\, T^{0} = 0$*, and this decomposition is unique.*

Proof. In dimension four, $g^{\mu\nu}g_{\mu\nu} = 4$. Therefore, $\mathrm{tr}_g(P_{\mathrm{tr}}S) = \mathrm{tr}_g\, S$, which proves idempotence of P_{tr}, vanishing of the trace of P_0S and, by expansion, the other identities. If $S = fg + Q$ with $\mathrm{tr}_g\, Q = 0$, taking the trace gives $f = \mathrm{tr}_g\, S/4$. This also determines $Q = S - fg$, and with it uniqueness. □

Theorem 136.3 (Covariant exchange between the two sectors)**.** *On the domain of* (136.1)*, let* $b_\nu := \mathring{\nabla}^{\mu}T_{b,\mu\nu}$*. The trace-exchange current is the one-form*

$$\mathcal{J}^{\mathrm{tr}}_{\nu} := \tfrac{1}{4}\partial_\nu\theta = \mathring{\nabla}^{\mu}T^{\Lambda}_{\mu\nu}. \tag{136.5}$$

The complete balances are

$$\mathring{\nabla}^{\mu}T^{0}_{\mu\nu} = -b_\nu - \mathcal{J}^{\mathrm{tr}}_{\nu}, \qquad \mathring{\nabla}^{\mu}(T^{\Lambda} + T^{0})_{\mu\nu} = -b_\nu. \tag{136.6}$$

If $b = 0$*, the two sectors exchange exactly opposite currents. In that case they are conserved separately if and only if* θ *is constant in the connected chart.*

Proof. Metric compatibility allows the calculation

$$\mathring{\nabla}^{\mu}\big(\tfrac{1}{4}\theta g_{\mu\nu}\big) = \tfrac{1}{4}g^{\mu\alpha}(\partial_\alpha\theta)g_{\mu\nu} = \tfrac{1}{4}\partial_\nu\theta.$$

Subtracting this equality from (136.2) produces (136.6). With $b = 0$, separate conservation is equivalent to $\mathrm{d}\theta = 0$; a smooth function with zero differential is constant on each connected component. □

The one-form $\mathcal{J}^{\mathrm{tr}}$ measures the exchange of the tensor partition. It retains a type distinct from the transport covector of section 128 and from the spin current $\sigma_{IJ} \in \Omega^3(\mathcal{U}; \Lambda^2E^*)$ of (131.4).

136.3 Link to the effective Cartan–Holst source

On a solution of (131.19), in the same units $c = 1$, the residual has expression

$$T^{\mathrm{res}} = T^{\mathrm{phys,LC}} + U^{\mathrm{phys,spin}} - T_b + \frac{\Lambda_{\mathrm{ref}} - \Lambda_{\mathrm{int}}}{\kappa_x}\, g. \tag{136.7}$$

Indeed, substituting $\mathsf{G} + \Lambda_{\mathrm{int}}g = \kappa_x(T^{\mathrm{phys,LC}} + U^{\mathrm{phys,spin}})$ into (136.1) gives equality term by term. When the material transducer publishes precisely $T_b = T^{\mathrm{phys,LC}}$ and $\Lambda_{\mathrm{ref}} = \Lambda_{\mathrm{int}}$, we obtain $T^{\mathrm{res}} = U^{\mathrm{phys,spin}}$. The separate conservation condition of T_b in theorem 136.1 retains its own function: (131.20) establishes the balance of the total source, and the partition among its components depends on the material action used.

136.4 Trace sector and acceleration of the selected charts

The TRIT reader of state x publishes $s = \tau(x) \in \{+1, 0, -1\}$, and its positive clock publishes

$$H_* = \frac{2\pi}{108 t_0(x)} > 0. \tag{136.8}$$

Let h_s be a three-dimensional metric of constant sectional curvature s. The three charts of the selected realization are written

$$g_s = -\mathrm{d}t^2 + a_s(t)^2 h_s, \qquad a_+(t) = H_*^{-1}\cosh(H_* t), \quad a_0(t) = \exp(H_* t), \quad a_-(t) = H_*^{-1}\sinh(H_* t). \tag{136.9}$$

The closed and flat charts are considered for $t \in \mathbb{R}$; the open chart, for $t > 0$. The sign s and the clock H_* retain their antecedents in the state selecting the realization.

Lemma 136.4 (Curvature of the three charts of the same clock)**.** *For the functions in* (136.9)*,*

$$\frac{\ddot{a}_s}{a_s} = H_*^2, \qquad \left(\frac{\dot{a}_s}{a_s}\right)^2 + \frac{s}{a_s^2} = H_*^2, \qquad \mathrm{Ric}[g_s] = 3H_*^2 g_s. \tag{136.10}$$

Consequently, $R[g_s] = 12H_*^2$ *and* $\mathsf{G}[g_s] = -3H_*^2 g_s$*.*

Proof. The first two equalities are obtained by differentiating the three functions: in the closed and open charts we use $\cosh^2 z - \sinh^2 z = 1$, and in the flat chart $\dot{a}_0 = H_* a_0$. To verify the tensor identification, the connection coefficients involving the time coordinate are

$$\Gamma^0{}_{ij} = a_s \dot{a}_s (h_s)_{ij}, \qquad \Gamma^i{}_{0j} = (\dot{a}_s/a_s)\delta^i_j,$$

and the purely spatial ones coincide with those of h_s. Contraction of the curvature gives

$$\mathrm{Ric}_{00} = -3\ddot{a}_s/a_s, \qquad \mathrm{Ric}_{0i} = 0, \qquad \mathrm{Ric}_{ij} = (a_s\ddot{a}_s + 2\dot{a}_s^2 + 2s)(h_s)_{ij}.$$

Substituting the first two identities of (136.10) produces $\mathrm{Ric}_{00} = -3H_*^2$ and $\mathrm{Ric}_{ij} = 3H_*^2 a_s^2 (h_s)_{ij}$. Taking the trace and forming the Einstein tensor proves the other statements, with the curvature convention of (131.19). □

Theorem 136.5 (Density, pressure, and sign of the selected acceleration)**.** *In the preceding charts, for* $\Lambda_{\mathrm{ref}} = 0$ *and* $T_b = 0$*, the residual tensor is*

$$T^{\mathrm{res}} = -\rho_* g_s, \qquad \rho_* := \frac{3H_*^2}{8\pi G_{\mathrm{int}}} > 0, \qquad T^\Lambda = T^{\mathrm{res}}, \quad T^0 = 0. \tag{136.11}$$

For a unit comoving observer $u = \partial_t$*, the density and isotropic pressure are* ρ_* *and* $p_* = -\rho_*$*. The acceleration equation evaluates to*

$$\frac{\ddot{a}_s}{a_s} = -\frac{4\pi G_{\mathrm{int}}}{3}(\rho_* + 3p_*) = H_*^2 > 0. \tag{136.12}$$

Proof. The preceding lemma and (136.1) give $T^{\mathrm{res}} = -(3H_*^2/\kappa_x) g_s$. Since $g_s(u,u) = -1$, $T^{\mathrm{res}}_{\mu\nu} u^\mu u^\nu = \rho_*$. The spatial projector $q^{\mu\nu} = g_s^{\mu\nu} + u^\mu u^\nu$ satisfies $q^{\mu\nu}(g_s)_{\mu\nu} = 3$, and therefore

$$p_* := \tfrac{1}{3} q^{\mu\nu} T^{\mathrm{res}}_{\mu\nu} = -\rho_*.$$

The trace equals $-4\rho_*$, so (136.3) produces the two sectors of (136.11). Finally, $\rho_* + 3p_* = -2\rho_*$, and substitution into the right-hand side of (136.12) gives $\kappa_x \rho_*/3 = H_*^2$, equal to the geometric value already calculated. Positivity uses exactly $G_{\mathrm{int}} > 0$ and $H_* > 0$. □

In this realization, the trace sector publishes a constant density and an opposite pressure, and the internal clock fixes positive acceleration. In a general chart, T^{Λ} describes the pointwise isotropic component, while T^0 preserves the traceless residual and both obey the exchange of (136.6). Identification of these sectors with observational dark sources requires a single realization to reproduce expansion, lensing, rotation, structure growth and the cosmic background jointly. The present construction fixes the tensor, its balances and the acceleration of the selected charts entering that comparison.

Chapter 137

Pure bipartite states, reduced entropy and horizon area laws

The enriched state preserves routes and their records during evolution; a subsystem reading determines which part of that information remains accessible. The construction of this section uses the angular channels of section 14.7, the incidence and the area operator of section 122, and the action, area and temperature units of sections 121 and 123. These are prior publications of the same APP–TRIT–TPK construction. Each here transmits its domain, orientation, normalization and realization conditions.

Three spaces are distinguished: the binary sheet record, the tritic factors of sectorial incidence, and the geometric realization of the exterior regions of a horizon. Purification, transport of observables, and dimensional conversion specify their relationships. This distinction preserves the structure accompanying each entropic value.

137.1 Sheet record and bipartite quantum purification

Write $a = A_{\mathrm{rad}}$ and $y = C^{*}_{\mathrm{rad}}$ for the angular coordinates already constructed. The channels and their probabilities are

$$\mathfrak{q}_{\pm} = e^{-a\mp y}, \qquad p_{\pm} = \frac{\mathfrak{q}_{\pm}}{\mathfrak{q}_{+}+\mathfrak{q}_{-}} = \frac{e^{\mp y}}{2\cosh y}. \tag{137.1}$$

The signs index these two channels. On $\mathcal{H}_{\mathrm{h}} = \mathrm{span}\{|+\rangle, |-\rangle\}$, define

$$\rho_{\mathrm{h}}(y) = p_{+}|+\rangle\langle+| + p_{-}|-\rangle\langle-|.$$

The auxiliary space $\mathcal{H}_{\mathrm{ar}} = \mathrm{span}\{|\mathrm{adv}\rangle, |\mathrm{ret}\rangle\}$ carries two orthonormal labels. With a phase χ_0 fixed, the joint state is

$$|\Psi_{\mathrm{ar}}\rangle = \sqrt{p_{+}}|+\rangle \otimes |\mathrm{adv}\rangle + e^{i\chi_0}\sqrt{p_{-}}|-\rangle \otimes |\mathrm{ret}\rangle. \tag{137.2}$$

Proposition 137.1 (Accessible information in the sheet record)**.** *The vector* (137.2) *is normalized, and its partial trace over* $\mathcal{H}_{\mathrm{ar}}$ *is* $\rho_{\mathrm{h}}(y)$*. Its entanglement entropy, expressed in nats, is*

$$s_{\mathrm{h}}(y) = \log(2\cosh y) - y\tanh y. \tag{137.3}$$

Exchange of the labels $+$ *and* $-$ *conjugates* $\rho_{\mathrm{h}}(y)$ *to* $\rho_{\mathrm{h}}(-y)$ *and preserves* s_{h}*.*

Proof. The squared norm is $p_{+} + p_{-} = 1$. Orthogonality of the auxiliary labels eliminates the cross terms in the partial trace. The squared Schmidt coefficients are precisely $p_{\pm}$. By (137.1), $p_{+} - p_{-} = -\tanh y$; substituting $\log p_{\pm} = \mp y - \log(2\cosh y)$ into $-\sum_{\pm} p_{\pm}\log p_{\pm}$ proves the formula. Exchange of signs permutes the two eigenvalues and preserves entropy. □

The common factor e^{-a} cancels upon normalization of the binary record. The relative phase χ_0 remains in the joint state even though this diagonal reduction does not express it. Logical reversibility concerns transport of the complete record, with its labels and correlations; phase return preserves the memory advance described in Section 125. A thermal implementation additionally has its own energy and dissipation law. In particular, s_{h} measures this binary subsystem and its concrete bipartition.

137.2 Quantum purification of the incidence sectors

The boundary incidence constructed earlier distinguishes two sets of eighteen links, with sectorial labels $m = 90, 120$. Each link carries $\mathbb{C}^3$, with basis $|0\rangle, |1\rangle, |2\rangle$ and occupation operator $\mathrm{diag}(0, 1, 2)$. Let N_m be their sectorial sums, and

$$\mathsf{N} = \mathsf{N}_{90} + \mathsf{N}_{120}, \qquad \mathsf{D}_\partial = 90\mathsf{N}_{90} + 120\mathsf{N}_{120}. \tag{137.4}$$

In the positive branch, γ_{BI} denotes the Barbero–Immirzi parameter already obtained, and $a_0 > 0$ its angular areal factor. This notation distinguishes the parameter from the connection and its holonomies. Area is published as

$$\widehat{\mathcal{A}}_m = \ell_P^2 \gamma_{\mathrm{BI}} a_0 \mathsf{N}_m, \qquad \widehat{\mathcal{A}} = \widehat{\mathcal{A}}_{90} + \widehat{\mathcal{A}}_{120}. \tag{137.5}$$

For the finite parameter $\Delta = \Delta_{\mathrm{mod}} \in \mathbb{R}$ of the reduced state, set

$$x_m = e^{-m\Delta}, \qquad z_m = 1 + x_m + x_m^2, \qquad Z(\Delta) = z_{90}^{18} z_{120}^{18}, \tag{137.6}$$

$$\rho_\partial(\Delta) = Z(\Delta)^{-1} e^{-\Delta \mathsf{D}_\partial}. \tag{137.7}$$

The trace in this section is the ordinary matrix trace, and the states have trace one. The parameter Δ preserves the selection and normalization of the antecedent state.

Proposition 137.2 (Tensor purification of the boundary)**.** *For a finite real weight w, let*

$$|\,\mathrm{TFD}(w)\rangle = \frac{|0\rangle|0\rangle + e^{-w/2}|1\rangle|1\rangle + e^{-w}|2\rangle|2\rangle}{\sqrt{1 + e^{-w} + e^{-2w}}}. \tag{137.8}$$

The product

$$|\Psi_\Delta\rangle = |\,\mathrm{TFD}(90\Delta)\rangle^{\otimes 18} \otimes |\,\mathrm{TFD}(120\Delta)\rangle^{\otimes 18} \tag{137.9}$$

is a normalized purification of $\rho_\partial(\Delta)$.

Proof. At each link, the sum of the squares of the three coefficients is one. The partial trace eliminates terms with different indices and returns $\mathrm{diag}(1, e^{-w}, e^{-2w})/(1 + e^{-w} + e^{-2w})$. Partial trace commutes with the tensor product. Multiplying the thirty-six reductions gives the exponential and partition function of (137.7). □

137.3 Informational equation and areal fluctuations

The occupation means and variances per link are

$$f_m = \frac{x_m + 2x_m^2}{z_m}, \qquad v_m = \frac{x_m + 4x_m^2}{z_m} - f_m^2. \tag{137.10}$$

Write $s(\rho) = -\mathrm{Tr}(\rho \log \rho)$, with $0 \log 0 = 0$. In the preceding purification, $s(\rho_\partial)$ is its entanglement entropy.

Proposition 137.3 (Mean area, entropy, and sectorial response)**.** *For real finite Δ, one has*

$$\langle \mathsf{N} \rangle_\Delta = 18(f_{90} + f_{120}), \tag{137.11}$$

$$\langle \mathsf{D}_\partial \rangle_\Delta = 18(90 f_{90} + 120 f_{120}), \tag{137.12}$$

$$\langle \widehat{\mathcal{A}} \rangle_\Delta = 18 \ell_P^2 \gamma_{\mathrm{BI}} a_0 (f_{90} + f_{120}), \tag{137.13}$$

$$s(\Delta) = \log Z(\Delta) + \Delta \langle \mathsf{D}_\partial \rangle_\Delta, \tag{137.14}$$

$$\frac{\mathrm{d}\langle \mathsf{D}_\partial \rangle_\Delta}{\mathrm{d}\Delta} = -\,\mathrm{Var}_\Delta(\mathsf{D}_\partial) = -18(90^2 v_{90} + 120^2 v_{120}), \tag{137.15}$$

$$\frac{\mathrm{d}s}{\mathrm{d}\Delta} = -\Delta\,\mathrm{Var}_\Delta(\mathsf{D}_\partial), \tag{137.16}$$

$$\frac{\mathrm{d}\langle\widehat{\mathcal{A}}\rangle_\Delta}{\mathrm{d}\Delta} = -18\ell_P^2\gamma_{\mathrm{BI}}a_0(90v_{90}+120v_{120}). \tag{137.17}$$

Proof. The local weights are $(1, x_m, x_m^2)/z_m$, whose first two moments give f_m and v_m. Additivity of occupation operators and factorization of the state give the first three formulas. Taking the trace of $-\rho_\partial\log\rho_\partial = \rho_\partial(\log Z + \Delta\mathsf{D}_\partial)$ gives the entropy. Differentiation of the finite sums yields $\partial_\Delta\log Z = -\langle\mathsf{D}_\partial\rangle_\Delta$ and $\partial_\Delta^2\log Z = \mathrm{Var}_\Delta(\mathsf{D}_\partial)$. Independent factors add their variances. In the derivative of (137.14), the two first-moment terms cancel. Finally, $\partial_\Delta f_m = -mv_m$ gives the area response. □

Loss of uniformity is quantified in dimension $d_\partial = 3^{36}$:

$$I_\partial(\Delta) = 36\log 3 - s(\Delta) = \mathcal{D}\big(\rho_\partial(\Delta)\|I/d_\partial\big) \geq 0. \tag{137.18}$$

Here $\mathcal{D}$ denotes relative entropy, defined and justified below. Substituting $\log(I/d_\partial) = -36\log 3\, I$ proves the identity. Convexity of $t\log t$ gives positivity; equality corresponds to the uniform distribution, which in this family occurs exactly at $\Delta = 0$. The deficit measures dimensionless information; the scale $\gamma_{\mathrm{BI}}a_0\ell_P^2$ measures area per occupation.

Theorem 137.4 (Occupation inversion and complementary area)**.** *Let $\mathcal{R}_\partial$ be the involution exchanging $|0\rangle$ and $|2\rangle$ and fixing $|1\rangle$ in each factor. Then*

$$\mathcal{R}_\partial\mathsf{N}\mathcal{R}_\partial^* = 72I - \mathsf{N}, \tag{137.19}$$

$$\mathcal{R}_\partial\mathsf{D}_\partial\mathcal{R}_\partial^* = 7560I - \mathsf{D}_\partial, \tag{137.20}$$

$$\rho_\partial(-\Delta) = \mathcal{R}_\partial\rho_\partial(\Delta)\mathcal{R}_\partial^*, \tag{137.21}$$

$$s(-\Delta) = s(\Delta), \tag{137.22}$$

$$\langle\widehat{\mathcal{A}}\rangle_{-\Delta} + \langle\widehat{\mathcal{A}}\rangle_\Delta = 72\gamma_{\mathrm{BI}}a_0\ell_P^2, \tag{137.23}$$

$$\mathcal{D}\big(\rho_\partial(\Delta)\|\rho_\partial(-\Delta)\big) = \Delta\big(7560 - 2\langle\mathsf{D}_\partial\rangle_\Delta\big). \tag{137.24}$$

Proof. Each occupation transforms by $n \mapsto 2-n$. In each sector, $\mathsf{N}_m \mapsto 36I - \mathsf{N}_m$, so the constant term of the weighted reader is $36(90+120) = 7560$. Moreover, $z_m(-\Delta) = e^{2m\Delta}z_m(\Delta)$ and $Z(-\Delta) = e^{7560\Delta}Z(\Delta)$. Substitution into the normalized state proves conjugation. Its spectrum preserves entropy; the transformation of N gives complementary area. Subtracting the logarithms of the two states and taking the expectation in $\rho_\partial(\Delta)$ gives the final equality. □

At $\Delta = 0$, $s = 36\log 3$. As $\Delta \to +\infty$, zero occupation concentrates the state and entropy tends to zero; inversion carries that limit to maximal occupation. For $\Delta > 0$, entropy decreases strictly because the weighted reader has more than one eigenvalue and the state has full rank. The complete configuration, area, and entropy thus preserve differentiated roles.

137.4 Modular law and finite area variations

Fix $\rho_0 = \rho_\partial(\Delta_0)$, with real finite Δ_0, and hold $\Delta_0, \gamma_{\mathrm{BI}}, a_0, \ell_P$ constant. The normalized areas $\mathcal{A}_m^{\mathrm{n}} = \gamma_{\mathrm{BI}}a_0\mathsf{N}_m$ express the modular operator as

$$\mathsf{K}_0 = -\log\rho_0 = c_0 I + \sum_{m=90,120}\beta_m\mathcal{A}_m^{\mathrm{n}}, \quad \beta_m = \frac{m\Delta_0}{\gamma_{\mathrm{BI}}a_0}, \quad c_0 = \log Z(\Delta_0). \tag{137.25}$$

One has $\beta_{120} = \frac{4}{3}\beta_{90}$, including when both are zero. The ratio is $4/3$ when $\Delta_0 \neq 0$. K_0 is the logarithm of this reference state; its type distinguishes it from temporal transport and the gravitational connection.

Theorem 137.5 (First modular law and finite correction)**.** *For any normalized state σ on the same space, let $\Delta_\sigma\langle B\rangle = \mathrm{Tr}[(\sigma-\rho_0)B]$. Then*

$$s(\sigma)-s(\rho_0)=\sum_{m=90,120}\beta_m\Delta_\sigma\langle\mathcal{A}^{\mathrm{n}}_m\rangle-\mathcal{D}(\sigma\|\rho_0), \tag{137.26}$$

where $\mathcal{D}(\sigma\|\rho_0)=\mathrm{Tr}[\sigma(\log\sigma-\log\rho_0)]$ is finite and nonnegative. If $\sigma(\varepsilon)$ is differentiable, normalized, and $\sigma(0)=\rho_0$, one obtains

$$\left.\frac{\mathrm{d}s(\sigma(\varepsilon))}{\mathrm{d}\varepsilon}\right|_0=\sum_{m=90,120}\beta_m\left.\frac{\mathrm{d}\langle\mathcal{A}^{\mathrm{n}}_m\rangle_{\sigma(\varepsilon)}}{\mathrm{d}\varepsilon}\right|_0. \tag{137.27}$$

Proof. By definition, $\mathcal{D}(\sigma\|\rho_0)=-s(\sigma)+\mathrm{Tr}(\sigma\mathsf{K}_0)$, whereas $s(\rho_0)=\mathrm{Tr}(\rho_0\mathsf{K}_0)$. Subtraction and (137.25) prove the finite identity, since normalization cancels the term $c_0 I$.

For positivity, diagonalize σ with eigenvalues p_i and vectors u_i, and ρ_0 with eigenvalues $q_j>0$ and vectors v_j. If $r_i=\langle u_i,\rho_0 u_i\rangle>0$, concavity of the logarithm gives

$$\sum_j|\langle u_i,v_j\rangle|^2\log q_j\le\log r_i.$$

Consequently, $\mathcal{D}(\sigma\|\rho_0)\ge\sum_i p_i\log(p_i/r_i)\ge 0$. The final inequality is convexity of $t\log t$ with weights r_i, whose sum is one. Faithfulness of ρ_0 keeps its logarithms finite; zero eigenvalues of σ are interpreted through $0\log 0=0$.

For the infinitesimal formula, in a neighbourhood of the faithful state ρ_0, the derivative of the trace of $-\sigma\log\sigma$ is $-\mathrm{Tr}[\dot\sigma(\log\rho_0+I)]$. This identity follows from the spectral derivative of the trace. The condition $\mathrm{Tr}\,\dot\sigma=0$ leaves $\mathrm{Tr}(\dot\sigma\mathsf{K}_0)$, and substitution of (137.25) completes the proof. □

For physical areas, the coefficients are β_m/ℓ_P^2. Applying the common transducer k_B gives $S=k_B s$ and preserves the term $k_B\mathcal{D}$ in full. The derivative in (137.17) varies Δ; law (137.27) holds Δ_0 fixed and varies the state around the reference. These are different operations.

137.5 Action, decision energy, and areal unit

We fix the same oriented action section and its gravitational and thermal sections, on the positive branch. The antecedents provide

$$\begin{aligned}\ell_P^2&=\frac{\hbar G}{c^3},&\qquad t_P&=\frac{\ell_P}{c},\\ T_{108}&=2\pi t_P,&\qquad h&=2\pi\hbar,\\ E_P&=\frac{\hbar}{t_P}=k_B\Theta_{\mathrm{clk}}\log 3.\end{aligned} \tag{137.28}$$

The thermal equality preserves the previously constructed pair and normalization; changing its dimensional basis transports k_B and Θ_{clk} jointly. The elementary area increment is $\delta\mathcal{A}=\gamma_{\mathrm{BI}}a_0\ell_P^2$.

Proposition 137.6 (Conversion coefficient between decision and area)**.** *The ratio of decision energy to the areal increment satisfies*

$$\begin{aligned}\mathcal{T}_{I/A}:=\frac{E_P}{\delta\mathcal{A}}&=\frac{k_B\Theta_{\mathrm{clk}}\log 3}{\gamma_{\mathrm{BI}}a_0\ell_P^2}\\ &=\frac{c^3}{\gamma_{\mathrm{BI}}a_0 G t_P}=\frac{c^4}{\gamma_{\mathrm{BI}}a_0 G\ell_P}.\end{aligned} \tag{137.29}$$

Its dimension is energy per area.

Proof. Substitution of $E_P = \hbar/t_P$ and $\ell_P^2 = \hbar G/c^3$ cancels the common action section. The identity $\ell_P = ct_P$ gives the final expression. Positivity of the factors ensures that all quotients are defined. □

The capacity 81 of the cubic cut and the bipartition of the thirty- six sectoral links correspond to different domains. In the cubic graph $\{0, \dots, N-1\}^3$, the N^2 parallel lines between two opposite faces are edge-disjoint: every cut intercepts the N^2, and a transverse layer attains that bound. For $N = 9$, the uniform product state of the trits of that layer has $s_{\mathrm{corte}} = 81 \log 3$. Assigning a decision of the same cycle to each link publishes

$$S_{\mathrm{corte}}^{\mathrm{fis}} = 81 k_B \log 3, \qquad E_{\mathrm{corte}}^{(1)} = 81 E_P = \Theta_{\mathrm{clk}} S_{\mathrm{corte}}^{\mathrm{fis}}. \tag{137.30}$$

The proof combines the minimum cut, additivity of product-state entropy, and (137.28). In the sectorial state, the relevant finite equation remains (137.26), with its relative entropy.

137.6 Cosmological normalization of the horizon

For an areal radius $R > 0$, the cosmological realization considered uses the triad

$$E_\Lambda(R) = \frac{c^4 R}{2G}, \qquad S_{\mathrm{H}}(R) = \frac{k_B \pi R^2}{\ell_P^2}, \qquad T_{\cos}(R) = \frac{\hbar c}{2\pi k_B R}. \tag{137.31}$$

The factor 2π belongs to this temperature normalization. The area $\mathcal{A}_{\mathrm{H}} = 4\pi R^2$ expresses the same entropy as $S_{\mathrm{H}}/k_B = \mathcal{A}_{\mathrm{H}}/(4\ell_P^2)$.

Theorem 137.7 (Gravitational half-action cell)**.** *In* (137.31)*,*

$$T_{\cos}(R) S_{\mathrm{H}}(R) = E_\Lambda(R), \tag{137.32}$$

$$T_{\cos}(\ell_P) S_{\mathrm{H}}(\ell_P) = \frac{E_P}{2} = \frac{k_B \Theta_{\mathrm{clk}} \log 3}{2}, \tag{137.33}$$

$$E_\Lambda(\ell_P) t_P = \frac{\hbar}{2}, \qquad E_\Lambda(\ell_P) T_{108} = \frac{h}{2}. \tag{137.34}$$

Keeping the chart constants fixed and varying R,

$$T_{\cos}\, \mathrm{d}S_{\mathrm{H}} = 2\, \mathrm{d}E_\Lambda, \qquad S_{\mathrm{H}}\, \mathrm{d}T_{\cos} = -\mathrm{d}E_\Lambda. \tag{137.35}$$

Proof. Multiplying temperature and entropy gives $\hbar c R/(2\ell_P^2) = c^4 R/(2G)$. At $R = \ell_P$, the result is $\hbar/(2t_P) = E_P/2$. The two durations in (137.28) give the action relations. For the differentials, $\mathrm{d}E_\Lambda = c^4 \mathrm{d}R/(2G)$, $\mathrm{d}S_{\mathrm{H}} = 2\pi k_B R \mathrm{d}R/\ell_P^2$, and $\mathrm{d}T_{\cos} = -T_{\cos} \mathrm{d}R/R$. Substitution proves both identities. □

The product $E_\Lambda = T_{\cos} S_{\mathrm{H}}$ expresses a state identity. Its radial variations preserve both terms of (137.35); a complete thermodynamic interpretation additionally specifies the corresponding time generator, work and material realization. In this chart, action and gravitation fix the unit ℓ_P^2, while Boltzmann converts information into dimensional entropy. Transport of sections with $G\hbar$ fixed preserves that unit when the section c is kept fixed, and preserves S_{H}/k_B at fixed radius.

In the cell $R = \ell_P$, $S_{\mathrm{H}}/k_B = \pi$, so the effective multiplicity is $\Omega_{\mathrm{eff}} = e^\pi$. For the tritic hyperbolic involution H_{trit}, with spectrum $\{1, -1\}$, diagonalization gives $\mathrm{spec}(e^{\pi H_{\mathrm{trit}}}) = \{e^\pi, e^{-\pi}\}$. This spectral relation specifies the expansive value; a microscopic identification of both spaces additionally preserves their realization map and the state.

137.7 Kruskal chart and exchange of the exteriors

Let us fix a published state x, its radius $R = R(x) > 0$ and the positive sections c, G. In the following chart, x, R, c, G are fixed parameters; t, r are the spacetime coordinates. The exterior realization of the two sheets is specified by

$$f(r) = 1 - \frac{R}{r}, \qquad \mathrm{d}s^2 = -f(r)c^2\mathrm{d}t^2 + f(r)^{-1}\mathrm{d}r^2 + r^2\mathrm{d}\Omega_2^2, \qquad r > R. \tag{137.36}$$

For the sheet reader $\sigma \in \{+1, -1\}$, define

$$r_* = r + R\log\left(\frac{r}{R} - 1\right), \tag{137.37}$$

$$\mathcal{U} = -\sigma\exp\left(-\frac{ct - r_*}{2R}\right), \qquad \mathcal{V} = \sigma\exp\left(\frac{ct + r_*}{2R}\right). \tag{137.38}$$

The symbols $\mathcal{U}, \mathcal{V}$ are coordinates and are distinguished from route-transport operators.

Theorem 137.8 (Geometric realization and sheet involution)**.** *Chart* (137.38) *realizes the two sheets as the two exteriors $\mathcal{U}\mathcal{V} < 0$ of the Kruskal extension. We have*

$$-\mathcal{U}\mathcal{V} = \left(\frac{r}{R} - 1\right)e^{r/R}, \qquad \mathrm{d}s^2 = -\frac{4R^3}{r}e^{-r/R}\,\mathrm{d}\mathcal{U}\,\mathrm{d}\mathcal{V} + r^2\mathrm{d}\Omega_2^2. \tag{137.39}$$

The involution

$$C_{\mathrm{H}} : (\sigma, \mathcal{U}, \mathcal{V}) \longmapsto (-\sigma, -\mathcal{U}, -\mathcal{V}) \tag{137.40}$$

is isometric, fixes the areal radius, exchanges the exterior regions, and preserves the horizon set $\mathcal{U}\mathcal{V} = 0$. The metric is regular at $r = R$ and satisfies the vacuum equations on the regular domain $r > 0$.

Proof. Multiplying the coordinates gives the first identity. Since $\mathrm{d}r_*/\mathrm{d}r = f^{-1}$,

$$\mathrm{d}\mathcal{U} = \frac{\mathcal{U}}{2R}(-c\,\mathrm{d}t + f^{-1}\mathrm{d}r), \qquad \mathrm{d}\mathcal{V} = \frac{\mathcal{V}}{2R}(c\,\mathrm{d}t + f^{-1}\mathrm{d}r).$$

Their product reproduces the radial and temporal part of the metric. The function $z \mapsto (z-1)e^z$ has derivative $ze^z > 0$ for $z > 0$, determining r from the product. In the exterior region one recovers $\sigma = \operatorname{sign}\mathcal{V}$ and $t = (R/c)\log(-\mathcal{V}/\mathcal{U})$. At $r = R$, the radial coefficient of the extended metric is $-4R^2/e$, finite and nonzero. Inversion preserves the product, radius, and $\mathrm{d}\mathcal{U}\mathrm{d}\mathcal{V}$, and exchanges the two sign choices.

To check the vacuum in the exterior chart, the independent mixed components of the Einstein tensor reduce to

$$\mathsf{G}^t{}_t = \mathsf{G}^r{}_r = \frac{rf' + f - 1}{r^2}, \qquad \mathsf{G}^\theta{}_\theta = \mathsf{G}^\phi{}_\phi = \frac{f'}{r} + \frac{f''}{2}.$$

With $f' = R/r^2$ and $f'' = -2R/r^3$, all vanish. The regular expression (137.39) prolongs the result through the horizons on the domain $r > 0$. □

This realization preserves $E_{\mathrm{H}} = c^4R/(2G)$ and $S_{\mathrm{H}} = k_B\pi R^2/\ell_P^2$, which depend on the areal radius. The temperature $T_{\cos}$ of (137.31) belongs to the cosmological realization declared there. The exterior chart (137.36) preserves its own temporal normalization. The geometric map establishes the exchange of exteriors; comparison with formation by collapse, radiation spectra or quasinormal modes additionally uses their respective physical conditions.

137.8 Transport of states, observables, and refinements

Let F_x denote the chart of the two exteriors just constructed. In each regular angular chart, its temporal and radial Jacobian is

$$\left|\frac{\partial(\mathcal{U},\mathcal{V})}{\partial(t,r)}\right| = \frac{cr}{2R^3}e^{r/R} > 0. \tag{137.41}$$

The identity follows from the two differentials of the preceding proof. The remaining state records are transported as fiber coordinates; the chart preserves its identity and memory.

Proposition 137.9 (Unitary equivalence of sheet and exterior readings)**.** *Let μ be a sheet measure and $\nu = (F_x)_*\mu$ its transported measure. Change of variables defines a unitary map*

$$W_{\mathrm{H}} : L^2(\mu) \longrightarrow L^2(\nu), \qquad (W_{\mathrm{H}}\psi)(z) = \psi(F_x^{-1}z). \tag{137.42}$$

With densities relative to fixed coordinate measures, the same map incorporates the corresponding Jacobian factor. Conjugation

$$\mathfrak{I}_{\mathrm{H}}(B) = W_{\mathrm{H}}BW_{\mathrm{H}}^* \tag{137.43}$$

is a unital completely positive $*$-isomorphism. It transports the state, its expectations, and its modular functional calculus. For the symmetric measure of the two sheets, it also intertwines their involutions.

Proof. The definition of pushforward measure gives $\int |\psi \circ F_x^{-1}|^2 \,\mathrm{d}\nu = \int |\psi|^2 \,\mathrm{d}\mu$. The chart is invertible in the exteriors, so the isometry is surjective. Unitary conjugation preserves product, adjoint and identity; in each matrix amplification it conjugates by $I_n \otimes W_{\mathrm{H}}$, proving complete positivity. For a state ρ, cyclicity of the trace gives $\mathrm{Tr}(W_{\mathrm{H}}\rho W_{\mathrm{H}}^* \,\mathfrak{I}_{\mathrm{H}}(B)) = \mathrm{Tr}(\rho B)$. Functional calculus implies $g(W_{\mathrm{H}}\rho W_{\mathrm{H}}^*) = W_{\mathrm{H}}g(\rho)W_{\mathrm{H}}^*$ on its domain, in particular for the logarithm and modular flow of a state faithful on its support. The equality $F_x \circ C_{\mathrm{hoja}} = C_{\mathrm{H}} \circ F_x$ proves the intertwining of involutions; the symmetric measure realizes them unitarily. □

If the refinements preserve (σ, t, r, R) and use the same isometry on the fiber registers, the inclusions j_m^{hoja} and j_m^{ext} satisfy

$$W_{\mathrm{H},m+1} j_m^{\mathrm{hoja}} = j_m^{\mathrm{ext}} W_{\mathrm{H},m}. \tag{137.44}$$

Indeed, change of variables acts on the preserved geometric coordinates and inclusion on the same fibre coordinates; evaluating both compositions gives the same function and the same transported density. The equality extends the map to the refined system without resetting the records.

For a bipartition $\mathcal{H}_A \otimes \mathcal{H}_B$, a factorized transport $U_A \otimes U_B$ satisfies

$$\rho'_A = U_A \rho_A U_A^*, \qquad s(\rho'_A) = s(\rho_A).$$

The first identity follows from partial trace and the second from the spectrum. For a general unitary, the equivalent formulation also transports the subsystem algebra: $\mathcal{B}(\mathcal{H}_A) \otimes I \mapsto U(\mathcal{B}(\mathcal{H}_A) \otimes I)U^*$. This preserves the same operational question concerning accessible information. Entropy equalities apply to states with the relevant finite entropies.

The relation between information and gravitation is expressed through distinct composable operations: the state and bipartition fix accessible information; incidence and Barbero–Immirzi fix areal resolution; action and gravitation construct the unit ℓ_P^2; Boltzmann realizes entropy dimensionally; the horizon chart transports geometry and its observables. The finite laws preserve relative entropy, and the geometric realizations preserve their measures, temporal normalization and domains.

Chapter 138

Observer, memory, and relational response

The reading operation acts on the transport published by the APP–TRIT–TPK enriched state. Its domain preserves route, sheet, orientation, carry and memory; the result accessible to the observer is a map of that domain. This distinction allows three operations to be described together: choosing an oriented representative of transport, inscribing its record in an apparatus and the local response induced by the global boundary. Each operation transmits different information. The following construction specifies their maps and the conditions allowing them to be composed.

138.1 Boundary transport and intrinsic reading

The finite publication of transport provides an oriented cycle C_n, with $n \geq 1$, vertices $0, \dots, n-1$ and edges $e_j : j \longrightarrow j+1 \pmod n$. We fix the lattice realization L of the record, its vector space $W = L \otimes_{\mathbb{Z}} \mathbb{R}$, a positive inner product and the nonzero primitive representative $v \in L$ of the distinguished residual class, in the declared chamber. The APP sum and product sheets provide the transport polarizations; TRIT orientation and TPK updating precede this linear publication. The cycle, the lattice realization and the residual class are the data this section receives from the transport constructed previously.

A 0-cochain is a function on the vertices with values in W; a 1-cochain assigns a vector to each oriented edge. The coboundary acts by

$$(d\Phi)(e_j) = \Phi(j+1) - \Phi(j).$$

We denote by $\mathbf{1}_E$ the constant scalar cochain on the edges. For a transport $\omega \in C^1(C_n; W)$, its accumulated publication Γ_ω is obtained through partial sums and affine interpolation; its closure defect is

$$B(\omega) = \Gamma_\omega(1) - \Gamma_\omega(0) = \sum_{j=0}^{n-1} \omega(e_j). \tag{138.1}$$

Theorem 138.1 (Exact decomposition and harmonic mode of the cycle)**.** *Every transport has a unique decomposition*

$$\omega = d\Phi + u\mathbf{1}_E, \qquad \Phi(0) = 0, \qquad u = \frac{1}{n}\sum_j \omega(e_j).$$

In particular, $B(d\Phi) = 0$ *and* $B(\omega) = nu$.

Proof. Let $y_j = \omega(e_j) - u$. The sum of the y_j is zero. We define $\Phi(k) = \sum_{j=0}^{k-1} y_j$; the value for $k = n$ coincides with $\Phi(0) = 0$, so the function is well defined on the cycle, including its final edge. We have $d\Phi(e_j) = y_j$. If $d\Phi + u\mathbf{1}_E = d\Psi + w\mathbf{1}_E$, the sum over edges gives $nu = nw$. Then $d(\Phi - \Psi) = 0$; connectivity of the cycle and normalization at the initial vertex imply $\Phi = \Psi$. The identity $\sum_j (\Phi(j+1) - \Phi(j)) = 0$ directly proves telescopic cancellation. □

Definition 138.2 (Admissible transport and normalized reading)**.** Transport is admissible when its harmonic mode belongs to $\mathbb{R}v$. On the space $\mathcal{T}_{\mathrm{adm}}$ of those transports we define the global reading and its density per phase:

$$\mathfrak{a}_v(\omega) = \frac{\langle B(\omega), v\rangle}{\langle v, v\rangle}, \qquad \mu_v(\omega) = \frac{\mathfrak{a}_v(\omega)}{n}. \tag{138.2}$$

We have $B(\omega) = \mathfrak{a}_v(\omega)v$.

Theorem 138.3 (Determination of the intrinsic reading)**.** *The quotient* $\mathcal{T}_{\mathrm{adm}}/dC^0(C_n; W)$ *is one-dimensional, generated by the class of* $v\mathbf{1}_E$*. The map* μ_v *is the unique linear functional on* $\mathcal{T}_{\mathrm{adm}}$ *satisfying*

$$F(\omega + d\Phi) = F(\omega), \qquad F(v\mathbf{1}_E) = 1.$$

Proof. The preceding decomposition writes each admissible transport as $d\Phi + \lambda v\mathbf{1}_E$. Its class modulo exact cochains is determined by λ. The class of $v\mathbf{1}_E$ is nonzero because its sum is $nv \neq 0$. By linearity and invariance, every functional with the indicated characteristics equals $F(\omega) = \lambda F(v\mathbf{1}_E) = \lambda$. The formula (138.2) gives precisely $\mu_v = \lambda$. The global reading is $\mathfrak{a}_v = n\mu_v$. □

Definition 138.4 (Edge Observer)**.** Given the polarization of the published transport, a boundary observer is a pair $O = (\varepsilon, \Psi)$, where $\varepsilon \in \{+1, -1\}$ and $\Psi \in C^0(C_n; W)$. Its reading action is

$$\omega^O = \varepsilon\omega + d\Psi. \tag{138.3}$$

The sign fixes the orientation and the cochain fixes the trivialization. Two polarization choices are compared through their published transport; the relation (138.3) describes exactly the representatives differing by orientation and coboundary.

Proposition 138.5 (Covariance of the reading)**.** *The action preserves admissibility and satisfies*

$$B(\omega^O) = \varepsilon B(\omega), \quad \mathfrak{a}_v(\omega^O) = \varepsilon\mathfrak{a}_v(\omega), \quad \mu_v(\omega^O) = \varepsilon\mu_v(\omega).$$

Trivializations with the same orientation publish the same closure.

Proof. The sum of the coboundary $d\Psi$ is zero. Therefore, the harmonic mode transforms into εu, which still belongs to $\mathbb{R}v$, and the defect transforms into εB. The two scalar equalities are obtained by normalized projection. □

The unit of this reading is the residual representative v, with its internal normalization. Trivialization freedom acts on the cochain representative; closure information resides in its class. The description allows observers to be compared while keeping genealogical transport, its linear publication and the scalar evaluating the residual class separate.

138.2 Boundary identity and cellular interior

Let K be a finite oriented two-dimensional cell complex, equipped with a fundamental 2-chain c_K such that $\partial c_K = [C_n]$. This condition expresses that interior incidences cancel with their multiplicities and orientations. Let $\Omega \in C^1(K; W)$ be an extension of ω and $F_\Omega = d\Omega$ its additive discrete curvature.

Theorem 138.6 (Cellular holographic identity of the reading)**.** *With the preceding data,*

$$B(\omega) = \langle \Omega, \partial c_K \rangle = \langle d\Omega, c_K \rangle, \qquad \mathfrak{a}_v(\omega) = \frac{\langle \langle d\Omega, c_K \rangle, v \rangle}{\langle v, v \rangle}. \tag{138.4}$$

In particular, $F_\Omega = 0$ implies $\mathfrak{a}_v(\omega) = 0$. If closure is nonzero, every extension of transport on that complex has nonzero discrete curvature.

Proof. The pairing between chains and cochains satisfies $\langle d\Omega, c_K \rangle = \langle \Omega, \partial c_K \rangle$. When it is expanded, each interior edge appears with the sum of the incidence numbers of the adjoining cells; the condition on ∂c_K leaves exactly the oriented boundary cycle. The restriction of Ω to that cycle is ω, whose sum is $B(\omega)$. Projection onto v gives the second formula and the flatness implication. □

The equality compares two evaluations of the same extended cochain: the accumulated boundary defect and the coboundary integrated over the interior. Its curvature takes values in W and uses the additive cellular coboundary. Realization by a geometric connection additionally preserves its own coefficient space and transport law.

138.3 Isometric inscription of the record in the apparatus

Let $\mathfrak{M}_n$ be a nonempty finite set of records published at level n, with result, route, carry, sheet and memory. Its record space is $\mathcal{H}_n = \ell^2(\mathfrak{M}_n)$, with basis $|m\rangle$. The visible update F_n has a record lift

$$\widehat{F}_n(m) = (F_n(m), m).$$

We take as records of the next level a set containing those pairs. The second coordinate preserves the antecedent and makes the lift injective. This construction specifies the memory retained by inscription, including when the visible update merges values.

In a finite apparatus space $\mathcal{K}_{A,n}$, we fix an orthonormal family $(|a_m\rangle)_{m\in\mathfrak{M}_n}$ and define

$$\iota_n : \mathcal{H}_n \longrightarrow \mathcal{K}_{A,n}, \quad \iota_n|m\rangle = |a_m\rangle, \qquad V_n : \mathcal{H}_n \longrightarrow \mathcal{H}_{n+1}, \quad V_n|m\rangle = |\widehat{F}_n(m)\rangle.$$

Let $\mathcal{P}_n = \iota_n\mathcal{H}_n$ be the pointer subspace.

Theorem 138.7 (Compatibility between updating and inscription)**.** *The maps ι_n and V_n are isometries. There exists a unique isometry $W_n^0 : \mathcal{P}_n \longrightarrow \mathcal{K}_{A,n+1}$ satisfying*

$$W_n^0\iota_n = \iota_{n+1}V_n. \tag{138.5}$$

If $\dim\mathcal{K}_{A,n+1} \geq \dim\mathcal{K}_{A,n}$, it extends to an isometry of all of $\mathcal{K}_{A,n}$. Through auxiliary complements equalizing dimensions, a unitary extension between the enlarged spaces is obtained.

Proof. The basis images under ι_n are orthonormal; the images under V_n are also orthonormal by injectivity of $\widehat{F}_n$. For $z = \iota_n\psi$ we define $W_n^0 z = \iota_{n+1}V_n\psi$. The representation of z is unique, and the three maps preserve its norm. The formula determines the map on a basis of $\mathcal{P}_n$ and proves the square. If the target dimension is sufficient, the orthogonal complement of $W_n^0\mathcal{P}_n$ contains an orthonormal family of the size of $\mathcal{P}_n^{\perp}$; sending a basis of the latter to that family constructs the extension. Finally both spaces are enlarged to the same dimension and the two prescribed families are completed to orthonormal bases. The map between those bases is unitary. □

When the record includes labels of a finite instrument, this same inscription realizes its environment: if the operators $K_{y,a} : \mathcal{H}_S \to \mathcal{H}'_S$ satisfy $\sum_{y,a} K_{y,a}^* K_{y,a} = I$, the map

$$V_{\mathrm{inst}}\psi = \sum_{y,a} K_{y,a}\psi \otimes |y, a\rangle$$

is isometric, since its squared norm is $\sum_{y,a}\langle\psi, K_{y,a}^* K_{y,a}\psi\rangle = \|\psi\|^2$. The inscription ι transports each label to its corresponding pointer. The model thus fixes the premeasurement record; a material device additionally requires its interaction and calibration.

138.4 Outcome map and record conditioning

The preceding inscription is composed with the apparatus reader on the complete genealogical state. Coupling preserves the operation that produces the correlation; the output map publishes the result; conditioning restricts the prolongations compatible with that register.

Let (Ω_S, Σ_S) and (Ω_M, Σ_M) be the measurable spaces of genealogical states of the system and apparatus, obtained from their compatible cylinders and records. A measurement context is written $\mathfrak{M}_a = (U_a, q_a, C_a, m_0)$: m_0 is the apparatus preparation, C_a specifies compatibilities and the extension horizon, and

$$U_a : \Omega_S \times \Omega_M \longrightarrow \Omega_S \times \Omega_M, \qquad q_a : \Omega_M \longrightarrow Y_a$$

are, respectively, the measurable coupling and the measurable readout into the outcome space (Y_a, Σ_{Y_a}). In a reversible regime, U_a is bijective and bimeasurable; the corresponding unitarity is also required in the Hilbert-space realization. The individual outcome is the composition

$$\mathrm{Out}_a(x) = q_a\big(\mathrm{pr}_M\, U_a(x, m_0)\big). \tag{138.6}$$

It is determined by the complete state, apparatus preparation, and context. The statistical description of a preparation is another readout of the set of compatible states.

Proposition 138.8 (Outcome measure and restriction to a fibre)**.** *Let $\Omega_a \in \Sigma_S$ be the set of states compatible with the preparation and context, endowed with a probability measure μ_a on $(\Omega_a, \Sigma_S|_{\Omega_a})$. The map $\mathrm{Out}_a|_{\Omega_a}$ is measurable and induces the outcome probability measure*

$$\nu_a(B) = \mu_a\big(\mathrm{Out}_a^{-1}(B) \cap \Omega_a\big), \qquad B \in \Sigma_{Y_a}. \tag{138.7}$$

For an outcome y whose singleton is measurable, the fibre $\Omega_{a,y} = \{x \in \Omega_a : \mathrm{Out}_a(x) = y\}$ is measurable. Whenever $\mu_a(\Omega_{a,y}) > 0$,

$$\mu_{a,y}(E) = \frac{\mu_a(E \cap \Omega_{a,y})}{\mu_a(\Omega_{a,y})} \tag{138.8}$$

for $E \in \Sigma_S|_{\Omega_a}$ defines a probability measure concentrated on that fibre.

Proof. The inclusion $x \mapsto (x, m_0)$, coupling, apparatus projection, and readout are measurable; their composition is therefore measurable. Preimages preserve disjoint unions and the whole space, so (138.7) is countably additive with total mass one. The fibre is the preimage of the singleton $\{y\}$ within Ω_a. Restricting μ_a to that fibre gives a positive, countably additive measure; division by its positive mass normalizes it. □

For a prefix s of depth n, the extensions following the record form

$$\mathrm{Fut}_{a,y}(s) = \{x \in \Omega_{a,y} : x|_n = s\}.$$

The restriction preserves the prefix already traversed and the coupling register; it modifies the family of admitted prolongations. The output (138.6) acts on complete states; the outcome law (138.7) is obtained by transporting the preparation measure through that map. Representation by a density matrix expresses the statistical level of the quantum realization and retains its conditions of correspondence with the genealogical fibers.

138.5 Accessible algebra and pointer complement

Define $P_m = |a_m\rangle\langle a_m|$, $P = \sum_m P_m$, and $P_\perp = I - P$. The algebra of the pointer subspace is $P\mathcal{B}(\mathcal{K}_{A,n})P$, whose unit is P. On it, the diagonal reader is

$$\mathcal{D}_P(A) = \sum_m P_m A P_m.$$

Proposition 138.9 (Record expectation and unital extension)**.** *The map $\mathcal{D}_P$ is a completely positive, trace-preserving and idempotent conditional expectation onto the pointer algebra, with $\mathcal{D}_P(P) = P$. Its image is $\bigoplus_m \mathbb{C}P_m$. On the algebra of the complete apparatus, the map*

$$\mathcal{E}_P(A) = \sum_m P_m A P_m + P_\perp A P_\perp \tag{138.9}$$

is a unital, completely positive, trace-preserving conditional expectation. Its image is

$$\left(\bigoplus_m \mathbb{C}P_m\right) \oplus \mathcal{B}(\mathcal{P}_n^{\perp}).$$

Both maps are orthogonal projections for the Hilbert–Schmidt inner product. Their spectra are contained in $\{0,1\}$. *When eliminated components exist, the separation between the fixed subspace and the kernel is one.*

Proof. The projectors P_m are mutually orthogonal. In the complete apparatus, the family that adds $P_\perp$ sums to I. Each term $A \mapsto P_m A P_m$ is completely positive: after any matrix amplification it acts by congruence with $I \otimes P_m$. The same holds for $P_\perp$. The sum preserves that property. Cyclicity of the trace and the sum of the projectors prove trace preservation on each domain. Orthogonality annihilates the cross terms when the map is iterated, so it is idempotent. Its fixed elements are precisely the stated blocks and the map is bimodular with respect to that subalgebra. Moreover, $\operatorname{Tr}(A^* P_m B P_m) = \operatorname{Tr}((P_m A P_m)^* B)$, from which self-adjointness for Hilbert–Schmidt follows. A self-adjoint projection has only the spectral values zero and one when both subspaces are nonzero. If only the fixed subspace remains, the map is the identity. □

The sum $\sum_m P_m A P_m$, applied without the complement to the entire apparatus, sends I to P. Its natural unit belongs to the pointer algebra. The formula (138.9) also preserves the auxiliary block; an orthonormal resolution of that block provides, when desired, diagonalization of the entire apparatus.

If U is a unitary transformation of the apparatus and $P'_m = U P_m U^*$, then

$$\mathcal{E}_{P'}(UAU^*) = U \mathcal{E}_P(A) U^*.$$

The equality is obtained by substituting each projector into the sum. Likewise, for $A \in \mathcal{B}(\mathcal{P}_n)$, the square (138.5) implies

$$\mathcal{D}_{P_{n+1}}(W_n^0 A (W_n^0)^*) = W_n^0 \mathcal{D}_{P_n}(A)(W_n^0)^*.$$

Indeed, each matrix unit $|a_m\rangle\langle a_{m'}|$ is transported to the unit corresponding to the distinct records $\widehat{F}_n(m), \widehat{F}_n(m')$; the diagonal reader preserves exactly the terms with $m = m'$.

138.6 Factorization of the response through the effective boundary

Let $\mathcal{X}_\infty^{\mathrm{enr}}$ be the domain of compatible enriched states and let $b : \mathcal{X}_\infty^{\mathrm{enr}} \longrightarrow B_\infty$ be its global boundary trace. We denote by $B_{\mathrm{ef}} = \operatorname{im} b$ the effectively published boundary. For a response space $\mathcal{V}_\nu$, we consider $I_\nu : \mathcal{X}_\infty^{\mathrm{enr}} \longrightarrow \mathcal{V}_\nu$.

Theorem 138.10 (Relational response criterion)**.** *There exists a unique map* $\overline{I}_\nu : B_{\mathrm{ef}} \to \mathcal{V}_\nu$ *such that* $I_\nu = \overline{I}_\nu \circ b$ *if and only if*

$$b(x) = b(x') \quad \Longrightarrow \quad I_\nu(x) = I_\nu(x'). \tag{138.10}$$

Uniqueness extends to all of B_∞ *when* b *is surjective.*

Proof. A composition through b is constant on each fiber. Conversely, for $\beta \in B_{\mathrm{ef}}$, we define $\overline{I}_\nu(\beta) = I_\nu(x)$ with $b(x) = \beta$. Condition (138.10) guarantees independence of the representative; every value of $\overline{I}_\nu$ is determined by some state of that fiber. If b is surjective, $B_{\mathrm{ef}} = B_\infty$. Otherwise, the composition equation determines only the values on B_{ef}. □

This criterion is set-theoretic. For a continuous factorization, the topologies and the quotient-map condition of $b : \mathcal{X}_\infty^{\mathrm{enr}} \to B_{\mathrm{ef}}$ are also preserved; with it, continuity of I_ν is equivalent to that of its factor. This specification separates determination of a reading from its regularity properties.

Let $\ell_v : \mathcal{X}_\infty^{\mathrm{enr}} \to \mathcal{L}_v$ be the restriction accessible to a local neighborhood. The effective global character is verified through two states satisfying

$$\ell_v(x) = \ell_v(x'), \qquad I_v(x) \neq I_v(x'). \tag{138.11}$$

If I_v factors through b, these two states have distinct boundaries. Moreover, the response cannot be written as a function of ℓ_v, since that function would assign the same value to both states. The difference of boundaries thus acquires a verifiable operator effect.

138.7 Ambivalence weights and self-inertial response

For the general formula we consider positive readers A_∂, V_∂ and an energy reader Q defined on the declared effective boundary domain. A realization on $B_{\mathrm{av}} \subseteq B_{\mathrm{ef}}$ is made explicit below. Memory remains as a boundary coordinate $\mathsf{M}_\partial(\beta)$. The ambivalence hinge provides the two orientations

$$r_{\mathrm{av}} = \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2}, \qquad r_{\mathrm{av}}^J = \frac{\varphi_{\mathrm{HMT}}}{\pi_{\mathrm{HMT}}^2}.$$

The superscript J expresses interchange of the generating pair.

138.7.0.1 Normalized realization of the boundary readers.

The APP–TRIT–TPK modal incidence, developed in the regional construction, provides the path envelope with matrix

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

The positive vector $v = (1, \varphi_{\mathrm{HMT}})^{\mathsf{T}}$ satisfies $F_{\mathrm{av}} v = \varphi_{\mathrm{HMT}} v$. Its normalized transition and stationary weights are

$$P_{ij} = \frac{(F_{\mathrm{av}})_{ij} v_j}{\varphi_{\mathrm{HMT}} v_i}, \qquad P = \begin{pmatrix} 0 & 1 \\ \varphi_{\mathrm{HMT}}^{-2} & \varphi_{\mathrm{HMT}}^{-1} \end{pmatrix}, \qquad (\mu_{\mathrm{ar}}, \mu_{\mathrm{vol}}) = \frac{(1, \varphi_{\mathrm{HMT}}^2)}{1 + \varphi_{\mathrm{HMT}}^2}. \tag{138.12}$$

The rows of P sum to one by $\varphi^{-2} + \varphi^{-1} = 1$; the stationary equation $\mu_{\mathrm{ar}} = \mu_{\mathrm{vol}} \varphi^{-2}$, together with normalization, determines the stated weights. This Parry measure weights the paths of the envelope; the chronological frequency $(1/3, 2/3)$ counts the instants of the modal calendar and retains its distinct function.

The common memory-amplitude realization is defined on the sector $\mathcal{X}_{\mathrm{av}} \subset \mathcal{X}_\infty^{\mathrm{enr}}$ in which the readers of the two sheets have the factorization

$$\mathcal{L}_{\mathrm{vol}}(x) = \mu_{\mathrm{vol}} \mathcal{M}(x), \qquad \mathcal{L}_{\mathrm{ar}}(x) = \pi_{\mathrm{HMT}} \mu_{\mathrm{ar}} \mathcal{M}(x), \qquad \mathcal{M}(x) > 0. \tag{138.13}$$

Both values belong to the same positive memory-amplitude line. The areal publication carries the regional mark π_{HMT} of the return. The dimensionless reading is obtained by dividing both sections by $\mathcal{L}_{\mathrm{vol}}(x)$. Thus the sums of the weights are performed between quantities of the same type; a possible realization as physical areas and volumes is then composed with their respective dimensional transducers.

Proposition 138.11 (Descent of the normalized readers and the energy clock)**.** *On $B_{\mathrm{av}} = b(\mathcal{X}_{\mathrm{av}})$, the rules*

$$A_\partial(b(x)) = \frac{\mathcal{L}_{\mathrm{ar}}(x)}{\mathcal{L}_{\mathrm{vol}}(x)} = r_{\mathrm{av}}, \qquad V_\partial(b(x)) = 1 \tag{138.14}$$

define two positive readers independent of the representative. Given an orientation a of the action section, the decision energy of (123.6) *determines on states*

$$q_a(x) = \frac{h_a(x)}{108 t_0(x)} = \frac{\hbar_a(x)}{t_P(x)} \in \mathcal{L}_E^+, \qquad t_P = \frac{54}{\pi_{\mathrm{HMT}}} t_0. \tag{138.15}$$

We write

$$b_{\mathrm{av}} := b|_{\mathcal{X}_{\mathrm{av}}} : \mathcal{X}_{\mathrm{av}} \longrightarrow B_{\mathrm{av}}.$$

There exists a unique reader $Q : B_{\mathrm{av}} \to \mathcal{L}_E^+$ such that $q_a|_{\mathcal{X}_{\mathrm{av}}} = Q \circ b_{\mathrm{av}}$ if and only if, for any $x, x' \in \mathcal{X}_{\mathrm{av}}$,

$$b(x) = b(x') \implies \frac{h_a(x)}{t_0(x)} = \frac{h_a(x')}{t_0(x')}. \tag{138.16}$$

In particular, a chart of fixed action and duration satisfies this condition and publishes $Q = h_a/(108 t_0)$.

Proof. The quotient of (138.13) cancels the common amplitude and equals $\pi_{\mathrm{HMT}} \mu_{\mathrm{ar}}/\mu_{\mathrm{vol}} = \pi_{\mathrm{HMT}}/\varphi_{\mathrm{HMT}}^2$. The result is the same for all representatives of a fiber of b_{av} within $\mathcal{X}_{\mathrm{av}}$ and is positive. Dividing both readers by the same positive factor preserves the weighted quotients in which they enter. For the clock, the factorization criterion of theorem 138.10, applied to q_a, is exactly equivalent to (138.16). Fixed sections satisfy it; so do variable sections whose quotient descends through b_{av}. Uniqueness follows from the definition of B_{av} as the effective image. □

The value of the reader Q belongs to the energy line $\mathcal{L}_E = \mathcal{L}_{\mathrm{act}} \otimes \mathcal{L}_T^*$. With E_1, E_2 on that same line, $E_1 E_2/Q^2$ is dimensionless and independent of the positive energy basis. The symbol Q in this section exclusively designates that clock energy; the quantity of heat in the Landauer realization belongs to the corresponding thermal balance. Common-amplitude factorization specifies a concrete realization of the general readers. Every other domain preserves its boundary maps and its own verification of descent. Complete memory remains in the enriched history and in its boundary publication, even though it cancels in the normalized quotient.

We define

$$w_A(\beta) = \frac{A_\partial(\beta) r_{\mathrm{av}}}{A_\partial(\beta) r_{\mathrm{av}} + V_\partial(\beta) r_{\mathrm{av}}^J}, \qquad w_V(\beta) = 1 - w_A(\beta).$$

In realization (138.14), the weights admit the exact evaluation

$$w_A = \frac{r_{\mathrm{av}}^2}{r_{\mathrm{av}}^2 + r_{\mathrm{av}}^J} = \frac{\pi_{\mathrm{HMT}}^4}{\pi_{\mathrm{HMT}}^4 + \varphi_{\mathrm{HMT}}^5}, \qquad w_V = \frac{\varphi_{\mathrm{HMT}}^5}{\pi_{\mathrm{HMT}}^4 + \varphi_{\mathrm{HMT}}^5}. \tag{138.17}$$

Substitution preserves both factors of the definition: the normalized areal reader and its subsequent weighting by r_{av}. Generation and regional marking of the return are performed once. Multiplying numerator and denominator by $\varphi_{\mathrm{HMT}}^4 \pi_{\mathrm{HMT}}^2$ gives the last expression. These weights remain constant in the common-amplitude sector. With fixed projectors and energies, a variation of the response may come from the reader Q; the witness to global dependence additionally requires the states with equal local restriction and different response of (138.11).

Let P_A, P_V be complementary orthogonal projections of a response space $\mathcal{H}_v$, and let $E_1, E_2 > 0$ be two internal energies fixed for comparison. The response is

$$\overline{I}_v(\beta) = \frac{E_1 E_2}{Q(\beta)^2} \big(w_A(\beta) P_A + w_V(\beta) P_V \big), \qquad I_v = \overline{I}_v \circ b. \tag{138.18}$$

Energies that are explicit readers of β are also admitted; in that case they are preserved as such in the formula.

Proposition 138.12 (Positivity and covariance of the response)**.** *The response* (138.18) *is self-adjoint, positive definite and constant on the fibers of b. Its eigenvalues in the nonzero sectors are*

$$\lambda_A = E_1 E_2 Q^{-2} w_A, \qquad \lambda_V = E_1 E_2 Q^{-2} w_V.$$

The complete exchange $(A_\partial, r_{\mathrm{av}}, P_A) \leftrightarrow (V_\partial, r^J_{\mathrm{av}}, P_V)$ *permutes the two summands and preserves the operator. A unitary transport of the response space conjugates that operator and preserves its spectrum with multiplicities.*

Proof. Positivity of the readers gives $0 < w_A, w_V < 1$. For $z = z_A + z_V$, with $z_A = P_A z$, $z_V = P_V z$,

$$\langle z, \bar{I}_v z\rangle = \frac{E_1 E_2}{Q^2}(w_A\|z_A\|^2 + w_V\|z_V\|^2) > 0$$

if $z \neq 0$. The orthogonal decomposition proves the spectral expressions. Exclusive dependence on β gives constancy on the fibers and factorization. Exchanging the triples exchanges numerators, weights and projectors at once, so it preserves the sum. For a unitary transport U, substitution $P_A \mapsto UP_AU^*$, $P_V \mapsto UP_VU^*$ produces $U\bar{I}_vU^*$, with the same spectrum. □

An exchange implemented within the same space by a unitary sending P_A to P_V requires the two sectors to have equal dimension. Exchanging the two triples as labels of the formula is valid with each sector's own dimension.

Condition (138.11) requires comparing the responses. For example, simultaneously multiplying A_∂, V_∂ by the same positive factor preserves both weights; if E_1, E_2, Q are also preserved, the response remains identical. In contrast, with equal weights and energies, replacing Q by qQ, $q > 0$, multiplies the response by q^{-2}. If two states with equal local restriction realize that change with $q \neq 1$, they provide the global witness. These are operator separation conditions; existence of the states is checked in the boundary domain being realized.

138.8 Connection of the joint state and acceleration reading

In a differentiable realization of the joint state $\Gamma = (x, O, m)$, let N be its realization space, ∇^{tot} its connection and $\pi_S : N \to N_S$ a differentiable projection onto the subsystem, equipped with connection ∇^S. The connection and projection preserve the provenance of the transports and readers they realize. The observed frame is obtained through

$$\mathfrak{F}_O(\Gamma) = \mathrm{Res}_O(\Gamma, \nabla^{\mathrm{tot}}).$$

For a twice-differentiable trajectory $\Gamma(t)$, write $\gamma_S = \pi_S \circ \Gamma$.

Definition 138.13 (Self-inertia and projection defect)**.** A trajectory is self-inertial with respect to the declared connection when $\nabla^{\mathrm{tot}}_{\dot\Gamma}\dot\Gamma = 0$. The defect of its projection is

$$\mathcal{K}_{\pi_S}(\dot\Gamma, \dot\Gamma) = \nabla^S_{\dot\gamma_S}\dot\gamma_S - d\pi_S\big(\nabla^{\mathrm{tot}}_{\dot\Gamma}\dot\Gamma\big). \tag{138.19}$$

Proposition 138.14 (Subsystem acceleration)**.** *For a self-inertial trajectory,*

$$a_S = \nabla^S_{\dot\gamma_S}\dot\gamma_S = \mathcal{K}_{\pi_S}(\dot\Gamma, \dot\Gamma).$$

In coordinates x^B *on* N *and* y^a *on* N_S*, the defect is quadratic in velocity, with components*

$$(\mathcal{K}_{\pi_S})^a{}_{BC} = \partial_B\partial_C\pi^a_S + (\Gamma^S)^a{}_{bc}\,\partial_B\pi^b_S\partial_C\pi^c_S - \partial_D\pi^a_S(\Gamma^{\mathrm{tot}})^D{}_{BC}. \tag{138.20}$$

Proof. The chain rule gives $\ddot\gamma^a_S = \partial_B\pi^a_S\ddot\Gamma^B + \partial_B\partial_C\pi^a_S\dot\Gamma^B\dot\Gamma^C$. Adding the contribution of ∇^S and subtracting the image of the total acceleration, the terms containing $\ddot\Gamma$ cancel. The remaining ones are (138.20) contracted with $\dot\Gamma^B\dot\Gamma^C$. When total acceleration vanishes, the stated equality results. □

The geometric response factors through b when the connection, projection and kinematic datum used in evaluation are determined by $\beta = b(x)$, or when those kinematic data are kept fixed in the comparison.

Under those conditions, (138.20) gives the same result for any representatives of a fiber; theorem 138.10 constructs its unique factor on B_{ef}. If the kinematic datum varies, it forms part of the domain of the response being compared.

The self-inertial formulation describes a geodesic trajectory of the joint state and an effective acceleration of its subsystem due to the projection defect. The gravitational realization then identifies that connection, the frame and the response with their physical observables. Observer covariance, memory inscription and factorization through the boundary thus retain complementary functions: the first determines the reading class, the second retains the genealogy and the third specifies the relational dependence of the response.

Chapter 139

Bidirectional propagation and absorption condition at the boundary

APP–TRIT–TPK transport preserves the route orientation and memory distinguishing its outward journey from its return. On that antecedent, three successive operations are constructed: the action of an enriched history, propagation of a source through the realized transports and imposition of a common condition at the boundary. Inscription of the record in an apparatus space, established in section 138.3, allows those operations to be expressed through linear maps without confusing the observable publication with the state preserving its antecedents.

Temporal direction, sheet conjugation and spatial reflection have distinct domains. Their relations are first fixed on the record; the intertwinings used by a dynamical realization are then specified. This order determines which information remains in the propagators and which condition is introduced by choosing an absorbing boundary.

139.1 Route action and record involutions

A finite route contains its positions, cardinal directions $(d_1, \dots, d_N)$, sheets $(\mu_1, \dots, \mu_N)$, with $\mu_t \in \{\Sigma, \Pi\}$, and the increments of its enriched record. For $\lambda \geq 0$, we define

$$\mathcal{A}_\lambda(\gamma) = \sum_{t=2}^{N} \big(1 - \delta_{d_t, d_{t-1}} + \lambda(1 - \delta_{\mu_t, \mu_{t-1}})\big). \tag{139.1}$$

The first term counts changes of direction; the second counts sheet changes with the declared weight. For $N \leq 1$, the sum is zero. The sector's general action additionally preserves its boundary and memory terms:

$$\mathcal{S}[\gamma] = \sum_{a \subset \gamma} \mathcal{L}_a + \mathcal{B}_{\mathrm{in}}(\partial_- \gamma) + \mathcal{B}_{\mathrm{out}}(\partial_+ \gamma) + \mathcal{M}(\gamma). \tag{139.2}$$

The terms of this expression specify a variational class with typed endpoints and preserved memory. The sector's dynamics determines $\mathcal{L}_a$, the admissible conditions and the term $\mathcal{M}$.

On the records we define C_{r} as the global exchange of sheets; $\mathcal{P}_{\mathrm{r}}$ as a reflection of the spatial chart and its directions; and $\mathcal{T}_{\mathrm{r}}$ as reversal of temporal order, replacement of each direction by its opposite and exchange of endpoints. Cardinal reflection is an involutive permutation. Inversion of the signed record is $C_M(a_1, \dots, a_N) = (-a_N, \dots, -a_1)$. The sheets, quotients and memory are transported with the complete route; the scalar reading of a word constitutes a subsequent operation.

Proposition 139.1 (Invariance of the local route action)**.** *On the preceding domain of records,*

$$\mathcal{A}_\lambda(\gamma) = \mathcal{A}_\lambda(C_{\mathrm{r}}\gamma) = \mathcal{A}_\lambda(\mathcal{P}_{\mathrm{r}}\gamma) = \mathcal{A}_\lambda(\mathcal{T}_{\mathrm{r}}\gamma) = \mathcal{A}_\lambda(C_{\mathrm{r}}\mathcal{P}_{\mathrm{r}}\mathcal{T}_{\mathrm{r}}\gamma).$$

Proof. A global permutation of the direction alphabet preserves equality between consecutive directions. Exchanging the two sheets preserves equality between consecutive sheets. Reversing the order establishes a bijection between adjacent pairs, and replacement by opposite directions also preserves their equalities. Each operation preserves the two counts of (139.1); their composition preserves them. □

The proposition determines the symmetry of the local functional. For a transformation also to act on a dynamical problem with fixed boundaries, it must transport its class of admissible prolongations and the other three terms of (139.2). In particular, two routes with the same final publication may retain distinct variational classes through their memory.

An interaction between two sections realized at a boundary B uses a coupling map

$$\mathcal{I}_B : E_B^{(1)} \otimes E_B^{(2)} \longrightarrow E_B^{(12)}$$

and a term $\mathcal{S}_{\mathrm{int}}$ in the same action. Its covariance is expressed by

$$\rho_{12}(g)\mathcal{I}_B = \mathcal{I}_B\big(\rho_1(g) \otimes \rho_2(g)\big).$$

When composing the histories, the shared boundary record is counted once; memories on both sides are transported to the composite section. The coupling and this covariance condition are data of the concrete interaction, distinct from the record involutions.

139.2 Reversal of the realized evolution

The realization of the record provides a real Hilbert space $\mathcal{H}_{\mathbb{R}}$, an orthogonal complex structure J_E, with $J_E^2 = -I$, and a self-adjoint Hamiltonian H. Its domain $\mathcal{D}(H)$ is invariant under J_E; we assume $HJ_E = J_E H$ on that domain. Let C be an orthogonal involution preserving $\mathcal{D}(H)$ and satisfying

$$CJ_EC^{-1} = -J_E, \qquad CHC^{-1} = H. \tag{139.3}$$

The internal action constant $\hbar_{\mathrm{int}} > 0$ retains its prior generation; here it normalizes the evolution group.

Theorem 139.2 (Temporal conjugation of the group)**.** *The generator $A = -J_E H/\hbar_{\mathrm{int}}$, with domain $\mathcal{D}(H)$, is skew-adjoint. Its orthogonal group, unitary with respect to J_E, satisfies*

$$U(t) = \exp(-J_E H t/\hbar_{\mathrm{int}}), \qquad CU(t)C^{-1} = U(-t). \tag{139.4}$$

Proof. Orthogonality and $J_E^2 = -I$ imply $J_E^* = -J_E$. Domain invariance and commutation with H allow H to be treated as a complex self-adjoint operator in the space defined by J_E. By functional calculus, $-J_E H/\hbar_{\mathrm{int}}$ is skew-adjoint and generates the stated group. In finite dimension, the same statements follow from the exponential series and from $A^* = -A$. The hypotheses on C give $CAC^{-1} = -A$. Conjugating the group, or its functional calculus, produces $\exp(-tA) = U(-t)$. □

The map realizing the signed involution C_M as C preserves temporal orientation and the preceding domain. Charge conjugation of a field representation additionally preserves its own characters and operators. The equality (139.4) fixes the time reversal of the group under the stated conditions.

139.3 Propagators of the chronological channel of a route

Isometric inscription of the record admits the unitary extensions of theorem 138.7. Let us fix a bilateral realization of a chronological channel with fibers $\mathcal{H}_n$, $n \in \mathbb{Z}$, and unitary transports $U_n : \mathcal{H}_n \to \mathcal{H}_{n+1}$. The choice includes auxiliary spaces when required; it preserves the route and its preceding records. We denote by $U(n,k)$ the transport from k to n, with

$$U(k,k) = I, \qquad U(n,k)U(k,l) = U(n,l), \qquad U(n,k)^* = U(k,n).$$

For $n > k$, it is the product $U_{n-1}\cdots U_k$; for $n < k$, it is the inverse of the transport from n to k.

The source space $\mathcal{J}_c$ contains finitely supported sections. The response space $\mathcal{F}$ contains all sections of those fibers. The route's quadratic action, in internal units of this realization, uses

$$(D\psi)_n = \psi_{n+1} - U_n\psi_n, \qquad E[\psi] = \sum_n \|(D\psi)_n\|^2$$

for finitely supported sections. The kinetic operator with the sign convention $L = -D^*D$ is

$$(L\psi)_n = U_n^*\psi_{n+1} - 2\psi_n + U_{n-1}\psi_{n-1}. \tag{139.5}$$

Indeed, $(D^*z)_n = z_{n-1} - U_n^* z_n$, and substitution produces that formula. The variation of E is $-2\,\mathrm{Re}\langle \delta\psi, L\psi\rangle$.

Theorem 139.3 (Exact Green operators of the covariant difference)**.** *Writing $x_+ = \max(x, 0)$, the maps*

$$(G_{\mathrm{ret}} j)_n = \sum_k (n-k)_+ U(n,k)\, j_k, \tag{139.6}$$

$$(G_{\mathrm{adv}} j)_n = \sum_k (k-n)_+ U(n,k)\, j_k \tag{139.7}$$

from $\mathcal{J}_c$ to $\mathcal{F}$ satisfy

$$L G_{\mathrm{ret}} j = L G_{\mathrm{adv}} j = j.$$

The retarded response vanishes before and at the first position of the source; the advanced one vanishes after and at the last. They are the unique solutions vanishing sufficiently far in the respective direction. For finitely supported ψ, we also have $G_{\mathrm{ret}} L\psi = G_{\mathrm{adv}} L\psi = \psi$. With respect to the pairing with finitely supported sources, $G_{\mathrm{adv}} = G_{\mathrm{ret}}^\dagger$.

Proof. Each sum has as many terms as source positions. Composition of transports reduces application of L to the scalar identity

$$(n+1-k)_+ - 2(n-k)_+ + (n-1-k)_+ = \delta_{nk}.$$

The function $(k-n)_+$ satisfies the same identity. This proves the two equations and the supports. For uniqueness, a homogeneous solution vanishing at two consecutive positions is determined as zero at the next by (139.5); the recurrence, using invertible transports, extends in both directions. The difference of two solutions with the same causal support vanishes sufficiently far away and is therefore zero. Applying this argument to ψ and to $G_{\mathrm{ret}} L\psi$, or to the advanced case, gives the right-hand identities on finite support. Finally, for two compactly supported sources, the double sum of the pairing is finite and

$$\big((k-n)_+ U(k,n)\big)^* = (k-n)_+ U(n,k).$$

Interchanging the indices proves the stated adjoint relation. □

The responses belong to $\mathcal{F}$; in general they grow linearly outside the source and do not belong to ℓ^2. The finite-support domain and its pairing specify the Green adjunction used here. The result constructs propagation of the chronological channel from its action; a realization of spatial fields additionally preserves its own operator, geometry and propagation domain.

Let C_0 be an isometric involution, linear or antilinear, of $\mathcal{H}_0$. Transport from the reference fiber constructs

$$C_n = U(-n, 0) C_0 U(0, n) : \mathcal{H}_n \longrightarrow \mathcal{H}_{-n}, \qquad (C\psi)_n = C_{-n}\psi_{-n}.$$

We have $C_{-n}C_n = I$ and $C_{n+1}U_n = U_{-n-1}^* C_n$, by composition. In particular, $C^2 = I$, and the change $(n,k) \mapsto (-n,-k)$ in the preceding sums proves

$$CL = LC, \qquad C G_{\mathrm{ret}} C^{-1} = G_{\mathrm{adv}}. \tag{139.8}$$

When C_0 is the declared realization of the record involution, these formulas transport that involution throughout the channel, without identifying phase return with a memory reset.

139.4 Causal interchange and decomposition of the response

In a field realization, let $L : \mathcal{D}(L) \subset \mathcal{H}_\Phi \to \mathcal{H}_J$ be an operator with unique retarded and advanced Green operators on the domains of admissible sources. We denote by $J^+(S)$ and $J^-(S)$ the future and past causal domains of a support S. The conditions are

$$\begin{aligned} LG_{\mathrm{ret}} = LG_{\mathrm{adv}} = I, \\ \operatorname{supp}(G_{\mathrm{ret}} j) \subset J^+(\operatorname{supp} j), \\ \operatorname{supp}(G_{\mathrm{adv}} j) \subset J^-(\operatorname{supp} j). \end{aligned}$$

Let C_Φ, C_J be isometric involutions, both linear or both antilinear, that transport these domains, reverse their causal orientations, and satisfy $LC_\Phi = C_J L$. A Green pairing is retained for which $G_{\mathrm{adv}} = G_{\mathrm{ret}}^\dagger$.

Proposition 139.4 (Conjugation of the Green operators and parities)**.** *Under the preceding conditions,*

$$C_\Phi G_{\mathrm{ret}} C_J^{-1} = G_{\mathrm{adv}}.$$

On the real space of response operators, define

$$\Theta(K) = C_\Phi K C_J^{-1}, \qquad \mathsf{P}_\pm(K) = \tfrac{1}{2}(K \pm \Theta(K)).$$

These are complementary projections for the involution Θ*, and*

$$G_{\mathrm{sym}} = \tfrac{1}{2}(G_{\mathrm{ret}} + G_{\mathrm{adv}}), \qquad G_{\mathrm{rad}} = \tfrac{1}{2}(G_{\mathrm{ret}} - G_{\mathrm{adv}}) \tag{139.9}$$

satisfy $\Theta(G_{\mathrm{sym}}) = G_{\mathrm{sym}}$, $\Theta(G_{\mathrm{rad}}) = -G_{\mathrm{rad}}$, $G_{\mathrm{sym}}^\dagger = G_{\mathrm{sym}}$, *and* $LG_{\mathrm{rad}} = 0$*. The support of the symmetric response is contained in the union of the two causal domains.*

Proof. The conjugated operator solves the source equation, since

$$LC_\Phi G_{\mathrm{ret}} C_J^{-1} = C_J L G_{\mathrm{ret}} C_J^{-1} = I.$$

Orientation reversal takes its support to the advanced domain; its uniqueness identifies it with G_{adv}. The involutions give $\Theta^2 = I$; expanding $(I \pm \Theta)^2$ and $(I + \Theta)(I - \Theta)$ proves the statements about projections. Green adjunction proves symmetry and the difference of the two source equations gives $LG_{\mathrm{rad}} = 0$. The sum of two responses has support contained in the union of their supports. □

If the involutions are antilinear, Θ is linear over real scalars; this is the operator space used in the proposition. Conjugation of the temporal group, the support condition, and the Green adjoint relation are three distinct properties, transmitted by the corresponding realization maps.

The causal reduction of the bilateral update, developed in the section on conservation and recovery, identifies another precise origin of retardation: elimination of a memory component that continues to participate in the joint evolution. Its discrete kernel preserves the order of interactions and the contribution of the initial memory. The Green operators constructed here instead solve a propagation equation with support conditions. The two constructions can be composed once the realization map identifying their domains and operators is specified; their respective causality conditions remain explicit.

139.5 Projection of absorbed currents and variation of the action

Let $\gamma_\partial : \mathcal{H}_\Phi \to \mathcal{H}_\partial$ be the boundary reading, distinct from the route γ. We define

$$\mathcal{A}_\partial = \gamma_\partial G_{\mathrm{rad}}.$$

The finite window fixes the source space and boundary reading; the Green operators preserve their response domain. At a finite level with fixed inner products, its Moore–Penrose inverse determines

$$P_{\mathrm{abs}} = I - \mathcal{A}_\partial^+ \mathcal{A}_\partial. \tag{139.10}$$

Theorem 139.5 (Absorption condition and boundary response)**.** *P_{abs} is the orthogonal projection onto* $\ker \mathcal{A}_\partial$*. For any source j, the source $j_{\mathrm{abs}} = P_{\mathrm{abs}} j$ satisfies*

$$\gamma_\partial G_{\mathrm{ret}} j_{\mathrm{abs}} = \gamma_\partial G_{\mathrm{adv}} j_{\mathrm{abs}}. \tag{139.11}$$

It is the absorbed source closest to j for the declared inner product. Nonzero absorbed sources exist exactly when $\operatorname{rank} \mathcal{A}_\partial < \dim \mathcal{H}_J$.

Proof. The orthogonal decomposition $\mathcal{H}_J = \ker \mathcal{A}_\partial \oplus \operatorname{im} \mathcal{A}_\partial^*$ shows that $\mathcal{A}_\partial^+ \mathcal{A}_\partial$ is the identity on the second summand and zero on the first. Its complement is the stated projector. By definition, $2\mathcal{A}_\partial j_{\mathrm{abs}} = \gamma_\partial (G_{\mathrm{ret}} - G_{\mathrm{adv}}) j_{\mathrm{abs}} = 0$. For $k \in \ker \mathcal{A}_\partial$, the orthogonal decomposition gives

$$\|j - k\|^2 = \|(I - P_{\mathrm{abs}}) j\|^2 + \|P_{\mathrm{abs}} j - k\|^2.$$

The unique minimum corresponds to $k = P_{\mathrm{abs}} j$. The final equivalence is the rank–nullity theorem. □

In the source-field pairing, the symmetric action is $\mathcal{S}[j] = \frac{1}{2}\langle j, G_{\mathrm{sym}} j\rangle$. The declared adjunction makes that quantity real. For real variations,

$$\mathrm{d}\mathcal{S}[j](h) = \mathrm{Re}\langle h, G_{\mathrm{sym}} j\rangle.$$

The action restricted to absorbed sources is written on the ambient space as

$$\mathcal{S}_{\mathrm{abs}}[j] = \tfrac{1}{2}\langle P_{\mathrm{abs}} j, G_{\mathrm{sym}} P_{\mathrm{abs}} j\rangle, \quad \mathrm{d}\mathcal{S}_{\mathrm{abs}}[j](h) = \mathrm{Re}\langle P_{\mathrm{abs}} h, G_{\mathrm{sym}} P_{\mathrm{abs}} j\rangle. \tag{139.12}$$

When the pairing identifies field and source through Riesz, the ambient gradient is $P_{\mathrm{abs}} G_{\mathrm{sym}} P_{\mathrm{abs}} j$. The field response remains $G_{\mathrm{sym}} j_{\mathrm{abs}}$; the gradient projector expresses restriction of the variations. The formulas are proved by expanding the quadratic term of $j + th$ and using self-adjointness. A material term $\mathcal{S}_{\mathrm{matter}}$ adds its own differential.

139.6 Explicit absorbing condition in a chronological window

The channel of subsection 139.3 allows that condition to be evaluated without introducing a target response. Let there be a source supported on $0, \dots, N$, with $N \geq 1$. Let us transport its components to the reference fiber:

$$\widehat{j}_k = U(0,k) j_k, \qquad M_0(j) = \sum_{k=0}^{N} \widehat{j}_k, \qquad M_1(j) = \sum_{k=0}^{N} k\, \widehat{j}_k.$$

The identity $(n-k)_+ - (k-n)_+ = n - k$ gives

$$(G_{\mathrm{ret}} - G_{\mathrm{adv}}) j\big|_n = U(n,0)\big(n M_0(j) - M_1(j)\big). \tag{139.13}$$

Proposition 139.6 (Absorption and two transported moments)**.** *If the trace evaluates the field at two distinct positions a, b, equality of the advanced and retarded traces is equivalent to $M_0(j) = M_1(j) = 0$. In a fibre of dimension d, the space of such sources has dimension $(N-1)d$. For $N \geq 2$, it contains the nonzero sources with transported components $(z, -2z, z, 0, \dots, 0)$, $z \neq 0$.*

Proof. Invertibility of $U(a,0)$ and $U(b,0)$ reduces the boundary condition to $aM_0 - M_1 = 0$ and $bM_0 - M_1 = 0$. Subtracting the equations gives $(a-b)M_0 = 0$, and then $M_1 = 0$. The two-moment map is surjective: for prescribed values (z_0, z_1), it suffices to take $\widehat{j_0} = z_0 - z_1$, $\widehat{j_1} = z_1$, and all other components zero. Its rank is $2d$; its nullity is $(N+1)d - 2d$. The indicated source has both moments zero. □

In the transported frame, write

$$B_N = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 0 & 1 & \cdots & N \end{pmatrix}.$$

The absorbing projector is evaluated by

$$P_N = \left[I - B_N^{\mathsf{T}} (B_N B_N^{\mathsf{T}})^{-1} B_N \right] \otimes I_{\mathcal{H}_0}. \tag{139.14}$$

The two rows of B_N are independent for $N \geq 1$, so the inverse exists. Unitary transport between frames takes this projector to the original fibers. For $N = 2$,

$$P_2 = \frac{1}{6} \begin{pmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{pmatrix} \otimes I_{\mathcal{H}_0}.$$

This construction produces a nonzero absorbed class and its projection operation in the chronological channel. Its amplitude and identification with a material current preserve the rules of the corresponding physical sector.

139.7 Refinement and continuity of the boundary condition

Let $R_m^{\Phi}, R_m^{J}, R_m^{\partial}$ be the refinement maps of fields, sources and boundaries. Suppose they intertwine the two Green operators and the trace. Composition gives

$$R_m^{\Phi} G_{\mathrm{sym},m} = G_{\mathrm{sym},m+1} R_m^{J}, \qquad R_m^{\partial} \mathcal{A}_{\partial,m} = \mathcal{A}_{\partial,m+1} R_m^{J}. \tag{139.15}$$

The second equality takes an absorbed source to another absorbed source. Naturality of the projectors additionally preserves the orthogonal structure of the realizations.

Proposition 139.7 (Naturality of the absorbing projection)**.** *For a source isometry $R = R_m^{J}$, we have $P_{\mathrm{abs},m+1} R = R P_{\mathrm{abs},m}$ exactly when*

$$R(\ker \mathcal{A}_{\partial,m}) \subset \ker \mathcal{A}_{\partial,m+1}, \qquad R((\ker \mathcal{A}_{\partial,m})^{\perp}) \subset (\ker \mathcal{A}_{\partial,m+1})^{\perp}.$$

With these conditions at every level, the projectors define an orthogonal projection in the completion of the isometric inductive limit of the source spaces.

Proof. Each source is decomposed into its components in the kernel and its orthogonal complement. The two inclusions imply that applying the projection before or after refinement respectively preserves the first component and annihilates the second. Conversely, equality of operators applied to each component gives the two inclusions. In the inductive limit, the equality allows an operator to be defined on the union of the levels without dependence on the representative. All projectors have norm at most one, so the operator extends to the completion. Its self-adjointness and idempotence identities, valid on the dense union, pass by continuity to the completion. □

To extend a Green response to a completion as well, its domains and corresponding continuity estimates are preserved. Finite-support chronological propagation and factorization of the response through the boundary image, proved in 138.10, thus retain their respective domains during refinement.

The physical realization of emission and absorption composes these operators with the material currents, action and causal conditions of the sector. The intertwining must preserve the route involution, the source-field pairing, the transported action and the boundary conditions at each level. The preceding construction provides reversal of the group, the Green operators of the chronological channel, temporal decomposition and an explicit absorbing projection; spatial realizations use their own maps and causality conditions.

Chapter 140

Action realization and metrological comparison

140.1 Comparison of the reduced Planck constant

The International System in force since 20 May 2019 fixes $h = 6.62607015 \times 10^{-34}\,\mathrm{J\,s}$ exactly [162]. Its reduced constant is therefore

$$\hbar_{\mathrm{SI}} = \frac{6.62607015}{2\pi}\,10^{-34}\,\mathrm{J\,s} = 1.0545718176461563912\ldots \times 10^{-34}\,\mathrm{J\,s}.$$

The ellipsis corresponds to the expansion of an exact expression, not to an experimental uncertainty of h in the current SI. This reference value is used here after evaluation of the HMT readers; its function is comparative.

The two return coordinates of the article are

$$H_5 = 1.0545718176461544696\ldots,$$
$$\eta_{\mathrm{ret}} = 1.0545718169150390611\ldots.$$

In identifying $\mathcal{U}_S$ with J s, the relative difference of the preceding section with respect to $\hbar_{\mathrm{SI}}$ is approximately -1.82×10^{-15}. That of the subsequent section is approximately -6.93×10^{-10}. The second difference contains the effect of the return used by the electronic corrector; it is not equivalent to the evaluation error of the first section.

The precision of these comparisons allows the quantitative proximity of the construction of H_5 to be recognized while distinguishing proximity from exact identity. In this manuscript the displayed expressions do not provide an exact equality with $h/(2\pi)$. The proved result concerning Σ_H is its decade -34; the analytic result concerning H_5 is evaluation of the declared character. Physical identification is examined in this chart and with the explicit differences just given. The SI value does not enter into the definition of the determinantal norm or the regional recurrence.

Chapter 141

Covariance, normal form, and Gaussian spectrum

The positive action section $\hbar_a$ and the preceding canonical realization determine the normalization used here. The correspondence between covariance, spectrum, purity and entropy is proved for a finite number of modes; Fock space retains all its occupations. Symplectic reduction and Gaussian formulas are established results reproduced to make their application to memory transport self-contained.

141.1 Canonical coordinates and characteristic function

On $L^2(\mathbb{R}^d)$, let $\widehat{q}_j = \widehat{Q}_j/\sqrt{\hbar_a}$, $\widehat{p}_j = \widehat{P}_j/\sqrt{\hbar_a}$, and $\widehat{z} = (\widehat{q}_1, \widehat{p}_1, \dots, \widehat{q}_d, \widehat{p}_d)$. On Schwartz space,

$$[\widehat{z}_j, \widehat{z}_k] = i J_{jk} I, \qquad J = \bigoplus_{j=1}^{d} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

This convention fixes the sign of the canonical form. The Weyl operators $\mathcal{W}(\xi) = \exp(i\xi^{\mathsf{T}}\widehat{z})$ are defined through the self-adjoint closures of the indicated linear combinations. For a centred state ρ with finite second moments, the normalized covariance and its characteristic function are

$$W_{jk} = \operatorname{Tr}\rho(\widehat{z}_j\widehat{z}_k + \widehat{z}_k\widehat{z}_j), \qquad \chi_\rho(\xi) = \operatorname{Tr}(\rho\mathcal{W}(\xi)), \qquad V = \frac{\hbar_a}{2} W.$$

The state is centred Gaussian when

$$\chi_\rho(\xi) = \exp(-\xi^{\mathsf{T}} W \xi/4). \tag{141.1}$$

The expectation of $(c^{\mathsf{T}}\widehat{z})^\dagger(c^{\mathsf{T}}\widehat{z})$ is $c^\dagger(W + iJ)c/2$; thus $W + iJ \geq 0$. Moreover $W > 0$: if v is real and $v^{\mathsf{T}}Wv = 0$, positivity implies $(W + iJ)v = 0$, hence $Jv = 0$ and $v = 0$.

Lemma 141.1 (Uniqueness of the characteristic function)**.** *Two trace-class operators with the same characteristic function are equal.*

Proof. Regrouping positions and momenta as $\xi = (u, v)$,

$$(\mathcal{W}(u, v)f)(x) = e^{iu\cdot(x+v/2)} f(x + v).$$

For a finite-rank operator with Schwartz vectors and kernel K_A,

$$\operatorname{Tr}(A\mathcal{W}(u, v)) = \int K_A(x + v/2, x - v/2) e^{iu\cdot x}\, dx.$$

Plancherel and the change of variables with absolute Jacobian one give

$$\|A\|_{\mathrm{HS}}^2 = (2\pi)^{-d} \int_{\mathbb{R}^{2d}} |\operatorname{Tr}(A\mathcal{W}(\xi))|^2\, d\xi.$$

The preceding operators are dense in the trace class. Convergence in that norm implies Hilbert–Schmidt convergence and uniform convergence of the characteristic functions, because their difference is bounded by $\|A - B\|_1$. The equality extends in L^2. Applying it to the difference of the operators proves the assertion. □

141.2 Symplectic reduction of the positive form

Theorem 141.2 (Williamson normal form)**.** *For a real symmetric matrix $W > 0$ of order $2d$, there exist $S \in \mathrm{Sp}(2d, \mathbb{R})$ and $\nu_j > 0$ such that*

$$W = SDS^{\mathsf{T}}, \qquad D = \bigoplus_j \nu_j I_2.$$

The eigenvalues of JW are $\pm i\nu_j$. The condition $W + iJ \geq 0$ is equivalent to $\nu_j \geq 1$ for every j.

Proof. The matrix $B = W^{1/2} J W^{1/2}$ is skew-symmetric and invertible. If $-B^2 u = \nu^2 u$ and $\|u\| = 1$, the vector $v = -Bu/\nu$ is a unit vector orthogonal to u; it satisfies $Bu = -\nu v$, $Bv = \nu u$. The complement of this pair is invariant. Iteration produces an orthogonal matrix O with $O^{\mathsf{T}} B O = JD$. Defining $S = W^{1/2} O D^{-1/2}$,

$$S^{\mathsf{T}} J S = J, \qquad SDS^{\mathsf{T}} = W.$$

The similarity $W^{1/2}(JW)W^{-1/2} = B$ determines the values ν_j, with multiplicity. Congruence by S^{-1} transforms $W + iJ$ into $D + iJ$; each block has eigenvalues $\nu_j + 1$ and $\nu_j - 1$. □

Lemma 141.3 (Unitary implementation in a finite number of modes)**.** *Every symplectic matrix S admits a unitary U_S satisfying $U_S^\dagger \mathcal{W}(\xi) U_S = \mathcal{W}(S^{\mathsf{T}}\xi)$. The conjugation $\rho \mapsto U_S \rho U_S^\dagger$ transports covariance by $W \mapsto SWS^{\mathsf{T}}$.*

Proof. In the polar decomposition $S = OP$, the identity $J^{-1}(S^{\mathsf{T}}S)J = (S^{\mathsf{T}}S)^{-1}$, applied by spectral calculus to the positive root, proves $PJP = J$. Then O is orthogonal symplectic. The eigenspaces of P are paired through J with eigenvalues λ, λ^{-1}; that of eigenvalue one is J-invariant. Hence an orthogonal symplectic matrix K reduces P to $\bigoplus_j \mathrm{diag}(e^{r_j}, e^{-r_j})$.

Dilation is implemented by

$$(\mathcal{D}_r f)(x) = e^{-\frac{1}{2}\sum_j r_j} f(e^{-r_1} x_1, \dots, e^{-r_d} x_d).$$

A change of variables proves its unitarity and the conjugation of $(\widehat{q}_j, \widehat{p}_j)$ to $(e^{r_j}\widehat{q}_j, e^{-r_j}\widehat{p}_j)$. An orthogonal symplectic transformation commutes with J and corresponds to a matrix $u \in U(d)$ on $a_j = (\widehat{q}_j + i\widehat{p}_j)/\sqrt{2}$. Its second quantization $\Gamma(u) = \bigoplus_{n\geq 0} u^{\otimes_s n}$ is unitary on $\mathcal{F}_s(\mathbb{C}^d)$, and the identity on finite symmetric products gives the transformation of the a_j.

The Hermite basis identifies this space with $L^2(\mathbb{R}^d)$. To justify its completeness, if f is orthogonal to all polynomials multiplied by $e^{-|x|^2/2}$, the entire function $F(z) = \int f(x) e^{-|x|^2/2} e^{z\cdot x}\, dx$ has all its derivatives at zero equal to zero. Thus $F = 0$; restricting to $z = i\xi$, Fourier uniqueness gives $f = 0$. Normalizations and orthogonality follow from creation and annihilation. Composition of the preceding implementations produces U_S. The Weyl equality implies $\chi_{U_S \rho U_S^\dagger}(\xi) = \chi_\rho(S^{\mathsf{T}}\xi)$, which proves the covariance law. □

141.3 Calculation of the spectrum of the isotropic density

In one mode, the normalized coherent states are

$$|z\rangle = e^{-|z|^2/2} \sum_{n\geq 0} \frac{z^n}{\sqrt{n!}} |n\rangle.$$

Its position function, $\pi^{-1/4} \exp[-(x - q_0)^2/2 + i p_0 (x - q_0/2)]$, where $q_0 = \sqrt{2}\,\Re z$, $p_0 = \sqrt{2}\,\Im z$, gives by integration

$$\langle z | \mathcal{W}(u, v) | z \rangle = e^{-(u^2+v^2)/4} e^{i(u q_0 + v p_0)}.$$

For $s > 0$, the integral

$$\rho^{(s)} = \frac{1}{\pi s}\int_{\mathbb{C}} e^{-|z|^2/s}|z\rangle\langle z|\, d^2z$$

converges in the trace class, is positive and has trace one. Angular integration annuls its entries off the occupation diagonal; radial integration gives

$$\langle n|\rho^{(s)}|n\rangle = \frac{s^n}{(1+s)^{n+1}}, \qquad \chi_{\rho^{(s)}}(u,v) = e^{-(2s+1)(u^2+v^2)/4}.$$

With $\nu = 2s+1$, the unique Gaussian state of covariance νI_2 is, by Lemma 141.1,

$$\rho_\nu = (1-t)\sum_{n\ge 0} t^n |n\rangle\langle n|, \qquad t = \frac{\nu-1}{\nu+1}. \tag{141.2}$$

The case $\nu = 1$ is the projector onto the vacuum.

Theorem 141.4 (Spectrum, purity, and entropy)**.** *A centred Gaussian state with covariance W and symplectic eigenvalues $\nu_1,\dots,\nu_d$ is unitarily equivalent to $\bigotimes_j \rho_{\nu_j}$. Its eigenvalues are*

$$\lambda_{\boldsymbol{n}} = \prod_j (1-t_j)t_j^{n_j}, \qquad t_j = \frac{\nu_j-1}{\nu_j+1}, \quad \boldsymbol{n}\in\mathbb{N}^d.$$

Con $0\log 0 = 0$*,*

$$\operatorname{Tr}\rho^2 = \prod_j \nu_j^{-1} = (\det W)^{-1/2}, \tag{141.3}$$

$$-\operatorname{Tr}(\rho\log\rho) = \sum_j \left[\frac{\nu_j+1}{2}\log\frac{\nu_j+1}{2} - \frac{\nu_j-1}{2}\log\frac{\nu_j-1}{2}\right]. \tag{141.4}$$

Proof. Williamson gives $W = SDS^{\mathsf T}$. The product of the isotropic densities has characteristic function $e^{-\xi^{\mathsf T}D\xi/4}$. Unitary implementation and Weyl uniqueness prove the equivalence and spectral formula. In one mode,

$$\sum_{n\ge0}(1-t)^2t^{2n} = \frac{1-t}{1+t} = \nu^{-1},$$

and

$$-\sum_{n\ge0}(1-t)t^n\log((1-t)t^n) = -\log(1-t) - \frac{t}{1-t}\log t.$$

Substitution of $t = (\nu-1)/(\nu+1)$ proves the formulas; the limit at $t=0$ is zero. In the finite product, Tonelli justifies summing the nonnegative entropy terms, and $\det W = \prod_j \nu_j^2$ gives the determinantal expression. □

If the state is the reduction of a pure bipartite vector, its eigenvalues are the squares of the Schmidt coefficients. The singular value decomposition of the Hilbert–Schmidt operator defined by that vector proves equality of the nonzero spectra of both reductions. The calculated entropy is then the entanglement entropy.

141.4 Preparation and transport of the two covariances

The subsequent application uses two identical pure Gaussian inputs. In general, for $M \in \mathrm{Sp}(2d, \mathbb{R})$, their covariance is $V_0 = (\hbar_a/2)MM^{\mathsf{T}}$, and the joint input has covariance $V_0 \oplus V_0$. Physical transport is obtained by conjugating U_H by $M \oplus M$. In the balanced chart, the normalized input covariance is I_{4d}; with

$$T = \frac{8I + H}{9}, \qquad D = \frac{\sqrt{8}(H - I)}{9},$$

the first reduction has

$$W = TT^{\mathsf{T}} + DD^{\mathsf{T}} = \frac{8I + HH^{\mathsf{T}}}{9}.$$

Its characteristic function is obtained by setting the arguments of the second group of modes to zero. It therefore remains Gaussian and the preceding theorems calculate its entire spectrum, without truncating the occupation register.

Chapter 142

Physical actions and the spectrum of reduced memory

142.1 Genealogy of the dimensional sections

The coordinates π, φ, e, α are realizations of the correlated outputs constructed in Sections 11.6 and 16.2. This section uses $\alpha = \alpha_A$, with the orientation and periodization of Definition 11.59:

$$\alpha = \pi + e - \varphi - 4 - \kappa_{\mathrm{per}}, \qquad \kappa_{\mathrm{per}} = \frac{N_K}{1000^{12} - 1}, \quad N_K = 234543140729659824621058914794146601. \tag{142.1}$$

Identity (11.126) follows from the recurrence with compatible carry and its telescoping, proved in Proposition 11.57. Period twelve corresponds to the register: a window of twelve blocks with a zero boundary condition and a compatible infinite section are different objects. Equality (142.1) uses the latter. The ordered register K preserves its incidences in addition to admitting the scalar reading κ_{per}.

The action reader, constructed in Section 14.6, determines

$$\begin{aligned} H_5 &= \tfrac{1}{2}\sqrt{1000\alpha/\varphi} - \alpha + \tfrac{9}{16}\alpha^2 - \tfrac{5}{9}\alpha^3 + \tfrac{7}{48}\alpha^4 - \tfrac{1}{54}\alpha^5, \\ D_A &= \exp(-100\pi\alpha/9), \\ C_\pi &= 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \\ \eta_{\mathrm{ret}} &= H_5 - \frac{90}{\pi}C_\pi, \\ \hbar_{\mathrm{pre}} &= H_5\mathcal{A}_0, \qquad \hbar_{\mathrm{ret}} = \eta_{\mathrm{ret}}\mathcal{A}_0, \qquad R_{\mathrm{act}} = H_5/\eta_{\mathrm{ret}}, \end{aligned} \tag{142.2}$$

Here $\mathcal{A}_0 = 10^{-34}\mathcal{U}_S$. The decade comes from the determinantal reader of $\varphi\alpha^{16}$, whose construction and bounds appear in Theorem 14.19; H_5 and η_{ret} are the mantissas of the two positive sections of reduced action. In each section $a \in \{\mathrm{pre}, \mathrm{ret}\}$, we have $h_a = 2\pi\hbar_a$.

The construction of the central electron, developed in Section 16.2, determines the coordinate

$$\begin{aligned} \Delta_4 &= \frac{\pi - e\log\pi}{270}\left(1 + \frac{\pi}{729}\right), \\ B_e &= \frac{\sqrt{3}}{4}\left[\exp(\varphi/\pi^2) - 22\alpha^3\right](1 + 15\Delta_4), \\ \kappa_e &= 80 - 54\alpha + 6\Delta_4, \qquad \varepsilon_e^\beta = B_e R_{\mathrm{act}}^{\kappa_e}. \end{aligned} \tag{142.3}$$

The prefactor $\sqrt{3}/4$ comes from the norm of the commutator of the sum and product projectors on the fibers specified in (16.12). The central route and its registers determine the return coefficients of (16.43) and (16.44). The dimensional realization is $E_e^\beta = \varepsilon_e^\beta\mathcal{U}_E$, $m_e^\beta = E_e^\beta/c^2$, with the energy chart of (16.47).

The dodecaphasic clock determines the energy basis of (16.48):

$$\omega_{108} = \frac{2\pi}{108 t_0}, \quad E_{\mathrm{clk}} = \hbar_{\mathrm{ret}}\omega_{108}, \quad b_e = B_e\frac{\mathcal{U}_E}{E_{\mathrm{clk}}}.$$

Substitution of (142.3), preserving the same physical energy, proves

$$\boxed{\frac{m_e^\beta c^2}{\hbar_{\mathrm{ret}}} = b_e R_{\mathrm{act}}^{\kappa_e}\omega_{108}, \qquad \frac{m_e^\beta c^2}{\hbar_{\mathrm{pre}}} = b_e R_{\mathrm{act}}^{\kappa_e - 1}\omega_{108}.} \tag{142.4}$$

Division by h_a, rather than $\hbar_a$, converts angular frequency into cyclic frequency. The conversion $\mathcal{U}_E/E_{\mathrm{clk}}$ remains explicit: a coordinate expressed in MeV changes when its energy basis is replaced by the clock quantum.

142.2 Normalization of the Maxwell–Dirac action

The constitutive response in Section 118 produces the speed c, impedance Z, permittivity ε, and permeability μ, all positive, with $\varepsilon c = Z^{-1}$ and $\mu c = Z$. The positive charge section in (119.1) satisfies

$$e_a^2 = \frac{2\alpha h_a}{Z} = 4\pi\alpha\varepsilon\hbar_a c. \tag{142.5}$$

Here $e_a > 0$ denotes the elementary charge of Section a; e in (142.1)–(142.3) denotes the Euler coordinate. We fix a homogeneous chart with positive constants $c, Z, \varepsilon, \mu, \hbar_a, e_a$, the mass $m = m_e^{\beta} > 0$ and the charge $q \in \{-e_a, e_a\}$; then $\sigma = q/e_a \in \{-1, 1\}$. In the Minkowski realization of signature $(+,-,-,-)$, with $x^0 = ct$, the Maxwell–Dirac action is

$$\mathcal{I} = \int d^4x \left[-\frac{F_{\mu\nu}F^{\mu\nu}}{4\mu c} + \bar{\psi}\left(i\hbar_a\gamma^\mu(\partial_\mu + iqA_\mu/\hbar_a) - mc\right)\psi \right]. \tag{142.6}$$

The component A_0 is the scalar potential divided by c, and $F = dA$. The spinor representation is fixed by the matrices g^I of (133.1), which satisfy $\{g^I, g^J\} = 2\eta^{IJ} I_4$ for $\eta = \mathrm{diag}(-1,1,1,1)$. Define $\Gamma^I = -ig^I$, so that $\{\Gamma^I, \Gamma^J\} = -2\eta^{IJ} I_4$, and, in this flat chart, $\gamma^\mu = \delta^\mu_I \Gamma^I$, $\bar{\psi} = \psi^\dagger \Gamma^0$. Signature, adjoint and imaginary factor are thus fixed jointly; the matrices γ^μ are distinguished from the scalar Holst parameter used later. Smooth fields and compactly supported variations are considered, or boundary conditions that annul boundary terms. The action realizes the already generated coefficients and the material representation of Section 133.1.

We define

$$\ell_a = \frac{\hbar_a}{mc}, \quad X = x/\ell_a, \quad \Psi = \ell_a^{3/2}\psi, \quad \mathsf{a}_\mu = \frac{e_a \ell_a}{\hbar_a} A_\mu, \quad \mathsf{f} = d\mathsf{a}.$$

Then

$$\boxed{\frac{\mathcal{I}}{\hbar_a} = \int d^4X \left[-\frac{\mathsf{f}_{\mu\nu}\mathsf{f}^{\mu\nu}}{16\pi\alpha} + \bar{\Psi}\big(i\gamma^\mu(\partial_\mu + i\sigma\mathsf{a}_\mu) - 1\big)\Psi \right].} \tag{142.7}$$

Proof. The volume change contributes ℓ_a^4, each derivative contributes ℓ_a^{-1}, and the product of fermionic fields contributes ℓ_a^{-3}. The coefficient of the mass term is $mc\ell_a/\hbar_a = 1$. The electromagnetic term contributes $\hbar_a/(4\mu c e_a^2)$. Since $\mu c = Z$ and $Ze_a^2 = 4\pi\alpha\hbar_a$, this coefficient is $1/(16\pi\alpha)$. Both terms and their normalizations are thereby determined. □

Variation with respect to a_ν, followed by substitution of (142.1), gives

$$\partial_\mu \mathsf{f}^{\mu\nu} = 4\pi\left(\pi + e - \varphi - 4 - \kappa_{\mathrm{per}}\right)\sigma\bar{\Psi}\gamma^\nu\Psi. \tag{142.8}$$

The cancellation composes the action sections, electronic construction, and constitutive response; the regional and dodecaphasic reader determines the remaining coupling. Mass persists in the length ℓ_a, and action in the dimensional conversion. Compton normalization is a subsequent realization of these outputs. In the Coulomb realization of the same electron, the Bohr length is ℓ_a/α: substituting $e_a^2 = 4\pi\alpha\varepsilon\hbar_a c$ into $4\pi\varepsilon\hbar_a^2/(me_a^2)$ gives exactly this expression.

The deduction uses fixed homogeneous coefficients. For spacetime-dependent coefficients, the product rule introduces derivatives of the scales and coupling into the transformed equations. Variation of an action with variable parameters must preserve these terms.

142.3 Areal scale of gravitational action and entropy

The circular realization of the cycle of 108 steps, constructed in Section 121, determines

$$t_P = \frac{54}{\pi} t_0, \qquad G_a = \frac{c^5 t_P^2}{\hbar_a}, \qquad \ell_P^2 = \frac{\hbar_a G_a}{c^3} = c^2 t_P^2. \tag{142.9}$$

The condition for coincidence of radii in that realization is $G_a m_{108,a}/c^2 = R_{108}$, with $m_{108,a} = \hbar_a \omega_{108}/c^2$. Its unique solution fixes the factor $(54/\pi)^2$; the radius used is $G_a m/c^2$. The reciprocity $G_{\mathrm{pre}}/G_{\mathrm{ret}} = R_{\mathrm{act}}^{-1}$ preserves ℓ_P^2 between the two sections.

Section 127 constructs the cotetrad and connection from the transport already realized. In the action of Section 131, fix a nondegenerate cotetrad e^I, the orientation, signature $\eta = \mathrm{diag}(-1, 1, 1, 1)$, and a metric connection ω, with curvature $F^{IJ} = d\omega^{IJ} + \omega^I{}_K \wedge \omega^{KJ}$. Take $\gamma \in \mathbb{R} \setminus \{0\}$ and hold $\gamma, \Lambda, c, G_a, \hbar_a$ constant during variation. Put $\Sigma^{IJ} = e^I \wedge e^J$ and $\mathsf{P}_\gamma = \star + \gamma^{-1} I$; internal Lorentzian duality satisfies $\star^2 = -I$ on bivectors. Variations have compact support or boundary conditions that cancel the surface term. Substitution of (142.9) into (131.3) gives

$$\boxed{\frac{\mathcal{I}_{\mathrm{grav}}}{\hbar_a} = \frac{1}{16\pi\ell_P^2}\left[\int \Sigma_{IJ} \wedge (\mathsf{P}_\gamma F)^{IJ} - 2\Lambda \int \mathrm{vol}_e\right].} \tag{142.10}$$

With the cotetrad and material fields fixed during variation of the connection, let σ_{IJ} be the material current defined by $\delta_\omega \mathcal{I}_m = -\frac{1}{2}\int \delta\omega^{IJ} \wedge \sigma_{IJ}$, according to (131.4). The same substitution in Theorem 131.1 produces

$$D_\omega(\mathsf{P}_\gamma \Sigma) = 8\pi\ell_P^2 \frac{\sigma}{\hbar_a}. \tag{142.11}$$

Proof. In section a, the original coefficient is $1/(2\kappa_a c)$, with $\kappa_a = 8\pi G_a/c^4$. Division by $\hbar_a$ converts it into $1/(16\pi\ell_P^2)$; the cosmological term retains its factor two. In the connection equation, $\kappa_a c \hbar_a = 8\pi\ell_P^2$ holds. Both identities use the same dimensional section. □

The Barbero functional of Section 120 retains its complete expression. Its channels are $q_\pm = e^{-x \mp y}$, with $x = \pi A/180$, $y = \pi C^*/180$, where A and C^* are the angular coordinates expressed in degrees. The domain $x > |y|$ ensures $0 < q_\pm < 1$:

$$\gamma = \frac{180}{\pi} xy + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)], \qquad S_m(q) = \frac{q^m}{1 - q^{3m}}. \tag{142.12}$$

Hexadic and octadic incidence fixes the multiplicities 90 and 120, and the oriented chain fixes the coefficients 12 and -1, as proved by (120.1). The areal construction of Section 122 additionally gives

$$a_0 = \frac{1 + 2\cos(\pi/18)}{3} \frac{y}{6}, \qquad \widehat{\mathcal{A}} = \gamma a_0 \ell_P^2 (N_{90} + N_{120}). \tag{142.13}$$

The inversion $y \mapsto -y$ changes the signs of γ and a_0; their product is even. The operator P_γ, by contrast, retains its orientation dependence. Invariance of the areal product is thus distinguished from invariance of each operator coefficient. The areal spectrum is realized in the positive section $\gamma a_0 > 0$, with the occupation operators $N_{90}, N_{120} \geq 0$ constructed in Section 122.

For the spherical cosmological horizon in (137.31), of radius $R > 0$ and area $\mathcal{A}_H = 4\pi R^2$, fix a section a and abbreviate $\hbar = \hbar_a$, $G = G_a$ during this calculation and the following bounce calculation. Then

$$\frac{S_H}{k_B} = \frac{\mathcal{A}_H}{4\ell_P^2}, \qquad T_{\cos} = \frac{\hbar c}{2\pi k_B R}, \qquad E_\Lambda = \frac{c^4 R}{2G}, \qquad T_{\cos} S_H = E_\Lambda. \tag{142.14}$$

The areal scale of (142.10), (142.11) and (142.14) is the same. The temperature corresponds to the spherical cosmological horizon; normalization of the Schwarzschild horizon constitutes another realization. On varying R while holding $c, G, \hbar, k_B$ fixed, differentiation of (142.14) preserves both terms: $T_{\cos}\, dS_H = 2\, dE_\Lambda$ and $S_H\, dT_{\cos} = -\, dE_\Lambda$, as in (137.35). What is obtained is the differential of an identity between states of that family, not a replacement of the dynamic balance with other variable parameters.

142.3.1 An explicit dynamical application

Theorem 133.4 determines the following amplitude for a spinor state with conserved second moment $q_\varrho > 0$. Conservation is expressed through (133.18); Corollary 133.5 provides a concrete realization in the three-occupation sector:

$$s_0 = \frac{3\pi G\hbar^2}{2c^2 V_c^2} \frac{\gamma^2}{1+\gamma^2} q_\varrho. \tag{142.15}$$

In the spatially flat realization with pressureless matter and $\Lambda = 0$, the constant $\rho_0 > 0$ is an energy density, $V_c > 0$ is the reference comoving volume, and $E_0 = \rho_0 V_c > 0$. The solution of Theorem 135.1 is

$$a_b^3 = s_0/\rho_0, \qquad \tau^2 = \frac{s_0 c^2}{6\pi G \rho_0^2}, \qquad a(t) = a_b \left(1 + \frac{(t-t_b)^2}{\tau^2}\right)^{1/3}.$$

Restoring c in (134.2) uses the coefficient $8\pi G/(3c^2)$ when the density is an energy density. In particular, $(\dot a/a)^2 = (8\pi G/(3c^2))(\rho_0 a^{-3} - s_0 a^{-6})$. Substitution of (142.15) into the solution parameters proves

$$\boxed{\left(\frac{\tau E_0}{\hbar}\right)^2 = \frac{q_\varrho}{4}\frac{\gamma^2}{1+\gamma^2}, \qquad \mathcal{V}_b = \frac{3\pi}{2} q_\varrho \frac{\gamma^2}{1+\gamma^2} \ell_P^2 \frac{\hbar c}{E_0}.} \tag{142.16}$$

Here $\mathcal{V}_b = V_c a_b^3$ is the physical volume at the bounce. The quantities G and V_c cancel in the first equality; E_0 remains as material energy. In the conserved sector $\Lambda^3\mathbb{C}^4$, Proposition 133.3 proves $\widehat{\mathcal{Q}}_{\mathrm{sp}} = 4I$, so $q_\varrho = 4$ for every normalized state supported in that sector, and $\tau E_0/\hbar = |\gamma|/\sqrt{1+\gamma^2}$. This yields a dynamical scale for the specified spinor state and cosmological realization.

142.4 Prolongation of memory transport: quantum realization

The quantum realization uses the reversible coupling of Section 113. We first consider a real canonical plane with alternating matrix $\Omega = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The nonadic partition determines

$$Q = \frac{1}{3}\begin{pmatrix} \sqrt{8}I & -I \\ I & \sqrt{8}I \end{pmatrix}, \qquad U_H = Q^{\mathsf{T}} \operatorname{diag}(I, H) Q = \frac{1}{9}\begin{pmatrix} 8I + H & \sqrt{8}(H - I) \\ \sqrt{8}(H - I) & I + 8H \end{pmatrix}. \tag{142.17}$$

The operator $H \in \mathrm{Sp}(2, \mathbb{R})$ transports that plane. The matrix Q preserves the Euclidean product and the form $\Omega \oplus \Omega$; thus U_H is symplectic. The first component defines the observed system and the second the memory register; the complete transformation acts on both.

The ordered loop constructed in Section 114 is

$$C=\begin{pmatrix}0 & -1\\ 1 & -1\end{pmatrix},\quad P=\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix},\quad H_n=C^{-1}P^{-n}CP^n=\begin{pmatrix}1+n & n^2\\ n & n^2-n+1\end{pmatrix}. \tag{142.18}$$

For each $n\in\mathbb{Z}$, $\det H_n=1$ holds. In particular, for $n=1$,

$$H_1=\begin{pmatrix}2 & 1\\ 1 & 1\end{pmatrix}=F_1^2,\quad F_1=\begin{pmatrix}1 & 1\\ 1 & 0\end{pmatrix}=PF_{\mathrm{av}}P^{-1}. \tag{142.19}$$

Its spectrum is $\{\varphi^2,\varphi^{-2}\}$, by subsequent recognition of the already generated golden mode. The integer n specifies the oriented number of iterations in the chosen loop. The family of transports is determined algebraically; its realization as a physical protocol also preserves the corresponding preparation and evolution data.

142.4.1 Initial state, transport and reading

Proposition 115.1 constructs the Gaussian realization of the action ellipse. In the positive section $\hbar_a>0$, with $A>0$ and $|C^*/A|<1$, its covariance is

$$V_\eta=\frac{\hbar_a}{2}M_\eta M_\eta^{\mathsf{T}},\qquad M_\eta=\mathrm{diag}(e^{\eta/2},e^{-\eta/2}),\qquad \eta=\operatorname{artanh}(C^*/A). \tag{142.20}$$

In the Schrödinger representation on $L^2(\mathbb{R},dQ)$, with $\widehat{Q}$ multiplication by Q and $\widehat{P}=-i\hbar_a\,d/dQ$, the relation $[\widehat{Q},\widehat{P}]=i\hbar_a I$ holds on Schwartz space. The normalized pure state is

$$\psi_\eta(Q)=(\pi\hbar_a e^{\eta})^{-1/4}\exp[-Q^2/(2\hbar_a e^{\eta})].$$

The preparation is the product $\psi_\eta\otimes\psi_\eta$. The transport of (142.17) is expressed in the same covariance chart by

$$\widetilde{U}=(M_\eta\oplus M_\eta)U_H(M_\eta^{-1}\oplus M_\eta^{-1}). \tag{142.21}$$

Lemma 141.3 provides a unitary implementation of this symplectic matrix in two modes. Implementations that differ by a global phase produce the same conjugation of the state. The covariance transforms by $V\mapsto\widetilde{U}V\widetilde{U}^{\mathsf{T}}$; the joint state remains pure and its partial trace over the second mode determines the visible reading, according to the construction in Section 141.

Theorem 142.1 (Reduced mixing through relative deformation)**.** *The covariance of the visible reading is*

$$\boxed{V_{\mathrm{vis}}=\frac{\hbar_a}{2}M_\eta\frac{8I+HH^{\mathsf{T}}}{9}M_\eta^{\mathsf{T}}.} \tag{142.22}$$

Its normalized symplectic eigenvalue is

$$\boxed{\nu(H):=\frac{2\sqrt{\det V_{\mathrm{vis}}}}{\hbar_a}=\sqrt{1+\frac{8}{81}\left[\mathrm{tr}(HH^{\mathsf{T}})-2\right]}.} \tag{142.23}$$

Proof. Let $T=(8I+H)/9$ and $D=\sqrt{8}(H-I)/9$. The two initial modes are independent and have the same covariance. In the balanced chart of the initial state, the normalized visible covariance is $TT^{\mathsf{T}}+DD^{\mathsf{T}}$. Expansion gives

$$TT^{\mathsf{T}} + DD^{\mathsf{T}} = \frac{64I + 8(H + H^{\mathsf{T}}) + HH^{\mathsf{T}} + 8[HH^{\mathsf{T}} - H - H^{\mathsf{T}} + I]}{81}$$
$$= \frac{8I + HH^{\mathsf{T}}}{9}.$$

This proves (142.22). Since $\det M_\eta = 1$ and $\det(HH^{\mathsf{T}}) = 1$, $\det(8I + HH^{\mathsf{T}}) = 65 + 8\,\mathrm{tr}(HH^{\mathsf{T}})$. Division by 81 proves (142.23). The two positive eigenvalues of HH^{T} are reciprocal; their sum is at least two. Consequently, $\nu \geq 1$, with equality exactly when $HH^{\mathsf{T}} = I$. □

The scale $\hbar_a$ and deformation η are retained until the determinant is taken. Their cancellation is covariant because state and operator are transported jointly. The expression HH^{T} uses the Euclidean product of the balanced chart of the initial state; a chart change also transports that metric product.

An orthogonal and symplectic transport H can have $D \neq 0$ while preserving the isotropic Gaussian product, with $\nu = 1$. The operatorial complement and state entanglement therefore have different criteria. In this preparation, the non-isometric relative deformation characterizes exactly the case $\nu > 1$.

142.4.2 Exact evaluation of the golden loop

For (142.18),

$$\mathrm{tr}(H_n H_n^{\mathsf{T}}) - 2 = n^2(2n^2 - 2n + 5), \qquad \nu_n^2 = 1 + \frac{8}{81}n^2(2n^2 - 2n + 5). \tag{142.24}$$

The identity follows by summing the squares of the four entries. For $n = 1$, the sum is seven and

$$\boxed{\nu_1 = \frac{11}{9}, \quad \mathrm{Tr}\,\rho_{\mathrm{vis}}^2 = \frac{9}{11}, \quad \bar{n}_{\mathrm{W}} = \frac{1}{9}, \quad \frac{S_{\mathrm{vis}}}{k_B} = \frac{10}{9}\log 10 - \log 9.} \tag{142.25}$$

Theorem 141.4 proves that the centred one-mode Gaussian state with symplectic value ν is unitarily equivalent to the geometric density of (141.2), whose mean occupation is $\bar{n}_{\mathrm{W}} = (\nu - 1)/2$. For $\nu = 11/9$, its eigenvalues are

$$\lambda_j = \frac{9}{10}\,10^{-j}, \qquad j = 0, 1, 2, \ldots. \tag{142.26}$$

The series sums to one. Its square sums to $(9/10)^2/(1 - 10^{-2}) = 9/11$. Finally,

$$-\sum_j \lambda_j \log \lambda_j = -\log(9/10) + \log 10 \sum_j j\lambda_j = \frac{10}{9}\log 10 - \log 9.$$

The Gaussian reduction and its spectrum are proved in Section 141; the two preceding series evaluate the purity and entropy of this realization exactly. Since the joint state is pure, S_{vis}/k_B is also its dimensionless entanglement entropy. The global transformation is unitary, while the restriction to the first mode has positive entropy.

The result corresponds to the specified preparation and transport. Its identification with a vacuum measurement requires the experimental realization of that protocol. The entropy of a reduction and the transferred heat remain distinct quantities.

142.4.3 Composition with the thermal transducer

In the Williamson normal representation, choose a frequency $\omega > 0$ and define the energy reader $\widehat{H}_{\mathrm{W}} = \hbar_a\omega(\widehat{N} + \frac{1}{2})$. The frequency is part of this energy realization; covariance determines the populations. With this choice, (142.26) is a Gibbs state whose population ratio satisfies

$$e^{-\hbar_a\omega/(k_B T_{\mathrm{W}})} = \frac{1}{10}, \qquad T_{\mathrm{W}} = \frac{\hbar_a\omega}{k_B \log 10}.$$

If the clock frequency of the same section is taken, $\omega = 1/t_P = 2\pi/T_{108}$, the thermal-transducer identity (123.8), $\hbar_a/t_P = k_B\Theta_{\mathrm{clk},a}\log 3$, gives

$$\boxed{\frac{T_{\mathrm{W}}}{\Theta_{\mathrm{clk},a}} = \frac{\log 3}{\log 10}.} \tag{142.27}$$

The action section and thermal transducer cancel in the ratio. The index a of $\Theta_{\mathrm{clk},a}$ preserves compatibility with the section $\hbar_a$, as in (123.10). The temperature T_{W} is the Gibbs-equivalent temperature for the declared Hamiltonian; the isolated unitary evolution retains its reversible character. This spectral identification determines a thermal ratio and does not postulate thermalization dynamics.

142.5 Extension to arbitrary finite canonical dimension

Let $H \in \mathrm{Sp}(2d,\mathbb{R})$, with $d \geq 1$ finite. Two pure, centered and identical Gaussian states are prepared, each of d modes, with covariance $V_0 = (\hbar_a/2)MM^{\mathsf{T}}$, where $M \in \mathrm{Sp}(2d,\mathbb{R})$. Section 141 specifies this preparation and the joint chart transport by $M \oplus M$. In the balanced chart, the same coupling U_H produces $W = (8I + HH^{\mathsf{T}})/9$. The product $A = HH^{\mathsf{T}}$ is positive, symmetric and symplectic.

There exists an orthonormal symplectic basis in which

$$A = \mathrm{diag}(e^{2r_1}, e^{-2r_1}, \ldots, e^{2r_d}, e^{-2r_d}), \qquad r_j \geq 0.$$

The identity $A\Omega A = \Omega$, with $\Omega = \bigoplus_{j=1}^{d} \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, implies that Ω takes the eigenspace of λ to that of λ^{-1}. Paired orthonormal bases are chosen in these spaces. The eigenspace of $\lambda = 1$ is Ω-invariant and admits orthonormal canonical pairs. Their union constructs the required basis.

The matrix W is diagonal in that basis. The products of each pair determine its symplectic eigenvalues:

$$\boxed{\nu_j^2 = \frac{(8 + e^{2r_j})(8 + e^{-2r_j})}{81} = 1 + \frac{32}{81}\sinh^2 r_j.} \tag{142.28}$$

Theorem 141.4 composes partial-trace reduction with these symplectic eigenvalues:

$$\boxed{\mathrm{Tr}\,\rho_{\mathrm{vis}}^2 = \prod_{j=1}^{d} \nu_j^{-1}, \qquad \frac{S_{\mathrm{vis}}}{k_B} = \sum_{j=1}^{d} \left[\frac{\nu_j + 1}{2} \log \frac{\nu_j + 1}{2} - \frac{\nu_j - 1}{2} \log \frac{\nu_j - 1}{2} \right].} \tag{142.29}$$

Adopt $0 \log 0 = 0$ by continuity. The proof of Theorem 141.4 unitarily transforms the state into a product of d geometric density operators: the purity of the product is the product of the purities, and its entropy is the sum of the entropies. The result holds for every finite d and every transport H in the domain, retaining all occupations of Fock space.

For the loop (142.19), $r = 2\log\varphi$ and $\sinh^2 r = 5/4$, reproducing (142.25). Thus the transport spectrum and nonadic partition $8 + 1$ determine the spectrum of the reduced reading; the action sections fix its dimensional realization.

The dimension of an exceptional realization and its transport come from the corresponding constructor. Equation (142.28) determines the spectral law after realizing its canonical modes and the specified Gaussian preparation.

142.5.1 Operator identity between memory and complex structure

In the balanced canonical chart, let $J = \Omega$, with $J^2 = -I$ and $J^{\mathsf{T}} = -J$. This matrix realizes the canonical complex structure of phase space, and $D = \sqrt{8}(H - I)/9$ is the memory block. For the same preparation and the same chart transport of Section 141, we have

$$\boxed{-(JW)^2 = I + [D, J]\,[D, J]^{\mathsf{T}}, \qquad W = \frac{2}{\hbar_a} M^{-1} V_{\mathrm{vis}} M^{-\mathsf{T}}.} \tag{142.30}$$

Proof. The symplectic identities $HJH^{\mathsf{T}} = J$ and $H^{\mathsf{T}}JH = J$, together with $A = HH^{\mathsf{T}}$, give $JAJ^{\mathsf{T}} = A^{-1}$. If $C = HJ - JH$, multiplication of the four summands produces

$$CC^{\mathsf{T}} = A + JAJ^{\mathsf{T}} - HJH^{\mathsf{T}}J^{\mathsf{T}} - JHJ^{\mathsf{T}}H^{\mathsf{T}} = A + A^{-1} - 2I.$$

On the other hand,

$$-(JW)^2 = \frac{(8I + A^{-1})(8I + A)}{81} = I + \frac{8}{81}(A + A^{-1} - 2I).$$

Since $[D, J] = \sqrt{8}\,C/9$, we obtain (142.30). The identity holds in every finite canonical dimension and for every H in the domain, including transports that mix the modes. □

The eigenvalues of this operator are the ν_j^2, each with multiplicity two. Formula (141.3) then expresses purity directly in terms of the memory block:

$$\boxed{\operatorname{Tr}\rho_{\mathrm{vis}}^2 = \det\Big(I + [D, J]\,[D, J]^{\mathsf{T}}\Big)^{-1/4}.} \tag{142.31}$$

In one mode,

$$\nu^2 = 1 + \frac{1}{2}\|[D, J]\|_F^2. \tag{142.32}$$

In this protocol, D commutes with J exactly when the reduced reading preserves purity. For the loop H_1, $\|[D, J]\|_F^2 = 80/81$, and (142.32) gives $\nu^2 = 121/81$, yielding the purity $9/11$.

The commutator uses the complex structure compatible with the initial covariance and its metric product. A canonical change jointly transports the three data; the transpose of (142.30) refers to the balanced chart. The prior construction of the imaginary element in the formal kernel precedes this representation, which specifies its canonical action.

The joint evolution preserves purity through its unitary implementation; the commutator determines the loss of purity of the partial reading. The information of the joint state remains in the two factors and their correlations. The formula determines the purity of the reduction, which is a nonlinear functional of its density operator.

The real decomposition relative to J distinguishes its complex-linear and complex-antilinear components:

$$D_{\mathrm{lin}} = \frac{D - JDJ}{2}, \qquad D_{\mathrm{anti}} = \frac{D + JDJ}{2}.$$

The first component commutes with J and the second anticommutes with J. The identities $[D, J] = 2D_{\mathrm{anti}}J$ and $JJ^{\mathsf{T}} = I$ imply

$$\boxed{-(JW)^2 = I + 4D_{\mathrm{anti}}D_{\mathrm{anti}}^{\mathsf{T}}, \qquad \operatorname{Tr}\rho_{\mathrm{vis}}^2 = \det(I + 4D_{\mathrm{anti}}D_{\mathrm{anti}}^{\mathsf{T}})^{-1/4}.} \tag{142.33}$$

The complex-antilinear component of the memory block, defined relative to the complex structure of the initial state, determines exactly the purity and mixedness spectrum of this protocol. The complex-linear component can transport memory while preserving purity. The distinction concerns linearity relative to J; combinatorial complexity and the arithmetic classification of constants are properties of other objects.

Chapter 143

Conclusions on conservation, recovery and physical realization

The generating structure preserves operations and history before its numerical realization. In realized transport, this conservation is formulated as an identity of forms. A partial reading distributes charge between the visible component and the complement; including both defines a recoverable transformation. Refinement prolongs the same identity without identifying different depths or replacing the register by its intensity.

143.1 Conservation, reconstruction, and dimension

The equality $\mathcal{U}^{*}(b' \oplus b')\mathcal{U} = b \oplus b$ brings reading and memory together in a reversible operation. For compressions with a separate register, telescoping preserves the positive terminal component and the complete series of complements. For a fixed unitary holonomy, the terminal component is the projection onto its fixed-vector space; for a variable sequence, it is determined by the strong limit of the Gram operators.

Dodecaphasic memory allows the centred register to be recovered through two specified observations, with inverse norm $9/4$. Incidence transports this recovery to the dimensional projector. Its exterior prolongation preserves each degree through Gram determinants and requires the inclusion of mixed components. Thus the relations between length, area, and volume are expressed through maps defined on the same register, with their respective invariants.

143.2 Action and physical consequences

Action, clock, and energy basis retain their types throughout composition. The electronic ratio $m_e c^2/\hbar$ is expressed through the clock frequency and electronic character, preserving the basis-conversion factor. Maxwell–Dirac normalization removes the chosen dimensional scales and leaves the dimensionless coupling. Gravitational transport preserves the areal scale $\hbar G/c^3$; the homogeneous bounce solution retains its conditions on matter, torsion, and cosmological constant.

The Noether relation is manifested when the transported observable is a charge of the declared action. Change of form adds a connection term to the Hamiltonian generator and preserves oriented action exactly. The holonomy of a return also preserves the information distinguishing phase return from return of the complete state.

143.3 Memory and quantum spectrum

Identity (142.30) determines the symplectic spectrum of the reduced covariance through the antilinear component of the memory operator. For the loop

$$H_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad W_1 = \frac{1}{9}\begin{pmatrix} 13 & 3 \\ 3 & 10 \end{pmatrix},$$

we obtain $\det W_1 = 121/81$, symplectic eigenvalue $11/9$, and purity $9/11$. The spectrum of the reduced density operator is

$$\lambda_j = \frac{9}{10}\,10^{-j}, \qquad j = 0, 1, 2, \dots,$$

and its entropy satisfies

$$\frac{S}{k_B} = \frac{10}{9}\log 10 - \log 9.$$

These figures come from the transport and state fixed in the theorem. Spectral equivalence with a Gibbs density operator assigns a temperature to the chosen Hamiltonian; that equivalence retains its distinction from a dynamical thermalization process.

143.4 Contenido fundacional

The articulation between number, incidence, and dynamics resides in the conservation of a common antecedent and in the operations relating its realizations. The article makes this articulation explicit through reconstructions, bounds, spectra, and actions. Generation of values and transport of symmetries retain the same provenance; their observable consequences are calculated subsequently with the reader and state that actually participate. This order provides a precise mathematical formulation of dynamical memory and its physical realizations.

Chapter 144

The holographic arithmetic-geometric coupling constant

The dodecaphasic register links two realizations of the same state: the positional reading that enters into arithmetic closure and the reading that preserves incidence, orientation, and direction. The following development determines the operations made possible by the register already produced, the information retained by its memories, and the errors under which its relations can be recovered. Its reference function admits a precise formulation: the phase orbit of the register is a complete basis of its carrier, and isometric memory preserves exactly the Gram matrix of that basis.

The starting point remains the composition

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{enriched state} \longrightarrow \text{joint discrete structure of the continuum.}$$

APP provides the sheets and their residues and quotients; TRIT preserves regime and orientation; TPK selects, transports and updates while preserving carry, route, boundary, survival and memory. The five constructions of the continuum belong to that joint production, developed in the preceding body. Here the work takes place after it, on its dodecaphasic integer publication and its typed readers. Neither the subsequent linear carrier nor the candidate set of a decoder replaces the domain of generated histories.

144.1 Definition of the register and realization of its coupling coordinate

Definition 11.30 fixes the generated object and its readers. We now develop its reversible encoding, its capacity for operator identification, and the stability of its incidence realizations.

The name expresses a constitutive function of the object, not an automatic identification with a physical interaction coefficient. Canonicity refers to the declared profile, origin and orientation. A chart transformation and an update incorporating new events are not the same operation. Nor is it asserted that the same numerical vector must be a necessary invariant of every system.

The calendar of one hundred and eight positions consists of twelve nonadic windows. The extension

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0$$

preserves phase carry; its non-splitting and the distinction between phase return and state return are proved in proposition 11.31. In each window, the register of a history retains the subroute, its orientation, the carry, and the incidence register. The causal order of extraction is

$$x_{\mathrm{term}} \longmapsto \mathcal{R}_{12}(x_{\mathrm{term}}) \longmapsto (b^{90}, b^{120}, Q_{\mathrm{TPK}}) \longmapsto U_{\mathrm{sgn}} \longmapsto K. \tag{144.1}$$

The regional coordinates and the signed register are two readings of the same terminal state; it is not postulated that the visible ternary matrix alone determines the signed coordinate.

To specify the final composition explicitly, the channels produced are

$$b^{90} = (-495, -116, -684, 108, 601, -90, -173, -88, 313, 560, -397, 461),$$
$$b^{120} = (-425, -281, -481, 671, -255, 30, 475, -543, 680, 251, 6, -128), \qquad Q_{\mathrm{TPK}} = 6263.$$

Let $B_r = \sum_{j=0}^{3} b^{120}_{r+3j}$ and $d_{r,j} = b^{90}_{r+3j}$, for $r = 1, 2, 3$. The harmonic reader gives

$$s_1 = \frac{Q_{\mathrm{TPK}} + 2B_1 + B_2}{3}, \qquad s_2 = s_1 - B_1, \qquad s_3 = s_2 - B_2,$$

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}).$$

In this compatible output, $(B_1, B_2, B_3) = (972, -1073, 101)$, $(s_1, s_2, s_3) = (2378, 1406, 2479)$, and

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Division by three belongs to the reader's three orbits, and the subsequent division by four to the Hadamard inverse; these are not coefficients fitted to a physical output. On the orbits $(1, 4, 7, 10)$, $(2, 5, 8, 11)$, $(3, 6, 9, 12)$, we apply

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \qquad H_4^2 = 4I.$$

The products $H_4 U^{(r)}$ are, respectively,

$$(936, 2916, 2484, 3176), \quad (2172, 2636, 232, 584), \quad (560, 3296, 3656, 2404).$$

Their divisibility by four verifies membership in the Hadamard sublattice. After dividing and restoring the natural order, we obtain

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601), \qquad \mathbf{1}^* K = 6263. \tag{144.2}$$

The inversion and its congruences, including the index 4096 of the signed image, are proved in lemma 11.39 y proposition 11.41. Reconstruction is unique because $H_4^{-1} = H_4/4$, not because a register close to a known value of alpha was sought.

To avoid a collision of symbols, we shall write $\Delta_j = I - S^j$ for the antecedent integer extractors, with $(Sx)_i = x_{i+1}$ and indices modulo twelve. Then

$$\Delta_3 K = b^{90}, \qquad \Delta_4 K = b^{120}, \qquad Q(K) = 6263.$$

The extractor (Δ_3, Δ_4, Q) is injective, with invariant factors $1, \dots, 1, 12$, according to proposition 11.42. Its integral-image conditions and accumulation law remain those of the antecedent register. The normalized memory complements to be denoted by D_j are different operators, derived from those shifts and with a declared metric codomain.

144.2 Periodic encoding, closure, and exact inversion

Fix $B = 1000$, twelve positions, and

$$N(k) = \sum_{j=1}^{12} k_j B^{12-j}, \qquad d_B = B^{12} - 1, \qquad \kappa(k) = \frac{N(k)}{d_B}.$$

In the canonical chart,

$$\begin{aligned} \kappa_{\mathrm{ag}} &= \frac{234543140729659824621058914794146601}{10^{36} - 1}, \\ \alpha_A &= A_{reg} - \kappa_{\mathrm{ag}}, \qquad A_{reg} = \pi_{\mathrm{HMT}} + e_{\mathrm{HMT}} - \varphi_{\mathrm{HMT}} - 4. \end{aligned} \tag{144.3}$$

The three regional coordinates are published before acting in this reader; their common origin with the dodecaphasic closure does not require confusing the order of evaluation of the correlated coordinates.

The equality is a subsequent reader of the generated state. Solving it once its outputs are known is a legitimate inverse operation, distinct from using a target value to produce the register.

Proposition 144.1 (Reversibility and radius of the encoding)**.** *On the domain* $\{0,\dots,999\}^{12}$*, the fraction* $\kappa(k)$*, with declared base and length, determines* k *exactly. The same holds for the exact pair* (A_{reg},α_A) *when it satisfies* (144.3)*. If*

$$|\widetilde{\kappa}-\kappa(k)|<\frac{1}{2d_B},$$

rounding $d_B\widetilde{\kappa}$ *to the nearest integer recovers* $N(k)$ *without error. If the two terms of the closure are approximated separately, it suffices that the sum of their absolute errors be less than* $1/(2d_B)$*.*

Proof. Multiplication by d_B recovers the integer $N(k)$. Twelve Euclidean divisions by B, retaining the remainders and the number of positions, recover its blocks, including leading zero blocks. The error hypothesis places the true integer at a distance strictly less than half a unit. For the closure, the triangle inequality is applied to the error in the difference. Two consecutive integers show that the radius cannot be increased uniformly over the entire code domain: their midpoint is ambiguous. □

If a reduced fraction p/q is received, $N=d_Bp/q$ is calculated and integrality and $0\le N\le d_B$ are checked. The reduced denominator does not replace the base, length and chart metadata. The code contains 10^{36} words and requires $\lceil\log_2(10^{36})\rceil=120$ bits in a fixed-length binary representation. Twelve ten-bit blocks also occupy 120 bits: no additional compression, authentication or cryptographic integrity has been demonstrated. An exact fraction or the integer N avoids losing the precision required for this inversion.

The periodic reading is not to be confused with the truncation N/B^{12}. If $\kappa_j=\kappa(S^jK)$, positional division gives

$$\kappa_{j+1}=1000\kappa_j-K_{j+1},\qquad K_{j+1}=1000\kappa_j-\kappa_{j+1},\qquad \sum_{j=0}^{11}\kappa_j=\frac{6263}{999}. \tag{144.4}$$

Thus, the scalar orbit preserves all phase differences and allows $B_K=\{j+1:1000\kappa_j-\kappa_{j+1}\ge 729\}$ to be read. The reversibility of these publications does not imply that a scalar reconstructs the entire enriched history that produced the register.

144.3 Representation of the register through its complete orbit

In $V=\mathbb{R}^{12}$, with its canonical Euclidean inner product, consider

$$O_K=(K,SK,\dots,S^{11}K):\mathbb{R}^{12}\longrightarrow V.$$

The shift S acts on register positions. Periodicity alone does not identify it with the complete TPK evolution or with a subsequent exceptional symmetry.

Theorem 144.2 (Complete cyclic reference and exact Gram matrix)**.** *The matrix* O_K *is invertible. Its Gram matrix has the eigenvalues*

value	*multiplicity*
$39225169=6263^2$	1
$1248025-345420\sqrt{3}$	2
589609	2
1407529	2
1053157	2
$1248025+345420\sqrt{3}$	2
697225	1

and satisfies

$$589609\|c\|^2 \le \|O_K c\|^2 \le 39225169\|c\|^2, \quad \|O_K^{-1}\| = \frac{1}{\sqrt{589609}}, \quad \operatorname{cond}(O_K) = \frac{6263}{\sqrt{589609}}. \tag{144.5}$$

The orbit of the centered register has rank eleven.

Proof. The entries of the Gram matrix are the autocorrelations $\langle K, S^j K\rangle$. In order $j = 0, \dots, 11$ they are

$$(4251257, 2999323, 3163385, 3287920, 3216553, 3344743,$$
$$2950064, 3344743, 3216553, 3287920, 3163385, 2999323).$$

Since S is orthogonal, the Gram matrix is circulant. Substituting the twelve characters $\exp(2\pi i m/12)$ into that row gives the preceding real table. All values are positive. The smaller of the two values with $\sqrt{3}$ is greater than 589609, since $658416 > 345420\sqrt{3}$, an inequality checked by squaring its two positive sides. The other values in the table are greater than or equal to 589609. The uniform mode has value $(\sum K_i)^2 = 6263^2$; the triangle inequality for the Fourier transform, using $K_i \ge 0$, bounds all the others by that same value. The spectral theorem gives (144.5). Centering the register removes only the uniform mode; the remaining eleven retain the nonzero values in the table. □

The determinant, with the column order and orientation of S fixed here, is

$$\det O_K = 548311925921402018385039793000828362 5.$$

The absence of a proper period of the vector would not, by itself, prove invertibility: what is decisive is that none of its Fourier modes vanishes. The condition number is approximately 8.15643. If K is normalized to unit norm, the extrema of the Gram matrix are divided by $\|K\|^2 = 4251257$; the amplitude of its components must not be confused with an intrinsic improvement over every reference of equal norm.

Corollary 144.3 (Polar separation of form and orientation)**.** *Let*

$$G_K = O_K^* O_K, \qquad P_K = G_K^{1/2}, \qquad F_K = O_K G_K^{-1/2}.$$

Then P_K is positive definite, F_K is orthogonal and $O_K = F_K P_K$. An isometric realization $J : V \to \mathcal{Y}$ preserves G_K and P_K, and transports the isometric part to JF_K.

Proof. The positivity of the Gram matrix allows its root and inverse to be defined through the spectral theorem. The identity $F_K^* F_K = G_K^{-1/2} G_K G_K^{-1/2} = I$ proves the assertion. Moreover, $(JO_K)^*(JO_K) = O_K^* O_K$. □

This polar decomposition is defined from the complete Gram matrix: it is not obtained solely from the list of its eigenvalues or from the norm of K. The symbol P_K in this corollary is not the rank-ten dimensional projector to be constructed below. This separation is a consequence of the cyclic reference, not an identification of all its readers.

144.4 Operator identification and relative holonomy

Theorem 144.4 (Twelve general responses or one cyclic response)**.** *For any linear operator $H : V \to V$, its twelve responses $Y = (HK, HSK, \dots, HS^{11}K)$ determine*

$$H = YO_K^{-1}.$$

If $HS = SH$, a single complete vector response $y = HK$ determines

$$H = O_y O_K^{-1}, \qquad H = \sum_{j=0}^{11} h_j S^j, \qquad h = O_K^{-1} y. \tag{144.6}$$

With errors E in the response matrix or δy in the vector, respectively,

$$\|\widehat{H} - H\| \le \frac{\|E\|}{\sqrt{589609}}, \tag{144.7}$$

$$\|\widehat{h} - h\| \le \frac{\|\delta y\|}{\sqrt{589609}}, \qquad \|\widehat{H} - H\| \le \sqrt{\frac{12}{589609}}\,\|\delta y\|. \tag{144.8}$$

Proof. The first equality is $Y = HO_K$. If H commutes with S, then $HS^jK = S^j y$, so that $Y = O_y$. Commutation with the cycle determines all columns of H from one of them; the twelve powers $I, S, \dots, S^{11}$ are independent and span that operator space. Therefore $y = O_K h$, which gives the second formula. The first two bounds use $\|O_K^{-1}\| = 1/\sqrt{589609}$. The last uses $\|O_{\delta y}\| \le \|O_{\delta y}\|_{\mathrm{F}} = \sqrt{12}\|\delta y\|$. □

A response here means a vector of twelve components, not a scalar measurement. As an example, if $H = 2I + 3S - S^5$, the response $y = HK$ returns exactly the coefficients $2, 3, -1$ in their positions through (144.6); the inverter requires y, K, S, not the filter coefficients. By contrast, the nonzero operator whose first row is $(543, -234, 0, \dots, 0)$ and whose remaining rows are zero annihilates K. A single response does not identify an arbitrary operator.

The reference function admits an exact generalization. In a cycle of n positions, $H \mapsto Hg$ identifies all operators that commute with the cycle if and only if $O_g = (g, Sg, \dots, S^{n-1}g)$ is invertible. Sufficiency is the preceding proof. If O_g is singular, a nonzero vector in its kernel gives a polynomial of degree less than n that annihilates g; the operator is nonzero because the powers of the cycle are independent. The unit impulse also has an orthonormal orbit and serves as a reference. The specific consequence for K is that the same object already used in closure and incidence satisfies that condition, without having been chosen anew to optimize the test. No identification of nonlinear dynamics or arbitrary hidden states follows; an application to a linearization requires that it have been defined and justified.

Corollary 144.5 (Recovery of relative holonomy from the complement)**.** *Let H_B be known and invertible, and let H_C be another transport of the same type. The complement*

$$D_\gamma = \frac{\sqrt{8}}{9}(H_C - H_B)$$

determines

$$H_B^{-1} H_C = I + \frac{9}{\sqrt{8}} H_B^{-1} D_\gamma.$$

If D_γ commutes with S, in particular if both transports do, $y = D_\gamma K$ suffices to recover

$$H_B^{-1} H_C = I + \frac{9}{\sqrt{8}} H_B^{-1} O_y O_K^{-1}. \tag{144.9}$$

The error in this reconstruction is bounded by

$$\frac{9}{\sqrt{8}}\|H_B^{-1}\|\sqrt{\frac{12}{589609}}\,\|\delta y\|.$$

Proof. We solve for H_C in the definition of the complement and apply theorem 144.4. The bound follows from submultiplicativity. □

For $H_B = S^3$ and $H_C = S^4$, the rational datum $y/\sqrt{8} = (S^4 - S^3)K/9$ allows the relative holonomy S to be reconstructed without providing H_C to the inverter. If H_B is unitary, the error factor is approximately 0.01435510. Without commutation, the twelve general responses are retained, not a fictitious symmetry hypothesis. The selection of a physical loop and its representation still belong to its generating dynamics; this protocol does not replace them.

144.5 Preservation of references and observables through the complete register

Let C_n be unitary operators on a Hilbert space H,

$$T_n = \frac{8I + C_n}{9}, \qquad D_n = \frac{\sqrt{8}(C_n - I)}{9}, \qquad R_0 = I, \quad R_{n+1} = T_n R_n.$$

This formulation includes nonstationary sequences. It does not require commutation between C_n at different stages.

Theorem 144.6 (Isometric register, terminal component, and transport)**.** *For every N,*

$$R_N^* R_N + \sum_{n=0}^{N-1} R_n^* D_n^* D_n R_n = I.$$

The positive strong limit $A_\infty = \lim_N R_N^* R_N$ exists, and

$$\mathcal{V}x = \big(A_\infty^{1/2}x, (D_n R_n x)_{n\ge 0}\big) \quad \textit{satisfies} \quad \mathcal{V}^*\mathcal{V} = I. \tag{144.10}$$

Its image $\mathcal{M}$ is closed, and the inverse on it is $\mathcal{V}^*$, of norm one. In the finite case, the first component is replaced by $R_N x$ and the N complements are retained. For any bounded operator A,

$$A^{\mathcal{V}} = \mathcal{V}A\mathcal{V}^*\big|_{\mathcal{M}}$$

preserves products, adjoints, norm, and spectrum on that image, and

$$\langle \mathcal{V}x, A^{\mathcal{V}}\mathcal{V}y\rangle = \langle x, Ay\rangle.$$

Proof. Expanding the two terms of one stage gives $T_n^*T_n + D_n^*D_n = I$. Hence $R_n^*R_n - R_{n+1}^*R_{n+1} = R_n^*D_n^*D_nR_n$, and the telescoping sum proves the finite identity. The sequence of positive operators $R_N^*R_N$ is decreasing and bounded below; its strong limit exists. Taking inner products with x, the sum of the squared norms of the complements is $\|x\|^2 - \langle x, A_\infty x\rangle$. This proves that the series belongs to the Hilbert direct sum and that $\mathcal{V}$ is an isometry. An isometry has closed image and an adjoint inverse on it. Inserting $\mathcal{V}^*\mathcal{V} = I$ into the compositions proves the transport properties. Neither convergence of $R_N x$ nor the assertion that A_∞ is a projector has been assumed. □

Applying this result to the carrier of the register,

$$(\mathcal{V}O_K)^*(\mathcal{V}O_K) = O_K^*O_K. \tag{144.11}$$

The twelve directions and all their correlations are preserved, with the same bounds as in theorem 144.2, at any depth of complete memory. It is not asserted that the entire continuum has dimension twelve: this

twelve-dimensional reference is preserved in its image. The correct operators there are $S^{\mathcal{V}} = \mathcal{V}S\mathcal{V}^*$ and $H^{\mathcal{V}} = \mathcal{V}H\mathcal{V}^*$. An ambient shift of storage slots is not presumed to be $S^{\mathcal{V}}$.

If $z = \mathcal{V}y + \delta z$, the estimate

$$\widehat{h} = O_K^{-1}\mathcal{V}^* z$$

retains the first bound of (144.8) with $\|\delta z\|$ in place of $\|\delta y\|$. Depth does not worsen this constant because the isometry and its terminal component are preserved. A partial selection of channels has a different Gram matrix and requires its own bounds. The exterior powers $\bigwedge^p \mathcal{V}$ preserve Gram determinants; finite-rank traces and spectral measures of the transported observables are also preserved. Squared norms are not thereby interpreted automatically as Born probabilities: that reading requires its subsequent realization.

144.6 Exceptional incidence and selected direction

Let $e = \mathbf{1}$, $\Pi = I - ee^*/12 = P_{11}$. In the already constructed icosahedral chart A_5/C_5, the orthogonal representation ρ and the three-dimensional character determine

$$P_3 = \frac{3}{60}\sum_{g\in A_5} \chi_3(g^{-1})\rho(g), \quad P_3^* = P_3 = P_3^2, \quad \operatorname{rank} P_3 = 3, \quad P_3 e = 0. \tag{144.12}$$

This projector is received together with its representation and markings; it is not chosen to fit a direction to the register. Put

$$k = \Pi K, \qquad u = P_3 k, \qquad E_u = \frac{uu^*}{\|u\|^2}, \qquad Q_K = \Pi - E_u = P_{10}(K).$$

The exact evaluations are

$$\|k\|^2 = \frac{11789915}{12}, \qquad \|u\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20}, \tag{144.13}$$

$$\eta_K := \frac{\|u\|^2}{\|k\|^2} = \frac{3(6638585 + 2275584\sqrt{5})}{5 \cdot 11789915} \simeq 0.5967954228259091. \tag{144.14}$$

In particular, $u \neq 0$. Since $u^*k = \|u\|^2$,

$$E_u k = u, \qquad Q_K k = k - u, \qquad k = u + Q_K k, \qquad \|k\|^2 = \|u\|^2 + \|Q_K k\|^2. \tag{144.15}$$

The reduction $12 \to 11 \to 10$ removes first the uniform direction and then a selected line. It does not remove all of $\operatorname{im} P_3$: in $\operatorname{im} Q_K$, the two directions of that subspace orthogonal to u remain. These dimensions are those of the algebraic spaces written above; identifying them with physical dimensions requires their realization maps.

Proposition 144.7 (Exact transport of direction and descriptor)**.** *For a complete isometry $\mathcal{V}$, let $\Pi^{\mathcal{V}} = \mathcal{V}\Pi\mathcal{V}^*$, $P_3^{\mathcal{V}} = \mathcal{V}P_3\mathcal{V}^*$, and $u^{\mathcal{V}} = \mathcal{V}u$. On $\mathcal{M}$,*

$$Q_K^{\mathcal{V}} = \mathcal{V}Q_K\mathcal{V}^* = \Pi^{\mathcal{V}} - \frac{u^{\mathcal{V}}(u^{\mathcal{V}})^*}{\|u^{\mathcal{V}}\|^2}, \quad \operatorname{rank} Q_K^{\mathcal{V}} = 10, \quad \eta_K^{\mathcal{V}} = \eta_K.$$

Moreover, $\mathcal{V}^*u^{\mathcal{V}} = u$ and $\mathcal{V}^*Q_K^{\mathcal{V}}\mathcal{V} = Q_K$.

Proof. Substitute the definitions and $\mathcal{V}^*\mathcal{V} = I$. Norms and rank are preserved by the isometry. The identity on the image is $\mathcal{V}\mathcal{V}^*$, not the identity of a larger ambient storage space. Extension by zero outside the image adds no physical directions. □

The complete incidence screen provides another exact realization. If $\mathcal{I}$ is the incidence operator already constructed and $E_{\mathrm{inc}} = \mathcal{I}(e^{\perp})$, then

$$F : V \longrightarrow Y = \mathbb{R} \oplus E_{\mathrm{inc}}, \quad Fx = \left(\mu(x), \frac{\mathcal{I}\Pi x}{6}\right), \quad \mu(x) = \frac{\mathbf{1}^* x}{12}, \qquad \langle (\mu, y), (\nu, z)\rangle_Y = 12\mu\nu + \langle y, z\rangle \tag{144.16}$$

is a surjective isometry, since $\Pi\mathcal{I}^*\mathcal{I}\Pi = 36\Pi$. For the raw incidence of the 132 hexads, the complete identity is $\mathcal{I}^*\mathcal{I} = 36I + 30\mathbf{11}^*$; centring removes exactly its uniform component. The incidence image has dimension eleven; Y is not all of $\mathbb{R} \oplus \mathbb{R}^{132}$. Applying F term by term to memory preserves both the Gram matrix and the preceding operators and their domains. This composition introduces no Monster action on K.

144.7 Continuous stability and the need to preserve correlations

Theorem 144.8 (Direction error from complete memory)**.** *Let $z = \mathcal{V}K + \delta$, $\|\delta\| \le \varepsilon$, and reconstruct $\widehat{K} = \mathcal{V}^* z$, $\widehat{k} = \Pi\widehat{K}$, $\widehat{u} = P_3\widehat{k}$. All three errors are less than or equal to ε. If $\varepsilon < \|u\|$,*

$$\|Q_{\widehat{K}} - Q_K\| \le \frac{\varepsilon}{\|u\|}.$$

Si $\varepsilon < \|k\|$,

$$|\eta_{\widehat{K}} - \eta_K| \le \frac{\varepsilon}{\|k\|}.$$

Proof. The adjoint of the isometry and the projectors have norm one. Noise orthogonal to $\mathcal{M}$ disappears upon applying $\mathcal{V}^*$. For two nonzero lines, the norm of the difference of their projectors is the sine of the angle between them. The distance from u to the line of $\widehat{u}$ does not exceed $\|u - \widehat{u}\|$, so this sine is less than or equal to $\varepsilon/\|u\|$. The hypothesis prevents $\widehat{u} = 0$.

For the second assertion, normalize k and $\widehat{k}$. The difference of their rank-one projectors has eigenvalues $\pm\sin\theta$ in their plane. When paired with $0 \le P_3 \le I$, its absolute value does not exceed $\sin\theta$: in an eigenvector basis one obtains $\sin\theta(a - b)$, with $a, b \in [0, 1]$. The same distance argument gives $\sin\theta \le \varepsilon/\|k\|$. If the lines coincide, the difference of the ratios is zero. □

Here $\|u\| \simeq 765.733$, so the factor for the projector is approximately 0.00130594. In the reader with only two memories, the inverse has norm $9/4$ and the corresponding factor is $9/(4\|u\|) \simeq 0.00293836$. These are different observed data; the difference is not a cost-free improvement of the same channel. The integer correction in theorem 145.5 will allow the register to be recovered first without error within its radius, instead of propagating a continuous error to the projector.

Proposition 144.9 (Cross blocks are part of the observable)**.** *Write $\mathcal{V}x = (W_a x)_a$, including the terminal component. Then the (a, b) block of $A^{\mathcal{V}}$ is*

$$(A^{\mathcal{V}})_{ab} = W_a A W_b^*.$$

Its quadratic form is computed as a limit of finite truncations

$$\langle y, A^{\mathcal{V}} y\rangle = \lim_{F} \sum_{a,b\in F} \langle y_a, W_a A W_b^* y_b\rangle.$$

General conservation is not obtained by discarding all terms $a \neq b$.

Proof. The block formula follows from multiplying $\mathcal{V}A\mathcal{V}^*$. The truncation projectors converge strongly to the identity and $A^{\mathcal{V}}$ is bounded, proving the limit without asserting absolute convergence of an arbitrary

double series. For a counterexample, take $H = \mathbb{R}$, $C = -I$, $T = 7/9$, $D = -2\sqrt{8}/9$, $\mathcal{V}x = (Tx, Dx)$, and $A = I$. The diagonal part, evaluated at $\mathcal{V}x$, equals

$$(T^4 + D^4)x^2 = \frac{3425}{6561}x^2.$$

The cross terms contribute $3136x^2/6561$, and the sum is x^2, as required by the isometry. □

144.8 Discrete correction before incidence selection

The canonical two-memory readers are

$$T_j = \frac{8I + S^j}{9}, \qquad D_j = \frac{\sqrt{8}(S^j - I)}{9}, \qquad Mx = (D_3x, D_4T_3x).$$

The second memory acts after the first compression; the first complement is not reinjected in this protocol. The general theorem in theorem 145.5 will prove, without using the register as a target, the strict radii

$$r_0 = \frac{2\sqrt{146}}{81}, \qquad r_q = \frac{4\sqrt{73}}{81}.$$

The first radius recovers ΠK from MK, modulo uniform integer translation. The second recovers K in full when $q = \mathbf{1}^*K$ is also preserved. If a real-valued charge is observed with error less than $1/2$, rounding first recovers its integer charge channel. None of these procedures requires the terminal component T_4T_3K.

The high-block selection in the marked chart is

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\}. \tag{144.17}$$

The fourth component is exactly at the threshold. This circumstance separates continuous stability from discrete stability.

Proposition 144.10 (Zero continuous margin and integer restoration)**.** *The direct selector in* (144.17)*, applied to the real-valued reconstruction of the memories, has no positive Euclidean stability radius at MK, even with exact charge. By contrast, prior integer decoding with error less than r_q restores B_K exactly without changing the threshold.*

Proof. Let L be the least-squares inverse, satisfying $LM = \Pi$. For $t > 0$, put

$$e_t = -tM\Pi e_4, \qquad \widetilde{K}_t = \frac{q}{12}\mathbf{1} + L(MK + e_t) = K - t\Pi e_4.$$

We have $\|e_t\| \to 0$, but the fourth component is $729 - 11t/12 < 729$. For small t, only this membership changes. The second assertion follows by first recovering K exactly through theorem 145.5 and then applying the same selector. □

If the remaining marked data $P_\pi, H_e, H_\varphi, H_{\mathrm{APP}}$ are kept exact or transported according to their law, uniqueness of the exceptional construction returns the same flag $B_K \subset H_\alpha^-$ and the same marked origin. Correcting the register is not equivalent to correcting independent errors in these markings. The witness

$$K' = K - e_4 + e_6, \qquad Q(K') = 6263, \qquad B_{K'} = \{6, 9, 10\}$$

also belongs to $\{0, \dots, 999\}^{12}$ and satisfies $\|MK' - MK\|^2 = 4672/6561$. The midpoint of the two memories is at distance r_q from each. This proves that the strict radius is uniformly optimal on the fixed-charge algebraic domain, not that K' has a constructed HMT terminal history.

Without charge, $M(K-\mathbf{1}) = MK$ and the high-block selections differ. The two memories can recover Q_K and η_K exactly, since these ignore the mean, without recovering that absolute selector. The condition $P_3\Pi K \neq 0$ remains necessary for the projector; integer correction does not remove this defining condition. A concrete finite test uses $z = MK/\sqrt{8}$ and adds $1/10$ to its first component. The error in the norm of M is $\sqrt{8}/10 < r_q$. The floor-and-ceiling algorithm in proposition 145.6 leaves 924 candidates of charge 6263 and recovers K and B_K exactly.

Applying the isometry (144.16) gives $M_{\mathrm{inc}} = (F \oplus F)M$. All distances between words and all noise norms are preserved, so the same radii and decoders apply in $Y \oplus Y$. This is the quantitative connection between memory and incidence; it does not depend on attributing an additional exceptional symmetry to the shift S.

144.9 Covariance, refinement, and variational charge

If G is orthogonal and $K' = GK$, $\Pi' = G\Pi G^*$, and $P_3' = GP_3G^*$ are transported jointly, then

$$u' = Gu, \qquad Q'_{K'} = GQ_KG^*, \qquad \eta'_{K'} = \eta_K.$$

In a fixed chart, $K \mapsto aK + b\mathbf{1}$, $a \neq 0$, leaves Q_K and η_K invariant; it does not leave the entire object invariant. For example, $\kappa(K+\mathbf{1}) - \kappa(K) = 1/999$, and the reading α_A would change by $-1/999$ if the regions were kept fixed. This is an algebraic counterexample to identifying the descriptor with the entire register, not an assertion of admissibility of that update as an HMT history.

Nor can one move only the register and arbitrarily keep the projector fixed. The finite formula (144.12) gives exactly

$$\|P_3S^3K\|^2 = \frac{369035}{4} - \frac{281861\sqrt{5}}{10}, \qquad \|P_3S^4K\|^2 = \frac{1327717}{4} + \frac{694618\sqrt{5}}{5}.$$

These differ from (144.13); in particular, $[P_3, S^3]$ and $[P_3, S^4]$ do not vanish. Transporting P_3 by the same conjugation does preserve η_K.

Decoding is also covariant. For a permutation R,

$$S' = RSR^{-1}, \qquad M' = (R \oplus R)MR^{-1}, \qquad M'R = (R \oplus R)M.$$

The lattice, charge and distances are preserved. In the uniqueness regime, the decoder output transforms by R; outside it, the set of minimizers is transported, without breaking a tie arbitrarily. A general orthogonal frame requires also transporting $\mathbb{Z}^{12}$ to $R\mathbb{Z}^{12}$ and the uniform direction to $R\mathbf{1}$. Rounding in the old lattice after changing the chart does not represent the same problem.

Proposition 144.11 (Naturality under refinement with local frames)**.** *Let z be a cylinder with children z', compatible positive weights $\sum_{\tau z'=z} \mu(z') = \mu(z)$, and orthogonal frames G_z. The refinement*

$$\mathcal{R}_n(e_z \otimes x) = \sum_{\tau z'=z} \sqrt{\frac{\mu(z')}{\mu(z)}}\, e_{z'} \otimes G_{z'}G_z^*x$$

is isometric. For the projector field $Q_z = G_zQ_KG_z^*$ and its block operator Q_n,

$$Q_{n+1}\mathcal{R}_n = \mathcal{R}_nQ_n, \qquad \mathcal{R}_n^*Q_{n+1}\mathcal{R}_n = Q_n.$$

The identities compose between levels and pass to the inductive limit.

Proof. The children have orthogonal supports and the weights sum to one; the frame changes are orthogonal. In each child, $Q_{z'}G_{z'}G_z^* = G_{z'}G_z^*Q_z$, which proves the first square. Multiplying by the

adjoint of the refinement gives the second. Compatibility by composition defines the operator in the limit of the isometric inclusions. The same argument holds for E_u. □

The statement preserves the publication of a prefix under compatible refinements. New events can produce another register; no temporal constancy of K follows. The ranks of fields over tensor fibers are those of these fibers, not a count of physical dimensions.

Proposition 144.12 (A transport charge with an explicit action)**.** *Let $U_n : E_n \to E_{n+1}$ be invertible transports. The auxiliary discrete action with cotangent variables is*

$$\mathcal{A}(q,p) = \sum_n p_{n+1}^*(q_{n+1} - U_n q_n).$$

Finitely supported variations give $q_{n+1} = U_n q_n$ and $p_n = U_n^* p_{n+1}$. If $G_0 = I$, $G_{n+1} = U_n G_n$, $A_0 = Q_K$, and $A_n = G_n A_0 G_n^{-1}$, the quantity

$$J_n = p_n^* A_n q_n$$

is independent of n. The action is invariant under the continuous group $q_n \mapsto \exp(tA_n)q_n$, $p_n \mapsto \exp(-tA_n^*)p_n$.

Proof. The indicated equations follow by varying p_{n+1} and q_n. The definition of A_n implies $A_{n+1}U_n = U_n A_n$. Hence

$$p_{n+1}^* A_{n+1} q_{n+1} = p_{n+1}^* U_n A_n q_n = p_n^* A_n q_n.$$

The same intertwining holds for the exponentials and makes every term of the action invariant. It is not required that A_n be antisymmetric: the covector transformation is contragredient. □

For orthogonal transports one may choose $p_n = q_n$; then $J_n = \|A_n q_n\|^2$, and the complement provides the remaining charge. In a symplectic realization without a positive form, the pairing written above is preserved and is not called positive energy. If the fibres are archived through isometries $\mathcal{V}_n$, the definitions

$$\widehat{U}_n = \mathcal{V}_{n+1} U_n \mathcal{V}_n^*, \qquad \widehat{A}_n = \mathcal{V}_n A_n \mathcal{V}_n^*, \qquad \widehat{q}_n = \mathcal{V}_n q_n,$$

with $\widehat{p}_n(y) = p_n(\mathcal{V}_n^* y)$ preserve the intertwining and the same charge on the images. This composition provides a concrete continuous action and symmetry for a conservation law; it attributes no automatic Noether currents to a finite group, nor does it identify the auxiliary action with every physical action.

144.10 Encoding, operator identification, and incidence stability

The contribution assembled here is an operational chain: the generated register has a reversible encoding; its orbit is a complete reference; terminal component and complements preserve this reference and the transported observables; the integer separation of two memories allows incidence and direction to be restored even when the real-valued selector has zero margin. The proofs concerning linearity, projection, and lattices are used as the language of subsequent realization, without changing the production of the object.

Each hypothesis has a witness that prevents it from being inadvertently extended. An isolated response does not identify an operator that does not commute with the cycle; only the complete memory automatically preserves the original Gram matrix; cross terms cannot be omitted; absolute selection requires charge; the projector requires a nonzero direction; noise bounds depend on normalization; a chart transformation requires transporting the form, lattice and readers. A reading perturbation does not become an admissible history by decree. These delimitations preserve the functions of K, without reducing it to twelve disconnected numbers or promoting a canonical property to universal invariance.

Chapter 145

Ternary algebraic norm and optimal recovery radii

The recovery radius of the two memories contains the factor $\sqrt{73}$. Its origin can be traced without introducing a target magnitude: it comes from the Gram matrix of the nonadic reader and, in another realization of the ternary cycle already constructed, is the scale factor of a similitude. Its square, 73, is the index of the rank-two lattice inclusion. These occurrences are brought together through explicit operations; they are not identified through a decimal resemblance to alpha. The subsequent comparison with $10^4\alpha_A$ will be treated as an inverse problem, with its data, permitted transformations, and exact conclusions.

The raw material preserves its APP–TRIT–TPK production and the subsequent cut described in section 144.1. The integer register, the orientation of shifts, the charge and the readers are received before distances are calculated. The canonical case uses $8+1$ and the denominator nine. The parameter $a>0$ introduced below is a subsequent algebraic family, not a new seed or a physical parameter selected by alpha. Errors belong to the reading channel; they are not identified with admissible events or histories.

145.1 Parametric double-memory operators and Gram matrix

In $V=\mathbb{R}^{12}$, let $(Sx)_i=x_{i+1}$, with indices modulo twelve. For $a>0$ and $j=3,4$, define

$$T_{j,a}=\frac{aI+S^j}{a+1},\qquad D_{j,a}=\frac{\sqrt{a}(S^j-I)}{a+1},\qquad M_a x=(D_{3,a}x,D_{4,a}T_{3,a}x). \tag{145.1}$$

The second memory is produced after the first compression. Archiving both is not equivalent to reinjecting the first complement into a reversible two-channel evolution. Let

$$p(a)=a^2+a+1,\quad A(a)=a(a^2+1),\quad B(a)=a^2,\qquad \Pi=I-\frac{\mathbf{1}\mathbf{1}^*}{12},\quad P_u=I-\Pi.$$

Proposition 145.1 (Balance and difference form)**.** *For every $a>0$,*

$$T_{j,a}^*T_{j,a}+D_{j,a}^*D_{j,a}=I,\qquad G_a:=M_a^*M_a=I-(T_{4,a}T_{3,a})^*(T_{4,a}T_{3,a}),$$

and

$$\begin{aligned}(a+1)^4G_a&=4ap(a)I-A(a)(S^3+S^4+S^8+S^9)\\&\quad-B(a)(S+S^5+S^7+S^{11}).\end{aligned} \tag{145.2}$$

Equivalently,

$$\begin{aligned}(a+1)^4\|M_ah\|^2&=A(a)\sum_i\big((h_i-h_{i+3})^2+(h_i-h_{i+4})^2\big)\\&\quad+B(a)\sum_i\big((h_i-h_{i+1})^2+(h_i-h_{i+5})^2\big).\end{aligned} \tag{145.3}$$

Proof. Orthogonality of S gives

$$T_{j,a}^*T_{j,a}=\frac{(a^2+1)I+a(S^j+S^{-j})}{(a+1)^2},\qquad D_{j,a}^*D_{j,a}=\frac{a(2I-S^j-S^{-j})}{(a+1)^2}.$$

Their sum is I; applying the identity to the two stages cancels the intermediate term $T_{3,a}^*T_{3,a}$. Since all operators are polynomials in S, the product of the two individual Gram matrices has diagonal $(a^2+1)^2$, single

terms of weight $a(a^2+1)$, and cross terms of weight a^2. The identity $(a+1)^4-(a^2+1)^2=4a(a^2+a+1)$ gives (145.2). Expanding the squared differences gives (145.3); each unoriented edge of each step appears exactly once, and the weighted degree is $4A+4B=4ap(a)$. □

At $a=8$ we obtain

$$\begin{aligned}6561G_8 = 2336I &- 520(S^3+S^4+S^8+S^9)\\ &- 64(S+S^5+S^7+S^{11}),\end{aligned} \tag{145.4}$$

whose first row is $(2336,-64,0,-520,-520,-64,0,-64,-520,-520,0,-64)/6561$. Also,

$$T_{4,8}T_{3,8}=\frac{64I+8S^3+8S^4+S^7}{81}.$$

The four distinct shifts produce the diagonal contribution

$$1-\frac{64^2+8^2+8^2+1}{81^2}=\frac{2336}{6561}=\frac{32(8^2+8+1)}{9^4}.$$

Theorem 145.2 (Kernel, spectrum, and complete inverse)**.** *For every $a>0$,* $\ker M_a=\mathbb{R}\mathbf{1}$. *The eigenvalues of G_a, indexed by the twelfth-root modes m, are*

m	$\lambda_m(G_a)$	*multiplicity*
0	0	1
$3,9$	$2a/(a+1)^2$	2
$4,8$	$3a/(a+1)^2$	2
6	$4a/(a+1)^2$	1
$1,5,7,11$	$a(5a^2+4a+5)/(a+1)^4$	4
$2,10$	$a(7a^2+2a+7)/(a+1)^4$	2

In particular,

$$L_a^{\mathrm{rec}}=(G_a+P_u)^{-1}M_a^*,\qquad L_a^{\mathrm{rec}}M_a=\Pi,\qquad \|L_a^{\mathrm{rec}}\|=\frac{a+1}{\sqrt{2a}}. \tag{145.5}$$

Proof. If $M_ah=0$, the first memory implies $S^3h=h$, hence $T_{3,a}h=h$; the second implies $S^4h=h$. Since $S=S^4(S^3)^{-1}$, the vector is uniform. The converse is immediate. In mode m, the balance gives

$$\lambda_m(G_a)=1-\frac{[a^2+1+2a\cos(\pi m/2)][a^2+1+2a\cos(2\pi m/3)]}{(a+1)^4}.$$

The twelve cosine pairs give the table. Relative to $2a/(a+1)^2$, the differences of the last two values are

$$\frac{3a(a^2+1)}{(a+1)^4}>0,\qquad \frac{a(5a^2-2a+5)}{(a+1)^4}>0.$$

The other positive values are larger multiples. Hence the minimum on $\mathbf{1}^\perp$ is $2a/(a+1)^2$. The operator G_a+P_u equals one on the uniform direction and is positive and invertible on its complement. The formula displayed is the least-squares inverse, and its norm is the reciprocal of the smallest nonzero singular value of M_a. □

At $a=1$, $T_{3,1}=(I+S^3)/2$ is singular, but (145.5) remains valid. Singularity of an individual compression does not eliminate recovery through its two memories. At $a=8$, the norm of the inverse is $9/4$.

145.2 Minimum distance in the domain of integer registers

The algebraic decoding domains are

$$C_0(a) = M_a\mathbb{Z}^{12}, \qquad \Lambda_q = \{k \in \mathbb{Z}^{12} : \mathbf{1}^* k = q\}, \qquad C_q(a) = M_a\Lambda_q.$$

The first image is identified with the lattice $\Pi\mathbb{Z}^{12}$ or with classes modulo $\mathbb{Z}\mathbf{1}$; the second preserves a fixed integer charge. These are domains containing the integer publications of the indicated type. Their use does not assert that each candidate corresponds to a produced HMT history.

Lemma 145.3 (Minimum cut of the memory graph)**.** *Every nonempty proper cut of the twelve positions has at least four edges of steps* $1, 5$ *and at least four of steps* $3, 4$*. Its weight in* (145.3) *is therefore at least* $4A(a) + 4B(a) = 4ap(a)$*; a cut consisting of one position attains this bound.*

Proof. Each step 1 and 5 forms a connected cycle of twelve positions; each cycle crosses a nontrivial cut at least twice. For the steps $3, 4$, if the set is invariant under neither, each family contributes at least two edges. If it is invariant under S^3, it is a union of classes modulo three; each of the four triangles of S^4 crosses it twice, giving eight edges. If it is invariant under S^4, it is a union of classes modulo four; each of the three length-four cycles of S^3 crosses it at least twice, giving at least six. Simultaneous invariance would force invariance under S, impossible for a nonempty proper set. All weights are positive; the cut of a single position has exactly four edges of each weight. □

Theorem 145.4 (Exact distances and fixed-charge minimizers)**.** *For every* $a > 0$ *and every integer charge* q*,*

$$d_0(a)^2 = \min_{h \notin \mathbb{Z}\mathbf{1}} \|M_a h\|^2 = \frac{4ap(a)}{(a+1)^4}, \qquad d_q(a)^2 = \frac{8ap(a)}{(a+1)^4}. \tag{145.6}$$

In the first minimum, h *ranges over* $\mathbb{Z}^{12}$*; the second is the minimum distance between distinct words of* $C_q(a)$*. The differences attaining the second minimum are exactly* $e_i - e_j$ *with* $j - i \equiv 2, 6, 10 \pmod{12}$*.*

Proof. For a nonconstant integer vector h, $(h_i - h_j)^2 \geq |h_i - h_j|$. Each absolute difference decomposes as a sum of cuts of the level sets $\{i : h_i > t\}$. There is at least one nontrivial level; by lemma 145.3, the weighted energy is at least $4ap(a)$. The vector e_i attains this energy. Division by $(a+1)^4$ proves the first minimum over all integer vectors, not only over indicators of subsets.

For fixed charge, the difference $h \neq 0$ is integral and has zero sum. Its squared norm is even because $h_i^2 \equiv h_i \pmod 2$. If $\|h\|^2 \geq 4$, the spectral minimum gives

$$\|M_a h\|^2 \geq \frac{8a}{(a+1)^2} > \frac{8ap(a)}{(a+1)^4},$$

since $(a+1)^2 - p(a) = a > 0$. Only $\|h\|^2 = 2$ remains, that is, $h = e_i - e_j$. Its energy is $2((G_a)_{ii} - (G_a)_{ij})$. The off-diagonal entries are zero at separations $2, 6, 10$, negative with weight $B(a)/(a+1)^4$ at $1, 5, 7, 11$, and negative with weight $A(a)/(a+1)^4$ at $3, 4, 8, 9$. The diagonal is $4ap(a)/(a+1)^4$; this proves the minimum and the exhaustive classification over the entire half-line $a > 0$. □

Theorem 145.5 (Strict correction radii of the memories)**.** *For a datum* $y = M_a k + e$*, with error measured in the joint norm of the two memories, the optimal uniform radii are*

$$r_0(a) = \frac{\sqrt{ap(a)}}{(a+1)^2}, \qquad r_q(a) = \frac{\sqrt{2ap(a)}}{(a+1)^2}. \tag{145.7}$$

If $\|e\| < r_0(a)$*, the nearest neighbour in* $C_0(a)$ *recovers* Πk *exactly. With known charge* q *and* $\|e\| < r_q(a)$*, the nearest neighbour in* $C_q(a)$ *recovers* k *exactly.*

Proof. The lattice $\Pi\mathbb{Z}^{12} \subset (1/12)\mathbb{Z}^{12}$ is discrete, and M_a is bounded below on $\mathbf{1}^{\perp}$. Its images are discrete and admit a nearest neighbour. If c' is another codeword and d is the corresponding minimum distance, $\|y - c'\| \geq d - \|e\| > \|e\|$ when $\|e\| < d/2$. The minimum is unique, and the preimage is unique in each indicated domain. The midpoint of two words at minimum distance lies at $d/2$ from both: no decoder can uniformly distinguish both inputs from that same datum. Optimality refers to the complete algebraic domains, not to an additional subset of histories that might have greater separation. □

In the canonical case,

$$r_0(8) = \frac{2\sqrt{146}}{81} \simeq 0.298346814162829, \qquad r_q(8) = \frac{4\sqrt{73}}{81} \simeq 0.421926110879878. \tag{145.8}$$

The 4 comes from $\sqrt{2 \cdot 8}$ when taking half the distance between fixed-charge registers; $73 = 8^2 + 8 + 1$ comes from the Gram matrix; $81 = 9^2$ comes from the two nonadic normalizations. The same product 9×9 counts the APP cells, but these two roles are not identified without their typed map. Nor does the factor four alone prove an assertion about spatial directions.

If the memories are stored as $z = M_a k/\sqrt{a}$, the noise norm changes. At $a = 8$, the radii for this rational datum are $r_0/\sqrt{8} = \sqrt{73}/81$ and $r_q/\sqrt{8} = \sqrt{146}/81$. Using the radii in (145.8) without this conversion changes the problem. The application to Q_K, η_K, B_K, and the incidence screen has been proved in section 144.8; in particular, the exact threshold $K_4 = 729$ does not prevent its discrete restoration.

145.3 Finite decoder and parameter duality

Proposition 145.6 (Floor and ceiling without a target table)**.** *With known charge, below radius $r_q(a)$, it suffices to examine at most $2^{12} = 4096$ integer candidates. Without charge, below $r_0(a)$, it suffices to examine at most $12 \cdot 2^{12} = 49152$ candidates distributed among twelve charge residues. The residual minimum returns k or its uniform class, respectively.*

Proof. With charge q, form $\widetilde{k} = L_a^{\mathrm{rec}} y + (q/12)\mathbf{1}$. Then

$$\|\widetilde{k} - k\| < \|L_a^{\mathrm{rec}}\| r_q(a) = \frac{\sqrt{p(a)}}{a+1} < 1.$$

Each true integer coordinate belongs to $\{\lfloor \widetilde{k}_i \rfloor, \lceil \widetilde{k}_i \rceil\}$. These products are enumerated, those whose sum differs from q are removed, and $\|M_a k' - y\|$ is minimized. The correct candidate is present, and theorem 145.5 ensures its uniqueness. Coordinates that are already integers reduce the number of candidates.

Without charge, write $\rho \in \{0, \dots, 11\}$ for the charge residue. Each class modulo $\mathbb{Z}\mathbf{1}$ has a unique representative whose sum is ρ, obtained by subtracting $\lfloor q/12 \rfloor \mathbf{1}$. For each ρ, form $\widetilde{k}_\rho = L_a^{\mathrm{rec}} y + (\rho/12)\mathbf{1}$. At the correct residue,

$$\|\widetilde{k}_\rho - k_\rho\| < \|L_a^{\mathrm{rec}}\| r_0(a) = \frac{\sqrt{p(a)}}{\sqrt{2}(a+1)} < 1.$$

Apply floor and ceiling, retain the sum ρ, and minimize the residual across the twelve groups. Uniqueness of the code class gives a unique centred output. These procedures use the reader and integrality; they do not receive K as a comparison table. □

For rational a, rational arithmetic can be retained by writing $R_a = M_a/\sqrt{a}$, $z = y/\sqrt{a}$. The centred reconstruction is

$$(R_a^* R_a + P_u)^{-1} R_a^* z = L_a^{\mathrm{rec}} y.$$

The residual can be compared through its rational squares. These guarantees concern twelve coordinates: they assert neither polynomial complexity for arbitrary dimension, cryptographic authentication, nor quantum advantage. The perturbed example in section 144.8 illustrates the canonical algorithm.

Proposition 145.7 (Reciprocal duality and maximum radius)**.** *The substitution $a \mapsto 1/a$ preserves the Gram matrix, distances, radii, and norm of the inverse. We have*

$$D_{j,1/a} = D_{j,a}, \qquad T_{j,1/a} = S^j T_{j,a}^{*}.$$

The maximum radii of this family are attained only at $a = 1$:

$$r_q(a) \le \frac{\sqrt{6}}{4}, \qquad r_0(a) \le \frac{\sqrt{3}}{4}.$$

Proof. The two operator identities follow by multiplying numerator and denominator by a. The individual Gram matrices and formula (145.1) are invariant under reciprocity. Put $t = a/(a+1)^2 \in (0, 1/4]$. Then

$$r_q(a)^2 = 2t(1-t), \qquad r_0(a)^2 = t(1-t).$$

Both expressions increase for $0 < t \le 1/4$, and $t = 1/4$ is equivalent to $a = 1$. Reciprocity leaves t fixed. As a tends to zero or infinity, t and the radii tend to zero. □

Equality of Gram matrices does not mean that $M_a = M_{1/a}$: the second complement includes a different compression. The endpoints $a = 0$ and $a = \infty$ do not belong to the domain of the theorem. Optimizing the family at $a = 1$, or using the reciprocal representative $1/8$, does not retroactively change the canonical origin $a = 8$.

145.4 The norm on the ternary augmentation module

The cycle $r \mapsto 4r \pmod 9$, with orbits $(1, 4, 7)$ and $(2, 8, 5)$, provides the augmentation module whose chart and form are

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad G_{A_2} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad C^2 + C + I = 0, \qquad C^T G_{A_2} C = G_{A_2}.$$

The action and its contragredient orientation belong to the antecedent ternary fibres. On this already constructed module, we compose

$$Q_-(a) = aI - C, \qquad Q_+(a) = (a+1)I + C.$$

Proposition 145.8 (Ternary similitude and integer index)**.** *For $a > 0$,*

$$Q_- Q_+ = p(a) I, \qquad Q_-^{\dagger_{G_{A_2}}} = Q_+, \qquad Q_-^T G_{A_2} Q_- = p(a) G_{A_2}, \qquad \det Q_- = p(a).$$

At $a = 8$, both conjugate charts are

$$Q_- = \begin{pmatrix} 8 & 1 \\ -1 & 9 \end{pmatrix}, \qquad Q_+ = \begin{pmatrix} 9 & -1 \\ 1 & 8 \end{pmatrix}, \qquad Q_- Q_+ = 73 I.$$

Each integer quotient is cyclic of order 73. On the quotient of Q_-, C acts by 8; on that of Q_+, by $-9 \equiv 64 \equiv 8^{-1} \pmod{73}$.

Proof. Expanding the product and substituting $C^2 = -C - I$ gives $p(a)I$. Invariance of the form implies $C^{\dagger_{G_{A_2}}} = C^{-1} = C^2 = -I - C$, from which the adjoint follows. Composing product and adjoint gives the

similitude. The determinant of $\left(\begin{smallmatrix} a & 1 \\ -1 & a+1 \end{smallmatrix}\right)$ is $p(a)$. At $a = 8$, the greatest common divisor of the entries of each matrix is one and its determinant is 73; the Smith normal form is $\operatorname{diag}(1, 73)$. The relation $8I - C = 0$ in the first quotient and $9I + C = 0$ in the second gives the indicated actions. Since $8^3 \equiv 1 \pmod{73}$ and $8 \not\equiv 1$, both actions have order three and opposite orientations. □

Thus $\sqrt{73}$ is a similitude scale factor in the inherited ternary metric, not merely a number inside a radius. Its subsequent complex recognition is $|8 - \omega|^2 = 73$, with $\omega^2 + \omega + 1 = 0$. The complex unit describes the realization of the operator; it was not used to select the register. These equalities do not identify the operator Q_- with K, with the entire TPK dynamics, or with the cokernel of the action-scale reader.

145.5 Invariant contribution of the complete cycle

On the twelve positions, let $U = S^4$. There are now four cycles of length three. Their projectors are

$$P_0 = \frac{I + U + U^2}{3}, \qquad P = I - P_0.$$

The rank of P_0 is four and that of P is eight. In particular, $P_0 \neq P_u$, since the latter preserves only the global uniform direction; nor is P to be confused with the exceptional projector P_3. We shall now use $Q_{-,a} = aI - U$ and $Q_{+,a} = (a + 1)I + U$ for the operators of the complete cycle.

Theorem 145.9 (Ternary Gram matrix with a fixed sector)**.** *We have*

$$Q_{-,a}^* Q_{-,a} = (a - 1)^2 P_0 + p(a)P = p(a)I - 3aP_0. \tag{145.9}$$

For $v \neq 0$, let

$$w(v) = \frac{\|P_0 v\|^2}{\|v\|^2}, \qquad s_{-,a}(v) = \frac{\|Q_{-,a} v\|^2}{\|v\|^2}.$$

Then $s_{-,a}(v) = p(a) - 3aw(v)$. In the canonical case,

$$Q_-^* Q_- = 49P_0 + 73P = 73I - 24P_0, \qquad s(v) = 73 - 24w(v) \in [49, 73]. \tag{145.10}$$

Proof. From $U^3 = I$ and $U^* = U^2$ it follows that P_0 is the orthogonal projector onto $\ker(U - I)$. Expanding the Gram matrix,

$$Q_{-,a}^* Q_{-,a} = (a^2 + 1)I - a(U + U^2).$$

The relation $U + U^2 = 3P_0 - I$ gives (145.9). Evaluating on v and dividing by its squared norm proves the remaining identities. □

The term 24 is exactly $3 \cdot 8$: it is the difference $p(8) - (8 - 1)^2$ between the augmentation reading and the fixed reading. It already appears in a single cycle of three positions. For m cycles, the fixed rank is m, the dimension is $3m$, and

$$\operatorname{tr}(Q_{-,a}^* Q_{-,a}) = 3m(a^2 + 1).$$

At $m = 4, a = 8$, the trace is 780 and the total trace correction is $24 \cdot 4 = 96$. The numerical equality $24 = 2 \cdot 12$ is not the operator origin of the coefficient.

The two complete charts also distinguish their determinants. In a single cycle,

$$\det(8I - U) = 8^3 - 1 = 511 = 7 \cdot 73, \qquad \det(9I + U) = 9^3 + 1 = 730 = 10 \cdot 73.$$

The fixed factor is 7 and 10, respectively; the augmentation module contributes 73. On the twelve positions, $\det(8I - U) = 511^4 = 68184176641$. The identity

$$73 = 8^2 + 8 + 1 = 9^2 - 9 + 1 = \frac{9^3 + 1}{9 + 1} = \frac{729 + 1}{10}. \tag{145.11}$$

is preserved. Its interpretation as an index and determinant factor precedes any subsequent completion or refinement reading.

145.6 Six received states and conservation of the complete reading

We again fix $a = 8$ and omit that subscript in the readers. The six objects being compared are

$$K, \quad k = \Pi K, \quad m_0 = D_3 K, \quad m_1 = D_4 T_3 K, \quad u = P_3 k, \quad q_K = Q_K K = k - u.$$

None is selected from the target of the test. To give the evaluations without replacing them by decimals, write $a_v = \|v\|^2$, $b_v = \|P_0 v\|^2$. We obtain

v	a_v	b_v
K	4251257	10684363/3
k	11789915/12	1170761/4
m_0	15413392/81	16838032/243
m_1	1105759232/6561	0
u	$1327717/4 + (568896/5)\sqrt{5}$	$3029657/20 + (145120/3)\sqrt{5}$
q_K	$1951691/3 - (568896/5)\sqrt{5}$	$2292404/15 - (145120/3)\sqrt{5}$

(145.12)

The table is obtained by applying $P_0 = (I + S^4 + S^8)/3$, the formulas for D_3, D_4T_3, and the finite character projector (144.12) to (144.2). All its entries belong to $\mathbb{Q}(\sqrt{5})$; the first four are rational. In particular, the first three weights and readings are

$$w(K) = \frac{10684363}{12753771}, \qquad s(K) = \frac{224866857}{4251257},$$
$$w(k) = \frac{3512283}{11789915}, \qquad s(k) = \frac{776369003}{11789915},$$
$$w(m_0) = \frac{1052377}{2890011}, \qquad s(m_0) = \frac{61904585}{963337}.$$

For u and q_K, the exact expressions are $w = b_v/a_v$, $s = 73 - 24b_v/a_v$, with the numerators and denominators in (145.12). The subsequent evaluations are

v	$w(v)$, approximately	$s(v)$, approximately
K	0.8377414805	52.89420446705527
k	0.2979057101	65.85026295779062
m_0	0.3641429046	64.26057028848679
m_1	0	73 exact
u	0.4428244530591254	62.37221312658099
q_K	0.1127385160275117	70.29427561533972

The equality for the second memory is forced: $P_0 D_4 T_3 = \sqrt{8} P_0 (U - I) T_3 / 9 = 0$, while $\|m_1\|^2 > 0$. It is not an evaluational coincidence.

Proposition 145.10 (Complete memory does not alter the fixed weight)**.** *The map*

$$V_2 v = (t(v), m_0(v), m_1(v)) = (T_4 T_3 v, D_3 v, D_4 T_3 v)$$

is isometric and intertwines P_0 and $Q_- = 8I - U$ with their diagonal copies. It separately preserves $\|v\|^2$, $\|P_0 v\|^2$, and $\|Q_- v\|^2$. Hence the joint readings w and s do not change.

Proof. The two-stage balance gives $V_2^* V_2 = I$. Each channel is a polynomial in S, so it commutes with U, P_0, Q_-. Applying the isometry to v, $P_0 v$, and $Q_- v$ gives the three norm equalities. After division, the total reading is the average of the channel readings weighted by their squared norms, not their unweighted arithmetic mean. Zero channels do not contribute to the total ratio. □

The terminal component explicitly confirms this partition. For K,

$$\|t(K)\|^2 = \frac{25538253193}{6561}, \qquad \|P_0 t(K)\|^2 = \frac{848595371}{243},$$

$$w(t(K)) = \frac{22912075017}{25538253193}, \qquad s(t(K)) = \frac{1314402682681}{25538253193} \simeq 51.46799480558349.$$

The sum of the three squared norms is 4251257, and that of the fixed-sector norms is $(848595371 + 16838032)/243 = 10684363/3$. For $k = \Pi K$,

$$\|t(k)\|^2 = \frac{16367568169}{26244}, \qquad \|P_0 t(k)\|^2 = \frac{217142795}{972}.$$

The complements do not change under centring, and their sums recover the norms in the second row of (145.12). Discarding the terminal component or a complement alters the partial reading; retaining them introduces no hidden correction.

In a complete infinite memory, the same conclusion holds when the channels commute with U: the positive limit and its square root also commute with it. Without this commutation, the transport $\mathcal{V} P_0 \mathcal{V}^*$ is used on the image, in accordance with theorem 144.6, not an assumed diagonal copy.

145.7 Inversion, charts and bilateral balance

Proposition 145.11 (What inversion or a frame change preserves)**.** *Replacing U by U^{-1} exchanges each $Q_{\pm,a}$ with its adjoint and preserves P_0, their Gram matrices, singular values, determinants, and induced distances. For every $v \neq 0$, it preserves $w(v)$ and $s_{-,a}(v)$. This is not the same as inverting $Q_{-,a}$, whose inverse, for $a \neq 1$, is*

$$Q_{-,a}^{-1} = \frac{1}{a-1} P_0 + \frac{1}{p(a)} Q_{+,a} P. \tag{145.13}$$

For an invertible chart R, transporting $U' = RUR^{-1}$, $v' = Rv$ and $g' = (R^{-1})^* R^{-1}$ preserves these norm ratios in the metric g'.

Proof. Since $U^* = U^{-1} = U^2$, inversion produces the adjoints. The operators are normal polynomials in U; their Gram matrices depend on $U + U^2$, which does not change. The written inverse is verified by sectors: $Q_{-,a}$ acts by $a - 1$ on the fixed sector and the product with $Q_{+,a}$ equals $p(a) I$ on the augmentation. For the chart, $Q' v' = RQv$ and $R^* g' R = I$; numerator and denominator recover exactly their original values. The adjoint with respect to g' is also the transport of the original adjoint. □

Reversing the order of the twelve positions satisfies $RSR^{-1} = S^{-1}$ and preserves P_0. The translations S^j, including the antipodal translation S^6, commute with U; a global negation also leaves the norms unchanged. None of these operations alone turns 73 into a different reading.

With the Euclidean metric fixed, an orthogonal frame applied only to the state preserves $s_{-,a}(v)$ for all v if and only if $R^* P_0 R = P_0$, by (145.9) and polarization. The group of these transformations is $O(\operatorname{im} P_0) \times O(\operatorname{im} P) \cong O(4) \times O(8)$, larger than the commutant of U. By contrast, a state filter $c_0 P_0 + c_1 P$, when its output is nonzero, produces

$$w' = \frac{|c_0|^2 w}{|c_0|^2 w + |c_1|^2 (1 - w)}. \tag{145.14}$$

The reading may change. Choosing these coefficients from a target is a declared adjustment of the filter, not a chart change of the same reading.

Theorem 145.12 (Bilateral balance and explicit recovery)**.** *For every $a > 0$, the complete cycle satisfies*

$$\begin{aligned} Q_{+,a} - Q_{-,a}^* &= 3P_0, & Q_{+,a}Q_{-,a} &= p(a)I - 3P_0, \\ Q_{+,a}^* Q_{+,a} &= p(a)I + 3(a+1)P_0, \end{aligned} \tag{145.15}$$

and

$$(a+1)Q_{-,a}^* Q_{-,a} + aQ_{+,a}^* Q_{+,a} = (2a+1)p(a)I. \tag{145.16}$$

The map

$$W_a v = \frac{(\sqrt{a+1}\, Q_{-,a}v, \sqrt{a}\, Q_{+,a}v)}{\sqrt{(2a+1)p(a)}} \tag{145.17}$$

is isometric and is inverted on its image by W_a^*, of norm one. For the unnormalized channels $x = Q_{-,a}v$, $y = Q_{+,a}v$,

$$v = \frac{x+y}{2a+1}, \qquad Uv = \frac{ay - (a+1)x}{2a+1}. \tag{145.18}$$

A pair (x, y) belongs to this unnormalized image if and only if $Q_{+,a}x = Q_{-,a}y$.

Proof. The expression $Q_+ - Q_-^*$ contains $I + U + U^2 = 3P_0$. Expanding Q_+Q_-, $Q_+^*Q_+$, and using $U^2 = 3P_0 - I - U$ gives (145.15). The weighted combination cancels the P_0 terms in (145.9) and (145.15); this proves $W_a^*W_a = I$. Since $Q_- + Q_+ = (2a+1)I$, the sum of the channels gives v; their combination $ay - (a+1)x$ gives $(2a+1)Uv$.

The image condition is necessary because the two operators commute. If it holds, define $v = (x+y)/(2a+1)$. Then

$$Q_-v = \frac{Q_-x + Q_-y}{2a+1} = \frac{(Q_- + Q_+)x}{2a+1} = x,$$

and similarly $Q_+v = y$. The adjoint inverse formula is that of any isometry; for the raw pair, the linear inverse $(x, y) \mapsto (x+y)/(2a+1)$ has norm $\sqrt{2}/(2a+1)$ on the product space. □

At $a = 8$, the fixed and augmentation sectors have readings $(49, 100)$ and $(73, 73)$, respectively, and

$$9\|Q_-v\|^2 + 8\|Q_+v\|^2 = 17 \cdot 73\|v\|^2 = 1241\|v\|^2, \qquad v = \frac{Q_-v + Q_+v}{17}.$$

For $s_{+,a}(v) = \|Q_{+,a}v\|^2/\|v\|^2$,

$$s_{+,a} = p(a) + 3(a+1)w, \qquad (a+1)s_{-,a} + as_{+,a} = (2a+1)p(a).$$

Cancellation holds even with a fixed component. The bilateral composition is a formalization on antecedent operators, not an assertion that every TPK return is automatically this pair.

If $\mathcal{V} : H \to \mathcal{M}$ is a complete isometric memory, $U^{\mathcal{V}} = \mathcal{V}U\mathcal{V}^*$ satisfies $(U^{\mathcal{V}})^3 = \Pi_{\mathcal{M}}$, where $\Pi_{\mathcal{M}} = \mathcal{V}\mathcal{V}^*$ is the identity on the image. There the transported Gram matrix is

$$(Q_{-,a}^{\mathcal{V}})^* Q_{-,a}^{\mathcal{V}} = p(a)\Pi_{\mathcal{M}} - 3a\mathcal{V}P_0\mathcal{V}^*.$$

It preserves the readings and bilateral recovery. Using diagonal copies of U is legitimate only when the channels intertwine that action; otherwise transport by $\mathcal{V}$ is the correct definition on the image.

145.8 Fine-structure coordinate and integer iterations

The complete alpha reader, with carry, supplies the block prefix

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \ldots).$$

The zero-boundary window ending in 663 is a different object; it is not interchanged with the complete output. The inherited closed interval suffices for the test:

$$\frac{729735256928380099728510547238 0662}{10^{36}} \le \alpha_A \le \frac{729735256928380099728510547238 0663}{10^{36}}. \tag{145.19}$$

The constant is not identified with either of its rational endpoints. Put $s_\alpha = 10^4 \alpha_A$, so that

$$\begin{aligned} \ell \le s_\alpha \le h, \\ \ell := 72.973525692838009972851054723806 62, \\ h := 72.973525692838009972851054723806 63. \end{aligned} \tag{145.20}$$

The inverse condition for the fixed reader is exactly

$$s(v) = s_\alpha \quad \Longleftrightarrow \quad w(v) = \frac{73 - s_\alpha}{24} \simeq 0.00110309613175. \tag{145.21}$$

None of the six states in section 145.6 satisfies this condition. The second memory has $s = 73$, whereas the other five have $s < 72$. This last inequality is verified exactly as $24b_v > a_v$. In the three rational rows, multiplying integers suffices; for u and q_K, the differences are

$$24b_u - a_u = \frac{66073183}{20} + \frac{5235904}{5}\sqrt{5} > 0,$$

$$24b_{q_K} - a_{q_K} = \frac{45259241}{15} - \frac{5235904}{5}\sqrt{5} > 0.$$

For the second, $\sqrt{5} < 5/2$ already gives a positive lower bound. The exclusion concerns this reader and these states; it does not declare a global absence of relations with alpha.

Since $0 < s_\alpha < 73$, the radius

$$r_\alpha = \frac{4\sqrt{s_\alpha}}{81} < \frac{4\sqrt{73}}{81}$$

is a valid conservative guarantee, but it is not the optimal radius of the canonical reader. Any number between zero and 73 would likewise give a smaller guarantee. If $a^2 + a + 1 = s_\alpha$ were imposed by changing a, then $\sqrt{2a}$ and $(a+1)^2$ would also change; one cannot replace only 73 and retain 4 and 81 as though the operator had not changed.

Theorem 145.13 (Exact weight iteration and reading recurrence)**.** *Let $v_n = Q_{-,a}^n v_0$, $a > 0$, and $b = (a-1)^2$. When $v_n \neq 0$,*

$$w_n = \frac{b^n w_0}{b^n w_0 + p(a)^n (1 - w_0)}. \tag{145.22}$$

For $a \neq 1$ and $0 < w_0 < 1$,

$$\frac{w_n}{1 - w_n} = \left(\frac{b}{p(a)}\right)^n \frac{w_0}{1 - w_0}, \quad w_n \downarrow 0, \quad s_n \uparrow p(a),$$

and

$$s_{n+1} = p(a) + b - \frac{bp(a)}{s_n}, \qquad s_{n+1} - s_n = \frac{(s_n - b)(p(a) - s_n)}{s_n} > 0. \tag{145.23}$$

Proof. The operator and its adjoints preserve the two orthogonal sectors. By (145.9), the squared norm in the fixed sector is multiplied by b at each step, and that in the complementary sector by $p(a)$. This gives (145.22) and the weight ratio. Since $p(a) - b = 3a > 0$, monotonicity and the limit are strict in the mixed case. Finally, $s_n = \|v_{n+1}\|^2/\|v_n\|^2$; the sequence of norms is a combination of b^n and $p(a)^n$, satisfying $a_{n+2} = (p(a) + b)a_{n+1} - bp(a)a_n$. Division by a_{n+1} gives the recurrence, and factoring its difference gives (145.23). □

If $a = 1$, the fixed sector vanishes at the first step. When the complement is nonzero, the subsequent reading is exactly three; a purely fixed state is transformed into zero and its quotient ceases to be defined. For $a \neq 1$, the inverse iteration satisfies $s_{n-1} = bp(a)/(p(a) + b - s_n)$ and decreases the mixed reading. Reversing the orientation $U \mapsto U^{-1}$, by contrast, leaves the sequence of norms of the forward iterations unchanged. Powers of $Q_{+,a}$ multiply the weight ratio by $((a+2)^2/p(a))^n > 1$ and decrease $s_{-,a}$ in mixed states. These are actual filters of the state, not chart changes or a TPK temporal trajectory identified by definition.

Proposition 145.14 (Complete test of all integer iterations)**.** *For $a = 8$, no integer iteration $Q_-^n v$, $n \geq 0$, with $v = K, k, m_0$, satisfies $s = s_\alpha$. The only crossings of the target interval occur between indices* 21, 22*,* 14, 15*, and* 15, 16*, respectively. For $v = m_1$, the reading remains exactly 73 in every iteration.*

Proof. Define the rational endpoints of the target weight:

$$w_L = \frac{73 - h}{24} = \frac{2647430716199002714894527619337}{240000000000000000000000000000000},$$

$$w_H = \frac{73 - \ell}{24} = \frac{4412384526998337858157546032 23}{400000000000000000000000000000000}.$$

For $w_0 = A/(A + B)$, the exact formula is $w_n = A49^n/(A49^n + B73^n)$. The data and crossings are

v	A	B	N
K	10684363	2069408	21
k	3512283	8277632	14
m_0	1052377	1837634	15

and satisfy, in each row,

$$\frac{A49^N}{A49^N + B73^N} > w_H > w_L > \frac{A49^{N+1}}{A49^{N+1} + B73^{N+1}}.$$

These are integer inequalities: for example, if $w_H = p_H/q_H$, the first is equivalent to $(q_H - p_H)A49^N > p_H B73^N$, and the last is verified analogously with w_L. Substitution of the three rows and the displayed integers proves the strict crossings without rounding alpha. Strict monotonicity in theorem 145.13 excludes all preceding and subsequent indices; between N and $N + 1$ there is no other integer. For m_1, $w_0 = 0$ remains zero by (145.22). □

For numerical reference, the readings on either side of the crossings are

v	s_N	s_{N+1}
K	72.97136264842803	72.98077012438719
k	72.96167997982852	72.97426483341773
m_0	72.96527892867391	72.97668298511348

The proof covers all $n \geq 0$, not only a finite sample. The inverse orientation gives the same crossings; positive powers of Q_+ and inverse powers of Q_- decrease the initial readings, already less than 72, and

likewise do not reach the target in these three families. The mixed products $Q_-^n Q_+^m$ constitute another two-index class, with weight ratio $(49/73)^n(100/73)^m w_0/(1-w_0)$; they are not excluded by the preceding one-parameter proof, nor are indices chosen here to fit alpha.

145.9 Angular reciprocity and the existing regional correction

The chart changes of proposition 145.11 include the reciprocity and elliptic deformation already constructed. In a plane realized by M_η, the transported form is $M_\eta^{-T} M_\eta^{-1}$; similarity does not change the eigenvalues of the cycle. Retaining the previous Euclidean product after changing the chart would define another observable. The involution $q_+ \leftrightarrow q_-$ preserves A, reverses C^* and sends η_{el} to $-\eta_{\mathrm{el}}$. The polar relation of the ellipse $\tan\phi = e^{-\eta_{\mathrm{el}}}\tan\theta$ changes the parametrization of the curve, not the spectral phase of a similar transport. It does not add a small phase to S^4.

Incidence conjugation has a different type. Its block law preserves

$$z_m^{\dagger} = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)}\neq 0\}} - 20c_{\rho(m)}, \qquad c_m = [C_m]_{10},$$

with the boundary terms before decimal reduction. In general, it is not an isometry of the block vector. Applying it to the register requires its connection to the incidence register and its endpoints; it is not replaced by an invented angular correction.

Proposition 145.15 (Regional correction of the counter-angle)**.** *The antecedent action and angle coordinates satisfy*

$$A_{\mathrm{deg}} = 1000\alpha_A, \qquad C^*_{\mathrm{ret,deg}} = 2(\eta_{\mathrm{ret}} + \alpha_A), \qquad C^*_{\mathrm{pre,deg}} = 2(H_5 + \alpha_A),$$

$$\eta_{\mathrm{ret}} = H_5 - \frac{90}{\pi}C_\pi(\alpha_A), \quad C_\pi(\alpha_A) = 169\alpha_A^6 + \frac{D_A\alpha_A^7}{1-D_A\alpha_A}, \qquad D_A = \exp(-100\pi\alpha_A/9).$$

The difference in radians is exactly

$$\theta_{C,\mathrm{pre}} - \theta_{C,\mathrm{ret}} = C_\pi(\alpha_A).$$

Proof. The difference in degrees is $2(H_5 - \eta_{\mathrm{ret}}) = 180C_\pi/\pi$. Multiplication once by $\pi/180$ gives the result. □

Here π and α_A are the coordinates already generated; H_5 and the remaining readers retain their antecedent definitions, not a recalibration. The subsequent evaluations are

coordinate	radians, approximately
θ_A	0.127362829013
θ_C	0.0370662264658
$\theta_C/6$ in each antipodal pair	0.00617770441097
C_π	$2.55207421805 \cdot 10^{-11}$

The channels $q_\pm$ are positive real attenuations; their exponent does not automatically become a unitary phase through a factor i. No additional divisors are introduced to approximate the scales of the following test.

145.10 Angular inverse problem in the dodecaphasic register

For a general target $s \in [49, 81]$, the principal orientation of the pure sector is obtained from

$$|8 - e^{i\theta}|^2 = 65 - 16\cos\theta, \qquad \theta(s) = \arccos\frac{65 - s}{16} \in [0, \pi].$$

In particular, the subsequent datum s_α determines

$$\theta_\alpha = \theta(s_\alpha), \qquad \delta_\alpha = \frac{2\pi}{3} - \theta_\alpha.$$

This formula solves an inverse problem using an output already produced; it does not claim to produce alpha anew.

Lemma 145.16 (Angular structure of the ternary complement)**.** *The operator $J = (U - U^2)/\sqrt{3}$ satisfies*

$$J^* = -J, \quad J^2 = -P, \quad JP_0 = P_0J = 0, \quad JP = PJ = J, \qquad U = P_0 - \frac{1}{2}P + \frac{\sqrt{3}}{2}J.$$

Proof. The adjoint exchanges U, U^2. Moreover, $(U-U^2)^2 = U+U^2-2I = -3P$, and $(I+U+U^2)(U-U^2) = 0$. The remaining relations follow from $P = I - P_0$; adding $(U + U^2)/2$ and $(U - U^2)/2$ gives the final expression. □

Theorem 145.17 (Angular family and preserved properties)**.** *For $\theta \in \mathbb{R}$, define*

$$U_\theta = P_0 + \cos\theta\, P + \sin\theta\, J, \qquad Q_{-,\theta} = 8I - U_\theta.$$

One has

$$U_\theta^* U_\theta = I, \qquad U_\theta U_\varphi = U_{\theta+\varphi}, \qquad U_{2\pi/3} = U,$$
$$Q_{-,\theta}^* Q_{-,\theta} = 49P_0 + (65 - 16\cos\theta)P. \tag{145.24}$$

The readers $T_\theta = (8I + U_\theta)/9$, $D_\theta = \sqrt{8}(U_\theta - I)/9$ preserve $T_\theta^* T_\theta + D_\theta^* D_\theta = I$. By contrast,

$$U_\theta^3 = I \iff 3\theta \in 2\pi\mathbb{Z}, \qquad \det Q_{-,\theta} = 7^4(65 - 16\cos\theta)^4.$$

Proof. The relations in lemma 145.16 cancel the terms between sectors. On the complement, $J^2 = -I$, so that $U_\theta^* U_\theta = P_0 + (\cos^2\theta + \sin^2\theta)P = I$. Multiplying two expressions and using the addition formulas gives the composition law. The expression for U in the lemma gives the ternary value. Since $U_\theta + U_\theta^* = 2P_0 + 2\cos\theta P$, expanding the Gram matrix gives (145.24). The expansions

$$81T_\theta^* T_\theta = 65I + 8(U_\theta + U_\theta^*), \qquad 81D_\theta^* D_\theta = 16I - 8(U_\theta + U_\theta^*)$$

prove the balance. The composition law gives $U_\theta^3 = P_0 + \cos(3\theta)P + \sin(3\theta)J$, which is the identity exactly under the indicated condition.

For the determinant, $Q_{-,\theta}$ acts by 7 on the four fixed dimensions. On the complement, a unit vector e and Je form an orthonormal pair; their orthogonal complement is stable under J. Repeating this yields four planes, on each of which the determinant is $(8 - \cos\theta)^2 + \sin^2\theta = 65 - 16\cos\theta$. Multiplying the contributions proves the formula. □

Corollary 145.18 (Inverse solution and separation of corrections)**.** *On $\operatorname{im} P$, the reader Q_{-,θ_α} has quadratic ratio exactly s_α. The orientations $\pm\theta_\alpha$ give the same ratio, and*

$$73 - s_\alpha = 8(1 - \cos\delta_\alpha + \sqrt{3}\sin\delta_\alpha).$$

We have $U_{\theta_\alpha}^3 \neq I$, and $\delta_\alpha \neq C_\pi(\alpha_A)$.

Proof. The definition of the phase solves for the complementary coefficient in (145.24); cosine is even. Substituting $\theta_\alpha = 2\pi/3 - \delta_\alpha$ gives the stable identity for the difference. Since $72 < s_\alpha < 73$, $\pi/2 < \theta_\alpha < 2\pi/3$ and $0 < 3\delta_\alpha < \pi/2$; hence $3\theta_\alpha$ is not a multiple of 2π.

The separation of corrections can be proved without relying on approximate figures. The interval gives $73 - s_\alpha > 0.026$. If $\delta_\alpha \leq 10^{-3}$, the inequalities $1 - \cos t \leq t^2/2$, $\sin t \leq t$, and $\sqrt{3} < 2$ would give

$$73 - s_\alpha \leq 8(10^{-6}/2 + 2 \cdot 10^{-3}) = 0.016004,$$

a contradiction. Thus $\delta_\alpha > 10^{-3}$. On the other hand, $0 < \alpha_A < 10^{-2}$, $0 < D_A < 1$ imply

$$0 < C_\pi(\alpha_A) < 169 \cdot 10^{-12} + \frac{10^{-14}}{1 - 10^{-2}} < 170 \cdot 10^{-12} < 10^{-7}.$$

The two quantities are distinct. □

The diagnostic evaluation gives

$$\delta_\alpha \simeq 0.00190956706826 \text{ rad} \simeq 0.109410133709^\circ, \qquad \theta_\alpha \simeq 119.890589866291^\circ.$$

The regional correction is approximately $7.48 \cdot 10^7$ times smaller. The inverse family preserves orthogonality and balance, but loses order three at that phase and changes the original determinant. Its real chart does not by itself provide an integer matrix or the lattice quotient of index 73. Solving the prescribed datum does not prove $73 = 10^4 \alpha_A$.

Proposition 145.19 (Reachability with the state preserved)**.** *For $v \neq 0$, with $w = w(v)$ fixed,*

$$s_\theta(v) = 49w + (65 - 16\cos\theta)(1 - w).$$

If $w < 1$, its image is exactly $[49, 81 - 32w]$. Hence a phase attaining s_α exists if and only if

$$w \leq \frac{81 - s_\alpha}{32} \simeq 0.250827322099.$$

When it exists, its principal orientation is

$$\theta_v = \arccos\left[\frac{1}{16}\left(65 - \frac{s_\alpha - 49w}{1 - w}\right)\right].$$

If $w = 1$, the reading is always 49 and does not reach the target.

Proof. Evaluating (145.24) and dividing by $\|v\|^2$ gives the formula. Cosine ranges over $[-1, 1]$, so the complementary coefficient ranges over $[49, 81]$. Its combination with the fixed weight gives the entire interval, without gaps. Solving the equality and checking the domain of arccosine gives the condition and the phase. The case $w = 1$ is evaluated without dividing by $1 - w$. □

Corollary 145.20 (Four impossibilities and two possible phases)**.** *Keeping the state and the fixed sector, the angular family does not attain s_α on K, k, m_0, u. It does attain it on m_1 and q_K.*

Proof. From the exact table (145.12), we obtain

v	$32b_v - 9a_v$
K	$227115677/3$
k	$2094607/4$
m_0	$122655440/243$
u	$37201759/20 + (7859008/15)\sqrt{5}$

All entries are positive. Thus $w > 9/32$, and the maximum $81 - 32w$ is less than $72 < s_\alpha$. For m_1, $b_{m_1} = 0$ and $a_{m_1} > 0$, so the interval is $[49, 81]$. For q_K,

$$a_{q_K} - 4b_{q_K} = \frac{588839 + 1195712\sqrt{5}}{15} > 0.$$

Hence $w(q_K) < 1/4$ and its maximum is greater than $73 > s_\alpha$. In both cases, the target belongs to the reachable interval. □

The evaluations of the maxima and, where they exist, the phases are

v	$\max_\theta s_\theta(v)$	θ_v
K	54.1922726227	does not exist
k	71.4670172771	does not exist
m_0	69.3474270513	does not exist
u	66.8296175021	does not exist
m_1	81	119.8905898663°
q_K	77.3923674871	133.5296853010°

The phase of the pure sector cannot be used indiscriminately for a mixed state: its reading would be $49w + s_\alpha(1-w) = s_\alpha - (s_\alpha - 49)w$. The original readers of V_2 are polynomials in S, such as P_0, P, J, U_θ. They therefore also intertwine the angular family, and $s_\theta(V_2 v) = s_\theta(v)$. Archiving the complements preserves the joint reading, but does not remove the reachability restrictions.

145.11 Composition of readers and scope of inverse problems

The factor 73 has been determined by three compatible and distinct operations: the Gram matrix of two nonadic readers, the similarity of the ternary augmentation module and the integer determinants of the two conjugate charts. Bilateral balance preserves the complete state and admits explicit inversion; complete memory preserves the transported readings, while selecting a channel can change their weights. The correction radii are uniform over all integer registers in the algebraic domain and throughout the family $a > 0$, with cut and parity proofs that do not depend on sampling.

The inverse test adds verifiable conditions, not a hidden calibration. Orientation inversion and metric covariance do not change the reading. The specified integer iterations are resolved for all their indices by an exact crossing and monotonicity. Angular variation produces a family that realizes the target in the allowed states, but the datum s_α selects its phase and the original ternary index is lost. Its identification with a prior realization of Γ_9 would require a typed composition on the enriched state; this test neither provides such an identification nor declares its absence throughout the corpus.

The new formalizations assemble antecedent operators and questions; no claim of worldwide priority is made for encoding, the use of cyclic vectors, or polar decomposition. The decimal figures presented are subsequent evaluations of exact formulas. The conditions of nonvanishing, charge, metric, lattice, and normalization are part of the result, and their counterexamples remain alongside the proofs. The function of the holographic arithmetic-geometric coupling constant is thereby preserved: a produced object that articulates readers, memory, and incidence, without turning every numerical coincidence into a new physical law.

Chapter 146

Conservative refinement of unity and carry memory

Dimensional completion and the record of operations admit an explicit arithmetic composition. The APP–TRIT–TPK architecture first provides the enriched state, the oriented readings, and the completions of unity. From these outputs we construct an affine transport that preserves the normalized sum and permits recovery, by Euclidean division, of every finite word of canonical carries. Its limit has cylinders of controlled diameter; boundary marks distinguish the histories that a scalar evaluation identifies.

The central block, its concatenations, and the partition $1000 = 729 + 243 + 27 + 1$ are antecedents of this construction. The composition of these objects with the affine transport, the residue inverse, and its prolongation constitutes the formalization developed in this section. The map acts on a declared arithmetic domain. Each realization on a particular enriched history also retains the prospective rule that produces its carries.

146.1 Relation between quadratic norm and cubic completion

Let $b \geq 1$ be an integer and $B = b + 1 \geq 2$. Define

$$p_b = b^2 - b + 1, \qquad q_b = B^2 - p_b = 3b. \tag{146.1}$$

Proposition 146.1 (Transfer of the completion)**.** *The identities*

$$Bp_b = b^3 + 1, \qquad Bq_b + 1 = B^3 - b^3 \tag{146.2}$$

determine the transformation

$$(p_b, q_b) \longmapsto (Bp_b - 1, Bq_b + 1) = (b^3, B^3 - b^3). \tag{146.3}$$

The sum changes from B^2 to B^3, while the sum of the coordinates divided by their respective scales remains equal to one.

Proof. The factorization $b^3 + 1 = (b + 1)(b^2 - b + 1)$ yields the first identity. The second follows from $(b + 1)^3 - b^3 = 3b^2 + 3b + 1$. Adding them proves the change of scale and, after division by B^3, the conservation of normalized unity. □

Para $b = 9$,

$$(73, 27) \longmapsto (10 \cdot 73 - 1, 10 \cdot 27 + 1) = (729, 271). \tag{146.4}$$

The complement preserves the decomposition $271 = 243 + 27 + 1$. In the preceding completion, the last contribution corresponds to the added apex. The correction $(-1, +1)$ has zero sum, and multiplication by ten transforms the scale of one hundred into that of one thousand. For the complete family, the transport of proportions is

$$\left(\frac{p_b}{B^2}, \frac{q_b}{B^2}\right) \longmapsto \left(\frac{p_b}{B^2} - \frac{1}{B^3}, \frac{q_b}{B^2} + \frac{1}{B^3}\right). \tag{146.5}$$

The quantity preserved by this identity is a linear sum. Quadratic forms and physical quantities have their own readers, which will subsequently be composed with this transport.

146.2 Concatenations of the central block and restoration of unity

For $b \geq 2$, the digits $b-2$ and 2 belong to the alphabet of base $B = b+1$. Consider the algebraic family

$$B_c^{(b)} = \begin{pmatrix} b-2 & 2 \\ 2 & b-2 \end{pmatrix}, \qquad C_B(u,v) = Bu + v. \tag{146.6}$$

The case $b = 9$ recovers the documented central block $\left(\begin{smallmatrix} 7 & 2 \\ 2 & 7 \end{smallmatrix}\right)$, with eigenvalues 9 and 5. Its concatenations satisfy $72 + 27 = 77 + 22 = 99$. Concatenation, spectrum, and central position are distinct readings of the same antecedent and retain their domains.

Applying the reader to the rows of (146.6) gives

$$C_B(b-2, 2) = B(b-2) + 2 = b(b-1) = p_b - 1, \tag{146.7}$$

$$C_B(2, b-2) = 2B + (b-2) = 3b = q_b. \tag{146.8}$$

The sum is $B^2 - 1$. Restoring one unit in the first coordinate and performing the subsequent transfer yields

$$(p_b - 1, 3b) \xrightarrow{+(1,0)} (p_b, 3b) \xrightarrow{L_1} (b^3, B^3 - b^3). \tag{146.9}$$

In particular,

$$(72, 27) \longmapsto (73, 27) \longmapsto (729, 271). \tag{146.10}$$

The first arrow increases the sum by one. The second changes the scale and redistributes one unit through a zero-sum correction. To represent the contribution explicitly before incorporating it, the triple $(p_b - 1, 3b, 1)$ already has sum B^2, and the map $(x, y, r) \mapsto (x + r, y, 0)$ preserves that sum. Its inversion requires retaining the datum $r = 1$: on arbitrary triples the map is not injective.

Restitution is thus recorded without attributing to it a closed conservation of the pair before incorporating the unit. The family in b algebraically extends the received central case; the selection of atlas states remains that established by APP–TRIT–TPK. The composition (146.9) determines a carry equal to one in that transition, leaving open the prospective specification of subsequent operations.

146.3 Affine transport and the positivity domain

Definition 146.2 (Transport with carry)**.** For an integer base $B \geq 2$ and $c \in \mathbb{Z}$, define

$$L_c(x, y) = (Bx - c, By + c). \tag{146.11}$$

The integer c specifies the oriented transfer between the two coordinates. A positive sign reduces the first relative to Bx and increases the second by the same amount.

Proposition 146.3 (Inverse and positivity)**.** *In the subsequent affine realization $\mathbb{R}^2$, with B, c fixed,*

$$L_c^{-1}(X, Y) = \left(\frac{X + c}{B}, \frac{Y - c}{B} \right), \qquad X + Y = B(x + y). \tag{146.12}$$

The preimage is integral exactly when $X + c$ and $Y - c$ are divisible by B. For $x, y \geq 0$, the image is nonnegative exactly when

$$-By \leq c \leq Bx. \tag{146.13}$$

Proof. Direct substitution proves both identities in (146.12) and the integrality criterion. The inequalities $Bx - c \geq 0$, $By + c \geq 0$ are equivalent to (146.13). □

For the canonical alphabet $c \in \{0, \dots, B-1\}$, all prolongations are admissible if $x \geq 1$, $y \geq 0$. At the boundary $x = 0$, nonnegativity permits only $c = 0$. The positive domain and the general integer domain are thus distinguished by an explicit condition.

146.4 Canonical recovery through the residue

Theorem 146.4 (Canonical inverse by residue)**.** *Let $M \geq 1$ be an integer and let $X, Y \geq 0$ be integers such that $X + Y = BM$. There is a unique admissible triple (x, y, c) satisfying*

$$x + y = M, \qquad 0 \leq c \leq B-1, \qquad L_c(x, y) = (X, Y).$$

It is obtained from

$$c = Y \bmod B = (-X) \bmod B, \qquad y = \frac{Y - c}{B}, \qquad x = \frac{X + c}{B}. \tag{146.14}$$

Proof. Every preimage satisfies $Y \equiv c \pmod{B}$. The representative in $\{0, \dots, B-1\}$ is unique. The congruence $X + Y \equiv 0 \pmod{B}$ implies the divisibility of $X + c$. Euclidean division of Y gives $y \geq 0$, and $X + c \geq 0$ gives $x \geq 0$. If $c > 0$, the case $x = 0$ would require $X = -c < 0$, so the resulting triple belongs to the admissible domain. Its sum is M, and substitution into (146.11) recovers (X, Y). Uniqueness of the residue and the two quotients proves uniqueness of the triple. □

The integer pair and its scale contain the last canonical carry. Its recovery follows from the residue and does not require retaining another independent copy of that carry. There are $M + 1$ triples with $c = 0$ and M triples in each of the remaining $B - 1$ classes: the total is $BM + 1$, equal to the number of nonnegative pairs with sum BM.

For a general integer carry, write $c = r + B\ell$, with $0 \leq r \leq B - 1$. The residue determines r, whereas complete reconstruction also uses the lift ℓ:

$$y = \frac{Y - r}{B} - \ell, \qquad x = \frac{X + r}{B} + \ell. \tag{146.15}$$

Retaining the lift distinguishes the signed carry from its residual reduction.

146.5 Variable iteration and complete finite memory

Fix $(x_0, y_0) \in \mathbb{Z}^2$, with sum $M_0 > 0$, and a sequence of operations $c_1, c_2, \dots$. Define

$$(x_n, y_n) = L_{c_n} \cdots L_{c_1}(x_0, y_0), \qquad C_n = \sum_{j=1}^{n} c_j B^{n-j}, \qquad C_0 = 0. \tag{146.16}$$

The carries are supplied by the corresponding operational rule; the limit value does not participate in their selection.

Theorem 146.5 (Exact accumulated record)**.** *For every $n \geq 0$,*

$$\begin{aligned} x_n &= B^n x_0 - C_n, & y_n &= B^n y_0 + C_n, \\ x_n + y_n &= B^n M_0, & C_{n+1} &= BC_n + c_{n+1}. \end{aligned} \tag{146.17}$$

Proof. For $n = 0$, the identities are immediate. If they hold at n, applying $L_{c_{n+1}}$ multiplies all previous contributions by B and adds c_{n+1} to the accumulated carry. This gives the recurrence and, by substitution, both coordinates of the next stage. Their sum eliminates C_n. □

When the carries are canonical,

$$0 \le C_n \le B^n - 1. \tag{146.18}$$

In particular, $x_0 \ge 1$, $y_0 \ge 0$ guarantee $x_n \ge B^n(x_0 - 1) + 1$ and $y_n \ge 0$ at every stage.

Corollary 146.6 (Recovery of the complete word)**.** *Given B, n and the final pair, a finite canonical history recovers its input and all its carries through*

$$C_n = y_n \bmod B^n, \qquad y_0 = \left\lfloor \frac{y_n}{B^n} \right\rfloor, \qquad x_0 = \frac{x_n + C_n}{B^n}, \tag{146.19}$$

$$c_j = \left\lfloor \frac{C_n}{B^{n-j}} \right\rfloor \bmod B, \qquad 1 \le j \le n. \tag{146.20}$$

Proof. The bound (146.18) makes $y_n = B^n y_0 + C_n$ a Euclidean division. Its quotient and residue give y_0, C_n; the first coordinate gives x_0. Successive divisions of C_n by powers of B recover the digits. Equivalently, apply (146.14) from the last step to the first. □

The depth retains the leading zeros of the word, even though they do not change the integer C_n. If M_0 is stored instead of n, the exact equality $(x_n + y_n)/M_0 = B^n$ recovers the depth. Normalizing the pair to sum to one removes that scale counter unless it is retained by another reading.

146.6 Block composition and compatibility with the partition

A block of n operations, with accumulated carry C, acts by

$$L_C^{[n]}(x, y) = (B^n x - C, B^n y + C). \tag{146.21}$$

The composition of two blocks satisfies

$$L_D^{[m]} \circ L_C^{[n]} = L_{B^m C + D}^{[n+m]}. \tag{146.22}$$

Indeed, substituting the first image into the second multiplies its accumulated carry by B^m and adds D. The law

$$(n, C), (m, D) \longmapsto (n + m, B^m C + D)$$

is associative by composition of maps. It expresses the contribution of the first block at the scale of the second and preserves the output when the same ordered events are regrouped. Adding new events changes the accumulated carry according to this same law.

146.7 Convergence and quantitative control of the reading

Write $M_n = B^n M_0$ and

$$f_n = \frac{x_n}{M_n}, \quad g_n = \frac{y_n}{M_n}, \quad \theta_n = \frac{C_n}{B^n} = \sum_{j=1}^{n} \frac{c_j}{B^j}. \tag{146.23}$$

The exact identities are

$$f_n = \frac{x_0}{M_0} - \frac{\theta_n}{M_0}, \qquad g_n = \frac{y_0}{M_0} + \frac{\theta_n}{M_0}, \qquad f_n + g_n = 1. \tag{146.24}$$

Theorem 146.7 (Convergence of bounded carries)**.** *If $|c_j| \le C$ for every j, the series $\theta = \sum_{j \ge 1} c_j B^{-j}$ converges absolutely. The readings f_n, g_n converge to f, g, with $f + g = 1$, and*

$$|f - f_n| = |g - g_n| \le \frac{C}{M_0 (B - 1) B^n}. \tag{146.25}$$

Proof. The absolute tail satisfies

$$\sum_{j>n} \frac{|c_j|}{B^j} \le C \sum_{j>n} B^{-j} = \frac{C}{(B-1)B^n}.$$

Convergence of the geometric series and (146.24) prove all the assertions. □

For carries of both signs, a sufficient condition for positivity at every stage is $x_0, y_0 \ge C/(B-1)$. For the canonical alphabet, $x_0 \ge 1, y_0 \ge 0$ suffices. In this second case, $0 \le \theta \le 1$, f_n decreases and g_n increases, and

$$0 \le f_n - f = g - g_n \le \frac{1}{M_0 B^n}. \tag{146.26}$$

The sum remains fixed while the individual proportions evolve. The finite history remains recoverable when the integer coordinates and the scale are retained.

146.8 Compatible cylinders and boundary multiplicity

Consider the full canonical alphabet and an input $x_0 \ge 1, y_0 \ge 0$. A prefix of depth n, with accumulated carry C_n, admits exactly the limit values

$$I_n(C_n) = \left[\frac{C_n}{B^n}, \frac{C_n + 1}{B^n}\right] \tag{146.27}$$

of the reading θ.

Theorem 146.8 (Nested cylinders and distinction of histories)**.** *The B child cylinders of $I_n(C_n)$ cover the parent and overlap only at endpoints. The cylinders of an infinite history have diameter B^{-n} and a single-point intersection. Two different canonical histories produce the same value if and only if, at their first differing position, their digits differ by one and the subsequent tails are, respectively, all $B-1$ for the smaller digit and all zero for the larger.*

Proof. The child associated with c has endpoints $(BC_n + c)/B^{n+1}$ and $(BC_n + c + 1)/B^{n+1}$. These expressions prove inclusion, coverage and overlap only at endpoints. The tail of an expansion ranges over the interval $[0, B^{-n}]$; successively choosing one of the children produces an expansion of each point in that interval. The nested closed intervals have diameters tending to zero, which proves existence and uniqueness of the intersection for each history.

Let k be the first differing position and let $d_k > c_k$. If the readings coincide,

$$\frac{d_k - c_k}{B^k} = \sum_{j>k} \frac{c_j - d_j}{B^j} \le \sum_{j>k} \frac{B-1}{B^j} = \frac{1}{B^k}.$$

The positive integer difference $d_k - c_k$ must equal one. Equality in the tail bound requires $c_j = B-1$, $d_j = 0$ for every $j > k$. Conversely, those tails differ by exactly B^{-k}, which compensates for the first digit. This proves the complete characterization. □

The image of the cylinder for f has diameter $1/(M_0 B^n)$, matching (146.26). The histories $(1, 0, 0, \ldots)$ and $(0, B-1, B-1, \ldots)$ both produce $\theta = 1/B$, while retaining distinct prefixes and finite integer states. A boundary mark retains both provenances. A convention excluding tails eventually constant at $B-1$ selects one expansion for $\theta < 1$; the endpoint $\theta = 1$ is retained separately, since in the fractional alphabet it corresponds to the constant tail $B-1$.

The history of operations, the cylinder system, and the limit value are objects related by these maps. The uniqueness of a history's reading coexists with multiplicity of histories at boundary values.

146.9 Signed carries and prospective specification

Allowing carries outside the canonical alphabet requires retaining their lifts. Already in two steps, the words

$$(c_1, c_2) = (1, -1), \qquad (d_1, d_2) = (0, B - 1)$$

produce $C_2 = B - 1$ and the same final pair. Both are bounded by $B - 1$ and can preserve positivity, for example starting from $(73, 27)$ in base ten. The signed word, or the lifts in (146.15), distinguishes that information. The convergence theorem remains valid for both words; the canonical inverse, however, has the domain specified in its statement.

The transition (146.4) fixes $c_1 = 1$. An additional rule imposing $c_j = 1$ for every j gives

$$\theta_n = \frac{1 - B^{-n}}{B - 1}, \qquad \theta = \frac{1}{B - 1}. \tag{146.28}$$

In base ten, starting from $(73, 27)$, the sequence is

$$(73, 27) \longmapsto (729, 271) \longmapsto (7289, 2711) \longmapsto (72889, 27111),$$

and the first fraction converges to

$$\frac{73}{100} - \frac{1}{900} = \frac{164}{225} = 0{,}728888\ldots. \tag{146.29}$$

The conclusion belongs to that additional constant-carry rule. With only $c_1 = 1$ and nonnegative canonical prolongations, the first-step cylinder yields exactly

$$\frac{b^3 - 1}{B^3} \le f \le \frac{b^3}{B^3}. \tag{146.30}$$

In base ten it is $[0{,}728, 0{,}729]$. Changing the sign or the alphabet changes this family and requires preserving the corresponding additional memory. The presence of 729 in the completion determines the written arithmetic transition; identification with another reader of constants also uses its effective generation rule and its intertwining.

146.10 Operator realization and transport amplitudes

The resulting procedure comprises updating through (146.11), local residue checking, reconstruction of every canonical word through (146.20), control of the limit reading through (146.25), and regrouping of events through (146.22). Each operation retains the order and scale counter used by its inverse.

Conservation of the sum and quadratic conservation have different forms. For a normalized transfer $\delta = c/(BM)$,

$$(f - \delta)^2 + (g + \delta)^2 - (f^2 + g^2) = 2\delta(g - f) + 2\delta^2. \tag{146.31}$$

This expression determines the quadratic variation and is generally nonzero. The connection with an isometry will be obtained on the amplitudes of the admissible labels, through the residue bijection of chapter 148. Arithmetic transport and its unitary realization are thus composed while retaining their respective quantities, rather than being identified through a scalar coincidence.

Chapter 147

Bilateral conservation and arithmetic-geometric transduction

The holographic arithmetic-geometric coupling constant denotes the generated register K, with its ordered windows, phase origin, orientation and incidences. Its production through the signed register and Hadamard inversion is developed in section 11.4. The rational reading κ, the orbital frame O_K and the operators transporting its realizations are different objects. This distinction allows their functions to be composed: generation of the register, arithmetic encoding, information transport and recovery of the geometric readings.

The regional coordinates and α are received from the APP–TRIT–TPK genealogy, from the same enriched state and its joint discrete structure of the continuum. The following constructions act on these subsequent realizations. Their additional content is a dimension-independent bilateral conservation, a reversible extension admitting incoming memory and an archiving policy with uniformly controlled recovery.

147.1 Bilateral law for two isometric transports

Let H, H' be Hilbert spaces and $\mathcal{B}, C : H \to H'$ linear isometries, so that $\mathcal{B}^*\mathcal{B} = C^*C = I_H$. The calligraphic typography distinguishes these transports from the arithmetic base of the refinement. For $a > 0$, we define

$$\begin{aligned} p &= a^2 + a + 1 = a(a+1) + 1, \\ N &= (2a+1)p = (a+1)^3 + a^3, \\ Q_- &= a\mathcal{B} - C, \qquad Q_+ = (a+1)\mathcal{B} + C. \end{aligned} \tag{147.1}$$

Theorem 147.1 (Balance of two isometric realizations)**.** *For the maps* (147.1)*,*

$$(a+1)Q_-^*Q_- + aQ_+^*Q_+ = NI_H. \tag{147.2}$$

The identity holds in every dimension, including infinite Hilbert spaces, and admits rectangular transports with common domain and codomain.

Proof. Write $X = \mathcal{B}^*C + C^*\mathcal{B}$. The isometry of both transports gives

$$\begin{aligned} Q_-^*Q_- &= (a^2+1)I - aX, \\ Q_+^*Q_+ &= ((a+1)^2+1)I + (a+1)X. \end{aligned}$$

The weights $a+1$ and a exactly cancel the terms of X. The remaining coefficient is

$$(a+1)(a^2+1) + a((a+1)^2+1) = 2a^3 + 3a^2 + 3a + 1 = N.$$

The compositions are of bounded operators and the calculation uses only their products and adjoints. The proof therefore also applies in infinite dimension. □

For $a = 8$, received from the preceding nonadic partition,

$$9\|Q_-v\|^2 + 8\|Q_+v\|^2 = (9^3 + 8^3)\|v\|^2 = 17 \cdot 73\,\|v\|^2. \tag{147.3}$$

The factor $73 = 8 \cdot 9 + 1$ normalizes this general identity. In the ternary module it is also realized as a determinant and similarity factor; each of these functions retains its own hypotheses. The balance (147.2) uses the isometries and admits both transports of finite order and transports without that property.

147.2 Factorization, recovery and compatibility of the channels

Define

$$E_a v = \frac{1}{\sqrt{p}} \begin{pmatrix} \sqrt{a(a+1)}\, \mathcal{B} v \\ C v \end{pmatrix}, \qquad R_a = \frac{1}{\sqrt{2a+1}} \begin{pmatrix} \sqrt{a} & -\sqrt{a+1} \\ \sqrt{a+1} & \sqrt{a} \end{pmatrix}. \tag{147.4}$$

The matrix R_a acts on the two channels and is understood to be tensored with $I_{H'}$. We check $E_a^* E_a = I_H$, $R_a^* R_a = I_{H' \oplus H'}$, and

$$W_a v = R_a E_a v = \frac{1}{\sqrt{N}} \begin{pmatrix} \sqrt{a+1}\, Q_- v \\ \sqrt{a}\, Q_+ v \end{pmatrix}, \qquad W_a^* W_a = I_H. \tag{147.5}$$

In the case $a = 8$, the first distribution has weights $72/73$ and $1/73$; the second operation is a rotation between the channels. This factorization expresses the role of $72 + 1$ within the total norm, distinguishing it from the linear restitution of (146.9).

Proposition 147.2 (Bilateral recovery)**.** *The unnormalized channels $x = Q_- v$, $y = Q_+ v$ determine*

$$\mathcal{B} v = \frac{x + y}{2a+1}, \qquad C v = \frac{a y - (a+1) x}{2a+1}, \qquad v = \mathcal{B}^* \frac{x+y}{2a+1}. \tag{147.6}$$

From the normalized channels we recover $v = W_a^* W_a v$, with $\|W_a^*\| = 1$ when $H \neq \{0\}$.

Proof. Adding the two definitions of $Q_\pm$ eliminates Cv. The combination $aQ_+ - (a+1)Q_-$ eliminates $\mathcal{B}v$. The first isometry has left inverse $\mathcal{B}^*$, which proves the formulas. The last assertion follows from (147.5): the adjoint of an isometry has norm one. □

The image $\operatorname{im} W_a$ is closed and its orthogonal projector is $W_a W_a^*$. A normalized pair is compatible with a unique input exactly when it belongs to this image; its orthogonal component quantifies the compatibility defect.

If $\mathcal{B} = I$ and $C = U$ is unitary, the two $Q_\pm$ are polynomials in the same operator. For the unnormalized channels, the criterion simplifies to

$$Q_+ x = Q_- y. \tag{147.7}$$

Indeed, the equality is equivalent to $U(x+y) = ay - (a+1)x$. Defining $v = (x+y)/(2a+1)$ yields by substitution $Q_- v = x$, $Q_+ v = y$. Necessity follows from the commutation of the two polynomials. This simplified form belongs to the indicated unitary domain; the image criterion covers general transports between different spaces.

147.3 Transformation of readers and relative symmetry

Theorem 147.3 (Common reader in both channels)**.** *For every bounded self-adjoint operator F on H',*

$$W_a^* \operatorname{diag}(F, F) W_a = \frac{a(a+1)\mathcal{B}^* F \mathcal{B} + C^* F C}{a(a+1)+1}. \tag{147.8}$$

If $\mathcal{B}$ is unitary and $C = \mathcal{B} R$, with R unitary, the expression, for $G = \mathcal{B}^* F \mathcal{B}$, is

$$\mathcal{T}_a(G) = \frac{a(a+1) G + R^* G R}{a(a+1)+1}. \tag{147.9}$$

We have $\mathcal{T}_a(G) = G$ if and only if $GR = RG$.

Proof. The expansion of $(a+1)Q_-^* F Q_- + a Q_+^* F Q_+$ cancels the terms $\mathcal{B}^* F C + C^* F \mathcal{B}$. The coefficients of the remaining terms are $a(a+1)(2a+1)$ for $\mathcal{B}^* F \mathcal{B}$ and $2a+1$ for $C^* F C$. Dividing by N proves

(147.8). Substituting $C = \mathcal{B}R$ produces (147.9). Equality with G is equivalent to $R^{*}GR = G$, and the unitarity of R makes it equivalent to $GR = RG$. □

The reader $F = I$ recovers the norm. For a general reader, the identity determines its transformation, and the coincidence $\mathcal{B}^{*}F\mathcal{B} = C^{*}FC = G$ guarantees its conservation in all states. In the canonical case,

$$\mathcal{T}_{8}(G) = \frac{72G + R^{*}GR}{73}. \tag{147.10}$$

This formula relates conservation and symmetry through a specific commutator. The observable that exactly transports an operator A from the domain to the complete image is $W_{a}AW_{a}^{*}$; its ambient identity is $W_{a}W_{a}^{*}$. The diagonal observable of (147.8) is another choice, whose difference is calculated by the same theorem.

147.4 Reversible update with incoming memory

Now let $\mathcal{B} = I$, $C = U$, with U unitary on the same space, and write

$$A = \frac{\sqrt{a+1}}{\sqrt{N}}Q_{-}, \qquad D = \frac{\sqrt{a}}{\sqrt{N}}Q_{+}. \tag{147.11}$$

The operators A, D are normal and commute with one another and with their adjoints, being polynomials in U and U^{*}.

Theorem 147.4 (Unitary extension)**.** *The update*

$$\mathscr{R}_{a} = \begin{pmatrix} A & -D^{*} \\ D & A^{*} \end{pmatrix} : H \oplus H \to H \oplus H \tag{147.12}$$

is unitary. Its first column is W_{a}, and an input (v, m) preserves $\|v\|^{2} + \|m\|^{2}$.

Proof. The balance gives $A^{*}A + D^{*}D = I$. Normality provides $AA^{*} + DD^{*} = I$. The cross products of $\mathscr{R}_{a}^{*}\mathscr{R}_{a}$ and $\mathscr{R}_{a}\mathscr{R}_{a}^{*}$ vanish by the preceding commutations; their two diagonals are the identity. The first column reproduces (147.5), and conservation of the joint norm is the unitary identity applied to (v, m). □

For data $z = (z_{1}, z_{2})$, the explicit inversion is

$$v_{\mathrm{rec}} = A^{*}z_{1} + D^{*}z_{2}, \qquad m_{\mathrm{rec}} = -Dz_{1} + Az_{2}. \tag{147.13}$$

The hypothesis of zero incoming memory is checked through $m_{\mathrm{rec}} = 0$. If e is a perturbation of the data, unitarity gives

$$\|e\|^{2} = \|e_{\mathrm{rec}}\|^{2} + \|e_{\mathrm{mem}}\|^{2}. \tag{147.14}$$

The second component detects the part incompatible with that input. An error belonging to $\operatorname{im} W_{a}$ has zero memory residual and alters the recovered input. Correction of the integer register also uses its discrete separation distances.

147.5 Causal reduction of the bilateral update

The preceding unitary update realizes the transport of the state and its memory in a common space. Observation of the first component admits an equivalent description that explicitly retains the contribution of the record. This description follows from the transport already constructed, with the same coefficients and orientation: memory removed from the list of variables reappears as dependence on earlier states.

Fix the operators A, D of (147.11) and a realization $H \oplus H$. For $n \geq 0$, let

$$v_{n+1} = Av_n - D^* m_n, \qquad m_{n+1} = Dv_n + A^* m_n. \tag{147.15}$$

The joint update is $\mathscr{R}_a$ and preserves $\|v_n\|^2 + \|m_n\|^2$.

Proposition 147.5 (Exact elimination of the auxiliary record)**.** *For every $n \geq 0$, interpreting the empty sum as zero,*

$$m_n = (A^*)^n m_0 + \sum_{k=0}^{n-1} (A^*)^{n-1-k} D v_k, \tag{147.16}$$

$$v_{n+1} = Av_n - D^*(A^*)^n m_0 + \sum_{k=0}^{n-1} \mathscr{K}_{n-1-k} v_k, \tag{147.17}$$

$$\mathscr{K}_j = -D^*(A^*)^j D \quad (j \geq 0). \tag{147.18}$$

The reduced equation, together with (147.16), is equivalent to the joint recurrence for the same data (v_0, m_0).

Proof. The first identity holds for $n = 0$. If it holds for n, the second equation in (147.15) gives

$$m_{n+1} = (A^*)^{n+1} m_0 + \sum_{k=0}^{n-1} (A^*)^{n-k} D v_k + D v_n,$$

which is the identity at $n + 1$. Substitution at depth n into the first equation proves (147.17). Conversely, if v_n satisfies the reduced equation and m_n is defined by (147.16), the same difference of sums proves $m_{n+1} = Dv_n + A^* m_n$; substitution recovers the other equation. □

The kernel $\mathscr{K}_j$ describes a transfer into the record, j updates within it, and re-entry into the observed component. The term $-D^*(A^*)^n m_0$ preserves the initial memory. Omitting it corresponds to the particular case $m_0 = 0$, rather than an identity of the general transport. Equation (147.17) involves only the current and earlier states. The causal character of this reduction follows from the order of the joint recurrence.

The normalization of the balance also provides an explicit estimate. Since $Q_- = aI - U$, we have $\|A\|^2 \leq r_a = (a+1)^3/((a+1)^3 + a^3) < 1$, whereas $D^* D \leq I$. Consequently,

$$\|\mathscr{K}_j\| \leq r_a^{j/2}, \qquad \sum_{j \geq 0} \|\mathscr{K}_j\| \leq \frac{1}{1 - \sqrt{r_a}}. \tag{147.19}$$

This bound controls the weight of memory in the reduced equation; the norm of the joint state remains conserved by the unitary update.

The proposition is the temporal realization of the Schur reduction in proposition 1.198. For the formal series $V(z) = \sum_{n \geq 0} v_n z^n$ and $M(z) = \sum_{n \geq 0} m_n z^n$, the recurrences yield $V = v_0 + zAV - zD^* M$ and $M = m_0 + zDV + zA^* M$. Eliminating M,

$$V(z) = \left[I - zA + z^2 D^*(I - zA^*)^{-1} D\right]^{-1} \left[v_0 - zD^*(I - zA^*)^{-1} m_0\right]. \tag{147.20}$$

The formal inverses exist because their constant terms are identities; the state series and the joint resolvent also converge for $|z| < 1$. The initial source and the effective operator transform together. A readout that also acts on m_n retains its contribution through (147.16), in accordance with (1.182).

Re-entry here updates an incoming record. By contrast, the archiving policy below retains each complement in a distinct coordinate. Each operation has its own conservation law. The kernel $\mathscr{K}_j$, Green propagators, and a dimensional force have different types: the first eliminates a state component, Green propagators solve source equations with support conditions, and the force adds a dynamical realization and its units.

147.6 Nonstationary register and uniform convergence of the terminal component to zero

At each stage let H_n, H_{n+1} be Hilbert spaces and $\mathcal{B}_n, C_n : H_n \to H_{n+1}$ isometries. The parameter $a > 0$ remains fixed. Let A_n, D_n be the normalized blocks of (147.5); we continue through the first and record the second:

$$R_0 = I, \qquad R_{n+1} = A_n R_n, \qquad m_n = D_n R_n v. \tag{147.21}$$

This policy has a domain distinct from the reintroduction of memory in (147.12); each preserves its update.

Lemma 147.6 (Uniform transfer)**.** *At every stage,*

$$\|A_n\|^2 \le r_a := \frac{(a+1)^3}{(a+1)^3 + a^3} < 1, \quad D_n^* D_n \ge (1 - r_a) I, \quad \|R_n\|^2 \le r_a^n. \tag{147.22}$$

Proof. The triangle inequality gives $\|a\mathcal{B}_n - C_n\| \le a + 1$. Its normalization provides the first bound. The difference $N - (a+1)^3 = a^3 > 0$ proves $r_a < 1$. The local balance implies $D_n^* D_n = I - A_n^* A_n$, and multiplying the bounds of the continued channels produces the estimate of R_n. □

Theorem 147.7 (Complete memory at arbitrary depth)**.** *The finite map*

$$Z_N v = (R_N v, D_0 v, D_1 R_1 v, \dots, D_{N-1} R_{N-1} v) \tag{147.23}$$

is isometric. The infinite archive

$$Z_\infty v = (D_n R_n v)_{n \ge 0} \tag{147.24}$$

belongs to the Hilbert direct sum of the output spaces and satisfies

$$\|v\|^2 = \sum_{n \ge 0} \|m_n\|^2, \qquad v = \sum_{n \ge 0} R_n^* D_n^* m_n, \qquad Z_\infty^* Z_\infty = I. \tag{147.25}$$

The inversion series converges in norm and the adjoint has norm one.

Proof. The identity $A_n^* A_n + D_n^* D_n = I$, conjugated by R_n, produces

$$R_n^* R_n - R_{n+1}^* R_{n+1} = R_n^* D_n^* D_n R_n.$$

Summing from zero to $N - 1$ gives

$$I = R_N^* R_N + \sum_{n < N} R_n^* D_n^* D_n R_n. \tag{147.26}$$

This is the Gram identity of Z_N. By (147.22), the first term tends to zero in operator norm. The limit identity proves the quadratic convergence of the archive and its isometry. The adjoint of that isometry is bounded; the truncations of every square-summable archive converge in the Hilbert direct sum, and

their image under the adjoint proves convergence of the series in (147.25). Applying it to the exact data recovers v. □

With only the first N complements, the truncated adjoint has exact residual

$$v - \sum_{n<N} R_n^* D_n^* m_n = R_N^* R_N v, \qquad \left\| v - \sum_{n<N} R_n^* D_n^* m_n \right\| \le r_a^N \|v\|. \tag{147.27}$$

If the joint data error is at most ε, the total bound is $\varepsilon + r_a^N \|v\|$. With Z_N, including its terminal, or with Z_∞, the adjoint recovers with error at most ε.

The preceding policy exhausts the terminal through the proved uniform bound. The preceding evolutions that have a positive terminal $A_\infty \neq 0$ preserve in their archive the component $A_\infty^{1/2} v$. Arbitrary depth and removal of the terminal are therefore different conclusions and retain their respective hypotheses.

Proposition 147.8 (Inverse of finite memories without a terminal)**.** *For $N \ge 1$, let $M_N v = (D_n R_n v)_{n<N}$. Then*

$$M_N^* M_N = I - R_N^* R_N \ge (1 - r_a^N) I, \tag{147.28}$$

and the least-squares inverse

$$L_N = (I - R_N^* R_N)^{-1} M_N^*, \qquad L_N M_N = I, \qquad \|L_N\| \le (1 - r_a^N)^{-1/2} \tag{147.29}$$

recovers the input exactly with exact data.

Proof. The Gram identity is (147.26); the bound follows from (147.22). The positive operator is invertible by its strict lower bound. Composition proves $L_N M_N = I$, and $L_N L_N^* = (M_N^* M_N)^{-1}$ proves the stated norm. □

This finite inverse removes the bias of (147.27); its noise amplification is specified by (147.29). The truncated adjoint, the finite inverse and the adjoint of the complete archive are thus three operators with distinct behaviors.

For $a = 8$, the general bound is $r_a = 729/1241$. In the case $\mathcal{B} = I$, $C = S^4$, with S the cycle of twelve positions, the fixed projector $P_0 = (I + S^4 + S^8)/3$ gives

$$A^* A = \frac{a+1}{(2a+1)p} \big((a-1)^2 P_0 + p(I - P_0) \big). \tag{147.30}$$

Since $p - (a-1)^2 = 3a > 0$, its norm is $(a+1)/(2a+1)$. At $a = 8$ we obtain $9/17$; the eigenvalue of the fixed sector is $441/1241$. These rates belong to the declared transports and policy.

If both channels are refined in a tree, each complete level is also isometric by composition. Summing the norms of all complete layers would repeatedly count the same initial norm. Conservation is formulated through the refinement maps between levels or through the archive of complements in (147.24).

147.7 Geometric forms and conservation of their transport

Let b, b' be nondegenerate forms, symmetric or alternating, on the corresponding spaces, and suppose $\mathcal{B}^* b' \mathcal{B} = C^* b' C = b$. The same bilinear expansion proves

$$(a+1) Q_-^* b' Q_- + a Q_+^* b' Q_+ = N b. \tag{147.31}$$

Indeed, the mixed terms $\mathcal{B}^* b' C + C^* b' \mathcal{B}$ cancel with the same weights; the pure terms sum to Nb. The transported form is thus preserved without requiring positivity. The contraction and noise bounds of the

preceding subsection use, by contrast, positive Hilbert norms. A Lorentzian or alternating form preserves (147.31), with its own geometric interpretations, and does not replace positivity in those inequalities.

147.8 Channel correlation and inverse reading problem

For $v \neq 0$, define

$$\chi(v) = \frac{\operatorname{Re}\langle \mathcal{B}v, Cv\rangle}{\|v\|^2}, \qquad -1 \le \chi(v) \le 1. \tag{147.32}$$

The bound is Cauchy–Schwarz applied to the two isometric images. The normalized readings are

$$s_- = a^2 + 1 - 2a\chi, \qquad s_+ = (a+1)^2 + 1 + 2(a+1)\chi. \tag{147.33}$$

For $a = 8$,

$$s_- = 65 - 16\chi, \qquad s_+ = 82 + 18\chi, \qquad 9s_- + 8s_+ = 1241. \tag{147.34}$$

The two individual norms contain a common real correlation. The balance is an exact relation between them, not two independent arithmetic determinations of a constant.

Using the prior output α_A in the inverse test $s_- = 10^4\alpha_A$ determines the complementary reading

$$s_+ = 73 + \frac{9}{8}(73 - 10^4\alpha_A) \simeq 73{,}029783595557. \tag{147.35}$$

The equality is obtained by substitution into (147.34). The weighted average $(9s_- + 8s_+)/17$ remains exactly equal to 73. The comparison preserves the prior genealogy of α_A: it is a condition of the inverse test, not another independent producer of that output. A unitary change can alter χ and the two readings while maintaining the balance, even if a particular relation of ternary order changes.

147.9 Cyclic frame and transport of responses

On the carrier of the generated register, let $H = \mathbb{R}^{12}$, $(Sx)_i = x_{i+1}$, with cyclic indices, and

$$O_K = (K, SK, \dots, S^{11}K), \qquad 589609I \le O_K^* O_K \le 39225169I. \tag{147.36}$$

The bounds are those of the cyclic frame of the canonical register, proved in theorem 144.2. Let Z be one of the complete isometries Z_N, Z_∞, or a composition with the preceding archives that preserves its terminals and complements.

Theorem 147.9 (Preservation of frame conditioning)**.** *If $x = O_K\beta$, the datum $z = Zx$ determines*

$$\beta = O_K^{-1} Z^* z, \qquad \|\delta\beta\| \le \frac{\|\delta z\|}{\sqrt{589609}}. \tag{147.37}$$

The twelve columns of ZO_K form a basis of its image, even when the archive has infinitely many coordinates.

Proof. The equality $Z^*Z = I$ gives $(ZO_K)^*(ZO_K) = O_K^* O_K$. The Gram bounds are preserved and O_K is invertible by the strict lower bound. Composing their inverses proves recovery, and the norm $\|O_K^{-1}\| \le 589609^{-1/2}$, together with $\|Z^*\| = 1$, gives stability. The number of columns and their independence determine the dimension of the image, without identifying it with the entire storage space. □

For an operator H_0 commuting with S, the complete vector response $y = H_0K$ determines

$$H_0 = O_y O_K^{-1}. \tag{147.38}$$

Indeed, commutation implies $H_0 O_K = (y, Sy, \dots, S^{11}y) = O_y$. If only $z = Zy + e$ is received, define $\widehat{y} = Z^* z$ and $\widehat{H}_0 = O_{\widehat{y}} O_K^{-1}$. Then

$$\|\widehat{H}_0 - H_0\| \le \sqrt{\frac{12}{589609}}\,\|e\|. \tag{147.39}$$

The orbital matrix of δy has twelve columns of norm $\|\delta y\|$, so its operator norm does not exceed $\sqrt{12}\|\delta y\|$. This and (147.36) prove (147.39). For a general operator the protocol requires the twelve responses $H_0 S^j K$; a response in the cyclic case means its twelve components, not a scalar measurement.

In the transported image, the shift is $S_Z = ZSZ^*$, and satisfies $S_Z^j Z = ZS^j$. This equality determines the phase advance in the archive; a translation of cells in the ambient space represents that advance only when it coincides with the transported operator.

147.10 Positional reading and scalar precision

Fix $B_0 = 1000$, $d = B_0^{12} - 1$, and the row

$$\ell = \frac{(B_0^{11}, B_0^{10}, \dots, 1)}{d}. \tag{147.40}$$

Its linear extension acts on H, and in the word domain $\kappa(K) = \ell K$. The representative of the transported reader is $Z\ell^*$, so that

$$\kappa(K) = \langle Z\ell^*, ZK\rangle, \qquad |\delta\kappa| \le \|\ell\|\,\|e\|. \tag{147.41}$$

The geometric sum provides

$$\|\ell\|^2 = \frac{B_0^{12} + 1}{(B_0^2 - 1)(B_0^{12} - 1)}. \tag{147.42}$$

Indeed, the numerator of the sum of squares is $(B_0^{24} - 1)/(B_0^2 - 1)$, which divided by $(B_0^{12} - 1)^2$ gives (147.42).

With only N complements and the truncated adjoint, an additional error of at most $\|\ell\| r_a^N \|K\|$ appears. The finite inverse (147.29) removes that bias and retains its explicit noise amplification bound.

In the domain $\{0, \dots, 999\}^{12}$, the exact reading recovers $N_K = d\kappa$ and its twelve blocks through successive Euclidean divisions. If

$$|\widetilde{\kappa} - \kappa| < \frac{1}{2d}, \tag{147.43}$$

rounding $d\widetilde{\kappa}$ to the nearest integer recovers N_K: its distance to the true integer is less than half a unit. The base, length, origin and orientation accompany that representation. The scalar threshold of (147.43) is distinct from the vector bounds of the memory registers.

147.11 Incidence, charge and exact restitution of the register

Let $\mathbf{1} = (1, \dots, 1)^T$ and $\Pi = I - \mathbf{1}\mathbf{1}^*/12$. The incidence isometry of section 107.1 is

$$Fv = (\mu(v), \mathcal{I}\Pi v/6), \qquad \mu(v) = \frac{\mathbf{1}^* v}{12}, \qquad \|(\mu, y)\|_Y^2 = 12\mu^2 + \|y\|^2. \tag{147.44}$$

Its codomain is its declared image. The Gram matrix $\mathcal{I}^*\mathcal{I} = 36I + 30\mathbf{1}\mathbf{1}^*$ gives $\|\mathcal{I}\Pi v\|^2 = 36\|\Pi v\|^2$, and $12\mu(v)^2 + \|\Pi v\|^2 = \|v\|^2$, whence $F^*F = I$. Thus, from $z = Zv + e$,

$$\widehat{Fv} = FZ^* z, \qquad \|FZ^* e\|_Y \le \|e\|. \tag{147.45}$$

The unnormalized incidence $\mathcal{I}\Pi v$ has error at most $6\|e\|$; the signed register $H_{12} v$, whose normalization satisfies $H_{12}^* H_{12} = 4I$, has error at most $2\|e\|$.

The direction $u = P_3 \Pi K$ and the projector $P_{10} = \Pi - uu^*/\|u\|^2$, with $u \neq 0$, are recovered with their prior readers and chart. The absolute selection $\{j : K_j \geq 729\}$ has a discontinuity at $K_4 = 729$. Integer restitution is carried out before applying that threshold.

Proposition 147.10 (Decoding after the complete archive)**.** *Let K be integer and $z = ZK + e$, with $Z^*Z = I$. If $\|e\| < 1/2$, rounding each coordinate of Z^*z recovers K exactly. Composition restores κ, the signatures and the geometric readers defined on that register.*

Proof. The contractivity of Z^* gives $\|Z^*z - K\| \leq \|e\| < 1/2$. Each coordinate error is bounded by the joint norm, so the unique integer coordinate at a distance less than half a unit is the original one. Once K is recovered, evaluating any of its exact readers restores its value. □

In particular, the support $\{4, 6, 9, 10\}$ is recovered, including its boundary component. The other marks involved in an exceptional flag are preserved through their own channels: restitution of K exactly corrects the information whose sufficient arguments lie in it.

147.12 Preservation of the conditioning of two memories

Retain the preceding readers

$$T_j = \frac{8I + S^j}{9}, \qquad D_j = \frac{\sqrt{8}}{9}(S^j - I), \qquad Mv = (D_3 v, D_4 T_3 v). \tag{147.46}$$

The D_j in this equation are the preceding complements; the second bilateral block D of (147.11) retains its own definition. We have $\ker M = \mathbb{R}\mathbf{1}$ and an inverse L_M of norm $9/4$ on the centered sector, as proved in chapter 114.

Let $E = W_a \oplus W_a$ be in the space of the two memories. Its isometry implies

$$(EM)^*(EM) = M^*M, \qquad \ker(EM) = \mathbb{R}\mathbf{1}. \tag{147.47}$$

For $z = EMv + e$ and exact charge $q = \mathbf{1}^*v$, the reconstruction is

$$\widehat{v} = \frac{q}{12}\mathbf{1} + L_M E^* z, \qquad \|\widehat{v} - v\| \leq \frac{9}{4}\|e\|. \tag{147.48}$$

The proof consists in applying $E^*E = I$, using $L_M M = \Pi$ and $\|E^*\| = 1$. If the charge has error δq, the uniform and centered sectors are orthogonal and

$$\|\widehat{v} - v\|^2 \leq \frac{|\delta q|^2}{12} + \frac{81}{16}\|e\|^2. \tag{147.49}$$

The distances of all words in the integer image are preserved exactly by E. Therefore, the restitution radii of theorem 145.5 remain

$$r_0 = \frac{2\sqrt{146}}{81} \quad \text{uncharged}, \qquad r_q = \frac{4\sqrt{73}}{81} \quad \text{with exact charge}. \tag{147.50}$$

Within the first radius, ΠK is recovered, and with it the direction and its projector when they are defined. Within the second, preserving q, we recover K and the absolute threshold selector. Subsequently composing another complete isometry, finite or infinite, maintains the same guarantees: its adjoint is applied before the decoder. The radius is always expressed in the joint norm of the data with the fixed normalization.

This composition redistributes the information of M without degrading its conditioning and without creating the uniform channel that M removes. Gram conservation distinguishes this property from a numerical improvement obtained by changing the reader or the noise scale.

147.13 Orbital norm and polar normalization of the frame

The preceding inversion satisfies $K = H_{12}U_{\mathrm{sgn}}/4$, $H_{12}^{*}H_{12} = 4I$. Therefore

$$\|K\|^2 = \frac{1}{4}\|U_{\mathrm{sgn}}\|^2 = 4251257, \qquad \|U_{\mathrm{sgn}}\|^2 = 17005028. \tag{147.51}$$

Each translation S^j is orthogonal and preserves $\nu = \sqrt{4251257}$. Hence

$$\mathrm{tr}(O_K^{*}O_K) = 12\nu^2 = 51015084. \tag{147.52}$$

Division by ν normalizes each orbital column to norm one and preserves its order. It is uniform over the orbit and under its isometric transports. Different registers retain their own norms, and the normalized columns continue to have a nonscalar Gram matrix: its eigenvalues are the preceding ones divided by 4251257.

Operator normalization uses the complete Gram matrix:

$$G_K = O_K^{*}O_K, \qquad F_K = O_K G_K^{-1/2}, \qquad F_K^{*}F_K = I. \tag{147.53}$$

The inverse root exists by (147.36). The equality is checked by multiplying the Gram matrix by $G_K^{-1/2}$ on both sides. Thus ZF_K is isometric at any depth. The full recovery of the frame is

$$O_K = F_K G_K^{1/2}. \tag{147.54}$$

The pair (F_K, G_K) preserves the complete information. The unitary factor preserves an oriented orthonormal frame in the fixed charts, while the amplitudes also reside in G_K. Multiplying K by a positive scalar preserves F_K and changes G_K; K and $-K$ share a Gram matrix and have opposite polar factors, with different linear reading κ. Each example shows why an isolated factor does not replace the pair.

Translation of the origin preserves ν, but in the fixed positional chart

$$\kappa(SK) = 1000\kappa(K) - K_1. \tag{147.55}$$

This is obtained by shifting the positional numerator and using $d = 1000^{12} - 1$. To preserve the same reading under a chart change, the row ℓ is also transported by the inverse of that change. A general unitary chart may leave the lattice of integer blocks; decoding by Euclidean divisions is performed in the recovered block chart.

147.14 Exact example of transduction and error correction

Take $a = 8$, $U = S^4$ and the canonical register

$$\begin{aligned} K = (&234, 543, 140, 729, 659, 824,\\ &621, 58, 914, 794, 146, 601). \end{aligned} \tag{147.56}$$

Denoting by $(v)_{1:6}$ and $(v)_{7:12}$ its two blocks of six coordinates, the unnormalized readings are

$$\begin{aligned} (Q_{-}K)_{1:6} &= (1213, 3520, 499, 5774, 4358, 5798),\\ (Q_{-}K)_{7:12} &= (4822, -137, 7078, 5809, 1028, 4079),\\ (Q_{+}K)_{1:6} &= (2765, 5711, 1881, 6619, 6845, 8210),\\ (Q_{+}K)_{7:12} &= (5735, 1123, 8460, 7689, 1454, 6138). \end{aligned} \tag{147.57}$$

Their sum, divided by 17, recovers all blocks of K. The matrix identity $9Q_{-}^{*}Q_{-} + 8Q_{+}^{*}Q_{+} = 1241I$ precedes its evaluation on the specific vector.

To introduce a controlled error, the two memories are rationalized as $z_0 = MK/\sqrt{8}$, and $1/10$ is added to the first coordinate. Each block of twelve components is then transported through the two bilateral channels. The inverse by summation recovers the perturbed memory data, and the charge decoder receives $q = 6263$ together with these data, without receiving the sought register. The error norm in the original memories is $\sqrt{8}/10 < r_q$. Isometric normalization preserves that norm; the unnormalized channels merely facilitate exact rational calculation.

In this instance, the candidates of the floor-and-ceiling reconstruction satisfying the charge are 924. The minimum rationalized squared distance is $1/100$, and the decoder recovers (147.56). Uniqueness follows from the strict radius (147.50), not from the mere size of the enumeration. The recovered arithmetic reading is

$$\kappa = \frac{2345431407296598246210589147941466 01}{99999999999999999999999999999999999999}, \tag{147.58}$$

and the incidence selection is again $\{4, 6, 9, 10\}$, with the same threshold 729. The example introduces noise before transport. The proofs of contractivity of the adjoint also cover arbitrary subsequent errors whose joint norm lies within the declared radius.

The combined operational capacity is to transport and recover a generated register, maintain the conditioning of its responses and restore its arithmetic and incidence readings before applying discontinuous selections. Its effective premises are the prior production of K, the invertible cyclic frame, bilateral balance, the discrete separation of memories and the archive theorem. The register of twelve components preserves the scope of its readers; the complete enriched history retains its other registers and dependencies.

Chapter 148

Unitary lift of refinement and transport of readers

The residue bijection of theorem 146.4 and the bilateral balance of theorem 147.1 admit an explicit composition. An admissible arithmetic triple, containing a pair and its carry, determines a refined pair. The linearization of this bijection on orthonormal bases provides a unitary transport; two transports of this type can act as the two realizations of the bilateral balance.

The realization is established after the objects generated by APP–TRIT–TPK. Its combinatorial domain comprises the pairs and carries satisfying the conditions written below. The norm preserved by its linearization is the sum of squared amplitudes over those labels; the arithmetic conservation of $L_c(x, y) = (Bx - c, By + c)$ remains the sum of the two coordinates after rescaling. The composition relates both conservations through a specific map.

148.1 Bijection of admissible arithmetic states

Let $B \geq 2$ and $M \geq 1$ be integers. We define

$$\mathcal{D}_{B,M} = \big\{(x, y, c) \in \mathbb{Z}_{\geq 0}^2 \times \{0, \dots, B-1\} : x + y = M,\ c \leq Bx\big\}, \tag{148.1}$$

$$\mathcal{E}_{BM} = \big\{(X, Y) \in \mathbb{Z}_{\geq 0}^2 : X + Y = BM\big\}. \tag{148.2}$$

The condition $c \leq Bx$ maintains the nonnegativity of the first output coordinate. For $x = 0$ it only admits $c = 0$; for $x \geq 1$ it admits the entire canonical alphabet.

Theorem 148.1 (Exact correspondence of labels)**.** *The map*

$$\ell_{B,M} : \mathcal{D}_{B,M} \longrightarrow \mathcal{E}_{BM}, \qquad (x, y, c) \longmapsto (Bx - c, By + c) \tag{148.3}$$

is bijective. Both sets have cardinality $BM + 1$. Its inverse is given by

$$c = Y \bmod B, \qquad y = \frac{Y - c}{B}, \qquad x = \frac{X + c}{B}. \tag{148.4}$$

Proof. The domain condition and the identity $Bx - c + By + c = BM$ ensure that the image belongs to $\mathcal{E}_{BM}$. Given a pair in that set, the residue c is the unique alphabet element satisfying $Y \equiv c \pmod B$. Since $X + Y$ is divisible by B, so is $X + c$. Euclidean division of Y gives $y \geq 0$, and $X + c \geq 0$ gives $x \geq 0$. If $c > 0$, $x = 0$ would require $X = -c < 0$, so admissibility is preserved. The sum of the preimages is M, and substitution verifies the original image. Uniqueness of the residue and quotients proves the bijection.

There are $M + 1$ labels with $c = 0$ and M with each positive carry; therefore the cardinality is $(M + 1) + (B - 1)M = BM + 1$, equal to that of the output. □

The proof makes explicit the boundary endpoint of the domain, a necessary condition for equality of cardinalities. The new carry label is a typed part of the input to refinement.

148.2 Unitarity on label amplitudes

Take the real or complex Hilbert spaces with counting measure

$$\mathscr{H}_D = \ell^2(\mathcal{D}_{B,M}), \qquad \mathscr{H}_E = \ell^2(\mathcal{E}_{BM}), \tag{148.5}$$

and their labeled orthonormal bases. Define linearly

$$J_{B,M}|x, y, c\rangle = |Bx - c, By + c\rangle. \tag{148.6}$$

We shall write J when the parameters are fixed.

Proposition 148.2 (Unitary lift)**.** *The operator $J : \mathscr{H}_D \to \mathscr{H}_E$ is unitary: $J^*J = I_D$, $JJ^* = I_E$. The adjoint recovers each label through* (148.4)*.*

Proof. theorem 148.1 ensures that J maps two complete orthonormal bases bijectively. For $\psi = \sum_{d\in\mathcal{D}} \psi_d |d\rangle$,

$$\|J\psi\|^2 = \sum_{e\in\mathcal{E}} |\psi_{\ell_{B,M}^{-1}(e)}|^2 = \sum_{d\in\mathcal{D}} |\psi_d|^2 = \|\psi\|^2.$$

Surjectivity proves the two unitary identities and the inverse bijection describes the adjoint on each basis vector. □

The label $|73, 27, 1\rangle$ has norm one; the integers $73, 27, 1$ describe its arithmetic values, not its amplitudes. The construction represents reversible processing of labels or amplitudes. A specific physical realization additionally preserves the protocol preparing and transporting those amplitudes.

148.3 Diagonal proportion and carry readers

For a real function h on a finite set, denote by M_h its diagonal operator: each basis vector is multiplied by the value of h at its label. The bijection satisfies

$$J^* M_h J = M_{h\circ\ell_{B,M}}. \tag{148.7}$$

This is proved by acting on all basis vectors.

Define on the output

$$f_E(X,Y) = \frac{X}{BM}, \quad g_E(X,Y) = \frac{Y}{BM}, \quad \gamma_E(X,Y) = Y \bmod B, \tag{148.8}$$

and on the input $f_D = x/M$, $g_D = y/M$, $\gamma_D = c$. The identity (148.7) provides

$$\begin{aligned} J^* M_{f_E} J &= M_{x/M - c/(BM)}, \\ J^* M_{g_E} J &= M_{y/M + c/(BM)}, \\ J^* M_{\gamma_E} J &= M_c. \end{aligned} \tag{148.9}$$

The first two readers transport the unit transfer. Since $f_E + g_E = 1$, their sum is I_E and their compression is I_D. The third reader exactly preserves the carry as an integer observable, with its residual representation in the output.

148.4 Complete routes at arbitrary depth

Let $\mathcal{P}_n(B, M)$ be the set of routes of length n starting from all nonnegative pairs with sum M and, at each stage, satisfying the conditions of (148.1) for the corresponding sum. A route preserves its initial pair and its n admissible carries. Let Φ_n be the map assigning each route its final pair.

Theorem 148.3 (Recovery of routes at every finite depth)**.** *For every $n \geq 0$,*

$$\Phi_n : \mathcal{P}_n(B,M) \longrightarrow \mathcal{E}_{B^n M} \quad \textit{is bijective}, \qquad |\mathcal{P}_n(B,M)| = B^n M + 1. \tag{148.10}$$

Its linearization

$$J_n : \ell^2(\mathcal{P}_n(B,M)) \longrightarrow \ell^2(\mathcal{E}_{B^n M}), \qquad J_n|\pi\rangle = |\Phi_n(\pi)\rangle \tag{148.11}$$

is unitary.

Proof. From a pair with sum $B^n M$, the residue of (148.4) recovers the last carry and a unique preceding pair with sum $B^{n-1}M$. Repeating n times produces a unique route whose input sums to M. The direct composition recovers the original endpoint, proving bijection and inverse. The count is also obtained from the inputs: the M pairs with $x_0 \geq 1$ admit B^n words each, whereas $(0, M)$ admits only the all-zero word. The total is $MB^n + 1$. The bijective transformation of orthonormal bases proves the unitarity of J_n. □

With the depth known, the readers on the final pair are

$$c_j = \left\lfloor \frac{Y}{B^{n-j}} \right\rfloor \bmod B, \quad y_0 = \left\lfloor \frac{Y}{B^n} \right\rfloor, \quad x_0 = M - y_0, \qquad 1 \leq j \leq n. \tag{148.12}$$

These functions are diagonal and recover the complete memory of the finite operations. Compatibility with prolongation is

$$\Phi_{n+1}(\pi, c) = L_c(\Phi_n(\pi)). \tag{148.13}$$

On passing from n to $n + 1$, the new admissible carry label is incorporated. The spaces increase in dimension and the boundary label admits only $c = 0$; therefore the domain of the next step preserves admissibility, instead of replacing it with an unrestricted product with B digits.

148.5 Fixed input and exact image

If an initial pair (x_0, y_0) is fixed, with $x_0 \geq 1$, there are B^n routes and their image is exactly

$$\left\{(B^n x_0 - C, B^n y_0 + C) : 0 \leq C \leq B^n - 1\right\}. \tag{148.14}$$

corollary 146.6 proves that the accumulated B^n are distinct and recover the respective words. The linearization is unitary onto the subspace spanned by this image. If $x_0 = 0$, the image consists of a single label.

The cylinders of theorem 146.8 correspond to a fixed input. The complete domain of (148.10) brings together all pairs with sum M. Both constructions have explicit inverses, but their images and cardinalities differ. Arbitrary depth expresses the validity of the proof for every n; the limit of the reading and its boundaries remain as established in section 146.7.

148.6 Bilateral coupling of refinement

Let R be a previously declared permutation of $\mathcal{D}_{B,M}$, unitarily linearized on $\mathscr{H}_D$. Define the two transports

$$\mathcal{B} = J, \qquad C = JR : \mathscr{H}_D \longrightarrow \mathscr{H}_E. \tag{148.15}$$

The permutation is part of the specification of the procedure and is fixed before evaluating an output reading. For $a > 0$, $p = a^2 + a + 1$, $N = (2a + 1)p$,

$$Q_- = J(aI - R), \qquad Q_+ = J((a + 1)I + R). \tag{148.16}$$

Proposition 148.4 (Distribution and recovery of refined amplitudes)**.** *The operators* (148.16) *satisfy*

$$(a + 1)Q_-^* Q_- + aQ_+^* Q_+ = NI_D, \tag{148.17}$$

and the map

$$W_a\psi = \frac{1}{\sqrt{N}}\left(\sqrt{a + 1}\, Q_-\psi, \sqrt{a}\, Q_+\psi\right) \tag{148.18}$$

is isometric. For the unnormalized channels $u = Q_-\psi$, $v = Q_+\psi$,

$$J\psi = \frac{u+v}{2a+1}, \qquad \psi = J^* \frac{u+v}{2a+1}. \tag{148.19}$$

Proof. The two maps $\mathcal{B}, C$ are unitary. The cross terms of the Gram matrices of Q_- and Q_+ have coefficients $-a$ and $a+1$, respectively. The balance weights cancel them, while the diagonal terms sum to N, as in theorem 147.1. Normalization proves isometry. The sum of the channels is $(2a+1)J\psi$, and the unitarity of J gives the inverse. □

From the normalized channels the input is recovered directly by W_a^*, with inversion norm one; this adjoint already incorporates J^*. The alternative formula (148.19) separates recovery of $J\psi$ by summation from application of J^*, whose action on labels is calculated through residues. Reversible arithmetic refinement and bilateral amplitude transport are thereby composed.

Proposition 148.5 (Compatibility in the refined space)**.** *An unnormalized pair (u, v) belongs to the image of $\psi \mapsto (Q_-\psi, Q_+\psi)$ if and only if*

$$\big[(a+1)I_E + JRJ^*\big]u = \big[aI_E - JRJ^*\big]v. \tag{148.20}$$

Proof. The equality is equivalent to $JRJ^*(u+v) = av - (a+1)u$. If it holds, define $\psi = J^*(u+v)/(2a+1)$. On substitution into (148.16), this identity gives $Q_-\psi = u$ and $Q_+\psi = v$. Necessity follows from inserting the true channels. All compositions act on $\mathcal{H}_E$, so their types are defined. □

148.7 Law of observables on the two channels

Theorem 148.6 (Bilateral compression of a refined reader)**.** *For $F = F^*$ on $\mathcal{H}_E$, write $F_J = J^*FJ$. Then*

$$W_a^* \operatorname{diag}(F, F) W_a = \frac{a(a+1)F_J + R^*F_JR}{p}. \tag{148.21}$$

The reader F_J is preserved exactly if and only if it commutes with R.

Proof. Expanding $(a+1)Q_-^*FQ_- + aQ_+^*FQ_+$ cancels the mixed terms. The coefficient of $\mathcal{B}^*F\mathcal{B}$ is $a(a+1)(2a+1)$; that of C^*FC is $2a+1$. Dividing by N, substituting $C = JR$ and using $F_J = J^*FJ$ proves the formula. Equality with F_J is equivalent to $R^*F_JR = F_J$, or to $[F_J, R] = 0$ by unitarity. □

In the case $a = 8$, the weights are $72/73$ and $1/73$. The identity reader recovers the norm. A nonconstant reader may change even when the norm is preserved; its variation is calculated by (148.21). The operator that exactly preserves an original reader A in the complete image is $W_aAW_a^*$, whose image identity is $W_aW_a^*$. Reading $\operatorname{diag}(F, F)$ corresponds to the specific compression of the theorem and does not automatically replace that transported observable.

148.8 Arithmetic example of dimension one thousand and one

Take $B = 10$, $M = 100$. The label spaces have dimension 1001, and

$$d_1 = (73, 27, 1) \longmapsto e_1 = (729, 271). \tag{148.22}$$

The diagonal readers give

$$f_E(e_1) = \frac{729}{1000} = \frac{73}{100} - \frac{1}{1000}, \qquad g_E(e_1) = \frac{271}{1000}, \qquad \gamma_E(e_1) = 1. \tag{148.23}$$

Declare R to be the transposition of the admissible labels $d_1 = (73, 27, 1)$ and $d_2 = (73, 27, 2)$, leaving the others fixed. The second image is $e_2 = (728, 272)$. The transposition rule has been fixed on the labels, before any comparison with a constant.

For $a = 8$ and $\psi = |d_1\rangle$,

$$Q_-\psi = 8|e_1\rangle - |e_2\rangle, \qquad Q_+\psi = 9|e_1\rangle + |e_2\rangle. \tag{148.24}$$

The squared norms are 65 and 82, and

$$9 \cdot 65 + 8 \cdot 82 = 585 + 656 = 1241 = 17 \cdot 73. \tag{148.25}$$

The sum of the channels is $17|e_1\rangle$. The inverse (148.19) and the residue recover $|d_1\rangle$, including the carry one.

If $F = M_{f_E}$ is evaluated diagonally in both normalized channels, theorem 148.6 gives

$$\begin{aligned}\langle W_8\psi, \operatorname{diag}(F,F)W_8\psi\rangle &= \frac{72 \cdot 729 + 728}{73 \cdot 1000} \\ &= \frac{729}{1000} - \frac{1}{73000}.\end{aligned} \tag{148.26}$$

The complementary reading is one minus this quantity. For the diagonal carry reader we obtain

$$\frac{72 \cdot 1 + 2}{73} = \frac{74}{73}. \tag{148.27}$$

The original carry remains exactly recoverable: the observable $W_8 M_c W_8^*$ has on $W_8|d_1\rangle$ the eigenvalue one. The value $74/73$ is the response of the diagonal reader applied to both channels, whose relative transport is specified by (148.21). The example distinguishes conservation of information from variation of a particular reading.

148.9 Articulation with the generated register and application domains

The unitary J acts on $BM + 1$ labels, whereas the realization of K has twelve coordinates. In the example, 1001 counts refined pairs with sum one thousand, and twelve counts windows of the produced register. Both dimensions retain their specific combinatorial origin.

A particular composition between these two carriers can be formulated through a material selector of twelve distinct routes, with its provenance and operators. If that selector provides an isometry $J_{12} : \mathbb{R}^{12} \to \ell^2(\mathcal{D}_{B,M})$, then JJ_{12} is isometric, and the intertwining of the readers is checked on it. This is the hypothesis of that additional composition, not an implicit identification through the number of coordinates. The bilateral maps already defined directly on K in chapter 147 retain their complete domains without using that additional selector.

The result of this section brings together an arithmetic bijection, its unitary, intertwined diagonal readers, route recovery at any finite depth and an explicit bilateral composition. The route domain is that defined in (148.10). Applications to histories filtered by TPK retain their survival and prolongation rules; the label correspondence here proves the declared transport, without replacing those operators with an unrestricted arithmetic domain.

Chapter 149

Dimensional unity, relative symmetry, and recovery of structure

The holographic extreme and mean ratio articulates conservation of unity under evolution with its arithmetic and dimensional realizations. The composition developed here makes this principle precise through conserved quantities, domains, and inverses. The enriched state generated by APP–TRIT–TPK precedes the numerical and geometric readouts; its operations preserve the orientation, carry, incidence, and memory used by each realization.

149.1 Arithmetic conservation, unitarity, and isometric decomposition

Arithmetic refinement preserves normalized unity:

$$(x, y, c) \longmapsto (Bx - c, By + c), \qquad \frac{Bx - c}{B(x + y)} + \frac{By + c}{B(x + y)} = 1. \tag{149.1}$$

The residue inverse recovers the canonical carry and the preceding pair. Block composition preserves the contributions of events with their scale, and successive divisions recover every finite history. The limiting readout is controlled by decreasing cylinders; boundary marks retain the distinction between histories sharing a scalar evaluation.

The lift on labels transforms that bijection into a unitary J. It preserves the norm of amplitudes and transports the fraction and carry readers through $J^* M_h J = M_{h\circ\ell}$. The quantity $x^2 + y^2$ and the norm of $|x, y, c\rangle$ belong to different realizations. The proof of unitarity establishes the precise link between arithmetic reversibility and preservation of amplitudes.

The third conservation law corresponds to two isometric realizations:

$$(a + 1)Q_-^* Q_- + aQ_+^* Q_+ = (2a + 1)(a^2 + a + 1)I. \tag{149.2}$$

Cancellation of the mixed terms allows bilateral distribution and its norm-one reconstruction. The law holds for the transports on the domain specified in theorem 147.1. Its specialization to the ternary cycle provides additional norm and index properties; the general law remains valid for isometries without that finite order.

The chain formed by the three maps is effective: the bijection produces J; a previously declared permutation produces the second transport JR; the two transports satisfy the balance, and their channels recover the input through (148.19). The example in section 148.8 shows the complete procedure with $(73, 27, 1)$, $(729, 271)$, and 1001 admissible labels. The numbers in that realization are not introduced as substitutes for the structure generating the register K.

149.2 Complete memory and the logical contribution to the Landauer bound

The vanishing logical contribution to the Landauer bound, established in (123.4), applies to the recoverable transports constructed in this article. The bijection of theorem 148.1 preserves the carry through (148.4); its prolongation preserves the path through theorem 148.3. The archives of theorem 147.7 preserve the amplitudes and correlations needed to invert the transport.

Corollary 149.1 (Recovery and spectral conservation of the archive)**.** *Let $Z : H \to \mathcal{K}$ be one of the complete archives Z_N, Z_∞ of theorem 147.7, or the unitary lift J_n of theorem 148.3, on its respective domain. For a finite-rank density operator ρ on H, the output $\rho_Z = Z\rho Z^*$ satisfies*

$$\operatorname{Tr}\rho_Z = 1, \qquad Z^*\rho_Z Z = \rho, \tag{149.3}$$

$$s(\rho_Z) = s(\rho), \qquad s(\rho) = -\operatorname{Tr}(\rho \log \rho), \tag{149.4}$$

with $0 \log 0 = 0$. *If X is a random variable on the finite paths of theorem 148.3, then*

$$\mathsf{H}(X \mid \Phi_n(X)) = 0. \tag{149.5}$$

Proof. Write the finite spectral decomposition $\rho = \sum_j p_j |v_j\rangle\langle v_j|$, with the v_j orthonormal, $p_j > 0$, and $\sum_j p_j = 1$. The identity $Z^*Z = I$ implies that the Zv_j are orthonormal. Thus, $\rho_Z = \sum_j p_j |Zv_j\rangle\langle Zv_j|$ has the same nonzero eigenvalues, proving the trace and entropy identities. Multiplication by Z^* and Z recovers ρ. Finally, Φ_n^{-1} recovers each path from its final pair; the conditional distribution is concentrated on a single path and its entropy is zero. □

Conservation pertains to the complete archive. A visible reading q, accompanied by memory M, recovers the state under the hypothesis of (123.3); information not published by q remains in $\mathsf{H}(M(X) \mid q(X))$. Discarding a necessary complement or terminal defines another operation, whose recoverability is determined on the information actually retained. The preceding spectral equality preserves the distinction between entropy of the complete state, entropy of a partial reduction and transferred heat.

In the thermal model of Section 123, with degenerate logical states, an isothermal environment, and accounting for auxiliary registers, the term associated with erasure is $k_B \Theta \, \mathsf{H}(X \mid F(X))$, for a deterministic logical map F. Its contribution vanishes for the preceding recoverable transports; resetting a uniform TRIT satisfies $Q_{\mathrm{reset}} \geq k_B \Theta \log 3$. Preparation, control, finite operating speed, and coupling to the environment retain their own physical balances. The total heat of an implementation requires combining these contributions with the logical operation.

149.3 Conservation of observables and relative symmetry

Norm balance is the identity case of a more informative operator law. When the transports are unitary, with relative operator R, the effective reader is

$$\mathcal{T}_a(G) = \frac{a(a+1)G + R^*GR}{a(a+1)+1}. \tag{149.6}$$

The observable remains fixed exactly when $[G, R] = 0$. In the canonical case,

$$\mathcal{T}_8(G) - G = \frac{R^*GR - G}{73}. \tag{149.7}$$

This identity determines both conservation and variation: the defect measures the difference between two realizations of the same reader, weighted by the split $72 + 1$.

Complete memory also transports the observable through $A \mapsto ZAZ^*$ on $\operatorname{im} Z$. Products, adjoints, and pairings are preserved on that image. The diagonal channel readout and the transported observable are different operations; the carry example shows the difference between $74/73$ for the former and eigenvalue one for the latter. The information remains recoverable even if a particular diagonal readout changes value.

The variational charges of proposition 114.1 retain their foundation in an explicit action and intertwining. For $A_{n+1}U_n = U_nA_n$, the transport action gives the charge $p_n^*A_nq_n$. Its conservation and the norm identity are compatible results with different hypotheses. Likewise, preservation of an alternating or Lorentzian form in (147.31) retains its geometric meaning; the noise and contraction estimates use the Hilbert positivity declared in their proofs.

149.4 Role of the register in transduction

The holographic arithmetic–geometric coupling constant has a role defined by its compositions. The register K is obtained beforehand by oriented and Hadamard extraction. Its readout κ is reversible with the base, length, and origin fixed. Its frame O_K identifies cyclic responses because its Gram matrix has a strictly positive lower bound. Incidence and geometric direction use their declared readers on the same register.

The archive isometry preserves the Gram matrix of that frame and therefore retains the recovery bounds at every depth. The complete register admits norm-one inversion; the two partial memories retain their factor $9/4$ and their uniform kernel. The additional charge distinguishes recovery of the absolute register from that of its centred component. Integer decoding subsequently recovers the incidence selection even at a threshold lying exactly on a coordinate.

This capacity is more specific than preserving a total quantity: it allows recovery of operations and relations acting on a generated object. Its guarantees rest on the Gram matrix, the distances in the integer image, and the proven inverses, not on retaining another copy of the input by definition. At the same time, the finite number of windows of K remains distinct from the complete enriched history; the other registers and accompanying marks retain their own transports.

The orbit and its isometric transports preserve the norm $\nu = \|K\|$. Normalizing its columns to norm one requires preserving ν to invert that operation. Polar normalization uses the pair (F_K, G_K). The formula $O_K = F_K G_K^{1/2}$ preserves orientation and amplitudes. The positional reader must travel with the chart change, since the common orbital norm does not make κ invariant under an isolated renumbering of its components.

149.5 Depth, dimensions, and physical realizations

The bilateral policy that continues one channel and archives the other has uniform exhaustion of the terminal. Its limit is an isometry into the sum of all memories, and its inverse is the convergent adjoint series in (147.25). A finite number of complements has two distinct reconstructions: the truncated adjoint retains an explicit bias, and the least-squares inverse eliminates it with the computed noise amplification. The preceding evolutions with a positive terminal retain that component in the complete archive.

The dimension of the label space, the rank of a projector, and the dimension of a physical realization are related through their concrete maps. The unitary on 1001 labels and the twelve-window register are two explicit realizations of different domains. The exterior conservation law of the preceding development extends the Gram products to every finite degree, including the mixed contributions between terminal and memories. The exceptional and physical realizations remain linked to their already constructed representations and operators; numerical ranks alone do not replace that link.

149.6 Conclusions on the evolutionary conservation of unity

The physical actions and the Gaussian protocol of the manuscript preserve the couplings, forms, units and preparations involved in their proofs. The bilateral balance added here extends the family of conservative transports and their readings; its application to a physical realization is established by composing the operators of that realization, without replacing them with a coincidence of normalization.

The unity of the development thus resides in a proven sequence: generation and completion, refinement with memory, unitary realization, relative symmetry of the reader, and recovery of structure. The holographic extreme and mean ratio thereby receives an exposition in which change of scale, incidence, conservation, and transduction have explicit operations and jointly retain the information needed to invert their readouts.

Arithmetic formalization of bilateral balance

The nonadic specialization of the bilateral law admits an independent arithmetic formalization. The point of application lies after the construction of its coefficients and transports: the base-ten partition fixes $a = 8$, and the two registers are $8x - y$ and $9x + y$. On that domain, Lean 4.21.0 verifies seven theorems with its library `Std`. The formal sources and execution procedure are provided as a supplement. This complement states their mathematical content and relates it to the operator proofs of the article.

Polynomial identity and conservation of the norm

For any $x, y \in \mathbb{Z}$, we have

$$9(8x - y)^2 + 8(9x + y)^2 = 17(72x^2 + y^2).$$

Expansion of the two squares cancels the cross products. When $y^2 = x^2$, the same equality gives

$$9(8x - y)^2 + 8(9x + y)^2 = 17 \cdot 73\, x^2.$$

These are the first two formalized theorems. The first expresses a balance for arbitrary integer inputs; the second applies the explicit hypothesis of norm conservation. The integer $73 = 72 + 1$ occupies exactly the place it has in theorem 147.1.

Arbitrary finite dimension

Let $n \in \mathbb{N}$ and $x, y : \mathbb{N} \to \mathbb{Z}$. Define

$$S_n(x) = \sum_{j<n} x_j^2, \qquad E_n(x, y) = \sum_{j<n} \bigl(9(8x_j - y_j)^2 + 8(9x_j + y_j)^2\bigr).$$

The next two theorems establish

$$E_n(x, y) = 17\bigl(72 S_n(x) + S_n(y)\bigr)$$

and, under the hypothesis $S_n(y) = S_n(x)$,

$$E_n(x, y) = 17 \cdot 73\, S_n(x).$$

The formal proof proceeds by induction on n. The initial case is the empty sum; the induction step adds a coordinate and applies the already proved polynomial identity. The universal quantifier over n includes all finite dimensions. The hypothesis uses the total sum of squares, so it admits redistributions between coordinates.

Exact inversion of the bilateral channels

Write $q_- = 8x - y$ and $q_+ = 9x + y$. The fifth and sixth theorems are the exact equalities

$$q_- + q_+ = 17x, \qquad 8q_+ - 9q_- = 17y.$$

In the image of the map $(x, y) \mapsto (q_-, q_+)$, both left-hand sides are multiples of 17. The divisions recover the two inputs. The seventh theorem states their injectivity consequence:

$$\left.\begin{aligned} 8x - y &= 8x' - y', \\ 9x + y &= 9x' + y' \end{aligned}\right\} \quad \Longrightarrow \quad x = x' \text{ and } y = y'.$$

Cancellation in $\mathbb{Z}$ justifies this conclusion. The formal content coincides with recovery of the pair of registers, before normalizing them as channels of a metric realization.

Correspondence between algebraic proof and formalization

The formalization uses arbitrary integer variables and mathematical induction. It therefore constitutes a proof of the preceding statements, distinct from checking a finite set of examples. The system accepts the proof terms without added axiomatic declarations or deferred proofs. The supplement also records the logical dependencies reported by the tool for each declaration.

The operator law in the body of the article has a broader domain: isometries between Hilbert spaces, archives of arbitrary depth and conjugated readers. Its proofs are found in theorem 147.1 y sections 149.3 y 149.5. The Lean proof incorporated here certifies the stated arithmetic specialization, with its quantifiers and initial structure. The provenance of the coefficients, generation of the register and exceptional realizations retain the preceding proofs on which they depend.

Part VII

Recoverable geometry and reciprocity of coupling

Positive deformation, lattice duality, polar structure and oriented recovery of action.

The function of the holographic arithmetic–geometric coupling constant becomes precise by studying the transformations determined by its register. Its positional reading preserves twelve ordered components; its incidence realization distinguishes a direction; the bilateral archive allows it to be reconstructed. Composing these operations leads to a family of geometries whose fundamental volume remains constant and whose inversion produces the dual lattice. The register therefore participates both in the constitution of the geometry and in its recovery.

This development maintains the generative order established in the article. The additive and multiplicative evaluations of APP, the orientation of TRIT and the selection, transport and memory operations of TPK produce the enriched state and its joint discrete structure of the continuum. The regional coordinates and exceptional incidence are readings of that state. The vector, lattice and hyperbolic realizations that follow receive these outputs with their charts and orientations. The genealogy is preserved when the representation changes.

The distinction between scale and shape acquires an exact expression here. A deformation can dilate certain directions and contract others while preserving the total product of its factors. In the action ellipse, the product of the semiaxes preserves the action section. In the twelve radii determined by K, compensation preserves the lattice covolume. In the mass operator, a decomposition separates the uniform factor from the relative transformation of determinant one. Each conservation has its own object and proof; the readers allow them to be related without confusing their domains.

Radial reciprocity articulates two complementary constructions. In the chart of the spectral double circle, inverting the radius is exactly equivalent to reflecting the complex coordinate across the line of real part $1/2$. In the modular chart of the ellipse, exchanging semiaxes inverts the radius and changes the sign of the Klein coordinate. Both correspondences admit inverse formulas. The orientation register distinguishes the conjugate reading from a modification of the physical action section.

Substituting the full expressions for the angular readers reveals a specific arithmetic cancellation: the linear term of the preceding action section is compensated by the corresponding term of the counterangle. The regional continuation remains intact. The angular contrast, radius and hyperbolic coordinate are then expressed through the same publications involved in α, with the coupling constant within its dodecaphasic closure. Conversely, the oriented radial coordinate recovers the return section and allows the complete electronic expression to be rewritten.

The spectral realization in turn distinguishes two kinds of information. Lattice duality imposes a functional equation; covolume fixes the poles and their residues; deformation modifies the entire part of the completed function. The archive recovers the direction determining that variation and provides explicit bounds for the reconstruction error. Conservation and reversibility are thus expressed through operations encompassing the register, geometry and their analytic readings.

Chapter 150

Positive deformation and lattice duality of the generated register

150.1 Coupling direction and unimodular flow

Extraction of the register, its rational reading and the incidence projectors have been constructed in the preceding sections. Base one thousand, the phase origin, carry boundary and chart of the twelve positions are preserved. In that chart,

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), \qquad u = P_3 P_{11} K,$$

$$P_{11} = I - \frac{1}{12}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \qquad u \perp \mathbf{1}, \qquad \|u\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0.$$

The finite definition of P_3, the order of its classes and the exact evaluation of this norm appear in Lemma 21.62. The following realization uses that generated object. We write

$$\rho = \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2} > 0, \qquad h_i = (\log \rho)\frac{u_i}{\|u\|}, \qquad r_i(t) = e^{t h_i}, \qquad g_t = \operatorname{diag}(r_i(t)). \tag{150.1}$$

The regional coordinates are evaluated before this composition. The parameter $t \in \mathbb{R}$ indexes a mathematical deformation; its definition does not identify it with the TPK clock or with a refinement depth.

Proposition 150.1 (Scale compensation)**.** *The flow satisfies*

$$g_t g_s = g_{t+s}, \qquad g_0 = I, \qquad g_t^{-1} = g_{-t}, \qquad \det g_t = \prod_i r_i(t) = 1.$$

Proof. Orthogonality to the uniform mode gives $\sum_i h_i = 0$. Multiplication of exponentials proves the group law and inversion. The determinant is $\exp(t \sum_i h_i) = 1$. Each diagonal factor is positive. □

Conservation is multiplicative and global: each radius may vary. In logarithmic coordinates the zero sum is preserved. This distinction will allow the transverse projector of rank ten to be interpreted geometrically.

150.2 Exact reciprocity of the double circle

Consider the complex chart and its inverse

$$z(s) = \frac{s-1}{s}, \qquad s(z) = \frac{1}{1-z}.$$

For $r > 0$, $z = re^{i\theta} \neq 1$, define $s(r,\theta) = (1 - re^{i\theta})^{-1}$.

Proposition 150.2 (Conjugation of the spectral mirror)**.** *The following hold*

$$s(r^{-1}, \theta) = 1 - \overline{s(r,\theta)}, \tag{150.2}$$

$$\Re s(r,\theta) - \frac{1}{2} = \frac{1 - r^2}{2|1 - re^{i\theta}|^2}. \tag{150.3}$$

The unit circle, except for the excluded point $z = 1$, corresponds to the line $\Re s = 1/2$.

Proof. Radial inversion at a fixed angle is $z' = 1/\overline{z}$. Therefore,

$$\frac{1}{1-z'} = \frac{\overline{z}}{\overline{z}-1} = 1 - \frac{1}{1-\overline{z}}.$$

Moreover, $(1-z)^{-1} = (1-\overline{z})/|1-z|^2$. Subtracting $1/2$ from its real part produces (150.3). □

The relation to the radii of (150.1) is explicit: $r_i(-t) = r_i(t)^{-1}$. Equality of products preserves volume, while the proposition identifies an involution between complex charts. Locating the zeros of a function requires the properties of that function; it is not a consequence of the unit product of radii.

Three circular realizations in the corpus should be distinguished. The spectral chart takes $s = 1/2 + i\gamma$ to $z_\gamma = (\gamma + i/2)/(\gamma - i/2)$. The dual circle of integer memory takes an integer n to the character z^n. Nonadic compression takes z to $(8+z)/9$, a circle with center $8/9$ and radius $1/9$, tangent to the unit circle at 1. The modal ratio $5/9$ belongs, by contrast, to the eigenvalues $9, 5$ of the APP block $\left(\begin{smallmatrix} 7 & 2 \\ 2 & 7 \end{smallmatrix}\right)$. Their functions remain distinct when composed.

150.3 Positroid strata and balanced invariants

Let X be a real matrix of rank k, with twelve columns, and let $\Delta_I(X)$ be its maximal minor with columns I. The right action satisfies

$$\Delta_I(Xg_t) = e^{t\sum_{i\in I} h_i}\Delta_I(X). \tag{150.4}$$

Diagonal positivity preserves all signs and all vanishing minors. Thus, for $1 \le k \le 11$, a trajectory starting in $\mathrm{Gr}_{\ge 0}(k,12)$ remains in the same positroid stratum. Stratification by minors is used here as a subsequent realization; see Postnikov's general construction [175].

Proposition 150.3 (Positive duality of complementary ranks)**.** *Let* $A_\pm = \mathrm{diag}(1,-1,1,-1,\dots)$ *and* $\mathfrak{D}(V) = V^\perp A_\pm$*, with the positive orientation of their complementary minors. Then*

$$\mathfrak{D}(Vg_t) = \mathfrak{D}(V)g_{-t}.$$

Proof. Since g_t is diagonal and self-adjoint, $(Vg_t)^\perp = V^\perp g_t^{-1}$. The matrix $A_\pm$ commutes with g_t. For the sign of the minors, the orthogonal complement introduces, up to a global sign, $(-1)^{\sum_{j\in J} j}$. Multiplication of the columns by $(-1)^{j-1}$ leaves a sign independent of J; it is removed by a single global orientation. The complementary minors are then nonnegative. □

If $\mathbf{1}_I + \mathbf{1}_J = \mathbf{1}_L + \mathbf{1}_M$ and the denominator is nonzero, coordinatewise cancellation of the exponents proves

$$\frac{\Delta_I(Xg_t)\Delta_J(Xg_t)}{\Delta_L(Xg_t)\Delta_M(Xg_t)} = \frac{\Delta_I(X)\Delta_J(X)}{\Delta_L(X)\Delta_M(X)}. \tag{150.5}$$

In rank two, the four-index quotients and their exchange relations are recovered. By contrast, the product of complementary minors of the two paired trajectories receives the same factor twice and may vary. Nor does the circulant frame O_K, invertible for recovery, acquire the property of being totally nonnegative through this action: its minors preserve the signs established in its own realization.

150.4 Covolume, dual lattice and ternary symmetry

Let Λ be the previously constructed marked Leech lattice, self-dual and of covolume one in the metric of its twelve orthogonal planes of type A_2. The marked isometry

$$\sigma = \bigoplus_{i=1}^{12} P_i \mathsf{C} P_i^{-1}, \qquad \mathsf{C}^2 + \mathsf{C} + I = 0,$$

preserves the gluing and Λ, has order three and has no nonzero fixed vectors. The P_i are frames of the planes, distinct from the projectors P_3, P_{11}. Define

$$G_t = \bigoplus_{i=1}^{12} r_i(t) I_{A_{2,i}}, \qquad \Lambda_t = G_t \Lambda. \tag{150.6}$$

Theorem 150.4 (Reciprocal lattice deformation)**.** *For every $t \in \mathbb{R}$,*

$$\operatorname{covol}(\Lambda_t) = 1, \qquad \Lambda_t^* = \Lambda_{-t}, \qquad \sigma\Lambda_t = \Lambda_t.$$

Every isometry of Λ commuting with G_t also preserves Λ_t.

Proof. Each radius acts on two coordinates; thus, $\det G_t = \prod_i r_i(t)^2 = 1$. For an invertible operator G, a vector y belongs to $(G\Lambda)^*$ exactly when $G^* y \in \Lambda^*$. Therefore $(G\Lambda)^* = G^{-*}\Lambda^*$. Applying self-duality, self-adjointness and inversion yields $\Lambda_t^* = G_{-t}\Lambda$. Finally, G_t is scalar on each plane and commutes with σ, whence $\sigma G_t \Lambda = G_t \sigma \Lambda = G_t \Lambda$. The same calculation applies to the other commuting isometries. The operator σ does not change, so it preserves its order and absence of fixed vectors. □

The point $t = 0$ recovers the original lattice with its integrality, parity, self-duality and exceptional realization. The complete family is a family of unimodular Euclidean lattices in the sense of covolume one; away from the original point, neither integrality nor a Moonshine realization is assumed. The theorem identifies exactly the ternary symmetry that does persist and the duality relating the two directions of deformation.

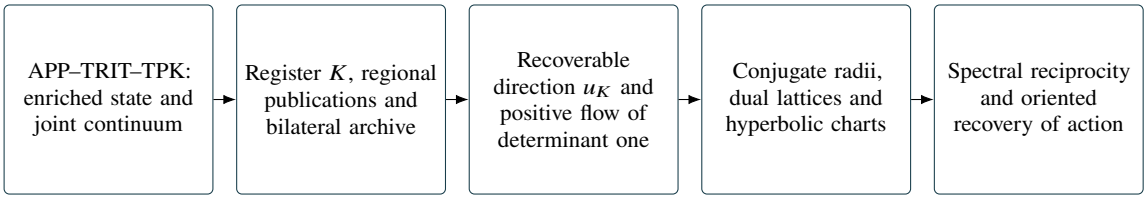


Figure 150.1: Dependencies of the extension. The arrows represent compositions specified in the proofs; recovery preserves the charts and orientation.

Chapter 151

Theta–Mellin reciprocity and conservation of the polar structure

151.1 Periodization of the generated lattice

The lattice Λ_t of Theorem 150.4 has an analytic reading that preserves its dependence on the register. For $x > 0$, define

$$\Theta_t(x) = \sum_{v \in \Lambda} \exp\Big(-\pi_{\mathrm{HMT}} x \|G_t v\|^2\Big). \tag{151.1}$$

In this realization, the previously generated coordinate π_{HMT} is identified with the circular normalization of Fourier analysis. That identification follows the construction of the lattice and the flow. To simplify the analytic notation, we shall use π with this meaning.

On every compact interval of t, the singular values of G_t are bounded above and below by positive constants. The lattice has $O(R^{24})$ points in a ball of radius R. Comparison over annuli with a Gaussian proves absolute and uniform convergence of (151.1) and its derivatives when x remains in a compact subset of $(0, \infty)$. The mode $v = 0$ contributes exactly one.

Theorem 151.1 (Reciprocity of the theta reading)**.** *For every $x > 0$ and $t \in \mathbb{R}$,*

$$\Theta_t(x) = x^{-12} \Theta_{-t}(1/x). \tag{151.2}$$

Proof. With the convention $\widehat{f}(\xi) = \int f(y) e^{-2\pi i \langle y, \xi \rangle}\, dy$, the Gaussian $f_x(y) = e^{-\pi x \|y\|^2}$ in dimension twenty-four has transform $\widehat{f_x}(\xi) = x^{-12} e^{-\pi \|\xi\|^2 / x}$. Periodize over Λ_t. Integrating the periodization over a fundamental domain shows that the Fourier coefficient indexed by $\xi \in \Lambda_t^*$ is $\widehat{f_x}(\xi)/\mathrm{covol}(\Lambda_t)$. Both Gaussian series converge absolutely; evaluating the expansion at the origin gives

$$\sum_{v \in \Lambda_t} f_x(v) = \mathrm{covol}(\Lambda_t)^{-1} \sum_{\xi \in \Lambda_t^*} \widehat{f_x}(\xi).$$

Substitute $\mathrm{covol}(\Lambda_t) = 1$ and $\Lambda_t^* = \Lambda_{-t}$. The proof retains the constant mode on both sides. □

The power x^{-12} comes from rank twenty-four. Inversion of the parameter follows from the dual lattice; it is not added as a symmetry independent of the reading.

151.2 Meromorphic continuation and functional equation

For $\Re s > 12$, let

$$\mathcal{E}_t(s) = \sum_{v \in \Lambda \setminus \{0\}} \|G_t v\|^{-2s}, \qquad \mathcal{L}_t(s) = \pi^{-s} \Gamma(s) \mathcal{E}_t(s). \tag{151.3}$$

The lattice-point count above proves absolute convergence in that half-plane. Applying the gamma integral term by term yields

$$\mathcal{L}_t(s) = \int_0^\infty (\Theta_t(x) - 1) x^{s-1}\, dx.$$

Theorem 151.2 (Polar decomposition and bilateral reciprocity)**.** *The function $\mathcal{L}_t$ extends meromorphically through*

$$\mathcal{L}_t(s) = \frac{1}{s - 12} - \frac{1}{s} + \int_1^\infty \Big[(\Theta_t(x) - 1) x^{s-1} + (\Theta_{-t}(x) - 1) x^{11 - s}\Big]\, dx. \tag{151.4}$$

Its only poles are simple, at $s = 0, 12$, with residues $-1, +1$, respectively. It satisfies

$$\mathcal{L}_t(s) = \mathcal{L}_{-t}(12 - s), \qquad \Xi_t(s) := s(s-12)\mathcal{L}_t(s) = \Xi_{-t}(12 - s), \tag{151.5}$$

and Ξ_t is entire.

Proof. Split the integral at $x = 1$. For $0 < x < 1$, theta reciprocity gives

$$\Theta_t(x) - 1 = x^{-12} - 1 + x^{-12}(\Theta_{-t}(1/x) - 1).$$

The first two terms integrate to $1/(s-12) - 1/s$. In the third, the change of variable $y = 1/x$ gives $\int_1^\infty (\Theta_{-t}(y) - 1)y^{11-s}\,dy$. The positive minimum length of the lattice ensures exponential decay of $\Theta_t(x) - 1$ as $x \to \infty$. On every compact set of s, that decay dominates every power of x and $\log x$; the integrals and their complex derivatives therefore converge uniformly and define entire functions. The residues are read from the rational part. Simultaneously replacing (t, s) by $(-t, 12 - s)$ exchanges the integrals and leaves the rational part invariant. Multiplication by $s(s-12)$ removes both poles. □

The center $s = 6$ results from the rank and exponent $-2s$ adopted in (151.3). The family is the Epstein completion of the deformed lattices. Its spectral variable and normalization remain distinct from those of the Riemann zeta function; the proved relation between the two constructions is the reciprocity of their circular and lattice realizations.

Corollary 151.3 (Polar conservation and antisymmetric component)**.** *The functions*

$$\mathcal{L}_t(s) - \mathcal{L}_0(s), \qquad \mathcal{A}_t(s) = \frac{\mathcal{L}_t(s) - \mathcal{L}_{-t}(s)}{2}$$

are entire. Moreover,

$$\mathcal{A}_t(12 - s) = -\mathcal{A}_t(s), \qquad \mathcal{A}_t(6) = 0.$$

Proof. The rational part of (151.4) is independent of t, so it cancels in both differences. The functional equation changes the sign of the second. Evaluation at its fixed point gives $2\mathcal{A}_t(6) = 0$. □

The deformation therefore resides in an entire difference and preserves the polar data exactly. The central zero belongs to the bilateral difference, without imposing a zero on each $\mathcal{L}_t$. The poles described are singularities of a Mellin transform; the physical connection with the mass operator is established later through the action section, with its own variable and reader.

151.3 Stable reconstruction from the bilateral archive

For the unitary cyclic shift S, the preceding recovery law provides

$$T_j = \frac{8I + S^j}{9}, \qquad D_j = \frac{\sqrt{8}(S^j - I)}{9}, \quad j = 3, 4,$$

$$m = (D_3K, D_4T_3K) = MK, \qquad LM = P_{11}, \qquad \|L\| = \frac{9}{4}. \tag{151.6}$$

Consequently,

$$u = P_3Lm. \tag{151.7}$$

With the incidence chart, marked lattice and regional ratio ρ fixed, this equality successively recovers $u/\|u\|$, g_t, G_t, Λ_t, Θ_t and $\mathcal{E}_t$. Reconstruction of this family requires neither the terminal T_4T_3K nor the uniform charge of the register. The geometric antecedents remain included among the fixed data.

Theorem 151.4 (Geometric and spectral stability of recovery)**.** *Suppose $m' = m + e$, $\|e\| \le \varepsilon$ and $9\varepsilon/4 < \|u\|$. Let $u' = P_3 L m'$, and reconstruct the flows with $u'/\|u'\|$, without varying ρ, P_3, L or the charts. If $a = |t \log \rho|$, then*

$$\|g_t' - g_t\|,\ \|G_t' - G_t\| \le \frac{9}{2}\, a e^a \frac{\varepsilon}{\|u\|}. \tag{151.8}$$

With $H_K = \operatorname{diag}(h_i)$, $b = |t|\|H_K' - H_K\|$, we have

$$b \le \frac{9a\varepsilon}{2\|u\|},$$

$$\Theta_t(e^{2b}x) \le \Theta_t'(x) \le \Theta_t(e^{-2b}x), \tag{151.9}$$

$$e^{-2\sigma b}\mathcal{E}_t(\sigma) \le \mathcal{E}_t'(\sigma) \le e^{2\sigma b}\mathcal{E}_t(\sigma), \qquad \sigma > 12. \tag{151.10}$$

Proof. The norm of the recovery operator gives $\|u' - u\| \le 9\varepsilon/4$, ensuring $u' \neq 0$. The triangle inequality and the reverse triangle inequality for norms imply

$$\left\| \frac{u'}{\|u'\|} - \frac{u}{\|u\|} \right\| \le \frac{2\|u' - u\|}{\|u\|}.$$

Every exponent lies between $-a$ and a. Applying the mean value theorem to the exponential and taking the diagonal operator norm proves (151.8); repeating each entry twice does not change that norm. The same estimate before exponentiation gives the bound on b.

Since the two generators are diagonal in the same chart,

$$e^{-2b}\|G_t v\|^2 \le \|G_t' v\|^2 \le e^{2b}\|G_t v\|^2.$$

Applying the decreasing exponential and summing proves (151.9). Raising to $-\sigma$ and summing within the domain of convergence proves (151.10). The bounds apply to all vectors, not to a finite truncation of the series. □

The error ε is measured in the normalized memories of (151.6). If the rational registers $z = m/\sqrt{8}$ are retained, their error is multiplied by $\sqrt{8}$ before applying the estimate. Independent errors in the regional ratio or in the charts require their own bounds. The result recovers a family of geometries and analytic readings, with quantitative control of the archive determining it.

Chapter 152

Hyperbolic geometry of the angular pair and recovery of action

152.1 Radion, action ellipse and marked charts

The oriented section of phase, action and memory called *radion* jointly preserves the angular increment $\theta = 1$, signed action $\sigma\hbar_{\mathrm{ret}}$, membership in the turn $2\pi_{\mathrm{HMT}}$, sheet, nonadic return and refinement cylinder. Its unity splice is expressed in the continuum chart by $1 = S_E - \Phi_T + \varepsilon_1$. The projection retaining only $\theta = 1$ produces the ordinary radian; the enriched section also preserves the register needed to transport and invert its other readings. In this section, *radius* will denote a positive shape coordinate, distinct from that enriched object.

To make the splice terms explicit, the dodecaphasic wheel has centers $30m - 10$ in its degree chart. The second phase contributes 50, and the publication $A = 1000\alpha$ determines

$$S_E = \frac{2\pi}{360}(50 + 1000\alpha).$$

The active sector of the tritic selector comprises $H_+ = \{1, 4, 7\}$ and $H_- = \{2, 5, 8\}$. The fifteen unordered pairs, read at the six active positions, produce $N_6 = 6\binom{6}{2} = 90$; their extension to eight positions produces $N_8 = 8\binom{6}{2} = 120$. The two sheets of the active sector, normalized by the ambient fibre of 729 elements, determine

$$\rho_{\mathrm{tw}} = \frac{2N_6}{729} = \frac{20}{81}, \qquad \Phi_T = \frac{20}{81}\Delta_4, \qquad S_T = 1 + \Phi_T, \qquad \varepsilon_1 = S_T - S_E,$$

where Δ_4 is the torsional coordinate in (153.1). The subtraction retains the preceding sections and directly proves $S_E - \Phi_T + \varepsilon_1 = S_E - \Phi_T + 1 + \Phi_T - S_E = 1$. The coefficient comes from the census and the two sheets, not from the value of the final difference.

The angular publications of the preceding sections satisfy, in the chart considered,

$$A > 0, \qquad |C^*| < A, \qquad \xi = \frac{C^*}{A}, \qquad \eta_{\mathrm{el}} = \operatorname{artanh}\xi.$$

The notation η_{el} distinguishes elliptic anisotropy from the return coefficient η_{ret}. For a positive action section $\hbar$, the semiaxes are

$$a_S = \sqrt{2\hbar}\, e^{\eta_{\mathrm{el}}/2}, \qquad b_S = \sqrt{2\hbar}\, e^{-\eta_{\mathrm{el}}/2}, \qquad a_S b_S = 2\hbar. \tag{152.1}$$

The product fixes the action per revolution, whereas the ratio fixes the shape. Define

$$R_\star = \sqrt{b_S/a_S} = e^{-\eta_{\mathrm{el}}/2}, \qquad \tau_{\mathrm{el}} = iR_\star^2.$$

Theorem 152.1 (Compatibility of the radial, Poincaré and Klein charts)**.** *Let*

$$w = \frac{\tau_{\mathrm{el}} - i}{\tau_{\mathrm{el}} + i}, \qquad x_K = \frac{2w}{1 + w^2}.$$

On the real diameter of the Poincare disk and its image in the Klein disk,

$$w = \frac{R_\star^2 - 1}{R_\star^2 + 1} = -\tanh\frac{\eta_{\mathrm{el}}}{2}, \tag{152.2}$$

$$x_K = \frac{R_\star^4 - 1}{R_\star^4 + 1} = -\tanh\eta_{\mathrm{el}} = -\frac{C^*}{A}. \tag{152.3}$$

All these transformations are invertible on their domains; in particular, $R_\star^2 = (1+w)/(1-w)$.

Proof. Substitute $\tau = iR_\star^2$ into the fractional transformation. The simplified ratio has denominator $R_\star^2 + 1 > 0$ and $-1 < w < 1$. Substituting this ratio into $2w/(1+w^2)$ and multiplying numerator and denominator by $(R_\star^2+1)^2$ gives $(R_\star^4-1)/(R_\star^4+1)$. The identity $R_\star^4 = e^{-2\eta_{\mathrm{el}}}$ yields the hyperbolic tangent. Finally, $\tanh\eta_{\mathrm{el}} = \xi$. The recovery maps follow by solving the ratios. □

The sign of (152.3) corresponds to the orientation chosen for the diameter. The opposite orientation changes both signs. It is a hyperbolic coordinate of the marked rectangular chart: it preserves the marking and is not confused with a modular form or with a coordinate of the modular quotient devoid of orientation.

Exchanging the semiaxes acts by

$$(R_\star, \eta_{\mathrm{el}}, \tau_{\mathrm{el}}, w, x_K) \longmapsto (R_\star^{-1}, -\eta_{\mathrm{el}}, -1/\tau_{\mathrm{el}}, -w, -x_K). \tag{152.4}$$

On this diameter, the hyperbolic metrics of curvature -1 satisfy

$$\frac{|d\tau|^2}{(\Im\tau)^2} = \frac{4\,dw^2}{(1-w^2)^2} = \frac{dx_K^2}{(1-x_K^2)^2} = d\eta_{\mathrm{el}}^2 = 4\frac{dR_\star^2}{R_\star^2}. \tag{152.5}$$

These identities are verified by differentiating (152.2)–(152.3). In particular, $d(0,w) = |\eta_{\mathrm{el}}|$ and $d(w,-w) = 2|\eta_{\mathrm{el}}|$. The transformation $w \mapsto -w$ is a rotation by π on the entire disk and reflection about the centre on the generated diameter.

The normalization $(Q/a_S, P/b_S)$ of the ellipse's interior produces another realization of the Klein disk, with position coordinates. Its metric is

$$ds_K^2 = \frac{(1-r^2)(du^2+dv^2) + (u\,du + v\,dv)^2}{(1-r^2)^2}, \qquad r^2 = u^2+v^2 < 1.$$

This physical chart and the marked shape chart retain different domains. On the focal branch $\eta_{\mathrm{el}} \geq 0$, the eccentricity $e_f = \sqrt{1-e^{-2\eta_{\mathrm{el}}}}$ gives depth $u_f = \operatorname{artanh} e_f$, whence $\eta_{\mathrm{el}} = \log\cosh u_f$. Focal depth is not identified with anisotropy.

Shape chart: $w \mapsto -w$ Spectral chart: radial inversion

Figure 152.1: Two typed reciprocities. The first remains inside the disk and transports shape orientation; the second exchanges the interior and exterior of the unit circle. The plotted positions are schematic.

152.2 Structural substitution of the angular coordinates

The already constructed dodecaphasic closure publishes the same coordinate α through

$$\alpha = \pi_{\mathrm{HMT}} + e_{\mathrm{HMT}} - \varphi_{\mathrm{HMT}} - 4 - \kappa_{\mathrm{ag}}, \qquad \kappa_{\mathrm{ag}} = \frac{K_{\mathrm{int}}}{1000^{12} - 1}.$$

The compatible base, phase origin and carry boundary are retained. The complete analytic recurrence of the article represents that correlated coordinate; its publication is not replaced by a finite jet. Abbreviating the HMT subscripts, the action readers of (14.64)–(14.67) are

$$\begin{aligned} H_5 &= \frac{1}{2}\sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \\ D_A &= e^{-100\pi\alpha/9}, \qquad C_\pi = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \\ \eta_{\mathrm{ret}} &= H_5 - \frac{90}{\pi}C_\pi, \qquad A = 1000\alpha, \qquad C^* = 2(\eta_{\mathrm{ret}} + \alpha). \end{aligned} \tag{152.6}$$

The rational continuation retains every term of the regional recurrence. Its convergence follows without truncation: for $\alpha > 0$,

$$0 < \alpha e^{-100\pi\alpha/9} \le \frac{9}{100\pi e} < 1,$$

since xe^{-cx} attains its maximum $1/(ce)$ at $x = 1/c$.

Proposition 152.2 (Structural cancellation in the angular contrast)**.** *The generated ratio $\xi = C^*/A$ satisfies exactly*

$$\begin{aligned} \xi = {} & \frac{1}{\sqrt{1000\varphi\alpha}} + \frac{9\alpha}{8000} - \frac{\alpha^2}{900} + \frac{7\alpha^3}{24000} - \frac{\alpha^4}{27000} \\ & - \frac{9}{50\pi}\left(169\alpha^5 + \frac{D_A\alpha^6}{1 - D_A\alpha}\right). \end{aligned} \tag{152.7}$$

Proof. Start from $\xi = (\eta_{\mathrm{ret}} + \alpha)/(500\alpha)$. The term $-\alpha$ in H_5 cancels against the $+\alpha$ of the counter-angle. Dividing each remaining term by 500α yields the expression: the square root gives $1/\sqrt{1000\varphi\alpha}$, and $90/500 = 9/50$. The regional continuation retains its complete denominator. □

Therefore, $x_K = -\xi$ and $R_\star = ((1-\xi)/(1+\xi))^{1/4}$ are explicit expressions of the regional publications and the register. Substitution of the closure of α completes the dependence on $\pi, \varphi, e, \kappa_{\mathrm{ag}}$, without promoting metrological recognitions to generators.

152.3 Recovery of the Planck coordinate and preservation of orientation

Theorem 152.3 (Radial recovery map for the return section)**.** *In the positive chart, if $q = R_\star^4$, then*

$$\eta_{\mathrm{ret}} = \alpha\frac{499 - 501q}{1 + q}, \tag{152.8}$$

$$\hbar_{\mathrm{ret}} = \alpha\frac{499 - 501q}{1 + q}\, 10^{-34}\mathcal{U}_S, \qquad \hbar_{\mathrm{pre}} = H_5\, 10^{-34}\mathcal{U}_S, \tag{152.9}$$

$$R_{\mathrm{act}} = \frac{\hbar_{\mathrm{pre}}}{\hbar_{\mathrm{ret}}} = \frac{H_5(1+q)}{\alpha(499 - 501q)}. \tag{152.10}$$

The branch $\eta_{\mathrm{ret}} > 0$ is equivalent to $q < 499/501$, within $q > 0$.

Proof. The angular identities give $\eta_{\mathrm{ret}} = \alpha(500\xi - 1)$. Substitute $\xi = (1-q)/(1+q)$; the numerator is $499 - 501q$. Both the denominator and α are positive. The dimensional realization and the ratio are those of the action sections already constructed. The decade -34 and the unit $\mathcal{U}_S$ retain their preceding determinantal derivation. □

This formula makes the Planck section within the counter-angle explicit. In terms of the action quantities, the angular relation is

$$C^* = 2\left(\frac{\hbar_{\mathrm{ret}}}{10^{-34}\mathcal{U}_S} + \alpha\right).$$

Division by the dimensional basis extracts the dimensionless coefficient; it does not add an action quantity to a dimensionless constant.

For the angular orientation $\varepsilon \in \{-1, +1\}$, define

$$C^*_\varepsilon = 2\varepsilon(\eta_{\mathrm{ret}} + \alpha), \quad \xi_\varepsilon = C^*_\varepsilon/A, \quad q_\varepsilon = \frac{1-\xi_\varepsilon}{1+\xi_\varepsilon} > 0.$$

The recovery map on both charts is

$$\eta_{\mathrm{ret}} = \alpha\left(500\varepsilon\frac{1-q_\varepsilon}{1+q_\varepsilon} - 1\right). \tag{152.11}$$

Corollary 152.4 (Invariance of the conjugate recovery map)**.** *The transformation* $(\varepsilon, q) \mapsto (-\varepsilon, q^{-1})$ *preserves* η_{ret}, $\hbar_{\mathrm{ret}}$, *and* R_{act}.

Proof. The ratio $(1-q)/(1+q)$ changes sign when q is inverted. The product by ε remains invariant. We apply (152.11) and then the action publications. □

If orientation is discarded and $\varepsilon = +1$ is incorrectly held fixed, the same expression gives $\eta_{\mathrm{ret}}(q^{-1}) = -\eta_{\mathrm{ret}}(q) - 2\alpha$. The register therefore distinguishes two algebraically different operations. Nor is the angular orientation ε automatically identified with the symplectic sign σ: exchanging the axes is antisymplectic, whereas the rotation that exchanges them preserves the area form.

152.4 Intertwining phase and anisotropy

Let

$$S_\eta = \mathrm{diag}(e^{\eta/2}, e^{-\eta/2}), \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J_\eta = S_\eta J_0 S_\eta^{-1} = \begin{pmatrix} 0 & -e^{\eta} \\ e^{-\eta} & 0 \end{pmatrix}.$$

Then $\det S_\eta = 1$, $J_\eta^2 = -I$, and $C_\eta(\theta) = e^{\theta J_\eta} = S_\eta e^{\theta J_0} S_\eta^{-1}$. The law $S_{\eta+\delta} = S_\delta S_\eta$ proves the exact intertwining identity

$$C_{\eta+\delta}(\theta) S_\delta = S_\delta C_\eta(\theta). \tag{152.12}$$

The transformation preserves $dQ \wedge dP$ and carries the phase evolution to the new ellipse. With the normalization of (152.1), the swept action is $\hbar\theta$, or $\sigma\hbar\theta$ on the oriented sheet. The phase θ, the anisotropy η_{el}, and the parameter t of the flow of K thus retain distinct functions, related by explicit operators.

152.5 Radial geodesic and differential realization of the direction of K

Each radius of (150.1) determines a rectangular chart and a hyperbolic coordinate:

$$\tau_i(t) = i r_i(t)^2, \qquad w_i(t) = \frac{r_i(t)^2 - 1}{r_i(t)^2 + 1} = \tanh(t h_i).$$

Theorem 152.5 (Geodesic of compensated radii)**.** *In the product of twelve hyperbolic diameters,*

$$d(w(s), w(t)) = 2|t - s|\,|\log \rho|, \qquad \sum_i \operatorname{artanh} w_i(t) = 0. \tag{152.13}$$

Composition of advances is given by

$$w_i(t + s) = \frac{w_i(t) + w_i(s)}{1 + w_i(t) w_i(s)}.$$

Proof. In coordinates $a_i = \log r_i$, the product metric is $4 \sum_i da_i^2$. The curve $a(t) = t h$ is a straight line, with speed $2\|h\| = 2|\log \rho|$, and $\sum_i a_i = 0$. Each diameter is totally geodesic. Distance in the product is the square root of the sum of squared distances in its factors, which are $2|t - s||h_i|$. The vanishing-sum identity and the composition law follow from $\operatorname{artanh} w_i = t h_i$ and the addition formula for the hyperbolic tangent. □

The regional ratio determines the speed of this realization; K determines its direction. If $\rho = 1$, the trajectory is constant. For $\rho \neq 1$, the space of radii with product one has dimension eleven, with tangent space $V_{11} = \mathbf{1}^{\perp}$, and the normal to the trajectory within it is

$$V_{10}(K) = V_{11} \cap u^{\perp}, \qquad P_{10}(K) = P_{11} - \frac{u u^{\mathsf{T}}}{\|u\|^2}.$$

The projector of Proposition 21.64 therefore acquires a differential realization: it removes the uniform scale and distinguishes the generated direction from its ten transverse variations. Its domain is this radial space, not the real twenty-four-dimensional space of all the disks.

With $D_t = \operatorname{diag}(\operatorname{sech}^2(t h_i))$, its expression in coordinates w is $D_t P_{10} D_t^{-1}$. Indeed, D_t is the differential of $a \mapsto \tanh a$, and transforms the metric $4I$ into $4 \operatorname{diag}((1 - w_i^2)^{-2})$. The transported projection is orthogonal for that metric.

152.6 Lattice lift and rectangular lift

The radii admit two realizations that should be retained explicitly. The operator $G_t = \bigoplus_i r_i I_2$ scales both axes of each A_2 plane simultaneously. It preserves the total covolume and commutes with the ternary isometry. The homothety of each plane preserves its internal complex modulus.

The chart $\tau_i = i r_i^2$ has another realization, of fixed area:

$$T_i = \operatorname{diag}(r_i^{-1}, r_i), \qquad (\omega_1, \omega_2) = (r_i^{-1}, i r_i), \qquad \omega_2/\omega_1 = i r_i^2.$$

This is the elliptic lift with $\eta = -2 \log r_i$. This convention transforms physical periods. The fractional linear action of the same matrix on i would give i/r_i^2; to obtain $i r_i^2$ under that other convention, one uses T_i^{-1}. Distinguishing the actions prevents an inadvertent inversion.

Both realizations are reconstructed from the same radii, but the ternary commutation of the homothety does not transfer to the anisotropic deformation. The state and its readers allow the two geometries to be compared without identifying their operators.

Chapter 153

Electron coordinate, operator scale, and reversibility of readings

153.1 Algebraic structure in the complete electronic expression

The expression in Section 16.2 is now composed with the radial recovery map, retaining its factors and dimensional realization. The electronic prefactor comes from the conditional projectors of addition and multiplication on $\ell^2(\{1,\dots,9\}^3)$. On the corresponding principal plane,

$$P=\begin{pmatrix}1&0\\0&0\end{pmatrix},\qquad Q=\begin{pmatrix}1/4&\sqrt{3}/4\\ \sqrt{3}/4&3/4\end{pmatrix},\qquad \|[P,Q]\|=\frac{\sqrt{3}}{4}.$$

The equality follows because $[P,Q]=(\sqrt{3}/4)\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$. Moreover,

$$S=2P-I,\qquad J=\frac{4}{\sqrt{3}}[P,Q],\qquad S^2=I,\quad J^2=-I,\quad SJ=-JS.$$

Hence $n_\pm=(S\pm J)/2$ satisfy $n_\pm^2=0$. The block retains hyperbolic, elliptic, and nilpotent generators within the algebraic realization already derived. The coefficient $\sqrt{3}/4$ and the normalization $4/\sqrt{3}$ serve distinct functions.

Using the previously constructed coordinates and regional integers, define

$$\begin{aligned}\Delta_4&=\frac{\pi-e\log\pi}{270}\left(1+\frac{\pi}{729}\right),\\ B_e&=\frac{\sqrt{3}}{4}\left[\exp\!\left(\frac{\varphi}{\pi^2}\right)-22\alpha^3\right](1+15\Delta_4),\qquad \kappa_e=80-54\alpha+6\Delta_4.\end{aligned}\tag{153.1}$$

Thus $\varepsilon_e^{\beta}=B_eR_{\mathrm{act}}^{\kappa_e}$. The pre-return action section and its return determine

$$\delta_{\mathrm{ret}}=H_5-\eta_{\mathrm{ret}}=\frac{90}{\pi}C_\pi,\qquad R_{\mathrm{act}}=1+\frac{\delta_{\mathrm{ret}}}{\eta_{\mathrm{ret}}}.$$

Corollary 153.1 (Radial realization of the electronic coordinate)**.** *On the branch $\alpha>0$, $\eta_{\mathrm{ret}}>0$, $|\xi|<1$,*

$$\begin{aligned}\varepsilon_e^{\beta}=\frac{\sqrt{3}}{4}&\left[\exp\!\left(\frac{\varphi}{\pi^2}\right)-22\alpha^3\right](1+15\Delta_4)\\ &\times\left[1+\frac{90C_\pi}{\pi\alpha}\,\frac{1+R_\star^4}{499-501R_\star^4}\right]^{80-54\alpha+6\Delta_4}.\end{aligned}\tag{153.2}$$

In the Klein coordinate of the marked chart,

$$R_{\mathrm{act}}=1+\frac{\delta_{\mathrm{ret}}}{\alpha(-500x_K-1)}.\tag{153.3}$$

The corresponding domains are

$$\frac{1}{500}<\xi<1,\qquad 0<R_\star^4<\frac{499}{501},\qquad -1<x_K<-\frac{1}{500}.$$

Proof. We substitute (152.8) into $R_{\mathrm{act}}=1+\delta_{\mathrm{ret}}/\eta_{\mathrm{ret}}$ and then into the complete electronic expression (16.7). The Klein chart gives $\xi=-x_K$. The intervals are obtained from $\eta_{\mathrm{ret}}=\alpha(500\xi-1)>0$ and the already proved monotone correspondences. □

The substitution preserves the regional deficit, torsional normalization, return character, and incompatibility norm. It expresses the same coordinate through a recoverable geometric reader. The radius and α remain correlated by their generation; they are not independent parameters fitted to a mass. The dimensional realization remains

$$E_e^{\beta} = \varepsilon_e^{\beta} \mathcal{U}_E, \qquad m_e^{\beta} = E_e^{\beta}/c_{\mathrm{int}}^2.$$

The action decade already incorporated into the Planck sections is not multiplied again onto a coordinate expressed in the energy chart.

A second exact form, useful for isolating the correction, is

$$\widehat{\delta}_{\mathrm{ret}} = \frac{90 C_\pi}{\pi H_5}, \qquad R_{\mathrm{act}} = (1 - \widehat{\delta}_{\mathrm{ret}})^{-1}, \qquad \varepsilon_e^{\beta} = B_e (1 - \widehat{\delta}_{\mathrm{ret}})^{-\kappa_e}.$$

Positivity of both sections ensures $1 - \widehat{\delta}_{\mathrm{ret}} > 0$. The three expressions are identities of the same return, not independent predictions.

153.2 Separation of scale and shape in a positive mass fibre

The operator law of return receives a finite positive fibre of routes together with its readers. To formulate its matrix consequence independently of species names, let $\mathcal{H}$ have dimension $d \geq 1$, let $B = B^*$ be the exponent operator, and let M^0 be positive definite. The return is the congruence

$$M^{\beta} = R_{\mathrm{act}}^{B/2} M^0 R_{\mathrm{act}}^{B/2}. \tag{153.4}$$

In the route basis of the diagonal reader, $Be_\gamma = \kappa_\gamma e_\gamma$ and $M^0 e_\gamma = m_e^{(0)} \chi_{\mathrm{ref}}(\sigma_\gamma - \sigma_e, \eta_\gamma - \eta_e) e_\gamma$. The signatures and character are those of the route construction; the following theorem applies to each positive fibre without replacing its selector.

Theorem 153.2 (Uniform and unimodular decomposition of return)**.** *If $R_{\mathrm{act}} > 0$, define*

$$\bar{\kappa} = \frac{\operatorname{tr} B}{d}, \qquad B_0 = B - \bar{\kappa} I, \qquad S_0 = R_{\mathrm{act}}^{B_0/2}.$$

Then

$$M^{\beta} = R_{\mathrm{act}}^{\bar{\kappa}} S_0 M^0 S_0, \qquad \det S_0 = 1, \tag{153.5}$$

$$\det M^{\beta} = R_{\mathrm{act}}^{\operatorname{tr} B} \det M^0. \tag{153.6}$$

With $\widehat{M} = M/(\det M)^{1/d}$,

$$\widehat{M}^{\beta} = S_0 \widehat{M}^0 S_0. \tag{153.7}$$

Proof. The scalar part of B commutes with B_0. Functional calculus gives $R_{\mathrm{act}}^{B/2} = R_{\mathrm{act}}^{\bar{\kappa}/2} S_0$. In an orthonormal eigenbasis of B,

$$\det S_0 = R_{\mathrm{act}}^{\frac{1}{2}\sum_\gamma (\kappa_\gamma - \bar{\kappa})} = 1.$$

Multiplication of determinants proves (153.6). Dividing (153.5) by the d-th root of that determinant cancels the scalar factor. The proof does not require B to commute with M^0; it does require positive definiteness for normalization by the determinant. □

In the joint diagonal basis of the source construction, the individual ratios are

$$q_\gamma = \frac{m_\gamma^{\beta}}{m_\gamma^{(0)}} R_{\mathrm{act}}^{-\bar{\kappa}} = R_{\mathrm{act}}^{\kappa_\gamma - \bar{\kappa}}, \qquad \prod_\gamma q_\gamma = 1.$$

The formal inversion $R_{\mathrm{act}} \mapsto R_{\mathrm{act}}^{-1}$ inverts each q_γ. It differs from oriented radial reciprocity, which preserves R_{act} by (152.11). The congruence preserves the normalized determinant, not the individual eigenvalues or the sum of masses. The null sector lies outside this positive normalization. For $d = 1$, $B_0 = 0$: the electron's return lies entirely in the uniform factor and remains in (153.2).

Substitution of (153.3) also acts on (153.4) through functional calculus. If

$$q(\xi) = 1 + \frac{\delta_{\mathrm{ret}}}{\alpha(500\xi - 1)} > 0,$$

then

$$M^\beta = e^{\frac{1}{2}(\log q(\xi))B}\, M^0\, e^{\frac{1}{2}(\log q(\xi))B}.$$

The genealogy of each exponent and each multisection is preserved. The structure shared with the flow of K consists of removing a uniform component and then applying a deformation of determinant one; B_0 is not identified with $\mathrm{diag}(u)$, nor R_{act} with ρ.

153.3 Circular compression and identification of transport

Theorem 153.3 (Bilateral recovery of a unitary operator)**.** *Let C be unitary on a Hilbert space, and let*

$$T = \frac{8I + C}{9}, \qquad D = \frac{\sqrt{8}(C - I)}{9}.$$

Then

$$T^*T + D^*D = I, \qquad v = Tv - \frac{Dv}{\sqrt{8}}, \qquad Cv = Tv + \sqrt{8}Dv. \tag{153.8}$$

Proof. Expansion gives

$$81T^*T = 65I + 8(C + C^*), \qquad 81D^*D = 16I - 8(C + C^*).$$

The sum is $81I$. The other identities follow by taking linear combinations of the two definitions. At a circular eigenvalue $|z| = 1$, the balance is $1 - |(8 + z)/9|^2 = (8/81)|1 - z|^2$. □

The archived pair recovers both the state and its image. For the invertible frame $O_K = (K, SK, \dots, S^{11}K)$ already constructed, the twelve responses form

$$Y = (CK, CSK, \dots, CS^{11}K), \qquad C = YO_K^{-1}.$$

If $CS = SC$, a single vector response $y = CK$ determines all the columns: $Y = (y, Sy, \dots, S^{11}y)$. The general case requires twelve vector trials. Recovery does not reduce that information to a single scalar.

The real diagonal generator $H_K = \mathrm{diag}(h_i)$ admits the unitary chart

$$C_K = (H_K + iI/2)(H_K - iI/2)^{-1}.$$

Each real eigenvalue is transformed into a number of modulus one. Moreover,

$$C_K - I = i(H_K - iI/2)^{-1}, \qquad H_K = \frac{i}{2}(C_K + I)(C_K - I)^{-1}.$$

Both inverses exist because H_K is self-adjoint and its spectrum is real. The archive of (153.8) therefore recovers this finite realization of the direction of K. Its dimension twelve and its domain remain distinct from those of a spectral operator for zeta.

153.4 Hierarchy of invariants and scope of the proofs

The completion $1000 = 729 + 271$, with normalized reading $1 = 729/1000 + 271/1000$, retains its generative function. The subsequent conservation laws are organized according to the object transported:

Realization	Invariant	Recovery or reciprocity
Archivo bilateral	Joint norm of reading and memory	State and image under the operator
Action ellipse	Product $a_S b_S = 2\hbar$	Radius and angular orientation
Flow of K	Product of twelve radii	Inversion $t \mapsto -t$
Deformed marked lattice	Covolume and ternary isometry	$\Lambda_t^* = \Lambda_{-t}$
Theta–Mellin reading	Poles and residues	Entire difference and functional equation
Positive mass fibre	Determinant after removing the common scale	Congruence by S_0

Conservation of volume does not imply conservation of each vector's norm. Polar conservation permits variation in the entire part. Recovery of action requires orientation to be retained. These distinctions express the operational content of evolutionary conservation: each transformation retains certain data, changes others, and preserves the register that allows the correspondence to be reconstructed.

The composition extends the function of K already proved in the article. The register participates in the reading of α; its direction is recovered from memory; that direction determines a radial geodesic and a reciprocal lattice deformation; the lattice family has a stable spectral reading. The action section provides the link to the electronic coordinate and to the positive congruence of the mass operator. This yields a chain of maps with explicit domains, inverses and invariants.

The supplement distinguishes written proofs from formalized ones. `FlujoReciproco.lean` contains ten theorems on diagonal flow, centering, determinants, inversion and lifting to twenty-four dimensions. `RadioKlein.lean` contains eleven rational theorems on charts, bounds, reciprocities and oriented recovery. The hyperbolic proofs, complete substitutions, separation of the mass operator and theta–Mellin argument appear in the text; their formalization is not attributed to compilation of those two files. Documentary receipts record provenance and reproduction, with a scope distinct from that of a mathematical proof.

Chapter 154

Carry and incidence registers and projective reconstruction of histories

The APP–TRIT–TPK construction fixes an enriched state before its numerical, incidence and operator realizations. The registers entering that construction are distinguished by the operation they allow to be inverted and by the relations they preserve. This section brings these distinctions together and composes recovery with refinement.

154.1 Carry, history, incidence and archive

The carry in Sections 146.4 and 146.5 is an integer arithmetic datum. Preserving it, together with depth and the final state, restores the antecedent through the explicit inverse (146.12). The transport history contains the ordered sequence of operations, sheets, orientations and returns. Within it, two stages with the same phase may have different accumulated registers. Holonomy acts on that history and on the frame in which its readings are performed.

The incidence register preserves the boundary relations used by the exceptional constructions. The ordered object K and its reading κ_{ag} enter with the normalizations and maps of Section 144.4. Sufficiency for recovering an operator is expressed there through the frame responses and the corresponding inversion. The archive memory of Section 147.6, in turn, is a sequence of components of a Hilbert space. The telescoping identity (147.26) preserves the norm between the remaining state and the archived components; the inverse (147.25) reconstructs the initial amplitude.

These four functions are composed while preserving their types. A carry integer, a transport word, an incidence and an archived amplitude constitute distinct data of the system. The recovery proofs specify which combination of them suffices for each reading. Figure 154.1 represents this functional hierarchy.

154.2 Refinement tree and local weights

Let H_k be the finite set of admissible prefixes of depth k, with truncations $p_{k+1,k}$, of the history system already constructed. An emission is counted with its multiplicity, and $d(\zeta) > 0$ denotes the number of emissions from ζ. The uniform seed distribution and the local weight $1/d(\zeta)$ define μ_k. The sum of the weights of the extensions of a prefix equals its weight: the measures are compatible and determine the cylinder measure μ. The propositions on fiber memory and naturality of the measure are proved below using these explicit definitions.

154.3 Enriched readers and fiber memory

The difference between history and value admits an operational criterion. At a finite level H_k, let $q_k : H_k \to Z_k$ be a reader and let $M_k : H_k \to \mathcal{M}_k$ be the preserved register. The enriched reader is $\widehat{q}_k = (q_k, M_k)$. Memory distinguishes states in the same reader fibre; its sufficiency is checked on the domain considered. This formulation incorporates into the article itself the complete development of the fibre information used below.

Proposition 154.1 (Recoverability at a finite level)**.** *An inverse of $\widehat{q}_k$ exists on its image if and only if $\widehat{q}_k$ is injective. For a random variable Z taking values in H_k with distribution μ_k, we have*

$$\mathsf{H}(Z) = \mathsf{H}(q_k(Z)) + \mathsf{H}(Z \mid q_k(Z)).$$

If $\widehat{q}_k$ is injective on the support of Z, then

$$\mathsf{H}(Z \mid q_k(Z), M_k(Z)) = 0.$$

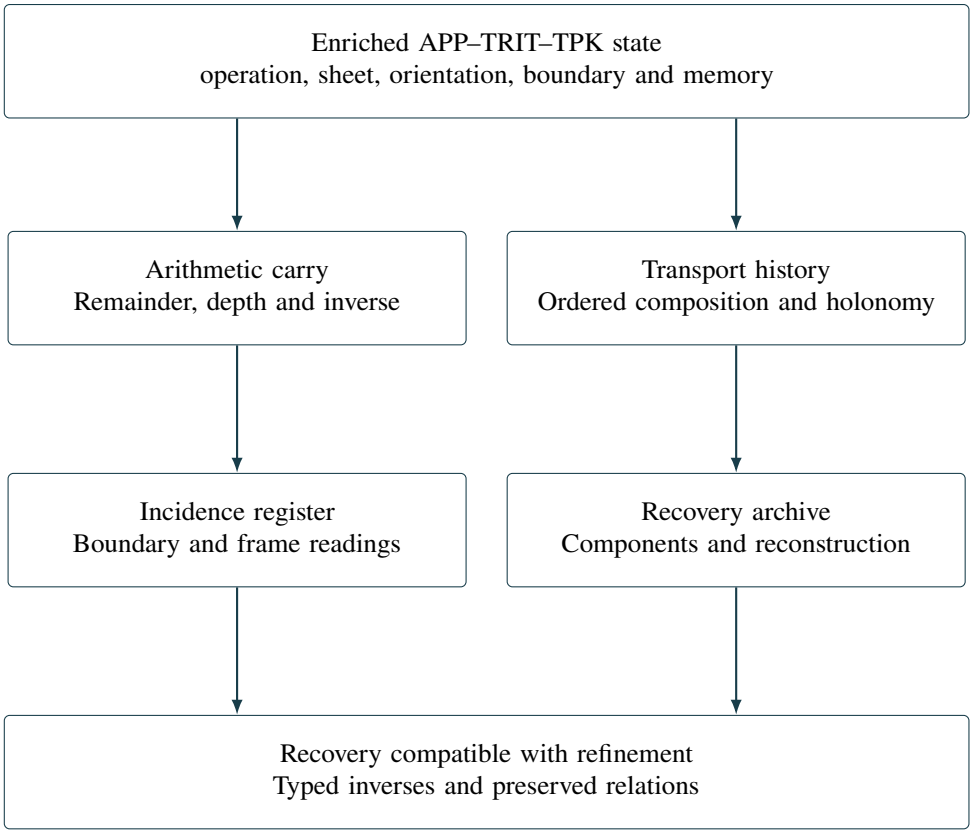


Figure 154.1: Organization of the registers by their role. The lower arrows indicate composition of recovery conditions, not identification of the four data types.

Here $\mathsf{H}(Z) = -\sum_z \Pr(Z = z) \log \Pr(Z = z)$, *with* $0 \log 0 = 0$, *is finite entropy in nats; conditional entropy is the average entropy of the conditional distributions.*

Proof. An inverse recovers a unique antecedent, which requires injectivity; conversely, injectivity allows each published pair to be assigned its unique antecedent. Since $q_k(Z)$ is a deterministic function of Z, grouping the sum defining $\mathsf{H}(Z)$ by fibres of q_k gives the chain rule. If the pair $(q_k(Z), M_k(Z))$ distinguishes the support, each corresponding conditional distribution is concentrated on a single state and its entropy is zero. □

Not publishing a coordinate and erasing the register that allows it to be recovered are distinct operations. This result does not identify entropy with the cardinality of the continuum or introduce a thermal interpretation: its domain, reader and distribution are those just fixed.

154.4 Equivariance of the measure under tree symmetries

Proposition 154.2 (Joint transport of the tree and its measure)**.** *Let* $g = (g_k)_{k \geq 0}$ *be a family of bijections* $g_k : H_k \to H_k$ *that commutes with truncations, fixes the root and bijectively transports the admissible emissions, preserving their multiplicities. Then*

$$d(g_k \zeta) = d(\zeta), \qquad (g_k)_* \mu_k = \mu_k, \qquad (g_\infty)_* \mu = \mu.$$

More generally, if μ_ζ *is the law of continuations of a history* $\zeta \in H_k$, *we have*

$$(g_\infty)_* \mu_\zeta = \mu_{g_k \zeta}.$$

Proof. The emission bijection preserves their number with multiplicities. The uniform seed distribution is also invariant. Each path and its image therefore receive the same product of local weights $1/d(\zeta)$. Equality of measures follows at each level and on each cylinder. Coherence with truncations defines g_∞, and uniqueness of the measure determined by cylinders proves equality in the limit. The same argument, starting at ζ, proves the assertion about continuations. □

This gives naturality of the tree and the measure: a coherent change of representation transports the weights and does not require choosing them anew. Preservation of emissions and multiplicities is an essential hypothesis of the proposition, not a datum added after the construction.

154.5 Recovery of histories and transport of membership

Proposition 154.3 (Compatibility of inverses and families)**.** *Let H_k be the preceding levels and $Y_k = \widehat{q}_k(H_k)$. Suppose that $\widehat{q}_k : H_k \to Y_k$ is bijective and that there exist truncations $s_{k+1,k}$ such that*

$$s_{k+1,k}\widehat{q}_{k+1} = \widehat{q}_k p_{k+1,k}.$$

Then the finite inverses induce a bijection

$$\widehat{q}_\infty : \varprojlim H_k \longrightarrow \varprojlim Y_k.$$

If $H = \varprojlim H_k$, $Y = \varprojlim Y_k$ and $\mathcal{T}(A) = \widehat{q}_\infty[A]$, the map $\mathcal{T} : \mathcal{P}(H) \to \mathcal{P}(Y)$ is bijective and preserves unions, intersections, relative complements and membership. Each restriction $A \to \mathcal{T}(A)$ is a bijection.

Proof. Composing the compatibility identity with the inverses gives

$$p_{k+1,k}\widehat{q}_{k+1}^{-1} = \widehat{q}_k^{-1} s_{k+1,k}.$$

Therefore, a compatible family $y = (y_k)$ determines the compatible family $x = (\widehat{q}_k^{-1}(y_k))$. The two coordinate maps are inverses. Direct image under a bijection has as its inverse direct image under the inverse bijection. For every family $(A_i)_{i\in I}$ of subsets of H,

$$\mathcal{T}\Big(\bigcup_i A_i\Big) = \bigcup_i \mathcal{T}(A_i),$$
$$\mathcal{T}\Big(\bigcap_i A_i\Big) = \bigcap_i \mathcal{T}(A_i),$$
$$\mathcal{T}(H \setminus A) = Y \setminus \mathcal{T}(A).$$

The first equality uses the definition of image. For the second and third, uniqueness of the antecedent allows the membership conditions to be transferred. Finally, $x \in A$ is equivalent to $\widehat{q}_\infty(x) \in \mathcal{T}(A)$, which also proves the assertion about restrictions. □

The route refinement theorem 148.3 provides a realization of this compatibility on arithmetic labels and their carries. The result distinguishes recovery of each state from preservation of operations on families. At the amplitude level, the unitary lift of 148.2 also transports the inner product. At the probabilistic level, Proposition 154.2 preserves the weights when emissions are transported with their multiplicities.

154.6 Connection with the genealogical closure of the continuum

Section 19.10 constructs a genealogical closure $\mathfrak{G}$ and establishes the cardinal identity $2^{\aleph_0} = \aleph_1$ over its internal power set. The closure, generator code and realized history form part of its definition; its set-theoretic representation is established after the genealogical construction through the operations and proofs included in that section.

The present articulation incorporates the proofs of memory, naturality and transport of families used by that relation to conservation. The proof chain is distinguished: inverses recover states; their compatibility recovers histories; the bijection transports membership operations; the genealogical closure of Section 19.10 fixes the domain of the cardinality statement. Thus, recovery and organization of families are integrated into the double projection with their respective objects, operations and conclusions.

Chapter 155

Census of residual-emitter signatures

This census specifies the finite fibers used in Section 4.3. Each initial condition is identified by

$$\nu = 4(9i + j) + d, \qquad 0 \le i, j \le 8, \quad d = 0, 1, 2, 3 \longleftrightarrow N, E, S, O.$$

Euclidean division recovers $d = [\nu]_4$, $j = [\lfloor \nu/4 \rfloor]_9$ and $i = \lfloor \nu/36 \rfloor$. Thus $0 \le \nu < 324$ enumerates each condition exactly once, and pairs of these identifiers enumerate the 104 976 conditions in Ω_{seed}.

Let $f_+(\nu)$ and $f_\times(\nu)$ be the six-component signatures obtained from the emission recurrence. At every active step the cell is read before the cursor advances; the direction vectors are

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

Coordinates are reduced modulo nine and the vector is reversed after steps 27 and 54. Zero-phase steps do not move either cursor. Together with (4.30), these rules determine f_+ and $f_\times$ without a target-value catalog.

The table gives their complete images. The multiplicity m_σ counts identifiers in the fiber $f_\sigma^{-1}(a)$; $\nu_{\min}$ is its least element. Each word retains its initial zeros. The complete fiber is determined by

$$f_\sigma^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_\sigma(\nu) = a\}.$$

f_+	m_+	$\nu_{\min}$	$f_\times$	$m_\times$	$\nu_{\min}$
122564	18	17	000000	108	8
122898	18	24	177393	8	1
212645	18	5	177555	8	54
212988	18	0	177771	8	11
221456	18	29	339555	8	6
221889	18	12	339771	8	15
456221	18	28	339933	8	59
465898	18	21	393177	8	0
546988	18	9	393393	8	45
564122	18	16	393555	8	40
645212	18	4	555177	8	52
654889	18	33	555339	8	4
889221	18	13	555393	8	41
889654	18	32	555555	24	84
898122	18	25	555717	8	14
898465	18	20	555771	8	18
988212	18	1	555933	8	24
988546	18	8	717555	8	12
			717717	8	21
			717933	8	19
			771177	8	9
			771339	8	13
			771555	8	16
			933339	8	57
			933555	8	26
			933717	8	17

Evaluating the recurrence for the 324 identifiers gives the two partitions in the table: $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. Their fiber product yields 432 fibers of size 144, 18 of size 432, and 18 of size

1944, by injectivity of (4.31). Finally, applying (4.32) to that product of images gives 243 distinct words. This finite census belongs to the residual emitter; subsequent coinductive refinement retains its own law.

Chapter 156

Nonadic holonomy, reciprocity and recovery of geometric readers

The nonadic connection, areal–volumetric incidence and radial reciprocity enter different realizations of the same enriched state. The connection preserves the oriented sequence of nine phases and extends memory on return. Incidence records transitions between two reading modes. Radial reciprocity exchanges the two orientations of a geometric form of fixed area. Their composition is expressed through three precise results: invariance of a mark quotient at every depth, factorization of a determinant-one deformation and reconstruction of anisotropy from a normalized length and its orientation.

The construction order remains

$$\begin{gathered}\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \text{enriched state},\\ \text{enriched state} \longrightarrow \text{joint discrete structure of the continuum},\\ \text{joint discrete structure of the continuum} \longrightarrow \text{readers composed here}.\end{gathered}$$

The coordinates π, φ, e, α and register K appearing in the formulas are outputs of the preceding regional generation. The tritic calendar already determines consecutive pairs of additive and multiplicative modes. The language of matrices, hyperbolic functions and exterior minors then makes explicit the realizations of that generated structure.

156.1 Modal incidence and regional marks on finite returns

156.1.1 Enumeration of returns between sheets

The temporal block $(+, -, 0)$ induces the modal sequence (Σ, Π, Π). Its consecutive pairs, including the return link, are (Σ, Π), (Π, Π) and (Π, Σ). The primitive incidence, in areal–volumetric order, is therefore

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \tag{156.1}$$

This matrix records the calendar pairs. The memory-one symbolic envelope it determines preserves that incidence; additional phase, orientation and register restrictions remain in the TPK state. The envelope count and survival of a complete enriched history are thus distinguished.

Let $f_0 = 0$, $f_1 = 1$ and $f_{n+1} = f_n + f_{n-1}$ for $n \geq 1$. These integers express here the counts produced by powers of the incidence. Conventional identification of this sequence remains subsequent to construction of the calendar.

Proposition 156.1 (Powers of incidence and diagonal returns)**.** *For every $n \geq 1$,*

$$F_{\mathrm{av}}^{n} = \begin{pmatrix} f_{n-1} & f_n \\ f_n & f_{n+1} \end{pmatrix}. \tag{156.2}$$

In particular, returns of length n to the areal and volumetric sheets have multiplicities f_{n-1} and f_{n+1}, respectively.

Proof. For $n = 1$, the formula is the definition of F_{av}, since $f_2 = 1$. Assuming validity for n, right multiplication gives

$$\begin{pmatrix} f_{n-1} & f_n \\ f_n & f_{n+1} \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} f_n & f_{n-1}+f_n \\ f_{n+1} & f_n + f_{n+1} \end{pmatrix} = \begin{pmatrix} f_n & f_{n+1} \\ f_{n+1} & f_{n+2} \end{pmatrix}.$$

Induction proves the identity. For an incidence matrix with entries zero or one, the multiplication rule sums sequences of compatible transitions between two positions. By induction on length, entry (i, j) of the power counts sequences of length n from i to j. The two diagonal entries count the stated returns. □

156.1.2 Regional marking of the return support

Let $e_{\mathrm{ar}} = (1, 0)^{\mathsf{T}}$ and $e_{\mathrm{vol}} = (0, 1)^{\mathsf{T}}$. For a positive coordinate c, produced before this evaluation, define the mark reader

$$D_c = c\, e_{\mathrm{ar}} e_{\mathrm{ar}}^{\mathsf{T}} + e_{\mathrm{vol}} e_{\mathrm{vol}}^{\mathsf{T}} = \mathrm{diag}(c, 1).$$

Diagonality preserves both sheets; the volumetric entry expresses its unit normalization. The areal mark c is applied once, after the return count. Its evaluation is

$$\Theta_{c,n} = \frac{\langle e_{\mathrm{ar}}, D_c F_{\mathrm{av}}^{n} e_{\mathrm{ar}} \rangle}{\langle e_{\mathrm{vol}}, F_{\mathrm{av}}^{n} e_{\mathrm{vol}} \rangle} = c\, \frac{f_{n-1}}{f_{n+1}}, \qquad n \geq 1. \tag{156.3}$$

The choice $c = \pi$ coincides with the boundary reader of the areal–volumetric return. The choice $c = e$ defines the complementary subsequent reader considered in this chapter. Both use the same return support and the regional coordinate already generated.

Theorem 156.2 (Depth-independent mark quotient)**.** *For any positive marks c, d and every $n \geq 2$,*

$$\frac{\Theta_{c,n}}{\Theta_{d,n}} = \frac{c}{d}. \tag{156.4}$$

In particular,

$$\frac{\Theta_{\pi,n}}{\Theta_{e,n}} = \frac{\pi}{e}, \qquad \lim_{n\to\infty} \Theta_{\pi,n} = \frac{\pi}{\varphi^2}, \qquad \lim_{n\to\infty} \Theta_{e,n} = \frac{e}{\varphi^2}. \tag{156.5}$$

Proof. The recurrence implies $f_j > 0$ for $j \geq 1$. For $n \geq 2$, the quotient $r_n = f_{n-1}/f_{n+1}$ is positive. By (156.3), the two evaluations are cr_n and dr_n, so division cancels exactly the same return factor. The preceding identification of the regional self-scaling coordinate with the stable ratio of this incidence gives $r_n \to \varphi^{-2}$. Multiplying by the constant marks π and e yields both limits. The finite quotient, by contrast, already has its exact value before passage to the limit. □

The first length has $f_0 = 0$: both evaluations vanish and their quotient lies outside the domain of (156.4). The restriction $n \geq 2$ expresses a property of the count and avoids extending a simplification to a zero denominator.

n	2	3	4	5	6
r_n	$1/2$	$1/3$	$2/5$	$3/8$	$5/13$
$\Theta_{\pi,n}$	$\pi/2$	$\pi/3$	$2\pi/5$	$3\pi/8$	$5\pi/13$
$\Theta_{e,n}$	$e/2$	$e/3$	$2e/5$	$3e/8$	$5e/13$

The table exhibits two simultaneous facts: each reader changes with depth, while their comparison remains fixed. Invariance pertains to a relationship between readers, not to immobility of the state supporting the reading.

156.1.3 Exact convergence bound to the stable ratio

The already established correspondence of the generated coordinate φ with the positive eigenvalue of F_{av} gives $\varphi^2 = \varphi + 1$ and $\varphi > 1$. The limit can be quantified without introducing a target decimal expansion.

Proposition 156.3 (Error of the return quotient)**.** *Setting $q = -\varphi^{-2}$, for $n \geq 1$ we have*

$$r_n = \varphi^{-2}\frac{1 - q^{n-1}}{1 - q^{n+1}}, \tag{156.6}$$

$$r_n - \varphi^{-2} = \varphi^{-2}\frac{q^{n-1}(q^2 - 1)}{1 - q^{n+1}}. \tag{156.7}$$

In particular,

$$|r_n - \varphi^{-2}| \leq \varphi^{-2}\frac{|q|^{n-1}(1 - q^2)}{1 - |q|^{n+1}}. \tag{156.8}$$

Proof. Let $\psi = -\varphi^{-1}$. From $\varphi^2 = \varphi + 1$ follows $\psi^2 = \psi + 1$. The sequence $b_n = (\varphi^n - \psi^n)/(\varphi - \psi)$ satisfies $b_0 = 0$, $b_1 = 1$ and the same recurrence as f_n. Induction gives $b_n = f_n$. Dividing the expression for f_{n-1} by that for f_{n+1}, and using $\psi/\varphi = q$, yields (156.6). Subtracting the limit produces (156.7). Finally, $|q| < 1$, $1 - q^2 > 0$ and $|1 - q^{n+1}| \geq 1 - |q|^{n+1} > 0$ give the bound. □

This formula is an exact reading of the convergence rate of the incidence already generated. The marks receive the error multiplied by c, but their quotient retains the exact cancellation of Theorem 156.2.

156.2 Multiplicative composition of the coupling flow

156.2.1 Generated direction and elimination of the uniform component

The dodecaphase register K, with its marking, produces the direction

$$u_K = P_3 P_{11} K, \qquad u_K \neq 0, \qquad \langle u_K, \mathbf{1} \rangle = 0.$$

The projectors are the operators already constructed in the corresponding realization of the register. This section fixes their result and writes

$$v = \frac{u_K}{\|u_K\|}, \qquad H_K = \operatorname{diag}(v_1, \ldots, v_{12}).$$

Then $\sum_i v_i = 0$, $\sum_i v_i^2 = 1$ and $\operatorname{tr} H_K = 0$. For a positive mark r and a real parameter t, define

$$g_{K,r}(t) = \exp\bigl(t \log(r) H_K\bigr) = \operatorname{diag}\bigl(r^{t v_1}, \ldots, r^{t v_{12}}\bigr). \tag{156.9}$$

The direction comes from the register; the mark determines the logarithmic scale of deformation. Parameter t parametrizes this geometric flow. Identification with a physical clock requires the corresponding temporal realization.

Theorem 156.4 (Mark homomorphism and determinant preservation)**.** *For $a, b > 0$ and $s, t \in \mathbb{R}$,*

$$g_{K,ab}(t) = g_{K,a}(t) g_{K,b}(t), \tag{156.10}$$

$$g_{K,a/b}(t) = g_{K,a}(t) g_{K,b}(t)^{-1}, \tag{156.11}$$

$$g_{K,r}(s + t) = g_{K,r}(s) g_{K,r}(t), \tag{156.12}$$

$$\det g_{K,r}(t) = 1. \tag{156.13}$$

Moreover, $g_{K,r}(t)^{-1} = g_{K,r}(-t) = g_{K,r^{-1}}(t)$.

Proof. All matrices are diagonal. Entry i of the first equality is $(ab)^{tv_i} = a^{tv_i}b^{tv_i}$, since $\log(ab) = \log a + \log b$ for positive bases. Every entry is strictly positive, so the matrix is invertible, and substitution $b \mapsto b^{-1}$ proves the second equality. The third follows from $r^{(s+t)v_i} = r^{sv_i}r^{tv_i}$. The determinant is the product of the twelve entries:

$$\prod_{i=1}^{12} r^{tv_i} = r^{t\sum_i v_i} = r^0 = 1.$$

The inverse formula again consists of inverting every entry. □

Corollary 156.5 (Composition of ambivalence and regional mirror)**.** *For every $n \geq 2$,*

$$g_{K,\Theta_{\pi,n}}(t)g_{K,\Theta_{e,n}}(t)^{-1} = g_{K,\pi/e}(t). \tag{156.14}$$

In the stable limit,

$$g_{K,\pi/\varphi^2}(t) = g_{K,\pi/e}(t)\, g_{K,e/\varphi^2}(t). \tag{156.15}$$

Proof. Both finite marks are positive for $n \geq 2$. Applying (156.11) and (156.4), their matrix quotient is the flow of mark π/e. For the second equality, it suffices to use $(\pi/e)(e/\varphi^2) = \pi/\varphi^2$ in (156.10). It also follows as the limit of the finite identity, since the exponential and logarithm are continuous on the domain of positive bases. □

The equality compares deformations with the same direction u_K. Internal mirror exchange of the regional channels retains its own domain on the enriched state; the ratio π/e is its subsequent scalar reading. The holonomy operator Γ_9 continues to act on phase and memory. These domains determine the meaning of the composition and preserve the register accompanying return.

156.2.2 Transport of minors and exterior invariants

Let X be a real matrix of rank k with twelve columns, and $\Delta_I(X)$ the column minor specified by a set I of k indices. Right action by $g_{K,r}(t)$ scales column i by r^{tv_i}.

Proposition 156.6 (Signs, strata and quotient invariants)**.** *We have*

$$\Delta_I\big(Xg_{K,r}(t)\big) = r^{t\sum_{i\in I} v_i}\Delta_I(X). \tag{156.16}$$

The signs and vanishing pattern of all minors are preserved. If I, J, P, Q satisfy $\mathbf{1}_I + \mathbf{1}_J = \mathbf{1}_P + \mathbf{1}_Q$, then, wherever $\Delta_P(X)\Delta_Q(X) \neq 0$, the quotient

$$\frac{\Delta_I\Delta_J}{\Delta_P\Delta_Q} \tag{156.17}$$

is invariant under the flow.

Proof. Multilinearity of the determinant multiplies the minor by the product of its column factors, proving (156.16). This product is positive and therefore preserves the minor's sign and vanishing. In (156.17), the total exponent is

$$t\sum_{i=1}^{12}(\mathbf{1}_I(i) + \mathbf{1}_J(i) - \mathbf{1}_P(i) - \mathbf{1}_Q(i))v_i = 0.$$

The denominator remains nonzero and cancellation is exact. □

For $k = 2$, a concrete application is

$$\frac{\Delta_{12}\Delta_{34}}{\Delta_{13}\Delta_{24}}.$$

Each of the four indices occurs once in the numerator and once in the denominator. In a configuration where the denominator minors are nonzero, this quantity remains fixed even though each individual minor changes. Exterior conservation thus possesses explicit observables in addition to the global determinant.

156.2.3 Lattice lifting and duality

The marked lattice realization of the preceding chapter uses the orthogonal sum of twelve planes. Write

$$G_{K,r}(t) = \bigoplus_{i=1}^{12} r^{t v_i} I_2.$$

If Λ is the reference self-dual Euclidean lattice already obtained, define $\Lambda_{r,t} = G_{K,r}(t)\Lambda$.

Proposition 156.7 (Covolume and reciprocity of deformed lattices)**.** *The covolume of $\Lambda_{r,t}$ equals that of Λ, and*

$$\Lambda_{r,t}^{*} = \Lambda_{r,-t}. \tag{156.18}$$

The factorization (156.15) *lifts to these twenty-four-dimensional operators.*

Proof. Each radius occurs twice in the determinant. Therefore,

$$\det G_{K,r}(t) = \prod_{i} r^{2 t v_i} = 1.$$

The linear change multiplies the covolume by the absolute value of the determinant, hence preserves it. For any invertible operator G, the condition $y \in (G\Lambda)^{*}$ is equivalent to $\langle y, Gx \rangle \in \mathbb{Z}$ for every $x \in \Lambda$. It is equivalent to $G^{\mathsf{T}} y \in \Lambda^{*}$, hence $(G\Lambda)^{*} = G^{-\mathsf{T}}\Lambda^{*}$. Here $\Lambda^{*} = \Lambda$, G is positive diagonal and $G^{-\mathsf{T}} = G_{K,r}(-t)$, proving (156.18). Factorization is verified on each two-dimensional block exactly as in (156.10). □

This conclusion preserves covolume and relates the deformation to its dual. Integrality of scalar products and evenness of norms are additional lattice properties. Theta–Mellin spectral reciprocity uses precisely the duality proved, together with Gaussian summability and the Poisson formula developed in their own chapter.

156.3 Normalized diagonal and restitution of radial orientation

156.3.1 Separation of scale and shape

The antecedent radial chart starts from the generated angular coordinates $A > 0$ and C^{*}, in the domain $|C^{*}| < A$. Define

$$\xi = \frac{C^{*}}{A}, \qquad \eta = \operatorname{artanh} \xi.$$

The variable η denotes shape anisotropy. Its meaning differs from that of the return mantissa η_{ret} entering the action section.

Let $s > 0$ be the areal scale produced by that section. In the action-ellipse realization, $s = 2\hbar$. The positive semiaxes are

$$a = \sqrt{s}\, e^{\eta/2}, \qquad b = \sqrt{s}\, e^{-\eta/2}, \qquad ab = s. \tag{156.19}$$

Reciprocity exchanges a and b. Its radial coordinate is $R_{*} = \sqrt{b/a} = e^{-\eta/2}$. The length considered below is the diagonal of the rectangle with sides a, b:

$$d = \sqrt{a^2 + b^2}, \qquad D = \frac{d}{\sqrt{ab}}.$$

Normalization removes the areal scale and preserves the rectangle's shape up to exchange of its sides.

Theorem 156.8 (Diagonal, anisotropy and reciprocity)**.** *In the preceding domain,*

$$D^2 = 2\cosh\eta = R_*^2 + R_*^{-2}, \qquad D = \frac{\sqrt{2}}{(1-\xi^2)^{1/4}}. \tag{156.20}$$

We have $D \geq \sqrt{2}$, and the following are equivalent

$$D = \sqrt{2}, \qquad \eta = 0, \qquad \xi = 0, \qquad R_* = 1, \qquad a = b.$$

For $D > \sqrt{2}$, the additional datum $\varepsilon = \operatorname{sgn}\eta \in \{-1, +1\}$ uniquely determines

$$\eta = \varepsilon\,\operatorname{arcosh}(D^2/2), \tag{156.21}$$

$$\xi = \varepsilon\sqrt{1 - 4/D^4}, \tag{156.22}$$

$$R_* = \exp\left[-\frac{\varepsilon}{2}\operatorname{arcosh}(D^2/2)\right]. \tag{156.23}$$

Proof. Dividing $a^2 + b^2$ by $ab = s$ in (156.19) yields

$$D^2 = e^{\eta} + e^{-\eta} = 2\cosh\eta.$$

Since $R_*^2 = e^{-\eta}$, the sum is also $R_*^2 + R_*^{-2}$. From $\xi = \tanh\eta$ and $1 - \tanh^2\eta = \cosh^{-2}\eta$, with $\cosh\eta > 0$, follows $\cosh\eta = (1-\xi^2)^{-1/2}$. The positive root supplies the second formula in (156.20).

The inequality $\cosh\eta \geq 1$, with equality only for $\eta = 0$, proves the bound and its equivalences through the definitions. For $D > \sqrt{2}$, cosh determines $|\eta| = \operatorname{arcosh}(D^2/2)$, and orientation ε determines its sign. Moreover,

$$\xi^2 = 1 - \cosh^{-2}\eta = 1 - \frac{4}{D^4}.$$

Monotonicity of tanh implies $\operatorname{sgn}\xi = \operatorname{sgn}\eta$, hence (156.22). Substitution of (156.21) into $R_* = e^{-\eta/2}$ proves the last formula. Every step is unambiguous in the stated domain. □

Corollary 156.9 (Exact recovery of the semiaxes)**.** *The data (s, D, ε), with $s > 0$ and $D > \sqrt{2}$, recover the ordered semiaxes through*

$$a^2 = \frac{s}{2}\left(D^2 + \varepsilon\sqrt{D^4 - 4}\right), \qquad b^2 = \frac{s}{2}\left(D^2 - \varepsilon\sqrt{D^4 - 4}\right). \tag{156.24}$$

If $D = \sqrt{2}$, recovery is $a = b = \sqrt{s}$.

Proof. The equations $a^2 + b^2 = sD^2$ and $a^2 b^2 = s^2$ determine a^2, b^2 as the positive roots of $z^2 - sD^2 z + s^2 = 0$. Its roots are the two quantities $s(D^2 \pm \sqrt{D^4 - 4})/2$. The sign of $a - b$ agrees with that of η, so ε assigns each root to its semiaxis. The roots are positive because $D^2 > \sqrt{D^4 - 4}$. Equality produces a double root s. □

156.3.2 Exact example of two reciprocal realizations

Take a test shape with $\xi = 3/5$. This example verifies the geometric reader; the concrete assignment of an HMT section is obtained from its own generated coordinates. We compute

$$\eta = \operatorname{artanh}(3/5) = \log 2, \qquad R_* = \frac{1}{\sqrt{2}}, \qquad D^2 = 2 + \frac{1}{2} = \frac{5}{2}.$$

For scale s, the semiaxes are $a = \sqrt{2s}$ and $b = \sqrt{s/2}$. The reciprocal shape has $\xi = -3/5$, $\eta = -\log 2$, $R_* = \sqrt{2}$ and exchanged semiaxes. Both have the same normalized diagonal $D = \sqrt{5/2}$ and the same product $ab = s$. The orientation datum distinguishes these two realizations exactly.

The diagonal therefore represents anisotropy intensity. The oriented register reconstructs the ordered shape. The areal scale, in turn, recovers its size. This separation specifies which datum each reader preserves and which datum the archive supplies.

156.3.3 Poincaré and Klein charts

The Poincaré chart of the radial diameter and its Klein chart are

$$w = \frac{R_*^2 - 1}{R_*^2 + 1} = -\tanh(\eta/2), \qquad x_{\mathrm{K}} = \frac{2w}{1 + w^2} = -\tanh \eta = -\xi. \tag{156.25}$$

The domains $R_* > 0$, $-1 < w < 1$ and $-1 < x_{\mathrm{K}} < 1$ correspond through these maps.

Proposition 156.10 (Radial reciprocity in both charts)**.** *Inversion $R_* \mapsto R_*^{-1}$ acts through*

$$\eta \mapsto -\eta, \qquad w \mapsto -w, \qquad x_{\mathrm{K}} \mapsto -x_{\mathrm{K}}.$$

The diagonal and areal scale remain invariant. The pair (D, ε) recovers both disk coordinates through (156.21) *and* (156.25)*.*

Proof. The identity $R_* = e^{-\eta/2}$ transforms inversion into a sign change of η. Both hyperbolic functions in (156.25) are odd, hence change sign. Formula $D^2 = R_*^2 + R_*^{-2}$ is invariant, and the product of the semiaxes remains s. The inverse formulas of Theorem 156.8 complete recovery. □

In the example $\xi = 3/5$, we obtain $w = -1/3$ and $x_{\mathrm{K}} = -3/5$. The reciprocal shape has $w = 1/3$ and $x_{\mathrm{K}} = 3/5$. Coincidence of the diagonal thus accompanies two opposite positions on the same diameter.

156.3.4 Radial inversion and reflection in the complex chart

The positive radius also enters the complex reciprocity chart. For $r > 0$ and an argument θ, let

$$z = re^{i\theta}, \qquad s(r, \theta) = \frac{1}{1 - z}, \qquad z \neq 1.$$

The domain excludes only the pole of this chart. Radial inversion preserves θ and replaces r by r^{-1}.

Proposition 156.11 (Reflection about the line of midpoint real part)**.** *In the preceding domain,*

$$s(r^{-1}, \theta) = 1 - \overline{s(r, \theta)}, \tag{156.26}$$

$$\operatorname{Re} s(r, \theta) - \frac{1}{2} = \frac{1 - r^2}{2|1 - re^{i\theta}|^2}. \tag{156.27}$$

Therefore, the circle $r = 1$, apart from the pole, maps to the line $\operatorname{Re} s = 1/2$*. Radial inversion is realized as reflection about that vertical line.*

Proof. The conjugate of $s(r, \theta)$ is $(1 - re^{-i\theta})^{-1}$. By direct calculation,

$$1 - \frac{1}{1 - re^{-i\theta}} = \frac{-re^{-i\theta}}{1 - re^{-i\theta}} = \frac{1}{1 - r^{-1}e^{i\theta}},$$

proving (156.26). For the second identity, multiply numerator and denominator of s by $1 - re^{-i\theta}$. Its real part is

$$\frac{1 - r\cos\theta}{1 - 2r\cos\theta + r^2}.$$

Subtracting $1/2$ reduces the numerator to $(1 - r^2)/2$. The denominator is positive on the domain. It follows that $\operatorname{Re} s = 1/2$ is equivalent to $r = 1$. Finally, if $s = x + iy$, then $1 - \overline{s} = (1 - x) + iy$, which is exactly reflection about the stated vertical line. □

For $\theta = \pi$, the radii $r = 2$ and $r = 1/2$ produce $s = 1/3$ and $s = 2/3$ respectively. The point $r = 1$ produces $s = 1/2$. This example makes reciprocity visible through three rational chart values.

The proved transformation belongs to the geometry of the complex chart. Applying it to a spectral operator requires preserving the operator and its domain as well as that chart. Likewise, the radial chart R_* preserves the angular section generating it. Equality of an inversion operation in two realizations supplies the stated correspondence; identification of their antecedent objects is maintained through the transports already constructed.

156.4 Recovery of the flow and compatibility of readers

Applying these readers to a flow direction requires preserving the generated direction as well as its scalar values. The reduced archive previously constructed has operators M and L with $LM = P_{11}$. If $m = MK$, then

$$u_K = P_3 L m. \tag{156.28}$$

This equality preserves the action of each operator and allows recovery to be composed with the flow.

Proposition 156.12 (Reconstruction of the flow from the reduced archive)**.** *With the preceding operators and marking, the archive m and a positive mark r determine*

$$g_{K,r}(t) = \operatorname{diag}\left(r^{\,t(P_3 L m)_1/\|P_3 L m\|}, \dots, r^{\,t(P_3 L m)_{12}/\|P_3 L m\|}\right).$$

All composition identities in (156.10)–(156.13) *are preserved.*

Proof. From $m = MK$ and $LM = P_{11}$ follows $P_3 L m = P_3 LMK = P_3 P_{11} K = u_K$. The direction is nonzero by the already proved property of the generated register. Normalization recovers v exactly, and its substitution into (156.9) recovers every flow entry. The identities depend only on this direction and positivity of marks, so remain valid. □

The archive identity recovers the direction of deformation. Regional readers supply their marks. The diagonal reader of the radial shape preserves its absolute anisotropy, and the oriented register recovers the sign. The resulting composition assigns invariance and reversibility to explicit data:

$$\begin{gathered}\text{sheet returns} \longrightarrow (\Theta_{\pi,n}, \Theta_{e,n}) \longrightarrow \pi/e,\\ \text{archive of } K \longrightarrow u_K \longrightarrow g_{K,r}(t) \longrightarrow \text{covolume and transported minors},\\ (s, D, \varepsilon) \longrightarrow (a, b) \longleftrightarrow (R_*, w, x_{\mathrm{K}}).\end{gathered}$$

The first line distinguishes the common support of two marks. The second expresses the register's operator function. The third specifies the geometric information needed for inversion. The enriched state preserves phase, turns, carries and incidences allowing these readers to be composed within their genealogy.

156.5 Temporal realization and domain of physical interpretation

Nonadic holonomy preserves its action on phase and memory; phase return increments the turn register. Counts of the modal envelope and mark ratios are evaluated on subsequent realizations. Radial reciprocity acts on a shape chart and its orientation. The proved articulation combines these operations through compatible readers, preserving the distinction between their parameters.

The relation π/φ^2 appears as a marked return ratio and as the scale of the conservative flow. The ratio π/e appears as a mark quotient on the same support. Both satisfy the exact factorization (156.15). Their involvement in a gravitational or cosmological realization preserves the conditions of the corresponding physical reader: clock identification, dimensional quantities, boundary response and geometric domain. This chapter's result concerns mathematical composition of the generated readers. Their physical applications subsequently use the corresponding realization.

Chapter 157

Positive geometry of refinement and conservation of canonical forms

Object and results

It is proved that reversible refinement with carry produced by the APP–TRIT–TPK architecture admits a stable canonical form: the sum of the forms of the refined cells coincides exactly with the form of the original cell, and all interior poles cancel. The identity extends exactly to the cuboidal product complex and preserves the iterated residues of the external boundary. The dodecaphase register K determines an invertible frame O_K, a Gram metric and an isometric archive preserving exterior volumes and allowing recovery of the history. An exhaustive calculation of minors shows that O_K is not totally nonnegative. The direction already constructed $u_K = P_3 P_{11} K$, combined with $\rho_A = \pi_{\mathrm{HMT}}/\varphi^2_{\mathrm{HMT}}$, defines a positive diagonal determinant-one flow preserving every Plücker sign and every positroid stratum. This separation permits formulation, under falsifiable conditions, of the intrinsic positroid atlas of the TPK, its cluster dynamics and its pushforward toward an amplituhedral section. The centrality of the electron is linked to the unique fixed point $(5, 5)$ of the APP atlas, without confusing it with the nonadic datum 111111 or the regional signature 555555.

157.1 Thesis and causal order

The starting object is not an externally chosen Grassmannian. The chain

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \widehat{x} \longrightarrow C^{\mathrm{disc}}_{\mathrm{cont}}$$

produces an enriched state preserving sheets, orientation, remainder, quotient, carry, route, boundary, survival, memory and return. The five constructions of the continuum emerge jointly. Positive geometry is an expression of their compatible refinement.

The article separates three assertions that must not be merged:

1. telescoping and the metric archive are exact results;
2. the moment curve, $D_K(s)$ and the centred flow $g_{K,\rho}(\tau)$ are exact new formalizations;
3. the intrinsic positroid atlas, cluster seed and equality with a complete amplituhedron require expressly declared additional maps.

This separation does not diminish the result: it identifies the proof of each arrow and avoids attributing positivity to a metric matrix that does not possess it.

157.2 Master theorem

157.3 Coinductive positive geometry of TPK refinement

The positive structure studied in this section is not introduced from a previously assumed Grassmannian. It arises from the same generating chain that constructs the enriched state:

$$\mathrm{APP} \longrightarrow \mathrm{TRIT} \longrightarrow \mathrm{TPK} \longrightarrow \widehat{x} \longrightarrow C^{\mathrm{disc}}_{\mathrm{cont}}.$$

APP simultaneously preserves both sheets, remainders and quotients; TRIT types the regime and orients each transition; TPK selects, transports and updates without losing carry, route, boundary, survival or memory. The five constructions of the continuum emerge jointly from that state. The following Grassmannian realization is a subsequent reader of its compatible refinement, not an input selecting the dynamics.

157.3.1 Carry, subdivision and ordered parameter

Fix an integer base $B \geq 2$. The elementary refinement operation with carry $c \in \{0, \dots, B-1\}$ is

$$L_c(x, y) = (Bx - c, By + c). \tag{157.1}$$

It preserves the rescaled sum, since

$$\frac{Bx - c}{B(x+y)} + \frac{By + c}{B(x+y)} = 1.$$

Its inverse recovers c through the declared residue class, then recovers (x, y) by Euclidean division. By composition, a carry word $c_1 \dots c_n$ determines a cell of depth n, and divisions in reverse order recover the complete word. Compatibility between truncations produces nested cylinders and preserves the history, not merely its limiting evaluation.

Let $M \geq 1$ be the number of intervals in the initial chart and

$$N_n = B^n M, \qquad t_{n,j} = \frac{j}{N_n}, \qquad 0 \leq j \leq N_n. \tag{157.2}$$

For $0 \leq j < N_n$, the subinterval of index c of point j has index

$$j' = Bj + c, \qquad t_{n+1,Bj+c} = t_{n,j} + \frac{c}{BN_n}. \tag{157.3}$$

The inequality $j < j' \Rightarrow t_{n,j} < t_{n,j'}$ makes visible the information required by positivity: carry does not merely refine; it inserts points while preserving the total order compatible with the linking maps.

157.3.2 Moment matrix and positive Grassmannian

Status. The following moment curve is a new formalization of the order already produced by refinement; it is not presented as an additional TPK operator or as an input to the generator.

For $1 \leq r \leq N_n + 1$, define the moment curve and its column matrix

$$v_r(t) = (1, t, \dots, t^{r-1})^{\mathsf{T}}, \qquad Z_n^{(r)} = (v_r(t_{n,0}) \ \cdots \ v_r(t_{n,N_n})).$$

Theorem 157.1 (Plücker positivity generated by carry)**.** *For every $I = \{j_1 < \cdots < j_r\}$,*

$$\Delta_I(Z_n^{(r)}) = \prod_{1 \leq a < b \leq r} (t_{n,j_b} - t_{n,j_a}) > 0. \tag{157.4}$$

Therefore, the row class of $Z_n^{(r)}$ belongs to $\mathrm{Gr}_{>0}(r, N_n + 1)$. Insertions (157.3) extend this system without altering the sign of any antecedent minor.

Proof. The minor is a Vandermonde determinant. Every factor of (157.4) is positive by the strict ordering of the points. An insertion maintains the order of the old columns and adds points within their intervals. Minors formed only from antecedent columns preserve their value exactly; those containing a new column are again products of positive differences. □

The weighted version preserves the same orientation. If $\lambda_{n,j} > 0$, put

$$C_{n;aj} = \lambda_{n,j} t_{n,j}^{a-1}.$$

Then

$$\Delta_I(C_n) = \left(\prod_{j \in I} \lambda_{n,j} \right) \Delta_I(Z_n^{(r)}) > 0. \tag{157.5}$$

Weights may receive a compatible cylindrical measure or the square root of a route multiplicity, provided they are declared before the external object is evaluated.

157.3.3 Survival masks and positroid realization

At every depth, the TPK preserves a survival mask $S_n \subseteq \{0, \dots, N_n\}$. To make it visible without changing the generator, allow

$$\lambda_{n,j} > 0 \quad (j \in S_n), \qquad \lambda_{n,j} = 0 \quad (j \notin S_n).$$

Status. Assigning a moment matrix to a survival mask is a *new formalization*: it uses data already generated by the TPK but is not presented as a theorem previously contained in the corpus. Its exact mathematical content is as follows.

Proposition 157.2 (Positroid realization of a mask)**.** *Suppose $|S_n| \geq r$. The weighted matrix satisfies*

$$\Delta_I(C_{n,S_n}) > 0 \iff I \subseteq S_n, \qquad \Delta_I(C_{n,S_n}) = 0 \iff I \not\subseteq S_n.$$

Its matroid is the uniform matroid of rank r on S_n, with the complementary positions as loops; in particular, it is a positroid. Realizations of two levels are compatible whenever their masks are compatible under the linking map and the old columns preserve their identification.

Proof. The factorization (157.5) shows that a minor vanishes exactly when it contains a zero weight. On S_n, all factors are positive and all subsets of cardinality r are bases. A uniform matroid realized by ordered points on the moment curve is positroidal; adding loops preserves this class. Compatibility follows because a refined mask inherits survivors and adds only the columns allowed by the compatibility hypothesis. □

The proposition certifies realization of each mask within the ordered-moment functor. It does not yet certify an intrinsic positroid atlas of the TPK or coverage of all cells of $\mathrm{Gr}_{\geq 0}(r, N_n + 1)$. That promotion requires a planar network or boundary measure derived from the TPK, together with its decorated permutation or Grassmann necklace, and naturality of those data under refinement. Absence of any of these components would falsify the global promotion, not the preceding proposition.

157.3.4 Plücker exchanges and cluster charts

For $r = 2$, write $z_j = \lambda_j(1, t_j)^{\mathsf{T}}$. If $a < b < c < d$, we have

$$\Delta_{ac}\Delta_{bd} = \Delta_{ab}\Delta_{cd} + \Delta_{ad}\Delta_{bc}. \tag{157.6}$$

Status. The following relation is exact for the ordered-moment realization. Reading it as a TPK cluster structure is a *new formalization*; it neither identifies carry with mutation nor claims that the corpus has already constructed a universal plabic graph.

Proposition 157.3 (Positive exchange in rank two)**.** *All coordinates of* (157.6) *are positive, and the mutation solving for either diagonal is subtraction-free. Insertion of a refined point through* (157.3) *commutes with evaluation of the Ptolemy relation. Triangulations of the ordered polygon and their flips therefore provide positive cluster charts for the ordered realization of each level.*

Proof. The identity follows by substituting $\Delta_{ij} = \lambda_i\lambda_j(t_j - t_i)$ and expanding. All its terms are positive. Insertion preserves the old coordinates and assigns the exact ordered parameter to the new column; evaluating the determinant first or the exchange relation first produces the same polynomial identity. □

In general rank, for $|I| = r - 2$ and $a < b < c < d$ outside I,

$$\Delta_{Iac}\Delta_{Ibd} = \Delta_{Iab}\Delta_{Icd} + \Delta_{Iad}\Delta_{Ibc}. \tag{157.7}$$

All terms are positive in the top cell and nonnegative in survival cells. Thus every short Plücker exchange is compatible with refinement within this realization. The correct assertion is that carry inserts data and mutations change charts on those data; a carry step is not identified with a flip. Intrinsic TPK cluster compatibility additionally requires deriving a seed, its quiver or plabic graph and the action of linking maps on mutations.

157.3.5 Canonical forms and cancellation of interior boundaries

For an oriented interval $I = [a, b]$, its canonical form is

$$\Omega_I = \frac{dt}{t-a} - \frac{dt}{t-b}. \tag{157.8}$$

If $a = c_0 < c_1 < \cdots < c_m = b$, then

$$\sum_{q=0}^{m-1} \Omega_{[c_q, c_{q+1}]} = \Omega_{[a,b]}. \tag{157.9}$$

Every interior pole occurs twice, with opposite residues.

The one-dimensional identity is exact and does not depend on an additional positroid interpretation. Its cubical extension is formulated below as an *exact new formalization*. Let $Q = \prod_{\ell=1}^{d} [a_\ell, b_\ell]$ with orientation $dx_1 \wedge \cdots \wedge dx_d$. Define

$$\Omega_Q = \bigwedge_{\ell=1}^{d} \left(\frac{dx_\ell}{x_\ell - a_\ell} - \frac{dx_\ell}{x_\ell - b_\ell} \right). \tag{157.10}$$

Theorem 157.4 (New formalization: multidimensional cubical telescoping)**.** *Subdivide each coordinate interval by the TPK refinement points, and denote the resulting cuboids, with induced orientation, by $Q_{\mathbf{j}}$. Then*

$$\boxed{\sum_{\mathbf{j}} \Omega_{Q_{\mathbf{j}}} = \Omega_Q.} \tag{157.11}$$

All interior-facet poles cancel. If $F_1 \supset F_2 \supset \cdots \supset F_s$ is a flag of oriented faces, then

$$\operatorname{Res}_{F_s} \cdots \operatorname{Res}_{F_1} \Omega_Q = \operatorname{sgn}(F_\bullet)\, \Omega_{F_s}. \tag{157.12}$$

In particular, the residue at each vertex has modulus one and the sign fixed by orientation.

Proof. The sum over the grid is the exterior product of the d one-dimensional sums. Applying (157.9) in each factor gives (157.11). An interior facet has two incident cells and opposite normal orientations, so their residues cancel. Taking a residue removes the corresponding normal factor and leaves the face's canonical form with its orientation sign. Iteration proves (157.12). □

Corollary 157.5 (Independence of refinement)**.** *The global form does not depend on how many carry levels have been inserted. Refining changes the triangulation or cubulation and preserves the sum of forms. TPK memory preserves which intermediate cells were traversed, even though the global form cancels their internal boundaries.*

Proposition 157.6 (General oriented cancellation criterion)**.** *Let $X = \bigcup_\sigma X_\sigma$ be a pure, oriented positive subdivision. Suppose each cell has a form Ω_σ and that, for every facet $F \subset \overline{X_\sigma}$,*

$$\operatorname{Res}_F \Omega_\sigma = [\sigma : F]\Omega_F.$$

For a chain $c = \sum_\sigma w_\sigma[\sigma]$,

$$\operatorname{Res}_F\left(\sum_\sigma w_\sigma \Omega_\sigma\right) = (\partial c)_F \Omega_F. \tag{157.13}$$

In particular, an interior facet cancels if and only if its coefficient in ∂c *is zero. Iterated residues on interior flags cancel by* $\partial^2 = 0$.

Proof. Linearity of the residue gives

$$\operatorname{Res}_F\left(\sum_\sigma w_\sigma \Omega_\sigma\right) = \sum_{\sigma \supset F} w_\sigma [\sigma : F]\Omega_F,$$

and the coefficient of Ω_F is precisely that of F in ∂c. Iteration identifies the sum of codimension-two incidences with the corresponding coefficient of $\partial^2 c$, which is zero. □

The proposition is an exact new formalization. Identifying the sum with the unique canonical form of X additionally requires excluding an ambiguous regular top form. The corpus already supplies the path-and-memory complex with $D^2 = 0$; positroid promotion requires constructing the morphism

$$\Omega_{\mathrm{TPK}} : C_\bullet^{\mathrm{TPK}} \longrightarrow \Omega^\bullet_{\log}, \qquad \operatorname{Res} \circ \Omega_{\mathrm{TPK}} = \Omega_{\mathrm{TPK}} \circ D.$$

On cylinders and products, this morphism is given by (157.8) and (157.10). For a general positroid atlas, strata, incidences, orientations, uniqueness and Plücker–cluster compatibility must be added. Nor can probabilistic history weights multiply the forms indiscriminately: a shared facet would retain a residue proportional to the weight difference. The factors $\sqrt{\mu}$ of a positive matrix belong to another operation.

157.3.6 The register K as a metric and archival frame

The dodecaphase register is produced by

$$x_{\mathrm{term}} \mapsto \mathcal{R}_{12} \mapsto (b^{90}, b^{120}, Q_{\mathrm{TPK}}) \mapsto U_{\mathrm{sgn}} \mapsto K$$

and equals

$$K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601).$$

Let S be the cyclic shift of twelve positions and

$$O_K = (K, SK, \dots, S^{11}K), \qquad G_K = O_K^* O_K.$$

The reference development proves

$$\det O_K = 548311925921402018385039793000828362 5 > 0, \qquad 589609 I \le G_K \le 6263^2 I. \tag{157.14}$$

In particular, O_K is invertible and defines a quantitatively controlled frame. Since its determinant is positive, its polar decomposition can be written

$$O_K = F_K P_K, \qquad P_K = G_K^{1/2} > 0, \qquad F_K \in SO(12).$$

Proposition 157.7 (Exact covariance of Plücker data)**.** *Let* $X \in \mathrm{Mat}_{r\times 12}(\mathbb{R})$ *have rank* r *and* $Y = XO_K^{\mathsf{T}}$. *Then*

$$\Delta_I(Y) = \sum_{\substack{J \subseteq \{1,\dots,12\} \\ |J|=r}} \Delta_J(X) \det\big((O_K)_{I,J}\big). \tag{157.15}$$

Equivalently, the Plücker vector transforms by $\bigwedge^r O_K$. In rank twelve,

$$\Delta_{\{1,\dots,12\}}(XO_K^{\mathsf{T}}) = (\det O_K)\Delta_{\{1,\dots,12\}}(X).$$

Proof. Formula (157.15) is the Cauchy–Binet identity applied to every column set of Y. The exterior formulation and rank-twelve case are its functorial and maximal expressions, respectively. □

Plücker covariance does not amount to preserving standard total positivity. Indeed, O_K has minors of both signs. Two explicit witnesses are

$$\det\begin{pmatrix} 234 & 543 \\ 543 & 140 \end{pmatrix} = -262089, \qquad \det\begin{pmatrix} 234 & 140 \\ 543 & 729 \end{pmatrix} = 94566. \tag{157.16}$$

For example, the image under O_K of the positive coordinate plane generated by the first two canonical vectors already contains the first negative minor of (157.16). Therefore,

$$\left\{[XO_K^{\mathsf{T}}] : [X] \in \mathrm{Gr}_{\geq 0}(r,12)\right\} \not\subseteq \mathrm{Gr}_{\geq 0}(r,12) \quad \text{in general.} \tag{157.17}$$

Thus, O_K is an invertible metric and archival frame; it is neither a boundary-measurement matrix nor a generator of standard total positivity.

Positive chart adapted to K. The exact image

$$\mathrm{Gr}^K_{\geq 0}(r,12) := \left\{[XO_K^{\mathsf{T}}] : [X] \in \mathrm{Gr}_{\geq 0}(r,12)\right\} \tag{157.18}$$

defines a positive chart adapted to the frame. If $\Phi_K([X]) = [XO_K^{\mathsf{T}}]$, its adapted coordinates are

$$\Delta_I^K([Y]) := \Delta_I\Big(Y(O_K^{\mathsf{T}})^{-1}\Big) \geq 0. \tag{157.19}$$

This positivity is exact by transport of structure; it must not be confused with positivity of the minors of Y in the standard ambient basis. Likewise, a canonical form Ω in the original chart is transported as $(\Phi_K)_*\Omega$.

New formalization: positive diagonal action of K. Since all components K_i are strictly positive, define

$$D_K(s) := \mathrm{diag}\big(e^{s\log K_1}, \dots, e^{s\log K_{12}}\big), \qquad s \in \mathbb{R}, \tag{157.20}$$

so that $D_K(1) = \mathrm{diag}(K_1, \dots, K_{12})$. For every matrix $X \in \mathrm{Mat}_{r\times 12}$,

$$\Delta_I(XD_K(s)) = e^{\,s\sum_{i\in I}\log K_i}\Delta_I(X). \tag{157.21}$$

This action exactly preserves $\mathrm{Gr}_{\geq 0}(r,12)$ and its stratification by vanishing minors. The diagonal action is a new formalization constructed from the generated register K; it is neither attributed to the original reference development of O_K nor used to conceal the mixed signs of (157.16).

New formalization: centred flow determined by K. The corpus constructs

$$u_K = P_3P_{11}K, \qquad \|u_K\|^2 = \frac{6638585 + 2275584\sqrt{5}}{20} > 0, \qquad u_K \perp \mathbf{1}. \tag{157.22}$$

Let $\widehat{u}_K = u_K/\|u_K\|$ and $\rho_A = \pi_{\mathrm{HMT}}/\varphi^2_{\mathrm{HMT}} > 0$. Then

$$g_{K,\rho}(\tau) = \operatorname{diag}\Big(\rho_A^{\tau\widehat{u}_{K,1}}, \dots, \rho_A^{\tau\widehat{u}_{K,12}}\Big) \in SL(12,\mathbb{R})_{>0}. \tag{157.23}$$

For every matrix X,

$$\Delta_I(X g_{K,\rho}(\tau)) = \rho_A^{\tau \sum_{i\in I} \widehat{u}_{K,i}} \Delta_I(X). \tag{157.24}$$

The flow therefore preserves signs and vanishing minors. The register K fixes the direction, and the generated ratio ρ_A normalizes the speed. The ratio ρ_A is evaluated here at a later stage: π_{HMT} and φ_{HMT} are two coordinates of the fourfold relation $(\pi, \varphi, e, \alpha)$ generated coinductively by the same APP–TRIT–TPK state. Neither their conventional values nor the already evaluated ratio select K, u_K or the refinement. Identifying τ with physical time requires a clock $\tau(t)$; extension to every depth requires a sectorial cocycle compatible with $j \mapsto Bj + c$.

The complete archive

$$\mathcal{V} v = \Big(A_\infty^{1/2} v, (D_j R_j v)_{j\ge 0}\Big), \qquad \mathcal{V}^*\mathcal{V} = I,$$

preserves this information in all degrees:

$$\left\langle \bigwedge^r \mathcal{V}\,\xi, \bigwedge^r \mathcal{V}\,\eta \right\rangle = \langle \xi, \eta\rangle, \qquad \| \wedge^r \mathcal{V}\,\xi\| = \|\xi\|. \tag{157.25}$$

It therefore preserves Gram determinants and Plücker norms. This isometry does not claim that $\mathcal{V}$ turns the minors of O_K into positive minors: it preserves the information archived in the space and orientation explicitly transported.

The rational reading

$$\kappa_{\mathrm{ag}} = \frac{234543140729659824621058914794146601}{10^{36} - 1}$$

is reversible once base, length and phase origin are fixed. The coupling function of K is therefore expressed in three distinct layers: its arithmetic reading κ_{ag}, its positive Gram form G_K and its incidence transport. All three are expressions of the generated register; none is used to choose the history retrospectively.

157.3.7 Areal–volumetric flow and marked ratio

The TPK calendar induces the primitive matrix

$$F_{\mathrm{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

whose powers obey the Fibonacci recurrence. The Parry measure of its symbolic envelope distinguishes the areal and volumetric modes and satisfies

$$\frac{\mu_{\mathrm{ar}}}{\mu_{\mathrm{vol}}} = \varphi_{\mathrm{HMT}}^{-2}. \tag{157.26}$$

The regional reading marks this ratio once:

$$\Theta_n = \pi_{\mathrm{HMT}} \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi_{\mathrm{HMT}}}{\varphi_{\mathrm{HMT}}^2}. \tag{157.27}$$

The reciprocal reader $R(x, y) = x/y^2$, conjugated by $J(x, y) = (y, x)$, produces the orientation $\varphi_{\mathrm{HMT}}/\pi_{\mathrm{HMT}}^2$. These ratios are typed readers of the flow; they are not identified with κ_{ag}, although all participate in the same coupling architecture.

The complex's canonical form is stable under refinement, while (157.26) determines the stationary proportion of its two modes. This closes the circle: arithmetic refinement generates subdivision; the holographic ratio governs its asymptotic distribution; K preserves metric and orientation; and the canonical form retains only the external boundary after cancelling internal walls.

157.3.8 Electronic centre and material section

Normalize the atlas 9×9 by

$$\nu(r, s) = \left(\frac{r-1}{8}, \frac{s-1}{8} \right).$$

Cell inversion $\rho_c(r, s) = (10 - r, 10 - s)$ becomes $(u, v) \mapsto (1 - u, 1 - v)$. Its unique fixed point is

$$\nu(5, 5) = \left(\frac{1}{2}, \frac{1}{2} \right). \tag{157.28}$$

The 81 states are points of a grid whose elementary cubulation has 64 rectangles; they are not presented as 81 two-dimensional cells.

Theorem 157.8 (Uniqueness of the equivariant point section)**.** *In a stable fibre on which the coarse label remains fixed, every equivariant point selection for ρ_c must take the value* $(5, 5)$*. The level-zero electronic fibre contains that unique point; no other fibre admits such a selection.*

Proof. If $a(y)$ is equivariant and y is fixed, then $a(y) = a(\rho_c y) = \rho_c a(y)$. The unique solution of $\rho_c(r, s) = (r, s)$ is $r = s = 5$. The atlas census places this cell in the level-zero electronic fibre. □

The theorem explains why the electron is a central point section and the other species are presented through orbits or multisections. In particular, the following typed objects must be kept distinct:

1. the fixed cell $(5, 5)$, belonging to the atlas 9×9;
2. the datum 111111, belonging to the fifth nonadic boundary;
3. the transverse multiplicative signature $\mathcal{E}_\times = 555555$, belonging to the regional reader of π;
4. the sign block $(+1, +1, +1)$, appearing within the dodecaphase word

$$\tau_e = (+1, +1, +1, -1, -1, -1, +1, +1, +1, -1, -1, -1).$$

The central APP structure, distinct from these three registers, is

$$\mathcal{A}(5, 5) = ((1, 1), (7, 2)), \qquad 5 + 5 = 1 + 9 \cdot 1, \qquad 5 \cdot 5 = 7 + 9 \cdot 2.$$

Coincidence or repetition of digits does not constitute identification: $(5, 5)$, 111111 and 555555 have different domains, operations and causal functions.

Material positivity is subsequent. If $M_M^{(0)} \geq 0$ and $R_{\mathrm{act}} > 0$, then

$$M_M^{\beta, r} = R_{\mathrm{act}}^{B_M^r/2} M_M^{(0)} R_{\mathrm{act}}^{B_M^r/2} \geq 0. \tag{157.29}$$

Congruence preserves the positive cone and realizes on the mass branch the positivity previously structured; it does not turn observed mass into an atlas input.

157.3.9 Positive multimorphology of the enriched state

Multimorphology does not denote an informal superposition of figures. It is a system of typed morphisms with a common source:

$$\begin{aligned}\mathfrak{P}_n &: \widehat{\mathcal{X}}_{\mathrm{TPK,n}} \longrightarrow \mathrm{Gr}_{\geq 0}(k, N_n),\\ \mathfrak{R}_n &: \widehat{\mathcal{X}}_{\mathrm{TPK,n}} \longrightarrow C^{+}_{d,n},\\ \mathfrak{C}_n &: \widehat{\mathcal{X}}_{\mathrm{TPK,n}} \longrightarrow Q^{+}_{\mathrm{con,n}},\\ \mathfrak{M}_n &: \widehat{\mathcal{X}}_{\mathrm{TPK,n}} \longrightarrow \mathcal{L}^{+}_{M,n}.\end{aligned}$$

Their images are, respectively, Plücker, lattice, constitutive and mass-operator positivities. Each morphism preserves the coordinates needed by its reader and commutes with the refinement link of its codomain. The four codomains are neither identified nor used to select one another.

Proposition 157.9 (Multimorphological coherence)**.** *If each morphism commutes with truncation, boundary and residue, cancellation of interior faces is transported to the four realizations. If, in addition, a kinematic map*

$$\mathfrak{A}_{Z,n} : C \longmapsto CZ^{\mathsf{T}}$$

has degree one, preserves orientation and covers its image with compatible boundaries, the induced amplituhedral form is independent of representation depth.

Proof. Proposition 157.6 applies before each realization. Compatibility squares transport the incidence coefficient and residue. The degree-one condition avoids interior multiplicities in the kinematic pushforward. □

The moment-curve realization $\mathfrak{P}_n$ is exact as an ordered formalization; its promotion to an intrinsic positroid atlas retains the conditions already declared. The lattice and operator branches have previous HMT–MD reference developments. The constitutive branch must specify its positive form and reader in each application. Multimorphology combines these maps; it does not add a fifth positivity.

157.3.10 Positive constitutive realization and spectral naturality

The constitutive realization of Proposition 157.9 receives an operator already constructed from the two oriented TPK sheets. Section 39.8 determines its response and inverse; the present composition makes explicit the positive form and refinement link used by that realization.

Let $\widehat{\mathcal{X}}_{\mathrm{con}}$ be the domain of enriched states whose angular reading belongs to chamber $x > y > 0$. For $\xi \in \widehat{\mathcal{X}}_{\mathrm{con}}$, the reading produces a two-sheet space E_ξ, an orthogonal involution $\mathcal{R}_\xi$ and the transport

$$P_{\pm,\xi} = \frac{I \pm \mathcal{R}_\xi}{2}, \qquad T_\xi = q_{+,\xi} P_{+,\xi} + q_{-,\xi} P_{-,\xi}, \qquad 0 < q_{+,\xi} < q_{-,\xi} < 1.$$

The incidences 90 and 120 arise from the calendar and act on that same transport. The response is

$$V_\xi = (I - T_\xi^{90})(I - T_\xi^{120})^{-1}, \qquad W_{m,\xi} = V_\xi \otimes I_{9^m}.$$

Amplification preserves every position of the new nonadic fibre. On $E_{m,\xi} = E_\xi \otimes \mathbb{C}^{9^m}$, define the form

$$\mathfrak{C}_m(\xi)(v, w) := \langle W_{m,\xi} v, W_{m,\xi} w \rangle = \langle v, W_{m,\xi}^2 w \rangle. \tag{157.30}$$

This is a concrete realization in the cone of positive-definite Hermitian forms. Squaring preserves the quadratic responses of the two sheets expressing the coordinates $\widehat{\varepsilon}$ and $\widehat{\mu}$ of (39.44).

Proposition 157.10 (Coercivity, recovery and nonadic link)**.** *The form* (157.30) *satisfies, for every nonzero* $v \in E_{m,\xi}$,

$$\frac{9}{16}\|v\|^2 < \mathfrak{C}_m(\xi)(v,v) < \|v\|^2.$$

If $A_{m,\xi} = W_{m,\xi}^2$ *and* $\mathbb{E}_m$ *is the normalized partial expectation over the last nine-dimensional factor, then*

$$\mathbb{E}_m(A_{m+1,\xi}) = A_{m,\xi}.$$

The form, together with the oriented sheet labels, recovers V_ξ*, the two channels* $q_{\pm,\xi}$ *and their angular coordinates. If a unitary transport* $J : E_\xi \to E_{\xi'}$ *satisfies* $JT_\xi = T_{\xi'}J$*, then*

$$(J \otimes I)^* A_{m,\xi'}(J \otimes I) = A_{m,\xi}.$$

The identity holds for any continuous function of the spectrum applied to the response before amplification.

Proof. The sheet projectors are orthogonal and sum to the identity. The spectral calculus of (39.42) gives

$$V_\xi = r_{+,\xi}P_{+,\xi} + r_{-,\xi}P_{-,\xi}, \qquad r_{\pm,\xi} = f(q_{\pm,\xi}^{30}), \qquad f(s) = \frac{1+s+s^2}{1+s+s^2+s^3}.$$

In $0 < s < 1$, we have

$$4(1+s+s^2) - 3(1+s+s^2+s^3) = (1-s)(3s^2+2s+1) > 0,$$

and the denominator exceeds the numerator by $s^3 > 0$. Therefore $3/4 < r_{\pm,\xi} < 1$. Orthogonal decomposition and squaring these bounds prove coercivity, which remains identical under amplification.

Since $A_{m+1,\xi} = A_{m,\xi} \otimes I_9$ and the normalized trace of the last factor satisfies $\mathrm{tr}_9(I_9) = 1$,

$$\mathbb{E}_m(A_{m+1,\xi}) = (\mathrm{id} \otimes \mathrm{tr}_9)(A_{m,\xi} \otimes I_9) = A_{m,\xi}.$$

The same equality holds for each polynomial in $W_{m,\xi}$. Uniform approximation on the compact spectrum and continuity of the expectation extend it to every continuous function.

The positive root of $A_{0,\xi}$ recovers V_ξ. The oriented projectors identify its two eigenvalues. The function f is strictly decreasing on $(0,1)$ by (39.43); its inverse determines the unique roots of (39.46). The positive thirtieth root restores $q_{\pm,\xi}$, and (39.47) restores x and y.

Finally, $JT_\xi = T_{\xi'}J$ implies the same identity for all its powers. Invertibility of $I - T^{120}$ gives

$$J(I - T_\xi^{120})^{-1} = (I - T_{\xi'}^{120})^{-1}J.$$

This yields $JV_\xi = V_{\xi'}J$. Amplification, squaring and unitarity of J give the stated congruence. The polynomial argument and its uniform extension prove the last assertion. □

The link in this construction is the normalized expectation preserving the amplified operator. Compatibility with a geometric transport uses the intertwining $JT_\xi = T_{\xi'}J$ expressed in the statement. Transport of canonical forms and residues, in turn, maintains the boundary squares of Proposition 157.9. The operations at each level are thus specified: angular response, positive constitutive form, fibre amplification and incidence transport. Angular restitution recovers this reading; the enriched archive additionally preserves the history producing it.

157.3.11 Conditional amplituhedral formalization

Status. What follows is a *new conditional formalization*. The corpus supplies refinement, routes, survival and ordered data; identifying its image with an amplituhedron and promoting the cellular sum to a canonical form require the explicitly stated geometric hypotheses.

Fix $k, m \geq 1$ and $N = N_n + 1 \geq k + m$. Let $Z_n = Z_n^{(k+m)}$ be the positive external datum constructed by carry and let $C_n \in \mathrm{Gr}_{\geq 0}(k, N)$ be a matrix of surviving routes. On the locus where $\mathrm{rank}(C_n Z_n^{\mathsf{T}}) = k$, define

$$Y_n = C_n Z_n^{\mathsf{T}} \in \mathrm{Gr}(k, k+m). \tag{157.31}$$

Definition 157.11 (Candidate amplituhedral image)**.** Let $\mathfrak{C}_n^{\mathrm{TPK}}$ be the positive complex of routes and carries at depth n. Its image

$$\mathcal{A}_n^{\mathrm{HMT}}(k, m) = \{C Z_n^{\mathsf{T}} : C \in \mathfrak{C}_n^{\mathrm{TPK}}\}$$

is called the candidate amplituhedral image at level n. Denote by

$$\mathcal{F}_{\mathrm{kin},n} : \mathfrak{C}_n^{\mathrm{TPK}} \longrightarrow \mathrm{Gr}(k, k+m), \qquad [C] \longmapsto [C Z_n^{\mathsf{T}}]$$

its kinematic map.

Proposition 157.12 (Canonical pushforward under degree and boundary hypotheses)**.** *Suppose $\mathfrak{C}_n^{\mathrm{TPK}}$ is an oriented pure-dimensional complex, that $\mathcal{F}_{\mathrm{kin},n}$ is proper on its closure, generically finite of degree one on each component, preserves orientation, covers its image without undeclared interior overlaps and maps boundaries to boundaries compatibly with residues. Then*

$$\Omega_n^{\mathrm{HMT}} = (\mathcal{F}_{\mathrm{kin},n})_* \left(\sum_{\sigma \in \mathfrak{C}_n^{\mathrm{TPK}}} \Omega_\sigma \right) \tag{157.32}$$

is well defined, has logarithmic poles only on the external boundary of $\mathcal{A}_n^{\mathrm{HMT}}$ and is independent of every further compatible subdivision. Its residues are the canonical forms of the face images.

Proof. Theorem 157.4 cancels interior facets before pushforward. Degree one and absence of interior overlaps prevent spurious multiplicities; boundary–residue compatibility transports logarithmic poles and their residues. Compatible refinement changes the cellular decomposition but not its sum. □

The proposition determines a structural and recoverable form only when its hypotheses are verified. If, additionally, $\mathfrak{C}_n^{\mathrm{TPK}} = \mathrm{Gr}_{\geq 0}(k, N)$ is proved on the relevant domain and the map has the required coverage, then $\mathcal{A}_n^{\mathrm{HMT}}(k, m)$ coincides with the standard amplituhedron of those data and (157.32) is its complete canonical form. Without derivation of the complex, proof of the pushforward hypotheses and coverage, the object remains a candidate formalization; it is neither a theorem attributed to the corpus nor a surjectivity assertion.

157.3.12 Synthesis

The resulting structural multimorphology can be written as the composition

$$\begin{array}{c} \text{TPK history} \longmapsto \text{ordered cell} \\ \longmapsto \text{ordered positroidal realization} \\ \longmapsto \text{canonical form} \longmapsto \text{archive oriented by } K. \end{array}$$

Each arrow preserves different information: the TPK preserves genealogy; the Grassmannian preserves minors; the canonical form preserves the boundary after cancelling internal walls; K preserves the metric frame and recovery. The discrete structure of the continuum thus appears as a system simultaneously

admitting infinite refinement, finite positivity at every level and an archived limit without loss of the history generating it. The positroid realization, cluster charts, cubical extension and amplituhedral image are added formalizations with the precise statuses stated in each subsection; they are not projected retrospectively as theorems already proved by the corpus.

157.4 Stability under refinement and compatibility of sectorial readers

Theorem 157.13 (Unity and distribution of the contribution)**.** *The refinement-stability theorem has its primary location in the joint construction of the continuum. Its applications to the Weil form, the five terminal realizations, arithmetic–geometric coupling and the mass operator are typed corollaries. None of these applications can select refinement retrospectively or replace its proof.*

Proof. The canonical form and its cancellation are defined before any sectorial reader is applied. Every reader receives the same cellular sum and must commute with boundary and residue. The common property is proved once; each sector's specificity lies in its realization map. If a reader selected cells from its result, it would reverse the causal chain. □

157.5 Consecuencias estructurales

157.5.1 Multimorphology of the Grassmannian and amplituhedron

The term *multimorphology* here denotes a precise mathematical structure. Let $\widehat{\mathcal{X}}_{\mathrm{TPK}}$ be the enriched-state space and $p_{n+1,n}$ its truncation system. Positive multimorphological data consist of a system of typed realizations

$$\begin{aligned}
\mathfrak{P}_n &: \widehat{\mathcal{X}}_{\mathrm{TPK,n}} \longrightarrow \mathrm{Gr}_{\geq 0}(k, N_n),\\
\mathfrak{R}_n &: \widehat{\mathcal{X}}_{\mathrm{TPK,n}} \longrightarrow C^{+}_{d,n},\\
\mathfrak{C}_n &: \widehat{\mathcal{X}}_{\mathrm{TPK,n}} \longrightarrow Q^{+}_{\mathrm{con,n}},\\
\mathfrak{M}_n &: \widehat{\mathcal{X}}_{\mathrm{TPK,n}} \longrightarrow \mathcal{L}^{+}_{M,n},
\end{aligned}$$

where the codomains represent, respectively, Plücker, lattice, constitutive and mass-operator positivity. The four maps must commute with their refinement links and preserve the state coordinates used by each reading. The four codomains are not required to be equal, nor is one required to select the others.

The kinematic map

$$\mathfrak{A}_{Z,n} : C \longmapsto CZ^{\mathsf{T}}$$

composes with $\mathfrak{P}_n$ and produces amplituhedral morphology. Thus the Grassmannian organizes charts, strata and their mutations; the amplituhedron organizes kinematic images and their canonical forms. Multimorphology is the diagram maintaining these realizations simultaneously as images of the same enriched state.

Proposition 157.14 (Multimorphological coherence under refinement)**.** *If every realization commutes with truncation and its map on forms commutes with boundary and residue, cancellation of interior faces is transported to the four codomains and every degree-one amplituhedral image. The expressed forms are depth-independent, although their archives preserve refined routes.*

Proof. The source state's form is the oriented sum of its cells. Commutation with truncation identifies the two representations of each refinement; commutation with boundary and residue transports cancellation of opposite incidences. A degree-one kinematic map preserves the form's multiplicity in each interior. Therefore, only external boundaries survive in each realization. □

The existence of $\mathfrak{R}_n$ and $\mathfrak{M}_n$ rests on the lattice and material reference constructions already obtained. The moment-curve realization $\mathfrak{P}_n$ is exact as an ordered formalization; promotion to an intrinsic positroid

atlas requires boundary measurement. The amplituhedral realization additionally requires rank, degree and coverage. This is multimorphology's evidentiary boundary, not a restriction of its definition.

157.5.2 Continuum

Passage to the limit does not add disconnected points: it stabilizes compatible forms of the same refined structure. Internal walls disappear from the global form but remain in the memory of histories. This yields an exact distinction between geometric invariance and genealogical conservation.

157.5.3 Riemann–Weil

Telescoping makes realization of the arithmetic form independent of the nonadic mesh. Weil positivity continues to depend on the gamma, prime and polar channels and on the mixed-moment identity turning them into a Gram form. The new result does not replace that proof: it proves that no interior refinement pole can introduce a spurious negative direction.

157.5.4 Five terminal realizations

A terminal reader transports cancellation if it commutes with boundary and if the residue of the transported form coincides with the face's form. This condition unifies the gluing mechanism without disaggregating the continuum or turning the five problems into five generators.

157.5.5 Coupling and flow

The holographic constant of arithmetic–geometric coupling recovers K, and K determines O_K, G_K and the archive. The general action $D_K(s)$ opens a positive route. The centred direction $u_K = P_3 P_{11} K$, satisfying $u_K \perp \mathbf{1}$, refines it to

$$g_{K,\rho}(\tau) = \operatorname{diag}\big(\rho_A^{\tau \widehat{u}_{K,1}}, \dots, \rho_A^{\tau \widehat{u}_{K,12}}\big) \in SL(12,\mathbb{R})_{>0}.$$

The ratios φ^{-2}, π/φ^2 and φ/π^2 can parametrize different flow sections after being generated and typed as expressions of the coinductive fourfold relation $(\pi, \varphi, e, \alpha)$; none selects the TPK retrospectively or is confused with κ_{ag}.

157.5.6 Particles and masses

The congruence

$$M_M^{\beta,r} = R_{\mathrm{act}}^{B_M^r/2} M_M^{(0)} R_{\mathrm{act}}^{B_M^r/2}$$

preserves the positive cone. The electronic exception precedes this evaluation: it arises from the unique fixed point $(5,5)$. The boundary 111111 and signature 555555 remain distinct registers of the same genealogy.

157.6 Criterion for promotion to a positroid atlas and amplituhedron

Promotion is completed when the following are concretely constructed:

1. a planar network $\mathcal{N}_z$ for each cylinder, with positive weights derived from the TPK;
2. the boundary-measurement matrix and its LGV formula for all minors;
3. the decorated permutation or Grassmann necklace of each stratum;
4. a seed or plabic graph whose mutation commutes with refinement and memory;
5. the kinematic map $C \mapsto CZ^{\mathsf{T}}$, with rank, dimension, degree, coverage and boundary correspondence;
6. the canonical-form pushforward identity.

These conditions are not a sociological list: they are the falsifiers distinguishing a complete realization from a collection of positive points.

157.7 Theorem of closure of genealogical spiral correspondences

The central contribution is a law of geometric stability under refinement: depth adds memory and resolution, while the canonical form preserves the total boundary. The register K makes this process recoverable and metric; its centred direction organizes a determinant-one flow respecting Plücker signs. The electron occupies the unique equivariant centre of the material branch. With these components, passage to the positive Grassmannian ceases to be an analogy: it has an exact ordered realization and a promotion programme whose maps and falsifiers are fully specified.

Chapter 158

Conclusions on the generative construction and its realizations

The development establishes a constructive relationship between modular arithmetic, oriented dynamics, value generation, exceptional incidence and information recovery. The unity of this relationship resides in the enriched state produced by APP–TRIT–TPK. Each realization receives a determined part of its structure and preserves the data required by the next operation. Numerical evaluation, geometric form and the record of an execution are thus articulated through explicit maps.

The principal conclusions concern three inseparable mathematical questions: how a value is produced, what structure its production preserves and under which operations it can be recovered. The realizations of Dimensional Mechanics add a fourth determination: how that structure is expressed in observables with declared units and coupling conditions.

158.1 Arithmetic restitution and memory of operations

The initial APP construction fixes two levels: position on the grid and the arithmetic evaluation performed there. Nonadic reduction preserves the integer exactly when it retains remainder and quotient. Balanced ternary reduction has the same property with its oriented representative and extension quotient. The composition identities prove that both registers can be transported between the two decompositions.

In particular, restitution $n = \rho_9(n) + 9q_9(n)$ and compatibility $c_3(n) = c_3(\rho_9(n)) + 3q_9(n)$ allow integer information to be tracked through changes of representation. Addition carries and cross-terms of multiplication are computed on those coordinates. Recoverable information thus has, from this stage onward, a determined algebraic content 1.3, 1.22.

Composition of trajectories adds ordered memory. If a transition acts through $U_\varepsilon(m) = C_\varepsilon m + \kappa_\varepsilon$, concatenation satisfies

$$C_{\beta\alpha} = C_\beta C_\alpha, \qquad \kappa_{\beta\alpha} = C_\beta \kappa_\alpha + \kappa_\beta.$$

The second trajectory transports the accumulated memory before adding its increment. This cocycle law explains why the chronological register contains more information than an aggregated carry. Its proof fixes the operation that must be preserved when composing realizations 6.22.

The nonadic calendar supplies another exact distinction. The return of nine phases increments the turn coordinate. The extension $C_{12} \to C_{108} \to C_9$, with its carry, expresses compatibility between the visible phase and the dodecaphase register. The enriched continuation also preserves successive turns and incidences. Periodicity of a reading and evolution of its memory therefore constitute two simultaneous properties of the same transport.

Transfer between quadratic and cubic completions supplies another exact conservation identity. In the nonadic case,

$$(72, 27) \longmapsto (73, 27) \longmapsto (10 \cdot 73 - 1, 10 \cdot 27 + 1) = (729, 271).$$

The first operation incorporates the registered unit; the second changes the scale and redistributes one unit through a zero-sum correction. The normalized sum remains equal to one. This construction gives precise content to the holographic extreme and mean ratio and its conservative extension across scales 146.1.

158.2 Determination of the regions and the coinductive fourfold relation

Finite enumeration identifies the admissible configurations, emissions and words on which the lifts act. The chain $w_6 \to w_{12} \to w_{18} \to w_{24} \to w_{30} \to R_{36} \to G_9$ preserves prefixes and transport conditions when extending the regional realizations. Cylindrical compatibility and contraction provide the passage to arbitrary depth.

The regional outputs π_{HMT}, φ_{HMT} and e_{HMT} combine a positional expansion with an operator provenance. The classification of the five regions of π, the incidence of the φ axis, the joint continuations and the mirror involution preserve their distinctions in the limit. Subsequent analytic identifications certify properties of those outputs and allow their comparison in established mathematical languages.

The nonadic connection relates regional generation to return structure. Modal incidence arises from the tritic selector and its sequence of operations. Its powers generate recurrences; regional marks are then applied to the return support. This precedence distinguishes the regional emitter, the recurrence of a modal realization and the evaluation of a boundary ratio. Composition makes their relationships computable without identifying objects of different types.

The register K is determined from the regional bands. The first zero advance of the conversion clock locates the nonadic antecedent; oriented defects and the phase–charge reader produce the descriptor. The terminal repertoire, ternary cylinders and exceptional incidences delimit the candidates. The heterotypy condition selects the register in that finite domain. Subsequent inversion recovers its order, windows and signed realization. Uniqueness expressly preserves the candidate domain and selector conditions.

The integer fine-structure closure composes regional readings with the rational coordinate of K. Its compatible carry determines a unique infinite section, and the tail identity distinguishes the boundary of an isolated window from the restriction of the limit. In the notation of its development,

$$\alpha_{\mathrm{HMT}} = \pi_{\mathrm{HMT}} + e_{\mathrm{HMT}} - \varphi_{\mathrm{HMT}} - 4 - \kappa_{\mathrm{per}}.$$

The equality preserves the origin of all its terms and the complete decimal normalization (11.135).

The analytic representation develops that same pre-coordinate through a function defined in all degrees. If a is the generated pre-coordinate and $q = 1/729$, the linking coefficient

$$\lambda = -\frac{E_9(a)(1 - qa^3)}{a^{10}}$$

determines the geometric extension $E_\infty(X) = E_9(X) + \lambda X^{10}/(1 - X^3/729)$. Subsequent coefficients satisfy $b_{10+3n} = q^n\lambda$ and $b_{11+3n} = b_{12+3n} = 0$. Uniqueness is proved within this class of extensions, and the root is identified with the common coordinate. The second route contributes a correlated analytic representation with convergence control and projective compatibility 11.6.4.2.

158.3 Archimedean realization and exceptional incidence of the continuum

The truncation systems preserve admissible histories and their registers. The proved extendibility guarantees the existence of compatible sections. Cylindrical partitions provide an Archimedean evaluation; the incidence structure remains in the conjugate projection. Contraction determines the real value, while orientation, boundary and memory preserve the distinctions of the state.

This double projection jointly organizes spectral transport, solenoidal balance, gauge incidence, cohomological coherence and the determinant line. Naturality identities allow refinements and realizations to be carried out in compatible orders. The discrete continuum thus receives a common operational structure, on which its subsequent evaluations act.

Exceptional incidence is developed through a sequence of concrete constructions. Codes and markings determine lattice gluings; the Leech neighbour is proved positive, even, unimodular and rootless in the declared metric and chamber. Lattice identification uses these properties jointly. The construction of

oscillators and the twisted lattice algebra, followed by the orbifold parity sector, determines the realization of $V^\natural$ and its Monster action 21.71.

The significance of this sequence lies in the related genealogy of value and symmetry and their precise objects of action. Exceptional symmetry acts on the corresponding code, lattice or algebra. Dimensional reductions and duality realizations receive those carriers with their conditions. Numerical refinement and exceptional extension thus retain their linking maps within the common state.

158.4 Bilateral conservation and reversibility at arbitrary depth

The bilateral law supplies an operator conclusion independent of dimension. For two isometries $\mathcal{B}, C : H \to H'$ and $a > 0$, define $Q_- = a\mathcal{B} - C$ and $Q_+ = (a+1)\mathcal{B} + C$. The identity

$$(a+1)Q_-^*Q_- + aQ_+^*Q_+ = \big((a+1)^3 + a^3\big)I$$

follows by cancelling the cross-terms. Its validity includes infinite-dimensional Hilbert spaces and transports between distinct carriers with common domain and codomain.

In the nonadic specialization, $a = 8$, the normalization factors through the weights $72/73$ and $1/73$, followed by a rotation of the two channels. The resulting isometry has a left inverse of norm one. Recovery therefore maintains optimal control of the joint error in the declared norm. The factor $73 = 72 + 1$ enters into that identity; its other realizations as a norm or lattice index retain their own proofs.

The transport of observables specifies the content of conservation. For a reader G and a unitary relative transport R, the induced reading is

$$\mathcal{T}_a(G) = \frac{a(a+1)G + R^*GR}{a(a+1)+1}.$$

The equality $\mathcal{T}_a(G) = G$ is equivalent to commutation $GR = RG$. This yields a relative-symmetry criterion distinguishing invariant observables from those that change during transduction 147.3.

The recording policy at arbitrary depth preserves the continued component and its complement at each stage. The telescopic balance produces a finite isometry and, under the proved contraction bound, an infinite-record isometry. For $a = 8$, the uniform rate is $r_8 = 729/1241 < 1$. The first N complements recover the state with truncation error bounded by $r_8^N \|v\|$; a joint error ε adds exactly ε to that bound. The complete record admits the convergent inverse of Theorem 147.7.

The evolutionary conservation of unity is therefore expressed through local identities, isometric compositions and recovery in the limit. Modification of form or carrier is compatible with restitution of the state when the channels and incidences specified by the theorem are preserved.

158.5 Geometric reciprocity and exterior conservation

The coupling register produces a direction orthogonal to the uniform mode. Its positive diagonal flow has determinant one. The radii vary compensatorily; Plücker minors preserve sign and vanishing, and balanced quotients preserve their value. The same direction generates a unit-covolume deformation on the lattice planes.

The reciprocity of this deformation is expressed by $\Lambda_t^* = \Lambda_{-t}$. The commuting ternary symmetry persists along the flow, while integrality and exceptional realization retain the specific domain in which they were proved. Theorem 150.4 determines these properties simultaneously.

In the chart $s = (1-z)^{-1}$, radial inversion $z \mapsto 1/\overline{z}$ becomes $s \mapsto 1 - \overline{s}$. The unit circle corresponds to the line $\Re s = 1/2$, with the singular point excluded. Theta–Mellin reciprocity, polar structure and the Poincaré and Klein realizations preserve the analytic and geometric conditions of their own readers. Recorded orientation allows the radial reading to be inverted wherever the scalar reading combines conjugate realizations.

These identities broaden the meaning of the coupling constant: the arithmetic register determines a recoverable frame, a deformation direction and transports preserving exterior invariants. The correspondence between realizations is obtained by composing the constructed maps.

158.6 Action, angular geometry and material observables

The Dimensional Mechanics realization preserves an explicit precedence. The action scale receives the fine-structure coordinate and the regional reading of φ; the local norm of order sixteen fixes its decimal valuation. The forward and return Planck sections preserve their quotient R_{act}. The angular pair and its conjugate channels are then evaluated, with their orientation and units 14.11.3.

The marked inclusion of a hexad in an octad produces incidences of cardinalities 90 and 120. The dodecaphase chain determines the multiplicities of the Barbero–Immirzi functional and the pole coefficient $1/24$. Its evaluation uses the angle, counter-angle and previously constructed channels. Parity under angular inversion preserves the orientation of that reading 15.5.

Angular transport toward Witt incidence supplies an explicit composition with the mixing reader. Its left inverse and Gram identity preserve the transported angular information 20.14. The electric and magnetic constitutive structure likewise receives its modes, return ratios and scales; its identities allow the angular coordinates to be reconstructed in the specified domain.

The contragredient gravitational section of action satisfies

$$G_a = \frac{c_{\mathrm{int}}^3 R_{108}^2}{\hbar_a}, \qquad \frac{G_a \hbar_a}{c_{\mathrm{int}}^3} = R_{108}^2.$$

The coincident-radii condition fixes this section within the declared class. Inversion of the action ratio compensates the gravitational ratio and preserves areal unity 80.3.8. The Boltzmann definition then relates energy per decision to the thermal section. For the clock of one hundred and eight events, $k_B \Theta_{\mathrm{clk},a} = h_a/(108 t_0 \log 3)$. Individualization of the transducer preserves the thermal section and unit bases it uses 48.5.2.

Material observables arise from the persistence of extensions and their route records. The electronic centre has a point selection determined by the atlas; the other classes use the operators of their fibres, orbits or multisections. The integer signature, character and towers transmit this information to the mass realization. Metrological comparison is carried out on the specified evaluations and couplings, preserving the distinction between operator, spectrum and individualized scalar.

158.7 Reconstruction and cardinal classification of the generated continuum

The cylinder construction proves coverage of values with recovery of registers. Cardinal closure adds a classification of those values in the genealogical closure $\mathfrak{G}$ determined by its term semantics and definability. Internal countability of the levels and the condensation lemma assign to each rational cut its first stage $\beta(r)$ and its first index $n(r)$ in that stage's enumeration.

The map

$$j_{\mathrm{HMT}}(r) = \omega\beta(r) + n(r)$$

is injective on the domain of the generated line $\mathbb{R}_{\mathrm{HMT}}^{\mathfrak{G}}$. Encoding countable well-orders constructs the injection of $\omega_1^{\mathfrak{G}}$ into that same line. Cantor–Bernstein concludes

$$\left|\mathbb{R}_{\mathrm{HMT}}^{\mathfrak{G}}\right| = \aleph_1^{\mathfrak{G}}, \qquad \mathfrak{G}_{\mathrm{HMT}}(h, m_\infty) \models \mathrm{CH}.$$

This is the affirmative closure of the continuum hypothesis established in Theorem 19.27.

The conclusion encompasses every internal subset of that line: each is countable or has the cardinality of the generated continuum. The representation $\mathfrak{G}_{\mathrm{HMT}}(h, m_\infty) = L[h, m_\infty]$ identifies the construction's

domain after developing its genealogy. Transfers to a different set-theoretic domain retain as hypotheses the coverage and coincidence of the objects on which the injections act. The formulation thus keeps the cardinal result, its semantics and its quantifiers together.

The spectral, solenoidal, gauge, cohomological and determinantal terminal realizations are articulated through the same discrete structure. Their theorems specify the operator, positivity, coercivity or descent used by each realization, together with the corresponding naturality identities. The joint factorization of Chapter 95 preserves that origin and the domain of each conclusion.

158.8 Demonstrative content and scope of the integration

The demonstrative strength of the work arises from composition between results of different types. Symbolic identities prove equalities for their quantified domains. Exhaustive enumerations resolve finite selectors and censuses. Rational bounds control approximations and roots. Compatibility, convergence and naturality arguments justify extensions and passages to the limit.

Lean formalizations verify the specific declarations they contain, with their types, premises and dependencies. Executable calculations certify the instances or finite domains defined by their algorithms. Correspondence between mathematical statement and certificate assigns each check its precise scope. The physical realization also preserves the dimensional basis and the coupling rule with the observable.

The overall result is an operational genealogy: discrete arithmetic produces oriented states; their extensions determine constants and a continuum with incidence; their registers admit conservative transformations and recovery; their dimensional realizations provide structured observables. The unity of the development is expressed in these demonstrated relationships. Value, symmetry and recoverable information are linked by the construction producing them.

Appendix A

Algebraic and Combinatorial Certificates of Finite Objects

The finite objects used in the proofs admit a second channel of verification. This channel reconstructs catalogs, operators, and invariants from the declared seeds and checks them against exact identities or directed bounds; it complements the proofs but does not replace them.

A.1 Finite Catalogs and Certificates of Reproducibility

The following finite objects allow the verifications to be repeated without introducing production notation into the mathematical exposition. Each control preserves its corresponding domain, operation, invariant, and falsifier.

A.1.1 Finite catalogs of the common state

Finite object	Verifiable mathematical content
Catalog of length-six emissions	The 468 emissions U_6, their multiplicities, orientation, and additive and multiplicative signatures.
Spectral partition of the emissions	The twenty-six classes of size 18, together with the explicit map that reconstructs each emission from its class and its internal position.
Dodecaphasic Atlas of Routes	The 468 minimal routes with sheet, phase, carry, boundary, and memory.
Archimedean Reader Catalog	The five regions of π, the conjugate regions of e, and the self-scaling axis of φ, with their cylinders and multiplicities.
Hensel Flow of Twenty Blocks	The unimodular matrix, its integer inverse, the 3-adic states, and the residue–quotient recurrence that prolongs the three channels.
Nonadic monodromy record	The twenty blocks, their phases, ternary prefixes, and decimal prefixes; includes the 43 complete returns and the 387 pairs $(K, K+9)$ per channel.
Ventana of survival $K = 5, \ldots, 13$	The local candidates, the compatible branch, and the three outputs of the transition $9 \to 1$, without identifying phase return with state reset.
Initial Archimedean reading	The twelve decimal triples obtained from the first thirteen blocks of each channel, separated from the certificate of arbitrary precision.

A.1.2 Algebraic, Operatorial, and Spectral Identities of Finite Certificates

Withtrol	Identities and Forgers
Monodromy and generation of constants	It verifies $w_6 \to w_{30} \to R_{36} \to G_9$, the selection $3 \to 1$, the change of state after nine phases, the five regions of π, and the compatibility of truncations. It fails if a target digit is introduced before the reader.

Withtrol	Identities and Forgers
Conjugate angular coordinate	Verify the regional selector 169, the determinant recurrence, the analytic sum and its remainder bound, and the uniqueness of C^*. The test fails if an SI constant intervenes in the selection.
Quotient dynamics $468 = 18 \cdot 26$	It certifies the bijection between emissions and classes, the uniform return, the measure of the averaged shadow, the mode cocycles, and the microscopic lift. It distinguishes the chronological frequency from the Parry measure.
Integral channels of the constants	Verifies the identities $(729, 271, 315, 161)$, the set-theoretic realizations of $271/315$, the dihedral orbits, and the factorization of visible parity and carry parity.
Witt Incidence	Reconstruct the point–hexad incidence, the module E_{30}, the positive transport of P_3, the Witt star, and the equivariance obstructions.
Variational selector	It reconstructs the normalizer S_3, the Reynolds average, the polar factorization, and the chain of rank-three projectors; positivity is proved exactly in $\mathbb{Q}(\sqrt{5})$.
Elevation $W_{12} \to W_{24}$	It checks the four- and five-point incidences, the unique lift of hexads, and the typed correspondence with the five regions of π.
Realizaciones de $-1/12$	It constructs the rational projectors of the step-three and step-four shifts, verifies symmetry and idempotence, and obtains $(3-4)/12 = -1/12$ as the difference of normalized traces.
Schläfli Holonomy	Traverses the plaquettes, certifies $A^2 = 5I - 4A + 5J$, the configuration of the twenty-seven lines, their 45 triangles and the partition $729 = 27 + 270 + 432$.
Spectral functionals	Recomputes Catalan, Stieltjes, Apéry, Bendersky–Glaisher–Kinkelin, the two Euler–Kronecker extensions, and the directed bounds on prime products.
Vacancies and Criticality	It checks the separations $21/22$, the returns $6/7$, the finite harmonic part, and the dodecaphase approximants of the critical mode.

A.1.3 Typed Dictionary of the Catalogues

`Field`	Meaning and Codomain
`Esig sig`	Signature of the multiplicative channel; it identifies one of the twenty-six classes in the factorization of the emitter.
`hplus_type`	Additive channel orientation class for the two pairs of declared cardinal directions.
`diag_c`	Integer index of the diagonal class of the additive cursor.
`px_size`	Cardinal of the fiber of the corresponding additive class.
`colw`	Sextet of column weights of the emitted block; it belongs to $\mathbb{Z}^6_{\geq 0}$.

A.1.4 Reproducibility of Certificates and Exact Obstructions of Finite Constructions

The controls use integer or rational arithmetic whenever the object permits it. The analytical bounds use intervals with directed rounding and separate the symbolic proof from the decimal comparison. Each run reconstructs the objects from the declared seeds, verifies the incidence and naturality identities, and checks the outputs against an independent arithmetic implementation.

The native-product control rejects every dependence on a decimal evaluation, an external factorization, a sieve, a table of zeros, or a target constant. The nonadic control rejects identification of the ambient fiber of 729 elements with the cardinality of a space of histories. The phase control rejects $g(K + 9) = g(K)$ as proof of equality of states. The metrological control rejects every route selection performed after an observed magnitude has been consulted.

A discrepancy among an identity in the text, the finite catalog, and its independent control refutes the corresponding certification. This rule replaces no proof: it fixes a second verification channel for the finite objects that intervene in it.

A.2 Global compatibility among types, domains, morphisms, and certificates

The integral control checks the causal chain, repeats the mathematical identities on the finite objects, and compares the exact realizations with their independent formulations. A divergence between a definition, its mathematical table, and the declared invariant falsifies the corresponding control.

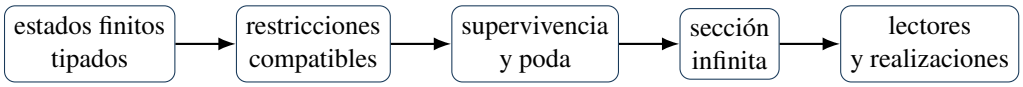
Appendix B

Exact Dependencies Between Generational Power, Reversible Information, and Gödel–Turing Obstruction

B.1 Logical Dependencies and Consistency Certificates

B.1.1 Causal chain of the certificates: finite states → restrictions → survival → infinite section → realizations

The logical chain of the integration does not begin from an Archimedean conclusion. It is organized in the following order:



A digit certificate verifies a restriction of the last arrow; it does not retrospectively create the states or replace the existence theorem for the limit.

B.1.2 Exact arithmetic identities of finite certificates

The censuses accompanying the nonadic filtration certificate satisfy

$$396 \cdot 6 = 2376, \qquad 43 \cdot 9 = 387, \qquad 432 + 4 \cdot 144 = 1008.$$

The first equality relates elementary boundary events and trits emitted per channel; the second counts the phase pairs $(K, K+9)$; the third reconstructs the total multiplicity of the five regions of π.

The two regional roles that cannot be permuted are

$$\begin{aligned} U_{\pi}^{\perp} &= (501, 614, 498, 169, 272, 272), \\ E_{\mathrm{sig}} &= 555555, \qquad \text{multiplicity } 432, \\ U_{\pi}^{--} &= (870, 923, 810, 418, 674, 521), \\ E_{\mathrm{sig}} &= 933717, \qquad \text{multiplicity } 144. \end{aligned}$$

B.1.3 Existence and uniqueness of the infinite branch in a finitely ramified tree of surviving prefixes

Lemma B.1 (Finitely branching tree)**.** *Let T be a tree with nonempty finite levels such that every surviving node has a surviving descendant. Then there is an infinite branch. If, moreover, each node of the selected subtower has exactly one compatible descendant, the branch is unique.*

Proof. Existence is König's lemma applied to the finitely branching tree. For uniqueness, two distinct branches would have a first level at which they diverge, yielding two compatible descendants of the same prefix, contrary to the hypothesis. □

Structural obstruction

An execution to depth N proves only what occurs in the truncated tree. The infinite claim requires the uniform law that preserves nonemptiness and compatibility at every level. The main text expressly separates the two statuses and, for the nonadic filtration, retains its certified uniform selector.

B.1.4 Closure and Full Power

For $A \subseteq X = \varprojlim X_n$, it has been proved

$$\overline{A} = \bigcap_n p_n^{-1}(p_n(A)).$$

The set of eventually null binary sequences is dense and proper in $2^{\mathbb{N}}$, yet it has all finite shadows. Therefore any algorithm that receives only $(p_n(A))_n$ cannot distinguish A from its closure. This example blocks the false identity

$$\varprojlim \mathcal{P}(X_n) = \mathcal{P}(\varprojlim X_n).$$

B.1.5 Typed compatibility of operators

The difference between the unital inclusion and the marked corner is verified directly:

$$\tau_{d+1}(a \otimes I_9) = \frac{\mathrm{Tr}(a)\,\mathrm{Tr}(I_9)}{9^{d+1}} = \tau_d(a),$$
$$\tau_{d+1}(a \otimes p_j) = \frac{\mathrm{Tr}(a)}{9^{d+1}} = \frac{1}{9}\tau_d(a).$$

The first branch is unital and tracial; the second preserves a marked corner and cannot be used to infer the factor II_1.

B.1.6 Cancellation of the Landauer cost in reversible state inversion

For a complete variable X and a projection $Y = q(X)$,

$$H(X) = H(Y) + H(X \mid Y).$$

If q is bijective, the fiber is a singleton and $H(X \mid Y) = 0$. If q is a reset of the uniform TRIT, the discarded information is $\log 3$ and the minimum cost is $k_B\Theta \log 3$. For the extension $\widehat{E}_N$ to L^1 of a normal trace-preserving conditional expectation E_N, and under the stated finiteness hypotheses,

$$S_\tau(\widehat{E}_N h) - S_\tau(h) = D_\tau(h \| \widehat{E}_N h) \geq 0.$$

The two formulas locate the same phenomenon in different domains: the cost belongs to the contraction of a fiber, not to the reversible evolution of the complete state.

B.1.7 Gödel–Turing Obstruction to a Uniform Projective Multiplicity Decider

Let T_e be the machine with index e. At level n, define

$$X_n^e = \{b_n\} \cup \{c_n : T_e \text{ did not halt before } n\}.$$

Each X_n^e is computable. The global fiber has two branches if T_e does not stop and one if it stops. Therefore

$$U_e : \quad \#q_\infty^{-1}(v) = 1 \quad \Longleftrightarrow \quad T_e \downarrow .$$

A uniform decider for U_e would decide HALT. A recursively axiomatizable theory that is sound and complete for every U_e would permit their decision by enumerating in parallel proofs of U_e and of its negation. This is the core of the *HMT theorem of uniform incompleteness for projective multiplicities*; its proof uses a Godel–Turing reduction. The uniform obstruction does not negate particular selectors constructed on calibrated HMT images.

Scope of the uniform incompleteness control

The Gödel–Turing reduction establishes the uniform obstruction for projective multiplicities. It does not negate the particular selectors constructed on calibrated HMT images, nor does it turn a global decision obstruction into indeterminacy of every concrete fiber.

B.2 Domain and Obstructions of Physical and Dimensional Realizations

The physical-dimensional realization is subject to ten families of control:

1. cylindrical concentration, unit mass, and finite moments;
2. spectra $(9, 5)$ of the APP block and $(7, 3)$ of the memory block;
3. identity $\tanh^2(2\log\varphi) = 5/9$;
4. antiunitarity in the stated finite model;
5. Poincare semidirect-product law;
6. constitutive identities of the vacuum;
7. split norm of the null modes and material closure;
8. selector $\theta_* = (54/\pi)^2$ and self-dual fixed point;
9. control CGLMP values;
10. invariants $(396, 729, 43, 387)$ of the nonadic filtration and domain $(81, 56, 13, 324)$, with separate null sector, of the General Law of Masses.

Exact scope of the controls

The ten families verify identities, finite domains, and reproducible examples. They do not replace complete functoriality of the four components, do not by themselves construct the common domain, do not prove the smooth Cartan prolongation, and do not promote open physical interfaces to internal theorems. The discrete realization retains its principal proof in section 62.2.1.

B.2.1 Functorial dependencies of the spectral, solenoidal, gauge, cohomological, and determinantal realizations

The preceding controls apply only after APP, TRIT, TPK, the common enriched state, the double projection, and the dimensional line have been constructed. The cylindrical concentration and finite spectrum prove properties of the realization; they do not retrospectively select the state. The periodicity of 108 phases, the atlas (81/56/13/324), the constitutive involution, and the cosmological programs retain separate domains and falsifiers.

The Dirac, von Neumann, CGLMP, Cartan, and Poincare identities are used as typed controls of the corresponding intertwiner. A numerical coincidence without the map specifying the domain, codomain, and action on generators does not constitute a physical realization.

Appendix C

Genealogical fidelity of the matrix bridge: domain, correspondence, and exact limits

C.1 Naturality, Positivity, and Compatibility Identities of the Genealogical–Matrix Bridge

The following controls are obtained by reconstructing the operators from their incidence data. Rational identities are calculated exactly; the floating-point comparisons state their error bound.

Verification	Status	Maximum error or identity
cuatro PVM qutrit	verified	4.72×10^{-16}
idempotency of the twelve projectors	verified	3.34×10^{-16}
solapamientos MUB	verified	$\lvert\mathrm{Tr}(PQ) - 1/3\rvert < 3.34 \times 10^{-16}$
QLS column obstruction	exact	minimum $1/3$
QMS aditivo $(9, 3)$	verified	exact rows and columns to tolerance
Three-context QMS $(9, 3)$	verified	numerical error $< 4.21 \times 10^{-16}$
noncommutativity of cells	verified	maximum norm $0.074074\ldots$
QMS atlas $(12, 3)$	verified	numerical error $< 4.24 \times 10^{-16}$
semiclassical decomposition	exact	zero reconstruction error
qutrit cube of order nine	verified	line error $< 4.21 \times 10^{-16}$
qutrit cube of order twelve	verified	line error $< 4.24 \times 10^{-16}$
$\mathbb{P}^1(\mathbb{Z}/9) \to \mathbb{P}^1(\mathbb{F}_3)$	exactly	$12 \to 4$ con fibras $(3, 3, 3, 3)$
truncation–compression	exact	conmutador cero
Direct reader $1/\pi$	exactly	unitary normalization
calibre MUB $3/\pi$	verified	probabilidad $1/\pi$, error $< 3.85 \times 10^{-16}$
Gram polynomial APP	exactly	$t^5(t-1)(t-5/7)(t-1/3)^2$
four Halmos intersections	exact sciences	$(64, 1, 5, 5)$
algebra C^* of APP in $d = 2$	exact	$\mathbb{C}^4 \oplus M_2 \oplus M_2$, dimension 12

The tolerance of the numerical controls is 2×10^{-10}. The rational assertions use no tolerance: they are obtained by exact fractions and Gaussian elimination over $\mathbb{Q}$.

C.2 Exact coefficients of the polynomial

The construction produces

$$\det(tI - CC^*) = t^9 - \frac{50}{21}t^8 + \frac{124}{63}t^7 - \frac{2}{3}t^6 + \frac{5}{63}t^5, \tag{GMC-A.1}$$

and coincides coefficient by coefficient with

$$t^5(t-1)(t-5/7)(t-1/3)^2. \tag{GMC-A.2}$$

The verification constructs CC^* from the exact cardinalities of the fibre intersections and applies Faddeev–LeVerrier over $\mathbb{Q}$; it does not depend on a numerical diagonalization.

C.3 Obstructions to the identification of undemonstrated equivalences in the matrix bridge

The finite certificate does not replace nonadic filtration to arbitrary depth, the General Mass Law, or the Planck derivations. Nor does it prove by itself APP–TRIT–TPK faithfulness of every cyclic QMS, complete equivariance of the calibration Ξ, identification of TPK memory with an auxiliary Stinespring system, infinite divisibility of an MPS channel, or categorical universality. Each of these assertions retains its principal residence and its own falsifiers.

Appendix D

Domain of seeds, emissions, and verifiable expansions

This block preserves the two levels supporting numerical publication: the exhaustive census of initial conditions and TPK emissions, and the extensive expansions produced by the rational readers. The complete catalog fixes the passage

$$104\,976 \longrightarrow 468 \longrightarrow 243,$$

whereas the ten-thousand-digit expansions document the effective prolongation of the sections of π, e, φ, and α_{HMT}.

The catalog of the (468) emissions and the (243) visible words is published in full below; the four expansions of ten thousand digits are published in full in the genealogy of constants. This terminal section links both developments without reducing their proofs, tables, or labels.

D.1 APP Seed Domain and Canonical Emission Catalogue

D.1.1 Exhaustive domain of initial conditions

Let

$$I_9 = \{0, 1, \ldots, 8\}, \qquad D = \{N, E, S, O\}, \qquad \mathcal{S} = I_9^2 \times D.$$

An element of $\mathcal{S}$ fixes the initial position and orientation of a cursor. The minimal emitter uses two cursors on the same APP support: one for additive evaluation and the other for multiplicative evaluation. Its complete domain is

$$\Omega_{\mathrm{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times \cong \mathcal{S}^2, \qquad |\mathcal{S}| = 9^2 \cdot 4 = 324, \qquad |\Omega_{\mathrm{seed}}| = 324^2 = 104\,976.$$

Definition D.1 (Complete seed of the APP–TPK emitter)**.** A seed is an ordered pair.

$$\omega = (s_+, s_\times), \qquad s_+ = (i_+, j_+, d_+), \quad s_\times = (i_\times, j_\times, d_\times), \qquad s_+, s_\times \in \mathcal{S}.$$

The 104 976 seeds are exactly the elements of Ω_{seed}. The catalog rows are of emission type, and each represents a fiber of this seed domain.

The fifty-four-step recurrence defines the finite application.

$$T_{\min} : \Omega_{\mathrm{seed}} \longrightarrow \mathcal{U}_6^{\mathrm{cat}}, \qquad \omega \longmapsto U_6 = (U_1, \ldots, U_6).$$

Its image contains 468 distinct decimal sextets. The componentwise reduction then produces 243 distinct words in the ambient $\mathbb{F}_3^6$, of cardinal 729. Therefore, the three levels are typed as

$$\underbrace{104\,976}_{\text{seeds } \omega} \xrightarrow{T_{\min}} \underbrace{468}_{\text{emissions } U_6} \xrightarrow{\rho_3} \underbrace{243}_{\text{visible words}} \subset \underbrace{\mathbb{F}_3^6}_{\text{729 words}}.$$

Theorem D.2 (Census of the domain and of the image)**.** *The canonical catalog contains* 468 *distinct emissions* U_6*, and* $\#T_{\min}^{-1}(U_6)$ *is the cardinality of the fiber* $T_{\min}^{-1}(U_6)$*. Their multiplicities satisfy*

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976,$$

so that the sum of all fibers exactly recovers the domain Ω_{seed}*. The direct projection* $\rho_3(U_6)$ *has* 243 *distinct values.*

Proof of the exhaustive census $104\,976 \to 468 \to 243$ **and of the multiplicities of the fibres of** $T_{\min}$ The recurrence defining $T_{\min}$ resides in algorithm 8.25; the factorization of its image and the computation of the fibres reside in theorem 8.34 y corollary 8.35. This appendix does not introduce a second enumeration: it realizes, with all its fields, the canonical catalog produced by those results.

D.1.2 Complete Catalog of the 468 Emissions

The following table incorporates the ten fields of the exhaustive catalogue. The ordinal in the first column establishes a reading order, with a type distinct from that of the seeds and mathematical identifiers. The abbreviated columns preserve `diag_c`, `hplus_type`, `px_size`, `px_res_type`, `delta_diag_subset`, `delta_orbit`, `Esig`, and $\rho_3(U_6)$.

Table D.1: Complete canonical catalog of U6 emissions.

#	U_6 / $\rho_3(U_6)$	$\#T_{\min}^{-1}(U_6)$ / diagonal_c	hplus_kind / px_dimension / px_resolution kind_type	delta_diag_subset and delta_orbit	E-sign
1	252|109|252|903|850|850 / 010011	144 / 0	NW / 6 / A	0,1,4,5 / 0,1,4,5	3,9,3,1,7,7
2	232|189|292|923|870|810 / 101200	144 / 0	NW / 6 / A	0,1,4,5 / 0,1,4,5	1,7,7,3,9,3
3	923|870|810|232|189|292 / 200101	144 / 0	EN / 6 / A	0,1,4,5 / 0,1,4,5	3,9,3,1,7,7
4	903|850|850|252|109|252 / 011010	144 / 0	EN / 6 / A	0,1,4,5 / 0,1,4,5	1,7,7,3,9,3
5	272|169|272|923|810|870 / 212200	144 / 0	NW / 8 / A	0,1,4,5 / 0,1,4,5	5,5,5,3,3,9
6	252|149|212|943|830|830 / 022122	144 / 0	NW / 8 / A	0,1,4,5 / 0,1,4,5	3,3,9,5,5,5
7	943|830|830|252|149|212 / 122022	144 / 0	EN / 8 / A	0,1,4,5 / 0,1,4,5	5,5,5,3,3,9
8	923|810|870|272|169|272 / 200212	144 / 0	EN / 8 / A	0,1,4,5 / 0,1,4,5	3,3,9,5,5,5
9	227|114|227|998|885|885 / 202200	1944 / 0	NO / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
10	292|189|232|903|850|850 / 101011	144 / 0	NO / 8 / B	2,3,6,8 / 0,1,4,6	7,7,1,1,7,7
11	232|189|292|963|850|890 / 101012	144 / 0	NO / 8 / B	2,3,6,8 / 0,1,4,6	1,7,7,7,7,1
12	998|885|885|227|114|227 / 200202	1944 / 0	ES / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
13	963|850|890|232|189|292 / 012101	144 / 0	ES / 8 / B	2,3,6,8 / 0,1,4,6	7,7,1,1,7,7
14	903|850|850|292|189|232 / 011101	144 / 0	ES / 8 / B	2,3,6,8 / 0,1,4,6	1,7,7,7,7,1
15	292|129|292|943|830|830 / 101122	144 / 0	NW / 8 / A	1,2,3,4 / 0,1,2,3	7,1,7,5,5,5
16	292|189|232|923|810|870 / 101200	144 / 0	NW / 6 / A	2,3,7 / 0,1,5	7,7,1,3,3,9
17	272|169|272|963|890|850 / 212021	144 / 0	NW / 8 / A	1,2,3,4 / 0,1,2,3	5,5,5,7,1,7
18	252|149|212|963|850|890 / 022012	144 / 0	NW / 6 / A	2,3,7 / 0,1,5	3,3,9,7,7,1
19	963|890|850|272|169|272 / 021212	144 / 0	EN / 8 / A	1,2,3,4 / 0,1,2,3	7,1,7,5,5,5
20	963|850|890|252|149|212 / 012022	144 / 0	EN / 6 / A	2,3,7 / 0,1,5	7,7,1,3,3,9
21	943|830|830|292|129|292 / 122101	144 / 0	EN / 8 / A	1,2,3,4 / 0,1,2,3	5,5,5,7,1,7
22	923|810|870|292|189|232 / 200101	144 / 0	EN / 6 / A	2,3,7 / 0,1,5	3,3,9,7,7,1
23	292|189|232|943|830|830 / 101122	144 / 0	NW / 6 / A	1,4,6,8 / 0,2,4,6	7,7,1,5,5,5
24	212|149|252|963|890|850 / 220021	144 / 0	NW / 8 / A	1,4,6,8 / 0,2,4,6	9,3,3,7,1,7
25	272|169|272|963|850|890 / 212012	144 / 0	NW / 6 / A	1,4,6,8 / 0,2,4,6	5,5,5,7,7,1
26	292|129|292|983|810|810 / 101200	144 / 0	NW / 8 / A	1,4,6,8 / 0,2,4,6	7,1,7,9,3,3
27	963|850|890|272|169|272 / 012212	144 / 0	EN / 6 / A	1,4,6,8 / 0,2,4,6	7,7,1,5,5,5
28	983|810|810|292|129|292 / 200101	144 / 0	EN / 8 / A	1,4,6,8 / 0,2,4,6	9,3,3,7,1,7
29	943|830|830|292|189|232 / 122101	144 / 0	EN / 6 / A	1,4,6,8 / 0,2,4,6	5,5,5,7,7,1
30	963|890|850|212|149|252 / 021220	144 / 0	EN / 8 / A	1,4,6,8 / 0,2,4,6	7,1,7,9,3,3
31	292|129|292|963|890|850 / 101021	144 / 0	NO / 4 / B	0,5 / 0,4	7,1,7,7,1,7
32	963|890|850|292|129|292 / 021101	144 / 0	ES / 4 / B	0,5 / 0,4	7,1,7,7,1,7
33	272|169|272|983|810|810 / 212200	144 / 0	NW / 6 / A	6,7,8 / 0,1,2	5,5,5,9,3,3
34	212|149|252|943|830|830 / 220122	144 / 0	NW / 6 / A	6,7,8 / 0,1,2	9,3,3,5,5,5
35	943|830|830|212|149|252 / 122220	144 / 0	EN / 6 / A	6,7,8 / 0,1,2	5,5,5,9,3,3
36	983|810|810|272|169|272 / 200212	144 / 0	EN / 6 / A	6,7,8 / 0,1,2	9,3,3,5,5,5
37	252|109|252|943|830|830 / 010122	144 / 0	NW / 6 / A	1,2,3,4 / 0,1,2,3	3,9,3,5,5,5
38	272|169|272|923|870|810 / 212200	144 / 0	NW / 6 / A	1,2,3,4 / 0,1,2,3	5,5,5,3,9,3
39	923|870|810|272|169|272 / 200212	144 / 0	EN / 6 / A	1,2,3,4 / 0,1,2,3	3,9,3,5,5,5
40	943|830|830|252|109|252 / 122010	144 / 0	EN / 6 / A	1,2,3,4 / 0,1,2,3	5,5,5,3,9,3
41	252|109|252|923|870|810 / 010200	144 / 0	NO / 4 / B	2,3 / 0,1	3,9,3,3,9,3
42	923|870|810|252|109|252 / 200010	144 / 0	ES / 4 / B	2,3 / 0,1	3,9,3,3,9,3
43	272|169|272|903|850|850 / 212011	144 / 0	NW / 6 / A	0,5,7 / 0,2,4	5,5,5,1,7,7
44	232|189|292|943|830|830 / 101122	144 / 0	NW / 6 / A	0,5,7 / 0,2,4	1,7,7,5,5,5
45	943|830|830|232|189|292 / 122101	144 / 0	EN / 6 / A	0,5,7 / 0,2,4	5,5,5,1,7,7
46	903|850|850|272|169|272 / 011212	144 / 0	EN / 6 / A	0,5,7 / 0,2,4	1,7,7,5,5,5
47	212|149|252|923|810|870 / 220200	144 / 0	NO / 8 / B	0,5,6,8 / 0,1,3,4	9,3,3,3,3,9
48	252|149|212|983|810|810 / 022200	144 / 0	NO / 8 / B	0,5,6,8 / 0,1,3,4	3,3,9,9,3,3
49	983|810|810|252|149|212 / 200022	144 / 0	ES / 8 / B	0,5,6,8 / 0,1,3,4	9,3,3,3,3,9
50	923|810|870|212|149|252 / 200220	144 / 0	ES / 8 / B	0,5,6,8 / 0,1,3,4	3,3,9,9,3,3
51	272|169|272|943|830|830 / 212122	432 / 0	NO / 12 / B	0,2,3,5,6,8 / 0,1,3,4,6,7	5,5,5,5,5,5
52	943|830|830|272|169|272 / 122212	432 / 0	ES / 12 / B	0,2,3,5,6,8 / 0,1,3,4,6,7	5,5,5,5,5,5
53	694|438|581|232|189|292 / 102101	144 / 1	NW / 6 / A	0,3,4,8 / 0,1,4,5	3,9,3,1,7,7
54	674|418|521|252|109|252 / 212010	144 / 1	NW / 6 / A	0,3,4,8 / 0,1,4,5	1,7,7,3,9,3
55	252|109|252|674|418|521 / 010212	144 / 1	EN / 6 / A	0,3,4,8 / 0,1,4,5	3,9,3,1,7,7
56	232|189|292|694|438|581 / 101102	144 / 1	EN / 6 / A	0,3,4,8 / 0,1,4,5	1,7,7,3,9,3
57	614|498|501|252|149|212 / 200022	144 / 1	NW / 8 / A	0,3,4,8 / 0,1,4,5	5,5,5,3,3,9

Continued on the next page.

Table D.1 (continued)

#	U_6 / $\rho_3(U_6)$	$\#T_{\min}^{-1}(U_6)$ / diagonal_c	hplus_kind / px_dimension / px_resolution kind_type	delta_diag_subset and delta_orbit	E-sign
58	694|478|541|272|169|272 / 111212	144 / 1	NW / 8 / A	0,3,4,8 / 0,1,4,5	3,3,9,5,5,5
59	272|169|272|694|478|541 / 212111	144 / 1	EN / 8 / A	0,3,4,8 / 0,1,4,5	5,5,5,3,3,9
60	252|149|212|614|498|501 / 022200	144 / 1	EN / 8 / A	0,3,4,8 / 0,1,4,5	3,3,9,5,5,5
61	669|443|556|227|114|227 / 021202	1944 / 1	NO / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
62	634|418|561|232|189|292 / 110101	144 / 1	NO / 8 / B	1,2,5,7 / 0,1,4,6	7,7,1,1,7,7
63	674|418|521|292|189|232 / 212101	144 / 1	NO / 8 / B	1,2,5,7 / 0,1,4,6	1,7,7,7,7,1
64	227|114|227|669|443|556 / 202021	1944 / 1	ES / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
65	292|189|232|674|418|521 / 101212	144 / 1	ES / 8 / B	1,2,5,7 / 0,1,4,6	7,7,1,1,7,7
66	232|189|292|634|418|561 / 101110	144 / 1	ES / 8 / B	1,2,5,7 / 0,1,4,6	1,7,7,7,7,1
67	634|458|521|272|169|272 / 122212	144 / 1	NW / 8 / A	0,1,2,3 / 0,1,2,3	7,1,7,5,5,5
68	634|418|561|252|149|212 / 110022	144 / 1	NW / 6 / A	1,2,6 / 0,1,5	7,7,1,3,3,9
69	614|498|501|292|129|292 / 200101	144 / 1	NW / 8 / A	0,1,2,3 / 0,1,2,3	5,5,5,7,1,7
70	694|478|541|292|189|232 / 111101	144 / 1	NW / 6 / A	1,2,6 / 0,1,5	3,3,9,7,7,1
71	292|129|292|614|498|501 / 101200	144 / 1	EN / 8 / A	0,1,2,3 / 0,1,2,3	7,1,7,5,5,5
72	292|189|232|694|478|541 / 101111	144 / 1	EN / 6 / A	1,2,6 / 0,1,5	7,7,1,3,3,9
73	272|169|272|634|458|521 / 212122	144 / 1	EN / 8 / A	0,1,2,3 / 0,1,2,3	5,5,5,7,1,7
74	252|149|212|634|418|561 / 022110	144 / 1	EN / 6 / A	1,2,6 / 0,1,5	3,3,9,7,7,1
75	634|418|561|272|169|272 / 110212	144 / 1	NW / 6 / A	0,3,5,7 / 0,2,4,6	7,7,1,5,5,5
76	654|478|581|292|129|292 / 012101	144 / 1	NW / 8 / A	0,3,5,7 / 0,2,4,6	9,3,3,7,1,7
77	614|498|501|292|189|232 / 200101	144 / 1	NW / 6 / A	0,3,5,7 / 0,2,4,6	5,5,5,7,7,1
78	634|458|521|212|149|252 / 122220	144 / 1	NW / 8 / A	0,3,5,7 / 0,2,4,6	7,1,7,9,3,3
79	292|189|232|614|498|501 / 101200	144 / 1	EN / 6 / A	0,3,5,7 / 0,2,4,6	7,7,1,5,5,5
80	212|149|252|634|458|521 / 220122	144 / 1	EN / 8 / A	0,3,5,7 / 0,2,4,6	9,3,3,7,1,7
81	272|169|272|634|418|561 / 212110	144 / 1	EN / 6 / A	0,3,5,7 / 0,2,4,6	5,5,5,7,7,1
82	292|129|292|654|478|581 / 101012	144 / 1	EN / 8 / A	0,3,5,7 / 0,2,4,6	7,1,7,9,3,3
83	634|458|521|292|129|292 / 122101	144 / 1	NO / 4 / B	4,8 / 0,4	7,1,7,7,1,7
84	292|129|292|634|458|521 / 101122	144 / 1	ES / 4 / B	4,8 / 0,4	7,1,7,7,1,7
85	614|498|501|212|149|252 / 200220	144 / 1	NW / 6 / A	5,6,7 / 0,1,2	5,5,5,9,3,3
86	654|478|581|272|169|272 / 012212	144 / 1	NW / 6 / A	5,6,7 / 0,1,2	9,3,3,5,5,5
87	272|169|272|654|478|581 / 212012	144 / 1	EN / 6 / A	5,6,7 / 0,1,2	5,5,5,9,3,3
88	212|149|252|614|498|501 / 220200	144 / 1	EN / 6 / A	5,6,7 / 0,1,2	9,3,3,5,5,5
89	694|438|581|272|169|272 / 102212	144 / 1	NW / 6 / A	0,1,2,3 / 0,1,2,3	3,9,3,5,5,5
90	614|498|501|252|109|252 / 200010	144 / 1	NW / 6 / A	0,1,2,3 / 0,1,2,3	5,5,5,3,9,3
91	252|109|252|614|498|501 / 010200	144 / 1	EN / 6 / A	0,1,2,3 / 0,1,2,3	3,9,3,5,5,5
92	272|169|272|694|438|581 / 212102	144 / 1	EN / 6 / A	0,1,2,3 / 0,1,2,3	5,5,5,3,9,3
93	694|438|581|252|109|252 / 102010	144 / 1	NO / 4 / B	1,2 / 0,1	3,9,3,3,9,3
94	252|109|252|694|438|581 / 010102	144 / 1	ES / 4 / B	1,2 / 0,1	3,9,3,3,9,3
95	614|498|501|232|189|292 / 200101	144 / 1	NW / 6 / A	4,6,8 / 0,2,4	5,5,5,1,7,7
96	674|418|521|272|169|272 / 212212	144 / 1	NW / 6 / A	4,6,8 / 0,2,4	1,7,7,5,5,5
97	272|169|272|674|418|521 / 212212	144 / 1	EN / 6 / A	4,6,8 / 0,2,4	5,5,5,1,7,7
98	232|189|292|614|498|501 / 101200	144 / 1	EN / 6 / A	4,6,8 / 0,2,4	1,7,7,5,5,5
99	654|478|581|252|149|212 / 012022	144 / 1	NO / 8 / B	4,5,7,8 / 0,1,3,4	9,3,3,3,3,9
100	694|478|541|212|149|252 / 111220	144 / 1	NO / 8 / B	4,5,7,8 / 0,1,3,4	3,3,9,9,3,3
101	212|149|252|694|478|541 / 220111	144 / 1	ES / 8 / B	4,5,7,8 / 0,1,3,4	9,3,3,3,3,9
102	252|149|212|654|478|581 / 022012	144 / 1	ES / 8 / B	4,5,7,8 / 0,1,3,4	3,3,9,9,3,3
103	614|498|501|272|169|272 / 200212	432 / 1	NO / 12 / B	1,2,4,5,7,8 / 0,1,3,4,6,7	5,5,5,5,5,5
104	272|169|272|614|498|501 / 212200	432 / 1	ES / 12 / B	1,2,4,5,7,8 / 0,1,3,4,6,7	5,5,5,5,5,5
105	923|870|810|561|418|634 / 200011	144 / 2	NW / 6 / A	2,3,7,8 / 0,1,4,5	3,9,3,1,7,7
106	903|850|850|581|438|694 / 011201	144 / 2	NW / 6 / A	2,3,7,8 / 0,1,4,5	1,7,7,3,9,3
107	581|438|694|903|850|850 / 201011	144 / 2	EN / 6 / A	2,3,7,8 / 0,1,4,5	3,9,3,1,7,7
108	561|418|634|923|870|810 / 011200	144 / 2	EN / 6 / A	2,3,7,8 / 0,1,4,5	1,7,7,3,9,3
109	943|830|830|581|478|654 / 122210	144 / 2	NW / 8 / A	2,3,7,8 / 0,1,4,5	5,5,5,3,3,9
110	923|810|870|501|498|614 / 200002	144 / 2	NW / 8 / A	2,3,7,8 / 0,1,4,5	3,3,9,5,5,5
111	501|498|614|923|810|870 / 002200	144 / 2	EN / 8 / A	2,3,7,8 / 0,1,4,5	5,5,5,3,3,9
112	581|478|654|943|830|830 / 210122	144 / 2	EN / 8 / A	2,3,7,8 / 0,1,4,5	3,3,9,5,5,5
113	998|885|885|556|443|669 / 200120	1944 / 2	NO / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
114	963|850|890|561|418|634 / 012011	144 / 2	NO / 8 / B	0,1,4,6 / 0,1,4,6	7,7,1,1,7,7
115	903|850|850|521|418|674 / 011212	144 / 2	NO / 8 / B	0,1,4,6 / 0,1,4,6	1,7,7,7,7,1
116	556|443|669|998|885|885 / 120200	1944 / 2	ES / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
117	521|418|674|903|850|850 / 212011	144 / 2	ES / 8 / B	0,1,4,6 / 0,1,4,6	7,7,1,1,7,7
118	561|418|634|963|850|890 / 011012	144 / 2	ES / 8 / B	0,1,4,6 / 0,1,4,6	1,7,7,7,7,1
119	963|890|850|501|498|614 / 021002	144 / 2	NW / 8 / A	0,1,2,8 / 0,1,2,3	7,1,7,5,5,5
120	963|850|890|581|478|654 / 012210	144 / 2	NW / 6 / A	0,1,5 / 0,1,5	7,7,1,3,3,9
121	943|830|830|521|458|634 / 122221	144 / 2	NW / 8 / A	0,1,2,8 / 0,1,2,3	5,5,5,7,1,7
122	923|810|870|521|418|674 / 200212	144 / 2	NW / 6 / A	0,1,5 / 0,1,5	3,3,9,7,7,1
123	521|458|634|943|830|830 / 221122	144 / 2	EN / 8 / A	0,1,2,8 / 0,1,2,3	7,1,7,5,5,5
124	521|418|674|923|810|870 / 212200	144 / 2	EN / 6 / A	0,1,5 / 0,1,5	7,7,1,3,3,9
125	501|498|614|963|890|850 / 002021	144 / 2	EN / 8 / A	0,1,2,8 / 0,1,2,3	5,5,5,7,1,7
126	581|478|654|963|850|890 / 210012	144 / 2	EN / 6 / A	0,1,5 / 0,1,5	3,3,9,7,7,1
127	963|850|890|501|498|614 / 012002	144 / 2	NW / 6 / A	2,4,6,8 / 0,2,4,6	7,7,1,5,5,5
128	983|810|810|521|458|634 / 200221	144 / 2	NW / 8 / A	2,4,6,8 / 0,2,4,6	9,3,3,7,1,7
129	943|830|830|521|418|674 / 122212	144 / 2	NW / 6 / A	2,4,6,8 / 0,2,4,6	5,5,5,7,7,1
130	963|890|850|541|478|694 / 021111	144 / 2	NW / 8 / A	2,4,6,8 / 0,2,4,6	7,1,7,9,3,3
131	521|418|674|943|830|830 / 212122	144 / 2	EN / 6 / A	2,4,6,8 / 0,2,4,6	7,7,1,5,5,5
132	541|478|694|963|890|850 / 111021	144 / 2	EN / 8 / A	2,4,6,8 / 0,2,4,6	9,3,3,7,1,7

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Table D.1 (continued)

#	U_6 / $\rho_3(U_6)$	$\#T_{\min}^{-1}(U_6)$ / diagonal_c	hplus_kind / px_dimension / px_resolution kind_type	delta_diag_subset and delta_orbit	E-sign
133	501|498|614|963|850|890 / 002012	144 / 2	EN / 6 / A	2,4,6,8 / 0,2,4,6	5,5,5,7,7,1
134	521|458|634|983|810|810 / 221200	144 / 2	EN / 8 / A	2,4,6,8 / 0,2,4,6	7,1,7,9,3,3
135	963|890|850|521|458|634 / 021221	144 / 2	NO / 4 / B	3,7 / 0,4	7,1,7,7,1,7
136	521|458|634|963|890|850 / 221021	144 / 2	ES / 4 / B	3,7 / 0,4	7,1,7,7,1,7
137	943|830|830|541|478|694 / 122111	144 / 2	NW / 6 / A	4,5,6 / 0,1,2	5,5,5,9,3,3
138	983|810|810|501|498|614 / 200002	144 / 2	NW / 6 / A	4,5,6 / 0,1,2	9,3,3,5,5,5
139	501|498|614|983|810|810 / 002200	144 / 2	EN / 6 / A	4,5,6 / 0,1,2	5,5,5,9,3,3
140	541|478|694|943|830|830 / 111122	144 / 2	EN / 6 / A	4,5,6 / 0,1,2	9,3,3,5,5,5
141	923|870|810|501|498|614 / 200002	144 / 2	NW / 6 / A	0,1,2,8 / 0,1,2,3	3,9,3,5,5,5
142	943|830|830|581|438|694 / 122201	144 / 2	NW / 6 / A	0,1,2,8 / 0,1,2,3	5,5,5,3,9,3
143	581|438|694|943|830|830 / 201122	144 / 2	EN / 6 / A	0,1,2,8 / 0,1,2,3	3,9,3,5,5,5
144	501|498|614|923|870|810 / 002200	144 / 2	EN / 6 / A	0,1,2,8 / 0,1,2,3	5,5,5,3,9,3
145	923|870|810|581|438|694 / 200201	144 / 2	NO / 4 / B	0,1 / 0,1	3,9,3,3,9,3
146	581|438|694|923|870|810 / 201200	144 / 2	ES / 4 / B	0,1 / 0,1	3,9,3,3,9,3
147	943|830|830|561|418|634 / 122011	144 / 2	NW / 6 / A	3,5,7 / 0,2,4	5,5,5,1,7,7
148	903|850|850|501|498|614 / 011002	144 / 2	NW / 6 / A	3,5,7 / 0,2,4	1,7,7,5,5,5
149	501|498|614|903|850|850 / 002011	144 / 2	EN / 6 / A	3,5,7 / 0,2,4	5,5,5,1,7,7
150	561|418|634|943|830|830 / 011122	144 / 2	EN / 6 / A	3,5,7 / 0,2,4	1,7,7,5,5,5
151	983|810|810|581|478|654 / 200210	144 / 2	NO / 8 / B	3,4,6,7 / 0,1,3,4	9,3,3,3,3,9
152	923|810|870|541|478|694 / 200111	144 / 2	NO / 8 / B	3,4,6,7 / 0,1,3,4	3,3,9,9,3,3
153	541|478|694|923|810|870 / 111200	144 / 2	ES / 8 / B	3,4,6,7 / 0,1,3,4	9,3,3,3,3,9
154	581|478|654|983|810|810 / 210200	144 / 2	ES / 8 / B	3,4,6,7 / 0,1,3,4	3,3,9,9,3,3
155	943|830|830|501|498|614 / 122002	432 / 2	NO / 12 / B	0,1,3,4,6,7 / 0,1,3,4,6,7	5,5,5,5,5,5
156	501|498|614|943|830|830 / 002122	432 / 2	ES / 12 / B	0,1,3,4,6,7 / 0,1,3,4,6,7	5,5,5,5,5,5
157	252|212|149|890|850|963 / 022210	144 / 3	NW / 6 / A	1,2,6,7 / 0,1,4,5	3,9,3,1,7,7
158	232|292|189|810|870|923 / 110002	144 / 3	NW / 6 / A	1,2,6,7 / 0,1,4,5	1,7,7,3,9,3
159	810|870|923|232|292|189 / 002110	144 / 3	EN / 6 / A	1,2,6,7 / 0,1,4,5	3,9,3,1,7,7
160	890|850|963|252|212|149 / 210022	144 / 3	EN / 6 / A	1,2,6,7 / 0,1,4,5	1,7,7,3,9,3
161	272|272|169|810|810|983 / 221002	144 / 3	NW / 8 / A	1,2,6,7 / 0,1,4,5	5,5,5,3,3,9
162	252|252|109|830|830|943 / 001221	144 / 3	NW / 8 / A	1,2,6,7 / 0,1,4,5	3,3,9,5,5,5
163	830|830|943|252|252|109 / 221001	144 / 3	EN / 8 / A	1,2,6,7 / 0,1,4,5	5,5,5,3,3,9
164	810|810|983|272|272|169 / 002221	144 / 3	EN / 8 / A	1,2,6,7 / 0,1,4,5	3,3,9,5,5,5
165	227|227|114|885|885|998 / 220002	1944 / 3	NO / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
166	292|292|129|890|850|963 / 110210	144 / 3	NO / 8 / B	0,3,5,8 / 0,1,4,6	7,7,1,1,7,7
167	232|292|189|850|850|903 / 110110	144 / 3	NO / 8 / B	0,3,5,8 / 0,1,4,6	1,7,7,7,7,1
168	885|885|998|227|227|114 / 002220	1944 / 3	ES / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
169	850|850|903|232|292|189 / 110110	144 / 3	ES / 8 / B	0,3,5,8 / 0,1,4,6	7,7,1,1,7,7
170	890|850|963|292|292|129 / 210110	144 / 3	ES / 8 / B	0,3,5,8 / 0,1,4,6	1,7,7,7,7,1
171	292|232|189|830|830|943 / 110221	144 / 3	NW / 8 / A	0,1,7,8 / 0,1,2,3	7,1,7,5,5,5
172	292|292|129|810|810|983 / 110002	144 / 3	NW / 6 / A	0,4,8 / 0,1,5	7,7,1,3,3,9
173	272|272|169|850|890|963 / 221120	144 / 3	NW / 8 / A	0,1,7,8 / 0,1,2,3	5,5,5,7,1,7
174	252|252|109|850|850|903 / 001110	144 / 3	NW / 6 / A	0,4,8 / 0,1,5	3,3,9,7,7,1
175	850|890|963|272|272|169 / 120221	144 / 3	EN / 8 / A	0,1,7,8 / 0,1,2,3	7,1,7,5,5,5
176	850|850|903|252|252|109 / 110001	144 / 3	EN / 6 / A	0,4,8 / 0,1,5	7,7,1,3,3,9
177	830|830|943|292|232|189 / 221110	144 / 3	EN / 8 / A	0,1,7,8 / 0,1,2,3	5,5,5,7,1,7
178	810|810|983|292|292|129 / 002110	144 / 3	EN / 6 / A	0,4,8 / 0,1,5	3,3,9,7,7,1
179	292|292|129|830|830|943 / 110221	144 / 3	NW / 6 / A	1,3,5,7 / 0,2,4,6	7,7,1,5,5,5
180	212|252|149|850|890|963 / 202120	144 / 3	NW / 8 / A	1,3,5,7 / 0,2,4,6	9,3,3,7,1,7
181	272|272|169|850|850|903 / 221110	144 / 3	NW / 6 / A	1,3,5,7 / 0,2,4,6	5,5,5,7,7,1
182	292|232|189|870|810|923 / 110002	144 / 3	NW / 8 / A	1,3,5,7 / 0,2,4,6	7,1,7,9,3,3
183	850|850|903|272|272|169 / 110221	144 / 3	EN / 6 / A	1,3,5,7 / 0,2,4,6	7,7,1,5,5,5
184	870|810|923|292|232|189 / 002110	144 / 3	EN / 8 / A	1,3,5,7 / 0,2,4,6	9,3,3,7,1,7
185	830|830|943|292|292|129 / 221110	144 / 3	EN / 6 / A	1,3,5,7 / 0,2,4,6	5,5,5,7,7,1
186	850|890|963|212|252|149 / 120202	144 / 3	EN / 8 / A	1,3,5,7 / 0,2,4,6	7,1,7,9,3,3
187	292|232|189|850|890|963 / 110120	144 / 3	NO / 4 / B	2,6 / 0,4	7,1,7,7,1,7
188	850|890|963|292|232|189 / 120110	144 / 3	ES / 4 / B	2,6 / 0,4	7,1,7,7,1,7
189	272|272|169|870|810|923 / 221002	144 / 3	NW / 6 / A	3,4,5 / 0,1,2	5,5,5,9,3,3
190	212|252|149|830|830|943 / 202221	144 / 3	NW / 6 / A	3,4,5 / 0,1,2	9,3,3,5,5,5
191	830|830|943|212|252|149 / 221202	144 / 3	EN / 6 / A	3,4,5 / 0,1,2	5,5,5,9,3,3
192	870|810|923|272|272|169 / 002221	144 / 3	EN / 6 / A	3,4,5 / 0,1,2	9,3,3,5,5,5
193	252|212|149|830|830|943 / 022221	144 / 3	NW / 6 / A	0,1,7,8 / 0,1,2,3	3,9,3,5,5,5
194	272|272|169|810|870|923 / 221002	144 / 3	NW / 6 / A	0,1,7,8 / 0,1,2,3	5,5,5,3,9,3
195	810|870|923|272|272|169 / 002221	144 / 3	EN / 6 / A	0,1,7,8 / 0,1,2,3	3,9,3,5,5,5
196	830|830|943|252|212|149 / 221022	144 / 3	EN / 6 / A	0,1,7,8 / 0,1,2,3	5,5,5,3,9,3
197	252|212|149|810|870|923 / 022002	144 / 3	NO / 4 / B	0,8 / 0,1	3,9,3,3,9,3
198	810|870|923|252|212|149 / 002022	144 / 3	ES / 4 / B	0,8 / 0,1	3,9,3,3,9,3
199	272|272|169|890|850|963 / 221210	144 / 3	NW / 6 / A	2,4,6 / 0,2,4	5,5,5,1,7,7
200	232|292|189|830|830|943 / 110221	144 / 3	NW / 6 / A	2,4,6 / 0,2,4	1,7,7,5,5,5
201	830|830|943|232|292|189 / 221110	144 / 3	EN / 6 / A	2,4,6 / 0,2,4	5,5,5,1,7,7
202	890|850|963|272|272|169 / 210221	144 / 3	EN / 6 / A	2,4,6 / 0,2,4	1,7,7,5,5,5
203	212|252|149|810|810|983 / 202002	144 / 3	NO / 8 / B	2,3,5,6 / 0,1,3,4	9,3,3,3,3,9
204	252|252|109|870|810|923 / 001002	144 / 3	NO / 8 / B	2,3,5,6 / 0,1,3,4	3,3,9,9,3,3
205	870|810|923|252|252|109 / 002001	144 / 3	ES / 8 / B	2,3,5,6 / 0,1,3,4	9,3,3,3,3,9
206	810|810|983|212|252|149 / 002202	144 / 3	ES / 8 / B	2,3,5,6 / 0,1,3,4	3,3,9,9,3,3
207	272|272|169|830|830|943 / 221221	432 / 3	NO / 12 / B	0,2,3,5,6,8 / 0,1,3,4,6,7	5,5,5,5,5,5
208	830|830|943|272|272|169 / 221221	432 / 3	ES / 12 / B	0,2,3,5,6,8 / 0,1,3,4,6,7	5,5,5,5,5,5
209	581|654|478|129|292|292 / 201011	144 / 4	NW / 6 / A	0,1,5,6 / 0,1,4,5	3,9,3,1,7,7

Continued on the next page.

Table D.1 (continued)

#	U_6 / $\rho_3(U_6)$	$\#T_{\min}^{-1}(U_6)$ / diagonal_c	hplus_kind / px_dimension / px_resolution kind_type	delta_diag_subset and delta_orbit	E-sign
210	561|634|418|149|212|252 / 011220	144 / 4	NW / 6 / A	0,1,5,6 / 0,1,4,5	1,7,7,3,9,3
211	149|212|252|561|634|418 / 220011	144 / 4	EN / 6 / A	0,1,5,6 / 0,1,4,5	3,9,3,1,7,7
212	129|292|292|581|654|478 / 011201	144 / 4	EN / 6 / A	0,1,5,6 / 0,1,4,5	1,7,7,3,9,3
213	501|614|498|149|252|212 / 020202	144 / 4	NW / 8 / A	0,1,5,6 / 0,1,4,5	5,5,5,3,3,9
214	581|694|438|169|272|272 / 210122	144 / 4	NW / 8 / A	0,1,5,6 / 0,1,4,5	3,3,9,5,5,5
215	169|272|272|581|694|438 / 122210	144 / 4	EN / 8 / A	0,1,5,6 / 0,1,4,5	5,5,5,3,3,9
216	149|252|212|501|614|498 / 202020	144 / 4	EN / 8 / A	0,1,5,6 / 0,1,4,5	3,3,9,5,5,5
217	556|669|443|114|227|227 / 102022	1944 / 4	NO / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
218	521|634|458|129|292|292 / 212011	144 / 4	NO / 8 / B	2,4,7,8 / 0,1,4,6	7,7,1,1,7,7
219	561|634|418|189|292|232 / 011011	144 / 4	NO / 8 / B	2,4,7,8 / 0,1,4,6	1,7,7,7,7,1
220	114|227|227|556|669|443 / 022102	1944 / 4	ES / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
221	189|292|232|561|634|418 / 011011	144 / 4	ES / 8 / B	2,4,7,8 / 0,1,4,6	7,7,1,1,7,7
222	129|292|292|521|634|458 / 011212	144 / 4	ES / 8 / B	2,4,7,8 / 0,1,4,6	1,7,7,7,7,1
223	521|674|418|169|272|272 / 221122	144 / 4	NW / 8 / A	0,6,7,8 / 0,1,2,3	7,1,7,5,5,5
224	521|634|458|149|252|212 / 212202	144 / 4	NW / 6 / A	3,7,8 / 0,1,5	7,7,1,3,3,9
225	501|614|498|189|232|292 / 020011	144 / 4	NW / 8 / A	0,6,7,8 / 0,1,2,3	5,5,5,7,1,7
226	581|694|438|189|292|232 / 210011	144 / 4	NW / 6 / A	3,7,8 / 0,1,5	3,3,9,7,7,1
227	189|232|292|501|614|498 / 011020	144 / 4	EN / 8 / A	0,6,7,8 / 0,1,2,3	7,1,7,5,5,5
228	189|292|232|581|694|438 / 011210	144 / 4	EN / 6 / A	3,7,8 / 0,1,5	7,7,1,3,3,9
229	169|272|272|521|674|418 / 122221	144 / 4	EN / 8 / A	0,6,7,8 / 0,1,2,3	5,5,5,7,1,7
230	149|252|212|521|634|458 / 202212	144 / 4	EN / 6 / A	3,7,8 / 0,1,5	3,3,9,7,7,1
231	521|634|458|169|272|272 / 212122	144 / 4	NW / 6 / A	0,2,4,6 / 0,2,4,6	7,7,1,5,5,5
232	541|694|478|189|232|292 / 111011	144 / 4	NW / 8 / A	0,2,4,6 / 0,2,4,6	9,3,3,7,1,7
233	501|614|498|189|292|232 / 020011	144 / 4	NW / 6 / A	0,2,4,6 / 0,2,4,6	5,5,5,7,7,1
234	521|674|418|109|252|252 / 221100	144 / 4	NW / 8 / A	0,2,4,6 / 0,2,4,6	7,1,7,9,3,3
235	189|292|232|501|614|498 / 011020	144 / 4	EN / 6 / A	0,2,4,6 / 0,2,4,6	7,7,1,5,5,5
236	109|252|252|521|674|418 / 100221	144 / 4	EN / 8 / A	0,2,4,6 / 0,2,4,6	9,3,3,7,1,7
237	169|272|272|521|634|458 / 122212	144 / 4	EN / 6 / A	0,2,4,6 / 0,2,4,6	5,5,5,7,7,1
238	189|232|292|541|694|478 / 011111	144 / 4	EN / 8 / A	0,2,4,6 / 0,2,4,6	7,1,7,9,3,3
239	521|674|418|189|232|292 / 221011	144 / 4	NO / 4 / B	1,5 / 0,4	7,1,7,7,1,7
240	189|232|292|521|674|418 / 011221	144 / 4	ES / 4 / B	1,5 / 0,4	7,1,7,7,1,7
241	501|614|498|109|252|252 / 020100	144 / 4	NW / 6 / A	2,3,4 / 0,1,2	5,5,5,9,3,3
242	541|694|478|169|272|272 / 111122	144 / 4	NW / 6 / A	2,3,4 / 0,1,2	9,3,3,5,5,5
243	169|272|272|541|694|478 / 122111	144 / 4	EN / 6 / A	2,3,4 / 0,1,2	5,5,5,9,3,3
244	109|252|252|501|614|498 / 100020	144 / 4	EN / 6 / A	2,3,4 / 0,1,2	9,3,3,5,5,5
245	581|654|478|169|272|272 / 201122	144 / 4	NW / 6 / A	0,6,7,8 / 0,1,2,3	3,9,3,5,5,5
246	501|614|498|149|212|252 / 020220	144 / 4	NW / 6 / A	0,6,7,8 / 0,1,2,3	5,5,5,3,9,3
247	149|212|252|501|614|498 / 220020	144 / 4	EN / 6 / A	0,6,7,8 / 0,1,2,3	3,9,3,5,5,5
248	169|272|272|581|654|478 / 122201	144 / 4	EN / 6 / A	0,6,7,8 / 0,1,2,3	5,5,5,3,9,3
249	581|654|478|149|212|252 / 201220	144 / 4	NO / 4 / B	7,8 / 0,1	3,9,3,3,9,3
250	149|212|252|581|654|478 / 220201	144 / 4	ES / 4 / B	7,8 / 0,1	3,9,3,3,9,3
251	501|614|498|129|292|292 / 020011	144 / 4	NW / 6 / A	1,3,5 / 0,2,4	5,5,5,1,7,7
252	561|634|418|169|272|272 / 011122	144 / 4	NW / 6 / A	1,3,5 / 0,2,4	1,7,7,5,5,5
253	169|272|272|561|634|418 / 122011	144 / 4	EN / 6 / A	1,3,5 / 0,2,4	5,5,5,1,7,7
254	129|292|292|501|614|498 / 011020	144 / 4	EN / 6 / A	1,3,5 / 0,2,4	1,7,7,5,5,5
255	541|694|478|149|252|212 / 111202	144 / 4	NO / 8 / B	1,2,4,5 / 0,1,3,4	9,3,3,3,3,9
256	581|694|438|109|252|252 / 210100	144 / 4	NO / 8 / B	1,2,4,5 / 0,1,3,4	3,3,9,9,3,3
257	109|252|252|581|694|438 / 100210	144 / 4	ES / 8 / B	1,2,4,5 / 0,1,3,4	9,3,3,3,3,9
258	149|252|212|541|694|478 / 202111	144 / 4	ES / 8 / B	1,2,4,5 / 0,1,3,4	3,3,9,9,3,3
259	501|614|498|169|272|272 / 020122	432 / 4	NO / 12 / B	1,2,4,5,7,8 / 0,1,3,4,6,7	5,5,5,5,5,5
260	169|272|272|501|614|498 / 122020	432 / 4	ES / 12 / B	1,2,4,5,7,8 / 0,1,3,4,6,7	5,5,5,5,5,5
261	810|983|810|458|634|521 / 020212	144 / 5	NW / 6 / A	0,4,5,8 / 0,1,4,5	3,9,3,1,7,7
262	890|963|850|478|654|581 / 201102	144 / 5	NW / 6 / A	0,4,5,8 / 0,1,4,5	1,7,7,3,9,3
263	478|654|581|890|963|850 / 102201	144 / 5	EN / 6 / A	0,4,5,8 / 0,1,4,5	3,9,3,1,7,7
264	458|634|521|810|983|810 / 212020	144 / 5	EN / 6 / A	0,4,5,8 / 0,1,4,5	1,7,7,3,9,3
265	830|943|830|478|694|541 / 212111	144 / 5	NW / 8 / A	0,4,5,8 / 0,1,4,5	5,5,5,3,3,9
266	810|923|870|498|614|501 / 020020	144 / 5	NW / 8 / A	0,4,5,8 / 0,1,4,5	3,3,9,5,5,5
267	498|614|501|810|923|870 / 020020	144 / 5	EN / 8 / A	0,4,5,8 / 0,1,4,5	5,5,5,3,3,9
268	478|694|541|830|943|830 / 111212	144 / 5	EN / 8 / A	0,4,5,8 / 0,1,4,5	3,3,9,5,5,5
269	885|998|885|443|669|556 / 020201	1944 / 5	NO / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
270	850|963|890|458|634|521 / 102212	144 / 5	NO / 8 / B	1,3,6,7 / 0,1,4,6	7,7,1,1,7,7
271	890|963|850|418|634|561 / 201110	144 / 5	NO / 8 / B	1,3,6,7 / 0,1,4,6	1,7,7,7,7,1
272	443|669|556|885|998|885 / 201020	1944 / 5	ES / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
273	418|634|561|890|963|850 / 110201	144 / 5	ES / 8 / B	1,3,6,7 / 0,1,4,6	7,7,1,1,7,7
274	458|634|521|850|963|890 / 212102	144 / 5	ES / 8 / B	1,3,6,7 / 0,1,4,6	1,7,7,7,7,1
275	850|903|850|498|614|501 / 101020	144 / 5	NW / 8 / A	5,6,7,8 / 0,1,2,3	7,1,7,5,5,5
276	850|963|890|478|694|541 / 102111	144 / 5	NW / 6 / A	2,6,7 / 0,1,5	7,7,1,3,3,9
277	830|943|830|418|674|521 / 212122	144 / 5	NW / 8 / A	5,6,7,8 / 0,1,2,3	5,5,5,7,1,7
278	810|923|870|418|634|561 / 020110	144 / 5	NW / 6 / A	2,6,7 / 0,1,5	3,3,9,7,7,1
279	418|674|521|830|943|830 / 122212	144 / 5	EN / 8 / A	5,6,7,8 / 0,1,2,3	7,1,7,5,5,5
280	418|634|561|810|923|870 / 110020	144 / 5	EN / 6 / A	2,6,7 / 0,1,5	7,7,1,3,3,9
281	498|614|501|850|903|850 / 020101	144 / 5	EN / 8 / A	5,6,7,8 / 0,1,2,3	5,5,5,7,1,7
282	478|694|541|850|963|890 / 111102	144 / 5	EN / 6 / A	2,6,7 / 0,1,5	3,3,9,7,7,1
283	850|963|890|498|614|501 / 102020	144 / 5	NW / 6 / A	1,3,5,8 / 0,2,4,6	7,7,1,5,5,5
284	870|923|810|418|674|521 / 020122	144 / 5	NW / 8 / A	1,3,5,8 / 0,2,4,6	9,3,3,7,1,7

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Table D.1 (continued)

#	U_6 / $\rho_3(U_6)$	$\#T_{\min}^{-1}(U_6)$ / diagonal_c	hplus_kind / px_dimension / px_resolution kind_type	delta_diag_subset and delta_orbit	E-sign
285	830|943|830|418|634|561 / 212110	144 / 5	NW / 6 / A	1,3,5,8 / 0,2,4,6	5,5,5,7,7,1
286	850|903|850|438|694|581 / 101012	144 / 5	NW / 8 / A	1,3,5,8 / 0,2,4,6	7,1,7,9,3,3
287	418|634|561|830|943|830 / 110212	144 / 5	EN / 6 / A	1,3,5,8 / 0,2,4,6	7,7,1,5,5,5
288	438|694|581|850|903|850 / 012101	144 / 5	EN / 8 / A	1,3,5,8 / 0,2,4,6	9,3,3,7,1,7
289	498|614|501|850|963|890 / 020102	144 / 5	EN / 6 / A	1,3,5,8 / 0,2,4,6	5,5,5,7,7,1
290	418|674|521|870|923|810 / 122020	144 / 5	EN / 8 / A	1,3,5,8 / 0,2,4,6	7,1,7,9,3,3
291	850|903|850|418|674|521 / 101122	144 / 5	NO / 4 / B	0,4 / 0,4	7,1,7,7,1,7
292	418|674|521|850|903|850 / 122101	144 / 5	ES / 4 / B	0,4 / 0,4	7,1,7,7,1,7
293	830|943|830|438|694|581 / 212012	144 / 5	NW / 6 / A	1,2,3 / 0,1,2	5,5,5,9,3,3
294	870|923|810|498|614|501 / 020020	144 / 5	NW / 6 / A	1,2,3 / 0,1,2	9,3,3,5,5,5
295	498|614|501|870|923|810 / 020020	144 / 5	EN / 6 / A	1,2,3 / 0,1,2	5,5,5,9,3,3
296	438|694|581|830|943|830 / 012212	144 / 5	EN / 6 / A	1,2,3 / 0,1,2	9,3,3,5,5,5
297	810|983|810|498|614|501 / 020020	144 / 5	NW / 6 / A	5,6,7,8 / 0,1,2,3	3,9,3,5,5,5
298	830|943|830|478|654|581 / 212102	144 / 5	NW / 6 / A	5,6,7,8 / 0,1,2,3	5,5,5,3,9,3
299	478|654|581|830|943|830 / 102212	144 / 5	EN / 6 / A	5,6,7,8 / 0,1,2,3	3,9,3,5,5,5
300	498|614|501|810|983|810 / 020020	144 / 5	EN / 6 / A	5,6,7,8 / 0,1,2,3	5,5,5,3,9,3
301	810|983|810|478|654|581 / 020102	144 / 5	NO / 4 / B	6,7 / 0,1	3,9,3,3,9,3
302	478|654|581|810|983|810 / 102020	144 / 5	ES / 4 / B	6,7 / 0,1	3,9,3,3,9,3
303	830|943|830|458|634|521 / 212212	144 / 5	NW / 6 / A	0,2,4 / 0,2,4	5,5,5,1,7,7
304	890|963|850|498|614|501 / 201020	144 / 5	NW / 6 / A	0,2,4 / 0,2,4	1,7,7,5,5,5
305	498|614|501|890|963|850 / 020201	144 / 5	EN / 6 / A	0,2,4 / 0,2,4	5,5,5,1,7,7
306	458|634|521|830|943|830 / 212212	144 / 5	EN / 6 / A	0,2,4 / 0,2,4	1,7,7,5,5,5
307	870|923|810|478|694|541 / 020111	144 / 5	NO / 8 / B	0,1,3,4 / 0,1,3,4	9,3,3,3,3,9
308	810|923|870|438|694|581 / 020012	144 / 5	NO / 8 / B	0,1,3,4 / 0,1,3,4	3,3,9,9,3,3
309	438|694|581|810|923|870 / 012020	144 / 5	ES / 8 / B	0,1,3,4 / 0,1,3,4	9,3,3,3,3,9
310	478|694|541|870|923|810 / 111020	144 / 5	ES / 8 / B	0,1,3,4 / 0,1,3,4	3,3,9,9,3,3
311	830|943|830|498|614|501 / 212020	432 / 5	NO / 12 / B	0,1,3,4,6,7 / 0,1,3,4,6,7	5,5,5,5,5,5
312	498|614|501|830|943|830 / 020212	432 / 5	ES / 12 / B	0,1,3,4,6,7 / 0,1,3,4,6,7	5,5,5,5,5,5
313	149|212|252|890|963|850 / 220201	144 / 6	NW / 6 / A	3,4,7,8 / 0,1,4,5	3,9,3,1,7,7
314	129|292|292|810|983|810 / 011020	144 / 6	NW / 6 / A	3,4,7,8 / 0,1,4,5	1,7,7,3,9,3
315	810|983|810|129|292|292 / 020011	144 / 6	EN / 6 / A	3,4,7,8 / 0,1,4,5	3,9,3,1,7,7
316	890|963|850|149|212|252 / 201220	144 / 6	EN / 6 / A	3,4,7,8 / 0,1,4,5	1,7,7,3,9,3
317	169|272|272|810|923|870 / 122020	144 / 6	NW / 8 / A	3,4,7,8 / 0,1,4,5	5,5,5,3,3,9
318	149|252|212|830|943|830 / 202212	144 / 6	NW / 8 / A	3,4,7,8 / 0,1,4,5	3,3,9,5,5,5
319	830|943|830|149|252|212 / 212202	144 / 6	EN / 8 / A	3,4,7,8 / 0,1,4,5	5,5,5,3,3,9
320	810|923|870|169|272|272 / 020122	144 / 6	EN / 8 / A	3,4,7,8 / 0,1,4,5	3,3,9,5,5,5
321	114|227|227|885|998|885 / 022020	1944 / 6	NO / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
322	189|292|232|890|963|850 / 011201	144 / 6	NO / 8 / B	0,2,5,6 / 0,1,4,6	7,7,1,1,7,7
323	129|292|292|850|963|890 / 011102	144 / 6	NO / 8 / B	0,2,5,6 / 0,1,4,6	1,7,7,7,7,1
324	885|998|885|114|227|227 / 020022	1944 / 6	ES / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
325	850|963|890|129|292|292 / 102011	144 / 6	ES / 8 / B	0,2,5,6 / 0,1,4,6	7,7,1,1,7,7
326	890|963|850|189|292|232 / 201011	144 / 6	ES / 8 / B	0,2,5,6 / 0,1,4,6	1,7,7,7,7,1
327	189|232|292|830|943|830 / 011212	144 / 6	NW / 8 / A	4,5,6,7 / 0,1,2,3	7,1,7,5,5,5
328	189|292|232|810|923|870 / 011020	144 / 6	NW / 6 / A	1,5,6 / 0,1,5	7,7,1,3,3,9
329	169|272|272|850|903|850 / 122101	144 / 6	NW / 8 / A	4,5,6,7 / 0,1,2,3	5,5,5,7,1,7
330	149|252|212|850|963|890 / 202102	144 / 6	NW / 6 / A	1,5,6 / 0,1,5	3,3,9,7,7,1
331	850|903|850|169|272|272 / 101122	144 / 6	EN / 8 / A	4,5,6,7 / 0,1,2,3	7,1,7,5,5,5
332	850|963|890|149|252|212 / 102202	144 / 6	EN / 6 / A	1,5,6 / 0,1,5	7,7,1,3,3,9
333	830|943|830|189|232|292 / 212011	144 / 6	EN / 8 / A	4,5,6,7 / 0,1,2,3	5,5,5,7,1,7
334	810|923|870|189|292|232 / 020011	144 / 6	EN / 6 / A	1,5,6 / 0,1,5	3,3,9,7,7,1
335	189|292|232|830|943|830 / 011212	144 / 6	NW / 6 / A	0,2,4,7 / 0,2,4,6	7,7,1,5,5,5
336	109|252|252|850|903|850 / 100101	144 / 6	NW / 8 / A	0,2,4,7 / 0,2,4,6	9,3,3,7,1,7
337	169|272|272|850|963|890 / 122102	144 / 6	NW / 6 / A	0,2,4,7 / 0,2,4,6	5,5,5,7,7,1
338	189|232|292|870|923|810 / 011020	144 / 6	NW / 8 / A	0,2,4,7 / 0,2,4,6	7,1,7,9,3,3
339	850|963|890|169|272|272 / 102122	144 / 6	EN / 6 / A	0,2,4,7 / 0,2,4,6	7,7,1,5,5,5
340	870|923|810|189|232|292 / 020011	144 / 6	EN / 8 / A	0,2,4,7 / 0,2,4,6	9,3,3,7,1,7
341	830|943|830|189|292|232 / 212011	144 / 6	EN / 6 / A	0,2,4,7 / 0,2,4,6	5,5,5,7,7,1
342	850|903|850|109|252|252 / 101100	144 / 6	EN / 8 / A	0,2,4,7 / 0,2,4,6	7,1,7,9,3,3
343	189|232|292|850|903|850 / 011101	144 / 6	NO / 4 / B	3,8 / 0,4	7,1,7,7,1,7
344	850|903|850|189|232|292 / 101011	144 / 6	ES / 4 / B	3,8 / 0,4	7,1,7,7,1,7
345	169|272|272|870|923|810 / 122020	144 / 6	NW / 6 / A	0,1,2 / 0,1,2	5,5,5,9,3,3
346	109|252|252|830|943|830 / 100212	144 / 6	NW / 6 / A	0,1,2 / 0,1,2	9,3,3,5,5,5
347	830|943|830|109|252|252 / 212100	144 / 6	EN / 6 / A	0,1,2 / 0,1,2	5,5,5,9,3,3
348	870|923|810|169|272|272 / 020122	144 / 6	EN / 6 / A	0,1,2 / 0,1,2	9,3,3,5,5,5
349	149|212|252|830|943|830 / 220212	144 / 6	NW / 6 / A	4,5,6,7 / 0,1,2,3	3,9,3,5,5,5
350	169|272|272|810|983|810 / 122020	144 / 6	NW / 6 / A	4,5,6,7 / 0,1,2,3	5,5,5,3,9,3
351	810|983|810|169|272|272 / 020122	144 / 6	EN / 6 / A	4,5,6,7 / 0,1,2,3	3,9,3,5,5,5
352	830|943|830|149|212|252 / 212220	144 / 6	EN / 6 / A	4,5,6,7 / 0,1,2,3	5,5,5,3,9,3
353	149|212|252|810|983|810 / 220020	144 / 6	NO / 4 / B	5,6 / 0,1	3,9,3,3,9,3
354	810|983|810|149|212|252 / 020220	144 / 6	ES / 4 / B	5,6 / 0,1	3,9,3,3,9,3
355	169|272|272|890|963|850 / 122201	144 / 6	NW / 6 / A	1,3,8 / 0,2,4	5,5,5,1,7,7
356	129|292|292|830|943|830 / 011212	144 / 6	NW / 6 / A	1,3,8 / 0,2,4	1,7,7,5,5,5
357	830|943|830|129|292|292 / 212011	144 / 6	EN / 6 / A	1,3,8 / 0,2,4	5,5,5,1,7,7
358	890|963|850|169|272|272 / 201122	144 / 6	EN / 6 / A	1,3,8 / 0,2,4	1,7,7,5,5,5
359	109|252|252|810|923|870 / 100020	144 / 6	NO / 8 / B	0,2,3,8 / 0,1,3,4	9,3,3,3,3,9
360	149|252|212|870|923|810 / 202020	144 / 6	NO / 8 / B	0,2,3,8 / 0,1,3,4	3,3,9,9,3,3
361	870|923|810|149|252|212 / 020202	144 / 6	ES / 8 / B	0,2,3,8 / 0,1,3,4	9,3,3,3,3,9

Continued on the next page.

Table D.1 (continued)

#	U_6 / $\rho_3(U_6)$	$\#T_{\min}^{-1}(U_6)$ / diagonal_c	hplus_kind / px_dimension / px_resolution kind_type	delta_diag_subset and delta_orbit	E-sign
362	810|923|870|109|252|252 / 020100	144 / 6	ES / 8 / B	0,2,3,8 / 0,1,3,4	3,3,9,9,3,3
363	169|272|272|830|943|830 / 122212	432 / 6	NO / 12 / B	0,2,3,5,6,8 / 0,1,3,4,6,7	5,5,5,5,5,5
364	830|943|830|169|272|272 / 212122	432 / 6	ES / 12 / B	0,2,3,5,6,8 / 0,1,3,4,6,7	5,5,5,5,5,5
365	478|541|694|232|292|189 / 111110	144 / 7	NW / 6 / A	2,3,6,7 / 0,1,4,5	3,9,3,1,7,7
366	458|521|634|252|212|149 / 221022	144 / 7	NW / 6 / A	2,3,6,7 / 0,1,4,5	1,7,7,3,9,3
367	252|212|149|458|521|634 / 022221	144 / 7	EN / 6 / A	2,3,6,7 / 0,1,4,5	3,9,3,1,7,7
368	232|292|189|478|541|694 / 110111	144 / 7	EN / 6 / A	2,3,6,7 / 0,1,4,5	1,7,7,3,9,3
369	498|501|614|252|252|109 / 002001	144 / 7	NW / 8 / A	2,3,6,7 / 0,1,4,5	5,5,5,3,3,9
370	478|581|654|272|272|169 / 120221	144 / 7	NW / 8 / A	2,3,6,7 / 0,1,4,5	3,3,9,5,5,5
371	272|272|169|478|581|654 / 221120	144 / 7	EN / 8 / A	2,3,6,7 / 0,1,4,5	5,5,5,3,3,9
372	252|252|109|498|501|614 / 001002	144 / 7	EN / 8 / A	2,3,6,7 / 0,1,4,5	3,3,9,5,5,5
373	443|556|669|227|227|114 / 210220	1944 / 7	NO / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
374	418|521|674|232|292|189 / 122110	144 / 7	NO / 8 / B	1,4,5,8 / 0,1,4,6	7,7,1,1,7,7
375	458|521|634|292|292|129 / 221110	144 / 7	NO / 8 / B	1,4,5,8 / 0,1,4,6	1,7,7,7,7,1
376	227|227|114|443|556|669 / 220210	1944 / 7	ES / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
377	292|292|129|458|521|634 / 110221	144 / 7	ES / 8 / B	1,4,5,8 / 0,1,4,6	7,7,1,1,7,7
378	232|292|189|418|521|674 / 110122	144 / 7	ES / 8 / B	1,4,5,8 / 0,1,4,6	1,7,7,7,7,1
379	418|561|634|272|272|169 / 101221	144 / 7	NW / 8 / A	3,4,5,6 / 0,1,2,3	7,1,7,5,5,5
380	418|521|674|252|252|109 / 122001	144 / 7	NW / 6 / A	0,4,5 / 0,1,5	7,7,1,3,3,9
381	498|501|614|292|232|189 / 002110	144 / 7	NW / 8 / A	3,4,5,6 / 0,1,2,3	5,5,5,7,1,7
382	478|581|654|292|292|129 / 120110	144 / 7	NW / 6 / A	0,4,5 / 0,1,5	3,3,9,7,7,1
383	292|232|189|498|501|614 / 110002	144 / 7	EN / 8 / A	3,4,5,6 / 0,1,2,3	7,1,7,5,5,5
384	292|292|129|478|581|654 / 110120	144 / 7	EN / 6 / A	0,4,5 / 0,1,5	7,7,1,3,3,9
385	272|272|169|418|561|634 / 221101	144 / 7	EN / 8 / A	3,4,5,6 / 0,1,2,3	5,5,5,7,1,7
386	252|252|109|418|521|674 / 001122	144 / 7	EN / 6 / A	0,4,5 / 0,1,5	3,3,9,7,7,1
387	418|521|674|272|272|169 / 122221	144 / 7	NW / 6 / A	1,3,6,8 / 0,2,4,6	7,7,1,5,5,5
388	438|581|694|292|232|189 / 021110	144 / 7	NW / 8 / A	1,3,6,8 / 0,2,4,6	9,3,3,7,1,7
389	498|501|614|292|292|129 / 002110	144 / 7	NW / 6 / A	1,3,6,8 / 0,2,4,6	5,5,5,7,7,1
390	418|561|634|212|252|149 / 101202	144 / 7	NW / 8 / A	1,3,6,8 / 0,2,4,6	7,1,7,9,3,3
391	292|292|129|498|501|614 / 110002	144 / 7	EN / 6 / A	1,3,6,8 / 0,2,4,6	7,7,1,5,5,5
392	212|252|149|418|561|634 / 202101	144 / 7	EN / 8 / A	1,3,6,8 / 0,2,4,6	9,3,3,7,1,7
393	272|272|169|418|521|674 / 221122	144 / 7	EN / 6 / A	1,3,6,8 / 0,2,4,6	5,5,5,7,7,1
394	292|232|189|438|581|694 / 110021	144 / 7	EN / 8 / A	1,3,6,8 / 0,2,4,6	7,1,7,9,3,3
395	418|561|634|292|232|189 / 101110	144 / 7	NO / 4 / B	2,7 / 0,4	7,1,7,7,1,7
396	292|232|189|418|561|634 / 110101	144 / 7	ES / 4 / B	2,7 / 0,4	7,1,7,7,1,7
397	498|501|614|212|252|149 / 002202	144 / 7	NW / 6 / A	0,1,8 / 0,1,2	5,5,5,9,3,3
398	438|581|694|272|272|169 / 021221	144 / 7	NW / 6 / A	0,1,8 / 0,1,2	9,3,3,5,5,5
399	272|272|169|438|581|694 / 221021	144 / 7	EN / 6 / A	0,1,8 / 0,1,2	5,5,5,9,3,3
400	212|252|149|498|501|614 / 202002	144 / 7	EN / 6 / A	0,1,8 / 0,1,2	9,3,3,5,5,5
401	478|541|694|272|272|169 / 111221	144 / 7	NW / 6 / A	3,4,5,6 / 0,1,2,3	3,9,3,5,5,5
402	498|501|614|252|212|149 / 002022	144 / 7	NW / 6 / A	3,4,5,6 / 0,1,2,3	5,5,5,3,9,3
403	252|212|149|498|501|614 / 022002	144 / 7	EN / 6 / A	3,4,5,6 / 0,1,2,3	3,9,3,5,5,5
404	272|272|169|478|541|694 / 221111	144 / 7	EN / 6 / A	3,4,5,6 / 0,1,2,3	5,5,5,3,9,3
405	478|541|694|252|212|149 / 111022	144 / 7	NO / 4 / B	4,5 / 0,1	3,9,3,3,9,3
406	252|212|149|478|541|694 / 022111	144 / 7	ES / 4 / B	4,5 / 0,1	3,9,3,3,9,3
407	498|501|614|232|292|189 / 002110	144 / 7	NW / 6 / A	0,2,7 / 0,2,4	5,5,5,1,7,7
408	458|521|634|272|272|169 / 221221	144 / 7	NW / 6 / A	0,2,7 / 0,2,4	1,7,7,5,5,5
409	272|272|169|458|521|634 / 221221	144 / 7	EN / 6 / A	0,2,7 / 0,2,4	5,5,5,1,7,7
410	232|292|189|498|501|614 / 110002	144 / 7	EN / 6 / A	0,2,7 / 0,2,4	1,7,7,5,5,5
411	438|581|694|252|252|109 / 021001	144 / 7	NO / 8 / B	1,2,7,8 / 0,1,3,4	9,3,3,3,3,9
412	478|581|654|212|252|149 / 120202	144 / 7	NO / 8 / B	1,2,7,8 / 0,1,3,4	3,3,9,9,3,3
413	212|252|149|478|581|654 / 202120	144 / 7	ES / 8 / B	1,2,7,8 / 0,1,3,4	9,3,3,3,3,9
414	252|252|109|438|581|694 / 001021	144 / 7	ES / 8 / B	1,2,7,8 / 0,1,3,4	3,3,9,9,3,3
415	498|501|614|272|272|169 / 002221	432 / 7	NO / 12 / B	1,2,4,5,7,8 / 0,1,3,4,6,7	5,5,5,5,5,5
416	272|272|169|498|501|614 / 221002	432 / 7	ES / 12 / B	1,2,4,5,7,8 / 0,1,3,4,6,7	5,5,5,5,5,5
417	810|870|923|674|521|418 / 002221	144 / 8	NW / 6 / A	1,2,5,6 / 0,1,4,5	3,9,3,1,7,7
418	890|850|963|694|541|478 / 210111	144 / 8	NW / 6 / A	1,2,5,6 / 0,1,4,5	1,7,7,3,9,3
419	694|541|478|890|850|963 / 111210	144 / 8	EN / 6 / A	1,2,5,6 / 0,1,4,5	3,9,3,1,7,7
420	674|521|418|810|870|923 / 221002	144 / 8	EN / 6 / A	1,2,5,6 / 0,1,4,5	1,7,7,3,9,3
421	830|830|943|694|581|438 / 221120	144 / 8	NW / 8 / A	1,2,5,6 / 0,1,4,5	5,5,5,3,3,9
422	810|810|983|614|501|498 / 002200	144 / 8	NW / 8 / A	1,2,5,6 / 0,1,4,5	3,3,9,5,5,5
423	614|501|498|810|810|983 / 200002	144 / 8	EN / 8 / A	1,2,5,6 / 0,1,4,5	5,5,5,3,3,9
424	694|581|438|830|830|943 / 120221	144 / 8	EN / 8 / A	1,2,5,6 / 0,1,4,5	3,3,9,5,5,5
425	885|885|998|669|556|443 / 002012	1944 / 8	NO / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
426	850|850|903|674|521|418 / 110221	144 / 8	NO / 8 / B	0,3,4,7 / 0,1,4,6	7,7,1,1,7,7
427	890|850|963|634|521|458 / 210122	144 / 8	NO / 8 / B	0,3,4,7 / 0,1,4,6	1,7,7,7,7,1
428	669|556|443|885|885|998 / 012002	1944 / 8	ES / 45 / H	0,1,2,3,4,5,6,7,8 / 0,1,2,3,4,5,6,7,8	0,0,0,0,0,0
429	634|521|458|890|850|963 / 122210	144 / 8	ES / 8 / B	0,3,4,7 / 0,1,4,6	7,7,1,1,7,7
430	674|521|418|850|850|903 / 221110	144 / 8	ES / 8 / B	0,3,4,7 / 0,1,4,6	1,7,7,7,7,1
431	850|890|963|614|501|498 / 120200	144 / 8	NW / 8 / A	2,3,4,5 / 0,1,2,3	7,1,7,5,5,5
432	850|850|903|694|581|438 / 110120	144 / 8	NW / 6 / A	3,4,8 / 0,1,5	7,7,1,3,3,9
433	830|830|943|634|561|418 / 221101	144 / 8	NW / 8 / A	2,3,4,5 / 0,1,2,3	5,5,5,7,1,7
434	810|810|983|634|521|458 / 002122	144 / 8	NW / 6 / A	3,4,8 / 0,1,5	3,3,9,7,7,1
435	634|561|418|830|830|943 / 101221	144 / 8	EN / 8 / A	2,3,4,5 / 0,1,2,3	7,1,7,5,5,5
436	634|521|458|810|810|983 / 122002	144 / 8	EN / 6 / A	3,4,8 / 0,1,5	7,7,1,3,3,9

Continued on the next page.

Table D.1 (continued)

#	U_6 / $\rho_3(U_6)$	$\#T_{\min}^{-1}(U_6)$ / diagonal_c	hplus_kind / px_dimension / px_resolution kind_type	delta_diag_subset and delta_orbit	E-sign
437	614|501|498|850|890|963 / 200120	144 / 8	EN / 8 / A	2,3,4,5 / 0,1,2,3	5,5,5,7,1,7
438	694|581|438|850|850|903 / 120110	144 / 8	EN / 6 / A	3,4,8 / 0,1,5	3,3,9,7,7,1
439	850|850|903|614|501|498 / 110200	144 / 8	NW / 6 / A	0,2,5,7 / 0,2,4,6	7,7,1,5,5,5
440	870|810|923|634|561|418 / 002101	144 / 8	NW / 8 / A	0,2,5,7 / 0,2,4,6	9,3,3,7,1,7
441	830|830|943|634|521|458 / 221122	144 / 8	NW / 6 / A	0,2,5,7 / 0,2,4,6	5,5,5,7,7,1
442	850|890|963|654|581|478 / 120021	144 / 8	NW / 8 / A	0,2,5,7 / 0,2,4,6	7,1,7,9,3,3
443	634|521|458|830|830|943 / 122221	144 / 8	EN / 6 / A	0,2,5,7 / 0,2,4,6	7,7,1,5,5,5
444	654|581|478|850|890|963 / 021120	144 / 8	EN / 8 / A	0,2,5,7 / 0,2,4,6	9,3,3,7,1,7
445	614|501|498|850|850|903 / 200110	144 / 8	EN / 6 / A	0,2,5,7 / 0,2,4,6	5,5,5,7,7,1
446	634|561|418|870|810|923 / 101002	144 / 8	EN / 8 / A	0,2,5,7 / 0,2,4,6	7,1,7,9,3,3
447	850|890|963|634|561|418 / 120101	144 / 8	NO / 4 / B	1,6 / 0,4	7,1,7,7,1,7
448	634|561|418|850|890|963 / 101120	144 / 8	ES / 4 / B	1,6 / 0,4	7,1,7,7,1,7
449	830|830|943|654|581|478 / 221021	144 / 8	NW / 6 / A	0,7,8 / 0,1,2	5,5,5,9,3,3
450	870|810|923|614|501|498 / 002200	144 / 8	NW / 6 / A	0,7,8 / 0,1,2	9,3,3,5,5,5
451	614|501|498|870|810|923 / 200002	144 / 8	EN / 6 / A	0,7,8 / 0,1,2	5,5,5,9,3,3
452	654|581|478|830|830|943 / 021221	144 / 8	EN / 6 / A	0,7,8 / 0,1,2	9,3,3,5,5,5
453	810|870|923|614|501|498 / 002200	144 / 8	NW / 6 / A	2,3,4,5 / 0,1,2,3	3,9,3,5,5,5
454	830|830|943|694|541|478 / 221111	144 / 8	NW / 6 / A	2,3,4,5 / 0,1,2,3	5,5,5,3,9,3
455	694|541|478|830|830|943 / 111221	144 / 8	EN / 6 / A	2,3,4,5 / 0,1,2,3	3,9,3,5,5,5
456	614|501|498|810|870|923 / 200002	144 / 8	EN / 6 / A	2,3,4,5 / 0,1,2,3	5,5,5,3,9,3
457	810|870|923|694|541|478 / 002111	144 / 8	NO / 4 / B	3,4 / 0,1	3,9,3,3,9,3
458	694|541|478|810|870|923 / 111002	144 / 8	ES / 4 / B	3,4 / 0,1	3,9,3,3,9,3
459	830|830|943|674|521|418 / 221221	144 / 8	NW / 6 / A	1,6,8 / 0,2,4	5,5,5,1,7,7
460	890|850|963|614|501|498 / 210200	144 / 8	NW / 6 / A	1,6,8 / 0,2,4	1,7,7,5,5,5
461	614|501|498|890|850|963 / 200210	144 / 8	EN / 6 / A	1,6,8 / 0,2,4	5,5,5,1,7,7
462	674|521|418|830|830|943 / 221221	144 / 8	EN / 6 / A	1,6,8 / 0,2,4	1,7,7,5,5,5
463	870|810|923|694|581|438 / 002120	144 / 8	NO / 8 / B	0,1,6,7 / 0,1,3,4	9,3,3,3,3,9
464	810|810|983|654|581|478 / 002021	144 / 8	NO / 8 / B	0,1,6,7 / 0,1,3,4	3,3,9,9,3,3
465	654|581|478|810|810|983 / 021002	144 / 8	ES / 8 / B	0,1,6,7 / 0,1,3,4	9,3,3,3,3,9
466	694|581|438|870|810|923 / 120002	144 / 8	ES / 8 / B	0,1,6,7 / 0,1,3,4	3,3,9,9,3,3
467	830|830|943|614|501|498 / 221200	432 / 8	NO / 12 / B	0,1,3,4,6,7 / 0,1,3,4,6,7	5,5,5,5,5,5
468	614|501|498|830|830|943 / 200221	432 / 8	ES / 12 / B	0,1,3,4,6,7 / 0,1,3,4,6,7	5,5,5,5,5,5

The table is an enumeration of the image of $T_{\min}$. The specification of the domain remains in $\Omega_{\text{seed}} = \mathcal{S}^2$; to recover the individual initial conditions of an emission, one takes its fiber $T_{\min}^{-1}(U_6)$, whose cardinality is listed as $\#T_{\min}^{-1}(U_6)$.

D.2 Certification at depth 10^4 of the coinductive expansions of π, e and φ

Mathematical scope. The preceding uniform theorem quantifies over every positive integer N and constructs the infinite object as the inverse limit of its compatible cylinders. This chapter runs the case $N = 10^4$ as a particularly demanding finite certificate; 10^4 is not a bound on the generator, just as a million-digit run would be another instance of the same algorithm, not a modification of its law.

The present execution fixes a computational depth and checks that the resulting cylinder is contained in a single decimal cell of ten thousand positions. The smallest number of blocks compatible with the nonadic monodromy and with

$$3^{6B} > 10^{10000}$$

is

$$B = 3501 = 389 \cdot 9, \qquad 6B = 21006 \text{ trits}.$$

For each of the three channels, the 729 subcylinders are exhaustively traversed at every stage, and the unique subcylinder compatible with the reader's rational bracket is retained. The calculation yields

quantity calculated per channel	value
complete partitions $729 \to 1$	3501
transported trits	21006
pairs of equal phase $(K, K+9)$	3492
complete nonadic turns	388

and does not require an additional nonad. The first 396 blocks coincide exactly with the preceding thousand-digit witness.

The closure, propagation, and self-scaling readers are evaluated by means of exact rational inequalities. Computational acceleration reorganizes the sums through integer arithmetic and does not open libraries of constants or target decimal prefixes. Only after the outputs have been written is a second, independent computation performed. The following are verified, in this order: the rational generation of the three channels, the G9–T1 composition of α_{HMT}, and the agreement of all ten thousand digits with the reference expansions consulted only at the end. The three comparisons are exact at all ten thousand computed positions. This finite verification does not replace the uniform argument for prolongation to arbitrary depth.

The expanded proof also makes the meaning of return precise. Across the 3492 pairs per channel, the phase is preserved and the complete state always changes, since depth, prefix, and cylinder advance. A reduced local projection may recur in some pairs without the TPK state returning. This distinction replaces the stronger claim of local aperiodicity with the property actually proved: phase monodromy with memory of the enriched state.

The four complete expansions are printed in appendix E. The recurrences, rational intervals, and preceding counts determine the calculation; the printed expansions are its finite truncations and make it possible to compare each digit without becoming premises of the generator.

Appendix E

Arbitrary precision accuracy certificates and finite reproducible decimal samples

This appendix prints the complete outputs of the execution described in appendix D.2 y section 31.6. The line breaks are exclusively typographical. Each expansion constitutes a finite witness of the coinductive law of arbitrary precision proved in Part III; it neither defines the constant nor limits the depth of the generator.

E.1 Coinductive expansion of π and compatible truncation certificate

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3.14159265358979323846264338327950288419716939937510582097494459230781640628620899862803482534211
70679821480865132823066470938446095505822317253594081284811174502841027019385211055596446229489
54930381964428810975665933446128475648233786783165271201909145648566923460348610454326648213393 6
07260249141273724587006606315588174881520920962829254091715364367892590360011330530548820466521 3
84146951941511609433057270365759591953092186117381932611793105118548074462379962749567351885752 7
24891227938183011949129833673362440656643086021394946395224737190702179860943702770539217176293 1
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98372978049951059731732816096318595024459455346908302642522308253344685035261931188171010000313 78
38752886587533208381420617177669147303598253490428755468731159562863882353787593751957818577805
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01852968995773622599413891249721775283479131515574857242454150695950829533116861727855889075098 3
81754637464939319255060400927701671139009848824012858361603563707660104710181942955596198946767 8
37449448255379774726847104047534646208046842590694912933136770289891521047521620569660240580381
50193511253382430035587640247496473263914199272604269922796782354781636009341721641219924586315 0
30286182974555706749838505494588586926995690927210797509302955321165344987202755960236480665499 1
19881834797753566369807426542527862551818417574672890977772793800081647060016145249192173212714 7
72350141441973568548161361157352552133475741849468435233239073941433345477624168625189835694855
62099219222184272550254256887671790494601653466804988627232791786085784383827967976681454100953 8
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45047123713786960956364371917287467764657573962413890865832645995813390478027590099465764078951 2
69468398352595709825822620522489407726719478268482601476990902640136394437455305068203496252541 7
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7464915995054973742562690104903778198683593814657412680492564879855614537234786733039046883834363465537949864192705638729317487233208376011230299113679386270894387993620162951541337142489283072201269014754668476533761647737946752004907571555278196536213239264061601363581559074222020203187277605277219005561484255518792530343513984425322341576233610642506390497500865627109535919465897514131034822769306247435363256916078154781811528436679570611086153315044521274739245449454236828860613408414863776700961207151249140430272538607648236341433462351897576645216413767969031495019108575984423919862916421939949072362346468441173940326591840443780513338945257423995082965912285085582157250310712570126683024029295252201187267675622041542051618416348475651699981161410100029960783869092916030288400269104140792886215078424516709087000699282120660418371806535567252532567532861291042487761825829765157959847035622262934860034158722980534989650226291748788202734209222245339856264766914905562842503912757710284027998066365825488926488025456610172967026640765590429099456815065265305371829412703369313785178609040708667114965583434347693857817113864558736781230145876871266034891390956200993936103102916161528813843790990423174733639480457593149314052976345757481193567091101377517210080315590248530906692037671922033229094334676851422144773793937517034366199104033751117354719185504644902636551281622882446257591633303910722538374218214088350865739171509682887478265699595974490661758344137522397096834080053559849175417381883999446974867626525165827658483588453142775687900290951702835297163445621296404352311760066510124120065975585127617858382920419748442360800719304576189323492292796501987518721272675079812554709589045563579212210333466974992356302549478024901141952123828153091140790738602515227429958180724716259166854513331239480494707911915326734302824418604142636395480004480026704962482017928964766975831832713142517029692348896276684403232609275249603579964692565049368183609003238092934595889706953653494060340216654437558900456328822505452556405644824651518754711962184439658253375438856909411303150952617937800297412076651479394259029896959469955657612186561967337862362561252163208628692221032748892184536480229678070576561514463204692790682120738837781423356282360896320806822246801224861177185896381409183903673672220888321513755600372798394004152970028783076670944474560134556417254370906979396122571429894671543578468788614445812314593571984922528471605049221242470141214780573455105008019086996033027634787081081754501193071412233908663938339529425786905076431006383519834389341596131854347546495569781038293097164651438407007073604112373599843452251610507027056235266012764848308407611830130527932054274628654036036745328651057065874882256981579367897669742205750596834408697350201410206723585020072452256326513410559240190274216248439140359989535394590944070469120914093870012645600162374288021092764579310657922955249887275846101264836999892256959688159205600101655256375678
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E.2 Coinductive expansion of e and compatible truncation certificate

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decimals

E.3 Coinductive expansion of φ and compatible truncation certificate

decimal file


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86784432382482189306175703195638032308197193635672741964387262587061543307296370381275151704060 0
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509231327977113803
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E.4 Expansion of α_{HMT}

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0.00729735256928380099728510472380662999564172539642709185404682532485972586311002762549975056 8
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836572260534209515
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Appendix F

Formal Verified Formalization of the Algebraic Core and Certification of Finite Identities

The formalization checks the finite identities on matrices, algebras, and typed dodecaphase states. Its function is to isolate the algebraic obligations amenable to mechanical reduction and to keep the provenance of the data, finite enumeration, and formal proof separate.

F.1 Formal Verification of Finite Algebraic Identities

The following finite equalities are formalized in Lean and decided by reduction of the logical core, without additional external axioms. The theorems prove:

Theorem	Contents
`aw_symmetric`	symmetry of the Paley–Witt matrix A_W;
`aw_squared_minus_identity`	$A_W^2 = -I_6$ over $\mathbb{F}_3$;
`x_sq_add_one_has_no_root` criterion	irreducibility of $x^2 + 1$ over $\mathbb{F}_3$;
`dual_has_nonzero_nilpotent` record	existence of a nilpotent in the parabolic algebra;
`split_has_nontrivial_idempotent`	existence of a nontrivial idempotent in the hyperbolic algebra;
`canonical_charge`	$\sum_{m=1}^{12} K_m = 6263$;
identifier	The three Hadamard orbits produce the published
`hadamard_orbit_one/two/three`	dodecaphasic state.

The formalization takes the matrices and the dodecaphase vector as typed data and verifies their algebraic consequences. The construction of the signed state and the enumeration of the (104 976) seeds belong to the preceding combinatorial theorems and are not replaced by this formal reduction.

Appendix G

Correspondence between statements, proofs, certificates, and primary scientific sources

The following table identifies the principal mathematical residence of each result and the negative control that distinguishes its derivation from a mere coincidence. No row replaces the statement or the principal proof.

Table G.1: Results, reference proofs and reproducible controls.

Result	Reference proof	Control reproducible
Construction of APP, double evaluation, and spectral jump	APP chapters and genealogical arithmetic	Support, spectrum, and reconstruction identities of the two leaves
TRIT state, orientation, and three quadratic algebras	TRIT Chapter and Algebraic Classification	Conservation of residue, quotient, orientation, and carry
Complete operational inventory of the TPK	Chapters of transport, memory, and fundamental operators	Commutativity, naturality, and compatible prolongation
Extrema and Holographic Mean Ratio	Nonadic cubic completion	Identity $1000 = 729 + 243 + 27 + 1$ and conservation of the unit
Extendible generation of π, e, φ	Coinductive Biographies and Nonadic Monodromy	Nested cylinders, compatible truncations, and oracle falsifier
Two internal constructions of α_{HMT}	Dodecaphase wheel and reversible cards	Equivalence of outputs and separation of CODATA recognition
Dodecaphase projectors and defect $-1/12$	Hensel–Witt Kernel	Symmetry, idempotence, ranks $3/4$, and normalized trace
Archimedean and exceptional double projection	Cylinder reader and incidence reader	Same input state, typed codomains and non-identification of outputs
Leech lattice, VOA and Moonshine	Complete exceptional runner	Incidence Entanglers and Equivariant Obstructions
Genealogy of the Number and Spectral Functionals	Closure of the concept of number and spectral arithmetic	Euler Products, Characters, and Directed Bounds
Action line and decimal order -34	Bidirectional action and conjugate angular coordinates	Cokernel of order 16 and separation of the relative corrector
Barbero-Immirzi, area operator and spectrum	Hexad-octad transition and modular Hamiltonian	Channels $90/120$, area, first law, and relative entropy
CKM Angles and CP Violation	Chapter of Rational Mixing and Reflection	Unitarity, Jarlskog and caution regarding physical reflection
Bidirectional electron and refined mass	Definition of particle and first electronic implementation	Central fixed point, readers K_D/K_Ω, and rector beta value
Atlas $81/56/13/324$ and mass operators	General Mass Law and route-by-route atlas	Full coverage, null sector and prospective selection

Result	Reference proof	Control reproducible
Permittivity and permeability of the vacuum	Bilayer constitutive response	Separate reconstruction of Z, μ, ε, c
Variational and Hamiltonian Formulation of Gravity	Einstein–Cartan–Holst Gravity	Euler–Lagrange/Hamilton equivalence and torsional controls
Navier–Stokes and Yang–Mills	Terminal Closures of the Fluid and Caliber Realizations	PF84 matrix, coercivity, and transfer

The preceding reduced formulations do not govern the theorems of Part IV. The scope of each specialization is determined there by its object, its domain, its maps, and its comparison identities. In particular, a historical label cannot by itself declare a conclusion open or closed, just as a documentary classification does not replace the commutativity of a diagram. The finite constructions in the preceding table are preserved as antecedents; their prolongation is decided through the explicit comparators of each chapter.

Appendix H

Domains, charts and realization interfaces

This block separates the internal theorems, evaluation charts, physical interfaces, and extension programs. The separation permits each equality to be read in its precise domain and keeps visible which data, gauges, orientations, or boundary conditions enter a realization.

H.1 Domains, codomains, and hypotheses of the results

1. The census $104\,976 \to 468 \to 243$, the image of 42 triples, the factorization $468 = 26 \times 18$, the five regions, and the five cycles are exhaustive theorems of the finite transport system specified in this work.

2. The enriched catalogue produces, through finite fiber predicates and without decimal consultation, the unique closure orbit, the unique propagation orbit and the unique self-scaling orbit; the internal gauge orients the representatives 010211, 201101, 121200 within them. Their characters are constructed, respectively, by the five-region closure with unit return, the semigroup recurrence $(n+1)a_{n+1} = a_n$ and the unique positive ray of the incidence F_{av}. Archimedean evaluation then publishes the generated values and conventional recognition names them π, e, φ. For each depth N, TPK extension produces a unique prefix compatible with that of depth $N-1$; the inverse limit of these prefixes fixes the complete expansion. Thus words, characters and values are successive stages of an internal construction, not labels chosen through external tables.

3. The dodecaphase theorem has as its causal domain the enriched terminal state x_{term} produced by APP–TRIT–TPK. The signed chart publishes from it $x_{\mathrm{term}} \xrightarrow{R_{\mathrm{sgn}}} U_{12}^{\mathrm{sgn}}$, and integral inversion then publishes $U_{12}^{\mathrm{sgn}} \xrightarrow{\mathcal{H}_{12}^{-1}} K$. In this domain, the correspondences with (D_3K, D_4K, Q), the Hadamard representation and the dodecaphase closure are exact and reversible. Neither U nor K are generating inputs. The reduced trace of a positional trajectory and the complete bivariate system are distinct objects, as established in the section 4.1.

4. Equality with the 2018 central string is an external recognition. The internal mathematical word and its proof are independent of the subsequent metrological recommendations.

5. The finite instance of the double projection proves a common flag, W_{12} and M_{12}. In the lattice branch, the gluing of A_2^{12}, the radial chamber and the origin are fixed; the resulting 3-neighbor is certified even, unimodular and rootless, and is classically identified with Leech. The construction of $V^\natural$ and the Monster is a classical consequence. The arrow $W_{12} \to W_{24}$ is realized through the residual design of a dodecad and the unique lift of its hexads, relative to a choice (D, ℓ); the identifications are made through maps, not through cardinality equalities. The projective compatibility of these data and their double realization on the common enriched state are proved in the theorems 17.8 y 17.9.

6. The cylinder theorem produces a unique real for each history in its domain. The extension law coherently covers each compact set and the directed exhaustion covers the entire line, as proved by the theorems 18.3 y 18.9 y proposition 18.5. Therefore, evaluation is surjective and its Archimedean quotient is $\mathbb{R}$; coverage is a conclusion of the HMT generator, not an additional premise.

7. The native product is proved in the canonical sector of words. The monoidal extension to the complete space of histories and the intertwining with the survival filtrations are formulated as distinct maps in the chapter 27.

H.2 Causal Invariance of Interfaces with Respect to External References

H.2.1 Criterion of causal independence with respect to external references: directed acyclic graph and prior fixation of the output

Let P be the internal primitives, R the declared rules, readers, gauges, and normalizations, G a generator, $o = G(\mathsf{P}, \mathsf{R})$ its output, and y an external reference.

Definition H.1 (Prospectively independent interface)**.** The interface is prospectively independent of y if:

1. $y \notin \mathrm{Anc}(o)$;
2. $\mathsf{P}, \mathsf{R}, G$ and the negative controls are fixed before y is opened;
3. the execution has no direct or indirect access to y;
4. o and its trace are sealed before computing a comparison $\mathrm{cmp}(o, y)$.

Definition H.2 (Selection independence)**.** The blindness of each execution is not enough if, after producing candidates $G_j(\mathsf{P}, \mathsf{R}_j)$, one publishes

$$j_*(y) = \arg\min_j \mathrm{cmp}\big(G_j(\mathsf{P}, \mathsf{R}_j), y\big)\,.$$

Selection independence prohibits this operation from determining the published output.

Definition H.3 (Multivalued prediction)**.** If the non-metrological physical descriptors and the internal constraints admit several multisections, the prospective output is the complete set

$$\widehat{\mathfrak{R}}(d) = \{\mathcal{M} : \mathcal{M} \text{ satisfies all the constraints preregistered for } d\}.$$

Nonuniqueness is an output of the procedure. Subsequently choosing an element by proximity to a magnitude is prohibited.

H.2.2 No-Interference Theorem

Definition H.4 (Directed Acyclic Faithful and Complete Causal Graph)**.** A directed acyclic graph D is faithful if every node represents the same function and the same inputs as the cited construction. It is complete for an output if every causally relevant dependency appears as an edge or a declared root. Faithfulness and completeness are semantic hypotheses; the acyclicity of the dataset does not prove them.

Theorem H.5 (Prospective independence conditional on the causal graph)**.** *Let a deterministic system of structural equations be represented by a finite directed acyclic graph D that is faithful and complete for o. If $y \notin \mathrm{Anc}_D(o)$, all ancestral roots of o remain frozen and the execution has no hidden access to y, then for any y_0, y_1,*

$$o_{\mathrm{do}(y=y_0)} = o_{\mathrm{do}(y=y_1)}.$$

Proof. The predecessors of o are topologically ordered. Their roots are internal and remain fixed. By induction on the order, each node receives the same arguments under both interventions and, by determinism, returns the same value. Equality propagates to o. The cryptographic digest does not create the independence; it merely makes a subsequent alteration detectable. □

Corollary H.6 (Permutation of the external comparison)**.** *If names, rows, or external values are permuted after sealing, the internal output does not change, although its score may change:*

$$o_{\mathrm{do}(y=\sigma y_0)} = o_{\mathrm{do}(y=y_0)}, \qquad \mathrm{cmp}(o, \sigma y_0) \neq \mathrm{cmp}(o, y_0).$$

Proposition H.7 (Calibrated branch separation)**.** *If an external reference intervened in a historical selection, the dependence does not contaminate internal objects that do not descend from that selection. The branch can be isolated and subjected to a new prospective protocol without modifying the kernel.*

Proof. To be a predecessor is a local property of the graph. The branches

$$Z_0 \longrightarrow p = 5 \longrightarrow \text{electronic scale}, \qquad \{\text{masses}\} \longrightarrow \{\text{calibrated signatures}\} \longrightarrow \text{table}$$

remain outside the protected subgraph. The towers, the formal character, the atlas, and the center $(5, 5)$ receive no edges from those references. □

H.2.3 Counterfactual Invariance on the Directed Causal Graph and Scope of the Conclusion

The published causal graph contains 71 nodes, 89 edges, 16 interfaces, and 29 protected outputs. The causal check establishes acyclicity, counts, declared routes, existence of the registered sources, absence of external references among the recorded ancestors, and counterfactual invariance of the symbolic model. The result is

$$\#\{\text{external predecessors of protected outputs}\} = 0.$$

The certificate does not by itself prove that no edge is missing, nor does it audit every data reading of each scientific program. Its strength is therefore PROVED FROM EXPLICIT STARTING STRUCTURE: the conclusion is obtained on the expressly declared directed acyclic graph and remains conditional on its fidelity and completeness.

Simultaneously replacing the CODATA, Planck, Barbero–Immirzi, CKM, mass, gravity and time-crystal references can alter comparisons and calibrated branches. Within the declared graph, it does not alter the catalogue of 468 emissions, the five regions of π, the correlated coinductive generation, α_{12}, C^*, the decade -34, the towers, the atlas, the electronic center, the double projection or the exceptional corridor.

H.2.4 Multivalued trial without access to the target value and identification criterion

Algorithm H.8 (Future assignment without magnitudes)**.** The physical interface is executed in four irreversible phases:

1. *Freezing*: authority, atlas, involutions, restrictions, code, priority, and negative controls are sealed.
2. *Permitted input*: only sector, generation, charge, color, conjugation, spin/statistics, composition, and metrological type are received; masses, refined signatures, residues, angles, and units are excluded.
3. *Output*: all compatible multisections are enumerated; for each extension of each multisection, the observable route, record, signature, χ_{form}, specialization, and observable are computed. In composite objects, the extensions and records are spliced first. The complete multiset is sealed.
4. *Opening*: a distinct custodian opens the magnitudes and computes residuals, uncertainties, and covariances without modifying any candidate.

The enriched multisection, rather than a point extension selected retrospectively, is the output of the physical matching. The subsequent evaluation may be performed fiber by fiber. The preceding algorithm is a falsifiable specification; it is not presented as an already executed multiset prediction.

Comparison Criterion H.9 (Failure of the blind protocol)**.** The protocol fails if the code can read a magnitude, if a calibrated historical signature reappears as an input, if a candidate is discarded after opening the reference, if positive mass is confused with exactly zero mass, or if the sealed multiset changes when the external catalog is permuted.

H.2.5 Formal character and reconstructive completeness

For $v = (n_A, s_C, k, b_{120}, b_\zeta) \in \mathbb{Z}^5$, the formal character prior to any specialization is

$$\chi_{\mathrm{form}}(v; x_A, x_C, x_\Delta, x_{120}, x_{270}) = \exp\bigl(n_A x_A + s_C x_C + k x_\Delta + b_{120} x_{120} + b_\zeta x_{270}\bigr)\,.$$

The declared specialization

$$\Theta_0 = \left(A_{\mathrm{rad}}, \frac{C^*_{\mathrm{rad}}}{6}, \Delta_4, s_{120}, \zeta_{270}\right), \qquad \zeta_{270} = 135\Delta_4^2,$$

produces $\chi_0(v) = \chi_{\mathrm{form}}(v; \Theta_0)$. The raw signature, the formal character, its central specialization, the global refinement $\eta \in \mathcal{E}_{\mathfrak{S}}$, and a calibrated table are five different objects.

Theorem H.10 (Positive reconstructive completeness)**.** *Let*

$$\mathcal{N} = \langle 90, 120\rangle_{>0}$$

and let $(n_j)_{j\ge1}$ be its increasing enumeration. Given $0 < q_+ < q_- < 1$, define

$$a_j = q_-^{n_j} - q_+^{n_j} > 0.$$

Then $a_j \to 0$ and $\sum_j a_j < \infty$. Having fixed a deterministic rule for resolving half-integer ties, for every $r \in \mathbb{R}$ the recurrence

$$r_0 = r, \qquad \nu_j = \mathrm{nint}\left(\frac{r_{j-1}}{a_j}\right), \qquad r_j = r_{j-1} - \nu_j a_j$$

satisfies

$$|r_j| \le \frac{a_j}{2} \longrightarrow 0, \qquad r = \sum_{j\ge1} \nu_j a_j, \qquad \sum_{j\ge1} |\nu_j a_j| < \infty.$$

Therefore every $R > 0$, with $r = \log R$, is represented exactly in the completion by $\exp(\sum_j \nu_j a_j)$.

Proof. The indices of $\mathcal{N}$ are 90, 120 and all multiples of 30 from 180 onward. Consequently,

$$\sum_{n\in\mathcal{N}} q_-^n = q_-^{90} + q_-^{120} + \frac{q_-^{180}}{1 - q_-^{30}} < \infty.$$

As $0 < a_j < q_-^{n_j}$, the sequence is sumable and converge to zero. The definition of integer mas proximo implica $|r_j| \le a_j/2$. By telescopado, $\sum_{i=1}^{N} \nu_i a_i = r - r_N \to r$. For $j \ge 2$,

$$|\nu_j a_j| = |r_{j-1} - r_j| \le \frac{a_{j-1} + a_j}{2},$$

and the summability of (a_j) proves absolute convergence. □

The theorem establishes completeness of the dictionary on $\mathbb{R}_{>0}$. If the coefficients ν_j are obtained from the residue of a known mass, their use is CALIBRATED RECONSTRUCTION; it does not provide prospective selection. Nor does it represent exactly zero mass. The photon, gluon, or a prospective spin-two mode require exact symmetry protection in the physical representation.

H.2.6 Prospective invariance protocols for constant families

Barbero–Immirzi. S_{90}, S_{120}, the weights $(12, -1)$, and their normalization are frozen before a physical reference is opened. The evaluation of $\Gamma_{6\to 8}$ is blind to that reference, but the prior uniqueness of the series and weights and the intertwiner with the area operator are distinct problems.

CKM and CP. The four equations, their coefficients, and the angular convention are sealed; the four parameters, $|V|$ and J, are published simultaneously. The comparison uses a covariance fixed in advance.

Constitutive. The dimensionless identities precede Z_{ref} and c_{ref}. A change of chart transforms the realization, not the identities, and does not create four independent predictions.

Gravity and inertia. The ratios $G_{\mathrm{tor}}/I_{\mathrm{tor}} = 1$ and $G_{\mathrm{av}}/I_{\mathrm{av}} = \pi/\varphi^2$ belong to different layers. Their physical interpretation requires a morphism and a prospective contrast; a cosmological phenomenon does not select the ratio.

H.2.7 Evidentiary stratification: internal identities, deductions, calibrated reconstructions, and interfaces subject to mathematical closure

Table H.1: Levels that causal independence does not allow to collapse.

Level	Example	Allowed Statement
Certified internal generator	Regions, α_{12}, C^*, -34, atlas	Output obtained in the declared domain.
Blind conditional evaluation	$\Gamma_{6\to 8}$, fixed CKM, $\chi_0(v)$	Do not consult the white, but it depends on the functional or signature.
Blind protocol not executed	Multivalued specification	There exists a falsifiable procedure; not a sealed output.
Calibrated Reconstruction	Historical signatures and refinement from $r = \log R$	Exact evaluation conditioned on a reference.
Pending physical interface	Particle–antiparticle, area, zero mass, gravity	Domain defined; lack of representation or contrast.

Appendix I

Reproducibility of the atlas and comparison of mass realizations

The theory of masses is generated over the complete domain of 81 positions, 56 route families, 13 multisections, 324 oriented routes, and the null sector. The structural phase computes

$$\begin{aligned}\text{position or fibre} &\longrightarrow \text{family} \longrightarrow \text{oriented route}\\ &\longrightarrow \text{record} \longrightarrow \text{nominal signature} \longrightarrow \text{character and tower},\end{aligned}$$

and, when the fiber possesses the corresponding operator reader, adds the refinement branch

$$\text{record} \longrightarrow (K_D, K_\Omega) \longrightarrow \text{fiber spectrum}.$$

The second branch refines the first and does not replace it: in particular, W, Z, and H preserve their CT–108 routes, their angular signatures, and their scalar characters even though they do not belong to the operatorial atlas of thirteen multisections. These invariants are fixed before the external comparison; a metrological reference does not intervene in the selection of route, position, signature, or coefficient.

I.1 Finite invariants of classification

- complete atlas of 81 cells with their readers;
- 56 route families with their spectra;
- 13 multisection operators;
- 324 oriented routes with ledger and character;
- complete inventories of masses and widths;
- chart comparativa posterior to the selection genealogica.

I.2 Conditions of accuracy, stability, and reproducibility of the certificates

The integral control requires the coverage $81/56/13/324$, the null sector, the 14 profiles of K_D, the 9 profiles of K_Ω, the 40 joint pairs, and the 18 reader coincidences. It further verifies that the refined electron mass is evaluated from a single expression, that the null sectors retain κ as not applicable, that an operator row does not publish a scalar residual, that the twelve nominal routes $e, \mu, \tau, u, d, s, c, b, t, W, Z, H$ retain their scalar character, and that the physical realization preserves the proper codomain of each output. The absence of a K_D/K_Ω pair in a sector cannot degrade or erase its native scalar reader.

The reproduced electronic output is

$$m_e^{\beta} = m_e^{(0)} R_{\mathrm{act}}^{80-54\alpha_{\mathrm{HMT}}+6\Delta_4} = 0.510998950690000808608257165\ldots\ \mathrm{MeV}/c^2.$$

Higher-rank fibers publish their operator pair, their spectra, and the corresponding realization morphism. Eigenvalue selection is determined by that morphism before external comparison.

I.3 Scientific domain of classification

The mass domain is the preceding complete atlas. The construction receives exclusively positions, families, multisections, routes, records, and typed operators; provenance data remain separate from the scientific load.

Appendix J

Complementary Proofs of the Algebraic, Spectral, Modular, and Electroweak Identities

This appendix gathers the proofs that can be verified without reconstructing the entire narration. The hypotheses appear in each statement. The classical results used to interpret an operator that has already been constructed are distinguished from the internal identities.

J.1 Positivity of the vacuum operator

Lemma J.1. *If $T = T^*$ and $\mathrm{Spec}(T) \subset (0,1)$, then $I - T^n > 0$ for every $n \geq 1$, $I - T^{120}$ is invertible and*

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1} > 0.$$

Proof. For $t \in (0,1)$, $1 - t^n > 0$. By functional calculus, $I - T^n$ is positive. Since the spectrum of $I - T^{120}$ is separated from zero in finite dimension, it has a positive inverse. Both factors are functions of T, commute, and their product is positive. □

Lemma J.2 (Monotonicity of the completion)**.** *The function*

$$r(q) = \frac{1 - q^{90}}{1 - q^{120}}$$

is strictly decreasing in $(0,1)$.

Proof. With $s = q^{30}$,

$$r(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}.$$

A derivation yields

$$\frac{dr}{ds} = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0.$$

As $ds/dq = 30q^{29} > 0$, also $dr/dq < 0$. □

J.2 Traces that recover the constituent characters

From $\log \mathcal{V} = (\log r_+)P_+ + (\log r_-)P_-$ and $\tau(P_\pm) = 1/2$,

$$2\tau(\log \mathcal{V}) = \log r_+ + \log r_- = \log(r_+ r_-).$$

Moreover, $RP_+ = P_+$, $RP_- = -P_-$; hence

$$-2\tau(R \log \mathcal{V}) = -\log r_+ + \log r_- = \log(r_-/r_+).$$

Exponentiating yields $\widehat{\mu}, \widehat{\varepsilon}, \widehat{Z}, \widehat{c}$. There is no approximation in this step.

J.3 UHF Tower Compatibility

Be $\mathcal{A}_m = M_2 \otimes M_{9^m}$ and the inclusion $\iota_m(x) = x \otimes I_9$. The expectativa

$$E_m = \mathrm{id}_{M_2} \otimes \mathrm{id}_{M_{9^m}} \otimes \tau_9$$

satisfies $E_m \iota_m = \mathrm{id}$. Therefore,

$$E_m(\mathcal{V} \otimes I_{9^{m+1}}) = \mathcal{V} \otimes I_{9^m}.$$

The same equality holds for every continuous function of $\mathcal{V}$, in particular $\log \mathcal{V}$. The normalized trace is compatible, and the characters do not change under refinement.

J.4 Coefficients of the analytic criticality profile

Let

$$b_m = \frac{2(3m-1)}{\sqrt{899}}.$$

The expansion $1 - \cos(b_m x) = b_m^2 x^2/2 - b_m^4 x^4/24 + O(x^6)$ gives

$$u_0(x) = \frac{1}{24} \sum_{m=1}^{12} b_m^2 x^2 - \frac{1}{288} \sum_{m=1}^{12} b_m^4 x^4 + O(x^6).$$

Consequently,

$$\sum b_m^2 = \frac{4}{899} \sum (3m-1)^2 = 24,$$

whence the quadratic coefficient is one. The integer calculation of the fourth moment reduces the coefficient to

$$-\frac{715769}{2424603}.$$

This fraction provides an algebraic control more informative than its decimal value.

J.5 Closed trigonometric sum

The identity

$$\sum_{m=1}^{M} \cos(a + (m-1)d) = \frac{\sin(Md/2)}{\sin(d/2)} \cos\!\left(a + \frac{M-1}{2} d\right)$$

with $M = 12$, $a = 4y$, $d = 6y$ produces

$$\sum_{m=1}^{12} \cos(2(3m-1)y) = \frac{\sin 36y}{\sin 3y} \cos 37y.$$

Dividing by 12 proves the first form of equation (45.6); $2 \sin 36y \cos 37y = \sin 73y - \sin y$ proves the second.

J.6 Schwarziano negativo

For $x \in (0,1]$, each $b_m x \in (0,\pi)$. Therefore

$$u_0'(x) = \frac{1}{12}\sum b_m \sin(b_m x) > 0,$$

$$u_0'''(x) = -\frac{1}{12}\sum b_m^3 \sin(b_m x) < 0.$$

Since $f_\lambda' = -\lambda u_0'$, the factors $-\lambda$ cancel in the Schwarzian quotients, and

$$S(f_\lambda) = \frac{u_0'''}{u_0'} - \frac{3}{2}(u_0''/u_0')^2 < 0.$$

Parity extends the conclusion to $[-1,0)$. The origin is critical quadratic by $u_0(x) = x^2 + O(x^4)$.

J.7 Levy integral

The dominated convergence allows

$$\begin{aligned} -\int_0^1 \log x \, d\mu_G(x) &= \frac{1}{\log 2}\int_0^1 \frac{-\log x}{1+x}\,dx \\ &= \frac{1}{\log 2}\sum_{n=0}^{\infty}(-1)^n \int_0^1 x^n(-\log x)\,dx \\ &= \frac{1}{\log 2}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n^2} \\ &= \frac{\eta(2)}{\log 2} = \frac{\pi^2}{12\log 2}. \end{aligned}$$

J.8 Values of the dodecaphase character

The product $\chi_{12} = \chi_{-3}\chi_{-4}$ vanishes if $\gcd(n,12) > 1$. In the four unit classes, multiplication yields $(+,-,-,+)$. The series

$$L(s,\chi_{12}) = \sum_{k\ge 0}\left[\frac{1}{(12k+1)^s} - \frac{1}{(12k+5)^s} - \frac{1}{(12k+7)^s} + \frac{1}{(12k+11)^s}\right]$$

can be evaluated at $s=1$ by means of Euler products or the field-unit formula for $\mathbb{Q}(\sqrt{3})$, yielding equation (41.8). At $s=2$, the generalized Bernoulli formula yields equation (41.9). Direct multiplication of the character provides a falsification criterion independent of both evaluations.

J.9 Reconstruction of the areal and modular sectors

Let

$$N = N_{90} + N_{120}, \qquad D = 90N_{90} + 120N_{120}.$$

The system matrix is

$$M = \begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}, \qquad \det M = 30.$$

Therefore,

$$M^{-1} = \frac{1}{30}\begin{pmatrix} 120 & -1 \\ -90 & 1 \end{pmatrix},$$

and

$$N_{90} = 4N - D/30, \qquad N_{120} = D/30 - 3N.$$

The determinant 30 explains why the fundamental modular step reappears in the infinite classification.

J.10 Purification through a thermofield double state

For a qutrit with weight $w > 0$,

$$Z(w) = 1 + \mathrm{e}^{-w} + \mathrm{e}^{-2w}, \qquad p_n = \frac{\mathrm{e}^{-nw}}{Z(w)}.$$

Define

$$|\mathrm{TFD}(w)\rangle = \sum_{n=0}^{2} \sqrt{p_n}\,|n\rangle_A |n\rangle_{\bar{A}}.$$

Then

$$\mathrm{Tr}_{\bar{A}}\,|\mathrm{TFD}\rangle\langle\mathrm{TFD}| = \sum_n p_n |n\rangle\langle n| = \rho_A.$$

The equality of the entropies of the two subsystems is immediate from the Schmidt decomposition. The product of eighteen copies of each channel purifies the total boundary state.

J.11 1st areal modular law

For a differentiable family $\rho(t)$, $\mathrm{Tr}\,\rho(t) = 1$, with faithful reference state ρ_0 and $K_0 = -\log \rho_0$,

$$\frac{d}{dt} S(\rho(t))\Big|_{t=0} = \mathrm{Tr}(\dot{\rho}_0 K_0),$$

porque $\mathrm{Tr}\,\dot{\rho}_0 = 0$. Si

$$K_0 - cI = \beta_{90}\mathcal{A}_{90} + \beta_{120}\mathcal{A}_{120},$$

the scalar term vanishes, and

$$\delta S = \beta_{90}\delta\langle\mathcal{A}_{90}\rangle + \beta_{120}\delta\langle\mathcal{A}_{120}\rangle.$$

As $\beta_m = m\Delta_4^{\mathrm{mod}}/(\gamma_{\mathrm{HMT}} a_0)$, the ratio is exactly $120/90 = 4/3$.

J.12 Reflexion racional CKM

Let $D = \mathrm{diag}(1,-1,1)$, $\boldsymbol{\theta} = M\mathbf{x}$, and $R = MDM^{-1}$. Immediately, $R^2 = I$, $\det R = \det D = -1$, and $RM = MD$. In the determined coordinates,

$$c_h = (-5,7,-20)^T, \qquad \ell_h = (75,176,-150)/3857, \qquad \ell_h c_h = 1.$$

The odd projector is $c_h \ell_h$, of rank one. Hence

$$R = I - 2c_h \ell_h.$$

Multiplying fractions verifies

$$R = \frac{1}{3857}\begin{pmatrix} 4607 & 1760 & -1500 \\ -1050 & 1393 & 2100 \\ 3000 & 7040 & -2143 \end{pmatrix}.$$

The phase $\delta_{\mathrm{CP}} = 9A$ remains fixed because D does not change A. CP conjugation in the physical realization is another operator.

J.13 Classification of the modular realization in the infinite limit

Suppose an infinite succession of qutrit factors with faithful states, whose logarithms of eigenvalue ratios belong to

$$90\Delta_4^{\mathrm{mod}}\mathbb{Z} \quad \text{or} \quad 120\Delta_4^{\mathrm{mod}}\mathbb{Z},$$

and suppose that both types appear with positive density. The additive group generated is

$$90\Delta_4^{\mathrm{mod}}\mathbb{Z} + 120\Delta_4^{\mathrm{mod}}\mathbb{Z} = 30\Delta_4^{\mathrm{mod}}\mathbb{Z}.$$

The asymptotic ratio is then

$$\{0\} \cup \exp(-30\Delta_4^{\mathrm{mod}}\mathbb{Z}) = \{0\} \cup \lambda_\partial^{\mathbb{Z}}, \qquad \lambda_\partial = \mathrm{e}^{-30\Delta_4^{\mathrm{mod}}}.$$

The Araki–Woods theorem for ITPFI factors classifies the GNS closure as the hyperfinite factor III_{λ_∂}. The application is conditional on the hypotheses of faithfulness, compatibility, and persistence; the weights entering the classification are internal.

Remark J.3. If only the channel 90 persists, the generating group changes to $90\Delta_4^{\mathrm{mod}}\mathbb{Z}$; if only 120 persists, it changes to $120\Delta_4^{\mathrm{mod}}\mathbb{Z}$. If the local states are not of the declared compatible product type, the criterion must be recomputed. These cases are refutation criteria, not stylistic notes.

J.14 Electroweak angular identity

For the character

$$\chi(n_A, s_C) = \exp(n_A A + s_C C^*/6)$$

and the routes $W = (94, -1)$, $Z = (95, -1)$,

$$\frac{m_W}{m_Z} = \frac{\chi(94, -1)}{\chi(95, -1)} = \mathrm{e}^{-A}.$$

Since $D_A = \mathrm{e}^{-2A} = \det T$,

$$(m_W/m_Z)^2 = D_A, \qquad \sin^2\theta_W^{\mathrm{OS}} = 1 - D_A.$$

The proof belongs to the angular chart. Its complete interpretation of gauge requires the declared basis intertwiner in the body of the volume.

Appendix K

Mathematical and Computational Certification of Intrinsic Derivations

K.1 Domain, magnitudes, and logical scope of the computational certificate of intrinsic derivations

Block	Controles ejecutados
Matrix	58 rows, 15 fields, unique identifiers and declared scope
Vacío	order of $q_{\pm}$, positivity, eigenvalues, four characters, constitutive identities, and mode 30 completion
Electrodébil	$D_A = q_+q_-$ y $\sin^2\theta_W = 1 - D_A$
Chaos	quadratic sum, $899 = 30^2 - 1 = 29 \cdot 31$, quartic coefficient, closed profile and numerical residues
Gauss map	Levy, entropy, and Lochs for bases 10 and 729
Vacancies	exhaustive window of 10^8 positions, gaps 21/22, returns 6/7, certified interval, and traces of the multiprecision verifier
Spectrum	character χ_{12}, $L(1)$, $L(2)$, hyperbolic unit and Bendersky tower
Modular Hamiltonian	Δ_4^{mod}, weights 90/120, entropy, oriented defect, area–memory inversion, TFD and first law
Type III	group of ratios $30\Delta_4^{\mathrm{mod}}\mathbb{Z}$ and $\lambda_\partial = \exp(-30\Delta_4^{\mathrm{mod}})$ under declared hypotheses
CKM	coordinate inversion, odd projector, rational reflection, determinant, and intertwining
Cosmología	elimination of n and the exact sixth-power law

K.2 Certified numerical values of the constitutive, spectral, and modular invariants

Object	Certified value
$\widehat{\mu}$	0.99944835047932693508998155590607589 5 . . .
$\widehat{\varepsilon}$	0.99999925709663670276044161554232369 3 . . .
$\widehat{Z}$	0.99972450853893721545555434661361139 5 . . .
$\widehat{c}$	1.00027631048615752610638626892957626 8 . . .
D_A	0.77512911924715643018888409648020278 4 . . .
$1 - D_A$	0.22487088075284356981111590351979721 6 . . .
$L(1, \chi_{12})$	0.76034599630094634753109425488040582 4 . . .
$L(2, \chi_{12})$	0.94970312629400939526349849174574151 6 . . .
$\gamma_{\mathbb{Q}(\sqrt{3})}$	1.0539656082484825800 . . .
L of Levy	1.18656911041562545282172297594723712 1 . . .
h_G	2.37313822083125090564344595189447424 1 . . .
I_{or}	0.00166918323386894065563122591299641 1 . . . in nats
λ_∂	0.99666964646895737861082997688516473 1 . . .

K.3 Typing Obstructions and Equivalence Separation Between Certified Domains

The certificate does not promote the following statements:

1. $I_{\mathrm{or}} = \gamma_{\mathrm{HMT}} a_0$; the values are proximos but distintos.
2. The vacuum operator is identical to the Feigenbaum renormalizer.
3. Cofinality of cylinders implies TPK–Gauss conjugation.
4. The closure III_{λ_∂} holds without persistence of both channels.
5. CKM sheet inversion is physical CP conjugation.
6. The realization φ and the chronology cyclic of 108 states representan the same cosmology.
7. The angular relation in the scheme of physical masses includes by itself all electroweak corrections.

These negative controls do not weaken the volume: they establish exactly what has been proved and what additional map would change the status.

Appendix L

Independent reproduction of the coinductive certificates at depth 10^4: $\pi, e, \varphi, \alpha_{\mathrm{HMT}}$

This appendix prints the complete outputs of the execution described in appendix D.2 y section 31.6. The line breaks are exclusively typographical. Each expansion constitutes a finite witness of the coinductive law of arbitrary precision proved in Part III; it neither defines the constant nor limits the depth of the generator.

This residence reproduces, from the independent owner in Part III, the four coinductive outputs published in the cumulative corpus. The exact identity of the listings constitutes a cross-check of reproducibility and provenance; it does not introduce a second generation of the constants or alter their cylinders, truncations, or certificates.

L.1 Independent reproduction of the coinductive expansion of π and verification of the compatible truncation

decimal file

```
3.14159265358979323846264338327950288419716939937510582097494459230781640628620899862803482534211
70679821480865132823066470938446095505822317253594081284811174502841027019385211055596446229489
54930381964428810975665933446128475648233786783165271201909145648566923460348610454326648213936
07260249141273724587006606315588174881520920962829254091715364367892590360011330530548820466521 3
84146951941511609433057270365759591953092186117381932611793105118548074462379962749567358185727
24891227938183011949129833673362440656643086021394946395224737190702179860943702770539217176293 1
76752384674818467669405132000568127145263560827785771342757789609173637178721468440901224953430 1
46549585371050792279689258923542019956112129021960864034418159813629774771309960518707211349999 9
98372978049951059731732816096318595024459455346908302642522308253344685035261931188171010003137 8
38752886587533208381420617177669147303598253490428755468731159562863882353787593751957781857780 5
32171226806613001927876611195909216420198938095257201065485863278865936153381827968230301952035 3
01852968995773622599413891249721775283479131515574857242454150695950829533116861727855889075098 3
81754637464939319255060400927701671139009848824012858361603563707660104710181942955596198946767 8
37449448255379774726847104047534646208046684259069491293313677028989152104752162056966024058038 1
50193511253382430035587640247496473263914199272604269922796782354781636009341721641219924586315 0
30286182974555706749838505494588586926995690927210797509302955321165344987202755960236480665499 1
19881834797753566369807426542527862551818417574672890977772793800081647060016145249192173212714 7
72350141441973568548161361157352552133475741849468435232323907394143334547762416862518983569485 5
62099219222184272550254256887671790494601653466804988627232791786085784383827967976681454100953 8
83786360950680064225125205117392984896084128488626945604241965285022210661186306744278622039194 9
45047123713786960956364371917287467764657573962413890865832645995813390478027590099465764078951 2
69468398352595709825822620522489407726719478268482601476990902640136394437455305068203496252451 7
49399651431429809190659250937221696461515709858387410597885959772975498930161753928468138268683 8
68942774155991855925245953959431049972524680845987273644695848653836736222626099124608051243884 3
90451244136549762780797715691435997700129616089441694868555848406353422072225828488648158456028 5
06016842739452267467678895252138522549954666727823986456596116354886230577456498035593634568174 3
24112515076069479451096596094025228879710893145669136867228748940560101503308617928680920874760 9
17824938589009714909675985261365549781893129784821682998948722658804857564014270477555132379641 4
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L.2 Independent reproduction of the coinductive expansion of e and verification of the compatible truncation

Decimal-digit file:

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L.3 Independent reproduction of the coinductive expansion of φ and verification of the compatible truncation

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1.6180339887498948482045868343656381177203091798057628621354486227052604628189024497072072041893 9113748475408807538689175212663386222353693179318006076672635443338908659593958290563822661399 2829026788067520876689250171169620703222104321626954862629631361443814975870122034080858879544547 4924618569536486444924104432077134494704956584678850874339442212544877066478091588460749887124 0076521705751797883416625624940758906970400028121042762177111777805315317141011704666599146697987 7317613560067087480710131795236894275219484353056783002287856997829778347845878228911097625003022 6961561700250464338243776486102838312683303724292675263116533924731671112115881863851331620384000 5222165791286675296459068113171599343235973494985090409476213222981017261070596116456299098162900 0555208524790352406020172799747175342777592778625619432082750513121815628551222480939471234145170 0223735805772786160086883829523045926478780178899219902707769038953219681986151437803149974110690 2608867429622675756052317277752035361393621076738937645560606059216589466759551900400555908959022 9530942312482355212212415444006470340565734797663972394949946584578873039623090375033993856210240 2369025138680414577995698122445747178034173126453220416397232134044449487302315417676893752103060 8737880344170093954409627955898678723209512426893557309704509595684401755519881921802064052905510 8934947592600734852282101088194644544222318891319294689622002301443770269923007803085261188075451 9288770502109684249362713592518760777884665836150238913493333122310533923213624319263728910670500 3399282265263556209029798642472759772565508615487543574826471814145127000602389016207773224499430 5308899909501680328112194320481964387675863314798571911397815397807476150772211750826945863932040 5652098969855567814106968372884058746103378105444390943683583581381131168993855576975484149144530 4150912954070050194775486163075422641729394680367319805861833918328599130396072014455950449779210 2076124785645916160837059498786006970189409886400764436170933417270919143365013715766011480381430 0626238051432117348151005590134561011800790506381421527093085880928757034505078081454588199066336 1298279814117453392731208092897279222132980642946878242748701745055406778757083237310975915117 7
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03
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decimals

L.4 Independent reproduction of the expansion of α_{HMT} and verification of compatible truncation

decimal file

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0.007297352569283800997285105472380662999564172539642709185404682532485972586311002762549975056833443130713789135355334087124799722897174078685912408079209664095202723675027634816553917299241492231308474268521602557532653978277821237086455081569242134313064385801449399419234807287901899795857265724875201590634168628401471781981272735601821931516084729067989300802451048585440023515516391794670253774858710348204249048193716380082266023130631463436707101750502152424699033053810998690460568332114100965066466578217401799422125040984809475964589443818169449435683443195305554573171332490452697178942552598379448561212084114919656629207472943924426062341846398703143478849122843506830016612206927240154198143555440307072613797032056273809800318991399313487817760078316896817460599105075828850327548669354583936564420235506171410889744378255394251699566529181632967019129029621940108026903658533248454344304985085037209746973807598842039900046616947132407800251339656974472406789733975913253198782383335304674439886073394095045541432499395052816541544137292736259803111134602366392355309284778870476055728588543787998424636746633060160082441699139740220308735292918136991488865598990865192327151516074896486713983265783339803629338082043809072951367229574526004299255873120086497815593020709149625802339160670676885148865467418462265824362479327683335464244465200659461263461225605029163884405323071947316240551838323045085346359363786749486222950866000634688903182716318009331093733512465264033486368005714656070247452272820348215849374997531908489858228925179202956576736232357414178137453408601578996157548082473588965845948212185793125373480165077668059242943005930713473541969084948105188925870848216440276813711150577414165289692977073392744771977785757470164334384021692260625761401090402714453182809004731459050700969219304526584936693658045923630890674774407733066099470274555587952980734315786890844476326734720675575556963388402892306740952910738610262678122580512941803981939998160971634574121863865421584998472950109772769868832105015504151832654262976288080817856977497396875910936526343423596428028628014618018088852909615000845217616345621921581919683464761656757199587889029943880887997872577998372817391724953759475460221142612049917302785573745573908444381666722072873198878481796342039159543588284919795540103680931256125220319524192977854891833474015705182924717091886534570540233646600029248535698528995753169204829245872413970572241626000700858896803335663552415722095692295960320669870113081509580986892364545324775742637441528171303113641520379595960187265597181632289358038396309097718054983307117662618727599049948063679293977315325029791815114575528353796414591068730109654696210306015413379250471171276643309923055810326158287183350710385105443419682034572290864719947733394934439948691913197848539945509439613761668505609729811826031879868406293551006972048747417056667989848666221834964893412738287768831865693022860002163473409049432334795781069116349879457358384827506183578361392284808617824709630266353175245708252444711255709068743929097132402923397653155911648717585035473429935556293549732460035936975103430764672317367285583586792378921496335220765116574781863500634830661454106037961300849456931907984133937243073196598064797018251688657084194561957047412105098936899328907924829675283055582884607393114467188427439310131055668980337713060084185079918653073168926359161403537793428274381903561227039130348553503381032804569322334517117288112007558879069858238948239825454427974988024868093328191759246939266029987207170769522051022985497999725332169628047520015542678264856751756507643084623989085654880477697508385409042387292036297357620685378928298818811856684001303157016896570482983256576266432628450902344529281391905774298740408490731031691657898514663653777867562388870506901499559117945888146633192523458106252849504623915003014189308397272222775871730119537505639375166973451402384834665077464417324152195902855211021136782262375268886441144478710756465106908913289041421621887233763213331365394973682696429632567684436560498392836768473595683953377433233894526022405334555571388815164082189764696960436240541944603810386117539205512861359140741333090772700385045129194206831702404128113410008618956068215757132369047553373296396369845443282336049403529042518060696772652376750173556139444143536469892918317626348757202047457495056883297733783440475487252715671645542010351508881160070591149861894919417242658467117715150117382927796155646371947102772469470438831684638494604641257751329087027050589909057408105765252303588095723963237361785047543043463068613340497691508140855736675017742606623306930922189467915532962108454837432428682928380002609109973797808005202582860638160863320818920733946855014858088415091190778465820984013989534930950503070503674437148459206772100034168386475726289066159833257069783128652065486524475562628219297051666179055563592306231112373594594190592526788472497463329664600940622731238688764822750712124027379522468927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74830147984236102610954322627621755737339607271352350384299599234305191233992739432266352710051 3
13291163184076021186516651335578401111328268424461858850779039837402271411584805059255419110559 0
00761011629535258233517417948564065863322625675018288073534968173432895192873769177494454076917 4
77445583285758667025485841605047763257582554218259501074564409360078076477453362347602684030705 3
77390455454465822981125879139867924952547682375095390256140535179588077260246689169198224687862 0
14754537522259930216174749646412626120839002535247851087409718525083744709914781168587962170425 6
31456452783367961530955764981814466483890858329592304309205676069826969728161227194406292680074 9
72441007643807622984836432607360840836763897774957555783782033624171993010674776621002545124049 7
17499569337038604628372462422411421916513168236357367936879554510087795525308056380382759967523 4
69209162629637000794589540861754726694091905602119390383320449145478220047804652284217760774341 8
74510934913644614751506704102821067071139869789572060636398541475342223969199529408196590204443 8
61761779895976392277114517839892673327869364953374758666723431532103797455486337816327941970031 7
63840868591017352052929218702143525009601199199984401506673588105681486282374619226793934736861 2
836572260534209515
```

Appendix M

Exhaustive classification of states, signatures, and mass realizations

M.1 Mathematical Conventions and Data Representation

This catalogue does not take a column of masses as the definition of the theory. It superimposes four layers without conflating them:

$$\begin{array}{rcl} \mathcal{C}_{81} \to \mathcal{R}_{56} \to \Sigma_{13} & : & \text{internal domain of cells, routes and multisections,} \\ \Gamma_{108}^{\mathrm{enr}} \to \mathcal{L}_{108} \to \mathbb{Z}^5 & : & \text{route genealogy, record and signature,} \\ \mathbb{Z}^5 \xrightarrow{\chi_{\mathrm{mass}}} \mathbb{R}_{>0} & : & \text{exact evaluation of a fixed signature,} \\ \mathbb{R}_{>0} \to \mathcal{U}_E & : & \text{energy chart and metrological comparison.} \end{array} \tag{C1}$$

The atlas is complete in its domain $81/56/13$. The compositional grammar is complete for the identities, mesonic expressions, baryonic expressions, and transitions that it constructs. Those cardinalities do not count physical particles. The numerical tables below declare, in every case, which coordinates are fixed and what role the comparison performs. In this way, all species and all localized values are displayed without passing off an inverse decoding as the genealogy it must explain.

M.2 Combinatorial domain prior to physical identification

The exact partition is

$$81 = 25_{\ell^-} + 20_{\nu} + 20_{d} + 16_{u}, \qquad 36_{\mathrm{quark}} = 12_R + 12_G + 12_B. \tag{C2}$$

The thirteen multisections have cardinalities

$$(1, 8, 16)_{\ell^-}, \quad (2, 4, 6, 8)_{\nu}, \quad (2, 4, 6, 8)_{d}, \quad (4, 12)_{u}. \tag{C3}$$

The electron is the central one-point multisection. The other twelve preserve the orbit and orientation of the entire fibre. A physical row is therefore not obtained by arbitrarily selecting a cell: the species is the enriched multisection, and the particle is a persistent section realized within it.

Table M.1: The thirteen multisections of the simple domain.

Internal family	Level	Cardinality	Lectura estructural
ℓ^-	$G0$	1	electronic center $(5, 5)$
ℓ^-	$G2$	8	family of second depth charge
ℓ^-	$G4$	16	loaded family exterior
ν	$G1$	2	first hidden layer
ν	$G2$	4	second neutral layer
ν	$G3$	6	third neutral layer
ν	$G4$	8	fourth neutral layer, including the sterile/exotic branch
d	$G1$	2	quark type d, first depth
d	$G2$	4	quark type d, second depth
d	$G3$	6	quark type d, third depth
d	$G4$	8	quark type d, fourth depth
u	$G1$	4	quark type u, first depth
u	$G3$	12	quark type u, third depth

The cardinal projections, fine signatures, and partial angular charts are specialized controls of the general law. The development that follows first fixes the grammar of composition, the null sector, and cellular identifiability; the exhaustive atlas then publishes all routes without turning a historical sample into the definition of the domain.

M.3 Composition of Mesonic and Baryonic Records

The theorem (92.17) constructs the complete grammar prior to every mass:

class	cardinal	internal condition
identity	81	dual of the same cell
mesonic expression	432	colour-neutral quark–antiquark pair
baryonic expression	1728	RGB triple with oriented positions
charged transition	320 + 52	leptonic or quark compatibility.

(C7)

Each expression receives a mass only after being physically realized, its records spliced, and its new signature extracted. The numbers 432, 1728, 372 count admissible combinations of the grammar, not physical species or entries in the resonance table.

M.4 Neutrinos, the null sector, and gauge representations

Neutrinos occupy four multisections of cardinals.

$$2, \quad 4, \quad 6, \quad 8.$$

The structure distinguishes the three active layers and the sterile/exotic layer. Its primary metrological observable is not an isolated absolute mass, but rather the quadratic differences, ordering, and scale bounds; those data are presented with their type and are never replaced by an invented row of χ_{mass}.

Photon and gluon belong to the null or gauge sector. The operator

$$L_{\text{TPK}} = 3(I - P_0)$$

has spectrum $\{0, 3, 3\}$; its central mode carries no compactified memory of its own. Quark color rests on the ternary residue $r - c \bmod 3$, whose balance is $12 + 12 + 12$, and the qutrit chart is used to construct $\mathfrak{sl}_3(\mathbb{C})$ and its compact form. Absence from $\mathbb{R}_{>0}$ does not eliminate these sectors: it proves that positive mass is not the atlas's only reader.

M.5 Fibonacci, Ising, and Majorana

The topological sectors are compared by fusion, quantum dimension, spin, and braiding, not by a supposed universal mass. The Fibonacci envelope produces the based ring

$$\tau^2 = 1 + \tau, \qquad \text{FPdim}(\tau) = \varphi. \tag{C8}$$

The involution C_M, CAR parity, the relativistic Majorana field, and the zero mode of an Ising realization are four distinct levels. Their separation explains which part follows from the route, which from fermionic statistics, and which from a concrete topological platform.

M.6 Cellular identifiability and boundary of the nominal link

Once the oriented APP frame has been fixed, each cell x determines its positive diagonal route γ_x^+. The dodecaphase record of this route satisfies, for the 81 cells,

$$\Sigma_+(x) = L_{\text{tick}}\big(\mathcal{L}_{\gamma_x^+}\big) = A(x) = (n_A, s_C, k_{\Delta_4}, b_{120}, b_\zeta). \tag{C9}$$

The winding

$$\omega_D(x) = \left(\frac{\Delta n_A}{2}, \frac{\Delta s_C}{6}, \frac{\Delta k_{\Delta_4}}{2}, \frac{\Delta b_{120}}{2}, \frac{\Delta b_\zeta}{4}\right) \in \mathbb{Z}^5 \tag{C10}$$

together with τ_{12}, recovers the 56 route families; the nonadic boundary class β_0 separates the 81 positions.

Exact scope of the result

The chain internal cell → oriented route → register → signature is constructed. The finite nominal realization is already closed by sabor(P) and by $\mathcal{S}_{\text{phys}}$ in (92.8)– (92.9): generation, branch, fine trace, and color orbit publish the route before any mass evaluation. For noncentral fibers, the species is a multisection, not an isolated cell; requiring the name to select a unique point would be a type error. The NSEO traces, fine signatures, and nominal tables preserve the material evidence of this map and do not constitute postselection by an external value.

Once the simple sections and the binding record have been constructed independently, the composites are formed before the signature is evaluated:

$$\begin{aligned} \mathcal{L}_{q\bar{q}} &= \text{Comp}_2(\mathcal{L}_q, \mathcal{L}_{C_M\bar{q}}; g), \\ \mathcal{L}_{q_1q_2q_3} &= \text{Comp}_3(\mathcal{L}_{q_R}, \mathcal{L}_{q_G}, \mathcal{L}_{q_B}; g), \\ v_X &= L_{\text{tick}}(\mathcal{L}_X), \\ m_X &= m_e^{\text{HMT}} \exp\langle \Theta_{\text{mass}}, v_X - v_e \rangle . \end{aligned} \tag{C11}$$

The composition does not query a target mass; neither does it by itself identify an external species with its internal components.

M.7 Heterogeneity of Physical Domains and Exhaustiveness of the Catalogue

The catalog collects all the layers located without converting them into the same observable:

- the 81 cells, 56 families, and 13 multisections form the simple domain;
- 81 + 432 + 1728 + 372 counts identities, expressions, and arrows of the compositional grammar, not physical particles;
- the complete signatures, the N/E/S/O projections, and the reduced charts remain as specialized controls of the evaluation, without limiting the exhaustive domain or retrospectively selecting its routes;
- neutrinos, the photon, the gluon, Fibonacci, Ising, and Majorana retain their own observables.

Scientific completeness consists in omitting no domain and in misrepresenting the type of no row. The character law is unique; each simple section or composite register is evaluated through proposition 92.25. The external value appears at the end, together with the object class and the reader that evaluates it. The comparison thus ceases to be a list of successes and becomes a verifiable reading of the genealogy.

M.8 Public projection of the internal record of the routes

Each route has 53 fields in the unified register and 57 fields in the enriched register. The scientific record publishes 49 fields. This projection does not eliminate information: the consistency states of the route, of the gluing, and of tower prolongation, together with the 5 raw components of the signature, remain in the complete registers and are gathered in the published invariants. Therefore, the publication map is a typed projection

$$\mathcal{R}_{57} \longrightarrow \mathcal{F}_{49},$$

accompanied by the complete register $\mathcal{R}_{53}$ and by the enriched register $\mathcal{R}_{57}$; it is not a reduction of the scientific domain.

M.9 Correspondence between the genealogical domain and the metrological inventories

The material domain is constructed from 81 cells, 56 route families, 13 multisections, and 324 oriented routes. Its complete deployment is found in the atlas of 324 routes, with a complete card and 49 fields by route. The external censuses of contrast are preserved in the inventory of 471 masses and 384 widths. The atlas constructs the fibers and their operators; the censuses do not retrospectively select a route or redefine the domain of the law.

M.10 Exact correspondence between the canonical record of 53 fields, the enriched record of 57 fields, and the 49 invariants of each card

The 324 oriented routes have two complete proprietary registers and one scientific publication per record. The registers of 53 and 57 fields overlap without being identified: their union contains 65 distinct identifiers. The publication column selects exactly 49 invariants for each record, while the remaining data remain preserved in the reproducibility layers. Therefore, projection onto the record neither eliminates information from the domain nor reduces the atlas.

Field	$\mathcal{R}_{53}$	$\mathcal{R}_{57}$	$\mathcal{F}_{49}$	Public interpretation or conservation residency
`candidate_id`	Yes	Yes	Yes	Canonical identifier
`cell_id`	Yes	Yes	Yes	Cell
`r`	Yes	Yes	Yes	APP row
`c`	Yes	Yes	Yes	APP column
`h0`	Yes	Yes	Yes	Horizontal orientation
`v0`	Yes	Yes	Yes	Vertical orientation
`is_B1_route`	Yes	—	—	Data preserved in the canonical record of reproducibility.
`coarse_class`	Yes	—	—	Data preserved in the canonical record of reproducibility.
`family`	Yes	—	—	Data preserved in the canonical record of reproducibility.
`sector`	Yes	—	—	Data preserved in the canonical record of reproducibility.
`generation`	Yes	—	—	Data preserved in the canonical record of reproducibility.
`color`	Yes	Yes	Yes	Colored orbit
`structural_multisection`	Yes	Yes	Yes	Multisection
`physical_family`	Yes	Yes	Yes	Physical family
`generation_depth`	Yes	Yes	Yes	Generational depth
`rho_orbit`	Yes	Yes	Yes	Cell Inversion Orbit
`first_class_phase_return`	Yes	Yes	Yes	First class return
`first_orbit_phase_return`	Yes	Yes	Yes	First return of orbit

Field	$\mathcal{R}_{53}$	$\mathcal{R}_{57}$	$\mathcal{F}_{49}$	Public interpretation or conservation residency
first_cell_phase_return	Yes	Yes	Yes	First cell return
first_kinematic_state_return_within_108	Yes	Yes	Yes	Kinematic return of the 108-phase calendar
first_kinematic_state_return_extended	Yes	Yes	Yes	Extended kinematic return to 216 phases
turns	Yes	Yes	Yes	Rotations
sheet_switches	Yes	Yes	Yes	Sheet changes
carry_debt_total	Yes	Yes	Yes	Total carry debt
carry_debt_final	Yes	Yes	Yes	Final carry debt
delayed_carry_blocks	Yes	Yes	Yes	Delayed carry blocks
stable_decimal_triads_final	Yes	Yes	Yes	Stabilized Decimal Triples
w6_repeat_K_to_Kplus9	Yes	Yes	Yes	Coincidence of w_6 between K and $K+9$
prefix_memory_changes_K_to_Kplus9	Yes	Yes	Yes	Memory changes between K and $K+9$
nA_raw	Yes	Yes	—	Data preserved in the canonical records and enriched with reproducibility.
sC_raw	Yes	Yes	—	Data preserved in the canonical records and enriched with reproducibility.
kappa0_raw	Yes	Yes	—	Data preserved in the canonical records and enriched with reproducibility.
nu120_raw	Yes	Yes	—	Data preserved in the canonical records and enriched with reproducibility.
nu270_raw	Yes	Yes	—	Data preserved in the canonical records and enriched with reproducibility.
tau12	Yes	Yes	Yes	Dodecaphonic Word
w6_words	Yes	Yes	Yes	Visible words w_6
D1	Yes	Yes	Yes	D_1
D3	Yes	Yes	Yes	D_3
D4	Yes	Yes	Yes	D_4
D27	Yes	Yes	Yes	D_{27}
D36	Yes	Yes	Yes	D_{36}
Omega90_120	Yes	Yes	Yes	Channels $\Omega_{90/120}$
P6	Yes	Yes	Yes	P_6
K_D	Yes	Yes	Yes	Reader K_D
K_Omega	Yes	Yes	Yes	Reader K_Ω
kappa_D	Yes	Yes	Yes	Exponent κ_D
kappa_Omega	Yes	Yes	Yes	Exponent κ_Ω
Ract_factor_D	Yes	Yes	Yes	Factor R_{act}, reading D
Ract_factor_Omega	Yes	Yes	Yes	Factor R_{act}, reading Ω
route_status	Yes	Yes	—	Data preserved in the canonical records and enriched with reproducibility.
route_genealogy_sha256	Yes	—	—	Data preserved in the canonical record of reproducibility.
reader_sha256	Yes	—	—	Data preserved in the canonical record of reproducibility.
cell_ledger_108_sha256	Yes	—	—	Data preserved in the canonical record of reproducibility.
route_family_R	—	Yes	Yes	Route family

Field	$\mathcal{R}_{53}$	$\mathcal{R}_{57}$	$\mathcal{F}_{49}$	Public interpretation or conservation residency
prefix_ambiguity_area	—	Yes	Yes	Ambiguity area of the prefix
prefix_log2_ambiguity_area	—	Yes	Yes	Binary logarithm of ambiguity
first_orbit_separation_block	—	Yes	Yes	First orbit separation block
first_cell_separation_block	—	Yes	Yes	First cell separation block
first_route_separation_block	—	Yes	Yes	First route separation block
final_ambiguity	—	Yes	Yes	Final Ambiguity
stable_triads_by_block	—	Yes	Yes	Block Stabilization Chronology
historic_signature_Z5	—	Yes	Yes	Integer signature in $\mathbb{Z}^5$
pareto_persistence_orbit	—	Yes	Yes	Orbit on the persistence boundary
join_status	—	Yes	—	Data preserved in the enriched reproducibility record.
tower_forward_status	—	Yes	—	Data preserved in the enriched reproducibility record.

This correspondence is applied uniformly to the 324 records. The censuses of 471 masses and 384 widths remain inventories of external recognition and do not participate in selecting routes, families, or atlas multisections.

Appendix N

Comprehensive genealogical atlas of the 324 mass routes

Return to the scientific portal of the atlas and its readers. Each entry brings together cell identity, orientation, dodecaphase word, bidirectional readers, returns, memory, carry, and signature. No metrological magnitude participates in this reconstruction.

The census satisfies

$$81 \text{ cells} \times 4 \text{ orientations} = 324 \text{ routes}.$$

N.1 Complete unfolding of the 324 oriented routes: correspondence among 53 canonical fields, 57 enriched fields, and 49 invariants published per section

Route 1 r1c1_h+1v+1

Canonical identifier	r1c1_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	1	**Cell Inversion Orbit**	1,1;9,9
APP row	1	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	24|0|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 2 r1c1_h+1v-1

Canonical identifier	r1c1_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	1	**Cell Inversion Orbit**	1,1;9,9
APP row	1	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	24|0|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585	**Final Ambiguity**	3
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	2
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 3 r1c1_h-1v+1

Canonical identifier	r1c1_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	1	**Cell Inversion Orbit**	1,1;9,9
APP row	1	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	24|0|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510	**Final Ambiguity**	3
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	0
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 4 r1c1_h-1v-1

Canonical identifier	r1c1_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	1	**Cell Inversion Orbit**	1,1;9,9
APP row	1	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	24|0|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 5 r1c2_h+1v+1

Canonical identifier	r1c2_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	2	**Cell Inversion Orbit**	1,2;9,8
APP row	1	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	40
Generational depth	G4	**Binary logarithm of ambiguity**	20.339850002884624
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	0
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|4|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 6 r1c2_h+1v-1

Canonical identifier	r1c2_h+1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	2	**Cell Inversion Orbit**	1,2;9,8
APP row	1	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|6|-2|4|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 7 r1c2_h-1v+1

Canonical identifier	r1c2_h-1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	2	**Cell Inversion Orbit**	1,2;9,8
APP row	1	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+0+—+0+—	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-6|2|-4|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 8 r1c2_h-1v-1

Canonical identifier	r1c2_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	2	**Cell Inversion Orbit**	1,2;9,8
APP row	1	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-4|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 9 r1c3_h+1v+1

Canonical identifier	r1c3_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	3	**Cell Inversion Orbit**	1,3;9,7
APP row	1	**First class return**	18
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 10 r1c3_h+1v-1

Canonical identifier	r1c3_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	3	**Cell Inversion Orbit**	1,3;9,7
APP row	1	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 11 r1c3_h-1v+1

Canonical identifier	r1c3_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	3	**Cell Inversion Orbit**	1,3;9,7
APP row	1	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|-2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 12 r1c3_h-1v-1

Canonical identifier	r1c3_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	3	**Cell Inversion Orbit**	1,3;9,7
APP row	1	**First class return**	9
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 13 r1c4_h+1v+1

Canonical identifier	r1c4_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	4	**Cell Inversion Orbit**	1,4;9,6
APP row	1	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|-4|-2|-8|0++—0++—	**First cell separation block**	1
Dodecaphasic word	0++—0++—	**First route separation block**	1
Visible words w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|-6|2|-4|-8
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 14 r1c4_h+1v-1

Canonical identifier	r1c4_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	4	**Cell Inversion Orbit**	1,4;9,6
APP row	1	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|-4|-2|-8|0++—0++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	2
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|-2|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 15 r1c4_h-1v+1

Canonical identifier	r1c4_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	4	**Cell Inversion Orbit**	1,4;9,6
APP row	1	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|-4|-2|-8|0++—0++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	0
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|2|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 16 r1c4_h-1v-1

Canonical identifier	r1c4_h-1v-1	Factor R_{act}, reading Ω	1.000000038550972354141645
Cell	4	Cell Inversion Orbit	1,4;9,6
APP row	1	First class return	27
Column APP	4	First return of orbit	27
Horizontal orientation	-1	First cell return	27
Vertical orientation	-1	Kinematic return of the calendar of 108 phases	54
Multisection	neutrino_G4	Kinematic return extended to 216 phases	216
Physical family	neutrino family	Ambiguity area of the prefix	26
Generational depth	G4	Binary logarithm of ambiguity	6.339850002884624
Color orbit	—	First orbit separation block	5
Route family	30|-6|-4|-2|-8|0++—0++—	First cell separation block	5
Dodecaphasic word	+++0–+++0–	First route separation block	5
Visible words w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243	Final Ambiguity	1
D_1	54	Rotations	71
D_3	56	Sheet changes	107
D_4	70	Total carry debt	2
D_{27}	48	Final carry debt	1
D_{36}	32	Delayed carry blocks	2
Channels $\Omega_{90/120}$	56	Stabilized Decimal Triples	16
P_6	5	Coincidence of w_6 between K and $K+9$	9
Reader K_D	(80,54,7)	Memory changes between K and $K+9$	9
Reader K_Ω	(56,54,5)	Integer signature in $\mathbb{Z}^5$	30|6|-2|4|-8
Exponent κ_D	79.60672133621751972652496	Orbit on the persistence boundary	yes
Exponent κ_Ω	55.60649894337213544641686	Block Stabilization Chronology	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 17 r1c5_h+1v+1

Canonical identifier	r1c5_h+1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	5	**Cell Inversion Orbit**	1,5;9,5
APP row	1	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	30
Generational depth	G4	**Binary logarithm of ambiguity**	8.0
Color orbit	—	**First orbit separation block**	5
Route family	34|0|-8|0|-8|+0+—+0+—	**First cell separation block**	5
Dodecaphasic word	+0+—+0+—	**First route separation block**	5
Visible words w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	0
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446419 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 18 r1c5_h+1v-1

Canonical identifier	r1c5_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	5	**Cell Inversion Orbit**	1,5;9,5
APP row	1	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|-8|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 19 r1c5_h-1v+1

Canonical identifier	r1c5_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	5	**Cell Inversion Orbit**	1,5;9,5
APP row	1	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|-8|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 20 r1c5_h-1v-1

Canonical identifier	r1c5_h-1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	5	**Cell Inversion Orbit**	1,5;9,5
APP row	1	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-8|0|-8|+0+—+0+—	**First cell separation block**	1
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	1
Visible words w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 21 r1c6_h+1v+1

Canonical identifier	r1c6_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	6	**Cell Inversion Orbit**	1,6;9,4
APP row	1	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 22 r1c6_h+1v-1

Canonical identifier	r1c6_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	6	**Cell Inversion Orbit**	1,6;9,4
APP row	1	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	0++—0++—	**First route separation block**	1
Visible words w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 23 r1c6_h-1v+1

Canonical identifier	r1c6_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	6	**Cell Inversion Orbit**	1,6;9,4
APP row	1	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++0–+++0–	**First route separation block**	1
Visible words w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 24 r1c6_h-1v-1

Canonical identifier	r1c6_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	6	**Cell Inversion Orbit**	1,6;9,4
APP row	1	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 25 r1c7_h+1v+1

Canonical identifier	r1c7_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	7	**Cell Inversion Orbit**	1,7;9,3
APP row	1	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	30|6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|6|0|0|-12
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 26 r1c7_h+1v-1

Canonical identifier	r1c7_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	7	**Cell Inversion Orbit**	1,7;9,3
APP row	1	**First class return**	9
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	30|6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585	**Final Ambiguity**	3
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	2
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 27 r1c7_h-1v+1

Canonical identifier	r1c7_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	7	**Cell Inversion Orbit**	1,7;9,3
APP row	1	**First class return**	18
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	30|6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510	**Final Ambiguity**	3
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	0
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 28 r1c7_h-1v-1

Canonical identifier	r1c7_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	7	**Cell Inversion Orbit**	1,7;9,3
APP row	1	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	30|6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|-6|0|0|-12
Exponent κ_D	79.606721336217519726524 96	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 29 r1c8_h+1v+1

Canonical identifier	r1c8_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	8	**Cell Inversion Orbit**	1,8;9,2
APP row	1	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	40
Generational depth	G4	**Binary logarithm of ambiguity**	20.339850002884624
Color orbit	—	**First orbit separation block**	—
Route family	28|6|2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	0
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 30 r1c8_h+1v-1

Canonical identifier	r1c8_h+1v-1	**Factor R_{act}, reading Ω**	1.000000024685259569118152
Cell	8	**Cell Inversion Orbit**	1,8;9,2
APP row	1	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	0++–00++–0	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	36	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(36,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	35.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 31 r1c8_h-1v+1

Canonical identifier	r1c8_h-1v+1	**Factor R_{act}, reading Ω**	1.000000024685259569118152
Cell	8	**Cell Inversion Orbit**	1,8;9,2
APP row	1	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	++00–++00–	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	36	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(36,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	35.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 32 r1c8_h-1v-1

Canonical identifier	r1c8_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	8	**Cell Inversion Orbit**	1,8;9,2
APP row	1	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 33 r1c9_h+1v+1

Canonical identifier	r1c9_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	9	**Cell Inversion Orbit**	1,9;9,1
APP row	1	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-2|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 34 r1c9_h+1v-1

Canonical identifier	r1c9_h+1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	9	**Cell Inversion Orbit**	1,9;9,1
APP row	1	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+0+—+0+—	**First route separation block**	—
Visible words w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 35 r1c9_h-1v+1

Canonical identifier	r1c9_h-1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	9	**Cell Inversion Orbit**	1,9;9,1
APP row	1	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	—
Visible words w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 36 r1c9_h-1v-1

Canonical identifier	r1c9_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	9	**Cell Inversion Orbit**	1,9;9,1
APP row	1	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|2|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 37 r2c1_h+1v+1

Canonical identifier	r2c1_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	10	**Cell Inversion Orbit**	2,1;8,9
APP row	2	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|12|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-4|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 38 r2c1_h+1v-1

Canonical identifier	r2c1_h+1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	10	**Cell Inversion Orbit**	2,1;8,9
APP row	2	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|12|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|6|-2|4|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 39 r2c1_h-1v+1

Canonical identifier	r2c1_h-1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	10	**Cell Inversion Orbit**	2,1;8,9
APP row	2	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|12|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+0+—+0+—	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-6|2|-4|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 40 r2c1_h-1v-1

Canonical identifier	r2c1_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	10	**Cell Inversion Orbit**	2,1;8,9
APP row	2	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	40
Generational depth	G4	**Binary logarithm of ambiguity**	20.339850002884624
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|12|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	1
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|4|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 41 r2c2_h+1v+1

Canonical identifier	r2c2_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	11	**Cell Inversion Orbit**	2,2;8,8
APP row	2	**First class return**	18
Column APP	2	**First return of orbit**	18
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	24|0|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486	**Final Ambiguity**	4
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 42 r2c2_h+1v-1

Canonical identifier	r2c2_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	11	**Cell Inversion Orbit**	2,2;8,8
APP row	2	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	24|0|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258	**Final Ambiguity**	4
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|2|0|0|-12
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 43 r2c2_h-1v+1

Canonical identifier	r2c2_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	11	**Cell Inversion Orbit**	2,2;8,8
APP row	2	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	24|0|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333	**Final Ambiguity**	4
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	36	**Final carry debt**	1
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|-2|0|0|-12
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 44 r2c2_h-1v-1

Canonical identifier	r2c2_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	11	**Cell Inversion Orbit**	2,2;8,8
APP row	2	**First class return**	9
Column APP	2	**First return of orbit**	9
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	24|0|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24	**Final Ambiguity**	4
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	3
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 45 r2c3_h+1v+1

Canonical identifier	r2c3_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	12	**Cell Inversion Orbit**	2,3;8,7
APP row	2	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|4|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 46 r2c3_h+1v-1

Canonical identifier	r2c3_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	12	**Cell Inversion Orbit**	2,3;8,7
APP row	2	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|-2|0|-2|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 47 r2c3_h-1v+1

Canonical identifier	r2c3_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	12	**Cell Inversion Orbit**	2,3;8,7
APP row	2	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|2|0|2|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 48 r2c3_h-1v-1

Canonical identifier	r2c3_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	12	**Cell Inversion Orbit**	2,3;8,7
APP row	2	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-4|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 49 r2c4_h+1v+1

Canonical identifier	r2c4_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	13	**Cell Inversion Orbit**	2,4;8,6
APP row	2	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 50 r2c4_h+1v-1

Canonical identifier	r2c4_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	13	**Cell Inversion Orbit**	2,4;8,6
APP row	2	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	4
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 51 r2c4_h-1v+1

Canonical identifier	r2c4_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	13	**Cell Inversion Orbit**	2,4;8,6
APP row	2	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	4
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 52 r2c4_h-1v-1

Canonical identifier	r2c4_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	13	**Cell Inversion Orbit**	2,4;8,6
APP row	2	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	1
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 53 r2c5_h+1v+1

Canonical identifier	r2c5_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	14	**Cell Inversion Orbit**	2,5;8,5
APP row	2	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486	**Final Ambiguity**	2
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|-6|0|-4|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 54 r2c5_h+1v-1

Canonical identifier	r2c5_h+1v-1	**Factor R_{act}, reading Ω**	1.000000013592751151887498
Cell	14	**Cell Inversion Orbit**	2,5;8,5
APP row	2	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	00+—00+—	**First route separation block**	—
Visible words w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258	**Final Ambiguity**	2
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	20	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(20,54,4)	**Integer signature in $\mathbb{Z}^5$**	50|-4|4|0|-4
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	19.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 55 r2c5_h-1v+1

Canonical identifier	r2c5_h-1v+1	**Factor R_{act}, reading Ω**	1.000000013592751151887498
Cell	14	**Cell Inversion Orbit**	2,5;8,5
APP row	2	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++00-+++00-	**First route separation block**	—
Visible words w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333	**Final Ambiguity**	2
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	36	**Final carry debt**	1
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	20	**Stabilized Decimal Triples**	16
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(20,54,4)	**Integer signature in $\mathbb{Z}^5$**	50|4|-4|0|-4
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	19.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 56 r2c5_h-1v-1

Canonical identifier	r2c5_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	14	**Cell Inversion Orbit**	2,5;8,5
APP row	2	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24	**Final Ambiguity**	2
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	3
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|6|0|4|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 57 r2c6_h+1v+1

Canonical identifier	r2c6_h+1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	15	**Cell Inversion Orbit**	2,6;8,4
APP row	2	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-16|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+0+—+0+—	**First route separation block**	—
Visible words w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 58 r2c6_h+1v-1

Canonical identifier	r2c6_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	15	**Cell Inversion Orbit**	2,6;8,4
APP row	2	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-16|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 59 r2c6_h-1v+1

Canonical identifier	r2c6_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	15	**Cell Inversion Orbit**	2,6;8,4
APP row	2	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-16|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 60 r2c6_h-1v-1

Canonical identifier	r2c6_h-1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	15	**Cell Inversion Orbit**	2,6;8,4
APP row	2	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-16|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	—
Visible words w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.60649894337213544641686	**Block Stabilization Chronology**	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 61 r2c7_h+1v+1

Canonical identifier	r2c7_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	16	**Cell Inversion Orbit**	2,7;8,3
APP row	2	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|0|2|-8|+++0–+++0–	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 62 r2c7_h+1v-1

Canonical identifier	r2c7_h+1v-1	**Factor R_{act}, reading Ω**	1.000000024685259569118152
Cell	16	**Cell Inversion Orbit**	2,7;8,3
APP row	2	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|0|2|-8|+++0–+++0–	**First cell separation block**	—
Dodecaphasic word	0++–00++–0	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	36	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(36,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	35.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 63 r2c7_h-1v+1

Canonical identifier	r2c7_h-1v+1	**Factor R_{act}, reading Ω**	1.000000024685259569118152
Cell	16	**Cell Inversion Orbit**	2,7;8,3
APP row	2	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|0|2|-8|+++0–+++0–	**First cell separation block**	—
Dodecaphasic word	++00–++00–	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	36	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(36,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	35.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 64 r2c7_h-1v-1

Canonical identifier	r2c7_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	16	**Cell Inversion Orbit**	2,7;8,3
APP row	2	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|0|2|-8|+++0–+++0–	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	1
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 65 r2c8_h+1v+1

Canonical identifier	r2c8_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	17	**Cell Inversion Orbit**	2,8;8,2
APP row	2	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	30|6|-8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486	**Final Ambiguity**	4
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|6|0|2|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 66 r2c8_h+1v-1

Canonical identifier	r2c8_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	17	**Cell Inversion Orbit**	2,8;8,2
APP row	2	**First class return**	9
Column APP	8	**First return of orbit**	9
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	30|6|-8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258	**Final Ambiguity**	4
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 67 r2c8_h-1v+1

Canonical identifier	r2c8_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	17	**Cell Inversion Orbit**	2,8;8,2
APP row	2	**First class return**	18
Column APP	8	**First return of orbit**	18
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	30|6|-8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333	**Final Ambiguity**	4
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	36	**Final carry debt**	1
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 68 r2c8_h-1v-1

Canonical identifier	r2c8_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	17	**Cell Inversion Orbit**	2,8;8,2
APP row	2	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	30|6|-8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24	**Final Ambiguity**	4
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	3
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|-6|0|-2|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 69 r2c9_h+1v+1

Canonical identifier	r2c9_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	18	**Cell Inversion Orbit**	2,9;8,1
APP row	2	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	40
Generational depth	G4	**Binary logarithm of ambiguity**	20.339850002884624
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 70 r2c9_h+1v-1

Canonical identifier	r2c9_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	18	**Cell Inversion Orbit**	2,9;8,1
APP row	2	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 71 r2c9_h-1v+1

Canonical identifier	r2c9_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	18	**Cell Inversion Orbit**	2,9;8,1
APP row	2	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 72 r2c9_h-1v-1

Canonical identifier	r2c9_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	18	**Cell Inversion Orbit**	2,9;8,1
APP row	2	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 73 r3c1_h+1v+1

Canonical identifier	r3c1_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	19	**Cell Inversion Orbit**	3,1;7,9
APP row	3	**First class return**	18
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-10|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.518708319695345554341 34	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 74 r3c1_h+1v-1

Canonical identifier	r3c1_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	19	**Cell Inversion Orbit**	3,1;7,9
APP row	3	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-10|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	3
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 75 r3c1_h-1v+1

Canonical identifier	r3c1_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	19	**Cell Inversion Orbit**	3,1;7,9
APP row	3	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-10|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|-2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 76 r3c1_h-1v-1

Canonical identifier	r3c1_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	19	**Cell Inversion Orbit**	3,1;7,9
APP row	3	**First class return**	9
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-10|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.518708319695345554341 34	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 77 r3c2_h+1v+1

Canonical identifier	r3c2_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	20	**Cell Inversion Orbit**	3,2;7,8
APP row	3	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|-4|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|-6|2|-4|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 78 r3c2_h+1v-1

Canonical identifier	r3c2_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	20	**Cell Inversion Orbit**	3,2;7,8
APP row	3	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|-4|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|-2|0|-2|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 79 r3c2_h-1v+1

Canonical identifier	r3c2_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	20	**Cell Inversion Orbit**	3,2;7,8
APP row	3	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|-4|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|2|0|2|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 80 r3c2_h-1v-1

Canonical identifier	r3c2_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	20	**Cell Inversion Orbit**	3,2;7,8
APP row	3	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|-4|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|6|-2|4|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 81 r3c3_h+1v+1

Canonical identifier	r3c3_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	21	**Cell Inversion Orbit**	3,3;7,7
APP row	3	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	22
Generational depth	G2	**Binary logarithm of ambiguity**	4.0
Color orbit	—	**First orbit separation block**	5
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	5
Dodecaphasic word	+++—+++—	**First route separation block**	5
Visible words w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	2
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	39.606610139794827586470 91	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 82 r3c3_h+1v-1

Canonical identifier	r3c3_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	21	**Cell Inversion Orbit**	3,3;7,7
APP row	3	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-6|0|0|-12
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 83 r3c3_h-1v+1

Canonical identifier	r3c3_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	21	**Cell Inversion Orbit**	3,3;7,7
APP row	3	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|6|0|0|-12
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 84 r3c3_h-1v-1

Canonical identifier	r3c3_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	21	**Cell Inversion Orbit**	3,3;7,7
APP row	3	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	22
Generational depth	G2	**Binary logarithm of ambiguity**	4.0
Color orbit	—	**First orbit separation block**	5
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	5
Dodecaphasic word	+++—+++—	**First route separation block**	5
Visible words w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 85 r3c4_h+1v+1

Canonical identifier	r3c4_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	22	**Cell Inversion Orbit**	3,4;7,6
APP row	3	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|4|-12
Exponent κ_D	79.518708319695345554344134	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 86 r3c4_h+1v-1

Canonical identifier	r3c4_h+1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	22	**Cell Inversion Orbit**	3,4;7,6
APP row	3	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+0+—+0+—	**First route separation block**	1
Visible words w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	3
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 87 r3c4_h-1v+1

Canonical identifier	r3c4_h-1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	22	**Cell Inversion Orbit**	3,4;7,6
APP row	3	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	1
Visible words w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 88 r3c4_h-1v-1

Canonical identifier	r3c4_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	22	**Cell Inversion Orbit**	3,4;7,6
APP row	3	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-4|-12
Exponent κ_D	79.518708319695345554341 34	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 89 r3c5_h+1v+1

Canonical identifier	r3c5_h+1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	23	**Cell Inversion Orbit**	3,5;7,5
APP row	3	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|6|0|-8|+0+—+0+—	**First cell separation block**	1
Dodecaphasic word	+0+—+0+—	**First route separation block**	1
Visible words w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|-6|2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 90 r3c5_h+1v-1

Canonical identifier	r3c5_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	23	**Cell Inversion Orbit**	3,5;7,5
APP row	3	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|6|0|-8|+0+—+0+—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 91 r3c5_h-1v+1

Canonical identifier	r3c5_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	23	**Cell Inversion Orbit**	3,5;7,5
APP row	3	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|6|0|-8|+0+—+0+—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.43380880300855130367114	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 92 r3c5_h-1v-1

Canonical identifier	r3c5_h-1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	23	**Cell Inversion Orbit**	3,5;7,5
APP row	3	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	22
Generational depth	G2	**Binary logarithm of ambiguity**	4.0
Color orbit	—	**First orbit separation block**	5
Route family	28|-6|6|0|-8|+0+—+0+—	**First cell separation block**	5
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	5
Visible words w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|6|-2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 93 r3c6_h+1v+1

Canonical identifier	r3c6_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	24	**Cell Inversion Orbit**	3,6;7,4
APP row	3	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|0|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	++0—++0—	**First route separation block**	1
Visible words w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	2
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|-6|2|-2|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 94 r3c6_h+1v-1

Canonical identifier	r3c6_h+1v-1	**Factor R_{act}, reading Ω**	1.000000002500088676796314
Cell	24	**Cell Inversion Orbit**	3,6;7,4
APP row	3	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|0|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	00+-0000+-00	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	4	**Stabilized Decimal Triples**	17
P_6	2	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(4,54,2)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|0
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	3.606165354104059026254707	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 95 r3c6_h-1v+1

Canonical identifier	r3c6_h-1v+1	**Factor R_{act}, reading Ω**	1.000000002500088676796314
Cell	24	**Cell Inversion Orbit**	3,6;7,4
APP row	3	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|0|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	+0000-+0000-	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	4	**Stabilized Decimal Triples**	17
P_6	2	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(4,54,2)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|0
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	3.606165354104059026254707	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 96 r3c6_h-1v-1

Canonical identifier	r3c6_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	24	**Cell Inversion Orbit**	3,6;7,4
APP row	3	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|0|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	+++–0+++–0	**First route separation block**	1
Visible words w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|6|-2|2|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 97 r3c7_h+1v+1

Canonical identifier	r3c7_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	25	**Cell Inversion Orbit**	3,7;7,3
APP row	3	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.518708319695345554344134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 98 r3c7_h+1v-1

Canonical identifier	r3c7_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	25	**Cell Inversion Orbit**	3,7;7,3
APP row	3	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	++0—++0—	**First route separation block**	1
Visible words w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	3
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 99 r3c7_h-1v+1

Canonical identifier	r3c7_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	25	**Cell Inversion Orbit**	3,7;7,3
APP row	3	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++–0+++–0	**First route separation block**	1
Visible words w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 100 r3c7_h-1v-1

Canonical identifier	r3c7_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	25	**Cell Inversion Orbit**	3,7;7,3
APP row	3	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 101 r3c8_h+1v+1

Canonical identifier	r3c8_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	26	**Cell Inversion Orbit**	3,8;7,2
APP row	3	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|0|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 102 r3c8_h+1v-1

Canonical identifier	r3c8_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	26	**Cell Inversion Orbit**	3,8;7,2
APP row	3	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|0|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 103 r3c8_h-1v+1

Canonical identifier	r3c8_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	26	**Cell Inversion Orbit**	3,8;7,2
APP row	3	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|0|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 104 r3c8_h-1v-1

Canonical identifier	r3c8_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	26	**Cell Inversion Orbit**	3,8;7,2
APP row	3	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|0|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 105 r3c9_h+1v+1

Canonical identifier	r3c9_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	27	**Cell Inversion Orbit**	3,9;7,1
APP row	3	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	30|6|0|2|-8|+++–0+++–0	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	2
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|6|-2|4|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 106 r3c9_h+1v-1

Canonical identifier	r3c9_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	27	**Cell Inversion Orbit**	3,9;7,1
APP row	3	**First class return**	9
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	108
Generational depth	G4	**Binary logarithm of ambiguity**	46.52932501298079
Color orbit	—	**First orbit separation block**	—
Route family	30|6|0|2|-8|+++–0+++–0	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	6
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|6|-2|4|-8
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858 10851337881		

Return to the domain, the signature, and the route operators.

Route 107 r3c9_h-1v+1

Canonical identifier	r3c9_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	27	**Cell Inversion Orbit**	3,9;7,1
APP row	3	**First class return**	18
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	108
Generational depth	G4	**Binary logarithm of ambiguity**	46.52932501298079
Color orbit	—	**First orbit separation block**	—
Route family	30|6|0|2|-8|+++–0+++–0	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	6
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-6|2|-4|-8
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.999999999878581085133788 1		

Return to the domain, the signature, and the route operators.

Route 108 r3c9_h-1v-1

Canonical identifier	r3c9_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	27	**Cell Inversion Orbit**	3,9;7,1
APP row	3	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	30|6|0|2|-8|+++–0+++–0	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|-6|2|-4|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 109 r4c1_h+1v+1

Canonical identifier	r4c1_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	28	**Cell Inversion Orbit**	4,1;6,9
APP row	4	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	26
Generational depth	G4	**Binary logarithm of ambiguity**	6.339850002884624
Color orbit	R	**First orbit separation block**	5
Route family	30|6|8|0|-12|+++—+++—	**First cell separation block**	5
Dodecaphasic word	+++—+++—	**First route separation block**	5
Visible words w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|6|0|4|-12
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 110 r4c1_h+1v-1

Canonical identifier	r4c1_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	28	**Cell Inversion Orbit**	4,1;6,9
APP row	4	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	2
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|-2|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 111 r4c1_h-1v+1

Canonical identifier	r4c1_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	28	**Cell Inversion Orbit**	4,1;6,9
APP row	4	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	0
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|2|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 112 r4c1_h-1v-1

Canonical identifier	r4c1_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	28	**Cell Inversion Orbit**	4,1;6,9
APP row	4	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|-6|0|-4|-12
Exponent κ_D	79.606721336217519726524 96	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 113 r4c2_h+1v+1

Canonical identifier	r4c2_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	29	**Cell Inversion Orbit**	4,2;6,8
APP row	4	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	0
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 114 r4c2_h+1v-1

Canonical identifier	r4c2_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	29	**Cell Inversion Orbit**	4,2;6,8
APP row	4	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	4
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 115 r4c2_h-1v+1

Canonical identifier	r4c2_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	29	**Cell Inversion Orbit**	4,2;6,8
APP row	4	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	4
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 116 r4c2_h-1v-1

Canonical identifier	r4c2_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	29	**Cell Inversion Orbit**	4,2;6,8
APP row	4	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 117 r4c3_h+1v+1

Canonical identifier	r4c3_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	30	**Cell Inversion Orbit**	4,3;6,7
APP row	4	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-4|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 118 r4c3_h+1v-1

Canonical identifier	r4c3_h+1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	30	**Cell Inversion Orbit**	4,3;6,7
APP row	4	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+0+—+0+—	**First route separation block**	1
Visible words w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 119 r4c3_h-1v+1

Canonical identifier	r4c3_h-1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	30	**Cell Inversion Orbit**	4,3;6,7
APP row	4	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	1
Visible words w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 120 r4c3_h-1v-1

Canonical identifier	r4c3_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	30	**Cell Inversion Orbit**	4,3;6,7
APP row	4	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|4|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 121 r4c4_h+1v+1

Canonical identifier	r4c4_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	31	**Cell Inversion Orbit**	4,4;6,6
APP row	4	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 122 r4c4_h+1v-1

Canonical identifier	r4c4_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	31	**Cell Inversion Orbit**	4,4;6,6
APP row	4	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	2
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 123 r4c4_h-1v+1

Canonical identifier	r4c4_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	31	**Cell Inversion Orbit**	4,4;6,6
APP row	4	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	0
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 124 r4c4_h-1v-1

Canonical identifier	r4c4_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	31	**Cell Inversion Orbit**	4,4;6,6
APP row	4	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.606721336217519726524 96	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 125 r4c5_h+1v+1

Canonical identifier	r4c5_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	32	**Cell Inversion Orbit**	4,5;6,5
APP row	4	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	0
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 126 r4c5_h+1v-1

Canonical identifier	r4c5_h+1v-1	**Factor R_{act}, reading Ω**	1.000000027458386692651400
Cell	32	**Cell Inversion Orbit**	4,5;6,5
APP row	4	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+0+-0-+0+-0-	**First route separation block**	1
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	1
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	40	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(40,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	39.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 127 r4c5_h-1v+1

Canonical identifier	r4c5_h-1v+1	**Factor R_{act}, reading Ω**	1.000000027458386692651400
Cell	32	**Cell Inversion Orbit**	4,5;6,5
APP row	4	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+0+-0-+0+-0-	**First route separation block**	1
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	1
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	40	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(40,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	39.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 128 r4c5_h-1v-1

Canonical identifier	r4c5_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	32	**Cell Inversion Orbit**	4,5;6,5
APP row	4	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 129 r4c6_h+1v+1

Canonical identifier	r4c6_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	33	**Cell Inversion Orbit**	4,6;6,4
APP row	4	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|10|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 130 r4c6_h+1v-1

Canonical identifier	r4c6_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	33	**Cell Inversion Orbit**	4,6;6,4
APP row	4	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|10|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	++0—++0—	**First route separation block**	1
Visible words w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 131 r4c6_h-1v+1

Canonical identifier	r4c6_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	33	**Cell Inversion Orbit**	4,6;6,4
APP row	4	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|10|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++–0+++–0	**First route separation block**	1
Visible words w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 132 r4c6_h-1v-1

Canonical identifier	r4c6_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	33	**Cell Inversion Orbit**	4,6;6,4
APP row	4	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|10|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 133 r4c7_h+1v+1

Canonical identifier	r4c7_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	34	**Cell Inversion Orbit**	4,7;6,3
APP row	4	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	++0—++0—	**First route separation block**	1
Visible words w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|-6|2|0|-8
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 134 r4c7_h+1v-1

Canonical identifier	r4c7_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	34	**Cell Inversion Orbit**	4,7;6,3
APP row	4	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	2
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 135 r4c7_h-1v+1

Canonical identifier	r4c7_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	34	**Cell Inversion Orbit**	4,7;6,3
APP row	4	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	0
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 136 r4c7_h-1v-1

Canonical identifier	r4c7_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	34	**Cell Inversion Orbit**	4,7;6,3
APP row	4	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	+++–0+++–0	**First route separation block**	1
Visible words w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|6|-2|0|-8
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 137 r4c8_h+1v+1

Canonical identifier	r4c8_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	35	**Cell Inversion Orbit**	4,8;6,2
APP row	4	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-20|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	0
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 138 r4c8_h+1v-1

Canonical identifier	r4c8_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	35	**Cell Inversion Orbit**	4,8;6,2
APP row	4	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-20|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	4
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|6|-2|2|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 139 r4c8_h-1v+1

Canonical identifier	r4c8_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	35	**Cell Inversion Orbit**	4,8;6,2
APP row	4	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-20|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	4
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-6|2|-2|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 140 r4c8_h-1v-1

Canonical identifier	r4c8_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	35	**Cell Inversion Orbit**	4,8;6,2
APP row	4	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-20|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 141 r4c9_h+1v+1

Canonical identifier	r4c9_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	36	**Cell Inversion Orbit**	4,9;6,1
APP row	4	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	34|0|-8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 142 r4c9_h+1v-1

Canonical identifier	r4c9_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	36	**Cell Inversion Orbit**	4,9;6,1
APP row	4	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	34|0|-8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.00000002730861739670870 5		

Return to the domain, the signature, and the route operators.

Route 143 r4c9_h-1v+1

Canonical identifier	r4c9_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	36	**Cell Inversion Orbit**	4,9;6,1
APP row	4	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	34|0|-8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|-2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 144 r4c9_h-1v-1

Canonical identifier	r4c9_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	36	**Cell Inversion Orbit**	4,9;6,1
APP row	4	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	34|0|-8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 145 r5c1_h+1v+1

Canonical identifier	r5c1_h+1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	37	**Cell Inversion Orbit**	5,1;5,9
APP row	5	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-16|0|-8|+++-0-+++-0-	**First cell separation block**	1
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	1
Visible words w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 146 r5c1_h+1v-1

Canonical identifier	r5c1_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	37	**Cell Inversion Orbit**	5,1;5,9
APP row	5	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|-16|0|-8|+++-0-+++-0-	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 147 r5c1_h-1v+1

Canonical identifier	r5c1_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	37	**Cell Inversion Orbit**	5,1;5,9
APP row	5	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|-16|0|-8|+++-0-+++-0-	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 148 r5c1_h-1v-1

Canonical identifier	r5c1_h-1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	37	**Cell Inversion Orbit**	5,1;5,9
APP row	5	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	30
Generational depth	G4	**Binary logarithm of ambiguity**	8.0
Color orbit	—	**First orbit separation block**	5
Route family	34|0|-16|0|-8|+++-0-+++-0-	**First cell separation block**	5
Dodecaphasic word	+0+—+0+—	**First route separation block**	5
Visible words w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	1
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 149 r5c2_h+1v+1

Canonical identifier	r5c2_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	38	**Cell Inversion Orbit**	5,2;5,8
APP row	5	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486	**Final Ambiguity**	2
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|6|0|4|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 150 r5c2_h+1v-1

Canonical identifier	r5c2_h+1v-1	**Factor R_{act}, reading Ω**	1.000000013592751151887498
Cell	38	**Cell Inversion Orbit**	5,2;5,8
APP row	5	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+00—+00—	**First route separation block**	—
Visible words w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258	**Final Ambiguity**	2
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	20	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(20,54,4)	**Integer signature in $\mathbb{Z}^5$**	50|-4|4|0|-4
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	19.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 151 r5c2_h-1v+1

Canonical identifier	r5c2_h-1v+1	**Factor R_{act}, reading Ω**	1.000000013592751151887498
Cell	38	**Cell Inversion Orbit**	5,2;5,8
APP row	5	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++-00+++-00	**First route separation block**	—
Visible words w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333	**Final Ambiguity**	2
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	36	**Final carry debt**	1
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	20	**Stabilized Decimal Triples**	16
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(20,54,4)	**Integer signature in $\mathbb{Z}^5$**	50|4|-4|0|-4
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	19.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 152 r5c2_h-1v-1

Canonical identifier	r5c2_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	38	**Cell Inversion Orbit**	5,2;5,8
APP row	5	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24	**Final Ambiguity**	2
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	3
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|-6|0|-4|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 153 r5c3_h+1v+1

Canonical identifier	r5c3_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	39	**Cell Inversion Orbit**	5,3;5,7
APP row	5	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	22
Generational depth	G2	**Binary logarithm of ambiguity**	4.0
Color orbit	—	**First orbit separation block**	5
Route family	28|6|2|0|-12|+++—+++—	**First cell separation block**	5
Dodecaphasic word	+++—+++—	**First route separation block**	5
Visible words w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 154 r5c3_h+1v-1

Canonical identifier	r5c3_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	39	**Cell Inversion Orbit**	5,3;5,7
APP row	5	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|2|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 155 r5c3_h-1v+1

Canonical identifier	r5c3_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	39	**Cell Inversion Orbit**	5,3;5,7
APP row	5	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|2|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.43380880300855130367114	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 156 r5c3_h-1v-1

Canonical identifier	r5c3_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	39	**Cell Inversion Orbit**	5,3;5,7
APP row	5	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|2|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 157 r5c4_h+1v+1

Canonical identifier	r5c4_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	40	**Cell Inversion Orbit**	5,4;5,6
APP row	5	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G1	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 158 r5c4_h+1v-1

Canonical identifier	r5c4_h+1v-1	**Factor R_{act}, reading Ω**	1.000000027458386692651400
Cell	40	**Cell Inversion Orbit**	5,4;5,6
APP row	5	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G1	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+0+-0-+0+-0-	**First route separation block**	1
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	1
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	40	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(40,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	39.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 159 r5c4_h-1v+1

Canonical identifier	r5c4_h-1v+1	Factor R_{act}, reading Ω	1.000000027458386692651400
Cell	40	Cell Inversion Orbit	5,4;5,6
APP row	5	First class return	27
Column APP	4	First return of orbit	27
Horizontal orientation	-1	First cell return	27
Vertical orientation	1	Kinematic return of the calendar of 108 phases	54
Multisection	neutrino_G1	Kinematic return extended to 216 phases	216
Physical family	neutrino family	Ambiguity area of the prefix	18
Generational depth	G1	Binary logarithm of ambiguity	0.0
Color orbit	—	First orbit separation block	1
Route family	28|-6|4|0|-12|+++—+++—	First cell separation block	1
Dodecaphasic word	+0+-0-+0+-0-	First route separation block	1
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	Final Ambiguity	1
D_1	24	Rotations	71
D_3	20	Sheet changes	107
D_4	24	Total carry debt	0
D_{27}	24	Final carry debt	0
D_{36}	16	Delayed carry blocks	0
Channels $\Omega_{90/120}$	40	Stabilized Decimal Triples	17
P_6	4	Coincidence of w_6 between K and $K+9$	9
Reader K_D	(40,24,2)	Memory changes between K and $K+9$	9
Reader K_Ω	(40,54,4)	Integer signature in $\mathbb{Z}^5$	20|0|0|0|-8
Exponent κ_D	39.82508593118257305617326	Orbit on the persistence boundary	yes
Exponent κ_Ω	39.60638774694944330636281	Block Stabilization Chronology	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 160 r5c4_h-1v-1

Canonical identifier	r5c4_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	40	**Cell Inversion Orbit**	5,4;5,6
APP row	5	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G1	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	1
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 161 r5c5_h+1v+1

Canonical identifier	r5c5_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	41	**Cell Inversion Orbit**	5,5
APP row	5	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G0	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G0_center	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 162 r5c5_h+1v-1

Canonical identifier	r5c5_h+1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	41	**Cell Inversion Orbit**	5,5
APP row	5	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G0	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G0_center	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+0+—+0+—	**First route separation block**	1
Visible words w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 163 r5c5_h-1v+1

Canonical identifier	r5c5_h-1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	41	**Cell Inversion Orbit**	5,5
APP row	5	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G0	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G0_center	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	1
Visible words w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	36	**Final carry debt**	1
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 164 r5c5_h-1v-1

Canonical identifier	r5c5_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	41	**Cell Inversion Orbit**	5,5
APP row	5	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G0	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G0_center	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	3
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 165 r5c6_h+1v+1

Canonical identifier	r5c6_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	42	**Cell Inversion Orbit**	5,4;5,6
APP row	5	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G1	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 166 r5c6_h+1v-1

Canonical identifier	r5c6_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	42	**Cell Inversion Orbit**	5,4;5,6
APP row	5	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G1	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 167 r5c6_h-1v+1

Canonical identifier	r5c6_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	42	**Cell Inversion Orbit**	5,4;5,6
APP row	5	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G1	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 168 r5c6_h-1v-1

Canonical identifier	r5c6_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	42	**Cell Inversion Orbit**	5,4;5,6
APP row	5	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G1	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 169 r5c7_h+1v+1

Canonical identifier	r5c7_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	43	**Cell Inversion Orbit**	5,3;5,7
APP row	5	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-4|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 170 r5c7_h+1v-1

Canonical identifier	r5c7_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	43	**Cell Inversion Orbit**	5,3;5,7
APP row	5	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|6|-2|2|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 171 r5c7_h-1v+1

Canonical identifier	r5c7_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	43	**Cell Inversion Orbit**	5,3;5,7
APP row	5	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-6|2|-2|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 172 r5c7_h-1v-1

Canonical identifier	r5c7_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	43	**Cell Inversion Orbit**	5,3;5,7
APP row	5	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	22
Generational depth	G2	**Binary logarithm of ambiguity**	4.0
Color orbit	—	**First orbit separation block**	5
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	5
Dodecaphasic word	+++—+++—	**First route separation block**	5
Visible words w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	1
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|4|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 173 r5c8_h+1v+1

Canonical identifier	r5c8_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	44	**Cell Inversion Orbit**	5,2;5,8
APP row	5	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486	**Final Ambiguity**	2
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|-6|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 174 r5c8_h+1v-1

Canonical identifier	r5c8_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	44	**Cell Inversion Orbit**	5,2;5,8
APP row	5	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258	**Final Ambiguity**	2
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|2|0|0|-12
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 175 r5c8_h-1v+1

Canonical identifier	r5c8_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	44	**Cell Inversion Orbit**	5,2;5,8
APP row	5	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333	**Final Ambiguity**	2
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	36	**Final carry debt**	1
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|-2|0|0|-12
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 176 r5c8_h-1v-1

Canonical identifier	r5c8_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	44	**Cell Inversion Orbit**	5,2;5,8
APP row	5	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|-6|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24	**Final Ambiguity**	2
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	3
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|6|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 177 r5c9_h+1v+1

Canonical identifier	r5c9_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	45	**Cell Inversion Orbit**	5,1;5,9
APP row	5	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	30
Generational depth	G4	**Binary logarithm of ambiguity**	8.0
Color orbit	—	**First orbit separation block**	5
Route family	34|0|-12|-2|-8|0++—0++—	**First cell separation block**	5
Dodecaphasic word	0++—0++—	**First route separation block**	5
Visible words w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 178 r5c9_h+1v-1

Canonical identifier	r5c9_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	45	**Cell Inversion Orbit**	5,1;5,9
APP row	5	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|-2|-8|0++—0++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|-2|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 179 r5c9_h-1v+1

Canonical identifier	r5c9_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	45	**Cell Inversion Orbit**	5,1;5,9
APP row	5	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|-2|-8|0++—0++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|2|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 180 r5c9_h-1v-1

Canonical identifier	r5c9_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	45	**Cell Inversion Orbit**	5,1;5,9
APP row	5	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|-2|-8|0++—0++—	**First cell separation block**	1
Dodecaphasic word	+++0–+++0–	**First route separation block**	1
Visible words w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446 41686	**Block Stabilization Chronology**	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 181 r6c1_h+1v+1

Canonical identifier	r6c1_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	46	**Cell Inversion Orbit**	4,9;6,1
APP row	6	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 182 r6c1_h+1v-1

Canonical identifier	r6c1_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	46	**Cell Inversion Orbit**	4,9;6,1
APP row	6	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	++0—++0—	**First route separation block**	1
Visible words w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	3
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 183 r6c1_h-1v+1

Canonical identifier	r6c1_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	46	**Cell Inversion Orbit**	4,9;6,1
APP row	6	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++–0+++–0	**First route separation block**	1
Visible words w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 184 r6c1_h-1v-1

Canonical identifier	r6c1_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	46	**Cell Inversion Orbit**	4,9;6,1
APP row	6	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	34|0|-4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 185 r6c2_h+1v+1

Canonical identifier	r6c2_h+1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	47	**Cell Inversion Orbit**	4,8;6,2
APP row	6	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|0|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+0+—+0+—	**First route separation block**	—
Visible words w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 186 r6c2_h+1v-1

Canonical identifier	r6c2_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	47	**Cell Inversion Orbit**	4,8;6,2
APP row	6	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|0|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 187 r6c2_h-1v+1

Canonical identifier	r6c2_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	47	**Cell Inversion Orbit**	4,8;6,2
APP row	6	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|0|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 188 r6c2_h-1v-1

Canonical identifier	r6c2_h-1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	47	**Cell Inversion Orbit**	4,8;6,2
APP row	6	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|0|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	—
Visible words w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 189 r6c3_h+1v+1

Canonical identifier	r6c3_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	48	**Cell Inversion Orbit**	4,7;6,3
APP row	6	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|0|2|-8|+++0–+++0–	**First cell separation block**	1
Dodecaphasic word	+++0–+++0–	**First route separation block**	1
Visible words w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	2
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|6|-2|2|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 190 r6c3_h+1v-1

Canonical identifier	r6c3_h+1v-1	**Factor R_{act}, reading Ω**	1.000000002500088676796314
Cell	48	**Cell Inversion Orbit**	4,7;6,3
APP row	6	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|0|2|-8|+++0–+++0–	**First cell separation block**	1
Dodecaphasic word	+0000-+0000-	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	4	**Stabilized Decimal Triples**	17
P_6	2	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(4,54,2)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|0
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	3.606165354104059026254707	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 191 r6c3_h-1v+1

Canonical identifier	r6c3_h-1v+1	**Factor R_{act}, reading Ω**	1.000000002500088676796314
Cell	48	**Cell Inversion Orbit**	4,7;6,3
APP row	6	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|0|2|-8|+++0–+++0–	**First cell separation block**	1
Dodecaphasic word	00+-0000+-00	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	4	**Stabilized Decimal Triples**	17
P_6	2	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(4,54,2)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|0
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	3.606165354104059026254707	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 192 r6c3_h-1v-1

Canonical identifier	r6c3_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	48	**Cell Inversion Orbit**	4,7;6,3
APP row	6	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|0|2|-8|+++0–+++0–	**First cell separation block**	1
Dodecaphasic word	0++—0++—	**First route separation block**	1
Visible words w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|-6|2|-2|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 193 r6c4_h+1v+1

Canonical identifier	r6c4_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	49	**Cell Inversion Orbit**	4,6;6,4
APP row	6	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 194 r6c4_h+1v-1

Canonical identifier	r6c4_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	49	**Cell Inversion Orbit**	4,6;6,4
APP row	6	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	0++—0++—	**First route separation block**	1
Visible words w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	3
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943373213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 195 r6c4_h-1v+1

Canonical identifier	r6c4_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	49	**Cell Inversion Orbit**	4,6;6,4
APP row	6	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++0–+++0–	**First route separation block**	1
Visible words w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 196 r6c4_h-1v-1

Canonical identifier	r6c4_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	49	**Cell Inversion Orbit**	4,6;6,4
APP row	6	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.518708319695345554341 34	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 197 r6c5_h+1v+1

Canonical identifier	r6c5_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	50	**Cell Inversion Orbit**	4,5;6,5
APP row	6	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|4|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	++0—++0—	**First route separation block**	1
Visible words w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|-6|2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 198 r6c5_h+1v-1

Canonical identifier	r6c5_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	50	**Cell Inversion Orbit**	4,5;6,5
APP row	6	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|4|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 199 r6c5_h-1v+1

Canonical identifier	r6c5_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	50	**Cell Inversion Orbit**	4,5;6,5
APP row	6	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|4|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.43380880300855130367114	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 200 r6c5_h-1v-1

Canonical identifier	r6c5_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	50	**Cell Inversion Orbit**	4,5;6,5
APP row	6	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	G	**First orbit separation block**	1
Route family	28|-6|4|-2|-8|++0—++0—	**First cell separation block**	1
Dodecaphasic word	+++–0+++–0	**First route separation block**	1
Visible words w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|6|-2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 201 r6c6_h+1v+1

Canonical identifier	r6c6_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	51	**Cell Inversion Orbit**	4,4;6,6
APP row	6	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	2
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 202 r6c6_h+1v-1

Canonical identifier	r6c6_h+1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	51	**Cell Inversion Orbit**	4,4;6,6
APP row	6	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G1	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|6|-2|0|-8
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 203 r6c6_h-1v+1

Canonical identifier	r6c6_h-1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	51	**Cell Inversion Orbit**	4,4;6,6
APP row	6	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G1	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+0+—+0+—	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-6|2|0|-8
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 204 r6c6_h-1v-1

Canonical identifier	r6c6_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	51	**Cell Inversion Orbit**	4,4;6,6
APP row	6	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G1	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	18
Generational depth	G1	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 205 r6c7_h+1v+1

Canonical identifier	r6c7_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	52	**Cell Inversion Orbit**	4,3;6,7
APP row	6	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|-8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.518708319695345554341 34	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 206 r6c7_h+1v-1

Canonical identifier	r6c7_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	52	**Cell Inversion Orbit**	4,3;6,7
APP row	6	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|-8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	3
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 207 r6c7_h-1v+1

Canonical identifier	r6c7_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	52	**Cell Inversion Orbit**	4,3;6,7
APP row	6	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|-8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|-2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 208 r6c7_h-1v-1

Canonical identifier	r6c7_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	52	**Cell Inversion Orbit**	4,3;6,7
APP row	6	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G2	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	B	**First orbit separation block**	1
Route family	28|6|-8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.518708319695345554343134	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 209 r6c8_h+1v+1

Canonical identifier	r6c8_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	53	**Cell Inversion Orbit**	4,2;6,8
APP row	6	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|-2|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|-6|2|-2|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 210 r6c8_h+1v-1

Canonical identifier	r6c8_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	53	**Cell Inversion Orbit**	4,2;6,8
APP row	6	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|-2|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|-2|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 211 r6c8_h-1v+1

Canonical identifier	r6c8_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	53	**Cell Inversion Orbit**	4,2;6,8
APP row	6	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|-2|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|2|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 212 r6c8_h-1v-1

Canonical identifier	r6c8_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	53	**Cell Inversion Orbit**	4,2;6,8
APP row	6	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|-2|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|6|-2|2|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 213 r6c9_h+1v+1

Canonical identifier	r6c9_h+1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	54	**Cell Inversion Orbit**	4,1;6,9
APP row	6	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|0|-8|+0+—+0+—	**First cell separation block**	1
Dodecaphasic word	+0+—+0+—	**First route separation block**	1
Visible words w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	2
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|-6|2|0|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 214 r6c9_h+1v-1

Canonical identifier	r6c9_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	54	**Cell Inversion Orbit**	4,1;6,9
APP row	6	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|0|-8|+0+—+0+—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-6|0|0|-12
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858 10851337881		

Return to the domain, the signature, and the route operators.

Route 215 r6c9_h-1v+1

Canonical identifier	r6c9_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	54	**Cell Inversion Orbit**	4,1;6,9
APP row	6	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|-6|0|0|-8|+0+—+0+—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|6|0|0|-12
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.999999999878581085133788 1		

Return to the domain, the signature, and the route operators.

Route 216 r6c9_h-1v-1

Canonical identifier	r6c9_h-1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	54	**Cell Inversion Orbit**	4,1;6,9
APP row	6	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	26
Generational depth	G4	**Binary logarithm of ambiguity**	6.339850002884624
Color orbit	R	**First orbit separation block**	5
Route family	30|-6|0|0|-8|+0+—+0+—	**First cell separation block**	5
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	5
Visible words w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|6|-2|0|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 217 r7c1_h+1v+1

Canonical identifier	r7c1_h+1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	55	**Cell Inversion Orbit**	3,9;7,1
APP row	7	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	30|-6|-8|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+0+—+0+—	**First route separation block**	—
Visible words w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|-6|2|0|-8
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446419 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 218 r7c1_h+1v-1

Canonical identifier	r7c1_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	55	**Cell Inversion Orbit**	3,9;7,1
APP row	7	**First class return**	18
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	30|-6|-8|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585	**Final Ambiguity**	3
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	2
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 219 r7c1_h-1v+1

Canonical identifier	r7c1_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	55	**Cell Inversion Orbit**	3,9;7,1
APP row	7	**First class return**	9
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	30|-6|-8|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510	**Final Ambiguity**	3
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	0
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 220 r7c1_h-1v-1

Canonical identifier	r7c1_h-1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	55	**Cell Inversion Orbit**	3,9;7,1
APP row	7	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	30|-6|-8|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	—
Visible words w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|6|-2|0|-8
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 221 r7c2_h+1v+1

Canonical identifier	r7c2_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	56	**Cell Inversion Orbit**	3,8;7,2
APP row	7	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|-4|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	0
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 222 r7c2_h+1v-1

Canonical identifier	r7c2_h+1v-1	**Factor R_{act}, reading Ω**	1.000000024685259569118152
Cell	56	**Cell Inversion Orbit**	3,8;7,2
APP row	7	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|-4|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	++00–++00–	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	36	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(36,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	35.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 223 r7c2_h-1v+1

Canonical identifier	r7c2_h-1v+1	**Factor R_{act}, reading Ω**	1.000000024685259569118152
Cell	56	**Cell Inversion Orbit**	3,8;7,2
APP row	7	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|-4|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	0++–00++–0	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	36	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(36,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	35.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 224 r7c2_h-1v-1

Canonical identifier	r7c2_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	56	**Cell Inversion Orbit**	3,8;7,2
APP row	7	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	34|0|-4|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 225 r7c3_h+1v+1

Canonical identifier	r7c3_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	57	**Cell Inversion Orbit**	3,7;7,3
APP row	7	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 226 r7c3_h+1v-1

Canonical identifier	r7c3_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	57	**Cell Inversion Orbit**	3,7;7,3
APP row	7	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	0++—0++—	**First route separation block**	1
Visible words w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943373213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 227 r7c3_h-1v+1

Canonical identifier	r7c3_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	57	**Cell Inversion Orbit**	3,7;7,3
APP row	7	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++0–+++0–	**First route separation block**	1
Visible words w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 228 r7c3_h-1v-1

Canonical identifier	r7c3_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	57	**Cell Inversion Orbit**	3,7;7,3
APP row	7	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 229 r7c4_h+1v+1

Canonical identifier	r7c4_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	58	**Cell Inversion Orbit**	3,6;7,4
APP row	7	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|6|0|0|-12
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 230 r7c4_h+1v-1

Canonical identifier	r7c4_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	58	**Cell Inversion Orbit**	3,6;7,4
APP row	7	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	2
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 231 r7c4_h-1v+1

Canonical identifier	r7c4_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	58	**Cell Inversion Orbit**	3,6;7,4
APP row	7	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	0
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 232 r7c4_h-1v-1

Canonical identifier	r7c4_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	58	**Cell Inversion Orbit**	3,6;7,4
APP row	7	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|-6|0|0|-12
Exponent κ_D	79.606721336217519726524 96	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 233 r7c5_h+1v+1

Canonical identifier	r7c5_h+1v+1	Factor R_{act}, reading Ω	1.000000055189812531859140
Cell	59	Cell Inversion Orbit	3,5;7,5
APP row	7	First class return	27
Column APP	5	First return of orbit	27
Horizontal orientation	1	First cell return	27
Vertical orientation	1	Kinematic return of the calendar of 108 phases	54
Multisection	charged_lepton_G2	Kinematic return extended to 216 phases	216
Physical family	charged_lepton	Ambiguity area of the prefix	22
Generational depth	G2	Binary logarithm of ambiguity	4.0
Color orbit	—	First orbit separation block	5
Route family	28|6|-2|0|-12|+++—+++—	First cell separation block	5
Dodecaphasic word	+++—+++—	First route separation block	5
Visible words w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264	Final Ambiguity	1
D_1	66	Rotations	71
D_3	64	Sheet changes	107
D_4	64	Total carry debt	0
D_{27}	60	Final carry debt	0
D_{36}	40	Delayed carry blocks	0
Channels $\Omega_{90/120}$	80	Stabilized Decimal Triples	17
P_6	6	Coincidence of w_6 between K and $K+9$	9
Reader K_D	(100,66,0)	Memory changes between K and $K+9$	9
Reader K_Ω	(80,54,6)	Integer signature in $\mathbb{Z}^5$	28|6|0|4|-12
Exponent κ_D	99.51837473042726913417918	Orbit on the persistence boundary	yes
Exponent κ_Ω	79.606610139794827586470 91	Block Stabilization Chronology	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 234 r7c5_h+1v-1

Canonical identifier	r7c5_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	59	**Cell Inversion Orbit**	3,5;7,5
APP row	7	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|6|-2|2|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 235 r7c5_h-1v+1

Canonical identifier	r7c5_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	59	**Cell Inversion Orbit**	3,5;7,5
APP row	7	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	36
Generational depth	G2	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-6|2|-2|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 236 r7c5_h-1v-1

Canonical identifier	r7c5_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	59	**Cell Inversion Orbit**	3,5;7,5
APP row	7	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-4|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 237 r7c6_h+1v+1

Canonical identifier	r7c6_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	60	**Cell Inversion Orbit**	3,4;7,6
APP row	7	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 238 r7c6_h+1v-1

Canonical identifier	r7c6_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	60	**Cell Inversion Orbit**	3,4;7,6
APP row	7	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 239 r7c6_h-1v+1

Canonical identifier	r7c6_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	60	**Cell Inversion Orbit**	3,4;7,6
APP row	7	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|-2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 240 r7c6_h-1v-1

Canonical identifier	r7c6_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	60	**Cell Inversion Orbit**	3,4;7,6
APP row	7	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G2	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	28|-6|8|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.518597123272653414287 29	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 241 r7c7_h+1v+1

Canonical identifier	r7c7_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	61	**Cell Inversion Orbit**	3,3;7,7
APP row	7	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	22
Generational depth	G2	**Binary logarithm of ambiguity**	4.0
Color orbit	—	**First orbit separation block**	5
Route family	24|0|4|0|-12|+++—+++—	**First cell separation block**	5
Dodecaphasic word	+++—+++—	**First route separation block**	5
Visible words w_6	657 24 243 657 18 15 486 423 15 657 24 243 657 18 15 486 423 15	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.60672133621751972652496	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 242 r7c7_h+1v-1

Canonical identifier	r7c7_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	61	**Cell Inversion Orbit**	3,3;7,7
APP row	7	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	24|0|4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	666 672 510 666 666 585 507 504 585 666 672 510 666 666 585 507 504 585	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	2
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|-2|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 13 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 243 r7c7_h-1v+1

Canonical identifier	r7c7_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	61	**Cell Inversion Orbit**	3,3;7,7
APP row	7	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G2	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	24|0|4|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	507 504 585 507 510 510 666 672 510 507 504 585 507 510 510 666 672 510	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	78	**Total carry debt**	0
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,78,25)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|2|0|0|-12
Exponent κ_D	59.43358641016316702356304	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000041204222405344905		

Return to the domain, the signature, and the route operators.

Route 244 r7c7_h-1v-1

Canonical identifier	r7c7_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	61	**Cell Inversion Orbit**	3,3;7,7
APP row	7	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G2	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	22
Generational depth	G2	**Binary logarithm of ambiguity**	4.0
Color orbit	—	**First orbit separation block**	5
Route family	24|0|4|0|-12|+++—+++—	**First cell separation block**	5
Dodecaphasic word	+++—+++—	**First route separation block**	5
Visible words w_6	486 423 15 486 429 243 657 24 243 486 423 15 486 429 243 657 24 243	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	70	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,7)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.606721336217519726524 96	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000055189889622315337		

Return to the domain, the signature, and the route operators.

Route 245 r7c8_h+1v+1

Canonical identifier	r7c8_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	62	**Cell Inversion Orbit**	3,2;7,8
APP row	7	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	24 243 657 24 246 264 90 570 264 24 243 657 24 246 264 90 570 264	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	0
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 246 r7c8_h+1v-1

Canonical identifier	r7c8_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	62	**Cell Inversion Orbit**	3,2;7,8
APP row	7	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 247 r7c8_h-1v+1

Canonical identifier	r7c8_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	62	**Cell Inversion Orbit**	3,2;7,8
APP row	7	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 248 r7c8_h-1v-1

Canonical identifier	r7c8_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	62	**Cell Inversion Orbit**	3,2;7,8
APP row	7	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G3	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|6|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	90 570 264 90 567 657 24 243 657 90 570 264 90 567 657 24 243 657	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	99.51837473042726913417918	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 17
Factor R_{act}, reading D	1.000000068994276645021895		

Return to the domain, the signature, and the route operators.

Route 249 r7c9_h+1v+1

Canonical identifier	r7c9_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	63	**Cell Inversion Orbit**	3,1;7,9
APP row	7	**First class return**	9
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	327 498 99 327 501 486 423 15 486 327 498 99 327 501 486 423 15 486	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-4|-12
Exponent κ_D	79.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 7 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 250 r7c9_h+1v-1

Canonical identifier	r7c9_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	63	**Cell Inversion Orbit**	3,1;7,9
APP row	7	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	333 255 252 333 258 414 420 258 414 333 255 252 333 258 414 420 258 414	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 251 r7c9_h-1v+1

Canonical identifier	r7c9_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	63	**Cell Inversion Orbit**	3,1;7,9
APP row	7	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	420 258 414 420 255 252 333 255 252 420 258 414 420 255 252 333 255 252	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 252 r7c9_h-1v-1

Canonical identifier	r7c9_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	63	**Cell Inversion Orbit**	3,1;7,9
APP row	7	**First class return**	18
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 12 99 327 498 99 423 15 486 423 12 99 327 498 99	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	64	**Total carry debt**	1
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|4|-12
Exponent κ_D	79.518597123272653414287 29	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000055128794709248880		

Return to the domain, the signature, and the route operators.

Route 253 r8c1_h+1v+1

Canonical identifier	r8c1_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	64	**Cell Inversion Orbit**	2,9;8,1
APP row	8	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|10|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 254 r8c1_h+1v-1

Canonical identifier	r8c1_h+1v-1	**Factor R_{act}, reading Ω**	1.000000024685259569118152
Cell	64	**Cell Inversion Orbit**	2,9;8,1
APP row	8	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|10|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	++00–++00–	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	36	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(36,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	35.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 255 r8c1_h-1v+1

Canonical identifier	r8c1_h-1v+1	**Factor R_{act}, reading Ω**	1.000000024685259569118152
Cell	64	**Cell Inversion Orbit**	2,9;8,1
APP row	8	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|10|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	0++–00++–0	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	36	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(36,54,4)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	35.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 256 r8c1_h-1v-1

Canonical identifier	r8c1_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	64	**Cell Inversion Orbit**	2,9;8,1
APP row	8	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	40
Generational depth	G4	**Binary logarithm of ambiguity**	20.339850002884624
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|10|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	1
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 257 r8c2_h+1v+1

Canonical identifier	r8c2_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	65	**Cell Inversion Orbit**	2,8;8,2
APP row	8	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	30|-6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486	**Final Ambiguity**	4
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|-6|0|-2|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 258 r8c2_h+1v-1

Canonical identifier	r8c2_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	65	**Cell Inversion Orbit**	2,8;8,2
APP row	8	**First class return**	18
Column APP	2	**First return of orbit**	18
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	30|-6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258	**Final Ambiguity**	4
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 259 r8c2_h-1v+1

Canonical identifier	r8c2_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	65	**Cell Inversion Orbit**	2,8;8,2
APP row	8	**First class return**	9
Column APP	2	**First return of orbit**	9
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	30|-6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333	**Final Ambiguity**	4
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	36	**Final carry debt**	1
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 260 r8c2_h-1v-1

Canonical identifier	r8c2_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	65	**Cell Inversion Orbit**	2,8;8,2
APP row	8	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	30|-6|4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24	**Final Ambiguity**	4
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	3
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|6|0|2|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 261 r8c3_h+1v+1

Canonical identifier	r8c3_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	66	**Cell Inversion Orbit**	2,7;8,3
APP row	8	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-8|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 262 r8c3_h+1v-1

Canonical identifier	r8c3_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	66	**Cell Inversion Orbit**	2,7;8,3
APP row	8	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-8|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 263 r8c3_h-1v+1

Canonical identifier	r8c3_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	66	**Cell Inversion Orbit**	2,7;8,3
APP row	8	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-8|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 264 r8c3_h-1v-1

Canonical identifier	r8c3_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	66	**Cell Inversion Orbit**	2,7;8,3
APP row	8	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	34|0|-8|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446 41686	**Block Stabilization Chronology**	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 265 r8c4_h+1v+1

Canonical identifier	r8c4_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	67	**Cell Inversion Orbit**	2,6;8,4
APP row	8	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|-4|2|-8|+++–0+++–0	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 266 r8c4_h+1v-1

Canonical identifier	r8c4_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	67	**Cell Inversion Orbit**	2,6;8,4
APP row	8	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|-4|2|-8|+++–0+++–0	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	4
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|6|-2|2|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 267 r8c4_h-1v+1

Canonical identifier	r8c4_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	67	**Cell Inversion Orbit**	2,6;8,4
APP row	8	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|-4|2|-8|+++–0+++–0	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	4
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-6|2|-2|-8
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 268 r8c4_h-1v-1

Canonical identifier	r8c4_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	67	**Cell Inversion Orbit**	2,6;8,4
APP row	8	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	34|0|-4|2|-8|+++–0+++–0	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	1
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 269 r8c5_h+1v+1

Canonical identifier	r8c5_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	68	**Cell Inversion Orbit**	2,5;8,5
APP row	8	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|-4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486	**Final Ambiguity**	2
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|6|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 270 r8c5_h+1v-1

Canonical identifier	r8c5_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	68	**Cell Inversion Orbit**	2,5;8,5
APP row	8	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|-4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258	**Final Ambiguity**	2
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|2|0|0|-12
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 271 r8c5_h-1v+1

Canonical identifier	r8c5_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	68	**Cell Inversion Orbit**	2,5;8,5
APP row	8	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|-4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333	**Final Ambiguity**	2
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	36	**Final carry debt**	1
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|-2|0|0|-12
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 272 r8c5_h-1v-1

Canonical identifier	r8c5_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	68	**Cell Inversion Orbit**	2,5;8,5
APP row	8	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	R	**First orbit separation block**	1
Route family	30|6|-4|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24	**Final Ambiguity**	2
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	3
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	30|-6|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 273 r8c6_h+1v+1

Canonical identifier	r8c6_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	69	**Cell Inversion Orbit**	2,4;8,6
APP row	8	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|2|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 274 r8c6_h+1v-1

Canonical identifier	r8c6_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	69	**Cell Inversion Orbit**	2,4;8,6
APP row	8	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|-2|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 275 r8c6_h-1v+1

Canonical identifier	r8c6_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	69	**Cell Inversion Orbit**	2,4;8,6
APP row	8	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|2|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 276 r8c6_h-1v-1

Canonical identifier	r8c6_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	69	**Cell Inversion Orbit**	2,4;8,6
APP row	8	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|-6|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-2|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 277 r8c7_h+1v+1

Canonical identifier	r8c7_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	70	**Cell Inversion Orbit**	2,3;8,7
APP row	8	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	99 327 498 99 324 423 15 486 423 99 327 498 99 324 423 15 486 423	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	0
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 278 r8c7_h+1v-1

Canonical identifier	r8c7_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	70	**Cell Inversion Orbit**	2,3;8,7
APP row	8	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	81 3 0 81 0 3 0 81 3 81 3 0 81 0 3 0 81 3	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 6 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 279 r8c7_h-1v+1

Canonical identifier	r8c7_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	70	**Cell Inversion Orbit**	2,3;8,7
APP row	8	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	0 81 3 0 84 0 81 3 0 0 81 3 0 84 0 81 3 0	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	20	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,24,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|6|0|0|-12
Exponent κ_D	39.82508593118257305617326	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000027610006159514215		

Return to the domain, the signature, and the route operators.

Route 280 r8c7_h-1v-1

Canonical identifier	r8c7_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	70	**Cell Inversion Orbit**	2,3;8,7
APP row	8	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G3	**Binary logarithm of ambiguity**	18.0
Color orbit	G	**First orbit separation block**	—
Route family	28|-6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	15 486 423 15 489 498 99 327 498 15 486 423 15 489 498 99 327 498	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	64	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	60	**Final carry debt**	1
D_{36}	40	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(100,66,2)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	99.51859712327265341428729	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 16
Factor R_{act}, reading D	1.000000068994430825936417		

Return to the domain, the signature, and the route operators.

Route 281 r8c8_h+1v+1

Canonical identifier	r8c8_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	71	**Cell Inversion Orbit**	2,2;8,8
APP row	8	**First class return**	9
Column APP	8	**First return of orbit**	9
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	24|0|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 663 486 423 15 486 243 657 24 243 663 486 423 15 486	**Final Ambiguity**	4
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 282 r8c8_h+1v-1

Canonical identifier	r8c8_h+1v-1	**Factor R_{act}, reading Ω**	1.000000016365878244659810
Cell	71	**Cell Inversion Orbit**	2,2;8,8
APP row	8	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	24|0|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	0+0—0+0—	**First route separation block**	—
Visible words w_6	255 252 333 255 258 258 414 420 258 255 252 333 255 258 258 414 420 258	**Final Ambiguity**	4
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	36	**Final carry debt**	0
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	24	**Stabilized Decimal Triples**	17
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(24,54,4)	**Integer signature in $\mathbb{Z}^5$**	50|-4|4|-2|-4
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	23.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 15 17
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 283 r8c8_h-1v+1

Canonical identifier	r8c8_h-1v+1	**Factor R_{act}, reading Ω**	1.000000016365878244659810
Cell	71	**Cell Inversion Orbit**	2,2;8,8
APP row	8	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	24|0|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++0-0+++0-0	**First route separation block**	—
Visible words w_6	414 420 258 414 414 333 255 252 333 414 420 258 414 414 333 255 252 333	**Final Ambiguity**	4
D_1	84	**Rotations**	71
D_3	28	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	36	**Final carry debt**	1
D_{36}	24	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	24	**Stabilized Decimal Triples**	16
P_6	4	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(60,84,28)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(24,54,4)	**Integer signature in $\mathbb{Z}^5$**	50|4|-4|2|-4
Exponent κ_D	59.39013588401554063774148	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	23.60638774694944330636281	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 16
Factor R_{act}, reading D	1.000000041174098946741575		

Return to the domain, the signature, and the route operators.

Route 284 r8c8_h-1v-1

Canonical identifier	r8c8_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	71	**Cell Inversion Orbit**	2,2;8,8
APP row	8	**First class return**	18
Column APP	8	**First return of orbit**	18
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_up_G3	**Kinematic return extended to 216 phases**	216
Physical family	quark_up	**Ambiguity area of the prefix**	72
Generational depth	G3	**Binary logarithm of ambiguity**	36.0
Color orbit	R	**First orbit separation block**	—
Route family	24|0|8|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	423 15 486 423 9 24 243 657 24 423 15 486 423 9 24 243 657 24	**Final Ambiguity**	4
D_1	54	**Rotations**	71
D_3	56	**Sheet changes**	107
D_4	68	**Total carry debt**	3
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	79.606610139794827586470 91	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 13 14 14 15 17
Factor R_{act}, reading D	1.000000055189812531859140		

Return to the domain, the signature, and the route operators.

Route 285 r8c9_h+1v+1

Canonical identifier	r8c9_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	72	**Cell Inversion Orbit**	2,1;8,9
APP row	8	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	40
Generational depth	G4	**Binary logarithm of ambiguity**	20.339850002884624
Color orbit	B	**First orbit separation block**	—
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	657 24 243 657 21 90 570 264 90 657 24 243 657 21 90 570 264 90	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|0|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 286 r8c9_h+1v-1

Canonical identifier	r8c9_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	72	**Cell Inversion Orbit**	2,1;8,9
APP row	8	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	672 510 666 672 507 504 585 507 504 672 510 666 672 507 504 585 507 504	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 287 r8c9_h-1v+1

Canonical identifier	r8c9_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	72	**Cell Inversion Orbit**	2,1;8,9
APP row	8	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	585 507 504 585 510 666 672 510 666 585 507 504 585 510 666 672 510 666	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 288 r8c9_h-1v-1

Canonical identifier	r8c9_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	72	**Cell Inversion Orbit**	2,1;8,9
APP row	8	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	quark_down_G4	**Kinematic return extended to 216 phases**	216
Physical family	quark_down	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	B	**First orbit separation block**	—
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	570 264 90 570 267 243 657 24 243 570 264 90 570 267 243 657 24 243	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	68	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,4)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|0|-12
Exponent κ_D	79.51881951611803769439539	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 1 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128948890161264		

Return to the domain, the signature, and the route operators.

Route 289 r9c1_h+1v+1

Canonical identifier	r9c1_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	73	**Cell Inversion Orbit**	1,9;9,1
APP row	9	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|2|-12
Exponent κ_D	79.518708319695345554341 34	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 290 r9c1_h+1v-1

Canonical identifier	r9c1_h+1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	73	**Cell Inversion Orbit**	1,9;9,1
APP row	9	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+0+—+0+—	**First route separation block**	—
Visible words w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	3
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 291 r9c1_h-1v+1

Canonical identifier	r9c1_h-1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	73	**Cell Inversion Orbit**	1,9;9,1
APP row	9	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	—
Visible words w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	47.606498943372135446 41686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 292 r9c1_h-1v-1

Canonical identifier	r9c1_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	73	**Cell Inversion Orbit**	1,9;9,1
APP row	9	**First class return**	27
Column APP	1	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|0|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-2|-12
Exponent κ_D	79.518708319695345554341 34	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 293 r9c2_h+1v+1

Canonical identifier	r9c2_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	74	**Cell Inversion Orbit**	1,8;9,2
APP row	9	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	++0—++0—	**First route separation block**	—
Visible words w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|-6|2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 294 r9c2_h+1v-1

Canonical identifier	r9c2_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	74	**Cell Inversion Orbit**	1,8;9,2
APP row	9	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 295 r9c2_h-1v+1

Canonical identifier	r9c2_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	74	**Cell Inversion Orbit**	1,8;9,2
APP row	9	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.43380880300855130367114	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 296 r9c2_h-1v-1

Canonical identifier	r9c2_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	74	**Cell Inversion Orbit**	1,8;9,2
APP row	9	**First class return**	27
Column APP	2	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	40
Generational depth	G4	**Binary logarithm of ambiguity**	20.339850002884624
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|6|-2|-8|++0—++0—	**First cell separation block**	—
Dodecaphasic word	+++–0+++–0	**First route separation block**	—
Visible words w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|6|-2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 297 r9c3_h+1v+1

Canonical identifier	r9c3_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	75	**Cell Inversion Orbit**	1,7;9,3
APP row	9	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	30|-6|0|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	2
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|-6|2|-4|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 298 r9c3_h+1v-1

Canonical identifier	r9c3_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	75	**Cell Inversion Orbit**	1,7;9,3
APP row	9	**First class return**	18
Column APP	3	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	108
Generational depth	G4	**Binary logarithm of ambiguity**	46.52932501298079
Color orbit	—	**First orbit separation block**	—
Route family	30|-6|0|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	6
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|6|-2|4|-8
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858 10851337881		

Return to the domain, the signature, and the route operators.

Route 299 r9c3_h-1v+1

Canonical identifier	r9c3_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	75	**Cell Inversion Orbit**	1,7;9,3
APP row	9	**First class return**	9
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	108
Generational depth	G4	**Binary logarithm of ambiguity**	46.52932501298079
Color orbit	—	**First orbit separation block**	—
Route family	30|-6|0|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	6
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-6|2|-4|-8
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.999999999878581085133788 1		

Return to the domain, the signature, and the route operators.

Route 300 r9c3_h-1v-1

Canonical identifier	r9c3_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	75	**Cell Inversion Orbit**	1,7;9,3
APP row	9	**First class return**	27
Column APP	3	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	30|-6|0|-2|-8|0++—0++—	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|6|-2|4|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 301 r9c4_h+1v+1

Canonical identifier	r9c4_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	76	**Cell Inversion Orbit**	1,6;9,4
APP row	9	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.518708319695345554344134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 302 r9c4_h+1v-1

Canonical identifier	r9c4_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	76	**Cell Inversion Orbit**	1,6;9,4
APP row	9	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	3
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 303 r9c4_h-1v+1

Canonical identifier	r9c4_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	76	**Cell Inversion Orbit**	1,6;9,4
APP row	9	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255	**Final Ambiguity**	1
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	8|-2|0|0|-12
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 304 r9c4_h-1v-1

Canonical identifier	r9c4_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	76	**Cell Inversion Orbit**	1,6;9,4
APP row	9	**First class return**	27
Column APP	4	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-12|0|-12|+++—+++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	34|0|0|0|-12
Exponent κ_D	79.518708319695345554343134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 305 r9c5_h+1v+1

Canonical identifier	r9c5_h+1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	77	**Cell Inversion Orbit**	1,5;9,5
APP row	9	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-8|-2|-8|0++—0++—	**First cell separation block**	1
Dodecaphasic word	0++—0++—	**First route separation block**	1
Visible words w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 306 r9c5_h+1v-1

Canonical identifier	r9c5_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	77	**Cell Inversion Orbit**	1,5;9,5
APP row	9	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-8|-2|-8|0++—0++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|-2|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 307 r9c5_h-1v+1

Canonical identifier	r9c5_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	77	**Cell Inversion Orbit**	1,5;9,5
APP row	9	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	34|0|-8|-2|-8|0++—0++—	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	1
Visible words w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672	**Final Ambiguity**	1
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	44|2|0|0|-12
Exponent κ_D	39.43380880300855130367114	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 308 r9c5_h-1v-1

Canonical identifier	r9c5_h-1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	77	**Cell Inversion Orbit**	1,5;9,5
APP row	9	**First class return**	27
Column APP	5	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	30
Generational depth	G4	**Binary logarithm of ambiguity**	8.0
Color orbit	—	**First orbit separation block**	5
Route family	34|0|-8|-2|-8|0++—0++—	**First cell separation block**	5
Dodecaphasic word	+++0–+++0–	**First route separation block**	5
Visible words w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327	**Final Ambiguity**	1
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	34|0|-2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 309 r9c6_h+1v+1

Canonical identifier	r9c6_h+1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	78	**Cell Inversion Orbit**	1,4;9,6
APP row	9	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	26
Generational depth	G4	**Binary logarithm of ambiguity**	6.339850002884624
Color orbit	—	**First orbit separation block**	5
Route family	30|6|0|0|-8|+++-0-+++-0-	**First cell separation block**	5
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	5
Visible words w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	2
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|6|-2|0|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 310 r9c6_h+1v-1

Canonical identifier	r9c6_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	78	**Cell Inversion Orbit**	1,4;9,6
APP row	9	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-8|+++-0-+++-0-	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-6|0|0|-12
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858 10851337881		

Return to the domain, the signature, and the route operators.

Route 311 r9c6_h-1v+1

Canonical identifier	r9c6_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	78	**Cell Inversion Orbit**	1,4;9,6
APP row	9	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-8|+++-0-+++-0-	**First cell separation block**	1
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	2
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|6|0|0|-12
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 312 r9c6_h-1v-1

Canonical identifier	r9c6_h-1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	78	**Cell Inversion Orbit**	1,4;9,6
APP row	9	**First class return**	27
Column APP	6	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	18
Generational depth	G4	**Binary logarithm of ambiguity**	0.0
Color orbit	—	**First orbit separation block**	1
Route family	30|6|0|0|-8|+++-0-+++-0-	**First cell separation block**	1
Dodecaphasic word	+0+—+0+—	**First route separation block**	1
Visible words w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657	**Final Ambiguity**	1
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	30|-6|2|0|-8
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.60649894337213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 313 r9c7_h+1v+1

Canonical identifier	r9c7_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	79	**Cell Inversion Orbit**	1,3;9,7
APP row	9	**First class return**	9
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	243 657 24 243 657 570 264 90 570 243 657 24 243 657 570 264 90 570	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|6|0|4|-12
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 314 r9c7_h+1v-1

Canonical identifier	r9c7_h+1v-1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	79	**Cell Inversion Orbit**	1,3;9,7
APP row	9	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	0++—0++—	**First route separation block**	—
Visible words w_6	252 333 255 252 333 420 258 414 420 252 333 255 252 333 420 258 414 420	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	3
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	3
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	16
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|-4|2|-2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943373213544641686	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 12 14 14 16 16
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 315 r9c7_h-1v+1

Canonical identifier	r9c7_h-1v+1	**Factor R_{act}, reading Ω**	1.000000038550972354141645
Cell	79	**Cell Inversion Orbit**	1,3;9,7
APP row	9	**First class return**	27
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++0–+++0–	**First route separation block**	—
Visible words w_6	258 414 420 258 414 255 252 333 255 258 414 420 258 414 255 252 333 255	**Final Ambiguity**	3
D_1	84	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	84	**Total carry debt**	2
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	56	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,84,30)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(56,54,5)	**Integer signature in $\mathbb{Z}^5$**	50|4|-2|2|-8
Exponent κ_D	39.39035827686092491784959	**Orbit on the persistence boundary**	no
Exponent κ_Ω	55.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 11 12 14 15 16 17
Factor R_{act}, reading D	1.000000027308617396708705		

Return to the domain, the signature, and the route operators.

Route 316 r9c7_h-1v-1

Canonical identifier	r9c7_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	79	**Cell Inversion Orbit**	1,3;9,7
APP row	9	**First class return**	18
Column APP	7	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	54
Generational depth	G4	**Binary logarithm of ambiguity**	28.5293250129808
Color orbit	—	**First orbit separation block**	—
Route family	28|6|-2|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	264 90 570 264 90 24 243 657 24 264 90 570 264 90 24 243 657 24	**Final Ambiguity**	3
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	2
D_{27}	48	**Final carry debt**	1
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	28|-6|0|-4|-12
Exponent κ_D	79.518708319695345554341 34	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 16
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 317 r9c8_h+1v+1

Canonical identifier	r9c8_h+1v+1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	80	**Cell Inversion Orbit**	1,2;9,8
APP row	9	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|4|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+0+—+0+—	**First route separation block**	—
Visible words w_6	498 99 327 498 99 15 486 423 15 498 99 327 498 99 15 486 423 15	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	0
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|-6|2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 318 r9c8_h+1v-1

Canonical identifier	r9c8_h+1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	80	**Cell Inversion Orbit**	1,2;9,8
APP row	9	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|4|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	510 666 672 510 666 507 504 585 507 510 666 672 510 666 507 504 585 507	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 319 r9c8_h-1v+1

Canonical identifier	r9c8_h-1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	80	**Cell Inversion Orbit**	1,2;9,8
APP row	9	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	36
Generational depth	G4	**Binary logarithm of ambiguity**	18.0
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|4|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	504 585 507 504 585 672 510 666 672 504 585 507 504 585 672 510 666 672	**Final Ambiguity**	2
D_1	78	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	78	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,78,27)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	14|-4|0|0|-12
Exponent κ_D	39.433808803008551303671 14	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	79.606610139794827586470 91	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 14 16 17
Factor R_{act}, reading D	1.000000027338740854894358		

Return to the domain, the signature, and the route operators.

Route 320 r9c8_h-1v-1

Canonical identifier	r9c8_h-1v-1	**Factor R_{act}, reading Ω**	1.000000033004718053243084
Cell	80	**Cell Inversion Orbit**	1,2;9,8
APP row	9	**First class return**	27
Column APP	8	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	neutrino_G4	**Kinematic return extended to 216 phases**	216
Physical family	neutrino family	**Ambiguity area of the prefix**	40
Generational depth	G4	**Binary logarithm of ambiguity**	20.339850002884624
Color orbit	—	**First orbit separation block**	—
Route family	28|-6|4|0|-8|+0+—+0+—	**First cell separation block**	—
Dodecaphasic word	+++-0-+++-0-	**First route separation block**	—
Visible words w_6	486 423 15 486 423 327 498 99 327 486 423 15 486 423 327 498 99 327	**Final Ambiguity**	2
D_1	66	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	66	**Total carry debt**	1
D_{27}	48	**Final carry debt**	0
D_{36}	32	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	48	**Stabilized Decimal Triples**	17
P_6	5	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(80,66,3)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(48,54,5)	**Integer signature in $\mathbb{Z}^5$**	28|6|-2|0|-8
Exponent κ_D	79.51870831969534555434134	**Orbit on the persistence boundary**	yes
Exponent κ_Ω	47.606498943372135446416 86	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 10 12 13 14 15 16 17
Factor R_{act}, reading D	1.000000055128871799705072		

Return to the domain, the signature, and the route operators.

Route 321 r9c9_h+1v+1

Canonical identifier	r9c9_h+1v+1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	81	**Cell Inversion Orbit**	1,1;9,9
APP row	9	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	24 243 657 24 243 423 15 486 423 24 243 657 24 243 423 15 486 423	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	2
D_{27}	24	**Final carry debt**	1
D_{36}	16	**Delayed carry blocks**	2
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	16
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 13 15 16 16
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Route 322 r9c9_h+1v-1

Canonical identifier	r9c9_h+1v-1	**Factor R_{act}, reading Ω**	1.0000000163657240637534 03
Cell	81	**Cell Inversion Orbit**	1,1;9,9
APP row	9	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	108
Generational depth	G4	**Binary logarithm of ambiguity**	46.52932501298079
Color orbit	—	**First orbit separation block**	—
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	0+00-00+00-0	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	6
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	24	**Stabilized Decimal Triples**	17
P_6	2	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(24,54,2)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	no
Exponent κ_Ω	23.60616535410405902625471	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 323 r9c9_h-1v+1

Canonical identifier	r9c9_h-1v+1	**Factor R_{act}, reading Ω**	1.000000016365724063753403
Cell	81	**Cell Inversion Orbit**	1,1;9,9
APP row	9	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	108
Generational depth	G4	**Binary logarithm of ambiguity**	46.52932501298079
Color orbit	—	**First orbit separation block**	—
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	0+00-00+00-0	**First route separation block**	—
Visible words w_6	3 0 81 3 0 81 3 0 81 3 0 81 3 0 81 3 0 81	**Final Ambiguity**	6
D_1	24	**Rotations**	71
D_3	24	**Sheet changes**	107
D_4	24	**Total carry debt**	0
D_{27}	0	**Final carry debt**	0
D_{36}	0	**Delayed carry blocks**	0
Channels $\Omega_{90/120}$	24	**Stabilized Decimal Triples**	17
P_6	2	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(0,24,0)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(24,54,2)	**Integer signature in $\mathbb{Z}^5$**	20|0|0|0|-4
Exponent κ_D	-0.17513646166281122393484 25	**Orbit on the persistence boundary**	no
Exponent κ_Ω	23.60616535410405902625471	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17
Factor R_{act}, reading D	0.99999999987858108513378 81		

Return to the domain, the signature, and the route operators.

Route 324 r9c9_h-1v-1

Canonical identifier	r9c9_h-1v-1	**Factor R_{act}, reading Ω**	1.000000055189812531859140
Cell	81	**Cell Inversion Orbit**	1,1;9,9
APP row	9	**First class return**	27
Column APP	9	**First return of orbit**	27
Horizontal orientation	-1	**First cell return**	27
Vertical orientation	-1	**Kinematic return of the calendar of 108 phases**	54
Multisection	charged_lepton_G4	**Kinematic return extended to 216 phases**	216
Physical family	charged_lepton	**Ambiguity area of the prefix**	58
Generational depth	G4	**Binary logarithm of ambiguity**	30.189475010096174
Color orbit	—	**First orbit separation block**	—
Route family	24|0|12|0|-12|+++—+++—	**First cell separation block**	—
Dodecaphasic word	+++—+++—	**First route separation block**	—
Visible words w_6	15 486 423 15 486 657 24 243 657 15 486 423 15 486 657 24 243 657	**Final Ambiguity**	3
D_1	54	**Rotations**	71
D_3	60	**Sheet changes**	107
D_4	72	**Total carry debt**	1
D_{27}	24	**Final carry debt**	0
D_{36}	16	**Delayed carry blocks**	1
Channels $\Omega_{90/120}$	80	**Stabilized Decimal Triples**	17
P_6	6	**Coincidence of w_6 between K and $K+9$**	9
Reader K_D	(40,54,6)	**Memory changes between K and $K+9$**	9
Reader K_Ω	(80,54,6)	**Integer signature in $\mathbb{Z}^5$**	24|0|0|0|-12
Exponent κ_D	39.60661013979482758647091	**Orbit on the persistence boundary**	no
Exponent κ_Ω	79.60661013979482758647091	**Block Stabilization Chronology**	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 15 17
Factor R_{act}, reading D	1.000000027458540873559518		

Return to the domain, the signature, and the route operators.

Appendix O

Exhaustive inventory of 471 masses and 384 widths: identification, provenance, and correspondence with the atlas

Return to the scientific portal of comparability and metrology.

O.1 Exhaustive inventory domain: 471 masses, 384 widths, and object encoding

The inventory distinguishes the HMT–MD generative domain from the external catalogue of observables. The former is organized by

$$\mathcal{C}_{81} \longrightarrow \mathcal{R}_{56} \longrightarrow \Sigma_{13} \longrightarrow \Gamma_{108}^{\mathrm{enr}} \longrightarrow \mathcal{L}_{108} \longrightarrow \mathbb{Z}^5,$$

and continues with the evaluation of signatures and their typed realizations. The second contains 1170 identities grouped into 438 groups and 855 summary-table observables, with the exact decomposition

$$855 = 471_{\mathrm{masa}} + 384_{\mathrm{anchura}}. \tag{I1}$$

Each observable appears exactly once within its owning group and retains the charge and antiparticle identities linked to that group. These cardinalities do not count HMT–MD particles or select routes, signatures, towers, or couplings.

Construction, evaluation, and comparison

The HMT–MD construction first produces the physical object as a simple section, multisection, or composite register. It then extracts its signature and applies the corresponding reader. Only then is metrological comparison opened. Mass, width, and catalogue name are distinct objects: mass evaluates a persistent register; width reads duration and decay; the external name identifies the comparison register.

O.2 Data provenance, record uniqueness, and validity domains

The external reference is Particle Data Group 2026, cutoff January 15, 2026, consulted July 26, 2026, and distributed under the CC BY 4.0 license. The 1170 identities remain separate, while mass and width properties are deduplicated by group. The 855 observables cover 403 of the 438 groups; the remaining 35 retain catalogue identity, but do not receive a numerical row not supplied by the source.

The cut declares neither scheme nor scale for any of the 855 observables. No pole mass, $\overline{\mathrm{MS}}$ mass, renormalization scale, or width convention is inferred from it. The group identifier, property identifier, uncertainty type, value type and bound, confidence level where one exists, and the active or historical status of already established comparisons are preserved. In every case, the external value is used only after the internal construction has been frozen, and never as a selector.

O.3 Identification of objects and application of the HMT–MD reader

The structural assignment of each row identifies the domain on which its reader must act:

$$\begin{array}{c|l} \mathsf{E} & \text{simple section or elementary representation,} \\ \mathsf{N} & \text{null or gauge sector,} \\ \mathsf{Q} & \text{quark multisection with generation, orientation and colour,} \\ \mathsf{M} & \text{mesonic or bosonic exotic composite record,} \\ \mathsf{B} & \text{baryonic or fermionic exotic composite record,} \\ \mathsf{G} & \text{external gravitational record, not an elementary HMT graviton.} \end{array} \tag{I2}$$

The subscript m refers to the mass character; the subscript Γ, to the width. The active and historical statuses qualify comparisons already established and do not alter the class of the object.

Table O.1: Complete partition of the cataloged observables.

Catalogue family	Groups with observable	Masses	Widths
General case	46	53	9
Quark sector	6	8	1
Mesons	201	243	225
Baryons	149	166	149
External gravitational record	1	1	0
Grand total	403	471	384

The gravitational row reproduces an external record and does not introduce an elementary graviton into the HMT–MD construction. Its realization requires the typed gravitational-representation map declared in the row itself.

O.4 Inventory coverage and semantics of the tables

None of the 471 masses or the 384 widths is reduced to a sample. Each row publishes exactly once its catalogue identity, species, value, unit, uncertainty, limit, confidence, HMT–MD sector, and required realization map. The absence of a closed individual link is declared as such and is not replaced by an invented correspondence.

O.5 Comprehensive inventory of evaluated masses and correspondence with the atlas realizations

Table O.2: Complete inventory of 471 masses: identifier, observable, convention, and HMT–MD realization.

Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B000 B000M prop. 349891	**Sigma(1580) MASS VALUE** Sigma(1580)- | Sigmabar(1580)+ | Sigma(1580)0 | Sigmabar(1580)0 | Sigma(1580)+ | Sigmabar(1580)-	~1580 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B004 B004M property: 348705	**N(1895) BREIT-WIGNER MASS** N(1895)0 | Nbar(1895)0 | N(1895)+ | Nbar(1895)-	1870 to 1920 (~1895) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B005 B005M property: 348836	**N(2060) BREIT-WIGNER MASS** N(2060)0 | Nbar(2060)0 | N(2060)+ | Nbar(2060)-	2030 to 2200 (~2100) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B007 B007M prop. 349780	**Lambda(2585) MASS (BUMPS):** Lambda(2585)0 | Lambdabar(2585)0	~2585 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B010 B010M property: 349097	**Delta(1700) Breit–Wigner resonance resonance mass** Delta(1700)- | Deltabar(1700)+ | Delta(1700)0 | Deltabar(1700)0 | Delta(1700)+ | Deltabar(1700)- | Delta(1700)++ | Deltabar(1700)–	1690 to 1730 (~1710) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B011 B011M property: 349151	**Delta(1905) Breit–Wigner resonance resonance mass** Delta(1905)- | Deltabar(1905)+ | Delta(1905)0 | Deltabar(1905)0 | Delta(1905)+ | Deltabar(1905)- | Delta(1905)++ | Deltabar(1905)–	1855 to 1910 (~1880) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B012 B012M prop. 349182	**Delta(1910) Breit–Wigner resonance resonance mass** Delta(1910)- | Deltabar(1910)+ | Delta(1910)0 | Deltabar(1910)0 | Delta(1910)+ | Deltabar(1910)- | Delta(1910)++ | Deltabar(1910)–	1850 to 1950 (~1900) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B013 B013M property: 349249	**Delta(1930) Breit–Wigner resonance resonance mass** Delta(1930)- | Deltabar(1930)+ | Delta(1930)0 | Deltabar(1930)0 | Delta(1930)+ | Deltabar(1930)- | Delta(1930)++ | Deltabar(1930)–	1900 to 2000 (~1950) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B014 B014M property: 348535	**N(1710) BREIT-WIGNER MASS** N(1710)0 | Nbar(1710)0 | N(1710)+ | Nbar(1710)-	1680 to 1740 (~1710) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B015 B015M prop. 348569	**N(1720) BREIT–WIGNER MASS:** N(1720)0 | Nbar(1720)0 | N(1720)+ | Nbar(1720)-	1680 to 1750 (~1720) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B016 B016M property: 348633	**N(1875) BREIT-WIGNER MASS** N(1875)0 | Nbar(1875)0 | N(1875)+ | Nbar(1875)-	1850 to 1920 (~1875) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B017 B017M property: 348778	**N(1990) BREIT–WIGNER MASS:** N(1990)0 | Nbar(1990)0 | N(1990)+ | Nbar(1990)-	1950 to 2100 (~2020) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B018 B018M prop. 348499	**N(1700) BREIT–WIGNER MASS:** N(1700)0 | Nbar(1700)0 | N(1700)+ | Nbar(1700)-	1650 to 1800 (~1720) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

Continuation of the inventory

Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B019 B019M property: 349050	**Delta(1600) Breit–Wigner resonance resonance mass** Delta(1600)- | Deltabar(1600)+ | Delta(1600)0 | Deltabar(1600)0 | Delta(1600)+ | Deltabar(1600)- | Delta(1600)++ | Deltabar(1600)–	1500 to 1640 (~1570) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B021 B021M property: 350410	**Xi(1620) MASS** Xi(1620)- | Xibar(1620)+ | Xi(1620)0 | Xibar(1620)0	~1620 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B022 B022M prop. 350457	**Xi(2250) MASS:** Xi(2250)- | Xibar(2250)+ | Xi(2250)0 | Xibar(2250)0	~2250 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B024 B024M property: 348906	**N(2120) BREIT-WIGNER MASS** N(2120)0 | Nbar(2120)0 | N(2120)+ | Nbar(2120)-	2060 to 2160 (~2120) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B026 B026M property: 350202	**Sigma(2100) resonance mass** Sigma(2100)- | Sigmabar(2100)+ | Sigma(2100)0 | Sigmabar(2100)0 | Sigma(2100)+ | Sigmabar(2100)-	~2100 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B027 B027M prop. 349731	**Lambda(2085) MASS:** Lambda(2085)0 | Lambdabar(2085)0	~2020 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B030 B030M property: 349126	**Delta(1900) Breit–Wigner resonance resonance mass** Delta(1900)- | Deltabar(1900)+ | Delta(1900)0 | Deltabar(1900)0 | Delta(1900)+ | Deltabar(1900)- | Delta(1900)++ | Deltabar(1900)–	1840 to 1920 (~1860) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B032 B032M property: 349902	**Sigma(1620) resonance mass** Sigma(1620)- | Sigmabar(1620)+ | Sigma(1620)0 | Sigmabar(1620)0 | Sigma(1620)+ | Sigmabar(1620)-	1600 to 1650 (~1620) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B033 B033M prop. 349034	**Mixed-charge states** Delta(1232)- | Deltabar(1232)+ | Delta(1232)0 | Deltabar(1232)0 | Delta(1232)+ | Deltabar(1232)- | Delta(1232)++ | Deltabar(1232)–	1230 to 1234 (~1232) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B034 B034M property: 350186	**Sigma(2070) MASS** Sigma(2070)- | Sigmabar(2070)+ | Sigma(2070)0 | Sigmabar(2070)0 | Sigma(2070)+ | Sigmabar(2070)-	2020 to 2100 (~2060) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B035 B035M property: 349763	**Lambda(2110) MASS:** Lambda(2110)0 | Lambdabar(2110)0	2050 to 2130 (~2090) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B036 B036M prop. 349555	**Lambda(1800) MASS:** Lambda(1800)0 | Lambdabar(1800)0	1750 to 1850 (~1800) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B037 B037M1 property: 349427	**PRODUCTION EXPERIMENTS:** Lambda(1405)0 | Lambdabar(1405)0	1405.1+1.3-1.0 MeV	asymmetric; asymmetric: +1.3119693539575821/-0.96455487987763788; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B038 B038M property: 349437	**Lambda(1520) MASS:** Lambda(1520)0 | Lambdabar(1520)0	1518 to 1520 (~1519) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B039 B039M prop. 349594	**Lambda(1820) MASS:** Lambda(1820)0 | Lambdabar(1820)0	1815 to 1825 (~1820) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

Continuation of the inventory

Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B040 B040M property: 349494	**Lambda(1670) MASS** Lambda(1670)0 | Lambdabar(1670)0	1670 to 1678 (~1674) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B041 B041M property: 349741	**Lambda(2100) MASS:** Lambda(2100)0 | Lambdabar(2100)0	2090 to 2110 (~2100) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B042 B042M prop. 349775	**Lambda(2350) MASS:** Lambda(2350)0 | Lambdabar(2350)0	2340 to 2370 (~2350) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B043 B043M+ property: 349876	**Sigma(1385)+ MASS** Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)-	1382.83+-0.34 MeV	asymmetric; asymmetric: +0.33905435517601018/-0.34419099315609542; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B043 B043M0 property: 349877	**Sigma(1385)0 resonance mass** Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)-	1383.8+-0.9 MeV	symmetric; symmetric: ±0.9351440440323722 8; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B043 B043M- prop. 349878	**Sigma(1385)- MASS VALUE** Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)-	1387.2+-0.5 MeV	symmetric; symmetric: ±0.5465370328051361; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B044 B044M property: 349946	**Sigma(1670) MASS** Sigma(1670)- | Sigmabar(1670)+ | Sigma(1670)0 | Sigmabar(1670)0 | Sigma(1670)+ | Sigmabar(1670)-	1665 to 1685 (~1675) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B045 B045M property: 349983	**Sigma(1775) resonance mass** Sigma(1775)- | Sigmabar(1775)+ | Sigma(1775)0 | Sigmabar(1775)0 | Sigma(1775)+ | Sigmabar(1775)-	1770 to 1780 (~1775) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B046 B046M prop. 350080	**Sigma(1915) MASS VALUE** Sigma(1915)- | Sigmabar(1915)+ | Sigma(1915)0 | Sigmabar(1915)0 | Sigma(1915)+ | Sigmabar(1915)-	1900 to 1935 (~1915) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B047 B047M property: 350166	**Sigma(2030) MASS** Sigma(2030)- | Sigmabar(2030)+ | Sigma(2030)0 | Sigmabar(2030)0 | Sigma(2030)+ | Sigmabar(2030)-	2025 to 2040 (~2030) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B048 B048M property: 350277	**Sigma(2250) resonance mass** Sigma(2250)- | Sigmabar(2250)+ | Sigma(2250)0 | Sigmabar(2250)0 | Sigma(2250)+ | Sigmabar(2250)-	2210 to 2280 (~2250) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B049 B049M0 prop. 350399	**Xi(1530)0 MASS:** Xi(1530)- | Xibar(1530)+ | Xi(1530)0 | Xibar(1530)0	1531.80+-0.32 MeV	symmetric; symmetric: ±0.31575272391181958; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B049 B049M- property: 350401	**Xi(1530)- MASS** Xi(1530)- | Xibar(1530)+ | Xi(1530)0 | Xibar(1530)0	1535.0+-0.6 MeV	symmetric; symmetric: ±0.63202166316087116; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B050 B050M property: 350420	**Xi(1820) MASS:** Xi(1820)- | Xibar(1820)+ | Xi(1820)0 | Xibar(1820)0	1823+-5 MeV	symmetric; symmetric: ±5; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B052 B052M prop. 350437	**Xi(1950) MASS:** Xi(1950)- | Xibar(1950)+ | Xi(1950)0 | Xibar(1950)0	1950+-15 MeV	symmetric; symmetric: ±15; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B053 B053M property: 350283	**Sigma(2455) MASS** Sigma(2455)- | Sigmabar(2455)+ | Sigma(2455)0 | Sigmabar(2455)0 | Sigma(2455)+ | Sigmabar(2455)-	~2455 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B054 B054M property: 350284	**Sigma(2620) resonance mass** Sigma(2620)- | Sigmabar(2620)+ | Sigma(2620)0 | Sigmabar(2620)0 | Sigma(2620)+ | Sigmabar(2620)-	~2620 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B055 B055M prop. 349518	**Lambda(1690) MASS:** Lambda(1690)0 | Lambdabar(1690)0	1685 to 1695 (~1690) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B056 B056M property: 349614	**Lambda(1830) MASS** Lambda(1830)0 | Lambdabar(1830)0	1820 to 1830 (~1825) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B057 B057M property: 349966	**Sigma(1750) resonance mass** Sigma(1750)- | Sigmabar(1750)+ | Sigma(1750)0 | Sigmabar(1750)0 | Sigma(1750)+ | Sigmabar(1750)-	1700 to 1800 (~1750) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B059 B059M prop. 350285	**Sigma(3000) MASS VALUE** Sigma(3000)- | Sigmabar(3000)+ | Sigma(3000)0 | Sigmabar(3000)0 | Sigma(3000)+ | Sigmabar(3000)-	~3000 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B060 B060M property: 349635	**Lambda(1890) MASS** Lambda(1890)0 | Lambdabar(1890)0	1870 to 1910 (~1890) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B061 B061M property: 348326	**N(1440) BREIT-WIGNER MASS** N(1440)0 | Nbar(1440)0 | N(1440)+ | Nbar(1440)-	1410 to 1470 (~1440) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B062 B062M prop. 348349	**N(1520) BREIT–WIGNER MASS:** N(1520)0 | Nbar(1520)0 | N(1520)+ | Nbar(1520)-	1510 to 1520 (~1515) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B063 B063M property: 348383	**N(1535) BREIT-WIGNER MASS** N(1535)0 | Nbar(1535)0 | N(1535)+ | Nbar(1535)-	1515 to 1545 (~1530) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B064 B064M property: 348435	**N(1675) BREIT–WIGNER MASS:** N(1675)0 | Nbar(1675)0 | N(1675)+ | Nbar(1675)-	1665 to 1680 (~1675) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B065 B065M prop. 348467	**N(1680) BREIT–WIGNER MASS:** N(1680)0 | Nbar(1680)0 | N(1680)+ | Nbar(1680)-	1680 to 1690 (~1685) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B066 B066M property: 348410	**N(1650) BREIT-WIGNER MASS** N(1650)0 | Nbar(1650)0 | N(1650)+ | Nbar(1650)-	1635 to 1665 (~1650) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B067 B067M property: 349999	**Sigma(1880) resonance mass** Sigma(1880)- | Sigmabar(1880)+ | Sigma(1880)0 | Sigmabar(1880)0 | Sigma(1880)+ | Sigmabar(1880)-	1820 to 1940 (~1880) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B068 B068M prop. 350443	**Xi(2030) MASS:** Xi(2030)- | Xibar(2030)+ | Xi(2030)0 | Xibar(2030)0	2025+-5 MeV	symmetric; symmetric: ±5; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B071 B071M property: 348953	**N(2190) BREIT-WIGNER MASS** N(2190)0 | Nbar(2190)0 | N(2190)+ | Nbar(2190)-	2140 to 2220 (~2180) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B077 B077M property: 349576	**Lambda(1810) MASS:** Lambda(1810)0 | Lambdabar(1810)0	1740 to 1840 (~1790) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B079 B079M prop. 349920	**Sigma(1660) MASS VALUE** Sigma(1660)- | Sigmabar(1660)+ | Sigma(1660)0 | Sigmabar(1660)0 | Sigma(1660)+ | Sigmabar(1660)-	1640 to 1680 (~1660) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B082 B082M property: 349078	**Delta(1620) Breit–Wigner resonance resonance mass** Delta(1620)- | Deltabar(1620)+ | Delta(1620)0 | Deltabar(1620)0 | Delta(1620)+ | Deltabar(1620)- | Delta(1620)++ | Deltabar(1620)–	1590 to 1630 (~1610) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B083 B083M property: 349308	**Delta(1950) Breit–Wigner resonance resonance mass** Delta(1950)- | Deltabar(1950)+ | Delta(1950)0 | Deltabar(1950)0 | Delta(1950)+ | Deltabar(1950)- | Delta(1950)++ | Deltabar(1950)–	1915 to 1950 (~1930) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B084 B084M prop. 349359	**Delta(2420) Breit–Wigner resonance resonance mass** Delta(2420)- | Deltabar(2420)+ | Delta(2420)0 | Deltabar(2420)0 | Delta(2420)+ | Deltabar(2420)- | Delta(2420)++ | Deltabar(2420)–	2300 to 2600 (~2450) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B087 B087M property: 348671	**N(1880) BREIT-WIGNER MASS** N(1880)0 | Nbar(1880)0 | N(1880)+ | Nbar(1880)-	1830 to 1930 (~1880) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B088 B088M property: 350188	**Sigma(2080) resonance mass** Sigma(2080)- | Sigmabar(2080)+ | Sigma(2080)0 | Sigmabar(2080)0 | Sigma(2080)+ | Sigmabar(2080)-	2060 to 2120 (~2090) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B089 B089M prop. 349646	**Lambda(2000) MASS:** Lambda(2000)0 | Lambdabar(2000)0	~2000 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B090 B090M property: 348986	**N(2220) BREIT-WIGNER MASS** N(2220)0 | Nbar(2220)0 | N(2220)+ | Nbar(2220)-	2200 to 2300 (~2250) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B098 B098M property: 350050	**Sigma(1910) resonance mass** Sigma(1910)- | Sigmabar(1910)+ | Sigma(1910)0 | Sigmabar(1910)0 | Sigma(1910)+ | Sigmabar(1910)-	1870 to 1950 (~1910) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B099 B099M prop. 350464	**Xi(2500) MASS:** Xi(2500)- | Xibar(2500)+ | Xi(2500)0 | Xibar(2500)0	~2500 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B101 B101M property: 349472	**Lambda(1600) MASS** Lambda(1600)0 | Lambdabar(1600)0	1570 to 1630 (~1600) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B102 B102M property: 350934	**Lambda_c(2625)+ MASS:** Lambda_c(2625)+ | Lambdabar_c(2625)-	2628.00+-0.15 MeV	symmetric; symmetric: ±0.14735968814977221; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B103 B103M prop. 350454	**Xi(2120) MASS:** Xi(2120)- | Xibar(2120)+ | Xi(2120)0 | Xibar(2120)0	~2120 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B104 B104M++ property: 350985	**Sigma_c(2455)++ resonance resonance mass** Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)–	2453.97+-0.14 MeV	symmetric; symmetric: ±0.14100978554010379; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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B104 B104M+ property: 350986	**Sigma_c(2455)+ resonance resonance mass** Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)–	2452.65+0.22-0.16 MeV	asymmetric; asymmetric: +0.2179325720612677/-0.15750625023659159; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B104 B104M0 prop. 350987	**Sigma_c(2455)0 resonance resonance mass** Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)–	2453.75+-0.14 MeV	symmetric; symmetric: ±0.14100937832457261; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B105 B105M property: 350412	**MIXED CHARGES** Xi(1690)- | Xibar(1690)+ | Xi(1690)0 | Xibar(1690)0	1690+-10 MeV	symmetric; symmetric: ±10; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B112 B112M property: 349774	**Lambda(2325) MASS:** Lambda(2325)0 | Lambdabar(2325)0	~2325 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B113 B113M prop. 349004	**N(2250) BREIT–WIGNER MASS:** N(2250)0 | Nbar(2250)0 | N(2250)+ | Nbar(2250)-	2250 to 2320 (~2280) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B115 B115M++ property: 351001	**Sigma_c(2520)++ resonance resonance mass** Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)–	2518.41+-0.22 MeV	symmetric; symmetric: ±0.22441420873039311; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B115 B115M+ property: 351002	**Sigma_c(2520)+ resonance resonance mass** Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)–	2517.4+0.7-0.5 MeV	asymmetric; asymmetric: +0.68505102093303338/-0.52419384686487569; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B115 B115M0 prop. 351003	**Sigma_c(2520)0 resonance resonance mass** Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)–	2518.48+-0.21 MeV	symmetric; symmetric: ±0.20795724066290969; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B117 B117M property: 349208	**Delta(1920) Breit–Wigner resonance resonance mass** Delta(1920)- | Deltabar(1920)+ | Delta(1920)0 | Deltabar(1920)0 | Delta(1920)+ | Deltabar(1920)- | Delta(1920)++ | Deltabar(1920)–	1870 to 1970 (~1920) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B118 B118M property: 350286	**Sigma(3170) resonance mass (PRODUCTION EXPERIMENTS)** Sigma(3170)- | Sigmabar(3170)+ | Sigma(3170)0 | Sigmabar(3170)0 | Sigma(3170)+ | Sigmabar(3170)-	~3170 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B119 B119M prop. 350917	**Lambda_c(2595)+ MASS:** Lambda_c(2595)+ | Lambdabar_c(2595)-	2592.25+-0.28 MeV	symmetric; symmetric: ±0.28142494558940517; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B120 B120M property: 349022	**N(2600) BREIT-WIGNER MASS** N(2600)0 | Nbar(2600)0 | N(2600)+ | Nbar(2600)-	2550 to 2750 (~2600) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B122 B122M property: 350979	**Lambda_c(2940)+ MASS** Lambda_c(2940)0 | Lambdabar_c(2940)0	2939.6+1.3-1.5 MeV	asymmetric; asymmetric: +1.261876684475213/-1.467184985071428; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B128 B128M prop. 349027	**N(~3000) BREIT-WIGNER MASS VALUE** N(~3000)0 | Nbar(~3000)0 | N(~3000)+ | Nbar(~3000)-	~3000 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B130 B130M+ property: 351271	**Xi_c(2970)+ MASS** Xi_c(2970)0 | Xibar_c(2970)0	2965.2+-1.0 MeV	symmetric; symmetric: ±1.0156810603859041; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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B130 B130M0 property: 351273	**Xi_c(2970)0 MASS:** Xi_c(2970)0 | Xibar_c(2970)0	2967.3+-1.9 MeV	symmetric; symmetric: ±1.850084394808527; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B131 B131M prop. 350458	**Xi(2370) MASS:** Xi(2370)- | Xibar(2370)+ | Xi(2370)0 | Xibar(2370)0	~2370 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B132 B132M property: 348871	**N(2100) BREIT-WIGNER MASS** N(2100)0 | Nbar(2100)0 | N(2100)+ | Nbar(2100)-	2050 to 2150 (~2100) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B135 B135M property: 349339	**Delta(2200) Breit–Wigner resonance resonance mass** Delta(2200)- | Deltabar(2200)+ | Delta(2200)0 | Deltabar(2200)0 | Delta(2200)+ | Deltabar(2200)- | Delta(2200)++ | Deltabar(2200)–	2150 to 2250 (~2200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B136 B136M prop. 349272	**Delta(1940) Breit–Wigner resonance resonance mass** Delta(1940)- | Deltabar(1940)+ | Delta(1940)0 | Deltabar(1940)0 | Delta(1940)+ | Deltabar(1940)- | Delta(1940)++ | Deltabar(1940)–	1940 to 2060 (~2000) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B141 B141M property: 350516	**Omega(2250)- MASS** Omega(2250)- | Omegabar(2250)+	2252+-9 MeV	symmetric; symmetric: ±8.8341044409654419; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B142 B142M property: 350520	**Omega(2380)- MASS:** Omega(2380)- | Omegabar(2380)+	~2380 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B143 B143M prop. 350523	**Omega(2470)- MASS:** Omega(2470)- | Omegabar(2470)+	2474+-12 MeV	symmetric; symmetric: ±12; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B144 B144M property: 348740	**N(1900) BREIT-WIGNER MASS** N(1900)0 | Nbar(1900)0 | N(1900)+ | Nbar(1900)-	1890 to 1950 (~1920) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B146 B146M+ property: 351204	**Xi_c(2645)+ resonance resonance mass** Xi_c(2645)0 | Xibar_c(2645)0 | Xi_c(2645)+ | Xibar_c(2645)-	2645.17+-0.27 MeV	asymmetric; asymmetric: +0.2628121352240531 1/-0.275224526580374; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B146 B146M0 prop. 351206	**Xi_c(2645)0 resonance resonance mass** Xi_c(2645)0 | Xibar_c(2645)0 | Xi_c(2645)+ | Xibar_c(2645)-	2646.24+-0.18 MeV	asymmetric; asymmetric: +0.17792728963027479/-0.1809798944958548; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B147 B147M+ property: 351307	**Xi_c(3080)+ MASS** Xi_c(3080)0 | Xibar_c(3080)0	3077.2+-0.4 MeV	symmetric; symmetric: ±0.38652749298648748; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B147 B147M0 property: 351308	**Xi_c(3080)0 MASS:** Xi_c(3080)0 | Xibar_c(3080)0	3079.9+-1.4 MeV	symmetric; symmetric: ±1.3603458501256609; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B148 B148M+ prop. 351232	**Xi_c(2815)+ resonance resonance mass** Xi_c(2815)0 | Xibar_c(2815)0 | Xi_c(2815)+ | Xibar_c(2815)-	2816.60+-0.16 MeV	asymmetric; asymmetric: +0.1551712678400561/-0.157672307586397; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B148 B148M0 property: 351234	**Xi_c(2815)0 resonance resonance mass** Xi_c(2815)0 | Xibar_c(2815)0 | Xi_c(2815)+ | Xibar_c(2815)-	2819.85+-0.27 MeV	asymmetric; asymmetric: +0.26809134636591098/-0.28025492882096131; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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B149 B149M+ property: 351217	**Xi_c(2790)+ resonance resonance mass** Xi_c(2790)0 | Xibar_c(2790)0 | Xi_c(2790)+ | Xibar_c(2790)-	2792.0+-0.5 MeV	asymmetric; asymmetric: +0.52973897333377062/-0.53585270924116923; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B149 B149M0 prop. 351218	**Xi_c(2790)0 resonance resonance mass** Xi_c(2790)0 | Xibar_c(2790)0 | Xi_c(2790)+ | Xibar_c(2790)-	2794.0+-0.5 MeV	symmetric; symmetric: ±0.49234423994343779; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B150 B150M property: 350951	**Lambda_c(2765)+ MASS** Lambda_c(2765)0 | Lambdabar_c(2765)0	2766.6+-2.4 MeV	symmetric; symmetric: ±2.404079865561874; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B151 B151M property: 350960	**Lambda_c(2880)+ MASS:** Lambda_c(2880)+ | Lambdabar_c(2880)-	2881.63+-0.24 MeV	asymmetric; asymmetric: +0.23613203643473499/-0.245558684143336; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B155 B155M++ prop. 351015	**Sigma_c(2800)++ MASS:** Sigma_c(2800)0 | Sigmabar_c(2800)0	2801+4-6 MeV	asymmetric; asymmetric: +4.4067675227993526/-5.7999655171389852; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B155 B155M+ property: 351016	**Sigma_c(2800)+ MASS** Sigma_c(2800)0 | Sigmabar_c(2800)0	2792+14-5 MeV	asymmetric; asymmetric: +13.690127829936l1/-5.01792785918669; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B155 B155M0 property: 351017	**Sigma_c(2800)0 MASS** Sigma_c(2800)0 | Sigmabar_c(2800)0	2806+5-7 MeV	asymmetric; asymmetric: +4.8590019921640533/-7.3871442628294739; type FC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B156 B156A03 prop. 351263	**Xi_c(2930)+ MASS:** Xi_c(2930)0 | Xibar_c(2930)0	2942+-5 MeV	symmetric; symmetric: ±4.6486557196677838; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B156 B156A04 property: 351264	**Xi_c(2930)0 MASS** Xi_c(2930)0 | Xibar_c(2930)0	2938.55+-0.30 MeV	symmetric; symmetric: ±0.30185805977630881; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B157 B157M+ property: 351297	**Xi_c(3055)+ MASS:** Xi_c(3055)0 | Xibar_c(3055)0	3055.2+-0.7 MeV	symmetric; symmetric: ±0.68999235826894623; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B157 B157M prop. 351298	**Xi_c(3055)0 MASS:** Xi_c(3055)0 | Xibar_c(3055)0	3061.0+-0.8 MeV	symmetric; symmetric: ±0.83240615086627034; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B158 B158M+ property: 351325	**Xi_c(3123)+ MASS** Xi_c(3123)0 | Xibar_c(3123)0	3122.9+-1.3 MeV	symmetric; symmetric: ±1.334166406412634; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B161 B161M property: 351844	**Xi_b(5945)0 MASS:** Xi_b(5945)0 | Xibar_b(5945)0	5952.3+-0.6 MeV	symmetric; symmetric: ±0.57765272773906229; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B162 B162M prop. 351694	**Lambda_b(5912)0 MASS:** Lambda_b(5912)0 | Lambdabar_b(5912)0	5912.16+-0.16 MeV	symmetric; symmetric: ±0.16007003100330261; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B163 B163M property: 351698	**Lambda_b(5920)0 MASS** Lambda_b(5920)0 | Lambdabar_b(5920)0	5920.07+-0.16 MeV	symmetric; symmetric: ±0.15862170338385659; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B164 B164M property: 349532	**Lambda(1710) MASS:** Lambda(1710)0 | Lambdabar(1710)0	1713+-13 MeV	symmetric; symmetric: ±13; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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B165 B165M prop. 349651	**Lambda(2050) MASS:** Lambda(2050)0 | Lambdabar(2050)0	2056+-22 MeV	symmetric; symmetric: ±22; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B166 B166M property: 349991	**Sigma(1780) MASS** Sigma(1780)- | Sigmabar(1780)+ | Sigma(1780)0 | Sigmabar(1780)0 | Sigma(1780)+ | Sigmabar(1780)-	1730 to 1830 (~1780) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B167 B167M property: 350019	**Sigma(1900) resonance mass** Sigma(1900)- | Sigmabar(1900)+ | Sigma(1900)0 | Sigmabar(1900)0 | Sigma(1900)+ | Sigmabar(1900)-	1900 to 1950 (~1925) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B168 B168M prop. 350102	**Sigma(1940) MASS VALUE** Sigma(1940)- | Sigmabar(1940)+ | Sigma(1940)0 | Sigmabar(1940)0 | Sigma(1940)+ | Sigmabar(1940)-	1920 to 1960 (~1940) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B169 B169M property: 351840	**Xi_b'(5935)- MASS** Xi_b^'(5935)- | Xibar_b^'(5935)+	5934.9+-0.4 MeV	symmetric; symmetric: ±0.4135242970782354; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B170 B170M property: 351848	**Xi_b(5955)- MASS:** Xi_b(5955)- | Xibar_b(5955)+	5955.5+-0.4 MeV	symmetric; symmetric: ±0.41457381951292599; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B171 B171M prop. 351951	**P_c_cbar(4380)+ MASS:** P_c_cbar(4380)+	4380+-30 MeV	symmetric; symmetric: ±30.083217912982651; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B172 B172M property: 351977	**P_c_cbar(4457)+ MASS** P_c_cbar(4457)+	4457.3+4.0-1.8 MeV	asymmetric; asymmetric: +4.1436698710201316/-1.802775637731995; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B173 B173M property: 351376	**Omega_c(3000)0 MASS** Omega_c(3000)0 | Omegabar_c(3000)0	3000.46+-0.25 MeV	asymmetric; asymmetric: +0.2431765836192144/-0.26399637894382499; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B174 B174M prop. 351380	**Omega_c(3050)0 MASS:** Omega_c(3050)0 | Omegabar_c(3050)0	3050.17+-0.19 MeV	asymmetric; asymmetric: +0.18820356358563539/-0.19089950264325409; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B175 B175M property: 351384	**Omega_c(3065)0 MASS** Omega_c(3065)0 | Omegabar_c(3065)0	3065.58+-0.21 MeV	asymmetric; asymmetric: +0.21345733449248469/-0.2139147501137518; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B176 B176M property: 351388	**Omega_c(3090)0 MASS:** Omega_c(3090)0 | Omegabar_c(3090)0	3090.15+-0.26 MeV	asymmetric; asymmetric: +0.25239209275859231/-0.26305043429981828; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B177 B177M prop. 351392	**Omega_c(3120)0 MASS:** Omega_c(3120)0 | Omegabar_c(3120)0	3118.98+0.27-0.35 MeV	asymmetric; asymmetric: +0.27459060435491961/-0.34669871646719441; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B178 B178M property: 350956	**Lambda_c(2860)+ MASS** Lambda_c(2860)0 | Lambdabar_c(2860)0	2856.1+2.3-6.0 MeV	asymmetric; asymmetric: +2.336664289109585/-5.873670062235365; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B179 B179M property: 350506	**Omega(2012)- MASS:** Omega(2012)- | Omegabar(2012)+	2012.9+-0.4 MeV	symmetric; symmetric: ±0.44536259664740679; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B180 B180M prop. 351864	**Xi_b(6227)- MASS:** Xi_b(6227)- | Xi_b(6227)+	6227.9+-0.9 MeV	symmetric; symmetric: ±0.90815128746094542; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B181 B181M property: 351734	**Sigma_b(6097)+ MASS** Sigma_b(6097)+ | Sigmabar_b(6097)-	6095.8+-1.7 MeV	symmetric; symmetric: ±1.7464249196572981; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B182 B182M property: 351738	**Sigma_b(6097)- MASS:** Sigma_b(6097)- | Sigmabar_b(6097)+	6098.0+-1.8 MeV	symmetric; symmetric: ±1.772004514666935; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B183 B183M prop. 350228	**Sigma(2110) MASS VALUE** Sigma(2110)- | Sigmabar(2110)+ | Sigma(2110)0 | Sigmabar(2110)0 | Sigma(2110)+ | Sigmabar(2110)-	2110+-50 MeV	symmetric; symmetric: ±51.991387525520771; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B184 B184M property: 350253	**Sigma(2230) MASS** Sigma(2230)- | Sigmabar(2230)+ | Sigma(2230)0 | Sigmabar(2230)0 | Sigma(2230)+ | Sigmabar(2230)-	2240+-27 MeV	symmetric; symmetric: ±27; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B185 B185M property: 351935	**P_c_cbar(4312)+ MASS:** P_c_cbar(4312)+	4311.9+7.0-0.9 MeV	asymmetric; asymmetric: +6.8359344642850397/-0.92195444572928864; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B186 B186M prop. 351955	**P_c_cbar(4440)+ MASS:** P_c_cbar(4440)+	4440+4-5 MeV	asymmetric; asymmetric: +4.3011626335213133/-4.876474136094644; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B187 B187M property: 351706	**Lambda_b(6146)0 MASS** Lambda_b(6146)0 | Lambdabar_b(6146)0	6146.2+-0.4 MeV	symmetric; symmetric: ±0.41759268734261917; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B188 B188M property: 351711	**Lambda_b(6152)0 MASS:** Lambda_b(6152)0 | Lambdabar_b(6152)0	6152.5+-0.4 MeV	symmetric; symmetric: ±0.35727006911778508; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B190 B190M prop. 349676	**Lambda(2070) MASS:** Lambda(2070)0 | Lambdabar(2070)0	2070+-24 MeV	symmetric; symmetric: ±24; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B191 B191M property: 349711	**Lambda(2080) MASS** Lambda(2080)0 | Lambdabar(2080)0	2082+-13 MeV	symmetric; symmetric: ±13; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B192 B192M0 property: 351255	**Xi_c(2923)0 MASS:** Xi_c(2923)0 | Xibar_c(2923)0	2923.2+-0.4 MeV	symmetric; symmetric: ±0.39743785839914991; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B192 B192MP prop. 351256	**Xi_c(2923)+ MASS:** Xi_c(2923)0 | Xibar_c(2923)0	2922.8+-0.6 MeV	symmetric; symmetric: ±0.61644140029689765; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B193 B193M property: 351905	**Omega_b(6316)- MASS** Omega_b(6316)- | Omegabar_b(6316)+	6315.6+-0.6 MeV	symmetric; symmetric: ±0.58048724929847628; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B194 B194M property: 351909	**Omega_b(6330)- MASS:** Omega_b(6330)- | Omegabar_b(6330)+	6330.3+-0.6 MeV	symmetric; symmetric: ±0.56503579231594825; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B195 B195M prop. 351913	**Omega_b(6340)- MASS:** Omega_b(6340)- | Omegabar_b(6340)+	6339.7+-0.6 MeV	symmetric; symmetric: ±0.55323181994360315; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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B196 B196M property: 351917	**Omega_b(6350)- MASS** Omega_b(6350)- | Omegabar_b(6350)+	6349.8+-0.6 MeV	symmetric; symmetric: ±0.60080400015155633; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B197 B197M property: 351702	**Lambda_b(6070)0 MASS** Lambda_b(6070)0 | Lambdabar_b(6070)0	6072.3+-2.9 MeV	symmetric; symmetric: ±2.9469880829275459; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B199 B199M prop. 351870	**Xi_b(6227)0 MASS:** Xi_b(6227)0 | Xibar_b(6227)0	6226.8+-1.6 MeV	asymmetric; asymmetric: +1.543777840252847/-1.6350076513752909; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B200 B200M property: 351860	**Xi_b(6100)- MASS** Xi_b(6100)- | Xibar_b(6100)+	6099.8+-0.4 MeV	symmetric; symmetric: ±0.44830769038414819; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B201 B201M property: 351874	**Xi_b(6327)0 MASS:** Xi_b(6327)0 | Xibar_b(6327)0	6327.28+-0.35 MeV	asymmetric; asymmetric: +0.35468295701936398/-0.34205262752974142; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B202 B202M prop. 351878	**Xi_b(6333)0 MASS:** Xi_b(6333)0 | Xibar_b(6333)0	6332.69+-0.28 MeV	asymmetric; asymmetric: +0.27802877548915689/-0.28425340807103788; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B203 B203M0 property: 351251	**Xi_c(2882) MASS** Xi_c(2882)0 | Xibar_c(2882)0	2882+-9 MeV	symmetric; symmetric: ±9.0476516290140179; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B204 B204M property: 351945	**P_c_cbar_s(4338) 0 MASS** P_c_cbar_s(4338)+	4338.2+-0.8 MeV	symmetric; symmetric: ±0.80622577482985491; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B205 B205M prop. 351999	**P_c_cbar(4312)+ MASS:** P_c_cbar(4312)+	4466+6-5 MeV	asymmetric; asymmetric: +6.3937962412886078/-4.5877765420030174; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B206 B206M property: 350973	**Lambda_c(2910)+ MASS** Lambda_c(2910)0 | Lambdabar_c(2910)0	2914+-7 MeV	symmetric; symmetric: ±6.7675697262754522; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B207 B207M property: 351852	**Xi_b(6087)0 MASS:** Xi_b(6087)	6087.0+-0.5 MeV	symmetric; symmetric: ±0.463167553682577; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B208 B208M prop. 351856	**Xi_b(6095)0 MASS:** Xi_b(6095)	6095.1+-0.4 MeV	symmetric; symmetric: ±0.44082216684770159; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B209 B209M property: 351396	**Omega_c(3185)0 MASS** Omega_c(3185)0 | Omegabar_c(3185)0	3185+7.6-1.9 MeV	asymmetric; asymmetric: +7.5953933407032981/-1.933907960581372; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B210 B210M property: 351400	**Omega_c(3327)0 MASS** Omega_c(3327)0 | Omegabar_c(3327)0	3327.1+1.2-1.8 MeV	asymmetric; asymmetric: +1.2206555615733701/-1.7804493814764859; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B211 B211M prop. 350514	**Omega(2109)- MASS:** Omega(2109)- | Omegabar(2109)+	2108+-5 MeV	symmetric; symmetric: ±5.2773099207835052; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
G033 G033M property: 332463	**graviton MASS:** graviton	<1.76E-23 eV	not applicable; not applicable; type L; limit U; confidence not reported	gravitational representation or typed hypothetical sector *exhaustive record without comparative promotion*	typed gravitational-representation map. external record accounted for; individual HMT–MD realization not yet linked.

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M001 M001M property: 333932	**omega(782) MASS:** omega(782)0	782.66+-0.13 MeV	asymmetric; asymmetric: +0.1290126855075981/-0.12899704603475121; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M002 M002M prop. 334006	**eta'(958) MASS:** eta^'(958)0	957.78+-0.06 MeV	symmetric; symmetric: ±0.062631506813104018; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M003 M003M1 property: 334163	**f_0(980) MASS** f_0(980)0	990+-20 MeV	symmetric; symmetric: ±20; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M004 M004M property: 334182	**phi(1020) MASS:** phi(1020)0	1019.460+-0.016 MeV	symmetric; symmetric: ±0.0155652966373344; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M005 M005M prop. 334354	**f_2(1270) MASS:** f_2(1270)0	1275.4+-0.8 MeV	asymmetric; asymmetric: +0.83226160812299765/-0.83243720122632603; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M006 M006M2 property: 334554	**f_1(1420) MASS** f_1(1420)0	1427.7+1.7-1.4 MeV	asymmetric; asymmetric: +1.669286801641032/-1.4373229857894489; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M008 M008M property: 334403	**f_1(1285) MASS:** f_1(1285)0	1281.8+-0.5 MeV	asymmetric; asymmetric: +0.45740251642395002/-0.45001100251184373; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009M0 prop. 333862	**neutral state only; e+ e- channel** rho(770)0 | rho(770)+ | rho(770)-	775.26+-0.23 MeV	symmetric; symmetric: ±0.23429187483587821; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009M5 property: 333863	**CHARGED ONLY, tau DECAYS and e+ e-** rho(770)0 | rho(770)+ | rho(770)-	775.11+-0.34 MeV	symmetric; symmetric: ±0.34073877998562979; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009M7 property: 333864	**MIXED CHARGES; OTHER REACTIONS** rho(770)0 | rho(770)+ | rho(770)-	763.0+-1.2 MeV	symmetric; symmetric: ±1.236931687685298; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009M2 prop. 333865	**charged states only; produced hadronically** rho(770)0 | rho(770)+ | rho(770)-	766.5+-1.1 MeV	symmetric; symmetric: ±1.106853523344296; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009M0P property: 333866	**NEUTRAL ONLY, PHOTOPRODUCED** rho(770)0 | rho(770)+ | rho(770)-	769.6+-0.8 MeV	asymmetric; asymmetric: +0.84746260647359306/-0.82807525956483763; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009M0R property: 333867	**NEUTRAL ONLY; OTHER REACTIONS** rho(770)0 | rho(770)+ | rho(770)-	769.0+-0.9 MeV	symmetric; symmetric: ±0.85061998301285813; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M010 M010M prop. 334331	**a_1(1260) MASS VALUE** a_1(1260)0 | a_1(1260)+ | a_1(1260)-	1230+-40 MeV	symmetric; symmetric: ±40; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M011 M011M property: 334309	**b_1(1235) MASS** b_1(1235)0 | b_1(1235)+ | b_1(1235)-	1229.5+-3.2 MeV	symmetric; symmetric: ±3.169161296684385; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M012 M012M0 property: 334469	**a_2(1320) resonance mass** a_2(1320)0 | a_2(1320)+ | a_2(1320)-	1318.2+-0.6 MeV	asymmetric; asymmetric: +0.60920186906481122/-0.60699292355954471; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M012 M012M1 prop. 334470	**3pi MODE VALUE** a_2(1320)0 | a_2(1320)+ | a_2(1320)-	1318.6+-1.3 MeV	asymmetric; asymmetric: +1.3253834593734439/-1.308467953256981; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M012 M012M2 property: 334471	**K Kbar MODE** a_2(1320)0 | a_2(1320)+ | a_2(1320)-	1318.1+-0.7 MeV	symmetric; symmetric: ±0.69776876905725704; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M012 M012M3 property: 334472	**eta pi MODE** a_2(1320) 0 | a_2(1320)+ | a_2(1320)-	1317.7+-1.4 MeV	symmetric; symmetric: ±1.44182788627946; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M012 M012M4 prop. 334473	**eta' pi MODE VALUE** a_2(1320)0 | a_2(1320)+ | a_2(1320)-	1322+-7 MeV	asymmetric; asymmetric: +6.6766122233790837/-7.0763164332234849; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013MX property: 334665	**f_2'(1525) MASS** f_2^'(1525)0	1517.3+-2.4 MeV	asymmetric; asymmetric: +2.434665632478874/-2.3241526996293991; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013M2 property: 334666	**PRODUCED BY K+- BEAM:** f_2^'(1525)0	1518.0+-2.7 MeV	asymmetric; asymmetric: +2.757522740444236/-2.6326327611784479; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013M3 prop. 334667	**PRODUCED IN e+ e- ANNIHILATION AND PARTICLE DECAYS:** f_2^'(1525)0	1514+-4 MeV	asymmetric; asymmetric: +4.4245859881758847/-4.089715458632266 4; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013M9 property: 334668	**PRODUCED IN pbar p ANNIHILATION** f_2^'(1525)0	1512+-4 MeV	symmetric; symmetric: ±3.6552461944810291; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013M4 property: 334669	**CENTRAL PRODUCTION:** f_2^'(1525)0	1515+-15 MeV	symmetric; symmetric: ±15; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013M10 prop. 334670	**PRODUCED IN e p COLLISIONS:** f_2^'(1525)0	1512+-3.2 MeV	asymmetric; asymmetric: +3.3105890714493702/-3.0413812651491101; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M014 M014M property: 333857	**f_0(500) BREIT-WIGNER MASS** f_0(500)0	400 to 800 MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015M property: 334842	**rho_3(1690) resonance mass** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	1688.8+-2.1 MeV	symmetric; symmetric: ±2.0985516723695361; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015M1 prop. 334843	**2pi MODE VALUE** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	1686+-4 MeV	symmetric; symmetric: ±3.9750025108542628; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015M2 property: 334844	**K Kbar AND K Kbar pi MODES** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	1696+-4 MeV	symmetric; symmetric: ±3.566236013260232; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015M3 property: 334845	**(4pi)± MODE** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	1686+-5 MeV	symmetric; symmetric: ±4.6428485030430204; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015M5 prop. 334846	**omega pi MODE VALUE** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	1681+-7 MeV	symmetric; symmetric: ±6.5243685482430793; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M015 M015M6 property: 334847	**eta pi+ pi- MODE** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	1682+-12 MeV	symmetric; symmetric: ±12.456821978061001; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M016 M016M property: 335130	**f_4(2050) MASS:** f_4(2050)0	2018+-11 MeV	asymmetric; asymmetric: +11.36985618415634/-11.505626326952621; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M017 M017M prop. 335090	**a_4(1970) MASS VALUE** a_4(1970)0 | a_4(1970)+ | a_4(1970)-	1967+-16 MeV	asymmetric; asymmetric: +16.038876268934711/-15.89171260745924; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M018 M018TMP property: 335789	**K*(892) T-Matrix Pole sqrt(s)** K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)-	890+-14 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M018 M018M1 property: 335790	**CHARGED STATES ONLY; HADRONICALLY PRODUCED** K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)-	891.88+-0.23 MeV	symmetric; symmetric: ±0.2311072255904166; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M018 M018MCT prop. 335791	**CHARGED STATES ONLY; PRODUCTION IN tau-LEPTON DECAYS** K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)-	895.5+-0.8 MeV	symmetric; symmetric: ±0.76655071586947199; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M018 M018M2 property: 335792	**NEUTRAL STATE ONLY** K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)-	895.56+-0.20 MeV	asymmetric; asymmetric: +0.2035793881115591/-0.2050633570239769 1; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M019 M019M property: 335878	**K_0*(1430) resonance resonance mass** K_0^*(1430)0 | Kbar_0^*(1430)0 | K_0^*(1430)+ | K_0^*(1430)-	1425+-50 MeV	symmetric; symmetric: ±50; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M020 M020M prop. 335146	**pi_2(2100) MASS VALUE** pi_2(2100)0 | pi_2(2100)+ | pi_2(2100)-	2090+-29 MeV	symmetric; symmetric: ±29.4174202707276; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M022 M022M1 property: 335888	**Charged states only, with final state K pi** K_2^*(1430)0 | Kbar_2^*(1430)0 | K_2^*(1430)+ | K_2^*(1430)-	1427.3+-1.5 MeV	asymmetric; asymmetric: +1.50306501170133/-1.5176464822711819; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M022 M022M4 property: 335889	**Neutral states only** K_2^*(1430)0 | Kbar_2^*(1430)0 | K_2^*(1430)+ | K_2^*(1430)-	1432.4+-1.3 MeV	symmetric; symmetric: ±1.2576940808242421; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M023 M023M prop. 335963	**K_2(1770) MASS VALUE** K_2(1770)0 | Kbar_2(1770)0 | K_2(1770)+ | K_2(1770)-	1773+-8 MeV	asymmetric; asymmetric: +7.9841554588538504/-7.9644541424158186; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M025 M025M property: 346343	**psi(4160) MASS:** psi(4160)	4191+-5 MeV	asymmetric; asymmetric: +5.1959983332711133/-4.9802084626429108; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M026 M026M property: 343286	**eta_c(1S) MASS:** eta_c(1S)	2984.09+-0.30 MeV	asymmetric; asymmetric: +0.2967710477123273 2/-0.3002103199855122 9; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M027 M027MX prop. 334525	**eta(1405) MASS:** eta(1405)0	1408.0+1.8-1.2 MeV	asymmetric; asymmetric: +1.817274601784945/-1.231728539767738; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M027 M027M1 property: 334526	**eta pi pi MODE:** eta(1405)0	1405.5+-2.3 MeV	asymmetric; asymmetric: +2.2617401948054172/-2.353950701102808; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M027 M027M4 property: 334527	**K Kbar pi MODE: (a_0(980) pi or direct K Kbar pi)** eta(1405)0	1413.5+2.0-0.9 MeV	asymmetric; asymmetric: +1.9652283255103979/-0.88864461966865371; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M027 M027M2 prop. 334528	**pi pi gamma MODE:** eta(1405)0	1403+-17 MeV	symmetric; symmetric: ±16.617414327881129; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M027 M027M7 property: 334529	**phi gamma MODE:** eta(1405)0	1422+6-8 MeV	asymmetric; asymmetric: +6.2625873247404709/-8.0777472107017552; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M028 M028MX property: 335817	**K_1(1270) resonance mass** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	1256+-6 MeV	symmetric; symmetric: ±6.4239824760278736; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M028 M028M2 prop. 335818	**PRODUCTION BY K-, BACKWARD SCATTERING, AND HYPERON EXCHANGE** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	1275+-10 MeV	symmetric; symmetric: ±10; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M028 M028M3 property: 335819	**GENERATED BY K BEAMS** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	1270+-10 MeV	symmetric; symmetric: ±10; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M028 M028M1 property: 335820	**PRODUCTION BY BEAMS OTHER THAN K MESONS** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	1252+-9 MeV	symmetric; symmetric: ±8.8239123358673819; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M028 M028MT prop. 335821	**PRODUCTION IN tau-LEPTON DECAYS** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	1250+-50 MeV	symmetric; symmetric: ±47.381430961928537; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M029 M029M property: 346244	**X(3940) MASS:** X(3940)	3942+-9 MeV	asymmetric; asymmetric: +9.0427689971016019/-8.2141639308556176; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M030 M030M property: 334306	**h_1(1170) MASS:** h_1(1170)0	1166+-6 MeV	symmetric; symmetric: ±5.8309518948453007; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M033 M033M3 prop. 335259	**decay channel: pi- p –> omega pi0 n** rho_5(2350)0 | rho_5(2350)+ | rho_5(2350)-	2330+-35 MeV	symmetric; symmetric: ±35; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M034 M034M property: 334783	**pi_2(1670) MASS** pi_2(1670)0 | pi_2(1670)+ | pi_2(1670)-	1670.6+2.9-1.2 MeV	asymmetric; asymmetric: +2.92487285188664/-1.1987692041830089; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M035 M035M property: 336016	**K_4*(2045) resonance resonance mass** K_4^*(2045)0 | Kbar_4^*(2045)0 | K_4^*(2045)+ | K_4^*(2045)-	2048+8-9 MeV	asymmetric; asymmetric: +8.0644653972191538/-8.9483400524220826; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M036 M036MX prop. 334171	**a_0(980) MASS VALUE** a_0(980)0 | a_0(980)+ | a_0(980)-	980+-20 MeV	symmetric; symmetric: ±20; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M037 M037M property: 334451	**eta(1295) MASS:** eta(1295)0	1294+-4 MeV	symmetric; symmetric: ±3.892110114334967; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M038 M038M property: 335002	**f_2(1810) MASS:** f_2(1810)0	1815+-12 MeV	asymmetric; asymmetric: +12.52864229025908/-11.9803740701169; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M040 M040M prop. 336032	**K_2(2250) MASS VALUE** K_2(2250)0 | Kbar_2(2250)0 | K_2(2250)+ | K_2(2250)-	2247+-17 MeV	symmetric; symmetric: ±16.84303842133038; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M041 M041M4 property: 335244	**p p CENTRAL PRODUCTION:** f_4(2300)0	2320+-60 MeV	symmetric; symmetric: ±60; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M042 M042M property: 335158	**f_2(2150) MASS: COMBINED MODES** f_2(2150)0	2157+-12 MeV	symmetric; symmetric: ±11.76667948645094; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M042 M042M3 prop. 335159	**eta eta MODE:** f_2(2150)0	2157+-12 MeV	symmetric; symmetric: ±11.76667948645094; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M045 M045M property: 334778	**omega_3(1670) MASS** omega_3(1670)0	1667+-4 MeV	symmetric; symmetric: ±3.9911056080338438; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M047 M047M property: 347744	**Upsilon(4S) MASS:** Upsilon(4S)	10579.4+-1.2 MeV	symmetric; symmetric: ±1.1896060082538451; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M048 M048M prop. 347659	**Upsilon(3S) MASS:** Upsilon(3S)	10355.1+-0.5 MeV	symmetric; symmetric: ±0.5; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M049 M049M property: 346837	**Upsilon(1S) MASS:** Upsilon(1S)	9460.40+-0.10 MeV	symmetric; symmetric: ±0.09842982316061092 7; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M050 M050M property: 346228	**chi_c2(3930) MASS:** chi_c2(3930)	3922.5+-1.0 MeV	symmetric; symmetric: ±1.0198944899855971; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M052 M052M prop. 347286	**Upsilon(2S) MASS:** Upsilon(2S)	10023.4+-0.5 MeV	symmetric; symmetric: ±0.5; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M053 M053M property: 345866	**psi(3770) MASS:** psi(3770)	3773.7+-0.7 MeV	symmetric; symmetric: ±0.6957511640714941 8; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M054 M054M property: 335023	**phi_3(1850) MASS:** phi_3(1850)0	1854+-7 MeV	asymmetric; asymmetric: +6.8824720161168527/-6.666666666666667; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M055 M055M prop. 344598	**chi_c1(1P) MASS:** chi_c1(1P)	3510.67+-0.05 MeV	symmetric; symmetric: ±0.050739750770671888; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M056 M056M property: 344304	**chi_c0(1P) MASS** chi_c0(1P)	3415.50+-0.19 MeV	asymmetric; asymmetric: +0.1931182921033357/-0.19240527880424149; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M057 M057M property: 344926	**chi_c2(1P) MASS:** chi_c2(1P)	3556.17+-0.07 MeV	symmetric; symmetric: ±0.07161203246291189 5; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M058 M058M prop. 334463	**mass of pi(1300)** pi(1300)0 | pi(1300)+ | pi(1300)-	1300+-100 MeV	symmetric; symmetric: ±100; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M059 M059M property: 345228	**eta_c(2S) MASS** eta_c(2S)	3637.8+-0.6 MeV	asymmetric; asymmetric: +0.6175976364914121 9/-0.6200487678843437 1; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M060 M060M property: 335975	**K_3*(1780) resonance resonance mass** K_3^*(1780)0 | Kbar_3^*(1780)0 | K_3^*(1780)+ | K_3^*(1780)-	1779+-8 MeV	asymmetric; asymmetric: +7.821524428452 43/-7.545019387046 4446; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M061 M061M prop. 337777	**mass of D*(2007)0** D^*(2007)0 | Dbar^*(2007)0	2006.86+-0.05 MeV	symmetric; symmetric: ±0.053769014750598178; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M062 M062M property: 337794	**D*(2010)+- MASS** D^*(2010)+ | D^*(2010)-	2010.27+-0.04 MeV	symmetric; symmetric: ±0.044491245593188788; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M064 M064M property: 335845	**K_1(1400) resonance mass** K_1(1400)0 | Kbar_1(1400)0 | K_1(1400)+ | K_1(1400)-	1403+-7 MeV	symmetric; symmetric: ±6.8731168328374306; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M065 M065M0 prop. 334880	**eta rho0 and pi+ pi- decay modes** rho(1700)0 | rho(1700)+ | rho(1700)-	1720+-20 MeV	symmetric; symmetric: ±20; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M066 M066M1 property: 334577	**f_2(1430) MASS** f_2(1430)0	~1430 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M067 M067M1 property: 334825	**e+ e- PRODUCTION:** phi(1680)0	1680+-20 MeV	symmetric; symmetric: ±20; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M068 M068M prop. 334927	**f_0(1710) Breit-Wigner MASS:** f_0(1710)0	1723+7-6 MeV	asymmetric; asymmetric: +6.7390813885535801/-5.5617775425136911; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M070 M070M property: 343505	**J/psi(1S) MASS:** J/psi(1S)	3096.900+-0.006 MeV	symmetric; symmetric: ±0.0061858811299374487; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M071 M071M property: 345299	**psi(2S) MASS:** psi(2S)	3686.097+-0.011 MeV	symmetric; symmetric: ±0.01055310115798376; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M072 M072M prop. 346249	**psi(4040) MASS:** psi(4040)	4040+-4 MeV	symmetric; symmetric: ±4.2999999999999998; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M073 M073M property: 346684	**psi(4415) MASS:** psi(4415)	4415+-5 MeV	symmetric; symmetric: ±4.5278832533731954; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M074 M074M property: 346460	**psi(4230) MASS:** psi(4230)	4221.7+-2.5 MeV	asymmetric; asymmetric: +2.4568685085110582/-2.454400623477099; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M075 M075M prop. 334966	**mass of pi(1800)** pi(1800)0 | pi(1800)+ | pi(1800)-	1810+9-11 MeV	asymmetric; asymmetric: +9.4045949221033531/-11.027513999271919; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M076 M076M property: 347129	**chi_b0(1P) MASS** chi_b0(1P)	9859.44+-0.42+-0.31 MeV	symmetric; symmetric: ±0.41999999999999998; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M077 M077M property: 347175	**chi_b1(1P) MASS:** chi_b1(1P)	9892.78+-0.26+-0.31 MeV	symmetric; symmetric: ±0.26000000000000001; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M078 M078M prop. 347234	**chi_b2(1P) MASS:** chi_b2(1P)	9912.21+-0.26+-0.31 MeV	symmetric; symmetric: ±0.26000000000000001; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M079 M079M property: 347497	**chi_b0(2P) MASS** chi_b0(2P)	10232.5+-0.4+-0.5 MeV	symmetric; symmetric: ±0.40000000000000002; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M080 M080M property: 347543	**chi_b1(2P) MASS:** chi_b1(2P)	10255.46+-0.22+-0.50 MeV	symmetric; symmetric: ±0.22; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M081 M081M prop. 347610	**chi_b2(2P) MASS:** chi_b2(2P)	10268.65+-0.22+-0.50 MeV	symmetric; symmetric: ±0.22; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M082 M082M property: 335203	**f_J(2220) MASS** f_J(2220)0	2231.1+-3.5 MeV	asymmetric; asymmetric: +3.5335248107562318/-3.5214610779100788; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M084 M084M property: 334658	**f_1(1510) MASS:** f_1(1510)0	1518+-5 MeV	symmetric; symmetric: ±5.2162953426963403; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M085 M085M prop. 335011	**X(1835) MASS:** X(1835)0	1834.0+4.0-3.0 MeV	asymmetric; asymmetric: +4.1178707457023851/-2.9844075950960041; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M089 M089M property: 335277	**f_6(2510) MASS** f_6(2510)0	2470+-50 MeV	symmetric; symmetric: ±44.999997965452351; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M090 M090M property: 336034	**K_3(2320) resonance mass** K_3(2320)0 | Kbar_3(2320)0 | K_3(2320)+ | K_3(2320)-	2324+-24 MeV	symmetric; symmetric: ±24; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M091 M091M prop. 336040	**K_4(2500) MASS VALUE** K_4(2500)0 | Kbar_4(2500)0 | K_4(2500)+ | K_4(2500)-	2490+-20 MeV	symmetric; symmetric: ±20; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M092 M092M property: 347804	**Upsilon(10860) MASS:** Upsilon(10860)	10885.2+2.6-1.6 MeV	asymmetric; asymmetric: +2.5658471171469732/-1.5686101402474319; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M093 M093M property: 347905	**Upsilon(11020) MASS:** Upsilon(11020)	11000+-4 MeV	asymmetric; asymmetric: +4.0234281589264329/-4.0242307387762244; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M094 M094M prop. 335865	**K*(1410) MASS VALUE** K^*(1410)0 | Kbar^*(1410)0 | K^*(1410)+ | K^*(1410)-	1414+-15 MeV	symmetric; symmetric: ±14.53984543474516; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M095 M095M property: 335946	**K*(1680) MASS** K^*(1680)0 | Kbar^*(1680)0 | K^*(1680)+ | K^*(1680)-	1718+-18 MeV	asymmetric; asymmetric: +17.07872691654519/-18.698162672594339; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M098 M098M property: 336036	**K_5*(2380) resonance resonance mass** K_5^*(2380)0 | Kbar_5^*(2380)0 | K_5^*(2380)+ | K_5^*(2380)-	2382+-24 MeV	symmetric; symmetric: ±23.60084744241189; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M099 M099M prop. 335944	**K_1(1650) MASS VALUE** K_1(1650)0 | Kbar_1(1650)0 | K_1(1650)+ | K_1(1650)-	1650+-50 MeV	symmetric; symmetric: ±50; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M101 M101M property: 335032	**eta_2(1870) MASS** eta_2(1870)0	1842+-8 MeV	symmetric; symmetric: ±8.0430753075960109; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M103 M103M property: 335180	**phi(2170) MASS:** phi(2170)0 | phi(2170)+ | phi(2170)-	2164+-5 MeV	symmetric; symmetric: ±5.1132736001610546; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M104 M104M prop. 336003	**K_2*(1980) resonance resonance mass** K_2^*(1980)0 | Kbar_2^*(1980)0 | K_2^*(1980)+ | K_2^*(1980)-	1990+60-50 MeV	asymmetric; asymmetric: +59.105152569706007/-45.405772247239312; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M105 M105M0 property: 334592	**rho(1450) MASS** rho(1450)0 | rho(1450)+ | rho(1450)-	1465+-25 MeV	symmetric; symmetric: ±25; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M106 M106M property: 335112	**f_2(2010) MASS:** f_2(2010)0	2010+60-80 MeV	asymmetric; asymmetric: +62/-76; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M107 M107M prop. 335234	**f_2(2300) MASS:** f_2(2300)0	2297+-28 MeV	symmetric; symmetric: ±28; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M108 M108M property: 335252	**f_2(2340) MASS** f_2(2340)0	2346+21-10 MeV	asymmetric; asymmetric: +21.3005624897101181/-9.7423326379963484; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M109 M109M property: 334552	**h_1(1415) MASS:** h_1(1415)0	1409+9-8 MeV	asymmetric; asymmetric: +9.4194068122733494/-7.776226726894202; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M112 M112M prop. 335199	**f_0(2200) MASS:** f_0(2200)0	2187+-14 MeV	asymmetric; asymmetric: +13.533041759519 27/-14.057750535181979; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M114 M114M property: 334950	**eta(1760) MASS:** eta(1760)0	1751+-15 MeV	asymmetric; asymmetric: +14.814874939752929/-14.980145590239159; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M115 M115M property: 335232	**eta(2225) MASS:** eta(2225)0	2221+13-10 MeV	asymmetric; asymmetric: +13.27133387053455/-9.5202460778307483; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M117 M117M prop. 334752	**f_2(1640) MASS:** f_2(1640)0	1639+-6 MeV	symmetric; symmetric: ±5.8313784211153239; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M121 M121M property: 338481	**D_s1(2536)+- MASS** D_s1(2536)+ | D_s1(2536)-	2535.12+-0.06 MeV	symmetric; symmetric: ±0.06035097747089856 3; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M123 M123M property: 334703	**f_2(1565) MASS:** f_2(1565)0	1571+-13 MeV	symmetric; symmetric: ±13.379294632448991; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M125 M125M prop. 334567	**omega(1420) MASS:** omega(1420)0	1410+-60 MeV	symmetric; symmetric: ±60; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M126 M126M property: 334768	**omega(1650) MASS:** omega(1650)0	1670+-30 MeV	symmetric; symmetric: ±30; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M129 M129M property: 336041	**K(3100) MASS** K(3100)0 | Kbar(3100)0 | K(3100)+ | K(3100)-	~3100 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M129 M129M1 prop. 336042	**3-BODY DECAYS:** K(3100)0 | Kbar(3100)0 | K(3100)+ | K(3100)-	3054+-11 MeV	symmetric; symmetric: ±10.68148842445026; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M129 M129M2 property: 336043	**4-BODY DECAYS** K(3100)0 | Kbar(3100)0 | K(3100)+ | K(3100)-	3059+-11 MeV	symmetric; symmetric: ±10.641252264801439; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M134 M134M property: 335999	**K_0*(1950) resonance resonance mass** K_0^*(1950)0 | Kbar_0^*(1950)0 | K_0^*(1950)+ | K_0^*(1950)-	1957+-14 MeV	symmetric; symmetric: ±14.320221981073329; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M135 M135M prop. 335075	**f_2(1950) MASS:** f_2(1950)0	1936+-12 MeV	symmetric; symmetric: ±11.73086748275699; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M142 M142M2 property: 335057	**f_2(1910) omega omega MODE:** f_2(1910)0	1900+-9 MeV	symmetric; symmetric: ±9.266557494580514; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M144 M144M property: 344851	**h_c(1P) MASS:** h_c(1P)	3525.37+-0.14 MeV	symmetric; symmetric: ±0.14473463841322931; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M146 M146M prop. 335990	**K_2(1820) MASS VALUE** K_2(1820)0 | Kbar_2(1820)0 | K_2(1820)+ | K_2(1820)-	1819+-12 MeV	asymmetric; asymmetric: +12.067628581227879/-12.471842992923371; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M147 M147M property: 334506	**Mass (Breit-Wigner)** f_0(1370)0	1200 to 1500 MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M148 M148M property: 338509	**D_s2*(2573) MASS** D_s2^*(2573) + | D_s2^*(2573)-	2569.1+-0.8 MeV	asymmetric; asymmetric: +0.8278063017675835 2/-0.8264983149616976 8; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M149 M149M prop. 334579	**a_0(1450) MASS VALUE** a_0(1450)0 | a_0(1450)+ | a_0(1450)-	1439+-34 MeV	symmetric; symmetric: ±34.426230116735738; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M151 M151M property: 342551	**B_J*(5732) resonance resonance mass** B_J^*(5732)0 | Bbar_J^*(5732)0 | B_J^*(5732)+ | B_J^*(5732)-	5698+-8 MeV	asymmetric; asymmetric: +7.5869468243446372/-7.6930254688839934; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M152 M152M property: 334628	**f_0(1500) MASS:** f_0(1500)0	1522+-25 MeV	symmetric; symmetric: ±25; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M153 M153M prop. 343151	**B_sJ*(5850) MASS VALUE** B_sJ^*(5850)0 | Bbar_sJ^*(5850)0	5853+-15 MeV	symmetric; symmetric: ±15; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M154 M154M property: 334758	**eta_2(1645) MASS** eta_2(1645)0	1617+-5 MeV	symmetric; symmetric: ±5.2613147962640197; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M156 M156M property: 335120	**f_0(2020) MASS:** f_0(2020)0	1982+54.1-3.0 MeV	asymmetric; asymmetric: +54.083269131959838/-3; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M158 M158M prop. 337854	**D*(2640)+- MASS VALUE** D^*(2640)0 | Dbar^*(2640)0 | D^*(2640)+ | D^*(2640)-	2637+-6 MeV	symmetric; symmetric: ±6.324555320336759; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M159 M159M property: 346204	**chi_c0(3915) MASS** chi_c0(3915)	3922.1+-1.8 MeV	asymmetric; asymmetric: +1.7730705038425429/-1.7603618343080081; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M160 M160M property: 335942	**K(1630) MASS** K(1630)0 | Kbar(1630)0 | K(1630)+ | K(1630)-	1629+-7 MeV	symmetric; symmetric: ±7; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M161 M161M prop. 334740	**a_1(1640) MASS VALUE** a_1(1640)0 | a_1(1640)+ | a_1(1640)-	1655+-16 MeV	asymmetric; asymmetric: +15.615362458792911/-16.742904840572461; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M162 M162M property: 334903	**a_2(1700) MASS** a_2(1700)0 | a_2(1700)+ | a_2(1700)-	1706+-14 MeV	asymmetric; asymmetric: +13.50144093159982/-14.839074467455429; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M164 M164TMP property: 334723	**pi_1(1600) T-matrix pole in sqrt(s)** pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)-	(1480–1680)-i(150–300) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M164 M164MEP prop. 334724	**pi_1(1600) MASS VALUE (eta pi mode)** pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)-	1354+-25 MeV	asymmetric; asymmetric: +25.620371427579411/-24.3777786955451; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M164 M164M property: 334725	**pi_1(1600) MASS (non-eta pi mode)** pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)-	1645+40-17 MeV	asymmetric; asymmetric: +39.719756226750718/-16.575679876304068; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M166 M166M property: 334720	**h_1(1595) MASS:** h_1(1595)0	1594+18-60 MeV	asymmetric; asymmetric: +18.02775637731995/-61.846584384264908; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M168 M168M prop. 335156	**f_0(2100) MASS:** f_0(2100)0	2095+17-19 MeV	asymmetric; asymmetric: +17.060496688737981/-18.979661007438018; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M171 M171M property: 346822	**eta_b(1S) MASS** eta_b(1S)	9398.7+-2.0 MeV	asymmetric; asymmetric: +2.081610145729452/-2.00237037046477; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M172 M172M property: 338429	**D_s0*(2317)± MASS** D_s0^*(2317)+ | D_s0^*(2317)-	2317.8+-0.5 MeV	symmetric; symmetric: ±0.51003296350202032; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M173 M173M prop. 338448	**D_s1(2460)+- MASS:** D_s1(2460)+ | D_s1(2460)-	2459.5+-0.6 MeV	symmetric; symmetric: ±0.63443027023618292; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M174 M174TMP property: 335785	**K_0*(700) pole of the T-matrix sqrt(s)** K_0^*(700)0 | Kbar_0^*(700)0 | K_0^*(700)+ | K_0^*(700)-	(630–730)-i(260–340) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M174 M174M property: 335786	**K_0*(700) Breit–Wigner resonance resonance mass** K_0^*(700)0 | Kbar_0^*(700)0 | K_0^*(700)+ | K_0^*(700)-	838+-11 MeV	asymmetric; asymmetric: +11.387904079019 41/-11.2973395209359; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M175 M175M5 prop. 334616	**eta(1475) MASS:** eta(1475)0	1476+-4 MeV	asymmetric; asymmetric: +3.9290750275045658/-4.0103127348477026; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M176 M176M property: 346130	**chi_c1(3872) MASS FROM J/psi X MODE** chi_c1(3872)	3871.64+-0.06 MeV	asymmetric; asymmetric: +0.057098859021772717/-0.057095838437930757; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M177 M177M property: 347489	**Upsilon_2(1D) MASS:** Upsilon_2(1D)	10163.7+-1.4 MeV	symmetric; symmetric: ±1.4386010588789799; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M180 M180M prop. 337827	**D_1(2430)0 MASS:** D_1(2430)0 | Dbar_1(2430)0	2412+-9 MeV	symmetric; symmetric: ±9.1747943402670877; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M181 M181M property: 346635	**psi(4360) MASS:** psi(4360)	4373+-7 MeV	asymmetric; asymmetric: +7.0267029998418007/-7.1263389859977933; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M182 M182M property: 338524	**D_s1*(2700) + MASS** D_s1^*(2700)+ | D_s1^*(2700)-	2714+-5 MeV	asymmetric; asymmetric: +5.3375261860918934/-4.5780472675086896; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M185 M185M prop. 335043	**pi_2(1880) MASS:** pi_2(1880)0	1874+26-5 MeV	asymmetric; asymmetric: +25.897946058811009/-4.7988156638370416; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M186 M186M property: 343137	**B_s2*(5840)0 MASS** B_s2^*(5840)0 | Bbar_s2^*(5840)0	5839.88+-0.12 MeV	symmetric; symmetric: ±0.115984227750511 8; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M187 M187M property: 343129	**B_s1(5830)0 MASS:** B_s1(5830)0 | Bbar_s1(5830)0	5828.73+-0.20 MeV	symmetric; symmetric: ±0.19651551951131191; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M188 M188M prop. 334714	**rho(1570) MASS:** rho(1570)0	1570+-70 MeV	symmetric; symmetric: ±71.693793315739683; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M189 M189M property: 346767	**psi(4660) MASS:** psi(4660)	4623+-10 MeV	asymmetric; asymmetric: +10.20904494225131/-10.127972763403161; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M190 M190M property: 346451	**X(4160) MASS:** X(4160)	4153+23-21 MeV	asymmetric; asymmetric: +23.03765481878029/-20.816561780400072; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M191 M191M prop. 347990	**T_c_cbar(4050)+ MASS:** T_c_cbar(4050)+	4051+24-40 MeV	asymmetric; asymmetric: +24.413111231467401/-43.324358044868937; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M192 M192M property: 348026	**T_c_cbar(4250)+ MASS** T_c_cbar(4250)+	4250+190-50 MeV	asymmetric; asymmetric: +185.29975715040749/-45.453272709454048; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M193 M193M property: 346336	**chi_c1(4140) MASS:** chi_c1(4140)	4146.5+-3.0 MeV	asymmetric; asymmetric: +2.988542411834247/-3.0016153327920829; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M194 M194M prop. 346628	**X(4350) MASS:** X(4350)	4351+-5 MeV	asymmetric; asymmetric: +4.6529560496527367/-5.1478150704935004; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M195 M195M property: 348030	**T_c_cbar_1(4430)+ MASS** T_c_cbar_1(4430)+	4478+15-17 MeV	asymmetric; asymmetric: +14.76917016091541/-17.453443521646189; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M196 M196M property: 338533	**D_s1*(2860) + MASS** D_s1^*(2860)+ | D_s1^*(2860)-	2859+-27 MeV	symmetric; symmetric: ±26.832815729997481; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M197 M197M prop. 338544	**D_sJ(3040)+ MASS:** D_sJ(3040)+ | D_sJ(3040)-	3044+31-9 MeV	asymmetric; asymmetric: +31.048349392520048/-9.4339811320566032; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M198 M198M property: 337843	**D_0(2550)0 MASS** D_0(2550)0 | Dbar_0(2550)0 | D_0(2550)+ | D_0(2550)-	2549+-19 MeV	symmetric; symmetric: ±19.004002841336341; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M199 M199M property: 337846	**D_1*(2600)0 resonance resonance mass** D_1^*(2600)0 | Dbar_1^*(2600)0 | D_1^*(2600)+ | D_1^*(2600)-	2627+-10 MeV	symmetric; symmetric: ±9.8128244436310279; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M200 M200M prop. 347281	**eta_b(2S) MASS:** eta_b(2S)	9999+-4 MeV	asymmetric; asymmetric: +4.4821869662029936/-3.982461550347975; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M203 M203M property: 337860	**D_3*(2750) resonance resonance mass** D_3^*(2750)0 | Dbar_3^*(2750)0 | D_3^*(2750)+ | D_3^*(2750)-	2763.1+-3.2 MeV	symmetric; symmetric: ±3.1908533146781788; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M204 M204M property: 347229	**h_b(1P) MASS:** h_b(1P)	9899.3+-0.8 MeV	symmetric; symmetric: ±0.75076896116034797; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M205 M205M prop. 347593	**h_b(2P) MASS:** h_b(2P)	10259.8+-1.2 MeV	symmetric; symmetric: ±1.2083045973594571; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M206 M206M property: 347734	**chi_b1(3P) MASS** chi_b1(3P)	10513.4+-0.7 MeV	symmetric; symmetric: ±0.67007462271003815; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M207 M207M property: 348052	**T_b_bbar_1(10610)± MASS** T_b_bbar_1(10610)+	10607.2+-2.0 MeV	symmetric; symmetric: ±2; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M207 M207M0 prop. 348053	**T_b_bbar_1(10610)0 MASS:** T_b_bbar_1(10610)+	10609+-6 MeV	symmetric; symmetric: ±5.6568542494923806; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M208 M208M property: 348075	**T_b_bbar_1(10650)+ MASS** T_b_bbar_1(10650)+	10652.2+-1.5 MeV	symmetric; symmetric: ±1.5; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M210 M210M property: 347938	**T_c_cbar_1(3900) MASS** T_c_cbar_1(3900) +	3886.7+-2.3 MeV	asymmetric; asymmetric: +2.293369197414477/-2.273282289576287; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M212 M212M prop. 346100	**psi_2(3823) MASS:** psi_2(3823)	3823.51+-0.34 MeV	symmetric; symmetric: ±0.33904706932929501; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M213 M213M property: 347976	**T_c_cbar(4020) MASS** T_c_cbar(4020)+	4024.1+-1.9 MeV	asymmetric; asymmetric: +1.824690634658487/-1.967162323895744 1; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M216 M216M property: 348022	**T_c_cbar_0(4240) + MASS** T_c_cbar_0(4240)0	4239+50-21 MeV	asymmetric; asymmetric: +48.466483264210538/-20.591260281974002; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M217 M217M prop. 343283	**B_c(2S)+- MASS:** B_c(2S)+ | B_c(2S)-	6871.2+-1.0 MeV	symmetric; symmetric: ±1.018454581927803; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M223 M223M property: 348000	**T_c_cbar(4055)+ MASS** T_c_cbar(4055)+	4054+-3.2 MeV	symmetric; symmetric: ±3.16227766016838; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M226 M226M property: 338535	**D_s3*(2860) + MASS** D_s3^*(2860)+ | D_s3^*(2860)-	2860+-7 MeV	symmetric; symmetric: ±7.0007142492748553; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M227 M227M prop. 335070	**a_0(1950) MASS:** a_0(1950)0	1931+-26 MeV	symmetric; symmetric: ±26.0768096208106; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M228 M228M property: 337857	**D_2(2740)0 MASS** D_2(2740)0 | Dbar_2(2740)0 | D_2(2740)+ | D_2(2740)-	2747+-6 MeV	symmetric; symmetric: ±6.3932930732238127; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M229 M229M property: 337872	**D_2*(3000)0 resonance resonance mass** D_2^*(3000)0 | Dbar_2^*(3000)0 | D_2^*(3000)+ | D_2^*(3000)-	3210+-60 MeV	symmetric; symmetric: ±56.938563381947027; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M231 M231M prop. 348012	**T_c_cbar_1(4200)+ MASS:** T_c_cbar_1(4200)+	4242+-26 MeV	asymmetric; asymmetric: +26.306839009567629/-25.567261845351091; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M232 M232M property: 348036	**T_b_sbar(5568)+ MASS** T_b_sbar(5568)+	5566.9+-3.3 MeV	asymmetric; asymmetric: +3.255764119219942/-3.324154027718933; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M233 M233M property: 346624	**chi_c1(4274) MASS:** chi_c1(4274)	4290+-6 MeV	asymmetric; asymmetric: +5.682397906375208 6/-6.461081982391395; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M234 M234M prop. 346755	**chi_c0(4500) MASS:** chi_c0(4500)	4484+-12 MeV	asymmetric; asymmetric: +12.002820072457739/-11.985284550375541; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M235 M235M property: 346816	**chi_c0(4700) MASS** chi_c0(4700)	4708+6-4 MeV	asymmetric; asymmetric: +5.9690361844477042/-3.9408505465575612; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M237 M237M property: 346122	**chi_c0(3860) MASS:** chi_c0(3860)	3862+50-35 MeV	asymmetric; asymmetric: +47.707441767506253/-34.539832078341092; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M238 M238M prop. 347741	**chi_b2(3P) MASS:** chi_b2(3P)	10524.0+-0.8 MeV	symmetric; symmetric: ±0.77833154889160183; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M239 M239M property: 335106	**pi_2(2005) MASS** pi_2(2005)0	1963+17-27 MeV	asymmetric; asymmetric: +16.663540525911429/-27.418323182256639; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M240 M240M property: 348006	**T_c_cbar(4100)+ MASS:** T_c_cbar(4100)+	4096+-28 MeV	asymmetric; asymmetric: +26.90724809414742/-29.732137494637009; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M241 M241M prop. 346116	**psi_3(3842) MASS:** psi_3(3842)	3842.71+-0.20 MeV	symmetric; symmetric: ±0.20000000000000001; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M243 M243M property: 347800	**Upsilon(10753) MASS:** Upsilon(10753)	10756.6+-2.8 MeV	symmetric; symmetric: ±2.846049894151542; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M244 M244M+ property: 342539	**B_1(5721)+ resonance mass** B_1(5721)0 | Bbar_1(5721)0 | B_1(5721)+ | B_1(5721)-	5726.0+2.5-2.7 MeV	asymmetric; asymmetric: +2.455967779831584/-2.7291514082802459; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M244 M244M0 prop. 342542	**B_1(5721)0 mass value** B_1(5721)0 | Bbar_1(5721)0 | B_1(5721)+ | B_1(5721)-	5726.1+-1.2 MeV	asymmetric; asymmetric: +1.2330295992331299/-1.262925411947059; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M245 M245M+ property: 342556	**B_2*(5747)+ resonance mass** B_2^*(5747)0 | Bbar_2^*(5747)0 | B_2^*(5747)+ | B_2^*(5747)-	5737.3+-0.7 MeV	asymmetric; asymmetric: +0.7106016585520667/-0.73206597699796105; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M245 M245M0 property: 342559	**B_2*(5747)0 resonance mass** B_2^*(5747)0 | Bbar_2^*(5747)0 | B_2^*(5747)+ | B_2^*(5747)-	5739.6+-0.7 MeV	symmetric; symmetric: ±0.6534656805868519; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M246 M246M+ prop. 342571	**B_J(5840)+ MASS VALUE** B_J(5840)0 | Bbar_J(5840)0 | B_J(5840)+ | B_J(5840)-	5851+-19 MeV	symmetric; symmetric: ±19.105159421307778; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M246 M246M0 property: 342574	**B_J(5840)0 MASS** B_J(5840)0 | Bbar_J(5840)0 | B_J(5840)+ | B_J(5840)-	5863+-9 MeV	symmetric; symmetric: ±8.6026189907829647; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M247 M247M property: 335261	**eta(2370) MASS:** eta(2370)0	2377+-9 MeV	asymmetric; asymmetric: +8.807211582813899/-9.65401633399925; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M248 M248M+ prop. 342583	**B_J(5970)+ MASS VALUE** B_J(5970)0 | Bbar_J(5970)0 | B_J(5970)+ | B_J(5970)-	5965+-5 MeV	symmetric; symmetric: ±4.5053723109770578; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M248 M248M0 property: 342586	**B_J(5970)0 MASS** B_J(5970)0 | Bbar_J(5970)0 | B_J(5970)+ | B_J(5970)-	5971+-5 MeV	symmetric; symmetric: ±5.3479868586791888; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M249 M249M property: 337868	**D_1*(2760)0 resonance resonance mass** D_1^*(2760)0 | Dbar_1^*(2760)0 | D_1^*(2760)+ | D_1^*(2760)-	2781+-22 MeV	symmetric; symmetric: ±22.203603311174518; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M250 M250M prop. 347916	**T_cbar_sbar_0*(2870)0 MASS VALUE** T_cbar_sbar_0^*(2870)0	2874+-11 MeV	symmetric; symmetric: ±10.737127438402171; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M251 M251M property: 347922	**T_cbar_sbar_1*(2900)0 MASS** T_cbar_sbar_1^*(2900)0	2904+-5 MeV	symmetric; symmetric: ±5.0990195135927836; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M252 M252M property: 337813	**D_0*(2300) resonance resonance mass** D_0^*(2300)0 | Dbar_0^*(2300)0 | D_0^*(2300)+ | D_0^*(2300)-	2343+-10 MeV	symmetric; symmetric: ±9.6544420794461843; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M253 M253M prop. 337817	**D_1(2420) MASS VALUE** D_1(2420)0 | Dbar_1(2420)0 | D_1(2420)+ | D_1(2420)-	2422.1+-0.6 MeV	symmetric; symmetric: ±0.57502282665337301; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M254 M254M property: 337830	**D_2*(2460) resonance resonance mass** D_2^*(2460)0 | Dbar_2^*(2460)0 | D_2^*(2460)+ | D_2^*(2460)-	2461.1+-0.8 MeV	symmetric; symmetric: ±0.79751842909338577; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M255 M255M property: 334941	**X(1750) MASS:** X(1750)0	1753.8+-2.7 MeV	asymmetric; asymmetric: +2.676722056187673/-2.7341242708038349; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M256 M256M prop. 338520	**D_s0(2590)+ MASS:** D_s0(2590)+ | D_s0(2590)-	2591+-9 MeV	symmetric; symmetric: ±9.2195444572928871; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M257 M257M property: 343153	**B_sJ(6063)0 MASS** B_sJ(6063)0 | Bbar_sJ(6063)0	6063.5+-1.4 MeV	symmetric; symmetric: ±1.442220510185596; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M258 M258M property: 343157	**B_sJ(6114)0 MASS:** B_sJ(6114)0 | Bbar_sJ(6114)0	6114+-6 MeV	symmetric; symmetric: ±5.8309518948453007; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M259 M259M prop. 347963	**T_c cbar sbar 1(4000) MASS:** T_c_cbar_sbar_1(4000)0	3980–4010 MeV	not applicable; not applicable; type V; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M260 M260M property: 348018	**T_c cbar sbar 1(4220)+ MASS** T_c_cbar_sbar_1(4220)+	4220+50-40 MeV	asymmetric; asymmetric: +49.244289008980523/-38.418745424597091; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M261 M261M property: 346810	**chi_c1(4685) MASS:** chi_c1(4685)	4677+13-15 MeV	asymmetric; asymmetric: +13.28238296721786/-15.144963476323969; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M262 M262M prop. 346761	**X(4630) MASS:** X(4630)	4636+29-100 MeV	asymmetric; asymmetric: +29.19090743149394/-102.4064255252877; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M263 M263M property: 334917	**a_0(1710) MASS** a_0(1710)0	1713+-19 MeV	symmetric; symmetric: ±18.627574622936329; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M264 M264M property: 334992	**f_0(1810) Breit-Wigner MASS:** f_0(1810)0	1785+-13 MeV	asymmetric; asymmetric: +13.478123022453611/-13.30532375830475; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M265 M265TMG prop. 347934	**T_c_c(3875)+ T-Matrix Pole sqrt(s)** T_c_c(3875)+	3874.75+-0.10 MeV	not reported; not reported; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M266 M266M property: 335273	**f_0(2470) MASS** f_0(2470)0	2470+6-7 MeV	asymmetric; asymmetric: +5.6568542494923806/-7.2111025509279782; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M267 M267M property: 335028	**eta_1(1855) MASS:** eta_1(1855)0	1855+11-9 MeV	asymmetric; asymmetric: +10.81665382639197/-9.0553851381374173; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M268 M268M prop. 348044	**T_c_c_cbar_cbar(6900)0 MASS VALUE** T_c_c_cbar_cbar(6900)+	6898+-12 MeV	asymmetric; asymmetric: +11.88793882015602/-11.51955409419207; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M269 M269M property: 347926	**T_c_sbar_0*(2900)0 MASS** T_c_sbar_0^*(2900)+	2892+-21 MeV	symmetric; symmetric: ±20.518284528683189; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M270 M270M property: 335221	**omega(2220) MASS:** omega(2220)0	2188+-21 MeV	symmetric; symmetric: ±20.57966470955094; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M272 M272M prop. 348040	**T_c c cbar cbar(6600) MASS:** T_c c cbar cbar(6600)+	6646+28-25 MeV	asymmetric; asymmetric: +28.34909296526099/-24.501265528141591; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M273 M273M property: 348048	**T_c c cbar cbar(7100) MASS** T_c c cbar cbar(7100)+	7134+60-29 MeV	asymmetric; asymmetric: +63.126856408346519/-29.154759474226498; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
Q001 Q001M property: 333589	**d-QUARK MASS:** d | dbar	4.70+-0.07 MeV	symmetric; symmetric: ±0.070000000000000007; type V; limit NONE; confidence 90%	down-type quark, first generation *historical comparison*	flavour–colour coupling, spectral projection and subsequent transport of scheme and scale. exact colour-fibre operator; conditional nominal signature; open scheme.
Q002 Q002M prop. 333588	**u-QUARK MASS:** u | ubar	2.16+-0.07 MeV	symmetric; symmetric: ±0.070000000000000007; type V; limit NONE; confidence 90%	up-type quark, first generation *historical comparison*	flavour–colour coupling, spectral projection and subsequent transport of scheme and scale. exact colour-fibre operator; conditional nominal signature; open scheme.
Q003 Q003M property: 333590	**s-QUARK MASS** s | sbar	92.9+-0.7 MeV	symmetric; symmetric: ±0.69999999999999996; type V; limit NONE; confidence 90%	down-type quark, second generation *historical comparison*	flavour–colour coupling, spectral projection and subsequent transport of scheme and scale. exact colour-fibre operator; conditional nominal signature; open scheme.
Q004 Q004M property: 333604	**c-QUARK MASS:** c | cbar	1.2729+-0.0045 GeV	symmetric; symmetric: ±0.0044999999999999997; type V; limit NONE; confidence 90%	up-type quark, second generation *historical comparison*	flavour–colour coupling, spectral projection and subsequent transport of scheme and scale. exact colour-fibre operator; conditional nominal signature; open scheme.
Q005 Q005M prop. 333611	**b-QUARK MASS:** b | bbar	4.186+-0.006 GeV	symmetric; symmetric: ±0.0060000000000000001; type V; limit NONE; confidence 90%	down-type quark, third generation *historical comparison*	flavour–colour coupling, spectral projection and subsequent transport of scheme and scale. exact colour-fibre operator; conditional nominal signature; open scheme.
Q007 Q007TP property: 333614	**t-Quark Mass (Direct Measurements)** t | tbar	172.60+-0.27 GeV	asymmetric; asymmetric: +0.27137500202757259/-0.2730478067418560 3; type AC; limit NONE; confidence not reported	up-type quark, third generation *historical comparison*	flavour–colour coupling, spectral projection and subsequent transport of scheme and scale. exact colour-fibre operator; conditional nominal signature; open scheme.
Q007 Q007TP2 property: 333615	**t-Quark Mass from Cross-Section Measurements:** t | tbar	162.5+2.1-1.5 GeV	asymmetric; asymmetric: +2.1316137087859799/-1.5324334305524001; type AC; limit NONE; confidence not reported	extended quark multisection *exhaustive record without comparative promotion*	Quark identity with declared scheme and scale. External record accounted for; individual HMT–MD realization not yet linked.
Q007 Q007TP4 prop. 333616	**t-Quark Pole Mass from Cross-Section Measurements:** t | tbar	172.1+-0.6 GeV	asymmetric; asymmetric: +0.61408644700147419/-0.5896776206425237 1; type AC; limit NONE; confidence not reported	extended quark multisection *exhaustive record without comparative promotion*	Quark identity with declared scheme and scale. External record accounted for; individual HMT–MD realization not yet linked.
S000 S000M property: 332461	**gamma MASS:** gamma	<1E-18 eV	not applicable; not applicable; type L; limit U; confidence not reported	electromagnetic gauge sector *exhaustive record without comparative promotion*	no additional map: closed mass nullity. closed null section.

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S003 S003AMU property: 332837	**e MASS: (atomic mass units u)** e- | e+	548.579909044+- 0.000000010 u	symmetric; symmetric: ±9.6999999999999991e-15; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S003 S003M prop. 332838	**e MASS** e- | e+	0.51099895069+- 0.00000000016 MeV	symmetric; symmetric: ±1.5999999999999999e-10; type AC; limit NONE; confidence not reported	charged lepton, first generation *active comparison*	no additional map. electronic beta closed scale.
S004 S004AMU property: 332848	**mu MASS (atomic mass units u)** mu- | mu+	0.1134289257+- 0.0000000025 u	symmetric; symmetric: ±2.5000000000000001e-09; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S004 S004M property: 332849	**mu MASS** mu- | mu+	105.6583755+- 0.0000023 MeV	symmetric; symmetric: ±2.3e-06; type AC; limit NONE; confidence not reported	charged lepton, second generation *active comparison*	Generational coupling and spectral projection without a target mass. Exact fiber operator; conditional nominal signature; open scalarization.
S008 S008M prop. 333660	**pi+- MASS** pi+ | pi-	139.57039+- 0.00018 MeV	symmetric; symmetric: ±0.0001820071604826009; type FC; limit NONE; confidence not reported	elementary section or typed extension *active comparison*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S009 S009M property: 333693	**pi0 MASS** pi0	134.9768+-0.0005 MeV	symmetric; symmetric: ±0.00048567928228423351; type FC; limit NONE; confidence not reported	elementary section or typed extension *active comparison*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S010 S010M property: 335281	**K+- MASS** K+ | K-	493.677+-0.015 MeV	symmetric; symmetric: ±0.015405665226071509; type FC; limit NONE; confidence not reported	elementary section or typed extension *active comparison*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S011 S011M prop. 335501	**K0 MASS:** K0 | Kbar0	497.611+-0.013 MeV	symmetric; symmetric: ±0.0129901997564242; type FC; limit NONE; confidence not reported	elementary section or typed extension *active comparison*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S014 S014M property: 333729	**eta MASS:** eta	547.862+-0.017 MeV	symmetric; symmetric: ±0.017334213711914789; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S016 S016AMU property: 348093	**p MASS: (atomic mass units u)** p | pbar	1.0072764665789+- 0.0000000000083 u	symmetric; symmetric: ±8.2999999999999998e-12; type V; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S016 S016M prop. 348095	**p MASS (MeV):** p | pbar	938.27208943+- 0.00000029 MeV	symmetric; symmetric: ±2.8999999999999998e-07; type V; limit NONE; confidence not reported	elementary section or typed extension *active comparison*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S017 S017AMU property: 348291	**n MASS (atomic mass units u)** n | nbar	1.0086649161+- 0.0000000004 u	symmetric; symmetric: ±4.0000000000000001e-10; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S017 S017M property: 348292	**n MASS: (MeV)** n | nbar	939.5654219+- 0.0000005 MeV	symmetric; symmetric: ±4.7999999999999996e-07; type AC; limit NONE; confidence not reported	elementary section or typed extension *active comparison*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S018 S018M prop. 349367	**Lambda MASS:** Lambda0 | Lambdabar0	1115.683+-0.006 MeV	symmetric; symmetric: ±0.0064376377786219867; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S019 S019M property: 349781	**Sigma+ MASS** Sigma+ | Sigmabar-	1189.37+-0.07 MeV	symmetric; symmetric: ±0.069622885612207835; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.

Continuation of the inventory

Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
S020 S020M property: 349844	**Sigma- MASS:** Sigma- | Sigmabar+	1197.449+-0.029 MeV	symmetric; symmetric: ±0.02915850307660851; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S021 S021M prop. 349828	**Sigma0 MASS:** Sigma0 | Sigmabar0	1192.642+-0.024 MeV	symmetric; symmetric: ±0.023595691425754781; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S022 S022M property: 350344	**Xi- MASS** Xi- | Xibar+	1321.71+-0.07 MeV	symmetric; symmetric: ±0.066535273622543412; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S022 S022M1 property: 350346	**Xibar+ MASS:** Xi- | Xibar+	1321.71+-0.07 MeV	symmetric; symmetric: ±0.066535273622543412; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S023 S023M prop. 350293	**Xi0 MASS:** Xi0 | Xibar0	1314.86+-0.20 MeV	symmetric; symmetric: ±0.20343194946851781; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S024 S024M property: 350469	**Omega- MASS** Omega- | Omegabar+	1672.45+-0.29 MeV	symmetric; symmetric: ±0.29314631976651251; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S026 S026M+ property: 351717	**Sigma_b+ resonance resonance mass** Sigma_b()- | Sigmabar_b()+ | Sigma_b()0 | Sigmabar_b()0 | Sigma_b()+ | Sigmabar_b()-	5810.56+-0.25 MeV	asymmetric; asymmetric: +0.25274062207162651/-0.25263562022485619; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S026 S026M- prop. 351718	**Sigma_b- resonance resonance mass** Sigma_b()- | Sigmabar_b()+ | Sigma_b()0 | Sigmabar_b()0 | Sigma_b()+ | Sigmabar_b()-	5815.64+-0.27 MeV	asymmetric; asymmetric: +0.27460654053367178/-0.27449500386905717; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S031 S031M property: 336044	**D+- MASS** D+ | D-	1869.66+-0.05 MeV	symmetric; symmetric: ±0.046806022411609027; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S032 S032M property: 336689	**D0 MASS:** D0 | Dbar0	1864.84+-0.04 MeV	symmetric; symmetric: ±0.044458705619801281; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S033 S033M prop. 350527	**Lambda_c+ MASS:** Lambda_c()+ | Lambdabar_c()-	2286.46+-0.14 MeV	symmetric; symmetric: ±0.1399999999999989; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S034 S034M property: 337875	**D_s+- MASS** D_s()+ | D_s()-	1968.35+-0.07 MeV	symmetric; symmetric: ±0.067861209529596483; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S035 S035M property: 332893	**tau MASS:** tau- | tau+	1776.93+-0.09 MeV	asymmetric; asymmetric: +0.0849149539132305 48/-0.08758019173411 4063; type AC; limit NONE; confidence not reported	charged lepton, third generation *active comparison*	Generational coupling and spectral projection without a target mass. Exact fiber operator; conditional nominal signature; open scalarization.

Continuation of the inventory

Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
S040 S040M prop. 351420	**Mass of Lambda_b0:** Lambda_b()0 | Lambdabar_b()0	5619.57+-0.16 MeV	symmetric; symmetric: ±0.15728560301875261; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S041 S041M property: 338546	**B+- MASS** B+ | B-	5279.41+-0.07 MeV	symmetric; symmetric: ±0.071087977743437336; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S042 S042M property: 340312	**B0 MASS:** B0 | Bbar0	5279.72+-0.08 MeV	symmetric; symmetric: ±0.084359430922022555; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S043 S043M prop. 332464	**W MASS** W+ | W-	80.3625+-0.0077 GeV	symmetric; symmetric: ±0.0077000000000000002; type V; limit NONE; confidence not reported	charged electroweak gauge sector *active comparison*	linking of the nominal boson to the enriched path and subsequent complex-pole operator. conditional nominal signature; path and pole realization still open.
S044 S044M property: 332513	**Z MASS** Z0	91.1879+-0.0020 GeV	symmetric; symmetric: ±0.0019668990326291501; type AC; limit NONE; confidence not reported	electroweak neutral gauge sector *active comparison*	linking of the nominal boson to the enriched path and subsequent complex-pole operator. conditional nominal signature; path and pole realization still open.
S045 S045M property: 351029	**Xi_c+ MASS:** Xi_c()+ | Xibar_c()-	2467.79+-0.15 MeV	asymmetric; asymmetric: +0.14783797163412299/-0.1515211169302543; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S047 S047M prop. 351327	**Omega_c0 MASS:** Omega_c()0 | Omegabar_c()0	2695.3+-0.4 MeV	symmetric; symmetric: ±0.40393344242041429; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S048 S048M property: 351116	**Xi_c0 MASS** Xi_c()0 | Xibar_c()0	2470.50+-0.25 MeV	asymmetric; asymmetric: +0.2460900310756039/-0.2593629370448782 8; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S053 S053M property: 351372	**Omega_c(2770)0 MASS:** Omega_c(2770)0 | Omegabar_c(2770)0	2766.0+0.9-1.0 MeV	asymmetric; asymmetric: +0.9044279377866114/-1.024667347651941; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S058 S058M prop. 351195	**Xi_c'+ MASS:** Xi_c^'()+ | Xibar_c^'()-	2578.3+-0.4 MeV	symmetric; symmetric: ±0.43863752697991182; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S059 S059M property: 351200	**Xi_c'0 MASS** Xi_c^'()0 | Xibar_c^'()0	2578.8+-0.5 MeV	asymmetric; asymmetric: +0.4802287168835209 2/-0.48697631770634259; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S062 S062M+ property: 351724	**Sigma_b*+ resonance resonance mass** Sigma_b^*()- | Sigmabar_b^*()+ | Sigma_b^*()0 | Sigmabar_b^*()0 | Sigma_b^*()+ | Sigmabar_b^*()-	5830.32+-0.27 MeV	asymmetric; asymmetric: +0.27472915359118399/-0.27501743053510241; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S062 S062M- prop. 351725	**Sigma_b*- resonance resonance mass** Sigma_b^*()- | Sigmabar_b^*()+ | Sigma_b^*()0 | Sigmabar_b^*()0 | Sigma_b^*()+ | Sigmabar_b^*()-	5834.74+-0.30 MeV	asymmetric; asymmetric: +0.2981607994575306/-0.29855805603668217; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S063 S063M property: 351884	**Omega_b- MASS** Omega_b()- | Omegabar_b()+	6045.8+-0.8 MeV	symmetric; symmetric: ±0.76541057086097497; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.

Continuation of the inventory

Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
S065 S065M property: 351404	**Xi_cc+ MASS:** Xi_cc()+ | Xibar_cc()-	3518.9+-0.9 MeV	symmetric; symmetric: ±0.94868329805051377; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S068 S068M prop. 351409	**Xi_cc++ MASS:** Xi_cc()++ | Xi_cc()–	3621.6+-0.4 MeV	symmetric; symmetric: ±0.378021163428716; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S069 S069M- property: 351742	**Xi_b- MASS** Xi_b()- | Xibar_b()+	5797.0+-0.4 MeV	symmetric; symmetric: ±0.43674265363571568; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S070 S070M0 property: 351793	**Xi_b0 MASS:** Xi_b()0 | Xibar_b()0	5791.7+-0.4 MeV	symmetric; symmetric: ±0.41342961949753559; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S074 S074M prop. 338415	**D_s*+- MASS:** D_s^*()+ | D_s^*()-	2112.2+-0.4 MeV	symmetric; symmetric: ±0.39518670911609538; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S085 S085M property: 342531	**B* MASS** B^*0 | Bbar^*0 | B^*+ | B^*-	5324.75+-0.20 MeV	symmetric; symmetric: ±0.20338563693989911; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S086 S086M property: 342595	**B_s0 MASS:** B_s()0 | Bbar_s()0	5366.93+-0.10 MeV	symmetric; symmetric: ±0.09935167389588262; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S087 S087M prop. 343123	**B_s* MASS:** B_s^*0 | Bbar_s^*0	5415.4+-1.4 MeV	symmetric; symmetric: ±1.435389239191782; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S091 S091M property: 343161	**B_c+ MASS** B_c()+ | B_c()-	6274.47+-0.32 MeV	symmetric; symmetric: ±0.31906112267087638; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S126 S126M property: 332733	**H MASS** H	125.13+-0.11 GeV	symmetric; symmetric: ±0.11172144776548799; type AC; limit NONE; confidence not reported	Higgs scalar sector *historical comparison*	linking of the nominal boson to the enriched path and subsequent complex-pole operator. conditional nominal signature; path and pole realization still open.

O.6 Comprehensive inventory of widths and transition rates of the atlas realizations

Table O.3: Complete inventory of 384 widths: identifier, observable, convention, and HMT–MD realization.

Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B004 B004W prop. 348706	**N(1895) BREIT–WIGNER WIDTH:** N(1895)0 | Nbar(1895)0 | N(1895)+ | Nbar(1895)-	80 to 200 (~120) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B005 B005W property: 348837	**N(2060) BREIT-WIGNER WIDTH** N(2060)0 | Nbar(2060)0 | N(2060)+ | Nbar(2060)-	300 to 450 (~400) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B010 B010W property: 349098	**Delta(1700) Breit–Wigner decay decay width** Delta(1700)- | Deltabar(1700)+ | Delta(1700)0 | Deltabar(1700)0 | Delta(1700)+ | Deltabar(1700)- | Delta(1700)++ | Deltabar(1700)–	220 to 380 (~300) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B011 B011W prop. 349152	**Delta(1905) Breit–Wigner decay decay width** Delta(1905)- | Deltabar(1905)+ | Delta(1905)0 | Deltabar(1905)0 | Delta(1905)+ | Deltabar(1905)- | Delta(1905)++ | Deltabar(1905)–	270 to 400 (~330) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B012 B012W property: 349183	**Delta(1910) Breit–Wigner decay decay width** Delta(1910)- | Deltabar(1910)+ | Delta(1910)0 | Deltabar(1910)0 | Delta(1910)+ | Deltabar(1910)- | Delta(1910)++ | Deltabar(1910)–	200 to 400 (~300) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B013 B013W property: 349250	**Delta(1930) Breit–Wigner decay decay width** Delta(1930)- | Deltabar(1930)+ | Delta(1930)0 | Deltabar(1930)0 | Delta(1930)+ | Deltabar(1930)- | Delta(1930)++ | Deltabar(1930)–	200 to 400 (~300) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B014 B014W prop. 348536	**N(1710) BREIT–WIGNER WIDTH:** N(1710)0 | Nbar(1710)0 | N(1710)+ | Nbar(1710)-	80 to 200 (~140) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B015 B015W property: 348570	**N(1720) BREIT-WIGNER WIDTH** N(1720)0 | Nbar(1720)0 | N(1720)+ | Nbar(1720)-	150 to 400 (~250) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B016 B016W property: 348634	**N(1875) BREIT-WIGNER WIDTH** N(1875)0 | Nbar(1875)0 | N(1875)+ | Nbar(1875)-	120 to 250 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B017 B017W prop. 348779	**N(1990) BREIT–WIGNER WIDTH:** N(1990)0 | Nbar(1990)0 | N(1990)+ | Nbar(1990)-	200 to 400 (~300) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B018 B018W property: 348500	**N(1700) BREIT-WIGNER WIDTH** N(1700)0 | Nbar(1700)0 | N(1700)+ | Nbar(1700)-	100 to 300 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B019 B019W property: 349051	**Delta(1600) Breit–Wigner decay decay width** Delta(1600)- | Deltabar(1600)+ | Delta(1600)0 | Deltabar(1600)0 | Delta(1600)+ | Deltabar(1600)- | Delta(1600)++ | Deltabar(1600)–	200 to 300 (~250) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B021 B021W prop. 350411	**Xi(1620) WIDTH:** Xi(1620)- | Xibar(1620)+ | Xi(1620)0 | Xibar(1620)0	32+8-9 MeV	asymmetric; asymmetric: +8.1499070124541788/-9.4119789269834886; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B024 B024W property: 348907	**N(2120) BREIT-WIGNER WIDTH** N(2120)0 | Nbar(2120)0 | N(2120)+ | Nbar(2120)-	260 to 360 (~300) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B030 B030W property: 349127	**Delta(1900) Breit–Wigner decay decay width** Delta(1900)- | Deltabar(1900)+ | Delta(1900)0 | Deltabar(1900)0 | Delta(1900)+ | Deltabar(1900)- | Delta(1900)++ | Deltabar(1900)–	180 to 320 (~250) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B032 B032W prop. 349903	**Sigma(1620) DECAY WIDTH** Sigma(1620)- | Sigmabar(1620)+ | Sigma(1620)0 | Sigmabar(1620)0 | Sigma(1620)+ | Sigmabar(1620)-	40 to 100 (~70) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B033 B033W property: 349035	**Mixed-charge states** Delta(1232)- | Deltabar(1232)+ | Delta(1232)0 | Deltabar(1232)0 | Delta(1232)+ | Deltabar(1232)- | Delta(1232)++ | Deltabar(1232)–	114 to 120 (~117) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B034 B034W property: 350187	**Sigma(2070) decay width** Sigma(2070)- | Sigmabar(2070)+ | Sigma(2070)0 | Sigmabar(2070)0 | Sigma(2070)+ | Sigmabar(2070)-	100 to 300 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B035 B035W prop. 349764	**Lambda(2110) WIDTH:** Lambda(2110)0 | Lambdabar(2110)0	200 to 300 (~250) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B036 B036W property: 349556	**Lambda(1800) WIDTH** Lambda(1800)0 | Lambdabar(1800)0	150 to 250 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B037 B037W1 property: 349428	**PRODUCTION EXPERIMENTS:** Lambda(1405)0 | Lambdabar(1405)0	50.5+-2.0 MeV	symmetric; symmetric: ±1.96116135138184; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B038 B038W prop. 349439	**Lambda(1520) WIDTH:** Lambda(1520)0 | Lambdabar(1520)0	15 to 17 (~16) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B039 B039W property: 349595	**Lambda(1820) WIDTH** Lambda(1820)0 | Lambdabar(1820)0	70 to 90 (~80) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B040 B040W property: 349495	**Lambda(1670) WIDTH:** Lambda(1670)0 | Lambdabar(1670)0	25 to 35 (~30) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B041 B041W prop. 349742	**Lambda(2100) WIDTH:** Lambda(2100)0 | Lambdabar(2100)0	100 to 250 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B042 B042W property: 349776	**Lambda(2350) WIDTH** Lambda(2350)0 | Lambdabar(2350)0	100 to 250 (~150) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B043 B043W+ property: 349879	**Sigma(1385)+ decay width** Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)-	36.2+-0.7 MeV	asymmetric; asymmetric: +0.69132395371450484/-0.70656746664477266; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B043 B043W0 prop. 349880	**Sigma(1385)0 DECAY WIDTH** Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)-	44+-8 MeV	symmetric; symmetric: ±8.1680421550261997; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B043 B043W- property: 349881	**Sigma(1385)- WIDTH** Sigma(1385)- | Sigmabar(1385)+ | Sigma(1385)0 | Sigmabar(1385)0 | Sigma(1385)+ | Sigmabar(1385)-	39.4+-2.1 MeV	symmetric; symmetric: ±2.080864718865612; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B044 B044W property: 349947	**Sigma(1670) decay width** Sigma(1670)- | Sigmabar(1670)+ | Sigma(1670)0 | Sigmabar(1670)0 | Sigma(1670)+ | Sigmabar(1670)-	40 to 100 (~70) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B045 B045W prop. 349984	**Sigma(1775) DECAY WIDTH** Sigma(1775)- | Sigmabar(1775)+ | Sigma(1775)0 | Sigmabar(1775)0 | Sigma(1775)+ | Sigmabar(1775)-	105 to 135 (~120) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B046 B046W property: 350081	**Sigma(1915) WIDTH** Sigma(1915)- | Sigmabar(1915)+ | Sigma(1915)0 | Sigmabar(1915)0 | Sigma(1915)+ | Sigmabar(1915)-	80 to 160 (~120) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B047 B047W property: 350167	**Sigma(2030) decay width** Sigma(2030)- | Sigmabar(2030)+ | Sigma(2030)0 | Sigmabar(2030)0 | Sigma(2030)+ | Sigmabar(2030)-	150 to 200 (~180) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B048 B048W prop. 350278	**Sigma(2250) DECAY WIDTH** Sigma(2250)- | Sigmabar(2250)+ | Sigma(2250)0 | Sigmabar(2250)0 | Sigma(2250)+ | Sigmabar(2250)-	60 to 150 (~100) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B049 B049W0 property: 350405	**Xi(1530)0 WIDTH** Xi(1530)- | Xibar(1530)+ | Xi(1530)0 | Xibar(1530)0	9.1+-0.5 MeV	symmetric; symmetric: ±0.48058271927942953; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B049 B049W- property: 350406	**Xi(1530)- WIDTH:** Xi(1530)- | Xibar(1530)+ | Xi(1530)0 | Xibar(1530)0	9.9+1.7-1.9 MeV	asymmetric; asymmetric: +1.731350237665344/-1.930203247244187; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B050 B050W prop. 350422	**Xi(1820) WIDTH:** Xi(1820)- | Xibar(1820)+ | Xi(1820)0 | Xibar(1820)0	24+15-10 MeV	asymmetric; asymmetric: +15/-10; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B052 B052W property: 350438	**Xi(1950) WIDTH** Xi(1950)- | Xibar(1950)+ | Xi(1950)0 | Xibar(1950)0	60+-20 MeV	symmetric; symmetric: ±20; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B055 B055W property: 349519	**Lambda(1690) WIDTH:** Lambda(1690)0 | Lambdabar(1690)0	60 to 80 (~70) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B056 B056W prop. 349615	**Lambda(1830) WIDTH:** Lambda(1830)0 | Lambdabar(1830)0	60 to 120 (~90) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B057 B057W property: 349967	**Sigma(1750) WIDTH** Sigma(1750)- | Sigmabar(1750)+ | Sigma(1750)0 | Sigmabar(1750)0 | Sigma(1750)+ | Sigmabar(1750)-	100 to 200 (~150) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B060 B060W property: 349636	**Lambda(1890) WIDTH:** Lambda(1890)0 | Lambdabar(1890)0	80 to 160 (~120) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B061 B061W prop. 348327	**N(1440) BREIT–WIGNER WIDTH:** N(1440)0 | Nbar(1440)0 | N(1440)+ | Nbar(1440)-	250 to 450 (~350) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B062 B062W property: 348350	**N(1520) BREIT-WIGNER WIDTH** N(1520)0 | Nbar(1520)0 | N(1520)+ | Nbar(1520)-	100 to 120 (~110) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B063 B063W property: 348384	**N(1535) BREIT-WIGNER WIDTH** N(1535)0 | Nbar(1535)0 | N(1535)+ | Nbar(1535)-	125 to 175 (~150) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B064 B064W prop. 348436	**N(1675) BREIT–WIGNER WIDTH:** N(1675)0 | Nbar(1675)0 | N(1675)+ | Nbar(1675)-	130 to 160 (~145) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B065 B065W property: 348468	**N(1680) BREIT-WIGNER WIDTH** N(1680)0 | Nbar(1680)0 | N(1680)+ | Nbar(1680)-	115 to 130 (~120) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B066 B066W property: 348411	**N(1650) BREIT–WIGNER WIDTH:** N(1650)0 | Nbar(1650)0 | N(1650)+ | Nbar(1650)-	100 to 150 (~125) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B067 B067W prop. 350000	**Sigma(1880) DECAY WIDTH** Sigma(1880)- | Sigmabar(1880)+ | Sigma(1880)0 | Sigmabar(1880)0 | Sigma(1880)+ | Sigmabar(1880)-	100 to 300 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B068 B068W property: 350445	**Xi(2030) WIDTH** Xi(2030)- | Xibar(2030)+ | Xi(2030)0 | Xibar(2030)0	20+15-5 MeV	asymmetric; asymmetric: +15/-5; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B071 B071W property: 348954	**N(2190) BREIT-WIGNER WIDTH** N(2190)0 | Nbar(2190)0 | N(2190)+ | Nbar(2190)-	300 to 500 (~400) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B077 B077W prop. 349577	**Lambda(1810) WIDTH:** Lambda(1810)0 | Lambdabar(1810)0	50 to 170 (~110) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B079 B079W property: 349921	**Sigma(1660) WIDTH** Sigma(1660)- | Sigmabar(1660)+ | Sigma(1660)0 | Sigmabar(1660)0 | Sigma(1660)+ | Sigmabar(1660)-	100 to 300 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B082 B082W property: 349079	**Delta(1620) Breit–Wigner decay decay width** Delta(1620)- | Deltabar(1620)+ | Delta(1620)0 | Deltabar(1620)0 | Delta(1620)+ | Deltabar(1620)- | Delta(1620)++ | Deltabar(1620)–	110 to 150 (~130) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B083 B083W prop. 349309	**Delta(1950) Breit–Wigner decay decay width** Delta(1950)- | Deltabar(1950)+ | Delta(1950)0 | Deltabar(1950)0 | Delta(1950)+ | Deltabar(1950)- | Delta(1950)++ | Deltabar(1950)–	235 to 335 (~285) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B084 B084W property: 349360	**Delta(2420) Breit–Wigner decay decay width** Delta(2420)- | Deltabar(2420)+ | Delta(2420)0 | Deltabar(2420)0 | Delta(2420)+ | Deltabar(2420)- | Delta(2420)++ | Deltabar(2420)–	300 to 700 (~500) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B087 B087W property: 348672	**N(1880) BREIT-WIGNER WIDTH** N(1880)0 | Nbar(1880)0 | N(1880)+ | Nbar(1880)-	200 to 400 (~300) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B088 B088W prop. 350189	**Sigma(2080) DECAY WIDTH** Sigma(2080)- | Sigmabar(2080)+ | Sigma(2080)0 | Sigmabar(2080)0 | Sigma(2080)+ | Sigmabar(2080)-	100 to 240 (~170) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B090 B090W property: 348987	**N(2220) BREIT-WIGNER WIDTH** N(2220)0 | Nbar(2220)0 | N(2220)+ | Nbar(2220)-	350 to 500 (~400) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B098 B098W property: 350051	**Sigma(1910) decay width** Sigma(1910)- | Sigmabar(1910)+ | Sigma(1910)0 | Sigmabar(1910)0 | Sigma(1910)+ | Sigmabar(1910)-	150 to 300 (~220) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B101 B101W prop. 349473	**Lambda(1600) WIDTH:** Lambda(1600)0 | Lambdabar(1600)0	150 to 250 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B102 B102W property: 350937	**Lambda_c(2625)+ WIDTH** Lambda_c(2625)+ | Lambdabar_c(2625)-	<0.52 MeV	not applicable; not applicable; type L; limit U; confidence 90%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B104 B104W++ property: 350997	**Sigma_c(2455)++ decay decay width** Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)–	1.89+0.09-0.18 MeV	asymmetric; asymmetric: +0.085968740877688063/-0.17741442138916921; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B104 B104W+ prop. 350998	**Sigma_c(2455)+ decay decay width** Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)–	2.3+-0.4 MeV	symmetric; symmetric: ±0.42426406871192851; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B104 B104W0 property: 350999	**Sigma_c(2455)0 decay decay width** Sigma_c(2455)0 | Sigmabar_c(2455)0 | Sigma_c(2455)+ | Sigmabar_c(2455)- | Sigma_c(2455)++ | Sigmabar_c(2455)–	1.83+0.11-0.19 MeV	asymmetric; asymmetric: +0.1113419499366561/-0.19347936378811631; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B105 B105W property: 350413	**MIXED CHARGES** Xi(1690)- | Xibar(1690)+ | Xi(1690)0 | Xibar(1690)0	20+-15 MeV	symmetric; symmetric: ±15; type E; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B113 B113W prop. 349005	**N(2250) BREIT–WIGNER WIDTH:** N(2250)0 | Nbar(2250)0 | N(2250)+ | Nbar(2250)-	300 to 600 (~500) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B115 B115W++ property: 351011	**Sigma_c(2520)++ decay decay width** Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)–	14.78+0.30-0.40 MeV	asymmetric; asymmetric: +0.3022109829484174 2/-0.37821726734358602; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B115 B115W+ property: 351012	**Sigma_c(2520)+ decay decay width** Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)–	17.2+4.0-2.2 MeV	asymmetric; asymmetric: +3.8600518131237571/-2.213594362117866; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B115 B115W0 prop. 351013	**Sigma_c(2520)0 decay decay width** Sigma_c(2520)0 | Sigmabar_c(2520)0 | Sigma_c(2520)+ | Sigmabar_c(2520)- | Sigma_c(2520)++ | Sigmabar_c(2520)–	15.3+0.4-0.5 MeV	asymmetric; asymmetric: +0.43805327113015552/-0.49172579862174309; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B117 B117W property: 349209	**Delta(1920) Breit–Wigner decay decay width** Delta(1920)- | Deltabar(1920)+ | Delta(1920)0 | Deltabar(1920)0 | Delta(1920)+ | Deltabar(1920)- | Delta(1920)++ | Deltabar(1920)–	240 to 360 (~300) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B119 B119W property: 350920	**Lambda_c(2595)+ WIDTH:** Lambda_c(2595)+ | Lambdabar_c(2595)-	2.6+-0.6 MeV	symmetric; symmetric: ±0.55758407437802593; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B120 B120W prop. 349023	**N(2600) BREIT–WIGNER WIDTH:** N(2600)0 | Nbar(2600)0 | N(2600)+ | Nbar(2600)-	500 to 800 (~650) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B122 B122W property: 350980	**Lambda_c(2940)+ WIDTH** Lambda_c(2940)0 | Lambdabar_c(2940)0	20+6-5 MeV	asymmetric; asymmetric: +5.9819637952003841/-5.2285381312943766; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B130 B130W+ property: 351281	**Xi_c(2970)+ WIDTH** Xi_c(2970)0 | Xibar_c(2970)0	27.2+2.8-3.3 MeV	asymmetric; asymmetric: +2.7691726997636179/-3.3147478336925231; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B132 B132W prop. 348872	**N(2100) BREIT–WIGNER WIDTH:** N(2100)0 | Nbar(2100)0 | N(2100)+ | Nbar(2100)-	200 to 320 (~260) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B135 B135W property: 349340	**Delta(2200) Breit–Wigner decay decay width** Delta(2200)- | Deltabar(2200)+ | Delta(2200)0 | Deltabar(2200)0 | Delta(2200)+ | Deltabar(2200)- | Delta(2200)++ | Deltabar(2200)–	200 to 500 (~350) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B136 B136W property: 349273	**Delta(1940) Breit–Wigner decay decay width** Delta(1940)- | Deltabar(1940)+ | Delta(1940)0 | Deltabar(1940)0 | Delta(1940)+ | Deltabar(1940)- | Delta(1940)++ | Deltabar(1940)–	300 to 500 (~400) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B141 B141W prop. 350517	**Omega(2250)- WIDTH:** Omega(2250)- | Omegabar(2250)+	55+-18 MeV	symmetric; symmetric: ±17.698364447639651; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B143 B143W property: 350524	**Omega(2470)- WIDTH** Omega(2470)- | Omegabar(2470)+	72+-33 MeV	symmetric; symmetric: ±33; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B144 B144W property: 348741	**N(1900) BREIT-WIGNER WIDTH** N(1900)0 | Nbar(1900)0 | N(1900)+ | Nbar(1900)-	100 to 320 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B146 B146W+ prop. 351213	**Xi_c(2645)+ decay decay width** Xi_c(2645)0 | Xibar_c(2645)0 | Xi_c(2645)+ | Xibar_c(2645)-	2.14+-0.19 MeV	symmetric; symmetric: ±0.1898989601311741; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B146 B146W0 property: 351214	**Xi_c(2645)0 decay decay width** Xi_c(2645)0 | Xibar_c(2645)0 | Xi_c(2645)+ | Xibar_c(2645)-	2.35+-0.22 MeV	symmetric; symmetric: ±0.22203603311174519; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B147 B147W+ property: 351309	**Xi_c(3080)+ WIDTH:** Xi_c(3080)0 | Xibar_c(3080)0	3.8+-0.9 MeV	symmetric; symmetric: ±0.93945766227499938; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B147 B147W0 prop. 351310	**Xi_c(3080)0 WIDTH:** Xi_c(3080)0 | Xibar_c(3080)0	5.6+-2.2 MeV	symmetric; symmetric: ±2.179860108626912; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B148 B148W+ property: 351240	**Xi_c(2815)+ decay decay width** Xi_c(2815)0 | Xibar_c(2815)0 | Xi_c(2815)+ | Xibar_c(2815)-	2.15+-0.15 MeV	symmetric; symmetric: ±0.15193272999421351; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B148 B148W0 property: 351241	**Xi_c(2815)0 decay decay width** Xi_c(2815)0 | Xibar_c(2815)0 | Xi_c(2815)+ | Xibar_c(2815)-	2.54+-0.25 MeV	symmetric; symmetric: ±0.24758836806279891; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B149 B149W+ prop. 351224	**Xi_c(2790)+ decay decay width** Xi_c(2790)0 | Xibar_c(2790)0 | Xi_c(2790)+ | Xibar_c(2790)-	8.9+-1.0 MeV	symmetric; symmetric: ±1; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B149 B149W0 property: 351225	**Xi_c(2790)0 decay decay width** Xi_c(2790)0 | Xibar_c(2790)0 | Xi_c(2790)+ | Xibar_c(2790)-	10.0+-1.1 MeV	symmetric; symmetric: ±1.063014581273465; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B150 B150W property: 350954	**Lambda_c(2765)+ WIDTH:** Lambda_c(2765)0 | Lambdabar_c(2765)0	50 MeV	symmetric; symmetric: ±0; type L; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B151 B151W prop. 350964	**Lambda_c(2880)+ WIDTH:** Lambda_c(2880)+ | Lambdabar_c(2880)-	5.6+0.8-0.6 MeV	asymmetric; asymmetric: +0.77200097671302059/-0.62289222418902057; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B155 B155W++ property: 351025	**Sigma_c(2800)++ WIDTH** Sigma_c(2800)0 | Sigmabar_c(2800)0	75+22-17 MeV	asymmetric; asymmetric: +21.633307652783941/-17.029386365926399; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B155 B155W+ property: 351026	**Sigma_c(2800)+ WIDTH** Sigma_c(2800)0 | Sigmabar_c(2800)0	60+60-40 MeV	asymmetric; asymmetric: +63.820059542435402/-44.418464629025621; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B155 B155W0 prop. 351027	**Sigma_c(2800)0 WIDTH:** Sigma_c(2800)0 | Sigmabar_c(2800)0	72+22-15 MeV	asymmetric; asymmetric: +22.093560797468029/-14.82346078512144l; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B156 B156A05 property: 351265	**Xi_c(2930)+ WIDTH** Xi_c(2930)0 | Xibar_c(2930)0	15+-9 MeV	symmetric; symmetric: ±9.1482238713315276; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B156 B156A06 property: 351266	**Xi_c(2930)0 WIDTH:** Xi_c(2930)0 | Xibar_c(2930)0	10.2+-1.4 MeV	symmetric; symmetric: ±1.360147050873544; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B157 B157W+ prop. 351299	**Xi_c(3055)+ WIDTH:** Xi_c(3055)0 | Xibar_c(3055)0	8.0+-0.8 MeV	symmetric; symmetric: ±0.76391812853200158; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B157 B157W property: 351300	**Xi_c(3055)0 WIDTH** Xi_c(3055)0 | Xibar_c(3055)0	12.4+-2.3 MeV	symmetric; symmetric: ±2.282542442102665; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B158 B158W+ property: 351326	**Xi_c(3123)+ WIDTH:** Xi_c(3123)0 | Xibar_c(3123)0	4+-4 MeV	symmetric; symmetric: ±3.801315561749643; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B161 B161W prop. 351845	**Xi_b(5945)0 WIDTH:** Xi_b(5945)0 | Xibar_b(5945)0	0.87+-0.07 MeV	asymmetric; asymmetric: +0.075069672174583921/-0.074706305477537885; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B162 B162W property: 351695	**Lambda_b(5912)0 WIDTH** Lambda_b(5912)0 | Lambdabar_b(5912)0	<0.25 MeV	not applicable; not applicable; type L; limit U; confidence 90%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B163 B163W property: 351699	**Lambda_b(5920)0 WIDTH:** Lambda_b(5920)0 | Lambdabar_b(5920)0	<0.19 MeV	not applicable; not applicable; type L; limit U; confidence 90%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B164 B164W prop. 349533	**Lambda(1710) WIDTH:** Lambda(1710)0 | Lambdabar(1710)0	180+-40 MeV	symmetric; symmetric: ±42; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B165 B165W property: 349652	**Lambda(2050) WIDTH** Lambda(2050)0 | Lambdabar(2050)0	490+-60 MeV	symmetric; symmetric: ±61; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B166 B166W property: 349992	**Sigma(1780) decay width** Sigma(1780)- | Sigmabar(1780)+ | Sigma(1780)0 | Sigmabar(1780)0 | Sigma(1780)+ | Sigmabar(1780)-	100 to 300 (~200) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B167 B167W prop. 350020	**Sigma(1900) DECAY WIDTH** Sigma(1900)- | Sigmabar(1900)+ | Sigma(1900)0 | Sigmabar(1900)0 | Sigma(1900)+ | Sigmabar(1900)-	140 to 190 (~165) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B168 B168W property: 350103	**Sigma(1940) WIDTH** Sigma(1940)- | Sigmabar(1940)+ | Sigma(1940)0 | Sigmabar(1940)0 | Sigma(1940)+ | Sigmabar(1940)-	100 to 400 (~250) MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B169 B169W property: 351841	**Xi_b'(5935)- WIDTH** Xi_b^'(5935)- | Xibar_b^'(5935)+	0.03+-0.032 MeV	symmetric; symmetric: ±0.031622776601683791; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B170 B170W prop. 351849	**Xi_b(5955)- WIDTH:** Xi_b(5955)- | Xibar_b(5955)+	1.43+-0.11 MeV	symmetric; symmetric: ±0.11313708498984761; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B171 B171W property: 351952	**P_c_cbar(4380)+ WIDTH** P_c_cbar(4380)+	210+-90 MeV	symmetric; symmetric: ±87.863530545955186; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B172 B172W property: 351978	**P_c_cbar(4457)+ WIDTH:** P_c_cbar(4457)+	6.4+6.0-2.8 MeV	asymmetric; asymmetric: +6.0406953242155828/-2.758622844826744; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B173 B173W prop. 351377	**Omega_c(3000)0 WIDTH:** Omega_c(3000)0 | Omegabar_c(3000)0	3.8+1.6-0.4 MeV	asymmetric; asymmetric: +1.606549096666517/-0.37013511046643488; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B174 B174W property: 351381	**Omega_c(3050)0 WIDTH** Omega_c(3050)0 | Omegabar_c(3050)0	<1.8 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B175 B175W property: 351385	**Omega_c(3065)0 WIDTH:** Omega_c(3065)0 | Omegabar_c(3065)0	3.4+0.7-0.8 MeV	asymmetric; asymmetric: +0.68496637416672579/-0.79385852123859457; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B176 B176W prop. 351389	**Omega_c(3090)0 WIDTH:** Omega_c(3090)0 | Omegabar_c(3090)0	8.5+0.8-1.7 MeV	asymmetric; asymmetric: +0.75213030785895074/-1.678689965419464; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B177 B177W property: 351393	**Omega_c(3120)0 WIDTH** Omega_c(3120)0 | Omegabar_c(3120)0	<2.5 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B178 B178W property: 350957	**Lambda_c(2860)+ WIDTH** Lambda_c(2860)0 | Lambdabar_c(2860)0	68+12-22 MeV	asymmetric; asymmetric: +11.78049234964312/-21.62336699036484; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B179 B179W prop. 350507	**Omega(2012)- WIDTH:** Omega(2012)- | Omegabar(2012)+	6.3+-2.0 MeV	asymmetric; asymmetric: +2.074289545754965/-1.919724985870655 1; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B180 B180W property: 351865	**Xi_b(6227)- WIDTH** Xi_b(6227)- | Xi_b(6227)+	19.9+-2.6 MeV	symmetric; symmetric: ±2.5806975801127878; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B181 B181W property: 351735	**Sigma_b(6097)+ WIDTH:** Sigma_b(6097)+ | Sigmabar_b(6097)-	31+-6 MeV	symmetric; symmetric: ±5.5443665102516446; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B182 B182W prop. 351739	**Sigma_b(6097)- WIDTH:** Sigma_b(6097)- | Sigmabar_b(6097)+	29+-4 MeV	symmetric; symmetric: ±4.2953463189829062; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B183 B183W property: 350229	**Sigma(2110) WIDTH** Sigma(2110)- | Sigmabar(2110)+ | Sigma(2110)0 | Sigmabar(2110)0 | Sigma(2110)+ | Sigmabar(2110)-	310+120-50 MeV	asymmetric; asymmetric: +122.34963012126789/-54.761074523641852; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B184 B184W property: 350254	**Sigma(2230) decay width** Sigma(2230)- | Sigmabar(2230)+ | Sigma(2230)0 | Sigmabar(2230)0 | Sigma(2230)+ | Sigmabar(2230)-	350+-50 MeV	symmetric; symmetric: ±50; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
B185 B185W prop. 351936	**P_c_cbar(4312)+ WIDTH:** P_c_cbar(4312)+	10+-5 MeV	asymmetric; asymmetric: +4.5803929962395147/-5.2478567053607703; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B186 B186W property: 351956	**P_c_cbar(4440)+ WIDTH** P_c_cbar(4440)+	21+10-11 MeV	asymmetric; asymmetric: +9.9849887330932923/-11.225862995778989; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B187 B187W property: 351708	**Lambda_b(6146)0 WIDTH:** Lambda_b(6146)0 | Lambdabar_b(6146)0	2.9+-1.3 MeV	symmetric; symmetric: ±1.3341664064126 34; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B188 B188W prop. 351714	**Lambda_b(6152)0 WIDTH:** Lambda_b(6152)0 | Lambdabar_b(6152)0	2.1+-0.9 MeV	symmetric; symmetric: ±0.8544003745317531; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B190 B190W property: 349677	**Lambda(2070) WIDTH** Lambda(2070)0 | Lambdabar(2070)0	370+-50 MeV	symmetric; symmetric: ±50; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B191 B191W property: 349712	**Lambda(2080) WIDTH:** Lambda(2080)0 | Lambdabar(2080)0	181+-29 MeV	symmetric; symmetric: ±29; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B192 B192W0 prop. 351257	**Xi_c(2923)0 WIDTH:** Xi_c(2923)0 | Xibar_c(2923)0	5.8+-1.3 MeV	symmetric; symmetric: ±1.30796716242523; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B192 B192WP property: 351258	**Xi_c(2923)+ WIDTH** Xi_c(2923)0 | Xibar_c(2923)0	5.3+-1.7 MeV	symmetric; symmetric: ±1.6643316977093241; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B193 B193W property: 351906	**Omega_b(6316)- WIDTH:** Omega_b(6316)- | Omegabar_b(6316)+	<4.2 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B194 B194W prop. 351910	**Omega_b(6330)- WIDTH:** Omega_b(6330)- | Omegabar_b(6330)+	<4.7 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B195 B195W property: 351914	**Omega_b(6340)- WIDTH** Omega_b(6340)- | Omegabar_b(6340)+	<1.8 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B196 B196W property: 351918	**Omega_b(6350)- WIDTH** Omega_b(6350)- | Omegabar_b(6350)+	<3.2 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B197 B197W prop. 351703	**Lambda_b(6070)0 WIDTH:** Lambda_b(6070)0 | Lambdabar_b(6070)0	72+-11 MeV	symmetric; symmetric: ±11.180339887498951; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B199 B199W property: 351871	**Xi_b(6227)0 WIDTH** Xi_b(6227)0 | Xibar_b(6227)0	19+5-4 MeV	asymmetric; asymmetric: +5.1923019942988677/-4.3324358044868934; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B200 B200W property: 351861	**Xi_b(6100)- WIDTH:** Xi_b(6100)- | Xibar_b(6100)+	0.94+-0.31 MeV	symmetric; symmetric: ±0.31048349392520053; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B201 B201W prop. 351875	**Xi_b(6327)0 WIDTH:** Xi_b(6327)0 | Xibar_b(6327)0	<2.56 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B202 B202W property: 351879	**Xi_b(6333)0 WIDTH** Xi_b(6333)0 | Xibar_b(6333)0	<1.92 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.

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B203 B203W0 property: 351252	**Xi_c(2882) WIDTH:** Xi_c(2882)0 | Xibar_c(2882)0	12+-8 MeV	symmetric; symmetric: ±7.8568441501661468; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B204 B204W prop. 351946	**P_c_cbar_s(4338)0 WIDTH:** P_c_cbar_s(4338)+	7.0+-1.8 MeV	symmetric; symmetric: ±1.769180601295413; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B205 B205W property: 352000	**P_c_cbar(4312)+ WIDTH:** P_c_cbar(4312)+	19+8-7 MeV	asymmetric; asymmetric: +8.1568830165262654/-7.255202813989527; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B206 B206W property: 350974	**Lambda_c(2910)+ WIDTH:** Lambda_c(2910)0 | Lambdabar_c(2910)0	52+-27 MeV	symmetric; symmetric: ±27.448861542876418; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B207 B207W prop. 351853	**Xi_b(6087)0 WIDTH:** Xi_b(6087)	2.4+-0.5 MeV	symmetric; symmetric: ±0.5197114584074513 7; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B208 B208W property: 351857	**Xi_b(6095)0 WIDTH** Xi_b(6095)	0.50+-0.35 MeV	symmetric; symmetric: ±0.3478505426185217 2; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B209 B209W property: 351397	**Omega_c(3185)0 WIDTH** Omega_c(3185)0 | Omegabar_c(3185)0	50+12-21 MeV	asymmetric; asymmetric: +12.2065556157337/-21.189620100417091; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B210 B210W prop. 351401	**Omega_c(3327)0 WIDTH:** Omega_c(3327)0 | Omegabar_c(3327)0	20+14-5 MeV	asymmetric; asymmetric: +13.928388277184119/-5.0990195135927836; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
B211 B211W property: 350515	**Omega(2109)- WIDTH** Omega(2109)- | Omegabar(2109)+	18+-17 MeV	symmetric; symmetric: ±17.362315513778679; type AC; limit NONE; confidence not reported	composite baryonic record *exhaustive record without comparative promotion*	nominal baryonic binding map. External record accounted for; individual HMT–MD realization not yet linked.
M001 M001W property: 333933	**omega(782) WIDTH:** omega(782)0	8.68+-0.13 MeV	symmetric; symmetric: ±0.1320000870776446; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M002 M002W prop. 334007	**eta'(958) WIDTH:** eta^'(958)0	0.188+-0.006 MeV	symmetric; symmetric: ±0.0064442212265340033; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M003 M003W1 property: 334164	**f_0(980) WIDTH** f_0(980)0	10 to 100 MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M004 M004W property: 334183	**phi(1020) WIDTH:** phi(1020)0	4.249+-0.013 MeV	symmetric; symmetric: ±0.01311655783805327; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M005 M005W prop. 334355	**f_2(1270) WIDTH:** f_2(1270)0	186.6+2.8-2.2 MeV	asymmetric; asymmetric: +2.802312637678575/-2.196115628622218 1; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M006 M006W property: 334555	**f_1(1420) WIDTH** f_1(1420)0	55.4+-3.2 MeV	asymmetric; asymmetric: +3.1572143311904868/-3.199679790915769; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M008 M008W property: 334404	**f_1(1285) WIDTH:** f_1(1285)0	23.0+-1.0 MeV	asymmetric; asymmetric: +1.045581016397475/-1.0256159102073861; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M009 M009W0 prop. 333870	**neutral state only; e+ e- channel** rho(770)0 | rho(770)+ | rho(770)-	147.4+-0.8 MeV	symmetric; symmetric: ±0.82096324211264493; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009W5 property: 333871	**CHARGED ONLY, tau DECAYS and e+ e-** rho(770)0 | rho(770)+ | rho(770)-	149.1+-0.8 MeV	symmetric; symmetric: ±0.84906762313788198; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009W7 property: 333873	**MIXED CHARGES; OTHER REACTIONS** rho(770)0 | rho(770)+ | rho(770)-	149.5+-1.3 MeV	symmetric; symmetric: ±1.3; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009W2 prop. 333874	**charged states only; produced hadronically** rho(770)0 | rho(770)+ | rho(770)-	150.2+-2.4 MeV	symmetric; symmetric: ±2.3925241925869849; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009W0P property: 333876	**NEUTRAL ONLY, PHOTOPRODUCED** rho(770)0 | rho(770)+ | rho(770)-	152.3+-1.7 MeV	asymmetric; asymmetric: +1.6144196138431619/-1.7476550950891081; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M009 M009W0R property: 333877	**NEUTRAL ONLY; OTHER REACTIONS** rho(770)0 | rho(770)+ | rho(770)-	150.9+-1.7 MeV	symmetric; symmetric: ±1.7341880321909591; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M010 M010W prop. 334333	**a_1(1260) WIDTH VALUE** a_1(1260)0 | a_1(1260)+ | a_1(1260)-	250 to 600 MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M011 M011W property: 334310	**b_1(1235) WIDTH** b_1(1235)0 | b_1(1235)+ | b_1(1235)-	142+-9 MeV	symmetric; symmetric: ±8.5057630105759703; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M012 M012W property: 334474	**a_2(1320) decay width** a_2(1320)0 | a_2(1320)+ | a_2(1320)-	107+-5 MeV	symmetric; symmetric: ±5; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M012 M012W1 prop. 334476	**3pi MODE VALUE** a_2(1320)0 | a_2(1320)+ | a_2(1320)-	105.0+1.7-1.9 MeV	asymmetric; asymmetric: +1.672423941919887/-1.8523961583384709; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M012 M012W2 property: 334477	**K Kbar MODE** a_2(1320)0 | a_2(1320)+ | a_2(1320)-	109.8+-2.4 MeV	symmetric; symmetric: ±2.3748902427360412; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M012 M012W3 property: 334478	**eta pi MODE** a_2(1320) 0 | a_2(1320)+ | a_2(1320)-	111.1+-2.4 MeV	symmetric; symmetric: ±2.448965935023065; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M012 M012W4 prop. 334479	**eta' pi MODE VALUE** a_2(1320)0 | a_2(1320)+ | a_2(1320)-	119+-25 MeV	symmetric; symmetric: ±25.063151645472001; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013WX property: 334671	**f_2'(1525) WIDTH** f_2^'(1525)0	72+7-6 MeV	asymmetric; asymmetric: +6.8828574538793053/-5.7886890416736358; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013W2 property: 334673	**PRODUCED BY K+- BEAM:** f_2^'(1525)0	82+-6 MeV	asymmetric; asymmetric: +6.0690031263411646/-5.743699645561148; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013W3 prop. 334674	**PRODUCED IN e+ e- ANNIHILATION AND PARTICLE DECAYS:** f_2^'(1525)0	86+-4 MeV	asymmetric; asymmetric: +4.2724130046702271/-3.9578983915293739; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M013 M013W9 property: 334675	**PRODUCED IN pbar p ANNIHILATION** f_2^'(1525)0	77+-5 MeV	symmetric; symmetric: ±4.7999999999999998; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013W4 property: 334676	**CENTRAL PRODUCTION:** f_2^'(1525)0	70+-25 MeV	symmetric; symmetric: ±25; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M013 M013W10 prop. 334677	**PRODUCED IN e p COLLISIONS:** f_2^'(1525)0	83+-10 MeV	asymmetric; asymmetric: +10.295630140987001/-9.8488578017961039; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M014 M014W property: 333858	**f_0(500) BREIT-WIGNER WIDTH** f_0(500)0	100 to 800 MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015W property: 334848	**2pi, K Kbar, and K Kbar pi decay modes** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	161+-10 MeV	asymmetric; asymmetric: +9.7014800918649904/-9.6537321264976139; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015W1 prop. 334849	**2pi MODE VALUE** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	186+-14 MeV	symmetric; symmetric: ±14.046724973634189; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015W2 property: 334850	**K Kbar AND K Kbar pi MODES** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	204+-18 MeV	symmetric; symmetric: ±17.888543819998318; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015W3 property: 334851	**(4pi)± MODE** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	129+-10 MeV	asymmetric; asymmetric: +10.284400967958939/-10.16157273699389; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015W5 prop. 334852	**omega pi MODE VALUE** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	190+-40 MeV	symmetric; symmetric: ±35.527838423396418; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M015 M015W6 property: 334853	**eta pi+ pi- MODE** rho_3(1690)0 | rho_3(1690)+ | rho_3(1690)-	130+-40 MeV	symmetric; symmetric: ±43.467351876049257; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M016 M016W property: 335131	**f_4(2050) WIDTH:** f_4(2050)0	237+-18 MeV	asymmetric; asymmetric: +17.880299432527 17/-18.179532837552909; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M017 M017W prop. 335091	**a_4(1970) WIDTH VALUE** a_4(1970)0 | a_4(1970)+ | a_4(1970)-	324+15-18 MeV	asymmetric; asymmetric: +14.533393315985499/-18.100116797853431; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M018 M018W1 property: 335795	**CHARGED STATES ONLY; HADRONICALLY PRODUCED** K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)-	48.5+-1.2 MeV	symmetric; symmetric: ±1.189202181804293; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M018 M018W5 property: 335797	**CHARGED STATES ONLY; PRODUCTION IN tau-LEPTON DECAYS** K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)-	46.2+-1.3 MeV	symmetric; symmetric: ±1.3416407864998741; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M018 M018W2 prop. 335798	**NEUTRAL STATE ONLY** K^*(892)0 | Kbar^*(892)0 | K^*(892)+ | K^*(892)-	47.1+-0.5 MeV	symmetric; symmetric: ±0.49347496850837202; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M019 M019W property: 335879	**K_0*(1430) decay decay width** K_0^*(1430)0 | Kbar_0^*(1430)0 | K_0^*(1430)+ | K_0^*(1430)-	270+-80 MeV	symmetric; symmetric: ±80; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M020 M020W property: 335147	**pi_2(2100) decay width** pi_2(2100)0 | pi_2(2100)+ | pi_2(2100)-	620+-50 MeV	symmetric; symmetric: ±52.400001025245267; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M022 M022W1 prop. 335890	**Charged states only, with final state K pi** K_2^*(1430)0 | Kbar_2^*(1430)0 | K_2^*(1430)+ | K_2^*(1430)-	100.0+-2.2 MeV	asymmetric; asymmetric: +2.2364281468619742/-2.1785326327209291; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M022 M022W4 property: 335892	**Neutral states only** K_2^*(1430)0 | Kbar_2^*(1430)0 | K_2^*(1430)+ | K_2^*(1430)-	109+-5 MeV	symmetric; symmetric: ±5.4778171395550794; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M023 M023W property: 335964	**K_2(1770) decay width** K_2(1770)0 | Kbar_2(1770)0 | K_2(1770)+ | K_2(1770)-	186+-14 MeV	asymmetric; asymmetric: +13.978028299828869/-13.96323582174179; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M025 M025W prop. 346344	**psi(4160) WIDTH:** psi(4160)	69+-10 MeV	asymmetric; asymmetric: +10.537685232070819/-9.6022233874145346; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M026 M026W property: 343287	**eta_c(1S) WIDTH** eta_c(1S)	30.0+-0.5 MeV	symmetric; symmetric: ±0.46062178744488053; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M027 M027WX property: 334530	**eta(1405) WIDTH:** eta(1405)0	49.8+2.3-2.0 MeV	asymmetric; asymmetric: +2.2980645831526521/-1.9821207426120699; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M027 M027W1 prop. 334531	**eta pi pi MODE:** eta(1405)0	51.1+2.8-2.1 MeV	asymmetric; asymmetric: +2.7979258032466028/-2.0581322794155499; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M027 M027W4 property: 334532	**K Kbar pi MODE: (a_0(980) pi or direct K Kbar pi)** eta(1405)0	49+-4 MeV	asymmetric; asymmetric: +3.9129589597273662/-4.1089639617502112; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M027 M027W2 property: 334533	**pi pi gamma MODE:** eta(1405)0	89+-17 MeV	symmetric; symmetric: ±17.307932545234081; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M027 M027W7 prop. 334534	**phi gamma MODE:** eta(1405)0	86+7-18 MeV	asymmetric; asymmetric: +7.1309185944028277/-17.60823670899503; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M028 M028WX property: 335822	**K_1(1270) WIDTH** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	90+-20 MeV	symmetric; symmetric: ±20; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M028 M028W2 property: 335824	**PRODUCTION BY K-, BACKWARD SCATTERING, AND HYPERON EXCHANGE** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	75+-15 MeV	symmetric; symmetric: ±15; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M028 M028W3 prop. 335825	**GENERATED BY K BEAMS** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	90+-8 MeV	symmetric; symmetric: ±8; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M028 M028W1 property: 335826	**PRODUCTION BY BEAMS OTHER THAN K MESONS** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	126+-16 MeV	symmetric; symmetric: ±16.081957783066429; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M028 M028WT property: 335827	**PRODUCTION IN tau-LEPTON DECAYS** K_1(1270)0 | Kbar_1(1270)0 | K_1(1270)+ | K_1(1270)-	260+120-110 MeV	asymmetric; asymmetric: +120.41594578792299/-106.30145812734651; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M029 M029W prop. 346245	**X(3940) WIDTH:** X(3940)	43+28-18 MeV	asymmetric; asymmetric: +28.102438182886321/-17.56265521056736; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M030 M030W property: 334307	**h_1(1170) WIDTH** h_1(1170)0	375+-35 MeV	symmetric; symmetric: ±34.525353003264136; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M033 M033W3 property: 335260	**decay channel: pi- p –> omega pi0 n** rho_5(2350)0 | rho_5(2350)+ | rho_5(2350)-	400+-100 MeV	symmetric; symmetric: ±100; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M034 M034W prop. 334784	**pi_2(1670) WIDTH VALUE** pi_2(1670)0 | pi_2(1670)+ | pi_2(1670)-	258+8-9 MeV	asymmetric; asymmetric: +7.9945231964062842/-9.0326516323147512; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M035 M035W property: 336017	**K_4*(2045) decay decay width** K_4^*(2045)0 | Kbar_4^*(2045)0 | K_4^*(2045)+ | K_4^*(2045)-	199+27-19 MeV	asymmetric; asymmetric: +26.508871026900351/-19.300507563547299; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M036 M036W1 property: 334172	**a_0(980) decay width** a_0(980)0 | a_0(980)+ | a_0(980)-	50 to 100 MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M037 M037W prop. 334452	**eta(1295) WIDTH:** eta(1295)0	55+-5 MeV	symmetric; symmetric: ±5.3662816306481913; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M038 M038W property: 335003	**f_2(1810) WIDTH** f_2(1810)0	197+-22 MeV	asymmetric; asymmetric: +21.979426688512731/-22.153970300247611; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M040 M040W property: 336033	**K_2(2250) decay width** K_2(2250)0 | Kbar_2(2250)0 | K_2(2250)+ | K_2(2250)-	180+-30 MeV	symmetric; symmetric: ±29.99999948657187; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M041 M041W4 prop. 335245	**p p CENTRAL PRODUCTION:** f_4(2300)0	250+-80 MeV	symmetric; symmetric: ±80; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M042 M042W property: 335160	**f_2(2150) WIDTH, COMBINED MODES** f_2(2150)0	152+-30 MeV	symmetric; symmetric: ±30.187002348708841; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M042 M042W3 property: 335161	**eta eta MODE:** f_2(2150)0	152+-30 MeV	symmetric; symmetric: ±30.187002348708841; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M042 M042W1 prop. 335162	**pbar p –> pi pi:** f_2(2150)0	250 MeV	not reported; not reported; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M045 M045W property: 334779	**omega_3(1670) WIDTH** omega_3(1670)0	168+-10 MeV	symmetric; symmetric: ±10.0179720379382; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M047 M047W property: 347745	**Upsilon(4S) WIDTH:** Upsilon(4S)	20.5+-2.5 MeV	symmetric; symmetric: ±2.473041036470411; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M048 M048W prop. 347661	**Upsilon(3S) WIDTH:** Upsilon(3S)	20.32+-1.85 keV	symmetric; symmetric: ±1.8500000000000001; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M049 M049W property: 346838	**Upsilon(1S) WIDTH:** Upsilon(1S)	54.02+-1.25 keV	symmetric; symmetric: ±1.25; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M050 M050W property: 346229	**chi_c2(3930) WIDTH:** chi_c2(3930)	35.2+-2.2 MeV	symmetric; symmetric: ±2.2286730591186932; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M052 M052W prop. 347288	**Upsilon(2S) WIDTH:** Upsilon(2S)	31.98+-2.63 keV	symmetric; symmetric: ±2.6299999999999999; type V; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M053 M053W property: 345870	**psi(3770) WIDTH:** psi(3770)	27.2+-1.0 MeV	symmetric; symmetric: ±0.9592482270788617; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M054 M054W property: 335024	**phi_3(1850) WIDTH:** phi_3(1850)0	87+28-23 MeV	asymmetric; asymmetric: +27.578792949203059/-23.092499274786309; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M055 M055W prop. 344599	**chi_c1(1P) WIDTH:** chi_c1(1P)	0.82+-0.04 MeV	symmetric; symmetric: ±0.041623417330381482; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M056 M056W property: 344305	**chi_c0(1P) WIDTH** chi_c0(1P)	12.40+-0.27 MeV	symmetric; symmetric: ±0.26507716522265817; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M057 M057W property: 344927	**chi_c2(1P) WIDTH:** chi_c2(1P)	1.94+-0.09 MeV	symmetric; symmetric: ±0.089048410157366226; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M058 M058W prop. 334464	**width of pi(1300)** pi(1300)0 | pi(1300)+ | pi(1300)-	200 to 600 MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M059 M059W property: 345229	**eta_c(2S) WIDTH** eta_c(2S)	11.6+-1.4 MeV	asymmetric; asymmetric: +1.4614052143583021/-1.424822737204585; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M060 M060W property: 335976	**K_3*(1780) decay decay width** K_3^*(1780)0 | Kbar_3^*(1780)0 | K_3^*(1780)+ | K_3^*(1780)-	161+-17 MeV	asymmetric; asymmetric: +16.775777759141921/-17.687369336790258; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M061 M061W prop. 337780	**width of D*(2007)0** D^*(2007)0 | Dbar^*(2007)0	<2.1 MeV	not applicable; not applicable; type L; limit U; confidence 90%	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M062 M062W property: 337799	**D*(2010)+- WIDTH** D^*(2010)+ | D^*(2010)-	83.4+-1.8 keV	symmetric; symmetric: ±1.8376713944967871; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M064 M064W property: 335846	**K_1(1400) decay width** K_1(1400)0 | Kbar_1(1400)0 | K_1(1400)+ | K_1(1400)-	174+-13 MeV	asymmetric; asymmetric: +12.611123655153699/-12.601160968977361; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M065 M065W0 prop. 334881	**eta rho0 and pi+ pi- decay modes** rho(1700)0 | rho(1700)+ | rho(1700)-	250+-100 MeV	symmetric; symmetric: ±100; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M067 M067W1 property: 334826	**e+ e- PRODUCTION:** phi(1680)0	150+-50 MeV	symmetric; symmetric: ±50; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M068 M068W property: 334928	**f_0(1710) Breit-Wigner WIDTH:** f_0(1710)0	149+-10 MeV	asymmetric; asymmetric: +10.92532406271617/-9.8917234176496365; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M070 M070W prop. 343506	**J/psi(1S) WIDTH:** J/psi(1S)	92.6+-1.7 keV	asymmetric; asymmetric: +1.7182511605720709/-1.7119132837100099; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M071 M071W property: 345302	**psi(2S) WIDTH:** psi(2S)	293+-9 keV	symmetric; symmetric: ±9.1216171327147304; type FC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M072 M072W property: 346250	**psi(4040) WIDTH:** psi(4040)	84+-12 MeV	symmetric; symmetric: ±12.300000000000001; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M073 M073W prop. 346685	**psi(4415) WIDTH:** psi(4415)	110+-13 MeV	symmetric; symmetric: ±13.28782789930157; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M074 M074W property: 346461	**psi(4230) WIDTH:** psi(4230)	51+-7 MeV	asymmetric; asymmetric: +6.9972903898004102/-6.8824149029832364; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M075 M075W property: 334967	**pi(1800) WIDTH:** pi(1800)0 | pi(1800)+ | pi(1800)-	215+7-8 MeV	asymmetric; asymmetric: +6.706577420969178/-8.1998993000281608; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M082 M082W prop. 335204	**f_J(2220) WIDTH:** f_J(2220)0	23+8-7 MeV	asymmetric; asymmetric: +8.2907233968204679/-6.995398154183702; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M084 M084W property: 334659	**f_1(1510) WIDTH** f_1(1510)0	73+-25 MeV	symmetric; symmetric: ±25.12449045628555; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M085 M085W property: 335012	**X(1835) WIDTH:** X(1835)0	130+-50 MeV	asymmetric; asymmetric: +46.521424857963467/-48.805761882116528; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M089 M089W prop. 335278	**f_6(2510) WIDTH:** f_6(2510)0	260+-40 MeV	symmetric; symmetric: ±42.426406871192853; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M090 M090W property: 336035	**K_3(2320) WIDTH** K_3(2320)0 | Kbar_3(2320)0 | K_3(2320)+ | K_3(2320)-	150+-30 MeV	symmetric; symmetric: ±30; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M092 M092W property: 347805	**Upsilon(10860) WIDTH:** Upsilon(10860)	37+-4 MeV	asymmetric; asymmetric: +4.2665620718053896/-3.9767365312397489; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M093 M093W prop. 347906	**Upsilon(11020) WIDTH:** Upsilon(11020)	24+8-6 MeV	asymmetric; asymmetric: +7.7077065645708283/-6.4571558257821966; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M094 M094W property: 335866	**K*(1410) WIDTH** K^*(1410)0 | Kbar^*(1410)0 | K^*(1410)+ | K^*(1410)-	232+-21 MeV	symmetric; symmetric: ±21.38237216053956; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M095 M095W property: 335947	**K*(1680) decay width** K^*(1680)0 | Kbar^*(1680)0 | K^*(1680)+ | K^*(1680)-	320+-110 MeV	asymmetric; asymmetric: +107.33834598069571/-107.80756648292591; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M098 M098W prop. 336037	**K_5*(2380) decay decay width** K_5^*(2380)0 | Kbar_5^*(2380)0 | K_5^*(2380)+ | K_5^*(2380)-	180+-50 MeV	symmetric; symmetric: ±48.918299234540036; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M099 M099W property: 335945	**K_1(1650) WIDTH** K_1(1650)0 | Kbar_1(1650)0 | K_1(1650)+ | K_1(1650)-	150+-50 MeV	symmetric; symmetric: ±50; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M101 M101W property: 335033	**eta_2(1870) WIDTH:** eta_2(1870)0	225+-14 MeV	symmetric; symmetric: ±14.06500754800965; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M103 M103W prop. 335181	**width of phi(2170)** phi(2170)0 | phi(2170)+ | phi(2170)-	88+26-21 MeV	asymmetric; asymmetric: +25.84561826322167/-20.63586514647379; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M104 M104W property: 336004	**K_2*(1980) decay decay width** K_2^*(1980)0 | Kbar_2^*(1980)0 | K_2^*(1980)+ | K_2^*(1980)-	348+50-30 MeV	asymmetric; asymmetric: +47.995015852124311/-29.822700392743499; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M105 M105W0 property: 334593	**rho(1450) WIDTH:** rho(1450)0 | rho(1450)+ | rho(1450)-	400+-60 MeV	symmetric; symmetric: ±60; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M106 M106W prop. 335113	**f_2(2010) WIDTH:** f_2(2010)0	200+-60 MeV	asymmetric; asymmetric: +67/-62; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M107 M107W property: 335235	**f_2(2300) WIDTH** f_2(2300)0	150+-40 MeV	symmetric; symmetric: ±41; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M108 M108W property: 335253	**f_2(2340) WIDTH:** f_2(2340)0	331+27-18 MeV	asymmetric; asymmetric: +27.40609495783065/-17.599089776866538; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M109 M109W prop. 334553	**h_1(1415) WIDTH:** h_1(1415)0	78+-11 MeV	asymmetric; asymmetric: +10.8947684184199 31/-10.35476102684496; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M112 M112W property: 335200	**f_0(2200) WIDTH** f_0(2200)0	210+-40 MeV	asymmetric; asymmetric: +41.638656568206237/-42.173251605545943; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M114 M114W property: 334951	**eta(1760) WIDTH:** eta(1760)0	240+-30 MeV	asymmetric; asymmetric: +30.76700000392665/-28.819843599945798; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M115 M115W prop. 335233	**eta(2225) WIDTH:** eta(2225)0	185+40-20 MeV	asymmetric; asymmetric: +36.894036523364868/-19.6088060392476; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M117 M117W property: 334753	**f_2(1640) WIDTH** f_2(1640)0	100+60-40 MeV	asymmetric; asymmetric: +55.007270965809873/-40.999999045433803; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M121 M121W property: 338489	**D_s1(2536)+- WIDTH:** D_s1(2536)+ | D_s1(2536)-	0.92+-0.05 MeV	symmetric; symmetric: ±0.049965313904947829; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M123 M123W prop. 334704	**f_2(1565) WIDTH:** f_2(1565)0	132+-23 MeV	symmetric; symmetric: ±22.499999212159828; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M125 M125W property: 334568	**omega(1420) WIDTH:** omega(1420)0	290+-190 MeV	symmetric; symmetric: ±190; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M126 M126W property: 334769	**omega(1650) WIDTH:** omega(1650)0	150+-50 MeV	symmetric; symmetric: ±50; type E; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M134 M134W prop. 336000	**K_0*(1950) decay decay width** K_0^*(1950)0 | Kbar_0^*(1950)0 | K_0^*(1950)+ | K_0^*(1950)-	170+-50 MeV	symmetric; symmetric: ±49.977634624979267; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M135 M135W property: 335076	**f_2(1950) WIDTH** f_2(1950)0	464+-24 MeV	asymmetric; asymmetric: +24.233608393520051/-23.858175111592871; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M142 M142W2 property: 335059	**f_2(1910) omega omega MODE:** f_2(1910)0	167+-21 MeV	symmetric; symmetric: ±20.593210779464879; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M142 M142W3 prop. 335060	**f_2(1910) eta eta' MODE:** f_2(1910)0	140+-40 MeV	symmetric; symmetric: ±41; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M144 M144W property: 344852	**h_c(1P) WIDTH** h_c(1P)	0.78+-0.28 MeV	asymmetric; asymmetric: +0.29546573405388321/-0.26832815729997478; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M146 M146W property: 335991	**K_2(1820) decay width** K_2(1820)0 | Kbar_2(1820)0 | K_2(1820)+ | K_2(1820)-	264+-34 MeV	asymmetric; asymmetric: +33.911162815562037/-33.585154163911888; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M147 M147W prop. 334507	**Full width (Breit-Wigner):** f_0(1370)0	200 to 500 MeV	not applicable; not applicable; type E; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M148 M148W property: 338511	**D_s2*(2573) WIDTH** D_s2^*(2573)+ | D_s2^*(2573)-	16.9+-0.7 MeV	asymmetric; asymmetric: +0.7397201342644392 8/-0.73758653054042789; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M149 M149W property: 334580	**a_0(1450) decay width** a_0(1450)0 | a_0(1450)+ | a_0(1450)-	258+-14 MeV	symmetric; symmetric: ±14.000000098926851; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M151 M151W prop. 342552	**B_J*(5732) decay decay width** B_J^*(5732)0 | Bbar_J^*(5732)0 | B_J^*(5732)+ | B_J^*(5732)-	128+-18 MeV	symmetric; symmetric: ±18.222158456767119; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M152 M152W property: 334629	**f_0(1500) WIDTH** f_0(1500)0	108+-33 MeV	symmetric; symmetric: ±33; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M153 M153W property: 343152	**B_sJ*(5850) WIDTH** B_sJ^*(5850) 0 | Bbar_sJ^*(5850)0	47+-22 MeV	symmetric; symmetric: ±22; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M154 M154W prop. 334759	**eta_2(1645) WIDTH:** eta_2(1645)0	181+-11 MeV	asymmetric; asymmetric: +10.785744799202121/-10.49166410643209; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M156 M156W property: 335121	**f_0(2020) WIDTH** f_0(2020)0	440+-50 MeV	asymmetric; asymmetric: +46.173585522460783/-49.162994213127419; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M158 M158W property: 337855	**D*(2640)+- decay width** D^*(2640)0 | Dbar^*(2640)0 | D^*(2640)+ | D^*(2640)-	<15 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M159 M159W prop. 346205	**chi_c0(3915) WIDTH:** chi_c0(3915)	20+-4 MeV	asymmetric; asymmetric: +3.923709318376599/-3.8485732714118699; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M160 M160W property: 335943	**K(1630) WIDTH** K(1630)0 | Kbar(1630)0 | K(1630)+ | K(1630)-	16+19-16 MeV	asymmetric; asymmetric: +19/-16; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M161 M161W property: 334741	**a_1(1640) decay width** a_1(1640)0 | a_1(1640)+ | a_1(1640)-	250+-40 MeV	asymmetric; asymmetric: +38.768026106973267/-38.004464447238831; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M162 M162W prop. 334904	**a_2(1700) WIDTH VALUE** a_2(1700)0 | a_2(1700)+ | a_2(1700)-	380+60-50 MeV	asymmetric; asymmetric: +56.802156616143733/-49.91978684730767; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M164 M164WEP property: 334726	**pi_1(1600) WIDTH (eta pi mode)** pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)-	330+-35 MeV	asymmetric; asymmetric: +34.727689315250451/-34.693151629682987; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M164 M164W property: 334727	**pi_1(1600) decay width (non-eta pi mode)** pi_1(1600)0 | pi_1(1600)+ | pi_1(1600)-	370+50-60 MeV	asymmetric; asymmetric: +50.324516391842117/-56.454714048381128; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M166 M166W prop. 334721	**h_1(1595) WIDTH:** h_1(1595)0	380+90-120 MeV	asymmetric; asymmetric: +92.195444572928878/-116.61903789690599; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M168 M168W property: 335157	**f_0(2100) WIDTH** f_0(2100)0	287+32-24 MeV	asymmetric; asymmetric: +31.594915701364421/-24.048675270055991; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M171 M171W property: 346826	**eta_b(1S) WIDTH:** eta_b(1S)	10+5-4 MeV	asymmetric; asymmetric: +4.7684157959520306/-3.6148198760107491; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M172 M172W prop. 338433	**D_s0*(2317)+- WIDTH VALUE** D_s0^*(2317)+ | D_s0^*(2317)-	<3.8 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M173 M173W property: 338454	**D_s1(2460)+- WIDTH** D_s1(2460)+ | D_s1(2460)-	<3.5 MeV	not applicable; not applicable; type L; limit U; confidence 95%	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M174 M174W property: 335787	**K_0*(700) Breit–Wigner decay decay width** K_0^*(700)0 | Kbar_0^*(700)0 | K_0^*(700)+ | K_0^*(700)-	463+-27 MeV	asymmetric; asymmetric: +28.018020010586849/-25.969205598480212; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M175 M175W5 prop. 334617	**eta(1475) WIDTH:** eta(1475)0	96+-9 MeV	asymmetric; asymmetric: +9.1647781281613359/-8.6589105601013205; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M176 M176W property: 346132	**chi_c1(3872) WIDTH** chi_c1(3872)	1.19+-0.21 MeV	asymmetric; asymmetric: +0.2144614101505273 9/-0.21212796092607489; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M180 M180W property: 337828	**D_1(2430)0 WIDTH:** D_1(2430)0 | Dbar_1(2430)0	314+-29 MeV	asymmetric; asymmetric: +28.68695276836397/-28.327914686041321; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M181 M181W prop. 346636	**psi(4360) WIDTH:** psi(4360)	124+-13 MeV	asymmetric; asymmetric: +13.061124706808121/-13.28650418227447; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M182 M182W property: 338525	**D_s1*(2700)+ WIDTH** D_s1^*(2700)+ | D_s1^*(2700)-	122+-10 MeV	asymmetric; asymmetric: +10.543417520412129/-10.06094705301401; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M185 M185W property: 335044	**pi_2(1880) WIDTH:** pi_2(1880)0	237+33-30 MeV	asymmetric; asymmetric: +33.317197463990517/-29.502658037886111; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M186 M186W prop. 343141	**B_s2*(5840)0 WIDTH VALUE** B_s2^*(5840)0 | Bbar_s2^*(5840)0	1.49+-0.27 MeV	symmetric; symmetric: ±0.26660479598163422; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M187 M187W property: 343133	**B_s1(5830)0 WIDTH** B_s1(5830)0 | Bbar_s1(5830)0	0.5+-0.4 MeV	symmetric; symmetric: ±0.42426406871192851; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M188 M188W property: 334715	**rho(1570) WIDTH:** rho(1570)0	140+-90 MeV	symmetric; symmetric: ±86.452298986203942; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M189 M189W prop. 346768	**psi(4660) WIDTH:** psi(4660)	55+-9 MeV	asymmetric; asymmetric: +8.889722918699217/-8.409913961865500 6; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M190 M190W property: 346452	**X(4160) WIDTH:** X(4160)	136+60-35 MeV	asymmetric; asymmetric: +56.539220923817354/-34.625230725662142; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M191 M191W property: 347991	**T_c_cbar(4050)+ WIDTH:** T_c_cbar(4050)+	82+50-28 MeV	asymmetric; asymmetric: +51.478150704934997/-27.802877548915689; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M192 M192W prop. 348027	**T_c_cbar(4250)+ WIDTH:** T_c_cbar(4250)+	180+320-70 MeV	asymmetric; asymmetric: +320.58072306363027/-72.401657439591816; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M193 M193W property: 346337	**chi_c1(4140) WIDTH** chi_c1(4140)	19+7-5 MeV	asymmetric; asymmetric: +6.5586673407753064/-5.2758202510488337; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M194 M194W property: 346629	**X(4350) WIDTH:** X(4350)	13+18-10 MeV	asymmetric; asymmetric: +18.439088914585771/-9.8488578017961039; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M195 M195W prop. 348031	**T_c_cbar_1(4430)+ WIDTH:** T_c_cbar_1(4430)+	189+-29 MeV	asymmetric; asymmetric: +28.960862620750699/-29.212281692432189; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M196 M196W property: 338534	**D_s1*(2860)+ WIDTH** D_s1^*(2860)+ | D_s1^*(2860)-	160+-80 MeV	symmetric; symmetric: ±80.361682411457764; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M197 M197W property: 338545	**D_sJ(3040)+ WIDTH:** D_sJ(3040)+ | D_sJ(3040)-	240+-60 MeV	asymmetric; asymmetric: +57.801384066473702/-54.671747731346578; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M198 M198W prop. 337844	**D_0(2550)0 DECAY WIDTH** D_0(2550)0 | Dbar_0(2550)0 | D_0(2550)+ | D_0(2550)-	165+-24 MeV	symmetric; symmetric: ±23.747553397709549; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M199 M199W property: 337847	**D_1*(2600)0 decay decay width** D_1^*(2600)0 | Dbar_1^*(2600)0 | D_1^*(2600)+ | D_1^*(2600)-	141+-23 MeV	symmetric; symmetric: ±22.836524044183388; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M200 M200W property: 347283	**eta_b(2S) WIDTH:** eta_b(2S)	<24 MeV	not applicable; not applicable; type L; limit U; confidence 90%	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M203 M203W prop. 337861	**D_3*(2750) decay decay width** D_3^*(2750)0 | Dbar_3^*(2750)0 | D_3^*(2750)+ | D_3^*(2750)-	66+-5 MeV	symmetric; symmetric: ±4.6026930917312541; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M207 M207W property: 348054	**T_b_bbar_1(10610)+- WIDTH** T_b_bbar_1(10610)+	18.4+-2.4 MeV	symmetric; symmetric: ±2.3999999999999999; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M208 M208W property: 348076	**T_b_bbar_1(10650) + WIDTH** T_b_bbar_1(10650)+	11.5+-2.2 MeV	symmetric; symmetric: ±2.2000000000000002; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M210 M210W prop. 347939	**T_c_cbar_1(3900) WIDTH:** T_c_cbar_1(3900)+	29.5+-2.4 MeV	asymmetric; asymmetric: +2.425749286587477/-2.4174014844353149; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M212 M212W property: 346101	**psi_2(3823) WIDTH** psi_2(3823)	<2.9 MeV	not applicable; not applicable; type L; limit U; confidence 90%	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M213 M213W property: 347977	**T_c_cbar(4020) WIDTH:** T_c_cbar(4020)+	13+-5 MeV	symmetric; symmetric: ±5.2742104610962652; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M216 M216W prop. 348023	**T_c_cbar_0(4240)+ WIDTH:** T_c_cbar_0(4240)0	220+120-90 MeV	asymmetric; asymmetric: +117.78370006074699/-87.664131775772461; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M223 M223W property: 348001	**T_c_cbar(4055)+ WIDTH** T_c_cbar(4055)+	45+-13 MeV	symmetric; symmetric: ±12.529964086141669; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M226 M226W property: 338536	**D_s3*(2860) + WIDTH** D_s3^*(2860)+ | D_s3^*(2860)-	53+-10 MeV	symmetric; symmetric: ±9.8994949366116654; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M227 M227W prop. 335071	**a_0(1950) WIDTH:** a_0(1950)0	270+-40 MeV	symmetric; symmetric: ±36.400549446402593; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M228 M228W property: 337858	**D_2(2740)0 WIDTH** D_2(2740)0 | Dbar_2(2740)0 | D_2(2740)+ | D_2(2740)-	88+-19 MeV	symmetric; symmetric: ±19.435218021172709; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M229 M229W property: 337873	**D_2*(3000)0 decay decay width** D_2^*(3000)0 | Dbar_2^*(3000)0 | D_2^*(3000)+ | D_2^*(3000)-	190+-80 MeV	symmetric; symmetric: ±81.412529748190479; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M231 M231W prop. 348013	**T_c_cbar_1(4200)+ WIDTH:** T_c_cbar_1(4200)+	310+-40 MeV	asymmetric; asymmetric: +35.260965800649629/-36.587057375401727; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M232 M232W property: 348037	**T_b_sbar(5568)+ WIDTH** T_b_sbar(5568)+	19+9-7 MeV	asymmetric; asymmetric: +8.6406018308911801/-7.1867934435323786; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M233 M233W property: 346625	**chi_c1(4274) WIDTH:** chi_c1(4274)	54+-8 MeV	asymmetric; asymmetric: +8.1968525494933324/-7.7933837739246821; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M234 M234W prop. 346756	**chi_c0(4500) WIDTH:** chi_c0(4500)	87+30-24 MeV	asymmetric; asymmetric: +29.927160405061439/-24.399652435665001; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M235 M235W property: 346817	**chi_c0(4700) WIDTH** chi_c0(4700)	78+15-11 MeV	asymmetric; asymmetric: +14.53972996408401/-10.852931317094249; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M237 M237W property: 346123	**chi_c0(3860) WIDTH:** chi_c0(3860)	200+180-110 MeV	asymmetric; asymmetric: +177.36967046256811/-105.89145385724009; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M239 M239W prop. 335107	**pi_2(2005) WIDTH:** pi_2(2005)0	370+16-90 MeV	asymmetric; asymmetric: +15.91185264772162/-94.13731621266399; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M240 M240W property: 348007	**T_c_cbar(4100)+ WIDTH** T_c_cbar(4100)+	150+80-70 MeV	asymmetric; asymmetric: +83.450584180100265/-67.742158217759794; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M241 M241W property: 346117	**psi_3(3842) WIDTH:** psi_3(3842)	2.8+-0.6 MeV	symmetric; symmetric: ±0.61854668376768462; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M243 M243W prop. 347801	**Upsilon(10753) WIDTH:** Upsilon(10753)	29+-9 MeV	symmetric; symmetric: ±8.8814413244698081; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M244 M244W+ property: 342547	**B_1(5721)+ width** B_1(5721)0 | Bbar_1(5721)0 | B_1(5721)+ | B_1(5721)-	31+-6 MeV	asymmetric; asymmetric: +5.84704469385892/-6.0921042006092243; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

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M244 M244W0 property: 342548	**B_1(5721)0 decay width** B_1(5721)0 | Bbar_1(5721)0 | B_1(5721)+ | B_1(5721)-	27.5+-3.4 MeV	symmetric; symmetric: ±3.4222778451865001; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M245 M245W+ prop. 342564	**B_2*(5747)+ decay width** B_2^*(5747)0 | Bbar_2^*(5747)0 | B_2^*(5747)+ | B_2^*(5747)-	20+-5 MeV	symmetric; symmetric: ±5.4684229158432851; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M245 M245W0 property: 342565	**B_2*(5747)0 decay width** B_2^*(5747)0 | Bbar_2^*(5747)0 | B_2^*(5747)+ | B_2^*(5747)-	24.2+-1.7 MeV	asymmetric; asymmetric: +1.6959086634280349/-1.7095264315461871; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M246 M246W+ property: 342577	**B_J(5840)+ decay decay width** B_J(5840)0 | Bbar_J(5840)0 | B_J(5840)+ | B_J(5840)-	220+-80 MeV	symmetric; symmetric: ±83.522452071284405; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M246 M246W0 prop. 342578	**B_J(5840)0 DECAY WIDTH** B_J(5840)0 | Bbar_J(5840)0 | B_J(5840)+ | B_J(5840)-	130+-40 MeV	symmetric; symmetric: ±38.013155617496423; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M247 M247W property: 335262	**eta(2370) WIDTH:** eta(2370)0	148+80-28 MeV	asymmetric; asymmetric: +77.222243700673161/-28.36717772063778; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M248 M248W+ property: 342589	**B_J(5970)+ decay width** B_J(5970)0 | Bbar_J(5970)0 | B_J(5970)+ | B_J(5970)-	62+-20 MeV	asymmetric; asymmetric: +20.648092630154419/-20.22152574413818; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M248 M248W0 prop. 342590	**B_J(5970)0 decay decay width** B_J(5970)0 | Bbar_J(5970)0 | B_J(5970)+ | B_J(5970)-	81+-12 MeV	asymmetric; asymmetric: +11.58405073203963/-11.42146343467506; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M249 M249W property: 337869	**D_1*(2760)0 decay decay width** D_1^*(2760)0 | Dbar_1^*(2760)0 | D_1^*(2760)+ | D_1^*(2760)-	180+-40 MeV	symmetric; symmetric: ±38.275318418009277; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M250 M250W property: 347917	**T_cbar_sbar_0*(2870)0 decay width** T_cbar_sbar_0^*(2870)0	68+-17 MeV	asymmetric; asymmetric: +16.282833932268769/-17.2087901693615; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M251 M251W prop. 347923	**T_cbar_sbar_1*(2900)0 WIDTH VALUE** T_cbar_sbar_1^*(2900)0	106+-10 MeV	symmetric; symmetric: ±10.39615906316193; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M252 M252W property: 337814	**D_0*(2300) decay decay width** D_0^*(2300)0 | Dbar_0^*(2300)0 | D_0^*(2300)+ | D_0^*(2300)-	229+-16 MeV	symmetric; symmetric: ±16.014886897955751; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M253 M253W property: 337822	**D_1(2420) decay width** D_1(2420)0 | Dbar_1(2420)0 | D_1(2420)+ | D_1(2420)-	31.3+-1.9 MeV	asymmetric; asymmetric: +1.939893766488169/-1.931511994340549; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M254 M254W prop. 337837	**D_2*(2460) decay decay width** D_2^*(2460)0 | Dbar_2^*(2460)0 | D_2^*(2460)+ | D_2^*(2460)-	47.3+-0.8 MeV	asymmetric; asymmetric: +0.7743134766834105 5/-0.7736295767464896 9; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M255 M255W property: 334942	**X(1750) WIDTH:** X(1750)0	120+-10 MeV	asymmetric; asymmetric: +9.2901225622471255/-9.8222008027276093; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

Continuation of the inventory

Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M256 M256W property: 338521	**D_s0(2590)+ WIDTH:** D_s0(2590)+ | D_s0(2590)-	89+-20 MeV	symmetric; symmetric: ±20; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M257 M257W prop. 343154	**B_sJ(6063)0 WIDTH:** B_sJ(6063)0 | Bbar_sJ(6063)0	26+-6 MeV	symmetric; symmetric: ±5.6568542494923806; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M258 M258W property: 343158	**B_sJ(6114)0 WIDTH** B_sJ(6114)0 | Bbar_sJ(6114)0	66+-28 MeV	symmetric; symmetric: ±27.658633371878661; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M259 M259W property: 347965	**T_c cbar sbar 1(4000) WIDTH:** T_c_cbar_sbar_1(4000)0	5–150 MeV	not applicable; not applicable; type V; limit R; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M260 M260W prop. 348019	**T_c cbar sbar 1(4220)+ WIDTH:** T_c_cbar_sbar_1(4220)+	230+110-90 MeV	asymmetric; asymmetric: +110.05907504608609/-89.627004859026727; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M261 M261W property: 346811	**chi_c1(4685) WIDTH** chi_c1(4685)	180+-50 MeV	asymmetric; asymmetric: +49.868992214526223/-51.68253103772755; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M262 M262W property: 346762	**X(4630) WIDTH:** X(4630)	350+140-100 MeV	asymmetric; asymmetric: +137.97165629659631/-102.5281698672277; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M263 M263W prop. 334918	**a_0(1710) WIDTH:** a_0(1710)0	107+-15 MeV	symmetric; symmetric: ±14.82500717857968; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M264 M264W property: 334993	**f_0(1810) Breit-Wigner WIDTH** f_0(1810)0	137+15-17 MeV	asymmetric; asymmetric: +15.027393581054231/-16.785983289291; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M265 M265W property: 347935	**T_c_c(3875)+ WIDTH:** T_c_c(3875)+	0.048+0.0020-0.0141 MeV	asymmetric; asymmetric: +0.002/-0.014142135623730951; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M266 M266W prop. 335274	**f_0(2470) WIDTH:** f_0(2470)0	75+14-12 MeV	asymmetric; asymmetric: +14.2126704035519/-12.041594578792299; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M267 M267W property: 335029	**eta_1(1855) WIDTH** eta_1(1855)0	188+-19 MeV	asymmetric; asymmetric: +18.248287590894659/-19.697715603592211; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M268 M268W property: 348045	**T_c_c_cbar_cbar(6900)0 decay width** T_c_c_cbar_cbar(6900)+	161+-26 MeV	asymmetric; asymmetric: +27.00038424522608/-25.460154585260579; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M269 M269W prop. 347928	**T_c_sbar_0*(2900)0 WIDTH VALUE** T_c_sbar_0^*(2900)+	119+-29 MeV	symmetric; symmetric: ±29.068883707497271; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M269 M269W++ property: 347929	**T_c_sbar_0*(2900)++ WIDTH** T_c_sbar_0^*(2900)+	140+-40 MeV	symmetric; symmetric: ±36.235341863986868; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M270 M270W property: 335222	**omega(2220) WIDTH:** omega(2220)0	105+-34 MeV	symmetric; symmetric: ±33.807267697768303; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.

Continuation of the inventory

Identifier	Object and species	Value	Uncertainty and Convention	Sector and status	Structural realization
M272 M272W prop. 348041	**T_c c cbar cbar(6600) WIDTH:** T_c c cbar cbar(6600)+	440+-70 MeV	asymmetric; asymmetric: +74.676988436660039/-68.966189416634236; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
M273 M273W property: 348049	**T_c c cbar cbar(7100) WIDTH** T_c c cbar cbar(7100)+	100+50-40 MeV	asymmetric; asymmetric: +49.406477308142499/-38.948684188300888; type AC; limit NONE; confidence not reported	composite mesonic record *exhaustive record without promotion to comparison*	nominal mesonic linking map. external record accounted for; individual HMT–MD realization not yet linked.
Q007 Q007W property: 333618	**t-quark DECAY WIDTH:** t | tbar	1.42+0.19-0.15 GeV	asymmetric; asymmetric: +0.19051342973542351/-0.15256041171179069; type AC; limit NONE; confidence not reported	extended quark multisection *exhaustive record without comparative promotion*	Quark identity with declared scheme and scale. External record accounted for; individual HMT–MD realization not yet linked.
S014 S014W prop. 333730	**eta WIDTH:** eta	1.31+-0.05 keV	symmetric; symmetric: ±0.045198908079192503; type FC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S026 S026W+ property: 351720	**Sigma_b+ decay decay width** Sigma_b()- | Sigmabar_b()+ | Sigma_b()0 | Sigmabar_b()0 | Sigma_b()+ | Sigmabar_b()-	5.0+-0.5 MeV	asymmetric; asymmetric: +0.4791980101675620 3/-0.47660443013974058; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S026 S026W- property: 351721	**Sigma_b- decay decay width** Sigma_b()- | Sigmabar_b()+ | Sigma_b()0 | Sigmabar_b()0 | Sigma_b()+ | Sigmabar_b()-	5.3+-0.5 MeV	asymmetric; asymmetric: +0.55180022333948409/-0.54475371590490007; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S043 S043W prop. 332468	**W WIDTH** W+ | W-	2.14+-0.05 GeV	symmetric; symmetric: ±0.053656225003149413; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S044 S044W property: 332514	**Z WIDTH** Z0	2.4955+-0.0023 GeV	symmetric; symmetric: ±0.0023; type V; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S062 S062W+ property: 351729	**Sigma_b*+ decay decay width** Sigma_b^*()- | Sigmabar_b^*()+ | Sigma_b^*()0 | Sigmabar_b^*()0 | Sigma_b^*()+ | Sigmabar_b^*()-	9.4+-0.5 MeV	asymmetric; asymmetric: +0.5280128552731296 9/-0.52651649982212589; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S062 S062W- prop. 351730	**Sigma_b*- decay decay width** Sigma_b^*()- | Sigmabar_b^*()+ | Sigma_b^*()0 | Sigmabar_b^*()0 | Sigma_b^*()+ | Sigmabar_b^*()-	10.4+-0.8 MeV	asymmetric; asymmetric: +0.8458982463825464/-0.84310730525944599; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S074 S074W property: 338419	**D_s*+- WIDTH** D_s^*()+ | D_s^*()-	<1.9 MeV	not applicable; not applicable; type L; limit U; confidence 90%	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.
S126 S126W property: 332734	**H DECAY WIDTH** H	3.0+1.5-0.7 MeV	asymmetric; asymmetric: +1.4529736900176831/-0.71501191940991071; type AC; limit NONE; confidence not reported	elementary section or typed extension *exhaustive record without comparative promotion*	identity towards the published sector or towards a new typed realization. external record accounted for; individual HMT–MD realization not yet linked.

Appendix P

Demonstrative appendices and classification tables

P.1 Ordered affine calculation of the APP leaves

P.1.1 Affine monoid $\mathrm{End}(U_{\mathrm{APP}}) \ltimes U_{\mathrm{APP}}$ and ordered composition of routes

Let U_{APP} be the radial module prior to its Archimedean evaluation. The correct envelope is the monoid

$$\mathrm{AffEnd}(U_{\mathrm{APP}}) = \mathrm{End}(U_{\mathrm{APP}}) \ltimes U_{\mathrm{APP}},$$

because the threshold or folding leaves may contain noninvertible endomorphisms. The action of (Λ, C) on $u \in U_{\mathrm{APP}}$ is

$$(\Lambda, C) \cdot u = \Lambda u + C.$$

The composition is calculated without commuting factors:

$$\begin{aligned}(\Lambda_2, C_2)(\Lambda_1, C_1) \cdot u &= \Lambda_2(\Lambda_1 u + C_1) + C_2 \\ &= (\Lambda_2\Lambda_1)u + (C_2 + \Lambda_2 C_1).\end{aligned} \tag{P.1}$$

Therefore,

$$(\Lambda_2, C_2)(\Lambda_1, C_1) = (\Lambda_2\Lambda_1, C_2 + \Lambda_2 C_1).$$

The second coordinate is not an ordinary sum of shifts: it is a cocycle twisted by the accumulated linear action.

Proposition P.1 (Prefix Formula)**.** *For a route with N factors $h_j = (\lambda_j, C_j)$, ordered from $j = 0$ through $j = N - 1$, the prefix holonomy is*

$$H_N = h_{N-1} \cdots h_1 h_0 = (\Lambda_N, C_N),$$

where

$$\Lambda_N = \prod_{j=0}^{N-1} \lambda_j$$

and

$$C_N = C_{N-1} + \lambda_{N-1} C_{N-2} + \lambda_{N-1}\lambda_{N-2} C_{N-3} + \cdots + \lambda_{N-1} \cdots \lambda_1 C_0.$$

Proof. For $N = 1$ the formula is immediate. If it holds for N, then

$$h_N H_N = (\lambda_N \Lambda_N, C_N + \lambda_N C_N),$$

which yields exactly the expression of length $N + 1$. The induction preserves the order of the route and does not identify paths with the same endpoints. □

P.1.2 Difference sum–product as curvature of order

Let $A_a = (I, a)$ and $M_\lambda = (\lambda, 0)$ be given. In the invertible sector,

$$A_a M_\lambda A_a^{-1} M_\lambda^{-1} = A_{a(1-\lambda)}.$$

The direct proof for u is

$$u \xmapsto{M_\lambda^{-1}} \lambda^{-1}u \xmapsto{A_a^{-1}} \lambda^{-1}u - a \xmapsto{M_\lambda} u - \lambda a \xmapsto{A_a} u + a(1 - \lambda).$$

Thus, the distinction between accumulation and folding is published as a translation. When $\lambda = 1$, the commutator vanishes; when $\lambda \neq 1$, the route retains a holonomy that cannot be reconstructed from the endpoint alone.

Example P.2 (Two routes with the same visible data)**.** Fix $u_0 = 0$, $a = 1$, $\lambda = 2$. The routes

$$A_1 M_2 : u_0 \longmapsto 1, \qquad M_2 A_{1/2} : u_0 \longmapsto 1$$

publish the same final value. However, their ledgers contain distinct factors, shifts, and prefixes. Pointwise evaluation identifies the routes; enriched affine holonomy distinguishes them.

P.1.3 Exhaustive enumeration of the local chart $(\tau_+, \tau_-) \mapsto (p, a, r_3, c_3)$

The following table does not posit that every global radial degree arises from a single pair. It records the elementary chart obtained by composing two TRIT orientations. The degree of a general route is the integer cocycle of equation (18.47); this chart provides its nine local generators.

Table P.1: The 9 oriented pairs of the local TRIT chart.

τ_+	τ_-	p	a	r_3	c_3	Lectura local
1	1	2	0	−1	1	double radial opening with positive carry
1	0	1	1	1	0	opening and threshold
1	−1	0	2	0	0	positive orientation equilibrium
0	1	1	−1	1	0	threshold and aperture
0	0	0	0	0	0	umbral doble
0	−1	−1	1	−1	0	threshold and contraction
−1	1	0	−2	0	0	negative orientation equilibrium
−1	0	−1	−1	−1	0	contraction and threshold
−1	−1	−2	0	1	−1	double contraction with negative carry

Here $p = r_3 + 3c_3$, with balanced residue $r_3 \in \{-1, 0, 1\}$. The carry distinguishes $p = 2$ from its residue -1 and $p = -2$ from its residue 1. The pair (p, a) likewise distinguishes the three local genealogies of degree 0.

P.2 Nonadic Certificate and Geometry of Memory

P.2.1 Typed distinction between decimal catalog, ternary words and ambient fiber

Table P.2: Strata of the certificate of the joint nonadic connection.

Object	Cardinality	Mathematical function
Catalog of source triples	104 976	Finite domain prior to the joint selection; it is not a decimal table.
Decimal emissions U_6	468	Compatible emissions of length 6 in the decimal publication.

Object	Cardinality	Mathematical function
Visible words w_6	243	Published ternary words after typing; they are not the complete environment.
Ambient space $\mathbb{F}_3^6$	729	Space of all ternary words of length 6; it contains the 243 visible words.
Certified levels	396	Finite prolongation up to the declared boundaries.
Trits recorridos	2 376	Longitud total $396 \cdot 6$.
Vueltas completas	43	Nonadic cycles in the certified segment.
Pairs $(K, K+9)$	387	Pairs of same-phase by channel; the ledger may differ.

The prolongation chain is preserved in the order

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9.$$

None of these objects can be replaced by the subsequent modal quotient. The holonomy Γ_9 acts on the enriched dynamics; the reduced-memory publication preserves only the information specified by its codomain.

P.2.2 Phase return, memory advance, and complete state

With

$$\rho_9(N) = 1 + ((N-1) \bmod 9), \qquad q_9(N) = \left\lfloor \frac{N-1}{9} \right\rfloor,$$

we have

$$\rho_9(N+9) = \rho_9(N), \qquad q_9(N+9) = q_9(N) + 1.$$

Therefore,

$$H_9(N+9) = H_9(N) + (0, 0, 1).$$

The equality of the first phase coordinate does not imply equality of the state:

$$g(K+9) = g(K), \qquad B_{K+9} \neq B_K$$

in general. The helical formula is precisely the minimal representation of this distinction.

P.2.3 Non-Split Dodecaphase Extension

The sequence

$$0 \longrightarrow C_{12} \longrightarrow C_{108} \longrightarrow C_9 \longrightarrow 0$$

couples the dodecaphasic calendar to the nonadic return. If a global homomorphic section $s : C_9 \to C_{108}$ existed, the generator of C_9 would have to lift to an element of order dividing 9 that projected onto the generator. The active temporal lift, however, preserves the accumulated dodecaphasic displacement and does not identify the state after one phase turn. The seam is recorded as carry; geometrically, it manifests as axial translation of the helix.

P.2.4 Decomposition of the microfiber of π into 5 regions and multiplicity of Archimedean evaluation

The microfiber of π contains five oriented regions with total multiplicity

$$432 + 4 \cdot 144 = 1008.$$

The regions publish the same real value, but retain distinct genealogies. This result provides the internal precedent for spiral classification: a visible curve does not by itself determine the region, the sheet, the carry, or the memory that produced it.

P.3 Algebra of L_p and classification of the 5 canonical families $p \in \{2, 1, 0, -1, -2\}$

P.3.1 The coordinate L_p and its inverse

For $x > 0$, define

$$L_p(x) = \begin{cases} (x^p - 1)/p, & p \neq 0, \\ \log x, & p = 0. \end{cases}$$

Its inverse is

$$\exp_p(u) = \begin{cases} (1 + pu)^{1/p}, & p \neq 0,\ 1 + pu > 0, \\ e^u, & p = 0. \end{cases}$$

The derivative

$$\frac{\mathrm{d}}{\mathrm{d}x} L_p(x) = x^{p-1} > 0$$

proves that L_p is strictly increasing on its domain and is therefore invertible.

Proposition P.3 (Sum–Product Law)**.** *For $x, y > 0$,*

$$L_p(xy) = L_p(x) + L_p(y) + pL_p(x)L_p(y).$$

Proof. Si $p \neq 0$,

$$\begin{aligned} L_p(x) + L_p(y) + pL_p(x)L_p(y) &= \frac{x^p - 1}{p} + \frac{y^p - 1}{p} + \frac{(x^p - 1)(y^p - 1)}{p} \\ &= \frac{x^p y^p - 1}{p} = L_p(xy). \end{aligned}$$

For $p = 0$, the equality reduces to $\log(xy) = \log x + \log y$. □

Corollary P.4 (Deformed group in the positive domain)**.** *The operation*

$$u \boxplus_p v = u + v + puv$$

satisfies

$$1 + p(u \boxplus_p v) = (1 + pu)(1 + pv).$$

Its neutral element is 0*, and the inverse of u is*

$$\ominus_p u = -\frac{u}{1 + pu},$$

si $1 + pu \neq 0$.

P.3.2 Evaluation of $\exp_p(u_0 + \beta m)$ in the 5 canonical families

Let m be the published depth, and suppose

$$L_p(r/r_0) = \beta m, \qquad \theta = \theta_0 + \omega m, \qquad \omega \neq 0.$$

Eliminando $m = (\theta - \theta_0)/\omega$,

$$r(\theta) = r_0 \left[1 + p\frac{\beta}{\omega}(\theta - \theta_0)\right]^{1/p}, \qquad p \neq 0,$$

and

$$r(\theta) = r_0 \exp\left[\frac{\beta}{\omega}(\theta - \theta_0)\right]$$

for $p = 0$. Positive changes of origin and rescalings yield the conventional forms:

Table P.3: Canonical publications in the translational sector.

p	Normal equation	Conventional name	Domain and feature
2	$r^2 = a^2\theta$	Fermat spiral	$\theta \geq 0$; linear increment of the quadratic radial area.
1	$r = a\theta$	Archimedes	$\theta \geq 0$; constant radial increment per angular unit.
0	$r = r_0 e^{b\theta}$	logarithmic	$\theta \in \mathbb{R}$; constant dilation per angular unit.
-1	$r = a/\theta$	hyperbolic	$\theta > 0$; the product $r\theta$ is constant.
-2	$r^2 = a^2/\theta$	lituus spiral	$\theta > 0$; product $r^2\theta$ is constant.

P.3.3 Conjugation $p \leftrightarrow -p$ of the power–logarithmic sheets

Para $p \neq 0$,

$$L_{-p}(x^{-1}) = \frac{x^p - 1}{-p} = -L_p(x).$$

The identity also holds for $p = 0$. Consequently, radial inversion interchanges p with $-p$ and reverses the sign of the linearized coordinate. This relation pairs the Fermat spiral with the lituus and the Archimedean spiral with the hyperbolic spiral as conjugate orientations of the reader, without identifying them as identical curves.

P.3.4 Classification of the affine sector by the pair $(p, [\Lambda, C])$

If the radial coordinate satisfies

$$u_{n+1} = \Lambda u_n + C,$$

the exact internal iteration is

$$u_n = \Lambda^n u_0 + \sum_{k=0}^{n-1} \Lambda^k C.$$

If $I - \Lambda$ is invertible, $u_* = (I - \Lambda)^{-1} C$ and $u_n - u_* = \Lambda^n (u_0 - u_*)$. Only in a positive scalar realization, $\chi\Lambda = \lambda\chi$ with $\lambda > 0$, does one define $u_* = C/(1 - \lambda)$. With $\theta_n = \theta_0 + n\omega$, $\omega \neq 0$, the interpolation published in that realization is

$$r(\theta) = r_0 \exp_p \left[u_* + \lambda^{(\theta - \theta_0)/\omega} (u_0 - u_*) \right].$$

The complete taxonomy is therefore not a list of five names. It contains at least the degree p, the orientation a, the affine class $[\Lambda, C]$, the carry cocycle, the boundary, the angular increment, and the memory ledger.

P.4 Naturality under truncation, nonadic equivariance, and Archimedean fidelity defect

P.4.1 Affine prefix reader and enriched record conservation

For a truncated history

$$\widehat{\gamma}_N = \widehat{e}_N \cdots \widehat{e}_1,$$

The enriched reader records

$$\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},N}(\widehat{\gamma}_N) = \left((H_0, \mathrm{Led}_0); \left(p(\widehat{e}_j), a(\widehat{e}_j), P_j, A_j, h_{e_j}, H_j, \mathrm{Led}_j \right)_{j=1}^{N} \right).$$

The term $H_j = h_{e_j} \cdots h_{e_1}$ preserves the prefixes. The ledger Led_j preserves, according to the route, sheet, residue, quotient, carry, boundary, orientation, cylinder, signature, survival, and memory. The real-valued coordinates are obtained subsequently:

$$\mathcal{P}^{\mathrm{esp}}_{\mathrm{arq}} = \mathrm{ev}_{\mathbb{R}^2} \circ \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}}.$$

P.4.2 Complete proof of naturality under truncation

Let $\pi_{N,M}$ denote deletion of the suffix of length $N - M$, with $M < N$. Let $\mathrm{pr}_{N,M}$ denote the projection that preserves the initial record and the first M edge records. Then

$$\mathrm{pr}_{N,M}\ \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},N}(\widehat{\gamma}_N) = \Big((H_0, \mathrm{Led}_0); \big(p(\widehat{e}_j), a(\widehat{e}_j), P_j, A_j, h_{e_j}, H_j, \mathrm{Led}_j\big)_{j=1}^{M} \Big).$$

By definition of truncation,

$$\pi_{N,M}(\widehat{\gamma}_N) = \widehat{e}_M \cdots \widehat{e}_1,$$

and upon reapplying the reader the same sequence is obtained. Therefore,

$$\mathrm{pr}_{N,M} \circ \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},N} = \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad},M} \circ \pi_{N,M}.$$

The proof does not use the value of the curve. It uses prefix identity, which is the information that the reduced seven-coordinate reader would have lost.

P.4.3 Compatibility criterion for general refinements of paths

For a refinement $r : \gamma \to \gamma'$, naturality requires a map $\widehat{r}$ on ledgers and the local equality

$$h_e = h_{e'_k} \cdots h_{e'_1}$$

for each edge e replaced by the refined chain $(e'_1, \ldots, e'_k)$. If, in addition, the carry, boundary, and memory of the refined chain compose the records of the original step, then

$$\widehat{r} \circ \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}} = \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}} \circ r.$$

This condition is a verifiable commutative square; it must not be assumed for an arbitrary refinement without exhibiting the factors.

P.4.4 Equivariance of the radial cocycle under the holonomy Γ_9

The action of Γ_9 returns the phase and prolongs the state. In the codomain of the reader, $\widehat{\Gamma}^{\mathrm{rad}}_9$ acts by shifting the records and updating the last prefix. The correct identity is

$$\mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}}(\Gamma_9 \gamma) = \widehat{\Gamma}^{\mathrm{rad}}_9 \mathcal{P}^{\mathrm{enr}}_{\mathrm{rad}}(\gamma).$$

Requiring invariance would erase precisely the memory that the connection preserves. In the reversible sector, route inversion satisfies

$$H_{C_{\mathrm{rev}}\gamma} = H_\gamma^{-1}$$

when all factors belong to the invertible affine subgroup. This typed assertion does not automatically extend to a global involution on an arbitrary subsequent quotient.

P.4.5 3 explicit obstructions to the faithfulness of Archimedean evaluation

Proposition P.5 (Intrinsic circle and projected circle)**.** *There exist two histories with the same visible curve.*

$$c(\theta) = (\cos\theta, \sin\theta)$$

and different ledgers.

Proof. The first history has constant axial memory $m = 0$. The second is the helix

$$h(\theta) = (\cos\theta, \sin\theta, \theta/2\pi),$$

whose ledger records $m(\theta) = \theta/2\pi$. Both satisfy $\pi_{xy} \circ h = c$. The projection is therefore non-injective. □

Proposition P.6 (Same endpoint, different affine order)**.** *If $a(1-\lambda) \neq 0$, the paths $A_a M_\lambda$ and $M_\lambda A_a$ are not equivalent under the enriched reader, even though they may be calibrated to publish the same endpoint.*

Proof. Its commutator is $A_{a(1-\lambda)} \neq I$. The commutator datum belongs to the prefix ledger and disappears when only the endpoint is retained. □

Proposition P.7 (Same curve, different genealogical region)**.** *Equality of the published real value does not identify the five regions of the microfiber of π; consequently, it is likewise insufficient to identify the spiral class when the reader preserves region, boundary, or carry.*

P.5 Stratified Taxonomy of Spiral Correspondences

P.5.1 Domain stratification: history, reader, Archimedean evaluation, and physical realization

Table P.4: Levels that must not collapse into a single notion of curvature.

Level	Invariant	Meaning
APP	$F(S,P) = 1,\ F(P,S) = -1$	Discrete curvature of the order of accumulation and folding.
TRIT	$\tau \in \{-1,0,1\}, J_\tau^2 = -\tau I$	Local elliptic, parabolic, or hyperbolic regime of the transport.
TPK	Γ_9, carry, boundary, route, and ledger	Phase holonomy and memory over the enriched state.
Radial level	$p, a, [\Lambda, C]$	Characters that determine the radial law before recognizing a curve.
Archimedean	$r(\theta)$	Curve published in $\mathbb{R}^2$; it may forget genealogy.
Frenet level	$\kappa_{\mathrm{F}}, \mathcal{T}_{\mathrm{F}}$	Differential curvature and torsion of the visible curve or suspension.
Dimensional level	fields, scales and representations	Posterior physical realization, with its own premises.

P.5.2 Classification by the increments (ω, β, μ)

Table P.5: Complete kinematic taxonomy of the minimal degenerate cases.

$\Delta\theta$	Δu_{rad}	Δm	Salida visible	Genealogical Information
0	0	0	point	published steady state
0	$\neq 0$	0	recta radial	scale change
0	0	$\neq 0$	fibra axial	non-transverse displacement memory
0	$\neq 0$	$\neq 0$	radial sheet suspended	scale and memory
$\neq 0$	0	0	circle	intrinsic angular closure
$\neq 0$	0	$\neq 0$	cylindrical helix	closed phase, high memory
$\neq 0$	$\neq 0$	0	espiral plana	radial character visible
$\neq 0$	$\neq 0$	$\neq 0$	radial helical suspension	turn, radius and depth preserved

P.5.3 Affine holonomy classification

The following table belongs to a real scalar realization of the linear part, $\chi\Lambda = \lambda\chi$; for ease of reading, we write Λ in the conditions column. It does not classify abstract endomorphisms by numerical inequalities.

Table P.6: Types of holonomy over the radial coordinate.

Condition	Orbit of u	Interpretation
$\Lambda = 1,\ C = 0$	$u_n = u_0$	constant radius; the memory may be zero or hidden by projection.
$\Lambda = 1,\ C \neq 0$	$u_n = u_0 + nC$	translational sector that publishes the family L_p.
$0 < \Lambda < 1$	$u_n \to u_*$	contraction towards a fixed section.
$\Lambda > 1$	$\lvert u_n - u_* \rvert$ grows	expansion from a fixed section.
$\Lambda < 0$	orientation alternation	requires control of the domain of $\exp_p$ at each step.
$\Lambda = 0$	$u_{n+1} = C$	noninvertible folding; it belongs to the monoid, not to the affine group.

P.5.4 Genealogical Equivalence Criterion

Two routes are not identified merely because they publish the same curve. A typed isomorphism preserving the following term by term is required:

$$\big((x_0, H_0, \mathrm{Led}_0); (\widehat{e}_j, p(\widehat{e}_j), a(\widehat{e}_j), P_j, A_j, h_{e_j}, H_j, \mathrm{Led}_j)_{j=1}^{N}\big)$$

and that commutes with the admissible truncations and refinements. The resulting classification is finer than the conventional taxonomy because it incorporates the history preceding the emergence of $\mathbb{R}^2$.

P.6 Electromagnetic and Material Realizations with Domain and Codomain Separation

P.6.1 Circular electromagnetic mode

Let

$$\zeta = \frac{z - c_{\mathrm{vac}} t}{\lambda},$$

and

$$\mathbf{E}_\sigma = E_0(\mathbf{e}_1 \cos 2\pi\zeta + \sigma \mathbf{e}_2 \sin 2\pi\zeta), \qquad \mathbf{B}_\sigma = c_{\mathrm{vac}}^{-1} \widehat{\mathbf{k}} \times \mathbf{E}_\sigma.$$

Then

$$\mathbf{E}_\sigma \perp \mathbf{B}_\sigma \perp \widehat{\mathbf{k}}, \quad |E| = c_{\mathrm{vac}}|B|, \quad \omega = c_{\mathrm{vac}} k.$$

At fixed time, the normalized endpoint of the field in the phase–position bundle describes

$$(\cos 2\pi\zeta, \sin 2\pi\zeta, \zeta).$$

This statement does not assert that every spatial field line is a helix: it describes the trajectory of the tip of the polarization vector together with the marked event.

P.6.2 Loxodromic compatibility

For

$$\Phi(r, \varphi, z, t) = \ell\varphi + \nu \log(r/r_0) + k_z z - \omega t,$$

the transformation

$$(r, \varphi) \longmapsto ((2 + \sqrt{3})r, \varphi + \pi/2)$$

preserves the phase modulo 2π exactly when

$$\nu \log(2 + \sqrt{3}) + \ell \frac{\pi}{2} \in 2\pi\mathbb{Z}.$$

Dilation is interpreted as a change of scale between solutions; not as spontaneous energy production.

P.6.3 Compatibility of the radial reader with the electronic fixed point $(5, 5)$ and atlas $81/56/13/324$

The electron occupies the unique fixed point $(5, 5)$ of the material atlas. This centrality is not derived anew in the present work. The prior operator compatibility includes

$$C^{(3)} m_{\mathrm{ph}} = -\frac{1}{2} m_{\mathrm{ph}}, \qquad 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4} = \frac{90}{120},$$

and

$$\|[P_{\Sigma,3}, P_{\Pi,3}]\| = \frac{\sqrt{3}}{4}.$$

The theory of spiral correspondences adds a route reader; it does not replace the General Mass Law or use the electron to select p, Λ, or C retrospectively.

P.6.4 Matrix of validity conditions by causal level

Table P.7: Falsifiers and type errors excluded by the construction.

Failure	Consequence
The classical curve selects p, Λ, C.	Circularity: the reader ceases to be constructive–generative.
Only the final holonomy is preserved.	Loss of prefixes, carry, and boundary; naturality is not established.

Failure	Consequence
Invariance under Γ_9 is required.	The memory advance is erased; the correct relation is equivariance.
U_6, w_6, and $\mathbb{F}_3^6$ are conflated.	Collapse of objects with cardinalities 468, 243, and 729.
The term phase return is identified with state return.	Contradiction with $B_{K+9} \neq B_K$ in general.
τ, p, and κ_{F} are identified.	Regime collapse, radial character and visible curvature.
A general helix is called any variable radius.	Differential statement unverified Lancret condition.
The φ axis is identified with the electron.	Mixture of the fiber of constants and the material realization.
Pell is interpreted as energy amplification.	Violation of scale reading and physical conservation.
Electromagnetic helicity is identified with spin $1/2$.	Confusion between distinct representations.
A point cell is imposed on every species.	Contradiction with the character of a fibre or multisection of the noncentral species.
The five constructions of the continuum are separated.	Breakdown of the previous unique genealogy prior to any geometric publication.

P.7 Mathematical Glossary

Table P.8: Symbols and Key Terms.

Symbol or term	Definition
Σ, Π	Additive and multiplicative evaluations on the common support APP.
τ	Local TRIT orientation regime; it is neither the radial degree nor Frenet torsion.
M_{ph}	Tritic operational-phase selector.
Γ_9	Nonadic holonomy of enriched dynamics.
H_N	Accumulated affine holonomy up to the prefix N.
Λ_N, C_N	Linear part and translational cocycle of H_N.
p	Integer cocycle of the radial character.
a	Local orientation character that separates genealogies of the same local degree.
$L_p, \exp_p$	Power-logarithmic coordinate and its positive inverse.
$\boxplus_p$	Deformed law that publishes the sum-product interaction.
Led_N	Enriched ledger of sheet, carry, boundary, route, and memory.
$\mathcal{P}_{\mathrm{rad}}^{\mathrm{enr}}$	Radial reader enriched prior to the curve.
$\mathcal{P}_{\mathrm{arq}}^{\mathrm{esp}}$	Subsequent evaluation on a curve in $\mathbb{R}^2$.
$\kappa_{\mathrm{F}}, \mathcal{T}_{\mathrm{F}}$	Curvature and torsion of Frenet of the visible curve.
Radial helical suspension	Variable-radius surface or sheet that preserves phase, depth, and path.
Unfaithful publication	Assessment that can identify distinct histories.

Symbol or term	Definition
Constructivo–generativo	Method that constructs the object and its genealogy before recognizing its conventional appearance.

Notation, types, and symbol disambiguation

Symbol	Object
$\mathcal{A}_9$	Nonadic arithmetic torus of paths with two internal evaluations.
TRIT	Oriented ternary coordinate of the transport; the visible output does not exhaust the enriched lift with carry.
TPK	Topological Phase Kernel with enriched state, transitions, routes, prolongations, and memory.
M_{ph}	Tritic operational-phase selector, distinct from cardinal transport and stroboscopic sampling.
τ_3	Primitive phase block $(+1, -1, 0)$.
$\mathcal{N}, \mathcal{E}, \mathcal{S}, \mathcal{O}$	Generators of transport cardinal.
Obs_3	Stroboscopic sampling of every third step.
$\mathcal{R}_{12}$	Internal TPK-oriented dodecaphasic temporal system.
$\Theta_{108} = \mathbb{Z}/108\mathbb{Z}$	Observable calendar 12×9; it is not the complete TPK state.
C_{12}	Rotation of the twelve temporal sectors.
Γ_{108}	Oriented transport orbit of 108 steps.
$\mathcal{G}_9$	Unitary pruning within the extension and boundary relation; it does not by itself constitute the monodromy.
$\mathfrak{M}_9$	Nonadic monodromy with memory: it returns the phase and changes the state.
π_{term}	Projection of the enriched terminal state onto the three selected visible blocks.
R_{sgn}	Reading of the signed integer coordinate of the same terminal enriched state.
U_{12}^{cat}	Visible concatenation $U_6 \| U_6$.
U_{12}^{sgn}	Signed coordinate of the enriched state, reversible with K by means of $\mathcal{H}_{12}$; it is not derived from U_{12}^{cat}, the calendrical projection, or the three terminal blocks in isolation.
$\mathcal{H}_{12}$	Reversible dodecaphase transformation between U_{12}^{sgn} and K.
$\mathcal{E}_{\mathrm{dod}}$	Integral extractor $(D_3, D_4, Q) : \mathbb{Z}^{12} \to \mathbb{Z}^{25}$ that reconstructs K.
$\overline{C}_9^{\,3}$	Nonadic cubic completion.
A, C^*	Intrinsic angular and antiangular coordinates.
$q_{\pm}$	Oriented channels of the double-layer reader.
$D_{\mathrm{raw}}^{90,120}$	Raw diagnostic difference of series in a fixed chart; it is not a canonical extractor.
$\iota_{6,8}$	Inclusion $\mathcal{F}_6 \hookrightarrow \mathcal{F}_8$ of hexad–octad incidences.
$K_{6\to 8}$	Angular functional of the two Lambert subtails, subsequent to $q_{\pm}$, the series, the pole, positivity, and minimality.
$\langle 90, 120 \rangle$	Spectral semigroup governing the hierarchy of towers.
$\varepsilon_{\mathrm{res}}^{\circ}$	Residue of the exact angular identity; it is not a finite extraction from the preceding 90/120 objects.
Δ_4	Minimal torsion or global torsional invariant prior to the species; it is not Cartan torsion.
$\mathsf{T}_{\alpha,\infty}$	Remote prolongation of the carry along the entire path of α, typed separately from the orbital generator $T_1^{\mathrm{orb}} = L_1$, the neutral counter, and the calendar.

Symbol	Object
N_1^{neu}	Number of neutral positions in the first window of the transport example; it equals 3 and is not an operator.
$T_1^{\mathrm{orb}} := L_1$	Second generator in the local orbital gauge; in its reference section it is abbreviated T_1.
$\varepsilon_{\mathrm{clk}}$	Energy quantum of the clock, $\hbar_{\mathrm{int}} 2\pi/(108 t_0)$; it is not a canonical energy basis unless the temporal frame is specified. The symbol ε_0 is reserved for the vacuum permittivity at the SI constitutive interface.
CLK1–CLK6	Labels of the six equations of the finite clock, from the Hamiltonian to the Schrödinger equation.
$\mathcal{K}_{27}$	Compound return of twenty-seven updates.
X_∞^{cal}	Common enriched domain of global evaluations.

Repeated cardinalities with different sources

Equality of cardinalities does not automatically transport domain, operation, memory, or evidentiary force. The following table sets out the collisions that most directly affect interpretation.

Index	Objeto tipado	Function and separation
108	Θ_{108}	The 12×9 calendar closes the observable clock.
108	Π_{108}^{cal}	Calendrical projection of a cursor with 324 seeds; it forgets the second cursor, orientation, carry, boundary, enriched chronological record, and survival.
108	w_{108}	Word of $108 = 18 \times 6$ trits; it is not the calendar.
$104,976$	$X_{\mathrm{TPK}}^{\mathrm{seed}}$	Two-cursor engine, $104,976 = 324^2$; it preserves a domain distinct from that of the calendar.
1080	w_{1080}	Length of a word.
1080	$\Gamma_0(1080)$	Level of a modular subgroup; requires its own realization.
729	$\mathbb{F}_3^6$	Ternary word environment.
729	$9^{\widehat{3}}$	Cubic cell/interior EMRH.
729	$\mathrm{Ext}(X_K)$	Number of subcylinders examined at each prolongation stage.
729	bulk EMRH	Inner layer of the base completion mil.
27	$\mathcal{K}_{27}$	Composite return operator; not the name of an autonomous theory.
27	cubic stratum	Contribution of two boundary coordinates in $999 = 729 + 243 + 27$.
27	module or classifier	Each occurrence requires its map before identifying itself with another.

Six distinct sources of 90/120

Object	Domain and operation	Exit and type prohibition
$\mathcal{E}_{\mathrm{dod}}$	$K \in \mathbb{Z}^{12}$; stepwise differences $3, 4$ and sum Q	Transverse dodecaphasic coordinates; no incidence nor angular series.
$D_{\mathrm{raw}}^{90,120}$	A q chart; raw difference $S_{90}(q) - S_{120}(q)$	Scale diagnostic, distinct from the Barbero–Immirzi functional and the incidence operators.
$\iota_{6,8}$	$H \subset O$; inclusion of $H \times \binom{H}{2}$ in $O \times \binom{H}{2}$	Cardinals 90, 120 and defect $-1/12$; neither an extractor nor a series.
$K_{6\to 8}$	Channels $q_{\pm}$ and Lambert subtails; combination $12\Delta S_{90} - \Delta S_{120}$	Angular functional relative to its premises; it is not transported as a mass block.
$\mathfrak{S} = \langle 90, 120 \rangle$	Moment heights $s_n = q_-^n - q_+^n$	Global tower semigroup; does not end at eight heights.

Four realizations of bidirectional reversibility

Operation	Domain	Action
$P_{\pm} = (I \pm R)/2$	Split Algebra	Project the two spectral sectors.
U-turn	Route and composite calendar	Invert the orientation of the transport without erasing its word.
$\mathrm{Ext}^{\pm}$	Extension Tree	It compares future and past valences; the balance may be read through $\log(\#\mathrm{Ext}^{+}/\#\mathrm{Ext}^{-})$.
$C_M(\tau) = -\mathrm{rev}(\tau)$	Dodecaphonic Word	It conjugates order and sign; it is not the cellular involution ρ_c.

The common function of these four realizations is to preserve reversible orientation. It does not authorize the identification of phase return, route return, word return, and state return.

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